IRCLASS
Indian Register of Shipping

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RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

July 2024

General Information

1. This consolidated version of the ‘Rules and Regulations for the Construction and Classification of Steel Ships’ (July 2024) supersedes the July 2023 edition of the Rules and includes the amendments published in the following:

a) Rules Change Notice No.1 dated September 2023

b) Rules Change Notice No.2 dated December 2023

c) Rules Change Notice No.3 dated February 2024

d) Rules Change Notice No.4 dated March 2024

2. For ease of reference by the users, a summary of additions and amendments incorporated in the various Rules Change Notices (issued since July 2023) along with their effective dates are indicated in the respective tables.

3. This edition of the Rules also consists of amendments carried out subsequent to the publication of Rules Change Notice No. 4 of March 2024. Such amendments are indicated in **Table 1**.

4. Corrigenda issued subsequent to the publication of Rules Change Notice No.4 of March 2024 are given in **Table 2**.

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF STEEL SHIPS – July 2023

RULES CHANGE NOTICE No. 1 – September 2023

TABLE – AMENDMENTS INCORPORATED IN THIS NOTICE

These amendments will come into force as indicated in the Table

Section / Clause	**Subject/ Amendments**
Part 3 Chapter 4: Design Loads *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
3/ 3.3.2	Amendments are made to specify that the parameter 'p_0' is not to be taken less than the design vapour pressure for cargo oil tanks and fuel tanks.
Part 3 Chapter 15: Anchoring and Mooring Equipment *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
3/ 3.2.5	Additional requirements for Offshore Support Vessels are deleted as they are shifted to Part 5, Chapter 8.
Part 3 Chapter 17: Welding *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
2/ 2.4.3	It is specified that alternative welding sequences may be accepted provided distortions are kept to a minimum in accordance with recognized fabrication standards.
Part 3 Chapter 18: Hull Inspection, Workmanship & Testing *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
3/ 3.1.2	Editorial changes are made for better clarity
3/ 3.3.3 (new)	The expression 'top of the overflow' is defined along with better clarity on overflow systems.
3/ 3.4.2.1	It is specified that alternative equivalent testing procedures may be considered for tanks manufactured from composite materials based on manufacturer's recommendations.
3/ 3.4.4.1.2	Test pressure restrictions for tanks designed for cargo densities greater than water are clarified.
Table 3.4.1	The requirement to test tanks based on the pressure relief valve setting has been replaced with a requirement to test tanks based on the design vapour pressure. Also, testing requirements for fuel oil overflow tanks not intended to hold fuel are now specified. Other editorial changes are made in the footnotes for better clarity.
Table 3.4.2	The requirement to test tanks based on the pressure relief valve setting has been replaced with a requirement to test tanks based on the design vapour pressure. Other editorial changes are made in the footnotes for better clarity.
4/ 4.1	Editorial changes are made for better clarity.
4/ 4.2.5 (new)	It is clarified that for tanks which are less than 2 m^3 in volume, structural testing may be replaced by leak testing.
4/ 4.2.6	Testing requirements for tanks and spaces in sister ships are clarified better.
Section 5 (new)	A new section is introduced that provides requirements for testing of tanks and tight boundaries for non-SOLAS ships.

Section / Clause	Subject/ Amendments
Part 4 Chapter 2: Piping Design Requirements *The amendments are applicable from 1 October 2023.*	
5/ 5.3.6 and Table 5.3.1	Amendments are made to delete certain requirements, as these are not relevant for plastic pipes.
Part 4 Chapter 3: Pumping and Piping *The amendments are applicable to ships constructed on or after 1 January 2024.*	
2/ 2.4.3, 2.4.4 (new) & 2.4.5 (new)	Arrangements to be provided (valve material, valve control etc.) if in case, the collision bulkhead is pierced by a pipe or by two pipes below the bulkhead deck/ freeboard deck of passenger/ cargo ships respectively, are specified.
MSC.1/ Circ. 864, Rev. 1 is already applicable from May 2019.	
4/ 4.3.11	Better clarity is provided on requirements for sampling points by aligning with MEPC.1/ Circ. 864, Rev.1.
Part 4 Chapter 4: Prime Movers and Propulsion Shafting Systems	
6/ 6.20.2.3	Version of the referred ISO standard is updated.
Part 5 Chapter 1: Dry Bulk Cargo Carriers *The amendments are applicable to ships constructed on or after 1 January 2024.*	
2/ IR 2.14.1.2	Relevant cross-references are updated as a result of changes made in Pt.4, Ch.3.
Part 5 Chapter 4: Liquefied Gas Carriers *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
2/ 2.7.1	Amendments are made to align with revised Regulation 27(13)(a) of the 1988 Load Lines Protocol accepting hinged watertight doors.
6/ 6.5.3.5	The testing requirements are extended to materials other than Aluminium alloys.
18/ 18.10.2.4 (new)	Requirements for fail-close action of the hydraulically or pneumatically actuated ESD valves are specified.
Part 5 Chapter 8: Offshore Support Vessels *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
2/ 2.7.1	Additional requirements for anchoring and mooring equipment in Offshore Support Vessels operating in the North Sea and not provided with dynamic positioning systems/ thrusters are specified.
Part 5 Chapter 35: Gaseous Fuel or other Low Flash Point Fueled Ships *The amendments are applicable to ships contracted for construction on or after 1 January 2024.*	
5/ 5.3.4	An error is corrected in the definition for f_v.
6/ 6.7.1.1	The requirement for pressure relief systems for tank cofferdams is deleted.
6/ 6.8.3 (new)	The option of allowing a higher loading limit based on tank insulation and tank location is given.
9/ 9.5.3 to 9.5.6 (all new)	Amendments are made to clarify that secondary enclosure and ventilation requirements (in Cl. 9.5.1) are only applicable to gaseous fuel pipes. It is also specified that the requirements in Cl. 9.5.4 need not be applied to fully welded fuel gas vent pipes led through mechanically vented spaces. A new Clause is added to clarify application of the secondary enclosure requirements for pipes with fuel in the liquefied state.
10/ 10.3.1.1.1 (new)	It is specified that the exhaust system is to be equipped with explosion relief systems unless designed to accommodate the worst-case overpressure due to ignited gas leaks or justified by the safety concept of the engine.

Section / Clause	Subject/ Amendments
11/ 11.3.3	Requirement to have a cofferdam or A 60 class division on the boundary between spaces containing the fuel containment system is deleted.
11/ 11.3.3.1 (new)	Qualifying requirements for a fuel storage hold space to be considered as a cofferdam are specified.
11/ 11.8 (new)	Fixed fire-fighting requirements for fuel preparation rooms containing pumps, compressors or other potential ignition sources are indicated.
16/ 16.3.3.5	The testing requirements are extended to materials other than Aluminium alloys.
Part 6 Chapter 2: Prevention of Fire and Explosion	
2/ 2.3.2.2	Amendments are made to align with SOLAS text.
Part 6 Chapter 8: Fire Safety Systems Code *The amendments are applicable to ships constructed on or after 1 January 2024.*	
9/ 9.2.1.8 (new)	Requirements for isolator modules in individually identifiable fire detection systems fitted in cargo and passenger ship balconies are specified.
15/ 15.2.2.3.2.1, 15.2.2.3.2.6, 15.2.2.4.2.1	The terminology is corrected (‘forward’ replaced by ‘downstream’).

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF STEEL SHIPS – July 2023

RULES CHANGE NOTICE No. 2 – December 2023

TABLE – AMENDMENTS INCORPORATED IN THIS NOTICE

These amendments will come into force as indicated in the Table

Section / Clause	**Subject/ Amendments**
Part 1 Chapter 1: General *The amendments are applicable from 1 January 2024.*	
Appendix 1	Additional Class Notation '**SHUTTLE TANKER**' is added to the list of notations.
Part 1 Chapter 2: Periodical Surveys *The amendments are applicable to surveys carried out on or after 1 July 2024.*	
2/ 2.2.25	Requirement for examination of double skin void spaces for bulk carriers exceeding 20 years of age and ≥ 150m in length, as a consequence of the results of special and intermediate surveys is introduced.
2/ 2.3.2.1, 2.4.5.1	The criteria for examination of ballast tanks are revised from 'POOR' to 'less than GOOD' during annual and special surveys.
2/ 2.4.5.4	Conditions when double-side skin void spaces bounding cargo holds for bulk carriers exceeding 20 years of age and of 150 m in length and upwards, need to be examined at annual intervals, are specified.
3/ 3.4.7.6	Conditions for acceptance (by the Surveyor) of cargo tank testing carried out by the ship's crew are revised.
The amendments are applicable to surveys carried out on or after 1 January 2024.	
2/ 2.2.4.15, 3/ 3.2.3.16, 4/ 4.2.3.16, 5/ 5.2.3.15, 6/ 6.2.4.15.	It is specified that all non-metallic expansion joints fitted in piping systems which penetrate the ship's side and both the penetration and the expansion joint being located below the deepest load waterline, are to be visually examined and their service records are to be checked.
Part 4 Chapter 3: Pumping and Piping *The amendments are applicable to ships constructed on or after 1 January 2024.*	
3/ Note to 3.6.3	The title of the referred Classification Note on water level detectors is revised.
3/ 3.7 (new)	New subsection providing requirements for water level detectors on multiple hold cargo ships other than bulk carriers and tankers, is introduced.
Part 5 Chapter 1: Dry Bulk Cargo Carriers *The amendments are applicable to ships constructed on or after 1 January 2024.*	
2/ Note to 2.13.1	The title of the referred Classification Note on water level detectors is revised.
Part 5 Chapter 4: Liquefied Gas Carriers *The amendments are applicable to ships constructed on or after 1 January 2024.*	
4/ IR 4.20.4	Verifications and examinations at gas trials and /or at first full cargo loading and unloading are revised.

Section / Clause	Subject/ Amendments
Part 5 Chapter 6: Passenger Ships *The amendments are applicable to ships constructed on or after 1 January 2024.*	
4/ 4.1.5 & 4.1.6 (both new)	New Clauses are introduced to align with SOLAS Ch. II-1, Reg. 17 and 17-1.
4/ 4.3.1	Amendments are made to align with SOLAS Ch. II-1, Reg. 17, to clearly specify requirements for ships constructed before 1 January 2024.
4/ 4.3.2 & 4.3.3 (both new)	Requirements for internal watertight subdivision arrangements and doors in such arrangements in new ships are specified.
4.4 (new)	New sub-section providing requirements for watertight integrity below the bulkhead deck, in alignment with SOLAS Ch. II-1, Reg. 13 is introduced.
Part 5 Chapter 24: Dynamic Positioning *The amendments are applicable from 1 January 2024.*	
1/ 1.2.7	The definition for 'dynamically positioned vessel' is amended for better clarity.
Part 5 Chapter 26: Diving Support Vessels and Diving Systems *The amendments are applicable to ships constructed on or after 1 January 2024.*	
Complete Revision	The Chapter is revised in order to align with the requirements in the International Code of Safety for Diving Operations, 2023.
Part 6 Chapter 1: General	
1/ Table 1	The referenced IMO Circulars (i.e. MSC.1/Circulars 1276, 1395 & 1430) are updated with their latest versions.
Part 6 Chapter 3: Suppression of Fire *MSC.1/ Circ 1276, Rev.1 and MSC.1/Circ. 1395, Rev.6 are already applicable from June 2023.*	
4/ 4.5.6.2	Reference to the latest version of MSC.1/ Circ. 1276 is updated.
4/ 4.7.1.4 & IR 4.7.2.1	Reference to the latest version of MSC.1/ Circ. 1395 is updated.
Part 6 Chapter 7: Special Requirements *The amendments are applicable to ships constructed on or after 1 January 2024.*	
3/ 3.6.1.3	Reference to the latest version of MSC.1/ Circ. 1430 is updated.
Part 6 Chapter 8: Fire Safety Systems Code *The amendments are applicable to ships constructed on or after 1 January 2024.*	
7/ 7.2.4	Reference to the latest version of MSC.1/ Circ. 1430 is updated.

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF STEEL SHIPS – July 2023

RULES CHANGE NOTICE No. 3 – February 2024

TABLE – AMENDMENTS INCORPORATED IN THIS NOTICE

Chapter No.	Title of Existing Chapter	Title of Revised Chapters
1	General, Definitions, Documentation	General, Definitions, Documentation
2	Materials of Construction	Materials of Construction and Corrosion Additions
3	Principles for Scantlings and Structural Details	General Arrangement and Structural Design Principles (1)
4	Design Loads	Stability (3)
5	Longitudinal Strength	Design Loads
6	Bar Keel, Stem and Stern frames (1)	Hull Girder Strength
7	Bottom Structure (1)	Hull Local Scantlings (1)
8	Side Structure (1)	Finite Element Analysis (3)
9	Deck Structure (1)	Buckling (3)
10	Bulkheads (1)	Fatigue (3)
11	Superstructures, Deckhouses and Bulwarks (1)	Special Hull Requirements (1)
12	Openings and Closing Appliances (2)	Openings and Closing Appliances (2)
13	Ventilators, Air Pipes and Discharges (2)	Rudders (4)
14	Rudders	Anchoring and Mooring Equipment (4)
15	Anchoring and Mooring Equipment	Masts and Rigging (4)
16	Masts and Rigging	Welding (4)
17	Welding	Hull Inspection, Workmanship and Testing (4)
18	Hull Inspection, Workmanship and Testing	

Note:
(1) Some contents of existing Chapters 6 to 11 moved to new Chapters 3, 7 and 11.
(2) Some contents of existing Chapters 12 and 13 moved to new Chapter 12.
(3) Entirely New Chapter.
(4) Unaltered Chapter: The Chapter number has been updated, but its contents remain unchanged. It is omitted from this Notice since there are no modifications to the Chapter compared to the previous Rules Change Notice/ Consolidated Edition.

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF STEEL SHIPS – July 2023

RULES CHANGE NOTICE No. 4 – March 2024

TABLE – AMENDMENTS INCORPORATED IN THIS NOTICE

These amendments will come into force as indicated in the Table

Section / Clause	**Subject/ Amendments**
Part 1 Chapter 1: General *The amendments are applicable from 1 July 2024.*	
2/ 2.1.4	The Industrial Personnel Code (IP Code, effective from 1st July 2024) and the Polar Code, are added to the list of Conventions/ Codes that are a prerequisite for Classification.
2/ 2.18.11 & 2.18.15	Rule provisions related to laying up of vessels after being suspended are deleted.
Appendix 1	Additional Class Notations '**CyS-E**', '**FC (Main)**' and '**FC (Auxiliary)**' are added to the list of notations.
Part 1 Chapter 2: Periodical Surveys *The amendments are applicable to surveys on vessels contracted for construction on or after 1 July 2024.*	
25 (new)	A new section pertaining to '**Cyber Resilience Surveys**' is added.
Part 2 Chapter 2: Mechanical Testing Procedures *The amendments are applicable to application for certification of manufacturer approval for EH47 and brittle crack arrest steels dated on or after 1 July 2024.*	
Section 5	This Section is completely revised, wherein detailed requirements have been deleted and instead references are made to ISO 20064:2019 *"Metallic materials –Steel- Method of test for the determination of brittle crack arrest toughness, Kca".*
Section 6	Existing references to Section 5 are replaced with references to ISO 20064:2019.
Part 2 Chapter 3: Rolled Plates, Steels and Bars *The amendments are applicable to applications for certification of manufacturer approval for EH47 and brittle crack arrest steels dated on or after 1 July 2024.*	
3/ 3.15.2.1.2 (new) & 10/ 10.2.3.1.2 (new)	A new Clause is added specifying that the products tested must be at their maximum approved thickness. If the chemical composition varies with thickness, testing is to cover each composition at its maximum thickness.
3/ 3.15.2.2 (b)	Reference to industry standard for hydrogen crack test is revised to ISO 17642:2005.
10/ 10.2.3.1.1	It is clarified that for EH47 steels with brittle crack arrest properties, certain requirements of the IRS Classification Note: '*Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'* are not applied.
10/ 10.2.3.3.4	Subject Clause is amended to clarify that if the initial Kca brittle crack arrest test fails, further tests can be done.
Annex 1 (new)	The new Annex outlines the approval process for small-scale test methods used in batch release testing of brittle crack arrest steels listed in Table 10.1.3.

Section / Clause	Subject/ Amendments
Part 3 Chapter 13: Rudders *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
2/ 2.1.4	It is clarified that a seal or stuffing box is to be fitted above the waterline at scantling draft (without trim). When the top of the rudder is below the water line at scantling draught, two separate seals or stuffing boxes are to be provided.
3/ 3.1.1	It is clarified that the maximum astern speed as defined in SOLAS Regulation II-1/3.15 is to be used for astern condition.
4/ 4.1.2	A Figure showing the details of use of steel backing bar in way of full penetration welding of rudder side plating is introduced.
4/ 4.2.1	The formula for rudder plating thickness 't' is amended to cater for the scantling draught.
5/ 5.1.4	It is clarified that for a spade rudder with trunk extending inside the rudder, the rudder stock scantlings are to be checked against the two cases defined in Section 8.3.
5/ 5.2.3	The formulae for required push-up pressures for pintles in case of dry fitting or oil injection fitting are indicated. The method to calculate the required push up length is also specified.
5/ 5.3.1.1	It is clarified that for rudder stocks and pintles having diameter less than 200 [mm], liners in way of bushes may be provided optionally.
6/ 6.4.2	Editorial correction is made to the formula for push-up pressure. It is clarified that for a spade rudder with trunk extending inside the rudder, the rudder coupling is to be checked against the two cases defined in Section 8.3.
7/ 7.3.1	Better clarity is provided on the formula for the fillet shoulder radius of the rudder trunk.
8/ 8.2.2	It is clarified that the maximum moment, M_C, in top of the cone coupling as shown in Fig. 8.2.2 is applicable for the connection between the rudder and the rudder stock.
8/ Fig. 8.3.2 (a) & (b)	Two Figures are introduced that depict the strength checks for a spade rudder with trunk extending inside the rudder.
Part 4 Chapter 3: Pumping and Piping *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
4/ Fig. 4.2.4 (A) & (B)	In the context of equivalent arrangements, it is clarified that any fuel oil which requires post service tank heating to achieve the required injection viscosity is not regarded as MDO.
Part 4 Chapter 4: Prime Movers and Propulsion Shafting Systems *The amendments are applicable to EGCS:* *i) when an application for installation, i.e. submission of plans, is made on or after 1 July 2024; or* *ii) which is installed in ships contracted for construction on or after 1 July 2024.*	
4/ 4.15.2.5, 4.15.2.6, 4.15.2.10 & 4.15.2.15	Editorial changes are made for better clarity.
4/ 4.15.3 (new)	Construction requirements for discharge water pipeline of EGCS are specified.
Part 4 Chapter 7: Control Engineering *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
Section 6	This Section is completely revised to better clarify requirements for computer-based systems, including those for system integrator.
Section 7 (old)	Requirements for integrated systems are deleted as they are covered in Section 6.
Section 7 (new)	A new Section specifying the minimum set of requirements for cyber resilience of ships is introduced.

Section / Clause	Subject/ Amendments
Section 8 (new)	A new Section specifying the minimum set of requirements for cyber resilience of on-board systems and equipment is introduced.
Part 5 Chapter 3: Chemical Carriers *The amendments are applicable to ships constructed on or after 1 July 2024.*	
2/ 2.9.2	Amendments are made to align with revised Regulation 27(13)(a) of the 1988 Load Lines Protocol accepting hinged watertight doors.
Part 5 Chapter 8: Offshore Support Vessels *The amendments are applicable from 1 July 2024.*	
5/ 5.1.6	Reference to SOLAS Chapter XV (Safety Measures for Ships carrying Industrial Personnel) is made. Further, the term 'Industrial Personnel' is defined for better clarity.
Part 5 Chapter 32: Polar Class *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
Section 3	Comprehensive amendments are undertaken in this Section, including introduction of requirements for icebreaker vessels.
Part 5 Chapter 35: Gaseous Fuel or other Low Flash Point Fueled Ships *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
7/ 7.3.6.3, 9/ 9.2.1 & 9.6.1	Requirements for single common flange in fuel transfer piping are better clarified.
Part 6 Chapter 3: Suppression of Fire *The amendments are applicable to ships constructed on or after 1 July 2024.*	
3/ IR3.7.3.1.2	The requirement of 'fire insulation' in the referred Clause is better clarified.
3/ IR3.7.3.2	Requirements for ventilation ducts passing through 'B' class bulkheads are better clarified.

TABLE – CORRIGENDA INCORPORATED IN THIS NOTICE

Section / Clause	Subject/ Corrigenda
Part 5 Chapter 3: Chemical Carriers	
17/17.1.1	Requirements for "*Bio-fuel blends of Diesel/gas oil and FAME (>25% but <99% by volume)*", which were inadvertently missed out are now included in the table in accordance with the IBC Code.

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF STEEL SHIPS – July 2024

TABLE 1 – AMENDMENTS INCORPORATED IN THIS EDITION

These amendments will come into force as indicated in the Table

Section / Clause	**Subject/ Amendments**
Part 1 Chapter 1: General *The amendments are applicable from 1 July 2024.*	
2/ 2.1.1	It is specified that verification of cyber resilience, where applicable, is also part of Ship Classification. Further, this will also include verification of cyber resilience, where applicable, of equipment and systems required by statutory regulations.
Appendix 1	Additional Class Notation '**SUPPLY-PLUS**' is added to the list of notations.
Part 1 Chapter 2: Periodical Surveys *The amendments are applicable to surveys carried out on or after 1 January 2025.*	
8/ 8.1.7	A cross reference to the new requirements in Pt.4 Ch.1, Sec. 3.8, 'Testing of the control system of controllable pitch propellers', is added.
Part 3 Chapter 12: Opening and Closing Appliances *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
2/ 2.4.1.5 (new)	It is clarified that the requirements of this sub-section do not apply for small hatches on container ships giving access to a cargo hold, if these comply with certain conditions for non-weathertight hatch covers as outlined in Part 5, Chapter 5, Cl. 3.2 of the Rules.
Part 3 Chapter 14: Anchoring and Mooring Equipment *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
1/ 1.2.7, 1.2.8 (both new)	Scope of the requirements is further amplified.
3/ 3.1.3	Permissible requirements for tugs less than 45m in length are specified.
3/ 3.1.4 (new)	It is clarified that requirements in sub-section 3.3 are not applicable to dredgers with an unusual design of the underwater part of the hull.
3/ 3.2.6 (new)	Requirements for fishing vessels are clarified.
3/ 3.2.7 (new)	Requirements for vessels other than fishing vessels are clarified.
3/ 3.3 (new)	This new sub-section provides requirements (for ships of length < 90m) for direct force calculation for anchoring equipment, as an alternative to the prescriptive approach.
6/ 6.2.4 & 6.2.6 (both deleted)	Requirements are deleted based on the introduction of Line Design Breaking force (LDBF) requirements in Cl. 6.3.1.
6/ 6.3.1	Requirements for Line Design Breaking force (LDBF) are clarified.
Part 4 Chapter 1: General Requirements for the Design and Construction of Machinery *The amendments are applicable to ships contracted for construction on or after 1 January 2025 except otherwise specified.*	
1/ 1.7.1	Ship accelerations and motions are included in the list of environmental conditions.

Section / Clause	Subject/ Amendments
1/ 1.7.5 (new)	The new sub-section details ship accelerations, motions and provides requirements to ensure that essential machinery operates effectively under the effects of ship accelerations and motions.
1/ 1.10	Cross reference is provided to the new subsection 3.2.8 'Testing of the control system of controllable pitch propellers'.
3/ 3.2.8 (new)	Requirements for testing of the pitch control system of controllable pitch propellers are specified. *These requirements are also applicable to vessels contracted for construction before 1 January 2025 and where testing is carried out according to Pt. 1 Ch.2, Cl. 8.1.7, on or after 1 January 2025.*
Part 4 Chapter 4: Prime Movers and Propulsion Shafting Systems	
The amendments are applicable to ships contracted for construction on or after 1 January 2025.	
4/ 4.9.2.1	It is clarified that the capacity of one independent compressor or the combined capacity of independent compressors is not to be less than 50% of the total required capacity.
4/ 4.13.1	Scope of application is expanded to marine engines using high pressure natural gas, including specifics for Dual Fuel and Gas Fuel engines, and integration with requirements of Pt. 5, Ch. 4 and Pt. 5, Ch. 35.
4/ 4.13.2	Definitions related to gas engines added/ amended such as 'Certified safe type equipment', 'Explosion Relief Device', 'High Pressure Gas', etc.
4/ 4.13.4.1	New text added which specifies that the strength of components related to explosion relief devices must be documented or demonstrated to effectively handle the maximum expected explosion scenario.
4/ 4.13.4.2	General requirements for gas piping are further clarified including aspects such as classification of various gas piping (Class I, Class II, etc.), flexible bellows, design pressures for gas pipes, etc.
4/ 4.13.4.4	Amendments made indicating that design requirements of "charge air system" in 4.3.4.1.2 are also applicable to "exhaust gas system".
4/ 4.13.4.5	Requirements for explosion relief devices are indicated.
4/ 4.13.4.6	Requirements for crankcase ventilation, where arranged, are added.
4/ 4.13.4.9	Amendments are made to differentiate requirements for electrically operated and hydraulically operated valves.
4/ Table 4.13.4.8	'Failure of crankcase ventilation system, if applicable' is added to the list of parameters to be monitored. Few notes are added in the Table for better clarity.
4/ 4.13.5.4	Requirements addressing two stroke engines (for scavenge air system and crankcase) are added.
4/ 4.13.6.2	It is clarified that 'pilot fuel temperature and pressure' is to be recorded during factory acceptance tests.
4/ 4.13.6.3	Requirements for shipboard trials are further amplified.
4/ 4.13.7 (new)	The certification process for engine components is clarified.
The amendments are applicable to ships contracted for construction on or after 1 July 2025.	
4/ 4.17 (new)	Requirements for the capacity and availability of compressed air required by systems, machinery and equipment providing essential services, other than for engine starting are specified.
Part 5 Chapter 1: Dry Bulk Cargo Carriers *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
1/ 1.1	Editorial amendments are made for better clarity.
1/ 1.3 (new)	Design loads in cargo holds are specified.
1/ 1.4 (new)	The requirements in this sub-section are shifted from erstwhile 2.4
1/ 1.5 (new)	Requirements for transverse vertically corrugated watertight bulkheads in flooded conditions are specified.

Section / Clause	Subject/ Amendments
1/ 1.6 (new)	Design load scenarios and design load sets are specified.
1/ 1.12 (new)	Requirements for protection against loading/ discharge equipment are specified.
2/ 2.2 & 2.3	Editorial changes are made to provide reference to relevant requirements in Part 3.
2/ 2.4 (new)	Detailed requirements for double side structure are specified.
2/ 2.5, 2.6, 2.7, 2.8 & 2.9	Amendments are made to provide better clarity in structural requirements for single side structure, hopper side tank structure, topside tank structure, deck structure and corrugated bulkheads respectively.
3	Cross-references to new part 3 are updated and other editorial amendments are made for better clarity.
4	Section 4 is completely revised to provide reference to relevant requirements in new Part 3 of the Rules.
Section 5 (deleted)	This section is deleted, as its contents are covered in Chapter 12, Sections 5 to 10 of the new Part 3 of the Rules.
Part 5 Chapter 2: Oil Tankers *The amendments are applicable to ships constructed on or after 1 July 2024.*	
Sections 4 & 5	The sections are completely revised to provide cross-references to relevant requirements in Part 3 as well as provide design load sets for PSMs and finite element analysis criteria, as relevant and applicable.
Part 5 Chapter 4: Liquefied Gas Carriers *The amendments are applicable to ships contracted for construction on or after 1 January 2025.*	
4/ IR4.3.3.1 & IR4.3.3.2	Safety margins for the design of cargo containment systems are explicitly specified.
Part 5 Chapter 5: Container Ships *The amendments are applicable to ships contracted for construction on or after 1 July 2024*	
2 & 5	Cross-references to new part 3 are updated along with some editorial changes.
Part 5 Chapter 6: Passenger Ships *The amendments are applicable to ships contracted for construction on or after 1 July 2024*	
1, 3 & 4	Cross-references to new part 3 are updated along with some editorial changes.
Part 5 Chapter 8: Offshore Support Vessels *The amendments are applicable to ships contracted for construction on or after 1 January 2025.*	
1/ Table 1.2.2.	New notation **SUPPLY-PLUS** for vessels intended to operate in the North Sea is added. Further, some cross-references are updated.
Section 2	This section is completely revised to provide additional requirements for Offshore Support Vessels operating in the North Sea. Further, amendments are also made to provide cross-references to relevant requirements in Part 3 of the Rules.
Part 5 Chapter 10: Dredgers *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
2	The thickness requirements for boundaries of hopper spaces and wells is deleted and instead cross-reference is provided to the relevant requirements of Part 3 of the Rules. Other cross-references to new part 3 are updated.
4	Cross-references to new part 3 are updated.
Part 5 Chapter 11: Barges and Pontoons *The amendments are applicable to ships contracted for construction on or after 1 July 2024*	

Section / Clause	Subject/ Amendments
2, 3 & 4	Cross-references to new part 3 are updated along with some editorial changes.
Part 5 Chapter 12: Floating Dry Docks *The amendments are applicable to ships contracted for construction on or after 1 July 2024.*	
2	Cross-references to new part 3 are updated.
Part 5 Chapter 35: Gaseous Fuel or other Low Flash Point Fueled Ships *The amendments are applicable to ships contracted for construction on or after 1 January 2025.*	
Introduction/ IR3.3	Amendments are made to provide reference to the IRS *Guidelines on LPG Fuelled Vessels.*
6/ IR6.4.1.5 (new)	Safety margins for the design of fuel containment systems are explicitly specified.

TABLE 2 – CORRIGENDA INCORPORATED IN THIS EDITION

Section / Clause	Subject/ Corrigenda
Part 3 Chapter 2: Materials of Construction and Corrosion Additions	
2/Table 2.2.1(e)	In Notes 1 & 2, nomenclature for span of the side frame is corrected from *h* to *l.*.
4/4.3.1	The term "*Hatches*" is deleted, as there is a separate chapter covering requirements for hatches.
4/ Note to 4.3.1	Editorial changes are made for better clarity.
5/Table 5.1.1	Unit [mm] is specified for Corrosion Addition (t_{c1} or t_{c2}) for clarity.
Part 3 Chapter 5: Design Loads	
2/Table 2.4.2.3	Typo error is corrected in LCF associated with Q_{WV} for load component OST-2P and OST-2S.
2/Table 5.1.2	Footnote 3 added in alignment with the existing footnote 8 of Table 3.4.2 of Part 3, Ch.17 "Hull Inspection, Workmanship and Testing".
Part 3 Chapter 12: Opening and Closing Appliances	
1/1.1.4 & 5/5.1.3	Subject clauses are deleted as the requirements are not covered here.
6/6.2.2.2 & 9/9.11.3.4	Cross reference associated with P_{Coam} corrected.
Part 5 Chapter 1: Dry Bulk Cargo Carriers	
1/1.3.3.1 & 1.4.3.4	Cross references corrected.

IRCLASS
Indian Register of Shipping

50 YEARS OF EXCELLENCE

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 1
REGULATIONS

July 2024

Indian Register of Shipping

Part 1

Regulations

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Chapter 1

General

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Section 1

General Information

1.1 Indian Register of Shipping

1.1.1 Indian Register of Shipping (hereinafter referred to as "IRS") was incorporated in 1975 as a Public Limited Company under the Indian Companies Act for the purpose of providing amongst other things a faithful and accurate classification of mercantile shipping classed with it, to approve designs of, to survey and to issue reports on mercantile and non mercantile ships, hovercrafts, hydrofoils etc; all within the scope of classification described in the Rules. This Section contains General Regulations which have been adopted by IRS for its governance. The Classification Regulations are given in Section 2.

1.1.2 The management of the affairs of IRS are carried out by the Managing Director (MD) and Jt. Managing Director (JMD), under the direction and control of the Board of Directors (hereinafter referred to as the 'Board'), in accordance with the provisions of its Memorandum and Articles of Association.

1.1.3 The Board of Directors shall consist of representative of the interests of various members of the Company and those concerned with shipping in general as under:

3	Directors representing Indian Shipowners
2	Directors representing Indian Shipbuilders
1	Director representing General Insurance Corporation of India and other Indian underwriters
1	Director being the Director General of Shipping, Ministry of Surface Transport, Govt. of India
1	Director representing Ship Design Research and Development Institutions
1	Director representing Manufacturers of Marine Engines/General Engineering Goods
1	Director representing Indian Navy/Coast Guard
1	Director being a person of eminence from the field of Law
3	Directors being persons of eminence from any industry allied with maritime activities
1	Managing Director being full-time employee appointed by the Board of Directors
1	Jt. Managing Director, where so appointed, being full-time employee, appointed by the Board of Directors.

The composition of the Board as above is to be in accordance with the Articles of Association of IRS (as may be amended from time to time).

1.1.4 The Board shall consist of not less than six and not more than fifteen Directors. If the actual representation on the Board of Directors exceeds 15 Directors, the provisions of the Indian Companies Act would apply to approve the increase in the sanctioned strength of the Board beyond the limit specified in the Articles of Association of IRS. The Board of Directors shall elect one of its members to be Chairman of the Board of Directors. The reference to Chairman shall include Executive Chairman, where so appointed by the Board of Directors.

1.1.5 The Board is to appoint a Sub-Committee of Classification representing concerned interests.

1.1.6 The Board is to appoint the Chairman of the Sub-Committee of Classification and the Jt. Managing Director, IRS to be 'ex- officio' member of the Sub-Committee of Classification.

1.1.7 The employees of IRS are to be appointed by and be under the direction of the MD, except for Key Managerial Personnel (KMP), who would be appointed by the Board on the recommendation of the Nomination & Remuneration Committee.

1.1.8 The Surveyors of IRS are not to be permitted without the special sanction of the Board of Directors to receive any fee, gratuity or reward whatsoever, for their own use or benefit, for any service performed by them in their capacity as Surveyors to IRS, except on pain of immediate dismissal.

1.1.9 The Funds and Accounts are to be under the authority and control of the Board of Directors.

1.2 Fees

1.2.1 Fees will be charged for all surveys and for other services rendered by IRS or any of its publications in accordance with established scales. Traveling expenses incurred by the Surveyors in connection with such services are also chargeable.

1.3 Technical committee

1.3.1 The Board is to appoint a Technical Committee whose function will be to consider:-

a) Formulation of Technical Rules for Classification Surveys, building of ships, their machinery and equipment.

b) Important alterations to Rules once framed as may be required from time to time.

1.3.2 All decisions of the Technical Committee including amendments and/or additions to the Rules for classification surveys and building of ships' hull, their machinery and equipment to be reported to the Board of Directors.

1.3.3 The Technical Committee to be constituted as follows:

Number of Members of	**Nominees/Representatives**
3	Board of Directors of IRS
1	Marine Engine Unit of M/s. Garden Reach Ship-builders and Engineers Ltd.
1	Other Marine Engine Builders
6	Shipbuilders
2	Indian Institution of Naval Architects
2	Institute of Marine Engineers (India)
1	Company of Master Mariners
2	Directorate General of Shipping
1	IMU (Earlier NSDRC)
4	Indian National Shipowners Association
1	Institution of Engineers (India)
1	Ex-Officio – Executive Chairman of IRS or his nominee
1	Indian Coastal Conference Shipping Association
1	Oil Industry Safety Directorate
5	Indian Navy

2	Indian Coast Guard
1	Research Institutes
2	Indian Institute of Technology/ National Institute of Technology
2	Maritime Training Institutes
3	Other Flag Administrations
1	Inland Waterways Authority of India.

1.3.4 In addition to the foregoing, the Technical Committee may co-opt to the main body other members of high managerial positions in Shipping, Ship Building, Marine Engineering, Naval Architecture, Marine Insurance, Steel Making, etc.

1.3.5 Nomination of all members to the Technical Committee to be subject to confirmation by the Board.

1.3.6 The Technical Committee can appoint panels from amongst its body to which representatives of any organisation or industry or individuals specialised in relevant disciplines could be co-opted for the purpose of considering any particular Technical problem or area of Rules.

1.3.7 The Board of Directors to appoint biennially, the Chairman of the Technical Committee and the Technical Committee to appoint from their own body biennially a Vice-Chairman. The appointment of Vice-Chairman is to be confirmed by the Board of Directors.

1.3.8 The terms of office of all members is to be not more than four years, one-fourth of all members (including those co-opted) to retire at the end of each calender year. The members so retiring being those who have been longest in office since their last nomination and such members to be eligible for re-nomination for a second term. Unless specially so authorised by the Board of Directors, no member other than Chairman and/or Vice- Chairman, who has served for two periods of nomination, to be eligible for re-nomination. In the event of any vacancy occurring before the expiration of the normal term of office, a representative to be nominated to fill the vacancy from the same group/body/institution and for such nominee the date of his nomination by the respective body is to be considered as date of his joining the Technical Committee for purposes of his retirement by rotation.

1.3.9 The meeting of the Technical Committee is to be convened as often and at such time and place as may appear necessary, but there shall be at least two meetings in each year.

1.3.10 The members desiring to propose alterations in, or additions to the Rules for the classification, survey or building of ship (hull and machinery) shall give notice of such proposals to the Secretary. Every meeting to be convened by notice from the Secretary, if possible one month before the date of the meeting and the Secretary to send to each member an Agenda paper as soon as possible thereafter.
Proposals for changes to rules may also be given by Flag Administrations, shipowners, shipbuilders and other interested parties who may not be represented in the Technical Committee.

1.3.11 The quorum for any meeting of Technical committee will be six members, with at least 50% of the members present being those who do not represent shipowners, shipbuilders or others engaged commercially in the manufacture, equipping, repair or operation of ships.

1.3.12 In the event that any matter is not decided by unanimity, the same may be decided by a majority of votes cast in favor, with each member, including co-opted members, having one vote only. In the event of a parity of votes, the Chairman of the Technical Committee would be entitled to an additional casting vote.

1.3.13 When any discussion relates to an item of interest to those representing shipowners, shipbuilders or others engaged commercially in the manufacture, equipping, repair or operation of ships, such representatives would not be entitled to vote, if such matter is to be decided by voting.

1.3.14 In the event that any member of the Technical Committee absents himself for 3 consecutive meetings of the Technical Committee without seeking leave of absence, he would be deemed to have vacated office and his vacancy would be filled by seeking fresh nomination from concerned interest represented.

1.3.15 In the absence of the Chairman and the Vice Chairman of the Technical Committee, the members of the Technical committee shall elect a Chairman, by majority vote, to preside over that particular meeting only.

1.3.16 The Board of Directors reserves to themselves the right of altering, adding to or rescinding any/or all of the above terms of reference including the dissolution of the Technical Committee.

1.4 Survey reports

1.4.1 All reports of survey are to be made by the Surveyors according to the form prescribed and submitted for consideration of the Board or the Sub-Committee of Classification, but the character and notations assigned by the latter is to be reported to the Board. The Board may, in specified instances, vest in the Managing Director / Jt. Managing Director discretionary powers to act on its behalf, and all such actions being reported to the Board at its subsequent meeting.

1.4.2 The reports of the Surveyors shall, subject to the approval of the Managing Director / Jt. Managing Director, be open to inspection of the Owner and any other person authorised in writing by the Owner. Copies of the reports will, subject to the approval of the Managing Director / Jt. Managing Director, be supplied to Owners or their representatives.

1.5 Register of Ships

1.5.1 A Register of Ships is available on-line on IRCLASS Website which contains the names of ships, character of class and notations assigned together with other relevant useful information for ships classed with IRS.

1.6 Liability

1.6.1 Whilst Indian Register of Shipping, a Classification Society, along with its subsidiaries and associates (hereinafter referred to as the Society) and its Board/Committees use their best endeavours to ensure that the functions of the Society are properly carried out, in providing services, information or advice, neither the Society nor any of its servants or agents warrants the accuracy of any information or advice supplied. Except as set out herein, neither the Society nor any of its servants or agents (on behalf of each of whom the Society has agreed this clause) shall be liable for any loss damage or expense whatever sustained by any person due to any act or omission or error of whatsoever nature and howsoever caused of the Society, its servants or agents or due to any inaccuracy of whatsoever nature and howsoever caused in any information or advice given in any way whatsoever by or on behalf of the Society, even if held to amount to a breach of warranty. Nevertheless, if any person uses services of the Society, or relies on any information or advice given by or on behalf of the Society and suffers loss, damage or expenses thereby which is proved to have been due to any negligent act omission or error of the Society its servants or agents or any negligent inaccuracy in information or advice given by or on behalf of the Society then the Society will pay compensation to such person for his proved loss up to but not exceeding the amount of the fee charged by the Society for that particular service, information or advice.

1.6.2 Any notice of claim for loss, damage or expense as referred to in 1.6.1 shall be made in writing to Head Office within six months of the date when the service, information or advice was first provided, failing which all the rights to any such claim shall be forfeited and the Society shall be relieved and discharged from all liabilities.

1.7 Audits and assessments by external organizations

1.7.1 The surveys required by the regulations, and conducted by IRS may be subject to Audit by an independent Accredited Certification Body (ACB) as per the requirements of ISO 9001:2015 standard and Quality Management System Certification Scheme (QSCS) of IACS. For this purpose, ACB auditors are to be given the necessary access to the ship, shipyard or works when requested by IRS.

Access is also to be given to auditors or inspectors accompanying the Surveyors as required by other external organizations.

1.8 Access of Surveyor to ships, shipyards or works

1.8.1 The Surveyors are to be given free access to ships classed with IRS as well as to shipyards, works, etc. so as to perform their duties, and are to receive adequate assistance for this purpose.

1.9 Requirements for service suppliers

1.9.1 In general, the following categories of service suppliers are to be approved:

a) Classification and / or Statutory services

- Firms engaged in thickness measurements on ships.
- Firms engaged in tightness testing of closing appliances such as hatches, doors, etc. with ultrasonic equipment.
- Firms carrying out an in-water survey on ships and mobile offshore units by diver or Remotely Operated Vehicle (ROV).
- Firms engaged in the examination of Ro-ro ships, bow, stern, side and inner doors.
- Firms engaged in measurement of noise levels onboard ships.
- Firms engaged in tightness testing of primary and secondary barriers of gas carriers with membrane cargo containment systems for vessels in service.
- Firms engaged in inspections and maintenance of fire extinguishing equipment and systems.
- Firms engaged in testing of coating systems in accordance with the requirements of IMO performance standards for protective coatings.
- Firms engaged in survey using Remote Inspection Techniques (RIT) as an alternative means for close-up survey of the structure of ships and mobile offshore units. See Ch.2, 1.2.20 for definition of RIT.
- Firms providing non-destructive testing (NDT) services on ship and offshore structures/ components (Refer IRS Classification Note – *Requirements for NDT Suppliers*).
- Firms engaged in cable transit seal systems inspection on ships and mobile offshore units.

b) Statutory services

- Firms engaged in servicing inflatable liferafts, inflatable lifejackets, hydrostatic release units, marine evacuation systems, etc.
- Firms engaged in the inspections and testing of radio communication equipment
- Firms engaged in inspections and maintenance of self contained breathing apparatus
- Firms engaged in the annual performance testing of voyage data recorder and simplified voyage data recorders
- Firms engaged in the inspections of low location lighting systems using photo luminescent materials and evacuation guidance systems used as an alternative to low location lighting systems
- Firms engaged in sound pressure level measurements of public address and general alarm systems on board ships
- Firms engaged in maintenance, thorough examination, operational testing, overhaul and repair of lifeboats and rescue boats, launching appliances, and release gear
- Firms engaged in the inspection, performance testing and maintenance of Automatic Identification Systems (AIS)
- Firms engaged in commissioning testing of Ballast Water Management System (BWMS).

Note : In general, suppliers providing services other than those listed above are not required to be approved.

1.9.2 Where the results of the following service providers are used by a Surveyor of IRS in making decisions affecting Classification and/or Statutory services, then that service provider is to be approved and service performance verified by IRS:

- Firms engaged in thickness measurements on ships.

- Firms carrying out an in-water survey on ships and mobile offshore units by diver or Remotely Operated Vehicle (ROV).

- Firms engaged in tightness testing of closing appliances such as hatches, doors etc. with ultrasonic equipment.

- Firms engaged in survey using Remote Inspection Techniques (RIT) as an alternative means for close-up survey of the structure of ships and mobile offshore units.

IRS may accept the services of firms approved by other IACS member classification societies for such services other than those listed above.

1.9.3 Where services are used by the Surveyors in making decisions affecting statutory certification and services, the firms are subject to approval and performance verification by IRS, where IRS is so authorized by the relevant Flag Administration (i.e. the Flag of the ship on which the servicing is to be done or the service equipment is to be used). For such services, IRS may accept approvals done by:

I. The Flag Administration itself,
II. Duly authorized organisation acting on behalf of the Flag Administration, or
III. Other organizations those are acceptable to the Flag Administration (e.g. other governments etc).

1.9.4 Use of the approved service suppliers is not mandatory for the following services, unless instructed otherwise by the flag Administration with respect to statutory certification:

- Firms engaged in the inspections of low location lighting systems using photo luminescent materials and evacuation guidance systems used as an alternative to low location lighting systems

- Firms engaged in sound pressure level measurements of public address and general alarm systems on board ships

- Firms engaged in measurement of noise levels onboard ships

- Firms engaged in testing of coating systems in accordance with the requirements of IMO performance standards for protective coatings.

- Firms engaged in the examination of bow, stern, side and inner doors.

1.9.5 Requirements to be complied with, as a minimum, by various categories of service suppliers are specified in the IRS Classification Note '*Approval of Service Supplier*s'.

1.10 Responding to Port State Control

1.10.1 When requested by Port State and upon concurrence by the vessel's owner/master IRS Surveyors would attend onboard a ship in order to assist in the rectification of reported deficiencies or other discrepancies that affect or may affect classification or the statutory certificates issued by IRS. The owner and the vessel's flag state will be notified of such attendance and survey. IRS Surveyors will also cooperate with Port States by providing inspectors with background information.

Section 2

Classification Regulations

2.1 General

2.1.1 The objective of ship classification is to verify the structural strength and integrity of essential parts of the ship's hull and its appendages; stability of the ship; the reliability and function (including cyber resilience, where applicable) of the propulsion and steering systems, power generation and those other features and auxiliary systems which have been built into the ship in order to maintain essential services on board. This also includes verification of cyber resilience, where applicable, of equipment and systems required by statutory regulations.

2.1.2 When a ship is assigned a specific Character of Class by Indian Register of Shipping, it implies that IRS has been satisfied that the said ship meets, for this particular class, with these Rules and Regulations or requirements equivalent thereto. The ship will continue to be classed with IRS so long as she is found, upon examination at the prescribed annual and periodical surveys, to be maintained in a fit and efficient condition and in accordance with the Periodical Survey requirements of these Rules. Classification will be conditional upon compliance with IRS requirements and assignment of character of class for both hull and machinery.

2.1.3 The Rules are framed on the understanding that ships will be properly loaded and handled; they do not, unless stated in the class notation, provide for special distributions or concentrations of loading and that ships will not be operated in environmental conditions more severe than those agreed for design basis and approval.

2.1.4 Compliance with the following IMO conventions, codes and resolutions, as amended, and as applicable, is a pre-requisite of classification:

- International Convention on Loadlines (ILLC)
- International Convention for the Safety of Life at Sea (SOLAS)
- International Convention for the Prevention of Pollution from Ships (MARPOL)
- The International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004 (BWM 2004)
- International Convention on the Control of Harmful Antifouling Systems on Ships (AFS Convention)
- International Bulk Chemical Code (IBC Code) or Bulk Chemical Code (BCH Code)
- International Gas Carrier Code (IGC Code) or Gas Carrier Code (GC Code)
- Fire Test Procedures Code (FTP Code) of the International Maritime Organization (IMO)
- Codes on noise levels on board ships.
- International Code for Ships Operating in Polar Waters (Polar Code)
- International Code of Safety for Ships Carrying Industrial Personnel (IP Code)
- Related Resolutions of the IMO Assembly, the Maritime Safety Committee (MSC) and the Marine Environment Protection Committee (MEPC) of IMO.

In addition, the Unified Interpretations (UIs) of IACS (International Association of Classification Societies) related to the above IMO Conventions and Codes are also to be complied with unless provided with written instruction to apply a different interpretation by the flag Administration.

Note: In general, IACS UIs for the above IMO Conventions and Codes related to various aspects covered by the Rules are addressed in relevant Parts/Chapters. Those IACS UIs which are not included in the Rules are indicated on the IRS Website under the webpage for "*Rules, Classification Notes and Guidelines*". This list also includes the IACS UIs to other IMO Conventions (International Tonnage Convention and COLREG Convention), which are to be complied with for statutory certification, unless flag Administration has provided written instructions to apply a different interpretation.

2.1.5 Where a vessel holds dual classification with IRS and the periodical survey requirements of the corresponding Society differ from those of the Rules of IRS, IRS may permit the requirements of the corresponding Society being applied, in so far as they are equivalent in purpose or are no less stringent than the IRS rule requirement.

2.1.6 The classification of a ship with IRS does not exempt the owners from compliance with any additional and/or more stringent requirements issued by the Administration of the state whose flag the ship is entitled to fly and provision for their application.

2.1.7 It is the responsibility of the Owners to ensure that the operating and maintenance instructions/manuals for the ship machinery equipment essential to the safe operation of the ship are available in a language understandable by those officers and crew members who are required to understand such

information / instructions in the performance of their duties.

2.1.8 When a ship is detained by Port State Control, IRS is to be immediately notified by the owner / operator for arranging attendance by a Surveyor.

2.2 Application of Rules

2.2.1 Unless directed otherwise by IRS, no new Regulations or amendments to the Rules relating to the character of classification or class notation is to be applied to the existing vessels.

2.2.2 Unless directed otherwise by IRS, no new Rules and Regulations or amendments to the existing Rules & Regulations become applicable within 6 months after the date of issue.

2.3 Scope and process of classification

2.3.1 Classification covers ship's hull, appendages and machinery including electrical systems to the extent as specified in these Rules & Regulations. Classification does not guarantee the design or performance of a vessel except for those aspects covered by the Rule requirements and subject to the conditions of operation of the vessel mentioned in 2.1.3.

2.3.2 The classification process consists of :

- A technical review of the design plans and related documents for a new vessel to verify compliance with the applicable Rules;

- Attendance at the construction of the vessel in the shipyard by IRS surveyor(s) to verify that the vessel is constructed in accordance with the approved design plans and classification Rules;

- Attendance by IRS surveyor(s) at the relevant production facilities that provide key components such as the steel, engine, generators and castings to verify that the component conforms to the applicable Rule requirements;

- Attendance by IRS surveyor(s) at the sea trials and other trials relating to the vessel and its equipment prior to delivery to verify conformance with the applicable Rule requirements;

- Upon satisfactory completion of the above, the builder's/ shipowner's request for the issuance of a class certificate will be considered by IRS and, if deemed satisfactory, the assignment of class may be approved and a certificate of classification issued;

- Once in service, the owner is to submit the vessel to a clearly specified programme of periodical class surveys, carried out onboard the vessel, to verify that the ship continues to meet the relevant Rule requirements for continuation of class. A classification survey is a visual examination that normally consists of an overall examination of the items identified in the Rules for survey, detailed check of selected parts on a sampling basis and witnessing tests, measurements and trials where applicable.

- When a Surveyor identifies corrosion, structural defects or damage to hull, machinery and/ or equipment which, based on the Rules and in the opinion of the Surveyor, affects the Ship's class, remedial measures and/ or appropriate conditions of class are specified in order to retain class. Conditions of class are requirements to the effect that specific measures, repairs, surveys etc. are to be carried out within a specified time limit in order to retain class.

2.3.3 On application by Builder or Owner, certain installation, e.g. refrigerating machinery may be classed by IRS.

2.4 Interpretations of the Rules

2.4.1 The correct interpretation of the requirements contained in the Rules and other Regulations is the sole responsibility and at the sole discretion of IRS.

2.5 Definitions

2.5.1 **Clear water** : Water having sufficient depth to permit the normal development of wind generated waves.

2.5.2 **Fetch** : The extent of clear water across which a wind has blown before reaching the ship.

2.5.3 **Sheltered water** : Water where the fetch is six nautical miles or less.

2.5.4 **Reasonable weather** : Reasonable weather is assumed to exclude winds exceeding Beaufort force six associated with sea states resulting in green water being frequently taken on board the ship's deck. However, it is realised that this is largely a matter of judgment and good seamanship and can vary for particular ships.

2.5.5 **Type notation** : A notation indicating that the ship has been designed and constructed with applicable Rules to that type of ship, e.g. "Bulk Carrier", "Oil Tanker", etc.

2.5.6 **Cargo notation** : A notation indicating that the ship has been designed, modified or arranged to carry one or more particular cargoes, e.g. "Phosphoric Acid". Ships with one or more particular cargo notations are not thereby prevented from carrying other cargoes for which they are suitable.

2.5.7 **Special feature notation** : A notation indicating that the ship incorporates special features which significantly affect the design, e.g. "movable decks".

2.5.8 **Service restriction notation** : A notation indicating that a ship has been classed on the understanding that it will be operated only in suitable areas or conditions which have been agreed to by IRS e.g. "Sheltered Water Service".

2.5.9 **Description**: A ‘description’ may be assigned to indicate the purpose or role of a vessel. In general, there would be no specific rule requirements to be complied with for assignment of a description. E.g. “Cutter Suction Dredger” can be assigned to describe the vessel type dredger.

2.6 Character of classification

2.6.1 The following Characters and symbols are assigned by IRS to indicate classification of Steel Ships. (For explanation of abbreviations, see Appendix I).

2.6.2 **Character SUL** assigned to sea-going vessels indicates that the hull and its appendages and equipment (i.e. anchors, chain cable and hawsers) meet the Rule requirements for assignment of this Character of Class.

Guidance Note : Appendages to the hull referred to in 2.6.2, 2.6.3 and 2.6.4 means the rudder & rudder stock, rudder horn, sole pieces, propeller nozzles, shaft brackets, skeg etc. which are covered by the rule requirements.

2.6.3 **Character SU (-)** assigned to sea-going vessels indicates that the hull and its appendages meet the Rule requirements but equipment (i.e. anchors, chain cable and hawsers) is not supplied or maintained as per the relevant Rules but is considered by IRS to be acceptable for particular service.

2.6.4 **Character SU** assigned to sea-going vessels indicates that the hull and its appendages meet the Rule requirements but where special consideration has been given for reason of particular purpose of service and normal equipment may be unnecessary. In such cases letter 'L' is omitted from the Character SUL.

2.6.5 **Character IY** assigned to self-propelled seagoing vessels indicates that the machinery meets the Rule requirements for assignment of this Character of Class.

2.6.6 **The distinguishing mark** 卐 inserted before a Character of Class (SUL, SU (-), SU, IY as appropriate) is assigned to new ships where the hull and its appendages, equipment and the machinery, as appropriate, are constructed under special survey of IRS in compliance with the Rules to the satisfaction of IRS.

2.6.7 The distinguishing mark 卐 inserted before a Character of Class (SUL, SU (-), SU, IY as appropriate), is assigned to vessels admitted in to IRS Class during the course of construction and surveyed by an IACS Society.

2.6.8 The distinguishing mark 卐 inserted before a Character of Class (SUL, SU (-), SU, IY as appropriate), is assigned to vessels admitted into IRS Class at the time of delivery of the vessel and constructed under the survey of an IACS Society.

2.7 Class notations - Hull

2.7.1 When requested by an Owner and agreed to by IRS or when considered necessary by IRS, a class notation will be appended to the character of classification. This class notation will consist of one of, or a combination of - a type notation, a cargo notation, a special duties notation, a special features notation and/or a service restriction

notation, e.g. 卐 SUL CHEMICAL TANKER, "Sulphuric Acid", ESP, "Indian Coastal Service".

2.7.2 Details of the ship types and additional class notations are given in Appendix 1 and applicable Chapters in Pt.5 of the Rules.

2.7.3 Service restriction notation will generally be assigned in one of the forms given below, but this does not preclude the Owners or Shipbuilders from requesting special consideration for other forms of restrictions.

a) Sheltered Water Service : Service in sheltered water adjacent to sand banks, reefs, breakwaters or other coastal features, and in sheltered water between islands. The geographical limits will form part of the Class Notation.

b) Restricted Water Service : Service in sheltered waters and also for short distances (generally less than 15 nautical miles) beyond sheltered waters in 'reasonable weather'. The geographical limits will form part of the Class Notation.

c) Specified Coastal Service : Service along a coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land and may cross gulfs or similar features recognised by the local Administration as a part of the coastal service. The geographical limits will form part of the Class Notation, e.g. "Indian Coastal Service".

d) Specified Route Service : Service between two or more points or other geographical features which will form part of the Class Notation.

e) Specified Operating Area Service : Service within one or more geographical area(s) which will form part of the Class Notation.

2.8 Class notations - Machinery

2.8.1 The class notations that may be assigned by IRS are given in Appendix 1. IRS may prescribe additional notations as found necessary / expedient from time to time.

2.9 Materials, components, equipment and machinery

2.9.1 The materials used in the construction of hull and machinery intended for classification, or in the repair of ships already classed, are to be of good quality and free from defects and are to be tested in accordance with the relevant Rules. The steel is to be manufactured by an approved process at works recognized by IRS. Alternatively, tests to the satisfaction of IRS will be required to demonstrate the suitability of the steel.

Consideration may be given by IRS to accept the works approved by IACS Member Societies with whom IRS currently has Cooperation Agreements for this purpose.

2.9.2 Certification of materials, components, equipment and machinery is carried out on basis of the following, considering IRS and/or IMO requirements, as applicable:

a) Type approval carried out by IRS.

b) Unit certification by IRS.

c) Alternative Certification Scheme by IRS (see Section 4).

2.9.3 Mutual recognition of certificates, if type approved by an IACS Member Society or European Union recognized organization based on commonly agreed design requirements under Mutual Recognition Scheme between IRS and the recognized organization, may also be used as basis for certification of materials, components, equipment and machinery.

2.10 Request for surveys

2.10.1 It is the responsibility of the Builders or Owners, as applicable, to inform the Surveyors of IRS in the port at which the surveys for supervision during new construction or ships in service are to be undertaken and to ensure that all surveys for issue of class certificate for new construction, and maintenance of class for ships in service are carried out.

2.10.2 It is the responsibility of the Owner to ensure that all surveys necessary for the maintenance of class are carried out by due date. IRS will notify an Owner of upcoming surveys and outstanding Conditions of Class before their becoming overdue. The non-receipt of such notification, however, does not absolve the Owner from his responsibility to comply with survey requirements for maintenance of class.

2.11 Repairs

2.11.1 Any repairs to the hull, machinery and equipment either as a result of damage or wear and tear which are required for the maintenance of ship's class are to be carried out under the inspection of and to the satisfaction of the Surveyors.

2.11.2 Where a vessel is damaged to an extent resulting in towage outside port limits, it shall be the Owners' responsibility to notify IRS at the first practicable opportunity.

2.11.3 Where such repairs are effected at a port where there is no Surveyor of IRS, the ship is to be surveyed by one of its Surveyors at the earliest opportunity.

2.11.4 Where repairs to hull, machinery or equipment, which affect or may affect classification, are to be carried out by a riding crew, they are to be planned in advance. A complete repair procedure including the extent of proposed repairs and the need for Surveyor's attendance during the voyage is to be submitted to and agreed upon by the Surveyor reasonably in advance. Failure to notify IRS, in advance of the repairs, may result in suspension of the vessel's class.

Where in any emergency circumstance, emergency repairs are to be effected immediately, the repairs should be documented in the ship’s log and submitted thereafter to IRS for use in determining further survey requirements.

2.12 Alterations

2.12.1 Any alterations proposed to be carried out to approved scantlings and arrangements of the hull, machinery or equipment are to meet with the approval of IRS and for this purpose plans and technical particulars are to be submitted for approval in advance. Such approved alterations are to be carried out under the inspection of, and to the satisfaction of, the Surveyors. If such alterations are carried out on items which may affect the classification of the ship without informing IRS, the class of the vessel will be liable to be suspended except in the case of emergency repairs mentioned in 2.11.4.

2.13 Classification of new constructions

2.13.1 The request for classification of new constructions is to be submitted to IRS by the shipyard or shipowner in the form provided by IRS. The request is to include complete details regarding class notation and statutory certificates required, where applicable.

The IRS Rules in force on the date of contract for construction of the vessel (See 2.14) will be applicable for classification, in general. However, statutory requirements coming into force after the date of contract for construction may have to be complied with if they become applicable based on any other criteria such as the date on which vessel is constructed (keel laid).

2.13.2 Where orders for major machinery and equipment are placed on manufacturer or suppliers, IRS will have to be informed. Responsibility for compliance with IRS Rules and Regulations shall be with the manufacturers/suppliers.

Where relevant, the date of application for certification of specific major machinery will also be considered in addition to the date of contract for construction of the vessel, for determining the applicable rules for such machinery.

2.13.3 Plans and particulars as specified in the Rules will have to be submitted to IRS in triplicate sufficiently in advance of commencement of construction. One copy with stamp of approval will be returned. Any deviation from approved drawings will require to be approved by IRS prior to execution of work.

IRS reserves the right to request for additional plans, information or particulars to be submitted.

Where it is proposed to use existing previously approved plans for a new contract, written application is to be made to IRS.

Approval of plans and calculations by IRS does not relieve the Builders of their responsibility for the design, construction and installation of the various parts, nor does it absolve the Builders from their duty of carrying out any alterations or additions to the various parts on board deemed necessary by IRS during construction or installation on board or trials.

2.13.4 IRS will assess the production facilities and procedures of the shipyard and other

manufacturers as to whether they meet the requirements of the construction Rules.

Review of the construction facilities prior to any steel work or construction shall be carried out under the following circumstances:

a) Where IRS has none or no recent experience of the construction facilities – typically after a one year lapse – or when significant new infrastructure has been added.

b) Where there has been a significant management or personnel re-structuring having an impact on the ship construction process, or

c) Where the shipbuilder contracts to construct a vessel of a different type or substantially different in design.

2.13.5 During construction of a vessel, IRS will ensure by surveys that parts of hull and machinery requiring approval have been constructed in compliance with approved drawings, all required tests and trials are performed satisfactorily, workmanship is in compliance with current engineering practices and welded parts are produced by qualified welders.

2.13.6 All hull, machinery and electrical installations will be subjected to operational trials in the presence of IRS Surveyor.

2.13.7 On completion of the ship copies of as fitted plans showing the ship as built, essential certificates and records, loading manual etc. are to be submitted by the Builder generally prior to issuance of the Interim Certificate of Class.

2.13.8 For each new construction the shipbuilder is required to prepare and deliver a ship construction file containing documents / plans / manuals etc. for facilitating the future inspection of survey, repair and maintenance as detailed in Pt.3, Ch.1, Sec.3. Some of these documents may be directly supplied by other parties e.g. shipowner, for inclusion in the ship construction file. The ship construction file is to be maintained onboard each ship.

2.14 Date of contract for construction

2.14.1 The date of “contract for construction” of a vessel is the date on which the contract to build the vessel is signed between the prospective owner and the shipbuilder. This date and the construction numbers (i.e. hull numbers) of all the vessels included in the contract are to be declared to IRS by the party applying for the assignment of class to a new building.

2.14.2 The date of “contract for construction” of a series of vessels, including specified optional vessels for which the option is ultimately exercised, is the date on which the contract to build the series is signed between the prospective owner and the shipbuilder.
For the purpose of this requirement, vessels built under a single contract for construction are considered a “series of vessels” if they are built to the same approved plans for classification purposes. However, vessels within a series may have design alterations from the original design provided:

a) Such alterations do not affect matters related to classification, or

b) If the alterations are subject to classification requirements, these alterations are to comply with the classification requirements in effect on the date on which the alterations are contracted between the prospective owner and the shipbuilder or, in the absence of the alteration contract, comply with the classification requirements in effect on the date on which the alterations are submitted to IRS for approval.

The optional vessels will be considered part of the same series of vessels if the option is exercised not later than 1 year after the contract to build the series was signed.

2.14.3 If a contract for construction is later amended to include additional vessels or additional options, the date of “contract for construction” for such vessels is the date on which the amendment to the contract, is signed between the prospective owner and the shipbuilder. The amendment to the contract is to be considered as a “new contract” to which 2.14.1 and 2.14.2 above apply.

2.14.4 If a contract for construction is amended to change the ship type, the date of “contract for construction” of this modified vessel, or vessels, is the date on which revised contract or new contract is signed between the Owner, or Owners, and the shipbuilder.

2.15 Date of build

2.15.1 The date of completion of the special survey inspection will normally be taken as the date of build to be entered in the Register Book.

Where there is a substantial delay between completion of construction survey and the ship commencing service, the date of commissioning may be specified on the classification certificate.

When modifications are carried out on a ship, the initial date of build remains assigned to the ship.

2.15.2 When a complete replacement or addition of a major portion of the ship (e.g. fwd. section, midship section or aft section) is involved, the following applies:

- Date of build assigned to each portion of the ship will be indicated on the classification certificate, and the date of modification will be indicated in the Register Book.

- Survey requirements shall be based on the date of build associated with each major portion of the ship.

2.16 Appeal from Surveyors' recommendations

2.16.1 If the recommendations of the Surveyors are considered in any case to be unnecessary or unreasonable, appeal may be made to IRS, who may direct a special examination to be held.

2.17 Certificates

2.17.1 Certificates of Class will be issued to Builders or Owners when the required reports on completion of Special Surveys of new ships or of existing ships submitted for classification have been received from the Surveyors and approved by IRS.

2.17.2 The Surveyors are permitted to issue Interim Certificates to enable a ship, classed with IRS, to proceed on her voyage provided that, in their opinion, she is in a fit and efficient condition. Such Certificates will contain Surveyors' recommendations for continuance of Class, but in all cases are subject to confirmation by IRS.

2.17.3 Individual Certificates can also be issued for propelling machinery, boilers, equipments and fittings which have been manufactured under IRS Survey and in accordance with these Regulations.

2.17.4 It is the responsibility of the owner to ensure that the validity of class and statutory certificates is maintained.

2.18 Suspension, withdrawal and deletion of class

2.18.1 Suspension

2.18.1.1 The class of a vessel will be automatically suspended from the expiry date of the Certificate of Class or by the expiry date of any extension granted, if the special survey has not been completed by the due date and an extension (See Ch.2, 1.11) has not been agreed to, or the vessel is not under attendance by the Surveyor with a view to complete the surveys prior to resuming service.

2.18.1.2 The class of a vessel will also be automatically suspended if the annual, intermediate survey become overdue unless the vessel is under attendance by the surveyor for completion of survey. (See Pt.1 Ch.2, Table 1.1.1 for due dates and window period). Where any ship proceeds to sea without having completed Surveyor’s recommendations which were required to be dealt with before leaving port, the ship’s class will be suspended.

2.18.1.3 When the surveys relating to specific additional notations of hull or equipment or machinery have not been complied with and thereby the ship is not entitled to retain that notation, then the specific notation will be suspended till the related surveys are completed.

2.18.1.4 The class of a vessel will be subject to a suspension procedure if an item of continuous survey is overdue at the time of annual survey, unless the item is dealt with or postponed by agreement.

2.18.1.5 The class of the vessel will also be subject to a suspension procedure if conditions of class are not dealt with by the due date or postponed by agreement, by the due date.

2.18.1.6 The class of a ship is liable to be withheld or, if already granted, may be withdrawn in case of any non-payment of fees or expenses chargeable for the service rendered.

2.18.1.7 Where any ship proceeds to sea with less freeboard than that approved by IRS or when the freeboard marks are placed higher on the ship's sides than the position assigned or approved by IRS, the ship's class will be suspended.

2.18.1.8 When it is found that a ship is being operated in a manner contrary to that agreed at the time of classification, or is being operated in conditions or in areas more onerous than those agreed, the class will be suspended.

2.18.1.9 The class of a vessel will be liable to be suspended if the Owner fails to notify IRS of any damage to the ship‘s hull, machinery or equipment, which may adversely affect classification of the vessel or subsequently fails to arrange for the survey as may be advised by IRS.

2.18.1.10 The class of a vessel will be suspended after a major casualty to the ship, such as grounding, sinking or breaking up, if the Owner is unable to arrange for the ship’s survey by IRS and commence repairs within a reasonable period of the occurrence of the casualty, unless otherwise agreed to with IRS. Procedures for withdrawal of class could be initiated unless the Owner is able to arrange for the ship’s survey by IRS and commence repairs within the agreed time period.

2.18.1.11 Vessels laid up in accordance with the Rules prior to surveys becoming overdue will not be suspended when surveys addressed above become overdue.

2.18.1.12 The ship’s class may be suspended if the owner does not implement the lay-up maintenance program and /or lay-up surveys (Ch.2, Sec. 1.3) are not carried out by due date.

2.18.1.13 When a vessel is intended for a demolition voyage with any periodical survey overdue, the vessel’s class will not be suspended till completion of a single direct ballast voyage from the lay up or final discharge port to the demolition yard.

2.18.1.14 Force Majeure : If, due to circumstances defined in Chapter 2, 1.11.3, the vessel is in a port where the overdue surveys cannot be completed at the expiry of the periods allowed above, IRS may allow the vessel to sail, in class, directly to an agreed discharge port, and if necessary, thereafter in ballast, to an agreed port at which the survey will be completed, provided:

a) Re-examination of the ship’s record is carried out by IRS,

b) IRS carries out the due and/or overdue surveys and examination of Conditions of Class at the first port of call when there is an unforeseen inability of IRS to attend the vessel in the present port, and

c) IRS is satisfied that the vessel is in condition to sail for one trip to a discharge port and subsequent ballast voyage to a repair facility if necessary (where there is unforeseen inability of IRS to attend the vessel in the present port, the master is to confirm that the ship is in condition to sail to the nearest port of call).

The scope of the overdue surveys will be based on the survey requirements applicable to the vessel at the original due date and not based on the age of the vessel when the survey is carried out. Such surveys will be credited from the date originally due.

If class has already been automatically suspended in such cases, it may be reinstated subject to the conditions prescribed above.

2.18.1.15 When a vessel is intended for a single voyage from laid-up position to a repair yard or another place of lay-up with any periodical survey overdue, the vessel’s class suspension may be held in abeyance and consideration may be given to allow the vessel to proceed on a single direct ballast voyage from the site of lay up to a repair yard or another place of lay-up, upon agreement with the Flag Administration and Port authorities, provided IRS finds the vessel in satisfactory condition after surveys, the extent of which are to be based on surveys overdue and duration of lay-up. A short-term Class Certificate with conditions for the intended voyage may be issued.

2.18.1.16 Classification will be reinstated upon satisfactory completion of overdue survey. The scope of the overdue surveys will be based on the survey requirements applicable to the vessel at the original due date and not based on the age of the vessel when the survey is

carried out. Such surveys will be credited from the date originally due. Such surveys will be credited from the date originally due. However, the vessel will remain dis-classed from the date of suspension until the date class is reinstated.

2.18.1.17 The Owners and the Flag State, where applicable, would be informed in writing, of the suspension and reinstatement of Classification.

2.18.2 Withdrawal

2.18.2.1 Ship's class will be withdrawn, at the end of six months of suspension, if the Owner has not commenced any action to reinstate the ship's class. A longer suspension period may be granted in the following cases when the vessel is:

a) not trading and awaiting attendance for reinstatement of class; or

b) laid up; or

c) awaiting disposal, in the event of a casualty.

2.18.2.2 When the class of a ship holding IRS class, is withdrawn by IRS in consequence of a request from the Owners, the notation "Class withdrawn at Owners' request" (with date) will be made in the subsequent reprints of the Register Of Ships. This entry will continue till the ship's class is reinstated or deleted.

2.18.2.3 When the Regulations as regards surveys on the hull or equipment or machinery have not been complied with and the ship thereby is not entitled to retain her class, the class will be withdrawn and the notation "Class withdrawn" (with date) will be made in the subsequent reprints of the Register Of Ships. This entry will continue till the ship's class is reinstated or deleted.

2.18.2.4 In case the vessel is not maintained satisfactorily and there are repeated failures, e.g. multiple PSC/FSI detentions, procedure for withdrawal of class could be initiated. In addition, procedure for withdrawal of class could also be initiated subsequent to a detention, if the vessel is not offered for additional examination(s), as requested by IRS.

Withdrawal of class may be extended to other vessels of a particular Owner based on overall failures/ PSC performance of the fleet. In case it is decided to withdraw the class of the vessel, the Owner will be provided with a time period not exceeding 3 months to enable vessel to change class. During this period, IRS may impose additional conditions such as restricting the area of operations, additional examinations, etc.

2.18.2.5 Procedure for withdrawal of class may be initiated if the vessel is found to be in breach of international sanctions or engaged in unlawful trading activities.

2.18.2.6 The withdrawal of a vessel will be confirmed in writing to the Owner and the Flag State, where applicable.

2.18.3 Deletion of Class

2.18.3.1 A ship will be considered to "cease to exist" when it is destroyed by scrapping or by sinking to unsalvageable depths or abandoned by the owner.

2.18.3.2 A ship can also be considered to "cease to exist" when it is broken up either by grounding or due to structural failure or due to actions of war or sabotage.

2.18.3.3 Ship's class will be deleted when it ceases to exist.

2.19 Reclassification of ships

2.19.1 When Owners request for reclassification of a ship for which the class previously assigned has been withdrawn, IRS will require a Special Survey for Reclassification to be held by the Surveyors. The extent of the survey will depend upon the age of the ship and the circumstances of each case.

2.19.2 If the ship is found or placed in good and efficient condition in accordance with the requirements of the Rules and Regulations at the Special Survey for Reclassification, IRS may decide to reinstate her original class or assign such other class as considered appropriate.

2.19.3 The date of reclassification will appear in the supplement to the Register Of Ships and the subsequent issue of Register Of Ships.

2.20 Fleet Quality Monitoring

2.20.1 The Fleet Quality Monitoring system of IRS is aimed at maintaining and improving the

quality of the classed fleet by according special attention to certain vessels. .

2.20.2 Vessels requiring special attention are identified either for 'condition monitoring' or 'condition improvement' based on multiple risk criteria through a computer program. Vessels may also be identified manually based upon feedback from the attending surveyors.

2.20.3 Vessels identified for 'condition monitoring' would not be subjected to any additional survey, but would be monitored during the class/ statutory surveys.

2.20.4 Vessels identified for 'condition improvement' would be subjected to additional surveys for which owners are to make necessary arrangements. Additional surveys are intended to identify areas requiring attention and to ensure that corrective actions are taken to improve the condition of the vessel and brought back to satisfactory condition to the satisfaction of the surveyor.

2.20.5 Vessels which have undergone additional surveys towards 'condition improvement' would be kept under 'condition monitoring', till the next class/ statutory periodical survey and removed from the list, once effectiveness of upkeep and maintenance is verified.

2.20.6 Procedures for withdrawal of class could be initiated in the cases mentioned in 2.18.2.4.

2.21 Transparency of classification and statutory information

2.21.1 The classification and statutory information which may be released to Shipowners, Flag State, Port state, Insurance company and Shipyards as relevant and the conditions for their release are indicated in Table 2.21.1. For oil tankers and bulk carriers subject to SOLAS Chapter II-1, Part A-1, Regulation 3-10, refer to IRS Rules for Bulk Carriers and Oil Tankers, Volume 1, Part 1, Chapter 1. In general, information on items which are not restricted are freely available on the IRS website 'www.irclass.org'.

Note : The European Commission (EC)/ European Maritime Safety Agency (EMSA) may be provided with Classification and Statutory information, when requested, as available to the flag state (as in Table 2.21.1.).

Table 2.21.1 : Transparency of information

		Information available to:				
	Information in Question	**Owners**	**Flag State**	**Port State**	**Insurance Company***	**Ship Yards**
1.	**Class Societies Standing Documents:**					
	Rules and Guidelines (Class and statutory requirements)	1	1	1	1	1
	Instructions to Surveyors		1			
	Quality Manual	1	1	1	1	1
	Register Book	1	1	1	1	1
2.	**Ship Related Information:**					
A.	**New Buildings**					
	Approved Drawings	6	1			7
	Formal Approval Letters	1				7
	Certificates of Important Equipment	2				7
B.	**Ships in Operation:**					
	Date (month and year) of all class surveys	7	1	1	1	
	Expiry Date of Class Certificate	7	7	1	1	
	Certificates/Reports	7	1	6	5	
	Overdue Surveys	7	7	1	1	
	Text of Conditions of Class	7	1	1	5	
	Text of Overdue Conditions of Class	7	1	1	1	
	Executive Hull Summary	7	3	3	3	
	Statutory Services					
	Due Dates of Statutory Surveys	7	7	1	1	
	Expiry Date of Statutory Certificates	7	7	1	1	
	Registered Statutory Conditions	7	7	1	5**	
	Overdue Statutory Conditions	7	7	1	1**	
3.	**Other Information:**					
	Correspondence File with Yard and/or Owner	6	6		5&6	
	Audit of Class Societies QA System	4	4	4	4	
	Class Transfer Reporting	7	7	7	7	
	Class Withdrawal Information	7	7	7	7	

* Insurance Company means P&I Clubs and Hull Underwriters

** Unless prevented by the agreement with the Flag State

KEY:

1. Available upon request
2. At delivery of the ship by Shipyard
3. Available under visit on board
4. Result of audit available on request
5. When accepted by Owners – or through special clause in insurance contract
6. When accepted by Owner (Master) or Shipyard as applicable
7. Automatically available

Section 3

Classification of Ships not Built under the Survey of Indian Register of Shipping

3.1 General procedure for classification of ships not built under survey of IRS

3.1.1 Plans of hull and machinery, as listed in 3.2.1 to 3.2.5 or equivalent, together with torsional vibration calculations, where applicable, are to be submitted for approval. Alternative technical data in lieu of specific plans or items may be accepted. It is preferable to have the plans approved before the classification survey is commenced. In case of transfer of class from an IACS member society/ adding dual class with an IACS member society, only copies of plans or equivalent (as listed in 3.2.1 to 3.2.5) are to be submitted to IRS.

3.1.2 Full special classification surveys would require to be carried out by IRS Surveyors in order to satisfy themselves regarding the workmanship and condition of the ship and to verify the approved scantlings and arrangements. The scope of these surveys may, however, be modified in the case of vessels built under the Special Survey and holding valid certificates of class of classification societies subject to verification of compliance with IACS QSCS and the vessels are in full compliance with all applicable and relevant IACS resolutions. Prior to commencement of survey by IRS, documentary evidence of all hull and machinery classification surveys held by the other society subsequent to last special survey carried out are to be produced. In such cases, a special survey notation will not be assigned in conjunction with the classification survey. The next special survey therefore would become due five years from the special survey held by the other society and not five years from classification with IRS.

In cases of transfer of class from another society to single class of IRS, the interim certificate of class or any other documents enabling the ship to trade, can be issued :

a) In case of vessels less than 15 years of age

   – only after IRS has completed all overdue surveys and overdue conditions of class issued earlier, as specified to the Owner by the previous society.

b) In case of vessels 15 years of age and over

   – only after the previous society has completed all overdue surveys and all overdue conditions of class issued earlier.

3.1.3 For vessels not built under survey of IRS but subsequently taken in class with the above procedure, the mark signifying the survey during construction will be omitted.

3.1.4 Once a vessel has been taken into IRS class, periodical surveys are subsequently to be held as per these rules.

3.2 Plans and data to be furnished

3.2.1 Plans of hull and equipment showing the main scantlings and arrangements of the actual ship and any proposed alterations are to be submitted. These should normally comprise of the following plans:

For information

- General arrangement
- Capacity plan
- Hydrostatic Curves

For approval

- Midship section
- Longitudinal section and decks
- Shell expansion plan
- Transverse Bulkheads
- Sternframe
- Rudder and Rudder Stock
- Loading manual, where required

- Loading instrument details
- Stability booklet
- Damage stability calculations, where required
- Hatch Covers
- Fire detection, fighting and extinction and such other plans as may be requested.

3.2.2 It would normally be expected that particulars of the process of manufacture and testing of material of construction are furnished. Consideration will however be given to waiving this where such particulars are not readily available, provided it can be established that the relevant vessel has been originally built under special survey of an established classification society and continues to be so classed with an established classification society. In the case of vessels which have been originally built under the special survey of an established classification society but subsequently not maintaining class, it should additionally be possible to reasonably ascertain that no changes that would significantly affect the material specifications have taken place.

3.2.3 Following machinery plans together with the particulars of the materials used in the construction of the boilers, air receivers and important forgings should be furnished:

For information

- General machinery lay-out

For approval

- Thrust, intermediate and propeller shafting
- Propeller
- Main engines, propulsion gears and clutch systems (or manufacturer make, model and rating information)
- For steam turbine engines, main boilers, superheaters and economisers (or manufacturer make, model and rating information) and steam piping
- Bilge and ballast piping system
- Wiring diagram
- Steering gear systems, piping and arrangements and steering gear manufacturer make and model information

- Additional plans for oil tankers
  - Pumping arrangements at the forward and after ends and drainage of cofferdams and pump rooms
  - General arrangement of cargo piping in tanks and on decks
- Additional plans for vessels with Unattended Machinery Spaces
  - Description and/or block diagram of method of operation of the control system
  - Line diagrams of the control system for :

    main propelling machinery and essential auxiliaries

    bilge level systems

    cargo pumping for tankers

    boiler controls

    fire detection

    fire prevention, including details in way of fuel oil pressure pipes

    overall alarm system including test schedule
- Additional plans for vessels with Ice Class notations:
  - Plans for flexible couplings and/or torque limiting shafting devices in the propulsion line shafting (or manufacturer make, model and rating information)

3.2.4 Calculations of torsional vibration characteristics of the main propelling machinery are to be furnished specially for ships which have been in service for less than about 2 years.

3.2.5 In addition to the requirements of 3.2.1 to 3.2.4, additional plans would require to be submitted in accordance with applicable Chapters of Pt.5 of the Rules for vessels with additional class notations.

3.2.6 In cases where the vessel has been previously classed by IRS or a Classification Society subject to verification of compliance with IACS QSCS and the vessel is in full compliance with all applicable and relevant IACS resolutions, the approval of plans may be specially considered subject to confirmation of no alteration/ modification to the vessel. In such cases, it is to be also verified that the vessel complies with any retroactively applicable classification or statutory requirements which came into effect subsequently.

Section 4

Alternative Certification Scheme based upon Quality Management Systems

4.1 General

4.1.1 Alternative Certification Scheme (ACS) is a certification scheme involving a manufacturer (and associated sub-suppliers, if needed) in the inspection, testing and certification of the manufacturer's products.

4.1.2 An ACS will clarify:

- The extent of the required inspection and testing;

- To which extent and under which conditions the manufacturer may perform all or parts of the required inspection and testing without the presence of a Surveyor from IRS when an IRS Certificate is required.

4.1.3 The extent to which the manufacturer is given permission to carry out inspections and testing without the presence of a Surveyor would be agreed on a case by case basis, e.g. for a specific product production line or for specific parts.

4.2 Scope

4.2.1 An ACS may be arranged with product manufacturers and/ or sub-suppliers.

4.2.2 An ACS with a manufacturer would define the handling of subcontracted parts (those that require IRS or work certificates or in any other way are addressed in the Rules). The sub-supplier may be included in the ACS of the manufacturer or have his own ACS or deliver parts that are inspected and certified by the IRS.

4.2.3 An ACS that permits the manufacturer to carry out all or parts of required inspection and testing without the presence of a Surveyor may be arranged in two versions with regard to traceability:

- The ACS describes inspection, testing and certification additional to the manufacturer's standard quality control procedures in order to meet the Rules. The components are to be stamped with a special stamp supplied by IRS or identified as required by IRS.

- The manufacturer has a standard quality control that covers all required inspection, testing and certification in compliance with the Rules. Traceability and the required type of product document for components or products will be defined in the ACS.

4.3 Conditions for approval

4.3.1 The conditions for the manufacturer to be granted the permission to carry out inspection and testing without the presence of a Surveyor are as follows:

- The manufacturer has an implemented Quality System according to a national or international standard (like ISO 9000 series of standards) approved by an accredited certification body or recognized by IRS.

- The manufacturer has a quality control system, current drawings, and

Rules and standards that cover the product to be certified.

- The inspection and testing required by the Rules are either standard procedures in the Quality System and recognized by IRS or specified in detail in the ACS.

- The manufacturer is to demonstrate that the firm has experience consistent with technology and complexity of the product for which approval is sought and that firm's products have been of a consistently high standard.

- IRS would initially ascertain the manufacturer's compliance with the ACS requirements by verifying the required product and process approvals and performing an initial audit. Follow-up and renewal audits would be conducted by IRS on a regular basis to verify that conditions of the ACS are continuously maintained by the manufacturer.

- If work certificates (W) or test reports (TR) are found not to fulfill the standards agreed with IRS, the component may not be accepted.

- The agreed ACS may be suspended or cancelled when/ if found justified by IRS.

- IRS may carry out unscheduled inspections at the manufacturer and/or subcontractor at its own discretion.

- The manufacturers (and designers, if producing under license) are to commit themselves to involve IRS when changes to the design, manufacturing process or testing are made as well as when any major production problems or any major product delivery problems have occurred.

4.4 Information to be submitted

4.4.1 For admission to an alternative certification scheme for a product, the manufacturer is to submit an application enclosing the following documentation/ information:

- Description of the products for which certification is required including, where applicable, model or type number;

- Applicable plans, details of materials used and procedures relevant to the manufacturing process;

- An outline description of all important manufacturing equipment and plant;

- A list of suppliers of materials with an indication of their class approval (as far as required by the Rules) and the type of material certification in each case;

- The list of main components of the product, including certificates.

- Existing class approvals of the manufacturer's products as far as required;

- Quality control plans relevant to the products and relevant components to be certified through the alternative certification scheme. Said plans are to detail the inspections and tests required by the Rules with an indication of which inspections and tests are delegated to the manufacturer and which are to be done in the presence of an IRS Surveyor.

- The procedures relevant to the quality control and inspections, their methods, frequency and certification;

- The quality system details;

- The system used for identification and traceability;

- List of nominated personnel for:

  - Marking/stamping of products;

  - Tests and Inspection (responsible);

  - Provision of data and information (e.g. declaration of conformity, test reports etc.).

- Any other additional documents that IRS may require in order to evaluate the manufacturing processes and product quality control.

4.5 Approval and maintenance of approval

4.5.1 After receipt and appraisal of the information required by 4.4, an initial audit of the Works would be carried out by the Surveyors. This audit is to verify that the manufacture of the product and the relevant controls are performed in accordance with the documents submitted and are in compliance with the requirements laid down in the Rules.

4.5.2 The ACS validity is for 5 years. At least one intermediate audit would be carried out by IRS during the validity of the ACS. Annual surveillance audits would be carried out as necessary, depending upon the complexity of the products and experience with the firms. The ACS may be renewed subject to an audit. The scope of the renewal audit would be as follows:

- verify the conditions of the ACS are still met;

- verify that the current products and processes are appropriately controlled.

4.5.3 A Quality Assurance Certificate would be issued by IRS on satisfactory completion of the initial/ renewal audit.

Appendix 1 **Table of characters of class and type notations of IRS, their expanded form and significance**		
Abbreviation	**Expanded Form**	**Significance**
Characters of Class		
SUL	SARVOUTAM LANGER	Denotes vessels which are classed with Indian Register of Shipping where the hull and its appendages and equipment (i.e. anchors, chain cables, hawsers) meet the Rule requirements.
SU (-)	SARVOUTAM (-)	Denotes vessels which are classed with IRS where the hull and its appendages meet the rule requirements but when the equipment of ship is not supplied or maintained as per the relevant Rules but is considered by IRS to be acceptable for particular service
SU	SARVOUTAM	Denotes vessels which are classed with IRS where the hull and its appendages meet the rule requirements but where for reason of their particular purpose or service normal equipment may be unnecessary
IY	INDIAN YANTRA	Denotes that for self propelled seagoing vessels, the machinery installation complies with the applicable requirements of Indian Register of Shipping
卐	SWASTIKA	This distinguishing mark inserted before a Character of Class is assigned to new ships where the hull and its appendages, equipment and the machinery as appropriate, are constructed under special survey of IRS in compliance with the Rules to the satisfaction of IRS
卐		The distinguishing mark inserted before a Character of Class (SUL, SU (-), SU, IY as appropriate), is assigned to vessels admitted in to IRS Class during the course of construction and surveyed by an IACS Society.
卐		The distinguishing mark inserted before a Character of Class (SUL, SU (-), SU, IY as appropriate), is assigned to vessels admitted into IRS Class at the time of delivery of the vessel and constructed under the survey of an IACS Society.
[ ]		When a Class Notation is enclosed within brackets, it indicates that applicable arrangements exist on board but the notation has been temporarily suspended

Abbreviation	Expanded Form	Significance
Class Notations - Hull		
Sheltered Water Service		Service in sheltered water adjacent to sand banks, reefs, breakwaters or other coastal features, and in sheltered water between islands
Restricted Water Service		Service in sheltered waters and also for short distances (generally less than 15 nautical miles) beyond sheltered waters in 'reasonable weather'
Specified Coastal Service		Service along a coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land and may cross gulfs or similar features recognised by the local Administration as a part of the coastal service. The geographical limits will form part of the Class Notation, e.g. "Indian Coastal Service"
Specified Route Service		Service between two or more points or other geographical features which will form part of the Class Notation
Specified Operating Area Service		Service within one or more geographical area(s) which will form part of the Class Notation
HAT(B)	HIMA ATITIVRA (B)	Denotes that the ship is strengthened for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 1.0 metre
HT(B)	HIMA TIVRA (B)	Denotes that the ship is strengthened for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 0.8 metre
HM(B)	HIMA MADHYAM (B)	Denotes that the ship is strengthened for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 0.6 metre
Ha(B)	HIMA ALPA (B)	Denotes that the ship is strengthened for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 0.4 metre
Ha	HIMA ALPA	Denotes that the ship is strengthened for light first year ice conditions in areas other than Northern Baltic
PC1	Polar Class 1	Denotes that the ship is strengthened for year-round operation in all polar waters according to Pt.5, Ch.32
PC2	Polar Class 2	Denotes that the ship is strengthened for year-round operation in moderate multi-year ice conditions according to Pt.5, Ch.32
PC3	Polar Class 3	Denotes that the ship is strengthened for year-round operation in second-year ice which may include multi-year ice inclusions according to Pt.5, Ch.32
PC4	Polar Class 4	Denotes that the ship is strengthened for year-round operation in thick first-year ice which may include old ice inclusions according to Pt.5, Ch.32

Abbreviation	Expanded Form	Significance
PC5	Polar Class 5	Denotes that the ship is strengthened for year-round operation in medium first-year ice which may include old ice inclusions according to Pt.5, Ch.32
PC6	Polar Class 6	Denotes that the ship is strengthened for summer / autumn operation in medium first-year ice which may include old ice inclusions according to Pt.5, Ch.32
PC7	Polar Class 7	Denotes that the ship is strengthened for summer / autumn operation in thin first-year ice which may include old ice inclusions according to Pt.5, Ch.32
ESP		Vessel complies with requirements of enhanced survey programme
STS	SAMATARANA STHIRATVA	Denotes that the ship complies with the requirements for survival capability (floatability in damaged condition) in accordance with Chapter 3 of IMO Resolution MSC 235(82)
"Strengthened for heavy cargoes"		This will be entered in the Register of Ships where the scantlings and arrangements have been approved for heavier cargo loadings in any hold filled up to the top of the hatch coaming with cargo at a stowage rate of $\leq$ 1 [m^3/tonne], when the draught in way of the hold is 80 per cent of the maximum permissible draught or more
"Hold(s) ...(to be specified)..... may be empty"		This will be entered in the Register of Ships when in association with the carriage of heavy cargoes, specified holds are permitted to be empty with the ship in the fully loaded condition
HELDK		This will be entered in the Register Of Ships for vessels with erected platforms or landing area designed for helicopter landings in accordance with applicable requirements of Part 5, Chapter 27 of the Rules
ETA	EMERGENCY TOWING ARRANGEMENT	Denotes that the ship is equipped with emergency towing arrangements in accordance with the requirements given in Part.5, Chapters 2, 3 or 4 of the Rules
Load Comp (1)	LOADING COMPUTER (1)	Denotes that the ship is equipped with a loading computer capable of ascertaining and comparing with the permissible limits, the hull girder loads in still water in any loading condition as required by Pt.3, Ch.5 and Pt.5, as applicable, of the Rules
Load Comp (2)	LOADING COMPUTER (2)	Denotes that in addition to the requirements for Load Comp (1) mentioned above, the ship's loading computer is capable of performing intact stability calculations for any loading condition to verify compliance with the applicable requirements indicated in Pt.3, Ch.1 of the Rules
Load Comp (3)	LOADING COMPUTER (3)	Denotes that in addition to the requirements for Load Comp (2) mentioned above, the ship's loading computer is capable of performing the stability calculations in damaged condition

Abbreviation	Expanded Form	Significance
Load Comp (4)	LOADING COMPUTER (4)	Denotes that in addition to the requirement for Load Comp (2) mentioned above, the ship's loading computer is capable of performing both stability and strength calculations in damaged condition
INWATER SURVEY		Denotes that the examination of the ship's bottom and related items may be carried out while the ship is afloat in accordance with the applicable requirements indicated in Pt.1, Ch.2, Sec.7.2
EDD (X)		Denotes that the vessel complies with the requirements of the IRS Classification Note “Extended Dry-Docking Scheme”, and is fit for service with an extended interval of ‘X’ years between successive dry-dockings.
"For carriage of cement in bulk"		This will be entered in the Register of Ships for vessels which are designed and constructed solely for carriage of cement in bulk and the scantlings and arrangements have been approved accordingly
DWA		This notation will be assigned to ships complying with the requirements of Pt.3, Ch.15, Sec.8 for anchoring in deep and unsheltered waters.
CLS	CONTAINER LASHING SOFTWARE	This additional notation will be assigned to Container Ships provided with IRS approved container lashing software in accordance with Part 5, Chapter 5 of the Rules.
CM	CONSTRUCTION MONITORING	This additional class notation will be assigned to ships, whose quality of construction is closely monitored by IRS in accordance with the relevant requirements of Pt. 3, Ch. 18.
Class Notations - Machinery		
SPM	SINGLE POINT MOORING	Denotes that the vessel is fitted with standard arrangements for single point mooring as per Pt.5, Ch.28 of the Rules
DOUBLE HULL		When bulk carriers, oil tankers, oil or ore carriers and oil or bulk carriers are of double hull construction, in the entire cargo area (See Ch.2, Sec.1), the main class notations described above will be appended by notation ‘DOUBLE HULL’
CCS	CENTRALIZED CONTROL STATION	Denotes that the propulsion and auxiliary machinery can be controlled and monitored with continuous supervision from a Centralized Control Station as detailed in Pt.4, Ch.7. It also denotes that the control engineering equipment has been arranged, installed and tested in accordance with Rules.
SYJ	SWACHALIT YANTRIK JAHAZ	Denotes that the ship can be operated with the machinery spaces unattended in accordance with the applicable requirements of Part 5, Chapter 22 of the Rules
HY	HIMIKAR YANTRA	Denotes that the refrigerated cargo installation of a ship complies with the applicable requirements of Part 5, Chapter 23 of the Rules

Abbreviation	Expanded Form	Significance
HY*	HIMIKAR YANTRA*	Denotes that the refrigerated cargo installation is capable of cooling down fruit in general or a catch of fish and complies with the applicable requirements of Part 5, Chapter 23 of the Rules
HY(LGC)	HIMIKAR YANTRA (LGC)	Denotes that the reliquefaction or refrigeration installation fitted on a classed liquefied gas carrier for cargo temperature and pressure control complies with the applicable requirements of Part 5, Chapter 4 of the Rules
NV	NIPRABHAV VASHPA	This notation is assigned to ships classed with Indian Register of Shipping intended for (i) carriage of oil in bulk or (ii) carriage of liquid chemicals in bulk. It denotes that the cargo spaces can be inerted by means of an inert gas system which has been approved, installed and tested in accordance with the relevant Rules
AGNI 1	AGNI SHAMAK	Denotes that the ship is equipped for early stage fire fighting and rescue operations close to structures including means for self protection of the vessel
AGNI 2		Denotes that the ship is equipped for continuous fighting of large fires and for cooling of structures on fire including means for self protection of the vessel
AGNI 3		Denotes that the ship is equipped for continuous fighting of large fires and for cooling of structures on fire and of greater capacity than for "AGNI 2"
DP (0)	DYNAMIC POSITIONING (0)	Denotes that the ship is fitted with automatic controls for position keeping and/or heading without joystick system back-up as per Pt. 5, Ch. 24 of the Rules
DP (1)	DYNAMIC POSITIONING (1)	Denotes that the ship is fitted with automatic controls for position keeping and/or heading, an independent joystick system back-up and a position reference back-up as per Pt.5, Ch.24 of the Rules
DP (2)	DYNAMIC POSITIONING (2)	Denotes that the ship is fitted with automatic controls of position keeping and/or heading with automatic standby controls and redundancy in design and equipment as required by Pt.5, Ch.24 of the Rules
DP (3)	DYNAMIC POSITIONING (3)	Denotes that the ship is fitted with automatic controls for position keeping and/or heading with automatic standby controls, redundancy in design and equipment and physical separation of components in different compartments as per Pt.5, Ch.24 of the Rules
BD	Bridge Design	The additional class notation ‘BD’ will be assigned to ships which comply with the requirements of ergonomic bridge design and layout in accordance with Pt.5 Ch.31 of the Rules
BDS	Bridge Design and Systems	The additional class notation ‘BDS’ will be assigned to ships which comply with the requirements of ergonomic bridge design and layout, navigation equipment and related systems in accordance with Pt.5 Ch.31 of the Rules

Abbreviation	Expanded Form	Significance
IBS	Integrated Bridge System	The additional class notation 'IBS' will be assigned to ships fitted with an Integrated Bridge System in accordance with the requirements given in Pt.5, Ch.31 of the Rules. Integrated Bridge System allows simplified and centralized bridge operation of the main functions of navigation, manoeuvring and communication, as well as monitoring from the bridge of other functions such as: - Passage execution - Communication system - Monitoring of the machinery installation - Pollution monitoring - Monitoring of HVAC for passenger ships - Safety and security.
DSV	DIVING SUPPORT VESSEL	Denotes that the ship has diving support capability and is equipped with diving systems in accordance with the requirements of Pt. 5, Ch. 26 of the Rules.
DIV-SURF	DIVING SYSTEMS-SURFACE	The additional notation DIV-SURF will be assigned to surface diving systems in accordance with Pt. 5, Ch. 26 of the Rules.
DIV-SAT	DIVING SYSTEMS-SATURATION	The additional notation DIV-SAT will be assigned to saturation diving systems in accordance with Pt. 5, Ch. 26 of the Rules.
PMS	PLANNED MAINTENANCE SYSTEM	Denotes that the machinery is subject to a system of planned maintenance and surveys in accordance with Part 1, Chapter 2, Section 14 of the Rules
PMS-CM	PLANNED MAINTENANCE SYSTEM – CONDITION MONITORING	Denotes that the machinery is subject to condition monitoring as part of the planned maintenance system and surveys in accordance with Part 1, Chapter 2, Section 21 of the Rules
PMS-CBM	PLANNED MAINTENANCE SYSTEM – CONDITION BASED MAINTENANCE	Denotes that the machinery is subject to condition based maintenance as part of the planned maintenance system and surveys in accordance with Part 1, Chapter 2, Section 21 of the Rules
TCM	TAILSHAFT CONDITION MONITORING	Denotes that the tail shaft condition is regularly monitored based on service records, oil and fresh water sample analysis, as applicable, in accordance with the survey requirements of Pt.1, Ch.2 and arrangements are provided for this purpose in accordance with Pt.4, Ch.4 of the Rules.
FT		This notation will be assigned to ships fitted with fuel oil treatment system and equipment complying with the relevant requirements of IRS Classification Note "Fuel Oil Treatment System".
BATTERY PROP	BATTERY PROPULSION	The additional notation BATTERY PROP will be assigned to vessels where the battery systems are used for propulsion and are in accordance with the *'Guidelines on Battery Powered Vessels'*.

Abbreviation	Expanded Form	Significance
LFPF	LOW FLASH POINT FUEL	This additional class notation will be assigned to a vessel, which is designed and constructed primarily for using low flash point fuel, in accordance with the applicable requirements of Part 5, Chapter 35 of the Rules. The following additional qualifiers may be assigned to the vessel depending upon the type of fuel or engine being used : (NG) – Natural Gas (EG) – Ethane (LPG) – Liquefied Petroleum Gas (ML) – Methanol (EL) - Ethanol (AM) - Ammonia (SFE) – Single Fuel Engines (DFE) – Dual Fuel Engines
LFPF BUNKERING VESSEL ([fuel name]) LFPF BUNKERING BARGE ([fuel name])		This additional class notation will be assigned to a vessel, which is designed and constructed in accordance with the applicable requirement of Part 5, Chapter 14, and primarily intended for carrying low flash point fuel to transfer to ships using such fuel. 'LFPF BUNKERING BARGE ([fuel name])' will be assigned to nonself- propelled vessels. The following qualifiers may be assigned to the vessel carrying LNG bunker and equipped with additional features complying with the relevant requirements in Section 2 of the Chapter: RE - where the vessel is designed to receive LNG from a gas fueled vessel whose LNG fuel tanks have to be emptied. ICD - where the vessel is designed for initial cooling down of the gas fueled vessel's LNG fuel tank. IGS - where the vessel is designed to supply inert gas and dry air, to ensure gas freeing and aeration, to a gas fueled vessel complying with Pt.5, Ch.35, 6.10.4. VR - where the vessel is designed to recover and manage the boiloff gas generated during the bunkering operation.
FC (Main) FC (Auxiliary)	Fuel Cell (Main) Fuel Cell (Auxiliary)	Vessels provided with fuel cell installations for propulsive power and auxiliary purposes in accordance with IRS Guidelines on Vessels with Fuel Cell Power Installations (Provisional), may be assigned additional class notation FC (Main) and FC (Auxiliary) respectively. A qualifier indicating the fuel (e.g. methanol, ethanol, hydrogen, etc) used in the fuel cell will be added to the notation, e.g: FC (Main) – Hydrogen.

Abbreviation	Expanded Form	Significance
RV(LNG)	REGASSIFICATI ON VESSEL (LIQUIEFIED NATURAL GAS)	This notation will be assigned to LNG Carriers fitted with regasification systems, complying with the requirements of Part 5, Chapter 38 of the Rules. These ships are also to comply with the applicable requirements of Part 5, Chapter 4 for liquefied gas carriers.
Pollution Prevention Notations		
VCS1 VCS2 VCS1-B VCS2-B	VAPOUR CONTROL SYSTEM	The additional notation VCS1 will be assigned to oil tankers, combination carriers (ore or oil/oil or bulk) and chemical tankers equipped with vapour control system complying with the requirements of Part 5, Chapter 29 of the Rules, except those given in 3.2.2 and Section 4 of that Chapter. The additional notation VCS2 will be assigned to oil tankers, combination carriers (ore or oil/oil or bulk and chemical tankers equipped with vapour control system meeting the requirements of USCG's regulations CFR46 Part 39 and those in Part 5, Chapter 29 of the Rules, except Section 4 of that Chapter. Vessels complying with requirements of Section 4 "Vapour Balancing" of Part 5, Chapter 29, in addition to other requirements in that Chapter will be eligible to be assigned class notation 'VCS1-B' or 'VCS2-B', as relevant and applicable.
CLEAN-SEA		The additional class notation CLEAN-SEA is assigned to ships provided with construction and procedural means to prevent pollution of sea. This is achieved by complying with the applicable requirements of Annexes I, II, III, IV, V of MARPOL Convention, as well as additional requirements related to prevention of sea pollution as follows: • Prevention of accidental pollution by locating fuel and lub.oil tanks above the double bottom and away from side shell. • Prevention of operational pollution by means of bilge water separation and filtering, holding tanks for treated sewage and grey water • Prevention of transfer of harmful organisms and pathogens in the ballast water • Prevention of pollution by tributyltin by means of TBT free antifouling paints • Prevention of pollution by solid garbage (resulting from the compacting device and incinerators) by means of proper storage of such waste The requirements for the assignment of this notation are given in Part 5, Chapter 30 of Rules.

Abbreviation	Expanded Form	Significance
CLEAN-AIR		The additional class notation CLEAN-AIR is assigned to ships provided with construction and procedural means to prevent pollution of the air. This is achieved by complying with the applicable requirements of Annex VI of MARPOL convention as well as following additional requirements related to emissions to the air: • Prevention of air pollution by exhaust gas (NOx, SOx, etc.) by means of low emission engines • Use of low sulphur content fuels and incinerators • Use of refrigerants and fixed fire fighting means with zero ozone depleting potential and low global warming potential • Control of release of refrigerants to the atmosphere by means of leak detection and evacuation systems • Recovery of vapours emitted from cargo systems of ships carrying dangerous liquid cargoes in bulk Note 1 : For ships with the service notation of oil tanker, oil or bulk carrier, ore or oil carrier, chemical tanker or liquefied gas carrier, the assignment of the notation VCS1 or VCS2 is a pre-requisite for assigning the notation CLEAN-AIR. The requirements for the assignment of this notation are given in Part 5, Chapter 30 of Rules.
EP	Environmental Protection	The additional class notation EP is assigned to ships provided with equipment and procedural means to prevent pollution of the sea and of air. This is achieved by complying with the requirements of Annexes I to VI of MARPOL Convention and additional requirements relevant to ship's liquid, solid and gas releases as detailed in Part 5, Chapter 30 of Rules. Additionally the vessel should have developed an Environmental Management Plan considering the IMO requirements for recycling. Note 1 : For ships with the service notation of oil tanker, oil or bulk carrier, ore or oil carrier, chemical tanker or liquefied gas carrier, the assignment of the notation VCS1 or VCS2 is a prerequisite for assigning the notation Environment Protection – EP. The requirements for the assignment of this notation are given in Part 5, Chapter 30 of Rules.
BFM	Biofouling Management	Ships complying with the requirements in IRS '*Guidelines on Biofouling Management*' will be assigned additional class notation BFM.

Ballast Water Convention		
Abbreviation	**Expanded Form**	**Significance**
BWE ( )	BALLAST WATER EXCHANGE	The additional class notation BWE is assigned to ships complying with the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, the relevant IMO Guidelines, and performing ballast water exchange in accordance with Part 5, Chapter 37 of the Rules.
BWT	BALLAST WATER TREATMENT	The additional class notation BWT is assigned to ships fitted with an IMO Member State approved ballast water treatment system complying with the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, the relevant IMO Guidelines and Part 5, Chapter 37 of the Rules.

Ship Type Notation	
Ship Type Notation	**Significance**
ASPHALT CARRIER (INTEGRAL TANK, MAX. CARGO TEMP ≤ XXX℃),	This additional notation will be assigned to oil tankers intended primarily to carry asphalt/ bitumen in integral cargo tanks in accordance with the applicable requirements of Part 5, Chapter 2 of the Rules. Such vessels will also be assigned notation ESP.
ASPHALT CARRIER (INDEPENDENT TANKS, MAX. CARGO TEMP ≤ XXX℃),	This additional notation will be assigned to oil tankers intended primarily to carry asphalt/ bitumen in independent cargo tanks in accordance with the applicable requirements of Part 5, Chapter 2 of the Rules. Structural analysis for such vessels is to be carried out in accordance with the requirements of '*Guidelines on Structural Assessment of Ships carrying Asphalt in Independent Cargo Tanks'*
BULK CARRIER	This notation will be assigned to ships designed primarily for the carriage of dry cargo in bulk and built in accordance with the applicable requirements of Part 5, Chapter 1 of the Rules for carriage of cargoes of density of at least 0.8 t/m^3
BULK CARRIER(BC-XII)	This notation will be assigned to Bulk carriers which are in accordance with the definition of SOLAS Ch XII and its construction features not being as per the definition of SOLAS Ch IX, Reg 1.6. (i.e. not fitted with hopper and topside tanks)
SELF UNLOADER	This notation will be assigned to Bulk carriers with single deck, double bottom, hopper side tanks and topside tanks and with single or double side skin construction in cargo length area and intended to carry and self-unload dry cargoes in bulk.
ORE CARRIER	This notation will be assigned to ships specially designed primarily to carry ore and built in accordance with applicable requirements of Part 5, Chapter 1 of the Rules
OIL TANKER	This notation will be assigned to tankers intended primarily to carry oil in bulk and built in accordance with applicable requirements of Part 5, Chapter 2 of the Rules. Where the scantlings and arrangements have been approved for the carriage of oil having a flash point of 60°C or above (closed cup test) or other liquid cargoes in bulk, the class notation will be suitably modified to show the nature of the cargo

Ship Type Notation	Significance
SHUTTLE TANKER	This notation will be assigned to an oil tanker which is equipped with dynamic positioning (in accordance with the applicable requirements of Part 5, Chapter 24) and specialized cargo handling equipment making it capable of loading crude oil at offshore installations (in accordance with Part 5, Chapter 29).
ORE OR OIL CARRIER OIL OR BULK CARRIER	These notations will be assigned to combination carriers specially designed primarily to carry dry bulk cargoes (bulk carriers and ore carriers) and alternatively, oil in bulk, in accordance with Pt.5, Ch.2, Sec.12 of the Rules. In such ships, simultaneous carriage of dry bulk cargo and oil is prohibited.
CHEMICAL TANKER	This notation will be assigned to ships specially designed primarily to carry chemicals in bulk and built in accordance with applicable requirements of Part 5, Chapter 3 of the Rules
LIQUEFIED GAS CARRIER	This notation will be assigned to ships specially designed for the carriage of liquefied petroleum, natural or other gases and built in accordance with applicable requirements of Part 5, Chapter 4 of the Rules
CONTAINER SHIP	This notation will be assigned to vessels built for the exclusive carriage of containers in holds and on deck in accordance with the applicable requirements of Part 5, Chapter 5 of the Rules
PASSENGER SHIP	This notation will be assigned to vessels intended for carrying more than 12 passengers and built in accordance with Part 5, Chapter 6 of the Rules
FERRY	This notation will be assigned to vessels intended for carriage of passengers and/or vehicles on regular scheduled service and built in accordance with Part 5, Chapter 6 of the Rules
Ro-Ro FERRY TRAIN FERRY	This notation will be assigned to ferries intended for carriage of passengers, vehicles and cargo in pallet form or in containers and loaded/unloaded by wheeled vehicles and built in accordance with Part 5, Chapter 6 of the Rules
TUG	This notation will be assigned to all ships built in accordance with applicable requirements of Part 5, Chapter 7 of the Rules
STERN TRAWLER	This notation will be assigned to ships built in accordance with applicable requirements of Part 5, Chapter 9 of the Rules and engaged in stern trawling
TRAWLER	This notation will be assigned to ships built in accordance with applicable requirements of Part 5, Chapter 9 of the Rules and engaged in side trawling
FISHING VESSEL	This notation will be assigned to ships built in accordance with applicable requirements of Part 5, Chapter 9 of the Rules and not equipped with trawling gear
DREDGER, HOPPER DREDGER, RECLAMATION CRAFT, HOPPER BARGE, SPLIT HOPPER BARGE	These notations will be assigned to self propelled or non self propelled vessels engaged in dredging or reclamation operation in accordance with applicable requirements of Part 5, Chapter 10 of the Rules

Ship Type Notation	Significance
BARGE	This notation will be assigned to non self- propelled, manned or unmanned ships carrying dry cargo in cargo holds and built in accordance with applicable requirements of Part 5, Chapter 11 of the Rules. For special purpose vessels, the Notation will be suitably modified, e.g. Shipborne Barge
OIL BARGE	This notation will be assigned to non self- propelled, manned or unmanned ships intended to carry oil in bulk and built in accordance with the applicable requirements of Part 5, Chapters 2 & 11 of the Rules. Where the scantlings and arrangements have been approved by IRS for the carriage of oil having a flash point of 60°C or above (closed cup test), or for other liquid cargoes in bulk, the class notation affixed to the Character will be suitably modified to show the nature of the cargo (e.g. water barge, molasses barge, etc.)
PONTOON	This notation will be assigned to non self- propelled, manned or unmanned ships designed specifically for the carriage of non-perishable cargo or equipment on deck and built in accordance with the applicable requirements of Part 5, Chapter 11 of the Rules. For special purpose vessels, the Notation will be suitably modified, e.g. Crane Pontoon
FLOATING DOCK	This notation will be assigned to floating dry docks built in accordance with the applicable requirements of Part 5, Chapter 12 of the Rules.
WELL STIMULATION VESSEL	This notation will be assigned to well stimulation vessels built in accordance with the applicable requirements of Part 5, Chapter 33.
OIL RECOVERY VESSEL	This notation will be assigned to oil recovery vessels built in accordance with the applicable requirements of Part 5, Chapter 34.
Ro-Ro CARGO SHIP	This notation will be assigned to a vessel designed and constructed to carry roll-on/roll-off cargoes, equipped for carriage of vehicles, on exposed or enclosed single deck or multiple exposed/ enclosed decks. The requirements of Part 3, Chapters 9 & 12 of the Rules are to be complied with for decks with wheel loading and opening and closing appliances (bow doors) respectively. The requirements of Part 6 are also to be complied with for fire safety requirements of ro-ro spaces and vehicle spaces.
SPECIAL PURPOSE SHIP	This notation will be assigned to vessels that comply with the requirements in the IMO Code of Safety for Special Purpose Ships (SPS Code).
MOORED OIL STORAGE TANKER at (Location)	This notation would be assigned to an existing Double Hull Oil Tanker that is intended to be used as a storage vessel moored at a specified fixed location, with the IY notation either maintained or suspended and the ESP notation suspended.
LIVESTOCK CARRIER	This notation will be assigned to vessels that comply with the requirements in Part 5, Chapter 40 of the Rules.

Ship Type Notation		**Significance**
OFFSHORE SUPPORT VESSEL		This notation will be assigned to all ships built in accordance with applicable requirements of Part 5, Chapter 8 of the Rules.
MULTIPURPOSE OFFSHORE SUPPORT VESSEL		Vessels intended for several roles/services related to offshore support will be assigned notation MULTIPURPOSE OFFSHORE SUPPORT VESSEL
	SUPPLY	This additional notation will be assigned to Offshore Support Vessels engaged in the transport of stores, materials, equipment and/or personnel to, from and between offshore installations. This notation is not applicable to Crew boats.
	SUPPLY-PLUS	This additional notation will be assigned to Offshore Support Vessels designed for operating in North Sea, engaged in the transport of stores, materials, equipment and/or personnel to, from and between offshore installations. This notation is not applicable to Crew boats.
	STS	This additional notation will be assigned to Offshore Support Vessels complying with the requirements for Survival Capability (Floatability in damaged condition) of Ch.3 of the IMO Resolution MSC 235(82).
	LHNS	This additional notation will be assigned to Offshore Support Vessels complying with the requirements of Pt.5, Ch.8, Sec. 2.8, intended to carry limited amounts of hazardous and noxious liquid substances in bulk.
	OCC	This additional notation will be assigned to Offshore Support Vessels complying with Pt. 5, Ch.8, Sec. 2.8; and intended to transport, handle hazardous and noxious liquid substances in bulk in compliance with the OSV Chemical Code (IMO Resolution A.1122 (30)).
	AH	This additional notation will be assigned to Offshore Support Vessels equipped for anchor handling operations and complying with the requirements of Pt.5, Ch.8, Sec.3.
	TOW	This additional notation will be assigned to Offshore Support Vessels equipped for towing operations and complying with the requirements of Pt.5, Ch.7
	SSR	This additional notation will be assigned to Offshore Support Vessels equipped to carry out rescue and standby services for offshore installations and complying with the requirements of Pt.5, Ch.8, Sec.4
	AGNI 1 AGNI 2 AGNI 3	These additional notations will be assigned to Offshore Support Vessels equipped for firefighting operations and complying with the relevant requirements of Pt.5, Ch.25.
	DSV	This additional notation will be assigned to Offshore Support Vessels equipped for providing diving support operations and complying with the relevant requirements of Pt.5, Ch.26
	WSV	This additional notation will be assigned to Offshore Support Vessels equipped for performing well stimulation operations and complying with the relevant requirements of Pt.5, Ch.33
	OIL RECOVERY	This additional notation will be assigned to Offshore Support Vessels equipped to recover oil spills from sea at a safe distance, handle, store and transport the recovered oil, and complying with the relevant requirements of Pt.5, Ch.34

Ship Type Notation		Significance
	WFSV	This additional notation will be assigned to Offshore Support Vessels equipped to serve and support wind farm service activities and complying with the requirements of Pt. 5, Ch. 8, Sec. 5.
	WFSV – HSC	This additional notation will be assigned to wind farm service vessels which comply with the requirements applicable to high speed crafts, as given in the Rules and Regulations for Construction and Classification of High Speed Crafts and Light Crafts, in addition to the requirements of Pt. 5, Ch. 8, Sec. 5.
PASSENGER SUBMERSIBLE WITH MAX. OPERATING DEPTH OF ___[m]		This notation will be assigned to submersibles designed and constructed for carriage of passengers (normally tourists) in accordance with MSC/Circ.981. The maximum safe operating depth of the submersible is to be indicated in the notation.
PIPE LAYING VESSEL		This notation will be assigned to vessels that are primarily engaged in installation of pipelines and are in compliance with Part 5 Chapter 13.
PIPE LAYING BARGE		This notation will be assigned to non self-propelled pipe laying vessels complying with the requirements for the class notation BARGE (i.e. Part 5, Chapter 11) and which comply with the relevant requirements of Part 5 Chapter 13
CABLE LAYING VESSEL		This notation will be assigned to vessels that are primarily engaged in installation of cables and are in compliance with Part 5 Chapter 13.
CABLE LAYING BARGE		This notation will be assigned to non self-propelled cable laying vessels complying with the requirements for the class notation BARGE (Part 5, Chapter 11) and which comply with the relevant requirements of Part 5 Chapter 13
ESCORT (Fs, Fb, V)		This notation will be assigned to tugs mainly intended for escort services such as for steering, braking and assisting in the maneuvering of escorted ships. The values of design maximum steering force (Fs), design maximum braking force (Fb), and design maximum escort speed (V), are to be supplemented to the class notation.
ACCOMMODATION BARGE ACCOMMODATION PONTOON		These notations will be assigned to non self-propelled vessels used for accommodation of personnel engaged in offshore work on other vessels, rigs, platforms etc. For ACCOMMODATION PONTOONS, all accommodation/storage/auxiliary machinery spaces are provided above main deck. For ACCOMMODATION BARGES, spaces below the deck may also be used for accommodation / storage / auxiliary machinery etc. In addition to the rule requirements for assignment of class notations 'BARGE'/ 'PONTOON', these vessels are to additionally comply with the requirements applicable to vessels carrying industrial personnel, as specified by the Flag Administration. Additional class notations for comfort, station-keeping, etc., may be provided at the request of the owner.

Passenger and Crew Comfort Notations		
Abbreviation	**Expanded Form**	**Significance**
CMF(Px) CMF(Px-c) CMF(Px-l) CMF(Px-c-l)	PASSENGER COMFORT	Denotes that passenger spaces have been evaluated and certified in accordance with the applicable requirements of Part 5, Chapter 39 of the Rules, for passenger comfort. '**x**' denotes the comfort rating related to noise and vibration in passenger spaces. Ships which additionally comply with the requirements for indoor climate and/ or lighting in passenger spaces will be assigned additional qualifiers '**c**' and/ or '**l**', respectively.
CMF(Cy) CMF(Cy-c) CMF(Cy-l) CMF(Cy-c-l)	CREW COMFORT	Denotes that crew spaces have been evaluated and certified in accordance with the applicable requirements of Part 5, Chapter 39 of the Rules, for crew comfort. '**y**' denotes the comfort rating related to noise and vibration in crew spaces. Ships which additionally comply with the requirements for indoor climate and/ or lighting in crew spaces will be assigned additional qualifiers '**c**' and/ or '**l**', respectively.
CMF(Px, Cy) CMF(Px-c-l, Cy-c-l) etc.	PASSENGER AND CREW COMFORT	Denotes that both passenger and crew spaces have been evaluated and certified in accordance with the applicable requirements of Part 5, Chapter 39 of the Rules, for passenger and crew comfort. '**x**' and '**y**' denote the comfort rating related to noise and vibration for passenger spaces and crew spaces respectively. Ships which additionally comply with the requirements for indoor climate and/ or lighting will be assigned additional qualifiers '**c**' and/ or '**l**', respectively.
Cyber Safety Notations		
CyS-I	INFORMED CYBER SAFETY	Vessels complying with the applicable requirements of 'Guidelines on Maritime Cyber Safety' may be assigned this notation. This notation is assigned where the organisation assesses the issues related to cyber safety, cyber risks and have documented policies. The organization implements minimum cyber safety practices.
CyS-E	ESSENTIAL CYBER SAFETY	The additional class notation CyS-E will be assigned to vessels that are in compliance with the requirements for essential cyber safety specified in Part 4, Chapter 7, Section 7 of the Rules.
CyS-II	ADVANCED CYBER SAFETY	This notation will be assigned to vessels which comply with the requirements for CyS-E notation and additionally also comply with the applicable requirements of 'Guidelines on Maritime Cyber Safety'. This notation is assigned where the cyber safety policies and procedures are reviewed periodically for continual improvement. The organisation implements advanced cyber safety practices.

Class Notation : General		
Abbreviation	**Expanded Form**	**Significance**
ADE	ALTERNATIVE DESIGN EVALUATION	The additional class notation ADE will be assigned to vessels where alternative design is used in lieu of complying with the prescriptive requirements of the Rules. The relevant details, aspects where alternative design is used would be suitably indicated in the Class Certificate. Alternative design approach may be used in accordance with the '*Guidelines on Alternative and Risk Based Design Evaluation'.*
RA	RISK ASSESSMENT	The additional class notation RA will be assigned to vessels where the design is risk based and the same has been accepted based on risk assessment. However, if risk based design approach has been used for a particular aspect, then the relevant details, aspects would be suitably indicated in the Class Certificate. Risk based design approach may be used in accordance with the '*Guidelines on Alternative and Risk Based Design Evaluation'.*
SFA ( )	SPECTRAL FATIGUE ANALYSIS (NUMBER OF YEARS)	The additional class notation SFA ( ) will be assigned to vessels where the design fatigue life has been assessed in accordance with the *'Guidelines on Spectral Fatigue Analysis of Ship Structures'*. The figures in parentheses would indicate the number of years for which the design fatigue life has been assessed.
HVSCS	HIGH VOLTAGE SHORE CONNECTION SYSTEMS	The additional class notation HVSCS would be assigned to vessels designed to be powered with shore power alone at port/ berth in accordance with the '*Guidelines on High Voltage Shore Connection Systems' (applicable to voltages > 1 kV)*.
DSA	DIRECT STRENGTH ASSESSMENT	The additional class notation DSA will be assigned to vessels where a direct strength assessment has been carried out in accordance with the 'Guidelines on Application of Direct Seakeeping Loads in Structural Analysis of Ships' and the 'Guidelines on Structural Assessment of Ships based on Finite Element Method'.
FDA	FATIGUE DESIGN ASSESSMENT	The additional class notation FDA will be assigned to vessels where a fatigue design assessment has been carried out in accordance with the relevant requirements of the Rules and the 'Guidelines on Fatigue Design Assessment of Ship Structures'. FDA notation is mandatory for ships of length greater than or equal to 150 m and may be optionally applied for ships of length greater than 90 m. If fatigue assessment is carried out for vessels for a number of years >25 and/ or restricted service, then the class notation FDA (N, specific area of operation), would be assigned; where 'N' denotes the number of years.
ERS	EMERGENCY RESPONSE SERVICE	This additional class notation will be assigned to ships which have opted for the emergency response service of IRS, under a prior agreement with the Ship Owner / Ship Manager. IRS will provide a computer assisted analysis of a damaged ship's stability and longitudinal strength in the event of a casualty to the ship.

Abbreviation	Expanded Form	Significance
REMOTE SURVEY		This additional class notation will be assigned to ships which are in compliance with the relevant requirements of Pt. 1, Ch. 2, Sec. 24 for remote surveys

End of Chapter

Chapter 2

Periodical Surveys

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Section 1

General Requirements

1.1 General

1.1.1 All vessels are to be subjected to Periodical Surveys for the purpose of maintenance of class. Survey notations and Survey intervals are given in Table 1.1.1 for main class Surveys. Where additional class notations have been assigned, Surveys are to be carried out at intervals given in Table 1.1.2.

1.1.2 Vessels with additional class notations for which there are no specific Survey requirements defined in this Chapter are to have the equipment and/or construction related to this additional class notation examined to the Surveyor's satisfaction at each Special Survey. However, at the time of continued effectiveness, of operational features, safety devices and control systems are to be verified.

1.1.3 Annual, intermediate and special surveys are to be held concurrently with the relevant statutory surveys under the various IMO conventions, in accordance with the Harmonised System of Survey and Certification of IMO. For cargo ships not subject to enhanced surveys under SOLAS Ch. XI-1/2, notwithstanding any other provisions, the intermediate and renewal surveys may be carried out and completed over the corresponding periods as specified in the 2011 ESP Code, as amended.

1.1.4 Special consideration may be given in the application of relevant requirements of this chapter to commercial vessels owned or chartered by Governments, which are utilized in support of military operations or service.

Table 1.1.1 : Periodical survey intervals for class survey notations

(Any specific requirements of the flag Administration are also to be complied with)

Survey	Class survey notation	Survey interval in years
Hull : Special Survey	SSH	5
Hull : Continuous Survey	CSH	5[5]
Machinery : Special Survey	SSM	5
Machinery : Continuous Survey	CSM	5
Intermediate Survey	IS	See Note 4, 9
Annual Survey	AS	1[1, 9,10]
Docking Survey	DS	2.5[2]
Shaft Survey		
Oil Lubricated	SH (OL)	
Fresh Water Lubricated Closed Loop System	SH (FW-C)	
Single Shaft Fresh Water Lubricated Open System	SH (S-FW-O)	
Single Shaft Corrosion Protected or Corrosion Resistant Material Water Lubricated Open System	SH (S-CP-O)	
Multiple Shaft, Water Lubricated Open System	SH (M-O)	See Note 3, 7
Single Shaft, Water Lubricated Open System	SH (S-O)	
Tailshaft Condition Monitoring	TCM	See Note 8

Table 1.1.1 (Contd.)		
Survey	**Class survey notation**	**Survey interval in years**
Directional propellers, water jet units, or athwartship thrust propellers	DPROP	5[3]
Main boilers	MBS	2.5[6]
Auxiliary boilers	ABS	2.5[6]
Thermal oil heating systems	TOH	2.5[6]
Exhaust gas steam generators and economisers	EGES	2.5[6]

Notes:

1 Survey is to be carried out within 3 months on either side of the anniversary date.

2 At least 2 Surveys are to be carried out within any 5 year special survey cycle, but the interval between two consecutive surveys is not to exceed 3 years and one of these two docking surveys should coincide with the Special Survey. Proposals for alternative means for providing underwater inspection equivalent to drydocking survey would be considered by IRS as detailed in Sec.7. However, for vessels operating solely in fresh-water only one docking survey coinciding with the special survey need be carried out till special survey no. IV provided the interval between consecutive docking surveys does not exceed 5 years. Survey requirements for floating dry docks are indicated in Section 23.

3 Upon request, IRS may extend the survey period to harmonise with docking survey and in accordance with the requirements of Section 11.

4 Survey is to be carried out either at or between the second or third Annual Survey. Those items, which are additional to the requirements of the Annual Surveys, may be surveyed either at or between the second and third Annual Survey.

5 At the request of the Owner, IRS may agree that the special survey of the hull, for ships other than bulk carriers, oil tankers, combination carriers, chemical tankers, liquefied gas carriers and general dry cargo ships, be carried out on the continuous basis.

6 At least 2 surveys are to be carried out within any 5 year special survey cycle. The interval between two consecutive surveys is not to exceed 3 years.

7 See Tables 11.2.3 and 11.3.3 for survey intervals of closed and open systems respectively.

8 Condition monitoring records are to be verified at the time of annual surveys. Survey intervals are to be as per Table 11.2.3.

9 If an annual / intermediate survey is completed before the period specified in Note 1 / 4 :

(a) The anniversary date shown on the class certificate will be amended by endorsement to a date which is not more than 3 months later than the date on which the survey was completed.

(b) The subsequent annual / intermediate surveys are to be completed at the prescribed intervals using the new anniversary date.

(c) The expiry date of the class certificate may remain unchanged provided one or more annual / intermediate surveys, as appropriate, are carried out so that the prescribed maximum intervals between the surveys are not exceeded.

10 (a) Annual surveys are not applicable to floating dry docks.

(b) Survey requirements for floating dry docks are indicated in Section 23.

Table 1.1.2 : Periodical survey intervals for additional class notations

Survey of	Additional class notation / corresponding class survey notation	Survey interval in years
Planned Maintenance System of Machinery	**PMS** **PMS-CM** **PMS-CBM**	See Note 5
Annual Audit	AA(PMS) AA (PMS-CM) AA (PMS-CBM)	1[3,4,6]
Diving support vessels	**DSV**	
Annual Survey Special Survey	AS(DSV) SS(DSV)	1[1] 5
Diving system	**DIV-SAT, DIV-SURF**	
Annual Survey Special Survey	AS(DIV) SS(DIV)	1[1] 5
Cargo refrigerating plant	**HY**	
Annual Survey Special Survey Continuous Survey	AS(HY) SS(HY) CS(HY)	1[1] 5 5
Dynamic positioning system	DP(1), (2) and (3)	
Annual Survey Special Survey	AS(DP) SS(DP)	1[1] 5
External fire fighting	**AGNI 1, 2 & 3**	
Annual survey Special survey	AS(AGNI) SS(AGNI)	1[1] 5
Inert gas installation	**NV**	
Annual Survey Special Survey	AS(NV) SS(NV)	1[1] 5
Un-attended Machinery Spaces	**SYJ**	
Annual Survey Special Survey	AS(SYJ) SS(SYJ)	1[1] 5

Notes:

1 May be carried out within 3 months on either side of anniversary date.

2 Survey may be carried out either at or between the second or third annual survey.

3 To be carried out in conjunction with annual survey.

4 See Section 14 for details of Planned Maintenance System

5 PMS denotes vessel with approved Planned Maintenance System (PMS) only. PMS-CM denotes vessel with approved PMS and Condition Monitoring (CM). PMS-CBM denotes vessel with approved PMS, CM and carrying out Condition Based Maintenance (CBM).

6 See Section 21 for details of CM and CBM systems. In general, requirements for CM/ CBM can be applied only to vessels operating on approved PMS survey scheme.

1.2 Definitions

1.2.1 A **Ballast tank** is a tank which is being used primarily for salt water ballast.

In the case of oil tankers and chemical tankers, combined cargo/ballast tanks which are used for the carriage of cargo or ballast water as a routine part of the vessel's operation, are to be treated as ballast tanks. Cargo tanks in which water may be carried only in exceptional cases as per MARPOL I/18(3), are to be treated as cargo tanks. In any case, when substantial corrosion is found in a tank used for both cargo and ballast, the tank is to be considered as a ballast tank for the purpose of application of survey requirements

In the case of bulk carriers, a space used for both cargo and salt water ballast is to be treated as a ballast tank for the purpose of application of survey requirements when substantial corrosion is found in that space. Further in the case of double skin bulk carrier a double side tank is to be considered as a separate tank even if it is in connection to either the topside tank or the hopper side tank.

1.2.2 A **Close-up survey** is a survey where the details of structural components are within the close visual inspection range of the Surveyor, i.e. normally within reach of hand.

1.2.3 An **Overall survey** is a survey intended to report on the overall condition of the hull structure and determine the extent of additional Close-up Surveys.

1.2.4 A **Transverse section** includes all longitudinal members such as plating, longitudinals and girders at the deck, side, bottom, inner bottom and hopper side plating, longitudinal bulkheads and bottom plating in top wing tanks. Where transversely framed, the transverse section is to include adjacent transverse stiffeners and their end connections.

1.2.5 **Representative spaces/tanks** are those which are expected to reflect the condition of other spaces/tanks of similar type and service and with similar corrosion prevention systems. When selecting Representative Spaces/Tanks account is to be taken of the service and repair history onboard and identifiable Critical Structural Areas and/or Suspect Areas.

1.2.6 **Suspect areas** are locations showing Substantial Corrosion and/or considered by the Surveyor to be prone to rapid wastage.

1.2.7 **Critical structural areas** are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or sister ships (if available) to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

1.2.8 **Substantial corrosion** is an extent of corrosion such that assessment of corrosion

pattern indicates a wastage in excess of 75 per cent of allowable margins, but within acceptable limits.

For vessels built under the IACS Common Structural Rules, substantial corrosion is an extent of corrosion such that the assessment of the corrosion pattern indicates a gauged (or measured) thickness between t_{ren} + 0.5 [mm] and t_{ren}.

't_{ren}' is the renewal thickness, which is defined as the minimum allowable thickness [mm], below which renewal of structural members is to be carried out.

1.2.9 **Corrosion prevention system** is normally to be of a full hard protective coating. Hard protective coatings are to usually be epoxy coating or equivalent. Soft or semi-hard coatings are not acceptable. Other coating systems may be considered acceptable as alternatives provided they are applied and maintained in compliance with the manufacturer's specifications.

1.2.10 **Coating condition** is defined as follows :

GOOD - Condition with only minor spot rusting;
FAIR - Condition with local breakdown at edges and weld connections and/or light rusting over 20 per cent or more of areas under consideration, but less than as defined for POOR condition;
POOR - Condition with general breakdown of coating over 20 per cent or more of areas or hard scale at 10 per cent or more of areas under consideration.

1.2.11 **Cargo area** in case of oil tankers and chemical tankers is that part of the ship which contains cargo tanks, slop tanks and cargo/ballast pump rooms, cofferdams, ballast tanks and void spaces adjacent to cargo tanks and also deck areas throughout the entire length and breadth of the part of the ship over the above mentioned spaces.

In the case of liquefied gas carriers, cargo area is that part of the ship which contains the cargo containment system, cargo/ballast pump rooms and compressor rooms, cofferdams, ballast tanks and void spaces adjacent to the cargo containment spaces and also deck areas throughout the entire length and breadth of the part of the ship over the above mentioned spaces.

In the case of bulk carriers and general dry cargo ships, cargo area is that part of the ship which contains all cargo holds and adjacent areas including fuel tanks, cofferdams, ballast tanks and void spaces etc. within the longitudinal extent from the aft bulkhead of the aft-most cargo hold to the fwd bulkhead of the foremost cargo hold.

Note: The definitions for cargo area given above are applicable only for the purpose of identifying spaces for examination during hull surveys as required by this chapter of the Rules. For the purpose of design and construction requirements, the definitions given in respective chapters of Part 5 of the rules would apply.

1.2.12 **Spaces** are separate compartments including holds, tanks, cofferdams and void spaces bounding cargo holds, decks and the outer hull.

1.2.13 **A prompt and thorough repair** is a permanent repair completed at the time of survey to the satisfaction of the Surveyor, therein removing the need for the imposition of any associated condition of classification.

1.2.14 **Anniversary date** means the day and month of each year corresponding to the expiry date of the classification certificate.

1.2.15 **A Double Hull Oil Tanker** is a ship which is constructed primarily for the carriage of oil (See Pt.5, Ch.2, for list of oils) in bulk, which has the cargo tanks protected by a double hull which extends for the entire length of the cargo area, consisting of double sides and double bottom spaces for the carriage of water ballast or as void spaces.

1.2.16 For the purpose of application of these requirements, a '**double skin bulk carrier'** is a ship which is constructed generally with single deck, top side tanks, hopper side tanks & double bottom in cargo spaces and is intended primarily to carry dry cargo in bulk, including such types as ore carriers and combination carriers, in which all cargo holds are bounded by double-side skin (regardless of the width of the wing space).

1.2.17 'General dry cargo ship' means ships carrying solid cargoes other than:

a) bulk carriers and ore carriers which are subject to the requirements of Sec.2;

b) following ship types which are subject to the requirements of Sec. 6

- dedicated Container carriers;
- ro-ro cargo ships;
- refrigerated cargo ships;
- dedicated wood chip carriers;
- dedicated Cement carriers;
- livestock carriers;
- ships carrying exclusively deck cargo;
- General dry cargo ships of double skin construction, with double skin extending for the length of the cargo area and for the height of the cargo hold to the upper deck. (Note: For ships with hybrid cargo hold arrangements, also see Section 6.1.2).

1.2.18 **Special consideration or specially considered** (in connection with close-up surveys and thickness measurements) means that sufficient close-up inspection and thickness measurements are to be taken to confirm the actual average condition of the structure under the coating.

1.2.19 **Air pipe heads** installed on the exposed decks are those extending above the freeboard deck or superstructure decks.

1.2.20 **Remote Inspection Techniques (RIT)** is a means of survey that enables visual examination of any part of the structure without the need for direct physical access of the surveyor.

1.2.21 A **ro-ro Ship** is a ship which utilizes a loading ramp to enable wheeled vehicles to be rolled-on and rolled-off the ship,

1.2.22 **Pitting corrosion** is defined as scattered corrosion spots/areas with local material reductions which are greater than the general corrosion in the surrounding area.

1.2.23 **Edge corrosion** is defined as local corrosion at the free edges of plates, stiffeners, primary support members and around openings. An example of edge corrosion is shown in Fig.1.2.23.

1.2.24 **Grooving corrosion** is typically local material loss adjacent to weld joints along abutting stiffeners and at stiffener or plate butts or seams. An example of groove corrosion is shown in Fig.1.2.24.

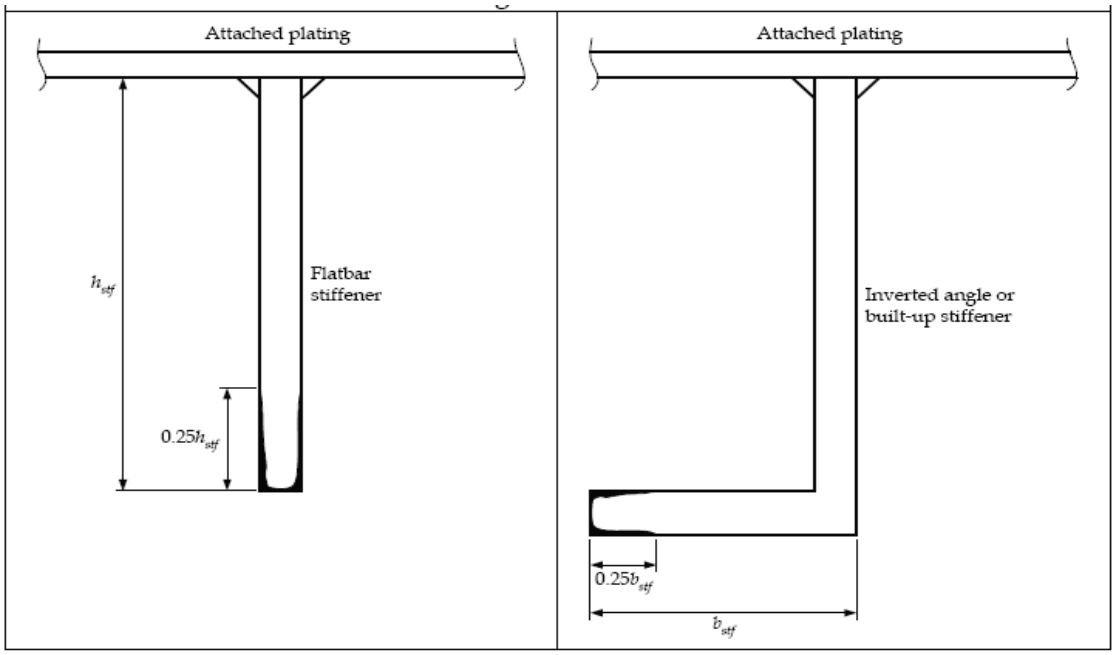

Fig.1.2.23 : Edge corrosion

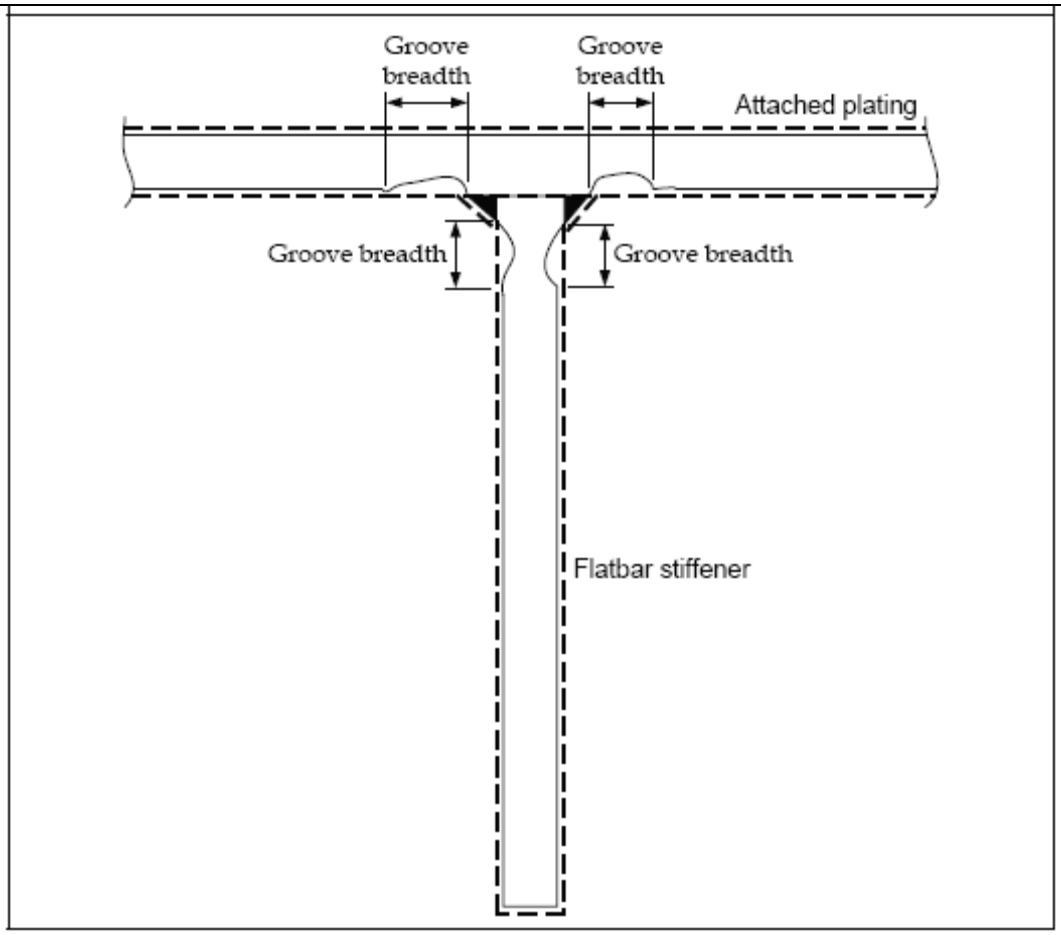

Fig. 1.2.24 : Grooving corrosion

1.3 Laid up vessels

1.3.1 Survey during lay-up

1.3.1 General

1.3.1.1 Vessels may be laid up when not carrying any cargo and not engaged in any commercial operation.

1.3.1.2 '*Hot lay-up*' means that the vessel is non-operational, with some of the machinery kept in operation for fast re-activation. This lay-up condition is relatively short term and may extend upto 12 months.

1.3.1.3 '*Cold lay-up*' means that the vessel is non-operational, with the machinery being shut down and the vessel being unable to be reactivated or brought to a state of readiness at short notice. This lay-up condition is relatively long term and may extend over 12 months.

1.3.1.4 When a ship is laid-up, the normal survey requirements may no longer apply provided that IRS is notified by the Owner in advance and requirements for lay up as advised by IRS are complied with. The Owner is also to submit a lay-up maintenance program and details to IRS for approval.

1.3.1.5 A record of lay-up preparations, maintenance and preservation actions is to be maintained throughout the vessel's lay-up and re-activation. Humidity levels in machinery spaces are to be recorded on a regular basis during the lay-up period, together with scheduled equipment maintenance and operation.

1.3.1.6 The requirements of flag Administrations and port or maritime authorities regarding lay-up are also to be complied with

1.3.1.7 During lay-up, the vessel is also to comply with the following:

a) The vessel is to be adequately manned in accordance with the statutory

requirements prevailing at the location of lay-up.

b) Adequate power is to be available on board to meet the following requirements, as applicable: -

- firefighting;
- bilge pumping;
- lighting;
- communication with shore;
- needs of crew on board, and
- operating anchor and mooring winches within a reasonable time.

The listed items are generally applicable for hot lay-up. For cold lay-up, arrangements may be specially considered.

1.3.2 Surveys during lay-up

1.3.2.1 Lay-up survey

.1 A lay-up survey is to be carried out at the beginning of the lay-up period to verify that the safety conditions, preservation measures, lay-up site and mooring arrangements are in accordance with the lay-up program approved by IRS.

.2 On successful completion of the survey, a lay-up status will be assigned by IRS.

1.3.2.2 Annual lay-up condition surveys

.1 A general examination of the hull and machinery is to be carried out in lieu of annual survey. An underwater examination is to be carried out in lieu of docking survey coinciding with special survey.

The purpose of the annual lay-up condition survey is to ascertain that the lay-up maintenance program is being continuously complied with and recorded in the lay-up record.

.2 The annual lay-up condition survey is to cover examination of hull integrity, firefighting arrangements, mooring arrangements, and other equipment in use depending on the type of lay-up.

1.3.3 Survey at re-activation

1.3.3.1 Vessels are to be surveyed and tested before re-entering service. On completion of the lay-up period, all outstanding class and statutory surveys are to be completed before the vessel returns to normal operation.

1.3.3.2 For vessels in hot lay-up, the scope of the reactivation survey is to include:

- a general examination of the hull, deck fittings, safety systems, machinery installations (including boilers whose survey is not due) and steering gear;

- all periodical surveys due at the date of re-activation or which became overdue during the lay-up period;

- dealing with the Conditions of Class due at the date of re-activation or which became due during the lay-up period.

Function testing of equipment would be specially considered.

1.3.3.3 The extent of the surveys and tests for vessels in cold lay-up will be considered in each case by IRS depending upon the time out of service, the maintenance and preservative measures taken during lay-up and the extent of surveys carried out during this time but, will at least include a sea trial for function testing of the machinery installation.

1.3.3.4 Upon satisfactory completion of the re-activation survey, the Class of the vessel will be revalidated.

1.4 Surveys by Chief Engineers

1.4.1 Continuous machinery Surveys of some machinery items may be carried out, subject to certain conditions, under the supervision of the Chief Engineer of the ship at ports where IRS is not represented and where practicable, at sea.

1.4.2 A limited confirmatory Survey should be carried out at the next port of call by an exclusive Surveyor of IRS. Where IRS Surveyor is not available, such Survey may be carried out by the exclusive Surveyor of a Society with whom IRS has collaboration agreement in force with prior information given to IRS.

1.4.3 Detailed procedure of this scheme may be obtained from IRS.

1.5 Surveys for damage or alteration

1.5.1 At any time when a ship is undergoing alterations or damage repairs, any exposed parts of the structure normally difficult for access are to be specially examined by the Surveyor.

1.6 Unscheduled surveys

1.6.1 In the event that IRS has reason to believe that its Rules and Regulations are not being complied with, IRS reserves the right to perform unscheduled surveys of the hull or machinery.

1.7 Provision for surveys

1.7.1 The Surveyors are to be provided with necessary facilities for a safe execution of survey.

In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access, are to be agreed between the owner and IRS in accordance with confined space safe entry procedure of IRS and IMO Resolution A1050(27) 'Revised recommend-dations for entering enclosed spaces abroad ships', as amended.

Details of the means of access are to be provided in the survey planning questionnaire.

In cases where the provisions of safety and required access are judged by the attending surveyor(s) as not adequate, the survey of the spaces concerned would not be carried out.

1.7.2 The surveys at sea or at anchorages may be carried out provided necessary precautions are taken while carrying out the survey and adequate assistance is provided.

1.7.3 Cargo holds, tanks and spaces are to be safe for access. Cargo holds, tanks and spaces are to be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it is to be verified that the atmosphere in the tank is free from hazardous gas and contains sufficient oxygen. All spaces are to be cleaned and loose accumulated corrosion scale is to be removed from all surfaces. They are to be sufficiently clean and free from water, scale, dirt, oil residues, etc. to reveal any significant corrosion, deformation, fractures, damages and other structural deterioration as well as the condition of the coating. However, those areas of structure whose renewal has already been decided by the owner need only be cleaned and descaled to the extent necessary to determine the limits of the renewal areas.

1.7.4 Sufficient illumination is to be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating.

1.7.5 Where soft coatings have been applied, safe access is to be provided for the Surveyor to verify the effectiveness of the coating and to carry out an assessment of the conditions of the internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft coating is to be removed.

1.7.6 For overall examination, means are to be provided to enable the Surveyor to examine the structure in a safe and practical way.

1.7.7 For Close-up Examination of the hull structure, other than cargo hold shell frames of bulk carriers, one or more of the following means of access to Surveyors satisfaction is to be provided, in general:

- Permanent staging and passages through structures
- Temporary staging and passages through structures
- Hydraulic arm vehicles such as conventional cherry pickers, lifts and movable platforms
- Portable ladders
- Boats or rafts
- Other equivalent means.

1.7.8 For Surveys conducted by use of a remote inspection technique, one or more of the following means for access, acceptable to the Surveyor, is to be provided:

- Unmanned robot arm
- Remotely Operated Vehicles (ROV)
- Unmanned Aerial Vehicles / Drones
- Other means acceptable to IRS.

1.7.9

a) For close-up surveys of the cargo hold shell frames of bulk carriers less than cape size (i.e. less than 100,000 dwt) one or more of the following means for access, acceptable to the Surveyor, is to be provided:

    - Permanent staging and passage through structures;

- Temporary staging and passages through structures;
- Portable ladder restricted to not more than 5 [m] in length may be accepted for surveys of lower section of a shell frame including bracket;
- Hydraulic arm vehicles such as conventional cherry pickers, lifts and movable platforms;
- Boats or rafts, provided the structural capacity of the hold is sufficient to withstand static loads at all levels of water;
- Other equivalent means.

b) For close-up surveys of the cargo hold shell frames of cape size bulk carriers (100,000 dwt and above), the use of portable ladders is not accepted and one or more of the following means for access, acceptable to the surveyor, is to be provided:

i) Annual surveys, intermediate survey under 10 years of age and Special Survey No.1

- permanent staging and passage through structures;
- temporary staging and passages through structures;
- hydraulic arm vehicles such as conventional cherry pickers, lifts and movable platforms;
- boats or rafts provided the structural capacity of the hold is sufficient to withstand static loads at all levels of water;
- other equivalent means.

ii) Subsequent intermediate surveys and special surveys:

- either permanent or temporary staging and passage through structures for close-up surveys of at least the upper part of hold frames;
- hydraulic arm vehicles such as conventional cherry pickers for surveys of lower and middle part of shell frames as alternative to staging:
- lifts and moveable platforms;
- boats or rafts provided the structural capacity of the hold is sufficient to withstand static loads at all levels of water;
- other equivalent means.

Notwithstanding the above requirements:

c) The use of a portable ladder fitted with a mechanical device to secure the upper end of the ladder is acceptable for the “close-up examination of sufficient extent, minimum 25% of frames, to establish the condition of the lower region of the shell frames including approx. lower one third length of side frame at side shell and side frame end attachment and the adjacent shell plating of the forward cargo hold” at annual survey, required in 2.2.2.9b) and the “one other selected cargo hold” required in 2.2.2.10.b).

d) The use of hydraulic arm vehicles or aerial lifts (“Cherry picker”) may be accepted by the attending surveyor for the close-up survey of the upper part of side shell frames or other structures in all cases where the maximum working height is not more than 17 [m].

1.7.10 For bulk carriers and oil tankers constructed on or after 1 Jan 2005, the access requirements for surveys in the cargo area and forward spaces are given in Pt.5, Ch.1, Sec.1.6 & 1.7 and Pt.5, Ch.2, Sec.2.3 & 2.4 respectively. The means of access are to be surveyed prior to, or in conjunction with its use for carrying out surveys.

1.7.11 During surveys of oil tankers, ore or oil carriers, oil or bulk carriers, chemical tankers and liquefied gas carriers, explosimeter, oxygen-meter, breathing apparatus, life line and whistles are to be at hand during the survey.

1.7.12 In order to ascertain the structural strength of ship's hull and machinery, thickness measurements as specified in subsequent sections, as applicable, are to be carried out normally by means of ultrasonic equipment. The accuracy of the equipment is to be proven to the Surveyor as required.

1.7.13 One or more of the following fracture detection procedures may be required if deemed necessary by the Surveyor:

- radiographic equipment
- ultrasonic equipment
- magnetic particle equipment
- dye penetrant.

1.7.14 Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use are to be made available during the survey. A safety check-list should be provided.

1.7.15 Adequate and safe lighting is to be provided for the safe and efficient conduct of the survey.

1.7.16 Adequate protective clothing is to be made available and used (e.g. safety helmet, gloves, safety shoes, etc.) during the survey.

1.7.17 If breathing apparatus and/or other equipment is used as “Rescue and emergency response equipment” then the equipment is to be suitable for the configuration of the space being surveyed.

1.7.18 Remote Inspection Techniques (RIT) for Surveys

1.7.18.1 The RIT is to provide the information normally obtained from a close-up survey. RIT surveys are to be carried out in accordance with the following requirements. These considerations are to be included in the proposals for use of a RIT which are to be submitted in advance of the survey so that satisfactory arrangements can be agreed with IRS.

1.7.18.2 The equipment and procedure for observing and reporting the survey using a RIT are to be discussed and agreed with the parties involved prior to the RIT survey, and suitable time is to be allowed to set-up, calibrate and test all equipment beforehand.

1.7.18.3 When using a RIT as an alternative to close-up survey, if not carried out by IRS itself, it is to be conducted by a firm approved as a service supplier and is to be witnessed by an attending surveyor of IRS.

1.7.18.4 The structure to be examined using a RIT is to be sufficiently clean to permit meaningful examination. Visibility is to be sufficient to allow for a meaningful examination. IRS is to be satisfied with the methods of orientation on the structure.

1.7.18.5 The Surveyor is to be satisfied with the method of data presentation including pictorial representation, and a good two-way communication between the Surveyor and RIT operator is to be provided.

1.7.18.6 If the RIT reveals damage or deterioration that requires attention, the Surveyor may require traditional survey to be undertaken without the use of a RIT.

1.7.18.7 Remote inspection techniques may include the use of:

- Divers
- Unmanned robot arm
- Remote Operated Vehicles (ROV)
- Climbers
- Drones
- Other means acceptable to IRS

1.7.18.8 When permitted remote inspection techniques may be used to facilitate the required external and internal examination, including close-up surveys and gauging. The methods applied for remote inspection techniques are to provide the survey results normally obtained for/by the surveyor. The results of the surveys by remote inspection techniques when being used towards the crediting of surveys are to be acceptable to the attending Surveyor. Inspection should be carried out in the presence of the Surveyor.

1.7.18.9 Confirmatory surveys/close-up surveys may be carried out by the Surveyor at selected locations to verify the results of the remote inspection technique. Confirmatory thickness measurements may be requested by the attending Surveyor appropriately.

1.7.18.10 An inspection plan for the use of remote inspection technique(s), including any confirmatory survey/close-up survey/thickness measurements, is to be submitted for review and acceptance in advance of the survey.

1.7.18.11 Conditions

a) Use of remote inspection technique may be restricted or limited where there is a record or indication of abnormal deterioration or damage to structure or to items to be inspected. The remote inspection technique may not be applicable if there are recommendations for repairs.

b) It may also be inapplicable if conditions, affecting the class of the vessel, are found during the course of the inspection.

c) If the remote inspection technique reveals damage or deterioration that requires attention, the Surveyor may require close-up survey/thickness measurements without the

use of remote inspection technique to be undertaken.

1.7.18.12 Procedures

a) The inspection is to be carried out by a qualified technician with adequate knowledge of the items to be inspected. Prior to the commencement of surveys a pre-meeting is to be held between the technician(s), the owner's representative(s) and the attending Surveyors(s) for the purpose to ascertain that all arrangements in the inspection plan are in place, so as to ensure the safe and efficient conduct of the inspection work to be carried out.

b) Means of thickness gauging and non-destructive testing may be required in conjunction with the use of remote inspection technique.

c) Items to be examined using remote inspection technique are to be sufficiently clean to permit meaningful examination.

d) Visibility is to be sufficient to allow for a meaningful examination.

1.8 Surveys at sea or at anchorage

1.8.1 Survey at sea or at anchorage may be accepted provided the Surveyor is given the necessary assistance from the personnel onboard. Necessary precautions and procedures for carrying out the survey are to be in accordance with 1.7.

1.8.2 A communication system is to be arranged between the survey party in the tank and the responsible officer on deck. This system must also include the personnel in charge of ballast pump handling if boats or rafts are used.

1.8.3 Surveys of tanks or applicable holds by means of boats or rafts may only be undertaken with the agreement of the Surveyor, who is to take into account the safety arrangements provided, including weather forecasting and ship response under foreseeable conditions and provided the expected rise of water within the tank due to ship motion does not exceed 0.25 [m].

1.8.4 When rafts or boats will be used for close-up survey the following conditions are to be observed:

a) Only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, are to be used;

b) The boat or raft is to be tethered to the access ladder and an additional person is to be stationed down the access ladder with a clear view of the boat or raft;

c) Appropriate lifejackets are to be available for all participants;

d) The surface of water in the tank or hold is to be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level stationary. On no account is the level of the water to be rising while the boat or raft is in use;

e) The tank, hold or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable; and

f) At no time is the water level to be allowed to be within 1 m of the deepest under deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverses is only to be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered.

g) If the tanks (or spaces) are connected by a common venting system, or inert gas system, the tank in which the boat or raft is used should be isolated to prevent a transfer of gas from other tanks (or spaces).

   The tank being examined is to be isolated also to prevent transfer of liquids/gas through cargo or ballast piping.

1.8.5 For the under deck areas in tanks, survey by means of boats or rafts will be allowed if the depth of the webs is 1.5 [m] or less.

If the depth of the webs is more than 1.5 [m], boats or rafts may be allowed only when:

i) the coating of the under deck structure is in GOOD condition and there is no evidence of wastage; or

ii) a permanent means of access is provided in each bay to allow safe entry and exit, which means the following:

a) access direct from the deck via a vertical ladder and a small platform fitted approximately 2 [m] below the deck in each bay; or

b) access to deck from a longitudinal permanent platform having ladders to deck in each end of the tank. The platform shall, for the full length of the tank, be arranged in level with, or above, the maximum water level needed for rafting of under deck structure. For this purpose, the ullage corresponding to the maximum water level is to be assumed not more than 3 [m] from the deck plate measured at the midspan of deck tranverses and in the middle length of the tank.

If neither of the above conditions are met, then staging or an “other equivalent means” is to be provided for the survey of the under deck areas.

See also IRS Classification Notes “Guidelines for approval / acceptance of alternative means of access to spaces in oil tankers, bulk carriers, ore carriers and combination carriers”.

1.8.6 The requirements regarding the usage of boats or rafts in 1.8.5 does not preclude their use for moving about within a tank during a survey.

1.9 Repairs

1.9.1 Any damage in association with wastage over the allowable limits (including buckling, grooving, detachment or fracture), or extensive areas of wastage over the allowable limits, which affects or, in the opinion of the Surveyor, will affect the vessel's structural, watertight or weathertight integrity, is to be repaired promptly and thoroughly as defined in 1.2.13. Areas to be considered as relevant, include:

- side structure and side plating;
- deck structure and deck plating;
- bottom structure and bottom plating;
- inner bottom structure and inner bottom plating;
- inner side structure and inner side plating;
- longitudinal bulkhead(s) plating and structure;
- transverse watertight or oiltight bulkheads, plating and structure; and
- hatch covers and hatch coamings.
- weld connection between air pipes and deck plating;
- air pipe heads;
- ventilators including closing devices;
- bunker and vent piping systems in bulk carriers, ore carriers and combination carriers.

Where adequate repair facilities are not available, consideration may be given to allow the vessel to proceed directly to a repair facility. This may require discharging the cargo and/or temporary repairs for the intended voyage.

1.9.2 Additionally, when a survey results in the identification of corrosion or structural defects, either of which, in the opinion of the Surveyor, will impair the vessel's fitness for continued service, remedial measures are to be implemented before the ship continues in service.

1.9.3 Where the damage found on structure mentioned in 1.9.1 is isolated and of a localized nature which does not affect the ship’s structural integrity, consideration may be given by the surveyor to allow an appropriate temporary repair to restore watertight or weather tight integrity and impose a Condition of Class with a specific time limit.

1.9.4 Extended thickness measurements when required by the rules are to be carried out and necessary repairs completed before the survey is credited as complete.

1.10 Thickness measurements and close-up surveys

1.10.1 In any kind of survey, i.e. special, intermediate, annual or any other surveys having the scope of the foregoing ones thickness measurements of structures in areas where close-up surveys are required are to be carried out simultaneously with close-up surveys.

1.10.2 Consideration may be given by the attending Surveyor to allow use of Remote Inspection Techniques (RIT) as an alternative to close-up survey. Surveys conducted using a RIT are to be completed to the satisfaction of the attending Surveyor. When RIT is used for a close-up survey, temporary means of access for the corresponding thickness measurements is to be provided unless such RIT is also able to carry out the required thickness measurements.

1.10.3 In case of vessels to which IMO ESP Code is applicable, usage of Remote Inspection Techniques (RIT) as an alternative to close-up survey is subject to prior approval from Flag Administration.

1.11 Extension of special surveys

1.11.1 Under “exceptional circumstances”, IRS may grant an extension not exceeding three (3) months to allow for completion of the special survey provided that the vessel is attended and the attending surveyor(s) so recommend(s) after the following has been carried out:

a) Annual Survey;
b) Re-examination of Conditions of Class;
c) In the case where dry docking is due prior to the end of the class extension, an underwater examination is to be carried out by an approved diving company. An underwater examination by an approved company may be dispensed with in the case of extension of dry-docking survey not exceeding 36 months interval provided the ship is without outstanding Condition of Class regarding underwater parts.

“Exceptional circumstances” means unavailability of dry-docking facilities; unavailability of repair facilities; unavailability of essential materials, equipment or spare parts; or delays incurred by action taken to avoid severe weather conditions. It is the responsibility of the Owner to sufficiently plan in advance for repairs and/ or dry-docking to avoid occurrences of such “exceptional circumstances”.

1.11.2 In the case that the Class Certificate will expire when the vessel is expected to be at sea, an extension to allow for completion of the Special Survey may be granted provided there is documented agreement to such an extension prior to the expiry date of the certificate and provided that positive arrangements have been made for attendance of the Surveyor at the first port of call and provided that IRS is satisfied that there is technical justification for such an extension. Such an extension is to be granted only until arrival at the first port of call after the expiry date of the certificate. However, if owing to “exceptional circumstances” the special survey cannot be completed at the first port of call, 1.11.1 may be followed, but the total period of extension shall in no case be longer than three months after the original due date of the special survey.

1.11.3 In case of force majeure as defined below, the completion of survey may be specially considered.

“Force Majeure” means damage to the ship; unforeseen inability of IRS to attend the vessel due to the governmental restrictions on right of access or movement of personnel; unforeseeable delays in port or inability to discharge cargo due to unusually lengthy periods of severe weather, strikes or civil strife; acts of war; or other similar circumstances.

Section 2

Surveys - Bulk Carriers and Ore Carriers

2.1 Scope

2.1.1 This section gives the requirements for periodical surveys of vessels which have been assigned the class notations :

Bulk Carrier ESP
BC-A, B or C ESP (from July 2003)
Ore Carrier ESP

2.1.2 The requirements that are applicable only to double skin bulk carriers and those applicable only to other bulk carriers have been indicated in the relevant clauses. For bulk carrier with hybrid cargo hold arrangements, e.g. with some cargo holds of single side skin and others of double side skin, the appropriate requirements are to applied to the individual cargo holds depending on their type of side skin construction.

2.2 Annual surveys

2.2.1 General

2.2.1.1 Annual Surveys are to be carried out within 3 months before or after the anniversary date each year. (Also refer to Note 9 of Sec.1, Table 1.1.1). These should be held concurrently with statutory annual or other relevant statutory Surveys, where practicable.

2.2.1.2 At Annual Surveys, the Surveyor is to examine the hull and machinery, so far as necessary and practicable, in order to be satisfied as to their general condition.

2.2.1.3 It is to be confirmed that no new installation of material containing asbestos was carried out since last survey.

2.2.1.4 Where applicable, examination of alternative design and arrangement for machinery or electrical installations or fire safety, in accordance with the test, inspection and maintenance requirements, if any, specified in the approved documentation is to be carried out.

2.2.2 Hull

2.2.2.1 The survey is to consist of an examination for the purpose of ensuring, as far as practicable, that the hull, weather decks; hatch coamings, hatch covers including their securing arrangement, other closing appliances, equipment and related piping are maintained in satisfactory and efficient condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file. Special attention is to be paid to the following :

a) Weather deck, shipside plating above waterline.

b) Hatchways on freeboard and superstructure decks; exposed casings; skylights and fiddley openings; deck houses; companionways and superstructure bulkheads; side scuttles and dead lights; flush deck scuttles; ash shoots and other openings.

c) Weld connection between air pipes and deck plating; air pipe heads on exposed decks (external examination); flame screens on vents to all bunker tanks; ventilators and closing devices.

d) Scuppers and sanitary discharges as far as practicable together with valves and their controls.

e) Guard rails, bulwarks, freeing ports, gangways, walkways and life lines, fittings and appliances for timber deck cargo.

f) Watertight bulkheads and their penetrations as far as practicable.

2.2.2.2 Cargo hatch covers and coamings are to be examined to ensure that no alterations have been made to the approved arrangements :

A thorough survey of cargo hatch covers and coamings is to be carried out by examination in the open as well as closed positions and should include verification of proper opening and closing operation. The hatch cover sets within the forward 25% of the ship’s length and at least one additional set, such that all sets on the ship are assessed at least once in every 5-year period, are to be surveyed open, closed

and in operation to the full extent on each direction at each annual survey, including:

i) stowage and securing in open condition;

ii) proper fit and efficiency of sealing in closed condition; and

iii) operational testing or hydraulic and power components, wires, chains and link drives.

The closing of the covers is to include the fastening of all peripheral and cross joint cleats or other securing devices. Particular attention is to be paid to the condition of the hatch covers in the forward 25% of the ship's length, where sea loads are normally greatest.

2.2.2.3 If there are indications of difficulty in operating and securing hatch covers, sets in addition to those required by 2.2.2.2, at the discretion of the surveyor, are to be tested in operation. Where hatch covers or coamings undergo substantial repairs, the strength of securing devices should be upgraded to comply with Pt.5, Ch.1, 5.5

Where the cargo hatch securing system does not function properly, repairs are to be carried out under the supervision of the Surveyors.

2.2.2.4 For each cargo hatch cover set the following items are to be surveyed:

a) Cover panels, including side plates and stiffener attachments that may be accessible in the open position by close-up survey (for corrosion, cracks, deformation);

b) Sealing arrangements of perimeter and cross joints (gaskets for condition and permanent deformation, flexible seals on combination carriers, gasket lips, compression bars, drainage channels and non return valves);

c) Clamping devices, retaining bars, cleating (for wastage, adjustment and condition of rubber components);

d) Closed cover locating devices (for distortion and attachments);

e) Chain or rope pulleys;

f) Guides;

g) Guide rails and track wheels;

h) Stoppers;

i) Wires, chains, tensioners and gypsies;

j) Hydraulic system, electrical safety devices and interlocks; and

k) End and interpanel hinges, pins and stools where fitted.

2.2.2.5 At each hatchway the coamings, with plating stiffeners and brackets are to be subjected to a close-up survey and checked for corrosion, cracks and deformation, especially of the coaming tops.

2.2.2.6 Where considered necessary, the effectiveness of sealing arrangements may be proved by hose or chalk testing supplemented by dimensional measurements of seal compressing components.

2.2.2.7 Cargo hatch covers of the portable type (i.e. wood or steel pontoons) are to be examined to confirm the satisfactory condition of:

- wooden covers and portable beams, carriers or sockets for the portable beams and their securing devices;
- steel pontoons;
- tarpaulins;
- cleats, battens and wedges;
- hatch securing bars and their securing devices;
- loading pads/bars and the side plate edge;
- guide plates and chocks;
- compression bars, drainage channels and drain pipes, if any.

2.2.2.8 All watertight doors in watertight bulkheads, to be examined and tested (locally and remotely) as far as practicable.

2.2.2.9 Suspect areas identified at previous special or intermediate surveys are to be close-up surveyed. Thickness measurements

are to be taken of the area of substantial corrosion identified at previous surveys.

For vessels built under the IACS Common Structural Rules, the annual thickness gauging may be omitted where a protective coating has been applied in accordance with the coating manufacturer's requirements and is maintained in good condition. In this case, it is recommended that the necessary documentation of coating is available on board.

Where annual gauging is required as per 2.4.12.3, thickness measurement of aft bulk-head of the forward cargo hold is to be carried out.

2.2.2.10 Examination of ballast tanks is to be carried out when required as a consequence of the results of the special and intermediate surveys. (See 2.4.5.1, 2.4.5.2 and 2.3.2.1). When considered necessary by the Surveyor or where extensive corrosion exists, thickness measurement is to be carried out. If the results of these thickness measurements indicate substantial corrosion additional thickness measurements are to be carried out to the extent given in Table 2.4.10.2(a) or Table 2.4.10.2(b).

2.2.2.11 Anchoring and mooring equipment is to be examined as far as is practicable. It is to be confirmed that the towing and mooring equipment is properly marked with any restriction associated with its safe operation.

2.2.2.12 Where applicable Surveyor should satisfy himself regarding the freeboard marks on the ship's side.

2.2.2.13 The Surveyor is to confirm that, where required, an approved loading instrument together with its operation manual is available on board, See Pt.3, Ch.5. It is to be verified by the Surveyor that the loading instrument is checked for accuracy at regular intervals by the ship's staff by applying test loading conditions.

2.2.2.14 On bulk carriers of over 10 years of age the following are to be carried out in addition to the requirements in 2.2.2.1 to 2.2.2.13:

a) Overall survey of two selected cargo holds in double skin bulk carriers and all cargo holds in the case of other bulk carriers.

b) For single side skin bulk carriers, close-up survey of sufficient extent, minimum 25 per cent of frames, to establish the condition of the lower region of the shell frames including approximately lower one third length of side frame at side shell and side frame end attachment and the adjacent shell plating in a forward cargo hold. Where this level of survey reveals the need for remedial measures, the survey is to be extended to include a close-up survey of all of the shell frames and adjacent shell plating of that cargo hold as well as a close-up survey of sufficient extent of all remaining holds.

c) When considered necessary by the Surveyor, thickness measurement is to be carried out. If the results of these thickness measurements indicate substantial corrosion additional thickness measure-ments are to be carried out to the extent given in Table 2.4.10.2(a) or Table 2.4.10.2(b).

d) Where a hard protective coating in cargo holds is found to be in GOOD condition, the extent of close-up surveys may be specially considered.

e) All piping and penetrations in cargo holds, including overboard piping are to be examined.

2.2.2.15 On bulk carriers of over 15 years of age the following are to be carried out in addition to the requirements given in 2.2.2.1 to 2.2.2.14:

a) Overall survey of all cargo holds.

b) Close-up examination of a selected cargo hold in addition to the forward cargo hold as required in 2.2.2.14b) and c).

c) Examination of all piping and penetrations in cargo holds, including overboard piping.

d) Where a hard protective coating in cargo holds is found to be in GOOD condition, the extent of close-up surveys may be specially considered.

2.2.2.16 Accommodation ladders are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) steps;
b) platforms;
c) all support points such as pivots, rollers, etc.;
d) all suspension points such as lugs, brackets, etc.;
e) stanchions, rigid handrails, hand ropes and turntables;
f) davit structure, wire and sheaves, etc.

2.2.2.17 Gangways are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) treads;
b) side stringers, cross-members, decking, deck plates, etc.;
c) all support points such as wheel, roller, etc.;
d) stanchions, rigid handrails, hand ropes.

2.2.2.18 Winches of accommodation ladders and gangways are to be examined to verify the satisfactory condition of the following items:

a) brake mechanism including condition of brake pads and band brake, if fitted;
b) remote control system, and
c) power supply system for electric motor.

2.2.2.19 Davits and fittings on the ship's deck associated with accommodation ladders and gangways are to be examined for satisfactory condition at annual surveys. Fittings or structures for means of access to deck such as handholds in a gateway or bulwark ladder and stanchions are also to be examined.

2.2.2.20 The maintenance and inspection records of accommodation ladders and gangways are to be verified. It is to be confirmed that supporting wires are being renewed at intervals not exceeding 5 years.

2.2.2.21 When examining internal spaces as far as practicable, the permanent means of access where appropriate, are to be verified that they remain in good condition.

2.2.2.22 For vessels subject to IMO PSPC (See Pt.3, Ch.2, 3.6), it is to be confirmed that the maintenance, repair and partial recoating of dedicated ballast tanks and double side skin spaces, as appropriate, are recorded in the coating technical file.

2.2.2.23 Watertight cable transit seal systems register is to be reviewed to confirm that it is being maintained and as far as practicable the transits are to be examined to confirm their satisfactory condition. Where there are records entered since last annual survey of any disruption to the cable transits or installation of new cable transits, the satisfactory condition of those transits is to be confirmed by review of records and, if deemed necessary, by examination. The results are to be recorded in the Register against the specific cable transit.

2.2.2.24 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

2.2.2.25 Examination of double-side skin void spaces, for bulk carriers exceeding 20 years of age and of 150 m in length and upwards, are to be carried out when required as a consequence of the results of the renewal survey (as required by para 2.4.5.4) and intermediate survey (as required by 2.3.4.1). When considered necessary, or when extensive corrosion exists, thickness measurements are to be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with Table 2.4.10.2b. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys are to be examined. Areas of substantial corrosion identified at previous surveys are to have thickness measurements taken. The annual thickness gauging may be omitted where a protective coating has been applied in accordance with the coating manufacturer's requirements and is maintained in good condition.

2.2.3 Additional annual survey requirements for ships subject to SOLAS XII/9:

2.2.3.1 For ships subject to SOLAS XII/9 i.e.:

- Bulk Carriers of 150 [m] in length and upwards of single side skin construction as defined in Pt.5, Ch.1, 1.1.4; and

- Carrying solid bulk cargoes having a density of 1780 [kg/m3] and above; and

- Contracted for construction before 1 July, 1999 and

- Constructed with an insufficient number of transverse watertight bulkheads to enable them to withstand flooding of the foremost cargo hold in all loading conditions and remain afloat in a satisfactory condition of equilibrium as specified in SOLAS XII/4.3.

2.2.3.2 In accordance with SOLAS XII/9.1 for the foremost cargo hold of such ships, the additional survey requirements listed in 2.2.3.3 to 2.2.3.6 shall apply.

2.2.3.3 For bulk carriers of 5 - 15 years of age:

a) An overall survey of the foremost cargo hold, including close-up survey of sufficient extent, minimum 25% of frames, is to be carried out to establish the condition of:

    - Shell frames including their upper and lower end attachments, adjacent shell plating, and transverse bulkheads.

    - Suspect areas found at the previous special survey.

b) Where considered necessary by the surveyor as a result of the overall and close-up survey as described in a) above, the survey is to be extended to include a close-up survey of all of the shell frames and adjacent shell plating of the cargo hold.

2.2.3.4 For bulk carriers exceeding 15 years of age:

a) An overall survey of the foremost cargo hold, including close-up survey is to be carried out to establish the condition of:

    - All shell frames including their upper and lower end attachments, adjacent shell plating and transverse bulkheads.

    - Suspect areas found at the previous special survey.

2.2.3.5 Extent of thickness measurement:

a) Thickness measurement is to be carried out to an extent sufficient to determine both general and local corrosion levels at areas subject to close-up survey, as described in 2.2.3.3a) and 2.2.3.4a) above. The minimum requirement for thickness measurements are areas found to be suspect areas at the previous special survey.

    Where substantial corrosion is found, the extent of thickness measurements is to be increased with the requirements of Table 2.4.10.2(a) or Table 2.4.10.2(b).

b) The thickness measurement may be dispensed with provided the surveyor is satisfied by the close-up survey, that there is no structural diminution and the protective coating where provided remains effective.

2.2.3.6 Where the protective coating in the foremost cargo hold is found to be in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

For existing bulk carriers, where owners may elect to coat or recoat cargo holds as noted above, consideration may be given to the extent of the close-up and thickness measurement surveys. Prior to the coating of cargo holds of existing ships, scantlings should be ascertained in the presence of a surveyor.

2.2.3.7 Bilge well alarms to all cargo holds and conveyor tunnels are to be verified to confirm their functionality.

2.2.4 Machinery and systems

2.2.4.1 A general examination of the machinery, boilers, all pressurised systems (steam, pneumatic, hydraulic) and their associated fittings, propulsion system and auxiliary machinery to see whether they are being properly maintained and with particular attention to the fire and explosion hazards.

2.2.4.2 Confirmation that machinery, boilers and other pressure vessels, associated piping systems and fittings are so installed and protected as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards.

2.2.4.3 Confirm that Periodical Surveys of boilers and other pressure vessels have been carried out as required by the Rules and the safety devices have been tested.

2.2.4.4 Confirmation that the normal operation of the propulsion machinery can be sustained

or restored even though one of the essential auxiliaries becomes inoperative.

2.2.4.5 Confirmation that means are provided so that machinery can be brought into operation from the dead ship condition without external aid.

2.2.4.6 All main and auxiliary steering arrangements and their associated equipment and control systems are to be examined and tested. Where applicable, Surveyors are to verify that log entries have been made in accordance with statutory requirements.

2.2.4.7 Steering chains are to be cleaned for ascertaining wear and tear and lengths of chain worn in mean diameter by more than 12 per cent of the original rule diameter are to be renewed.

2.2.4.8 All the means of communication between the navigating bridge and the machinery control positions, as well as the bridge and the main alternative steering position, if fitted, are to be tested. It is to be confirmed that means of indicating the angular position of the rudder are operating satisfactorily.

2.2.4.9 Confirmation that with ships having emergency steering positions there are means of relaying heading information and, when appropriate, supplying visual compass readings to the emergency steering positions.

2.2.4.10 Confirmation that various alarms required for hydraulic power operated, electric and electro-hydraulic steering gears are, operating satisfactorily and that the recharging arrangements for hydraulic power operated steering gears are being maintained.

2.2.4.11 Examining the means for the operation of the main and auxiliary machinery essential for propulsion and the safety of the ship, including when applicable, the means of remotely controlling the propulsion machinery from the navigating bridge and the arrangements to operate the main and other machinery from a machinery control room.

2.2.4.12 Confirmation that the engine room telegraph, the second means of communication between the navigation bridge and the machinery space and the means of communication with any other positions from which the engines are controlled are operating satisfactorily.

2.2.4.13 Examination and testing of the general emergency alarm system and confirmation that the engineer's alarm is clearly audible in the engineer's accommodation.

2.2.4.14 It is to be confirmed that ballasting and de-ballasting arrangements are satisfactory. The bilge pumping systems and bilge wells including operation of each bilge pump, extended spindles and level alarms, where fitted, are to be examined as far as is practicable. It is also to be confirmed that bilge pumping system for each watertight compartment is satisfactory.

It is also to be confirmed that drainage from enclosed cargo spaces situated on freeboard deck is satisfactory.

2.2.4.15 Examining visually the condition of non-metallic expansion joints where fitted in piping systems which penetrate the ship’s side, with both the penetration and the expansion joint located below the deepest load waterline, and checking the service record.

2.2.4.16 General examination visually and in operation, as feasible, of the main electrical machinery, the emergency sources of electrical power, the switch gear, other electrical equipment including the lighting system is to be carried out.

2.2.4.17 Confirmation as far as practicable, the operation of the emergency source(s) of electrical power, including their starting arrangement, the systems supplied, and when appropriate, their automatic operation.

2.2.4.18 Examining in general, that the precautions provided against shock, fire and other hazards of electrical origin are being maintained.

2.2.4.19 A General Examination of automation equipment is to be carried out. Satisfactory operation of safety devices, bilge level detection and alarm systems and control systems is to be verified.

2.2.4.20 Examination of bunker and vent piping systems.

2.2.4.21 For ships complying with the requirements of SOLAS XII/12 for hold, ballast and dry space water level detectors (See 2.4.14), the annual survey is to include an examination and a test, at random, of the

water ingress detection systems and of their alarms.

2.2.4.22 For ships complying with the requirements of SOLAS XII/13 for the availability of pumping systems (See 2.4.15), the annual survey is to include an examination and a test, of the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold and of their controls.

2.2.4.23 Confirmation that machinery space ventilation systems are in good working condition.

2.2.4.24 For Ships constructed before 01 July 2017 and fitted with harmonic filters; following are to be verified at each survey:

a) Records of annual measurement of harmonic distortion levels of main bus bar under seagoing conditions and log entries showing which equipment was running and / or filters in service at the time of measurement. Harmonic distortion readings are to be carried out when the greatest amount of distortion is indicated by the measuring equipment.

b) Records of measurement of distortion levels, after any modification to the ship's electrical distribution system.

2.2.4.25 For Ships constructed on or after 01 July 2017 and fitted with harmonic filters; or for ships retrofitted with harmonic filters, on or after 01 July 2017; following are to be verified at each survey:

a) Engine log book containing distortion monitoring readings/ automatic records of same

b) Documentation of any approved relaxation on allowable distortion limits.

2.2.4.26 For ships contracted for construction on or after 01 Jan 2017 following are to be verified in emergency generator rooms at each survey:

a) Hand operated ventilation louvres, closing appliances, if fitted, are kept open and corresponding instruction plates are visible near the louvres.

b) Power operated ventilation louvres and closing appliances, if fitted, are fail-to-open type and capable of opening automatically whenever emergency generator is starting / in operation.

c) It is to be possible to close ventilation openings by a manual operation from a clearly marked safe position outside the space where the closing operation can be easily confirmed. Further, such closing is not possible from any other remote position. The louvre status (open/closed) is indicated at this position.

2.2.4.27 For vessels fitted with ICCP system, proper functioning of the system is to be checked and records of system operation are to be verified. It is to be verified that all maintenance as per manufacturer's instructions are carried out.

2.2.5 Fire protection, detection and extinction

2.2.5.1 The arrangements for fire protection, detection and extinction are to be examined and are to include confirmation that no changes have been made in the structural fire protection. Following are to be examined / verified:

a) verification that fire control plans are properly posted;

b) examination as far as possible and testing as feasible of, any fire detection and alarm system and any sample extraction smoke detection system;

c) examination of the fire main system and verification that each fire pump including the emergency fire pump can be operated separately so that the two required powerful jets of water can be produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main.;

d) verification that fire hoses, nozzles, applicators and spanners are in good working condition and situated at their respective locations;

e) examination of fire-fighting system (including fixed fire-fighting system for the

machinery spaces) controls, piping, instructions and marking, checking for evidence of proper maintenance and servicing including date of last systems tests;

f) verification that all semi-portable and portable fire extinguishers are in their stowed positions, checking for evidence of proper maintenance and servicing, conducting random check for evidence of discharged containers;

g) verification, as far as practicable, that the remote controls for stopping fans and machinery in machinery spaces are in working order. Examination of the arrangements for oil fuel, lubricating oil and other flammable oils and testing the remote closing of valves for oil fuel, lubricating oil and other flammable oils and, the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils;

h) examination of the closing arrangements of ventilators, funnel annular spaces, skylights, doorways and tunnel, where applicable;

i) confirmation that the fire fighters' outfits including its self-contained compressed air breathing apparatus and emergency escape breathing devices (EEBDs) are complete and in good condition and that the cylinders, including the spare cylinders, of any required self-contained breathing apparatus are suitably charged and that on board means of recharging apparatus cylinders used during drills or a suitable number of spare cylinders to replace those used are provided, and provision of two-way portable radiotelephone apparatus of an explosion-proof type or intrinsically safe.

j) examination of any manual and automatic fire doors and proving their operations.

k) examination of the fire-extinguishing systems for spaces containing paint and/or flammable liquids and deep fat cooking equipment in accommodation and service spaces;

l) examination of the fire safety requirements of helicopter facilities, including foam firefighting appliances ;

m) examination of the fire protection arrangements in cargo spaces, as far as practicable and as appropriate, the operation of the means of control provided for closing the various openings;

n) examination, when appropriate of the special arrangements for carrying dangerous goods, including checking the electrical equipment and wiring, ventilation, provision of protective clothing and portable appliances and testing of the water supply, bilge pumping and any water spray system.

o) Examination of the fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space.

p) testing the fire dampers of ventilation ducts and the means of closing the main inlets and outlets of all ventilation systems and testing the means of stopping power ventilation systems from outside the space served.

2.2.5.2 For ships designed to carry containers on or above weather deck, as applicable, examining the water mist lance, and as appropriate, the mobile water monitors and all necessary hoses, fittings and required fixing hardware.

Surveys carried out by the National Authority of the country in which the ship is registered would normally be accepted as meeting these requirements, at the discretion of the Surveyor.

2.2.5.3 Confirmation that the means of escape from accommodation, machinery spaces and other spaces are satisfactory.

2.2.5.4 Examination of the arrangements for gaseous fuel for domestic purposes.

2.3 Intermediate Surveys

2.3.1 General

2.3.1.1 Intermediate surveys are to be carried out at or between the second or third Annual

Survey. (Also refer to Note 9 of Sec.1, Table 1.1.1).

However, only those items which are additional to the requirements of annual survey may be examined between the second or third Annual Survey.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

2.3.1.2 The following requirements are applicable for vessels over five years of age. For vessels below 5 years of age additional examination over and above the requirements of Annual survey may be required at the discretion of the Surveyors.

2.3.1.3 For vessels over 10 years of age a specific survey program is to be worked out by the owner in cooperation with the Surveyors considering the requirements of the previous special survey, executive hull summary of that survey, later relevant survey records and taking account of any amendments to the survey requirements after the last special survey. The survey program is to be submitted in written format for approval and kept on board until the intermediate survey has been completed. (See 2.4.2 for guidance on preparation of the survey program).

2.3.1.4 Prior to the commencement of any part of the intermediate survey, a survey planning meeting is to be held between the attending Surveyor(s), the owner's representative in attendance and where involved, the thickness measurement company representative and the master of ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

2.3.2 Vessels of between 5 and 10 years of age

2.3.2.1 Ballast tanks

a) An overall survey of representative ballast tanks is to be carried out. The selection of representative tanks is to include the fore and aft peak tanks and a number of other tanks, taking into account the total number and type of ballast tanks. When extensive corrosion is found, thickness measurements are to be carried out. If the overall survey reveals no visible structural defects, the examination may be limited to a verification that the corrosion prevention system remains efficient.

b) Where a hard protective coating is found to be in less than GOOD condition, corrosion or other defects are found in ballast tanks or where a hard protective coating was not applied from the time of construction, the examination is to be extended to other ballast tanks of the same type.

c) For ballast tanks other than double bottom tanks, where a hard protective coating is found to be in less than GOOD condition and is not renewed or where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question are to be examined and thickness measurements carried out as necessary at subsequent annual surveys.

When such breakdown of hard protective coating is found in double bottom ballast tanks, or where a soft or semi-hard coating has been applied or where a hard protective coating has not been applied, the tanks in question may be examined at annual intervals. When considered necessary by the Surveyor, or where extensive corrosion exists, thickness measurements are to be carried out.

d) Suspect areas identified at previous surveys are to be overall and close-up surveyed.

2.3.2.2 Cargo holds

a) Overall survey of all cargo holds. For single side skin bulk carriers, close-up survey of sufficient extent, minimum 25 per cent of frames, is to be carried out to establish the condition of :

- Shell frames including their end attachments, adjacent shell plating and transverse bulkheads in the forward cargo hold and one other selected cargo hold.

- Areas found suspect at the previous Special Survey.

b) For single side skin bulk carriers, where considered necessary by the Surveyor as a result of the overall and close-up survey, the survey is to be extended to include a close-up survey of all of the shell frames and adjacent shell plating of that cargo hold as well as a close-up survey of sufficient extent of all remaining cargo holds.

For double skin bulk carriers, as a result of overall survey, the survey may be extended to include a close-up survey of those areas of structure in the cargo holds selected by the Surveyor.

2.3.2.3 Extent of thickness measurements

a) Thickness measurements are to be carried out to an extent sufficient to determine both general and local corrosion levels at areas subject to close-up survey as per 2.3.2.2a). The minimum requirements for thickness measurement are the areas found to be suspect areas at the previous survey. The extent of thickness measurement may be specially considered provided the Surveyor is satisfied by the close-up survey, that there is no structural diminution and the hard protective coatings are found to be in GOOD condition.

b) Where substantial corrosion is found, the extent of thickness measurements is to be increased in accordance with the requirements of Table 2.4.10.2. Suspect areas identified at previous surveys are to be examined. Areas of substantial corrosion identified at previous surveys are to have thickness measurements taken.

For vessels built under IACS Common Structural Rules, the identified substantial corrosion areas may be:

i) protected by coating applied in accordance with the coating manufacturer's requirements and examined at annual intervals to confirm the coating in way is still in good condition, or alternatively

ii) required to be gauged at annual intervals.

In this case of i) above, it is recommended that the necessary documentation of coating is available on board.

c) Where the hard protective coating in cargo holds is found to be in GOOD condition, the extent of close-up surveys may be specially considered.

For existing bulk carriers, where owners may elect to coat or recoat cargo holds as noted above, consideration may be given to the extent of the close-up and thickness measurement surveys. Prior to the coating of cargo holds of existing ships, scantlings should be ascertained in the presence of a surveyor.

2.3.3 Vessels of between 10 and 15 years of age

2.3.3.1 The requirements of the intermediate survey are to be to the same extent as the previous special survey as required in 2.4 for hull structure and piping systems in way of the cargo holds cofferdams, pipe tunnels, void spaces and fuel oil tanks in the cargo area and all ballast tanks. However, tank testing specified in 2.4.8, survey of automatic air pipe heads specified in 2.4.6.9 and internal examination of fuel oil, lub.oil and fresh water tanks specified in Table 2.4.6.1 need not be carried out unless deemed necessary by the attending Surveyor. Thickness measurement is to be carried out for items 1) to 6) of Table 2.4.10.1.

A survey programme as per 2.4.2 is to be worked out prior to commencement of survey.

2.3.3.2 The intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion by the third annual survey in lieu of the application of 2.4.1.4.

2.3.3.3 An in-water survey complying with the requirements of Sec.7.2 may be accepted in lieu of docking survey required by 2.4.1.9.

2.3.4 Vessels of over 15 years of age

2.3.4.1 The requirements of the intermediate survey are to be to the same extent as the previous special survey as required in 2.4 for hull structure and piping systems in way of the

cargo holds cofferdams, pipe tunnels, void spaces and fuel oil tanks in the cargo area and all ballast tanks. However, tank testing specified in 2.4.8, survey of automatic air pipe heads specified in 2.4.6.9 and internal examination of fuel oil, lub.oil and fresh water tanks specified in Table 2.4.6.1 need not be carried out unless deemed necessary by the attending Surveyor. Thickness measurement is to be carried out for items 1) to 6) of Table 2.4.10.1.

A survey programme as per 2.4.2 is to be worked out prior to commencement of survey.

2.3.4.2 The intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion by the third annual survey in lieu of the application of 2.4.1.4.

2.3.4.3 A survey in dry dock is to be part of the intermediate survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of cargo holds and ballast tanks (i.e. parts below light ballast water line), are to be completed in the dry-dock.

2.4 Special Surveys - Hull

2.4.1 General

2.4.1.1 All ships classed with IRS are to undergo Special Surveys at 5 yearly intervals. The first Special Survey is to be completed within 5 years from the date of the initial classification survey and thereafter 5 years from the assigned date of the previous Special Survey. However, an extension of class of 3 months maximum beyond the 5th year may be granted in exceptional circumstances in accordance with 1.11. In such cases, the next period of class will start from the expiry date of the Special Survey before extension was granted.

2.4.1.2 The interval between the Special Surveys may be reduced at the request of the parties concerned or by IRS if considered appropriate.

2.4.1.3 For surveys completed within 3 months before the expiry date of the Special Survey, the next period of class will start from the expiry date of the Special Survey. For surveys completed more than 3 months before the expiry date of the Special Survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the special survey. If the owner elects to carry out the next due special survey, the period of class will start from the survey completion date. Any requirement of the Flag Administration in this regard is also to be complied with.

2.4.1.4 The Special Survey may be commenced at the 4th Annual Survey and be progressed with a view to completion by the 5th anniversary date. When the special survey is commenced prior to the fourth annual survey, the entire survey is to be completed within 15 months if such work is to be credited to the special survey and in this case the next period of class will start from the survey completion date.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

2.4.1.5 As part of the preparation for Special Survey, the proposed Survey Programme (See 2.4.2) including the schedule for thickness measurements (See 2.4.10) are to be submitted at least 3 months in advance of the intended commencement of the Special Survey.

2.4.1.6 Record of Special Survey will not be assigned until the Machinery Survey has been completed or postponed in agreement with IRS.

2.4.1.7 Ships which have satisfactorily passed a Special Survey will have a record entered in the Supplement to the Register Book indicating the assigned date of Special Survey. In addition a notation "ESP" will be entered for bulk carriers, ore carriers and combination carriers.

2.4.1.8 The special survey is to include, in addition to the requirements of the Annual Survey, examination, tests and checks of sufficient extent to ensure that the hull, equipment and related piping as indicated in 2.4.6.3 are in satisfactory condition and that the ship is fit for its intended purpose for the new period of class of five years to be assigned subject to proper maintenance and

operation and the periodical surveys being carried out at the due dates.

2.4.1.9 A Docking Survey in accordance with the requirements of Sec.7 is to be carried out as part of the Special Survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of cargo holds and ballast tanks (i.e. parts below light ballast water line) are to be completed in the dry-dock.

2.4.2 Planning and preparation for survey

2.4.2.1 A specific Survey Programme is to be worked out in advance of the Survey by the Owner in cooperation with the Surveyors and submitted to IRS for approval. The Survey Programme is to be in a written format based on IMO Resolution MSC197(80). The survey is not to commence until the survey programme has been agreed. Prior to the development of the survey programme, the survey planning questionnaire based on IMO Resolution MSC197(80) is to be completed by the owner and forwarded to IRS. The Survey Programme at Intermediate Survey may consist of the Survey Programme at the previous Special Survey supplemented by the Executive Hull Summary of that Special Survey and later relevant survey reports. The survey program is to be worked out taking into account any amendments to the survey requirements after the last special survey.

2.4.2.2 In developing the Survey Programme, the following documentation is to be collected and consulted with a view to selecting tanks, areas and structural areas for examination :

- Survey status and basic ship information
- Documentation on board as per 2.4.3
- Main structural plans (scantling drawings), including information regarding use of high strength steels, clad steel and stainless steel
- Relevant previous survey and inspection reports of IRS and the Owners
- Information regarding the use of the ship's holds and tanks, typical cargoes and other relevant data
- Information regarding corrosion prevention level at the time of construction
- Information regarding the relevant maintenance level during operation.

2.4.2.3 The Survey Programme submitted for approval is to account for and comply, as a minimum, with the requirements of close-up survey, thickness measurement and tank testing respectively, and to include relevant information including at least :

- Basic ship information and particulars
- Main structural plans (scantling drawings), including information regarding use of high strength steels, clad steel and stainless steel
- Plan of holds and tanks
- List of holds and tanks with information on use, corrosion prevention and condition of coating
- Conditions for survey (e.g. information like holds and tank cleaning, gas freeing, ventilation, lighting, etc.)
- Provisions and methods of access to structures
- Equipment for survey
- Nomination of holds and tanks and areas for close-up survey (As per Table 2.4.9.1)
- Nomination of sections for thickness measurement (As per Table 2.4.10.1)
- Nomination of tanks for testing
- Damage experience related to the ship in operation.

2.4.2.4 IRS will advise the Owner of the maximum acceptable structural corrosion diminution levels applicable to the vessel.

2.4.2.5 The ship is to be prepared for overall survey in accordance with the requirements of Table 2.4.2.5. The preparation is to be of sufficient extent to facilitate an examination to ascertain any excessive corrosion, deformation, fractures, damages and other structural deterioration.

2.4.2.6 Proper preparation and the close co-operation between the attending surveyor(s) and the owner’s representatives onboard prior to and during the survey are an essential part

in the safe and efficient conduct of the survey. During the survey on board safety meetings are to be held regularly.

Table 2.4.2.5 : Survey preparation			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
1) The holds, deep tanks, peaks, bilges and drain wells, engine and boiler spaces and other spaces are to be cleared out and cleaned as necessary for examination. Floor plates in engine and boiler spaces are to be lifted as may be necessary for examination of the structure underneath. Where necessary ceiling, lining, casings and loose insulation are to be removed as required by the Surveyor for examination of the structure. Compositions on the plating are to be examined and sounded, but need not be disturbed if found satisfactorily adhering to the plating.	1) Requirements of Special Survey I to be complied with 2) The chain locker is to be cleaned internally. The chain cables are to be ranged for inspection. The anchors are to be cleaned and placed in an accessible position for inspection.	1) Requirements of Special Survey II to be complied with 2) Portions of wood sheathing, or other covering, on steel decks are to be removed, as considered necessary by the Surveyor, in order to ascertain the condition of the plating.	1) Requirements of Special survey III to be complied with
2) The steelwork is to be exposed and cleaned as may be required for its proper examination by the Surveyor and close attention is to be paid to the parts of the structure which are particularly liable to excessive corrosion or to deterioration due to other causes. 3) All tanks are to be cleaned as necessary to permit examination. 4) Casings or covers of air, sounding, steam and other pipes, spar ceiling and lining in way of the side scuttles are to be removed, as required by the Surveyor.			

2.4.2.7 Prior to commencement of any part of the special survey, a survey planning meeting is to be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement firm representative and the master of the ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

2.4.2.8 The following is an indicative list of items that are to be addressed in the meeting:

a) Schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);

b) Provisions and arrangements for thickness measurements (i.e. access, cleaning/de-scaling, illumination, ventilation, personal safety);

c) Extent of the thickness measurements;

d) Acceptance criteria (refer to the list of minimum thicknesses);

e) Extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;

f) Execution of thickness measurements;

g) Taking representative readings in general and where uneven corrosion / pitting is found;

h) Mapping of areas of substantial corrosion; and

i) Communication between attending surveyor(s) the thickness measurement firm operator(s) and owner representative(s) concerning findings.

2.4.3 Documentation on board

2.4.3.1 The Owners are to obtain, supply and maintain on board documentation as specified in 2.4.3.3, which is to be readily available for the Surveyor.

2.4.3.2 The documentation is to be kept on board for the life time of the ship.

2.4.3.3 A Survey Report File is to be a part of the documentation on board consisting of

- Reports of structural surveys
- Executive Hull Summary
- Thickness measurement reports.

2.4.3.4 The Survey Report File is to be available also in the Owners and the Classification Society's management offices.

2.4.3.5 The following additional supporting documentation is to be available on board:

Main structural plans of cargo holds and ballast tanks (for CSR ships these plans are to include for each structural element both the as-built and renewal thickness. Any thickness for voluntary addition is also to be clearly indicated on the plans. The midship section plan to be supplied on board the ship is to include the minimum allowable hull girder sectional properties for hold transverse section in all cargo holds)

- Previous repair history
- Cargo and ballast history
- Inspection by ship's personnel with reference to
    - structural deterioration in general
    - leakages in bulkheads and piping
    - condition of coating or corrosion prevention system, if any
- Any other information that will help identify critical structural areas and/or Suspect Areas requiring inspection.
- Survey Programme as required by 2.4.2 until such time as the Special Survey has been completed.

2.4.3.6 Prior to survey, the Surveyor is to examine the completeness of the documentation on board, and its contents as a basis for the survey.

2.4.4 Surveys at sea or at anchorage

2.4.4.1 See Sec.1.8.

2.4.5 Space protection

2.4.5.1 Where provided, the condition of coating or corrosion protection of cargo holds and ballast tanks is to be examined. For ballast tanks excluding double bottom tanks where a hard protective coating is found to be in less than GOOD condition and it is not renewed or the tanks in question are to be examined at annual surveys. Thickness measurements are to be carried out as deemed necessary by the Surveyor.

2.4.5.2 When such breakdown of hard protective coating is found in double bottom ballast tanks and it is not renewed, where a soft or semi-hard coating has been applied, or where a hard protective coating has not been applied from the time of construction, the tanks in question may be examined at annual intervals. When considered necessary by the Surveyor or where extensive corrosion exists, thickness measurements are to be carried out.

2.4.5.3 Areas in spaces where a hard protective coating is provided and found in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

Areas in tanks where a hard protective coating is provided and found in GOOD condition, the extent of thickness measurements may be specially considered.

2.4.5.4 For double-side skin void spaces bounding cargo holds for bulk carriers exceeding 20 years of age and of 150 m in length and upwards, where provided, the condition of the corrosion prevention system of void spaces are to be examined. Where a hard protective coating is found to be in POOR condition, and it is not renewed, or where a soft or semi-hard coating has been applied, or where a hard protective coating has not been applied from the time of construction, the void spaces in question are to be examined at annual intervals. Thickness measurements are to be carried out as deemed necessary by the surveyor.

2.4.6 Survey and examination

2.4.6.1 All spaces within the hull and superstructure are to be examined. The special survey is to include examination of underwater parts as per Section 7.

2.4.6.2 All tanks other than water ballast tanks are to be examined internally in accordance with the requirements of Table 2.4.6.1.

2.4.6.3 All spaces including cargo holds and their ‘tween decks where fitted; double bottom, deep, ballast, peak tanks; pipe tunnel, duct keels, machinery spaces, dry spaces, cofferdams and voids are to be internally examined including the plating and framing, bilge and drain wells, sounding, venting, pumping and drainage arrangements. This examination is to be supplemented by testing and thickness measurement as required by 2.4.8 and 2.4.10 to ensure that the structural integrity remains effective. The aim of the examination is to discover Substantial Corrosion, significant deformation, fractures, damages or other structural deterioration and if deemed necessary by the Surveyor, a suitable non-destructive examination may be required.

All piping systems within the above spaces are to be examined and operationally tested to working pressure to attending Surveyors satisfaction to ensure that the tightness and condition remains satisfactory. At special survey No.3 and subsequent special surveys, structural downflooding ducts and structural ventilation ducts are to be internally examined.

Table 2.4.6.1 : Requirements for internal examination of tanks

Tank	Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequent Age > 15
Fuel oil bunker tanks				
- Engine room	None	None	One	One
- Cargo area	None	One	Two	Half the number of tanks, minimum 2
- If no tanks in Cargo Area, additional fuel tank(s) outside of Engine Room (if fitted)	None	One	One	Two
Lub.oil	None	None	None	One
Fresh water	None	One	All	All

Notes:

1) These requirements apply to tanks of integral (structural) type.
2) If a selection of tanks is accepted to be examined, then different tanks are to be examined at each special survey, on a rotational basis.
3) Peak tanks (all uses) are subject to internal examination at each special survey.
4) At special survey No.III and subsequent special surveys, one deep tank for fuel oil in the cargo area is to be included, if fitted.

2.4.6.4 Where ballast tanks have been converted to void spaces, the survey extent is to be based upon ballast tank requirements.

2.4.6.5 All watertight bulkheads and watertight doors are to be examined. All decks, casings and superstructures are to be examined. Attention is to be given to the corners of openings and other discontinuities in way of the strength decks and top sides.

2.4.6.6 The masts, standing rigging and anchors are to be examined.

The Surveyor should satisfy himself that there are sufficient mooring ropes on board and also that a tow line is provided when this is a Rule requirement.

2.4.6.7 The steering gear, and its connections and control systems (main and alternative) are to be examined. The auxiliary steering gear with its various parts are to be examined in working condition.

2.4.6.8 The hand pumps and suctions, air and sounding pipes are to be examined. The Surveyors are to ensure that striking plates are fitted under the sounding pipes whilst examining the tanks internally

Automatic air pipe heads are to be internally examined at special surveys as indicated in Table 2.4.6.8. For designs where the inner parts cannot be properly inspected from outside, the head is to be removed from the air pipe. Particular attention is to be paid to the condition of the zinc coating in heads constructed from galvanised steel.

2.4.6.9 The Surveyor should satisfy himself regarding the efficient condition of the following :

- Means of escape from machinery spaces, crew and passenger spaces and spaces in which crew are normally employed;
- Means of communication between bridge and engine room and between bridge and alternative steering position;
- Helm indicator;
- Protection to the aft steering wheel and the gear.

Table 2.4.6.8 : Requirements for internal examination of automatic air pipe heads

Location	Special survey No.I Age ≤ 5	Special survey No.II 5 < Age ≤ 10	Special survey No.III and subsequent Age > 10
forward 0.25L	Two air pipe heads[1,2] one port and one starboard on exposed decks	All air pipe heads on exposed decks	All air pipe heads[3] on exposed decks
aft of 0.25L from the forward perpendicular	Two air pipe heads[1,2] one port and one starboard on exposed decks	At least 20% of air pipe heads[1,2] on exposed decks	

Notes:

1. Preferably air pipes serving ballast tanks.
2. The selection of air pipe heads is to be made by the attending Surveyor. According to the results of the inspection, the Surveyor may require additional air pipe heads to be examined.
3. When there is substantiated evidence of replacement within the previous five years, individual air pipe heads may not be examined.

2.4.6.10 The chain cables are to be ranged and the anchors and the chain cables are to be examined. At special survey no. II and subsequent special surveys, the chain cables are to be gauged. Any length of chain cable which is found to have reduced in mean diameter at its most worn part by more than 12 per cent of its original rule diameter is to be renewed.

2.4.6.11 The windlass is to be examined.

2.4.6.12 The chain locker, hold fasts, hawse pipes and chain stoppers are to be examined and pumping arrangements of the chain locker tested.

2.4.6.13 Engine room structure is to be examined. Particular attention being given to tank tops, shell plating in way of tank tops, brackets connecting side shell frames and tank tops and engine room bulkheads in way of tank tops and bilge wells. Where excessive areas of wastage are found, thickness measurements are to be carried out and renewals of repairs made when wastage exceeds allowable limits.

2.4.6.14 The loading instrument is to be checked for accuracy by applying test load conditions in presence of the Surveyor.

2.4.6.15 For ships complying with the requirements of SOLAS XII/12 for hold, ballast and dry space water level detectors, the special survey is to include an examination and a test of the water ingress detection systems and of their alarms.

2.4.6.16 For ships complying with the requirements of SOLAS XII/13 for the availability of pumping systems, the special survey is to include an examination and a test of the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold and of their controls.

2.4.6.17 Examination of accommodation ladders, gangways and their winches are to be carried out as required for annual surveys. In addition, the accommodation ladders and gangways are to be operationally, tested with the specified maximum operation load.

The tests are to be carried out with the load applied as uniformly as possible along the length of the accommodation ladder or gangway, at an angle of inclination corresponding to the maximum bending moment on the accommodation ladder or gangway.

Accommodation ladder winch is to be operationally tested at special surveys. The brake system of the winch is to be tested for holding the maximum operational load on the ladder.

For existing installations on board ships constructed prior to 01 Jan 2010 where the

maximum operational load is not known, load nominated by the shipowner or operator may be considered as the test load.

2.4.7 Hatch covers and coamings

The hatch covers and coamings are to be surveyed as follows:

2.4.7.1 A thorough inspection of the items listed in 2.2.2.2 is to be carried out in addition to all hatch covers and coamings.

2.4.7.2 Checking of the satisfactory operation of all mechanically operated hatch covers is to be made, including:

- stowage and securing in open condition;
- proper fit and efficiency of sealing in closed condition;
- operational testing of hydraulic and power components, wires, chains and link drives.

2.4.7.3 Checking the effectiveness of sealing arrangements of all hatch covers by hose testing or equivalent.

2.4.7.4 Close up survey and thickness measurement of the hatch cover and coaming plating and stiffeners is to be carried out as given in the relevant tables.

2.4.8 Tank testing

2.4.8.1 Boundaries of double bottom, deep, ballast, peak and other tanks including holds adapted for the carriage of water ballast, are to be tested with a head of liquid to the top of air pipes or nearabout to the top of hatches for ballast/cargo holds. Boundaries of fuel oil, lub.oil and fresh water tanks are to be tested with a head of liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil, lub.oil and fresh water tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the Master stating that the pressure testing has been carried out according to the requirement with satisfactory results.

2.4.8.2 The testing of double bottom tanks and other spaces not designed for the carriage of liquid may be omitted, provided a satisfactory internal examination together with an examination of the tank top is carried out.

2.4.9 Close-up surveys

2.4.9.1 The minimum requirements for Close-up survey are given in Table 2.4.9.1(a) or Table 2.4.9.1(b) or Table 2.4.9.1(c).

2.4.9.2 The Surveyor may extend the close-up survey as deemed necessary taking into account the maintenance of the spaces under survey, the condition of the corrosion prevention system and where spaces have structural arrangements or details which have suffered defects in similar spaces or on similar ships according to available information.

2.4.9.3 For areas in tanks and cargo holds where hard protective coatings are found in GOOD condition, the extent of Close-up survey may be specially considered.

Table 2.4.9.1a : Requirements of close-up survey - Dry bulk cargo ships other than double skin bulk carriers

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
A) 25% of frames in the forward cargo hold at representative positions	A) All shell frames in the forward cargo hold and 25% of frames in remaining cargo holds including upper and lower end attachments and adjacent shell plating	A) All shell frames in the forward cargo hold and one other selected cargo hold and 50% of frames in remaining cargo holds including upper and lower end attachments and adjacent shell plating	A) All shell frames including their upper and lower end attachments and adjacent shell plating in all cargo holds.
A) Selected frames in all cargo holds	For bulk carriers of 100,000 dwt and above all shell frames in the forward cargo hold and 50% of shell frames in each of the remaining cargo holds, including upper and lower end attachments and adjacent shell plating		
B) One transverse web with associated plating and longitudinals in two representative ballast tanks of each type (i.e. topside or hopper side tank)	B) One transverse web with associated plating and longitudinals in each ballast tank	B) All transverse webs with associated plating and longitudinals in each ballast tank	B) All transverse webs with associated plating and longitudinals in each ballast tank
	B) Forward and aft transverse bulkhead in one ballast tank, including stiffening system	B) All transverse bulkheads in ballast tanks, including stiffening system	B) All transverse bulkheads in ballast tanks, including stiffening system
C) Two selected cargo hold transverse bulk-heads including internal structure of upper and lower stools, where fitted	C) All cargo hold trans-verse bulkheads including internal structure of upper and lower stools, where fitted	C) All cargo hold transverse bulkheads including internal structure of upper and lower stools, where fitted	C) All cargo hold transverse bulkheads including internal structure of upper and lower stools, where fitted
D) All cargo hold hatch covers and coamings (plating and stiffeners)	D) All cargo hold hatch covers and coamings (plating and stiffeners)	D) All cargo hold hatch covers and coamings (plating and stiffeners)	D) All cargo hold hatch covers and coamings (plating and stiffeners)
	E) All deck plating and underdeck structure inside line of hatch openings between all cargo hold hatches	E) All deck plating and underdeck structure inside line of hatch openings between all cargo hold hatches	E) All deck plating and underdeck structure inside line of hatch openings between all cargo hold hatches

Table 2.4.9.1a : (Contd.)

A) Cargo hold transverse frames	B) Transverse web frame or watertight transverse bulkhead in ballast tanks	C) Cargo hold transverse bulkheads plating, stiffeners and girders
D) Cargo hold hatch covers and coamings. Close up survey/thickness measurement is to be carried out for accessible parts of hatch cover structures. For cargo hold hatch covers of approved design which have no access to internal structures, close up survey/thickness measurements of such inaccessible structural members need not be carried out.	E) Deck plating and underdeck structure inside line of hatch openings, between cargo hold hatches	

Note : Close-up survey of transverse bulkheads to be carried out at four levels -

Level (a)	Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.
Level (b)	Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools) and immediately above the line of the shedder plates.
Level (c)	About mid-height of the bulkhead.
Level (d)	Immediately below the upper deck plating and immediately adjacent to the upper wing tank and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

Table 2.4.9.1b : Minimum requirements for close-up survey at special hull survey of double skin bulk carriers other than ore carriers			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
One transverse web with associated plating and longitudinals in two representative ballast tanks of each type (This is to include the foremost topside and double side ballast tanks on either side) **(A)**	One transverse web with associated plating and longitudinals as applicable in each ballast tank **(A)**	All transverse webs with associated plating and longitudinals as applicable in each ballast tank **(A)**	All transverse webs with associated plating and longitudinals as applicable in each ballast tank **(A)**
	Forward and aft transverse bulkheads including stiffening system in the topside, hopper side and double side ballast tanks on one side of the ship (i.e. port or starboard) that fall in one transverse section **(A)**	All transverse bulkheads including stiffening system in each ballast tank **(A)**	All transverse bulkheads including stiffening system in each ballast tank **(A)**
	25% of ordinary transverse frames for transverse framing system or 25% of longitudinals for longitudinal framing system on side shell and inner side plating at forward, middle and aft parts,in the foremost double side tanks **(B)**	25% of ordinary transverse frames for transverse framing system or 25% of longitudinals for longitudinal framing system on side shell and inner side plating at forward, middle and aft parts,in all double side tanks **(B)**	All ordinary transverse frames for transverse framing system or all of longitudinals for longitudinal framing system on side shell and inner side plating at forward, middle and aft parts,in all double side tanks **(B)**
Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted **(C)**	One transverse bulkhead in each cargo hold, including internal structure of upper and lower stools, where fitted **(C)**	All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted **(C)**	Areas **(C)** - **(E)** as for SS No.III

Table 2.4.9.1b (Contd.)

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
All cargo hold hatch covers and coamings (plating and stiffeners) **(D)**	All cargo hold hatch covers and coamings (plating and stiffeners) **(D)**	All cargo hold hatch covers and coamings (plating and stiffeners) **(D)**	
	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches **(E)**	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches **(E)**	

(A), (B), (C), (D) and **(E)** are areas to be subjected to close-up surveys and thickness measurements.

(A) : Transverse web frame or watertight transverse bulkhead in topside, hopper side and double side ballast tanks. In fore and aft peak tanks transverse web frame means a complete transverse web frame ring including adjacent structural members.
(B) : Ordinary transverse frame in double side tanks.
(C) : Cargo hold transverse bulkheads plating, stiffeners and girders.
(D) : Cargo hold hatch covers and coamings. Close up survey/thickness measurement is to be carried out for accessible parts of hatch cover structures. For cargo hold hatch covers of approved design which have no access to internal structures, close up survey/thickness measurements of such inaccessible structural members need not be carried out.
(E) : Deck plating inside line of hatch openings between cargo hold hatches.

Note : Close-up survey of transverse bulkheads to be carried out at four levels.

Level (a) : Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.

Level (b) : Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools) and immediately above the line of the shedder plates.

Level (c) : About mid-height of the bulkhead.

Level (d) : Immediately below the upper deck plating and immediately adjacent to the upper wing tank and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

Table 2.4.9.1c : Minimum requirements for close-up survey at special hull survey of ore carriers			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
One web frame ring complete including adjacent structural members in a ballast wing tank **(A)** One transverse bulkhead lower part – including girder system and adjacent structural members – in a ballast tank **(A)**	All web frame rings complete including adjacent structural members in a ballast wing tank **(A)** One deck transverse including adjacent deck structural members in each remaining ballast tank **(A)** Forward and aft transverse bulkheads complete – including girder system and adjacent structural members – in a ballast wing tank **(A)**	All web frame rings complete including adjacent structural members in each ballast wing tank **(A)** All transverse bulkheads complete – including girder system and adjacent structural members – in each ballast tank **(A)** One web frame ring complete including adjacent structural members in each wing void space **(A)**	As for special survey for age from 10 to 15 years
	One transverse bulkhead lower part – including girder system and adjacent structural members – in each remaining ballast tank **(A)**	Additional web frame rings in void spaces as deemed necessary by the Surveyor **(A)**	
Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted **(C)**	One transverse bulkhead in each cargo hold, including internal structure of upper and lower stools, where fitted **(C)**	All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted **(C)**	Areas **(C) – (E)** as for age interval 10 to 15 years
All cargo hold hatch covers and coamings (plating and stiffeners) **(D)**	All cargo hold hatch covers and coamings (plating and stiffeners) **(D)**	All cargo hold hatch covers and coamings (plating and stiffeners) **(D)**	
	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches **(E)**	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches **(E)**	
(A), (C), (D) and **(E)** are areas to be subjected to close-up surveys and thickness measurements			
(A) :	Transverse web frame or watertight transverse bulkhead in ballast wing tanks and void spaces. In fore and aft peak tanks transverse web frame means a complete transverse web frame ring including adjacent structural members		
(C) :	Cargo hold transverse bulkheads plating, stiffeners and girders		
(D) :	Cargo hold hatch covers and coamings. Close up survey/thickness measurement is to be carried out for accessible parts of hatch cover structures. For cargo hold hatch covers of approved design which have no access to internal structures, close up survey/thickness measurements of such inaccessible structural members need not be carried out.		
(E) :	Deck plating and under deck structure inside line of hatch openings between cargo hold hatches		

Table 2.4.9.1c (Contd.)	
Note : Close-up survey of transverse bulkheads to be carried out at four levels:	
Level (a) :	Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.
Level (b) :	Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools) and immediately above the line of the shedder plates.
Level (c) :	About mid-height of the bulkhead.
Level (d) :	Immediately below the upper deck plating and immediately adjacent to the upper wing tank and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

2.4.10 Thickness measurement

2.4.10.1 The minimum requirement for thickness measurement at special survey are given in Table 2.4.10.1.

2.4.10.2 Provisions for extended measurements for areas with Substantial Corrosion in case of dry bulk cargo ships are given in Table 2.4.10.2(a) and Table 2.4.10.2(b) and may be additionally specified in the survey programme as required by 2.4.2. Areas of substantial corrosion identified at the previous surveys are to have thickness measurements taken.

For vessels built under IACS Common Structural Rules, the identified substantial corrosion areas may be:

a) Protected by coating applied in accordance with the coating manufacturer's requirements and examined at annual intervals to confirm the coating in way is still in good condition, or alternatively

b) Required to be gauged at annual intervals.

In this case of a) above, it is recommended that the necessary documentation of coating is available on board.

The Surveyor may further extend the thickness measurements as deemed necessary.

2.4.10.3 Transverse sections are to be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements, one of which is to be in the amidships area.

2.4.10.4 Representative thickness measurement to determine both general and local levels of corrosion in the shell frames and their end attachments in all cargo holds and ballast tanks is to be carried out.

Thickness measurement is also to be carried out to determine the corrosion levels on the transverse bulkhead plating.

The extent of thickness measurements may be specially considered provided the Surveyor is satisfied by the close-up survey, that there is no structural diminution and the hard protective coating where applied remains efficient.

Table 2.4.10.1 : Thickness measurement at special hull survey of dry bulk cargo ships (including double skin bulk carriers)

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel
	2) Within the cargo area: a) 2 transverse sections of deck plating outside line of cargo hatch opening b) Wind and water strakes in way of the transverse sections considered in item (a) above	2) Within the cargo area: a) Each deck plate outside line of cargo hatch openings b) 2 transverse sections, one in the amidship area, outside line of cargo hatch openings c) All wind and water strakes	2) Within the cargo area: a) Each deck plate outside line of cargo hatch openings b) 3 transverse sections, one in the amidship area, outside line of cargo hatch openings c) All wind and water strakes d) Each bottom plate e) Duct keel or pipe tunnel plating and internals
	3) Measurement, for general assessment and recording of corrosion pattern, of those structural members subject to Close-up Survey according to Table 2.4.9.1(a) or Table 2.4.9.1(b), as applicable	3) Measurement, for general assessment and recording of corrosion pattern, of those structural members subject to Close-up Survey according to Table 2.4.9.1(a) or Table 2.4.9.1(b), as applicable	3) Measurement, for general assessment and recording of corrosion pattern, of those structural members subject to Close-up Survey according to Table 2.4.9.1(a) or Table 2.4.9.1(b), as applicable
	4) For vessels where UR S31 applies as mentioned in Table 2.4.13.1, additional thickness measurement of side shell frames and brackets for verifying compliance with UR S31.	4) Same as for special survey II	4) Same as for special survey II
	5) Selected wind and water strakes outside the cargo area	5) Selected wind and water strakes outside the cargo area	5) All wind and water strakes outside the cargo area
		6) For vessels subject to compliance of IMO standards as mentioned in 2.4.12 thickness measurement applicable to the vertically corrugated aft transverse bulkhead of the foremost cargo hold is to be carried out as per Table 2.4.12 and Fig.2.4.12	6) Same as for Special Survey III

Table 2.4.10.1 (Contd.)			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
		7) Internals in forepeak and after peak ballast tanks	7) Internals in forepeak and after peak ballast tanks
			8) All exposed main deck plating outside cargo area
			9) Representative exposed superstructure deck plating (poop, bridge, forecastle deck)
			10) All keel plates, additional bottom plates in way of machinery space, aft end of tanks and cofferdams outside cargo area
			11) a) Plating of sea chests. b) Shell plating in way of overboard discharges as considered necessary by the Surveyor.

Table 2.4.10.2a : Requirements for extent of thickness measurement on areas of substantial corrosion in dry bulk cargo ships other than double skin bulk carriers

Structural Member		Extent of Measurement	Pattern of Measurement
Deck Structure including Cross Strips, Main Cargo Hatchways, Hatch Covers, Coamings and Topside Tanks			
1	Cross deck strip plating	Suspect cross deck strip plating	5 point pattern between underdeck stiffeners over 1 metre in length
2	Underdeck stiffeners	Transverse members	5 point pattern at each end and mid span
		Longitudinal members	5 point pattern on both web and flange
3	Hatch covers	Side and end skirts, each 3 locations	5 point pattern at each location
		3 longitudinal bands, 2 outboard strakes and centreline strake	5 point measurement each band
4	Hatch coaming	Each side and end of coaming, one band lower 1/3, one band upper 2/3 of coaming	5 point measurement of each band, i.e. end or side coaming
5	Topside ballast tanks	a) Watertight transverse bulkheads	
		i) lower 1/3 of bulkhead	i) 5 point pattern over 1 sq. metre of plating
		ii) upper 2/3 of bulkhead	ii) 5 point pattern over 1 sq. metre of plating
		iii) stiffeners	iii) 5 point pattern over 1 metre length
		b) 2 Representative swash transverse bulkheads	
		i) lower 1/3 of bulkhead	i) 5 point pattern over 1 sq. metre of plating
		ii) upper 2/3 of bulkhead	ii) 5 point pattern over 1 sq. metre of plating
		iii) stiffeners	iii) 5 point pattern over 1 metre length
		c) 3 Representative bays of slope plating	
		i) lower 1/3 of tank	i) 5 point pattern over 1 sq. metre of plating
		ii) upper 2/3 of tank	ii) 5 point pattern over 1 sq. metre of plating
		d) Longitudinals, suspect and adjacent areas	5 point pattern both web and flange over 1 metre length
6	Main deck plating	Suspect plates and adjacent areas	5 point pattern over 1 metre length of plating
7	Main deck longitudinals	Minimum of 3 longitudinals where plating measured	5 point pattern both web and flange over 1 metre length
8	Web frames/transverses	Suspect plates	5 point pattern over 1 sq. metre
Double Bottom and Hopper Structure			
1	Inner/double bottom plating	Suspect plates plus all adjacent plates	5 point pattern for each panel between longitudinals over 1 metre length
2	Inner/double bottom longitudinals	Three longitudinals where plates measured	3 measurements in line across web and 3 measurements on flange
3	Longitudinal girders or transverse floors	Suspect plates	5 point pattern over about 1 sq. metre
4	Watertight bulkheads (WT floors)	a) lower 1/3 of tank	a) 5 point pattern over 1 sq. metre of plating
		b) upper 2/3 of tank	b) 5 point pattern alternate plates over 1 sq. metre of plating
5	Web frames	Suspect plates	5 point pattern over 1 sq. metre of plating
6	Bottom/side shell longitudinals	Minimum of three longitudinals in way of suspect areas	3 measurements in line across web
			3 measurements on flange

Table 2.4.10.2a (Contd.)			
Structural Member		**Extent of Measurement**	**Pattern of Measurement**
Cargo Holds			
1	Side shell frames	Suspect stiffener and each adjacent area	a) At each end and mid span: 5 point pattern of both web and flange
			b) 5 point pattern within 25 mm of welded attachment to both shell and lower slope plate
Shell Structures			
1	Bottom and side shell plating	a) Suspect plate, plus four adjacent plates	a) 5 point pattern for each panel between longitudinal
		b) See other tables for particulars on gauging in way of tanks and cargo holds	
2	Bottom/side shell longitudinals	Minimum of three longitudinals in way of suspect areas	3 measurements in line across web 3 measurements on flange
Transverse Bulkheads in Cargo Holds			
1	Lower stool	a) Transverse band within 25 mm of welded connection to inner bottom	a) 5 point between stiffeners over 1 metre length
		b) Transverse band within 25 mm of welded connection to shelf plate	b) 5 point between stiffeners over 1 metre length
2	Transverse bulkhead	a) Transverse band at approximately mid height	a) 5 point between stiffeners over 1 sq. metre of plating
		b) Transverse band at part of bulkhead adjacent to upper deck or below upper stool shelf plate (for those ships fitted with upper stools)	b) 5 point between stiffeners over 1 sq. metre of plating

Table 2.4.10.2b : Requirements for extent of thickness measurements on areas of substantial corrosion of double skin bulk carriers		
Structural member	**Extent of measurement**	**Pattern of measurement**
Bottom, inner bottom and hopper structure		
Bottom, inner bottom and hopper structure plating	Minimum of three bays across double bottom tank, including aft bay	5-point pattern for each panel between longitudinals and floors
	Measurements around and under all suction bell mouths	
Bottom, inner bottom and hopper structure longitudinals	Minimum of three longitudinals in each bay where bottom plating measured	Three measurements in line across flange and three measurements on vertical web
Bottom girders, including the watertight ones	At fore and aft watertight floors and in centre of tanks	Vertical line of single measurements on girder plating with one measurement between each panel stiffener or a minimum of three measurements
Bottom floors, including the watertight ones	Three floors in bays where bottom plating measured, with measurements at both ends and middle	5-point pattern over two square meter area
Hopper structure web fame ring	Three floors in bays where bottom plating measured	5-point pattern over one square metre of plating. Single measurements on flange

Table 2.4.10.2b : (Contd.)		
Structural member	**Extent of measurement**	**Pattern of measurement**
Hopper structure transverse watertight bulkhead or swash bulkhead	Lower 1/3 of bulkhead	5-point pattern over one square metre of plating
	Upper 2/3 of bulkhead	5-point pattern over two square metre of plating
	Stiffeners (minimum of three)	For web, 5-point pattern over span (two measurements across web at each end and one at centre of span). For flange, single measurements at each end and centre of span
Panel stiffening	Where applicable	Single measurements
Deck structure including cross strips, main cargo hatchways, hatch covers, coamings and topside tanks		
Structural member	**Extent of measurement**	**Pattern of measurement**
Cross deck strip plating	Suspect cross deck strip plating	5 point pattern between underdeck stiffeners over 1 metre length
Underdeck stiffeners	Transverse members	5 point pattern at each end and mid span
	Longitudinal members	5 point pattern on both web and flange
Hatch covers	Side and end skirts, each 3 locations	5 point pattern at each location
	3 longitudinal bands, 2 outboard strakes and centerline strake	5 point measurement each band
Hatch coamings	Each side and end of coaming, one band lower 1/3, one band upper 2/3 of coaming	5 point measurement each band i.e. end or side coaming
Topside ballast tanks	a) watertight transverse bulkheads:	
	- Lower 1/3 of bulkhead	5 point pattern over 1 sq. metre of plating
	- Upper 2/3 of bulkhead	5 point pattern over 1 sq. metre of plating
	- Stiffeners	5 point pattern over 1 metre length
	b) two representative swash transverse bulkheads:	
	- Lower 1/3 of bulkhead	5 point pattern over sq. metre of plating
	- Upper 2/3 of bulkhead	5 point pattern over 1 sq. metre of plating
	- stiffeners	5 point pattern over 1 metre length
	c) three representative bays of slope plating:	
	- Lower 1/3 of tank	5 point pattern over 1 sq. metre of plating
	- Upper 2/3 of tank	5 point pattern over 1 sq. metre of plating
	d) Longitudinals, suspect and adjacent	5 point pattern on both web and flange over 1 metre length

Table 2.4.10.2b (Contd.)		
Structural member	**Extent of measurement**	**Pattern of measurement**
Main deck plating	Suspect plates and four adjacent plates	5 point pattern over 1 sq. metre of plating
Main deck longitudinals	Suspect plates	5 point pattern on both web and flange over 1 metre length
Web frames / transverses	Suspect plates	5 point pattern over 1 sq. metre
Structure in double side spaces of double skin bulk carriers including wing void spaces of ore carriers		
Structural member	**Extent of measurement**	**Pattern of measurement**
Side shell and inner side plating:	a) upper strake and strakes in way of horizontal girders plating between each pair of transverse frames / longitudinals in a minimum of three bays (along the tank)	single measurement
	b) all other strakes plating between every third pair of longitudinals in same three bays	single measurement
Side shell and inner side transverse frames / longitudinals on:	a) upper strake each transverse frame / longitudinal in same three bays	3 measurements across web and 1 measurement on flange
	b) all other strakes every third transverse frame / longitudinal in same three bays	3 measurements across web and 1 measurement on flange
Transverse frames / longitudinals – brackets	Minimum of three at top, middle and bottom of tank in same three bays	5-point pattern over area of brackets
Vertical web and transverse bulkheads:	a) strakes in way of horizontal girders minimum of two webs and both transverse bulkheads	5-point pattern over approx. two square metre area
	b) other strakes minimum of two webs and both transverse bulkheads	two measurements between each pair of vertical stiffeners
Horizontal girders	Plating on each girder in a minimum of three bays	Two measurements between each pair of longitudinal girder stiffeners
Panel stiffening	Where applicable	Single measurements

Table 2.4.10.2b (Contd.)

Structural member	Extent of measurement	Pattern of measurement
Transverse bulkheads in cargo holds		
Structural member	**Extent of measurement**	**Pattern of measurement**
Lower stool, where fitted	Transverse band within 25 [mm] of welded connection to inner bottom	5-point pattern between stiffeners over one metre length
	Transverse band within 25 [mm] of welded connection to shelf plate	5-point pattern between stiffeners over one metre length
Transverse bulkheads	Transverse band at approximately mid height	5-point pattern over one square metre of plating
	Transverse band at part of bulkhead adjacent to upper deck or below upper stool shelf plate (for those ships fitted with upper stools)	5-point pattern over one square metre of plating

2.4.10.5 The thickness measurements are to be carried out by a qualified company certified by IRS.

2.4.10.6 In order to ensure necessary control during the process of thickness measurements, these are normally to be carried out under the supervision of the Surveyor. The Surveyor has the right to re-check the measurements as deemed necessary to ensure acceptable accuracy.

2.4.10.7 In all cases the extent of thickness measurements is to be sufficient as to represent the actual average condition.

2.4.10.8 A thickness measurement report is to be prepared. The report is to give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and has to be signed by the operator. The Surveyor is to review the report of the final thickness measurement after repairs have been carried out and countersign the cover page.

2.4.11 Reporting and evaluation of survey

2.4.11.1 The data and information on the structural condition of the vessel collected during the survey is to be evaluated for acceptability and continued structural integrity of the vessel.

2.4.11.2 For CSR bulk carriers, the ship's longitudinal strength is to be evaluated by using the thickness of structural members measured, renewed and reinforced, as appropriate, during the special surveys carried out after the ship reached 15 years of age (or during the special survey No.3, if this is carried out before the ship reaches 15 years) in accordance with the criteria for longitudinal strength of the ship's hull girder for CSR bulk carriers specified in Ch.13 of CSR.

2.4.11.3 The final result of evaluation of the ship's longitudinal strength required in 2.4.11.2 after renewal or reinforcement work of structural members, if carried out as a result of initial evaluation, is to be reported as a part of the Executive Hull Summary.

2.4.11.4 An Executive Hull Summary of the survey and results is to be issued to the Owner and placed on board the vessel for reference at future surveys. The Executive Hull Summary is to be endorsed by IRS Head Office.

2.4.12 Compliance with IMO standards for scantlings of aft transverse bulkhead of the foremost cargo hold and allowable hold loading for foremost cargo hold

2.4.12.1 These requirements apply to all bulk carriers of 150 [m] in length and above, intending to carry cargoes having bulk density of 1.78 [t/m^3] or above where:

a) The foremost hold is bounded by the side shell only in ships which were contracted for construction prior to 1 July 1998 and

b) The foremost hold is of double skin construction of less than 760 [mm] in breadth, measured perpendicular to the side shell, in ships, the keel of which were laid or were at a similar stage of construction, before 1 July 1999.

2.4.12.2 For existing bulk carriers of the type as mentioned in 2.4.12.1 a) and b) above which have not yet complied with the IMO standards, the compliance is to be achieved within the following time limits:

i) for ships which were 15 years of age or more on 1 July 1998, immediately before continuance in service;

ii) for ships which were 10 years of age or more but less than 15 years of age on 1 July 1998, by the due date of the first intermediate or special survey to be held after the date on which the ship reaches 15 years of age but not later than the date on which the ship reaches 17 years of age;

iii) for ships which were 5 years of age or more but less than 10 years of age on 1 July 1998, on or before the due date, after 1 July 2003, of the first intermediate or special survey after the date on which the ship reaches 10 years of age; and

iv) for ships which were less than 5 years of age on 1 July 1998, on or before the date on which the ship reaches 10 years of age.

Completion prior to 1 July 2003 of an intermediate or special survey with a due date after 1 July 2003, can not be used to postpone compliance.

2.4.12.3 For bulk carriers described in 2.4.12.2, thickness measurement of vertically corrugated aft transverse bulkhead of the foremost cargo hold is to be carried out to determine the general condition of the structure and to define the extent of possible repairs and/or reinforcements required to meet the above mentioned IMO standards. As a minimum, the thickness measurement is to be carried out at the critical locations given in Table 2.4.12 and Fig.2.4.12.

In addition, thickness measurement is to be carried out at subsequent intermediate surveys (for ships over 10 years of age) and special surveys for the purpose of verifying continuing compliance with the above standards.

Where annual gauging has been adopted as an alternative to steel renewal in accordance with the IMO standards, thickness measurement is also to be carried out at subsequent annual surveys.

2.4.13 Compliance with IACS requirements for existing vessels

2.4.13.1 Compliance with the standards mentioned in Table 2.4.13.1, as applicable, is to be achieved by the due date indicated in 2.4.13.2.

2.4.13.2 Compliance with the requirements indicated in Table 2.4.13.1, as applicable is to be achieved within the following time limits:

i) For ships which will be 15 years of age or more on 1 January 2004 by the due date of the first intermediate or special survey after that date;

ii) For ships which will be 10 years of age or more on 1 January 2004 by the due date of the first special survey after that date;

iii) For ships which will be less than 10 years of age on 1 January 2004 by the date on which the ship reaches 10 years of age.

Completion prior to 1 January 2004 of an intermediate or special survey with a due date after 1January 2004 cannot be used to postpone compliance. However, completion prior to 1 January 2004 of an intermediate survey the window for which straddles 1 January 2004 can be accepted.

2.4.13.3 Ships which are required to comply with UR S31 as indicated in Table 2.4.13.1 are subject to additional thickness measurements with respect to the side shell frames and brackets for the purpose of determining compliance, prior to the due dates as mentioned in 2.4.13.2 and at subsequent intermediate and special surveys for verifying continuing compliance with UR S31.

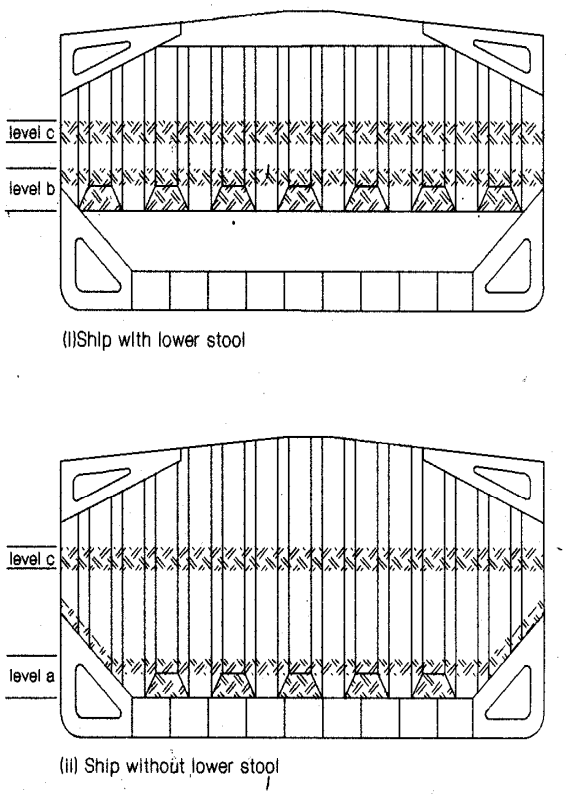

Fig.2.4.12 : Locations for thickness measurement for verification of IMO standard compliance

Table 2.4.12 : Critical levels at which thickness measurement is to be carried out for the vertically corrugated aft transverse bulkhead of the foremost cargo hold of existing bulk carriers to determine compliance with IMO standard

Level as per Fig.2.4.12	**Locations where thickness is to be measured in Zone 2 (P&S) and Zone 1**
	Ships with or without lower stool
Level (a) OR Level (b)	1) mid breadth of each corrugation flange at about 200 [mm] above line of shedder plates
	2) middle of gusset plates between corrugation flanges, where fitted
	3) middle of each shedder plate
	4) mid breadth of each corrugation web at approximately 200 [mm] above the line of each shedder plate
Level (c)	1) mid breadth of each corrugation flange and web at about the mid-height of the corrugation

Notes:

1. To adequately assess the scantlings of each individual vertical corrugation, each corrugation flange, web, shedder plate and gusset plate within each of the levels given above are to be gauged.
2. Where the thickness changes within the horizontal levels, the thinner plate is to be gauged.
3. Steel renewal/reinforcement is to comply with IMO standard.

2.4.14 Compliance with SOLAS regulation 12, Ch.XII : Hold, ballast and dry space water level detectors

Bulk carriers constructed before 1st July, 2004 are to be fitted with water level detectors as described in Pt.5, Ch.1, 2.13 of the Rules, not later than the date of the annual, intermediate or special survey of the ship to be carried out after 1st July, 2004, whichever comes first.

2.4.15 Compliance with SOLAS regulation 13, Ch.XII : Availability of pumping systems (for dewatering of forward spaces)

Bulk carriers constructed before 1st July, 2004 are to comply with this SOLAS regulation in accordance with details given in Pt.5, Ch.1, 2.14 of the Rules, not later than the date of the first intermediate or special survey of the ship to be carried out after 1st July, 2004, but in no case later than 1st July, 2007.

2.4.16 Watertight Cable Transits

2.4.16.1 The requirements for Special Survey may be undertaken by the attending Surveyor or by a firm approved as a service supplier according to Pt.1 Ch.1, Sec 1.9.

2.4.16.2 All transits are to be examined to confirm their satisfactory condition and the Cable Transit Seal System Register is to be reviewed to confirm that it is being maintained. The Special Survey is to be recorded in the Register, in which a single record entry will be sufficient to record the survey of all transits.

2.4.16.3 From review of the Register, where there are records entered since the last special survey of any disruption to the cable transits or installation of new cable transits (except which are reviewed and examined at previous annual surveys), the satisfactory condition of those transits is to be confirmed by the attending Surveyor by review of records and examination of the transits; the results are to be recorded in the Register against each of those cable transits.

2.4.16.4 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

2.5 Special Surveys – Machinery

2.5.1 Requirements for examination of machinery and piping are given in Sec.8.

Table 2.4.13.1 : Compliance with new IACS unified requirements

Item	Applicable types of vessels	Reference IACS unified requirement	Applicable locations on the ship
Renewal criteria for side shell frames and brackets	Bulk carriers with side frames and/or brackets not meeting the requirements of Pt.5, Ch.1 of the 1998 or subsequent rules	S31	Cargo holds bounded by single side shell

Section 3

Surveys - Oil Tankers, Ore or Oil Carriers, Oil or Bulk Carriers

3.1 Scope

3.1.1 This section gives the requirements for periodical surveys of vessels which have been assigned the following class notations, in general:

Oil Tanker ESP
Ore or Oil Carrier ESP
Oil or Bulk Carrier ESP

3.1.2 For vessels with class notation Ore or Oil Carrier ESP and Oil or Bulk Carrier ESP, the relevant requirements given in Sec.2 are also to be applied.

3.1.3 For vessels with notation "Moored Oil Storage Tanker at (location)", the requirements of this section apply except that special consideration may be given for carrying out in- water surveys instead of docking surveys, taking into account the condition of the vessel. In this case the IY notation may be either maintained or suspended and the ESP notation will be suspended. The mooring arrangements are to be approved considering the environmental parameters and water depth at the specified location.

This is subject to the acceptance and compliance with any additional requirements of the Flag Administration.

3.1.4 For vessels with additional class notation **ASPHALT CARRIER (INTEGRAL TANK, MAX. CARGO TEMP ≤ XXX℃),ESP** requirements of 3.6 are to be applied.

3.1.5 For vessels with additional class notation **ASPHALT CARRIER (INDEPENDENT TANK, MAX. CARGO TEMP ≤ XXX℃),** requirements of 3.7 are to be applied.

3.2 Annual surveys

3.2.1 General

3.2.1.1 Annual Surveys are to be carried out within 3 months before or after the anniversary date each year. (Also refer to Note 9 of Sec.1, Table 1.1.1). These should be held concurrently with statutory annual or other relevant statutory Surveys, where practicable.

3.2.1.2 At Annual Surveys, the Surveyor is to examine the hull and machinery, so far as necessary and practicable, in order to be satisfied as to their general condition.

3.2.1.3 Access to cargo tanks or other spaces within the cargo area, necessitating gas freeing will normally not be required unless necessary for checking items of equipment and installations for correct functioning.

3.2.1.4 It is to be confirmed that no new installation of material containing asbestos was carried out since last survey.

3.2.1.5 Where applicable, examination of alternative design and arrangement for machinery or electrical installations or fire safety, in accordance with the test, inspection and maintenance requirements, if any, specified in the approved documentation is to be carried out.

3.2.2 Hull and weather deck

3.2.2.1 The survey is to consist of an examination for the purpose of ensuring, as far as practicable, that the hull, equipment, closing appliances and related piping are maintained in satisfactory and efficient condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file. Special attention is to be paid to the following:

a) Weather deck, ship side plating above water line.

b) Openings on freeboard and superstructure decks; exposed casings; skylights and fiddley openings; deck houses; companionways and superstructure bulkheads; side scuttles and dead lights; flush deck scuttles; ash shoots and other openings.

c) Weld connection between air pipes and deck plating; air pipe heads on exposed

decks (external examination); ventilators and closing devices.

d) Watertight bulkheads and their penetrations as far as practicable.

e) Scuppers and sanitary discharges as far as practicable together with valves and their controls.

f) Guard rails, bulwarks, freeing ports, gangways, walkways and life lines.

g) Cargo tank openings including gaskets, covers, coamings and flame screens.

h) Cargo tanks pressure/vacuum valves, secondary means to prevent over or under pressure and devices to prevent passage of flame.

i) Examination of devices to prevent the passage of flame on vents to all bunker, oily-ballast and oily-slop tanks and void spaces, as far as practicable

j) Examination of cargo, crude oil washing, bunker and vent piping systems, including vent masts and headers.

3.2.2.2 Examination of pump room and pipe tunnel, if fitted, is to be carried out including:

a) Examination of all pump room bulkheads for signs of oil leakage or fractures and, in particular, the sealing arrangements of all penetrations of pump room bulkheads.

b) Examination of the condition of all piping systems.

3.2.2.3 All watertight doors in watertight bulkheads, to be examined and tested (locally and remotely) as far as practicable.

3.2.2.4 Suspect areas identified at previous special or intermediate surveys are to be close-up surveyed. Thickness measurements are to be taken in the area of substantial corrosion identified at previous surveys.

3.2.2.5 Examination of Ballast Tank is to be carried out when required as a consequence of the results of the Special Survey or Intermediate Survey. (See 3.4.5.1 and 3.3.3.3 respectively). When considered necessary by the Surveyor or where extensive corrosion is found, thickness measurement is to be carried out. When examination of ballast tanks reveals substantial corrosion, additional thickness measurements are to be carried out to the extent given in Table 3.4.9.2a) or Table 3.4.9.2b).

For vessels built under IACS Common Structural Rules, the identified substantial corrosion areas are required to be examined and additional thickness measurements are to be carried out.

3.2.2.6 Anchoring and mooring equipment is to be examined as far as is practicable. A general examination of emergency towing arrangements is to be carried out to ensure their ready availability. It is to be confirmed that the towing and mooring equipment is properly marked with any restriction associated with its safe operation.

For vessels with single point mooring arrangements where 'SPM' notation is assigned, the following are to be generally examined:

a) Components of the single point mooring system (bow chain stoppers, bow fairleads, pedestal roller fairleads, winches and capstans), to verify their satisfactory condition.

b) Hull structures supporting and adjacent to the components of the single point mooring system, to verify that there is no deformation or fracture.

3.2.2.7 Where applicable Surveyor should satisfy himself regarding the freeboard marks on the ship's side.

3.2.2.8 The Surveyor is to confirm that, where required, an approved loading instrument together with its operation manual are available on board, See Pt.3, Ch.5. It is to be verified by the Surveyor that the loading instrument is checked for accuracy at regular intervals by the ship's staff by applying test loading conditions.

3.2.2.9 Accommodation ladders are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) steps;
b) platforms;
c) all support points such as pivots, rollers, etc.;

d) all suspension points such as lugs, brackets, etc.;
e) stanchions, rigid handrails, hand ropes and turntables;
f) davit structure, wire and sheaves, etc.

3.2.2.10 Access to bow arrangements are to be examined. Gangways are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) treads;
b) side stringers, cross-members, decking, deck plates, etc.;
c) all support points such as wheel, roller, etc.;
d) stanchions, rigid handrails, hand ropes.

3.2.2.11 Winches of accommodation ladders and gangways are to be examined to verify the satisfactory condition of the following items:

a) brake mechanism including condition of brake pads and band brake, if fitted;

b) remote control system, and

c) power supply system for electric motor.

3.2.2.12 Davits and fittings on the ship's deck associated with accommodation ladders and gangways are to be examined for satisfactory condition at annual surveys. Fittings or structures for means of access to deck such as handholds in a gateway or bulwark ladder and stanchions are also to be examined.

3.2.2.13 The maintenance and inspection records of accommodation ladders and gangways are to be verified. It is to be confirmed that supporting wires are being renewed at intervals not exceeding 5 years.

3.2.2.14 When examining internal spaces, as far as practicable, the permanent means of access where appropriate, are to be verified that they remain in good condition.

3.2.2.15 For vessels subject to IMO PSPC (See Pt.3, Ch.2, 3.6) it is to be confirmed that the maintenance, repair and partial re-coating of dedicated ballast tanks when appropriate are recorded in the coating technical file.

3.2.2.16 Confirmation that in-service maintenance and repair activities of coating systems in cargo oil tanks are recorded in the coating technical file.

3.2.2.17 Watertight cable transit seal systems register is to be reviewed to confirm that it is being maintained and as far as practicable the transits are to be examined to confirm their satisfactory condition. Where there are records entered since last annual survey of any disruption to the cable transits or installation of new cable transits, the satisfactory condition of those transits is to be confirmed by review of records and, if deemed necessary, by examination. The results are to be recorded in the Register against the specific cable transit.

3.2.2.18 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

3.2.3 Machinery and systems

3.2.3.1 A general examination of the machinery, boilers, all pressurised systems (steam, pneumatic, hydraulic) and their associated fittings, propulsion system and auxiliary machinery to see whether they are being properly maintained and with particular attention to the fire and explosion hazards.

3.2.3.2 Confirmation that machinery, boilers and other pressure vessels, associated piping systems and fittings are so installed and protected as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards.

3.2.3.3 Confirmation that Periodical Surveys of boilers and other pressure vessels have been carried out as required by the Rules and the safety devices have been tested.

3.2.3.4 Confirmation that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative.

3.2.3.5 Confirmation that means are provided so that machinery can be brought into operation from the dead ship condition without external aid.

3.2.3.6 All main and auxiliary steering arrangements and their associated equipment and control systems are to be examined and tested. Where applicable, Surveyors are to verify that log entries have been made in accordance with statutory requirements.

3.2.3.7 Confirming, when appropriate, that requisite arrangements to regain steering capability in the event of the prescribed single failure are being maintained.

3.2.3.8 Steering chains are to be cleaned for ascertaining wear and tear and lengths of chain worn in mean diameter by more than 12 per cent of the original rule diameter are to be renewed.

3.2.3.9 All the means of communication between the navigating bridge and the machinery control positions, as well as the bridge and the main alternative steering position, if fitted, are to be tested. It is to be confirmed that means of indicating the angular position of the rudder are operating satisfactorily.

3.2.3.10 Confirmation that with ships having emergency steering positions there are means of relaying heading information and, when appropriate, supplying visual compass readings to the emergency steering positions.

3.2.3.11 Confirmation that various alarms required for hydraulic power operated, electric and electro-hydraulic steering gears are, operating satisfactorily and that the recharging arrangements for hydraulic power operated steering gears are being maintained.

3.2.3.12 Examining the means for the operation of the main and auxiliary machinery essential for propulsion and the safety of the ship, including when applicable, the means of remotely controlling the propulsion machinery from the navigating bridge and the arrangements to operate the main and other machinery from a machinery control room.

3.2.3.13 Confirmation that the engine room telegraph, the second means of communication between the navigation bridge and the machinery space and the means of communication with any other positions from which the engines are controlled are operating satisfactorily.

3.2.3.14 Examination and testing of the general emergency alarm system and confirmation that the engineer's alarm is clearly audible in the engineer's accommodation.

3.2.3.15 It is to be confirmed that ballasting and de-ballasting arrangements are satisfactory. The bilge pumping systems and bilge wells including operation of each bilge pump, extended spindles and level alarms, where fitted, are to be examined as far as is practicable. It is also to be confirmed that bilge pumping system for each watertight compartment is satisfactory.

It is also to be confirmed that drainage from enclosed cargo spaces situated on freeboard deck is satisfactory.

3.2.3.16 Examining visually the condition of non-metallic expansion joints where fitted in piping systems which penetrate the ship's side, with both the penetration and the expansion joint located below the deepest load waterline, and checking the service record.

3.2.3.17 General examination visually and in operation, as feasible, of the main electrical machinery, the emergency sources of electrical power, the switch gear, other electrical equipment including the lighting system is to be carried out.

3.2.3.18 Confirmation as far as practicable, the operation of the emergency source(s) of electrical power, including their starting arrangement, the systems supplied, and when appropriate, their automatic operation.

3.2.3.19 Examining in general, that the precautions provided against shock, fire and other hazards of electrical origin are being maintained.

3.2.3.20 General Examination of automation equipment is to be carried out. Satisfactory operation of safety devices, bilge level detection and alarm systems and control systems is to be verified.

3.2.3.21 Confirmation that machinery space ventilation systems are in good working condition.

3.2.3.22 Examination so far as is possible of cargo, bilge, ballast and stripping pumps for excessive gland seal leakage, verification of proper operation of electrical and mechanical remote operating and shutdown devices and operation of pump room bilge system, and checking that pump foundations are intact.

3.2.3.23 Checking the protection of cargo pump room and in particular:

a) Checking temperature sensing devices for bulkhead glands and alarms;

b) Checking interlock between lighting and ventilation.

c) Checking gas detection system.

d) Checking bilge level monitoring devices and alarms.

3.2.3.24 Verification that installed pressure gauges on cargo discharge lines and level indicating systems are operational.

3.2.3.25 Examination of emergency lighting in all cargo pump rooms of tankers constructed after 1 July 2002.

3.2.3.26 For Ships constructed before 01 July 2017 and fitted with harmonic filters; following are to be verified at each survey:

a) Records of annual measurement of harmonic distortion levels of main bus bar under seagoing conditions and log entries showing which equipment was running and / or filters in service at the time of measurement. Harmonic distortion readings are to be carried out when the greatest amount of distortion is indicated by the measuring equipment.

b) Records of measurement of distortion levels, after any modification to the ship's electrical distribution system.

3.2.3.27 For Ships constructed on or after 01 July 2017 and fitted with harmonic filters; or for ships retrofitted with harmonic filters, on or after 01 July 2017; following are to be verified at each survey:

a) Engine log book containing distortion monitoring readings/ automatic records of same

b) Documentation of any approved relaxation on allowable distortion limits.

3.2.3.28 For ships contracted for construction on or after 01 Jan 2017 following are to be verified in emergency generator rooms at each survey:

a) Hand operated ventilation louvres, closing appliances, if fitted, are kept open and corresponding instruction plates are visible near the louvres.

b) Power operated ventilation louvres and closing appliances, if fitted, are fail-to-open type and capable of opening automatically whenever emergency generator is starting / in operation.

c) It is to be possible to close ventilation openings by a manual operation from a clearly marked safe position outside the space where the closing operation can be easily confirmed. Further, such closing is not possible from any other remote position. The louvre status (open/closed) is indicated at this position.

3.2.3.29 For vessels fitted with ICCP system, proper functioning of the system is to be checked and records of system operation are to be verified. It is to be verified that all maintenance as per manufacturer's instructions are carried out.

3.2.4 Fire protection, detection and extinction

3.2.4.1 The arrangements for fire protection, detection and extinction are to be examined and are to include confirmation that no changes have been made in the structural fire protection. Following are to be examined / verified:

a) verification that fire control plans are properly posted;

b) examination as far as possible and testing as feasible, any fire detection and alarm system and any sample extraction smoke detection system;

c) examination of the fire main system and verification that each fire pump including the emergency fire pump can be operated separately so that the two required powerful jets of water can be produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main;

d) verification that fire hoses, nozzles, applicators and spanners are in good working condition and situated at their respective locations;

e) examination of fire-fighting system (including fixed fire-fighting system for the machinery spaces) controls, piping,

instructions and marking, checking for evidence of proper maintenance and servicing including date of last systems tests;

f) verification that all semi-portable and portable fire extinguishers are in their stowed positions, checking for evidence of proper maintenance and servicing, conducting random check for evidence of discharged containers;

g) verification, as far as practicable, that the remote controls for stopping fans and machinery in machinery spaces are in working order. Examination of the arrangements for oil fuel, lubricating oil and other flammable oils and testing the remote closing of valves for oil fuel, lubricating oil and other flammable oils and , the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils;

h) examination of the closing arrangements of ventilators, funnel annular spaces, skylights, doorways and tunnel, where applicable;

i) confirmation that the fire fighters' outfits including its self-contained compressed air breathing apparatus and emergency escape breathing devices (EEBDS) are complete and in good condition and that the cylinders, including the spare cylinders, of any required self-contained breathing apparatus are suitably charged and that on board means of recharging breathing apparatus cylinders used during drills or a suitable number of spare cylinders to replace those used are provided, and provision of two-way portable radiotelephone apparatus of an explosion-proof type or intrinsically safe;

j) examination of any manual and automatic fire doors and proving their operations;

k) verification that the pump room ventilation system is operational, ducting intact, dampers operational and screens are clean;

l) external examination of piping and cut-out valves of cargo tank and cargo pump room fixed fire-fighting system. Confirming, as far as practicable and when appropriate, the operation of the remote means for closing the various openings;

m) verification that the deck foam system and deck sprinkler system are in good operating condition so far as is practicable;

n) verification that all electrical equipment in dangerous zones is in good condition and has been properly maintained so far as is practicable;

o) checking the deck foam system, including the supplies of foam concentrate and testing that minimum number of jets of water at the required pressure in the fire main is obtained when the system is in operation;

p) examination of the fire-extinguishing systems for spaces containing paint and/or flammable liquids and deep fat cooking equipment in accommodation and service spaces;

q) examination of the fire safety requirements of helicopter facilities, including foam firefighting appliances ;

r) Examination of the fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space;

s) Examination of the fire protection arrangements in cargo spaces and confirming, as far as practicable and as appropriate, the operation of the means of control provided for closing the various openings;

t) Checking the provision of at least one portable instrument for measuring oxygen and one for measuring flammable vapour concentrations, together with a sufficient set of spares and suitable means for the calibration of these instruments;

u) Examination of arrangements for gas measurement in double-hull spaces and double bottom spaces, including the fitting of permanent gas sampling lines, where appropriate;

v) Examination, as far as possible, and testing the fixed hydrocarbon gas detection system;

w) Confirmation that potential sources of ignition in or near the cargo pump room are eliminated, such as loose gear, combustible materials, etc., that there are no signs of undue leakage of cargo and that access ladders are in good condition;

x) Testing the fire dampers of ventilation ducts and the means of closing the main inlets and outlets of all ventilation systems and testing the means of stopping power ventilation systems from outside the space served.

Surveys carried out by the National Authority of the country in which the ship is registered would normally be accepted as meeting these requirements, at the discretion of the Surveyor.

3.2.4.2 For tankers fitted with inert gas system, examination of the inert gas system as detailed in Section 12, Para 12.2.1 is to be carried out.

3.2.4.3 Confirmation that the means of escape from accommodation, machinery spaces and other spaces are satisfactory.

3.2.4.4 Examination of the arrangements for gaseous fuel for domestic purposes.

3.3 Intermediate Surveys

3.3.1 General

3.3.1.1 Intermediate surveys are to be carried out at or between the second or third Annual Survey. (Also refer to Note 9 of Sec.1, Table 1.1.1). However, only those items which are additional to the requirements of annual survey may be examined between the second or third annual survey.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

3.3.1.2 The following requirements are applicable for vessels over five years of age. For vessels below 5 years of age additional examination over and above the requirements of Annual survey may be required at the discretion of the Surveyors.

3.3.1.3 For vessels over 10 years of age a specific survey program is to be worked out by the owner in cooperation with the Surveyors considering the requirements of the previous special survey, executive hull summary of that survey, later relevant survey records and taking account of any amendments to the survey requirements after the last special survey. The survey program is to be submitted in written format for approval and kept on board until the intermediate survey has been completed. (See 3.4.2 for guidance on preparation of survey program).

3.3.1.4 Prior to the commencement of any part of the intermediate survey, a survey planning meeting is to be held between the attending Surveyor(s), the owner's representative in attendance and where involved, the thickness measurement firm representative and the master of ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

3.3.2 Examination and testing

3.3.2.1 The following are to be carried out:

a) The survey to the extent specified in 3.3.3 to 3.3.5 depending on the age and type of the tanker.

b) For weather decks, an examination as far as applicable of cargo, crude oil washing, bunker, ballast, steam and vent piping systems as well as vent masts and headers. If upon examination there is any doubt as to the condition of the piping, the piping may be required to be pressure tested, thickness measured or both.

c) For vessels built under IACS Common Structural Rules, the identified substantial corrosion areas are required to be examined and additional thickness measurements are to be carried out.

d) A general examination of electrical equipment in dangerous zones and testing of insulation resistance of the circuits. In cases where a proper record of testing is maintained, consideration may be given to accepting recent readings. These measurements are not to be attempted until the ship is in gas free condition and

are to be carried out within an acceptable time period.

3.3.3 Oil tankers between 5 and 10 years of age

3.3.3.1

a) Single hull oil tankers:

All ballast tanks are to be examined. When considered necessary by the Surveyor, thickness measurement and testing are to be carried out to ensure that the structural integrity remains effective.

b) Double hull oil tankers:

For tanks used for water ballast, an Overall Survey of Representative Tanks selected by the Surveyor is to be carried out. If such inspections reveal no visible structural defects, the examination may be limited to a verification that the hard Protective Coating remains in GOOD condition.

3.3.3.2 In addition to the requirements above, suspect areas identified at previous surveys are to be close-up surveyed.

3.3.3.3 A ballast tank is to be examined at subsequent annual intervals where:

a) a hard protective coating has not been applied from the time of construction, or

b) a soft or semi-hard coating has been applied, or

c) substantial corrosion is found within the tank, or

d) the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the Surveyor.

3.3.4 Oil tankers between 10 and 15 years of age

3.3.4.1 The requirements of the intermediate survey are to be to the same extent as the previous special survey as required in 3.4 for hull structure and piping systems in way of cargo tanks, pump rooms, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks. However, pressure testing of cargo and ballast tanks, survey of automatic air pipe heads and specified in 3.4.6.10 and the requirements for longitudinal strength evaluation of hull girder required in 3.4.10.2 are not required unless deemed necessary by the attending surveyor. Thickness measurement is to be carried out for items 1) to 4) of Table 3.4.9.1.

3.3.4.2 The intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion by the third annual survey in lieu of the application of 3.4.1.4.

3.3.4.3 An in-water survey complying with the requirements of Sec.7.2 may be accepted in lieu of the requirements of 3.4.1.9.

3.3.5 Oil tankers exceeding 15 years of age the following is to apply:

3.3.5.1 The requirements of the intermediate survey are to be to the same extent as the previous special survey as required in 3.4 for hull structure and piping systems in way of cargo tanks, pump rooms, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks. However, pressure testing of cargo and ballast tanks, survey of automatic air pipe heads as specified in 3.4.6.10 and the longitudinal strength evaluation of hull girder required in 3.4.10.2 need not be carried out unless deemed necessary by the attending surveyor. Thickness measurement is to be carried out for items 1) to 4) of Table 3.4.9.1.

3.3.5.2 The intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion by the third annual survey in lieu of the application of 3.4.1.4.

3.3.5.3 A survey in dry dock is to be part of the intermediate survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of cargo tanks and ballast tanks (i.e. parts below light ballast water line) are to be completed in the dry-dock.

3.4 Special Surveys - Hull

3.4.1 General

3.4.1.1 All ships classed with IRS are to undergo Special Surveys at 5 yearly intervals. The first Special Survey is to be completed within 5 years from the date of the initial classification survey and thereafter 5 years

from the assigned date of the previous Special Survey. However, an extension of class of 3 months maximum beyond the 5th year may be granted in exceptional circumstances in accordance with 1.11. In such cases, the next period of class will start from the expiry date of the Special Survey before extension was granted.

3.4.1.2 The interval between the Special Surveys may be reduced at the request of the parties concerned or by IRS if considered appropriate.

3.4.1.3 For surveys completed within 3 months before the expiry date of the Special Survey, the next period of class will start from the expiry date of the Special Survey. For surveys completed more than 3 months before the expiry date of the Special Survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the special survey. If the owner elects to carry out the next due special survey, the period of class will start from the survey completion date. Any requirement of the Flag Administration in this regard is also to be complied with.

3.4.1.4 The Special Survey may be commenced at the 4th Annual Survey and be progressed with a view to completion by the 5th anniversary date. When the special survey is commenced prior to the fourth annual survey, the entire survey is to be completed within 15 months if such work is to be credited to the special survey and in this case the next period of class will start from the survey completion date.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

3.4.1.5 As part of the preparation for Special Survey, the proposed Survey Programme (See 3.4.2) including the schedule for thickness measurements (See 3.4.9) are to be submitted in advance of the Special Survey.

3.4.1.6 Record of Special Survey will not be assigned until the Machinery Survey has been completed or postponed in agreement with IRS.

3.4.1.7 Ships which have satisfactorily passed a Special Survey will have a record entered in the Supplement to the Register Book indicating the assigned date of Special Survey. In addition, a notation "ESP" will be entered for oil tankers.

3.4.1.8 The special survey is to include, in addition to the requirements of the Annual Survey, examination, tests and checks of sufficient extent to ensure that the hull, equipment and related piping are in satisfactory condition and that the ship is fit for its intended purpose for the next five (5) year class period, subject to proper maintenance and operation and the periodical surveys being carried out at the due dates.

3.4.1.9 A Docking Survey in accordance with the requirements of Sec.7 is to be carried out as part of the Special Survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of cargo tanks and ballast tanks (i.e. parts below light ballast water line) are to be completed in the dry-dock.

3.4.2 Planning and preparation for survey

3.4.2.1 A specific Survey Programme is to be worked out in advance of the Survey by the Owner in cooperation with the Surveyors and submitted to IRS for approval. The Survey Programme is to be in a written format based on IMO Resolution MSC197(80). The survey is not to commence until the survey programme has been agreed. Prior to the development of the survey programme, the survey planning questionnaire based on IMO Resolution MSC197(80) is to be completed by the owner and forwarded to IRS. The Survey Programme at Intermediate Survey may consist of the Survey Programme at the previous Special Survey supplemented by the Executive Hull Summary of that Special Survey and later relevant survey reports. The survey program is to be worked out taking into account any amendments to the survey requirements after the last special survey.

3.4.2.2 In developing the Survey Programme, the following documentation is to be collected and consulted with a view to selecting tanks, areas and structural areas for examination :

a) Survey status and basic ship information;

b) Documentation on board as per 3.4.3;

c) Main structural plans (scantling drawings), including information regarding use of high strength steels, clad steel and stainless steel;

d) Executive Hull Summary;

e) Relevant previous damage and repair history;

f) Relevant previous survey and inspection reports of IRS and the Owners;

g) Cargo and ballast history for the last 3 years, including carriage of cargo under heated conditions;

h) Details of the inert gas plant and tank cleaning procedures;

i) Information and other relevant data regarding conversion or modification of the ship's cargo and ballast tanks since the time of construction.

j) Description and history of the coating and corrosion protection system (including previous class notations), if any;

k) Inspections by the Owner's personnel during the last 3 years with reference to structural deterioration in general, leakages in tank boundaries and piping and condition of the coating and corrosion protection system if any;

l) Information regarding the relevant maintenance level during operation including port state control reports of inspection containing hull related deficiencies. Safety Management System non-conformities relating to hull maintenance, including the associated corrective action(s); and

m) Any other information that will help identify suspect areas and critical structural areas.

3.4.2.3 The Survey Programme submitted for approval is to account for and comply, as a minimum, with the requirements of close-up survey, thickness measurement and tank testing respectively, and to include relevant information including at least :

a) Basic ship information and particulars;

b) Main structural plans (scantling drawings), including information regarding use of high strength steels, clad steel and stainless steel;

c) Plan of tanks;

d) List of tanks with information on use, extent of coatings and corrosion protection systems;

e) Conditions for survey (e.g. information like tank cleaning, gas freeing, ventilation, lighting, etc.);

f) Provisions and methods of access to structures;

g) Equipment for survey; and

h) Identification of tanks and areas for close-up survey (As per Table 3.4.8.1a or Table 3.4.8.1b)

i) Identification of sections for thickness measurement (As per Table 3.4.9.1)

j) Identification of tanks for testing

k) Identification of the thickness measurement firm.

l) Damage experience related to the ship in question.

m) Critical structural areas and suspect areas, where relevant.

3.4.2.4 IRS will advise the Owner of the maximum acceptable structural corrosion diminution levels applicable to the vessel.

3.4.2.5 The ship is to be prepared for overall survey in accordance with the requirements of Table 3.4.2.5. The preparation is to be of sufficient extent to facilitate an examination to ascertain any excessive corrosion, deformation, fractures, damages and other structural deterioration.

3.4.2.6 Proper preparation and the close co-operation between the attending surveyor(s) and the owner's representatives onboard prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings are to be held regularly

3.4.2.7 Prior to commencement of any part of the special survey, a survey planning meeting is to be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement firm representative and the master of the ship or an

appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

Table 3.4.2.5 : Survey preparation

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) All tanks, peaks, bilges and drain wells, engine and boiler spaces and other spaces are to be cleared out and cleaned as necessary for examination. Floor plates in engine and boiler spaces are to be lifted as may be necessary for examination of the structure underneath. Where necessary ceiling, lining, casings and loose insulation are to be removed as required by the Surveyor for examination of the structure. Compositions on the plating are to be examined and sounded, but need not be disturbed if found satisfactorily adhering to the plating 2) The steelwork is to be exposed and cleaned as may be required for its proper examination by the Surveyor and close attention is to be paid to the parts of the structure which are particularly liable to excessive corrosion or to deterioration due to other causes. 3) All tanks are to be cleaned as necessary to permit examination. 4) Casings or covers of air, sounding, steam and other pipes, spar ceiling and lining in way of the side scuttles are to be removed, as required by the Surveyor.	1) Requirements of Special Survey I to be complied with 2) The chain locker is to be cleaned internally. The chain cables are to be ranged for inspection. The anchors are to be cleaned and placed in an accessible position for inspection.	1) Requirements of Special Survey II to be complied with 2) Portions of wood sheathing, or other covering, on steel decks are to be removed, as considered necessary by the Surveyor, in order to ascertain the condition of the plating.	1) Requirements of Special survey III to be complied with

3.4.2.8 The following is an indicative list of items that are to be addressed in the meeting:

a) Schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);

b) Provisions and arrangements for thickness measurements (i.e. access, cleaning/de-scaling, illumination, ventilation, personal safety);

c) Extent of the thickness measurements;

d) Acceptance criteria (refer to the list of minimum thicknesses);

e) Extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;

f) Execution of thickness measurements;

g) Taking representative readings in general and where uneven corrosion / pitting is found;

h) Mapping of areas of substantial corrosion; and

Communication between attending surveyor(s) the thickness measurement firm operator(s) and owner representative(s) concerning findings.

3.4.3 Documentation on board

3.4.3.1 The Owners are to obtain supply and maintain on board documentation as specified in 3.4.3.3, which is to be readily available for the Surveyor.

3.4.3.2 The documentation is to be kept on board for the life time of the ship.

3.4.3.3 A Survey Report File is to be a part of the documentation on board consisting of

- Reports of structural surveys
- Executive Hull Summary
- Thickness measurement reports.

3.4.3.4 The Survey Report File is to be available also in the Owners management offices and with IRS.

3.4.3.5 The following additional supporting documentation is to be available on board:

- Main structural plans of cargo and ballast tanks (for CSR ships these plans are to include for each structural element both the as-built and renewal thickness. Any thickness for voluntary addition is also to be clearly indicated on the plans. The midship section plan to be supplied on board the ship is to include the minimum allowable hull girder sectional properties for the tank transverse section in all cargo tanks)
- Previous repair history
- Cargo and ballast history
- Extent of use of inert gas plant and tank cleaning procedures
- Inspection by ship's personnel with reference to
  - structural deterioration in general
  - leakages in bulkheads and piping
  - condition of coating or corrosion prevention system, if any
- Any other information that will help identify critical structural areas and/or Suspect Areas requiring inspection.
- Survey Programme as required by 3.4.2 until such time as the Special Survey has been completed.

3.4.3.6 Prior to survey, the Surveyor is to examine the completeness of the documentation on board, and its contents as a basis for the survey.

3.4.4 Surveys at sea or at anchorage

3.4.4.1 See Sec.1.8.

3.4.5 Space protection

3.4.5.1 Where provided, the condition of coating or corrosion protection of cargo tanks and ballast tanks is to be examined.

A ballast tank is to be examined at subsequent annual intervals where:

a) a hard protective coating has not been applied from the time of construction, or

b) a soft or semi-hard coating has been applied, or

c) substantial corrosion is found within the tank, or

d) the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the Surveyor.

3.4.5.2 Where the hard protective coating in tanks is found to be in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

3.4.6 Survey and examination

3.4.6.1 All spaces within the hull and superstructure are to be examined. The special survey is to include examination of underwater parts as per Section 7.

3.4.6.2 All tanks other than cargo and water ballast tanks are to be examined internally in accordance with the requirements of Table 3.4.6.2.

3.4.6.3 All spaces including double bottom, deep, ballast, peak and cargo tanks; pump rooms, pipe tunnels, duct keels, machinery spaces, dry spaces, cofferdams and voids are

to be internally examined including the plating and framing, bilges and drain wells, sounding, venting pumping and drainage arrangements. This examination is to be supplemented by testing and thickness measurement as required by 3.4.7 and 3.4.9 to ensure that the structural integrity remains effective. The aim of the examination is to discover Substantial Corrosion, significant deformation, fractures, damages or other structural deterioration and if deemed necessary by the Surveyor, a suitable non-destructive examination may be required. At special survey No.3 and subsequent special surveys, structural downflooding ducts and structural ventilation ducts are to be internally examined.

3.4.6.4 Cargo piping on deck including crude oil washing (COW) piping and cargo and ballast piping systems within the above tanks and spaces are to be examined and operationally tested to working pressure to attending Surveyor's satisfaction to ensure that tightness and condition remain satisfactory. Special attention is to be given to ballast piping in cargo tanks and cargo piping in Ballast Tanks and void spaces and Surveyors are to be advised on all occasions when the piping, including valves and fittings are open during repair periods and can be examined internally.

Table 3.4.6.2 : Requirements for internal examination of tanks

Tank	Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequent Age > 15
Fuel oil bunker tanks				
- Engine room	None	None	One	One
- Cargo area	None	One	Two	Half the number of tanks, minimum 2
- If no tanks in Cargo Area, additional fuel tank(s) outside of Engine Room (if fitted)	None	One	One	Two
Lub.oil	None	None	None	One
Fresh water	None	One	All	All

Notes:

1) These requirements apply to tanks of integral (structural) type.

2) If a selection of tanks is accepted to be examined, then different tanks are to be examined at each special survey, on a rotational basis.

3) Peak tanks (all uses) are subject to internal examination at each special survey.

4) At special survey No.III and subsequent special surveys, one deep tank for fuel oil is to be included, if fitted.

3.4.6.5 Where ballast tanks have been converted to void spaces, the survey extent is to be based upon ballast tank requirements.

3.4.6.6 All watertight bulkheads and watertight doors are to be examined. All decks, casings and superstructures are to be examined. Attention is to be given to the corners of openings and other discontinuities in way of the strength decks and top sides.

3.4.6.7 The masts, standing rigging and anchors are to be examined.

The Surveyor should satisfy himself that there are sufficient mooring ropes on board and also

that a tow line is provided when this is a Rule requirement.

3.4.6.8 The steering gear, and its connections and control systems (main and alternative) are to be examined. The auxiliary steering gear with its various parts are to be examined in working condition.

3.4.6.9 The hand pumps and suctions, air and sounding pipes are to be examined. The Surveyors are to ensure that striking plates are fitted under the sounding pipes whilst examining the tanks internally.

Automatic air pipe heads are to be internally examined at special surveys as indicated in Table 3.4.6.10. For designs where the inner parts cannot be properly inspected from outside, the head is to be removed from the air pipe. Particular attention is to be paid to the condition of the zinc coating in heads constructed from galvanised steel.

3.4.6.10 The Surveyor should satisfy himself regarding the efficient condition of the following :

- Means of escape from machinery spaces, crew and passenger spaces and spaces in which crew are normally employed;

- Means of communication between bridge and engine room and between bridge and alternative steering position;

- Helm indicator;

- Protection to the aft steering wheel and the gear.

3.4.6.11 The chain cables are to be ranged and the anchors and the chain cables are to be examined. At special survey no. II and subsequent special surveys, the chain cables are to be gauged. Any length of chain cable which is found to have reduced in mean diameter at its most worn part by more than 12 per cent of its original rule diameter is to be renewed.

3.4.6.12 The windlass is to be examined.

Table 3.4.6.10 : Requirements for internal examination of automatic air pipe heads

Location	Special survey No.I Age ≤ 5	Special survey No.II 5 < Age ≤ 10	Special survey No.III and subsequent Age > 10
forward 0.25L	Two air pipe heads[1,2] one port and one starboard on exposed decks	All air pipe heads on exposed decks	All air pipe heads[3] on exposed decks
aft of 0.25L from the forward perpendicular	Two air pipe heads[1,2] one port and one starboard on exposed decks	At least 20% of air pipe heads[1,2] on exposed decks	

Notes:

1. Preferably air pipes serving ballast tanks.
2. The selection of air pipe heads is to be made by the attending Surveyor. According to the results of the inspection, the Surveyor may require additional air pipe heads to be examined.
3. When there is substantiated evidence of replacement within the previous five years, individual air pipe heads may not be examined.

3.4.6.13 A thorough examination of aft and forward emergency towing arrangements is to be carried out to ensure its ready availability and satisfactory condition.

- examination of pick-up gear, towing pennant and chafing gear for possible deterioration;

- examination of strong points, fairleads, pedestal roller and their attachment to hull structure.

The survey is to confirm that one of the towing arrangements (aft or forward) are pre-rigged.

3.4.6.14 For vessels with single point mooring arrangements where 'SPM' notation is assigned, the following is to be carried out:

a) A thorough examination of the components of the single point mooring system (bow chain stoppers, bow fairleads, pedestal roller fairleads, winches and capstans) to verify their satisfactory condition.
b) A close-up examination of the hull structures supporting and adjacent to the components of the single point mooring system to verify that there is no deformation or fracture. Thickness determination and non-destructive tests are to be carried out if required by the Surveyor.

3.4.6.15 The chain locker, hold fasts, hawse pipes and chain stoppers are to be examined and pumping arrangements of the chain locker tested

3.4.6.16 Engine room structure is to be examined. Particular attention being given to tank tops, shell plating in way of tank tops, brackets connecting side shell frames and tank tops and engine room bulkheads in way of tank tops and bilge wells. Where excessive areas of wastage are found, thickness measurements are to be carried out and renewals of repairs made when wastage exceeds allowable limits.

3.4.6.17 The attachment to the structure and condition of anodes in tanks is to be examined.

3.4.6.18 Where fitted, the strums of the cargo suction pipes are to be removed or lifted to facilitate examination of the shell plating and bulkheads in the vicinity, unless other means for visual inspection of these parts are provided.

3.4.6.19 The loading instrument is to be checked for accuracy by applying test load conditions in presence of the Surveyor.

3.4.6.20 Examination of accommodation ladders, gangways and their winches are to be carried out as required for annual surveys. In addition, the accommodation ladders and gangways are to be operationally, tested with the specified maximum operation load.

The tests are to be carried out with the load applied as uniformly as possible along the length of the accommodation ladder or gangway, at an angle of inclination corresponding to the maximum bending moment on the accommodation ladder or gangway.

Accommodation ladder winch is to be operationally tested at special surveys. The brake system of the winch is to be tested for holding the maximum operational load on the ladder.

For existing installations on board ships constructed prior to 01 Jan 2010 where the maximum operational load is not known, load nominated by the shipowner or operator may be considered as the test load.

3.4.7 Extent of tank testing

3.4.7.1 The minimum requirements for tank testing for oil tankers (including ore/oil and ore/bulk/oil ships) in the cargo area are given in Table 3.4.7.1.

Table 3.4.7.1 : Table of minimum requirements for tank testing at special survey of oil tankers, ore/oil ships and ore/bulk/oil ships (including double hull oil tankers)

Special Survey No. I Age ≤ 5	Special Survey No. II and subsequent Age > 5
1 All ballast tank boundaries	1 All ballast tank boundaries
2 Cargo tank boundaries facing ballast tanks, void spaces, pipe tunnels, pump rooms or cofferdams	2 All Cargo tank bulkheads

3.4.7.2 All double bottom and peak tanks are to be tested in addition. The Surveyor may extend the tank testing as deemed necessary.

3.4.7.3 Boundaries of ballast tanks are to be tested with a head of liquid to the top of air pipes. Boundaries of cargo tanks are to be tested to the highest point that liquid will rise under service conditions.

3.4.7.4 Boundaries of fuel oil, lub.oil and fresh water tanks are to be tested with a head of

liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil, lub.oil and fresh water tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the Master stating that the pressure testing has been carried out according to the requirement with satisfactory results

3.4.7.5 For double hull oil tankers the testing of double bottom tanks and other spaces not designed for the carriage of liquid may be omitted, provided a satisfactory internal examination together with an examination of the tank top is carried out.

3.4.7.6 Cargo tank testing carried out by the ship's crew under the direction of the Master may be accepted by the surveyor provided the following conditions are complied with:

a) the tank testing procedure, specifying fill heights, tank being filled and bulkheads being tested, taking into account the guidelines provided in the ESP Planning document, has been submitted by the owner and reviewed by IRS prior to the testing being carried out;

b) the tank testing is carried out prior to overall survey and close-up survey;

c) the tank testing is carried out within the special survey window and not more than 3 months prior to the date on which the overall and close up survey is completed;

d) the tank testing has been satisfactorily carried out and there is no record of leakage, distortion or substantial corrosion that would affect the structural integrity of the tank;

e) the satisfactory results of the testing is recorded in the vessel's logbook; and

f) the internal and external condition of the tanks and associated structure are found satisfactory by the surveyor at the time of the overall and close up survey.

3.4.8 Close-up survey requirements

3.4.8.1 The minimum requirements for Close-up Surveys at Special Survey are given in Table 3.4.8.1a and Table 3.4.8.1b as applicable.

3.4.8.2 The Surveyor may extend the Close-up Survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases:

a) In particular, tanks having structural arrangements or details which have suffered defects on similar tanks or on similar ships according to available information.
b) In tanks which have structures approved with reduced scantlings due to an approved corrosion control system.

3.4.8.3 For areas in tanks where coatings are found to be in a GOOD condition, the extent of Close-up Surveys according to Tables 3.4.8.1a and Table 3.4.8.1b may be specially considered by the Surveyor.

Table 3.4.8.1a : Minimum requirements to close-up surveys at special survey of oil tankers, ore/oil ships, ore/bulk/oil ships other than double hull oil tankers

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
A) One Web Frame Ring - in a ballast wing tank, if any, or a cargo wing tank used primarily for water ballast	**A) All Web Frame Rings** - in a ballast wing tank, if any, or a cargo wing tank, used primarily for water ballast	**A) All Web Frame Rings** - in all ballast tanks	As Special Survey No.3
B) One Deck Transverse - in a cargo oil tank	B) **One Deck Transverse** - in each of the remaining ballast tanks, if any	**A) All Web Frame Rings** - in a cargo wing tank	Additional transverses as deemed necessary by the Surveyor
D) **One Transverse Bulkhead** - in a ballast tank	B) **One Deck Transverse** - in a cargo wing tank	**A)**. A minimum of 30% of web frame rings [1)] in each remaining cargo wing tank	
D) **One Transverse Bulkhead** - in a cargo oil wing tank	B) **One Deck Transverse** - in two cargo centre tanks	**C) All Transverse Bulkheads** - in all cargo and ballast tanks	
D) **One Transverse Bulkhead** - in a cargo oil centre tank	C) **Both Transverse Bulkheads** - in a wing ballast tank, if any, or a cargo wing tank used primarily for water ballast	**E**) A minimum of 30% of deck and bottom transverses [1)] including adjacent structural members in each cargo centre tank	
	D) **One Transverse Bulkhead** - in each remaining ballast tank	**F)** As considered necessary by the Surveyor	
	D) **One Transverse Bulkhead** - in a cargo oil wing tank D) **One Transverse Bulkhead** - in two cargo centre tanks		

A) Complete transverse web frame ring including adjacent structural members

B) Deck transverse including adjacent deck structural members

C) Transverse bulkhead complete - including girder system and adjacent members

D) Transverse bulkhead lower part - including girder system and adjacent structural members

E) Deck and bottom transverse including adjacent structural members

F) Additional complete transverse web frame ring

Note : 1) The 30% is to be rounded up to the next whole integer.

Table 3.4.8.1b : Minimum requirements for close-up survey at special survey of Double Hull Oil Tankers			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
One web frame **(1)**, in a complete ballast tank (see Note 1)	All web frames **(1)**, in a complete ballast tank (see Note 1) The knuckle area and the upper part (5 metres approximately) of one web frame in each remaining ballast tank **(6)**	All web frames **(1)**, in all ballast tanks	As for Special Survey for age from 10 to 15 years Additional transverse areas as deemed necessary by IRS
One deck transverse, in a cargo oil tank **(2)**	One deck transverse, in two cargo oil tanks **(2)**	All web frames **(7)**, including deck transverse and cross ties, if fitted, in a cargo oil tank One web frame **(7)**, including deck transverse and cross ties, if fitted, in each remaining cargo oil tank	
One transverse bulkhead **(4)**, in a complete ballast tank (see Note 1)	One transverse bulkhead **(4)**, in each complete ballast tank (see Note 1)	All transverse bulkheads, in all cargo oil **(3)** and ballast **(4)** tanks	
One transverse bulkhead **(5)** in a cargo oil centre tank One transverse bulkhead **(5)**, in a cargo oil wing tank (see Note 2)	One transverse bulkhead **(5)**, in two cargo oil centre tanks One transverse bulkheads **(5)**, in a cargo oil wing tank (see Note 2)		

(1), (2), (3), (4), (5), (6) and **(7)** are areas to be subjected to close-up surveys and thickness measurements (see Fig.3.4.8).

(1) : Web frame in a ballast tank means vertical web in side tank, hopper web in hopper tank, floor in double bottom tank and deck transverse in double deck tank (where fitted), including adjacent structural members. In fore and aft peak tanks web frame means a complete transverse web frame ring including adjacent structural members.

(2) : Deck transverse, including adjacent deck structural members (or external structure on deck in way of the tank, where applicable).

(3) : Transverse bulkhead complete in cargo tanks, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal structure of lower and upper stools, where fitted.

(4) : Transverse bulkhead complete in ballast tanks, including girder system and adjacent structural members, such as longitudinal bulkheads, girders in double bottom tanks, inner bottom plating, hopper side, connecting brackets.

(5) : Transverse bulkhead lower part in cargo tank, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal structure of lower stool, where fitted.

(6) : The knuckle area and the upper part (5 metres approximately), including adjacent structural members. Knuckle area is the area of the web frame around the connections of the slope hopper plating to the inner hull bulkhead and the inner bottom plating, upto 2 metres from the corners both on the bulkhead and the double bottom.

(7) : Web frame in a cargo oil tank means deck transverse, longitudinal bulkhead structural elements and cross ties, where fitted, including adjacent structural members.

Note 1 : Ballast tank : Apart from the fore and aft peak tanks, the term “ballast tank” has the following meaning:

.1 all ballast compartments (hopper tank, side tank and double-deck tank, if separate from double-bottom tank) located on one side, i.e. portside or starboard side, and additionally double-bottom tank on portside plus starboard side, when the longitudinal central girder is not watertight and, therefore, the doublebottom tank is a unique compartment from portside to starboard side; or

.2 all ballast compartments (double-bottom tank, hopper tank, side tank and double-deck tank) located on one side, i.e. portside or starboard side, when the longitudinal central girder is watertight and, therefore, the portside double-bottom tank separate from the starboard-side double-bottom tank.

Note 2 : Where no centre cargo tanks are fitted (as in the case of centre longitudinal bulkhead), transverse bulkheads in wing tanks are to be surveyed.

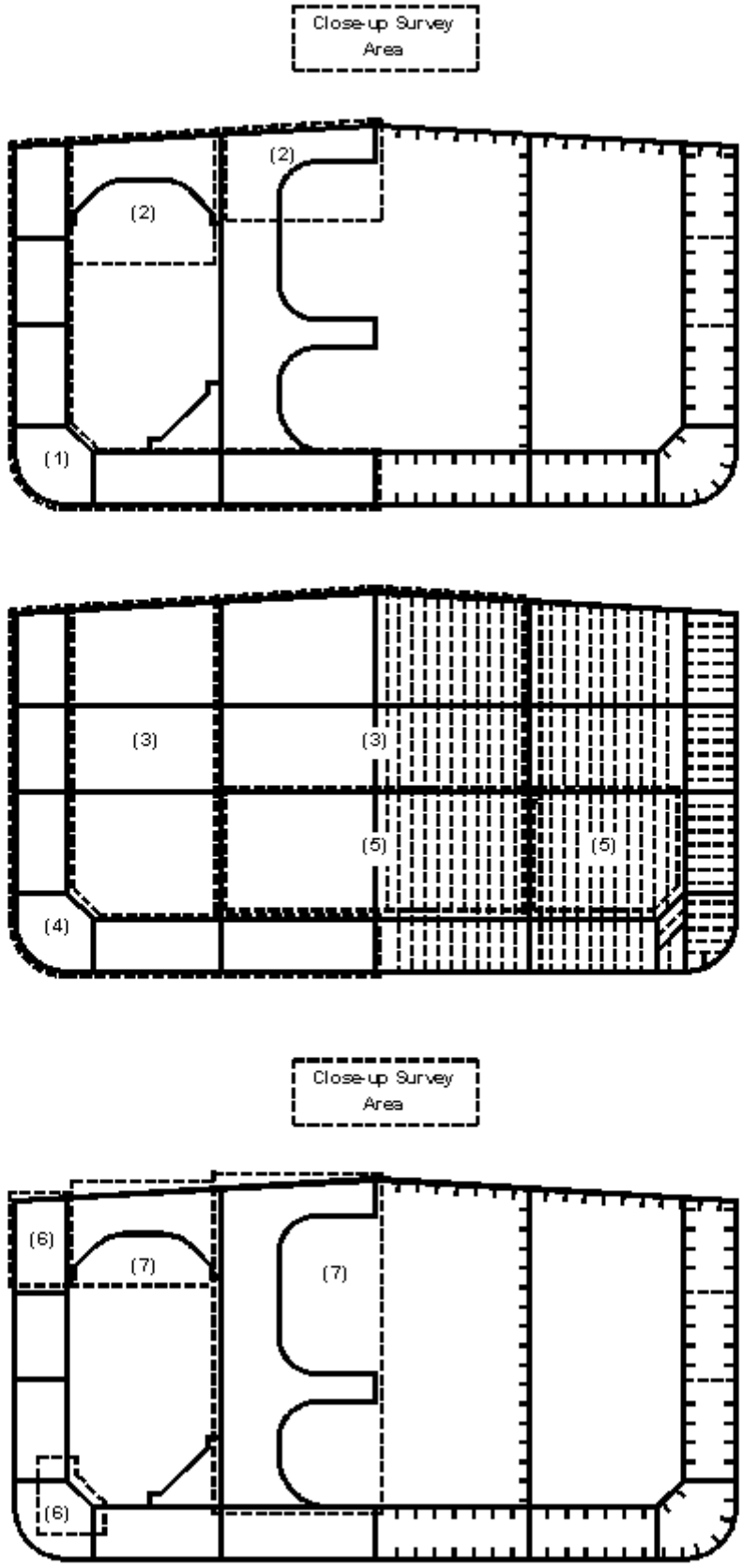

Fig.3.4.8 : Close-up survey requirements for double hull oil tankers areas (1) to (7)

3.4.9 Thickness measurement

3.4.9.1 The minimum requirements for thickness measurements at Special Survey are given in Table 3.4.9.1.

3.4.9.2 Provisions for extended measurements for areas with Substantial Corrosion in case of Oil Tankers, Ore/oil and Ore/Bulk/Oil Ships are given in Table 3.4.9.2a) and Table 3.4.9.2b).

For vessels built under IACS Common Structural Rules, the identified substantial corrosion areas are required to be examined and additional thickness measurements are to be carried out at annual and intermediate surveys.

The Surveyor may further extend the thickness measurements as deemed necessary.

3.4.9.3 In cases where two or three sections are to be measured in Oil Tankers, Ore/oil and Ore/Bulk/Oil Ships at least one is to include a Ballast Tank within 0.5L amidships.

Transverse sections are to be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

3.4.9.4 The thickness measurements are to be carried out by a qualified firm certified by IRS.

3.4.9.5 In order to ensure necessary control during the process of thickness measurements these are normally to be carried out under the supervision of the Surveyor. The Surveyor has the right to re-check the measurements as deemed necessary to ensure acceptable accuracy.

3.4.9.6 In all cases the extent of thickness measurements is to be sufficient as to represent the actual average condition.

3.4.9.7 A thickness measurement report is to be prepared. The report is to give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and has to be signed by the operator. The Surveyor is to review the report of the final thickness measurement after repairs have been carried out and countersign the cover page.

Table 3.4.9.1 : Table of minimum requirements to thickness measurements at special survey of oil tankers, ore/oil ships and ore/bulk/oil ships (including double hull oil tankers)

Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequently Age > 15
1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel
2) Within the cargo area: One section of deck plating for the full beam of the ship (in way of a ballast tank, if any, or a cargo tank used primarily for water ballast) (See Note 1)	2) Within the cargo area: a) Each deck plate b) One Transverse section	2) Within the cargo area: a) Each deck plate b) Two Transverse sections c) All wind and water strakes	2) Within the cargo area: a) Each deck plate b) Three Transverse sections c) All wind and water strakes d) Each bottom plate
3) Measurements of structural members subject to close-up survey according to Table 3.4.8.1a, for general assessment and recording of corrosion pattern(See Note 1)	3) Measurements of structural members subject to close-up survey according to Table 3.4.8.1a and Table 3.4.8.1b, for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 3.4.8.1a and Table 3.4.8.1b, for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 3.4.8.1a and Table 3.4.8.1b, for general assessment and recording of corrosion pattern

Table 3.4.9.1 (Contd.)

Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequently Age > 15
	4) Selected wind and water strakes outside the cargo area	4) Selected wind and water strakes outside the cargo area	4) All wind and water strakes outside the cargo area
		5) Internals in forepeak and after peak ballast tanks	5) Internals in forepeak and after peak ballast tanks
			6) All exposed main deck plating outside cargo area
			7) Representative exposed superstructure deck plating (poop, bridge, forecastle deck)
			8) All keel plates, additional bottom plates in way of machinery space, aft end of tanks and cofferdams outside cargo area
			9) a) Plating of sea chests. b) Shell plating in way of overboard discharges as considered necessary by the Surveyor.

Note:

1. Not applicable to Double Hull Oil Tankers

Table 3.4.9.2a : Requirements for extent of thickness measurement on areas of substantial corrosion in oil tankers, ore/oil ships and ore/bulk/oil ships other than double hull oil tankers

Structural Member		Extent of Measurement	Pattern of Measurement
Bottom Structure			
1	Bottom plating	Minimum of 3 bays across tank including aft bay. Measurements around and under all bell mouths	5 point pattern between stiffeners over 1 metre in length
2	Bottom longitudinal	Minimum of 3 longitudinals in each bay where bottom plating measured	3 measurements in line across flange and 3 measurements on vertically web
3	Bottom girders and brackets	At fore and after transverse bulkhead bracket toes and in centre of tanks	Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across face flat. 5 point pattern on girder/bulkhead brackets
4	Bottom transverse webs	3 webs in bays where bottom plating measured, with measurements at both ends and middle	5 point pattern over 2 square metre area. Single measurements on face flat
5	Panel stiffening	Where provided	Single measurements

Table 3.4.9.2a (Contd.)

Structural Member		Extent of Measurement	Pattern of Measurement
Deck Structure			
1	Deck plating	Two bands across tank	Minimum of three measurements per plate per band
2	Deck longitudinals	Minimum of 3 longitudinals in each of two bays	3 measurements in line vertically on webs and 2 measurements on flange (if fitted)
3	Deck girders and brackets	At fore and after transverse bulkhead bracket toes and in centre of tanks	Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across face flat. 5 point pattern on girder/bulkhead brackets
4	Deck transverse webs	Minimum of two webs with measurements at middle and both ends of span	5 points pattern over 2 square metre area. Single measurements on face flat
5	Panel stiffening	Where provided	Single measurements
Shell and Longitudinal Bulkheads			
1	Deckheads and bottom strakes and strakes in way of stringer platforms	Plating between each pair of longitudinals in a minimum of 3 bays	Single measurement
2	All other strakes	Plating between every 3rd pair of longitudinals in same 3 bays	Single measurement
3	Longitudinals - deckhead and bottom strakes	Each longitudinal in same 3 bays	3 measurements across web and 1 measurement on flange
4	Longitudinals - all others	Every 3rd longitudinal in same 3 bays	3 measurements across web and 1 measurement on flange
5	Longitudinals - brackets	Minimum of 3 at top, middle and bottom of tank in same 3 bays	5 point pattern over area of bracket
6	Web frames and cross ties	3 webs with minimum of three locations on each web, including in way of cross tie connections	5 points pattern over about 2 sq. metre area, plus single measurements on web frame and across tie face flats
Transverse Bulkheads and Swash Bulkheads			
1	Deckhead and bottom strakes and strakes in way of stringer platforms	Plating between pair of stiffeners at three locations - approx. 1/4, 1/2 and 3/4 width of tank	5 points pattern between stiffeners over 1 metre length
2	All other strakes	Plating between pair of stiffeners at middle location	Single measurement
3	Strakes in corrugated bulkheads	Plating for each change of scantling at centre of panel and at flange or fabricated connection	5 point pattern over about 1 sq. metre of plating
4	Stiffeners	Minimum of three typical stiffeners	For web, 5 point pattern over span between bracket connections (2 measurements across web at each bracket connection and one at centre of span). For flange, single measurements at each bracket toe and at centre of span
5	Brackets	Minimum of three at top, middle and bottom of tank	5 point pattern over areas of bracket
6	Deep webs and girders	Measurements at toe of bracket and at centre of span	For web, 5 point pattern over about 1 sq. metre. 3 measurements across face flat
7	Stringer platforms	All stringers with measurements at both ends and middle	5 point pattern over 1 sq. metre of area plus single measurements near bracket toes and on face flats

Table 3.4.9.2b : Requirements for extent of thickness measurements at those areas of substantial corrosion - Special Survey of Double Hull Oil Tankers

Structural Member		Extent of Measurement	Pattern of Measurement
Bottom, Inner Bottom and Hopper Structure			
1	Bottom, inner bottom and hopper structure plating	Minimum of 3 bays across double bottom tank including aft bay.	5 point pattern for each panel between longitudinals and floors
		Measurements around and under all suction bell mouths	
2	Bottom, inner bottom and hopper structure longitudinals	Minimum of 3 longitudinals in each bay where bottom plating measured	3 measurements in line across flange and 3 measurements on vertical web
3	Bottom girders, including the watertight ones	At fore and aft watertight floors and in centre of tanks	Vertical line of single measurements on girder plating with one measurement between each panel stiffener, or a minimum of three measurements
4	Bottom floors, including the watertight ones	3 floors in bays where bottom plating measured, with measurements at both ends and middle	5 points pattern over 2 square metre area
5	Hopper structure web frame ring	3 floors in bays where bottom plating measured	5 point pattern over one square metre of plating. Single measurements on flange
6	Hopper structure transverse watertight bulkhead or swash bulkhead	Lower 1/3 of bulkhead	5 point pattern over two square metre of plating
		Upper 2/3 of bulkhead	5 point pattern over two square metre of plating
		Stiffeners (minimum of three)	For web, 5 point pattern over span (two measurements across web at each end and one at centre of span). For flange, single measurements at each end and centre of span
7	Panel stiffening	Where provided	Single measurements
Deck Structure			
1	Deck plating	Two transverse bands across tank	Minimum of three measurements per plate per band
2	Deck longitudinals	Every third longitudinal in each of two bands with a minimum of one longitudinal	3 measurements in line vertically on webs and 2 measurements on flange (if fitted)
3	Deck girders and brackets (usually in cargo tanks only)	At fwd and aft transverse bulkhead, bracket toes and in centre of tanks	Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across flange. 5 point pattern on girder/bulkhead brackets
4	Deck transverse webs	Minimum of two webs with measurements at both ends and middle of span	5 points pattern over one square metre area. Single measurements on face flat
5	Vertical web and transverse bulkhead in wing ballast tank (two metres from deck)	Minimum of two webs and both transverse bulkheads	5 point pattern over one square metre area
6	Panel stiffening	Where applicable	Single measurements

Table 3.4.9.2b (Contd.)

	Structural Member	Extent of Measurement	Pattern of Measurement
Structure in Wing Ballast Tanks			
1	Side shell and longitudinal bulkhead plating: - Upper strake and strakes in way of horizontal girders - All other strakes	 - Plating between each pair of longitudinals in a minimum of 3 bays (along the tank) - Plating between every third pair of longitudinals in same 3 bays	 - Single measurement - Single measurement
2	Side shell and longitudinal bulkhead longitudinals on: - Upper strake - All other strakes	 - Each longitudinal in same 3 bays - Every third longitudinal in same 3 bays	 - 3 measurements across web and 1 measurement one flange - 3 measurements across web and 1 measurement on flange
3	Longitudinals - brackets	Minimum of three at top, middle and bottom of tank in same three bays	5 point pattern over area of bracket
4	Vertical web and transverse bulkheads (excluding deckhead area): - Strakes in way of horizontal girders - Other strakes	 - Minimum of two webs and both transverse bulkheads - Minimum of two webs and both transverse bulkheads	 - 5 point pattern over approx. two square metre area - two measurements between each pair of vertical stiffeners
5	Horizontal girders	Plating on each girder in a minimum of three bays	2 measurements between each pair of longitudinal girder stiffeners
6	Panel stiffening	Where applicable	Single measurements
Longitudinal Bulkheads in Cargo Tanks			
1	Deckheads and bottom strakes and strakes in way of the horizontal stringer of transverse bulkheads	Plating between each pair of longitudinals in a minimum of 3 bays	Single measurement
2	All other strakes	Plating between every 3rd pair of longitudinals in same 3 bays	Single measurement
3	Longitudinals on deckhead and bottom strakes	Each longitudinal in same 3 bays	3 measurements across web and 1 measurement on flange
4	All other longitudinals	Every third longitudinal in same three bays	3 measurements across web and 1 measurement on flange
5	Longitudinals - brackets	Minimum of 3 at top, middle and bottom of tank in same 3 bays	5 point pattern over area of bracket
6	Web frames and cross ties	3 webs with minimum of three locations on each web, including in way of cross tie connections	5 points pattern over approximately 2 sq. metre area of webs, plus single measurements on flanges of web frame and cross ties
7	Lower end brackets (opposite side of web frame)	Minimum of three brackets	5 points pattern over approximately 2 sq. metre area of brackets, plus single measurements on bracket flanges
Transverse Watertight and Swash Bulkheads in Cargo Tanks			
1	Upper and lower stool, where fitted	- Transverse band within 25 mm of welded connection to inner bottom/deck plating - Transverse band within 25 mm of welded connection to shelf plate	5 point pattern between stiffeners over one metre length
2	Deckhead and bottom strakes and strakes in way of horizontal stringers	Plating between pair of stiffeners at three locations - approx. 1/4, 1/2 and 3/4 width of tank	5 points pattern between stiffeners over 1 metre length
3	All other strakes	Plating between pair of stiffeners at middle location	Single measurement
4	Strakes in corrugated bulkheads	Plating for each change of scantling at centre of panel and at flange of fabricated connection	5 point pattern over about 1 sq. metre of plating

Table 3.4.9.2b (Contd.)			
Structural Member		**Extent of Measurement**	**Pattern of Measurement**
5	Stiffeners	Minimum of three typical stiffeners	For web, 5 point pattern over span between bracket connections (2 measurements across web at each bracket connection and one at centre of span). For flange, single measurements at each bracket toe and at centre of span
6	Brackets	Minimum of three at top, middle and bottom of tank	5 point pattern over areas of bracket
7	Horizontal stringer	All stringers with measurements at both ends and middle	5 point pattern over 1 sq. metre of area plus single measurements near bracket toes and on flange

3.4.10 Reporting and evaluation of survey

3.4.10.1 The data and information on the structural condition of the vessel collected during the survey is to be evaluated for acceptability and continued structural integrity of the vessel.

3.4.10.2 In case of oil tankers of 130 [m] in length and upwards (as defined in ILLC), the ship's longitudinal strength is to be evaluated using the thickness of the structural members measured, renewed, reinforced, as appropriate, during the renewal survey of cargo ship safety construction certificate or the cargo ship safety certificate carried out after the ship reached 10 years of age in accordance with the criteria for longitudinal strength of the ship's hull girder

3.4.10.3 Where any structural renewal or reinforcement work is carried out as a result of the initial evaluation of the ship's longitudinal strength as required by 3.4.10.2, the final result of the evaluations is to be included as a part of the Executive hull summary.

3.4.10.4 An Executive Hull Summary of the survey and results is to be issued to the Owner and placed on board the vessel for reference at future surveys. The Executive Hull Summary is to be endorsed by IRS Head Office.

3.4.11 Watertight Cable Transits

3.4.11.1 The requirements for Special Survey may be undertaken by the attending Surveyor or by a firm approved as a service supplier according to Pt.1 Ch.1, Sec 1.9.

3.4.11.2 All transits are to be examined to confirm their satisfactory condition and the Cable Transit Seal System Register is to be reviewed to confirm that it is being maintained. The Special Survey is to be recorded in the Register, in which a single record entry will be sufficient to record the survey of all transits.

3.4.11.3 From review of the Register, where there are records entered since the last special survey of any disruption to the cable transits or installation of new cable transits (except which are reviewed and examined at previous annual surveys), the satisfactory condition of those transits is to be confirmed by the attending Surveyor by review of records and examination of the transits; the results are to be recorded in the Register against each of those cable transits.

3.4.11.4 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

3.5 Special Surveys - Machinery

3.5.1 Requirements for examination of machinery and systems are given in Sec.8.

3.6 Survey Requirements for vessels carrying Asphalt in Integral tanks

3.6.1 General

3.6.1.1 In addition to the relevant requirements in subsection 3.1 to 3.5 (as applicable), the following requirements are applicable to vessels with class notation**:**

OIL TANKER, ESP, ASPHALT CARRIER (INTEGRAL TANK, MAX. CARGO TEMP ≤ XXX℃)

3.6.2 Annual Surveys

3.6.2.1 Hull

.1 The Survey is to include:

a) confirmation that wheelhouse doors and windows, side scuttles and windows in superstructure and deckhouse ends facing the cargo area are in satisfactory condition

3.6.2.2 Cargo Machinery

.1 The survey is to include:

a) confirmation that electrical equipment in cargo pump rooms and other spaces is in satisfactory condition and has been properly maintained

b) confirmation that the remote operation of the cargo pump room bilge system is satisfactory

c) examination of the cargo heating system (Refer Sec 16)

d) examination of the cargo-transfer arrangement and confirmation that any hoses are suitable for their intended purpose

e) confirmation that devices provided for measuring the temperature of the cargo, if any, operate satisfactorily.

3.6.3 Intermediate Surveys

3.6.3.1 The survey (in addition to Annual Survey Requirements) is to include:

a) examination, so far as is possible, bunker, ballast, steam and vent piping systems. If upon examination, there is any doubt as to the condition of the piping, pressure testing and/ or, thickness measurement may be required.

3.6.4 Special Surveys

3.6.4.1 The Survey (in addition to the Intermediate Survey Requirements) is to include:

.1 All safety valves on cargo piping and of cargo tanks are to be dismantled for examination, adjusted and, as applicable, resealed.

.2 A general examination of all cargo pump room boundaries is to be carried out. All gas-tight shaft-sealing devices are to be examined. The bottom of cargo pump rooms is to be presented clean for the examination of stripping devices and gutters.

.3 Ballast and stripping pumps are to be internally examined and their prime movers checked. A functional test of the pumps is to be carried out. Maintenance records of cargo pumps are to be made available to the Surveyor.

.4 The satisfactory condition of the cargo heating system including clamping and piping is to be verified and, the system is to be pressure tested, if deemed necessary by the Surveyor.

3.7 Survey Requirements for vessels carrying Asphalt in Independent tanks

3.7.1 General

3.7.1.1 In addition to the relevant requirements in subsection 3.1 to 3.5 (as applicable), the following requirements are applicable to vessels with class notation**:**

OIL TANKER, ASPHALT CARRIER (INDEPENDENT TANK, MAX. CARGO TEMP ≤ XXX°C)

3.7.1.2 ESP requirements are not applicable to these vessels. Accordingly, hull survey requirements, for the cargo region, in sub-section 3.1 to 3.5 are not applicable to such vessels. In general, hull surveys are to be carried out in accordance with the requirements in Section 6 for “other ship types’.

3.7.1.3 Requirements in 3.6.2 to 3.6.4 are also to be complied with, for these vessels.

3.7.2 Independent Cargo Tanks

3.7.2.1 Annual Survey

.1 The survey is to include, as far as practicable, a general external examination of the structure surrounding the independent cargo tanks for damage, cracks or wastage, including the thermal isolating material and elements supporting and/ or securing the cargo tanks.

3.7.2.2 Special Survey

.1 All independent cargo tanks are to be internally examined including internal mountings and equipment. The tanks are to be thoroughly cleaned and made gas-free before entry, and all precautions are to be taken to ensure safety during examination.

.2 Foundations, chocks, sway braces and anti-flotation arrangements are to be examined.

.3 Venting systems for the cargo containment systems are to be examined. All relief valves are to be opened, examined, tested, and readjusted as necessary. Liquid-level indicators are to be proven to be satisfactory. Where a proper record of continuous overhaul and retesting of individually identifiable relief valves is maintained, consideration will be given to acceptance on the basis of opening, internal examination, and testing of a representative sampling of valves including each size of each type of relief valve in use, provided there is logbook evidence that the remaining valves have been overhauled and tested since the crediting of the previous Special Survey. The testing and setting of relief valves may be carried out in-situ or after removal.

.4 All piping, machinery, and equipment for loading, venting, heating or otherwise handling the cargo are to be generally examined. All valves in the cargo piping systems are to be examined and tested.

.5 Insulation is to be removed in way of any distorted or suspect part in the cargo containment system and elsewhere, as required by the Surveyor. Insulation may be required to be removed to carry out any survey, at the discretion of the Surveyor.

.6 Thickness gauging of the independent tank is to be carried as deemed necessary by the Surveyor.

.7 Tanks are to be tested with a head of liquid to the overflow or by an alternative method with the approval of the attending surveyor. Testing may be omitted if the tanks are found without corrosion and other damages and in acceptable condition.

.8 At Special Surveys after Special Survey No. 3, the plating of all independent cargo tanks (shell, heads, and domes) is to be thickness gauged.

Section 4

Surveys - Chemical Tankers

4.1 Scope

4.1.1 This section gives the requirements for periodical surveys of vessels which have been assigned the class notation:

Chemical Tanker ESP

For vessels which also carry products covered by the IGC Code it may be necessary in accordance with Pt.5, Ch.4, Sec.1 to apply the survey requirements given in Sec.5 of this chapter.

4.2 Annual surveys

4.2.1 General

4.2.1.1 Annual Surveys are to be carried out within 3 months before or after the anniversary date each year. (Also refer to Note 9 of Sec.1, Table 1.1.1). These should be held concurrently with statutory annual or other relevant statutory Surveys, where practicable.

4.2.1.2 At Annual Surveys, the Surveyor is to examine the hull and machinery, so far as necessary and practicable, in order to be satisfied as to their general condition.

4.2.1.3 Access to cargo tanks or other spaces within the cargo area, necessitating gas freeing will normally not be required unless necessary for checking items of equipment and installations for correct functioning.

4.2.1.4 It is to be confirmed that no new installation of material containing asbestos was carried out since last survey.

4.2.1.5 Where applicable, examination of alternative design and arrangement for

machinery or electrical installations or fire safety, in accordance with the test, inspection and maintenance requirements, if any, specified in the approved documentation is to be carried out.

4.2.2 Hull and weather deck

4.2.2.1 The survey is to consist of an examination for the purpose of ensuring, as far as practicable, that the hull, equipment, closing appliances and related piping are maintained in satisfactory and efficient condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file. Special attention is to be paid to the following:

a) Weather decks, ships side plating above waterline.

b) Openings on freeboard and superstructure decks; exposed casings; skylights and fiddley openings; deck houses; companionways and superstructure bulkheads; side scuttles and dead lights; flush deck scuttles; ash shoots and other openings.

c) Weld connection between air pipes and deck plating; air pipe heads on exposed decks (external examination); ventilators and closing devices.

d) Watertight bulkheads and their penetrations as far as practicable.

e) Scuppers and sanitary discharges as far as practicable together with valves and their controls.

f) Guard rails, bulwarks, freeing ports, gangways, walkways and life lines.

g) Examination of cargo tank openings including gaskets, covers, coamings and flame screens.

h) Examination of cargo tank pressure/vacuum valves, secondary means to prevent over or under pressure and devices to prevent passage of flame and the arrangements of cargo tank purging with inert gas, as applicable.

i) Examination of devices to prevent the passage of flame on vents to all bunker, oily-ballast and oily-slop tanks and void spaces, as far as practicable.

j) Examination of cargo, bunker and vent piping systems, including vent masts and headers.

k) Verification that wheelhouse doors and windows, side scuttles and windows in superstructure and deckhouse ends facing the cargo area are in good condition.

4.2.2.2 Examination of pump room and pipe tunnel, if fitted:

a) Examination of all pump room bulkheads for signs of chemical leakage or fractures and in particular, the sealing arrangements of all penetrations of pump room bulkheads.

b) Examination of the condition of all piping systems.

4.2.2.3 All watertight doors in watertight bulkheads, to be examined and tested (locally and remotely) as far as practicable.

4.2.2.4 Suspect areas identified at previous special or intermediate surveys are to be close-up surveyed. Thickness measurements are to be taken in the area of substantial corrosion identified at previous surveys.

4.2.2.5 Examination of Ballast Tanks is to be carried out when required as a consequence of the results of the Special Survey or Intermediate Survey. (See 4.4.5.1 and 4.3.3.3 respectively). When considered necessary by the Surveyor or where extensive corrosion is found, thickness measurement is to be carried out. When examination of ballast tanks reveals substantial corrosion, additional thickness measurements is to be carried out to the extent given in Table 4.4.9.2.

4.2.2.6 Anchoring and mooring equipment is to be examined as far as is practicable. A general examination of the emergency towing arrangements is to be carried out to ensure their ready availability. It is to be confirmed that the towing and mooring equipment is properly marked with any restriction associated with its safe operation

For vessels with single point mooring arrangements where 'SPM' notation is assigned, the following are to be generally examined:

a) Components of the single point mooring system (bow chain stoppers, bow

fairleads, pedestal roller fairleads, winches and capstans), to verify their satisfactory condition.

b) Hull structures supporting and adjacent to the components of the single point mooring system, to verify that there is no deformation or fracture.

4.2.2.7 Where applicable, Surveyor should satisfy himself regarding the freeboard marks on the ship's side.

4.2.2.8 The Surveyor is to confirm that, where required, an approved loading instrument together with its operation manual are available on board, See Pt.3, Ch.5. It is to be verified by the Surveyor that the loading instrument is checked for accuracy at regular intervals by the ship's staff by applying test loading conditions.

4.2.2.9 Accommodation ladders are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) steps;
b) platforms;
c) all support points such as pivots, rollers, etc.;
d) all suspension points such as lugs, brackets, etc.;
e) stanchions, rigid handrails, hand ropes and turntables;
f) davit structure, wire and sheaves, etc.

4.2.2.10 Access to bow arrangements are to be examined. Gangways are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) treads;

b) side stringers, cross-members, decking, deck plates, etc.;

c) all support points such as wheel, roller, etc.;

d) stanchions, rigid handrails, hand ropes.

4.2.2.11 Winches of accommodation ladders and gangways are to be examined to verify the satisfactory condition of the following items:

a) brake mechanism including condition of brake pads and band brake, if fitted;

b) remote control system, and

c) power supply system for electric motor.

4.2.2.12 Davits and fittings on the ship's deck associated with accommodation ladders and gangways are to be examined for satisfactory condition at annual surveys. Fittings or structures for means of access to deck such as handholds in a gateway or bulwark ladder and stanchions are also to be examined.

4.2.2.13 The maintenance and inspection records of accommodation ladders and gangways are to be verified. It is to be confirmed that supporting wires are being renewed at intervals not exceeding 5 years.

4.2.2.14 When examining internal spaces, as far as practicable, the permanent means of access where appropriate, are to be verified that they remain in good condition.

4.2.2.15 For vessels subject to IMO PSPC (See Pt.3, Ch.2, 3.6) it is to be confirmed that the maintenance, repair and partial re-coating of dedicated ballast tanks, as appropriate, are recorded in the coating technical file.

4.2.2.16 Watertight cable transit seal systems register is to be reviewed to confirm that it is being maintained and as far as practicable the transits are to be examined to confirm their satisfactory condition. Where there are records entered since last annual survey of any disruption to the cable transits or installation of new cable transits, the satisfactory condition of those transits is to be confirmed by review of records and, if deemed necessary, by examination. The results are to be recorded in the Register against the specific cable transit.

4.2.2.17 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

4.2.3 Machinery and systems

4.2.3.1 A general examination of the machinery, boilers, all pressurised systems (steam, pneumatic, hydraulic) and their associated fittings, propulsion system and auxiliary machinery to see whether they are being properly maintained and with particular attention to the fire and explosion hazards.

4.2.3.2 Confirmation that machinery, boilers and other pressure vessels, associated piping systems and fittings are so installed and protected as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards.

4.2.3.3 Confirmation that Periodical Surveys of boilers and other pressure vessels have been carried out as required by the Rules and the safety devices have been tested.

4.2.3.4 Confirmation that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative.

4.2.3.5 Confirmation that means are provided so that machinery can be brought into operation from the dead ship condition without external aid.

4.2.3.6 All main and auxiliary steering arrangements and their associated equipment and control systems are to be examined and tested. Where applicable, Surveyors are to verify that log entries have been made in accordance with statutory requirements.

4.2.3.7 Confirming, when appropriate, that requisite arrangements to regain steering capability in the event of the prescribed single failure are being maintained.

4.2.3.8 Steering chains are to be cleaned for ascertaining wear and tear and lengths of chain worn in mean diameter by more than 12 per cent of the original rule diameter are to be renewed.

4.2.3.9 All the means of communication between the navigating bridge and the machinery control positions, as well as the bridge and the main alternative steering position, if fitted, are to be tested. It is to be confirmed that means of indicating the angular position of the rudder are operating satisfactorily.

4.2.3.10 Confirmation that with ships having emergency steering positions there are means of relaying heading information and, when appropriate, supplying visual compass readings to the emergency steering positions.

4.2.3.11 Confirmation that various alarms required for hydraulic power operated, electric and electro-hydraulic steering gears are, operating satisfactorily and that the recharging arrangements for hydraulic power operated steering gears are being maintained.

4.2.3.12 Examining the means for the operation of the main and auxiliary machinery essential for propulsion and the safety of the ship, including when applicable, the means of remotely controlling the propulsion machinery from the navigating bridge and the arrangements to operate the main and other machinery from a machinery control room.

4.2.3.13 Confirmation that the engine room telegraph, the second means of communication between the navigation bridge and the machinery space and the means of communication with any other positions from which the engines are controlled are operating satisfactorily.

4.2.3.14 Examination and testing of the general emergency alarm system and confirmation that the engineer's alarm is clearly audible in the engineer's accommodation.

4.2.3.15 It is to be confirmed that ballasting and de-ballasting arrangements are satisfactory. The bilge pumping systems and bilge wells including operation of each bilge pump, extended spindles and level alarms, where fitted, are to be examined as far as is practicable. It is also to be confirmed that bilge pumping system for each watertight compartment is satisfactory.

It is also to be confirmed that drainage from enclosed cargo spaces situated on freeboard deck is satisfactory.

4.2.3.16 Examining visually the condition of non-metallic expansion joints where fitted in piping systems which penetrate the ship's side, with both the penetration and the expansion joint located below the deepest load waterline, and checking the service record.

4.2.3.17 General examination visually and in operation, as feasible, of the main electrical machinery, the emergency sources of electrical power, the switch gear, other electrical equipment including the lighting system is to be carried out.

4.2.3.18 Confirmation as far as practicable, the operation of the emergency source(s) of electrical power, including their starting arrangement, the systems supplied, and when appropriate, their automatic operation.

4.2.3.19 Examining in general, that the precautions provided against shock, fire and other hazards of electrical origin are being maintained.

4.2.3.20 General Examination of automation equipment is to be carried out. Satisfactory operation of safety devices, bilge level detection and alarm systems and control systems is to be verified.

4.2.3.21 Examination so far as is possible of cargo, bilge, ballast and stripping pumps for excessive gland seal leakage, verification of proper operation of electrical and mechanical remote operating and shutdown devices and remote operation of pump room bilge system, and checking that pump foundations are intact.

4.2.3.22 Verification that installed pressure gauges on cargo discharge lines including those fitted outside the cargo pump room and level indicating systems are operational.

4.2.3.23 Verification that pumps, valves and pipelines are identified and distinctively marked.

4.2.3.24 Confirmation that machinery space ventilation systems are in good working condition.

4.2.3.25 Emergency lightning in all cargo pump rooms are to be examined.

4.2.3.26 For Ships constructed before 01 July 2017 and fitted with harmonic filters; following are to be verified at each survey:

a) Records of annual measurement of harmonic distortion levels of main bus bar under seagoing conditions and log entries showing which equipment was running and / or filters in service at the time of measurement. Harmonic distortion readings are to be carried out when the greatest amount of distortion is indicated by the measuring equipment.

b) Records of measurement of distortion levels, after any modification to the ship's electrical distribution system.

4.2.3.27 For Ships constructed on or after 01 July 2017 and fitted with harmonic filters; or for ships retrofitted with harmonic filters, on or after 01 July 2017; following are to be verified at each survey:

a) Engine log book containing distortion monitoring readings/ automatic records of same

b) Documentation of any approved relaxation on allowable distortion limits.

4.2.3.28 For ships contracted for construction on or after 01 Jan 2017 following are to be verified in emergency generator rooms at each survey:

a) Hand operated ventilation louvres, closing appliances, if fitted, are kept open and corresponding instruction plates are visible near the louvres.

b) Power operated ventilation louvres and closing appliances, if fitted, are fail-to-open type and capable of opening automatically whenever emergency generator is starting / in operation.

c) It is to be possible to close ventilation openings by a manual operation from a clearly marked safe position outside the space where the closing operation can be easily confirmed. Further, such closing is not possible from any other remote position. The louvre status (open/closed) is indicated at this position.

4.2.3.29 For vessels fitted with ICCP system, proper functioning of the system is to be checked and records of system operation are to be verified. It is to be verified that all maintenance as per manufacturer's instructions are carried out.

4.2.4 Fire protection, detection and extinction

4.2.4.1 The arrangements for fire protection, detection and extinction are to be examined and are to include confirmation that no changes have been made in the structural fire protection. Following are to be examined / verified:

a) verification that fire control plans are properly posted;

b) examination as far as possible and testing as feasible, any fire detection and alarm system and any sample extraction smoke detection system;

c) examination of the fire main system and verification that each fire pump including the emergency fire pump can be operated separately so that the two required powerful jets of water can be produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main;

d) verification that fire hoses, nozzles, applicators and spanners are in good working condition and situated at their respective locations;

e) examination of fire-fighting system (including fixed fire-fighting system for the machinery spaces) controls, piping, instructions and marking, checking for evidence of proper maintenance and servicing including date of last systems tests;

f) verification that all semi-portable and portable fire extinguishers are in their stowed positions, checking for evidence of proper maintenance and servicing, conducting random check for evidence of discharged containers;

g) verification, as far as practicable, that the remote controls for stopping fans and machinery in machinery spaces are in working order. Examination of the arrangements for oil fuel, lubricating oil and other flammable oils and testing the remote closing of valves for oil fuel, lubricating oil and other flammable oils and, the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils;

h) examination of the closing arrangements of ventilators, funnel annular spaces, skylights, doorways and tunnel, where applicable;

i) confirmation that the fire fighters' outfits including its self-contained compressed air breathing apparatus and emergency escape breathing devices (EEBDs) are complete and in good condition and that the cylinders, including the spare cylinders, of any required self-contained breathing apparatus are suitably charged and that on board means of recharging breathing apparatus cylinders used during drills or a suitable number of spare cylinders to replace those used are provided, and provision of two-way portable radiotelephone apparatus of an explosion-proof type or intrinsically safe;

j) examination of any manual and automatic fire doors and proving their operations;

k) Verification that the pump room ventilation system is operational, ducting intact, dampers operational and screens are clean;

l) External examination of piping and cut-out valves of cargo tank and cargo pump room fixed fire fighting system;

m) Verification that all electrical equipment in dangerous zones is suitable for such locations, is in good condition and has been properly maintained so far as is practicable;

n) Checking the deck foam system, including the supplies of foam concentrate and testing that minimum number of jets of water at the required pressure in the fire main is obtained when the system is in operation;

o) Examining the fixed fire-fighting system for the cargo pump room and the deck foam system for the cargo area and confirming that their means of operation are clearly marked;

p) Confirming that potential sources of ignition in or near the cargo pump room are eliminated, such as loose gear, combustible materials etc., that there are no signs of undue leakage and that access ladders are in satisfactory condition;

q) Confirming that the condition of portable fire-fighting equipment for the cargoes to be carried in the cargo area is satisfactory;

r) examination of the fire-extinguishing systems for spaces containing paint and/or flammable liquids and deep fat cooking equipment in accommodation and service spaces;

s) examination of the fire safety requirements of helicopter facilities, including foam firefighting appliances ;

t) Confirming, as far as practicable, that the intrinsically safe systems and circuits used for measurement, monitoring, control and communication purposes in all hazardous locations are being properly maintained.

u) Confirming that the system for continuous monitoring of the concentration of flammable vapours is satisfactory. (Refer Pt.5, Ch.3, Cl. 11.1.4).

v) Confirming that sampling points or detector heads are located in suitable positions in order that potentially dangerous leakages are readily detected. (Refer Pt.5, Ch.3, Cl. 11.1.4).

w) Examination of the fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space.

x) Examination of the fire protection arrangements in cargo spaces and confirming, as far as practicable and as appropriate, the operation of the means of control provided for closing the various openings.

y) Testing the fire dampers of ventilation ducts and the means of closing the main inlets and outlets of all ventilation systems and testing the means of stopping power ventilation systems from outside the space served.

Surveys carried out by the National Authority of the country in which the ship is registered would normally be accepted as meeting these requirements, at the discretion of the Surveyor.

4.2.4.2 For tankers fitted with inert gas system, examination of the inert gas system as detailed in Section 12, Para 12.2.1 is to be carried out.

4.2.4.3 Confirmation that the means of escape from accommodation, machinery spaces and other spaces are satisfactory.

4.2.4.4 Examination of the arrangements for gaseous fuel for domestic purposes.

4.2.5 Other safety arrangements related to cargo

4.2.5.1 The following are to be dealt with where applicable:

a) Examination of gauging devices, high level alarms and valves associated with overflow control.

b) Verification that any devices provided for measuring the temperature of the cargo and any associated alarms are satisfactory.

c) Examination of the cargo heating/cooling system sampling arrangements where fitted.

d) Verification that removable pipe lengths or other approved equipment necessary for cargo separation are available, and satisfactory.

e) Verification that the ventilation system including portable equipment, if any, of all spaces in the cargo area is operational.

f) Verification that arrangements are made for sufficient inert/padding/drying gas to be carried or generated to compensate for normal losses and that means are provided for monitoring of ullage spaces.

g) Verification that arrangements are made for sufficient medium to be carried where drying agents are used on air inlets to cargo tanks.

h) Examining the cargo transfer arrangements and confirming that any hoses are suitable for their intended purpose and where appropriate, marked with date of testing.

i) Verification that any special arrangements made for bow or stern loading and unloading are in good condition and testing the means of communication and remote shutdown of the cargo pump.

j) Verification that, if applicable, the provisions made for chemicals which have special requirements listed in relevant Ch.3 of Pt.5 of the Rules are in order.

4.2.6 Personnel protection

a) Verification that cargo pump room rescue arrangements are in order.

b) Verification that suitable protective clothing is available for crew engaged in loading

and discharging operations and that suitable storage is maintained.

c) Verification that the requisite safety equipment and associated breathing apparatus with requisite air supplies and emergency escape respiratory and eye protection, if required, are in good condition and are properly stowed.

d) Verification that medical first aid equipment including stretchers and oxygen resuscitation is in good condition and that satisfactory arrangements are made for antidotes for cargoes actually carried to be on board.

e) Verification that decontamination arrangements are operational.

f) Verification that the requisite gas detection instruments are on board and that satisfactory arrangements are made for the supply of any required vapour detection tubes.

g) Verification that the cargo sample stowage arrangements are in good condition.

4.3 Intermediate Surveys

4.3.1 General

4.3.1.1 Intermediate surveys are to be carried out at or between the second or third Annual Survey. (Also refer to Note 9 of Sec.1, Table 1.1.1). However, only those items which are additional to the requirements of annual survey may be examined between the second or third annual survey.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

4.3.1.2 The following requirements are applicable for vessels over five years of age. For vessels below 5 years of age additional examination over and above the requirements of Annual survey may be required at the discretion of the Surveyors.

4.3.1.3 For vessels over 10 years of age a specific survey program is to be worked out by the owner in cooperation with the Surveyors considering the requirements of the previous special survey, executive hull summary of that survey, later relevant survey records and taking account of any amendments to the survey requirements after the last special survey. The survey program is to be submitted in written format for approval and kept on board until the intermediate survey has been completed. (See 4.4.2 for guidance on preparation of survey program).

4.3.1.4 Prior to the commencement of any part of the intermediate survey, a survey planning meeting is to be held between the attending Surveyor(s), the owner's representative in attendance and where involved, the thickness measurement company representative and the master of ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

4.3.2 Examination tanks and spaces

4.3.2.1 The survey of integral cargo tanks and ballast tanks depending on the age of the vessel is to be carried out as specified in 4.3.3 to 4.3.5.

4.3.3 Chemical tankers between 5 and 10 years of age

4.3.3.1 The following are to be carried out:

For ballast tanks, an Overall Survey of Representative Tanks selected by the Surveyor is to be carried out. If such inspections reveal no visible structural defects, the examination may be limited to a verification that the hard Protective Coating remains in GOOD condition.

4.3.3.2 In addition to the requirements above, suspect areas identified at previous surveys are to be examined.

4.3.3.3 A ballast tank is to be examined at subsequent annual intervals where:

a) a hard protective coating has not been applied from the time of construction, or
b) a soft or semi-hard coating has been applied, or
c) substantial corrosion is found within the tank, or
d) the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the Surveyor.

4.3.4 Chemical tankers between 10 and 15 years of age

4.3.4.1 The requirements of the intermediate survey are to be to the same extent as the previous special survey as required in 4.4 for hull structure and piping systems in way of cargo holds, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks. However, testing of cargo and ballast tanks and survey of automatic air pipe heads specified in 4.4.6.2 are not required unless deemed necessary by the attending surveyor. Thickness measurement is to be carried out for items 1) to 4) of Table 4.4.9.1.

4.3.4.2 The intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 4.4.1.4.

4.3.4.3 An in-water survey complying with the requirements of Sec.7.2 may be accepted in lieu of the requirements of 4.4.1.10.

4.3.5 Chemical tankers exceeding 15 years of age the following is to apply:

4.3.5.1 The requirements of the intermediate survey are to be to the same extent as the previous special survey as required in 4.4 for hull structure and piping systems in way of cargo holds, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks. However, testing of cargo and ballast tanks is not required unless deemed necessary by the attending surveyor. Thickness measurement is to be carried out for items 1) to 4) of Table 4.4.9.1.

4.3.5.2 For the application of 4.3.5.1 the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 4.4.1.4.

4.3.5.3 A survey in dry dock is to be part of the intermediate survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of cargo tanks and water ballast tanks (i.e. parts below light ballast waterline) are to be completed in the dry-dock.

4.3.6 Safety arrangements related to cargo

4.3.6.1 The following are to be dealt with as applicable:

a) Examination of vent line drainage arrangements.

b) Verification that the cargo heating/cooling system is in good condition.

c) Verification that the ship's cargo hoses are approved and in good condition.

d) Verification that where applicable, pipelines and independent cargo tanks are electrically bonded to the hull.

e) An examination of cargo, cargo washing, bunker, ballast, steam and vent piping on the weather decks, as well as vent masts and headers. If upon examination there is any doubt as to the condition of the piping, the pipe may require to be pressure tested, gauged or both.

f) A General Examination of the electrical equipment and cables in dangerous zones such as cargo pump-rooms, and areas adjacent to cargo tanks for defective certified safe type equipment, improperly installed wiring, non-approved lighting and fixtures and dead end wiring. Testing insulation resistance of the circuits except in cases where a proper record of the testing is maintained, consideration is to be given to accepting recent readings by the crew. If any of the readings are marginal or if the condition of the cables, fixtures or equipments appears defective in any way, verification measurements may be required. These measurements are not to be attempted until the ship is in a gas free condition and are to be carried out within an acceptable time period.

g) Confirmation that spares are provided for cargo area mechanical ventilation fans.

h) Examining the equipment for personal protection.

i) Examining externally and confirming that the pumping and piping systems, including a stripping system if fitted and associated equipment remain as approved.

j) Examining externally the tank washing piping and confirming that the type,

capacity, number and arrangement of the tank washing machines are as approved.

k) Examining externally the wash water heating system.

l) Examining externally, as far as practicable the underwater discharge arrangement.

m) Confirming that the means of controlling the rate of discharge of the residue is as approved.

n) Confirming that flow rate indicating device is operable.

o) Confirming that the ventilation equipment for residue removal is as approved and satisfactory.

p) Examining externally, as far as is accessible, the heating system required for solidifying and high viscosity substances.

q) Confirming that any cargo tank high level alarms are operable.

r) Verifying from the Cargo Record Book that the pumping and stripping arrangements have been emptying the tanks efficiently and all are in working order.

s) Confirming, if possible, that the discharge outlets are in good condition.

t) Confirming the satisfactory operation of the recording device, as fitted and verifying by an actual flow test that it has an accuracy of $\pm$ 15% or better.

Surveys carried out by the National Authority of the country in which the ship is registered may be accepted at the discretion of the Surveyor as meeting these additional requirements.

4.4 Special Surveys - Hull

4.4.1 General

4.4.1.1 All ships classed with IRS are to undergo Special Surveys at 5 yearly intervals. The first Special Survey is to be completed within 5 years from the date of the initial classification survey and thereafter 5 years from the assigned date of the previous Special Survey. However, an extension of class of 3 months maximum beyond the 5th year may be granted in exceptional circumstances in accordance with 1.11. In such cases, the next period of class will start from the expiry date of the Special Survey before extension was granted.

4.4.1.2 The interval between the Special Surveys may be reduced at the request of the parties concerned or by IRS if considered appropriate.

4.4.1.3 For surveys completed within 3 months before the expiry date of the Special Survey, the next period of class will start from the expiry date of the Special Survey. For surveys completed more than 3 months before the expiry date of the Special Survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the special survey. If the owner elects to carry out the next due special survey, the period of class will start from the survey completion date. Any requirement of the Flag Administration in this regard is also to be complied with.

4.4.1.4 The Special Survey may be commenced at the 4th Annual Survey and be progressed with a view to completion by the 5th anniversary date. When the special survey is commenced prior to the fourth annual survey, the entire survey is to be completed within 15 months if such work is to be credited to the special survey and in this case the next period of class will start from the survey completion date.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

4.4.1.5 As part of the preparation for Special Survey, the proposed Survey Programme (See 4.4.2) including the schedule for thickness measurements (See 4.4.9) are to be submitted at least 3 months in advance of the intended commencement of the Special Survey.

4.4.1.6 Record of Special Survey will not be assigned until the Machinery Survey has been completed or postponed in agreement with IRS.

4.4.1.7 Ships which have satisfactorily passed a Special Survey will have a record entered in the Supplement to the Register Book indicating the assigned date of Special Survey. In addition, a notation "ESP" will be entered for chemical tankers.

4.4.1.8 The special survey is to include, in addition to the requirements of the Annual Survey, examination, tests and checks of sufficient extent to ensure that the hull, equipment and related piping are in satisfactory condition and that the ship is fit for its intended purpose for the next five (5) year class period, subject to proper maintenance and operation and the periodical surveys being carried out at the due dates.

4.4.1.9 The special survey requirements to be applied in respect of independent cargo tanks are given in 4.4.11.

4.4.1.10 A Docking Survey in accordance with the requirements of Sec.7 is to be carried out as part of the Special Survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of cargo tanks and water ballast tanks (i.e. parts below light ballast water line) are to be completed in the dry-dock.

4.4.2 Planning and preparation for survey

4.4.2.1 A specific Survey Programme is to be worked out in advance of the Survey by the Owner in cooperation with the Surveyors and submitted to IRS for approval. The Survey Programme shall be in a written format prior to the development of survey programme, the survey planning questionnaire in prescribed format is to be completed by the owner and forwarded to IRS. The survey program is to be worked out taking into account any amendments to the survey requirements after the last special survey.

4.4.2.2 In developing the Survey Programme, the following documentation is to be collected and consulted with a view to selecting tanks, areas and structural areas for examination :

a) Survey status and basic ship information

b) Documentation on board as per 4.4.3

c) Main structural plans (scantling drawings), including information regarding use of high strength steels, clad steel and stainless steel

d) Executive hull summary

e) Relevant previous damage and repair history

f) Relevant previous survey and inspection reports of IRS and the Owners

g) Information regarding the use of the ship's holds and tanks, typical cargoes and other relevant data

h) Details of the inert gas plant and tank cleaning procedures

i) Information and other relevant data regarding conversion or modification of the ship’s cargo and ballast tanks since the time of construction

j) Description and history of the coating and corrosion protection system (previous class notations), if any

k) Inspections by the Owner’s personnel during the last 3 years with reference to structural deterioration in general, leakages in tank boundaries and piping and condition of the coating and corrosion protection system, if any

l) Information regarding the relevant maintenance level during operation including port state control reports of inspection containing hull related deficiencies. Safety Management System non-conformities relating to hull maintenance, including the associated corrective action(s); and

m) Any other information that will help identify suspect areas and critical structural areas.

4.4.2.3 The Survey Programme submitted for approval is to account for and comply, as a minimum, with the requirements of close-up survey, thickness measurement, tank testing and pipe testing respectively, and to include relevant information including at least :

a) Basic ship information and particulars
b) Main structural plans (scantling drawings), including information regarding use of high strength steels, clad steel and stainless steel
c) Plan of tanks

d) List of tanks with information on use, corrosion prevention and condition of coating
e) Conditions for survey (e.g. information like tank cleaning, gas freeing, ventilation, lighting, etc.)
f) Provisions and methods of access to structures
g) Equipment for survey
h) Nomination of tanks and areas for close-up survey (As per Table 4.4.8.1a) and Table 4.4.8.1b))
i) Nomination of sections for thickness measurement (As per Table 4.4.9.1
j) Nomination of tanks for testing and the pipes that are to undergo pipe testing as per 4.4.10
k) Identification of the thickness measurement company
l) Damage experience related to the ship in operation.
m) Critical structural areas and suspect areas, where relevant.

4.4.2.4 IRS will advise the Owner of the maximum acceptable structural corrosion diminution levels applicable to the vessel.

4.4.2.5 The ship is to be prepared for overall survey in accordance with the requirements of Table 4.4.2.5. The preparation is to be of sufficient extent to facilitate an examination to ascertain any excessive corrosion, deformation, fractures, damages and other structural deterioration.

4.4.2.6 Proper preparation and close co-operation between the attending surveyor(s) and the owner's representatives onboard prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings are to be held regularly.

4.4.2.7 Prior to the commencement of any part of the Special and Intermediate Survey a survey planning meeting is to be held between the attending Surveyor(s), the Owner's representative in attendance and the thickness measurement firm representative, where involved, and the master of the ship or an appropriately qualified representative appointed by the master or Company for the purpose of ascertaining that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

Table 4.4.2.5 : Survey preparation			
Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) All tanks, peaks, bilges and drain wells, engine and boiler spaces and other spaces are to be cleared out and cleaned as necessary for examination. Floor plates in engine and boiler spaces are to be lifted as may be necessary for examination of the structure underneath. Where necessary ceiling, lining, casings and loose insulation are to be removed as required by the Surveyor for examination of the structure. Compositions on the plating are to be examined and sounded, but need not be disturbed if found satisfactorily adhering to the plating 2) The steelwork is to be exposed and cleaned as may be required for its proper examination by the Surveyor and close attention is to be paid to the parts of the structure which are particularly liable to excessive corrosion or to deterioration due to other causes. 3) All tanks are to be cleaned as necessary to permit examination. 4) Casings or covers of air, sounding, steam and other pipes, spar ceiling and lining in way of the side scuttles are to be removed, as required by the Surveyor.	1) Requirements of Special Survey I to be complied with 2) The chain locker is to be cleaned internally. The chain cables are to be ranged for inspection. The anchors are to be cleaned and placed in an accessible position for inspection.	1) Requirements of Special Survey II to be complied with 2) Portions of wood sheathing, or other covering, on steel decks are to be removed, as considered necessary by the Surveyor, in order to ascertain the condition of the plating.	1) Requirements of Special survey III to be complied with

4.4.2.8 The following is an indicative list of items that are to be addressed in the meeting:

a) Schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations etc.);
b) Provisions and arrangements for thickness measurements (i.e. access, cleaning / de-scaling, illumination, ventilation, personal safety);
c) Extent of the thickness measurements;
d) Acceptance criteria (refer to the list of minimum thicknesses);
e) Extent of close-up survey and thickness measurement considering the coating condition and suspect areas / areas of substantial corrosion;
f) Execution of thickness measurements;
g) Taking representative readings in general and where uneven corrosion / pitting is found;
h) Mapping of areas of substantial corrosion; and
i) Communication between attending surveyor(s) the thickness measurement, firm operator(s) and owners representative(s) concerning findings.

4.4.3 Documentation on board

4.4.3.1 The Owners are to obtain supply and maintain on board documentation as specified in 4.4.3.3, which is to be readily available for the Surveyor.

4.4.3.2 The documentation is to be kept on board for the life time of the ship.

4.4.3.3 A Survey Report File is to be a part of the documentation on board consisting of

- Reports of structural surveys
- Executive Hull Summary
- Thickness measurement reports.

4.4.3.4 The Survey Report File is to be available also in the Owners management offices and with IRS.

4.4.3.5 The following additional supporting documentation is to be available on board:

- Main structural plans of cargo holds, cargo and ballast tanks
- Previous repair history
- Cargo and ballast history
- Extent of use of inert gas plant and tank cleaning procedures
- Inspection by ship's personnel with reference to
    - structural deterioration in general
    - leakages in bulkheads and piping
    - condition of coating or corrosion prevention systems, if any
- Any other information that will help identify critical structural areas or Suspect Areas requiring inspection
- Survey Programme as required by 4.4.2 until such time as the Special Survey has been completed.

4.4.3.6 Prior to survey, the Surveyor is to examine the completeness of the documentation on board, and its contents as a basis for the survey.

4.4.4 Surveys at sea or at anchorage

4.4.4.1 See Sec.1.8.

4.4.5 Space protection

4.4.5.1 Where provided, the condition of the corrosion prevention system of cargo tanks is to be examined.

A ballast tank is to be examined at subsequent annual intervals where:

a) a hard protective coating has not been applied from the time of construction, or
b) a soft or semi-hard coating has been applied, or
c) substantial corrosion is found within the tank, or
d) the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the Surveyor.

4.4.5.2 Where the hard protective coating in cargo tanks is found to be in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

4.4.6 Survey and examination

4.4.6.1 All spaces within the hull and superstructure are to be examined. The special survey is to include examination of underwater parts as per Section 7.

4.4.6.2 All tanks other than cargo and ballast tanks are to be examined internally in accordance with the requirements of Table 4.4.6.2.

4.4.6.3 All spaces including double bottom, deep,ballast, peak and cargo tanks; pump rooms, pipe tunnels, duct keels, machinery spaces, dry spaces, cofferdams and voids are to be internally examined including the plating and framing, bilges and drain wells, sounding, venting, pumping and drainage arrangements. This examination is to be supplemented by testing and thickness measurement as required by 4.4.7 and 4.4.9 to ensure that the structural integrity remains effective. The aim of the examination is to discover Substantial Corrosion, significant deformation, fractures, damages or other structural deterioration and if deemed necessary by the Surveyor, a suitable non-destructive examination may be required. At special survey No.3 and subsequent special surveys, structural downflooding ducts and structural ventilation ducts are to be internally examined.

Table 4.4.6.2 : Requirements for internal examination of tanks

Tank	Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequent Age > 15
Fuel oil bunker tanks				
- Engine room	None	None	One	One
- Cargo area	None	One	Two	Half the number of tanks, minimum 2
- If no tanks in Cargo Area, additional fuel tank(s) outside of Engine Room (if fitted)	None	One	One	Two
Lub.oil	None	None	None	One
Fresh water	None	One	All	All

Notes:

1) These requirements apply to tanks of integral (structural) type.

2) If a selection of tanks is accepted to be examined, then different tanks are to be examined at each special survey, on a rotational basis.

3) Peak tanks (all uses) are subject to internal examination at each special survey.

4) At special survey No.III and subsequent special surveys, one deep tank for fuel oil is to be included, if fitted.

4.4.6.4 Cargo piping on deck and cargo and ballast piping systems within the above tanks and spaces are to be examined and operationally tested to working pressure to attending Surveyor's satisfaction to ensure that tightness and condition remain satisfactory. Special attention is to be given to ballast piping in cargo tanks and cargo piping in Ballast Tanks and void spaces and Surveyors are to be advised on all occasions when this piping, including valves and fittings are open during repair periods and can be examined internally.

4.4.6.5 Where ballast tanks have been converted to void spaces, the survey extent is to be based upon ballast tank requirements.

4.4.6.6 All watertight bulkheads and watertight doors are to be examined. All decks, casings and superstructures are to be examined. Attention is to be given to the corners of openings and other discontinuities in way of the strength decks and top sides.

4.4.6.7 The masts, standing rigging and anchors are to be examined.

The Surveyor should satisfy himself that there are sufficient mooring ropes on board and also that a tow line is provided when this is a Rule requirement.

4.4.6.8 The steering gear, and its connections and control systems (main and alternative) are to be examined. The auxiliary steering gear with its various parts are to be examined in working condition.

4.4.6.9 The hand pumps and suctions, air and sounding pipes are to be examined. The Surveyors are to ensure that striking plates are fitted under the sounding pipes whilst examining the tanks internally.

Automatic air pipe heads are to be internally examined at special surveys as indicated in Table 4.4.6.9. For designs where the inner parts cannot be properly inspected from

outside, the head is to be removed from the air pipe. Particular attention is to be paid to the condition of the zinc coating in heads constructed from galvanised steel.

Table 4.4.6.9 : Requirements for internal examination of automatic air pipe heads

Location	Special survey No.I Age ≤ 5	Special survey No.II 5 < Age ≤ 10	Special survey No.III and subsequent Age > 10
forward 0.25L	Two air pipe heads[1,2] one port and one starboard on exposed decks	All air pipe heads on exposed decks	All air pipe heads[3] on exposed decks
aft of 0.25L from the forward perpendicular	Two air pipe heads[1,2] one port and one starboard on exposed decks	At least 20% of air pipe heads[1,2] on exposed decks	

Notes:

1. Preferably air pipes serving ballast tanks.
2. The selection of air pipe heads is to be made by the attending Surveyor. According to the results of the inspection, the Surveyor may require additional air pipe heads to be examined.
3. When there is substantiated evidence of replacement within the previous five years, individual air pipe heads may not be examined.

4.4.6.10 The Surveyor should satisfy himself regarding the efficient condition of the following :

- Means of escape from machinery spaces, crew and passenger spaces and spaces in which crew are normally employed;
- Means of communication between bridge and engine room and between bridge and alternative steering position;
- Helm indicator;
- Protection to the aft steering wheel and the gear.

4.4.6.11 The chain cables are to be ranged and the anchors and the chain cables are to be examined. At special survey no. II and subsequent special surveys, the chain cables are to be gauged. Any length of chain cable which is found to have reduced in mean diameter at its most worn part by more than 12 per cent of its original rule diameter is to be renewed.

4.4.6.12 The windlass is to be examined.

4.4.6.13 The chain locker, hold fasts, hawse pipes and chain stoppers are to be examined and pumping arrangements of the chain locker tested.

4.4.6.14 A thorough examination of aft and forward emergency towing arrangements is to be carried out to ensure its ready availability and satisfactory condition.

- examination of pick-up gear, towing pennant and chafing gear for possible deterioration;

- examination of strong points, fairleads, pedestal roller and their attachment to hull structure.

The survey is to confirm that one of the towing arrangements (aft or forward) are pre-rigged.

4.4.6.15 For vessels with single point mooring arrangements where 'SPM' notation is assigned, the following is to be carried out:

a) A thorough examination of the components of the single point mooring system (bow chain stoppers, bow fairleads, pedestal roller fairleads, winches and capstans) to verify their satisfactory condition.

b) A close-up examination of the hull structures supporting and adjacent to the components of the single point mooring system to verify that there is no

deformation or fracture. Thickness determination and non-destructive tests are to be carried out if required by the Surveyor.

4.4.6.16 Engine room structure is to be examined. Particular attention being given to tank tops, shell plating in way of tank tops, brackets connecting side shell frames and tank tops and engine room bulkheads in way of tank tops and bilge wells. Where excessive areas of wastage are found, thickness measurements are to be carried out and renewals of repairs made when wastage exceeds allowable limits.

4.4.6.17 The attachment to the structure and condition of anodes in tanks is to be examined.

4.4.6.18 Where fitted, the strums of the cargo suction pipes are to be removed or lifted to facilitate examination of the shell plating and bulkheads in the vicinity, unless other means for visual inspection of these parts are provided.

4.4.6.19 The loading instrument is to be checked for accuracy by applying test load conditions in presence of the Surveyor.

4.4.6.20 Examination of accommodation ladders, gangways and their winches are to be carried out as required for annual surveys. In addition, the accommodation ladders and gangways are to be operationally, tested with the specified maximum operation load.

The tests are to be carried out with the load applied as uniformly as possible along the length of the accommodation ladder or gangway, at an angle of inclination corresponding to the maximum bending moment on the accommodation ladder or gangway.

Accommodation ladder winch is to be operationally tested at special surveys. The brake system of the winch is to be tested for holding the maximum operational load on the ladder.

For existing installations on board ships constructed prior to 01 Jan 2010 where the maximum operational load is not known, load nominated by the shipowner or operator may be considered as the test load.

4.4.7 Extent of tank testing

4.4.7.1 The minimum requirements for tank testing for chemical tankers are given in Table 4.4.7.1.

4.4.7.2 All double bottom and peak tanks are to be tested in addition. The Surveyor may extend the tank testing as deemed necessary.

Table 4.4.7.1 : Table of minimum requirements for tank testing at special survey of chemical tankers

Special Survey No. I Age ≤ 5	Special Survey No. II and subsequent Age > 5
1. All ballast tank boundaries	1. All ballast tank boundaries
2. Cargo tank boundaries facing ballast tanks, void spaces, pipe tunnels, pump rooms or cofferdams	2. All cargo tank bulkheads

4.4.7.3 Boundaries of ballast tanks are to be tested with a head of liquid to the top of air pipes. Boundaries of cargo tanks are to be tested to the highest point that liquid will rise under service conditions.

4.4.7.4 Boundaries of fuel oil, lub.oil and fresh water tanks are to be tested with a head of liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil, lub.oil and fresh water tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the Master stating that the pressure testing has been carried out according to the requirement with satisfactory results.

4.4.7.5 The testing of double bottom tanks and other spaces not designed for the carriage of liquid may be omitted, provided a satisfactory internal examination together with an examination of the tank top is carried out.

4.4.7.6 Cargo tank testing carried out by the vessel's crew under the direction of the Master may be accepted by the surveyor subject to the approval of the Administration, provided the following conditions are complied with:

a) a tank testing procedure, specifying fill heights, tanks being filled and bulkheads being tested, taking into account the guidelines provided in the ESP Planning document, has been submitted by the owner and reviewed by IRS prior to the testing being carried out;

b) there is no record of leakage, distortion or substantial corrosion that would affect the structural integrity of the tank;

c) the tank testing has been satisfactorily carried out, in accordance with the reviewed procedure and the guidelines mentioned in a) above, within special survey window not more than 3 months prior to the date of the survey on which the overall or close up survey is completed;

d) the satisfactory results of the testing is recorded in the vessel's logbook;

e) the internal and external condition of the tanks and associated structure are found satisfactory by the surveyor at the time of the overall and close up survey.

4.4.8 Close-up survey requirements

4.4.8.1 The minimum requirements for Close-up Surveys at Special Survey are given in Table 4.4.8.1a) and Table 4.4.8.1b).

The survey of stainless steel tanks may be carried out as an overall survey supplemented by close-up survey as deemed necessary by the Surveyor.

4.4.8.2 The Surveyor may extend the Close-up Survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion protection system and also in the following cases:

a) In particular, tanks having structural arrangements or details which have suffered defects on similar tanks or on similar ships according to available information.

b) In tanks which have structures approved with reduced scantlings due to an approved corrosion control system.

4.4.8.3 For areas in tanks where coatings are found to be in a GOOD condition, the extent of Close-up Surveys according to Table 4.4.8.1a) and Table 4.4.8.1b) may be specially considered by the Surveyor.

4.4.9 Thickness measurement

4.4.9.1 The minimum requirements for thickness measurements at Special Survey are given in Table 4.4.9.1.

4.4.9.2 Provisions for extended measurements for areas with Substantial Corrosion in case of Chemical Tankers are given in Table 4.4.9.2.

The Surveyor may further extend the thickness measurements as deemed necessary.

4.4.9.3 Transverse sections are to be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

In cases where two or three sections are to be measured at least one is to include a Ballast Tank within 0.5L amidships.

4.4.9.4 The thickness measurements are to be carried out by a qualified firm certified by IRS.

Table 4.4.8.1a) : Minimum requirements for close-up survey at special survey of single skin chemical tankers

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
A) **One Web Frame Ring** - in a ballast wing tank	A) **All Web Frame Rings** - in a ballast wing tank or double bottom ballast tank	A) **All Web Frame Rings** - in all ballast tanks	As Special Survey No.3
B) **One Deck Transverse** - in a cargo tank or on deck	B) **One Deck Transverse** - in each of the remaining ballast tank or on deck	A) **All Web Frames Rings** - in a cargo wing tank	Additional transverse areas as deemed necessary by the Surveyor
D) **One Transverse Bulkhead** – lower part in a ballast tank	B) **One Deck Transverse** - in a cargo wing tank or on deck	A) **One Web Frames Rings** - in each remaining cargo tank	
D) **One Transverse Bulkhead** - lower part in a cargo wing tank	B) **One Deck Transverse** – in two cargo center tanks or on deck	C) **All Transverse Bulkheads** - in all cargo and tanks	
D) **One Transverse Bulkhead** – lower part in a cargo center tank (See Note 1)	C) **Both Transverse Bulkhead** - in a ballast wing tank	C) **All Transverse Bulkheads** – in all ballast tanks	
	D) **One Transverse Bulkhead** – lower part in each remaining ballast tank D) **One Transverse Bulkhead** - lower part in a cargo wing tank D) **One Transverse Bulkhead** - lower part in two cargo centre tanks (See Note 1)		

Note 1) Where no center cargo tanks are fitted (as in the case of center longitudinal bulkhead), transverse bulkheads in wing tanks are to be surveyed.

A-D are areas subjected to close-up surveys and thickness measurements (See Fig.4.4.8.1 to 4.4.8.3)

A) Complete transverse web frame ring including adjacent structural members
B) Deck transverse including adjacent deck structural members
C) Transverse bulkhead complete - including girder system and adjacent members
D) Transverse bulkhead lower part - including girder system and adjacent structural members

Table 4.4.8.1b) : Minimum requirements for close-up survey at special survey of double skin chemical tankers

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
1) **One Web Frame Ring** – in a ballast double hull tank (See Note 1)	1) **All Web Frame Rings** – in a ballast wing tank or ballast double hull tank (See Note 1)	1) **All Web Frame Rings** – in all ballast tanks	As special survey No.3
2) **One Deck Transverse** – in a cargo tank or on deck	6) **The Knuckle Area and the Upper Part** (3 metres approx.) of one web frame in each remaining ballast tank	7) **All Web Frame Rings** – in a cargo wing tank	Additional transverse areas as deemed necessary by the Surveyor
4) **One Transverse Bulkhead** – in a ballast tank (See Note 1)		7) **One Web Frame Ring** – in each remaining cargo tank	
5) **One transverse bulkhead** – in a cargo wing tank	2) **One Deck Transverse** – in two cargo tanks	3) **All Transverse Bulkheads** – in all cargo tanks	
5) **One Transverse Bulkhead** – in a cargo center tank (See Note II)	4) **One Transverse Bulkhead** – in each ballast tank (See Note I)	4) **All Transverse Bulkheads** – in all ballast tanks	
	5) **One Transverse Bulkhead** – in two cargo centre tanks (See Note II)		
	5) **One Transverse Bulkhead** – in cargo wing tank		

(1), (2), (3), (4), (5), (6) and (7) are areas to be subjected to close-up surveys and thickness measurements (See Figs. 4.4.8.1 to 4.4.8.3).

(1) Web frame in a ballast tank means vertical web in side tank, hopper web in hopper tank, floor in double bottom tank and deck transverse in double deck tank (where fitted), including adjacent structural members. In fore and aft peak tanks web frame means a complete transverse web frame ring including adjacent structural members.

(2) Deck transverse, including adjacent deck structural members (or external structure on deck in way of the tank, where applicable).

(3) Transverse bulkhead complete in cargo tanks, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal structure of lower and upper stools, where fitted.

(4) Transverse bulkhead complete in ballast tanks, including girder system and adjacent structural members, such as longitudinal bulkheads, girders in double bottom tanks, inner bottom plating, hopper side, connecting brackets.

(5) Transverse bulkhead lower part in cargo tank, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal structure of lower stool, where fitted.

(6) The knuckle area and the upper part (3 metres approximately), including adjacent structural members. Knuckle area is the area of the web frame around the connections of the slope hopper plating to the inner hull bulkhead and the inner bottom plating, upto 2 metres from the corners both on the bulkhead and the double bottom.

(7) Web frame in a cargo tank means deck transverse, longitudinal bulkhead structural elements and cross ties, where fitted, including adjacent structural members.

Note I : Ballast double hull tank : means double bottom tank plus double side tank plus double deck tank, as applicable, even if these tanks are separate.

Note II : Where no center cargo tanks are fitted (as in the case of center longitudinal bulkhead), transverse bulkheads in wing tanks are to be surveyed.

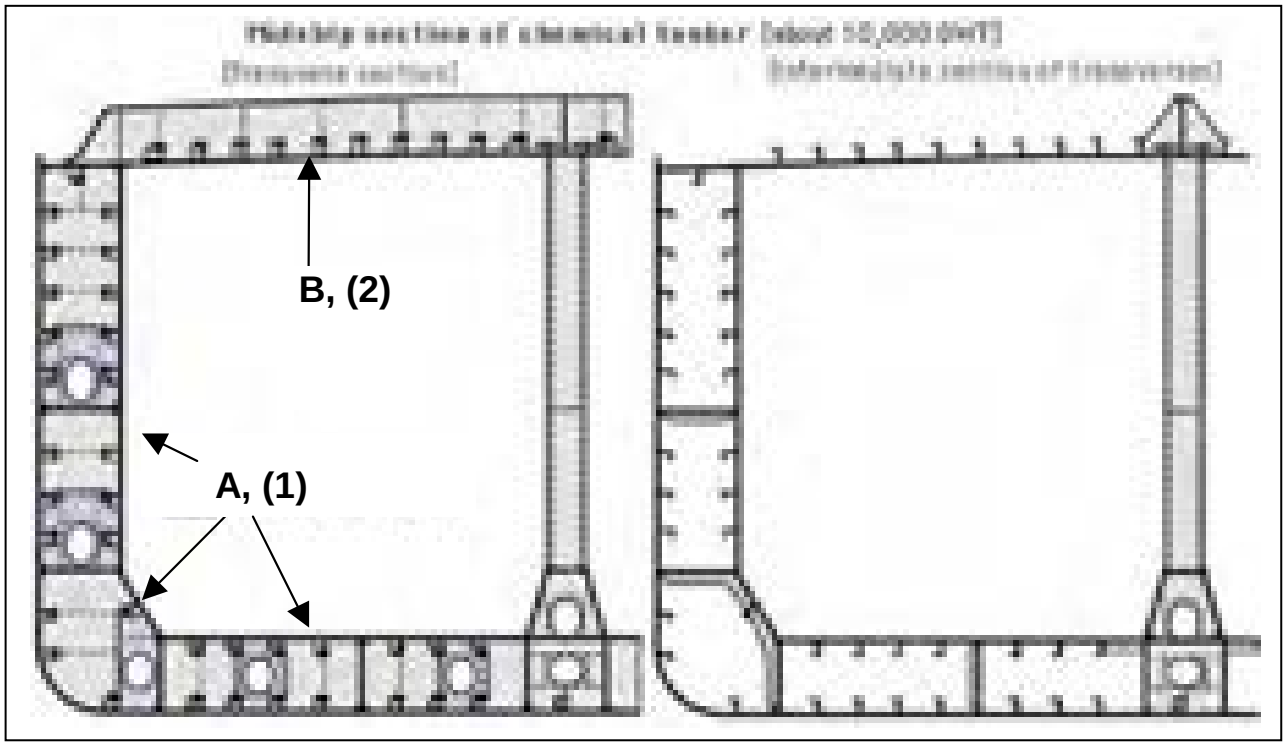

Fig. 4.4.8.1 : Representative transverse section of chemical tanker. Areas 1 and 2

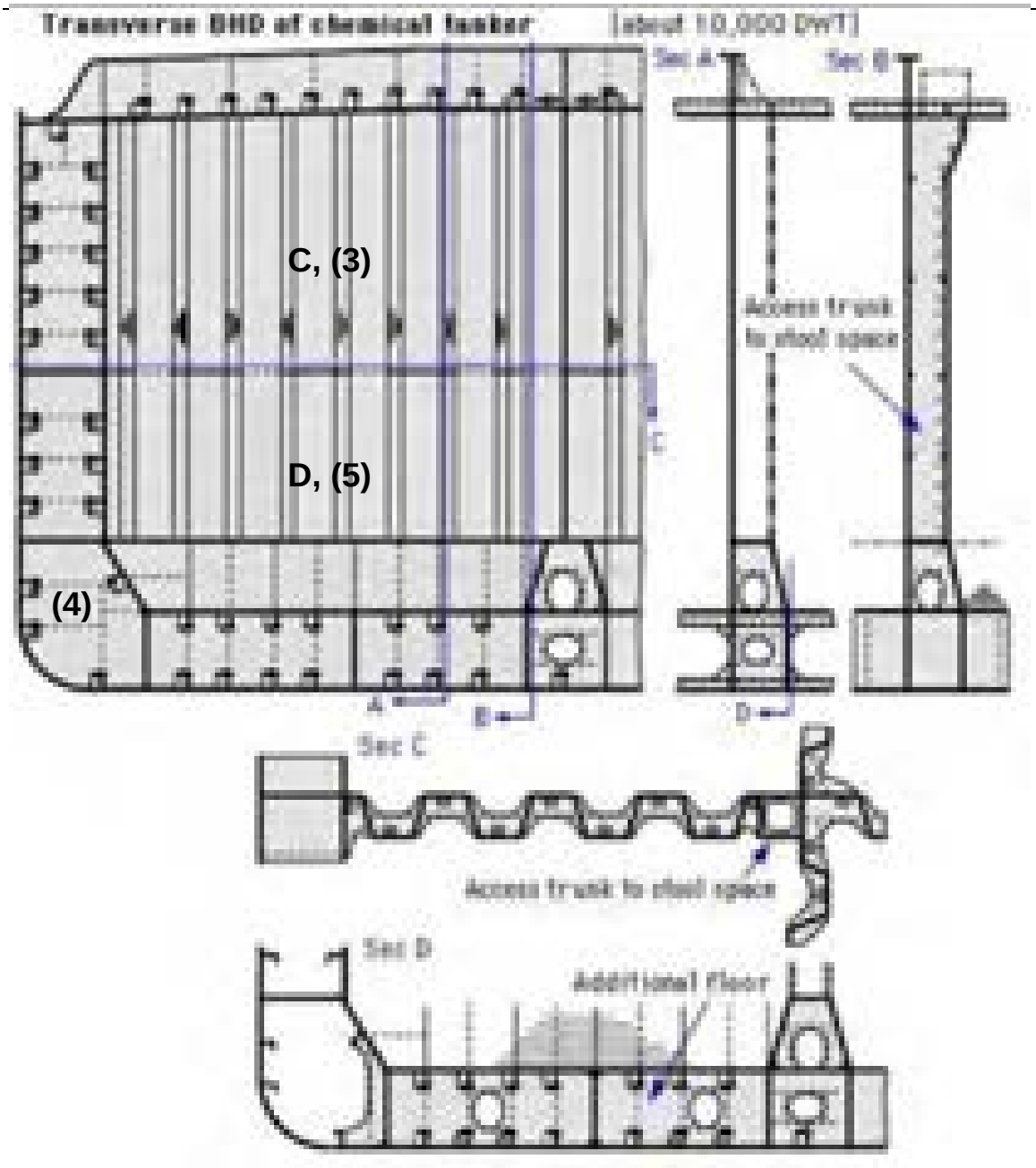

Fig. 4.4.8.2 : Representative transverse section of chemical tanker. Areas 3, 4 and 5

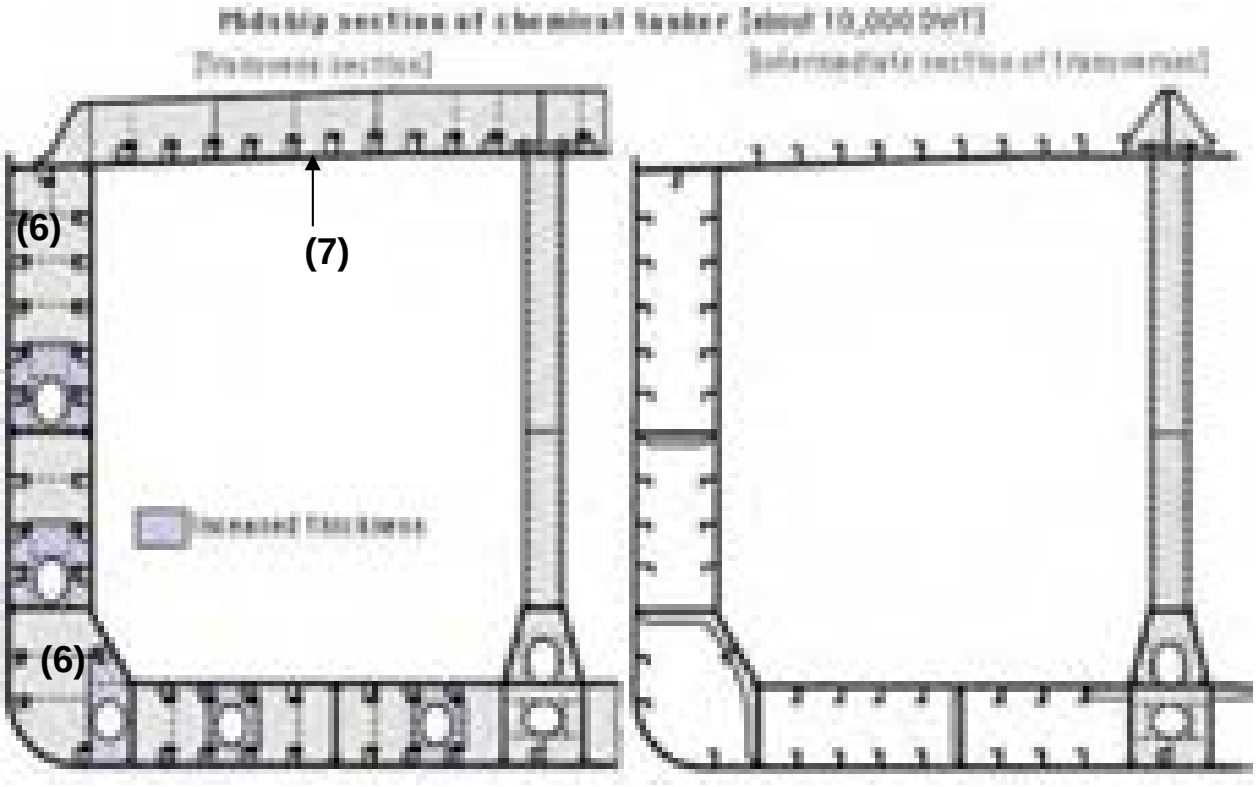

Fig. 4.4.8.3 : Representative transverse section of chemical tanker. Areas 6 and 7

Table 4.4.9.1 : Table of minimum requirements for thickness measurements at special survey of chemical tankers

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel
	2) Within the cargo area: a) Each deck plate b) One Transverse section	2) Within the cargo area: a) Each deck plate b)Two Transverse sections(1) c) All wind and water strakes	2) Within the cargo area: a) Each deck plate b) Three Transverse sections(1) c) All wind and water strakes d) Each bottom plate
	3) Measurements of structural members subject to close-up survey according to Table 4.4.8.1a) and Table 4.4.8.1b), for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 4.4.8.1a) and Table 4.4.8.1b), for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 4.4.8.1a) and Table 4.4.8.1b), for general assessment and recording of corrosion pattern

Table 4.4.9.1 : (Contd.)			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
	4) All wind and water strakes outside the cargo area	4) All wind and water strakes outside the cargo area	4) All wind and water strakes outside the cargo area
		5) Internals in forepeak and after peak ballast tanks	5) Internals in forepeak and after peak ballast tanks
			6) All exposed main deck plating outside cargo area
			7) Representative exposed superstructure deck plating (poop, bridge, forecastle deck)
			8) All keel plates, additional bottom plates in way of machinery space, aft end of tanks and cofferdams outside cargo area
			9) a) Plating of sea chests. b) Shell plating in way of overboard discharges as considered necessary by the Surveyor.
(1) At least one section is to include a ballast tank within 0.5L amidships			

Table 4.4.9.2 : Requirements for extent of thickness measurements at those areas of substantial corrosion, special survey of chemical tankers within the cargo area length		
Structural member	**Extent of measurement**	**Pattern of measurement**
Bottom, inner bottom and hopper structure		
Bottom, inner bottom and hopper structure plating	Minimum of three bays across tank, including aft bay Measurements around and under all suction bell mouths	5-point pattern for each panel between longitudinals and floors
Bottom, inner bottom and hopper structure longitudinals	Minimum of three longitudinals in each bay where bottom plating measured	Three measurements in line across the flange and three measurements on vertical web
Bottom girders, including the watertight ones	At fore and aft watertight floors and in center of tanks	Vertical line of single measurements on girder plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across face flat where fitted
Bottom floors, including the watertight ones	Three floors in bays where bottom plating measured, with measurements at both ends and middle	5-point pattern over two square metre area

Table 4.4.9.2 : Contd.)		
Structural member	**Extent of measurement**	**Pattern of measurement**
Hopper structure web frame ring	Three floors in bays where bottom plating measured	5-point pattern over one square metre of plating. Single measurements on flange
Hopper structure transverse watertight bulkhead or swash bulkhead	- Lower 1/3 of bulkhead	- 5-point pattern over one square metre of plating
	- Upper 2/3 of bulkhead	- 5-point pattern over two square metre of plating
	- Stiffeners (minimum of three)	- For web, 5-point pattern over span (two measurements across web at each end and one at center of span). For flange, single measurements at each end and center of span
Panel stiffening	Where applicable	Single measurements
Deck structure		
Deck plating	Two transverse bands across tank	Minimum of three measurements per plate per band
Deck longitudinals	Every third longitudinal in each of two bands with a minimum of one longitudinal	Three measurements in line vertically on webs and two measurements on flange (if fitted)
Deck girders and brackets	At fore and aft transverse bulkhead, bracket toes and in centre of tanks	Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across flange, 5-point pattern on girder/bulkhead brackets
Deck transverse webs	Minimum of two webs, with measurements at both ends and middle of span	5-point pattern over one square metre area. Single measurements on flange
Vertical web and transverse bulkhead in wing ballast tank for double hull design (two metres from deck)	Minimum of two webs and both transverse bulkheads	5-point pattern over one square metre area
Panel stiffening	Where applicable	Single measurements
Side shell and longitudinal bulkheads		
Side shell and longitudinal bulkhead plating: - Deckhead and bottom strakes and strakes in way of horizontal girders - All other strakes	- Plating between each pair of longitudinals in a minimum of three bays (along the tank) - Plating between every third pair of longitudinals in same three bays	Single measurement

Table 4.4.9.2 : Contd.)		
Structural member	**Extent of measurement**	**Pattern of measurement**
Side shell and longitudinal bulkhead longitudinals on: - Deckhead and bottom strakes - All other strakes	- Each longitudinal in same three bays - Every third longitudinal in same three bays	3 measurements across web and 1 measurement on flange
Longitudinals – brackets	- Minimum of three at top, middle and bottom of tank in same three bays	5-point pattern over area of brackets
Vertical web and transverse bulkheads of double side tanks (excluding deck area): - Strakes in way of horizontal girders - Other strakes	- Minimum of two webs and both transverse bulkheads - Minimum of two webs and both transverse bulkheads	- 5-point pattern over approx. two square metre area - Two measurements between each pair of vertical stiffeners
Web frames and cross ties for other tanks than double side tanks	Three webs with minimum of three locations on each web, including in way of cross tie connections and lower end brackets	5-point pattern over approximately two square metre area of webs, plus single measurements on flanges of web frame and cross ties
Horizontal girders	Plating on each girder in a minimum of three bays	Two measurements between each pair of longitudinal girder stiffeners
Panel stiffening	Where applicable	Single measurements
Transverse watertight and swash bulkheads		
Upper and lower stool, where fitted	- Transverse band within 25 [mm] of welded connection to inner bottom / deck plating - Transverse band within 25 [mm] of welded connection to shelf plate	5-point pattern between stiffeners over one metre length
Deckhead and bottom strakes and strakes in way of horizontal stringers	Plating between pair of stiffeners at three locations: approximately ¼, ½ and ¾ width of tank	5-point pattern between stiffeners over one metre length
All other strakes	Plating between pair of stiffeners at middle location	Single measurement
Strakes in corrugated bulkheads	Plating for each change of scantling at center of panel and at flange of fabricated connection	5-point pattern between stiffeners over one metre length
Stiffeners	Minimum of three typical stiffeners	For web, 5-point pattern over span between bracket connections (two measurements across web at each bracket connection and one at center of span). For flange, single measurements at each bracket toe and at center of span

Table 4.4.9.2 : Contd.)		
Structural member	**Extent of measurement**	**Pattern of measurement**
Brackets	Minimum of three at top, middle and bottom of tank	5-point pattern over areas of bracket
Horizontal stringers	All stringers with measurements at both ends and middle	5-point pattern over one square metre areas, plus single measurements near bracket toes and on flanges
Deep webs and girders	Measurements at toe of bracket and at center of span	For web, 5 point pattern over about 1 square metre, 3 measurements across face flat

4.4.9.5 In order to ensure necessary control during the process of thickness measurements, these are normally to be carried out under the supervision of the Surveyor. The Surveyor has the right to re-check the measurements as deemed necessary to ensure acceptable accuracy.

4.4.9.6 In all cases the extent of the thickness measurements are to be sufficient as to represent the actual average condition.

4.4.9.7 A thickness measurement report is to be prepared. The report is to give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and has to be signed by the operator. The Surveyor is to review the report of the final thickness measurement after repairs have been carried out and countersign the cover page.

4.4.10 Additional requirements for cargo and ballast piping

4.4.10.1 For chemical tankers exceeding 10 years of age selected steel cargo pipes outside cargo tanks and ballast pipes passing through cargo tanks are to be:

- Thickness measured at random or selected pipe lengths to be opened for internal inspection.

- Pressure tested to the maximum working pressure. Special attention is to be given to cargo/slop discharge piping through ballast tanks and void spaces.

4.4.11 Survey of independent tanks

4.4.11.1 All independent cargo tanks are to be surveyed internally and externally together with tank supports and the adjacent hull structure.

4.4.11.2 Thickness gauging of the independent cargo tanks is to be carried out as found necessary by the Surveyor.

4.4.11.3 All independent cargo tanks are to be hydraulically pressure tested to their MARVS (Maximum Allowable Relief Valve Setting). However, testing of gravity type tanks may be omitted if the tanks are found without corrosion and other damages and otherwise acceptable.

4.4.12 Reporting and evaluation of survey

4.4.12.1 The data and information on the structural condition of the vessel collected during the survey is to be evaluated for acceptability and continued structural integrity of the vessel.

4.4.12.2 An Executive Hull Summary of the survey and results is to be issued to the Owner and placed on board the vessel for reference at future surveys. The Executive Hull Summary is to be endorsed by IRS Head Office.

4.4.13 Watertight Cable Transits

4.4.13.1 The requirements for Special Survey may be undertaken by the attending Surveyor or by a firm approved as a service supplier according to Pt.1 Ch.1, Sec 1.9.

4.4.13.2 All transits are to be examined to confirm their satisfactory condition and the Cable Transit Seal System Register is to be reviewed to confirm that it is being maintained. The Special Survey is to be recorded in the

Register, in which a single record entry will be sufficient to record the survey of all transits.

4.4.13.3 From review of the Register, where there are records entered since the last special survey of any disruption to the cable transits or installation of new cable transits (except which are reviewed and examined at previous annual surveys), the satisfactory condition of those transits is to be confirmed by the attending Surveyor by review of records and examination of the transits; the results are to be recorded in the Register against each of those cable transits.

4.4.13.4 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

4.5 Special Surveys - Machinery

4.5.1 Requirements for examination of machinery and systems are given in Sec.8.

Section 5

Surveys - Liquefied Gas Carriers

5.1 Scope

5.1.1 This section gives the requirements for periodical surveys of vessels which have been assigned the class notation:

Liquefied gas carrier.

5.2 Annual surveys

5.2.1 General

5.2.1.1 Annual Surveys are to be carried out within 3 months before or after the anniversary date each year. (Also refer to Note 9 of Sec.1, Table 1.1.1). These should be held concurrently with statutory annual or other relevant statutory Surveys, where practicable.

5.2.1.2 At Annual Surveys, the Surveyor is to examine the hull and machinery, so far as necessary and practicable, in order to be satisfied as to their general condition.

5.2.1.3 The annual survey is preferably to be carried out during a loading or unloading operations.

5.2.1.4 It is to be confirmed that no new installation of material containing asbestos was carried out since last survey.

5.2.1.5 Where applicable, examination of alternative design and arrangement for segregation of the cargo area or machinery or electrical installations or fire safety, in accordance with the test, inspection and maintenance requirements, if any, specified in the approved documentation is to be carried out.

5.2.2 Hull and weather deck

5.2.2.1 The survey is to consist of an examination for the purpose of ensuring, as far as practicable, that the hull, equipment, closing appliances and related piping are maintained in satisfactory/efficient condition. Special attention is to be paid to the following:

a) Weather deck, shipside plating above waterline.

b) Openings on freeboard and superstructure decks; exposed casings; skylights and fiddley openings; deck houses; companionways and superstructure bulkheads; side scuttles and dead lights; flush deck scuttles; ash shoots and other openings.

c) Weld connection between air pipes and deck plating; air pipe heads on exposed decks (external examination); flame screens on vents to all bunker tanks; ventilators and closing devices.

d) Watertight bulkheads and their penetrations as far as practicable.

e) Scuppers and sanitary discharges as far as practicable together with valves and their controls.

f) Guard rails, bulwarks, freeing ports, gangways, walkways and life lines.

g) Examining the cargo, bunker, ballast and vent piping systems, including PRVs, vacuum relief valves, vent masts and devices to prevent the passage of flame on vents to all bunker, holds and void spaces, as far as practicable and confirming that the PRVs are type-approved or marked with date of testing.

h) Confirming that the wheel house doors and windows, non-opening type side scuttles and windows in superstructure and deck house ends facing the cargo area are in a satisfactory gastight condition.

i) Confirming that any special arrangements to survive conditions of damage are in order.

5.2.2.2 Examining the cargo machinery spaces and turret compartments, including their escape routes

Examination of the electric motor room and compressor room bulkheads for signs of leakage or fractures and in particular, the sealing arrangements of all penetrations on bulkheads.

5.2.2.3 Confirming that the sealing arrangements at the gas domes are satisfactory.

5.2.2.4 All watertight doors in watertight bulkheads, to be examined and tested (locally and remotely) as far as practicable.

5.2.2.5 Suspect areas identified at previous special or intermediate surveys are to be close-up surveyed. Thickness measurements are to be taken in the area of substantial corrosion identified at previous surveys.

5.2.2.6 Examination of ballast tanks is to be carried out when required as a consequence of the results of the Special Survey or Intermediate Survey. (See 5.4.3.5, 5.4.3.6, 5.3.2.1 and 5.3.2.2). When considered necessary by the Surveyor or where extensive corrosion is found, thickness measurement is to be carried out. When examination of ballast tanks reveals substantial corrosion, additional thickness measurements are to be carried out to determine the extent of the areas of substantial corrosion in accordance with Table 5.4.4.2.

5.2.2.7 Anchoring and mooring equipment is to be examined as far as is practicable. A general examination of the emergency towing arrangements is to be carried out to ensure their ready availability. It is to be confirmed that the towing and mooring equipment is properly marked with any restriction associated with its safe operation.

For vessels with single point mooring arrangements where 'SPM' notation is assigned, the following are to be generally examined:

a) Components of the single point mooring system (bow chain stoppers, bow fairleads, pedestal roller fairleads, winches and capstans), to verify their satisfactory condition.

b) Hull structures supporting and adjacent to the components of the single point mooring system, to verify that there is no deformation or fracture.

5.2.2.8 Where applicable Surveyor should satisfy himself regarding the freeboard marks on the ship's side.

5.2.2.9 The Surveyor is to confirm that, where required, an approved loading instrument together with its operation manual are available on board, See Pt.3, Ch.5. It is to be verified by the Surveyor that the loading instrument is checked for accuracy at regular intervals by the ship's staff by applying test loading conditions.

5.2.2.10 Accommodation ladders are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) steps;
b) platforms;
c) all support points such as pivots, rollers, etc.;
d) all suspension points such as lugs, brackets, etc.;
e) stanchions, rigid handrails, hand ropes and turntables;
f) davit structure, wire and sheaves, etc.

5.2.2.11 Access to bow arrangements are to be examined. Gangways are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular:

a) treads;
b) side stringers, cross-members, decking, deck plates, etc.;
c) all support points such as wheel, roller, etc.;
d) stanchions, rigid handrails, hand ropes.

5.2.2.12 Winches of accommodation ladders and gangways are to be examined to verify the satisfactory condition of the following items:

a) brake mechanism including condition of brake pads and band brake, if fitted;
b) remote control system, and
c) power supply system for electric motor.

5.2.2.13 Davits and fittings on the ship's deck associated with accommodation ladders and gangways are to be examined for satisfactory condition at annual surveys. Fittings or structures for means of access to deck such as handholds in a gateway or bulwark ladder and stanchions are also to be examined.

5.2.2.14 The maintenance and inspection records of accommodation ladders and gangways are to be verified. It is to be confirmed that supporting wires are being renewed at intervals not exceeding 5 years.

5.2.2.15 When examining internal spaces, as far as practicable, the permanent means of access where appropriate, are to be verified that they remain in good condition.

5.2.2.16 For vessels subject to IMO PSPC (See Pt.3, Ch.2, 3.6) it is to be confirmed that the maintenance, repair and partial re-coating of dedicated ballast tanks, as appropriate, are recorded in the coating technical file.

5.2.2.17 Watertight cable transit seal systems register is to be reviewed to confirm that it is being maintained and as far as practicable the transits are to be examined to confirm their satisfactory condition. Where there are records entered since last annual survey of any disruption to the cable transits or installation of new cable transits, the satisfactory condition of those transits is to be confirmed by review of records and, if deemed necessary, by examination. The results are to be recorded in the Register against the specific cable transit.

5.2.2.18 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

5.2.3 Machinery and systems

5.2.3.1 A general examination of the machinery, boilers, all pressurised systems (steam, hydraulic, pneumatic) and their associated fittings, propulsion system and auxiliary machinery to see whether they are being properly maintained and with particular attention to the fire and explosion hazards.

5.2.3.2 Confirmation that machinery, boilers and other pressure vessels, associated piping systems and fittings are so installed and protected as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards.

5.2.3.3 Confirmation that Periodical Surveys of boilers and other pressure vessels have been carried out as required by the Rules and the safety devices have been tested.

5.2.3.4 Confirmation that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative.

5.2.3.5 Confirmation that means are provided so that machinery can be brought into operation from the dead ship condition without external aid.

5.2.3.6 All main and auxiliary steering arrangements and their associated equipment and control systems are to be examined and tested. Where applicable, Surveyors are to verify that log entries have been made in accordance with statutory requirements.

5.2.3.7 Confirming, when appropriate, that requisite arrangements to regain steering capability in the event of the prescribed single failure are being maintained.

5.2.3.8 All the means of communication between the navigating bridge and the machinery control positions, as well as the bridge and the main alternative steering position, if fitted, are to be tested. It is to be confirmed that means of indicating the angular position of the rudder are operating satisfactorily.

5.2.3.9 Confirmation that with ships having emergency steering positions there are means of relaying heading information and, when appropriate, supplying visual compass readings to the emergency steering positions.

5.2.3.10 Confirmation that various alarms required for hydraulic power operated, electric and electro-hydraulic steering gears are, operating satisfactorily and that the recharging arrangements for hydraulic power operated steering gears are being maintained.

5.2.3.11 Examining the means for the operation of the main and auxiliary machinery essential for propulsion and the safety of the ship, including when applicable, the means of remotely controlling the propulsion machinery from the navigating bridge and the arrangements to operate the main and other machinery from a machinery control room.

5.2.3.12 Confirmation that the engine room telegraph, the second means of communication between the navigation bridge and the machinery space and the means of communication with any other positions from which the engines are controlled are operating satisfactorily.

5.2.3.13 Examination and testing of the general emergency alarm system and confirmation that the engineer's alarm is clearly audible in the engineer's accommodation.

5.2.3.14 It is to be confirmed that ballasting and de-ballasting arrangements are satisfactory. The bilge pumping systems and bilge wells including operation of each bilge pump, extended spindles and level alarms, where fitted, are to be examined as far as is practicable. It is also to be confirmed that bilge pumping system for each watertight compartment is satisfactory.

It is also to be confirmed that drainage from enclosed spaces in cargo areas situated on freeboard deck is satisfactory.

5.2.3.15 Examining visually the condition of non-metallic expansion joints where fitted in piping systems which penetrate the ship's side, with both the penetration and the expansion joint located below the deepest load waterline, and checking the service record.

5.2.3.16 General examination visually and in operation, as feasible, of the main electrical machinery, the emergency sources of electrical power, the switch gear, other electrical equipment including the lighting system is to be carried out.

5.2.3.17 Confirmation as far as practicable, the operation of the emergency source(s) of electrical power, including their starting arrangement, the systems supplied, and when appropriate, their automatic operation.

5.2.3.18 Examining in general, that the precautions provided against shock, fire and other hazards of electrical origin are being maintained.

5.2.3.19 General Examination of automation equipment is to be carried out. Satisfactory operation of safety devices, bilge level detection and alarm systems and control systems is to be verified.

5.2.3.20 Confirmation that machinery space ventilation systems are in good working condition.

5.2.3.21 For Ships constructed before 01 July 2017 and fitted with harmonic filters; following are to be verified at each survey:

a) Records of annual measurement of harmonic distortion levels of main bus bar under seagoing conditions and log entries showing which equipment was running and / or filters in service at the time of measurement. Harmonic distortion readings are to be carried out when the greatest amount of distortion is indicated by the measuring equipment.

b) Records of measurement of distortion levels, after any modification to the ship’s electrical distribution system.

5.2.3.22 For Ships constructed on or after 01 July 2017 and fitted with harmonic filters; or for ships retrofitted with harmonic filters, on or after 01 July 2017; following are to be verified at each survey:

a) Engine log book containing distortion monitoring readings/ automatic records of same

b) Documentation of any approved relaxation on allowable distortion limits.

5.2.3.23 For ships contracted for construction on or after 01 Jan 2017 following are to be verified in emergency generator rooms at each survey:

a) Hand operated ventilation louvres, closing appliances, if fitted, are kept open and corresponding instruction plates are visible near the louvres.

b) Power operated ventilation louvres and closing appliances, if fitted, are fail-to-open type and capable of opening automatically whenever emergency generator is starting / in operation.

c) It is to be possible to close ventilation openings by a manual operation from a clearly marked safe position outside the space where the closing operation can be easily confirmed. Further, such closing is not possible from any other remote position. The louvre status (open/closed) is indicated at this position.

5.2.3.24 For vessels fitted with ICCP system, proper functioning of the system is to be checked and records of system operation are to be verified. It is to be verified that all maintenance as per manufacturer's instructions are carried out.

5.2.4 Fire protection, detection and extinction

5.2.4.1 The arrangements for fire protection, detection and extinction are to be examined and are to include confirmation that no changes have been made in the structural fire protection. Following are to be examined / verified:

a) verification that fire control plans are properly posted;

b) examination as far as possible and testing as feasible, any fire detection and alarm system and any sample extraction smoke detection system;

c) examination of the fire main system and verification that each fire pump including the emergency fire pump can be operated separately so that the two required powerful jets of water can be produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main; testing the remote means for starting one main fire pump;

d) verification that fire hoses, nozzles, applicators and spanners are in good working condition and situated at their respective locations;

e) examination of fire-fighting system (including fixed fire fighting for the machinery spaces) controls, piping, instructions and marking, checking for evidence of proper maintenance and servicing including date of last systems tests;

f) verification that all semi-portable and portable fire extinguishers are in their stowed positions, checking for evidence of proper maintenance and servicing, conducting random check for evidence of discharged containers;

g) verification, as far as practicable, that the remote controls for stopping fans and machinery in machinery spaces are in working order. Examination of the arrangements for oil fuel, lubricating oil and other flammable oils and testing the remote closing of valves for oil fuel, lubricating oil and other flammable oils and , the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils;

h) examination of the closing arrangements of ventilators, funnel annular spaces, skylights, doorways and tunnel, where applicable;

i) confirmation that the fire fighters' outfits including its self-contained compressed air breathing apparatus and emergency escape breathing devices (EEBDs) are complete and in good condition and that the cylinders, including the spare cylinders, of any required self-contained breathing apparatus are suitably charged, and that on board means of recharging breathing apparatus cylinders used during drills or a suitable number of spare cylinders to replace those used are provided, and provision of two-way portable radiotelephone apparatus of an explosion-proof type or intrinsically safe ;

j) examination of any manual and automatic fire doors and proving their operations;

k) Examining the fixed fire fighting system for the enclosed cargo machinery spaces, and the enclosed cargo motor room within the cargo area, and confirming that its means of operation is clearly marked;

l) Examining the water spray system for cooling, fire protection and crew protection and confirming that its means of operation is clearly marked;

m) Confirming that all electrical equipment in gas dangerous spaces and zones is suitable for such locations, is in satisfactory condition and properly maintained;

n) Examining the dry chemical powder fire extinguishing system for the cargo area and confirming that its means of operation is clearly marked;

o) Examining the fixed fire fighting system for the cargo pump room and the deck foam system for the cargo area and confirming that their means of operation are clearly marked.

p) Examining the appropriate fire-extinguishing system for the enclosed cargo machinery spaces for ships that are dedicated to the carriage of a restricted number of cargoes and the internal water spray system for the turret compartments and confirming that their means of operation is clearly marked.

q) examination of the fire-extinguishing systems for spaces containing paint and/or flammable liquids and deep fat cooking equipment in accommodation and service spaces;

r) examination of the fire safety requirements of helicopter facilities, including foam firefighting appliances .

s) Examination of the fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space.

t) Examination of the fire protection arrangements in cargo spaces and confirming as far as practicable and as appropriate, the operation of the means of control provided for closing the various openings.

u) Testing the fire dampers of ventilation ducts and the means of closing the main inlets and outlets of all ventilation systems and testing the means of stopping power ventilation systems from outside the space served.

Surveys carried out by the National Authority of the country in which the ship is registered would normally be accepted as meeting these requirements, at the discretion of the Surveyor.

5.2.4.2 On ships where inert gas systems are installed, the requirements of Sec.12 are to be complied with.

5.2.4.3 Confirmation that the means of escape from accommodation, machinery spaces and other spaces are satisfactory.

5.2.4.4 Examination of the arrangements for gaseous fuel for domestic purposes.

5.2.5 Other safety arrangements related to cargo / cargo area

5.2.5.1 The ship's log and operational records for the cargo containment system covering the period from the previous survey are to be examined taking into consideration hours per day of the reliquefaction plant or the boil-off rate. Any malfunction of the system entered in the log is to be investigated, the cause ascertained, and that part of the system at fault is to be found or placed in good order.

5.2.5.2 Cargo liquid level indicating devices are to be generally examined. The low level, high level, and overfill alarms are to be examined and tested to ascertain that they are in working order. Consideration will be given to the acceptance of simulated tests, provided that they are carried out at the cargo temperature, or comprehensive maintenance records, including details of tests held, in accordance with the cargo plant instrumentation maintenance manual. The calibration status of the measuring instruments may also be verified.

5.2.5.3 Cargo gas leakage detection systems are to be examined and tested to ascertain that they are in working order and calibrated using sample gas.

The gas detection arrangements for cargo control rooms and the measures taken to

exclude ignition sources where such spaces are classified as hazardous, are to be examined.

5.2.5.4 The correct functioning of the cargo containment system temperature and pressure indicating equipment, including, when fitted, the thermal oxidation systems and any refrigeration system, together with any associated alarms, is to be verified.

5.2.5.5 Control devices for the cargo containment systems and cargo handling equipment, together with any associated shutdown and/or interlock, are to be checked under simulated working conditions and, if necessary, recalibrated.

5.2.5.6 The arrangements for manually operated emergency shutdown system together with the automatic shut down of the cargo pumps and compressors are to be checked to ascertain they are in working order.

Alternatively, the log books may be examined for verification that the emergency shutdown system has been tested.

5.2.5.7 Ventilation systems and air locks in working spaces are to be checked for satisfactory operation.

5.2.5.8 Cargo pipeline, valves and fittings are to be generally examined, with special reference to expansion bellows, supports and vapour seals on insulated pipes.

5.2.5.9 Examining the process piping including the expansion arrangements, insulation from hull structure, pressure relief and drainage arrangements and water curtain protection as appropriate.

5.2.5.10 Portable and/or fixed drip trays, or insulation for deck protection in the event of cargo leakage, are to be examined for condition.

5.2.5.11 Examining when applicable bow or stern loading and unloading arrangements with particular reference to the electrical equipment, fire- fighting arrangements and means of communications between the cargo control room and the shore location.

5.2.5.12 Confirming that the cargo tank and interface pressure and relief valves, including safety systems and alarms are satisfactory.

5.2.5.13 Confirming that all liquid and vapour hoses are suitable for their intended purpose.

5.2.5.14 Confirming that any air drying system and any inter barrier and hold space purging inert gas system are satisfactory.

For membrane containment systems, normal operation of the nitrogen control system for insulation and inter barrier spaces is to be confirmed by the Master.

5.2.5.15 Confirming that two sets of portable gas detection equipment suitable for the cargoes to be carried and a suitable instrument for measuring oxygen levels have been provided.

5.2.5.16 Electrical bonding of the cargo pipes and tanks to the hull is to be verified.

5.2.5.17 Confirming that when applicable arrangements are made for sufficient inert gas to be carried to compensate for normal losses and that means are provided for monitoring the spaces.

5.2.5.18 Confirming that the use of inert gas has not increased beyond that needed to compensate for normal losses by examining records of inert gas usage.

5.2.5.19 Examining the emergency lighting in all cargo compressor rooms and electric motor rooms for cargo compressors.

5.2.5.20 Examining the cargo control room.

5.2.6 Reliquefaction/refrigeration equipment

5.2.6.1 Where reliquefaction or refrigeration equipment for cargo temperature and pressure control is fitted and notation HY(LGC) assigned, the following are to be examined, so far as is practicable:

a) The machinery under working conditions.

b) The shells of all pressure vessels in the system, externally, insulation need not be removed for this examination, but any deterioration of insulation or evidence of dampness which could lead to external corrosion of the vessels of their connections, is to be investigated.

c) Primary refrigerant gas and liquid pipes, cargo vapour and liquid condensate pipes and condenser cooling water pipes. Insulation need not be removed, but any deterioration or evidence of dampness is to be investigated.

d) The reliquefaction/refrigeration plant spare gear.

5.2.7 Methane burning equipment and other equipment components

5.2.7.1 The following components are to be generally examined externally. If insulation is fitted, this need not be removed, but any deterioration of insulation, or evidence of dampness which could lead to external corrosion of the vessels or their connections, is to be investigated.

a) Heat exchangers and pressure vessels for use with methane burning in boilers or machinery.

b) Cargo heaters, vaporizers, masthead heaters and other miscellaneous pressure vessels.

5.2.7.2 Controls and interlocks are to be checked.

5.2.7.3 Alarm systems are to be checked to ascertain they are in working order.

5.2.7.4 Exhaust fans and/or pressurizing system for gas trunking are to be tested.

5.2.7.5 The relevant instruction and information material such as cargo handling plans, filling limit information, cooling down procedures etc. are to be verified as being onboard.

5.2.8 Cargo containment systems

5.2.8.1 At the first Annual Survey after initial commissioning of the ship, the operating records of the primary gas detection system are to be examined.

5.2.8.2 Where the insulation arrangement is such that the insulation cannot be examined during special surveys (See 5.4.6.9) the surrounding structures of wing tanks, double bottom tanks and cofferdams are to be examined for cold spots, prior to the survey. This examination is to be held at a convenient cargo discharge operation with the cargo tanks loaded at approximately the minimum notation temperature.

5.2.8.3 On application by the Owner, consideration will be given to the cold spot examination, where applicable, being carried out by the ship's staff.

5.2.8.4 When tests are required after repairs, independent cargo tanks, other than independent tanks, type C, are to be tested by hydraulic or hydropneumatic means as appropriate. Test heads and pressures should be as defined in Pt.5, Ch.4. Cargo tanks of the membrane or semi-membrane type are to be tested by means of a detectable gas in the inter-barrier spaces and discolouring paint on the weld seams of the cargo tanks wall, or other suitable means. Independent cargo tanks of type C are to be tested hydraulically at 1.25 times the approved maximum vapour pressure.

5.2.9 Personnel protection

a) Verification that the requisite safety equipment and associated breathing apparatus with requisite air supplies and emergency escape respiratory and eye protection, if required, are in good condition and are properly stowed.

b) Verification that medical first air equipment including stretchers and oxygen resuscitation is in god condition and that satisfactory arrangements are made for antidotes for cargoes actually carried to be on board.

c) Verification that decontamination arrange-ments are operational.

5.3 Intermediate Surveys

5.3.1 General

5.3.1.1 Intermediate surveys are to be carried out at or between the second or third Annual Survey. (Also refer to Note 9 of Sec.1, Table 1.1.1). However, only those items which are additional to the requirements for annual surveys may be examined between the second or third annual surveys.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

5.3.1.2 The following requirements are applicable for vessels over five years of age. For vessels below 5 years of age additional examination over and above the requirements of Annual survey may be required at the discretion of the Surveyors.

5.3.1.3 For ships carrying liquefied gases in bulk the intermediate survey is preferably to be carried out with the ship in a gas free condition. The purpose of the intermediate survey is to supplement the annual survey by testing the cargo handling installations with related automatic control, alarm and safety systems for their correct functioning. The extent of the testing required for the intermediate survey will normally be such that the survey cannot be carried out during loading or discharging operation.

5.3.1.4 Prior to the commencement of any part of the intermediate survey, a survey planning meeting is to be held between the attending Surveyor(s), the owner's representative in attendance and where involved, the thickness measurement company representative and the master of ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

5.3.2 Ballast tanks

5.3.2.1 Vessels of age more than 5 years but not exceeding 10 years

a) An overall survey of representative ballast tanks is to be carried out. When extensive corrosion is found, thickness measurements are to be carried out. If such examination reveals no visible structural defects, the examination may be limited to a verification that the corrosion prevention system remain efficient.

b) Where POOR coating condition, corrosion or other defects are found in ballast tanks or where hard protective coating was not applied from the time of construction, the examination is to be extended to other ballast tanks of the same type.

c) For ballast tanks, other than double bottom tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question are to be examined at subsequent annual surveys.

d) For double bottom ballast tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question may be examined at subsequent Annual surveys.

5.3.2.2 Vessels of age more than 10 years

a) An overall survey of all ballast tanks is to be carried out. When extensive corrosion is found, thickness measurements are to be carried out. If such examination reveals no visible structural defects, the examination may be limited to a verification that the corrosion prevention system remain efficient.

b) For ballast tanks, other than double bottom tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question are to be examined at subsequent annual surveys.

c) For double bottom ballast tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question may be examined at subsequent Annual surveys.

d) The Surveyor is to carry out a Close-up survey and thickness measurement of structure identified at the previous Special Survey as having substantial corrosion. The minimum requirements for close-up surveys at intermediate survey are given in Table 5.3.2.2.

Table 5.3.2.2 : Table of the minimum requirements for close-up survey of hull at intermediate surveys of liquefied gas carriers

10 < age ≤ 15	**age > 15**
Close-up survey of: - all web frames and both transverse bulkheads in a representative ballast tank (1) and (2) - the upper part of one web frame in another representative ballast tank - one transverse bulkhead in another representative ballast tank (2)	Close-up survey of : - all web frames and both transverse bulkheads in two representative ballast tanks (1) and (2)
(1) Complete transverse web frame including adjacent structural members	
(2) Transverse bulkhead complete, including girder system and adjacent members and adjacent longitudinal bulkhead structure	
Note 1 : Ballast tanks include topside, double hull side, double bottom, hopper side or any combined arrangement of the aforementioned and peak tanks where fitted.	
Note 2 : For areas in tanks where protective coating is found to be in good condition, the extent of close-up survey may be specially considered.	
Note 3 : For ships having independent tanks of type C, with a midship section similar to that of a general cargo ship, the extent of close-up surveys may be specially considered.	
Note 4 : The extent of close-up surveys may be extended by the Surveyor as deemed necessary, taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases: - in particular, in tanks having structural arrangements or details which have suffered defects in similar tanks, or on similar ships according to available information; - in tanks having structures approved with reduced scantlings.	

5.3.3 Safety arrangements related to cargo

5.3.3.1 The following are to be dealt with as applicable:

a) Examination of means for draining the vent piping system.

b) Verification that the heating arrangements, if any, for steel structures are satisfactory.

c) With regard to the Cargo containment systems examination of the items in accordance with 5.2.8.1 and 5.2.8.2.

d) Confirmation, where applicable that pipelines and independent cargo tanks are electrically bonded to the hull.

e) Generally examining the electrical equipment and cables in hazardous areas and zones such as cargo machinery spaces and areas adjacent to cargo tanks to check for defective equipment, fixtures and wiring. The insulation resistance of the circuits is to be tested and in cases where a proper record of testing is maintained, consideration should be given to accepting recent readings. Examination with reference to the following is to be carried out:

- Protective earthing (spot check).

- Integrity of flame proof enclosures.

- Damage of outer sheath of cables.

- Function testing of pressurized equipment and associated alarms.

- Testing of systems for de-energizing non-certified safe electrical equipment located in spaces protected by air

locks, such as electrical motor-rooms, cargo control rooms, etc.

f) Confirming that spares are provided for cargo area mechanical ventilation fans.

g) Checking the provision of equipment for personnel protection.

h) Examining when applicable, the arrangements for the use of cargo as fuel and testing as far as practicable, that the gas supply to the machinery space is cut off should the exhaust ventilation not be functioning correctly and that the master gas fuel valve may be remotely closed from within the machinery space.

i) The emergency shutdown system is to be tested without flow in the pipelines, to verify that the trip system stops the cargo pumps and compressors.

5.4 Special Surveys - Hull

5.4.1 General

5.4.1.1 All ships classed with IRS are to undergo Special Surveys at 5 yearly intervals. The first Special Survey is to be completed within 5 years from the date of the initial classification survey and thereafter 5 years from the assigned date of the previous Special Survey. However, an extension of class of 3 months maximum beyond the 5th year may be granted in exceptional circumstances in accordance with 1.11. In such cases, the next period of class will start from the expiry date of the Special Survey before extension was granted.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

5.4.1.2 The interval between the Special Surveys may be reduced at the request of the parties concerned or by IRS if considered appropriate.

5.4.1.3 For surveys completed within 3 months before the expiry date of the Special Survey, the next period of class will start from the expiry date of the Special Survey. For surveys completed more than 3 months before the expiry date of the Special Survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the special survey. If the owner elects to carry out the next due special survey, the period of class will start from the survey completion date. Any requirement of the Flag Administration in this regard is also to be complied with.

5.4.1.4 The Special Survey may be commenced at the 4th Annual Survey and be progressed with a view to completion by the 5th anniversary date. When the special survey is commenced prior to the fourth annual survey, the entire survey is to be completed within 15 months if such work is to be credited to the special survey and in this case the next period of class will start from the survey completion date.

5.4.1.5 For the purpose of special survey, results of thickness measurement carried out during or after the fourth annual survey only would be considered.

5.4.1.6 Record of Special Survey will not be assigned until the Machinery Survey has been completed or postponed in agreement with IRS.

5.4.1.7 Ships which have satisfactorily passed a Special Survey will have a record entered in the Supplement to the Register Book indicating the assigned date of Special Survey.

5.4.1.8 The special survey is to include, in addition to the requirements of the Annual Survey, examination, tests and checks of sufficient extent to ensure that the hull, equipment and related piping are in satisfactory condition and that the ship is fit for its intended purpose for the new period of class of five years to be assigned subject to proper maintenance and operation and the periodical surveys being carried out at the due dates.

5.4.1.9 The examinations of the hull are to be supplemented by thickness measurements (See 5.4.4) and testing as deemed necessary, to ensure that the structural integrity remains effective. The aim of the examination is to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration, that may be present.

5.4.1.10 A Docking Survey in accordance with the requirements of Sec.7 is to be carried out

as part of the Special Survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements and repairs applicable to the lower portions of ballast tanks (i.e. parts below light ballast waterline) are to be completed in the dry dock.

5.4.2 Planning and preparation for survey

5.4.2.1 The ship is to be prepared for overall survey in accordance with the requirements of Table 5.4.2.1. The preparation is to be of sufficient extent to facilitate an examination to ascertain any excessive corrosion, deformation, fractures, damages and other structural deterioration.

5.4.2.2 Prior to commencement of any part of the special survey, a survey planning meeting is to be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company representative and the master of the ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

5.4.3 Examination and testing

5.4.3.1 All spaces within the hull and superstructure are to be examined. The special survey is to include examination of underwater parts as per Section 7.

5.4.3.2 All tanks other than cargo and ballast tanks are to be examined internally in accordance with the requirements of Table 5.4.3.2.

All bilge and ballast piping systems are to be examined and operationally tested to working pressure to attending Surveyor's satisfaction to ensure that tightness remains satisfactory.

Table 5.4.2.1 : Survey preparation

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) All tanks, peaks, bilges and drain wells, engine and boiler spaces and other spaces are to be cleared out and cleaned as necessary for examination. Floor plates in engine and boiler spaces are to be lifted as may be necessary for examination of the structure underneath. Where necessary ceiling, lining, casings and loose insulation are to be removed as required by the Surveyor for examination of the structure. Compositions on the plating are to be examined and sounded, but need not be disturbed if found satisfactorily adhering to the plating 2) The steelwork is to be exposed and cleaned as may be required for its proper examination by the Surveyor and close attention is to be paid to the parts of the structure which are particularly liable to excessive corrosion or to deterioration due to other causes. 3) All tanks are to be cleaned as necessary to permit examination. 4) Casings or covers of air, sounding, steam and other pipes, spar ceiling and lining in way of the side scuttles are to be removed, as required by the Surveyor.	1) Requirements of Special Survey I to be complied with 2) The chain locker is to be cleaned internally. The chain cables are to be ranged for inspection. The anchors are to be cleaned and placed in an accessible position for inspection.	1) Requirements of Special Survey II to be complied with 2) Portions of wood sheathing, or other covering, on steel decks are to be removed, as considered necessary by the Surveyor, in order to ascertain the condition of the plating.	1) Requirements of Special survey III to be complied with

Table 5.4.3.2 : Requirements for internal examination of tanks

Tank	Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequent Age > 15
Fuel oil bunker tanks				
- Engine room	None	None	One	One
- Cargo area	None	One	Two	Half the number of tanks, minimum 2
- If no tanks in Cargo Area, additional fuel tank(s) outside of Engine Room (if fitted)	None	One	One	Two
Lub.oil	None	None	None	One
Fresh water	None	One	All	All

Notes:

1) These requirements apply to tanks of integral (structural) type.

2) If a selection of tanks is accepted to be examined, then different tanks are to be examined at each special survey, on a rotational basis.

3) Peak tanks (all uses) are subject to internal examination at each special survey.

4) At special surveys No.III and subsequent special surveys one deep tank for fuel oil is to be included, if fitted.

5.4.3.3 All spaces including double bottom, deep, ballast, peak and cargo tanks; pump rooms, cargo compressor and electrical motor rooms, pipe tunnels, duct keels, machinery spaces, dry spaces, cofferdams and voids are to be internally examined including the plating and framing, bilges and drain wells, sounding, venting, pumping and drainage arrangements. This examination is to be supplemented by thickness measurement and testing as deemed necessary, to ensure that the structural integrity remains effective. The aim of the examination is to discover Substantial Corrosion, significant deformation, fractures, damages or other structural deterioration and if deemed necessary by the Surveyor, a suitable non-destructive examination may be required. At special survey No.3 and subsequent special surveys, structural downflooding ducts and structural ventilation ducts are to be internally examined.

5.4.3.4 Where ballast tanks have been converted to void spaces, the survey extent is to be based upon ballast tank requirements.

5.4.3.5 For ballast tanks, excluding double bottom tanks, where a hard protective coating is found in POOR condition and it is not renewed, where soft or semi-hard coating has been applied, or where a protective coating was not applied from the time of construction, the space in question is to be internally examined at Annual Surveys. Thickness measurement is to be carried out as deemed necessary by the Surveyor.

5.4.3.6 For double bottom ballast tanks where a protective coating is found in POOR condition and it has not been renewed or where soft or semi-hard coating has been applied, or where a protective coating was not applied from the time of construction, the spaces in question may be examined at Annual Surveys. When considered necessary by the Surveyor, thickness measurement is to be carried out.

5.4.3.7 All watertight bulkheads and watertight doors are to be examined. All decks, casings and superstructures are to be examined.

Attention is to be given to the corners of openings and other discontinuities in way of the strength decks and top sides.

5.4.3.8 The masts, standing rigging and anchors are to be examined.

The Surveyor should satisfy himself that there are sufficient mooring ropes on board and also that a tow line is provided when this is a Rule requirement.

5.4.3.9 The steering gear, and its connections and control systems (main and alternative) are to be examined. The auxiliary steering gear with its various parts are to be examined in working condition.

5.4.3.10 The hand pumps and suctions, air and sounding pipes are to be examined. The Surveyors are to ensure that striking plates are fitted under the sounding pipes whilst examining the tanks internally.

Automatic air pipe heads are to be internally examined at special surveys as indicated in Table 5.4.3.10. For designs where the inner parts cannot be properly inspected from outside, the head is to be removed from the air pipe. Particular attention is to be paid to the condition of the zinc coating in heads constructed from galvanised steel.

Table 5.4.3.10 : Requirements for internal examination of automatic air pipe heads

Location	Special survey No.I Age ≤ 5	Special survey No.II 5 < Age ≤ 10	Special survey No.III and subsequent Age > 10
forward 0.25L	Two air pipe heads[1,2] one port and one starboard on exposed decks	All air pipe heads on exposed decks	All air pipe heads[3] on exposed decks
aft of 0.25L from the forward perpendicular	Two air pipe heads[1,2] one port and one starboard on exposed decks	At least 20% of air pipe heads[1,2] on exposed decks	

Notes:

1. Preferably air pipes serving ballast tanks.
2. The selection of air pipe heads is to be made by the attending Surveyor. According to the results of the inspection, the Surveyor may require additional air pipe heads to be examined.
3. When there is substantiated evidence of replacement within the previous five years, individual air pipe heads may not be examined.

5.4.3.11 The Surveyor should satisfy himself regarding the efficient condition of the following :

- Means of escape from machinery spaces, crew and passenger spaces and spaces in which crew are normally employed;

- Means of communication between bridge and engine room and between bridge and alternative steering position;

- Helm indicator;

- Protection to the aft steering wheel and the gear.

5.4.3.12 The chain cables are to be ranged and the anchors and the chain cables are to be examined. At special survey no. II and subsequent special surveys, the chain cables are to be gauged. Any length of chain cable which is found to have reduced in mean diameter at its most worn part by more than 12 per cent of its original rule diameter is to be renewed.

5.4.3.13 The windlass is to be examined.

5.4.3.14 The chain locker, hold fasts, hawse pipes and chain stoppers are to be examined and pumping arrangements of the chain locker tested.

5.4.3.15 A thorough examination of aft and forward emergency towing arrangements is to be carried out to ensure its ready availability and satisfactory condition.

- examination of pick-up gear, towing pennant and chafing gear for possible deterioration;
- examination of strong points, fairleads, pedestal roller and their attachment to hull structure.

The survey is to confirm that one of the towing arrangements (aft or forward) are pre-rigged.

5.4.3.16 For vessels with single point mooring arrangements where 'SPM' notation is assigned, the following is to be carried out:

a) A thorough examination of the components of the single point mooring system (bow chain stoppers, bow fairleads, pedestal roller fairleads, winches and capstans) to verify their satisfactory condition.

b) A close-up examination of the hull structures supporting and adjacent to the components of the single point mooring system to verify that there is no deformation or fracture. Thickness determination and non-destructive tests are to be carried out if required by the Surveyor.

5.4.3.17 Engine room structure is to be examined. Particular attention being given to tank tops, shell plating in way of tank tops, brackets connecting side shell frames and tank tops and engine room bulkheads in way of tank tops and bilge wells. Where excessive areas of wastage are found, thickness measurements are to be carried out and renewals or repairs made when wastage exceeds allowable limits.

5.4.3.18 Boundaries of double bottom, deep, ballast, peak and other tanks are to be tested with a head of liquid to the top of air pipes. Boundaries of fuel oil, lub.oil and fresh water tanks are to be tested with a head of liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil, lub.oil and fresh water tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the Master stating that the pressure testing has been carried out according to the requirement with satisfactory results.

5.4.3.19 The loading instrument is to be checked for accuracy by applying test load conditions in presence of the Surveyor.

5.4.3.20 Examination of accommodation ladders, gangways and their winches are to be carried out as required for annual surveys. In addition, the accommodation ladders and gangways are to be operationally, tested with the specified maximum operation load.

The tests are to be carried out with the load applied as uniformly as possible along the length of the accommodation ladder or gangway, at an angle of inclination corresponding to the maximum bending moment on the accommodation ladder or gangway.

Accommodation ladder winch is to be operationally tested at special surveys. The brake system of the winch is to be tested for holding the maximum operational load on the ladder.

For existing installations on board ships constructed prior to 01 Jan 2010 where the maximum operational load is not known, load nominated by the shipowner or operator may be considered as the test load.

5.4.4 Thickness measurement

5.4.4.1 The minimum requirements for thickness measurement are given in Table 5.4.4.1. The surveyor may extend the thickness measurements as deemed necessary.

5.4.4.2 Where thickness measurements indicate substantial corrosion, the number of thickness measurements are to be increased to determine the extent of substantial corrosion. Table 5.4.4.2 may be used as guidance for additional measurements.

5.4.4.3 For areas in tanks where coating are found to be in a GOOD condition, the extent of thickness measurements may be specially considered, by the Surveyor.

5.4.4.4 Transverse sections are to be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

5.4.4.5 The thickness measurements are to be carried out by a qualified firm certified by IRS.

5.4.4.6 In order to ensure necessary control during the process, thickness measurements are normally to be carried out under the supervision of the Surveyor. The Surveyor has the right to re-check the measurements as deemed necessary to ensure acceptable accuracy.

5.4.4.7 A thickness measurement report is to be prepared. The report is to give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurements were carried out, type of measurement equipment, names of personnel and their qualifications and has to be signed by the operator. The Surveyor is to review the report of the final thickness measurement after repairs have been carried out and countersign the cover page.

Table 5.4.4.1 : Table of minimum requirements to thickness measurements of liquefied gas carriers

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel
2) Within the cargo area : One section of deck plating for the full beam of the ship within 0.5L amidships in way of a ballast tank, if any.	2) Within the cargo area: a) Each deck plate b) One Transverse section within 0.5L amidships in way of a ballast tank, if any	2) Within the cargo area: a) Each deck plate b) Two Transverse sections (1) c) All wind and water strakes	2) Within the cargo area: a) Each deck plate b) Three Transverse sections (1) c) All wind and water strakes d) Each bottom plate e) Duct keel or pipe tunnel plating and internals
3) Measurements of structural members subject to close-up survey according to Table 5.4.5.1, for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 5.4.5.1, for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 5.4.5.1, for general assessment and recording of corrosion pattern	3) Measurements of structural members subject to close-up survey according to Table 5.4.5.1, for general assessment and recording of corrosion pattern
	4) Selected wind and water strakes outside the cargo area	4) Selected wind and water strakes outside the cargo area	4) All wind and water strakes outside the cargo area
		5) Internals in forepeak and after peak ballast tanks	5) Internals in forepeak and afterpeak ballast tanks
			6) All exposed main deck plating outside cargo area
			7) Representative exposed superstructure deck plating (poop, bridge, forecastle deck)
			8) All keel plates, additional bottom plates in way of machinery space, aft end of tanks and cofferdams outside cargo area
			9) a) Plating of sea chests. b) Shell plating in way of overboard discharges as considered necessary by the Surveyor.
(1) at least one section is to include a ballast tank within 0.5L amidships, if any.			

Table 5.4.4.2 : Guidance for additional thickness measurements in way of substantial corrosion		
Structural Member	**Extent of Measurement**	**Pattern of Measurement**
Plating	Suspect area and adjacent plates	5 point pattern over 1 square meter
Stiffeners	Suspect area	3 measurements each in line across web and flange

5.4.5 Close-up survey requirements

5.4.5.1 The minimum requirements for Close-up Surveys at Special Survey are given in Table 5.4.5.1.

5.4.5.2 The Surveyor may extend the Close-up Survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases:

a) In particular, tanks having structural arrangements or details which have suffered defects on similar tanks or on similar ships according to available information.

b) In tanks which have structures approved with reduced scantlings due to an approved corrosion control system.

5.4.5.3 For areas in tanks where hard protective coatings are found to be in a GOOD condition, the extent of Close-up Surveys according to Table 5.4.5.1 may be specially considered by the Surveyor.

Table 5.4.5.1 : Table of minimum requirements for close-up surveys at special surveys of liquefied gas carriers

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **subsequent age > 10**
One web frame in a representative ballast tank of the topside, hopper side and double hull side type (1) One transverse bulkhead in a ballast tank (3)	All web frames in a ballast tank, which is to be a double hull side tank or a topside tank. If such tanks are not fitted, another ballast tank is to be selected (1) One web frame in each remaining ballast tank (1) One transverse bulkhead in each ballast tank (2)	All web frames in all ballast tanks (1) All transverse bulkheads in all ballast tanks (2)

(1) Complete transverse web frame including adjacent structural members.
(2) Transverse bulkhead complete, including girder system and adjacent members and adjacent longitudinal bulkhead structure.
(3) Transverse bulkhead lower part including girder system and adjacent structural members.

Notes :

a) Ballast tanks include topside, double hull side, double bottom, hopper side, or any combined arrangement of the aforementioned and peak tanks where fitted.

b) For ships having independent tanks of type C, with a midship section similar to that of a general cargo ship, the extent of close-up surveys may be specially considered by the Surveyor.

5.4.6 Cargo containment and systems

5.4.6.1 All cargo tanks are to be examined internally, also externally so far as is practicable, particular attention being paid to the plating in way of supports of securing arrangements, tower structures, seatings and pipe connections. Attention is also to be paid to the sealing arrangements in way of the deck penetrations and the antiflotation arrangements for independent tanks. The tightness of all cargo tanks is to be verified by an appropriate procedure. The ships gas detection equipment may be used for tightness test of independent tanks below deck. Provided that the structural examination is satisfactory, that the gas leakage monitoring systems have been found to be operating satisfactorily and that the voyage records have not shown any abnormal operation, cargo tanks do not require to be hydraulically tested.

5.4.6.2 Non-destructive testing is to supplement cargo tank inspection with special attention to be given to the integrity of the main structural members, tank shell and highly stressed parts, including welded connections as deemed necessary by the Surveyor. However, for Type C independent tanks non-destructive testing can not be dispensed with totally. The following items are, inter alia, considered as highly stressed parts:

- Cargo tank supports and anti-rolling/anti-pitching devices.
- Web frames or stiffening rings.
- Swash bulkhead boundaries.
- Dome and stump connections to tank shell.
- Foundations for pumps, towers, ladders etc.
- Pipe connections.
- Y-connections between tank shell and a longitudinal bulkhead of bilobe tanks.

5.4.6.3 For independent tanks of type B, the extent of non-destructive testing is to be given in a programme specially prepared for cargo tank design.

5.4.6.4 If findings of 5.4.6.1 to 5.4.6.2 or examination of voyage records do not reveal a satisfactory structural integrity of the cargo tanks, a hydraulic or hydro-pneumatic test is to be carried out. For integral tanks and for independent tanks of type A and B, the test pressure is to be atleast equal to MARVS. It is to be ensured that the maximum primary membrane stress or maximum bending stress in a primary membrane under the test conditions does not exceed 90% of the yield strength of the material (as fabricated) at the test temperature.

5.4.6.5 At every other special survey (i.e. 2nd, 4th, 6th etc.) independent cargo tanks of type C are to be either:

- hydraulically or hydro-pneumatically tested to 1.25 MARVS followed by non-destructive testing in accordance with para 5.4.6.2;

OR

- subjected to a thorough planned non-destructive testing. The testing programme is to be specially prepared for the tank design. Alternatively the items detailed in para 5.4.6.2 are to be subjected to non-destructive testing. The extent of radiographic examination with insulation removed as necessary is to be atleast 10% of the welded connections in each of the above mentioned areas, internally and externally as applicable.

5.4.6.6 Deck mounted cargo storage tanks are to be examined in the same manner as main cargo tanks.

5.4.6.7 Secondary barriers are to be examined for their effectiveness, visually whenever possible, or by means of pressure/vacuum tests on the inter-barrier spaces over a period of 36 hours. Alternative means of checking the secondary barriers will be considered. For membrane and semi-membrane tank systems, inspection and testing are to be carried out in accordance with programmes specially prepared in accordance with an approved method for the actual tank system.

For membrane containment systems, a tightness test of the primary and secondary barrier is to be carried out in accordance with the system designers procedures and acceptance criteria as approved by IRS. Low differential pressure may be used for monitoring the cargo containment system performance, however, is not considered an

acceptable test for the tightness of the secondary barrier.

For membrane containment systems with glued secondary barriers if the designer's threshold values are exceeded, an investigation is to be carried out and additional testing such as thermographic or acoustic emissions testing is to be carried out.

5.4.6.8 Where cargo containment systems have secondary barriers which cannot be examined or tested and have been approved on the basis of extensive prototype testing, the barriers will be considered to remain efficient provided a cold spot examination of the adjacent steel work is satisfactory and records of the steel work temperature readings are verified as acceptable.

5.4.6.9 Where a cargo tank or the hull structure is insulated and the insulation is accessible, the insulation should be examined externally, together with any vapour or protective barrier, and sections removed for examination, if considered necessary by the Surveyor. Special attention should be given to insulation in way of chocks, supports and keys. Portions of the insulation are also to be removed, if required by the Surveyor, to enable the condition of the plating to be ascertained. Where the insulation is not accessible, see 5.2.8.2.

5.4.6.10 Cargo tank internal pipes and fittings are to be examined, and all valves and cocks in direct communication with the interiors of the tanks are to be opened out for inspection and the connection pipes are to be examined internally, so far as is practicable.

5.4.6.11 Pressure relief valves and vacuum relief valves on cargo tanks and inter-barrier spaces are to be opened out for inspection. Pressure relief valves are subsequently to be adjusted to lift at a pressure not more than the percentage given below, above the maximum vapour pressure for which the tank have been approved:

For 0 to 1.5 bar – 10%
For 1.5 to 3.0 bar – 6%
For pressure exceeding 3.0 bar – 3%.

Relief valve harbour settings are to be checked, if applicable. Valves may be removed from the shell for the purpose of making this adjustment under pressure of air or other suitable gas. Relief valves on cargo gas and liquid pipelines are to have their pressure settings checked. The valves may be removed from the pipelines for the purpose.

If the cargo tanks are equipped with relief valves with non-metallic membranes in the main or pilot valves, then such non-metallic membranes are to be replaced. If a proper record of continuous overhaul and retesting of individually identifiable relief valves is maintained, then consideration will be given to acceptance on the basis of opening, internal examination and testing of a representative sampling of valves, including each size and type of liquefied gas or vapour relief valve in use, provided there is a logbook evidence that the remaining valves have been overhauled and tested since crediting of the previous special survey.

5.4.6.12 All cargo pumps, cargo booster pumps and cargo vapour pumps are to be opened out for examination. If requested by the Owner, these items may be examined on a Continuous Survey basis provided the interval between examination of each item does not exceed five years. Pumping systems for inter-barrier spaces are to be checked and verified to be in good working order.

5.4.6.13 Where considered necessary, insulated cargo gas and liquid pipelines are to have sections of insulation removed to ascertain the condition of the pipes. If visual examination raises doubt as to the integrity of the pipe lines, then the pipe lines are to be pressure tested to 1.25 times MARVS. Care is to be taken that in the replacement of insulation the outer vapour seal is made good.

5.4.6.14 Equipment for the production of inert gas is to be examined and shown to be operating satisfactorily within the gas specification limits. Pipelines, valves, etc., for the distribution of the inert gas are to be generally examined. Pressure vessels for the storage of inert gas are to be examined internally and externally and the securing arrangements are to be specially examined. Pressure relief valves are to be demonstrated to be in good working order. Liquid nitrogen storage vessels are to be examined, so far as is practicable, and all control equipment, alarms and safety devices are to be verified as operational.

5.4.6.15 Gas tight bulkhead shaft seals are to be opened out so that the sealing arrangements may be checked.

5.4.6.16 Sea connections associated with the cargo handling equipment are to be opened out when the ship is in dry-dock.

5.4.6.17 The arrangements for discharging the cargo overboard in an emergency are to be checked.

5.4.6.18 Special Survey I (Ships five years old)- reliquefaction / refrigeration equipment

a) Each reciprocating compressor is to be opened out. Cylinder bores, pistons, piston rods, connecting rods, valves and seats, glands, relief devices, suction filters and lubricating arrangements are to be examined. Crankshafts are to be examined but crankcase glands and the lower half of main bearings need not be exposed if the Surveyor is satisfied with the alignment and wear.

b) Where other than reciprocating type compressors are fitted, or where there is a program of replacement instead of surveys on board, alternative survey arrangements will be considered. Each case will be given individual consideration.

c) The water end covers of condensers are to be removed for examination of the tubes, tube plates and covers.

d) Refrigerant condenser cooling water pumps, including standby pump(s) which may be used on other services, are to be opened out for examination.

e) Where a pressure vessel is insulated, sufficient insulation is to be removed, especially in way of connections and supports, to enable the vessel's condition to be ascertained.

f) Insulated pipes are to have sufficient insulation removed to enable their condition to be ascertained. Vapour seals are to be specially examined for condition.

g) The Surveyor is to satisfy himself that all pressure relief valves and/or safety discs throughout the system are in good order. No attempt, however, is to be made to test primary refrigerant pressure relief valves on board ship.

h) The items covered by (a) to (d) may, at the request of the Owner, be examined on a Continuous Survey basis provided the interval between examination of each item does not exceed five years.

5.4.6.19 Special Survey I (ships five years old)- Methane burning equipment

a) Where methane is used as fuel for main propulsion purposes, the associated compressors and heat exchangers are to be opened out and examined as for reliquefaction/refrigeration equipment. The steam side of steam heaters is to be hydraulically tested to 1.5 times the design pressure.

b) Methane gas pipe trunks or casings are to be generally examined and the exhaust or inerting arrangements for these trunks are to be verified.

c) All alarms associated with the methane burning systems are to be verified.

5.4.6.20 Special Survey II and Special Surveys thereafter (ships 10 years old and over)

a) The requirements of 5.4.6.1 to 5.4.6.19 are to be complied with.

b) Water cooled condensers in which the primary refrigerant is in contact with the shell are to have the end covers removed and the shell pneumatically tested to a pressure equal to the designed working pressure.

c) All other pressure vessels in the reliquefaction/refrigeration system, methane burning system and other handling systems are to be pneumatically tested to a pressure equal to the designed working pressure.

d) Liquid cargo pipes are to be tested by approved means, to a pressure equal to 1.25 times the working pressure. Alternatively, selected representative lengths may be removed for internal examination and hydraulic testing.

e) At Special Survey III and at each alternate Special Survey thereafter, plating of independent cargo tanks, and also the hull structural plating surrounding cargo tanks, where it is insulated on one side, is to be measured for thickness by a non-destructive method.

5.4.7 Watertight Cable Transits

5.4.7.1 The requirements for Special Survey may be undertaken by the attending Surveyor or by a firm approved as a service supplier according to Pt.1 Ch.1, Sec 1.9.

5.4.7.2 All transits are to be examined to confirm their satisfactory condition and the Cable Transit Seal System Register is to be reviewed to confirm that it is being maintained. The Special Survey is to be recorded in the Register, in which a single record entry will be sufficient to record the survey of all transits.

5.4.7.3 From review of the Register, where there are records entered since the last special survey of any disruption to the cable transits or installation of new cable transits (except which are reviewed and examined at previous annual surveys), the satisfactory condition of those transits is to be confirmed by the attending Surveyor by review of records and examination of the transits; the results are to be recorded in the Register against each of those cable transits.

5.4.7.4 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register inorder to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

5.5 Special Surveys - Machinery

5.5.1 Requirements for examination of machinery and systems are given in Sec.8.

5.6 Additional survey requirements for Liquefied Gas Carriers with RV(LNG) Notation

5.6.1 Application

5.6.1.1 The requirements of this subsection are applicable to liquefied gas carriers assigned with additional class notation **RV(LNG)** *(Refer Part 5, Chapter 38)*

5.6.2 Annual Survey

5.6.2.1 Annual survey of the regasification system is to be carried out during unloading operation with the system in operation.

5.6.2.2 Logbooks and operating records pertaining to the regasification system (since the last survey) are to be examined in order to establish whether certain parts have shown any irregularities in operation.

5.6.2.3 The annual survey is to include the following:

5.6.2.3.1 Vessel Arrangement and Structure

a) General examination of the areas dedicated to regasification equipment

b) Examination of closing devices of air intakes and openings into dedicated spaces for regasification, as applicable

c) Examination of drip trays or insulation for deck protection and recesses against cargo leakage in regasification area

d) Confirmation that relevant shipboard instructions and procedures of the regasification installation are available

5.6.2.3.2 Systems and Equipment

a) External examination of vaporizers and their fittings

b) External examination of heat exchangers

c) General examination of high pressure pumps as far as practicable (including gland leakage and vibration indications)

d) Examination of the suction drum(s) and their sealing arrangements, as applicable

e) General examination of cargo and process piping, including the expansion arrangements, insulation from the hull structure, pressure control valves, inlet/ outlet valves, drainage arrangements and high pressure manifolds

f) Confirmation that all accessible cargo piping systems and components (suction drum(s), vaporizers, high-pressure pumps) are electrically bonded to the hull. Bonding straps, where fitted are to be examined.

5.6.2.3.3 Safety System

a) Examination of pressure gauges, control valves, metering unit, temperature and vibration indicating equipment. Particular attention is to be given to the instrumentation of high-pressure pumps and safety devices.

b) Confirmation that the certificate for the relief valves opening/closing pressures is on board. Examination of the relief valves, their sealing and associated safety systems and alarms.

c) Examination and testing of fixed gas detection equipment, as appropriate.

d) Examination of the logbooks for confirmation that the shutdown systems (emergency shutdown and process shutdown) have been tested and are satisfactory.

e) Examination of operational readiness and maintenance of firefighting system and confirmation that means of operation is clearly marked.

5.6.2.3.4 Control, Alarm and Monitoring

a) Confirmation that the instrumentation fitted on regasification equipment and in contact with gas and associated alarms are satisfactory.

b) Confirmation that the regasification control station is in satisfactory working condition.

5.6.3 Intermediate Surveys

5.6.3.1 In addition to the requirements of Annual surveys, the intermediate survey is to include the following:

5.6.3.1.1 Vessel Arrangement and Structure

a) Examination, as far as practicable, of stripping and vent piping system and high pressure manifolds of cargo and process. If upon examination there is any doubt as to the condition of the piping, pressure testing, thickness measurement or both may be required.

b) Examination of vent line drainage arrangement

5.6.3.1.2 Systems and Equipment

a) Review of the maintenance records of vaporizers.

b) If found necessary by the surveyor, thickness measurement, pressure test and/or opening up and internal examination of equipment may be required. Particular attention is to be given to rupture device, if fitted.

c) Examination of lifting arrangements of high pressure pumps.

d) Thickness measurement of suction drum(s) if deemed necessary by the surveyor.

e) Examination of electrical equipment

5.6.3.1.3 Safety Systems

a) Examination of pressure gauges, control valves, metering unit, liquid level gauges, temperature and vibration indicating equipment and associated alarms. Particular attention is to be given to high-pressure pumps instrumentation and safety devices. Instrumentation is to be tested by changing pressure, level and temperature as far as practicable and comparing with test instruments. Simulated tests may be accepted for sensors, which are not accessible. The test is to include alarm and safety functions.

b) Fixed firefighting system is to be tested.

5.6.3.1.4 Control, Alarm and Monitoring

a) Operational testing of the emergency shutdown system.

5.6.4 Special Survey

5.6.4.1 In addition to the requirements of Annual surveys and Intermediate Surveys, the special survey is to include the following:

5.6.4.1.1 Systems and Equipment

5.6.4.1.1.1 Suction Drum(s)

a) Suction drums are to be presented in clean and examined internally.

b) When accessible, the outer surface of uninsulated suction drum or the outer surface of regasification unit insulation together with any vapour or protective barrier is to be examined. Particular attention is to be given to the stocking unit and insulation in way of supports. Removal of insulation, in part or in entirety, may be required in order to verify the condition of the unit or the insulation itself if deemed necessary by the surveyor.

c) Thickness measurements or non-destructive testing may be required if deemed necessary by the surveyor

d) The tightness of suction drums is to be verified by an appropriate procedure.

e) Where the results of the above examination or the examination of the operational records raise doubts as to the structural integrity of the unit, a hydraulic or hydro pneumatic test is to be carried out. The test pressure is not to be less than 1.25 times the MARVS.

f) When the ship is 10 years old, at every alternate special survey, suction drum is to be either

- Hydraulically tested and thereafter nondestructively tested or,
- Subjected to thorough non-destructive testing as per 5.4.6.5

g) Suction drum supporting structures are to be visually examined, as far as practicable.

h) Fittings, valves and safety devices of suction drums are to be opened up, as deemed necessary by the surveyor, for visual examination and pressure tested as appropriate.

5.6.4.1.1.2 Gas handling equipment and Equipment in hazardous areas

a) Piping for cargo and process, stripping, venting systems high-pressure manifolds are to be examined. Insulation is to be removed as deemed necessary to ascertain the condition of the pipes. If the visual examination raises doubt as to the integrity of the pipelines, a pressure test at 1.25 times the MARVS for the pipeline is to be carried out. After reassembly, the complete piping systems are to be tested for leaks.

b) Maintenance records and logbooks of high-pressure pumps including starting pumps are to be made available to the Surveyor. High-pressure pumps are to be opened up for visual examination and their parts and components pressure tested, as deemed necessary by the Surveyor. The lifting device is to be tested.

c) The bottom of the high-pressure pump recess and vaporizer spaces are to be presented clean for the examination of gutters and stripping devices.

d) The pressure relief valves are to be opened for examination, adjusted, function tested and sealed. The valves including inlet/outlet valves, pressure control valves and discharge valves are to be tested for proper operation.

e) The maintenance records of vaporizers are to be reviewed. The Surveyor may require as considered necessary:

- external and internal examination of vaporizers which are to be presented clean
- opening up of their fittings, valves and safety devices

for visual examination and pressure tests

- thickness measurements and/or pressure test
- Confirmation that cargo pipes and regasification components are electrically bonded to the hull.

Particular attention is to be taken in way of the vaporizer rupture device if fitted.

f) Vaporizers, heat exchangers are to be opened up for visual examination as deemed necessary by the Surveyor and their parts and components are to be pressure tested, as appropriate. A working test is also to be carried out, including testing of alarms and safety devices.

5.6.4.1.2 Safety Systems

a) Examination of pressure gauges, control valves, metering unit, liquid level gauges, temperature and vibration indicating equipment, and associated alarms. Particular attention is to be given to high-pressure pumps instrumentation and safety devices. Instrumentation is to be tested by changing pressure, level and temperature as applicable and comparing with test instruments. Simulated tests may be accepted for sensors, which are not accessible. The test is to include alarms and safety functions.

b) For water spraying system:

- the associated pumps are to be opened up and examined at the Surveyor's discretion
- a working test of the system is to be carried out.

5.6.4.1.3 Control, Alarm and Monitoring

a) The shutdown systems are to be functionally tested.

Section 6

Surveys - Other Ship Types

6.1 Scope

6.1.1 This section gives the requirements for periodical surveys of all ship types other than bulk carriers, oil tankers, chemical tankers and liquefied gas carriers which are covered in Sections 2 to 5 of this chapter. The requirements for general dry cargo ships defined in 1.2.17, are also included.

6.1.2 For General Dry Cargo Ships with hybrid cargo hold arrangements, e.g. with some cargo holds of single-side skin and others of double-side skin, the requirements specified for general dry cargo ships are to be applied only to structure in way of the single-side skin cargo hold region. Special consideration may also be given to ships that are of double side-skin construction but with single skin in way of several frame spaces e.g. in way of a cargo hold entrance or in way of fore body hull form at the forward end of the foremost cargo hold.

6.2 Annual surveys

6.2.1 General

6.2.1.1 Annual Surveys are to be carried out within 3 months before or after the anniversary date each year. (Also refer to Note 9 of Sec.1, Table 1.1.1). These should be held concurrently with statutory annual or other relevant statutory Surveys, where practicable.

6.2.1.2 At Annual Surveys, the Surveyor is to examine the hull and machinery, so far as necessary and practicable, in order to be satisfied as to their general condition.

6.2.1.3 It is to be confirmed that no new installation of material containing asbestos was carried out since last survey.

6.2.1.4 Where applicable, examination of alternative design and arrangement for machinery or electrical installations or fire safety, in accordance with the test, inspection and maintenance requirements, if any,

specified in the approved documentation is to be carried out.

6.2.2 Hull - General

6.2.2.1 The survey is to consist of an examination for the purpose of ensuring, as far as practicable, that the hull, equipment, hatch coamings, hatch covers including their securing arrangement, other closing appliances and related piping are maintained in satisfactory/efficient condition.

a) Weather deck, shipside plating above water line.

b) Hatchways on freeboard and superstructure decks; exposed casings; skylights and fiddley openings; deck houses; companionways and superstructure bulkheads; side, bow and stern doors; side scuttles and dead lights; flush deck scuttles; ash shoots and other openings.

c) Weld connection between air pipes and deck plating, air pipe heads on exposed decks (external examination), flame screens on vents to all bunker tanks; ventilators and closing devices.

d) Scuppers and sanitary discharges as far as practicable together with valves and their controls.

e) Guard rails, bulwarks, freeing ports, gangways, walkways and life lines, fittings and appliances for timber deck cargo.

f) Watertight bulkheads and their penetrations as far as practicable.

6.2.2.2 Cargo hatch covers and coamings are to be examined to ensure that no alterations have been made to the approved arrangements :

a) Mechanically operated hatch covers are to be examined for satisfactory condition of:

- hatch covers;
- tightness devices of longitudinal, transverse and intermediate cross junctions, gaskets, gasket lips, compression bars, drainage channels;
- clamping devices, retaining bars, cleating;
- chain or rope pulleys;
- guides;
- guide rails and track wheels;
- stoppers, etc.;
- wires, chains, gypsies, tensioning devices;
- hydraulic system essential to closing and securing;
- safety locks and retaining devices.

b) Cargo hatch covers of the portable type (i.e. wood or steel pontoons) are to be examined to confirm the satisfactory condition of:

- wooden covers and portable beams, carriers or sockets for the portable beams and their securing devices;
- steel pontoons;
- tarpaulins;
- cleats, battens and wedges;
- hatch securing bars and their securing devices;
- loading pads/bars and the side plate edge;
- guide plates and chocks;
- compression bars, drainage channels and drain pipes, if any.

c) Checking the satisfactory condition of hatch coamings plating and their stiffeners.

d) Random checking of the satisfactory operation of mechanically operated hatch covers :

- stowage and securing in open position;
- proper fit, locking and efficiency of sealing in closed condition;
- operational testing of hydraulic and power components, wires, chains and link drives.

In the case of hatch covers / steel pontoons of general dry cargo ships, a close-up survey of hatch cover plating is to be carried out.

6.2.2.3 All watertight doors in watertight bulkheads, to be examined and tested (locally and remotely) as far as practicable.

6.2.2.4 Anchoring and mooring equipment is to be examined as far as is practicable. It is to be confirmed that the towing and mooring equipment is properly marked with any restriction associated with its safe operation.

6.2.2.5 Where applicable Surveyor should satisfy himself regarding the freeboard marks on the ship's side.

6.2.2.6 Suspect areas identified at previous special or intermediate surveys are to be close-up surveyed. Thickness measurements are to be taken in the area of substantial corrosion identified at previous surveys.

6.2.2.7 Examination of Ballast Tanks when required as a consequence of the results of the Special Survey or Intermediate Survey (See 6.4.3.1, 6.4.3.2 and 6.3.2, 6.3.3 respectively). When extensive corrosion is found, thickness measurement is to be carried out. If the results of these thickness measurements indicate substantial corrosion then the extent of thickness measurements are to be increased to determine the extent of areas of substantial corrosion in accordance with Table 6.4.8.2.

6.2.2.8 The Surveyor is to confirm that, where required, an approved loading instrument together with its operation manual are available on board, (See Pt.3, Ch.5). It is to be verified by the Surveyor that the loading instrument is checked for accuracy at regular intervals by the ship's staff by applying test loading conditions.

6.2.2.9 Accommodation ladders are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) steps;
b) platforms;
c) all support points such as pivots, rollers, etc.;
d) all suspension points such as lugs, brackets, etc.;
e) stanchions, rigid handrails, hand ropes and turntables;
f) davit structure, wire and sheaves, etc.

6.2.2.10 Gangways are to be examined at annual surveys. Satisfactory condition of the following items is to be checked, in particular :

a) treads;
b) side stringers, cross-members, decking, deck plates, etc.;
c) all support points such as wheel, roller, etc.;
d) stanchions, rigid handrails, hand ropes.

6.2.2.11 Winches of accommodation ladders and gangways are to be examined to verify the satisfactory condition of the following items:

a) brake mechanism including condition of brake pads and band brake, if fitted;
b) remote control system, and
c) power supply system for electric motor.

6.2.2.12 Davits and fittings on the ship’s deck associated with accommodation ladders and gangways are to be examined for satisfactory condition at annual surveys. Fittings or structures for means of access to deck such as handholds in a gateway or bulwark ladder and stanchions are also to be examined.

6.2.2.13 The maintenance and inspection records of accommodation ladders and gangways are to be verified. It is to be confirmed that supporting wires are being renewed at intervals not exceeding 5 years.

6.2.2.14 When examining internal spaces, as far as practicable, the permanent means of access where appropriate, are to be verified that they remain in good condition.

6.2.2.15 For vessels subject to IMO PSPC (See Pt.3, Ch.2, 3.6) it is to be confirmed that the maintenance, repair and partial re-coating of dedicated ballast tanks, as appropriate, are recorded in the coating technical file.

6.2.2.16 Watertight cable transit seal systems register is to be reviewed to confirm that it is being maintained and as far as practicable the transits are to be examined to confirm their satisfactory condition. Where there are records entered since last annual survey of any disruption to the cable transits or installation of new cable transits, the satisfactory condition of

those transits is to be confirmed by review of records and, if deemed necessary, by examination. The results are to be recorded in the Register against the specific cable transit.

6.2.2.17 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

6.2.3 Examination of cargo holds of general dry cargo ships.

6.2.3.1 For cargo ships of 10 – 15 years of age:

a) Overall survey of one forward and one after cargo hold and their associated tween deck spaces.

b) When considered necessary by the Surveyor or where extensive corrosion exists, thickness measurement is to be carried out. If the results of these thickness measurements indicate substantial corro-sion, then the extent of thickness measurements is to be increased to determine the extent of areas of substantial corrosion in accordance with Table 6.4.8.2.

6.2.3.2 For cargo ships over 15 years of age:

a) Overall survey of all cargo holds and tween deck spaces.

b) In a forward lower cargo hold and one other selected lower cargo hold:

- Close-up examination of sufficient extent, minimum 25% of frames, to establish the condition of the lower region of the shell frames including approx. lower one third length of side frame at side shell, side frame end attachment and the adjacent shell plating.

- Where this level of survey reveals the need for remedial measures, the survey is to be extended to include a close-up survey of all of the shell frames and adjacent shell plating of those cargo holds and associated tween deck spaces (as applicable) as well as a close-up survey of sufficient extent of all remaining cargo holds and tween deck spaces (as applicable).

c) When considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement is to be carried out. If the results of these thickness measurements indicate substantial corro-sion, then the number of thickness measurements are to be increased to determine the extent of substantial corrosion in accordance with Table 6.4.8.2.

d) Where the protective coating in cargo holds, as applicable, is found to be in a GOOD condition, the extent of close-up surveys may be specially considered.

e) All piping and penetrations in cargo holds, including overboard piping are to be examined.

6.2.4 Machinery and systems

6.2.4.1 A general examination of the machinery, boilers, all pressurised systems (steam, pneumatic, hydraulic) and their associated fittings, propulsion system and auxiliary machinery to see whether they are being properly maintained and with particular attention to the fire and explosion hazards.

6.2.4.2 Confirmation that machinery, boilers and other pressure vessels, associated piping systems and fittings are so installed and protected as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards.

6.2.4.3 Confirmation that Periodical Surveys of boilers and other pressure vessels have been carried out as required by the Rules and the safety devices have been tested.

6.2.4.4 Confirmation that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative.

6.2.4.5 Confirmation that means are provided so that machinery can be brought into operation from the dead ship condition without external aid.

6.2.4.6 All main and auxiliary steering arrangements and their associated equipment and control systems are to be examined and

tested. Where applicable, Surveyors are to verify that log entries have been made in accordance with statutory requirements.

6.2.4.7 Steering chains are to be cleaned for ascertaining wear and tear and lengths of chain worn in mean diameter by more than 12 per cent of the original rule diameter are to be renewed.

6.2.4.8 All the means of communication between the navigating bridge and the machinery control positions, as well as the bridge and the main alternative steering position, if fitted, are to be tested. It is to be confirmed that means of indicating the angular position of the rudder are operating satisfactorily.

6.2.4.9 Confirmation that with ships having emergency steering positions there are means of relaying heading information and, when appropriate, supplying visual compass readings to the emergency steering positions.

6.2.4.10 Confirmation that various alarms required for hydraulic power operated, electric and electro-hydraulic steering gears are, operating satisfactorily and that the recharging arrangements for hydraulic power operated steering gears are being maintained.

6.2.4.11 Examining the means for the operation of the main and auxiliary machinery essential for propulsion and the safety of the ship, including when applicable, the means of remotely controlling the propulsion machinery from the navigating bridge and the arrangements to operate the main and other machinery from a machinery control room.

6.2.4.12 Confirmation that the engine room telegraph, the second means of communication between the navigation bridge and the machinery space and the means of communication with any other positions from which the engines are controlled are operating satisfactorily.

6.2.4.13 Examination and testing of the general emergency alarm system and confirmation that the engineer's alarm is clearly audible in the engineer's accommodation.

6.2.4.14 It is to be confirmed that ballasting and de-ballasting arrangements are satisfactory. The bilge pumping systems and bilge wells including operation of each bilge pump, extended spindles and level alarms, where fitted, are to be examined as far as is practicable. It is also to be confirmed that bilge pumping system for each watertight compartment is satisfactory.

It is also to be confirmed that drainage from enclosed cargo spaces situated on freeboard deck is satisfactory.

The drainage facilities are to be examined visually for blockage or other damage and the provision of means to prevent blockage of drainage arrangements are to be confirmed, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used

6.2.4.15 Examining visually the condition of non-metallic expansion joints where fitted in piping systems which penetrate the ship's side, with both the penetration and the expansion joint located below the deepest load waterline and checking the service record.

6.2.4.16 General examination visually and in operation, as feasible, of the main electrical machinery, the emergency sources of electrical power, the switch gear, other electrical equipment including the lighting system is to be carried out.

6.2.4.17 Confirmation as far as practicable, the operation of the emergency source(s) of electrical power, including their starting arrangement, the systems supplied, and when appropriate, their automatic operation.

6.2.4.18 Examining in general, that the precautions provided against shock, fire and other hazards of electrical origin are being maintained.

6.2.4.19 General Examination of automation equipment is to be carried out. Satisfactory operation of safety devices, bilge level detection and alarm systems and control systems is to be verified.

6.2.4.20 For single hold cargo ships which require fitment of hold water level detectors as per Pt.4, Ch.3, Sec.3.6, the annual survey is to include an examination and a test of the water ingress detection system and their alarms.

6.2.4.21 Confirmation that machinery space ventilation systems are in good working condition.

6.2.4.22 For Ships constructed before 01 July 2017 and fitted with harmonic filters; following are to be verified at each survey:

a) Records of annual measurement of harmonic distortion levels of main bus bar under seagoing conditions and log entries showing which equipment was running and / or filters in service at the time of measurement. Harmonic distortion readings are to be carried out when the greatest amount of distortion is indicated by the measuring equipment.

b) Records of measurement of distortion levels, after any modification to the ship's electrical distribution system.

6.2.4.23 For Ships constructed on or after 01 July 2017 and fitted with harmonic filters; or for ships retrofitted with harmonic filters, on or after 01 July 2017; following are to be verified at each survey:

a) Engine log book containing distortion monitoring readings/ automatic records of same

b) Documentation of any approved relaxation on allowable distortion limits.

6.2.4.24 For ships contracted for construction on or after 01 Jan 2017 following are to be verified in emergency generator rooms at each survey:

a) Hand operated ventilation louvres, closing appliances, if fitted, are kept open and corresponding instruction plates are visible near the louvres.

b) Power operated ventilation louvres and closing appliances, if fitted, are fail-to-open type.

c) It is to be possible to close ventilation openings by a manual operation from a clearly marked safe position outside the space where the closing operation can be easily confirmed. The louvre status (open/closed) is to be indicated at this position.

6.2.4.25 For vessels fitted with ICCP system, proper functioning of the system is to be checked and records of system operation are to be verified. It is to be verified that all maintenance as per manufacturer's instructions are carried out.

6.2.5 Fire protection, detection and extinction

6.2.5.1 The arrangements for fire protection, detection and extinction are to be examined and are to include confirmation that no changes have been made in the structural fire protection. Following are to be examined / verified:

a) verification that fire control plans are properly posted;

b) examination as far as possible and testing as feasible, any fire detection and alarm system and any sample extraction smoke detection system;

c) examination of the fire main system and verification that each fire pump including the emergency fire pump can be operated separately so that the two required powerful jets of water can be produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main;

d) verification that fire hoses, nozzles, applicators and spanners are in good working condition and situated at their respective locations;

e) examination of fire-fighting system (including fixed fire-fighting system of the specific machinery, closed vehicle, special category and ro-ro spaces, as appropriate) controls, piping, instructions and marking, checking for evidence of proper maintenance and servicing including date of last systems tests;

f) verification that all semi-portable and portable fire extinguishers are in their stowed positions, checking for evidence of proper maintenance and servicing, conducting random check for evidence of discharged containers;

g) verification, as far as practicable, that the remote controls for stopping fans and machinery in machinery spaces are in working order. Examination of the arrangements for oil fuel, lubricating oil and other flammable oils and testing the remote closing of valves for oil fuel, lubricating oil and other flammable oils and , the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils;

h) examination of the closing arrangements of ventilators, funnel annular spaces, skylights, doorways and tunnel, where applicable;

i) confirmation that the fire fighters' outfits including its self-contained compressed air breathing apparatus and emergency escape breathing devices (EEBDs) are complete and in good condition and that the cylinders, including the spare cylinders, of any required self-contained breathing apparatus are suitably charged and that on board means of recharging breathing apparatus cylinders used during drills or a suitable number of spare cylinders to replace those used are provided, and provision of two-way portable radiotelephone apparatus of an explosion-proof type or intrinsically safe;

j) examination of any manual and automatic fire doors and proving their operations;

k) examination of the fire-extinguishing systems for spaces containing paint and/or flammable liquids and deep fat cooking equipment in accommodation and service spaces;

l) examination of the fire safety requirements of helicopter facilities, including foam firefighting appliances ;

m) examination of the fire protection arrangement in cargo, vehicle and ro-ro spaces, including the fire safety arrangements for vehicle carriers carrying motor vehicles with compressed hydrogen or natural gas in their tanks for their own propulsion as cargo, as applicable and confirmation, as far as practicable and as appropriate, the operation of the means of control provided for closing the various openings;

n) examination and testing the portable gas detectors suitable for the detection of the gas fuel, for vehicle carriers carrying motor vehicles with compressed hydrogen or natural gas in their tanks for their own propulsion as cargo;

o) Examination, when appropriate, of the special arrangements for carrying dangerous goods, including checking the electrical equipment and wiring, the ventilation, protective clothing and portable appliances and testing of water supply, bilge pumping and any water spray system.

p) Examination of the fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space.

q) Testing the fire dampers of ventilation ducts and the means of closing the main inlets and outlets of all ventilation systems and testing the means of stopping power ventilation systems from outside the space served.

Surveys carried out by the National Authority of the country in which the ship is registered would normally be accepted as meeting these requirements, at the discretion of the Surveyor.

6.2.5.2 Confirmation that the means of escape from accommodation, machinery spaces and other spaces are satisfactory.

6.2.5.3 Examination of the arrangements for gaseous fuel for domestic purposes.

6.2.5.4 For ships designed to carry containers on or above weather deck, as applicable, examining the water mist lance, and as appropriate, the mobile water monitors and all necessary hoses, fittings and required fixing hardware.

6.2.6 Shell and inner doors of ro-ro ships

6.2.6.1 The survey is to consist of an examination to verify, as far as is practicable, that the bow, inner, side shell and stern doors are maintained in a satisfactory condition.

6.2.6.2 Confirmation is to be obtained that no unapproved changes have been made to the bow, inner, side shell and stern doors since the last survey.

6.2.6.3 Documents

If an Operating and Maintenance Manual (OMM) is required, it is to be verified that a copy approved by IRS is on board and any possible modifications are included. It is to be

verified that documented operating procedures for closing and securing doors are kept on board and posted at an appropriate place. The OMM is to be examined with special attention to the register of inspections and its contents as a basis for the survey.

6.2.6.4 Structural examination

Bow, inner, side shell and stern doors are to be examined with particular attention paid to:

- Structural arrangement of doors including plating, secondary stiffeners, primary structure, hinging arms and welding;
- Shell structure surrounding the opening of the doors and the securing, supporting and locking devices (refer Pt.3, Ch.12, Sections 5 and 6) including shell plating, secondary stiffeners, primary structure and welding;
- Hinges and bearings, thrust bearings;
- Hull and door side supports for securing, supporting and locking devices;
- Close-up survey of securing, supporting and locking devices including welding, refer to Table 6.2.6.4 for minimum requirements.

Whenever a crack is found, an examination with NDT is to be carried out in the surrounding areas and for similar items as considered necessary by the Surveyor.

Table 6.2.6.4 : Minimum requirements for close-up surveys of doors, locking, securing and supporting devices and fittings

List of the devices and fittings and associated welding to be subject to close-up survey by the attending Surveyor:	
i)	Cylinder securing pins, supporting brackets, back-up brackets (where fitted) and their welding connections;
ii)	Hinge pins, supporting brackets, back-up brackets (where fitted) and their welded connections;
iii)	Locking hooks, securing pins, supporting brackets, back-up brackets (where fitted) and their welded connections;
iv)	Locking pins, supporting brackets, back-up brackets (where fitted) and their welded connections;
v)	Locating and stopper devices and their welded connections.

6.2.6.5 Measurement of clearances

Clearances of hinges, bearings and thrust bearings are to be taken, where no dismantling is required. Where the function test is not satisfactory, dismantling may be required to measure the clearances. If dismantling is carried out, a visual examination of hinge pins and bearings together with NDT of the hinge pin is to be carried out. Clearances of securing, supporting and locking devices are to be measured, where indicated in the OMM

6.2.6.6 Sealing arrangement

An examination of packing material / rubber gaskets and retaining bars or channels, including welding is to be carried out.

6.2.6.7 Drainage arrangement

An examination of drainage arrangement, including bilge wells and drain pipes is to be carried out.

6.2.6.8 Function test of doors

Checking of the satisfactory operation of the bow, inner, side shell and stern doors during a complete opening and closing operation is to be carried out, as applicable, including:

- Proper working of the hinging arms and hinges;
- Proper engagement of the thrust bearings;
- Device for locking the door in the open position;
- Securing, supporting and locking devices;

- Proper sequence of the interlock system for the opening / closing system and the securing and locking devices;

- Mechanical lock of the securing devices;

- Proper locking of hydraulic securing devices in the event of a loss of the hydraulic fluid, according to the procedure provided by the OMM;

- Correct indication of open / closed position of doors and securing / locking devices at navigation bridge and other control stations;

- Isolating of the hydraulic securing / locking devices from other hydraulic systems;

- Confirmation that the operating panels are inaccessible to unauthorized persons;

- Verification that a notice plate giving instructions to the effect that all securing devices are to be closed and locked before leaving harbour is placed at each operating panel and supplemented by warning indicator lights;

- Examination of electrical equipment for opening, closing and securing the doors.

6.2.6.9 Function test of the indicator system

Checking of the satisfactory operation of the indicator system, where fitted, is to be carried out, as applicable, including:

- Proper visible indication and audible alarm on the navigation bridge panel, according to the selected function “harbour / sea voyage” and on the operating panel;

- Lamp test function on both panels;

- Verification that it is not possible to turn off the indicator light on both panels;

- Verification of failsafe performance, according to the procedure provided by the OMM;

- Confirmation that power supply for indicator system is supplied by the emergency source or other secure power_supply and independent of the power supply for operating the doors;

- Verification of proper condition of sensors and protection from water, ice formation and mechanical damage.

6.2.6.10 Test of water leakage detection system (where fitted)

The water leakage detection system is to be tested including proper audible alarm on the navigation bridge panel and on the engine control room panel, according to the procedure provided by the OMM.

6.2.6.11 Test of television surveillance system (where fitted)

The television surveillance system is to be tested including proper indication on the navigation bridge monitor and on the engine control room monitor.

6.2.6.12 Tightness test

A hose test or equivalent is to be carried out. If the visual examination and function test have shown satisfactory results, the tightness test of shell doors on ro-ro cargo ships need not be carried out unless considered necessary by the attending surveyor.

6.2.6.13 NDT and thickness measurements

When considered necessary by the Surveyor, NDT and thickness measurements may be required after visual examination and function test.

6.2.7 Container ships with container lashing software

6.2.7.1 For Container ships provided with approved lashing software and assigned CLS notation, confirmation that the approved version of the software is installed on-board and that user manual is available.

6.3 Intermediate surveys

6.3.1 General

6.3.1.1 Intermediate surveys are to be carried out at or between the second or third Annual Survey. (Also refer to Note 9 of Sec.1, Table 1.1.1).

Those items which are additional to the requirements of annual survey may be examined at or between the second and third annual survey.

6.3.1.2 The following requirements are applicable for vessels over five years of age.

For vessels below 5 years of age additional examination over and above the requirements of Annual survey may be required at the discretion of the Surveyors.

6.3.1.3 Prior to the commencement of any part of the intermediate survey, a survey planning meeting is to be held between the attending Surveyor(s), the owner's representative in attendance and where involved, the thickness measurement company representative and the master of ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

6.3.1.4 Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

6.3.2 Vessels of age between 5 and 10 years

6.3.2.1 Ballast tanks

a) An overall survey of representative ballast tanks is to be carried out. When extensive corrosion is found, thickness measurements are to be carried out. If such examination reveals no visible structural defects, the examination may be limited to a verification that the corrosion prevention system remain efficient.

b) Where POOR coating condition, corrosion or other defects are found in ballast tanks or where hard protective coating was not applied from the time of construction, the examination is to be extended to other ballast tanks of the same type.

c) For ballast tanks, other than double bottom tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question are to be examined at subsequent annual surveys.

d) For double bottom ballast tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question may be examined at subsequent Annual surveys.

6.3.2.2 Cargo hold of general dry cargo ships

a) An overall survey of one forward and one after cargo hold and their associated tween deck spaces.

b) Areas found suspect at previous surveys are to be surveyed in accordance with 6.2.2.6.

6.3.3 Vessels of age between 10 and 15 years

6.3.3.1 Ballast tanks:

a) An overall survey of all ballast tanks is to be carried out. When extensive corrosion is found, thickness measurements are to be carried out. If such examination reveals no visible structural defects, the examination may be limited to a verification that the corrosion prevention system remain efficient.

b) For ballast tanks, other than double bottom tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question are to be examined at subsequent annual surveys.

c) For double bottom ballast tanks, where a protective coating is found to be in POOR condition and is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating was not applied from the time of construction, the tank(s) in question may be examined at subsequent Annual surveys.

6.3.3.2 Cargo holds of general dry cargo ships:

a) An overall survey of all cargo holds and tween deck spaces.

b) Areas found suspect at previous surveys are to be surveyed in accordance with 6.2.2.6.

c) When considered necessary by the Surveyor or where extensive corrosion exists, thickness measurement is to be carried out. If the results of these thickness measurements indicate substantial corrosion, then the extent of thickness measurements are to be increased to determine the extent of areas of substantial corrosion in accordance with Table 6.4.8.2.

6.3.3.3 In the case of ships other than those engaged in the carriage of dry cargoes only, an internal examination of selected cargo spaces is to be carried out.

6.3.4 Vessels of age more than 15 years

a) For vessels other than general dry cargo ships, in addition to the requirements given in 6.3.3, an internal examination of selected cargo holds is to be carried out.

b) For general dry cargo ships, the requirements of the intermediate survey is to be to the same extent as the previous special survey as required in 6.4 for hull structure and piping systems in way of the cargo holds, cofferdams, pipe tunnels, void spaces and fuel oil tanks in the cargo area and all ballast tanks. However, tank testing specified in 6.4.6, survey of automatic air pipe heads specified in 6.4.4.7 and internal examination of fuel oil, lub.oil and freshwater tanks specified in Table 6.4.4.1 need not be carried out unless deemed necessary by the Surveyor. In water survey complying with the requirements of 7.2 may be accepted in lieu of docking survey required by 6.4.1.11. Thickness measurement is to be carried out for items 1 to 4 of Table 6.4.8.1b except for item 2d).

In lieu of the application of 6.4.1.4, the intermediate survey may be commenced at the second annual survey and be progressed with a view to completion at the third annual survey.

6.4 Special surveys - Hull

6.4.1 General

6.4.1.1 All ships classed with IRS are to undergo Special Surveys at 5 yearly intervals. The first Special Survey is to be completed within 5 years from the date of the initial classification survey and thereafter 5 years from the assigned date of the previous Special Survey. However, an extension of class of 3 months maximum beyond the 5th year may be granted in exceptional circumstances in accordance with 1.11. In such cases, the next period of class will start from the expiry date of the Special Survey before extension was granted.

6.4.1.2 The interval between the Special Surveys may be reduced at the request of the parties concerned or by IRS if considered appropriate.

6.4.1.3 For surveys completed within 3 months before the expiry date of the Special Survey, the next period of class will start from the expiry date of the Special Survey. For surveys completed more than 3 months before the expiry date of the Special Survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the special survey. If the owner elects to carry out the next due special survey, the period of class will start from the survey completion date. Any requirement of the Flag Administration in this regard is also to be complied with.

6.4.1.4 The Special Survey may be commenced at the 4th Annual Survey and be progressed with a view to completion by the 5th anniversary date. When the special survey is commenced prior to the fourth annual survey, the entire survey is to be completed within 15 months if such work is to be credited to the special survey and in this case the next period of class will start from the survey completion date.

Concurrent crediting to both Intermediate survey and Special survey for surveys and thickness measurements of spaces is not acceptable.

6.4.1.5 For the purpose of special survey, results of thickness measurement carried out during or after the fourth annual survey only would be considered.

6.4.1.6 Record of Special Survey will not be assigned until the Machinery Survey has been completed or postponed in agreement with IRS.

6.4.1.7 Ships which have satisfactorily passed a Special Survey will have a record entered in the Supplement to the Register Book indicating the assigned date of Special Survey.

6.4.1.8 For vessels other than general dry cargo ships, IRS may, at the request of the Owners, accept a Special Survey of the hull on a continuous basis spread over a period of 5 years. Proposals for such continuous Surveys are to be submitted for the consideration of IRS. In general, approximately one-fifth of the Special Survey is to be completed every year. All compartments of the hull should be opened for Survey and testing in rotation such that not more than 5 years elapse between consecutive examination of each part.

The Surveyor may extend the inspection at his discretion, to other items if the inspections carried out reveal any defects.

The agreement for surveys to be carried out on a continuous survey system basis may be withdrawn at the discretion of IRS or on the request of Owners.

Ships on continuous survey system are not exempt from other periodical survey requirements. For ships more than 10 years of age, the ballast tanks are to be internally examined twice in each five-year class period, i.e. once within the scope of the intermediate survey and once within the scope of the continuous system for the hull special survey.

Ships which have completed satisfactorily the Survey of all the items on such basis will have a record entered in the supplement to the Register Of Ships indicating the date of completion of the Survey.

6.4.1.9 The special survey is to include, in addition to the requirements of the Annual Survey, examination, tests and checks of sufficient extent to ensure that the hull, equipment and related piping as required in 6.4.4.2 and 6.4.4.3 are in satisfactory condition and that the ship is fit for its intended purpose for the new period of class of five years to be assigned subject to proper maintenance and operation and the periodical surveys being carried out at the due dates.

6.4.1.10 The examinations of the hull are to be supplemented by thickness measurements. (See 6.4.8) and testing as deemed necessary, to ensure that the structural integrity remains effective. The aim of the examination is to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration, that may be present.

6.4.1.11 A Docking Survey in accordance with the requirements of Sec.7 is to be carried out as part of the Special Survey. Any remaining work in respect of the overall and close-up surveys and thickness measurements, as applicable, of the lower portions of cargo holds and ballast tanks (i.e. parts below light ballast water line) are to be completed in dry dock.

6.4.2 Preparation for survey

6.4.2.1 The ship is to be prepared for overall survey in accordance with the requirements of Table 6.4.2.1. The preparation is to be of sufficient extent to facilitate an examination to ascertain any excessive corrosion, deformation, fractures, damages and other structural deterioration.

6.4.2.2 Prior to commencement of any part of the special survey, a survey planning meeting is to be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company representative and the master of the ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

6.4.3 Tank protection

6.4.3.1 For ballast tanks, excluding double bottom tanks, where a hard protective coating is found in POOR condition and it is not renewed or where soft or semi-hard coating has been applied, or where a hard protective coating was not applied from the time of construction, tank in question is to be internally

examined at annual surveys. Thickness measurements are to be carried out as deemed necessary by the Surveyor.

6.4.3.2 For double bottom ballast tanks where a hard protective coating is found in POOR condition and it has not been renewed or where soft or semi-hard coating has been applied, or where a hard protective coating was not applied from the time of construction, the tanks in question may be examined at Annual Surveys. When considered necessary by the surveyor, thickness measurements are to be carried out.

Table 6.4.2.1 : Survey preparation

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) The holds, tween decks, deep tanks, peaks, bilges and drain wells, engine and boiler spaces, coal bunkers and other spaces are to be cleared out and cleaned as necessary for examination. Floor plates in engine and boiler spaces are to be lifted as may be necessary for examination of the structure underneath. Where necessary ceiling, lining, casings and loose insulation are to be removed as required by the Surveyor for examination of the structure. Compositions on the plating are to be examined and sounded, but need not be disturbed if found satisfactorily adhering to the plating.	1) Requirements of Special Survey I to be complied with	1) Requirements of Special Survey II to be complied with	1) Requirements of Special survey III to be complied with
2) In ships with single bottom, a sufficient amount of close ceiling is to be lifted to enable examination of the structure below. The ceilings to be lifted is to comprise of at least two strakes on each side of centreline fore and aft and one of these strakes is to be in way of the bilges	2) In ships having a single bottom, a sufficient amount of ceiling is to be lifted to allow the examination of the structure underneath. The lifting of the ceiling is to comprise of at least three strakes all fore and aft on each side and one such strake one each side to be in way of the bilges. Where the ceiling is fitted in hatches, the whole of the hatches and at least one strake of planks in way of the bilges on each side are to be lifted. If the Surveyor considers it necessary the whole of the ceiling and the limber boards are to be lifted	2) Ceiling in the holds is to be removed in order to ascertain that the steel work is in good condition, free from rust and coated. If the Surveyor is satisfied, after removal of portions of the ceiling, than it need not all be removed	2) Where holds are insulated for the purpose of carrying refrigerated cargoes, limbers and hatches are to be lifted and a sufficient additional amount of insulation is to be removed in each compartment to enable the Surveyor to ascertain the condition of the structure in way and to enable the thickness of the shell plating to be ascertained

Table 6.4.2.1 : (Contd.)

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
3) In ships having double bottom, a sufficient amount of ceiling is to be lifted from the tank top and the bilges to enable the condition of plating underneath to be ascertain. If the condition of the plating is found to be satisfactory, lifting of the remainder of the ceiling may be dispensed with. All bilges are to be cleaned for examination. Where the inner bottom plating is covered with cement or asphalt the removal of such covering may be dispensed with provided it is found to be adhering properly to the plating when carefully examined by hammering and chipping	3) In ships with double bottom, a sufficient amount of ceiling in the holds and other spaces is to be removed from the bilges and the inner bottom to enable the condition of the structure in the bilges, the inner bottom plating, pillar feet and the bottom plating of bulkheads and tunnel sides to be examined. If the Surveyor considers it necessary, the whole of the ceiling is to be removed	3) Portions of wood sheathing, or other covering, on steel decks are to be removed, as considered necessary by the Surveyor, in order to ascertain the condition of the plating	
4) Where holds are insulated for the carriage of refrigerated cargoes and the hull in way was examined by IRS Surveyors prior to the fitting of the insulation, it will be sufficient to remove the limbers and hatches for examination of the structure in way. In all other cases additional insulation will require to be removed as considered necessary to enable the Surveyor to satisfy himself regarding condition of the structure	4) The chain locker is to be cleaned internally. The chain cables are to be ranged for inspection. The anchors are to be cleaned and placed in an accessible position for inspection	4) Where the holds are insulated for the purpose of carrying refrigerated cargoes, the limbers and hatches are to be lifted and sufficient insulation is to be removed in each of the chambers to enable the Surveyor to satisfy himself of the condition of the framing and plating	
5) The steel work is to be exposed and cleaned as may be required for its proper examination by the Surveyor and close attention is to be paid to the parts of the structure which are particularly liable to excessive corrosion or to deterioration due to other causes			
6) All tanks are to be cleaned as necessary to permit examination required by Table 6.4.3.1			
7) Casings or covers of air, sounding, steam and other pipes, spar ceiling and lining in way of the side scuttles are to be removed, as required by the Surveyor			

Table 6.4.2.1 : (Contd.)			
Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
8) In refrigerated cargo spaces the condition of the coating behind the insulation is to be examined at representative locations. The examination may be limited to verification that the protective coating remains effective and that there are no visible structural defects. Where POOR coating condition is found, the examination is to be extended as deemed necessary by the Surveyor. The condition of the coating is to be reported. If indents, scratches, etc., are detected during surveys of shell plating from the outside, insulations in way are to be removed as required by the Surveyor, for further examination of the plating and adjacent frames.			

6.4.4 Survey and examination

6.4.4.1 All spaces within the hull and superstructure are to be examined. The special survey is to include examination of underwater parts as per Section 7.

6.4.4.2 All tanks are to be examined internally in accordance with the requirements of Table 6.4.4.1.

All bilge and ballast piping systems are to be examined and operationally tested to working pressure to attending Surveyor's satisfaction to ensure that tightness and condition remain satisfactory.

6.4.4.3 All spaces including holds and their 'tween decks where fitted; double bottom, deep, ballast, peak and cargo tanks; pumprooms, pipe tunnels, duct keels, machinery spaces, dry spaces, cofferdams and voids are to be internally examined including the plating and framing, bilges and drain wells, sounding, venting, pumping and drainage arrangements. At special survey No.3 and subsequent special surveys, structural downflooding ducts and structural ventilation ducts are to be internally examined.

For general dry cargo ships, the following requirements are also to be applied:

- An overall survey of all cargo holds, pipe tunnels, cofferdams and void spaces bounding cargo holds, decks and outer hull is to be carried out.

- All piping systems within the above spaces and in ballast tanks are to be examined and operationally tested to working pressure to attending Surveyor's satisfaction to ensure that tightness and condition remains satisfactory.

- The survey extent of ballast tanks converted to void spaces is to be based on the requirement for ballast tanks.

Table 6.4.4.1 : Requirements for internal examination of tanks				
Tank	**Special Survey No. I Age ≤ 5**	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequent Age > 15**
Fuel oil bunker tanks				
- Engine room	None	None	One	One
- Cargo area	None	One	Two	Half the number of tanks, minimum 2
- If no tanks in Cargo Area, additional fuel tank(s) outside of Engine Room (if fitted)	None	One	One	Two
Lub.oil	None	None	None	One
Fresh water	None	One	All	All
Water ballast	All	All	All	All

Notes:

1) These requirements apply to tanks of integral (structural) type.

2) If a selection of tanks is accepted to be examined, then different tanks are to be examined at each special survey, on a rotational basis.

3) Peak tanks (all uses) are subject to internal examination at each special survey.

4) At special survey No.III and subsequent special surveys one deep tank for fuel oil in the cargo area is to be included, if fitted.

6.4.4.4 All watertight bulkheads and watertight doors are to be examined.

6.4.4.5 All decks, casings and superstructures are to be examined. Attention is to be given to the corners of openings and other discontinuities in way of the strength decks and top sides.

Wooden decks or sheathings are to be examined and if decay or rot is found or the wood is excessively worn, the wood should be renewed.

Attention is to be given to the condition of the plating under wood decks, sheathing or other deck coverings. Removal of such coverings may be dispensed with if they are found to be sound and adhering satisfactorily to the plating.

6.4.4.6 Engine room structure is to be examined. Particular attention being given to tank tops, shell plating in way of tank tops, brackets connecting side shell frames and tank tops and engine room bulkheads in way of tank tops and bilge wells. Where excessive areas of wastage are found, thickness measurements are to be carried out and renewals or repairs made when wastage exceeds allowable limits.

6.4.4.7 The hand pumps and suctions, air and sounding pipes are to be examined. The Surveyors are to ensure that striking plates are fitted under the sounding pipes whilst examining the tanks internally.

For vessel other than passenger ships, automatic air pipe heads are to be internally examined at special surveys as indicated in Table 6.4.4.7. For designs where the inner parts cannot be properly inspected from outside, the head is to be removed from the air pipe. Particular attention is to be paid to the condition of the zinc coating in heads constructed from galvanised steel.

Table 6.4.4.7 : Requirements for internal examination of automatic air pipe heads

Location	Special survey No.I Age ≤ 5	Special survey No.II 5 < Age ≤ 10	Special survey No.III and subsequent Age > 10
forward 0.25L	Two air pipe heads[1,2] one port and one starboard on exposed decks	All air pipe heads on exposed decks	All air pipe heads[3] on exposed decks
aft of 0.25L from the forward perpendicular	Two air pipe heads[1,2] one port and one starboard on exposed decks	At least 20% of air pipe heads[1,2] on exposed decks	

1. Preferably air pipes serving ballast tanks.
2. The selection of air pipe heads is to be made by the attending Surveyor. According to the results of the inspection, the Surveyor may require additional air pipe heads to be examined.
3. When there is substantiated evidence of replacement within the previous five years, individual air pipe heads may not be examined.

6.4.4.8 The steering gear, and its connections and control systems (main and alternative) are to be examined. The auxiliary steering gear with its various parts are to be examined in working condition.

6.4.4.9 The Surveyor should satisfy himself regarding the efficient condition of the following :

- Means of escape from machinery spaces, crew and passenger spaces and spaces in which crew are normally employed;

- Means of communication between bridge and engine room and between bridge and alternative steering position;

- Helm indicator;

- Protection to the aft steering wheel and the gear.

6.4.4.10 The masts, standing rigging and anchors are to be examined.

The Surveyor should satisfy himself that there are sufficient mooring ropes on board and also that a tow line is provided when this is a Rule requirement.

6.4.4.11 The chain cables are to be ranged and the anchors and the chain cables are to be examined. At special survey no. II and subsequent special surveys, the chain cables are to be gauged. Any length of chain cable which is found to have reduced in mean diameter at its most worn part by more than 12 per cent of its original rule diameter is to be renewed.

6.4.4.12 The windlass is to be examined.

6.4.4.13 The chain locker, hold fasts, hawse pipes and chain stoppers are to be examined and pumping arrangements of the chain locker tested.

6.4.4.14 The loading instrument is to be checked for accuracy by applying test load conditions in presence of the Surveyor.

6.4.4.15 For single hold cargo ships which require fitment of hold water level detectors as per Pt.4, Ch.3, Sec.3.6, the special survey is to include an examination and a test of the water ingress detection system and their alarms.

6.4.4.16 Examination of accommodation ladders, gangways and their winches are to be carried out as required for annual surveys. In addition, the accommodation ladders and gangways are to be operationally, tested with the specified maximum operation load.

The tests are to be carried out with the load applied as uniformly as possible along the length of the accommodation ladder or gangway, at an angle of inclination corresponding to the maximum bending moment on the accommodation ladder or gangway.

Accommodation ladder winch is to be operationally tested at special surveys. The brake system of the winch is to be tested for holding the maximum operational load on the ladder.

For existing installations on board ships constructed prior to 01 Jan 2010 where the maximum operational load is not known, load nominated by the shipowner or operator may be considered as the test load.

6.4.5 Hatch covers and coamings

6.4.5.1 In addition to the requirements of annual survey, the following examination/testing is to be carried out:

a) checking of the satisfactory operation of the mechanically operated hatch covers:

   - stowage and securing in open condition;

   - proper fit, locking and efficiency in closed condition;

   - operational testing of hydraulic and power components, wires, chains and link drives.

b) checking the effectiveness of sealing arrangements of all hatch covers by hose testing or equivalent.

c) checking the residual thickness of coamings, steel pontoon or hatch cover plating and stiffening members as deemed necessary by the Surveyor.

For general dry cargo ships, close-up survey and thickness measurement of plating and stiffeners of hatch cover and coaming is to be carried out as required by Table 6.4.7.1 and Table 6.4.8.1b. Close up survey/thickness measurement is to be carried out for accessible parts of hatch cover structures. For cargo hold hatch covers of approved design which have no access to internal structures, close up survey/thickness measurements of such inaccessible structural members need not be carried out.

6.4.6 Tank testing

6.4.6.1 Boundaries of double-bottom, deep, ballast, peak and other tanks, including holds adapted for the carriage of water ballast are to be tested with a head of liquid to the top of air pipes or to near the top of hatches for ballast/cargo holds. Boundaries of fuel oil, lub.oil and fresh water tanks are to be tested with a head of liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil lub.oil and fresh water tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the master stating that the pressure testing has been carried out according to the requirements with satisfactory results.

6.4.6.2 The Surveyor may extend the tank testing as may be necessary.

6.4.7 Close-up surveys of general dry cargo ships

6.4.7.1 The minimum requirements for close-up survey for general dry cargo ships are given in Table 6.4.7.1.

The surveyor may extend the close-up survey as deemed necessary taking into account the maintenance of the spaces under survey, the condition of the corrosion prevention system and where spaces have structural arrangements or details which have suffered defects in similar spaces or on similar ships according to available information.

6.4.7.2 For areas in tanks and cargo holds where coatings are found in GOOD condition, the extent of close-up examination may be specially considered.

Table 6.4.7.1 : Requirements of close-up survey – General dry cargo ships

Special Survey No. I Age ≤ 5	Special Survey No. II 5 < Age ≤ 10	Special Survey No. III 10 < Age ≤ 15	Special Survey No. IV and subsequent Age > 15
(A) Selected shell frames in one forward and one aft cargo hold and associated tween deck spaces. (B) One selected cargo hold transverse bulkhead. (D) All cargo hold hatch covers and coamings (plating and stiffeners).	(A) Selected shell frames in all cargo holds and tween deck spaces. (B) One transverse bulkhead in each cargo hold. (B) Forward and aft transverse bulkhead in one side ballast tank, including stiffening system. (C)One transverse web with associated plating and framing in two representative ballast tanks of each type (i.e. topside, hopper side, side tank or double bottom tank). (D) All cargo hold hatch covers and coamings (plating and stiffeners). (E) Selected areas of all deck plating and underdeck structure inside line of hatch openings between cargo hold hatches. (F) Selected areas of inner bottom plating.	(A) All shell frames in the forward lower cargo hold and 25% frames in each of the remaining cargo holds and tween deck spaces including upper and lower end attachments and adjacent shell plating. (B) All cargo hold transverse bulkheads. (B) All transverse bulkheads in ballast tanks, including stiffening system. (C) All transverse webs with associated plating and framing in each ballast tank (D) All cargo hold hatch covers and coamings (plating and stiffeners). (E) All deck plating and underdeck structure inside line of hatch openings between cargo hold hatches. (F) All areas of inner bottom plating.	(A) All shell frames in all cargo holds and tween deck spaces including upper and lower end attachments and adjacent shell plating. Areas (B—F) as for Special Survey No.III

(A) Cargo hold transverse frames.

(B) Cargo hold transverse bulkhead plating, stiffeners and girders.

(C) Transverse web frame or watertight transverse bulkhead in ballast tanks.

(D) Cargo hold hatch covers and coamings. Close up survey/thickness measurement is to be carried out for accessible parts of hatch cover structures. For cargo hold hatch covers of approved design which have no access to internal structures, close up survey/thickness measurements of such inaccessible structural members need not be carried out.

(E) Deck plating and underdeck structure inside line of hatch openings between cargo hold hatches.

(F) Inner bottom plating.

Note : Close-up survey of cargo hold transverse bulkheads to carried out at the following levels:

- Immediately above the inner bottom and immediately above the tween decks, as applicable.
- Mid-height of the bulkheads for holds without tween decks.
- Immediately below the main deck plating and tween deck plating.

6.4.8 Thickness measurement

6.4.8.1 The minimum requirements for thickness measurement are given in Table 6.4.8.1a and Table 6.4.8.1b.

The Surveyor may extend the thickness measurements as deemed necessary.

Where thickness measurements indicate substantial corrosion, the number of thickness measurements are to be increased to determine the extent of substantial corrosion. Table 6.4.8.2 may be used as guidance for additional measurements.

Table 6.4.8.1a : Thickness measurement : General Ship Types (other than general dry cargo ships)

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel
	2) One transverse section of deck plating within the amidships 0.5L(in way of a cargo space, where provided)	2) Two transverse sections within the amidships 0.5L(in way of two cargo spaces, where provided)	2) A minimum of three transverse sections within the amidships 0.5L (in way of cargo spaces, where provided)
		3) Internals in forepeak and after peak ballast tanks	3) Internals in forepeak and after peak ballast tanks
		4) All cargo hold hatch covers and coamings (plating and stiffeners)	4) All cargo hold hatch covers and coamings (plating and stiffeners)
			5) All exposed main deck plating full length
			6) Representative exposed superstructure deck plating (poop, bridge and forecastle deck)

Table 6.4.8.1a : (Contd.)

Special Survey No. I Age ≤ 5	**Special Survey No. II 5 < Age ≤ 10**	**Special Survey No. III 10 < Age ≤ 15**	**Special Survey No. IV and subsequently Age > 15**
			7) Lowest strake and strakes in way of tween decks of all transverse bulkheads in cargo spaces together with internals in way
			8) All wind and water strakes, port and starboard, full length
			9) All keel plates full length. Also additional bottom plates in way of cofferdams, machinery space and aft end of tanks
			10) a) Plating of sea chests. b) Shell plating in way of overboard discharges as considered necessary by the Surveyor

Notes:

1) Thickness measurement locations are to be selected to provide the best representative sampling of areas likely to be most exposed to corrosion considering cargo and ballast history and arrangement and condition of protective coatings.
2) Thickness measurements of internals may be specially considered by the Surveyor if the hard protective coating is in GOOD condition.
3) For ships less than 100 [m] in length, the number of transverse sections required at Special Survey No.III may be reduced to one (1) and the number of transverse sections required at Subsequent Special Surveys may be reduced to two (2).
4) For ships equal to or more than 100 [m] in length, at special survey No.III, additional thickness measurements of exposed main deck plating within amidship 0.5L may be required.
5) Close up survey/thickness measurement is to be carried out for accessible parts of hatch cover structures. For cargo hold hatch covers of approved design which have no access to internal structures, close up survey/thickness measurements of such inaccessible structural members need not be carried out.

Table 6.4.8.1b : Thickness measurement : General dry cargo ships

Special Survey No. I **Age ≤ 5**	**Special Survey No. II** **5 < Age ≤ 10**	**Special Survey No. III** **10 < Age ≤ 15**	**Special Survey No. IV and subsequently** **Age > 15**
1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel	1) Suspect areas throughout the vessel
	2) Within the cargo area: One transverse section of deck plating in way of cargo space within the amidships 0.5L	2) Within the cargo area: a) Two transverse sections within the amidships 0.5L in way of two different cargo spaces b) Each deck plate outside line of cargo hatch openings c) All wind and water strakes	2) Within the cargo area: a) A minimum of three transverse sections within the amidships 0.5L b) Each deck plate c) All wind and water strakes d) Each bottom plate, including lower turn of bilge. e) Duct keel or pipe tunnel plating and internals
	3) Measurement for general assessment and recording of corrosion pattern of those structural members subject to close-up survey according to Table 6.4.7.1	3) Measurement for general assessment and recording of corrosion pattern of those structural members subject to close-up survey according to Table 6.4.7.1	3) Measurement for general assessment and recording of corrosion pattern of those structural members subject to close-up survey according to Table 6.4.7.1
		4) Selected wind and water strakes outside the cargo length area	4) All wind and water strakes outside cargo area
		5) Internals in forepeak and afterpeak ballast tank	5) Internals in forepeak and afterpeak ballast tank
			6) All exposed main deck plating outside cargo area
			7) Representative exposed super structure deck plating (poop, bridge and fore castle deck)
			8) All keel plates, additional bottom plates in way of machinery space, aft end of tanks and cofferdams outside cargo area
			9) a) Plating of sea chests. b) Shell plating in way of overboard discharges as considered necessary by the Surveyor

Notes:

1. Thickness measurement locations should be selected to provide the best representative sampling of areas likely to be most exposed to corrosion, considering cargo and ballast history and arrangement and condition of protective coatings.
2. For ships less than 100 metres in length, the number of transverse sections required at Special Survey No. III may be reduced to one and the number of transverse sections at Special Survey No. IV and subsequent surveys may be reduced to two.

6.4.8.2 For areas in spaces where hard protective coatings are found to be in a GOOD condition, the extent of thickness measurements may be specially considered by the Surveyor.

6.4.8.3 Transverse sections are to be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

6.4.8.4 The thickness measurements are to be carried out by a qualified firm certified by IRS.

6.4.8.5 In order to ensure necessary control during the process of thickness measurements, these are normally to be carried out under the supervision of the Surveyor. The Surveyor has the right to re-check the measurements as deemed necessary to ensure acceptable accuracy.

Table 6.4.8.2 : Guidance for additional thickness measurements in way of substantial corrosion

Structural Member	Extent of Measurement	Pattern of Measurement
Plating	Suspect area and adjacent plates	5 point pattern over 1 square meter
Stiffeners	Suspect area	3 measurements each in line across web and flange

6.4.8.6 A thickness measurement report is to be prepared. The report is to give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurements were carried out, type of measurement equipment, names of personnel and their qualifications and has to be signed by the operator. The Surveyor is to review the report of the final thickness measurement after repairs have been carried out and countersign the cover page.

6.4.9 Reporting and evaluation of survey of general dry cargo ships

6.4.9.1 The data and information on the structural condition of the vessel collected during the survey is to be evaluated for acceptability and continued structural integrity of the vessel.

6.4.9.2 The survey report would be issued to the owner and the same along with a copy of the thickness measurement report, verified and countersigned by the surveyor, is to be placed on board the vessel for reference at future surveys.

6.4.10 Shell and inner doors of ro-ro ships

6.4.10.1 The special survey is to include, in addition to the requirements of the Annual Survey as required in 6.2.6, examination, tests and checks of sufficient extent to verify that the bow, inner, side shell and stern doors, are in satisfactory condition and considered able to remain in compliance with applicable requirements, subject to proper maintenance and operation in accordance with the Operation and Maintenance Manual (OMM) or manufacturer's recommendations and the periodical surveys being carried out at the due dates for the five year period until the next Special Survey.

6.4.10.2 The examinations of the doors are to be supplemented by thickness measurements and testing to verify compliance with applicable requirements so that the structural and weathertight integrity remain effective. The aim of the examination is to identify corrosion, significant deformation, fractures, damages or other structural deterioration, that may be present.

6.4.10.3 The bow, inner, side shell and stern doors are to be surveyed as follows:

i) A survey of the items listed in 6.2.6.4, including close-up survey of securing, supporting and locking devices, together with welding, according to Table 6.2.6.4.

ii) Non-destructive testing and thickness measurements are to be carried out on securing, supporting and locking devices, including welding, to the extent considered necessary by the Surveyor. Whenever a crack is found, an examination with NDT is to be carried out in the surrounding area and for similar items as considered necessary by the Surveyor.

iii) The maximum thickness diminution of hinging arms, securing, supporting and locking devices is not to be more than 15% of the as-built thickness, in general.

iv) Effectiveness of sealing arrangements is to be verified by hose testing or equivalent.

v) Clearances of hinges, bearings and thrust bearings are to be taken. Unless otherwise specified in the OMM or by manufacturer's recommendation, the measurement of clearances on ro-ro cargo ships may be limited to representative bearings where dismantling is needed in order to measure the clearances. If dismantling is carried out, a visual examination of hinge pins and bearings together with NDT of the hinge pin is to be carried out.

vi) The non-return valves of the drainage system are to be dismantled and examined.

6.4.11 For general dry cargo ships compliance with new IACS requirements for strength of air pipes, ventilators and their closing devices and of small hatches and their securing devices is to be achieved as indicated in 2.4.13.

6.4.12 Watertight Cable Transits

6.4.12.1 The requirements for Special Survey may be undertaken by the attending Surveyor or by a firm approved as a service supplier according to Pt.1 Ch.1, Sec 1.9.

6.4.12.2 All transits are to be examined to confirm their satisfactory condition and the Cable Transit Seal System Register is to be reviewed to confirm that it is being maintained. The Special Survey is to be recorded in the Register, in which a single record entry will be sufficient to record the survey of all transits.

6.4.12.3 From review of the Register, where there are records entered since the last special survey of any disruption to the cable transits or installation of new cable transits (except which are reviewed and examined at previous annual surveys), the satisfactory condition of those transits is to be confirmed by the attending Surveyor by review of records and examination of the transits; the results are to be recorded in the Register against each of those cable transits.

6.4.12.4 In case the cable transits have been examined by an approved service supplier, the attending surveyor is to review the Register in order to ascertain that it has been properly maintained by the owner and correctly endorsed by the service supplier.

6.4.13 Container ships with container lashing software

6.4.13.1 In addition to the annual survey requirements, the functionality of the software is to be tested and verified with approved test cases/ samples.

6.5 Special surveys - Machinery

6.5.1 Requirements for examination of machinery and piping at special survey are given in Sec.8.

6.6 Surveys of Livestock Carriers

6.6.1 General

6.6.1.1 In addition to the relevant requirements of this Section, the following are to be complied with for the surveys of livestock carriers.

6.6.2 Annual Surveys

6.6.2.1 Confirmation that no unapproved changes have been made to livestock pens, passageways as well as the arrangement and protection of livestock since the last survey; confirmation that passageways and exits of the escape route of pen area are unblocked; confirming that two means of egress in each livestock space are not separated by the pen or other obstructions.

6.6.2.2 Confirmation that ventilation arrangements in livestock space are satisfactory and the operation of fans, including fan starters, failure indicators and alarms at different positions is normal. If only one fan is provided, confirmation that spare parts are adequate.

6.6.2.3 Confirming that fodder and drinking water supply, fresh water supply system, water washing system and sewage drainage and discharge system of livestock space are in good condition. The fodder tanks and the fresh water tanks in the livestock spaces are to be examined.

6.6.2.4 Confirming that all fire hydrants and fire hose connections arranged in the pen area are in satisfactory condition.

6.6.2.5 Verification that instructions for the changeover between main and secondary

sources of power and vice-versa are available and confirming the function of the changeover between main and secondary sources of power and vice-versa.

6.6.2.6 General visual examination of lamp casings on light fittings for the livestock spaces.

6.6.3 Special Surveys

6.6.3.1 In addition to the requirements in 6.6.2, the special survey is to include the examination, tests and checks specified in 6.6.3.2 to 6.6.3.5.

6.6.3.2 All fresh water pumps, sewage pumps and system components are to be opened up and examined.

6.6.3.3 Performance test of fodder system, fresh water supply system, water washing system and sewage drainage and discharge system.

6.6.3.4 Ventilation fan motor in the livestock space together with its control and operational mechanism are to be inspected and if necessary, operationally tested in working condition.

6.6.3.5 Interlocking devices of fan starters in the livestock space are to be tested.

6.7 Surveys of Towing Winch Emergency Release Systems

6.7.1 General

6.7.1.1 The requirements in this Section are applicable to surveys of winch emergency release systems that are used on towing ships within close quarters, ports or terminals, including those ships normally not intended for towing operations in transverse direction.

6.7.2 Annual Surveys

6.7.2.1 Operation of the towing winch emergency release system is to be confirmed with reference to the documented instructions for surveys provided by the manufacturer. Operation of the winch emergency release system under no load condition is to be verified. Where practical, activation of the emergency release system may be confirmed by observation of the winch brake.

6.7.2.2 The function of the alarms associated with the emergency release system is to be verified, as far as practicable and reasonable.

6.7.2.3 The condition of the emergency release system is to be visually examined to confirm it remains in satisfactory condition.

6.7.2.4 The means of emergency release of the towline in the event of a blackout is to be examined, and where additional sources of energy are arranged for this purpose, the sources of energy are to be visually inspected and operationally tested.

6.7.2.5 It is to be verified that the performance capabilities and operating instructions of the emergency release system are documented and made available on board the ship on which the winch has been installed.

6.7.3 Special Surveys

6.7.3.1 In addition to the annual survey requirements, instructions for special survey provided by the manufacturer, as appropriate, are to be followed.

6.7.3.2 The full functionality of the emergency release system is to be tested to the satisfaction of the surveyor. Testing may be conducted either during a bollard pull test or by applying the load against a strong point on the deck of the tug or the shore that is certified to the appropriate load.

6.7.3.3 The emergency release system is to be tested at a towline load that is equal to the lesser of 30% of the maximum design load or 80% of vessel bollard pull in both a normal power condition and power blackout condition to the satisfaction of the surveyor.

Section 7

Docking Surveys

7.1 General

7.1.1 The requirements for docking survey intervals are given in Table 1.1.1. Attention should also be given to any relevant statutory requirements of the National Authority of the country in which the ship is registered.

An extension of examination of the ship's bottom of 3 months beyond the due date can be granted in exceptional circumstances due to any of the following reasons:

a) Non-availability of drydocking facilities, repair facilities, essential materials, equipment or spare parts.

b) Delays incurred by action taken to avoid severe weather conditions or delays caused by severe weather conditions.

7.1.2 The ship is to be placed on blocks of sufficient height in a drydock or on a slipway and proper staging is to be erected as may be necessary. Shell plating, stern frame or stern post, rudder and sea inlet and discharge openings are to be examined. Particular attention is to be given to the bilge keels and their connection to the bilge strake, shell plating in way of side, bow and stern doors, ash shoots, and other openings and to parts of the structure particularly liable to excessive corrosion or to deterioration from causes such as chafing and lying on the ground and to any undue unfairness of the bottom plating. Important plate unfairness or other deterioration which do not necessitate immediate repairs are to be recorded.

Visible parts of rudder, rudder pintles, rudder shafts and couplings are to be examined. The clearances in the rudder bearings are to be ascertained. The rudder is to be lifted for examination of the pintles, if considered necessary by the Surveyor. Where applicable, pressure test of the rudder may be required as considered necessary by the Surveyor.

For vessels fitted with ICCP system, the condition of anodes, reference electrodes, and dielectric shields is to be examined.

7.1.3 When the chain cables are ranged the anchors and cables should be examined by the Surveyor.

7.1.4 Sea chests and their gratings, sea connections and overboard discharge valves and cocks and their fastenings to the hull or sea chests are to be examined. Valves and cocks need not be opened up more than once in a special survey period unless considered necessary by the Surveyor.

7.1.5 The Surveyor should examine the ship so far as is practicable in order to satisfy himself as to her general condition.

7.1.6 The propeller should be examined for erosion, pitting, cracking of blades or possible contact damage. The clearance in the stern bush should be measured. In the case of approved oil glands, measurements by poker gauges or other devices for ascertaining the stern bush weardown may be accepted, provided the sealing arrangements appear satisfactory.

7.1.7 For controllable pitch propellers, the Surveyor is to be satisfied with the fastenings and tightness of hub and blade sealing. Dismantling need not be carried out unless considered necessary by the Surveyor.

7.1.8 Exposed parts of side thrusters are to be examined. Propulsion systems which also have maneuvering characteristics (such as directional propellers, vertical axis propellers, water jets) are to be examined externally with focus on the condition of gear housing, propeller blades, bolt locking and other fastening arrangements. Sealing arrangement of propeller blades, propeller shaft and steering column is to be verified.

7.1.9 For vessels with class notation, EDD (X), an inspection of the outside of the vessel's bottom carried out as detailed in the IRS Classification Note "*Extended Dry-Docking Scheme*" may be credited as equivalent to the relevant requirements of a special survey in dry-dock.

7.2 In-water surveys

7.2.1 Eligibility

7.2.1.1 For ships less than 15 years in age the examination of the outside of ship's bottom and related items of ships may be carried out while the ship is afloat (inwater survey) in lieu of the required intermediate docking between special surveys provided following conditions are satisfied.

a) The ship has been assigned the class notation INWATER SURVEY as defined in Ch.1. However, on application by the owner and in special circumstances, such inwater survey may be considered for ships not assigned with the class notation INWATER SURVEY.

b) The ship does not have any outstanding Conditions of Class which require repair work in dry dock to the underwater part of the shell plating, the rudder, the propeller or the propeller shaft.

7.2.1.2 Acceptance of inwater surveys as above may be specially considered for ships other than bulk carriers, oil tankers and chemical tankers which are 15 years of age and over.

7.2.2 Requirements for assignment of INWATER SURVEY notation

7.2.2.1 Detailed plans of the hull and hull attachments below the water line are to be submitted to IRS in triplicate for approval. These plans are to indicate the location and/or the general arrangement of:

- all shell openings
- stem
- rudder and fittings
- sternpost
- propeller, including the means used for identifying each blade
- anodes, including securing arrange-ments
- bilge keels
- welded seams and butts.

The plans are also to include the necessary instructions to facilitate the divers' work, especially for taking clearance measurements.

7.2.2.2 The plans for approval are also to include the procedure for measurement or verification, as the case may be, of the following:

- rudder pintle/bush clearance
- stern bush clearance
- pintle securing arrangement in the socket.

7.2.2.3 As far as practicable, a photographic documentation, used as a reference during the in-water surveys, of the following hull parts is to be submitted to IRS:

- propeller boss
- rudder pintles, where slack is measured
- typical connections to the sea
- directional propellers, if any
- other details, as deemed necessary by IRS on a case by case basis.

7.2.2.4 The Owner is to put on board the ship the plans and documents given in 7.2.2.1 and 7.2.2.2 and they are to be made available to the Surveyor and the divers when an in-water survey is carried out.

7.2.3 Survey requirements

7.2.3.1 The in-water survey is to provide all the information normally obtained from a docking survey. However, special consideration may be given to ascertaining rudder bearing clearances and stern bush clearances of oil stern bearings based on a review of the operating history, on board testing and stern tube oil sample reports. These details are to be included in the proposals for in-water surveys which are to be submitted in advance of the survey so that satisfactory arrangements can be agreed with IRS.

7.2.3.2 The in-water survey is to be carried out with the ship in as light condition as possible in sheltered water preferably with weak tidal streams and currents. The in-water visibility

and the cleanliness of the hull below the waterline is to be good enough to permit a meaningful examination which allows the Surveyor and the in-water survey firm to determine the condition of the plating, appendages and the welding. IRS is to be satisfied with the methods of orientation of the divers or Remotely Operated Vehicle (ROV) on the plating, which should make use where necessary of permanent markings on the plating at selected points.

7.2.3.3 The in-water survey is to be carried out under surveillance of a Surveyor by an in-water survey firm approved as a service supplier.

The equipment, procedure for observing and reporting the survey are to be discussed with the parties involved prior to the in-water survey and suitable time is to be allowed to enable the in-water survey firm to test all equipment beforehand.

7.2.3.4 When professional divers are employed, the Surveyor is to be satisfied with the method of pictorial representation, and a good two-way communication between the Surveyor and divers is to be provided.

7.2.3.5 If the in-water survey reveals damage or a deterioration that requires early attention, the Surveyor may require that the ship to be drydocked in order that a detailed survey can be undertaken and the necessary repairs carried out.

7.2.3.6 The condition of the coating is to be confirmed at every drydocking for continuance of the INWATER SURVEY notation.

Section 8

Machinery - Special Surveys

8.1 General

8.1.1 Special surveys are to be carried out at 5 years intervals to renew the Class Certificate.

The first Special survey is to be completed within 5 years from the initial classification survey and thereafter 5 years from the assigned date of the previous special survey. However, an extension of class of 3 months maximum beyond the 5th year can be granted in exceptional circumstances.

In this case, the next period of class will start from the expiry date of the special survey before the extension was granted.

8.1.2 For surveys completed within 3 months before the expiry date of the special survey, the next period of class will start from the expiry date of the special survey. For surveys completed more than 3 months before the expiry date of the special survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the special survey. If the owner elects to carry out the next due special survey, the period of class will start from the survey completion date.

Any requirement of the Flag Administration in this regard is also to be complied with.

8.1.3 The special survey may be commenced at the 4th annual survey and be progressed with a view to completion by the 5th anniversary date. When the special survey is commenced prior to the 4th annual survey, the entire survey is to be completed within 15 months if such work is to be credited to the special survey.

8.1.4 Upon satisfactory completion of the Machinery Special Survey an appropriate record will be entered in the supplement of register of ships.

8.1.5 Further parts of machinery may require to be opened should any defects be found during the course of Survey of any item and the defects made good to the satisfaction of the Surveyor.

8.1.6 As part of the Special Survey of Machinery, a dock trial is to be carried out to attending Surveyor's satisfaction to confirm satisfactory operation of main and auxiliary machinery. If significant repairs are carried out to main or auxiliary machinery or steering

gear, consideration should be given to a sea trial to attending Surveyor's satisfaction.

8.1.7 If the significant repairs as stated in 8.1.6, is considered by IRS to have any impact on response characteristics of the propulsion systems, then the scope of sea trial is also to include a test plan for ahead and astern response characteristics based on those required for such an equipment or systems when fitted to the new ship. (*Refer to Pt.4, Ch.1, 1.10 and 3.2.8*). The tests are to demonstrate the satisfactory operation of the equipment or system under realistic service conditions at least over the manoeuvring range of the propulsion plant, for both ahead and astern directions. Depending on the actual extent of the repair, IRS may accept a reduction of the test plan.

8.2 Continuous survey of machinery (CSM)

8.2.1 At the request of the Owners and upon approval of the proposed arrangement a system of continuous Survey of Machinery (CSM) may be undertaken whereby the requirements of Special Survey of machinery are completed within a 5 year period. The Survey cycle should be such as would ensure that the interval between consecutive examination of each item does not exceed five years and it is expected that approximately an equal proportion of the machinery would be subjected for Survey each year.

8.2.2 Upon satisfactory completion of the continuous Machinery Survey cycle, a record indicating the date of completion of the cycle will be shown in the supplement to the Register Of Ships.

8.3 Survey requirements: Auxiliary engines

8.3.1 All auxiliary engines driving the generators and other essential machinery together with their coolers and attached pumps are to be opened up and examined as considered necessary by the Surveyor. Alarms and safety devices fitted on these units are to be included in this Survey.

8.4 Survey requirements: Air compressors, receivers and starting air pipes

8.4.1 All air receivers and other pressure vessels for essential services together with their mountings and safety devices are to be cleaned internally and examined internally and externally. If an internal examination of an air receiver is not practicable it is to be tested hydraulically to 1.3 times the working pressure.

8.4.2 Air compressors are to be opened up and coolers tested as considered necessary by the Surveyor. Selected pipes in the starting air systems are to be removed for internal examination and hammer tested. If an appreciable amount of lubricating oil is found in the pipes the starting air system is to be thoroughly cleaned by steaming or other suitable means. Some of the pipes selected are to be those adjacent to the starting air valves at the cylinders and to the discharges from the air compressors.

8.5 Survey requirements: Fresh water generators and evaporators

8.5.1 To the extent applicable, these are to be opened up and examined. After such Survey their relief valves are to be checked under working conditions, where practicable.

8.6 Survey requirements: Fuel tanks

8.6.1 Fuel tanks which do not form part of the ship's structure are to be examined externally and, if considered necessary by the Surveyor also internally. All mountings, fittings and remote control devices are to be examined as far as practicable. All such tanks are to be tested by filling to maximum working level.

8.7 Survey requirements: Pumps, heat exchangers, forced draught fans, etc.

8.7.1 All pumps, heat exchangers, forced draught fans, etc. used for essential purposes are to be opened up and examined as considered necessary by the Surveyor.

8.8 Survey requirements: Pumping and piping system

8.8.1 The valves, cocks and strainers of the bilge system including bilge injection are to be opened up as considered necessary by the Surveyor and, together with pipes, are to be examined and tested under working conditions. If non-return valves are fitted in hold bilges, these be opened up for examination.

8.8.2 The oil fuel, feed and lubricating systems and ballast connections and blanking arrangements to deep tanks which may carry liquid or dry cargoes, together with all pressure

filters, heaters and coolers used for essential service, are to be opened up and examined or tested as considered necessary by the Surveyor. All safety devices for the foregoing are to be examined.

8.8.3 Non-metallic expansion joints in piping systems, if located in system which penetrates the ship's side and both the penetration and the non-metallic expansion joint are located below the deepest load waterline, are to be examined and replaced as necessary or at an interval recommended by the manufacturer.

8.9 Survey requirements: Reduction gears, flexible couplings and clutch arrangements

8.9.1 Reduction gears, flexible couplings and clutch arrangements are to be opened as considered by the Surveyor in order to permit the examination of the gears, gear teeth, spiders, pinions, shafts and bearings, reversing gears, etc. Essential parts of other power transmission arrangements are to be opened up and examined as considered necessary by the Surveyor.

8.10 Survey requirements: Securing arrangements

8.10.1 Holding down bolts and chocks of main and auxiliary engines, gear cases, thrust blocks and tunnel bearings are to be checked.

8.11 Survey requirements: Shafting

8.11.1 Intermediate shafts and bearings, thrust bearings and their seating are to be examined. The lower halves of bearings need not be exposed if alignment and wear are found acceptable.

8.12 Survey requirements: Sea connections

8.12.1 All openings to the sea including sanitary and other overboard discharges in the machinery spaces and pump rooms together with valves and cocks are to be examined internally and externally. The fastenings of valves and cocks to the hull are to be examined and are to be renewed when considered necessary by the Surveyor. Particular attention is to be given to the sea suctions and sea water cooling pipes.

8.13 Survey requirements: Windlass and steering machinery

8.13.1 These are to be examined to ascertain that they are in good working order. Any relief valves fitted are to be included in the above examination.

8.14 Survey requirements: Internal combustion engines for propulsion

8.14.1 All working parts of the engines and their attached pumps are to be opened and examined. These should include all cylinders, cylinder heads, valves and valve gear, pistons, piston rods, cross heads, guides, connecting rods, crankshafts, vibration dampers and all bearings, camshafts and driving gear, fuel pumps and fittings, scavenge pumps, scavenge blowers and their prime movers, superchargers, air compressors, inter coolers, clutches, reverse gears, crankcase door fastenings and explosion relief devices and such other parts of the machinery as may be considered necessary. Integral piping systems are to be examined. The maneuvering of engines is to be tested under working condition.

8.15 Survey requirements: Steam turbines for propulsion

8.15.1 Upper halves of turbine casings are to be opened up rotors to be lifted up and inside of casings, rotors and bearings (including thrust) and governors to be examined.

8.15.2 At the first Special Periodical Survey only, for vessels having more than one main propulsion ahead turbine with emergency steam crossover arrangement, the turbine casings need not be opened provided approved vibration indicators and rotor position indicators are fitted and that the Surveyor considers the operating records to be satisfactory. An operational test of the turbines may be required if considered necessary by the Surveyor.

8.15.3 Essential valves attached to the turbines and flexible couplings are to be examined. The maneuvering of the turbines is to be tested under working conditions.

8.15.4 Exhaust steam turbines supplying power for main propulsion purposes in conjunction with reciprocating engines together with their gearing and appliances, steam compressors or electrical machinery are to be examined as far as practicable. Where cone connections to internal gear shafts are fitted, the coned ends are to be examined as far as practicable. The maneuvering of engines is to be tested under working conditions.

8.16 Survey requirements: Steam reciprocating engines for propulsion

8.16.1 Working parts of main engines and attached pumps including bulkhead stop valves, maneuvering valves, cylinders, pistons, valves and valve gear, piston rods, connecting rods, crankshaft and bearings and governor are to be opened up and examined. The maneuvering of engines is to be tested under working conditions.

8.17 Survey requirements: Gas turbines and free piston gas generators for propulsion

8.17.1 The Survey should include opening and examination of the following parts :-

- The blading, rotors, and casings of the turbines, the impellers or blading, rotors and casing of air compressors, the combustion chambers, burners, inter coolers, heat exchangers, gas and air pressure piping and fittings and reversing arrangements. When gas turbines operate in conjunction with free piston gas generators, the following parts of the free piston gas generators are to be opened and examined: the gas and air compressor cylinders and pistons and the compressor end covers, the valves and valve gear, fuel pumps and fittings, synchronising and control gear, cooling system explosion relief devices, gas and air piping, receivers and valves including by-pass. The maneuvering of engines is to be tested under working conditions.

8.18 Survey requirements: Unattended machinery spaces/Remote control systems

8.18.1 Where remote and/or automatic controls such as bridge controls, bilge controls and bilge level alarms, local hand controls, fire detection and prevention, alarms warning systems and shut-offs, electric supply, main controls station, are fitted for essential machinery, they are to be examined and tested to demonstrate that they are in good working order.

8.18.2 During such trials the proper operation of the safety devices will be checked, in particular, such as emergency stops, emergency astern movement, standby control of the propelling gear, fire alarm.

8.18.3 The log recording the operating conditions should be checked. If such scrutiny reveals that certain portion of the automated equipment has behaved abnormally the cause of such failure is to be investigated and appropriate remedies determined.

8.19 Survey requirements : Electrical equipment survey

8.19.1 Electrical installations including auxiliary and emergency equipment are to be examined in accordance with the following during each Survey cycle.

8.19.2 Switch boards (including for emergency use) and their accessories including section-boards and sub-division fuse boards are to be examined as far as possible and over current protective devices and fuses inspected to verify that they provide suitable protection for their respective circuits. In the case of diving support vessels using the emergency generator as the independent source of power for the diving system, the test should be carried out for the total load including diving system load.

8.19.3 All generator circuit breakers are to be tested, as far as practicable, to verify that the protective devices including preference tripping relays, if fitted, operate satisfactorily. The generators are to be run under load either separately or in parallel, and the governing of the engines to be tested.

8.19.4 The insulation resistance of cables, switch gear, generators, motors, heaters, lighting and other fittings is to be tested and should not be less than 100,000 ohms between all insulated circuits and earth. The installation may be subdivided to any desired extent by opening switches, removing fuses or disconnecting appliances for the purpose of this test.

The electric cables are to be examined as far as possible without undue disturbance of fixtures or casings unless deemed necessary by the Surveyor.

8.19.5 Transformers are to be examined. Samples of oil are to be taken and tested for breakdown voltage, acidity and moisture in case of oil immersed transformers or electrical apparatus associated with supplies to essential services. The testing is to be carried

out by a competent testing authority and a certificate giving the test results is to be furnished to the Surveyor.

8.19.6 Motors used for essential services including their starters are to be examined, and under working conditions if considered necessary by the Surveyors.

8.19.7 Generators and steering gear motors are to be examined under working conditions. Air gaps are to be checked for excessive wear down.

8.19.8 In the case of electromagnetic couplings, the air gaps are to be measured and reported and any excessive eccentricity corrected. Switch gear and couplings for same are to be examined and tested.

8.19.9 The emergency source of power and its associated circuits are to be tested. In the case of passenger ships, the temporary source of power and its automatic arrangements (if fitted) are also to be tested.

8.19.10 Navigation light indicators are to be tried under working conditions, and correct operation on the failure of supply or failure of navigation lights verified.

8.19.11 In tankers, a General examination of electrical equipment located in dangerous zones and spaces is to be made to ensure that the integrity of the safe type electrical equipment has not been impaired owing to corrosion, missing bolts, etc. Cable runs are to be examined for sheath and armouring defects, where practicable, and to ensure that the means of supporting the cable are in good order. Tests are to be carried out to demonstrate the effectiveness of earth bonding straps. Alarms and interlocks associated with pressurized equipment or spaces are to tested for correct operation.

Insulation resistance is to be measured for circuits terminating in, or passing through dangerous zones or spaces.

8.20 Survey requirements : Electrical propelling machinery

8.20.1 On ships which are electrically propelled, the main propulsion motors, generators, cables, together with all ancillary electrical gear, exciters and ventilating plant (including coolers) are to be examined and their insulation resistance to earth to be measured. Protective gear and alarm devices are to be checked as far as practicable and special attention should be given to windings, commutators and slip rings. Safety interlocks intended to prevent unsafe operation or unauthorised access are to be checked to verify that they are functioning correctly. Emergency over speed governors are to be tested. Where insulating oil is used, samples of oil are to be taken and tested for breakdown voltage, acidity and moisture by a competent testing authority and a certificate giving the test results is to be furnished to the Surveyor.

8.21 In service testing of large permanently installed breathing gas containers onboard diving vessels - Special Requirements

8.21.1 At the first special survey, following is to be carried out:

- External and internal survey, by intrascope if necessary.

- If internal survey is not possible or if corrosion or other items of concern are found, hydraulic test to 1.25 x design pressure to be carried out.

At subsequent special surveys, following to be carried out:

- External and internal survey, by intrascope if necessary.

- Hydraulic test to 1.25 x design pressure.

8.22 Surveys Requirements: Fuel installations on ships utilizing Gaseous Fuels or Other Low Flash Point Fuels

8.22.1 Surveys of Fuel installations on ships using Gaseous fuels or other low flash point fuels are to be carried out in accordance with the requirements of Section 20, as applicable.

Section 9

Boiler Surveys

9.1 General

9.1.1 All main and auxiliary boilers, exhaust gas steam generators and economisers are to be surveyed at intervals specified in Sec.1 of this Chapter. An extension of examination of the boiler of up to 3 months beyond the due date can be granted in exceptional circumstances. “Exceptional Circumstances” means un-availability of repair facilities, unavailability of essential materials, equipment or spare parts, or delays incurred by action taken to avoid severe weather conditions.

9.1.2 At each Survey, the boilers, superheaters, economisers, air heaters, desuperheaters, and other equipment are to be examined internally (water/steam side) and externally (fire side) as considered necessary.

In exhaust gas heated economizers of the shell type, all accessible welded joints are to be subjected to a visual examination for detection of cracks. Nondestructive testing may be required for this purpose.

When direct visual internal inspection is not feasible due to the limited size of the internal spaces, such as for small boilers and/or narrow internal spaces, this may be replaced by a hydrostatic pressure test or by alternative verifications as determined by IRS.

9.1.3 Boiler safety valve and its relieving gear are to be examined and tested to verify satisfactory operation. The adjustment of the safety valves is to be verified during each boiler internal survey. Safety valves are to be adjusted to a pressure not greater than 3 percent of the approved working pressure.

9.1.4 Review of the following records since the last boiler survey is to be carried out as part of the survey:

a) Operation
b) Maintenance
c) Repair history
d) Feed water chemistry

9.1.5 External survey of boilers including test of safety and protective devices and test of safety valve using its relieving gear, is to be carried out annually, within the window of the Annual Survey of a ship. For exhaust gas heated economizers, the safety valves are to be tested by the Chief Engineer at sea within the annual survey window. This test is to be recorded in the log book for review by the attending Surveyor prior to crediting the Annual Survey of Machinery.

9.1.6 Principal boiler mountings and safety valves are to be examined at each Survey. The remaining mountings are to be opened if considered necessary by the Surveyor. Manhole and hand hole doors, are to be examined to ensure that the joining faces are in good condition and that the clearances at the spigot are satisfactory.

9.1.7 In case where it is considered necessary, the parts subjected to pressure are to be hydraulically tested and the thickness of plates and size of stays ascertained to determine the safe working pressure. Collision chocks, rolling stays and boiler stools are to be examined and maintained in efficient condition. The shell plating in way of welded lugs or fabricated feet are to be carefully examined at each Survey. Insulation and sheathing in way are to be removed as considered necessary for this purpose.

In fired boilers employing forced circulation the pumps used for this purpose are to be opened and examined at each boiler Survey.

9.1.8 The proper operation of the water level indicators are to be confirmed at each Survey. The oil fuel burning system is to be examined under working conditions and a general examination made of the fuel tank valves, pipe, deck control gear and oil discharge pipes between pumps and burners.

GUIDANCE :

Every effort should be made to complete the boiler Survey once it has been commenced at the same port if possible or immediately afterwards on the first occasion when steam is raised.

9.1.9 An extension of survey may be granted by IRS, on the basis of 9.1.1, after the following is satisfactorily carried out:

a) External examination of the boiler

b) Boiler safety valve relieving gear (easing gear) is examined and operationally tested.

c) Boiler protective devices operationally tested.

d) Review of the records since the last boiler survey as mentioned in 9.1.4.

9.2 Steam heated steam generator

9.2.1 Steam heated steam generators are to be surveyed at two and half yearly intervals.

9.2.2 At each steam heated steam generator survey the mountings and the safety valves are to be examined. Manhole and hand hole doors are to be examined to ensure that joining faces are in good condition and that the clearances at the spigot are satisfactory.

9.2.3 Where it is considered necessary, parts subjected to pressure are to be hydraulically tested and the thickness of plates and the sizes of stays ascertained to determine the safe working pressure.

9.2.4 The proper operation of the water level indicators is to be confirmed at each Survey. The safety valves of the steam heated steam generator are to be adjusted to a pressure not greater than 3 per cent above the approved working pressure.

Section 10

Steam Pipes Surveys

10.1 General

10.1.1 Steam pipes are to be surveyed as part of special surveys of machinery or continuous surveys of machinery (CSM).

10.1.2 At each Survey a selected number of main and auxiliary steam pipes over 75 [mm] bore supplying steam for essential services are to be examined internally and tested hydraulically to 1.5 times the working pressure. If these are found satisfactory, the remaining need not be tested. In cases of pipes having welded joints, the lagging in way of the welds is to be removed and the welds examined and if considered necessary crack detected.

10.2 Cylindrical boilers having smoke tube superheaters

10.2.1 Where the saturated steam pipes adjoining the saturated steam headers are situated partly in the boiler smoke boxes, all such pipes adjoining and cross connecting these headers in the smoke boxes are required to be included in the pipes required for examination and testing for the steam pipe Survey.

10.3 Copper pipes

10.3.1 At 10 years from the date of build, or of installation and thereafter at 5 yearly intervals, all copper pipes over 75 [mm] bore and supplying steam for essential services at sea are to be hydraulically tested to twice the working pressure. Pipes which are subjected to bending and/or vibration are to be annealed before testing.

Section 11

Surveys of Propeller Shafts, Tube Shafts and Propellers

11.1 General

11.1.1 At shaft surveys, propeller shafts and tube shafts, if any, are to be sufficiently drawn to permit entire examination at intervals as detailed in Sec.1 of this Chapter.

Unless alternative means are provided to ensure the condition of the propeller shaft assembly, these requirements apply to all vessels with conventional shafting fitted with a propeller as follows:

.1 from 1 Jan 2016 for ships delivered on or after 1 Jan 2016;

.2 after the first shaft survey scheduled on or after 1 Jan 2016, for ships delivered before 1 Jan 2016. Upon completion of the first shaft survey scheduled on or after 1 Jan 2016, the designation of dates for the next shaft survey is to be made based on the requirements in this section.

11.1.2 For the purpose of these requirements, the following definitions are applicable:

.1 Shaft is a general definition that includes:

- Propeller shaft
- Tube shaft

The definition does not include the intermediate shaft(s) which is (are) considered part of the propulsion shafting inside the vessel.

.2 *Propeller Shaft.* Propeller shaft is the part of the propulsion shaft to which the propeller is fitted. It may also be called screwshaft or tailshaft. Where a separate tube shaft is not fitted, the propeller shaft runs through the stern tube and is connected to the intermediate shaft within the ship. In this case, only the propeller shaft and intermediate shaft are fitted and there is no tube shaft.

.3 *Tube Shaft.* Tube shaft is a shaft placed between the intermediate shaft and propeller shaft, normally arranged within a stern tube or running in open water. It may also be called Stern Tube Shaft.

.4 *Stern Tube*. Tube or pipe fitted in the shell of a ship at the stern (or rear part of the ship), below the water-line, through which passes the tube shaft or the propeller-shaft. Stern tube is the housing of the shaft bearings, generally two (one aft and one fore), that sustain the shaft and allows its rotation with less frictional resistance. The stern tube also accommodates the shaft sealing arrangement.

.5 *Close Loop (system) Oil Lubricated Bearing*. Closed loop oil lubricating systems use oil to lubricate the bearings and are sealed against the environment (seawater) by adequate sealing / gland devices.

.6 *Water Lubricated Bearing*. Water lubricated bearings are bearings cooled / lubricated by water (fresh or salt).

.7 *Closed Loop System Fresh Water Lubricated Bearing*. Closed loop water lubricating systems use fresh water to lubricate the bearings and are sealed against the environment (such as seawater) by adequate sealing / gland devices.

.8 *Open Systems (water)*. Open water lubricating systems use water to lubricate the bearings and are exposed to the environment.

.9 *Adequate means for protection against corrosion*. An adequate means for protection against corrosion is an approved means for full protection of the core shaft against sea water intrusion and subsequent corrosion attack. Such means are used for the protection of common steel material against corrosion particularly in combination with water lubricated bearings. Typical means are for example:

- continuous metallic, corrosion resistant liners,
- continuous cladding,
- multiple layer synthetic coating,
- multiple layer of fiberglass,
- combinations of above mentioned,
- rubber / elastomer covering coating.

The means for protection against corrosion are installed / applied according to IRS approved procedures.

.10 *Corrosion Resistant Shaft*. Corrosion resistant shaft is made in approved corrosion resistant steel as core material for the shaft.

.11 *Sterntube Sealing System*. Sterntube sealing system is the equipment installed on the inboard extremity and, for closed systems, at outboard extremity of the sterntube.

Inboard Seal is the device fitted on the fore part of the sterntube that achieve the sealing against the possible leakage of the lubricant media in to the ship internal.
Outboard seal is the device fitted on the aft part of the sterntube that achieve the sealing against the possible sea water ingress and the leakage of the lubricant media.

.12 *Service Records*. Service records are regularly recorded data showing in-service conditions of the shaft(s) and may include, as applicable: lubricating oil temperature, bearing temperature and oil consumption records (for oil lubricated bearings) or water flow, water temperature, salinity, pH, make-up water and water pressure (for closed loop fresh water lubricated bearings depending on design).

.13 *Oil sample examination*. An oil sample examination is a visual examination of the stern tube lubricating oil taken in presence of the surveyor with a focus on water contamination.

.14 *Lubricating oil analysis*. Lubricating oil analysis is to be carried out at regular intervals not exceeding six (6) months. The documentation on lubricating oil analysis is to be available on board.

Oil samples, to be submitted for the analysis, are to be taken under service conditions.

.15 *Fresh water sample test*. Fresh water sample test is to be carried out at regular intervals not exceeding six (6) months
Samples are to be taken under service conditions and are to be representative of the water circulating within the sterntube. Analysis results are to be retained on board and made available to the surveyor. At time of survey the sample for the test is to be taken in the presence of the surveyor.

Fresh water sample test is to include the following parameters:

- chlorides content,
- pH value,
- presence of bearing particles or other particles (only for laboratory analysis, not required for tests carried out in presence of the surveyor).

.16 *Keyless Connection*. Keyless connection is the forced coupling methodology between the shaft and the propeller without a key achieved through interference fit of the propeller boss on the shaft tapered end.

.17 *Keyed Connection*. Keyed connection is the forced coupling methodology between the shaft and the propeller with a key and keyway achieved through the interference fit of the propeller boss on the shaft tapered end.

.18 *Flanged Connection.* Flanged connection is the coupling methodology, between the shaft and the propeller, achieved by a flange, built in at the shaft aft end, bolted to propeller boss.

.19 *Alternative Means*. Shafting arrangements such as, but not limited to, an approved Condition Monitoring Scheme and/ or other reliable approved means for assessing and monitoring the condition of the tail shaft, bearings, sealing devices and the stern tube lubricant system capable to assure the condition of the propeller shaft assembly with an equivalent level of safety as obtained by survey methods as indicated in this Section. “TCM” notation requirements in Pt.4, Ch.4 of the Rules are considered as alternative means. Condition monitoring records, as applicable, are to be verified at annual surveys. For shaft surveys, methods and extensions based on the type of lubrication and connection, as indicated in Table 11.2.3 will be applicable.

11.2 Oil Lubricated Shafts or Closed Loop System Fresh Water Lubricated Shafts (Closed System)

11.2.1 Shaft Survey Methods

11.2.1.1 **METHOD 1**

.1 The survey is to consist of:

- Drawing the shaft and examining the entire shaft, seals system and bearings;

- For keyed and keyless connections:

- Removing the propeller to expose the forward end of the taper,

- Performing a non-destructive examination (NDE) by an approved surface crack-detection method all around the shaft in way of the forward portion of the taper section, including the keyway (if fitted).

  For shaft provided with liners the NDE is to extend to the after edge of the liner.

- For flanged connection:

- Whenever the coupling bolts of any type of flange-connected shaft are removed or the flange radius is made accessible in connection with overhaul, repairs or when deemed necessary by the surveyor, the coupling bolts and flange radius are to be examined by means of an approved surface crack detection method.

- Checking and recording the bearing clearances;

- Verification that the propeller is free of damages which may cause the propeller to be out of balance;

- Verification of the satisfactory conditions of inboard and outboard seals during the re-installation of the shaft and propeller;

- Recording the bearing wear down measurements (after re-installation)

11.2.1.2 **METHOD 2**

.1 The following are to be verified and found satisfactory as a pre-requisite for application of Method 2:

- Review of service records;

- Review of test records of:

- Lubricating Oil analysis (for oil lubricated shafts), or

- Fresh water sample test (for closed system fresh water lubricated shafts);

- Oil sample examination (for oil lubricated shafts), or fresh water sample test (for closed system fresh water lubricated);

- Verification that there are no repairs by grinding or welding of shaft and/or propeller.

.2 The survey is to consist of:

- For keyed and keyless connections:

- Removing the propeller to expose the forward end of the taper;

- Performing a non-destructive examination (NDE) by an approved surface crack-detection Method all around the shaft in way of the forward portion of the taper section, including the keyway (if fitted).

- For flanged connection:

- Whenever the coupling bolts of any type of flange-connected shaft are removed or the flange radius is made accessible in connection with overhaul, repairs or when deemed necessary by the surveyor, the coupling bolts and flange radius are to be examined by means of an approved surface crack detection Method.

- Checking and recording the bearing wear down measurements;

- Visual Inspection of all accessible parts of the shafting system;

- Verification that the propeller is free of damages which may cause the propeller to be out of balance;

- Seal liner found to be or placed in a satisfactory condition;

- Verification of the satisfactory re-installation of the propeller including verification of satisfactory conditions of inboard and outboard seals.

11.2.1.3 **METHOD 3**

.1 The pre-requisites for application of Method 3, are the same as those given in 11.2.1.2.1

.2 The survey is to consist of

- Checking and recording the bearing wear down measurements;

- Visual Inspection of all accessible parts of the shafting system;

- Verification that the propeller is free of damage which may cause the propeller to be out of balance;

- Seal liner found to be or placed in a satisfactory condition;

- Verification of the satisfactory conditions of inboard and outboard seals.

11.2.2 Shaft Extension Surveys – Extension Types

11.2.2.1 *Extension up to 2.5 years*

.1 In addition to the pre-requisites listed in 11.2.1.2.1, the Chief Engineer is to confirm that the shafting arrangement is in good working condition, for extension up to 2.5 years to be applied.

.2 The survey is to consist of:

- Checking and recording the bearing wear down measurements, as far as practicable;

- Visual inspection of all accessible parts of the shafting system;

- Verification that the propeller is free of damages which may cause the propeller to be out of balance;

- Verification of the effectiveness of inboard seal and outboard seals.

11.2.2.2 *Extension up to 1 year*

.1 In addition to the pre-requisites listed in 11.2.2.1.1, the previous wear down and/ or clearance recordings are also to be reviewed, for extension up to 1 year to be applied.

.2 The survey is to consist of:

- Visual inspection of all accessible parts of the shafting system;

- Verification that the propeller is free of damage which may cause the propeller to be out of balance;

- Verification of the effectiveness of inboard seal and outboard seals.

11.2.2.3 *Extension up to 3 months*

.1 The pre-requisites listed in 11.2.2.2.1 above are to be verified and found satisfactory for extension up to 3 months to be applied.

.2 The survey is to consist of:

- Visual inspection of all accessible parts of the shafting system;

- Verification of the effectiveness of the inboard seal.

11.2.3 Methods for oil lubricated shafts and fresh water lubricated shafts (closed loop systems) and shaft survey intervals.

Details of the applicable methods for flanged propeller connection, keyless propeller connection and keyed propeller connection and survey intervals are listed in Table 11.2.3. The types of survey extensions applicable, are also listed in Table 11.2.3.

Table 11.2.3 : Survey Intervals (Closed Systems)

Oil Lubricated [Survey Notation SH (OL)]			
	Flanged Propeller Coupling	Keyless Propeller Coupling	Keyed Propeller Coupling[b]
Every five years[a]	Method 1 or Method 2 or Method 3	Method 1 or Method 2 or Method 3[c]	Method 1 or Method 2
Extension 2.5 Y	Yes[d]	Yes[d]	Yes[d]
Extension 1 Y	Yes[e]	Yes[e]	Yes[e]
Extension 3 M	Yes[f]	Yes[f]	Yes[f]
Fresh Water Lubricated [Survey Notation SH (FW-C)]			
	Flanged Propeller Coupling	Keyless Propeller Coupling	Keyed Propeller Coupling[b]
Every five years[a]	Method 1[g] or Method 2 or Method 3	Method 1[g] or Method 2 or Method 3	Method 1[g] or Method 2
Extension 2.5 Y	Yes[d]	Yes[d]	Yes[d]
Extension 1 Y	Yes[e]	Yes[e]	Yes[e]
Extension 3 M	Yes[f]	Yes[f]	Yes[f]

General Notes :

For surveys (Method 1, or Method 2, or Method 3) completed within 3 months before the shaft survey due date, the next period is to start from the shaft survey due date.

The extension survey is normally to be carried out within 1 month of the shaft survey due date and the extension counts from the shaft survey due date. If the extension survey is carried out more than 1 month prior to the shaft survey due date, then the period of extension counts from the date on which the extension survey was completed.

Methods 1, 2 & 3 are described in 11.2.1.1, 11.2.1.2 and 11.2.1.3 respectively. The pre-requisites for each Method are to be satisfactorily verified, prior surveys.

Survey methods 1, 2 and 3 would normally require drydocking of the vessel. Extension surveys may be carried out without dry-docking by in water surveys.

a) Unless an Extension type (Extension 2.5 Y, Extension 1 Y, Extension 3 M) is applied in between.

b) Method 3 is not allowed.

c) The maximum interval between two surveys carried out according to Method 1 or Method 2 is not to exceed 15 years, except in the case when one extension for no more than three months is granted.

d) No more than one extension can be granted. No further extension of other type can be granted.

e) No more than two consecutive “one year extensions” can be granted.In the event an additional extension is requested the requirements of the “2.5 years extension” are to be carried out and the shaft survey due date, prior to the previous extension(s), is extended for a maximum of 2.5 years.

f) No more than one “three months extension” can be granted. In the event an additional extension is requested, the requirements of “one year extension” or “2.5 years extension” are to be carried out and the shaft survey due date prior to the previous extension is to be extended for a maximum of one year or 2.5 years.

g) The maximum interval between two surveys carried out according to Method 1 is not to be more than 15 years. An extension for no more than three months can be granted.

11.3 Water Lubricated Shafts (Open Systems)

11.3.1 Shaft Survey Methods

11.3.1.1 **METHOD 4**

The survey is to consist of

- Drawing the shaft and examining the entire shaft (including liners, corrosion protection system and stress reducing features, where provided), inboard seal system and bearings.

- For keyed and keyless connections:

- removing the propeller to expose the forward end of the taper,
- performing a non-destructive examination (NDE) by an approved surface crack detection method all around the shaft in way of the forward portion of the taper section, including the keyway (if fitted). For shaft provided with liners the NDE is to be extended to the after edge of the liner

- For flanged connection:

- Whenever the coupling bolts of any type of flange-connected shaft are removed or the flange radius is made accessible in connection with overhaul, repairs or when deemed necessary by the surveyor, the coupling bolts and flange radius are to be examined by means of an approved surface crack detection method.

- Checking and recording the bearing clearances.

- Verification that the propeller is free of damage which may cause the propeller to be out of balance.

- Verification of the satisfactory conditions of inboard seal during re-installation of the shaft and propeller.

11.3.2 Shaft Extension Surveys – Extension Types

11.3.2.1 *Extension up to 1 year*

.1 Pre-requisites to be verified and found satisfactory in order to apply extension up to 1 year are as follows:

- Review of the previous clearance recordings;

- Service records;

- Verification that there are no repairs by grinding or welding of shaft and/or propeller;

- Confirmation from the Chief Engineer that the shafting arrangement is in good working condition.

.2 The survey is to consist of:

- Visual Inspection of all accessible parts of the shafting system;

- Verification that the propeller is free of damages which may cause the propeller to be out of balance;

- Checking and recording the clearances of bearing;

- Verification of the effectiveness of the inboard seal.

11.3.2.2 *Extension up to 3 months*

.1 The pre-requisites listed in 11.3.2.1.1 are to be satisfactorily verified, for extension up to 3 months to be applied.

.2 The survey is to consist of:

- Visual Inspection of all accessible parts of the shafting system;

- Verification that the propeller is free of damage which may cause the propeller to be out of balance;

- Verification of the effectiveness of the inboard seal.

11.3.3 Methods for water lubricated shafts (open loop systems) and shaft survey intervals, along with applicable details are given in Table 11.3.3.

11.3.3.1 For single shaft operating exclusively in fresh water, single shaft provided with adequate means of corrosion protection, corrosion resistant shafts and multiple shaft arrangements; survey intervals up to 5 years may be allowed. Applicable details along with maximum permissible survey intervals are given in Table 11.3.3.

11.3.3.2 Shaft arrangements, other than the configurations listed in 11.3.3.1, are to be surveyed according to Method 4, every 3 years. Applicable details along with maximum permissible survey intervals are also given in Table11.3.3.

Table 11.3.3 : Survey Intervals (Open Systems)

Single shaft operating exclusively in fresh water [Survey Notation SH (S-FW-O)] **Single shaft provided with adequate means of corrosion protection, single corrosion resistant shaft** [Survey Notation SH (S-CP-O)] **All kinds of multiple shaft configurations** [Survey Notation SH (M-O)]		**Other Shaft Configurations** [Survey Notation SH (S-O)]	
All kinds of propeller couplings[d]			
Every five years[a]	Method 4	Every three years[a]	Method 4
Extension 1 Y	Yes[b]	Extension 1 Y	Yes[b]
Extension 3 M	Yes[c]	Extension 3 M	Yes[c]

General Notes :

For surveys (Method 4) completed within 3 months before the shaft survey due date, the next period will start from the shaft survey due date.

The extension survey is to be normally carried out within 1 month of the shaft survey due date and the extension counts from the shaft survey due date. If the extension survey is carried out more than 1 month prior to the shaft survey due date, then the period of extension is to count from the date on which the extension survey was completed.

Method 4 is described in 11.3.1.1. The pre-requisites for the Method are to be satisfactorily verified, prior survey.

Survey method 4 would normally require dry docking of the vessel. Extension surveys may be carried out without dry-docking by in water surveys.

In case of multiple shaft configurations, if there is a failure to one shaft system, vessel is to be dry-docked for carrying out surveys of all shaft systems, and repairs as necessary.

Notes :

a) Unless an Extension type (Extension 1 Y, Extension 3 M) is applied in between.

b) No more than one extension can be granted. No further extension, of other type, can be granted.

c) No more than one extension can be granted. In the event an additional extension is requested the requirements of the one year extension are to be carried out and the shaft survey due date prior to the previous extension is to be extended for a maximum of one year.

d) For keyless propeller connections the maximum interval between two consecutive dismantling and verifications of the shaft cone by means of non-destructive examination (NDE) is not to exceed 15 years.

11.4 Steerable and azimuth thrusters

The survey is to normally comprise of examination of the following:

a) Exposed parts including attachment to the hull.

b) The following items upon removal of propeller:

- propeller shaft threaded end and nut;
- cone, key and keyway including examination by an efficient crack detection method of fore part of the shaft cone;
- sealing glands.

c) Lubricating oil analysis (to include wear particle analysis) records to detect possible wear of internal gears and bearings.

d) Internal gears and control gears as far as practicable through hand holes or limited opening of controlling device.

If the above checks are not satisfactory, complete dismantling of the internal parts may be required.

11.5 Vertical axis propellers

The survey is to normally comprise of examination of the following:

a) Exposed parts.

b) Tightness of the oil glands and the backlash of the gears from outside by action on the blades.

c) Gears, as far as practicable through hand holes and observation ports.

d) Control gear for proper functioning.

e) Lubricating oil analysis (to include wear particle analysis) records to detect possible wear of internal gears and bearings.

If the above checks are not satisfactory, complete dismantling of the internal parts may be required.

11.6 Water jet systems

The survey is to normally comprise of examination of the following:

a) Impeller, shaft and bearing clearances.

b) Sealing glands.

c) Nozzle assembly.

d) Control and reversing gear.

e) Suction grids.

If the above checks are not satisfactory further dismantling may be required.

11.7 Controllable pitch propellers

Where controllable pitch propellers are fitted, in addition to the survey requirements of the shafts, the working parts and control gear are to be opened up sufficiently to enable Surveyors to examine them at each shaft survey. The survey is to include the following:

a) Analysis of hydraulic oil including wear particle analysis.

b) Propeller blades and hub including crack detection of blade root, flange and blade securing arrangements.

c) Examination of seals, carrier bearings, crank pin ring, fillets, blade openings in the boss and blade bolts, upon removal of at least one blade.

d) Examination of distribution box seal and bearings.

e) Verification upon re-assembly of, servomechanism and hydraulic test of hub and hydraulic piping including pitch controls together with limit stops.

Section 12

Surveys of Inert Gas Systems

12.1 General

12.1.1 Inert gas systems installed on board ships intended for the carriage of oil or liquid chemicals in bulk or liquefied gases are to be surveyed periodically as detailed in Sec.1 of this Chapter.

12.2 Annual surveys

12.2.1 Following are to be examined at each Annual Survey:

a) external examination of the condition of all piping and components for signs of corrosion or gas/effluent leakage.

b) verification of the proper operation of both inert gas blowers.

c) checking the operation of scrubber room ventilation system.

d) checking of deck water seal for automatic filling and draining and checking for the presence of water carry over and checking the condition of non-return valve.

e) examination of the operation of all remotely or automatically operated valves and, in particular, the flue gas isolating valve(s).

f) checking the interlocking feature of soot blowers.

g) checking that the gas pressure regulation valve automatically closes when the inert gas blowers are secured.

h) checking as far as practicable, the following alarms and safety devices of the inert gas systems using simulated conditions where necessary:

- high oxygen content of gas in the inert gas main;
- low gas pressure in the inert gas main;
- low pressure in the supply to the deck water seal;
- high temperature of gas in the inert gas main;
- low water pressure to the scrubber;
- accuracy of portable and fixed oxygen measuring equipment by means of calibration gas;
- high water level in scrubber;
- failure of the inert gas blowers;
- failure of the power supply to the automatic control system for the gas regulating valve and to the instrumentation for continuous indication and permanent recording of pressure and oxygen content in the inert gas main;
- high pressure of gas in the inert gas main.
- Surveys carried out by the National Authority of the country in which the ship is registered would normally be accepted as meeting these requirements, at the discretion of the Surveyor.

12.2.2 Checking when practicable, the proper operation of the inert gas system on completion of the checks listed in 12.2.1 h).

12.3 Special surveys

12.3.1 At each Special Survey of the inert gas system, the inert gas generator, scrubber and blower are to be opened out as considered necessary and examined. Gas distribution lines and shut off valves, including soot blower interlocking devices are to be examined as considered necessary. The deck seal and non-return valve is to be examined. Cooling water systems including the effluent piping and overboard discharge from scrubbers are to be examined. All automatic shut down devices and alarms are to be tested. The complete installation is to be tested under working conditions on completion of Survey.

12.3.2 When, at the request of an Owner, it has been agreed by IRS that Complete Survey of the inert gas system may be carried out on the continuous basis, the various items of the system are to be opened up for survey in rotation, so far as practicable, to ensure that the interval between consecutive examinations of each item will not exceed five years. In general one-fifth of the machinery is to be examined each year.

12.3.3 If any examination during Continuous Survey reveals defects, further parts are to be opened up and examined as considered necessary by the Surveyor, and the defects are to be made good to his satisfaction.

Section 13

Surveys of Vessels with Refrigerated Cargo Installations

13.1 General

13.1.1 Refrigerated Cargo Installation, on ships which have been assigned the character HY, are to be surveyed periodically as indicated in Sec.1 of this Chapter.

13.1.2 Where a Refrigerated Cargo Installation holds dual classification with IRS, additional Periodical Survey requirements, if any, of the corresponding Society would also be applicable.

13.2 Periodical surveys

13.2.1 A Special Survey is to be held at 5 yearly intervals. When due to special circumstances a Special Survey is commenced prior to its due date, the Survey is to be completed within a period not exceeding 9 months and not later than the expiry date of the classification certificate including any postponement that may be granted by IRS.

13.2.2 If a vessel at the time classification certificate expires is not in a port in which it is to be surveyed, IRS may upon Owners request extend the validity of the classification certificate for a period not exceeding 3 months. Such requests will only be considered to enable the vessel to complete its voyage to the port of survey and only in cases where it is considered proper and reasonable to do so.

13.3 Continuous special survey of refrigerating installation

13.3.1 At the request of the Owners and upon receipt of a satisfactory proposed arrangement, IRS may give consideration to a

system of Continuous Survey of refrigerating installation (CS HY) being carried out whereby the requirements of Special Surveys as detailed in 13.5 and 13.6 are completed within a 5 year period. In such case the various items of machinery should be opened out for Survey in rotation, so far as practicable, to ensure that the interval between consecutive examination of each item does not exceed five years and it is expected that approximately an equal proportion of the surveyable items would be subjected to Survey each year.

13.3.2 In case any defects are revealed during these examinations, further parts may have to be opened up as considered necessary by the Surveyor.

13.4 Annual survey requirements

13.4.1 Log Books or other records are to be examined to ascertain that the installation has been working satisfactorily. Any indications of breakdowns or defects, during the previous 12 months, are to be noted and reported. Whenever possible, an examination of the refrigerating plant is to be made upon arrival of the ship at the port of discharge before refrigerated cargo is unloaded.

13.4.2 Refrigerated chambers/holds are to be examined after careful cleaning. A detailed examination is to be made of these spaces. For this purpose, removable panels and covers are to be removed and if appropriate, spot checks of the insulation are to be carried out for condition and shrinkage. If there is trace of damp or leakage of the insulation from whatever source (fuel tank, pipe work, pipe coils, scuppers, etc.) or any trace of deterioration in the insulation or its lining, of the air ducts or air coolers, a more thorough examination is to be made and necessary measures taken to correct the defects thus ascertained.

13.4.3 Where the cargo is refrigerated by circulating air, the proper air tightness of the air ducts, air cooler casings, hatch covers, doors, butterfly dampers in ventilation ducts is to be checked. The insulation of the lining joints should also be carefully checked.

13.4.4 The bilges are to be cleaned and suction pipes, sounding pipes and scupper non-return valves examined. The Surveyor should satisfy himself that all scuppers draining the chambers and cooler trays are in good working order.

13.4.5 Air cooler coils, cooling grids and valves are to be examined and the Surveyor should satisfy himself that no pipe is partially or completely choked and that the valves are in good working order.

13.4.6 Brine coils are to be examined whilst under a pressure of 1.5 times working pressure or 3 bar whichever is greater.

13.4.7 Primary refrigerant cooler coils and grids are to be examined whilst under the refrigerant pressure prevailing in the system at the time of the Survey with the plant at rest and the regulating valves opened just sufficient to obtain an approximate balance of pressure throughout the system and to avoid accumulation of liquid in the coils or grids.

13.4.8 The shells of condensers, evaporators, separators, receivers, and other pressure vessels are to be examined as far as practicable. Any evidence of excessive corrosion of water end covers of "shell and tube" and "double-pipe" type condensers is to be investigated.

13.4.9 Primary refrigerant gas and liquid pipes, condenser cooling water piping and valves are to be examined as far as possible.

13.4.10 Any evidence of dampness or deterioration of the insulation which could lead to external corrosion of the vessels or other parts mentioned, is to be investigated and necessary measures are to be taken to correct the defects thus ascertained.

13.4.11 A general examination is to be made of the fans, their motors, control gear and the insulation resistance is to be measured. The insulation resistance is not to be less than 100,000 ohms. The generating plant supplying electric power is to be examined generally with a view to ascertaining that the plant is being efficiently maintained.

13.4.12 The thermometers for measuring the chamber air suction and air delivery temperatures are to be examined. If repairs and renewals are carried out, the thermometers are afterwards to be checked for accuracy.

13.4.13 A Survey book or other permanent record is to be kept on board the ship to show the date of examination of various parts. This is to be available to the Surveyor at all times and is to be signed by the Surveyor on each occasion after the Survey.

13.5 Requirements of first special survey

13.5.1 Each reciprocating compressor, including those provided for sub- cooling the primary refrigerant, is to be opened up. Cylinder bores, pistons, piston rods, connecting rods, valves and seats, glands, relief devices, suction filters and lubricating arrangements are to be examined. Crankshafts are to be examined, but crankcase glands and the lower halves of main bearings need not be exposed if the Surveyor is satisfied as to alignment and wear.

13.5.2 For screw-type compressors, the period before opening up may be extended to 6 years or 25000 running hours, whichever is the earlier.

13.5.3 Where there is a programme of replacement instead of Surveys on board, alternative Survey arrangements will be considered. Each case will be given individual consideration.

13.5.4 Refrigerant condenser cooling water pumps, including standby pump(s) which may be used on other services, are to be opened up and their working parts exposed.

13.5.5 Brine and primary refrigerant pumps are to be opened up and their working parts exposed. Special consideration will be given to Survey requirements for primary refrigerant pumps of the hermetically sealed type.

13.5.6 The water end covers of "shell and tube" and "double-pipe" type condensers are to be removed for examination of the tubes, tube plates and covers.

13.5.7 The shells and connections of "shell-and-tube" and "double-pipe" type condensers and evaporators, separators, receivers, driers, filters and other pressure vessels, and the coil terminals of "coil-in-casing" type condensers and evaporators, are to be examined as far as practicable.

13.5.8 In the case of pressure vessels covered by insulation, any evidence of dampness or deterioration of the insulation which could lead to external corrosion of the vessels or their connections is to be investigated.

13.5.9 Sufficient insulation is to be stripped from insulated pressure vessels to allow the condition of the vessels and their connections to be ascertained. Care is to be taken that in replacement of the insulation, the vapour sealing of the outer covering is made good.

13.5.10 Sufficient insulation is to be stripped from pipes carrying the refrigerant at various points of the system both outside and inside the insulated chambers to permit the condition of the pipes to be ascertained. Sections of piping exposed are to include locations where lengths of piping have been connected by screwed couplings or butt welding. Care is to be taken that when ungalvanized portions of the piping in way of joints have been exposed they be suitably coated and taped, after pressure testing, to prevent corrosion. On replacement of the insulation, the vapour sealing of the outer covering is to be made good.

13.5.11 The Surveyor is to satisfy himself that all pressure relief valves and/or safety discs throughout the refrigerating plant are in good order. However, no attempt is to be made to test primary refrigerant pressure relief valves on board ship.

13.5.12 Sea connections to refrigerant condensers cooling water pumps are to be opened up on the occasion of the hull and/or main machinery Special Survey.

13.5.13 The electric motors driving refrigerant compressors, pumps and fans, together with their control gear and cables, are to have their insulation resistance tested and this is to be not less than100000 ohms between all insulated circuits and earth. The installation may be subdivided to any desired extent by opening switches, removing fuses or disconnecting appliances for the purpose of this test.

13.5.14 The fittings on switchboards and Section boards are to be examined, and over-current protective devices and fuses are to be inspected to verify that they provide suitable protection for their respective circuits.

13.5.15 Any arrangements fitted to disconnect automatically the excess non- essential load when the electrical generators are overloaded are to be examined to ascertain that the circuits for cargo refrigerating machinery are included in the last group to be disconnected.

13.5.16 All automatic controls and alarms are to be tested.

13.5.17 Sufficient air trunking and insulation lining is to be stripped from the chamber's overhead and vertical surfaces to allow the condition of the insulation, insulation linings, grounds, supports, hangers and fixtures which support the insulation, grids, meat rails, etc., to be ascertained. Care is to be taken that on replacement the ducts and linings are sealed against air blowing into the insulation, or against moisture ingress from refrigerated cell or space atmosphere.

13.5.18 Sufficient tank top insulation is to be stripped to allow the condition of the grounds and inner insulation lining to be ascertained.

13.5.19 Due consideration is to be given to the type of insulation used in the holds and chambers when determining the amount of insulation lining to be removed as detailed in 13.5.17 and 13.5.18. Where organic foam insulants have been used, including foamed "in situ", or other insulants in slab form, the Surveyor should use his discretion regarding the removal of linings if he is able to satisfy himself that the condition of the insulation is good by means of test bore holes.

13.5.20 Under normal circumstances, the condition of hold and chamber insulation, grounds, etc., can be ascertained when the Special Survey of the ship's steel structure is being held.

13.5.21 Arrangements made for defrosting air coolers, and for draining condensate from trays below coolers, are to be examined to ascertain that they are in working order.

13.5.22 Any air refreshing arrangements are to be examined.

13.6 Subsequent special surveys

13.6.1 In addition to the requirements for first Special Survey the following items, as mentioned in Table 13.6.1, are to be tested and examined.

13.7 Loading port surveys

13.7.1 Where the Owner or his authorised representative requests for a Loading Port Survey, a Survey as detailed below is to be carried out at the Loading Port.

13.7.2 In the case of ships engaged on voyages of less than two months duration, a loading port certificate will be considered as valid for two months, provided the cargoes carried are of such a nature as not to damage the insulation or appliances in the insulated chambers, nor to affect by taint or mould the refrigerated cargoes loaded during that period.

13.7.3 If a vessel loads at more than one port, Loading Port Survey at the first Loading Port would only be required provided all the chambers which are intended to be loaded with refrigerated cargo during the voyage are offered for examination and no general cargo is subsequently loaded in these chambers prior to loading of refrigerated cargo.

13.7.4 The refrigerated cargo spaces are to be examined in an empty state prior to loading the cargo. The Surveyor is to satisfy himself that the spaces are clean and free from odour which may adversely affect the cargo to be loaded. The brine or other refrigerant pipe grids, cooler coils and connections should show no trace of leakage. The Surveyor should also satisfy himself that the fixed cargo battens on the vertical surfaces are in good order, that cargo gratings or dunnage battens are provided for the floors or decks and that no damage has been sustained to the insulation or its lining prior to the loading of the refrigerated cargo.

13.7.5 Any indication of defective insulation not considered to warrant immediate attention should be noted and specially reported.

13.7.6 The scuppers and bilges in the refrigerated cargo spaces are to be clean and dry and the liquid seals should be primed.

13.7.7 The Surveyor will check whether the entire refrigerated cargo installation operates satisfactorily and he will record the temperatures in the cargo spaces.

13.7.8 The proper operation of air duct couplings for connecting refrigerated containers on the ship's own refrigerating installation has to be checked. If refrigerated containers are coupled to the air ducts during the on board Survey, the tight sealing effect of the couplings is also to be checked.

13.7.9 It is to be clearly understood that the certificate issued for the carrying out of a Loading Port Survey is not in respect of the cargo to be loaded or the manner in which it is to be stowed.

Table 13.6.1 : The requirements of special surveys			
Item	**Dichlorodifluoro-methane (R12)**	**Ammonia (NH_3) or Monochlorodifluoro-methane R22)**	**Carbondioxide (CO_2)**
"Shell and tube" type gas condensers or gas evaporators (brine coolers) (primary refrigerant in the shell)	7 bar	14 bar	-
	Water or brine end covers to be removed and shell pneumatically tested with the refrigerant or air or a mixture of inert gas and refrigerant to the above pressures		
"Coil in casing" type gas condensers	17 bar	70 bar	140 bar
	Where it is impracticable to remove the coils they may be examined and tested in place		
"Coil in casing" type gas evaporators (brine coolers)	14 bar	35 bar	105 bar
	Where it is impracticable to remove the coils they may be examined and tested in place		
Primary refrigerant chamber grids or air cooler coils	7 bar	10 bar	70 bar
"Shell and tube" type gas evaporators (brine coolers) (brine is in the shell)	Primary refrigerant end covers are to be removed and tested		
	Shell to be hydraulically tested to twice the design pressure but not less than 3 bar		

Section 14

Planned Maintenance System

14.1 General

14.1.1 An approved Planned Maintenance System on board a ship may be accepted in lieu of the regular surveys by IRS Surveyors on CSM basis provided the requirements of this section are complied with.

Any item not covered by PMS is to be surveyed and credited in the usual way. In general the intervals for PMS is not to exceed those specified for CSM survey. However, for components where the maintenance is based on running hours longer intervals may be accepted as long as the intervals are based on the manufacturer's recommendations. PMS is to be programmed and maintained by a computerised system.

14.1.2 A vessel complying with the requirements of section will be assigned the PMS survey notation.

14.1.3 The survey system (PMS) is to be approved by IRS before being implemented. When the system is implemented machinery inspection may be based on calendar or running hours calling for items to be opened for inspection and overhaul at specified periods.

14.1.4 Other items of machinery and components, which are not subjected to an approved condition monitoring system and part of PMS are required to be surveyed at intervals not greater than those specified for CSM.

14.1.5 Chief Engineer of the vessel will be responsible and in-charge of PMS and is to be approved by IRS as per Pt.1, Ch.2, 1.4.3.

14.1.6 The survey of machinery and components (defined at 14.2) & covered by the PMS may be carried out by the Chief Engineer. Survey of machinery and components defined in 14.3 is not eligible to be surveyed by the Chief Engineer.

14.1.7 At the time of the Annual Classification Survey an audit of the planned maintenance system will be carried out to ensure that the

system is being correctly operated in accordance with the conditions of approval.

14.1.8 When the annual audit is held, a confirmatory survey will be carried out for those items to be credited which have been examined by the Chief Engineer under the planned maintenance system during the preceding year.

14.2 Machinery acceptable for survey by Chief Engineers under Planned Maintenance System

14.2.1 The following machinery may be surveyed by the Chief Engineer under PMS:

- Main engine cylinder covers.
- Main engine valves and valve gears.
- Main engine cylinder liners.
- Main engine pistons and piston rods.
- Main engine connecting rods, crossheads, top end bearings, guides, gudgeon pins and bushes.
- Main engine crankshafts and bearings (multiple engine installations only).
- Main engine fuel injection pumps and fuel booster pumps.
- M.E. Scavenge pumps, blowers and air coolers.
- Main engine detuners, dampers and balancer units.
- Main engine camshaft and camshaft drive.
- Main propulsion steam turbines (casing, rotor and blading) at alternate surveys provided the monitoring is provided and vibration measurements and full power trials are carried out at the time of the survey in the presence of a Surveyor.
- M.E. driven pumps, e.g. bilge, lubricating oil, cooling water.
- Independently driven pumps and associated motors and cables where insulation resistance readings are supplied e.g. bilge, ballast, fresh water cooling, sea water cooling, lubricating oil, oil fuel transfer.
- M.E. fresh water and lubricating oil coolers.
- Low pressure heaters used in fuel oil systems of internal combustion engines
- Main and auxiliary condensers/drain coolers.
- Air compressors including their safety devices.
- Windlass and windlass machinery.
- Forced or induced draught fans.
- Auxiliary oil and steam engines including their coolers and pumps (provided the number of generating sets is such that all services essential to the propulsion and safety of the ship, also the preservation of refrigerated cargo, can be supplied when any two sets are not working. One of these sets can be overhauled while the other remains as "stand-by").
- Intermediate shafts.
- Main engine thrust bearing.

  Note : In cases where torsional vibration characteristics indicate that there is no susceptibility to damage as a result of uneven firing and condition monitoring equipment is installed, a special arrangement may be granted whereby the Chief Engineer is permitted to survey the main engine crankshafts and bearings on single engine installations provided a modified confirmatory survey is carried out by IRS Surveyors at the time of the annual audit as follows:

  - Check condition-monitoring records.
  - Check bearing clearances where possible.
  - Check for signs of wiped or broken white metal in crankcase.
  - Check witness marks of shrink fits.

- Check bed plate structure inside and outside.
- Obtain Chief Engineer's statements regarding crankpins, journals and bearings.

Ships fitted with Approved Inert Gas Systems

- Scrubber units
- Blowers
- Independent gas generators.

14.3 Machinery not acceptable for Survey by Chief Engineers

- Main engine crankshaft and bearings in single engine installations where special requirements in 14.2.1 are not complied with
- Reduction/increase gearing, flexible coup-lings and clutches.
- Holding down bolts and chocks
- Crankcase doors, crankcase and scavenge relief devices
- Boilers and all other pressure vessels
- Boiler fuel oil heaters
- Steam pipes and starting air pipes
- Maneuvering valves and bulkhead stop valves
- Steering machinery

- Pumping arrangements for Bilge/Ballast/ Fuel Oil/Fresh Water/Sea water/Lub.Oil/ Fire
- Electrical equipment other than that defined in 14.2.1
- Propellers
- Screwshafts
- Sea connections
- M.E. controls, bridge, centralised or automatic and controls in unmanned machinery spaces
- Engine trial
- First start arrangements trial.

Ships fitted with Approved Inert Gas Systems

- All other I.G. system components not listed in 14.2.1.

14.4 System Administration

14.4.1 The Owner is to make a formal request to IRS providing the documentation and information detailed in 14.4.2 below for approval of the system.

14.4.2 The documentation and information to be submitted is to include the following:

a) A description of the system and its application onboard and organizational interface identifying the areas of responsibility ashore and the people responsible for the PMS onboard.

b) The list of items of machinery, equipment and components to be considered for inclusion in the PMS. The list is to be same in terms of description and identification with the identification system adopted by IRS.

c) Time schedules and scope of the maintenance procedures for each item listed in b), including acceptable limit conditions of the parameters to be monitored on items of opened up machinery, based on the manufacturer's recommendations or recognized standards. These are to be laid down in appropriate PMS sheets.

d) The documentation flow and filing procedure.

   This is to include a system for reporting to Owners, records to be maintained onboard and at Owner’s head quarters.

e) A list of all personnel likely to be in-charge of the PMS system.

14.4.3 Computerized system requirement

14.4.3.1 The access to and updating the maintenance documentation and the maintenance program is to be permitted by the Chief Engineer or other authorized person only.

14.4.3.2 The system is to be suitably protected by suitable password access w.r.t. alterations to maintenance schedules, list of items under PMS and noting of damages.

14.4.3.3 The computerized system is to include a backup procedure, which is activated at regular intervals.

14.4.3.4 The functional applications of these computerized systems are to be approved by IRS.

14.4.4 Information to be available onboard.

a) All the documentation listed in 14.4.2, duly updated.

b) Maintenance instructions for each machinery, as applicable (supplied by the manufacturer or the shipyard).

c) Reference documentation (trend investi-gation procedures etc.).

d) The records of the maintenance performed, including conditions found, repairs and renewals carried out.

14.5 System Implementation

14.5.1 After the PMS documentation has been approved, an Implementation survey is to be carried out by IRS Surveyors.

14.5.2 Upon successful completion of the Implementation Survey, the PMS is considered approved and the survey notation PMS is assigned to the vessel and entered in the Register of Ships.

14.5.3 The PMS is retained throughout the class period of the vessel provided that :

a) An annual report covering the year’s service is submitted to IRS detailing the list of items of machinery and components which were subjected to preventive

maintenance in the period under review, together with preventive maintenance sheets and any changes to PMS documentation.

b) An annual audit of the PMS is carried out.

c) Any change to the PMS is submitted to IRS for approval.

14.5.4 The survey agreement for machinery according to PMS will be withdrawn by IRS if the PMS is not satisfactorily operated in terms of improperly maintained records or unsatisfactory condition of machinery or failure to observe the agreed intervals between overhauls.

14.5.5 The Owner may discontinue the PMS at any time by informing IRS in writing. In such a case the items that have been inspected under the PMS since last annual audit may be credited for IRS records at the discretion of the attending surveyor carrying out confirmatory surveys.

14.5.6 In case of change of management of the ship or transfer of class approval is to be reconsidered.

14.6 System Surveys

14.6.1 **Implementation Survey** is to be carried out by an IRS Surveyor within one year from the date of the documentation approval. The scope of this survey is to verify that :

a) The PMS is implemented as per the approved documentation and is suitable to the type and complexity of machinery and systems onboard.

b) The documentation required for the annual audit is available and the adopted system is able to produce such a report.

c) The requirements of surveys and testing for continuing the class status are complied with.

d) The shipboard personnel are familiar with the PMS procedures including documentation.

14.6.2 **Annual Audit** is to be carried out once the PMS is implemented and approved, to verify the continued compliance with the documented PMS. The annual audit is carried out in conjunction with the annual class surveys. The scope of the audit is to be as given in the following :

a) The Surveyor is to verify that the PMS is correctly operated and that all items (due for survey in the relevant period) have actually been surveyed in due time.

b) The Surveyor is to verify that the machinery has been functioning satisfactorily upon review of the maintenance and performance records since the previous survey or audit and where needed necessary measures have been taken in response to machinery operating parameters exceeding acceptable limits and that the overhaul intervals have been observed.

c) A report detailing overhaul/repairs carried out and spare parts used on items in the list of surveyable items is to be presented by the Chief Engineer. Any machinery part or component, which has been replaced by a spare due to damage, is to be retained onboard and submitted to the attending surveyor's examination.

d) The attending surveyor after verification of records on board for the identification details of the Chief Engineers who have undertaken the maintenance activity and prepared the reports given in c), for compliance with the approved PMS, and upon satisfactory general examination and confirmatory surveys will credit the items for survey.

e) Written reports of break down or malfunction are to be made available.

f) If the surveyor is not satisfied with results of the PMS i.e. with degree of accuracy as regards the maintenance records and/or the general condition of the machinery, a report will be forwarded to IRS recommending that the special arrangements dealing with machinery surveys be suspended.

g) Upon satisfactory completion of the annual audit the surveyor confirms the validity of the PMS by crediting the PMS Annual audit.

14.7 Damage and repairs

14.7.1 Damage to components or items of machinery covered by the PMS which may

affect the class is to be reported to IRS immediately. A surveyor will attend on board, survey the damage and on the basis of the survey results decide whether condition of class is to be recommended.

14.7.2 All parts of machinery or components, which need to undergo substantial repairs, are to be surveyed by IRS before, during and after the repairs, as deemed appropriate by the attending surveyor.

14.7.3 In the case of outstanding conditions of class or records of unrepaired damage, which may affect the PMS, the relevant items are to be taken out of the PMS until the conditions of class are dealt with or the repairs are carried out.

Section 15

Surveys - Vapour Control Systems

15.1 For tankers and combination carriers with notations VCS1 and VCS2

15.1.1 Annual Surveys

At each annual survey, the vapour emission control system is to be generally examined for ensuring its satisfactory condition. The survey is also to include:

a) Components and piping

External examination of all components and piping, including isolating / relief valves, means of collecting and draining condensate, means for separation of vapours from non-compatible cargoes and electrical continuity / bonding arrangements.

b) Manifold isolating valve

Confirmation of the proper operation of the vapour manifold isolating valve, including manual operation.

c) Vapour connection flanges

Confirmation of the continuing effectiveness of the “lug and hole” construction of the vapour connection flange(s), or other approved means of preventing misconnection of the loading hose to the vapour collection system.

d) Hoses

Confirmation that any hose used for the conveyance of vapour are in accordance with section 2.1.2 of Pt.5, Ch.29.

e) Inert gas piping

Where inert gas distribution piping is used for vapour collection, confirmation of the continuing effectiveness of the inert gas main isolating valve(s).

f) Cargo gauging system

Correct operation of the closed cargo gauging system for each tank, which is connected to the vapour collection system. Where portable gauging devices are used, the number of devices available are to be equal to the number of tanks that can be simultaneously loaded plus two (2) additional units.

g) Cargo tank venting system

Examination of the cargo tank venting system, including flame screens, where fitted.

h) Alarms and safety devices

Verification of the satisfactory operation of the following alarms and safety devices using simulated conditions, where necessary.

- High vapour pressure in main vapour collection line(s) (not required for tank barges).
- Low vapour pressure in main vapour collection line(s) (not required for tank barges).
- Cargo tank high liquid level.
- Cargo tank overfill (high-high liquid level), where fitted.
- Loss of power to alarm system or, alternatively, failure of tank level sensor circuitry.

- Automatic shutdown system, where fitted.

i) Operation and maintenance records

Examination of the permanent records to verify the satisfactory operation and maintenance of the system. Consideration may be given by the Surveyor for crediting certain items whose satisfactory operation has been properly documented and recorded.

15.1.2 Special surveys

At each Special Survey – Machinery, the following items of the vapour emission control system are to be examined for ensuring its satisfactory condition; in addition to the requirements of annual survey in 15.1.1:

a) Valves

All valves, including cargo tank isolating valves, main vapour line cross-over valves (where fitted for vapour segregation), condensate drain valves, manifold isolating valves, pressure / vacuum relief valves and spill valves / rupture disks (where fitted as additional overfill control devices) are to be examined.

b) Gauging system

The closed gauging system, including portable gauging devices where applicable, is to be examined.

c) Alarms

The independent cargo tank overfill alarms, where fitted, are to be examined.

d) Hoses

Vapour collection system hoses are to be tested for electrical continuity or non-conductivity, as applicable.

e) Inert gas piping

Where inert gas distribution piping is used for vapour collection, deck seals or double block and bleed assemblies are to be examined.

15.2 For tankers and combination carriers with notations VCS1-B and VCS2-B

15.2.1 Annual surveys

At each annual survey, in addition to the requirements of 15.1.1, the following items of the vapour emission control system are to be examined for ensuring its satisfactory condition.

a) Detonation flame arrester

Confirmation that the detonation flame arrester (where fitted) is in satisfactory condition.

b) Vapour manifold

Confirmation that a means of electrical insulation (insulating flange or non-conductive hose, etc.) is provided for the vapour manifold connection.

c) Fixed oxygen analyzer

Confirmation of the accuracy of the fixed oxygen analyzer (required to be fitted within 3 [m] (10 feet) of the vessel's vapour manifold connection) by means of a calibration gas.

d) Vapour blowers / compressors

General examination of any devices (such as compressors or blowers) used to increase the vapour flow rate.

15.2.2 Special surveys

At each Special Survey – Machinery, in addition to the requirements of 15.1.2 and 15.2.1, the following items of the vapour emission control system are to be examined for ensuring its satisfactory condition:

a) Vapour Blowers / Compressors and detonation flame arresters

Vapour blowers / compressors and detonation flame arresters, where fitted, are to be examined, including isolating and relief valves, as applicable.

Section 16

Surveys of Thermal Oil Heating Systems

16.1 General

16.1.1 Thermal Oil Heating systems are to be surveyed at intervals specified in Sec.1, Table 1.1.1 of this Chapter.

16.1.2 At each Survey, the following components are to be examined as applicable.

- Oil fired thermal oil heater/s and Economiser/s are to be examined externally including the heat exchanger coils, insulation and fuel oil burning arrangement
- Circulating pumps
- Dump cooler
- Piping arrangement including control valves and insulation
- Expansion tank and temperature blocking pipe
- Alarms and safety devices
- Thermal oil (analysis results).

16.2 Oil fired thermal oil heater and Economiser

16.2.1 These are to be examined externally for condition of the insulation, cleanliness and their attachment to the ship's structure.

16.2.2 Heat exchanger coils are to be examined for any signs of overheating or corrosion and that they are in a state to ensure proper heat exchange.

16.2.3 Heat exchanger coils are to be subjected to a leakage test, at working pressure and with working fluid.

16.2.4 All mountings are to examined externally and internally, if considered necessary by the surveyor.

16.2.5 Fuel oil burning arrangements are to be examined for proper operation including operation of safety cut-outs.

16.3 Circulating pumps

16.3.1 Circulating pumps are to be examined externally and internally, if considered necessary by the surveyor. The pumps are to be examined for proper operation including automatic change over.

16.4 Dump cooler

16.4.1 Dump coolers are to be examined internally and externally including pressure relief devices.

16.5 Piping arrangement

16.5.1 Complete piping arrangement is to be externally examined including condition of insulation and tested for any leaks.

16.5.2 Pipe fittings and valves are to be examined externally and internally, if considered necessary by the surveyor.

16.6 Expansion tank

16.6.1 Expansion tank including associated save-alls is to be visually examined.

16.6.2 Arrangement to limit the temperature of thermal oil in the expansion tanks (in open vent systems) such as 'Temperature Blocking Pipe" are to be examined for proper operation.

16.6.3 De-aeration arrangement and vent is to be examined for proper operation.

16.7 Alarms and safety devices

16.7.1 Following alarms and safety devices are to be tested:

- Thermal oil high temperature alarm and safety cut out (both in Oil fired heater and Economiser)
- Minimum flow control device
- Minimum flow control alarm and safety cut out (both in Oil fired heater and Economiser)
- Flue gas high temperature alarm and safety cut out (both in Oil fired heater and Economiser)
- Thermal oil leakage alarm and safety cut out (both in Oil fired heater and Economiser)

- Expansion tank low level and low-low level alarm and safety cut out
- Fire Extinguishing arrangements.

16.8 Thermal oil analysis

16.8.1 Thermal oil is to be analysed at regular intervals – atleast annually.

16.8.2 The thermal oil shall be free from harmful contaminants and signs of oxidation or deterioration.

16.9 System operation test

16.9.1 On completion of survey the entire system is to be examined under working conditions with particular attention to signs of leakage and functioning of control and safety devices.

16.10 Repairs

16.10.1 Any defects to alarms and safety devices the proper operation of which may pose a fire risk are to be promptly repaired to the satisfaction of the surveyor.

16.10.2 When repairs or renewals are carried out to system parts or components, these are to be subjected to a pressure test to 1.5 times the working pressure, prior to putting them into service.

Section 17

Surveys of External Fire Fighting Systems

17.1 General

17.1.1 External Fire Fighting Systems on ships assigned with class notations **Agni 1, Agni 2** or **Agni 3** are to be surveyed at intervals specified in Sec. 1 Table 1.1.2 of this chapter.

17.2 Annual survey

17.2.1 The annual survey is to consist of general examination / verification of the items indicated in 17.2.2 to 17.2.7.

17.2.2 Hull

General examination of:

- Maneuvering arrangements including side thrusters and power management system, where fitted
- Floodlights.

17.2.3 Self Protection of the Vessel

Agni 1 :

- General examination and functional verification of fixed water-spraying system.

Agni 2 & Agni 3 :

- General examination of dead lights and external steel shutters
- General examination / functional verification of insulation / water spray system, where fitted.

17.2.4 Water monitors and controls

- General examination of water monitors and their securing arrangements
- Functional verification of water monitors
- General examination and functional verification of remote and local/manual controls.

Additional for Agni 3:

- General examination of foam monitors, foam concentrate tank and associated pumping and piping arrangements including their securing arrangements
- Condition of foam concentrate
- Functional test of the foam monitors
- General examination and functional verification of remote and local/manual controls.

17.2.5 Pumping and piping

- General examination of fire pumps and water spray pumps, associated piping from sea inlets to monitors and required instru-mentation.
- Examination and test of water spray piping over pressure preventing arrangement, where fire pumps also supply to water spray piping.
- Verification of arrangement for prevention of overheating of fire pumps at low delivery rates.

17.2.6 Portable fire fighting equipment

- General examination of hose stations and fire hydrants.
- Ensuring ready availability of hoses and nozzles for the hose stations and their condition.
- Examination of pressure reducing arrangement at hose stations.

Additional for Agni 2 & Agni 3:

- Examination of portable high expansion foam generator including condition of the foam forming liquid.

17.2.7 Fireman's Outfits

- Ensuring ready availability and condition of fireman's outfit including self contained breathing apparatus with spare air bottle.
- General examination of compressor for recharging air bottles.

17.3 Special survey

17.3.1 In addition to the requirements for annual survey the following need to be examined at each special survey.

17.3.2 Internal examination and testing of the following:

- Fire and water spray pumps
- Fire and water spray pump prime movers
- One section of fire and water spray piping to assess the condition of corrosion prevention measures
- Electrical equipment associated with the external fire fighting systems
- Foam tank to the extent possible.

Section 18

Surveys of Diving Support Vessels and Diving Systems

18.1 General Requirements for Diving Support Vessels

18.1.1 Requirements for initial surveys and classification pertaining to diving related interfaces and facilities in diving support vessels are indicated in Pt. 5, Ch. 26, Sec. 2.

18.1.2 All diving related interfaces and facilities are required to be maintained in conditions for which they are approved and installed.

18.1.3 Periodical surveys of the diving support vessel and diving systems may be carried out independent of each other.

18.1.4 Periodical survey requirements for diving facilities in diving support vessels are covered in 18.2 and 18.3.

18.2 Annual Surveys

18.2.1 Documentation that the Vessel meets the stability requirements of Class Notation 'STS' is to be verified.

18.2.2 Operational availability of position keeping capability i.e. Dynamic Positioning / Anchoring System is to be verified.

18.2.3 Communication system provided between position control station and dive control station is to be checked. Operational availability of manually operated alarm is to be confirmed.

18.2.4 Securing arrangements and supporting structure of the diving systems are to be examined for any deterioration/ damage.

18.2.5 Moon pool structure (if fitted), is to be examined as far as practicable for visible signs of damage/ deterioration.

18.2.6 It is to be confirmed that diving systems, if situated on open deck are

- located away from vicinity of ventilation outlet from machinery spaces and galley;
- not located in hazardous area zone 0, and
- no additions/ alterations to the system have been carried out.

18.2.7 It is to be verified that

- exhaust, venting and pressure relief gases from diving system are released away from source of ignition;

- no high pressure piping carrying mixed gas or oxygen is passing through the accommodation and machinery space of Category A, and is protected from mechanical damage.

18.2.8 It is to be confirmed that oxygen bottles are placed in separate storage room and ventilation system of room is effective.

18.2.9 Operational verification of oxygen sensor fitted in storage room for initiating audio- visual alarm in manned control station. Operational verification of sensor located in upper level and lower level of the room to confirm satisfactory alarm initiation.

18.2.10 Verification that the emergency source of power is located outside the machinery spaces and that this independent source of power is available for safe termination of diving operation in case of failure of main supply.

18.2.11 General examination of the visible parts of items forming structural fire protection, such as bulkheads, decks, doors and trunks with due attention being given to integrity of insulating materials.

18.2.12 General condition of fire protection system of outer area, self- closing doors on outer areas and the ventilation system for the outer areas is to be examined.

18.2.13 Satisfactory operation of fire detection and alarm system is to be confirmed by Surveyor. Verification of satisfactory functioning of repeater panel fitted in dive control and machinery control is to be carried out.

18.2.14 Verification that sufficient number of portable fire extinguishers are provided and fire-fighters outfit is provided for each person required for operation of diving system during fire.

18.2.15 Verification that control stations are provided with breathing apparatus for use during bell recovery or launching of hyperbaric evacuation unit and they are in satisfactory condition.

18.3 Special Surveys

18.3.1 In addition to the scope of annual surveys indicated above, the special survey of spaces containing the diving system is to include the following:

18.3.1.1 Verification that diving equipment/ divers are not directly exposed to any sea intake and such opening if any are protected by mesh or other structure.

18.3.1.2 Verification of accuracy of oxygen sensors by means of calibration gas, as far as practicable.

18.4 General requirements for diving systems

18.4.1 Survey requirements for diving systems permanently installed on diving support vessels, stand-alone diving systems and diving systems temporarily installed on diving support vessels are covered in the succeeding paragraphs.

18.4.2 Requirements for initial survey and classification of diving systems are laid down in Pt. 5, Ch. 26, Sec. 3.

18.4.3 The surveyor may require assistance of technically qualified personnel during surveys and repairs. Assistance may be required for the following:

- Life supports systems and monitoring equipment

- Hyperbaric Systems

- Electronic equipment and communication facilities

- Emergency operations after system malfunction due to following:

- Failure of components;
- Loss of breathing gas pressure;
- Loss of power; etc.

18.4.4 All diving equipment/ systems and hyperbaric facilities are required to be maintained in conditions for which they were approved and installed.

18.4.5 A log book is to be maintained by responsible personnel detailing the dive depths, dates and times, dive durations and other details. The log book is to contain maintenance records based on inspections and checks by the crew/ other specialized personnel. Inspection procedures and check lists are also to be part of the log book. The log book is to be available for verification by the Surveyor.

18.5 Annual Surveys

18.5.1 Annual Surveys of the hull, machinery and equipment are to be made within three months either way of anniversary date each year.

18.5.2 The Surveyor is to review the Operating and Maintenance Manuals, operational and maintenance records, log books including all recorded malfunctions or repairs carried out since the last survey. Summarized procedures for normal and emergency operations are to be verified on board. The Surveyor is to record the date of last use and the extent of use of the facilities since last survey.

The Annual Survey is to include surveys/ examination/ verification of the following, as applicable:

18.5.2.1 Pressure Equipment

.1 Pressure Vessels for Human Occupancy

- Structural integrity and corrosion status of chambers and bell are to be examined by internal and external survey. Special attention is to be paid to bilges (especially floor plating contact with shell), equipment locks, location of battery packs etc. for damage and possible corrosion. Examination with NDE may be carried out, as necessary. Condition of external anodes (if fitted) is to be reviewed;
- Shell penetrations are to be examined with special attention to hot water penetrations for sign of corrosion;
- External fairings are to be removed as far as practicable to provide better access to the entire surface of the pressure boundaries. Removal of protective coatings and insulation is not required unless deterioration or trace of rust or corrosion is evident or as otherwise required by the Surveyor for inspection or NDE;
- Careful examination of doors, safe locking mechanisms, appendages, attachments, framework, sealing surfaces, supporting structures for chambers and systems, bunk supports, insulation, paintwork; etc. Presence of safe locking mechanisms is to be checked on all hatches and function tested with the relevant chamber under pressure;
- The windows/ viewports are to be examined for de-colouration, possible deterioration, signs of damage, cracks or scratches; integrity of sealing arrangement and protective shield. Windows exceeding 10 years of age are normally to be replaced;
- Adequacy of over pressure-relief valve and alarm;
- Function testing of Built in Breathing Systems (BIBS) in chambers, bells and HES.

.2 Gas Storage & Pressure Vessels in Supply and Return Lines

- Provision of hazard warning signs, Colour coding, marking and stamping of testing date on cylinders and pressure vessels is to be verified;
- Structural integrity and corrosion status of gas containers and pressure vessels is to be inspected, with NDE, if necessary. Attachments and framework, supporting steelwork are to be examined for signs of corrosion;
- In the case of gas storage compartments, high/ low Oxygen alarms fitted external to the door and

in repeated position(s) are to be verified.

.3 Pumps and Compressors

- Verification that maintenance of equipment is being carried out in accordance with manufacturer's instructions;
- Quality of breathing air and gas to be verified by specialist/ competent agency;
- Verification of solenoid switches for auto-stopping facility of compressors on overheating.

.4 System components including piping, umbilicals, hoses, valves, filters etc.

- System cleanliness is to be examined by visual survey and checked for possible contamination;
- Verification of any possible chafing/ twisting of umbilicals and leaks in gas piping or in distribution boxes. Proof of pressure test of the system components by crew is to be verified.

18.5.2.2 Controls

.1 Gas Supply and Distribution

- System cleanliness to be examined by visual survey and checked for possible contamination by grease, oil, etc.;
- Function test of gas-regeneration units and Oxygen make-up system to be carried out;
- Arrangements for each diver's gas supply line are to be confirmed such that if one fails, the failure does not interfere with another diver's supply.

.2 Depth / Pressure Controls

- Calibration of internal gauges, external gauges, depth and life support system gauges is to be verified;
- Life support system gauges (which indicate pressure critical to life support functions) are to be examined for functionality.

.3 Temperature and Humidity Control

- Calibration evidence of all available instrumentation is to be examined;
- Functional tests of the following is to be carried out:
  - Heaters in chambers, bells and HES;
  - Cooling systems for divers (if fitted);
  - Environment conditioning units (by temperature monitoring);
  - Standby power supply to the heating/ cooling system to maintain operations for the period of diver recovery (in case of main power failure);

.4 Gas Contamination Analysis

- Verification of calibration of gas analyzers and arrangements for monitoring of Carbon dioxide, Oxygen, temperature, relative humidity in chamber;
- Primary and secondary means of Carbon dioxide scrubbing arrangements are to be examined with special attention to check that the secondary Carbon dioxide scrubbing arrangement provided is independent of any surface power supply and has minimum endurance of 24 [hrs];

.5 Domestic/ Sanitary Systems

- Internal inspection of sewage tanks;
- Functional test of toilet flush (special attention to safe working under pressure) and bilge drains in chambers (special attention to chamber shell under deck plating. Ultrasonic measurements may be carried out at the discretion of the Surveyor).

.6 Auxiliary Services to Life Support Equipment

- Main and emergency power supplies for following are to be examined/ tested;
  - Lighting of chambers and bell;
  - Monitoring devices;
  - Control consoles;

- Normal and emergency life support systems;
- Alarm systems including fire loops;
- DP alarms

- General condition of switchboards, breakers and cables including electrical hygiene, earth leakage detection systems and alarms is to be examined;

- Changeover from main to emergency supplies is to be tested.

18.5.2.3 Communication

.1 The listed alarms and record of their checks as part of their planned maintenance by crew is to be verified.

.2 Functional checks of the following are to be carried out:

- CCTV systems;
- sound powered telephone;
- recording instruments.

.3 Operational checks of primary and secondary means of communication are to be undertaken:

- between bridge and dive control station;
- dive control and
  - bells
  - chambers
  - divers in water
  - diving machinery areas
  - winch/ crane operator

.4 Ultrasound underwater telephones may be checked by simulation or during test dive.

18.5.2.4 Launch and Recovery Systems

.1 General

- Bell lifting arrangements, especially load bearing components are to be carefully examined, with NDE as required;

- Attachments and framework, supporting steelwork are to be examined for signs of corrosion, with NDE as necessary;

- General condition of wire rope and its ends is to be visually inspected. Lubrication records and records of exchange of ends are to be checked;

- Dynamic load test of system is to be carried out on main and emergency power supply. The alarm and safety functions are to be verified;

- Simulated power failure test of launch and recovery system in accordance with laid down procedures for emergency retrieval is to be performed in presence of Surveyor;

- Hydraulic power system during launch and recovery is to be examined for

  - unusually high vibration levels
  - oil leakage
  - pressure peaks and jerking

- Oil hygiene analysis is to be carried out, as necessary;

- Bell emergency release mechanism is to be tested.

.2 Controls

- The main electrical power supplies to the following are to be checked:

  - Monitoring devices
  - Alarm systems
  - Control consoles
  - Switchboards, breakers

- The correct operation of change-over from main to emergency supply is to be verified.

.3 Auxiliary Services to Launch and Recovery Systems

- Close examination of umbilical winch systems, including brake and carry out function tests;

- Examine condition of umbilical and safety provisions to stop umbilicals from jumping out of sheaves;

- Examine breathing apparatus provided for winch operator in case of fire related emergency, requiring bell recovery.

18.5.2.5 Safety Equipment

- The condition of A-60 insulation towards any enclosed spaces from the outer area of the diving system is to be examined;

- The alarms, safety limits and failure conditions of the fixed fire-fighting systems is to be verified;

- The quantity and efficacy of portable fire-extinguishers (as per approved plan), is to be verified.

18.6 Additional requirements for Third Annual Survey

18.6.1 In addition to the annual survey requirements stipulated in 18.5, the requirements indicated in the succeeding paragraphs are to be additionally undertaken at every third annual survey, as applicable:

18.6.1.1 Pressure Equipment

.1 Pressure Vessels for Human Occupancy

- Leak test of bell and chamber to Maximum Allowable Working Pressure (MAWP) with minimum 20% Helium. Leak test of pressure equipment is to be carried out at low and high pressures by increasing the pressure to the working pressure, stepwise. Leak test is to be carried out as per approved procedure, adopting appropriate safety precautions listed in the procedure. The procedure is to include an inspection for leaks at low pressure before increasing to MAWP. The test duration is to be minimum 6 [hrs].

.2 Gas Storage & Pressure Vessels in Supply and Return Lines

- Leak test of gas containers and pressure vessels to minimum 10% Helium. Precautions and test duration to be as per 18.6.1.1.1;

- Function test of valves on permanently installed gas bottles is to be undertaken.

Note: Leak tests are normally to be carried out with a minimum of 10% Helium, if the pressure vessel or piping is designed to hold Heliox mixtures or pure Helium. Due to the inherent fire risk with the use of air at high pressure, it is recommended that leak tests may be carried out incorporating Nitrogen, where applicable.

.3 Pumps and Compressors

- Running condition test of pumps and compressors is to be carried out;

- Safety valves are to be examined/ tested as per manufacturer's instructions.

.4 System components including piping, umbilicals, hoses, valves, filters etc.

- Leak test of pipe work to MAWP with minimum 10% Helium;

- Manual/ automatic control valves are to be randomly tested and recorded ;

- Examine and function test the main bell umbilical to MAWP of hoses.

18.6.1.2 Controls

.1 Gas Supply and Distribution

- Life support systems including pipe work and reclaim units are to be examined and tested.

.2 Depth/ Pressure Control

- As per scope of Annual Survey

.3 Temperature and Humidity Control

- As per scope of Annual Survey

.4 Gas Contamination Analysis

- As per scope of Annual Survey

.5 Domestic/ Sanitary Systems

- As per scope of Annual Survey

.6 Auxiliary Services to Life Support Equipment

- As per scope of Annual Survey

18.6.1.3 Communication

- Scope of annual survey is to be applied for primary, secondary and emergency means of communication;

- Listed alarms and record of their checks/ routine tests as part of planned maintenance system by crew are to be verified, as applicable;

- At least the following alarms, if fitted, are to be function tested:

    - High/ low pressure alarm – chamber
    - Temperature alarm – chamber
    - Humidity alarm – chamber
    - Divers hot water temperature alarm
    - Divers hot water pressure alarm
    - High pressure alarm – gas delivery compressor
    - Low pressure alarm – compressor cooling system
    - Low pressure lub.oil alarm – compressor
    - High level alarm for gas reclaim bag
    - High/low alarm – Oxygen analyzers
    - Oxygen sensors/ detectors in control stations

18.6.1.4 Launch and Recovery Systems

.1 General

(a) For submergence launch and recovery system:

- The working weight of the bell is to be determined, compared with previous weight and recorded;

- The following alarms for hydraulic systems for bell launch and recovery are to be tested:

    - Low pressure alarm
    - High temperature alarm
    - Low oil level in tank alarm

.2 Controls

- The power failure alarms for controls of launch and recovery systems, including (for hydraulic system) are to be tested:

    - Low pressure alarm
    - High temperature alarm
    - Low oil in tank alarm

- All alarms are to be function tested.

18.6.1.5 Safety Equipment

- The fire detection and alarm system is to be tested;

- One portable extinguisher at random, is to be function tested;

- Blow through with air of water based fixed extinguishing system is to be carried out to clear the lines;

- Records of last inspection of the fixed extinguishing system are to be reviewed by the Surveyor.

18.7 Special Surveys

18.7.1 Scope

18.7.1.1 Special surveys are to be carried out at 5 yearly intervals before the anniversary date. In addition to the requirements of the third annual survey, the scope of special survey is to be as described in 18.7.2 to 18.7.7.

18.7.2 Pressure Tests

.1 Pressure tests are to be carried out in accordance with an approved procedure, based on a recognized national/ international standard for the relevant pressure vessels. Procedures for testing are to be submitted to IRS, well in advance for review.

.2 The pressure test is to be repeated when the modifications to the pressure boundaries have been carried out and the test requirements apply as given for new pressure vessels.

.3 All pressure tests are to be carried out in presence of IRS Surveyor.

.4 A thorough internal and external visual inspection is to be carried out, prior to hydro tests.

.5 If diving bells are used for observation diving, they are to be hydro-tested to the external pressure test pressure, as per the design code.

18.7.3 Pressure Equipment

.1 Pressure Vessels for Human Occupancy

- Bell insulation is to be examined and it is to be determined if affected areas require renewal because of damage/ corrosion. Ultrasonic examination may be carried out, as necessary;
- Areas around windows are to be subject to critical examination;
- Chamber and bell are to be hydro-tested as per the manual and recorded.

.2 Gas Storage & Pressure Vessels in Supply and Return Lines

- Gas storage tubes, air and gas tanks, sewage tanks, hot water tanks are to be hydro tested and recorded.

.3 Pumps and Compressors

- As per scope of third annual survey

.4 System components including piping, umbilicals, hoses, valves, filters etc.

- Pressure test of piping systems is to be carried out.

18.7.4 Controls

Survey of the following items is to be carried out with the scope of survey same as that of third annual survey:

.1 Gas Supply and Distribution

.2 Depth/ Pressure Control

.3 Temperature and Humidity Control

.4 Gas Contamination Analysis

.5 Domestic/ Sanitary Systems

.6 Auxiliary Services to Life Support Equipment

18.7.5 Communication

- Primary, secondary and emergency means of communication are to be examined as per scope of third annual survey.

18.7.6 Launch and Recovery Systems

.1 General

(a) For submergence launch and recovery system:

- Bell ballast weight release is to be load tested and NDE to be carried out on load bearing members;
- Emergency retrieval of bell to be tested.

(b) For surface launch and recovery:

- Various systems are to be load tested as per manual and NDE to be carried out on load bearing members;

.2 Controls

- As per scope of third annual survey.

.3 Auxiliary Services to Launch and Recovery Systems

- As per scope of third annual survey.

18.7.7 Safety Equipment

- As per scope of third annual survey.

18.8 Surveys of Hyperbaric Evacuation Systems (HES)

18.8.1 Initial Surveys

.1 The requirements for initial surveys and classification of Hyperbaric Evacuation Systems (HES) are laid down in Pt.5, Ch. 26, Sec. 4.

.2 Statutory requirements of national authorities/ flag administrations are to be taken into account whilst approving and surveying HES.

.3 The interface between the HES and the installed diving system is to be examined carefully so that the HES does not become a hazard to the diving system operations.

.4 HES are connected to the diving system and are to be surveyed in the same manner as the main diving system. Corrosion on the chamber exterior is to be examined carefully due to high humidity levels.

18.8.2 Annual and Special Surveys of HES

18.8.2.1 The initial survey requirements for HES, stipulated in Cl. 4.4.1 of Pt.5, Ch.26, Sec. 4 are to be applied for all annual surveys in addition to the scope laid down below.

18.8.2.2 Annual and Special Survey requirements for components of the HES are indicated in 18.8.2.3 to 18.8.2.7

18.8.2.3 Pressure Equipment

.1 Pressure Vessels for Human Occupancy

- Annual and special survey requirements as applicable to diving systems may be followed.

.2 Gas Storage & Pressure Vessels in Supply and Return Lines

- Examine if gas cylinders exposed to sea water are certified and tested every two years;
- For other components, periodical survey requirements as applicable to diving systems are to be applied.

.3 Pumps and Compressors

- Annual and special survey requirements as applicable to diving systems are to be applied

.4 System components including piping, umbilicals, hoses, valves, filters etc.

- Examine, if all valves are firmly secured to prevent any movement due to ship motions in rough weather;
- Examine, if all external valves on the chambers are secured “open” to prevent accidental operation during deployment of HEU.

18.8.2.4 Controls

Annual and special survey requirements are to be as applicable for diving systems, for items .1 to .6, with additional requirements where indicated for each item:

.1 Gas Supply and Distribution

- The following are to be additionally verified at each annual survey:
  - Oxygen supply is fitted with a device to ensure Oxygen injection at controlled rate;
  - Availability of sufficient Oxygen for metabolic make-up for the maximum complement of divers at a rate of 0.5 [lit/min/diver], for a minimum duration of 24 [hrs];
  - There is a minimum amount of Helium/ Oxygen carried, sufficient to compensate for the use of medical locks, as specified by the equipment manufacturer.

.2 Depth/ Pressure Control

.3 Temperature and Humidity Control

- Annual and special survey requirements are to be as applicable to diving systems may be followed. In addition, the following are to be verified at each annual survey:
  - Confirm that there are adequate means of maintaining the diver’s thermal balance in the compression chamber to the required level, for a minimum of 24 [hrs];
  - Function test of the external fired diesel heater (if fitted).

.4 Gas Contamination Analysis

.5 Domestic/ Sanitary Systems

.6 Auxiliary Services to Life Support Equipment

- The following are to be additionally verified at each annual survey:
  - The safety provisions applied to reduce potential hazards from battery charging;
  - Ducted hydrogen gas ventilation for safe dumping;
  - Housing of rechargeable batteries in separate well ventilated area, located outside the crew area;

- Precautions as laid down by the manufacturer being followed for in-situ charging of batteries.

18.8.2.5 Communication

Annual and special survey requirements are to be as applicable to diving systems for items .1 to .4, with additional requirements where indicated for each item:

.1 Primary Means:

- Additionally, the following are to be verified at each annual survey:
    - Examine, if the hard wired communication system linking HES launch control/ HES internal/ saturation life support control is available and effective.

.2 Secondary Means:

.3 Emergency Means:

- Additionally, at each annual survey, examine if the HEU is fitted with relocation devices including:
    - Radar reflector
    - Strobe Light
    - Radio distress beacon

.4 Alarms:

18.8.2.6 Launch and Recovery Systems

.1 General

- Manufacturer's instructions for periodical surveys are to be followed;
- Examine HES release arrangements;
- Launch test the HES with a full load during surveys. In case of a float away chamber, a floatation test is to be carried out with full load;
- Examine if a suitable method of assisting an injured/ unconscious diver in to the chamber is available;
- Examine if, safety locking mechanism is fitted between the saturation diving complex and HES, and the same is effective;
- Examine fall length and confirm that the same allows the HEU to be fully supported in water when the ship/ installation is in the lightest sea-going state. The same is to be determined at the worst angle of list and trim on the side where the HEU is located;
- Examine falls for turning and deterioration. Falls are to be turned end to end at 30 month intervals and renewed when deteriorated, or at intervals not exceeding 5 years;
- Load test of HEU launch system to be carried out, as applicable on annual basis;
- Where secondary means of launch is provided, the launch of the HEU by secondary means is to be tested under same conditions as primary means of launch.

2 Controls

- Annual and special survey requirements as applicable to diving systems may be followed.

.3 Auxiliary Services to Launch and Recovery Systems

- Annual and special survey requirements as applicable to diving systems may be followed.

18.8.2.7 Safety Equipment

- As per the relevant Statutory requirements.

18.9 Surveys for Transit Damage

18.9.1 During transportation of the diving system or hyperbaric facility, precautions are to be taken to protect the pressure hull, external structures, acrylic windows, batteries, etc. from undue deterioration or damage.

18.9.2 A survey for transit damage is to be conducted after the transportation of diving system or hyperbaric facility from the manufacturer's or the assembler's plant to the operational location or from one location of operations to another.

18.9.3 This survey is to include but may not be limited to the following:

- Pressure boundary and its components are to be visually checked for damage.
- Systems and components are to be inspected, and recalibration of instruments may be required if deemed necessary by the Surveyor.
- External structures, fairings and skins are to be checked for damage.
- Further in-depth examination may be required by and at the discretion of the Surveyor.

18.9.4 If the survey is carried out by the Owner and damage is found which affects or may affect classification, IRS is to be advised of the details and provisions are to be made for examination by the Surveyor at first available opportunity.

Section 19

Surveys – Ballast Water Management

19.1 General

19.1.1 Periodical survey requirements for ballast water management techniques (Ballast Water Exchange and Ballast Water Treatment) used onboard vessels are covered in the succeeding paragraphs. Requirements for design and installation of Ballast Water Management Systems are given in Part 5, Chapter 37 of the Rules.

19.2 Vessels Using Ballast Water Exchange Methods

19.2.1 Annual Survey

19.2.1.1 The following documentation is to be checked onboard during the survey:

a) Approved ballast water management plan

b) Ballast water record book stating type of exchange (sequential, flow-through, dilution).

19.2.1.2 The sediment management arrangement and control of all valves and each pump required for ballast water exchange are to be examined and verified.

19.2.1.3 The function of control system used for dilution method is to be verified, if applicable.

19.2.2 Intermediate Survey

19.2.2.1 Verification of the accessibility of sampling points.

19.2.2.2 Verification of satisfactory operation of tank level indicating system, valve position indicating system, draught indicating system and communication to local control station at central ballast control station.

19.2.3 Special Survey

19.2.3.1 Verification of satisfactory operation of local and remote control at each pump used during ballast water exchange.

19.2.3.2 Verification that manually operated independent means of control of all valves required for ballast water exchange are in satisfactory condition.

19.2.3.3 Examine and verify function of control, monitoring and communication systems on the ballast control station, including:

a) tank level indicating system

b) valve position indicating system

c) draught indicating system

d) communication to local control station.

19.3 Vessels Using Ballast Water Treatment Systems

19.3.1 Annual Survey

19.3.1.1 The following documentation is to be verified onboard and in practical use:

a) approved ballast water management plan

b) ballast water record book

c) type approval certificate for treatment system

d) calibration procedures and certificates for sensors on the treatment system

e) equipment manuals for major components on the treatment system

f) maintenance log for treatment system

g) operational manual for treatment system including maintenance procedure

h) material safety data sheet of active substances, if relevant.

19.3.1.2 Confirming that no unauthorized alteration or modification has been done to the original arrangement including no changes to electrical, controls & monitoring system or hardware and software (version) for computer based systems;

19.3.1.3 Checking whether any new equipment has been fitted and, if so, confirm that it has been approved before installation and that any changes are reflected in the certificate;

19.3.1.4 General external examination of the structure, equipment, systems, fittings, arrangements and material/ process associated with the ballast water management plan, in order to confirm that the BWMS has been maintained and remain in compliance with the requirements of the Convention.

19.3.1.5 Examination of satisfactory condition of non-metallic expansion joints (if any) in the piping system which penetrates the ship's side.

19.3.1.6 Verification to confirm the satisfactory operation of the control and monitoring equipment including examination of records of the proper functioning or failure of the BWMS.

19.3.1.7 Confirmation that the records of the recording device fitted are satisfactory and adequate supply of consumables for the recording device(s) are available on board.

19.3.1.8 For systems using active substances:

.1 Confirmation that active substance(s) in accordance with the manufacturer's recommendations are provided on board

.2 Confirmation that dosage instruction for active substance(s) or preparations are available on board

.3 Confirmation that storage of the active substance(s) is satisfactory taking into account the risks/hazards involved with the substance as provided in the safety data sheet

.4 Confirmation of the satisfactory condition of the chemical storage tanks if provided, piping and fittings by external examination.

19.3.1.9 The sediment management arrangement and control of all valves and each ballast pumps are to be examined.

19.3.2 Intermediate Survey

19.3.2.1 In addition to the scope of annual surveys, examination of the ballast water management system for obvious defects, deterioration or damage including examination of associated pumps, piping and fittings for wear and corrosion is to be undertaken and confirmed that these are in satisfactory condition.

19.3.3 Special Survey

19.3.3.1 In addition, to the scope of intermediate surveys, indicated above, it is to be confirmed that the operation (by simulation test or equivalent) of the ballast water management system is satisfactory.

Section 20

Surveys – Fuel Installations on Ships Utilizing Gaseous Fuel or Other Low Flash Point Fuels

20.1 Application

20.1.1 Requirements of this section apply to ships, other than those covered by Section 5, which utilize gaseous fuel or other low flash point fuels as a fuel for propulsion prime mover/auxiliary power generation arrangements and associated systems. These requirements are in addition to the requirements of Section 8.

20.2 Annual Survey

20.2.1 Schedule

20.2.1.1 Annual Surveys are to be held within 3 months before or after each anniversary date of the date of the initial classification survey or of the date credited for the last Special Survey. They will normally be performed at the same time as an Annual Hull survey.

20.2.2 Scope

20.2.2.1 The following is to be carried out during the survey of the Fuel Storage, Fuel Bunkering
System, and Fuel Supply System:

20.2.2.1.1 Logbooks/Records

i) The logbooks and operating records are to be examined with regard to correct functioning of the systems, such as, gas detection systems and fuel supply/gas systems. The hours per day of the reliquefaction plant, gas combustion unit, as applicable, the boil-off rate, and nitrogen consumption (for membrane containment systems) are to be considered together with gas detection records.

20.2.2.1.2 Operating and Maintenance Instruction Manuals

i) The manufacturer/builder instructions and manuals covering the operations, safety and maintenance requirements and occupational health hazards relevant to fuel storage, fuel bunkering, and fuel supply and associated systems for the use of the fuel, are to be confirmed as being aboard the vessel.

20.2.2.1.3 Control, Monitoring and Safety Systems

i) Gas detection and other leakage detection equipment in compartments containing fuel storage, fuel bunkering, and fuel supply equipment or components or associated systems, including indicators and alarms, are to be confirmed in satisfactory operating condition. Recalibration of the gas detection systems is to be verified in accordance with the manufacturers' recommendations.

ii) Verification of the satisfactory operation of the control, monitoring and automatic shutdown systems as far as practicable of the fuel supply and bunkering systems.

iii) Operational test, as far as practicable, of the shutdown of ESD protected machinery spaces.

20.2.2.1.4 Fuel Handling Piping, Machinery and Equipment

i) Piping, hoses, emergency shut-down valves, remote operating valves, relief valves, machinery and equipment for fuel storage, fuel bunkering, and fuel supply such as venting, compressing, refrigerating, liquefying, heating, cooling or otherwise handling the fuel are to be examined, as far as practicable. Means for inerting is to be examined. Stopping of pumps and compressors upon emergency shut-down of the system is to be confirmed as far as practicable.

20.2.2.1.5 Ventilating System

i) Examination of the ventilation system, including portable ventilating equipment where fitted, is to be made for spaces containing fuel storage, fuel bunkering, and fuel supply units or components or associated systems, including air locks, pump rooms, compressor rooms, fuel

preparation rooms, fuel valve rooms, control rooms and spaces containing gas burning equipment. Where alarms, such as differential pressure and loss of pressure alarms, are fitted, these should be operationally tested as far as practicable.

20.2.2.1.6 Drip Trays

i) Portable and fixed drip trays and insulation for the protection of the ship's structure in the event of leakage are to be examined.

20.2.2.1.7 Hazardous Areas

i) Electrical equipment and bulkhead/deck penetrations including access openings in hazardous areas are to be examined for continued suitability for their intended service and installation area.

20.2.2.1.8 Electrical Bonding.

i) Electrical bonding arrangements in hazardous areas, including bonding straps where fitted, are to be examined.

20.2.2.1.9 Fuel Storage, Bunkering and Supply Systems

i) The following are to be examined, so far as applicable. Insulation need not be removed, but any deterioration or evidence of dampness is to be investigated:

a. Fuel Storage

i. External examination of the storage tanks including secondary barrier if fitted and accessible.

ii. General examination of the fuel storage hold place.

iii. Internal examination of tank connection space.

iv. External examination of tank and relief valves.

v. Verification of satisfactory operation of tank monitoring system.

vi. Examination and testing of installed bilge alarms and means of drainage of the compartment.

vii. Testing of the remote and local closing of the installed main tank valve.

b. Fuel Bunkering System

i. Examination of bunkering stations and the fuel bunkering system.

ii. Verification of satisfactory operation of the fuel bunkering control, monitoring and shutdown systems.

c. Fuel Supply System

i. Examination of the fuel supply system during working condition as far as practicable.

ii. Verification of satisfactory operation of the fuel supply system control, monitoring and shut-down systems.

iii. Testing of the remote and local closing of the master fuel valve for each engine compartment.

20.3 Intermediate Survey

20.3.1 Schedule

20.3.1.1 The Intermediate Survey is to be held at or between either the 2nd or 3rd Annual Survey.

20.3.1.2 Those items which are additional to the requirements of the Annual Surveys may be surveyed either at or between the 2nd and 3rd Annual Survey.

20.3.2 Scope

20.3.2.1 In addition to the applicable requirements of the Annual Survey, the Intermediate Survey is to include the following:

20.3.2.2 Safety Systems

20.3.2.2.1 Gas detectors, temperature sensors, pressure sensors, level indicators, and other equipment providing input to the fuel safety system are to be randomly tested to confirm satisfactory operating condition. Proper response of the fuel safety system upon fault conditions is to be verified.

20.4 Special Survey

20.4.1 Schedule

20.4.1.1 Special Surveys are to be carried out at 5 years intervals to renew the Classification Certificate.

20.4.1.2 The first Special Survey is to be completed within 5 years from the date of the initial classification survey and thereafter within 5 years from the credited date of the previous Special Survey. However, an extension of class of 3 months maximum beyond the 5th year can be granted in exceptional circumstances. In this case, the next period of class will start from the expiry date of the Special Survey before the extension was granted.

20.4.1.3 For surveys completed within 3 months before the expiry date of the Special Survey, the next period of class will start from the expiry date of the Special Survey. For surveys completed more than 3 months before the expiry date of the Special Survey, the period of class will start from the survey completion date. In cases where the vessel has been laid up or has been out of service for a considerable period because of a major repair or modification and the owner elects to only carry out the overdue surveys, the next period of class will start from the expiry date of the Special Survey. If the owner elects to carry out the next due Special Survey, the period of class will start from the survey completion date.

20.4.1.4 The Special Survey may be commenced at the 4th Annual Survey and be progressed with a view to completion by the 5th anniversary date. When the Special Survey is commenced prior to the 4th Annual Survey, the entire survey is to be completed within 15 months if such work is to be credited to the Special Survey.

20.4.1.5 Special Surveys may be carried out on a continuous survey basis. In this case, the interval between consecutive examinations of each item is not to exceed five (5) years.

20.4.2 Scope

20.4.2.1 The Special Survey is to include, in addition to the requirements of the Annual and Intermediate Surveys, examination, tests and checks of sufficient extent to ensure that the fuel installations are in a satisfactory condition and are fit for their intended purpose for the new period of class of 5 years to be assigned, subject to proper maintenance and operation and to periodical surveys being carried out at the due dates.

20.4.2.2 Fuel Handling and Piping

i) All piping for fuel storage, fuel bunkering, and fuel supply such as venting, compressing, refrigerating, liquefying, heating storing, burning or otherwise handling the fuel and liquid nitrogen installations are to be examined. Removal of insulation from the piping and opening for examination may be required. Where deemed suspect, a hydrostatic test to 1.25 times the Maximum Allowable Relief Valve Setting (MARVS) for the pipeline is to be carried out. After reassembly, the complete piping is to be tested for leaks. Where water cannot be tolerated and the piping cannot be dried prior to putting the system into service, the Surveyor may accept alternative testing fluids or alternative means of testing.

20.4.2.3 Fuel Valves

i) All emergency shut-down valves, check valves, block and bleed valves, master gas valves, remote operating valves, isolating valves for pressure relief valves in the fuel storage, fuel bunkering, and fuel supply piping systems are to be examined and proven operable. A random selection of valves is to be opened for examination.

20.4.2.4 Pressure Relief Valves

i) Fuel Storage Tank Pressure Relief Valves: The pressure relief valves for the fuel storage tanks are to be opened for examination, adjusted, and function tested. If the tanks are equipped with relief valves with non-

metallic membranes in the main or pilot valves, such non-metallic membranes are to be replaced.

ii) Fuel Supply and Bunkering Piping Pressure Relief Valves: Pressure relief valves for the fuel supply and bunkering piping are to be opened for examination, adjusted, and function tested. Where a proper record of continuous overhaul and retesting of individually identifiable relief valves is maintained, consideration will be given to acceptance on the basis of opening, internal examination, and testing of a representative sampling of valves, including each size and type of liquefied gas or vapor relief valve in use, provided there is logbook evidence that the remaining valves have been overhauled and tested since crediting of the previous Special Survey.

iii) Pressure/Vacuum Relief Valves: The pressure/vacuum relief valves, rupture disc and other pressure relief devices for interbarrier spaces and hold spaces are to be opened, examined, tested and readjusted as necessary, depending on their design.

20.4.2.5 Fuel Handling Equipment

i) Fuel pumps, compressors, process pressure vessels, inert gas generators, heat exchangers and other components used in connection with fuel handling are to be examined as per the requirements in Sec 5, 8 & 12, as applicable.

20.4.2.6 Electrical Equipment

i) Examination of electrical equipment to include the physical condition of electrical cables and supports, intrinsically safe, explosion proof, or increased safety features of electrical equipment.

ii) Functional testing of pressurized equipment and associated alarms.

iii) Testing of systems for de-energizing electrical equipment which is not certified for use in hazardous areas.

iv) An electrical insulation resistance test of the circuits terminating in, or passing through, the hazardous zones and spaces is to be carried out.

20.4.2.7 Safety Systems

i) Gas detectors, temperature sensors, pressure sensors, level indicators, and other equipment providing input to the fuel safety system are to be tested to confirm satisfactory operating condition.

ii) Proper response of the fuel safety system upon fault conditions is to be verified.

iii) Pressure, temperature and level indicating equipment are to be calibrated in accordance with the manufacturer's requirements.

20.4.2.8 Fuel Storage Tanks

i) Fuel storage tanks are to be examined in accordance with an approved survey plan. In developing the inspection/survey plan, the requirements for the survey of liquefied gas fuel containment systems are to be in accordance with the relevant requirements of Section 5, 5.4.6, except as noted below:

a. The tank insulation and tank support arrangements should be visually examined. Non-destructive testing may be required if conditions raise doubt to the structural integrity.

b. Vacuum insulated independent fuel storage tanks of type C without access openings need not be examined internally. Where fitted, the vacuum monitoring system should be examined and records should be reviewed.

Section 21

Condition Monitoring and Condition Based Maintenance

21.1 General

21.1.1 Application

.1 The requirements in this Section are applicable to the approved Condition Monitoring and Condition Based Maintenance schemes where the condition monitoring results are used to influence the scope and/ or frequency of Class surveys.

.2 A condition monitoring system may be used to provide a greater understanding of equipment condition, and a condition based maintenance scheme may be used to obtain maintenance efficiency. IRS approval is required where owners wish to change the survey cycle based on CM/ CBM.

.3 This scheme may be applied to components and systems covered by the Continuous Survey of Machinery (CSM), and other components and systems as requested by the Owner.

.4 The extent of condition based maintenance and associated monitoring of equipment to be included in the PMS is to be decided by the owner, based on the recommendations of the OEM and availability of suitable condition monitoring equipment. The CM/ CBM scheme is to be submitted to IRS for approval.

.5 In general, these requirements can be applied only to vessels operating on approved PMS survey scheme.

.6 The scheme may be applied to any individual items and systems. Any items not covered by the scheme are to be surveyed and credited in accordance with the requirements of Section 14 or traditional survey methods for machinery.

.7 Software systems can use complex algorithms, machine learning and knowledge of global equipment populations/ defect data in order to identify acceptability for continued service or the requirement for maintenance. These systems may be independent of the OEM recommended maintenance and condition monitoring suggested limits. Approval of this type of software would be based on OEM recommendations, industry standards and IRS experience.

21.1.2 Definitions

21.1.2.1 The following definitions (from ISO 13372:2012) are applicable for the requirements of this Section.

21.1.2.2 *Condition Monitoring*: acquisition and processing of information and data that indicate the state of a machine over time. The machine state deteriorates if faults or failures occur.

21.1.2.3 *Diagnostic*: examination of symptoms and syndromes to determine the nature of faults or failures.

21.1.2.4 *Condition Based Maintenance:* **m**aintenance performed as governed by condition monitoring programmes.

21.1.3 Condition Monitoring

21.1.3.1 Where an approved condition monitoring system is fitted, credit for survey may be based on acceptable condition monitoring results. The condition monitoring results are to be reviewed during the annual audit.

21.1.3.2 Limiting parameters are to be based on the Original Equipment Manufacturers guidelines, or a recognized international standard.

21.1.3.3 The condition monitoring system is to provide an equivalent or greater degree of confidence in the condition of the machinery to traditional survey techniques.

21.1.3.4 The condition monitoring system is to be approved in accordance with the methodology in 21.2 and 21.3.

21.1.3.5 IRS retains the right to test or open-up the machinery, irrespective of the CM results, if deemed necessary.

21.1.4 Condition Based Maintenance (CBM)

21.1.4.1 Where an owner wishes to base their equipment maintenance on a CBM approach, this is to meet the requirements of the ISM Code.

21.1.4.2 Where an agreed planned maintenance and CBM scheme is in operation, the CSM and other survey intervals may be extended based on OEM maintenance recommendations and acceptable condition monitoring results.

21.1.4.3 Limiting parameters (alarms and warnings) are to be based on the OEM guidelines, or a recognized international standard.

21.1.4.4 The CBM scheme is to provide an equivalent or greater degree of confidence in the condition of the machinery to traditional maintenance techniques.

21.2 Procedures and Conditions for approval of CM and CBM

21.2.1 Onboard Responsibility

21.2.1.1 The Chief Engineer is to be the responsible person on board in charge of the CM and CBM.

21.2.1.2 Documentation on the overhaul of items covered by CM and CBM schemes is to be reported by the Chief Engineer.

21.2.1.3 Access to computerized systems for updating of the maintenance documentation and maintenance program is only to be permitted by the chief engineer or other authorized person.

21.2.1.4 All personnel involved in CM and CBM are to be appropriately qualified.

Note: CM does not replace routine surveillance or the chief engineer's responsibility for taking decisions in accordance with his judgement.

21.2.2 Equipment and System Requirements

21.2.2.1 The CM/ CBM scheme and its extent would be approved by IRS.

21.2.2.2 The CBM scheme is to be capable of producing a condition report, and maintenance recommendations.

21.2.2.3 A system is to be provided to identify where limiting parameters (alarms and warnings) are modified during the operation of the scheme.

21.2.2.4 Where CM and CBM schemes use remote monitoring and diagnosis (i.e. data is transferred from the vessel and analyzed remotely), the system is to meet the applicable standards for Cyber Safety and Security. The system is to be capable of continued onboard operation in the event of loss of the communication function.

21.2.2.5 CBM schemes are to identify defects and unexpected failures that were not prevented by the CM system.

21.2.2.6 Systems are to include a method of backing up data at regular intervals.

21.2.3 Documentation and Information

21.2.3.1 The following documentation is to be made available to IRS for approval of the scheme:

(i) Procedure for changes to software system and CM parameters

(ii) Listing of equipment to be included in the scheme

(iii) Listing of acceptable condition monitoring parameters

(iv) Description of CBM scheme

(v) Listing, specifications and maintenance procedures for condition monitoring equipment

(vi) Baseline data for equipment with condition monitoring

(vii) Qualification of personnel and company responsible for analyzing CM results

21.2.3.2 In addition to the above documentation the following information is to be available on board:

(i) All requirements in 21.2.3.1 in an up-to-date manner

(ii) Maintenance instructions (manufacturer's and shipyard's)

(iii) Condition monitoring data including all data since last opening of the machine and the original base line data

(iv) Reference documentation (trend investigation procedures etc.)

(v) Records of maintenance including repairs and renewals carried out

(vi) Records of changes to software systems and parameters

(vii) Sensors calibration records / certification / status

21.2.4 Approval validity

21.2.4.1 An Annual Audit is to be carried out to maintain the validity of the CM/CBM scheme.

21.2.4.2 The survey arrangement for machinery under CM/ CBM may be cancelled by IRS if it is evident that the scheme is not being satisfactorily carried out either from the maintenance records or the general condition of the machinery.

21.2.4.3 In case of sale or change of management of the ship or transfer of class the approval is to be reconsidered.

21.2.4.4 The owner may, at any time, cancel the survey arrangement for machinery under the scheme by informing IRS in writing and for this case the items which have been inspected under the scheme since the last annual Audit can be credited for class at the discretion of the attending surveyor.

21.3 Surveys

21.3.1 Installation Survey

21.3.1.1 Condition monitoring equipment is to be installed and surveyed, and a set of base line readings is to be taken, as part of Installation Survey.

21.3.2 Implementation Surveys

21.3.2.1 The Implementation Survey is to be carried out by IRS surveyor no earlier than 6 months after installation survey and no later than the first Class annual survey.

21.3.2.2 During the Implementation survey the following is to be verified by the surveyor:

(i) the CM/ CBM scheme is implemented according to the approval documentation, including a comparison with baseline data;

(ii) the scheme is producing the documentation required for the Annual Audit and the requirements of surveys and testing for the maintenance of class are complied with;

(iii) the onboard personnel are familiar with operating the scheme;

(iv) records of any limiting parameters (alarms and warnings) that have been modified during the operation of the scheme;

(v) records of any failures of monitored equipment are to be reviewed to ensure that the condition monitoring scheme is effective/ sufficient.

21.3.2.3 When this survey is carried out and the implementation is found in order, a report describing the scheme is to be submitted to IRS and the scheme may be put into service.

21.3.3 Annual Audit

21.3.3.1 An annual audit of the CM/ CBM scheme is to be carried out concurrently with the annual class surveys.

21.3.3.2 The purpose of this audit is to verify that the scheme is being correctly operated and that the machinery has been functioning satisfactorily since the previous audit. This is to include any limiting parameters (alarms and warnings) that have been modified since the last audit. A general examination of the items concerned is to be carried out.

21.3.3.3 The performance, condition monitoring and maintenance records are to be examined to verify that the machinery has functioned satisfactorily since the previous survey, or action has been taken in response to machinery operating parameters exceeding acceptable tolerances.

21.3.3.4 Written details of break-down or malfunction are to be made available by the Chief Engineer.

21.3.3.5 At the discretion of the surveyor, function tests, confirmatory surveys and random check readings, where Condition Monitoring / Condition Based Maintenance equipment is in use, are to be carried out as far as practicable and reasonable.

21.3.3.6 The familiarity of the Chief Engineer and other personnel involved with the CM system is to be verified.

21.3.3.7 Calibration status of sensors and equipment is to be verified.

21.3.3.8 Verification that the suitability of the CM/ CBM scheme has been reviewed following defects and failures is to be carried out.

21.3.4 Damage and repairs

21.3.4.1 Damage to components or items of machinery is to be reported to IRS by the Chief Engineer. The repairs of such damaged components or items of machinery are to be carried out to the satisfaction of the Surveyor.

21.3.4.2 Details of repairs and maintenance carried out are to be examined. Any machinery part, which has been replaced by a spare one, due to damage, is to be retained on board where possible until examined by IRS Surveyor.

21.3.4.3 Defects and failure data are to be reviewed in order to ensure the system output is appropriate. Where necessary, following review of the failure data, there is to be a method of amending the CM and CBM scheme.

Section 22

Surveys – Comfort On-board Ships

22.1 General

22.1.1 The requirements in this Section are applicable to surveys of noise and vibration aspects in Ships which are assigned additional Class notations for Comfort in accordance with the requirements in Part 5, Chapter 39 of the Rules.

22.1.2 It is the responsibility of the Owner to advise IRS regarding any modifications, alterations or repairs, including replacement of major machinery and equipment, which may lead to significant changes in the noise and vibration levels on board, compared to the initial surveys.

22.2 Annual Surveys

22.2.1 The following information is to be provided by the Owner and reviewed by the attending Surveyor for issues that could affect the Comfort notation, since previous survey:

- Modifications/ repairs carried out in crew or passenger spaces;
- HVAC/ duct routing modifications;
- Major repairs, modifications, replacement of machinery;
- Collision and grounding reports and repairs thereof;
- Fire, repair and damage reports

22.2.2 Based on above information, the attending Surveyor would review whether changes or alterations have taken place that could affect the Comfort notation. As a result of the review, if the observations indicate that the noise and/ or vibration levels may exceed the stipulated criteria and those measured at the initial survey, a measurement program would be agreed between the Owner and IRS; and measurements taken again in accordance with the Rules.

22.3 Special Surveys

22.3.1 Information is to be provided by the Owner, as indicated in 22.2.1 for review by the attending Surveyor.

22.3.2 Noise and vibration can increase due to general wear and tear. The noise and vibration measurements are to be carried out during special surveys.

22.3.3 The measurements can be limited in comparison to initial survey measuring points depending on ship type as follows;

- For passenger ships, 30% of initial survey measuring points (excluding passenger cabins) and 10% of initial survey measuring points in passenger cabins.
- For Cargo ships, 30% of initial survey measuring points.

Section 23

Surveys of Floating Dry Docks

23.1.General

23.1.1 This Section is applicable to Periodical Surveys of floating dry docks in normal service for maintenance of Classification, in accordance with the intervals and window period specified in Section 1 of this Chapter.

23.1.2 General requirements for Surveys of Ships would be applied as relevant and applicable to floating dry docks, unless otherwise specified in this Section.

23.1.3 Floating dry docks are subjected to Intermediate Surveys and Special Surveys. Annual Surveys are not applicable to floating dry docks. Further, floating docks may be subjected to Unscheduled Surveys or Surveys for damage or alterations.

23.1.4 Attention should also be given to any relevant requirements of the statutory authority or flag Administration with which the floating dry dock is registered.

23.2 Intermediate Surveys

23.2.1 At each Intermediate Survey, the following are to be examined and placed in good condition:

- Pontoon, safety and top decks, steel structures externally above the light waterline, keel and side blocks and their foundations, swing bridges (where provided), access gangways
- Crane foundations
- Vents and overflow pipes and overboard scuppers
- Hatchways, manholes and their closing appliances
- Casings, skylights, companionways, ladders, and handrails and other means of protection that might be provided for access to all spaces
- Ventilators and other deck openings
- Overboard scuppers and discharges
- Means of escape from machinery spaces, crew accommodation and work spaces
- Deflection and stress monitoring system
- Draught marks
- Fire protection and extinguishing arrangements
- Machinery including boilers (where fitted), bilge and ballast system, electrical equipment, pumps and other equipment

23.3 Special Surveys

23.3.1 In addition to the requirements for Intermediate Surveys, the following are to be carried out:

.1 Pontoon and wing wall tanks are to be cleaned, examined internally, and hydraulically tested to the satisfaction of the Surveyor. For floating docks over 15 years of age, oil fuel tanks forming part of the main structure are to be examined internally.

.2 Examination of condition of means of protection to openings.

.3 Spaces above safety deck are to be examined internally, removing linings, etc. where necessary for survey.

.4 Where fitted, sounding pipes are to be examined. The Surveyors are to ensure that striking plates are fitted under the sounding pipes whilst examining the tanks internally.

.5 Where the surface of plating is covered with cement, composition, or wood sheathing, the covering is to be removed as may be required for examination of the plating.

.6 Thickness measurement of any part of the structure where signs of deterioration are evident may be required by the Surveyor. The structure is to be renewed, where necessary.

.7 Anchoring and mooring arrangements are to be examined for satisfactory condition. For areas which may be inaccessible /not readily accessible, Owner is to submit to IRS a plan highlighting procedures for surveys and verification of conditions for such areas.

.8 The efficient condition of the means of communication between control position and machinery spaces is to be verified.

23.3.2 At the Special Survey after the dock is 20 years old, and at 10 years intervals thereafter, in addition to the requirements in 23.3.1, thickness measurements of the structure are to be carried out to assess the overall general condition of the floating dry dock. Two sections of gauging are to be made within the 0.4 L mid-length of the dock. In addition, the Surveyor may require additional thickness measurements to be taken in those areas where it is suspected that corrosion and wastage exceed the acceptable limits.

23.3.3 Surveys of boilers are to be in accordance with the requirements of Section 9 of this Chapter.

23.3.4 Surveys of machinery, piping systems, electrical equipment and control engineering systems are to be in accordance with the requirements of Section 8 of this Chapter, as relevant and applicable.

23.3.5 Bottom Surveys

23.3.5.1 General

.1 Bottom survey means the examination of the outside of the floating dry dock's bottom, related items and part of the dry dock below the light waterline. This examination may be carried out with the floating dock either in dry dock (or on a slipway) or afloat.

.2 Bottom surveys are to be carried out along with Special Surveys.

23.3.5.2 In-water Survey

.1 Floating dry docks may be assigned INWATER SURVEY notation in compliance with the requirements of Section 7.2.2 of this Chapter, as relevant to such dry-docks.

.2 Survey of the outer bottom below the light waterline may be carried out by some combination of:

- Ultrasonic measurement of plate thickness
- Heeling/ tilting of the dock for partial examination of the bottom
- Examination by remote operated vehicle (ROV)
- Examination by diver(s)

.3 The methods and extent of the examination is to be agreed with IRS.

.4 Unless dry docked, the owner's proposal for survey of that part of the dry dock below the light waterline is to be submitted for consideration. This proposal is to include such information on the dry dock as the area of operations, type of coatings and method of corrosion control, if any.

.5 Where it is intended to tilt the dry dock, consideration is to be given to the stability and longitudinal strength of the dry dock. Such information is to be included in the operating manual, where applicable.

.6 For the extension of intervals of the examination below the waterline, special consideration may be given by IRS based on a risk analysis taking into account the age, overall condition of the dock, coating condition and method of corrosion protection.

.7 Grids for sea chests are to be removed. Valves and cocks with their fastenings are to be examined externally. If considered

necessary by the Surveyor, valves and cocks are to be dismantled for internal inspection.

23.3.5.3 Bottom survey in dry-dock

.1 When a floating dock is dry-docked, it is to be placed on blocks of sufficient height and with necessary staging to permit the examination of elements such as shell plating including bottom and bow plating, stern frame, sea chests and valves, appendages.

.2 The outer shell plating is to be visually examined for excessive corrosion, or deterioration due to chafing or contact with the ground or for any undue deformation or buckling.

.3 Significant plate unevenness or other deterioration which does not necessitate immediate repairs is to be recorded.

.4 Sea chests and their gratings, sea connections and overboard discharge valves and cocks and their fastenings to the hull or sea chests are to be examined.

Section 24

Remote Surveys

24.1 Application

24.1.1 This Section indicates requirements to be complied with for satisfactory conduct of remote surveys.

Remote Survey Notation

24.1.2 The additional class notation **REMOTE SURVEY** will be assigned to ships considered capable by IRS, for carrying out classification surveys remotely, subject to the following:

a) provided with suitable equipment/ hardware/ software that are acceptable to IRS

b) vessel not under condition improvement/ condition monitoring programme of IRS

c) procedures available for training and familiarity of on-board personnel/ crew in use of equipment/ hardware/ software

d) satisfactory completion of sample remote survey(s)

24.1.3 IRS reserves the right, at its sole discretion, to either suspend or withdraw this additional class notation in case the above-mentioned evaluation is no longer valid, e.g. due to a deterioration in the ship's condition of maintenance, or change of Management, non-availability of suitable equipment/ hardware/ software on board.

24.1.4 As far as statutory surveys are concerned, requirements of the relevant Flag Administration on remote surveys are to be complied with.

24.1.5 Unmanned non-self-propelled barges/ pontoons are not eligible to be assigned **REMOTE SURVEY** notation.

24.1.6 For ships assigned **REMOTE SURVEY** notation, the satisfactory functioning of equipment/ hardware/ software and, procedures for training of on-board personnel/ crew in the use of equipment/ hardware/ software, is to be verified by the Surveyor during Annual Surveys.

24.1.7 Ships assigned **REMOTE SURVEY** notation indicate that capability to undertake remote surveys has been verified by IRS in accordance with the relevant requirements, thereby facilitating use of this capability during various surveys, as indicated in Table 24.4.1.

24.1.8 Conduct of remote surveys on ships not assigned **REMOTE SURVEY** notation will be considered on case-to-case basis, based on specific request from the Owners, and compliance with relevant requirements of this Section.

Conduct of Remote Surveys

24.2 Definitions

24.2.1 *Remote survey* is a process of verifying that a ship and its equipment are in compliance with the Rule requirements, where the verification is partially or fully undertaken by applying digital and technological means, in an efficient and secure manner, without attendance on-board by a Surveyor.

Note: ‘Attendance on-board by a Surveyor’ means physical attendance on-board the ship

by a Surveyor. Remote classification activities not requiring a survey, such as some administrative tasks, are not considered as remote surveys.

24.2.2 *Administrative task* is a task where a survey decision is not being made, for example re-issue of a certificate or record following a correction, or an update to the ship's records held by IRS or a document review.

24.2.3 *Documentary verification* means any verification that can be undertaken based on examination of documents.

24.2.4 *Owner's Representative* means Master/ Chief Engineer or crew member designated by the Master as responsible for remote surveys.

24.2.5 *Information and Communication Technology (ICT)* are the technologies used in the scope of remote surveys for gathering, storing, retrieving, processing, analyzing and transmitting information which includes both hardware and software.

24.3 Requirements for equivalency

The requirements for equivalency of a remote survey, to a survey attended on board by a Surveyor include:

- eligibility of the remote survey
- qualification of Surveyors
- planning of the remote survey
- performance of the remote survey
- assessment of the remote survey
- reporting

Equivalency is obtained when, with the use of available ICT, a Surveyor can perform a survey remotely being able to (with same level of assurance as obtained with physical attendance on-board):

- obtain the supporting and technical evidence required according to the applicable requirements and;
- verify applicable survey items and relevant tests,

24.3.1 Eligibility of the remote survey

.1 Eligibility of the remote survey is to be decided based on type and scope of the requested survey, in accordance with Cl. 24.4.1 and, if applicable, flag Administration acceptance and possible instructions, when the Classification Survey is also related to a statutory item, and IRS is carrying out the statutory survey on behalf of the flag Administration.

.2 A remote survey is deemed eligible when it provides the same level of assurance, according to the requirements for equivalency, as if it was conducted with attendance on board by a Surveyor.

24.3.2 Qualification of Surveyors

.1 Remote surveys will be carried out by appropriately trained and qualified IRS Surveyors.

24.3.3 Planning of the remote survey

.1 Request for remote surveys is to be made by the Owner, well in advance of the survey falling due, with details of the technologies proposed to facilitate remote surveys. Planning of the remote survey is required to ensure that the remote survey is carried out in accordance with the applicable requirements. The planning is to be based on the scope of the remote survey (Also refer 24.4.1).

.2 To ensure that the Surveyor can properly plan the remote survey and communicate with personnel/ crew, so that the survey is carried out according to the applicable rules, adequate means are to be available enabling the Surveyor and allowing IRS to:

- properly interact with personnel/ crew involved in the remote survey, before and during the survey process;
- agree on the ICT means to be used;
- verify that personnel/ crew involved in the remote survey are suitably skilled to use the electronic devices and/or software used by IRS to perform the remote survey;
- acquire as deemed necessary information on identity and ranking of personnel/ crew involved in the remote survey;

- provide the survey item/scope to the personnel/ crew involved in facilitating the remote surveys, including the tests that will be performed;

- communicate, during the remote survey, additional actions depending on the evidence to be collected.

.3 One or more of the following means is to be provided for planning the remote survey:

- live-streaming video and audio connection

- exchange of data/ electronic documents through use of ICT

.4 The owner is to provide the necessary facilities for the safe execution of the survey

24.3.4 Performance of the remote survey

.1 To ensure that the Surveyor can properly perform the remote survey according to the applicable Rules, the available evidence is to allow the attending Surveyor to:

- examine and assess a survey item and/ or a group of items and/ or supporting documents,

- verify and assess applicable tests and/ or services.

.2 The evidence provided to the Surveyor is subject to technical evaluation and final acceptance by the Surveyor with respect to the completeness and accuracy, necessary to perform the requested survey according to the applicable requirements.

.3 One or more of the following types of evidence is to be provided for performing the remote survey:

- live-streaming video and audio
- recorded videos provided by the Owner's representative
- photos provided by the Owner's representative
- other data and/or supporting documents acceptable to IRS (such as ship's logbook, master's/ chief engineer's statement, owner's confirmation, ship's maintenance records, service reports issued by manufacturer's, service suppliers or service providers).

24.3.5 Assessment of the remote survey

.1 The Surveyor is to evaluate all evidence received and accept them before crediting the remote survey.

.2 The means used for the remote survey are to allow the Surveyor to collect the necessary evidence that will be examined according to the Surveyor's professional judgement in order to satisfactorily complete and credit the relevant survey items.

.3 If the remote survey does not provide the same level of assurance as a survey with attendance on board, then, a physical visit onboard by the Surveyor will be required.

24.4 Scope and Procedures of remote surveys

24.4.1 Scope of Remote Surveys

.1 A remote survey may be proposed, as an alternative to a survey attended on-board by the Surveyor for surveys listed in Table 24.4.1:

.2 Remote surveys are generally to be carried out with internet connection allowing a live streaming visual examination, although, at the discretion of the Surveyor, a combination of remote survey methods (see 24.3.4) may be used. For simple/ limited verifications, other types of ICT may be accepted by the Surveyor.

.3 When the classification survey is also related to a statutory item, and IRS is carrying out the statutory survey on behalf of the flag Administration, then the flag Administration acceptance is required, and additional requirements (if any) are to be complied with.

Table 24.4.1: Eligible Remote Survey Items

SNo.	Surveys and related items eligible for remote surveys	Live streaming required (See Notes)
1	Postponement, issuance, deletion of Condition of Class	X (1)
2	Postponement of Class surveys (Boiler Surveys, Tailshaft surveys for a maximum period of one year, Docking Surveys and CSM)	X (1)
3	Items of Continuous Survey for Machinery or Planned Maintenance Scheme	X (1)
4	Occasional survey for change of ship's name	X (1)
5	Occasional survey for loss of anchor	X (1)
6	Occasional survey for minor machinery or equipment damage	X (1)
7	Occasional survey for minor hull damage	X (1)
8	Occasional survey for minor deficiencies/ defects not subject to a Condition of Class	X (1)
9	In-water bottom survey	X
10	Specified items of a class periodical survey (excluding additional specific items of initial or special surveys), including completion of remaining items of a part held class periodical survey	X (1),(2)
11	Non-propelled / un-manned barges/pontoon – annual surveys when no survey of hull compartments is due	X
12	Minor retrofit / installation/upgrade of equipment	X (1)
13	Documentary or data based initial / periodical / renewal / occasional verifications and surveys	(2)

Notes:

(a) X - Yes

(b) (1): means that live streaming may not be required for minor survey scope or that a combination remote survey method, as listed in 24.3.4, may be used at the sole discretion of IRS.

(c) (2): means that pure documentary verifications are eligible in accordance with item 13.

(d) (3) Live streaming may be required for surveys not marked X in the Table, depending on the survey scope at the sole discretion of IRS.

(e) "Minor" in the items 6, 7, 8 and 12 means that the item can be surveyed remotely according to requirements for equivalency given in 24.3.

24.4.2 Procedures for Remote Surveys

24.4.2.1 Verification that eligibility requirements have been met (Refer 24.3.1)

24.4.2.2 Digital information quality, completeness and accuracy

.1 Verification of the quality of digital information by the Surveyor, and that the content and quality of digital information collected, and the survey carried out is satisfactory, allowing the Surveyor to confirm its completion.

.2 The Owner is to provide complete and accurate digital information. The digital information submitted by the Owner to the Surveyor is to reflect the real situation of the surveyed item. The date and time, when a photo or video was taken are to be made available to the Surveyor or identifiable from its metadata.

.3 Digital information is required to be collected and stored as evidence of the survey. It is not necessary to store all of digital information received; the exact digital

information stored is to support the survey decision and is to be decided by the Surveyor crediting the survey in accordance with the procedures of IRS.

.4 The remote survey is to be carried out under the supervision and upon instructions of the Surveyor, who is in charge of crediting the remote surveys. A Surveyor attendance on board may be required to complete the survey, upon the Surveyor's request and at his/ her discretion.

24.4.2.3 Requirements for a remote survey when live streaming is not used.

.1 When live streaming is not used, communication and digital information collection are to be performed through an ICT channel (such as emails, data streams and clouds), which is to be accepted by IRS prior to the survey.

.2 The Owner’s representative is to confirm the identity of the ship at the commencement of the survey.

24.4.2.4 Requirements for a remote survey when live streaming is used

.1 The Owner is to ensure that:

- the Owner’s representative is attending onboard and has access to the areas intended to be surveyed.

- the Owner’s representative has at his disposal a 2-ways visual and audible communication means complying with the requirements in 24.5.

- ICT solution is available on the communication means and meets the requirements indicated in 24.5.

In case these requirements cannot be fulfilled, the remote survey may be rejected. The Surveyor is to verify the identity of the ship at the commencement of the survey by live streaming.

24.5 Hardware and ICT Solution

24.5.1 Hardware

.1 The Owner is responsible for ensuring that all hardware installations on board used for the remote survey will comply with the applicable requirements relevant for use and location on board, including hazardous areas. The ICT will typically consist of:

- A host computer device onshore, to receive the streaming of images/data/video. This is usually a laptop or desktop computer compatible with the software application used for the remote survey

- On board standalone device which may include digital cameras capable of capturing videos/ photos/ data

- On board smart device compatible with the applicable software/technology

- Communication accessories like headphones and microphone for the noisy environment as applicable and as deemed necessary.

Note: The smart device may be a smartphone, tablet, computer, wearable device, smart glass, digital camera, or any other device which can be connected to the network and capable of transmitting the necessary data/ images to shore.

.2 The communication equipment used for live streaming is to have the following minimum functionalities:

- both ends are to simultaneously see the same image/videos in near real-time (i.e., live streaming)

- two-way direct voice communication

- possibility to take screenshots

.3 When using a portable device on board for live streaming, the movement of the handheld device may affect the stability of the video and the image, leading to lower quality outputs. When necessary, a suitable anti-shake device is to be used to provide proper stability.

Note:
1. The host computer screen is to be able to present an image quality that is sufficient to enable a survey decision to be made.

2. Portable equipment on board is to be equipped with a power capacity suitable for the intended scope and time of the survey.

24.5.2 Requirements for Internet Connectivity

.1 The Owner's representative is to ensure that internet connectivity tests are carried out before the survey and that proper connectivity is available and maintained during the survey.

.2 When live streaming communication is applied, the internet connection is to have sufficient and stable bandwidth capacity to ensure quality (such as resolution and frame rate) of the direct colour image/video and voice communication to the remote survey location to the satisfaction of the Surveyor.

.3 When remote survey by live streaming is being undertaken, a connection that enables live streaming between the Surveyor and the Owner's representative attending on board is required. The quality of the live streaming connection (audio and video) is to ensure proper communication and to allow the Surveyor to carry out the survey remotely, to the Surveyor's satisfaction. In case where a live streaming connection with the Surveyor is not possible or is not continuous at the place of the survey (e.g. Engine Room), partly online sequences (where the Owner is able to capture pictures and videos offline of those items not covered by live streaming) may be accepted by the Surveyor.

.4 The on-board smart devices are to have the capability of transmitting the images/ video/ data over a cellular, wi-fi or satellite connection to the remote Surveyor.

24.5.3 Software and Data Security

The audio/ video software or application selected for use for the remote survey is to be acceptable to IRS. The overall function and ability of the software used to ensure the security of data is to be evaluated prior to use as per the requirements given below.

.1 The Surveyor is to normally control the live video call, providing instructions to the on-site personnel/crew and supervising survey activities for capturing relevant information. The on-board device is to have the capability of transmitting the data over a cellular, wi-fi, or satellite connection to the Surveyor.

.2 The software used to perform the remote survey may also be provided with technologies that support the Surveyor in the process of making a decision, such as:

- Artificial Intelligence (AI) for the recognition and the classification of defects,
- Internet of things (IoT) for collecting parameters and evaluating acceptability/ working condition of machinery and equipment,
- Data driven verification or other means considered acceptable by IRS.

.3 When considering the use of software/ applications and other technologies, data protection is to be considered in accordance with applicable requirements of IRS (IRS *Guidelines on Maritime Cyber Safety* may be referred) before the remote survey is commenced. The software/ application used to perform the remote survey is to be compatible with the technical requirements detailed in this paragraph; in addition, the software used is to comply with IRS' applicable requirements for:

- Cybersecurity
- Data protection and confidentiality for the transmitted data

.4 During the survey preparation, it is the Owner's responsibility to ensure that their data security policies are implemented as per the Company's Safety Management System.

24.6 Recording of evidence (also see 24.3.4) and reporting of surveys

24.6.1 In principle, live streaming video and audio are applied to remote surveys as primary means (refer to Table 24.4.1).

24.6.2 Additionally, and/or alternatively, one or more of the following types of evidence may be submitted or verified as requested by the Surveyor during remote survey so that the Surveyor is able to verify conditions of survey items:

- Recorded video and audio
- Photos
- Ship's logbook
- Owner's confirmation

Note: Recorded videos/ photos provided by the Owner's representative is to be supplemented with a statement signed by the Master and/ or the Chief Engineer confirming the condition of the items shown in the evidence.

24.6.3 For the recorded videos/ photos, the following information is to be available:

- confirmation that they were actually taken on the ship by the Owner's representative
- date and time when they were taken
- identity of the personnel/ crew responsible for taking evidence

24.6.4 The Surveyor may request the Owner's representative to submit other supporting documents such as ship's maintenance reports and record for the operation of machinery, and equipment and service reports issued by manufacturers, service suppliers or service providers.

24.6.5 The Owner's representative is to confirm the correctness and completeness of the provided information and evidence (if any) relevant to the condition of the items requested to be surveyed.

24.6.6 The Master is to make entries into ship's logbook on the following occasions and submit copies of the relevant pages when requested by the Surveyor:

- when a remote survey is carried out by the Surveyor
- when videos/photos are taken and submitted to the Surveyor with the Master's/ Chief Engineer's statement and additional documents as applicable.

24.6.7 The final evaluation of the remote survey by the Surveyor is to be based on all of the provided evidence, and it does not delegate the responsibility to the Master/ Chief Engineer's statement only.

Section 25

Cyber Resilience Surveys

25.1 General

25.1.1 The requirements in this Section are applicable to periodical surveys for verifying compliance with cyber resilience requirements for ships assigned CyS-E Class Notation.

25.1.2 Construction and initial survey requirements for ships assigned CyS-E notation are specified in Part 4, Chapter 7, Section 7 of the Rules.

25.1.3 Ships will be subjected to annual and special surveys for verification of compliance with cyber resilience requirements.

Note: In the context of this Section, the term 'first annual survey' refers to the first annual survey post the delivery of the ship.

25.1.4 The general requirements to be complied with during the operation life, for ships assigned CyS-E notation are as follows:

.1 Once the cyber resilient ship is delivered, the Owner/Company is to manage technical and organizational security countermeasures by establishing and implementing processes as specified in Part 4, Chapter 7, Section 7 of the Rules.

.2 Modifications to the Computer Based Systems (CBSs) are to be carried out in accordance with the management of change (MoC) requirements in Part 4, Chapter 7, Section 6 of the Rules. This includes maintaining documentation of the CBSs up to date.

.3 The Owner/Company, with the support of suppliers, is to keep the Ship cyber resilience test procedure up to date and aligned with the CBSs onboard the ship and the networks connecting such systems to each other and to other CBSs not onboard (e.g. ashore).

.4 The Owner/Company is to update the Ship cyber resilience test procedure considering the changes occurred on CBSs and networks onboard, possible emerging risks related to such changes, new threats, new vulnerabilities and other possible changes in the ship's operational environment.

.5 The Owner/Company is to prepare and implement operational procedures, provide

periodic training and carry out drills for the onboard personnel and other concerned personnel ashore to familiarize them with the CBSs onboard the ship and the networks connecting such systems to each other and to other CBSs not onboard (e.g. ashore), and to properly manage the measures adopted for the fulfilment of requirements.

.6 The Owner/Company, with the support of supplier, is to keep the measures adopted for the fulfilment of requirements up to date, e.g. by periodic maintenance of hardware and software of CBSs onboard the ship and the networks connecting such systems.

.7 The Owner/Company is to retain onboard a copy of results of execution of tests and an updated Ship cyber resilience test procedure and make them available to IRS.

25.1.5 First Annual Survey

.1 the Owner/Company is to submit a Ship cyber security and resilience program to IRS for approval, documenting management of cyber security and cyber resilience of the CBSs, as relevant and applicable, well in advance of the first annual survey.

.2 The Ship cyber security and resilience program is to include policies, procedures, plans and/or other information documenting the processes/ activities specified in subsections “demonstration of compliance” in Part 4, Chapter 7, Section 7 of the Rules.

.3 The Owner/Company is to demonstrate compliance by presenting records or other documented evidence of implementation of the processes described in the approved Ship cyber security and resilience program.

.4 Change of vessel management company will require a new verification of the Ship cyber security and resilience program.

25.1.6 Subsequent Annual Surveys

In the subsequent annual surveys of the ship, the Owner/Company is to demonstrate implementation of the Ship cyber security and resilience program, when requested by IRS.

25.1.7 Special Surveys

The Owner/Company is to carry out testing witnessed by the Surveyor in accordance with the Ship cyber resilience test procedure. Certain security safeguards are to be demonstrated at Special survey whereas others need only be carried out upon request by the Surveyor based on modifications to the CBSs as specified in subsections “Operation phase” in Part 4, Chapter 7, Section 7 of the Rules.

25.2 Vessel asset inventory

25.2.1 First Annual Survey

.1 The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- The approved management of change process has been adhered to.

- Known vulnerabilities and functional dependencies have been considered for the software in the CBSs.

- The vessel asset inventory has been kept updated.

25.2.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.2.3 Special Surveys

The Owner/Company is to demonstrate to IRS the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.1.1.3 c) of the Rules) as per the Ship cyber resilience test procedure.

25.3 Security Zones and Network Segmentation

25.3.1 First annual survey

The Owner/Company is to demonstrate to the Surveyor that the Zones and conduit diagram has been kept updated and present records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that security zone boundaries are managed in accordance with the above requirements.

25.3.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.3.3 Special surveys

The Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.2.1.3 c) of the Rules) as per the Ship cyber resilience test procedure.

25.4 Network protection safeguards

25.4.1 Special surveys

Subject to modifications of the CBSs, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.2.2.3 a) of the Rules) as per the Ship cyber resilience test procedure.

25.5 Antivirus, antimalware, antispam and other protections from malicious code

25.5.1 First annual survey

The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- Any anti-malware software has been maintained and updated.

- Procedures for use of portable, mobile or removable devices have been followed.

- Policies and procedures for access control have been followed.

- Physical safeguards are maintained.

25.5.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.5.3 Special surveys

The Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.2.3.3 c) of the Rules) as per the Ship cyber resilience test procedure.

25.6 Access control

25.6.1 First annual survey

The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- Personnel are authorized to access the CBSs in accordance with their responsibilities.

- Only authorized devices are connected to the CBSs.

- Visitors are given access to the CBSs according to relevant policies and procedures.

- Physical access controls are maintained and applied.

- Credentials, keys, secrets, certificates, relevant CBS documentation, and other sensitive information is managed and kept confidential according to relevant policies and procedures.

25.6.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.7 Wireless communication

25.7.1 Special surveys

Subject to modifications of the wireless networks, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning

phase, refer Part 4, Chapter 7, Section 7.6.2.5.3 c) of the Rules) as per the Ship cyber resilience test procedure.

25.8 Remote access control and communication with untrusted networks

25.8.1 First annual survey

The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- Remote access sessions have been recorded or logged and carried out as per relevant policies and user manuals.

- Installation of security patches and other software updates have been carried out in accordance with Management of change procedures and in cooperation with the supplier.

25.8.2 Subsequent Annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.8.3 Special surveys

The Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.2.6.3 c) of the Rules) as per the Ship cyber resilience test procedure.

25.9 Use of Mobile and Portable Devices

25.9.1 First annual survey

The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- The use of mobile, portable or removable media is restricted to authorized personnel and follows relevant policies and procedures.

- Only authorized devices are connected to the CBSs.

- Means to restrict use of physical interface ports are implemented as per approved design documentation.

25.9.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.9.3 Special surveys

The Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.2.7.3 c) of the Rules) as per the Ship cyber resilience test procedure.

25.10 Network operation monitoring

25.10.1 First annual survey

The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- The CBSs are routinely monitored for anomalies by inspection of security audit records and investigation of alerts in the CBSs.

25.10.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.10.3 Special surveys

Subject to modifications of the CBS, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.3.1.3 a) of the Rules) as per the Ship cyber resilience test procedure.

25.11 Verification and diagnostic functions of CBS and networks

25.11.1 First annual survey

The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that the security functions in the CBSs are periodically tested or verified.

25.11.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.12 Incident Response Plan

25.12.1 First annual survey
The Owner/Company is to present to the Surveyor records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- The incident response plans are available for the responsible personnel onboard.

- Procedures or instructions for local/manual controls are available for responsible personnel onboard.

- Procedures or instructions for disconnection/isolation of security zones are available for responsible personnel onboard.

- Any cyber incidents have been responded to in accordance with the incident response plans.

25.12.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.13 Local, independent and/ or manual operation

25.13.1 Special surveys

Subject to modifications of the CBSs, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.4.2.3 b) of the Rules) as per the Ship cyber resilience test procedure.

25.14 Network isolation

25.14.1 Special surveys

Subject to modifications of the CBSs, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.4.3.3 b) of the Rules) as per the Ship cyber resilience test procedure.

25.15 Fallback to a minimal risk condition

25.15.1 Special surveys

Subject to modifications of the CBSs, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.4.4.3 b) of the Rules) as per the Ship cyber resilience test procedure.

25.16 Recovery plan

25.16.1 First annual survey

The Owner/Company is to present to the Surveyor, records or other documented evidence demonstrating implementation of the Ship cyber security and resilience program, i.e., that:

- Instructions and/or procedures for incident recovery are available for the responsible personnel onboard.

- Equipment, tools, documentation, and/or necessary software and data needed for recovery is available for the responsible personnel onboard.

- Backup of the CBSs have been taken in accordance with the policies and procedures.

- Manuals and procedures for shutdown, reset, restore and restart

are available for the responsible personnel onboard.

25.16.2 Subsequent annual surveys

The Owner/Company is to upon request by the Surveyor demonstrate implementation of the Ship cyber security and resilience program by presenting records or other documented evidence as specified for the first annual survey.

25.17 Backup and restore capability

25.17.1 Special surveys

Subject to modifications of the CBSs, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.5.2.3 a) of the Rules) as per the Ship cyber resilience test procedure.

25.18 Controlled shutdown, reset, roll-back and restart

25.18.1 Special surveys

Subject to modifications of the CBSs, the Owner/Company is to demonstrate to the Surveyor the relevant activities (pertaining to the commissioning phase, refer Part 4, Chapter 7, Section 7.6.5.3.3 b) of the Rules) as per the Ship cyber resilience test procedure.

25.19 Appendix 1 may be referred for summary of actions to be taken and documents to be maintained by various stakeholders.

Appendix I – Summary of actions and documents

Legend:

Submit	The stakeholder is to submit the document to IRS for verification and approval of compliance with requirements in Part 4, Chapter 7, Section 7 of the Rules.
Maintain	The stakeholder is to keep the document updated in accordance with procedure for management of change (MoC). Updated document and change management records are to be submitted to IRS as per Part 4, Chapter 7, Section 6 of the Rules.
Demonstrate	The stakeholder is to demonstrate compliance to IRS in accordance with the approved document.
1st AS	First annual survey
AS	Annual survey
SS	Special survey

Document (All referred Clauses in Part 4, Chapter 7, Section 7)	**Systems integrator**			**Owner/Company**			
	Design	**Construction**	**Commissioning**	**Operation**	**1st AS**	**AS**	**SS**
Approved supplier documentation [7.7]		Maintain	Maintain	Maintain			
Zones and conduit diagram [7.7.1.1]	Submit	Maintain	Maintain	Maintain			
Cyber security design description [7.7.1.2]	Submit	Maintain	Maintain	Maintain			
Vessel asset inventory [7.7.1.3]	Submit	Maintain	Maintain	Maintain			
Risk assessment for the exclusion of CBSs [7.7.1.4] Note1	Submit	Maintain	Maintain	Maintain			
Description of compensating countermeasures [7.7.1.5]Note 1	Submit	Maintain	Maintain	Maintain			
Ship cyber resilience test procedure [7.7.2.1]		Submit	Demonstrate	Maintain			Demonstrate
Ship cyber security and resilience program [7.7.3.1] - Management of change (MoC) [7.6.1.1.3 d)] - Management of software updates [7.6.1.1.3 d)] - Management of firewalls [7.6.2.1.3 d)] - Management of malware protection [7.6.2.3.3 d)] - Management of access control [7.6.2.4.3 d)] - Management of confidential information [7.6.2.4.3 d)] - Management of remote access [7.6.2.6.3 d)] - Management of mobile and portable devices [7.6.2.7.3 d)] - Detection of security anomalies [7.6.3.1.3 b)] - Verification of security functions [7.6.3.2.3.b)] - Incident response plans [7.6.4.1.3 b)] - Recovery plans [7.6.5.1.3 c)]				Maintain	Submit	Demonstrate	
Note 1: If applicable							

End of Chapter

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RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 2
INSPECTION AND TESTING OF MATERIALS

July 2024

Indian Register of Shipping

Part 2

Inspection and Testing of Materials

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Chapter 1

General Requirements

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Section 1

Conditions for Manufacture, Survey and Certification

1.1 Scope

1.1.1 Materials, used for the construction or repair of the hull and machinery of ships which are classed or intended to be classed with IRS, are to be manufactured, tested and inspected in accordance with the requirements of this Part.

1.1.2 Materials complying with recognized national or international standards with specifications equivalent to the requirements of this Part may be accepted.

1.2 Information to be supplied to the manufacturer

1.2.1 The ship or machinery builder is to provide the manufacturer with such information as is necessary to ensure that inspection and testing can be carried out in accordance with these Rules.

1.3 Manufacture

1.3.1 Materials used for the construction or repair of the hull and machinery of ships which are classed or intended to be classed with IRS are to be made at works which have been approved by IRS for the type of the product being supplied.

1.3.2 Where this part of the Rules requires materials or products to be manufactured at works approved by IRS, consideration may be given by IRS for the acceptance of works approved by other IACS Member Societies.

1.3.3 The manufacturer should demonstrate to the satisfaction of IRS that necessary manufacturing and testing facilities are available and are supervised by qualified personnel.

1.3.4 Approval of manufacturers with respect to the materials and grades covered by this Part will be considered by IRS on the basis of a detailed description of the manufacturing process and inspection routines, results from testing of materials and a report made by IRS Surveyors confirming the information given by the works and results.

1.3.5 Where the manufacturer has more than one works, approval for individual works would be required.

1.4 Survey procedure

1.4.1 The Surveyors are to be allowed access to all the relevant parts of the works and are to be provided with necessary facilities and information to enable them to verify that manufacture is being carried out in accordance with the approved procedure. Adequate facilities are also to be provided for the selection of test materials, the witnessing of mechanical tests and the examination of materials, as required by these Rules.

1.4.2 Prior to the submission of material for acceptance, manufacturers are to provide the Surveyors with details of the order specification and any special conditions additional to the Rule requirements.

1.4.3 Before final acceptance, all materials are to be submitted to specified tests and examinations under conditions acceptable to the Surveyors. The results are to comply with Rules and all materials are to be to the satisfaction of the Surveyors.

1.4.4 The specified tests and examinations are to be carried out prior to the dispatch of all finished materials from the manufacturer's works. Where materials are supplied in the rough or unfinished condition, as many as possible of the specified tests are to be carried out by the manufacturer and any tests or examinations not completed are to be carried out in consultation with the Surveyors, at a subsequent stage of manufacture.

1.4.5 In the event of any material proving unsatisfactory, during subsequent working, machining or fabrication, it is to be rejected, not withstanding any previous certification.

1.5 Chemical composition

1.5.1 The chemical composition of the ladle samples is to be determined by the manufacturer in an adequately equipped and competently staffed laboratory. The manufacturer's analysis will be accepted, but may be subject to occasional independent checks if required by the Surveyors.

1.5.2 At the discretion of the Surveyors, a check chemical analysis of suitable samples from products may also be required. These samples are to be taken from the material used for mechanical tests, but where this is not practicable an alternative procedure for obtaining a representative sample is to be agreed with the manufacturer.

1.6 Heat treatment

1.6.1 Materials are to be supplied in the condition specified in, or permitted by the Rules. Heat treatment is to be carried out in properly constructed furnaces which are efficiently maintained and have adequate means for control and recording of temperature. The furnace dimensions are to be such as to allow the whole item to be uniformly heated to the necessary temperature. In the case of very large components which require heat treatment, alternative methods will be specially considered.

1.7 Test material

1.7.1 Sufficient test material is to be provided for the preparation of the tests detailed in the specific requirements. It is, however, in the interests of manufacturers to provide additional material for any retests which may be necessary, as insufficient or unacceptable test material may be a cause for rejection.

1.7.2 The test material is to be representative of the item or batch and is not to be separated until all the specified heat treatment has been completed, except where provision for an alternative procedure is made in the subsequent chapters of this Part.

In case of castings where separately cast test samples are accepted, the test samples are to be cooled down under the same conditions as the castings.

1.7.3 All test material is to be selected by the surveyor and identified by suitable markings which are to be maintained during the preparation of the test specimen.

1.8 Mechanical tests

1.8.1 The number and direction of test specimens and their dimensions are to be in accordance with the requirements of subsequent chapters of this Part and the specific requirements for the product.

1.8.2 Where Charpy impact tests are required, a set of three test specimens are to be prepared and the average energy value is to comply with the requirements of subsequent Chapters of this part. One individual value may be less than the required average value provided that it is not less than 70 per cent of that value.

1.8.3 Where metric or imperial units are to be used for acceptance testing, the specified values are to be converted in accordance with the appropriate conversions given in Table 1.8.1.

Table 1.8.1 : **Conversion of SI units to metric and imperial units**

1 N/mm^2 or Mpa	= 0.102 kgf/mm^2
1 N/mm^2 or Mpa	= 0.0647 tonf/in^2
1 N/mm^2 or Mpa	= 0.145 x 10^3 lbf/in^2
1 J	= 0.102 Kgf m
1 J	= 0.738 ft lbs
1 Kgf/mm^2	= 9.81 N/mm^2 or MPa
1 tonf/in^2	= 15.45 N/mm^2 or MPa
1 lbf/in^2	= 6.89 x 10^{-3} MPa
1 kgf m	= 9.81 J
1 ft lbf	= 1.36 J

Notes :

The conversions may be rounded to the nearest multiples as follows :

1	For tensile strength values at ambient temperature	1 Kgf/mm^2 0.5 tonf/in^2 1 x 10^3 lbf.in^2 10N/mm^2
2	For yield and proof stress values at ambient temperature	0.5 kgf/mm^2 0.2 tonf/in^2 0.5 x 10^3 lbf/in^2 5 N/mm^2
3	For lower yield or proof stress values at elevated temperatures and stress to rupture.	0.1 kgf/mm^2 0.05 tonf/in^2 0.1 x 10^3 lbf/in^2 1 N/mm^2
4	For impact energy values	0.1 kgf m 1 ft lbf 1 J

1.9 Definitions

1.9.1 The following definitions are applicable to this Part:

Item A single forging, casting, plate, tube or other rolled product as delivered.

Piece The rolled product from a single slab or billet or from a single ingot if this is rolled directly into plates, strips, sections or bars.

Batch A number of similar items or pieces presented as a group for acceptance testing.

1.10 Retest procedures

1.10.1 Where the result of any test, other than an impact test, does not comply with the requirements, two additional tests of the same type may be taken. For acceptance of the material satisfactory results are to be obtained from both of these tests.

1.10.2 Where the results from a set of three impact test specimens do not comply with the requirements, an additional set of three impact test specimens may be tested provided that not more than two individual values are less than the required average value and, of these, not more than one is less than 70 per cent of this average value. The results obtained are to be combined with the original results to form a new average which, for acceptance, is not to be less than the required average value. Additionally, for these combined results, not more than two individual values are to be less than the required average value and, of these, not more than one is to be less than 70 per cent of this average value.

1.10.3 The additional tests detailed in 1.10.1 and 1.10.2 are, where possible, to be taken from material adjacent to the original tests. For castings, however, where insufficient material remains in the original test samples, the

additional tests may be prepared from other test samples representative of the castings.

1.10.4 When unsatisfactory results are obtained from tests representative of a batch of material, the item or piece from which the tests were taken is to be rejected. The remainder of the batch may be accepted provided that two further items or pieces are selected and tested with satisfactory results. If the tests from one or both of these additional items or pieces give unsatisfactory results, the batch is to be rejected.

1.10.5 When a batch is rejected, the remaining items or pieces in the batch may be re-submitted individually for test, and those which give satisfactory results may be considered for acceptance by the Surveyors.

1.10.6 At the option of the manufacturer, rejected material may be re-submitted as another grade and may then be considered for acceptance by the Surveyors, provided that the test results comply with the appropriate requirements.

1.10.7 When material which is intended to be supplied in the "as rolled" or "hot finished" condition fails test, it may be suitably heat treated and re-submitted for test, with the prior concurrence of the ship or machinery builder. Similarly materials supplied in the heat-treated condition may be re-heat treated and re-submitted for test.

1.11 Visual and non-destructive examination

1.11.1 Prior to the final acceptance of materials, surface inspection, verification of dimensions and non-destructive examination are to be carried out in accordance with the requirements detailed in subsequent chapters of this Part.

1.11.2 When there is visible evidence to doubt the soundness of any material or component, such as flaws in test specimens or suspicious surface marks, the manufacturer is expected to prove the quality of the material by any acceptable method.

1.12 Rectification of defective material

1.12.1 Small surface imperfections may be removed by mechanical means provided that, after such treatment, the dimensions are acceptable, the area is proved free from defects and the rectification has been completed in accordance with applicable requirements of subsequent chapters of this Part and to the satisfaction of Surveyors.

1.12.2 The repair of defects by welding can be accepted only when permitted by the appropriate specific requirements and provided that the agreement of the Surveyor is obtained before the work is commenced. When a repair has been agreed, it is necessary in all cases to prove by suitable methods of non-destructive examination that the defects have been completely removed before welding is commenced. Welding procedures and inspection on completion of the repair are to be in accordance with the appropriate specific requirement and are to be to the satisfaction of the Surveyor.

1.13 Identification of materials

1.13.1 The manufacturer is to adopt a system of identification which will enable all finished material to be traced to the original cast, and the Surveyors are to be given all facilities for so tracing the material when required. When any item has been identified by the personal mark of a Surveyor, or his deputy, this is not to be removed until an acceptable new identification mark has been made. Failure to comply with this condition will render the item liable to rejection.

1.13.2 Before any item is finally accepted it is to be clearly marked by the manufacturer in at least one place with the particulars detailed in the appropriate specific requirements.

1.13.3 Hard stamping is to be used except where this may be detrimental to the material, in which case stenciling, painting or electric etching is to be used. Paints used to identify alloy steels are to be free from lead, copper, zinc or tin, i.e., the dried film is not to contain any of these elements in quantities more than 250 ppm.

1.13.4 Where a number of identical items are securely fastened together in bundles, the manufacturer need only brand the top of each bundle. Alternatively a durable label giving the required particulars may be attached to each bundle.

Section 2

Certification of Materials Based on Alternative Certification Scheme

2.1 General

2.1.1 Alternative procedures for survey and testing may be accepted by IRS at works where materials are manufactured under closely control rolled conditions by semi-continuous or continuous processes under the Alternative Certification Scheme (ACS), as detailed in Part1, Chapter 1, Section 4 of the Rules.

2.1.2 Where it is considered that compliance with Rule requirements can be satisfactorily achieved, IRS will issue a Quality Assurance Approval Certificate, based on the ACS to the manufacturer.

2.1.3 The quality system procedures and practices of a manufacturer who has been granted approval will be kept under continuous review and audited as per the ACS.

2.2 Requirements for approval

2.2.1 The conditions for approval are broadly outlined in Pt.1, Ch.1, Sec. 4, Cl. 4.3 of the Rules.

2.3 Information required for approval

2.3.1 Manufacturers applying for approval under this scheme are to submit the information required by 2.3.2 to 2.3.6, in addition to the requirements of the ACS (Pt.1, Ch.1, Sec. 4, Cl. 4.4).

2.3.2 A detailed specification for each product.

2.3.3 An outline description of all important manufacturing plant and equipments. This is to include a production flow chart indicating all stages where testing and inspection are carried out along with details of equipments used for measuring and testing.

2.3.4 The system used for the identification and traceability of raw materials, semi-finished and finished Products.

2.3.5 Information on the system of procurement and acceptance of materials e.g. ingots, billets or blooms for further processing where the manufacturer does not produce such raw materials.

2.3.6 Consolidated test results, physical, chemical, non-destructive tests etc. for a period of preceding three months of products, if possible, covering the full range of thickness, weight range and grades for which approval is sought. The data is to include the number of samples, minimum, maximum, average value and standard deviation. For high strength ship steels, the carbon equivalent values are also required. The data is to also include numbers of rejections during manufacture as well as after delivery and reasons thereof.

2.4 Assessment and approval, maintenance of approval and certification of products would be generally based on the ACS detailed in Pt.1, Ch.1, Sec.4 of the Rules.

End of Chapter

Chapter 2

Mechanical Testing Procedures

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1 General Requirements

2 Tensile Testing

3 Impact Tests

4 Ductility Testing of Pipes and Tubes

5 The Brittle Crack Arrest Toughness Test

6 Isothermal Crack Arrest Temperature (CAT) Test

Section 1

General Requirements

1.1 General

1.1.1 All tests are to be carried out by competent personnel. The machines are to be maintained in satisfactory and accurate condition and are to be recalibrated at approximately annual intervals. This calibration is to be carried out by a nationally recognized Authority or other organization of standing and is to be carried out to the satisfaction of Surveyors. The accuracy of test machines is to be within ± one per cent. A record of all calibrations is to be kept available in the test house.

Testing machines are to be calibrated in accordance with the following or other equivalent recognized standards:

a) Tensile / compression testing : ISO 7500-1:2018
b) Impact testing : ISO 148-2:2016

1.2 Selection of test samples

1.2.1 Test samples are to be selected by the Surveyor unless otherwise agreed.

1.2.2 All materials in a batch presented for testing are to be of the same product form (e.g. plates, sections, bars). Normally, the materials are to be from the same cast and in the same condition of heat treatment.

1.3 Preparation of test specimens

1.3.1 If test samples are cut from material by flame cutting or shearing, a reasonable margin is required to enable sufficient material to be removed from the cut edges during final machining.

1.3.2 Test specimens are to be cut and prepared in a manner which does not affect their properties, i.e. not subjected to any significant cold straining or heating.

1.3.3 Where possible, test specimens from rolled materials are to retain their rolled surface on both sides.

1.4 Discarding of test specimens

1.4.1 If a test specimen fails because of faulty manufacture, visible defects, or incorrect operation of the testing machine, it may be discarded at the Surveyor's discretion and replaced by a new test specimen prepared from material adjacent to the original test.

Section 2

Tensile Testing

2.1 Dimensions of tensile test specimens

2.1.1 Generally, proportional test specimens with a gauge length of $5.65\sqrt{S_o}$ (where S_o is the cross-sectional area of the test length) are to be used. Where it is not possible to use such specimens, non-proportional specimens may be considered.

2.1.2 For the purpose of determining the different parameters related to tensile testing, three different types of test specimens may be used :

- Round test specimens;
- Flat test specimens; and
- Full cross-section test specimens.

See also Fig.2.1.1.

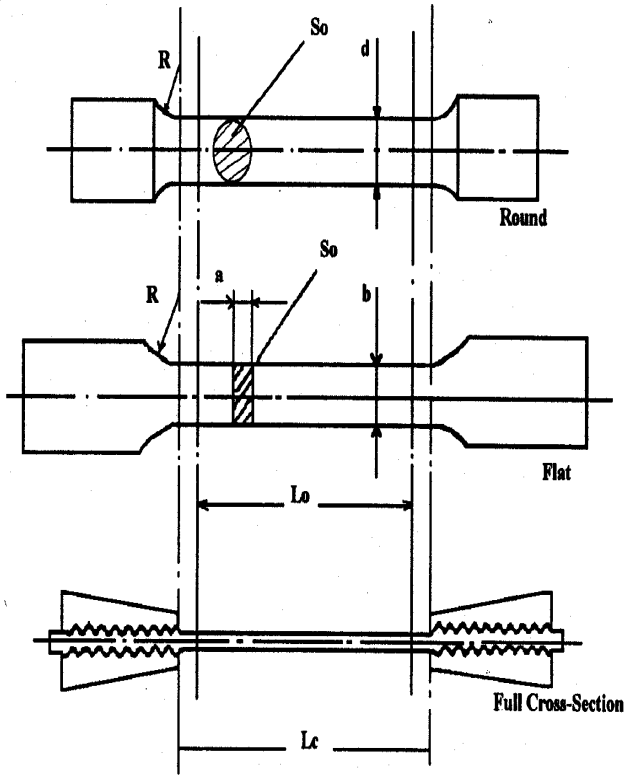


Fig. 2.1.1 : Tensile test specimens

2.1.2.1 The following symbols have been used in the figure and in subsequent paragraphs:-

d =diameter
a =thickness of specimen
b =width
Lo = Original gauge length
Lc = Parallel length
So = Original cross-sectional area
R = Transition radius
D =External tube diameter
t = plate thickness

2.1.2.2 The gauge length may be rounded off to the nearest 5 [mm] provided that the difference between this length and Lo is less than 10% of Lo.

2.1.2.3 For plates with thickness equal to and greater than 3 [mm], test specimen according to alternatives A or B given below are to be used. Where the capacity of the available testing machine is insufficient to allow the use of a test specimen of full thickness, this may be

reduced by machining one of the rolled surfaces.

Alternatively for materials over 40 [mm] thick, proportional round test specimens with dimensions as specified in C below may be used.

Alternative A, Non-proportional flat test specimen

a = t
b =25 [mm]
Lo =200 [mm]
Lc ≥ 212.5 [mm]
R =25 [mm]

Alternative B, Proportional flat test specimen

a = t

b =25 [mm]

$Lo = 5.65 \sqrt{S_o}$

$Lc \approx Lo + 2 \sqrt{S_o}$

R =25 [mm]

Alternative C, round test specimen

d =14 [mm] in general, but in no case less than 10 mm nor more than 20 [mm].

Lo = 5d [mm

Lc ≥ Lo + d/2 [mm]

R =10 [mm], in general
≥ 1.5 d [mm], for nodular cast iron and materials with a specified elongation of less than 10%.

2.1.2.4 The round test specimen is to be located with its center t/4 from the plate surface or as close to this position as possible.

2.1.2.5 For sheets and strips with thickness less than 3 [mm]

a = t
b =12.5 [mm]
Lo =50 [mm]
Lc ≈75 [mm]
R =25 [mm]

2.1.2.6 Wires: Full cross sectional test specimen with the following dimensions is to be used:

Lo =200 [mm]

Lc =Lo + 50 [mm].

2.1.2.7 For forgings, castings (excluding grey cast iron) and bars round test specimens with dimensions as specified in alternative C of 2.1.2.3 are usually to be used.

2.1.2.8 If for special reasons, other dimensions are to be used, they will have to conform with the following geometric relationship:

Lo =5d;

Lc =Lo + d:

R=10 [mm], except for materials with a specified minimum elongation A ≤ 10 per cent, where R is to be 1.5 x d.

2.1.2.9 For tubes, test specimen according to alternative A or B below are to be used:

Alternative A :- Full cross-section test specimens with plugged ends -

$Lo = 5.65 \sqrt{S_o}$

$Lc \approx Lo + D/2$

Lc is the distance between the grips or the plugs, whichever is smaller.

Alternative B :- Strip
a =wall thickness of tube
b ≥ 12 [mm]
$Lo = 5.65 \sqrt{S_o}$
Lc = Lo + 2b

The parallel test length is not to be flattened. but the enlarged ends may be flattened for gripping in the testing machine.

Round test specimens may also be used provided that the wall thickness is sufficient to allow the machining of such specimens to the dimensions in alternative C in 2.1.2.3 above with their axes located at the midwall thickness.

2.1.2.10 The above is subject to any specific dimensions or minimum cross-sectional area

requirements, with respect to test specimens, given in any subsequent Chapters of this Part.

2.1.2.11 Tensile test specimens for grey cast iron are to be machined to the dimensions shown in Fig.2.1.2. Usually test specimens are machined from separately cast standard test coupons with 30 [mm] diameter.

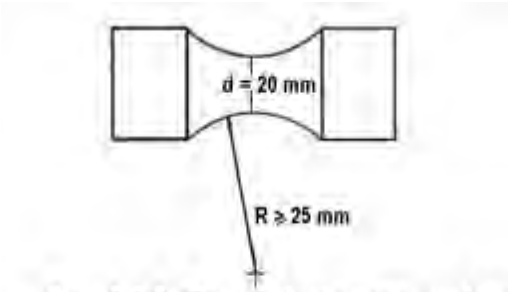


Fig : 2.1.2 : Tensile test specimen for grey cast iron

2.1.2.12 The tolerances on specimen dimensions are to be in accordance with ISO 6892-1:2019, ISO 6892-2:2018 or other recognised standards as appropriate.

2.1.2.13 Aluminium Alloys: Flat tensile test specimens are to be used for specified thicknesses up to and including 12.5 [mm]. The tensile test specimens are to be prepared so that both rolled surfaces are maintained. For thicknesses exceeding 12.5 [mm], round tensile specimens are to be used. For thicknesses up to and including 40 [mm], the longitudinal axis of the round tensile test specimen is to be located at a distance from the surface equal to half of the thickness. For thicknesses over 40 [mm], the longitudinal axis is to be located at a distance from one of the surfaces equal to one quarter of the thickness.

2.2 Fracture elongation

2.2.1 Unless otherwise specified, the elongation values in this part correspond to those required for proportional test specimens over a gauge length 5.65 $\sqrt{S_o}$.

If any part of the fracture takes place outside of the middle one-third of the original gauge length, the elongation value obtained may not be representative of the material. In such cases if the elongation measured is less than the minimum requirements, the test result may be discarded and a retest carried out.

2.2.2 If the material is ferritic steel of low or medium strength and not cold worked the elongation may also be measured on a non-proportional gauge length after agreement with IRS.

In that case the elongation required is to be calculated from the following formula:

$$A_o = 2 \text{ x } A_s \left[\frac{\sqrt{S_o}}{L_o} \right]^{0.4}$$

where,

A_o = Required elongation for the non-proportional test specimen

A_s = Specified elongation on a gauge length of $5.65\sqrt{S_o}$

S_o = Cross-sectional area of test specimen

L_o = Gauge length of test specimen.

For tables and graphs, refer ISO 2566-1:1984, ISO 2566-2:1984.

2.3 Definition of yield stress

2.3.1 The yield phenomenon is not exhibited by all the steels detailed in this Part but, for simplification the term "Yield Stress" is used throughout when requirements are specified for acceptance testing at ambient temperature.

2.3.2 Where reference is made to "Yield Stress" in the requirements for carbon, carbon-manganese and alloy steel products and in the requirements for the approval of welding consumables, either the upper yield stress or the 0.2 per cent proof stress under load is to be determined.

2.3.3 For austenitic and duplex stainless steel products and welding consumables, both the 0.2 per cent and 1.0 per cent proof stresses are to be determined.

2.4 Procedure for tensile testing at ambient temperature

2.4.1 Unless otherwise specified, the test is to be carried out at ambient temperature between 10^0C and 35^0C.

2.4.2 Yield stress (Yield point) is to be taken as the value of stress measured at the commencement of plastic deformation at yield or the value of the stress measured at the first peak obtained during yielding even when the peak is equal to or less than any subsequent peaks observed during plastic deformation at

yield. The tensile test is to be carried out with an elastic stress rate within the limits indicated in Table 2.4.2.

Table 2.4.2

Modulus of elasticity of the material (E) [N/mm^2]	Rate of stressing [N/mm^2] per second	
	Min.	Max.
< 150 000	2	20
≥ 150 000	6	60

2.4.3 After reaching the yield or proof load, the straining rate may be increased to a maximum of 0.008 per second for the determination of tensile strength.

2.4.4 For steel, the upper yield stress is to be calculated from :

a) the load immediately prior to a distinct drop in the testing machine lever; or

b) the load immediately prior to a fall back in the movement of the pointer or the load at a marked hesitation of this pointer; or

c) a load/extension diagram using the value of load measured either at the commencement of plastic deformation or yield or at the first peak obtained during yielding even when that peak is equal to or less than any subsequent peaks observed.

2.4.5 The 0.2 or 1.0 per cent proof stress (non-proportional elongation) is to be determined from an accurate load/extension diagram by drawing a line parallel to the straight elastic portion and distant from it by an amount representing 0.2 or 1.0 per cent of extensometer gauge length. The point of intersection of this line with the plastic portion of the diagram represents the proof load, from which 0.2 or 1.0 per cent proof stress can be calculated.

2.5 Procedure for tensile testing at elevated temperatures

2.5.1 The test specimens used for the determination of lower yield or 0.2 per cent proof stress at elevated temperatures are to have an extensometer gauge length of not less than 50 [mm] and a cross sectional area of not less than 65 [mm^2]. Where, however, this is precluded by the dimensions of the product or by the test equipment available, the test specimen is to be of the largest practical dimensions.

2.5.2 The heating apparatus is to be such that the temperature of the specimen during testing does not deviate from that specified by more than ± 5°C.

2.5.3 The straining rate when approaching the lower yield or proof load is to be controlled within the range 0.1 to 0.3 per cent of the extensometer gauge length per minute.

2.5.4 The time intervals used for estimation of strain rate from measurements of strain are not to exceed 6 seconds.

2.6 Bend Test Specimen

2.6.1 Flat bend test specimen, as given in the following, is to be used (Ref. Fig. 2.6.1). Edges on tension side to be rounded to a radius of 1 to 2 mm.

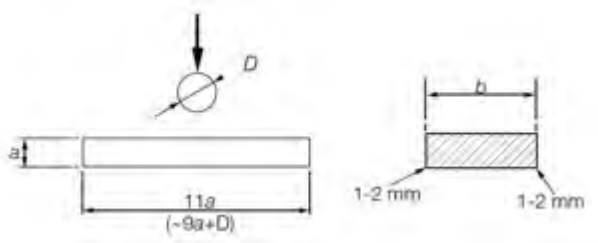


Fig. 2.6.1: Bend Test Specimen

2.6.2 Forgings, castings and semi-finished products

a = 20 mm
b = 25 mm

2.6.3 Plates, structural sections, sheets:

a = t
b = 30 mm

2.6.4 Butt welds, transverse specimen

a) face and root bend

a = t
b = 30 mm

If the as rolled thickness t is greater than 25 mm, it may be reduced to 25 mm by machining on the compression side of the bend specimen.

The surfaces of the weld are to be machined (ground) flush with the surface of the plate.

b) side bend

a = 10 mm
b = t

If t ≥ 40 mm, the side-bend specimen may be subdivided, each part being at least 20 mm wide.

2.6.5 Butt weld, longitudinal specimens

The test specimens, for longitudinal face and root test, are to be in accordance with an appropriate recognized standard.

Section 3

Impact Tests

3.1 Dimensions of test pieces

3.1.1 Impact tests are to be of either the charpy V-notch or the charpy U-notch type as required by the subsequent Chapters. The test specimens are to be machined to the dimensions and tolerances given in Table 3.1.1 and Table 3.1.2 and are to be carefully checked for dimensional accuracy.

3.1.2 For material under 10 [mm] in thickness the largest possible size of standard subsidiary charpy V-notch is to be prepared with the notch cut in the narrow face. Generally, impact tests are not required when the thickness of material is less than 5 [mm] (less than 6 [mm] for pipes and tubes).

Table 3.1.1 : Dimensions and tolerances for charpy V-notch impact test specimens

Dimensions	Nominal	Tolerance
Length [mm]	55	± 0.60
Width [mm]		
- standard specimen	10	± 0.11
- subsize specimen	7.5	± 0.11
- subsize specimen	5	± 0.06
Thickness [mm]	10	± 0.06
Angle of notch	45°	± 2°
Depth below notch [mm]	8	± 0.06
Root radius [mm]	0.25	± 0.025
Distance of notch from end of specimen [mm]	27.5	± 0.42
Angle between plane of symmetry of notch and longitudinal axis of test specimen	90°	± 2°
Ref. Fig.3.1.1		

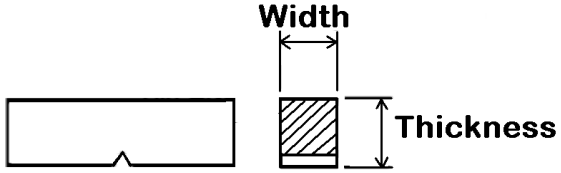


Fig.3.1.1

Table 3.1.2 : Dimensions and tolerances for charpy U-notch impact test specimens		
Dimensions	**Nominal**	**Tolerance**
Length [mm]	55	± 0.60
Width [mm]	10	± 0.11
Thickness [mm]	10	± 0.11
Depth of notch [mm]	5	± 0.09
Root radius [mm]	1	± 0.07
Distance of notch from end of test specimen [mm]	27.5	± 0.42
Angle between plane of symmetry of notch and longitudinal axis of test specimen	90°	± 2°
Ref. Fig.3.1.1		

3.2 Testing procedure

3.2.1 All impact tests are to be carried out on Charpy machines having a striking energy of not less than 150J and complying with following requirements:-

a) Distance between supports 40 + 5 [mm] - 0

b) Radius of curvature of supports 1 -1.5 [mm]

c) Taper of supports 1 in 5

d) Angle at tip of hammer 30 ± 1°

e) Radius of curvature of hammer 1.0-2.5[mm]

f) Speed of hammer at the instant of striking 4.5-7 [m/sec].

Note: Charpy machines complying with the requirements of recognized national or international standards such as ISO 148-2:2016 are acceptable.

3.2.2 Charpy U-notch impact tests are generally to be carried out at ambient temperature. Charpy V- notch impact tests may be carried out at ambient or lower temperatures in accordance with specific requirements given in subsequent Chapters. Where the test temperature is other than ambient, the temperature of the test specimen is to be controlled to within ± 2°C for sufficient time to ensure uniformity throughout the cross section of the test specimen, and suitable precautions are to be taken to prevent any significant change in temperature during the actual test. In cases of dispute, ambient temperature is to be considered as 18°C -27°C.

3.2.3 When reporting results, the units used for expressing the energy absorbed and the testing temperature are to be clearly stated. It is preferred that energy values for both charpy V-notch and charpy U-notch impact tests be expressed in Joules and not [J/cm^2].

3.2.4 The minimum average values for specimens are as given in Table 3.2.4.

3.3 Charpy Re-test procedure

3.3.1 When the average value of the three initial Charpy V-notch impact specimens fails to meet the stated requirements, or the value for more than one specimen is below the required average value, or when the value of any one specimen is below 70% of the specified average value, three additional specimens from the same material may be tested and the results added to those previously obtained to form a new average. If this new average complies with the requirements and if not more than two individual results are lower than the required average and of these, not more than one result is below 70% of the specified average value the piece or batch (as specified for each product) may be accepted.

3.4 Drop Weight Specimens

3.4.1 Drop weight specimens for determination of no–break performance according to ASTM E-208:2019 are to comply with this ASTM

standard and have one of the following dimensions (mm):

Type P-1: 25 by 90 by 360
Type P-2: 19 by 50 by 130
Type P-3: 16 by 50 by 130

The following is to be noted if not otherwise specified: the specimen sides are to be saw-cut or machined (minimum 25 mm to flame-cut surface) the machining of the plate to prescribed specimen thickness is to be on one side only the specimens may be of any orientation, but the orientation is to be the same for all specimens.

Table 3.2.4

Charpy V-notch specimen size	Minimum energy, average of 3-specimens
10 mm x 10 mm	E
10 mm x 7.5 mm	5E/6
10 mm x 5.0 mm	2E/3

Notes:

E = the values of energy specified for full thickness 10 mm x 10 mm specimens
All other dimensions and tolerances are to be as specified in Table 3.1.1.
Only one individual value may be below the specified average value provided it is not less than 70% of that value.
In all cases, the largest size Charpy specimens possible for the material thickness shall be machined

Section 4

Ductility Testing of Pipes and Tubes

4.1 Bend tests

4.1.1 The test specimens are to be cut as circum- ferential strips of full wall thickness and with a width of not less than 40 [mm]. For thick walled pipes, the thickness of the test specimens may be reduced to 20 [mm] by machining. The edges of specimen may be rounded to a radius of 1.6 [mm].

4.1.2 Testing is to be carried out at ambient temperature, and the specimens are to be doubled over, in the direction of the original curvature, around a former. The diameter of the former is to be in accordance with the specific requirements for the material. The test is to be considered satisfactory if, after bending, the specimens are free from cracks and laminations. Small cracks at the edges of the test specimen are to be disregarded.

4.2 Flattening tests

4.2.1 The test specimens are to be cut with the ends perpendicular to the axis of the pipe or tube. The length of the specimen is to be not less than 10 [mm] or greater than 100 [mm].

4.2.2 Testing is to be carried out at ambient temperature and is to consist of flattening the specimens in a direction perpendicular to the longitudinal axis of the pipe. (Reference is made to ISO 8492:2013). Flattening is to be carried out between two plain parallel and rigid platens which extend over both the full length and width after flattening of the test specimen. Flattening is to be continued until the distance between the platens, measured under load, is not greater than the value given by the formula:-

$$H = \frac{t(1+C)}{C + \frac{t}{D}}$$

where,

H = distance between platens [mm];

t = specified thickness of the pipe [mm];

D = Specified outside diameter [mm];

C = a constant dependent on the steel type and detailed in the specific requirements.

After flattening, the specimens are to be free from cracks or other flaws. Small cracks at the ends of the test specimens may be disregarded.

4.2.3 For welded pipes or tubes, the weld is to be placed at an angle of 90° to the direction of the pressure.

4.3 Drift expanding test

4.3.1 The test specimens are to be cut with the ends perpendicular to the axis of the tube. The edges of the end to be tested may be rounded by filing.

Metallic tubes: The length 'L' equal to twice the external diameter 'D' of the tube if the angle of the drift is 30° and L equal to 1.5D if the angle of the drift is 45° or 60°. (Reference ISO 8493:1998). The test piece may be shorter if after testing the remaining cylindrical portion is not less than 0.5D.

The rate of penetration of the mandrel is not to exceed 50 [mm]/minute.

4.3.2 Testing is to be carried out at ambient temperature and is to consist of expanding the end of the tube symmetrically by means of a hardened conical steel mandrel having a total included angle of 45° or 60°. The mandrel is to be forced into the test specimen until the percentage increase in the outside diameter of the end of the test specimen is not less than the value given in the specific requirements for boiler and superheater tubes. The mandrel is to be lubricated, but there is to be no rotation of the tube or mandrel during the test. The expanded portion of the tube is to be free from cracks or other flaws.

4.4 Flanging tests

4.4.1 The test specimens are to be cut with the ends perpendicular to the axis of the tube. The length of the specimens is to be approximately 1.5D. The length may be shorter provided that after testing the remaining cylindrical portion is not less than O.5D (Reference ISO 8494:2013).

The edges of the end to be tested may be rounded by filing. The rate of penetration shall not exceed 50 [mm]/minute.

4.4.2 Testing is to be carried out at ambient temperature and is to consist of flanging the end of the tube symmetrically by means of hardened conical steel mandrels.

4.4.3 The first stage of flanging is to be carried out with a conical angled mandrel having an included angle of approximately 90° (See Fig.4.4.3(a)) The completion of the test is achieved with a second forming tool as shown in Fig.4.4.3(b). The mandrels are to be lubricated and there is to be no rotation of the tube or mandrels during the test. The test is to continue until the drifted portion has formed a flange perpendicular to the axis of the test specimens. The percentage increase in the external diameter of the end of specimens is not to be less than the value given in the specific requirements for boiler and superheater tubes. The cylindrical and flanged portion of the tube is to be free from cracks or other flaws.

4.5 Ring expanding test

4.5.1 The test piece consists of a ring having a length of between 10 to 16 [mm]. (Reference ISO 8495:2013). The rate of penetration of the mandrel is not to exceed 30 [mm]/second.

4.6 Ring tensile test

4.6.1 The ring is to have a length of about 15 [mm] with plain and smoothed ends cut perpendicular to the tube axis. The ring is to be drawn to fracture by means of two mandrels placed inside the ring and pulled in tensile testing machine. The rate shall not exceed 5 [mm]/second. (Reference ISO 8496:2013).

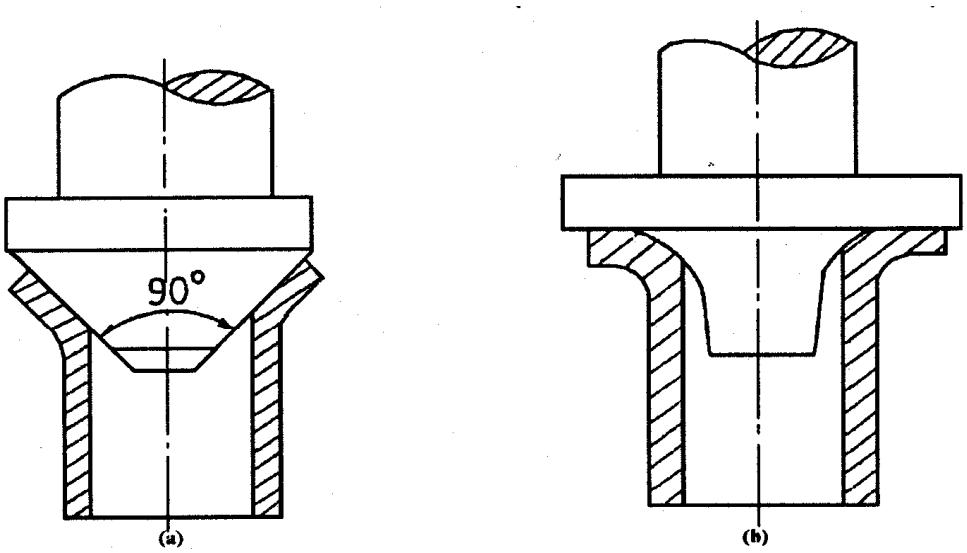


Fig.4.4.3 : Flanging test

Section 5

The Brittle Crack Arrest Toughness Test

5.1 Scope

5.1.1 This Section is applicable to hull structural steels with the thickness over 50 [mm] and not greater than 100 [mm] specified in Chapter 3. ISO20064: 2019 provides a test method for the determination of brittle crack arrest toughness of steel by using wide plates with a temperature gradient. The test procedures for brittle crack arrest toughness (i.e. K_{ca}) of steel using fracture mechanics parameter and determination method of Kca at a specific temperature (as specified in ISO 20064:2019) are provided. Additionally, the method of evaluation of Kca of test plate is also specified ..

5.2 Test Procedures

5.2.1 The test procedures including testing equipment, test specimens, test methods, determination of arrest toughness, reporting of test results, etc. are to be in accordance with ISO 20064: 2019. As a method for initiating a brittle crack, a secondary loading mechanism can be used in accordance with Annex D of ISO 20064: 2019, except that the first sentence in Annex B.2.4 of ISO 20064: 2019 is revised to "Obtain the value $\{K_{ca} /[K_0 *\exp(-c/T_{caK})]\}$ for each data point" (where T_K [K] (= T [C]+273), c and K_0 are constants).

5.3 Determination of Kca at a specific temperature and its evaluation

5.3.1 Method

5.3.1.1 The method for conducting multiple tests to obtain Kca value at a specific temperature is to be in accordance with Annex B of ISO 20064: 2019.

5.3.2 Evaluation

5.3.2.1 The straight-line approximation of Arrhenius plot for valid K_{ca} data by interpolation method are to comply with either the following (1) or (2):

(1) The evaluation temperature of K_{ca} (i.e. - 10 degree C) is located between the upper and lower limits of the arrest temperature, with the K_{ca} corresponding to the evaluation temperature not lower than the required K_{ca} (e.g. 6,000 N/mm$^{3/2}$ or 8,000 N/mm$^{3/2}$), as shown in Fig.5.3.2.1(a).

(2) The temperature corresponding to the required K_{ca} (e.g. 6,000 N/mm$^{3/2}$ or 8,000 N/mm$^{3/2)}$ is located between the upper and lower limits of the arrest temperature, with the temperature corresponding to the required K_{ca} not higher than the evaluation temperature (i.e. -10 degree C), as shown in Fig. 5.3.2.1(b).

If both of (1) and (2) above are not satisfied, conduct additional tests to satisfy this condition.

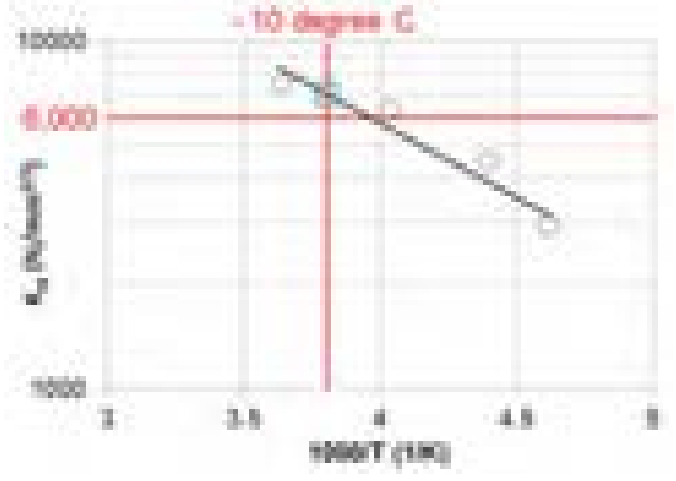


Fig. 5.3.2.1(a) Example for evaluation of K_{ca} at - 10 degree C

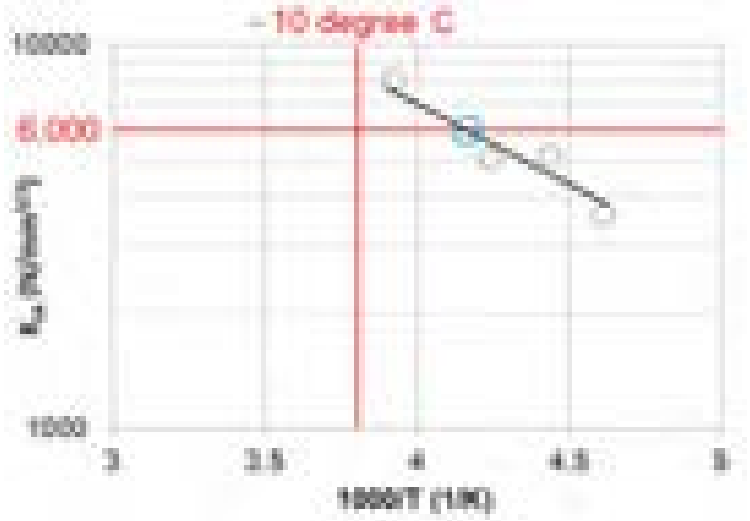


Fig. 5.3.2.1(b) Example for evaluation of temperature corresponding to the required K_{ca}

Section 6

Isothermal Crack Arrest Temperature (CAT) Test

6.1 Scope

6.1.1 This Section is applicable to steels with thickness over 50 [mm] and not greater than 100 [mm] for evaluating brittle crack arrest properties by the Crack Arrest Temperature (CAT) method.

6.1.2 Requirements for test procedures and test conditions when using the isothermal crack arrest test to determine a valid test result under isothermal conditions and in order to establish the crack arrest temperature (CAT) are specified in this Section.

6.1.3 This method uses an isothermal temperature in the test specimen being evaluated. Unless otherwise specified in this Section, the other test parameters are to be in accordance with ISO 20064:2019.

6.1.4 Chapter 3, Section 10, Table 10.1.3 specifies the relevant requirements for the brittle crack arrest property described by the crack arrest temperature (CAT).

6.1.5 The manufacturer is to submit the test procedure to IRS for review prior to testing.

6.2 Symbols

Table 6.2 supplements Table 1 in ISO 20064:2019 with specific symbols for the isothermal test

Table 6.2: Symbols used and their meanings (These are supplementary to Table 1 in ISO 20064:2019)		
Symbol	Unit	Significance
t	mm	Test specimen thickness
L	mm	Test specimen length
W	mm	Test specimen width
a_{MN}	mm	Machined notch length on specimen edge
L_{SG}	mm	Side groove length on side surface from the specimen edge. L_{SG} is defined as a groove length with constant depth except a curved section in depth at side groove end.
d_{SG}	mm	Side groove depth in section with constant depth
$L_{EB\ -\ min}$	mm	Minimum length between specimen edge and electron beam re-melting zone front
$L_{EB\text{-}s1,\ -s2}$	mm	Length between specimen edge and electron beam re-melting zone front appeared on both specimen side surfaces
L_{LTG}	mm	Local temperature gradient zone length for brittle crack runway
a_{arrest}	mm	Arrested crack length
T_{target}	°C	Target test temperature
T_{test}	°C	Defined test temperature
T_{arrest}	°C	Target test temperature at which valid brittle crack arrest behaviour is observed
σ	N/mm^2	Applied test stress at cross section of W x t
SMYS	N/mm^2	Specified minimum yield strength of the tested steel grade to be approved
CAT	°C	Crack arrest temperature, the lowest temperature, T_{arrest}, at which running brittle crack is arrested

6.3 Testing equipment

6.3.1 The test equipment to be used is to be of the hydraulic type of sufficient capacity to provide a tensile load equivalent to ⅔ of SMYS of the steel grade to be approved.

6.3.2 The temperature control system is to be equipped to maintain the temperature in the specified region of the specimen within ±2ºC from T_{target}.

6.3.3 Methods for initiating the brittle crack may be of drop weight type, air gun type or double tension tab plate type.

6.3.4 The detailed requirements for testing equipment are to be in accordance with ISO 20064:2019.

6.4 Test specimens

6.4.1 Impact type crack initiation

6.4.1.1 Test specimens are to be in accordance with ISO 20064:2019 unless otherwise specified in this Section.

6.4.1.2 Specimen dimensions are indicated in Fig. 6.4.1.2. The test specimen width is to be 500 [mm]. The test specimen length, L is to be equal or greater than 500 [mm].

6.4.1.3 V-shape notch for brittle crack initiation is machined on the specimen edge of the impact side. The whole machined notch length is to be equal to 29 [mm] with a tolerance range of ±1 [mm].

6.4.1.4 The requirements for side grooves are specified in 6.4.4.

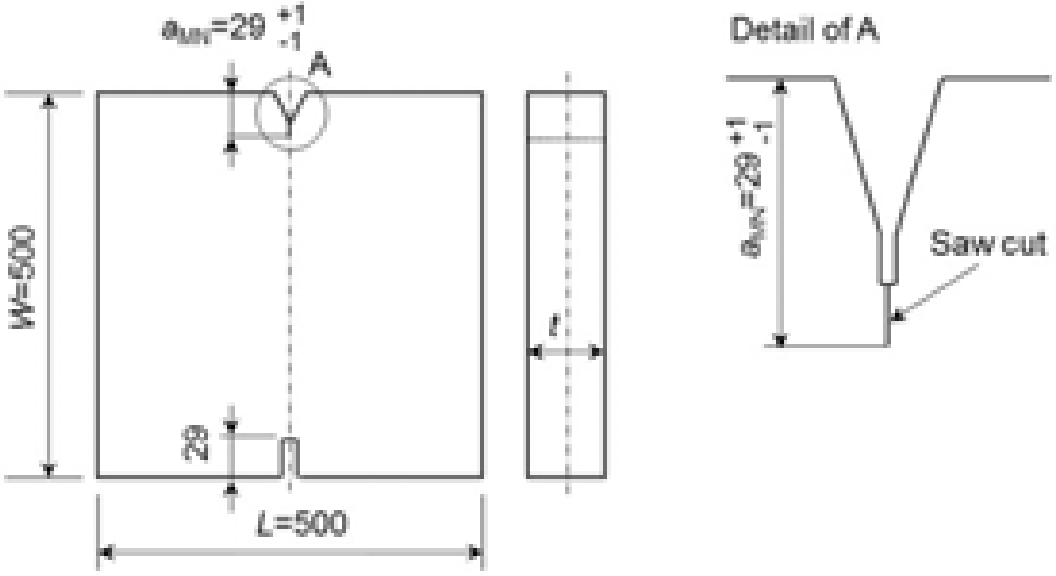


Fig. 6.4.1.2: Test specimen dimensions for an impact type specimen

Note: Saw cut notch radius may be machined in the range 0.1mmR and 1mmR in order to control a brittle crack initiation at test.

6.4.2 Double tension type crack initiation

6.4.2.1 Annex D of ISO 20064:2019 may be referred for the shape and sizes in secondary loading tab and secondary loading method for brittle crack initiation.

6.4.2.2 In a double tension type test, the secondary loading tab plate may be subject to further cooling to enhance an easy brittle crack initiation.

6.4.3 Embrittled zone setting

6.4.3.1 An embrittled zone is to be applied to ensure the initiation of a running brittle crack. Either Electron Beam Welding (EBW) or Local Temperature Gradient (LTG) may be adopted to facilitate the embrittled zone.

6.4.3.2 In EBW embrittlement, electron beam welding is applied along the expected initial crack propagation path, which is the centre line of the specimen in front of the machined V-notch.

6.4.3.3 The complete penetration through the specimen thickness is required along the embrittled zone. One side EBW penetration is preferable, but dual sides EB penetration may also be adopted when the EBW power is not enough to achieve the complete penetration by one side EBW.

6.4.3.4 The EBW embrittlement is recommended to be prepared before specimen contour machining.

6.4.3.5 In EBW embrittlement, zone is to be of an appropriate quality.

Note: EBW occasionally behaves in an unstable manner at start and end points. EBW line is recommended to start from the embrittled zone tip side to the specimen edge with an increasing power control or go/ return manner at start point to keep the stable EBW.

6.4.3.6 In LTG system, the specified local temperature gradient between machined notch tip and isothermal test region is regulated after isothermal temperature control. LTG temperature control is to be achieved just before brittle crack initiation, nevertheless the steady temperature gradient through the thickness shall be ensured.

6.4.4 Side grooves

6.4.4.1 Side grooves on side surface can be machined along the embrittled zone to keep brittle crack propagation straight. Side grooves

are to be machined in the specified cases as specified in this sub-section.

6.4.4.2 In EBW embrittlement, side grooves are not necessarily mandatory. Use of EBW avoids the shear lips. However, when shear lips are evident on the fractured specimen, e.g. shear lips over 1 [mm] in thickness in either side then side grooves are to be machined to suppress the shear lips.

6.4.4.3 In LTG embrittlement, side grooves are mandatory. Side grooves with the same shape and size are to be machined on both side surfaces.

6.4.4.4 The length of side groove, L_{SG} is to be no shorter than the sum of the required embrittled zone length.

6.4.4.5 When side grooves would be introduced, the side groove depth, the tip radius and the open angle are not regulated, but are adequately selected in order to avoid any shear lips over 1mm thickness in either side. An example of side groove dimensions is shown in Fig. 6.4.4.5.

6.4.4.6 Side groove end is to be machined to make a groove depth gradually shallow with a curvature larger than or equal to groove depth, d_{SG}. Side groove length, L_{SG} is defined as a groove length with constant depth except a curved section in depth at side groove end.

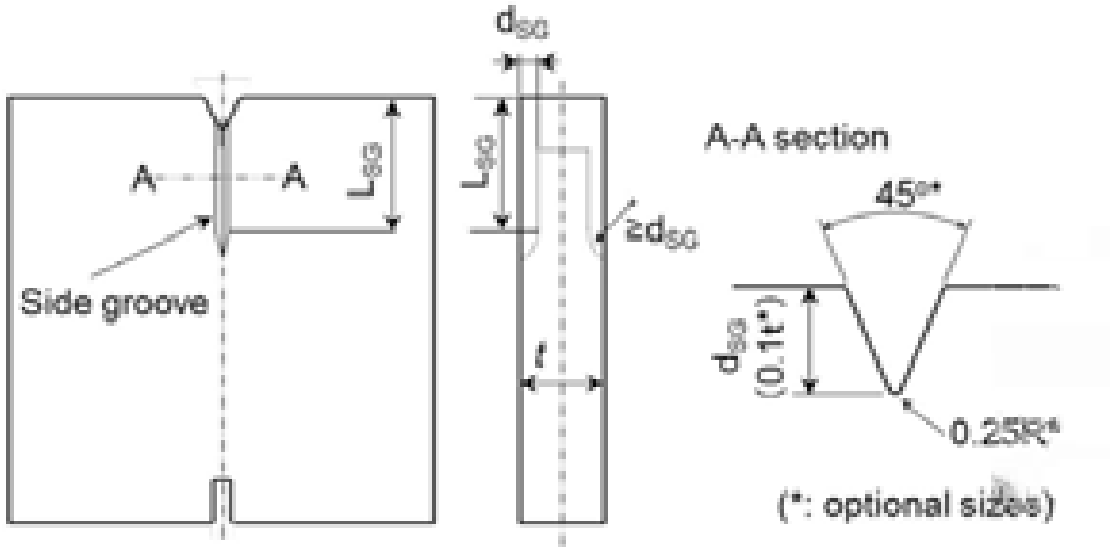


Figure 6.4.4.5: Side groove configuration and dimensions

6.4.5 Nominal length of embrittled zone

6.4.5.1 The length of embrittled zone is to be at least 150 [mm] .

6.4.5.2 EBW zone length is regulated by three measurements on the fracture surface after test as shown in Figure 6.4.5.2, $L_{EB\text{-}min}$ between specimen edge and EBW front line, and $L_{EB\text{-}s1}$ and $L_{EB\text{-}s2}$.

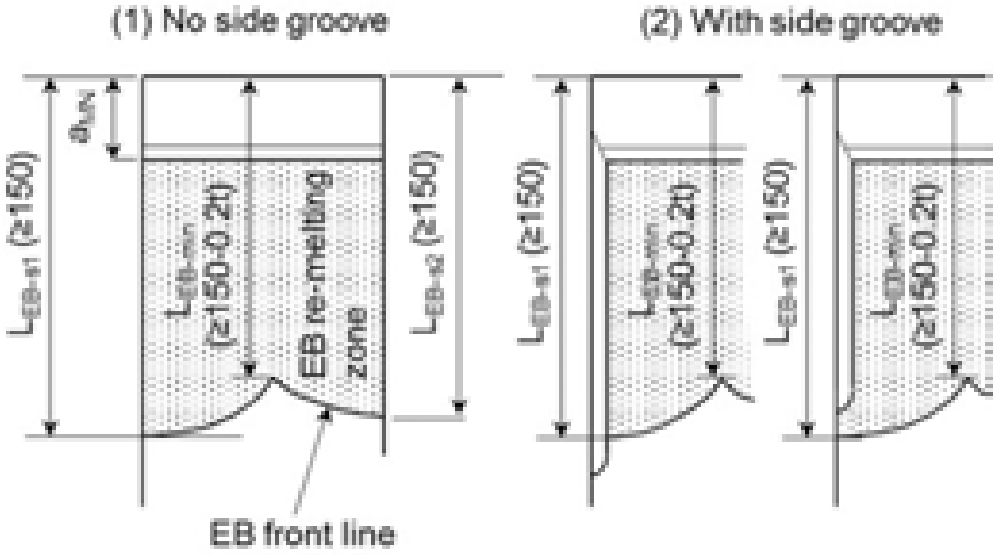


Fig. 6.4.5.2: Definition of EBW length

6.4.5.3 The minimum length between specimen edge and EBW front line, $L_{EB\text{-}min}$ should be no smaller than 150 [mm]. However, it can be acceptable even if $L_{EB\text{-}min}$ is no smaller than 150mm-0.2*t*, where *t* is specimen thickness. When $L_{EB\text{-}min}$ is smaller than 150 [mm], a temperature safety margin is to be considered into T_{test} (See Cl. 6.8.1.2).

6.4.5.4 Another two are the lengths between specimen edge and EBW front appeared on both side surfaces, as denoted with $L_{EB\text{-}s1}$ and $L_{EB\text{-}s2}$. Both of $L_{EB\text{-}s1}$ and $L_{EB\text{-}s2}$ are to be no smaller than 150 [mm].

6.4.5.5 In LTG system, L_{LTG} is set as 150 [mm].

6.4.6 Tab plate / pin chuck details and welding of test specimen to tab plates

6.4.6.1 The configuration and size of tab plates and pin chucks may be referred to ISO 20064:2019. The welding distortion in the integrated specimen, which is welded with specimen, tab plates and pin chucks, is to be also within the requirements in ISO 20064:2019.

6.5 Test Method

6.5.1 Preloading

6.5.1.1 Preloading at room temperature can be applied to avoid unexpected brittle crack initiation at test. The applied load value is to be no greater than the test stress. Preloading can be applied at higher temperature than ambient temperature when brittle crack initiation is expected at preloading process. However, the specimen is not to be subjected to temperature higher than 100 [^{o}C].

6.5.2 Temperature measurement and control

6.5.2.1 Temperature control plan showing the number and position of thermocouples is to be in accordance with this sub-section.

6.5.2.2 Thermocouples are to be attached to both sides of the test specimen at a maximum interval of 50 [mm] in the whole width and in the longitudinal direction at the test specimen centre position (0.5 *W*) within the range of ±100 [mm] from the centreline in the longitudinal direction, refer to Fig. 6.5.2.2.

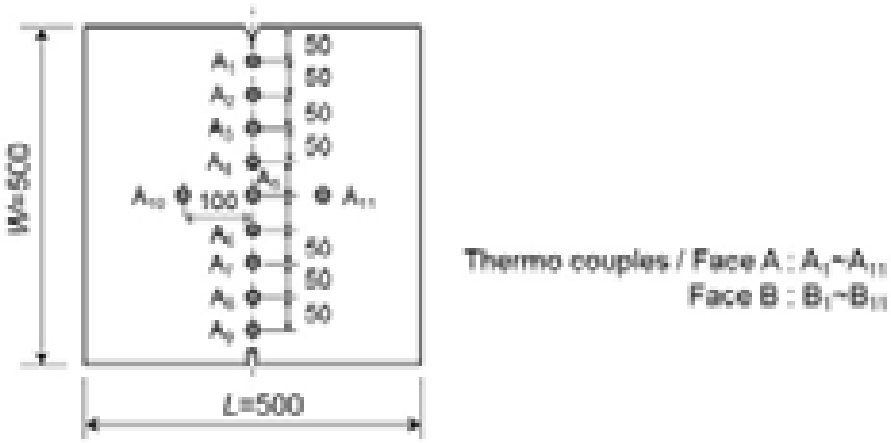


Fig. 6.5.2.2: Locations of temperature measurement

6.5.2.3 For EBW embrittlement

.1 The temperatures of the thermocouples across the range of 0.3*W*~0.7*W* in both width and longitudinal directions are to be controlled within ± 2°C of the target test temperature, T_{target}.

.2 When all measured temperatures across the range of 0.3*W*~0.7*W* have reached T_{target}, steady temperature control is to be kept at least for 10 + 0.1 x *t* [mm] minutes to ensure a uniform temperature distribution into mid-thickness prior to applying test load.

.3 The machined notch tip can be locally cooled to easily initiate brittle crack. Nevertheless, the local cooling is not to disturb the steady temperature control across the range of 0.3*W*~0.7*W*.

6.5.2.4 For LTG embrittlement

.1 In LTG system, in addition to the temperature measurements shown in Fig. 6.5.2.2, the additional temperature measurement at the machine notch tip, A_0 and B_0 is required. Thermocouples positions within LTG zone are indicated in Fig. 6.5.2.4.1.

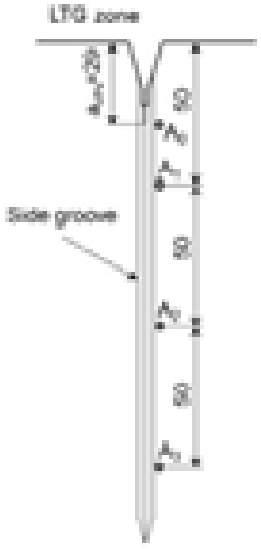


Figure 6.5.2.4.1: Detail of LTG zone and additional thermocouple A_0

.2 The temperatures of the thermocouples across the range of 0.3*W*~0.7*W* in both width and longitudinal directions are to be controlled within ± 2°C of the target test temperature, T_{target}. However, the temperature measurement at 0.3*W* (location of A_3 and B_3) is to be in accordance with 6.5.2.4.6 below.

.3 Once the all measured temperatures across the range of 0.3*W*~0.7*W* have reached T_{target}, steady temperature control is to be kept at least for 10 + 0.1 x *t* [mm] minutes to ensure a uniform temperature distribution into mid-thickness, then the test load is to be applied.

.4 LTG is controlled by local cooling around the machined notch tip. LTG profile is to be recorded by the temperature measurements from A_0 to A_3 shown in Fig. 6.5.2.4.4 below.

.5 LTG zone is established by temperature gradients in three zones, Zone I, Zone II and Zone III. The acceptable range for each temperature gradient is listed in Table 6.5.2.4.5.

.6 Temperature measurements at A_2, B_2 and A_3, B_3 are to satisfy the following requirements:

T at A_3, T at $B_3 < T_{target} - 2°C$
T at $A_2 < T$ at $A_3 - 5°C$
T at $B_2 < T$ at $B_3 - 5°C$

.7 There are no requirements for T at A_0 and T at A_1 temperatures when T at A_3 and T at A_2 satisfy the requirements above. Similar requirements are applicable for Face B.

.8 The temperatures from A_0, B_0 to A_3, B_3 are to be decided at test planning stage. Table 6.5.2.4.5 may be referred for the recommended temperature gradients in three zones, Zone I, Zone II and Zone III in LTG zone.

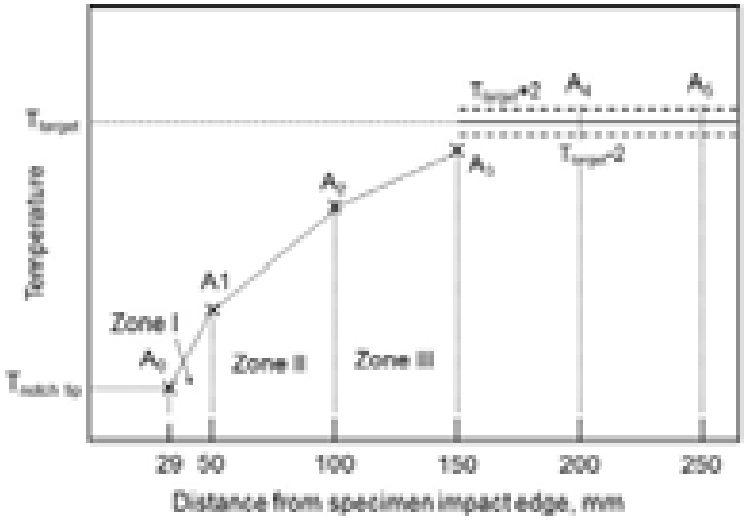


Fig. 6.5.2.4.4: Schematic temperature gradient profile in LTG zone

Table 6.5.2.4.5: Acceptable LTG range

Zone	Location from edge	Acceptable range of temperature gradient
Zone I	29mm – 50mm	2.00 ˚C/mm – 2.30 ˚C/mm
Zone II	50mm – 100mm	0.25 ˚C/mm – 0.60 ˚C/mm
Zone III[1)]	100mm – 150mm	0.10 ˚C/mm – 0.20 ˚C/mm

Note: The zone III arrangement is mandatory

.9 The temperature profile in LTG zone mentioned above is to be ensured after holding time at least for 10 + 0.1 x *t* [mm] minutes to ensure a uniform temperature distribution into mid-thickness before brittle crack initiation.

.10 The acceptance of LTG in the test is to be decided from Table 6.5.2.4.5 based on the measured temperatures from A_0 to A_3.

6.5.2.5 For double tension type crack initiation specimen:

.1 Temperature control and holding time at steady state is to be the same as the case of EBW embrittlement specified in 6.5.2.3 or the case of LTG embrittlement specified in Section 6.5.2.4.

6.5.3 Loading and brittle crack initiation

6.5.3.1 Prior to testing, a target test temperature (T_{target}) is to be selected.

6.5.3.2 Test procedures are to be in accordance with ISO 20064:2019 except that the applied stress is to be ⅔ of SMYS of the steel grade tested.

6.5.3.3 The test load is to be held at the test target load or higher for a minimum of 30 seconds prior to crack initiation.

6.5.3.4 Brittle crack can be initiated by impact or secondary tab plate tension after all of the temperature measurements and the applied force are recorded.

6.6 Measurements after test and test validation judgement

6.6.1 Brittle crack initiation and validation

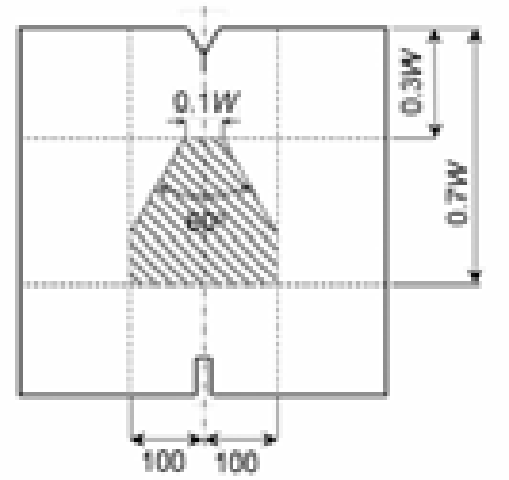
6.6.1.1 If brittle crack spontaneously initiates before the test force is achieved or the specified hold time at the test force is not achieved, the test would be considered as invalid.

6.6.1.2 If brittle crack spontaneously initiates without impact or secondary tab tension but after the specified time at the test force is achieved, the test would be considered as a valid initiation. The following validation judgments of crack path and fracture appearance are to be examined.

6.6.2 Crack path examination and validation

6.6.2.1 When brittle crack path in embrittled zone deviates from EBW line or side groove in LTG system due to crack deflection and/ or crack branching, the test would be considered as invalid.

6.6.2.2 All of the crack path from embrittled zone end is to be within the range shown in Fig. 6.6.2.2. If not, the test would be considered as invalid.



Fig, 6.6.2.2: Allowable range of main crack propagation path

6.6.3 Fracture surface examination, crack length measurement and their validation

6.6.3.1 Fracture surface is to be observed and examined. The crack "initiation" and "propagation" are to be checked for validity and judgements recorded. The crack "arrest" positions are to be measured and recorded.

6.6.3.2 When crack initiation trigger point is clearly detected at side groove root, other than the V-notch tip, the test would be invalid.

6.6.3.3 In EBW embrittlement setting, EBW zone length is quantified by three measurements of $LEB\text{-}_{s1}$, $LEB\text{-}_{s2}$ and $LEB\text{-}_{min}$, which are defined in 6.4.5. When either or both of LEB_{-s1} and LEB_{-s2} are smaller than 150 [mm], the test would be invalid. When LEB_{-min} is smaller than 150mm-0.2t, the test would be invalid.

6.6.3.4 When the shear lip with thickness over 1 [mm] in either side near side surfaces of embrittled zone are visibly observed independent of the specimens with or without side grooves, the test would be invalid.

6.6.3.5 In EBW embrittlement setting, the penetration of brittle crack beyond the EBW front line is to be visually examined. When any brittle fracture appearance area continued from the EB front line is not detected, the test would be invalid.

6.6.3.6 The weld defects in EBW embrittled zone are to be visually examined. If detected, they are to be quantified. A projecting length of defect on the thickness line through EB weld region along brittle crack path is to be measured, and the total occupation ratio of the projected defect part to the total thickness is defined as defect line fraction (See Fig. 6.6.3.6). When the defects line fraction is larger than 10 %, the test would be invalid.

6.6.3.7 In EBW embrittlement by dual sides' penetration, if a gap on embrittled zone fracture surface which is induced by miss meeting of dual fusion lines, is visibly detected at an overlapped line of dual side penetration, the test would be invalid.

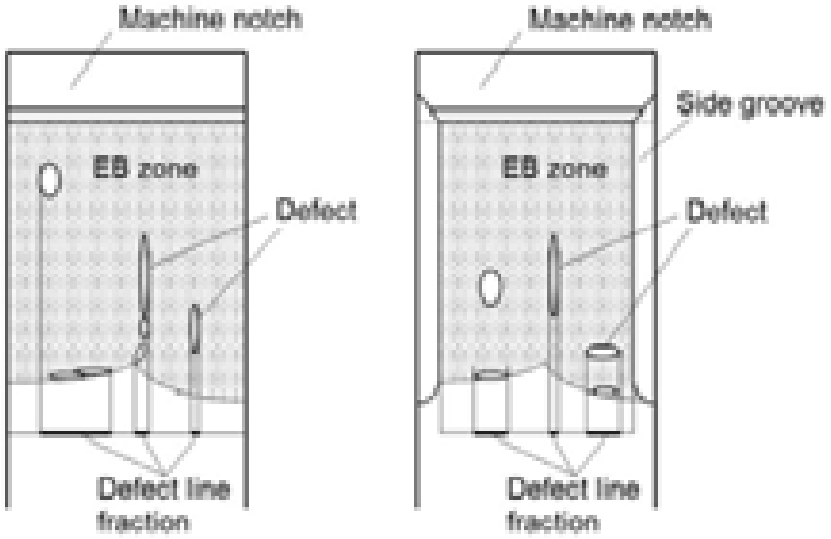


Fig. 6.6.3.6: Counting procedure of defect line fraction

6.7 Judgement of 'arrest' or 'propagate'

6.7.1 The judgment of "arrest", "propagate" or "invalid" would be decided based on the requirements indicated in 6.7.2 through 6.7.6.

6.7.2 If initiated brittle crack is arrested and the tested specimen is not broken into two pieces, the fracture surfaces should be exposed with the procedures specified in ISO 20064:2019.

6.7.3 When the specimen is not broken into two pieces during testing, the arrested crack length, a_{arrest} is to be measured on the fractured surfaces. The length from the specimen edge of impact side to the arrested crack tip (the longest position) is defined as a_{arrest}.

6.7.4 For LTG and EBW, a_{arrest} is to be greater than L_{LTG} and $L_{EB\text{-}s1}$, $L_{EB\text{-}s2}$ or $L_{EB\text{-}min}$. If not, the test would be considered as invalid.

6.7.5 Even when the specimen is broken into two pieces during testing, it can be considered as "arrest" when brittle crack re-initiation is clearly evident. Even in the fracture surface all occupied by brittle fracture, when a part of brittle crack surface from embrittled zone is continuously surrounded by thin ductile tear line, the test can be judged as re-initiation behaviour. If so, the maximum crack length of the part surrounded tear line can be measured as a_{arrest}. If re-initiation is not visibly evident, the test is judged as "propagate".

6.7.6 The test would be judged as "arrest" when the value of a_{arrest} is no greater than 0.7W. If not, the test would be judged as "propagate".

6.8 T_{test}, T_{arrest} and CAT determination

6.8.1 T_{test} determination

6.8.1.1 It shall be ensured on the thermocouple measured record that all temperature measurements across the range of 0.3W ~ 0.7W in both width and longitudinal direction are in the range of T_{target} ±2°C at brittle crack initiation. If not, the test shall be invalid. However, the temperature measurement at 0.3W (location of A_3 and B_3) in LTG system shall be exempted from this requirement.

6.8.1.2 If $L_{EB\text{-}min}$ in EBW embrittlement is no smaller than 150 [mm], T_{test} can be defined to equal with T_{target}. If not, T_{test} shall be equaled with T_{target} + 5°C.

6.8.1.3 In LTG embrittlement, T_{test} can be equaled with T_{target}.

6.8.1.4 The final arrest judgment at T_{test} is concluded by at least two tests at the same test condition which are judged as "arrest".

6.8.2 T_{arrest} **determination**

6.8.2.1 When at least repeated two "arrest" tests appear at the same T_{target}, brittle crack arrest behaviour at T_{target} will be decided (T_{arrest} = T_{target}).When a "propagate" test result is included in the multiple test results at the same T_{target}, the T_{target} cannot to be decided as T_{arrest}.

6.8.3 CAT determination

6.8.3.1 When CAT is determined, one "propagate" test is needed in addition to two "arrest" tests. The target test temperature, T_{target} for "propagate" test is recommended to select 5°C lower than T_{arrest}. The minimum temperature of T_{arrest} is determined as CAT.

6.8.3.2 With only the "arrest" tests, without "propagation" test, it is decided only that CAT is lower than T_{test} in the two "arrest" tests, i.e. not deterministic CAT.

6.9 Reporting

6.9.1 The following items are to be reported:

(i) Test material: grade and thickness

(ii) Test machine capacity

(iii) Test specimen dimensions: thickness t; width W and length L; notch details and length a_{MN}, side groove details if machined;

(iv) Embrittled zone type: EBW or LTG embrittlement

(v) Integrated specimen dimensions: Tab plate thickness, tab plate width, integrated specimen unit length including the tab plates, and distance between the loading pins, angular distortion and linear misalignment

(vi) Brittle crack trigger information: impact type or double tension. If impact type, drop weight type or air gun type, and applied impact energy.

(vii) Test conditions; Applied load; preload stress, test stress
- Judgements for preload stress limit, hold time requirement under steady test stress.

(viii) Test temperature: complete temperature records with thermocouple positions for measured temperatures (figure and/or table) and target test temperature.

- Judgements for temperature scatter limit in isothermal region.

- Judgement for local temperature gradient requirements and holding time requirement after steady local temperature gradient before brittle

crack trigger, if LTG system is used.

(ix) Crack path and fracture surface: tested specimen photos showing fracture surfaces on both sides and crack path side view; Mark at “embrittled zone tip” and “arrest” positions.

- Judgment for crack path requirement.

- Judgment for cleavage trigger location (whether side groove edge or V-notch edge).

(x) Embrittled zone information:

When EBW is used: $L_{EB\text{-}s1}$, $L_{EB\text{-}s2}$ and $L_{EB\text{-}min}$

- Judgement for shear lip thickness requirement

- Judgment whether brittle fracture appearance area continues from the EBW front line

- Judgement for EBW defects requirement

- Judgement for EBW lengths, $L_{EB\text{-}s1}$, $L_{EB\text{-}s2}$ and $L_{EB\text{-}min}$ requirements

When LTG is used: L_{LTG}

- Judgment for shear lip thickness requirement

Test results:

When the specimen did not break into two pieces after brittle crack trigger, arrested crack length a_{arrest}

When the specimen broke into two pieces after brittle crack trigger,

- judgement whether brittle crack re-initiation or not.

If so, arrested crack length a_{arrest}:

- Judgement for a_{arrest} in the valid range ($0.3W < a_{arrest} \leq 0.7W$)

- Final judgement either “arrest”, “propagate” or “invalid”

(xi) Dynamic measurement results: History of crack propagation velocity, and strain change at pin chucks, if needed

6.10 Use of test for material qualification testing

Where required, the method can also be used for determining the lowest temperature at which a steel can arrest a running brittle crack (the determined CAT) as the material property characteristic in accordance with 6.8.3.

End of Chapter

Chapter 3

Rolled Steel Plates, Strips, Sections and Bars

Contents

Section	
1	*General Requirements*
2	*Normal Strength Steels for Ship Structures*
3	*Higher Strength Steels for Ship Structures*
4	*High Strength Steels for Welded Structures*
5	*Steel for Low Temperature Service*
6	*Steels for Boilers and Pressure Vessels*
7	*Steels for Machinery Structures*
8	*Plates with Specified minimum through Thickness Properties ('Z' quality)*
9	*Austenitic and Duplex Stainless Steels*
10	*Brittle Crack Arrest Steels*

Section 1

General Requirements

1.1 Scope

1.1.1 This Chapter gives general requirements for hot rolled plates, wide flats, strips and sections intended for use in the construction of ships, boilers, pressure vessels and machinery structures. These requirements are also applicable to hot rolled bars, except where such materials are intended for the manufacture of bolts, shafts, etc. by machining operations only. When used for this purpose hot rolled bars are to comply with the requirements of Ch. 5.

1.2 Manufacture

1.2.1 The steel is to be manufactured at the approved works by the open hearth, electric furnace or one of the basic oxygen processes or by other processes specially approved by IRS.

The approval of the steel works is to be carried out in accordance with IRS Classification Notes: 'Approval scheme for the manufacturing process of normal and higher strength hull structural steels'. Also refer Chapter 1, Section 1, Cl. 1.3.2.

1.2.2 The suitability of each grade of steel for forming and welding is to be demonstrated during the initial approval tests at the steel works. The type and the extent of testing required is at the discretion of IRS.

1.2.3 It is the manufacturer's responsibility to assure that effective process and production controls in operation are adhered to in accordance with the manufacturing specifications. Where control imperfection that may lead to inferior quality of product occurs, the manufacturer is to identify the cause and establish counter measure to prevent its occurrence. Also the complete investigation report is to be submitted to the Surveyor. Each affected piece considered for further usage is to be tested to the Surveyor's satisfaction.

The frequency of testing may be increased to gain confidence for subsequent products as considered necessary.

1.3 Quality of materials

1.3.1 Defects not prejudicial to the proper application of steel are not, except by special agreement, to be grounds for rejection. Where necessary, suitable methods of non-destructive examination may be used for the detection of harmful surface and internal defects. The extent of this examination, together with appropriate acceptance standards, is to be agreed between the purchaser, manufacturer and Surveyors.

1.4 Thickness tolerance of plates and wide flats with width ≥ 600 [mm]

1.4.1 Following requirements are applicable to the tolerance on thickness of steel plates and wide flats with widths of 600 [mm] or greater (herein after referred to as: product or products) with thickness of 5 [mm] and over, covering the following :

(i) Normal and higher strength hull structural steels (Refer Sec 2, and Sec 3)
(ii) High strength steels for welded structure (Refer Sec 4)
(iii) Steel for machinery structures (Refer Sec 7)

These requirements do not apply to products intended for the construction of boilers, pressure vessels and independent tanks, e.g. for the transportation of liquefied gases or chemicals.

These requirements do not apply to products intended for the construction of lifting appliances.

1.4.2 The tolerance on thickness of a given product are defined as follows:

a) Minus tolerance is the lower limit of the acceptable range below the nominal thickness.
b) Plus tolerance is the upper limit of the acceptable range above the nominal thickness.

Note : Nominal thickness is stated by the purchaser at the time of enquiry and order.

1.4.3 The minus tolerance for products for normal strength, higher strength and high strength quenched and tempered steels is 0.3 [mm] irrespective of nominal thickness.

1.4.4 The minus tolerance for products intended for machinery structures are to be in accordance with Table 1.4.4.

1.4.5 The tolerance for thickness below 5 [mm] is to be in accordance with a national or international standard, e.g. Class B of ISO 7452:2013. However, the minus tolerance is not to exceed 0.3 [mm].

Table 1.4.4

Nominal thickness [mm]	Minus tolerance on nominal thickness [mm]
≥ 3 to < 5	-0.3
≥ 5 to < 8	-0.4
≥ 8 to < 15	-0.5
≥ 15 to < 25	-0.6
≥ 25 to < 40	-0.7
≥ 40 to < 80	-0.9
≥ 80 to < 150	-1.1
≥ 150 to < 250	-1.2
≥ 250	-1.3

1.4.6 The plus tolerance on nominal thickness is to be in accordance with a recognized national or international standard or as specified.

1.4.7 The tolerance on sections (except for wide flats) are to be in accordance with the requirements of recognized international or national standard.

1.4.8 The tolerances on nominal thickness are not applicable to areas repaired by grinding. For areas repaired by grinding, the requirements of Sec.2, 2.7.4.1 are to be applied, unless stricter requirements as per a recognized standard are specified by the purchaser.

1.4.9 For materials intended for applications as detailed in Sec. 5 and 6, no minus tolerance is permitted in the thickness of plates and strip.

1.4.10 The responsibility for verification and maintenance of the production within the required tolerance rests with the manufacturer. The Surveyor may require to witness some measurements.

1.4.11 The responsibility for storage and maintenance of the delivered products with acceptable level of surface conditions rests with

the shipyard before the products are used in fabrication.

1.4.12 Where zero minus tolerance is applied in accordance with Class C of ISO 7452-2013 or equivalent national or international standards, the requirements of 1.4.13 to 1.4.15 need not be applied.

Additionally, if Class C of ISO 7452-2013 is applied, it is required that the steel mill demonstrates to the satisfaction of IRS that the number of measurements and measurement distribution is appropriate to establish that the mother plates produced are at or above the specified nominal thickness.

1.4.13 Average thickness

1.4.13.1 The average thickness of products is defined as the arithmetic mean of the measurements made in accordance with the requirements of 1.4.14.

1.4.13.2 The average thickness of products for hull structural steels is not to be less than the nominal thickness.

1.4.14 Thickness measurements

1.4.14.1 The thickness is to be measured at locations of products as defined in 1.4.15.

1.4.14.2 Automated method or manual method may be applied to the thickness measurements.

1.4.14.3 The procedure and the records of measurements are to be made available to the Surveyor and copies provided on request.

1.4.15 Thickness measuring locations

1.4.15.1 The requirements of 1.4.15.2 are to be applied to the thickness measuring locations for the thickness tolerance and the average thickness of the product.

1.4.15.2 At least two lines among Line 1, Line 2 or Line 3 as shown in Fig.1.4.15.2 are to be selected for the thickness measurements and at least three points on each selected line are to be selected for thickness measurement. If more than three points are taken on each line the number of points are to be equal on each line.

Note : The measurement locations apply to a product rolled directly from one slab or steel ingot even if the product is to be later cut by the manufacturer. Examples of the original measurements relative to later cut products are shown in Fig.1.4.15.2b), It is to be noted that the examples shown are not representative of all possible cutting scenarios.

For automated methods, the measuring points at sides are to be located not less than 10 [mm] but not greater than 300 [mm] from the transverse or longitudinal edges of the product.

For manual methods, the measuring points at sides are to be located not less than 10 [mm] but not greater than 100 [mm] from the transverse or longitudinal edges of the product.

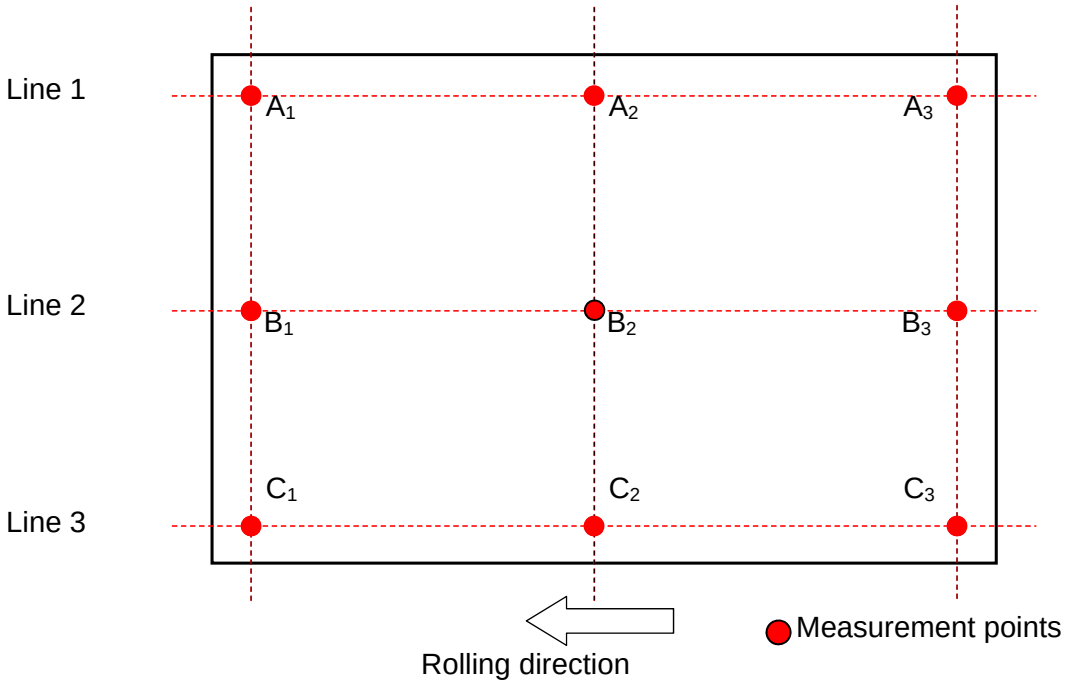
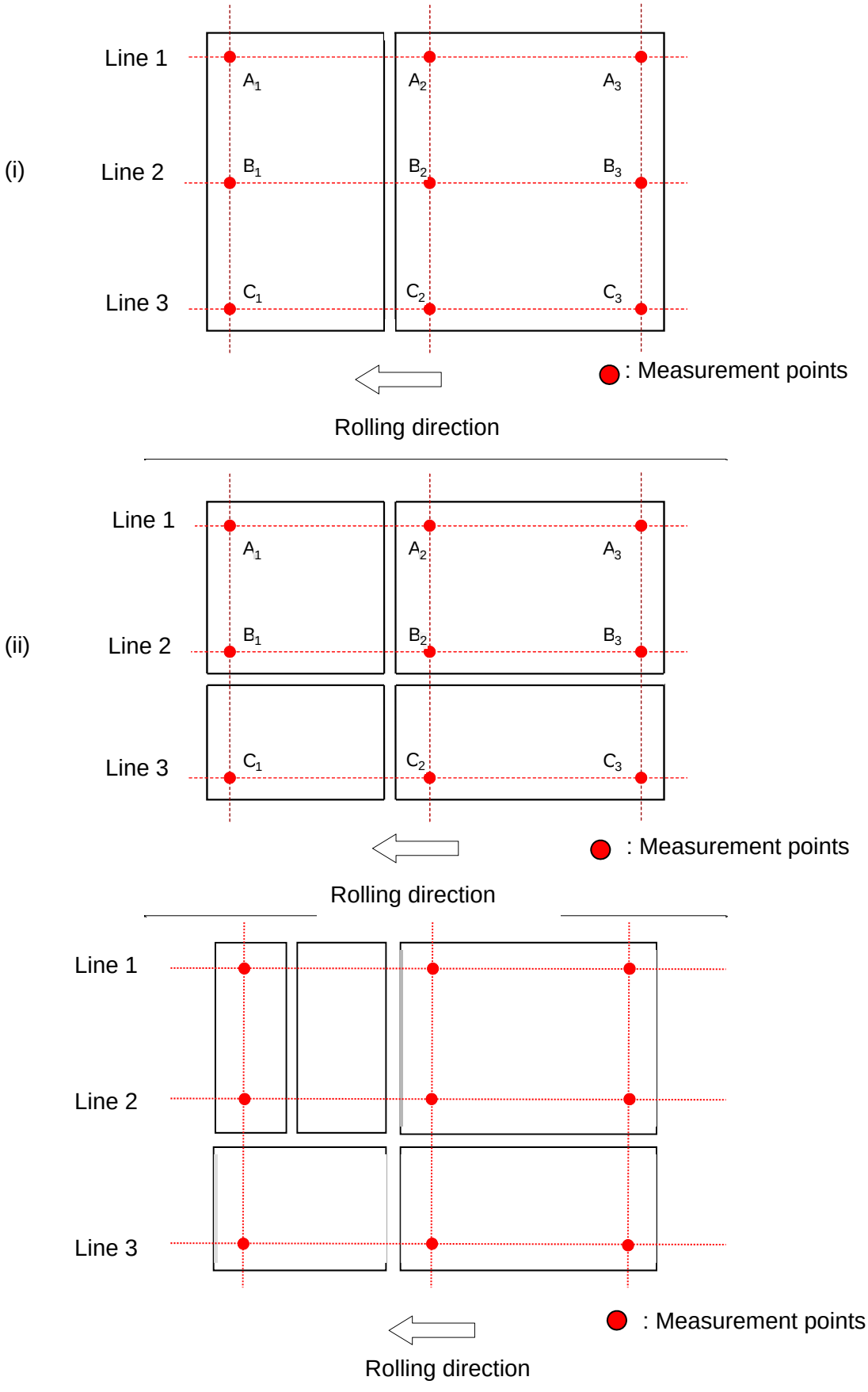


Fig. 1.4.15.2a) : Locations of Thickness Measuring Points for the Original Steel Plates

Fig. 1.4.15.2b) : Locations of Thickness Measuring Points for the Cut Steel Products



1.5 Heat treatment, condition of supply

1.5.1 All materials are to be supplied in the heat treated conditions described in the subsequent sections of this chapter unless supply in the as-rolled condition is allowed.

1.5.2 Where the material is supplied in the as rolled condition and intended for subsequent hot forming, the manufacturer is to carry out any heat treatment which may be necessary to prevent hydrogen cracking or make the material in a safe condition for transit and Surveyors are to be advised of any such heat treatment carried out. This requirement is applicable mainly to carbon and carbon-manganese steel products over 50 [mm] thick and to alloy steel products.

1.5.3 Where controlled rolling or thermo-mechanical processing is permitted as an alternative to normalising, these procedures may be used subject to full details being submitted and a test program being carried out under the supervision of the Surveyors and the test results being found satisfactory by IRS. These rolling processes are defined as follows:

(See Fig.1.5.3).

a) As Rolled, AR - this procedure involves steel being cooled as it is rolled with no further heat treatment. The rolling and finishing temperatures are typically in the austenite recrystalisation region and above the normalising temperature. The strength and toughness properties of steel produced by this process are generally less than steel heat treated after rolling or than steel produced by advanced processes.

b) Normalising, N - normalising involves heating rolled steel above the critical temperature, Ac3 and in the lower end of the austenite recrystalisation region for a specific period of time, followed by air-cooling. The process improves the mechanical properties of as rolled steel by refining the grain size and homogenizing the microstructure.

c) Controlled rolling, CR (Normalizing Rolling, NR) - A rolling procedure in which the final deformation is carried out in the normalising temperature range, allowed to cool in air, resulting in a material condition generally equivalent to that obtained by normalising.

d) Quenching and Tempering, QT – Quenching involves a heat treatment process in which steel is heated to an appropriate temperature above the Ac3, held for specific period of time and then cooled with an appropriate coolant for the purpose of hardening the microstructure. Tempering subsequent to quenching is a process in which the steel is reheated to an appropriate temperature not higher than the Ac1, maintained at that temperature for a specific period of time to restore toughness properties by improving the microstructure and reduce the residual stress caused by the quenching process.

e) Thermo-mechanical Rolling, TM – Thermo-mechanical controlled processing - this is a procedure which involves the strict control of both the steel temperature and the rolling reduction. Generally, a high proportion of the rolling reduction is carried out close to or below the AR3 transition temperature and may involve rolling towards the lower end of the temperature range of the inter critical duplex phase region thus permitting little if any recrystallisation of the austenite. Unlike controlled rolled (normalised rolling) the properties conferred by TM (TMCP) cannot be reproduced by subsequent normalising or other heat treatment.

   The use of accelerated cooling on completion of TM-rolling may also be accepted subject to the special approval of IRS. The same applies for use of tempering after completion of the TM-rolling.

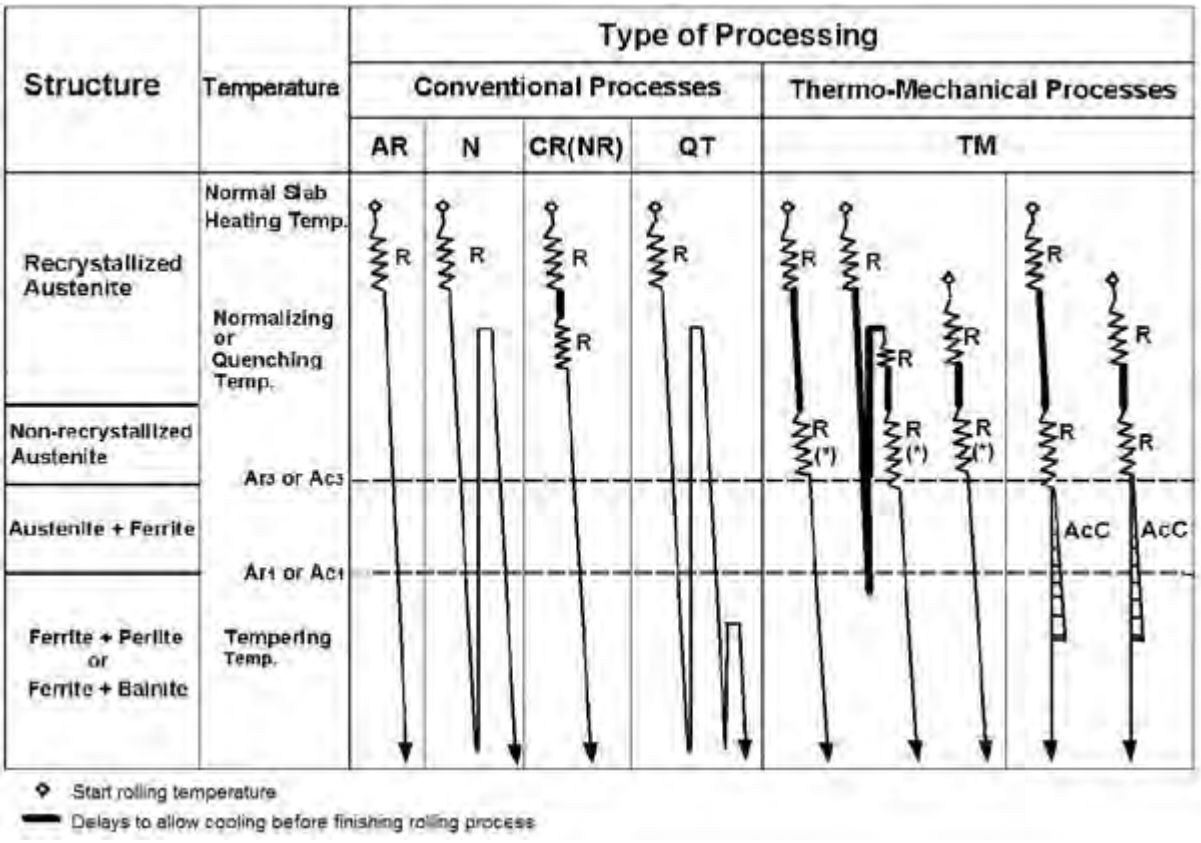
f) Accelerated cooling AcC - accelerated cooling is a process, which aims to improve mechanical properties by controlled cooling with rates higher than air cooling immediately after the final TM-rolling operation. Direct quenching is excluded from the accelerated cooling.

   The material properties conferred by TM and AcC cannot be reproduced by subsequent normalising or other heat treatment.

1.5.3.1 Where NR (CR) and TM with/without AcC are applied, the programmed rolling schedules are to be verified by IRS at the steel works and are to be made available when required by the attending Surveyor. On the manufacturer's responsibility, the programmed

rolling schedules are to be adhered to during the rolling operation. (Refer 1.2.3). To this effect, all the records of actual rolling are to be reviewed by the manufacturer and occasionally by the Surveyor.

When deviation from the programmed rolling schedules or normalizing or quenching and tempering procedures occurs, the manufacturer shall take further measures required in 1.2.3 to the Surveyor's satisfaction.



Notes:

AR:	As Rolled
N:	Normalizing
CR(NR):	Controlled Rolling (Normalizing Rolling)
QT:	Quenching and Tempering
TM:	Thermo-Mechanical Rolling (Thermo-Mechanical Controlled Process)
R:	Reduction
(*):	Sometimes rolling in the dual-phase temperature region of austenite and ferrite
AcC:	Accelerated Cooling

Fig.1.5.3 : Schematic Diagrams of Thermo-Mechanical and Conventional Processes

1.5.3.2 The conditions of supply and the impact test requirements are detailed in subsequent sections of the Chapter.

1.6 Test material

1.6.1 All material in a batch presented for acceptance tests are to be of the same product form e.g. plates, flats, sections. etc., from the same cast and in the same condition of supply.

1.6.2 Test samples

a) The test samples are to be fully representative of the material and, where appropriate, are not to be cut from the

material until heat treatment has been completed.

b) The test specimens are not to be separately heat treated in any way.

1.6.3 Unless otherwise agreed, the test samples are to be taken from the following position :

1.6.3.1 **Plates and flats with a width ≥ 600 [mm]** : The test samples are to be taken from one end at a position approximately midway between the axis in the direction of the rolling and the edge of the rolled product (See Fig.1.6.1 a). Unless otherwise agreed the tensile test specimens are to be prepared with their longitudinal axis transverse to the final direction of rolling.

1.6.3.2 **Flats with a width < 600 [mm], bulb flats and other sections** : For flats having a width of 600 [mm] or less, bulb flats and other sections the test specimens are to be taken from one end at a position approximately one third from the outer edge (See Figs.1.6.1 b,c,d), or in the case of small sections as near as possible to this position. In the case of channels, beams or bulb angles the test samples may alternatively be taken from a position approximately one quarter of the width from the web centreline or axis (See Fig.1.6.1 c). The tensile test specimens may be prepared with their longitudinal axis either parallel or transverse to the final direction of rolling.

1.6.3.3 **Bars and other similar products**: The test specimens are to be taken so that the axis of the test specimen is parallel to the direction of rolling. For small sizes, the test specimen may consist of a suitable length of the full cross section of the product (the impact test specimen receiving nevertheless the necessary machining). For larger sizes, the test samples are to be taken so that the axis of the test specimen lies as near as possible to the following :

a) for non-cylindrical sections, at one third of the half diagonal from the outside.

b) for cylindrical sections, at one third of the radius from outside (See Fig.1.6.1 e).

1.6.3.4 For plates and flats with thicknesses in excess of 40 [mm], full thickness specimens may be prepared, but when instead a machined round specimen is used then the axis is to be located at a position lying one-quarter of the product thickness from the surface as shown in Fig.1.6.1.f.

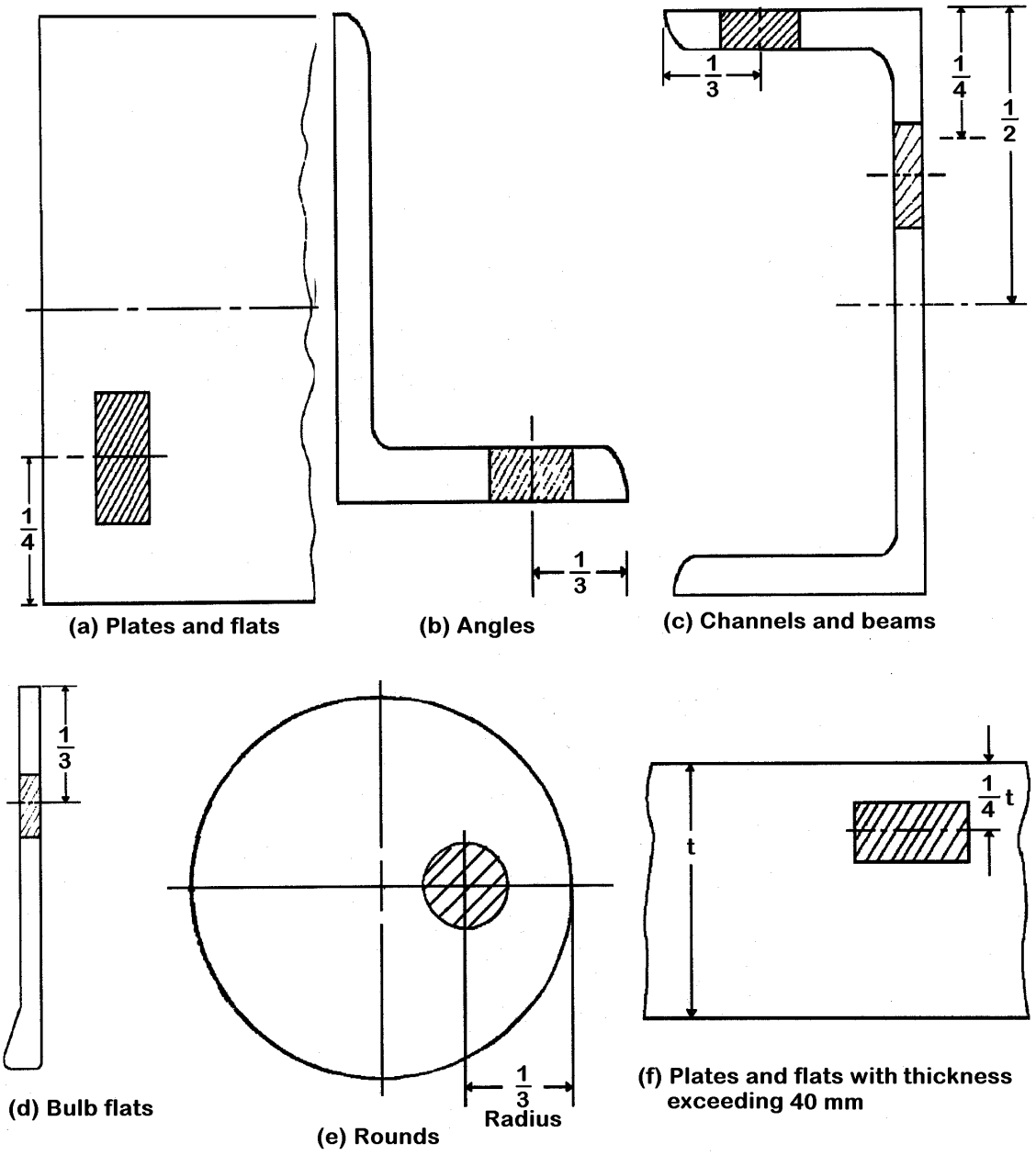


Fig.1.6.1 : Samples for testing

1.7 Mechanical test specimens

1.7.1 The tensile test specimens are to be machined to the dimensions detailed in Ch. 2.

1.7.2 Impact test specimens: The impact test specimens are to be of the charpy V-notch type machined to the dimensions detailed in Ch. 2 and cut with their longitudinal axis either parallel or transverse to the final direction of rolling of the material. They are to be taken from a position close to one of the rolled surfaces, except that for plates and sections over 40 [mm] thick the axis of test specimens are to be one quarter of the thickness from one of the rolled surfaces. For bars and other similar products the axis of the test specimens are to be as specified in 1.6.3.3. The notch is to be cut in a face of the test specimen which was originally perpen-dicular to the rolled surface. The position of the notch is to be not nearer than 25 [mm] to a flame-cut or sheared edge.

1.8 Surface inspection and dimensions

1.8.1 Surface inspection and verification of dimensions are the responsibility of the steel-maker, and acceptance by the Surveyors of material later found to be defective shall not absolve the steel maker from this responsibility.

The manufacturer is also responsible for compliance with the general requirements concerning freedom from harmful internal defects.

1.9 Freedom from defects

1.9.1 All products must have a workmanlike finish and must be free from defects and imperfections which may impair their proper workability and use. This may however, include some discontinuties of a harmless nature, minor imperfections e.g. pittings, rolling in scale, indentations, roll marks, scratches and grooves which cannot be avoided completely despite proper manufacturing and which will not be objected to provided they do not exceed the acceptable limits contained herein.

1.9.2 **Imperfections** : Notwithstanding this, the products may have imperfections exceeding the discontinuities inherent to the manufacturing process, as defined under 1.9.1. In such cases, limits for their acceptability are to be agreed with IRS, taking the end use of the product into consideration.

1.9.3 **Defects** : Cracks, shells, sand patches and sharp edged seams are always considered defects which would impair the end use of the product and which require rejection or repair, irrespective of their size and number. The same applies to other imperfection exceeding the acceptable limits.

1.10 Special quality plate material ('z' quality)

1.10.1 When plate material, intended for welded construction, will be subject to significant strains in a direction perpendicular to the rolled surfaces, it is recommended that consideration be given to the use of special plate material with specified through thickness properties. These strains are usually associated with thermal contraction and restraint during welding, particularly for full penetration "T"- butt welds, but may also be associated with loads applied in service or during construction. Requirements for these materials are detailed in Sec. 8 and it is the responsibility of shipbuilder or fabricator to make provision for the use of this material.

1.11 Branding of materials

1.11.1 Every finished item is to be clearly marked by the manufacturer in at least one place with IRS brand IR and the following particulars:

a) The manufacturer's name or trade mark;

b) Identification mark for the grade of steel, (material supplied in the thermo-mechanically controlled process condition is to have the letter TM added after the identification mark) ;

c) Cast or identification number and/or initials which enable the full history of the item to be traced;

d) If required by the purchaser, his order number or other identification marks.

e) Steels, which have been specially approved and which differ from the requirements given in this Chapter are to have the letter "S" marked after the agreed identification mark.

f) Steel plates that have complied with the requirements for corrosion resistant steel will be identified by adding a corrosion designation to the unified identification mark for the grade of steel. The corrosion resistant steel is to be designated according to its area of application as follows:

- Lower surface of strength deck and surrounding structures; **RCU**
- Upper surface of inner bottom plating and surrounding structures; **RCB**
- For both strength deck and inner bottom plating; **RCW**

1.11.2 Products complying with the requirements of Sec. 8 are to be marked "Z 25" or 'Z 35' as appropriate, in addition to the material grade designation e.g. 'EH36Z25' or 'EH36Z35'.

1.11.3 The above particulars, but excluding the manufacturer's name or trade mark where this is embossed on finished products, are to be encircled with paint or otherwise marked so as to be easily recognizable.

1.11.4 In the event of any material bearing IRS brand failing to comply with the test requirements, the brand name is to be unmistakably defaced.

1.12 Test certificates or other documentation

1.12.1 The Surveyor is to be supplied, in duplicate, copies of the test certificates or other documentation for all accepted materials, IRS may require separate documents for each grade of steel. These documents are to contain, in addition to the description, dimensions, etc. of the material at least the following particulars:

a) Purchaser's order number and if known the ship number for which the material is intended;

b) Identification number and/or initials;

c) Identification of steel works;

d) Identification of the grade of steel;

e) Cast number and ladle analysis;

f) For steel with a corrosion resistant steel designation the weight percentage of each element added or intentionally controlled for improving corrosion resistance.

g) Condition of supply when other than as rolled e.g. normalized or controlled rolled;

h) If the material is of rimming quality, this should be stated;

i) Test results.

In the case of 'Z' quality steel, notation 'Z25' or 'Z35' as appropriate, is to be indicated with the steel grade and test results are to include through thickness reduction in area (%).

1.12.2 Before the test certificates or shipping statements are signed by the Surveyor, the manufacturer is required to furnish him with a written declaration stating that the material has been made by an approved process and that it has been subjected to and has withstood satisfactorily the required tests in the presence of the Surveyor or his authorized deputy. The following form of declaration will be accepted if stamped or printed on each test certificate or shipping statement with the name of steelworks and initialed by the makers or an authorized deputy:
"We hereby certify that the material has been made by an approved process in accordance with the Rules of Indian Register of Shipping and has been tested satisfactorily in the presence of the surveyors of Indian Register of Shipping".

1.12.3 When steel is not produced at the works at which it is rolled a certificate is to be supplied to the Surveyor at the rolling mill stating the process by which it was manufactured and the name of the manufacturer, the number of cast from which it was made and the ladle analysis. The Surveyors are to have access to the works at which the steel was produced and the works must be approved by IRS. Also refer Chapter 1, Section1, Cl. 1.3.2.

Section 2

Normal Strength Steels for Ship Structures

2.1 General

2.1.1 Requirements of this section are applicable to weldable normal strength hot-rolled steel plates, wide flats, sections and bars intended for use in hull construction. Steel differing in chemical composition, deoxidation practice, heat treatment or mechanical properties may be accepted, subject to special agreement by IRS.

2.1.2 These requirements are primarily intended to apply to steel plates and wide flats not exceeding 100 [mm] in thickness and sections and bars not exceeding 50 [mm] in thickness. For greater thickness, certain variations in the requirements may be allowed or required in particular cases after consideration of the technical circumstances involved.

2.1.3 Requirements of this section also apply to normal strength Corrosion Resistant steels when such steel is used as the alternative means of corrosion protection for cargo oil tanks as specified in the performance standard MSC.289 (87) of Regulation 3-11, Part A-1, Chapter II-1 of the SOLAS Convention (Corrosion protection of cargo oil tanks of crude oil tankers). Corrosion Resistant steels as defined within this section, are steels whose corrosion resistance performance in the bottom

or top of the internal cargo oil tank is tested and approved to satisfy the requirements in MSC.289 (87) in addition to other relevant requirements for hull structural steels, structural strength and construction. It is not intended that such steels be used for corrosion resistant applications in other areas of a vessel that are outside of those specified in the performance standard MSC.289 (87) of Regulation 3-11, Part A-1, Chapter II-1 of the SOLAS Convention. These requirements apply to plates, wide flats, sections and bars in all grades up to a maximum thickness of 50 [mm].

2.2 Approval

2.2.1 Normal strength steel for ship hull structure is to be approved in accordance with requirements given in Section 1.

2.3 Method of Manufacture

2.3.1 Steel is to be manufactured by the basic oxygen, electric furnace or open hearth processes or by other processes specially approved by IRS.

2.3.2 The definitions of applicable rolling procedures and the schematic diagrams are given in Sec 1.

2.3.3 The de-oxidation practice used for each grade is to comply with the appropriate requirements of Table 2.4.1.

2.3.4 The rolling practice applied for each grade is to comply with the appropriate condition of supply of Table 2.5.1

2.4 Chemical composition

2.4.1 The chemical composition of samples taken from each ladle of each cast is to be determined by the manufacturer in an adequately equipped and competently staffed laboratory and is to comply with appropriate requirements of Table 2.4.1. For steel plates and wide flats over 50 [mm] thick, slight deviations in the chemical composition may be allowed as approved by IRS.

2.4.2 The manufacturer's declared analysis will be accepted subject to occasional checks if required by the surveyor.

2.5 Condition of supply

2.5.1 All materials are to be supplied in a condition complying with Table 2.5.1 and Table 2.5.2. Where alternative arrangements are permitted these are at the option of the steelmaker, unless otherwise expressly stated in the order for the material.

2.6 Mechanical Properties

2.6.1 For tensile test either the upper yield stress (ReH) or where ReH cannot be determined, the 0.2 percent proof stress (Rp 0.2) is to be determined and the material is considered to comply with the requirements if either value meets or exceeds the specified minimum value for yield strength (Re).

2.6.2 Results obtained from tensile tests are to comply with the appropriate requirements of Table 2.6.1.

2.6.3 Minimum average energy values are specified for Charpy V-notch impact test specimens taken in either the longitudinal or transverse directions. Generally, only longitudinal test specimens need be prepared and tested except for special applications where transverse test specimens may be required. Transverse test results are to be guaranteed by the manufacturer. The tabulated values are for standard specimens 10 [mm] x 10 [mm]. For plate thicknesses lower than 10 [mm], sub-size specimens may be used with reduced requirements as follows:

Specimen 10 x 7.5 [mm] : 5/6 of tabulated energy
Specimen 10 x 5 [mm] : 2/3 of tabulated energy.

2.6.4 For impact tests, the average value obtained from one set of three impact tests is to comply with the requirements given in Table 2.6.1. One individual value may be less than the required average value provided that it is not less than 70 per cent of this average value. See also Chapter 1.

2.6.5 Generally, impact tests are not required when the nominal plate thickness is less than 6 [mm].

Table 2.4.1 : Deoxidation and chemical composition

Grade	A	B	D	E
Deoxidation practice	For t ≤ 50 mm Any method except rimmed steel[1]	For t ≤ 50 mm Any method except rimmed steel	For t ≤ 25 [mm] killed,	Killed and fine grain treated
	For t > 50 mm Killed	For t > 50 mm Killed	For t > 25 mm Killed and fine grain treated	
Chemical composition per cent [4,7,8] (ladle samples)				
Carbon max.	0.21[2]	0.21	0.21	0.18
Manganese min	2.5 x Carbon %	0.80[3]	0.60	0.70
Silicon max	0.50	0.35	0.35	0.10 - 0.35
Phosphorus max	0.035	0.035	0.035	0.035
Sulphur max	0.035	0.035	0.035	0.035
Aluminium (acid soluble min)	-	-	0.015[5,6]	0.015[6]
Carbon + 1/6 of the manganese content is not to exceed 0.40 per cent				

Notes :

1 Grade A sections up to thickness of 12.5 mm may be accepted in rimmed steel subject to the special approval of IRS.
2 Max. 0.23% for sections.
3 When Grade B steel is impact tested the minimum manganese content may be reduced to 0.60%
4 When any grade of steel is supplied in the thermo-mechanically rolled condition variations in the specified chemical composition may be allowed or required by IRS.
5 Aluminium is required for thickness above 25 [mm].
6 The total aluminium content may be determined instead of the acid soluble content. In such cases the total aluminium content is to be not less than 0.020 per cent.
7 IRS may limit the amount of residual/trace elements which may have an adverse effect on the working and use of the steel, e.g. copper and tin.
8 Where additions of any other element have been made as part of the steelmaking practice, the content is to be specified.

Table 2.5.1 : Condition of supply for normal strength steel (1)

Grades	Thickness	Condition of supply
A	≤ 50 mm	Any
	> 50 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled (2)
B	≤ 50 mm	Any
	> 50 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled (2)
D	≤ 35 mm	Any
	> 35 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled (3)
E	≤ 100 mm	Normalized or thermo-mechanically rolled (3)

Notes:

1) These conditions of supply and the impact test requirements are summarised in Table 2.5.2

2) Subject to the special approval of IRS, Grades A and B steel plates may be supplied in the as rolled condition. See 2.14.2.2.

3) Subject to the special approval of IRS, sections in Grade D steel may be supplied in the as rolled condition provided satisfactory results are consistently obtained from Charpy V-notch impact tests. Similarly sections in Grade E steel may be supplied in the as rolled or controlled rolled condition. For the frequency of impact tests see 2.14.3.2 and 2.14.3.3.

Table 2.5.2 : Required condition of supply and number of impact tests for normal strength steels

Grade	Deoxidation Practice	Pro-ducts	Condition of supply (batch for impact tests) (1)(2)							
			Thickness [mm]							
			10	12.5	20	25	30	35	40	50	100
A	Rimmed	Sections	A(-)	Not applicable							
	For t ≤ 50 mm Any method except rimmed For t > 50 mm Killed	Plates	A(-)								N(-) TM(-)[3] CR(50), AR*(50)
		Sections	A(-)								Not applicable
B	For t ≤ 50 mm Any method except rimmed For t > 50 mm Killed	Plates	A(-)			A(50)					N(50) TM(50) CR(25), AR*(25)
		Sections	A(-)			A(50)					Not applicable
D	Killed	Plates Sections	A(50)								Not applicable
D	Plates Killed and fine grain treated	Plates	A(50)						N(50) CR(50) TM(50)		N(50) TM(50) CR(25)
D		Sections	A(50)						N(50) CR(50) TM(50) AR*(25)		Not applicable
E	Killed and fine grain treated	Plates	N(Each piece) TM(Each piece)								
		Sections	N(25) TM(25) AR*(15), CR*(15)								Not applicable

Remarks:

1. Condition of Supply

A - Any (Not Specified)
N - Normalised Condition
CR - Controlled Rolled Condition
TM - Thermo-Mechanical Rolling
AR* - As Rolled Condition subject to special approval of IRS
CR* - Controlled Rolled Condition subject to special approval of IRS.

2. Number of Impact Tests

One set of impact tests is to be taken from each batch of the specified weight in ( ) in tones or fraction thereof.

3. See Note 5 of Table 2.6.1

Table 2.6.1 : Mechanical properties for normal strength steels

Grade	Yield strength ReH [N/mm²] min.	Tensile strength Rm [N/mm²]	Elongation 5.65 $\sqrt{S_o}$ A5 (%)	Impact Test					
				Test Temp. °C	Average impact energy (J) min.				
					t ≤ 50 mm		50 < t ≤ 70 mm		70 < t ≤ 100 mm	
					Long (3)	Trans (3)	Long (3)	Trans (3)	Long (3)	Trans (3)
A	235	400/520 (1)	22(2)	+20	-	-	34(5)	24(5)	41(5)	27(5)
B				0	27(4)	20(4)	34	24	41	27
D				-20	27	20	34	24	41	27
E				-40	27	20	34	24	41	27

Notes:

1) For all thicknesses of Grade A sections the upper limit for the specified tensile strength range may be exceeded at the discretion of IRS.

2) For full thickness flat tensile test specimens with a width of 25 mm and a gauge length of 200 mm the elongation is to comply with the following minimum values:

Thickness [mm]		> 5	> 10	> 15	> 20	> 25	> 30	> 40	
	≤ 5	≤ 10	≤ 15	≤ 20	≤ 25	≤ 30	≤ 40	≤ 50	
Elongation	14	16	17	18	19	20	21	22	

3) See 2.6.3.

4) Charpy V-notch impact tests are generally not required for Grade B steel with thickness of 25 mm or less.

5) Impact tests for Grade A over 50 mm thick are not required when the material is produced using fine grain practice and furnished normalised. TM rolling may be accepted without impact testing at the discretion of IRS.

2.7 Surface Quality

2.7.1 The steel is to be free from surface defects prejudicial to the use of the material for the intended application. The finished material is to have a surface quality in accordance with a recognized standard such as EN 10163 parts 1, 2 and 3, or an equivalent standard accepted by IRS, unless otherwise specified in this section.

2.7.2 The responsibility for meeting the surface finish requirements rests with the manufacturer of the material, who is to take the necessary manufacturing precautions and is to inspect the products prior to delivery. At that stage, however, rolling or heat treatment scale may conceal surface discontinuities and defects. If, during the subsequent descaling or working operations, the material is found to be defective, IRS may require materials to be repaired or rejected.

2.7.2.1 The surface quality inspection method is to be in accordance with recognized national or international standards agreed between purchaser and manufacturer, accepted by IRS.

2.7.2.2 If agreed between the manufacturer and purchaser, steel may be ordered with improved surface finish over and above these requirements.

2.7.3 Acceptance Criteria

2.7.3.1 Imperfections

2.7.3.1.1 Imperfections of a harmless nature, for example pitting, rolled-in scale, indentations, roll marks, scratches and grooves, regarded as being inherent of the manufacturing process, are permissible irrespective of their number, provided the maximum permissible limits of Class A of EN 10163-2 or limits specified in a recognized equivalent standard accepted by IRS, are not exceeded and the remaining plate or wide flat thickness remains within the average allowable minus thickness tolerance specified in Sec 1, 1.4. Total affected area with imperfection not exceeding the specified limits are not to exceed 15 % of the total surface in question.

2.7.3.2 Defects

2.7.3.2.1 Affected areas with imperfections with a depth exceeding the limits of Class A of EN 10163-2 or the maximum permissible limits specified in a recognized equivalent standard accepted by IRS, are to be repaired irrespective of their number. Cracks, injurious surface flaws, shells (over lapping material with non-metallic inclusion), sand patches, laminations and sharp edged seams (elongated defects) visually evident on surface and/or edge of plate are considered defects, which would impair the end use of the product and which require rejection or repair, irrespective of their size and number.

2.7.4 Repair

2.7.4.1 Grinding repair

2.7.4.1.1 Grinding may be applied provided all the conditions below are adhered to:

(a) The nominal product thickness will not be reduced by more than 7% or 3 mm, whichever is the less.

(b) Each single ground area does not exceed 0.25 [m^2].

(c) All ground areas do not exceed 2% of the total surface in question.

(d) Ground areas lying in a distance less than their average breadth to each other are to be regarded as one single area.

(e) Ground areas lying opposite each other on both surfaces are not to decrease the product thickness by values exceeding the limits as indicated in (a).

Defects or unacceptable imperfections are to be completely removed by grinding and the remaining plate or wide flat thickness is to remain within the average allowable minus thickness tolerance specified in Sec.1, 1.4. The ground areas are to be a smooth transition to the surrounding surface of the product. Complete elimination of the defect is to be verified by magnetic particle or by liquid penetrant testing.

2.7.4.2 Welding repair

2.7.4.2.1 Weld repair procedures and the method are to be reported and be approved by the IRS. Repair of defects such as unacceptable imperfections, cracks, shells or seams are to be followed by magnetic particle or liquid penetrant testing. Local defects which cannot be repaired by grinding as stated in 2.7.4.1 may be repaired by welding with the agreement of IRS subject to the following conditions:

(a) Any single welded area is not to exceed 0.125 [m^2] and the sum of all areas is not to exceed 2% of the surface side in question.

(b) The distance between two welded areas is not to be less than their average width.

(c) The weld preparation is not to reduce the thickness of the product below 80% of the nominal thickness. For occasional defects with depths exceeding the 80% limit, special consideration at the Surveyor's discretion will be necessary.

(d) If weld repair depth exceeds 3 [mm], UT may be requested by IRS. If required, UT is to be carried out in accordance with an approved procedure.

(e) The repair is to be carried out by qualified welders using an approved procedure for the appropriate steel grade. The electrodes are to be of low hydrogen type and are to be dried in accordance with the manufacturer's requirements and protected against re-humidification before and during welding.

2.7.5 The surface quality and condition requirement herein are not applied to products in forms of bars and tubulars, which will be subject to manufacturers' conformance standards.

2.8 Internal soundness

2.8.1 If plates and wide flats are ordered with ultrasonic inspection, this is to be made in accordance with an accepted standard at the discretion of IRS.

2.8.2 Verification of internal soundness is the responsibility of the manufacturer. The acceptance of internal soundness by surveyor does not absolve the manufacturer from this responsibility.

2.9 Tolerances

2.9.1 Unless otherwise agreed or specially required the thickness tolerances as per Sec.1, 1.4 are applicable.

2.10 Identification of Materials

2.10.1 The steelmaker is to adopt a system for the identification of ingots, slabs and finished pieces which will enable the material to be traced to its original cast.

2.10.2 The Surveyor is to be given full facilities for so tracing the material when required.

2.11 Testing and Inspection

2.11.1 Facilities for Inspection

2.11.1.1 The manufacturer is to afford the Surveyor all necessary facilities and access to all relevant parts of the works to enable him to verify that the approved process is adhered to, for the selection of test materials, and the witnessing of tests, as required by the Rules, and for verifying the accuracy of the testing equipment.

2.11.2 Testing Procedures

2.11.2.1 The prescribed tests and inspections are to be carried out at the place of manufacture before dispatch. The test specimens and procedures are to be in accordance with Ch.2. All the test specimens are to be selected and stamped by the Surveyor and tested in his presence, unless otherwise agreed.

2.11.3 Through Thickness Tensile Tests

2.11.3.1 If plates and wide flats with thickness of 15 [mm] and over are ordered with through thickness properties, the through thickness tensile test in accordance with Sec.8 is to be carried out.

2.11.4 Dimensions

2.11.4.1 Verification of dimensions are the responsibility of the steel maker. The acceptance by Surveyor does not absolve the steel maker from this responsibility.

2.12 Test Material

2.12.1 Definitions

2.12.1.1 Refer to Ch1 for the definitions of piece and batch.

2.12.2 Test Samples

2.12.2.1 Refer to Sec 1, for the requirements related to test samples.

2.13 Mechanical Test specimens

2.13.1 Tensile Test Specimens. The dimensions of the tensile test specimens are to be in accordance with Ch.2. Generally, for plates, wide flats and sections flat test specimens of full product thickness are to be used. Round test specimens may be used when the product thickness exceeds 40 [mm] or for bars and other similar products. Alternatively, for small sizes of bars, etc. test specimens may consist of a suitable length of the full cross section of the product.

2.13.2 Impact Test Specimens. The impact test specimens are to be of the Charpy V-notch type cut with their edge within 2 [mm] from the “as rolled” surface with their longitudinal axes either parallel (indicated “Long” in Table 2.6.1) or transverse (indicated "Trans" in Tables 2.6.1) to the final direction of rolling of the material. The notch is to be cut in a face of the test specimen which was originally perpendicular to the rolled surface. The position of the notch is not to be nearer than 25 [mm] to a flame cut or sheared edge (see also 2.6.3). Where the product thickness exceeds 40 [mm], the impact test specimens are to be taken with their longitudinal axis at a quarter thickness position.

2.14 Number of Test Specimens

2.14.1 Number of Tensile Tests

2.14.1.1 For each batch presented. except where specially agreed by IRS. one tensile test is to be made from one piece unless the weight of finished material is greater than 50 tonnes in which case one extra test piece is to be made from a different piece from each 50 tonnes or fraction thereof. Additional tests are to be made for every variation of 10 [mm] in thickness of plate or diameter of products from the same cast. For sections the thickness to be considered is the thickness of the product at the point at which samples are taken for mechanical tests.

2.14.2 Number of Impact Tests (except for Grade E)

2.14.2.1 For each batch presented, except where specially agreed by IRS at least one set of three Charpy V-notch test specimens is to be made from one piece unless the weight of finished material is greater than 50 tonnes in which case one extra set of three test specimens is to be made from a different piece from each 50 tonnes or fraction thereof. The piece selected for the preparation of test specimen is to be the thickest of each batch. Where steel plates except for Grade 'A' steel over 50 [mm] in thickness is supplied in the controlled rolled condition, the frequency of impact test is to be made from a different piece from each 25 tonnes or fraction thereof.

2.14.2.2 When subject to the special approval of IRS, material is supplied in the as rolled condition the frequency of impact tests is to be increased to one set from each batch of 25 tonnes or fraction thereof. However, for Grade 'A' steel over 50 [mm] thickness when supplied in the "as rolled" condition, one set of three charpy V-notch test specimens may be taken from each batch of 50 tonnes or fraction thereof.

2.14.2.3 The piece selected for the preparation of the test specimens is to be the thickest in each batch.

2.14.3 Number of Impact Tests (for Grade E)

2.14.3.1 For steel plates supplied in the normalized or TM condition, one set of impact test specimens is to be taken from each piece.

2.14.3.2 For sections, one set of impact tests is to be taken from each batch of 25 tonnes or fraction thereof.

2.14.3.3 When, subject to the special approval of IRS, sections are supplied in the as-rolled or controlled rolled condition.one set of impact tests is to be taken from each batch of 15 tonnes or fraction thereof.

2.14.3.4 For 2.14.3.2 and 2.14.3.3 above the piece selected for the preparation of the test specimens is to be the thickest in each batch.

2.15 Retest Procedures

2.15.1 Retest procedures are to be as per the requirements provided in Chapter 1.

Section 3

Higher Strength Steels for Ship Structures

3.1 General

3.1.1 Requirements of this section is applicable to weldable higher strength hot-rolled steel plates, wide flats, sections and bars intended for use in hull construction.

3.1.2 The requirements of this section are primarily intended to apply to plates and wide flats not exceeding 100 [mm] in thickness in general, and sections and bars not exceeding 50 [mm] in thickness. For greater thickness, these requirements may be applied with certain variations, as may be agreed by IRS.

3.1.3 Steel differing in chemical composition, deoxidation practice, heat treatment or mechanical properties may be accepted, subject to special approval by IRS. Such steel is to be given special designation.

3.1.4 Provision is made for three strength levels (315, 355 and 390 N/mm^2) each subdivided into four grades, AH, DH, EH and FH based on impact test temperature.
The additional requirements for high strength plates having specified minimum yield point of 460 [N/mm^2] with thickness over 50 [mm] and not greater than 100 [mm] for use in longitudinal structural members in the upper deck region of container ships (such as hatch side coaming, hatch coaming, hatch coaming top and attached longitudinals) and denoted by Grade EH47 are also given in this section.

Brittle Crack Arrest Steels: Brittle crack arrest steels are defined as steel plate with the specified brittle crack arrest properties measured by either the brittle crack arrest toughness Kca or Crack Arrest Temperature (CAT). Additional requirements for brittle crack arrest steels are specified in Section 10. Brittle crack arrest steels are to be used for container ships, as indicated in Pt. 5, Ch. 5 of the Rules.

3.1.5 Requirements of this section also apply to high strength Corrosion Resistant steels when such steel is used as the alternative means of corrosion protection for cargo oil tanks as specified in the performance standard MSC.289 (87) of Regulation 3-11, Part A-1, Chapter II-1 of the SOLAS Convention (Corrosion protection of cargo oil tanks of crude oil tankers). Corrosion Resistant steels as defined within this section, are steels whose corrosion resistance performance in the bottom or top of the internal cargo oil tank is tested and approved to satisfy the requirements in MSC.289 (87) in addition to other relevant requirements for hull structural steels, structural strength and construction. It is not intended that such steels be used for corrosion resistant applications in other areas of a vessel that are outside of those specified in the performance standard MSC.289 (87) of Regulation 3-11, Part A-1, Chapter II-1 of the SOLAS Convention. These requirements apply to plates, wide flats, sections and bars in all grades up to a maximum thickness of 50 [mm].

3.2 Approval

3.2.1 Higher strength steel for ship hull structure is to be approved in accordance with requirements given in Section 1.

3.2.2 It should be noted that when fatigue loading is present, the effective fatigue strength of a welded construction of higher strength steels may not be greater than that of a construction fabricated from the normal strength steels. Precautions against corrosion fatigue may also be necessary.

Note: Before subjecting steels produced by thermo-mechanical rolling to further heating for forming or stress relieving or using high heat-input welding, special consideration must be given to the possibility of a consequent reduction in mechanical properties.

3.3 Method of Manufacture

3.3.1 Steel is to be manufactured by the basic oxygen, electric furnace or open hearth processes or by other processes specially approved by IRS.

3.3.2 The definitions of applicable rolling procedures and the schematic diagrams are given in Sec 1.

3.3.3 The deoxidation practice used for each grade is to comply with the appropriate requirements of Table 3.4.1.

3.3.4 The rolling practice applied for each grade is to comply with the appropriate condition of supply of Table 3.5.1

3.4 Chemical composition

3.4.1 The chemical composition of samples taken from each ladle of each cast is to be determined by the manufacturer in an adequately equipped and competently staffed laboratory and is to comply with appropriate requirements of Table 3.4.1. For steel plates and wide flats over 50 [mm] thick, slight deviations in the chemical composition may be allowed as approved by IRS.

The chemical composition of EH 47 steel plates would be specially considered.

3.4.2 The manufacturer's declared analysis will be accepted subject to occasional checks if required by the surveyor.

3.4.3 When required, the carbon equivalent value is to be calculated from the ladle analysis using the following formula.

$$\text{Carbon eq.} = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15} \ \%$$

Note: This formula is applicable only to steels which are basically of the carbon manganese type and gives a general indication of the weldability of the steel.

Table 3.4.1 : Chemical composition and deoxidation practice for higher strength steels			
Grade[1)]	**AH32/DH32/EH32 AH36/DH36/EH36 AH40/DH40/EH40**	**FH32 FH36 FH40**	**EH47**
Deoxidation practice	Fully killed and fine grain refined		
Chemical Composition per cent (Ladle sample)[5),6)]			
C max. Mn Si max. P max. S max.	0.18 0.90 - 1.60[2)] 0.50 0.035 0.035	0.16 0.90 - 1.60 0.50 0.025 0.025	0.18 0.9 – 2.00 0.55 0.020 0.020
Grain refining elements[5)]			
Al (acid soluble) min.	0.015[3),4)]		
Nb V Ti max. Total (Nb + V + Ti)	0.02 - 0.05[4)] 0.05 - 0.10[4)] 0.02 0.12 max.		
Residual elements			
Cu max. Cr max. Ni max. Mo max. N max.	0.35 0.20 0.40 0.08 -	0.35 0.20 0.80 0.08 0.009 (0.012 if Al is present)	0.35 0.25 1.0 0.08 -

Notes:

1) The number following the grade designation indicates the yield point to which the steel is ordered or produced in [Kg/mm^2].

2) For thickness upto and including 12.5 [mm] the minimum manganese content may be reduced to 0.70 percent.

3) The total aluminium content may be determined instead of the acid soluble content. In such cases the total aluminium content is to be not less than 0.020 percent

4) The steel is to contain aluminium, niobium, vanadium or other suitable grain refining elements, either single or in combination. When used singly the steel is to contain the specified minimum content of the grain refining element. When used in combination, the specified minimum content of a fine graining element is not applicable.

5) When any grade of higher strength steel is supplied in the thermo-mechanically rolled condition variations in the specified chemical composition may be allowed or required by IRS

6) Where additions of any other element have been made as part of the steel making practice, the content is to be indicated.

3.4.4 For TM (TMCP) steels the following special requirements apply:

i) The carbon equivalent value is to be calculated from the ladle analysis using the following formula and to comply with the requirements of the following table:

Carbon equivalent for higher strength steels upto 100 mm in thickness produced by TM		
Grade	**Carbon equivalent Max. (%)[1)]**	
	t ≤ 50	**50 < t ≤ 100**
AH32, DH32, EH32, FH32	0.36	0.38
AH36, DH36, EH36, FH36	0.38	0.40
AH40, DH40, EH40, FH40	0.40	0.42
EH47	NA	0.49
t = thickness [mm]		
Note: 1) It is a matter for the manufacturer and shipbuilder to mutually agree in individual cases as to whether they wish to specify a more stringent carbon equivalent.		

$$\text{Carbon eq.} = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15} \ \%$$

ii) Other means such as cold cracking susceptibility P_{cm} (in %), may be considered instead of the carbon equivalent for evaluating the weldability.

$$P_{cm} = C + \frac{Si}{30} + \frac{Mn}{20} + \frac{Cu}{20} + \frac{Ni}{60} + \frac{Cr}{20} + \frac{Mo}{15} + \frac{V}{10} + 5B$$

3.4.5 The carbon equivalent of EH47 grade steel calculated as per 3.4.3 is not to exceed 0.49%. The cold cracking susceptibility P_{cm} calculated using the formula mentioned in 3.4.4.(ii) is not to exceed 0.22%.

3.5 Condition of supply

3.5.1 All materials are to be supplied in a condition complying with the requirements given in Table 3.5.1.

Table 3.5.1 : Condition of supply for Higher strength steel [1)]

Grades	Grain Refining Elements Used	Thickness	Condition of supply
A32 A36	Nb and/or V	≤ 12.5 mm	Any
		> 12.5 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled [(3)]
	Al alone or with Ti	≤ 20 mm	Any
		> 20 mm ≤ 35 mm	Any, as rolled subject to special approval of IRS [(2)]
		> 35 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled [(3)]
A40	Any	≤ 12.5 mm	Any
		> 12.5 mm ≤ 50 mm	Normalized, controlled rolled or thermo-mechanically rolled
		> 50 mm ≤ 100 mm	Normalized, thermo-mechanically rolled or quenched and tempered
D32 D36	Nb and/or V	≤ 12.5 mm	Any
		> 12.5 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled [(3)]
	Al alone or with Ti	≤ 20 mm	Any
		> 20 mm ≤ 25 mm	Any, as rolled subject to special approval of IRS [(2)]
		> 25 mm ≤ 100 mm	Normalized, controlled rolled or thermo-mechanically rolled [(3)]

Table 3.5.1 : (Contd.)

Grades	Grain Refining Elements Used	Thickness	Condition of supply
D40	Any	≤ 50 mm	Normalized, controlled rolled or thermo-mechanically rolled
		> 50 mm ≤ 100 mm	Normalized, thermo-mechanically rolled or quenched and tempered
E32 E36	Any	≤ 50 mm	Normalized or thermo-mechanically rolled (3)
		> 50 mm ≤ 100 mm	Normalized, thermo-mechanically rolled
E40	Any	≤ 50 mm	Normalized, thermo-mechanically rolled or quenched and tempered
		> 50 mm ≤ 100 mm	Normalized, thermo-mechanically rolled or quenched and tempered
F32 F36 F40	Any	≤ 50 mm	Normalized, thermo-mechanically rolled or quenched and tempered (4)
		> 50 mm ≤ 100 mm	Normalized, thermo-mechanically rolled or quenched and tempered

Notes:

(1) These conditions of supply and the requirements for impact tests are summarised in Table 3.5.2.
(2) The frequency of impact tests is to be in accordance with 3.14.2 (ii).
(3) Subject to the special approval of IRS, sections in Grades AH32, AH36, DH32 and DH36 steels may be supplied in the as rolled condition provided satisfactory results are consistently obtained from Charpy V-notch impact tests. Similarly sections in Grades EH32 and EH36 steels maybe supplied in the as rolled or controlled rolled condition. The frequency of impact tests is to be in accordance with 3.14.2 (ii) and 3.14.2 (iii) respectively.
(4) Subject to the special approval of IRS, sections in Grades FH32 and FH36 steels may be supplied in the controlled rolled condition. The frequency of impact tests is to be in accordance with 3.14.3 (iii).

Table 3.5.2 : Required condition of supply and number of impact tests for higher strength steels

Grade	Deoxi-dation Practice	Grain Refining Elements	Pro-ducts	Condition of supply (batch for impact tests) (1)(2)								
				Thickness [mm]								
				10	12.5	20	25	30	35	40	50	100
AH32 AH36	Killed and fine grain treated	Nb and/or V	Plates	A(50)	N(50) CR(50),TM(50)						N(50),CR(25), TM(50)	
			Sections	A(50)	N(50) CR(50),TM(50) AR*(25)						Not applicable	
		Al alone or with Ti	Plates	A(50)	AR*(25)					Not applicable		
					N(50), CR(50) TM(50)						N(50), CR(25) TM(50)	
			Sections	A(50)	N(50) CR(50) TM(50) AR*(25)						Not applicable	
AH40	Killed and fine grain treated	Any	Plates	A(50)	N(50) CR(50) TM(50)						N(50) TM(50) QT (Each length as heat treated)	
			Sections	A(50)	N(50) CR(50) TM(50)						Not applicable	
DH32 DH36	Killed and fine grain treated	Nb and/or V	Plates	A(50)	N(50) CR(50),TM(50)						N(50),CR(25), TM(50)	
			Sections	A(50)	N(50) CR(50),TM(50) AR*(25)						Not applicable	
		Al alone or with Ti	Plates	A(50)	AR*(25)					Not applicable		
					N(50), CR(50) TM(50)						N(50), CR(25) TM(50)	
			Sections	A(50)	N(50) CR(50) TM(50) AR*(25)						Not applicable	
DH40	Killed and fine grain treated	Any	Plates	N(50) CR(50) TM(50)							N(50) TM(50) QT (Each length as heat treated)	
			Sections	N(50) CR(50) TM(50)							Not applicable	
EH32 EH36	Killed and fine grain treated	Any	Plates	N(Each piece) TM(Each piece)								
			Sections	N(25) TM(25) AR*(15) CR*(15)							Not applicable	
EH40	Killed and fine grain treated	Any	Plates	N(Each piece) TM(Each piece) QT(Each length as heat treated)								
			Sections	N(25) TM(25) QT(25)							Not applicable	

Table 3.5.2 : (Contd.)

Grade	Deoxi-dation practice	Grain refining elements	Products	Condition of supply (Batch for impact tests [(1)(2)] Thickness [mm] 10 12.5 20 25 30 35 40 50 100	
FH32 FH36	Killed and fine grain treated	Any	Plates	N(Each piece) TM(Each piece) QT(Each length as heat treated)	Not applicable
			Section	N(25) TM(25) QT(25) CR*(15)	Not applicable
FH40	Killed and fine grain treated	Any	Plates	N(Each piece) TM(Each piece) QT(Each length as heat treated)	Not applicable
			Sections	N(25) TM(25) QT(25)	Not applicable

Remarks

1. Condition of Supply

A - Any (Not Specified)

N - Normalised Condition

CR - Controlled Rolled Condition

TM - Thermo-Mechanical Rolling

QT - Quenched and Tempered Condition

AR* - As Rolled Condition subject to special approval of IRS

CR* - Controlled Rolled Condition subject to special approval of IRS.

2. Number of Impact Tests

One set of impact tests is to be taken from each batch of the "specified weight" in ( ) in tones or fraction thereof.

For Grades A32 and A36 steels charpy impact tests are not generally required provided that satisfactory results are obtained from occasional check tests selected by the Surveyor.

3.6 Mechanical Properties

3.6.1 For tensile test either the upper yield stress (ReH) or where ReH cannot be determined, the 0.2 percent proof stress (Rp 0.2) is to be determined and the material is considered to comply with the requirements if either value meets or exceeds the specified minimum value for yield strength (Re).

3.6.2 Results obtained from tensile tests are to comply with the appropriate requirements of Table 3.6.1.

Table 3.6.1 : Mechanical properties for higher strength steels

Grade	Yield stren-gth ReH [N/mm^2] min.	Tensile strength Rm [N/mm^2]	Elon-gation 5.65 $\sqrt{S_o}$ A5 (%)	Impact Test						
				Test Temp. °C	Average impact energy (J) min.					
					t ≤ 50 mm		50 < t ≤ 70 mm		70 < t ≤ 100 mm	
					Long (2)	Trans (2)	Long (2)	Trans (2)	Long (2)	Trans (2)
AH32	315	440/570	22(1)	0	31(3)	23(3)	38	26	46	31
DH32				-20	31	22	38	26	46	31
EH32				-40	31	22	38	26	46	31
FH32				-60	31	22	38	26	46	31
AH36	355	490/630	21(1)	0	34(3)	24(3)	41	27	50	34
DH36				-20	34	24	41	27	50	34
EH36				-40	34	24	41	27	50	34
FH36				-60	34	24	41	27	50	34
AH40	390	510/660	20(1)	0	39	27	46	31	55	37
DH40				-20	39	27	46	31	55	37
EH40				-40	39	27	46	31	55	37
FH40				-60	39	27	46	31	55	37

t = thickness [mm]

NOTES:

1) For full thickness flat tensile test specimens with a width of 25 [mm] and a gauge length of 200 [mm] the elongation [%] is to comply with the following minimum values:

Grade	Thickness [mm]							
	≤ 5	> 5 ≤ 10	> 10 ≤ 15	> 15 ≤ 20	> 20 ≤ 25	> 25 ≤ 30	> 30 ≤ 40	> 40 ≤ 50
AH32, DH32, EH32 & FH32	14	16	17	18	19	20	21	22
AH36, DH36, EH36 & FH36	13	15	16	17	18	19	20	21
AH40, DH40, EH40 & FH40	12	14	15	16	17	18	19	20

2) See 3.6.3.

3) For Grades AH32 and AH36 steels a relaxation in the number of impact tests for acceptance purposes may be permitted by special agreement with IRS provided that satisfactory results are obtained from occasional check tests.

Table 3.6.2 : Conditions of supply, grade and mechanical properties for EH47 steel plates

Supply Condition	Grade	Mechanical Properties			Impact Test			
		Yield Strength [N/mm^2]min	Tensile Strength [N/mm^2]	Elongation (%) min.	Test Temp. (^{o}C)	Average Impact Energy [J] min.		
						50<t≤ 70	70<t≤8 5	85<t≤ 100
TMCP	EH 47	460	570/720	17	-40^{o}C	53	64	75

Note : t: Thickness [mm]

1). The additional requirements for EH47 steel with brittle crack arrest properties are specified in Section 10.

3.6.3 Minimum average energy values are specified for Charpy V-notch impact test specimens taken in either the longitudinal or transverse directions. Generally, only longitudinal test specimens need be prepared and tested except for special applications where transverse test specimens may be required. Transverse test results are to be guaranteed by the manufacturer. The tabulated values are for standard specimens 10 [mm] x 10 [mm]. For plate thicknesses lower than 10 [mm], sub-size specimens may be used with reduced requirements as follows :

Specimen 10 x 7.5 [mm] : 5/6 of tabulated energy
Specimen 10 x 5 [mm] : 2/3 of tabulated energy.

3.6.4 For impact tests, the average value obtained from one set of three impact tests is to comply with the requirements given in Table 3.6.1. One individual value may be less than the required average value provided that it is not less than 70 per cent of this average value. See also Chapter 1.

3.6.5 Generally, impact tests are not required when the nominal plate thickness is less than 6 [mm].

3.7 Surface Quality

3.7.1 Requirements of 2.7 are applicable.

3.8 Internal Soundness

3.8.1 Requirements of 2.8 are applicable.

3.9 Tolerances

3.9.1 Requirements of 2.9 are applicable.

3.10 Identification of Materials

3.10.1 Requirements of 2.10 are applicable

3.11 Testing and Inspection

3.10.1 Requirements of 2.11 are applicable

3.12 Test Material

3.12.1 Requirements of 2.12 are applicable

3.13 Mechanical tests specimens

3.13.1 Tensile Test Specimens. The dimensions of the tensile test specimens are to be in accordance Ch.2. Generally for plates, wide flats and sections flat test specimens of full product thickness are to be used. Round test specimens may be used when the product thickness exceeds 40 [mm] or for bards and other similar products. Alternatively for small sizes of bars, etc. test specimens may consist of a suitable length of the full cross section of the product.

3.13.2 Impact Test Specimens. The impact test specimens are to be of the Charpy V-notch type cut with their edge within 2 [mm] from the “as rolled” surface with their longitudinal axes either parallel (indicated “Long” in Table 3.6.1) or transverse (indicated "Trans" in Table 3.6.1) to the final direction of rolling of the material. The notch is to be cut in a face of the test

specimen which was originally perpendicular to the rolled surface. The position of the notch is not to be nearer than 25 [mm] to a flame cut or sheared edge (see also 3.6.3). Where the product thickness exceeds 40 [mm], the impact test specimens are to be taken with their longitudinal axis at a quarter thickness position.

3.14 Number of Test Specimens

3.14.1 Number of Tensile Tests. For each batch presented, except where specially agreed by IRS, one tensile test is to be made from one piece unless the weight of finished material is greater than 50 tonnes in which case one extra test piece is to be made from a different piece from each 50 tonnes or fraction thereof. Additional tests are to be made for every variation of 10 [mm] in thickness of plate or diameter of products from the same cast. For sections, the thickness to be considered is the thickness of the product at the point at which samples are taken for mechanical tests.

3.14.2 Number of Impact Tests (except for Grades EH32, EH36, EH40, EH47, FH32, FH36 and FH40):

i) Except where otherwise specified or specially agreed by IRS, for each batch presented, at least one set of three Charpy V-notch impact test specimen is to be made from one piece unless the weight of finished material is greater than 50 tonnes, in which case one extra set of three test specimens is to be made from a different piece from each 50 tonnes or fraction thereof. When steel plates over 50 [mm] in thickness is supplied in the controlled rolled condition, the frequency of impact test is to be made from a different piece from each 25 tonnes or fraction therof.

ii) For steel plates of Grades AH40 and DH40 with thickness over 50 [mm] in normalized or TM condition, one set of impact test specimens is to be taken from each batch of 50 tonnes or fraction thereof. For those in QT condition, one set of impact test specimens is to be taken from each length as heat treated.

iii) When, subject to special approval of IRS, material is supplied in the as rolled condition, the frequency of impact tests is to be increased to one set from each batch of 25 tonnes or fraction thereof.

iv) The piece selected for the preparation of test specimens is to be the thickest in each batch.

3.14.3 Number of Impact Tests (Grades EH32, EH36, EH40, FH32, FH36 and FH40):

i) For plates supplied in the normalized or TM condition one set of three Charpy V-notch impact test specimens is to be taken from each piece. For quenched and tempered steel plates one set of impact test specimens is to be taken from each length as heat treated.

ii) For sections one set of impact tests is to be taken from each batch of 25 tonnes or fraction thereof.

iii) When, subject to special approval of IRS, sections other than Grades EH40 and FH40 are supplied in the as-rolled or controlled rolled condition, one set of impact tests is to be taken from each batch of 15 tonnes or fraction thereof.

iv) For (ii) and (iii) above the piece selected for the preparation of test specimens is to be the thickest in each batch.

3.15 Manufacturing approval scheme for EH47 steels

3.15.1 Approval scheme

3.15.1.1 Manufacturing approval scheme for EH47 steels is to be in accordance with Part 2 of the IRS Classification Note: '*Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels',* unless otherwise specified in this sub-section.

3.15.2 Approval tests

3.15.2.1 *Extent of Approval tests* :

.1 Cl. 2.1 (c) and (d) of Part 2 of subject Classification Note are not to be applied to manufacturing approval of EH47 steels.

.2 The products for testing are to represent the maximum thickness for approval. If the target chemical composition changes with the thickness, the maximum thickness for each specified chemical composition specification is to be tested.

3.15.2.2 *Type of tests*

(a) *Brittle fracture initiation test*: Deep notch test or Crack Tip Opening Displacement (CTOD) test is to be carried out. The test method is to be acceptable to IRS.

(b) *Y-groove weld cracking test (Hydrogen crack test):* The test method is to be in accordance with recognized national standards such as ISO 17642-2:2005.

In addition to the requirements specified in (a) and (b) above, the approval tests specified in Part 2 of the IRS Classification Note: '*Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels',* are also to be carried out. Additional tests may be required when deemed necessary by IRS.

3.16 Retest Procedures

3.16.1 Retest procedures are to be as per the requirements provided in Chapter 1.

Section 4

High Strength Steels for Welded Structures

4.1 General

4.1.1 These requirements apply to hot-rolled, fine-grain, weldable high strength structural steels, intended for use in marine and offshore structural applications. These requirements do not apply to steels intended for hull structure of commercial ships whose requirements are specified in previous sections.

4.1.2 The steel covered by the scope of these requirements are specified in yield strength levels of 420, 460, 500, 550, 620, 690,890 and 960 [N/mm^2]. For each yield strength level grades AH, DH, EH and FH are specified, based on the impact test temperature, except for yield strength level of 890 and 960 [N/mm^2] for which grade F is not applicable. The full list of grades are:

AH420	**DH420**	**EH420**	**FH420**
AH460	**DH460**	**EH460**	**FH460**
AH500	**DH500**	**EH500**	**FH500**
AH550	**DH550**	**EH550**	**FH550**
AH620	**DH620**	**EH620**	**FH620**
AH690	**DH690**	**EH690**	**FH690**
AH890	**DH890**	**EH890**	
AH960	**DH960**	**EH960**	

4.1.3 Steels covered by the scope may be delivered in Normalized (N)/Normalized rolled (NR); Thermo-mechanical controlled rolled (TM) or Quenched and Tempered (QT) condition.

Note: TM is a generic delivery condition that may not include accelerated cooling, and may or may not include direct quenching followed by tempering after TM-rolling.

4.1.4 Product forms include plates, wide flats, sections bars and seamless tubulars.

4.1.5 Steels with a thickness beyond the maximum thicknesses as given in Table 4.5.3 may be approved at the discretion of the IRS.

4.1.6 Steels differing in chemical composition, deoxidation practice, delivery condition and mechanical properties may be accepted, subject to the special approval of the IRS. Such steels are to be given a special designation.

4.2 Approval

4.2.1 For applications subjected to Classification, all steels are to be manufactured at steel works which have been approved by IRS for the type and grade of steel which is being supplied. The procedure for approval is available in Classification Note "*Manufacturing Approval Scheme of High Strength Steels for Welded Structures"*. Also refer Chapter 1, Section 1, Cl. 1.3.2.

4.2.2 It is the steelmaker's responsibility to assure that effective quality, process and production controls during manufacturing are adhered to within the manufacturing specification. The manufacturing specification is to be submitted to IRS at the time of initial approval.

4.2.3 Where non-conformities arise, the manufacturer is to identify the root cause and establish countermeasures to prevent its recurrence. The non-conformities and the countermeasures are to be documented and reported to IRS.

4.2.4 When the semi-finished products were not manufactured by the approved manufacturer of the finish rolled and heat treated products, the manufacturer of the semi-finished product is also to be subject to approval by IRS.

Note 1: The attention of the users must be drawn to the fact that when fatigue loading is present, the effective fatigue strength of a welded joint of high strength steel may not be greater than that of a welded joint in normal strength steels.

Note 2: Before subjecting steels produced by thermos-mechanical rolling or quenched and tempered after rolling to further heating for forming or stress relieving, or using high heat-input welding, special consideration must be given to the possibility of a consequent reduction in mechanical properties

4.3 Method of manufacture

4.3.1 Steel making process

4.3.1.1 The steel is to be manufactured by the basic oxygen, basic electric arc furnace or by processes specially approved by IRS.

4.3.1.2 Vacuum degassing is to be used for any of the following:

a) all steels with enhanced through-thickness properties, and

b) all steels of grade H690, H890 and H960.

4.3.2 Deoxidation

4.3.2.1 The steel shall be fully killed .

4.3.3 Grain size

4.3.3.1 The steel is to be fine grain treated, and is to have a fine grain structure. The fine grain practice is to be as detailed in the manufacturing specification.

Note: A fine grain structure has an equivalent index ≥ 6 determined by micrographic examination in accordance with ISO 643 or alternative test method.

4.3.4 Nitrogen control

4.3.4.1 The steels are to contain nitrogen binding elements as detailed in the manufacturing specification. Also see note 4 in Table 4.4.1.

4.4 Chemical composition

4.4.1 The chemical composition is to be determined by the steel maker, in an adequately equipped competently staffed laboratory. The method of sampling is to follow that carried out for the initial approval tests, either from the ladle, the tundish or the mould in the case of continuous casting. The aim analysis is to be in accordance with the manufacturing specification. All the elements listed in the Table 4.4.1 are to be reported.

4.4.2 Elements used for alloying, nitrogen binding, and fine grain treatment, and as well as the residual elements are to be as detailed in the manufacturing specification, e.g. When boron is deliberately added for enhancement of hardenability of the steels, the maximum content of the boron content is not to be higher than 0.005%; and the analysis result is to be reported.

4.4.3 The carbon equivalent value is to be calculated from the ladle analysis. Maximum values are specified in Table 4.4.2.

a) For all steel grades the following formula of IIW may be used:

$$Ceq = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}\ (\%)$$

b) For steel grades H460 and higher, CET may be used instead of Ceq at the discretion of the manufacturer, and is to be calculated according to the following formula:

$$CET = C + \frac{(Mn + Mo)}{10} + \frac{(Cr + Cu)}{20} + \frac{Ni}{40}\ (\%)$$

Note: The CET is included in the standard EN 10011-2:2001 used as one of the parameters for pre-heating temperature determination which is necessary for avoiding cold cracking.

c) For TM and QT steels with carbon content not more than 0.12%, the cold cracking susceptibility *P*cm for evaluating weldability may be used instead of carbon equivalent of *Ceq* or *CET* at manufacturer's discretion and is to be calculated using the following formula:

$$P_{cm} = C + \frac{Si}{30} + \frac{Mn}{20} + \frac{Cu}{20} + \frac{Ni}{60} + \frac{Cr}{20} + \frac{Mo}{15} + \frac{V}{10} + 5B$$

Table 4.4.1 : Chemical composition						
Delivery condition[1)]	**N/NR**		**TM**		**QT**	
Steel grade / **Chemical Composition[2)]**	**AH420 DH420 AH460 DH460**	**EH420 EH460**	**AH420 DH420 AH460 DH460 AH500 DH500 AH550 DH550 AH620 DH620 AH690 DH690 AH890**	**EH420 FH420 EH460 FH460 EH500 FH500 EH550 FH550 EH620 FH620 EH690 FH690 DH890 EH890**	**AH420 DH420 AH460 DH460 AH500 DH500 AH550 DH550 AH620 DH620 AH690 DH690 AH890 AH960**	**EH420 FH420 EH460 FH460 EH500 FH500 EH550 FH550 EH620 FH620 EH690 FH690 DH890 EH890 DH960 EH960**
Carbon % max	0.20	0.18	0.16	0.14	0.18	
Manganese %	1.0~1.70		1.0~1.70		1.70	
Silicon % max	0.60		0.60		0.80	
Phosphorus % max[3)]	0.030	0.025	0.025	0.020	0.025	0.020
Sulphur % max[3)]	0.025	0.020	0.015	0.010	0.015	0.010
$Aluminium_{total}$ % min[4)]	0.02		0.02		0.018	
Niobium % max[5)]	0.05		0.05		0.06	
Vanadium % max[5)]	0.20		0.12		0.12	
Titanium % max[5)]	0.05		0.05		0.05	
Nickel % max[6)]	0.80		2.00[6)]		2.00[6)]	
Copper % max	0.55		0.55		0.50	
Chromium % max[5)]	0.30		0.50		1.50	
Molybdenum % max[5)]	0.10		0.50		0.70	
Nitrogen % max	0.025		0.025		0.015	
Oxygen ppm max[7)]	Not applicable		Not applicable	50	Not applicable	30

Note 1 See section 4.5 for definition of delivery conditions

Note 2 The chemical composition is to be determined by ladle analysis and is to meet the approved manufacturing specification at the time of approval.

Note 3 For sections the P and S content can be 0.005 % higher than the value specified in the table.

Note 4 The total aluminium to nitrogen ratio shall be a minimum of 2:1. When other nitrogen binding elements are used, the minimum Al value and Al/N ratio do not apply.

Note 5 Total Nb+V+Ti ≤ 0.26 % and Mo+Cr ≤0.65%, not applicable for QT steels.

Note 6 Higher Ni content may be approved at the discretion of IRS.

Note 7 The requirement on maximum Oxygen content is only applicable to DH890; EH890; DH960 and EH960.

Table 4.4.2 : Maximum Ceq, *CET* and *P*cm values								
Steel grade & delivery condition	**Carbon Equivalent (%)**							
	Ceq						***CET***	***P*cm**
	Plates			**Sections**	**Bars**	**Tubulars**	**all**	**all**
	t≤50 (mm)	**50<t≤100 (mm)**	**100<t≤250 (mm)**	**t≤50 (mm)**	**t≤250 or d≤250 (mm)**	**t≤65 (mm)**	**all**	**all**
H420N/NR	0.46	0.48	0.52	0.47	0.53	0.47	N.A	N.A
H420TM	0.43	0.45	0.47	0.44	N.A	N.A	N.A	N.A
H420QT	0.45	0.47	0.49	N.A	N.A	0.46	N.A	N.A
H460N/NR	0.50	0.52	0.54	0.51	0.55	0.51	0.25	N.A
H460TM	0.45	0.47	0.48	0.46	N.A	N.A	0.30	0.23
H460QT	0.47	0.48	0.50	N.A	N.A	0.48	0.32	0.24
H500TM	0.46	0.48	0.50	N.A	N.A	N.A	0.32	0.24
H500QT	0.48	0.50	0.54	N.A	N.A	0.50	0.34	0.25
H550TM	0.48	0.50	0.54	N.A	N.A	N.A	0.34	0.25
H550QT	0.56	0.60	0.64	N.A	N.A	0.56	0.36	0.28
H620TM	0.50	0.52	N.A	N.A	N.A	N.A	0.34	0.26
H620QT	0.56	0.60	0.64	N.A	N.A	0.58	0.38	0.30
H690TM	0.56	N.A	N.A	N.A	N.A	N.A	0.36	0.30
H690QT	0.64	0.66	0.70	N.A	N.A	0.68	0.40	0.33
H890TM	0.60	N.A	N.A	N.A	N.A	N.A	0.38	0.28
H890QT	0.68	0.75	N.A	N.A	N.A	N.A	0.40	N.A
H960QT	0.75	N.A	N.A	N.A	N.A	N.A	0.40	N.A
Note: N.A = Not applicable								

4.5 Delivery Condition - Rolling Process and Heat Treatment

4.5.1 Steel is to be delivered in accordance with the processes approved by IRS. These processes include:

- Normalized (N)/Normalized rolled (NR)
- Thermo-mechanical controlled rolled (TM)/with Accelerated cooling (TM+AcC)/with direct quenching followed by tempering (TM+DQ), or
- Quenched and Tempered condition (QT)

The definition of these delivery conditions are defined in previous sections.

Note: Direct quenching after hot-rolling followed by tempering is considered equivalent to conventional quenching and tempering.

4.5.2 Rolling reduction ratio

4.5.2.1 The rolling reduction ratio of slab, billet, bloom or ingot is not to be less than 3:1 unless agreed at the time of approval.

4.5.3 Thickness limits for approval

4.5.3.1 The maximum thickness of slab, billet or bloom from the continuous casting process is to be at the manufacturer's discretion.

4.5.3.2 Maximum thickness of plates, sections, bars and tubulars over which a specific delivery condition is applicable are shown in Table 4.5.3.

Table 4.5.3 : Maximum thickness limits				
Delivery condition	**Maximum thickness (mm)**			
	Plates	**Sections**	**Bars**	**Tubulars**
N	250[2)]	50	250	65
NR	150	[1)]		
TM	150	50	Not applicable	Not applicable
QT	150[2)]	50	Not applicable	50

Note 1 The maximum thickness limits of sections, bars and tubulars produced by NR process route are less than those manufactured by N route, and are to be at the discretion of IRS.

Note 2 Approval for N steels with thickness larger than 250 [mm] and QT steels with thickness larger than 150 [mm] is subject to the special consideration of IRS.

4.6 Mechanical Properties

4.6.1 Test specimens and test procedures for mechanical properties are in accordance with Chapter 2 and Section 3 of this chapter.

4.6.2 Tensile test

4.6.2.1 Test specimens are to be cut with their longitudinal axes transverse to the final direction of rolling, except in the case of sections, bars, tubulars and rolled flats with a finished width of 600 [mm] or less, where the tensile specimens may be taken in the longitudinal direction.

4.6.2.2 Full thickness flat tensile specimens are to be prepared. The specimens are to be prepared in such a manner as to maintain the rolling scale at least at one side. When the capacity of the test machine is exceeded by the use of a full thickness specimen, sub-sized flat tensile specimens representing either the full thickness or half of the product thickness retaining one rolled surface are to be used. Alternatively, machined round test specimens may be used. The specimens are to be located at a position lying at a distance of t/4 from the surface and additionally at t/2 for thickness above 100 [mm] or as near as possible to these positions.

4.6.2.3 The results of the tests are to comply with the appropriate requirements of Table 4.6.2.3. In the case of product forms other than plates and wide flats where longitudinal tests are agreed, the elongation values are to be 2 percentage units above those transverse requirements as listed in Table 4.6.2.3.

Table 4.6.2.3 : Tensile properties at ambient temperature for all steel grades

Mechanical properties / Steel grade & delivery condition		Minimum yield strength *R*eH[1] (N/mm²)			Ultimate tensile strength *R*m (N/mm²)		Minimum Percentage elongation After fracture (%) $L_0=5.65\sqrt{S_0}$ [2]		Charpy V-notch impact test		
		Nominal thickness (mm)[4]			Nominal thickness (mm)[4]				Test temp (°C)	Minimum (Joules)	
		≥3 ≤50	>50 ≤ 100	>100 ≤250	≥3 ≤100	>100 ≤250	T	L[3]		T	L
H420N/NR H420TM H420QT	A D E F	420	390	365	520~680	470~650	19	21	0 -20 -40 -60	28	42
H460N/NR H460TM H460QT	A D E F	460	430	390	540~720	500~710	17	19	0 -20 -40 -60	31	46
H500TM H500QT	A D E F	500	480	440	590~770	540~720	17	19	0 -20 -40 -60	33	50
H550TM H550QT	A D E F	550	530	490	640~820	590~770	16	18	0 -20 -40 -60	37	55
H620TM H620QT	A D E F	620	580	560	700~890	650~830	15	17	0 -20 -40 -60	41	62
H690TM H690QT	A D E F	690	650	630	770~940	710~900	14	16	0 -20 -40 -60	46	69
H890TM H890QT	A D E	890	830	Not applicable	940~1100	Not applicable	11	13	0 -20 -40	46	69
H960QT	A D E	960	Not applicable	Not applicable	980~1150	Not applicable	10	12	0 -20 -40	46	69

Note 1 For tensile test either the upper yield stress (R_{eH}) or where R_{eH} cannot be determined, the 0.2 percent proof stress (*R*p0.2) is to be determined and the material is considered to comply with the requirement if either value meets or exceeds the specified minimum value of yield strength.

Note 2 For full thickness flat test specimens with a width of 25 [mm] and a gauge length of 200 [mm] the elongation is to comply with the minimum values shown in Table 4.6.2.4.

Note 3 In the case that the tensile specimen is parallel to the final rolling direction, the test result shall comply with the requirement of elongation for longitudinal (L) direction.

Note 4 For plates and sections for applications, such as racks in offshore platforms etc, where the design requires that tensile properties are maintained through the thickness, a decrease in the minimum specified tensile properties is not permitted with an increase in the thickness.

Table 4.6.2.4 : Elongation minimum values for a width of 25 [mm] and a 200 [mm] gauge length[1)]

Strength Grade	Thickness [mm]						
	≤ 10	**> 10 ≤ 15**	**> 15 ≤ 20**	**> 20 ≤ 25**	**> 25 ≤ 40**	**> 40 ≤ 50**	**> 50 ≤ 70**
H420	11	13	14	15	16	17	18
H460	11	12	13	14	15	16	17
H500	10	11	12	13	14	15	16
H550	10	11	12	13	14	15	16
H620	9	11	12	12	13	14	15
H690	9[2)]	10[2)]	11[2)]	11	12	13	14

Note 1 The tabulated elongation minimum values are the requirements for testing specimen in transverse direction. H890 and 960 specimens and specimens which are not included in this table is to be proportional specimens with a gauge length of $L_0=5.65\sqrt{S_0}$.

Note 2 For H690 plates with thickness ≤ 20 [mm], round specimen in accordance with Chapter 2 may be used instead of the flat tensile specimen. The minimum elongation for testing specimen in transverse direction is 14%.

4.6.3 Impact test

4.6.3.1 The Charpy V-notch impact test specimens for plates and wide flats over 600 [mm] in width are to be taken with their axes transverse to the final rolling direction and the results should comply with the appropriate requirements for transverse direction of Table 4.6.2.3. For other product forms, the impact tests are to be in the longitudinal direction, the results of the tests are to comply with the appropriate requirements for longitudinal direction of Table 4.6.2.3.

4.6.3.2 Sub-surface test specimens will be taken in such a way that one side is not further away than 2 [mm] from a rolled surface, however, for material with a thickness in excess of 50 [mm], impact tests are to be taken at the quarter thickness (t/4) location and mid-thickness (t/2).

4.6.3.3 Impact test for a nominal thickness less than 6 [mm] are normally not required.

4.6.4 Test frequency

4.6.4.1 Tensile test sample is to be randomly selected from each batch, as defined in section 3, that is to be less than or equal to 25 tonnes, and to be from the same cast, in the same delivery condition and of the same thickness.

4.6.4.2 Impact test

a) For steels plates in N/NR or TM condition test sample is to be taken from each piece.

b) For steels in QT condition test sample is to be taken from each individually heat treated part thereof.

c) For sections, bars and tubulars, test sample is to be taken from each batch of 25 tonnes or fraction thereof.

Note 1: If the mass of the finished material is greater than 25 tonnes, one set of tests from each 25 tonnes and/or fraction thereof is required. (e.g. for consignment of 60 tonnes would require 3 plates to be tested).

Note 2: For continuous heat treated product special consideration may be given to the number and location of test specimens required by the manufacturer to be agreed by IRS.

4.6.5 Traceability

4.6.5.1 Traceability of test material, specimen sampling and test procedures including test equipment with respect to mechanical properties testing, is to be in accordance with section 3.

4.6.6 Re-test procedures

4.6.6.1 Re-test procedures for tensile tests and Charpy impact tests are to be in accordance with Chapter 2.

4.6.7 Through thickness tensile test

4.6.7.1 For steels designated with improved through thickness properties, through thickness tensile tests are to be performed in accordance with Section 8.

4.6.7.2 Subject to the discretion of IRS, through thickness tensile strength may be required to be not less than 80% of the specified minimum tensile strength.

4.7 Tolerances

4.7.1 Unless otherwise agreed or specially required, the thickness tolerances in Sec 1, 1.4 are applicable.

4.8 Surface Quality

4.8.1 All materials are to be free from cracks, injurious surface flaws, injurious laminations and similar defects.

4.8.2 The surface quality inspection method is to be in accordance with recognised national or international standards agreed between purchaser and manufacturer.

a) Welding repair procedures and the method for reporting repairs are to be approved by the IRS.

b) Where repair by grinding is carried out then the remaining plate thickness below the ground area must be within the allowable under thickness tolerance.

4.8.3 Surface finish requirement are to be in accordance with the relevant requirements in Section 3.

4.8.4 Surface inspection is the responsibility of the manufacturer. The acceptance by IRS's Surveyor of material later found to be defective shall not absolve the manufacturer of this responsibility.

4.9 Internal Soundness

4.9.1 Verification of internal soundness is the responsibility of the manufacturer. The acceptance by the IRS's Surveyor shall not absolve the manufacturer of this responsibility.

4.9.2 Ultrasonic examination

4.9.2.1 If required by the IRS, ultrasonic examination should be carried out in accordance with Section 2 for the requirement of internal soundness, and is to be performed in accordance with an approved standard.

4.10 Stress relieving heat treatment and other heat treatments

4.10.1 Steels approved by the procedures given in Classification Note "*Manufacturing Approval Scheme of High Strength Steels for Welded Structures"* with respect to Heat Treatment are suitable for stress relieving heat treatment such as post-weld heat treatment and stress relieving heat treatment after cold forming for the purpose of reducing the risk of brittle fracture, increasing the fatigue lifetime and dimensional stability for machining.

Note: Products can be susceptible to deterioration in mechanical strength and toughness if they are subjected to incorrect post-weld heat treatment procedures or other processes involving heating such as flame straightening, rerolling, etc. where the heating temperature and the holding time exceed the limits given by the manufacturer.

4.11 Facilities for Inspection

4.11.1 Testing is to be carried out under the witness of the Surveyor, or an authorised deputy, in order to verify whether the test results meet the specified requirements.

4.11.2 The manufacturer is to afford the Surveyor all necessary facilities and access to all relevant parts of the steel works to enable him to verify the approved process is adhered to, for the selection of test materials, and the witnessing of tests, as required by this Section. Also for verifying the accuracy of the testing, calibration of inspection equipment and traceability of materials.

4.12 Identification of Materials

4.12.1 The manufacturer is to adopt a system for the identification of ingots, slabs, billet or bloom and finished products, which will enable

the material to be traced to its original cast. The Surveyor is to be given full facilities for so tracing the material when required.

4.13 Branding

4.13.1 Each finished piece is to be clearly marked by the manufacturer with the following particulars:

a) IRS's brand mark

b) Unified identification mark for the grade of steel (e.g. EH620)

c) Name or initials to identify the steelworks

d) Cast number/Heat number, plate number or equivalent identification mark

e) Delivery condition (N/NR, TM/TM+AcC/TM+DQ or Q&T)

The entire markings are to be encircled with paint or otherwise marked so as to be easily recognised. Steels which have been specially approved by IRS and which differ from these requirements (see 4.1.6) are to have the letter "S" after the identification mark (e.g. EH620S)

4.14 Documentation of Inspection Tests

4.14.1 The Surveyor is to be supplied with two copies, of the test certificates or shipping statements for all accepted materials. In addition to the description, dimensions, etc., of the material, the following particulars are to be included:

a) Purchaser's order number

b) Identification of the cast and piece

c) Manufacturer's identification

d) Identification of the grade of steel

e) Chemical analysis, *Ceq*, *CET* or *P*cm value

f) Delivery condition with heat treatment temperatures

g) Mechanical properties test results, including traceable test identification

h) Surface quality and inspection results

i) UT result, where applicable

4.14.2 Before the test certificates are signed by the Surveyor, the steelmaker is required to provide a written declaration stating that the material has been made by an approved process, and that it has been subjected to and has withstood satisfactorily the required tests in the presence of the Surveyor, or an authorised deputy. The following form of declaration will be accepted if stamped or printed on each test certificate with the name of the steelworks and signed by an authorised representative of the manufacturer:
"We hereby certify that the material has been made by an approved process and has been satisfactorily tested in accordance with the Rules of IRS".

Section 5

Steel for Low Temperature Service

5.1 General

5.1.1 This section gives specific requirements for carbon-manganese and nickel alloy steels with toughness properties at low temperatures and intended for use in the construction of cargo tanks and process pressure vessels for liquefied gases.

5.1.2 The requirements of this section are also applicable for other types of pressure vessels where the use of steels with guaranteed impact properties at low temperature is required.

5.1.3 Provision is made for plates and sections up to 40 [mm] thick.

5.1.4 Steel differing in chemical composition, condition of supply or mechanical properties may be accepted, subject to special agreement by IRS.

5.2 Deoxidation and chemical composition

5.2.1 All steels are to be in the fully killed and fine grain refined condition.

5.2.2 The chemical composition of carbon-manganese steels are to comply with the appropriate requirements of grades AH, DH, EH and FH strength levels 32, 36 and 40 (See Table 3.2.1). However, these grades are to be designated as L T -AH, L T -DH, L T -EH and L T -FH respectively for the uses defined in 5.1.1.

5.2.3 The chemical compositions of nickel alloy steels are to comply with the appropriate requirements of Table 5.2.1.

Table 5.2.1 : Chemical composition of nickel alloy steels

Elements	1.5 Ni	3.5 Ni	5 Ni	9 Ni
C max.	0.18	0.15	0.12	0.10
Si	0.10 - 0.35	0.10 - 0.35	0.10 - 0.35	0.10 - 0.35
Mn	0.30 - 1.50	0.30 - 0.90	0.30 - 0.90	0.30 - 0.90
Ni	1.30 - 1.70	3.20 - 3.80	4.70 - 5.30	8.50 - 10.0
P max.	0.025	0.025	0.025	0.025
S max.	0.020	0.020	0.020	0.020
Al min. (acid soluble)[1)]	0.015	0.015	0.015	0.015
Residual elements				
Cr max.	0.25	0.25	0.25	0.25
Cu max.	0.35	0.35	0.35	0.35
Mo max.	0.08	0.08	0.08	0.08
Total of residual elements max.	0.60	0.60	0.60	0.60

1) The total aluminium content may be determined by other methods instead of the acid soluble method. In such cases the total aluminium content is to be not less than 0.020 percent.

5.3 Heat treatment

5.3.1 All materials are to be supplied in a condition complying with Table 5.3.1.

Table 5.3.1 : Conditions of supply

Grade	Plates	Sections and Bars
LT-AH	N, TMCP	Any
LT-DH	N, TMCP	Any
LT-EH	N[2], TMCP, QT	N, TMCP
LT-EH	N[2], TMCP, QT	N, TMCP
1.5 Ni	N[2], QT, normalized and tempered	
3.5 Ni	N[2], QT, normalized and tempered	
5 Ni	N[2], QT, normalized and tempered	
9 Ni	QT, Double normalized and tempered	

1 N = Normalized
TMCP = Thermo-mechanically controlled process
QT = Quenched and tempered
2 The term "Normalized" does not include normalized rolling.

5.4 Mechanical tests

5.4.1 Test pieces for tensile testing of plates are to be cut with their principal axes transverse to the final direction of rolling.

5.4.2 For each batch of plate presented, one tensile test is to be made from one end of each piece unless the mass and length of the piece exceeds 5 tonnes and 15 m in which case test pieces are to be taken from both ends of each piece.

5.4.3 Sections and bars are to be presented for acceptance test in batches containing not more than 50 lengths, as supplied. The material in each batch is to be of the same section size, from the same cast and in the same condition of supply. One tensile test specimen is to be taken from material representative of each batch, except that additional tests are to be taken when the mass of a batch exceeds 10 tonnes.

5.4.4 One set of three Charpy V-notch impact test specimens are to be taken for each tensile test specimen required. For plates, these are to be cut with their principal axis perpendicular to the final direction of rolling and for sections. these are to be taken longitudinally.

5.4.5 The results of all tensile tests are to comply with appropriate requirements given in Table 5.4.1. The ratio between the yield stress and the tensile strength is not to exceed 0.9 for normalized and TMCP steels and 0.94 for Q & T steels.

5.4.6 The average energy value from each set of three impact tests are to comply with appropriate requirements given in Table 5.4.1.

5.4.7 When standard subsidiary impact specimens are necessary (See Sec. 2).

5.4.8 When steel with improved through thickness properties is required or specified in the order, the materials are to be tested as detailed in Sec. 8.

Table 5.4.1 : Mechanical properties for acceptance purposes

Grade of Steel		Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Elongation on 5.65$\sqrt{S_o}$ % min.	Charpy V-notch impact test	
					Test temp.°C	Impact Energy min.
LT-AH	32	315	440 - 590	22	0	Plates transverse tests Average energy 27 J Sections and bars longitudinal tests Average energy 41 J
	36	355	490 - 620	21		
	40	390	510 - 650	20		
LT-DH	32	315	440 - 590	22	-20	
	36	355	490 - 620	21		
	40	390	510 - 650	20		
LT-EH	32	315	440 - 590	22	-40	
	36	355	490 - 620	21		
	40	390	510 - 650	20		
LT-FH	32	315	440 - 590	22	-60	
	36	355	490 - 620	21		
	40	390	510 - 650	20		
1.5 Ni		275	490 - 640	22	-80	
3.5 Ni		285	450 - 610	21	-95	
5 Ni		390	540 - 740	21	-110	
9 Ni		490	640 - 790	18	-196	

Notes:

1 These requirements are applicable to products not exceeding 40 [mm] in thickness. The requirements for thicker products are subject to agreement.

2 The minimum design temperatures at which plates of different thicknesses in the above grades may be used are given in Pt.3, Ch.2, Table 2.4.1 and Pt.5, Ch.4, Table 6.1.2 and Table 6.1.3. Consideration will be given to the use of thicknesses greater than those in the table or to the use of temperatures below - 165°C

Section 6

Steels for Boilers and Pressure Vessels

6.1 General

6.1.1 The following requirements are for carbon, carbon-manganese and alloy steels intended for use in the construction of boilers and pressure vessels. In addition to specifying mechanical properties at ambient temperature for the purpose of acceptance testing, these requirements also give details of appropriate mechanical properties at elevated temperatures which may be used for design purposes.

6.1.2 Where it is proposed to use a carbon or carbon-manganese steel with a specified minimum tensile strength intermediate to the following specified properties, corresponding minimum values for yield and elongation and mechanical properties at elevated temperatures may be obtained by interpolation.

6.1.3 Carbon and carbon-manganese steels with a specified minimum tensile strength of greater than 490 [N/mm^2] but not exceeding 520 [N/mm^2] may be accepted provided that details of proposed specifications are submitted for approval.

6.1.4 Where it is proposed to use alloy steels other than those specified herein, details of the specifications are to be submitted for approval. In such cases the specified minimum tensile strength is not to exceed 600 [N/mm^2].

6.1.5 Materials intended for use in the construction of the cargo tanks and process pressure vessels, storage tanks for liquefied gases and for other low temperature applications are to comply with the requirements of Sec. 5.

6.2 Deoxidation and chemical composition

6.2.1 The method of deoxidation and the chemical analysis of ladle samples is to comply with the requirements of Table 6.2.1.

Table 6.2.1 : Deoxidation and chemical composition

Grade of steel	Deoxidation	Chemical composition per cent						
C and C-Mn steel		**C max.**	**Si**	**Mn**	**P max.**	**S max.**	**Al**	**Residual elements**
360 AR	Any method except rimmed steel	0.18	0.50 max.	0.40-1.30	0.040	0.04	-	Cr 0.25 max. Cu 0.30 max. Mo 0.10 max. Ni 0.30 max. Total 0.70 max.
410 AR		0.21	0.50 max.	0.40-1.30	0.040	0.040	-	
460 AR		0.23	0.50 max.	0.80-1.50	0.040	0.040	-	
360	Any method except rimmed steel	0.17	0.35 max.	0.40-1.20	0.035	0.035	-	
410		0.20	0.35 max.	0.50-1.30	0.035	0.035	-	
460		0.20[1]	0.40 max.	0.80-1.40	0.035	0.035	-	
490	Killed	0.20[1]	0.10-0.50	0.90-1.60	0.035	0.035		
360 FG	Killed fine grained	0.17	0.35 max.	0.40-1.20	0.035	0.035	See note 2	
410 FG		0.20	0.35 max.	0.50-1.30	0.035	0.035	See note 2	
460 FG		0.20[1]	0.40 max.	0.80-1.50	0.035	0.035	See note 2	
490 FG		0.20[1]	0.10-0.50	0.90-1.60	0.035	0.035	See note 2	

Alloy steels	**Deoxi-dation**	**C**	**Si**	**Mn**	**P max.**	**S max.**	**Al**	**Cr**	**Mo**	**Residual elements**
1 Cr 1/2 Mo 470	Killed	0.10-0.18	0.15-0.35	0.4-0.8	0.035	0.035	See note 3	0.70-1.30	0.40-0.60	Cu 0.30 max. Ni 0.30 max.
2 1/4 Cr 1 Mo 480		0.08-0.18	0.15-0.50	0.4-0.8	0.035	0.035	See note 3	2.00-2.50	0.90-1.10	

Notes:

1 For thickness greater than 30 [mm], carbon 0.22 percent max.

2 Aluminium (acid soluble) 0.015 per cent min, or Aluminium (total) 0.018 percent min. Niobium, Vanadium or other suitable grain refining elements may be used either in place of or in addition to aluminium.

3 Aluminium (acid soluble or total) 0.020 percent max.

6.3 Heat treatment, condition of supply

6.3.1 All materials are to be supplied in a condition complying with the requirements of Table 6.3.1.

However, when agreed, material intended for hot forming may be supplied in the as rolled condition.

Table 6.3.1 : Heat treatment

Grade of steel	Condition of supply
Carbon and carbon- manganese 360 AR to 460 AR	As rolled, maximum thickness or diameter is 40 [mm]
Carbon and carbon- manganese 360 to 490	Normalized or controlled rolled
Carbon and carbon- manganese 360 FG to 490 FG	Normalized or controlled rolled
1Cr 1/2 Mo 470	Normalized and tempered
2 1/4 Cr 1 Mo 480	Normalized and tempered

6.4 Mechanical tests

6.4.1 For plates a tensile test specimen is to be taken from one end of each piece when the weight does not exceed 5 tonnes and the length does not exceed 15 [m]. When either of these limits is exceeded, tensile test specimens are to be taken from both ends of each piece. A piece is to be regarded as the rolled product from a single slab or a single ingot, if this is rolled directly into plates.

6.4.2 For strips, tensile test specimens are to be taken from both ends of each coil.

6.4.3 Sections and bars are to be presented for acceptance tests in batches containing not more than 50 lengths, as supplied. The material in each batch is to be of the same section size, from the same cast and in the same condition of supply. One tensile test specimen is to be taken from material representative of each batch, except that additional tests are to be taken when the weight of a batch exceeds 10 tonnes.

6.4.4 Where plates are required for hot forming and it has been agreed that the heat treatment will be carried out by the fabricator, the tests at the steel works are to be made on material which has been cut from the plates and given a normalizing or normalizing and tempering heat treatment in a manner simulating the treatment which will be applied to the plates.

6.4.5 If required by the Surveyors or by the fabricator test material may be given a simulated stress relieving heat treatment prior to the preparation of the test specimens. This has to be stated on the order together with agreed details of the simulated heat treatment and the mechanical properties which can be accepted.

6.4.6 The results of the tensile tests are to comply with the appropriate requirements given in Table 6.4.1, Table 6.4.2 and Table 6.4.3.

Table 6.4.1 : Mechanical properties for acceptance purposes : carbon and carbon-manganese steels - as rolled

Grade of steel	Thick-ness [mm]	Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Elongation on 5.65$\sqrt{So}$ % min.
360 AR	$\leq$ 40	190	360-480	24
410 AR	$\leq$ 40	215	410-530	22
460 AR	$\leq$ 40	240	460-580	21

Table 6.4.2 : Mechanical properties for acceptance purposes : carbon and carbon-manganese steels-normalized or controlled rolled

Grade of steel	Thickness [mm] (see Note)	Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Elongation on 5.65√So % min.
360	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	205 195 185	360 - 480	26 26 25
410	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	235 225 215	410 - 530	24 24 23
460	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	285 255 245	460 - 580	22 22 21
490	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	305 275 265	490 - 610	21 21 20
360 FG	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	235 215 195	360 - 480	26 26 25
410 FG	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	265 245 235	410 - 530	24 24 23
460 FG	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	295 285 275	460 - 580	22 22 21
490 FG	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	315 315 305	490 - 610	21 21 21

Note:

For thicknesses greater than 63 [mm], the minimum values for yield stress may be reduced by 1 per cent for each 5 [mm] increment in thickness over 63 [mm]. The minimum elongation values may also be reduced one unit, e.g. 20 percent reduced to 19 percent for all thicknesses over 63 [mm]. For thicknesses over 100 [mm], the above values are to be agreed.

Table 6.4.3 : Mechanical properties for acceptance purposes : alloy steels-normalized and tempered

Grade of steel	Thickness [mm] (see Note)	Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Elongation on 5.65√So % min.
1 CR 1/2 Mo 470	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	305 305 305	470 - 620	20 20 19
2 1/4 Cr 1 Mo 480	> 3 ≤ 16 > 16 ≤ 40 > 40 ≤ 63	275 265 265	480 - 630	18 18 17

Table 6.4.3 (Contd.)

Note:

For thicknesses greater than 63 [mm], the minimum values for yield stress may be reduced by 1 per cent for each 5 [mm] increment in thickness over 63 [mm]. The minimum elongation values may also be reduced one unit, e.g. 20 percent reduced to 19 percent for all thicknesses over 63 [mm]. For thicknesses over 100 [mm], the above values are to be agreed.

6.5 Mechanical properties for design purposes at elevated temperatures

6.5.1 Nominal values for the minimum lower yield or 0.2 per cent proof stress at temperatures of 50°C and higher are given in the following tables :

Table 6.5.1 Carbon and carbon manganese steels - As rolled (applicable only when the design temperature does not exceed 350°C).

Table 6.5.2 Carbon and carbon-manganese steels normalized or controlled rolled.

Table 6.5.3 Alloy steels. Normalized and tempered.

6.5.2 These values are intended for design purposes only and verification is not required except for materials complying with National or proprietary specifications where the elevated temperature properties used for design purposes are higher than those given in Table 6.5.1 to Table 6.5.3. The extent of testing in such cases would have to be specially agreed by IRS.

6.5.3 Values for the estimated average stress to rupture in 100,000 hours are given in Table 6.5.4 and may be used for design purposes.

Table 6.5.1 : Mechanical properties for design purposes - Carbon and carbon - manganese steels - as rolled

Grade of steel	Thickness [mm]	Design temperature°C (See Note)						
		50	**10**	**150**	**200**	**250**	**300**	**350**
		Nominal minimum lower yield or 0.2 percent proof stress [N/mm^2]						
360 AR	≤ 40	154	153	152	145	128	108	102
410 AR		186	183	181	174	155	134	127
460 AR		218	213	210	203	182	161	153

Note : Maximum permissible design temperature is 350°C

Table 6.5.2 : Mechanical properties for design purposes - carbon and carbon - manganese steels - normalized or controlled rolled

Grade of steel	Thickness [mm] (see Note)	Design temperature °C								
		50	100	150	200	250	300	350	400	450
		Nominal minimum lower yield or 0.2 percent proof stress [N/mm^2]								
360	> 3 ≤ 16	183	175	172	168	150	128	117	115	113
	> 16 ≤ 40	173	171	169	162	144	124	117	115	113
	> 40 ≤ 63	166	162	158	152	141	124	117	115	113
410	> 3 ≤ 16	220	211	208	201	180	150	142	138	136
	> 16 ≤ 40	204	201	198	191	171	150	142	138	136
	> 40 ≤ 63	196	192	188	181	168	150	142	138	136
460	> 3 ≤ 16	260	248	243	235	210	176	168	162	158
	> 16 ≤ 40	235	230	227	220	198	176	168	162	158
	> 40 ≤ 63	227	222	218	210	194	176	168	162	158
490	> 3 ≤ 16	280	270	264	255	228	192	183	177	172
	> 16 ≤ 40	255	248	245	237	214	192	183	177	172
	> 40 ≤ 63	245	240	236	227	210	192	183	177	172
360 FG	> 3 ≤ 16	214	204	185	165	145	127	116	110	106
	> 16 ≤ 40	200	196	183	164	145	127	116	110	106
	> 40 ≤ 63	183	179	172	159	145	127	116	110	106
410 FG	> 3 ≤ 16	248	235	216	194	171	152	141	134	130
	> 16 ≤ 40	235	228	213	192	171	152	141	134	130
	> 40 ≤ 63	222	215	204	188	171	152	141	134	130
460 FG	> 3 ≤ 16	276	262	247	223	198	177	167	158	153
	> 16 ≤ 40	217	260	242	220	198	177	167	158	153
	> 40 ≤ 63	262	251	236	217	198	177	167	158	153
490 FG	> 3 ≤ 16	297	284	265	240	213	192	182	173	168
	> 16 ≤ 40	293	279	260	237	213	192	182	173	168
	> 40 ≤ 63	286	272	256	234	213	192	182	173	168

Note : For thicknesses greater than 63 [mm], the values for lower yield or 0.2 percent stress are to be reduced by 1 percent for each 5 [mm] increment in thickness upto 100 [mm]. For thicknesses over 100 [mm], the values are to be agreed and verified by test.

Table 6.5.3 : Mechanical properties for design purposes : alloy steels-normalized tempered

Grade of steel	Thickness [mm] (see Note)	Design temperature °C									
		50	100	200	300	350	400	450	500	550	600
		Nominal minimum lower yield or 0.2 percent proof stress [N/mm^2]									
1 Cr 1/2 Mo 470	3 ≤ 63	284	270	248	216	203	199	194	188	181	174
2 1/4 Cr 1 Mo 480	3 ≤ 63	255	249	233	219	212	207	194	180	160	137

Note : For thicknesses greater than 63 [mm], the values for lower yield or 0.2 percent stress are to be reduced by 1 percent for each 5 [mm] increment in thickness upto 100 [mm]. For thicknesses over 100 [mm], the values are to be agreed and verified by test.

Table 6.5.4 : Mechanical properties for design purposes : estimated average values for stress to rupture in 100,000 hours [N/mm^2]

Temp. °C	Grades of steel				
	Carbon and carbon-manganese			Alloy Steels	
	360 FG 410 FG 460 FG	360 410 460	490 490 FG 510 FG	1 Cr 1/2 Mo 470	2 1/4 Cr 1 Mo 480
380	171	219	227	-	-
390	155	196	203	-	-
400	141	173	179	-	-
410	127	151	157	-	-
420	114	129	136	-	-
430	102	109	117	-	-
440	90	92	100	-	-
450	78	78	85	-	221
460	67	67	73	-	204
470	57	57	63	-	186
480	47	48	55	210	170
490	36	-	47	177	153
500	-	-	-	146	137
510	-	-	-	121	122
520	-	-	-	99	107
530	-	-	-	81	93
540	-	-	-	67	79
550	-	-	-	54	69
560	-	-	-	43	59
570	-	-	-	35	51
580	-	-	-	-	44

Section 7

Steels for Machinery Structures

7.1 General

7.1.1 Steel plates, strips, sections or bars intended for use in the construction of welded machinery structures are to comply with one of the following alternatives:

a) Any grade of normal strength structural steel or high strength structural steel as detailed in Sec. 2 and 3.

b) Any grade of carbon or carbon-manganese steel as detailed in Sec. 6 except that for this application batch testing is acceptable and the same is to be carried out in accordance with the requirements of Sec. 2.

Section 8

Plates with Specified minimum through Thickness Properties ('Z' quality)

8.1 General

8.1.1 Following requirements are for special quality plate material with improved ductility in the through thickness or "Z direction.

8.1.2 The use of this material known as 'Z' quality steel, is recommended when plate material, intended for welded construction, will be subject to significant strain in a direction perpendicular to the rolled surfaces. These strains are usually associated with thermal contraction and restraint during welding, particularly for full penetration "T"- butt welds but may also be associated with loads applied in service or during construction. Where these strains are of sufficient magnitude, lamellar tearing may occur. Two 'Z' quality steels are specified; Z25 for normal ship applications and 'Z35' for more severe applications.

Through thickness properties are characterized by specified values for reduction of area in a through thickness tensile test.

8.1.3 This special quality material is to comply with the requirements of Sec. 2, 3, 4, 5, 6 and 7 as appropriate and the following additional requirements.

8.2 Manufacture

8.2.1 All plates are to be manufactured at works which have been approved by IRS for this quality of material. Also refer Chapter 1, Section 1, Cl. 1.3.2.

8.2.2 The sulphur content is not to exceed 0.008 per cent, as determined by ladle analysis. It is recommended that the steel should be efficiently vacuum de-gassed.

8.3 Test material

8.3.1 Unless otherwise agreed, through thickness tensile tests are only required for plate materials where the thickness exceeds 15 [mm]. A test sample large enough to provide six test specimens are to be cut from the centre of one end of each rolled piece representing the batch. (See Fig.8.3.1). Where appropriate the end selected should be representative of the top end of an ingot or the start of a concast strand. Generally three through thickness tensile test specimens are to be prepared while the rest of the sample remains for possible retests.

8.3.2 The batch size is to be determined depending on the product and sulphur content as given in Table 8.3.2.

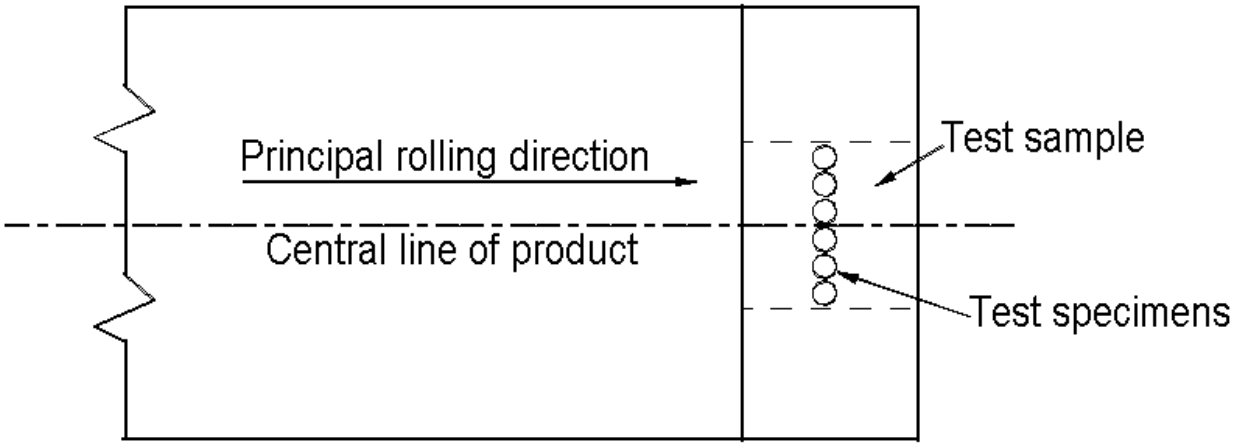


Fig.8.3.1 : Plate and wide flat sampling position

Table 8.3.2 : Batch size dependent on product and sulphur content

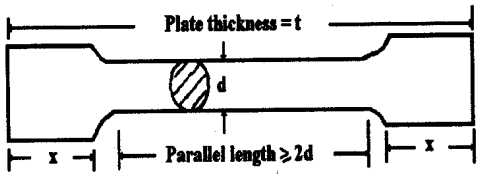
Product	S > 0.005%	S ≤ 0.005%
Plates	Each piece (parent plate)	Maximum 50t of products of the same cast, thickness and heat treatment
Wide flats of nominal thickness ≤ 25 mm	Maximum 10t of products of the same cast, thickness and heat treatment	Maximum 50t of products of the same cast, thickness and heat treatment
Wide flats of nominal thickness > 25 mm	Maximum 20t of products of the same cast, thickness and heat treatment	Maximum 50t of products of the same cast, thickness and heat treatment

8.4 Dimensions of through thickness tensile test specimens

8.4.1 At the option of the steel maker test specimens (Fig.8.4.1a) or test specimens with welded extensions (Fig.8.4.1b) may be used. For both types of test specimens, the diameter of the parallel portion is not to be less than 6 [mm] when plate thickness is less than or equal to 25 [mm] and 10 [mm] when the plate thickness is greater than 25 [mm].

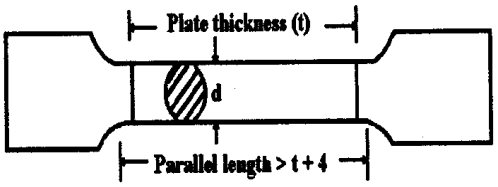
Alternatively, round test specimens, including those with welded extensions, may be prepared in accordance with a recognized standard.

8.4.2 The tolerances on specimen dimensions are to be in accordance with ISO 6892-98 or other recognised standards as appropriate.



t = thickness of plate [mm]
d = 6 mm; when 15 mm ≤ t ≤ 25 [mm]
d = 10 mm min. when t > 25 [mm]
x = 5 mm max. when d = 6 [mm]
x = 8 mm max. when d = 10 [mm]

Fig 8.4.1 (a) : Plain test specimen



t = Thickness of plate [mm]
d = 6 mm; when 15 mm ≤ t ≤ 25 [mm]
d = 10 mm when t 25 [mm]

Fig 8.4.1 (b) : Test specimen with welded extension

8.5 Mechanical tests

8.5.1 The acceptable minimum average value for the reduction of area of the three tensile test specimens taken in the through thickness direction are given in Table 8.5.1. Only one individual value may be below the minimum average, but not less than the minimum individual value for the appropriate grade.

Table 8.5.1 : Reduction of area acceptance values

Grade	Z25	Z35
Minimum average	25%	35%
Minimum individual	15%	25%

8.5.2 A value less than minimum individual value will require rejection of the piece. However, in case of batch testing each remaining piece in the batch may be individually tested.

8.5.3 Depending on the test results, retest may be permitted in the cases shown in Fig.8.5.3. In these instances, three more tensile tests are to be taken from the remaining test sample. The average of all 6 tensile tests is to be greater than the required minimum average with not more than two results below the minimum average.

In case of failure after retest, either the batch represented by the piece is rejected or each piece within batch may be retested.

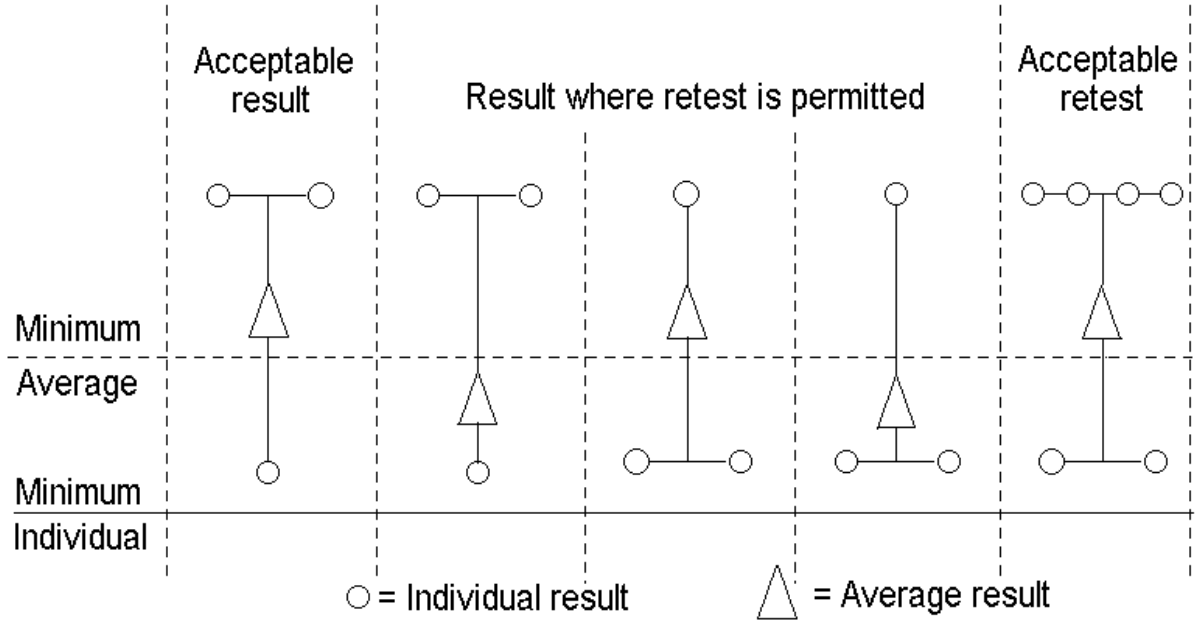


Fig.8.5.3 : Diagram showing acceptance / rejection and retest criteria

8.6 Non-destructive examination

8.6.1 All special 'Z' quality plates are to be ultrasonically tested in the final supply condition, with a probe of frequency 4 MHz. The ultrasonic testing is to be carried out in accordance with either EN 10160:1999 Level S1/E1 or ASTM A578:2017 Level C.

Section 9

Austenitic and Duplex Stainless Steels

9.1 Scope

9.1.1 This section gives the requirements for rolled products in austenitic and duplex (austenite plus ferrite) stainless steels intended for use in the construction of cargo tanks, storage tanks and process pressure vessels for chemicals and liquefied gases.

9.1.2 Austenitic stainless steels are suitable for applications where the lowest design temperature is not lower than –165°C.

9.1.3 Austenitic stainless steels are also suitable for service at elevated temperatures and for such applications the proposed specification should contain, in addition to the requirements of 9.1.6, minimum values for 0.2 and 1.0 per cent proof stresses at the design temperature.

9.1.4 Duplex stainless steels are suitable for applications where the lowest design temperature is above 0°C. Any requirement to use duplex stainless steels below 0°C will be subject to special consideration.

9.1.5 Duplex stainless steels are also suitable for service at temperatures upto 300°C and for such applications the proposed specification should include, in addition to the requirements of 9.1.6, a minimum value for 0.2 per cent proof stress at the design temperature.

9.1.6 A specification giving details of the chemical composition, heat treatment and

mechanical properties, including for the austenitic grades, both the 0.2 and 1.0 percent proof stresses, is to be submitted for consideration and approval.

9.2 Chemical composition

9.2.1 The chemical composition of ladle samples is to comply with the requirements given in Table 9.2.1.

9.3 Heat treatment

9.3.1 All materials are to be supplied in the solution treated condition.

Table 9.2.1 : Chemical composition

Type and grade of steel	Chemical composition % (see Note)									
	C max	Si max	Mn max	P max	S max	Cr	Ni	Mo	N	Other
Austenitic										
304L	0.03	}	}	}	}	17.0-20.0	8.0-13.0	-	0.10	-
304LN	"	}	}	}	}		3.0-12.0	-	0.10-0.22	-
316L	"	}	}	}	}	17.0-20.0	10.0-15.0	2.0-3.0	0.10	-
316LN	"	1.0	2.0	0.045	0.03			2.0-3.0	0.10-0.22	-
317L	"	}	}	}	}	16.0-18.5	10.0-14.5	3.0-4.0	0.10	-
317LN	"	}	}	}	}			3.0-4.0	0.10-0.22	-
321	0.06	}	}	}	}	16.0-18.5	11.0-15.0	-	0.10	5xC≤Ti≤0.7
347	0.06	}	}	}	}			-	0.10	10xC≤Nb≤1.0
						18.0-20.0	12.5-15.0			
						18.0-20.0	9.0-12.0			
						17.0-19.0	9.0-13.0			
						17.0-19.0				
Duplex										
UNS S31803	0.03	1.0	2.0	0.03	0.02	21.0-23.0	4.5-6.5	2.5-3.5	0.08-0.20	-
UNS S32750	0.03	0.80	1.2	0.035	0.02	24.0-26.0	6.0-8.0	3.0-5.0	0.24-0.32	Cu 0.50 max.

9.4 Mechanical tests

9.4.1 Tensile test specimens are to be taken in accordance with the appropriate requirements of 5.4 and 6.4.1.

9.4.2 For the duplex grades, one set of three Charpy V-notch impact test specimens machined in the longitudinal direction from each tensile test piece is to be tested at –20°C. The average energy value of the three specimens is to be not less than 41 Joules.

For austenitic grades of steel, impact tests are only required for design temp. below –105°C. In such cases, impact tests carried out at a temperature of –196°C on a set of three charpy V-notch specimens are to comply with the following:

a) Plates : Transverse test pieces; minimum average energy value 27 Joules.
b) Strips, sections and bars : Longitudinal test pieces, minimum average energy value 41 Joules.

9.4.3 Where standard subsidiary Charpy V-notch test specimens are necessary, see Chapter 2, Sec.3.1.2.

9.4.4 The results of all tensile tests are to comply with the requirements of Table 9.4.1 or the approved specification.

9.5 Through thickness tests

9.5.1 Where material will be strained in a through thickness direction during welding or in service, through thickness tests are required on

plates over 10 [mm] thick in all the grades of steels listed in Table 9.2.1, apart from Grades 304L, 304LN, 321 and 347.

9.5.2 Testing is to conform with the requirements of Section 8, with the exception given in 9.5.3.

9.5.3 When the reduction in area is less than 35 per cent, metallographic or other evidence is required to show that no significant amount of any detrimental phase, such as sigma, is present.

Table 9.4.1 : Mechanical properties for acceptance purposes

Type and grade of steel	0.2% proof stress [N/mm^2] minimum	1% proof stress [N/mm^2] minimum	Tensile strength [N.mm^2] minimum	Elongation on 5.65 $\sqrt{S_o}$ % minimum
Austenitic				
304L	170	210	485	40
304LN	205	245	515	40
316L	170	210	485	40
316LN	205	245	515	40
317L	205	245	515	40
317LN	240	280	550	40
321	205	245	515	40
347	205	245	515	40
Duplex				
UNS S 31803	450	-	620	25
UNS S 32750	550	-	795	15

9.6 Intergranular corrosion tests

9.6.1 For certain specific applications such as storage tanks for chemicals, it may be necessary to demonstrate that the material used is not susceptible to intergranular corrosion resulting from grain boundary precipitation of chromium-rich carbides.

9.6.2 When required, one test of this type is to be carried out for each tensile test. The testing is to be carried out in accordance with ASTM A262, practice E, copper-copper sulphate-sulphuric acid or another recognized standard. The bent specimen is to be free from cracks indicating the presence of intergranular attack. The material for the test is to be taken adjacent to that for the tensile test.

9.7 Dimensional tolerances

9.7.1 The minimum tolerance on thickness is to be as given in Table 1.4.1.

9.8 Clad plates

9.8.1 Carbon or carbon-manganese steel plates, clad on one or both surfaces with a suitable grade of austenitic or duplex stainless steel, may be used for the construction of cargo or storage tanks for chemicals.

9.8.2 The carbon or carbon-manganese steel base plates are to comply with the requirements of Section 6 and the austenitic or duplex cladding material generally with the requirements of this section.

9.8.3 The process of manufacture is to be specially approved and may be either by roll cladding or by explosive bonding.

9.8.4 Where the use of clad materials is proposed, the material specification is to be submitted for consideration, together with details of the extent and the acceptance standards for non-destructive examination.

9.9 Identification of materials

9.9.1 The particulars detailed in 1.12 are to be marked on all materials which have been accepted.

9.10 Certification of materials

9.10.1 Each test certified or shipping statement is to give the information detailed in 1.13, together with general details of heat treatment and where applicable, the results obtained from intercrystalline corrosion tests. The chemical composition is to include the content of all the elements detailed in Table 9.2.1.

Section 10

Brittle Crack Arrest Steels

10.1 General

10.1.1 Brittle crack arrest steels are defined as steel plate with the specified brittle crack arrest properties measured by either the brittle crack arrest toughness Kca or Crack Arrest Temperature (CAT).

10.1.2 In addition to the required mechanical properties indicated in Section 3 for EH36, EH40 and EH47 steels, brittle crack arrest steels are to comply with the requirements specified in Table 10.1.3 and Table 10.1.4 of this Section.

10.1.3 The brittle crack arrest properties specified in Table 10.1.3 are to be evaluated and approved by IRS. Test specimens are to be taken from each piece (means "the rolled product from a single slab or ingot if this is rolled directly into plates" as defined in Chapter 1, Section 1.9.1).

10.1.4 The chemical composition and deoxidation practice for brittle crack arrest steels is indicated in Table 10.1.4.

Table 10.1.3 : Requirement of brittle crack arrest properties for brittle crack arrest steels

Suffix to the steel grade[1)]	Thickness range [mm]	Brittle crack arrest properties[2), 6)]	
		Brittle Crack Arrest Toughness K_{ca} at -10 °C [N/mm$^{3/2}$] [3)]	Crack Arrest Temperature CAT [°C] [4)]
BCA1	50 < t ≤ 100	6,000 min.	-10 or below
BCA2	80 < t ≤ 100[(7)]	8,000 min.	See Note 5.

t : thickness [mm]

Notes:

1) Suffix "BCA1" or "BCA2" is to be affixed to the steel grade designation (for e.g. EH40-BCA1, EH47-BCA1, EH47-BCA2, etc.).

2) Brittle crack arrest properties for brittle crack arrest steels are to be verified by either the brittle crack arrest toughness Kca or Crack Arrest Temperature (CAT).

3) Kca value is to be obtained by the brittle crack arrest test specified in Chapter 2, Section 5.

4) CAT is to be obtained by the test method specified in Chapter 2, Section 6.

5) Criterion of CAT for brittle crack arrest steels corresponding to Kca=8,000 N/mm$^{3/2}$ is to be approved by IRS.

6) Where small-scale tests are used for product testing (batch release testing), these test methods are to be approved by IRS in accordance with Annex 1.

7) Approval for lower thicknesses would be specially considered.

Table 10.1.4 : Chemical composition and deoxidation practice for brittle crack arrest steels

Grade	**EH36-BCA**	**EH40-BCA**	**EH47-BCA**
Deoxidation Practice	Killed and fine grain treated		
Chemical Composition % [1), 7)] (ladle samples)			
C max.	0.18		0.18
Mn	0.90 – 2.00		0.90 – 2.00
Si max.	0.50		0.55
P max.	0.020		0.020
S max.	0.020		0.020
Al (acid soluble min)	0.015 [2), 3)]		0.015 [2), 3)]
Nb	0.02 – 0.05 [3), 4)]		0.02 – 0.05 [3), 4)]
V	0.05 – 0.10 [3), 4)]		0.05 – 0.10 [3), 4)]
Ti max.	0.02[(4)]		0.02 [4)]
Cu max.	0.50		0.50
Cr max.	0.25		0.50
Ni max.	2.0		2.0
Mo max.	0.08		0.08
C_{eq} max.[5)]	0.47	0.49	0.55
P_{cm} max.[6)]	-		0.24

Notes :

1. Chemical composition of **brittle crack arrest steels** is to comply with this Table, regardless of chemical composition for EH36, EH40 and EH47 steels specified in Section 3.

2. The total aluminium content may be determined instead of the acid soluble content. In such cases the total aluminium content is to be not less than 0.020%.

3. The steel is to contain aluminium, niobium, vanadium or other suitable grain refining elements, either singly or in any combination. When used singly the steel is to contain the specified minimum content of the grain refining element. When used in combination, the specified minimum content of a fine graining element is not applicable.

4. The total niobium, vanadium and titanium content is not to exceed 0.12%.

5. The carbon equivalent C_{eq} value is to be calculated from the ladle analysis using the following formula:

$$C_{eq} = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}(\%)$$

6. Cold cracking susceptibility P_{cm} value is to be calculated using the following formula:

$$P_{cm} = C + \frac{Si}{30} + \frac{Mn}{20} + \frac{Cu}{20} + \frac{Ni}{60} + \frac{Cr}{20} + \frac{Mo}{15} + \frac{V}{10} + 5B(\%)$$

7. Where additions of any other element have been made as part of the steelmaking practice, the content is to be indicated.

10.2 Manufacturing Approval Scheme

10.2.1 Scope

The Manufacturing approval scheme for brittle crack steels is to be in accordance with Part 2 of the IRS Classification Note: '*Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'* and requirements of this sub-section.

10.2.2 Approval application

10.2.2.1 Documents to be submitted

The manufacturer is to submit to IRS, the following documents together with those required in Cl. 1.1 of Part 2 of the IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'*:

a) In-house test reports of the brittle crack arrest properties of the steels intended for approval
b) Approval test program for the brittle crack arrest properties (see 3.1 below)
c) Production test procedure for the brittle crack arrest properties.

10.2.3 Approval tests

10.2.3.1 Extent of Approval Tests

.1 The extent of the test program is specified in 10.2.3.2 and 10.2.3.3. Cl. 2.1 of Part 2 of the IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'* is to be complied with for the extent of the approval tests, if the manufacturing process and mechanism to ensure the brittle crack arrest properties for the steels intended for approval are the same. For EH47 steels with brittle crack arrest properties, Cl. 2.1 c) and d) of Part 2 the IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'* are not applied.

.2 The products for testing are to represent the maximum thickness for approval. If the target chemical composition changes with the thickness, the maximum thickness for each specified chemical composition specification is to be tested.

.3 The number of test samples and test specimens may be increased when deemed necessary by IRS, based on the in-house test reports of the brittle crack arrest properties of the steels intended for approval specified in 10.2.2.1 a).

10.2.3.2 Type of tests

.1 Brittle crack arrest tests are to be carried out in accordance with 10.2.3.3, in addition to the approval tests specified in Part 2 of the IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'* and/ or Cl. 3.15.2 of Section 3.

.2 In the case of applying for addition of the specified brittle crack arrest properties for EH36, EH40 and EH47 steels of which, manufacturing process has been approved by IRS (i.e. The aim analyses and method of manufacture are similar and the steelmaking process, deoxidation and fine grain practice, casting method and condition of supply are the same), brittle crack arrest tests, chemical analyses, tensile test and Charpy V-notch impact test are to be carried out in accordance with this sub-section and Part 2 of the IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'*:

10.2.3.3 Testing procedure and test specimens for brittle crack arrest tests

.1 The test specimens of the brittle crack arrest tests are to be taken with their longitudinal axis parallel to the final rolling direction of the test plates.

.2 The loading direction of brittle crack tests is to be parallel to the final rolling direction of the test plates.

.3 The thickness of the test specimens of the brittle crack arrest tests is to be the full thickness of the test plates.

.4 The test specimens and repeat test specimens are to be taken from the same steel plate. Where the brittle crack arrest properties are evaluated by Kca, and the brittle crack arrest test result fails to meet the requirement, further brittle crack arrest tests may be carried out. In this case, the judgment of acceptance is to be made on the arrest toughness value Kca of all test specimens (results of the initial test, failed tests and additional tests are to be included in the testing report.).

.5 The thickness of the test specimen is to be the maximum thickness of the steel plate requested for approval.

.6 In the case where the brittle crack arrest properties are evaluated by K_{ca}, the brittle crack arrest test method is to be in accordance with Chapter 2, Section 5. In the case where the brittle crack arrest properties are evaluated by CAT, the test method is to be in accordance with Chapter 2, Section 6.

10.2.4 Results

10.2.4.1 Part 2, of the IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'* is to be complied for the results.

10.2.4.2 Additionally, results of test items and the procedures are to comply with the test program approved by IRS. In the case where the brittle crack arrest properties are evaluated by K_{ca} or CAT, the manufacturer is to submit to IRS, the brittle crack arrest test reports in accordance with Chapter 2, Section 5 for Kca and Chapter 2 Section 6 for CAT respectively.

10.2.5 Approval and Certification

10.2.5.1 Upon satisfactory completion of the survey and tests, approval would be granted by IRS with the grade designation having the suffix "BCA1" or "BCA2" (e.g. EH40-BCA1, EH47-BCA1, EH47-BCA2, etc.).

10.2.5.2 General conditions for approval and certification would be in accordance with IRS Classification Note: *'Approval Scheme for the Manufacturing Process of Normal and Higher Strength Hull Structural Steels'.*

10.2.6 Renewal of approval

10.2.6.1 The manufacturer is also to submit to IRS actual manufacturing records of the approved brittle crack arrest steels within the term of validity of the manufacturing approval certificate.

Note: Chemical composition, mechanical properties, brittle crack arrest properties (e.g. brittle crack arrest test results or small-scale test results) and nominal thickness are to be described in the form of histogram or statistics

Annex 1

Approval Scheme of Small-scale Test Methods for Brittle Crack Arrest Steels

A1.1. Scope

A1.1.1 This Annex specifies the approval scheme of small-scale test methods which are used for product testing (batch release testing) of brittle crack arrest steels specified as Table 10.1.3 .

A1.1.2 Unless otherwise specified in this Annex, 3.15 and/or 10.2 of this Chapter are to be followed.

A1.2. Approval Application

A1.2.1 The manufacturer is to submit the following documents:

a) Application for approval of small-scale test procedure specification

b) Small-scale test procedure specification including the following items at least the following items:

- Applicable material grades, thickness range, deoxidation practice, heat treatment, etc.
- Types and methods of small-scale tests
- Sampling positions in plate thickness direction and final rolling direction of test specimens
- Size and dimension of test specimens
- Number of test specimens
- Test conditions, such as test temperature
- Acceptance criterion
- Example of format of test report
- Example of product inspection certificate including small-scale test results
- Handling of the products when small-scale test results do not satisfy the criterion

c) Mechanism of achieving the brittle crack arrest properties of brittle crack arrest steels

d) Technical background for enabling the evaluation of brittle crack arrest properties by small-scale test methods considering the mechanism specified in above c).

e) Procedure of the evaluation for the brittle crack arrest properties of brittle crack arrest steels by small-scale test results.

f) Data records which validate the correlation between small-scale test results and the large brittle crack arrest test results of brittle crack arrest steels whose number can satisfy the requirement for minimum data number given in A1.3.3

g) Proposed test plan for approval

A1.2.2 Small-scale test procedure specification is to be prepared in accordance with A1.3.

A1.2.3 Where the manufacturer proposes to change any part of the approved small-scale test procedure specification, then the manufacturer is to submit the documents which can cover all items specified in A1.2.1.

A1.2.4 The documents confirming the reason for the change is to be submitted to identify the impact of those changes on the existing procedure, and the proposed actions to address any such impacts.

A1.3. Establishment of Procedure Specification for Small-scale Testing

A1.3.1 General

A1.3.1.1 Small-scale test methods are to be determined based on the manufacturer's own technical philosophy with regard to achieving the brittle crack arrest properties of brittle crack arrest steels. Furthermore, description of an appropriate correlation between large scale brittle crack arrest properties and small-scale test results is to be required, and the acceptance criterion of the small-scale test are to be determined, based on the followings:

- Mechanism of achieving the suitable brittle crack arrest properties
- Sampling position and direction
- Frequency of sampling
- Small-scale test methodology
- Demonstrated correlation between brittle crack arrest test results and small-scale test results
- Derivation of small scale testing acceptance criterion based on the statistical analysis

A1.3.1.2 The manufacturer is to prepare the small-scale test procedure specification in accordance with the following A1.3.2 through A1.3.5.

A1.3.2 Types and Methods of Testing

A1.3.2.1 Types, methods, dimension and positions as well as direction of test specimens, etc. of small-scale tests are to be specified by the manufacturer, and approved.

A1.3.2.2 In general, the test method is to reproduce the crack initiation, propagation and arrest feature by such as the following test method.

- Combination of test methods, e.g. NRL drop weight test and V-notch Charpy impact test

- One test method, e.g. press-notch Charpy impact test or side-section drop weight test

A1.3.2.3 In general, brittle crack arrest properties of brittle crack arrest steels are to be predicted using a regression equation on the relationship between small scale test result (e.g. transition temperature obtained by small scale tests) and large scale brittle crack arrest test result (e.g. K_{ca} or temperature corresponding to the specific brittle crack arrest properties).

Note: Table A1.3.2.3.1, Table A1.3.2.3.2 and Table A1.3.2.3.3 give the examples of small scale test methods.

A1.3.2.4 For determination of test methods, the manufacturer is to confirm the applicability of these test methods to their brittle crack arrest steels theoretically taking into account the methodology of test methods, their own mechanism of achieving the brittle crack arrest properties, and sampling positions of test specimens (See A1.3.1.1). Then, the manufacturer is also to submit the technical background for determination of small-scale test methods as given in A1.2.1.

A1.3.3 Testing Data

A1.3.3.1 Selection of test plates

A1.3.3.1.1 Brittle crack arrest tests and small-scale tests are to be conducted for each material grade (including all suffixes) of brittle crack arrest steels in accordance with A1.3.3.

A1.3.3.1.2 Brittle crack arrest tests and small-scale tests are to be carried out on at least 12 test plates, in accordance with A1.3.3.1.3, by which these test results can reliably estimate brittle crack arrest properties of brittle crack arrest steels.

Note: "*One test plate*" means "*the rolled product from a single slab or ingot if this is rolled directly into plates*" as defined in Section 2 and 3.

A1.3.3.1.3 In order to ensure appropriate correlation between small-scale test results and brittle crack arrest properties with various manufacturing conditions of steel plates, the steel plates are to be representative for each combination of thickness range and heat sample to include:

- The intended maximum and minimum plate thickness;
- Different heats are to be chosen for each thickness.

Furthermore, the above test plates are to include a fixed number of steel plate(s) whose brittle crack arrest properties (i.e. brittle crack arrest test results) do not comply with the requirements specified in Table 10.1.3. Such a number is to be at least one, but not exceeding one quarter of all test plates. Manufacturing process of these test plates can be different (or intentionally altered from the approved manufacturing process) from that of the brittle crack arrest steels to which the small-scale test method is applied. It is recommended that the strength grade of these test plates (non-compliant with the relevant requirements of brittle crack arrest properties) are similar to that of the brittle crack arrest steels.

Where the manufacturer has requested approval for only a single thickness, the thickness of test plates can be only a single thickness. In this case, at least four steel plates for each combination of thickness (single thickness) and heats (three different heats) is to be used, and the applicable thickness of the small scale test is only that single thickness condition.

A1.3.3.1.4 Brittle crack arrest steels used for the approval test of manufacturing process of these steels (and its approval test results) can also be used as the test plates specified in A1.3.3.1.3

A1.3.3.1.5 Brittle crack arrest test specimens and small-scale test specimens are to be taken from the same test plate.

A1.3.3.1.6 A decrease of the total of the indicated number of test plates may be accepted in the following (a) or (b) cases:

(a) When the manufacturer applies a small-scale test procedure specification to multiple material grades, and the manufacturing process and mechanism to ensure the brittle crack arrest properties of these different material grades are the same.
(b) When a small-scale test procedure specification is already approved by IRS for one or some material grades, and the manufacturer applies similar small-scale test procedure specification to the other material grade(s), and the manufacturing process and mechanism to ensure the brittle crack arrest properties of these different material grades are same.

A1.3.3.2 Brittle crack arrest tests

A1.3.3.2.1 Brittle crack arrest tests are to be carried out for each test plate in accordance with 10.2.3.3.

A1.3.3.2.2 Where brittle crack arrest tests are carried out for evaluation of K_{ca}, K_{ca} at a specific temperature is to be obtained in accordance with Ch.2, Sec.5.3.

A1.3.3.2.3 Where brittle crack arrest tests are carried out for evaluation of CAT, deterministic (actual) CAT is to be obtained in accordance with Ch.2, Sec.6.8.3.

A1.3.3.3 Small-scale tests

A1.3.3.3.1 Small-scale tests are to be carried out in accordance with small-scale test procedure specification to be approved for each test plate.

A1.3.3.3.2 In general, the test specimens of small-scale tests are to be taken with their

longitudinal axis parallel to the final rolling direction of the test plates.

A1.3.3.3.3 The test specimens of small-scale tests are to be taken from the specified positions in plate thickness direction of the test plates, as given in A1.3.2.3.

A1.3.4 Validation of Correlation

A1.3.4.1 A regression equation on the relationship between brittle crack arrest property obtained from brittle crack arrest test and single or multiple small-scale test results is to be established. For brittle crack arrest properties, a specific temperature (e.g. $T_{Kca6000}$ in BCA1, $T_{Kca8000}$ in BCA2 or CAT) or the K_{ca} value at -10℃ may be used.

A1.3.4.2 The validity of the regression equation is to be examined to predict brittle crack arrest properties with enough accuracy. The correlation in brittle crack arrest properties between the calculated values from small scale tests and the brittle crack arrest test results is to be assured by using the value of twice the standard deviation (2σ). When using temperature for brittle crack arrest property, 2σ is not to be greater than 20℃. In other cases (e.g. K_{ca} value at -10℃), an upper limit of 2σ is to be established with the agreement of IRS.

NOTE:
Calculation procedure of the standard deviation (σ) is given as follows:

$$\sigma = \sqrt{\frac{1}{(n-1)}\sum_{i=1}^{n}(y_i - x_i)^2}$$

n: number of test plates
y_i: brittle crack arrest property obtained from brittle crack arrest test for one test plate
x_i: brittle crack arrest property estimated from small scale tests for one test plate

A1.3.5 Acceptance Criterion

A1.3.5.1 Acceptance criterion of brittle crack arrest steels by the small-scale tests is to be proposed by the manufacturer based on the regression equation which is assured in the correlation with brittle crack arrest properties in A1.3.4 above. The criterion is to be determined so that regression equation can predict brittle crack arrest properties on safety side, considering the scatter of brittle crack arrest properties from the predicted value by the regression equation.

A1.3.5.2 Unless otherwise agreed by IRS, an acceptance criterion of small-scale tests is to be determined by following procedures:

(a) For correlation by means of temperature

(i) The required temperature (see Fig A1.3.5.2 a) is obtained by subtracting 2σ (℃) from the brittle crack arrest steel specification in Table 10.1.3 , that is -10-2σ (℃), where 2σ is given in A1.3.4.2.
$T_{Kca6000}$ and $T_{Kca8000}$ in Fig. A5-1 are the temperatures at which the K_{ca} value of steel plates equals 6000N/mm$^{3/2}$ and 8000N/mm$^{3/2}$, respectively.

(ii) The temperature predicted from the small-scale test results through the regression equation is to be no higher than the value of -10-2σ(℃).

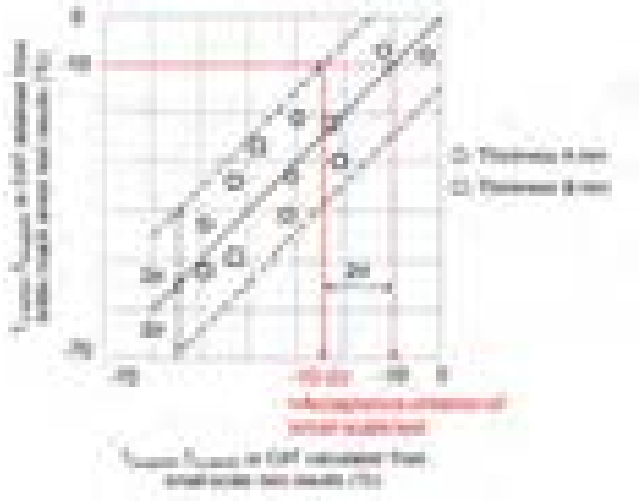


Fig. A1.3.5.2 a): Example for determination of acceptance criterion of small-scale test for correlation by means of temperature (Note: This is only a schematic and may not represent the actual data obtained)

(b) For correlation by means of brittle crack arrest toughness (K_{ca}):

(i) The required K_{ca} (see Fig. A1.3.5.2 b) is obtained by adding 2σ (N/mm$^{3/2}$) to the brittle crack arrest steel specification in Table 10.1.3, that is either 6000+2σ(N/mm$^{3/2}$) in BCA1 or 8000+2σ(N/mm$^{3/2}$) in BCA2, where 2σ is given in A1.3.4.2.

(ii) The K_{ca} value predicted from the small-scale test results through the regression equation is to be no smaller than the value

of $6000+2\sigma(N/mm^{3/2})$ for BCA1, or $8000+2\sigma(N/mm^{3/2})$ for BCA2.

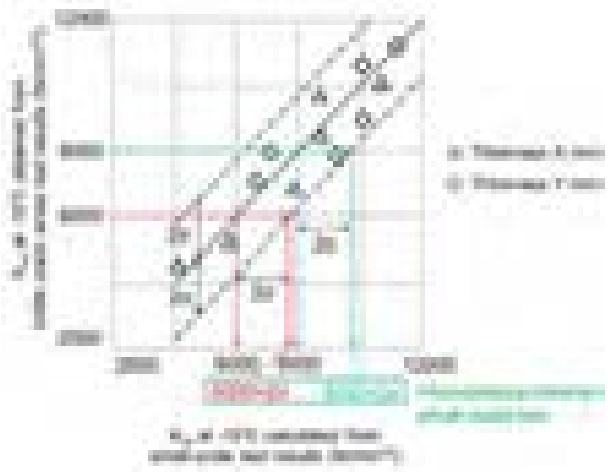


Fig. A1.3.5.2 b) Example for determination of acceptance criteria of small-scale test for correlation by means of brittle crack arrest toughness (K_{ca}) (Note: This is only a schematic and may not represent the actual data obtained)

A1.4. Approval Tests

A1.4.1 General

A1.4.1.1 In order to confirm the validity of the submitted technical documents specified in A1.2.1, approval tests are to be carried out.

A1.4.1.2 Approval test plan is to be approved by IRS prior to testing.

A1.4.1.3 Considering the contents of the submitted technical documents specified in A1.2.1, IRS may require additional tests in the following cases:

a) When IRS determines that the number of brittle crack arrest tests or small-scale tests is too few to adequately confirm the validity of the acceptance criterion of small-scale tests (See A1.3.3.1);

b) When IRS determines that the testing data obtained for setting the acceptance criterion of small-scale tests varies too widely (See A1.3.4.2), or that the data is clustered producing a biased correlation curve;

c) When IRS determines that the validity of brittle crack arrest test results or small-scale test results for setting the acceptance criterion of small-scale tests is insufficient, or has some flaws during tests and/or for test results (See A1.3.3.2 and A1.3.3.3); and

d) Others as deemed necessary by IRS.

A1.4.2 Extent of the approval tests

A1.4.2.1 Extent of the approval tests is to be in accordance with 3.15.2.1 and 10.2.3.1.

A1.4.3 Type of tests

A1.4.3.1 Brittle crack arrest tests

A1.4.3.1.1 Brittle crack arrest tests are to be carried out in accordance with 10.2.3.3.

A1.4.3.1.2 Where brittle crack arrest tests are carried out for evaluation of K_{ca}, K_{ca} at a specific temperature ($T_{Kca6000}$ or $T_{Kca8000}$) is to be obtained in accordance with Ch.2, 5.3.

A1.4.3.1.3 Where brittle crack arrest tests are carried out for evaluation of CAT, deterministic CAT is to be obtained in accordance with Ch.2, 6.8.3.

A1.4.3.2 Small-scale tests

A1.4.3.2.1 Small-scale tests are to be carried out in accordance with A1.3.3.3.

A1.5. Results

A1.5.1 Results of test items and the procedures are to comply with the test program approved by IRS.

A1.5.2 For the brittle crack arrest test results, the manufacturer is to submit to IRS the brittle crack arrest test reports in accordance with Ch.2, Sec.5 for K_{ca} and Ch.2, Sec.6 for CAT.

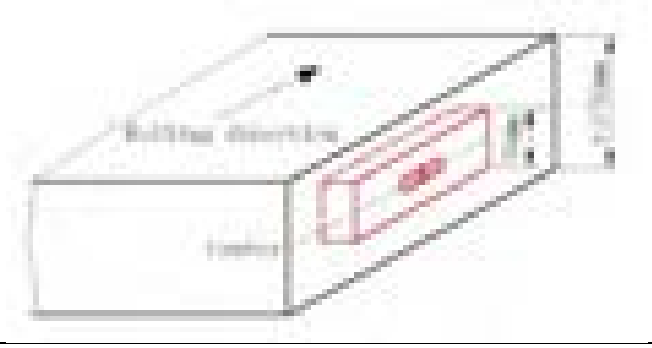
A1.5.3 For small-scale test results, the manufacturer is to submit to IRS the small-scale test reports in accordance with the example of format of test reports submitted as specified in A1.2.1 b).

A1.6. Approval

Upon satisfactory completion of the survey and tests, and satisfactory confirmation of the submitted technical documents, the approval for small scale test procedure specification is granted.

Table A1.3.2.3.1 : Example of small-scale test method using NRL drop weight test and V-notch Charpy impact test (Informative)	
Test type:	NRL drop weight test and V-notch Charpy impact test
Standard:	ASTM E208:2020 and ISO 148-1:2016
Sampling positions of test specimens:	NRL drop weight test: at surface V-notch charpy impact test: 1/4 of thickness
Length direction of test specimen:	Parallel to the final rolling direction of test plate
Regression equation:	$T_{Kca} = \alpha \cdot (NDTT + 10) + \beta \cdot {}_{v}Trs + 153(t-5)^{1/13} - 170.5$ T_{Kca}: Temperature at K_{ca} of 6,000N/mm$^{3/2}$ or K_{ca} of 8,000N/mm$^{3/2}$, (℃) *NDTT*: Nil-ductility transition temperature (℃) ${}_{v}T_{rs}$: Transition temperature of the absorbed energy (℃) *t*: thickness α, β[(1)]: constant
Notes:	(1) α and β are determined by comparing small-scale test results with brittle crack arrest test results.

Table A1.3.2.3.2 : Example of small-scale test method using pressed-notch Charpy impact test (Informative)	
Test type:	Pressed-notch Charpy impact test
Standard:	Dimension, shape, introducing method of notch: Manufacturer's proposal Others: ISO148-1:2016
Sampling position of test specimen:	1/2 of thickness
Length direction of test specimen:	Parallel to the final rolling direction of test plate
Regression equation:	$T_{Kca} = \alpha\, {}_{p}T_{E\gamma J} + \beta$ T_{Kca}: Temperature at K_{ca} of 6,000N/mm$^{3/2}$ or K_{ca} of 8,000N/mm$^{3/2}$, (℃) ${}_{p}T_{E\gamma J}$: Test temperature at absorbed energy of γ (J), (℃) α and β: Constant γ: Absorbed energy at brittle fracture surface ratio of δ (%),(J)
Notes:	(1) α, β, γ and δ are determined by comparing small-scale test results with brittle crack arrest test results.

Table A1.3.2.3.3 : Example of small-scale test method using Side-section drop weight test (Informative)	
Test type:	Side-section drop weight test
Standard:	Dimension: P-2 type of ASTM E 208 2020
Sampling positions of test specimens:	1/2 of thickness and side-section 
Length direction of test specimen:	Parallel to the final rolling direction of test plate
Regression equation :	$T_{Kca} = \alpha + \beta \cdot T_{NDT}^{side} + \gamma \cdot t^{1.5}$ T_{Kca}: Temperature at K_{ca} of 6000N/mm$^{3/2}$ or K_{ca} of 8000N/mm$^{3/2}$, (℃) T_{NDT}^{side} : Nil-ductility transition temperature obtained by side-section drop weight test, (℃) t : thickness α, β,γ(1): constant
Notes:	(1) α, β and γ are to be determined by comparing small-scale test results with brittle crack arrest test results.

End of Chapter

Chapter 4

Steel Castings

Contents

Section	
1	*General Requirements*
2	*Hull and Machinery Steel Castings for General Applications*
3	*Ferritic Steel Castings for Low Temperature Services*
4	*Steel Castings for Propellers*
5	*Austenitic Stainless Steel Castings*
6	*Castings for other applications*

Section 1

General Requirements

1.1 Scope

1.1.1 All important steel castings, as defined in the relevant construction rules are to be manufactured and tested in accordance with the requirements of this Chapter.

1.1.2 Where required by the relevant Rules dealing with design and construction, castings are to be manufactured and tested in accordance with Ch.1 and Ch.2, together with the general requirements given in this Section and the appropriate specific requirements given in Sec.2 to 5.

1.1.3 As an alternative to 1.1.3, castings which comply with national or proprietary specifications may be accepted provided that these specifications give reasonable equivalence to the requirements of this chapter or alternatively are approved for a specific application. Generally, survey and certification are to be carried out in accordance with the requirements of Ch.1.

1.2 Manufacture

1.2.1 Castings are to be made by manufacturer approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

1.2.2 The steel is to be manufactured by a process approved by IRS.

1.2.3 All flame cutting, scarfing or arc-air gouging to remove surplus metal is to be undertaken in accordance with recognized good practice and is to be carried out before the final heat-treatment. Preheating is to be employed when necessitated by the chemical composition and/or thickness of the castings. If necessary, the affected areas are to be either machined or ground smooth.

1.2.4 For certain components including steel castings subjected to surface hardening process, the proposed method of manufacture may require special approval by IRS.

1.2.5 When two or more castings are joined by welding to form a composite the proposed welding procedure is to be submitted for approval. Welding procedure qualification tests may be required.

1.3 Quality of castings

1.3.1 All castings are to be free from surface or internal defects which would be prejudicial to their proper application in service. The surface finish is to be in accordance with good practice and any specific requirements of the approved plan.

1.3.2 The surfaces are not to be treated in any way which may obscure defects.

1.4 Chemical composition

1.4.1 All castings are to be made from killed steel and the chemical composition is to be appropriate for the type of steel and the mechanical properties specified for the castings. The chemical composition of each heat is to be determined by the manufacturer on a sample taken preferably during the pouring to the heat. When multiple heats are tapped into a common ladle, the ladle analysis shall apply.

1.5 Inspection

1.5.1 All castings are to be cleaned and adequately prepared for examination; suitable methods include pickling, caustic cleaning, wire brushing, local grinding, shot or sand blasting. The surfaces are not to be hammered, peened or treated in any way which may obscure defects.

1.5.2 Before acceptance all castings are to be presented to the Surveyors for visual examination. Where applicable, this is to include the examination of internal surfaces. Unless otherwise agreed, the verification of dimensions is the responsibility of the manufacturer.

1.5.3 When required by the relevant construction Rules, or by the approved procedure for welded composite components appropriate non-destructive testing is also to be carried out before acceptance and the results are to be reported by the manufacturer. The castings to be examined, extent of testing and acceptance criteria are detailed in IRS Classification Notes on “Guidelines for Non-Destructive Testing of Steel Castings for Marine Application.

1.5.4 When required by the relevant construction Rules castings are to be pressure tested before final acceptance. These tests are to be carried out in the presence of the Surveyors and are to be to their satisfaction.

1.5.5 In the event of any casting proving defective during subsequent machining or testing, it is to be rejected notwithstanding any previous certification.

1.6 Hydraulic pressure testing

1.6.1 When required by the relevant construction Rules, castings are to be pressure tested before final acceptance. These tests are to be carried out in the presence of the Surveyors and are to be to their satisfaction.

1.7 Rectification of defective castings

1.7.1 General

i) Steel casting defects are to be removed with or without weld repair before considering suitable for use subject to approval of IRS. Where castings are to be repaired, the manufacturer is to exercise adequate control of all activities with respect to dimensions, heat treatment, inspection and quality control (as considered necessary).

ii) Where the defective area is to be repaired by welding, the excavations are to be suitably shaped to allow good access for welding. The resulting grooves are to be subsequently ground smooth and complete elimination of the defective material is to be verified by MT or PT.

iii) Defects and unacceptable indications are to be repaired as indicated below:

Defective parts of material may be removed by grinding, or by chipping and grinding, or by arc air-gouging and grinding. Thermal methods of metal removal are only allowed before the final heat treatment. All grooves are to have a bottom radius of approximately three times the groove depth and are to be smoothly blended to the surface area with a finish equal to that of the adjacent surface.

iv) For NDT of steel castings after repair, see 1.5.3.

v) Shallow grooves or depressions resulting from the removal of defects may be accepted provided that they will cause no appreciable reduction in the strength of the casting or affect the intended use, and the depth of defect removal is not over 15mm or 10% of wall thickness, whichever is less. The resulting grooves or depressions are to be subsequently ground smooth and complete elimination of the defective material is to be verified by MT or PT. Small surface irregularities sealed by welding are to be treated as weld repairs, see 1.7.2.

1.7.2 Weld repairs

In addition to the requirements given in 1.7.1, the following apply for weld repairs:

i) For C and C-Mn steel castings, weld repairs are to be suitably classified as major or minor. For alloy steel castings, repair requires approval from IRS.

a. Major repairs are those where:

- the depth is greater than 25% of the wall thickness or 25mm whichever is less, or

- the total weld area on a casting exceeds 0.125 m^2 of the casting surface noting that where a distance between two welds is less than their average width, they are to be considered as one weld.

b. Minor weld repairs: Weld repairs not classified as major are considered as minor and need to be carried out in accordance with a qualified welding procedure.

c. All weld repairs (major and minor) are to be carried out before the final delivery heat treatment condition.

ii) The following is required for major repairs:

a. To comply with requirements in (iv) below

b. Before welding is started, full details of the extent and location of the repair, the proposed welding procedure, heat treatment and subsequent inspection procedures are to be submitted for approval.

(iii) The following is required for minor repairs:

a. With the exception of alloy steels, do not require prior approval by IRS, except as given in (b)

b. IRS may request minor repairs in critical areas to be treated as major repairs.

iv) The following requirements apply for all weld repairs (major and minor):

a. All castings in alloy steels and all castings for crankshafts are to be suitably pre-heated prior to welding. Castings in carbon or carbon-manganese steel may also require to be pre-heated depending on their chemical composition and the dimensions and position of the weld repairs.

b. Welding procedures are to be qualified and are to match the delivery condition of the casting. Qualification of welding procedures is subject to agreement with IRS or in accordance with a recognized standard such as ISO 11970:2016.

c. Welding is to be done under cover in positions free from draughts and adverse weather conditions by qualified welders with adequate supervision. As far as possible, all welding is to be carried out in the downhand (flat) position.

d. The welding consumables used are to be of an appropriate composition, giving a weld deposit with mechanical properties similar and in no way inferior to those of the parent castings. Welding procedure tests are to be carried out by the manufacturer to demonstrate that satisfactory mechanical properties can be obtained after heat treatment as detailed in Sec.2.

e. After welding has been completed the castings are to be given either a suitable heat treatment in accordance with the requirements of Sec.2 or a stress relieving heat treatment at a temperature of not less than 550°C for C and C-Mn steel castings. For alloy steel castings, the heat treatment has to be agreed with IRS. The type of heat treatment employed will be dependent on the chemical composition of the casting and the dimensions, positions and nature of the repairs and should not affect the properties of the casting.

Subject to the prior agreement of IRS special consideration may be given to the omission of postweld heat treatment or to the acceptance of local stress relieving heat treatment where the repaired area is small and machining of the casting has reached an advanced stage.

f. On completion of heat treatment the weld repairs and adjacent material are to be ground smooth and examined by magnetic particle or liquid penetrant testing. Supplementary examination by ultrasonics or radiography may also be repaired depending on the dimensions and nature of the original defect. Satisfactory results are to be obtained from all forms of non-destructive testing used.

g. The manufacturer is to maintain full records detailing the extent and location of repairs made to each casting and details of weld procedures and heat treatments applied for repairs. These records are to be available to the Surveyor and copies provided on request.

1.7.3 Recommendation for welding

For steels with C ≥ 0.23 or Ceq ≥ 0.45, the WPQT on which the WPS is based, is to be qualified on a base material having a Ceq as follows:

the Ceq of the base material is not to fall below more than 0.02 of the material to be welded. (Example: WPQT for a material with actual Ceq = 0.50 may be qualified on a material with Ceq ≥ 0.48.)

1.8 Identification of castings

1.8.1 The manufacturer is to adopt a system of identification which will enable all finished castings to be traced to the original cast and Surveyors are to be given full facilities for so tracing the castings when required.

1.8.2 Before acceptance, all castings which have been tested and inspected with satisfactory results are to be clearly marked by the manufacturer with the following particulars:

i) Steel quality.

ii) Identification number, cast number or other marking which will enable the full history of the casting to be traced.

iii) Manufacturer’s name or trade mark.

iv) The IRS brand name ‘IR’.

v) Abbreviated name of the IRS local office.

vi) Personal stamp of Surveyors responsible for inspection.

vii) Where applicable, test pressure.

1.8.3 When small castings are manufactured in large numbers, modified arrangements for identification may be specially agreed with IRS.

1.9 Certification

1.9.1 The manufacturer is to provide the Surveyor with a test certificate or shipping statement giving the following particulars for each casting or batch of castings which has been accepted:-

a) Purchaser's name and order number;

b) Description of castings and steel quality;

c) Identification number;

d) Steel making process, cast number and chemical analysis of ladle samples;

e) Results of mechanical testing;

f) General details of heat treatment;

g) Where applicable, test pressure.

Section 2

Hull and Machinery Steel Castings for General Applications

2.1 Scope

2.1.1 The requirements given in this section are applicable to C, C-Mn and alloy steel castings intended for hull and machinery applications specified in the relevant Chapters of the Rules. Grades that are intended for fabrication by welding, as well as grades not intended for welding are also considered in this Section.

2.1.2 Additional requirements may be necessary, especially when the castings are intended for service at low or elevated temperatures, e.g. for ships with ice-class or for boilers. Additional requirements will typically be required for castings for offshore units depending on applicable service temperature and environment.

2.1.3 Similarly, C and C-Mn steel castings and alloy steel castings which comply with national or proprietary specifications may be accepted provided such specifications give reasonable equivalence to these requirements or are otherwise specially approved or required by IRS.

2.2 Manufacture

2.2.1 Castings are to be made at a manufacturer approved by IRS.

2.2.2 The steel is to be manufactured by a process approved by IRS.

2.2.3 All flame cutting, scarfing or arc-air gouging to remove surplus metal is to be undertaken in accordance with recognized good practices and is to be carried out before the final heat treatment. Preheating is to be employed when necessitated by the chemical composition and/or thickness of the castings. If necessary, the affected areas are to be either machined or ground smooth.

2.2.4 For certain components including steel castings subjected to surface hardening process, the proposed method of manufacture may require special approval by IRS.

2.2.5 Joining of two or more castings by welding to form a composite component: Welding procedure qualification tests are subject to agreement with IRS. Welders for hull structural steel castings are to be qualified in accordance with IRS Classification Notes: *'Qualification Scheme for Welders of Hull Structural Steel'*. Requirements for other WPS and qualification thereof, for welder certification and for type approval of welding consumables are at the discretion of IRS.

2.2.6 Temporary welds made for operations such as lifting, handling, staging, etc., are to be in accordance with approved welding procedures and qualified welders, and are to be removed, ground, and inspected using suitable NDT methods.

2.3 Quality of Castings

2.3.1 All castings are to be free from surface or internal defects, which would be detrimental to their proper application in service. The surface finish is to be in accordance with good practice and any specific requirements of the approved plan.

2.4 Chemical composition

2.4.1 All castings are to be made from killed steel and the chemical composition is to be appropriate for the type of steel and the mechanical properties specified for the castings.

2.4.2 The chemical composition of each heat is to be determined by the manufacturer on a sample taken preferably during the pouring of the heat. When multiple heats are tapped into a common ladle, the ladle analysis shall apply.

2.4.3 The chemical composition is to comply with the overall limits given in Table 2.4.1 and Table 2.4.2, respectively, or, where applicable, the requirements of the approved specification.

2.4.4 Suitable grain refining elements such as aluminium may be used at the discretion of the manufacturer or as agreed with IRS.

Table 2.4.1 : Chemical composition limits for hull and machinery steel castings (%): C, C-Mn steels

Steel type	Applications	C (max.)	Si (max.)	Mn	S (max.)	P (max.)	Residual elements (max.)				Total residuals (max.)
							Cu	Cr	Ni	Mo	
C, C-Mn	Castings for non-welded construction	0.40	0.60	0.50 – 1.60	0.035	0.035	0.30	0.30	0.40	0.15	0.80
	Castings for welded construction	0.23	0.60	0.5- 1.60	0.035	0.035	0.30	0.30	0.40	0.15	0.80

Table 2.4.2 : Chemical composition limits for hull and machinery steel castings (%) : Alloy steels

Steel type	Applications	C (max.)	Si (max.)	Mn	S (max.)	P (max.)	Alloying elements[1)] (min.)			
							Cu	Cr	Ni	Mo
Alloy	Castings for non-welded construction	0.45	0.60	0.50 – 1.60	0.030	0.035	0.30	0.40	0.40	0,15
	Castings for welded construction	Alloying element values to be agreed with IRS								

1) At least one of the elements is to comply with the minimum content.

2.5 Heat treatment (including straightening)

2.5.1 Castings are to be supplied in one of the following delivery conditions:

(a) Carbon and carbon-manganese steels:
Fully annealed
Normalised
Normalised and tempered
Quenched and tempered

(b) Alloy steels:
Normalized
Normalized and tempered
Quenched and tempered

For all types of steel, the tempering temperature is to be not less than 550°C.

The delivery condition is to meet the design and application requirements. It is the manufacturers responsibility to select the appropriate heat treatment method to obtain the required mechanical properties.

2.5.2 Castings for component such as crankshafts and engine bedplates, where dimensional stability and freedom from internal stresses are important are to be given a stress relief heat treatment. This is to be carried out at a temperature of not less than 550°C followed by furnace cooling to 300°C or lower.

2.5.3 Heat treatment is to be carried out in properly constructed furnaces which are efficiently maintained and have adequate means for control and recording of temperature. The furnace dimensions are to be such as to allow the whole casting to be uniformly heated to the necessary temperature. In the case of very large castings alternative methods for heat treatment will be specially considered by IRS. Sufficient thermocouples are to be connected to the furnace charge to measure and record that its temperature is adequately uniform unless the temperature uniformity of the furnace is verified at regular intervals.

2.5.4 If a casting is locally reheated or any straightening operation is performed after the final heat treatment, a subsequent stress relieving heat treatment may be required in order to avoid the possibility of harmful residual stresses. The manufacturer is to have

strict control of this temperature in order to avoid any detrimental effects to the final heat treatment and resultant microstructure and mechanical properties of the casting.

2.5.5 The manufacturer's works is to maintain records of heat treatment identifying the furnace used, furnace charge, date, temperature and time at temperature. The records are to be presented to the Surveyor on request.

2.6 Mechanical tests

2.6.1 Test material, sufficient for the required tests and for possible retest purposes is to be provided for each casting or batch of castings.

2.6.2 At least one test block is to be provided for each casting. Unless otherwise agreed these test blocks are to be either integrally cast or gated to the castings.

2.6.3 The size of the test blocks for mechanical testing is to be such that the heat treatment and microstructure is representative for the section of the casting with the ruling section, i.e. the section for which the specified mechanical properties apply, (see also ISO 683-1:2018 and ISO 683-2:2018).

For C, C-Mn steel castings this is in general to be achieved as follows:

The test block is to have a thickness (t_S) of not less than the ruling section of the casting, or 30 mm, whichever is larger.

For large thickness castings other than stern tube, stern frame, anchor and rudder horn, t_S normally need not exceed 150 mm. Length and width of the test block is normally to be at least three times t_S, unless otherwise agreed with IRS, as shown in Fig. 2.6.1 (Note that longer or wider test blocks may be necessary to accommodate the required test specimens).

For castings for stern tube, stern frame, anchor and rudder horn the test block thickness t_S is to represent the ruling section.

Guidance:

Shorter width or length may be accepted for test blocks where actual casting width or length (t_A) is in the range between t_S and $3t_S$.

Example 1: For a general casting with dimensions 140 x 160 x 1250 mm the required test block size would typically be 140 x 160 x 420 mm (that is: t_S x t_A x $3t_S$).

Example 2: For a stern tube casting with ruling section t_S = 170 mm and width/height/length $t_{A2}/t_{A1}/t_{A3}$ = 1000/600/1800 mm, the required test block size would typically be 170 x 510 x 510 mm (i.e. t_S x $3t_S$ x $3t_S$) see Fig. 2.6.2.

For alloy steel castings, the manufacturer is to propose dimensions for the test block and demonstrate the representative nature of it.

2.6.4 For test blocks with thickness ≤ 56 mm, the longitudinal axis of the test specimens is to be located at ≥ 14 mm from the surface in the thickness direction. For test blocks with thickness > 56 mm, the longitudinal axis of the test specimens is to be located at ≥ ¼ t_S from the surface. Test specimens are to be taken in such a way that no part of the gauge length is machined from material closer than t_S to any of the other surfaces. For impact testing, this requirement applies to the complete test specimen (refer Fig. 2.6.1 for location of test specimens in relation to the test block).

2.6.5 Where the casting is of complex design or where the finished mass exceeds 10 tonnes, two cast on test blocks are to be provided from the heaviest section, located as far as practicable from each other.

2.6.6 Where large castings are made from two or more casts, which are not mixed in a ladle prior to pouring, two or more test blocks are to be provided corresponding, the number of the casts involved. These are to be integrally cast at locations as widely separated as possible.

2.6.7 For castings where the method of manufacture has been specially approved by IRS in accordance with 1.2.4, the number and position of test blocks is to be agreed with IRS having regard to the method of manufacture employed.

2.6.8 As an alternative to 2.6.2, where several small castings of about the same size, each of which is under 1000 [kg] in mass are made from one cast and heat treated in the same furnace charge, a batch testing procedure may be adopted using separately cast test blocks of suitable dimensions. At least one test block is to be provided for each batch of castings.

2.6.9 The test blocks are not to be detached from the casting until the specified heat

treatment has been completed and they have been properly identified.

2.6.10 One tensile test specimen and one set impact tests are to be taken from each test block.

2.6.11 The preparation of test specimens and the procedures used for mechanical testing are to comply with the relevant requirements of Ch.2. Unless otherwise agreed all tests are to be carried out in the presence of the Surveyors.

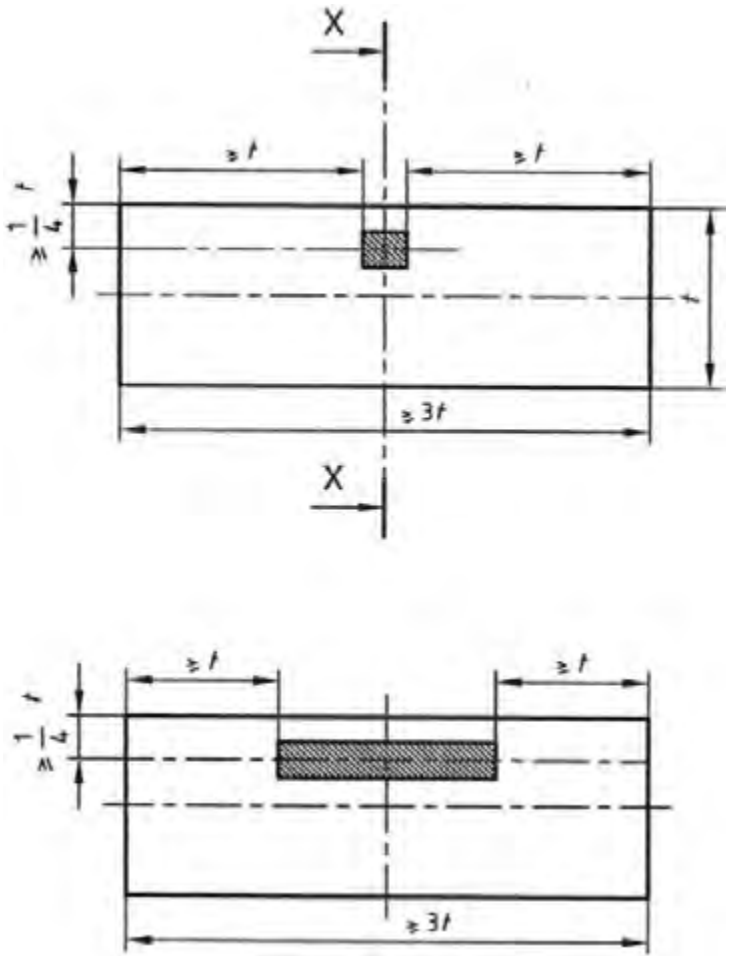


Fig. 2.6.1: Specimen positions relative to the test block in accordance with ISO 4990:2015*

***** Figure taken from ISO 4990:2015, *Steel castings - General technical delivery requirements*, is reproduced with the permission of the International Organization for Standardization, ISO. Copyright remains with ISO.

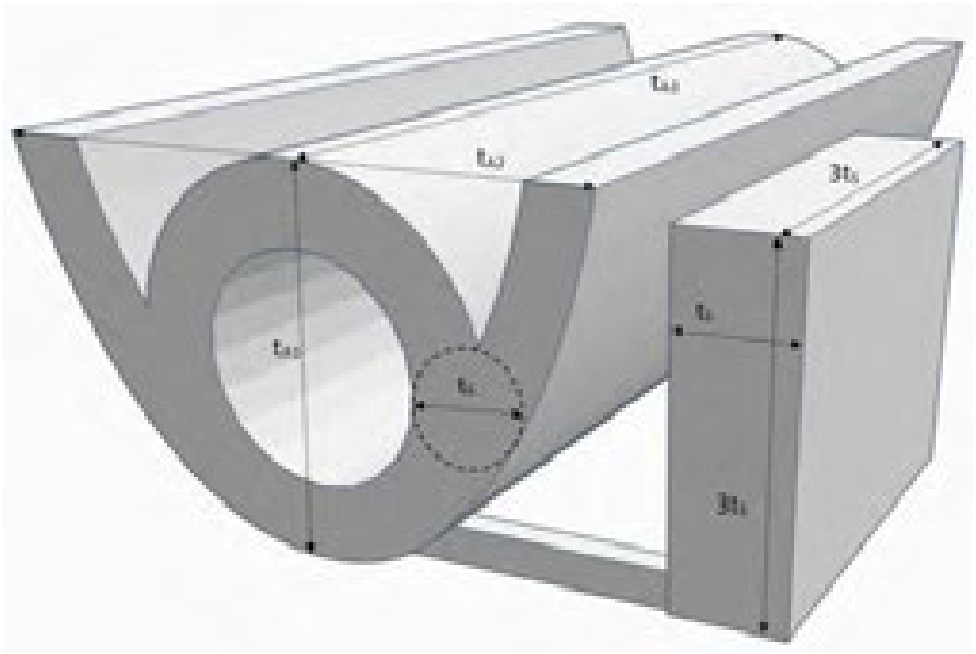


Fig. 2.6.2: Example 2: test block gated to stern tube casting

2.7 Mechanical properties

2.7.1 Table 2.7.1 and Table 2.7.2 give the minimum requirements for yield stress, elongation, reduction of area and impact test energy values corresponding to steel types and different strength levels. Where it is proposed to use a steel with a specified minimum tensile strength intermediate to those given, corresponding minimum values for the other properties may be obtained by interpolation.

2.7.2 Castings may be supplied to any specified minimum tensile strength selected within the general limits detailed in Table 2.7.1 and Table 2.7.2, respectively but subject to any additional requirements of the relevant construction rules.

2.7.3 The mechanical properties are to comply with the requirements of Table 2.7.1 and Table 2.7.2, respectively, appropriate to the specified minimum tensile strength or, where applicable, the requirements of the approved specification.

2.7.4 Re-test requirements for tensile tests are to be in accordance Chapter 2.

2.7.5 The additional tests detailed in 2.7.4 are to be taken, preferably from the same, but alternatively from another, test sample representative of the casting or batch of castings.

2.7.6 At the option of the manufacturer, when a casting or batch of castings has failed to meet the test requirements, it may be reheat treated and re-submitted for acceptance tests.

Table 2.7.1: Mechanical properties for steel castings intended for welding

Steel Type	Specified minimum tensile strength (1) [N/mm^2]	Yield stress [N/mm^2] min.	Elongation on 5.65 √So (%) min.	Reduction of area (%) min.	Charpy V-notch impact test2)	
					Test temperature(°C)	Minimum average energy(J)
C, C-Mn	400	200	25	40	0	27
	440	220	22	30		
	480	240	20	27		
	520	260	18	25		
	560	300	15	20		
	600	320	13	20		
Alloy	550	355	18	30	0	27
	600	400	16	30		
	650	450	14	30		
	700	540	12	28		

Note:
(1) A tensile strength range of 150 [N/mm^2] may additionally be specified.

(2) Special consideration may be given to alternative requirements for Charpy V-notch impact test, depending on design and application, and subject to agreement with IRS.

Table 2.7.2: Mechanical properties for steel castings not intended for welding

Steel Type	Specified minimum tensile strength (1) [N/mm^2]	Yield stress [N/mm^2] min.	Elongation on 5.65 √So (%) min.	Reduction of area (%) min.	Charpy V-notch impact test2)	
					Test temperature(°C)	Minimum average energy(J)
C, C-Mn	400	200	25	40	AT3)	27
	440	220	22	30		
	480	240	20	27		
	520	260	18	25		
	560	300	15	20		
	600	320	13	20		
Alloy	550	340	16	35	AT3)	27
	600	400	16	35		
	650	450	14	32		
	700	540	12	28		

Note:
(1) A tensile strength range of 150 [N/mm^2] may additionally be specified.

(2) Special consideration may be given to alternative requirements for Charpy V-notch impact test, depending on design and application, and subject to agreement with IRS.

3) AT refers to Ambient Temperature (i.e. 23°C±5°C), which is specified in ISO 148-1:2016

Section 3

Ferritic Steel Castings for Low Temperature Services

3.1 General

3.1.1 This Section gives the requirements for castings in carbon-manganese and nickel alloy steels intended for use in liquefied gas piping systems where the design temperature is lower than 0°C and for other applications where guaranteed impact properties at low temperatures is required.

3.1.2 Other steel types may also be accepted upon consideration in each case.

3.2 Chemical composition

3.2.1 The chemical composition of ladle samples is to comply with the overall limits given in Table 3.2.1. The carbon-manganese steel is to be fine grain treated.

Table 3.2.1 : Chemical composition of ferritic steel castings for low temperature service

Type of steel	Chemical composition %						
	C max.	Si max.	Mn	S max.	P max.	Ni	Residual elements max.
Carbon-manganese	0.25	0.60	0.70-1.60	0.030	0.030	0.80 max.	
$2\frac{1}{4}$ Ni	0.25	0.60	0.50-0.80	0.025	0.030	2.00-3.00	Cr 0.25 Cu 0.30 Mo 0.15 V 0.03 Total 0.60
$3\frac{1}{2}$ Ni	0.15	0.60	0.50-0.80	0.020	0.025	3.00-4.00	

3.3 Heat treatment

3.3.1 Castings are to be supplied in one of the following conditions :

a) normalized.
b) normalized and tempered.
c) quenched and tempered.

3.4 Mechanical tests

3.4.1 The mechanical properties of steel castings are to comply with requirements given in Table 3.4.1.

3.4.2 The tensile test is to be carried out at ambient temperature and the impact tests are to be carried out at the temperature specified in the table.

3.4.3 The average energy value from a set of three charpy V-notch impact test specimens is not to be lower than the required average value given in Table 3.4.1. One individual value may be less than the required average value provided that it is not less than 70 per cent of this average value.

3.5 Non-destructive testing

3.5.1 The non-destructive testing of castings is to be carried out in accordance with the appropriate requirements of 1.7 and additionally agreed between the manufacturer, purchaser and Surveyor.

Table 3.4.1 : Mechanical properties for acceptance purposes : ferritic steel castings for low temperature service

Type of steel	Grade	Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Elongation on 5.65√So% min.	Reduction of area % min.	Charpy V-notch impact test	
						Test temp.^{o}C	Average energy J min.
Carbon-manganese	400 430 450	200 215 230	400 - 550 430 - 580 460 - 610	25 23 22	40 35 30	-60 (see Note)	27
$2\frac{1}{4}$ Ni	490	275	490 - 640	20	35	-70	34
$3\frac{1}{2}$ Ni	490	275	490 - 640	20	35	-95	34

Note : The temperature for carbon-maganese steels may be 5^{o}C below the design temperature if the latter is above -55^{o}C, with a maximum test temperature of -20^{o}C.

Section 4

Steel Castings for Propellers

4.1 Scope

4.1.1 These requirements are applicable to the manufacture, inspection and repair procedures of cast steel propellers, blades and bosses.

4.1.2 Where the use of alternative alloys is proposed, particulars of chemical composition, mechanical properties and heat treatment are to be submitted for approval.

4.1.3 These requirements may also be used for the repair of propellers damaged in service, subject to prior approval of IRS.

4.2 Foundry Approval

4.2.1 All propellers, blades and bosses are to be manufactured by foundries approved in accordance with Ch.1. Also refer Chapter 1, Section 1, Cl. 1.3.2. The castings are to be manufactured and tested in accordance with the requirements of this Section.

4.2.2 It is the manufacturer's responsibility to assure that effective quality, process and production controls during manufacturing are adhered to within the manufacturing specification. The manufacturing specification is to be submitted to IRS at the time of initial approval, and is to at least include the following particulars:

a) description of the foundry facilities,

b) steel material specification,

c) runner and feeder arrangements,

d) manufacturing procedures,

e)non-destructive testing and repair procedures.

4.2.3 The scope of the approval test is to be agreed with IRS. This is to include the presentation of cast test coupons of the propeller materials in question for approval testing in order to verify that the chemical composition and the mechanical properties of these materials comply with this section.

4.2.4 The foundry is to have an adequately equipped laboratory, manned by experienced personnel, for the testing of moulding materials chemical analyses, mechanical testing, microstructural testing of metallic materials and non-destructive testing. Where testing activities are assigned to other companies or other laboratory, additional information required by IRS is to be included.

4.3 Quality of castings

4.3.1 Freedom from defects

4.3.1.1 All castings are to have a workmanlike finish and are to be free from imperfections and defects which would be prejudicial to their proper application in service. Minor casting defects which may still be visible after machining such as small sand and slag inclusions, small cold shuts and scabs are to be trimmed off by the manufacturer in accordance with 4.11.

4.3.2 Removal of defects

4.3.2.1 Casting defects which may impair the service performance of the castings, e.g. major non-metallic inclusions, shrinkage cavities, blow holes and cracks, are not permitted. They may be removed by one of the methods described in 4.11 and repaired within the limits and restrictions for the severity zones. Full description and documentation must be available for the surveyor.

Table 4.2.1 : Typical chemical composition for steel propeller castings

Alloy type	C max. (%)	Mn max. (%)	Cr (%)	Mo[1)] max. (%)	Ni (%)
Martensitic (12 Cr 1 Ni)	0.15	2.0	11.5 - 17.0	0.5	Max. 2.0
Martensitic (13 Cr 4 Ni)	0.06	2.0	11.5 - 17.0	1.0	3.5 - 5.0
Martensitic (16 Cr 5 Ni)	0.06	2.0	15.0 - 17.5	1.5	3.5 - 6.0
Austenitic (19 Cr 11 Ni)	0.12	1.6	16.0 - 21.0	4.0	8.0 - 13.0

Note 1) Minimum values are to be in accordance with recognised national or international standards

Table 4.2.2 : Mechanical properties for steel propeller castings

Alloy type	Proof stress $R_{p0.2}$ min. [N/mm^2]	Tensile strength R_m min. [N/mm^2]	Elongation A_5 min. (%)	Red. Of area Z min. (%)	Charpy V-notch[1)] Energy min. (J)
(12 Cr 1 Ni)	440	590	15	30	20
(13 Cr 4 Ni)	550	750	15	35	30
(16 Cr 5 Ni)	540	760	15	35	30
(19 Cr 11 Ni)	180[2)]	440	30	40	-

Notes:

1) Not required for general service and the lowest ice class notations. For other ice class notations, tests are to be made -10^{o}C.

2) $R_{p1.0}$ value is 205 [N/mm^2].

4.4 Dimensions, dimensional and geometrical tolerances

4.4.1 The verification of dimensions, the dimensional and geometrical tolerances is the responsibility of the manufacturer. The report on the relevant examinations is to be submitted to the Surveyor, who may require checks to be made in his presence.

4.4.2 Static balancing is to be carried out on all propellers in accordance with the approved drawing. Dynamic balancing may be necessary for propellers running above 500 rpm.

4.5 Chemical Composition

4.5.1 Typical cast steel propeller alloys are grouped into four types depending on their chemical composition as given in Table 4.2.1. Cast steel whose chemical composition deviate from the typical values of Table 4.2.1 must be specially approved by IRS.

4.5.2 The manufacturer is to maintain records of the chemical analyses of the production casts, which are to be made available to the Surveyor so that he can satisfy himself that the chemical composition of each casting is within the specified limits.

4.6 Heat treatment

4.6.1 Martensitic castings are to be austenitized and tempered. Austenitic castings should be solution treated.

4.7. Mechanical properties

4.7.1 The mechanical properties are to comply with values given in Table 4.2.2. These values refer to the test specimens machined from integrally cast test coupons attached to the hub or on the blade. The thickness of test coupon is to be in accordance with a recognized standard.

4.7.2 Where possible, the test coupons attached on blades are to be located in an area between 0.5 to 0.6R, where R is the radius of the propeller.

4.7.3 The test bars are not to be detached from the casting until the final heat treatment has been carried out. Removal is to be by non-thermal procedures.

4.7.4 Separately cast test bars may be used subject to prior approval of IRS. The test bars are to be cast from the same heat as the castings represented and heat treated with the castings represented.

4.7.5 At least one set of mechanical tests is to be made on material representing each casting in accordance with Ch.2.

4.7.6 As an alternative to 4.7.5, where a number of small propellers of about the same size, and less than 1[m] in diameter, are made from one cast and heat treated in the same furnace charge, a batch testing procedure may be adopted using separately cast test samples of suitable dimensions. At least one set of mechanical tests is to be provided for each multiple of five castings in the batch.

4.8 Definition of skew, severity zones

4.8.1 In order to relate the degree of inspection to the criticality of imperfections in propeller blades and to help reduce the risk of failure by fatigue cracking after repair, propeller blades are divided into three severity zones designated A, B and C. Definition of skew, and, severity zones are given in Ch.8, 3.9.

4.9 Non-destructive examination

4.9.1 Qualification of personnel involved in NDT

4.9.1.1 Refer to IRS Classification Note "*Requirements for NDT Suppliers*".

4.9.2 Visual Testing

4.9.2.1 All finished castings are to be 100% visually inspected by the manufacturer. Castings are to be free from cracks, hot tears or other imperfections which, due to their nature, degree or extent, will interfere with the use of the castings. A general visual examination is to be carried out by the Surveyor.

4.9.3 Liquid penetrant testing

4.9.3.1 Liquid penetrant testing procedure is to be submitted to IRS and is to be in accordance with ISO 3452-1:2013 or a recognized standard. The acceptance criteria are specified in 4.10.

4.9.3.2 For all propellers, separately cast blades and hubs, the surfaces covered by severity zones A, B and C are to be liquid penetrant tested. Testing of zone A is to be undertaken in the presence of the Surveyor, whilst testing of zone B and C may be witnessed by the Surveyor upon his request.

4.9.3.3 If repairs have been made either by grinding or by welding, the repaired areas are additionally to be subjected to the liquid penetrant testing independent of their location and/or severity zone. Weld repairs are, independent of their location, always to be assessed according to zone A.

4.9.4 Magnetic particle testing

4.9.4.1 Magnetic particle testing may be used in lieu of liquid penetrant testing for examination of martensitic stainless steels

castings. Magnetic particle testing procedure is to be submitted to IRS and is to be in accordance with ISO 9934-1:2016 or a recognized standard.

4.9.5 Radiographic and ultrasonic testing

4.9.5.1 When required by IRS or when deemed necessary by the manufacturer, further non-destructive testing (e.g. radiographic and/or ultrasonic testing) are to be carried out. The acceptance criteria or applied quality levels are then to be agreed between the manufacturer and IRS in accordance with a recognized standard.
Note: due to the attenuating effect of ultrasound within austenitic steel castings, ultrasonic testing may not be practical in some cases, depending on the shape/type/thickness, and grain-growth direction of the casting.

4.10 Acceptance criteria for liquid penetrant testing and magnetic particle testing

4.10.1 Definitions of liquid penetrant indications

4.10.1.1 **Indication:** In the liquid penetrant testing an indication is the presence of detectable bleed-out of the penetrant liquid from the material discontinuities appearing at least 10 minutes after the developer has been applied.

4.10.1.2 **Relevant indication:** only indications which have any dimension greater than 1.5mm shall be considered relevant for the categorization of indications.

4.10.1.3 **Non-linear indication:** an indication with a largest dimension less than three times its smallest dimension (i.e. $l < 3\ w$).

4.10.1.4 **Linear indication:** an indication with a largest dimension three or more times its smallest dimension (i.e. $l \geq 3\ w$).

4.10.1.5 **Aligned indications**:

a) Non-linear indications form an alignment when the distance between indications is less than 2 [mm] and at least three indications are aligned. An alignment of indications is considered to be a unique indication and its length is equal to the overall length of the alignment.

b) Linear indications form an alignment when the distance between two indications is smaller than the length of the longest indication.

Illustration of liquid penetrant indications is given in Fig. 4.10.1.

4.10.2 Acceptance standard

4.10.2.1 The surface to be inspected is to be divided into reference areas of 100 [cm^2]. Each reference area may be square or rectangular with the major dimension not exceeding 250 [mm]. The area is to be taken in the most unfavourable location relative to the indication being evaluated.

4.10.2.2 The relevant indications detected with respect to their size and number, are not to exceed the values given in the Table 4.10.1. Areas which are prepared for welding are independent of their location always to be assessed according to zone A. The same applies to the welded areas after being finished machined and/or grinded.

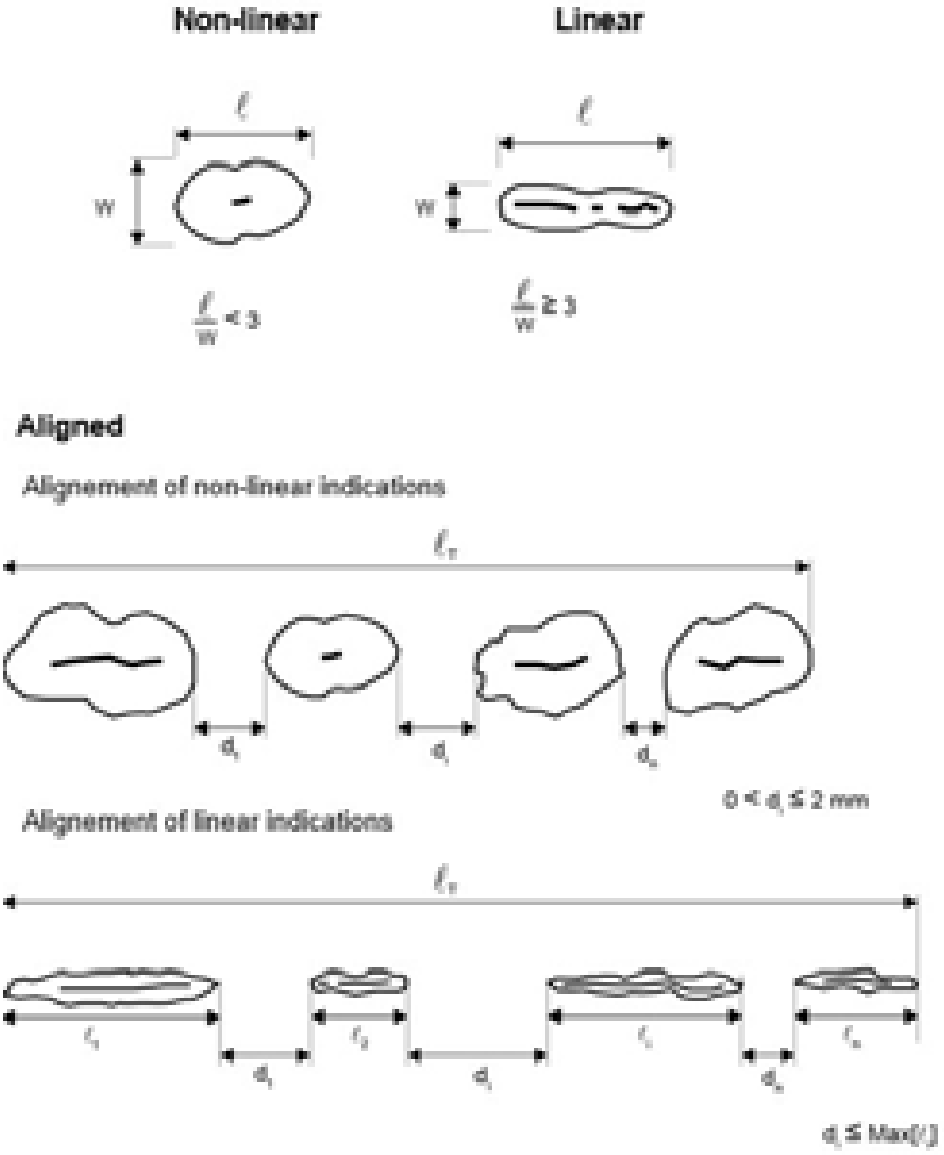


Fig.4.10.1 : Shape of indications

Table 4.10.1 : Allowable number and size of relevant indications in a reference area of 100 cm², depending on severity zones

Severity zone	Max. total number of indications	Type of Indication	Max. number for each type[1),2)]	Max. dimension of indication [mm]
A	7	Non-linear	5	4
		Linear	2	3
		Aligned	2	3
B	14	Non-linear	10	6
		Linear	4	6
		Aligned	4	6
C	20	Non-linear	14	8
		Linear	6	6
		Aligned	6	6

Table 4.10.1 (Contd.)

Notes:

1) Single non-linear indications less than 2 [mm] in Zone A and less than 3 [mm] for the other zones are not considered relevant.

2) The total number of non-linear indications may be increased to the maximum total number, or part thereof, represented by the absence of linear or aligned indications.

4.11 Repair of defects

4.11.1 Defective castings are to be repaired in accordance with the requirements given in 4.11.2 to 4.11.7 and, where applicable, the requirements of 4.12.

4.11.2 In general the repairs are to be carried out by mechanical means, e.g. by grinding, chipping or milling. The resulting grooves are to be blended into the surrounding surface so as to avoid any sharp contours. Complete elimination of the defective material is to be verified by liquid penetrant testing, or magnetic particle testing, if applicable.

4.11.3 Weld repairs are to be undertaken only when they are considered to be necessary and have prior approval of the Surveyor.

4.11.4 The excavations are to be suitably shaped to allow good access for welding. The resulting grooves are to be subsequently ground smooth and complete elimination of the defective material is to be verified by liquid penetrant testing. Welds having an area less than 5 [cm^2] are to be avoided.

4.11.5 Grinding in severity Zone A may be carried out to an extent that maintains the blade thickness. Repair welding is generally not permitted in severity Zone A and will only be allowed after special consideration.

4.11.6 Defects in severity Zone B that are not deeper than t/40 [mm] ("t" is the minimum local thickness according to the Rules) or 2 [mm], whichever is greatest, are to be removed by grinding. Those defects that are deeper may be repaired by welding subject to prior approval from IRS.

4.11.7 Repair welding is generally permitted in severity Zone C.

4.11.8 Repair documentation

4.11.8.1 The foundry is to maintain records of inspections, welding, and any subsequent heat treatment, traceable to each casting. Before welding is started, full details of the extent and location of the repair, the proposed welding procedure, heat treatment and subsequent inspection procedures are to be submitted to the IRS for approval.

4.12 Welding repair procedure

4.12.1 Before welding is started, manufacturer is to submit to IRS a detailed welding procedure specification covering the weld preparation, welding positions, welding parameters, welding consumables, preheating, post weld heat treatment and inspection procedures.

4.12.2 All weld repairs are to be carried out in accordance with qualified procedures, and by welders who are qualified to a recognized standard. Welding Procedure Qualification Tests are to be carried out in accordance with 4.15 and witnessed by the Surveyor. Defects to be repaired by welding are to be ground to sound material according to 4.10. The welding grooves are to be prepared in such a manner which will allow a good fusion of the groove bottom. The resulting ground areas are to be examined in the presence of the Surveyor by liquid penetrant testing in order to verify the complete elimination of defective material.

4.12.3 Welding is to be done under controlled conditions free from draughts and adverse weather.

4.12.4 Metal arc welding with electrodes or filler wire used in the procedure tests is to be used. The welding consumables are to be stored and handled in accordance with the manufacturer's recommendations.

4.12.5 Slag, undercuts and other imperfections are to be removed before depositing the next run.

4.12.6 The martenistic steels are to be furnace re-tempered after weld repair. Subject to prior approval, however, local stress relieving may be considered for minor repairs.

4.12.7 On completion of heat treatment the weld repairs and adjacent material are to be ground smooth. All weld repairs are to be liquid penetrant tested.

4.13 Identification and marking

4.13.1 The manufacturer is to adopt a system for the identification of all castings, which enable the material to be traced to its original cast. The Surveyor is to be given full facilities for so tracing the castings when required. Each finished casting propeller is to be marked by the manufacturer at least with the following particulars:

a) Heat number or other marking which will enable the full history of the casting to be traced;

b) Grade of cast material or corresponding abbreviated designation

c) The IRS certificate number and abbreviated name of local IRS office;

d) Ice class symbol, where applicable;

e) Skew angle for high skew propellers;

f) Date of final inspection.

4.13.2 The IR stamp is to be put on when the casting has been accepted.

4.14 Document and Certification

4.14.1 The manufacturer is to provide the Surveyor with an inspection certificate giving the following particulars for each casting which has been accepted:

a) Purchaser's name and order number;

b) Vessel identification, where known;

c) Description of the casting with drawing number;

d) Diameter, number of blades, pitch, direction of turning;

e) Skew angle for high skew propellers;

f) Final weight;

g) Alloy type, heat number and chemical composition;

h) Casting identification number;

i) Details of time and temperature of heat treatment;

j) Results of the mechanical tests.

k) Results of non-destructive tests and details of test procedure where applicable.

4.15 Welding procedure qualification test for repair of cast steel propeller

4.15.1 General

4.15.1.1 This sub-section provides requirements for qualification tests of welding procedures intended for the repair of cast steel propellers.

4.15.1.2 For the welding procedure approval the welding procedure qualification tests are to be carried out with satisfactory results. The qualification tests are to be carried out with the same welding process, filler metal, preheating and stress-relieving treatment as those intended applied by the actual repair work. Welding procedure specification is to refer to the test results achieved during welding procedure qualification testing.

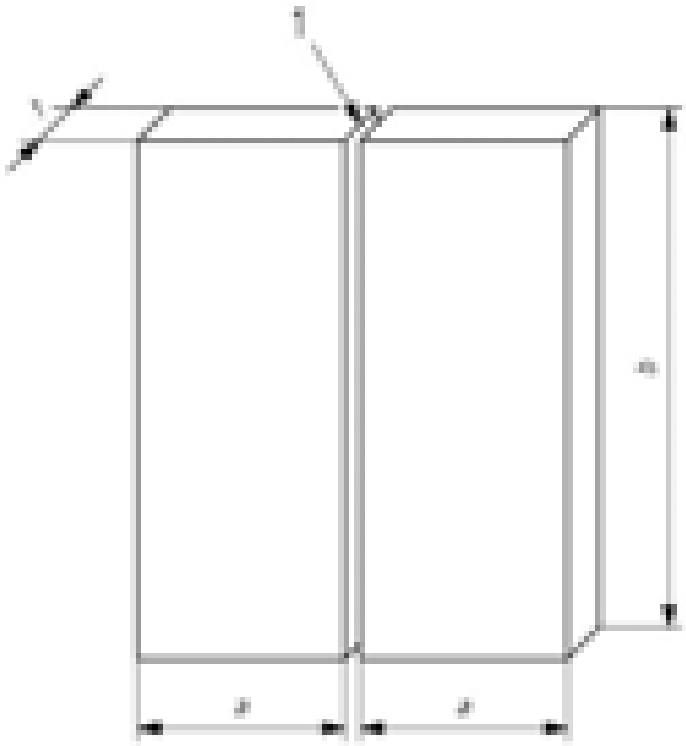
4.15.1.3 Welding procedures qualified at a manufacturer are valid for welding in workshops under the same technical and quality management.

4.15.2 Test piece and welding of sample

4.15.2.1 The test assembly, consisting of cast samples, is to be of a size sufficient to ensure a reasonable heat distribution and according to Fig. 4.15.2.1 with the minimum dimensions. The dimensions and shape of the groove is to be representative of the actual repair work.

4.15.2.2 Preparation and welding of test pieces are to be carried out in accordance with the general condition of repair welding work which it represents.

4.15.2.3 Welding of the test assemblies and testing of test specimens are to be witnessed by the Surveyor.



1: Joint preparation and fit-up as detailed in the preliminary Welding Procedure Specification
a: minimum value 150mm
b: minimum value 350mm
t: material thickness

Fig.4.15.2.1 : Test piece for welding repair procedure

4.15.3 Examinations and tests

4.15.3.1 Test assembly is to be examined non-destructively and destructively in accordance with Table 4.15.3.1 and Fig. 4.15.3.1.

4.15.3.2 Non-destructive testing

.1 Test assembly is to be examined by visual and liquid penetrant testing, or magnetic particle testing if applicable, prior to the cutting of test specimen. In case, that any post-weld heat treatment is required or specified, non-destructive testing is to be performed after heat treatment. No cracks are permitted. Imperfections detected by liquid penetrant testing, or magnetic particle testing if applicable, are to be assessed in accordance with 4.10.

4.15.3.3 Tensile test

.1 Two flat transverse tensile test specimens are be prepared. Testing procedures are to be in accordance with Ch.2. Alternatively tensile test specimens according to recognized standards acceptable to IRS may be used. The tensile strength is to meet the specified minimum value of the base material. The location of fracture is to be reported, i.e. weld metal, HAZ or base material.

4.15.3.4 Bend test

.1 Transverse bend tests for butt joints are to be in accordance with Ch.2, or, according to a recognized standard. The mandrel diameter shall be 4 x thickness except for austenitic steels, in which case the mandrel diameter is to be 3 x thickness. The bending angle is to be 180°. After testing, the test specimens are not to reveal any open defects in any direction

greater than 3 [mm]. Defects appearing at the corners of a test specimen during testing are to be investigated case by case. Two root and two face bend specimens are to be tested. For thickness 12 [mm] and over, four side bend specimens may alternatively be tested.

4.15.3.5 Macro-examination

.1 Two macro-sections are to be prepared and etched on one side to clearly reveal the weld metal, the fusion line, and the heat affected zone. Cracks and lack of fusion are not permitted. Imperfections such as slag inclusions, and pores greater than 3 [mm] are not permitted.

4.15.3.6 Impact test

.1 Impact test is required, where the base material is impact tested. Charpy V-notch test specimens are to be in accordance with Ch.2. Two sets are to be taken, one set with the notch positioned in the center of the weld and one set with the notch positioned in the HAZ (i.e. the mid-point of the notch is to be at 1 [mm] to 2 [mm] from the fusion line), respectively. The test temperature, and impact energy are to comply with the requirement specified for the base material.

4.15.3.7 Hardness test

.1 The macro-section representing the start of welding is to be used for HV 10 hardness testing. Indentations are to traverse 2 [mm] below the surface. At least three individual indentations are to be made in the weld metal, the HAZ (both sides) and in the base metal (both sides). The values are to be reported for information.

4.15.3.8 Re-testing

.1 If the test piece fails to comply with any of the requirements for visual or non-destructive testing one further test piece is to be welded and subjected to the same examination. If this additional test piece does not comply with the relevant requirements, the pWPS (preliminary welding procedure specification) is to be regarded as not capable of complying with the requirements without modification.

.2 If any test specimens fail to comply with the relevant requirements for destructive testing due to weld imperfections only, two further test specimens are to be obtained for each one that failed. These specimens can be taken from the same test piece if there is sufficient material available or from a new test piece, and are to be subjected to the same test. If either of these additional test specimens does not comply with the relevant requirements, the pWPS is to be regarded as not capable of complying with the requirements without modification.

.3 If a tensile test specimen fails to meet the requirements, the re-testing is to be in accordance with Ch.2.

.4 If there is a single hardness value above the maximum values allowed, additional hardness tests are to be carried out (on the reverse of the specimen or after sufficient grinding of the tested surface). None of the additional hardness values is to exceed the maximum hardness values required.

.5 The re-testing of Charpy impact specimens are to be carried out in accordance with Ch.2.

.6 Where there is insufficient welded assembly remaining to provide additional test specimens, a further assembly is to be welded using the same procedure to provide the additional specimens.

Table 4.15.3.1 Type of tests and extent of testing

Type of test	Extent of testing
Visual testing	100% as per 4.15.3.2
Liquid penetrant testing (1)	100% as per 4.15.3.2
Transverse tensile test	Two specimens as per 4.15.3.3
Bend test (2)	Two root and two face specimens as per 4.15.3.4
Macro examination	Three specimens as per 4.15.3.5
Impact test	Two sets of three specimens as per 4.15.3.6
Hardness test	As per 4.15.3.7

(1) Magnetic particle testing may be used in lieu of liquid penetrant testing for martensitic stainless steels.
(2) For t≥12mm, the face and root bend may be substituted by 4 side bend test specimens.

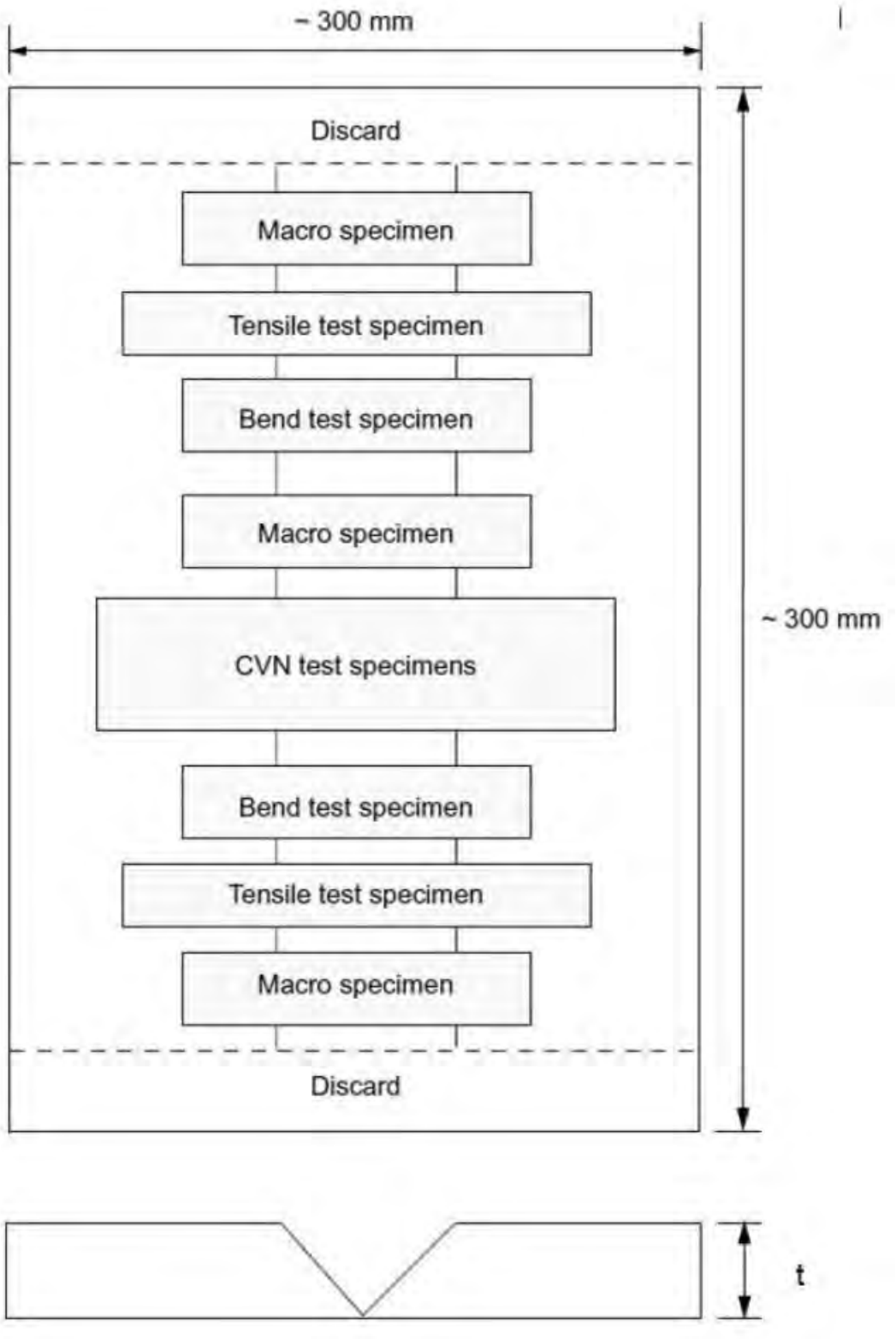


Fig.4.15.3.1 : Weld test assembly

4.15.4 Test record

4.15.4.1 Welding conditions for test assemblies and test results are to be recorded in welding procedure qualification. Forms of welding procedure qualification records may be in accordance with recognised standards.

4.15.4.2 A statement of the results of assessing each test piece, including repeat

tests, is to be made for each welding procedure qualification records. The relevant items listed for the WPS are to be included.

4.15.4.3 The welding procedure qualification record is to be signed by the Surveyor witnessing the test and is to include IR identification.

4.15.5 Range of approval

4.15.5.1 General

.1 All the conditions of validity stated below are to be met independently of each other. Changes outside of the ranges specified are to require a new welding procedure test. A qualification of a WPS obtained by a manufacturer is valid for welding in workshops or sites under the same technical and quality control of that manufacturer.

4.15.5.2 Base metal

.1 Range of approval for steel cast propeller is limited to steel grade tested.

4.15.5.3 Thickness

.1 The qualification of a WPS carried out on a weld assembly of thickness t is valid for the thickness range given in Table 4.15.5.3.

Table 4.15.5.3 Range of qualification for thickness

Thickness of the test piece, t (mm)	Range of approval
15<t≤30	3mm to 2t
t>30	0.5t to 2t or 200mm, whichever is the greater

4.15.5.4 Welding position

.1 Approval for a test made in any position is restricted to that position.

4.15.5.5 Welding process

.1 The approval is only valid for the welding process used in the welding procedure test. Single run is not qualified by multi-run butt weld test used in this Section.

4.15.5.6 Filler metal

.1 The approval is only valid for the filler metal used in the welding procedure test.

4.15.5.7 Heat input

.1 The upper limit of heat input approved is 15% greater than that used in welding the test piece. The lower limit of heat input approved is 15% lower than that used in welding the test piece.

4.15.5.8 Preheating and interpass temperature

.1 The minimum preheating temperature is not to be less than that used in the qualification test. The maximum interpass temperature is not to be higher than that used in the qualification test.

4.15.5.9 Post-weld heat treatment

.1 The heat treatment used in the qualification test is to be specified in pWPS. Holding time may be adjusted as a function of thickness.

Section 5

Austenitic Stainless Steel Castings

5.1 Scope

5.1.1 This section gives the requirements for castings in austenitic stainless steels for piping systems in ships for liquefied gases where the design temperature is not lower than –165°C and in bulk chemical carriers.

5.1.2 Where it is proposed to use alternative steels, particulars of the specified chemical composition, mechanical properties and heat treatment are to be submitted for approval.

5.2 Chemical composition

5.2.1 The chemical composition of ladle samples is to comply with the requirements given in Table 5.2.1.

Table 5.2.1 : Chemical composition of austenitic stainless steel castings

Type of steel	Chemical composition %								
	C max.	Si	Mn	S	P	Cr	Mo	Ni	Others
304L	0.03	0.20-1.5	0.50-2.0	0.40 max.		17.0-21.0	-	8.0-12.0	-
304	0.08						-	8.0-12.0	-
316L	0.03						2.0-3.0	9.0-13.0	-
316	0.08						2.0-3.0	9.0-13.0	-
317	0.08						3.0-4.0	9.0-12.0	-
347 (see Note)	0.06						-	9.0-12.0	Nb ≥8xC≤0.90

Note: When guaranteed impact values at low temperature are not required, the maximum carbon content may be 0.08% and the maximum niobium may be 1.00%.

5.3 Heat treatment

5.3.1 All castings are to be solution treated at a temperature of not less than 1000°C and cooled rapidly in air, oil or water.

5.4 Mechanical tests

5.4.1 One tensile test specimen is to be prepared from material representing each casting or batch of castings. In addition, where the castings are intended for liquefied gas applications, where the design temperature is lower than –55°C, one set of three Charpy V-notch impact test specimens is to be prepared.

5.4.2 The tensile test is to be carried out at ambient temperature and the results are to comply with the requirements given in Table 5.4.2.

5.4.3 The average value for impact test specimens is to comply with the appropriate requirements given in Table 5.4.2. One individual value may be less than the required average value provided that it is not less than 70 percent of this average value. See Ch.1, 1.10 for re-test procedures.

Table 5.4.2 : Mechanical properties for acceptance purposes : austenitic stainless steel castings

Type of steel	Tensile strength [N/mm^2] minimum	1.0% proof stress [N/mm^2] minimum	Elongation on 5.65 $\sqrt{S_o}$ % minimum	Reduction of area % minimum	Charpy V-notch impact tests	
					Test temp. °C	Average energy J minimum
304L	430	215	26	40	-196	41
304	480	220				
316L	430	215	26	40	-196	41
317	480	240				
347	480	215	22	35	-196	41

5.5 Intergranular corrosion tests

5.5.1 Where corrosive conditions are anticipated in service, intergranular corrosion tests are required on castings in grades 304, 316 and 317. Such tests may not be required for grades 304L, 316L and 347.

5.5.2 Where an intergranular corrosion test is specified, it is to be carried out in accordance with the standard referred in 9.6.2 of Chapter 3.

5.6 Non-destructive examination

5.6.1 The non-destructive examination of castings is to be carried out in accordance with the appropriate requirements of Classification Notes “Guidelines for non-destructive testing of steel castings for marine application” and additionally agreed between the manufacturer, purchaser and Surveyor.

Section 6

Castings for other Applications

6.1 General

6.1.1 Details of chemical composition, heat treatment, mechanical properties of steel castings for crankshafts and those intended for elevated temperature service are to be submitted for approval of IRS.

End of Chapter

Chapter 5

Steel Forgings

Contents

Section

1 *General Requirements*

2 *Hull and Machinery Steel Forgings for General Applications*

3 *Ferritic Steel Forgings for Low Temperature Service*

4 *Austenitic Stainless Steel Forgings*

Section 1

General Requirements

1.1 Scope

1.1.1 All important steel forgings, as defined in the relevant construction Rules, are to be manufactured and tested in accordance with the requirements of this Chapter.

1.1.2 Where required by the relevant Rules dealing with design and construction, forgings are to be manufactured and tested in accordance with Ch.1 and 2, together with the general requirements given in this Chapter.

1.1.3 Alternatively, forgings which comply with National or proprietary specifications may be accepted provided such specifications give reasonable equivalence to these requirements or are otherwise specially approved for a specific application by IRS.

1.2 Manufacture

1.2.1 Forgings are to be made at the works approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

1.2.2 The steel used in the manufacture of forgings is to be made by a process approved by IRS. The works at which the steel was produced is to be approved by IRS. Where the steel is produced at a separate works to the forging, the steel manufacturer is also to be approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

1.2.3 Adequate top and bottom discards are to be made to ensure freedom from piping and harmful segregations in the finished forgings.

1.2.4 The plastic deformation is to be such as to ensure soundness, uniformity of structure and satisfactory mechanical properties after heat treatment. The reduction ratio is to be in accordance with the following Table:

Method of manufacture	**Total reduction ratio (See Notes 1, 2 & 3)**
Made directly from ingots or forged blooms or billets	3:1 where L > D 1.5:1 where L ≤ D
Made from rolled products	4:1 where L > D 2:1 where L ≤ D

Notes

1 L and D are the length and diameter respectively of the part of the forging under consideration.

2 the reduction ratio is to be calculated with reference to the average cross-sectional area of the ingot. Where an ingot is initially upset, this reference area may be taken as the average cross-sectional area after this operation.

3 For rolled bars used as a substitute for forgings (see 1.1.1) the reduction ratio is not to be less than 6 : 1

4 For forgings made by upsetting, the length after upsetting is to be not more than one-third of the length before upsetting or, in the case of an initial forging reduction of at least 1.5:1, not more than one half of the length before upsetting.

1.2.5 For crankshafts, where grain flow is required in the most favourable direction having regard to the mode of stressing in service, the proposed method of manufacture may required special approval by IRS. In such cases, tests may be required to demonstrate that a satisfactory structure and grain flow are obtained.

1.2.6 The shaping of forgings or rolled slabs and billets by flame cutting, scarfing or arc-air gouging is to be undertaken in accordance with recognized good practice and unless otherwise approved, is to be carried out before the final heat treatment. Preheating is to be employed when necessitated by the composition and/or thickness of the steel.

1.2.7 For certain components, subsequent machining of all flame cut surfaces may be required.

1.2.8 When two or more forgings are joined by welding to form a composite component the proposed welding procedure specification is to be submitted for approval. Welding procedure qualification tests are to be carried out.

1.2.9 Welders intended to be engaged in fusion welding of steel forgings for hull structures are to be qualified in accordance with IRS Classification Note "*Qualification Scheme for Welders of Hull Structural Steel*".

1.3 Quality of forgings

1.3.1 All forgings are to be free from surface or internal defects which would be prejudicial to their proper application in service.

1.4 Chemical composition

1.4.1 All forgings are to be made from killed steel, and the chemical composition is to be appropriate for the type of steel, dimensions and required mechanical properties of the forgings being manufactured.

1.4.2 The chemical composition of each heat is to be determined by the manufacturer on a sample taken preferably during the pouring of the heat. When multiple heats are tapped into a common ladle, the ladle analysis is applicable.

1.5 Heat treatment (including surface hardening and straightening)

1.5.1 At an appropriate stage of manufacture, after completion of all hot working operations, forgings are to be suitably heat treated to refine the grain structure and to obtain the required mechanical properties. Heat treatment is to be carried out in properly constructed furnaces which are efficiently maintained and have adequate means for control and recording of temperature. The furnace dimensions are to be such as to allow the whole furnace charge to be uniformly heated to the necessary temperature. In the case of very large forgings alternative methods of heat treatment will be specially considered by IRS.

1.5.2 Except as provided in 1.5.7 and 1.5.8 forgings are to be supplied in one of the following conditions:

a) Carbon and carbon-manganese steels

Fully annealed
Normalized
Normalized and tempered
Quenched and tempered

b) Alloy steels

Normalized
Normalized and tempered
Quenched and tempered

For all types of steel the tempering temperature is not less than 550°C.

The delivery condition is to meet the design and application requirements, it is the manufacturers responsibility to select the appropriate heat treatment method to obtain the required mechanical properties. Where forgings for gearing are not intended for surface hardening, lower tempering temperature may be allowed.

1.5.3 Sufficient thermocouples are to be connected to the furnace charge to measure and record that its temperature is adequately uniform unless the temperature uniformity of the furnace is verified at regular intervals.

1.5.4 If for any reasons a forging is subsequently heated for further hot working the forging is to be re-heat treated.

1.5.5 If any straightening operation is performed after the final heat treatment, a subsequent stress relieving heat treatment to avoid harmful residual stresses is to be carried out, unless otherwise agreed.

1.5.6 Where it is intended to surface harden forgings, full details of the proposed procedure and specification are to be submitted for the approval of IRS. For the purpose of this approval, the manufacturer may be required to demonstrate by test that the proposed procedure gives a uniform surface layer of the required hardness and depth and that it does not impair the soundness and properties of the steel.

1.5.7 Where induction hardening or nitriding is to be carried out after machining, forgings are to be heat treated at an appropriate stage to a condition suitable for this subsequent surface hardening.

1.5.8 Where carburizing is to be carried out after machining, forgings are to be heat treated at an appropriate stage (generally either by full annealing or by normalising and tempering) to a condition suitable for subsequent machining and carburizing.

1.5.9 If a forging is locally reheated or any straightening operation is performed after the final heat treatment, consideration is to be given to a subsequent stress relieving heat treatment. The manufacturer is to have strict control of this temperature to avoid any detrimental effects to the final heat treatment and resultant microstructure and mechanical properties of the forging.

1.5.10 The manufacturer is to maintain records of heat treatment identifying the furnace used, furnace charge, date, temperature and time at the beginning and end of heat treatment cycle. The records are to be presented to the Surveyor on request.

1.6 Mechanical tests

1.6.1 The requirements of Mechanical tests and mechanical properties are given in Section 2 and 3.

1.7 Inspection

1.7.1 All forgings should be subjected to a 100% visual examination of all accessible surfaces by the manufacturer and made available to the Surveyor. Where applicable, this visual examination is to include the examination of internal surfaces and bores. Unless otherwise agreed, the verification of the dimensions is the responsibility of the manufacturer.

1.7.2 When required by the relevant construction Rules, or by the approved procedure for welded composite components appropriate non-destructive testing is also to be carried out before acceptance and the results are to be reported by the manufacturer.

1.7.3 Where required by appropriate part of the rules or IRS Classification Notes *“Guidelines for Non- Destructive Testing of Hull and Machinery Steel Forgings*”, ultrasonic examination is to be carried out after the forgings have been machined to a condition suitable for this type of examination and after

the final heat treatment. Both radial and axial scanning are to be carried out where appropriate for the shape and the dimensions of the forgings being examined.

1.7.4 The method and the extent of inspection, NDT and acceptance criteria are to be agreed with IRS. IRS Classification Notes *"Guidelines for Non- Destructive Testing of Hull and Machinery Steel Forgings"* is regarded as an example of an acceptable standard.
For mass produced forgings the extent of examination will be specially considered in each case.

1.7.5 Unless otherwise agreed, examinations are to be carried out by the manufacturer, although Surveyors may be present in order to verify that the examination is being carried out in accordance with the agreed procedures.

1.7.6 If the forging is supplied in the 'as forged' condition for machining at a separate works, the manufacturer is to ensure that a suitable ultrasonic examination is carried out to verify the internal quality of the forging.

1.7.7 Where advanced ultrasonic testing methods are applied, e.g. PAUT or TOFD, reference is made to IRS Classification Notes *"Requirements for Non-Destructive Testing",* Section 2 "Advanced non-destructive testing of materials and welds".
In such cases, acceptance levels regarding accept/reject criteria may be as per the applicable section in the IRS Classification Notes "*Guidelines for Non- Destructive Testing of Hull and Machinery Steel Forgings*".

1.7.8 When required by the conditions of approval for surface hardened forgings, (1.5.6) additional test samples are to be processed at the same time as the forgings which they represent. These test samples are subsequently to be sectioned in order to determine the hardness, shape and depth of the locally hardened zone and which are to comply with the requirements of the approved specification.

1.7.9 In the event of any forging proving defective during subsequent machining or testing, it is to be rejected notwithstanding any previous certification.

1.8 Rectification of defective forgings

1.8.1 Defects may be removed by grinding or chipping and grinding provided the component dimensions are acceptable. The resulting grooves are to have a bottom radius of approximately three times the groove depth and are to be blended into the surrounding surface so as to avoid any sharp contours. Complete elimination of the defective material is to be verified by magnetic particle testing or liquid penetrant testing.

1.8.2 Repair welding of forgings subjected to torsional fatigue, such as crankshaft forgings and propeller shaft forgings is not permitted. In the case of other forgings repair welding may be allowed subject to prior approval of IRS. In such cases, full details of the extent and location of the repair, the proposed welding procedure, heat treatment and subsequent inspection procedures are to be submitted for the approval.

1.8.3 The forging manufacturer is to maintain records of repairs and subsequent inspections traceable to each forging repaired. The records are to be presented to the Surveyor on request.

a) Purchaser's name and order number;

b) Description of forgings and steel quality identification number;

c) Steel making process, cast number and chemical analysis of ladle sample;

d) Results of mechanical tests;

e) General details of heat treatment;

f) Identification number.

1.9 Identification of forgings

1.9.1 Before acceptance, all forgings, which have been tested and inspected with satisfactory results, are to be clearly marked in at least one place with the IRS brand IR and the following particulars:

a) The manufacturer's name or trade mark;

b) Identification mark for the grade of steel;

c) Identification number and/or initials which enable the full history of the forging to be traced;

d) Personal stamp of Surveyor responsible for inspection;

e) Test pressure, where applicable;

f) Date of final inspection;

g) The 'IRS' name;

h) Abbreviated name of IRS local office.

1.9.2 Where small forgings are manufactured in large numbers, modified arrangements for identification may be specially agreed with IRS.

1.10 Certification

1.10.1 The manufacturer is to provide the Surveyor, in duplicate, with a test certificate or shipping statement giving the following particulars for each forging or batch of forgings which has been accepted:

a) Purchaser's name and order number;

b) Description of forgings and steel quality identification number;

c) Steel making process, cast number and chemical analysis of ladle sample;

d) Results of mechanical tests;

e) General details of heat treatment;

f) Identification number.

Section 2

Hull and Machinery Steel Forgings for General Applications

2.1 Scope

2.1.1 The requirements given in this section are applicable to steel forgings intended for hull and machinery applications such as rudder stocks, pintles, propeller shafts, crankshafts, connecting rods, piston rods, gearing etc. Where relevant, these requirements are also applicable to material for forging stock and to rolled bars intended to be machined into components of simple shape.

2.1.2 These requirements are applicable only to steel forgings where the design and acceptance tests relate to mechanical properties at ambient temperature. For other applications, additional requirements may be necessary especially when the forgings are intended for service at low or elevated temperatures.

2.2 Chemical Composition

2.2.1 The chemical composition is to comply with the overall limits given in Tables 2.2.1 and Table 2.2.2 or, where applicable, the requirements of the approved specification.

2.2.2 At the option of the manufacturer, suitable grain refining elements such as aluminium, niobium or vanadium may be added. The content of such elements is to be reported.

2.2.3 Elements designated as residual elements in the individual specifications are not to be intentionally added to the steel. The content of such elements is to be reported.

Table 2.2.1 : Chemical composition limits [1] for hull steel forgings [6]

Steel type	C	Si	Mn	P	S	Cr [4]	Mo [4]	Ni [4]	Cu [4]	Total residuals
C, C-Mn	0.23 [2], [3]	0.45	0.20-1.50	0.035	0.035	0.30	0.15	0.40	0.30	0.85
Alloy	[5]	0.45	[5]	0.035	0.035	[5]	[5]	[5]	0.30	-

[1] Composition in percentage mass by mass maximum unless shown as a range.

[2] The carbon content may be increased above this level provided that the carbon equivalent (Ceq) is not more than 0.41%, calculated using the following formula:

$$Ceq = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15} (\%)$$

[3] The carbon content of C and C-Mn steel forgings not intended for welded construction may be 0.65 maximum.

[4] Elements are considered as residual elements.

[5] Specification is to be submitted for approval.

[6] Rudder stocks and pintles should be of weldable quality.

Table 2.2.2 : Chemical composition limits [1] for machinery steel forgings

Steel type	C	Si	Mn	P	S	Cr [4]	Mo [4]	Ni [4]	Cu [4]	Total residuals
C, C-Mn	0.23 [2], [3]	0.45	0.30-1.50	0.035	0.035	0.30	0.15	0.40	0.30	0.85
Alloy [5]	0.45	0.45	0.30-1.00	0.035	0.035	Min 0.40 [6]	Min 0.15 [6]	Min 0.40 [6]	0.30	-

[1] Composition in percentage mass by mass maximum unless shown as a range or as a minimum.

[2] The carbon content may be increased above this level provided that the carbon equivalent (Ceq) is not more than 0.41%.

[3] The carbon content of C and C-Mn steel forgings not intended for welded construction may be 0.65 maximum.

[4] Elements are considered as residual elements unless shown as a minimum.

[5] Where alloy steel forgings are intended for welded constructions, the proposed chemical composition is subject to approval by IRS.

[6] One or more of the elements is to comply with the minimum content.

2.3 Mechanical tests

2.3.1 Adequate number of test coupons are to be provided for carrying out tests including for retest purposes, with a cross-sectional area of not less than that part of the forging which it represents. This test material is to be integral with each forging except as provided in 2.3.8 and 2.3.11. Where batch testing is permitted according to 2.3.11 the test material may alternatively be a production part or separately forged. Separately forged test material is to have a reduction ratio similar to that used for the forgings represented.

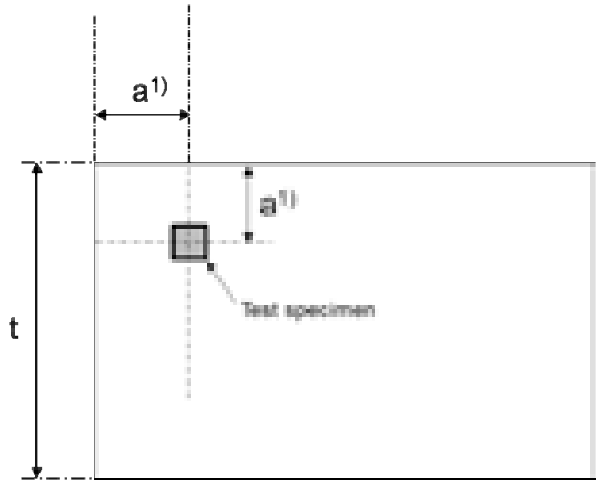
2.3.2 For the purpose of these requirements a set of tests is to consist of one tensile test specimen and when required in other sections of Rules three Charpy V-notch impact test specimens.

2.3.3 Test specimens are normally to be cut with their axes either parallel (longitudinal test) or tangential (tangential test) to the principal axial direction of each product.

2.3.4 The test specimen is to be positioned as follows:

a) For forgings having a thickness, t, or diameter D up to maximum 50mm, the longitudinal axis of the test specimen is to be located at a distance of t/2 or D/2 below the heat treated surfaces.
b) For forgings having a thickness, t, or diameter D greater than 50mm, the longitudinal axis of the test specimen is to be located at a distance of t/4 or D/4 (mid-radius) or 80mm, whichever is less, below any heat treated surface. Test specimen is to be located with its longitudinal axis at a distance from any heat treated surface as shown in Fig. 2.3.4.
c) For ring and disc forgings (noting that the test specimen locations for these shaped forgings may be different to elongated or free form forgings), tangential sample is to be taken at t/2 for thickness ≤ 25mm and 12.5mm below the surface for thickness >25mm, in both the vertical and horizontal direction. Where achievable, for thickness >25mm, no part of the test material is to be closer than 12.5 mm to any heat treated surface, as shown in Fig. 2.3.4.

2.3.5 Where the manufacturer can demonstrate that a proposed testing location or orientation is more representative of the required mechanical properties of a component, this may be agreed with IRS. In such cases, the heat treatment process, a proposed testing location or orientation, and technical justification is to be submitted to IRS for approval.



1) "a" is the distance from the test specimen to heat treated surface based on the above b) or c).

Fig.2.3.4 : Position of the test specimen

2.3.6 Except as provided in 2.3.11 the number and direction of tests is to be as follows:

a) Hull components such as rudder stocks, pintles etc. General machinery components such as shafting, connecting rods, etc.
One set of tests is to be taken from the end of each forging in a longitudinal direction except that, at the discretion of the manufacture the alternative directions or positions as shown in Fig.2.3.5a, Fig.2.3.5b and Fig.2.3.5c may be used. Where a forging exceeds both 4 tonnes in mass and 3 [m] in length one set of tests is to be taken from each end. These limits refer to the 'as forged' mass and length but excluding the test material.

b) Pinions - Where the finished machined diameter of the toothed portion exceeds 200 [mm] one set of tests is to be taken from each forging in a tangential direction adjacent to the toothed portion (test position B in Fig.2.3.5d). Where the dimensions preclude the preparation of tests from this position, tests in a tangential direction are to be taken from the end of the journal (test position C in Fig.2.3.5d). If however, the journal diameter is 200 [mm] or less the tests are to be taken in a longitudinal direction (test position A in Fig.2.3.5d). Where the finished length of the toothed portion exceed 1.25 [m], one set of tests is to be taken from each end.

c) Small pinions - Where the finished diameter of the toothed portion is 200 [mm] or less one set of tests is to be taken in a longitudinal direction (test position A in Fig.2.3.5d).

d) Gear wheels - One set of tests is to be taken from each forging in tangential direction (test position A or B in Fig.2.3.5e).

e) Gear wheel rims (made by expanding)
One set of tests is to be taken from each forging in a tangential direction (test position A or B in Fig.2.3.5.f). Where the finished diameter exceeds 2.5 [m] or the mass (as heat treated including test material) exceeds 3 tonnes, two sets of tests are to be taken from diametrically opposite positions (test positions A and B in Fig. 2.3.5f). The mechanical properties for longitudinal test are also to be applied.

f) Pinion sleeves - One set of tests is to be taken from each forging in tangential direction (test position A or B in Fig.2.3.5g). Where the finished length exceeds 1.25 [m] one set of tests is to be taken from each end.

g) Crankwebs
One set of tests is to be taken from each forging in a tangential direction.

h) Solid open die forged crankshafts
One set of tests is to be taken in a longitudinal direction from the driving shaft end of each forging (test position A in Fig.2.3.5h).
Where the mass (as heat treated but excluding test material) exceeds 3 tonnes tests in a longitudinal direction are to be taken from each end (test positions A and B in Fig.2.3.5h). Where, however, the crankthrows are formed by machining or flame cutting, the second set of tests is to be taken in a tangential direction from material removed from the crankthrow at the end opposite the driving shaft end (test position C in Fig.2.3.5h).

i) Forged Rings (such as slewing rings)
One set of tests is to be taken from each forging in a tangential direction (test positions are shown in Fig. 2.3.5i). Where the finished diameter exceeds 2.5 [m] or the mass (as heat treated, including test material) exceeds 3 tonnes then two sets of tests are to be taken in diametrically opposite positions.

2.3.7 For closed die crankshaft forgings and crankshaft forgings where the method of manufacture has been specially approved in accordance with 1.2.5, the number and position of test specimens is to be agreed with IRS having regard to the method of manufacture employed.

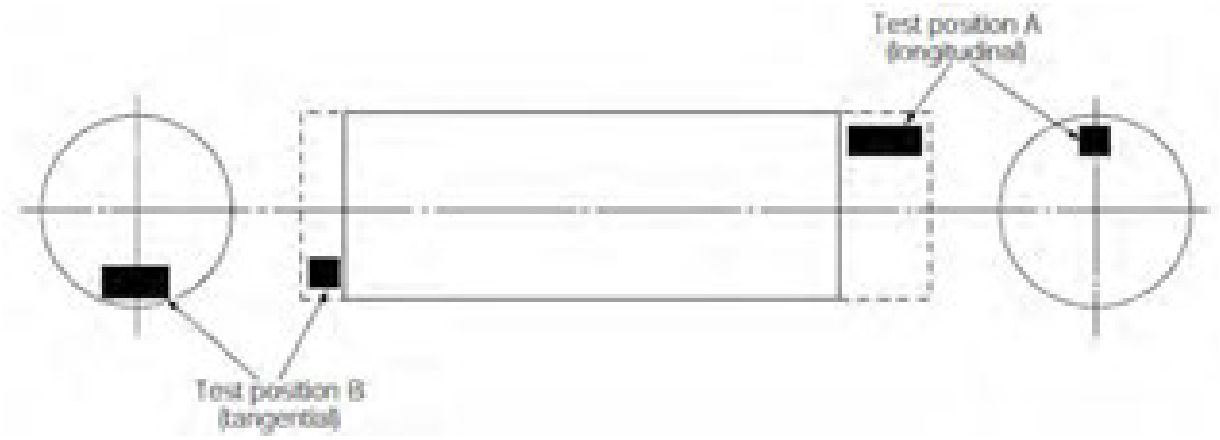


Fig. 2.3.5a Plain shaft

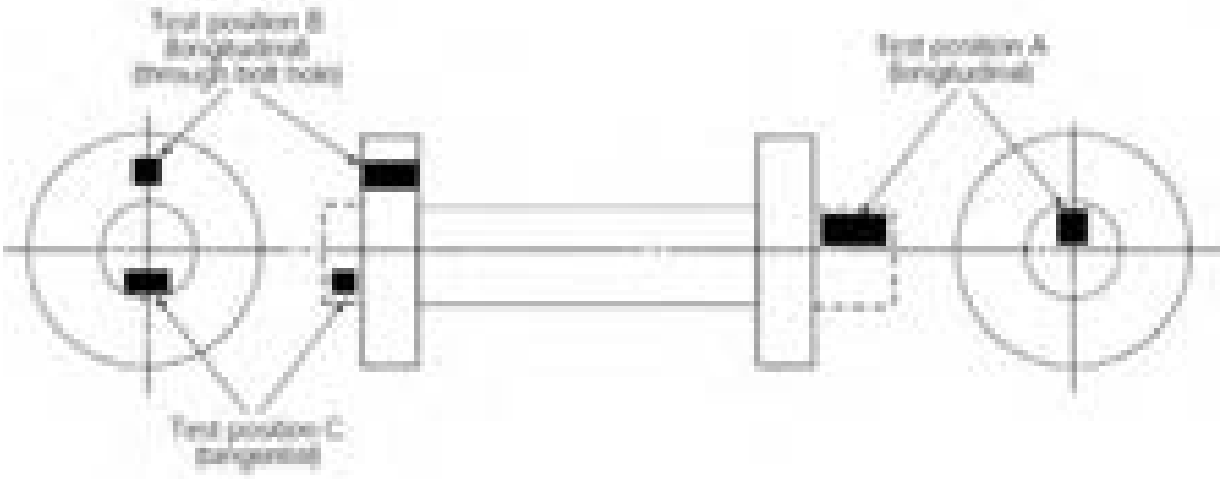


Fig.2.3.5b Flanged shaft

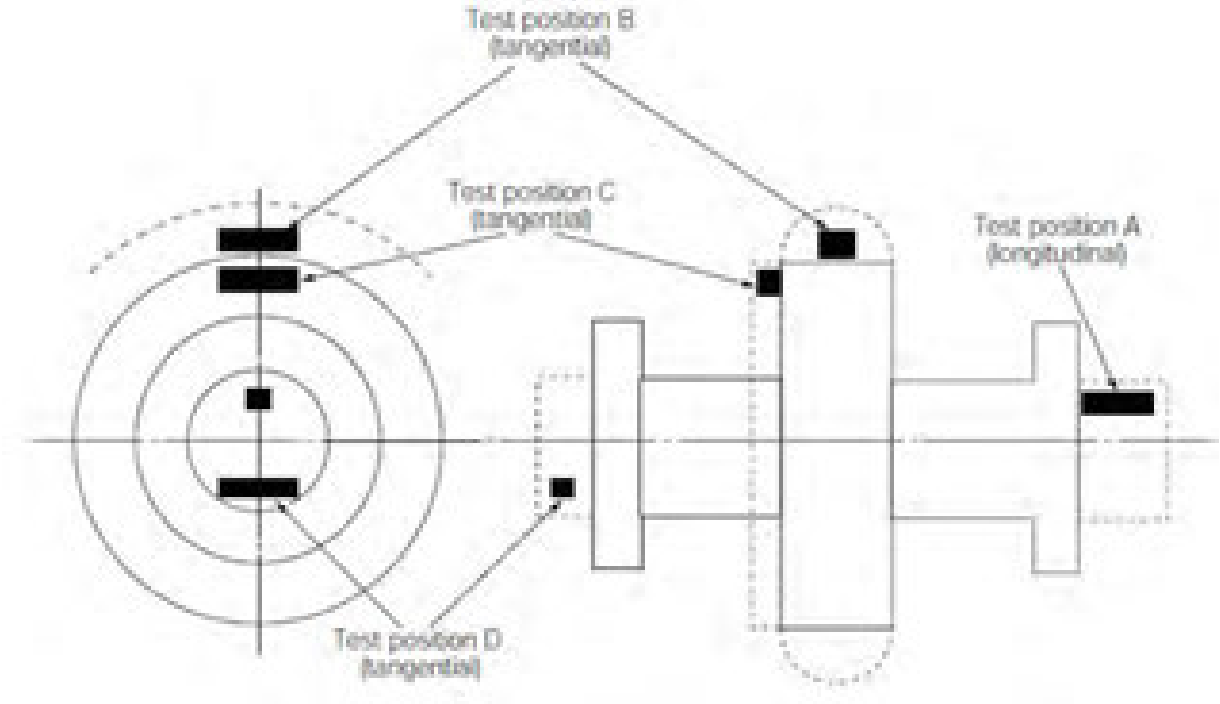


Fig. 2.3.5c Flanged shaft with collar

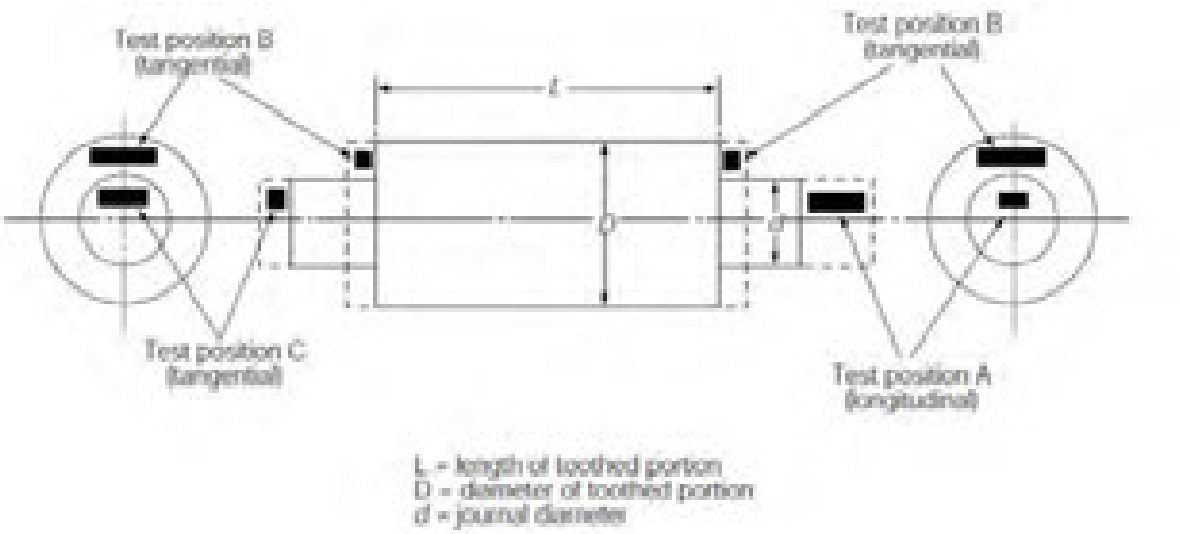


Fig. 2.3.5d Pinion

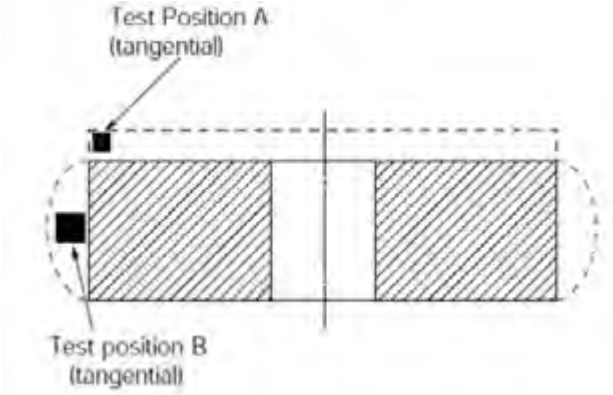


Fig.2.3.5e Gear wheel

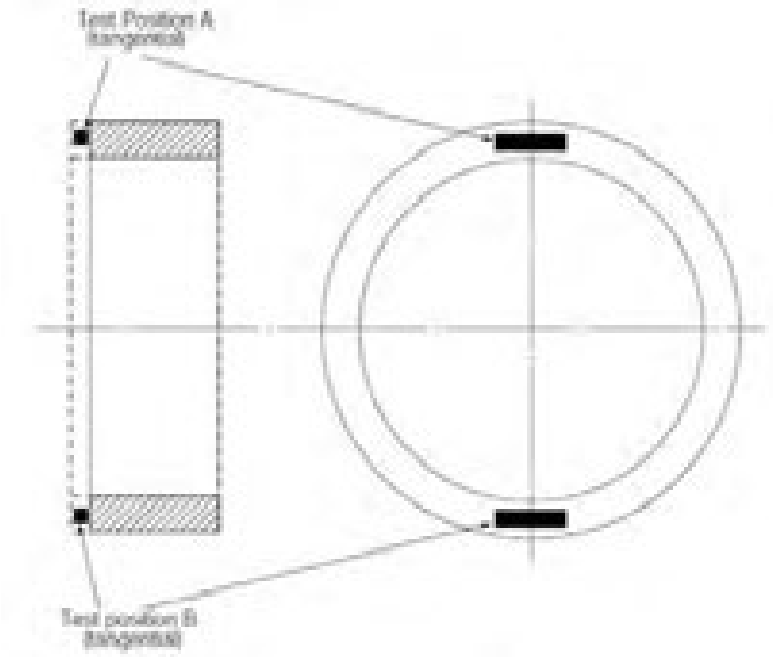


Fig. 2.3.5f Gear rim (made by expanding)

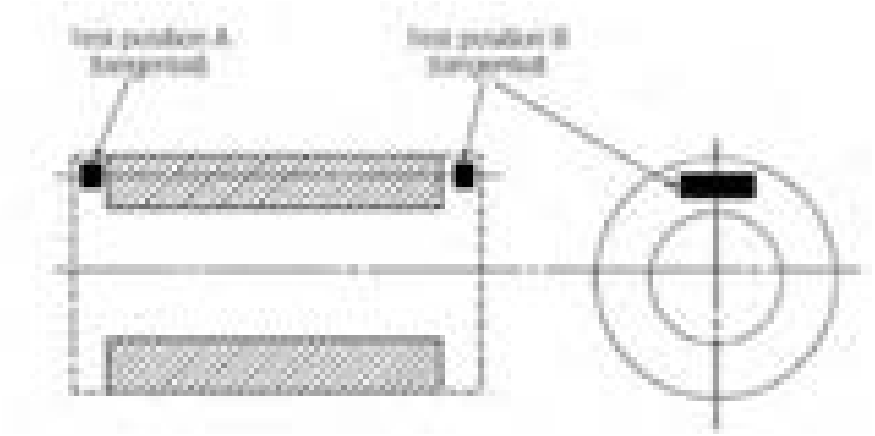


Fig. 2.3.5g Pinion sleeve

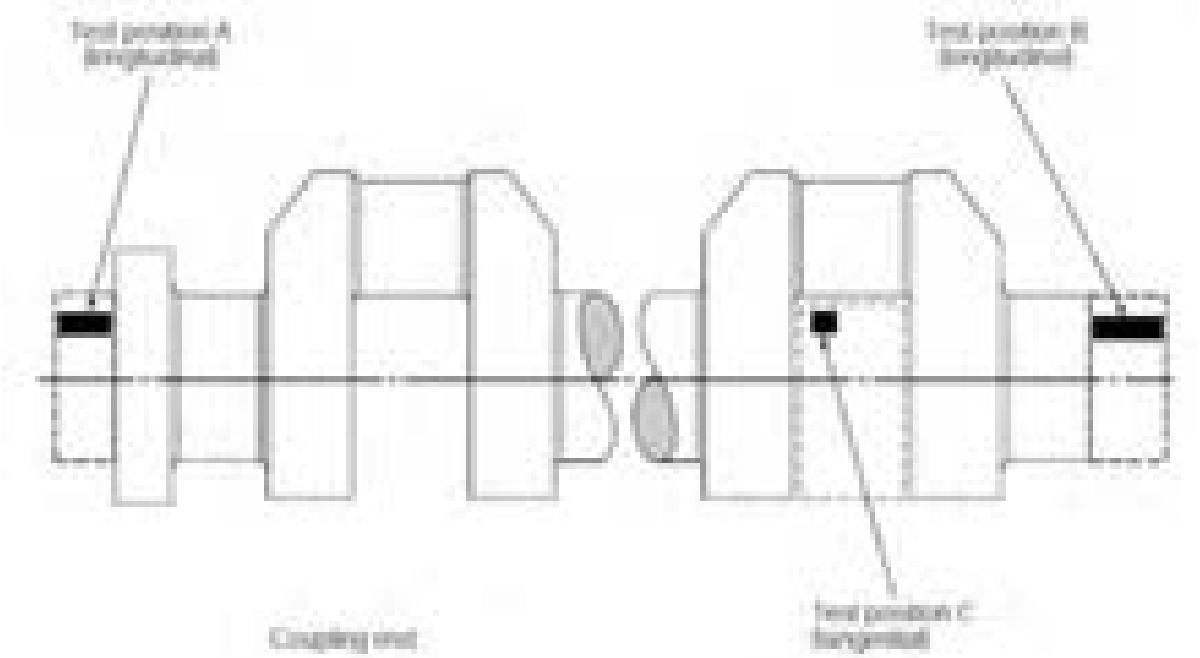


Fig. 2.3.5h Solid forged crankshaft

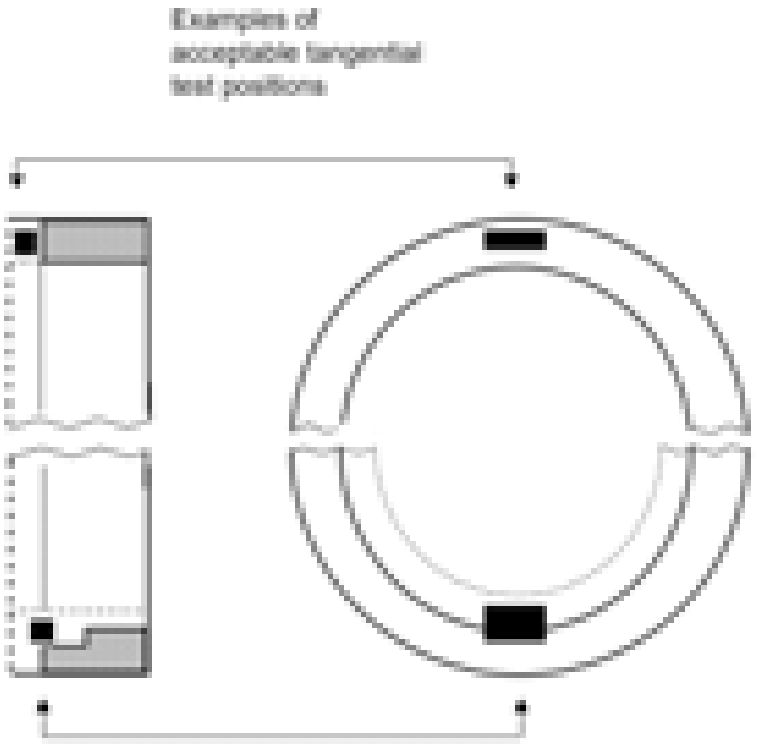


Fig.2.3.5i Forged rings

2.3.8 When a forging is subsequently divided into a number of components, all of which are heat treated together in the same furnace charge, for test purposes this may be regarded as one forging and the number of tests required is to be related to the total length and mass of the original multiple forging.

2.3.9 Except for components which are to be carburized or for hollow forgings where the ends are to be subsequently closed, test material is not to be cut from a forging until all heat treatment has been completed.

2.3.10 When forgings are to be carburized sufficient test material is to be provided for both preliminary tests in the as forged condition and for final tests after completion of carburizing.

For this purpose duplicate sets of test material are to be taken from positions as detailed in 2.3.6, except that irrespective of the dimensions or mass of the forging, tests are required from one position only and in the case of forgings with integral journals, are to be cut in a longitudinal direction.

This test material is to be machined to a diameter of D/4 or 60 [mm], whichever is less, where D is the finished diameter of the toothed portion.

For preliminary tests, one set of test material should be given a blank carburizing and it should undergo same heat treated cycle which the forged material will be subjected to.

For final acceptance tests, the second set of test material is to be blank carburized and heat treated along with the forgings which they represent.

At the discretion of the forge master or gear manufacturer test samples of larger cross section may be either carburized or blank carburized, but these are to be machined to the required diameter prior to the final quenching and tempering heat treatment.

Alternative procedures for testing of forgings which are to be carburized may be specially agreed with IRS.

2.3.11 Normalized forgings with mass upto 1000 [kg] each and quenched and tempered forgings with mass upto 500 [kg] each may be batch tested. A batch is to consist of forgings of similar shape and dimensions, made from the same heat of steel, heat treated in the

same furnace charge and with a total mass not exceeding 6 tonnes for normalized forgings and 3 tonnes for quenched and tempered forgings respectively.

2.3.12 A batch testing procedure may also be used for hot rolled bars. A batch is to consist of either:

i) material from the same rolled ingot or bloom provided that where this is cut into individual lengths, these are all heat treated in the same furnace charge, or

ii) bars of the same diameter and heat, heat treated in the same furnace charge and with a total mass not exceeding 2.5 tonnes.

2.3.13 The preparation of test specimens and the procedures used for mechanical testing are to comply with the relevant requirements of Pt.2, Ch.2. Unless otherwise agreed all tests are to be carried out in the presence of the Surveyor.

2.4 Mechanical properties

2.4.1 Table 2.4.1 and Table 2.4.2 gives the minimum requirements for yield stress, elongation, reduction of area and impact test energy values corresponding to different strength levels but it is not tended that these should necessarily be regarded as specific grades. Where it is proposed to use a steel with a specified minimum tensile strength intermediate to those given, corresponding minimum values for the other properties may be obtained by interpolation.

2.4.2 Forgings may be supplied to any specified minimum tensile strength selected within the general limits detailed in Table 2.4.1 and Table 2.4.2 but subject to any additional requirements of the relevant construction rules.

2.4.3 The mechanical properties are to comply with the requirements of Table 2.4.1 and Table 2.4.2 appropriate to the specified minimum tensile strength or, where applicable the requirements of the approved specification.

2.4.4 At the discretion of IRS hardness tests may be required in the following cases:

i) Gear forgings after completion of heat treatment and prior to machining the gear teeth:

The hardness is to be determined at four positions equally spaced around the circumference of the surface where teeth will subsequently be cut. Where the finished diameter of the toothed portion exceeds 2.5 [m], the above number of test positions is to be increased to eight. Where the width of a gear wheel rim forging exceeds 1.25 [m], the hardness is to be determined at eight positions at each end of the forging.

ii) Small crankshaft and gear forgings which have been batch tested:

In such cases at least one hardness test is to be carried out on each forging.

The results of hardness tests are to be reported and, for information purposes, typical Brinell hardness values are given in Table 2.4.2.

2.4.5 Hardness tests may also be required on forgings which have been induction hardened, nitrided or carburized. For gear forgings these tests are to be carried out on the teeth after, where applicable, they have been ground to the finished profile. The results of such tests including depth of hardening are to comply with the approved specifications. (See 1.5.6).

2.4.6 Where the result of a tensile test does not comply with the requirements, two additional tests may be taken. If satisfactory results are obtained from both of these additional tests the forging or batch of forgings is acceptable. If one or both retests fail the forging or batch of forgings is to be rejected.

2.4.7 Where the results from a set of three impact test specimens do not comply with the requirements an additional set of three impact test specimens may be taken provided that not more than two individual values are less than the required average value and of these not more than one is less than 70% of this average value. The results obtained are to be combined with the original results to form a new average which, for acceptance of the forgings or batch forgings, is to be not less than the required average value.

Additionally, for these combined results not more than two individual values are to be less than the required average value and of these not more than one is to be less than 70% of this average value.

2.4.8 The additional tests detailed in 2.4.6 and 2.4.7 are to be taken, preferably from material adjacent to the original tests, but alternatively from another test position or sample representative of the forging or hatch of forgings.

2.4.9 At the option of the manufacturer, when a forging or a batch of forgings has failed to meet the test requirements, it may be re- heat treated and re-submitted for acceptance tests.

Table 2.4.1 : Mechanical properties for hull steel forgings

Steel type	Tensile strength [1] R_m min. [N/mm^2]	Yield stress R_e min. [N/mm^2]	Elongation as min. %		Reduction of area Z min. %		Charpy V-notch impact test [2]		
							Test Temperature (°C)	Minimum average energy (J)	
			Long.	Tang.	Long.	Tang.		Long.	Tang.
C and C-Mn	400	200	26	19	50	35	0	27	18
	440	220	24	18	50	35			
	480	240	22	16	45	30			
	520	260	21	15	45	30			
	560	280	20	14	40	27			
	600	300	18	13	40	27			
Alloy	550	350	20	14	50	35			
	600	400	18	13	50	35			
	650	450	17	12	50	35			

1) The following ranges for tensile strength may be additionally specified:
specified minimum tensile strength : < 600 [N/mm^2] ≥ 600 [N/mm^2]
tensile strength range : 120 [N/mm^2] 150 [N/mm^2]
2) Special consideration may be given to alternative requirements for Charpy V-notch test, depending on design and application, and subject to agreement by IRS.

Table 2.4.2 : Mechanical properties for machinery steel forgings [2]

Steel type	Tensile strength [1] Rm min. [N/mm^2]	Yield stress Re min. [N/mm^2]	Elongation As min %		Reduction of area Z min. %		Hardness [3] (Brinell)	Charpy V-notch impact test [2], [4]		
								Test Temperature (°C)	Minimum average energy (J)	
			Long.	Tang.	Long.	Tang.			Long.	Tang.
C and C-Mn	400	200	26	19	50	35	110-150	Ambient Temperature [5]	27	18
	440	220	24	18	50	35	125-160			
	480	240	22	16	45	30	135-175			
	520	260	21	15	45	30	150-185			
	560	280	20	14	40	27	160-200			
	600	300	18	13	40	27	175-215			
	640	320	17	12	40	27	185-230			
	680	340	16	12	35	24	200-240			
	720	360	15	11	35	24	210-250			
	760	380	14	10	35	24	225-265			
Alloy	600	360	18	14	50	35	175-215			
	700	420	16	12	45	30	205-245			
	800	480	14	10	40	27	235-275			
	900	630	13	9	40	27	260-320			
	1000	700	12	8	35	24	290-365			
	1100	770	11	7	35	24	320-385			

Table 2.4.2 (Contd.)

[1)] The following ranges for tensile strength may be additionally specified:

specified minimum tensile strength :	< 900 [N/mm^2]	≥ 900 [N/mm^2]
tensile strength range :	150 [N/mm^2]	200 [N/mm^2]

[2)] For materials used for machinery exposed to sea water temperature, such as propeller shafts and shaft bolts, intended for ships with ice class notation HAT(B), HT(B), HM(B) and Ha(B), Charpy V-notch impact testing is to be carried out for all steel types at –10°C and the average energy value is to be minimum 20J (longitudinal test). One individual value may be less than the required average value provided that it is not less than 70% of this average value.

[3)] The hardness values are typical and are given for information purposes only.

[4)] Special consideration may be given to alternative requirements for Charpy V-notch test, depending on design and application, and subject to agreement by IRS.

[5)] Ambient Temperature (i.e. 23°C±5°C), which is specified in ISO 148-1:2016.

Section 3

Ferritic Steel Forgings for Low Temperature Service

3.1 Scope

3.1.1 The requirements for carbon-manganese and nickel steels suitable for low temperature service are detailed in this section. They are applicable to all forgings with material thickness up to and including 50 [mm] used for the construction of cargo tanks, storage tanks and process pressure vessels for liquefied gases and where the design temperature is less than 0°C, to forgings for the piping systems.

3.1.2 The requirements are also applicable to forgings for other pressure vessels and pressure piping systems where the use of steels with guaranteed impact properties at low temperatures is required.

3.2 Chemical composition

3.2.1 The chemical composition of ladle samples is, in general, to comply with the requirements given in Table 3.2.1 of Ch.3.

3.3 Heat treatment

3.3.1 Forgings are to be normalized, normalized and tempered or quenched and tempered in accordance with the approved specification.

3.4 Mechanical tests

3.4.1 At least one tensile and three V-notch impact test specimens are to be taken from each forging or each batch of forgings. Where the dimensions and shape allow, the test specimens are to be cut in a longitudinal direction.

3.4.2 The impact tests are to be carried out at a temperature appropriate to the type of steel and for the proposed application. Where forgings are intended for ships for liquefied gases the test temperature is to be in accordance with the requirements given in Table 5.4.1 of Ch.3, Sec.5.

3.4.3 The results of all tensile tests are to comply with the approved specification.

3.4.4 The average energy values for impact tests are also to comply with the approved specification and generally with the requirements of Ch.3, Sec.5. See also Ch.2.

3.4.5 For material thickness above 50 [mm], the material properties are to be agreed.

3.5 Pressure tests

3.5.1 When applicable, pressure tests are to be carried out in accordance with the requirements of the relevant construction Rules.

Section 4

Austenitic Stainless Steel Forgings

4.1 General

4.1.1 Forgings in austenitic stainless steels are acceptable for use in the construction of cargo tanks, storage tanks and piping systems for chemicals and liquefied gases. They may also be accepted for elevated temperature service in boilers.

4.1.2 Where it is proposed to use forgings in these types of steels, details of the chemical composition, heat treatment and mechanical properties are to be submitted for approval. These are to comply in general, with the requirements of Chapter 3, Section 9 for austenitic steel plates.

4.1.3 Unless otherwise specified, impact tests are not required for acceptance purposes. Where they are required tests are to be made on longitudinal specimens at minus 196°C and the minimum average energy requirements is to be 41J.

4.2 Mechanical properties for design purposes

4.2.1 Where austenitic stainless steel forgings are intended for service at elevated temperatures, the nominal values for the minimum one per cent proof stress at temperatures of 100°C and higher given in Table 4.2.1 may be used for design purposes. Verification of these values is not required except for material complying with a National or proprietary specification in which the elevated temperature properties proposed for design purposes are higher than those given in Table 4.2.1.

4.3 Non-destructive testing

4.3.1 Non-destructive testing is to be carried out in accordance with the requirements of Classification Notes “Guidelines for non-destructive testing of hull and machinery steel forgings” or as otherwise agreed between the manufacturer, purchaser and IRS.

4.4 Intergranular corrosion tests

4.4.1 Where corrosive conditions are anticipated in service, intergranular corrosion tests are required on forgings in Grades 304, 316 and 317. Such tests may not be required for Grades 304L, 316L, 321 and 347.

4.4.2 When an intergranular corrosion test is specified, it is to be carried out in accordance with the standard referred in Section 9.6.2 of Chapter 3.

Table 4.2.1 : Mechanical properties for design purposes : austenitic stainless steels

Grade	Nominal 1% proof stress [N/mm^2] at a temperature												
	100°C	150°C	200°C	250°C	300°C	350°C	400°C	450°C	500°C	550°C	600°C	650°C	700°C
304L	168	150	137	128	122	116	110	108	106	102	100	96	93
316L	177	161	149	139	133	127	123	119	115	112	110	107	105
316LN	238	208	192	180	172	166	161	157	152	149	144	142	138
321	192	180	172	164	158	152	148	144	140	138	135	130	124
347	204	192	182	172	166	162	159	157	155	153	151	-	-

End of Chapter

Chapter 6

Steel Pipes and Tubes

Contents

Section	
1	*General Requirements*
2	*Seamless Pressure Pipes*
3	*Welded Pressure Pipes*
4	*Boiler and Superheater Tubes*
5	*Tubes and Pipes for Low Temperature Services*
6	*Austenitic Stainless Steel Pressure Pipes*

Section 1

General Requirements

1.1 Scope

1.1.1 The requirements of this Chapter are applicable to boiler tubes, superheater tubes and pipes intended for use in the construction of boilers, pressure vessels and ship and machinery pressure piping systems.

1.1.2 In addition to specifying mechanical properties for the purpose of acceptance testing, these requirements give details of appropriate mechanical properties at elevated temperatures to be used for design purposes.

1.1.3 Except for pipes for Class III pressure systems (as defined in Pt.4, Ch.2) all pipes and tubes are to be manufactured and tested in accordance with the requirements of Ch.1 and 2 of this Part, the general requirements of this Section and the appropriate requirements given in Sec.1 to 5.

1.1.4 Steels, intended for the cargo and process piping systems of ships for liquefied gases where the design temperature is less than 0°C, are to comply with specific requirements of Sec.5.

1.1.5 Pipes and tubes, which comply with national or proprietary specifications may be accepted provided that these specifications give reasonable equivalence to the requirements of this Section or are otherwise specially approved for a specific application and provided that survey is carried out in accordance with Ch.1 of this Part.

1.1.6 At the discretion of the Surveyor, a modified testing procedure may be adopted for small quantities of materials. In such cases, these may be accepted on the manufacturer's declared chemical composition and hardness tests or other evidence of satisfactory properties.

1.1.7 Pipes for Class III pressure systems are to be manufactured and tested in accordance with the requirements of an acceptable national specification. The manufacturer's test certificate will be acceptable and is to be provided for each consignment of material. Forge butt welded pipes are not acceptable for certain applications as detailed in Pt.4, Ch.2 and Pt.5, Ch.23.

1.2 Manufacture

1.2.1 Pipes for Class I and II pressure systems, boilers and superheater tubes are to be manufactured at works approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2. The steel used is to be manufactured in accordance with Ch.3, Sec.1.

1.2.2 Unless a particular method is requested by the purchaser, pipes and tubes may be manufactured by any of the following methods:-

a) hot finished seamless;

b) cold finished seamless;

c) electric resistance or induction welded;

d) cold finished electric resistance or induction welded;

e) electric fusion welded.

1.2.3 Care is to be taken during manufacture that the pipe or tube surfaces coming in contact with any non-ferrous metals or their compounds are not contaminated to such an extent as could prove harmful during subsequent fabrication and operation.

1.3 Quality

1.3.1 All pipes and tubes are to have a workmanlike finish and are to be clean and free from such surface and internal defects as can be established by the specified tests.

1.3.2 All pipes and tubes are to be reasonably straight. The ends are to be cut nominally square with the axis of the pipes or tubes, and are to be free from excessive burrs.

1.3.3 The tolerances on the wall thickness and diameter of pipes and tubes are to be in accordance with an acceptable national / international standards.

1.4 Chemical composition

1.4.1 The requirements for the chemical composition of the ladle sample and the acceptable method of de-oxidation is to comply with the requirements detailed in the relevant Section of this Chapter.

1.5 Heat treatment

1.5.1 All pipes and tubes are to be supplied in the condition detailed in the relevant specific requirements.

1.6 Test material

1.6.1 Pipes and tubes are to be presented for test in batches. The size of a batch and the number of tests to be performed are dependent on the application.

1.6.2 Where heat treatment has been carried out, a batch is to consist of pipes or tubes of the same size, manufactured from the same type of steel and subjected to the same finishing treatment in a continuous furnace, or heat treated in the same furnace charge in a batch type furnace.

1.6.3 Where no heat treatment has been carried out, a batch is to consist of pipes or tubes of the same size manufactured by the same method from material of the same type of steel.

1.6.4 For pipes for class I pressure systems and boiler and superheater tubes, at least 2 per cent of the number of lengths in each batch is to be selected at random for the preparation of tests at ambient temperature.

1.6.5 For pipes for class II pressure systems, each batch is to contain not more than the number of lengths given in the following Table. Tests are to be carried out on at least one pipe selected at random from each batch or part thereof.

Outside diameter [mm]	Number of pipes in a batch
$\leq$ 323.9	200 pipes as made
> 323.9	100 pipes as made

1.7 Test specimens and testing procedures

1.7.1 The procedures for mechanical tests and the dimensions of the test specimens are to be in accordance with Ch.2.

1.8 Visual and non-destructive testing

1.8.1 All pipes for Class I and II pressure systems, boiler and superheater tubes are to be presented for visual examination and verification of dimensions. The manufacturer is to provide adequate lighting conditions to enable an internal and external examination of the pipes and tubes to be carried out.

1.8.2 For welded pipes and tubes the manufacturer is to employ suitable non-destructive methods for the quality control of the welds. It is preferred that this examination is carried out on a continuous basis.

1.9 Hydraulic tests

1.9.1 Each pipe and tube is to be subjected to a hydraulic test at the manufacturer's works.

1.9.2 The hydraulic test pressure is to be determined by the following formula, except that the maximum test pressure need not exceed 14 [N/mm^2].

$$P = \frac{2st}{D}$$

where,

P = test pressure [N/mm^2];

D = nominal outside diameter [mm];

t = nominal wall thickness [mm];

s = 80 per cent of the specified minimum yield stress [N/mm^2], for ferritic steels and 70 per cent of the specified minimum 1.0 per cent proof stress [N/mm^2] for austenitic steels. These relate to the values specified for acceptance testing at ambient temperature.

1.9.3 The test pressure is to be maintained for sufficient time to permit proof and inspection. Unless otherwise agreed, the manufacturer's certificate of satisfactory hydraulic test will be accepted. Where it is proposed to adopt a test pressure other than determined as in 1.9.2, the proposal will be subject to special consideration.

1.10 Rectification of defects

1.10.1 Surface imperfections may be removed by grinding provided that the thickness of the pipe or tube after dressing is not less than the required minimum thickness. The dressed area is to be blended into the contour of the tube.

1.10.2 By agreement with the Surveyor, the repair of minor defects by welding can be accepted. Welding procedures, including preheating, post weld heat treatment and inspection, are to be to the complete satisfaction of the Surveyor. In all cases, the area is to be tested by magnetic particle examination, or in case of austenitic steels, by liquid penetrant examination on completion of welding, heat treatment and surface grinding.

1.11 Identification

1.11.1 Pipes and tubes are to be clearly marked by the manufacturer in accordance with the requirements of Ch.1. The following details are to be shown on all materials which have been accepted:-

a) IR;

b) Manufacturer's name or trade mark;

c) Identification mark for the specification or grade of steel;

d) Identification number and/or initials which will enable the full history of the item to be traced;

e) The personal stamp of the Surveyor responsible for the final inspection.

1.11.2 It is recommended that hard stamping be restricted to the end face, but it may be accepted in other positions in accordance with national standards and practices.

1.12 Certification

1.12.1 The manufacturer is to provide the Surveyor with copies of test certificate or shipping statement for all material which has been accepted.

1.12.2 Each test certificate is to contain the following particulars:-

a) Purchaser's name and order number;

b) The yard number for which the material is intended, if known;

c) Address to which material is despatched;

d) Specification or the grade of material;

e) Description and dimensions;

f) Identification number and/or initials;

g) Cast number and chemical composition of ladle samples;

h) Mechanical test results, and results of the intercrystalline corrosion tests where applicable;

i) Condition of supply.

1.12.3 The chemical composition stated on the certificate is to include the content of all the elements detailed in the specific requirements. Where rimmed steel is supplied, this is to be stated on the certificate.

1.12.4 When steel is not produced at the pipe or tube mill, a certificate is to be supplied by the steelmaker stating the process of manufacture, the cast number and the ladle analysis.

1.12.5 The works at which the steel was produced must be approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

Section 2

Seamless Pressure Pipes

2.1 General

2.1.1 Following requirements are applicable for seamless pressure pipes in carbon, carbon-manganese and low alloy steels.

2.1.2 Where pipes are used for the manufacture of pressure vessel shells and headers, the requirements of forgings in Ch.5 are applicable where the wall thickness exceeds 40 [mm].

2.2 Manufacture and chemical composition

2.2.1 Tubes are to be manufactured by a seamless process and may be hot or cold finished.

2.2.2 The method of de-oxidation and the chemical composition of ladle samples are to comply with the appropriate requirements given in Table 2.2.1.

2.3 Heat treatment

2.3.1 Pipes are to be supplied in the condition given in Table 2.3.1.

Table 2.3.1 : Heat treatment

Type of steel	Condition of supply
Carbon and carbon-manganese	
Hot finished	Not finished[1]
	Normalized[2]
Cold finished	Normalized[2]
Alloy steels	
1 Cr $\frac{1}{2}$ Mo	Normalized and tempered
2 $\frac{1}{4}$ Cr 1 Mo	
Grade 410	Fully annealed
Grade 490	Normalized and tempered 650-750°C
$\frac{1}{2}$ Cr $\frac{1}{2}$ Mo $\frac{1}{4}$ V	Normalized and tempered

Notes:

1. Provided that the finishing temperature is sufficiently high to soften the material.
2. Normalized and tempered at the option of the manufacturer.

Table 2.2.1 : Chemical composition of seamless pressure pipes

Type of steel	Grade	Method of deoxi-dation	Chemical composition of ladle samples %											
			C	Si	Mn	S max	P max	Residual elements						
Carbon and carbon-manganese	320	Semi-killed or killed	≤ 0.16	-	0.40 - 0.70	0.050	0.050	Ni 0.30 max., Cr 0.25 max., Mo 0.10 max., Cu 0.30 max. Total 0.70 max.						
	360		≤ 0.17	≤ 0.35	0.40 - 0.80	0.045	0.045							
	410	Killed	≤ 0.21	≤ 0.35	0.40 - 1.20	0.045	0.045							
	460	Killed	≤ 0.22	≤ 0.35	0.80 - 1.40	0.045	0.045							
	490	Killed	≤ 0.23	≤ 0.35	0.80 - 1.50	0.045	0.045							
								Ni	Cr	Mo	Cu	Sn	V	Al
1 Cr $\frac{1}{2}$ Mo	440	Killed	0.10-0.18	0.10-0.35	0.40-0.70	0.040	0.040	0.30 max.	0.70-1.10	0.45-0.65	0.25 max.	0.03 max.	-	≤ 0.020
2 $\frac{1}{4}$ Cr 1 Mo	410 490	Killed	0.08-0.15	0.10-0.50	0.40-0.70	0.040	0.040	0.30 max.	2.0-2.5	0.90-1.20	0.25 max.	0.03 max.	-	≤ 0.020
$\frac{1}{2}$ Cr $\frac{1}{2}$ Mo $\frac{1}{4}$ V	460	Killed	0.10-0.18	0.10-0.35	0.40-0.70	0.040	0.040	0.30 max.	0.30-0.60	0.50-0.70	0.25 max.	0.03 max.	0.22 - 0.32	≤ 0.020

2.4 Mechanical tests

2.4.1 All pipes are to be presented in batches as defined in Sec.1.

2.4.2 Each pressure pipe selected for test is to be subjected to tensile and flattening or bend tests.

2.4.3 The results of all mechanical tests are to comply with the appropriate requirements given in Table 2.4.1.

2.5 Mechanical properties for design

2.5.1 Values for nominal minimum lower yield or 0.2 per cent proof stress at 50°C and higher are given in Table 2.5.1 and are intended for design purposes only. Verification of these values is not required, except for materials complying with national or proprietary specifications where the elevated temperature properties used for design are higher than those given in Table 2.5.1.

2.5.2 In such cases, at least one tensile test at the proposed design or other agreed temperature is to be made on each cast. The test specimen is to be taken from material adjacent to that used for tests at ambient temperature and tested in accordance with the procedures given in Ch.2. If tubes or pipes of more than one thickness are supplied from one cast, the test is to be made on the thickest tube or pipe.

2.5.3 As an alternative to 2.5.2, a manufacturer may carry out an agreed comprehensive test program for a stated grade of steel to demonstrate that the specified minimum mechanical properties at elevated temperatures can be consistently obtained. This test program is to be carried out under the supervision of the Surveyors, and the results submitted for assessment and approval. When a manufac-turer is approved on this basis, tensile tests at elevated temperatures are not required for acceptance purposes, but at the discretion of the Surveyors, occasional check tests of this type may be requested.

2.5.4 Values for the estimated average stress to rupture in 100,000 hours are given in Table 2.5.2 and may be used for design purposes.

Table 2.4.1 : Mechanical properties for acceptance purposes : Seamless pressure pipes (maximum wall thickness 40 mm)

Type of steel	Grade	Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Elongation on $5.65\sqrt{S_o}$% min.	Flattening test constant C	Bend test diameter of former (t = thickness)
Carbon and carbon-manganese	320	195	320-440	25	0.10	4t
	360	215	360-480	24	0.10	4t
	410	235	410-530	22	0.08	4t
	460	265	460-580	21	0.07	4t
	490	285	490-610	21	0.07	4t
1 Cr $\frac{1}{2}$ Mo	440	275	440-590	22	0.07	4t
2 $\frac{1}{4}$ Cr 1 Mo	410 [1]	135	410-560	20	0.07	4t
	490 [2]	275	490-640	16	0.07	4t
$\frac{1}{2}$ Cr $\frac{1}{2}$ Mo $\frac{1}{4}$ V	460	275	460-610	15	0.07	4t

Notes:

1 Annealed condition
2 Normalized and tempered condition

Table 2.5.1 : Mechanical properties for design purposes : Seamless pressure pipes

Type of steel	Grade	Nominal minimum lower yield or 0.2% proof stress [N/mm^2] Temperature °C											
		50	100	150	200	250	300	350	400	450	500	550	600
Carbon and carbon-manganese	320	172	168	158	147	125	100	91	88	87	-	-	-
	360	192	187	176	165	145	122	111	109	107	-	-	-
	410	217	210	199	188	170	149	137	134	132	-	-	-
	460	241	234	223	212	195	177	162	159	156	-	-	-
	490	256	249	237	226	210	193	177	174	171	-	-	-
1 Cr $\frac{1}{2}$ Mo	440	254	240	230	220	210	183	169	164	161	156	151	-
2 $\frac{1}{4}$ Cr 1 Mo	410 [1]	121	108	99	92	85	80	76	72	69	66	64	62
	490 [2]	268	261	253	245	236	230	224	218	205	189	167	145
$\frac{1}{2}$ Cr $\frac{1}{2}$ Mo $\frac{1}{4}$ V	460	266	259	248	235	218	192	184	177	168	155	148	-

Notes:

1 Annealed condition

2 Normalized and tempered condition

Table 2.5.2 : Mechanical properties for design purposes : Seamless pressure pipes - estimated values for stress to rupture in 100 000 hours (units [N/mm^2])

Temperature °C	Carbon and carbon-manganese		1 Cr ½ Mo	2¼ Cr 1 Mo		½ Cr ½ Mo ¼ V
	Grade 320, 360, 410	Grade 460, 490	Grade 440	Grade 410 Annealed	Grade 490 Normalized and tempered (see Note)	Grade 460
380	171	227	-	-	-	-
390	155	203	-	-	-	-
400	141	179	-	-	-	-
410	127	157	-	-	-	-
420	114	136	-	-	-	-
430	102	117	-	-	-	-
440	90	100	-	-	-	-
450	78	85	-	196	221	-
460	67	73	-	182	204	-
470	57	63	-	168	186	-
480	47	55	210	154	170	218
490	36	47	177	141	153	191
500	-	41	146	127	137	170
510	-	-	121	115	122	150
520	-	-	99	102	107	131
530	-	-	81	90	93	116
540	-	-	67	78	79	100
550	-	-	54	69	69	85
560	-	-	43	59	59	72
570	-	-	35	51	51	59
580	-	-	-	44	44	46

Note : When tempering temperature exceeds 750°C, the values for Grade 410 are to be used.

Section 3

Welded Pressure Pipes

3.1 General

3.1.1 Following requirements are applicable to welded pressure pipes in carbon, carbon-manganese and low alloy steels.

3.1.2 Where it is proposed to use submerged arc longitudinally welded pipes, details of the specification are to be submitted.

3.2 Manufacture and chemical composition

3.2.1 Pipes are to be manufactured by the electric or induction welding process and, if required, may be subsequently hot reduced or cold finished.

3.2.2 The method of de-oxidation and the chemical composition of ladle samples are to comply with the appropriate requirements given in Table 3.2.1.

3.2.3 Where rimmed steel is used, the strips are to be rolled in single widths and not slit longitudinally, except to trim the edges.

Table 3.2.1 : Chemical composition of welded pressure pipes

Type of steel	Grade	Method of deoxi-dation	Chemical composition of ladle samples %										
			C	Si	Mn	S max.	P max.	Residual elements					
Carbon and carbon-manga-nese	320	Any method (see Note)	≤ 0.16	-	0.30-0.70	0.050	0.050	Ni 0.30 max., Cr 0.25 max., Mo 0.10 max., Cu 0.30 max. Total 0.70 max.					
	360		≤ 0.17	≤ 0.35	0.40-1.00	0.045	0.045						
	410	Killed	≤ 0.21	≤ 0.35	0.40-1.20	0.045	0.045						
	460	Killed	≤ 0.22	≤ 0.35	0.80-1.40	0.045	0.045						
								Ni	Cr	Mo	Cu	Sn	Al
1 Cr $\frac{1}{2}$ Mo	440	Killed	0.10-0.18	0.10-0.35	0.40-0.70	0.040	0.040	0.30 max.	0.70-1.10	0.45-0.65	0.25 max.	0.03 max.	≤ 0.020
Note : For rimmed steels the carbon content may be increased to 0.19% max.													

3.3 Heat treatment

3.3.1 Pipes are to be supplied in the heat treated condition given in Table 3.3.1.

Table 3.3.1 : Heat treatment : Welded pressure pipes

Type of steel	Condition of supply
Carbon and carbon-manganese	Normalized (Normalized and tempered at the option of the manufacturer)
1 Cr $\frac{1}{2}$ Mo	Normalized and tempered

3.4 Mechanical tests

3.4.1 All pipes are to be presented in batches as defined in Sec.1.

3.4.2 Each pressure pipe selected for test is to be subjected to tensile and flattening or bend tests.

3.4.3 The results of all mechanical tests are to comply with the appropriate requirements given in Table 3.4.1.

3.5 Mechanical properties for design

3.5.1 The mechanical properties at elevated temperature for carbon and carbon-manganese steels in Grades 320 [N/mm^2] to 460 [N/mm^2] and 1 Cr $\frac{1}{2}$ Mo steel can be taken from the appropriate Tables in Sec.2.

3.5.2 Where rimmed steel is used, the design temperature is limited to 400°C.

Table 3.4.1 : Mechanical properties for acceptance purposes : Welded pressure pipes

Type of steel	Grade	Yield stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation on 5.65$\sqrt{S_o}$% minimum	Flattening test constant C
Carbon and carbon-manganese	320	195	320 - 440	25	0.10
	360	215	360 - 480	24	0.10
	410	235	410 - 530	22	0.08
	460	265	460 - 580	21	0.07
1 Cr $\frac{1}{2}$ Mo	440	275	440 - 590	22	0.07

Section 4

Boiler and Superheater Tubes

4.1 General

4.1.1 The following requirements are applicable for boiler and superheater tubes in carbon, carbon-manganese and low alloy steels.

4.1.2 Austenitic stainless steels may also be used for this type of service. Where such applications are proposed, details of the chemical composition, heat treatment and mechanical properties are to be submitted for consideration and approval.

4.2 Manufacture and chemical composition

4.2.1 Tubes are to be seamless or welded and are to be manufactured in accordance with the requirements of Sec.2 and 3 respectively.

4.2.2 The method of de-oxidation and the chemical composition of ladle samples are to comply with the requirements given in Table 2.2.1 or Table 3.2.1, as appropriate.

4.3 Heat treatment

4.3.1 All tubes are to be supplied in accordance with the requirements given in Table 2.3.1 or Table 3.3.1 as appropriate, except that 1 Cr $\frac{1}{2}$ Mo steel may be supplied in the normalized only condition when the carbon content does not exceed 0.15 per cent.

4.4 Mechanical tests

4.4.1 Tubes are to be presented for test in batches as defined in Sec.1.

4.4.2 Each boiler and superheater tube selected for test is to be subjected to at least the following:

a) Tensile test;

b) Flattening or bending tests at the manufacturer's option;

c) Expanding or flanging tests at the manufacturer's option.

4.4.3 The results of all mechanical tests are to comply with the appropriate requirements given in Table 4.4.1.

4.5 Mechanical properties for design

4.5.1 The mechanical properties at elevated temperature for carbon and carbon-manganese steels in Grades 320 [N/mm^2] to 460 [N/mm^2], 1 Cr $\frac{1}{2}$ Mo and 2 $\frac{1}{4}$ Cr 1 Mo steels can be taken from the appropriate Tables in Sec.2.

4.5.2 Where rimmed steel is used, the design temperature is limited to 400°C.

Table 4.4.1 : Mechanical properties for acceptance purposes : Boilers and superheater tubes

Type of steel	Grade	Yield stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation on $5.65\sqrt{S_o}$ % minimum	Flattening test constant C	Bend test diameter of former (t=thickness)	Drift expanding and flanging test minimum % increase in outside diameter		
							Ratio : Inside diameter / Outside diameter		
							≤ 0.6	> 0.6 ≤ 0.8	> 0.8
Carbon and carbon-manganese	320	195	320-440	25	0.10	4t	12	15	19
	360	215	360-480	24	0.10	4t	12	15	19
	410	235	410-530	22	0.08	4t	10	12	17
	460	265	460-580	21	0.07	4t	8	10	15
1 Cr $\frac{1}{2}$ Mo	440	275	440-590	22	0.07	4t	8	10	15
2 $\frac{1}{4}$ Cr 1 Mo	410 [1]	135	410-560	20	0.07	4t	8	10	15
	490 [2]	275	490-640	16	0.07	4t	8	10	15

Notes:

1. Annealed condition
2. Normalized and tempered condition

Section 5

Tubes and Pipes for Low Temperature Services

5.1 Scope

5.1.1 This Section gives the requirements for seamless and welded carbon, carbon-manganese and nickel alloy steel pipes not exceeding 25 [mm] in thickness intended for use in liquefied gas piping systems where the design temperature is lower than 0°C and also for other pressure piping systems where guaranteed impact properties at low temperature is required.

5.2 Manufacture

5.2.1 The pipes are to be manufactured seamless or by a welding process, and may be hot or cold finished.

5.3 Chemical composition

5.3.1 The chemical composition of ladle samples is in general to comply with the requirements given in Table 5.3.1. Steels for the production of tubes and pipes are to be killed.

Table 5.3.1 : Chemical composition

Type of steel	Grade	Method of deoxi-dation	Chemical composition of ladle sample %							
			C max.	Si	Mn	P max.	S max.	Ni	Al met	Residual Elements
Carbon	360	Fully killed	0.17	0.10-0.35	0.40-1.00	0.045	0.045	-	≥ 0.015 see note	Cr 0.25 Cu 0.30 Mo 0.10 Ni 0.30 Total 0.70
Carbon-manganese	410 & 460		0.2	0.10-0.35	0.60-1.40	0.045	0.045	-	≥ 0.015 see note	
3.5 Ni	440		0.15	0.15-0.35	0.30-0.90	0.040	0.040	3.25-3.75	-	Cr 0.25 Cu 0.30 Mo 0.10 Total 0.60
9 Ni	690		0.3	0.15-0.30	0.30-0.90	0.040	0.040	8.50-9.50	-	

Note : Where a minimum Al_{met} of 0.015% is specified, the determination of the total aluminium is acceptable provided that the result is not less than 0.018%.

5.4 Heat treatment

5.4.1 Pipes are to be supplied in the condition given in Table 5.4.1.

Table 5.4.1 : Heat treatment

Type of Steel	Condition of Supply
Carbon and Carbon-manganese	Hot finished Normalized Normalized and tempered
3.5 Ni	Normalized Normalized and tempered
9 Ni	Double normalized and tempered Quenched and tempered

5.5 Mechanical tests

5.5.1 All pipes are to be presented for test in batches as defined in Sec.1 pressure piping systems.

5.5.2 At least two percent of the number of lengths in each batch is to be selected at random for the preparation of the tests.

5.5.3 Each pipe or tube selected for test is to be subjected to following tests:

- Seamless pipes and tubes:
  - one tensile test
  - one set of impact tests
  - one flattening test or bend test or ring tensile test
  - one drift or one ring expanding test where appropriate.
- Welded tubes and pipes:
  - one tensile test on the base material
  - one tensile test on the weld for pipes with D ≥ 508 [mm]
  - one set of impact tests
  - two flattening tests or bend tests or one ring tensile test (ERW and IW)
  - one drift or one ring expanding test (ERW and IW) -two bend tests (SAW).

5.5.4 Ring tensile test may be carried out in conformity with ISO 8495 or other equivalent standard.

5.5.5 The impact tests are to consist of a set of three Charpy V-notch test specimens cut in the longitudinal direction with the notch perpendicular to the original surface of the pipe. The dimension of the test specimens are to be in accordance with the requirements of Ch.2. Impact testing is not required for wall thickness below 6 [mm].

5.5.6 The impact values are to be determined at the lowest test temperature specified for the steel grade and the wall thickness in question.

5.5.7 The results of all mechanical tests are to comply with the appropriate values given in Table 5.5.1.

5.5.8 The energy value from a set of three Charpy V-notch impact test specimens is not to be lower than the required average value given in Table 5.5.1. One individual value may be less than the required average value provided that it is not less than 70 per cent of this average value.

Table 5.5.1 : Mechanical properties for acceptance purposes

Type of steel	Grade	Yield stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation on 5.65 $\sqrt{S_o}$% min.	Flattening test constant C	Bend test diameter of former (t=thickness)	Charpy V-notch impact tests	
							Test temperature °C	Average energy J minimum
Carbon	360	210	360-480	24	0.10	4t	-40	27
Carbon-manganese	410	235	410-530	22	0.08	4t	-50	27
	460	260	460-580	21	0.07		-50	27
3.5 Ni	440	245	440-590	16	0.08	4t	-100	27
9 Ni	690	510	690-840	15	0.08	4t	-196	39

Section 6

Austenitic Stainless Steel Pressure Pipes

6.1 Scope

6.1.1 This section gives the requirements for austenitic stainless steel pipes suitable for use in the construction of the piping systems for chemicals and for liquefied gases where the design temperature is not less than –165°C.

6.1.2 Austenitic stainless steels are also suitable for service at elevated temperatures. Where such applications are proposed, details of the chemical composition, heat treatment and mechanical properties are to be submitted for consideration and approval. See also Pt.4, Ch.2, 1.9.5.

6.1.3 Where it is intended to supply seamless pipes in the direct quenched condition, a programme of tests for approval is to be carried out under the supervision of the Surveyors, and the results are to be to the satisfaction of IRS.

6.2 Manufacture and chemical composition

6.2.1 Pipes are to be manufactured by a seamless or a continuous automatic electric fusion welding process.

6.2.2 Welding is to be in a longitudinal direction, with or without the addition of filler metal.

6.2.3 The chemical composition of the ladle samples is to comply with the appropriate requirements of Table 6.2.1.

Table 6.2.1 : Chemical composition										
Type of steel	**Grade**	**Chemical composition of ladle sample %**								
		C max.	**Si**	**Mn**	**P max.**	**S max.**	**Cr**	**Mo**	**Ni**	**Others**
304L	490	0.03	<1.00	<2.00	0.045	0.030	17.0-19.0	-	9.0-13.0	-
316L	490	0.03	<1.00	<2.00	0.045	0.030	16.0-18.5	2.0-3.0	11.0-14.5	-
321	510	0.08	<1.00	<2.00	0.045	0.030	17.0-19.0	-	9.0-13.0	Ti $\geq$ 5 xC $\leq$ 0.80
347	510	0.08	<1.00	<2.00	0.045	0.030	17.0-19.0	-	9.0-13.0	Nb$\geq$10 x C$\leq$1.00

6.3 Heat treatment

6.3.1 Pipes are generally to be supplied by the manufacturer in the solution treated condition over their full length.

6.3.2 Alternatively, seamless pipes may be direct quenched immediately after hot forming, while the temperature of the pipes is not less than the specified minimum solution treatment temperature.

6.4 Mechanical tests

6.4.1 All pipes are to be presented in batches as defined in Section 1 for Class I and II piping systems.

6.4.2 Each pipe selected for test is to be subjected to tensile and flattening or bend tests.

6.4.3 The results of all mechanical tests are to comply with the appropriate requirements given in Table 6.4.1.

Where the design temperature is less than –105°C, impact tests are to be carried out on a set of three Charpy V-notch specimens. The tests are to be made on longitudinal specimens at –196°C and the average energy is to be not less than 41 Joules.

Table 6.4.1 : Mechanical properties for acceptance purposes							
Type of steel	**Grade**	**0.2% proof stress [N/mm^2] (see Note)**	**1.0% proof stress [N/mm^2]**	**Tensile strength [N/mm^2]**	**Elongation on 5.65 $\sqrt{S_o}$ % minimum**	**Flattening test constant**	**Bend test diameter of former (t=thickness)**
304L	490	175	205	490-690	30	0.09	3t
316L	490	185	215	490-690	30	0.09	3t
321	510	195	235	510-710	30	0.09	3t
347	510	205	245	510-710	30	0.09	3t
Note : The 0.2% proof stress values given for information purposes and unless otherwise agreed are not required to be verified by test.							

6.5 Intergranular corrosion tests

6.5.1 For materials used for piping systems for chemicals, intercrystalline corrosion tests are to be carried out on one per cent of the number of pipes in each batch, with a minimum of one pipe.

6.5.2 For pipes with an outside diameter not exceeding 40 [mm], the test specimens are to consist of a full cross section. For larger pipes, the test specimens are to be cut as circumferential strips of full wall thickness and having a width of not less than 12.5 [mm]. In both cases, the total surface areas is to be between 15 and 35 [cm^2].

6.5.3 When required, one test of this type is to be carried out for each tensile test. The testing is to be carried out in accordance with ASTM A262, practice E, copper-copper sulphate-sulphuric acid or another recognized standard. The bent specimen is to be free from cracks indicating the presence of intergranular attack. The material for the test is to be taken adjacent to that for the tensile test.

6.5.4 After immersion, the full cross-section test specimens are to be subjected to a flattening test in accordance with the requirements of Chapter 2. The strip test specimens are to be subjected to a bend test through 90° over a mandrel of diameter equal to twice the thickness of the test specimen.

6.6 Fabricated pipework

6.6.1 Fabricated pipework is to be produced from material manufactured in accordance with 6.2, 6.3, 6.4 and 6.5.

6.6.2 Welding is to be carried out in accordance with an approved and qualified procedure by suitably qualified welders.

6.6.3 Fabricated pipework may be supplied in the as-welded condition without subsequent solution treatment provided that welding procedure tests have demonstrated satisfactory material properties including resistance to intercrystalline corrosion.

6.6.4 In addition, butt welds are to be subjected to 5 per cent radiographic examination for Class I and 2 per cent for Class II pipes.

6.6.5 Fabricated pipework in the as-welded condition and intended for systems located on deck is to be protected by a suitable corrosion control coating.

End of Chapter

Chapter 7

Iron Castings

Contents

Section

1 *General Requirements*

Section 1

General Requirements

1.1 Scope

1.1.1 This Chapter gives the requirements for both grey and spheroidal or nodular graphite iron castings intended for ship and machinery construction.

1.1.2 All important iron castings, as defined in the relevant parts of the Rules dealing with design and construction, are to be manufactured and tested in accordance with the requirements of Ch.1 and 2 and the requirements given in the following paragraphs.

1.1.3 As an alternative to 1.1.2, castings which comply with National or Proprietary specifications may be accepted, provided that such specifications give reasonable equivalence to these requirements or otherwise are specially approved or required by IRS.

1.1.4 Where small castings are produced in large quantities, the manufacturer may adopt alternative procedure for testing and inspection, subject to the approval of IRS.

1.1.5 These requirements are applicable only to castings where the design and acceptance tests are related to mechanical properties at ambient temperature. For other applications additional requirements may be necessary, especially when the castings are intended for service at low or elevated temperatures.

1.2 Manufacture

1.2.1 All castings, as designated in1.1.2, are to be manufactured at foundries approved by IRS.

1.2.2 Suitable mechanical methods are to be employed for the removal of surplus material from the castings. Thermal cutting processes are not acceptable, except as a preliminary operation to the mechanical methods.

1.2.3 Where castings of the same type are regularly produced in quantity, the manufacturer is to make any tests necessary to prove the quality of the prototype castings and is also to make periodical examinations to verify the continued efficiency of the manufacturing technique. The Surveyor is to be given the opportunity to witness these tests.

1.3 Quality of castings

1.3.1 Castings are to be free from surface or internal defects which could be prejudicial to their proper application in service. The surface finish is to be in accordance with good practice and any specific requirements of the approved plan.

1.4 Chemical composition

1.4.1 The chemical composition of the iron used is left to the discretion of the manufacturer, who is to ensure that it is suitable to obtain the mechanical properties specified for the castings.

1.5 Heat treatment

1.5.1 Except as required by 1.5.2, castings may be supplied in either the as cast or heat treated condition.

1.5.2 For some applications, such as high temperature service or where dimensional stability is important, castings may be required to be given a suitable tempering or stress relieving heat treatment. This is to be carried out after any refining heat treatment and before machining. The special qualities with 350 [N/mm^2] and 400 [N/mm^2] nominal tensile strength and impact test are to undergo ferritizing heat treatment.

1.5.3 Where it is proposed to locally harden the surface of castings, full details of the proposed procedure and specifications are to be submitted for approval by IRS.

1.6 Mechanical tests

1.6.1 Separately cast test samples are to be used unless otherwise agreed between the manufacturer and the purchaser. The test samples are generally to be one of the standard types detailed in Fig.1.6.1, Fig.1.6.2 and Fig.1.6.3 with a thickness of 25 [mm]. Test samples of dimensions, other than as detailed in Fig.1.6.1 to Fig.1.6.3 may, however, be specially required for some components. For grey cast iron the test samples are to be in the form of cylindrical bars of 30 [mm] diameter and of suitable length. When two or more test samples are cast simultaneously in a single mould, the bars are to be at least 50 [mm] apart as indicated in Fig.1.6.4.

1.6.2 Integrally cast samples may be used when a casting is more than 20 [mm] thick and its mass exceeds 200 [kgs] subject to agreement between the manufacturer and the purchaser. The type and location of the test sample are to be selected to provide approximately the same cooling conditions as for the casting it represents.

1.6.3 At least one test sample is to be provided for each casting or batch of castings. A batch consists of castings poured from a single ladle of metal provided they are all of similar type and dimensions. A batch should not normally exceed two tonnes of fettled castings and a single casting will constitute a batch if its mass is two tonnes or more.

1.6.4 For continuous melting of same grade of cast iron in large tonnages the mass of the batch may be increased to the output of two hours of pouring. If production is carefully monitored by systematic checking of the melting process, such as chill testing, chemical analysis or thermal analysis, test samples may be taken at longer intervals.

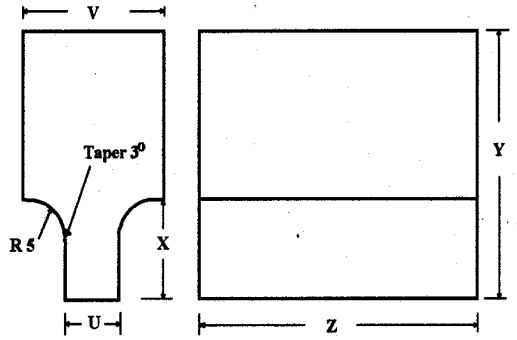
1.6.5 For large castings where more than one ladle of treated metal is used, additional test samples are to be provided so as to be representative of each ladle used.

1.6.6 All test samples are to be suitably marked to identify them with the castings which they represent.

1.6.7 Where castings are supplied in the heat treated condition, the test samples are to be heat treated together with the castings which they represent.

1.6.8 The test samples are to be cast in moulds made from the same type of material as used for the castings and are not to be stripped from the moulds until the metal temperature is below 500°C.

1.6.9 One tensile test specimen is to prepared from each test sample. The dimensions of the test specimens and the testing procedures used are to be in accordance with Ch.2.



Standardsample		Alternative samples, when required		
U	25	12	50	75
V	55	40	90	125
X	40	30	60	65
Y	100	80	150	185
Z	To suit testing machine (approx. 150 [mm])			

Fig.1.6.1 : Standard test specimen

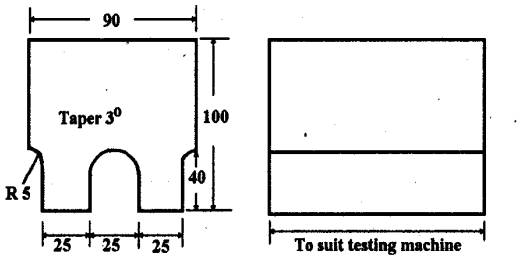
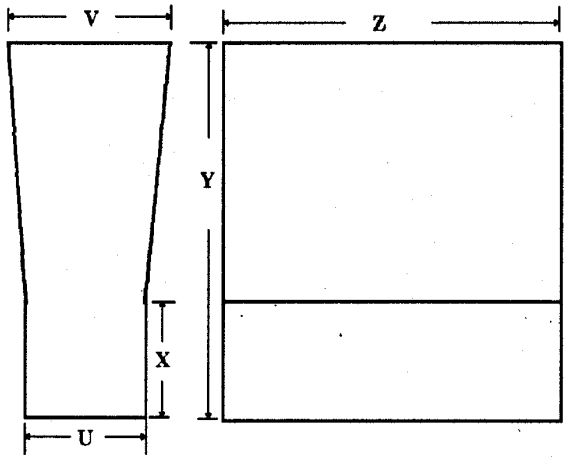


Fig.1.6.2 : Standard test specimen



Standardsample		Alternative samples, when required		
U	25	12	50	75
V	55	40	90	125
X	40	25	50	65
Y	140	135	150	175
Z	To suit testing machine (approx. 150 [mm])			

Fig.1.6.3 : Standard test specimen

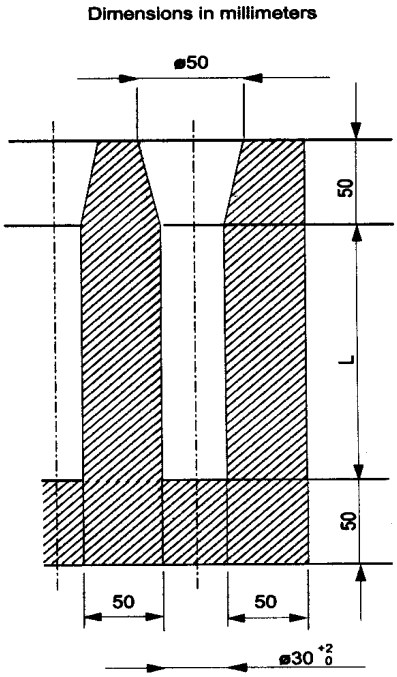


Fig.1.6.4 : Test sample for grey cast iron

1.7 Mechanical properties

1.7.1 For grey iron castings, only the tensile strength is to be determined and the results obtained are to comply with the minimum value specified for the castings being supplied. The specified minimum tensile strength is to be not less than 200 [N/mm^2] and not more than 350 [N/mm^2]. The fractured surfaces of all tensile test specimens are to be granular and grey in appearance.

1.7.2 For spheroidal or nodular graphite iron castings the tensile strength and elongation are to be determined. The results of all tests are to comply with the requirements of Table 1.7.1, but subject to any additional requirements of the relevant construction rules. Typical ranges of hardness values are also given in Table 1.7.1 and are intended for information purposes.

1.7.3 Retest requirements for tensile tests are to be in accordance with 1.10 of Chapter 1.

1.8 Visual and non-destructive examination

1.8.1 All castings are to be cleaned and adequately prepared for examination. The surfaces are not to be hammered, peened or treated in any way which may obscure defects.

Table 1.7.1 : Mechanical properties for acceptance purposes (spheroidal or nodular graphite iron)

Specified Minimum Tensile Strength [N/mm²]		0.2% proof stress (see Note) [N/mm²]	Elongation on $5.65\sqrt{S_o}$ % min.	Typical Hardness Brinell (see para 1.7.2)	Impact Energy		Typical structure of matrix (see para 1.9.2)
					Test Temp. °C	J Min.[2]	
Ordinary qualities	370	230	17	120 - 180	-	-	Ferrite
	400	250	12	140 - 200	-	-	Ferrite
	500	320	7	170 - 240	-	-	Ferrite/Pearlite
	600	370	3	190 - 270	-	-	Ferrite/Pearlite
	700	420	2	230 - 300	-	-	Pearlite
	800	480	2	250 - 350	-	-	Pearlite or Tempered Structure
Special qualities	350	220	22[3]	110 - 170	20	17(14)	Ferrite
	400	250	18[3]	140 - 200	20	14(11)	Ferrite

Notes:

1. For intermediate values of specified minimum tensile strength, the minimum values for 0.2% proof and elongation may be obtained by interpolation.
2. The average value measured on 3 Charpy V-notch specimens. One result may be below the average value but not less than the minimum shown in brackets.
3. In the case of integrally cast samples, the elongation may be 2 percentage points less.

1.8.2 Before acceptance, all castings are to be visually examined including, where applicable, the examination of internal surfaces. Unless otherwise agreed the verification of dimensions is the responsibility of the manufacturer.

1.8.3 Supplementary examination of castings by suitable non-destructive testing procedures is generally not required except in circumstances where there is reason to suspect the soundness of the casting.

1.8.4 When required by the relevant construction Rules, castings are to be pressure tested before final acceptance.

1.8.5 Cast crankshafts are to be subjected to a magnetic particle inspection. Crack like indications are not permitted.

1.9 Metallographic examination

1.9.1 For spheroidal or nodular graphite iron castings, a representative sample from each ladle of treated metal is to be prepared for metallographic examination. These samples may conveniently be taken from the tensile test specimens but alternative arrangement for the provision of the samples may be adopted provided that they are taken from the ladle towards the end of the casting period.

1.9.2 Examination of the samples is to show that at least 90 per cent of the graphite is in a dispersed spheroidal or nodular form. Details of the typical matrix structure are given in Table 1.7.1 and are intended for information purposes only.

1.10 Rectification of defective castings

1.10.1 At the discretion of the Surveyor, small surface blemishes may be removed by local grinding.

1.10.2 Subject to the prior approval of the Surveyor, castings containing local porosity may be rectified by impregnation with a suitable plastic filler, provided that the extent of the porosity is such that it does not adversely affect the strength of the castings.

1.10.3 Repairs by welding are generally not permitted, but may be considered in special circumstances. In such cases, full details of

the proposed repair procedure are to be submitted for approval prior to commencement of the proposed rectification.

1.11 Identification of castings

1.11.1 The manufacturer is to adopt a system of identification which will enable all finished castings to be traced to the original ladle of treated metal and the Surveyor is to be given full facilities for so tracing the castings when required.

1.11.2 Before acceptance, all castings which have been tested and inspected with satisfactory results are to be clearly marked by the manufacturer with the following particulars:-

a) Grade of cast iron;

b) Identification number, or other marking which will enable the full history of the casting to be traced;

c) Manufacturer's name or trade mark;

d) IR and the abbreviated name of the local office of IRS;

e) Personal stamp of the Surveyor responsible for inspection;

f) Where applicable, test pressure;

g) Date of final inspection.

1.11.3 Where small castings are manufactured in large numbers, modified arrangements for identification may be specially agreed with IRS.

1.12 Certification

1.12.1 The manufacturer is to provide the Surveyor with a written statement giving the following particulars for each casting or batch of castings which has been accepted:-

a) Purchaser's name and order no;

b) Description of castings and quality of cast iron;

c) Identification number;

d) Results of mechanical tests;

e) Where applicable, details of heat treatment;

f) Where specially required, the chemical analysis of the ladle sample;

g) Where applicable, test pressure.

End of Chapter

Chapter 8

Copper Alloys

Contents

Section

1 *General Requirements*

2 *Castings for Valves and Fittings*

3 *Castings for Propellers*

4 *Tubes*

Section 1

General Requirements

1.1 Scope

1.1.1 The Rules in this Chapter apply to copper alloys used in castings for valves and fittings, propeller castings and tubes.

1.1.2 When required by the relevant parts of the Rules, dealing with design and construction, tubes and castings are to be manufactured and tested in accordance with the appropriate requirements of Ch.1 and 2 and the requirements of this Chapter.

1.1.3 Alternatively, tubes and castings which comply with National or proprietary specifications may be accepted provided these specifications give reasonable equivalence to the requirements of this Chapter and provided that survey is carried out in accordance with the requirements of Ch.1.

1.1.4 Where it is proposed to use an alloy which is not specified in this Chapter, details of chemical composition, heat treatment and mechanical properties are to be submitted for approval.

Section 2

Castings for Valves and Fittings

2.1 Scope

2.1.1 Following requirements make provision for copper alloy castings for valves, liner bushes and other fittings intended for use in ship and machinery construction.

2.2 Manufacture

2.2.1 Approval of Works, as required by Ch.1, for the manufacture of castings, covered by this Section, is not required.

2.3 Quality of castings

2.3.1 All castings are to be free from surface or internal defects, which could be prejudicial to their proper application in service.

2.4 Chemical composition

2.4.1 The chemical composition is to comply with the appropriate requirements of Table 2.4.1.

2.4.2 Where a cast is wholly prepared from ingots for which an analysis is already available, and provided that no significant alloy additions are made during melting, the ingot maker's certified analysis may be accepted subject to occasional check tests as requested by the Surveyors.

Table 2.4.1 : Chemical composition

Designation	Chemical composition %								
	Cu	**Sn**	**Zn**	**Pb**	**Ni**	**Mn**	**P**	**Fe**	**Al**
90/10 Cu-Sn Phosphor-bronze	Remainder	9.0-11.0	0.5 max.	0.75 max.	0.5 max.	-	0.50 max.	-	-
85/5/10 Leaded bronze	Remainder	4.0-6.0	2.0 max.	9.0-11.0	2.0 max.	-	0.10 max.	-	-
88/10/2 Gunmetal	Remainder	8.5-11.0	1.0-3.0	1.5 max.	1.0 max.	-	-	-	-
87/7/3/3 Leaded Gunmetal	Remainder	6.0-8.0	1.5-3.0	2.5-3.5	2.0 max.	-	-	-	-
85/5/5/5 Leaded Gunmetal	Remainder	4.0-6.0	4.0-6.0	4.0-6.0	2.0 max.	-	-	-	-
70/30 Cu-Ni-Fe	Remainder	-	-	-	29.0-32.0	0.5-1.50	-	0.4-1.0	-
90/10 Cu-Ni-Fe	Remainder	-	-	-	9.0-11.0	0.5-1.0	-	1.0-1.8	-
Ni-Al-bronze	Remainder	0.10 max.	1.0 max.	0.03 max.	3.0-6.0	0.5-4.0	-	2.0-6.0	7.0-11.0

2.5 Heat treatment

2.5.1 At the option of the manufacturer castings may be supplied in the 'as cast' or heat treated condition.

2.6 Mechanical tests

2.6.1 The test material may be separately cast as a keel block sample in accordance with Fig.3.6.1 or as otherwise agreed with the Surveyor. For liners and bushes, the test material may be cut from the ends of the casting.

2.6.2 Where castings are supplied in a heat treated condition, the test samples are to be similarly heat treated prior to the preparation of the tensile specimens.

2.6.3 The results of all tests are to comply with the appropriate requirements given in Table 2.6.1.

Table 2.6.1 : Mechanical properties for acceptance purposes

Designation	0.2% proof stress [N/mm^2] minimum (see Note)	Tensile Strength [N/mm^2] minimum	Elongation on $5.65\sqrt{S_o}$% minimum
90/10 Cu-Sn Phosphor-bronze	120	250	15
85/5/10 Leaded bronze	100	200	16
88/10/2 Gunmetal	130	270	13
87/7/3/3 Leaded Gunmetal	130	250	16
85/5/5/5 Leaded Gunmetal	100	200	16
70/30 Cu-Ni-Fe	220	420	20
90/10 Cu-Ni-Fe	160	320	20
Ni-Aluminium bronze	240	590	16

Note:

The 0.2% proof stress values are given for information purposes only and, unless otherwise agreed, are not required to be verified by test.

2.7 Visual examination

2.7.1 All castings must be supplied in a clean fettled condition.

2.7.2 Before acceptance, all castings are to be presented for visual examination by the Surveyor. This is to include the examination of internal surfaces where applicable.

2.7.3 The accuracy and verification of dimensions are the responsibility of the manufacturer, unless otherwise agreed.

2.8 Pressure testing

2.8.1 Where required by the relevant construction Rules, castings are to be pressure tested before final acceptance. Unless otherwise agreed, these tests are to be carried out in the presence of the Surveyors and are to be to their satisfaction.

2.9 Rectification of defective castings

2.9.1 Minor surface defects may be removed by grinding provided that the dimensional tolerances are not exceeded.

2.9.2 Proposal to repair a defective casting by welding are to be submitted to the Surveyor for approval before this work is commenced. Such proposals are to include details of the extent and positions of all defects. The Surveyor is to satisfy himself the number and size of the defects are such that castings can be efficiently repaired.

2.9.3 A statement and/or sketch detailing the extent and position of all weld repairs is to be prepared by the manufacturer as permanent record.

2.9.4 Weld repairs to liners in copper alloys containing more than 0.5 per cent lead are not permitted.

2.10 Identification

2.10.1 Before acceptance, all castings which have been tested and inspected with satisfactory results are to be clearly marked with the following details:

a) Identification number, cast number or other markings which will enable the full history of the casting to be traced;

b) IR and the abbreviated name of the IRS local office;

c) Personal stamp of the Surveyor responsible for inspection;

d) Test pressure, where applicable;

e) Date of final inspection.

2.10.2 Where small castings are manufactured in large numbers, modified arrangements for identification may be specially agreed with the Surveyor.

2.11 Certification

2.11.1 The manufacturer is to provide the Surveyor with a written statement giving the following particulars for each casting or batch of castings which has been accepted:-

a) Purchaser's name and order no.;

b) Description of castings and alloy type;

c) Identification number

d) Type of heat treatment, where applicable;

e) Ingot or cast analysis.

2.11.2 In addition to 2.11.1 the manufacturer is to provide a signed statement and/or sketch detailing the extent and position of all weld repairs made to each casting.

Section 3

Castings for Propellers

3.1 Scope

3.1.1 These requirements are applicable to the manufacture, inspection and repair procedures of cast copper alloys propellers, blades and bosses.

3.1.2 These requirements may also be used for the repair of propellers damaged in service, subject to prior agreement with IRS

3.1.3 Where the use of alternative alloys is proposed, particulars of chemical composition, mechanical properties and heat treatment are to be submitted for approval.

3.2 Foundry approval

3.2.1 Approval

3.2.1.1 All propellers and propeller components are to be manufactured by foundries approved in accordance with Ch.1. Also refer Ch.1, Sec.1, Cl.1.3.2. The castings are to be manufactured and tested in accordance with the requirements of these rules.

3.2.2 Application for approval

3.2.2.1 It is the manufacturer's responsibility to assure that effective quality, process and production controls during manufacturing are adhered to within the manufacturing specification. The manufacturing specification is to be submitted to IRS at the time of initial approval, and is to at least include the following particulars:

a) description of the foundry facilities,

b) copper alloy material specification,

c) runner and feeder arrangements,

d) manufacturing procedures,

e) non-destructive testing

f) inspection procedures, and

g) repair procedures.

3.2.3 Scope of the approval test

3.2.3.1 The scope of the approval test is to be agreed upon with IRS. This is to include the presentation of cast test coupons of the propeller materials in question for approval testing in order to verify that the chemical composition and the mechanical properties of these materials comply with these rules.

3.2.4 Inspection facilities

3.2.4.1 The foundry is to have an adequately equipped laboratory, manned by experienced personnel, for the testing of moulding materials chemical analyses, mechanical testing, and microstructural testing of metallic materials and non-destructive testing. Where testing activities are assigned to other companies or other laboratory, additional information required by IRS is to be included.

3.3. Moulding and casting

3.3.1 Pouring

3.3.1.1 The pouring is to be carried out into dried moulds using degassed liquid metal. The pouring is to be controlled as to avoid turbulences of flow. Special devices and/or procedures must prevent slag flowing into the mould.

3.3.2 Stress relieving

3.3.2.1 Subsequent stress relieving heat treatment may be performed to reduce the residual stresses. For this purpose, the manufacturer is to submit a specification containing the details of the heat treatment to IRS for approval. For stress relieving temperatures and holding times see Tables 3.12.3(a) and (b).

3.4 Quality of castings

3.4.1 Freedom from defects

3.4.1.1 All castings must have a workmanlike finish and are to be free from defects which would be prejudicial to their proper application in service. Minor casting defects which may still be visible after machining such as small

sand and slag inclusions, cold shuts and scabs are to be trimmed off by the manufacturer in accordance with 3.11.

3.4.2 Removal of defects

3.4.2.1 Casting defects which may impair the serviceability of the castings, e.g. major non-metallic inclusions, shrinkage cavities, blow holes and cracks, are not permitted. They may be removed by one of the methods described in 3.11 and repaired within the limits and restrictions for the severity zones. Full description and documentation are to be available for the surveyor.

3.5 Dimensions, dimensional and geometrical tolerances

3.5.1 The verification of dimensions, the dimensional and geometrical tolerances is the responsibility of the manufacturer. The report on the relevant examinations is to be submitted to the Surveyor, who may require checks to be made in his presence.

3.5.2 Static balancing is to be carried out on all propellers in accordance with the approved drawing. Dynamic balancing is necessary for propellers running above 500 rpm.

3.6 Chemical composition and metallurgical characteristics

3.6.1 Chemical composition

3.6.1.1. Typical copper propeller alloys are grouped into the four types CU 1, CU 2, CU 3 and CU 4 depending on their chemical composition as given in Table 3.6.1. Copper alloys whose chemical composition deviate from the typical values of Table 3.6.1 are to be specially approved by IRS. The manufacturer is to maintain records of the chemical analyses of the production casts, which are to be made available to the Surveyor.

3.6.2 Metallurgical characteristics

3.6.2.1 The main constituents of the microstructure in the copper-based alloys categories CU 1 and CU 2 are alpha and beta phase. Important properties such as ductility and resistance to corrosion fatigue are strongly influenced by the relative proportion of beta phase (too high a percentage of beta phase having a negative effect on these properties). To ensure adequate cold ductility and corrosion fatigue resistance, the proportion of beta phase is to be kept low. The concept of the zinc equivalent is to be used as control since it summarizes the effect of the tendency of various chemical elements to produce beta phase in the structure.

3.6.2.2 The structure of CU 1 and CU 2 type alloys must contain an alpha phase component of at least 25 % as measured on a test bar by the manufacturer. The zinc equivalent defined by the following formula is not to exceed a value of 45 %:

$$\text{Zinc equivalent} = 100 - \frac{100 \text{ x } \%\text{Cu}}{100 + \text{A}}$$

where A is the algebraic sum of the following:

1 x % Sn

5 x % Al

- 0.5 x % Mn

- 0.1 x % Fe

- 2.3 x % Ni.

The negative sign in front of the elements Mn, Fe and Ni signifies that these elements tend to reduce the proportion of beta phase.

The micro structure of alloy types CU 1 and CU 2 is to be verified by determining the proportion of alpha phase. For this purpose, at least one specimen is to be taken from each heat. The proportion of alpha phase is to be determined as the average value of 5 counts.

Table 3.6.1 : Typical chemical compositions of cast copper alloy for propellers								
Alloy Designation	**Chemical composition of ladle samples %**							
	Cu	**Sn**	**Zn**	**Pb**	**Ni**	**Fe**	**Al**	**Mn**
Grade CU1 Manganese Bronze (high strength brass)	52 - 62	1.5 max.	35 - 40	0.5 max.	1.0 max.	0.5-2.5	0.5-3.0	0.5-4.0
Grade CU2 Ni-Manganese Bronze (high strength brass)	50 - 57	1.5 max.	33 - 38	0.5 max.	3.0-8.0	0.5-2.5	0.5-2.0	1.0-4.0
Grade CU3 Ni-Aluminium Bronze	77 - 82	0.1 max.	1.0 max.	0.03 max.	3.0-6.0	2.0-6.0	7.0-11.0	0.5-4.0
Grade CU4 Mn - Aluminium Bronze	70 - 80	1.0 max.	6.0 max.	0.05 max.	1.5-3.0	2.0-5.0	6.5-9.0	8.0-20.0

3.7 Mechanical properties and tests

3.7.1 The mechanical properties are to comply with the values in Table 3.7.1. These values are applicable to test specimens taken from separately cast samples in accordance with Figure 3.7.1 or with a recognised standard.

Note : These properties are a measure of the mechanical quality of the metal in each heat and they are generally not representative of the mechanical properties of the propeller casting itself which may be upto 30% lower than that of a separately cast test coupon. For integrally cast test specimens the requirements are to be specially agreed with IRS.

3.7.2 Other alloys

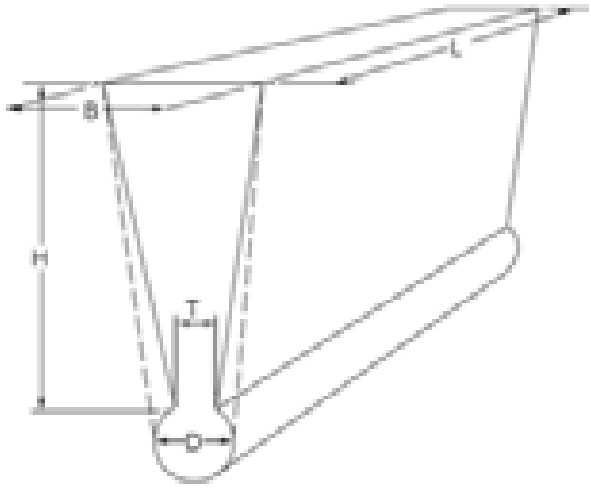
3.7.2.1 The mechanical properties of alloys not meeting the minimum values of Table 3.7.1 are to comply with a specification approved by IRS.

3.7.3 Tensile tests and specimens

3.7.3.1 Tensile tests and specimens are to be in accordance with Ch.2.

3.7.3.2 Generally, the specimens are to be taken from separately cast sample pieces in accordance with 3.7.1. The test samples are to be cast in moulds made of the same material as the mould for the propeller and they are to be cooled down under the same conditions as the propeller. At least one tensile test specimen is to be taken from each ladle. If propellers are subjected to a heat treatment the test samples are to be heat treated together with them.

3.7.3.3 Where test specimens are to be taken from integrally cast test samples, these are to be the subject of special agreement with IRS. Wherever possible, the test samples are to be located on the blades in an area lying between 0.5 to 0.6 R, where R is the radius of the propeller. The test sample material is to be removed from the casting by non thermal procedures.



H= 100 mm, B=50 mm, L>150 mm, T=15 mm and D = 25 mm

Fig.3.7.1 : Test sample casting

Table 3.7.1 : Mechanical properties of cast copper alloys for propellers (separately cast test coupons)			
Alloy Designation	**0.2% proof stress [N/mm^2] minimum**	**Tensile Strength [N/mm^2] minimum**	**Elongation on 5.65$\sqrt{S_o}$% minimum**
Grade CU1 Manganese bronze	175	440	20
Grade CU2 Ni-Manganese bronze	175	440	20
Grade CU3 Ni-Aluminium bronze	245	590	16
Grade CU4 Mn-Aluminium bronze	275	630	18

3.8 Definition of skew, severity zones

3.8.1 Definition of skew

3.8.1.1 The skew of a propeller is defined as follows:

The maximum skew angle of a propeller blade is defined as the angle, in projected view of the blade, between a line drawn through the blade tip and the shaft centreline and a second line through the shaft centreline which acts as a tangent to the locus of the mid-points of the helical blade section. See Fig.3.8.1.

High skew propellers have a skew angle greater than 25°, low skew propellers a skew angle of up to 25°.

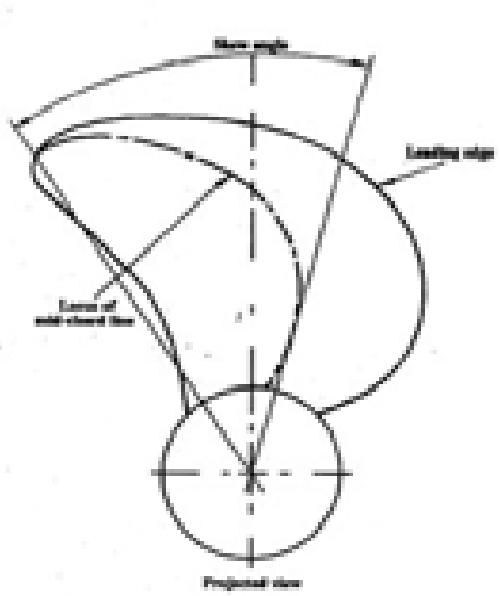


Fig.3.8.1 : Definition of skew angle

3.8.2 Severity zones

In order to relate the degree of inspection to the criticality of defects in propeller blades and to help reduce the risk of failure by fatigue cracking after repair, propeller blades are divided into the three severity zones designated A, B and C.

Zone A is the region carrying the highest operating stresses and which, therefore, requires the highest degree of inspection. Generally, the blade thicknesses are greatest in this area giving the greatest degree of restraint in repair welds and this in turn leads to the highest residual stresses in and around any repair welds. High residual tensile stresses frequently lead to fatigue cracking during subsequent service so that relief of these stresses by heat treatment is essential for any welds made in this zone. Welding is generally not permitted in Zone A and will only be allowed after special consideration. Every effort is to be made to rectify a propeller which is either defective or damaged in this area without recourse to welding even to the extent of reducing the scantlings, if this is acceptable. If a repair using welding is agreed, postweld stress relief heat treatment is mandatory.

Zone B is a region where the operating stresses may be high. Welding is to preferably be avoided but generally is allowed subject to prior approval from IRS. Complete details of the defect / damage and the intended repair procedure are to be submitted for each instance in order to obtain such approval.

Zone C is a region in which the operating stresses are low and where the blade thicknesses are relatively small so that repair welding is safer and, if made in accordance with an approved procedure is freely permitted.

3.8.2.1 Low-skew propellers

Zone A is in the area on the pressure side of the blade, from and including the fillet to 0.4R, and bounded on either side by lines at a distance 0,15 times the chord length Cr from the leading edge and 0.2 times Cr from the trailing edge, respectively (see Fig. 3.8.2).Where the hub radius (Rb) exceeds 0.27R, the other boundary of zone A is to be increased to 1.5Rb.

Zone A also includes the parts of the separate cast propeller hub which lie in the area of the windows as described in Fig. 3.8.4 and the flange and fillet area of controllable pitch and built-up propeller blades as described in Fig. 3.8.5.

Zone B is on the pressure side the remaining area up to 0.7R and on the suction side the area from the fillet to 0.7R (see Fig. 3.8.1).

Zone C is the area outside 0.7R on both sides of the blade. It also includes the surface of the hub of a monoblock propeller and all the surfaces of the hub of a controllable pitch propeller other than those designated Zone A above.

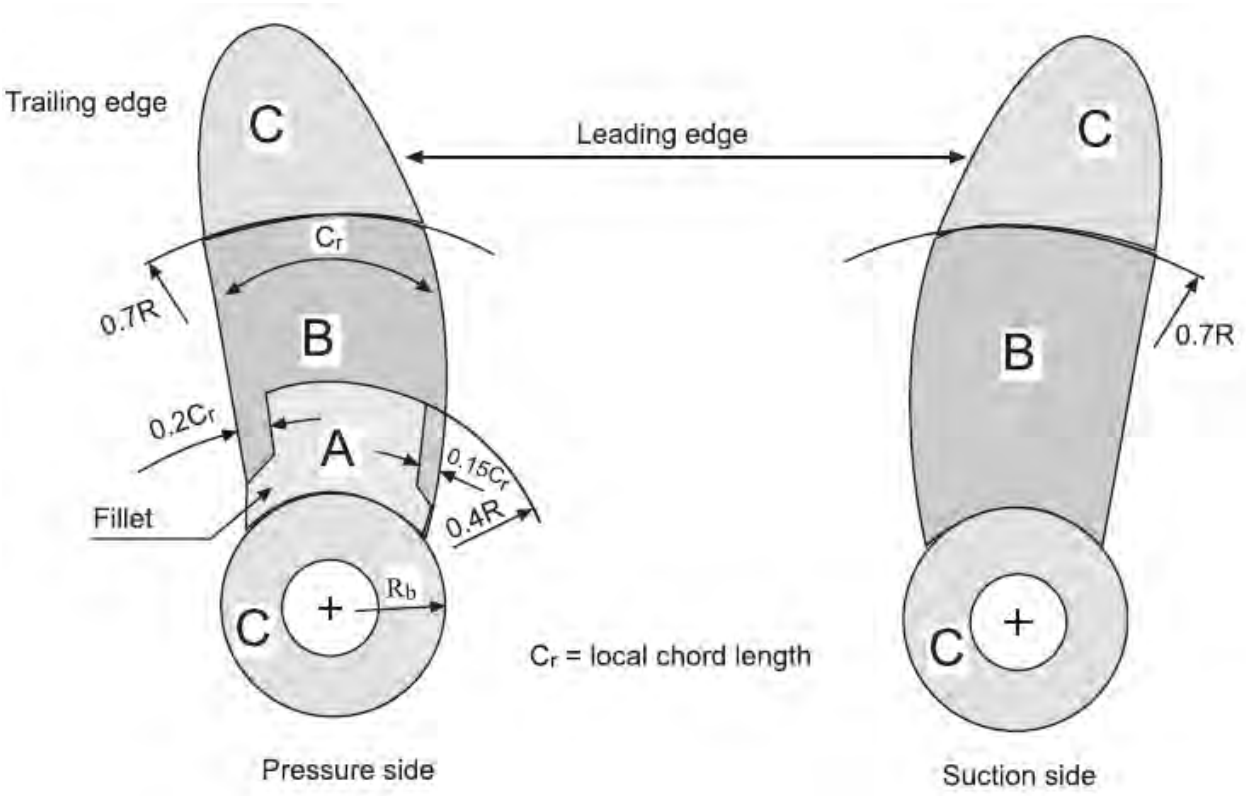


Fig.3.8.2 : Severity zones for integrally cast low skew propellers

3.8.2.2 High-skew propellers

Zone A is the area on the pressure face contained within the blade root-fillet and a line running from the junction of the leading edge with the root fillet to the trailing edge at 0.9 R and at passing through the mid-point of the blade chord at 0.7 R and a point situated at 0.3 of the chord length from the leading edge at 0.4 R. It also includes an area along the trailing edge on the suction side of the blade from the root to 0.9 R and with its inner boundary at 0.15 of the chord lengths from the trailing edge. Zone B constitutes the whole of the remaining blade surfaces. Zone A and B are illustrated in Fig. 3.8.3.

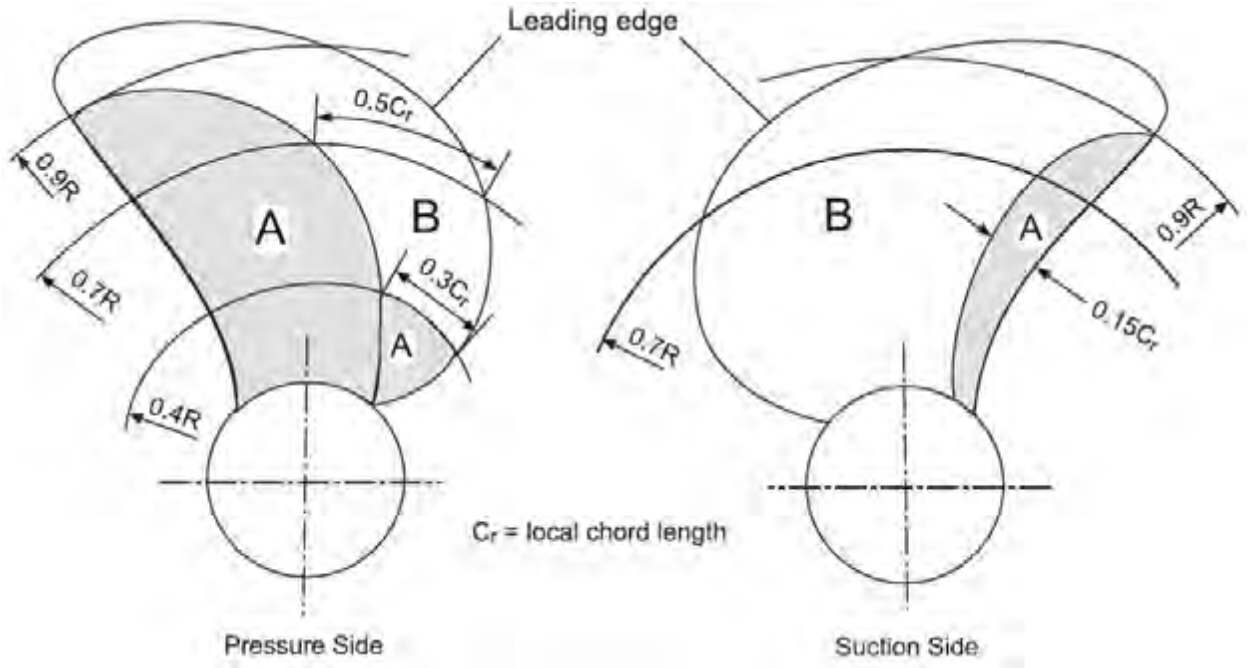


Fig.3.8.3 : Severity zones in blades with skew angles greater than 25º

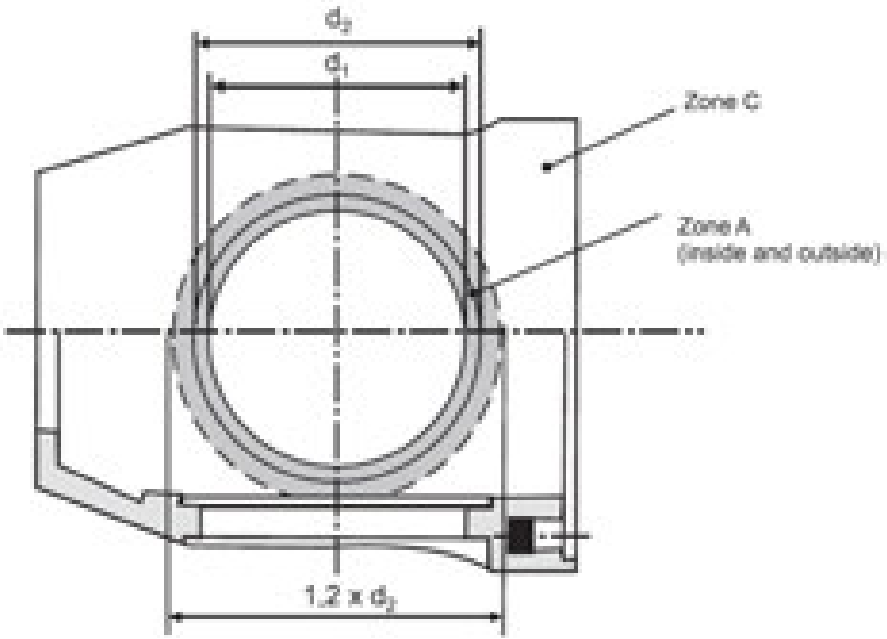


Fig.3.8.4 : Severity zones for controllable pitch propeller boss

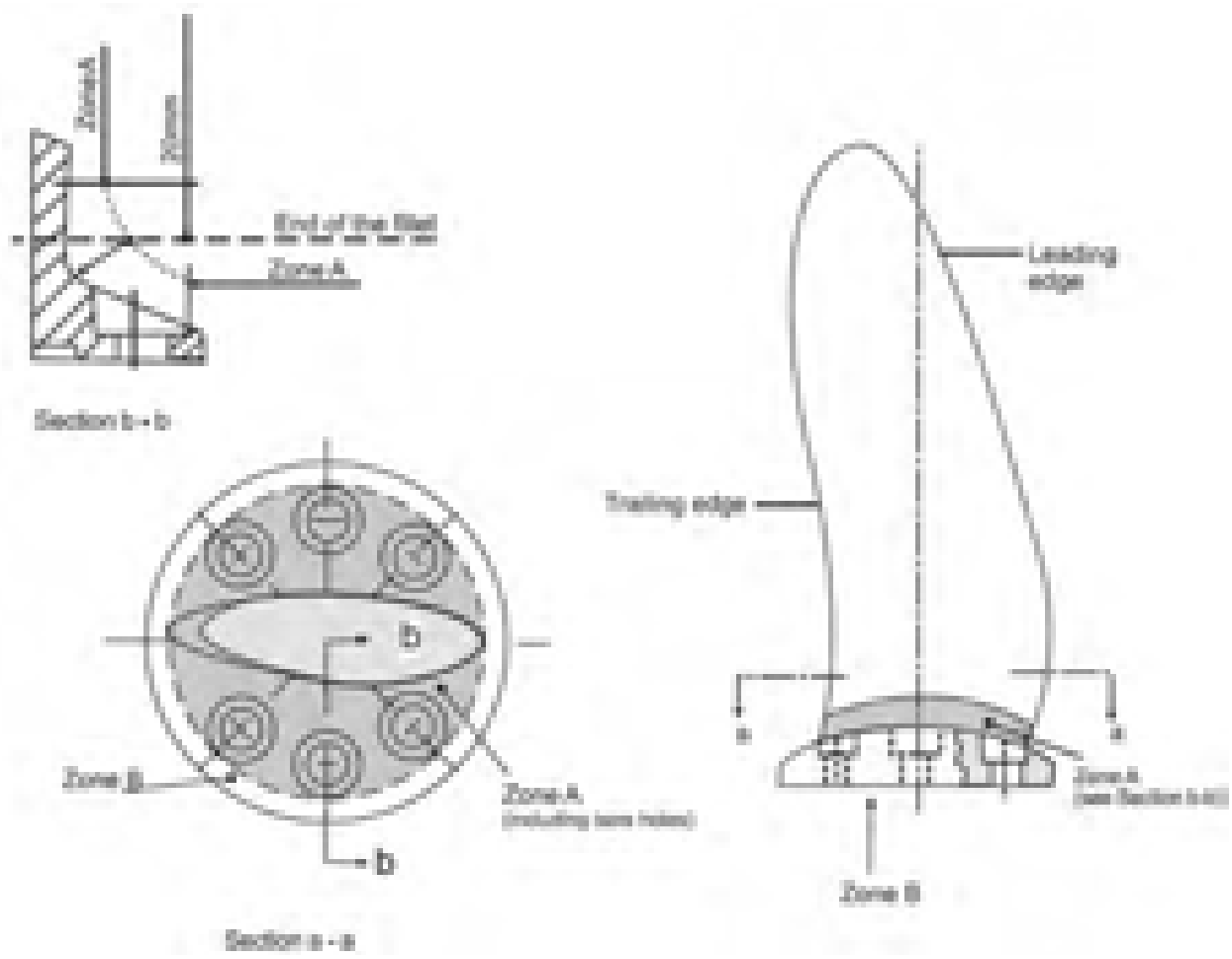


Fig.3.8.5 : Severity zones for controllable pitch and built-up propeller

Note:
The remaining surface of the propeller blades is to be divided into the severity zones as given for solid cast propellers (Fig. 3.8.2 and Fig. 3.8.3)

3.9 Non-destructive testing

3.9.1 Qualification of personnel involved in NDT

3.9.1.1 Refer to IRS Classification Note: "*Requirements for NDT Suppliers*".

3.9.2 Visual testing

3.9.2.1 All finished castings are to be 100% visually inspected by the manufacturer. Castings are to be free from cracks, hot tears or other imperfections which, due to their nature, degree or extent, will interfere with the use of the castings. A general visual examination is to be carried out by the Surveyor.

3.9.3 Liquid penetrant testing

3.9.3.1 Liquid penetrant testing procedure is to be submitted to IRS and is to be in accordance with ISO 3452-1:2013 or a recognized standard. The acceptance criteria are specified in 3.10. The severity zone A is to be subjected to a liquid penetrant testing in the presence of the Surveyor.
In zones B and C the liquid penetrant testing is to be performed by the manufacturer and may be witnessed by the Surveyor upon his request. If repairs have been made either by grinding, straightening or by welding the repaired areas are additionally to be subjected to the liquid penetrant testing independent of their location and/or severity zone.

3.9.4 Radiographic and ultrasonic testing

3.9.4.1 When required by IRS or when deemed necessary by the manufacturer, further non-destructive testing (e.g. radiographic and/or ultrasonic testing) are to be carried out. The acceptance criteria or applied quality levels are to be agreed between the manufacturer and IRS in accordance with a recognized standard.
Note: due to the attenuating effect of ultrasound within cast copper alloys, ultrasonic testing may not be practical in some cases, depending on the shape/type/thickness, and grain-growth direction of the casting.

In such cases, effective ultrasound penetration into the casting is to be practically demonstrated on the item. This would normally be determined by way of back-wall reflection, and/or target features within the casting.

3.10 Acceptance criteria for liquid penetrant testing

3.10.1 Definitions of liquid penetrant indications

Indication: In the liquid dye penetrant inspection testing an indication is the presence of detectable bleed-out of the penetrant liquid from the material discontinuities appearing at least 10 minutes after the developer has been applied.

Relevant indication: Only indications which have any dimension greater than 1.5 mm are to be considered relevant for the categorization of indications.

Non-linear indication: an indication with a largest dimension less than three times its smallest dimension (i.e. $l < 3\ w$).

Linear indication: an indication with a largest dimension three or more times its smallest dimension (i.e. $l \geq 3\ w$).

Aligned indications:

a) Non-linear indications form an alignment when the distance between indications is less than 2 [mm] and at least three indications are aligned. An alignment of indications is considered to be a unique indication and its length is equal to the overall length of the alignment.

b) Linear indications form an alignment when the distance between two indications is smaller than the length of the longest indication.

Illustration of liquid penetrant indication is given in Fig. 3.10.1.

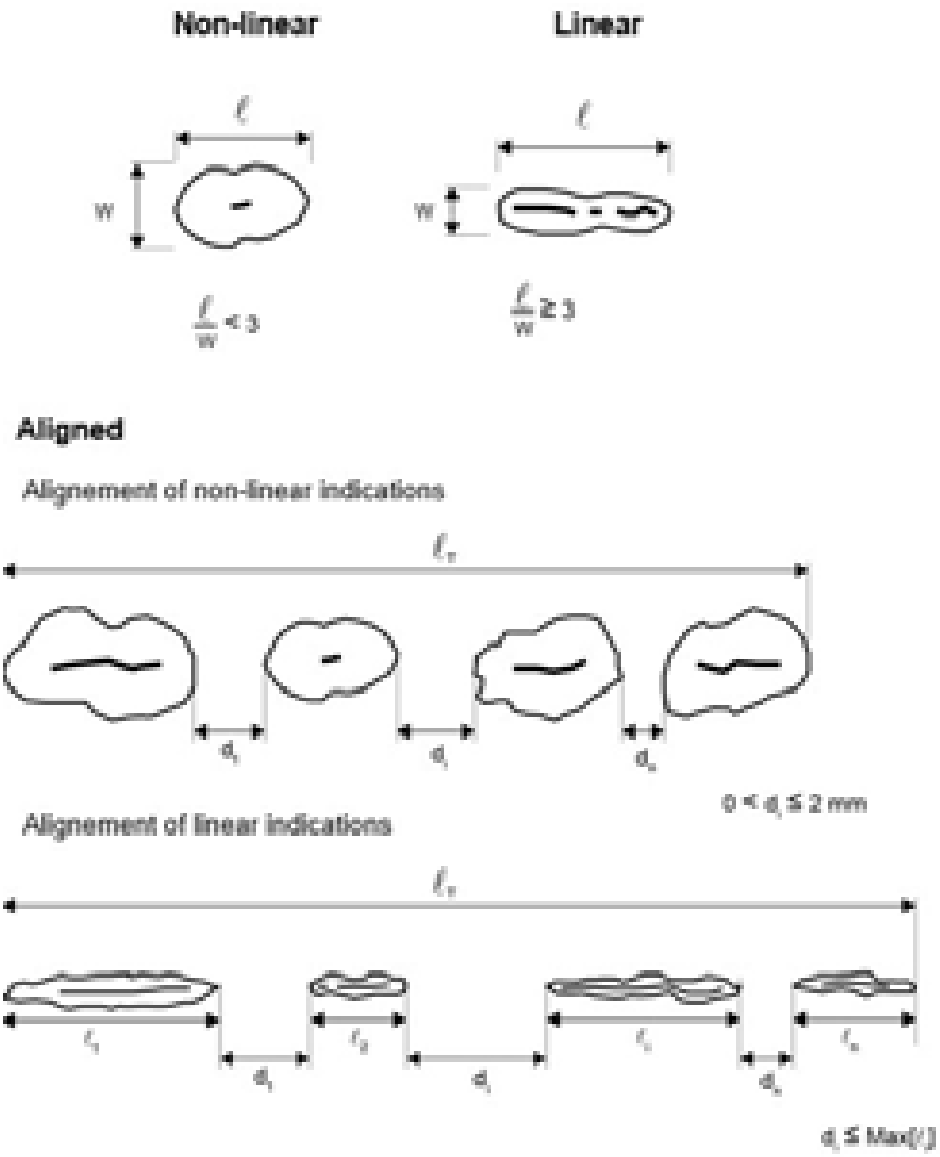


Fig.3.10.1 : Shape of indications

Table 3.10.2 : Allowable number and size of relevant indications in a reference area of 100 [cm^2], depending on severity zones[1)]

Severity zones	Max. total number of indications	Type of indication	Max. number of each type [1)2)]	Max. acceptable value for "a" or "l" of indications [mm]
A	7	Non-Linear Linear Aligned	5 2 2	4 3 3
B	14	Non-Linear Linear Aligned	10 4 4	6 6 6
C	20	Non-Linear Linear Aligned	14 6 6	8 6 6

Notes:

1) Singular non-linear indications less than 2 [mm] for zone A and less than 3 [mm] for the other zones are not considered relevant.

2) The total number of non-linear indications may be increased to the max. total number, or part thereof, represented by the absence of linear or aligned indications.

3.10.2 Acceptance standard

3.10.2.1 The surface to be inspected is to be divided into reference areas of 100 [cm^2] . Each reference area may be square or rectangular with the major dimension not exceeding 250mm.

The area is to be taken in the most unfavourable location relative to the indication being evaluated.

The relevant indications detected, with respect to their size and number, are not to exceed the values given in the Table 3.10.2.

3.10.2.2 Areas which are prepared for welding are, independent of their location, always to be assessed according to Zone A. The same applies to the welded areas after being finished machined and/or ground.

3.11 Repair of defects

3.11.1 Definitions

3.11.1.1 Indications exceeding the acceptance standard of Table 3.10.2, cracks, shrinkage cavities, sand, slag and other non-metallic inclusions, blow holes and other discontinuities which may impair the safe service of the propeller are defined as defects and must be repaired.

3.11.2 Repair procedures

a) In general the repairs are to be carried out by mechanical means, e.g. by grinding, chipping or milling. Welding may be applied subject to the agreement with IRS if requirements of 3.11.3, 3.11.4 and/or 3.11.5 will be complied with.

b) After milling or chipping grinding is to be applied for such defects which are not to be welded. Grinding is to be carried out in such a manner that the contour of the ground depression is as smooth as possible in order to avoid stress concentrations or to minimise cavitation corrosion. Complete elimination of the defective material is to be verified by liquid penetrant testing.

c) Welding of areas less than 5 [cm^2] is to be avoided.

3.11.3 **Repair of defects in zone A**

a) In zone A, repair welding will generally not be allowed unless specially approved by IRS. In some cases the propeller designer

may submit technical documentation to propose a modified zone A based on detailed hydrodynamic load and stress analysis for consideration by IRS.

b) Grinding may be carried out to an extent which maintains the blade thickness of the approved drawing.

c) The possible repair of defects which are deeper than those referred to above will be specially considered by IRS.

3.11.4 **Repair of defects in zone B**

a) Defects that are not deeper than dB = (t/40) [mm] (t = minimum local rule thickness [mm]) or 2 [mm] (whichever is greater) below minimum local rule thickness is to be removed by grinding.

b) Those defects that are deeper than allowable for removal by grinding may be repaired by welding.

3.11.5 **Repair of defects in zone C**

In zone C, repair welds are generally permitted.

3.11.6 **Repair documentation**

3.11.6.1 The foundry is to maintain records of inspections, welding, and any subsequent heat treatment, traceable to each casting. Before welding is started, full details of the extent and location of the repair, the proposed welding procedure, heat treatment and subsequent inspection procedures are to be submitted to IRS for approval.

3.12 Welding repair procedure

3.12.1 General requirements

3.12.1.1 Before welding is started, manufacturer is to submit to IRS a detailed welding procedure specification covering the weld preparation, welding parameters, filler metals, preheating and post weld heat treatment and inspection procedures.

3.12.1.2 All weld repairs are to be carried out in accordance with qualified procedures, and, by welders who are qualified to a recognized standard. Welding Procedure Qualification Tests are to be carried out in accordance with 3.16 and witnessed by the Surveyor.

3.12.2 Defects to be repaired by welding are to be ground to sound material according to 3.11.2.

3.12.2.1 The welding grooves are to be prepared in such a manner which will allow a good fusion of the groove bottom. The resulting ground areas are to be examined in the presence of the Surveyor by liquid penetrant testing in order to verify the complete elimination of defective material.

3.12.3 Welding repair procedure

3.12.3.1 Metal arc welding is recommended to be used for all types of welding repair on cast copper alloy propellers.

Arc welding with coated electrodes and gas-shielded metal arc process (GMAW) are generally to be applied. Argon-shielded tungsten welding (GTAW) is to be used with care due to the higher specific heat input of this process.

Recommended filler metals, pre-heating and stress relieving temperatures are listed in Table 3.12.3(a).

3.12.3.2 All propeller alloys are generally to be welded in down-hand (flat) position. Where this cannot be done, gas-shielded metal arc welding is to be carried out.

The section to be welded is to be clean and dry. Flux-coated electrodes are to be dried before welding according to the maker's instructions.

To minimize distortion and the risk of cracking, interpass temperatures are to be kept low especially in the case of CU3 alloys.

Slag, undercuts and other defects are to be removed before depositing the next run.

Table 3.12.3(a) : Recommended filler metals and heat treatments

Alloy type	Filler metal	Preheat temp.°C [min]	Interpass temp.°C [max]	Stress relief temp.°C	Hot straightening temp.°C
CU1	Al-bronze [1] Mn-bronze	150	300	350 - 500	500 - 800
CU2	Al-bronze Ni-Mn-bronze	150	300	350 - 550	500 - 800
CU3	Al-bronze Ni-Al-bronze [2] Mn-Al-bronze	50	250	450 - 500	700 - 900
CU4	Mn-Al-bronze	100	300	450 - 600	700 - 850

Notes:

1) Ni-Al-bronze and Mn-Al-bronze are acceptable.

2) Stress relieving not required, if filler metal Ni-Al-bronze is used.

Table 3.12.3(b) : Soaking times for stress relief heat treatment of copper alloy propellers

Stress relief temp.°C	Alloy grade CU1 and CU2		Alloy grade CU3 and CU4	
	Hours per 25 [mm] thickness	Max. recommended total time hours	Hours per 25 [mm] thickness	Max. recommended total time hours
350	5	15	-	-
400	1	5	-	-
450	1/2	2	5	15
500	1/4	1	1	5
550	1/4	1.2	1/2 [1]	2 [1]
600	-	-	1/4 [1]	1 [1]

Note 1) 550°C and 600°C only applicable for CU4 alloys

3.12.3.3 All welding work is to be carried out preferably in the shop free from draughts and influence of the weather.

3.12.3.4 With the exception of alloy CU3 (Ni-Al-bronze) all weld repairs are to be stress relief heat treated, in order to avoid stress corrosion cracking. However, stress relief heat treatment of alloy CU3 propeller castings may be required after major repairs in zone B (and specially approved welding in Zone A) or if a welding consumable susceptible to stress corrosion cracking is used. In such cases the propeller is to be either stress relief heat treated in the temperature 450 to 500°C or annealed in the temperature range 650 - 800°C, depending on the extent of repair, see Table 3.12.3(a).

3.12.3.5 The soaking times for stress relief heat treatment of copper alloy propellers is to be in accordance with Table 3.12.3(b). The heating and cooling is to be carried out slowly under controlled conditions. The cooling rate after any stress relieving heat treatment is not to exceed 50°C/hr until the temperature of 200°C is reached.

3.13 Straightening

3.13.1 **Application of load**

For hot and cold straightening purposes, static loading only is to be used.

3.13.2 **Hot straightening**

Weld repaired areas may be subject to hot straightening, provided it can be demonstrated that weld properties are not impaired by the hot straightening operations.

Straightening of a bent propeller blade or a pitch modification is to be carried out after heating the bent region and approximately 500 [mm] wide zones on either side of it to the suggested temperature range given in Table 3.12.3.(a).

The heating is to be slow and uniform and the concentrated flames such as oxy-acetylene and oxy-propane are not to be used. Sufficient time is be allowed for the temperature to become fairly uniform through the full thickness of the blade section. The temperature is to be maintained within the suggested range throughout the straightening operation. A thermocouple instrument or temperature indicating crayons are to be used for measuring the temperature.

3.13.3 **Cold straightening**

Cold straightening is to be used for minor repairs of tips and edges only. Cold straightening on CU1, CU2 and CU4 bronze is always to be followed by a stress relieving heat treatment, See Table 3.12.3(a).

3.14 Identification and marking

3.14.1 Identification

3.14.1.1 The manufacturer is to adopt a system for the identification of all castings, which enable the material to be traced to its original cast. The Surveyor is to be given full facilities for so tracing the castings when required.

3.14.2 Marking

3.14.2.1 Each finished casting propeller is to be marked by the manufacturer at least with the following particulars:

a) Grade of cast material or corresponding abbreviated designation

b) Manufacturer’s mark

c) Heat number, casting number or another mark enabling the manufacturing process to be traced back

d) Date of final inspection

e) IRS certificate number

f) Ice class symbol, where applicable

g) Skew angle for high skew propellers.

3.15 Manufacturer’s Certificates

3.15.1 For each casting propeller the manufacturer is to supply to the Surveyor a certificate containing the following details:

a) Purchaser and order number

b) Shipbuilding project number, if known

c) Description of the casting with drawing number

d) Diameter, number of blades, pitch, direction of turning

e) Grade of alloy and chemical composition of each heat

f) Heat or casting number

g) Final weight

h) Results of non-destructive tests and details of test procedure where applicable

i) Portion of alpha-structure for CU 1 and CU 2 alloys

k) Results of the mechanical tests

l) Casting identification Number.

m) Skew angle for high skew propellers, see 3.8.1.

3.16 Welding procedure qualification tests for repair of cast copper alloy propeller

3.16.1 General

3.16.1.1 This sub-section includes the requirements for qualification tests of welding procedures intended for the repair of cast copper alloy propellers.

3.16.1.2 For the welding procedure approval the welding procedure qualification tests are to be carried out with satisfactory results. The qualification tests are to be carried out with the same welding process, filler metal, preheating and stress-relieving treatment as those intended applied by the actual repair work. Welding procedure specification (WPS) is to refer to the test results achieved during welding procedure qualification testing.

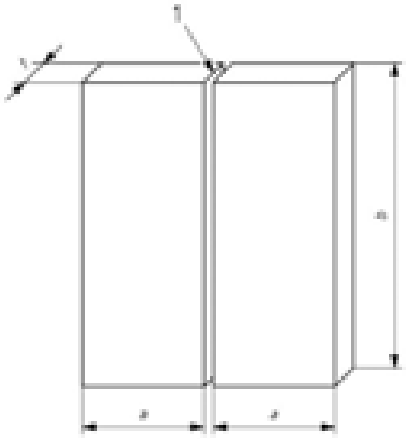
3.16.1.3 Welding procedures qualified at a manufacturer are valid for welding in workshops under the same technical and quality management.

3.16.2 Test piece and welding of sample

3.16.2.1 The test assembly, consisting of cast samples, is to be of a size sufficient to ensure a reasonable heat distribution and according to Fig. 3.16.2.1 with the minimum dimensions. A test sample of minimum 30 mm thickness is to be used.

3.16.2.2 Preparation and welding of test pieces are to be carried out in accordance with the general condition of repair welding work which it represents.

3.16.2.3 Welding of the test assemblies and testing of test specimens are to be witnessed by the Surveyor.



1: Joint preparation and fit-up as detailed in the preliminary welding procedure specification
a: minimum value 150mm
b: minimum value 300mm
t: material thickness.

Fig.3.16.2.1 : Test piece for welding repair procedure

3.16.3 Examination and tests

3.16.3.1 Test assembly is to be examined non-destructively and destructively in accordance with the Table 3.16.3.1 and Fig. 3.16.3.1.

Table 3.16.3.1 : Type of tests and extent of testing

Type of test [1)]	Extent of testing
Visual testing	100% as per 3.16.2
Liquid penetrant testing	100% as per 3.16.2
Transverse tensile test	Two specimens as per 3.16.3
Macro examination	Three specimens as per 3.16.4
Note 1: bend or fracture test are at the discretion of IRS.	

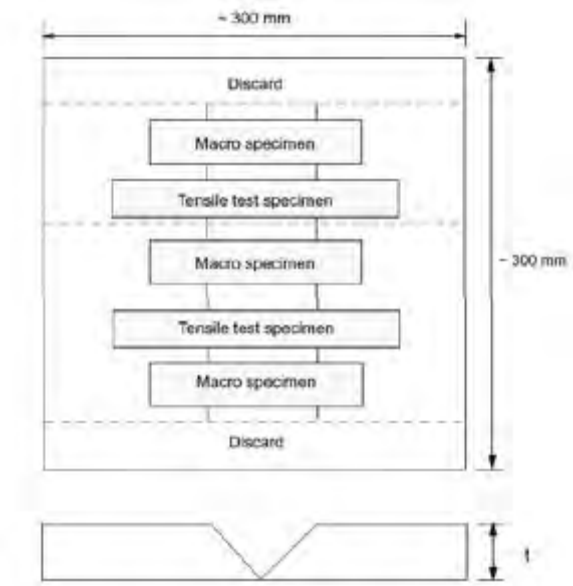


Fig.3.16.3.1 : Test Specimen

3.16.3.2 Non-destructive testing

.1 The test assembly is to be examined by visual and liquid penetrant testing prior to the cutting of test specimen. In case, that any post-weld heat treatment is required or specified, non-destructive testing is to be performed after heat treatment. No cracks are permitted. Imperfections detected by liquid penetrant testing are to be assessed in accordance with 3.10.

3.16.3.3 Tensile test:

.1 Two tensile tests are to be prepared as

shown in Ch.2. Alternatively tensile test specimens according to recognized standards acceptable to IRS may be used. The tensile strength is to meet the values given in Table 3.16.3.3.

3.16.3.4 Macroscopic examination

Three test specimens are to be prepared and etched on one side to clearly reveal the weld metal, the fusion line and the heat affected zone (Figure 3.16.3.1). A suitable etchant for this purpose is:

5 g iron (III) chloride
30 ml hydrochloric acid (cone)
100 ml water.

The test specimens are to be examined for imperfections present in the weld metal and the heat affected zone. Cracks and lack of fusion are not permitted. Imperfections such as pores, or slag inclusions, greater than 3 [mm] are not permitted.

Table 3.16.3.3 : Required tensile strength values

Alloy Type	Tensile strength [N/mm^2] min
CU 1	370
CU 2	410
CU 3	500
CU 4	550

3.16.3.5 Re-testing

.1 If the test piece fails to comply with any of the requirements for visual or non-destructive testing one further test piece is to be welded and subjected to the same examination. If this additional test piece does not comply with the relevant requirements, the pWPS (preliminary welding procedure specification) is to be regarded as not capable of complying with the requirements without modification.

.2 If any test specimens fail to comply with the relevant requirements for destructive testing due to weld imperfections only, two further test specimens are to be obtained for each one that failed. These specimens can be taken from the same test piece if there is sufficient material available or from a new test piece, and are to be subjected to the same test. If either of these additional test specimens does not comply with the relevant requirements, the pWPS is to be regarded as not capable of complying with the requirements without modification.

.3 If a tensile test specimen fails to meet the requirements, the re-testing is to be in accordance with Ch.2.

.4 If there is a single hardness value above the maximum values allowed, additional hardness tests are to be carried out (on the reverse of the specimen or after sufficient grinding of the tested surface). None of the additional hardness values is to exceed the maximum hardness values required.

.5 The re-testing of Charpy impact specimens are to be carried out in accordance with Ch.2.

.6 Where there is insufficient welded assembly remaining to provide additional test specimens, a further assembly is to be welded using the same procedure to provide the additional specimens.

3.16.4 Test record

3.16.4.1 Welding conditions for test assemblies and test results are to be recorded in welding procedure qualification record. Forms of welding procedure qualification records may be in accordance with recognised standards.

3.16.4.2 A statement of the results of assessing each test piece, including repeat tests, is to be made for each welding procedure qualification records. The relevant items listed for the WPS are to be included.

3.16.4.3 The welding procedure qualification record is to be signed by the Surveyor witnessing the test and is to include the IR identification.

3.16.5 Range of approval

3.16.5.1 General

.1 All the conditions of validity stated below are to be met independently of each other. Changes outside of the ranges specified are to require a new welding procedure test. A qualification of a WPS obtained by a manufacturer is valid for welding in workshops or sites under the same technical and quality control of that manufacturer.

3.16.5.2 Base metal

.1 The range of qualification related to base metal is given in Table 3.16.5.2.

Table 3.16.5.2 : Range of qualification for base metal

Copper alloy material grade used for qualification	Range of approval
CU1	CU1
CU2	CU1& CU2
CU3	CU3
CU4	CU4

3.16.5.3 Thickness

.1 The qualification of a WPS carried out on a weld assembly of thickness t is valid for the thickness range given in Table 3.16.5.3.

Table 3.16.5.3 : Range of qualification for thickness

Thickness of the test piece, t (mm)	Range of approval
30≤t	≥3 mm

3.16.5.4 Welding position

.1 Approval for a test made in any position is restricted to that position.

3.16.5.5 Welding process

.1 The approval is only valid for the welding process used in the welding procedure test. Single run is not qualified by multi-run butt weld test used in this section

3.16.5.6 Filler metal

.1 The approval is only valid for the filler metal used in the welding procedure test.

3.16.5.7 Heat input

.1 The upper limit of heat input approved is 25% greater than that used in welding the test piece. The lower limit of heat input approved is 25% lower than that used in welding the test piece.

3.16.5.8 Preheating and interpass temperature

.1 The minimum preheating temperature is not to be less than that used in the qualification test. The maximum interpass temperature is not to be higher than that used in the qualification test.

3.16.5.9 Post-weld heat treatment

The heat treatment used in the qualification test is to be specified in pWPS. Soaking time may be adjusted as a function of thickness.

Section 4

Tubes

4.1 Scope

4.1.1 Following requirements make provision for copper and copper alloy tubes intended for use in heat exchangers, condensers and pressure piping systems.

4.1.2 Except for pipes for Class lll pressure systems (as defined in Pt.4, Ch.2) all pipes and tubes are to be manufactured and tested in accordance with the requirements of Ch.1 and 2 of this Part and the requirements of this Section.

4.1.3 Pipes and tubes which comply with national/international or proprietary specifica-ions may be accepted provided that these specifications give reasonable equivalence to the requirements of this Section or are otherwise specifically approved for a specific application and provided that survey is carried out in accordance with Ch.1 of this Part.

4.1.4 At the discretion of the Surveyor, modified testing procedure may be adopted for small quantities of materials. In such cases, these may be accepted on the manufacturer's declared chemical composition and hardness tests or other evidence of satisfactory properties.

4.1.5 Pipes for Class lll pressure systems are to be manufactured and tested in accordance with the requirements of an acceptable national/ international specification. The manufacturer's test certificate will be acceptable and is to be provided for each consignment of material.

4.2 Manufacture

4.2.1 Approval of Works, as required by Ch.1, for the manufacture of copper and copper alloy tubes is generally not required.

4.2.2 Unless otherwise agreed tubes shall be solid drawn.

4.3 Quality

4.3.1 Tubes are to have a workmanlike finish and are to be clean and free from such surface and internal defects as can be established by the specified tests.

4.3.2 The tubes are to be supplied in straight lengths, and the ends are to be cut clean and square with the axis of the tube.

4.3.3 The tolerance on wall thickness and diameter of pipes and tubes are to be in accordance with an acceptable national/ international standard.

4.4 Chemical composition

4.4.1 The chemical analysis is to comply with the requirements of Table 4.4.1. Residual elements are not to be present in amounts greater than specified in an acceptable national/international standard.

4.5 Heat treatment

4.5.1 All tubes are to be supplied in the annealed condition. Aluminium brass tubes may additionally be required to be given a suitable stress relieving heat treatment when subjected to a cold straightening operation after annealing.

4.6 Mechanical tests

4.6.1 The tubes are to be presented in lots of 600 tubes or 900 [Kg], whichever is greater. Each lot is to contain tubes of the same dimensions, material grade and in the same state of heat treatment. From each lot 2 tubes are to be selected for testing.

4.6.2 Following tests are to be carried out on each tube selected for testing in accordance with the requirements of Ch.2:

a) Tensile test;

b) Flattening test;

c) Drift Expanding test.

4.6.3 Flattening test is to be carried out until the interior surfaces of the tube meet.

4.6.4 For the drift-expanding test, the mandrel is to have an included angle of 45°.

4.6.5 The results of all mechanical tests are to comply with the requirements of Table 4.6.1.

Table 4.4.1 : Chemical composition of tubes [1)]

Designation	Chemical composition %								
	Cu	Fe	Ni	Zn	As	Al	Mn	P	Pb
Phosphorus deoxidised non-arsenical copper	99.90 [2)] min.	-	-	-	-	-	-	0.013-0.050	-
Phosphorus deoxidised arsenical copper	99.20 [2)] min.	-	-	-	0.30-0.50	-	0.013-0.050	-	-
Al-brass	76.0-79.0	-	-	Remainder	0.02-0.06	1.8-2.3	-	-	-
Copper-nickel 90/10	Remainder	1.0-1.8	9.0-11.0	-	-	-	0.5-1.0	-	-
Copper-nickel 70/30	Remainder	0.4-1.0	30.0-32.0	-	-	-	0.5-1.5	-	-

Notes:

1) Table shows essential alloying elements only
2) Includes silver also.

Table 4.6.1 : Mechanical properties for acceptance purposes

Designation	0.2% proof stress [N/mm^2] minimum	Tensile strength [N/mm^2] minimum	$5.65\sqrt{S_o}$% minimum	Drift expansion test % minimum
Phosphorus deoxidised non-arsenical copper	100	220	35	30
Phosphorus deoxidised arsenical copper	100	220	35	30
Al-brass	110	320	35	30
Copper-nickel 90/10	100	270	30	30
Copper-nickel 70/30	120	360	30	30

4.7 Visual examination

4.7.1 All pipes are to be presented for visual examination and verification of dimensions. The manufacturer is to provide adequate lighting conditions to enable an internal and external examination of the tubes to be carried out.

4.8 Stress cracking test

4.8.1 This test is applicable to aluminium brass only. Mercurous Nitrate Test or alternatively at the express agreement between purchaser and manufacturer Ammonia Vapour Cracking Test are to be carried out on test specimen to prove that the tubes are free from internal stresses. The tests are to be carried out in accordance with an acceptable national/international standard.

4.8.2 Should a specimen reveal cracks when tested, the manufacturing batch shall be rejected. The manufacturer shall be free to submit the batch to renewed heat treatment before presenting it for retesting.

4.9 Hydraulic test

4.9.1 All tubes are to be hydraulically tested by the manufacturer to the following pressure:

$$P = \frac{5 \text{ x } t \text{ x } Rm}{D}$$

where,

P = Test pressure;

t = nominal wall thickness;

D = nominal outside diameter;

Rm = Tensile strength in accordance with Table 4.6.1.

Unless otherwise stated the pressure need not be greater than 7.0 [N/mm^2].

4.9.2 The test pressure is to be maintained for sufficient time to permit proof and inspection. Unless otherwise agreed, the manufacturer's certificate of satisfactory hydraulic test will be accepted subject to 10 per cent of the tubes being retested in the presence of the Surveyor. If one of the tubes in a batch does not pass the test, it will be rejected, and all other tubes in the batch are to be retested.

4.10 Identification

4.10.1 Tubes are to be clearly marked by the manufacturer in accordance with the requirements of Ch.1, with at least the following details:

a) IR;

b) Manufacturer's name or trade mark;

c) Grade of material.

4.10.2 Identification is to be by rubber stamp or stencil. Hard stamping is not to be used.

4.11 Certification

4.11.1 The manufacturer is to provide the Surveyor with a written statement giving the following particulars for each lot of material accepted:-

a) Purchaser's name and Order no.;
b) Grade of material;
c) Description and dimensions;
d) Cast number and chemical composition;
e) Mechanical test results and results of stress cracking tests where applicable.

End of Chapter

Chapter 9

Aluminium Alloys

Contents

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1 *General*

2 *Wrought Aluminium Alloys*

3 *Aluminium Alloy Castings*

4 *Aluminium/Steel Transition Joints*

Section 1

General

1.1 Scope

1.1.1 This Chapter specifies the requirements for wrought aluminium alloys for structural applications, aluminium alloy castings and aluminium/steel transition joints intended for use in ship and machinery construction.

1.1.2 This Chapter is not applicable to aluminium alloys for forgings and to the use of aluminium alloys at low temperature for cryogenic applications. For these products suitable alloys which comply with recognized standards may be used.

1.1.3 These requirements are applicable to wrought aluminium alloy products within a thickness range of 3 [mm] and 50 [mm] inclusive. The application of aluminium alloys products outside this thickness range requires prior agreement of IRS.

1.1.4 The numerical designation (grade) of aluminium alloys and the temper designation are based on those of the Aluminium Association.

Temper conditions (delivery heat treatment) are as defined in EN 515:2017 Or ANSI H35.1:2017.

1.1.5 When required by the relevant Chapters of the Rules dealing with design and construction, structural aluminium alloys, aluminium alloy castings and aluminium/steel transition joints are to be manufactured and tested in accordance with the appropriate requirements of Ch.1 and 2 and those detailed in this Chapter.

1.1.6 Consideration may be given to aluminium alloys not specified in this chapter and to alternative temper conditions, complying with recognized national or international standards with specifications equivalent to the requirements of this chapter.

Section 2

Wrought Aluminium Alloys

2.1 Scope

2.1.1 This Section deals with wrought aluminium alloys for structural applications including plates, sections, tubes, bars and rivet bars and rivets.

2.1.2 Wrought aluminium alloys are to have a satisfactory resistance to corrosion in marine environment. Grades for welded structures are to be weldable, applying one of the welding methods approved by IRS.

2.1.3 The alloy grades 6005A, 6061 of the 6000 series should not be used in direct contact with sea water unless protected by anodes and/or paint system.

2.2 Manufacture

2.2.1 Aluminium alloys are to be manufactured at Works approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

2.2.2 The alloys may be cast either in ingot moulds or by an approved continuous casting process. Plates are to be formed by rolling and may be hot or cold finished. Bars and sections may be formed by rolling, extrusion or drawing.

2.3 Quality of materials

2.3.1 Materials are to be free from surface or internal defects of such a nature as would be harmful in service.

2.4 Dimensional tolerances

2.4.1 The dimensional tolerances are to be in accordance with Table 2.4.1, Table 2.4.2 and Table 2.4.3 and are minimum requirements.

2.4.2 Dimensional tolerances other than those given in Table 2.4.1, Table 2.4.2 and Table 2.4.3 are to comply with a recognized national or international standard.

2.5 Chemical composition

2.5.1 Samples for chemical analysis are to be taken representative of each cast, or the equivalent where a continuous melting process is involved.

2.5.2 The chemical composition of these samples is to comply with the requirements of Table 2.5.1.

Table 2.4.1 : Under thicknesses tolerances for rolled products

	Thickness tolerances for nominal width [mm]		
Nominal thickness [t] [mm]	$w \leq 1500$	$1500 < w \leq 2000$	$2000 < w \leq 3500$
$3.0 \leq t < 4.0$	0.10	0.15	0.15
$4.0 \leq t < 8.0$	0.20	0.20	0.25
$8.0 \leq t < 12.0$	0.25	0.25	0.25
$12.0 \leq t < 20.0$	0.35	0.40	0.50
$20.0 \leq t < 50.0$	0.45	0.50	0.65

Table 2.4.2 : Under thicknesses tolerances for extruded open profiles

	Thickness tolerances for nominal thicknesses for a diameter of the circumscribing circle [mm]		
Nominal thickness [mm]	Upto 250	From 250 to 400	Above 400
From 3 to 6	0.25	0.35	0.40
From 6 to 50	0.30	0.40	0.45

Table 2.4.3 : Under thicknesses tolerances for extruded closed profiles

Nominal thickness [mm]	Thickness tolerances [mm]
From 3 to 6	0.25
From 6 to 50	0.30

Table 2.5.1 : Chemical composition

Grade	Al	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Oth-ers (2)	Oth-ers (2)	Misc.
	%	%	%	%	%	%	%	%	%	Each %	Total %	
5059	Remain der	0.45	0.50	0.25	0.6-1.2	5.0-6.0	0.25	0.40-0.90	0.20	0.05 [3)]	0.15 [4)]	
5083	Remain der	≤ 0.40	≤ 0.40	≤ 0.10	0.4-1.0	4.0-4.9	0.05-0.25	≤ 0.25	≤ 0.15	≤ 0.05	≤ 0.15	
5086	Remain der	≤ 0.40	≤ 0.50	≤ 0.10	0.20-0.7	3.5-4.5	0.05-0.25	≤ 0.25	≤ 0.15	≤ 0.05	≤ 0.15	
5383	Remain der	0.25	0.25	0.20	0.7-1.0	4.0-5.2	0.25	0.40	0.15	0.05 [3)]	0.15 [3)]	
5754	Remain der	≤ 0.40	≤ 0.40	≤ 0.10	≤ 0.50	2.6-3.6	≤ 0.30	≤ 0.20	≤ 0.15	≤ 0.05	≤ 0.15	0.10 ≤ Mn + Cr ≤ 0.60
5456	Remain der	0.25	0.40	0.10	0.50 – 1.0	4.7 – 5.5	0.05 – 0.20	0.25	0.20	0.05	0.15	
6005-A	Remain der	0.50-0.9	≤ 0.35	≤ 0.30	≤ 0.50	0.040-0.7	≤ 0.30	≤ 0.20	≤ 0.10	≤ 0.05	≤ 0.15	0.12 ≤ Mn + Cr ≤ 0.50
6061	Remain der	0.40-0.8	≤ 0.7	0.15-0.40	≤ 0.15	0.8-1.2	0.04-0.35	≤ 0.25	≤ 0.15	≤ 0.05	≤ 0.15	
6082	Remain der	0.7-1.3	≤ 0.50	≤ 0.10	0.40-1.0	0.6-1.2	≤ 0.25	≤ 0.20	≤ 0.10	≤ 0.05	≤ 0.15	

Notes:

1. Slight variations in the content of some elements, compared with values indicated in this Table may be accepted with IRS's agreement.
2. Other metallic elements such as Ni, Ga.V are considered as impurities. The regular analysis need not be made for these elements.
3. Zr: maximum 0.20. The total for other elements does not include Zirconium.
4. Zr: 0.05-0.25. The total for other elements does not include Zirconium.

2.5.3 The manufacturer's declared analysis will be accepted subject to occasional checks if required by IRS Surveyor, particularly, product analysis may be required where the final product chemistry is not well represented by the analysis from the cast.

2.5.4 When the aluminium alloys are not cast in the same works in which they are manufactured into semi finished products, the works is to give a certificate detailing the chemical composition and heat number.

2.6 Heat treatment

2.6.1 Temper conditions (delivery heat treatment) are defined in Table 2.8.1.

2.7 Test material

2.7.1 All materials in a lot forwarded for sampling are to be of the same alloy, production batch and product form (plates, sections etc.). The materials in one lot are to be of the same dimensions and in the same delivery condition. Artificially aged grades are to be from the same furnace batch.

2.7.2 Wherever practicable, the tensile test pieces for rolled and extruded sections are to be of full section of material. Otherwise, the pieces are to be taken in the range one third to half the distance from the edge to center of the predominant or thickest part of the section.

2.8 Testing and inspection

2.8.1 Testing procedures

The test specimens and procedures are to be in accordance with Ch.2.

2.8.2 Verification of proper fusion of press welds for closed profiles.

The Manufacturer has to demonstrate by macrosection tests or drift expansion tests of closed profiles performed on each batch of closed profiles that there is no lack of fusion at the press welds.

2.8.3 Drift expansion tests

2.8.3.1 Every fifth profile is to be sampled after final heat treatment.

One sample is to be selected from the batches of five profiles or less.

Every profile is to be selected if the length exceeds 6 [m].

2.8.3.2 Two samples are to be cut from the front and back end of each production profile.

2.8.3.3 The test specimens are to be cut with the ends perpendicular to the axis of the profile. The edges of the end may be rounded by filing.

2.8.3.4 The length of the specimen is to be in accordance with details given in Chapter 2.

2.8.3.5 Testing is to be carried out at ambient temperature and is to consist of expanding the end of the profile by means of a hardened conical steel mandrel having an included angle of at least 60°.

2.8.3.6 The sample is considered to be unacceptable if the sample fails with a clean split along the weld line which confirms lack of fusion.

2.8.4 Requirements of mechanical properties for rolled products in different delivery conditions are given in Table 2.8.1 and are applicable for thickness within the range 3 [mm] to 50 [mm]. For thickness above 10 [mm], however, lower mechanical properties may be accepted.

2.8.5 Requirements of mechanical properties for extruded products in different delivery conditions are given in Table 2.8.2 and are applicable for thickness within the range 3 [mm] to 50 [mm].

2.8.6 Requirements of mechanical properties and delivery conditions for extruded closed profiles are given in Table 2.8.3.

2.8.7 Other delivery conditions with related mechanical properties may be accepted by IRS, in each particular case.

Table 2.8.1 : Mechanical properties for rolled products 3 [mm] ≤ t ≤ 50 [mm]

Grade	Temper [3] condition	Thickness, [t]	0.2% proof stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation % minimum [1]	
					On gauge length of 50 [mm]	On gauge length of 5 x dia
5083	O	3 ≤ t ≤ 50 mm	125	275-350	16	14
	H111	3 ≤ t ≤ 50 mm	125	275-350	16	14
	H112	3 ≤ t ≤ 50 mm	125	275	12	10
	H116	3 ≤ t ≤ 50 mm	215	305	10	10
	H321	3 ≤ t ≤ 50 mm	215-295	305-385	12	10
5383	O	3 ≤ t ≤ 50 mm	145	290	-	17
	H111	3 ≤ t ≤ 50 mm	145	290	-	17
	H116	3 ≤ t ≤ 50 mm	220	305	10	10
	H321	3 ≤ t ≤ 50 mm	220	305	10	10
5059	O	3 ≤ t ≤ 50 mm	160	330	24	24
	H111	3 ≤ t ≤ 50 mm	160	330	24	24
	H116	3 ≤ t ≤ 20 mm	270	370	10	10
		20 < t ≤ 50 mm	260	360	-	10
	H321	3 ≤ t ≤ 20 mm	270	370	10	10
		20 < t ≤ 50 mm	260	360	-	10
5086	O	3 ≤ t ≤ 50 mm	95	240-305	16	14
	H111	3 ≤ t ≤ 50 mm	95	240-305	16	14
	H112	3 ≤ t ≤ 12.5 mm	125	250	8	
		12.5 < t ≤ 50 mm	105	240		9
	H116	3 ≤ t ≤ 50 mm	195	275	10 [2]	9
5754	O	3 ≤ t ≤ 50 mm	80	190-240	18	17
	H111	3 ≤ t ≤ 50 mm	80	190-240	18	17

Table 2.8.1 : (Contd.)

Grade	Temper [3] condition	Thickness, t	0.2% proof stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation % minimum [1]	
					On gauge length of 50 [mm]	On gauge length of 5 x dia
5456	O	3 ≤ t ≤ 6.3 mm	130-205	290-365	16	
		6.3 < t ≤ 50 mm	125-205	285-360	16	14
	H116	3 ≤ t ≤ 30 mm	230	315	10	10
		30 < t ≤ 40 mm	215	305		10
		40 < t ≤ 50 mm	200	285		10
	H321	3 ≤ t ≤ 12.5 mm	230-315	315-405	12	
		12.5 < t ≤ 40 mm	215-305	305-385		10
		40 < t ≤ 50 mm	200-295	285-370		10

Notes:

1) Elongation in 50 mm apply for thicknesses upto and including 12.5 mm and in 5d for thicknesses over 12.5 mm.
2) 8% for thicknesses upto and including 6.3 mm.
3) The mechanical properties for the O and H111 tempers are the same. However, they are separated to discourage dual certification as these tempers represent different processing.

Designation	Condition
F	As fabricated
O	Annealed, soft
H1	Strain hardened only
H2	Strain hardened and partially annealed
H3	Strain hardened and thermally stabilized
H321	Strain hardened and stabilized
H11	Strain hardened to specified strength
H12	Strain hardened to specified strength
H13	Strain hardened to specified strength
H111	Less strain hardened than H11 e.g. by straightening or stretching
H112	No controlled strain hardening, but there are mechanical property limits
H116	Treatment against exfoliation corrosion
T5	Cooled from an elevated temperature shaping process and then artificially aged
T6	Solution heat treated and then artificially aged.

Table 2.8.2 : Mechanical properties for extruded products 3 [mm] ≤ t ≤ 50 mm

Grade	Temper condition	Thickness, t	0.2% proof stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation % minimum 1)	
					On gauge length of 50 [mm]	On gauge length of 5 x dia
5083	O	3 ≤ t ≤ 50 mm	110	270-350	14	12
	H/111	3 ≤ t ≤ 50 mm	165	275	12	10
	H112	3 ≤ t ≤ 50 mm	110	270	12	10
5383	O	3 ≤ t ≤ 50 mm	145	290	17	17
	H111	3 ≤ t ≤ 50 mm	145	290	17	17
	H112	3 ≤ t ≤ 50 mm	190	310		13
5059	H112	3 ≤ t ≤ 50 mm	200	330		10
5086	O	3 ≤ t ≤ 50 mm	95	240-315	14	12
	H111	3 ≤ t ≤ 50 mm	145	250	12	10
	H112	3 ≤ t ≤ 50 mm	95	240	12	10
6005A	T5	3 ≤ t ≤ 50 mm	215	260	9	8
	T6	3 ≤ t ≤ 10 mm	215	260	8	6
		10 ≤ t ≤ 50 mm	200	250	8	6
6061	T6	3 ≤ t ≤ 50 mm	240	260	10	8
6082	T5	3 ≤ t ≤ 50 mm	230	270	8	6
	T6	3 ≤ t ≤ 50 mm	250	290	6	
		3 ≤ t ≤ 50 mm	260	310	10	8

Notes:

1) The values are applicable for longitudinal and transverse tensile test specimens as well.
2) Elongation in 50 mm applies for thicknesses upto and including 12.5 mm and in 5d for thicknesses over 12.5 mm.

Table 2.8.3 : Mechanical properties for extruded closed profiles (testing transverse to extruding direction)

Grade	Temper condition	0.2% proof stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation % min on gauge length of 5 x dia
6061	T5/T6	205	245	4
6005A	T5/T6	215	250	5
6082	T5/T6	240	290	5

2.9 Freedom from defects

2.9.1 The finished material is to have a good finish and is to be free from internal and surface defects prejudicial to the use of the concerned material for the intended application.

2.9.2 Slight surface imperfections may be removed by smooth grinding or machining as long as the thickness of the material remains within the tolerances given in 2.4.

2.10 Corrosion testing

2.10.1 Rolled 5xxx-alloys of type 5083, 5383, 5059, 5086 and 5456 in the H116 and H321 tempers intended for use in marine hull construction or in marine applications where frequent direct contact with seawater is expected, are to be corrosion tested with respect to exfoliation and intergranular corrosion resistance.

2.10.2 The manufacturers are to establish the relationship between microstructure and resistance to corrosion when the above alloys are approved. A reference photomicrograph taken at 500x under the conditions specified in ASTM B928:2015, Section 9.4.1, is to be established for each of the alloy-tempers and relevant thickness ranges. The reference photographs are to be taken from samples which have exhibited no evidence of exfoliation corrosion and a pitting rating of PB or better, when subjected to the test described in ASTM G66:2018 “Standard test method for visual assessment of exfoliation, corrosion susceptability of 5xxx series aluminium alloys” (ASSET Test). The samples are also to have exhibited resistance to intergranular corrosion at a mass loss not greater than 15 [mg/cm^2], when subjected to tests described in ASTM G67:2018 “Standard test method for determining the susceptibility to intergranular corrosion of 5xxx series aluminium alloys by mass loss after exposure to nitric acid” (NAMLT). Upon satisfactory establishment of the relationship between microstructure and resistance to corrosion, the master photomicrographs and the results of the corrosion tests are to be approved by IRS. Production practices are not to be changed after approval of the reference micrographs.

Other test methods may also be accepted at the discretion of IRS.

2.10.3 For batch acceptance of 5xxx-alloys in the H116 and H321 tempers, metallographic examination of one sample selected from mid width at one end of a coil or random sheet or plate is to be carried out. The microstructure of the sample is to be compared to the reference photomicrograph of acceptable material in the presence of the Surveyor. A longitudinal section perpendicular to the rolled surface is to be prepared for metallographic examination under the conditions specified in ASTM B928:2015, Section 9.6.1. If the microstructure shows evidence of continuous grain boundary network of aluminium-magnesium precipitate in excess of the reference photomicrographs of acceptable material, the batch is either to be rejected or tested for exfoliation-corrosion resistance and intergranular corrosion resistance subject to the agreement of the Surveyor. The corrosion tests are to be in accordance with ASTM G66:2018 and G67:2018 or equivalent standards. Acceptance criteria are as noted below:

i) The sample is to exhibit no evidence of exfoliation corrosion
ii) The pitting rating of the sample is to be PB or better when subjected to ASTM G66:2018 ASSET test
iii) The sample is to exhibit resistance to intergranular corrosion at a mass loss no greater than 15 [mg/cm^2] when

subjected to ASTM G67:2018 NAMLT test.

If the results from testing satisfy the acceptance criteria the batch is accepted, else it is to be rejected.
As an alternative to metallographic examination, each batch may be tested for exfoliation-corrosion resistance and intergranular corrosion resistance, in accordance with ASTM G66:2018 and G67:2018 under the conditions specified in ASTM B928:2015 or equivalent standards. If this alternative is used, then the results of the test must satisfy the acceptance criteria stated above.

2.11 Test materials

2.11.1 Definition of batches

Each batch is made up of products:

- of the same alloy grade and from the same cast
- of the same product form and similar dimensions (for plates, the same thickness)
- manufactured by the same process
- having been submitted simultaneously to the same temper condition.

2.11.2 The test samples are to be taken

- at one third of the width from a longitudinal edge of rolled products.
- in the range 1/3 to 1/2 of the distance from the edge to the centre of the thickest part of extruded products.

2.11.3 Test samples are to be taken so that the orientation of test specimens is as follows:

a) Rolled products

Normally, tests in the transverse direction are required. If the width is insufficient to obtain transverse test specimen, or in the case of strain hardening alloys, tests in the longitudinal direction will be permitted.

b) Extruded products

The extruded products are tested in longitudinal direction.

2.11.4 After removal of test samples, each test specimen is to be marked in order that its original identity, location and orientation is maintained.

2.12 Mechanical test specimens

2.12.1 Type and location of tensile test specimens are to be in accordance with details given in Ch.2.

2.13 Number of test specimens

2.13.1 Tensile test

a) Rolled products

- One tensile test specimen is to be taken from each batch of the product. If the weight of one batch exceeds 2000 [kg], one extra tensile test specimen is to be taken from every 2000 [kg] of the product or fraction thereof, in each batch.
- For single plates or for coils weighing more than 2000 [kg] each, only one tensile test specimen per plate or coil shall be taken.

b) Extruded products

- For the products with a nominal weight of less than 1 [kg/m], one tensile test specimen is to be taken from each 1000 [kg], or fraction thereof, in each batch. For nominal weights between 1 and 5 [kg/m], one tensile test specimen is to be taken from each 2000 [kg] or fraction hereof, in each batch. If the nominal weight exceeds 5 [kg/m], one tensile test specimen is to be taken for each 3000 [kg] of the product or fraction thereof, in each batch.

2.13.2 Corrosion tests

For rolled plates of grade 5083, 5383, 5059 and 5086 delivered in the tempers H116 or H321, one sample is to be tested per batch.

2.14 Retest procedures

2.14.1 When the tensile test from the first piece selected in accordance with Sec.11 fails to meet the requirements, two further tensile tests may be made from the same piece. If both of these additional tests are satisfactory,

this piece and the remaining pieces from the same batch may be accepted.

2.14.2 If one or both the additional tests referred to above are unsatisfactory, the piece is to be rejected, but the remaining material from the same batch may be accepted provided that two of the remaining pieces in the batch selected in the same way, are tested with satisfactory results. If unsatisfactory results are obtained from either of these two pieces then the batch of material is to be rejected.

2.14.3 In the event of any material bearing the Classification Society's brand failing to comply with the test requirements, the brand is to be unmistakably defaced by the manufacturer.

2.15 Visual and non-destructive examination

2.15.1 Surface inspection and verification of dimensions are the responsibility of the manufacturer, and acceptance by the Surveyors of material later found to be defective shall not absolve the manufacturer from this responsibility.

2.15.2 In general, the non-destructive examination of materials is not required for acceptance purposes. Manufacturers are expected, however to employ suitable methods of non-destructive examination for the general maintenance of quality standards.

2.15.3 For applications where the non-destructive examination of materials is considered to be necessary, the extent of this examination, together with appropriate acceptance standards, are to be agreed between the purchaser, manufacturer and Surveyor.

2.16 Rectification of defects

2.16.1 Local surface defects may be removed by machining or grinding, provided the thickness of the material remains within the tolerances given in para 2.4. The extent of repairs is to be agreed upon with the Surveyor, and all repairs are to be carried out under Surveyor's supervision, unless otherwise arranged.

2.16.2 Surface defects which cannot be dealt with as in 2.12.1 are not allowed to be repaired, unless it can be ensured that repair by welding does not affect the strength and stability of the piece for the intended purpose. Any case of repair by welding is to be specified in detail for consideration and approval by the Surveyor. Prior to any such repair welding, the defect is to be removed by machining or grinding. After complete removal of the defect and before welding the thickness of the piece at no place is to be reduced by more than 20 per cent. The welding is to be carried out by approved welders. The weld is to be ground flush with the surrounding piece surface. Before repair welding is commenced and after grinding the weld bead, suitable non destructive testing may be required at the discretion of the Surveyor.

2.17 Identification

2.17.1 The manufacturer is to adopt a system of identification which will ensure that all finished material in a batch presented for test is of the same nominal chemical composition.

2.17.2 Products are to be clearly marked by the manufacturer in accordance with the requirements of Ch.1. The following details are to be shown on all materials which have been accepted:

a) Manufacturer's name or trade mark;

b) Grade of alloy;

c) Identification mark which will enable the full history of the item to be traced;

d) Abbreviated designation of temper condition in accordance with para 2.6;

e) Personal stamp of the Surveyor responsible for the final inspection and also IRS’s stamp.

f) Tempered grades that are corrosion tested in accordance with 2.12 are to be marked “M” after the temper condition, e.g. 5083 H321 M.

2.17.3 When extruded products are bundled together or packed in crates for delivery, the marking specified in para 2.17.2 are to be affixed by a securely fastened tag or label.

2.18 Certification

2.18.1 Each test certificate or shipping statement is to include the following particulars:

a) Purchaser's name and order number;

b) Contract number;

c) Address to which material is to be dispatched;

d) Description and dimensions;

e) Specification or grade of alloys;

f) Identification mark which will enable the full history of the item to be traced;

g) Chemical composition;

h) Mechanical test results (Not required on shipping statement);

i) Details of heat treatment, where applicable; and

j) Corrosion test results (if any).

2.18.2 Where the alloy is not produced at the works at which it is wrought, a certificate is to be supplied by the Manufacturer of the alloy stating the cast number and chemical composition. The works at which alloys are produced must be approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

Section 3

Aluminium Alloy Castings

3.1 Scope

3.1.1 Provision is made in this section for aluminium alloy castings intended for use in the construction of ships, ships for liquid chemicals and other marine structures, liquefied gas piping systems where the design temperature is not lower than minus 165°C. These materials should not be used for piping outside cargo tanks except for short lengths of pipes attached to cargo tanks in which case fire resisting insulation should be provided.

3.1.2 Castings are to be manufactured and tested in accordance with Ch.1 and Ch.2 and also with the requirements of this Section.

3.1.3 As an alternative to 3.1.2, castings which comply with National/International and proprietary specifications may be accepted provided that these specifications give reasonable equivalence to the requirements of this Section or are approved for a specific application. Generally survey and certification are to be carried out in accordance with the requirements of Ch.1.

3.2 Manufacture

3.2.1 Castings are to be manufactured at foundries approved by IRS.

3.3 Quality of castings

3.3.1 All castings are to be free from surface or internal defects which would be prejudicial to their proper application in service.

3.4 Chemical composition

3.4.1 The chemical composition of a sample from each cast is to comply with the requirements given in Table 3.4.1. Suitable grain refining elements may be used at the discretion of the Manufacturer. The content of such elements is to be reported in ladle analysis.

3.4.2 Where it is proposed to use alloys not specified in Table 3.4.1 details of chemical composition, heat treatment and mechanical properties are to be submitted for approval.

3.4.3 When a cast is wholly prepared from ingots for which an analysis is already available, and provided that no significant alloy additions are made during melting, the ingot maker's certified analysis can be accepted subject to occasional checks as required by the Surveyor.

Table 3.4.1 : Chemical composition for aluminium alloy castings

Alloy Element %	Grade			
	AlMg3	**AlSi12**	**AlSi10Mg**	**AlSi7 High purity**
Copper	0.1 max	0.1 max.	0.1 max.	0.1 max.
Magnesium	2.5 - 4.5	0.1 max.	0.15 - 0.4	0.25 - 0.45
Silicon		0.5 max. 11.0 - 13.5	9.0 - 11.0	6.5 - 7.5
Iron	0.5 max.	0.7 max.	0.6 max.	0.2 max.
Manganese	0.6 max.	0.5 max.	0.6 max.	0.1 max.
Zinc	0.2 max.	0.1 max.	0.1 max.	0.1 max.
Chromium	0.1 max.	-	-	-
Titanium	0.2 max.	0.2 max.	0.2 max.	0.2 max.
Others				
each	0.05 max.	0.05 max.	0.05 max.	0.05 max.
Total	0.15 max.	0.15 max.	0.15 max.	0.15 max.
Aluminium	Remainder	Remainder	Remainder	Remainder

3.5 Heat treatment

3.5.1 Castings are to be supplied in the following conditions:

Grade Al-Mg 3 — As manufactured

Grade Al-Si 12 — As manufactured

Grade Al-Si 10 Mg — As manufactured or solution heat treated and precipitation hardened

Grade Al-Si 7 Mg — Solution heat treated and precipitation (high purity) hardened

3.6 Mechanical tests

3.6.1 At least one tensile specimen is to be tested from each cast, where heat treatment is involved, for each treatment batch from each cast. Where continuous melting is employed 500 [kgs] of fettled castings may be regarded as a cast.

3.6.2 The test samples are to be separately cast in moulds made from the same type of material as used for the castings. These moulds should conform to National Standards.

3.6.3 The methods and procedures for the identification of the test specimens, and the castings they represent, are to be agreed with the Surveyor. The identification marks are to be maintained during the preparation of test specimens.

3.6.4 Where castings are supplied in the heat treated condition, the test samples are to be heat treated together with the castings which they represent prior to testing.

3.6.5 The results of all tensile tests are to comply with the appropriate requirements given in Table 3.6.1 and/or Table 3.6.2.

Table 3.6.1 : Minimum mechanical properties for acceptance purpose of sand cast and investment cast reference test pieces

Alloy	Temper (see Note)	Tensile strength [N/mm^2]	Elongation %
AlMg3	M	150	5
AlSi12	M	150	3
AlSi10Mg	M	150	2
AlSi10Mg	TF	220	1
AlSi7Mg	TF	230	5

Note
M - As cast condition
TF - Solution heat treated and precipitation hardened condition

Table 3.6.2 : Minimum mechanical properties for acceptance purpose of chill cast reference test pieces

Alloy	Temper (see Note)	Tensile strength [N/mm^2]	Elongation %
AlMg3	M	150	5
AlSi12	M	170	3
AlSi10Mg	M	170	3
AlSi10Mg	TF	240	1.5
AlSi7Mg	TF	250	5

Note
M - As cast condition
TF - Solution heat treated and precipitation hardened condition

3.7 Visual examination

3.7.1 All castings are to be cleaned and adequately prepared for inspection.

3.7.2 The accuracy and verification of dimensions are the responsibility of the manufacturer, unless otherwise agreed.

3.7.3 Before acceptance, all castings are to be presented to the Surveyor for visual examination.

3.8 Rectification of defective castings

3.8.1 At the discretion of the Surveyor, small surface blemishes may be removed by local grinding.

3.8.2 Where appropriate, repair by welding may be accepted at the discretion of the Surveyor. Such repair is to be made in accordance with an approved procedure.

3.9 Pressure testing

3.9.1 Where required by the relevant construction rules, castings are to be pressure tested before final acceptance. Unless otherwise agreed, these tests are to be carried out in the presence and to the satisfaction of the Surveyor.

3.10 Identification

3.10.1 The manufacturer is to adopt a system of identification which will enable all finished castings to be traced to the original cast and

the Surveyor is to be given full facilities for tracing the casting when required.

3.10.2 All castings which have been tested and inspected with satisfactory results are to be clearly marked with following details:

a) Identification number, cast number or other numbers which will enable the full history of the casting to be traced;

b) IR and the abbreviated name of local office of IRS;

c) Personal stamp of the surveyor responsible for the inspection;

d) Test pressure where applicable; and

e) Date of final inspection.

3.10.3 Where small castings are manufactured in large numbers, modified arrangements for identification may be specially agreed with the Surveyor.

3.11 Certification

3.11.1 The manufacturer is to provide the Surveyor with a written statement giving the following particulars for each casting or batch of castings which have been accepted :

a) Purchaser name and order number;

b) Description of castings and alloy type;

c) Identification number;

d) Ingot or Cast analysis;

e) General details of heat treatment where applicable;

f) Results of mechanical tests; and

g) Test pressure, where applicable.

Section 4

Aluminium/Steel Transition Joints

4.1 Scope

4.1.1 Provision is made in this section for explosion bonded composite aluminium/steel transition joints used for connecting aluminium structures to steel plating.

4.1.2 Each design is to be separately approved by IRS.

4.2 Manufacture

4.2.1 Transition joints are to be manufactured by an approved producer in accordance with an approved specification which is to include the maximum temperature allowable at the interface during welding.

4.2.2 The aluminium material is to comply with the requirements of Sec.1 and the steel is to be of an appropriate grade complying with the requirements of Ch.3.

4.2.3 Alternative materials which comply with International, National or proprietary specifications may be accepted provided that they give reasonable equivalence to the requirements of 4.2.2 or are approved for a specific application.

4.2.4 Intermediate layers between aluminium and steel may be used, in which case the material of any such layer is to be specified by the manufacturer and will be recorded in the approval certificate. Any such intermediate layer is then to be used in all production joints.

4.3 Visual and non-destructive examination

4.3.1 Each composite plate is to be subjected to 100 per cent visual and ultrasonic examination in accordance with a relevant National/ International standard to determine the extent of any unbounded areas. The unbounded areas are unacceptable and any such area and the surrounding 25 [mm] area is to be discarded.

4.4 Mechanical tests

4.4.1 Two shear test specimens and two tensile test specimens are to be taken from each end of each composite plate for tests to be made on bond strength. One shear and one tensile test specimen from each end are to be tested at ambient temperature after heating to the maximum allowable interface temperature; the other two specimens are to be tested without heat treatment.

4.4.2 Shear tests may be made on a specimen as shown in Fig.4.4.1 or an appropriate equivalent. Tensile tests may be made across the interface by welding extension pieces to each surface or by the ram method shown in Fig.4.4.2 or by an appropriate alternative method.

4.4.3 The shear and tensile strengths of all the test specimens are to comply with the requirements of the manufacturing specification.

4.4.4 If either the shear or tensile test strength of the bond is less than the specified minimum but not less than 70 per cent of the specified minimum, two additional shear and two tensile test specimens from each end of the composite plate are to be tested and, in addition bend tests as described in 4.4.6 and Table 4.4.1 are to be conducted.

4.4.5 If either the shear or the tensile strength of the bond is less than 70 per cent of the specified minimum the case is to be investigated. After evaluation of the results of this investigation IRS will consider the extent of composite plate which is to be rejected.

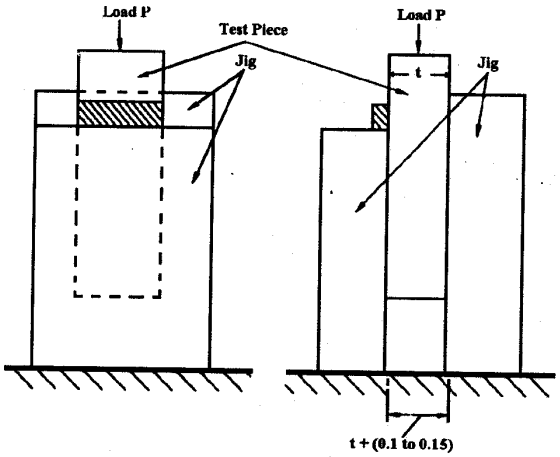


Fig.4.4.1 : Shear tests

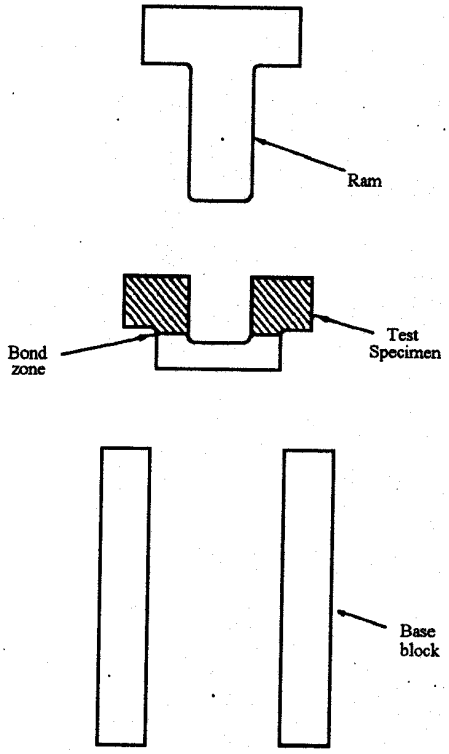


Fig.4.4.2 : Tensile tests

4.4.6 Bend tests, when required, are to be made under the following conditions, as listed in Table 4.4.1 :

a) the aluminium plate is in tension;

b) the steel plate is in tension; and

c) a side bend is applied.

Table 4.4.1 : Bend tests on explosion bonded aluminium / steel transition joints

Type of test	Minimum bend, degrees	Diameter of former
Aluminium in tension	90	3T
Steel in tension	90	3T
Side bend	90	6T

4.5 Identification

4.5.1 Each acceptable transition strip is to be clearly marked with IRS brand IR and the following particulars:

a) Manufacturers name or trade mark;

b) Identification mark for the grade of aluminium; and

c) Identification mark for the grade of steel.

The particulars are to be stamped on the aluminium surface at one end of the strip.

4.6 Certification

4.6.1 Each test certificate or shipping statement is to include the following particulars:

a) Purchaser's name and order number;

b) The contract number for which the material is intended, if known;

c) Address to which the material is dispatched;

d) Description and dimensions of the material;

e) Specification or grades of both the aluminium alloy and the steel and any intermediate layer;

f) Cast numbers of steel and aluminium plates;

g) Identification number of the composite plate; and

h) Mechanical test results (not required on the shipping statement).

End of Chapter

Chapter 10

Equipment

Contents

Section

1 *Anchors*

2 *Stud Link Chain Cables*

3 *Short Link Chain Cables*

4 *Steel Wire Ropes*

5 *Offshore Mooring Chains*

Section 1

Anchors

1.1 Scope

1.1.1 The following paragraphs give requirements for cast, forged or fabricated steel anchor heads, shanks and anchor shackles. The requirements given in this section are applicable to the following types of anchors:

a) Ordinary stockless and stocked anchors
b) High Holding Power (HHP) anchors, and
c) Super High Holding Power (SHHP) anchors not exceeding 1500 [kg] in mass.

1.2 Manufacture

1.2.1 Cast steel anchor heads, shanks and shackles are to be manufactured and tested in accordance with the relevant requirements for castings for welded construction of Ch.4. The steel is to be fine grain treated with aluminium. The toughness of steel castings for SHHP anchors is to be not less than charpy V-notch energy average of 27 J at 0°C.

1.2.2 Forged steel anchor heads, shanks, shackles and anchor crown pins are to be manufactured in accordance with the requirements for forgings of weldable quality of Ch.5.

1.2.3 Plate material and bars used for the manufacture of fabricated parts of steel anchors are to comply with the requirements of Ch.3.

For welded super high holding power (SHHP) anchors, the base steel grades are to be selected with respect to the material grade requirements for Class II in Part 3, Chapter 2 'Materials of Construction' Section 2 'Use of Steel Grades'.

1.2.4 The welding consumables are to meet the toughness for the base steel grades in accordance with Part 2, Chapter 11 'Approval of Welding Consumables for Use in Ship Construction'.

1.2.5 Fabrication is to be carried out by qualified welders using approved welding procedure.

1.2.6 The toughness of the anchor shackles for SHHP anchors is to meet that for Grade CC3 anchor chain given in Section 2. The toughness of steel castings for SHHP anchors is to be not less than charpy V-notch energy average of 27 J at 0°C.

1.2.7 Hardness values of mating parts are to be such that the more easily replaceable part wears faster.

1.3 Dimensions and tolerances

1.3.1 Anchors are to be manufactured as per approved drawings or as per internationally recognised designs meeting the tolerances specified in such documents. In addition the following dimensional tolerances are also to be applied:

- the clearance either side of the shank within the shackle jaws is not to be more than 3 [mm] for anchors upto 3000 [kg] mass, 4 [mm] for anchors upto 5000 [kg] mass, 6 [mm] for anchors upto 7000 [kg] mass and 12 [mm] for larger anchors.

- the shackle pin is to be push fit in the eyes of the shackle, which are to be chamfered on the outside to ensure tightness when the pin is clenched over. The shackle pin to hole clearance is not to be more than 0.5 [mm] for pins upto 57 [mm] and not more than 1 [mm] for pins of larger diameter

- the anchor crown pin is to be snug fit within the chamber and long enough to prevent horizontal movement. The gap is not to be more than 1% of the chamber length.

- The lateral movement of the shank should not exceed 3 degrees.

1.4 Proof test of anchors

1.4.1 Anchors of all sizes are to be proof load tested with the load specified in Table 1.4.1. Anchors inclusive of stock, having a mass of 75 [kgs] or more (56 [kgs] in case of high holding power anchors) are to be tested at a proving establishment recognized by IRS.

1.4.2 The proof test load is to be as given in Table 1.4.1. The mass to be used in the Table is to be as follows:-

a) For stockless anchors - the total mass of the anchor;

b) For stocked anchors - the mass of the anchor excluding the stock;

c) For high holding power anchors - a nominal mass equal to 1.33 times the actual mass of the anchor;

d) For super high holding power anchors - a nominal mass equal to 2.0 times the actual mass of the anchor.

1.4.3 The proof load is to be applied on the arm or on the palm at a spot which, measured from the extremity of the bill, is one-third of the distance between it and the centre of the crown.

In the case of the stockless anchors, both arms are to be tested at the same time, first on one side of the shank, then reversed and tested on the other.

1.4.4 Before application of proof test load the anchors are to be examined to be sure that castings are reasonably free of surface imperfections of harmful nature.

On completion of the proof load tests the anchors are to be examined for cracks and other defects and for anchors made in more than one piece, the anchors are to be examined for free rotation of their heads over the complete angle.

In every test the difference between the gauge lengths (shown in Fig.1.4.4), where one-tenth of the required load was applied first and where the load has been reduced to one-tenth of the required load from the full load, is not to exceed one percent (1%).

1.4.5 In addition to the requirements given in this Chapter attention must be given to any relevant statutory requirements of the National Authority of the country in which the ship is to be registered.

Table 1.4.1 : Proof loads for anchors					
Mass of anchor [kg] (1)	**Proof load* [kN] (2)**	**Mass of anchor [kg] (1)**	**Proof load* [kN] (2)**	**Mass of anchor [kg] (1)**	**Proof load* [kN] (2)**
50	23.2	2000	349.0	7000	804.0
55	25.2	2100	362.0	7200	818.0
60	27.1	2200	376.0	7400	832.0
65	28.9	2300	388.0	7600	845.0
70	30.7	2400	401.0	7800	861.0
75	32.4	2500	414.0	8000	877.0
80	33.9	2600	427.0	8200	892.0
90	36.3	2700	438.0	8400	908.0
100	39.1	2800	450.0	8600	922.0
120	44.3	2900	462.0	8800	936.0
140	49.0	3000	474.0	9000	949.0
160	53.3	3100	484.0	9200	961.0
180	57.4	3200	495.0	9400	975.0
200	61.3	3300	506.0	9600	987.0
225	65.8	3400	517.0	9800	998.0
250	70.4	3500	528.0	10000	1010.0
275	74.9	3600	537.0	10500	1040.0
300	79.5	3700	547.0	11000	1070.0
325	84.1	3800	557.0	11500	1090.0
350	88.8	3900	567.0	12000	1110.0
375	93.4	4000	577.0	12500	1130.0
400	97.9	4100	586.0	13000	1160.0
425	103.0	4200	595.0	13500	1180.0
450	107.0	4300	604.0	14000	1210.0
475	112.0	4400	613.0	14500	1230.0
500	116.0	4500	622.0	15000	1260.0
550	125.0	4600	631.0	15500	1270.0
600	132.0	4700	638.0	16000	1300.0
650	140.0	4800	645.0	16500	1330.0
700	149.0	4900	653.0	17000	1360.0
750	158.0	5000	661.0	17500	1390.0
800	166.0	5100	669.0	18000	1410.0
850	175.0	5200	677.0	18500	1440.0
900	182.0	5300	685.0	19000	1470.0
950	191.0	5400	691.0	19500	1490.0
1000	199.0	5500	699.0	20000	1520.0
1050	208.0	5600	706.0	21000	1570.0
1100	216.0	5700	713.0	22000	1620.0
1150	224.0	5800	721.0	23000	1670.0
1200	231.0	5900	728.0	24000	1720.0
1250	239.0	6000	735.0	25000	1770.0
1300	247.0	6100	740.0	26000	1800.0
1350	255.0	6200	747.0	27000	1850.0
1400	262.0	6300	754.0	28000	1900.0
1450	270.0	6400	760.0	29000	1940.0
1500	278.0	6500	767.0	30000	1990.0
1600	292.0	6600	773.0	31000	2030.0
1700	307.0	6700	779.0	32000	2070.0
1800	321.0	6800	786.0	34000	2160.0
1900	335.0	6900	794.0	36000	2250.0

Table 1.4.1 : Proof loads for anchors (Contd.)					
Mass of anchor [kg] (1)	**Proof load* [kN] (2)**	**Mass of anchor [kg] (1)**	**Proof load* [kN] (2)**	**Mass of anchor [kg] (1)**	**Proof load* [kN] (2)**
38000	2330.0	42000	2490.0	46000	2650.0
40000	2410.0	44000	2570.0	48000	2730.0
Proof loads for intermediate masses are to be determined by linear interpolation					

Notes

Where ordinary anchors have a mass exceeding 48 000 [kg], the proof loads are to be taken as 2.059 (mass of anchor in kg)$^{2/3}$ [kN].

Where high holding power anchors have a mass exceeding 38 000 [kg], the proof loads are to be taken as 2.452 (actual mass of anchor in kg)$^{2/3}$ [kN].

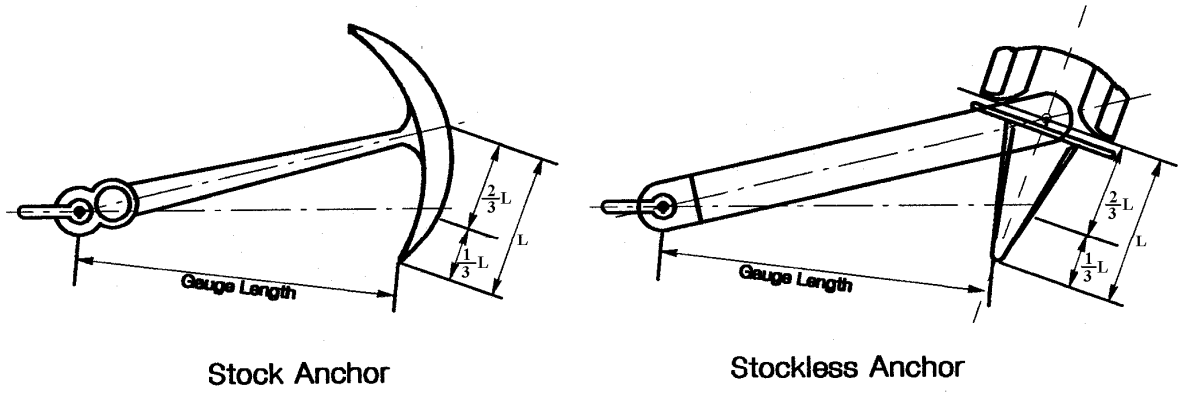


Fig.1.4.4 : Gauge Length

1.5 Inspections and other tests

1.5.1 Inspection and testing of anchor components is to be carried out as per the following:

a) Cast components are to be tested as per

Test Programme A

or

Test Programme B, where the Charpy V notch energy average of the cast material at 0°C is not less than 27J.

b) Forged / fabricated components are to be tested as per Test Programme B.

Test Programme A is to consist of Drop Test, Hammering Test, Visual Inspection and General NDE as described below.

Test Programme B is to consist of Visual Inspection, general NDE and Extended NDE as described in 1.5.5 and 1.5.6 below.

1.5.2 Drop test is to be carried out by dropping each anchor component individually from a height of 4 [m] to an iron or steel slab. The iron or steel slab should be able to resist the impact. The component under test should not fracture.

1.5.3 Hammering test is to be carried on each fluke and shank, after the drop test, by hammering the component, hung clear off the ground using a non-metallic sling, with a hammer of not less than 3 [kg] mass, to check the soundness.

1.5.4 Visual inspection is to be carried out of all accessible surfaces after the proof load test.

1.5.5 General non-destructive examination is to be carried out, after proof load testing, as per Table 1.5.5.

1.5.6 Extended non-destructive examination is to be carried out, after proof load testing, as per Table 1.5.6.

Table 1.5.5 : General NDE for Anchors		
Location	**Method of NDE**	
	SHHP	**Ordinary / HHP**
Feeders of castings	DP or MP and UT	DP or MP
Risers of castings	DP or MP and UT	DP or MP
All surfaces of castings	DP or MP	Not required
Weld repairs	DP or MP	DP or MP
Forged components	Not required	Not required
Fabrication welds	DP or MP	DP or MP
DP : Dye Penetrant Test		
MP : Magnetic Particle Test		
UT : Ultrasonic Testing		

Table 1.5.6 : Extended NDE for ordinary, HHP and SHHP anchors	
Location	**Method of NDE**
Feeders of castings	DP or MP and UT
Risers of castings	DP or MP and UT
All surfaces of castings	DP or MP
Random areas of castings	UT
Weld repairs	DP or MP
Forged components	Not required
Fabrication welds	DP or MP
DP : Dye Penetrant Test	
MP : Magnetic Particle Test	
UT : Ultrasonic Testing	

1.6 Identification

1.6.1 All identification marks are to be stamped on one side of the anchor, on the shank and the fluke, at locations reserved solely for this purpose.

1.6.2 The following details are to be marked on all the anchors:-

a) IR and abbreviated name of local office of IRS issuing the certificate;

b) Number of the certificate;

c) Month and year of test;

d) Mass (also the letters 'HHP/SHHP', when approved for as high holding power anchor/super high holding power anchor);

e) Mass of stock (in case of stocked anchors);

f) Personal stamp of Surveyor responsible for inspection.

g) Manufacturer’s mark

h) Unique cast identification number of shank and fluke, if applicable.

1.6.3 In addition to the markings detailed in 1.6.2, each important part of the anchor is to be plainly marked with the words 'forged steel' or 'cast steel' as appropriate.

1.7 Painting

1.7.1 Anchors are to be painted only on completion of all inspections and tests.

Section 2

Stud Link Chain Cables

2.1 Scope

2.1.1 The following requirements apply to the materials, design, manufacture and testing of stud link anchor chain cables and accessories used for ships.

2.1.2 Depending upon the nominal tensile strength of the chain cable steel used for manufacture, stud link chain cables are subdivided in to three grades, namely CC1, CC2 and CC3.

2.2 Manufacture

2.2.1 Chain cables and accessories are to be manufactured at Works approved by IRS for the pertinent type of chain cable, size and method of manufacture. Also refer Chapter 1, Section 1, Cl. 1.3.2.

2.2.2 Chain cables are to be preferably manufactured by flash butt welding using material suitable for CC1, CC2 or CC3 grades of chain cables. Chain cables may also be manufactured by drop forging or casting.

Accessories such as shackles, swivels and swivel shackles are to be forged or cast in steel of at least grade CC2 material. The welded construction of these components will be specially considered.

2.2.3 Details of the method of manufacture and the specification of the steel are to be submitted for approval.

2.2.4 All materials used for the manufacture of chain cables and accessories are to be supplied by manufacturer's works approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2. However, for Grade CC1 steel bars, approval of material manufacturer is not required.

For Grade CC3 steel bars, detailed material specifications including manufacturing procedure, deoxidation practice, specified chemical composition, heat treatment and mechanical properties are to be submitted.

2.3 Design and tolerances

2.3.1 The form and proportion of chain cable links and shackles are to be in accordance with ISO 1704:2008 (see Figs.2.3.1 to 2.3.6). All dimensions in the figures are shown in multiples of the nominal diameter d of the common link. The dimensions in brackets may be chosen for studless links in outboard end swivel pieces. Where designs do not comply with this and where accessories are of welded construction, plans giving full details of the design, manufacturing process and heat treatment are to be submitted for approval.

2.3.2 The following tolerances are applicable to links with the provision that the plus tolerance may be up to 5 per cent of the nominal diameter:

a)

Nominal diameter Measured at the Crown (see note)	**Max. minus tolerance**
Upto 40 [mm]	1 [mm]
Over 40 and upto 84 [mm]	2 [mm]
Over 84 and upto 122 [mm]	3 [mm]
Over 122 and upto 152 [mm]	4 [mm]
Over 152 and upto 184 [mm]	6 [mm]
Over 184 and upto 222 [mm]	7.5 [mm]

Note : Two measurements are to be taken at the same location: one in the plane of the link (see d_p in Fig.2.3.7) and one perpendicular to the plane of the link. The cross sectional area at the crown is to be calculated using the average of the diameters with negative and plus tolerance

The cross sectional area at the crown must not have any negative tolerance. For diameters of 20 [mm] or greater, the plus tolerance may be upto 5 per cent of the nominal diameter. For diameters less than 20 [mm] the plus tolerance is to be agreed with IRS at the time of approval;

Diameter measured at locations other than the crown is to have no negative tolerance. Plus tolerance may be up to 5 per cent of the nominal diameter except at the butt weld where it is to be in accordance to manufacturer's specification, which is to be agreed with IRS. For diameters less than 20[mm], the plus tolerance is to be agreed with IRS at the time of approval.

a) The maximum allowable tolerance on assembly measured over a length of 5 links may equal +2.5 per cent but may not be negative (measured with the chain under tension after proof load test);

b) All other dimensions are subject to a manufacturing tolerance of ± 2.5 per cent, provided that all of the final link parts fit together properly;

c) Studs must be located in the links centrally and at right angles to the sides of the link, although the studs at each end of any length may also be located off-centre to facilitate the insertion of the joining shackle. The following tolerances are regarded as being inherent in the method of manufacture and will not be objected to provided the stud fits snugly and its ends lie practically flush against the inside of the link.

Maximum off-centre distance 'X'	10 per cent of the nominal diameter d
Maximum deviation "α" from the 90° position	4°

The tolerances are to be measured in accordance with Fig.2.3.7.

2.3.3 The following tolerances are applicable to accessories :

Nominal diameter : + 5 per cent, - 0 per cent
Other diameter : ± 2.5 per cent.

2.4 Material for welded chain cables and accessories

2.4.1 Bar material intended for the manufacture of welded chain cables is to be in accordance with the appropriate requirements of Ch.3. Rimming steel is not acceptable for this application.

2.4.2 Bars of the same nominal diameter are to be presented for test in batches of 50 tonnes or fraction thereof from the same cast. A suitable length from one bar in each batch is to be selected for test purposes.

2.4.3 In order to evaluate the suitability of the bar material the sample selected from each batch is to be tested in a heat treatment condition equivalent to that of the finished chain cable and accessories. For this purpose only the sample need be heat treated.

2.4.4 For all grades, one tensile test is to be taken from each sample selected. Additionally one set of three Charpy V-notch impact test specimens is to be prepared and tested as required in Table 2.4.1.

2.4.5 Where the dimensions allow, the test specimens are to be taken at approximately one-third of the radius from the outer surface as shown in Fig.2.4.1. For smaller diameters the test specimens are to be taken as close as possible to these positions.

2.4.6 The cross-sectional area of the tensile test specimen is to be not less than 150 [mm^2]. Alternatively, the tensile test specimen may be a suitable length of bar tested in full cross-section.

2.4.7 The impact test specimens are to be notched in the radial direction as shown in Fig.2.4.1.

2.4.8 The results of all the mechanical testing are to comply with the requirements of Table 2.4.1.

2.4.9 The average value obtained from one set of three impact test specimens is to comply with the requirements given in Table 2.4.1. One individual value only may be below the specified average value provided it is not less than 70% of that value.

If the Charpy V-notch impact test requirements are not achieved, a retest of three further specimens selected from the same sample as per 1.10.2 of Chapter 1 shall be permissible. Failure to meet the requirements will result in rejection of the test unit represented unless it can be clearly attributable to improper simulated heat treatment.

If the tensile test requirements are not achieved, a retest of two further specimens selected from the same sample shall be permissible. Failure to meet the specified requirements in either of the additional tests will result in rejection of the test unit represented unless it can be clearly attributable to improper simulated heat treatment.

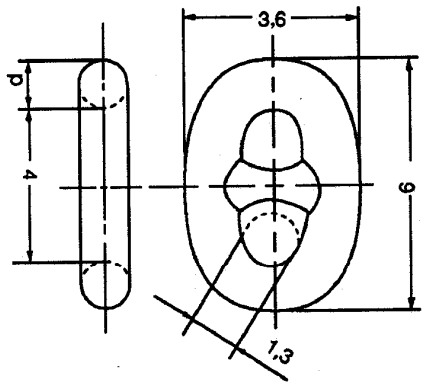


Fig. 2.3.1 : Common link

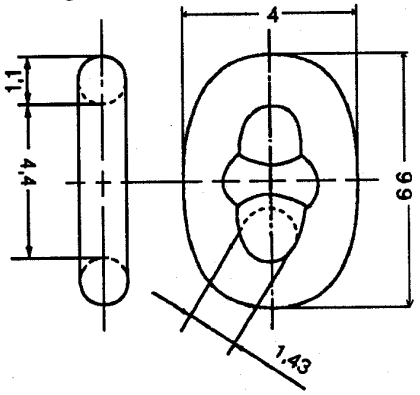


Fig. 2.3.2 : Enlarged link

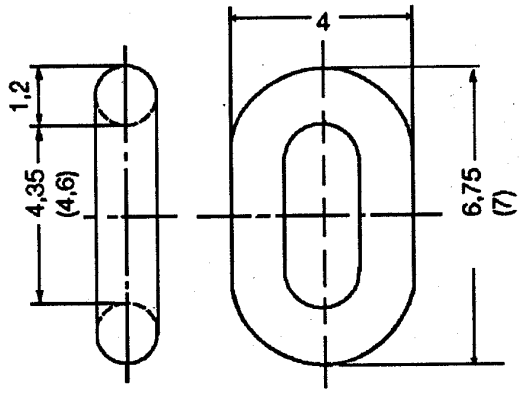


Fig. 2.3.3 : End link

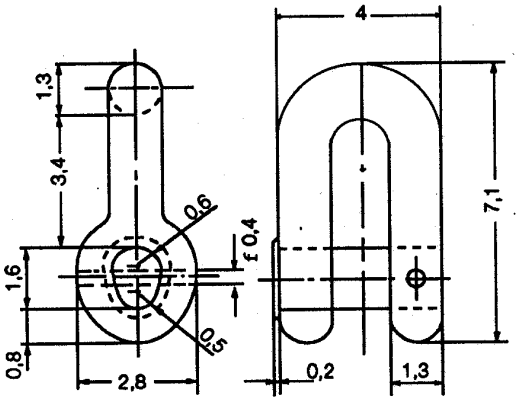


Fig. 2.3.4 : Joining shackle with shackle pin

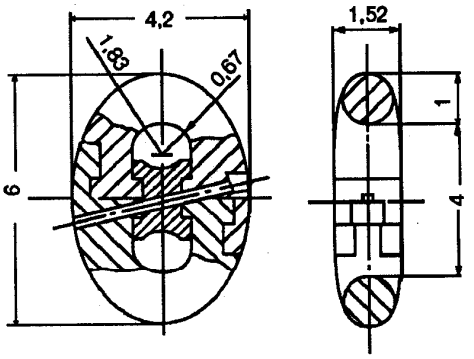


Fig. 2.3.5 : Kenter type lugless joining shackle

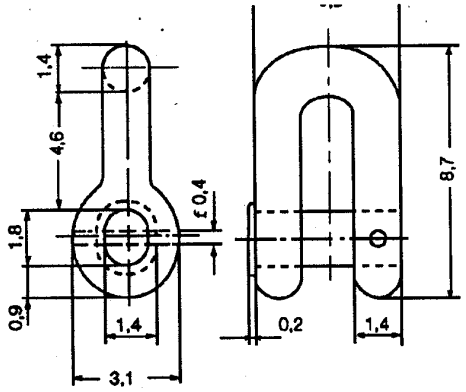


Fig. 2.3.6 : End shackle

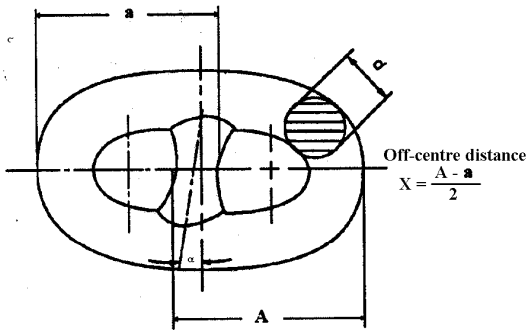


Fig. 2.3.7 : Link-stud tolerance

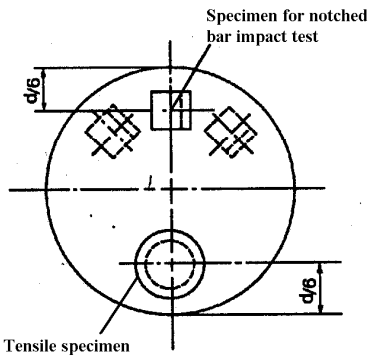


Fig. 2.4.1 : Position of test pieces

Table 2.4.1 : Mechanical properties of rolled steel bars for acceptance purposes

Designation	Tensile strength [N/mm²]	Yield strength [N/mm²] min.	Elongation on 5.65 $\sqrt{S_o}$% min.	Reduction of area % min.	Impact Tests	
					Test temp.°C	Average energy J min.[1]
Grade CC1	370 - 490	-	25	-	-	-
Grade CC2	490 - 690	295	22	-	0	27[1]
Grade CC3	Min. 690	410	17	40	0[2]	60
					-20	35

1 The impact test of grade CC2 material may be waived, if the chain is to be supplied in a heat treated condition as per Table 2.9.1.

2 The impact testing is normally to be carried out at 0°C.

Table 2.4.2 : Chemical composition of rolled steel bars

Designation	Chemical Composition					
	C max.	Si	Mn	P max.	S max.	Al (Total)[1] min.
Grade CC1	0.20	0.15 - 0.35	min. 0.40	0.040	0.040	-
Grade CC2[2]	0.24	0.15 - 0.55	max. 1.60	0.035	0.035	0.020
Grade CC3 [3]	To be specially considered in each case					

1 Aluminium may be replaced partly by other grain refining elements.

2 Subject to special consideration, additional alloying elements may be added.

3 To be killed and of fine grain.

Table 2.4.3 : Dimensional tolerance of rolled steel bars

Nominal diameter [mm]	Tolerance on diameter [mm]	Tolerance on roundness (dmax - dmin) [mm]
Less than 25	-0 + 1.0	0.6
25 – 35	-0 +1.2	0.8
36 – 50	-0 + 1.6	1.1
51 – 80	-0 + 2.0	1.5
81 – 100	-0 +2.6	1.95
101 – 120	-0 + 3.0	2.25
121 – 160	-0 + 4.0	3.0
161 – 210	-0 + 5.0	4.0

2.4.10 If failure to pass the tensile test or the Charpy V-notch impact test is definitely attributable to improper heat treatment of the test sample, a new test sample may be taken from the same piece and re-heat treated. The complete test (both tensile and impact test) is to be repeated and the original results obtained may be disregarded.

2.4.11 The chemical composition of the steel bars is to be generally within the limits given in Table 2.4.2.

2.4.12 The tolerances on diameter and roundness of rolled steel bars are to be within the limits specified in Table 2.4.3 unless otherwise agreed.

2.4.13 The minimum markings required for the steel bars are the manufacturers' brandmark, the steel grade and an abbreviated symbol of the heat. Steel bars having diameters of up to and including 40 mm and combined into bundles, may be marked on permanently affixed labels.

2.4.14 Material certification : Bar material for Grade 2 or Grade 3 is to be certified by IRS. For each consignment, manufacturers shall forward to the Surveyor a certificate containing at least the following data:

- Manufacturer's name and/or purchaser's order no.
- Number and dimensions of bars and weight of consignment
- Steel specification and chain grade
- Heat number
- Manufacturing procedure
- Chemical composition
- Details of heat treatment of the test sample (where applicable)
- Results of mechanical tests (where applicable)
- Number of test specimens (where applicable).

2.5 Material for cast chain cables and accessories

2.5.1 Manufacture of cast steel chain cables is generally to be in accordance with Ch.4, as appropriate.

2.5.2 All castings must be properly heat treated i.e. normalized, normalized and tempered or quenched and tempered, as specified in Table 2.7.1 for the relevant grade of steel.

2.6 Material for forged chain cables and accessories

2.6.1 The procedure for the manufacture of drop forgings for chain cables will be specially considered, but is generally to be in accordance with the appropriate requirements of Ch.5.

2.6.2 The stock material may be supplied in the as rolled condition. Finished forgings are to be properly heat treated, i.e. normalized, normalized and tempered or quenched and tempered, as specified for the relevant grade of steel in Table 2.7.1.

2.7 Heat treatment of completed chain cables

2.7.1 The completed chain cable and accessories are to be heat treated in accordance with Table 2.7.1, for the appropriate grade of cable.

2.7.2 In all cases, heat treatment is to be carried out prior to the proof, load test, breaking load test and all mechanical testing.

Table 2.7.1 : Condition of supply of chain cables and accessories		
Grade	**Chain cables**	**Accessories**
CC1	As welded or normalized	NA
CC2	As welded or normalized [1)]	Normalized
CC3	Normalized, Normalized and tempered or Quenched and tempered	Normalized, normalized and tempered or Quenched and tempered
1) Grade CC2 chain cables made by forging or casting are to be supplied in the normalized condition NA = Not Applicable.		

2.8 Materials and welding of studs

2.8.1 The studs are to be made of steel corresponding to that of the chain cable or from rolled, cast or forged mild steels. The use of other materials, e.g. grey or nodular cast iron is not permitted.

2.8.2 The welding of studs is to be in accordance with an approved procedure subject to following :

a) The studs being of weldable steel;

b) The studs being welded at one end only, i.e. opposite to the weldment of the link. The stud ends must fit inside of the link without appreciable gap;

c) The welds, preferably in the horizontal position, are to be executed by qualified welders using suitable welding consumables;

d) All the welds are to be completed before the final heat treatment of the chain cable; and

e) The welds are to be free from defects liable to impair the proper use of the chain.

2.9 Testing of completed chain cables

2.9.1 Finished chain cables are to be tested in the presence of a Surveyor, at a proving establishment recognized by IRS. For this purpose the chain cables must be free from paint and anti-corrosive media. Special attention would be given to the visual inspection of the flash-butt-weld, if present. In addition to the requirements of this Chapter, attention must be given to any relevant statutory requirements of the National Authority of the country in which the ship is to be registered.

2.9.2 The design and/or standard breaking loads and proof loads of stud link chain cables are given in Table 2.9.1 (a). The test loads rounded off from the loads in 2.9.1 (a) to be used for testing and acceptance of chain cables, are given in Table 2.9.1 (b). Each length of chain is to be subjected to a proof loading test in an approved testing machine and is to withstand the load given in Table 2.9.1 for the appropriate grade and size of cable. On completion of the test, each length of cable is to be examined and is to be free from significant defects.

Should a proof load test fail, the defective link(s) is (are) to be replaced, a local heat treatment to be carried out on the new link(s) and the proof load test is to be repeated. In addition, an investigation is to be made to identify the cause of the failure.

Table 2.9.1(a) : Formulae for proof loads and breaking loads of stud link chain cables			
Test	**Grade CC1**	**Grade CC2**	**Grade CC3**
Proof load (kN)	$0.00686d^2$ (44 – 0.08d)	$0.00981d^2$ (44 – 0.08d)	$0.01373d^2$ (44 – 0.08d)
Breaking load (kN)	$0.00981d^2$ (44 – 0.08d)	$0.01373d^2$ (44 – 0.08d)	$0.01961d^2$ (44 – 0.08d)
Note d = nominal diameter [mm]			

Table 2.9.1(b) : Test load values for stud link chain cables

Chain cable diameter (mm)	Grade CC1		Grade CC2		Grade CC3	
	Proof load (kN)	Breaking load (kN)	Proof load (kN)	Breaking load (kN)	Proof load (kN)	Breaking load (kN)
1	2	3	4	5	6	7
11	36	51	51	72	72	102
12.5	46	66	66	92	92	132
14	58	82	82	116	116	165
16	76	107	107	150	150	216
17.5	89	127	127	179	179	256
19	105	150	150	211	211	301
20.5	123	175	175	244	244	349
22	140	200	200	280	280	401
24	167	237	237	332	332	476
26	194	278	278	389	389	556
28	225	321	321	449	449	642
30	257	368	368	514	514	735
32	291	417	417	583	583	833
34	328	468	468	655	655	937
36	366	523	523	732	732	1050
38	406	581	581	812	812	1160
40	448	640	640	896	896	1280
42	492	703	703	981	981	1400
44	583	769	769	1080	1080	1540
46	585	837	837	1170	1170	1680
48	635	908	908	1270	1270	1810
50	686	981	981	1370	1370	1960
52	739	1060	1060	1480	1480	2110
54	794	1140	1140	1590	1590	2270
56	851	1220	1220	1710	1710	2430
58	909	1290	1290	1810	1810	2600
60	969	1380	1380	1940	1940	2770
62	1030	1470	1470	2060	2060	2940
64	1100	1560	1560	2190	2190	3130
66	1160	1660	1660	2310	2310	3300
68	1230	1750	1750	2450	2450	3500
70	1290	1840	1840	2580	2580	3690
73	1390	1990	1990	2790	2790	3990
76	1500	2150	2150	3010	3010	4300
78	1580	2260	2260	3160	3160	4500
81	1690	2410	2410	3380	3380	4820
84	1800	2580	2580	3610	3610	5160
87	1920	2750	2750	3850	3850	5500
90	2050	2920	2920	4090	4090	5840
92	2130	3040	3040	4260	4260	6080
95	2260	3230	3230	4510	4510	6440

Table 2.9.1(b) : Test load values for stud link chain cables

Chain cable diameter (mm)	Grade CC1		Grade CC2		Grade CC3	
	Proof load (kN)	Breaking load (kN)	Proof load (kN)	Breaking load (kN)	Proof load (kN)	Breaking load (kN)
1	2	3	4	5	6	7
97	2340	3340	3340	4680	4680	6690
100	2470	3530	3530	4940	4940	7060
102	2560	3660	3660	5120	5120	7320
105	2700	3850	3850	5390	5390	7700
107	2790	3980	3980	5570	5570	7960
111	2970	4250	4250	5940	5940	8480
114	3110	4440	4440	6230	6230	8890
117	3260	4650	4650	6510	6510	9300
120	3400	4850	4850	6810	6810	9720
122	3500	5000	5000	7000	7000	9990
124	3600	5140	5140	7200	7200	10280
127	3750	5350	5350	7490	7490	10710
130	3900	5570	5570	7800	7800	11140
132	4000	5720	5720	8000	8000	11420
137	4260	6080	6080	8510	8510	12160
142	4520	6450	6450	9030	9030	12910
147	4790	6840	6840	9560	9560	13660
152	5050	7220	7220	10100	10100	14430
157	5320	7600	7600	10640	10640	15200
162	5590	7990	7990	11170	11170	15970

Table 2.9.2 : Number of mechanical test specimens for finished chain cables and accessories

Grade	Manufacturing method	Condition of supply 1)	Number of test specimens		
			Tensile test for base metal	Charpy V-notch impact test	
				Base metal	Weldment
CC1	Flash-butt welded	AW N	NR	NR	NR
CC2	Flash-butt welded	AW --- N	1 --- NR	3 --- NR	3 --- NR
	Forged or Cast	N	1	3 [2)]	NA
CC3	Flash-butt welded	N NT QT	1	3	3
	Forged or Cast	N NT QT	1	3	NA

1) AW = As Welded, N = Normalized, NT = Normalized and Tempered, QT = Quenched and Tempered

2) For chain cables, Charpy V-notch impact test is not required.

NR = Not required
NA = Not applicable

Table 2.9.3 : Mechanical properties of finished chain cables and accessories

Grade	Yield strength [N/mm^2] min.	Tensile strength [N/mm^2] min.	Elongation on 5.65 √So % min.	Reduction of area % min.	Charpy V-notch impact test		
					Test temperature, in C	Absorbed energy, in Joules min.	
						Base metal	Weldment
CC1	NR	NR	NR	NR	NR	NR	NR
CC2	295	490 – 690	22	NR	0	27	27
CC3	410	690 min.	17	40	0 [1)]	60	50
					-20	35	27

1) Testing is normally to be carried out at 0°C.
NR = Not required.

2.9.3 Sample lengths comprising of at least three links are to be taken from every four lengths or fraction of chain cables and tested at the breaking loads given in Table 2.9.1. The breaking load is to be maintained for a minimum of 30 seconds. The links concerned are to be made in a single manufacturing cycle together with the chain cable and must be welded and heat treated together with it. Only after this these may be separated from the chain cable in the presence of the Surveyor.

2.9.4 Where a breaking load test specimen fails, a further specimen is to be cut from the same length of cable and subjected to test. If this re-test fails, the length of cable from which it was taken is to be rejected. When this test is also representative of other lengths, each of the remaining lengths in the batch is to be individually tested and is to meet the requirements of the breaking load test.

2.9.5 For large diameter cables where the required breaking load is greater than the capacity of the testing machines, special consideration will be given to acceptance of other alternative testing procedure.

2.9.6 Mechanical test specimens required in Table 2.9.2 are to be taken from every four lengths in accordance with 2.9.7. For forged or cast chain cables where the batch size is less than four lengths, the sampling frequency is to be by heat treatment charge. Mechanical tests are to be carried out in the presence of the Surveyor. The test specimens and their location are to be according to 2.4.5 to 2.4.7 and Fig.2.4.1. Testing and re-testing are to be carried out as given in 2.4.9.

2.9.7 An additional link (or where the links are small, several links) for mechanical test specimen removal is to be provided in a length of chain cable not containing the specimen for the breaking test. The specimen link must be manufactured and heat treated together with the length of chain cable.

2.10 Accessories for chain cables

2.10.1 End and joining shackles, attachment links, adapter pieces, swivels and other fittings are to be subjected to the proof and breaking loads appropriate to the grade and size of cable for which they are intended in accordance with the requirements of Table 2.9.1.

2.10.2 The breaking load is to be applied to at least one item out of every 25 (one in 50 for lugless shackles). The items need not necessarily be representative of each heat of steel or individual purchase order. Enlarged links and end links need not be tested provided that they are manufactured and heat treated together with the chain cable. The tested item is to be destroyed and not used as part of an outfit, in general. However, the accessories, which have been successfully tested at the prescribed breaking load appropriate to the chain, may be used in service at the discretion of IRS where the accessories are manufactured with the following:

a) material having higher strength characteristics than those specified for the part in question (e.g. grade 3 material for accessories for grade 2 chain).

b) or alternatively, same grade material as the chain but with increased dimensions and it is verified by procedure tests that such accessories are so designed that the breaking strength is not less than 1.4 times the prescribed breaking load of the chain for which they are intended.

2.10.3 The breaking load test may be waived if –

a) The breaking load has been demonstrated on the occasion of the approval testing of parts of the same design, and

b) The mechanical properties of each manufacturing batch are approved, and

c) The parts are subjected to suitable non-destructive testing.

2.10.4 Unless otherwise specified, the forging or casting must at least comply with the mechanical properties given in Table 2.9.3, when properly heat treated. For test sampling, forgings or castings of similar dimensions originating from the same heat treatment charge and the same heat of steel are to be combined into one test unit. From each test unit, one tensile test specimen and the Charpy V-notch impact test specimens are to be taken in accordance with Table 2.9.2. Mechanical tests are to be carried out in the presence of the Surveyor. Location of test specimens and test procedure are to be as given in 2.4.5 to 2.4.7 and Fig.2.4.1. Testing / re-testing is to be carried out as per 2.4.9. Enlarged links and end links need not be tested provided they are manufactured and heat treated together with the chain cable.

2.11 Identification

2.11.1 All lengths of cables and accessories are to be stamped with the following identification marks:-

a) IR and the abbreviated name of the local office of IRS issuing the certificate;
b) Number of certificate;
c) Date of test;
d) Proof load and grade of chain;
e) Personal stamp of the Surveyor responsible for inspection.

Section 3

Short Link Chain Cables

3.1 General

3.1.1 Details regarding the form and proportions of short link chain cable, materials, method of manufacture and testing are to be submitted for special consideration by IRS.

3.1.2 In general the short link chain cables are to comply with the grades L(3) and M(4) of ISO 1834.

3.2 Testing and inspection of chain cables

3.2.1 All chain cable of 12.5 [mm] diameter and above, and all steering chains irrespective of diameter are to be tested at a proving establishment recognised by IRS. In addition to the requirements stated in this Chapter, attention is to be given to any relevant statutory requirements of the National Authority of the country in which the ship is to be registered.

3.2.2 For chain of diameter less than 12.5 [mm], other than steering chains, the manufacturer's tests will be accepted.

3.2.3 After completion of all manufacturing processes, including heat treatment and galvanising, the whole of the chain is to be subjected to the appropriate proof load specified in Table 3.2.1.

3.2.4 The whole of the chain is to be inspected after the proof load test and is to be free from significant defects.

3.2.5 At least one sample, consisting of seven or more links, is to be selected by the Surveyor from each 200 [m] or less of chain for breaking load tests. Two additional links may be required for engagement in the jaws of the testing machine. These extra links are not to be taken into account in determining the total elongation (See 3.2.7).

3.2.6 The breaking load is to comply with the appropriate requirements of Table 3.2.1.

3.2.7 The total elongation of the breaking load sample at fracture, expressed as a percentage of the original inside length of the sample after proof loading, is to be not less than 20 per cent.

Table 3.2.1 : Mechanical test requirements for short link chain cables

Chain diameter [mm]	Grade L(3)		Grade M(4)	
	Proof load [kN]	Breaking load min. [kN]	Proof load [kN]	Breaking load min. [kN]
5	-	-	7.9	15.8
6	9	18	-	-
6.3	-	-	12.5	25
7.1	-	-	15.9	31.8
8	16	32	20.2	40.4
9	-	-	25.5	51
10	25	50	31.5	63
11.2	-	-	39.5	79
12	35.5	71	-	-
12.5	-	-	49.1	98.2
14	-	-	63	126
16	-	-	81	162
18	-	-	102	204
20	-	-	126	252
22.4	-	-	158	316
25	-	-	197	394
28	-	-	247	494
32	-	-	322	644
36	-	-	408	816
40	-	-	503	1006
45	-	-	637	1274

Section 4

Steel Wire Ropes

4.1 General

4.1.1 Steel wire ropes are to be manufactured at Works approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2.

4.1.2 The wire ropes are to be of six strand type with minimum of 16 wires in each strand. In addition to complying with the requirements of this Chapter, the details regarding form of construction and minimum breaking strength are to be in accordance with IS: 2266-1989. Alternative type of wire ropes will be specially considered on the basis of an equivalent breaking load and the suitability of the construction for the purpose intended.

4.1.3 It is recommended that the wire ropes intended for stream wires, towlines and mooring lines be of fiber core construction and wire ropes for towlines and mooring lines used in association with mooring winches be of wire rope core.

4.2 Materials

4.2.1 The wire used in the manufacture of the rope is to be drawn from steel made in accordance with the requirements of Ch.3.

Table 4.2.1 : Torsion test - Speed of testing	
Diameter of coated wire [mm]	Maximum speed of testing twists per minute
< 1.5	90
≥ 1.5 < 3.0	60
≥ 3.0 < 4.0	30

4.2.2 The tensile strength is generally to be within the ranges 1420 to 1570 [N/mm^2]; 1570 to 1770 [N/mm^2] or 1770 to 1960 [N/mm^2].

4.2.3 The wire is to be galvanized by a hot dip or electrolytic process to give a uniform coating which may be any of the following grades:-

Grade 1 : heavy coating, drawn after galvanizing;

Grade 2 : heavy coating, finally galvanized;

Grade 3 : light coating, drawn after galvanizing.

4.2.4 Torsion and zinc coating tests are to be carried out on wire samples taken from a suitable length of the completed rope. After unstranding and straightening, six wires are to be subjected to both a torsion test and a wrap test for adhesion of coating. Additionally, tests to determine the uniformity of the zinc coating are to be carried out.

4.2.5 As an alternative to test specimens taken as detailed in 4.2.4, tests may be carried out on the wire before the rope is stranded.

4.2.6 For the torsion test, the length of the sample is to be such as to allow a length between the grips of 100 times the wire diameter or 300 [mm], whichever is less. The wire is to be twisted by causing one or both of the vices to be revolved until fracture occurs. The speed of testing is not to exceed, for a length equal to 100 times the diameter, that given in Table 4.2.1 (a tensile load not exceeding 2 per cent of the breaking load of the wire may be applied to keep the wire stretched). The wire is to withstand, without fracture on a length of 100 times the diameter of wire, the number of complete twists given in Table 4.2.2.

Table 4.2.2 : Torsion test - Minimum number of twists				
Diameter of coated wire [mm]	**Minimum number of twists**			
	Grade 2		**Grade 1 or 3**	
	Tested before stranding	**Tested after stranding**	**Tested before stranding**	**Tested after stranding**
< 1.3	15	13	27	24
≥ 1.3 < 2.3	15	13	26	23
≥ 2.3 < 3.0	14	12	23	20
≥ 3.0 < 4.0	12	10	21	18

4.3 Zinc coating tests

4.3.1 The mass per unit area of the zinc coating is to be determined in accordance with a recognised standard and is to comply with the minimum values given in Table 4.3.1.

Table 4.3.1 : Zinc coating		
Diameter of coated wire [mm]	**Zinc coating [grams/m^2]**	
	Grade 1 or 2	**Grade 3**
≥ 0.40 < 0.50	75	40
≥ 0.50 < 0.6	90	50
≥ 0.6 < 0.8	110	60
≥ 0.8 < 1.0	130	70
≥ 1.0 < 1.2	150	80
≥ 1.2 < 1.5	165	90
≥ 1.5 < 1.9	180	100
≥ 1.9 < 2.5	205	110
≥ 2.5 < 3.2	230	125
≥ 3.2 < 4.0	250	135

4.3.2 The uniformity of the zinc coating is to be determined by a dip test carried out in accordance with the requirements of a recognized standard.

4.3.3 The adhesion of the coating is to be tested by wrapping the wire round a cylindrical mandrel for 10 complete turns. The ratio between the diameter of the mandrel and that of the wire is to be as in Table 4.3.2. After wrapping on the appropriate mandrel the zinc coating is to have neither flaked nor cracked to such an extent that any zinc can be removed by rubbing with bare fingers.

Table 4.3.2 : Wrap test for adhesion of zinc coating		
Coating	**Diameter coated wire [mm]**	**Max. ratio of mandrel to wire diameter**
Grade 1 and 2	< 1.5	4
	≥ 1.5	6
Grade 3	< 1.5	2
	≥ 1.5	3

4.4 Test on completed ropes

4.4.1 The breaking load is to be determined by testing to destruction a sample cut from the completed rope. This sample is to be of sufficient length to provide a clear test length of at least 36 times the rope diameter between the grips.

4.4.2 The actual breaking load is not to be less than that given in the appropriate approved standard.

4.5 Identification

4.5.1 All completed ropes are to be identified with attached labels detailing the rope type, diameter and length.

Section 5

Offshore Mooring Chains

5.1 Scope

5.1.1 Following requirements apply to the materials, design, manufacture and testing of offshore mooring chain and accessories intended to be used for application such as: mooring of mobile offshore drilling units, mooring of floating production units, mooring of offshore loading systems and mooring of gravity based structures during fabrication.

5.1.2 Mooring equipment covered by these requirements are common stud and studless links, connecting common links (splice links), enlarged links, end links, detachable connecting links (shackles), end shackles, subsea connectors, swivels and swivel shackles.

5.1.3 Studless link chain is normally to be deployed only once, being intended for long-term permanent mooring systems with pre-determined design life.

5.2 Chain grades

5.2.1 Depending on the nominal tensile strength of the steels used for manufacture, chains are subdivided into five grades, namely R3, R3S, R4, R4S and R5.

5.2.2 Manufacturers propriety specifications for R4S and R5 may vary subject to design conditions and acceptance of IRS.

5.2.3 Each Grade is to be individually approved. Approval for a higher grade does not constitute approval of a lower grade. If it is demonstrated to the satisfaction of the IRS that the higher and lower grades are produced to the same manufacturing procedure using the same chemistry and heat treatment, consideration will be given to qualification of a lower grade by a higher. The parameters applied during qualification are then, not to be modified during production.

5.3 Approval of chain manufacturers

5.3.1 Offshore mooring chain are to be manufactured only by works approved by IRS. Also refer Chapter 1, Section 1, Cl. 1.3.2. For this purpose approval tests are to be carried out, the scope of which is to include proof and breaking load tests, measurements and mechanical tests including fracture mechanics tests.

5.3.2 Manufacturers are to submit for review and approval the sequence of all operations from receiving inspection of raw materials to the shipment of finished products along with details of the following manufacturing processes:

a) bar heating and bending including the method, temperatures, temperature control and recording;

b) flash welding including current, force, time and dimensional variables as well as control and recording of parameters, maintenance procedure and programme for welding machine;

c) flash removal including method and inspection;

d) stud insertion method for stud link chain;

e) heat treatment including furnace types, means of specifying, controlling and recording of temperature and chain speed and allowable limits, quenching bath and agitation, cooling method after exit;

f) proof and break loading including method/machinery, means of horizontal support (if applicable), method of measurement and recording;

g) non-destructive examination procedures;

h) the manufacturer's surface quality requirement of mooring components;

i) the manufacturer's procedure for removing and replacing defective links without heat treatment of the entire chain.

5.3.3 For initial approval CTOD (Crack Tip Opening Displacement) tests are to be carried out on the particular mooring grade of material. CTOD tests are to be conducted in accordance with a recognized standard such as BS 7448 Part 1 and BS EN ISO

15653:2010. The CTOD test piece is to be a standard 2 x 1 single edge notched bend piece, test location as shown in Fig.5.3.1. The notch of the CTOD specimen is to be located as close to the surface as practicable. The minimum cross section of the test piece is to be 50 x 25 [mm] for chain diameter less than 120 [mm] and 80 x 40 [mm] for diameter 120 [mm] and above. CTOD specimens are to be taken from both the side of the link containing the weld and from the opposite side. Three links are to be selected for testing, i.e, a total of six CTOD specimens. The tests are to be taken at minus 20°C and the lowest CTOD of each set of 3 specimens is to meet the minimum values indicated below in Table 5.3.3:

Table 5.3.3 Minimum CTOD test values for chain type

Chain Type	R3 [mm]		R3S [mm]		R4 [mm]		R4S and R5 [mm]	
	BM	WM	BM	WM	BM	WM	BM	WM
Stud link	0.20	0.10	0.22	0.11	0.24	0.12	0.26	0.13
Studless	0.20	0.14	0.22	0.15	0.24	0.16	0.26	0.17

Note : BM : Base metal WM : Welded metal

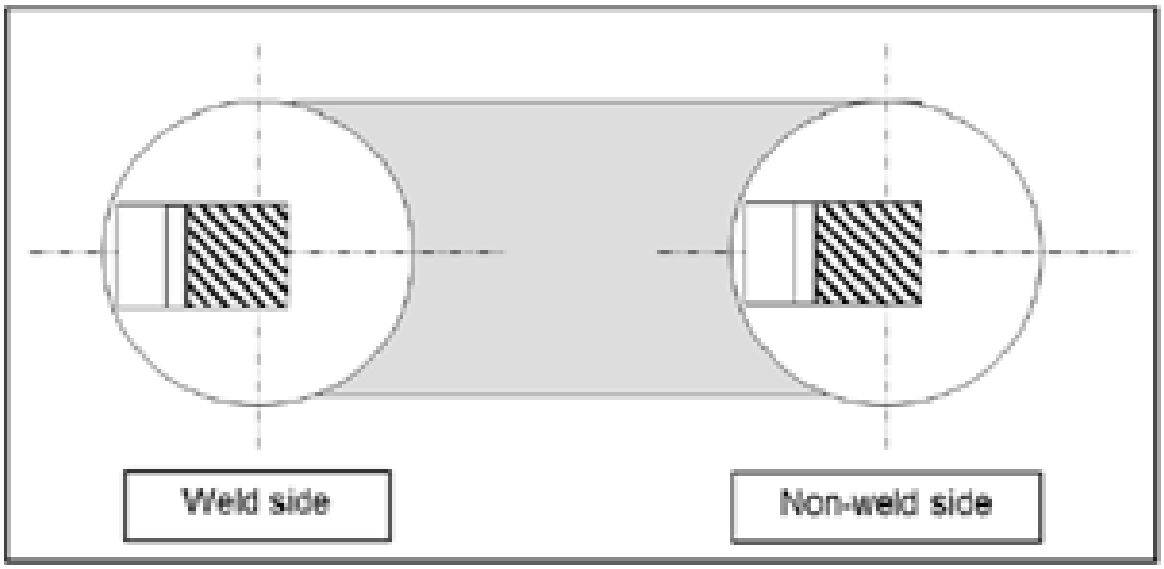


Fig. 5.3.1 : Location of CTOD test specimens for Chain

5.3.4 Calibration of furnaces is to be verified by measurement and recording of a calibration test piece with dimensions equivalent to the maximum size of link manufactured. The manufacturer is to submit a procedure for furnace temperature surveys which is to include the following requirements:

(a) The temperature uniformity of furnaces is to be surveyed whenever approval of manufacturer is requested and at least annually during normal operating conditions.

(b) Furnaces are to be checked by conveying a monitoring link instrumented with two thermocouples through the furnaces at representative travel speed.

(c) One thermocouple is to be attached to the surface of the straight part and one thermocouple is to be imbedded in a drilled hole located at the mid thickness position of the straight part of the calibration block. The time-temperature curves are to show that the temperatures

throughout the cross section and the soaking times are within specified limits as given in the heat treatment procedure.

5.3.5 For R4S and R5 chain and accessories, prior to approval, the manufacturer is to have undertaken experimental tests or have relevant supporting data to develop the chain and accessory material. The tests and data may include: fatigue tests, hot ductility tests (no internal flaws are to develop whilst bending in the link forming temperature range), welding parameter research, heat treatment study, strain age resistance, temper embrittlement study, stress corrosion cracking (SCC) data and hydrogen embrittlement (HE) study, using slow strain test pieces in hydrated environments. Reports indicating the results of experimental tests are to be submitted.

5.4 Approval of quality system at chain and accessory manufacturers

5.4.1 Chain and accessory manufacturers are to have a documented and effective quality system approved by IRS. The provision of such a quality system is required in addition to, and not in lieu of, the witnessing of tests by a Surveyor as specified in 5.7 to 5.14.

5.5 Approval of steel mills - rolled bar

5.5.1 Bar material intended for chain and accessories are to be manufactured only by works approved by IRS. The approval is limited to a nominated supplier of bar material. If a chain manufacturer wishes to use material from a number of suppliers, separate approval tests must be carried out for each supplier. Also refer Chapter 1, Section 1, Cl. 1.3.2.

Approval will be given only after successful testing of the completed chain. Each Grade is to be individually approved. Approval for a higher grade does not constitute approval of a lower grade. If it is demonstrated to the satisfaction of IRS that the higher and lower grades are produced to the same manufacturing procedure using the same chemistry and heat treatment, consideration will be given to qualification of a lower grade by a higher. The parameters applied during qualification are not to be modified during production. The approval will normally be limited upto the maximum diameter equal to that of the chain diameter tested. The rolling reduction ratio is to be recorded and is to be at least 5:1 for R3, R3S, R4, R4S and R5. The rolling reduction ratio used in production can be higher, but should not be lower than that qualified.

5.5.2 The bar manufacturer is to submit a specification of the chemical composition of the bar material, which must be approved by IRS and by the chain manufacturer. The bar maker is to confirm by analysis and testing that the specification is met.

For Grade R4, R4S and R5 chain the steel is to contain a minimum of 0.20 per cent molybdenum.

5.5.3 A heat treatment sensitivity study simulating chain production conditions is to be applied in order to verify mechanical properties and establish limits for temperature and time combinations. All test details and results are to be submitted to IRS.

5.5.4 The bar manufacturer is to provide evidence that the manufacturing process produces material that is resistant to strain ageing, temper embrittlement and for R3S, R4, R4S and R5, hydrogen embrittlement. All test details and results are to be submitted to IRS.

5.6 Approval of forge shops and foundries - accessories

5.6.1 Forge shops and foundries intending to supply finished or semi-finished accessories are to be approved by IRS. A description of manufacturing processes and process controls is to be submitted to IRS. The scope of approval is to be approved by IRS. The approval is to be limited to a nominated supplier of forged or cast material. If an accessory manufacturer wishes to use material from a number of suppliers, a separate approval must be carried out for each supplier.

5.6.2 Approval will be given only after successful testing of the completed accessory. Approval for a higher grade does not constitute approval of a lower grade. If it is demonstrated to the satisfaction of IRS that the higher and lower grades are produced to the same manufacturing procedure using the same steel specification, supplier and heat treatment, consideration will be given to qualification of a lower grade by a higher. The approval will normally be limited to the type of accessory and the designated mooring grade of material upto the maximum diameter or thickness equal to that of the completed accessory used for qualification. unless otherwise agreed by IRS. However, for the

different accessories that have the same geometry, the tests for initial approval are to be carried out on the one having the lowest reduction ratio. Qualification of accessory pins to maximum diameters is also required. Individual accessories of complex geometries will be specially considered.

5.6.3 For forgings – Forgings are to have wrought microstructure and the minimum reduction ratio is to be 3:1. The forging reduction ratio, used in the qualification tests, from cast ingot / slab to forged component is to be recorded. The forging reduction ratio used in production can be higher, but should not be lower than that qualified. The degree of upsetting during qualification is to be recorded and maintained during production. Heat cycling during forging and reheating is to be monitored by the manufacturer and recorded in the forging documentation. The manufacturer is to have a maintenance procedure and schedule for dies and tooling which is to be submitted to IRS.

5.6.4 The forge shop or foundry is to submit a specification of chemical composition of the forged or cast material, which must be approved by IRS. For Grade R4, R4S and R5 chain the steel should contain a minimum of 0.20 per cent molybdenum.

5.6.5 Forges and foundries are to provide evidence that the manufacturing process produces material that is resistant to strain ageing, temper embrittlement and for R4S and R5 grades, hydrogen embrittlement. A heat treatment sensitivity study simulating accessory production conditions shall be applied in order to verify mechanical properties and establish limits for temperature and time combinations. (Cooling after tempering shall be appropriate to avoid temper embrittlement). All test details and results are to be submitted to IRS.

5.6.6 For initial approval CTOD tests are to be carried out on the particular mooring grade of material. Three CTOD tests are to be tested in accordance with a recognized standard such as BS 7448 Part 1 and BS EN ISO 15653:2010. For rectangular accessories, the CTOD test piece is to be a standard 2 x 1 single edge notched bend specimen of thickness equal to full thickness of material to be tested. Subsized specimens may be used, subject to approval by IRS. For circular geometries, the minimum cross section of the test piece is to be 50 x 25 [mm] for accessory diameters less than 120 [mm], and 80 x 40 [mm] for diameters 120 [mm] and above. The notch of the CTOD specimen is to be located as close to the surface as practicable. The tests are to be taken at minus 20°C and the results submitted for review. The minimum values of each set of three specimens are to at least meet the requirements as indicated in Table 5.6.6 (same as that of the studless chain material shown in Table 5.3.3).

Table 5.6.6 : CTOD Test Values for Accessories

Grade of Accessory	R3 [mm]	R3S [mm]	R4 [mm]	R4S & R5 [mm]
CTOD	0.20	0.22	0.24	0.26

The geometry of accessories can vary. Figure 5.6.6 shows the CTOD location for circular and rectangular cross sections such as those of the D-shackle and accessories fabricated from rectangular sections. The orientation of the specimen is to consider the direction of the grain flow. Figure 5.6.6 (b) shows two possible sampling positions for CTOD test specimens with notch orientation for rectangular type accessories.

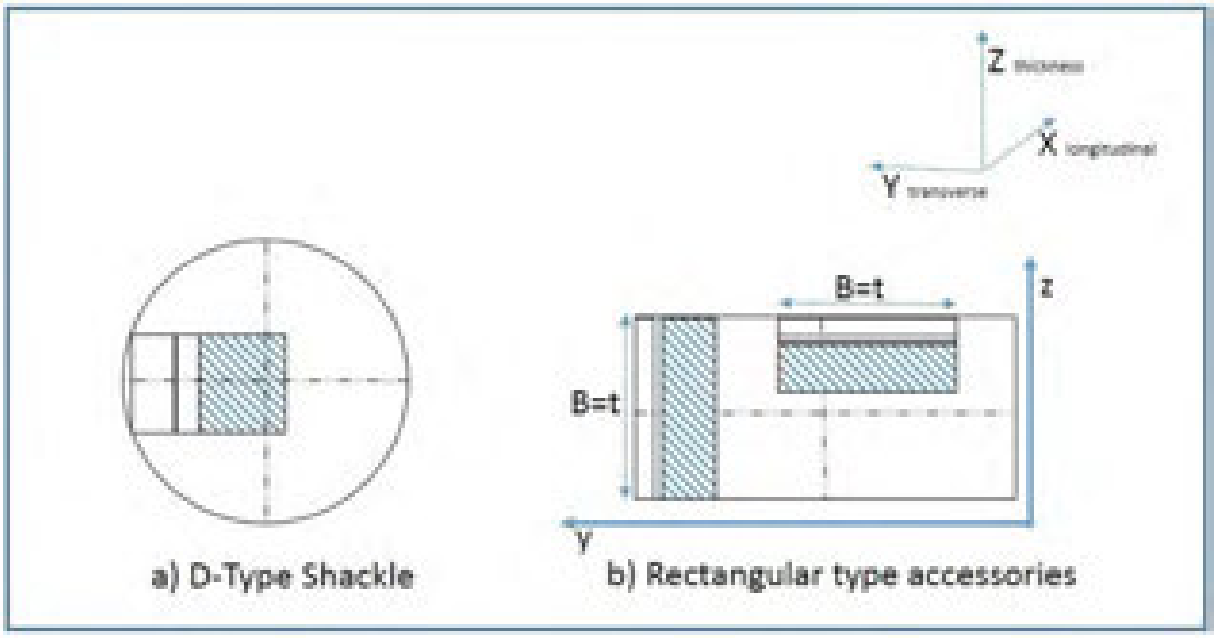


Figure 5.6.6 : Location of CTOD test specimens: a) Circular type accessory and b) rectangular type accessory, B corresponds to the thickness of material, the grain flow is considered in the longitudinal direction X

5.6.7 Calibration of furnaces shall be verified by measurement and recording of a calibration test piece with dimensions equivalent to the maximum size of link manufactured. Thermocouples are to be placed both on the surface and in a drilled hole located to the mid thickness position of the calibration block. The furnace dimensions are to be such as to allow the whole furnace charge to be uniformly heated to the necessary temperature. Temperature uniformity surveys of heat treatment furnaces for forged and cast components are to be carried out according to API Spec 6A/ISO 10423 Annex M or ASTM A991. The initial survey is to be carried out with maximum charge (load) in the furnace. Subsequent surveys are to be carried out annually and may be carried out with no furnace charge.

The quench bath maximum temperature and the maximum heat treatment transfer times from furnace to quench are to be established and documented. During production the established quenching parameters are to be followed and records are to be maintained of bath temperatures and transfer times.

5.6.8 For R4S and R5 refer to additional requirements in 5.3.5.

5.7 Rolled steel bars

5.7.1 Steel manufacture

5.7.1.1 The steels are to be manufactured by basic oxygen, electric furnace or such other process as may be specially approved. All steels are to be killed and fine grain treated. The austenitic grain size for R3, R3S and R4 is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at 1/3 radius.

5.7.1.2 Steel for bars intended for R4S and R5 chain is to be vacuum degassed. The austenitic grain size is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at 1/3 radius.

5.7.1.3 For R4S and R5 the following information is to be supplied by the bar manufacturer to the mooring chain manufacturer and the results included in the chain documentation:

a) Each heat is to be examined for non-metallic inclusions. The level of micro inclusions is to be quantified and assessed in accordance with national/ international standards; to be sure inclusion levels are acceptable for the final product.

b) A sample from each heat is to be macro etched according to ASTM E381 or equivalent to be sure there is no injurious segregation or porosity.

c) Hardenability data, according to ASTM A255, or equivalent, is to be supplied with each heat.

5.7.2 Chemical composition

For acceptance tests, the chemical composition of ladle samples of each heat is to be determined by the bar manufacturer and is to comply with the approved specification (see 5.5.2).

5.7.3 Mechanical tests

.1 Bars of the same nominal diameter are to be presented for test in batches of 50 tonnes or fraction thereof from the same heat. Test specimens are to be taken from material heat treated in the same manner as intended for the finished chain.

.2 Each heat of Grade R3S, R4, R4S and R5 is to be tested for hydrogen embrittlement. In case of continuous casting, test samples representing both the beginning and the end of the charge are to be taken. In case of ingot casting, test samples representing two different ingots are to be taken.

a) Two tensile test specimens are to be taken from the central region of bar material which has been subjected to the heat treatment cycle intended to be used in production. The specimens are to preferably have a diameter of 20 mm, alternatively 14 mm. One specimen is to be tested within max. 3 hours (1 1/2 hours for 14 mm diameter specimen) after machining. Where this is not possible, the specimen is to be cooled to -60°C immediately after machining and kept at that temperature for a period of maximum 5 days.

   The second specimen is to be tested after baking at 250°C for 4 hours (2 hours for 14 mm diameter specimen).

b) A slow strain rate < 0.0003 per second is to be used during the entire test, until fracture occurs. (This is approx. 10 minutes for a 20 mm diameter specimen).

c) Tensile strength, elongation and reduction of area are to be reported. The requirement for the test is:

$$\frac{Z_1}{Z_2} \geq 0.85$$

where,

Z_1 = Reduction of area without baking

Z_2 = Reduction of area after baking

d) If the requirement of $Z_1/Z_2 \geq 0.85$ is not achieved, the bar material may be subjected to a hydrogen degassing treatment after agreement with the Surveyor(s). New tests are to be performed after degassing.

.3 For all grades, one tensile and three Charpy V-notch specimens are to be taken from each sample selected. The test specimens are to be taken at approximately one-third radius below the surface, as shown in Sec. 2, Fig.2.4.1.
The results of all tests are to be in accordance with the appropriate requirements of Table 5.7.1.

.4 If the tensile test requirements are not achieved, a retest of two further specimens selected from the same sample will be permissible.
Failure to meet the specified requirements of either or both additional tests will result in rejection of the batch represented unless it can be clearly attributable to improper simulation of the heat treatment.

.5 If the impact test requirements are not achieved, a retest of three further specimens selected from the same sample will be permissible. The results are to be added to those previously obtained to form a new average. The new average is to comply with the requirements. No more than two individual results are to be lower than the required average and no more than one result is to be below 70 per cent of the specified average value.

Failure to meet the requirements will result in rejection of the batch represented. Where the failure can be clearly attributable to improper simulation of the heat treatment, new tests may be permitted at the discretion of the Surveyor.

5.7.4 Dimensional tolerances

The diameter and roundness are to be within the tolerances specified in Sec. 2, Table 2.4.3, unless otherwise agreed.

5.7.5 Non-destructive examination and repair

a) Non-destructive examination is to be performed in accordance with recognized standards such as those given below or equivalent:

   i. Magnetic Particle Testing (MT) of Bars – ASTM E1444 and ISO 9934
   ii. Magnetic Leakage Flux Testing (MLFT) – JIS Z2319
   iii. Eddy Current Testing (ET) of bars – ISO 15549

   Non-destructive examination procedures, together with rejection / acceptance criteria are to be submitted to IRS.

b) Manufacturers are to prepare written procedures for NDE. NDE personnel are to be qualified and certified according to ISO 9712, ACCP or equivalent. Personnel qualification to an employer or responsible agency based qualification scheme as SNT-TC-1A may be accepted if the employer's written practice is reviewed and found acceptable and the Level III is ASNT Level III, ISO 9712 Level III or ACCP Professional Level III and certified in the applicable method. NDE operators are to be qualified to at least level II.

c) The manufacturer is to ensure that 100 percent of bar material intended for either chain or fittings is subjected to ultrasonic examination at an appropriate stage of the manufacture to procedures approved by IRS and to the acceptance criteria required.. The bars are to be free of pipe, cracks and flakes. If the end length of the delivered bars is not subjected to UT then it is to be agreed between the bar supplier and the chain manufacturer of what length of bar is to be removed from the ends. The details are to be documented in the approval of each bar supplier. Phased array UT procedures may be applied, subject to approval by IRS.

d) 100 percent of the bar material is to be examined by magnetic particle (MT) or eddy current (ET) or Magnetic Leakage Flux Testing (MLFT) methods. The bars are to be free of injurious surface imperfections such as seams, laps and rolled-in mill scale. Provided that their depth is not greater than 1 per cent of the bar diameter, longitudinal discontinuities may be removed by grinding and blending to a smooth contour.

   All bars supplied in a machined (peeled) condition are to be 100% visually inspected. 10% inspection with magnetic particle testing (MT) or eddy current testing (ET) or Magnetic Leakage Flux Testing (MLFT), for longitudinal imperfections may be undertake at the discretion of the Surveyor. The maximum depth of peeling is to be agreed and documented in the approval of each supplier.

e) The frequency of NDE may be reduced at the discretion of the Surveyor provided it is verified by statistical means that the required quality is consistently achieved.

f) Weld repair of bars is not permitted.

5.7.6 Marking

Each bar is to be stamped with the steel grade designation and the charge number (or a code indicating the charge number) on one of the end surfaces. Other marking methods may be accepted subject to agreement.

Table 5.7.1 : Mechanical properties of offshore mooring chain and accessories

Grade	Yield stress [N/mm^2] min. (Note 1)	Tensile strength [N/mm^2] min. (Note 1)	Elongation % min.	Reduction of area % min. (Note 3)	Charpy V-notch impact test		
					Test temp. °C (Note 2)	Average energy J min.	Average energy flash weld J min.
R3	410	690	17	50	0 -20	60 40	50 30
R3S	490	770	15	50	0 -20	65 45	53 33
R4	580	860	12	50	-20	50	36
R4S (Note 4)	700	960	12	50	-20	56	40
R5 (Note 4)	760	1000	12	50	-20	58	42

Notes:

1 Aim value of yield to tensile ratio: 0.92 max.

2 At the option of IRS the impact test of Grade R3 and R3S may be carried out at either 0°C or minus 20°C.

3 Reduction of area of cast steel is to be for Grades R3 and R3S: min. 40%, for R4, R4S and R5 : min. 35% (see 5.9.4).

4 Aim maximum hardness for R4S is HB330 and R5, HB340.

5.8 Forged steel

5.8.1 Manufacture

5.8.1.1 Forged steels used for the manufacture of accessories are to be in compliance with approved specifications and the submitted test reports approved by IRS. Steel is to be manufactured by basic oxygen, electric furnace or such other process as may be specially approved. All steel is to be killed and fine grain treated. The austenitic grain size for R3, R3S and R4 is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at 1/3 radius. Measurements for non-circular sections are to be taken at 1/4t.

5.8.1.2 Steel for forgings intended for R4S and R5 chain is to be vacuum degassed. The austenitic grain size is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at 1/3 radius. Measurements for non-circular sections are to be taken at 1/4t.

5.8.1.3 For steel intended for R4S and R5 accessories the following information is to be supplied by the steel manufacturer to the mooring accessory manufacturer and the results included in the accessory documentation:

a) Each heat is to be examined for non-metallic inclusions. The level of micro inclusions is to be quantified and assessed in accordance with the national / international standards; to be sure inclusion levels are acceptable for the final product.

b) A sample from each heat is to be macroetched according to ASTM E381 or

equivalent, to be sure there is no injurious segregation or porosity.

c) Hardenability data, according to ASTM A255 or equivalent, is to be supplied with each heat.

5.8.2 Chemical composition

For acceptance tests, the chemical composition of ladle samples of each heat is to be determined by the bar manufacturer and is to comply with the approved specification (see 5.5.2).

5.8.3 Heat treatment

Finished forgings are to be properly heat treated in compliance with specifications approved by IRS.

5.8.4 Mechanical properties

The forgings are to comply with the mechanical properties given in Table 5.7.1, when properly heat treated.

5.8.5 Mechanical tests

5.8.5.1 For test sampling, forgings of similar dimensions (diameters do not differ by more than 25 mm) originating from the same heat treatment charge and the same heat of steel are to be combined into one test unit. From each test unit one tensile and three impact test specimens are to be taken and tested. For the location of the test specimens see Sec. 2, Fig.2.4.1.

5.8.5.2 Each heat of Grade R3S, R4, R4S and R5 is to be tested for hydrogen embrittlement. In case of continuous casting, test samples representing both the beginning and the end of the charge is to be taken. In case of ingot casting, test samples representing two different ingots are to be taken.

5.8.5.2.1 Two (2) tensile test specimens are to be taken from the central region of forged material which has been subjected to the heat treatment cycle intended to be used in production. A specimen with a diameter of 20 [mm] is preferred (consideration will be given to a diameter of 14 [mm]).

5.8.5.2.2 One of the specimens is to be tested within a maximum of 3 hours after machining (for a 14 [mm] diameter specimen, the time limit is 1½ hours). Where this is not possible, the specimen is to be immediately cooled to -60°C after machining and kept at that temperature for a maximum period of 5 days.

5.8.5.2.3 The second specimen is to be tested after baking at 250°C for 4 hours, alternatively 2 hours for 14 [mm] diameter specimen.

5.8.5.2.4 A slow strain rate < 0,0003 s^{-1} is to be used during the entire test, until fracture occurs (This is approximately 10 minutes for the 20 [mm] diameter specimen). Tensile strength, elongation and reduction of area are to be reported.

5.8.5.2.5 The acceptance requirement for the test is as follows:

$Z_1 / Z_2 \geq 0.85$

Where:

Z_1 = Reduction of area without baking
Z_2 = Reduction of area after baking

If the requirement $Z_1/Z_2 \geq 0.85$ is not achieved, the bar material may be subjected to a hydrogen degassing treatment after agreement with IRS. New tests are to be performed after degassing.

5.8.6 Non-destructive examination and repair

5.8.6.1 Non-destructive examination is to be performed in accordance with recognized standards such as those indicated below, or equivalent. The non-destructive examination procedures, together with rejection / acceptance criteria are to be submitted to IRS.

(a) Magnetic particle testing (MT) of forgings: - EN 10228-1, ASTM A275, using wet continuous magnetization technique

(b) Ultrasonic testing (UT) of forgings: - EN 10228-3, ASTM A388, ISO 13588

5.8.6.2 Manufacturers are to prepare written procedures for NDE. NDE personnel are to be qualified and certified according to ISO 9712, ACCP or equivalent. Personnel qualification to an employer or responsible agency based qualification scheme as SNT-TC-1A may be accepted if the employer's written practice is reviewed and found acceptable and the Level III is ASNT Level III, ISO 9712 Level III or ACCP Professional Level III and certified in the applicable method. NDE operators are to be qualified to at least level II.

5.8.6.3 The forgings are to be subjected to 100 percent ultrasonic examination at an appropriate stage of manufacture and in compliance with the standard approved by IRS.

5.8.6.4 Defects on non-machined surfaces may be removed by grinding to a depth of 5% of the nominal diameter. Grinding is not permitted on machined surfaces, except for slight inspection grinding on plane surfaces to a maximum depth of 0.8 [mm] in order to investigate spurious indications. Welding repairs are not permitted.

5.8.7 Marking

Marking is to be similar to that specified in 5.7.6.

5.9 Cast steel

5.9.1 Manufacture

5.9.1.1 Cast steels used for the manufacture of accessories are to be in compliance with approved specifications and the submitted test reports approved by IRS. Steel is to be manufactured by basic oxygen, electric furnace or such other process as may be specially approved. All steel is to be killed and fine grain treated. The austenitic grain size for R3, R3S and R4 is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at 1/3 radius. Measurements for non-circular sections are to be taken at 1/4t.

5.9.1.2 Steel for castings intended for R4S and R5 accessories is to be vacuum degassed. The austenitic grain size is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at 1/3 radius. Measurements for non-circular sections are to be taken at 1/4t.

5.9.1.3 For steel intended for R4S and R5 accessories the following information is to be obtained and the results included in the accessory documentation:

a) Each heat is to be examined for non-metallic inclusions. The level of micro inclusions is to be quantified and assessed in accordance with the national/international standards; to be sure inclusion levels are acceptable for the final product.

b) A sample from each heat is to be macro etched according to ASTM E381 or equivalent to be sure there is no injurious segregation or porosity.

c) Hardenability data, according to ASTM A255, or equivalent is to be supplied with each heat.

5.9.2 Chemical composition

For acceptance tests, the chemical composition of ladle samples of each heat is to be determined by the bar manufacturer and is to comply with the approved specification (see 5.5.2).

5.9.3 Heat treatment

All castings are to be properly heat treated in compliance with specifications approved by IRS.

5.9.4 Mechanical properties

The castings are to comply with the mechanical properties given in Table 5.7.1. The requirement for reduction of area is, however, reduced to 40 per cent for grades R3 and R3S and 35 per cent for grade R4, R4S and R5.

5.9.5 Mechanical tests

For test sampling, castings of similar dimensions originating from the same heat treatment charge and the same heat of steel are to be combined into one test unit. From each test unit one tensile and three impact test specimens are to be taken and tested. For the location of the test specimens see Sec. 2, Fig.2.4.1.

5.9.6 Non-destructive Examination and Repair

5.9.6.1 Non-destructive examination is to be performed in accordance with recognized standards, such as those indicated below, or equivalent. The non-destructive examination procedures, together with rejection / acceptance criteria are to be submitted to IRS.

(a) Magnetic particle testing (MT) of castings: - ASTM E709, using wet continuous magnetisation technique

(b) Ultrasonic testing (UT) of castings: - ASTM A609, ISO 13588

5.9.6.2 Manufacturers are to prepare written procedures for NDE. NDE personnel are to be qualified and certified according to ISO 9712, ACCP or equivalent. Personnel qualification to an employer or responsible agency based qualification scheme as SNT-TC-1A may be accepted if the employer's written practice is reviewed and found acceptable and the Level III is ASNT Level III, ISO 9712 Level III or ACCP Professional Level III and certified in the applicable method. NDE operators are to be qualified to at least level II.

5.9.6.3 The castings are to be subjected to 100 percent ultrasonic examination in compliance with the standard approved by IRS.

5.9.6.4 Defects on non-machined surfaces may be removed by grinding to a depth of 5% of the nominal diameter. Grinding is not permitted on machined surfaces, except for slight inspection grinding on plane surfaces to a maximum depth of 0.8 [mm] in order to investigate spurious indications.

5.9.6.5 Where the repair entails removal of more than 5% of the diameter or thickness, the defective area is to be repaired by welding. The excavations are to be suitably shaped to allow good access for welding. The resulting grooves are to be subsequently ground smooth and complete elimination of the defective material is to be verified by NDE.

5.9.6.6 Weld repairs are classified as major or minor. A weld repair is considered major when the depth of the groove prepared for welding exceeds 25% of the diameter/thickness or 25 [mm], whichever is smaller. All other weld repairs are considered minor.

5.9.6.7 Major weld repairs require approval before the repair is commenced. Proposals for major repairs are to be accompanied by sketches or photographs showing the extent and positions of the repairs. A grain refining heat treatment is to be given to the whole casting prior to major repairs. A post weld heat treatment or repeat of original heat treatment of castings is to be carried out.

5.9.6.8 Minor and major weld repairs are to be recorded on sketches or photographs showing the extent and positions of the repairs.

5.9.6.9 All weld repairs are to be done by qualified welders using qualified procedures. Welders are to be qualified according to ISO 9606, ASME IX, ASTM A488 or equivalent. Procedures are to be qualified according to ISO 15614, ASME IX, ASTM A488 or equivalent with the following additional requirements: Charpy V notch impact tests with notch locations in weld metal, fusion line and heat affected zone + 2 [mm] and + 5 [mm] from fusion line, respectively. Test results are to meet the requirements specified for the parent metal.

5.9.7 Marking

Marking is to be similar to that specified in 5.7.6.

5.10 Materials for studs

5.10.1 The studs intended for stud link chain cable are to be made of steel corresponding to that of the chain or in compliance with specifications approved by IRS. In general, the carbon content is not to exceed 0.25 per cent if the studs are to be welded in place.

5.11 Design and Manufacture

5.11.1 Plans with design calculations, giving detailed design of chain and accessories made by or supplied through the chain manufacturer are to be submitted for approval. Typical designs are given in ISO 1704:2008 (see Sec. 2, Fig.2.3.1 to 2.3.6). For studless chain the shape and proportions are to comply with the requirements of this section. Other studless proportions are to be specially approved. It should be considered that new or non-standard designs of chain, shackles or fittings, may require a fatigue analysis and possible performance, fatigue or corrosion fatigue testing.

5.11.2 In addition for stud link chain, plans showing the detailed design of the stud are to be submitted for information. The stud is to give an impression in the chain link which is sufficiently deep to secure the position of the stud, but the combined effect of shape and depth of the impression is to be such as not to cause any harmful notch effect or stress concentration in the chain link.

5.11.3 Machining of Kenter shackles is to result in fillet radius of minimum 3 per cent of nominal diameter.

5.12 Chain cable manufacturing process

5.12.1 General

Offshore mooring chains are to be manufactured in continuous lengths by flash butt welding and are to be heat treated in a continuous furnace; batch heat treatment is not permitted. The use of joining shackles to replace defective links is subject to the written approval of the end purchaser in terms of the number and type permitted, except in special circumstances where short lengths of chain are delivered, such as chafing chain (Ref. 5.15). The use of connecting common links is restricted to 3 links in each 100 m of chain.

5.12.2 Manufacturing process records

Records of bar heating, flash welding and heat treatment are to be made available for inspection by the Surveyor.

5.12.3 Bar heating

a) Bars for links are to be heated by electric resistance, induction or in a furnace.

b) For electric resistance heating or induction heating, the heating phase is to be controlled by an optical heat sensor. The controller is to be checked at least once every 8 hours and the records are to be maintained.

c) For furnace heating, the heat is to be controlled and the temperature continuously recorded using thermocouples in close proximity to the bars. The controls are to be checked at least once every 8 hours and the records are to be maintained.

5.12.4 Flash welding

The following welding parameters are to be controlled during welding of each link:

a) Platen motion;

b) Current as a function of time;

c) Hydraulic pressure.

The controls are to be checked at least every 4 hours and records are to be maintained.

5.12.5 Heat treatment

a) Chain is to be austenitized, above the upper transformation temperature, at a combina-tion of temperature and time within the limits established.

b) When applicable, chain is to be tempered at a combination of temperature and time within the limits established. Cooling after tempering is to be appropriate to avoid temper embrittlement.

c) Temperature and time or temperature and chain speed are to be controlled and continuously recorded.

d) Grain determination is to be made for the final product. The austenitic grain size for R3, R3S, R4, R4S and R5 is to be 6 or finer in accordance with ASTM E112 or equivalent grain size index in accordance to ISO 643. Measurements for circular sections are to be taken at surface, 1/3 radius and centre for the base material, HAZ and weld.

5.12.6 Mechanical properties

The mechanical properties of finished chain and accessories are to be in accordance with Table 5.7.1. For the location of test specimens see Sec. 2, Fig.2.4.1.

5.12.7 Proof and breaking test loads

Chains and accessories are to withstand the proof and break test loads given in Table 5.12.1.

5.12.8 Freedom from defects

All chains are to have a workmanlike finish consistent with the method of manufacture and be free from defects. Each link is to be examined in accordance with 5.15.5 using approved procedures.

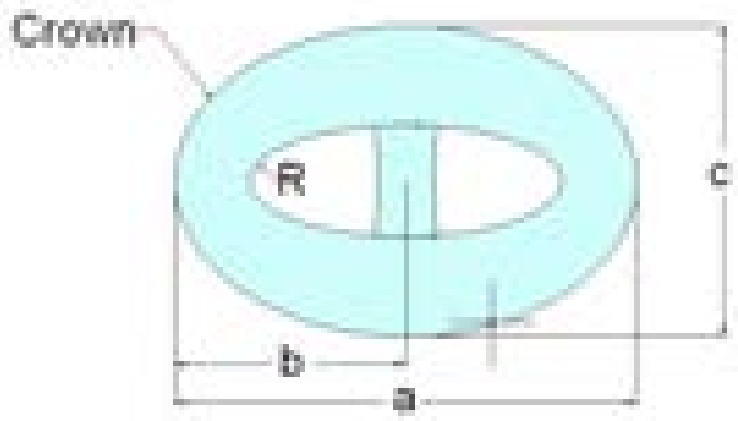
5.12.9 Dimensions and dimensional tolerances

The form and proportion of links and accessories are to be in accordance with the approved design. The tolerances given in Sec. 2, Cl. 2.3 are applicable, in general.

In addition, the tolerances for stud link and studless common links are to be measured in accordance with Fig.5.12.9a) and b).

Table 5.12.1 : Formulae for proof and break test loads, weight and length over 5 links		
Grade Type	**Proof load [kN]**	**Break load [kN]**
Grade R3 Stud Link	0.0148 d^2 (44 – 0.08d)	0.0223 d^2 (44 – 0.08d)
Grade R3S Stud Link	0.0180 d^2 (44 – 0.08d)	0.0249 d^2 (44 – 0.08d)
Grade R4 Stud Link	0.0216 d^2 (44 – 0.08d)	0.0274 d^2 (44 – 0.08d)
Grade R4S Stud Link	0.0240 d^2 (44 – 0.08d)	0.0304 d^2 (44 – 0.08d)
Grade R5 Stud Link	0.0251 d^2 (44 – 0.08d)	0.0320 d^2 (44 – 0.08d)
Grade R3 Studless	0.0148 d^2 (44 – 0.08d)	0.0223 d^2 (44 – 0.08d)
Grade R3S Studless	0.0174 d^2 (44 – 0.08d)	0.0249 d^2 (44 – 0.08d)
Grade R4 Studless	0.0192 d^2 (44 – 0.08d)	0.0274 d^2 (44 – 0.08d)
Grade R4S Studless	0.0213 d^2 (44 – 0.08d)	0.0304 d^2 (44 – 0.08d)
Grade R5 Studless	0.0223 d^2 (44 – 0.08d)	0.0320 d^2 (44 – 0.08d)
Chain Weight [kg/m]	Stud link = 0.0219 d^2	
Chain Weight [kg/m]	Studless chain weight calculations for each design are to be submitted	
Pitch Length :	Five link measure	
Minimum	22d	
Maximum	22.55d	

a) **Stud Link** – The internal link radii (R) and external radii should be uniform

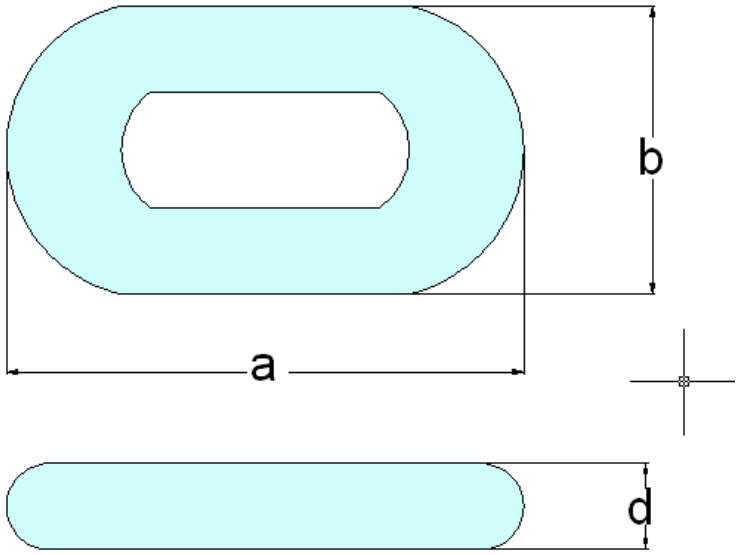


Designation [1)]	Description	Nominal Dimension of the Link	Minus Tolerance	Plus Tolerance
a	Link length	6d	0.15d	0.15d
b	Link half length	a*/2	0.1d	0.1d
c	Link width	3.6d	0.09d	0.09d
e	Stud angular misalignment	0 degrees	4 degrees	4 degrees
R	Inner radius	0.65d	0	----

Notes : 1 Dimension designation is shown in above figure
d = Nominal diameter of chain, (Refer Fig.2.3.1) a* = Actual link length

Fig.5.12.9a) : Stud Link common link, proportions dimensions and tolerances

b) **Studless** – The internal link radii (R) and external radii should be uniform.



Designation [1)]	Description	Nominal Dimension of the Link	Minus Tolerance	Plus Tolerance
a	Link length	6d	0.15d	0.15d
b	Link width	3.35d	0.09d	0.09d
R	Inner radius	0.60d	0	----

Notes : 1 Dimension designation is shown in above figure
d = Nominal diameter of chain
2 Other dimension ratios are subject to special approval.

Fig.5.12.9b) : Stud less common link, proportions dimensions and tolerances

5.12.10 Welding of studs

a) A welded stud may be accepted for grade R3 and R3S chains. Welding of studs in grade R4 chain is not permitted unless specially approved.

b) Where studs are welded into the links, the welding is to be completed before the chain is heat treated.

c) The stud ends must be a good fit inside the link and the weld is to be confined to the stud end opposite to the flash butt weld. The full periphery of the stud end is to be welded unless otherwise approved.

e) Welding of studs at both ends is not permitted unless specially approved.

f) The welds are to be made by qualified welders using an approved procedure and low-hydrogen approved consumables.

g) The size of the fillet weld is, as a minimum, to be as per API Specification 2F.

h) The welds are to be of good quality and free from defects such as cracks, lack of fusion, gross porosity and undercuts exceeding 1 mm.

i) All stud welds are to be visually examined. At least 10 per cent of all stud welds within each length of chain are to be examined by dye penetrant or magnetic particles after proof load testing. If cracks or lack of fusion are found, all stud welds in that length are to be examined.

5.12.11 Connecting common links (splice links)

.1 Single links to substitute for test links or defective links without the necessity for re-heat treatment of the whole length are to be made in accordance with an approved procedure. Separate approvals are required for each grade of chain and the tests are to be made on the maximum size of chain for which approval is sought.

.2 Manufacture and heat treatment of connecting common link is not to affect the properties of the adjoining links. The temperature reached by these links is nowhere to exceed 250°C.

.3 Each link is to be subjected to the appropriate proof load and non-destructive examination as detailed in Table 5.12.1 and 5.13.5 respectively. A second link is to be made identical to the connecting common link; the link is to be tested and inspected as per 5.13.4 and 5.14.5.

.4 Each connecting common link is to be marked either on the stud for stud link chain, or on the outer straight length on the side opposite the flash butt weld for studless chain. This marking is to be in accordance with 5.13.7 plus a unique number for the link. The adjoining links are also to be marked on the studs or straight length as above.

5.13 Testing and inspection of finished chain

5.13.1 General

a) This section applies to but is not limited to finished chain cable such as common stud and studless links, end links, enlarged end links and connecting common links (splice links).

b) All chains are to be subjected to proof load tests, sample break load tests and sample mechanical tests after final heat treatment in the presence of Surveyor. Where the manufacturer has a procedure to record proof loads and the Surveyor is satisfied with the adequacy of the recording system, he may not witness all proof load tests. The Surveyor is to satisfy himself that the testing machines are calibrated and maintained in a satisfactory condition.

c) Prior to inspection chains are to be free from scale, paint or other coating and are to have a suitably prepared surface as per the applied NDE testing standard. The chains are to be sand- or shot blast to meet this requirement.

5.13.2 Proof and break load tests

a) The entire length of chain is to withstand the proof load specified in Table 5.12.1 without fracture and

without crack in the flash weld. The load applied is not to exceed the proof load by more than 10 per cent when stretching the chain. Where plastic straining is used to set studs, the applied load is not to be greater than that qualified in approval tests.

b) A break-test specimen consisting of at least 3 links is to be either taken from the chain or produced at the same time and in the same manner as the chain. The test frequency is to be based on tests at sampling intervals according to Table 5.13.1 provided that every cast is represented. Each specimen is to be capable of withstanding the break load specified without fracture and without crack in the flash weld. It will be considered acceptable if the specimen is loaded to the specified value and maintained at that load for 30 seconds.

c) For chain diameters over 100 [mm], alternative break-test proposals to the above break-test will be considered whereby a one link specimen is used. Alternatives are to be approved by IRS. Every heat is to be represented, the test frequency is to be in accordance with Table 5.13.1 and it is to be demonstrated and proven that the alternative test represents an equivalent load application to the three link test.

d) If the loading capacity of the testing machine is insufficient, an alternative load testing machine is to be used that does have sufficient capacity (e.g. two loading machines in parallel) provided the testing and calibration procedure are agreed with IRS.

Table 5.13.1 : Frequency of break and mechanical tests

Nominal chain diameter (mm)	Maximum sampling interval (m)
Min - 48	91
49 - 60	110
61 - 73	131
74 - 85	152
86 - 98	175
99 - 111	198
112 - 124	222
125 - 137	250
138 - 149	274
150 - 162	297
163 - 175	322
176 – 186	346
187 – 198	370
199 - 210	395
211 – 222	420

5.13.3 Dimensions and dimensional tolerances

After proof load testing, measurements are to be taken on at least 5 per cent of the links in accordance with 5.12.9.

The entire chain is to be checked for the length, five links at a time. By the five link check the first five links are to be measured. In the next set of five links, at least two links from the previous five links set are to be included.

This procedure is to be followed for the entire chain length. The measurements are to be taken preferably while the chain is loaded to 5 - 10 per cent of the minimum proof load. The tolerances for the 5 link measurements are indicated in Table 5.12.1, any deviations from the 5 link tolerances are to be agreed by the client and IRS. The links held in the end blocks may be excluded from this measurement.

Chain dimensions are to be recorded and the information retained on file.

5.13.4 Mechanical tests

a) Links of samples detached from finished, heat treated chain are to be sectioned for determination of mechanical properties. A test unit is to consist of one tensile and nine impact specimens. The tensile specimen is to be taken in the side opposite the flash weld. Three impact specimens are to be taken across the flash weld with the notch centered in the middle. Three impact specimens are to be taken across the unwelded side and three impact specimens are to be taken from the bend region.

b) The test frequency is to be based on tests at sampling intervals according to Table 5.13.1 provided that every cast is represented. Mechanical properties are to be as specified in Table 5.7.1.

c) The frequency of impact testing in the bend may be reduced at the discretion of the Surveyor provided it is verified by statistical means that the required toughness is consistently achieved.

d) Hardness tests are to be carried out on finished chain. The frequency and locations are to be agreed with IRS. The recorded values are for information only and used as an additional check to verify that the heat treatment process has been stable during the chain production.

5.13.5 Non-destructive examination after proof load testing

a) All surfaces of every link are to be visually examined. Burrs, irregularities and rough edges are to be contour ground. Links are to be free from mill defects, surface cracks, dents and cuts, especially in the vicinity where gripped by clamping dies during flash welding. Studs are to be securely fastened. Chain is to be positioned in order to have good access to all surfaces. In order to allow optimal access to the surface area it is recommended that chain be hung in the vertical position, however access to inspect the interlink area may only be possible with the chain in the horizontal position.

b) Testing is to be performed in accordance with a recognized standard and the procedures, together with acceptance / rejection criteria are to be submitted to IRS for review. Manufacturers are to prepare written procedures for NDE. NDE personnel are to be qualified and certified according to ISO 9712, ACCP or equivalent. Personnel qualification to an employer or responsible agency based qualification scheme as SNT-TC-1A may be accepted if the employer's written practice is reviewed and found acceptable and the Level III is ASNT Level III, ISO 9712 Level III or ACCP Professional Level III and certified in the applicable method. NDE operators are to be qualified to at least level II.

c) Magnetic particle testing is to be employed to examine the flash welded area including the area gripped by the clamping dies. Procedures are to be submitted to IRS for approval. Approved procedures and equipment are to be used. Every link is to be examined. Additionally, 10% of links are to be tested on all accessible surfaces.

   Link surface at the flash weld is to be free from cracks, lack of fusion and gross porosity. Testing is to be performed in accordance with ASTM E709 or another recognized standard (e.g. ISO 9934) using wet continuous fluorescent magnetization technique. Non fluorescent techniques can be accepted in special cases where the standard inspection procedures are impractical.

   Links are to be free from:
   - relevant linear indications exceeding 1.6 mm in transverse direction
   - relevant linear indications exceeding 3.2 mm in longitudinal direction
   - relevant non-linear indications exceeding 4.8 mm.

d) Ultrasonic testing is to be employed to examine the flash weld fusion. Procedures are to be submitted to IRS for approval. Approved procedures and

equipment are to be used. On-site calibration standards for chain configurations are to be approved.

Every link is to be examined.

The flash weld is to be free from defects causing ultrasonic back reflections equal to or greater than the calibration standard. The flash butt welds are to be ultrasonic tested (UT) in accordance with ASTM E587 or another recognized standard using single probe, angle-beam shear waves in the range from 45 to 70°.

Single probe technique has limitations as far as testing of the central region is concerned and the flash weld imperfections such as flat spots may have poor reflectivity. Where it is deemed necessary, detectability of imperfections may need to be carried out by using a tandem technique, TOFD or phased array.

e) Stud welds, if used, are to be visually inspected. The toes of the fillets are to have a smooth transition to the link with no undercuts exceeding 1.0 [mm]. Additionally, at least 10% of the stud welds distributed through the length are to be dye penetrant tested according to ASTM E1417 or magnetic particle tested according to ASTM E1444 or equivalent. Cracks, lack of fusion or gross porosity are not acceptable. If defects are found, testing is to be extended to all stud welds in that length.

5.13.6 Retest, rejection and repair criteria

a) If the length over 5 links is short, the chain may be stretched by loading above the proof test load specified provided that the applied load is not greater than that approved and that only random lengths of the chain need stretching. If the length exceeds the specified tolerance, the overlength chain links are to be cut out and the requirements of 5.13.6(b) applied.

b) If single links are found to be defective or do not meet other applicable requirements, defective links may be cut out and a connecting common link inserted in their place. The individual heat treatment and inspection procedure of connecting common links is subject to the approval of the Surveyor. Other methods of repair are subject to approval of the Surveyor and the end purchaser. Weld repair of chain is not permitted.

c) If a crack, cut or defect in the flash weld is found by visual or magnetic particle examination, it is to be ground down no more than 5 per cent of the link diameter in depth and streamlined to avoid any sharp contours. However, the final dimensions are to conform to the agreed standard.

d) If indications of interior flash weld defects in reference to the accepted calibration standards are detected during ultrasonic examination, requirements of 5.13.6(b) are applied.

e) If link diameter, length, width and stud alignment do not conform to the required dimensions, these are to be compared to the dimensions of 40 more links; 20 on each side of the affected link. If a single particular dimension fails to meet the required dimensional tolerance in more than 2 of the sample links, all links are to be examined and requirements of 5.13.6(b) are applied.

f) If a break load test fails, a thorough examination is to be carried out to identify the cause of failure. Two additional break test specimens representing the same sampling length of chain are to be subjected to the break load test. Based upon satisfactory results of the additional tests and the results of the failure investigation, it will be decided what lengths of chain can be accepted. Failure of either or both additional tests will result in rejection of the sampling length of chain represented and requirements of 5.13.6(b) are applied.

g) If a link fails during proof load testing, a thorough examination is to be carried out to identify the probable cause of failure of the proof load test. In the event that two or more links in the proof loaded length fail, that section of proof loaded length is to be rejected.

The above mentioned failure investigation is to especially aim at ascertaining the presence in other lengths of factors or conditions thought to be causal to failure.

In addition, a break test specimen is to be taken from each side of the one

failed link and subjected to the breaking test. Where multiple chains are produced simultaneously it is recognized that the preceding flash butt welded link and subsequent flash butt welded link will be on an alternative chain length or the other end of the chain length. In such cases IRS may require that two additional break tests are to be taken from the lengths of chain that include the preceding and subsequent welded links. Based upon satisfactory results of both break tests and the results of the failure investigation, it will be decided what length of chain can be considered for acceptance. Failure of either or both breaking tests will result in rejection of the same proof loaded length.

Replacement of defective links is to be in accordance with the requirements of 5.13.6(b). If the investigation identifies defects in the flash butt weld or a lower strength flash weld “a glue-weld” is found, additional NDT such as phased array UT is to be carried out to identify if other links are affected. A full assessment of the flash butt welding machine is to be carried out, together with assessment of the condition of the bar ends prior to welding.

h) If the tensile test fails to meet the requirements, a retest of two further specimens selected from the same sample will be permissible. Failure to meet the specified requirements of either or both additional tests will result in rejection of the sampling length of chain represented and requirements of 5.13.6(b) are applied.

i) If the impact test requirements are not achieved, a retest of three further specimens selected from the same sample will be permissible. The results are to be added to those previously obtained to form a new average. The new average is to comply with the requirements. No more than two individual results are to be lower than the required average and no more than one result is to be below 70 per cent of the specified average value.

Failure to meet the requirements will result in rejection of the sampling length represented and requirements of 5.13.6(b) are applied.

5.13.7 Marking

.1 The chain is to be marked at the following places:

- At each end.
- At intervals not exceeding 100 m.
- On connecting common links.
- On links next to shackles or connecting common links.

All marked links are to be stated on the certificate, and the marking is to make it possible to recognize leading and tail end of the chain. In addition to the above required marking, the first and last common link of each individual charge used in the continuous length is to be traceable and adequately marked.

The marking is to be permanent and legible throughout the expected lifetime of the chain.

.2 The chain is to be marked on the studs as follows:

- Chain grade
- Certificate No.
- IRS stamp

The Certificate number may be exchanged against an abbreviation or equivalent, in which case the details are to be stated in the certificate.

The chain certificate will contain information on number and location of connecting common links. The certificate number and replacement link number may be exchanged against an abbreviation or equivalent, in which case, the details are to be stated in the certificate.

5.13.8 Documentation

a) A complete Chain Inspection and Testing Report in booklet form is to be provided by the chain manufacturer for each continuous chain length. This booklet is to include all dimensional checks, test and inspection reports, NDT reports, process records, photographs as well as any nonconformity, corrective action and repair work.

b) Individual certificates are to be issued for each continuous single length of chain.

c) All accompanying documents, appendices and reports are to carry reference to the original certificate number.

d) The manufacturer will be responsible for storing, in a safe and retrievable manner, all documentation produced for a period of at least 10 years.

5.14 Testing and inspection of accessories

5.14.1 General

a) Requirements in 5.14 apply to but is not limited to mooring equipment accessories such as detachable connecting links (shackles), detachable connecting plates (triplates), end shackles, swivels, swivel shackles and subsea connectors.

b) Prior to test and inspection the chain accessories are to be free from scale, paint or other coating.

c) All accessories are to be subjected to proof load tests sample, break load tests and sample mechanical tests after final heat treatment in the presence of a Surveyor. Where the manufacturer has a procedure to record proof loads and the Surveyor is satisfied with the adequacy of the recording system, he may not witness all proof load tests. The Surveyor is to satisfy himself that the testing machines are calibrated and maintained in a satisfactory condition.

d) For accessory production a Manufacturing Procedure Specification (MPS) is to be submitted to IRS that details all critical aspects of accessory production, casting, forging, heat treating (including arrangement and spacing of components in the heat treatment furnaces), quenching, mechanical testing, proof and break loading and NDE.

5.14.2 Proof and break load tests

a) All accessories are to be subjected to the proof load specified for the corresponding stud link chain.

b) Chain accessories are to be tested to the break test loads prescribed for the grade and size of chain for which they are intended. At least one accessory out of every 25 accessories is to be tested.

For individually produced, individually heat treated accessories or accessories produced in small batches (less than 5), alternative testing will be subject to special consideration and the following additional conditions may apply:

(i) Alternative testing is described in a written procedure and manufacturing procedure specification (MPS).

(ii) A finite element analysis is provided at the break load and demonstrates that the accessory has a safety margin over and above the break load of the chain.

(iii) Strain age testing (as per procedure approved by IRS) is carried out on the material grade produced to the same parameters at the time of qualification.

(iv) If an accessory is of a large size that will make heat treating in batches unfeasible or has a unique design, strain gauges are to be applied during the proof and break load tests during initial qualification and during production. The strain gauge results from production are to be comparable with the results from qualification.

A batch is defined as accessories that originate from the same heat treatment charge and the same heat of steel. Accessories which have been subjected to a break test are to be scrapped, in general. However, where the accessories are of increased dimensions or alternatively a material with higher strength characteristics is used, they may be included in the outfit at the discretion of IRS provided that:

v) the accessories are successfully tested at the prescribed breaking load appropriate to the chain for which they are intended, and

vi) it is verified by procedure tests that such accessories are so designed that the breaking strength is not less than 1.4 times the prescribed breaking load of the chain for which they are intended.

vii) strain age properties have been carried out on the material grade produced to the same parameters.

viii) strain gauges are to be applied during the break load test in the high stress locations to monitor that the strains stay within allowable limits.

5.14.3 Dimensions and dimensional tolerances

a) After proof load testing, at least one accessory (of the same type, size and nominal strength) out of 25 is to be checked for dimensions in accordance with 5.12.9. The manufacturer is to provide an adequate evidence indicating compliance with the purchaser's requirements.

b) The following tolerances are applicable to accessories:

- Nominal diameter : +5 percent, -0 percent
- Other dimensions : ± 2 ½ percent.

These tolerances do not apply to machined surfaces.

5.14.4 Mechanical tests

a) Accessories are to be subjected to mechanical testing as described in 5.8 and 5.9. Mechanical tests are to be taken from proof loaded full size accessories that have been heat treated with the production accessories they represent. At least one accessory out of every batch or every 25 accessories, whichever is less, is to be tested. Hardness tests are to be carried out on finished accessories. The frequency and locations are to be agreed with IRS. The recorded values are for information only and used as an additional check to verify that the heat treatment process has been stable during the accessory production. The use of separate representative coupons is not permitted except as indicated in e) below.

b) Test location of forged shackles. Forged shackle bodies and forged Kenter shackles are to have a set of three impact tests and a tensile test taken from the crown of the shackle. Tensile_tests on smaller diameter shackles can be taken from the straight part of the shackle, where the geometry does not permit a tensile specimen from the crown. The tensile properties and impact values are to meet the requirements of Table 5.7.1 in the locations specified in Fig.2.4.1 of Section 2 with the Charpy pieces on the outside radius.

c) The locations of mechanical tests of cast shackles and cast Kenter shackles can be taken from the straight part of the accessory. The tensile properties and impact values are to meet the requirements of Table 5.7.1 in the locations specified in Fig.2.4.1 of Section 2.

d) The locations of mechanical tests of other accessories with complex geometries are to be agreed with IRS.

For non-circular sections, 1/4*t* (thickness) from the surface is considered appropriate.

Rolled plates are to be tested to the Standard to which they are produced.

e) For individually produced (heat treated) accessories or accessories produced in small batches, (less than 5), alternative testing can be proposed to IRS. Each proposal for alternative testing is to be detailed by the manufacturer in a written procedure and submitted to IRS. The following additional conditions may apply:

i) If separately forged or cast coupons are used, they are to have a cross-section and, for forged coupon, a reduction ratio similar to that of the accessories represented, and are to be heat treated in the same furnace and quenched in the same tank at the same time, as the actual forgings or castings. Thermocouples are to be attached to the coupon and to the accessories.

ii) If separately forged or cast coupons are agreed, it is to be verified by procedure test that coupon properties are representative of accessory properties.

f) A batch is defined as accessories that originate from the same heat treatment charge and the same heat of steel.

g) Mechanical tests of pins are to be taken as per Fig.2.4.1 of Section 2 from the mid length of a sacrificial pin of the same diameter as the final pin. For oval pins the diameter taken is to represent the smaller dimension. Mechanical tests may be taken from an extended pin of the same diameter as the final pin that incorporates a test prolongation and a heat treatment buffer prolongation, where equivalence with mid length test values have

been established. The length of the buffer is to be at least equal to 1 pin diameter dimension which is removed after the heat treatment cycle is finished. The test coupon can then be removed from the pin. The buffer and test are to come from the same end of the pin as per Fig.5.14.4 below:

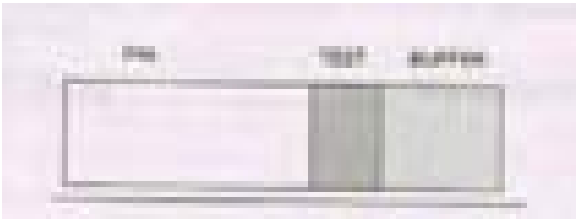


Fig.5.14.4 : Buffer and test piece location

5.14.5 Non-destructive examination after proof load testing

a) All chain accessories are to be subjected to a close visual examination. Special attention is to be paid to machined surfaces and high stress regions. Prior to inspection, chain accessories are to have a suitably prepared surface as per the applied NDE testing standard. All non-machined surfaces, are to be sand or shot blast to permit a thorough examination. Where applicable, accessories shall be dismantled for inspection of internal surfaces. All accessories are to be checked by magnetic particle or dye penetrant testing. UT of accessories may be required by the Surveyor. The acceptance /rejection criteria of UT established for the design is to be met.

b) Testing is to be performed in accordance with a recognized standard, such as those indicated below, or equivalent. The procedures, together with acceptance / rejection criteria are to be submitted to IRS for review.

   Magnetic particle testing (MT) of forgings:
   - EN 10228-1, ASTM A275, using wet continuous magnetization technique or equivalent standards such as ISO 4986, IACS Rec 69

   Ultrasonic testing (UT) of forgings:
   - EN 10228-3, ASTM A388, ISO 13588

   Magnetic particle testing (MT) of castings:
   - ASTM E709, using wet continuous magnetization technique

   Ultrasonic testing (UT) of castings: - ASTM A609, ISO 13588

   All surfaces are to be magnetic particle tested (MT). Testing is to be performed in accordance with standards referenced using the fluorescent technique. As a minimum surfaces are to be free from:

   - relevant linear indications exceeding 1.6 mm in transverse direction

   - relevant linear indications exceeding 3.2 mm in longitudinal direction

   - relevant non-linear indications exceeding 4.8 mm.

   When required by the Surveyor, ultrasonic testing is to be carried out on 100% of cast or forged accessories. The acceptance / rejection criteria established for the design is to be met.

   Manufacturers are to prepare written procedures for NDE. NDE personnel are to be qualified and certified according to ISO 9712, ACCP or equivalent. Personnel qualification to an employer or responsible agency based qualification scheme as SNT-TC-1A may be accepted if the employer's written practice is reviewed and found acceptable and the Level III is ASNT Level III, ISO 9712 Level III or ACCP Professional Level III and certified in the applicable method. NDE operators are to be qualified to at least level II.

c) The manufacturer is to provide adequate evidence that non-destructive examination has been carried out with satisfactory results together with details of technique used and to the operator's qualification.

d) Weld repair of finished accessories is not permitted.

5.14.6 Test failures

In the event of a failure of any test the entire batch represented is to be rejected unless the cause of failure is determined and it is demonstrated to the Surveyor's satisfaction that the condition causing the failure is not present in any of the remaining accessories.

5.14.7 Marking

.1 Each accessory is to be marked as follows:

- Chain grade
- Certificate No.
- IRS stamp

.2 All detachable component parts are to be stamped with a serial number to avoid mixing of components.

.3 The Certificate number may be exchanged against an abbreviation or equivalent, in which case, the details are to be stated in the certificate.

5.14.8 Documentation

.1 A complete Inspection and Testing Report in booklet form is to be provided by the manufacturer for each order. This booklet is to include all dimensional checks, test and inspection reports, NDT reports, process records and example photographs of components positioned in furnaces as well as any nonconformity, corrective action and repair work

.2 Each type of accessory is to be covered by separate certificates.

.3 All accompanying documents, appendices and reports are to carry reference to the original certificate number.

.4 The manufacturer will be responsible for storing, in a safe and retrievable manner, all documentation produced for a period of at least 10 years.

5.15 Chafing chain for single point mooring arrangements

5.15.1 Scope

These requirements apply to short lengths (approximately 8 m) of 76 [mm] diameter chain to be connected to hawsers for the tethering of oil tankers to single point moorings, FPSO's and similar uses.

5.15.2 Approval of manufacturing

The chafing chain is to be manufactured by works approved by IRS according to 5.3 of this chapter. Also refer Chapter 1, Section 1, Cl. 1.3.2.

5.15.3 Materials

The materials used for the manufacture of the chafing chain are to satisfy the requirements of 5.7 to 5.10 of this chapter.

5.15.4 Design, manufacturing, testing and certification

a) The chafing chain is to be designed, manufactured, tested and certified in accordance with 5.11 to 5.14 except that batch heat treatment is permitted.

b) The arrangement of the end connections is to be of an approved type.

c) The common link is to be of the stud link type and of Grade R3 or R4.

d) The chafing chain is to be capable of withstanding the breaking test loads of 4884 [kN] (Grade R3) and 6001 [kN] (Grade R4).

   Documented evidence of satisfactory testing of similar diameter mooring chain in the prior six month period may be used in lieu of break testing subject to agreement with IRS.

e) The chain lengths shall be proof load tested in accordance with 5.13.2. The proof test loads for Grade R3 and Grade R4 are to be 3242 [kN] and 4731 [kN] respectively.

f) The requirements herein are also applicable to other diameter chafing chains, such as 84 [mm] and 96 [mm], subject to compliance with the proof and break load requirements specified for the chain grade and diameters in Table 5.12.1

End of Chapter

Chapter 11

Approval of Welding Consumables for Use in Ship Construction

Contents

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2	*Electrodes for Normal Penetration Manual Welding*
3	*Deep Penetration Electrodes for Manual Welding*
4	*Wire-flux Combinations for Submerged Arc Automatic Welding*
5	*Wires and Wire-gas Combinations for Semi-automatic and Automatic Welding*
6	*Consumables for use in Electro-slag and Electro-gas Vertical Welding*
7	*Welding Consumables for High Strength Steels for Welded Structures*
8	*Consumables for Welding of Aluminium Alloys*

Section 1

General

1.1 Scope

1.1.1 This Chapter gives the requirements for approval and inspection of welding consumables such as electrodes, wires, fluxes etc. intended for welding of the following types of materials used in ship construction:

a) Normal strength steel for ship structures, Grades A, B, D and E (See Ch.3).

b) Higher strength steels for ship structures Grades AH32, DH32, EH32, AH36, DH36 and EH36 (See Ch.3).

c) Higher strength steels for ship structures with minimum yield strength 390 [N/mm^2] : Grades AH40, DH40 and EH40 (See Ch.3).

d) Higher strength steels for ship structures for low temperature application : Grades FH32, FH36 and FH40 (See Ch.3).

e) Higher strength steels for welded structures.

f) Aluminium alloys (See Ch.9).

1.2 Manufacture

1.2.1 The manufacturer's plant and method of production of welding consumables are to be such as to ensure reasonable uniformity in manufacture. IRS is to be notified of any alteration proposed to be made in the process of manufacture subsequent to approval.

1.3 Grading

1.3.1 Welding consumables for steel materials specified in 1.1.1 a) to d) above.

These consumables are divided into 3 strength groups each of which is further graded as per the Charpy V-notch impact test requirements as shown below:

Groups	**Grading**
Normal strength steel	1, 2, 3
Higher strength steel : yield strength upto 355 [N/mm^2]	1Y, 2Y, 3Y, 4Y
Higher strength steels : yield strength upto 390 [N/mm^2]	2Y40, 3Y40, 4Y40, 5Y40
Hydrogen marks Welding consumables of Grades 2 and 3; and of Grades 2Y, 3Y and 4Y and of Grades 2Y40, 3Y40, 4Y40 and 5Y40 for which hydrogen content has been controlled in accordance with Sec.2.5 are identified by the mark H15, H10 or H5	
The following suffixes are added after the Grade mark as applicable: S : Semi-automatic T : Two-run technique M : Multi-run technique TM : Both two-run and multi-run technique V : Vertical	

See Table 1.3.1 for correlation of welding consumables to hull structural steel grades.

1.3.2 For grading of consumables for welding higher strength quenched and tempered steels indicated in 1.1.1e) above, refer to Sec.7.

1.3.3 For grading of consumables for welding aluminium alloys indicated in 1.1.1f) above, refer to Sec.8.

Table 1.3.1 : Correlation of welding consumables to hull structural steel grades

Grades of Welding Consumables (see notes)	**Hull structural steel Grades [1)]**											
	A	**B**	**D**	**E**	**AH32 /36**	**DH32 /36**	**EH32 /36**	**FH32/ 36**	**AH40**	**DH40**	**EH40**	**FH40**
1, 1S, 1T, 1M, 1TM, 1V	X											
1YS, 1YT, 1YM, 1YTM, 1YV	X				X [3)]							
2, 2S, 2T, 2M, 2TM, 2V	X	X	X									
2Y, 2YS, 2YT, 2YM, 2YTM, 2YV	X	X	X	X	X							
2Y40, 2Y40S, 2Y40T, 2Y40M, 2Y40TM, 2Y40V	2)	2)	2)		X	X			X	X		
3, 3S, 3T, 3M, 3TM, 3V	X	X	X	X								
3Y, 3YS, 3YT, 3YM, 3YTM, 3YV	X	X	X	X	X	X	X					
3Y40, 3Y40S, 3Y40T, 3Y40M, 3Y40TM, 3Y40V	2)	2)	2)	2)	X	X	X		X	X	X	
4Y, 4YS, 4YT, 4YM, 4YTM, 4YV	X	X	X	X	X	X	X	X				
4Y40, 4Y40S, 4Y40T, 4Y40M, 4Y40TM, 4Y40V	2)	2)	2)	2)	X	X	X	X	X	X	X	X
5Y40, 5Y40S, 5Y40T, 5Y40M, 5Y40TM, 5Y40V	2)	2)	2)	2)	X	X	X	X	X	X	X	X

1) Requirements for other grades of steels given in ch.3 but not included here will be specially considered.
2) See Note d)
3) See Note e)

a) When joining normal to higher strength structural steel, consumables of the lowest acceptable grade for either material being joined may be used.
b) When joining steels of same strength level but of different toughness grade, consumables of the lowest acceptable grade for either material being joined may be used.
c) It is recommended that controlled low hydrogen type consumables are to be used when joining higher strength structural steels to the same or lower strength level, except that other consumables may be used when the carbon equivalent is below or equal to 0.41%. When other than controlled low hydrogen type electrodes are used appropriate procedure test for hydrogen cracking may be conducted subject to approval of IRS.
d) The welding consumables approved for steel Grades AH40, DH40, EH40 and/or FH40 may also be used for welding of the corresponding grades of normal strength steels subject to the approval of IRS.
e) When joining higher strength steels using Grade 1Y welding consumables, the material thicknesses is not to exceed 25 [mm].

1.4 Approval procedure

1.4.1 Approval of welding consumables will be considered on the basis of the manufacturer's description of the works and detailed description of the method of production control, satisfactory inspection of the works by the Surveyors and compliance with the test requirements detailed in subsequent paragraphs of this Chapter.

1.4.2 When a welding consumable is manufactured in several locations of the same company, the complete series of approval tests would be carried out in one Works only. In other locations, a reduced test programme based upon the requirements of annual testing may be accepted subject to the manufacturer certifying that the materials and the fabrication process used are identical with those of the main unit.

This requirement is also applicable to all manufacturers of welding consumables under license.

Note : In case of wire flux combination for submerged arc welding where a unique powder flux is combined with different wires coming from several factories belonging to the same firm only one test series may be carried out provided the wires conform to the same technical specification.

1.4.3 The test assemblies are to be prepared under the supervision of the Surveyor, and all tests are to be carried out in his presence.

1.4.4 IRS may require, in any particular case, such additional tests or spacing requirements as may be necessary.

1.5 Test assemblies

1.5.1 The test assemblies are to be prepared and tested under the supervision of the Surveyor(s).

1.5.2 When a welded joint is performed, the edges of the plates are to be bevelled either by mechanical machining or by oxygen cutting; in the latter case, a descaling of the edges is necessary.

1.5.3 The welding conditions used such as amperage, voltage, travel speed, etc. are to be within the range recommended by the manufacturer for normal and good welding practice. Where a welding consumable is suitable for both alternating current (AC) and direct current (DC), AC is to be used for the preparation of test assemblies.

1.6 Annual inspection and tests

1.6.1 All establishments, where approved welding consumables are manufactured, and the associated quality control procedures, are to be subjected to annual inspection. On these occasions, samples of the approved consumables are to be selected by the Surveyor and subjected to the tests detailed in the subsequent paragraphs of this Chapter.

1.7 Upgrading and uprating

1.7.1 Upgrading and uprating of welding consumables will be considered only at the manufacturer's request, preferably at the time of annual testing. Generally, for this purpose, tests from butt weld assemblies will be required in addition to the normal annual approval tests.

1.7.2 Upgrading refers to notch toughness of the welding consumable while uprating refers to extension to cover higher strength level steels.

1.7.3 Any alteration to the approved consumable which may result in a change in the chemical composition and the mechanical properties of the deposited metal, must be immediately notified by the manufacturer. Additional tests may be necessary.

1.8 Dimensions of test specimens

1.8.1 Deposited metal tensile test specimens are to be machined to the dimensions shown in Fig.1.8.1. Care is to be taken to ensure that the longitudinal axis of the test piece coincides with the centre of the weld and midthickness of the plates. The test piece may be heated to a temperature not exceeding 250°C for a period not exceeding 16 hours for hydrogen removal, prior to testing.

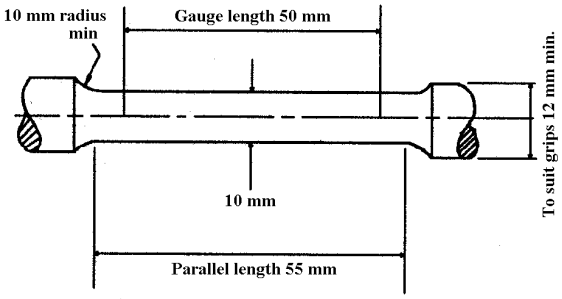


Fig.1.8.1 : Deposited metal tensile test

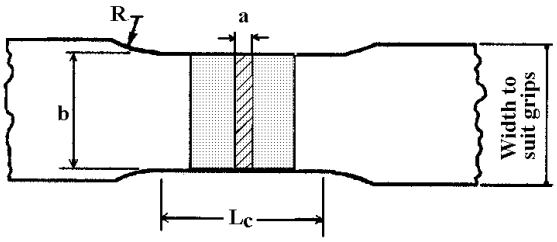


Fig.1.8.2 : Butt weld tensile test specimen

1.8.2 Butt weld tensile test specimens are to be machined to the following dimensions (see Fig.1.8.2).

a = thickness of plate 't'
b = 12 for $t \leq 2$ [mm]
= 25 for $t > 2$ [mm]
L_c = width of weld + 60 [mm]
$R > 25$ [mm].

The upper and lower surfaces of the weld are to be filed, ground or machined flush with the surface of the plates.

1.8.3 Butt weld bend test specimens are to be 30 [mm] in width. Upper and lower surfaces of the weld are to be filed, ground or machined flush with the surfaces of the plates and sharp corners of the specimens are to be rounded to a radius not exceeding 2 [mm].

1.8.4 All impact test specimens are to be of the standard 10 [mm] x 10 [mm] Charpy V-notch type, machined to the dimensions and tolerances detailed in Ch.2.

1.9 Testing procedures

1.9.1 The procedures used for all tensile and impact tests are to comply with the requirements of Ch.2.

1.9.2 Butt weld bend test specimens are to be tested at ambient temperature. The test specimens are to be capable of withstanding, without fracture, being bent through an angle of 120 degrees over a former having a diameter three times the thickness of the specimen. One specimen from each welded assembly is to be tested with the face of the weld in tension and the other with the root of the weld in tension.
The test pieces can be considered as complying with the requirements if, on completion of the test, no crack or defect at the outer surface of the test specimen can be seen.

1.9.3 Tensile Tests : On deposited metal test specimens, the values of tensile strength, yield stress and elongation are to be recorded. On butt weld specimens, the values of tensile strength and the position of fracture are to be recorded.

1.9.4 Charpy V-notch Impact Tests : A set of three test specimens is to prepared and tested. The average absorbed energy value is to comply with the requirements of subsequent sections. One individual value may be less than the required average value provided that it is not less than 70 per cent of this value.

The test temperature for Grades 2, 2Y, 2Y40, 3, 3Y, 3Y40, 4Y, 4Y40 and 5Y40 test pieces is to be controlled within ±2°C of the prescribed temperature.

1.10 Re-test procedures

1.10.1 Where the results of a tensile or bend test do not comply with the requirements, duplicate test specimens of the same type are to be prepared and are to be satisfactorily tested. Where insufficient original welded assembly is available, a new assembly is to be

prepared using welding consumables from the same batch. If the new assembly is made with the same procedure (particularly the number of runs) as the original assembly, only the duplicate re-test specimens need to be tested. Otherwise, all test specimens are to be prepared and re-tested.

1.10.2 Where the results from a set of three impact test specimens do not comply with the requirements, an additional set of three impact test specimens may be taken provided that not more than two individual values are less than the required average value and, of these, not more than one is less than 70 per cent of the average value. The results obtained are to be combined with the original results to form a new average which, for acceptance, is to be not less than the required value. Additionally, for these combined results not more than two individual values are to be less than the required average value, and of these, not more than one is to be less than 70 per cent of the average value. Further retests may be made at the Surveyor's discretion, but these must be made on a new welded assembly and must include all tests required for the original assembly, even those which were previously satisfactory.

1.11 Chemical composition

1.11.1 The chemical analysis of the weld metal made by the electrode is to be supplied by the manufacturer.

Section 2

Electrodes for Normal Penetration Manual Welding

2.1 General

2.1.1 Based on the results of the Charpy V-notch impact tests, electrodes are divided into the following grades:-

For normal strength steel - Grades 1, 2 and 3

For higher strength steel with minimum yield strength upto 355 [N/mm^2] - Grades 2Y, 3Y and 4Y. (Grade 1Y not applicable for manual welding).

For higher strength steels with minimum yield strength upto 390 [N/mm^2] - Grades 2Y40, 3Y40, 4Y40 and 5Y40.

2.1.2 If the electrodes are in compliance with the requirements of the hydrogen test given in 2.5, a suffix H15, H10 or H5 will be added to the grade mark.

2.1.3 For initial approval the tests specified in this Section including hydrogen test, if applicable, are to be carried out.

2.2 Deposited metal tests

2.2.1 Two deposited metal test assemblies are to be prepared in the downhand position as shown in Fig.2.2.1, one using 4 [mm] electrodes and the other using the largest size manufactured. If an electrode is manufactured in one diameter only, one test assembly is sufficient. Any grade of ship structural steel may be used for the preparation of these test assemblies.

2.2.2 The weld metal is to be deposited in single or multi-run layers according to normal practice and the direction of each layer is to alternate from each end of the plate, each run of the weld metal being not less than 2 [mm] and not more than 4 [mm] thick. Between each run the assembly is to be left in still air until it has cooled to 250°C but not less than 100°C, the temperature being taken in the centre of the weld, on the surface of the seam. After being welded the test assemblies are not to be subjected to any heat treatment.

2.2.3 One tensile and three impact test specimens are to be taken from each test assembly as shown in Fig.2.2.1. The impact test specimens are to be cut perpendicular to the weld, with their axes 10 [mm] from the upper surface of the plate. The notch is to be positioned in the centre of the weld and cut in the face of the test specimen perpendicular to the surface of the plate.

2.2.4 The chemical analysis of the deposited weld metal in each test assembly is to be supplied by the manufacturer and is to include the content of all significant alloying elements.

2.2.5 The results of all tests are to comply with the requirements of Table 2.2.1 as appropriate.

Table 2.2.1 : Requirements for deposited metal tests (covered electrodes)

Grade	Yield stress [N/mm^2] min.	Tensile strength [N/mm^2]	Minimum elongation on 50 mm gauge length [%]	Impact Tests	
				Test temp. °C	Average energy J min.
1 2 3	305	400 - 560	22	20 0 -20	47
2Y 3Y 4Y	375	490 - 660	22	0 -20 -40	47
2Y40 3Y40 4Y40 5Y40	400	510 - 690	22	0 -20 -40 -60	47

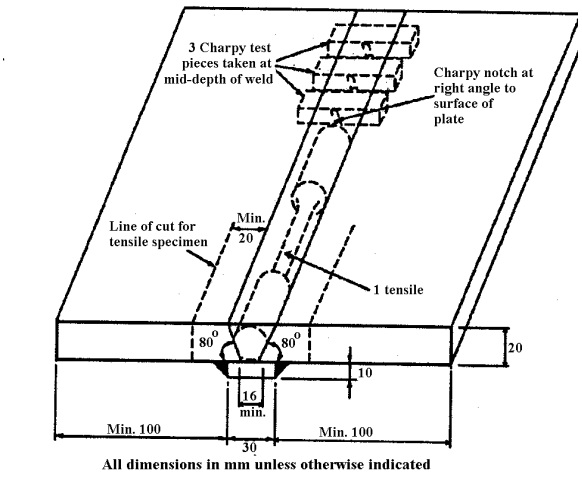


Fig.2.2.1 : Deposited metal test assembly

2.3 Butt weld tests

2.3.1 Butt weld assemblies as shown in Fig.2.3.1 are to be prepared for each welding position (downhand, horizontal-vertical, vertical-upward, vertical-downward and overhead) for which the electrode is recommended by the manufacturer, except that electrodes satisfying the requirements for downhand and vertical-upward positions will be considered as also complying with the requirements for the horizontal-vertical position.

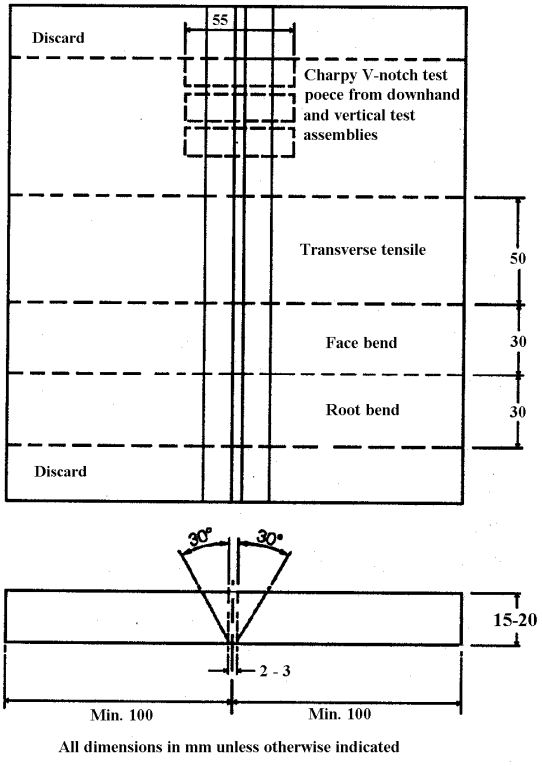


Fig.2.3.1 : Butt weld test assembly

Table 2.3.1 : Requirements for butt weld test (covered electrodes)

Grade	Tensile strength [N/mm^2] min. (transverse test)	Charpy V-notch impact test		
		Test temp. °C	Average energy J min.	
			Down-hand horizon-tal vertical, over-head	Vertical (upward and down-ward)
1 2 3	400	20 0 -20	47	34
2Y 3Y 4Y	490	0 -20 -40	47	34
2Y40 3Y40 4Y40 5Y40	510	0 -20 -40 -60	47	39

2.3.2 Where the electrode is only to be approved in the downhand position an additional test assembly is to be prepared in that position.

2.3.3 The grades of steels used for the preparation of the test assemblies are to be as follows:

Grade 1 electrodes	A
Grade 2 electrodes	A, B, D
Grade 3 electrode	A, B, D, E
Grade 2Y electrode	AH32, AH36, D32, D36
Grade 3Y electrode	AH32, AH36, DH32, DH36, EH32, EH36
Grade 4Y electrodes	AH32, AH36, DH32, DH36, EH32, EH36, FH32, FH36
Grade 2Y40 electrodes	AH40, DH40
Grade 3Y40 electrodes	AH40, DH40, EH40
Grade 4Y40 electrodes	AH40, DH40, EH40, FH40
Grade 5Y40 electrodes	AH40,DH40, EH40, FH40

2.3.4 Where higher strength steel with minimum yield strength 315 [N/mm^2] is used for grade 2Y, 3Y and 4Y electrodes, the actual tensile strength of the steel is to be not less than 490 [N/mm^2]. The chemical composition including the content of grain refining elements is to be reported.

2.3.5 The following welding procedure should be adopted in making the test assemblies:-

DOWNHAND (a)

First run with 4 [mm] diameter electrode. Remaining runs (except last two layers) with 5 [mm] diameter electrodes or above according to the normal welding practice with the electrodes. The runs of the last two layers with the largest size of electrode manufactured.

DOWNHAND (b)

Where a second downhand test is required:-

First run with 4 [mm] diameter electrode. Next run with an intermediate size electrode 5 [mm] or 6 [mm] diameter and the remaining runs with the largest size of electrode manufactured.

HORIZONTAL-VERTICAL

First run with 4 [mm] or 5 [mm] diameter electrode, subsequent runs with 5 [mm] diameter electrodes.

VERTICAL UPWARDS AND OVERHEAD

First run with 3.25 [mm] diameter electrode. Remaining runs with 4 [mm] diameter electrodes or possibly 5 [mm] diameter electrodes if this is recommended by the manufacturer for the positions concerned.

VERTICAL DOWNWARD

The method to be adopted is to be as recommended by the manufacturer.

2.3.6 In all cases the back sealing runs are to be made with 4 [mm] diameter electrodes in the welding position appropriate to each test sample after cutting out the root run to clean metal. For electrodes suitable for downhand welding only, the test assemblies may be turned over to carry out the back sealing run.

2.3.7 The butts are to be welded using normal welding practice and between each run the assembly is to be left in still air until it has cooled to 250°C but not below 100°C, the temperature being taken in the centre of the weld, on the surface of the seam.

2.3.8 After being welded, the test assemblies are not to be subjected to any heat treatment.

2.3.9 It is recommended that the welded assemblies be subjected to a radiographic examination to ascertain any defects in the weld prior to testing.

2.3.10 From each test assembly one tensile, one face and one root bend and a set of three Charpy V-notch test specimens are to be prepared, except that the impact test specimens need not be prepared for test assemblies welded in the overhead position.

2.3.11 The results of all mechanical testing are to comply with the requirements of Table 2.3.1. The position of the fracture in the transverse tensile test is to be reported. The bend test specimens can be considered as complying with the requirements if, after bending, no crack or defect having dimensions exceeding 3 [mm] can be seen on the outer surface of the test specimen.

2.4 Fillet weld tests

2.4.1 When an electrode is submitted for approval for fillet welding only and to which butt weld tests, as per 2.3, are not considered applicable, the initial approval tests are to consist of the fillet weld test, described herein, and the deposited metal test as per 2.2. When the electrode is submitted for approval for both butt and fillet welding, the initial approval is to include one fillet weld test as detailed hereunder and welded in the horizontal-vertical position in addition to the tests required by 2.2 and 2.3.

2.4.2 Fillet weld assemblies as shown in Fig.2.4.1 are to be prepared for each welding position (horizontal-vertical, vertical-upwards, vertical- downwards or overhead) for which the electrode is recommended by the manufacturer. The grade of steel used is to be in accordance with 2.3.3 as appropriate. The test assemblies are to be welded using an electrode of a diameter recommended by the manufacturer. The length of the test assembly, L, is to be sufficient to allow at least the deposition of the entire length of the electrode being tested.

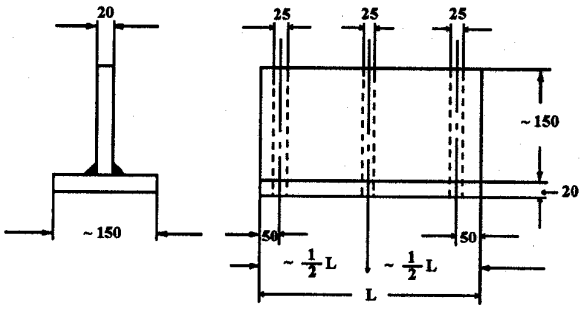


Fig.2.4.1 : Fillet weld test assembly

The first side is to be welded using the maximum size of electrode manufactured and the second side is to be welded using the minimum size of electrode manufactured and recommended for fillet welding. The fillet size will in general be determined by the electrode size and the welding current employed during testing.

2.4.3 The assembly is to be sectioned to form three macro sections each about 25 [mm] thick and the hardness readings are to be made in each section as indicated in Fig.2.4.2. The hardness of the weld is to be determined and is to meet the following listed equivalent values:-

a) Diamond Pyramid Hardness (98 N load) = 150 minimum

b) Rockwell (980 N load) B = 80 minimum

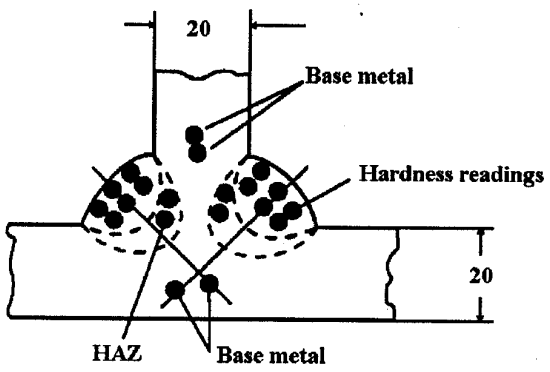


Fig.2.4.2 : Hardness tests for fillet weld test

2.4.4 The hardness of both heat affected zone and base metal is also to be determined and is to be reported for information.

2.4.5 One of the remaining sections of the assembly is to have the weld on the first side gouged or machined to facilitate breaking the fillet weld on the second side by closing the two plates together, subjecting the root of the weld to tension. On the other remaining section the weld on the second side is to be gouged or machined and the section fractured using the same procedure. The fractured surfaces are to be examined and there should be no evidence of incomplete penetrations, nor internal cracking and they should be reasonably free from porosity.

2.5 Hydrogen test

2.5.1 At the request of the manufacturer, electrodes may be submitted to a hydrogen test. A suffix H15, H10 or H5 will be added to

the grade number to indicate compliance with the requirements of this test.

2.5.2 The mercury method or thermal conductivity detector method according to standard ISO 3690:2018 is to be used. Four weld assemblies are to be prepared. The temperature of the specimens and minimum holding time are to be complied with following, according to the measuring method respectively:

Measuring Method		**Test Temp (° C)**	**Minimum Holding Time**
Thermal Conductivity Detector Method [1]	Gas Chromatography	45	72
		150	6
Note 1 - The use of hot carrier gas extraction method may be considered subject to verification of the testing procedure to confirm that collection and measurement of the hydrogen occurs continuously until all of the diffusible hydrogen is quantified.			

Alternatively, the glycerine method as described below is to be used.

2.5.3 Glycerine method

a) Four test specimens are to be prepared measuring 12 x 25 [mm] in cross- section by about 125 [mm] in length. The parent metal may be any grade of ship building steel and, before welding, the specimens are to be weighed to the nearest 0.1 gram. On the 25 [mm] surface of each test specimen a single bead of welding is to be deposited about 100 [mm] in length by a 4 [mm] electrode using about 150 [mm] of the electrode. The welding is to be carried out with as short an arc as possible and with a current of about 150 amperes. The electrode prior to welding, can be submitted to the normal drying process recommended by the manufacturer.

b) Within thirty seconds of the completion of welding of each specimen the slag is to be removed and the specimen quenched in water at approximately 20°C. After a further thirty seconds the specimens are to be cleaned and placed in an apparatus suitable for the collection of hydrogen by displacement of glycerine. The glycerine is to be kept at a temperature of 45°C during the test. All the four specimens are to be welded and placed in the hydrogen collecting apparatus within 30 minutes.

c) The specimens are to be kept immersed in the glycerine for a period of 48 hours and, after removal, are to be cleaned in water and spirit, dried and weighed to the nearest 0.1 gram to determine the amount of weld deposited. The amount of gas evolved is to be measured to the nearest 0.05 [cm^3] and corrected for temperature and pressure to 20C and 760 [mm] Hg.

2.5.4 The individual and average diffusible hydrogen contents of the four specimens are to be reported, and the average value in [cm^3] per 100 grams is not to exceed the following:

Mark	**Diffusible Hydrogen Contents**	**Measuring Method**
H15	15[1]	Mercury Method Thermal Conductivity Detector Method Glycerine Method
H10	10[2]	
H5	5	Mercury Method Thermal Conductivity Detector Method
1 – 10 cm^3 per 100 gms where the glycerine method is used		
2 – 5 cm^3 per 100 gms where the glycerine method is used		
Note :- The glycerine method is not to be used for welding consumables with H5 mark.		

2.6 Covered electrodes for gravity or contact welding

2.6.1 Where an electrode is submitted solely for approval for use in contact welding using automatic gravity or similar welding devices, deposited metal tests (see 2.2), fillet weld tests (see 2.4) and, where appropriate, butt weld tests (see 2.3) similar to those for normal manual electrodes are to be carried out using the process for which the electrode is recommended by the manufacturer.

2.6.2 Where an electrode is submitted for approval for use in contact welding using automatic gravity or similar welding devices in addition to normal manual welding, fillet weld tests (see 2.4) and, where appropriate, butt weld tests (see 2.3) similar to those for normal manual electrodes are to be carried out using the process for which the electrode is recommended by the manufacturer and these

tests are to be in addition to the normal approval tests.

2.6.3 In the case of approval of a fillet welding electrode using automatic gravity or similar contact welding devices, the fillet welding is to be carried out using the welding process recommended by the manufacturer, with the longest size of electrode manufactured. The manufacturer's recommended current range is to be reported for each electrode.

2.6.4 Where approval is requested for the welding of both normal strength and higher tensile steels, the assemblies are to be prepared using higher tensile steel.

2.7 Annual tests

2.7.1 For normal penetration electrodes, the annual tests are to consist of two deposited metal test assemblies. These are to be prepared and tested in accordance with 2.2. If an electrode is available in one diameter only, one test assembly is sufficient.

2.7.2 Where an electrode is approved solely for gravity or contact welding, the annual test is to consist of one deposited metal test assembly using the gravity or other contact device as recommended by the manufacturer.

2.8 Upgrading and uprating

2.8.1 Upgrading and uprating will be considered only at the manufacturer's request and preferably at the time of annual testing. Tests on butt weld assemblies, in addition to the requirements of annual testing, are to be carried out.

2.9 Certification

2.9.1 Each carton or package of approved electrode is to contain a certificate from the manufacturer generally in accordance with the following:-

"The ..... company certifies that composition and quality of these electrodes conform with those of the electrodes used in making the test pieces submitted to and approved by Indian Register of Shipping."

Section 3

Deep Penetration Electrodes for Manual Welding

3.1 General

3.1.1 Where an electrode is designed solely for the deep penetration welding of downhand butt joints and horizontal-vertical fillets, only the test detailed in 3.2 and 3.3 are required for initial approval purposes.

3.1.2 Deep penetration electrodes will only be approved as complying with Grade 1 requirements. The suffix D.P. will be added.

3.1.3 Where a manufacturer recommends that an electrode having deep penetrating properties can also be used for downhand butt welding of thicker plates with prepared edges, the electrode will be treated as normal penetration electrode, and the full series of tests in the downhand position is to be carried out as per normal penetration electrode, together with deep penetration tests given in 3.2 and 3.3.

3.1.4 Where a manufacturer desires to demonstrate that an electrode in addition to its use as normal penetration electrode also has deep penetrating properties when used for downhand butt welding and horizontal - vertical fillet welding, the additional tests given in 3.2 and 3.3 are to be carried out.

3.1.5 Where the manufacturer prescribes a different welding current and procedure for the electrode when used as a deep penetration electrode and a normal penetration electrode, the recommended current and procedure are to be used when making the test specimens in each case.

3.2 Deep penetration butt weld tests

3.2.1 Two plates of thickness equal to twice the diameter of the core of the electrode plus 2 [mm] are to be butt welded together with one downhand run of welding from each side. The

plates are to be not less than 100 [mm] wide and of sufficient length to allow the cutting out of the test specimens of the correct number and size as shown in Fig.3.2.1. Grade A steel is to be used for these test assemblies. The joint edges are to be prepared square and smooth and, after tacking, the gap is not to exceed 0.25 [mm].

3.2.2 The test assembly is to be welded using a 8 [mm] diameter electrode or the largest diameter size manufactured if this is less than 8 [mm].

3.2.3 After welding the test assembly is to be cut to form two transverse tensile test pieces, two bend test pieces and three Charpy V-notch test pieces as shown in Fig.3.2.1. The results of tensile and impact testing are to comply with the requirements of Table 2.3.1 for Grade 1 electrodes.

3.2.4 The discards at the end of the welded assemblies are to be not more than 35 [mm] wide. The joints of these discards are to be polished and etched and must show complete fusion and interpenetration of the welds. At each cut in the test assembly the joints are also to be examined to ensure that complete fusion has taken place.

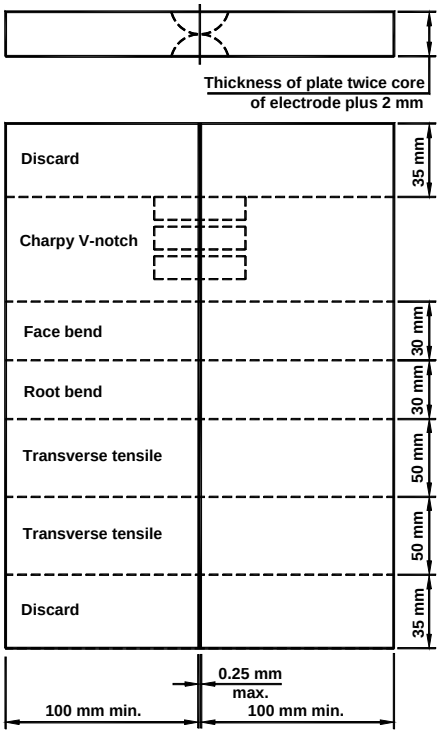


Fig. 3.2.1 : Deep penetration butt weld test assembly

3.3 Deep penetration fillet weld test

3.3.1 A fillet weld assembly is to be prepared as shown in Fig.3.3.1 with plates about 12.5 [mm] in thickness. The welding is to be carried out with one run for each fillet with plate A in the horizontal plane during the welding operations. The length of the fillet is to be 160 [mm] and the gap between the plates is to be not more than 0.25 [mm]. Grade A steel is to be used for these test assemblies.

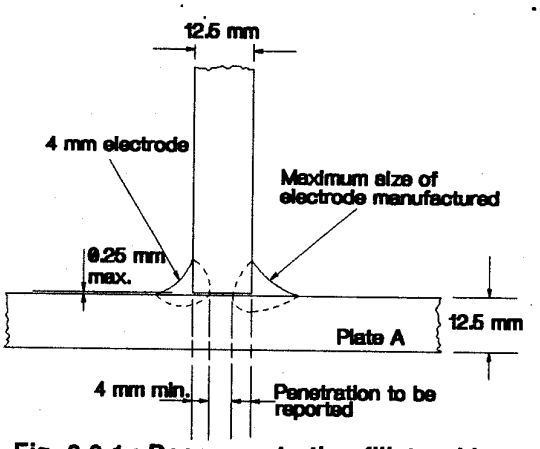


Fig. 3.3.1 : Deep penetration fillet weld test assembly

3.3.2 The fillet weld on one side of the assembly is to be carried out with 4 [mm] electrode and that on the other side with the maximum size of the electrode manufactured. The welding current used is to be within the range recommended by the manufacturer and the welding is to be carried out using normal welding practice.

3.3.3 The welded assembly is to be cut by sawing or machining within 35 [mm] of the ends of the fillet welds and the joints are to be polished and etched. The welding of the fillet made with a 4 [mm] electrode is to show a penetration of 4 [mm] (See Fig.3.3.1) and the corresponding penetration of the fillet made with the maximum size of electrode manufactured is to be reported.

3.4 Electrodes designed for gravity or contact welding

3.4.1 This type of approval is available for welding only normal strength and higher tensile steels with minimum specified yield strengths up to 345 [N/mm^2].

3.4.2 Where an electrode is submitted solely for approval for use in contact welding using automatic gravity or similar welding devices, deposited metal tests, and where appropriate, fillet weld tests similar to those for normal manual electrodes are to be carried out using the process for which the electrode is recommended by the manufacturer.

3.4.3 Where an electrode is submitted for approval for use in contact welding using automatic gravity or similar welding devices in addition to normal manual welding, butt weld and, where appropriate, fillet weld tests, using gravity or other contact device as recommended by the manufacturer, are to be carried out in addition to the normal approval tests.

3.5 Annual tests

3.5.1 Where an electrode is approved only for deep penetration welding, the annual test is to consist of one butt welded test assembly in accordance with 3.2.

3.5.2 Where an electrode is approved for both normal and deep penetration welding, annual tests are to consist of following:-

a) Two deposited metal test assemblies in accordance with 2.2; and

b) One butt welded test assembly in accordance with 3.2.

3.5.3 Where an electrode is approved solely for gravity or contact welding, the annual test is to consist of one deposited metal test assembly using the gravity or other contact device as recommended by the manufacturer.

3.6 Certification

3.6.1 Each carton or package of approved electrodes is to contain a certificate from the manufacturer generally in accordance with 2.9.

Section 4

Wire-flux Combinations for Submerged Arc Automatic Welding

4.1 General

4.1.1 Wire-flux combinations for single electrode submerged-arc automatic or semi-automatic welding are divided into following two categories:-

a) For use with multi-run technique;

b) For use with two-run technique.

Where wire-flux combinations are suitable for welding with both the techniques, tests are to be carried out for each technique.

Wire-flux combinations for multiple electrode submerged arc welding will be subject to separate approval tests. They are to be carried out generally in accordance with the requirements of this section.

4.1.2 Dependent on the results of impact tests, wire-flux combinations are divided into the following grades:-

For normal strength steel - Grades 1, 2 or 3;

For higher strength steel with minimum yield strength upto 355 [N/mm^2] - Grades 1Y, 2Y, 3Y or 4Y.

For higher strength steels with minimum yield strength upto 390 [N/mm^2] - Grades 2Y40, 3Y40, 4Y40 or 5Y40.

4.1.3 The suffixes T, M or TM will be added to the grade mark to indicate two-run technique, multi-run technique or both techniques respectively.

4.1.4 The welding current may be either a.c. or d.c. (electrode positive or negative) according to the recommendation of the manufacturer. If both a.c. and d.c. are recommended, a.c. is to be used for the tests.

4.2 Multi-run technique

4.2.1 When approval for use with multi-run technique is required, deposited metal and butt weld tests are to be carried out in accordance with 4.3 and 4.4 respectively.

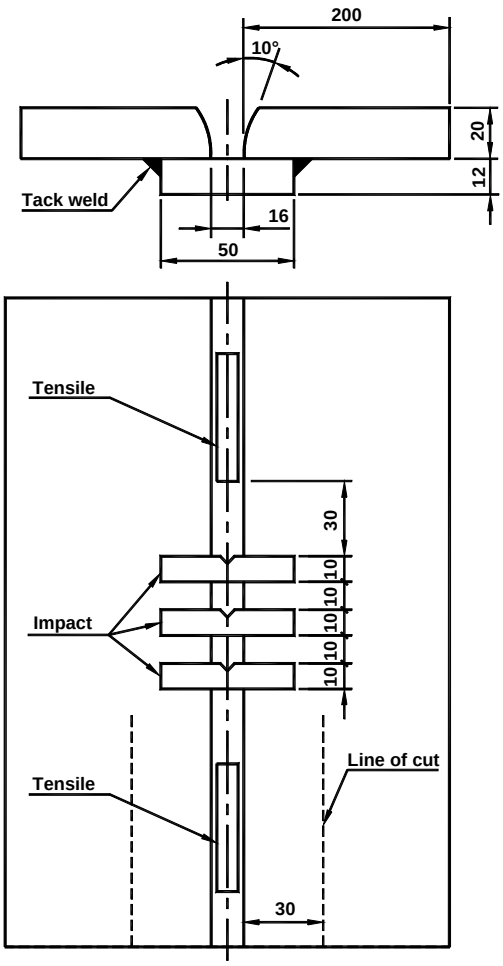
4.3 Deposited metal tests

4.3.1 An all weld metal test assembly is to be prepared in the downhand position as shown in Fig.4.3.1, using any grade of hull structural steel.

4.3.2 The bevelling of the plate edges is to be carried out by machining or mechanized gas cutting. In the latter case any remaining scale is to be removed from the bevelled edges.

4.3.3 The direction of deposition of each run is to alternate from each end of the plate and after completion of each run the flux and welding slag is to be removed. Between each run the assembly is to be left in still air until it has cooled to 250°C but not below 100°C, the temperature being taken in the centre of the weld, on the surface of the seam. The thickness of the layer is to be not less than the diameter of the wire but not less than 4 [mm].

4.3.4 The welding conditions (amperage, voltage and rate of travel) are to be in accordance with the recommendations of the manufacturer and are to conform with normal good welding practice for multi-run welding.



All dimensions in mm unless otherwise indicated

Fig. 4.3.1 : Deposited metal test assembly

4.3.5 The welded assembly is to be cut longitudinally at a distance of 30 [mm] from the edges of the weld and then cut transversely.

4.3.6 Two longitudinal tensile and three impact test specimens are to be taken from each test assembly as shown in Fig.4.3.1. Care is to be taken that the axes of the tensile test specimens coincide with the centre of the weld and the midthickness of the plates. The impact test specimens are to be cut perpendicular to the weld with their axes 10 [mm] from the upper surface. The notch is to be positioned in the centre of the weld and cut in the face of the test specimen perpendicular to the surface of the plate.

4.3.7 The results of all tests are to comply with requirements of Table 4.3.1 as appropriate. The chemical analysis of the deposited weld metal including the content of the significant alloying elements is to be submitted by the manufacturer.

Table 4.3.1 : Requirements for deposited metal tests (wire-flux combinations)

Grade	Yield stress [N/mm^2]	Tensile strength [N/mm^2]	Elongation on 50 mm gauge length % min.	Charpy V-notch impact	
				Test temp. °C	Avg. energy - J min.
1 2 3	305	400 - 560	22	20 0 -20	34
1Y 2Y 3Y 4Y	375	490 - 660	22	20 0 -20 -40	34
2Y40 3Y40 4Y40 5Y40	400	510 - 690	22	0 -20 -40 -60	39

4.4 Butt weld test (two-run technique)

4.4.1 Two welded assemblies for each grade of wire-flux combination are to be prepared in accordance with Fig.4.4.1, using the following plate thicknesses:-

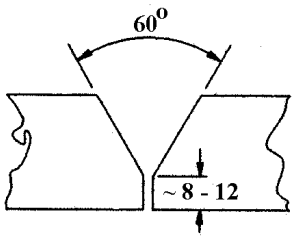
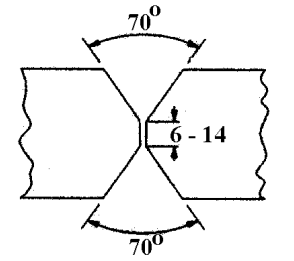
For Grades 1 and 1Y	12 to 15 [mm] and 20 to 25 [mm]
For Grades 2, 2Y, 3, 3Y and 4Y	20 to 25 [mm] and 30 to 35 [mm]
For Grades 2Y40, 3Y40, 4Y40 and 5Y40	20 to 25 [mm] and 30 to 35 [mm]

A limitation of the approval to the medium range (upto the maximum welded plate thickness) may be agreed in which case the test assemblies are to be welded using plates of 12 to 15 [mm] and 20 to 25 [mm] irrespective of the grade for which the approval is requested.

Where approval is requested for welding of both normal strength and higher tensile steel, two assemblies are to be prepared using higher tensile steel.

4.4.2 The maximum diameter of wire, grades of steel plate and edge preparation to be used are to be in accordance with Table 4.4.2. Small deviations in the edge preparation may be allowed if requested by the manufacturer. The bevelling of the plate edges is to be performed by machining or mechanized gas cutting. In the latter case any remaining scale is to be removed from the bevelled edges. The root gap should not exceed 1.0 [mm].

Table 4.4.2 : Butt weld test assemblies (two-run technique)

Plate thick-ness [mm]	Recommended preparation [mm]	Max. diameter of wire [mm]	Grade of wire-flux combi-nation	Grade of normal strength steel	Grade of higher strength steel
About 12 -15		5	1	A	-
			1Y	-	AH32 AH36
About 20 - 25		6	1	A	-
			1Y	-	AH32, AH36
			2	A,B or D	-
			2Y	-	AH32, AH36, DH32, DH36
			2Y40	-	AH40, DH40
			3	A,B,D or E	-
			3Y	-	AH32, AH36, DH32 DH36, EH32, EH36
			3Y40	-	AH40, DH40, EH40
			4Y	-	AH32, AH36, DH32, DH36 EH32, EH36, FH32, FH36
			4Y40	-	AH40, DH40, EH40, FH40
			5Y40	-	AH40, DH40, EH40, FH40
About 30 - 35		7	2	A, B or D	-
			2Y	-	AH32, AH36, DH32, DH36
			2Y40	-	AH40, DH40
			3	A, B, D or E	-
			3Y	-	H32, AH36, DH32, DH36, EH32, EH36
			3Y40	-	AH40, DH40, EH40
			4Y	-	AH32, AH36, DH32, DH36, EH32, EH36, FH32, FH36
			4Y40	-	AH40, DH40, EH40, FH40
			5Y40	-	AH40, DH40, EH40, FH40

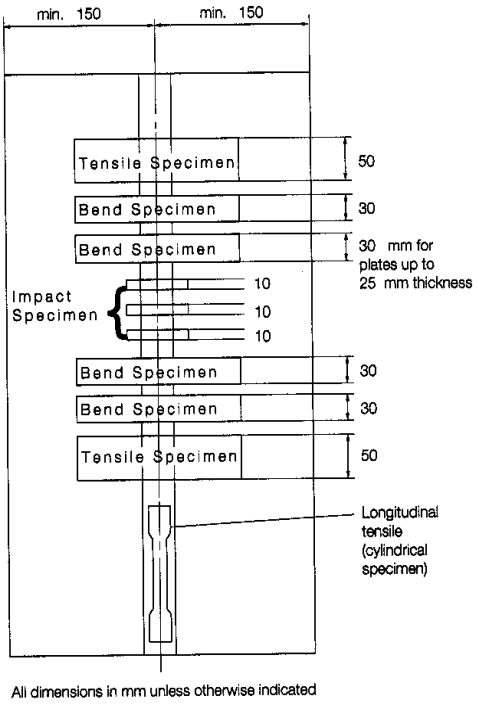


Fig.4.4.1 : Butt weld test assembly (two-run technique)

4.4.3 The welding current may be either a.c. or d.c. (electrode positive or negative) according to the recommendation of the manufacturer. If both a.c. and d.c. are recommended a.c. is to be used for test pieces.

4.4.4 Each butt weld is to be welded in two runs, one from each side, using amperages, voltages and travel speeds in accordance with the recommendations of the manufacturer and normal good welding practice.

4.4.5 After completion of the first run, the flux and welding slag are to be removed and the assembly is to be left in still air until it has cooled to 100°C, the temperature being taken in the centre of the weld, on the surface of the seam. After being welded the test assemblies are not to be subjected to any heat treatment.

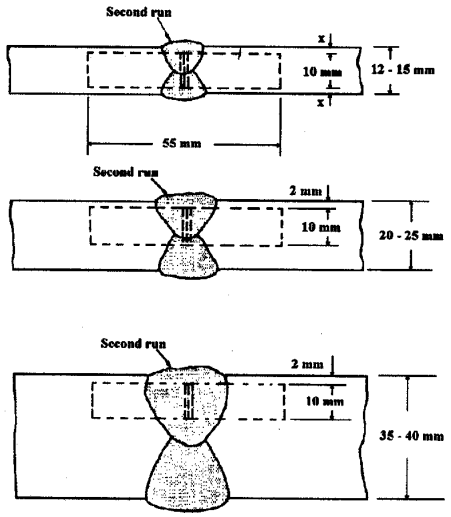


Fig. 4.4.3 : Butt weld test assemblies - two-run technique : position of impact test specimens

4.4.6 It is recommended that welded assemblies be subjected to radiographic examination to ascertain any defects in the weld prior to testing.

4.4.7 The assemblies are to be cut transversely, to form two tensile test pieces and two bend test pieces as shown in Fig.4.4.1, three impact test pieces as shown in Fig.4.4.1 and Fig.4.4.3. The edges of all test pieces and also the discards are to be examined to ensure complete fusion and interpenetration of welds.

4.4.8 Where the wire-flux combination is to be used for two-run technique only, a longitudinal test is also to be made in accordance with Fig.4.5.1 on the thicker plate tested.

4.4.9 The results of the transverse tensile and impact tests are to comply with the requirements of Table 4.5.1 as appropriate. The results of longitudinal tensile test are to comply with the requirements of Table 4.3.1 as appropriate except that for Grades 1Y, 2Y and 3Y the tensile strength is not to be less than 490 [N/mm^2].

4.5 Butt weld test (multi-run technique)

4.5.1 A butt weld assembly, as shown in Fig.4.5.1, is to be prepared in the downhand position by welding together two 20 [mm] thick plates of not less than 150 [mm] in width and of sufficient length to allow the cutting out of test specimens of the prescribed number and size.

4.5.2 The grade of steel used for the preparation of the test assembly is to be as follows:-

Grade 1 wire-flux combination	Grade A
Grade 2 wire-flux combinations	Grade A, B, D
Grade 3 wire-flux combinations	Grade A, B, D, E
Grade 1Y wire-flux combinations	Grade AH32, AH36
Grade 2Y wire-flux combinations	AH32, AH36, DH32, DH36
Grade 3Y wire-flux combinations	AH32, AH36, DH32, DH36, EH32, EH36
Grade 4Y wire-flux combinations	AH32, AH36, DH32, DH36, EH32, EH36, FH32, FH36
Grade 2Y40 wire-flux combinations	AH40, DH40
Grade 3Y40 wire-flux combinations	AH40, DH40, EH40
Grade 4Y40 wire-flux combinations	AH40, DH40, EH40, FH40
Grade 5Y40 wire-flux combinations	AH40, DH40, EH40, FH40

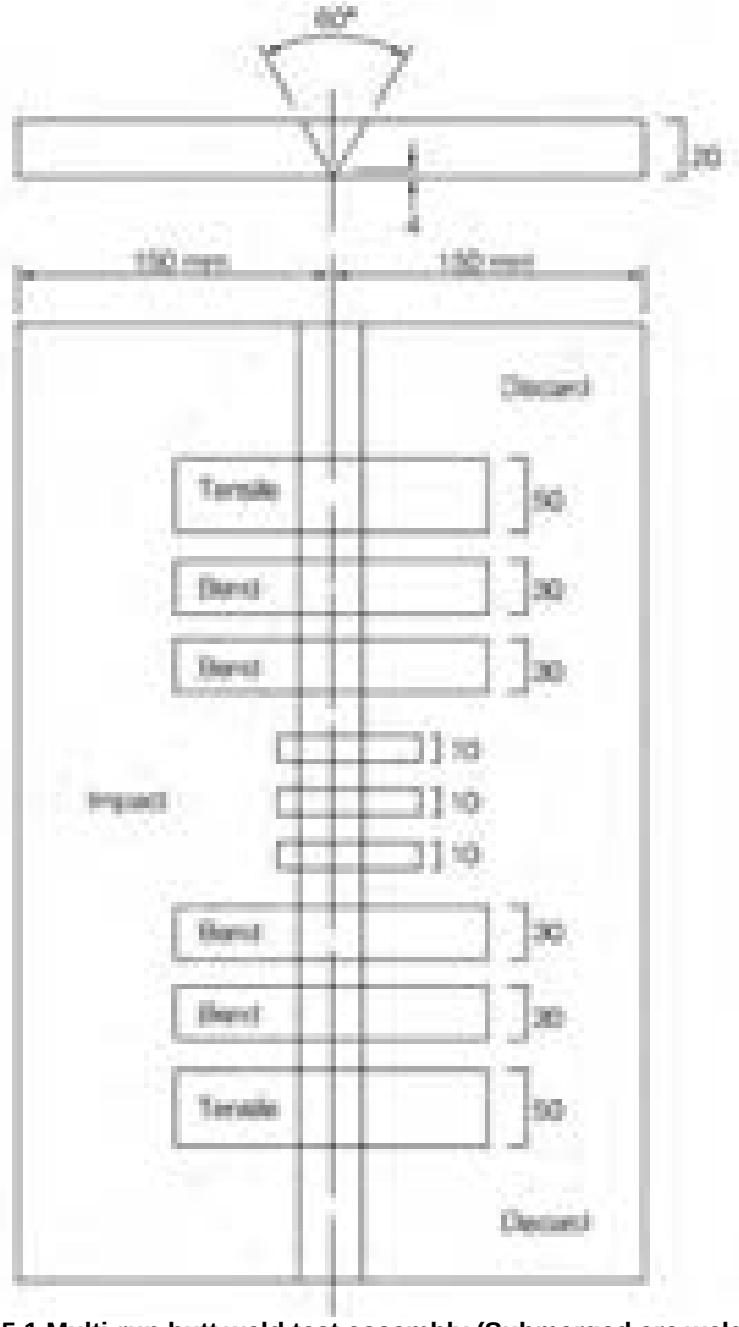


Fig 4.5.1 Multi-run butt weld test assembly (Submerged arc welding)

4.5.3 Welding is to be carried out in the downhand position, and the direction of deposition of each run is to alternate from each end of the plate. After completion of each run, the flux and welding slag is to be removed. Between each run the assembly is to be left in still air until it has cooled to less than 250°C but not below 100°C, the temperature being taken in the centre of the weld, on the surface of the seam. The thickness of the layer is to be not less than the diameter of the wire nor less than 4 [mm].

4.5.4 The plate edges are to be prepared to form a single vee joint, the included angle between the fusion faces being 60 degrees and the root face being 4 [mm]. The bevelling of the plate edges is to be carried out by machining or mechanized gas cutting. In the latter case, any remaining scale is to be removed from the bevelled edges.

4.5.5 The welding is to be carried out by the multi-run technique and the welding conditions are to be the same as those adopted for the deposited metal test assembly.

4.5.6 The back sealing run is to be applied in the downhand position after cutting out the root run to clean metal. After being welded the test assembly is not to be subjected to any heat treatment.

4.5.7 It is recommended that the welded assembly be subjected to radiographic examination to ascertain any defects in the weld prior to testing.

4.5.8 The test assembly is to be cut to form two tensile; two face bend; two root bend; three impact test pieces as shown in Fig.4.5.1.

4.5.9 The results of all tensile and impact test specimens are to comply with the requirements of Table 4.5.1 as appropriate. The position of the fracture of the transverse tensile test is to be reported.

Table 4.5.1 : Requirements for butt weld tests (wire flux-combination)

Grade	Tensile strength (transverse test) [N/mm^2] min.	Charpy V-notch impact test	
		Test temp. °C	Avg. energy J min. (See note)
1 2 3	400	20 0 -20	34
1Y 2Y 3Y 4Y	490	20 0 -20 -40	34
2Y40 3Y40 4Y40 5Y40	510	0 -20 -40 -60	39
Note : No individual impact test value is to be less than 23J			

4.6 Annual tests

4.6.1 Following tests on wire-flux combinations are to be carried out at the time of annual inspection:-

a) For two-run technique:- On butt weld assembly with 20 [mm] minimum plate thickness : One transverse tensile, two transverse bends and three impact tests. One longitudinal tensile test specimen is also to be prepared where the wire-flux combination is approved solely for the two-run technique.

b) For multi-run technique :- Deposited metal Tests - One tensile and three impact tests in accordance with 4.3.

4.6.2 The specimens are to be prepared and tested in accordance with, and on grades of steel specified for initial approval tests and the results are to comply with the results of the approved grade.

4.7 Upgrading and uprating

4.7.1 Requests for upgrading and uprating will generally be considered at the time of annual testing and additional tests in accordance with the requirements of 2.8 would be required.

Section 5

Wires and Wire-gas Combinations for Semi-automatic and Automatic Welding

5.1 General

5.1.1 Wire-gas combinations and flux-cored or flux-coated wires (for use with or without a shielding gas) are divided into following categories for the purposes of approval testing:-

a) For use in semi-automatic multi-run welding;

b) For use in single electrode multi-run automatic welding; and

c) For use in single electrode two-run automatic welding.

5.1.2 The term 'semi-automatic' is used to describe processes in which the weld is made manually by a welder holding a gun through which the wire is continuously fed. A suffix S will be added after the grade mark to indicate approval for semi-automatic multi-run welding.

5.1.3 Dependent on the results of impact tests, wires and wire-gas combinations are divided into the following grades:-

For normal strength steel	Grades 1, 2 and 3
For higher strength steel with minimum yield strength upto 355 [N/mm^2]	Grades 1Y, 2Y, 3Y and 4Y.
For higher strength steels with minimum yield strength upto 390 [N/mm^2]	Grades 2Y40, 3Y40, 4Y40 and 5Y40

5.1.4 For wires intended for automatic welding, the suffixes T, M or TM will be added after the grade mark to indicate approval for two-run, multi- run or both welding techniques, respectively.

5.1.5 For wires intended for both semi-automatic and automatic welding, the suffixes will be added in combination.

5.1.6 Where applicable, the composition of the shielding gas is to be reported. Unless otherwise agreed, additional approval tests are required when the shielding gas is different from that used for the original approval tests.

Where a wire in combination with any particular gas has been approved, usage of the same wire with another gas in the same group as defined in Table 5.1.6 may be considered.

5.1.7 Flux-cored or flux-coated wires which have satisfied the requirements for Grades 2, 2Y, 2Y40, 3, 3Y, 3Y40, 4Y, 4Y40 and 5Y40 may, at the option of the manufacturer, be submitted to the hydrogen test as detailed in 2.5 using the manufacturer's recommended welding condi-tions and adjusting the deposition rate to give a weight of weld deposit per sample similar to that deposited when using manual electrodes. A suffix H15, H10 or H5 will be added to the grade mark, in the same conditions as for manual arc welding electrodes to indicate compliance with the requirements of the test.

Table 5.1.6 : Compositional limits of designated groups of gas types and mixtures

Group		Gas Composition (Vol. %)			
		CO_2	O_2	H_2	Ar
M1	1	> 0 to 5	-	> 0 to 5	Rest 1) 2)
	2	> 0 to 5	-	-	
	3	-	> 0 to 3	-	
	4	> 0 to 5	> 0 to 3	-	
M1	1	> 5 to 25	-	-	Rest 1) 2)
	2	-	3 to 10	-	
	3	> 5 to 25	> 0 to 3	-	
M1	1	> 25 to 50	-	-	Rest 1) 2)
	2	-	> 10 to 15	-	
	3	> 5 to 50	> 8 to 15	-	
C	1	100	-	-	Rest 1) 2)
	2	Rest	> 0 to 30	-	

1) Argon may be substituted by Helium upto 95% of the Argon content.

2) Approval covers gas mixtures with equal or higher Helium contents only.

5.2 Approval tests for two-run automatic welding

5.2.1 Approval tests for two-run automatic welding are to be carried out generally in accordance with the requirements of Sec.4 using the two-run automatic welding technique for the preparation of all test assemblies.

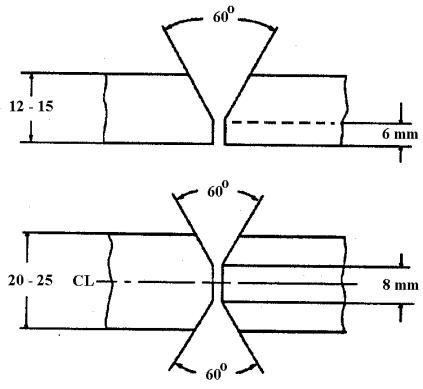


Fig.5.2.1 : Recommended edge preparation for two-run butt weld test assemblies

5.2.2 Two butt weld test assemblies are to be prepared generally as detailed in 4.4.1 and 4.4.2 using plates 12-15 [mm] and 20-25 [mm] in thickness.

5.2.3 If approval is requested for welding plates thicker than 25 [mm], one assembly is to be prepared using plates approximately 20 [mm] in thickness and the other using plates of maximum thickness for which approval is requested.

5.2.4 The edge preparation of test assemblies is to be as shown in Fig.5.2.1. Small deviations in the edge preparation may be allowed, if requested by the manufacturer. For assemblies using plates over 25 [mm] in thickness, the edge preparation is to be reported for information.

5.3 Approval tests for semi-automatic multi-run welding

5.3.1 Approval tests for semi-automatic multi-run welding are to be carried out generally in accordance with the requirements of Sec.2, using the semi-automatic multi-run technique for the preparation of all test assemblies.

5.3.2 Two deposited metal test assemblies are to be prepared in the downhand position as shown in Fig.2.2.1, one using the smallest diameter, and the other using the largest diameter of the wire intended for the welding of ship structures. The weld metal is to be deposited according to the practice recommended by the manufacturer, and the thickness of each layer of weld metal is to be between 2 [mm] and 6 [mm]. Where only one diameter is manufactured, only one deposited metal assembly is to be prepared.

5.3.3 Butt weld assemblies as shown in Fig.2.3.1 are to be prepared for each welding position (downhand, horizontal-vertical, vertical-upwards, vertical-downwards and overhead) for which the wire is recommended by the manufacturer.

5.3.4 The downhand assembly is to be welded using, for the first run, wire of 1.2 [mm] diameter or of the smallest diameter manufactured and, for the remaining runs, wire of 2.4 [mm] diameter or the largest diameter manufactured.

5.3.5 Where approval is requested only in the downhand position, an additional butt weld assembly is to be prepared in that position using wires of different diameter from those required by 5.3.4.

5.3.6 The butt weld assemblies, in positions other than downhand, are to be welded using for the first run, wire of 1.2 [mm] diameter or of the smallest diameter manufactured, and for the remaining runs, the largest diameter of wire recommended by the manufacturer for the position concerned.

5.3.7 Fillet weld test in accordance with Sec.2 is to be carried out.

5.4 Approval tests for multi-run automatic welding

5.4.1 Approval tests for multi-run automatic welding are to be carried out generally in accordance with the requirements of Sec.4 using the multi-run automatic welding technique for the preparation of all test assemblies.

5.4.2 One deposited metal test assembly is to be prepared as shown in Fig.4.3.1. Welding is to be as detailed in Sec.4 except that thickness of each layer is to be not less than 3 [mm].

5.4.3 A butt weld assembly is to be prepared, as shown in Fig.4.5.1.

5.5 Annual tests

5.5.1 The annual tests are to consist of at least the following:-

a) Wires approved for semi-automatic or for both semi-automatic and automatic multi-run welding: One deposited metal test assembly prepared in accordance with 5.3 using a wire of diameter within the range intended for the welding of the ship structures;

b) Wires approved for automatic multi-run welding: One deposited metal test assembly prepared in accordance with 5.4 using a wire of diameter within the range intended for the welding of the ship structure;

c) Wires approved for two-run automatic welding: One butt weld test assembly prepared in accordance with 5.2 using plates 20 to 25 [mm] in thickness. The diameter of the wire used is to be reported.

5.5.2 From the test assemblies prepared in accordance with 5.5.1, only the following tests are to be carried out:-

a) For deposited metal assemblies: One tensile and three impact tests;

b) For butt weld assemblies: One transverse tensile, two bend and three impact tests. One longitudinal tensile test is also required where the wire is approved solely for two-run automatic welding.

5.6 Upgrading and uprating

5.6.1 Requests for upgrading and uprating will generally be considered at the time of annual testing and additional tests in accordance with the requirements of 2.8 would be required.

Section 6

Consumables for use in Electro-slag and Electro-gas Vertical Welding

6.1 General

6.1.1 The requirements for the two-run technique as detailed in Sec.4 are applicable for the approval of special consumable used in electro-slag and electro-gas vertical welding with or without consumable nozzles except as otherwise required by the following requirements especially as regards the number and kind of the test-pieces used for the mechanical tests and taken from the butt welded assemblies.

6.1.2 For Grades 1Y, 2Y, 3Y, 4Y, 2Y40, 3Y40, 4Y40 and 5Y40 approval of the consumables may be restricted for use only with specific types of higher tensile steel. This is in respect of the content of grain refining elements, and if general approval is required, a niobium treated steel is to be used for the approval tests.

6.1.3 For these special welding consumables, the requirements of 1.3 may not be entirely applicable for technical reasons.

Where approval is requested for welding of both normal strength and higher tensile steel two assemblies are to be prepared using higher tensile steel. Two assemblies prepared using normal strength steel may also be required at the discretion of IRS.

6.2 Butt weld tests

6.2.1 Preparation of test assemblies

- Two butt weld test assemblies are to be prepared, one of them with plates 20/25 [mm] thick, the other with plates 35/40 [mm] thick or more. The grade of the steel to be used for each one of these assemblies must be selected according to the requirements given in the Table 4.4.2.

- The chemical composition of the plate, including the content of grain refining elements is to be reported.

- The welding conditions and the edges preparation are to be those recommended by the welding consumable manufacturer and are to be reported.

6.2.2 Radiographic examination

It is recommended that the welded assemblies be subjected to a radiographic examination to ascertain if there are any defects in the weld prior to the preparation of test specimens.

6.2.3 Test series

- Each assembly shall be cut to give test specimens according to Fig.6.2.1.

The length of the assembly should be sufficient to allow the selection of all the following test specimens :

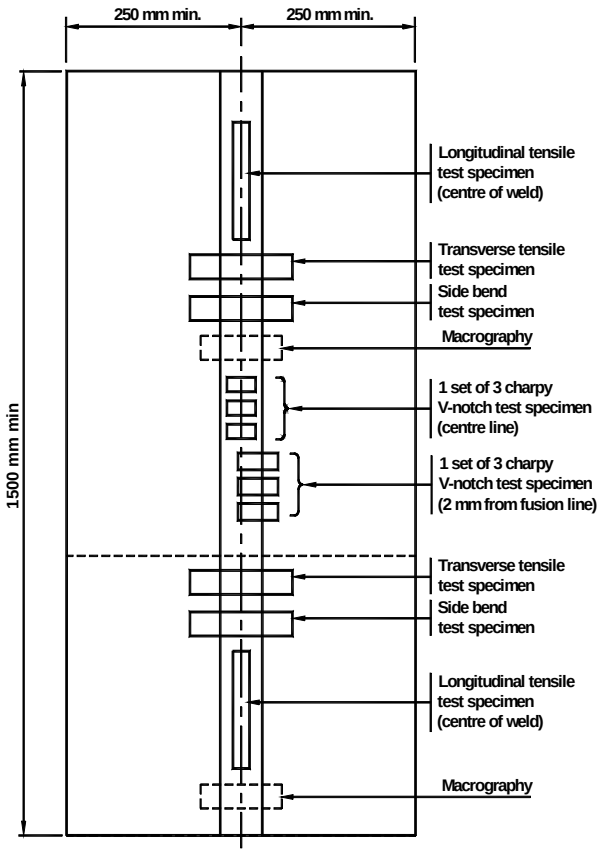


Fig. 6.2.1 : Butt weld test assembly

- 2 longitudinal tensile test specimens with their axis at the centre of the weld;

- 2 transverse tensile test specimens;
- 2 side bend test specimens;
- 2 sets of 3 Charpy-V notch impact test specimens in accordance with Fig.6.2.1 comprising of :
- 1 set with the notch in the axes of the weld;
- 1 set with the notch at 2 [mm] from the fusion line in the deposited metal; and
- 2 macro-sections of the weld (towards the middle of the weld and towards one end).

6.2.4 Results to be obtained

The results of the tensile, bend and impact tests are to comply with the requirements of 4.4 (two-run welding) for the class of filler product in question.

6.3 Annual tests

6.3.1 One test assembly must be prepared from plates 20/25 [mm] thick, and tested as indicated in 6.2.

The following specimens are to be selected :

- 1 longitudinal tensile specimen from the axis of the weld:
- 1 transverse tensile specimen;
- 2 side bend specimens;
- 3 Charpy-V specimens notched at the centre of the weld (position 1 Fig.6.3.1);
- 3 Charpy-V specimens cut out transverse to the weld with their notches at 2 [mm] from the fusion line, in the weld; and
- macro section.

6.3.2 The results to be obtained should meet the requirements given in 4.4 (two-run welding) for the class of the consumables in question.

6.4 Upgrading and uprating

6.4.1 Upgrading and uprating will be considered only at the manufacturers request, preferably at the time of annual testing. Generally, for this purpose, full tests from butt weld assemblies as indicated in 6.2 will be required, irrespective of the other tests requested if the concerned consumable is also approved (and possibly upgraded or uprated) according to Sec.4 or Sec.5.

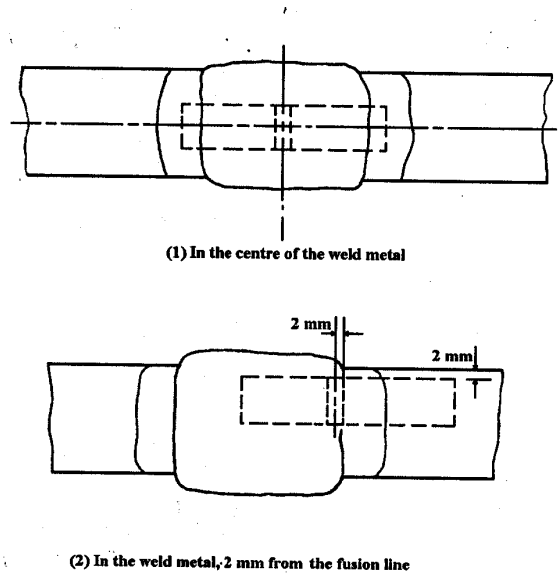


Fig.6.3.1 : Position of Charpy V-notch impact test specimens

Section 7

Welding Consumables for High Strength Steels for Welded Structures

7.1 General

7.1.1 Scope

7.1.1.1 These requirements supplement the requirements of Sections 1 to 6 and give the conditions of approval and inspection of welding consumables used for high strength steels for welded structures according to Ch.3, Sec.4 with yield strength levels from 420 [N/mm^2] upto 960 [N/mm^2] and impact grades AH, DH, EH and FH, except impact grade FH is not applicable for 890 [N/mm^2] and 960 [N/mm^2] yield strength levels.

Where no special requirements are given, those of Sections 1 to 6 apply in analogous manner.

7.1.1.2 The welding consumables preferably to be used for the steels concerned are divided into several categories as follows:

- covered electrodes for manual welding,
- wire-flux combinations for multirun sub-merged arc welding,
- solid wire-gas combinations for arc welding (including rods for gas tungsten arc welding),
- flux cored wire with or without gas for arc welding.

7.1.2 Grading, Designation

7.1.2.1 Based on the yield strength of the weld metal, the welding consumables concerned are divided into eight (yield) strength groups:

Y42 for welding steels with minimum yield strength 420 [N/mm^2]

Y46 for welding steels with minimum yield strength 460 [N/mm^2]

Y50 for welding steels with minimum yield strength 500 [N/mm^2]

Y55 for welding steels with minimum yield strength 550 [N/mm^2]

Y62 for welding steels with minimum yield strength 620 [N/mm^2]

Y69 for welding steels with minimum yield strength 690 [N/mm^2]

Y89 for welding steels with minimum yield strength 890 [N/mm^2]

Y96 for welding steels with minimum yield strength 960 [N/mm^2].

Wire-flux combinations for single or two-run technique are subject to special consideration of IRS.

7.1.2.2 Each of the eight (yield) strength groups is further divided into three main grades in respect of charpy V-notch impact test requirements (test temperatures):

Grade	Test temperature
3	- 20°C
4	- 40°C
5	- 60°C

7.1.2.3 Analogously to the designation scheme used in Section 1 to 6 the welding consumables for high strength steels are subject to additional designation and approval as follows:

- According to 7.1.2.2 with the quality grades 3, 4 or 5.
- With the added symbol Y and an appended code number designating the minimum yield strength of the weld metal corresponding to 7.1.2.1 : Y42, Y46, Y50, Y55, Y62, Y69, Y89 and Y96.
- With the added symbol H10 or H5 for controlled hydrogen content of the weld metal.
- With the added symbol S (= semi-automatic) for semi-mechanised welding.
- With the added symbol M designating multirun technique and is applicable only

to welding consumables for fully mechanised welding).

7.1.2.4 Each higher quality grade includes the one (or those) below, AH, DH steels according to Ch.3, Sec.4 are to be welded using welding consumables of at least quality grade, 3, grade EH steels using at least quality grade 4 and grade FH steels using at least quality grade 5, as per the following table:

Consumables Grade	Steel Grades covered
3Y..	DH..
4Y..	EH..and FH..
5Y..	FH.., EH.. and DH..

7.1.2.5 Welding consumables approved with grades.Y42, ..Y46 and ..Y50 are also considered suitable for welding steels in the two strength levels below that for which they have been approved. Welding consumables approved with grades ..Y55, ..Y62 and ..Y69 are also considered suitable for welding steels in the one strength level below that for which they have been approved. Welding consumables with grade Y89 are considered suitable for welding steels in the same strength level only. Welding consumables with grade Y96 are also considered suitable for welding steels in the one strength level below that for which they have been approved. For grade Y89 and Y96, where the design requirements permit undermatching weld joint, then welding consumables within the scope of this section can be considered subject to IRS's discretion and Manufacturer's recommendations.

7.1.2.6 IRS may, in individual cases, restrict the range of application in (up to) such a way, that approval for any one strength level does not justify approval for any other strength level.

7.1.3 Manufacture, testing and approval procedure

7.1.3.1 Manufacturer's plant, production methods and quality control measures shall be such as to ensure reasonable uniformity in manufacture, see also Sec.1.

7.1.3.2 Testing and approval procedure shall be in accordance with Sec.1 and as required in Section 1 to 6 for the individual categories (types) of welding consumables mentioned in 7.1.1.2 above.

7.2 Testing of the weld metal

7.2.1 For testing the deposited weld metal, test pieces analogous to those called for in Sections 1 to 6 respectively shall be prepared, depending on the type of the welding consumables (and according to the welding process). The base metal used shall be a fine-gained structural steel compatible with the properties of the weld metal, or the side walls of the weld shall be buttered with a weld metal of the same composition.

7.2.2 The chemical composition of the deposited weld metal shall be determined and certified in a manner analogous to that prescribed in Sec.2, Cl.2.2.4. The results of the analysis shall not exceed the limit values specified in the standards or by the manufacturer, the narrower tolerances being applicable in each case.

7.2.3 Depending on the type of the welding consumables (and according to the welding process), the test specimens prescribed in Sections 1 to 6 respectively shall be taken from the weld metal test pieces in a similar manner.

7.2.4 The mechanical properties must meet the requirements stated in Table 7.2.1 and Table 7.2.2. The provisions of Sections 1 to 6 apply in analogous manner to the performance of the tests, including in particular the maintenance of the test temperature in the notched bar impact test and the carrying out of results.

7.2.5 Specifications of welding consumables used for welding high strength extremely thick steel plates of thickness more than 50 [mm] but not exceeding 100[mm] of EH47 grade used in container carriers are to be in accordance with Table 7.2.3

7.2.6 Welding consumables for brittle crack arrest steels are to be in accordance with the relevant requirements for each steel grade excluding suffix “BCA1” or “BCA2” specified in Table 10.1.3 of Chapter 3, Section 10.

7.3 Testing on welded joints

7.3.1 Depending on the type of the welding consumables (and according to the welding process), the testing on the welded joints shall be performed on butt-weld test pieces in a manner analogous to that called for in Sections 1 to 6.

7.3.2 Depending on the type of the welding consumables (and according to the welding process), the butt-weld test pieces called for in para 7.3.1 shall be welded in a manner analogous to that prescribed in Sections 1 to 6. The base metal used shall be a high-strength fine-grained structural steel with a minimum yield strength and tensile strength matching the consumable grade being approved and compatible with the added symbol for which application is made.

7.3.3 Depending on the type of the welding consumables (and according to the welding process), the test specimens described in Sections 1 to 6 shall be taken from the butt-weld test pieces.

7.3.4 The mechanical properties must meet the requirements stated in Table 7.3.1. The provisions of Sections 1 to 6 apply in analogous manner to the performance of the tests, including in particular the maintenance of the test temperatures in the notched bar impact test and the requirements regarding the retest specimens.

Table 7.2.1 : Required toughness properties of the weld metal

Quality Grade	Test temp.°C	Min. notch impact energy [J][1)]
3	- 20	Y42: ≥ 47 Y46: ≥ 47 Y50: ≥ 50 Y55: ≥ 55 Y62: ≥ 62 Y69: ≥ 69 Y89: ≥ 69[2)] Y96: ≥ 69[2)]
4	- 40	
5	- 60	

1) Charpy V-notch impact test specimen, mean value of three specimens; for requirements regarding minimum individual values and retests, See Section 1, 1.10
2) Quality grade 5 is not applicable for Y89 and Y96 grade consumables.

Table 7.2.2 : Required strength properties of the weld metal

Symbols added to quality grade	Min. yield strength or 0.2% proof stress [N/mm^2]	Tensile Strength [N/mm^2]	Minimum elongation [%]
Y42	420	520 - 680	20
Y46	460	540 - 720	20
Y50	500	590 - 770	18
Y55	550	640 – 820	18
Y62	620	700 - 890	18
Y69	690	770 - 940	17
Y89	890	940 - 1100	14
Y96	960	980 - 1150	13

Table 7.2.3 : Required strength properties for deposited metal used to weld high strength extremely thick steel plates of thickness more than 50[mm] but not exceeding 100[mm], of EH47 grade used in container carriers,				
Mechanical Properties			Impact Test	
Yield Strength [N/mm^2] min.	Tensile Strength [N/mm^2]	Elongation (%) min	Test Temp. [^{o}C]	Average Impact Energy [J] min.
460	570 - 720	19	-20	64

7.3.5 Where the bending angle required in Table 7.3.1 is not achieved, the specimen may be considered as fulfilling the requirements, if the bending elongation on a gauge length Lo fulfills the minimum elongation requirements stated in Table 7.2.2. The gauge length Lo = Ls + t (Ls = width of weld, t = specimen thickness), see Fig.7.3.1.

7.3.6 Mechanical Properties for Butt weld tests for high strength extremely thick steel plates of thickness more than 50[mm] but not exceeding 100[mm], of EH47 grade used in container carriers are to be as per Table 7.3.2

Table 7.3.1 : Required properties of welded joints					
Quality Grade	**Added symbol**	**Min. tensile strength [N/mm^2]**	**Min. notch impact energy, test temperature**	**Minimum bending angle 1)**	**Bend ratio D/t 2)**
3 to 5 in accordance with Table 7.2.1	Y42 Y46 Y50 Y55 Y62 Y69 Y89 Y96	520 540 590 640 700 770 940 980	Depending on the quality grade and yield strength in accordance with Table 7.2.1	120°	4 4 4 5 5 5 6 7

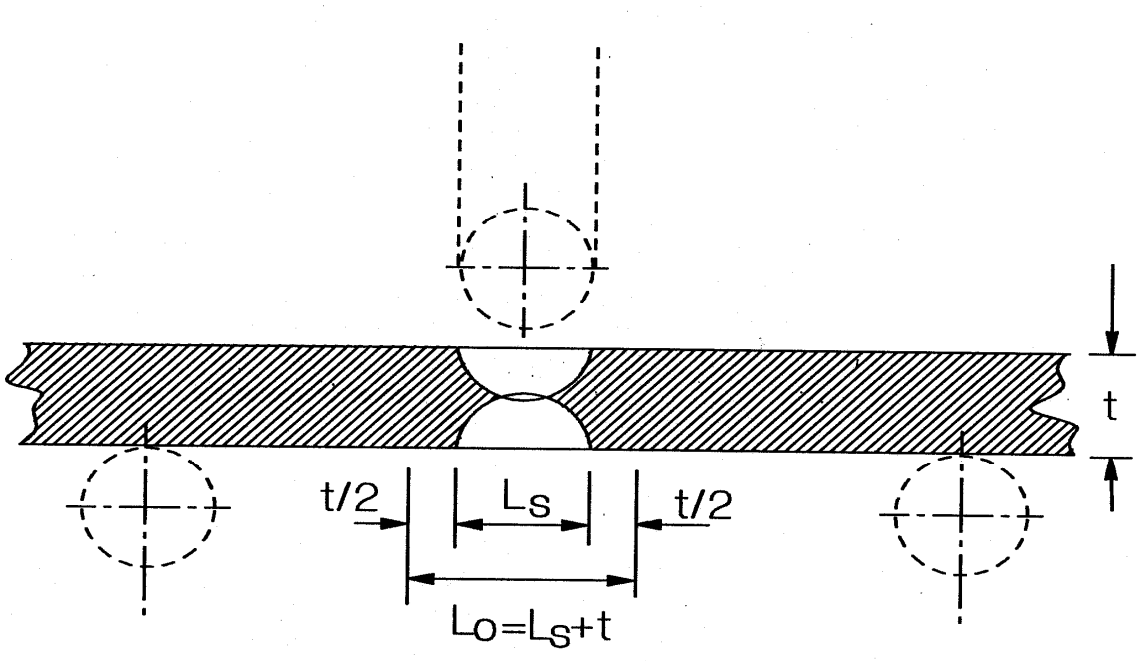


Fig.7.3.1 : Required proportion of welded joints

1) Bending angle attained before the first incipient crack, minor pore exposures upto a maximum length of 3 mm allowed.

2) D = Mandrel diameter, t = specimen thickness

Table 7.3.2 : Mechanical Properties for Butt weld tests for high strength extremely thick steel plates of thickness more than 50[mm] but not exceeding 100[mm], of EH47 grade used in container carriers

Tensile Strength [N/mm^2]	Bend Test Ratio: D/t	Charpy V-notch Impact Tests	
		Test Temperature (^oC)	Average Energy (J) min.
570-720	4	-20	64

7.4 Hydrogen test

7.4.1 The welding consumables, other than solid wire-gas combinations, shall be subjected to a hydrogen test in accordance with the mercury method to ISO 3690:2018, or any other method such as the gas chromatographic method which correlates with that method, in respect of cooling rate and delay times during preparation of the weld samples, and the hydrogen volume determinations.

7.4.2 The diffusible hydrogen content of the weld metal determined in accordance with the provisions of Sec.2, Para 2.5 shall not exceed the limits given in Table 7.4.1.

Table 7.4.1 : Allowable diffusible hydrogen content

Yield strength group	**Hydrogen symbol**	**Max. hydrogen content [cm^3/100 g deposited weld metal]**
Y42 Y46 Y50	H 10	10
Y55 Y62 Y69	H 5	5
Y89 Y96	H 5	5

7.5 Annual tests

7.5.1 The annual repeat tests specified in Sections 1 to 6 shall entail the preparation and testing of weld metal test pieces as prescribed under 7.2. For grades Y69 to Y96 annual hydrogen test is required. In special cases, IRS may require more extensive tests.

Section 8

Consumables for Welding of Aluminium Alloys

8.1 General

8.1.1 Tests for the approval of consumables intended for welding the aluminium alloys detailed in Ch.9 are to be carried out generally in accordance with the requirements of Secs.1,2 and 5, except as otherwise detailed in this Section.

8.1.2 The welding consumables are divided into two categories as follows:

W = wire electrode, wire - gas combinations for metal arc inert gas welding (MIG, 131 according to ISO 4063:2009), tungsten inert gas welding (TIG, 141) or plasma arc welding (15)

R = rod - gas combinations for tungsten inert gas arc welding (TIG, 141) or plasma arc welding (15)

8.1.3 Approval will be indicated by the grade as shown in Table 8.1.3.

Table 8.1.3 : Consumables grades and base materials for the approval test

Consumable quality grade (Symbol)	Base material for the tests	
	Alloy Designation	
	Numerical	Chem-Symbol
RA/WA	5754	AlMg3
RB/WB	5086	AlMg4
RC/WC	5083	AlMg4.5 Mn0.7
	5383	AlMg4.5 Mn0.9
	5456	AlMg5
	5059	-
RD/WD	6082	AlSi1MgMn
	6005A	AlSiMg(A)
	6061	AlMg1SiCu
Note: Approval on higher strength AlMg base materials covers also the lower strength AlMg grades and their combination with AlSi grades		

8.1.4 The welding technique will be indicated in the approval grading by a letter as under:

m - manual multi-run welding (GTAW);
S - semi-automatic multi-run welding (GMAW);
M - automatic multi-run welding (GTAW or GMAW);
T - automatic two-run welding (GMAW).

8.1.5 The compositions, of the shielding gas and the filler/electrode wire are to be reported.

8.1.6 Approval of a wire or a rod will be granted in conjunction with a specific shielding gas according to Table 8.1.6 or defined in terms of composition and purity of "special" gas to be designated with group sign "S". The composition of the shielding gas is to be reported. Where a wire in combination with any particular gas has been approved, usage of the same wire with another gas in the same group as defined in Table 8.1.6 may be considered.

Table 8.1.6 : Compositional limits of shielding gases and mixtures to be used

Group	Gas composition (Vol.%)[1])	
	Argon	**Helium**
I - 1	100	-
I - 2	-	100
I - 3	Rest	> 0 to 33
I - 4	Rest	> 33 to 66
I - 5	Rest	> 66 to 95
S	Special gas, composition to be specified, See 8.1.6	

1) Gases of other chemical composition (mixed gases) may be considered as "special gases" and covered by a separate test.

8.1.7 On completion of welding, assemblies must be allowed to cool naturally to ambient temperature. Welded test assemblies and test specimens must not be subjected to any heat treatment after welding except for the alloy Grades 6005A, 6061 and 6082. These are to be allowed to naturally age at ambient temperature for a period of 72 hours from the completion of welding, before the testing is carried out. A second solution heat treatment is not permitted. The time and temperature of any ageing treatment is to be reported in detail.

8.2 Initial approval tests for manual, semi-automatic and automatic multi-run techniques

8.2.1 Plate of the corresponding type of aluminium alloy and of appropriate thickness is to be used for the preparation of the weld test assemblies.

8.2.2 The welding current and power requirements are to be within the range recommended by the manufacturer and are to be reported.

8.2.3 Welded assemblies are to be prepared and tested in accordance with 8.3, 8.4 and 8.5.

8.3 Deposited metal test assemblies

8.3.1 One assembly is to be prepared in the downhand position as shown in Fig.8.3.1.

8.3.2 The chemical composition of the plate used for the assembly is to be compatible with the weld metal.

8.3.3 The thickness of the plate used and the length of the assembly are to be appropriate to the welding process. The plate thickness is to be not less than 12 [mm].

8.3.4 For the approval of filler wire/gas and electrode wire/gas combinations for manual or semi-automatic welding by GTAW or GMAW, one test assembly is to be welded using any size of wire within the range for which approval is sought.

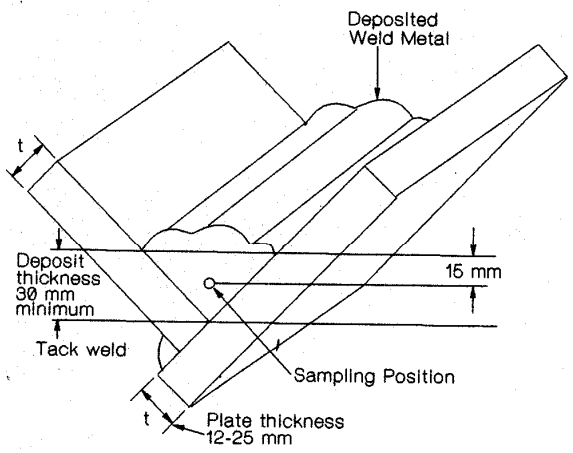


Fig.8.3.1 : Deposited metal test assembly

8.3.5 For automatic multi-run approval, one test assembly is to be welded by the respective process using the recommended diameter of wire.

8.3.6 The weld metal is to be deposited in multi-run layers in accordance with normal practice. The direction of deposition of each layer is to alternate from each end of the plate.

8.3.7 The deposited weld metal in each test assembly is to be analysed and reported including the contents of all significant elements. The elements reported will be dependent on the type of aluminium alloy for which approval of the consumables is requested. The results of the analysis are to be within the tolerances specified in the standards and by the manufacturer.

8.4 Butt weld test assemblies

8.4.1 Plate of the corresponding type of aluminium alloy and of an appropriate thickness is to be used for the preparation of the test assemblies.

Table 8.4.1 : Requirements for the transverse tensile and bend tests

Grade	Base material used for the test	Tensile strength R_m [N/mm^2] min.	Former diameter	Bending angle [1] [°] min.
RA/WA	5754	190	3t	180
RB/WB	5086	240	6t	
RC/WC	5083	275	6t	
	5383 or 5456	290		
	5059	330		
RD/WD	6061, 6005A or 6082	170	6t	

1) During testing, the test specimen shall not reveal any one single flaw greater than 3 [mm] in any direction. Flaws appearing at the corners of a test specimen shall be ignored in the evaluation, unless there is evidence that they result from lack fusion.

8.4.2 In order to ensure sound and representative welds, it is essential that test assemblies are cleaned and degreased prior to welding. Assemblies as shown in Fig.8.4.2 are to be prepared for each welding position (downhand, horizontal-vertical, vertical-upward, vertical-downward and overhead) for which the consumable is recommended by the manufacturer; except that consumables satisfying the requirements for downhand and vertical-upward positions will be considered as also complying with the requirements for the horizontal-vertical position.

Back sealing runs are allowed in single V weld assemblies. In case of double V assembly both sides shall be welded with the same welding position.

8.4.3 One additional assembly, as shown in Fig.8.4.3, is to be prepared for welding in the downhand position. The assembly is to be welded using, for the first run, wire of the smallest diameter recommended by the manufacturer and for the remaining runs, wire of the largest diameter to be approved.

8.4.4 The manufacturer's recommended procedures are to be used in making the welds and are to be reported.

8.4.5 The welded assemblies should be subjected to both radiographic and visual examination, aided where necessary by dye penetrant testing, to ensure that the welds are free from cracks and porosity.

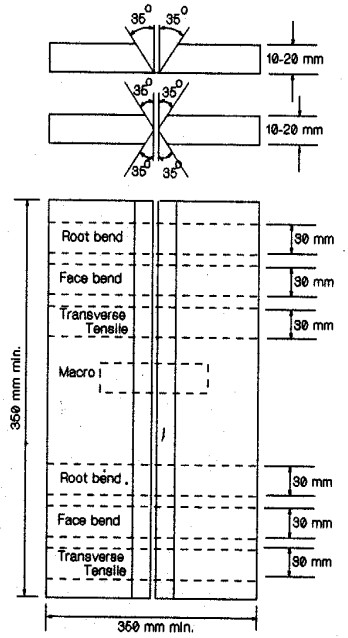


Fig.8.4.2 : Butt weld test assembly (positional welding)

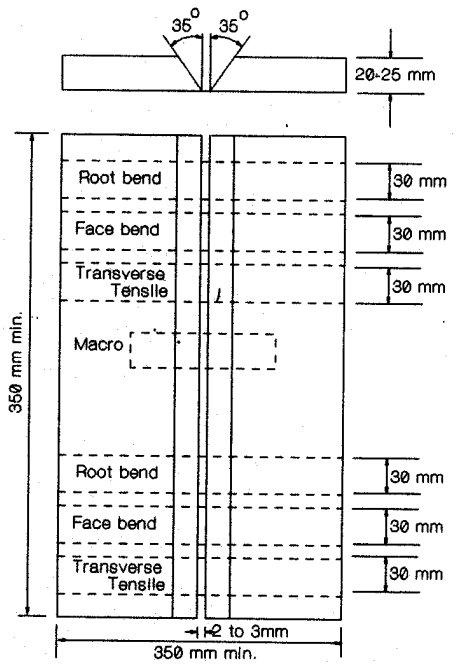


Fig.8.4.3 : Additional butt weld assembly (downhand)

8.4.6 The test specimens are to be taken from the welded assemblies as shown in Fig.8.4.2 and Fig.8.4.3. For each assembly they are to comprise:

2 transverse tensile specimens;
1 macro specimen;
2 face bend specimens; and
2 root bend specimens.

8.4.7 All tensile test specimens should have a tensile strength not less than the respective value shown in Table 8.4.1. The position of each fracture is to be reported.

8.4.8 The bend test specimens are to be bent around a former having a diameter not more than the number of times the thickness (t) of the test specimen as shown in Table 8.4.1.

8.5 Fillet weld test assemblies

8.5.1 Assemblies are to be prepared and tested in accordance with the appropriate requirements of 2.4 except that the plates are to be of the aluminium alloy for which approval is required, that no hardness tests are required and that for automatic multi-run approval only one fillet weld bead is to be made using the recommended wire diameter. In this case, the bead size should be as large as the maximum single bead size recommended by the manufacturer for fillet welding.

8.5.2 The results of examination of the macro specimens and the fractured fillet welds are to be reported in accordance with 2.4.3 and 2.4.5. Particular attention is to be given to the presence of any porosity.

8.6 Initial approval tests for two-run technique

8.6.1 Two butt weld test assemblies are to be prepared using the following plate thicknesses as shown in Fig.8.7.1:

a) one with the maximum thickness for which approval is requested;

b) one with a thickness approximately one half to two thirds that of the maximum thickness.

8.7 Annual tests

8.7.1 Annual repeat tests are to consist of preparation and testing of the deposited weld metal test assembly as prescribed in 8.3 (Fig.8.3.1) and of the downhand buttweld assembly according to 8.4 (Fig.8.4.2).

End of Chapter

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RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 3
GENERAL HULL REQUIREMENTS
VOLUME I

July 2024

Indian Register of Shipping

Part 3 : General Hull Requirements

This Part (Part 3) of the Rules is arranged in two Volumes (Volume I and Volume II). Chapters 1 to 9 are covered in this Volume. For Chapter 10 onwards, please refer Volume II)

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2 *Definitions*

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Section 1

General

1.1 Scope of application

1.1.1 In general, the Rules in this part apply to all-welded, single hull sea going steel ships of normal form, proportions and speed of Length equal to or greater than 24[m].

1.1.2 For additional class notations relating to various ship types, requirements as per Part 5 are to be complied with.

1.1.3 Ships of unconventional forms and proportions or intended for carriage of cargoes not covered by the Rules or to be engaged in special service will receive individual consideration based on the general principles of the Rules. In these cases, however, additional calculations and/or model testing may be required to be carried out and submitted for approval.

1.1.4 Proposals for use of alternative materials e.g. aluminum, wood, etc. for some parts of the ship will receive special consideration.

1.1.5 In general, these Rules are not applicable for vessels required to comply with IACS Common Structural Rules for Oil Tankers and Bulk Carriers. For ships with CSR notation, *IRS Rules for Bulk Carriers and Oil Tankers,* are applicable.

1.2 Equivalence

1.2.1 Alternative arrangements, scantlings and equipment may be accepted provided they can be shown to be equivalent to the overall safety and strength standard of the Rules. Direct calculations for the derivation of the scantlings as an alternative to those derived by the Rule formulae, may be accepted on special consideration. The calculation procedure and the assumptions made are to be submitted for approval.

1.3 National and international regulations

1.3.1 While the Rules cover requirements for the classification of ships, the attention of all concerned is drawn to requirements of various local, national or international Regulations, Codes and Recommendations which the vessel may also have to comply with.

1.4 Application

1.4.1 Rules for hull structure are provided in Part 3 and Part 5.

1.4.2 Part 3 contains the general hull requirements and includes application related to hull shape, arrangement, main dimensions, materials, etc.

1.4.3 Part 5 (Vol. I and II) covers the application and additional requirements related to special ship types and systems.

1.5 Design Basis

1.5.1 General assumptions:

This sub-section specifies the assumptions and the design parameters that are used as the basis of the design principles of the Rules:

1.5.1.1 Ships are to be designed to withstand, in the intact condition, the environmental loads anticipated during the design life, for the appropriate loading conditions. Structural strength is to be checked against buckling, yielding, fatigue and hull girder ultimate strength. Ships are to be designed to have sufficient strength to withstand the loads due to flooded conditions, as applicable.

1.5.1.2 Fatigue strength is to be assessed for being adequate for the specified fatigue life and design environments.

1.5.1.3 Vibrations in the hull structural elements are not considered in relation to the requirements for scantlings given in the rules. However, it is to be ensured by designer/shipyard that significant dynamic excitation of major orders from propellers and machinery items do not fall close to any natural frequency of the hull and that special considerations are made to avoid resonance, causing structural failures.

1.5.1.4 It is assumed that the ships will be competently handled and loaded as per the approved loading manuals.

1.5.1.5 For ships over 350[m] in length, special consideration would be made for the wave loads by IRS.

1.5.2 Transverse strength

1.5.2.1 Additional calculations are required for vessels, when total number of effective transverse bulkheads is less than that is given in Chapter 3.

1.5.3 Design life

1.5.3.1 The design life is the period that the ship is assumed to be exposed to operating and/or environmental conditions and minimum design life is taken as 25 years, unless specified otherwise.

1.5.4 Environmental conditions

1.5.4.1 The requirements are based on wave-induced loads covering the worst anticipated wave environment and related long-term sea state spectrum. The minimum environmental conditions to be used in the rules for unrestricted service are the North Atlantic wave environment according to IACS Rec.34 standard wave data.

1.5.4.2 The effects of ice are covered by Part 5, Chapter 21 (Ice Class Requirements) and Part 5, Chapter 32 (Polar Class).

1.5.5 Design temperatures

1.5.5.1 The rules assume that the structural assessment of hull strength members is valid for the following design temperatures:

- Lowest mean daily average temperature in air is above –10[°C]

- Lowest mean daily average temperature in sea water is 0[°C].

1.5.5.2 Ships intended to operate in areas with lower mean daily average temperature (below and including –10[°C]), e.g. regular service during winter season to Arctic or Antarctic waters is subject to the requirements in Chapter 2, Section 2.3.

1.5.6 Operating draughts

1.5.6.1 The design operating draughts are to be specified by the designer and to be used to derive the appropriate structural scantlings. All operational loading conditions in the loading manual are to comply with the specified design operating draughts. Following draughts are to be considered:

- Scantling draught for assessment of structure

- Maximum and minimum draughts amidships specified by the designer;

- Normal and heavy ballast draught, if applicable, for assessment of some particular structure.

1.5.7 Maximum service speed

1.5.7.1 Maximum service speed, is the speed with which the vessel is running ahead at maximum continuous rated shaft rpm and at the summer load waterline. Special consideration will be given to ships designed to operate at higher speed at a lower draft, and with full immersion of rudder. For ships with high speed and large flare (in particular L>120[m] and speed V>17[knots]), special consideration for the wave loads may be required based on the distribution of still water and vertical wave bending moments.

1.6 Design principles

1.6.1 Overall rule principles

1.6.1.1 In general, the Rules are based on the basic principles, as follows:

(a) The safety of the structure can be assessed by Limit State approach with respect to possible structural failure mode(s) when the ship is subjected to operational and environmental loads/ conditions.

(b) The design complies with the design basis, given in 1.5.

(c) Structural assessments are carried out based on consistent design load scenarios.

1.6.2 Ship structure design principles

1.6.2.1 In general, the ship's structure is to be designed such that:

(a) Ship structure has inherent redundancy. The ship's structure works in a hierarchical manner and the failure of structural elements lower down in the hierarchy do not result in immediate consequential failure of elements higher up in the hierarchy such as failure of plating will not lead to immediate consequent failure of stiffened panels.

(b) Permanent deformations are minimized. Permanent deformations of local plate panels or individual stiffened plate panels is acceptable provided that this does not affect the structural integrity, containment integrity and the performance of structural or other systems.

(c) Occurrence of in-service cracking is minimized, particularly in locations which affect the structural integrity, containment integrity and the performance of structural or other systems.

(d) Adequate structural redundancy to survive in the accidental damage in the event of flooding of any compartment.

(e) Good access conditions during construction to assure satisfactory quality of welding, coating and safe survey/ inspection.

(f) Good access conditions during the operational phase to allow adequate maintenance and safe periodical survey/ inspection to identify inspect and repair cracks by crew, IRS and relevant authorities.

1.6.3 Limit State Design Principles

1.6.3.1 Limit state design, on which the Rules are based, is a systematic approach wherein possible failure modes related to the design scenarios for each structural element is identified and evaluated. There may be one or more relevant limit states for each failure mode. The limit load for the structural element is found as the minimum limit load resulting from all the relevant limit states. The limit states are divided into four categories: serviceability limit state (SLS), ultimate limit state (ULS), fatigue limit state (FLS) and accidental limit state (ALS). The Rules include requirements for relevant limit states for the various modes of failure in ship structures and the same are indicated in Table 1.6.3.

Table 1.6.3: Failure modes in relation to the limit states to be considered				
Failure modes	**Limit States**			
	ULS	**SLS**	**ALS**	**FLS**
Yielding	Yes	Yes	Yes	-
Plastic collapse	Yes		Yes	-
Buckling	Yes	Yes	Yes	-
Rupture	Yes		Yes	-
Brittle fracture	Yes	Yes	Yes	-
Fatigue crack	-	-	-	Yes

1.7 Design load scenarios

1.7.1 The design and assessment of the ship structures is to be based on the design load scenarios encountered by the ship in its life time. A realistic set of design load scenarios corresponds to either normal operations (at sea or harbor) or accidental operations. The design load scenarios are based on operational loads and accidental loads as given below.

1.7.2 Operational Loads

1.7.2.1 The operational load scenarios include static and dynamic load components and are further subdivided into two distinct load groups

(a) Static or still water design loads scenario (S)

(i) Loads caused by the ship's loading condition, self-weight due to gravity, equipment loads, weight of cargo, buoyancy loads and mass, thermal loads etc.

(ii) Covers application of relevant static loads and typically covers load scenarios in harbour and sheltered water or tank testing.

(b) Environmental (dynamic or time varying) loads in seagoing conditions

(i) Static plus dynamic design load (S+D) covers application of relevant static loads and simultaneously occurring dynamic load components and typically cover load scenarios for seagoing operations

(ii) Fatigue or cyclic loads (F) caused by wave action, including pressure loading on side shell, inertial loads due to ship motions, and the induced shear forces and bending moments.

(iii) Impact loads (I) caused by wave impact, bottom slamming, stern slamming and liquid impact in tanks encountered during seagoing operations

(iv) Sloshing loads (SL) due to liquid sloshing in partially filled tanks encountered during seagoing operations

1.7.3 Accidental Load Scenario (A)

1.7.3.1 Accidental load scenarios include loads that do not occur during normal operations but as a result of an accident, such as flooding of compartments.

1.8 Structural capacity assessment

1.8.1 The capacity assessment method is capable of analyzing the failure mode in question to the required degree of accuracy. The possible failure modes are controlled by appropriate capacity assessment methods using relevant capacity models. The method may be different for each of the limit states even for the same failure mode, as the degree of the ultimate capacity utilization is different. The capacity models are based on a net thickness approach with corresponding scantling requirements given as net thickness values.

1.8.2 Failure modes to be assessed

1.8.2.1 The main failure modes in ship structures considered for the assessment of ship structural safety with the relation to the limit states (SLS, ULS, ALS, and FLS) are as follows:

(a) Yielding: The yielding failure mode is the mode in which plastic strain occurs locally in the structural member under combined in-plane and normal stresses. Local plastic strain is controlled in SLS, ULS and ALS models by checking the stress levels in the structural member being below the permissible value.

(b) Plastic collapse: The plastic collapse failure mode usually appears in the local structural members under large lateral impact pressure. In this failure mode, permanent lateral deflection in the local structural members will occur, but does not influence the global strength. This mode is controlled in ULS and ALS by using conventional plastic design method.

(c) Buckling: The buckling failure mode is the instability phenomena of structural members under compressive loads. Elastic (reversible) buckling occurs, when the structural members reach its elastic buckling stress. This buckling failure mode is controlled in SLS model. If the compressive load is increased, stress redistribution occurs due to buckling of the weakest structural member and the stress reaches the yield value. This buckling failure mode with large elastic deflection is controlled in ULS or ALS models. When compression is unloaded, no consequence of failure due to buckling is seen.
Plastic (irreversible) buckling occurs when the stress in structural members exceeds the yield stress. As a result, permanent deflections due to plastic buckling will occur. This irreversible buckling failure mode is controlled only in ULS or ALS models for global hull girder strength.

(d) Rupture: The rupture failure mode is the mode in which breaking occurs in the structural members under large tensile stress beyond the yield stress of the material. This failure mode is controlled in ULS or ALS models, but the assessment of this failure mode will be covered by controlling the yielding failure.

(e) Brittle Fracture: Brittle fracture occurs with very little plastic deformation under tensile stresses less than the yield stress of the material when small cracks are contained in the structural members due to undesirable material properties. Brittle fracture depends on the material tensile strength, temperature and thickness. Therefore, this mode is controlled by the material rule requirement of steel grade based on the impact test and addresses all limit states except FLS. The consequence of this failure mode is directed to the overall collapse of the ship hull.

(f) Fatigue cracking: This failure mode is different from the failure modes mentioned above and is controlled in FLS alone.
Failure modes in ship structures with relation to the limit states are shown in Table 1.6.3.

1.8.3 Capacity models for ULS, SLS and ALS

1.8.3.1 The structural capacity assessment methods are either prescriptive or require the use of more advanced calculations such as finite element analysis methods.

1.8.4 Capacity models for FLS

1.8.4.1 The fatigue assessment method provides requirements for assessing structural details against fatigue failure. The fatigue capacity model is based on a linear cumulative damage summation (Palmgren-Miner's rule) in combination with a design S-N curve, a reference stress range and an assumed long-term stress distribution curve.

1.8.4.2 The fatigue capacity assessment models are either prescriptive or require the use of more advanced calculations, such as finite element analysis methods. These methods account for the combined effects of global and local dynamic loads. The procedure for fatigue assessment of ship structures is indicated in Chapter 10.

1.8.5 Net Scantling Approach

The application of the net thickness approach to assess the structural capacity is specified in Chapter 2, Section 4.

1.8.6 Intact Structure

All strength calculations for ULS, SLS and FLS are based on the assumption that the structure is intact. But in ALS, the residual strength of the ship in a structurally damaged condition is assessed.

1.9 Rule design methods

1.9.1 General

1.9.1.1 The requirements associated with the assessment of the scantlings are based on the Working Stress Design (WSD) or Partial Safety Factor (PSF) design method. For both WSD and PSF, the design assessment conditions and corresponding acceptance criteria are given. These conditions are associated with the probability level of the combined 'expected' and 'extreme' loads covered by static and dynamic load combinations.

(i) Working Stress Design (WSD), also known as the permissible or allowable stress method

(ii) Partial Safety Factor (PSF), also known as Load and Resistance Factor Design (LRFD).

The safety factors for both WSD and PSF method are calibrated for the various rule requirements such that consistent and acceptable safety level for all combinations of static and dynamic load effects are achieved.

(a) The WSD method has the following composition:

$$W_{stat} + W_{dyn} \leq \eta R$$

where:

W_{stat} = simultaneously occurring static loads or load effects in terms of stresses.

W_{dyn} = simultaneously occurring dynamic loads or load effects. The dynamic loads are typically a combination of local and global load components.

R = characteristic capacity or material strength (e.g. yield stress or buckling capacity)

η = permissible utilization factor (resistance factor). The utilization factor includes consideration of uncertainties in loads, structural capacity and the consequence of failure

(b) The PSF method has the following composition:

$$\gamma_{stat} W_{stat} + \gamma_{dyn} W_{dyn} \leq \frac{R}{\gamma_R}$$

where:

W_{stat} = simultaneously occurring static loads or load effects in terms of stresses.

γ_{stat} = partial safety factor associated with the uncertainties related to static loads

W_{dyn} = simultaneously occurring dynamic loads or load effects. The dynamic loads are typically a combination of local and global load components.

γ_{dyn} = partial safety factor associated with the uncertainties related to dynamic loads

R = characteristic capacity or material strength (e.g. yield stress or buckling capacity)

γ_R = the partial factor associated with the capacity (resistance), or the resistance factors

1.9.1.2 In general, the working stress design (WSD) method is applied in these Rules, except for the hull girder ultimate strength criteria where the partial safety factor (PSF) method is applied. The partial safety factor format is applied for this highly critical failure mode to better account for uncertainties related to static loads, dynamic loads and capacity formulations.

1.9.1.3 The identified load scenarios are addressed in terms of design loads, design format and acceptance criteria set, as given in Table 1.10.1.2. The table is indicative and only intended to give an overview.

1.9.1.4 Load based prescriptive requirements provide scantling requirements for all plating, local supporting members, most primary supporting members and the hull girder and cover all structural elements including deckhouses, foundations for deck equipment.

1.9.1.5 In general, these requirements explicitly control one particular failure mode and hence several requirements may need to be applied to assess one particular structural member.

1.9.1.6 Minimum requirements indicate the minimum scantlings which are to be applied irrespective of all other requirements, hence thickness below the minimum is not allowed. The minimum requirements are usually in one of the following forms:

(i) Minimum thickness, which is independent of the specified minimum yield stress.

(ii) Minimum stiffness and proportion, which are based on buckling failure modes.

1.9.2 Design loads for SLS, ULS and ALS

1.9.2.1 The structural assessment of compartment boundaries, e.g. bulkheads, is based on loading condition relevant for the type of ship and the operation the ship is intended for.

1.9.2.2 To provide consistency of approach, standardized rule values for parameters, such as GM, R_{roll}, T_{SC} and C_b are to be applied to calculate the rule load values.

1.9.2.3 The probability level of the dynamic global, local and impact loads (Refer Table 1.10.1.1) is 10^{-8} and is derived using the long-term statistical approach.

1.9.2.4 The probability level of the sloshing loads (Refer Table 1.10.1.1) is 10^{-4}. These loads are considered to be frequent loads.

1.9.2.5 The design load scenarios for structural verification apply the applicable simultaneously acting local and global load components. The relevant design load scenarios are given in Chapter 5, Section 6.

1.9.2.6 The simultaneously occurring dynamic loads are specified by applying a dynamic load combination factor to the dynamic load values given in Chapter 5. The dynamic load combination factors that define the dynamic load cases are given in Chapter 5, Section 2.

1.9.2.7 Design load conditions for hull girder ultimate strength are indicated in Chapter 6, Section 4.

1.9.3 Design loads for FLS

1.9.3.1 The fatigue requirements are given in Chapter 10.

1.9.3.2 The expected load history for the design life is characterized at a probability level of 10^{-2} of the dynamic load value, the load history for each structural member is represented by Weibull probability distributions of the corresponding stresses.

1.9.3.3 The considered wave induced loads include:

(i) Hull girder loads (i.e. vertical and horizontal bending moments and torsional moments).

(ii) Dynamic wave pressures.

(iii)Dynamic pressure and forces from cargo, ballast etc.

1.9.4 Structural response analysis

1.9.4.1 In general, the following approaches are to be followed for determination of the structural response to the applied design load combinations:

(a) Beam theory: Used for prescriptive requirements

(b) FE analysis:

(i) Coarse mesh for cargo hold model
(ii) Fine mesh for local models
(iii) Very fine mesh for fatigue assessment.

1.10 Acceptance criteria

1.10.1 General

1.10.1.1 The acceptance criteria are categorized into four acceptance criteria sets. These are indicated in Table 1.10.1.1 to Table 1.10.1.3. The specific acceptance criteria set that is applied in the rule requirements is dependent on the probability level of the characteristic combined load.

1.10.1.2 The acceptance criteria set AC-1 is applied for operational static and sloshing design loads. The allowable stress for such loads is lower than that for an extreme load to take into account effects of:

- Repeated yield
- Allowance for some dynamics
- Margins for some selected limited operational mistakes.

1.10.1.3 The acceptance criteria set AC-II is applied for the static + dynamic design load combinations where considered loads are extreme loads with a low probability of occurrence.

1.10.1.4 The acceptance criteria set AC-III is typically applied for impact loads such as bottom slamming, bow impact and liquid impact in tanks.

1.10.1.5 The acceptance criteria set AC-IV is to be applied for accidental loads like flooding.

Table 1.10.1.1: Load scenarios and corresponding rule requirements			
Operation	**Load type**	**Design load scenario**	**Acceptance criteria**
Load Scenario 1: Seagoing operations			
Transit	Static and dynamic loads in heavy weather	S + D	AC-II
	Impact loads in heavy weather	Impact (I)	AC-III
	Internal sloshing loads	Sloshing (SL)	AC-I
	Cyclic wave loads	Fatigue (F)	**----**
Load Scenario 2: Harbour and sheltered operations			
Loading, unloading and ballasting	Typical maximum loads during loading, unloading and ballasting operations	S	AC-I
Tank testing	Typical maximum loads during tank testing operations	S	AC-I
Special conditions in harbour	Typical maximum loads during special operations in harbour, e.g. propeller inspection afloat or dry-docking loading conditions	S	AC-I
Load Scenario 3: Flow through ballast water exchange			
BWE by flow through	Static and dynamic loads in heavy weather	S + D	AC-II
Load Scenario 4: Sequential ballast water exchange			
BWE by sequential methods	Static and dynamic loads in heavy weather	S + D	AC-II
Load Scenario 5: Accidental condition			
Flooded conditions	Typically, maximum loads on internal watertight subdivision structure in accidental flooded conditions	A	AC-I for collision bulkheads AC-IV for other watertight boundaries.

Table 1.10.1.2: Acceptance criteria - prescriptive requirements						
Acceptance criteria	**Plate panels and local support members**		**Primary supporting members**		**Hull girder members**	
	Yield	Buckling	Yield	Buckling	Yield	Buckling
AC-I AC-II AC-IV	Permissible stress: Chapter 7 Chapter 11	Control of stiffness and proportions: Chapter 9	Permissible stress: Chapter 7 Chapter 11	Control of stiffness and proportions: Chapter 9	Permissible stress: Chapter 6	Allowable buckling utilization factor: Chapter 9
AC-III	Plastic criteria: Chapter 11	Control of stiffness and proportions: Chapter 9	Plastic criteria: Chapter 11	Control of stiffness and proportions: Chapter 9	N/A	N/A

Table 1.10.1.3: Acceptance criteria – Finite Element analysis			
Acceptance criteria	**Direct /Cargo hold analysis**		**Fine mesh analysis**
	Yield	Buckling	Yield
AC-I AC-II	Permissible stress: Chapter 8	Allowable buckling Utilization factor: Chapter 9	Permissible von Mises stress: Chapter 8

1.10.2 Acceptance criteria

1.10.2.1 Table 1.10.1.1 to Table 1.10.1.3 gives a general overview of the acceptance criteria for the different design load scenarios covered by these rules for the yield and buckling failure modes.

1.10.2.2 For the yield criteria the permissible stress is proportional to the specified minimum yield stress of the material.

1.10.2.3 For the buckling failure mode, the acceptance criteria are based on the control of stiffness and proportions as well as on the buckling utilization factor.

Section 2

Definitions

2.1 Principal particulars

2.1.1 Length

2.1.1.1 The forward perpendicular, F.P., [m], is the perpendicular drawn at the intersection of the water line at the scantling draught with the fore side of the stem, Refer Figure 2.1.1.1. In ships with unusual bow arrangement the position of the F.P. will be specially considered.

2.1.1.2 The after perpendicular, A.P., [m], is the perpendicular drawn at the intersection of the water line at the scantling draught with the after side of the rudder post or the centerline of the rudder stock if there is no rudder post, Refer Figure 2.1.1.1. In ships with unusual stern arrangement the position of the A.P. will be specially considered.

2.1.1.3 Rule length, L, is the distance, [m], between the forward and after perpendiculars. However, L is to be not less than 96%, and need not be greater than 97% of the extreme length on the waterline at the scantling draught. In ships with unusual bow and/or stern arrangement the Rule length, L, will be specially considered.

In ships without rudder stock, (e.g. ships fitted with azimuth thrusters), the rule length L is to be taken equal to 97% of the extreme length on the waterline at the scantling draught.

2.1.1.4 "Amidship" is at 0.5L aft of the F.P.

The midship part of a ship is the part extending 0.4L amidships, unless otherwise specified. Refer Figure 2.1.1.1.

2.1.1.5 Load line length, L_L, is to be taken as 96% of the total length [m], on a waterline at 85% of the least moulded depth measured from the top of the keel, or as the length from the fore side of the stem to the axis of the rudder stock on that waterline, if that is greater.

For ships without a rudder stock, the length L_L shall be taken as 96% of the waterline at 85% of the least moulded depth.

In ships designed with a rake of keel the waterline on which this length is measured is parallel to the designed waterline at 85% of the least moulded depth D_{min}, found by drawing a line parallel to the keel line of the ship (including skeg) tangent to the moulded sheer line of the freeboard deck.

The least moulded depth is the vertical distance measured from the top of the keel to the top of the freeboard deck beam at side at the point of tangency. Refer Figure 2.1.1.2.

2.1.1.6 Where the stem contour is concave above the waterline at 85% of the least moulded depth, and where the aftermost point of the stem is above the waterline, forward end of the length, L_L is to be taken at the aftermost point of the stem above that waterline, Refer Figure 2.1.1.3.

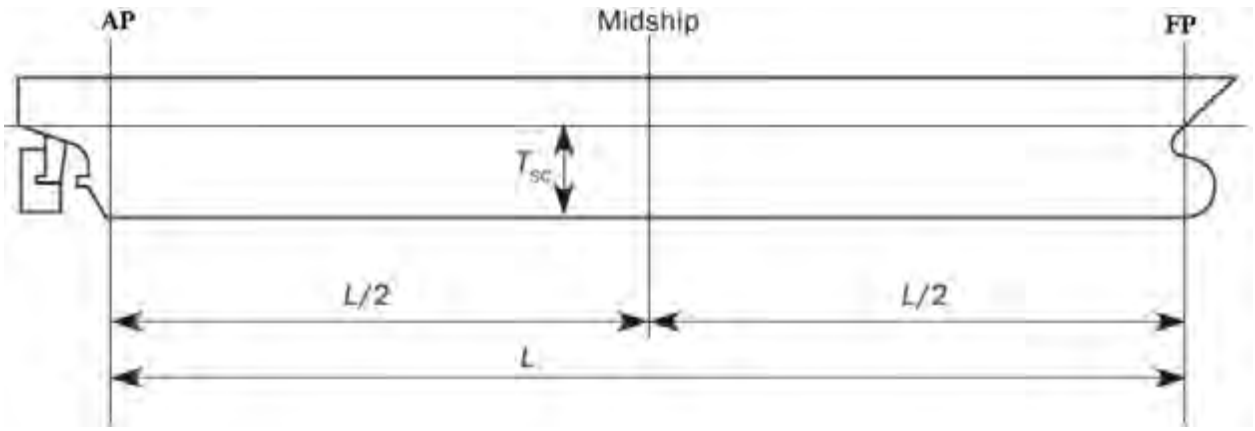


Figure 2.1.1.1: Ends and Midship

Figure 2.1.1.2: Ship with rake of keel

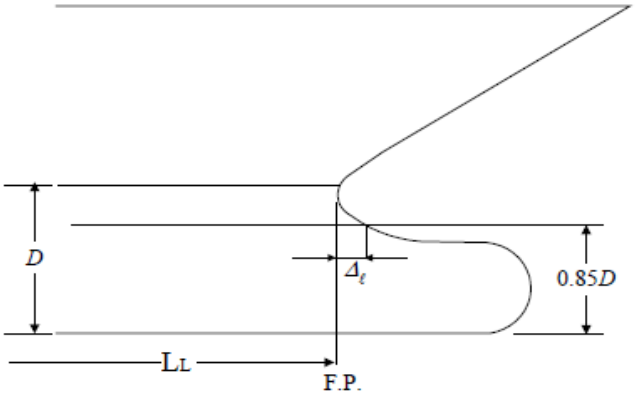


Figure 2.1.1.3: Concave stem contour

2.1.2 Breadth, B, [m], is the greatest moulded breadth measured amidship at the scantling draught, T_{SC}.

2.1.3 Depth, D, is the moulded depth [m], measured amidships from top of the keel to the moulded deck line of the uppermost continuous deck at side. When a rounded gunwale is arranged the depth is to be measured to the continuation of the moulded deck line as shown in Figure 2.1.3.

Where the freeboard deck is stepped and the raised part of the deck extends over the point at which the moulded depth is to be determined, the moulded depth is to be measured to a line of reference extending from the lower part of the deck along a line parallel with the raised part.

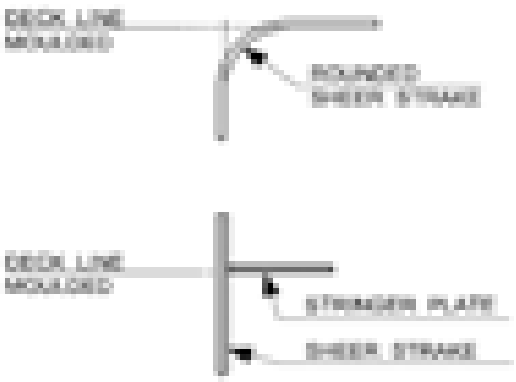


Figure 2.1.3: Deck corners

2.1.4 Load line Draught, T_L, [m], is the moulded draught amidships corresponding to the summer load waterline. Note this may be less than the maximum permissible scantling draught.

The scantling draught T_{SC} [m] is the draft at which the strength requirements for the scantlings of the ship are met and represents the full load condition. The scantling draught T_{SC} is not to be less than that corresponding to the assigned freeboard. (i.e. T_{SC} is not to be less than T_L)

The lowest design ballast draft, T_B is the minimum draught amidships in ballast conditions, including ballast water exchange operation, if any, for any ballast conditions in the loading manual including both departure and arrival conditions.

If the lowest design ballast draught amidships is not available in the initial design stage, T_B is taken from the Table 2.1.4. When T_B taken from the below table is used for strength assessment, it is to be confirmed that all ballast conditions in the final loading manual are covered.

T_{BH} is the minimum design heavy ballast draught, in [m], at which the strength requirements for the scantlings of the ship are met. This heavy ballast draught is to be considered for ships having heavy ballast condition.

Table 2.1.4: Minimum design ballast draught

Ship Type	T_B
Ship carrying dry cargo	$0.35T_{SC}$
Ship carrying liquid cargoes (Tankers)	2+0.02L
Ro-Ro ships	$0.5T_{SC}$
Passenger ships	$0.75T_{SC}$

2.1.5 Moulded displacement, [t], corresponds to the underwater volume of the ship, at a draught, in sea water having a density of 1.025[t/m^3].

2.1.6 Block coefficient, C_b, is the moulded block co-efficient corresponding to the waterline at the scantling draught, T_{SC} based on rule length, L and moulded breadth, B and calculated as follows: -

$$C_b = \frac{\text{Moulded displacement } (m^3) \text{ at } T_{SC}}{L\ B\ T_{SC}}$$

2.1.7 Design speed, V, [knots] means the speed with which the vessel is running ahead at the maximum continuous rated shaft rpm and at the summer load waterline.

2.1.8 Deadweight, [t], is the difference between the displacement at the summer draught in sea water having a density of 1.025[t/m^3] and the lightweight of the ship.

2.1.9 Lightweight, [t], is the displacement of a ship without cargo, fuel, lubricating oil, ballast water, fresh water and feed water in tanks, consumable, stores, and passengers and crew and their effects but including liquids in piping and machinery and mediums required for the fixed fire-fighting systems (e.g. fresh water, CO_2, dry chemical powder, foam concentrate, etc.).

2.2 Structural terms

2.2.1 The general terms used in the Rules for various structural parts of the ships are defined as under:

2.2.1.1 Strength Deck: The strength deck is, in general, the uppermost continuous deck. In the case of a superstructure or deckhouses of length greater than 0.15L, extending within the 0.4L amidships, the strength deck is the deck of the superstructure or the deck of the uppermost deckhouse.

2.2.1.2 Freeboard Deck: The freeboard deck is normally the uppermost complete deck exposed to weather and sea, which has permanent means of closing all openings in the weather part and below which all openings on the sides of the ship are fitted with means for watertight closing.

2.2.1.3 Bulkhead Deck: In a passenger ship, the uppermost deck at any point in the subdivision length (L_S), of the ship to which the main bulkheads and the ship's shell are carried watertight and the lowermost deck from which passenger and crew evacuation will not be impeded by water in any stage of flooding for

damage cases defined in SOLAS Ch II-1 Regulation 8 and in SOLAS Ch II-1 Part B-2. The bulkhead deck may be a stepped deck. In a cargo ship the freeboard deck may be taken as the bulkhead deck.

2.2.1.4 Superstructure: A decked structure on freeboard deck extending from side to side of the ship or with the side plating not inboard of shell plating by more than 4 per cent of the breadth B.

2.2.1.5 Deckhouse: A decked structure above the freeboard deck with the side plating being inboard of the shell plating by more than 4% of the breadth B.

2.2.1.6 Bottom Structure: Shell plating with stiffeners and girders below the upper turn of bilge and all other elements below and including the inner bottom plating in case of the double bottom. Sloping hopper tank top is to be regarded as a bulkhead.

2.2.1.7 Side Structure: Shell plating with stiffeners and girders between the upper turn of bilge and the uppermost continuous deck at side. A rounded gunwale is included in the side structure.

2.2.1.8 Deck Structure: Deck plating with stiffeners, girders, and supporting pillars.

(i) Girder: A collective term for the primary supporting members, other terms include:

- Transverses - transverse girders under the deck.

- Web frames - side vertical girders.

- Hatch end beams - transverse deck girders at the ends of the hatch.

- Stringers - horizontal girders.

- Cross-ties - girders connecting two vertical girders in a deep tank.

- Floor - bottom transverse girders.

(ii) Stiffener: A collective term for secondary supporting members; other terms being:

- Frames.

- Bottom, inner bottom, side or deck longitudinals.

- Reverse frame - transverse stiffener on the inner bottom.

- Horizontal or vertical bulkhead stiffeners.

2.2.2 Weathertight

A closing appliance is considered weathertight if it is designed to prevent the passage of water into the ship in any sea conditions. Generally, all openings on the freeboard deck and in enclosed superstructures are to be provided with weathertight closing appliances.

2.2.3 Watertight

A closing appliance is considered watertight if it is designed to prevent the passage of water in either direction under a head of water for which the surrounding structure is designed. Generally, all openings below the freeboard deck in the outer hull or main watertight bulkheads are to be fitted with permanent means of watertight closing appliances.

2.2.4 Position 1 and position 2 (Refer Figure 2.2.4)

2.2.4.1 Position 1includes:

- exposed freeboard and raised quarter decks for their entire length- exposed superstructure decks situated forward of $0.25L_L$ from FP_{LL}.

2.2.4.2 Position 2 includes:

- exposed superstructure decks situated aft of $0.25L_L$ from FP_{LL} and located atleast one standard height of superstructure above the freeboard deck.

- exposed superstructure decks situated forward of $0.25L_L$ from FP_{LL} and located at least two standard heights of superstructure above the freeboard deck.

Note: The upper limit of the vertical extent of Position 2 need not be considered to exceed two standard superstructure heights at locations abaft the $0.25L_L$ from the FP_{LL} and three standard superstructure heights forward of $0.25L_L$ from the FP_{LL}.

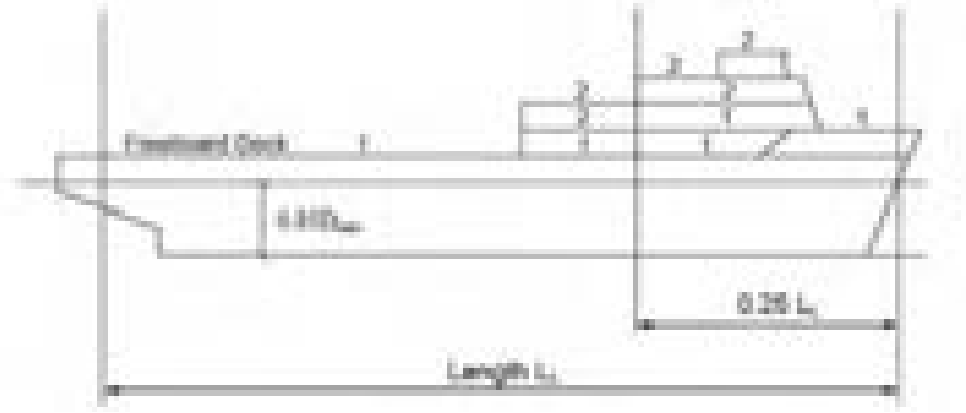


Figure 2.2.4: Position 1 and 2 for general ships

2.2.5 Standard height of superstructure

2.2.5.1 The height of a superstructure is the least vertical height measured at side from the top of the superstructure deck beams to the top of the freeboard deck beams. The standard height of superstructure is defined in Table 2.2.5.

Table 2.2.5: Standard height of superstructure		
Freeboard length L_L[m]	**Standard height h_s, [m]**	
	Raised quarter deck	**All other superstructures**
$L_L \leq 30$	0.9	1.8
75	1.2	1.8
$L_L > 125$	1.8	2.3
The standard heights at intermediate lengths of ships are obtained by linear interpolation.		

2.2.5.2 A tier is defined as a measure of the extent of a deckhouse. A deckhouse tier consists of a deck and external bulkheads. In general, the first tier is the tier situated on the freeboard deck.

2.3 Type A and Type B freeboard ships

2.3.1 Type 'A' ship

Type 'A' ship is one which:

(a) Is designed to carry only liquid cargoes in bulk.

(b) Has a high integrity of the exposed deck with only small access openings to cargo compartments, closed by watertight gasketed covers of steel or equivalent material.

(c) Has low permeability of loaded cargo compartments.

Type 'A' ship is to be assigned a freeboard in accordance with the requirements specified in the *International Convention on Load Lines* (ICLL).

2.3.2 Type 'B' ship

All ships which are not Type 'A' ships stated in 2.3.1 are to be considered as Type 'B' ships. Type 'B' ship is to be assigned a freeboard following the requirements specified in the ICLL.

2.3.3 Type 'B-60' ship

Type 'B-60' ship is any Type 'B' ship of over 100[m] in length which, according to applicable requirements of ICLL is assigned with a value of tabular freeboard which can be reduced up to 60% of the difference between the 'B' and 'A' tabular values for the appropriate ship lengths.

2.3.4 Type B-100 ship

Type ‘B-100’ ship is any Type ‘B’ ship of over 100[m] in length which, according to applicable requirements of ICLL is assigned with a value of tabular freeboard which can be reduced up to 100% of the difference between the ‘B’ and ‘A’ tabular values for the appropriate ship lengths.

2.3.5 Type 'B+' ship

Cargo ships with increased freeboard on account of hatch cover arrangement.

2.4 Material factor

2.4.1 Material factor ‘k’, a factor depending on material strength is defined in Chapter 2.

2.5 Reference coordinate system

2.5.1 The ship’s geometry, motions, accelerations and loads are defined with respect to the following right-hand coordinate system, (Refer Figure 2.5.1):

Origin = at the intersection among the longitudinal plane of symmetry of ship, the aft end (AP) of L and the baseline.

X axis = longitudinal axis, positive forward

Y axis = transverse axis, positive towards port side

Z axis = vertical axis, positive upwards.

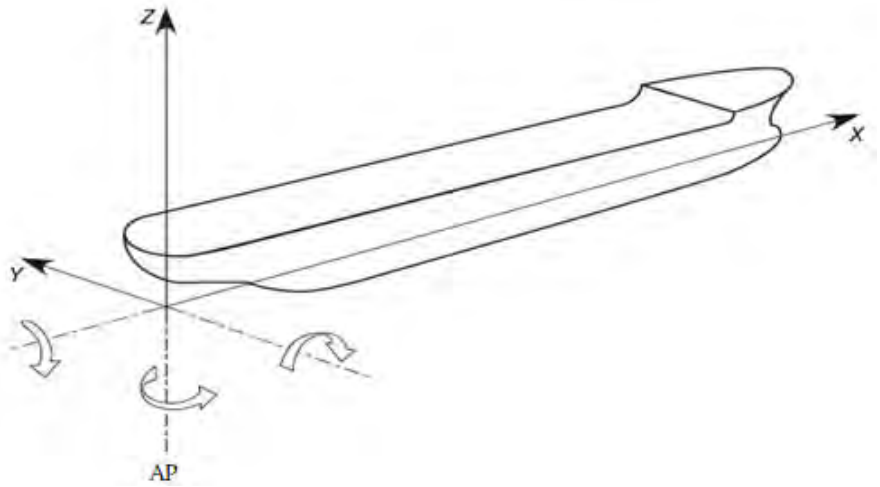


Figure 2.5.1: Reference coordinate system

2.6 Primary symbols and units

2.6.1 General

2.6.1.1 Unless otherwise specified, the general symbols and their units used in these rules are those defined in Table 2.6.1.

Table 2.6.1: Primary symbols and Units		
Symbol	**Definition**	**Units**
C	Correction factors	
F	Force and Concentrated loads	N
M	Bending moment	$\mathrm{kN-m}$
m	Mass	t
P	Pressure	$\mathrm{N/mm^2}$
$Q/F_s/F_w$	Shear force	kN
a	Acceleration	$\mathrm{m/s^2}$
g	Acceleration due to gravity, taken equal to $9.81[\mathrm{m/s^2}]$	$\mathrm{m/s^2}$
n	Number of items	
x	X coordinate along longitudinal axis, [Refer 2.5]	m
y	Y coordinate along transverse axis, [Refer 2.5]	m
z	Z coordinate along vertical axis, [Refer 2.5]	m
δ	Angle of repose of cargo	deg
∇	Moulded volume	$\mathrm{m^3}$
α	Angle	deg
ρ	Density of seawater, taken equal to $1.025[\mathrm{t/m^3}]$	$\mathrm{t/m^3}$
σ	Normal stress	$\mathrm{N/mm^2}$
τ	Shear stress	$\mathrm{N/mm^2}$
σ_b	Hull girder bending stress	$\mathrm{N/mm^2}$
Z	Section modulus	$\mathrm{cm^3}$
σ_c	Compressive stress	$\mathrm{N/mm^2}$
σ_e	Ideal elastic buckling stress	$\mathrm{N/mm^2}$
σ_{crb}	Critical buckling stress	$\mathrm{N/mm^2}$
r	Radius of curvature	mm
R_s	Service limitations [Refer Part 1, Chapter 1]	
t_D	Design temperature	$^{\circ}\mathrm{C}$

2.6.2 Ship's main data

2.6.2.1 Unless otherwise specified, symbols regarding ship's main data and their units used in these rules are those defined in Table 2.6.2.

Table 2.6.2: Ship's main data

Symbol	Definition	Units
L	Rule length	m
L_L	Freeboard length or Load line length	m
FP	Forward perpendicular	m
AP	Aft perpendicular	m
B	Moulded breadth of ship	m
D	Moulded depth of ship	m
T	Moulded draught amidships	m
T_{SC}	Scantling draught	m
T_B	Lowest design ballast draught amidships	m
T_{DAM}	Deepest equilibrium waterline in damage condition. For passenger vessels, deepest intermediate or final equilibrium damage waterline	m
T_{BH}	Heavy ballast draught	m
C_b	Block coefficient at draught, T_{SC}	
V	Maximum ahead service speed	knot
X, Y, Z	x, y, z coordinates of the calculation point with respect to the reference coordinate system	m

2.6.3 Materials

2.6.3.1 Unless otherwise specified, symbols regarding materials and their units used in these rules are those defined in Table 2.6.3.

Table 2.6.3: Materials

Symbol	Definition	Units
E	Modulus of Elasticity [Refer Part 3, Chapter 2, Sec.1]	N/mm^2
μ	Poisson's ratio [Refer Part 3, Chapter 2, Sec.1]	
k	Material factor [Refer Part 3, Chapter 2, Sec.1]	
R_{eH}	Specified minimum yield stress of material	N/mm^2
R_m	Specified tensile stress of material	N/mm^2
τ_{eH}	Specified shear yield stress, $\tau_{eH} = \frac{R_{eH}}{\sqrt{3}}$	N/mm^2
R_y	Nominal yield stress, taken equal to 235/k	N/mm^2
λ	Slenderness ratio	

2.6.4 Loads

2.6.4.1 Unless otherwise specified, symbols regarding loads and their units used in these rules are those defined in Table 2.6.4.

Table 2.6.4: Loads

Symbol	Definition	Units
C_w	Wave coefficient	
$\emptyset$	Roll angle	deg
θ	Pitch angle	deg
T_r	Roll period	s
T_p	Pitch period	s
k_r	Roll radius of gyration	m
GM	Metacentric height	m
a_o	Common acceleration parameter	
C_v	Wave parameter	
a_{surge}	Surge acceleration	m/s^2
a_{sway}	Sway acceleration	m/s^2
a_{heave}	Heave acceleration	m/s^2
M_{sw}	Permissible still water bending moment	kNm
Q_{sw}	Permissible still water shear force	kN
Q_{wv}	Vertical wave shear force	kN
M_{wv}	Vertical wave bending moment	kNm
M_{wh}	Horizontal wave bending moment	kNm
M_u	Hull girder ultimate bending moment capacity	kNm
P_{Bow}	Bow impact pressure	kN/m^2
P_{SL}	Design bottom slamming pressure	kN/m^2

2.6.5 Scantlings

2.6.5.1 Unless otherwise specified, symbols regarding scantlings and their units used in these rules are those defined in Table 2.6.5.

Table 2.6.5: Scantlings		
Symbol	**Definition**	**Units**
s	Stiffener spacing [Refer Part 3, Chapter 3, section 7]	mm
S	Primary supporting member spacing [PSM] [Refer Part 3, Chapter 3, section 7]	m
ℓ	Span of stiffeners or PSM	m
b_f	Face plate width of stiffener or PSM	mm
h_{stf}	Height of stiffener or PSM	mm
h_w	Web height of stiffeners or PSM	mm
t_w	Web thickness of stiffener or PSM	mm
t_f	Face plate thickness of stiffener or PSM	mm
t_p	Thickness of the plating attached to a stiffener or to a PSM	mm
t_c	Corrosion addition	mm
t_{off}	Net offered thickness	mm
t_{net}	Net required thickness	mm
$t_{as-built}$	As-built thickness taken as the actual thickness provided at the new building stage	mm
t_{gr-off}	Gross offered thickness	mm
t_{gr}	Gross required thickness	mm
$t_{vol-add}$	Thickness for voluntary addition taken as the thickness voluntarily added as the owner's extra margin or builder's extra margin for corrosion wastage in addition to tc	mm
t_{res}	Reserve thickness, taken equal to 0.5[mm]	mm
t_{c1}, t_{c2}	Corrosion addition on one side of the considered structural member as defined in [Refer Part 3, Chapter 2, Section 5,]	mm
ℓ_{bdg}	Effective bending span [Refer Part 3, Chapter 3, Section 7]	m

Section 3

Documentation

3.1 General

3.1.1 Documentation is to be submitted as per the following paragraphs. In case of certain ship types additional documentation may be required as per Part 5.

3.2 Plans for information

3.2.1 The following supporting plans and calculations are to be submitted for **information**:

- General arrangement.- Tank plan.- Capacity plan indicating the volume and position of the centre of gravity of all compartments and tanks.

- Lines plan and Hydrostatic curves or tables.

- Docking plan.

- Disposition of anodes (Refer Chapter 2, Section 3.5).

- Engine room arrangement

3.3 Plans for Approval

3.3.1 Plans as relevant are to be submitted to IRS for approval are listed in Table 3.3.1. These should as far as practicable be complete in all necessary details.

Table 3.3.1: Plans/Documents to be submitted for Approval

Plan Category	Drawing/Document	Including Information On
Hull structure drawings	Midship section drawing	- Main particulars (L, B, D, T, C_b, V) - Complete class notation applied for; - Stiffener and web frame spacing - Density of cargo - Deck loads, if other than those specified in the Rules - Openings on the deck - Minimum ballast draught(s) - Extent of flat part of bottom forward - Openings on the shell - Flat of bottom, flat of side - Material grades
	Shell expansion drawing	
	Transverse and Longitudinal sections drawing	
	Decks and inner bottom	
	Framing plans	
	Other transverse section drawings	
	Pillar arrangements	
	Watertight bulkheads / weather tight	
	Aft peak construction drawing	
	Fore peak construction drawing	
	Wash bulkheads	
	Double bottom	- Indication of access - Height and location of overflows - Loading on inner bottom
	Bilge keel, sea chest details	- Material grade - Details of connection to hull structure
	Standard construction details	

Loading Manual [1)]	Preliminary/Final loading manual	- Details of loading in all contemplated loading conditions - Resulting Still Water Bending Moment, Shear Force and Torsional Moments (TM) - Design values of SWBM, SWSF & TM
	Longitudinal strength calculation, if any	
	Freeboard calculation, if any	
Super structure, Deckhouses and Machinery casings	Structure drawings: a) Decks and profile plan b) Section drawings including Transverse, side and end bulkhead	- Height of sills from deck and closing appliances for companionways - Extension and mechanical properties of the aluminium alloy used (where applicable)
Welding	Welding tables and details	
Engine room area	Foundation and structural drawings for Main engine, DG sets etc.	- Type, power and R.P.M of propulsion machinery - Weight of machinery, boilers, etc. - Unbalance moments and forces if any
	Engine and thrust block seatings	
Aft end construction	Structural drawings	- Aft-end structure - Aft peak tank - Stern frame or sternpost - Steering gear arrangement
Fore end construction	Structural drawings, General arrangement of transverse thruster, tunnel structure, if any	- Fore-end structure - Fore peak tank details - Openings on non-watertight bulkheads and diaphragm plates - Height and location of overflows - Connections of bow thruster with tunnel and hull structures.
Propeller Details	Structural drawings for propeller shaft brackets, propeller nozzles, propeller bearings, struct palms, A-Bracket & P-Bracket details	- Type, R.P.M, no of blades, diameter of propeller - propeller outline - propeller thrust, torque - height and location of overflow
Rudder Details	Rudder, stock and tiller Steering gear arrangement	- speed of the ship (ahead & astern) - material of bearings, coupling bolts, stock and the locking device - rudder carrier

Anchoring arrangement	Arrangement Plan, Equipment list and Equipment number calculation	- It includes main dimensions and design loads. - List of equipment covering windlasses, anchors, anchor chain etc. - Geometric elements for calculation; grade of anchor chain, type and breaking load of chain - Construction and breaking load of steel wires - Material, construction, breaking load and relevant elongation of synthetic ropes
Anchor windlass supporting structure	Supporting structure and foundations for windlasses and chain stoppers.	- Design loads and directions of load actions - Reaction forces - Details of connection of the foundations to the deck, including specifications for holding down bolts for windlasses - Supporting structure below deck - Material specifications and welding
Mooring and towing arrangement	Mooring and towing arrangement plan	- Location on the ship - Fitting types - Dimensions - Safe working load - Maximum breaking strength of the towing and mooring lines the purpose (mooring / harbour towing etc.) - Manner of applying the lines including limiting fleet angles.
	Supporting structure and foundations for ship board fitting associated with mooring and towing operations.	- Design loads and directions of load actions - Rated pull and holding load for mooring winches - Reaction forces - Details of connection of the foundations to the deck, including specifications for holding down bolts for mooring winches - Supporting structure below deck - Material specifications and welding
	Emergency Towing Procedure	Covering SOLAS II-1/3-4.

Any Deck equipment/machinery supporting structure	Structural drawing	Foundation details above and below deck supporting structure and fixation (bolts, chocks and shear stoppers)
Masts and Derrick posts	Arrangement plan	- Safe working load - Information about the operation of derrick booms - Derrick length and loading - dimensions and positions of stays and shrouds - Material quality
	Structural drawing for masts, derrick posts and support structure	
Cranes	Arrangement and Structural drawing and support structure	- Safe working load and dynamic coefficients - Design loads and reaction forces during operation and in stowed position - Maximum sea state in offshore operation, if any connections to hull structure
Lifesaving appliances	Arrangement and Structural drawing of supporting structure	- Design loads, - Safe working load and self-weight, - Connections to hull structure.
Internal watertight doors/ ramps	Testing plan of tanks & bulkheads	- Test procedure for various compartments - Height of pipes for testing
	Watertight subdivision bulkheads & watertight tunnels	- Openings and their closing appliances
	Oil tight/water tight and partition bulkheads in cargo tanks, ballast tanks and deep tanks	- Electrical diagram of power control and position indication circuits. - Intended tank contents & their densities - Height and location of overflow/air pipes - Tanks intended to be partially filled - Corrosion protection; if any
Miscellaneous structural/arrangement plans	Weathertight/outer doors	- Position and type where it is located - Closing, opening, sealing, securing and locking arrangement - Spacing of bolts or wedges - Locations of hatch covers from summer load waterline and from fore end - Design pressure/load if different from those specified in the rules
	Hatchways	
	Hatches and Hatch covers	
	Shell doors (bow, side, stern as applicable)	
	Side ports	
	Windows and side scuttles	
	Manholes arrangement plan	
	Plan showing access to and escape from spaces	

	Bulwarks and freeing ports	- Arrangements and dimensions of bulwarks and freeing ports on the freeboard deck and the superstructure deck.
Air conditioning and Ventilation	Ventilation plan including ventilators and tank vents	- Height and location of air vents outlets of various compartments. - Use of spaces intended
Note: 1) Refer Chapter 1, Section 4. 2) One drawing may contain more than one of the items from each group. 3) Where other steering or propulsion systems are adopted (e.g. Steering nozzles or azimuth propulsion systems), the plans showing the relevant arrangement and structural scantlings are to be submitted. 4) The information is to be included as indicated in the 3rd column is required for approval of plans given in the 2nd column of the Table. However, approval of plans mentioned in the 2nd column does not necessarily mean that all the information given in 3rd column is also approved or need approval.		

3.4 Plans to be kept on board

3.4.1 A set of as built construction drawings and other plans showing any subsequent structural alterations are to be kept on board. The set of plans are to include the following:

(a) Main plan includes:

(i) General arrangement
(ii) Capacity plan
(iii) Hydrostatic curves
(iv) Loading manual, where required

(b) Structural plans include:

(i) Midship section
(ii) Scantling profile
(iii) Decks
(iv) Shell expansion
(v) Transverse bulkheads
(vi) Rudder and rudder stock
(vii) Cargo hatch covers, when applicable

(c) Bilge ballast and cargo piping diagrams.

An additional set of such drawings are to be kept ashore by the company responsible for the operation of the ship.

3.4.2 To facilitate the ordering of materials for repairs, the above plans are to show the disposition and extent of high strength steel and steel of grades other than Grade A, along with the information relating to their physical and mechanical properties, recommended working, treatment and welding procedures etc.

3.4.3 A ship construction file is required to be maintained on board the ship to facilitate inspection (survey) and repair and maintenance in future. The ship construction file to contain:

(a) Plans as mentioned in Cl.3.4.1;

(b) Essential certificates and records;

(c) Manuals required for classification and statutory requirements;

(d) Details of equipment forming part of the watertight and weathertight integrity of the ship;

(e) Tank testing plan including details of the test requirements;

(f) Corrosion protection specifications;

(g) Details for the in-water survey, if applicable, information for divers, clearances measurements instructions etc. tank and compartment boundaries;

(h) Docking plan and details of all penetrations normally examined at dry docking;

(i) Coating technical file, for ships subject to compliance with the IMO Coating Performance Standard (PSPC). (Refer Chapter 2, Section 3.6).

(j) Watertight cable transit seal systems register (See Part 4, Chapter 8, Section 3.21)

Section 4

Loading Guidance Information

4.1 General Requirements

4.1.1 Application

4.1.1.1 The requirements in this section apply to all ships of 65[m] in length and above.

4.1.1.2 All ships covered by Regulation 10 of the International Convention on Load Lines are to be provided with an approved loading manual except those belonging to Category II with length less than 90[m] for which the maximum deadweight does not exceed 30% of the displacement at the summer load line draft.

4.1.1.3 In addition to the Loading Manual, all ships of Category I of 100[m] in length and above are to be provided with an approved loading instrument for calculation and control of hull strength.

4.1.1.4 The requirements concerning the loading manual and the loading instruments are indicated in 4.3 and 4.4 respectively.

4.1.1.5 Additional requirements for bulk carriers, ore carriers and combination carriers of L ≥ 150[m] are indicated in Part.5, Chapter 1.

4.1.1.6 For chemical carriers and gas carriers existing analog computer may be retained.

4.2 Definitions

4.2.1 Loading Manual is a document which describes:

- The loading conditions on which the design of the ship has been based for seagoing and harbour/ sheltered waters, including permissible limits of still water bending moment and shear force. Additionally, where applicable, shear force correction values and permissible limits for still water torsional moment and lateral loads.

- The conditions specified in the ballast water management procedure and dry-docking procedure.

- The results of the calculations of still water bending moments, shear forces and where applicable, limitations due to torsional and lateral loads.

- The allowable local loading for the structure (hatch covers, decks, double bottom, etc.)

An operation manual is always to be provided with the loading instrument.

4.2.2 Loading Instrument is an instrument, which unless otherwise stated is digital, by means of which it can be easily and quickly ascertained that, at specified read-out points, the still water bending moments, shear forces, and the still water torsional and lateral loads, where applicable, in any loaded or ballast condition will not exceed the specified permissible values.

The loading instrument is a ship specific onboard equipment and the results of the calculations are only applicable to the ship for which it has been approved.

An approved loading instrument cannot replace an approved loading manual.

4.2.3 Category I Ships:

- Ships with large deck openings where combined stresses due to vertical and horizontal hull girder bending, torsional and lateral loads have to be considered; e.g. container ships.

- Ships liable to carry non-homogeneous loadings, where the cargo and/or ballast may be unevenly distributed; e.g. bulk carriers, tankers.

- Chemical tankers and gas carriers.

4.2.4 Category II Ships:

- Ships with arrangement giving small possibilities for variation in the distribution of cargo and ballast,

- Ships on regular and fixed trading pattern where the Loading Manual gives sufficient guidance,

- Ships of less than 120[m] in length, when their design takes into account uneven distribution of cargo or ballast, belong to Category II.

- In addition, those exceptions given under Category I.

4.3 Loading Manuals

4.3.1 Conditions of Approval of Loading Manuals

4.3.1.1 The approved Loading Manual is to be based on the final data of the ship. The Manual is to include the design loading and ballast conditions; sub divided into departure and arrival conditions, upon which the approval of the hull scantlings is based.

4.3.1.2 In case of modifications implying changes in the main data of the ship (e.g. lightship weight, buoyancy distribution, tank volumes or usage, etc.) a new Loading Manual is to be issued duly approved.

4.3.1.3 The Loading Manual is to be prepared in a language understood by the users. If this language is not English, a translation into English is to be included.

4.3.2 Loading conditions

4.3.2.1 The loading manual is to contain the design (cargo and ballast) loading conditions, subdivided into departure and arrival conditions, and ballast exchange at sea conditions, where applicable, upon which the approval of the hull scantlings is based, as defined in Chapter 5, Section 7.

4.3.2.2 Loading conditions for ship specific types are indicated in Part 5 of the Rules.

4.3.2.3 In general, the loading conditions, based on amount of bunkers, fresh water and stores at departure and arrival, should be included in the Loading Manual.

4.3.3 Operational limitations

4.3.3.1 The loading manual is to indicate relevant operational limitations, as follows:

- Scantling draught

- Load specifications for cargo decks

- Design minimum ballast draught at midships

- Minimum slamming draught forward with forward double bottom ballast tanks filled

- Minimum slamming draught forward with any of the forward double bottom ballast tanks empty

- Maximum allowable cargo density

- Maximum cargo density and filling heights in any loading condition in loading manual

- Restrictions in cargo mass- and angle of repose

- Maximum service speed

- Envelope results and permissible limits of still water bending moments and shear forces, and still water torsional moments, if applicable

- Restrictions to GM-value

- Restrictions to filling of tanks in seagoing conditions

- Restrictions to ballast holds which may not be partially filled in seagoing conditions

- Restrictions to double bottom ballast tanks which may be filled in alternate loaded seagoing conditions.

4.4 Loading Instruments

4.4.1 Conditions of Approval of Loading Instruments

4.4.1.1 The loading instrument and its operation manual are subject to approval. The approval includes the following:

- Verification of type approval, if any

- Verification that the final data of the ship has been used

- Acceptance of number and position of read-out points

- Acceptance of relevant limits for all read-out points

- Checking of proper installation and operation of the instrument on board in accordance with the agreed test conditions; and that a copy of the approved operation manual is available.

4.4.2 In case of modifications implying changes in the main data of the ship, the loading instrument is to be modified accordingly and approved.

4.4.3 The operation manual and the instrument output must be prepared in a language understood by the users. If this language is not English, a translation into English is to be included.

End of Chapter

Chapter 2

Materials of Construction and Corrosion Additions

Contents

Section	
1	*General*
2	*Use of Steel Grades*
3	*Corrosion Protection*
4	*Net Scantling Approach*
5	*Corrosion Addition*

Section 1

General

1.1 Scope

1.1.1 The Rules relate, in general, to the construction of steel ships. Consideration will however be given to the use of other materials also.

1.1.2 The materials used in the construction of the ship are to be manufactured and tested in accordance with the requirements of Part 2. Materials for which provision is not made in Part 2 may be accepted, provided that they comply with an approved specification and such tests as may be considered necessary.

1.2 Steel

1.2.1 Ordinary hull structural steel is a hull structural steel with a minimum yield stress of 235[N/mm^2] and a tensile strength generally in the range of 400-520[N/mm^2]. Steels having a higher specified minimum yield stress are regarded as higher strength steels.

For ordinary hull structural steel, the **material factor `k'** is to be taken as 1.0.

1.2.2 Where higher tensile steel is used, the hull girder section modulus and the local scantlings may be reduced in accordance with the relevant requirement of the Rules.

1.2.3 For this purpose, a material factor 'k', is to be taken from Table 1.2.3.

1.2.4 Where steel castings or forgings are used for stern frames, rudder frames, rudder stocks, propeller shaft brackets and other major structural items, they are to comply with Part 2 as appropriate.

1.2.5 Following material grades of hull structural steels are commonly used in the ship construction.

1.2.5.1 A, B, D, E and F denote normal strength steel grades.

1.2.5.2 AH, DH, EH and FH denote high strength steel grades, where ‘H’ indicates the material strength. Strength levels are supplied in four levels, 32, 36, 40 and 47, intended for use in hull construction. Table 1.2.5 gives specified yield stress and tensile strength for rolled steels generally used in construction of ships.

Table 1.2.3: Material factor (k)	
Specified minimum yield stress R_{eH}, in [N/mm^2]	**k**
235	1.00
315	0.78
355	0.72
390	0.66/0.68*
460	0.62 (#)
(*) The material factor k may be taken equal to 0.66 for steels with yield stress equal to 390[N/mm^2], provided that a fatigue assessment of the structure is performed to the satisfaction of IRS. (#) The application of high strength steel with yield stress equal to 460[N/mm^2] is limited to ships with the notation **Container Ship** as defined in Part 5, Chapter 5. The application of high strength steel with R_{eH} > 390[N/mm^2] for other ship types is specially considered.	
For intermediate values of yield stress, k is obtained by linear interpolation.	

Table 1.2.5: Mechanical properties of hull structural steels		
Steel Grades	**Specified minimum yield stress R_{eH} [N/mm^2]**	**Specified tensile stress R_m [N/mm^2]**
A-B-D-E	235	400/520
AH32, DH32, EH32 & FH32	315	440/570
AH36, DH36, EH36 & FH36	355	490/ 630
AH40, DH40, EH40 & FH40	390	510/660
EH47#	460	570/720
(#) For note, Refer Table 1.2.3, Material factor (k)		

1.3 Aluminium

1.3.1 Where seawater resisting aluminium alloys manufactured and tested in accordance with the requirements of Part 2 are used for super-structures, deckhouses, hatch covers, helicopter decks or other structural components, scantlings equivalent to steel are to be derived as follows:

plating thickness, $t_a = t_s \cdot \sqrt{k_a}$

section modulus of stiffeners, $z_a = z_s \cdot k_a$

Web area of stiffeners, $A_a = A_s \cdot k_a$

where,

k_a = Material factor for Aluminium alloy

t_a, t_s= plating thickness of aluminium and mild steel respectively.

z_a, z_s = section modulus of aluminium and mild steel stiffeners respectively.

A_a, A_s = Web area of stiffeners of aluminium and mild steel stiffeners respectively.

1.3.2 The material factor, k_a for aluminium alloys is to be obtained from the following formula:

$$k_a = \frac{235}{\sigma'_{lim}}$$

Where,

σ'_{lim} = Minimum guaranteed yield stress of the parent metal in welded condition $\sigma'_{p0.2}$, in [N/mm^2], but not to be taken greater than 70% of the minimum guaranteed tensile strength of

the parent metal in welded condition σ'_m, in [N/mm^2].

$\sigma'_{p0.2}$ = Minimum guaranteed yield stress, in [N/mm^2], of material in welded condition.

$$\sigma'_{p0.2} = \eta_1 \sigma_{p0.2}$$

σ'_m = Minimum guaranteed tensile strength, in [N/mm^2], of material in welded condition.

$$\sigma'_m = \eta_2 \sigma_m$$

$\sigma_{p0.2}$ = Minimum guaranteed yield stress, in [N/mm^2], of material in delivery condition.

σ_m = Minimum guaranteed tensile strength, in [N/mm^2], of material in delivery condition.

η_1, η_2 = Coefficients for welded construction for Aluminium alloys specified in Table 1.3.2.1.

Table 1.3.2.1: Coefficients for welded construction for Aluminium alloys

Aluminium alloy	η_1	η_2
Alloys without work-hardening treatment (series 5000 in annealed condition O or annealed flattened condition H111)	1	1
Alloys hardened by work hardening (series 5000 other than condition O or H111)	$\sigma'_{p0.2}/\sigma_{p0.2}$	σ'_m/σ_m
Alloys hardened by heat treatment (series 6000) (1)	$\sigma'_{p0.2}/\sigma_{p0.2}$	0.6

(1) When no information is available, coefficient η_1 is to be taken equal to the metallurgical efficiency coefficient β defined in Table 1.3.2.2.
Note: For $\sigma_{p0.2}$ and σ_m values, Refer Part 2, Chapter 9, Section 2.8, Table 2.8.1 and 2.8.2.

Table 1.3.2.2: Metallurgical efficiency coefficient β for Aluminium alloys

Aluminium alloy	Temper condition	Gross thickness [mm]	β
6005 A (Open sections)	T5 or T6	t ≤ 6	0.45
		t > 6	0.40
6005 A (Closed sections)	T5 or T6	all	0.50
6061 (Sections)	T6	all	0.53
6082 (Sections)	T6	all	0.45

1.3.3 The smaller modulus of elasticity of aluminium is to be taken into account, when determining the buckling strength of structural elements subjected to compression and the deflections, where relevant.

1.3.4 In the case of welding of two different aluminium alloys, the material factor, k_a to be considered for the scantlings is to be the greater material factor of the aluminium alloys of the assembly.

1.3.5 Welding heat input lowers locally the mechanical strength of aluminium alloys hardened by work hardening (series 5000 other than condition O or H111) or by heat treatment (series 6000).

1.3.6 For welded constructions in hardened aluminium alloys (series 5000 other than condition O or H111 and series 6000), greater characteristics than those in welded condition may be considered, provided that welded connections are located in areas where stress levels are acceptable for the alloy considered in annealed or welded condition.

1.3.7 The as-welded properties of aluminium alloys of series 5000 are in general those of condition O or H111. Higher mechanical characteristics may be considered, provided they are duly justified.

1.3.8 Extruded plating

1.3.8.1 Extrusions with built-in plating and stiffeners, referred to as extruded plating, may be used.

1.3.8.2 In general, the application of extruded plating is limited to decks, bulkheads, superstructures and deckhouses. Other uses may be permitted by IRS on a case-by-case basis.

1.3.8.3 Extruded plating to be oriented so that the stiffeners are parallel to the direction of main stresses.

1.3.8.4 Connections between extruded plating and primary members are to be given special attention.

1.4 Connections between steel and aluminium

1.4.1 Details of the proposed method of joining any aluminium and steel structures are to be submitted for approval.

1.4.2 To prevent galvanic corrosion a non-hygroscopic insulation material is to be applied between steel and aluminium when a bolted connection is used.

1.4.3 An aluminium-steel transition joint in accordance with the rules Part 2, Chapter 9, Section 4 may be used in a welded connection.

1.4.4 Any direct contact between steel and aluminium alloy is to be avoided (e.g. by means of zinc or cadmium plating of the steel parts and application of a suitable coating on the corresponding light alloy parts).

1.4.5 Bolts with nuts and washers are either to be of stainless steel or cadmium plated or hot galvanized steel. The bolts are to be fitted with sleeves of insulating material. The spacing is normally not to exceed 4 times the bolt diameter.

1.5 Young's modulus and Poisson's ratio

1.5.1 The Young's modulus and the Poisson's ratio to be used in the strength assessment are:

1.5.1.1 For carbon manganese steel materials

$E = 206\ 000[N/mm^2]$

$v = 0.3$

1.5.1.2 For aluminium alloys

$E = 70\ 000[N/mm^2]$

$v = 0.33$

Section 2

Use of Steel Grades

2.1 Grades of steel

2.1.1 The resistance to fracture is controlled, in part, by the notch toughness of the steel used in the structure. Steels with different levels of notch toughness are specified in Part 2. The grade of steel to be used is, in general, related to the air temperature in the area of operation, thickness of the material and the stress pattern associated with its location.

2.1.2 The grades of steel to be used for structures of ships in normal worldwide service and those exposed to low air temperatures (≤ -10°C), are given in Section 2.2 and 2.3 respectively. The grades of steel to be used for certain hull members in refrigerated spaces are given in Section 2.4.

2.1.3 Where tee or cruciform connections employ full penetration welds, and the plate material is subject to significant strains in a direction perpendicular to the rolled surfaces, it is recommended that consideration be given to the use of special plate material with specified through thickness properties, as detailed in Part 2.

2.2 Ships in normal worldwide service

2.2.1 Materials in the various strength members are not to be of lower grade than those corresponding to material classes and grades specified in Tables 2.2.1(a) to 2.2.1(f) and Table 2.2.2.

General requirements are given in Table 2.2.1(a), while additional minimum requirements are given in the following:

- Table 2.2.1(b): For ships, excluding liquefied gas carriers covered in Table 2.2.1(c), with length exceeding 150[m] and single strength deck,
- Table 2.2.1(c): For membrane type liquefied gas carriers with length exceeding 150[m].
- Table 2.2.1(d): For ships with length exceeding 250[m],
- Table 2.2.1(e): For single side bulk carriers subject to SOLAS regulation XII/6.4.3.
- Table 2.2.1(f): For ships with ice strengthening.

2.2.2 The material grade requirements for hull members of each class depending on thickness are defined in Table 2.2.2.

2.2.3 For strength members not mentioned in Tables 2.2.1(a) to 2.2.1(f), refer Table 2.2.3.

2.2.4 The steel grade is to correspond to the as-built plate thickness and material class.

2.2.5 Plating materials for stern-frames supporting the rudder and propeller boss, rudders, rudder horns and shaft brackets are in general not to be of lower grades than corresponding to Class II. For rudder and rudder body plates subjected to stress concentrations (e.g. in way of lower support of semi-spade rudders or at upper part of spade rudders) Class III is to be applied.

Table 2.2.1(a): Material classes and grades for ships in general	
Structural member category	**Material class/grade**
SECONDARY:	
Longitudinal bulkhead strakes, other than that belonging to the primary category	- Class I within 0.4L amidships - Grade A/AH outside 0.4L amidships
Deck plating exposed to weather, other than that belonging to the primary or special category	
Side plating	
PRIMARY:	
Bottom plating, including keel plate	- Class II within 0.4L amidships - Grade A/AH outside 0.4L amidships
Strength deck plating, excluding that belong to the special category	
Continuous longitudinal plating of strength members above strength deck, excluding hatch coamings	
Uppermost strake in longitudinal bulkhead	
Vertical strake (hatch side girder) and uppermost sloped strake in top wing tank	
SPECIAL:	
Sheer strake at strength deck (1)	- Class III within 0.4L amidships - Class II outside 0.4L amidships - Class I outside 0.6L amidships
Stringer plate in strength deck (1)	
Deck strake at longitudinal bulkhead, excluding deck plating in way of inner skin bulkhead of double hull ships (1)	
Strength deck plating at outboard corners of cargo hatch openings in container carriers and other ships with similar hatch opening configurations	- Class III within 0.4L amidships - Class II outside 0.4L amidships - Class I outside 0.6L amidships - Min. class III within cargo region
Strength deck plating at corners of cargo hatch openings in bulk carriers, ore carriers combination carriers and other ships with similar hatch opening configurations Trunk deck and inner deck plating at corners of openings for liquid and gas domes in membrane type liquefied gas carriers	- Class III within 0.6L amidships - Class II within rest of cargo region
Bilge strake in ships with double bottom over the full breadth and length less than 150[m] (1)	- Class II within 0.6L amidships - Class I outside 0.6L amidships
Bilge strake in other ships (1)	- Class III within 0.4L amidships - Class II outside 0.4L amidships - Class I outside 0.6L amidships
Longitudinal hatch coamings of length greater than 0.15L including coaming top plate and flange (Refer Figure 2.2.1).	- Class III within 0.4L amidships - Class II outside 0.4L amidships - Class I outside 0.6L amidships - Not to be less than Grade D/DH
End brackets and deck house transition of longitudinal cargo hatch coamings	
(1) Single strakes required to be of Class III within 0.4L amidships are to have breadths not less than 800+5L[mm], need not be greater than 1800[mm], unless limited by the geometry of the ship's design.	

Table 2.2.1(b): Minimum material grades for ships with length exceeding 150[m] and single strength deck	
Structural member category	**Material grade**
Longitudinal plating of strength deck where contributing to the longitudinal strength	Grade B/AH within 0.4L amidships
Continuous longitudinal plating of strength members above strength deck	Grade B/AH within 0.4L amidships
Single side strakes for ships without inner continuous longitudinal bulkhead(s) between bottom and the strength deck	Grade B/AH within cargo region

Table 2.2.1(c): Minimum material grades for membrane type liquefied gas carriers with length exceeding 150[m]		
Structural member category		**Material grade**
Longitudinal plating of strength deck where contributing to the longitudinal strength		Grade B/AH within 0.4L amidships
Continuous longitudinal plating of strength members above strength deck	Trunk deck plating	Class II within 0.4L amidships
	• Inner deck plating • Longitudinal strength member plating between the trunk deck and inner deck	Grade B/AH within 0.4L amidships
Note: Table 2.2.1(c) is applicable to membrane type liquefied gas carriers with deck arrangements as shown in Figure 2.2.2. Table 2.2.1(c) may also apply to similar ship types with a “double deck” arrangement above the strength deck.		

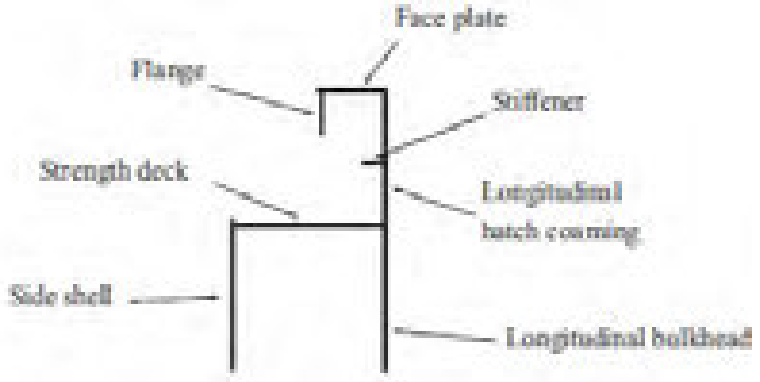


Figure 2.2.1: Typical cross section in longitudinal hatch coaming area

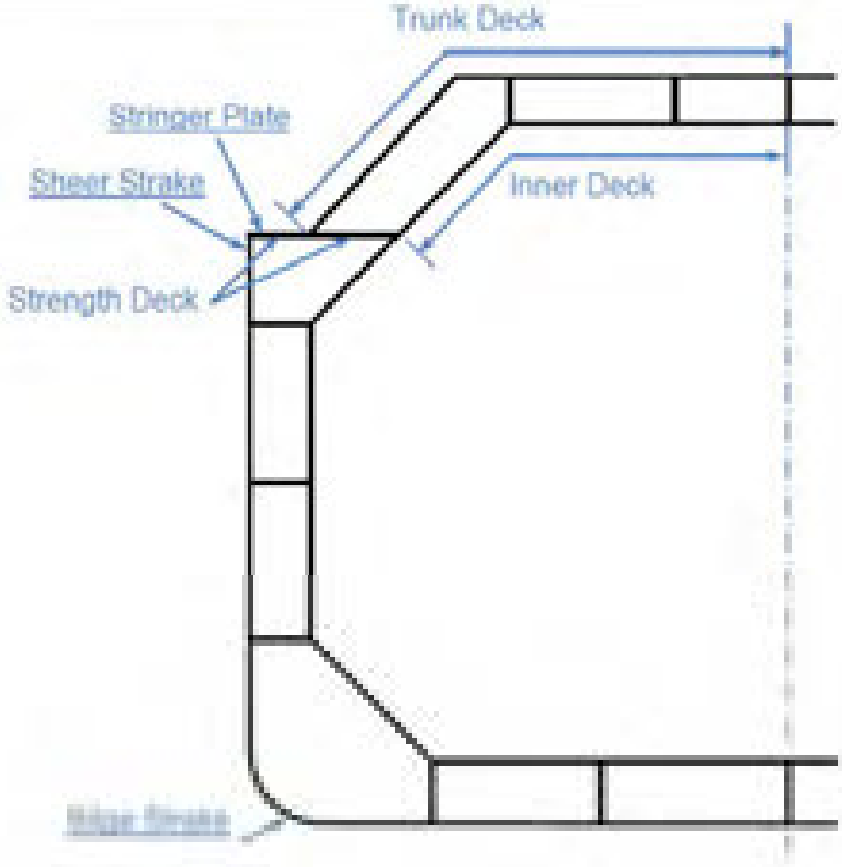


Figure 2.2.2: Typical deck arrangement for membrane type Liquefied Natural Gas Carriers

Table 2.2.1(d): Minimum material grades for ships with length exceeding 250[m]	
Structural member category	**Material grade**
Sheer strake at strength deck (1)	Grade E/EH within 0.4L amidships
Stringer plate at strength deck (1)	Grade E/EH within 0.4L amidships
Bilge strake (1)	Grade D/DH within 0.4L amidships
(1) Single strakes required to be of Grade D/DH or Grade E/EH as shown in the above table and within 0.4L amidships are to have breadths not less than 800+5L[mm], need not be greater than 1800[mm], unless limited by the geometry of the ship's design.	

Table 2.2.1(e): Minimum material grades for single-side skin bulk carriers subjected to SOLAS Regulation XII/6.4 (3)	
Structural member category	**Material grade**
Lower bracket of ordinary side frame (1), (2)	Grade D/DH
Side shell strakes included totally or partially between the two points located to 0.125*l* above and below the intersection of side shell and bilge hopper sloping plate or inner bottom plate (3)	Grade D/DH
(1) The term "lower bracket" means webs of lower brackets and webs of the lower part of side frames upto the point of 0.125*l* above the intersection of side shell and bilge hopper sloping plate or inner bottom plate. (2) The span of the side frame, *l*, is defined as the distance between the supporting structures. Refer Figure 2.5.5 of Part 5, Chapter 1, Section 2. (3) SOLAS Regulation XII/6.4 applies to bulk carriers of $L_L \geq 150$[m], carrying solid bulk cargoes having a density of 1000[kg/m^3] and above constructed on or after 1 July 2006.	

Table 2.2.1(f): Minimum material grades for ships with ice strengthening	
Structural member category	**Material grade**
Shell strakes in way of ice strengthening area for plates (Refer Part 5, Chapter 21)	Grade B/AH

Table 2.2.2: Material grade requirements for classes, I, II and III						
Class	**I**		**II**		**III**	
Thickness [mm]	**MS**	**HT**	**MS**	**HT**	**MS**	**HT**
$t \leq 15$	A	AH	A	AH	A	AH
$15 < t \leq 20$	A	AH	A	AH	B	AH
$20 < t \leq 25$	A	AH	B	AH	D	DH
$25 < t \leq 30$	A	AH	D	DH	D	DH
$30 < t \leq 35$	B	AH	D	DH	E	EH
$35 < t \leq 40$	B	AH	D	DH	E	EH
$40 < t \leq 50$	D	DH	E	EH	E	EH

Table 2.2.3: Minimum material grade requirements for strength members other than mentioned in Table 2.2.1(a) to (f)		
As built thickness [mm]	**Minimum**	
	MS	**HT**
$t \leq 40$	A	AH
$40 < t \leq 50$	B[1)]	AH
$50 < t \leq 150$	D[1)]	DH[1)]
(1) For heavy foundation plates in engine room: — Outside 0.6L amidship, grade A/AH is acceptable. — Within 0.6L amidship, grade B/AH to be used for plate thickness more than 30[mm].		

2.3 Ships exposed to low air temperatures

2.3.1 For ships intended to operate in areas with low air temperatures (below and including -10[°C]), e.g. regular service during winter seasons to Arctic or Antarctic waters, the materials in exposed structures are to be selected based on the design temperature t_D as defined in Cl.2.3.2.

2.3.2 The design temperature t_D is to be taken as the lowest mean daily average air temperature in the area of operation as illustrated in Figure 2.3.2.

Mean: Statistical mean over observation period (at least 20 years)

Average: Average during one day and night.

Lowest: Lowest during year.

MDHT = Mean Daily High (or maximum) Temperature

MDAT = Mean Daily Average Temperature

MDLT = Mean Daily Low (or minimum) Temperature.

For seasonally restricted service the lowest value within the period of operation would be applicable.

For the purpose of issuing a Polar Ship Certificate in accordance with the Polar Code, the design temperature t_D is to be no more than

13[°C] higher than the Polar Service Temperature (PST) of the ship.

In the Polar Regions, the statistical mean over observation period is to be determined for a period of at least 10 years.

2.3.3 Materials in the various strength members above the lowest ballast water line (BWL) exposed to air (including the structural members covered by the Note 5 of Table 2.3.3 and materials of cargo tank boundary plating for which 2.5 is applicable are not to be of lower grades than those corresponding to classes I, II and III as given in Table 2.3.3, depending on the categories of structural members (SECONDARY, PRIMARY and SPECIAL). For non-exposed structures (except as indicated in Note [5] of Table 2.3.3) and structures below the lowest ballast water line, 2.2 applies.

2.3.4 The material grade requirements for hull members of each class depending on thickness and design temperature are defined in Table 2.3.4. For design temperatures t_D<-55[°C], materials will be specially considered.

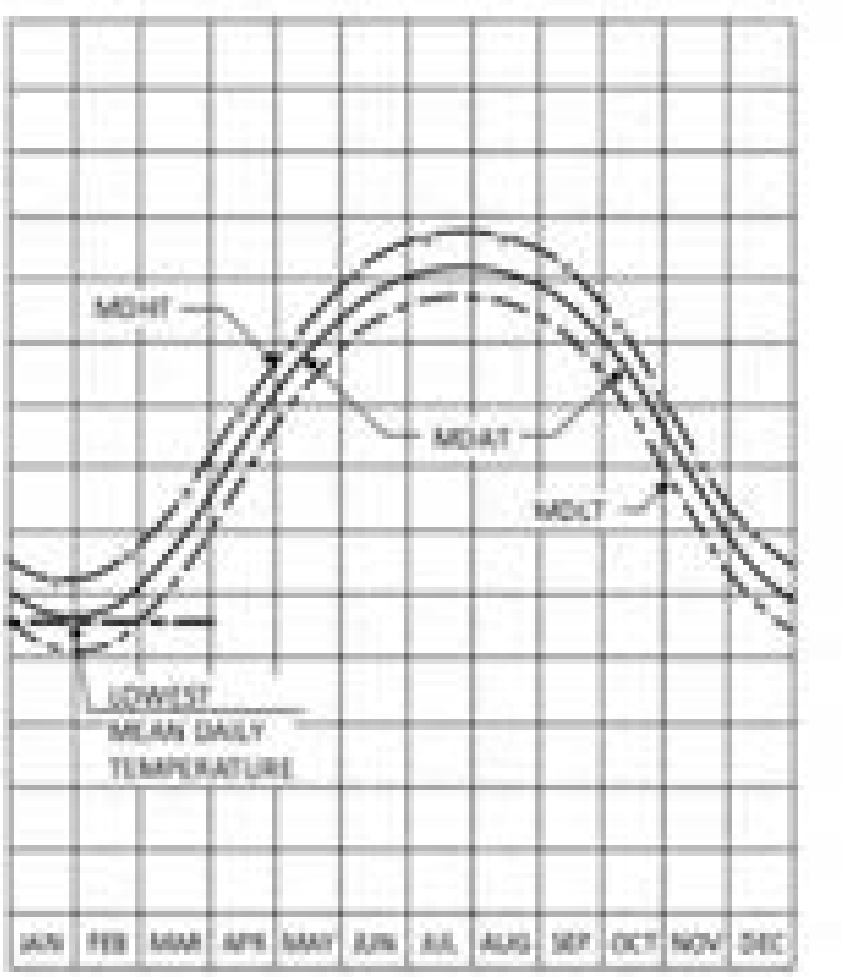


Fig 2.3.2: Commonly used definitions of temperatures

MDHT: Mean Daily High (or maximum) Temperature
MDAT: Mean Daily Average Temperature
MDLT: Mean Daily Low (or minimum) Temperature.

Table 2.3.3: Application of material classes and grades - Structures exposed to low temperatures		
	Material class/grade	
Structural member category	**Within 0.4L amidships**	**Outside 0.4L amidships**
SECONDARY: • Deck plating exposed to weather, in general • Side plating above BWL Transverse bulkheads above BWL (Refer Note 5) • Cargo tank boundary plating exposed to cold cargo (Refer Note 6)	I	I
SECONDARY: • Deck plating exposed to weather, in general • Side plating above BWL • Transverse bulkheads above BWL (Refer Note 5)	I	I
PRIMARY: • Strength deck plating (see Note 1) • Continuous longitudinal members above strength deck, excluding longitudinal hatch coamings • Longitudinal bulkhead above BWL (Refer Note 5) • Top wing tank bulkhead above BWL (Refer Note 5)	II	I
SPECIAL: • Sheer strake at strength deck (Refer Note 2) • Stringer plate in strength deck (Refer Note 2) • Deck strake at longitudinal bulkhead (Refer Note 3) • Continuous longitudinal hatch coamings (Refer Note 4)	III	II
Notes: 1. Plating at corners of large hatch openings to be specially considered. Class III or grade E/EH to be applied in positions where high local stresses may occur. 2. Not to be less than grade E/EH within 0.4L amidships in ships with length exceeding 250[m]. 3. In ships with breadth exceeding 70[m] at least three deck strakes to be class III. 4. Not to be less than grade D/DH. 5. Applicable to plating attached to hull envelope plating exposed to low air temperature. At least one strake is to be considered in the same way as exposed plating and the strake width is to be at least 600[mm]. 6. For cargo tank boundary plating exposed to cold cargo for ships other than liquefied gas carriers, Refer 2.5.		

Table 2.3.4: Material grade requirements for classes I, II and III at low temperatures

Class I

Plate thickness [mm]	-11/-15°C		-16/-25°C		-26/-35°C		-36/-45°C		-46/-55°C	
	MS	HT	MS	HT	MS	HT	MS	HT	MS	HT
t ≤ 10	A	AH	A	AH	B	AH	D	DH	D	DH
10 < t ≤ 15	A	AH	B	AH	D	DH	D	DH	D	DH
15 < t ≤ 20	A	AH	B	AH	D	DH	D	DH	E	EH
20 < t ≤ 25	B	AH	D	DH	D	DH	D	DH	E	EH
25 < t ≤ 30	B	AH	D	DH	D	DH	E	EH	E	EH
30 < t ≤ 35	D	DH	D	DH	D	DH	E	EH	E	EH
35 < t ≤ 45	D	DH	D	DH	E	EH	E	EH	X	FH
45 < t ≤ 50	D	DH	E	EH	E	EH	X	EH	X	FH

Class II

Plate thickness [mm]	-11/-15°C		-16/-25°C		-26/-35°C		-36/-45°C		-46/-55°C	
	MS	HT	MS	HT	MS	HT	MS	HT	MS	HT
t ≤ 10	A	AH	B	AH	D	DH	D	DH	E	EH
10 < t ≤ 20	B	AH	D	DH	D	DH	E	EH	E	EH
20 < t ≤ 30	D	DH	D	DH	E	EH	E	EH	X	FH
30 < t ≤ 40	D	DH	E	EH	E	EH	X	FH	X	FH
40 < t ≤ 45	E	EH	E	EH	X	FH	X	FH	X	X
45 < t ≤ 50	E	EH	E	EH	X	FH	X	FH	X	X

Class III

Plate thickness [mm]	-11/-15°C		-16/-25°C		-26/-35°C		-36/-45°C		-46/-55°C	
	MS	HT	MS	HT	MS	HT	MS	HT	MS	HT
t ≤ 10	B	AH	D	DH	D	DH	E	EH	E	EH
10 < t ≤ 20	D	DH	D	DH	E	EH	E	EH	X	FH
20 < t ≤ 25	D	DH	E	EH	E	EH	X	FH	X	FH
25 < t ≤ 30	D	DH	E	EH	E	EH	X	FH	X	FH
30 < t ≤ 35	E	EH	E	EH	X	FH	X	FH	X	X
35 < t ≤ 40	E	EH	E	EH	X	FH	X	FH	X	X
40 < t ≤ 50	E	EH	X	FH	X	FH	X	X	X	X

Notes:

1. X = Not applicable.
2. Single strakes required to be of class III or of grade E/EH of FH are to have breadths not less than 800 + 5.L[mm] maximum 1800[mm].
3. Plating materials for stern frames rudder horns, rudders and shaft brackets are not to be of lower grades than those corresponding to the material classes given in Section 2.2.

2.4 Refrigerated spaces

2.4.1 Where the minimum design temperature of the steel falls below 0[°C] in refrigerated spaces, in addition to the requirements of Cl.2.2.1, the grade of steel for the following items is to comply, in general, with the requirements of Table 2.4.1:

- Deck plating

- Webs of deck girders.

- Longitudinal bulkhead strakes attached to deck.

- Shelf plates and their face bars supporting hatch covers.

2.4.2 Unless a temperature gradient calculation is carried out to assess the design temperature in the items defined in Cl.2.4.1, the temperature to which the steel deck may be subjected is to be assessed as shown in Table 2.4.2.

Table 2.4.1: Grades of steel for minimum design temperatures below 0[°C]		
Min. design temp. in [°C]	**Thickness, in [mm]**	**Grades of steel**
0 to -10	t ≤ 12.5 12.5 ≤ t ≤ 25.5 t > 25.5	B/AH D/DH E/EH
-10 to -25	t ≤ 12.5 t > 12.5	D/DH E/EH
-25 to -40	t ≤ 12.5 t > 12.5	E/EH C-Mn LT 60 or 1/2 Ni steels

Table 2.4.2: Assessment of deck temperature		
Arrangement		**Deck temperature**
1)	Deck not covered with insulation in the refrigerated space	Temperature of the refrigerated space
2)	Deck covered with insulation in the refrigerated space and not insulated on the other side	Temperature of the space on the un-insulated side
3)	Deck covered with insulation on both sides	
	a) Temperature difference less than or equal to 11[°C] b) Temperature difference greater than 11[°C] but not greater than 33[°C] c) Temperature difference greater than 33[°C]	Mean of the temperatures of the spaces above and below the deck Mean of the temperatures of the spaces above and below the deck less 3[°C] Deck temperature will be specially assessed
Note: Where one of the internal spaces concerned is not refrigerated, the temperature of the space is to be taken as 5[°C].		

2.5 Cold cargo for ships other than liquefied gas carriers

2.5.1 For ships other than liquefied gas carriers, intended to be loaded with liquid cargo having a temperature below -10[°C], e.g. loading from cold onshore storage tanks during winter conditions, the material grade of cargo tank boundary plating is defined in Table 2.3.4 based on the following:

- t_c design minimum cargo temperature in [°C].

- steel grade corresponding to Class I as given in Table 2.3.3.

2.5.2 The design minimum cargo temperature, $t_{cargo,}$ is to be specified in the loading manual.

Section 3

Corrosion Protection

3.1 General

3.1.1 All steelwork, except inside tanks intended for the carriage of oil or bitumen, is to be protected against corrosion by application of suitable coating.

For protection required in salt water ballast spaces, Refer 3.6. For protection of cargo oil tanks of crude oil tankers, Refer 3.7. For protection required in holds of dry bulk cargo carriers, Refer Part 5, Chapter 1, Section 2, 2.11.

For the protection required in tanks carrying chemicals or other special cargoes, Refer Part 5, Chapter 3.

3.1.2 Where bimetallic connections are made, measures are to be incorporated to preclude galvanic corrosion.

3.1.3 Narrow spaces are generally to be protected by an efficient protective product, particularly at the ends of the ship where inspections and maintenance are not easily practicable due to their inaccessibility.

3.2 Surface preparation, prefabrication primers and paints or coatings

3.2.1 Steelwork is to be cleared of millscale and suitably cleaned before the application of surface paints and coatings. It is recommended that blast cleaning or other equally effective means be employed for this purpose.

3.2.2 Where a primer is used to coat steel after surface preparation and prior to fabrication, the composition of the coating is to be such that it will have no significant deleterious effect on subsequent welding work and that it is compatible with the paints or other coatings subsequently applied.

3.2.3 To determine the influence of the primer coating on the characteristics of welds, tests are to be made as detailed in Cl.3.2.4 to Cl.3.2.6.

3.2.4 Three butt weld assemblies are to be tested using plate material 20 to 25[mm] thick. A 'V' preparation is to be used and, prior to welding, the surfaces and edges are to be treated as follows:-

(a) Assembly 1 - Coated in accordance with the manufacturer's instructions.

(b) Assembly 2 - Coated to a thickness approximately twice the manufacturer's instructions.

(c) Assembly 3 - Uncoated.

3.2.5 Tests as follows are to be taken from each test assembly:-

(a) Radiographs - These are to have a sensitivity of better than 2 per cent of the plate thickness under examination, as shown by an image quality indicator.

(b) Photo-macrographs - These may be of actual size and are to be taken from near each end and from the centre of the weld.

(c) Face and reverse bend test - The test specimens are to be bent by pressure or hammer blows round a former of diameter equal to 3 times the plate thickness.

(d) Impact tests - These are to be carried out at ambient temperature, on three Charpy V-notch test specimens prepared in accordance with Part 2. The specimens are to be notched at the centreline of the weld, perpendicular to the plate surface.

3.2.6 The tests are to be carried out in the presence of a Surveyor of IRS or by an independent laboratory specializing in such work. A copy of the test report is to be submitted, together with radiographs and macrographs.

3.2.7 Paints or other coatings are to be suitable for the intended purpose in the locations where they are to be used. Unless previously agreed, at least two coats are to be applied.

3.2.8 The paint or coating is to be compatible with any previously applied primer, Refer Cl.3.2.2.

3.2.9 Paints, varnishes and similar preparations having a nitrocellulose or other highly flammable base, are not to be used in accommodation or machinery spaces.

3.2.10 In ships intended for the carriage of oil cargoes having a flash point below 60[°C] (closed cup test), paint containing aluminium should not in general be used in cargo tanks, adjacent ballast tanks, cofferdams, pump rooms as well as on deck above the mentioned spaces, nor in any other areas where cargo vapours may accumulate, unless it has been shown by appropriate tests that the paint to be used does not increase the incentive sparking hazard.

3.3 Internal cathodic protection

3.3.1 Impressed current cathodic protection systems are not permitted in any tank.

When a cathodic protection system is to be fitted in tanks for the carriage of liquid cargo with flash point not exceeding 60[°C], a plan showing details of the locations and attachment of anodes is to be submitted. The arrangements will be considered for safety against fire and explosion aspects only.

3.3.2 Particular attention is to be given to the locations of anodes in relation to the structural arrangements and openings of the tank.

3.3.3 Anodes are to be of approved design and sufficiently rigid to avoid resonance in the anode support. Weldable steel cores are to be fitted, and these are to be so designed as to retain the anode even when the anode is wasted.

3.3.4 Anodes are to be attached to the structure in such a way that they remain secure both initially and during service. The following methods of attachment would be acceptable:

(a) Steel core connected to the structure by continuous welding of adequate section.

(b) Steel core bolted to separate supports, provided that a minimum of two bolts with lock nuts are used at each support. The separate supports are to be connected to the structure by continuous welding of adequate section.

(c) Approved means of mechanical clamping.

3.3.5 Anodes are to be attached to stiffeners, or may be aligned in way of stiffeners on plane bulkhead plating, but they are not to be attached to the shell. The two ends are not to be attached to separate members which are capable of relative movement.

3.3.6 Where cores or supports are welded to the main structure, they are to be kept clear of the toes of brackets and similar stress raisers. Where they are welded to asymmetrical stiffeners, they are to be connected to the web with the welding kept at least 25[mm] away from the edge of the web. In the case of stiffeners or girders with symmetrical face plates, the connection may be made to the web or to the centreline of the face plate but well clear of the free edges. However, it is recommended that anodes are not fitted to face plates of high strength steel longitudinals.

3.4 Aluminium and magnesium anodes

3.4.1 Aluminium and aluminium alloy anodes are permitted in oil cargo tanks and also in tanks adjacent to oil cargo tanks only at locations where the potential energy does not exceed 275[J] (i.e. 28[kgf m]). The weight of the anode is to be taken as the weight at the time of fitting, including any inserts and fitting devices.

3.4.2 The height of the anode is, in general, to be measured from the bottom of the tank to the centre of the anode. Where the anode is located on or closely above a horizontal surface (such as a bulkhead girder) not less than 1[m] wide, provided with an upstanding flange or face plate projecting not less than 75[mm] above the horizontal surface, the height of the anode may be measured above that surface.

3.4.3 Aluminium anodes are not to be located under tank hatches or tank cleaning openings unless protected by adjacent structure.

3.4.4 Magnesium or magnesium alloy anodes are not permitted in oil cargo tanks and in tanks adjacent to oil cargo tanks.

3.5 External hull protection

3.5.1 Suitable protection for the underwater portion of the hull is to be provided.

3.5.2 Where an impressed current cathodic protection system is fitted, plans showing the proposed layout of anodes, reference cells, wiring diagram and the means of bonding-in of the rudder and propeller, are to be submitted.

3.5.3 The arrangements for glands, where cables pass through the shell, are to include a small cofferdam. Cables to anodes are not to be led through tanks intended for the carriage of low flash point oils. Where cables are led through cofferdams or clean ballast tanks of tankers, they are to be enclosed in a substantial steel tube of about 10[mm] thickness.

3.5.4 When Impressed Current Cathodic Protection (ICCP) systems or Sacrificial anodes are fitted, the system is to be capable of polarizing the underwater steel surface to a potential of about -750[Mv] versus the Ag/AgCl/seawater reference electrode, and to maintain the potential in this interval throughout

3.6 Corrosion protection coatings for salt water ballast spaces and double side skin spaces

3.6.1 The following spaces are to be coated during construction in accordance with the Performance Standards for Protective Coatings (PSPC) adopted by the IMO by Resolution MSC.215(82), as per applicability indicated in Cl.3.6.2.

(a) Dedicated salt water ballast tanks in all ships of 500GT and above.

(b) Double side skin spaces in bulk carriers of $L_L \geq 150$[m].

The following tanks are not considered to be dedicated salt water ballast tanks and therefore need not comply with the requirement of the Performance standard for protective coatings for dedicated salt water ballast tanks in all types of ships and double-side skin spaces of bulk carriers (MSC.215(82)), provided the coatings applied in the tanks described in ii) and iii) below are confirmed by the coating manufacturer to be resistant to the media stored in these tanks and provided such coatings are applied and maintained according to the coating manufacturer's procedures.

i) Salt water ballast tanks identified as "Spaces included in Net Tonnage" in the 1969 International Tonnage Convention Certificate;

ii) Salt water ballast tanks in passenger vessels also designated for the carriage of grey water;

iii) Salt water ballast tanks in livestock carriers also designated for the carriage of the livestock dung.

3.6.2 The IMO Performance Standards for Protective Coatings (PSPC) are to be applied to ships as follows:

i) In case of bulk carriers and tankers that are subject to the IACS common structural rules:

- which are contracted for construction on or after 8 December, 2006.

ii) In case of all ships other than those mentioned in i) above:

- Which are contracted for construction on or after 1 July 2008, or

- In the absence of a building contract, the keels of which are laid or which are at a similar stage of construction on or after 1 January 2009, or

- The delivery of which is on or after 1 July 2012.

3.6.3 For oil tankers and bulk carriers constructed on or after 1 July 1998 and for which Cl.3.6.1 and Cl.3.6.2 are not applicable, the following are to be complied with:

(a) All salt water ballast spaces are to have an efficient corrosion protection hard coating, epoxy or equivalent, applied in accordance with the manufacturer's requirements. The durability of the coatings could affect the frequency of survey of the spaces and light coloured coatings would assist in improving the effectiveness of subsequent surveys. It is therefore recommended that these aspects be taken into account by those agreeing to the specifications for the coatings and their application.

(b) The scheme for the selection, application and maintenance of the coating system to follow the requirements of IMO Resolution A.798(19) and contain, as a minimum the following documentation:

- Owner's, coating manufacturer's and shipyard's explicit agreement to the scheme for coating selection, application and maintenance.

- List of seawater ballast tanks identifying the coating system for each tank, including coating color and whether coating system is a hard coating.

- Details of anodes, if used.

- Manufacturer's technical product data sheet for each product.

- Manufacturer's evidence of product quality and ability to meet owner's requirements.

- Evidence of shipyard's and/or its subcontractor's experience in coating application.

- Surface preparation procedures and standards, including inspection points and methods.

- Application procedures and standards, including inspection points and methods.

- Format for inspection reports on surface preparation and coating application.

- Manufacturer's product safety data sheets for each product and owner's, coating manufacturer's and shipyard's explicit agreement to take all precautions to reduce health and other safety risks which are required by the authorities.

- Maintenance requirements for the coating system.

3.7 Corrosion protection for Cargo Oil tanks of Crude Oil Tankers

3.7.1 Cargo oil tanks of crude oil tankers are to be:

(i) Coated during construction in accordance with the Performance Standards for Protective Coating (PSPC-COT) adopted by the IMO by Resolution MSC. 288(87); or

(ii) Protected by alternative means of corrosion protection or utilization of corrosion resistance material to maintain required structural integrity for 25 years in accordance with the performance standard for alternative means of corrosion protection for cargo oil tanks of crude oil tankers, adopted by the IMO by Resolution MSC. 289(87).

3.7.2 The requirements given in Cl.3.7.1 are to be applied for crude oil tanker greater than 5000 tonnes deadweight where:

(i) Building contract is placed on or after 1st January 2013;

(ii) In absence of building contract, the keels of which are laid or which are at a similar stage of construction on or after 1st July 2013;

(iii) Or the delivery of which is on or after 1st January 2016.

3.7.3 The Administration may exempt a crude oil tanker from the requirements of Cl.3.7.1 to allow the use of prototype alternatives to the coating system specified in Cl.3.7.1 (i), for testing, provided they are subject to suitable controls, regular assessment and acknowledgement of the need of immediate remedial action if the system fails or is shown to be failing. Such exemption is to be recorded on an exemption certificate.

3.7.4 The Administration may exempt a crude oil tanker from the requirement of Cl.3.7.1 if the ship is built to be engaged solely in the carriage of cargoes and cargo handling operations not causing corrosion. Such exemption and conditions for which it is granted is to be recorded on an exemption certificate. (See MSC.1/Circ.1421 "Guidelines on exemption for crude oil tankers solely engaged in the carriage of cargoes and cargo handling operations not causing corrosion").

3.8 Protection against galvanic corrosion in tanks

3.8.1 Non-stainless steel is to be electrically insulated from stainless steel or from aluminium alloys.

3.8.2 Where stainless steel or aluminium alloys are fitted in the same tank as non-stainless steel, a protective coating is to cover both materials.

3.9 Protection of bottom by ceiling

3.9.1 Where ceilings are fitted it is to be laid over the inner bottom and lateral bilges.

3.9.2 Planks forming ceiling over the bilges and on the inner bottom are to be easily removable to permit access for maintenance.

3.9.3 Where the double bottom is intended to carry fuel oil, ceiling on the inner bottom is to be separated from the plating by means of battens, in order to facilitate the efficient drainage of oil leakages to the bilges.

3.9.4 Where the double bottom is intended to carry water, ceiling on the inner bottom may lie next to the plating, provided a suitable protective composition is applied prior to fitting of ceiling.

3.9.5 The attachment of ceiling is not to affect the tightness of the inner bottom.

3.10 Protection of decks by wood sheathing

3.10.1 Where plated decks are sheathed with wood or an approved composition, reductions in plate thickness may be allowed.

3.10.2 The steel deck is to be coated with a suitable material in order to prevent corrosive action, and the sheathing or composition is to be effectively secured to the deck.

3.10.3 Deck coverings in the following positions are to be of a type which will not readily ignite or give rise to smoke or toxic or explosive hazards at elevated temperatures in accordance with the requirements of the International Code for the Application of Fire Test Procedures:

(a)forming the crown of machinery or cargo spaces within accommodation spaces of cargo ships

(b)within accommodation spaces, control stations, stairways and corridors of passenger ships.

3.10.4 Before fitting the wood sheathing, deck plates are to be provided with suitable protective coating.

Section 4

Net Scantling Approach

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

t_c = Corrosion addition, in [mm].

4.1 Application

4.1.1 The net scantling approach is the method of obtaining the required thickness in accordance with the rules before adding corrosion margin. The scantlings obtained by applying these criteria are net scantlings which provide the structural strength required to sustain the design loads, giving certain allowance for corrosion that is likely to occur during the operation phase of the ship.

4.1.2 No credit is given in the assessment of structural capability for the presence of coatings or similar corrosion protection systems.

4.1.3 Any additional thicknesses specified by the owner, as Owner's extras, are not included in the assessment of the required new building thickness.

4.1.4 The strength is to be assessed using the net thickness approach on all scantlings which is the minimum scantling that must be kept throughout the service life of the ship.

4.2 Local and global corrosion

4.2.1 The net scantling approach distinguishes between local and global corrosion.

4.2.2 Local corrosion is defined as uniform corrosion of local structural elements, such as a single plate or stiffener.

4.2.3 Global corrosion is defined as the average corrosion of larger areas, such as primary supporting members and the hull girder. As the hull girder cross section does not corrode evenly, the reduction of the hull girder sectional properties will be less than the sum of the allowable local wastage allowance for members contributing to the longitudinal strength.

4.3 Exceptions in gross scantling

4.3.1 The scantlings of the below structures are directly determined in terms of gross scantlings and do not follow the net scantling approach. They already include additions for corrosion but without any owner's extra margin. They are:

- Bow doors and inner doors
- Deck equipment
- Massive pieces made of steel forgings, steel castings or iron castings

- Rudder structures and hull appendages

- Side doors and stern doors

- Scantlings of superstructures and deckhouses as given in Chapter 7, Section 8.

Note: The side plating of superstructure is based on net scantling approach as given in Chapter 7, Cl.8.3.1.2, similar to side shell plating. For aft and front plating of superstructures and all plating of deckhouses, gross scantling approach is followed. In cases where the deckhouses are effective in contributing to hull girder strength, the corrosion margins given in Chapter 2, Table 5.1.1 are to be used to determine the reduction in thickness due to corrosion. Half the stipulated corrosion margin is to be deducted from the gross scantlings to arrive at the net-50 scantlings for calculation of hull girder section modulus. For fatigue calculations, 50% corrosion margin is deducted whereas for buckling calculations, 100% corrosion margin is deducted.

4.4 Gross and net scantling definitions

4.4.1 Gross required thickness: The gross required thicknesses (t_{gr} in [mm]) are obtained by adding the corrosion addition t_c, as defined in Part 3, Chapter 2, Section 5 to the net required thickness (t_{net}), as follows:

$$t_{gr} = t_{net} + t_c$$

4.4.2 Gross offered thickness: The gross offered thickness (t_{gr_off} in [mm]) is the gross thickness at the new building stage, which is obtained by deducting any thickness for voluntary addition from the as-built thickness, as follows:

$$t_{gr_off} = t_{as_built} - t_{vol_add}$$

4.4.3 Net offered thickness: The net offered thickness (t_{off} in [mm]) is obtained by subtracting the corrosion addition (t_c) from the gross offered thickness (t_{gr_off}), as follows:

$$t_{off} = t_{gr_off} - t_c$$

4.5 Rounding off of scantlings

4.5.1 The rounding off of the gross required plate thickness, t_{gr}, calculated in accordance with the rule requirements is obtained by rounding to the nearest half millimeter.

For example:

- for $10.75 \leq t_{gr} < 11.25$[mm], the rule required gross thickness is 11.0[mm].

- for $11.25 \leq t_{gr} < 11.75$[mm], the rule required gross thickness is 11.5[mm].

4.6 Strength and scantling compliance

4.6.1 The required strength characteristics are:

4.6.1.1 Thickness, for plating including that which constitutes primary supporting members.

4.6.1.2 Section modulus, shear area, moments of inertia and local thickness, for ordinary stiffeners as well as primary supporting members

4.6.1.3 Section modulus, moments of inertia and first moment for the hull girder.

4.6.1.4 Grillage analysis, Buckling capacity of primary supporting members

4.6.1.5 Buckling/collapse capacity of hull girder

4.6.2 The required scantling characteristics are:

4.6.2.1 The net offered thickness of plating to be equal to or greater than the net required thickness of plating.

4.6.2.2 The required net section modulus, moment of inertia and shear area properties of local supporting members are to be calculated using the net thickness of the attached plate, web and flange. The net sectional dimensions of local supporting members are defined in Figure 4.6.2.

4.6.2.3 The offered net sectional properties of primary supporting members and the hull girder are to be equal to or greater than the required net sectional properties which are to be based on the gross offered scantling with a reduction of the applicable corrosion addition, as specified in Table 4.6.2, applied to all component structural members.

4.6.2.4 The strength assessment methods prescribed are to be assessed by applying the corrosion addition specified in Table 4.6.2 to the offered gross scantlings. Half of the applied corrosion addition specified in Table 4.6.2 is to be deducted from both sides of the structural members being considered.

4.6.2.5 Corrosion additions are not to be taken less than those given in Section 5, Table 5.1.1.

4.7 Minimum requirements

4.7.1 Minimum requirements specify the minimum scantling requirements which is to be applied irrespective of all other requirements. The minimum requirements are usually in one of the following forms:

— Minimum thickness

— Minimum stiffness and proportions.

4.8 Considered net scantling

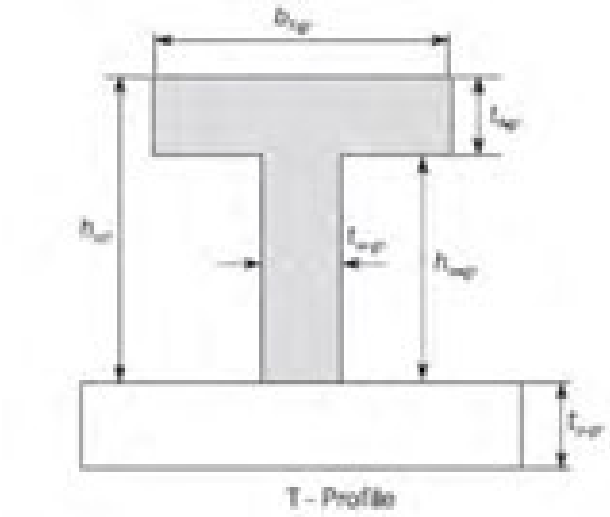
4.8.1 Yielding check of the hull girder: The net thickness of structural members to be considered for the yielding check of the hull girder, according to Chapter 6, Section 3, is to be obtained by deducting $0.5t_c$ from the gross thickness offered.

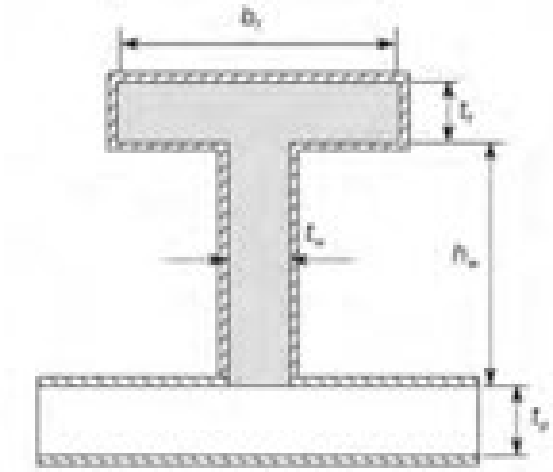
4.8.2 Buckling check of the hull girder: The net thickness of structural members to be considered for the buckling check, according to Chapter 9, Section 2, is to be obtained by deducting t_c from the gross thickness offered.

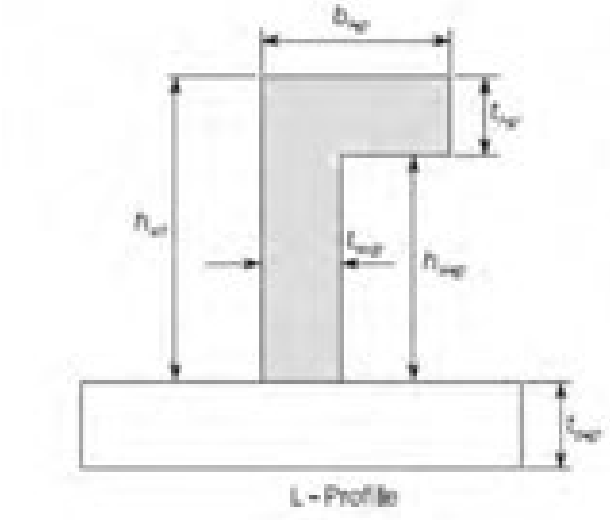
4.8.3 Ultimate strength check of the hull girder: The net thickness of structural members to be considered for the ultimate strength check of the hull girder, according to Chapter 6, Section 4, is to be obtained by deducting $0.5t_c$ from the gross thickness offered.

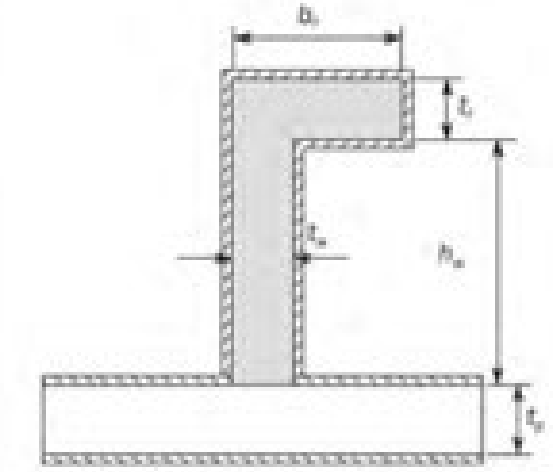
4.8.4 All buckling capacity assessments are to be based on corrosion addition, as defined in Table 4.6.2.

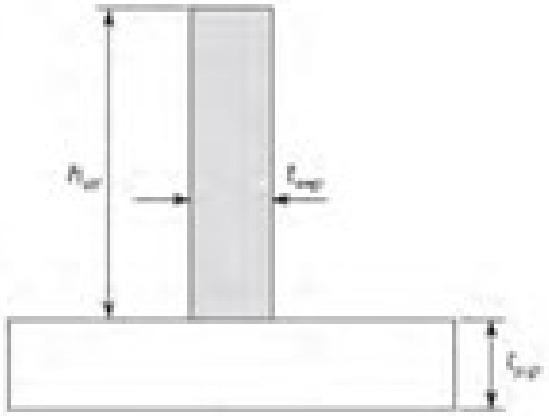
4.8.5 FE models for vibration analysis to be based on gross offered thickness.











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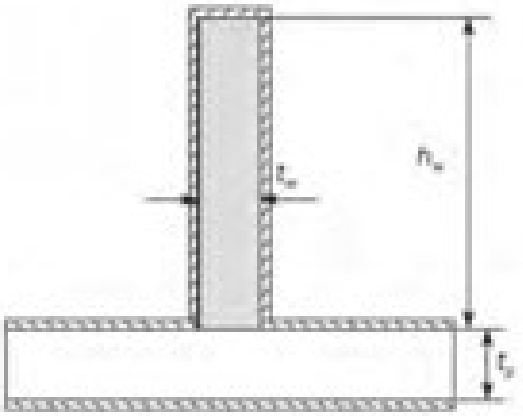

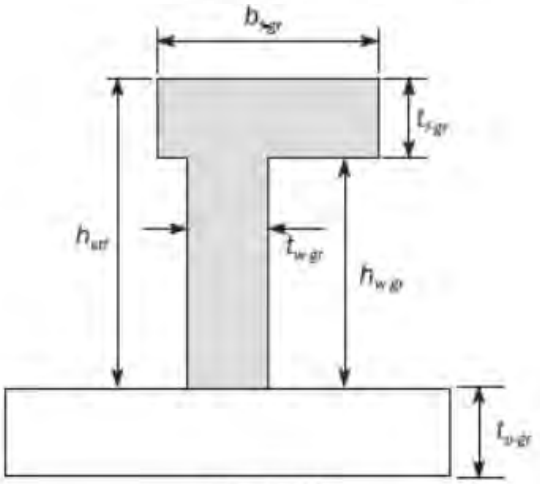

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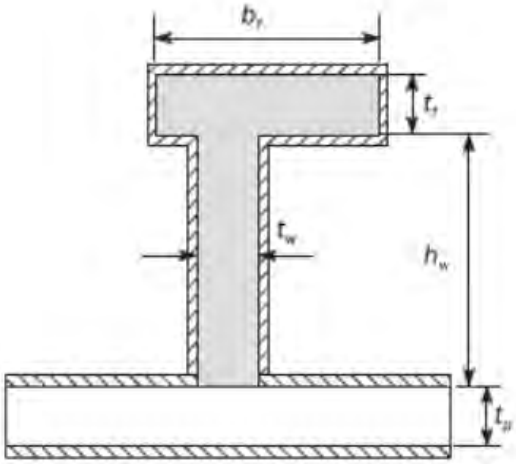

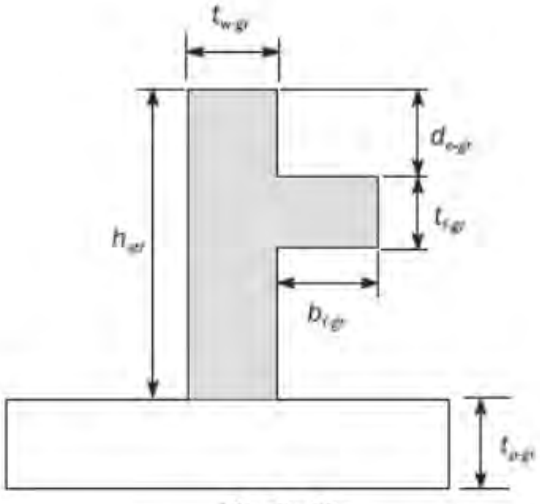

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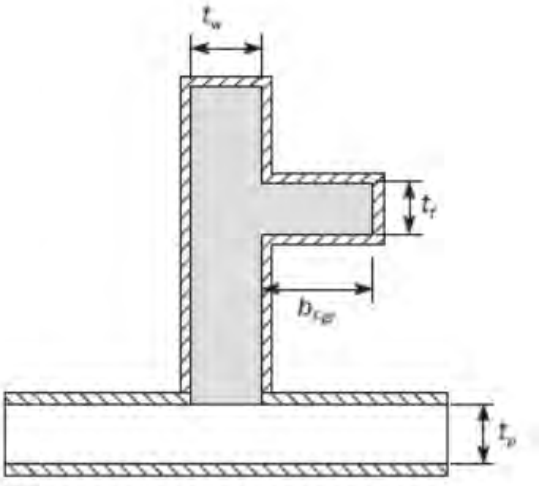


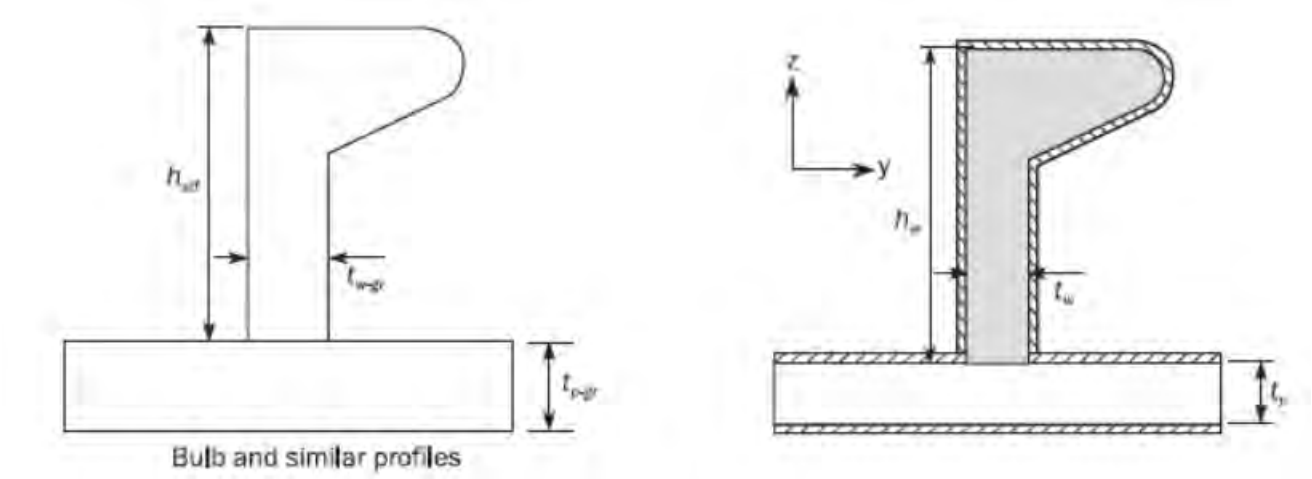


Figure 4.6.2: Net sectional properties of local supporting members

Table 4.6.2: Applied corrosion addition for structural assessment		
Structural requirement	**Property/analysis type**	**Applied corrosion addition for structural assessment**
Minimum thickness all members including primary supporting members (PSM)	Thickness	t_c
Local strength member check (Plates, stiffeners, and hold frames)	Thickness / Sectional properties	t_c
	Proportions / Buckling capacity	t_c
Primary supporting members (Prescriptive check)	Grillage analysis	$0.5t_c$
	Sectional properties	$0.5t_c$
	Proportions of web and flange Buckling capacity	t_c
Strength assessment by Finite Element Method (FEM)	Global FE Model	$0.5t_c$
	Local fine mesh finite element model	$0.5t_c$
	Cargo tank/cargo hold finite element model	$0.5t_c$
	Buckling capacity	t_c
Hull girder strength (Prescriptive check)	Sectional properties/Hull girder yield check	$0.5t_c$ [1)]
	Buckling capacity	t_c
Hull girder ultimate strength	Sectional properties	$0.5t_c$
	Buckling /collapse capacity	$0.5t_c$
Fatigue assessment (simplified stress analysis)	Hull girder section properties Local supporting member	$0.5t_c$
Fatigue assessment (Finite Element Stress analysis)	Very fine mesh Finite Element model	$0.5t_c$
1) For vertical hull girder bending and hull girder shear strength check, as per Chapter 6, Section 2, Cl.2.1.2, Cl.2.1.3 and Cl.2.2.2, gross offered thickness to be applied.		

Section 5

Corrosion Additions

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

t_{c1}, t_{c2} = Corrosion addition, in [mm], on each of the two sides of the considered structural member, as defined in Table 5.1.1.

t_{res} = Reserve thickness, taken as 0.5[mm], also to be applied in non-corrosive environment and for stainless steel and aluminium to account for possible thickness reductions, e.g. wear and tear, grinding or under tolerance thickness.

5.1 Application

5.1.1 The corrosion additions given in these Rules are applicable to carbon-manganese steels, stainless steels, stainless clad steels and aluminium alloys. Corrosion addition for the exposed carbon steel side of stainless clad structure to be taken from Table 5.1.1 for the corresponding compartment.

5.2 Corrosion addition determination

5.2.1 The corrosion addition for each of the two sides of a structural member, t_{c1} or t_{c2}, is specified in Table 5.1.1.

5.2.2 The total corrosion addition, t_c [mm], for both sides of the structural member is obtained by the following formula:

$$t_c = \text{Roundup}_{0.5}\,(t_{c1} + t_{c2}) + t_{res}$$

$\text{Roundup}_{0.5}$ (t) means that "t" is rounded to the upper half millimeter.

For L<90[m], $\text{Roundup}_{0.5}$ (t) need not be considered in the total corrosion addition.

5.2.3 For an internal member within a given compartment, the total corrosion addition, t_c [mm], is obtained from the following formula:

$$t_c = \text{Roundup}_{0.5}\,(2t_{c1}) + t_{res}$$

where t_{c1} is the value specified in Table 5.1.1 for one side exposure to that compartment.

For L<90[m], $\text{Roundup}_{0.5}$ (t) need not be considered in the total corrosion addition.

5.2.4 The total corrosion addition, t_c [mm], for compartment boundaries and internal members made from stainless steel, or aluminium is to be taken as:

$$\mathbf{t_c = t_{res} = 0.5}$$

5.2.5 In case of stainless clad steel, the corrosion additions, t_{c1}, for the carbon steel side and t_{c2}, for the stainless-steel side are respectively to be taken as:

$\mathbf{t_{c1}}$ as specified for the corresponding compartment in Table 5.1.1

$\mathbf{t_{c2}}$ = 0

5.3 Stiffener

5.3.1 The corrosion addition of a stiffener is determined according to the location of its connection to the attached plating.

5.3.2 When an elementary plate panel or a stiffener is affected by more than one value of corrosion addition, the largest value is to be applied.

5.3.3 The net strength characteristics are to be calculated for the net transverse section and are to be obtained by deducting the corrosion addition $\mathbf{t_c}$ from the gross thickness of the elements which constitute the stiffener profile.

Table 5.1.1: Corrosion addition for one side of a structural member				
Sl.No.	**Compartment Type**	**Structural Member**	**Corrosion Addition (t_{c1} or t_{c2}) [mm]**	
			General	Special case
1	Tanks for Ballast water[1)] /Sea water	All members[6)]	1.00	1.25[2)]
2	Tanks for Cargo Oil[1)] and Slop Tank	All members	1.00	1.25[2)]
3	Tanks for Liquid Chemical	All members	1.00	
4	Tanks for fuel oil, lube oil	All members	0.50	1.00[2)]
5	Tanks for Brine, Urea, Bilge Tank, Drain Storage, Chain Locker	Face plates of PSM's	1.00	
		Other members	1.00	
6	Tanks for fresh water	All members	0.50	
7	Other tanks - cargo tanks only intended to carry fresh water, fuel oil, lube oil, RSW or mud.	All members	0.50	
8	Dry Bulk Cargo[3), 7)]	General	1.00	
		Inner bottom plating	1.50	
		Sloping stool plate of hopper tanks and lower stool		
		Transverse bulkhead plating		
		Brackets, Frames, ordinary stiffeners and primary supporting members	1.00	1.5[4)]
9	Containers and General cargo hold	All members in hold	1.00	
10	Accommodation spaces	All members	0.00	
11	Void, dry spaces and other category spaces not mentioned above[5)].	Bottom plate of the compartment / Upper surface of horizontal stringers in double hull spaces	0.5	
		Elsewhere	0.0	
12	Stainless steel and aluminium (independent of compartment type)	All members	0.00	
13	External Surfaces (exposed to air/water)	All members	0.50	

(1) Ballast tank: also includes combined ballast and cargo oil tanks but does not include cargo oil tanks which may carry ballast according to Regulation 18 of MARPOL 73/78 as amended.
(2) This minimum corrosion addition is applied in the area within 1.5[m] below the top of the tank. This is applied to ballast or liquid cargo tanks with exposed weather deck as the tank top.
(3) Dry bulk cargo hold includes holds intended for the carriage of dry bulk cargoes, which may carry oil or water ballast.
(4) This minimum corrosion addition is applied in the area within 1.5[m] above the bottom of hold, e.g. inner bottom. The bottom of hold is defined as the lowest horizontal boundaries of the hold.
(5) Applicable for the spaces containing membrane or independent cargo tanks of gas carriers. Also applicable for semi-enclosed compartments, e.g. open ro-ro spaces.
(6) 0.5[mm] is to be added to the plate surface exposed to ballast for the plate boundary between water ballast and heated cargo oil tanks or heated fuel/lube oil tanks/slop tanks. 0.3[mm] is to be added to each surface of the web and face plate of a stiffener in a ballast tank and attached to the boundary between water ballast and heated cargo oil tanks or heated fuel/lube oil tanks/slop tanks. Heated oil tanks are defined as tanks/slop tanks arranged with any form of heating capability (the most common type is heating coils). For L<90[m], these additional requirements need not be considered.
(7) For cargo holds carrying cement, corrosion additions need not to be considered.

End of Chapter

Chapter 3

General Arrangement and Structural Design Principles

Contents

Section

1 General
2 Subdivision Arrangement
3 Compartment Arrangement
4 Access Arrangements
5 Structural Arrangements
6 Structural Detail Design
7 Structural Idealization

Section 1

General

1.1 Application

1.1.1 This chapter covers the arrangement design and structural design principles.

Section 2

Subdivision Arrangement

2.1 Watertight bulkhead arrangement

2.1.1 Number and disposition of watertight bulkheads

2.1.1.1 The following transverse watertight bulkheads are to be fitted in all ships:

- A collision bulkhead;

- An after peak bulkhead;

- One bulkhead forward of the machinery space, and one bulkhead at the aft end of the machinery space which may be the aft peak bulkhead.

2.1.1.2 In the case of ships with an electrical propulsion plant, both the generator room and the engine room are to be enclosed by watertight bulkheads.

2.1.1.3 In addition to the requirements of Cl.2.1.1.1 and Cl.2.1.1.2, the number and disposition of bulkheads are to be arranged to meet the requirements for subdivision, floodability and damage stability in accordance with the requirements of the National Authority of the country in which the ship is registered.

2.1.1.4 For ships without longitudinal bulkheads in the cargo region, additional transverse watertight bulkheads are to be fitted so that the total number of bulkheads is not less than that given in Table 2.1.1.1.

2.1.1.5 The ordinary transverse watertight bulkheads in the holds should be spaced at reasonably uniform intervals. Where non uniform spacing is unavoidable and the length of a hold is unusually large, the transverse strength of the ship is to be maintained by

providing additional web frames, increased framing etc. and details are to be submitted.

2.1.1.6 Proposals to dispense with one or more of these bulkheads will be considered, subject to suitable structural compensation, in case they interfere with the requirements of a special trade.

2.1.1.7 The watertight bulkheads are to extend to the bulkhead deck. In exceptional cases at the request of the Owner, IRS may allow transverse watertight bulkheads to terminate at a deck below that from which freeboard is measured, provided that this deck is at an adequate distance above the full load waterline.

2.1.1.8 Where it is not practicable to arrange a watertight bulkhead in one plane, a stepped bulkhead may be fitted. In this case, the part of the deck which forms the step is to be watertight and equivalent in strength to the bulkhead.

2.1.1.9 In ships with a raised quarter deck, the watertight bulkheads within the quarter deck region are to extend to this deck.

2.1.1.10 Bulkhead requirements for specific ships types, Refer to Part 5.

Table 2.1.1.1: Total number of bulkheads

Ship Length [m]	Engine Room Location	
	Aft*	Elsewhere
L ≤ 65	3	4
65 < L ≤ 85	4	4
85 < L ≤ 105	4	5
105 < L ≤ 125	5	6
125 < L ≤ 145	6	7
145 < L ≤ 165	7	8
165 < L ≤ 190	8	9
L > 190	Specially considered	

(*) With Aft peak bulkhead forming after boundary of machinery space.

2.2 Collision bulkhead

2.2.1 Arrangement of collision bulkhead

2.2.1.1 A collision bulkhead is to be fitted which is to be watertight up to the bulkhead deck of passenger ships and the freeboard deck of cargo ships. It is to extend, preferably in one plane, to the freeboard deck.

2.2.1.2 In the case of vessels having long superstructures at the fore end, it is to be extended weathertight to the deck next above the bulkhead deck of passenger ships and the freeboard deck of cargo ships. The extension need not be fitted directly over the bulkhead below, provided that the location of the extension meets the following requirements in Cl.2.2.2 and the part of the deck which forms the step is made effectively weathertight. Any recesses or steps in collision bulkheads are to fall within the freeboard deck, the extension need only be to weathertight standards. If a step occurs at that deck, the deck also need be to only weathertight standards in way of the step unless the step forms the crown of a tank.

2.2.1.3 In ships fitted with visor or bow doors, in which a sloping loading ramp forms part of the collision bulkhead above the bulkhead deck of passenger ships and the freeboard deck of cargo ships, that part of the ramp which is more than 2.3[m] above the freeboard deck may extend forward of the minimum limit specified below in Cl.2.2.2. Such a ramp is to be weathertight over its complete length. Refer Figure 2.2.1.1.

Ramps not meeting the requirements in Cl.2.2.2 is not considered as an extension of the collision bulkhead.

2.2.1.4 Doors, manholes, permanent access openings or ventilation ducts are not to be cut in the collision bulkhead below the bulkhead deck of passenger ships and the freeboard deck of cargo ships. Where the collision bulkhead is extended above the freeboard deck, the number of openings in the extension is to be kept to a minimum compatible with the design and proper working of the ship.

2.2.2 Position of collision bulkhead

2.2.2.1 The distance $[X_c]$, in [m] from perpendicular FP_{LL} to the collision bulkhead is to be between the following limits. Refer Figure 2.2.2.1 and Figure 2.2.2.2 for $X_{c,min}$ and $X_{c,max}$.

$$X_{c,min} \begin{cases} = 0.05L_L - X_R & \text{for } L_L < 200 \\ = 10 - X_R & \text{for } L_L \geq 200 \end{cases}$$

$$X_{c,max} \begin{cases} = 0.05L_L - X_R + 3 & \text{for } L_L < 100 \\ = 0.08L_L - X_R & \text{for } L_L \geq 100 \end{cases}$$

2.2.2.2 For ships having any part of the underwater body extending for'd of the forward perpendicular (ships with bulbous bows), the reference point, (RP) is to be measured from the forward end of L_L a distance X_R forward; where X_R is to be taken as the least of the following:

- 0.5G;
- $0.015L_L$;
- 3.0[m], where

G = the distance in [m] from forward perpendicular to the forward end of the protruded part

L_L = the load line length of the vessel in [m], as per International Load Line Convention (Refer Chapter 1, Section 2).

T_L = the draught in [m] at 85% of the least moulded freeboard depth, D_L. Refer Figure 2.2.2.1.

2.2.2.3 For ships without bulbous bows the reference point is to be taken where the forward end of L_L coincides with the forward side of stem, on the waterline which L_L is measured that is $X_R = 0$, Refer Figure 2.2.2.1.

2.2.2.4 Consideration will however be given to proposals for the collision bulkhead positioned aft of the limits given in Cl.2.2.2.1, provided that the application is accompanied by calculations showing that with the ship fully loaded to summer draught on even keel, flooding of space forward of the collision bulkhead will not result in any part of the bulkhead deck / freeboard deck becoming submerged, nor result in any unacceptable loss of stability.

Definition:

Forward load line perpendicular: The forward load line perpendicular, FP_{LL}, is to be taken at the forward end of the length L_L and is to coincide with the foreside of the stem on the waterline on which the length L_L is measured.

After load line perpendicular: The after load line perpendicular, AP_{LL}, is to be taken at the aft end of the length L_L.

Moulded freeboard depth: The moulded freeboard depth D_L is the least moulded depth taken as the vertical distance, in [m], from the top of the keel to the top of the freeboard deck beam at the side. Where the freeboard deck is stepped and the raised part of the deck extends over the point at which the moulded depth is to be determined, the moulded depth is to be measured to a line of reference extending from the lower part of the deck along a line parallel with the raised part.

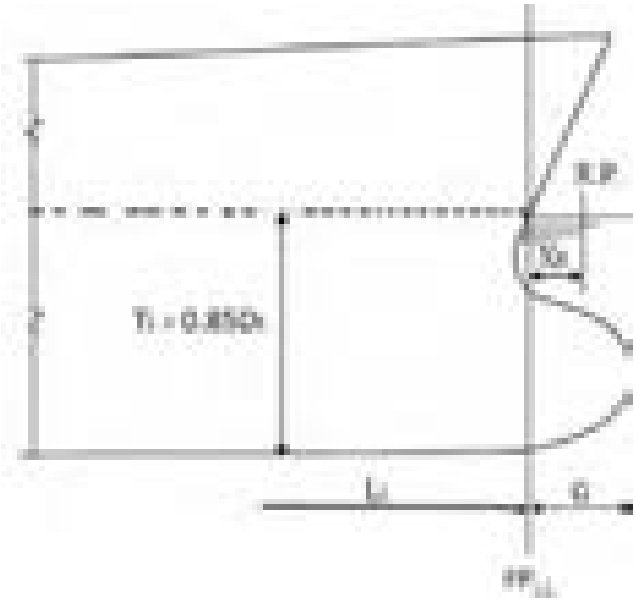


Figure 2.2.2.1: Reference Point for Vessels with Bulbous Bow

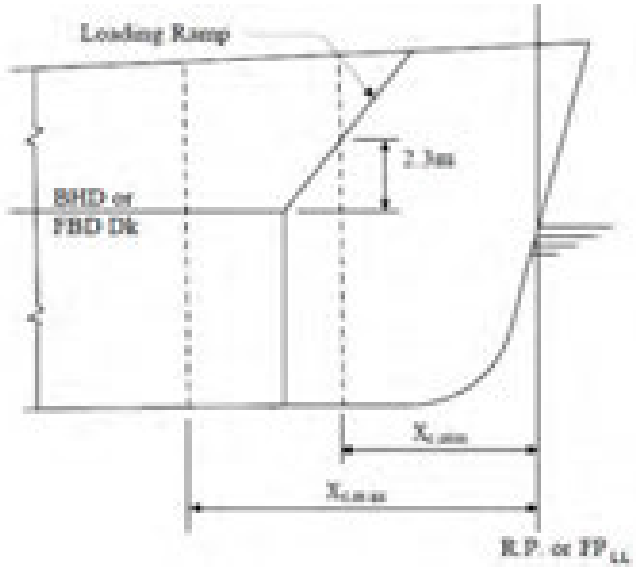


Figure 2.2.1.1: Collision Bulkhead in Vessels with Bow Door

2.3 After peak, Machinery space bulkheads and shaft tunnel

2.3.1 All ships are to have an after peak bulkhead generally enclosing the stern tube and rudder trunk in a watertight compartment. In twin screw ships where the bossing ends forward of the after peak bulkhead, the stern tubes are to be enclosed in suitable watertight spaces inside or aft of the shaft tunnels.

2.3.2 In passenger ships, the stern gland is to be situated in a watertight shaft tunnel or other watertight space separate from the stern tube compartment and of such volume that if flooded by leakage through the stern gland, the bulkhead deck will not be immersed.

2.3.3 In cargo ships, where the inboard end of the stern tube extends into the engine room, provision of an approved watertight/ oil tight gland system for sealing of the inboard end of the stern tube at the aft peak/engine room watertight bulkhead is considered sufficient to minimize danger of water penetrating into the ship in case of damage to stern tube arrangements. (Refer also Part 4, Chapter 4, Section 6, 6.16).

2.3.4 For ships less than 65[m], where the after peak bulkhead in way of the stern tube stuffing box is not provided, stern tubes are to be enclosed in watertight spaces of moderate volume.

2.3.5 In ships with engines situated amidships, a watertight shaft tunnel is to be arranged. Openings in the forward end of shaft tunnels are to be fitted with watertight sliding doors capable of being operated from a position above the load water line.

2.3.6 An afterpeak bulkhead also to be fitted and made watertight up to the bulkhead deck or the freeboard deck. The after peak bulkhead may be stepped below the bulkhead deck or the freeboard deck, provided the degree of safety of the ship as regards subdivision is not thereby diminished.

2.3.7 Bulkheads are to be fitted separating the machinery space from cargo and accommodation spaces forward and aft and made watertight up to the bulkhead deck of passenger ships and the freeboard deck of cargo ships.

2.4 Height of bulkheads

2.4.1 The watertight bulkheads are in general to extend to the bulkhead deck/freeboard deck. The afterpeak bulkhead however, may terminate at the first deck above the load waterline, provided that this deck is made watertight to the stern or to a watertight transom floor provided the degree of safety of the ship as regards the subdivision is not thereby diminished.

2.4.2 The collision bulkhead is normally to extend to the to the bulkhead deck of passenger ships and the freeboard deck of cargo ships or, in the case of ships with combined bridge and forecastle or a long superstructure that includes a forecastle, to the superstructure deck. However, if a ship is fitted with more than one complete superstructure deck, the collision bulkhead may be terminated at the deck next above the bulkhead deck / freeboard deck. Where the collision bulkhead extends above the bulkhead deck / freeboard deck, the extension need only be to weathertight standard.

2.4.3 In passenger ships of restricted draught and all ships of unusual design, the height of bulkheads will be specially considered.

2.5 Openings in watertight bulkheads and closing appliances – General

2.5.1 Openings may be accepted in watertight bulkheads except in that part of collision bulkhead which is situated below the bulkhead deck / freeboard deck. The number of openings in watertight subdivisions is to be kept to a minimum compatible with the design and proper working of the ship. Where penetrations of watertight bulkheads are necessary for access, piping, ventilation, electrical cables, etc., arrangements are to be made to maintain the watertight integrity. Relaxation in the water tightness of openings above the bulkhead deck / freeboard deck may be considered provided it is demonstrated that any progressive flooding can be easily controlled and that the safety of the ship is not impaired.

Ventilation penetrations through watertight subdivision bulkheads are to be avoided. Where penetrations are unavoidable, the ventilation ducting is to satisfy watertight bulkhead requirements or watertight closing appliances are to be installed at the bulkhead penetrations. For ventilation penetrations below the bulkhead deck or below damage equilibrium waterlines, the closing appliances are to be operable from the bridge. Where the penetration is located above the bulkhead deck and damage waterline, local manual controls may be provided at the closing appliances, on one or both sides of the bulkhead, so that the controls will be accessible in the prescribed flooded conditions.

2.5.2 Openings in the collision bulkhead above the bulkhead deck / freeboard deck are to have weathertight doors or an equivalent arrangement.

2.5.3 Doors, manholes, permanent access openings or ventilation ducts are not to be cut in the collision bulkhead below the bulkhead deck of passenger ships and the freeboard deck of cargo ships.

2.5.4 Where watertight bulkhead stiffeners are cut in way of watertight doors in the lower part of the bulkhead, the opening is to be suitably framed and reinforced. Where stiffeners are not cut but the spacing between the stiffeners is increased on account of watertight doors, the stiffeners at the sides of the doorways are to be increased in depth and strength so that the efficiency is at least equal to that of the unpierced bulkhead, without taking the stiffeners of the door-frame into consideration.

2.5.5 Lead or other heat sensitive materials may not be used in systems which penetrate watertight subdivision bulkheads, where

deterioration of such systems in the event of fire would impair the watertight integrity of the bulkheads.

2.5.6 Valves not forming part of a piping system are not permitted in watertight subdivision bulkheads.

2.6 Watertight ventilators and trunks

2.6.1 Watertight ventilators and trunks are to be carried at least up to the bulkhead deck in passenger ships and up to the freeboard deck in ships other than passenger ships.

2.6.2 In passenger ships, where a ventilation trunk passing through a structure penetrates a watertight area of the bulkhead deck, the trunk is to be capable of withstanding the water pressure that may be present within the trunk, after having taken into account the maximum heel angle during flooding, in accordance with regulation SOLAS Chapter II-1, Regulation 7-2.

2.6.3 In ro-ro passenger ships, where all or part of the penetration of the bulkhead deck is on the main ro-ro deck, the trunk is to be capable of withstanding impact pressure due to internal water motions (sloshing) of water trapped on the ro-ro deck.

2.7 Doors in watertight bulkheads for ships where subdivision / damage stability requirements are applicable

2.7.1 General

The requirements in Cl.2.7.1 to Cl.2.7.15 apply to doors located in way of the internal watertight subdivision boundaries and the external watertight boundaries necessary to ensure compliance with the relevant subdivision and damage stability regulations. These requirements do not apply to doors located in external boundaries above equilibrium or intermediate waterplanes.

2.7.2 Definitions

For the purpose of the requirements in this subsection 2.7 the following definitions apply:

2.7.2.1 *Watertight*: Capable of preventing the passage of water in any direction under a design head. The design head for any part of a structure is to be determined by reference to its location relative to the bulkhead deck or freeboard deck, as applicable, or to the most unfavorable equilibrium / intermediate waterplane, in accordance with the applicable subdivision and damage stability regulations, whichever is the greater. A watertight door is thus one that will maintain the watertight integrity of the subdivision bulkhead in which it is located.

2.7.2.2 *Equilibrium Waterplane*: The waterplane in still water when, taking account of flooding due to an assumed damage, the weight and buoyancy forces acting on a vessel are in balance. This relates to the final condition when no further flooding takes place or after cross flooding is completed.

2.7.2.3 *Intermediate Waterplane*: The waterplane in still water, which represents the instantaneous floating position of a vessel at some intermediate stage between commencement and completion of flooding when, taking account of the assumed instantaneous state of flooding, the weight and buoyancy forces acting on a vessel are in balance.

2.7.2.4 *Sliding Door or Rolling Door*: A door having a horizontal or vertical motion generally parallel to the plane of the door.

2.7.2.5 *Hinged Door*: A door having a pivoting motion about one vertical or horizontal edge.

2.7.3 Structural design

Doors are to be of approved design and are to be of a strength equivalent to that of the subdivision bulkheads in which they are fitted.

2.7.4 Operation mode, location and outfitting

Doors are to be fitted in accordance with all requirements regarding their operation mode, location and outfitting, i.e. provision of controls, means of indication etc. as shown in Table 2.7.4.

For passenger ships, in addition to the requirements given in the Table 2.7.4 the watertight doors and their controls are to be located in compliance with the following:

(a) The door is to be located inboard of the damage zone B/5 on P&S as per SOLAS II- 1 Regulation.13.7.

(b) The door controls including hydraulic piping and electric cables are to be kept as close as practicable to the bulkhead in which the doors

are fitted in order to minimize the likelihood of them being involved in any damage to the ship. The positioning of doors and controls is to be such that if the ship sustains damage within damage zone B/5 as mentioned above, the operation of the doors clear of the damaged portion of the ship is not impaired.

Table 2.7.4: Doors in Internal Watertight bulkheads and External Watertight Boundaries in Passenger ships and Cargo ships

A. Door in Internal Watertight Bulkheads

Position relative to bulkhead or freeboard deck	1. Regulation	2. Frequency of use while at sea	3. Type	4. Remote Closure	5. Remote Indication	6. Audible or Visual Alarm	7. Notice	8. Comments
I. Passenger Ships								
(1)Below	SOLAS II-1/10, 13.4, 13.5.1, 13.5.2,13.6, 13.7.1,13.8.1, 13.8.2, 16.2, 22.1, 22.3 and 22.4	Used	POS	Yes	Yes	Yes (local)	No	For doors that are used, see SOLAS II-1/22.3 and IMO MSC.1/Circ.1564
	SOLAS II-1/10, 13.9.1, 13.9.2, 14.2, 16.2, 22.2 and 22.5	Perm. Closed	S, H	No	No	No	Yes	See Notes 3, 4 & 6
(2)At or above	SOLAS II-1/10, 16.2, 17.1 and 22.3	Used	POS, POH	Yes	Yes	Yes (local)	No	See Note 7
	SOLAS II-1/17-1.1, 17-1.2, 17-1.3, 23.6 and 23.8		S, H	No	Yes	No	Yes	See Note 1
	SOLAS II-1/17-1.1.1, 17-1.1.2, 17-1.1.3, 23.6 and 23.8		S, H	No	Yes	Yes (remote)	Yes	Doors giving access to below Ro-Ro Deck
	SOLAS II-1/17-1.1.1, 17-1.1.2, 17-1.1.3, 22.7 and 23.3 to 23.5	Perm. Closed	S, H	No	Yes	Yes (remote)	Yes	See Notes 1, 3 & 4
II. Cargo Ships								
(1) Below	SOLAS II-1/10, 13-1.2, 16.2 and 22.3 MARPOL I/28.3 ICLL66+A.320 1988 Protocol to ICLL66, IBC, and IGC	Used	POS	Yes	Yes	Yes (local)	No	
	SOLAS II-1/10, 13-1.3, 16.2, 22.3 and 24.4	Norm. Closed	S, H	No	Yes	No	Yes	See Note 1
	SOLAS II-1/10, 13-1.4, 16.2, 24.3, and 24.4	Perm. Closed	S, H	No	No	No	Yes	See Notes 3 & 4
	SOLAS II-1/10, 13-1.4, 13-1.5, 16.2, 22.2, 24.3 and 24.4							

(2)At or above	SOLAS II-1/10, 13-1.2, 16.2 and 22.3 MARPOL I/28.3 ICLL66+A.320 1988 Protocol to ICLL66, IBC, and IGC	Used	POS	Yes	Yes	Yes (local)	No	See Notes 2 & 5
	SOLAS II-1/10, 13-1.3, 16.2, 22.3 and 24.4	Norm. Closed	S, H	No	Yes	No	Yes	See Note 1
	SOLAS II-1/10, 13-1.4, 13-1.5, 16.2, 24.3 and 24.4	Perm. Closed	S, H	No	No	No	Yes	See Notes 3 & 4

Notes:
1. If hinged, this door is to be of quick acting or single action type.
2. Under ICLL66, doors separating a main machinery space from a steering gear compartment may be hinged quick acting type provided the lower sill of such doors is above the Summer Load Line and the doors remain closed at sea whilst not in use.
3. The time of opening such doors in port and closing them before the ship leaves port is to be entered in the logbook, in case of doors in watertight bulkheads subdividing cargo spaces.
4. Doors are to be fitted with a device which prevents unauthorized opening.
5. Under MARPOL, hinged watertight doors may be acceptable in watertight bulkhead in the superstructure.
6. Passenger ships which have to comply with SOLAS II-1/14.2 require an indicator on the navigation bridge to show automatically when each door is closed and all door fastenings are secured.
7. Refer to the Explanatory Note to Regulation 17.1 of Res.MSC.429 (98) regarding sliding watertight doors with a reduced pressure head and sliding semi-watertight doors.

B. Door in External Watertight Boundaries below equilibrium or intermediate waterplane

Position relative to bulkhead deck or freeboard deck	**1. Regulation**	**2. Frequency of use while at sea**	**3. Type**	**4. Remote Closure**	**5. Remote Indication**	**6. Audible or Visual Alarm**	**7. Notice**	**8. Comments**
I. Passenger Ships								
(1) Below	SOLAS II-1/15.9, 22.6 and 22.12	Perm. Closed	S, H	No	No	No	Yes	See Notes 2 & 3
(2) At or above	SOLAS II-1/17.1 and 22.3 MSC.Circ.541	Norm. Closed	S, H	No	Yes	No	Yes	See Note 1
	SOLAS II-1/17-1.1,17-1.2, 17-1.3, 23.6 and 23.8		S, H	No	Yes	Yes (remote)	Yes	Doors giving access to below Ro-Ro Deck
	SOLAS II-1/17-1.1, 17-1.2, 17-1.3, 23.3 and 23.5	Perm. Closed	S, H	No	Yes	Yes (remote)	Yes	See Notes 2 & 3
II. Cargo Ships								
(1) Below	SOLAS II-1/15.9,15-1.2, 15-1.3, 15-1.4, 22.6, 22.12 and 24.1	Perm. Closed	S, H	No	Yes	No	Yes	See Notes 2 & 3
(2) At or above	SOLAS II-1/15-1.2	Norm. Closed	S, H	No	Yes	No	Yes	See note 1
	SOLAS II-1/15-1.2 and 15-1.4	Perm. Closed	S, H	No	Yes	No	Yes	See Notes 2 & 3

Notes:
1. If hinged, this door is to be of quick acting or single action type.
2. The time of opening such doors in port and closing them before the ship leaves port is to be entered in the logbook.
3. Doors are to be fitted with a device which prevents unauthorized opening.

2.7.4.1 Frequency of use whilst at sea

a) Normally closed
Kept closed at sea but may be used if authorised. To be closed again after use.

b) Permanently closed
The time of opening such doors in port and of closing them before the ship leaves port shall be entered in the log-book. Should such doors be accessible during the voyage, they shall be fitted with a device to prevent unauthorised opening.

c) Used
Kept closed, but may be opened during navigation to permit the passage of passengers or crew, or when work in the immediate vicinity of the door necessitates it being opened. The door is to be immediately closed after use. This is subject to prior approval by the Administration.

2.7.4.2 Types of doors in Table 2.7.4

- Power operated, sliding or rolling — POS
- Power operated, hinged — POH
- Sliding or rolling — S
- Hinged — H

2.7.4.3 Control

(a) Local

All doors, except those which are to be permanently closed at sea, are to be capable of being opened and closed by hand, (and by power, where applicable) locally, from both sides of the doors, with the ship listed to either side. In the case of passenger ships, it is to be possible to close the door from a position above the bulkhead deck with an all round crank motion or some other movement providing the same degree of safety.

For passenger ships, the angle of list at which operation by hand is to be possible is 15 degrees .

For cargo ships, the angle of list at which operation by hand is to be possible is 30 degrees.

(* Arrangements for passenger ships is to be in accordance with SOLAS II-I/Reg.13.7.1.4 and 13.7.1.5 and II-1/Reg.13-1.2 for cargo ships)

(b) Remote

Where indicated in Table 2.7.4, doors are to be capable of being remotely closed by power from the bridge* for all ships, and by hand also from a position above the bulkhead deck for passenger ships as required by SOLAS II-1/Regulation 13 7.1.4. Where it is necessary to start the power unit for operation of the watertight door, means to start the power unit is also to be provided at remote control stations. The operation of such remote control is to be in accordance with Part 4, Chapter 8, Cl.10.1.9 to Cl.10.1.12. For tankers, where there is a permanent access from a pipe tunnel to the main pump room, in accordance with SOLAS II-2/Reg.4.5.2.4 the watertight door is to be capable of being manually closed from outside the main pump room entrance in addition to the requirements above.

(* - Arrangements are to be in accordance with SOLAS II-1/ Regulation 13.7.1.5 for passenger ships and 13-1.2 for cargo ships.)

2.7.4.4 Indication*

(a) Where shown in Table 2.7.4, position indicators are to be provided at all remote operating positions for all ships and provided locally on both sides of the internal doors for cargo ships, to show whether the doors are open or closed and if applicable, with all dogs/cleats fully and properly engaged.

(b) The door position indicating system is to be of self-monitoring type and means for testing of the indicating system is to be provided at the position where the indicators are fitted.

(c) A diagram showing the location of the door and an indication to show its position is to be provided at the central operating console located at the navigating bridge. A red light is to indicate the door is in the open position and a green light is to indicate the door is in the closed position. When the door is closed from this remote position, the red light is to flash when the door is in an intermediate position.

This applies to passenger ships and cargo ships.

(d) Signboard/instructions should be placed in way of the door advising how to act when the door is in "doors closed" mode.

(* - Refer to SOLAS II-I/Regulation 13, 13-1, 15-1 and 17-1, IEC 60092-504, IMO International Code on Alarms and Indicators 2009)

2.7.4.5 Alarms*

a) For passenger ships, failure of the normal power supply of the required alarms is to be indicated by an audible and visual alarm at the central operating console at the navigation bridge. For cargo ships, failure of the normal power supply of the required alarms is to be indicated by an audible and visual alarm at the navigation bridge.

b) All door types including power-operated sliding watertight doors which are to be capable of being remotely closed are to be provided with an audible alarm, distinct from any other alarm in the area, which will sound whenever such a door is remotely closed. For passenger ships the alarm is to sound for at least 5[sec] but not more than 10[sec] before the door begins to move and is to continue sounding until the door is completely closed. In the case of remote closure by hand operation, an alarm is required to sound only while the door is actually moving.

In passenger areas and areas of high ambient noise, the audible alarms are to be supplemented by visual signals at both sides of the doors.

c) All watertight doors, including sliding doors, operated by hydraulic door actuators, either a central hydraulic unit or an independent hydraulic unit for each door is to be provided with a low fluid level alarm or low gas pressure alarm, as applicable or some other means of monitoring loss of stored energy in the hydraulic accumulators. For passenger ships this alarm is to be both audible and visible and is to be located on the central operating console at the navigation bridge. For cargo ships, this alarm is to be both audible and visible and should be located at the navigation bridge.

(***** - Refer to SOLAS II-1/ Reg.13, 13-1, 15-1 and 17-1, IEC 60092-504, IMO International Code on Alarms and Indicators 2009)

2.7.4.6 Notices

As shown in Table 2.7.4, doors which are normally closed at sea but not provided with means of remote closure, are to have notices fixed to both sides of the doors stating, 'To be kept closed at sea'. Doors which are to be permanently closed at sea are to have notices fixed to both sides stating, 'Not to be opened at sea'.

2.7.4.7 Location

For passenger ships the watertight doors and their controls are to be located in compliance with SOLAS II-1/Reg.13.5.3 and II-1/Reg.13.7.1.2.2.

2.7.5 Fire doors

Watertight doors may also serve as fire doors but need not be fire-tested notwithstanding the fire resistance of the division in which the watertight doors are fitted. However, such doors fitted above the bulkhead deck on passenger ships are to be tested to the FTP Code in accordance with the division they are fitted. If it is not practicable to ensure self-closing, means of indication on the bridge showing whether these doors are open or closed and a notice stating ‘To be kept closed at sea’ can be alternative of the self-closing.

Where a watertight door is located adjacent to a fire door, both doors are to be capable of independent operation, remotely if required by SOLAS ll-1/ Regulation 13.8.1 to 13.8.3 and from both sides of each door.

2.7.6 Testing

(a) Doors which become immersed by an equilibrium or intermediate waterplane are to be subjected to a hydrostatic pressure test. (Refer also Chapter 17, Table 3.3.1).

(b) For large doors intended for use in the watertight subdivision boundaries of cargo spaces, structural analysis may be accepted in lieu of pressure testing. Where such doors utilize gasket seals, a prototype pressure test

to confirm that the compression of the gasket material is capable of accommodating any deflection, revealed by the structural analysis, is to be carried out.

(c) Doors above freeboard or bulkhead deck, which are not immersed by an equilibrium or intermediate waterplane but become intermittently immersed at angles of heel in the required range of positive stability beyond the equilibrium position are to be hose tested.

2.7.6.1 Pressure testing

(a) The head of water used for the pressure test corresponds to at least the head measured from the lower edge of the door opening, at the location in which the door is to be fitted in the vessel, to the bulkhead deck or freeboard deck, as applicable, or to the most unfavorable damage waterplane, if that be greater. Testing may be carried out at the factory or other shore-based testing facility prior to installation in the ship.

(b) For doors on passenger ships which are normally open and used at sea or which become submerged by the equilibrium or intermediate waterplane, a prototype test is to be conducted, on each side of the door, to check the satisfactory closing of the door against a force equivalent to a water head of at least 1[m] above the sill on the centre line of the door *.

(* - Arrangements for passenger ships are to be in accordance with SOLAS Regulation II-1/13.5.2.)

2.7.6.2 Hose testing

After installation in a ship if pressure test is not carried out, all watertight doors are to be subject to a hose test in accordance with Part 3, Chapter 17. Hose testing is to be carried out from each side of the door unless, for a specific application, exposure to floodwater is anticipated only from one side. Where a hose test is not practicable because of possible damage to machinery, electrical equipment, insulation or outfitting items, it may be replaced by means such as an ultrasonic leak test or an equivalent test.

2.8 Minimum bow height

Attention is drawn to minimum bow height requirement of ships, in ICLL Regulation 39, as applicable.

Section 3

Compartment Arrangement

3.1 Cofferdams

3.1.1 Definition

A cofferdam means an empty space arranged so that compartments on each side have no common boundary; a cofferdam may be located vertically or horizontally. As a rule, a cofferdam is to be kept gastight and is to be properly ventilated, provided with drainage arrangement and of sufficient size to allow proper inspection, maintenance and safe evacuation.

3.1.2 Arrangement of cofferdams

3.1.2.1 Cofferdams are to be provided between compartments intended for liquid hydrocarbons (including fuel oil, lubricating oil) and those intended for feed water (water for propelling machinery and boilers) as well as tanks intended for the carriage of liquid foam for fire extinguishing and tanks for edible oil. Cofferdams are also to be provided between compartments intended for carrying fuel oil, lubricating oil and those intended for aviation fuel.

3.1.2.2 Furthermore, tanks carrying fresh water for human consumption are to be separated from other tanks containing substances hazardous to human health by cofferdams or other means as approved by IRS. Normally, tanks for fresh water and water ballast are considered non-hazardous.

3.1.2.3 Where a corner-to-corner situation occurs, tanks are not considered to be adjacent.

3.1.2.4 The cofferdams specified in Cl.3.1.2.1 may not be required when deemed impracticable or unreasonable by IRS in relation to the characteristics and dimensions of the spaces containing such tanks, provided that:

- the thickness of common boundary plates of adjacent tanks is increased, with respect to the thickness obtained according to Chapter 7, Section 3, Cl.3.1.1 by 2[mm] in the case of tanks carrying fresh water or boiler feed water, and by 1[mm] in all other cases,

- the sum of the throats of the weld fillets at the edges of these plates is not less than the thickness of the plates themselves,

- the structural test is carried out with a test pressure increased by 1[m] with respect to Chapter 17, Section 3, 3.4.

3.1.2.5 Tanks for lubricating oil are also to be separated by cofferdams from those carrying fuel oil. However, these cofferdams need not be fitted provided that the common boundaries have full penetration welds and the head of lubricating oil is not less than that in the adjacent fuel oil tanks. In this case, a permanent notice is to be displayed near the lubricating oil tank that the oil level is not to be less than that in the adjacent fuel oil tank at any time.

3.1.2.6 Vented cofferdam may be required to separate heated oil fuel tanks from enclosed spaces located directly above (Refer Part 4, Chapter 3, Section 4, Cl.4.5.3).

3.1.2.7 Cofferdam arrangements and requirements for specific ships types, refer to Part 5.

3.2 Double bottom

3.2.1 General

3.2.1.1 Except as specified in Cl.3.2.1.6, passenger ships and cargo ships other than tankers, are to be fitted with a double bottom extending from the collision bulkhead to the after peak bulkhead as far as this is practical and compatible with the design and proper working of the ship.

Note: For the purpose of this requirement, a “tanker” is a cargo ship constructed or adapted for the carriage in bulk of liquid cargoes of an inflammable nature.

3.2.1.2 For oil tankers, chemical carriers and liquefied gas carriers, the requirements indicated in Part 5, Chapter 2, Chapter 3 and Chapter 4 respectively regarding double bottom and location of wells above bottom shell are to be complied with.

3.2.1.3 A double bottom need not be fitted in way of watertight tanks, including dry tanks of moderate size provided the safety of the ship is not impaired in the event of bottom or side damage as regulated in SOLAS II-1, Regulation 9.

3.2.1.4 A double bottom need not be fitted in way of watertight compartments used exclusively for the carriage of liquids, provided the safety of the ship is not impaired in the event of bottom damage.

3.2.1.5 Any part of a passenger ship or a cargo ship other than tanker that is not fitted with a double bottom in accordance with Cl.3.2.1.1 or Cl.3.2.1.4 is to be capable of withstanding bottom damages as specified in SOLAS II-1, Paragraph 8 of Regulation 9.

3.2.1.6 A double bottom need not be fitted on the following ships:

- Cargo ships of less than 500 tons gross tonnage.

- Ships not propelled by mechanical means.
- Fishing vessels.

3.2.1.7 The engine room of all ships having L>80[m] is to have a double bottom.

3.2.1.8 For ships of L>90[m], single bottoms within the cargo region, where permitted, are normally to be longitudinally stiffened.

3.2.1.9 For ships of L>150[m] and for ships strengthened for heavy cargoes, double bottoms within the cargo region are normally to be longitudinally stiffened.

3.2.1.10 The continuity of the bottom, bilge and inner bottom longitudinal is to be maintained in accordance with Chapter 3, Section 5.

3.2.1.11 The weld connections are to comply with the requirements of Chapter 16.

3.2.1.12 For double bottom requirements for specific ships types, Refer to Part 5.

3.2.2 Height of double bottom

Where double bottom is required to be fitted, the inner bottom is to be continued out to the ship side in such a manner as to protect the bottom to the turn of the bilge. Such protection is to be deemed satisfactory if the inner bottom is not lower at any part than a plane parallel with the keel line and which is located not less than a vertical distance ‘d’ in [mm] measured from the keel line, as calculated by the formula:

$$d = 50B$$

However, in no case the value of d is to be less than 760[mm], and need not to be taken more than 2000[mm]. This is applicable for passenger ships and cargo ships other than tankers.

And also, the depth ’d’ of the centre girder is not to be less than:

$d = 250 + 20B + 50T$ with a minimum of 760[mm].

For ships with large rise of floor, the minimum height may have to be increased after special consideration.

3.2.3 Small wells in double bottom tank

Bilge / Small wells constructed in the double bottom, in connection with the drainage arrangement of holds, are not to extend in depth more than necessary. A well extending to the outer bottom, may be permitted at the after end of the shaft tunnel of the ship. In no case, the vertical distance from the bottom of such a well to the plane coinciding with the keel is not to be less than 500[mm] or half the required double bottom height whichever is greater, for passenger ships and cargo ships other than tankers unless compliance for that part of the ship with SOLAS II-1, Paragraph 8 of Regulation 9- Double bottoms in passenger ships and cargo ships other than tankers is demonstrated.

Other wells e.g. for lubricating oil under main engines may be considered provided the arrangements give protection equivalent to that offered by a double bottom. For a cargo ship of 80 [m] in length and upwards or for a passenger ship, proof of equivalent protection is to be shown by demonstrating that the ship is capable of withstanding bottom damages as specified in SOLAS II-1, Paragraph 8 of Regulation 9.

Alternatively, wells for lubricating oil below main engines can protrude into the double bottom below the boundary line defined by the distance ‘d’ provided that the vertical distance between the well bottom and a plane coinciding with the keel line is not less than

500[mm] or half the required double bottom height, whichever is greater.

For cargo ships of less than 80[m] in length, the arrangements provided are to be to the satisfaction of IRS.

3.3 Compartments forward of the collision bulkhead

In ships of 400GT and above, compartments forward of the collision bulkhead are not to be arranged for carriage of oil and other inflammable liquids.

3.4 Aft end compartments

3.4.1 Stern tube entry

Stern tubes are to be enclosed in a watertight space (or spaces) of moderate volume. In case the stern tube terminates at an aft peak bulkhead also being a machinery space bulkhead, a pressurized stern tube sealing system may be accepted as an alternative to the watertight enclosure. Other measures to minimize the danger of water penetrating into the ship in case of damage to the stern tube arrangement may be taken at the discretion of IRS.

3.4.2 Steering gear compartment

The steering gear compartment is to be readily accessible and separated from machinery spaces. Refer Part 4, Chapter 6 (Steering Gear), for additional requirements for the steering gear compartment.

3.5 Fuel oil tanks

3.5.1 Arrangement of fuel oil tanks

Fuel oil tanks are to be arranged in accordance with the requirements in SOLAS Chapter II-2, Regulation 4.2.

3.5.2 Protective location of fuel oil tanks

3.5.2.1 All ships with an aggregate oil fuel capacity of 600[m^3] and above are to satisfy the requirements for oil fuel tank protection as given in MARPOL Regulation I/12A. For this purpose, the distances of the oil fuel tanks above the bottom shell and inboard of the side shell are to be not less than the values specified in the above MARPOL Regulation. Alternatively, such ships may comply with the accidental oil outflow performance standard specified in paragraph 11 of MARPOL Regulation I/12A.

3.5.2.2 Small oil fuel tanks of capacity not greater than 30[m^3] are exempted from the requirements for protective location of the above MARPOL Regulation provided the aggregate capacity of such exempted tanks is not greater than 600[m^3].

Section 4

Access Arrangements

4.1 Access to tanks and compartments

4.1. General

4.1.1 All vessels are to be provided with means of access giving safe and practical access to the internal structure during operational phase.

4.1.2 All tanks are to be accessible for easy inspection and close-up survey (as relevant).

4.1.3 All closed spaces are to be accessible for easy inspection. Special measures for inspection and maintenance are to be put in place for small closed spaces for which the design causes impracticality for the access.

4.1.4 Access arrangements for specific ships types, Refer to Part 5.

4.2 Access to double bottom compartments

4.2.1 Inner bottom manholes

4.2.1.1 The number, opening size and location of inner bottom manholes are to be so arranged as to provide convenient access to any part of the double bottom.

4.2.1.2 Inner bottom manholes are to be closed by watertight plate covers. Doubling plates are to be fitted on the covers, where secured by bolts. Where no ceiling is fitted, covers are to be adequately protected from damage by the cargo.

4.2.2 Floor and girder manholes for access, air and drain holes for ventilation and drainage

4.2.2.1 Adequate access is to be provided to all parts of the double bottom. The vertical dimension of lightening holes is generally not to exceed 50% of the girder height. The diameter of lightening holes in the bracket floors is not to exceed 1/3 of the breadth of the brackets. Lightening holes or manholes are normally not to be cut in floors or girders towards their ends and under large pillars or supporting structures. Manholes in inner bottom are to have reinforcement rings, and the man hole covers in the inner bottom plating in cargo holds are to be effectively protected. The edges of all holes are to be smooth.

4.2.2.2 To ensure the free passage of air and water from all parts of the tanks to air pipes and suctions, air and drain holes are to be provided in all non-watertight members. The air holes are to be placed as near to the inner bottom as possible and their total area is to be greater than the area of the filling pipes. The drain holes are to be placed as near to the bottom as possible.

4.2.2.3 The access opening to pipe tunnel is to be visible above the floor plates and is to be fitted with a rigid watertight closing device. A notice board stating that the access opening to the pipe tunnel is to be kept closed, is to be fitted near the opening. The opening is to be regarded as an opening in watertight bulkhead.

4.3 Dimensions of access hatchways

4.3.1 The dimensions of access, through horizontal openings, hatches or manholes, are to be sufficient to allow a person wearing a self-contained air-breathing apparatus to ascend or descend the ladder without obstruction and also to provide a clear opening to facilitate the hoisting of an injured person from the bottom of the tank. In no case is the clear opening to be less than 600[mm] x 600[mm] which may have corner radii of not more than 100[mm].

4.3.2 The dimensions of any access, through vertical openings or manholes in swash bulkheads floors, girders and web frames providing passage through the length and breadth of the space, the minimum dimensions of 600[mm] x 800[mm] are to be arranged in the floors at a height of not more than 600[mm] from the bottom shell plating unless gratings or other foot holds are provided.

4.4 Access to shaft tunnel

4.4.1 Tunnels are to be large enough to ensure easy access to shafting.

4.4.2 Access to the tunnel is to be provided by a watertight door fitted on the aft bulkhead of

the engine room in compliance with Chapter 3, Section 2.6

4.4.3 Where two or more shafts are fitted, the tunnels are to be connected by an interconnecting passage. If two shafts are fitted, there is to be only one door between the machinery space and the tunnel spaces and only two doors where there are more than two shafts.

4.4.4 Where fore and aft under deck passageways are arranged at the ship's side, the after access thereto is to be by a watertight trunk led to the upper deck. Alternative arrangements to prevent the engine room being flooded, in the event of a collision or if the passageway doors are left open, will be considered.

4.4.5 When access to any machinery space of category A is provided at a lower level from an adjacent shaft tunnel, there must be provided in the shaft tunnel, near the watertight door, a light steel fire-screen door operable from both sides.

4.5 Access to steering gear compartment

4.5.1 The steering gear compartment are to be readily accessible and separated from machinery spaces.

4.5.2 The steering gear compartment are to be provided with suitable arrangements to ensure working access to steering gear machinery and controls. These arrangements are to include handrails and gratings or other non-slip surfaces to ensure suitable working conditions in the event of hydraulic fluid leakage

Section 5

Structural Arrangements

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

5.1 General

5.1.1 Frame spacing

5.1.1.1The standard frame spacing between aft peak and 0.2L from F.P. may be taken as:

- For transverse framing: 450 + 2L [mm]

- For longitudinal framing: 550 + 2L [mm]

However, it is generally not to exceed 1000[mm].

5.1.1.2 Elsewhere, the frame spacing is generally not to exceed the following:

- In peaks and cruiser sterns: 600[mm] or as in Cl.5.1.1.1, whichever is lesser.

- Between collision bulkhead and 0.2L from F.P.: 700[mm] or as in Cl.5.1.1.1, whichever is lesser.

5.1.1.3 Where the actual frame spacing is higher than that mentioned above, the minimum thicknesses of various structural members as given in the Rules may require to be increased.

5.2 Structural continuity

5.2.1 General

5.2.1.1 Attention is to be paid to the structural continuity, in particular in the following areas:

- In way of changes in the framing system

- At end connections of primary supporting members or ordinary stiffeners.

- In way of the transition zones between midship area and fore part, aft part and machinery space

- In way of side and end bulkheads of superstructures.

5.2.1.2 At the termination of a structural member, structural continuity is to be maintained by the fitting of suitable supporting

structure. Abrupt changes in transverse section properties of longitudinal members, sharp corners and points of stress concentration are to be avoided.

5.2.1.3 The variation of thickness of bottom, side and strength deck between the midship region and the fore and aft parts are to be gradual.

5.2.1.4 Where members abut on both sides of a bulkhead or similar structure, care is to be taken to ensure good alignment.

5.2.1.5 Continuity is to be maintained where primary members intersect and where the members are of the same depth, a suitable gusset plate is to be fitted.

5.2.1.6 End connections of structural members are to provide adequate end fixity and effective distribution of the load into the supporting structure.

5.2.1.7 The toes of brackets, etc. should not land on unstiffened plate panels. Special care should be taken to avoid notch effects at the toes of brackets, by making the toe concave or otherwise tapering it off.

5.2.1.8 Where stress concentrations may occur in way of structural discontinuities, adequate compensation and reinforcements are to be provided.

5.2.1.9 Openings are to be avoided, as far as practicable, in way of highly stressed areas. Where necessary, the shape of openings is to be specially designed to reduce the stress concentration factors. Openings are to be generally well rounded with smooth edges.

5.2.2 Primary supporting members

Primary supporting members are to be arranged in such a way that they ensure adequate continuity of strength. Abrupt changes in height or in cross-section are to be avoided.

5.2.3 Longitudinal members

5.2.3.1 Longitudinal members are to be arranged in such a way that continuity of strength is maintained in 0.4L amidship.

5.2.3.2 Longitudinal members contributing to the hull girder longitudinal strength are to extend continuously as far as practicable towards the ends of the ship.

5.2.3.3 For longitudinal bulkheads or deep girders, large transition brackets (e.g. scarfing brackets) fitted in line with the longitudinal member are a possible means to achieve such structural continuity.

5.2.4 Longitudinal stiffeners

5.2.4.1 Stiffeners are to be arranged in such a way that continuity of strength is maintained.

5.2.4.2 Ordinary stiffeners contributing to the hull girder longitudinal strength are to be continuous when crossing primary supporting members within the 0.4L amidships and as far as practicable outside 0.4L amidships. Proposals of longitudinals abutting at transverse members may be accepted outside 0.4L amidships region and in ships less than 150[m] in length, provided the brackets connecting the ends of the longitudinals are continuous and of adequate size.

5.2.4.3 Where stiffeners are terminated in way of large openings, foundations and partial girders, compensation is to be arranged to provide structural continuity in way of the end connection.

5.2.4.4 When the deck, bottom or inner bottom is longitudinally stiffened, the longitudinal is to be continuous through transverse members.

5.2.5 Transverse stiffeners

Transverse and vertical stiffeners are to be continuous through stringers/girders, if provided, or fitted with bracket end connections.

5.3 Bottom structure

5.3.1 Variation in height of double bottom

Any variation in the height of the double bottom is generally to be made gradually and over an adequate length; the knuckles of inner bottom plating are to be located in way of plate floors or girders. Where such arrangement is not possible, suitable longitudinal structures such as partial girders, longitudinal brackets or carlings, fitted across the knuckle are to be arranged.

5.3.2 Connection between inner hull and inner bottom plating

Connection between the inner hull plating and the inner bottom plating is to be designed such that stress concentration is minimized. The connection of inner hull plating or hopper plating with inner bottom is to be effectively supported, e.g. by a longitudinal girder or gusset plates.

5.3.3 Drainage of tank top

For ships designed to carry solid cargoes, effective arrangements are to be provided for draining water from the tank top. Where wells are provided for the drainage, Refer section 3, Cl.3.2.3, for bilge/small well requirements.

5.3.4 Framing system

For ships greater than 120[m] in length, the bottom shell, the inner bottom and the sloped bulkheads of hopper tanks, in general, are to be longitudinally framed.

5.3.5 Striking plate

Striking plates of adequate thickness or other equivalent arrangements are to be provided under sounding pipes to prevent the sounding rod from damaging the plating.

5.3.6 Duct keel

5.3.6.1 Where a duct keel is arranged, the centre girder may be replaced by two girders spaced no more than 3[m] apart. Otherwise, for a spacing wider than 3[m], the two girders are to be provided with support of adjacent structure and subject to IRS's approval.

5.3.6.2 The inner bottom and bottom shell within the duct keel are to be suitably stiffened. The primary stiffening in the transverse direction is to be suitably aligned with the floors in the adjacent double bottom tanks.

5.3.7 Keel plating

Keel plating is to extend over the bottom for the full length of the ship. The width of the keel plate in [mm] is not be less than 800+5L, and need not to be greater than 1800[mm]. The thickness is to be 2[mm] greater than that required for the adjacent bottom plating Refer to Chapter 7, Section 3, Cl.3.1.1.

5.3.8 Girders in single bottom

5.3.8.1 Transverse framing

- On all ships one centre girder is to be fitted and in addition side girders are to be fitted such that the spacing of girders does not exceed 2.5[m]. The girders are to extend as far forward and aft as practicable and where they are cut at transverse bulkheads the longitudinal strength is to be maintained.

- Where the single bottom structure changes into a double bottom structure, the bottom girders are to extend at least 3 frame spaces into double bottom structures. Structural continuity of the girders is to be maintained within the full length of the ship.

5.3.8.2 Longitudinal framing

The spacing of bottom transverses is normally not to exceed the greater of 0.02L [m] or 3.6[m]. The bottom transverses are to be supported by primary girders or longitudinal bulkheads. Where the design does not incorporate a centreline bulkhead, at least a docking girder is to be provided. The scantlings of a complex girder system are to be based on Finite element analysis.

5.3.9 Girders in double bottom

5.3.9.1 General

- Double bottom girders are to be arranged in line with longitudinal bulkheads.

- For longitudinal or transverse framing, the side bottom girders are to be extended as far forward and aft as practicable.

5.3.9.2 Transverse framing

- In double bottoms with transverse framing the longitudinal girders are to be stiffened at every transverse frame.

- The side girders are normally to be fitted at a spacing not exceeding 4.0[m] and are to be extended as far forward and aft as practicable. The girders are to be stiffened at every bracket floor by a vertical stiffener of depth same as that of reverse frame and thickness that of the girder.

- The bottom frames and reverse frames are to be attached to the centre girder and margin plate by means of brackets of same thickness as that of the plate floors. The breadth of the brackets is not to be less than 0.75 times the

depth of the centre girder and the brackets are to be flanged 75[mm] at their free edges.

5.3.9.3 Longitudinal framing

Other than peak and machinery spaces, the side girders are normally to be fitted at a spacing not exceeding 5.0[m] and are to be extended as far forward and aft as practicable.

5.3.10 Floors

5.3.10.1 General

- Plate floors are to be fitted under bulkheads, pillars, thrust seating, boiler bearers and in way of change of depth of double bottom.

- Floors are to be provided with web stiffeners in way of longitudinal stiffeners. Where the web stiffeners are not welded to the longitudinal stiffeners, design standard as given in in Chapter 10 (Fatigue), Appendix A may be applied.

5.3.10.2 Transversely framed single bottom other than peak spaces and engine room

Plate floors are to be fitted at every frame and the top of floors, in general, is to be level from side to side.

5.3.10.3 Transversely framed double bottom other than peak spaces and engine room

The spacing of plate floors other than peak spaces and engine room is not to exceed that given by Table 5.3.10.

Table 5.3.10: Spacing of plate floors (Transverse framing)

Draught T[m]	Under deep tanks (See Note 1)	Clear of deep tanks (maximum 3.0[m])
$T \leq 2$	Every 4th frame	Every 6th frame
$2 < T \leq 5.4$	Every 3rd frame	Every 5th frame
$5.4 < T \leq 8.1$	Every 3rd frame	Every 4th frame
$T > 8.1$	Every 2nd frame	Every 3rd frame
Note 1: Under tanks with height greater than 0.7 times the distance between inner bottom and main deck		

5.3.10.4 Longitudinally framed double bottom other than peak spaces and engine room.

- The spacing of floors is normally not to exceed 2.5[m] under deep tanks with height greater than 0.7 times the distance between inner bottom and main deck. Elsewhere, the spacing is normally not to exceed 3.6[m].

- The plate floors are to be stiffened at every longitudinal by a vertical stiffener having a depth not less than 150[mm] and thickness as that of the floor. For ships of length, L< 90[m], the depth is not to be less than 1.65L[mm], with a minimum of 50[mm].

- In general, intermediate brackets are to be fitted connecting either the margin plate or the centre girder to the nearest bottom and inner bottom ordinary stiffeners. Such brackets are to be stiffened at the edge with a flange having a width not less than 1/10 of the local double bottom height. If necessary, the Society may require a welded flat bar to be arranged in lieu of the flange.

5.3.11 Docking

5.3.11.1 Docking of the vessel is to be evaluated and considered at the design stage by the designer. The bottom structure is to withstand the forces imposed by dry-docking the ship.

5.3.11.2 Docking brackets connecting the centreline girder and margin girder to the bottom plating, are to be connected to the adjacent bottom longitudinals and are to be flanged 75[mm] at their free edges. Docking brackets are to be fitted between plate floors at every frame and at a spacing not exceeding 1.25[m] on both sides of the centre girder. The thickness of brackets is to be same as that of the plate floors.

5.3.11.3 Alternative arrangements require special consideration of local buckling strength

of centre girder and local strength of bottom longitudinal in way of docking block.

5.4 Side structure

5.4.1 General

5.4.1.1 The ship's side shell and inner hull bulkheads may be stiffened longitudinally or vertically.

5.4.1.2 Where the side shell is stiffened longitudinally, the continuity of the side longitudinals within a distance of 0.15D from bottom or from strength deck is to be maintained in accordance with Chapter 3, Section 5, Cl.5.2.3. The web frames are to be fitted in line with the bottom transverses or plate floors.

5.4.1.3 The position, shape and reinforcement of sea inlets or other openings in side shell are to be in accordance with the requirements of Chapter 3, Section 6, Cl.6.5.6.

5.4.1.4 The thickness of the shell plating is to be increased locally by 50% in way of stern frame, propeller brackets and rudder horn. For reinforcements in way of anchor pockets, hawse pipes etc. refer to Chapter 14, Section 2, 2.1.

5.4.1.5 The transversely framed side structures are built with transverse frames possibly supported by side girders.

5.4.1.6 The longitudinally framed side structures are built with longitudinal ordinary stiffeners supported by side vertical primary supporting members.

5.4.1.7 Unless otherwise specified, horizontal side stringers are to be fitted aft of the collision bulkhead up to 0.2L aft of the fore end, in line with fore peak girders.

5.4.1.8 Side vertical primary supporting members are to be fitted in way of hatch end beams or similar large deck opening supporting transverse structure.

5.4.2 Sheer strake

5.4.2.1 Where a rounded gunwale is adopted the radius in general, is not to be less than 15 times the plate thickness.

5.4.2.2 Where the sheer strake extends above the deck stringer plate, the top edge of the sheer strake is to be kept free from notches and isolated welded fittings, and is to be ground smooth with rounded edges. Drainage openings with smooth transition in the longitudinal direction may be allowed on special consideration. Fixtures such as bulwarks, eye plates are not to be directly welded on the upper edge of sheer strake within 0.6L amidships.

5.4.2.3 The welding of deck fittings to rounded sheer strake within 0.6L amidships is not permitted, unless approved in each case.

5.4.2.4 Sheer strakes are to have breadths not less than 0.8 + L/200[m], measured vertically, but need not be greater than 1.8[m]. For ships of L< 90[m], it need not exceed 200D[mm]

5.4.2.5 Where the thickness of the strength deck plating is greater than that required for side plating, the sheer strake thickness is not to be less than the mean of the two values.

5.4.2.6 Longitudinal seam welds of rounded sheer strake are to be located outside the bent area at a distance not less than 5 times the maximum net thicknesses of the sheer strake.

5.4.2.7 The transition from a rounded gunwale to angular gunwale is to be designed to avoid any discontinuities.

5.4.2.8 The net thickness of a rounded sheer strake is to be not less than the actual net thickness of the adjacent deck plating.

5.4.2.9 The net thickness of a welded sheer strake is to be not less than that of the adjacent side plating, taking into account of use of higher strength steel.

5.5 Deck structure

5.5.1 General

5.5.1.1 In tankers, the deck is normally to be stiffened longitudinally in the cargo tank region, however, where L does not exceed 75[m], consideration may be given to transversely stiffened decks.

5.5.1.2 In dry cargo ships of L>120[m], the strength deck outside the line of hatchway openings is normally to be longitudinally stiffened.

5.5.1.3 The continuity of the deck longitudinals is to be maintained in accordance with Chapter 3, Section 5, Cl.5.2.3.

5.5.1.4 The deck within the line of hatchway openings is preferably to be stiffened transversely or alternatively the arrangements are to provide adequate transverse buckling strength. Where the deck outside the line of hatchway openings is framed longitudinally, the transverse beams or buckling stiffeners between the hatchways are to extend at least up to the second longitudinal from the hatch side or equivalent.

5.5.1.5 In ships with large hatch openings, the effective cross-sectional area of the deck between the hatchways is to be sufficient to withstand the transverse load acting on the ship's sides. Bending and shear stresses arising as a result of loading on the transverse bulkhead supported by the deck area and also as a result of displacements caused by torsion of the hull girder, are to be considered and the necessary reinforcements are to be provided.

5.5.1.6 The following items may be included in the effective cross-sectional area of the decks; corrosion additions, if any, are to be deducted:

- deck plating
- transverse beams and deck transverses
- hatch end beams (upon special consideration)
- cross section of stool at top of transverse bulkhead
- cross section of transverse bulkhead (if plane or horizontally corrugated) down to base of top wing tank or to 0.15D from deck.

5.5.1.7 The compressive stress in the above-mentioned items is not to be greater than 120/k [N/mm2], nor 80% of the critical buckling stress found in accordance with Chapter 9.

5.5.1.8 Hatchway corners are to be of streamlined, elliptical or circular shape as given in Chapter 3, Section 6, Cl.6.5.4.

5.5.1.9 Deck supporting structures under deck machinery, cranes and king posts are to be adequately stiffened.

5.5.1.10 The deck supporting structure consists of ordinary stiffeners (beams or longitudinals), longitudinally or transversely arranged, supported by primary supporting members which may be sustained by pillars.

5.5.1.11 Pillars or other supporting structures are generally to be fitted under heavy concentrated cargoes.

5.5.2 Stringer plate

5.5.2.1 Stringer plates are to have breadths not less than 0.8 + L/200[m], measured parallel to the deck, but need not be greater than 1.8[m].

5.5.2.2 Stringer plates of lower decks not extending over the full ship's length are to be gradually tapered or terminated with adequately sized brackets.

5.5.2.3 The net thickness of the stringer plate is to be not less than the actual net thickness of the adjacent deck plating.

5.5.2.4 In way of ends of bridges, poops and forecastles, the thickness of the strength deck stringer strake is to be increased by 20% over four frame spaces fore and also aft of the end bulkheads.

5.5.3 Watertight deck arrangement

Watertight decks are to be of the same strength as watertight bulkheads at corresponding levels. The means used for making them watertight, and the arrangements adopted for closing openings in them, are to be to the satisfaction of IRS.

5.5.4 Connection of deckhouses and superstructures

Connection of deckhouses and superstructures to the strength deck and hatch coamings is to be designed such that loads are transmitted into the under deck supporting structure.

5.6 Bulkhead structure

5.6.1 General

5.6.1.1 The requirements of this Section apply to longitudinal or transverse bulkhead structures which may be plane or corrugated.

5.6.1.2 Bulkheads may be horizontally or vertically stiffened. Horizontally framed bulkheads consist of horizontal ordinary stiffeners supported by vertical primary supporting members. Vertically framed bulkheads consist of vertical ordinary stiffeners which may be supported by horizontal girders.

5.6.1.3 The web height of vertical PSMs on bulkheads may be gradually tapered from bottom to deck.

5.6.1.4 Bulkheads are to be stiffened in way of deck girders.

5.6.1.5 Bulkhead girders and deep transverses are to be arranged in line with other primary supporting structure to the adjoining deck, side shell and bottom so as to facilitate the formation of continuous ring structures. Otherwise, equivalent scarphing arrangement is to be provided

5.6.1.6 Floors are to be fitted in the double bottom in line with the plane transverse bulkhead.

5.6.1.7 The main strength members included in the calculation of hull moment of inertia and section modulus are to extend continuously through the cargo region and sufficiently far towards the ends of the ship. Longitudinal bulkheads are to terminate at effective transverse bulkheads and large transition brackets are to be fitted in line with the longitudinal bulkheads. Where significant changes in structural arrangement occur, adequate transitional structure is to be provided.

5.6.1.8 Where the longitudinal watertight bulkheads contribute to longitudinal strength, the plating thickness is to be uniform for a distance of at least 0.1D from the deck and bottom. Refer Chapter 2, for applicable material grades.

5.6.1.9 If cargo is carried in a compartment adjacent to an oil fuel settling tank which may be heated, the compartment side of the bulkhead (or deck) is to be insulated, or equivalent arrangements provided.

5.6.1.10 The continuity of bulkhead longitudinal within a distance of 0.15D from the bottom or the strength deck is to be maintained in accordance with Chapter 3, Section 5, Cl.5.2.3.

5.6.1.11 Bulkheads or parts thereof supporting deck structure are also to be designed as pillars. The permissible axial loads and buckling strength are to be calculated in accordance with Chapter 9. In calculating sectional properties, the width of attached plating is to be equal to 35 times and not to be taken more than 40 times the plate net thickness.

5.6.1.12 A centerline wash bulkhead is to be fitted in deep tanks of fuel oil intended for ship's use extending from side to side. These apply to transverse and longitudinal wash bulkheads whose main purpose is to reduce the sloshing effect.

5.7 Cantilevers

The scantlings of cantilever beams and supporting frame will be specially considered.

5.8 Pillars

5.8.1 General

5.8.1.1 Pillars and pillar bulkheads are to be fitted in the same vertical line wherever possible, and elsewhere arrangements are to be made to transmit the out of line forces satisfactorily. The load at head and heel of pillars is to be effectively distributed and arrangements are to be made to ensure the adequacy and lateral stability of the supporting members. Where pillars support eccentric loads, they are to be strengthened for the additional bending moment imposed upon them.

5.8.1.2 Pillars are to be provided in line with the double bottom girder and /or floors or as close thereto as practicable.

5.8.1.3 Where pillars connected to the inner bottom are not located in way of the intersection of plate floors and girders, partial floors or girders or equivalent structures such as intercostal girders are to be fitted as necessary to support the pillars. Lightening holes or manholes are not to be cut in the floors and girders below the heels of pillars.

5.8.1.4 Inside spaces intended for products which may produce explosive gases or in tanks, hollow pillars are not to be used and strengthening at the heads and heels of pillars is not be obtained by means of doubling plates.

5.8.1.5 Where the hydrostatic pressure may result in tensile stresses in the pillar, their sectional area 'A' in [cm^2], is not be less than,

$$A = 70.A_L.P.k$$

where,

p = Design pressure in [N/mm^2] that causes the tensile stress in pillar.

A_L = Load area of deck, in [m^2], being supported by the pillar.

The weld area is to be such that the shear stresses in the welds of end connection does not exceed 80/k [N/mm^2]. Also, where pillars are fitted inside tanks or under watertight flats, the end connection is to be such that the tensile stress in the weld does not exceed 108/k [N/mm^2].

5.8.1.6 Doubling or insert plates are generally to be fitted at the head and heel of hollow pillars.

5.8.2 Connections

5.8.2.1 Where the pillars are likely to be subjected to tensile loads, the head and heel of pillars are to be efficiently secured to withstand the tensile loads and the doubling plates are generally to be replaced by insert plate.

5.8.2.2 The net thickness of doubling plates, when fitted, is to be not less than 1.5 times the net thickness of the pillar, (except in case of solid pillars).

5.8.2.3 The pillars are to have a bearing fit and are to be attached to the head and heel plates by continuous welding.

5.9 Aft peak arrangement

5.9.1 Application

The requirements of this section apply for the arrangement of structures located aft of the aft peak bulkhead and below bulkhead deck.

5.9.2 Minimum thickness requirement

The net thickness of the aft peak bulkhead plating in way of the stern tube penetration is to be at least 1.6 times the required thickness for the bulkhead plating itself or a doubling plate (in way of tanks, insert plate is to be used) of same thickness as the corresponding strake is to be fitted.

5.9.3 Bottom structure

5.9.3.1 Longitudinal framing

(i) Floors

- In case of single bottom, floors are to be fitted at every four frame spacing but spaced not more than 2.5[m].

- In case of double bottom, the spacing of plate floors is not to be greater than 3.6[m]. However, under deep tanks with height greater than 0.7 times the distance between inner bottom and main deck, the spacing of floors is normally not to exceed 2.5[m].

- The minimum depth of the floor at the centreline is not to be less than the required depth of the double bottom. Refer Chapter 3, Section 3, Cl.3.2.2.

(ii) Bottom girders

- Centerline girder is to be extended up to the transom bulkhead.

- In case of single bottom, the spacing of bottom girders is not to exceed 3.5[m] and are to be extended as far forward and aft as practicable.

- In case of double bottom, the spacing of the bottom girders is not to exceed 5.0[m] and are to be extended as far forward and aft as practicable.

5.9.3.2 Transverse framing

(i) Floors

- In case of single bottom, plate floors are to be fitted at every frame.

– In the after peak of single screw ships, the height of the floors is to be increased such that their upper edge is well above the stern tube.

– Heavy plate floors are to be fitted in way of the aft face of the rudder horn and in line with the webs in the rudder horn. They may be required to be carried up to the first deck or flat. In this area, cut outs, scallops or other openings are to be kept to a minimum.

(ii) Bottom girders

– Centerline girder is to be extended up to the transom bulkhead.

– In case of single bottom, the spacing of bottom girders is not to be greater than 2.5[m].

– In case of double bottom, the spacing of bottom girders is not to be greater than 4.0 [m] and are to be extended as far forward and aft as practicable.

5.9.4 Side structure

5.9.4.1 Longitudinal framing

– Web frames are to be aligned with floors and are to extend to the upper deck.

– Adequate deep beams are to be provided on deck in line with the web frames.

5.9.4.2 Transverse framing

(i) Platforms and stringers

• Platforms and side girders within the peak are to be arranged in line with those located in the area immediately forward.

• Where this arrangement is not possible due to the shape of the hull and for access, structural continuity between the peak and the structures of the area immediately forward is to be ensured by means of scarphing brackets.

• Where the aft peak is adjacent to a machinery space with longitudinal framing, the side girders in the aft peak are to be fitted with tapering brackets.

• Above the floors, transverse side framing is to be supported by one of the following arrangements:

- Side stringers spaced not more than 2.6[m] apart, supported by web frames or transverse bulkheads. The lower ends of the web frames (supporting the stringers) are to be supported under the tank top by suitable deep beams or plate floors to form a ring structure.

– Perforated flats spaced not more than 2.6[m] apart. The area of perforations in each flat is to be not less than 10% of the total area of the flat.

– A combination of the above arrangements.

(ii) Vertical peak frames

• Web frames are to be arranged at every four frame spaces in transverse framing and the scantling requirements as specified in Chapter 7, Section 6 to be provided.

• Peak frames are to be extended up to a deck located above waterline corresponding to the scantling draught.

• Peak frames are to be bracketed at top and bottom and in way of side stringers, the connection is to provide adequate shear strength.

5.9.5 Longitudinal wash bulkheads

A centreline longitudinal non-tight bulkhead (wash bulkhead) is to be fitted,

- in the upper part of the peak, and stiffened at each frame spacing.

- in peak spaces used as tanks, where the breadth of the tank exceeds 0.5B and also in deep tanks used for fuel oil extending from side to side.

– Where either the stern overhang is very large or the maximum breadth of the space divided by watertight and wash bulkheads is greater than 20[m], additional longitudinal wash bulkheads may be required.

– The area of perforations is to be between 5% to 10% of the total area of bulkhead. The plating is to be suitably stiffened in way of the openings and the arrangement of openings is to be such as to maintain adequate shear rigidity.

5.9.6 Stern tube

The thickness of the stern tube provided is to be at least same as the required thickness of the side plating adjacent to the stern frame.

5.9.7 Alternative design verification

The spacing and arrangement, not complying with sections defined in Cl.5.9.3, Cl.5.9.4 and Cl.5.9.5, would be specially considered based on grillage analysis or FE analysis. The acceptance criteria to be applied for grillage analysis are defined in Chapter 7, Section 6. A FE analysis is to be performed in accordance with the requirements in Chapter 8.

5.9.8 Stiffening of floors and girders in aft peak

5.9.8.1 Stiffeners on the floors and girders in aft peak ballast or fresh water tanks above propeller are to be designed in accordance with Cl.5.9.8.2 and Cl.5.9.8.3. This applies for stiffeners located in an area extending longitudinally between the forward edge of the rudder and the after end of the propeller boss and transversely within the diameter of the propeller.

5.9.8.2 The height of stiffeners, h_{stf}, in [mm], on the floors and girders are not to be less than:

$h_{stf} = 80\ell_{stf}$ for flat bar stiffeners.

$h_{stf} = 70\ell_{stf}$ for bulb profiles and flanged stiffeners,

Where:

ℓ_{stf} = Span of stiffener, in [m], as shown in Figure 5.9.8. For this purpose, the length need not be taken greater than 5[m].

5.9.8.3 End brackets are to be provided as follows:

- Brackets are to be fitted at the lower and upper ends when ℓ_{stf-t} exceeds 4[m]

- Brackets are to be fitted at the lower end when ℓ_{stf-t} exceeds 2.5[m], where:

ℓ_{stf-t} = total length of stiffener, in [m], as shown in Figure 5.9.8.

5.9.8.4 Edge stiffener thickness not to be less than h_{stf}/15.

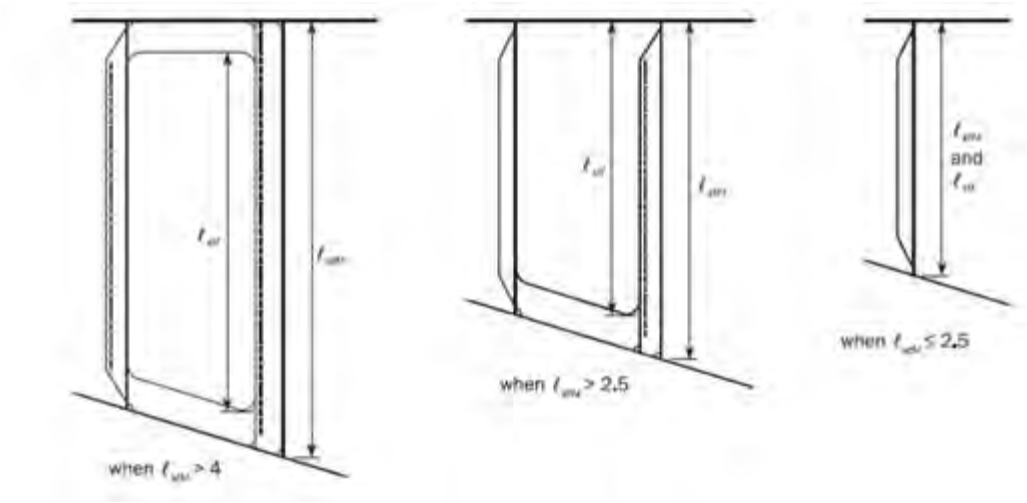


Figure 5.9.8: Floor and girder stiffening in the aft peak tank

5.10 Engine room arrangement

5.10.1 Application

5.10.1.1 The requirements of this section apply to the arrangement of structures located in the engine room. The requirements in Cl.5.10.2 to Cl.5.10.9 apply unless verification by means of direct analysis.

5.10.1.2 Requirements are given for engine room situated as follows:

- In the midship region.

- In the aft end region but with a compartment between it and the after peak bulkhead.

- In the aft end region where the after peak bulkhead forms the aft end of the machinery space.

Arrangements other than mentioned above are specially considered.

5.10.1.3 For midship engine room, if the shell and decks aft and forward of it are longitudinally framed, then longitudinal framing is to be adopted in engine room also.

5.10.1.4 For engine rooms situated aft, where the longitudinal framing terminates and is replaced by transverse framing, a suitable scarfing arrangement of the longitudinal framing extending at least three frame spaces to the adjacent structure is to be fitted.

5.10.1.5 Where openings in decks/bulkheads are provided in the engine room, the arrangements are to support the deck, side, and bottom structure.

5.10.1.6 All parts of the machinery, shafting, etc., are to be supported to distribute the loads into the ship's structure. The adjacent structure is to be suitably stiffened.

5.10.2 Double bottom structures

5.10.2.1 General

If the engine room is situated adjacent to the after peak, the double bottom is generally to be transversely framed. Elsewhere it may be transversely or longitudinally framed as appropriate.

5.10.2.2 Double bottom height

- The double bottom height at the centreline , irrespective of the location of the engine room, is not to be less than that defined in Chapter 3, Section 3, Cl.3.2.2.

- This height may need to be considerably increased in relation to the type and depth of main engine seatings. The above height is to be increased where the engine room is very large and where there is a considerable variation in draught between light ballast and full load conditions.

- Where the double bottom height in the engine room differs from that in adjacent spaces, structural continuity of longitudinal members is to be provided by sloping the inner bottom over an adequate e longitudinal extent. The knuckles in the sloped inner bottom are to be located in way of floors. Lesser double bottom height may be accepted in local areas provided that the overall strength of the double bottom structure is not thereby impaired.

5.10.2.3 Centreline girder

The double bottom is to be arranged with a centreline girder or side girders adjacent to centreline giving sufficient support for docking loads. Duct keel arrangement as per Cl.5.3.6 is acceptable instead of centerline girder.

5.10.2.4 Girders in way of machinery seatings

- Sufficient fore and aft girders are to be arranged in way of the main machinery to effectively distribute its weight and to ensure adequate rigidity of the structure. The girders are generally to extend over the full length of the engine room and are to be suitably scarfed into the bottom structure beyond. The height of the girders not be less than that of the floors in the engine room.

- Outboard of the engines, side girders are to be arranged, where practicable, to line up with the side girders in adjacent cargo spaces.

5.10.2.5 Longitudinal framing

- The bottom side girders are to be a continuation of any bottom longitudinals in the areas adjacent to the engine room and are generally to have a spacing not greater than 3 times that of longitudinals and in no case greater than 3[m].

- Plate floors are to be fitted at every second side frames. Additionally, under the main engine seatings, floors extending to the first side girder outside the engine seating, are to be fitted at intermediate frames.

5.10.2.6 Transverse framing

Where the double bottom in the engine room is transversely stiffened, plate floors are to be fitted at every frame.. In way of boilers, plate floors are to be fitted under the boiler bearers and other places as per requirements given in Cl.5.3.10.

5.10.3 Single bottom structures

5.10.3.1 General

For ships having a single bottom structure within the engine room, seatings are to be located above the floors and to be adequately connected to the latter and to the girders located below. Other arrangements will be specially considered..

5.10.3.2 Longitudinal framing

(i) Floors

- In way of main engine and thrust bearing, plate floors extending from side to side are to be fitted at every frame. In way of other areas of machinery spaces, the floors may be fitted at alternate frames.

• Additional plate floors are to be fitted under boiler foundations or other important machinery.

(ii) Girders

• The spacing and the arrangement of the centre line and side girders in a longitudinally framed single bottom are to be as per the requirements of Chapter 3, Section 5, Cl.5.3.8.1.

• The scantlings of the side girders are not to be less than that of the plate floors in the machinery space.

• Girders in way of engines are to be extended to the engine room bulkheads and suitably scarfed in to the bottom structure beyond. Alternatively, the engine girders are to be continued as side girder beyond the engine room.

5.10.3.3 Transverse framing

(i) Floors

• Plate floors are to be fitted at every frame.

• In the engine room, the thickness of floors webs and the face plate area are to be increased by 1[mm] and 50% respectively than the requirements as specified in Chapter 7, Section 3, Cl.3.3.1. The floors are not to be flanged.

• If the height of floors is reduced in way of crankcase, the face plate area is to be suitably increased, however the reduced height is normally not to be less than 2/3 of the depth of plate floor as specified in Chapter 7, Section 3, Cl.3.3.1 or B/16 whichever is greater.

• In way of engine room situated amidships the minimum depth of floors is to be at least 10% greater than that required elsewhere.

• In way of engine room situated aft, or where there is considerable rise of floor, the depth of the floors will be specially considered.

• Additional plate floors are to be fitted under boiler foundations or other important machinery.

(ii) Girders

• Sufficient fore and aft girders are to be arranged in way of machinery to effectively distribute its weight and ensure adequate rigidity of the structure.

• In midship engine room the girders are to be extended to the engine room bulkheads and are to be carried aft to support the foremost shaft tunnel bearing and in forward suitably scarfed in to the bottom structure.

• In engine room situated at the aft end, the girders are to be extended to the engine room bulkheads and suitably scarfed in to the bottom structure beyond. Alternatively, the engine girders are to be continued as side girder beyond the engine room.

5.10.4 Side structure

5.10.4.1 General

• In general, the engine room side shell stiffening is to be same as that of adjacent areas.

• In ships where the engine room is located aft and where the side shell is longitudinally framed, the longitudinal structure is to extend for the full length of the machinery space.

• Web frames are generally to be aligned with floors.

• End connections of side longitudinals at transverse bulkheads are to provide fixity, lateral support, and when not continuous are to be provided with soft-toe brackets. Brackets lapped onto the longitudinals are not to be fitted.

5.10.4.2 Longitudinal framing

• In case of side shell with longitudinal framing, the web frame spacing is not to be greater than four frame spacing.

• Web frames are to be connected at top and bottom to members of adequate stiffness and supported at lower decks by transverses or deep beams.

5.10.4.3 Transverse framing

• Where the engine room is situated aft, the spacing of web frames is generally not to exceed five transverse frame spaces. Web

frames extending from the tank top to the upper deck are to be connected at the top and bottom to members of suitable stiffness, and supported by deck transverses.

- In addition to the above if the span of the frames up to the lowest deck exceeds 6.5[m], a side stringer is to be provided to support the frames. These stringers are to be efficiently bracketed to bulkheads, and their connection to the web frames is to be such as to provide adequate continuity of face material.

- For two-stroke engines, web frames are generally to be fitted at the forward and aft ends of the engine. The depth of the girders should not be less than 2.5 times the depth of the cutout (if any) for the passage of secondary stiffener. The web frames are to be evenly distributed along the length of the engine.

- Greater web frame spacing may be accepted provided that cargo hold structural analysis in accordance with Chapter 8, Section 3 is carried out.

- Where the engine room is situated in the midship region, the spacing of web frames is generally not to exceed five transverse frame spaces and extending from the tank top to the level of the lowest deck above the load waterline.

5.10.5 Deck structure

5.10.5.1 Longitudinal framing

Where a longitudinal framing system is adopted, deck longitudinals are to be supported by deck transverses in line with web frames in association with pillars or pillar bulkheads. For the maximum spacing of deck transverses, refer Cl.5.10.4.2.

5.10.5.2 Transverse framing

Where a transverse framing system is adopted in engine room, deck beams are to be supported by a suitable arrangement of longitudinal girders in association with pillars or pillar bulkheads. Where fitted, deck transverses/deep beams are to be arranged in line with web frames to provide end fixity and transverse continuity of strength.

5.10.6 Inner bottom plating

5.10.6.1 Where main engines or thrust bearings are bolted directly to the inner bottom., the net thickness of the inner bottom plating is to be increased. Hold-down bolts are to be arranged as close as possible to floors and longitudinal girders. Manufacturer's recommendations are to be considered for plating thickness and the arrangements of hold-down bolts.

5.10.7 Manholes, lightening holes and wells

5.10.7.1 The number and size of manholes in floors/girders located in way of seatings and adjacent areas are to be kept minimum and permitted only where absolutely necessary for double bottom access and maintenance.

5.10.7.2 In general, manhole edges are to be stiffened with face plate; failing this, the floor plate is to be adequately stiffened with flat bars at manhole sides.

5.10.7.3 Drainage of the shaft tunnel is to be arranged through a well located at the aft end of the tunnel.

5.10.7.4 Lightening holes in engine foundations are to be kept as small as practicable and the edges are to be suitably reinforced.

5.10.8 Heavy equipment

5.10.8.1 Where heavy equipment is mounted directly on the inner bottom, the thickness of the floors and girders is to be suitably increased as per manufacturer's recommendation.

5.10.8.2 All parts of the machinery, shafting, etc., are to be supported to distribute the loads into the ship's structure. The adjacent structure is to be suitably stiffened

5.10.9 Machinery casings

Machinery casings are to be supported by a suitable arrangement of deck transverses and longitudinal girders in association with pillars or pillar bulkheads. In way of particularly large machinery casing openings, cross ties may be required and are to be arranged in line with deck transverses. Machinery casings are to be effectively framed and reinforced by deck transverses and girders around the openings.

5.11 Fore peak arrangement

5.11.1 Application

The requirements of this section apply for the arrangement of structures located forward of collision bulkhead and below bulkhead deck/freeboard deck.

5.11.2 Bottom structure

5.11.2.1 General

- The fore peak spaces are normally to have single bottom with transverse framing.

- The arrangement of the floors and centreline girder in the fore peak spaces are to be sufficient to give adequate stiffness to the structure. Webs and the upper edges of the floors and centreline girder are to be suitably stiffened.

5.11.2.2 Longitudinal framing

(i) Floors

- In case of single bottom, floors are to be fitted at every four frame spacing and generally spaced not more than 2.5[m].

- In case of double bottom, the spacing of plate floors is not to be greater than 3.6[m]. However, under deep tanks with height greater than 0.7 times the distance between inner bottom and main deck, the spacing of floors is normally not to exceed 2.5[m].

- The minimum depth of the floor at the centreline is not to be less than the required depth of the double bottom. Refer Chapter 3, Section 3, Cl.3.2.3.

(ii) Bottom girders

- In case of single bottom, the spacing of bottom girders is not to exceed 3.5[m] and are to be extended as far forward and aft as practicable.

- In case of double bottom, the spacing of the bottom girders is not to exceed 5.0[m] and are to be extended as far forward and aft as practicable.

- A supporting structure is to be provided at the centerline either by extending the centerline girder to the stem or by providing a deep girder or centerline bulkhead.

- In areas where the hull shape is very narrow, alternative arrangement, e.g. without centerline girder and with web fitted at every frame, may be accepted.

- Where a centerline girder is fitted, the minimum depth and thickness is not to be less than that required for the depth of the double bottom in the neighboring cargo hold region, and the upper edge is to be stiffened and to be connected to the collision bulkhead by means of a large end bracket.

5.11.2.3 Transverse framing

(i) Floors

- Plate floors are to be fitted at every frame. In case of ships of full form, the plate floors are to be supported by a centreline girder or a centreline wash bulkhead. In other cases, the centreline girder is to be carried as far forward as practicable.

(ii) Bottom girders

- In case of single bottom, the spacing of bottom girders is not to be greater than 2.5[m].

- In case of double bottom, the spacing of bottom girders is not to be greater than 4.0[m] and are to be extended as far forward and aft as practicable.

5.11.3 Side structure

5.11.3.1 Longitudinal framing

(i) Web frames

- Web frames are to be aligned with floors and are to extend to the upper deck.

- Adequate deep beams are to be provided on deck in line with the web frames.

(ii) Stringers

- Stringers aft of the collision bulkhead are to be continued as far forward as practicable.

- Where the depth of a tank exceeds 10[m], side transverses are generally to be supported

by one or more perforated flats or an arrangement of struts.

5.11.3.2 Transverse framing

(i) Peak frames

• Vertical peak frames of scantling as given in Chapter 7, Section 6, are to be provided at every frame and are to have the same vertical extension as the collision bulkhead(i.e. up to bulkhead deck). Peak frames are to be bracketed at top and bottom and in way of side stringers, the connection is to provide adequate shear strength.

• In way of a forward deep tank, web frames are to be fitted at every 5 frame spaces.

(ii) Stringers

• Above the floors, peak frames are to be supported by one of the following arrangements:

- Side stringers spaced about 2.6[m] apart and supported by (struts) panting beams (sections having the greater side vertically arranged) fitted at alternate frames.

The (struts) panting beams are to be bracketed to the frames and where the span is long, supported at the centreline by a complete or partial wash bulkhead or equally effective structure. Intermediate frames are to be bracketed to the stringer plates. The panting beams are to comply with buckling strength requirement given in Chapter 9, Section 5, 5.4.

– Side stringers spaced about 2.6[m] apart and supported by web frames. Web frames are to be in line with adequate deep beams on deck and floors in bottom to form a ring structure.

– Perforated flats/non-tight platforms spaced not more than 2.6[m] apart. The area of perforations in each flat is to be not less than 10 per cent of the total area of the flat. The plating is to be suitably framed in way of openings.

– A combination of the above arrangements.

• Where the depth of the fore peak space exceeds 10[m], a perforated flat is to be arranged at about mid-depth.

• Where the length of the peak spaces exceeds 10[m] and the side framing is supported as above, additional transverse strengthening in the form of transverse wash bulkheads or intermediate web frames are to be provided.

• The thickness of the plating and the section modulus of ordinary stiffeners in the perforated flats are not to be less than those required for the non-tight central longitudinal bulkhead.

5.11.4 Alternative design verification

The spacing and arrangement, other than those defined in Cl.5.11.2 and Cl.5.11.3 would be specially considered based on grillage analysis or FE analysis. The acceptance criteria to be applied for grillage analysis are defined in Chapter 7, Section 6. A FE analysis is to be performed in accordance with the requirements in Chapter 8.

5.11.5 Panting structures

5.11.5.1 Each panting beam is to be connected to the side frames and stringer by means of brackets whose arms are generally to be not less than twice the panting beam depth.

5.11.5.2 Side frames not supporting panting beams are to be connected to side stringer by means of brackets.

5.11.6 Breasthooks

Breasthooks are to have the same thickness of that required for platforms. They are to be arranged on the stem, in way of every side longitudinal in the case of longitudinal framing and at 1.0[m] in the case of transverse framing, extending aft for a length equal to approximately twice the breasthook spacing.

5.11.7 Wash bulkheads

- A centerline wash bulkhead is to be fitted in fore peak spaces used as tanks, where the breadth of the tank exceeds 0.5B.

- Where a centerline wash bulkhead is fitted, the lowest strake is to have thickness not less than required for a centerline girder.

- Webs of the primary supporting members are to be provided with end brackets and to be securely connected to the struts, if any.

- In general, openings in wash bulkheads are to have generous radii and the area of perforations is generally to be between 5% to 10% of the total area of bulkhead. The area of non-tight bulkhead is the whole cross-sectional area in one plane that covers the tank boundaries.

- The openings in the bulkhead plating is to be suitably stiffened to avoid high stress concentrations areas in way of connections with other structural members and the stiffeners are to be bracketed at top and bottom. Openings are to be located such as to affect as little as possible the plating sections adjacent to primary supporting members.

Section 6

Structural Detail Design

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

6.1 Local reinforcements

6.1.1 Reinforcements at knuckles

6.1.1.1 Knuckles are in general to be stiffened to achieve out-of-plane stiffness by fitting ordinary stiffeners or equivalent means in line with the knuckle.

6.1.1.2 Whenever a knuckle in a main member (shell, longitudinal bulkhead etc.) is arranged, stiffening in the form of webs, brackets or profiles is to be connected to the members which they are to transfer the load (in shear). See example of reinforcement at main member and upper hopper knuckle in Figure 6.1.1.1 and 6.1.1.2 respectively.

6.1.1.3 Where stiffeners intersect the knuckle as shown in Figure 6.1.1.1, effective support is to be provided for the stiffeners in way of the knuckle by fitting brackets.

6.1.1.4 When the stiffeners of the shell, inner shell or bulkhead intersect a knuckle at a narrow angle, it may be accepted to interrupt the stiffener at the knuckle, provided that proper end support in terms of carling, bracket or equivalent is fitted. Alternative design solution with, e.g. closely spaced carlings fitted across the knuckle between longitudinal members above and below the knuckle may be accepted.

6.1.1.5 For longitudinal shallow knuckles, i.e. angle less than 10[deg], closely spaced carlings are to be fitted across the knuckle, between longitudinal members above and below the knuckle. Carlings or other types of reinforcement need not be fitted in way of shallow knuckles that are not subject to high lateral loads and/or high in-plane loads across the knuckle, such as deck camber knuckles.

6.1.1.6 Generally, the distance between the knuckle and the support stiffening in line with the knuckle is not be greater than 50[mm] within 0.6L. For shallow knuckles, i.e. angles less than 10 deg, the distance of 75[mm] is acceptable. Alternative arrangements can be considered based on fatigue analysis in accordance with Chapter 10.

6.1.1.7 When a stiffener or primary supporting member is knuckled within the length of the span, effective support is to be provided by fitting tripping bracket or equivalent for the support of the face plate, and tripping bracket or equivalent for supporting the knuckled web section, Refer Figure 6.1.1.3.

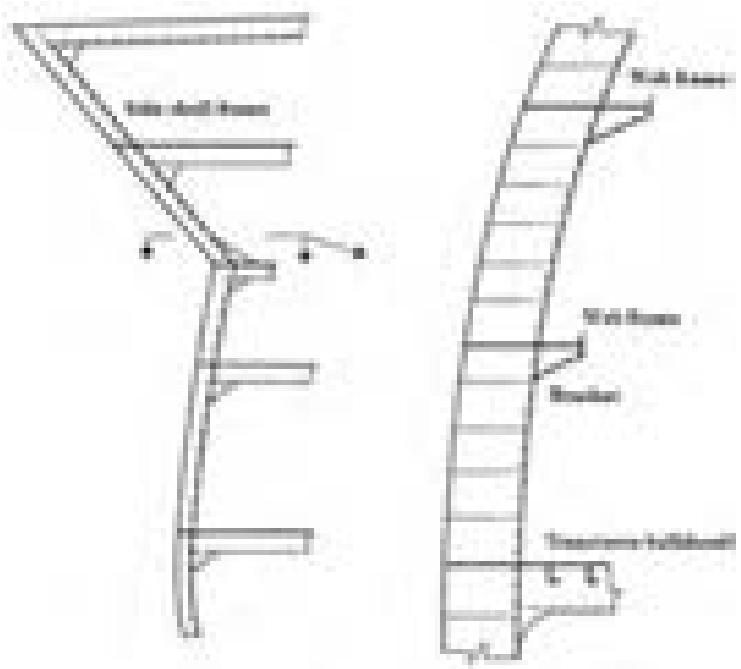


Figure 6.1.1.1: Reinforcement at main member knuckle

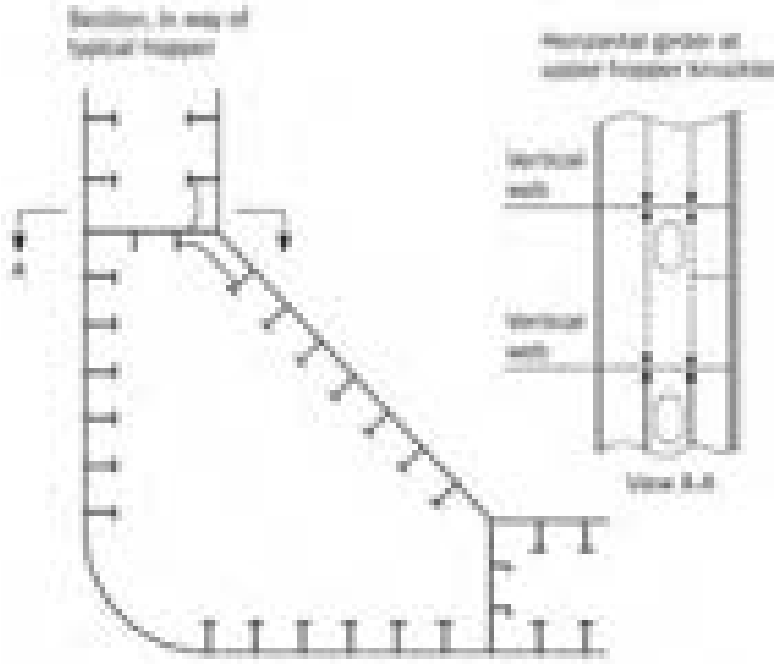


Figure 6.1.1.2: Reinforcement at upper hopper knuckle

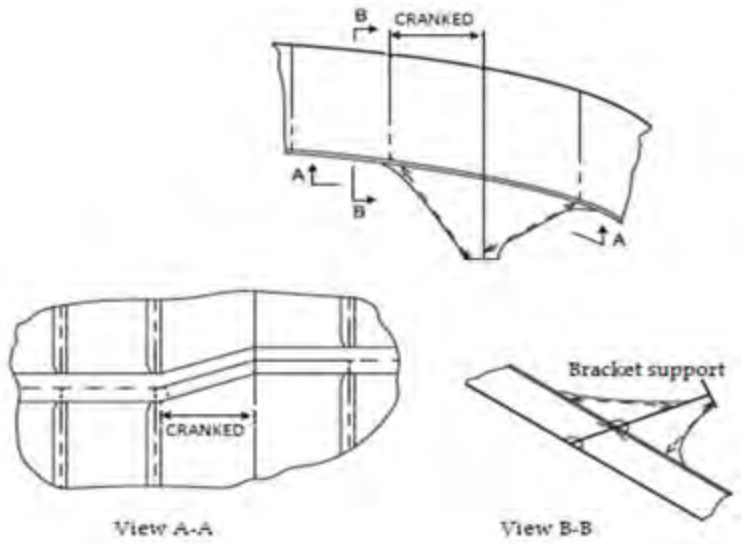


Figure 6.1.1.3: Reinforcement for knuckled stringer by bracket support

6.1.2 Knuckle support at integral bracket

If the flange transition between the stiffener and an integral bracket is knuckled, the flange is to be effectively supported in way of the knuckle. Alternatively, the flange may be curved with radius, in [mm], not less than,

$$r = \frac{0.4\, b_f^{\,2}}{t_f}$$

where:

b_f, t_f = Flange width and net thickness of the curved flange in [mm]. Refer Figure 6.1.2.

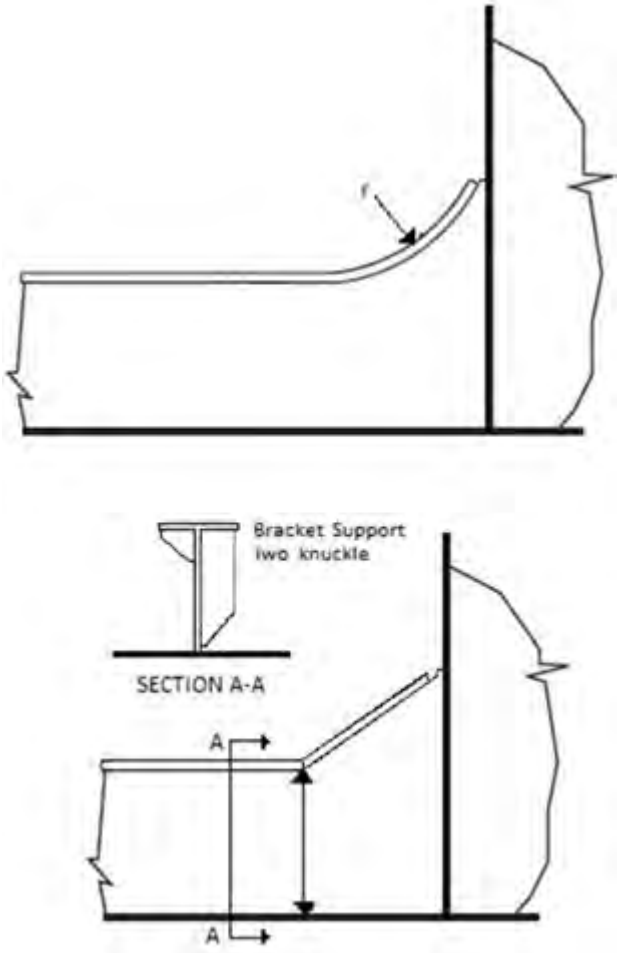


Figure 6.1.2: Reinforcement for knuckle support at integral bracket

6.1.3 Reinforcement in way of attachments for permanent means of access

Local reinforcement, considering location and strength, is to be provided in way of attachments to the hull structure for permanent means of access.

6.1.4 Reinforcement of deck structure in way of concentrated loads

The deck structure is to be reinforced in way of concentrated loads, such as anchor windlass, deck machinery, cranes, masts and derrick posts.

6.1.5 Reinforcement by insert plates

Insert plates are to be made of materials with, at least, the same specified minimum yield stress and the same grade as the plates to which they are welded.

6.2 Stiffeners

6.2.1 General

6.2.1.1 All type of stiffeners (excluding web stiffeners) in local areas are in general to be connected at their ends. However, in special cases sniped ends may be permitted. Requirements for the various types of connections (bracketed, bracketless or sniped ends) are given in Cl.6.2.2 to Cl.6.2.4.

6.2.1.2 Where the angle between the web plate of the stiffener and the attached plating is less than 50[deg], a tripping bracket is to be fitted spaced not more than 4 times the stiffener spacing. If the angle, α, between the web plate of an unsymmetrical stiffener and the attached plating is less than 50[deg], the face plate of the stiffener is to be fitted on the open angle side, Refer Figure 6.2.1.

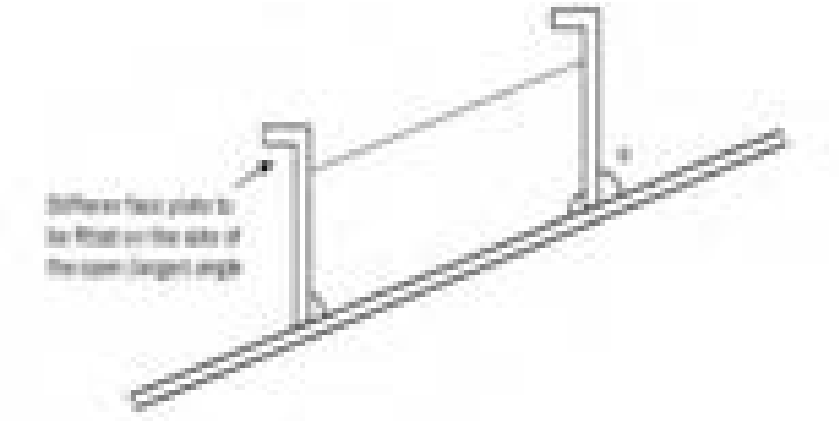


Figure 6.2.1: Stiffener on attached plating with an angle less than 50 deg

6.2.2 Bracketed end connections/attachments of stiffeners

6.2.2.1 General

- Where continuity of strength of longitudinal members is provided by brackets, the alignment of the brackets on each side of the primary supporting member is to be ensured, and the scantlings of the brackets are to be such that the combined stiffener/bracket section modulus and effective cross sectional area are not less than those of the member.

- At bracketed end connections, continuity of strength is to be maintained at the stiffener connection to the bracket and at the connection of the bracket to the supporting member.

6.2.2.2 Net web thickness of bracket

The net bracket web thickness, t_b, in [mm], is to comply with the following:

$$t_b \geq \left(2 + f_{bkt}\sqrt{Z}\right)\sqrt{\frac{R_{eH-stf}}{R_{eH-bkt}}}$$

and need not be greater than 13.5[mm].

where:

f_{bkt} = 0.2 for brackets with flange or edge stiffener. For edge stiffening, Refer 6.2.2.5.

= 0.3 for brackets without flange or edge stiffener.

Z = Net required section modulus, of the stiffener, in [cm^3]. In the case of two stiffeners connected, Z is the smaller net required section modulus of the two connected stiffeners.

R_{eH-stf} = Specified minimum yield stress of the stiffener material, in [N/mm^2].

R_{eH-bkt} = Specified minimum yield stress of the bracket material, in [N/mm^2].

Generally, thickness of the bracket not to be less than the lower of the web thickness of the two connected stiffeners.

6.2.2.3 Brackets size

Brackets are to be fitted at the ends of non-continuous stiffeners with arm lengths, l_{bkt} in [mm], taken as:

$$l_{bkt} = C_{bkt}\sqrt{\frac{Z}{t_b}}$$

l_{bkt} is not to be taken less than:

l_{bkt} = $1.8h_{stf}$ for connections where the end of the stiffener web is supported and the bracket is welded in line with the stiffener web or with offset necessary to enable welding, Refer item (c), (e), (f) in Figure 6.2.2.

l_{bkt} = $2.0\ h_{stf}$ for other cases, Refer items (a), (b) and (d) in Figure 6.2.2.

where:

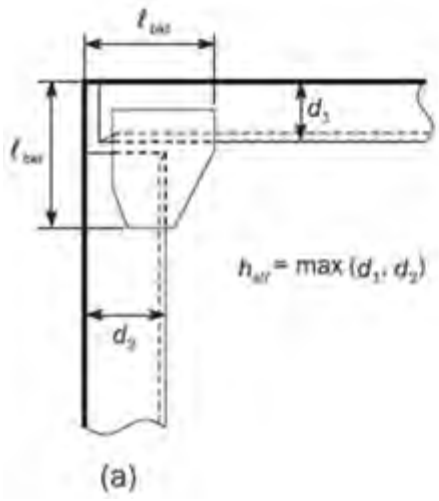
C_{bkt} = 65 for brackets with flange or edge stiffener

= 70 for brackets without flange or edge stiffener

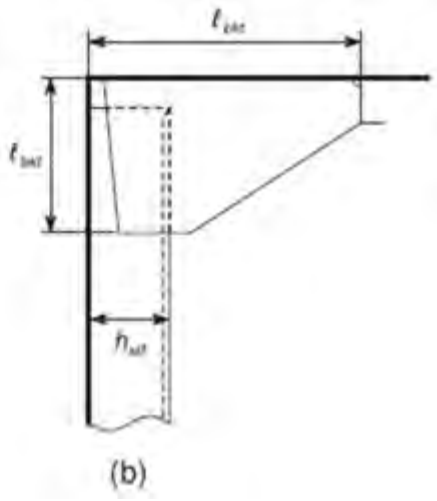
t_b = minimum net bracket thickness, in [mm].

For connections similar to item (b) in Figure 6.2.2, but not lapped, the bracket arm length is to comply with $l_{bkt} \geq h_{stf}$.

For connections similar to items (c) and (d) in Figure 6.2.2 where the smaller stiffener is connected to a primary supporting member or bulkhead, the bracket arm length is not be less than $2\ h_{stf}$.



(a)



(b)

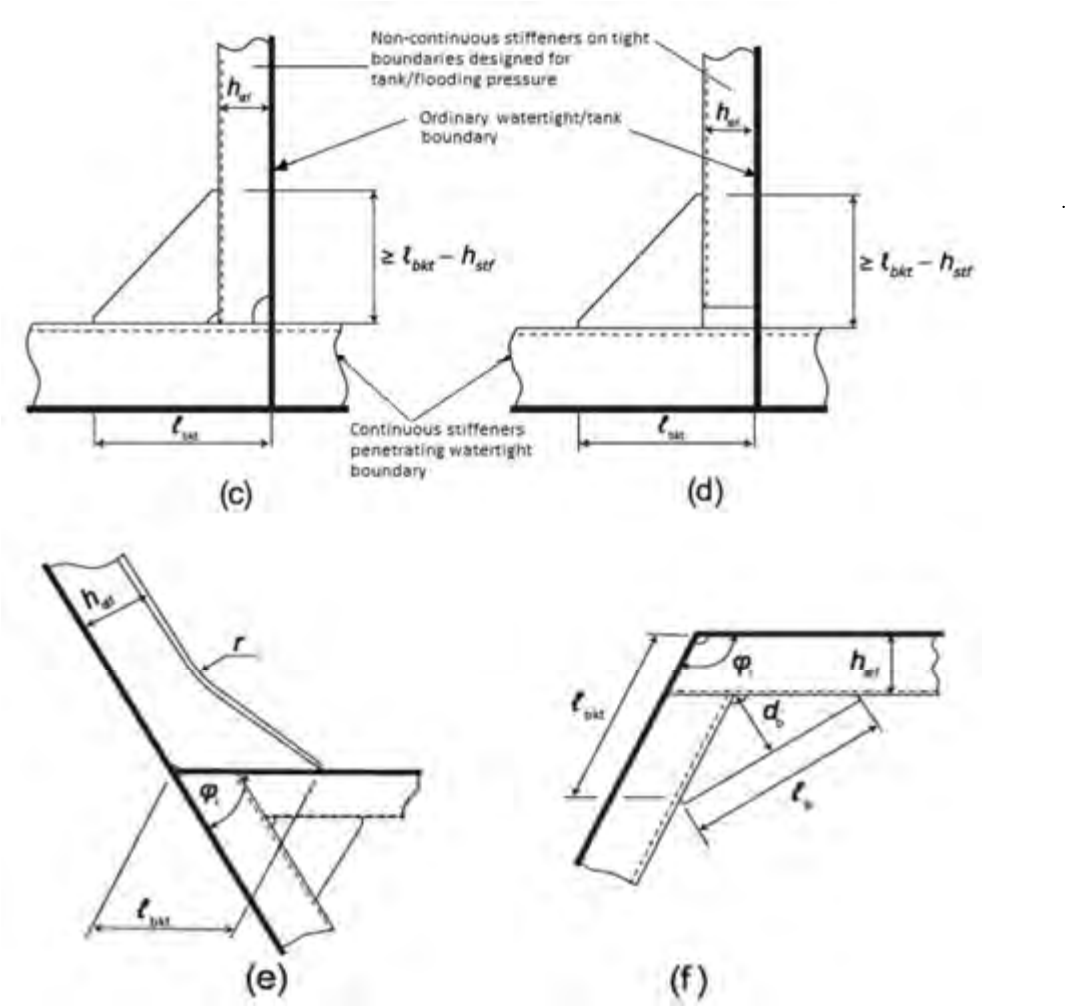


Figure 6.2.2: Bracket arm length of non-continuous stiffeners

6.2.2.4 Brackets with different arm lengths

The lengths of the arms, measured from the plating to the toe of the bracket, are to be such that:

$a, b \geq 0.8\ell_{bkt}$ and $a + b \geq 2.0\ell_{bkt}$ where ℓ_{bkt} is as defined in 6.2.2.3.

6.2.2.5 Edge stiffening of bracket

Where an edge stiffener is required, the web height of the edge stiffener, h_w in [mm], is not be less than:

$$h_w = 45\left(1 + \frac{Z}{2000}\right) \text{ but not less than } 50[\text{mm}].$$

where:

Z = Net required section modulus, of the stiffener, in [cm^3] as defined in 6.2.2.2.

Brackets subjected to compression is to fulfill the buckling requirement given in Chapter 9, Section 2 2.5.

6.2.3 Connection of continuous stiffeners

6.2.3.1 Stiffeners penetrating non-tight members: Connections for longitudinals and other stiffeners running continuously through girders (web frames, transverses, stringers, bulkheads etc.), may be without end brackets provided sufficient connection area is arranged for. Refer to Chapter 16, Section 3, Cl.3.5.3.

6.2.3.2 Stiffeners penetrating tight boundaries: Bracketed end connections are in general to be provided between non-continuous stiffeners on tight boundaries designed for tank pressure or flooding pressure and continuous stiffeners on adjacent boundaries, Refer item (c) and (d) in Figure 6.2.2.

6.2.3.3 Stiffeners penetrating tank boundaries: To prevent local plate bending due to relative deformation between the tank boundary stiffener and the stiffeners penetrating the tank boundary, brackets/stiffeners are to be fitted (Refer Figure 6.2.3).

For locations where the penetrating stiffeners are on a non-tight member e.g. a stringer or a girder, such brackets/stiffeners are omitted.

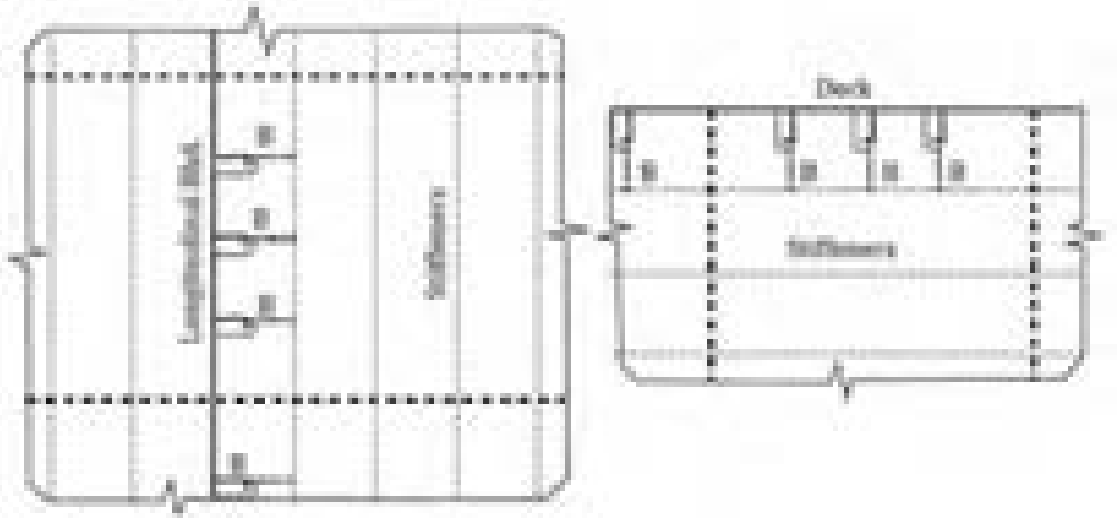


Figure 6.2.3: Brackets fitted in way of longitudinals penetrating a tank bulkhead (Transverse view)

6.2.4 Sniped ends

6.2.4.1 Sniped ends may be used where dynamic pressures are small, provided the net thickness of plating supported by the stiffener, in [mm], is not less than:

$$t_p = c_1 \sqrt{\left(1000\ell - \frac{s}{2}\right)\frac{s.P.k}{10^6}}$$

where:

P = Design pressure for the stiffener for the design load set being considered, in [kN/m^2]

c_1 = Coefficient for the design load set being considered, to be taken as:

= 1.2 for acceptance criteria set AC-I

= 1.1 for acceptance criteria set AC-II

ℓ = span of the sniped stiffener, in [m].

s = spacing of stiffeners, in [mm].

6.2.4.2 For sniped stiffeners fitted between stiffeners, the spacing s, in [mm], need not to be taken greater than 1000ℓ, where ℓ is the span of the sniped stiffener, in [m].

6.2.4.3 Sniped stiffeners are not to be used on structures in the vicinity of engines or generators in the machinery space or propeller impulse zone in the stern area nor on the shell envelope (at boundaries of sea chest).

6.2.4.4 Bracket toes and sniped stiffeners ends are to be terminated close to the adjacent member. The distance is not to exceed 40[mm] unless the bracket or member is supported by another member on the opposite side of the plating. Tapering of the sniped end is not to be more than 30[deg] in way of the toe where it is not practical to comply with this requirement, alternative arrangements are specially considered. The depth of toe or sniped end is, generally, not to exceed the thickness of the bracket toe or sniped end member, but need not be less than 15[mm].

6.2.5 Stiffeners on watertight bulkheads

6.2.5.1 Bulkhead stiffeners cut in way of watertight doors are to be supported by carlings extending to the adjacent stiffeners. Generally, the scantling of stiffeners adjacent to the door openings are to be designed to withstand this additional load. The carlings are to be located close to the door opening.

6.2.5.2 The watertight integrity of the bulkheads is to be ensured without taking strength of the doorframes into consideration.

6.2.5.3 The design pressure for ordinary watertight bulkheads is to be at least the pressure due to a head of water up to the bulkhead deck/ freeboard deck depending on the type of vessel. However, in damage conditions the head of water is to be considered in the worst situation at equilibrium, including intermediate stages of flooding. Refer Figure 6.2.5.

6.2.5.4 Where it is shown by damage stability calculations that the deepest equilibrium damaged waterline in way of the load point is below the freeboard deck, a reduced value of water head may be considered for bulkheads other than subdivision bulkheads.

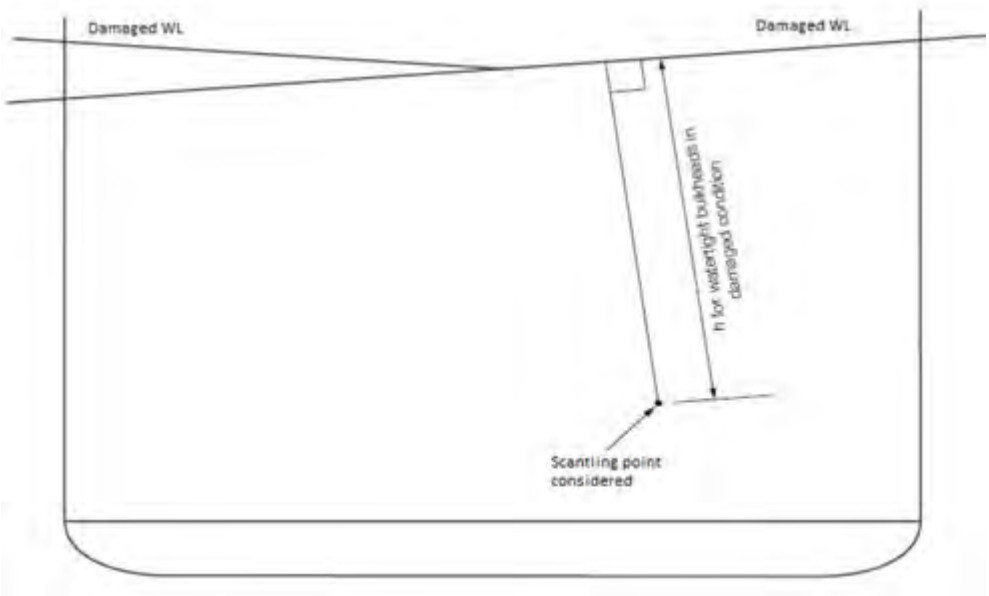


Figure 6.2.5: Heads for watertight bulkheads in damaged condition

6.3 Primary supporting members (PSM)

6.3.1 General

6.3.1.1 Primary supporting members web stiffeners, tripping brackets and end brackets are to comply with Cl.6.3.2 to Cl.6.3.4. Where the structural arrangement is such that these requirements cannot be complied with, adequate alternative arrangement with supporting calculations, if any, are to be submitted by the designer.

6.3.1.2 Abrupt changes of web height or cross section is to be avoided. Smooth transitions are to be provided.

6.3.2 Web stiffening arrangement

Web stiffeners arranged on primary supporting members are to comply with the requirements for scantlings of such stiffeners given in Chapter 9, Section 2, Cl.2.4.2.

6.3.3 Tripping bracket arrangement

6.3.3.1 In general girders are to be provided with tripping brackets and web stiffeners to obtain adequate lateral and web panel stability.

Tripping brackets welded to the face plate are generally to be fitted at every fourth stiffener in case of symmetrical sections and at every alternate stiffener in case of asymmetrical section , without exceeding 4[m].

6.3.3.2 Tripping brackets (Refer Figure 6.3.3) are generally to be fitted:

- at positions along the member span such that it satisfies the criteria of Chapter 9, Section 2, 2.5 for tripping bracket spacing and flange slenderness

- at the toes of end brackets

- at ends of continuous curved face plates not satisfying the radius criteria mentioned in Cl.6.1.2.

• in way of concentrated loads such as heels of pillars or cross ties.

• near a change of section

6.3.3.3 Tripping of flanges

Where the width of face plate on one side of the web exceeds $15t_f$, or 200[mm] whichever is lesser, the tripping brackets are to be connected to the face plate. In case of symmetrical face plate, backing brackets are to be fitted on other side of the tripping brackets.

6.3.3.4 Stiffeners are to be connected at ends. Where stiffener webs are not perpendicular to the shell plating adequate tripping brackets are to be fitted.

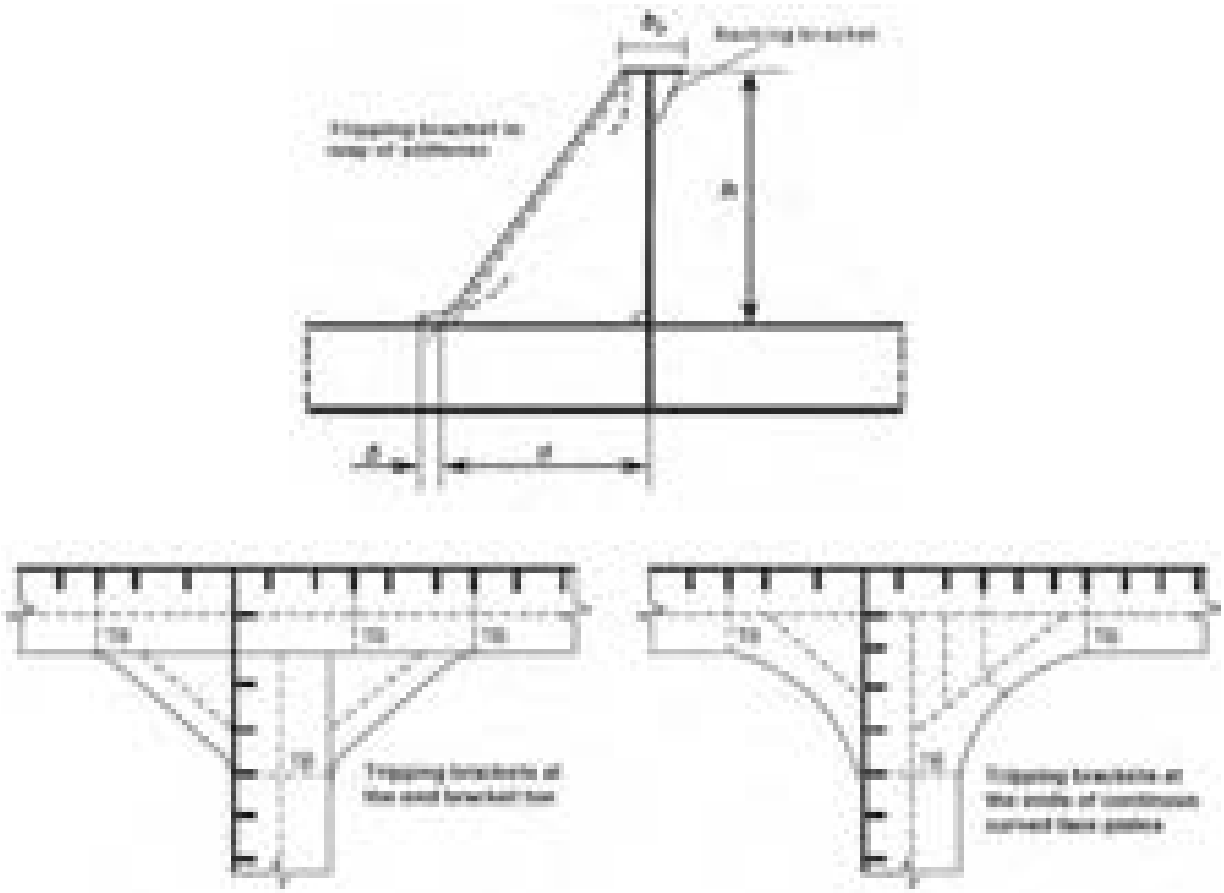


Figure 6.3.3: Tripping bracket arrangement for primary supporting members

6.3.4 Arm length and edge stiffening

6.3.4.1 The arm length, d in [m], of tripping brackets is not be less than the greater of the following values:

$$d = 0.38\,h$$

$$d = 0.85\,h\sqrt{\frac{s_t}{t_b}}$$

where:

h = height of tripping brackets, in [m], shown in Figure 6.3.3.

s_t = spacing of tripping brackets, in [m].

t_b = net thickness of tripping brackets, in [mm].

6.3.4.2 Tripping brackets on primary supporting members are to be stiffened by a flange or edge stiffener if the effective length of the edge, ℓ_b in [mm], is greater than:

$$\ell_b = 75t_b$$

where:

t_b = Bracket net web thickness, in [mm].

6.3.4.3 Additional stiffeners are to be fitted parallel to the free edge to ensure that the arm length of an unstiffened triangular end panel does not exceed 100t [mm].

6.3.4.4 The tripping brackets are to be adequately dimensioned at base and are to have a smooth transition to the adjoining stiffeners. It is recommended that the bracket toe should be designed as shown in Figure 6.3.3..

6.3.4.5 Where the span of frames exceeds 5[m] or the flange width is less than 1/20 of the span of stiffeners 'ℓ', tripping brackets are to be fitted at the middle of the span.

6.3.4.6 Forward of 0.15L from F.P., the tripping brackets are to be fitted on frames as shown in Figure 6.3.4. The spacing between tripping brackets is not to exceed 2.5[m] vertically.

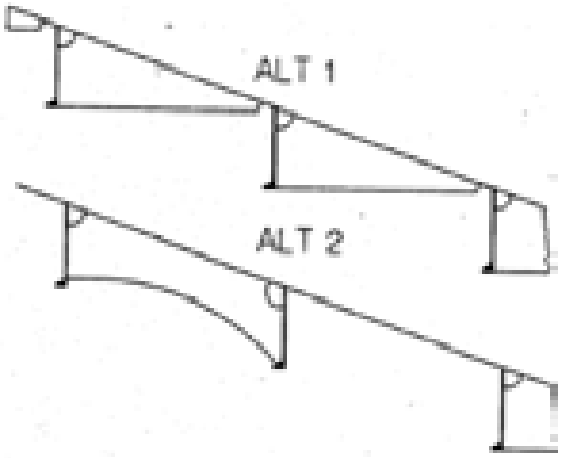


Figure 6.3.4: Alternative arrangements of tripping bracket

6.3.5 End connections/attachments

6.3.5.1 General

- Brackets or equivalent structure are to be provided at ends of primary supporting members.

- End brackets are generally to be made with soft toe in areas considered critical with respect to fatigue.

- Bracket less connections may be applied provided that there is adequate support of adjoining face plates, Refer Cl.6.3.6.

- The end attachments and supporting structure of the girders are to provide adequate resistance against rotation and displacement of the joint and effective distribution of the load from the member. Supporting members, to which the girders are being connected, may require additional strengthening to provide adequate stiffness to resist rotation of the joint.

- Where the end attachment provides only a low degree of restraint against rotation, the girder is generally to be extended beyond the point of support by at least two frame spaces before being gradually tapered.

- Connections between girders forming a ring system are to be such as to minimize stress concentrations at the junctions. Integral brackets are generally to be radiused or well rounded at the toes.

- Where the face plate of the girder is not continuous over the bracket, the free edge of the bracket is to be stiffened and the face plate of the girder is to be extended well beyond the toe of the bracket.

6.3.5.2 Scantling of end brackets

- In general, the arm lengths of brackets connecting PSMs, as shown in Figure 6.3.5.1 are not be less than the web depth of the member, and need not be taken greater than 1.5 times the web depth.

- The arm length 'a' including the depth of girder is not to be less than:

$$a = 83\sqrt{\frac{Z}{t}}$$

Z = The section modulus in [cm^3], of the girder to which the bracket is connected as defined in Cl.6.2.2.2.

- Direct strength calculations by means of grillage analysis or FE analysis will be considered as alternative basis for the scantlings. The acceptance criteria to be applied for grillage analysis are defined in Chapter 7, Section 6. A FE analysis is to be performed in accordance with the requirements in Chapter 8.

• The thickness of the bracket is, in general, not to be less than that of the PSM web plate.

• Scantlings of the end brackets are to be such that the section modulus of the PSM with end bracket, excluding face plate where it is sniped, is not to be less than that of the primary supporting member at mid-span. The net cross sectional area, A_f in [cm^2], of face plates of brackets is not be less than:

$$A_f = \ell_b \times t_b$$

where:

ℓ_b = length of bracket edge, in [m], Refer Figure 6.3.5.1. For brackets that are curved, the length of the bracket edge may be taken as the length of the tangent at the midpoint of the edge

t_b = minimum net bracket thickness, in [mm], as defined in Cl.6.2.2.2.

Moreover, the net thickness of the face plate is to be not less than that of the bracket web. Additional requirements with respect to buckling are given in Chapter 9, Section 2.

6.3.5.3 Arrangement of end brackets

• In addition, where the length of free edge of bracket, ℓ_b, is greater than 1.5[m], the web of the bracket is to be stiffened as follows:

- the net sectional area, in [cm^2], of web stiffeners is to be not less than the requirement for face plates given in Cl.6.3.5.2 not to be greater than 16.5ℓ, where ℓ is the span, in [m], of the stiffener.

– tripping flat bars are to be fitted. Where the width of the symmetrical face plate is greater than 400[mm], additional backing brackets are to be fitted.

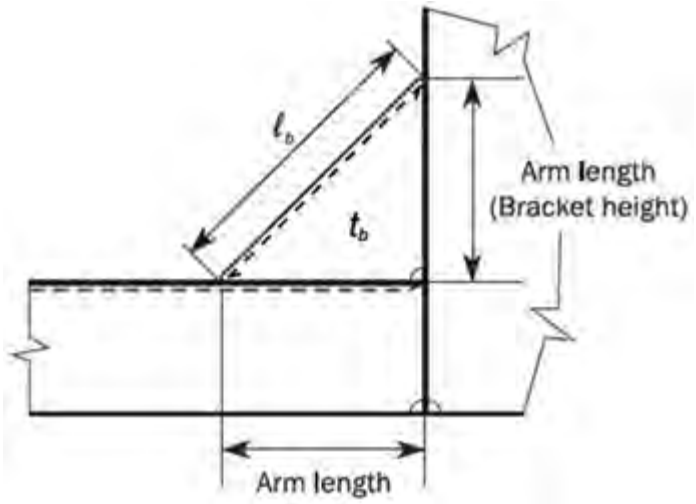


Figure 6.3.5.1: Dimensions of bracket

• For a ring system where the end bracket is integral with the webs of the members and the face plate is carried continuously along the edges of the members and the bracket, the full area of the largest face plate is to be maintained close to the mid-point of the bracket and gradually tapered to the smaller face plates.

• Butts in face plates are to be kept well clear of the bracket toes.

• Where a wide face plate abuts a narrower one, the taper is not to be greater than 1 to 4.

• The toes of brackets are not to land on unstiffened plating. The toe height is not to be greater than the thickness of the bracket toe, but need not be less than 15[mm]. In general, the end brackets of primary supporting members are to be soft-toed.

• Where primary supporting members are constructed of higher strength steel, particular attention is to be paid to the design of the end bracket toes in order to minimize stress concentrations.

• Where a face plate is welded onto the edge or welded adjacent to the edge of the end bracket, Refer Figure 6.3.5.2, the face plate is to be sniped and tapered at an angle not greater than 30[deg].

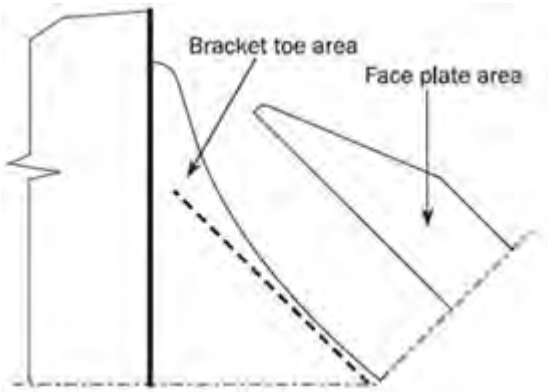


Figure 6.3.5.2: Bracket face plate adjacent to the edge

6.3.6 Bracketless end connections

6.3.6.1 Two-member connection

• The net thickness, t_b in [mm], of the common web at the joint of two-member bracketless connection, Refer Figure 6.3.6.1, is not to be less than the greater of the following:

$$t_b = \left(\frac{\sigma_1 A_1}{h_2} - \tau_2 t_2\right)\frac{k}{110}$$

$$t_b = \left(\frac{\sigma_2 A_2}{h_1} - \tau_1 t_1\right)\frac{k}{110}$$

and also, t_b is not to be less than t_1 and t_2.

where,

A_1, A_2 = minimum required flange area of member 1 and 2 respectively, in [mm^2].

h_1, h_2 = web height of member 1 and 2 respectively, in [mm].

t_1,t_2 = minimum required thickness of web of member 1 and 2 respectively, in [mm].

τ_1,τ_2 = average shear stress in member 1 and 2 respectively in [N/mm^2].

σ_1, σ_2 = bending stress at flange area of member 1 and 2 respectively, in [N/mm^2].

k = material factor for common web at corner, Refer Chapter 2.

• Bending and shear stress computation is to be carried out by considering each member separately as a beam with appropriate boundary conditions. The corner joint between the members is to be considered as fixed joint.

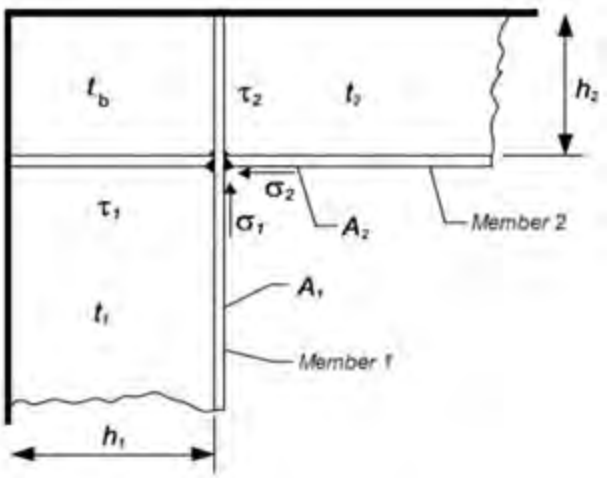


Figure 6.3.6.1: Bracketless end connection for two members

6.3.6.2 Three Member Connection

• In the case of three-member bracketless connection, when the flanges of member 1 and member 3 are continuous, Refer Figure 6.3.6.2, the thickness, t_b in [mm], of the common web at the joint is not to be less than the following:

$$t_b = \left(\frac{\sigma_2 A_2}{h_1} - \tau_1 t_1\right)\frac{k}{110}$$

and also, t_b is not to be less than t_1 and t_2.

• When the web height of member 1 and member 2 is more than $100t_b$, the common part of the corner web is to be stiffened.

• In the case of three-member bracketless connection, when the flanges of member 1 and member 3 are not continuous, the thickness, t_b of the common web at the joint is to be determined as given by Cl.6.3.6.1

• Welding of the extended face plates of the joining members to the common web at the corner is to be of sufficient length and size to transfer the axial loads from the face plates to the common web.

• Welding of the discontinuous and extended face plate to the continuous face plate is to be full penetration or deep penetration of equivalent strength.

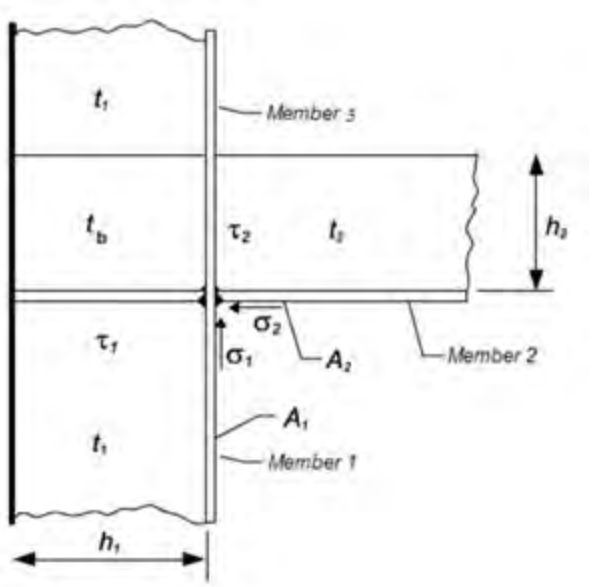


Figure 6.3.6.2: Bracketless end connection for three members

6.4 Corrugated bulkheads

6.4.1 Construction

6.4.1.1 The main dimensions a, R, c, d, t_f, t_w, S_c of corrugated bulkheads are defined in Figure 6.4.1.

6.4.1.2 The corrugation angle φ is generally not be less than 55[deg]. Corrugation angle between 45[deg] and 55[deg] may be accepted provided that the permissible stress/permissible utilization factor for overall bending is reduced by 10%, Refer Chapter 7, Section 5, Cl.5.1.1.5 for section modulus, Chapter 8, Section 3, Cl.3.4.2.4 for yield check and Chapter 9 Section 1, Cl.1.3.3 for buckling assessment.

6.4.1.3 Welds in a direction parallel to the bend axis are to be avoided in the zone of the bending.

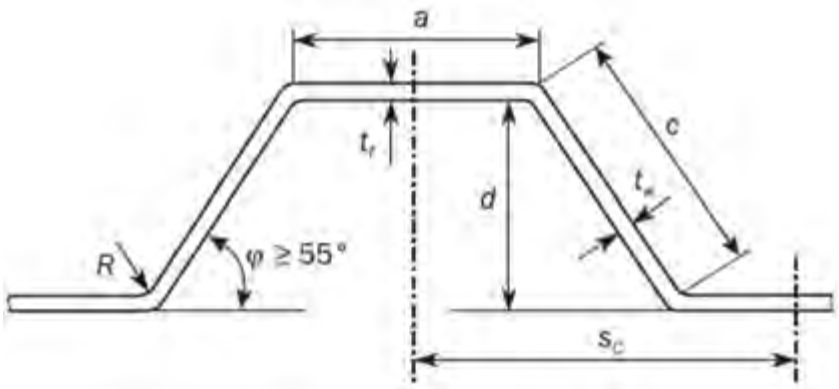


Figure 6.4.1: Dimensions of a corrugated bulkhead

6.4.2 Corrugated bulkhead depth

The depth of the corrugation, d, in [mm], is not to be less than:

$$d \geq \frac{1000\, l_c}{C}$$

where

l_c = Mean span of considered corrugation, in [m], as defined in 6.4.3.

C = Coefficient to be taken as:

15 for tank and water ballast cargo hold bulkheads.

18 for dry cargo hold bulkheads.

6.4.3 Span of corrugations

The length, l_c of the corrugations is to be taken as the distance shown in Figure 6.4.3. For the definition of l_c , the bottom of the upper stool is not be taken more than a distance from the deck at the centre line equal to:

- 3 times the depth of corrugation, for non-rectangular stool.

- 2 times the depth of corrugation, for rectangular stool.

l_{bdg} = effective bending span taken as the distance between supports, in [m]. For vertically corrugated bulkheads with stool tanks taken as the distance between the area centre of the upper stool A_U and the area centre of the lower stool A_L as shown in Figure 6.4.3.

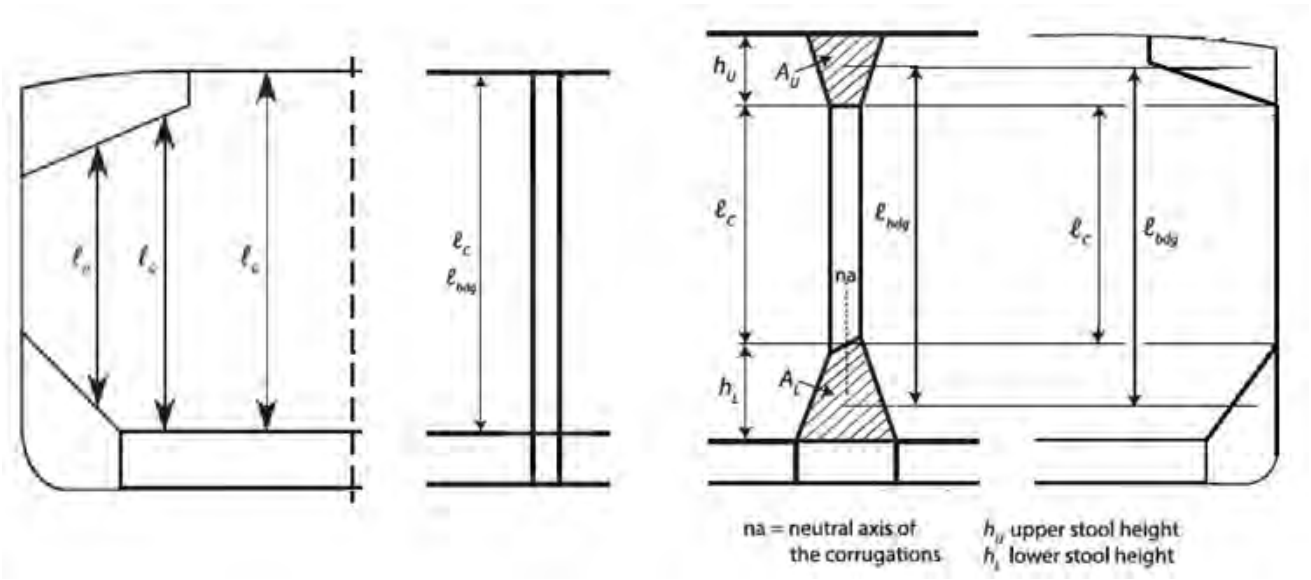


Figure 6.4.3: Span of the corrugations

6.4.4 Structural arrangements

Where corrugated bulkheads are cut in way of primary supporting members, corrugations on each side of the primary member are to be aligned with each other.

6.4.5 Corrugated bulkhead end supports

6.4.5.1 The strength continuity of corrugated bulkheads is to be maintained at the ends of corrugations.

6.4.5.2 Where a bulkhead is provided with a lower stool, floors or girders are to be fitted in line with both sides of the lower stool to ensure proper stress flow from the bulkhead to the plate floors.

Where a bulkhead is not provided with a lower stool, floors or girders are to be fitted in line with both flanges of the vertically corrugated transverse bulkhead. The supporting floors or girders are to be connected to each other by suitably designed shear plates.

6.4.5.3 At deck, if no upper stool is fitted, transverse or longitudinal stiffeners are to be fitted in line with the corrugation flanges.

6.4.5.4 The thickness and material properties of the floors, transverse or longitudinal deck stiffeners, in line with the corrugation flanges are to be at least equal to those provided for the corrugation flanges.

6.4.5.5 When the corrugation flange connected to the adjoining boundary structures (i.e. inner hull, side shell, longitudinal bulkhead, trunk, etc.) is smaller than 50% of the width of the typical corrugation flange, an advanced analysis of the connection is required.

6.4.6 Corrugation web support

For boundaries of tanks, brackets to be provided in line with the corrugation webs at lower end of vertical corrugations and at both ends for horizontal corrugations. Alternatively, supporting brackets in way of every knuckle of corrugation web may be fitted.

6.4.7 Bulkhead stool

Stool side plating is to be aligned with the corrugation flanges.

6.5 Openings

6.5.1 Openings and scallops in stiffeners

Air and drain holes as mentioned in Chapter 3, Section 4, Cl.4.2.2.2 are to comply with the following requirements:

6.5.1.1 Figure 6.5.1.1 shows examples of air holes, drain holes and scallops. In general, the ratio of a/b, as defined in Figure 6.5.1.1, is to be between 0.5 and 1.0. In fatigue sensitive areas further consideration may be required with respect to the details and arrangements of openings and scallops.

6.5.1.2 Openings and scallops are to be kept at least 200[mm] clear of the toes of end brackets, end connections and other areas of high stress concentration, measured along the length of the stiffener toward the mid-span and 50[mm] measured along the length in the opposite direction, Refer Figure 6.5.1.2. In areas where the shear stress is less than 60% of the permissible stress, alternative arrangements may be accepted.

The stopper holes/crack arrestor holes are also to be avoided in way of shaded area shown in Figure 6.5.1.2, since these areas are of high stress concentrations which lead to cracks and subsequently piping effect.

6.5.1.3 Closely spaced scallops or drain holes preferably of elliptical shape, i.e. where the distance between scallops/drain holes is less than twice the width b as shown in Figure 6.5.1.1, are not permitted in stiffeners contributing to the longitudinal strength. For other stiffeners, closely spaced scallops/drain holes are not permitted within 20% of the stiffener span measured from the end of the stiffener.

6.5.1.4 Openings are to be well rounded with smooth edges. Closely spaced scallops are not permitted in high strength steel members.

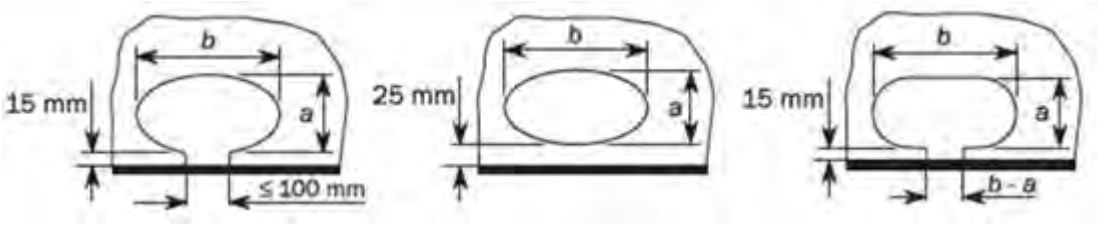


Figure 6.5.1.1: Examples of air holes, drain holes and scallops

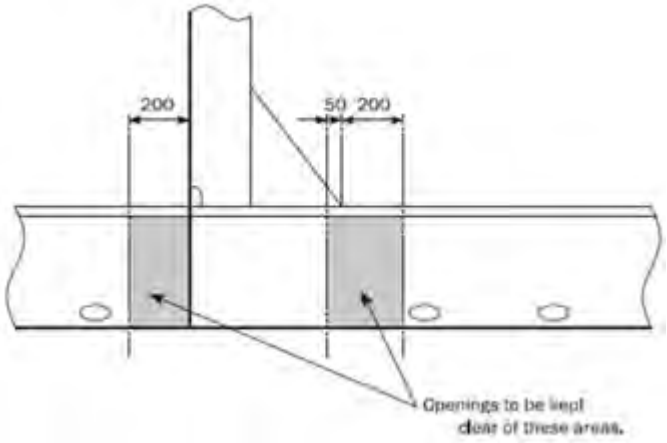


Figure 6.5.1.2: Location of air holes drain holes and crack arrestor holes

6.5.2 Openings in primary supporting members

6.5.2.1 General

• Manholes, lightening holes and other similar openings are to be avoided in way of concentrated loads and areas of high shear. In particular, manholes and similar openings are to be avoided in high stress areas unless the stresses in the plating and the panel buckling characteristics have been calculated and found satisfactory.

• Examples of high stress areas include:

– Vertical or horizontal diaphragm plates in narrow cofferdams/double plate bulkheads within one-sixth of their length from either end.

– Floors or double bottom girders close to their span ends.

– Primary supporting member webs in way of end bracket toes.

–Above the heads and below the heels of pillars.

• Where openings are arranged, the shape of openings is to be such that the stress concentration remains within acceptable limits. Openings are to be well rounded with smooth edges.

• Cut-outs for the passage of ordinary stiffeners are to be as small as possible and well-rounded with smooth edges.

• In general, the depth of cut-outs is to be not greater than 50% of the depth of the primary supporting member.

• Where openings are cut in the girder web, they are to be away from the girder ends and scallops for stiffeners; with their centre located as near to the neutral axis of the girder as practicable. Openings of depth exceeding 25% of the girder depth or 300[mm] and, of length exceeding the depth of the girder or 60% of the secondary stiffener spacing, are to be reinforced all around at the edge; or alternatively by providing horizontal and vertical stiffeners.

• The breadth of cut-out is to be kept as small as practicable and the top edge is to be rounded, or the corner radii made as large as practicable. The extent of direct connection of the web plating, or the scantlings of lugs or collars, is to be sufficient for the load to be transmitted from the secondary member.

6.5.2.2 Manholes and lightening holes

- Web openings as indicated below do not require reinforcement

– In single skin sections, having depth not exceeding 25% of the web depth and located so that the edges are not less than 40% of the web depth from the faceplate.

– In double skin sections, having depth not exceeding 50% of the web depth and located so that the edges are well clear of cut outs for the passage of stiffeners.

- For web openings without reinforcements of free edges, the length of openings is not be greater than:

– At the mid-span of primary supporting members: lesser of the distance between adjacent openings.

– At the ends of the span: 25% of the distance between adjacent openings.

- For openings cut in single skin sections, the length of opening is not to be greater than the web depth or 60% of the secondary stiffener spacing, whichever is greater.

- The ends of the openings are to be equidistant from the cut outs for stiffeners.

- Where lightening holes are cut in the brackets, the distance from the circumference of the hole to the free flange of brackets is not to be less than the diameter of the lightening hole.

- Openings which require reinforcement are according to Cl.6.5.3. Where larger openings are proposed, the arrangements and compensation are to be submitted to IRS.

6.5.2.3 Scallops, air- and drain holes

- Scallops are to be avoided in way of fatigue sensitive areas.

- Examples of fatigue sensitive areas include:

– Connections of transverse webs in double side or double bottom tanks to hopper tanks.

– Connections at horizontal stringer heel.

– Connections of vertically corrugated bulkhead to lower and upper stool diaphragms.

– Connections of vertically corrugated bulkhead without stool to inner bottom/hopper supporting structures.

– Connections of plain cofferdam bulkhead vertical frames to inner bottom longitudinal girders.

For spacing and dimensions of openings requirements given in Cl.6.5.1 to be followed.

6.5.3 Reinforcements around openings

6.5.3.1 Manholes and lightening holes are to be stiffened according to this requirement, except where alternative arrangements are demonstrated as satisfactory, i.e. the stresses in the plating and the panel buckling characteristics are to be calculated and proved to be satisfactory or with the analysis methods described in Chapter 8.

6.5.3.2 On members contributing to longitudinal strength, stiffeners are to be fitted along the free edges of the openings parallel to the vertical and horizontal axis of the opening.

6.5.3.3 Stiffeners may be omitted in one direction if the shortest axis is less than 400[mm] and in both directions if length of both axes is less than 300[mm]. Edge reinforcement may be used as an alternative to stiffeners, Refer Figure 6.5.3.1.

6.5.3.4 All openings are to be adequately framed; attention is to be paid to structural continuity, and abrupt changes of shape, section or plate thickness are to be avoided. Arrangements in way of corners and openings are to be such as to minimize the creation of stress concentrations.

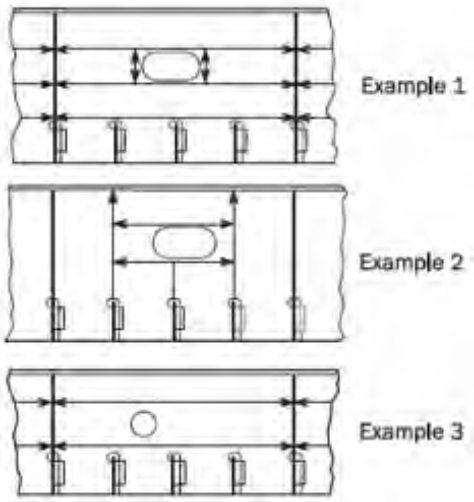


Figure 6.5.3.1: Web plate with openings

6.5.3.5 In the case of large openings in the web of PSMs (e.g. where a pipe tunnel is fitted in the double bottom), the secondary stresses in PSMs are to be considered for the reinforcement of these openings.

6.5.3.6 Where no FE analysis is performed, this may be carried out by assigning an equivalent net shear sectional area to the PSM obtained, in [cm^2], according from the following formula:

$$A_{sh} = \frac{A_{sh1}}{1 + \frac{32\, l_{shr}^2\, A_{sh1}}{I_1}} + \frac{A_{sh2}}{1 + \frac{32\, l_{shr}^2\, A_{sh2}}{I_2}}$$

where:

I_1, I_2 = Net moments of inertia, in [cm^4], of deep webs (1) and (2), respectively, with attached plating around their neutral axes parallel to the plating.

A_{sh1}, A_{sh2} = Net shear sectional areas, in [cm^2], of deep webs (1) and (2), respectively, taking account of the web height reduction by the depth of the cut out for the passage of the ordinary stiffeners, if any.

l_{shr} = Shear span, in [m], of deep webs (1) and (2) as defined in Chapter 3, Section 7. Deep web (1) and (2) are defined in Figure 6.5.3.2.

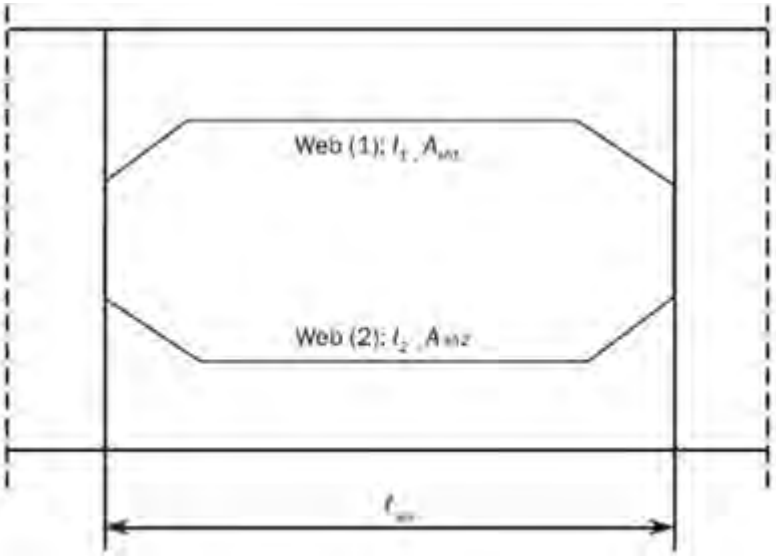


Figure 6.5.3.2: Large openings in the web of PSM

6.5.4 Openings in the deck contributing to longitudinal strength

6.5.4.1 General

• Openings in the strength deck are to be kept to a minimum and spaced as far as practicable from one another and from the ends of superstructures. Openings are to be located as far as practicable from high stress regions such as side shell platings, hatchway corners, or hatch side coamings.

• All openings in the strength deck are to have well rounded corners.

• Openings in the strength deck within 0.6L amidships (within the cargo hold region for ships with large hatch openings) are as far as practicable to be located inside the line of large hatch openings. Necessary openings outside this line are to be kept well clear of the ship's side and hatch corners as shown in Figure 6.5.4.1.

• Openings in lower decks are to be kept well clear of the main hatch corners and other areas of high stresses.

• Openings are generally to be located outside the limits as shown in Figure 6.5.4.1 in dashed area, defined by:

– The bent area of a rounded sheer strake, if any, or the side shell.

– $e = 0.25(B - b)$ from the edge of opening.

– $c = 0.07\ell + 0.1b$ or $0.25b$, whichever is greater.

where:

b = Width, in [m], of the hatchway considered, measured in the transverse direction, Refer Figure 6.5.4.1.

ℓ = Width, in [m], in way of the corner considered, of the cross deck strip between two consecutive hatchways, measured in the longitudinal direction, Refer Figure 6.5.4.1.

Transverse distance between the above limits and openings or between hatchways and openings as shown Figure 6.5.4.1 is not to be less than:

– $g_2 = 2a_2$ for circular openings.

– $g_1 = a_1$ for elliptical openings.

Transverse distance between openings as shown in Figure 6.5.4.1 is not to be less than:

– $2(a_1 + a_2)$ for circular openings.

– $1.5(a_1 + a_2)$ for elliptical openings.

where:

a_1 = Transverse dimension of elliptical openings, or diameter of circular openings.

a_2 = Transverse dimension of elliptical openings, or diameter of circular openings.

a_3 = Longitudinal dimension of elliptical openings, or diameter of circular openings.

Longitudinal distance between openings is not to be less than:

–$(a_1 + a_3)$ for circular openings.

– $0.75(a_1 + a_3)$ for elliptical openings and for an elliptical opening in line with a circular one.

• If the opening arrangements do not comply with these requirements, the hull girder longitudinal strength assessment is to be carried out by subtracting such opening areas, Refer Chapter 3, Section 6, Cl.6.5.4.6.

• Smaller openings in the strength deck within 0.6L amidships or within the cargo hold region for ships with large hatch openings, are to be reinforced as given in Cl.6.5.4.2 below. The area of these reinforcements is not to be included in the sectional areas used in the section modulus calculation.

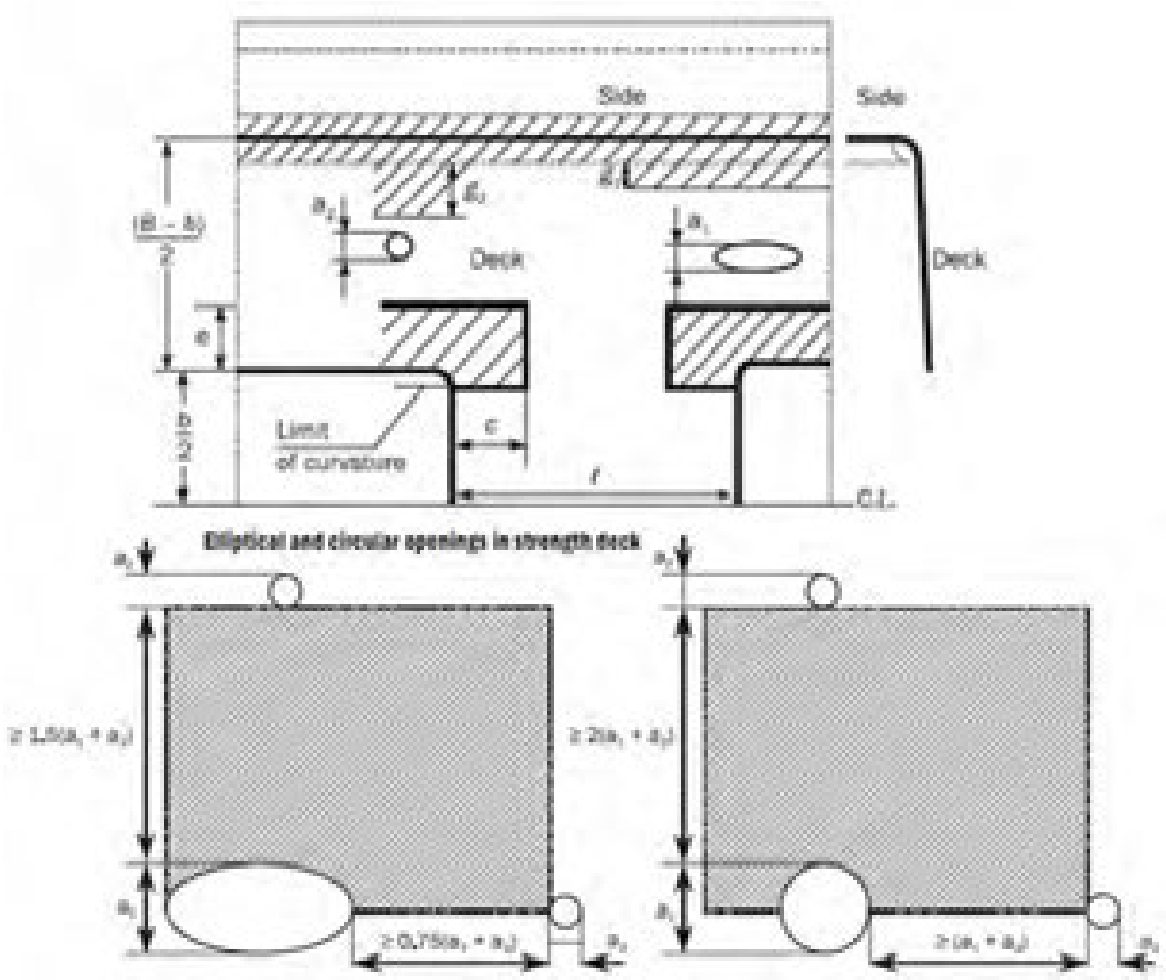


Figure 6.5.4.1: Position of openings in strength deck

6.5.4.2 Reinforcements

• Circular openings in strength deck within 0.6 L amidships and for ships with very large hatch openings within cargo hold region are to have edge reinforcement. The cross-sectional area of edge reinforcements, in [cm^2], is not to be less than:

$$A = 2.5\ d\ t$$

where,

d = diameter, in [m], of opening

t = net plate thickness of strength deck, in [mm].

The edge reinforcement of circular openings may be dispensed with, where the opening diameter is less than 300[mm] and the smallest distance from another opening is not less than 5 times the diameter of the smaller opening.

• The reinforcement is normally to be of a vertical ring welded to the plate edge. Alternative arrangements may be accepted provided the reinforcement is within a distance of 0.05d from the plate edge.

• Elliptical openings are to have their major axis in the fore and aft (longitudinal) direction. Where the ratio of the major axis to minor axis is less than 2 the openings are to be reinforced as circular openings taking d as the breadth of the opening (minor axis).

• Rectangular openings are to have their corners well rounded. Where corners are of circular shape the radius, (R) is not to be less than 20 % of the breadth of the opening and the edges are to be reinforced as circular openings taking d as the breadth of the opening. Special considerations with respect to longitudinal stresses and fatigue will be required in case of R < 0.2b.

• Where corners are of elliptical shape as given above or of streamlined shape, the reinforcement will generally not be required provided that the transverse extension of the curvature, a, shown in Figure 6.5.4.2 and Table 6.5.4 is not less than

$$a = 0.15b \text{ [m]}$$

• In way of the machinery spaces, special attention is to be paid to the maintenance of lateral stiffness by means of webs and heavy pillars in way of deck opening and casings.

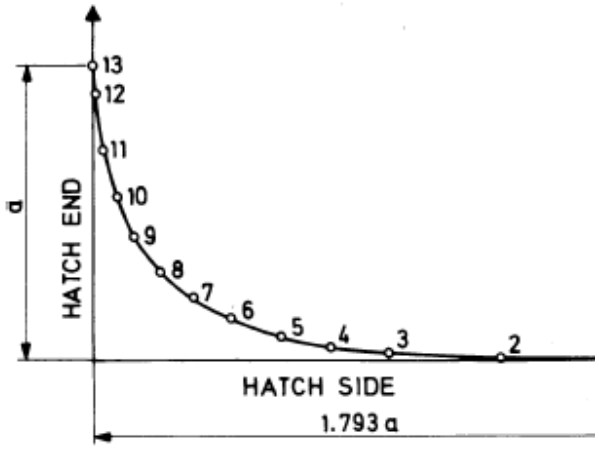


Figure 6.5.4.2: Streamlined deck corner

Table 6.5.4: Ordinates of streamlined corner		
Point	**Abscissa, x**	**Ordinate, y**
1	1.793a	0.0
2	1.381a	0.002a
3	0.987a	0.021a
4	0.802a	0.044a
5	0.631a	0.079a
6	0.467a	0.131a
7	0.339a	0.201a
8	0.224a	0.293a
9	0.132a	0.408a
10	0.065a	0.548a
11	0.022a	0.712a
12	0.022a	0.899a
13	0.0	1.00a

6.5.4.3 Corners of hatchways and Insert plates

• Where shapes other than the streamlined shape or equivalent are adopted, insert plates are to be fitted at the hatch corners in strength deck.

• Where insert plates are required, their net thickness is to be obtained, in [mm], from the following formula:

$$t_{INS} = \left(0.8 + 0.4\frac{b}{\ell}\right) t_{off}$$

without being taken less than t_{off} or greater than $1.6\, t_{off}$.

where:

t_{off} = offered net thickness, in [mm], of the deck at the side of the hatchways.

b = width, in [m], of the hatchway considered, measured in the transverse direction

ℓ = length, in [m], in way of the corner considered, of the cross-deck strip between two consecutive hatchways, measured in the longitudinal direction.

• For hatchways located within the cargo area, radiused insert plates, with thickness not less than determined above, are to be fitted in way of corners where the plating cut-out has a circular profile. The radius of circular corners is to be not less than:

– 5% of the hatch width, where a continuous longitudinal deck girder is fitted below the hatch coaming

– 8% of the hatch width, where no continuous longitudinal deck girder is fitted below the hatch coaming.

• Corner radius, in the case of the arrangement of two or more hatchways athwartship, is considered by IRS on a case-by-case basis.

• For hatchways located within the cargo area, insert plates are, in general, not required in way of corners where the plating cut-out has an elliptical or parabolic profile and the half axes of elliptical openings, or the half lengths of the parabolic arch, are not less than:

– 1/20 of the hatchway width or 600[mm], whichever is the lesser, in the transverse direction

– twice the transverse dimension, in the fore and aft direction (longitudinal direction)

• For the extreme corners of end hatchways, insert plates are required. The net thickness of these insert plates is to be 60 % greater than

the net offered thickness of the adjacent deck plating and the extent of the insert plate is shown in Figure 6.5.4.3 which d1, d2, d3 and d4 are to be greater than the stiffener spacing.

- A lower thickness may be accepted by IRS on the basis of calculations showing that stresses at hatch corners are lower than permissible values.

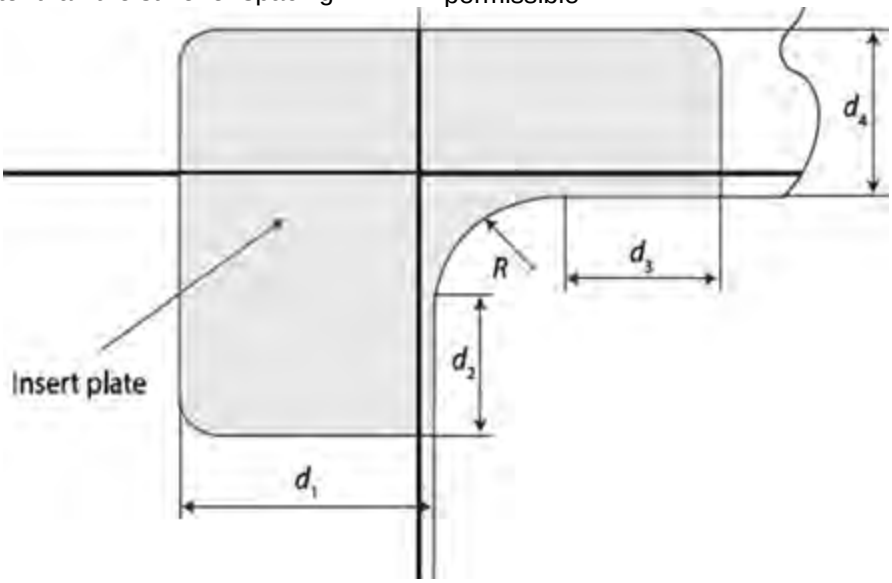


Figure 6.5.4.3: Hatch corner insert plate

- The butts of insert plates are to be well clear of those in coaming.

- For hatchways located outside the cargo area, a reduction in the thickness of the insert plates in way of corners may be considered by IRS on a case-by-case basis.

- For ships having length, L≥150[m], the corner radius, the thickness and the extent of insert plate may be determined by the results of a direct calculations according to Chapter 8, including buckling check and fatigue strength assessment of hatch corners according to Chapter 9 and Chapter 10 respectively.

6.5.4.4 Definitions of openings

The following definitions of opening are to be applied:

(i) Large openings are:

- Elliptical openings exceeding 2.5[m] in length or 1.2[m] in breadth.
- Circular openings exceeding 0.9[m] in diameter.

(ii) Small openings (i.e. lightening holes, single scallops in way of seams, etc.) are openings that are not large ones.

(iii) Manholes.

(iv) Isolated openings are openings spaced not less than 1[m] apart in the ship's transverse/vertical direction.

6.5.4.5 Large openings, manholes and nearby small openings

Large openings and manholes are to be deducted from the sectional area used in hull girder moment of inertia and section modulus. When small openings are spaced less than 1[m] apart in the ship's transverse/vertical direction to large openings or manholes, the total breadth of them is to be deducted from the sectional area.

Additionally, isolated small openings which do not comply with the arrangement requirements given in Chapter 3, Section 6, Cl.6.5.4.1 are to be deducted from the sectional areas included in the hull girder transverse sections.

Large openings and scallops where scallop-welding is applied, are always to be deducted from the sectional areas used in the hull girder moment of inertia and section modulus calculation.

6.5.4.6 Isolated small openings

Isolated small openings in one transverse section in the strength deck or bottom area need not be deducted from the sectional areas

included in the hull girder transverse sections, provided that:

$$\sum b_s \leq 0.06\left(B - \sum b\right)$$

where,

$\sum b_s$ = Total breadth of isolated small openings, in [m], in the strength deck or bottom area at the transverse section considered, determined as indicated in Figure 6.5.4.6, not deducted from the section area as per Cl.6.5.4.5.

$\sum b$ = Total breadth of large openings, in [m], at the transverse section considered, determined as indicated in Figure 6.5.4.6, deducted from the section area as defined in Cl.6.5.4.5.

Where the total breadth of isolated small openings Σb_s does not fulfil the above criteria, only the excess of breadth is to be deducted from the sectional areas included in the hull girder transverse sections.

Smaller openings (manholes, lightening holes, single scallops in way of seams etc.) need not to be deducted provided that the sum of their breadths or shadow area breadths in one transverse section does not reduce the section modulus at deck or bottom by more than 3% and provided that the height of lightening holes, draining holes and single scallops in longitudinals or longitudinal girders does not exceed 25% of the web depth, for scallops maximum 75[mm].

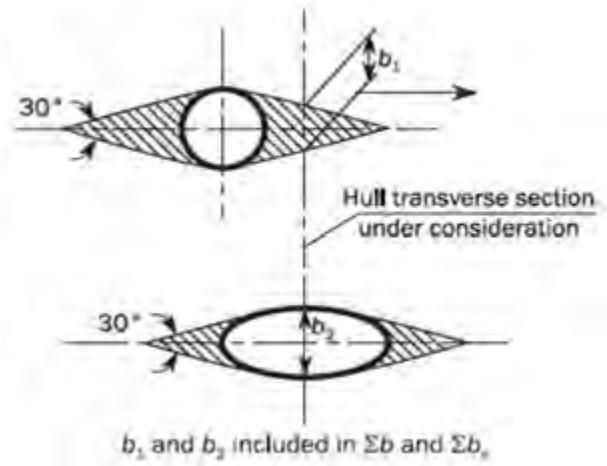


Figure 6.5.4.6: Calculation of $\sum b$ and $\sum b_s$

6.5.4.7 Lightening holes, draining holes and single scallops

Lightening holes, draining holes and single scallops in longitudinals need not be deducted if their height is less than $0.25h_w$, where h_w is the web height of the longitudinals, in [mm]. Otherwise, the excess is to be deducted from the sectional area or compensated.

6.5.4.8 Non-continuous decks and longitudinal bulkheads

When calculating the effective area in way of non-continuous decks and longitudinal bulkheads, the effective area is to be taken as shown in Figure 6.5.4.8. The shadow areas, which are not effective for hull girder bending but effective for hull girder shear force, are obtained by drawing two tangent lines with an angle of 15[deg] to the longitudinal axis of the ship.

The shadow area of an opening is the area forward and aft of the opening enclosed by lines drawn tangential to the corners of the opening and intersecting each other to form an included angle of 30[deg] as shown in Figure 6.5.4.8.

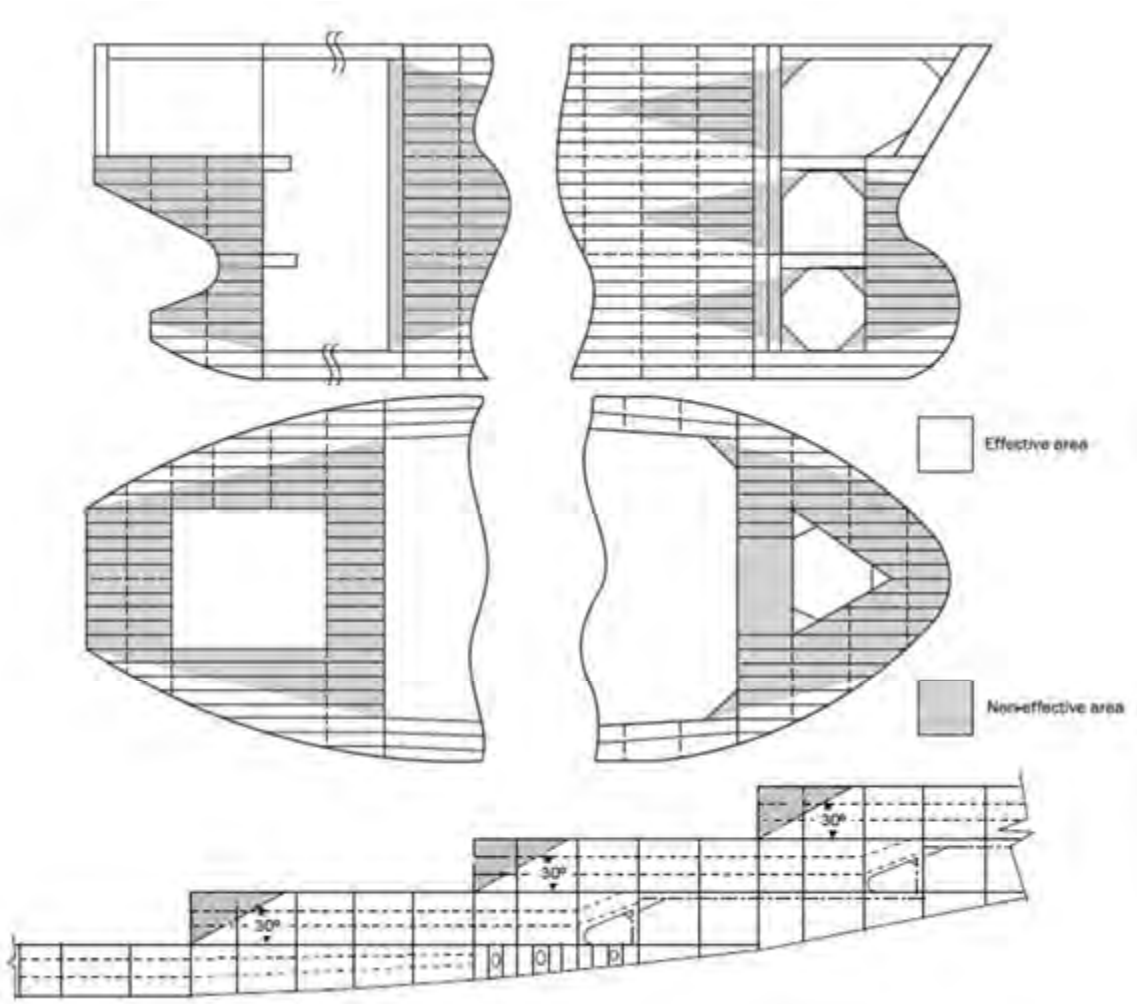


Figure 6.5.4.8: Effective area in way of non-continuous decks and bulkheads

6.5.5 Openings in side shell, longitudinal bulkheads and girders

6.5.5.1 Openings in the side shell, longitudinal bulkheads and longitudinal girders are to be located at a vertical distance below the strength deck or termination of rounded deck corner not less than.

– two times the opening breadth, in case of rectangular opening

– two times the opening diameter, in case of circular opening

– the opening minor axis, in case of elliptical openings.

6.5.5.2 Smaller openings in the outer bottom within 0.6L amidships or within the cargo hold region for ships with large hatch openings are to be reinforced as given in Cl.6.5.4.2. The area of these reinforcements is not to be included in the sectional areas used in the section modulus calculation.

6.5.5.3 Openings in side shell subjected to large shear stresses are to be of circular shape and are to be reinforced as given in Cl.6.5.4.2 irrespective of the size of opening.

6.5.5.4 Where openings are cut in the shell plating for windows or side scuttles, hawses, scuppers, sea valves etc., they are to have well rounded corners. If they exceed 700[mm], the openings are to be reinforced by framing, a thicker plate or a ring.

6.5.5.5 Openings in the ship sides, e.g. for cargo ports, are to be well rounded at the corners and located well clear of superstructure ends or any openings in the deck areas at sides of hatchways.

6.5.5.6 Openings for sea intakes are to be well rounded at the corners and, within 0.6L amidships, located outside the bilge strakes.

6.5.5.7 Where arrangements are such that sea intakes are unavoidably located in the curved zone of the bilge strakes, such openings are to be elliptical with the major axis in the longitudinal direction. Openings for stabilizer fins are considered by IRS on a case-by-case basis.

6.5.5.8 In general, compensation will not be required for holes in the steer strake which are located away from the gunwale or any deck openings situated outside the line of the main hatchways and whose depth does not exceed 20% of the depth of the sheer strake or 380 [mm], whichever is the lesser. Openings are not to be cut in a rounded gunwale.

6.5.5.9 The keel plate is normally not to have any openings.

6.5.5.10 In the bilge plate, within 0.6L amidships, openings are to be avoided as far as practicable. Any necessary openings in the bilge plate are to be kept well clear of the bilge keel.

Section 7

Structural Idealisation

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

PSM = Primary Supporting Member.

EPP = Elementary Plate Panel.

LCP = Load Calculation Point.

φ_w = Angle, in [deg], between the stiffener or PSM web and the attached plating, Refer Figure 7.4.3.

φ_w is to be taken equal to 90[deg] if the angle is greater than or equal to 75[deg].

l_{bdg} = Effective bending span, in [m], as defined in Cl.7.1.1 for stiffeners and Cl.7.1.7 for PSM.

l_{shr} = Effective shear span, in [m], as defined in Cl.7.1.3 for stiffeners and Cl.7.1.8 for PSM.

l = Full length of stiffener or of PSM, in [m], between their supports.

s = Stiffener spacing, in [mm], as defined in Cl.7.2.1.

S = Primary supporting member spacing, in [m], as defined in Cl.7.2.2.

a = Length, in [mm], of EPP as defined in Cl.7.5.1.

b = Breadth, in [mm], of EPP as defined in Cl.7.5.1.

h_{stf} = Stiffener height, including the face plate, in [mm].

t_p = Net thickness of attached plate, in [mm].

t_w = Net web thickness, in [mm].

b_f = Breadth of flange, in [mm].

t_f = Net thickness of flange, in [mm].

7.1 Structural idealization of stiffeners and Primary Supporting Members

7.1.1 Effective bending span of stiffeners

7.1.1.1 Where arrangements differ from those defined in this article, span definition may be specially considered.

7.1.1.2 The effective bending span l_{bdg} of stiffeners is to be measured as shown in Figure 7.1.1.1 for single skin structures and Figure 7.1.1.2 for double skin structures.

7.1.1.3 If the web stiffener is sniped at the end or not attached to the stiffener under consideration, the effective bending span is to be taken as the full lengths between PSMs unless a backing bracket is fitted.

7.1.1.4 The effective bending span may be reduced where brackets are fitted to the flange or free edge of the flat bar stiffeners. Brackets fitted on the side opposite to that of the stiffener with respect to attached plating are not to be considered as effective in reducing the effective bending span.

7.1.1.5 In single skin structures, the effective bending span of a stiffener supported by a bracket or by a web stiffener on one side only of the primary supporting member web, is to be taken as the total span between primary supporting members as shown in item (a) of Figure 7.1.1.1. If brackets are fitted on both sides of the primary supporting member, the effective bending span is to be taken as in items (b), (c) and (d) of Figure 7.1.1.1.

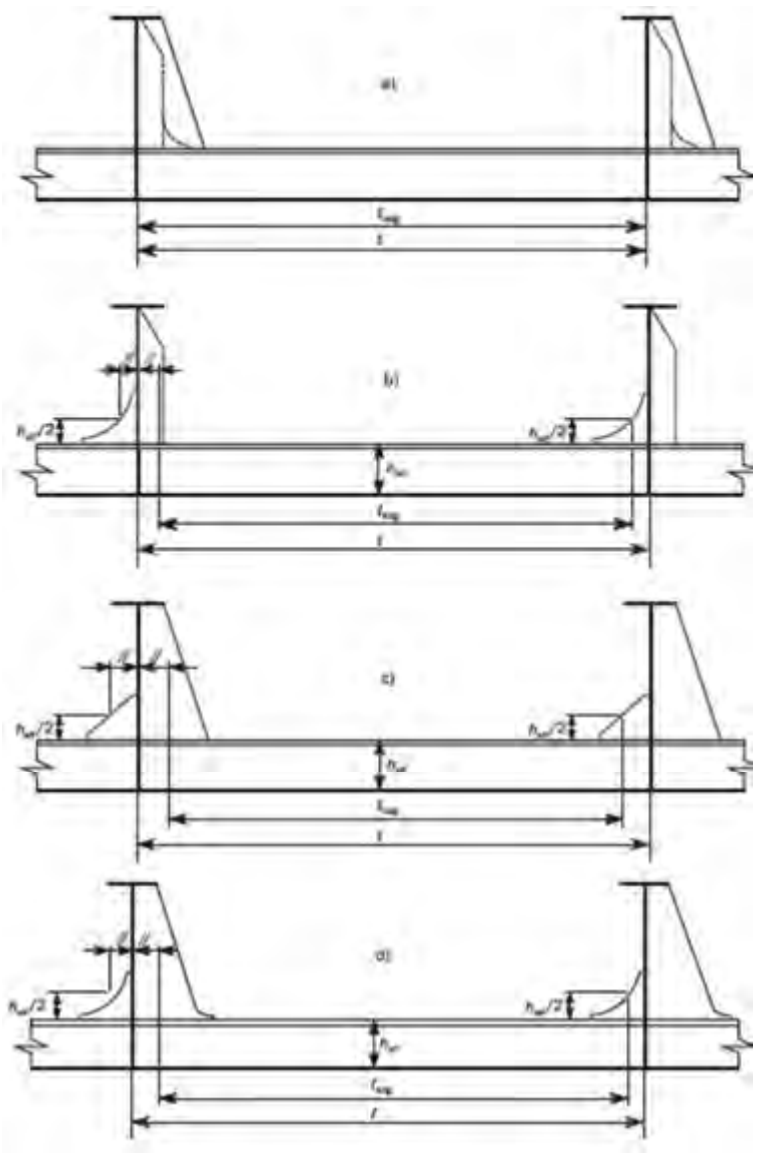


Figure 7.1.1.1: Effective bending span of stiffeners supported by web stiffeners (single skin construction)

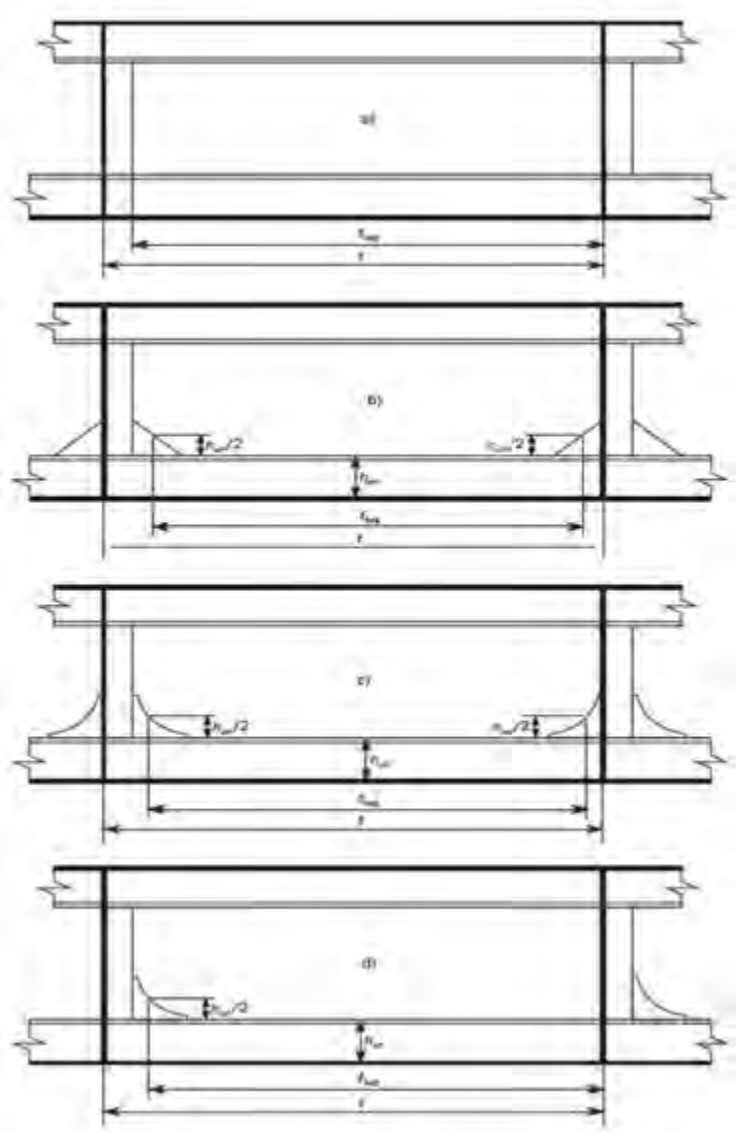


Figure 7.1.1.2: Effective bending span of stiffeners supported by web stiffeners (double skin construction)

7.1.2 Effective bending span of stiffeners with continuous flange along bracket edge

Where the flange of the stiffener is continuous along the edge of the bracket, the effective bending span is to be taken to the position where the depth of the bracket is equal to one quarter of the depth of the stiffener, Refer Figure 7.1.2.

7.1.3 Effective shear span of stiffeners

7.1.3.1 The effective shear span, l_{shr} in [m], of stiffeners is to be measured as shown in Figure 7.1.3.1 for single skin structures and Figure 7.1.3.2 for double skin structures.

7.1.3.2 The effective shear span may be reduced for brackets fitted on either the flange or the free edge of the flat bar stiffeners, or for brackets fitted to the attached plating on the side opposite to that of the stiffener.

7.1.3.3 If brackets are fitted at both the flange or free edge of the stiffener, and to the attached plating on the side opposite to the stiffener the effective shear span may be reduced using the longer effective bracket arm.

7.1.3.4 Regardless of support detail, the full length of the stiffener may be reduced by a minimum of s/4000 [m] at each end of the member, hence the effective shear span l_{shr}, is not be taken greater than:

$$l_{\mathrm{shr}} = l - \frac{\mathrm{s}}{2000}$$

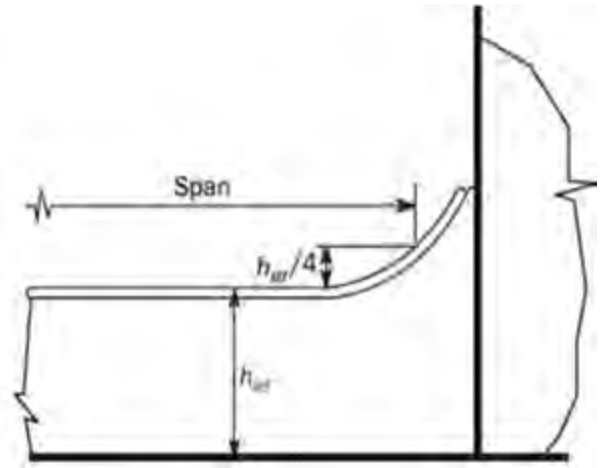


Figure 7.1.2: Effective bending span of stiffener with continuous flange along bracket edge

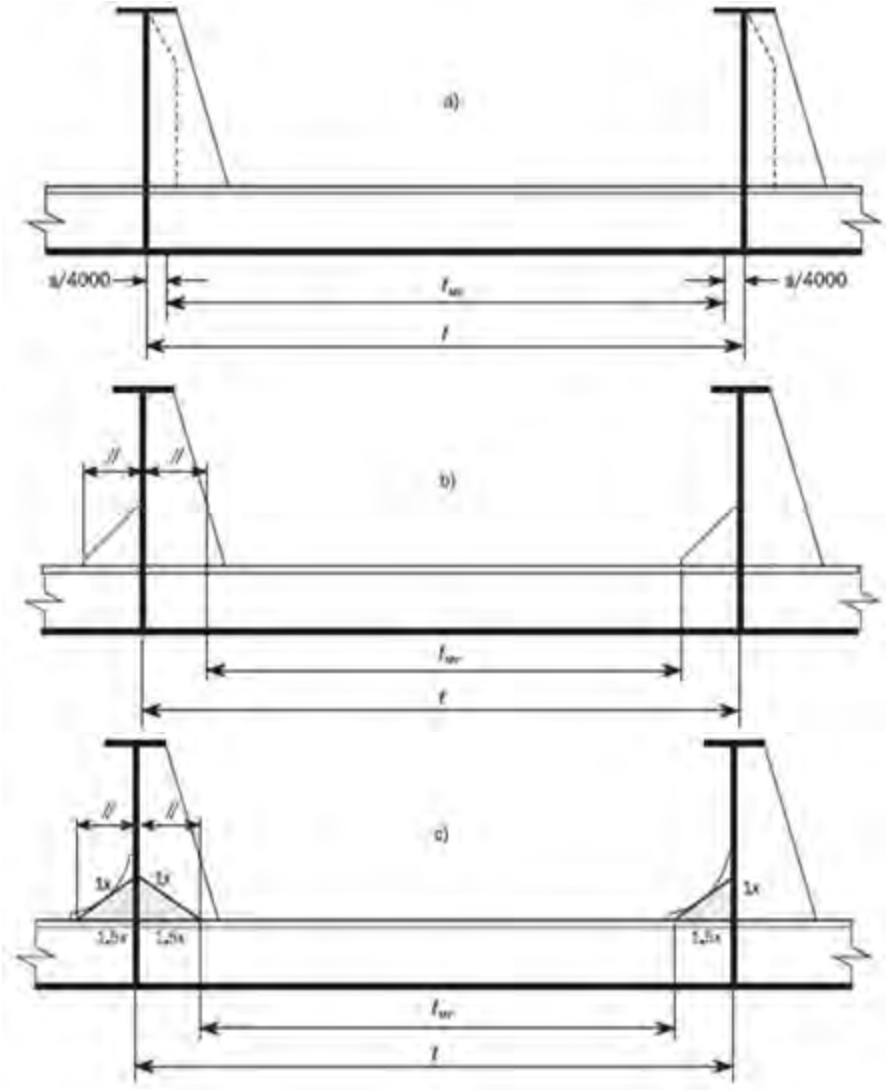


Figure 7.1.3.1: Effective shear span of stiffeners supported by web stiffeners (single skin construction)

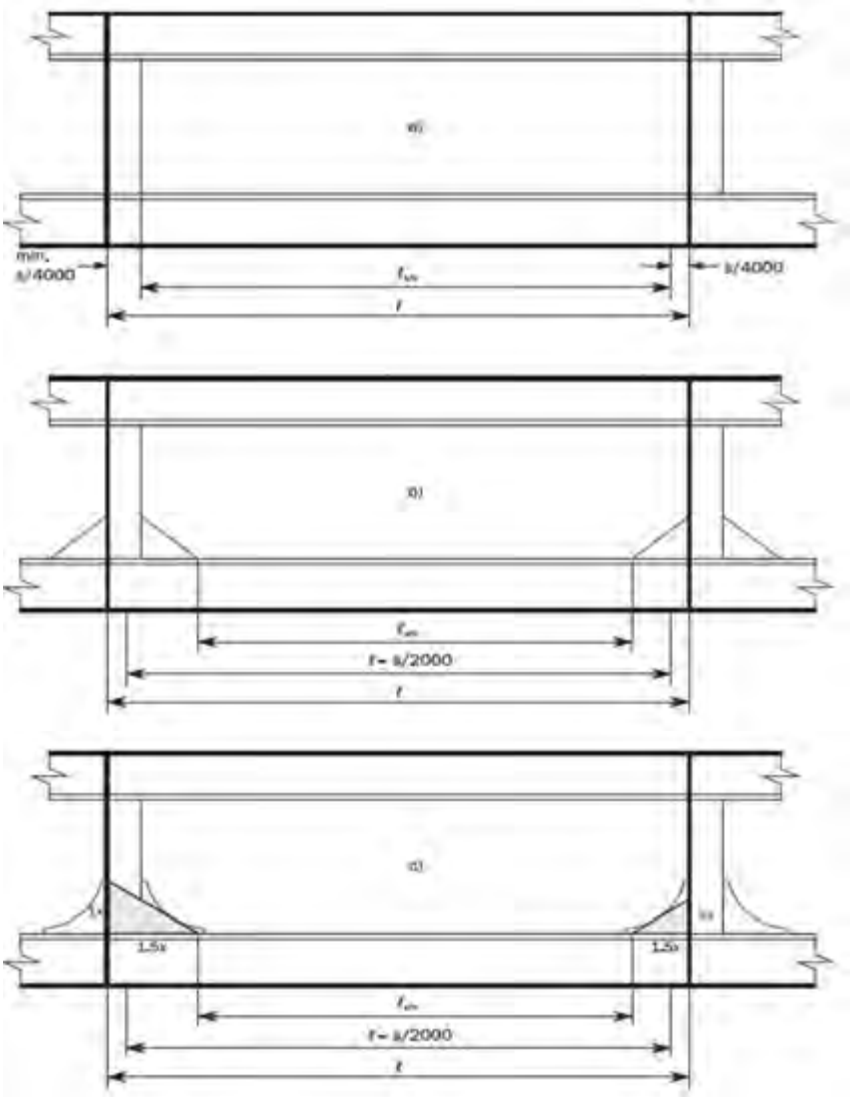


Figure 7.1.3.2: Effective shear span of stiffeners supported by web stiffeners (double skin construction)

7.1.4 Effective shear span of stiffeners with continuous flange along bracket edge

7.1.4.1 For curved and/or long brackets (high length/height ratio), the effective bracket length is to be taken as the maximum inscribed 1:1.5 right angled triangle as shown in item (c) of both Figure 7.1.3.1 and Figure 7.1.3.2.

7.1.4.2 Where the face plate of the stiffener is continuous along the curved edge of the bracket, the bracket length to be considered for determination of the span point location is not to be taken greater than 1.5 times the length of the bracket arm as shown in Figure 7.1.4.

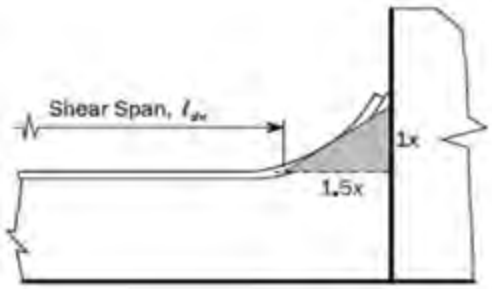


Figure 7.1.4: Effective shear span for local support members with continuous face plate along bracket edge

7.1.5 Effect of hull form shape on span of stiffeners

For curved stiffeners, the span is defined as the chord length between span points to be measured at the flange for stiffeners with a flange, and at the free edge for flat bar stiffeners, Refer Figure 7.1.5. The calculation of the effective span is to be in accordance with requirements given in Cl.7.1.1 and Cl.7.1.3.

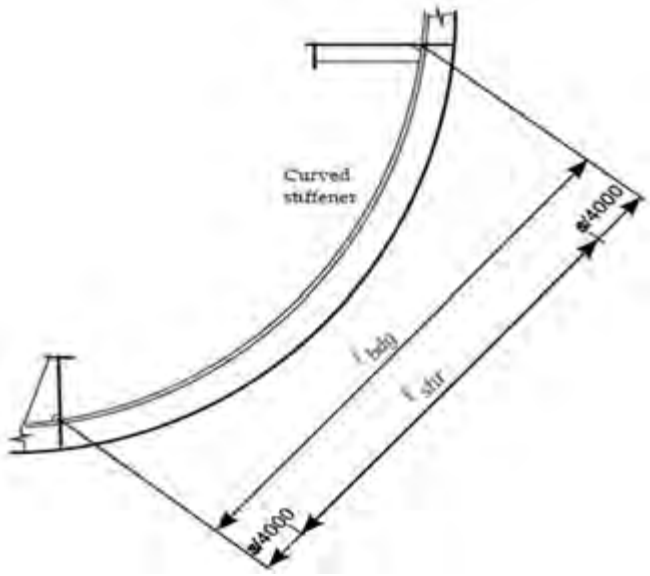


Figure 7.1.5: Effective span of curved stiffener

7.1.6 Effective span of stiffeners supported by struts

7.1.6.1 The arrangement of stiffeners supported by struts is not allowed for ships over 120[m] in length.

7.1.6.2 Stiffeners supported by struts are not allowed in way of boundaries to cargo tanks or fuel oil tanks.
Struts, in general, are not to be fitted in way where cargo is discharged by grabs, or heavy liquid cargoes are carried, or in the bottom forward slamming area;

Particular attention is to be paid to deflection of members and to the ability of the structure to resist buckling. Where necessary, additional calculations will be required.

7.1.6.3 The bending and shear spanl_{bdg} and l_{shr}, of stiffeners supported by one strut fitted at mid distance of the primary supporting members is to be taken as 0.8l, Refer Figure 7.1.6.1.

7.1.6.4 In case where two struts are fitted at 1/3 and 2/3 length between primary supporting members, the bending and shear span, l_{bdg} and l_{shr}, of stiffeners is to be taken as 0.7l, Refer Figure 7.1.6.2.

7.1.6.5 When the ratio between the net section modulus of the larger stiffeners and the smaller stiffeners connected by strut(s) exceeds 1.5, the above reduction in the bending span is not applicable.

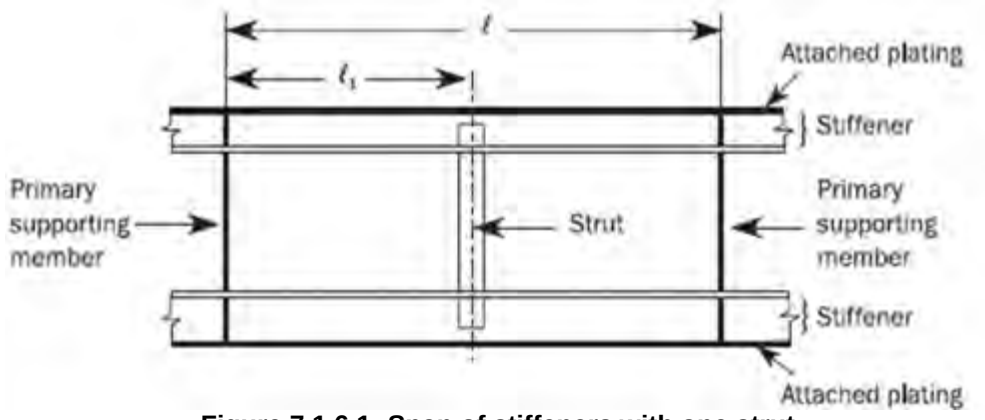


Figure 7.1.6.1: Span of stiffeners with one strut

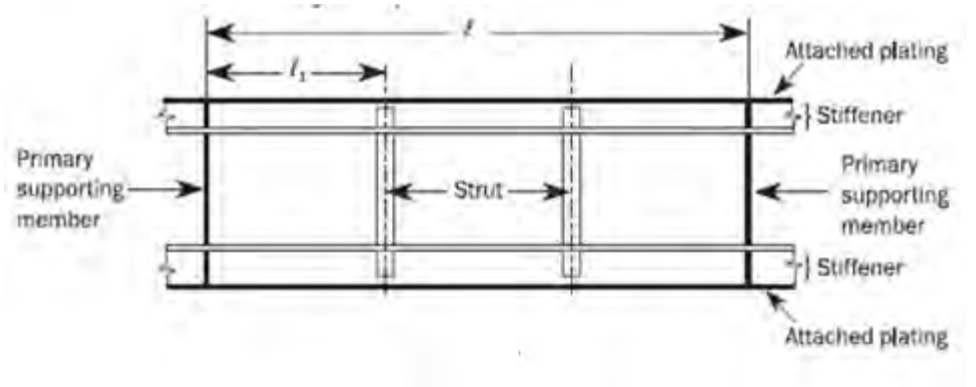


Figure 7.1.6.2: Span of stiffeners with two struts

7.1.7 Effective bending span of primary supporting members

7.1.7.1 The effective bending span, l_{bdg}, in [m], of a primary supporting member without end bracket is to be taken as the length of the member between supports.

7.1.7.2 The effective bending span, l_{bdg} of a primary supporting member may be taken as less than the full length of the member between supports provided that suitable end brackets are fitted.

7.1.7.3 The effective bending span l_{bdg}, in [m], of a primary supporting member with end brackets is taken between points where the depth of the bracket is equal to half the web height of the primary supporting member as shown in item (b) of Figure 7.1.7. The effective bracket used to define these span points are to be taken as given in Cl.7.1.10.

7.1.7.4 In case of brackets where the face plate of the member is continuous along the face of the bracket, as shown in items (a), (c) and (d) of Figure 7.1.7, the effective bending span l_{bdg}, in m, is taken between points where the depth of the bracket is equal to one quarter the web height of the primary supporting member. The effective bracket used to define these span points is to be taken as given in Cl.7.1.10.

7.1.7.5 For straight brackets with a length to height ratio greater than 1.5, the span point is to be taken to the effective bracket; otherwise, the span point is to be taken to the fitted bracket.

7.1.7.6 For curved brackets, for span positions above the tangent point between fitted bracket and effective bracket, the span point is to be taken to the fitted bracket; otherwise, the span point is to be taken to the effective bracket.

7.1.7.7 For arrangements where the primary supporting member face plate is carried on to the bracket and backing brackets are fitted, the span point need not be taken greater than to the position where the total depth reaches twice the depth of the primary supporting member. Arrangements with small and large backing brackets are shown in items (e) and (f) of Figure 7.1.7.

7.1.7.8 For arrangements where the height of the primary supporting member is maintained and the face plate width is increased towards the support; the effective bending span may be taken to a position where the face plate breadth reaches twice the nominal breadth. Attention is to be paid to tripping of the flanges as per Section 6, Cl.6.3.3.4.

7.1.8 Effective shear span of primary supporting members

7.1.8.1 The effective shear span of the primary supporting member may be reduced compared to effective bending span, and taken between the toes of the effective brackets supporting the member, where the toes of effective brackets are as shown in Figure 7.1.8. The effective bracket used to define the toe point is given in Cl.7.1.10.

7.1.8.2 For arrangements where the effective backing bracket is larger than the effective bracket in way of face plate, the shear span is to be taken as the mean distance between toes of the effective brackets as shown in item (f) of Figure 7.1.8.

7.1.9 Effect of hull form shape on span of primary supporting members

For curved primary supporting members, the span is defined as the chord length between span points. The calculation of the effective span is to be in accordance with requirements given in Cl.7.1.8.

7.1.10 Effective bracket definition

The effective bracket is defined as the maximum size of right angled triangular bracket with a length to height ratio of 1.5 that fits inside the fitted bracket. Refer Figure 7.1.7 for examples.

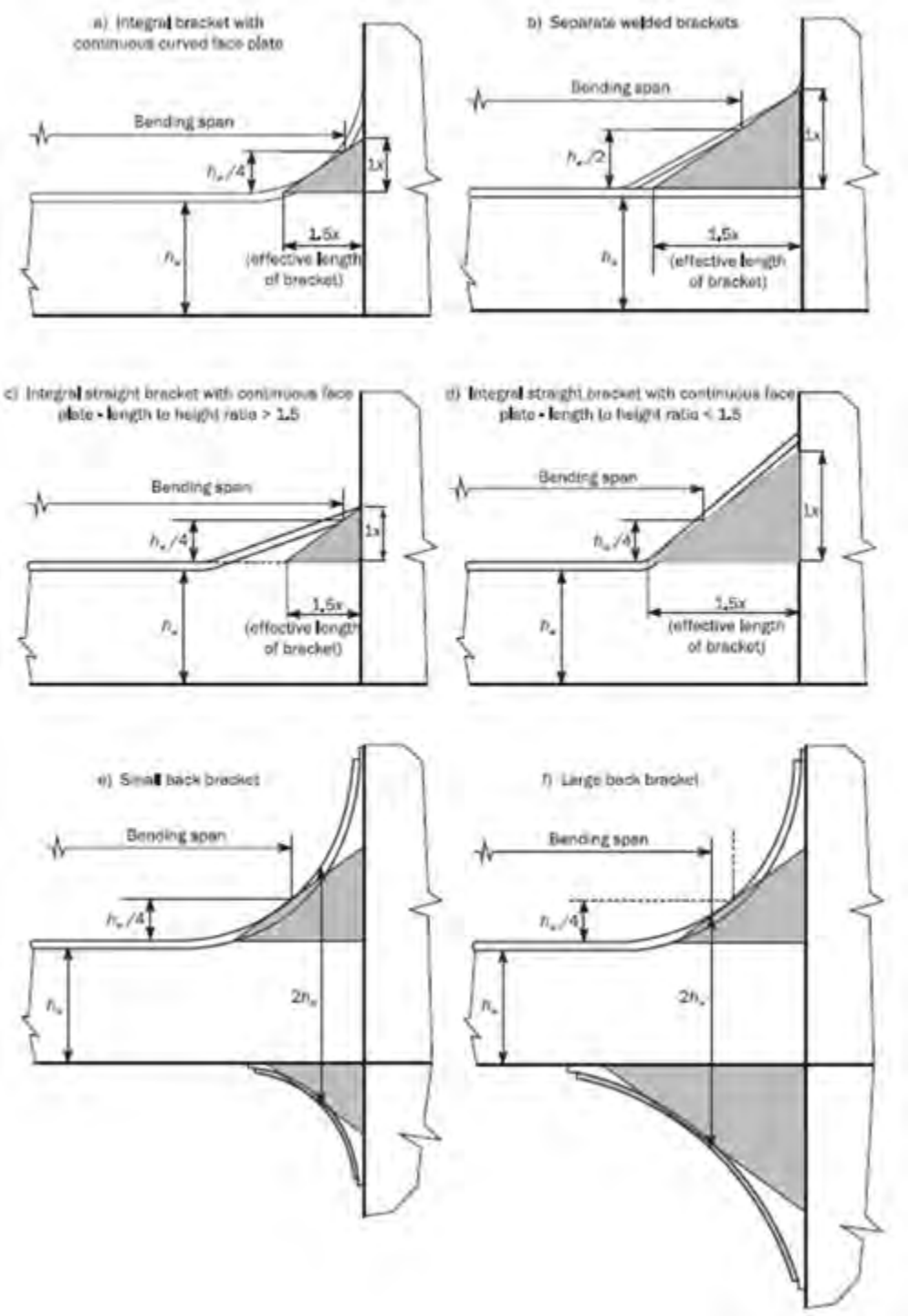


Figure 7.1.7: Effective bending span of primary supporting member

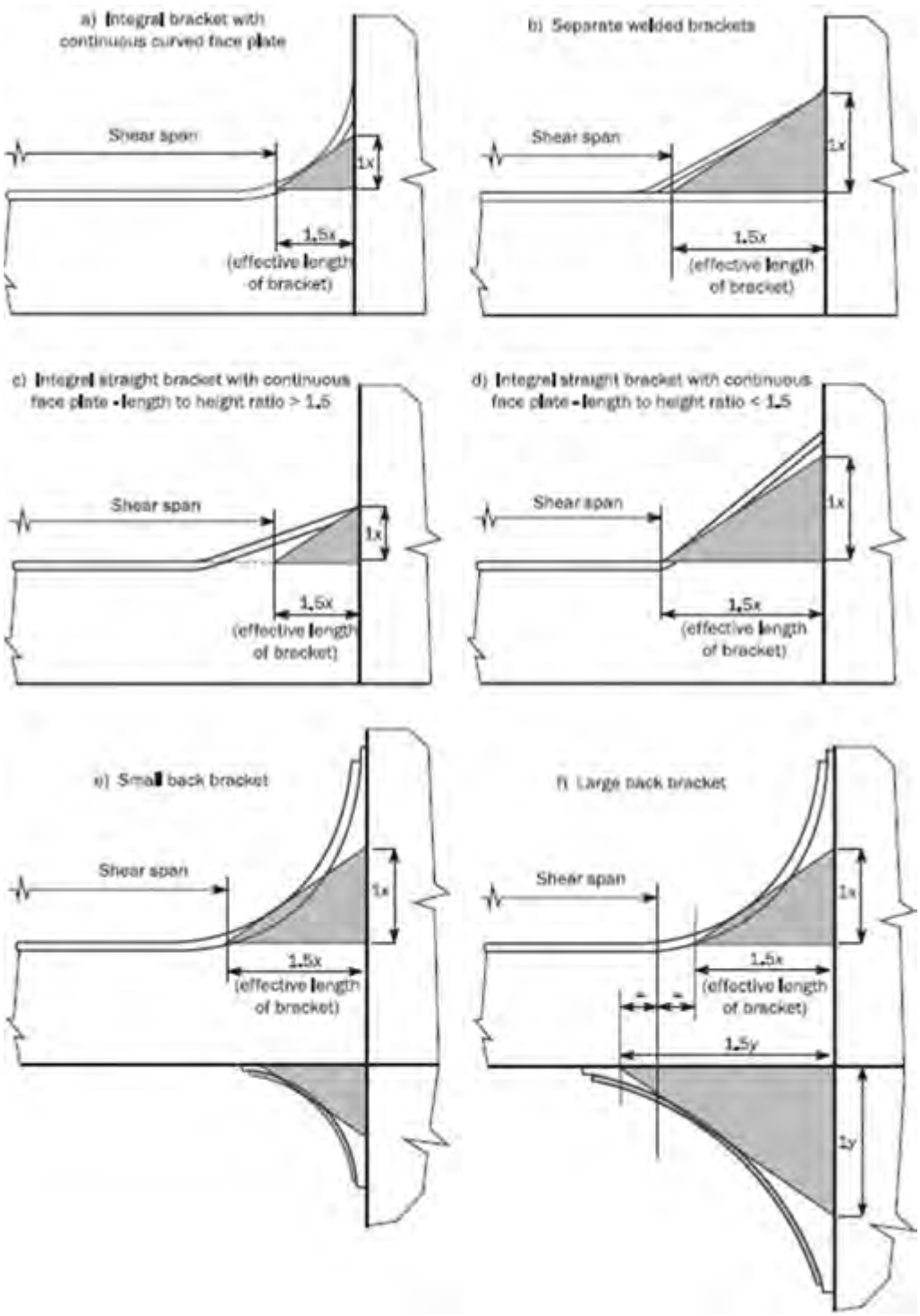


Figure 7.1.8: Effective shear span of primary supporting member

7.2 Spacing and load supporting breadth

7.2.1 Stiffeners

Stiffener spacing, s, in [mm], for the calculation of the effective attached plating of stiffeners is to be taken as the mean spacing between stiffeners and taken equal to, Refer Figure 7.2.1:

$$s = \frac{b_1 + b_2 + b_3 + b_4}{4}$$

where:

b_1, b_2, b_3, b_4 = Spacing between stiffeners at ends, in [mm].

In general, the loading breadth supported by stiffener is to be taken equal to s.

7.2.2 Primary supporting member

Primary supporting member spacing, S, for the calculation of the effective attached plating of primary supporting members is to be taken as the mean spacing between adjacent primary supporting members, and taken equal to, Refer Figure 7.2.1.

$$S = \frac{b_1 + b_2 + b_3 + b_4}{4}$$

where:

b_1, b_2, b_3, b_4 = Spacings between primary supporting members at ends, in [mm].

In general, the loading breadth supported by a primary supporting member is to be taken equal to S.

7.2.3 Spacing of curved plating

For curved plating, the stiffener spacing, s or the primary supporting member spacing, S is to be measured on the mean chord between members.

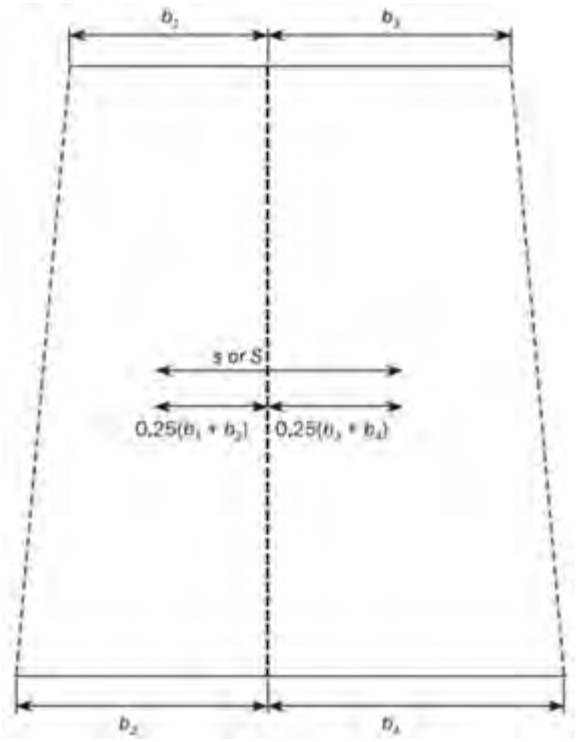


Figure 7.2.1: Spacing of plating

7.3 Effective breadth of attached plating

7.3.1 Stiffeners

The effective breadth, b_{eff}, in [mm], of the attached plating to be considered in the actual net section modulus for the yielding check of stiffeners is to be obtained from the following formulae:

7.3.1.1 where the plating extends on both sides of the stiffener:

$$b_{eff} = \min(200l\ ;\ s)$$

7.3.1.2 where the plating extends on one side of the stiffener, i.e. stiffeners bounding openings:

$$b_{eff} = \min(100l\ ;\ 0.5s)$$

However, where the attached plate net thickness is less than 8[mm], the effective breadth is not be taken greater than 600[mm]. The effective breadth, b_{eff}, in [mm], of the attached plating to be considered for the buckling check of stiffeners is given in Chapter 9, Section 5, Cl.5.2.3.5.

7.3.2 Primary supporting members

7.3.2.1 The effective breadth of attached plating, b_{eff}, in [m], for calculating the section modulus and/or moment of inertia of a primary supporting member with uniform load is to be taken as:

$$b_{eff} = S\,.\min\left[\frac{1.12}{1 + \frac{1.75}{\left(\frac{l_{bdg}}{S\sqrt{3}}\right)^{1.6}}}\ ;\ 1.0\right] \text{ for } \frac{l_{bdg}}{S\sqrt{3}} \geq 1$$

$$b_{eff} = 0.407\left(\frac{l_{bdg}}{\sqrt{3}}\right) \text{ for } \frac{l_{bdg}}{S\sqrt{3}} < 1$$

7.3.2.2 For double skin sections such as double bottom or double side structures with uniform load, full flange effectivity may normally be assumed for the members representing the floors or web frames. The effective breadth of attached plating, b_{eff} in [m], is to be taken as:

$$b_{eff} = S$$

7.3.2.3 Within $0.1l_{bdg}$ from the end of bending span or in way of concentrated point loads, the effective breadth b_{eff^*} in [m] of the attached plating is to be taken as:

$$b_{eff^*} = 0.5.b_{eff}$$

7.3.2.4 In case of girders on corrugated bulkheads which run across the corrugations, the effective width of attached plating is to be taken as 10% of that obtained from Cl.7.3.2.1.

7.3.2.5 The effective cross-sectional area of the attached plating is not to be less than that of the face plate.

7.3.3 Effective area of curved face plate and attached plating of primary supporting members

The effective net area given in Cl.7.3.3.1 and Cl.7.3.3.2 below is only applicable to curved face plates and curved attached plating of primary supporting members. It is not applicable for the area of web stiffeners parallel to the face plate.

The effective net area is applicable to primary supporting members for the following calculations:

- Actual net section modulus used for comparison with the scantling requirements in Chapter 7.

- Actual effective net area of curved face plates, modelled by beam elements, used in Chapter 8.

7.3.3.1 The effective net area, $A_{eff-n50}$, in [mm^2], is to be taken as:

$$A_{eff-n50} = C_f.t_{f-n50}.b_f$$

where:

C_f = Flange efficiency coefficient is to be obtained from the following formula but not to be greater than 1.0. Refer Figure 7.3.3.2:

$$C_f = C_{f1}\frac{1.285}{\beta k_1} \quad \text{for symmetrical face plate}$$

$$C_f = 0.18 + \frac{0.08}{\beta^2} \quad \text{for unsymmetrical face plate}$$

$$C_f = C_{f1}\frac{1.285}{\beta} \quad \text{for attached plating of box girders}$$

C_{f1} = Coefficient taken equal to:

- For symmetrical face plates,

$$C_{f1} = \frac{(\sinh k_1\beta\ \cosh k_1\beta + \sin k_1\beta\ \cos k_1\beta)}{(\cosh k_1\beta)^2 + (\cos k_1\beta)^2}$$

- For attached plating of box girders with two webs,

$$C_{f1} = \frac{0.78\ (\sinh\beta + \sin\beta)(\cosh\beta - \cos\beta)}{(\sinh\beta)^2 + \sin^2\beta}$$

- For attached plating of box girders with multiple webs,

$$C_{f1} = \frac{1.56\ (\cosh\beta - \cos\beta)}{\sinh\beta + \sin\beta}$$

β = Coefficient calculated as:

$$\beta = \frac{1.285\ b_1}{\sqrt{r_f\ t_f}},\ \text{in [rad]}$$

k_1 = Coefficient calculated as:

= 1.4+1.25$(1.4-\beta)^3$ for $\beta < 1.4$

= 1.4 for $\beta \geq 1.4$

b_1 = Breadth, in [mm], to be taken equal to:

- for symmetrical face plates, b_1 = 0.5(b_f – t_{w-n50})

- for unsymmetrical face plates, $b_1 = b_f$

- for attached plating of box girders, $b_1 = s_w - t_{w-n50}$

s_w = Spacing of supporting webs for box girders, in [mm].

t_{f-n50} = Net flange thickness in [mm].

For calculation of C_f and β of unsymmetrical face plates, t_{f-n50} is not be taken greater than t_{w-n50}.

t_{w-n50} = Net web plate thickness, in [mm].

r_f = Radius of curved face plate or attached plating, in [mm], Refer Figure 7.3.3.1 at mid thickness

b_f = Breadth of face plate or attached plating, in [mm], Refer Figure 7.3.3.1.

7.3.3.2 The effective net area, $A_{eff-n50}$ in [mm^2], of curved face plates supported by radial

brackets, or attached plating supported by radial stiffeners, is given by:

$$A_{eff-n50} = \left(\frac{3r_f\, t_f + C_f s_r^2}{3r_f\, t_f + s_r^2}\right) t_f . b_f$$

where:

s_r = Spacing of tripping brackets or web stiffeners or stiffeners normal to the web plating, in [mm], Refer Figure 7.3.3.1.

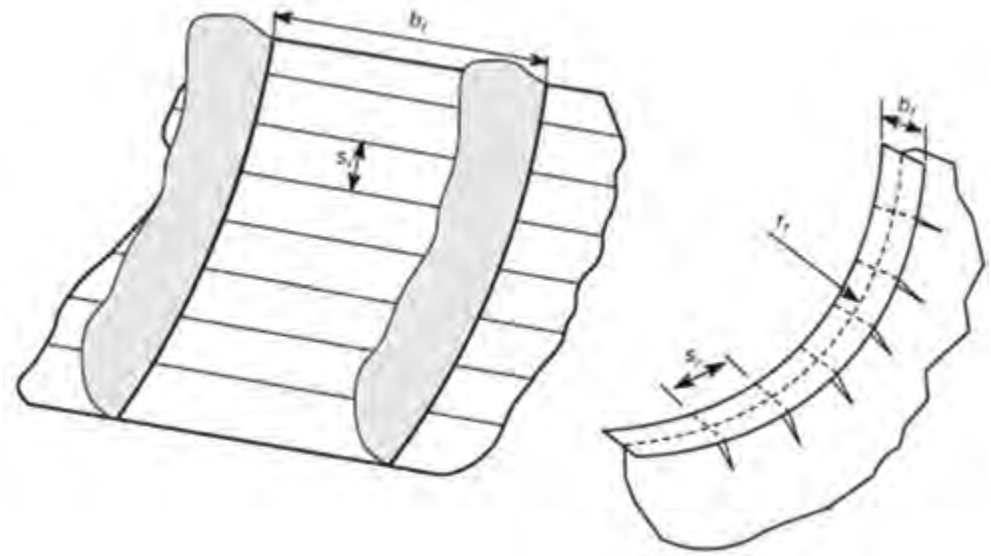


Figure 7.3.3.1: Curved shell panel and face plate

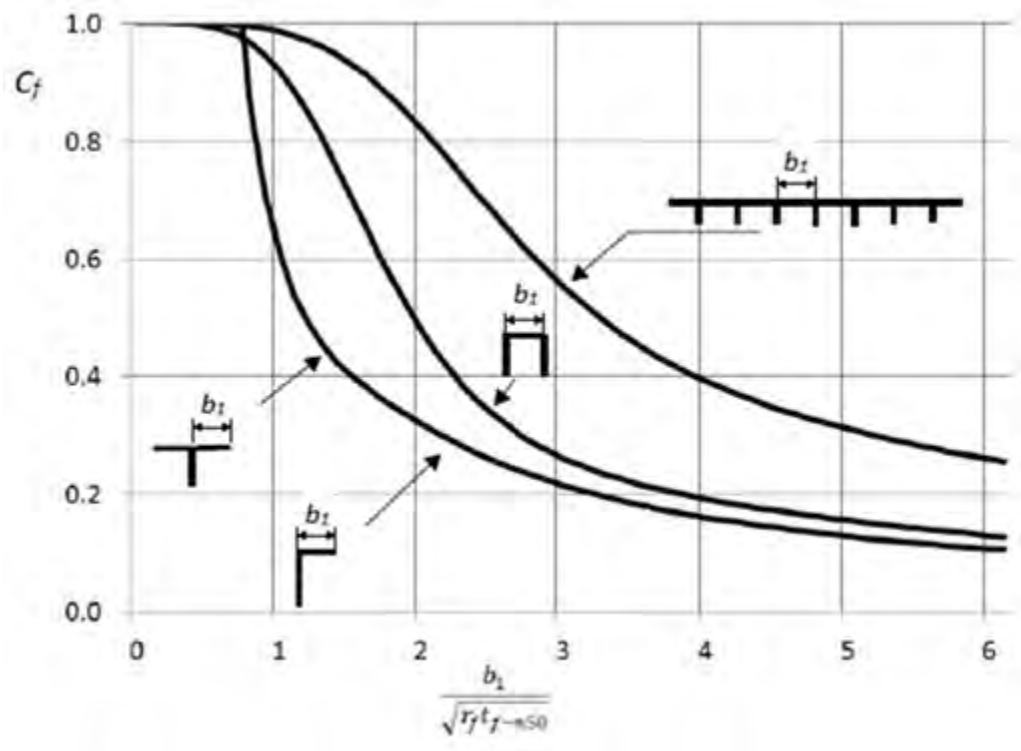


Figure 7.3.3.2: Flange efficiency coefficient for curved face plates

7.4 Geometrical properties of stiffeners and primary supporting members

7.4.1 Stiffener profile with a bulb section

The properties of bulb profile sections are to be determined by direct calculations. Where direct calculation of properties is not possible, a bulb section may be taken equivalent to an

angle section. The net dimensions of the equivalent angle section are to be obtained, in [mm], from the following formulae.

$$h_w = h'_w - \frac{h'_w}{9.2} + 2$$

$$b_f = \alpha\left(t'_w + \frac{h'_w}{6.7} - 2\right)$$

$$t_f = \frac{h'_w}{9.2} - 2$$

$$t_w = t'_w$$

where:

h'_w, t_w = Net height and net thickness of a bulb section, in [mm], as shown in Figure 7.4.1

α = Coefficient equal to:

$$\alpha = 1.1 + \frac{(120 - h'_w)^2}{3000} \quad \text{for } h'_w \leq 120$$

$$\alpha = 1.0 \quad \text{for } h'_w > 120$$

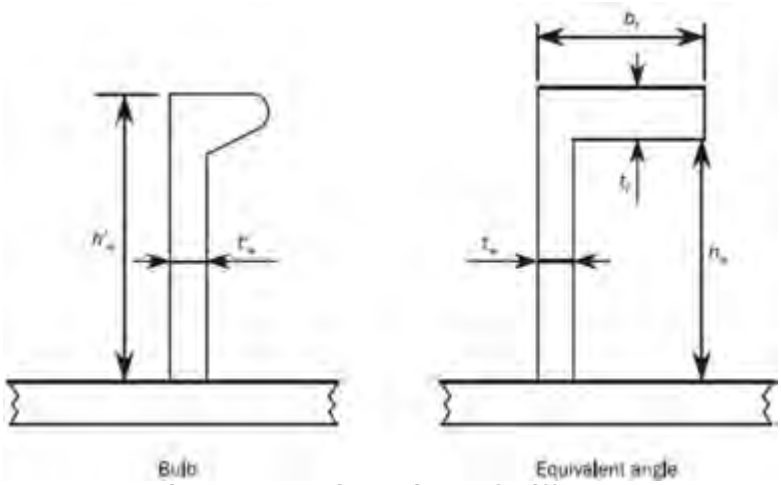


Figure 7.4.1: Dimensions of stiffeners

7.4.2 Net elastic shear area of stiffeners

The net elastic shear area, A_{shr}, in [cm^2], of stiffeners is to be taken as:

$$A_{shr} = d_{shr}\, t_w\, 10^{-2}$$

d_{shr} = Effective shear depth of stiffener, in [mm], as defined in Cl.7.4.3.

t_w = Net web thickness of the stiffener, in [mm] as defined in Chapter 2, Section 4, Figure 4.6.2.

7.4.3 Effective shear depth of stiffeners

The effective shear depth of stiffeners, d_{shr}, in [mm], is to be taken as:

$$d_{shr} = \left(h_{stf} - 0.5t_{c-stiff} + t_p + 0.5t_{c-pl}\right)$$

$$\text{for } 75° \leq \varphi_w \leq 90°$$

$$d_{shr} = \left(h_{stf} - 0.5t_{c-stiff} + t_p + 0.5t_{c-pl}\right) \sin\varphi_w$$

$$\text{for } \varphi_w < 75°$$

where:

h_{stf} = Height of stiffener, in [mm], as defined in Chapter 2, Section 4, Figure 4.6.2.

t_p = Net thickness of the stiffener attached plating, in [mm], as defined in Chapter 2, Section 4, Figure 4.6.2.

φ_w = Angle, in [deg], as defined in Figure 7.4.3.

φ_w is to be taken as 90[deg] if the angle is greater than or equal to 75[deg].

7.4.4 Elastic net section modulus and net moment of inertia of stiffeners

The elastic net section modulus, *Z* of stiffeners, in [cm^3], and the net moment of inertia, I, in [cm^4] of stiffeners is to be taken as:

$$Z = (Z_{stf}) \quad \text{for } 75° \leq \varphi_w \leq 90°$$

$$Z = (Z_{stf}) \sin\varphi_w \quad \text{for } \varphi_w < 75°$$

$$I = (I_{st}) \text{ for } 75° \leq \varphi_w \leq 90°$$

$$I = (I_{st}) \sin \varphi_w \text{ for } \varphi_w < 75^\circ$$

where:

Z_{stf} = Net section modulus of the stiffener, in [cm^3], considered perpendicular to its attached plate, i.e., with φ_w = 90[deg].

I_{st} = Net moment of inertia of the stiffener, in [cm^4], considered perpendicular to its attached plate, i.e., with φ_w = 90[deg].

φ_w = Angle, in [deg], as defined in Figure 7.4.3.

φ_w is to be taken as 90[deg] if the angle is greater than or equal to 75[deg].

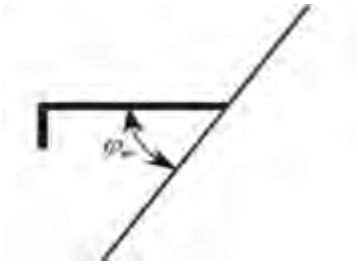


Figure 7.4.3: Angle between stiffener web and attached plating

7.4.5 Effective net plastic shear area of stiffeners

The net plastic shear area, A_{shr-pl}, of stiffeners, in [cm^2], which is used for assessment against impact loads is to be taken as:

$$A_{shr-pl} = A_{shr}$$

where:

A_{shr} = Net elastic shear area, in [cm^2], as defined in Cl.7.4.2.

7.4.6 Effective net plastic section modulus of stiffeners

The effective net plastic section modulus, Z_{pl}, of stiffeners, in [cm^3], with attached plating which is used for assessment against impact loads, is to be taken as:

$$Z_{pl} = \frac{h_w t_w (h_w + t_p)}{2000} + \frac{(2\gamma - 1) A_f (h_{f-ctr} + t_p/2)}{1000} \text{ for } 75^\circ \leq \varphi_w \leq 90^\circ$$

$$Z_{pl} = \frac{h_w t_w (h_w + t_p) \sin \varphi_w}{2000} + \frac{(2\gamma - 1) A_f \left((h_{f-ctr} + t_p/2) \sin \varphi_w - b_{f-ctr} \cos \varphi_w \right)}{1000} \text{ for } \varphi_w < 75^\circ$$

where:

t_p = Net thickness of the stiffener attached plating, in [mm], as defined in Chapter 2, Section 4, Figure 4.6.2.

t_w = Net web thickness of the stiffener, in [mm] as defined in Chapter 2, Section 4, Figure 4.6.2.

h_w = Depth of stiffener web, in [mm], taken equal to:

- For T, L (rolled and built-up) and flat bar profiles, as defined in Chapter 2, Section 4, Figure 4.6.2.

- For L2, L3 as defined in Chapter 2, Section 4, Figure 4.6.2.

- For bulb profiles, to be taken as defined in Cl.7.4.1.

γ = coefficient equal to:

$$\gamma = \frac{1 + \sqrt{3 + 12\beta}}{4}$$

β = coefficient equal to:

- For L profiles without a mid-span tripping bracket, but not to be taken greater than 0.5.

$$\beta = \frac{t_w^2 \, \ell_{bdg}^2}{80 \, b_f^2 \, t_f \, h_{f-ctr}} 10^6 + \frac{t_w}{2 b_f}$$

- For other cases, 0.5.

A_f = net cross sectional area of flange, in [mm^2]:

- For flat bar stiffeners, $A_f = 0$

- For other stiffeners, $A_f = b_f . t_f$

b_{f-ctr} = Distance from mid thickness of stiffener web to the centre of the flange area, in [mm].

- For rolled angle profiles and bulb profiles

$$b_{f-ctr} = 0.5(b_f - t_w)$$

- For T profiles

$$b_{f-ctr} = 0$$

h_{f-ctr} = Height of stiffener measured to the mid thickness of the flange, in [mm]:

- For profiles with flange of rectangular shape except for L3 profiles and for bulb profiles

$$h_{f-ctr} = h_w + 0.5t_f$$

- For L3 profiles as defined in Chapter 2, Section 4, Figure 4.6.2.

$$h_{f-ctr} = h_w - d_e - 0.5t_f$$

d_e = Distance from upper edge of web to the top of the flange, in [mm], for L3 profiles, Refer Chapter 2, Section 4, Figure 4.6.2.

t_f = Net flange thickness, in [mm]

- For flat bar stiffeners, $t_f = 0$

- For bulb profiles to be taken as defined in Cl.7.4.1.

7.4.7 Primary supporting member web not perpendicular to attached plating

Where the primary supporting member web is not perpendicular to the attached plating, the actual net shear area in [cm^2], and the actual net section modulus, in [cm^3], can be obtained from the following formulae:

- Actual net shear area:

$$A_{sh-n50} = A_{sh-0-n50} \sin\varphi_w \quad \text{for } \varphi_w < 75°$$

$$A_{sh-n50} = A_{sh-0-n50} \quad \text{for } 75° \leq \varphi_w \leq 90°$$

- Actual net section modulus:

$$Z_{n50} = Z_{perp-n50} \sin\varphi_w \quad \text{for } \varphi_w < 75°$$

$$Z_{n50} = Z_{perp-n50} \quad \text{for } 75° \leq \varphi_w \leq 90°$$

where:

$A_{sh-0-n50}$= Actual net shear area, in [cm^2], of the primary supporting member assumed to be perpendicular to the attached plating, to be taken equal to:

$$A_{sh-0-n50} = (h_w + t_{f-n50} + t_{p-n50})t_{w-n50}10^{-2}$$

h_w = Web height of PSM, in [mm].

t_{f-n50} = Net thickness of flange in [mm].

t_{p-n50} = Net thickness of plate in [mm].

t_{w-n50} = Net thickness of web in [mm].

$Z_{perp-n50}$ = Actual section modulus, in [cm^3], with its attached plating of the primary supporting member assumed to be perpendicular to the attached plating.

φ_w = angle, in [deg], between web and attached plating, Refer Figure 7.4.7.

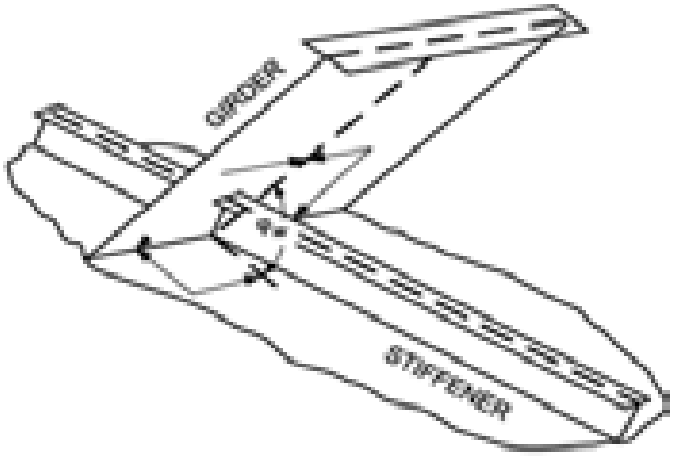


Figure 7.4.7: The web angle φ_w of primary supporting members

7.4.8 Shear area of primary supporting members with web openings

7.4.8.1 The effective web height, h_{eff}, in [mm], to be considered for calculating the effective net shear area, A_{sh-n50}, is to be taken as the lesser of:

$$h_{eff} = h_w$$

$$h_{eff} = h_{w3} + h_{w4}$$

$$h_{eff} = h_{w1} + h_{w2} + h_{w4}$$

where:

h_w = web height of PSM, in [mm]

$h_{w1}, h_{w2}, h_{w3}, h_{w4}$ = dimensions as shown in Figure 7.4.8.1.

Where an opening is located at a distance less than $h_w/3$ from the cross-section under consideration, h_{eff} is to be taken as the smaller of the net height and the net distance through the opening. Refer Figure 7.4.8.1.

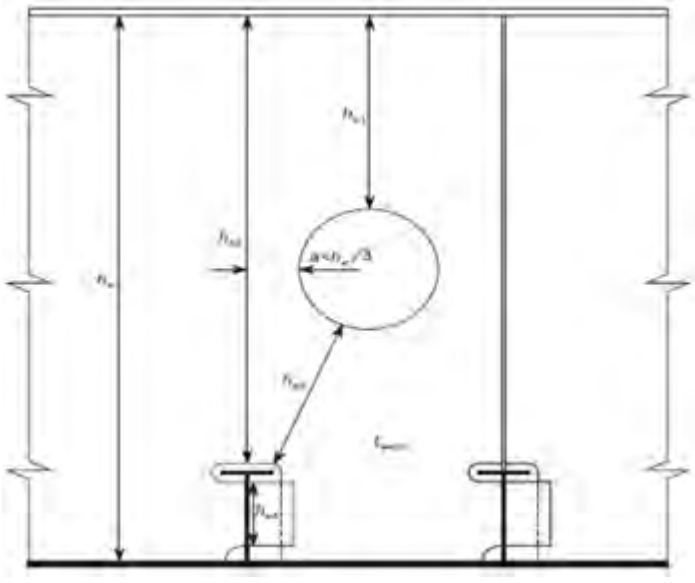


Figure 7.4.8.1: Effective shear area in way of web openings

7.4.8.2 The effective net cross-sectional area of the girder web, in [cm^2], is to be taken as:

$$A_{w-n50} = 0.01 h_{eff} . t_{w-n50}$$

where,

h_{eff} = Net web height, [mm], after deduction of cutouts at the section under consideration.

7.4.8.3 Where the PSM flange is not parallel to the attached plate (girder flange is at an angle $[\theta]$ to the length of the girder), the effective net web area, in [cm^2], is to be taken as:

$$A_{w-n50} = 0.01 h_{eff} \cdot t_{w-n50} + 1.3 A_{F-n50} \cdot \sin 2\theta \sin \theta$$

where:

h_{eff} = net web height, [mm], after deduction of cutouts at the considered section as shown in Figure 7.4.8.2.

A_{F-n50} = Actual net flange area in [cm^2].

θ = Angle of slope of continuous flange in [deg] as shown in Figure 7.4.8.2.

t_{w-n50} = Net thickness of web in [mm] as shown in Figure 7.4.8.2.

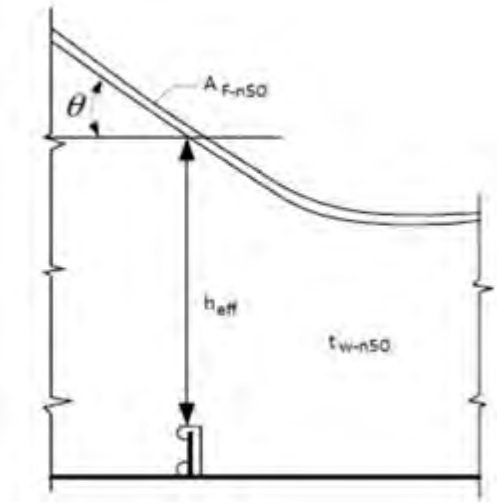


Figure 7.4.8.2: Effective web area in way of brackets

7.5 Plates

7.5.1 Idealisation of EPP

7.5.1.1 Elementary Plate Panel (EPP)

An elementary plate panel (EPP) is the unstiffened part of the plating between stiffeners and/or primary supporting members. The plate panel length, a, and breadth, b, of the EPP are defined as the longest and shortest plate edges, respectively, as shown in Figure 7.5.1.1.

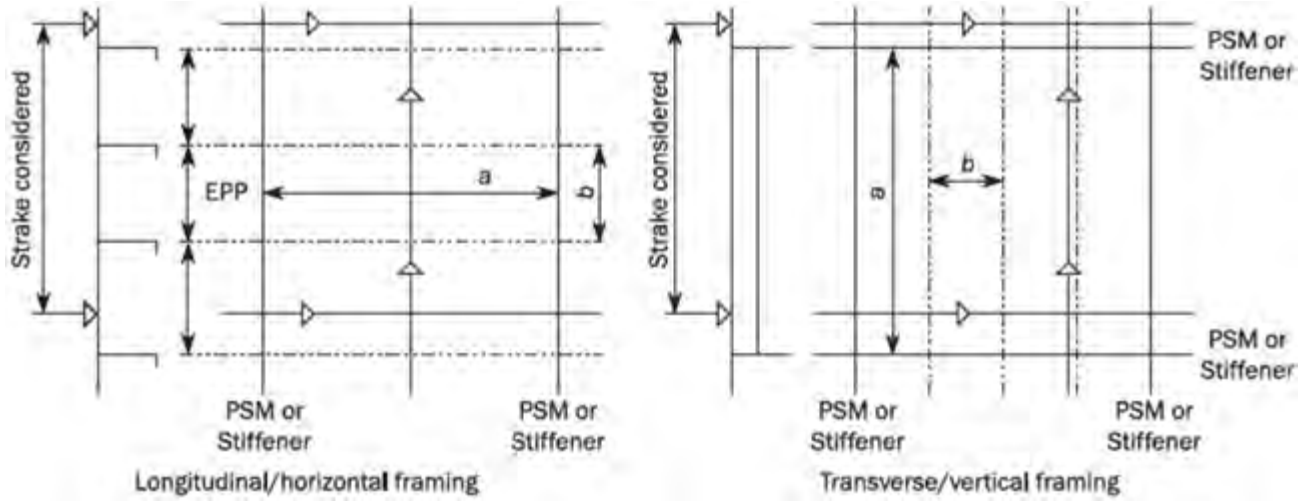


Figure 7.5.1.1: Elementary Plate Panel (EPP) definition

7.5.1.2 Strake required thickness

The required thickness of a plate strake is to be taken as the greatest value required for each EPP within that strake. The requirements given in Table 7.5.1 are to be applied for the selection of strakes to be considered as shown in Figure 7.5.1.2. The maximum corrosion addition within a strake is to be applied according to Chapter 3, Section 5.

Table 7.5.1: Strake considered in a given EPP		
	a/b > 2	**a/b < 2**
$a_1 > b/2$	All strakes (St1, St2, St3, St4)	All strakes (St1, St2, St3, St4)
$a_1 \leq b/2$	Strakes St2 and St4	All strakes (St1, St2, St3, St4)

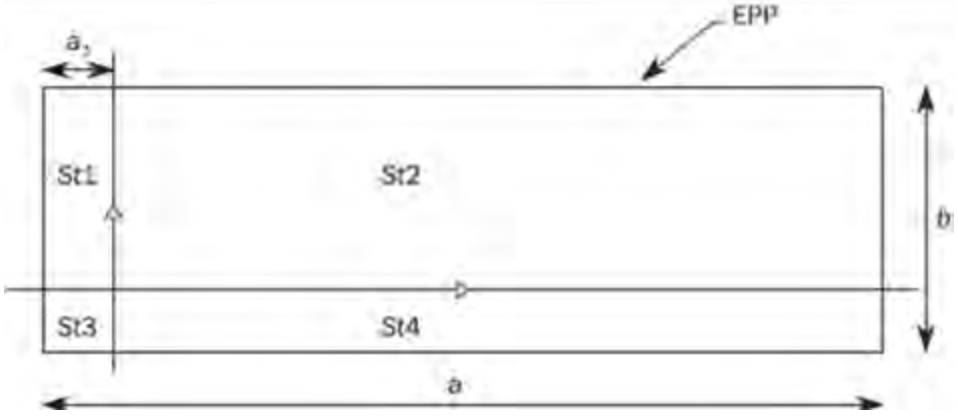


Figure 7.5.1.2: Strake considered in a given EPP

where:

a_1 = Distance, in [mm], measured inside the considered strake in the direction of the long edge of the EPP, between the strake boundary weld seam and the EPP edge.

For direct strength assessment, the EPP is idealised with the mesh arrangement in the finite element model.

7.5.2 Load calculation point

7.5.2.1 Yielding

For the yielding check, the local pressure and hull girder stress, used for the calculation of the local scantling requirements are to be taken at the Load Calculation Point (LCP) having coordinates x, y and z as defined in Table 7.5.2.1.

Table 7.5.2.1: LCP coordinates for yielding						
LCP coordinates	**General (1)**		**Horizontal plating**		**Vertical transverse structure and transverse stool plating**	
	Longitudinal framing Figure 7.5.2.1	**Transverse framing Figure 7.5.2.2**	**Longitudinal framing**	**Transverse framing**	**Horizontal framing Figure 7.5.2.3**	**Vertical framing Figure 7.5.2.4**
x coordinate	Mid-length of the EPP		Mid-length of the EPP		Corresponding to y and z values	
y coordinate	Corresponding to x and z coordinates		Outboard y value of the EPP		Outboard y value of the EPP, taken at z level (2)	
z coordinate	Lower edge of the EPP	The greater of lower edge of the EPP or lower edge of the strake	Corresponding to x and y values		Lower edge of the EPP	The greater of lower edge of the EPP or lower edge of the strake

1) All structures other than horizontal platings or vertical transverse structures.
2) For transom plate, the y coordinate of the LCP is to be taken corresponding to y value at side shell at z level of the load calculation point, for the external dynamic pressure calculation

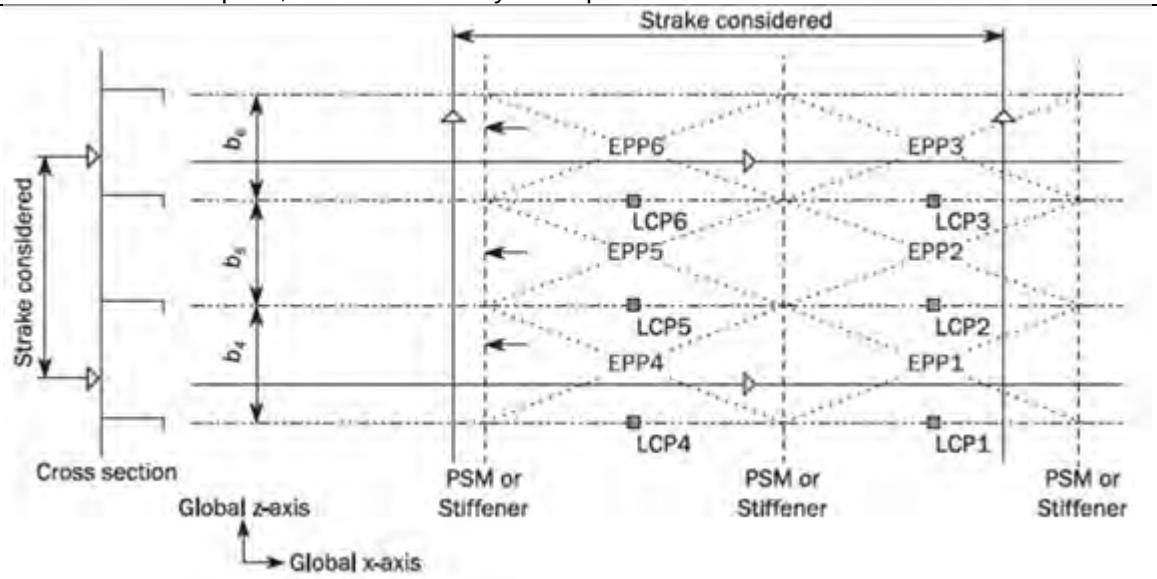


Figure 7.5.2.1: LCP for longitudinal plating with longitudinal framing

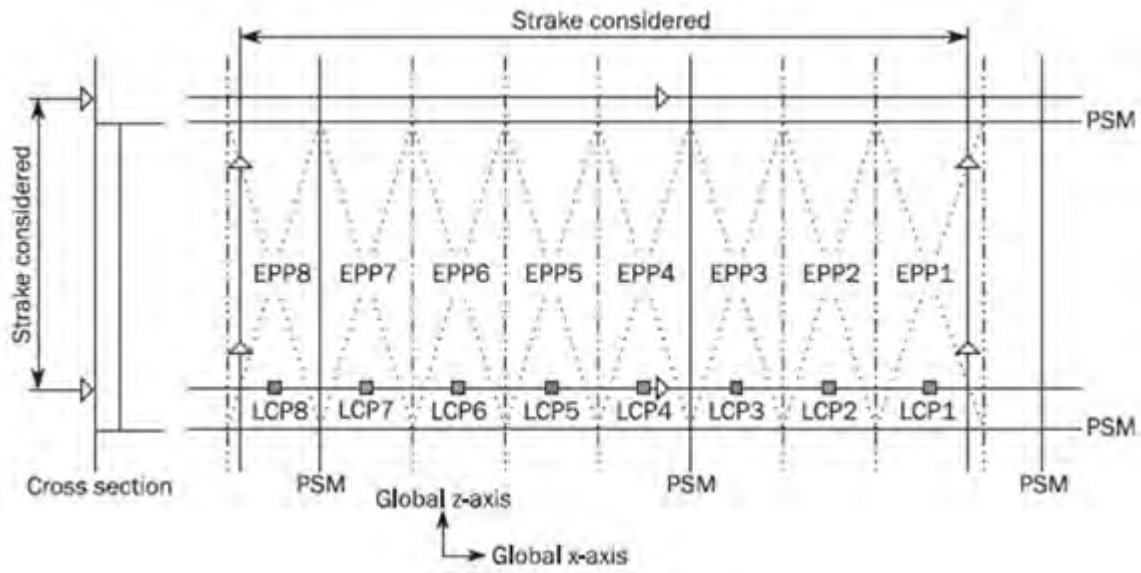


Figure 7.5.2.2: LCP for longitudinal plating with transverse framing

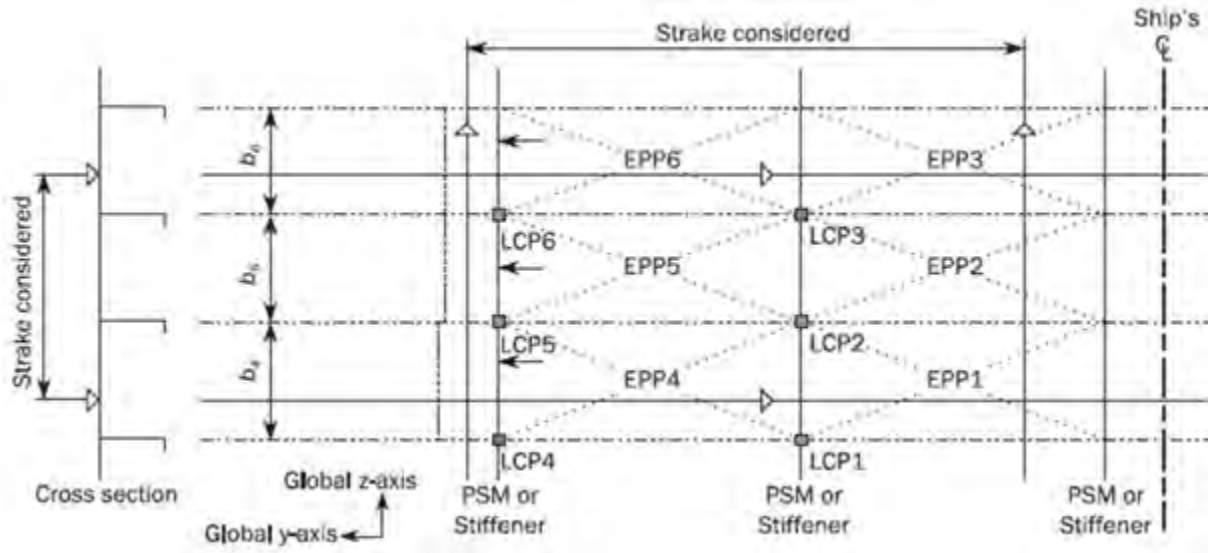


Figure 7.5.2.3: LCP for horizontal framing on transverse vertical structure

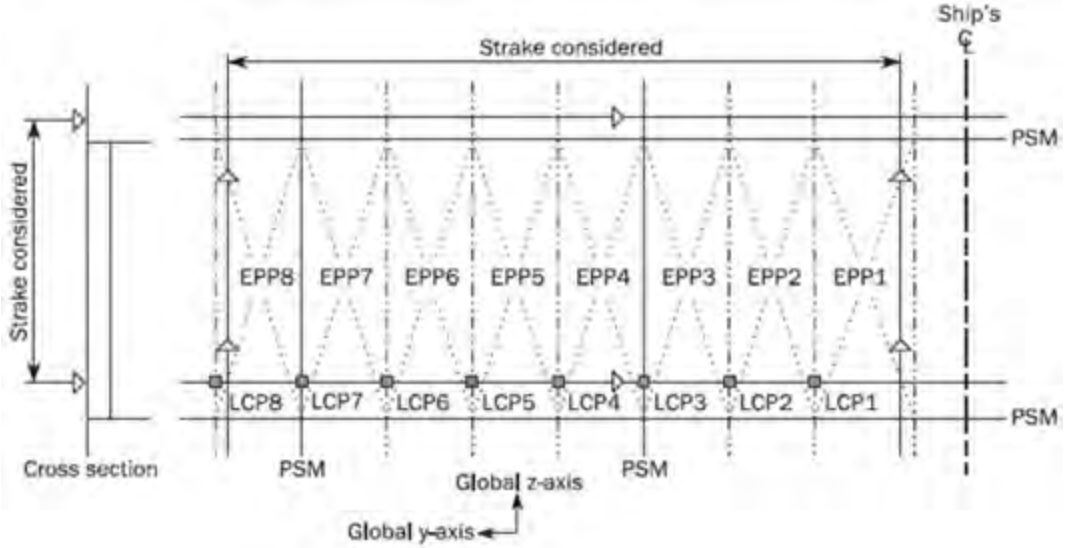


Figure 7.5.2.4: LCP for vertical framing on transverse vertical structure

7.5.2.2 Buckling

• For the prescriptive buckling check of the EPP according to Chapter 9, the LCP for the pressure and for the hull girder stresses are defined in Table 7.5.2.2.

• For the FE buckling check, Chapter 8 is applicable.

Table 7.5.2.2: LCP coordinates for plate buckling

LCP coordinates	**LCP for pressure**	**LCP for hull girder stresses Figure 7.5.2.5**		
		Bending stresses (1)		**Shear stresses**
		Non horizontal plate	Horizontal plate	
x coordinate	Same coordinates as LCP for yielding Refer Table 7.5.2.1	Mid-length of the EPP		
y coordinate		Both upper and lower ends of the EPP (points A1 and A2)	Outboard and inboard ends of the EPP (points A1 and A2)	Mid-point of EPP (point B)
z coordinate		Corresponding to x and y values		
(1) The bending stress for curved plate panel is the mean value of the stresses calculated at points A1 and A2.				

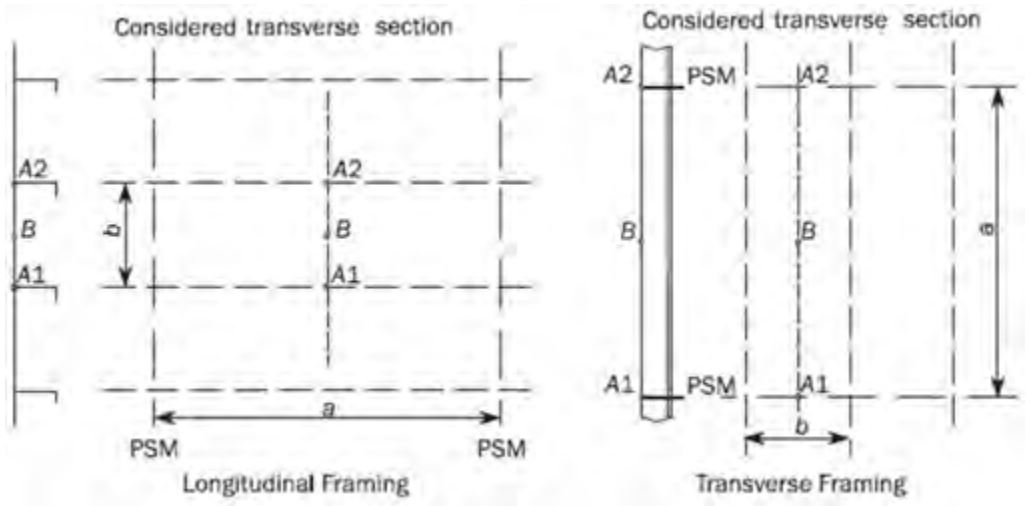


Figure 7.5.2.5: LCP for plate buckling – hull girder stresses

7.6 Stiffeners

7.6.1 Reference point

The requirements for section modulus for stiffeners relate to the reference point giving the minimum section modulus. This reference point is generally located as shown in Figure 7.6.1 for typical profiles.

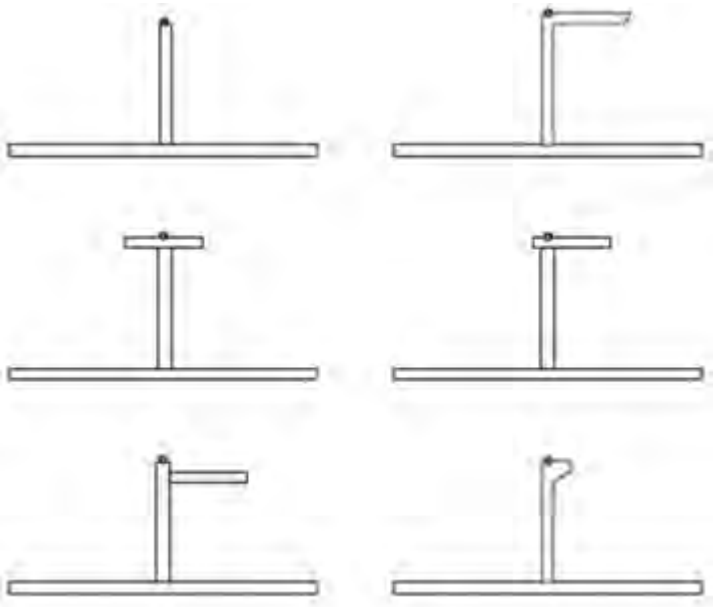


Figure 7.6.1: Reference point for calculation of section modulus and hull girder stress for local scantling assessment

7.6.2 Load calculation points

7.6.2.1 LCP for Pressure

The load calculation point for the pressure is located at:

- Middle of the full length, ℓ, of the considered stiffener.

- The intersection point between the stiffener and its attached plate.

For stiffeners located on transom plate, the y coordinate of the load calculation point is to be taken corresponding to y value at side shell at z level of the load calculation point, for the external dynamic pressure calculation.

7.6.2.2 LCP for hull girder bending stress

The load calculation point for the hull girder bending stresses is defined as follows:

- For yielding check, according to Chapter 7.

- At the middle of the full length, ℓ, of the considered stiffener

– At the reference point given in Figure 7.6.1.

- For prescriptive buckling check, according to Chapter 9.

– At the middle of the full length, ℓ, of the considered stiffener.

– At the intersection point between the stiffener and its attached plate.

7.6.3 Non-horizontal stiffeners

The lateral pressure, P is to be calculated as the maximum between the value obtained at middle of the full length, ℓ, and the value obtained from the following formulae:

7.6.3.1 When the upper end of the vertical stiffener is below the lowest zero pressure level,

$$P = \frac{P_u + P_L}{2}$$

7.6.3.2 When the upper end of the vertical stiffener is at or above the lowest zero pressure level, Refer Figure 7.6.3.

$$P = \frac{\ell_1}{\ell}\frac{P_L}{2}$$

where:

ℓ_1 = Distance, in [m], between the lower end of vertical stiffener and the lowest zero pressure level.

P_u, P_L = Lateral pressures at the upper and lower end of the vertical stiffener span ℓ, respectively.

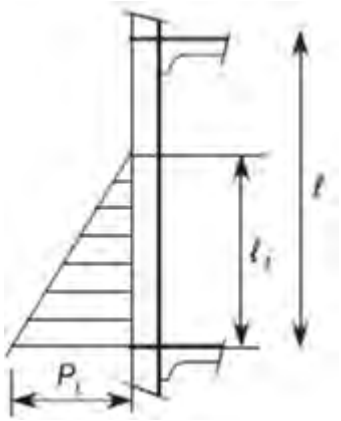


Figure7.6.3: Definition of pressure for vertical stiffeners

7.7 Primary supporting members

7.7.1 Load calculation point

The load calculation point is located at the middle of the full length, ℓ, at the attachment point of the primary supporting member with its attached plate.

For primary supporting members located on transom plate, the y coordinate of the load calculation point is to be taken corresponding to y value at side shell at z level of load calculation point for the external dynamic pressure calculation.

End of Chapter

Chapter 4

Stability

Contents

Section

1 *General*

2 *Intact Stability*

3 *Damage Stability*

4 *Damage Control Plan and Booklet*

Section 1

General

1.1 Application

1.1.1 All ships will be assigned class only after it has been demonstrated that their intact stability and damage stability (where applicable) are to be in compliance with standards laid down by the relevant Administration

1.1.2 The level of stability is not to be less than that provided by these Rules.

1.1.3 Special consideration may be given to the ships registered for a restricted service by IRS based on the acceptance of the Flag Administration.

1.1.4 Where other criteria for stability are accepted by the Administration concerned, such criteria may be used for the purpose of classification.

Section 2

Intact Stability

2.1 General

2.1.1 The level of intact stability in any case is not to be less than that required by Part A of the International Code on Intact Stability, 2008 (IMO Resolution MSC 267(85), as amended) as applicable to the type of ship being considered.

2.1.2 For pontoons of all sizes, in case the Administration concerned has not laid down criteria for intact stability, it is recommended that the criteria indicated below or equivalent are satisfied:

- Chapter 2, Section 2.2 of Part B of the International Code on Intact Stability, 2008 (Resolution MSC 267(85)).

2.1.3 Each ship is to be provided with a stability booklet approved by IRS and is to be made available onboard. The Intact stability booklet is to include information as is necessary to enable the Master by a rapid and simple process to obtain accurate guidance as to the Intact stability of the ship under varying conditions of service.

2.1.4 The format and content of the stability book is further described in Model Loading and Stability Manual (MSC/Circ.920). In addition, information such as design limitations, maximum speed, worst intended weather conditions or other information regarding the handling of the vessel that the Master needs to

operate the ship is also to be included to the booklet.

2.1.5 Stability data and associated plans are to be drawn up in the working language of the ship and any other language, the Administration may require. Reference is also made to the International Safety Management (ISM) Code, adopted by the IMO by resolution A.741(18). All translations of the stability booklet are to be approved.

2.1.6 If used, permanent ballast is to be located in accordance with a plan approved by IRS and in a manner that prevents shifting of position. Permanent ballast is not to be removed from the ship or relocated within the ship without the approval of IRS. Permanent ballast particulars are to be noted in the ship's stability booklet as a separate item. Permanent solid ballast is to be installed under the supervision of IRS.

2.2 Documentation

2.2.1 For the purpose of the examination of intact stability, plans/ booklets/ documents as indicated below are to be submitted:

a) Lines plan and offset tables
b) Draft Mark plan
c) Tank Capacity Plan
d) External Watertight and Weathertight Integrity plan / Freeboard plan
e) Preliminary Intact stability manual
f) Permanent Ballast Arrangement Plan (As Applicable)
g) Inclining test and lightship survey procedure
h) Inclining test or lightship survey report
i) Final Intact stability manual

2.2.2 Preliminary Intact stability manual

2.2.2.1 This manual provides general stability information, including documentation necessary to demonstrate compliance with the applicable Rules and Regulations where the light ship particulars are based on estimation.

2.2.3 Final Intact stability manual

2.2.3.1 Final stability manual based on the results of the inclining test or the lightship check is to be submitted for examination.

2.2.4 External watertight and weathertight integrity

2.2.4.1 The plan is to contain information on all external openings leading to compartments affecting stability calculations.

2.2.4.2 The following information is to be shown on the plan:

a) Superstructures and deckhouses contributing to buoyancy in intact or damage stability calculations

b) Position of each opening (longitudinal, transverse and vertical)

c) Type of closing appliance (watertight, weather tight or unprotected)

d) Alarms, indicators, remote controls and signboards fitted for each opening/closing appliance

e) Spaces that each opening lead to

Alternatively, the above information may be shown on the Damage Control Plan.

Section 3

Damage Stability

3.1 Application

3.1.1 The damage stability calculations and criteria to be applied depending on the ship type as described by its Class notation and corresponding requirements specified in Part 5 of the Rules and additional requirements laid down by the relevant Administration (if any).

3.1.2 For tankers (ships carrying oil, chemical and gas) assigned with a tropical freeboard, the corresponding loading conditions and damage stability calculations are to be submitted in the damage stability booklet.

3.1.3 All ships are required to comply with the applicable damage stability requirements in accordance with the following:

a) Part B and Part B-1 through B-4 of Chapter II-1 of SOLAS 1974, as amended for passenger ships regardless of length and cargo ships having load line length "L_L" of 80[m] and above.

Cargo ships which are shown to comply with any other damage stability regulations indicated in the following (3.1.3 b) to g)) are excluded from the applicability of the above.

b) Regulation 27 of ILLC 1966 as applied to tankers and dry cargo vessels in compliance with Resolutions A.320(IX) and A.514(13) provided that in the case of ships to which regulation 27(9) applies, main transverse watertight bulkheads, to be considered effective, are spaced according to paragraph 12(f) of resolution A.320(IX). Compliance with Reg.27 of ILLC Protocol 1988 is accepted as equivalent to the above.

In case ships for which Reg.27 of ILLC applies are intended to carry deck cargo also, then a limiting GM or KG curve is to be provided to the master in the stability manual based on compliance with the probabilistic damage stability analysis of SOLAS Chapter II-1 Part B-1. In such cases the KG used for demonstrating compliance with the deterministic damage stability requirements of ILLC Reg.27 should be same as that used for the probabilistic damage stability calculations required by SOLAS Chapter II-1 Part B-1 at the deepest subdivision loadline.

c) Annex I (Regulation 28) of MARPOL 1973/78, for Oil Tankers except that combination carriers with Type B freeboard are not excluded.

d) IBC Code, as amended for Chemical Tankers in general.

e) IGC Code, as amended for Liquefied Gas Carriers.

f) IGF Code, as amended for Low-flashpoint or gaseous fuel ships.

g) Resolution MSC 235(82) for Offshore Supply Vessels.

h) Resolution MSC 266(84) for Special Purpose Ships.

3.2 Documentation

3.2.1 For all ships to which damage stability requirements apply, damage stability calculations are to be submitted. The damage stability calculations are to include:

3.2.1.1 List of the characteristics (volume, center of gravity, permeability) of each compartment which can be damaged.

3.2.1.2 A table of openings in bulkheads, decks and side shell reporting all the information about:

- Identification of the opening

- Vertical, transverse and horizontal location

- Type of closure: sliding, hinged or rolling for doors

-Type of tightness: watertight, weathertight, semi watertight or unprotected

- Operating system: remote control, local operation, indicators on the bridge, television surveillance, water leakage detection, audible alarm, as applicable

- Foreseen utilization: open at sea, normally closed at sea, kept closed at sea

3.2.1.3 List of all damage cases corresponding to the applicable requirements.

3.2.1.4 Detailed results of damage stability calculations for all the loading conditions foreseen in the applicable requirements.

3.2.1.5 Limiting GM/KG curve, if foreseen in the applicable requirements.

3.2.1.6 Capacity plan.

3.2.1.7 Cross and down flooding devices and the calculations thereof according to MSC 362(92) with information about diameter, valves, pipe lengths and coordinates of inlet/outlet.

3.2.1.8 Watertight and weathertight door plan with pressure calculation.

3.2.1.9 Side contour and wind profile.

3.2.1.10 Pipes and damaged area when the destruction of these pipes results in progressive flooding.

3.2.2 Additional information for the probabilistic approach

In addition to the information listed in Cl.3.2.1, the following is to be provided:

- Subdivision length 'L_S'

- Initial draughts and the corresponding GM-values

- Required subdivision index 'R'

- Attained subdivision index 'A' with a summary table for all contributions for all damaged zones.

- Draught, trim, GM in each damaged condition

- Damage extent and definition of damage cases with probabilistic values ('p',' 'v' and 'r') *

- Righting lever (including GZ_{max} and range) with factor of survivability, 's'

- Critical weathertight and unprotected openings with their angle of immersion

- Details of sub-compartments with amount of in-flooded water/lost buoyancy with their centre of gravity

* For definition of symbols, SOLAS as amended is to be referred.

3.2.3 Internal watertight integrity plan

3.2.3.1 This plan is to contain items affecting damage stability calculations, such as internal subdivision, possibility of progressive flooding from one compartment to another through internal openings, pipes, tunnels or ventilation ducts and pipes, ducts, tunnels in the damage penetration zone specified in the applicable damage assumptions. The following information is to be shown on the plan:

(a) Internal watertight boundaries (bulkheads, decks and tunnels)

(b) Internal openings in these boundaries such as: doors, hatches, stairways, ventilation ducts and so forth

(c) Spaces each opening leads to (for ventilation ducts)

(d) Type of closing appliance (watertight, weather tight or unprotected)

(e) Alarms, indicators, remote controls and fitted for each opening/closing appliance

(f) Marking of the damage penetration zone, as specified in the applicable damage assumptions, on various waterlines/decks and cross sections

(g) Pipes, ducts and tunnels in the marked damage penetration zone and spaces where these lead

(h) Information on materials in piping if it is intended to use heat sensitive materials

3.2.4 The Damage stability booklet is to be revised whenever there is any major modification / conversion to the vessel resulting in the change of initial loading conditions used for computation.

Section 4

Damage Control Plan and Booklet

4.1 General

4.1.1 The damage control plan and damage control booklet are intended to provide ship's crew with clear information on the ship's watertight subdivision and equipment/ means related to maintaining the boundaries and effectiveness of the subdivision so that, in the event of damage to the ship causing flooding, proper precautions can be taken to prevent progressive flooding through openings therein and effective action can be taken quickly to mitigate and, where possible, recover the ship's loss of stability. The damage control plan and damage control booklet are to be clear and easy to understand.

4.2 Application

4.2.1 Damage Control Plan and Booklet are to be provided:

4.2.1.1 For all cargo ships (not less than 500 gross tonnage) and All passenger ships

a) Which are constructed on or after 01 January 2009

(Chapter II-1, Reg.19 of 2006 amendments to SOLAS 1974)

4.2.1.2 For dry cargo ships (not less than 500 gross tonnage)

a) Which are constructed on or after 01 February 1992 and before 01 January 2009

(Chapter II-1, Reg.23-1 of 1989 amendments to SOLAS 1974,)

4.2.1.3 For Passenger ships

a) Which are constructed on or after 25 May 1980 and before 01 September 1984

(Chapter II-1, Reg.20 of SOLAS 1974)

b) Which are constructed on or after 01 September 1984 and before 01 January 2009

(Chapter II-1, Reg.23 of 1981 amendments to SOLAS 1974)

4.3 Requirements

4.3.1 Damage Control Plan and Booklet is to include information in accordance with the following:

- IMO MSC.1/Circular 919 "*Guidelines for Damage Control Plans*" for vessels constructed before 01 January 2009,

- IMO MSC.1/Circular 1245 "*Guidelines for Damage Control Plans and Information to the master*" for vessels constructed on or after 01 January 2009,

- IMO MSC.1/Circular 1570 "*Amendments to section 3 of the Guidelines for Damage Control Plans and Information to the Master (MSC.1/Circular 1245)*" for passenger ships built on or after 09 June 2017

4.3.2 For passenger ships, the damage control plan is to be permanently exhibited or readily available on the navigating bridge, as well as in the ship's control station, safety centre or equivalent.

4.3.3 For cargo ships the damage control plan is to be permanently exhibited or readily available on the navigating bridge. Furthermore, the damage control plan is to be permanently exhibited or readily available in the cargo control room, all ship's office or other suitable location

End of Chapter

Chapter 5

Design Loads

Contents

Section

1 *General*
2 *Ship motions and Accelerations*
3 *Hull Girder Loads*
4 *External Loads*
5 *Internal Loads*
6 *Design Load Scenarios*
7 *Loading Conditions*
8 *Guidelines for Ballast Loading Conditions of Cargo Vessels Involving Partially Filled Ballast Tanks*

Section 1

General

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

1.1 Application

1.1.1 This chapter provides the design load for strength and fatigue assessments. The load combinations are to be derived for the design load scenarios specified in Section 6.

1.1.2 The scantlings of various hull members are to be based on the design values of ship motions, accelerations, lateral pressures and forces given in this Chapter. The design values of pressure caused by slamming, bow impact and sloshing are given in Chapter 11 dealing with specific structures.

1.1.3 Additional requirements applicable to specific ship types are defined in Part 5.

1.1.4 IRS may accept the values of wave induced dynamic loads derived from direct calculations in lieu of formulas given in Sections 4 and Section 5 provided the calculations are submitted for approval. In this regard, due consideration is to be given to the environmental conditions, probability of occurrence and uncertainties in load and response predictions.

1.2 Definitions

1.2.1 Still water loads

Still water loads are loads that are acting on the hull structure when a ship is floating in still (calm) water.

1.2.2 Wave loads/Dynamic loads

Wave loads are those due to wave pressures and ship motions.

1.2.3 Impact loads

Impact loads are those that have a duration much shorter than the period of the wave loads.

1.2.4 Local loads

Local loads are pressures and forces which are directly applied to the individual structural members such as plate panels, ordinary

stiffeners and primary supporting members. They consist of:

a) Still water local loads constitute of hydrostatic external sea pressures and the static pressures and forces induced by the weights carried by the ship.

b) Wave local loads constitutes of external sea pressures due to waves and the inertial pressures and forces induced by the ship accelerations.

c) Impact local loads due to Slamming (Bow and Bottom slamming) and Sloshing (tank sloshing) pressures.

1.2.5 Hull girder loads

Hull girder loads are (still water, wave and impact) forces and moments which result as effects of local loads acting on the ship as a whole and considered as a beam.

1.2.6 Load case

A load case is a combination of hull girder and local loads applied to the ship structure.

1.2.7 Coordinate system

The coordinate system is defined in Chapter 1, Section 2.5, Cl.2.5.1.

1.2.8 Sign convention for ship motions

The ship motions are defined with respect to the ship's centre of gravity (COG) as shown in Figure 1.2.8, where:

- positive surge is translation in the *X*-axis direction (positive forward)
- positive sway is translation in the Y-axis direction (positive towards port side of ship)
- positive heave is translation in the Z-axis direction (positive upwards)
- positive roll motion is positive rotation about a longitudinal axis through the COG (starboard down and port up)
- positive pitch motion is positive rotation about a transverse axis through the COG (bow down and stern up)
- positive yaw motion is positive rotation about a vertical axis through the COG (bow moving to port and stern to starboard).

1.2.9 Equivalent Design Wave (EDW)

The dynamic loads associated with each dynamic load case are based on the Equivalent Design Wave (EDW) concept. The EDW concept applies a consistent set of dynamic loads to the ship such that specified dominant load response is equivalent to the required long term response value.

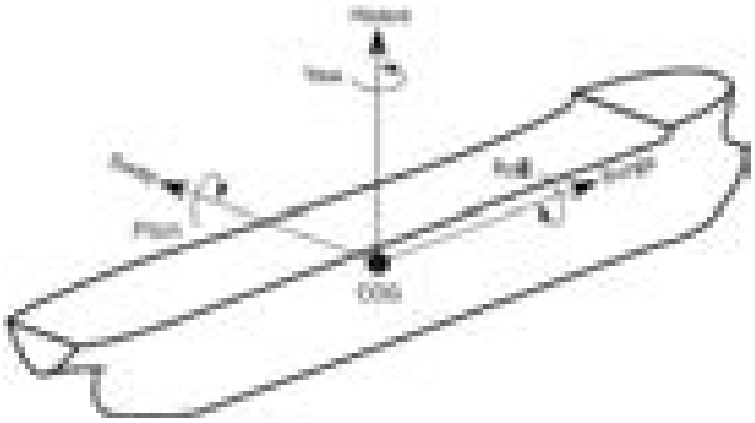


Figure 1.2.8: Definition of positive motions

1.3 Load definition criteria

1.3.1 Hull girder loads

The still water, wave and impact hull girder loads are used for the determination of:

a) the hull girder strength, according to the requirements of Chapter 6.

b) the structural scantling of plating, ordinary stiffeners and primary supporting members contributing to the hull girder strength, in combination with the local loads given in Section 4 and Section 5.

1.3.2 Local loads

The local loads defined in Cl.1.2.4 are to be calculated for each load cases considered. The wave local loads are to be calculated on the basis of the reference values of ship motions and accelerations as specified in Section 2, for sea pressures, Refer Section 4 and internal pressures and forces, Refer Section 5.

The local loads are used in the structural scantling of:

a) Plating, according to Chapter 7, Section 4

b) Ordinary stiffeners, according to Chapter 7, Section 5

c) Primary supporting members, according to Chapter 7, Section 6

1.4 Probability level for strength and fatigue assessments

The probability levels of loads are considered as follows:

- Wave induced dynamic loads for strength assessment excluding fatigue are at a probability level of 10^{-8}

- Wave induced dynamic loads for fatigue assessment are at a probability level of 10^{-2}.

1.5 Loads for strength assessment

The strength assessment is to be undertaken for all design load scenarios and the final assessment is to be made on the most onerous strength requirement.

Each design load scenario for strength assessment is composed of a Static (S) load case or a Static + Dynamic (S+D) load case, where the static and dynamic loads are dependent on the loading condition being considered.

The static loads are defined in the following sections:

- Still water hull girder loads in Section 3.

- External loads in Section 4.

- Internal loads in Section 5.

The EDWs for the strength assessment and the dynamic load combination factors for global loads are listed in Section 2, 2.4.

The dynamic load components are defined in the following sections:

- Dynamic hull girder load components in Section 3.

- External loads in Section 4.

- Internal loads in Section 5.

1.6 Loads for fatigue assessment

Each design load scenario for fatigue assessment is composed of a Static + Dynamic (S+D) load case, where the static and dynamic loads are dependent on the loading condition being considered.

The static loads are defined in the following sections:

- Still water hull girder loads in Section 3.

- External loads in Section 4.

- Internal loads in Section 5.

The EDWs for the fatigue assessment are listed in Section 2, 2.5.

The dynamic load components are defined in the following sections:

- Dynamic hull girder load components in Section 3.

- External loads in Section 4.

- Internal loads in Section 5.

Section 2

Ship motions and Accelerations

Symbols

For symbols not defined in this section, refer to Chapter 1, Section 2.

ρ = Density of liquid cargo or dry cargo, [t/m^3].

P = Design pressure [N/mm^2]

C_w = Wave Coefficient

$= 0.0856L \text{ for } L < 90[m]$

$= 10.75 - \left[\frac{300-L}{100}\right]^{1.5} \text{ for } 90 \leq L \leq 300[m]$

$= 10.75 \text{ for } 300 \leq L \leq 350[m]$

$= 10.75 - \left[\frac{L-350}{150}\right]^{1.5} \text{ for } 350 \leq L \leq 500[m]$

a_o = Common acceleration parameter

$$= (1.58 - 0.47C_b)\left(\frac{2.4}{\sqrt{L}} + \frac{34}{L} - \frac{600}{L^2}\right)$$

not to be taken more than 0.7

T_r = Roll period, in [s], as defined in Cl.2.1.2.1.

$\emptyset$ = Roll angle, in [deg], as defined in Cl.2.1.2.1.

T_p = Pitch period, in [s], as defined in Cl.2.1.2.2.

θ = Pitch angle, in [deg], as defined in Cl.2.1.2.2.

f_{ps} = Coefficient to be taken for strength assessment which is dependent on the applicable design load scenario specified in Section 6, and to be taken as:

For unrestricted service:

= 1.0 for extreme sea loads design load scenario.

= 0.8 for the ballast water exchange design load scenario.

= 0.8 for the accidental flooded design load scenario at sea.

= 0.4 for the harbour and tank testing design load scenario

For vessels with service restriction:

= $1.R_s$ for design load scenario specified in Section 6 except for harbour and tank testing design load scenario

= 0.4 for the harbour and tank testing design load scenario

where,

R_s = Service restriction factor

= 0.75 for coastal service

= 0.65 for restricted service

= 0.55 for sheltered water service.

For definitions of service limitations, Refer Part 1, Chapter 1.

f_{fa} = 0.9; fatigue coefficient

f_β = heading correction factor

For strength assessment:

= 1.05 in general

= 0.8 for BSR and BSP load cases for the extreme sea loads design load scenario

= 1.0 for HSA, OST and OSA load cases for the extreme sea loads design load scenario.

= 1.0 for ballast water exchange at sea, harbour and tank testing, accidental flooded design load scenarios.

For fatigue assessment: = 1.0.

R = vertical coordinate, in [m], of the ship rotation centre, is to be taken as:

$$R = \min\left[\left(\frac{D}{4} + \frac{T_{LC}}{2}\right), \frac{D}{2}\right]$$

a_{roll-y} = Transverse acceleration due to roll, in [m/s^2], as defined in Cl.2.1.4.2.

$a_{pitch-x}$ = Longitudinal acceleration due to pitch, in [m/s^2], as defined in Cl.2.1.4.1.

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2], as defined in Cl.2.1.4.3.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2], as defined in Cl.2.1.4.3.

T_{LC} = draught, in [m], amidships for the considered loading condition.

In case loading condition is not defined, T_{LC} = T_{SC} is to be applied.

f_T = Ratio between draught at a loading condition and scantling draught, is to be taken as:

$$f_T = \frac{T_{LC}}{T_{SC}} \quad \text{but is not to be taken less than } 0.5.$$

f_{xl} = Ratio between X-coordinate of the load point and L, to be taken as:

$$f_{xl} = \frac{x}{L}$$

but not to be taken less than 0 or greater than 1.0

x, y, z = X, Y and Z coordinates, in [m], of the considered point with respect to the coordinate system, as defined in Section 1, Cl.1.2.8

f_{lp} = Factor depending on longitudinal position along the ship, to be taken as:

$$f_{lp} = 1.0 \text{ for } x/L \leq 0.5$$

$$f_{lp} = -1.0 \text{ for} \frac{x}{L} > 0.5$$

f_{lp-OST} = Factor for the longitudinal distribution of the torsional moment for the OST load case, as defined in Section 3, Cl.3.6.4.

f_{lp-OSA} = Factor for the longitudinal distribution of the torsional moment for the OSA load case as defined in Section 3, Cl.3.6.4.

WS = Weather side, side of the ship exposed to the incoming waves

LS = Lee side, sheltered side of the ship away from the incoming waves

M_{wv} = Vertical wave bending moment, in [kN-m], defined in Section 3

Q_{wv} = Vertical wave shear force, in [kN], defined in Section 3

M_{Wh} = Horizontal wave bending moment, in [kN-m], defined in Section 3

M_{wt} = Torsional wave bending moment, in [kN-m], defined in Section 3

C_{WV} = Load combination factor to be applied to the vertical wave bending moment

C_{QW} = Load combination factor to be applied to the vertical wave shear force

C_{WH} = Load combination factor to be applied to the horizontal wave bending moment

C_{WT} = Load combination factor to be applied to the wave torsional moment

C_{XS} = Load combination factor to be applied to the surge acceleration

C_{XP} = Load combination factor to be applied to the longitudinal acceleration due to pitch

C_{XG} = Load combination factor to be applied to the longitudinal acceleration due to pitch motion

C_{YS} = Load combination factor to be applied to the sway acceleration

C_{YR} = Load combination factor to be applied to the transverse acceleration due to roll

C_{YG} = Load combination factor to be applied to the transverse acceleration due to roll motion

C_{ZH} = Load combination factor to be applied to the heave acceleration

C_{ZR} = Load combination factor to be applied to the vertical acceleration due to roll

C_{ZP} = Load combination factor to be applied to the vertical acceleration due to pitch

2.1 Ship motions and accelerations

2.1.1 General

The ship motions and accelerations are assumed to be sinusoidal. The motion values defined by the formulae in this section are

single amplitudes, i.e. half of the 'crest to trough' height.

2.1.2 Ship motions

2.1.2.1 Roll motions

The roll period T_r, in [s], to be taken as:

$$T_r = \frac{2.3\ \pi\ k_r}{\sqrt{g.\ GM}}$$

The roll angle ∅, in [deg], to be taken as:

$$\emptyset = \frac{9000(1.25 - 0.025\ T_r)f_p f_{bk}}{(B + 75)\pi}$$

where:

f_p = Coefficient to be taken as:

= f_{ps} for strength assessment.

= $f_{fa}[0.23 - 4f_T B \times 10^{-4}]$ for fatigue assessment

f_{bk} = 1.2 for ships without bilge keel.

= 1.0 for ships with bilge keel or fin stabilizers.

T_r is not be taken greater than 30[s]. Minimum value of T_r resulting from various possible combinations of k_r and GM is to be used.

k_r = Roll radius of gyration, in [m], in the considered loading condition. In case k_r has not been carried out, the values given in Table 2.1.2, may be used.

GM = Metacentric height, in [m], in the considered loading condition. In case GM is not available, the following values from Table 2.1.2, may be used for guidance.

Table 2.1.2: Table for k_r and GM Values

Loading condition (1) (2) (4)	Notation/ Ship Types	k_r	GM
Full load condition / Homogeneous loading	**OIL TANKER, ESP**	0.35 B	0.12 B
	BULK CARRIER, ESP, SELF-UNLOADING BULK CARRIER ESP		
	CONTAINER SHIP with B ≤ 32.2[m]		0.05 B
	CONTAINER SHIP with B ≥ 40.0[m]		0.11 B
	Other ships		0.07 B
	ORE CARRIER, ESP	0.25 B	0.12 B
Full load condition / Homogeneous heavy cargo	**BULK CARRIER, ESP, SELF-UNLOADING BULK CARRIER ESP**	0.42 B	0.25B
	ORE CARRIER, ESP	0.25 B	
Full load condition / Alternate heavy cargo	**BULK CARRIER, ESP, SELF-UNLOADING BULK CARRIER ESP**	0.40 B	0.20 B
	ORE CARRIER, ESP	0.20 B	
Full load condition / Alternate light cargo	**BULK CARRIER, ESP, SELF-UNLOADING BULK CARRIER ESP**	0.35 B	0.12 B
	ORE CARRIER, ESP	0.20 B	
Partial load condition	**OIL TANKER, ESP**	0.40 B	0.24 B
	ASPHALT CARRIER CHEMICAL TANKER, ESP COMBINATION CARRIER	0.35 B	0.07 B
Heavy ballast condition	**BULK CARRIER, ESP, SELF-UNLOADING BULK CARRIER ESP**	0.40 B	0.25 B
	ORE CARRIER, ESP	0.35 B	0.30 B
Normal ballast condition	**OIL TANKER, ESP**	0.45 B	0.33 B
	BULK CARRIER, ESP, SELF-UNLOADING BULK CARRIER ESP		
	ORE CARRIER, ESP	0.35 B	
	Other ships	0.45 B	0.18 B

Steel coil loading (3)	**All ships intended to carry steel coils**	0.42 B	0.25 B
Linear interpolation may be used for 32.2 [m] ≤ B ≤ 40.0 [m].			

(1) For flooded loading conditions, the values of k_r and GM are to be taken as for the full load condition unless provided in the loading manual.

(2) For Multiple port loading conditions of bulk carrier ESP, self-unloading bulk carrier ESP, ore carrier ESP and combination carrier (limited to direct strength analysis)

- loading conditions with draught greater than or equal to $0.9T_{sc}$, the values of k_r and GM, unless provided in the loading manual, are to be taken as those from the most appropriate full load condition.
- loading conditions with draught between T_{BH} and $0.9T_{sc}$, the values of k_r and GM, unless provided in the loading manual, are to be obtained by linear interpolation, based on the draught, between the heavy ballast condition and the most appropriate full load condition.
- loading conditions with a draught below T_{BH}, the values of k_r and GM for the heavy ballast condition are to be used.

(3) When steel coil loading condition is provided by the designer in the loading manual, this condition is to be assessed with the values of k_r and GM taken from the above Table 2.1.2.

(4) Block Loading conditions are to be assessed with k_r and GM values given in Table 2.1.2 for Homogeneous heavy cargo loading condition.

2.1.2.2 Pitch motions

The pitch period T_p, in [s], is to be taken as:

$$T_p = \sqrt{\frac{2\pi\lambda_\varphi}{g}}$$

where:

$$\lambda_\varphi = 0.6(1 + f_T)L$$

The pitch angle, in [deg], is to be taken as given in formula below and need not to be taken greater than 20[deg].

$$\theta = 1350\, f_p L^{-0.94}\left[1.0 + \left(\frac{2.57}{\sqrt{gL}}\right)^{1.2}\right]$$

where:

f_p = Coefficient to be taken as:

For strength assessment

$f_p = f_{ps}$

For fatigue assessment

$f_p = f_{fa}[(0.27 - 0.02f_T) - (13 - 5f_T)L \times 10^{-5}]$

2.1.3 Ship accelerations

The design values of various acceleration components are normally to be taken as:

2.1.3.1 Surge acceleration

The longitudinal acceleration due to surge, in [m/s^2], is to be taken as:

$$a_{surge} = 0.2\, f_p\, a_o\, g$$

where:

f_p = Coefficient to be taken as:

For strength assessment

$f_p = f_{ps}$

For fatigue assessment

$f_p = f_{fa}[(0.27) - (15 + 4f_T)L \times 10^{-5}]$

2.1.3.2 Sway acceleration

The transverse acceleration due to sway, in [m/s^2], is to be taken as:

$$a_{sway} = 0.3\, f_p\, a_o\, g$$

where:

f_p = Coefficient to be taken as:

For strength assessment

$f_p = f_{ps}$

For fatigue assessment

$f_p = f_{fa}[(0.24) - (6 - 2f_T)B \times 10^{-4}]$

2.1.3.3 Heave acceleration

The vertical acceleration due to heave, a_{heave} in [m/s^2], is to be taken as:

$$a_{heave} = \frac{0.7\, a_o\, g}{\sqrt{C_b}} f_p$$

where:

f_p = Coefficient to be taken as:

For strength assessment

$f_p = f_{ps}$

For fatigue assessment

$f_p = f_{fa}[(0.27 + 0.02f_T) - 17L \times 10^{-5}]$

2.1.3.4 Roll acceleration

The roll acceleration, a_{roll}, in [rad/s^2], is to be taken as:

$$a_{roll} = \emptyset \frac{\pi}{180}\left(\frac{2\pi}{T_r}\right)^2 f_p$$

f_p = Coefficient to be taken as:

For strength assessment

$f_p = f_{ps}$

For fatigue assessment

$f_p = f_{fa}[0.23 - 4f_T B \times 10^{-4}]$

$\emptyset$ = Roll angle using f_p equal to 1.0.

2.1.3.5 Pitch acceleration

The pitch acceleration, a_{pitch}, in [rad/s^2], is to be taken as:

$$a_{pitch} = \theta \frac{\pi}{180}\left(\frac{2\pi}{T_p}\right)^2 f_p$$

where:

θ = pitch angle using f_p equal to 1.0

f_p = Coefficient to be taken as:

For strength assessment

$f_p = f_{ps}$

For fatigue assessment

$f_p = f_{fa}[0.28 - (5 + 6f_T)L \times 10^{-5}]$

2.2 Accelerations at any position

2.2.1 General

The accelerations used to derive the inertial loads at any position are defined with respect to the ship fixed coordinate system. The acceleration values defined in Cl.2.2.2 and Cl.2.2.3 include the gravitational acceleration components due to the instantaneous roll and pitch angles.

The accelerations to be applied for the dynamic load cases defined in Section 2.3 are given in Cl. 2.2.2.

The combined accelerations as defined in Cl.2.2.3 are provided for advisory purposes and may be used for other design purpose when the maximum design acceleration values are required, for e.g. crane foundations, machinery foundations, etc.

2.2.2 Accelerations for dynamic load cases for L ≥ 90[m].

2.2.2.1 Longitudinal acceleration

The longitudinal acceleration at any position for each dynamic load case, in [m/s^2], is to be taken as:

$$a_x = \left[-C_{XG}\, g\, \sin\theta + C_{XS}\, a_{surge} + C_{XP}\, a_{pitch}(z - R)\right]$$

2.2.2.2 Transverse acceleration

The transverse acceleration at any position for each dynamic load case, in [m/s^2], is to be taken as:

$$a_y = \left[C_{YG}\, g\, \sin\varphi + C_{YS}\, a_{sway} - C_{YR}\, a_{roll}(z - R)\right]$$

2.2.2.3 Vertical acceleration

The vertical acceleration at any position for each dynamic load case, in [m/s^2], is to be taken as:

$$a_z = \left[C_{ZH}\, a_{heave} + C_{ZR}\, a_{roll} \cdot y - C_{ZP}\, a_{pitch}(x - 0.45L)\right]$$

2.2.3 Combined accelerations

The combined accelerations in the ship's vertical, transverse, and longitudinal directions are to be taken as below:

2.2.3.1 Longitudinal acceleration

The combined longitudinal acceleration, a_ℓ in [m/s^2] at any position, (combined effect of surge and pitch) is to be taken as:

$$a_\ell = \sqrt{a_{surge}^2 + \left[\left(g \sin\theta + a_{pitch-x}\right)\right]^2}$$

where,

a_{surge} = surge acceleration given in Cl.2.1.3.1.

$a_{pitch-x}$ = Longitudinal acceleration due to pitch, in [m/s^2]

= $a_{pitch}(z - R)$

2.2.3.2 Transverse acceleration

The combined transverse acceleration, a_t in [m/s^2] at any position, is to be taken as:

$$a_t = \sqrt{a_{sway}^2 + \left[\left(g \sin\varphi + a_{roll-y}\right)\right]^2}$$

where,

a_{sway} = sway/yaw acceleration given in Cl.2.1.3.2.

a_{roll-y} = Transverse acceleration due to roll, in [m/s^2]

= $a_{roll}(z - R)$

2.2.3.3 Vertical acceleration

The combined vertical acceleration, a_v in [m/s^2] at any position, (combined effect of heave, pitch and roll;)

$$a_v = \sqrt{\left(a_{heave}^2 + a_{roll-z}^2 + a_{pitch-z}^2\right)}$$

where,

a_{heave} = heave acceleration as given in Cl.2.1.3.3.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2]

= $a_{pitch}(x - 0.45L)$

= $a_{pitch}\left|(x - 0.45L)\right|$ for L<90[m].

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2]

= $a_{roll}\, y$

= $a_{roll}\left|y\right|$ for L<90[m].

2.3 Dynamic load cases

2.3.1 Definition of dynamic load cases

The following EDW is to be used to generate the dynamic load cases for structural assessment for L ≥ 90[m]:

- HSM load cases: HSM-1 and HSM-2: head sea EDWs that minimize and maximize the vertical wave bending moment amidships respectively.

- HSA load cases: HSA-1 and HSA-2: head sea EDWs that maximize and minimize the head sea vertical acceleration at FP respectively.

- FSM load cases: FSM-1 and FSM-2: following sea EDWs that minimize and maximize the vertical wave bending moment amidships respectively.

- BSR load cases: BSR-1P and BSR-2P: beam sea EDWs that minimize and maximize the roll motion downward and upward on the port side respectively with waves from the port side.

BSR-1S and BSR-2S: beam sea EDWs that maximize and minimize the roll motion downward and upward on the starboard side respectively with waves from the starboard side.

- BSP load cases: BSP-1P and BSP-2P: beam sea EDWs that maximize and minimize the hydrodynamic pressure at the waterline amidships on the port side respectively.

BSP-1S and BSP-2S: beam sea EDWs that maximize and minimize the hydrodynamic pressure at the waterline amidships on the starboard side respectively.

- OST load cases: OST-1P and OST-2P: oblique sea EDWs that minimize and maximize the torsional moment at 0.25L from the AE with waves from the port side respectively.

OST-1S and OST-2S: oblique sea EDWs that maximize and minimize the torsional moment at 0.25L from the AE with waves from the starboard side respectively.

- OSA load cases: OSA-1P and OSA-2P: oblique sea EDWs that maximize and minimize the pitch acceleration with waves from the port side respectively.

OSA-1S and OSA-2S: oblique sea EDWs that maximize and minimize the pitch acceleration with waves from the starboard side respectively.

HSA and OSA load cases are not be used for fatigue assessment.

Note 1: 1 and 2 denote the maximum or the minimum dominant load component for each EDW.

Note 2: P and S denote that the weather side is on port side or starboard side respectively.

2.3.2 Application

The dynamic load cases described in this section are to be used for determining the dynamic loads required by the design load scenarios described in Section 6. These dynamic load cases are to be applied to the following structural assessments:

2.3.2.1 Strength assessment:

- for plating, stiffeners and primary supporting members by prescriptive methods
- for hull girder strength
- for the direct strength method (FE analysis) assessment of structural members.

2.3.2.2 Fatigue assessment:

- for structural details covered by simplified stress analysis.
- for structural details covered by FE stress analysis.

2.4 Dynamic load cases for strength assessment

2.4.1 Description of dynamic load cases

Table 2.4.1.1 to Table 2.4.1.3 describe the ship motions responses and the global loads corresponding to each dynamic load case to be considered for the strength assessment for L ≥ 90[m].

Table 2.4.1.1: Ship responses for HSM, HSA and FSM load cases – Strength assessment

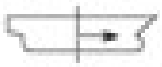
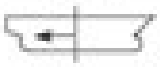
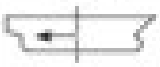
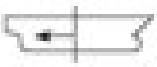
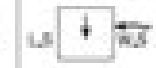
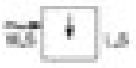
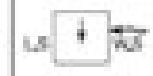
Loadcase	HSM-1	HSM-2	HSA-1	HSA-2	FSM-1	FSM-2
EDW	HSM		HSA		FSM	
Heading	Head		Head		Following	
Effect	Max. bending moment		Max. vertical acceleration		Max. bending moment	
VWBM	Sagging	Hogging	Sagging	Hogging	Sagging	Hogging
VWSF	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore
HWBM	-	-	-	-	-	-
TM	-	-	-	-	-	-
Surge	To stern	To bow	To stern	To bow	To bow	To stern
a_{surge}						
Sway	-	-	-	-	-	-
a_{sway}	-	-	-	-	-	-
Heave	Down	Up	Down	Up	-	-
a_{heave}					-	-
Roll	-	-	-	-	-	-
a_{roll}	-	-	-	-	-	-
Pitch	Bow down	Bow up	Bow down	Bow up	Bow up	Bow down
a_{pitch}						

Table 2.4.1.2: Ship responses for BSR and BSP load cases – Strength assessment

Load case	BSR-1P	BSR-2P	BSR-1S	BSR-2S	BSP-1P	BSP-2P	BSP-1S	BSP-2S
EDW	BSR		BSR		BSP		BSP	
Heading	Beam				Beam			
Effect	Max. roll				Max. pressure at waterline			
VWBM	Sagging	Hogging	Sagging	Hogging	Sagging	Hogging	Sagging	Hogging
VWSF	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore
HWBM	Stbd tensile	Port tensile	Port tensile	Stbd tensile	Stbd tensile	Port tensile	Port tensile	Stbd tensile
TM	-	-	-	-	-	-	-	-
Surge	-	-	-	-	-	-	-	-
A_{surge}	-	-	-	-	-	-	-	-
Sway	To starboard	To portside	To portside	To starboard	To portside	To starboard	To starboard	To portside
A_{sway}								
Heave	Down	Up	Down	Up	Down	Up	Down	Up
A_{heave}								
Roll	Portside down	Portside up	Starboard down	Starboard up	Portside down	Portside up	Starboard down	Starboard up
A_{roll}								
Pitch	-	-	-	-	Bow down	Bow up	Bow down	Bow up

Load case	BSR-1P	BSR-2P	BSR-1S	BSR-2S	BSP-1P	BSP-2P	BSP-1S	BSP-2S
A_{pitch}	-	-	-	-				

Table 2.4.1.3: Ship responses for OST and OSA load cases – Strength assessment

Load case	OST-1P	OST-2P	OST-1S	OST-2S	OSA-1P	OSA-2P	OSA-1S	OSA-2S
EDW	OST				OSA			
Heading	Oblique				Oblique			
Effect	Max. torsional moment				Max. pitch acceleration			
VWBM	Sagging	Hogging	Sagging	Hogging	Hogging	Sagging	Hogging	Sagging
VWSF	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore
HWBM	Port tensile	Stbd tensile	Stbd tensile	Port tensile	Stbd tensile	Port tensile	Port tensile	Stbd tensile
TM								
Surge	To bow	To stern	To bow	To stern	To bow	To stern	To bow	To stern
A_{surge}								
Sway	-	-	-	-	To portside	To starboard	To starboard	To portside
A_{sway}	-	-	-	-				
Heave	Down	Up	Down	Up	Up	Down	Up	Down
A_{heave}								
Roll	Portside down	Portside up	Starboard down	Starboard up	Portside down	Portside up	Starboard down	Starboard up

Load case	OST-1P	OST-2P	OST-1S	OST-2S	OSA-1P	OSA-2P	OSA-1S	OSA-2S
A_{roll}								
Pitch	Bow up	Bow down	Bow up	Bow down	Bow up	Bow down	Bow up	Bow down
A_{pitch}								

2.4.2 Load combination factors

The load combinations factors, LCFs for the global loads and inertia load components for strength assessment are defined in:

Table 2.4.2.1: LCFs for HSM, HSA and FSM load cases.

Table 2.4.2.2: LCFs for BSR and BSP load cases.

Table 2.4.2.3: LCFs for OST and OSA load cases.

Table 2.4.2.1: Load combination factors, LCF's for HSM, HSA and FSM load cases								
Load component		**LCF**	**HSM-1**	**HSM-2**	**HSA-1**	**HSA-2**	**FSM-1**	**FSM-2**
Hull girder loads	M_{wv}	C_{WV}	-1	1	-0.7	0.7	$-0.4f_T - 0.6$	$0.4f_T + 0.6$
	Q_{wv}	C_{QW}	$-1.f_{lp}$	$1.f_{lp}$	$-0.6.f_{lp}$	$0.6.f_{lp}$	$-1.f_{lp}$	$1.f_{lp}$
	M_{wh}	C_{WH}	0	0	0	0	0	0
	M_{wt}	C_{WT}	0	0	0	0	0	0
Longitudinal accelerations	a_{surge}	C_{XS}	$0.3 - 0.2f_T$	$0.2f_T - 0.3$	0.2	-0.2	$0.2 - 0.4f_T$	$0.4f_T - 0.2$
	$a_{pitch-x}$	C_{XP}	-0.7	0.7	$-0.4f_T - 0.4$	$0.4f_T + 0.4$	0.15	-0.15
	$g.\sin\theta$	C_{XG}	0.6	-0.6	$0.4f_T + 0.4$	$-0.4f_T - 0.4$	-0.2	0.2
Transverse accelerations	a_{sway}	C_{YS}	0	0	0	0	0	0
	a_{roll-y}	C_{YR}	0	0	0	0	0	0
	$g.\sin\varphi$	C_{YG}	0	0	0	0	0	0
Vertical accelerations	a_{heave}	C_{ZH}	$0.5f_T - 0.15$	$0.15 - 0.5f_T$	$0.4f_T - 0.1$	$0.1 - 0.4f_T$	0	0
	a_{roll-z}	C_{ZR}	0	0	0	0	0	0
	$a_{pitch-z}$	C_{ZP}	-0.7	0.7	$-0.4f_T - 0.4$	$0.4f_T + 0.4$	0.15	-0.15

Table 2.4.2.2: Load combination factors, LCF's for BSR and BSP load cases						
Load component		**LCF**	**BSR-1P**	**BSR-2P**	**BSR-1S**	**BSR-2S**
Hull girder loads	M_{wv}	C_{WV}	$0.1 - 0.2f_T$	$0.2f_T - 0.1$	$0.1 - 0.2f_T$	$0.2f_T - 0.1$
	Q_{wv}	C_{QW}	$(0.1 - 0.2f_T)f_{lp}$	$(0.2f_T - 0.1)f_{lp}$	$(0.1 - 0.2f_T)f_{lp}$	$(0.2f_T - 0.1)f_{lp}$
	M_{wh}	C_{WH}	$1.2 - 1.1f_T$	$1.1f_T - 1.2$	$1.1f_T - 1.2$	$1.2 - 1.1f_T$
	M_{wt}	C_{WT}	0	0	0	0
Longitudinal accelerations	a_{surge}	C_{XS}	0	0	0	0
	$a_{pitch-x}$	C_{XP}	0	0	0	0
	$g.\sin\theta$	C_{XG}	0	0	0	0
Transverse accelerations	a_{sway}	C_{YS}	$0.2 - 0.2f_T$	$0.2f_T - 0.2$	$0.2f_T - 0.2$	$0.2 - 0.2f_T$
	a_{roll-y}	C_{YR}	1	-1	-1	1
	$g.\sin\varphi$	C_{YG}	-1	1	1	-1
Vertical accelerations	a_{heave}	C_{ZH}	$0.7 - 0.4f_T$	$0.4f_T - 0.7$	$0.7 - 0.4f_T$	$0.4f_T - 0.7$
	a_{roll-z}	C_{ZR}	1	-1	-1	1
	$a_{pitch-z}$	C_{ZP}	0	0	0	0

Load component		LCF	BSP-1P	BSP-2P	BSP-1S	BSP-2S
Hull girder loads	M_{wv}	C_{WV}	$0.3 - 0.8f_T$	$0.8f_T - 0.3$	$0.3 - 0.8f_T$	$0.8f_T - 0.3$
	Q_{wv}	C_{QW}	$(0.3 - 0.8f_T)f_{lp}$	$(0.8f_T - 0.3)f_{lp}$	$(0.3 - 0.8f_T)f_{lp}$	$(0.8f_T - 0.3)f_{lp}$
	M_{wh}	C_{WH}	$0.7 - 0.7f_T$	$0.7f_T - 0.7$	$0.7f_T - 0.7$	$0.7 - 0.7f_T$
	M_{wt}	C_{WT}	0	0	0	0
Longitudinal accelerations	a_{surge}	C_{XS}	0	0	0	0
	$a_{pitch-x}$	C_{XP}	$0.1 - 0.3f_T$	$0.3f_T - 0.1$	$0.1 - 0.3f_T$	$0.3f_T - 0.1$
	$g.\sin\theta$	C_{XG}	$0.3f_T - 0.1$	$0.1 - 0.3f_T$	$0.3f_T - 0.1$	$0.1 - 0.3f_T$
Transverse accelerations	a_{sway}	C_{YS}	-0.9	0.9	0.9	-0.9
	a_{roll-y}	C_{YR}	0.3	-0.3	-0.3	0.3
	$g.\sin\varphi$	C_{YG}	-0.2	0.2	0.2	-0.2
Vertical accelerations	a_{heave}	C_{ZH}	1	-1	1	-1
	a_{roll-z}	C_{ZR}	0.3	-0.3	-0.3	0.3
	$a_{pitch-z}$	C_{ZP}	$0.1 - 0.3f_T$	$0.3f_T - 0.1$	$0.1 - 0.3f_T$	$0.3f_T - 0.1$

Table 2.4.2.3: Load combination factors, LCF's for OST and OSA load cases

Load component		LCF	OST-1P	OST-2P	OST-1S	OST-2S
Hull girder loads	M_{wv}	C_{WV}	$-0.3 - 0.2f_T$	$0.3 + 0.2f_T$	$-0.3 - 0.2f_T$	$0.3 + 0.2f_T$
	Q_{wv}	C_{QW}	$(-0.35 - 0.2f_T)f_{lp}$	$(0.35 + 0.2f_T)f_{lp}$	$(-0.35 - 0.2f_T)f_{lp}$	$(0.35 + 0.2f_T)f_{lp}$
	M_{wh}	C_{WH}	-0.9	0.9	0.9	-0.9
	M_{wt}	C_{WT}	$-f_{lp-OST}$	f_{lp-OST}	f_{lp-OST}	$-f_{lp-OST}$
Longitudinal accelerations	a_{surge}	C_{XS}	$0.1f_T - 0.15$	$0.15 - 0.1f_T$	$0.1f_T - 0.15$	$0.15 - 0.1f_T$
	$a_{pitch-x}$	C_{XP}	$0.7 - 0.3f_T$	$0.3f_T - 0.7$	$0.7 - 0.3f_T$	$0.3f_T - 0.7$
	$g.\sin\theta$	C_{XG}	$0.2f_T - 0.45$	$0.45 - 0.2f_T$	$0.2f_T - 0.45$	$0.45 - 0.2f_T$
Transverse accelerations	a_{sway}	C_{YS}	0	0	0	0
	a_{roll-y}	C_{YR}	$0.4f_T - 0.25$	$0.25 - 0.4f_T$	$0.25 - 0.4f_T$	$0.4f_T - 0.25$
	$g.\sin\varphi$	C_{YG}	$0.1 - 0.2f_T$	$0.2f_T - 0.1$	$0.2f_T - 0.1$	$0.1 - 0.2f_T$
Vertical accelerations	a_{heave}	C_{ZH}	$0.2f_T - 0.05$	$0.05 - 0.2f_T$	$0.2f_T - 0.05$	$0.05 - 0.2f_T$
	a_{roll-z}	C_{ZR}	$0.4f_T - 0.25$	$0.25 - 0.4f_T$	$0.25 - 0.4f_T$	$0.4f_T - 0.25$
	$a_{pitch-z}$	C_{ZP}	$0.7 - 0.3f_T$	$0.3f_T - 0.7$	$0.7 - 0.3f_T$	$0.3f_T - 0.7$

Load component		LCF	OSA-1P	OSA-2P	OSA-1S	OSA-2S
Hull girder loads	M_{wv}	C_{WV}	$0.75 - 0.5f_T$	$-0.75 + 0.5f_T$	$0.75 - 0.5f_T$	$-0.75 + 0.5f_T$
	Q_{wv}	C_{QW}	$(0.6 - 0.4f_T)f_{lp}$	$(-0.6 + 0.4f_T)f_{lp}$	$(0.6 - 0.4f_T)f_{lp}$	$(-0.6 + 0.4f_T)f_{lp}$
	M_{wh}	C_{WH}	$0.55 + 0.2f_T$	$-0.55 - 0.2f_T$	$-0.55 - 0.2f_T$	$0.55 + 0.2f_T$
	M_{wt}	C_{WT}	$-f_{lp-OSA}$	f_{lp-OSA}	f_{lp-OSA}	$-f_{lp-OSA}$
Longitudinal accelerations	a_{surge}	C_{XS}	$0.1f_T - 0.45$	$0.45 - 0.1f_T$	$-0.45 + 0.1f_T$	$0.45 - 0.1f_T$
	$a_{pitch-x}$	C_{XP}	1.0	-1.0	1.0	-1.0
	$g.\sin\theta$	C_{XG}	-1.0	1.0	-1.0	1.0
Transverse accelerations	a_{sway}	C_{YS}	$-0.2 - 0.1f_T$	$0.2 + 0.1f_T$	$0.2 + 0.1f_T$	$-0.2 - 0.1f_T$
	a_{roll-y}	C_{YR}	$0.3 - 0.2f_T$	$0.2f_T - 0.3$	$0.2f_T - 0.3$	$0.3 - 0.2f_T$
	$g.\sin\varphi$	C_{YG}	$0.1f_T - 0.2$	$0.2 - 0.1f_T$	$0.2 - 0.1f_T$	$0.1f_T - 0.2$
Vertical accelerations	a_{heave}	C_{ZH}	$-0.2f_T$	$0.2f_T$	$-0.2f_T$	$0.2f_T$
	a_{roll-z}	C_{ZR}	$0.3 - 0.2f_T$	$0.2f_T - 0.3$	$0.2f_T - 0.3$	$0.3 - 0.2f_T$
	$a_{pitch-z}$	C_{ZP}	1.0	-1.0	1.0	-1.0

2.5 Dynamic load cases for fatigue assessment

2.5.1 Description of dynamic load cases

Table 2.5.1.1 to Table 2.5.1.3 describe the ship motions responses and the global loads corresponding to each dynamic load case to be considered for the fatigue assessment.

Table 2.5.1.1: Ship responses for HSM and FSM load cases – Fatigue assessment

Table 2.5.1.2: Ship responses for BSR and BSP load cases– Fatigue assessment

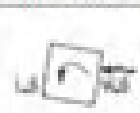
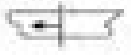
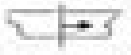
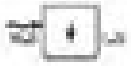
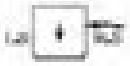
Load case	BSR-1P	BSR-2P	BSR-1S	BSR-2S	BSP-1P	BSP-2P	BSP-1S	BSP-2S
EDW	BSR		BSR		BSP		BSP	
Heading	Beam				Beam			
Effect	Max. roll				Max. pressure at waterline			
VWBM	Sagging	Hogging	Sagging	Hogging	Sagging	Hogging	Sagging	Hogging
VWSF	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore
HWBM	Stbd tensile	Port tensile	Port tensile	Stbd tensile	Stbd tensile	Port tensile	Port tensile	Stbd tensile
TM	-	-	-	-	-	-	-	-
Surge	+	+	+	+	+	+	+	+
A_{surge}	-	-	-	-	-	-	-	-
Sway	To starboard	To portside	To portside	To starboard	To portside	To starboard	To starboard	To portside
A_{sway}								
Heave	Down	Up	Down	Up	Down	Up	Down	Up
A_{heave}								
Roll	Portside down	Portside up	Starboard down	Starboard up	Portside down	Portside up	Starboard down	Starboard up
A_{roll}								
Pitch	-	-	-	-	-	-	-	-
A_{pitch}	+	+	+	+	+	+	+	+

Table 2.5.1.3: Ship responses for OST load cases– Fatigue assessment

Load case	OST-1P	OST-2P	OST-1S	OST-2S
EDW	OST			
Heading	Oblique			
Effect	Max. torsional moment			
VWBM	Sagging	Hogging	Sagging	Hogging
VWSF	Negative-aft Positive-fore	Positive-aft Negative-fore	Negative-aft Positive-fore	Positive-aft Negative-fore
HWBM	Port tensile	Stbd tensile	Stbd tensile	Port tensile
TM				
Surge	To bow	To stern	To bow	To stern
A_{surge}				
Sway	-	-	-	-
A_{sway}	-	-	-	-
Heave	Up	Down	Up	Down
A_{heave}				
Roll	Portside down	Portside up	Starboard down	Starboard up
A_{roll}				
Pitch	Bow up	Bow down	Bow up	Bow down
A_{pitch}				

2.5.2 Load combination factors

The load combinations factors, LCFs for the global loads and inertia load components for fatigue assessment are defined in:

Table 2.5.2.1: LCFs for HSM and FSM load cases.

Table 2.5.2.2: LCFs for BSR and BSP load cases.

Table 2.5.2.3: LCFs for OST load cases.

Table 2.5.2.1: Load combination factors, LCF's for HSM and FSM load cases						
Load component		**LCF**	**HSM-1**	**HSM-2**	**FSM-1**	**FSM-2**
Hull girder loads	M_{wv}	C_{WV}	-1	1	$-0.75 - 0.2f_T$	$0.75 + 0.2f_T$
	Q_{wv}	C_{QW}	$-1.f_{lp}$	$1.f_{lp}$	$(-0.75 - 0.2f_T)f_{lp}$	$(0.75 + 0.2f_T)f_{lp}$
	M_{wh}	C_{WH}	0	0	0	0
	M_{wt}	C_{WT}	0	0	0	0
Longitudinal accelerations	a_{surge}	C_{XS}	$0.3 - 0.2f_T$	$0.2f_T - 0.3$	$0.2 - 0.4f_T$	$0.4f_T - 0.2$
	$a_{pitch-x}$	C_{XP}	-0.9	0.9	0.1	-0.1
	$g.\sin\theta$	C_{XG}	$0.4f_T + 0.4$	$-0.4f_T - 0.4$	-0.15	0.15
Transverse accelerations	a_{sway}	C_{YS}	0	0	0	0
	a_{roll-y}	C_{YR}	0	0	0	0
	$g.\sin\varphi$	C_{YG}	0	0	0	0
Vertical accelerations	a_{heave}	C_{ZH}	$0.8f_T - 0.15$	$0.15 - 0.8f_T$	0	0
	a_{roll-z}	C_{ZR}	0	0	0	0
	$a_{pitch-z}$	C_{ZP}	-0.9	0.9	0.1	-0.1

Table 2.5.2.2: Load combination factors, LCF's for BSR and BSP load cases						
Load component		**LCF**	**BSR-1P**	**BSR-2P**	**BSR-1S**	**BSR-2S**
Hull girder loads	M_{wv}	C_{WV}	$0.1 - 0.2f_T$	$0.2f_T - 0.1$	$0.1 - 0.2f_T$	$0.2f_T - 0.1$
	Q_{wv}	C_{QW}	$(0.1 - 0.2f_T)f_{lp}$	$(0.2f_T - 0.1)f_{lp}$	$(0.1 - 0.2f_T)f_{lp}$	$(0.2f_T - 0.1)f_{lp}$
	M_{wh}	C_{WH}	$1.1 - f_T$	$f_T - 1.1$	$f_T - 1.1$	$1.1 - f_T$
	M_{wt}	C_{WT}	0	0	0	0
Longitudinal accelerations	a_{surge}	C_{XS}	0	0	0	0
	$a_{pitch-x}$	C_{XP}	0	0	0	0
	$g.\sin\theta$	C_{XG}	0	0	0	0
Transverse accelerations	a_{sway}	C_{YS}	$0.2 - 0.2f_T$	$0.2f_T - 0.2$	$0.2f_T - 0.2$	$0.2 - 0.2f_T$
	a_{roll-y}	C_{YR}	1	-1	-1	1
	$g.\sin\varphi$	C_{YG}	-1	1	1	-1
Vertical accelerations	a_{heave}	C_{ZH}	$0.7 - 0.4f_T$	$0.4f_T - 0.7$	$0.7 - 0.4f_T$	$0.4f_T - 0.7$
	a_{roll-z}	C_{ZR}	1	-1	-1	1
	$a_{pitch-z}$	C_{ZP}	0	0	0	0

Load component		LCF	BSP-1P	BSP-2P	BSP-1S	BSP-2S
Hull girder loads	M_{wv}	C_{WV}	$0.3 - 0.8f_T$	$0.8f_T - 0.3$	$0.3 - 0.8f_T$	$0.8f_T - 0.3$
	Q_{wv}	C_{QW}	$(0.3 - 0.8f_T)f_{lp}$	$(0.8f_T - 0.3)f_{lp}$	$(0.3 - 0.8f_T)f_{lp}$	$(0.8f_T - 0.3)f_{lp}$
	M_{wh}	C_{WH}	$0.6 - 0.6f_T$	$0.6f_T - 0.6$	$0.6f_T - 0.6$	$0.6 - 0.6f_T$
	M_{wt}	C_{WT}	0	0	0	0
Longitudinal accelerations	a_{surge}	C_{XS}	0	0	0	0
	$a_{pitch-x}$	C_{XP}	0	0	0	0
	$g.\sin\theta$	C_{XG}	0	0	0	0
Transverse accelerations	a_{sway}	C_{YS}	-0.95	0.95	0.95	-0.95
	a_{roll-y}	C_{YR}	0.3	-0.3	-0.3	0.3
	$g.\sin\varphi$	C_{YG}	-0.2	0.2	0.2	-0.2
Vertical accelerations	a_{heave}	C_{ZH}	1	-1	1	-1
	a_{roll-z}	C_{ZR}	0.3	-0.3	-0.3	0.3
	$a_{pitch-z}$	C_{ZP}	0	0	0	0

Table 2.5.2.3: Load combination factors, LCF's for OST load cases

Load component		LCF	OST-1P	OST-2P	OST-1S	OST-2S
Hull girder loads	M_{wv}	C_{WV}	-0.4	0.4	-0.4	0.4
	Q_{wv}	C_{QW}	$-0.4f_{lp}$	$0.4f_{lp}$	$-0.4f_{lp}$	$0.4f_{lp}$
	M_{wh}	C_{WH}	-0.9	0.9	0.9	-0.9
	M_{wt}	C_{WT}	$-f_{lp-OST}$	f_{lp-OST}	f_{lp-OST}	$-f_{lp-OST}$
Longitudinal accelerations	a_{surge}	C_{XS}	$-0.25 + 0.2f_T$	$0.25 - 0.2f_T$	$-0.25 + 0.2f_T$	$0.25 - 0.2f_T$
	$a_{pitch-x}$	C_{XP}	$0.4 - 0.2f_T$	$-0.4 + 0.2f_T$	$0.4 - 0.2f_T$	$-0.4 + 0.2f_T$
	$g.\sin\theta$	C_{XG}	$-0.4 + 0.2f_T$	$0.4 - 0.2f_T$	$-0.4 + 0.2f_T$	$0.4 - 0.2f_T$
Transverse accelerations	a_{sway}	C_{YS}	0	0	0	0
	a_{roll-y}	C_{YR}	$-0.4 + 0.6f_T$	$0.4 - 0.6f_T$	$0.4 - 0.6f_T$	$-0.4 + 0.6f_T$
	$g.\sin\varphi$	C_{YG}	$0.2 - 0.3f_T$	$-0.2 + 0.3f_T$	$-0.2 + 0.3f_T$	$0.2 - 0.3f_T$
Vertical accelerations	a_{heave}	C_{ZH}	-0.05	0.05	-0.05	0.05
	a_{roll-z}	C_{ZR}	$-0.4 + 0.6f_T$	$0.4 - 0.6f_T$	$0.4 - 0.6f_T$	$-0.4 + 0.6f_T$
	$a_{pitch-z}$	C_{ZP}	$0.4 - 0.2f_T$	$-0.4 + 0.2f_T$	$0.4 - 0.2f_T$	$-0.4 + 0.2f_T$

Section 3

Hull girder loads

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

f_β = heading correction factor as defined in Section 2.

For strength assessment:

= 1.05 in general

= 0.8 for BSR and BSP load cases for the extreme sea loads design load scenario for L≥90[m].

= 1.0 for HSA, OST and OSA load cases for the extreme sea loads design load scenario for L≥90[m].

= 1.0 for ballast water exchange at sea, harbour and tank testing design load scenario.

For fatigue assessment: = 1.0.

f_{fa} = 0.90; fatigue coefficient

3.1 Application

3.1.1 General

The hull girder loads for the static (S) design load scenarios is to be taken as the still water loads defined in 3.3.

The total hull girder loads for the static plus dynamic (S+D) design load scenarios are to be derived for each load case and are to be taken as the sum of the still water loads defined in 3.3 and the dynamic loads defined in 3.6.

In general, these Rules are valid for ships not exceeding 500[m] in length (L), having a breadth (B), not exceeding one-fifth of the length (L), nor 2.5 times the depth (D) to the strength deck and having a block coefficient, C_b >0.6. Vessels beyond these proportions will be specially considered.

For container ships the hull girder vertical wave bending moment and shear force defined in Part 5 is to apply in lieu of vertical bending moment and shear force defined in this section. (Refer UR S11A)

3.2 Sign convention for hull girder loads

3.2.1 The sign conventions of vertical bending moments, vertical shear forces, horizontal bending moments and torsional moments at any ship transverse section are as shown in Figure 3.2.1, below:

- The vertical bending moments M_{sw} and M_{wv} are positive when they induce tensile stresses in the strength deck (hogging bending moment) and negative when they induce tensile stresses in the bottom (sagging bending moment)

- The vertical shear forces Q_{sw} and Q_{wv} are positive in the case of downward resulting forces acting aft of the transverse section and upward resulting forces acting forward of the transverse section under consideration

- The horizontal bending moment M_{wh} is positive when it induces tensile stresses in the starboard side and negative when it induces tensile stresses in the port side

- The torsional moment M_{wt} is positive in the case of resulting moment acting aft of the transverse section following negative rotation around the X-axis, and of resulting moment acting forward of the transverse section following positive rotation around the *X*-axis.

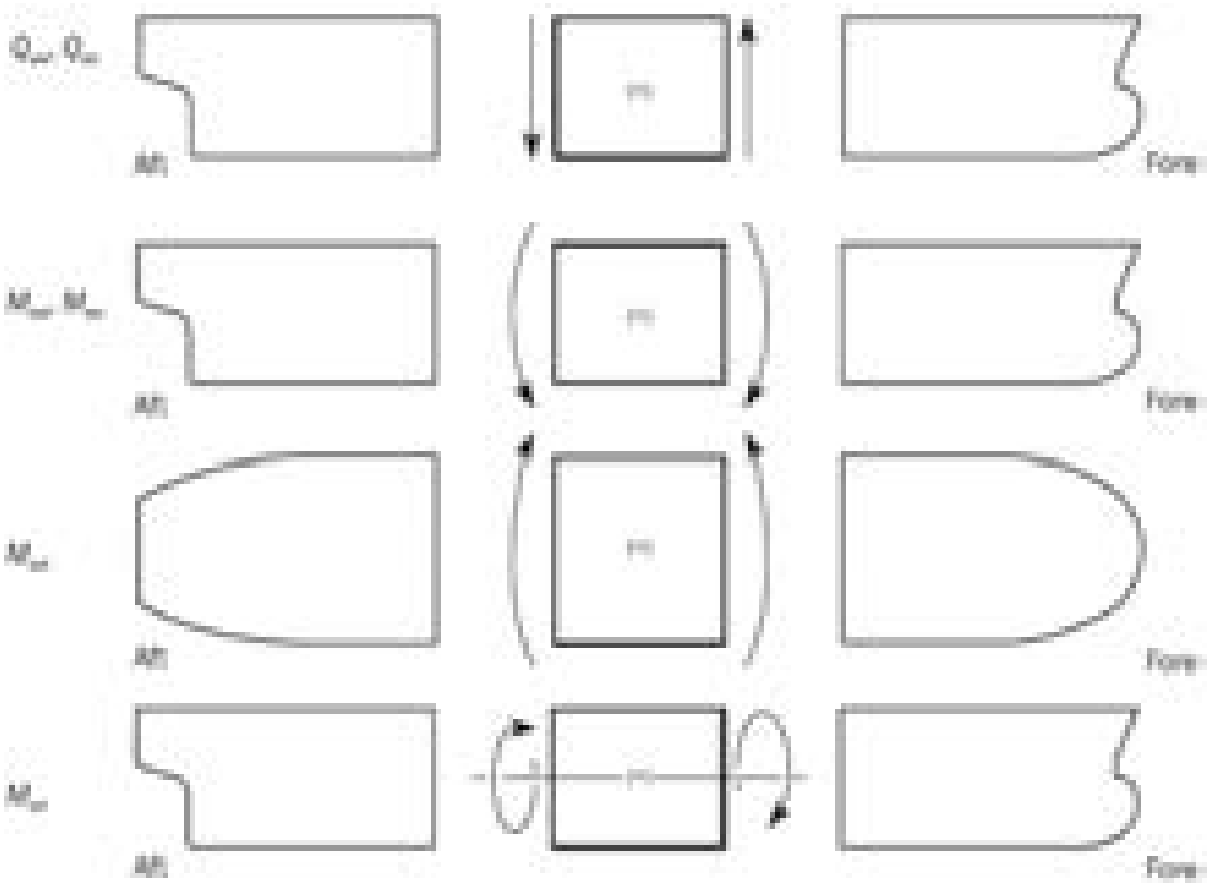


Figure 3.2.1: Sign conventions for shear force $\mathbf{Q_{sw}}, \mathbf{Q_{wv}}$ and bending moment $\mathbf{M_{sw}}, \mathbf{M_{wv}}, \mathbf{M_{wh}}, \mathbf{M_{wt}}$

3.3 Still water hull girder loads

3.3.1 General

3.3.1.1 Seagoing and harbour/sheltered water conditions

Permissible still water bending moments and shear forces for seagoing operations to be provided for ships with L>65[m], (Refer Cl.3.4.2, Cl.3.4.3, Cl.3.5.1 and Cl.3.5.2) and may upon consideration also be requested for smaller ships.

For ships with L≤65[m], still water bending moment in seagoing condition defined in Cl.3.4.1 is to be taken as the rule minimum value.

The permissible still water hull girder loads to be given at points of local maxima for the design loading conditions. For typical cargo vessels permissible values at the following points to be provided:

- at each transverse bulkhead in the cargo area
- at the middle of cargo compartments
- at the collision bulkhead
- at the engine room forward bulkhead
- at the mid-point between the forward and aft engine room bulkheads.

The permissible hull girder bending moments and shear forces at any other position may be obtained by linear interpolation.

3.3.1.2 Flooded condition

The designer is to provide the envelope of permissible still water bending moment and shear force in flooded condition.

3.3.2 Still water loads for the fatigue assessment

The still water bending moment and shear force values and distribution to be used for the fatigue assessment are to be taken as the most typical values applicable for the loading conditions that the ship will operate in for most of its life. The

definition of loading conditions to be used for specific ship types, refer Part 5.

3.4 Vertical still water bending moment

3.4.1 Minimum still water bending moment in seagoing condition

The minimum still water bending moment, $M_{sw-h-min}$ and $M_{sw-s-min}$, in [kN-m], in hogging and sagging condition, respectively in seagoing condition may be taken as:

Hogging conditions:

$$M_{sw-h-min} = f_{sw}((171C_wL^2B(C_b + 0.7)10^{-3}) - M_{wv-h-mid})$$

Sagging conditions:

$$M_{sw-s-min} = -0.85\, f_{sw}((171C_wL^2B(C_b + 0.7)10^{-3}) + M_{wv-s-mid})$$

where:

$M_{wv-h-mid}$ = Vertical wave bending moment for strength assessment in hogging condition, as defined in 3.6 using f_p and f_m equal to 1.0.

$M_{wv-s-mid}$ = Vertical wave bending moment for strength assessment in sagging condition, as defined in 3.6 using f_p and f_m equal to 1.0.

f_{sw} = Distribution factor along the ship length. To be taken as, Refer Figure 3.4.1:

= 0.0 for x ≤ 0

= 0.15 at x = 0.1 L

= 1.0 for 0.3 L ≤ x ≤ 0.7 L

= 0.15 at x = 0.9 L

= 0.0 for x ≥ 1L

Intermediate values of f_{sw} are to be obtained by linear interpolation.

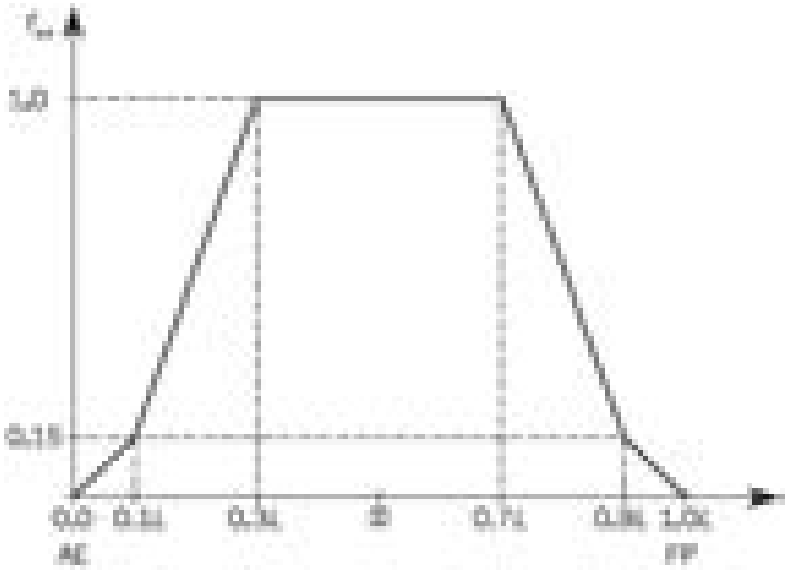


Figure 3.4.1: Distribution factor f_{sw}

3.4.2 Permissible vertical still water bending moments for sea going conditions

The permissible vertical still water bending moments, M_{sw-h} and M_{sw-s} in [kN-m], for hogging and sagging respectively, in seagoing condition at any longitudinal position are to envelop:

- The most severe still water bending moments calculated, in hogging and sagging conditions, respectively, for the seagoing loading conditions defined in Section 7.
- The most severe still water bending moments for the seagoing loading conditions defined in the loading manual.
- The minimum still water bending moment defined in Cl.3.4.1.

3.4.3 Permissible vertical still water bending moments for harbour and tank testing condition

The permissible vertical still water bending moments, $M_{sw-har-h}$ and $M_{sw-har-s}$ in [kN-m] for hogging and sagging respectively, in the

harbour r and tank testing condition at any longitudinal position are to envelop:

- The most severe still water bending moments, in hogging and sagging conditions, respectively, for the harbour loading conditions defined in Section 7

- The most severe still water bending moments for the harbour loading conditions defined in the loading manual.

- The permissible still water bending moment defined in Cl.3.4.2.

- The minimum still water bending moment defined in Cl.3.4.1 increased by 25%.

3.5 Vertical still water shear force

3.5.1 Permissible still water shear force in sea going conditions

The positive and negative permissible vertical still water shear forces, Q_{sw-pos} and Q_{sw-neg} in [kN], in seagoing condition at any longitudinal position are to envelop:

- The most severe still water shear forces, positive or negative, for the seagoing loading conditions defined in Section 7.

- The most severe still water shear forces for the seagoing loading conditions defined in the loading manual.

3.5.2 Permissible still water shear force in harbour and tank testing condition

The positive and negative permissible vertical still water shear forces, Q_{sw-har} in [kN], in the harbour and tank testing condition at any longitudinal position are to envelop:

- The most severe still water shear forces, positive or negative, for the seagoing loading conditions defined in Section 7

- The most severe still water shear forces for the harbour loading conditions defined in the loading manual

- The permissible still water shear forces defined in Cl.3.5.1.

3.6 Dynamic hull girder loads

3.6.1 Vertical wave bending moment

3.6.1.1 The vertical wave bending moments at any longitudinal position, in [kN-m], is to be taken as:

Hogging condition:

$$M_{wv-h} = +0.19\, f_m\, f_p\, f_{nl-vh}\, C_w L^2\, B\, C_b$$

Sagging condition:

$$M_{wv-s} = -0.19\, f_m\, f_p\, f_{nl-vs}\, C_w L^2\, B\, C_b$$

where:

f_m = Distribution factor for strength assessment for vertical wave bending moment along the ship's length, to be taken as:

= 0.0 for x ≤ 0

= 1.0 for 0.4 L ≤ x ≤ 0.65 L

= 0.0 for x ≥ L

Intermediate values of f_m are to be obtained by linear interpolation, Refer Figure 3.6.1(a1).

f_{nl-vh} = coefficient considering non-linear effects applied to hogging, is to be taken as:

= 1.0 for strength and fatigue assessment

f_{nl-vs} = coefficient considering non-linear effects applied to sagging, is to be taken as:

$0.58\frac{C_b + 0.7}{C_b}$ for strength assessment

= 1 for fatigue assessment

f_p = coefficient to be taken as

$f_p = f_{ps}$ for strength assessment

= $0.9(0.27 - (6 + 4f_T)L.10^{-5})$ for fatigue assessment

C_b = Not to be taken as less than 0.6

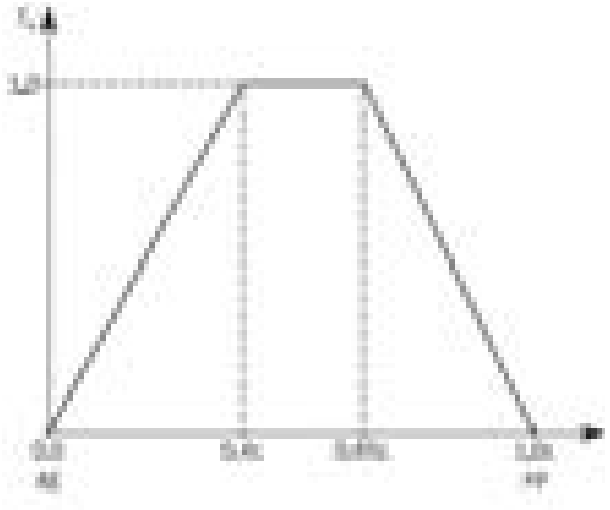


Figure 3.6.1(a1): Distribution factor f_m

3.6.1.2 However, for ships with high speed and/or large flare in the forebody, the value of f_m is to be modified as per Table 3.6.1 based on C_{AV} and C_{AF} given below, Refer Figure 3.6.1(b):

$$C_{AV} = \frac{C_v . V}{\sqrt{L}}$$

$$C_{AF} = \frac{C_v . V}{\sqrt{L}} + \frac{A_{dk} - A_{wp}}{L\, Z_{fp}}$$

$$C_v = \frac{\sqrt{L}}{50} \text{ for } L < 100[m]$$

$$= 0.2 \text{ for } L \geq 100[m]$$

A_{dk} = Projected area in horizontal plane of uppermost deck (including forecastle deck if any), forward of 0.2L from F.P. in [m^2]

A_{wp} = Area of waterplane at draught T_{SC}, forward of 0.2L abaft of F.P. in [m^2]

Z_{fp} = Vertical distance, in [m], measured at F.P., from the waterline at draught T_{SC}, to the uppermost deck (or forecastle deck).

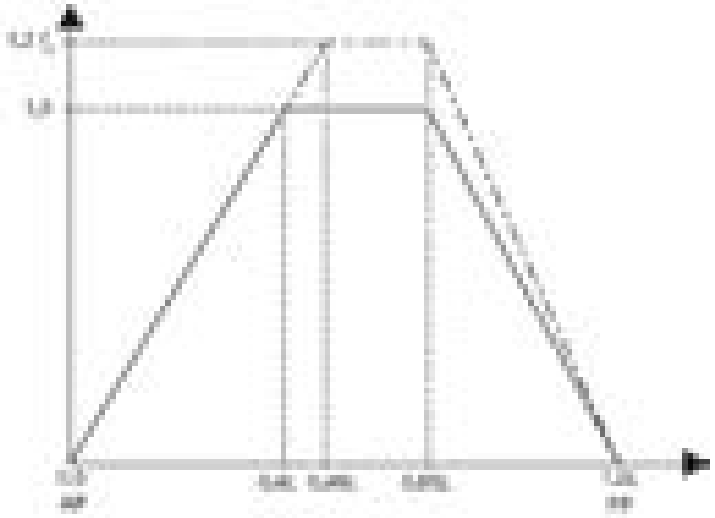


Figure 3.6.1(b): Distribution factor f_m with and without adjustment

Table 3.6.1: Adjustments to factor f_m values for high speed/flare				
	$C_{AV} \leq 0.28$		**$C_{AV} \geq 0.32$**	
	Load condition	**Curve**	**Load condition**	**Curve**
$C_{AF} \leq 0.4$	Hogging	No adjustments	Hogging	1.2 between 0.48L and 0.65L from AP. 0.0 at FP. and AP.
	Sagging	No adjustments	Sagging	No adjustments
$C_{AF} \geq 0.5$	Hogging	No adjustments	Hogging	No adjustments
	Sagging	1.2 between 0.48L and 0.65L from AP. 0.0 at FP. and AP.	Sagging	1.2 between 0.48L and 0.65L from AP. 0.0 at FP. and AP.
For intermediate values of C_{AV} and/or C_{AF}, values of f_m are to be obtained by linear interpolation.				

The adjustment is limited to the control for buckling as given in Chapter 8.

3.6.2 Vertical wave shear force

3.6.2.1 The rule vertical wave shear force Q_{wv} , in [kN] at any section along the length of the ship is to be taken as:

$$Q_{wv-pos} = 0.52\, f_{q-pos}\, f_p\, C_w L\, B\, C_b$$

$$Q_{wv-neg} = -(0.52\, f_{q-neg}\, f_p\, C_w L\, B\, C_b)$$

where,

f_{q-pos} = Distribution factor along the ship length for positive wave shear force, to be taken as:

= 0.0 for x ≤ 0

= 0.92 f_{nl-vh} for 0.2 L ≤ x ≤ 0.3 L

= 0.7 for 0.4 L ≤ x ≤ 0.6 L

= 1.0 f_{nl-vs} for 0.7 L ≤ x ≤ 0.85 L

= 0.0 for x ≥ L

Intermediate values of f_{q-pos} are to be obtained by linear interpolation, Refer Figure 3.6.2(a1).

f_{q-neg} = Distribution factor along the ship length for negative wave shear force, to be taken as:

= 0.0 for x ≤ 0

= 0.92 f_{nl-vs} for 0.2 L ≤ x ≤ 0.3 L

= 0.7 for 0.4 L ≤ x ≤ 0.6 L

= 1.0 f_{nl-vh} for 0.7 L ≤ x ≤ 0.85 L

= 0.0 for x ≥ L

Intermediate values of f_{q-neg} are to be obtained by linear interpolation, Refer Figure 3.6.2(a2).

f_{nl-vh}, f_{nl-vs} = Coefficient considering nonlinear effects defined in Cl.3.6.1.1.

f_p = coefficient to be taken as

= $f_p = f_{ps}$ for strength assessment

= $0.9.(0.27 - (17 - 8f_T)L.10^{-5})$ for fatigue assessment

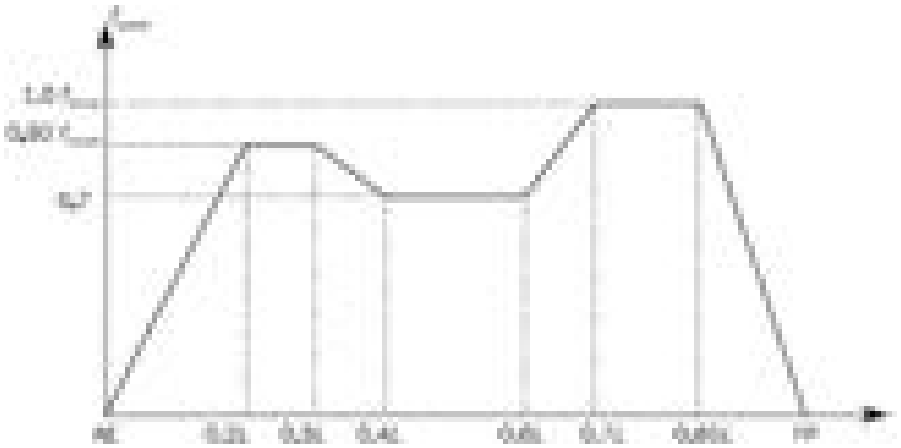


Figure 3.6.2(a1): Distribution factor of positive vertical shear force f_{q-pos}

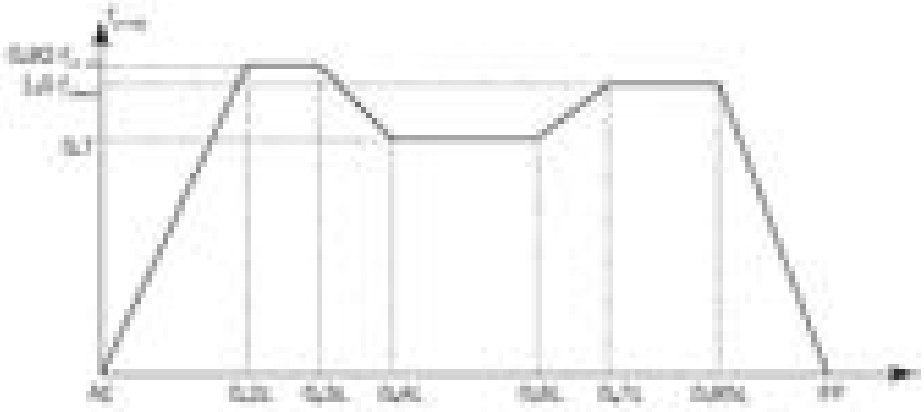


Figure 3.6.2(a2): Distribution factor of positive vertical shear force $\mathbf{f_{q-neg}}$

However, for ships with high speed and/or large flare in the forebody, the adjustments to $\mathbf{f_{q-pos}}$ and $\mathbf{f_{q-neg}}$ as given in Cl.3.6.2.2 apply for hull girder buckling strength check in accordance with Chapter 9, Section 3.

3.6.2.2 If required by Cl.3.6.2.1 $\mathbf{f_{q-pos}}$ and $\mathbf{f_{q-neg}}$ is to be adjusted according to Table 3.6.2. Refer Figure 3.6.2(b1) and Figure 3.6.2(b2).

Table 3.6.2: Adjustments to $\mathbf{f_{q-pos}}$ and $\mathbf{f_{q-neg}}$ for high speed/flare

	$\mathbf{C_{AV} \leq 0.28}$		$\mathbf{C_{AV} \geq 0.32}$	
	Load condition	**Curve**	**Load condition**	**Curve**
$\mathbf{C_{AF} \leq 0.4}$	Hogging	No adjustments	Hogging	1.2 between 0.7L and 0.85L from AP. 0.0 at FP. and AP.
	Sagging	No adjustments	Sagging	No adjustments
$\mathbf{C_{AF} \geq 0.5}$	Hogging	No adjustments	Hogging	No adjustments
	Sagging	1.2 between 0.7L and 0.85L from AP. 0.0 at FP. and AP.	Sagging	1.2 between 0.7L and 0.85L from AP. 0.0 at FP. and AP.
For intermediate values of $\mathbf{C_{AV}}$ and/or $\mathbf{C_{AF}}$, values of f_q are to be obtained by linear interpolation.				

$\mathbf{C_{AV}}$ **=** as defined in Cl.3.6.1

$\mathbf{C_{AF}}$ = as defined in Cl.3.6.1

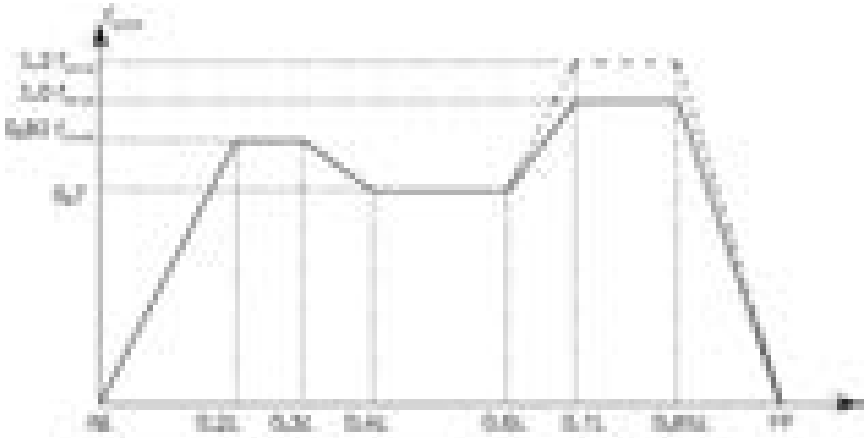


Figure 3.6.2(b1): Distribution factor of positive vertical shear force $\mathbf{f_{q-pos}}$, with and without adjustment

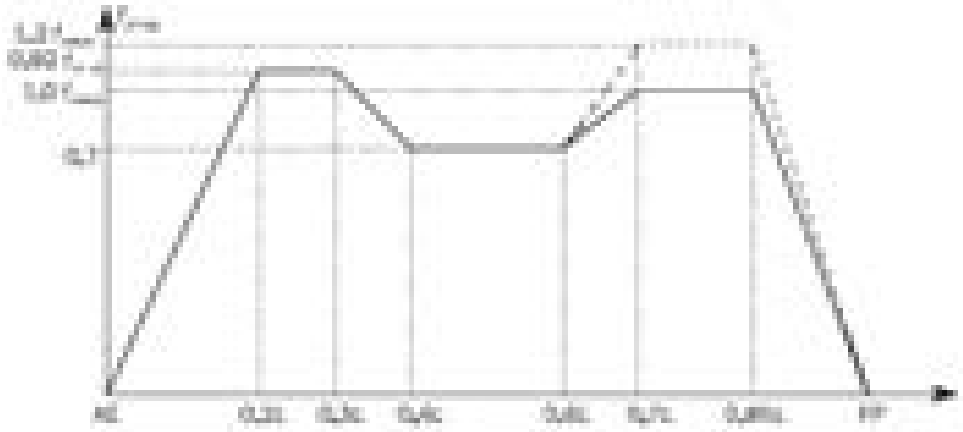


Figure 3.6.2(b2): Distribution factor of positive vertical shear force f_{q-neg}, with and without adjustment

3.6.3 Horizontal wave bending moment

The horizontal wave bending moment at any longitudinal position, in [kN-m], is to be taken as:

$$M_{wh} = \left(0.31 + \frac{L}{2800}\right) f_{nl-h}\, f_m\, f_p C_w\, L^2\, T_{LC}\, C_b$$

where,

f_m = Distribution factor defined in Cl.3.6.1.1.

f_{nl-h} = Coefficient considering nonlinear effect to be taken as:

= 0.9 for strength assessment

= 1.0 for fatigue assessment

f_p = coefficient to be taken as

$f_p = f_{ps}$ for strength assessment

$= 0.9.[(0.2 + 0.04f_T) + (11 - 8f_T)L.10^{-5}]$ for fatigue assessment

3.6.4 Wave torsional moment

3.6.4.1 The wave torsional moment at any longitudinal position with respect to the ship baseline, in [kN-m], is to be taken as:

$$M_{WT} = f_p(M_{wti} + M_{wt2})$$

where,

$$M_{wt1} = 0.4\, f_{t1} C_w \sqrt{\frac{L}{T_c}} B^2\, D\, C_b$$

$$M_{wt2} = 0.22\, f_{t2} C_w\, L\, B^2\, C_b$$

f_{t1}, f_{t2} = Distribution factors, taken as: -

f_{t1} = 0 for x < 0

$= \left|\sin\left(\frac{2\pi x}{L}\right)\right|$ for 0 ≤ x ≤ L

= 0 for x > L

f_{t2} = 0 for x < 0

$= \sin^2\left(\frac{\pi x}{L}\right)$ for 0 ≤ x ≤ L

= 0 for x > L

f_p = coefficient to be taken as

$= f_{ps}$ for strength assessment

$=0.9.(0.2 + (5f_T - 4.25)B.10^{-4})$

for fatigue assessment

3.6.4.2 The factors for the longitudinal distribution of the torsional moment for the OST load case is to be taken as follows:

f_{lp-OST} = 5 f_{xl} for x/L < 0.2

f_{lp-OST} = 1.0 for 0.2 ≤ x/L < 0.4

f_{lp-OST} = -7.6 f_{xl} +4.04 for 0.4 ≤ x/L < 0.65

f_{lp-OST} = -0.9 for 0.65 ≤ x/L < 0.85

f_{lp-OST} = 6 f_{xl} - 6.0 for 0.85 ≤ x/L

3.6.4.3 The factors for the longitudinal distribution of the torsional moment for the OSA load case is to be taken as follows:

f_{lp-OSA} = - (0.2+0.3 f_T) for x/L < 0.4

f_{lp-OSA} = - (0.2+0.3 f_T)$(5.6 - 11.5f_{xl})$ for 0.4 ≤ x/L < 0.6

f_{lp-OSA} = 1.3(0.2+0.3 f_T) for 0.6 ≤ x/L

3.7 Hull girder loads for dynamic load cases for L ≥ 90[m].

3.7.1 General

The dynamic hull girder loads to be applied for the dynamic load cases defined in Section 2, 2.3, are given in Cl.3.7.2 to Cl.3.7.5.

3.7.2 Vertical wave bending moment

The vertical wave bending moment, M_{wv-LC}, in [kN-m], to be used for each dynamic load case defined in Section 2, 2.3, is defined in Table 3.7.2.

Table 3.7.2: Vertical wave bending moment for dynamic load cases

Load combination factor	$\mathbf{M_{wv-LC}}$
$C_{WV} \geq 0$	$f_\beta\, C_{WV}\, M_{wv-h}$
$C_{WV} < 0$	$f_\beta\, C_{WV}\, \lvert M_{wv-s}\rvert$

where:

C_{WV} = Load combination factor for vertical wave bending moment, to be taken as specified in Section 2.

M_{wv-h}, M_{wv-s} = Hogging and sagging vertical wave bending moment taking into account the considered design load scenario, as defined in Cl.3.6.1.

3.7.3 Vertical wave shear force

The vertical wave shear force, Q_{wv-LC} , in [kN], to be used for each dynamic load case defined in Section 2, 2.3 is defined in Table 3.7.3.

Table 3.7.3: Vertical wave shear force for dynamic load cases

Load combination factor	$\mathbf{Q_{wv-LC}}$
$C_{QW} \geq 0$	$f_\beta\, C_{QW}\, Q_{wv-pos}$
$C_{QW} < 0$	$f_\beta\, C_{QW} \lvert Q_{wv-neg}\rvert$

where:

C_{QW} = Load combination factor for vertical wave shear force, to be taken as specified in Section 2.

M_{wv-pos}, M_{wv-neg} = Positive and negative vertical wave shear force taking into account the considered design load scenario, as defined in Cl.3.6.2.

3.7.4 Horizontal wave bending moment

The horizontal wave bending moment, M_{wh-LC}, in [kN-m], to be used for each dynamic load case defined in Section 2, 2.3, is to be taken as:

$$M_{wh-LC} = f_\beta\, C_{WH}\, M_{wh}$$

where:

C_{WH} = Load combination factor for horizontal wave bending moment, to be taken as specified in Section 2.

M_{wh} = Horizontal wave bending moment taking account of the appropriate design load scenario, as defined in Cl.3.6.3.

3.7.5 Wave torsional moment

The wave torsional moment, M_{wt-LC}, in [kN-m], to be used for each dynamic load case defined in Section 2, 2.3, is to be taken as:

$$M_{wt-LC} = f_\beta\, C_{WT}\, M_{wt}$$

where:

C_{WT} = Load combination factor for wave torsional moment, to be taken as specified in Section 2.

M_{wt} = Wave torsional moment taking account of the appropriate design load scenario, as defined in Cl.3.6.4.

Section 4

External Loads

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

λ = Wave length, in [m]

B_x = Moulded breadth at the waterline, in [m], at the considered cross section.

f_{yB} = Ratio between Y-coordinate of the load point and B_x, to be taken as:

$$f_{yB} = \left|\frac{2y}{B_x}\right| \text{ but not greater than } 1.0$$

= 1, when $B_x = 0$

f_{yB1} = Ratio between Y-coordinate of the load point and B, to be taken as:

$$f_{yB1} = \left|\frac{2y}{B}\right| \text{ but not greater than } 1.0$$

f_{xl} = Ratio between X-coordinate of the load point and L, to be taken as:

$$f_{xl} = \frac{x}{L}$$

but not be taken less than 0.0 or greater than 1.0

C_w = Wave coefficient defined in Section 2

f_T = Ratio as defined in Section 2

$P_{W,WL}$ = Wave pressure at the waterline, [kN/m^2], for the considered dynamic load case

$$P_{W,WL} = P_W$$

for $z = T_{LC}$, $\quad y = B_x/2$ when $y \geq 0$,

$$y = -B_x/2 \text{ when } y < 0$$

h_W = Water head equivalent to the pressure at waterline, in [m], to be taken as:

$$h_w = \frac{P_{W,WL}}{\rho . g}$$

f_β = heading correction factor as defined in Section 2.

4.1 Sea pressure

4.1.1 Total pressure

4.1.1.1 The external pressure, P_{ex} at any load point of the hull, in [kN/m^2], for the static (S) design load scenarios, is to be taken as:

$$P_{ex} = P_s \text{ but not less than 0.}$$

4.1.1.2 The total pressure P_{ex} at any load point of the hull for the static plus dynamic (S+D) design load scenario, is to be derived from each dynamic load case and is to be taken as:

$$P_{ex} = P_s + P_w$$

where:

P_s = Hydrostatic pressure, in [kN/m^2], defined in Cl.4.1.2.

P_w = Wave pressure, in [kN/m^2], is defined in Cl.4.1.3.2 to Cl.4.1.3.8 for L ≥ 90[m] and Cl.4.1.3.10 for L < 90[m].

4.1.2 Hydrostatic pressure

The hydrostatic pressure, P_s at any load point, in [kN/m^2], is obtained from Table 4.1.2. Refer also Figure 4.1.2.

Table 4.1.2: Hydrostatic pressure, P_s

Location	Hydrostatic pressure, P_s in [kN/m^2]
$z \leq T_{LC}$	$\rho\, g(T_{LC} - z)$
$z > T_{LC}$	0

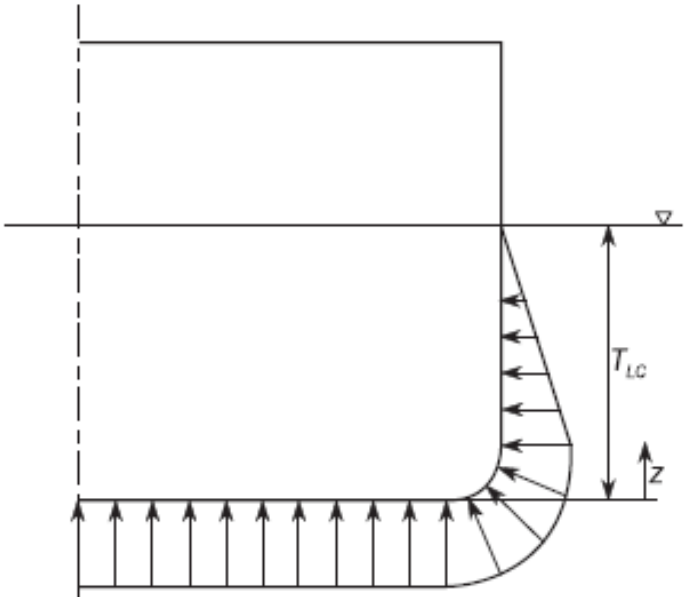


Figure 4.1.2: Hydrostatic pressure, P_s

4.1.3 External dynamic pressures for strength assessment

4.1.3.1 General

The hydrodynamic pressures for each dynamic load case defined in Section 2, 2.3 are defined in Cl.4.1.3.2 to Cl.4.1.3.8.

For ships with L ≥ 90[m], the wave pressure at any point, P_{w-max}, is to be taken as the greatest pressure obtained from any of the load cases determined by Cl.4.1.3.2 to Cl.4.1.3.8.

4.1.3.2 Hydrodynamic pressures for HSM load cases

The hydrodynamic pressures, P_W, for HSM-1 and HSM-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.3(a). Refer Figure 4.1.3(b1) and Figure 4.1.3(b2).

Table 4.1.3(a): Hydrodynamic pressures for HSM load cases

	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < z \leq h_W + T_{LC}}$	$\mathbf{z > h_W + T_{LC}}$
HSM-1	$P_W = \max(-P_{HS}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
HSM-2	$P_W = \max(P_{HS}, \rho.g(z - T_{LC}))$		

where:

$$P_{HS} = f_\beta f_{ps} f_{nl} f_h k_a k_p f_{yz} C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

f_{nl} = Coefficient considering non-linear effects, to be taken as:

For extreme sea loads design load scenario:

= 0.7 at $f_{xl} = 0$

= 0.9 at $f_{xl} = 0.3$

= 0.9 at $f_{xl} = 0.7$

= 0.6 at $f_{xl} = 1.0$

For ballast water exchange design load scenario:

= 0.85 at $f_{xl} = 0$

= 0.95 at $f_{xl} = 0.3$

= 0.95 at $f_{xl} = 0.7$

= 0.80 at $f_{xl} = 1.0$

Intermediate values are obtained by linear interpolation.

f_{yz} = Girth distribution coefficient, to be taken as:

$$f_{yz} = \frac{Z}{T_{LC}} + f_{yB} + 1$$

f_h = Coefficient to be taken as:

$$f_h = 3.0(1.21 - 0.66f_T)$$

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be taken as:

$$k_a = (0.5 + f_T)\left[\left(3 - 2\sqrt{f_{yB}}\right) - \frac{20}{9}f_{xl}\left(7 - 6\sqrt{f_{yB}}\right)\right] + \frac{2}{3}(1 - f_T) \text{ for } f_{xl} < 0.15$$

$$k_a = 1.0 \text{ for } 0.15 \leq f_{xl} < 0.7$$

$$k_a = 1 + (f_{xl} - 0.7)\left\{\left(\frac{40}{3}f_T - 5\right) + 2(1 - f_{yB})\left[\frac{18}{C_b}f_T(f_{xl} - 0.7) - 0.25(2 - f_T)\right]\right\} \text{ for } f_{xl} \geq 0.7$$

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.6L(1 + f_T)$$

k_p = Phase coefficient to be obtained from Table 4.1.3(b). Intermediate values are to be interpolated.

Table 4.1.3(b): Definition of phase coefficient k_p

f_{xl}	0	0.3-0.1f_T	0.35-0.1f_T	0.8-0.2f_T	0.9-0.2f_T	1.0
k_p	$-0.25f_T(1 + f_{yB})$	-1	1	1	-1	-1

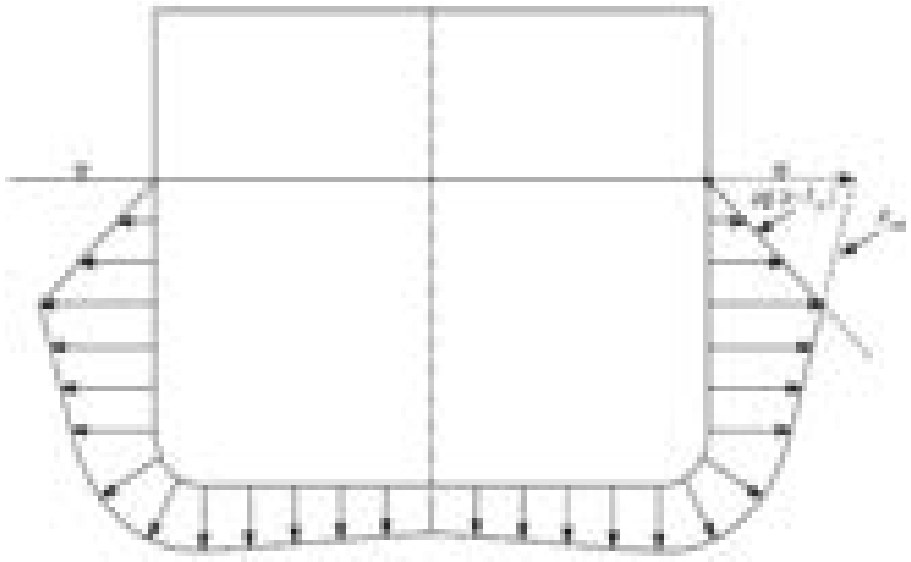


Figure 4.1.3(b1): Transverse distribution amidships of dynamic pressure for HSM-1, HSA-1 and FSM-1 load cases

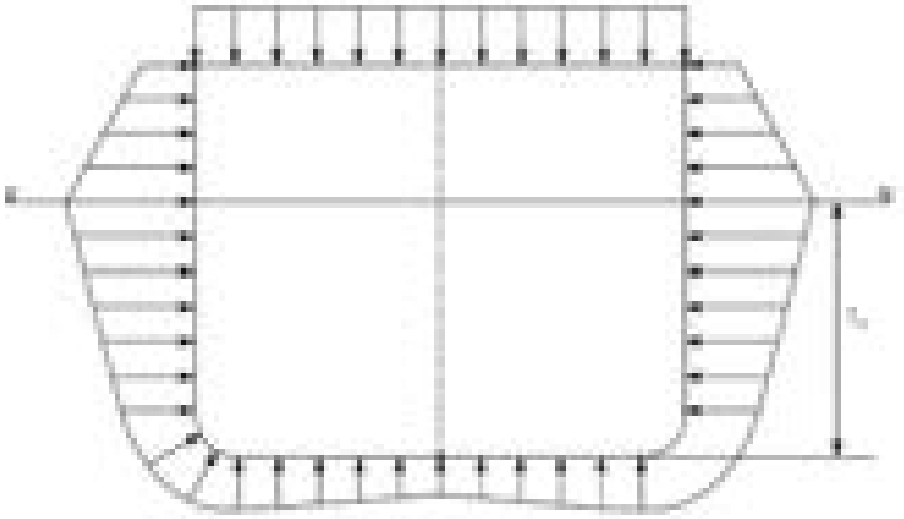


Figure 4.1.3(b2): Transverse distribution amidships of dynamic pressure for HSM-2, HSA-2 and FSM-2 load cases

4.1.3.3 Hydrodynamic pressures for HSA load cases

The hydrodynamic pressures, P_W, for HSA-1 and HSA-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.3(c). Refer Figure 4.1.3(b1) and Figure 4.1.3(b2).

Table 4.1.3(c): Hydrodynamic pressures for HSA load cases

	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < z \leq h_W + T_{LC}}$	$\mathbf{z > h_W + T_{LC}}$
HSA-1	$P_W = \max(-P_{HS}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
HSA-2	$P_W = \max(P_{HS}, \rho.g(z - T_{LC}))$		

where:

P_{HS} = As defined in Cl.4.1.3.2

f_{nl} = As defined in Cl.4.1.3.2

f_{yz} = As defined in Cl.4.1.3.2

f_h = Coefficient to be taken as:

$f_h = 2.4(1.21 - 0.66f_T)$

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be taken as defined in Cl.4.1.3.2

λ = Wave length of the dynamic load case, in m, as defined in Cl.4.1.3.2

k_p = Phase coefficient to be obtained from Table 4.1.3(c1). Intermediate values are to be interpolated.

Table 4.1.3(c1): Definition of phase coefficient k_p

f_{xl}	0	0.3-0.1f_T	0.5-0.2f_T	0.8-0.2f_T	0.9-0.2f_T	1.0
k_p	$1.5 - f_T - (0.5f_{yB})$	-1	1	1	-1	-1

4.1.3.4 Hydrodynamic pressures for FSM load cases

The hydrodynamic pressures, P_W, for FSM-1 and FSM-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.3(d). Refer Figure 4.1.3(b1) and Figure 4.1.3(b2).

Table 4.1.3(d): Hydrodynamic pressures for FSM load cases			
	Wave pressure, in [kN/m²]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
FSM-1	$P_W = \max(-P_{FS}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
FSM-2	$P_W = \max(P_{FS}, \rho.g(z - T_{LC}))$		

where:

$$P_{FS} = f_\beta\, f_{ps}\, f_{nl}\, f_h\, k_a\, k_p\, f_{yz}\, C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

f_{nl} = Coefficient considering non-linear effects, to be taken as:

= 0.9 for extreme sea loads design load scenario:

= 0.95 for ballast water exchange design load scenario:

f_{yz} = Girth distribution coefficient, to be taken as:

$$f_{yz} = \frac{Z}{T_{LC}} + f_{yB} + 1$$

f_h = Coefficient to be taken as, $f_h = 2.6$

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be taken as:

$$k_a = 1 + (3.75 - 2f_T)(1 - 5f_{xl})(1 - f_{yB}) \text{ for } f_{xl} < 0.2$$

$$k_a = 1.0 \text{ for } 0.2 \leq f_{xl} < 0.9$$

$$k_a = 1 + 20\,(1 - f_{yB})(f_{xl} - 0.9) \text{ for } f_{xl} \geq 0.9$$

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.6L(1 + 2/3\, f_T)$$

k_p= Phase coefficient to be obtained from Table 4.1.3(d1). Intermediate values are to be interpolated.

Table 4.1.3(d1): Definition of phase coefficient k_p						
f_{xl}	0	$0.35\text{-}0.1f_T$	$0.5\text{-}0.2f_T$	0.75	0.8	1.0
k_p	$-0.75\text{-}0.25f_{yB}$	-1	1	1	-1	$-0.75\text{-}0.25f_{yB}$

4.1.3.5 Hydrodynamic pressures for BSR load cases

The hydrodynamic pressures, P_W, for BSR-1 and BSR-2 load cases, at any load point, in [kN/m²], are to be obtained from Table 4.1.3(e). Refer Figure 4.1.3(e1) and Figure 4.1.3(e2).

Table 4.1.3(e): Hydrodynamic pressures for BSR load cases			
	Wave pressure, in [kN/m²]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
BSR-1P	$P_W = \max(P_{BSR}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
BSR-2P	$P_W = \max(-P_{BSR}, \rho.g(z - T_{LC}))$		
BSR-1S	$P_W = \max(P_{BSR}, \rho.g(z - T_{LC}))$		
BSR-2S	$P_W = \max(-P_{BSR}, \rho.g(z - T_{LC}))$		

where:

For BSR-1P and BSR-2P load cases.

$$P_{BSR} = f_{\beta}\, f_{nl} \left[10y \sin \emptyset + 0.88 f_{ps} C_w \sqrt{\frac{L_0 + \lambda - 125}{L}} \left(f_{yB1} + 1 \right) \right]$$

For BSR-1S and BSR-2S load cases.

$$P_{BSR} = f_{\beta}\, f_{nl} \left[-10y \sin \emptyset + 0.88 f_{ps} C_w \sqrt{\frac{L_0 + \lambda - 125}{L}} \left(f_{yB1} + 1 \right) \right]$$

f_{nl} = Coefficient considering non-linear effects, to be taken as:

= 1.0 for extreme sea loads design load scenario

= 1.0 for ballast water exchange design load scenario

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = \frac{g}{2\pi} T_R^2$$

where,

P_D = Green sea pressure on exposed deck as defined in 4.2.

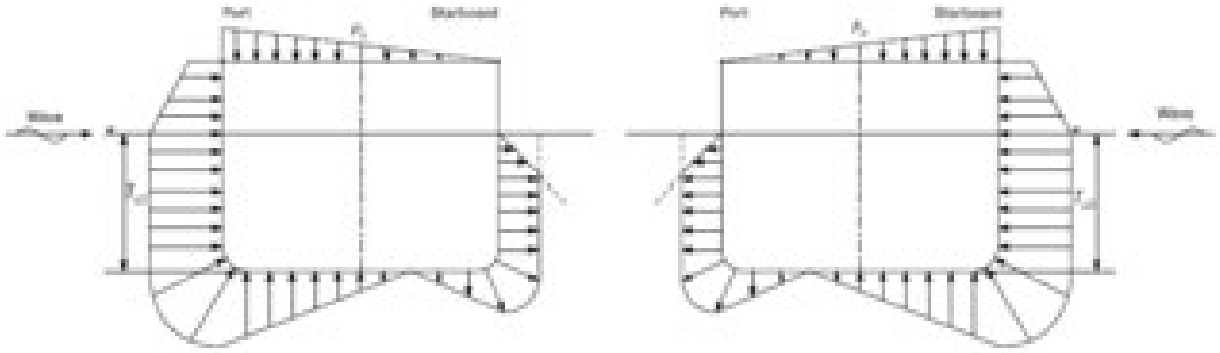


Figure 4.1.3(e1): Transverse distribution of dynamic pressure for BSR-1P (left) and BSR-1S (right) load cases

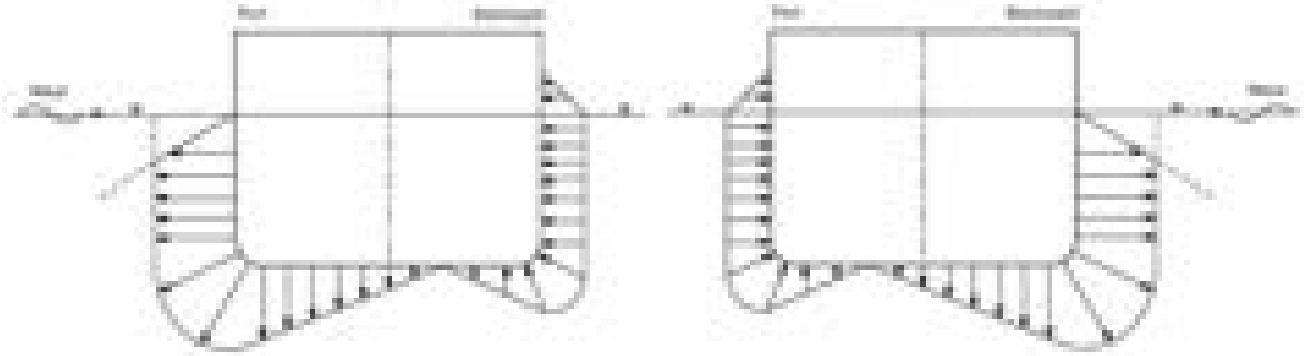


Figure 4.1.3(e2): Transverse distribution of dynamic pressure for BSR-2P (left) and BSR-2S (right) load cases

4.1.3.6 Hydrodynamic pressures for BSP load cases

The hydrodynamic pressures, P_W, for BSP-1 and BSP-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.3(f). Refer Figure 4.1.3(f1) and Figure 4.1.3(f2).

Table 4.1.3(f): Hydrodynamic pressures for BSP load cases			
	Wave pressure, in [kN/m²]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
BSP-1P	$P_W = \max(P_{BSP}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
BSP-2P	$P_W = \max(-P_{BSP}, \rho.g(z - T_{LC}))$		
BSP-1S	$P_W = \max(P_{BSP}, \rho.g(z - T_{LC}))$		
BSP-2S	$P_W = \max(-P_{BSP}, \rho.g(z - T_{LC}))$		

where:

$$P_{BSP} = 4.5\, f_\beta\, f_{ps}\, f_{nl}\, f_{yz}\, C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.2L(1 + 2f_T)$$

f_{yz} = Girth distribution coefficient, to be taken as:

For BSP-1P and BSP-2P

$$f_{yz} = 2\frac{Z}{T_{LC}} + 2.5\, f_{yB1} + 0.5 \text{ for } y \geq 0$$

$$f_{yz} = 2/3\frac{Z}{T_{LC}} + 1/2\, f_{yB1} + 0.5 \text{ for } y < 0$$

For BSP-1S and BSP-2S

$$f_{yz} = 2/3\frac{Z}{T_{LC}} + 1/2\, f_{yB1} + 0.5 \text{ for } y \geq 0$$

$$f_{yz} = 2\frac{Z}{T_{LC}} + 2.5\, f_{yB1} + 0.5 \text{ for } y < 0$$

f_{nl} = Coefficient considering non-linear effects, to be taken as:

For extreme sea loads design load scenario:

= 0.60 at $f_{xl} = 0$

= 0.80 at $f_{xl} = 0.3$

= 0.80 at $f_{xl} = 0.7$

= 0.60 at $f_{xl} = 1.0$

For ballast water exchange design load scenario:

= 0.60 at $f_{xl} = 0$

= 0.80 at $f_{xl} = 0.3$

= 0.80 at $f_{xl} = 0.7$

= 0.60 at $f_{xl} = 1.0$

Intermediate values are obtained by linear interpolation.

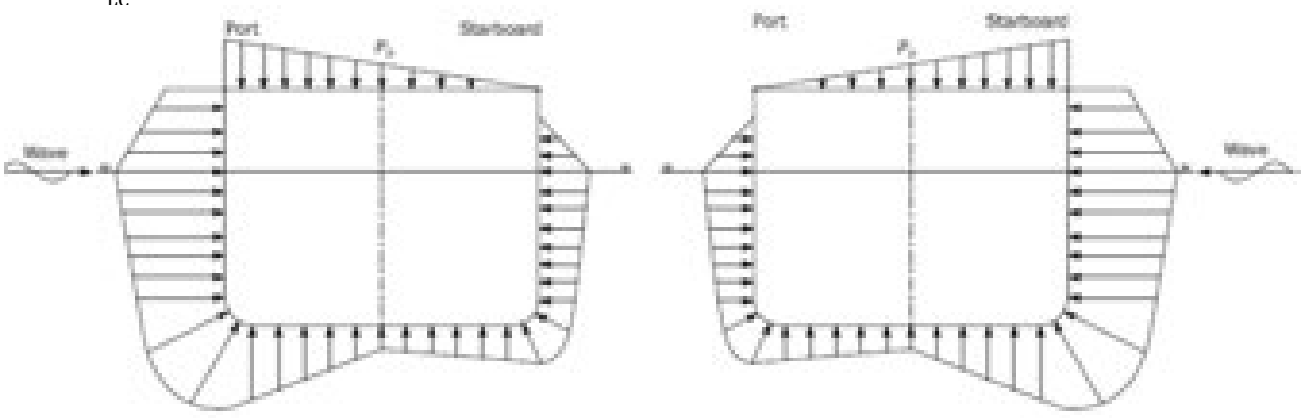


Figure 4.1.3(f1): Transverse distribution of dynamic pressure for BSP-1P (left) and BSP-1S (right) load cases

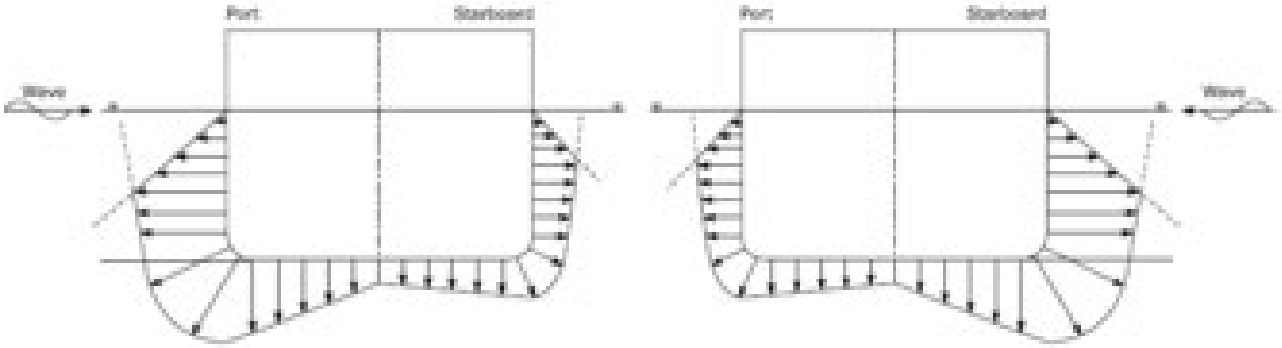


Figure 4.1.3(f2): Transverse distribution of dynamic pressure for BSP-2P (left) and BSP-2S (right) load cases

4.1.3.7 Hydrodynamic pressures for OST load cases

The hydrodynamic pressures, P_W, for OST-1 and OST-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.3(g). Refer Figure 4.1.3(g1) and Figure 4.1.3(g2).

Table 4.1.3(g): Hydrodynamic pressures for OST load cases			
	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
OST-1P	$P_W = \max(P_{OST}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
OST-2P	$P_W = \max(-P_{OST}, \rho.g(z - T_{LC}))$		
OST-1S	$P_W = \max(P_{OST}, \rho.g(z - T_{LC}))$		
OST-2S	$P_W = \max(-P_{OST}, \rho.g(z - T_{LC}))$		

where:

$$P_{OST} = 1.38\, f_{ps}\, f_{nl}\, k_a\, k_p\, f_{yz}\, C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

f_{yz} = Girth distribution coefficient, obtained from Table 4.1.3(g1).

λ = Wave length of the dynamic load case, in [m], to be taken as

$\lambda = 0.45L$

f_{nl} = Coefficient considering non-linear effects, to be taken as:

= 0.8 for extreme sea loads design load scenario

= 0.9 for ballast water exchange design load scenario

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be obtained from Table 4.1.3(g2).

k_p = Phase coefficient to be obtained from Table 4.1.3(g3). Intermediate values are to be interpolated.

Table 4.1.3(g1): Definition of girth distribution coefficient, f_{yz}		
Transverse position	**OST-1P - OST-2P**	**OST-1S - OST-2S**
$y \geq 0$	$5\frac{z}{T_{LC}} + 3.5f_{yB} + 1.5$	$1.5\frac{z}{T_{LC}} + 1.5$
$y < 0$	$1.5\frac{z}{T_{LC}} + 1.5$	$5\frac{z}{T_{LC}} + 3.5f_{yB} + 1.5$

Table 4.1.3(g2): Definition of amplitude coefficient , $\mathbf{k_a}$			
Transverse position	**Longitudinal position**	**OST-1P - OST-2P**	**OST-1S - OST-2S**
$y \geq 0$	$f_{xl} \leq 0.2$	$1.0 + 3.5(1 - f_{yB})(1 - 5f_{xl})$	$1.0 + \left[3.5 - (4f_T - 0.5)f_{yB}\right](1 - 5f_{xl})$
	$0.2 < f_{xl} \leq 0.8$	1.0	1.0
	$f_{xl} > 0.8$	1.0	$1.0 + 4(1 - f_T)(5f_{xl} - 4)f_{yB}$
$y < 0$	$f_{xl} \leq 0.2$	$1.0 + \left[3.5 - (4f_T - 0.5)f_{yB}\right](1 - 5f_{xl})$	$1.0 + 3.5(1 - f_{yB})(1 - 5f_{xl})$
	$0.2 < f_{xl} \leq 0.8$	1.0	1.0
	$f_{xl} > 0.8$	$1.0 + 4(1 - f_T)(5f_{xl} - 4)f_{yB}$	1.0

Table 4.1.3(g3): Definition of phase coefficient, $\mathbf{k_p}$			
Transverse position	$\mathbf{f_{xl}}$	**OST-1P - OST-2P**	**OST-1S - OST-2S**
$y \geq 0$	0.0	1.0	1.0
	0.2	1.0	$1.0 + (0.75 - 1.5f_T)f_{yB}$
	0.4	-1.0	$-1.0 + (1.75 - 0.5f_T)f_{yB}$
	0.5	-1.0	$-1.0 + (1.75 - 0.5f_T)f_{yB}$
	0.7	$-0.1 + (1.6f_T - 1.5)f_{yB}$	$-0.1 + (0.25 - 0.3f_T)f_{yB}$
	0.9	$0.8+0.2f_{yB}$	$0.8 - (0.9f_T + 0.85)f_{yB}$
	1.0	$f_{yB} - 1.0$	$-1.0 + (0.5 - 0.5f_T)f_{yB}$
$y < 0$	0.0	1.0	1.0
	0.2	$1.0 + (0.75 - 1.5f_T)f_{yB}$	1.0
	0.4	$-1.0 + (1.75 - 0.5f_T)f_{yB}$	-1.0
	0.5	$-1.0 + (1.75 - 0.5f_T)f_{yB}$	-1.0
	0.7	$-0.1 + (0.25 - 0.3f_T)f_{yB}$	$-0.1 + (1.6f_T - 1.5)f_{yB}$
	0.9	$0.8 - (0.9f_T + 0.85)f_{yB}$	$0.8+0.2f_{yB}$
	1.0	$-1.0 + (0.5 - 0.5f_T)f_{yB}$	$f_{yB} - 1.0$

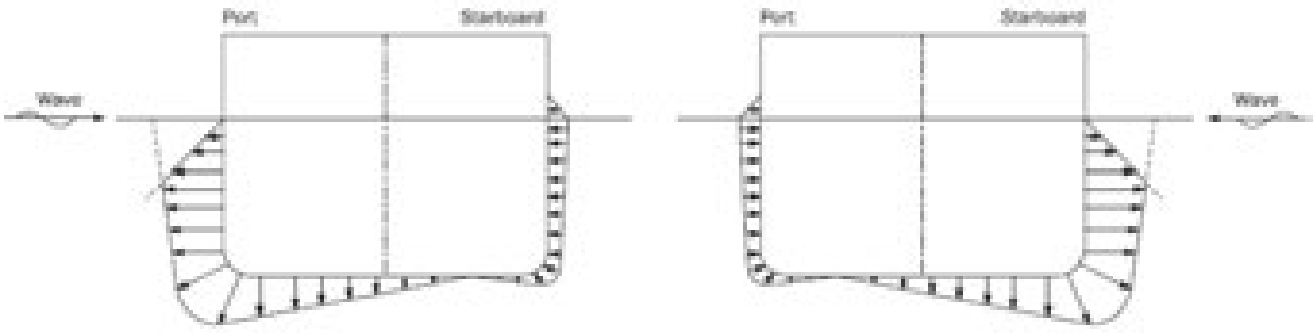


Figure 4.1.3(g1): Transverse distribution of dynamic pressure for OST-1P (left) and OST-1S (right) load cases

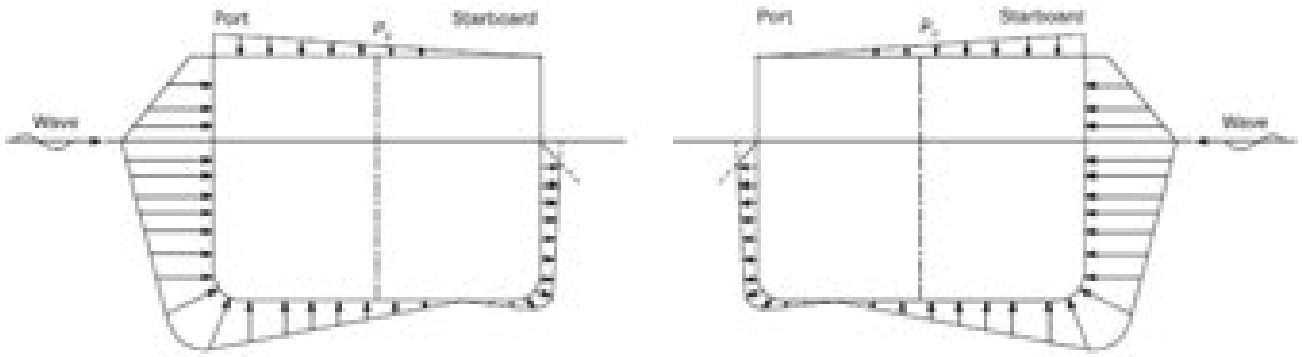


Figure 4.1.3(g2): Transverse distribution of dynamic pressure for OST-2P (left) and OST-2S (right) load cases

4.1.3.8 Hydrodynamic pressures for OSA load cases

The hydrodynamic pressures, P_W, for OSA-1 and OSA-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.3(h). Refer Figure 4.1.3(h1) and Figure 4.1.3(h2).

Table 4.1.3(h): Hydrodynamic pressures for OSA load cases

	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
OSA-1P	$P_W = \max(P_{OSA}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
OSA-2P	$P_W = \max(-P_{OSA}, \rho.g(z - T_{LC}))$		
OSA-1S	$P_W = \max(P_{OSA}, \rho.g(z - T_{LC}))$		
OSA-2S	$P_W = \max(-P_{OSA}, \rho.g(z - T_{LC}))$		

where:

$$P_{OSA} = 0.81\, f_{ps}\, f_{nl}\, k_a\, k_p\, f_{yz}\, C_w \sqrt{\frac{L_0 + \lambda - 125}{L}} (1 + 0.5 f_T)$$

f_{yz} = Girth distribution coefficient, obtained from Table 4.1.3(h1).

λ = Wave length of the dynamic load case, in [m], to be taken as

$\lambda = 0.7L$

f_{nl} = Coefficient considering non-linear effects, to be taken as:

For extreme sea loads design load scenario:

= 0.50 at $f_{xl} = 0$

= 0.80 at $f_{xl} = 0.3$

= 0.80 at $f_{xl} = 0.7$

= 0.60 at $f_{xl} = 1.0$

For ballast water exchange design load scenario:

= 0.75 at $f_{xl} = 0$

= 0.90 at $f_{xl} = 0.3$

= 0.90 at $f_{xl} = 0.7$

= 0.80 at $f_{xl} = 1.0$

Intermediate values are obtained by linear interpolation.

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be obtained from Table 4.1.3(h2).

k_p = Phase coefficient to be obtained from Table 4.1.3(h3). Intermediate values are to be interpolated.

Table 4.1.3(h1): Definition of girth distribution coefficient, f_{yz}		
Transverse position	**OSA-1P - OSA-2P**	**OSA-1S - OSA-2S**
$y \geq 0$	$5.5\frac{z}{T_{LC}} + 5.3f_{yB} + 2.2$	$0.9\frac{z}{T_{LC}} + 0.4f_{yB} + 2.2$
$y < 0$	$0.9\frac{z}{T_{LC}} + 0.4f_{yB} + 2.2$	$5.5\frac{z}{T_{LC}} + 5.3f_{yB} + 2.2$

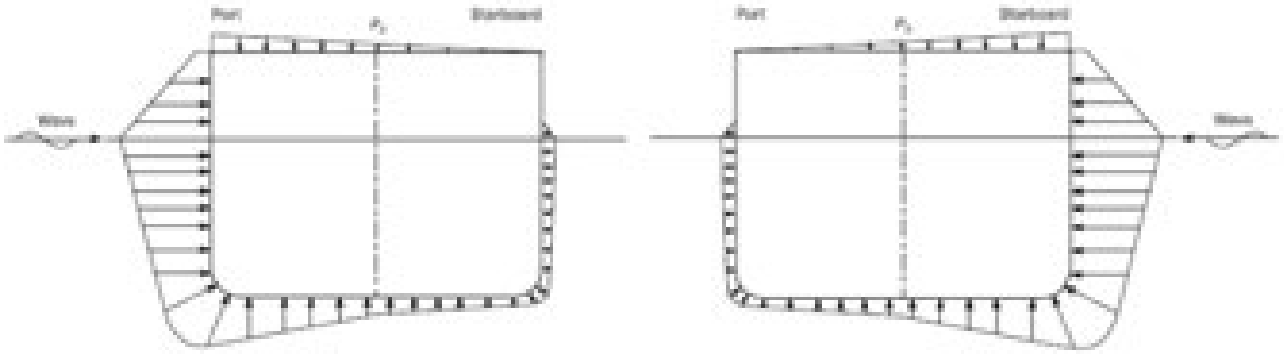


Figure 4.1.3(h1): Transverse distribution of dynamic pressure for OSA-1P (left) and OSA-1S (right) load cases

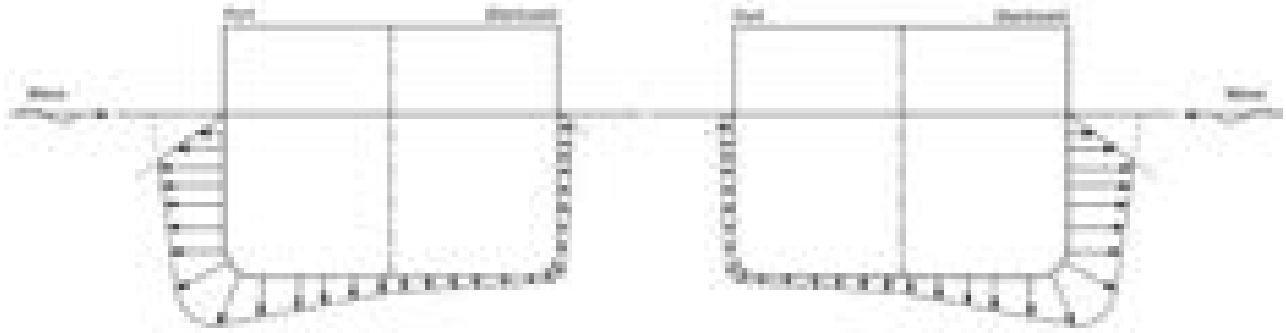


Figure 4.1.3(h2): Transverse distribution of dynamic pressure for OSA-2P (left) and OSA-2S (right) load cases

Table 4.1.3(h2): Definition of amplitude coefficient , k_a			
Transverse position	**f_{xl}**	**OSA-1P - OSA-2P**	**OSA-1S - OSA-2S**
$y \geq 0$	$f_{xl} \leq 0.2$	$1.0 + 3.0(2 - f_T)(1 - 5f_{xl})\big(1 - f_{yB}\big)$	$1.0 + 3.0(2 - f_T)(1 - 5f_{xl}) + \{(28f_{xl} - 5) + 3f_T(1 - 5f_{xl})\}f_{yB}$
	$0.2 < f_{xl} \leq 0.5$	1.0	$1.0 + (1 - 2f_{xl})f_{yB}$
	$0.5 < f_{xl} \leq 0.8$	1.0	$1.0 + 1.5(2f_{xl} - 1)f_{yB}$
	$f_{xl} > 0.8$	$1.0 + (f_{xl} - 0.8)\big(1 - f_{yB}\big)A$	$1.0 + \{1.5(2f_{xl} - 1) - (f_{xl} - 0.8)A\}f_{yB} + (f_{xl} - 0.8)A$
$y < 0$	$f_{xl} \leq 0.2$	$1.0 + 3.0(2 - f_T)(1 - 5f_{xl}) + \{(28f_{xl} - 5) + 3f_T(1 - 5f_{xl})\}f_{yB}$	$1.0 + 3.0(2 - f_T)(1 - 5f_{xl})\big(1 - f_{yB}\big)$
	$0.2 < f_{xl} \leq 0.5$	$1.0 + (1 - 2f_{xl})f_{yB}$	1.0
	$0.5 < f_{xl} \leq 0.8$	$1.0 + 1.5(2f_{xl} - 1)f_{yB}$	1.0
	$f_{xl} > 0.8$	$1.0 + \{1.5(2f_{xl} - 1) - (f_{xl} - 0.8)A\}f_{yB} + (f_{xl} - 0.8)A$	$1.0 + (f_{xl} - 0.8)\big(1 - f_{yB}\big)A$
Where, $A = 22 - 15f_T + 3[22(f_{xl} - 0.8) - 0.25(2 - f_T)]$			

Table 4.1.3(h3): Definition of phase coefficient, k_p			
Transverse position	**f_{xl}**	**OSA-1P - OSA-2P**	**OSA-1S - OSA-2S**
$y \geq 0$	0.0	$0.75 - 0.5\, f_{yB}$	0.75
	0.2	$f_T - 0.25 + (1.25 - f_T)f_{yB}$	$f_T - 0.25 + (0.35f_T - 0.47)f_{yB}$
	0.4	1.0	$1.0 + (2.7f_T - 3.2)f_{yB}$
	0.5	$1.25 - 0.5f_T + (0.5f_T - 0.25)f_{yB}$	$1.25 - 0.5f_T + (2.7f_T - 3.2)f_{yB}$
	0.6	$1.5 - f_T + (f_T - 1.07)f_{yB}$	$1.5 - f_T + (2.68f_T - 3.19)f_{yB}$
	0.85	$0.5f_T - 1.25 + (0.25 - 0.5f_T)f_{yB}$	$0.5f_T - 1.25 + (0.2 - 0.1f_T)f_{yB}$
	1.0	$0.5f_T - 1.25 + (0.25 - 0.5f_T)f_{yB}$	$0.5f_T - 1.25 + (0.2 - 0.1f_T)f_{yB}$
$y < 0$	0.0	0.75	$0.75 - 0.5\, f_{yB}$
	0.2	$f_T - 0.25 + (0.35f_T - 0.47)f_{yB}$	$f_T - 0.25 + (1.25 - f_T)f_{yB}$
	0.4	$1.0 + (2.7f_T - 3.2)f_{yB}$	1.0
	0.5	$1.25 - 0.5f_T + (2.7f_T - 3.2)f_{yB}$	$1.25 - 0.5f_T + (0.5f_T - 0.25)f_{yB}$
	0.6	$1.5 - f_T + (2.68f_T - 3.19)f_{yB}$	$1.5 - f_T + (f_T - 1.07)f_{yB}$
	0.85	$0.5f_T - 1.25 + (0.2 - 0.1f_T)f_{yB}$	$0.5f_T - 1.25 + (0.25 - 0.5f_T)f_{yB}$
	1.0	$0.5f_T - 1.25 + (0.2 - 0.1f_T)f_{yB}$	$0.5f_T - 1.25 + (0.25 - 0.5f_T)f_{yB}$

4.1.3.9 Envelope of dynamic pressure

The envelope of wave pressure at any point, P_{w-max} is to be taken as the greatest pressure obtained from any of the load cases determined by Cl.4.1.3.2 to Cl.4.1.3.8.

4.1.3.10 Envelope wave pressure for L< 90[m]

For ships with L< 90[m] the envelope wave pressure, P_w, as obtained from Table 4.1.3(j) is to be used in lieu of the pressures defined in Cl.4.1.3.2 to Cl.4.1.3.8.

The total pressure P_{ex} at any load point of the hull (bottom and side shell, including sides of the superstructure) for the static plus dynamic (S+D) design load scenario on, is to be derived from each dynamic load case and is to be taken as:

$$P_{ex} = P_s + P_w$$

where:

P_s = Hydrostatic pressure, in [kN/m^2], defined in Cl.4.1.2.

P_w = Wave pressure, in [kN/m^2], as defined in Table Cl.4.1.3(j).

P_{ex} also not to be taken less than

$\lceil \min\{(20 + 0.5(L - 90))R_s\}, 50R_s \rceil$ in [kN/m^2].

Table 4.1.3(j): Wave pressure in [kN/m^2], for ships with L $<$ 90[m]

Wave pressure on bottom shell, side shell and superstructure side in [kN/m^2]			
For bottom shell below waterline			$P_w = P_{w-head(bwl)}$
For side shell below waterline			$P_w =$ greater of $(P_{w-head(bwl)})$ and $P_{w-beam(bwl)}$
For side shell above waterline			$P_w =$ greater of $(P_{w-head(awl)})$ and $P_{w-beam(awl)}$
Dynamic pressure in head sea condition	$P_{w-head(bwl)}$	Below waterline	$1.8R_s\left[(3K_s + 0.8)C_w + 135\frac{\lvert y\rvert}{(B+75)} - 1.2(T_{LC} - z)\right]$
	$P_{w-head(awl)}$	Above waterline	$1.8R_s\left[(3K_s + 0.8)C_w + 135\frac{\lvert y\rvert}{(B+75)}\right] - 10(z - T_{LC})$
Dynamic pressure in beam sea condition	$P_{w-beam(bwl)}$	Below waterline	$17R_s\left[\lvert y\rvert\frac{\emptyset}{360}.\pi + C_B\left(\frac{\lvert y\rvert + 0.8C_w}{14}\right)\left(0.7 + 2\frac{z}{T_{LC}}\right)\right]$
	$P_{w-beam(awl)}$	Above waterline	$17R_s\left[\lvert y\rvert\frac{\emptyset}{360}.\pi + 2.7C_B\left(\frac{\lvert y\rvert + 0.8C_w}{14}\right)\right] - 10(z - T_{LC})$

where:

$P_{w-head(bwl)}, P_{w-head(awl)}, P_{w-beam(bwl)}, P_{w-beam(awl)}$ are dynamic pressures, in head sea or beam sea conditions as given above.

R_s = Reduction factor related to service restriction as defined in Section 2.

$$K_s = C_B + \frac{0.83}{\sqrt{C_B}} \text{ at and aft of rule AP}$$

$$K_s = C_B \text{ for } 0.2 \le x/L \le 0.6$$

$$K_s = C_B + \frac{1.33}{C_B} \text{ at and forward of FP.}$$

Intermediate values to be linearly interpolated.

C_w = as defined in Section 2.

$\emptyset$ = Roll angle, in [deg], Refer Section 2.

y = Offset of the load point in consideration in [m] not to be taken less than B/4.

z = Vertical distance [m] from the baseline to the load coordinate point.

4.1.4 External dynamic pressures for fatigue assessment for L≥ 90[m]

4.1.4.1 General

The external pressure P_{ex} at any load point of the hull for the fatigue static plus dynamic (S+D) design load scenario, is to be derived from each fatigue dynamic load case and is to be taken as:

$$P_{ex} = P_s + P_w \text{ but not less than 0.}$$

where:

P_s = Hydrostatic pressure, in [kN/m^2], defined in Cl.4.1.2.

P_w = Wave pressure, in [kN/m^2], is defined in Cl.4.1.4.2 to Cl.4.1.4.6.

4.1.4.2 Hydrodynamic pressures for HSM load cases

The hydrodynamic pressures, P_W, for HSM-1 and HSM-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.4(a).

Table 4.1.4(a): Hydrodynamic pressures for HSM load cases

	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \le T_{LC}}$	$\mathbf{T_{LC} < Z \le h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
HSM-1	$P_W = \max(-P_{HS}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
HSM-2	$P_W = \max(P_{HS}, \rho.g(z - T_{LC}))$		

where:

$$P_{HS} = f_p\, f_h\, k_a\, k_p\, f_{yz}\, C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

f_{yz} = Girth distribution coefficient, to be taken as:

$$f_{yz} = \frac{Z}{T_{LC}} + f_{yB} + 1$$

f_h = Coefficient to be taken as:

$$f_h = 2.75(1.21 - 0.66 f_T)$$

f_p = coefficient to be taken as

= $f_{fa}((0.21 + 0.02 f_T) + (6 - 4 f_T)L.10^{-5})$ for fatigue assessment

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be taken as:

$$k_a = 1.0 + 3f_T - (1.0 + f_T)f_{yB} + \left[5(1.0 + f_T)f_{yB} - 15f_T\right]f_{xl} \text{ for } f_{xl} < 0.2$$

$$k_a = 1.0 \text{ for } 0.2 \le f_{xl} < 0.6$$

$$k_a = 1 + (f_{xl} - 0.6)\{(13.5 - 3.5f_T)f_{yB} + (14.5f_T - 17) + 40(1 - f_{yB})(f_{xl} - 0.6)\} \text{for } f_{xl} \ge 0.6$$

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.6L(1 + f_T)$$

k_p = Phase coefficient to be obtained from Table 4.1.4(b). Intermediate values are to be interpolated.

Table 4.1.4(b): Definition of phase coefficient k_p

f_{xl}	0	0.3-0.1f_T	0.5-0.2f_T	0.9-0.4f_T	0.9-0.2f_T	1.0
k_p	$(1 - f_T) + (0.5 - f_T)f_{yB}$	-1	1	1	-1	-1

4.1.4.3 Hydrodynamic pressures for FSM load cases

The hydrodynamic pressures, P_W, for FSM-1 and FSM-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.4(c).

Table 4.1.4(c): Hydrodynamic pressures for FSM load cases			
	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
FSM-1	$P_W = \max(-P_{FS}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
FSM-2	$P_W = \max(P_{FS}, \rho.g(z - T_{LC}))$		

where:

$$P_{HS} = f_p\, f_h\, k_a\, k_p\, f_{yz}\, C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

f_h = Coefficient to be taken as, $f_h = 2.6$

f_p = coefficient to be taken as

= $f_{fa}((0.21 + 0.02f_T) + (6 - 4f_T)L.10^{-5})$ for fatigue assessment

f_{yz} = Girth distribution coefficient, to be taken as:

$$f_{yz} = \frac{Z}{T_{LC}} + f_{yB} + 1$$

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be taken as:

$k_a = 1 + (3.5 - 2f_T)(1 - 5f_{xl})(1 - f_{yB})$ for $f_{xl} < 0.2$

$k_a = 1.0$ for $0.2 \leq f_{xl} < 0.9$

$k_a = 1 + 15\,(1 - f_{yB})(f_{xl} - 0.9)$ for $f_{xl} \geq 0.9$

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.6L(1 + 2/3\, f_T)$$

k_p = Phase coefficient to be obtained from Table 4.1.4(c1). Intermediate values are to be interpolated.

Table 4.1.4(c1): Definition of phase coefficient k_p						
f_{xl}	0	$0.35\text{-}0.1f_T$	$0.5\text{-}0.2f_T$	0.75	$0.9\text{-}0.1f_T$	1.0
k_p	$-0.75\text{-}0.25f_{yB}$	-1	1	1	-1	$-0.5\text{-}0.5f_{yB}$

4.1.4.4 Hydrodynamic pressures for BSR load cases

The hydrodynamic pressures, P_W, for BSR-1 and BSR-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.4(d).

Table 4.1.4(d): Hydrodynamic pressures for BSR load cases			
	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \leq T_{LC}}$	$\mathbf{T_{LC} < Z \leq h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
BSR-1P	$P_W = \max(P_{BSR}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
BSR-2P	$P_W = \max(-P_{BSR}, \rho.g(z - T_{LC}))$		
BSR-1S	$P_W = \max(P_{BSR}, \rho.g(z - T_{LC}))$		
BSR-2S	$P_W = \max(-P_{BSR}, \rho.g(z - T_{LC}))$		

where:

For BSR-1P and BSR-2P load cases.

$$P_{BSR} = \left[10y \sin \emptyset + 0.88 f_p C_w \sqrt{\frac{L_0 + \lambda - 125}{L}} \left(f_{yB1} + 1\right)\right]$$

For BSR-1S and BSR-2S load cases.

$$P_{BSR} = \left[-10y \sin \emptyset + 0.88 f_p C_w \sqrt{\frac{L_0 + \lambda - 125}{L}} \left(f_{yB1} + 1\right)\right]$$

f_p = coefficient to be taken as

= $f_{fa}\left((0.21 + 0.04 f_T) - (12 f_T - 2)B.10^{-4}\right)$ for fatigue assessment

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = \frac{g}{2\pi} T_R^2$$

4.1.4.5 Hydrodynamic pressures for BSP load cases

The hydrodynamic pressures, P_W, for BSP-1 and BSP-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.4(e).

Table 4.1.4(e): Hydrodynamic pressures for BSP load cases

	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \le T_{LC}}$	$\mathbf{T_{LC} < Z \le h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
BSP-1P	$P_W = \max\left(P_{BSP}, \rho . g(z - T_{LC})\right)$	$P_W = P_{W,WL} - \rho . g(z - T_{LC})$	$P_W = 0$
BSP-2P	$P_W = \max\left(-P_{BSP}\ \rho . g(z - T_{LC})\right)$		
BSP-1S	$P_W = \max\left(P_{BSP}, \rho . g(z - T_{LC})\right)$		
BSP-2S	$P_W = \max\left(-P_{BSP}, \rho . g(z - T_{LC})\right)$		

where:

$$P_{BSP} = 4.5\ f_p\ f_{yz}\ C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.2L(1 + 2f_T)$$

f_{yz} = Girth distribution coefficient, to be taken as:

For BSP-1P and BSP-2P

$$f_{yz} = 2\frac{Z}{T_{LC}} + 2.5\ f_{yB1} + 0.5 \text{ for } y \ge 0$$

$$f_{yz} = 2/3\frac{Z}{T_{LC}} + 1/2\ f_{yB1} + 0.5 \text{ for } y < 0$$

For BSP-1P and BSP-2P

$$f_{yz} = 2/3\frac{Z}{T_{LC}} + 1/2\ f_{yB1} + 0.5 \text{ for } y \ge 0$$

$$f_{yz} = 2\frac{Z}{T_{LC}} + 2.5\ f_{yB1} + 0.5 \text{ for } y < 0$$

f_p = coefficient to be taken as

= $f_{fa}(0.2 + (8 + 16 f_T)10^{-3})$ for fatigue assessment

4.1.4.6 Hydrodynamic pressures for OST load cases

The hydrodynamic pressures, P_W, for OST-1 and OST-2 load cases, at any load point, in [kN/m^2], are to be obtained from Table 4.1.4(f).

Table 4.1.4(f): Hydrodynamic pressures for OST load cases

	Wave pressure, in [kN/m^2]		
Load case	$\mathbf{z \le T_{LC}}$	$\mathbf{T_{LC} < Z \le h_W + T_{LC}}$	$\mathbf{Z > h_W + T_{LC}}$
OST-1P	$P_W = \max(P_{OST}, \rho.g(z - T_{LC}))$	$P_W = P_{W,WL} - \rho.g(z - T_{LC})$	$P_W = 0$
OST-2P	$P_W = \max(-P_{OST}\ \rho.g(z - T_{LC}))$		
OST-1S	$P_W = \max(P_{OST}, \rho.g(z - T_{LC}))$		
OST-2S	$P_W = \max(-P_{OST}, \rho.g(z - T_{LC}))$		

where:

$$P_{OST} = 1.38\ f_p\ k_a\ k_p\ f_{yz}\ C_w \sqrt{\frac{L_0 + \lambda - 125}{L}}$$

f_{yz} = Girth distribution coefficient, obtained from Table 4.1.4(f1).

f_p = coefficient to be taken as

= $f_{fa}((0.25 - 0.02f_T) + (12f_T - 9)B.10^{-4})$ for fatigue assessment

λ = Wave length of the dynamic load case, in [m], to be taken as

$$\lambda = 0.45L$$

k_a = Amplitude coefficient in the longitudinal direction of the ship, to be obtained from Table 4.1.4(f2).

k_p = Phase coefficient to be obtained from Table 4.1.4(f3). Intermediate values are to be interpolated.

Table 4.1.4(f1): Definition of girth distribution coefficient, f_{yz}

Transverse position	**OST-1P - OST-2P**	**OST-1S - OST-2S**
$y \ge 0$	$5\frac{z}{T_{LC}} + 3.3f_{yB} + 1.7$	$\frac{z}{T_{LC}} + 0.3f_{yB} + 1.7$
$y < 0$	$\frac{z}{T_{LC}} + 0.3f_{yB} + 1.7$	$5\frac{z}{T_{LC}} + 3.3f_{yB} + 1.7$

Table 4.1.4(f2): Definition of amplitude coefficient , k_a

Transverse position	**Longitudinal position**	**OST-1P - OST-2P**	**OST-1S - OST-2S**
$y \ge 0$	$f_{xl} \le 0.2$	$1.0 + [(3.5 - 2f_T) + (10f_T - 17.5)f_{xl}](1 - f_{yB})$	$1.0 + (3.5 - 2f_T - 1.5f_{yB}) + (10f_T - 17.5 + 7.5f_{yB})f_{xl}$
	$0.2 < f_{xl} \le 0.8$	1.0	1.0
	$f_{xl} > 0.8$	1.0	$1.0 + 2(1 - f_T)(5f_{xl} - 4)f_{yB}$
$y < 0$	$f_{xl} \le 0.2$	$1.0 + (3.5 - 2f_T - 1.5f_{yB}) + (10f_T - 17.5 + 7.5f_{yB})f_{xl}$	$1.0 + [(3.5 - 2f_T) + (10f_T - 17.5)f_{xl}](1 - f_{yB})$
	$0.2 < f_{xl} \le 0.8$	1.0	1.0
	$f_{xl} > 0.8$	$1.0 + 2(1 - f_T)(5f_{xl} - 4)f_{yB}$	1.0

Table 4.1.4(f3): Definition of phase coefficient, k_p			
Transverse position	f_{xl}	**OST-1P - OST-2P**	**OST-1S - OST-2S**
$y \geq 0$	0.0	1.0	$1.0 + (0.5 - f_T)f_{yB}$
	0.2	1.0	$1.0 + 3(0.5 - f_T)f_{yB}$
	0.4	-1.0	$-1.0 + (2.7 - 2.4f_T)f_{yB}$
	0.5	-1.0	$-1.0 + (2.8 - 2.6f_T)f_{yB}$
	0.7	$(f_T - 0.62)f_{yB} - 0.38$	$(2.38 - 3f_T)f_{yB} - 0.38$
	0.9	$0.24+0.76f_{yB}$	$0.24 - (0.24 + f_T)f_{yB}$
	1.0	$0.5f_{yB} - 1.0$	-1.0
$y < 0$	0.0	$1.0 + (0.5 - f_T)f_{yB}$	1.0
	0.2	$1.0 + 3(0.5 - f_T)f_{yB}$	1.0
	0.4	$-1.0 + (2.7 - 2.4f_T)f_{yB}$	-1.0
	0.5	$-1.0 + (2.8 - 2.6f_T)f_{yB}$	-1.0
	0.7	$(2.38 - 3f_T)f_{yB} - 0.38$	$(f_T - 0.62)f_{yB} - 0.38$
	0.9	$0.24 - (0.24 + f_T)f_{yB}$	$0.24+0.76f_{yB}$
	1.0	-1.0	$0.5f_{yB} - 1.0$

4.2 Loads on exposed decks

4.2.1 Application

4.2.1.1 The external pressures and forces on exposed decks are only to be applied for strength assessment, i.e. yield and buckling assessment.

4.2.1.2 The green sea pressures defined in Cl.4.2.2 for exposed decks are to be considered independently of the pressures due to distributed cargo or other equipment loads and any concentrated forces due to cargo or other unit equipment loads, defined in Cl.4.2.3.1 and Cl.4.2.3.2 respectively.

4.2.2 Green sea loads

4.2.2.1 Pressure on exposed deck

The external dynamic pressure due to green sea loading, P_D, at any point of an exposed deck, in [kN/m^2], for the static plus dynamic (S + D) design load scenarios are to be derived for each dynamic load case and be taken as defined below.

The external dynamic pressure due to green sea loading, P_D, at any point of an exposed deck for the static (S) design load scenarios is zero.

4.2.2.2 Effect of wave breakers need not be considered for green sea pressure calculation.

4.2.2.3 HSM, HSA and FSM load cases for L≥ 90[m]

The external pressure, P_D, for HSM, HSA and FSM load cases, at any load point of an exposed deck is to be obtained, in [kN/m^2], from the following formula:

$P_D = \chi P_w$ but not to be taken less than χP_{D-min}.

where:

$P_w = P_{w,D}$.

$P_{w,D}$ = Pressure in [kN/m^2] obtained at side of the exposed deck for HSM, HSA and FSM load cases as defined in Cl.4.1.3.

P_{D-min} = Minimum pressure on exposed freeboard deck, in [kN/m^2], to be taken as:

P_{D-min} = 0 for global or partial ship finite element analysis according to Chapter 8

= 0 for PSM grillage according to Chapter 7
= as defined in Table 4.2.2.1, for other cases

χ = Coefficient defined in Table 4.2.2.2.

Table 4.2.2.1: Minimum pressures on exposed decks for HSM, HSA, FSM load cases		
Location	**Minimum pressure on exposed deck, P_{D-min}, in [kN/m^2]**	
	$L_{LL} \geq 100[m]$	$L_{LL} < 100[m]$
$X_{LL}/L_{LL} \leq 0.75$	34.3	$14.9 + 0.195L_{LL}$
$X_{LL}/L_{LL} > 0.75$	$34.3 + \left(14.8 + a(L_{LL} - 100)\right)\left(4\frac{X_{LL}}{L_{LL}} - 3\right)$	$12.2 + \frac{L_{LL}}{9}\left(5\frac{X_{LL}}{L_{LL}} - 2\right) + 3.6\frac{X_{LL}}{L_{LL}}$
where: a = coefficient taken equal to: a = 0.356 for type A, type B-60 and type B-100 freeboard ships a = 0.0726 for type B freeboard ships X_{LL} = X-coordinate of the load point measured from the aft end of the freeboard length L_{LL}.		

Table 4.2.2.2: Coefficient for pressure on exposed decks	
Exposed deck location	χ
Freeboard deck	1.00
Superstructure deck including forecastle deck	0.75
1st tier of deckhouse	0.56
2nd tier of deckhouse	0.42
3rd tier of deckhouse	0.32
4th tier of deckhouse	0.25
5th tier of deckhouse	0.20
6th tier of deckhouse	0.15
7th tier of deckhouse and above	0.10

4.2.2.4 BSR, BSP, OST and OSA load cases for L≥ 90[m]

The external pressure, P_D, for BSR,BSP,OST and OSA load cases, at any load point of an exposed deck is to be obtained, in [kN/m^2], by linear interpolation between the pressures at the port and starboard deck edges, Refer also Figure 4.1.3(e1), Figure 4.1.3(f1), Figure 4.1.3(g2) and Figure 4.1.3(h1):

$$P_{D,stbd} = \chi P_{w,D-stbd}$$

$$P_{D,pt} = \chi P_{w,D-pt}$$

where:

$P_{w,D-stbd}$ = Pressure obtained at starboard deck edge for BSR, BSP, OST or OSA load cases as defined in Cl.4.1.3, as appropriate.

$P_{w,D-pt}$ = Pressure obtained at port deck edge for BSR, BSP, OST and OSA load cases as defined in Cl.4.1.3, as appropriate.

χ = Coefficient defined in Table 4.2.2.2.

4.2.2.5 Envelope of dynamic pressures on exposed deck

The envelope of dynamic pressure at any point of an exposed deck, P_{D-max}, is to be taken as the greatest pressure obtained from any of the load cases determined by Cl.4.2.2.3 and Cl.4.2.2.4.

4.2.2.6 Envelope of dynamic pressures on exposed deck for L< 90[m]

For ships with L< 90[m] the envelope of dynamic pressure, on weather deck is to be obtained from Table 4.1.3(j) for side shell above waterline and not be less than minimum design pressure, as given below:

$P_D = P_w$ but not to be taken less than χP_{D-min}.

where:

P_w = Pressure in [kN/m^2] obtained as per Table 4.1.3(j) for side shell above waterline.

P_{D-min} = Minimum pressure on exposed freeboard deck, in [kN/m^2], as defined in Table 4.2.2.1.

χ = Coefficient defined in Table 4.2.2.2.

4.2.3 Loads carried on exposed deck

4.2.3.1 Pressure due to distributed load

If a distributed load is carried on an exposed deck or platform, for example deck cargo or other equipment, the static and dynamic pressures due to this distributed load are to be considered.

The total pressure P_{dl},in [kN/m^2], due to this distributed load for the static (S) design load scenario is to be taken as

$$P_{dl} = P_{dl-s}$$

The pressure P_{dl}, in [kN/m^2], due to this distributed load for the static plus dynamic (S+D) design load scenario is to be derived for each dynamic load case and is to be taken as:

$$P_{dl} = P_{dl-s} + P_{dl-d}$$

where:

P_{dl-s} = Static pressure, in [kN/m^2], due to the distributed load, minimum 3.5 [kN/m^2], including self-weight, unless a higher load is defined by the designer

P_{dl-d} = Dynamic pressure, in [kN/m^2], due to the distributed load, in [kN/m^2], to be taken as:

$$P_{dl-d} = f_\beta P_{dl-s} \frac{a_z}{g}$$

a_z = Vertical acceleration, in [m/s^2], at the centre of gravity of the distributed load, for the considered load case, to be obtained according to Section 2, Cl.2.2.2.3.

For ships with L< 90[m],

$$P_{dl-d} = P_{dl-s} \frac{a_v}{g}$$

where,

a_v = Combined vertical heave acceleration as given in Cl.2.2.3.3.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2]

$= a_{pitch} |(x - 0.45L)|$

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2]

$= a_{roll} |y|$

4.2.3.2 Concentrated force due to unit load for L≥ 90[m]

If a unit load, for example deck cargo, is carried on an exposed deck, the static and dynamic forces due to the unit load carried are to be considered.

The force F_U, in [kN], due to this concentrated load for the static (S) design load scenarios, is to be taken as:

$$F_U = F_{U-s}$$

The longitudinal force,F_{U-x} transverse force F_{U-y} and vertical force F_{U-z}, in [kN], due to this concentrated load for the static plus dynamic (S + D) design load scenarios is to be taken as:

$$F_{U-x} = m_u\, a_{x-u}$$

$$F_{U-y} = m_u\, a_{y-u}$$

$$F_{U-z} = F_{U-s} + m_u\, a_{z-u}$$

F_{U-s} = Static force, in [kN], due to the unit load to be taken equal to:

$= m_u . g$

m_u = mass of the unit load carried, in [t]

$a_{x-u}, a_{y-u}, a_{z-u}$ = Combined accelerations, in [m/s^2], as defined in Table 4.2.3.

a_ℓ, a_t, a_v = Combined longitudinal, transverse and vertical accelerations, in [m/s^2], as defined in Cl.2.2.3.

The combined vertical acceleration $a_{v-pitch}$ in [m/s^2] at any position, for load combination head sea, Refer Table 4.2.3, be taken as:

$$a_{v-pitch} = \sqrt{(a_{heave}^2 + a_{pitch-z}^2)}$$

The combined vertical acceleration a_{v-roll} in [m/s^2] at any position, for load combination beam sea, Refer Table 4.2.3, be taken as:

$$a_{v-roll} = \sqrt{(a_{heave}^2 + a_{roll-z}^2)}$$

where,

a_{heave} = heave acceleration as given in Cl.2.1.3.3.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2]

$= a_{pitch}(x - 0.45L)$

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2]

$= a_{roll}.y$

The accelerations acting on supporting structures and securing systems for heavy units of cargo, equipment or structural components are to be taken as given below in Table 4.2.3:

Table 4.2.3: Combined accelerations for heavy deck cargo units and equipment

Load combination	$\mathbf{a_{x-u}}$	$\mathbf{a_{y-u}}$,	$\mathbf{a_{z-u}}$
Head sea 1	a_l	0	$a_{v-pitch}$
Beam sea 1	0	a_t	a_{v-roll}
Oblique sea 1	$0.67\ a_l$	$0.67\ a_t$	a_v

Note:
1)For uplift conditions, the $a_{z-u} = -a_v$
2) The horizontal accelerations a_{x-u} and a_{y-u} are to be applied in the direction(s) giving maximum response.

Guidance note:

1) For heavy units with centre of gravity (COG) less than $b_c/4$ or $\ell_c/4$ above the deck it is attached to, the horizontal accelerations a_{x-u} and a_{y-u} ,can be neglected, where b_c and ℓ_c are the breadth and length, in [m] of the unit.

2) The longitudinal force,F_{U-x} transverse force F_{U-y} and vertical force F_{U-z}, in [kN], due to this concentrated load for the static plus dynamic (S + D) design load scenarios can be derived for each dynamic load case when these forces are considered in combination with hull girder loads.

4.2.3.3 Forces due to heavy units of cargo and equipment for L< 90[m]

For ships with L< 90[m], the gravity and acceleration forces in [kN] due to heavy units of cargo and equipment is to be taken as combination of the longitudinal force,F_{U-x} transverse force F_{U-y} and vertical force F_{U-z}, acting simultaneously. Worst among the following two sets of combination are to be considered..

$$F_{U-z} = m_u\left(g + 0.5a_{heave} + a_{pitch-z}\right)$$

$$F_{U-x} = m_u\left(g\sin\theta - 0.6a_{surge} + a_{pitch-x}\right)$$

$$F_{U-y} = 0$$

or

$$F_{U-z} = m_u\left(g + a_{heave} + a_{roll-z}\right)$$

$$F_{U-x} = 0$$

$$F_{U-y} = m_u\left(g\sin\emptyset + a_{roll-y}\right)$$

where,

m_u = as defined in Cl.4.2.3.2.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2]

$= a_{pitch}|(x - 0.45L)|$

$a_{pitch-x}$ = Longitudinal acceleration due to pitch, in [m/s^2]

$= a_{pitch}(z - R)$

a_{roll-y} = Transverse acceleration due to roll, in [m/s^2]

$= a_{roll}(z - R)$

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2]

$= a_{roll}\,|y|$

4.2.4 External pressure on hatch covers

The external pressures on hatch covers are only to be applied for strength assessment.

The green sea loads at any load point of a hatch cover, P_{HC}, in [kN/m^2], is to be taken as follows:

- For cargo hold analysis according to Chapter 8.

$P_{HC} = P_D - \rho g(z_{HC} - D)$ without being less than 0.

- For other cases: Refer Chapter 12, Section 6, Cl.6.2.

P_D = Green Sea pressure, in [kN/m^2], on the deck in way of the hatch cover obtained according to Cl.4.2.2, considering χ equal to 1.0.

z_{HC} = z coordinate of the top of the hatch cover, in [m].

If a distributed load or a unit load is carried on a hatch cover, the pressure is to be obtained according to Chapter 12, Section 6, Cl.6.2.3.

4.3 External pressures on superstructure and deckhouses

4.3.1 Application

4.3.1.1 The external pressures on superstructure and deckhouses are only to be applied for strength assessment. These pressures are to be considered as dynamic pressures and are to be applied to the appropriate structure without any static pressure load component.

4.3.1.2 The pressure defined in Cl.4.3.4 are only applicable to the requirements given in Chapter 7, Section 8.

4.3.2 Exposed superstructure and deckhouse tops

The lateral pressure for exposed superstructure and deckhouse tops, in [kN/m^2], is to be taken according to Cl.4.2.2.

4.3.3 Sides of superstructures

The design pressure for the external sides of superstructures, P_{sl}, in [kN/m^2], is to be taken as:

$$P_{sl} = 2.1R_s.C_w.C_F(C_b + 0.7)\frac{20}{10 + (Z_{SD} - T_{sc})}$$

where:

C_F = Distribution factor according to Table 4.3.3.

Z_{SD} = Z coordinate, in [m], of the midpoint of stiffener span, or of the middle of the elementary plate panel.

Table 4.3.3: Distribution factor C_F

Location	C_F
$x/L < 0.2$	$1.0 + \frac{5}{C_b}\left(0.2 - \frac{x}{L}\right)$ without taking x/L less than 0.1
$x/L \geq 0.2$	1.0

4.3.4 End bulkheads of superstructures and deckhouse walls

4.3.4.1 The external pressure, P_A, for the aft and forward external bulkheads of superstructures and deckhouse walls (front, side and aft bulkheads), in [kN/m^2], is to be taken as:

$$P_A = 10.f_n f_c[f_b f_d - (Z_{SD} - T_{sc})]$$

but is not to be less than P_{A-min}.

where:

f_n = Coefficient defined in Table 4.3.4.1.

f_c = Coefficient, to be taken as:

$$f_c = 0.3 + 0.7\frac{b_1}{B_1}$$

For exposed parts of machinery casings, f_c is to be taken equal to 1

The value of (b_1/B_1) is not to be taken less than 0.25.

b_1 = Breadth of deckhouse at the position considered.

B_1 = Actual maximum breadth of ship on the exposed weather deck at the position considered.

L, L_1 = Length of ship in [m], L_1 need not be taken greater than 300[m]

f_b = Coefficient defined in Table 4.3.4.2.

f_d = Coefficient defined in Table 4.3.4.3.

Values of "f_d' may be linearly interpolated from Table 4.3.4.3.

P_{A-min} = Minimum lateral pressure, in [kN/m^2], as defined in Table 4.3.4.4.

Z_{SD} = vertical distance, in [m], taken at mid-height of the bulkhead or wall with respect to the reference coordinate system defined in Chapter 1, Section 1.

Guidance note:

Longitudinal deckhouse walls in recesses with limited length and protected from green sea loads may be defined as superstructure side, hence Cl.4.3.3 applies.

Table 4.3.4.1: Values of coefficient 'f_n'

Type of bulkhead	Location	Values of 'f_n'	Maximum values of 'f_n'
Unprotected front bulkhead [1)]	Lowest tier [2)]	$2.0 + \frac{L_1}{120}$	4.5
	Second tier	$1.0 + \frac{L_1}{120}$	3.5
	Third tier	$0.5 + \frac{L_1}{150}$	2.5
	Fourth tier	$0.9\left(0.5 + \frac{L_1}{150}\right)$	2.25
	Fifth tier and above	$0.8\left(0.5 + \frac{L_1}{150}\right)$	2
Protected front bulkhead [1)]	Lowest, second and third tiers	$0.5 + \frac{L_1}{150}$	2.5
	Fourth tier	$0.9\left(0.5 + \frac{L_1}{150}\right)$	2.25
	Fifth tier and above	$0.8\left(0.5 + \frac{L_1}{150}\right)$	2
Side bulkheads [3)]	Lowest, second and third tiers	$0.5 + \frac{L_1}{150}$	2.5
	Fourth tier	$0.9\left(0.5 + \frac{L_1}{150}\right)$	2.25
	Fifth tier and above	$0.8\left(0.5 + \frac{L_1}{150}\right)$	2
Aft end bulkheads	All tiers: Abaft amidships	$0.7 - \frac{0.8x}{L} + \frac{L_1}{1000}$	$1 - \frac{0.8x}{L}$
	All tiers: Forward of amidships	$0.5 - \frac{0.4x}{L} + \frac{L_1}{1000}$	$0.8 - \frac{0.4x}{L}$

1) The front bulkhead of a superstructure or deckhouse may be considered as protected when it is located less than B_X behind another superstructure or deckhouse, and the width of the front bulkhead being considered is less than the width of the aft bulkhead of the superstructure or deckhouse forward of it. B_X is the local breadth of the ship at the front bulkhead.

2) The lowest tier is normally that tier which is directly situated above the uppermost continuous deck to which the moulded depth, D is measured. However, when (D-T_{SC}) exceeds the minimum non-corrected tabular freeboard (according to ILLC as amended) by at least one standard superstructure height (as defined in Chapter 1, Section 2, Cl.2.2.5), then this tier may be defined as the 2nd tier and the tier above as the 3rd tier.

3) Applicable only to side bulkheads of deckhouses

Table 4.3.4.2: Values of coefficient 'f_b'

Bulkhead Location (1)	Values of 'f_b'
$\frac{x}{L} \leq 0.45$	$1.0 + \left[\frac{\frac{x}{L} - 0.45}{C_b + 0.2}\right]^2$
$\frac{x}{L} > 0.45$	$1.0 + 1.5\left[\frac{\frac{x}{L} - 0.45}{C_b + 0.2}\right]^2$

where:
C_b= Block coefficient with 0.60<Cb<0.80. For aft deckhouse bulkheads located forward of amidships, C_b need not be taken less than 0.80.
Note (1) For deckhouse sides, the deckhouse is to be subdivided into parts of approximately equal length, not exceeding 0.15L each, and x is to be taken as the X-coordinate of the centre of each part considered.

Table 4.3.4.3: Values of coefficient 'f_d'

Length L of ship, in [m]	Values of 'f_d'
$L \leq 150$	$\frac{L}{10} \cdot e^{(-L/300)} + \left[\frac{L}{150}\right]^2 - 1.0$
$150 \leq L < 300$	$\frac{L}{10} \cdot e^{(-L/300)}$
$L \geq 300$	11.03

Table 4.3.4.4: Values of 'P_{A-min}'

Length L of ship, in [m]	Lowest tier of unprotected fronts	Elsewhere (1)
	'P_{A-min}' in [kN/m^2]	
$L \leq 50$	30	15
$50 < L < 250$	$25 + \frac{L}{10}$	$12.5 + \frac{L}{20}$
$L \geq 250$	50	25

(1) For the 4th tier and above, P_{A-min} is to be taken equal to 12.5[kN/m^2] for the front bulkhead, and 2.5[kN/m^2] for deckhouse side and aft wall.

Section 5

Internal Loads

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

∅ = Roll angle, in [deg], as defined in Cl.2.1.2.1.

θ = Pitch angle, in [deg], as defined in Cl.2.1.2.2.

a_{heave} = Vertical acceleration due to heave in [m/s^2], as defined in Cl.2.1.3.3.

a_{surge} = Longitudinal acceleration due to surge in [m/s^2], as defined in Cl.2.1.3.1.

a_{roll-y} = Transverse acceleration due to roll, in [m/s^2], as defined in Cl.2.1.4.2.

$a_{pitch-x}$ = Longitudinal acceleration due to pitch, in [m/s^2], as defined in Cl.2.1.4.1.

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2], as defined in Cl.2.1.4.3.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2], as defined in Cl.2.1.4.3.

x,y,z = X, Y and Z coordinates, in [m], of the load point with respect to the reference coordinate system defined in Section 1, Cl.1.2.7.

x_G, y_G, z_G = X, Y and Z coordinates, in [m], of the volumetric centre of gravity of the tank, considered with respect to the reference coordinate system defined in Section 1, Cl.1.2.7.

a_X, a_Y, a_Z = Longitudinal, transverse and vertical accelerations, in [m/s^2], at x_G, y_G, z_G, as defined in Section 2, Cl.2.2.2.

f_β = heading correction factor as defined in Section 2.

ρ_L = Density of liquid in the tank and ballast hold, in [t/m^3], normally not to be taken less than:

For strength assessment:

ρ_L = 1.025 for all liquids including oil and product cargoes. If a tank filled at 98% is intended to carry heavier liquid cargoes than 1.025 (i.e.$\rho_{max_-LM} > 1.025$), then $\rho_L = \rho_{max-LM}$.

For Fatigue assessment:

ρ_L = 0.9 for oil and oil product cargoes

ρ_L = 1.025 for ballast tanks

$\rho_L = \rho_{max-LM}$ for cargo tanks intended to carry liquid other than oil products and for cargo tanks intended to carry oil products with density exceeding 1.025[t/m^3]

ρ_{max-LM} = Maximum liquid cargo density in [t/m^3], associated with a full tank at 98%, from any section on in the ship's loading manual or value specified by the designer.

f_{cd} = Factor for joint probability of occurrence of liquid cargo density and maximum sea state in 25 years design life, to be taken as:

For strength assessment with FE analysis of cargo tanks filled with oil or oil products cargo

f_{cd} = 0.88 for $\rho_L \leq$ 1.025[t/m^3]

= 1.0 for ρ_L > 1.025[t/m^3]

For other cases

f_{cd} = 1.0

Z_{top} = Z coordinate of the highest point of tank, excluding small hatchways, in [m].

h_{air} = Height of air pipe or overflow pipe above the top of the tank, in [m].

P_{PV} = Design vapour pressure, in [kN/m^2], not to be taken less than 25[kN/m^2].

P_{drop} = Overpressure, in [kN/m^2], due to sustained liquid flow through air pipe or

overflow pipe in case of overfilling or filling during flow through ballast water exchange. It is to be defined by the designer, but not to be less than 25[kN/m^2].

5.1 **Pressures due to liquids**

5.1.1 Total pressure

5.1.1.1 Pressures for the strength and fatigue assessments in intact conditions

The internal pressure due to liquid acting on any load point of a watertight boundary (tank and ballast holds), in [kN/m^2], for the static (S) design load scenarios, given in Section 6, is to be taken as:

$$P_{in} = P_{ls} \text{ but not less than } 0$$

The internal pressure due to liquid acting on any load point of a watertight boundary (tank and ballast holds), in kN/m^2, for the static plus dynamic (S + D) design load scenarios is to be derived for each dynamic load case and is be taken as:

$$P_{in} = P_{ls} + P_{ld} \text{ but not less than } 0$$

where:

P_{ls} = Static pressure due to liquid in watertight boundaries (tank and ballast holds), in [kN/m^2], as defined in Cl.5.1.2.

P_{ld} = Dynamic inertial pressure due to liquid in watertight boundaries (tank and ballast holds), in [kN/m^2], as defined in Cl.5.1.3.1 for L≥ 90[m] and Cl.5.1.3.2 for L< 90[m].

5.1.1.2 Pressures for the strength assessments of flooded conditions

The internal pressure in flooded condition, in [kN/m^2], acting on any load point of the watertight boundary of a hold, tank or other space for the flooded static (S) design load scenario, given in Section 6, is to be taken as:

$$P_{in-f} = P_{fs} \text{ but not less than } \rho\, g\, d_o$$

where:

P_{fs} = Static pressure of seawater due to flooding in the compartment, in [kN/m^2], as defined in Cl.5.1.4.

d_0 = Distance, in m, to be taken as:

= 0.02L for L<120[m]

= 2.4 for L≥120[m]

5.1.2 Static liquid pressure in intact conditions

5.1.2.1 Static liquid pressure for strength assessment

(a) Normal operations at sea

The static pressure due to liquid in watertight boundaries (tank and ballast holds), P_{ls-1} during normal operations at sea, in [kN/m^2], is to be taken as:

- For cargo tanks filled with liquid cargo/for tanks arranged with pressure relief valves

$$P_{ls-1} = f_{cd}\, \rho_L\, g(Z_{top} - Z) + P_{PV}$$

- For other cases

$$P_{ls-1} = \rho_L\, g(Z_{top} - Z + 0.5\, h_{air})$$

(b) Normal operations at harbour operations

The static pressure, P_{ls-2} due to liquid in watertight boundaries (tank and ballast holds), for harbour operations, in [kN/m^2], is to be taken as:

- For cargo tanks filled with liquid cargo/for tanks arranged with pressure relief valves

$$P_{ls-2} = \rho_L\, g(Z_{top} - Z) + P_{PV}$$

- For other cases

$$P_{ls-2} = \rho_L\, g(Z_{top} - Z) + P_0$$

where:

P_0 = Static pressure, in [kN/m^2], to be taken as:

P_0 = 0.2L+ 6 for L < 90[m]

= 24 for L ≥ 90[m]

for ballast hold in dry cargo vessels:

P_0 = 0

However, in mechanically propelled cargo ships of 500GT and above and passenger ships, for tanks forming part of the watertight subdivision, P_0 is to be taken as not less than 24[kN/m^2]. For this purpose, tanks meeting any of the following conditions are to be considered as part of watertight subdivision:

(i) Tanks considered watertight in damage stability calculations.

(ii) Double bottom tanks, peak tanks or tanks bounded by main transverse watertight bulkheads required as per Chapter 3, Section 2, Cl.2.1.1.

(c) Flow through ballast water exchange

The static pressure, P_{ls-3} due to liquid in ballast tanks associated with flow through ballast water exchange operations, in [kN/m^2], is to be taken as:

$$P_{ls-3} = \rho_L \, g(Z_{top} - Z + h_{air}) + P_{drop}$$

(d) Sequential ballast water exchange

The static pressure, P_{ls-4} due to liquid in ballast tanks associated with sequential ballast water exchange operations, in [kN/m^2], is to be taken as:

$$P_{ls-4} = \rho_L \, g(Z_{top} - Z + 0.5 \, h_{air})$$

(e) Ballasting using ballast water treatment system

The static pressure, due to liquid in watertight boundaries (tank and ballast holds), associated with ballasting operations using a ballast water treatment system is to be taken as defined for sequential ballast exchange in Cl.5.1.2.1(d). The ship designer has to inform IRS, if the ballast water treatment system implies additional pressure to be considered as P_{drop}, etc in addition to the pressure defined in Cl.5.1.2.1(d).

(f) Tank testing

The static pressure P_{ls-ST} in [kN/m^2], acting on plates and stiffeners subject to tank testing, is to be taken as:

$$P_{ls-ST} = 10(Z_{ST} - Z)$$

where:

Z_{ST} = Testing load height, in [m], as defined in Table 5.1.2.

The actual strength testing is to be carried out in accordance with Chapter 17, Section 3.4, Table 3.4.1.

In order to assess the structure, static design pressures are to be applied. No inertial pressure is to be considered as acting on plates and stiffeners subject to tank testing.

Table 5.1.2: Design testing load height, Z_{ST}	
Compartment	**Load height**
Double bottom tanks	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ $Z_{ST} = Z_{bd}$
Double side tanks	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3) $Z_{ST} = Z_{bd}$
Hopper side tanks[1)], topside tanks[1)], double side tanks[1)]	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3) $Z_{ST} = Z_{bd}$
Deep tanks (other than those listed elsewhere in this table)	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3)
Fore and aft peaks used as tank	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3)
Cargo oil tanks [2)]	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3) $Z_{ST} = Z_{top} + 0.1P_{PV}$
Fuel Oil Tanks	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3) $Z_{ST} = Z_{top} + 0.1P_{PV}$ $Z_{ST} = Z_{bd}$
Ballast hold of bulk carriers	$Z_{ST} = Z_h$
Chain locker	$Z_{ST} = Z_c$
Independent tanks (Edible liquid Tanks)	The greater of the following: $Z_{ST} = Z_{top} + h_{air}$ $Z_{ST} = Z_{top} + 0.9$
Chemical carriers	The greater of the following: $Z_{ST} = Z_{top} + 0.1P_{PV}$ $Z_{ST} = Z_{top} + 2.4$ (See Note 3)
Ballast ducts	The greater of the following: • testing load height corresponding to ballast pump maximum pressure • stetting of any pressure relief valve

where:
Z_{bd} = Z coordinate, in [m], of the bulkhead deck
Z_h = Z coordinate, in [m], of the top of hatch coaming
Z_c = Z coordinate, in [m], of the top of chain pipe.

1) Applicable to double bottom tank connected with hopper side tanks, topside tanks or double side tanks.
2) Tank test load is not applicable for cargo tanks carrying LNG.
3) For vessels of L < 90[m], the head of water above highest point of tank may be (0.2L + 0.6) [m], but not less than 1.0[m]. However, in mechanically propelled cargo ships of 500GT and above and passenger ships, for tanks forming part of the watertight subdivision (See Pt.3, Ch.5, Cl.5.1.2.1(b)), head of water above the highest point of tank is to be 2.4[m].

5.1.2.2 Static liquid pressure for fatigue assessment

The static pressure due to liquid in watertight boundaries (tank and ballast holds), P_{ls} to be used for the fatigue assessment, in [kN/m^2], is to be taken as:

$$P_{ls} = \rho_L g(Z_{top} - Z)$$

for all tanks (cargo and water ballast tanks, ballast hold and other tanks).

5.1.3 Dynamic liquid pressure

5.1.3.1 The dynamic pressure, P_{ld} due to liquid in watertight boundaries (tank and ballast holds), in [kN/m^2] is to be taken as:

$$P_{ld} = f_\beta f_{cd} \rho_L \left[a_Z(z_0 - z) + f_{ull-l} a_x(x_0 - x) + f_{ull-t} a_y(y_0 - y)\right]$$

where:

f_{ull-l} = Longitudinal acceleration correction factor for the ullage space above the liquid in watertight boundaries (tank and ballast holds), taken as:

For strength assessment:

= 0.62 for cargo tanks filled with any liquids including water ballast

= 1.0 for other cases

For fatigue assessment:

For cargo tanks and ballast holds

$$f_{ull-l} = 0.5 + \frac{|z_0 - z|}{\ell_{fs}} \frac{180}{\theta\pi}$$

f_{ull-l} = 1.0 for other cases.

f_{ull-l} is not to be less than 0.0 nor greater than 1.0

f_{ull-t} = Transverse acceleration correction factor to account for the ullage space above the liquid in watertight boundaries (tank and ballast holds), taken as:

For strength assessment:

= 0.67 for cargo tanks filled with any liquids including water ballast.

= 1.0 for other cases.

For fatigue assessment:

For cargo tanks and ballast holds

$$f_{ull-t} = 0.5 + \frac{|z_0 - z|}{b_{top}} \frac{180}{\emptyset\pi}$$

f_{ull-t} = 1.0 for other cases.

f_{ull-t} is not to be less than 0 nor greater than 1.0

ℓ_{fs} = Cargo tank length at the top of the tank or length of the ballast hold hatch coaming, in [m].

b_{top} = Cargo tank breadth at the top of the tank or breadth of the ballast hold hatch coaming, in [m] determined at mid length of the tank or ballast hold hatch coaming.

x_0 = X coordinate, in [m], of the reference point.

y_0 = Y coordinate, in [m], of the reference point.

z_0 = Z coordinate, in [m], of the reference point.

The reference point is to be taken as the point with the highest value of V_j, calculated for all points that define the upper boundary of the tank or ballast hold as follows:

$$V_j = a_x(x_j - x_G) + a_y(y_j - y_G) + (a_z + g)(z_j - z_G)$$

where:

x_j = X coordinate, in [m], of the point j on the upper boundary of the tank or ballast hold.

y_j = Y coordinate, in [m], of the point j on the upper boundary of the tank or ballast hold.

z_j = Z coordinate, in [m], of the point j on the upper boundary of the tank or ballast hold.

The following simplified method of determination of the reference point assuming a rectangular shape with area equal A_{top} of the top of the tank or the ballast hold hatch coaming is acceptable, Refer Figure 5.1.3:

$$x_j = x_{top} \pm 0.5\ell_{fs}$$

$$y_j = y_{top} \pm 0.5b_{top}$$

where

x_{top} = X coordinate, in [m], of the centre of the rectangular area A_{top} at the top of the tank or the ballast hold hatch coaming.

y_{top} = Y coordinate, in [m], of the centre of the rectangular area A_{top} at the top of the tank or the ballast hold hatch coaming.

The area of a rectangular shape at the top of the tank or the ballast hold hatch coaming, in [m^2].

$A_{top} = \ell_{fs} \, . \, b_{top}$

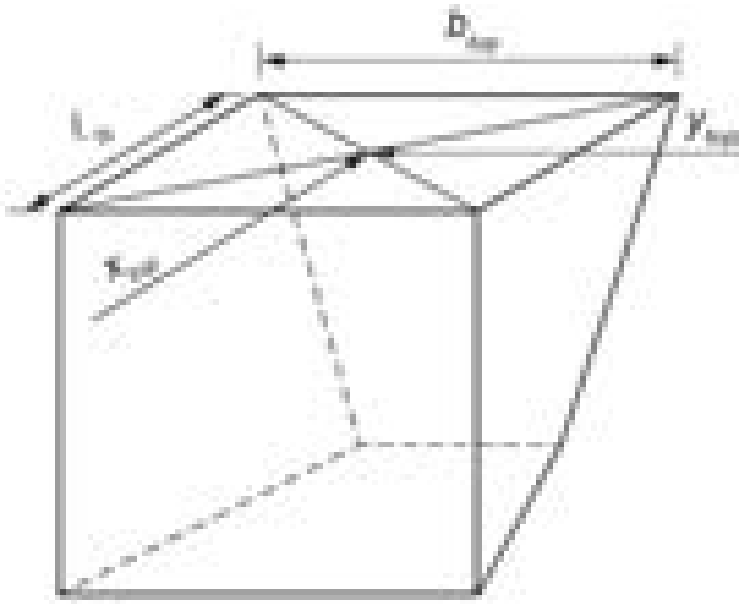


Figure 5.1.3: Area of a rectangular shape at the top of a tank

5.1.3.2 Dynamic liquid pressure for ships with L < 90[m]

The dynamic pressure, in [kN/m^2], due to liquid in watertight boundaries for ships with L< 90[m], may be taken as:

$$p_{ld} = \max\begin{pmatrix}(0.5a_{heave} + a_{pitch-z})\rho_L . h_s + (g\sin\theta - 0.6a_{surge} + a_{pitch-x})\rho_L . \ell \\ (a_{heave} + a_{roll-z})\rho_L . h_s + (g\sin\emptyset + a_{roll-y})\rho_L . b\end{pmatrix}$$

where:

ρ_L = Density of liquid carried in the tank and ballast hold, in [t/m^3].

h_s = Vertical distance, in [m] from the load point to the top of the tank.

ℓ = The largest longitudinal distance in [m], from the load point to the tank corner at the top of the tank which is the furthermost away from the load point. For tank tops with stepped contour, the uppermost tank corner will normally be decisive.

b = The largest athwartship distance in [m], from the load point to the tank corner at the top of the tank which is the furthermost away from the load point. For tank tops with stepped contour, the uppermost tank corner will normally be decisive.

$\emptyset$ = Roll angle, in [deg], as defined in Cl.2.1.2.1.

θ = Pitch angle, in [deg], as defined in Cl.2.1.2.2.

a_{heave} = heave acceleration as given in Cl.2.1.3.3.

a_{surge} = surge acceleration given in Cl.2.1.3.1.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2]

$= a_{pitch}\,|(x - 0.45L)|$

$a_{pitch-x}$ = Longitudinal acceleration due to pitch, in [m/s^2]

$= a_{pitch}(z - R)$

a_{roll-y} = Transverse acceleration due to roll, in [m/s^2]

$= a_{roll}(z - R)$

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2]

$= a_{roll}\,|y|$

5.1.4 Static pressure in flooded conditions

5.1.4.1 Static pressure in flooded compartments

The internal pressure in flooded condition, P_{fs} in [kN/m^2], on watertight bulkheads is to be taken as:

$$P_{fs} = \rho\, g\, h_{fs} \text{ but not less than 0.}$$

where:

h_{fs} = Pressure height, in [m], in flooded condition, to be taken as:

$$h_{fs} = \max\begin{Bmatrix} Z_{fd} - Z \\ |Y| \sin\theta_{dam} + (Z_{dam} - Z)\cos\theta_{dam} \end{Bmatrix}$$

where,

Z_{fd} = Z coordinate, in [m], of the freeboard deck at side in way of the transverse section considered or the deepest equilibrium waterline in the damaged condition (T_{dam}) whichever is the greater.

Z_{dam} = Z coordinate, in [m], of the deepest equilibrium waterline at centre line in the damaged condition (or in intermediate stages of flooding).

θ_{dam} = Angle, in [deg], between the deepest equilibrium waterline in the damaged condition (or in intermediate stages of flooding) and the base line.

For ships where damage stability calculations are not applicable,

$$h_{fs} = Z_{fd} - Z$$

where,

Z_{fd} = Z coordinate, in [m], of the freeboard deck/bulkhead deck at side in way of the transverse section considered.

5.2 Non-exposed decks and platforms

5.2.1 Application

The loads defined in Cl.5.2.2 and Cl.5.2.3 are applicable to non-exposed decks, accommodation decks, wheelhouse decks and platforms in machinery space.

5.2.2 Pressure due to distributed load

If a distributed load is carried on a deck, the static and dynamic pressures due to this distributed load are to be considered. The dynamic distributed loads are to be calculated according to Section 4, Cl.4.2.3.1.

The static distributed load, P_{dl-s} in [kN/m^2], including self-weight are to be defined by the designer without being less than:

- 2.5 [kN/m^2] (0.25 [t/m^2] distributed mass) for accommodation decks and platforms in general

- 3.5 [kN/m^2] (0.35 [t/m^2] distributed mass) for wheelhouse deck

- 10 [kN/m^2] (1.0 [t/m^2] distributed mass) for platforms in machinery space.

- 7H [kN/m^2] (0.7H[1)] [t/m^2] distributed mass) for tween deck, tween deck hatch covers and inner bottom in way of cargo and stores

where,

H = Tween deck height or height in [m], measured up to the top of hatchway coaming above

[1)] Note:
In case of ships where due to large volumetric capacity of cargo spaces the stowage rate is less than 0.7 [t/m^3], a reduction in the value of distributed load may be considered, if requested for.

5.2.3 Concentrated force due to unit load

Concentrated forces on non-exposed decks are to be calculated according to Section 4, Cl.4.2.3.2 or Cl.4.2.3.3 as applicable.

5.3 Pressure for internal structures in tanks

5.3.1 The pressure, in [kN/m^2], for internal structures in tanks, e.g. web of primary supporting members, is to be taken as:

$$P_{int} = 12$$

5.4 Dry bulk cargo pressures

5.4.1 Total pressure

The total pressure due to dry bulk cargo acting on any load point of a cargo hold boundary, in [kN/m^2], is to be:

$P_{in} = P_{bs}$ for strength assessment of intact conditions for static (S) design load scenarios as given in Section 6.

$P_{in} = P_{bs} + P_{bd}$ for strength assessment of intact conditions and fatigue assessment for static (S) plus dynamic (S+D) design load scenarios as given in Section 6.

where:

P_{bs}= Static pressure due to dry bulk cargo in [kN/m^2], as defined in Cl.5.4.2.

P_{bd} = Dynamic inertial pressure due to dry bulk cargo for each dynamic load case in [kN/m^2], as defined in Cl.5.4.3.1 for L≥ 90[m] and Cl. 5.4.3.2 for L< 90[m].

5.4.2 Static pressure

The dry bulk cargo static pressure P_{bs}, in [kN/m^2], is to be taken as:

$$P_{bs} = \rho_c \, g \, K_c \, (z_B - z) \text{ but not less than 0.}$$

where,

ρ_c = Density of bulk cargo, in [t/m^3] for the loading condition considered.

K_c = Coefficient taken equal to:

= $\cos^2\alpha + \sin^2\alpha(1 - \sin\Psi)$ for inner bottom, hopper tank, transverse and longitudinal bulkheads, lower stool, vertical upper stool, inner side and side shell

= 0 for topside tank, main deck and sloped upper stool.

α = Angle, in [deg], between panel considered and the horizontal plane.

Ψ = Assumed angle of repose, in [deg], of bulk cargo (considered drained and removed); to be taken as:

= 20° for light bulk cargo (e.g. coal, grain)

= 25° for bulk cement cargo

= 35° for heavy bulk cargo (e.g. ore)

z_B = Height of the upper surface of the cargo above the baseline in way of the load point, in [m], to be taken as, Refer Figure 5.4.2:

$$z_B = h_{DB} + h_c$$

h_{DB} = Height in [m], of the double bottom at the centreline, measured at mid-length of the cargo hold.

h_c = Height of bulk cargo, in [m], from the inner bottom to the upper surface of bulk cargo.

$$h_c = \frac{\frac{M}{\rho_c \ell_h} + \frac{V_{LS}}{\ell_h} + (h_{HT} - h_{DB}) b_{HT}}{2 Y_{HT}}$$

where:

M = Mass of cargo, in [t], in the hold depending on the loading condition considered.

ℓ_h = Length in [m], of the hold to be taken as the longitudinal distance between the transverse bulkheads which form boundaries of the hold considered.

V_{LS} = Volume in [m^3], of the transverse bulkhead lower stool (above the inner bottom), to be taken equal to zero in the case of bulkheads fitted without lower stool.

h_{HT} = Height in [m], of the hopper tank to be taken as the vertical distance from the baseline to the top of the hopper tank.

b_{HT} = Breadth in [m], of the hopper tank to be taken as the transverse distance from the outermost double bottom girder to the outermost point of the hopper tank.

Y_{HT} = Half breadth in [m], of the hold measured at the middle of ℓ_h and at a vertical level corresponding to the top of the hopper tank.

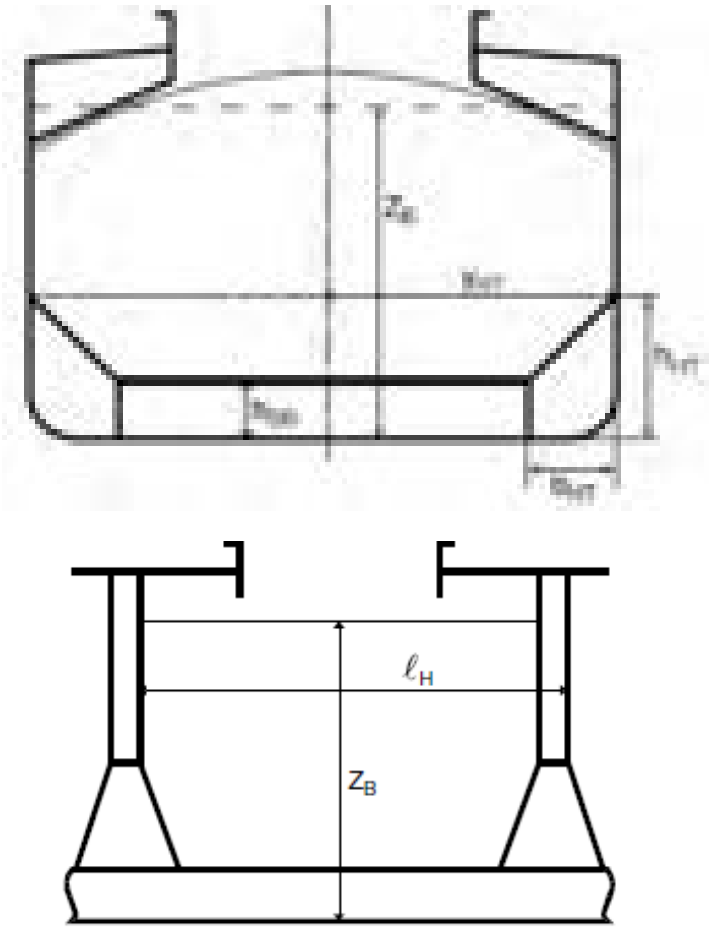


Figure 5.4.2: Rated upper surface of the bulk cargo

5.4.3 Dynamic dry bulk cargo pressure

5.4.3.1 The dynamic dry bulk cargo pressure P_{bd}, in [kN/m^2], for each load case is to be taken as:

$$P_{bd} = f_\beta \, \rho_c \, a_z \, K_c \, f_{dc} \, (z_B - z) \quad \text{for } z \leq z_B$$

$$P_{bd} = 0 \quad \text{for } z > z_B$$

where,

f_{dc} = Dry cargo factor:

= 1.0 for strength assessment

= 0.5 for fatigue assessment

For vessels designed to carry heavy bulk cargoes which are also required to carry lighter cargoes, the pressure 'p' based on maximum mass of cargo to be carried in the hold and filled up to the top of hatch coaming would also require to be considered.

5.4.3.2 Dynamic dry bulk cargo pressure for ships with L < 90[m]

The dynamic dry bulk cargo pressure P_{bd}, in [kN/m^2], for ships with L < 90[m], is to be taken as:

$$P_{bd} = \rho_c \, a_v \, K_c \, f_{dc} \, (z_B - z) \quad \text{for } z \leq z_B$$

$$P_{bd} = 0 \quad \text{for } z > z_B$$

where,

a_v = Combined vertical heave acceleration as given in Cl.2.2.3.3.

$a_{pitch-z}$ = Vertical acceleration due to pitch, in [m/s^2]

$= a_{pitch} \, |(x - 0.45L)|$

a_{roll-z} = Vertical acceleration due to roll, in [m/s^2]

$= a_{roll} \, |y|$

Section 6

Design Load Scenarios

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

VBM = Design vertical bending moment, in [kN-m].

M_{sw-h}, M_{sw-s} = Permissible hull girder hogging and sagging vertical still water bending moment in seagoing condition, in [kN-m], as defined in Section 3, Cl.3.4.2.

M_{sw-har} = Permissible hull girder hogging and sagging still water bending moment for harbour operation, in [kN-m], as defined in Section 3, Cl.3.4.3.

M_{wv} = Vertical wave bending moment in seagoing condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Section 3, Cl.3.6.1.

M_{wv-LC} = Vertical wave bending moment for a considered dynamic load case, in [kN-m], as defined in Section 3, Cl.3.7.2.

HBM = Design horizontal bending moment, in [kN-m].

M_{wh-LC} = Horizontal wave bending moment for a considered dynamic load case, in [kNm] as defined in Section 3, Cl.3.7.4.

M_{wT-LC} = Wave torsional moment for a considered dynamic load case, in [kN-m], as defined in Section 3, Cl.3.7.5.

VSF = Design vertical shear force, in [kN].

Q_{sw} = Permissible hull girder positive and negative still water shear force limits for seagoing operation, in [kN], as defined in Section 3, Cl.3.5.1.

Q_{sw-har} = Permissible hull girder positive and negative still water shear force limits for harbour operation, in [kN], as defined in Section 3, Cl.3.5.2.

Q_{wv-LC} = Vertical wave shear force for a considered dynamic load case, in in [kN] as defined in Section 3, Cl.3.7.3.

P_{ex} = Design external pressure, in [kN/m^2].

P_s = Static sea pressure at considered draught, in [kN/m^2], as defined in Section 4.

P_w = Dynamic pressure for a considered dynamic load case, in [kN/m^2], as defined in Section 4.

P_D = Green sea load for a considered dynamic load case, in [kN/m^2], as defined in Section 4.

P_{in} = Design internal pressure, in [kN/m^2], as defined in Section 5.

P_{in-f} = Design internal pressure in flooded condition, in [kN/m^2], as defined in Section 5.

P_{ls-ST} = Tank testing pressure, in [kN/m^2], as defined in Section 5.

P_{int} = Minimum pressure for internal structures in tanks as given in Section 5.

P_{ls-1} = Static tank pressure during normal operations at sea as given in Section 5.

P_{ls-2} = Static tank pressure during normal operations at harbour as given in Section 5.

P_{ls-3} = Static tank pressure during flow through ballast water exchange as given in Section 5.

P_{ls-4} = Static tank pressure during sequential ballast water exchange as given in Section 5.

P_{ld} = Dynamic liquid pressure in tank for a considered dynamic load case, in [kN/m^2], as defined in Section 5.

P_{fs} = Static pressure in compartments and tanks in flooded condition, in [kN/m^2], as defined in Section 5.

P_{dl-s} = Static pressure on non-exposed decks and platforms, in [kN/m^2], as defined in Section 5.

P_{dl-d} = Dynamic pressure on non-exposed decks and platforms for a considered dynamic load case, in [kN/m^2], as defined in Section 5.

P_{bs} = Static pressure due to dry bulk cargo in [kN/m^2], as defined in Section 5.

P_{bd} = Dynamic inertial pressure due to dry bulk cargo for each dynamic load case in [kN/m^2], as defined in Section 5.

F_{U-s} = Static load acting on supporting structures and securing systems for heavy units or cargo, equipment or structural components, in [kN], as defined in Section 4.

F_{U-d} = Dynamic load acting on supporting structures and securing systems for heavy units of cargo, equipment or structural components, in [kN], as defined in Section 4.

6.1 Application

6.1.1 This section gives the design load scenarios that are to be used for:

- Strength assessment by prescriptive and direct analysis (Finite Element Method, FEM) methods, as given in Cl.6.2.1.

- Fatigue assessment by prescriptive and direct analysis (FEM) methods, as given in 6.3.

6.1.2 For the strength assessment, the principal design load scenarios consist of either S (Static) loads or S+D (Static + Dynamic) loads. In some cases, the letter 'A' prefixes the S or S+D to denote that this is an accidental design load scenario. There are some additional design load scenarios to be considered which relate to impact (I) loads, sloshing (SL) loads and fatigue (F) load. Design load scenarios for impact loads (I) are given in Chapter 11, Section 2, Section 3 and Section 4. Design load scenarios for sloshing loads in tanks (SL) are given in Chapter 11, Section 7.

6.2 Design load scenarios for strength assessment

6.2.1 Principal design load scenarios

6.2.1.1 The principal design load scenarios are given in Table 6.2.1.1 for ships with L ≥ 90[m].

Table 6.2.1.1: Principal design load scenarios

	Design Load scenarios				
	1	**2**	**3**	**4**	**5**
Load Component	**Normal operation at sea**	**Normal operation at harbour and Tank testing**	**Flow through ballast water exchange [4)]**	**Sequential ballast water exchange [4)]**	**Flooding [4)]**
	Static + Dynamic (S+D)	Static (S)	Static + Dynamic (S+D)	Static + Dynamic (S+D)	Accidental (A)
Hull Girder Loads					
VBM	$M_{sw} + M_{wv-LC}$	M_{sw-har}	$M_{sw} + M_{wv-LC}$	$M_{sw} + M_{wv-LC}$	M_{sw}
HBM	M_{wh-LC}	-	M_{wh-LC}	M_{wh-LC}	-
VSF	$Q_{sw} + Q_{wv-LC}$	Q_{sw-har}	$Q_{sw} + Q_{wv-LC}$	$Q_{sw} + Q_{wv-LC}$	-
TM [2)]	M_{WT-LC}	-	M_{WT-LC}	M_{WT-LC}	-

Local Loads - P_{ex}					
Exposed decks	P_D	-	-	-	-
External shell	$P_s + P_w$	P_s	$P_s + P_w$	$P_s + P_w$	-
Superstructure sides	Max (P_w, P_{sl})	-	-	-	-
Superstructure end bulkheads and deckhouse walls	P_A	-	-	-	-
Local Loads - P_{in}					
Boundaries of water ballast tanks[1)]	$P_{ls-1} + P_{ld}$	Max (P_{ls-2}, P_{sl-ST})	$P_{ls-3} + P_{ld}$	$P_{ls-4} + P_{ld}$	
Boundaries of tanks other than water ballast tanks			-	-	
Watertight boundaries	-	-	-	-	P_{in-f}
Boundaries of bulk cargo holds	$P_{bs} + P_{bd}$	P_{bs}	-	-	-
Internal structures in tanks	-	P_{int}	-	-	-
Local Loads - P_{dl}					
Exposed decks and non-exposed decks and platforms	$P_{dl-s} + P_{dl-d}$	P_{dl-s}	-	-	-
Heavy units on internal and external decks	$F_{U-s} + F_{U-d}$	F_{U-s}	-	-	-
Decks and hatch covers/ Ro-Ro equipment	P_{wl-1}[3)]	P_{wl-2}[3)]	-	-	-

1) WB cargo hold is considered as ballast tank except for design load scenario "ballast water exchange".
2) Hull girder torsion to be considered for ships with large deck openings only.
3) Wheel load to be considered as per Chapter 11.
4) Applicable to prescriptive assessment only.

6.2.1.2 For ships with L < 90[m] the design load scenarios given in Table 6.2.1.2 may be used as an alternative to those given in Table 6.2.1.1.

Table 6.2.1.2: Principal design load scenarios for ships with L < 90[m]				
	Design Load scenarios			
	1	**2**	**4**	**5**
Load Component	**Normal operation at sea**	**Normal operation at harbour and Tank testing**	**Sequential ballast water exchange 2)**	**Flooding 2)**
	Static + Dynamic (S+D)	Static (S)	Static (S)	Accidental (A)
VBM	$M_{sw} + M_{wv}$	M_{sw}	$M_{sw} + M_{wv}$	M_{sw}
Exposed decks	P_D	-	-	-
External shell	$P_s + P_w$	P_s	$P_s + P_w$	-
Superstructure sides	Max (P_w, P_{sl})	-	-	-
Superstructure end bulkheads and deckhouse walls	P_A	-	-	-
Boundaries of water ballast tanks	$P_{ls-1} + P_{ld}$	Max (P_{ls-2}, P_{sl-st})	$P_{ls-4} + P_{ld}$	-
Boundaries of tanks other than water ballast tanks			-	-
Watertight boundaries	-	-	-	P_{fs}
Boundaries of bulk cargo holds	$P_{bs} + P_{bd}$	P_{bs}	$P_{bs} + P_{bd}$	-
Internal structures in tanks	-	P_{int}	-	-
Exposed decks and non-exposed decks and platforms	$P_{dl-s} + P_{dl-d}$	P_{dl-s}	-	-
Heavy units on internal and external decks	$F_{U-s} + F_{U-d}$	F_{U-s}	-	-
Decks and hatch covers/ Ro-Ro equipment	P_{wl-1} 1)	P_{wl-2} 1)	-	-

1) Wheel load to be considered as per Chapter 11.
2) Applicable to prescriptive assessment only.

6.3 Design load scenarios for fatigue assessment

The principal design load scenarios are given in Table 6.3.1.

6.3.1 Principal design load scenarios

Table 6.3.1: Principal design load scenarios for fatigue assessments

Design Load scenarios / **Load Components**	**Fatigue: Static + Dynamic (F: S+D)**
Hull Girder Loads	
VBM	$M_{sw} + M_{wv-LC}$
HBM	M_{wh-LC}
VSF	$Q_{sw} + Q_{wv-LC}$
TM [2)]	M_{WT-LC}
Local Loads - P_{ex}	
Exposed decks	--
External shell	$P_s + P_w$
Local Loads - P_{in}	
Boundaries of water ballast tanks[1)]	$P_{ls} + P_{ld}$
Boundaries of tanks other than water ballast tanks	
Watertight boundaries	
Boundaries of bulk cargo holds	$P_{bs} + P_{bd}$
Local Loads - P_{dl}	
Exposed decks and non-exposed decks and platforms	--
External deck for distributed loads	--
Heavy units on internal and external decks	--

1) WB cargo hold is considered as ballast tank except for design load scenario "ballast water exchange".
2) Hull girder torsion to be considered for ships with large deck openings only.

Section 7

Loading Conditions

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

7.1 Loading Conditions

7.1.1 Definition

7.1.1.1 Departure conditions

The departure conditions are to be based on bunker tanks not taken less than 95% full and other consumables taken at 100% capacity.

7.1.1.2 Arrival conditions

The arrival conditions are to be based on 10% of the maximum capacity of bunker, fresh water and stores.

7.1.2 General

Still water bending moments, M_{sw} [kN-m], and still water shear forces, Q_{sw} [kN], are to be calculated and submitted along the ship length for design cargo and ballast loading conditions as specified in Cl.2.3.3 together with the distribution of lightship weights.

For these calculations, downward loads are assumed to be taken as positive values, and are to be integrated in the forward direction from the aft end of L. The sign conventions of M_{sw} and Q_{sw} to be followed as given in Chapter 5, Section 3, 3.2.

All the design loading conditions are to be evaluated for at least departure and arrival conditions.

The severest loading condition from the loading manual, midship section drawing or otherwise specified by the Designer are to be considered for the longitudinal strength and for local strength check of plating, ordinary stiffeners and primary supporting members.

7.1.3 Seagoing Conditions

In general, the following loading conditions, based on amounts of bunkers, fresh water and stores at departure and arrival, are to be considered for the calculations of M_{sw} and Q_{sw}.

Where the amount and disposition of consumables at any intermediate stage of the voyage are considered more severe, calculations for such intermediate conditions are to be submitted in addition to those for departure and arrival conditions. Also, where any ballasting and/or deballasting is intended during voyage, calculations of the intermediate condition just before and just after ballasting and/or deballasting any ballast tank are to be submitted and where approved, included in the loading manual for guidance.

a) General cargo ships, Container Ships, passenger ships, roll on-roll off ships and refrigerated cargo carriers, Ore Carriers and Bulk Carriers:

- Homogeneous loading conditions at maximum draught
- Ballast conditions
- Special loading conditions, e.g. container or light load conditions at less than the maximum draught, heavy cargo, empty holds or non-homogeneous cargo conditions deck cargo conditions, etc., where applicable
- Short voyage or harbour conditions, where applicable
- Docking condition afloat
- Loading and unloading transitory conditions, where applicable

b) Oil Tankers:

- Homogeneous loading conditions (excluding dry and clean ballast tanks) and ballast or part-loaded conditions for both departure and arrival
- Any specified non-uniform distribution of loading

- Mid-voyage conditions relating to tank cleaning or other operations where these differ significantly from the ballast conditions

- Docking condition afloat

- Loading and unloading transitory conditions

c) Chemical Tankers:

- Conditions as specified for oil tankers

- Conditions for high density or heated cargo and segregated cargo where these are included in the approved cargo list

d) Liquefied Gas Carriers:

- Homogeneous loading conditions for all approved cargoes for both arrival and departure

- Ballast conditions for both arrival and departure

- Cargo condition where one or more tanks are empty or partially filled or where more than one type of cargo having significantly different densities is carried, for both arrival and departure

- Harbour condition for which an increased vapour pressure has been approved

- Docking condition afloat

e) Combination Carriers:

- Conditions as specified for oil tanker and cargo ships.

f) All ships:

- Any other loading condition likely to result in high bending moments and/or shear forces (including docking conditions, as appropriate).

7.1.4 Partially Filled Ballast Tanks in seagoing condition

For permissible use of partially filled ballast tanks in design loading conditions, Refer Cl.7.1.4.1, Cl.7.1.4.2, Cl.7.1.4.3.

7.1.4.1 Partially filled ballast tanks in ballast loading condition

(i) Ballast loading conditions involving partially filled peaks and/or other ballast tanks at departure, arrival or during intermediate conditions are not to be used as design conditions unless:

- permissible bending and shear stresses are not exceeded for all filling levels between empty and full; and

- for bulk carriers, the requirement for longitudinal strength of hull girder in flooded conditions as per Part.5, Chapter 1, Section 2.2 are complied with for all filling levels between empty and full.

(ii) To demonstrate the strength compliance with all filling levels between empty and full, it will be acceptable if, the still water bending moment, shear force and torsional moment, if applicable, are calculated and shown within the relevant permissible limits. These values are to be obtained for each condition at departure, arrival and where required by Cl.7.1.3 at any intermediate condition, with the tanks intended to be partially filled are assumed to be:

- empty
- full
- partially filled at intended level.

(iii) Where multiple tanks are intended to be partially filled, all combinations of empty, full or partially filled at intended level for those tanks are to be investigated.

(iv) However, for conventional ore carriers with large wing water ballast tanks in the cargo area, where empty or full ballast water filling levels of one or maximum two pairs of these tanks lead to the ship's trim exceeding one of the following conditions, it is sufficient to demonstrate compliance with maximum, minimum and intended partial filling levels of these one or maximum two pairs of ballast tanks such that the ship's condition does not exceed any of these trim limits. Filling levels of all other wing ballast tanks are to be considered between empty and full. The trim conditions mentioned above are:

- Trim by stern of 3% of the ship's length, or

- Trim by bow of 1.5% of the ship's length, or

- Any trim that cannot maintain propeller immersion (I/D) not less than 25%, where:

I = distance from propeller centerline to the waterline

D = propeller diameter. Refer Figure 7.1.4.

The maximum and minimum filling levels of the above mentioned pairs of wing water ballast tanks are to be indicated in the loading manual.

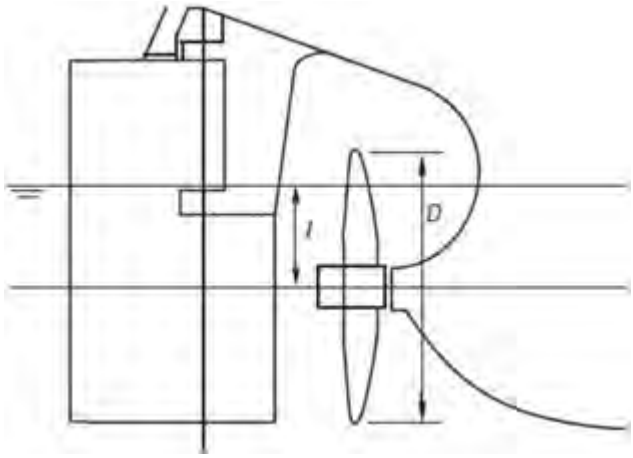


Figure 7.1.4: Propeller immersion

7.1.4.2 Partially filled ballast tanks in cargo loading condition

In cargo loading conditions, the requirements given in Cl.7.1.4.1 above needs to be applied to peak ballast tanks only.

7.1.4.3 Partially filled ballast tanks in Sequential ballast water exchange condition

Requirements of Cl.7.1.4.1 and Cl.7.1.4.2 above are not applicable to ballast water exchange using the sequential method. However, bending moment and shear force calculations for each deballasting or ballasting stage in the ballast water exchange sequence are to be included in the loading manual or ballast water management plan of any vessel that intends to employ the sequential ballast water exchange method.

7.1.5 Harbour conditions

The following harbour conditions are to be included in the loading manual:

- Conditions representing typical complete loading and unloading operations.

- Docking condition afloat.

- Propellers inspection afloat condition, in which the propeller shaft centreline is at least D/4 above the waterline in way of the propeller. Ships with podded propulsion system arrangements are to be specially considered.

Section 8

Guidelines for Ballast Loading Conditions of Cargo Vessels Involving Partially Filled Ballast Tanks

8.1. General

8.1.1 This Section is intended for guidance and interpretation of Cl.7.1.4.1 i.e. partially filled ballast tanks in ballast loading conditions.

8.1.2 Case A and B are generally applicable for ballast loading conditions for any cargo vessel which might have one Ballast Water (BW) Tank (or one pair of BW Tanks) partially filled.

8.1.3 Case C is showing the conditions necessary for checking longitudinal strength for a conventional ore carrier with two pairs of large wing water ballast tanks partly filled during the ballast voyage.

8.1.4 Where applicable, similar considerations are to be given to other cargo vessels covered where ballast loading conditions involving partially filled ballast tanks may cause concerns for the longitudinal strength of the vessels.

8.1.5 This Section does not apply to CSR Bulk Carriers and Oil Tankers or to Container Ships.

8.1.6 In the Figures, the conditions only intended for strength verification (not operational) are marked with a star (*).

8.2 Case A and B

8.2.1 Case A

Figure 1 shows Case A, with a cargo vessel where partial filling of BW Tank no. 6 (P/S) is permitted and may take place at any time during the ballast voyage. Intermediate condition(s) should be specified as shown in the Figure, however filling/partial filling of BW Tank no. 6 (P/S) may be done at any step to keep acceptable trim and propeller immersion during the ballast voyage.

To obtain full operational flexibility regarding the filling level of BW Tank no. 6 (P/S), loading conditions A2 (full at departure)* and A8 (empty at arrival)* are to be added for strength verification.

Additional conditions (full and empty BW Tank no. 6 (P/S)) related to the intermediate conditions A3-A6 are not necessary as A2* and A8* will be the most critical one.

8.2.2 Case B

Figure 2 shows Case B, with a cargo vessel where partial filling of BW Tank no. 6 (P/S) to a given level (f6-int%) will be done after a specified % consumables is reached, see conditions B2 and B3. Before this % consumables (shown as 50% in this Figure) is reached, BW Tank no. 6 (P/S) shall be kept empty. When reaching a given level of consumables (shown as 20% in Figure 2), BW Tank no. 6 (P/S) are to be kept full, see conditions B5 and B6. Two additional intermediate conditions (B4* and B7*) are to be added for longitudinal strength verification.

In order to categorize a vessel according to Case B, clear operational guidance for partial filling of ballast tanks, in association with the consumption level as shown in Figure 2, is to be given in the loading manual. If such operational guidance is not given, Case A is to be applied.

Case A has no limitation of consumables, whereas Case B has limitation of consumables.

8.3 Case C – Conventional (with usual arrangement of WBT) ore carrier with two pairs of partially filled ballast water tanks.

8.3.1 Figure 3(a) show the operational loading conditions, departure condition (C1), four intermediate conditions (C2-C5) and arrival condition (C6), for a conventional (with usual arrangement of WBT) ore carrier with partial filling of both BW tank no.1 (P/S) and 7 (P/S) during voyage. Refer Table 8.3.1.

8.3.2 Figure 3(b) and Figure 3(c) show the additional twelve loading conditions (C1-1 ~ C1-12) which are to be added for longitudinal strength verification of the departure condition (C1).

8.3.3 Figure 3(d) to Figure 3(i) show the additional 32 loading conditions (C2-1 ~ C2-12, C3-1 ~ C3-4, C4-1 ~ C4-12 and C5-1 ~ C5-4) which are to be added for longitudinal strength verification of the intermediate conditions (C2 ~ C5).

8.3.4 Figure 3(j) and Figure 3(k) show the additional twelve loading conditions (C6-1 ~ C6-12) which are to be added for longitudinal strength verification of the arrival condition (C6).

8.3.5 For the additional loading conditions, the maximum and the minimum filling level of BW tank are according to trim and propeller immersion limitations indicated in Cl.7.1.4.1:

Table 8.3.1: Filling level in partially filled BW tanks nos.1 (P/S) and 7 (P/S) for the operational conditions during ballast voyage.

Loading condition	Consumables	Filling level, WBT 1(P/S)	Filling level, WBT 7(P/S)
C1- Departure	100%	f_{1dep}%	f_{7dep}%
C2– Intermediate 1	50% [(i)]	f_{1dep}%	f_{7dep}%
C3 – Intermediate 2	50% [(i)]	f_{1int}%	f_{7int}%
C4 – Intermediate 3	20% [(i)]	f_{1int}%	f_{7int}%
C5 – Intermediate 4	20% [(i)]	f_{1arr}%	f_{7arr}%
C6 - Arrival	10% [(i)]	f_{1arr}%	f_{7arr}%

Note:
[(i)] % consumables to be specified, indicated to 50% and 20 %

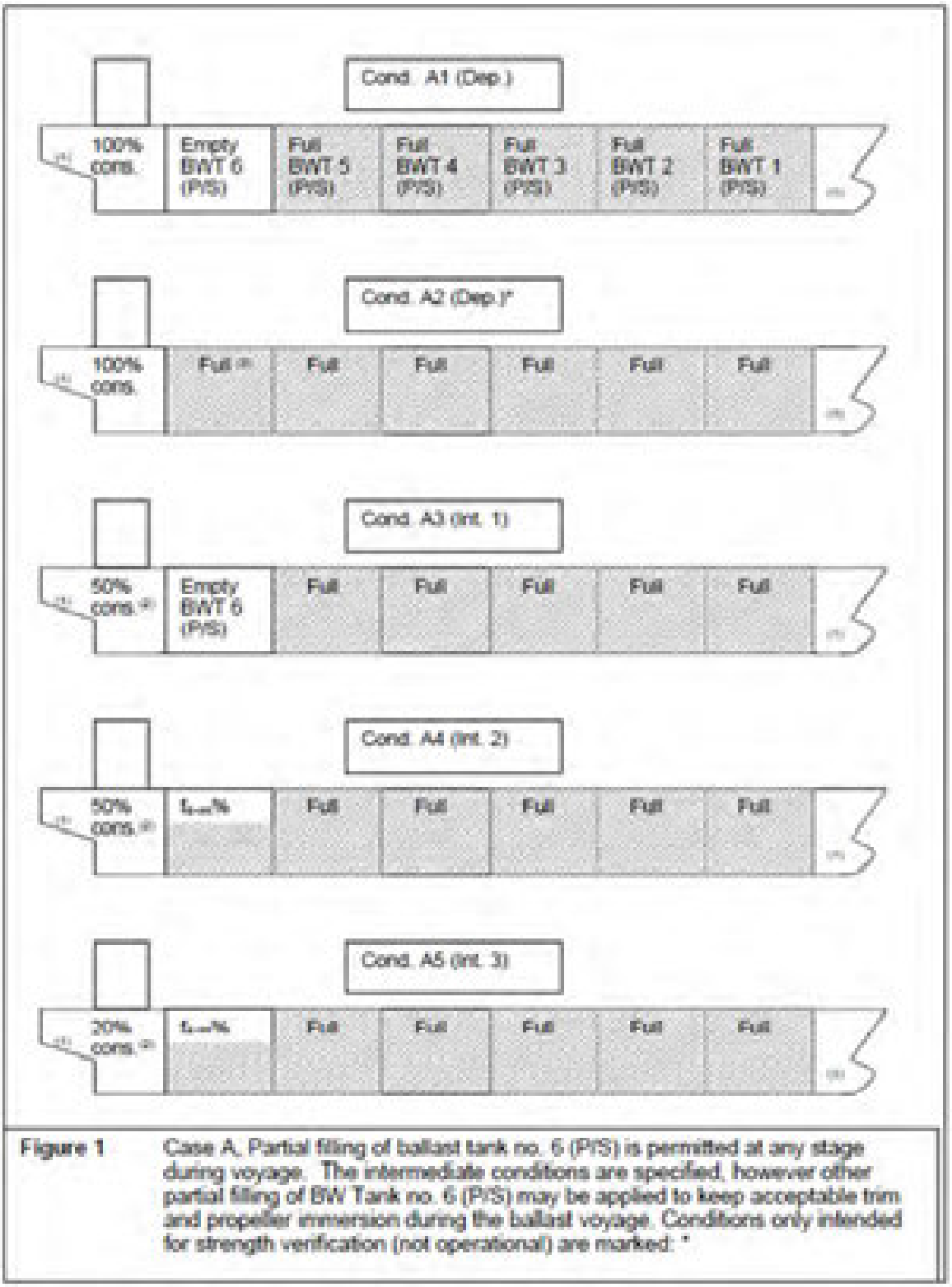
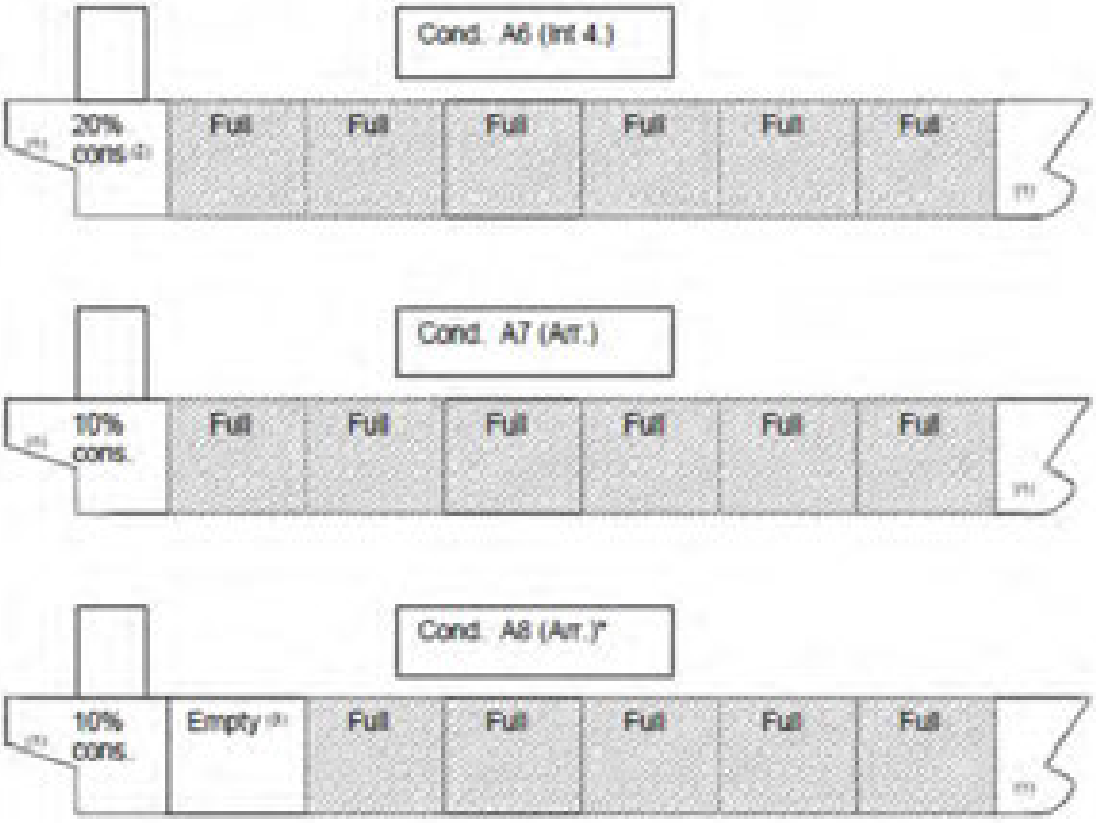


Figure 1 Case A, Partial filling of ballast tank no. 6 (P/S) is permitted at any stage during voyage. The intermediate conditions are specified, however other partial filling of BW Tank no. 6 (P/S) may be applied to keep acceptable trim and propeller immersion during the ballast voyage. Conditions only intended for strength verification (not operational) are marked: *



Notes

(1) For peak tanks intended to be partially filled, all combinations of full or partially filled at intended level for those tanks are to be investigated.

(2) The intermediate condition(s) to be specified incl. % consumables.

(3) For bulk carriers carrying ore and with large wing water ballast tanks full/empty may be replaced with maximum/minimum filling levels according to trim limitations given in 2.1.3.

Figure 1 (Continued)
Case A, Partial filling of ballast tank no. 6 (P/S) is permitted at any stage during voyage. The intermediate condition is specified, however other partial filling of BW Tank no. 6 (P/S) may be applied to keep acceptable trim and propeller immersion during the ballast voyage. Conditions only intended for strength verification (not operational) are marked: *

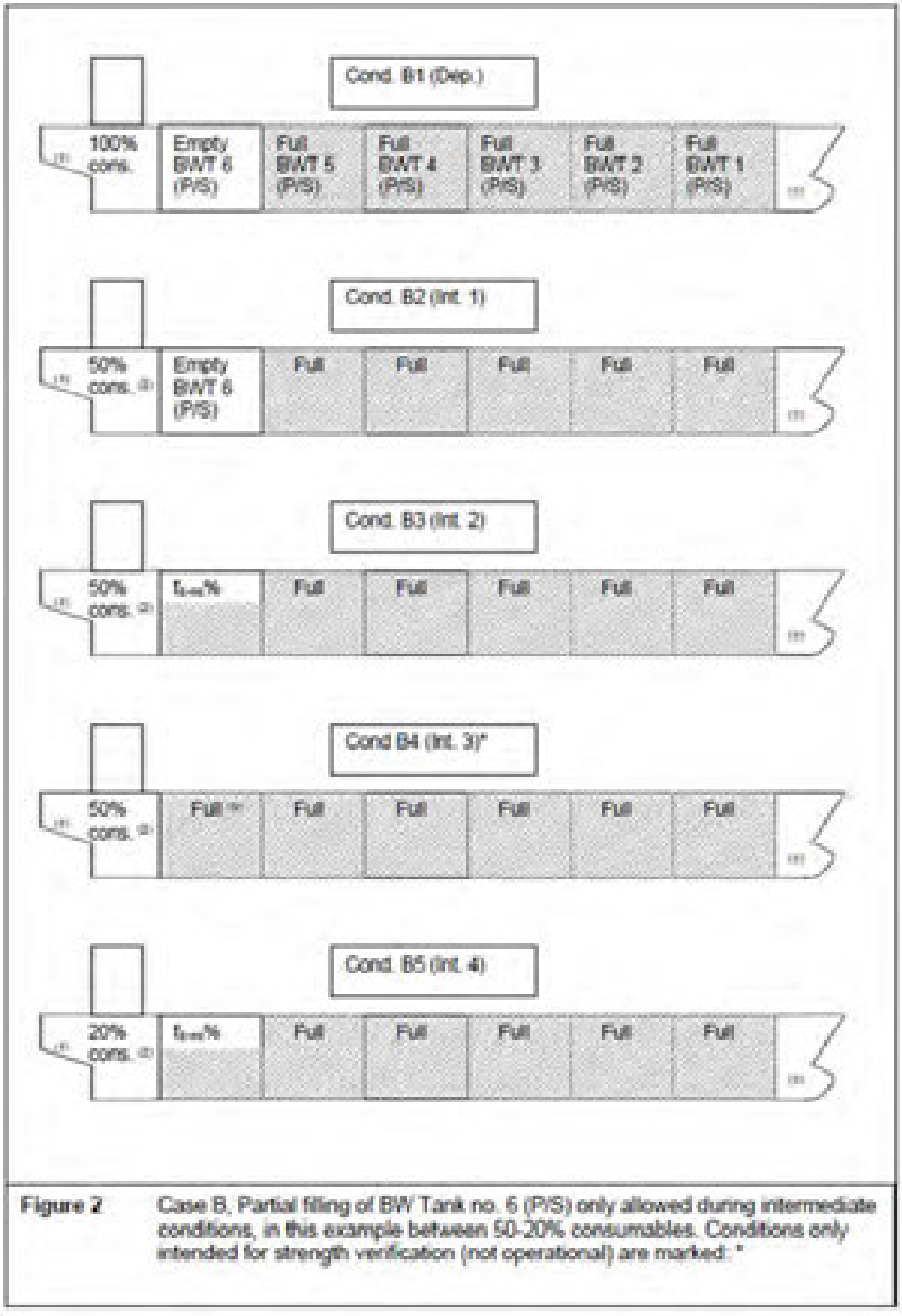
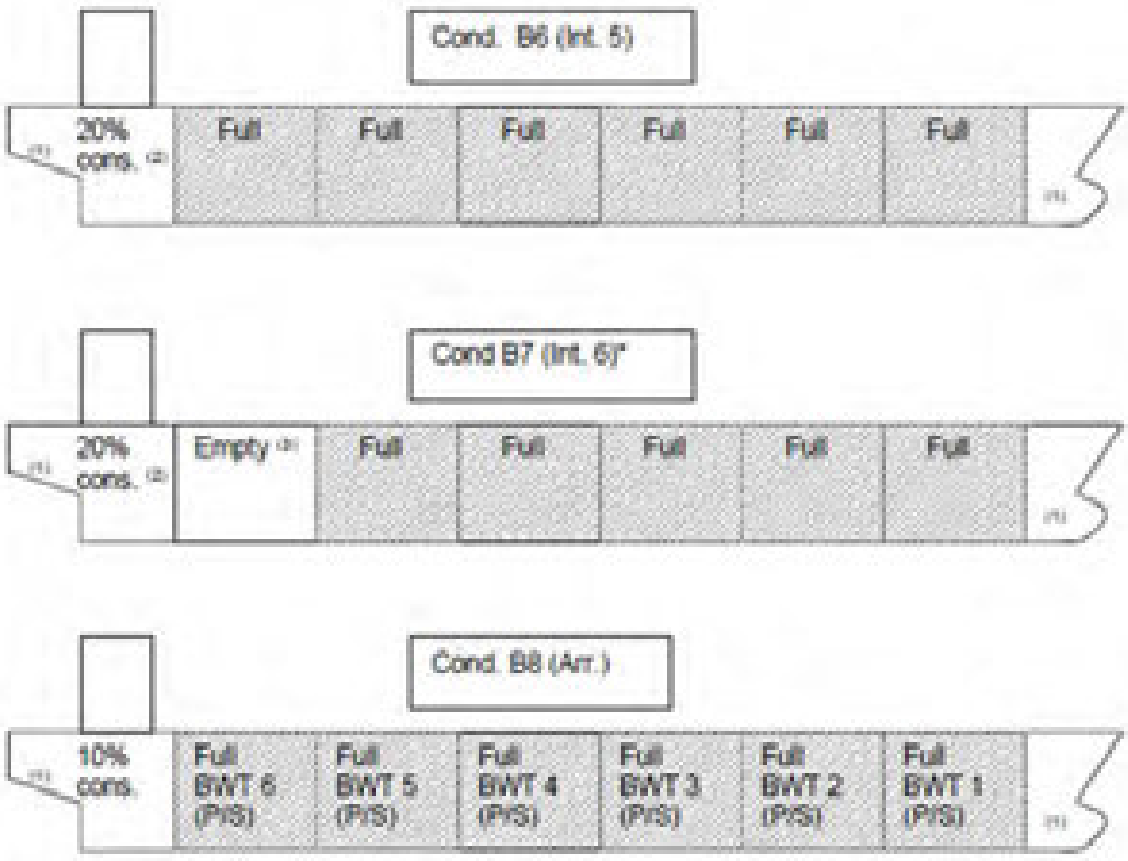


Figure 2 Case B, Partial filling of BW Tank no. 6 (P/S) only allowed during intermediate conditions, in this example between 50-20% consumables. Conditions only intended for strength verification (not operational) are marked: *



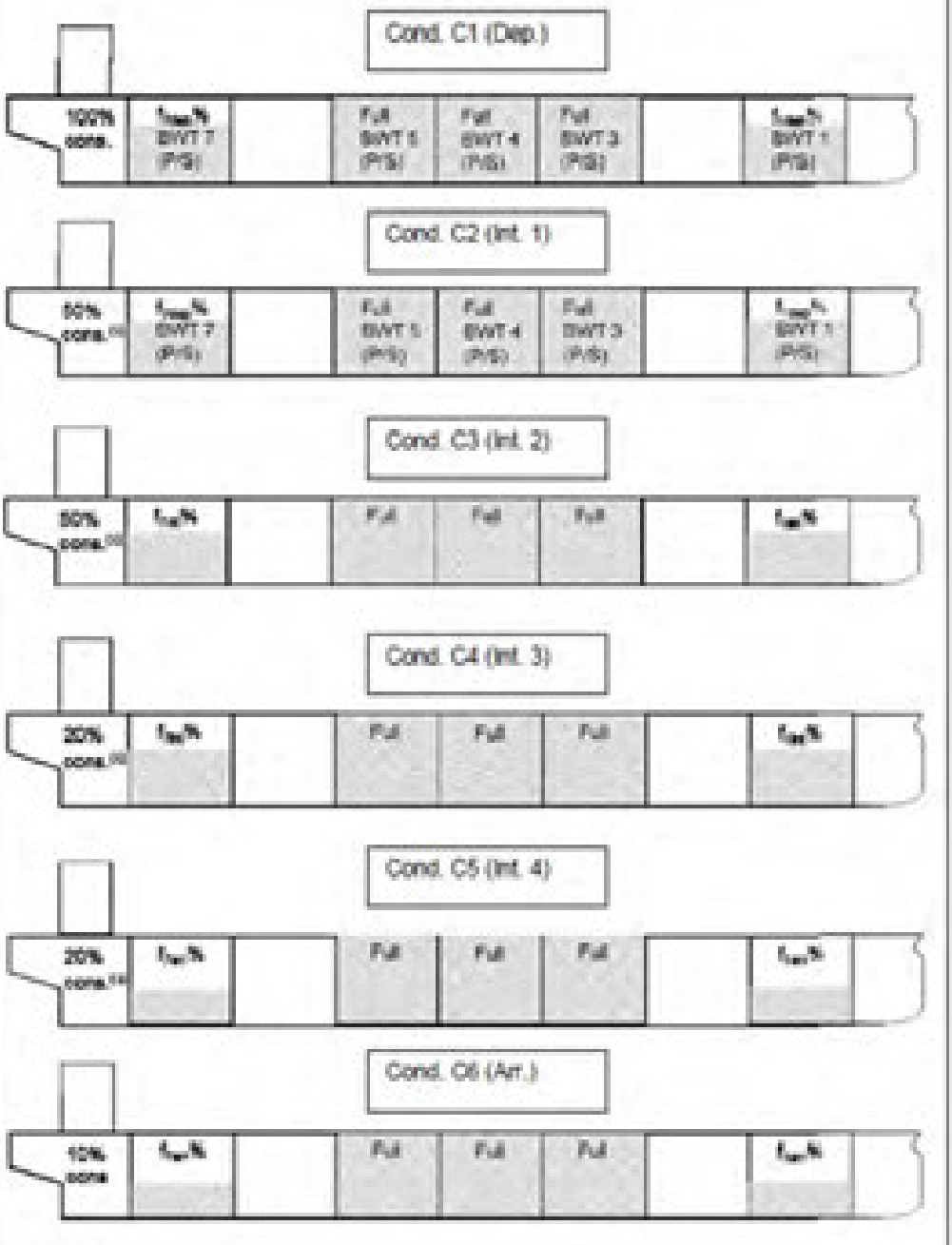
Notes

(1) For peak tanks intended to be partially filled, all combinations of full or partially filled at intended level for those tanks are to be investigated.

(2) The intermediate condition(s) to be specified incl. % consumables.

(3) For bulk carriers carrying ore and with large wing water ballast tanks full/empty may be replaced with maximum/minimum filling levels according to trim limitations given in 2.1.3.

Figure 2 (Continued)
Case B, Partial filling of BW Tank no. 6 (P/S) only allowed during intermediate conditions, in this example between 50-20% consumables. Conditions only intended for strength verification (not operational) are marked: *



Notes
(1) The intermediate condition(s) to be specified incl. % consumables.
(2) Figures 3(b)-3(k): Maximum and minimum filling level of BW tank according to trim and propeller immersion limitations given in 2.1.3.

Figure 3(a) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during ballast voyage, operational conditions C1-C6.

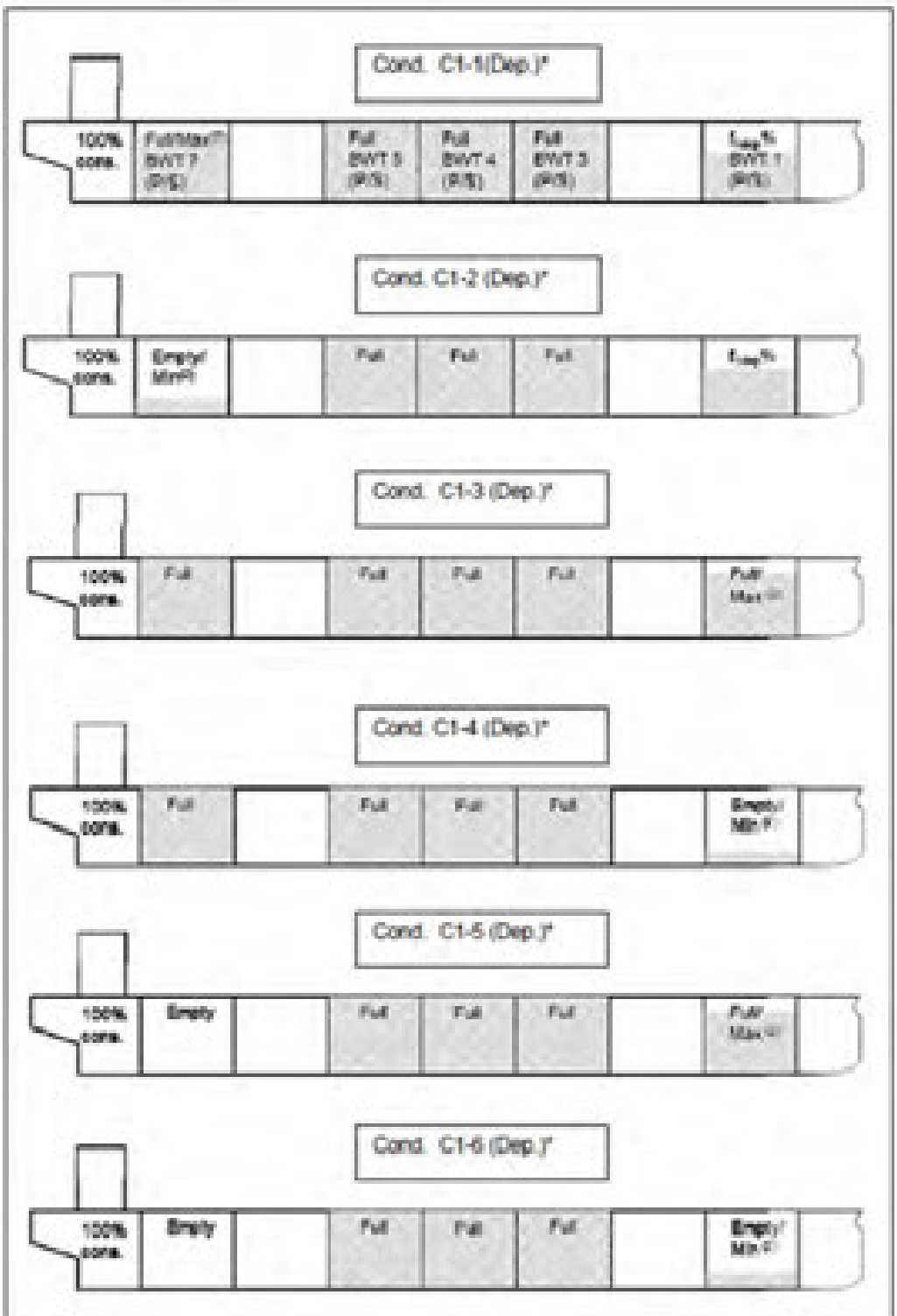


Figure 3(b) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Departure conditions C1-1–C1-6, only intended for strength verification (not operational) are marked: *

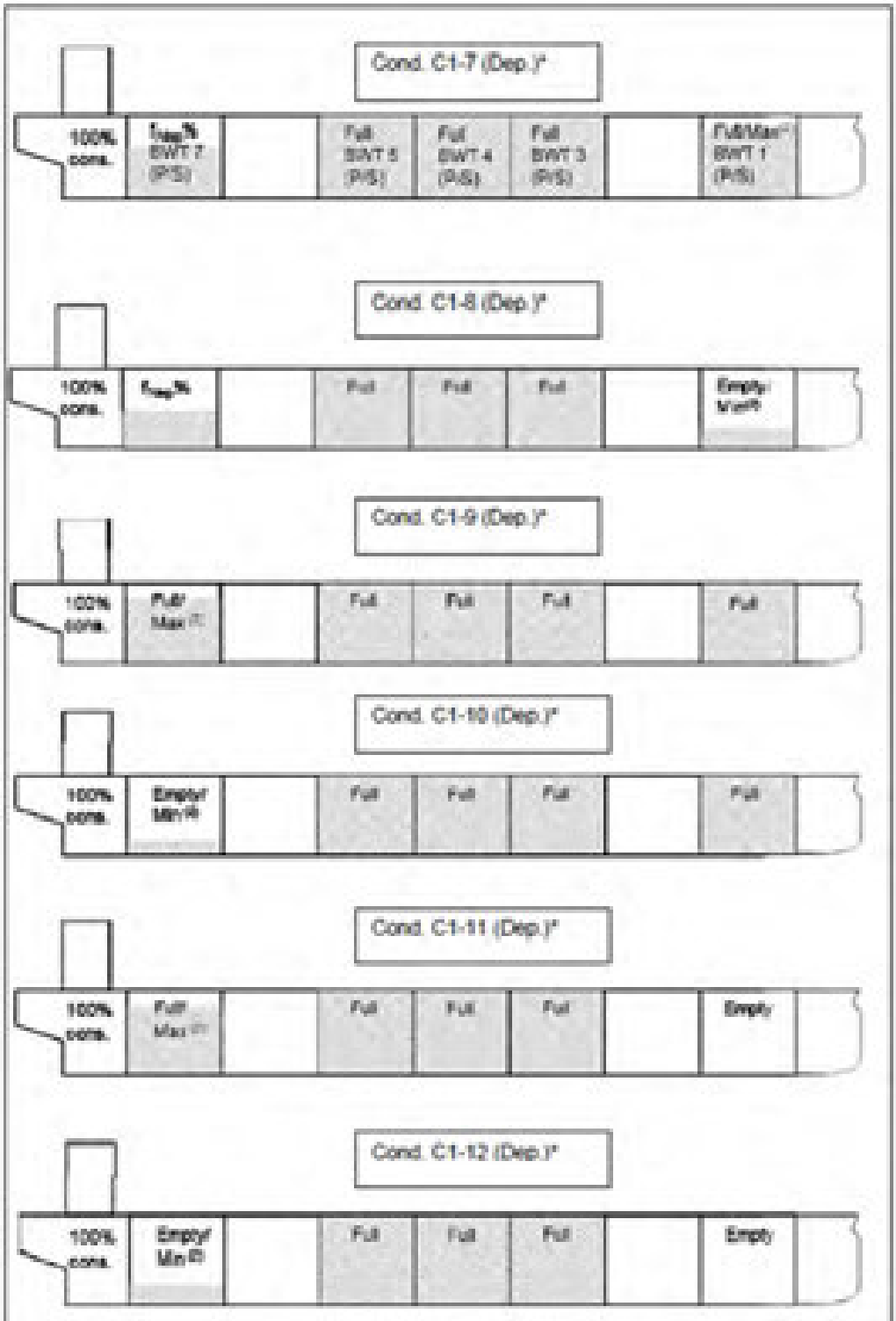


Figure 3(c) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Departure conditions CD1-7–C1-12, only intended for strength verification (not operational) are marked: *

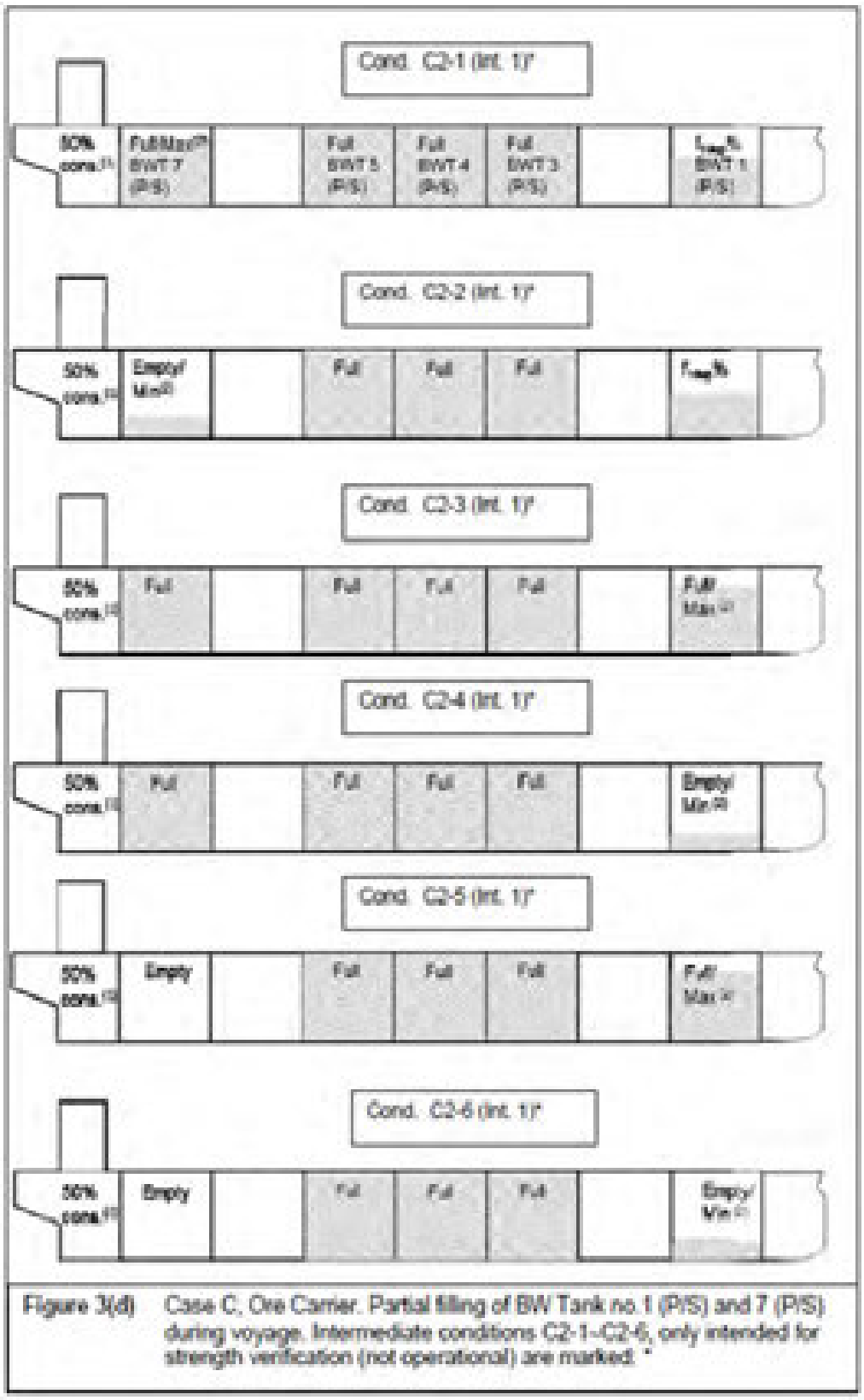


Figure 3(d) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Intermediate conditions C2-1-C2-6, only intended for strength verification (not operational) are marked. *

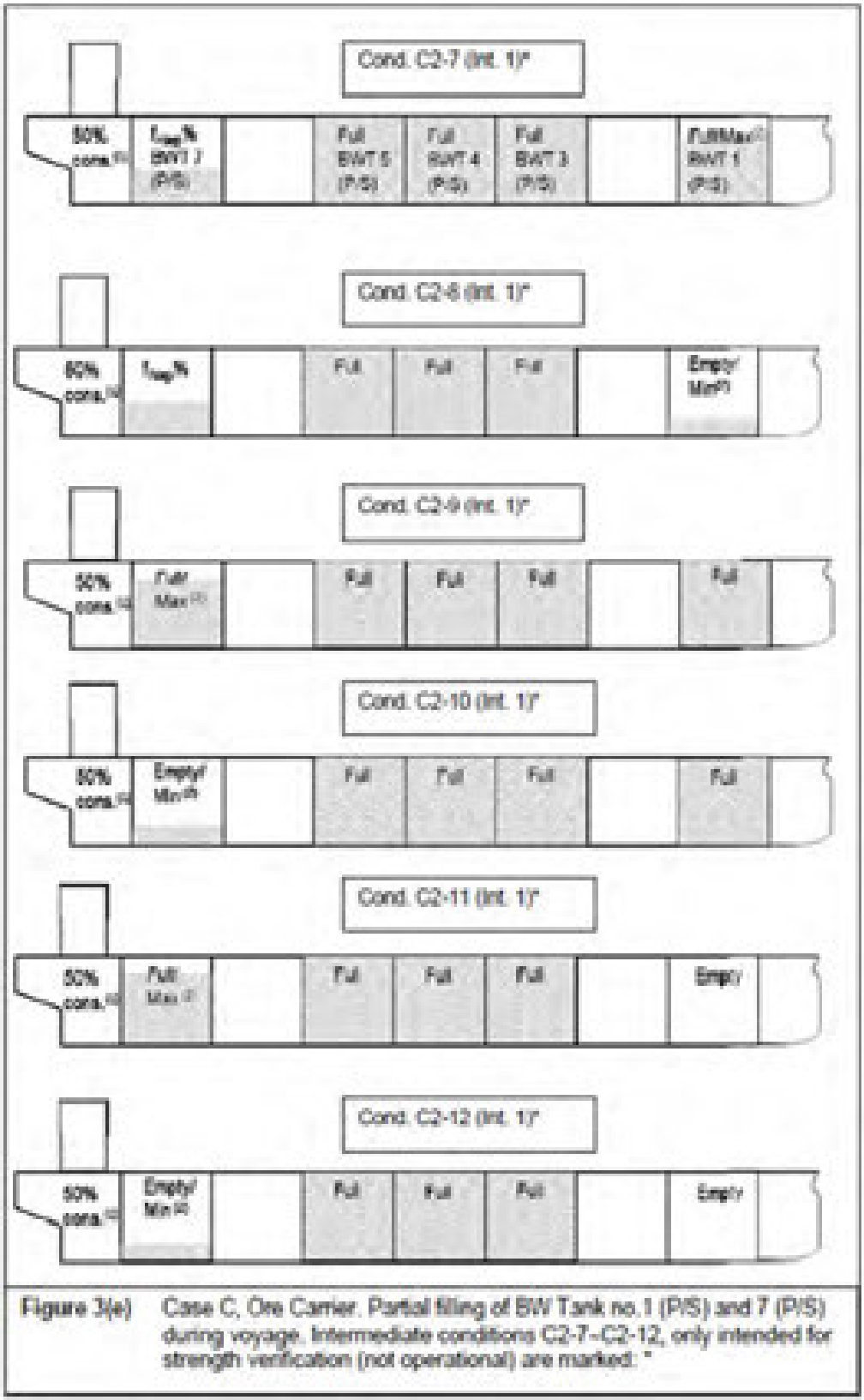


Figure 3(e) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Intermediate conditions C2-7~C2-12, only intended for strength verification (not operational) are marked: *

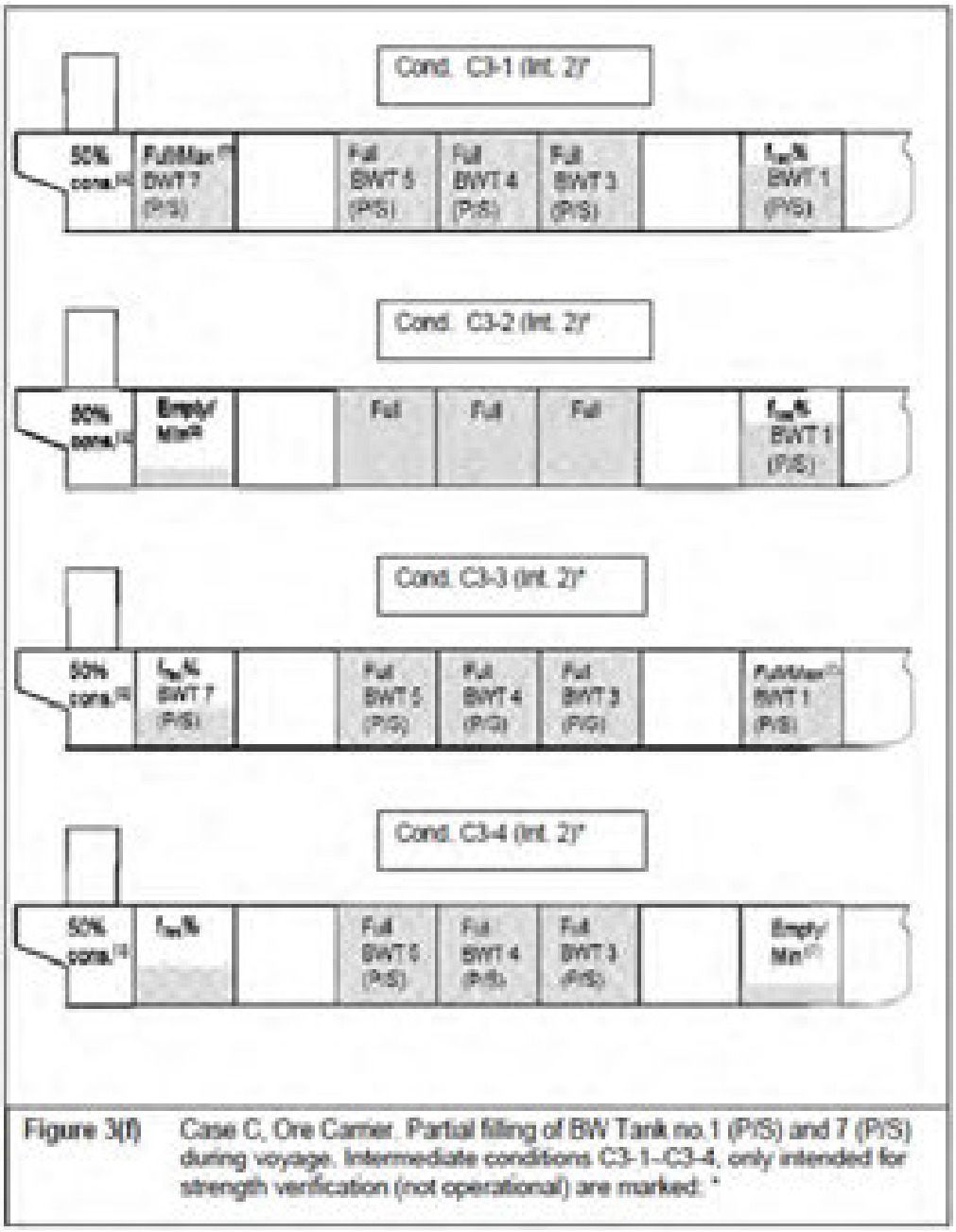


Figure 3(f) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Intermediate conditions C3-1–C3-4, only intended for strength verification (not operational) are marked: *

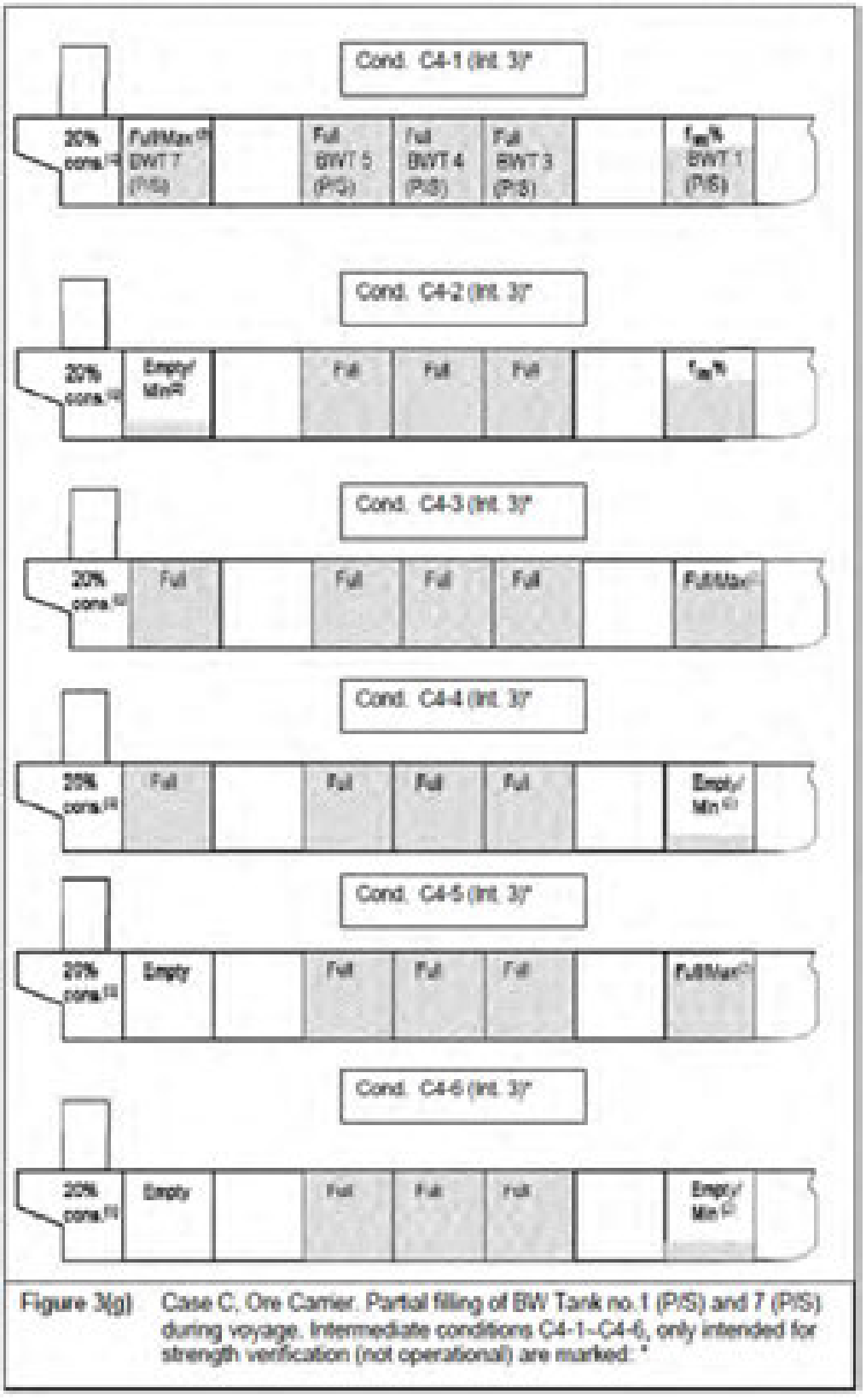


Figure 3(g) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Intermediate conditions C4-1–C4-6, only intended for strength verification (not operational) are marked: *

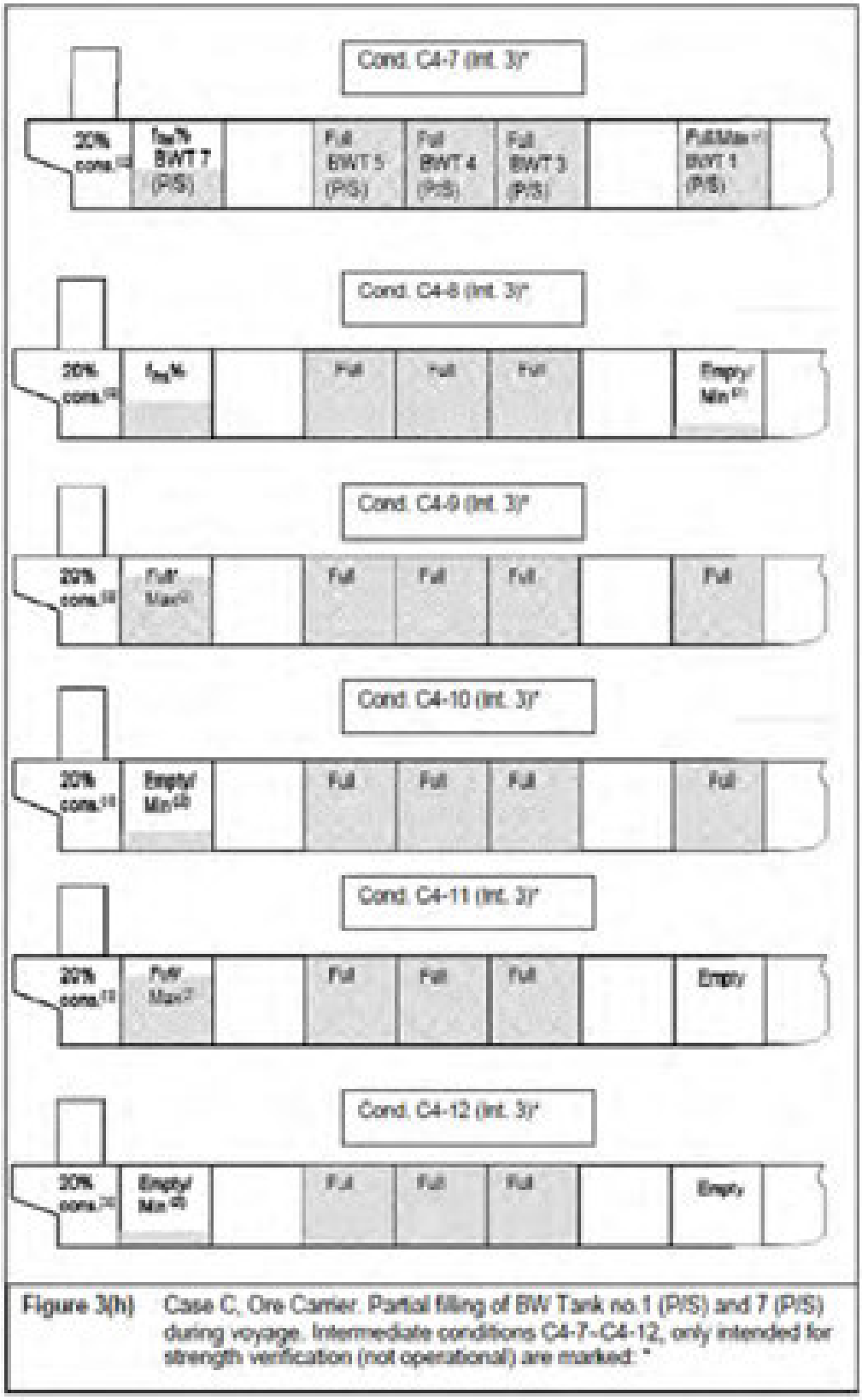


Figure 3(h) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Intermediate conditions C4-7 - C4-12, only intended for strength verification (not operational) are marked: *

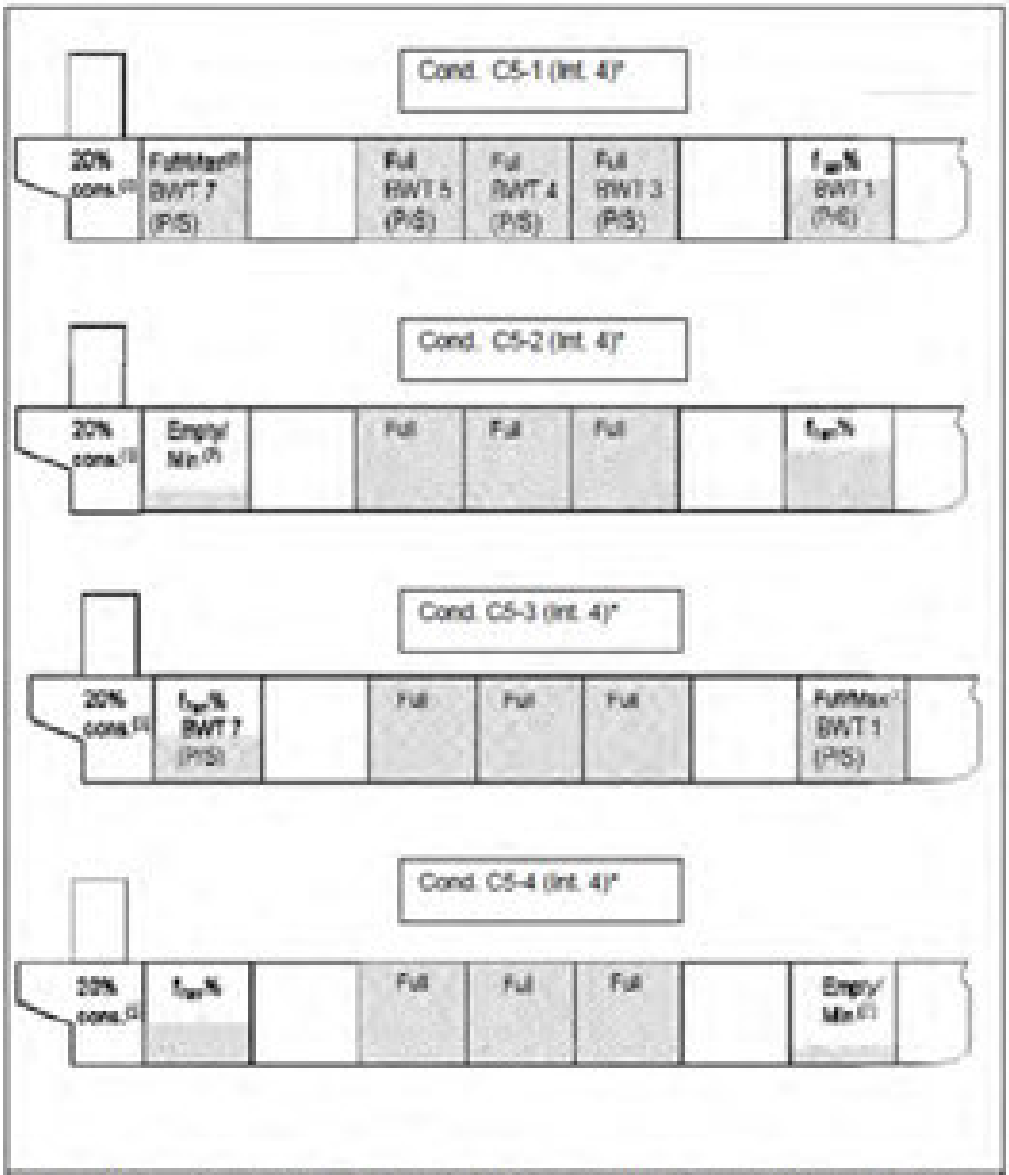


Figure 3(i) Case C, One Carrier. Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Intermediate conditions C5-1-C5-4, only intended for strength verification (not operational) are marked: *

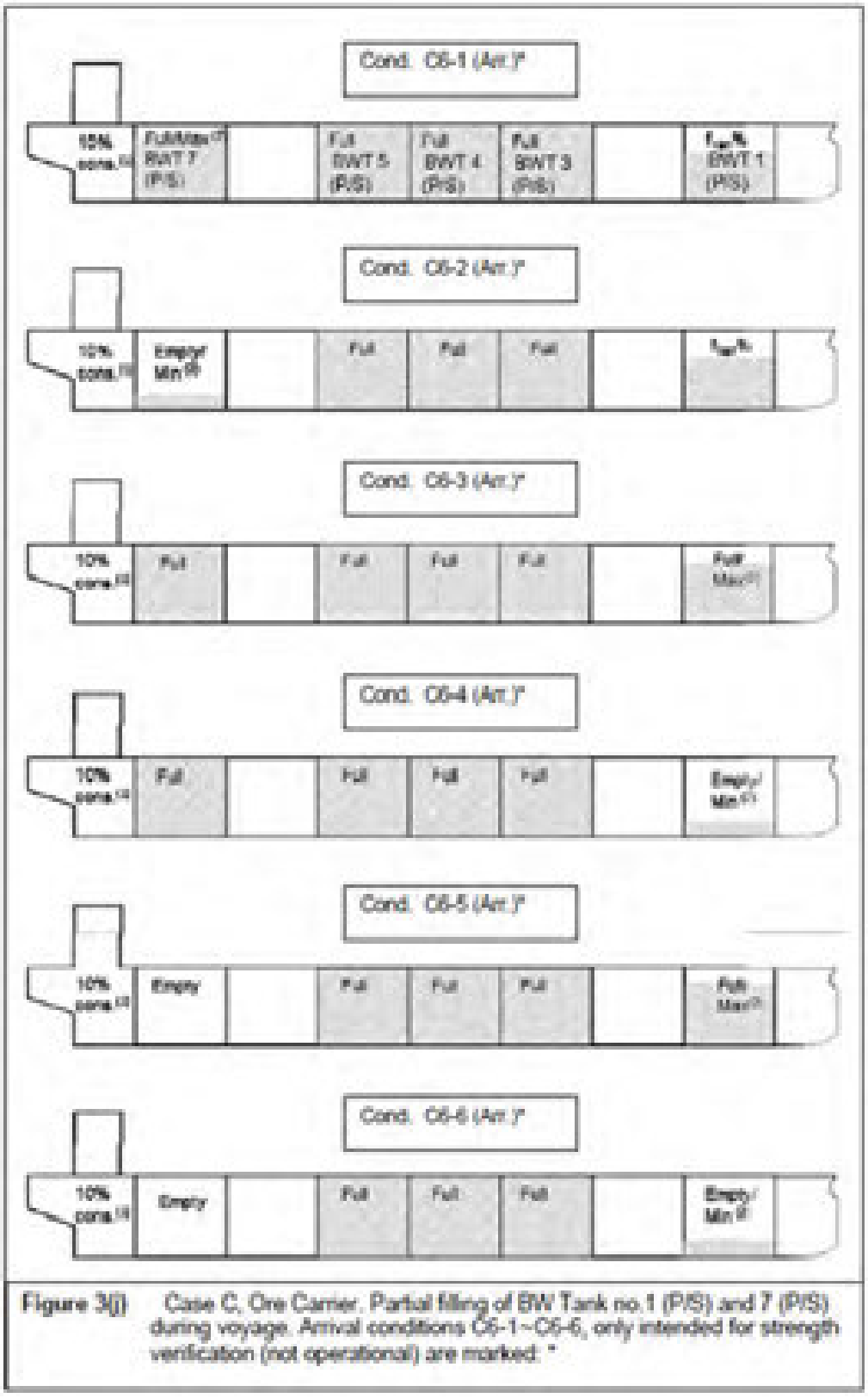


Figure 3(j) Case C, Ore Carrier, Partial filling of BW Tank no.1 (P/S) and 7 (P/S) during voyage. Arrival conditions C6-1~C6-6, only intended for strength verification (not operational) are marked: *

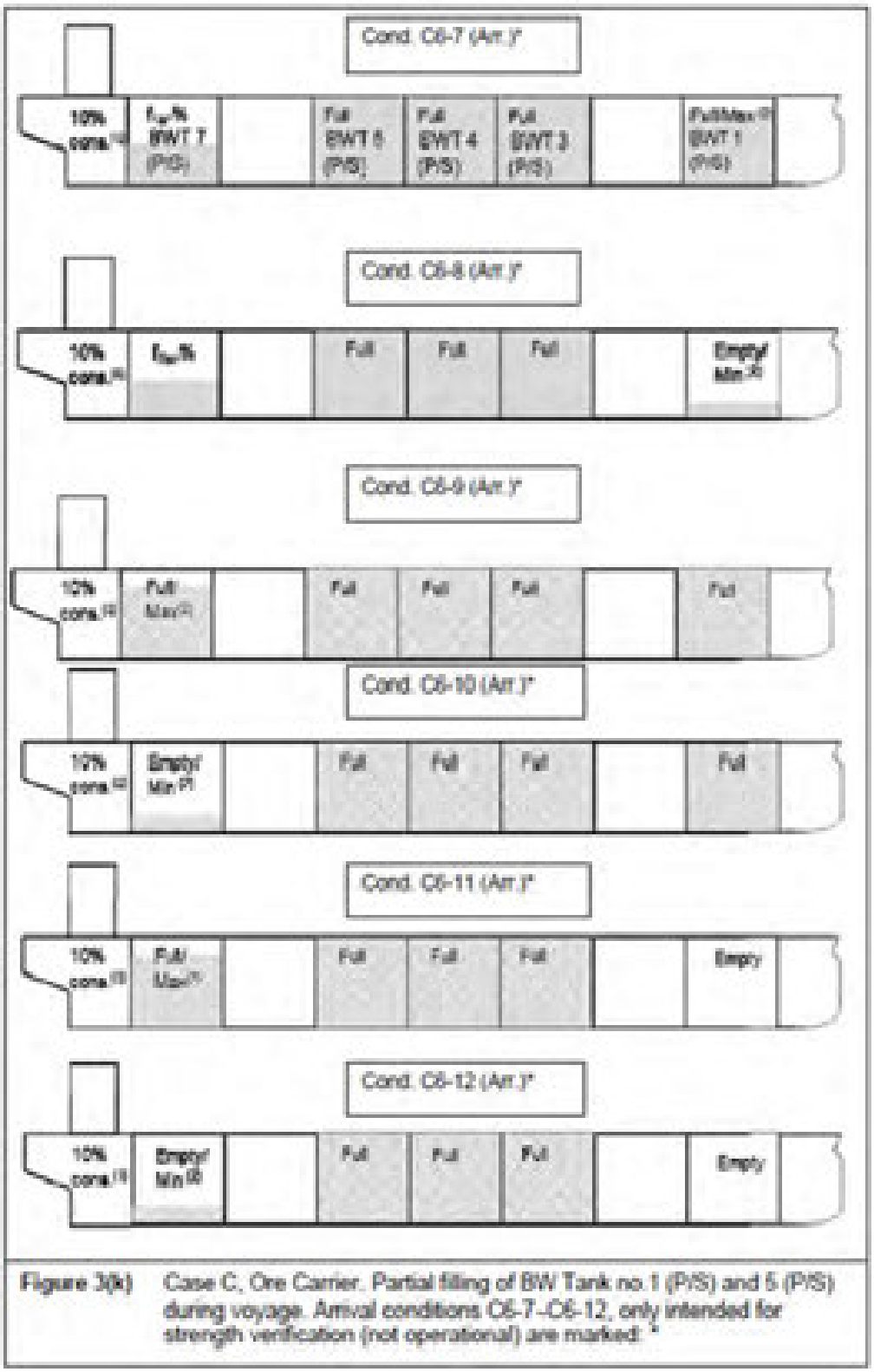


Figure 3(k) Case C, Ore Carrier. Partial filling of BW Tank no.1 (P/S) and 5 (P/S) during voyage. Arrival conditions C6-7-C6-12, only intended for strength verification (not operational) are marked: *

End of Chapter

Chapter 6

Hull Girder Strength

Contents

Section

1 *Strength characteristics of hull girder transverse sections*

2 *Vertical hull girder bending and shear strength*

3 *Hull girder yield check*

4 *Hull girder ultimate strength check*

Section 1

Strength characteristics of hull girder transverse sections

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

1.1 Application

1.1.1 The requirement of this section applies to steel ships, of normal form, proportions, speed and length L≥65[m]. However, ships having one or more of the following characteristics, will be specially considered.

- Proportion L/B ≤ 5 B/D ≥ 2.5
- Length L ≥ 500[m]
- Block coefficient C_b< 0.6
- Large deck opening
- Ships with large flare
- Carriage of heated cargoes
- Unusual type or design
- Unusual hull weight distribution

1.1.2 Scantlings of hull members contributing to longitudinal strength are to comply with the requirements given in this Chapter. These members are also to comply with the requirements of buckling strength given in Part 3, Chapter 9 and local strength requirements given in Chapter 7.

1.1.3 For ships with small block coefficient, high speed and large flare (in particular L>120[m] and speed V>17[knots]), the section modulus in the forebody is to be specially considered based on the distribution of still water and wave bending moments given in Chapter 5, Section 3.4, Cl.3.4.1 and Section 3.6, Cl.3.6.1 respectively.

1.1.4 For ships with narrow beams the combined effects of vertical and horizontal bending of hull girder may have to be specially considered as given in Cl.3.2.1.

1.1.5 For ships with large deck openings (Refer Cl.1.3.2) other than container ships the combined effects of hull girder bending and torsion related possible local bending and shear stresses may have to be specially considered as given in Cl.3.3.1.

1.1.6 Longitudinal strength requirements given in this section do not apply to Container Ships. Requirements given in Part 5, Chapter 5, Section 4 are applicable to container ships.

1.2 Longitudinal strength calculations

1.2.1 Longitudinal strength calculations are to be carried out for all ships where $L \geq 65[m]$, covering the design cargo and ballast loading conditions proposed, in order to determine the required hull girder strength. The calculations of still water shear forces and bending moments are to cover both departure and arrival conditions and any special mid-voyage conditions caused by changes in ballast distribution.

1.2.2 For ships where $L < 65[m]$, longitudinal strength calculations may be required, depending upon the design loading conditions or structural arrangement.

1.2.3 In general, where a long superstructure or deckhouse (*) of length greater than 0.15L is fitted, extending within the 0.4L amidships, the requirements for longitudinal strength in the hull and erection will be considered in each case.

*Note: For superstructure or deckhouses, if their contribution to be considered in longitudinal strength calculation, a finite element analysis of the full ship which takes into account the general arrangement of the longitudinal elements (side, decks, bulkheads) may be done.

In lieu of full ship analysis, the sectional area of effective continuous longitudinal members contributing to the longitudinal strength can be taken into account subject to the conditions given in Cl.1.2.3. Relevant parts of the shadow areas are to be deducted from sectional areas according to Figure 1.2.1.

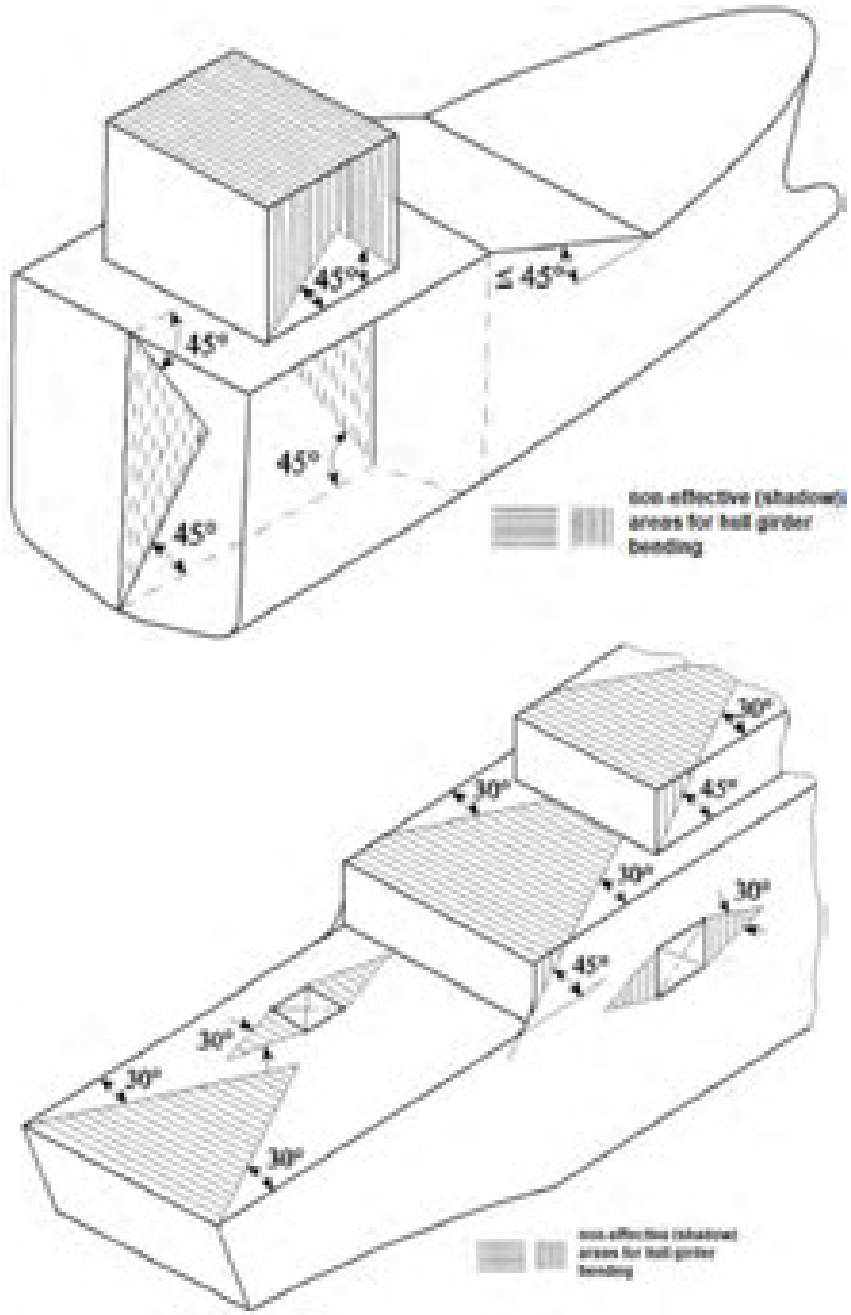


Figure 1.2.1: Representative figure showing shadow area concept for ship's hull and superstructures

1.3 Definitions

1.3.1 Effective longitudinal bulkhead

A longitudinal bulkhead extending from bottom to deck and which is connected to the ship's side by transverse bulkheads both forward and aft.

1.3.2 Ships with large openings in the strength deck

A ship is regarded as one with large deck openings in strength deck if any one of the following conditions applies:

$$\frac{b}{B_0} > 0.7$$

$$\frac{\ell_1}{\ell_0} > 0.89$$

$$\frac{b}{B_0} > 0.6 \text{ and } \frac{\ell_1}{\ell_0} > 0.7$$

where:

b = Breadth of the hatch opening, in [m]. Where there are multiple openings in the same hull transverse section, they are considered as a single opening, and b is calculated as the sum of the individual opening breadths. Refer Figure 1.3.2.

B_0 = Extreme breadth of deck measured, in [m] at the mid length of hatch opening. Refer Figure 1.3.2.

ℓ_1= Length of the hatch opening, in [m]. Refer Figure 1.3.2.

ℓ_0 = Distance, between centers of the deck strip at each end of the hatch opening, in [m]. Where there is no further hatch opening beyond the one under consideration, the point to which ℓ_0 is measured will be specially considered. Refer Figure 1.3.2.

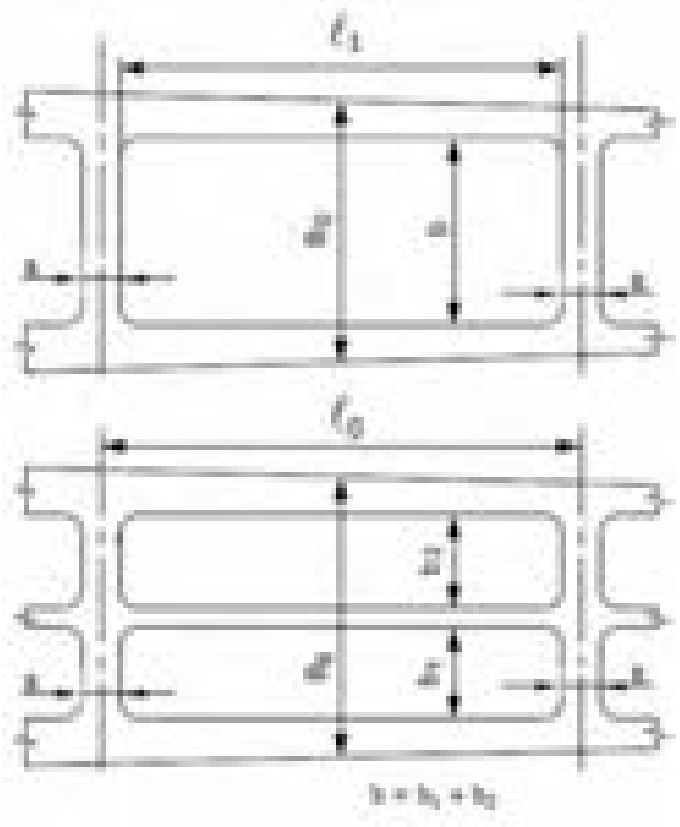


Figure 1.3.2: Ship with large openings

1.3.3 Strength deck

The strength deck is, in general, the uppermost continuous deck. In the case of a superstructure or deckhouses of length greater than 0.15L, extending within the 0.4L amidships, the strength deck is the deck of the superstructure or the deck of the uppermost deckhouse. Refer Cl.1.2.3.

1.4 Hull girder transverse sections

1.4.1 General

This section specifies the criteria for calculating the hull girder strength characteristics to be used for the bending and shear strength assessments, in association with the hull girder loads as specified in Chapter 5.

Hull girder transverse sections are to be considered as being constituted by the members contributing to the hull girder longitudinal strength, i.e. all continuous longitudinal members below and including the strength deck defined in Cl.1.3.3, taking into account the requirements in Cl.1.4.2 to Cl.1.4.7.

These members are to have:
- gross offered scantlings, when the hull girder strength characteristics calculated, are used for the yield checks as given in Section 2.

- net scantlings, when the hull girder strength characteristics calculated, are used for the ultimate strength checks as given in Section 4 and for calculating the hull girder stresses for the strength checks of plating, ordinary stiffeners and primary supporting members as given in Chapter 7.

1.4.2 Structural members not contributing to hull girder sectional area

The following members are not to be considered in the calculation as they are not considered contributing to the hull girder sectional area:

- Superstructures and Deckhouses which do not form a strength deck Refer Cl.1.3.3

- Vertically corrugated bulkheads, according to Cl.1.4.6.

- Bulwarks and gutter plates.

- Bilge keels.

- Sniped or non-continuous longitudinal stiffeners.

- Non-continuous hatch coaming.

1.4.3 Continuous trunks and continuous longitudinal hatch coamings

In case of ships with continuous trunks and continuous longitudinal hatch coamings their sectional area can be included in the hull girder section modulus calculations, provided that they are effectively supported by longitudinal bulkheads or primary supporting members. The section modulus at deck however, is then to be calculated as given in Cl.2.1.6.3.

1.4.4 Longitudinal stiffeners or girders welded above the strength deck

Longitudinal stiffeners or girders welded above the strength deck, including the deck of any trunk fitted as specified in Cl.1.4.3, are to be included in the hull girder transverse sections.

1.4.5 Longitudinal girders between hatchways, supported by longitudinal bulkheads

Where longitudinal girders, effectively supported by longitudinal bulkheads, are fitted between hatchways, the sectional area of these longitudinal girders are to be included in the hull girder transverse section. Refer Cl.1.5.1.

1.4.6 Longitudinal bulkheads with vertical corrugations

1.4.6.1 For longitudinal bulkheads with vertical corrugations, the vertical corrugations are not to be included in the hull girder transverse section. Longitudinal bulkheads with vertical corrugations are not effective for hull girder bending, but they are effective for hull girder shear force.

1.4.6.2 For the purpose of calculating the unit shear flow for hull girder vertical shear flow, q_{vi-gr}, as defined in Section 2 and q_{vi_n50}, as defined in Section 3 for vertical corrugations, the following reduction factor is used to obtain effective thickness of the corrugation:

$$C_{corr} = \frac{s_c}{a + c}$$

where s_c, a and c are defined in Chapter 3, Section 6, Cl.6.4.1.

But for the thickness used in the calculation of hull girder shear capacity as defined in Section 2, Cl.2.2.2 and in the calculation of shear stress as defined in Section 3, Cl.3.5.2, this thickness reduction is not applicable.

1.4.7 Members in materials other than steel

Where a member contributing to the longitudinal strength is made of a material other than steel having a Young's modulus, E different from 2.06×10^5[N/mm^2], the steel equivalent sectional area that may be included in hull girder transverse section is obtained, in [m^2], from the following formula:

$$A_{SE} = \frac{E}{2.06 \times 10^5} A_M$$

where:

A_{SE} = Steel equivalent sectional area in [m^2].

A_M = Sectional area of the member under consideration in [m^2].

1.5 Calculation of section properties

1.5.1 General

• When calculating the moment of inertia and section moduli, the effective sectional area (net area after deduction for openings as given in Chapter 3, Section 6, Cl.6.5.4 of all continuous longitudinal strength members is to be taken into account.

• The main strength members included in the calculation of hull moment of inertia and section modulus are to extend continuously through the cargo region and sufficiently far towards the ends of the ship. Longitudinal bulkheads are to terminate at effective transverse bulkheads and large transition brackets are to be fitted in line with the longitudinal bulkheads. Where significant changes in structural arrangement occur, adequate transitional structure is to be provided.

In case of ships, where longitudinal girders are fitted between hatchways, the sectional area included in the hull girder transverse sections is obtained, in [m^2], from the following formula:

$$A_{eff} = A_{lg} \times a$$

where:

A_{lg} = Sectional area of longitudinal girders in [m^2].

a = Coefficient:

= 1 for longitudinal girders effectively supported by longitudinal bulkheads or primary supporting members.

$= 0.6\left(\frac{s}{b} + 0.15\right)^{0.5}$ for longitudinal girders not effectively supported by longitudinal bulkheads or primary supporting members and having dimensions and scantlings such that $\ell_0/r \leq 60$.

= 0 for longitudinal girders not effectively supported by longitudinal bulkheads or primary supporting members and having dimensions and scantlings such that $\ell_0/r > 60$.

r = Minimum radius of gyration, in [m], of the longitudinal girder transverse section.

ℓ_0, s, b = Dimensions, in [m], as defined in Figure 1.3.2.

Section 2

Vertical hull girder bending and shear strength

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

R_s = 1.0 for unrestricted service
= 0.75 for coastal service
= 0.65 for restricted service
= 0.55 for sheltered water service.

Q_{sw} = Permissible positive or negative still water shear force for seagoing operation, in [kN], at the hull transverse section considered, as defined in Chapter 5, Section 3, Cl.3.6.2.

Q_{sw-har} = Permissible positive or negative still water shear force for harbour/sheltered operation, in [kN], at the hull transverse section considered, as defined in Chapter 5, Section 3, Cl.3.6.3.

Q_{wv} = Positive and negative vertical wave shear force, in [kN], for seagoing condition at the hull transverse section being considered, as defined in Chapter 5, Section 3, Cl.3.7.2.

ΔQ_s = Shear force correction at the transverse section considered, in [kN], if applicable. as defined in Cl.2.2.5.

Q_{sw_lcd} = Vertical still water shear force for considered loading condition in seagoing operation, in [kN], at the hull transverse section considered.

$Q_{sw_lcd_har}$ = Vertical still water shear force for considered loading condition in harbour/sheltered operation, in [kN], at the hull transverse section considered.

Q_R = Total vertical hull girder shear capacity, in [kN], at the hull transverse section being considered as defined in Cl.2.2.2.

q_{vi-gr} = Unit shear flow for hull girder vertical shear force, in [mm^{-1}], for the plate i based on gross thickness t_i, in [mm] obtained from a numerical calculation based on thin-walled beam theory as given in Appendix A.

M_{sw} = Permissible vertical still-water bending moment in [kN-m], for hogging and sagging, in seagoing condition at the hull transverse section being considered as defined in Chapter 5, Section 3, Cl.3.4.2.

M_{wv} = Vertical wave bending moment in [kN-m], for hogging and sagging, in seagoing condition at the hull transverse section being considered as defined in Chapter 5, Section 3, Cl.3.7.1.

V_D = Vertical distance to the equivalent deck line, in [m], as defined in Cl.2.1.6.4.

I_Y = Gross moment of inertia, in [cm^4], of the hull transverse section about its horizontal neutral axis, to be calculated according to Cl.2.1.7.

I_Z = Gross moment of inertia, in [cm^4], of the hull transverse section about its vertical neutral axis, to be calculated according to Cl.2.1.7.

x = X coordinate of the calculation point with respect to the reference coordinate system in [m], as defined in Chapter 1, Section 2, 2.5.

z = Z co-ordinate of the calculation point with respect to the reference co-ordinate system in [m], as defined in Chapter 1, Section 2, 2.5.

z_n = Z co-ordinate of horizontal neutral axis of the hull transverse sections with gross scantling defined in 1.4, in [m], with respect to the reference co-ordinate system defined in Chapter 1, Section 2, 2.5.

z_D = Z coordinate of strength deck at ship side, in [m].

Z_B = Gross section modulus, in [cm^3], at bottom, to be calculated according Cl.2.1.6.

Z_D = Gross section modulus, in [cm^3], at deck, to be calculated according Cl.2.1.6.

2.1 Vertical hull girder bending strength

2.1.1 General

• Scantlings of all continuous hull girder longitudinal strength members based on section modulus requirement in Cl.2.1.2 and moment of inertia requirement in Cl.2.1.3 are to

be maintained within 0.4L amidships. However, in special cases, based on consideration of ship type, hull form and loading conditions, the scantlings may be gradually reduced towards the end of the 0.4L amidships, bearing in mind the desire not to inhibit the vessel's loading flexibility.

• The k material factors are defined with respect to the materials used for the bottom and deck members contributing to the hull girder longitudinal strength according to Section 1, 1.4. When higher strength steels are used, the requirements of Cl.2.1.8 apply.

2.1.2 Minimum Hull Girder Section Modulus

The minimum gross hull girder section modulus, Z_{min} in [cm^3] about the transverse neutral axis at midship is not to be less than obtained from the following equation:

$$Z_{min} = k.C.L^2.B(C_b + 0.7)(0.5 + 0.5R_s)$$

where,

$C = C_1$, for new ships

$C = 0.9\ C_1$, for ships in service

$$C_1 = 4 + 0.0412L \text{ for } L < 90[m]$$

$$C_1 = 10.75 - \left(\frac{300 - L}{100}\right)^{\frac{3}{2}} \text{ for } 90 \leq L \leq 300[m]$$

$$C_1 = 10.75 \text{ for } 300 < L < 350[m]$$

$$C_1 = 10.75 - \left(\frac{L - 350}{150}\right)^{\frac{3}{2}} \text{ for } 350 \leq L \leq 500[m]$$

2.1.3 Minimum Hull Girder Moment of Inertia

The gross moment of inertia I_{min} of the hull section about the transverse neutral axis, at midship, is not to be less than:

$$I_{min} = 3.C_1.L^3B(C_b + 0.7)R_S\ [cm^4]$$

2.1.4 Hull Girder Section Modulus with in 0.4L amidships

At any transverse section, the required gross hull section modulus Z_{gr}, in [cm^3] with in 0.4L amidships about the transverse neutral axis for the still water bending moments $[M_{sw}]$ and wave bending moments $[M_{wv}]$ is not to be less than obtained from the following equation:

$$Z_{gr} = \left(\frac{M_{sw} + M_{wv}}{\sigma_L}\right) \times 10^3$$

where,

σ_L = Permissible hull girder bending stress

= 175/k [N/mm^2] within 0.4L amidships

= 125/k [N/mm^2] within 0.1L from A.P. and F.P.

Between the specified regions σ_L is to be obtained by linear interpolation.

In the region outside 0.4L amidships, the scantlings are to be gradually tapered to the local requirements at ends.

2.1.5 Hull Girder Strength Outside of 0.4L Amidships

2.1.5.1 The strength of the hull girder is to be checked at sections outside of 0.4L amidships. The required bending strength outside 0.4L amidships is to be determined as specified in Cl.2.1.4. As a minimum, hull girder bending strength checks are to be carried out specifically at the following locations:

• In way of the forward end of the engine room.

• In way of the forward end of the foremost cargo hold.

• At any locations where there are significant changes in hull cross-section.

• At any locations where there are changes in the framing system.

• At any locations where there is a change of material from Midship.

2.1.5.2 Buckling strength of members contributing to the longitudinal strength and subjected to compressive and shear stresses is to be checked, in particular in regions where changes in the framing system or significant changes in the hull cross-section occur. The buckling evaluation criteria is given in Chapter 9.

2.1.5.3 For ships with large deck openings in the strength deck, such as containerships, sections at or near to the aft and forward quarter length positions are to be checked. For such ships with cargo holds aft of the

superstructure, deckhouse or engine room, strength checks of sections in way of the aft end of the aft-most holds, and the aft end of the deckhouse or engine room are to be performed.

2.1.6 Section Modulus at any point

2.1.6.1 The gross section modulus, Z in [cm^3], at any point of a hull transverse section is obtained, from the following formula:

$$Z = \frac{I_Y}{|z - z_n|} \times 10^{-2}$$

2.1.6.2 At bottom

The gross section modulus at bottom, Z_B in [cm^3], is obtained from the following formulae:

$$Z_B = \frac{I_Y}{z_n} \times 10^{-2}$$

2.1.6.3 At deck

The gross section modulus at deck, Z_D in [cm^3], is obtained from the following formulae:

$$Z_D = \frac{I_Y}{V_D} \times 10^{-2}$$

2.1.6.4 Equivalent deck line

• In case of ships with effective longitudinal members, as specified in Cl.1.4.5 and Cl.1.4.6 are positioned above a line extending from strength deck at side to a position $((z_D - z_n)/0.9)$ from the neutral axis at the centerline, the equivalent deck line is obtained, in [m], from the following formula:

$$V_D = (z_T - z_n)\left(0.9 + 0.2\left(\frac{y_T}{B}\right)\right) \geq (z_D - z_n)$$

• When no effective longitudinal members as specified in Cl.1.4.5 and Cl.1.4.6 are positioned above a line extending from strength deck at side to a position $((z_D - z_n)/0.9)$ from the neutral axis at the centreline, the equivalent deck line is obtained, in [m], from the following formula:

$$V_D = (z_D - z_n)$$

where,

y_T, z_T = Y and Z coordinates of the top of continuous trunk, hatch coaming, longitudinal stiffeners or girders, to be measured for the point which maximizes the value of V_D, in [m].

2.1.7 Moments of inertia

The gross moment of inertia, I_Y and I_Z, in [cm^4], are those, calculated about the horizontal and vertical neutral axes, respectively, of the hull transverse sections defined in 1.4.

2.1.8 Extent of high strength steel

2.1.8.1 Where high strength steels are used in the main hull structure in order to reduce the section modulus requirement, the vertical and longitudinal extent of its use is to be such that adjacent structure made of ordinary hull structural steel is not stressed beyond the stress level permissible for ordinary steel.

2.1.8.2 Vertical Extent

The vertical extent of the high strength steel used in deck zone or bottom zone is to be as shown in the Figure 2.1.8.1, the distance z_0 in [m], is not to be more than:

$$z_0 = \frac{I_y}{Z} \times 10^{-2}$$

where,

Z = Rule midship section modulus required for ordinary steel (k=1.0) as given in Cl.2.1.2 and Cl.2.1.4.

For narrow beam ships the vertical extent of high strength steel may have to be increased after special consideration.

2.1.8.3 Longitudinal Extent

The longitudinal extent of the high strength steel is to be 0.4L amidships and beyond, such that at the junction with ordinary steel the scantlings are equal to those required for an identical ship built of ordinary steel. At the junction, continuity in scantlings is to be maintained. Refer Figure 2.1.8.2.

2.1.9 Permissible vertical still water bending moments for harbour/sheltered water condition

The permissible still water bending moments, M_{sw-har} in [kN-m], for harbour/sheltered operations in hogging and sagging is to comply with the following criteria:

$$|M_{sw-har}| \leq |M_{sw} + 0.5M_{wv}|$$

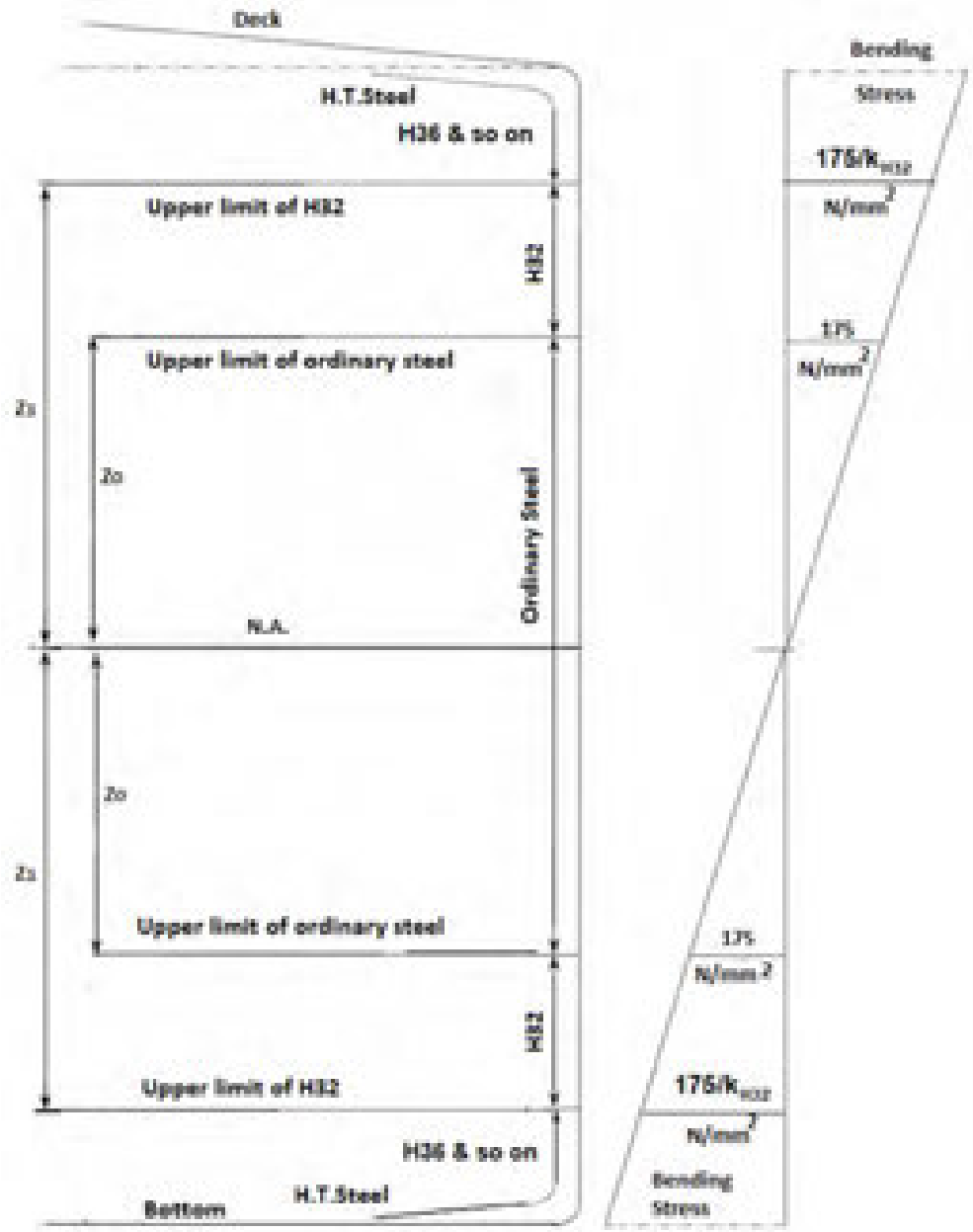


Figure 2.1.8.1: Vertical extent of higher strength steel

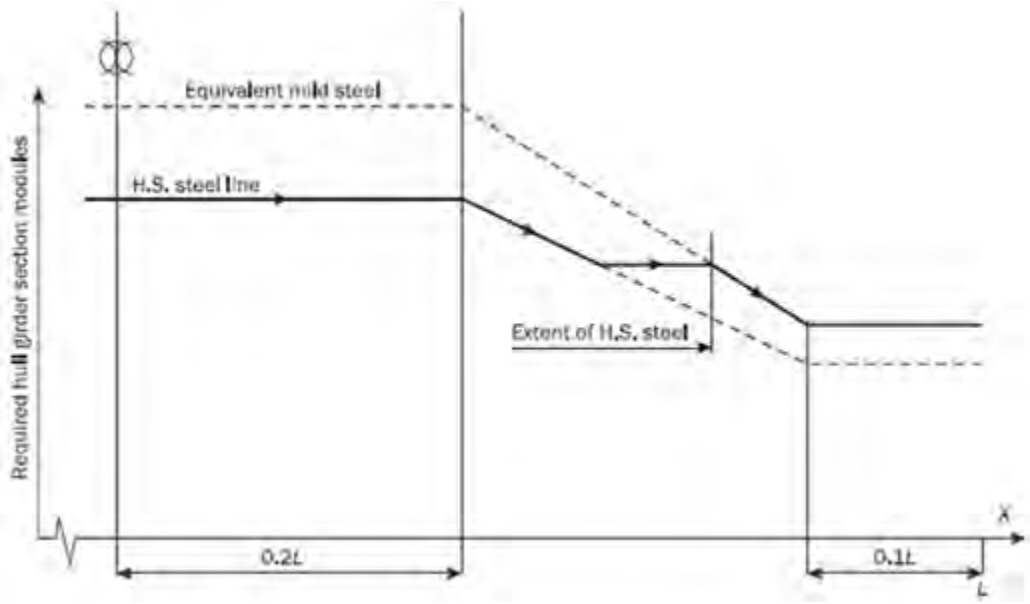


Figure 2.1.8.2: Longitudinal extent of higher strength steel

2.2 Vertical hull girder shear strength

2.2.1 General

2.2.1.1 Hull girder shear strength calculations are to be carried out for ships with length L≥ 65[m].

2.2.1.2 For passenger ships, the assessment of permissible still water shear forces is to take into consideration the effectiveness of the continuous superstructures and the sizes and arrangements of window and door openings.

2.2.1.3 Where longitudinal bulkheads are perforated by cut-outs, the assessment of permissible still water shear forces is to take into consideration the loss of material.

2.2.1.4 For calculating the total shear stress, $[\tau]$, due to still-water and wave-induced loads at a particular section, the maximum algebraic sum of the shear force due to still-water $[Q_{sw}]$ and wave $[Q_{wv}]$ at that particular section to be used.

2.2.1.5 The thickness of the side shell, inner shell and the longitudinal bulkheads, is to be such that the total shear stress $[\tau]$, based on the total shear force is not to exceed 110/k [N/mm^2].

2.2.2 Total hull girder shear capacity

The total vertical hull girder shear capacity, Q_R in [kN], is the minimum of the calculated values for all plates "i" contributing to the hull girder shear of the transverse section considered and is obtained by the following formula:

$$Q_R = \min_i \left(\frac{\tau_{perm} \cdot t_i}{q_{vi_gr}} \right) 10^{-3}$$

where:

τ_{perm} = Permissible shear stress, in [N/mm^2], for plate i, to be taken as:

= 110/k

t_i = Gross thickness, in [mm], for plate i considered.

2.2.3 Permissible still water shear force in sea going conditions

2.2.3.1 The positive and negative permissible vertical still water shear force, in [kN], for seagoing condition is to comply with the following criteria:

$$|Q_{sw}| \leq Q_R - |Q_{wv}|$$

The shear force Q_{wv} used above is to be taken with the same sign as the considered shear force Q_{sw}.

2.2.3.2 The vertical still water shear forces, in [kN], for all loading conditions for seagoing operations is to comply with the following criteria:

$$|Q_{sw-lcd} - \Delta Q_s| \leq |Q_{sw}|$$

The shear force Q_{sw} used above is to be taken with the same sign as the considered shear force Q_{sw_lcd}.

2.2.4 Permissible still water shear force in harbour/sheltered water condition

2.2.4.1 The positive and negative permissible vertical still water shear force, in [kN], for harbour/sheltered water operations is to comply with the following criteria:

$$|Q_{sw-har}| \leq |Q_{sw} + 0.5Q_{wv}|$$

The shear force Q_{wv} used above is to be taken with the same sign as the considered shear force Q_{sw-har}.

2.2.4.2 The vertical still water shear forces, in [kN], for all loading conditions for harbour/sheltered water operations is to comply with the following criteria:

$$|Q_{sw-lcd-har} - \Delta Q_s| \leq |Q_{sw-har}|$$

The shear force $Q_{sw-lcd-har}$ used above is to be taken with the same sign as the considered shear force Q_{sw-har}.

2.2.5 Shear force corrections

For hull configurations where there are no longitudinal bulkheads (excluding the inner shell), shear force correction, which takes into account the portion of loads transmitted by the double bottom longitudinal girders to the transverse bulkheads, is to be considered.

Shear force correction is defined as the difference between shear force on a shear panel determined from shear flow calculation and the actual shear force.

For the considered cargo hold, the shear force correction at the considered transverse section is to be obtained, in [kN], from the following formula:

$$\Delta Q_s = C_d \alpha \left| \frac{M}{B_H \ell_H} - \rho T_1 \right|$$

where:

ΔQ_S = shear correction due to uneven transverse load distribution and shear carrying longitudinal bottom members (girders or stiffeners).

= 0 for nearly uniform loading conditions.

C_d = Distribution coefficient taken as:

= -1 at the aft end of the considered cargo hold except for aftmost cargo hold.

= 1 at the fore end of the considered cargo hold except for foremost cargo hold.

= 0 at mid-length of the cargo hold.

= 0 at the aft bulkhead of the aftmost cargo hold.

= 0 at the fore bulkhead of the foremost cargo hold.

= Linearly distributed at other locations.

α = Coefficient taken as:

$$\alpha = g \frac{\ell_o b_o}{2 + \varphi \frac{\ell_o}{b_o}}$$

$$\varphi = 1.38 + 1.55 \frac{\ell_o}{b_o} \text{ but not graeater than } 3.7.$$

M = Mass, in [t], in the hold in way of the considered transverse section for the considered loading condition. M is to include the mass of ballast water and fuel oil located directly below the flat portion of the inner bottom, if any, excluding the portion under the bulkhead stool.

B_H = Breadth of the cargo hold, in [m], measured at mid-length of the cargo hold and at the mid height between the top of hopper tank and the bottom of topside tank, if fitted as given below in Figure 2.2.5.1.

ℓ_H = Length of the cargo hold, in [m], at the centreline between the transverse bulkheads. This is to be measured to the mid-depth of the corrugated bulkhead(s) if fitted as given below in Figure 2.2.5.1.

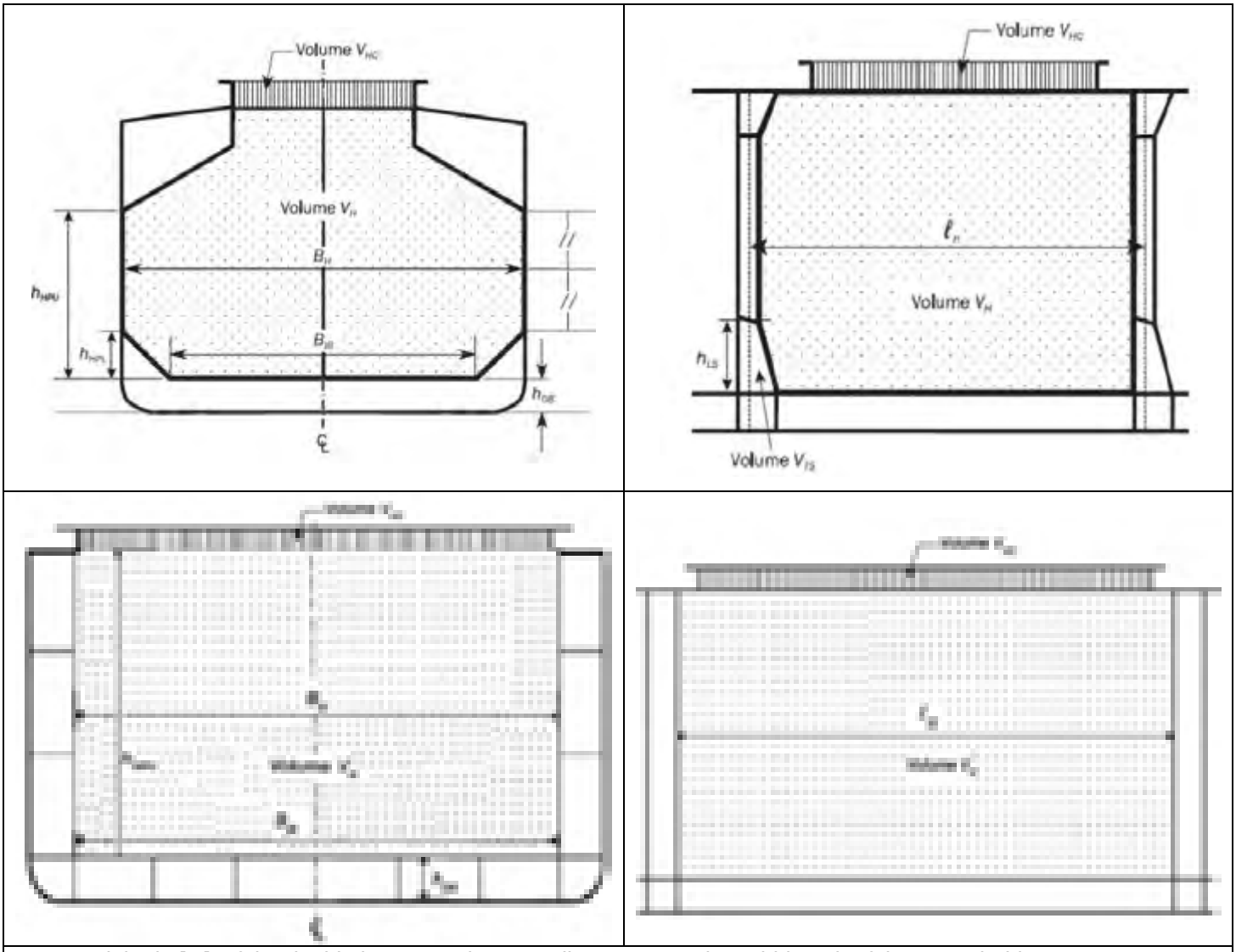
ℓ_o, b_o = Length and breadth, respectively, in [m], of the flat portion of the double bottom in way of the hold considered; b_o is to be measured on the hull transverse section at the middle of the hold.

T_1 = Draught, in [m], measured vertically on the hull transverse section at the middle of the hold

considered, from the moulded baseline to the waterline in the loading condition considered.

ΔQ_{CE} = Shear force correction for the empty hold as given below in Figure 2.2.5.2.

ΔQ_{CF} = Shear force correction for the loaded hold as given below in Figure 2.2.5.2.



h_{DB} = Height, in [m], of the double bottom at the centreline, measured at mid-length of the cargo hold.
h_{HPL} = Vertical distance, in [m], from the inner bottom at centreline to the upper intersection of hopper tank and side shell or inner side for double side bulk carriers, determined at mid length of the considered cargo hold.
h_{HPL} = 0 if there is no hopper tank.
h_{HPU} = Vertical distance, in [m], from the inner bottom at centreline to the lower intersection of topside tank and side shell or inner side for double side bulk carriers, determined at mid length of the cargo hold at midship.
h_{LS} = Mean height, in [m], of the lower stool, measured from the inner bottom.

Figure 2.2.5.1: Definition of cargo hold parameters for bulk carrier

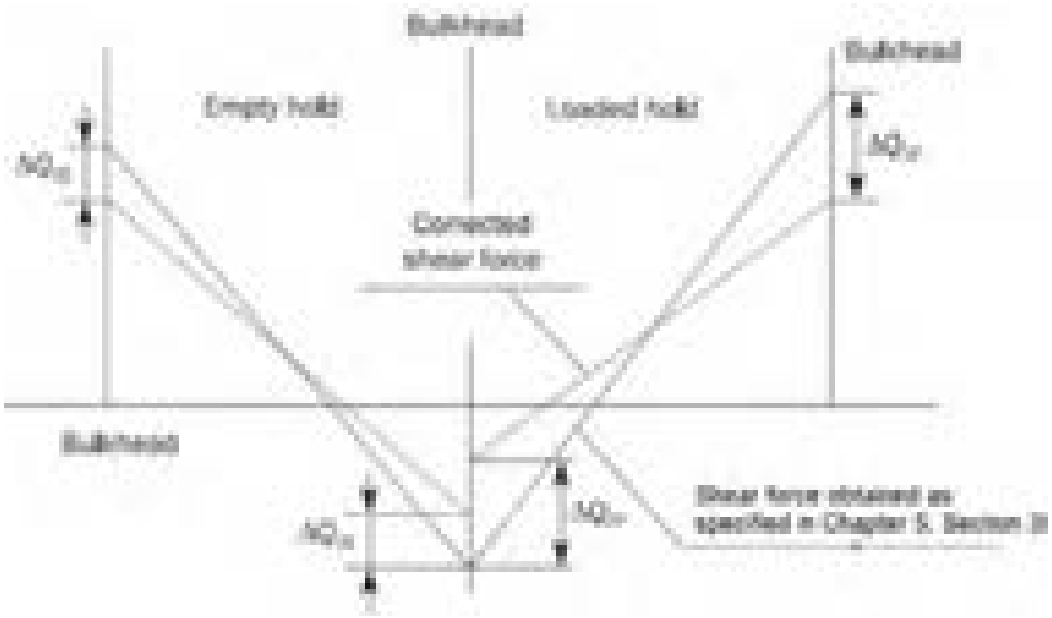


Figure 2.2.5.2: Shear force correction

Section 3

Hull Girder Yield Check

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

I_{y-n50} = Net moment of inertia, in [m^4], of the hull transverse section about its horizontal neutral axis.

I_{z-n50} = Net moment of inertia, in [m^4], of the hull transverse section about its vertical neutral axis

M_{sw-h}, M_{sw-s} = Permissible vertical still-water bending moment in [kN-m], for hogging and sagging respectively, in seagoing condition at the hull transverse section being considered as defined in Chapter 5, Section 3, Cl.3.4.2.

Q_{sw} = Permissible positive or negative still water shear force for seagoing operation, in [kN], at the hull transverse section considered, as defined in Chapter 5, Section 3, Cl.3.5.1.

M_{wv-_LC} = Vertical wave bending moment in [kN-m], for a dynamic load case, at the hull transverse section being considered as defined in Chapter 5, Section 3, Cl. 3.7.2.

For vessels L<90[m], $M_{wv_LC} = M_{wv}$ as defined in Chapter 5, Section 3, Cl.3.6.1 to be considered.

Q_{wv-_LC} = Positive and negative vertical wave shear force, in [kN], for seagoing Condition for a dynamic load case at the hull transverse section being considered, as defined in Chapter 5, Section 3, Cl.3.7.3.

For vessels L<90[m], $Q_{wv_LC} = Q_{wv_}$ as defined in Chapter 5, Section 3, Cl.3.6.2 to be considered.

M_{wh-LC} = Horizontal wave bending moment, in [kN-m], for a dynamic load case at the hull Transverse section being considered, as defined in Chapter 5, Section 3, Cl.3.7.4.

For vessels L<90[m], $M_{wh-LC} = M_{wh}$ as defined in Chapter 5, Section 3, Cl.3.6.3 to be considered.

q_{vi_n50} = Unit shear flow for hull girder vertical shear force, in [mm^{-1}], for the plate i based on net thickness t_{i_net}, in [mm].

x = X coordinate, in [m], of the calculation point with respect to the reference coordinate system defined in Chapter 1, Section 2, 2.5.

y = Y coordinate, in [m], of the calculation point with respect to the reference coordinate system defined in Chapter 1, Section 2, 2.5.

z = Z coordinate, in [m], of the calculation point with respect to the reference coordinate system defined in Chapter 1, Section 2, 2.5.

z_{n-n50} = Z coordinate, in [m], of horizontal neutral axis of the hull transverse section with net scantlings, with respect to the reference coordinate system defined in Chapter 1, Section 2, 2.5.

3.1 Application

3.1.1 General

This section gives the criteria for calculating the hull girder bending and shear strength with respect to the dynamic load cases as defined in Chapter 5, Section 2, 2.4. The hull girder stresses as defined in 3.2 for ships without large deck openings and in 3.3 for ships with large deck openings obtained are used for following local assessments:

- hull local scantling assessment according to Chapter 7
- buckling assessment according to Chapter 9
- fatigue assessment according to Chapter 10
- special requirements according to Chapter 11.

3.2 Ships without large deck openings

3.2.1 Hull girder longitudinal stresses

For seagoing condition, the hull girder longitudinal stress σ_{hg}, in [N/mm^2] at any

transverse section, about the transverse neutral axis for the still water bending moment $[M_{sw}]$ and wave bending moments (vertical and horizontal bending moments $[M_{wv}]$ and $[M_{wh}]$ respectively) is obtained from the following formula:

$$\sigma_{hg} = \sigma_{hg-sw} + \sigma_{hg-dyn}$$

where:

σ_{hg-sw} = $\begin{cases}\sigma_{sw-h}\\ \sigma_{sw-s}\end{cases}$

σ_{hg-dyn} = $\sigma_{wv-LC} + \sigma_{wh-LC}$

3.2.2 Hull girder shear stresses

For seagoing condition, the hull girder shear stress, τ_{hg} in [N/mm^2], induced by vertical shear forces for a dynamic load case at the transverse section considered is obtained from the following formula:

$$\tau_{hg} = \tau_{hg-sw} + \tau_{hg-dyn}$$

where:

τ_{hg-sw} = $\begin{cases}\tau_{sw-pos}\\ \tau_{sw-neg}\end{cases}$

τ_{hg-dyn} = τ_{wv-LC}

3.3 Ships with large deck openings

3.3.1 Hull girder longitudinal stresses

In case of ships with large deck openings, the additional longitudinal stresses caused by still water and wave torsional moments are also to be considered. Hence, for the seagoing condition, the hull girder longitudinal stress σ_{hg}, in [N/mm^2], for a dynamic load case at the transverse section being considered is obtained from the following formula:

$$\sigma_{hg} = \sigma_{hg-sw} + \sigma_{hg-dyn}$$

where:

σ_{hg-sw} = $\begin{cases}\sigma_{sw-h} + |\sigma_{st}|\\ \sigma_{sw-h} - |\sigma_{st}|\\ \sigma_{sw-s} + |\sigma_{st}|\\ \sigma_{sw-s} - |\sigma_{st}|\end{cases}$

σ_{hg-dyn} = $\sigma_{wv-LC} + C.\sigma_{wh-LC} + \sigma_{wt-LC}$

where:

C = reduction coefficient due to combination with warping stress

= 0.85

σ_{st} = The longitudinal warping stress in [N/mm^2], induced by still water torsional moment in sea going condition. Refer Cl.3.5.1.3.

σ_{wt-LC} = The longitudinal warping stress, in [N/mm^2], induced by wave torsional moment for a dynamic load case, at the transverse section considered in seagoing condition. Refer Cl.3.5.1.3.

3.3.2 Hull girder shear stresses

In case of ships with large deck openings, the additional shear stresses caused by still water and wave torsional moments are also to be considered. Hence, for the seagoing condition, the hull girder shear stress, τ_{hg} in [N/mm^2], for a dynamic load case at the transverse section being considered is obtained from the following formula:

$$\tau_{hg} = \tau_{hg-sw} + \tau_{hg-dyn}$$

where:

τ_{hg-sw} = $\begin{cases}\tau_{sw-pos} + \tau_{st}\\ \tau_{sw-pos} - \tau_{st}\\ \tau_{sw-neg} + \tau_{st}\\ \tau_{sw-neg} - \tau_{st}\end{cases}$

τ_{hg-dyn} = $\tau_{wv-LC} + \tau_{wt-LC}$

τ_{st} = The warping shear stress in [N/mm^2], induced by still water torsional moment in sea going condition. Refer Cl.3.5.2.3.

τ_{wt-LC} = The warping shear stress, in [N/mm^2], induced by wave torsional moment for a dynamic load case, at the transverse section considered in seagoing condition. Refer Cl.3.5.2.3.

3.4 Equivalent stress

3.4.1 The equivalent stress, in [N/mm^2], related to the hull girder longitudinal and shear stresses, σ_{hg} and τ_{hg}, defined in Cl.3.3.1 and Cl.3.3.3 respectively, at any point being considered is obtained from the following formula:

$$\sigma_v = \sqrt{(\sigma_{hg}^2 + 3\tau_{hg}^2)}$$

3.4.2 Design criteria

The equivalent stress, σ_v in [N/mm^2], at any point along the length L of the ship considered, is to comply with the following criteria:

$$\sigma_v \leq \sigma_{v-perm}$$

where:

σ_{v-perm} = Permissible equivalent stress, in [N/mm^2], to be taken as:

$= 220/k$

3.5 Definitions of hull girder stress components

3.5.1 Longitudinal stress

3.5.1.1 Longitudinal stresses induced by still water vertical hull girder bending moment

The longitudinal stresses, in [N/mm^2], induced by vertical still water bending moment (in hogging and sagging cases) in seagoing condition at the transverse section considered, are obtained from the following formula:

$$\sigma_{sw-h} = \frac{M_{sw-h}}{I_{y-n50}}(z - z_{n-n50})10^{-3}$$

$$\sigma_{sw-s} = \frac{M_{sw-s}}{I_{y-n50}}(z - z_{n-n50})10^{-3}$$

3.5.1.2 Longitudinal stresses induced by dynamic hull girder bending moment

The longitudinal stresses, in [N/mm^2], induced by vertical and horizontal wave bending moment for a dynamic load case in seagoing condition at the transverse section considered, are obtained from the following formula:

$$\sigma_{wv-LC} = \frac{M_{wv-LC}}{I_{y-n50}}(z - z_{n-n50})10^{-3}$$

$$\sigma_{wh-LC} = -\frac{M_{wh-LC}}{I_{z-n50}}\, y.10^{-3}$$

3.5.1.3 Longitudinal warping stresses induced by wave and still water hull girder torsion

The longitudinal warping stress, σ_{st} and σ_{wt-LC} in [N/mm^2], induced by still water torsional moment, M_{st} and wave torsional moment for a dynamic load case, M_{wt-LC}, respectively at the transverse section considered in seagoing condition, is to be determined by direct calculations. Refer to the IRS Guideline for "Evaluation of Warping Stresses in Ships with Wide Hatch Openings Due To Torsion Moments" for the calculations.

3.5.2 Shear stresses

3.5.2.1 Shear stresses induced by still water vertical hull girder shear force

The shear stresses, in [N/mm^2], induced by vertical still water shear force in seagoing condition at the transverse section considered, are obtained from the following formulae:

$$\tau_{sw-pos} = \frac{Q_{sw-pos} \cdot q_{vi-n50}}{t_{i-n50}}10^3$$

$$\tau_{sw-neg} = \frac{Q_{sw-neg} \cdot q_{vi-n50}}{t_{i-n50}}10^3$$

3.5.2.2 Shear stresses induced by dynamic vertical hull girder shear force

The shear stresses, in [N/mm^2], induced by dynamic vertical shear force for a dynamic load case at the transverse section considered is obtained from the following formula:

$$\tau_{wv-LC} = \frac{Q_{wv-LC} \cdot q_{vi-n50}}{t_{i-n50}}10^3$$

3.5.2.3 Warping shear stresses induced by wave and still water hull girder torsion

The warping shear stress, τ_{st} and τ_{wt-LC} in [N/mm^2], induced by still water torsional moment, M_{st} and wave torsional moment for a dynamic load case, $M_{\tau-wt-LC}$, respectively at the transverse section considered in seagoing condition, is to be determined by direct calculations. Refer to the IRS Guideline for "Evaluation of Warping Stresses in Ships with Wide Hatch Openings Due To Torsion Moments" for the calculations.

Section 4

Hull girder Ultimate Strength Check

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

M_{sw-h}, M_{sw-s} = Permissible vertical still water bending moment in seagoing condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Chapter 5, Section 3, Cl.3.4.2.

M_{wv} = Vertical wave bending moment in seagoing condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Chapter 5, Section 3, Cl.3.6.1.

4.1 Application

4.1.1 General

4.1.1.1 This section applies to all sea going ships with unrestricted service, length, L≥ 150[m] and ships having single deck.

4.1.1.2 The hull girder ultimate strength requirements given in Cl.4.2.1 applies mainly to the cargo space area and to the following locations:

- in way of the forward end of the engine room
- in way of the forward end of the foremost cargo hold
- at any locations where there are significant changes in hull cross-section
- at any locations where there are changes in the framing system
- At any locations where there is a change of material from Midship.

In addition, for ships with large deck openings like container ships following locations are also to be checked:

- sections at or near to 0.25L and 0.75L.
- sections in way of the aft end of the aft-most holds, and the aft end of the deckhouse or engine room, if the cargo holds are at the aft of the superstructure, deckhouse or engine room.

4.1.1.3 The hull girder ultimate bending capacity is to be checked to ensure that it satisfies the criteria given in 4.2.

4.1.1.4 The checking criteria given in 4.2 are applicable to intact ships only.

4.1.1.5 The hull girder ultimate bending capacity, M_U is defined as the maximum bending moment capacity of the hull girder beyond which the hull structure collapses.

4.2 Checking criteria

4.2.1 General

4.2.1.1 The vertical hull girder ultimate bending capacity is to be checked for hogging and sagging conditions for load components as defined in Table 4.2.1.

Table 4.2.1: Load Components

Load components	**Permissible still water bending moment, M_{sw-U}**
Seagoing condition (S+D)	M_{sw-h} or M_{sw-s}

4.2.1.2 The vertical hull girder ultimate bending capacity at any hull transverse section is to satisfy the following criteria:

$$M \leq \frac{M_U}{\gamma_R}$$

where:

M = Vertical bending moment in sea going condition in [kN-m], to be obtained as specified in 4.3.

M_U = Vertical hull girder ultimate bending capacity, in [kN-m], to be obtained as specified in 4.4.

γ_R = partial safety factor for the vertical hull girder ultimate bending capacity to be taken as:

$$\gamma_R = \gamma_M \gamma_{DB}$$

γ_M = Partial safety factor for the vertical hull girder ultimate bending capacity, covering material, geometric and strength prediction uncertainties; in general, to be taken as:

= 1.05 for container ships

= 1.1 for other ship types

γ_{DB} = Partial safety factor for the vertical hull girder ultimate bending capacity, covering the effect of double bottom bending, to be taken as:

For hogging conditions except flooded condition:

= 1.25 for empty cargo holds in alternate condition of bulk carrier having notation "BULK CARRIER Strengthened for heavy cargos, hold(s)… (to be specified) ...may be empty".

= 1.1 for (i) oil tankers (ii) bulk carriers having notation "BULK CARRIER" and "BULK CARRIER Strengthened for heavy cargos", for loaded cargo holds in alternate condition of bulk carrier having notation "BULK CARRIER Strengthened for heavy cargos, hold(s)…(to be specified)..may be empty".

= 1.0 for passenger ships.

= 1.15 for container ships.

For sagging conditions except flooded condition: = 1.0.

For hogging and sagging condition, for flooded condition: = 1.0

4.3 Hull girder ultimate bending loads

4.3.1 The vertical hull girder bending moment in hogging and sagging conditions, in [kN-m], to be considered in the ultimate strength check is to be taken as:

$$M = \gamma_S M_{sw-U} + \gamma_W M_{wv}$$

where:

M_{sw-U} = Permissible still water bending moment, in [kN-m], in hogging and sagging conditions at the hull transverse section considered as defined in Table 4.2.1.

γ_S = Partial safety factor for the still water bending moment, to be taken as:

= 1.0

γ_W = Partial safety factor for the vertical wave bending moment, to be taken as:

= 1.2.

4.4 Hull girder ultimate bending capacity

4.4.1 Curve $M - \chi$

The ultimate bending moment capacities of a hull girder transverse section, in hogging and sagging conditions, are defined as the maximum values of the curve of bending moment capacity [M] versus the curvature [χ] of the transverse section considered (Refer Figure 4.4.1). The curvature [χ] is positive for hogging condition and negative for sagging condition.

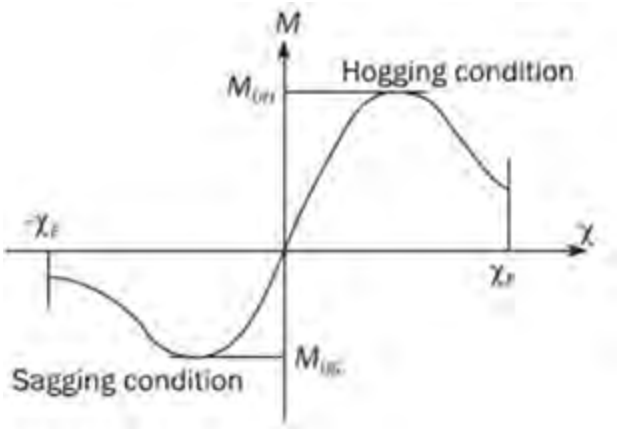


Figure 4.4.1: Bending moment capacity versus curvature [χ]

The hull girder ultimate bending moment capacity M_U, is to be calculated using the incremental-iterative method specified in Part 5, Chapter 5, Section 7.2 or using an alternative method specified in Part 5, Chapter 5, Section 7.3.

4.4.2 Hull girder transverse sections

The hull girder transverse sections are constituted by the elements contributing to the hull girder longitudinal strength, considered with their net scantlings, according to Section 1, Cl.1.4.1.

APPENDIX A

DIRECT CALCULATION OF SHEAR FLOW

A.1 Calculation Formula

A.1.1 General

This appendix describes the procedures of direct calculation of shear flow around a ship's cross section due to hull girder vertical shear force. Shear flow q_v, at each location in the cross section, is calculated by considering the cross section is subjected to a unit vertical shear force, 1[N], in the direction of z coordinate.

The unit shear flow per mm, q_v in [N/mm], is to be taken as:

$$q_v = q_D + q_I$$

where:

q_D = Determinate shear flow, as defined in A.1.2.

q_I = Indeterminate shear flow which circulates around the closed cells, as defined in A.1.3.

In the calculation of the unit shear flow, the longitudinal stiffeners are to be taken into account.

A.1.2 Determinate shear flow

A.1.2.1 The determinate shear flow, q_D in [N/mm], at each location in the cross section can be obtained from the following line integration:

$$q_D(s) = -\frac{1}{10^6\, I_y}\int_0^S (z - z_n)\, t\, ds$$

where:

s = Coordinate value of running coordinate along the cross section, in [m].

I_y = Moment inertia of the cross section, in [m^4].

t = Thickness of plating, in [mm], or equivalent thickness of corrugated plate as defined in Section 1, Cl.1.4.6.

A.1.2.1 It is assumed that the cross section is composed of line segments as shown in Figure A.1.1 (where each line segment has a constant plate net thickness), the determinate shear flow can be calculated by the following equation:

$$q_{Dk} = q_D(\ell) = -\frac{t.\ell}{2x10^6\, I_y}(z_k + z_i - 2z_n) + q_{Di}$$

where:

q_{Dk}, q_{Di} = Determinate shear flow at node k and node i respectively, in [N/mm].

ℓ = Length of line segments, in [m].

z_k, z_i = Z coordinate of the end points k and i of line segment, in [m], as defined in Figure A.1.1.

A.1.2.2 Where the cross section includes closed cells, the closed cell are to be cut with virtual slits, as shown in Figure A.1.2 in order to obtain the determinate shear flow. These virtual slits must not be located in walls which form part of another closed cell.

A.1.2.3 Calculations of the determinate shear flow at bifurcation points can be calculated by water flow calculations as shown in Figure A.1.2.

A.1.3 Indeterminate shear flow

A.1.3.1 The indeterminate shear flow around closed cells of a cross section is considered as a constant value within the same closed cell. The following system of equation for determination of indeterminate shear flows can be developed. In the equations, contour integrations of several parameters around all closed cells are performed.

$$q_{Ik}\oint_k \frac{1}{t}ds - \sum_{i=1}^{N_w}\left(q_{Ii}\oint_{k\&i}\frac{1}{t}ds\right) = -\oint_k \frac{q_D}{t}ds$$

where:

N_w = Number of common walls shared by cell k and all other cells.

$k\&i$ = Common wall shared by cells k and i

q_{Ik}, q_{Ii} = Indeterminate shear flow around the closed cell k and i respectively [N/mm].

A.1.3.2 With assuming assembly of line segments shown in Figure A.1.1, the equations in A.1.3.1 can be expressed as follows:

$$q_{Ik}\sum_{k=1}^{N_k}\left(\frac{\ell}{t}\right)_k - \sum_{i=1}^{N_w}\left\{q_{Ii}\left[\sum_{k=1}^{N_m}\left(\frac{\ell}{t}\right)_k\right]_i\right\} = -\sum_{k=1}^{N_k}\phi_k$$

$$\phi_k = \left[-\frac{\ell^2}{6\times 10^3 I_Y}(z_k + 2z_i - 3z_n) + \frac{\ell}{t}q_{Di}\right]$$

where:

N_k = Number of line segments in cell k.

N_m = Number of line segments on the common wall shared by cells k and I, Refer Figure A.1.3.

q_{Di} = Determinate shear flow, in [N/mm], calculated according to A.1.2.1.

A.1.3.2 The difference in the directions of running coordinates specified in A.1.2 and in this section has to be considered.

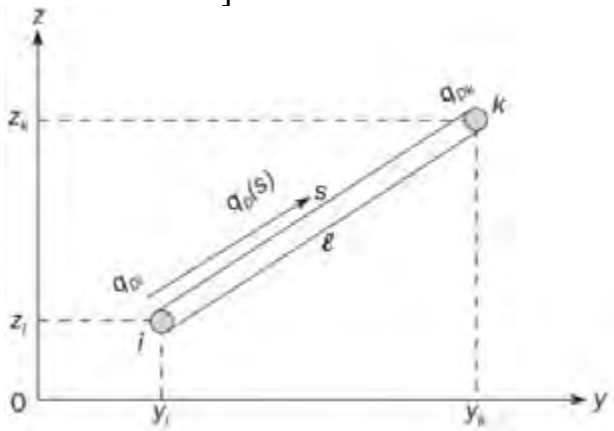


Figure A.1.1: Definition of line segment

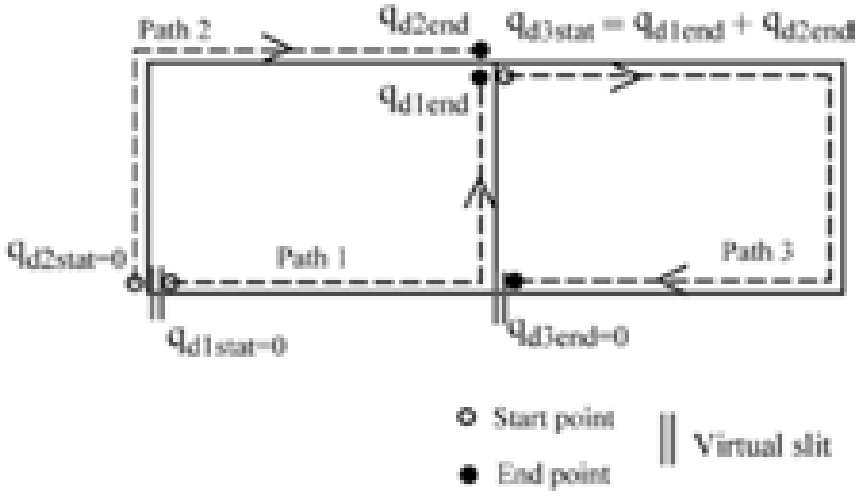


Figure A.1.2: Placement of virtual slits and calculation of determinate shear flow at bifurcation points

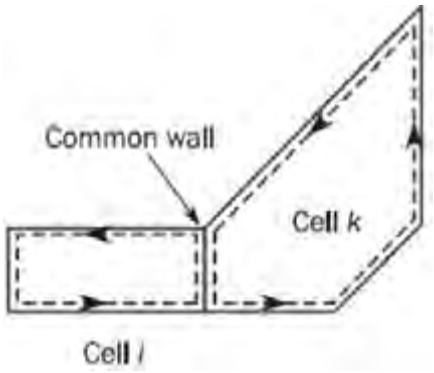


Figure A.1.3: Closed cells and common wall

A.2 Computation of sectional properties

A.2.1 Properties of the cross section are to be obtained by the following formulae where the cross section is assumed as the assembly of line segments:

$$\ell = \sqrt{(y_k - y_i)^2 + (z_k - z_i)^2}$$

$$a = \ell . t . 10^{-3}$$

$$A = \sum a$$

$$s_y = \frac{a}{2}(z_k - z_i)$$

$$S_y = \sum s_y$$

$$i_{y0} = \frac{a}{3}\left(z_k^2 + z_{k.} z_i + z_i^2\right)$$

$$I_{y0} = \sum i_{y0}$$

where:

a, A = Area of the line segment and the cross section respectively, in [m^2].

s_y, S_y = First moment of the line segment and the cross section about the baseline, in [m^3].

i_{y0}, I_{y0} = Moment inertia of the line segment and the cross section about the baseline, in [m^4].

A.2.2 The height of horizontal neutral axis, z_n in [m], can be obtained as follows:

$$z_n = \frac{S_y}{A}$$

A.2.3 Inertia moment about the horizontal neutral axis, in m4, can be calculated as follows:

$$I_y = I_{y0} - {z_n}^2 . A$$

End of Chapter

Chapter 7

Hull Local Scantlings

Contents

Section

1 *General*
2 *Load Application*
3 *Minimum Thickness*
4 *Plating*
5 *Stiffeners*
6 *Primary Supporting Members and Pillars*
7 *Intersection of Stiffeners and Primary Supporting Members*
8 *Superstructure, Deckhouse and Companionways*
9 *Bulwarks and Guard Rails*
10 *Means of Embarkation and Disembarkation*

Section 1

General

1.1 Application

1.1.1 This chapter applies to hull structure over the full length of the ship.

1.1.2 This chapter provides requirements for evaluation of plating, stiffeners and Primary Supporting Members (PSM) subject to lateral pressure, local loads and to hull girder loads, as applicable. Requirements are specified for:

- Load application in Section 2.

- Minimum thickness of plates, stiffeners and PSM in Section 3.

- Plating in Section 4

- Stiffeners in Section 5

- PSM and pillars in Section 6

- Intersection of stiffeners and PSM in Section 7

- Superstructure, deckhouse and Companionways in Section 8.

In addition, other requirements not related to defined design load sets, are provided.

1.1.3 Required scantlings

The offered net scantling is to be greater than or equal to the required scantlings based on requirements provided in this chapter.

1.1.4 Additional local strength requirements

- Additional local strength requirements are provided in Chapter 11, considering bow impact loads, bottom slamming loads, sloshing loads, wheel loading and additional local strength requirement for various structural members.

The design values of loads to be considered are given in Chapter 5.

- Scantlings of hull members contributing to the longitudinal strength are also to comply with the requirements of Chapter 6.

- The plating's, stiffeners and PSM's are also to comply with the requirements of buckling strength given in Chapter 9.

1.2 Acceptance criteria

1.2.1 Acceptance criteria set to be selected for scantling requirements are based on design load as follows:

- AC-I for design load S; static loads.

- AC-II for design load S+D; combination of static and dynamic loads.

- AC-IV for Accidental load like flooding.

Section 2

Load Application

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

2.1 Load combination

2.1.1 Hull girder stress

2.1.1.1 Hull girder longitudinal stress applicable for static loads 'S'

The hull girder longitudinal stress σ_{hg} in [N/mm^2], for static load 'S' to be taken as:

$$\sigma_{hg} = \sigma_{hg-sw}$$

where:

σ_{hg-sw} = Hull girder longitudinal stress, in [N/mm^2], due to vertical still water bending moment as defined in Chapter 6, Section 3, 3.2 for ships without large deck openings and in Chapter 6 section 3, 3.3 for ships with large deck openings.

2.1.1.2 Hull girder longitudinal stress applicable for loads 'S+D'

The hull girder longitudinal stress σ_{hg} in [N/mm^2], for loads 'S+D' to be taken as defined in Chapter 6, Section 3, 3.2 for ships without large deck openings and in Chapter 6 section 3,3.3 for ships with large deck openings.

2.2 Lateral pressures

2.2.1 Lateral pressure in intact conditions

The static and dynamic lateral pressures (defined in Chapter 5) induced by the sea and the various types of cargoes, ballast and other liquids are to be considered. Applied loads will depend on the location of the elements under consideration, and the adjacent type of compartments.

Sloshing pressures are to be applied to tank structures, according to Chapter 11, Section 7.

2.2.2 Lateral pressure in flooded conditions

Following compartments are subjected to lateral pressure in flooded conditions:

- Watertight boundaries of compartments not intended to carry liquids (excluding shell envelope).

- Watertight boundaries of compartments intended to carry liquids which have overflow height below freeboard deck.

- Plating located below the deepest equilibrium waterline (excluding side shell plating) which constitute boundaries intended to stop vertical and horizontal flooding.

2.3 Pressure combination

2.3.1 Elements of the outer shell

If the compartment adjacent to the outer shell is intended to carry liquids, the static and dynamic lateral pressures to be considered are the differences between the internal pressures and

the external sea pressures at the corresponding draught.

If the compartment adjacent to the outer shell is not intended to carry liquids, the internal pressures and the external sea pressures are considered individually.

2.3.2 Elements other than those of the outer shell

Except as specified in Cl.2.3.1, the static and dynamic lateral pressures on an element separating two adjacent compartments are those obtained considering the two compartments individually loaded.

2.4 Design load sets

2.4.1 Application

These requirements apply to:

- Plating and stiffeners along the full length of main hull of the ship.
- Primary supporting members, if scantlings are not determined by global FE analysis or cargo hold analysis
- Plating, stiffeners and primary supporting members in superstructures and deckhouses contributing to hull girder longitudinal strength.

2.4.2 Load components

The static and dynamic load components are to be determined in accordance with Chapter 5, Section 6, Cl.6.2.1.

Radius of gyration, K_r, and metacentric height, GM, are to be in accordance with Chapter 5, Section 2 for the considered loading conditions specified in the design load sets given in Table 2.4.3.

2.4.3 Design load sets for plating, stiffeners and PSM

Design load sets for plating, stiffeners and primary supporting members are given in Table 2.4.3. Impact and sloshing load sets are considered in Chapter 11.

Table 2.4.3 Design load sets

Structural member	Design load set	Design load scenario	Load component[4)]	Draught	Acceptance criteria	Loading condition
External shell and exposed deck	SEA-1	1	$P_S + P_W$, P_D	T_{SC}	AC-II	Full load condition
Superstructure side			max (P_W, P_{SI})			
External shell	SEA-2	2	P_S	T_{SC}	AC-I	Harbour condition
Boundaries of water ballast tanks and ballast holds	WB-1	1	$P_{ls-1} + P_{ld} - (P_S + P_W)$[1)]	T_B	AC-II	Normal ballast condition
	WB-2	2	$\max(P_{ls-2}, P_{ls-ST}) - P_S$[1)]	min (T_B, $0.25T_{SC}$)	AC-I	Harbour and Tank Testing condition
	WB-3	3	$P_{ls-3} + P_{ld} - (P_S + P_W)$[1)]	T_B	AC-II	Water ballast exchange
	WB-4	4	$P_{ls-4} + P_{ld} - (P_S + P_W)$[1)]			
Boundaries of tanks other than	TK-1	1	$P_{ls-1} + P_{ld} - (P_S + P_W)$[1)]	T_B	AC-II	Normal ballast condition

water ballast tanks	TK-2	2	$\max(P_{ls-2}, P_{ls-ST}) - P_S$ [1)]	$0.25T_{SC}$	AC-I	Harbour and Tank Testing condition
Bulk cargo hold	BC-1	1	$P_{bs} + P_{bd}$	T_{SC}	AC-II	Homogeneous loading, fully filled hold
	BC-2	2	P_{bs}		AC-I	
	BC-3	1	$P_{bs} + P_{bd}$	T_{SC}	AC-II	Homogeneous loading, Partially filled hold [5)]
	BC-4	2	P_{bs}		AC-I	
Cargo oil/Liquid cargo tank	LT-1	1	$P_{ls-1} + P_{ld} - (P_S + P_W)$ [1)]	T_{SC}	AC-II	Full Load condition
	LT-3			$0.6T_{SC}$		Partial load condition
	LT-2	2	$\max(P_{ls-2}, P_{ls-ST}) - P_S$ [1)]	T_{SC}	AC-I	Harbour and Tank Testing condition
Internal structures in tanks (e.g., Webs of PSM's)	INT-1		P_{int}	T_{SC}	AC-I	-
Collision bulkhead	FD-1	5	P_{in-f}	T_{DAM}	AC-I	Flooded condition
Watertight boundaries other than collision bulkhead					AC-IV	
Exposed decks and non-exposed decks and platforms with distributed load	UDL-1 [2)]	1	$P_{dl-s} + P_{dl-d}$ $F_{U-s} + F_{U-d}$	T_{SC} [3)]	AC-II	Full load condition
	UDL-2 [2)]	2	P_{dl-s} F_{U-s}	-	AC-I	
Decks and hatch covers/ RO/RO equipment with wheel loading	WL-1 [2)]	1	P_{wl-1}	T_{SC} [3)]	AC-II	Full load condition
	WL-2 [2)]	2	P_{wl-2}	-	AC-I	

1) $(P_S + P_W)$ and P_S is considered for external shell only.
2) Distributed or concentrated loads only. Need not to be combined with simultaneously occurring green sea pressure.
3) For cargo loads, draught, GM and K_r values may be based on actual cargo conditions in the loading manual.
4) Local loads:
P_S = Hydrostatic sea pressure as given in Chapter 5, Section 4, Cl.4.1.2.
P_W = Wave pressure as given in Chapter 5, Section 4, Cl.4.1.3.
P_D = External dynamic design pressure for exposed decks and wheelhouse top due to green sea loading as given in Chapter 5, Section 4, Cl.4.2.2 and Chapter 5, Section 4, Cl.4.3.2, respectively
P_A = Design pressure for end bulkheads of superstructure and deckhouse walls as given in Chapter 5, Section 4, Cl.4.3.4.
P_{SI} = External design pressure for external side of superstructure as given in Chapter 5, Section 4, Cl.4.3.3.
P_{ls-1} = Static tank pressure during normal operations at sea as given in Chapter 5, Section 5, Cl.5.1.2.1(a).

P_{ls-2} = Static tank pressure during normal operations at harbour/sheltered water as given in Chapter 5, Section 5, Cl.5.1.2.1(b).

P_{ls-3} = Static tank pressure during flow through ballast water exchange as given in Chapter 5, Section 5, Cl.5.1.2.1(c).

P_{ls-4} = Static tank pressure due to sequential ballast water exchange as given in Chapter 5, Section 5, Cl.5.1.2.1(d).

P_{ls-ST} = Static tank testing pressure as given in Chapter 5, Section 5, Cl.5.1.2.1(f).

P_{in-f} = Internal pressure in flooded condition as given in Chapter 5, Section 5, Cl.5.1.1.2.

P_{int} = Minimum pressure for internal structures in tanks as given in Chapter 5 Section 5, Cl.5.3.1.

P_{dl} = Pressure due to distributed deck load as given in Chapter 5, Section 4, Cl.4.2.3.1 or Chapter 5, Section 5, Cl.5.2.2.

F_U = Force on exposed and non-exposed decks and platforms due to unit loads as given in Chapter 5, Section 4, Cl.4.2.3.2.

P_{wl-1} = Design pressure for wheel loading during normal operation at sea as given in Chapter 11, Section 5.

P_{wl-2} = Design pressure for wheel loading during normal operation at harbour as given in in Chapter 11, Section 5.

5) Considered for heavy cargo densities.

Section 3

Minimum Thickness

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

M_{GRAB} = Mass of unladen grab, in [t].

3.1 Plating

3.1.1 The net thickness of plating in [mm], is to comply with the appropriate minimum thickness requirements given in Table 3.1.1 and given by:

$$t_{net} = a + bL\sqrt{k}$$

where:

a = coefficient as defined in Table 3.1.1.

b = coefficient as defined in Table 3.1.1.

L = Rule length, but need not be taken greater than 300[m].

Table 3.1.1: Minimum net thickness for plating in [mm]

Sl. No	Plating	Location	Coefficients	
			a	b
1	Keel plate	-	6.5	0.035
2	Bottom plate	-	4.5	0.035
3	Bilge plate	-	4.5	0.035
4	Side shell and super structure side	From upper end of bilge plating to T_{SC} + 4.6m	4.5	0.035
		From T_{SC} + 4.6 m to T_{SC} + 6.9m		0.025
		From T_{SC} + 6.9 m to T_{SC} + 9.2m		0.015
		Elsewhere		0.01
5	Deck plate	Weather deck, Strength deck[1)]	4.5	0.02
		Boundary for cargo tanks, water ballast tanks and hold intended for cargo in bulk	4.5	0.02
		Tween decks and first tier of erections	4.5	0.015
		Other decks[2),3)]	4.5	0.01
6	Inner bottom	General	5	0.02
		Cargo spaces loaded through cargo hatches except container holds	5.5	0.025
7	Bulkheads	Watertight (Transverse and longitudinal) /Tank bulkheads	4.5	0.015
		Bulkheads for cargo tanks		
		Peak bulkheads		
		Hold intended for bulk cargo		
		Non-tight bulkhead, Wash bulkheads, Bulkheads between dry spaces.	4.5	0.01
		Pillar bulkheads in fore and aft peaks	7.0	
		Pillar bulkheads in holds[5)]	7.0	
		Pillar bulkheads in tween decks[5)]	6.5	
		Walls in accommodation	4.5	

8	Other members	Other plates in general	4.5	0.01

1) Includes Forecastle deck also.
2) For the decks not contributing to hull girder longitudinal strength, b may be taken equal to 0.
3) Where deck coverings in accordance with Chapter 2, Section 3.10 are provided, the deck thickness may be reduced by 10 percent without being less than 4.5[mm].
4) For ships discharged by grabs, the net plating thickness 't' of the inner bottom and exposed parts of sloping bulkheads is not to be less than:

$$t = 0.62\sqrt{s.k}\left(\frac{M_{GRAB}}{20}\right)^{0.25} [mm]$$

Where ceiling is fitted, the above requirement may be reduced by 2[mm].
The slope bulkhead plating of increased thickness is to extend for a minimum distance corresponding to a vertical height of 1.5[m] above the line of the inner bottom.
5) For ships L<90[m], minimum thickness of Pillar bulkheads is to be 5.5[mm].

3.2 Stiffeners and Tripping brackets

3.2.1 Minimum thickness requirements

The net thickness of the web and face plate, if any, of stiffeners and tripping brackets in [mm], is to comply with the minimum net thickness given in Table 3.2.1 and given by,

$$t_{net} = a + bL$$

where,

a = coefficient as defined in Table 3.2.1.

b = coefficient as defined in Table 3.2.1.

L = Rule length, but need not be taken greater than 250[m].

Table 3.2.1: Minimum net thickness for stiffeners and tripping brackets

Sl. No	Element	Location	Coefficients	
			a	b
1	Stiffeners and attached end brackets	Watertight boundaries: Tank boundary, Cargo hold, boundary of dry bulk cargo, strength deck and shell up to freeboard deck	3.5	0.015
		Other structure	3	0.015
		Structures in deckhouse and superstructure and decks for vessels with more than 2 continuous decks above 0.7D from baseline	4	-
2	Tripping brackets		4.5	0.01

3.3 Primary supporting members

3.3.1 Minimum thickness requirements of PSM's

3.3.1.1 The net thickness of web plating and flange of primary supporting members in [mm], is to comply with the appropriate minimum thickness requirements given by:

$$t_{net} = a + bL\sqrt{k}$$

where:

a = coefficient as defined in Table 3.3.1.

b = coefficient as defined in Table 3.3.1.

L = Rule length, but need not be taken greater than 250[m].

Table 3.3.1: Minimum net thickness for PSM's in [mm]					
Sl. No	**Member**	**Location**	**Within 0.4L amidships (Including Machinery space Unless noted otherwise)**		**Remarks**
			Coefficients		
			a	**b**	
1	Double bottom	Center girder	5	0.03	
		Lower strake of centreline wash bulkhead			
		Side girder	5	0.02	
		Floors	5	0.02	
		Girders bounding a duct keel	5	0.025	Refer Notes 2&3
		Margin plate	5	0.02	Refer Notes 1&3
2	Single bottom	Center girder	6	0.04	Refer Note 1
		Side girder	5	0.04	
		Floors	5	0.025	Refer Note 4
	Other PSM's				
3	Peak Spaces	PSM's in Aft and Fore peak	5	0.05	Refer Note 5
4	PSM's not mentioned above	At tank boundaries, boundaries of holds intended for cargo in bulk, single strength deck and shell up to freeboard deck	5	0.015	
		Deckhouses and superstructures and decks for vessels with more than 2 continuous decks above 0.7D from baseline	4.5	0.015	Refer Note 6
		Elsewhere	4	0.015	

Note 1: Towards the fore and after ends, the scantlings of the bottom members may be reduced by 10%.

Note 2: The minimum net thickness is to be taken not less than that required for the centre girder.

Note 3: In the engine room, thickness of girders bounding duct keel and margin plate is to be increased by 1[mm].

Note 4: In the engine room, the thickness of floors webs is to be increased by 1[mm] and the face plate area of floor (as given in Table 3.3.2) is to be increased by 50%.

Note 5: May not be taken greater than 10[mm].

Note 6: May not be taken greater than 6.5[mm].

Table 3.3.2: Transversely framed Single bottom PSM's		
Item	**Depth 'd' in [mm]**	**Face plate area 'a' in [cm^2]**
Floor	40(B+T)	3.5 T. k^$^{0.5}$
Center girder	As for floors	5+0.6L k
Side girders	As for floors	5+0.3L k
1) The thickness of face plates is not to be less than 1/15 of the face width.		
(2) The top of floors, in general, is to be level from side to side. However, in ships having considerable rise of floor, the depth of web at 10 % of the span from ends, is not to be less than 0.5d [mm].		

3.3.2 Where single bottom in the cargo region is stiffened by transverse frames supported by longitudinal girders, the scantlings of the frames and longitudinal girders are to be determined in accordance with Section 6, Cl.6.1.2 for section modulus and Chapter 3, section 7, Cl.7.4.8.1 for girder web area.

3.3.3 The longitudinal girders are to be satisfactorily stiffened against buckling in accordance with the requirements given in Chapter 9. Tripping brackets are to be fitted in accordance with the requirements given in Chapter 3, Section 6.

Section 4

Plating

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

α_p = correction factor for the panel aspect ratio to be taken as below but not to be greater than 1.0:

$$\alpha_p = 1.2 - \frac{b}{2.1\,a}$$

a = Length of plate panel, in [mm], as defined in Chapter 3 Section 7, Cl.7.5.1.

b = Breadth of plate panel, in [mm], as defined in Chapter 3 Section 7, Cl.7.5.1.

P = Design pressure for the considered design load set, Refer Section 2, 2.4, calculated at the load calculation point defined in Chapter 3 Section 7, Cl.7.5.2, in [kN/m^2].

σ_{hg} = Hull girder longitudinal stress, in [N/mm^2], as defined in Section 2, 2.1, calculated at the load calculation point as defined in Chapter 3, Section 7, Cl.7.5.2.

4.1 Plating subjected to lateral pressure

4.1.1 General

4.1.1.1 The net thickness, in [mm], is not to be taken less than the greatest value for all applicable design load sets, as defined in Section 2, Cl.2.4.3, given by:

$$t_{net} = 0.0158.\alpha_p.b\sqrt{\frac{|P|}{\chi.C_a.R_{eH}}}$$

where:

C_a = Permissible bending stress coefficient for plate taken equal to:

$$C_a = \beta - \alpha\frac{|\sigma_{hg}|}{R_{eH}}$$

not to be taken greater than $C_{a\text{-}max}$

β = Coefficient as defined in Table 4.11.

α = Coefficient as defined in Table 4.11.

C_{a-max} = Maximum permissible bending stress coefficient as defined in Table 4.11.

χ = Coefficient taken equal to:

- In intact condition:

= 0.70 for inner bottom or bilge hopper tank plating in cargo holds of bulk carriers

= 1.0 for other cases.

- In flooded condition:

= 1.00 for collision bulkheads for acceptance criteria set AC-I,

= 1.15 for other watertight boundaries of compartments.

Table 4.1.1: Plating, definition of β, α and $C_{a\text{-}max}$

Acceptance criteria	Structural member		β	α	C_{a-max}
AC-I	Longitudinal Strength members	Longitudinal stiffened plating	0.9	0.5	0.8
		Transverse stiffened plating	0.9	1.0	0.8
	Other members		0.8	0.0	0.8
AC-II/ AC-IV	Longitudinal Strength members	Longitudinal stiffened plating	1.05	0.5	0.95
		Transverse stiffened plating	1.05	1.0	0.95
	Other members		1.0	0.0	1.0

4.1.2 Plating of corrugated bulkheads

4.1.2.1 Cold and hot formed corrugations

The net thicknesses, in [mm], of the web and flange plates of corrugated bulkheads are not to be taken less than the greatest value calculated for all applicable design load sets, as defined in Chapter 7, Section 2, Cl.2.4.3, given by:

$$t_{net} = 0.0158\, b_p \sqrt{\frac{|P|}{C_{CB}.\, R_{eH}}}$$

where:

b_p = Breadth of plane corrugation plating in [mm].

= a for flange plating, in [mm], as defined in Figure 4.1.2.

= c for web plating, in [mm], as defined in Figure 4.1.2.

C_{CB} = Permissible bending stress coefficient for corrugated bulkhead plating taken equal to:

$$C_{CB} = \beta_{CB} - \alpha_{CB} \frac{|\sigma_{hg}|}{R_{eH}},$$

not to be greater than $C_{CB-max.}$

β_{CB} = Coefficient as defined in Table 4.1.2.

α_{CB} = Coefficient as defined in Table 4.1.2.

C_{CB-max}= Maximum permissible bending stress coefficient as defined in Table 4.1.2.

Table 4.1.2: Definition of β_{CB}, α_{CB} and $C_{CB-max.}$

Acceptance criteria set	Structural member	β_{CB}	α_{CB}	C_{CB-max}
AC-I	a) Horizontally corrugated longitudinal bulkheads	0.90	0.50	0.75
	b) Other corrugated bulkheads			
	• Transverse	0.75	0.00	0.75
	• Vertically corrugated longitudinal bulkheads			
AC-II	a) Horizontally corrugated longitudinal bulkheads	1.05	0.50	0.90
	b) Other corrugated bulkheads			
	• Transverse	0.90	0.00	0.90
	• Vertically corrugated longitudinal bulkheads			

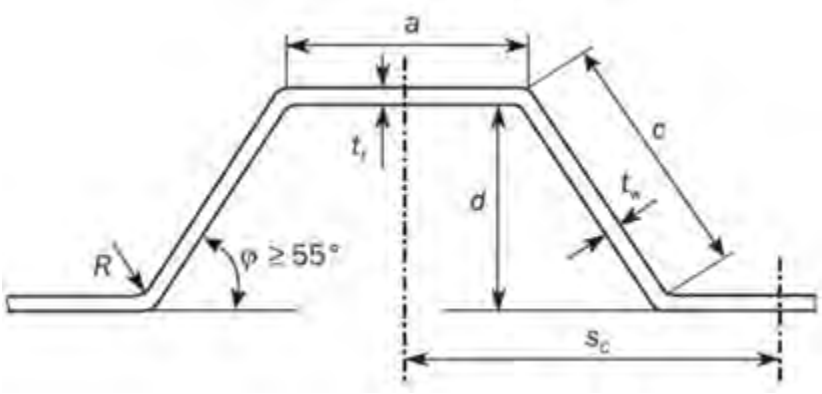


Figure 4.1.2: Dimensions of a corrugated bulkhead

4.1.2.2 Built-up corrugations

For built-up corrugations, with flange and web plate of different thickness, the net thickness, in [mm], is to be taken as the greatest value calculated for all applicable design load sets, as defined in Chapter 7, Section 2, Cl.2.4.3, given by:

$$t_1 = \sqrt{\frac{0.0005\, {b_p}^2.\, |P|}{C_{CB}.\, R_{eH}} - (t_2)^2}$$

where:

t_1 = Net thickness of the thicker plating, in [mm] either flange or web.

t_2 = Net required thickness of the thinner plating, in [mm] either flange or web, not to be less than required by Cl.4.1.2.1.

b_p = Breadth of thicker plate, in [mm] either flange or web.

C_{CB} = Permissible bending stress coefficient as defined in Cl.4.1.2.1.

4.1.3 Bilge plating

4.1.3.1 The thickness of plating in bilge area as defined in Chapter 1, Section 2, Cl.2.6.6 is to be equal or greater than the lesser of Section 3, Cl.3.1.1 and Cl.4.1.3.3, and also to satisfy Cl.4.1.3.2.

4.1.3.2 The net thickness of bilge plating is not to be taken less than the offered net thickness for the adjacent bottom shell or adjacent side shell plating, whichever is greater.

4.1.3.3 The net thickness of rounded bilge plating, in [mm], is not to be taken less than:

$$t_{net} = 6.45 \times 10^{-4}(P_{ex}.s_b)^{0.4}R^{0.6}$$

where:

P_{ex} = Design sea pressure in [kN/m^2] for the design load set SEA-1 calculated at the lower turn of the bilge.

R = Effective bilge radius in [mm].

$$R = R_0 + 0.5(\Delta s_1 + \Delta s_2)$$

R_0 = Radius of curvature, in [mm]. Refer Figure 4.5.8.

Δs_1 = Distance between the lower turn of bilge and the outermost bottom longitudinal, in [mm], Refer Figure 4.1.3. Where the outermost bottom longitudinal is within the curvature, this distance is to be taken as zero.

Δs_2 = Distance between the upper turn of bilge and the lowest side longitudinal, in [mm], Refer Figure 4.1.3. Where the lowest side longitudinal is within the curvature, this distance is to be taken as zero.

s_b = Distance, in [mm] between transverse stiffeners, webs or bilge brackets.

4.1.3.4 Longitudinally stiffened bilge plating is to be assessed as regular stiffened plating. The bilge thickness is not to be less than the lesser of the value obtained by Cl.4.1.1.1 and Cl.4.1.3.3. A bilge keel is not considered as an effective 'longitudinal stiffening' member.

4.1.3.5 Transverse extension of bilge minimum plate thickness

Where a plate seam is located in the straight plate just below the lowest stiffener on the sideshell, any increased thickness required for the bilge plating does not have to be extended to the adjacent plate above the bilge provided the plate seam is not more than $s_2/4$ below the lowest side longitudinal.

Similarly, for the flat part of adjacent bottom plating, any increased thickness for the bilge plating does not have to be extended to the adjacent plate provided that the plate seam is not more than $s_1/4$ beyond the outboard bottom longitudinal. For definition of s_1 and s_2, Refer Figure 4.1.3.

4.1.3.6 Hull envelope framing in bilge area

For transversely stiffened bilge plating, a longitudinal is to be fitted at the bottom and at the side close to the position where the curvature of the bilge plate starts. The scantling of those longitudinals are to be not less than the one of the closer adjacent stiffener. The distance between the lower turn of bilge and the outermost bottom longitudinal, Δs_1, is generally not to be greater than one-third of the spacing between the two outermost bottom longitudinals,s_1.

Similarly, the distance between the upper turn of the bilge and the lowest side longitudinal, Δs_2, is generally not to be greater than one-third of the spacing between the two lowest side longitudinals, s_2. Refer Figure 4.1.3.

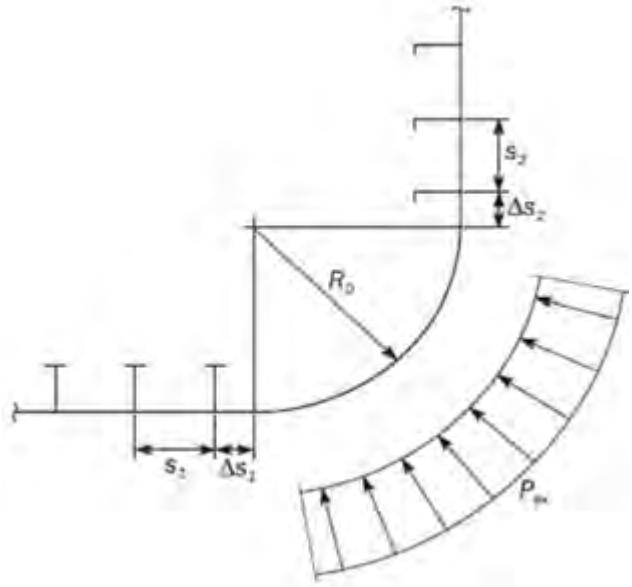


Figure 4.1.3: Transverse stiffened bilge plating

4.1.4 Local strength of plating for ships with L<90 [m]

4.1.4.1 Application

The yield check in Cl.4.1.1 are generally based on values of C_{a-max} and C_{CB-max} when considering envelope loads applied in accordance with Chapter 5, Section 6, Table 6.2.1.2 and provided:

- Scantlings of longitudinal members in midship region is checked and found in compliance with Cl.4.1.1 using stress coefficient calculated with applicable hull girder stress and also found in compliance with the buckling requirements in accordance with Chapter 9.

- Scantlings of longitudinal members are kept constant within 0.4L amidship and tapered toward fore and aft end.

Section 5

Stiffeners

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

P = Design pressure, in [kN/m^2], for the considered design load set, Refer Section 2, 2.4, calculated at the load calculation point as defined in Chapter 3, Section 7, Cl.7.5.2.

σ_{hg} = Hull girder longitudinal stress, in [N/mm^2], as defined in Section 2, 2.1, calculated at the load calculation point as defined in Chapter 3, Section 7, Cl.7.5.2.

ℓ_{bdg} = Effective bending span, in [m,] as defined in Chapter 3, Section 7, Cl.7.1.1.

ℓ_{shr} = Effective shear span, in [m], as defined in Chapter 3, Section 7, Cl.7.1.3 for stiffeners.

d_{shr} = Effective shear depth, in [mm], as defined in Chapter 3, Section 7, Cl.7.4.3 for stiffeners.

χ = Coefficient taken equal to:

- In intact condition:

= 0.90 for stiffeners attached to inner bottom or bilge hopper tank plating in cargo holds of bulk carriers,

= 1.0 for other cases.

- In flooded condition:

= 1.00 for collision bulkheads for acceptance criteria set AC-I.

= 1.15 for other watertight boundaries of compartments.

5.1 Stiffeners subjected to lateral pressure

5.1.1 Yielding check

5.1.1.1 Web plating

The minimum net web thickness, t_w in [mm], is not to be taken less than the greatest value calculated for all applicable design load sets as defined in Section 2, Cl.2.4.3, given by:

$$t_w = \frac{f_{shr}.|P|.s.\ell_{shr}}{d_{shr}.\chi.C_t.\tau_{eH}}$$

with $\chi.C_t$ not to be taken greater than 1.0.

where:

f_{shr} = Shear force distribution factor taken as:

For continuous stiffeners with fixed ends, f_{shr} is not to be taken less than:

= 0.5 for horizontal stiffeners and upper end of vertical stiffeners.

= 0.7 for lower end of vertical stiffeners.

For non-continuous stiffeners with simply supported ends

= 0.5 for all stiffeners.

For stiffeners with end fixity different from above, variable load or being part of grillage, the requirement in Cl.5.1.2 applies.

C_t = Permissible shear stress coefficient for the design load set being considered, taken as:

= 0.75 for acceptance criteria set AC-I.

= 0.90 for acceptance criteria set AC-II.

5.1.1.2 Section modulus

The minimum net section modulus, Z in [cm^3], is not to be taken less than the greatest value calculated for all applicable design load sets as defined in Section 2, Cl.2.4.3, given by:

$$Z = \frac{|P|.s.\ell_{bdg}^2}{f_{bdg}.\chi.C_s.R_{eH}}$$

with $\chi.C_s$ not to be taken greater than 1.0

where:

f_{bdg} = Bending moment factor taken as:

For all continuous horizontal and vertical stiffeners, f_{bdg} is to be taken from Table 5.1.1.1.

For all non-continuous stiffeners, f_{bdg} is taken as:

= 8 For simply supported ends

= 10 For fixed at both ends

For stiffeners with end fixity different from above, variable load or being part of grillage, the requirement in Cl.5.1.2 applies.

C_s = Permissible bending stress coefficient as defined in Table 5.1.1.1 for the design load set being considered.

β_s = Coefficient as defined in Table 5.1.1.2.

α_s = Coefficient as defined in Table 5.1.1.2.

C_{s-max}= Maximum permissible bending stress coefficient as defined in Table 5.1.1.2.

Table 5.1.1.1: Definition of C_s

Structural member	**Sign of hull girder bending stress, σ_{hg}**	**Lateral pressure acting on**	Stiffener boundary condition (1)	f_{bdg}	**Coefficient C_s**
For continuous stiffeners	Tension (positive)	Stiffener side	F - F	12	$C_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than C_{s-max}
			F - S	8	
			S - S	8	$C_s = C_{s-max}$
		Plate side	F - F	24	$C_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than C_{s-max}
			F - S	14	
			S - S	8	
	Compression (negative)	Stiffener side	F - F	24	$C_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than C_{s-max}
			F - S	14	
			S - S	8	
		Plate side	F - F	12	$C_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than C_{s-max}
			F - S	8	
			S - S	8	$C_s = C_{s-max}$
For non-continuous stiffeners	Tension (positive)	Plate side	Refer Cl. 5.1.1.2		$C_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than C_{s-max}
	Compression (negative)	Stiffener side			
	Tension (positive)	Stiffener side			$C_s = C_{s-max}$
	Compression (negative)	Plate side			

(1) F - F stands for both ends of the stiffener fixed against rotation
F - S stands for one end of the stiffener fixed and the other not fixed against rotation
S - S stands for both ends of the stiffener not fixed against rotation

Table 5.1.1.2: Definition of $\alpha_s, \beta_s, C_{s-max.}$				
Acceptance criteria set	**Structural member**	β_s	α_s	$C_{s-max.}$
AC-I	Longitudinal strength member	0.85	1.00	0.75
	Transverse or vertical member	0.75	0.00	0.75
AC-II/AC-IV	Longitudinal strength member	1.00	1.00	0.90
	Transverse or vertical member	0.90	0.00	0.90

5.1.1.3 Groups of equal ordinary stiffeners

Scantlings of stiffeners based on requirements in Cl.5.1.1.1 and Cl.5.1.1.2 may be decided based on the concept of grouping designated sequentially placed stiffeners of equal scantlings on a single stiffened panel between primary supporting members. The scantling of the group is to be taken as the greater of the following:

- The average of the required scantling of all stiffeners within a group.

- 90% of the maximum scantling required for any one stiffener within the group.

5.1.1.4 Plate and stiffener of different materials

When the minimum specified yield stress of a stiffener exceeds the minimum specified yield stress of the attached plate by more than 35%, the following criterion is to be satisfied:

$$\sigma_s \leq \left[\sigma_p - \frac{\alpha_s |\sigma_{hg}|}{\beta_s}\right]\frac{Z_p}{Z} + \frac{\alpha_s |\sigma_{hg}|}{\beta_s}$$

where:

σ_s = Minimum specified yield stress, in [N/mm^2], of the material of the stiffener.

σ_p = Minimum specified yield stress, in [N/mm^2], of the material of the attached plate.

σ_{hg} = Hull girder bending stress, in [N/mm^2], as defined in Section 2, Cl.2.1.1 with $|\sigma_{hg}|$ not to be taken less than $0.4\,\sigma_p$.

Z = Net section modulus, in [cm^3], in way of face plate/free edge of the stiffener.

Z_p = Net section modulus, in [cm^3], in way of the attached plate of stiffener.

α_s , β_s = Coefficients defined in Table 5.1.1.2.

5.1.1.5 Struts connecting ordinary stiffeners

The net sectional area,A_{SR} in [cm^2], and the net moment of inertia about the main axis, I_{SR} in [cm^4], of struts connecting stiffeners is not be less than the values obtained from the following formulae:

$$A_{SR} = \frac{P_{SR}.s.\ell}{20000}$$

$$I_{SR} = \frac{0.75.s.\ell(P_{SR1} + P_{SR2})A_{ASR}.\ell_{SR}^2}{47200.A_{ASR} - s.\ell(P_{SR1} + P_{SR2})}$$

where:

P_{SR} = Pressure in [kN/m^2], to be taken as the greater of the following values:

$$P_{SR} = 0.5(P_{SR1} + P_{SR2})$$

$$P_{SR} = P_{SR3}$$

P_{SR1} = External pressure in way of the strut, in [kN/m^2], acting on one side, outside the compartment in which the strut is located.

P_{SR2} = External pressure in way of the strut, in [kN/m^2], acting on the opposite side, outside the compartment in which the strut is located.

P_{SR3} = Internal pressure at mid-span of the strut, in [kN/m^2], in the compartment in which the strut is located.

s = Spacing, in [mm], of stiffeners, measured at mid-span along the chord.

ℓ = Span, in [m], of stiffeners connected by the strut.

ℓ_{SR} = Length of the strut, in [m].

A_{ASR} = Actual net sectional area of the strut, in [cm^2].

5.1.1.6 Net section modulus of corrugations

(a) The net section modulus of a corrugation may be obtained, in [cm^3], from the following formula:

$$Z = \left[d\left(\frac{3.a.t_f + c.t_w}{6}\right)\right] 10^{-3}$$

where:

t_f, t_w = Net thickness of the plating of the corrugation, in [mm], Refer Section 4, Figure 4.1.2.

d, a, c = Dimensions of the corrugation, in [mm], Refer Section 4, Figure 4.1.2.

Where the web continuity is not ensured at ends of the bulkhead, the net section modulus of a corrugation is to be obtained, in [cm^3], from the following formula:

$$Z = 0.5\, a.t_f.d\, 10^{-3}$$

(b) The net section modulus, in [cm^3], of corrugated bulkheads subjected to lateral pressure is not to be less than the value obtained from the following formula:

$$Z = \frac{|p|.S_c.\ell_{bdg}^2}{f_{bdg}.C_s.R_{eH}}$$

S_c = Half pitch length, in [mm], of the corrugation, as defined in Refer Section 4, Figure 4.1.2.

ℓ_{bdg} = Effective bending span, in [m,] as defined in Chapter 3, Section 6, Cl.6.4.3.

C_s = Permissible bending stress coefficient as defined in Table 5.1.1.1 for the design load set being considered.

f_{bdg} = Bending moment factor as defined in Cl.5.1.1.2.

5.1.2 Beam analysis

5.1.2.1 Application

The maximum bending stress, σ_b and shear stress, τ in a stiffener using net properties with reduced end fixity, variable load or being part of grillage are to be determined by direct calculations taking into account the following:

The distribution of static and dynamic pressures and forces, if any.

- The number and position of intermediate supports (e.g., decks, girders, etc.).

- The condition of fixity at the ends of the stiffener and at intermediate supports.

- The geometrical characteristics of the stiffener on the intermediate spans.

- The effective breadth, b_{eff}, for yield check.as given in Chapter 3, Section 7, Cl.7.4.4.

5.1.2.2 Stress criteria

The calculated stress, in [N/mm^2] is to comply with the following criteria where the coefficients C_t and C_s, are defined in Cl.5.1.1.1 and Cl.5.1.1.2.

$$\tau \leq \chi C_t \tau_{eH}$$

$$\sigma_b \leq \chi C_s R_{eH}$$

5.1.3 Yielding strength of stiffeners for ships with L<90 [m]

5.1.3.1 Application

The yield check in Cl.5.1.1 and Cl.5.1.2 are generally based on C_{s-max} when considering envelope loads applied in accordance with Chapter 5, Section 6, Table 6.2.1.2 and provided:

- Scantlings of longitudinal members in midship region is checked and found in compliance with Cl.5.1.1 or Cl.5.1.2 using stress coefficient calculated with applicable hull girder stress and also found in compliance with the buckling requirements in accordance with Chapter 9.

- Scantlings of longitudinal members are kept constant within 0.4L amidship and tapered toward fore and aft end.

Section 6

Primary Supporting Members and Pillars

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

P = Design pressure, in [kN/m^2], for the considered design load set, Refer Section 2, 2.4, calculated at the load calculation point as defined in Chapter 3 Section 7, Cl.7.5.2.

σ_{hg} = Hull girder longitudinal stress, in [N/mm^2], as defined in Section 2, 2.1, calculated at the load calculation point as defined in Chapter 3, Section 7, Cl.7.5.2.

ℓ_{bdg} = Effective bending span, in [m,] as defined in Chapter 3, Section 7, Cl.7.1.7.

ℓ_{shr} = Effective shear span, in [m], as defined in Chapter 3, Section 7, Cl.7.1.8 for PSM's.

χ = Coefficient taken equal to:

- In intact condition:

= 0.90 for primary supporting members attached to inner bottom or bilge hopper tank plating in cargo holds of bulk carriers,

= 1.0 for other cases.

- In flooded condition:

= 1.00 for collision bulkheads for acceptance criteria set AC-I.

= 1.15 for other watertight boundaries of compartments.

6.1 PSM subjected to lateral pressure

6.1.1 General

The requirements of this section apply to primary supporting members (PSM's) subjected to lateral pressure and concentrated loads and pillars subjected to compressive axial loads. The yielding and buckling check is to be carried out for such members subjected to specific loads. Alternatively, the scantlings of the members are to be verified by means of FE analysis as defined in Chapter 8.

6.1.2 Net section modulus

The section modulus, Z in [cm^3], of primary supporting members subjected to lateral pressure, calculated in accordance with Chapter 3 Section 7, Cl.7.4.7, is not to be taken less than the greatest value for all applicable design load sets defined in Section 2, Cl.2.4.3, given by:

$$Z = \frac{1000\,|P|.S.\ell_{bdg}^2}{\chi.f_{bdg}.C_s.R_{eH}}$$

where:

f_{bdg} = Bending moment distribution factor, as given in Table 6.1.2.2.

C_s = Permissible bending stress coefficient taken equal to

$$C_s = \beta_s - \alpha_s \frac{|\sigma_{hg}|}{R_{eH}}$$

β_s = Coefficient as defined in Table 6.1.2.1.

α_s = Coefficient as defined in Table 6.1.2.1.

C_{s-max}= Maximum permissible bending stress coefficient as defined in Table 6.1.2.1.

The section modulus is to be based on the effective breadth of attached plating, b_{eff}, as defined in Chapter 3 Section 7, Cl.7.3.2.

6.1.3 Net shear area

6.1.3.1 The net shear area, A_{sh} in [cm^2], of primary supporting members subjected to lateral pressure is not to be taken less than the greatest value for all applicable design load sets defined in Section 2, Cl.2.4.3, given by:

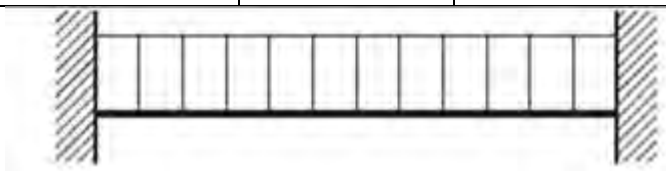
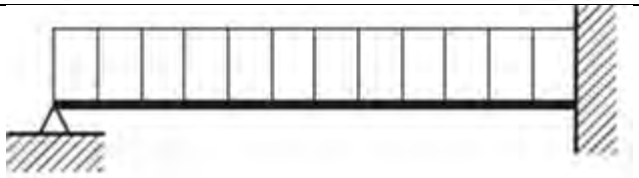
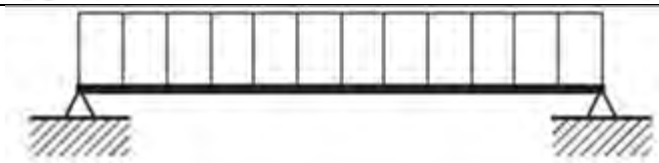
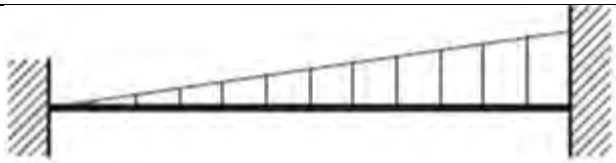
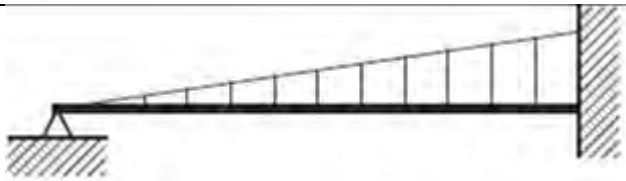
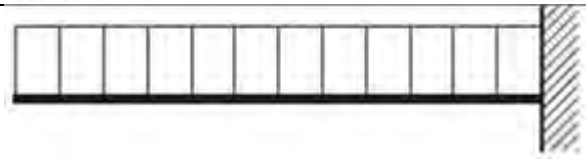
$$A_{sh} = \frac{10\,|P|.S.f_{shr}.\ell_{shr}}{\chi.C_t.\tau_{eH}}$$

where:

f_{shr} = Shear force distribution factor, as given in Table 6.1.2.2.

C_t = Permissible shear stress coefficient for the acceptance criteria set being considered, as given in Table 6.1.2.1.

Table 6.1.2.1: Permissible bending and shear stress coefficients for primary supporting members					
Acceptance criteria set	**Structure attached to PSM** (All boundaries, including decks and flats)	β_s	α_s	C_{s-max}	C_t
AC-I	Members subjected to hull girder loads	0.85	1.00	0.7	0.7
	Members not subjected to hull girder loads	0.70	0.00		
AC-II/AC-IV	Members subjected to hull girder loads	0.95	1.00	0.85	0.85
	Members not subjected to hull girder loads	0.85	0.00		

Table 6.1.2.2: Bending moment and shear force factors, f_{bdg} and f_{shr}						
Load and boundary condition			**Bending moment and shear force distribution factors (based on load at mid span, where load varies)**			
Position				1	2	3
Load model	**1 Support**	**2 Field**	**3 Support**	f_{bdg1} f_{shr1}	f_{bdg2}	f_{bdg3} f_{shr3}
A				12.0 0.50	24.0 -	12.0 0.50
B				- 0.38	14.2 -	8.0 0.63
C				- 0.50	8.0 -	- 0.50
D				15.0 0.3	23.3 -	10.0 0.70
E				- 0.20	16.8 -	7.5 0.80
F				- -	- -	2.0 1.0

Note 1 The bending moment distribution factor, f_{bdg} for the support positions is applicable for a distance of $0.2\ell_{bdg}$ from the end of the effective bending span of the primary supporting member.

Note 2 The shear force distribution factor, f_{shr} for the support positions is applicable for a distance of $0.2\ell_{shr}$ from the end of the effective shear span of the primary supporting member.

Note 3 Application of f_{bdg} and f_{shr}
The section modulus requirement within $0.2\ell_{bdg}$ from the end of the effective span is to be determined using the applicable f_{bdg1} and f_{bdg3}, however the f_{bdg} is not to be taken greater than 12.
The section modulus of mid-span area is to be determined using $f_{bdg} = 24$ or f_{bdg2} from the table if lesser.
The shear area requirement for end connections within $0.2\ell_{shr}$ from the end of the effective span is to be determined using $f_{shr} = 0.5$ or the applicable f_{shr1} or f_{shr3}, whichever is greater.
For models A through F, the value of f_{shr} may be gradually reduced outside of $0.2\ell_{shr}$ towards $0.5f_{shr}$ at mid-span, where f_{shr} is the greater value of f_{shr1} and f_{shr3}.

6.2 Advanced calculation methods

6.2.1 Direct strength - beam analysis

Where complex grillage structures are employed or cross ties are fitted inside shell primary supporting members or bulkhead girders, which are a part of a complex girder system in holds for heavy cargo or tanks, the scantlings are to be determined by direct calculation taking into account:

- The distribution of still water and wave pressure and forces, if any.

- The number and position of intermediate supports (e.g., decks, girders, etc.).

- The condition of fixity at the ends of the primary supporting members and at intermediate supports.

- The geometrical characteristics of the primary supporting members on the intermediate spans.

- The effective breadth, b_{eff}, for yield check as given in in Chapter 3, Section 7, Cl.7.3.2

- Hull girder loads, when $|\sigma_{hg}| \geq 0.15R_{eH}$

6.2.2 Analysis criteria

The calculated stress, in [N/mm^2], are to comply with the following criteria where the coefficients C_t and C_s, are defined in Table 6.1.2.1.

$$\tau \leq \chi C_t \tau_{eH}$$

$$\sigma \leq \chi C_s R_{eH}$$

where:

τ = Shear stress in member, in [N/mm^2].

σ = Normal stress in member, in [N/mm^2].

C_s = Permissible bending stress coefficient taken equal to

$$C_s = \beta_s - \alpha_s \frac{|\sigma_{hg}|}{R_{eH}}$$

β_s = Coefficient as defined in Table 6.1.2.1.

α_s = Coefficient as defined in Table 6.1.2.1.

$C_{s-max.}$= Maximum permissible bending stress coefficient as defined in Table 6.1.2.1.

C_t = Permissible shear stress coefficient as defined in Table 6.1.2.1.

6.2.3 Yield and buckling criteria for finite element analysis

When finite element analysis is used for analysis of PSM then all significant stress components is to be considered, to a similar extent as for cargo hold analysis as described in Chapter 8. The strength is to satisfy the yield criteria given in Chapter 8, Section 3 and buckling criteria given in Chapter 9.

6.2.4 Buckling criteria for beam analysis

Buckling assessment may be required for panels with high compressive stresses e.g., bottom/deck in midship area, with acceptance criteria given in Chapter 9, Section 4.

The normal stresses including hull girder longitudinal stress, σ_{hg}, and the shear stresses

taken from the strength assessment to be applied for buckling capacity calculation of plate panels, as given in Chapter 9, Section 4, is to be corrected as given in Chapter 9, Section 5, Cl.5.2.2.7.

6.3 Pillars

6.3.1 Pillars subjected to compressive axial load

The maximum applied compressive axial load on a pillar, F_{cp} , in [kN], is to be taken as the greatest value calculated for all applicable design load sets as defined in Section 2, Cl.2.4.3, and is given by the following formula:

$$F_{cp} = P.b.\ell + F_{a-upr}$$

where:

b = Mean breadth of area supported, in [m].

ℓ = Mean length of area supported, in [m].

F_{a-upr} = Axial load from pillar including axial load from pillars above, in [kN], if any.

The buckling check of the pillar is to be performed according to Chapter 9, Section 4,4.5, with σ_c in [N/mm^2], as

$$\sigma_c = 10\frac{F_{cp}}{A}$$

A = Net cross section area, in [cm^2], of the pillar.

6.3.2 Pillars subject to tensile axial load

Pillars and PSM members subjected to tensile axial load are to satisfy the criteria given in Cl.6.2.2.

$$\tau \leq \chi C_t \tau_{eH}$$

$$\sigma \leq \chi C_s R_{eH}$$

where:

σ = Normal stress in [N/mm^2], for the minimum cross section inclusive end connections

τ = Shear stress in [N/mm^2] the bracket/gusset end connections

C_s = Permissible normal stress coefficient as defined in Table 6.1.2.1.

C_t = Permissible shear stress coefficient as defined in Table 6.1.2.1.

End connections of the pillars are also to be verified for tensile loads as above.

6.3.3 Pillars subjected to combined axial load and bending

With reference to Cl.6.3.1, only compressive axial load is to be considered for the buckling check of pillars. For pillars subjected to combined axial load and bending, buckling utilization is to be checked in accordance with Chapter 9, Section 1.

Section 7

Intersection of Stiffeners and Primary Supporting Member

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

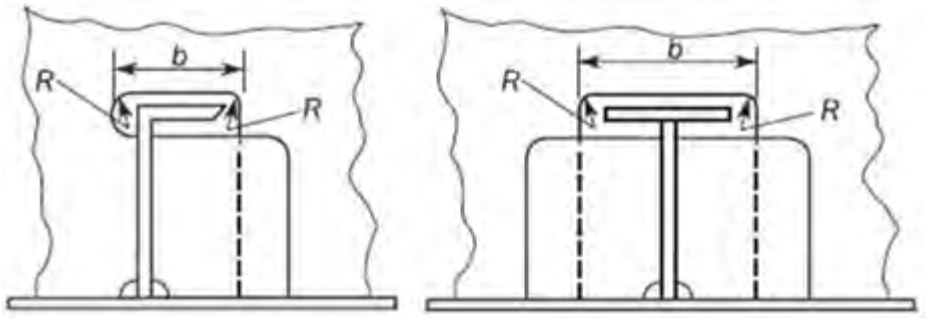
7.1 Cutouts

7.1.1 Cutouts for the passage of stiffeners through the web of primary supporting members, and the related collaring arrangements, are to be designed to minimize stress concentrations around the perimeter of the opening and on the attached web stiffeners.

7.1.2 The total depth of cut-outs without lug or collar plate is to be not greater than 50% of the depth of the primary supporting member. A larger depth of cut-outs may be accepted given that shear strength and buckling capacity are sufficient.

7.1.3 Cut-outs in way of cross tie ends and floors under bulkhead stools or in high stress areas are to be fitted with lug plates welded to the attached plating, Refer Figure 7.1.3.

Where ordinary stiffeners are continuous through primary supporting members, they are to be connected to the web plating so as to ensure proper transmission of loads.



$R \geq 0.2b$ but not less than 25mm.

Figure 7.1.3: Lug plates welded to the attached plating

7.1.4 Lug type collar plates are to be fitted in cut-outs where required for compliance with the requirements of 7.2, and in areas of high stress concentrations, e.g., in way of primary supporting member toes. Refer Figure 7.1.4 for typical lug arrangements.

7.1.5 At connection to shell envelope longitudinals below the scantling draught, T_{sc} and at connection to inner bottom longitudinals, a soft heel is to be provided in way of the heel of the primary supporting member web stiffeners when the calculated direct stress, σ_w, in the primary supporting member web stiffener according to 7.2 exceeds 80% of the permissible values. The soft heel is to have a keyhole, similar to that shown in item (c) in Figure 7.1.5.

A soft heel is not required at the intersection with watertight bulkheads and primary supporting members, where a back bracket is fitted or where the primary supporting member web is welded to the stiffener face plate.

When calculating the direct stress, σ_w, the bottom slamming or bow impact loads using the design pressures defined in Chapter 11, Section 3 and 2 respectively, need not be applied.

7.1.6 Cut-outs are to have rounded corners and the corner radii, R, are to be as large as practicable, with a minimum of 20% of the breadth, b, of the cut-out or 25 [mm], whichever is greater. The corner radii, R, does not need to be greater than 50[mm], Refer Figure 7.1.3. Consideration is to be given to other shapes on the basis of maintaining equivalent strength and minimizing stress concentration.

Consideration is to be given to the provision of adequate drainage and unimpeded flow of air and water when designing the cut-outs and connection details.

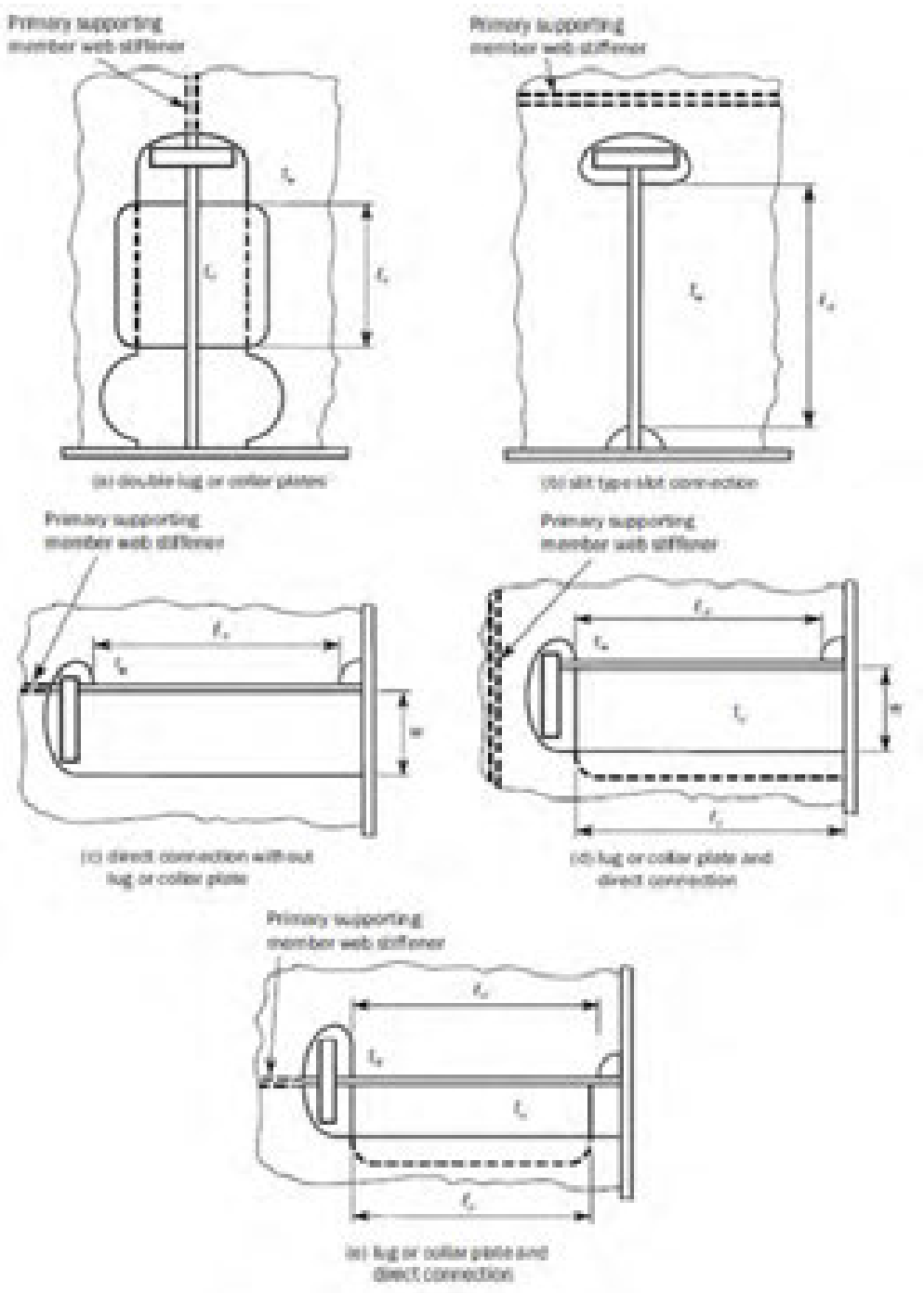


Figure 7.1.4: Symmetric and asymmetric cut-outs

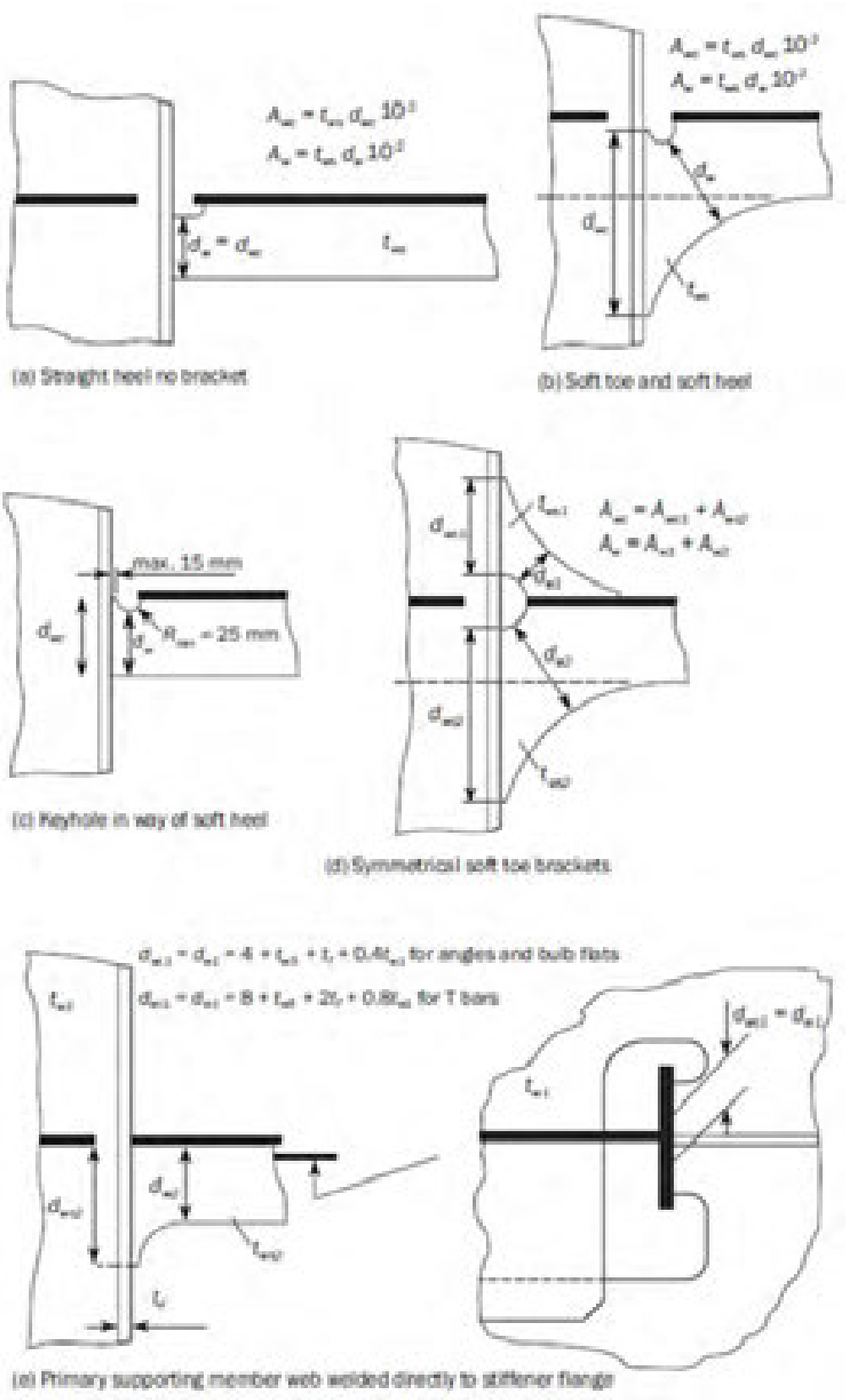


Figure 7.1.5: Primary supporting member web stiffener details

Symbols shown in Figure 7.1.5 is to be taken as:

t_w, t_{ws1}, t_{ws2} = Net thickness, in [mm], of the PSM web stiffener/backing bracket.

d_w, d_{w1}, d_{w2} = Minimum depth, in [mm], of the PSM web stiffener/backing bracket.

d_{wc}, d_{wc1}, d_{wc2} = Length of connection, in [mm], between the PSM web stiffener/backing bracket and the stiffener.

t_f = Net thickness of the flange, in [mm]. For bulb profile, t_f is to be obtained as defined in Chapter 3, Section 7, Cl.7.4.1.

7.2 Connection of stiffeners to PSM

7.2.1 Application

For ships of length L≥90[m] the calculation of connection of stiffeners to PSM, can be performed according to 7.2 and a simplified calculation method for ships of length L<90[m] according to 7.3.

7.2.2 General

For connection of stiffeners to PSM in case of lateral pressure, Cl.7.2.3 and Cl.7.2.4 are to be applied.

The cross-sectional areas of the connections are to be determined from the proportion of load transmitted through each component in association with its appropriate permissible stress.

7.2.3 The load, W_1, in [kN], transmitted through the shear connection is to be taken as follows.

7.2.3.1 If the web stiffener is connected to the intersecting stiffener:

$$W_1 = W\left(\alpha_a + \frac{A_1}{4f_c A_w + A_1}\right)$$

7.2.3.2 If the web stiffener is not connected to the intersecting stiffener:

$$W_1 = W$$

where:

W = Total load, in [kN], transmitted through the stiffener connection to the PSM taken equal to:

$$W = \frac{P_1 s_1\left(S_1 - \frac{s_1}{2000}\right) + P_2 s_2\left(S_2 - \frac{s_2}{2000}\right)}{2\sin(\varphi_{w1})\sin(\varphi_{w2})} 10^{-3}$$

P_1, P_2 = Design pressure applied on the stiffener for the design load set being considered, in [kN/m^2], on each side of the considered connection.

For bow impact loads or bottom slamming, P_1 and P_2 are the design pressure as defined in Chapter 11, Section 2 and Chapter 11, Section 3 respectively.

S_1, S_2 = Spacing between the considered and the adjacent PSM on each side of the considered connection, in [m].

s_1, s_2 = Spacing of the stiffener, in [mm], on each side of the considered connection.

α_a = Panel aspect ratio, not to be taken greater than 0.25.

$$\alpha_a = \frac{s}{1000\,S}$$

$$S = \frac{S_1 + S_2}{2}$$

$$s = \frac{s_1 + s_2}{2}$$

φ_{w1} = Angle, in [deg], between the PSM web and the attached plating, as defined in Chapter 3 Section 7.φ_{w1} is to be taken equal to 90 [deg] if the angle is greater than or equal to 75 [deg].

φ_{w2} = Angle, in [deg], between the stiffener web and the attached plating, as defined in Chapter 3 Section 7.φ_{w2} is to be taken equal to 90 [deg] if the angle is greater than or equal to 75 [deg].

A_1 = Effective net shear area, in [cm^2], of the connection, to be taken equal to:

$$A_1 = A_{1d} + A_{1c}$$

In case of a slit type slot connections area, A_1, is given by:

$$A_1 = 2A_{1d}$$

In case of a typical double lug or collar plate connection area, A_1, is given by:

$$A_1 = 2A_{1c}$$

A_{1d} = Net shear connection area, in [cm^2], excluding lug or collar plate, as given by:

$$A_{1d} = \ell_d t_w 10^{-2}$$

ℓ_d = Length of direct connection, in [mm], between stiffener and PSM web, in [mm].

t_w = Net web thickness, in [mm], of the primary supporting member,

A_{1c} = Net shear connection area, in [cm^2], with lug or collar plate, given by:

$$A_{1c} = f_1 \ell_c t_c 10^{-2}$$

ℓ_c = Length of connection in [mm], between lug or collar plate and PSM.

t_c = Net thickness of lug or collar plate, in [mm], not to be taken greater than the net thickness of the adjacent PSM web.

f_1 = Shear stiffness coefficient, taken as:

= 1.0, for stiffeners of symmetrical cross section.

= 140/w, not to be taken greater than 1.0, for stiffeners of asymmetrical cross section.

w = Width of the cut-out, in [mm], for an asymmetrical stiffener, measured from the cut-out side of the stiffener web, as indicated in Figure 7.1.4.

A_w = Effective net cross sectional area, in [cm^2], of the PSM web stiffener in way of the connection including backing bracket where fitted, as shown in Figure 7.1.5. If the PSM web stiffener incorporates a soft heel ending or soft heel and soft toe ending, A_w is to be measured at the throat of the connection, as shown in Figure 7.1.5.

f_c = Collar load factor taken equal to:

For intersecting stiffeners of symmetrical cross section:

$$f_c = 1.85 \text{ for } A_w \leq 14$$

$$f_c = 1.85 - 0.0441(A_w - 14) \text{ for } 14 < A_w \leq 31$$

$$f_c = 1.1 - 0.013(A_w - 31) \text{ for } 31 < A_w \leq 58$$

$$f_c = 0.75 \text{ for } A_w > 58$$

For intersecting stiffeners of asymmetrical cross section:

$$f_c = 0.68 + 0.0172 \frac{\ell_s}{A_w}$$

ℓ_s = Connection length equal to:

For a single lug or collar plate connection to the PSM:

$$\ell_s = \ell_c$$

For a single sided direct connection to the PSM:

$$\ell_s = \ell_d$$

In the case of a lug or collar plus a direct connection:

$$\ell_s = 0.5(\ell_c + \ell_d)$$

7.2.4 The load, W_2 in kN, transmitted through the PSM web stiffener is to be taken as:

7.2.4.1 if the web stiffener is connected to the intersecting stiffener:

$$W_2 = W\left(1 - \alpha_a - \frac{A_1}{4 f_c A_w + A_1}\right)$$

7.2.4.2 If the web stiffener is not connected to the intersecting stiffener:

$$W_2 = 0$$

where:

W = Load, in [kN], as defined in Cl.7.2.3.

f_c = Collar load factor as defined in Cl.7.2.3.

α_a = Panel aspect ratio, as defined in Cl.7.2.3.

A_1 = Effective net shear area, in [cm^2], as defined in Cl.7.2.3.

A_w = Effective net cross sectional area, in [cm^2], as defined in Cl.7.2.3.

The values of A_w, A_{wc} and A_1 are to be such that the calculated stresses satisfy the following criteria:

• For the connection to the PSM web stiffener not in way of the weld: $\sigma_w \leq \sigma_{perm}$

• For the connection to the PSM web stiffener in way of the weld: $\sigma_{wc} \leq \sigma_{perm}$

• For the shear connection to the PSM web: $\tau_w \leq \tau_{perm}$

σ_w = Direct stress, in [N/mm^2], in the PSM web stiffener at the minimum bracket area away from the weld connection:

$$\sigma_w = 10\frac{W_2}{A_w}$$

σ_{wc} = Direct stress, in [N/mm^2], in the PSM web stiffener in way of the weld connection:

$$\sigma_{wc} = 10\frac{W_2}{A_{wc}}$$

τ_w = Shear stress, in [N/mm^2], in the shear connection to the PSM web:

$$\tau_w = 10\frac{W_1}{A_1}$$

A_{wc} = effective net area, in [cm^2], of the PSM web stiffener in way of the weld as shown in Figure 7.1.5.

σ_{perm} = Permissible direct stress, in [N/mm^2], as given in Table 7.2.3 for AC-I, AC-II, AC-III.

τ_{perm} = Permissible shear stress, in [N/mm^2], as given in Table 7.2.3 for AC-I, AC-II, AC-III.

Table 7.2.3: Permissible stresses for connection between stiffeners and PSMs

Item	Direct stress, σ_{perm}, in [N/mm^2]			Shear stress, τ_{perm}, in [N/mm^2]		
	Acceptance criteria set[1)]			Acceptance criteria set[1)]		
	AC-I	AC-II	AC-III	AC-I	AC-II	AC-III
PSM web stiffener	$0.83R_{eH}$[(3)]	R_{eH}	R_{eH}	-	-	-
PSM web stiffener to intersecting stiffener in way of weld connection: • Double continuous fillet • Partial penetration weld	$0.58\ R_{eH}$[(3)] $0.83R_{eH}$[(2)(3)]	$0.70\sigma_y$[(3)] σ_y[(2)]	$R_{eH}(3)$ $R_{eH}(2)$	-	-	-
PSM stiffener to intersecting stiffener in way of lapped welding	$0.5R_{eH}$	$0.6R_{eH}$	R_{eH}	-	-	-
Shear connection including lugs or collar plates: • Single sided connection • Double sided connection	- -	- -	- -	$0.71\tau_{eH}$ $0.83\tau_{eH}$	$0.85\tau_{eH}$ τ_{eH}	τ_{eH} τ_{eH}

(1) Acceptance criteria for the design load sets are given in Section 2.
(2) The root face is not to be greater than one third of the gross thickness of the PSM stiffener.
(3) Permissible stresses may be increased by 5% where a soft heel is provided in way of the heel of the PSM web stiffener.

7.2.5 Where a backing bracket is fitted in addition to the PSM web stiffener, it is to be aligned with the web stiffener, and on the opposite side. The arm length of the backing bracket is not to be less than the depth of the web stiffener. The net cross sectional area through the throat of the bracket is to be included in the calculation of A_w as shown in Figure 7.1.5.

7.2.6 Lapped connections of PSM web stiffeners or tripping brackets to stiffeners are not permitted within 0.4L for vessels with single deck and with length L≥150 [m].

7.2.7 Where the web stiffener of the PSM is parallel to the web of the intersecting stiffener, but not connected to it, the offset PSM web stiffener is to be located in close proximity to the slot edge. The ends of the offset web stiffeners are to be suitably tapered and softened.

To facilitate access for welding the offset stiffeners are to be located 50[mm] from the slot edge furthest from the web of the secondary member.

Locations where the web stiffener of the PSM are not connected to the intersecting stiffeners as well as the detail arrangements are to be specially considered on the basis of their ability to transmit load with equivalent effectiveness to that of Cl.7.2.2 through Cl.7.2.7. Details of calculations made and/or testing procedures and results are to be submitted.

7.2.8 The size of the fillet welds for connections between stiffeners and PSM is to be calculated according to Chapter 17 based on the weld factors given in Table 7.2.8.

For the welding in way of the shear connection the size is not to be less than that required for the PSM web plate for the location under consideration.

Table 7.2.8: Weld factors, f_{weld}, for connection between stiffeners and PSM's

Item	Acceptance criteria	Weld Factor
Primary supporting member to intersecting stiffener	AC I	$0.6\,\sigma_{wc}/\sigma_{perm}$ not to be less than 0.38
	AC II	
	AC III	
Shear connection inclusive lug or collar plate	AC I	0.38
	AC II	
	AC III	
Shear connection inclusive of lug or collar plate, where the web stiffener of the PSM is not connected to the intersection stiffener	AC I	$0.6\ \tau_w/\tau_{perm}$ not to be less than 0.44
	AC II	
	AC III	

Note 1:
τ_w = Shear stress, in [N/mm^2], as defined in Cl.7.2.4.
σ_{wc} = Stress, in [N/mm^2], as defined in Cl.7.2.4.
τ_{perm} = Permissible shear stress, in [N/mm^2], Refer Table 7.2.3.
σ_{perm} = Permissible direct stress, in [N/mm^2], Refer Table 7.2.3.
W = Load, in kN, as defined in Cl.7.2.3.
A_1 : Effective net shear area, in [cm^2], defined in Cl.7.2.3.
A_w : Effective net cross sectional area, in [cm^2], defined in Cl.7.2.3.

7.3 Simplified calculation of connection of stiffeners to PSM

7.3.1 Application

The simplified calculation method given below can be applied for ships of length L<90[m].

7.3.2 General

For connection of stiffeners to PSM in case of lateral pressure, Cl.7.3.3 is to be applied.

The cross sectional areas of the connections are to be determined from the proportion of load transmitted through each component in association with its appropriate permissible stress.

7.3.3 The total load, in [kN], transmitted through the connection to the PSM is to be taken equal to:

$$W = \frac{P.s}{\sin(\varphi_{w1})\sin(\varphi_{w2})}\left(S - \frac{s}{2000}\right)10^{-3}$$

where:

P = Design pressure applied on the stiffener for the design load set being considered, in [kN/m^2].

For stiffeners subjected to different pressure, P_1 and P_2, at each side of the PSM, the average load for the two sides is to be applied to estimate the reaction force, e.g., vertical stiffeners at transverse bulkhead, taken as:

$$P = 0.5(P_1 + P_2)$$

S = Average spacing of the considered PSM crossed by the stiffener, in [m]. For stiffeners having different spans on each side of PSM (i.e., different PSM spacing, S_1 and S_2), the average span is taken as:

$$S = 0.5(S_1 + S_2)$$

A_1 = Effective net shear area, in [cm^2], of the connection to the PSM web, to be taken equal to the sum of the single or double sided connection:

$$A_1 \geq \frac{10W_1}{\tau_{perm}}$$

A_w = Effective net cross sectional area, in [cm^2], of the PSM web stiffener in way of the connection including backing bracket where fitted, as shown in Figure 7.1.5.

$$A_w \geq \frac{10W_2}{\sigma_{perm}}$$

W_1= The load transmitted through the shear connection, in [kN].

(a) If the web stiffener is connected to the intersecting stiffener:

$W_1 = 0.4W$ for symmetrical cross sections, e.g., T-profile.

$W_1 = 0.5W$ for asymmetrical cross sections, e.g., L-profile.

(b) If the web stiffener is not connected to the intersecting stiffener:

$$W_1 = W$$

W_2 = The load transmitted through the PSM web stiffener, in [kN].

$W_2 = W - W_1 = 0.6W$ for symmetrical cross sections, e.g., T-profile.

$W_2 = W - W_1 = 0.5W$ for asymmetrical cross sections, e.g., L-profile.

Where the web stiffener is not connected to the intersecting stiffener, $W_2 = 0$.

σ_{perm} = Permissible direct stress, in [N/mm^2], as given in Table 7.3.3.

τ_{perm} = Permissible shear stress, in [N/mm^2], as given in Table 7.3.3.

7.3.4 The size of the fillet welds for connections between stiffeners and PSM is to be calculated according to Chapter 16 based on the weld factors given in Table 7.3.4.

Table 7.3.3: Load distribution and permissible stresses						
Load	**Direct stress, σ_{perm}**			**Shear stress, τ_{perm}**		
	AC-I	AC-II	AC-III	AC-I	AC-II	AC-III
W_2	$0.52R_{eH}$	$0.63R_{eH}$	$0.9R_{eH}$			
W_1				$0.64\tau_{eH}$	$0.76\tau_{eH}$	$0.9\tau_{eH}$

Table 7.3.4: Weld factors, f_{weld}, for connection between stiffeners and PSM's		
Item	**Weld factor, f_{weld}**	
Primary supporting member web stiffener connection to intersecting stiffener	At tank boundary	0.4
	Elsewhere	0.3
Shear connection inclusive lug or collar plate	Without web stiffener on top of the intersecting stiffener	0.3
	With web stiffener on top of the intersecting stiffener	0.25

Section 8

Superstructure, Deckhouse and Companionways

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

8.1 General

8.1.1 Application

8.1.1.1 The requirements of this section are applicable to superstructures, deckhouses and companionways.

8.1.1.2 The minimum scantling requirements of the decks and sides of the superstructures and deckhouses are to be in accordance with Section 3.

8.1.1.3 Applicable rule requirements and design pressures and forces for plate, stiffener and primary supporting members are given in Table 8.1.1 for the different design load sets.

Exposed sides of superstructures, when in line with ship side, inclusive their supporting structure is to comply with the bow impact requirements given in Chapter 11, Section 2, if applicable.

8.1.1.4 With reference to Chapter 2, Section 4, Cl.4.3.1, all scantlings referred in 8.3 are gross, unless otherwise specified.

Table 8.1.1: Design load sets

Structural member	Design load set	Design load scenario	Load component [3]	Applicable rule requirements	Acceptance criteria	Loading condition
Superstructure side	SEA-1	1	$\max(P_W, P_{SI})$	Section 4, Section 5 and Section 6	AC-II	Full load condition
PSM's on - Superstructure front and aft bulkheads - Deckhouse side, front and aft bulkheads				Section 6		
Superstructure front and aft bulkheads [4]	SEA-1	1	P_A	Section 8	-	-
Deckhouse side, front and aft bulkheads [4]						
Exposed decks	SEA-1	1	P_D	Section 4, Section 5 and Section 6	AC-II	Full load condition
Exposed decks and non-exposed decks with distributed load	UDL-1 [1]	1	$P_{dl-s} + P_{dl-d}$ $F_{U-s} + F_{U-d}$			Normal ballast condition [2]
	UDL-2 [1]	2	P_{dl-s} F_{U-s}		AC-I	-

Exposed decks and non-exposed decks with wheel loading	WL-1[1)]	1	P_{wl-1}	Chapter 11	AC-II	Normal ballast condition[2)]
	WL-2[1)]	2	P_{wl-2}		AC-I	-

Notes:
1) Distributed or concentrated loads only. Need not to be combined with simultaneously occurring green sea pressure.
2) For cargo loads, draught, GM and k_r values may be based on actual cargo conditions in the loading manual.
3) Local loads:
P_W = Wave pressure as given in Chapter 5, Section 4, Cl.4.1.3.
P_D = External dynamic design pressure for exposed decks and wheelhouse top due to green sea loading as given in Chapter 5, Section 4, Cl.4.2.2 and Chapter 5, Section 4, Cl.4.3.2, respectively
P_A = Design pressure for end bulkheads of superstructure and deckhouse walls (side, front and aft bulkheads) as given in Chapter 5, Section 4, Cl.4.3.4.
P_{SI} = External design pressure for external side of superstructure as given in Chapter 5, Section 4, Cl.4.3.3.
P_{dl-s} = Static pressure due to distributed deck load as given in Chapter 5, Section 4,4.2.3.1 or Chapter 5, Section 5, 5.2.2.
P_{dl-d} = Dynamic pressure due to distributed deck load as given in Chapter 5, Section 4,4.2.3.1.
F_U = Force on exposed and non-exposed decks and platforms due to unit loads as given in Chapter 5, Section 4, Cl.4.2.3.2.
P_{wl-1} = Design pressure for wheel loading during normal operation at sea as given in Chapter 11, Section 5.
P_{wl-2} = Design pressure for wheel loading during normal operation at harbour as given in in Chapter 11, Section 5.
4) Applicable to plate and stiffeners only.

8.1.2 Definitions

8.1.2.1 For definitions of the terms 'Superstructure' and 'Deckhouse' refer to Chapter 1.

8.1.2.2 Superstructure deck is a deck forming the upper boundary of a superstructure.

Where the summer freeboard is increased such that the resulting draught is not more than that corresponding to the minimum summer freeboard for the same ship from an assumed freeboard deck located at a distance equal to or greater than the standard superstructure height below the actual freeboard deck;

The actual freeboard deck may be treated as the superstructure deck, while considering the requirements given in this chapter.

8.1.2.3 Where the freeboard is increased in accordance with Cl.8.1.2.2 above, the first tier above the uppermost continuous deck may be treated as the second tier. The lowest tier is normally the tier that is directly situated on the deck to which the rule depth 'D' is measured.

8.1.2.4 Long deckhouse is a deckhouse having more than 0.15L of its length within 0.4L amidships.

8.2 Structural Arrangement and Details

8.2.1 Structural continuity

8.2.1.1 Adequate transverse strength is to be provided to the deckhouses and superstructures by means of transverse bulkheads, girders and web frames.

8.2.1.2 The front and the after-end bulkheads of the superstructures and deckhouses should be effectively supported below by a transverse bulkhead or by a combination of partial bulkheads, girders and pillars.

Similarly, the exposed sides and internal bulkheads of various tiers of erections are to be

located, as far as practicable, above bulkheads or above deep girders below, or equivalent.

8.2.1.3 Local reinforcement is to be provided in way of large openings and areas supporting lifesaving appliances or high loads from other equipment, fittings, etc.

8.2.1.4 Arrangements are to be made to minimize the effect of discontinuities in erections. All openings cut on the sides are to be substantially framed and have well rounded corners. Continuous coamings or girders are to be fitted below and above doors and similar openings. Horizontal stiffeners are to be fitted at the upper and lower edge of large openings for windows.

8.2.1.5 At the ends of superstructures, which have no set-in from the ships' side, and at the ends of poop and bridge, the side plating is to extend beyond the ends of the superstructure, and is to be gradually reduced in height down to the sheer strake. The transition should be smooth and without any discontinuity. The extended plating is to be adequately stiffened, particularly at its upper edge.

8.2.1.6 In the case of superstructures exceeding 0.15L in length and ending within 0.4L amidships, the side plating of the superstructures is to be increased by 25% in way of the break.

8.2.1.7 The thickness of sheer strake is to be increased by 30% on each side of a superstructure end bulkhead located within 0.4L amidships if the superstructure deck is a partial strength deck.

8.2.1.8 Deck girders are to be fitted below the sides of long deckhouses and are to extend at least three frame spaces beyond the deckhouse ends. The depth of the girders is to be at least 100 [mm] greater than the depth of the beams and the girders are to be adequately stiffened at the lower edge.

8.2.1.9 The connection area between the corners of long deckhouses and deck plating should be increased locally. Deck beams under the corners of deck houses are not to be scalloped for a distance of 0.5[m] on either side of the corners.

8.2.1.10 Web frames, transverse partial bulkheads or other equivalent strengthening are to be fitted inside deckhouses of at least 0.5B in breadth extending more than 0.15L in length within 0.4L amidships. These transverse strengthening reinforcements are to be spaced approximately 9[m] apart and are to be arranged, where practicable, in line with the transverse bulkheads below.

8.2.2 End attachments

8.2.2.1 Webs of lower tier stiffeners are to be welded to the decks at their ends.

8.2.2.2 Deck stiffeners or transverse beams are to be connected to side frames by brackets according to Chapter 3, Section 6, Cl.6.2.2. Beams crossing longitudinal walls and girders may be attached to the stiffeners of longitudinal walls and the webs of girders respectively by welding without brackets.

8.2.2.3 Face plates of longitudinal and transverse deck girders are to be stiffened by tripping brackets according to Chapter 3, Section 6, Cl.6.3.3.

8.2.2.4 Vertical frames of superstructure are to be welded to the main frames below, or to the deck under provision of a sufficient supporting structure.

8.2.3 Navigation bridge visibility

Attention is drawn to Chapter V, Regulation 22 of SOLAS 1994 Amendments regarding Navigation Bridge Visibility requirements, applicable to ships of loadline length equal to or greater than 55[m].

8.3 Scantlings

For minimum thicknesses in general, Refer Section 3. For members not covered by Section 3, Refer Cl.8.3.1 to Cl.8.3.2.

8.3.1 Thickness of bulkhead and side plating

8.3.1.1 The gross thickness, in [mm], of front and aft bulkheads of superstructure and deck house side and bulkheads is not to be taken less than:

$$t = 0.95\ s\sqrt{k.P_A}$$

but in no case is the thickness to be less than given below:

$t = (5 + 0.01L_1)\sqrt{k}$ for first tier above freeboard deck

$t = (4 + 0.01L_1)\sqrt{k}$ for upper tiers but not less than 5.0[mm]

For ships with $L_1 < 65m$, the minimum thickness of plating should be as follows:

t_{min} = 5mm for the lowest unprotected front
= 4mm for all other cases

where:

P_A = Lateral pressure in [kN/m^2] defined in Chapter 5 Section 4, Cl.4.3.4.1.

L_1 = Length of the ship, but need not be taken greater than 300[m].

s = Spacing of stiffeners, in [m].

8.3.1.2 The net thickness, in [mm], of plating of sides of superstructures is to be taken not less than:

$$t_{net} = 0.0158.\alpha_p.b\sqrt{\frac{|P|}{\chi.C_a.R_{eH}}}$$

where:

P = Lateral pressure in [kN/m^2] to be taken as given in Table 8.1.1.

C_a = Permissible bending stress coefficient for plate as defined in Section 4, Cl.4.1.1.1.

χ = Coefficient as defined in Section 4, Cl.4.1.1.1

8.3.2 Stiffeners

8.3.2.1 The gross section modulus of stiffeners, in [cm^3], on fronts, sides and end bulkheads of all erections other than sides of superstructures is not to be taken less than:

$$Z = 0.35.k.s.\ \ell^2\ .P_A$$

where,

s = Spacing of stiffeners (m), measured along the plating.

ℓ = Unsupported span (m), which is to be taken as the 'tween deck height but in no case less than 2.0[m].

8.3.2.2 Webs of stiffeners on sides and after ends of the lowest tiers of all erections are to be welded to the deck at both ends. Scantlings for other types of end connections are to be specially considered.

8.3.2.3 The section modulus of deck house side stiffeners need not be greater than that of side frames on the deck situated directly below, taking account of spacing and span.

8.3.2.4 For longitudinals on sides of long deck house within 0.4L amidships and at strength deck level is given by:

$$Z = 0.625\ s\ \ell^2 P_A$$

The modulus may be linearly decreased to the general value at the first deck above the strength deck and at 0.1L from the perpendiculars.

8.3.2.5 Stiffeners on fronts are to be connected to deck at both ends with a connection area 'a' not less than

$$a = 0.07 s\ \ell\ P_A\ k$$

8.3.2.6 The net section modulus of stiffeners, in [cm^3], on sides of superstructures is to be taken not less than:

$$Z = \frac{|P|.s.\ell_{bdg}^2}{f_{bdg}.\chi.C_s.R_{eH}}$$

with $\chi.C_s$ not to be taken greater than 1.0

where:

P = Lateral pressure in [kN/m^2] to be taken as given in Table 8.1.1.

f_{bdg} = Bending moment factor as defined in Section 5, Cl.5.1.1.2.

C_s = As defined in Section 5, Cl.5.1.1.2

χ = Coefficient as defined in Section 5, Cl.5.1.1.2.

8.3.3 Thickness of protected casings and internal walls

8.3.3.1 The gross thickness, in [mm], of plate for the protected casing and internal wall is:

$t = (0.0085\ s)\sqrt{k}$ but not less than $6.0\sqrt{k}$ for bulkheads in way of cargo holds

$t = (0.0065\ s)\sqrt{k}$ but not less than $5.0\sqrt{k}$ for bulkheads in way of accommodations

where,

s = Spacing of stiffeners, in [mm] but not to be taken less than 760[mm].

8.3.3.2 The gross section modulus of stiffeners, in [cm^3], of protected machinery casings is not to be less than:

$$Z = 0.003.s.\ell^2.\sqrt{k}$$

where, ℓ is not to be taken less than 2.5[m].

8.3.3.3 Casings supporting one or more decks above are to be adequately strengthened.

8.4 Alternative grillage analysis for girders and transverses

8.4.1 Where arrangements of deck girders and transverses are such it acts as a grillage structure; additional analysis may be carried out with a structural model based on the gross scantling.

The resulting stresses are not to exceed the following permissible bending, shear and equivalent stresses, in [N/mm^2], taken as:

$$\sigma_b = 150/k$$

$$\tau = 100/k$$

$$\sigma_{eqv} = 180/k$$

8.4.2 In direct strength analysis of primary support members, for superstructure or deck house, the design pressure and forces are to be applied consistently with the application for the hull structure. The green sea loads are to be calculated according to Chapter 5 and interpolated in vertical direction.

8.5 Aluminium deckhouse or superstructure

8.5.1 Scantlings

8.5.1.1 Scantlings of plates and stiffeners is to be in accordance with 8.2 and 8.3.

8.5.1.2 For Aluminium deckhouse or superstructure, the material factor to be used is k_a instead of k. Refer Chapter 2, Section 1, 1.3 for the calculation of k_a.

8.6 Companionways

The scantlings of companionways are to be determined in accordance with 8.3.

Section 9

Bulwarks and Guard Rails

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

9.1 General

9.1.1 Application

Bulwarks or guard rails are to be fitted around all exposed decks.

9.1.2 General requirements

9.1.2.1 The height of the bulwarks or guard rails measured above the sheathing, if any, is not to be less than 1.0[m] and are to be constructed as required in 9.2 and 9.3.

9.1.2.2 Consideration will be given to cases where this height would interfere with the normal operation of the ship.

9.1.2.3 The freeing arrangements in bulwarks are to be in accordance with Cl.9.2.3.

9.2 Bulwark

9.2.1 Bulwark construction

9.2.1.1 Plate bulwarks are to be stiffened at the upper edge by a suitable rail and supported either by stays or plate brackets spaced not more than 2.0[m] apart. The free edge of the stay or the plate bracket is to be stiffened. The spacing of stays forward of 0.07L from F.P. is to be not more than 1.2[m] on Type 'A', Type 'B-60' and Type 'B-100' ships as defined in the ICLL.

9.2.1.2 In the case of ships intended for the carriage of timber deck cargoes, the specific provisions of the freeboard regulations are to be complied with.

9.2.1.3 Where bulwarks are cut in way of a gangway or other openings, stays of increased strength are to be fitted at the ends of the openings.

9.2.1.4 Bulwark stays are to be supported by, or are to be in line with, suitable underdeck stiffening, which is to be connected by double continuous fillet welds in way of the bulwark stay connection.

9.2.1.5 Bulwarks are to be adequately strengthened in way of the eye plates for cargo gear, and in way of the mooring pipes the plating is to be increased in thickness and also adequately stiffened.

9.2.1.6 Bulwarks are generally not to be welded to the top of the sheer strake within 0.6L amidships.

Within 0.6L amidships, bulwarks are to be arranged to ensure that they are free from hull girder stresses.

9.2.1.7 Bulwarks should not be cut in way of the breaks of the superstructures, and are also to be arranged to ensure their freedom from main structural stresses.

9.2.1.8 At the ends of the superstructures where the side plating is extended and tapered to align with the bulwark plating, the transition plating is to be suitably stiffened and supported. Where freeing ports or other openings are essential in this plate, they are to be suitably framed and kept well clear of the free edge

9.2.1.9 Where mooring fittings subject the bulwark to large forces, the stays are to be adequately strengthened.

9.2.2 Bulwark scantlings

9.2.2.1 Plating

The gross thickness of the bulwark plating is not to be less than that required for the superstructure side plating in the same location if the height of the bulwark is equal to or greater than 1.8[m]. Where the height of the bulwark is 1.0[m], the thickness need not exceed 6.0[mm]. For intermediate bulwark heights the thickness may be obtained by interpolation. For ships L ≤ 30[m], the thickness need not exceed 5.5[mm].

9.2.2.2 Stays

(a) The gross section modulus 'Z' in [cm^3] at the bottom of the bulwark stay is not to be less than:

$$Z = (33 + 0.44L)h^2 s$$

Where,

h = Height of the bulwark from the top of the deck plating to the top of the rail, in [m].

s = Spacing of bulwark stays, in [m].

L = Length of ship, in [m], but to be taken not greater than 100[m].

(b) In the calculation of section modulus 'Z', only the material connected to the deck is to be included.

The bulb or flange of the stay may be taken into account were connected to the deck, and where, at the ends of the ship, the bulwark plating is connected to the sheer strake, a width of plating not exceeding 600[mm] may also be included. The free edge of the stay is to be stiffened.

(c) When the bulwark is subjected to loads from any deck cargo, the scantlings will be specially considered.

9.2.3 Freeing Ports

9.2.3.1 Basic area

(a) Where bulwarks on the weather portion on freeboard or superstructure decks form wells, provision is to be made for rapidly freeing the decks of water and for draining them. In general, the minimum freeing area "A" in [m^2], on each side of the ship for each well on the freeboard deck is to be given by the following in cases where the sheer in way of the well is standard or greater than standard.

– Where length of the bulwark (ℓ) in the well is less than or equal to 20[m] - $A_1 = 0.7 + 0.035\ell$

- Where (ℓ) exceeds 20[m] $A_1 = 0.07\ell$

ℓ need not to be taken greater than $0.7L_L$, where L_L is the Load line length of the vessel (Refer Chapter 1)

(b) If the bulwark is more than 1.2[m] in average height, the required area is to be increased by 0.004 [m^2] per meter of length of the well for each 0.1[m] difference in height. If the bulwark is less than 0.9[m] in average height, the required area may be decreased in the same proportion.

(c) The minimum area for each well on superstructure decks is to be one half of the area obtained above. Wells on raised quarter decks should be treated as being on freeboard deck.

9.2.3.2 Vessels with Less than Standard Sheer

In ships with no sheer the area of freeing ports as calculated above is to be increased by 50%. Where the sheer is less than standard, percentage of increase to be obtained by linear interpolation.

9.2.3.3 Multiple wells

On a flush deck ship with a substantial deckhouse amidships, it is considered that the deckhouse provides sufficient break to form two wells and that each could be given the required freeing port area based upon the length of the 'well'. It would not then be allowed to base the area upon $0.7L_L$.

For a 'substantial' deckhouse, the breadth of the deckhouse should be at least 80% of the beam of the vessel and the passageways along the sides of the ship should not exceed 1.5[m] in width.

Where a screen bulkhead is fitted completely across the vessel, at the forward end of a midship deck house, this would effectively divide the exposed deck into wells and no limitation on the breadth of the deckhouse is considered necessary in this case.

9.2.3.4 Freeing port area for continuous trunk or continuous hatchway coaming

The effectiveness of the freeing port area in bulwarks required in 9.2.3.1 above depends on the free flow across the deck of a ship. Where a vessel is fitted with a trunk, and open rails are not fitted on weather parts of the freeboard deck in way of the trunk for at least half their length, or where continuous or substantially continuous hatchway side coamings are fitted between detached superstructures, the minimum area of the freeing-port openings is to be calculated from the following Table 9.2.3.

Table 9.2.3: Minimum area of freeing ports	
Breadth of hatchway or trunk in relation to the breadth of ship	Area of freeing ports in relation to the total area of the bulwarks
40% or less	20%
75% or more	10%

The area of freeing ports at intermediate breadths are to be obtained by linear interpolation.

9.2.3.5 The freeflow area on deck is the net area of gaps between hatchways, and between hatchways and superstructures and deckhouses up to the actual height of the bulwark.

The freeing port area in bulwarks should be assessed in relation to the net flow area as follows:

(i) If the freeflow area is not less than the freeing area calculated from Cl.9.2.3.4 above as if the hatchway coamings were continuous, then the minimum freeing port area calculated from Cl.9.2.3.1 and Cl.9.2.3.2 should be deemed sufficient.

(ii) If the freeflow area is equal to or less than the freeing area calculated from Cl.9.2.3.1 and Cl.9.2.3.2, the minimum freeing area in the bulwarks should be determined from Cl.9.2.3.4.

(iii) If the freeflow area is smaller than calculated from Cl.9.2.3.4 but greater than calculated from 9.2.3.1 and 9.2.3.2, the minimum freeing port area should be determined from the following formula.

$$A_3 = A_1 + A_2 - f_p$$

where,

A_1 = Minimum freeing area [m^2] calculated from Cl.9.2.3.1 and Cl.9.2.3.2.

A_2 = Minimum freeing area [m^2] calculated from Cl.9.2.3.4.

f_p = The total net area of passages and gaps between hatch ends and superstructures or deckhouses up to the actual height of bulwark [m^2].

9.2.3.6 Freeing ports in way of wells in combination with open superstructures

In the case of vessels having open superstructures on the freeboard or superstructure decks, which open to wells formed by bulwarks on the peripheries of the open decks, the minimum freeing port areas on each side of the ship for the open superstructure A_s and for the open well A_w are to be determined as follows:

(a) Area of the freeing port in [m^2], for the open well is not to be less than that obtained from the following formula:

$$A_w = (0.07\ell_w + A_c)S_c\left(\frac{0.5h_s}{h_w}\right) \text{ for } \ell_t \geq 20[\text{m}]$$

$$A_w = (0.7 + 0.035\ell_w + A_c)S_c\left(\frac{0.5h_s}{h_w}\right)$$

$$\text{for } \ell_t \leq 20[\text{m}]$$

where,

A_1 = Freeing port area, in [m^2], required for an open well of length ℓ_w, in accordance with Cl.9.2.3.1

(b) The freeing port area in [m^2], for the open superstructure is to be not less than that obtained from the following formula:

$$A_s = (0.07\ell_t)S_c\frac{b_0}{\ell_t}\left(1 - \left(\frac{\ell_w}{\ell_t}\right)^2\right)\left(\frac{0.5h_s}{h_w}\right)$$

$$\text{for } \ell_t \geq 20[\text{m}]$$

$$A_s = (0.7 + 0.035\ell_t)S_c\frac{b_0}{\ell_t}\left(1 - \left(\frac{\ell_w}{\ell_t}\right)^2\right)\left(\frac{0.5h_s}{h_w}\right)$$

$$\text{for } \ell_t \leq 20[\text{m}]$$

where:

ℓ_t = Total well length, in [m], to be taken equal to:

$$\ell_t = \ell_w + \ell_s$$

ℓ_w = The length of the open deck enclosed by bulwarks in [m]

ℓ_s = The length of the common space within the open superstructure, in [m]

S_c = Sheer correction factor

= 1.5 for no sheer

= 0 for standard sheer

For intermediate values of sheer S_c to be interpolated.

b_0 = Breadth of openings in the end bulkhead of the enclosed superstructure, in [m]

h_w = Height of the well deck above the freeboard deck, in [m]

h_s = Standard superstructure height, in [m] as defined in Chapter 1, Section 2, Cl.2.2.5, Table 2.2.5

h_b = Height of bulwark, in [m], not to be taken as greater than h_s

A_c = Bulwark height correction factor, in [m^2], taken as:

= 0 for bulwarks between 0.9 and 1.2 [m] in height

= 0.04 ℓ_w (h_b − 1.2) for bulwarks of height greater than 1.2 [m]

= 0.04 ℓ_w (h_b − 0.9) for bulwarks of height less than 0.9 [m].

To adjust the freeing port area for the distance of the well deck above the freeboard deck, for decks located more than $0.5h_s$ above the freeboard deck, multiply by the factor $0.5(h_s/h_w)$.

9.2.4 Freeing Port arrangement

9.2.4.1 The lower edges of the freeing ports are to be as near the deck as practicable.

9.2.4.2 Two-thirds of the freeing-port area required is to be provided in the half of the well nearest the lowest point of the sheer curve. With zero or little sheer on the exposed freeboard deck or an exposed superstructure deck, it is considered that the freeing port area should be spread along the length of the well. One third of the freeing port area required is to be evenly spread along the remaining length of the well

9.2.4.3 All freeing port openings in the bulwarks are be protected by rails or bars spaced not more than 230[mm] apart. If shutters are fitted to freeing ports, arrangements are provided to prevent jamming. Hinges are to have pins or bearings of non-corrodible material. If shutters are fitted with securing devices, these appliances are to be of approved construction.

9.2.4.4 Openings in bulwarks are to be arranged so that the protection of the crew is to be at least equivalent to that provided by the horizontal courses in Cl.9.3.2.5. For this purpose, vertical rails or bars spaced 230[mm] apart may be accepted in lieu of rails or bars arranged horizontally.

9.2.5 Special requirements

9.2.5.1 Freeing arrangement for Type ‘A’ and Type ‘B-100’ ships with trunks

(i) Type ‘A’ and Type ‘B-100’ ships with bulwarks/trunks are to have open rails fitted for at least half the length of the exposed parts of the freeboard deck or other effective freeing arrangements are to be provided. It is considered that a freeing port area in the lower part of the bulwarks, of 33% of the total area of the bulwark provides the 'other effective freeing arrangements' and may be considered equivalent to the requirement of 50% open rails.

However, for Type A ships which have superstructures connected by trunks open rails are to be fitted for the whole length of the exposed part of the freeboard deck.

(ii) The upper edge of the sheer strake is to be kept as low as possible.

9.2.5.2 Freeing arrangement for Type 'B-60'

For Type B ships with freeboards reduced by not more than 60% of the difference between the freeboards given in Tables B and A of the International Loadline Convention, 1966; there are to be freeing port area in the lower part of the bulwarks equal to at least 25% of the total area of the bulwarks. The upper edge of the sheer strake is to be kept as low as possible.

9.3 Guard rails

9.3.1 General

Guard rails fitted according to Cl.9.1.1 and Cl.9.1.2, should meet the following requirements given in Cl.9.3.2.

Guard rails fitted on superstructure and freeboard decks are to have at least 3 courses. The opening below the lowest course of guard rails is not to exceed 230[mm]. The other courses are to be spaced not more than 380[mm] apart. Where rounded gunwales are fitted the guard rail supports are to be placed on the flat of the deck, as close as possible to the beginning of the curvature of the gunwale. In other locations, guard rails with at least two courses are to be fitted.

In the case of ships with welded sheer strake, the stanchions are not to be attached to the sheer strake, upstand or a continuous gutter bar.

9.3.2 Guard rail construction

The guard rails fitted should meet the following requirements:

9.3.2.1 Fixed, removable or hinged stanchions are to be fitted about 1.5[m] apart. Removable or hinged stanchions are to be capable of being locked in the upright position.

9.3.2.2 At least every third stanchion is to be supported by a bracket or stay.

In lieu of at least every third stanchion supported by stay, alternatively any of the following options may be provided: (Refer Figure 9.3.2):

(i) At least every third stanchion of increased breadth: $k.\,b_s = 2.9\,b_s$

(ii) At least every second stanchion of increased breadth: $k.\,b_s = 2.4\,b_s$

(iii) Every stanchion of increased breadth: $k.\,b_s = 1.9\,b_s$

where,

b_s = Breadth of normal stanchion according to the design standard.

In any arrangement of (i), (ii) or (iii) above, the following detail is to be complied with:

Flat steel stanchions of increased breadth are to be aligned with member below deck which is to be minimum100 x 12 flat bar welded to deck by double continuous fillet welds with leg size of minimum 7[mm] or as specified by the design standard. The stanchions with increased breadth need not be aligned with under deck structure for deck plating exceeding 20[mm].

9.3.2.3 Wire ropes may only be accepted in lieu of guardrails in special circumstances and then only in limited lengths.

9.3.2.4 Lengths of chain may be accepted in lieu of guard rails if they are fitted between two fixed stanchions and/or bulwarks.

9.3.2.5 The clear opening below the lowest course of the guard rails is not to exceed 230[mm]. Where this course is not measured from the deck, but from the sheer strake or a waterway bar, which is not in the same vertical plane as the rails, the length of opening is the diagonal distance between the lowest of the rails and the top of the sheer strake or waterway bar. The other courses are not to be more than 380[mm] apart.

9.3.2.6 Wires are to be made taut by means of turnbuckles.

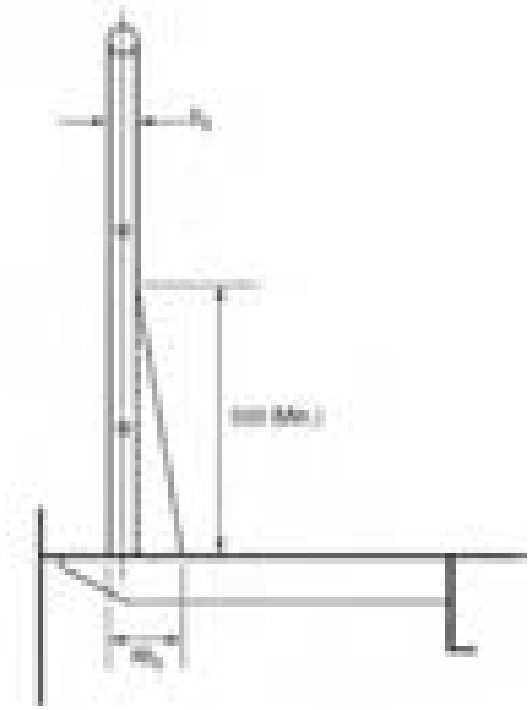


Figure 9.3.2: Guardrail Stanchion

9.4 Gangways, walkways and passageways

9.4.1 Satisfactory means (in the form of guard rails, life lines, gangways or under deck passages etc.) are to be provided for the protection of the crew in getting to and from their quarters, the machinery space and all other parts used in the necessary work of the ship in accordance with Table 9.4.1.

Protection for the crew in the form of guard rails or life lines are to be provided above the deck cargo if there is no convenient passage on or below the deck of the ship.

9.4.2 Protection of crew requirements for specific types

Note: Some Administrations may have more stringent requirements for protection of crew than those given below.

Protection of crew are to be provided by at least one of the means denoted in Table 9.4.1. Acceptable arrangements referred to in the table are defined as follows:

(a) A well-lighted and ventilated under-deck passageway (clear opening 0.8[m] wide, 2.0[m] high) as close as practicable to the freeboard deck, connecting and providing access to the locations in question.

(b) On or near the centreline of a ship a permanent and efficiently constructed gangway fitted at or above the level of the superstructure deck, providing a continuous platform at least 0.6[m] in width and a nonslip surface, with guard rails extending on each side throughout its length. Guard rails are to be at least 1[m] high with courses as required in Cl.9.3.1 and Cl.9.3.2.5 above and supported by stanchions spaced not more than1.5 [m]; a foot stop are to be provided.

(c) A permanent walkway at least 0.6[m] in width fitted at freeboard deck level consisting of two rows of guard rails with stanchions spaced not more than 3[m]. The number of courses of rails and their spacing are to be as required by Cl.9.3.1 and Cl.9.3.2.5. On type B ships, hatchway coamings not less than 0.6[m] in height may be regarded as forming one side of the walkway, provided that between the hatchways two rows of guardrails are fitted.

(d) A 10[mm] minimum diameter wire rope lifeline supported by stanchions about 10[m] apart, or A single handrail or wire rope attached to hatch coamings, continued and adequately supported between hatchways.

(e) A permanent and efficiently constructed gangway fitted at or above the level of the superstructure deck on or as near as practicable to the centre line of the ship:

– located so as not to hinder easy access across the working areas of the deck;

– providing a continuous platform at least 1.0[m] in width

– constructed of fire resistant and non-slip material;

Note: FRP gratings used in lieu of steel gratings for safe access to tanker bows are to possess:

- low flame spread characteristics and are not to generate excessive quantities of smoke and toxic products, as per the FTP Code; and

- adequate structural fire integrity as per national / international standards (for example, the standard specification for Fibre Reinforced Polymer (FRP) gratings used in marine construction and shipbuilding (ASTM F 3059, latest version)).

– fitted with guard rails extending on each side throughout its length; guard rails are to be at least 1.0[m] high with courses as required by Cl.9.3.2.5 and supported by stanchions spaced not more than 1.5[m].

– provided with a foot stop on each side;

– having openings, with ladders where appropriate, to and from the deck. Openings should not be more than 40[m] apart;

– having shelters of substantial construction set in way of the gangway at intervals not exceeding 45[m] if the length of the exposed deck to be traversed exceeds 70[m]. Every such shelter should be capable of accommodating at least one person and be so constructed as to afford weather protection on the forward, port and starboard sides.

(f) A permanent and efficiently constructed walkway fitted at freeboard deck level on or as near as practicable to the centre line of the ship having the same specifications as those for a permanent gangway listed in (e) except for foot-stops. On type B ships (certified for the carriage of liquids in bulk), with a combined height of hatch coaming and fitted hatch cover of together not less than 1[m] in height the hatchway coamings may be regarded as forming one side of the walkway, provided that between the hatchways two rows of guard rails are fitted.

Alternative transverse locations to (c), (d) and (f) above: (indicated as (1) to (5) in Table 9.4.1.)

(1) at or near centre line of ship; or fitted on hatchways at or near centre line of ship

(2) fitted on each side of the ship

(3) fitted on one side of the ship, provision being made for fitting on either side

(4) fitted on one side only

(5) fitted on each side of the hatchways as near to the centre line as practicable.

Additional requirements are also to be met:

– In all cases where wire ropes are fitted, adequate devices are to be provided to ensure their tautness.

– Wire ropes may only be accepted in lieu of guardrails in special circumstances and then only in limited lengths.

- Lengths of chain may only be accepted in lieu of guardrails if fitted between two fixed stanchions.

– Where stanchions are fitted, every 3rd stanchion to be supported by a bracket or stay.

– Removable or hinged stanchions to be capable of being locked in the upright position.

– A means of passage over obstructions, if any, such as pipes or other fittings of a permanent nature, should be provided.

– Generally, the width of the gangway or deck-level walkway should not exceed 1.5[m].

– For tankers less than 100[m] in length, the minimum width of the gangway platform or deck level walkway fitted in accordance with arrangement (e) or (f), respectively, may be reduced to 0.6[m].

9.4.3 A fore and aft permanent gangway, constructed in accordance with the provisions of regulation Cl.9.4.2(e), to be fitted on type 'A' ships at the level of the superstructure deck between the poop and the midship bridge or deckhouse where fitted. The arrangement contained in regulation Cl.9.4.2(a) is considered an equivalent means of access to carry out the purpose of the gangway.

9.4.4 Safe access from the gangway level are to be available between separate crew accommodations and also between crew accommodations and the machinery space.

Table 9.4.1: Protection of crew							
Type of Ship	**Location of access in ships**		**Assigned summer freeboard**	**Acceptable arrangements according to type of freeboard assigned:**			
				Type A	**Type B-100**	**Type B-60**	**Type B&B+**
All ships other than Oil Tankers**, Chemical Tankers** and Gas Carriers**	**1.1**	**Access to Midship Quarters**					
	1.1.1	Between poop and bridge, or Between poop and deckhouse containing living accommodation or navigating equipment, or both	$\leq 3000[\mathrm{mm}]$	a b e	a b e	a b c (1) e f (1)	a b c (1) c (2) c (4) d (1) d (2) d (3) e f (1) f (2) f (4)
	1.1.2		$> 3000[\mathrm{mm}]$	a b e	a b e	a b c (1) c (2) e f (1) f (2)	
	1.2	**Access to ends**					
	1.2.1	Between poop and bow (if there is no bridge)	$\leq 3000[\mathrm{mm}]$	a b c (1) e f (1)	a b c (1) c (2) e f (1) f (2)	a b c (1) c (2) e f (1) f (2)	
	1.2.2	Between bridge and bow, or					
	1.2.3	Between a deckhouse containing living accommodation or navigating equipment, or both, and bow, or	$> 3000[\mathrm{mm}]$	a b c (1) d (1) e f (1)	a b c (1) c (2) d (1) d (2) e f (1) f (2)	a b c (1) c (2) c (4) d (1) d (2) d (3) e f (1) f (2) f (4)	
	1.2.4	In the case of a flush deck vessel, between crew accommodation and the forward and aft end of ship***					
Oil Tankers**, Chemical Tankers** and Gas Carriers**	**2.1**	**Access to Bow**					
	2.1.1	Between poop and bow or	$\leq (\mathrm{A_f} + \mathrm{H_s})^*$	a e f (1) f (5)			
	2.1.2	Between a deckhouse containing living accommodation or navigating equipment, or both, and bow, or					
	2.1.3	In the case of a flush deck vessel, between crew accommodation and the	$> (\mathrm{A_f} + \mathrm{H_s})^*$	a e f (1)			

		forward end of ship		f (2)
	2.2 2.2.1	**Access to After End** In the case of a flush deck vessel, between crew accommodation and the after end of ship***		As required in 1.2.4 for other types of ships

* A_f = the minimum summer freeboard calculated as type A ship regardless of the type of freeboard actually assigned.
H_s = the standard height of superstructure as defined in ICLL Regulation 33, in [mm].

** Oil tankers, chemical tankers and gas carriers as defined in SOLAS Ch. II-1/2.12, VII/8.2 and VII/11.2, respectively, as amended.

*** Access to after end of ships is not applicable when crew accommodation is located aft.

Note: Deviations from some or all of these requirements or alternative arrangements for such cases as ships with very high gangways (i.e., certain gas carriers) may be allowed subject to agreement case-by-case with the relevant flag Administration.

Section 10

Means of Embarkation and Disembarkation

10.1 General

10.1.1 Ships constructed on or after 1 Jan 2010 are to be provided with means of embarkation and disembarkation such as gangways and accommodation ladders, for use in port and in port related operations in accordance with the requirements of this section. However, compliance with particular provisions may be waived by the Administration, if it is deemed to be unreasonable or impractical.

10.1.2 Where means of embarkation and disembarkation other than those specifically covered in the section are fitted, an equivalent level of safety is to be provided.

10.1.3 For all ships (i.e., constructed before, on or after 1 Jan 2010), means of embarkation and disembarkation are to be inspected and maintained in suitable condition for the intended purpose. Requirements for surveys of such means are given in Part 1, Chapter 2 of the rules.

10.2 Construction

10.2.1 Accommodation ladders and gangways for means of embarkation and disembarkation which are provided on board ships constructed on or after 1 Jan 2010 are to meet applicable international standards such as ISO 5488:1979, “Shipbuilding – accommodation ladders”, ISO 7061:1993, “Shipbuilding – aluminium shore gangways for seagoing vessels” and/or national standards and/or other requirements recognized by the Administration. Such accommodation ladders and gangways fitted on ships constructed before 1 Jan 2010, which are replaced after that date, should, in so far as is reasonable and practicable, comply with the requirements of this section.

10.2.2 The structure of the accommodation ladders and gangways and their fittings and attachments should be such as to allow regular inspection, maintenance of all parts and, if necessary, lubrication of their pivot pin. Special care should be taken to ensure that the welding work is carried out satisfactorily.

10.2.3 The construction and test of accommodation ladder winches are to be in accordance with applicable international standards such as ISO 7364:1983 “Shipbuilding and marine structures – deck machinery – accommodation ladder winches”.

10.3 Installation

10.3.1 Location

As far as practicable, the means of embarkation and disembarkation are to be sited clear of the working area and are not to be placed where cargo or other suspended loads may pass overhead.

10.3.2 Lighting

Adequate lighting is to be provided to illuminate the means of embarkation and disembarkation, the position on deck where persons embark or disembark and the controls of the arrangement.

10.3.3 Lifebuoy

A lifebuoy equipped with a self-igniting light and a buoyant lifeline is to be available for immediate use in the vicinity of the embarkation and disembarkation arrangement when in use.

10.3.4 Arrangement

10.3.4.1 Each accommodation ladder is to be of such a length to ensure that, at a maximum design operating angle of inclination, the lowest platform will be not more than 600[mm] above the waterline in the “lightest seagoing condition”, as defined in the following:

“Lightest seagoing condition” is the loading condition with the ship on even keel, without cargo, with 10% stores and fuel remaining and in the case of a passenger ship with the full number of passengers and crew and their luggage.

10.3.4.2 The arrangement at the head of the accommodation ladder is to provide direct access between the ladder and the ship’s deck by a platform securely guarded by handrails and adequate handholds. The ladder should be securely attached to the ship to prevent overturning.

10.3.4.3 For ships on which the height of the embarkation / disembarkation deck exceeds 20[m] above the waterline specified in Cl.10.3.4.1 and on other ships where compliance with the provisions of Cl.10.3.4.1 is considered impractical, an alternative means of providing safe access to the ship or supplementary means of safe access to the bottom platform of the accommodation ladder may be accepted.

10.3.5 Marking

Each accommodation ladder or gangway is to be clearly marked at each end with a plate showing the restrictions on the safe operation and loading, including the maximum and minimum permitted design angles of inclination, design load, maximum load on bottom end plate, etc. Where the maximum operational load is less than the design load, it should also be shown on the marking plate.

10.3.6 Test

10.3.6.1 After installation, the winch and the accommodation ladder are to be operationally tested to confirm proper operation and condition of the winch and the ladder after the test.

10.3.6.2 The winch is to be tested as a part of the complete accommodation ladder unit through a minimum of two times hoisting and lowering of the accommodation ladder in accordance with the onboard test requirement specified in applicable international standards such as ISO 7364:1983.

10.3.6.3 Every new accommodation ladder is to be subjected to a static load test of the specified maximum working load upon installation.

10.3.7 Positioning

10.3.7.1 Gangways are not to be used at an angle of inclination greater than 30[deg] from the horizontal and accommodation ladders are not to be used at an angle greater than 55[deg] from the horizontal, unless designed and constructed for use at angles greater than these and marked as such, as required by Cl.10.3.5.

10.3.7.2 Gangways are not to be secured to ship's guardrails unless they have been designed for that purpose. If positioned through an open section of bulwark or railings, any remaining gaps are to be adequately fenced.

10.3.7.3 Adequate lighting for means of embarkation and disembarkation and for the immediate approaches are to be ensured from the ship and/or the shore in hours of darkness.

10.3.8 Safety net

A safety net is to be mounted in way of the accommodation ladders and gangways where it is possible that a person may fall from them or between the ship and quayside.

10.3.9 Verification

Upon installation the entire arrangement of the means of embarkation and disembarkation will be verified for compliance with the requirements of this section.

10.4 Inspection and maintenance

10.4.1 The maintenance and inspection requirements for means of embarkation and disembarkation are to be addressed by the ship's safety management system.

10.4.2 The wires used to support the means of embarkation and disembarkation are to be renewed when necessary due to deterioration or at intervals of not more than 5 years, whichever is earlier.

10.4.3 Arrangements should also be made to examine the underside of gangways and accommodation ladders at regular intervals.

10.4.4 All inspections, maintenance work and repairs of accommodation ladders and gangways should be recorded in order to provide an accurate history for each appliance. The information to be recorded appropriately on board should include the date of the most recent inspection, the name of the person or body who carried out that inspection, the due date for the next inspection and the dates of renewal of wires used to support the embarkation and disembarkation arrangement.

End of Chapter

Chapter 8

Finite Element Analysis

Contents

Section

1 *General*
2 *Full Ship Strength Analysis*
3 *Cargo Hold Analysis*
4 *Local Structural Strength Analysis*

Section 1

General

1.1 Scope

1.1.1 This chapter provides requirements applicable to ships for assessing the structural strength and scantling of hull structures using finite element (FE) analysis. Aspects such as modeling techniques, loads and loading conditions, acceptance criteria and documentation required for FE analysis are also discussed.

1.2 Types of FE analysis

1.2.1 Rule covers the following types of FE analyses:

1.2.1.1 Full ship strength analysis for assessing the hull girder strength and scantlings and to provide the correct boundary conditions to partial ship structural models or to local FE models, wherever applicable.

1.2.1.2 Cargo hold analysis for assessing the strength of longitudinal hull girder structural members, primary supporting structural members and bulkheads.

1.2.1.3 Local structural strength analysis or fine mesh analysis for assessing highly stressed areas in local structural details.

1.2.1.4 Very fine mesh analysis is to assess the fatigue capacity of the structural details according to Chapter 10.

1.3 Loads and Loading conditions

1.3.1 FE calculations are to be based on the most severe, realistic loading conditions, as applicable, for the ship and are taken as specified in Section 2.3.

1.3.2 General design loads are specified in Section 2, Cl.2.3.1 and Section 3, Cl.3.3.6.

1.3.3 For specific type of ships, Part 5 is to be referred for design loads and FE design load combinations.

1.4 Net scantling approach

1.4.1 FE models for full ship strength analysis, cargo hold analysis, local fine mesh analysis and very fine mesh analysis, are to be based on the net scantling approach, as defined in Chapter 2, Section 4, Table 4.6.2.

1.4.2 Buckling capacity assessment is to be based on corrosion addition, as defined in Chapter 2, Section 4, Table 4.6.2.

1.5 Material properties

1.5.1 Steel and aluminum are the preferred materials as load carrying members. If special materials are used, e.g. composites, wood etc. such material may be included as anisotropic material in FE model. Such properties are to be defined on a case to case basis. Additionally,

Refer Part 3, Chapter 2, Section 1.2 and Section 1.3.

1.6 Type of Finite Elements used in analysis

1.6.1 The structural assessment is to be based on linear finite element analysis of three-dimensional structural models. The general types of finite elements to be used in the finite element analysis are given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method, Section 1, Cl.1.3.3 for finite element-based assessment.

1.6.2 In general, two-noded line elements and four-noded isotropic shell elements are considered to be sufficient to represent hull structures.

1.7 Submission of analysis report

1.7.1 A detailed report of the structural analysis is to be submitted to IRS in order to demonstrate the compliance with specified structural design criteria.

1.7.2 The major information to be included in the report are the following:

1.7.2.1 List of plans (drawings, loading manual etc.) used including their drawing numbers, names, revisions and dates.

1.7.2.2 Detailed description of structural modeling including all modeling assumptions, model compliance with actual structure and any deviations in geometry and arrangement of structure compared with plans.

1.7.2.3 Documentation related to the computer program used to transfer the hydrodynamic load to the 3D FE structural model.

1.7.2.4 Plots clearly demonstrating correct structural modeling and property assignment.

1.7.2.5 List of units used in the analysis.

1.7.2.6 Details of material properties, plate thickness and beam properties used in the model.

1.7.2.7 Details of the boundary condition.

1.7.2.8 Details of all loading conditions reviewed with calculated hull girder shear force, bending moment and torsional moment distributions.

1.7.2.9 All data relating to load application and its verification.

1.7.2.10 Plots/results showing the expected behavior of the structural model under the applied loads.

1.7.2.11 Summaries and plots of global and local deflections and reaction forces are to be shown to verify the load cases.

1.7.2.12 Summaries and plots of stress distribution to demonstrate that the design criteria are not exceeded in any member. Wherever otherwise, plots should clearly point out such members/locations.

1.7.2.13 Details and results of panel buckling analysis (unstiffened and stiffened).

1.7.2.14 Details and results of pillar buckling analysis, wherever applicable.

1.7.2.15 Proposed modifications and amendments to structure, including the revised estimation of stresses, deflections, buckling and fatigue showing compliance to design criteria.

1.7.2.16 Structural analysis results related to specific type of ships as detailed in Part 5

1.7.2.17 Reference of the Finite Element program including its version and date.

1.7.2.18 Any other necessary data or information pertaining to the analysis.

1.8 Computer program used for the analysis

1.8.1 Any finite element computation program recognized by IRS may be utilized for structural assessment to determine the stress and deflection of the hull structure, provided that the combined effects of bending, shear, axial and torsional deformations are considered.

1.8.2 Any computer program that has demonstrated to provide reliable results to the satisfaction of IRS is regarded as a recognized program. It is recommended that the designer consults IRS on the suitability of a computer program intended to be used prior to the commencement of work.

Section 2

Full Ship Strength Analysis

2.1 Scope

2.1.1 The purpose of the full ship strength analysis is to evaluate the overall hull girder stiffness and to assess the global stresses and deformations of primary hull members.

2.1.2 Full ship strength analysis covers the whole ship and is performed when the structural response of the hull girder cannot otherwise be sufficiently determined by prescriptive calculations for:

2.1.2.1 Ships with novel designs (e.g.: ships with huge openings in the side shell, where structural responses cannot be calculated solely based on beam theory).

2.1.2.2 Ships with large deck openings (e.g.: Container ships with large deck openings subjected to high torsional deformation and stresses.)

2.1.2.3 Ships with or without limited transverse bulkheads over the length of the ship (e.g., Ro-Ro ship or car carrier)

2.1.2.4 Large passenger ships (L>150[m]) with partly effective superstructure and/or partly effective upper part of hull girder, and

2.1.2.5 For specific class notation **FST** (Full Ship Strength Analysis)

2.1.3 According to the hull shape and ship type notation, the full ship strength analysis is to be performed to assess and verify compliance with the criteria as specified in Section 2.4. Additional requirements for specific ship types are given in Part 5.

2.2 Structural model

2.2.1 The full ship FE model represents the actual geometry of the ship and must cover the entire length, breadth and width of the vessel.

2.2.2 All members that contribute to the overall structural integrity of the hull girder such as primary supporting members, transverse members (watertight/non-watertight bulkheads, deck and cross deck structure etc.), double side structures (shell plating, inner hull, stringers and web frames) and double bottom structures (bottom plate, inner bottom plate, girders, floors etc.) are to be included in the FE model in order to simulate the stiffness of hull girder.

2.2.3 Mesh size, model idealization and boundary conditions are described in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method. Additional requirements for specific type of ships are given in Part 5.

2.3 Loading conditions

2.3.1 Loads for full ship strength analysis are to be determined by sea-keeping analysis/ direct hydrodynamic calculations.

2.3.2 The loading conditions are to be taken as per ship type. For specific ship types, refer to Part 5.

2.3.3 Load application is to be performed as described in IRS Guidelines on *Application of Direct Seakeeping Loads in Structural Analysis of Ships*.

2.4 Analysis criteria

2.4.1 General

The analysis criteria that apply for full ship FE model are described in this section. Structural adequacy of hull girder structural members and primary structural and secondary structural members are to be checked. FE analysis results for failure mode of material yielding, buckling and fatigue are to be checked. Wherever, the full ship FE model is partially or entirely refined to a mesh arrangement as used for partial ship analysis, the analysis criteria for partial ship analysis apply, as given in Section 3.4.

2.4.2 Yield strength

2.4.2.1 For all plates of the structural members, the von Mises stress, σ_{vm} [N/mm^2] is to be calculated based on the membrane normal and shear stresses of the shell element. The stresses are to be evaluated at the element centroid of the mid-plane (layer), as follows:

$$\sigma_{vm} = \sqrt{{\sigma_x}^2 - \sigma_x\sigma_y + {\sigma_y}^2 + 3{\tau_{xy}}^2}$$

where,

σ_x, σ_y = Element normal membrane stresses, in [N/mm^2].

τ_{xy} = Element shear stress, in [N/mm^2].

2.4.2.2 Nominal axial (normal), shear and von Mises stresses derived from a full ship analysis are to be checked according to the acceptance criteria given in Part 5 for specific ship type.

2.4.3 Deflection criteria for primary supporting members

2.4.3.1 The maximum relative deflection of primary supporting members such as girders is not to exceed the following criteria:

$$\delta_{max} = \frac{l_i}{325}$$

where,

δ_{max} = Maximum relative deflection, in [mm].

l_i = Span of primary girder, in [mm].

2.4.4 Buckling strength

2.4.4.1 All major structural elements need to be individually assessed against buckling requirements as per Chapter 9, Section 3.

2.4.5 Fatigue strength

2.4.5.1 Fatigue strength analysis of relevant structural details is to be performed as per IRS Guidelines on Fatigue Design Assessment of Ship Structures.

2.4.5.2 General assumptions, methodology and acceptance criteria for the fatigue strength assessment are given in IRS Guidelines on Fatigue Design Assessment of Ship Structures.

Section 3

Cargo Hold Analysis

3.1 General

3.1.1 Definitions

3.1.1.1 Cargo hold analysis: For ships with cargo hold or tank arrangement, the analysis performed to assess the strength of longitudinal hull girder structural members, primary supporting structural members and bulkheads.

3.1.1.2 Evaluation area: The evaluation area is an area in the part ship model, where the verification of results against the acceptance criteria are to be carried out. For a cargo hold structural analysis, evaluation area is defined in Cl.3.4.1.

3.1.2 The FE model extent for a cargo hold analysis is defined in Cl.3.3.2.

3.1.3 For ships without cargo hold arrangement or for evaluation areas outside cargo area, the analysis procedures given in this chapter can be used with special consideration.

3.1.4 The length of FE model is to be selected in such a way that boundary condition at the ends of the model are sufficiently away from the evaluation area so that it does not interfere with the results.

3.2 Application

3.2.1 This section gives the application of the cargo hold analysis which is mandatory for:

- Ships having rule length L >150[m]
- Ships with novel design
- Specific ship types as given in Part 5
- or whenever required by IRS.

3.2.2 Cargo hold strength analysis of mid-ship is mandatory to applicable ships given in Cl.3.2.1.

3.2.3 FE based strength assessment is to be performed outside mid-ship region if structural loads are substantially different from that of mid-ship region. This analysis is mandatory for specific ship types as given in Part 5.

3.2.4 Scantlings of primary supporting members outside the evaluation area should be

assessed for equivalency to those evaluated by FEA.

3.2.5 Primary supporting members outside cargo hold region not analyzed using FEM are to be assessed using the procedure given in Chapter 7, Section 6.

3.2.6 FE model extent, boundary conditions, applicable loading cases and acceptance criteria are to be as given in the following sections. For specific ship types, additional analysis requirements are given in Part 5.

3.3 Structural model

3.3.1 Model idealization

3.3.1.1 All main longitudinal and transverse structural elements are to be modelled.

3.3.1.2 All plates and stiffeners on the structure, including web stiffeners, are to be modelled. Brackets which contribute to primary supporting member strength and the size of which is not less than the typical mesh size (s × s), Refer Cl.3.3.4, are to be modelled.

3.3.1.3 Openings are to be modelled in accordance with IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

3.3.2 Model extent

3.3.2.1 For the purpose of the FE analysis, the mid-hold is defined as the middle hold(s) of the three cargo hold length FE model. In case of foremost and aft-most cargo hold assessment, the mid-hold represents the foremost and aft-most cargo hold including the slop tank if any, respectively. Additional details regarding modelling and extent of foremost and aft-most cargo hold are given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

3.3.2.2 The longitudinal extent of the cargo hold FE model is to cover three cargo hold lengths. The transverse bulkheads and the web frames at the ends of the model are to be modelled.

3.3.2.3 Transverse extent: Both port and starboard sides of the ship are to be modelled.

3.3.2.4 Vertical extent: The full depth of the ship is to be modelled including all primary and secondary members above the upper deck, trunks, forecastle and/or cargo hatch coaming, if any. The superstructure or deck house in way of machinery space and the bulwark are not required to be included in the model.

3.3.2.5 Additional details related to specific ship types are to be taken as per Part 5.

3.3.3 Finite element types

3.3.3.1 Plates are to be modelled using shell element and stiffeners are to be modelled as beam elements having axial, torsional, bi-directional shear and bending stiffness. The eccentricity of the neutral axis is to be modelled.

3.3.3.2 Sniped stiffeners, face plates of primary supporting members and brackets are to be modelled using rod or beam elements.

3.3.4 Mesh

3.3.4.1 The shell element mesh is to follow the stiffening system as far as practicable, hence representing the actual plate panels between stiffeners i.e. (s × s), where ‘s’ is the stiffener spacing.

3.3.4.2 Detailed guidelines on mesh configuration are given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

3.3.5 Boundary conditions

3.3.5.1 The boundary conditions given applicable to cargo hold analyses in cargo hold region are given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

3.3.6 Loads and Loading conditions

3.3.6.1 FE load combination definition

FE load combination is defined as a loading pattern, a draught, a value of still water bending and shear force, associated with a given dynamic load case.

3.3.6.2 Design load combination

(a) For cargo hold structural strength analysis, the design load combinations specified in Chapter 5, Section 6 are to be used for considered ship type and considered cargo hold regions.

(b) Each design load combination consists of a loading pattern and dynamic load cases as

given in Chapter 5, Section 2. Each load combination requires the application of the structural weight, internal and external pressures and hull girder loads.

(c) For seagoing condition with extreme sea loads, both static and dynamic load components (S+D) are applied.

(d) For harbour and tank testing condition, only static load components (S) are applied.

(e) Design load combinations for specific type of ship are to be taken as per Part 5.

3.3.6.3 Design loads

(a) Design loads are to be applied as given in Chapter 5. Loads for specific ship types are to be taken as given in Part 5. Design loads are to be combined with FE load combination as discussed in Cl.3.3.6.2.

(b) Where the loading conditions specified by the designer are not covered by the load combinations given in IRS rules, these additional loading conditions are to be examined according to the procedure in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

3.3.6.4 Load application

(a) All simultaneously acting global and local loads are to be applied on the FE model. Application of loads are to be performed as described in IRS Guidelines on Application of Direct Seakeeping Loads in Structural Analysis of Ships.

. Particular types of load for specific ships are to be applied as per the relevant class guideline. Load application includes the following loads:

- Structural weight
- External and Internal loads
- Hull girder loads
- Quasi-static and Inertial loads
– Target values for hull girder loads

(b) Effect of the weight of hull structure is to be included in cargo hold analysis. Density of steel may be taken as 7850[kg/m^3].

(c) External pressure is to be calculated in accordance with Chapter 5, Section 4. External pressures include static sea pressure, wave pressure and green sea pressure.

(d) Internal pressure is to be calculated for each load case in accordance with Chapter 5, Section 5 for design load scenarios as given in Chapter 5, Section 6. Internal pressure application is to be performed as per ship type described in IRS Guidelines on Application of Direct Seakeeping Loads in Structural Analysis of Ships.

(e) Hull girder loads in terms of shear force, bending moment and torsional moment are to be applied on the FE model corresponding to each loading condition.

(f) The hull girder loads are the combinations of still water hull girder loads and wave induced hull girder loads as specified in Chapter 5, Section 3. For each required FE load combination, the wave induced hull girder loads are to be calculated with the Load Combination Factors (LCFs), specified in Chapter 5, Section 2. When the FE load combinations are not specified in Part 5 for a considered ship type, the required component of hull girder loads will be decided by IRS on a case-to-case basis.

(g) The required target hull girder load values are to be achieved at given locations for each load combination. The target locations for hull girder shear force, bending moment and torsional moments is to be taken as given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method. For specific ship types, Part 5 and IRS Guidelines on Structural Assessment of Ships based on Finite Element Method are to be referred.

(h) Load application is to be carried out as per IRS Guidelines on Application of Direct Seakeeping Loads in Structural Analysis of Ships.

3.4 Analysis criteria

3.4.1 Evaluation areas

3.4.1.1 Mid-hold cargo analysis

Verification of results against the acceptance criteria is to be carried out within the longitudinal extent of the mid-hold for the following structural members:

- All hull girder longitudinal structural members
- All primary supporting structural members
- Bulkheads, including forward and aft transverse bulkheads of the mid-hold.

3.4.1.2 Fore- and aftmost cargo hold analyses

For the fore- and aftmost cargo hold analysis, the evaluation areas extend to the following structural elements, in addition to elements listed in Cl.3.4.1.1:

- Foremost Cargo hold: All structural members being part of the collision bulkhead, and extending to one web frame spacing forward of the collision bulkhead.

– Aftmost Cargo hold: All structural members being part of the transverse bulkhead of the aftmost cargo hold and all hull girder longitudinal structural members aft of this transverse bulkhead with the extent of 15% of the aftmost cargo hold length.

3.4.2 Yield Strength Assessment

3.4.2.1 Von Mises Stress

For all plates of the structural members within the evaluation area, the von Mises stress, σ_{vm} [N/mm^2] is to be calculated based on the membrane normal and shear stresses of the shell element. The stresses are to be evaluated at the element centroid of the mid- plane (layer), as follows:

$$\sigma_{vm} = \sqrt{{\sigma_x}^2 - \sigma_x\sigma_y + {\sigma_y}^2 + 3{\tau_{xy}}^2}$$

where,

σ_x, σ_y = Element normal membrane stresses, in [N/mm^2].

τ_{xy} = Element shear stress, in [N/mm^2].

3.4.2.2 Axial stress in beams and rod elements

For beams and rod elements, the axial stress, σ_{axial} [N/mm^2] is to be calculated based on axial force alone. The axial stress is to be evaluated at the middle of element length.

3.4.2.3 Yield criteria

(a) The structural elements given in Cl.3.4.1 are to comply with the following criteria:

$$\lambda_y \leq \lambda_{yperm}$$

where:

λ_y = Yield utilization factor

In general,

$\lambda_y = \frac{\sigma_{vm}}{R_Y}$ for shell elements

$\lambda_y = \frac{|\sigma_{axial}|}{R_Y}$ for rod or beam elements

$\lambda_y = \frac{|\sigma_{axial}|}{R_{eH}}$ for rod or beam elements representing curved flanges.

σ_{vm} = von Mises stress, in [N/mm^2].

σ_{axial} = Axial stress rod and beam element, in [N/mm^2].

λ_{yperm}= Coarse mesh permissible yield utilization factors defined in Table 3.4.2.3.

R_y = Nominal yield stress, taken equal to 235/k

(b) The yield check criteria are to be based on axial stress for the following members:

- The flange of primary supporting members
- The intersections between the flange and web of the corrugations, according to Cl.3.4.2.4.

(c) Where the von Mises stress of the elements in the cargo hold FE model in way of the area under investigation by fine mesh exceeds the yield criteria, average von Mises stress, obtained from the fine mesh analysis, calculated over an area equivalent to the mesh size of the cargo hold finite element model is to satisfy the yield criteria above.

(d) In way of cut-outs, yield utilization factor is to be obtained with shear stress correction, as given in Cl.3.4.2.5.

3.4.2.4 Coarse mesh permissible yield utilization factors

The coarse mesh permissible yield utilization factors, λ_{yperm}, given in Table 3.4.2.3 are based on the mesh sizes and element types described in Cl.3.3.3 to Cl.3.3.4.

The yield utilization factor resulting from element stresses of each structural component are not to exceed the permissible values as given in Table 3.4.2.3.

Table 3.4.2.3: Coarse mesh permissible yield utilization factor			
Structural Members	**Load Components**	**Acceptance criteria (1)**	λ_{yperm}
Plating of all longitudinal hull girder structural members, primary supporting structural members and bulkheads. Dummy rod of corrugated bulkhead. Face plate of primary supporting members modelled using shell or rod elements.	Load combination: S Load combination: S+D	AC-I AC-II	0.80 1.0
Corrugation of corrugated bulkheads under lateral pressure from liquid loads, for shell elements only. For corrugation angle between 45° and 55° the reduction in, λ_{yperm} as given in Chapter 3, Section 6, Cl.6.4.1.2 applies.	Load combination: S Load combination: S+D	AC-I AC-II	0.72 0.90
(1) Refer Chapter 1, Section 1, Cl.1.10 for acceptance criteria definition			

3.4.2.5 Corrugation of corrugated bulkhead

The stress in corrugation of corrugated bulkheads is to be evaluated based on:

- The von Mises stress, σ_{vm}, in shell elements on the flange and web of the corrugation.

- The axial stress, σ_{axial}, in dummy rod elements, modelled with unit cross sectional properties at the intersection between the flange and web of the corrugation.

3.4.2.6 Shear stress correction for cut-outs

Except as indicated in Cl.3.4.2.7, the element shear stress in way of cut-outs in webs is to be corrected for loss in shear area in accordance with the following formula. The corrected element shear stress is to be used to calculate the von Mises stress of the element for verification against the yield criteria.

$$\tau_{cor} = \frac{h\, t_{mod}}{A_{shr-n50}} \tau_{elem}$$

where:

τ_{cor} = Corrected element shear stress, in [N/mm^2].

t_{mod} = Modelled web thickness, in [mm], in way of opening.

h = Height of web of girder, in [mm], in way of opening, Refer Table 3.4.2.5. Where the geometry of the opening is modelled, h is to be taken as the height of web of the girder deducting the height of the modelled opening.

τ_{elem} = Element shear stress, in [N/mm^2] before correction.

$A_{shr-n50}$ = Effective net shear area of web [mm^2] taken as the web area deducting the area lost of all openings, including slots for stiffeners, calculated in accordance with Chapter 3, Section 7, Cl.7.4.8.

3.4.2.7 Exceptions for shear stress correction for openings

Correction of element shear stress due to presence of cut-outs is not required for cases given in Table 3.4.2.6 provided yield criteria given in Cl.3.4.2.3 is satisfied based on the following yield utilization factor:

$$\lambda_y = \frac{\lambda_{y,FE}}{C_r}$$

where:

$\lambda_{y,FE}$ = yield utilization factor from FE assessment with openings not reflected in the model.

C_r = reduction factor for yield criteria as given in Table 3.4.2.6.

If the above criterion is not satisfied then the criteria as given in Cl.3.4.2.6 needs to be applied.

3.4.3 Buckling strength assessment

All structural elements in FE analysis carried out in accordance with this section are to be assessed individually against the buckling requirements as defined in Chapter 9, Section 4.

Table 3.4.2.5: Representation of openings in primary supporting webs

Criteria	Modeling Requirement
$h_0/h < 0.5$ and $g_0 < 2.0$	Opening need not be modeled
$h_0/h \geq 0.5$ and $g_0 \geq 2.0$	Opening geometry needs to be represented in the model

$$g_o = \left(1 + \frac{{l_o}^2}{2.6\,(h - h_o)^2}\right)$$

l_o **=** Length of opening parallel to primary supporting member web direction, in [m], Refer Figure 3.4.2.5(a). For sequential openings where the distance, d_o between openings is less than 0.25h, the length l_o is to be taken as the length across openings as shown in Figure 3.4.2.5(b).

h_o = Height of opening parallel to depth of web, in m, Refer Figure 3.4.2.5(a) and Figure 3.4.2.5(b).

h = Height of web of primary supporting member in way of opening, in m, Refer Figure 3.4.2.5(a) and Figure 3.4.2.5(b).

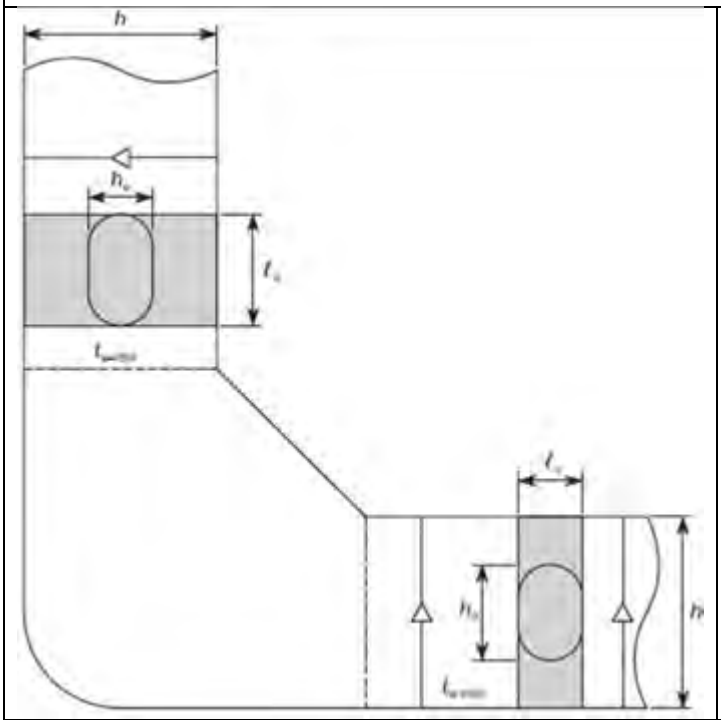


Figure 3.4.2.5(a): Openings in Primary Supporting member

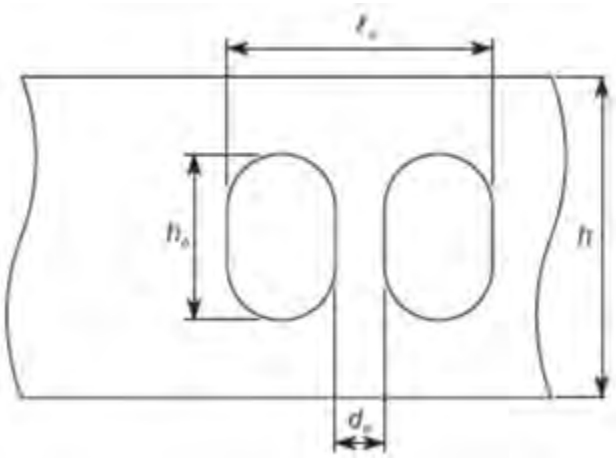
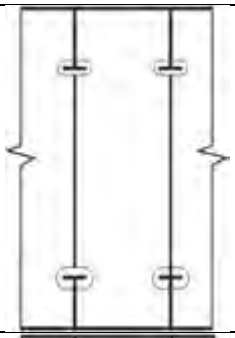
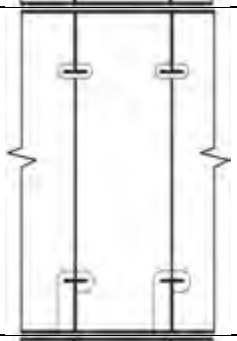
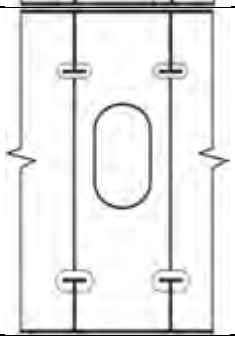


Figure 3.4.2.5(b): Length for Sequential openings with d_0 < 0.25h

Table 3.4.2.6: Exceptions for shear stress correction			
Identification	**Figure**	**Difference between modeled shear area and the net effective shear area in % of the modeled shear area** $\frac{\mathbf{A_{FEM} - A_{shr}}}{\mathbf{A_{FEM}}}.\mathbf{100}\%$	**Reduction factor for yield criteria,** C_r
Upper and lower slots for local support stiffeners fitted with lugs or collar plates		< 15 %	0.85
Upper and lower slots for local support stiffeners fitted with lugs and collar plates		< 20 %	0.80
In way of opening; upper and lower slots for local support stiffeners fitted with collar plates		< 40 %	0.60

Section 4

Local Structural Strength Analysis

4.1 Scope

4.1.1 All structural details which are considered critical are analyzed using local structural strength analysis. Additionally, local strength analysis is carried out on structural details which are required by specific ship types as given in Part 5.

4.1.2 The structural details are assessed by fine mesh FE analysis according to the requirements given in this section. Detailed procedures are given in IRS *Guidelines on Structural Assessment of Ships based on Finite Element Method*.

4.1.3 The selection of critical locations on the structural members and general procedure for screening of critical locations for fine mesh analysis is to be in accordance with the requirements given in IRS *Guidelines on Structural Assessment of Ships based on Finite Element Method,* in general.

4.1.4 The details to be assessed by fine mesh analysis are to be modelled according to the requirements given in IRS *Guidelines on Structural Assessment of Ships based on Finite Element Method*, under the FE load combinations defined in 4.3 and to comply with the acceptance criteria given in 4.4.

4.2 Structural model

4.2.1 General

Evaluation of detailed stresses requires the use of refined finite element mesh in way of areas of high stress. This fine mesh analysis can be performed using a separate local finite element model with fine mesh zones in conjunction with the boundary conditions obtained from the partial or global FE model. Alternately, fine mesh zones can be directly integrated into the partial ship model or global FE model, as applicable.

4.2.2 Model Extent

If a separate local fine mesh model is used, its extent is to be such that the calculated stresses at the evaluation area are not significantly affected by the imposed boundary conditions. The boundary of the fine mesh model is to coincide with primary supporting members, such as web frames, girders, stringers, floors etc.

4.2.3 Mesh size

4.2.3.1 The mesh size in the fine mesh zones is not to be greater than 50[mm] x 50[mm].

4.2.3.2 The extent of the fine mesh zone is not to be less than 10 elements in all directions from the area under investigation. A smooth transition of mesh density from fine mesh zone to the boundary of the fine mesh model is to be maintained.

4.3 FE Load combinations

4.3.1 General

4.3.1.1 The fine mesh analysis is to be carried out for all FE load combinations applied to the corresponding partial ship or full ship FE analysis.

4.3.2 Application of loads and boundary conditions

4.3.2.1 Where a separate local model is used for the fine mesh detailed stress analysis the displacements or forces from the cargo tank model (or full ship model) are to be applied to the corresponding boundary contour on the local model. All loads, including local loads and loads applied for hull girder bending moment and/or shear force adjustments, in way of the structure represented by the separate local finite element model are to be applied to the model.

4.4 Analysis criteria

4.4.1 Reference stress:

4.4.1.1 Reference stress, σ_{vm} in [N/mm^2] is based on the membrane direct axial and shear stresses of the plate element evaluated at the element centroid. Where shell elements are used, the stresses are to be evaluated at the mid plane of the element.

4.4.2 Permissible stress:

4.4.2.1 The maximum permissible stresses are based on the mesh size of 50[mm] × 50[mm].

4.4.2.2 Where a smaller mesh size is used, an area weighted von Mises stress calculated over an area equal to the specified mesh size may be used to compare with the permissible stresses and is calculated as follows:

$$\sigma_{vm-av} = \frac{\sum_1^n A_i \sigma_{vm-i}}{\sum_1^n A_i}$$

where,

σ_{vm-av} = Average von Mises stress in [N/mm^2].

A_i = Area of the mesh size, in [mm^2]

4.4.2.3 The averaging is to be based only on elements with their entire boundary located within the desired area. The average stress is to be calculated based on stresses at element centroid; stress values obtained by interpolation and/or extrapolation are not to be used. Stress averaging is not to be carried across structural discontinuities and abutting structure.

4.4.3 Acceptance criteria

4.4.3.1 Verification of stress results against the acceptance criteria is to be carried out in accordance with Cl.4.4.3.2.

4.4.3.2 The structural assessment is to demonstrate that the von Mises stresses obtained from the fine mesh analysis do not exceed the maximum permissible stress, as follows:

$$\lambda_f \leq \lambda_{fperm}$$

where,

λ_f = Fine mesh yield utilization factor.

In general,

$\lambda_y = \frac{\sigma_{vm}}{R_Y}$ for shell elements

$\lambda_y = \frac{|\sigma_{axial}|}{R_Y}$ for rod or beam elements

σ_{vm} = von Mises Stress, in [N/mm^2].

σ_{axial} = Axial Stress in rod element, in [N/mm^2].

λ_{fperm} = Permissible fine mesh utilization factor, as given in the Table 4.4.3.

R_y = Nominal yield stress, taken equal to 235/k

Table 4.4.3: Permissible fine mesh yield utilization factor, λ_{fperm}

Structural Member	Acceptance criteria	Load components	λ_{fperm}
Elements not adjacent to weld	AC-I	S	1.36 f_f
	AC-II	S+D	1.7 f_f
Elements adjacent to weld	AC-I	S	1.2 f_f
	AC-II	S+D	1.5 f_f

where,
f_f = fatigue factor taken as:
= 1.0 in general
= 1.2 for details where fatigue strength is verified by hot spot stresses based on very fine mesh finite element analysis, Refer IRS *Guidelines on Fatigue Design Assessment of Ship Structures*.

End of Chapter

Chapter 9

Buckling

Contents

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1 General

2 Slenderness Requirements

3 Buckling Requirements for Prescriptive Analysis

4 Buckling Requirements for Finite Element Analysis

5 Buckling Capacity

Appendix A - Stress-based reference stresses

Section 1

General

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

1.1 Assumption

1.1.1 This chapter contains the strength criteria for buckling and ultimate strength of local supporting members, primary supporting members, and other structures such as pillars, corrugated bulkheads, and brackets. These criteria are to be applied as specified in Chapter 7 (Hull Local Structures) and Chapter 8 (Finite Element Analysis).

1.1.2 For each structural member, the characteristic buckling strength is to be taken as the most unfavourable/critical buckling failure mode.

1.1.3 Unless otherwise specified, the buckling strength of structural members in this chapter is based on net scantling obtained by deducting t_c, from the gross offered thickness, where t_c is defined in Chapter 2, Section 5.

1.1.4 In this chapter, compressive and shear stresses are taken as positive whereas the tensile stresses are taken as negative.

1.2 Application

1.2.1 Scope

The buckling checks are to be carried out according to:

- Section 2 for the slenderness requirements of plates, longitudinal and transverse stiffeners, primary supporting members, brackets, and other structures such as pillars.

- Section 3 for the hull girder buckling requirements of plates, longitudinal and transverse stiffeners, primary supporting members, and other structures.

- Section 4 for the buckling requirements of the FE analysis for the plates, stiffened panels, and other structures. Buckling analysis according to Section 4 is also applicable for beam analysis of structure elements subject to compressive and shear stresses.

- Section 5, for the buckling capacity of prescriptive and FE buckling requirements.

1.2.2 Stiffeners:

The buckling check of the stiffeners referred to in this Chapter is applicable to the stiffener fitted along the long edge of the buckling panel.

1.2.3 Enlarged stiffeners:

Enlarged stiffeners, with or without web stiffening, used in platforms for Permanent Means of Access (PMA) are to be considered as primary supporting members (PSM) and are to comply with the following requirements:

1.2.3.1 Slenderness requirements for primary supporting members are as follows:

- For enlarged stiffener webs, Refer to Section 2, Cl. 2.4.1.1 (i).

- For enlarged stiffener flange, Refer to Section 2, Cl.2.4.1.1 (ii) and Section 2, Cl.2.5.1.

- For web stiffeners fitted on enlarged stiffeners, Refer to Section 2, Cl.2.3.1.1 and Section 2, Cl.2.3.1.3.

1.2.3.2 Buckling strength for prescriptive requirements is as follows:

- For enlarged stiffener webs, Refer to Section 3, Cl.3.2.

- For web stiffeners fitted on enlarged stiffeners, Refer to Section 3, Cl.3.1 and Section 3, Cl.3.3.

1.2.3.3 All structural elements used for PMA are to be comply with the buckling requirements of the FE analysis in Section 4, when applicable.

1.2.3.4 Buckling strength of longitudinal PMA platforms without stiffeners fitted on enlarged stiffener web is to be checked using the criteria for local supporting members in Section 3, Cl. 3.1 and Section 3, Cl. 3.3.

1.3 Definitions

1.3.1 Assessment methods

The buckling assessment is carried out according to one of the two methods by considering the type of boundary conditions on the plate panel, as follows:

1.3.1.1 Method A

All the edges of the elementary plate panel are forced to remain straight (but free to move in the in-plane directions) due to the surrounding structure/neighbouring plates.

1.3.1.2 Method B

The edges of the elementary plate panel are not forced to remain straight due to low in-plane stiffness at the edges and/or no surrounding structure/neighbouring plates.

1.3.2 Buckling utilization factor:

1.3.2.1 General

The utilisation factor, η is defined as the ratio between the applied loads and the corresponding ultimate capacity or buckling strength.

1.3.2.2 Combined load case

- For combined loads, the utilisation factor, η_{act} is defined as the ratio of the applied equivalent stress and the corresponding buckling capacity, as shown in Figure 1.3.2, and is taken as:

$$\eta_{act} = \frac{W_{act}}{W_u} = \frac{1}{\gamma_c}$$

where:

W_{act} = Equivalent applied stress, in [N/mm^2], due to the combined membrane stress, the actual applied stress are given in Section 3 and Section 4 for buckling assessment by prescriptive and finite element analysis respectively.

W_u = Equivalent buckling capacity, in [N/mm^2], for plates and stiffeners, their respective buckling or ultimate capacities are given in Section 5.

γ_c = Stress multiplier factor at failure.

For each typical failure mode, the corresponding capacity of the panel is calculated by applying the actual stress combination and then increasing or decreasing the stresses proportionally until collapse.

Figure 1.3.2 illustrates the buckling capacity and the buckling utilization factor of a structural member subject to σ_x and σ_y stresses.

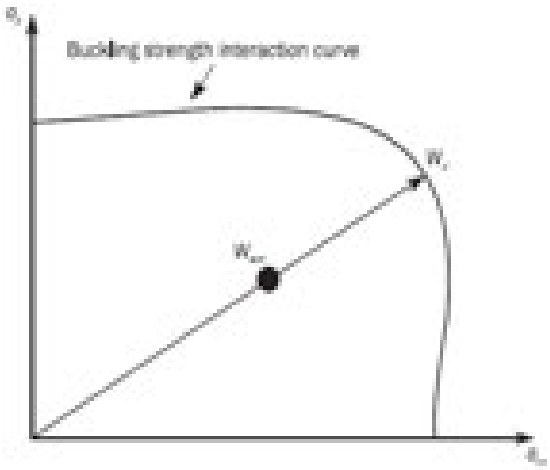


Figure 1.3.2: Example of buckling capacity and buckling utilisation factor

1.3.3 Allowable buckling utilisation factor

The allowable buckling utilisation factor is defined as per Table 1.3.3 (applicable for vessels L≥90[m]). For L< 90[m], Refer to Section 3, Cl 3.4.

1.3.4 Buckling acceptance criteria

1.3.4.1 A structural member is considered to have an acceptable buckling strength if it satisfies the following criterion

$$\eta_{act} \leq \eta_{all}$$

where,

η_{act} = Buckling utilisation factor based on the applied stress as per Cl.1.3.2.

η_{all} = Allowable buckling utilisation factor as per Cl.1.3.3.

Table 1.3.3: Allowable buckling utilisation factor (η_{all})

Structural member	Acceptance criteria	Design Load Scenario	η_{all}
Plates and stiffeners/ stiffened panels	AC-I	S	0.80
	AC-II, AC-III	S + D	1.00
	AC-IV[1)]	S	1.00
Struts, pillars, and crossties	AC-I	S	0.65
	AC-II, AC-III	S + D	0.75
	AC-IV[1)]	S	0.75
Corrugations of corrugated bulkheads under lateral pressure from liquid loads. For corrugation angles between 45[deg] and 55[deg] the reduction in η_{all} as given in Chapter 3, Section 6, Cl.6.4.1.2 applies.	AC-I	S	0.72
	AC-II, AC-III	S + D	0.90
	AC-IV[1)]	S	0.90
1) For members of the collision bulkhead, AC-I is to be used.			

Section 2

Slenderness Requirements

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

b_{f-out} = Maximum distance, in [mm], from the mid-thickness of the web to the flange edge, as shown in Figure 2.3.1.

h_w = Depth of stiffener web, in [mm], as shown in Figure 2.3.1.

l_b = Effective length of the edge of the bracket, in [mm], as defined in Table 2.5.1.

s_{eff} = Effective width of the attached plate of the stiffener, in [mm], taken equal to = 0.8s

t_f = Net flange thickness, in [mm].

t_p = Net thickness of plate, in [mm].

t_w= Net web thickness, in [mm].

2.1 Structural Elements

2.1.1 General

All structural elements are to comply with the applicable slenderness and proportion requirements given in Section 2.2 to Section 2.4. A lower specified minimum yield stress value than the actual value can be used for assessing the following slenderness criteria, provided that the requirements specified in Section 3 and Section 4 are satisfied with the same lower specified minimum yield stress value.

2.2 Plates

2.2.1 Application

Based on requirements in accordance with Section 3 and Section 4, slenderness requirements according to Section 2, Cl 2.2.2 may be specially considered. The plate slenderness requirement does not apply to transversely stiffened bilge plates and gunwales with radius.

2.2.2 Net thickness of plate panels

$$t_p \geq \frac{b}{C}\sqrt{\frac{R_{eH}}{235}}$$

where,

b = Distance, in [mm], between stiffeners at mid-length of plate field, as shown in Chapter 3, Section 7 (Figure 7.2.1).

R_{eH} = Specified minimum yield stress of the plate material, in [N/mm^2].

A lower specified minimum yield stress may be used in this slenderness criterion provided the requirements specified in Section 3 and Section 4 are satisfied for the strake assumed in the same lower specified minimum yield stress value.

C = Slenderness coefficient taken as:

= 100 for the outer shell including the strength deck except in superstructures, for vessels with length L≥ 90[m].

= 125 for the outer shell including the strength deck except in superstructures, for vessels with length L< 90[m].

= 175 for internal structural members, except tank boundaries, in vessels with more than three continuous decks and for members in deck house/superstructure

= 125 for other structures not mentioned above.

2.3 Stiffeners

2.3.1 Proportions of stiffeners (except for U-type stiffeners)

2.3.1.1 Net thickness of stiffeners

The net thickness of stiffeners is to satisfy the following criteria:

(i) Stiffener web plate

$$t_w \geq \frac{h_w}{C_w}\sqrt{\frac{R_{eH}}{235}}$$

(ii) Flange

$$t_f \geq \frac{b_{f-out}}{C_f}\sqrt{\frac{R_{eH}}{235}}$$

where,

C_w, C_f = Slenderness coefficients given in Table 2.3.1.

If requirement [2.3.1.1(ii)] is not fulfilled, the effective free flange out-stand, in [mm], used in strength assessment including the calculation of actual net section modulus is not be taken greater than:

$$b_{f-out-max} = C_f . t_f \sqrt{\frac{235}{R_{eH}}}$$

For built-up profile where the relevant yielding strength defined in Chapter 7 and Chapter 8 for the web of built-up profile without the edge stiffener is acceptable, as an alternative the web can be assessed according to the web requirements of Angle and L2 in Table 2.3.1 and the edge stiffener can be assessed as a flat bar stiffener according to Cl.2.3.1.1. The requirement to flange is to be taken as per Cl.2.3.1.2.

Table 2.3.1: Slenderness coefficients

Type of profile	C_w	C_f
Angle bars and L2	75	12[1)]
T-bars	75	12[1)]
Bulb bars	45	-
Flat bars	22	-
1) C_f = 22, non-continuous straight flanges with end		

2.3.1.2 Net dimensions of angle and T-bars

The total flange breadth, in [mm], for angle and T-bars needs to satisfy the following criterion:

$$b_f \geq 0.20\, h_w$$

2.3.1.3 Bending stiffness of stiffeners

The net moment of inertia, in [cm^4], of the stiffener with the effective width of the attached plate, s_{eff}, about the neutral axis parallel to the attached plating, is not to be less than the minimum value given by:

$$I_{st} \geq Cl^2 A_{eff} \frac{R_{eH}}{235}$$

where,

A_{eff} = Net sectional area of stiffener including the effective attached plate, s_{eff}, in [cm^2].

R_{eH} = Specified minimum yield stress of the material of the attached plate, in [N/mm^2].

C = Slenderness coefficient taken as:

= 1.43 for longitudinal stiffeners including sniped stiffeners.

= 0.72 for other stiffeners.

2.4 Primary Supporting Members

2.4.1 Proportions and stiffness

2.4.1.1 Proportions of web plate and flange

The net thicknesses of the web plates and flanges of primary supporting members are to satisfy the following criteria:

(i) Web Plate

$$t_w \geq \frac{s_w}{C_w}\sqrt{\frac{R_{eH}}{235}}$$

(ii) Flange

$$t_f \geq \frac{b_{f-out}}{C_f}\sqrt{\frac{R_{eH}}{235}}$$

where,

s_w = Plate breadth, in [mm], taken as the spacing of the web stiffeners. (See web-plate idealisation also)

C_w = Slenderness coefficient for the web plate taken as:

= 125 for double-skin construction

= 100 elsewhere

C_f = Slenderness coefficient for the flange taken as 12

If requirement [Cl.2.4.1.1 (ii)] is not fulfilled, the effective free flange out-stand, in [mm], used in strength assessment including the calculation of actual net section modulus, is not be taken greater than:

$$b_{f-out-max} = C_f . t_f \sqrt{\frac{235}{R_{eH}}}$$

2.4.1.2 Deck transverse primary supporting members

The net moment of inertia for deck transverse primary supporting members, in [cm^4], supporting deck longitudinals subject to axial compressive hull girder stress, within its central half of the bending span, is to comply with the following criterion:

$$I_{PSM-n50} \geq 300 \frac{\ell_{bdg}^4}{s\, S^3} I_{st}$$

where,

$I_{PSM-n50}$= Net moment of inertia, in [cm^4], of deck transverse primary supporting member, with an effective width of attached plate equal to 0.8S

l_{bdg} = Unsupported bending span of deck transverse primary supporting member in [m], as defined in Chapter 3, Section 7.

ℓ = Length between zero-points of bending moment curve of deck transverse primary supporting member with uniformly distributed load, in m:

$\ell = l_{bdg}$

for deck transverse primary supporting member arranged above the deck without strong rotational support at ends

$\ell = 0.7\ l_{bdg}$

for deck transverse primary supporting member arranged under the deck and connected to the side tank,

S = Spacing of deck transverse primary supporting members, in [m], as defined in Chapter 3, Section 7.

I_{st} = Moment of inertia in [cm^4] of deck stiffeners within the central half of the bending span as given in Cl.2.3.1.3.

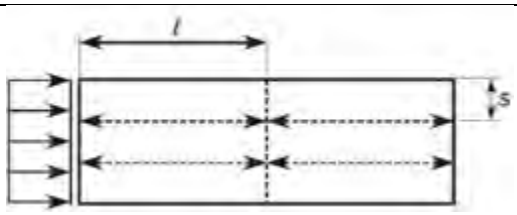
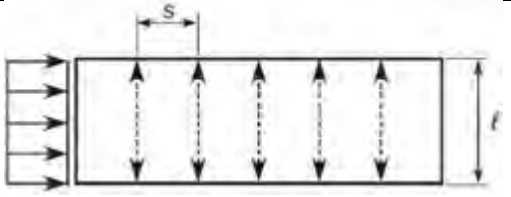
2.4.2 Web stiffeners of primary supporting members

2.4.2.1 Proportion of web stiffeners

The net thickness of the web and flange of web stiffeners fitted on primary supporting members is to satisfy the requirements specified in Cl.2.3.1.1 and Cl.2.3.1.2.

2.4.2.2 Bending stiffness of web stiffeners

The net moment of inertia, in [cm^4], of web stiffener, I_{st}, fitted on primary supporting members, with the effective attached plate, s_{eff}, should not be less than the minimum moment of inertia as defined in Table 2.4.2.

Table 2.4.2: Stiffness criteria for web stiffeners		
Stiffener arrangement		Minimum moment of inertia of web stiffeners, in [cm^4]
A		$I_{st} \geq C\, l^2 A_{eff} \frac{R_{eH}}{235}$
B		$I_{st} \geq 1.14\, l\, s^2 t_w \left(2.5\, \frac{1000\, l}{s} - 2\, \frac{s}{1000 l}\right) \frac{R_{eH}}{235} 10^{-5}$

where:
C = Slenderness coefficient to be taken as:
C = 1.43 for longitudinal stiffeners including sniped stiffeners.
C = 0.72 for other stiffeners.

l = Length of web stiffener, in [m].
For web stiffeners welded to local supporting members, the length is to be measured between the flanges of the local support members.

For sniped web stiffeners, the length is to be measured between the lateral supports, e.g., the total distance between the flanges of the primary supporting member as shown for stiffener arrangement B.

A_{eff} = Net section area of web stiffener including effective attached plate, S_{eff}, in [cm^2].

t_w = Net web thickness of the primary supporting member, in [mm].

R_{eH} = Specified minimum yield stress of the material of the web plate of the primary supporting member, in [N/mm^2].

2.5 Brackets

2.5.1 Tripping Brackets

2.5.1.1 Unsupported flange length

Dimensions of tripping brackets are to be in accordance with Chapter 3, Section 6, Cl. 6.3.3. The unsupported length of the flange of the primary supporting member, in [m], i.e., the distance between tripping brackets, is not to be greater than:

$$S_b = b_f C \sqrt{\frac{A_{f-n50}}{\left(A_{f-n50} + \frac{A_{w-n50}}{3}\right)} \left(\frac{235}{\sigma_y}\right)}$$

; but need not be less than S_{b-min}

where,

b_f = Flange breadth of primary supporting members, in [mm].

C = Slenderness coefficient taken as:

C = 0.022 for symmetrical flanges.

C = 0.033 for asymmetrical flanges.

A_{f-n50}= Net cross-sectional area of the flange, in [cm^2]

A_{w-n50}= Net cross-sectional area of the web plate, in [cm^2]

R_{eH} = Specified minimum yield stress of the PSM material, in [N/mm^2]

S_{b-min} = Minimum unsupported flange length is taken as:

S_{b-min} = 3.0 m for tank/hold boundaries, or hull envelope including external decks.

S_{b-min}= 4.0 m for other areas.

2.5.1.2 Edge Stiffening

Tripping brackets on primary supporting members are to be stiffened by a flange or edge stiffener if the effective length of the edge, l_b, as defined in Table 2.5.1, in [mm], is greater than:

$$l_b = 75t_b$$

where,

t_b = Bracket net web thickness, in [mm].

2.5.2 End Bracket:

2.5.2.1 Proportions

The net web thickness of end brackets, in [mm], subject to compressive stresses should not be less than:

$$t_b = \frac{d_b}{C}\sqrt{\frac{R_{eH}}{235}}$$

where,

d_b = Depth of brackets, in [mm], as defined in Table 2.5.1.

C = Slenderness coefficient as defined in Table 2.5.1.

R_{eH} = Specified minimum yield stress of the end bracket material, in [N/mm^2]. R_{eH} is not to be taken less than the value of the non-continuous member it is supporting but need not be taken greater than that value.

2.5.3 Edge reinforcement

2.5.3.1 Edge reinforcement of bracket edges

The depth of the stiffener web, in [mm], of edge stiffeners in way of bracket edges should not be less than:

$h_w = Cl_b/1000\sqrt{\frac{R_{eH}}{235}}$ or 50[mm] (whichever is greater)

where,

C = Slenderness coefficient taken as:

C = 75 for end brackets.

C = 50 for tripping brackets.

R_{eH} = Specified minimum yield stress of the stiffener material, in [N/mm^2], not to be taken less than the value of the bracket.

2.5.3.2 Proportions of edge stiffeners

The net thickness of the web plate and flange of the edge stiffener should satisfy the requirements specified in Cl.2.3.1.1 and Cl.2.3.1.2.

2.6 Pillars

2.6.1 Proportions of I-section pillars

For I-sections, the thickness of the web plate and the thickness of the flange are to comply with the requirements specified in Cl. 2.3.1.1 and Cl. 2.3.1.2.

2.6.2 Proportions of box section pillars

The thickness of thin-walled box sections is to comply with the requirements specified in Cl. 2.3.1.1 (i).

2.6.3 Proportions of circular section pillars

The net thickness, t , of circular section pillars, in [mm], is to comply with the following criterion:

$$t \geq \frac{r}{50}$$

where,

r = Mid-thickness radius of the circular section, in [mm].

2.7 Edge Reinforcement in way of openings

2.7.1 Depth of edge stiffener

When fitted as shown in Figure 2.7.1, the depth of the web, h_w, in [mm], of edge stiffeners in way of openings is not to be less than:

$h_w = Cl\sqrt{\frac{R_{eH}}{235}}$ or 50 [mm] (whichever is greater).

where,

C = Slenderness coefficient taken as 50

R_{eH} = Specified minimum yield stress of the edge stiffener material, in [N/mm^2].

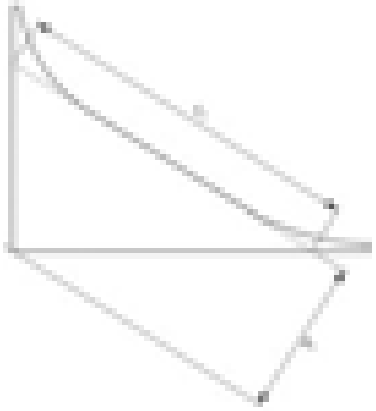
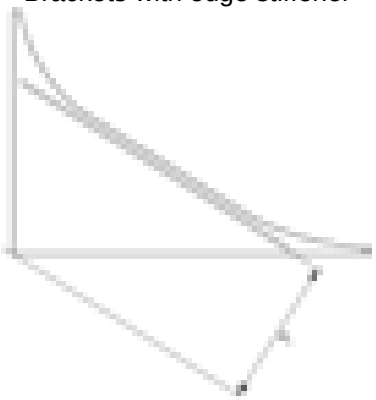
l = Length of edge stiffener in way of opening, in [m], as defined in Figure 2.7.1

2.7.2 Proportions of edge stiffeners

The net thickness of the web plate and flange of the edge stiffener needs to satisfy the requirements specified in Cl. 2.3.1.1 and Cl. 2.3.1.2

The net thickness of the web plate and flange of the edge stiffener needs to satisfy the requirements specified in Cl. 2.3.1.1 and Cl. 2.

Table 2.5.1: Buckling coefficient, C, for proportions of brackets

Mode	C
Brackets without edge stiffener 	$C = 20\left(\frac{d_b}{l_b}\right) + 16$ Where: $0.25 \leq \frac{d_b}{l_b} \leq 1.00$
Brackets with edge stiffener 	$C = 70$

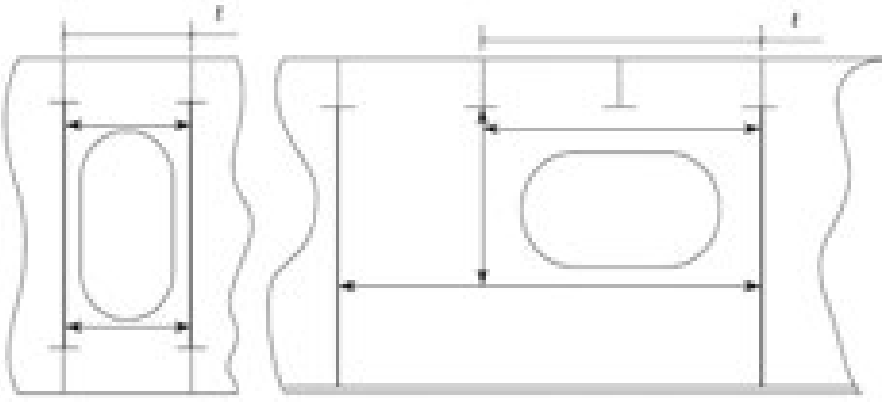


Figure 2.7.1: Typical edge reinforcements

Section 3

Buckling Requirements for Prescriptive Analysis

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

η_{all} = Allowable buckling utilisation factor, as defined in Section 1, Cl. 1.3.3.

EPP = Elementary Plate Panel as defined in Chapter 3, Section 7, Cl.7.5.1.

LCP = Load calculation point as defined in Chapter 3, Section 7, Cl.7.5.2.

3.1 General

3.1.1 Scope

3.1.1.1 This section applies to plate panels including curved plate panels and stiffeners subject to hull girder compression and shear stresses. Additionally, the following structural members are also to be checked:

- Corrugations of longitudinal corrugated bulkhead
- Strut
- Pillar
- Cross tie

3.1.1.2 The hull girder buckling strength requirements apply along the full length of the ship.

3.1.1.3 Design load sets

The buckling checks are to be performed for all design load sets defined in Chapter 7, Section 2, Cl. 2.4. For each design load set, for all dynamic load cases, the lateral pressure is to be determined according to Chapter 5 at the load calculation point (LCP) defined in Chapter 3, Section 7 and is to be applied together with the hull girder stress combinations given in Section 3, Cl. 3.2.2.

3.1.2 Equivalent Plate Panel

3.1.2.1 In longitudinal stiffening arrangement, when the plate thickness varies over the width *b*, of a plate panel, the buckling check is to be performed for an equivalent plate panel width, combined with the smaller plate thickness, t_1. The width of this equivalent plate panel, b_{eq}, in [mm], is defined by the following formula:

$$b_{eq} = l_1 + l_2 \left(\frac{t_1}{t_2}\right)^{1.5}$$

where,

l_1 = Width of the part of the plate panel with the smaller net plate thickness, t_1, in [mm], as defined in Figure 3.1.1.

l_2 = Width of the part of the plate panel with the greater net plate thickness, t_2, in [mm], as defined in Figure 3.1.1.

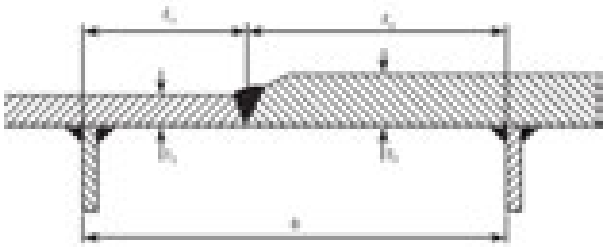


Figure 3.1.1: Plate thickness change over the width

3.1.2.2 In transverse stiffening arrangement, when an EPP is made with different thicknesses, the buckling check of the plate and stiffeners are to be made for each thickness considered constant on the EPP, the stresses, and pressures being estimated for the EPP at the LCP.

3.1.2.3 Materials

When the plate panel is made of different materials, the minimum yield strength is to be used for the buckling assessment.

3.2 Hull girder stress

3.2.1 General

3.2.1.1 Hull girder stress applicable for load components 'S'

The hull girder bending stresses, σ_{hg} in [N/mm^2] for load component S, is taken as:

$$\sigma_{hg} = \sigma_{hg-sw}$$

where,

σ_{hg-sw} = Hull girder bending stress, in [N/mm^2], due to vertical still water bending moment as defined in Chapter 6, Section 3 Cl. 3.2 for ships without large deck openings and in Chapter 6, Section 3, Cl. 3.3 for ships with large deck openings.

3.2.1.2 Hull girder stress applicable for load components 'S+D'

The hull girder bending stresses, σ_{hg} in [N/mm^2] for load component, 'S+D' is to be taken as defined in Chapter 6, Section 3, Cl. 3.2 for ships without large deck openings and in Chapter 6, Section 3, Cl.3.3 for ships with large deck openings.

3.2.1.3 The hull girder shear stresses, τ_{hg} in [N/mm^2], are determined according to Chapter 6, Section 3, Cl.3.2 and Cl.3.3.

3.2.2 Stress combinations

Each elementary plate panel and stiffeners are to satisfy the criteria defined in Section 3.3 with the following stress combinations:

3.2.2.1 Longitudinal stiffening arrangement:

(1) Stress combination 1 with:

$\sigma_x = \sigma_{hg}$
$\sigma_y = 0$
$\tau = 0.7\,\tau_{hg}$

(2) Stress combination 2 with:

$\sigma_x = 0.7\,\sigma_{hg}$
$\sigma_y = 0$
$\tau = \tau_{hg}$

3.2.2.2 Transverse stiffening arrangement

(1) Stress combination 1 with:

$\sigma_x = 0$
$\sigma_y = \sigma_{hg}$
$\tau = 0.7\,\tau_{hg}$

(2) Stress combination 2 with:

$\sigma_x = 0$
$\sigma_y = 0.7\,\sigma_{hg}$
$\tau = \tau_{hg}$

where,

σ_{hg} = Hull girder bending stress in the elementary plate panel or stiffener, as defined in Cl. 3.2.1.1 and Cl. 3.2.1.2, in [N/mm^2].

τ_{hg} = Hull girder shear stress, in [N/mm^2], in the elementary plate panel or stiffener attached plate as defined in Cl.3.2.1.3.

3.3 Buckling criteria

3.3.1 Overall stiffened panel

The buckling strength of overall stiffened panels is to satisfy the following criterion:

$$\eta_{overall} \leq \eta_{all}$$

where,

$\eta_{overall}$ = Maximum utilization factor as defined in Section 5, Cl.5.2.1.

3.3.2 Plates

The buckling strength of elementary plate panels is to satisfy the following criterion:

$$\eta_{plate} \leq \eta_{all}$$

where,

η_{plate} = Maximum plate utilization factor calculated according to SP-A, as defined in Section 5, Cl. 5.2.2.

For the determination of η_{plate} for vertically stiffened plating, Cases 12 and 16 of Section 5, Table 5.2.2.3 corresponding to the shorter edge of the plate panel clamped, may be considered together with a mean σ_y stress and ψ_y = 1, provided that the shorter edges are supported by longitudinal plating/decks and are located within $0.7\,z_1$ from the horizontal neutral axis.

where,

$'z_1'$ is the distance from the horizontal neutral axis to the equivalent deck line or baseline respectively as defined in Chapter 6, Section 2, Cl.2.1.8.

3.3.3 Stiffeners

The buckling strength of stiffeners is to satisfy the following criterion:

$$\eta_{\text{stiffener}} \leq \eta_{\text{all}}$$

where,

$\eta_{\text{stiffener}}$ = Maximum stiffener utilisation factor, as defined in Section 5, Cl.5.2.3.

Note 1: This capacity check can only be fulfilled when the overall stiffened panel capacity, as defined in Cl.3.1.1, is satisfied.

3.3.4 Vertically corrugated longitudinal bulkheads

The shear buckling strength of vertically corrugated longitudinal bulkheads is to satisfy the following criterion:

$$\eta_{\text{shear}} \leq \eta_{\text{all}}$$

where,

η_{shear} = Maximum shear utilisation factor corrugated bulkhead.

$$\eta_{\text{shear}} = \frac{\tau_{\text{bhd}}}{\tau_c}$$

τ_{bhd} = Hull girder shear stress, in [N/mm^2], in the longitudinal bulkheads taken as defined in Cl.3.2.1.3.

τ_c = Shear critical stress, in [N/mm^2], as defined in Section 5, Cl.5.2.2.

3.3.5 Horizontally corrugated longitudinal bulkheads

Each corrugation, within the extension of half flange, web, and half flange, is to satisfy the following criterion:

$$\eta \leq \eta_{\text{all}}$$

where,

η = Overall column utilisation factor, as defined in Section 5, Cl.5.3.1.

End constraints factor corresponding to pinned ends is to be applied except for fixed end support to be used in way of stool with the width exceeding two times the depth of the corrugation.

3.3.6 Struts, pillars, and cross ties

The compressive buckling strength of struts, pillars, and cross ties is to satisfy the following criterion:

$$\eta \leq \eta_{\text{all}}$$

where,

η = Maximum buckling utilisation factor of struts, pillars, or cross ties, defined in Section 5, Cl.5.4.1.

3.4 Application to vessels with L<90[m]

3.4.1 For ships with L< 90[m] the buckling requirement is to be obtained from Sections 3.4 to 3.8, in lieu of the requirements described in Sections 3.1 to 3.3. These requirements apply to plate panels and longitudinals subject to hull girder bending and shear stresses.

3.4.2 Buckling strength of members contributing to the longitudinal strength and subjected to compressive and shear stresses are to be checked, especially in regions where changes in the framing system or significant changes in the hull cross-section occur.

3.4.3 The hull girder buckling strength requirements apply along the full length of the ship.

3.5 Elastic buckling stresses

3.5.1 Elastic Buckling of Plates

3.5.1.1 Compression:

The ideal elastic buckling stress 'σ_E' for plating in [N/mm^2] may be taken as:

$$\sigma_E = 0.9\,K\,E\left[\frac{(t - t_c)}{s}\right]^2$$

where,

For plating with longitudinal stiffeners in the direction parallel to the compressive stress:

$$K = \frac{8.4}{\psi + 1.1}$$

For $0 \leq \psi \leq 1$

For plating with transverse stiffeners in the direction perpendicular to compressive stress:

$$K = C\left[1 + \left(\frac{s}{1000l}\right)^2\right]^2 \frac{2.1}{\psi + 1.1}$$

For $0 \leq \psi \leq 1$

ψ = Stress ratio of the smallest to the largest compressive stress assuming a linear variation (Refer Figure 3.5.1).

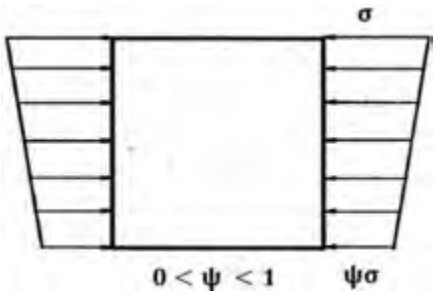


Figure 3.5.1: Schematic for stress ratio

E = Modulus of Elasticity, 2.06×10^5 [N/mm^2], for steel.

s = Shorter side of the plate panel in [mm].

l = Longer side of the plate panel in [m].

C = Coefficient for various structural members, Refer Table 3.5.1.

Table 3.5.1: Coefficient 'C' for various structural members

Structural Member	C
Plating stiffened by primary supporting members (floors or deep girders)	1.30
Stiffeners are angles or T-sections	1.21
Stiffeners are bulb flats	1.10
Stiffeners are flat bars	1.05

3.5.1.2 Shear:

The ideal elastic buckling stress for plate panels in shear in [N/mm^2] may be taken as:

$$\tau_E = 0.9\, K_S\, E \left[\frac{(t - t_c)}{s}\right]^2$$

where,

$$K_S = 5.34 + 4\left(\frac{s}{1000l}\right)^2$$

3.5.2 Elastic Buckling of Longitudinals

3.5.2.1 Column Buckling without rotation of the cross-section

Ideal elastic buckling stress $'\sigma_E'$ for the lateral column buckling mode of longitudinals (perpendicular to the plane of plating), in [N/mm^2], is given by:

$$\sigma_E = \frac{0.001\, E\, I_a}{A\, l^2}$$

where:

I_a = Net moment of inertia [cm^4], of the longitudinal including plate flange, about the neutral axis parallel to the plating and calculated with net thickness.

A = Net Cross-sectional area [cm^2], of the longitudinal including plate flange.

l = Span [m], of the longitudinal.

In the calculation of I_a and A, the attached plating area corresponding to the longitudinal spacing and net thickness is to be included.

3.5.2.2 Torsional Buckling Mode

The ideal elastic buckling stress value $'\sigma_E'$ for the torsional buckling mode of longitudinals, in [N/mm^2], may be taken as:

$$\sigma_E = \frac{\pi^2\, E\, I_w}{10^4\, I_p\, l^2}\left(m^2 + \frac{K}{m^2}\right) + 0.385\, E\, \frac{I_t}{I_p}$$

where,

m = Number of deformation half waves depending on K, as per Table 3.5.2.

I_t = Net St. Venant's moment of inertia, in [cm^4], of profile (without plate flange).

$= \frac{h_w\, t_w^3\, 10^{-4}}{3}$ For flat bars (slabs)

$= \frac{1}{3}\left[h_w\, t_w^3 + b_f\, t_f^3\, \left(1 - 0.63\, \frac{t_f}{b_f}\right)\right] 10^{-4}$ For flanged profiles

I_p = Net Polar moment of inertia, in [cm^4], of profile about the connection of stiffener to the plate.

$= \frac{t_w\, h_w^3\, 10^{-4}}{3}$ For flat bars (slabs)

$= \left[\frac{t_w\, h_w^3}{3} + b_f\, h_w^2\, t_f\right] 10^{-4}$ For flanged profiles

I_w= Net Sectorial moment of inertia of profile in [cm^6], about the connection of stiffener to the plate.

$= \frac{t_w{}^3\, h_w^3\, 10^{-6}}{36}$ For flat bars (slabs)

$= \frac{t_f\, b_f^3\, h_w^2\, 10^{-6}}{12}$ For T- profiles

$= \frac{b_f^3\, h_w^2 \left[t_f\left(b_f^2 + 2\, b_f\, h_w + 4\, h_w^2\right) + (3\, t_w\, b_f\, h_w)\right] 10^{-6}}{12\,(b_f + h_w)^2}$

For Angles and bulb sections

h_w = Web height, in [mm].

t_w = Net Web thickness, [in mm].

b_f = Flange width, in [mm].

t_f = Net Flange thickness, in [mm]. For bulb profiles, the mean thickness of the bulb may be used.

$K = \frac{C\, l^4\, 10^6}{\pi^4\, E\, I_w}$

C = Spring stiffness exerted by supporting plate panel.

$$C = \frac{k_p\, E\, t_p^3\, t_w^3}{3\, s\, t_w^3 + 4\, k_p\, h_w\, t_p^3}$$

l = Span of the profile, in [m].

s = Spacing of profiles, in [mm]

$k_p = (1 - \eta_p)$, not to be taken less than zero; for flanged profiles need not be taken less than 0.1.

t_p= Net plate thickness [mm]

η = Utilization factor

$$\eta = \frac{\sigma_c}{\sigma_{Ep}}$$

σ_c = Calculated compressive stress for longitudinals as per Cl.3.7.1.

σ_{Ep}= Elastic buckling stress of supporting plate calculated as per Cl.3.5.1.

Table 3.5.2: Number of deformation half waves depending on K

K	$0 < K < 4$	$4 < K < 36$	$36 < K < 144$	$144 < K < 400$	$(m-1)^2 m^2 \le K \le (m+1)^2 m^2$
m	1	2	3	4	m

3.5.2.3 Web and Flange Buckling

For web plates of longitudinals, the ideal buckling stress $'\sigma_E'$ value for the web buckling mode, in [N/mm^2], may be taken as:

$$\sigma_E = 3.8\, E\, \left(\frac{t_w}{h_w}\right)^2$$

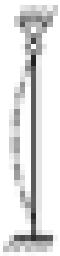
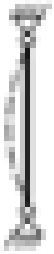
Buckling of flanges on angles and T-sections of longitudinals is taken care of if the as-built flange thickness is not less than 1/15 of the flange width on one side of the web.

3.5.3 Critical column buckling stress of axially loaded members:

The critical column buckling stress of axially loaded members $'\sigma_E'$, in [N/mm^2], may be taken as:

$$\sigma_E = 0.001\, C\, E\, \left[\frac{i}{l_m}\right]^2$$

C = Coefficient to be obtained as per Table 3.5.4.

Table 3.5.4: Coefficient 'C' for various boundary conditions	
Boundary conditions of the member	**C**
Both ends fixed	4
One end is fixed; one end pinned 	2
Both ends pinned 	1

I = Net moment of inertia of the member, [cm^4], about the axis perpendicular to the direction of buckling being considered.

a = Cross-sectional area of the member, [cm^2].

i = Radius of gyration of the member, [cm].

$$i = \sqrt{\frac{I}{a}}$$

l_m = length of the member, in [m].

Where end connections of a member are different for the two principal axes, σ_E is to be found out for both cases using appropriate values of 'C' and 'I'.

3.6 Critical Buckling Stresses

3.6.1 Compression

The critical compressive buckling stress 'σ_{cr}' may be determined as follows:

$\sigma_{cr} = \sigma_E$ when $\sigma_E \leq 0.5\ R_{eH}$

$\sigma_{cr} = R_{eH}\left(1 - \frac{R_{eH}}{4\sigma_E}\right)$ when $\sigma_E > 0.5\ R_{eH}$

where,

σ_E = Ideal elastic buckling stress, in [N/mm^2], as per Cl.3.5.1.1.

R_{eH} = Yield stress of the material, in [N/mm^2].

3.6.2 Shear

The critical shear buckling stress 'τ_{cr}' may be determined as follows:

$\tau_{cr} = \tau_E$ when $\tau_E \leq 0.5\ \tau_y$

$\tau_{cr} = \tau_y\left(1 - \frac{\tau_y}{4\tau_E}\right)$ when $\tau_E > 0.5\ \tau_y$

where,

τ_E = Ideal elastic buckling stress in shear, in [N/mm^2], as perCl.3.5.1.2.

τ_y = Yield stress in shear of the material, in [N/mm^2].

$$= \frac{R_{eH}}{\sqrt{3}}$$

3.7 Working Stress

3.7.1 Longitudinal compressive stresses

3.7.1.1 The compressive stress value 'σ_c', in [N/mm^2], for plate panels and stiffeners taking part in the longitudinal strength is to be taken as:

$$\sigma_c = (M_s + M_w)\frac{z_{n-n50}}{I_{y-n50}}$$

but not to be taken less than 30/k [N/mm^2].

where,

M_s = Still water bending moment [kN-m] as per Chapter 5, Section 3, Cl.3.4.1.

M_w = Wave bending moment [kN-m] as per Chapter 5, Section 3, Cl.3.6.1.

Values of sagging moments (M_s and M_w) are to be considered for members above the neutral axis and the hogging values for members below the neutral axis.

I_{y-n50} = Net moment of inertia of the hull girder (considering deduction of 0.5t_c from the gross offered thickness of structural members) about the neutral axis, [cm^4].

z_{n-n50} = Vertical distance [m], from the neutral axis of the hull girder to the member under consideration.

3.7.1.2 For ships with high speed and large flares in the forebody, σ_c as obtained from above is to be increased by σ_{ca} in the region forward of 0.3L abaft of F.P. The value of σ_{ca} in [N/mm^2] is to be taken as:

$\sigma_{ca} = 0$ For $C_{af} < 0.4$

$$\sigma_{ca} = \frac{50}{k}\left(1 - \frac{x}{0.3L}\right) \text{ For } C_{af} > 0.5$$

where,

x = Distance [m], from F.P., however, need not be taken as smaller than 0.1L.

C_{af} = Factor as defined in Chapter 5, Section 3, Cl.3.6.1.2.

Intermediate values of C_{af}, σ_{ca} are to be obtained by interpolation.

3.7.2 Shear Stresses

3.7.2.1 Hull girder shear stresses in plate panels of side shell and longitudinal bulkheads are obtained according to Chapter 6, Section 3, Cl.3.2 for ships without large deck openings, and Chapter 6, Section 3, Cl.3.3 for ships with large deck openings, respectively.

3.7.3 Axial compressive stresses

The compressive stress value 'σ_c', in [N/mm^2], for axially loaded members is to be taken as:

$$\sigma_c = \frac{10\,F}{a}$$

where,

a = Cross-sectional area of the member, in [cm^2].

F = Nominal axial force in the member, in [kN].
$= A_L\, p\, 10^3$

p = Design pressure [N/mm^2] as per Chapter 7, Table 2.4.3.

A_L= Load area, in [m^2], being supported by the member.

In the calculation of axial force 'F', for deck pillars, contributions, if any, from the decks above are also to be considered.

In the calculation of axial force 'F', for cross ties or panting beams, the larger of the design pressure 'p' at either end of the member is to be considered.

3.8 Scantling Criteria

3.8.1 Compression

The critical buckling stress 'σ_{cr}' of plate panels and stiffeners and other members (struts, pillars, and cross ties) to compressive loads is to be such that:

$$\sigma_{cr} \geq \frac{\sigma_c}{\eta}$$

where,

σ_c = Compressive stress to be considered as per Section 3, Cl.3.5.1.

η = Allowable buckling utilization factor as defined in Table 3.8.1.

Table 3.8.1: Utilization factors for various structural members	
Structural Member	**η**
Deck plating, longitudinally stiffened side shell plating, and single bottom plating	1.00
Bottom plating, inner bottom plating in the double bottom, and transversely stiffened side shell plating	0.90
Local plate panels where an extreme load level is applied. (E.g., impact pressure)	1.00
Local plate panels where a normal load level is applied	0.80
Lateral buckling mode of longitudinals	0.85
Torsional and web buckling mode of longitudinals	0.90
Axially loaded members such as pillars, crossties, panting beams, etc. (in general) **(Note 1)**	$= \frac{0.7}{1 + \left(\frac{l_m}{i}\right)}$ (Need not be taken lesser than 0.3)
Note 1: Where the loads are primarily dynamic in nature, to be reduced by 15%.	

3.8.2 Shear

The critical shear stress of plate panels, stiffeners, and other members is to be such that:

$$\tau_{cr} \geq \frac{\tau_c}{\eta}$$

where,

τ_c = Actual shear stress calculated for the plate panels.

Hull girder shear stresses in plate panels of side shell and longitudinal bulkheads are obtained according to Chapter 6, Section 3, Cl.3.2 for ships without large deck openings and Chapter 6, Section 3, Cl.3.3 for ships with large deck openings, respectively.

η = Allowable buckling utilization factor as defined in Table 3.8.2.

Table 3.8.2: Utilization factors for various structural members	
Structural Member	**η**
Ship's side and longitudinal bulkhead subject to hull girder shear forces	0.90
Local panels in girder webs when nominal shear stresses are calculated (τ_c= Q/A)	0.85
Local panels in girder webs where shear stresses are determined by finite element calculation.	0.90

Section 4

Buckling Requirements for Finite Element Analysis

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

η_{all} = Allowable buckling utilisation factor, as defined in Section 1, Cl.1.3.2.

4.1 General

4.1.1 The requirements given in this section apply to the buckling assessment using finite element analysis for structural elements subjected to compressive stress, shear stress, and lateral pressure.

4.1.2 All structural elements in the FE analysis carried out according to Chapter 8, need to be assessed individually for buckling. The checks are to be performed for the following structural elements:

- stiffened and unstiffened panels, including curved panels
- web plate in way of openings
- corrugated bulkheads
- struts, pillars and cross ties

4.1.4 Verification of buckling strength as per the requirements in this section is also applicable for the structure analysed by beam elements when structural elements are subjected to compressive and shear stresses.

4.2 Stiffened and unstiffened panels

4.2.1 General

4.2.1.1 Boundary condition A and boundary condition B as defined in Section 1, Cl.1.3.1.3 is to be used according to Table 4.2.1.

Table 4.2.1: Structural members

Structural elements	Assessment method	Normal panel definition
Longitudinal structures, Refer example Figure 4.2.1.1		
Longitudinally stiffened panels Shell envelope Deck Inner hull Hopper tank side Longitudinal bulkheads	SP-A	Length: between web frames Width: between primary supporting members
Double bottom longitudinal girders in line with longitudinal bulkhead or connected to the hopper tank side	SP-A	Length: between web frames Width: full web depth
Web of double bottom longitudinal girders not in line with longitudinal bulkhead or not connected to the hopper tank side	SP-B (3)	Length: between web frames Width: full web depth
Web of horizontal girders in double side space connected to hopper tank side	SP-A (3)	Length: between web frames Width: full web depth
Web of horizontal girders in double side space not connected to the hopper tank side	SP-B (3)	Length: between web frames Width: full web depth
Web of single-skin longitudinal girders or stringers	UP-B (3)	Plate between local stiffeners/ face plate/ PSM
Transverse structures, Refer Figure 4.2.1.2		

Web of transverse deck frames including brackets	UP-B (3)	Plate between local stiffeners/face plate/PSM
Vertical web in double-side space	SP-B (3)	Length: full web depth Width: between primary supporting members
Irregularly stiffened panels, e.g., web panels in way of the hopper tank and bilge	UP-B (3)	Plate between local stiffeners/face plate/PSM
Double bottom floors	SP-B (4)	Length: full web depth Width: between primary supporting members
Vertical web frame including brackets	UP-B (3)	Plate between vertical web stiffeners/face plate/ PSM
Cross-tie web plate	UP-B (3)	Plate between vertical web stiffeners/face plate/ PSM
Transverse bulkheads, Refer example Figure 4.2.1.3		
Regularly stiffened bulkhead panels inclusive of the secondary buckling stiffeners perpendicular to the regular stiffener (such as carlings)	SP-A	Length: between primary supporting members Width: between primary supporting members
Irregularly stiffened bulkhead panels, e.g. web panels in way of the hopper tank and bilge	UP-B	Plate between local stiffeners/ face plate
Web plate of bulkhead stringers including brackets	UP-B (3)	Plate between web stiffeners or face plate
Cross deck and transverse corrugated bulkheads, [Refer: Figures 4.2.1.4 and Figure 4.2.1.5]		
Upper/lower stool including stiffeners	SP-A	Length: between internal web diaphragms Width: length of stool side
Stool internal web diaphragm	UP-B (3)	Plate between local stiffeners/ face plate/ PSM
Cross deck	SP-A	Plate between local stiffeners/ PSM

Note 1: SP and UP stand for stiffened and unstiffened panels respectively.

Note 2: A and B stand for Method A and Method B respectively.

Note 3: For PSM web panels with one of the long edges along the face plate or along the attached plating without "inline support", i.e., the edge is free to pull in, boundary condition B (SP-B or UP-B) is to be applied. In other cases, boundary condition A (SP-A or UP-A) is applicable.

Note 4: Typically, the short plate edge is attached to the plate flanges and boundary condition A (SP-A or UP-A) is applicable. However, in case one of the long edges is without "in-line support" and is free to pull in, boundary condition B (SP-B or UP-B) is to be applied.

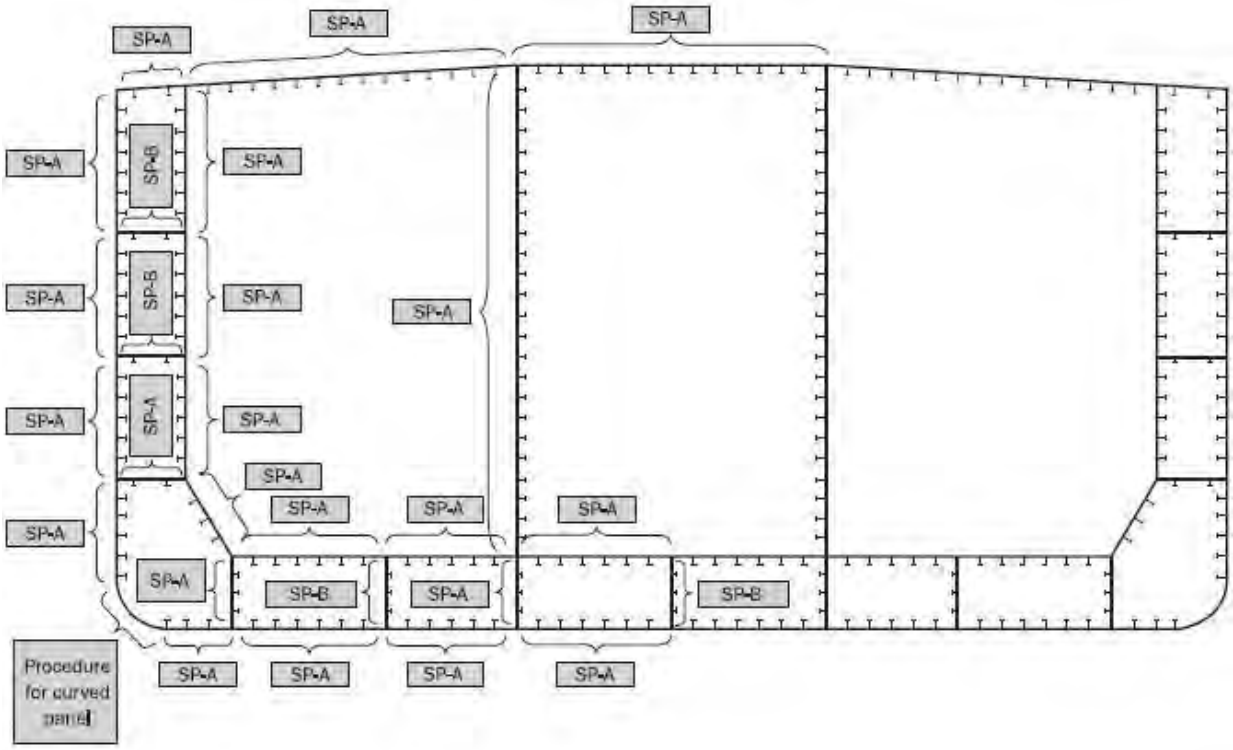


Figure 4.2.1.1: Longitudinal plates for oil tankers

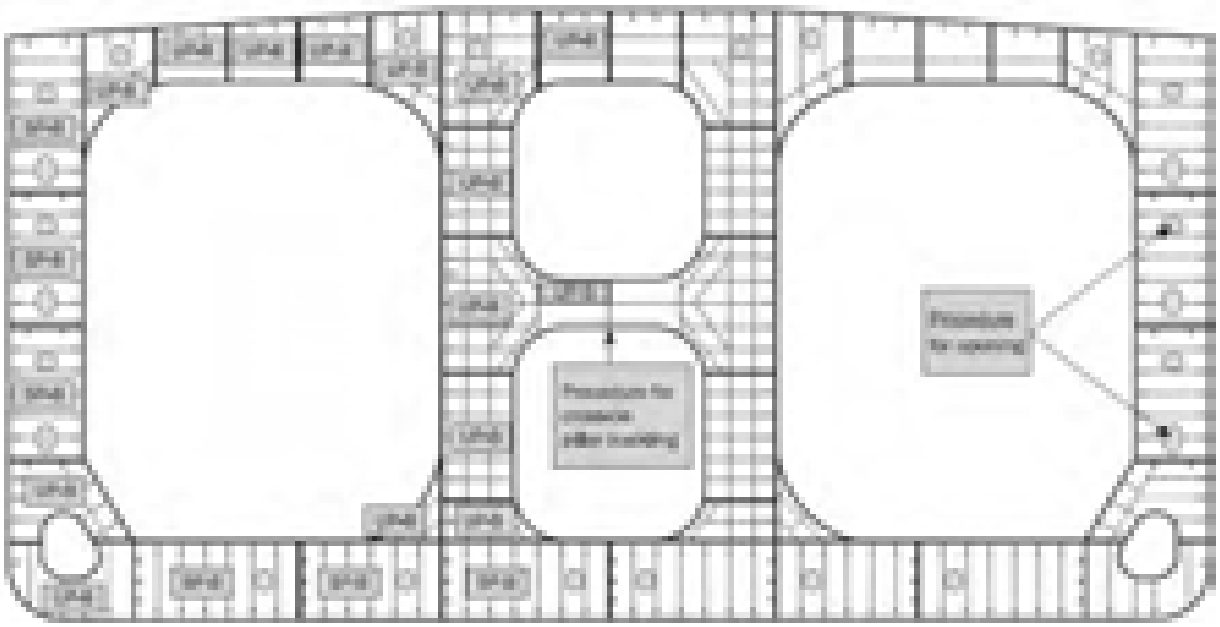


Figure 4.2.1.2: Transverse web frames for oil tankers

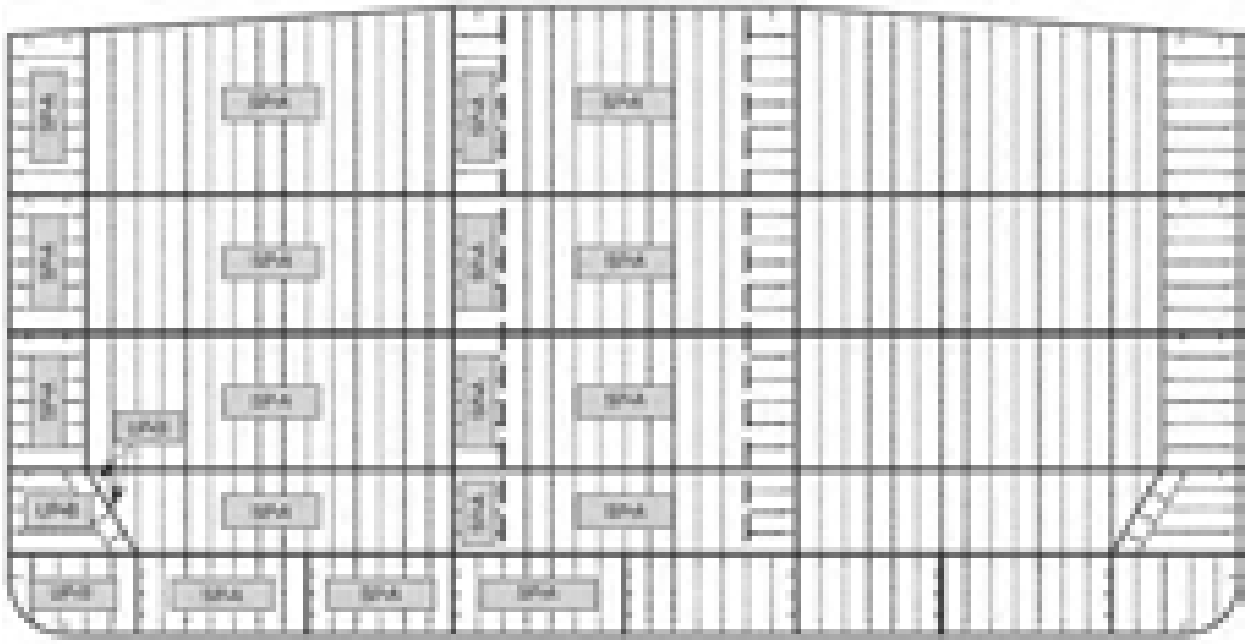


Figure 4.2.1.3: Transverse bulkhead for oil tankers

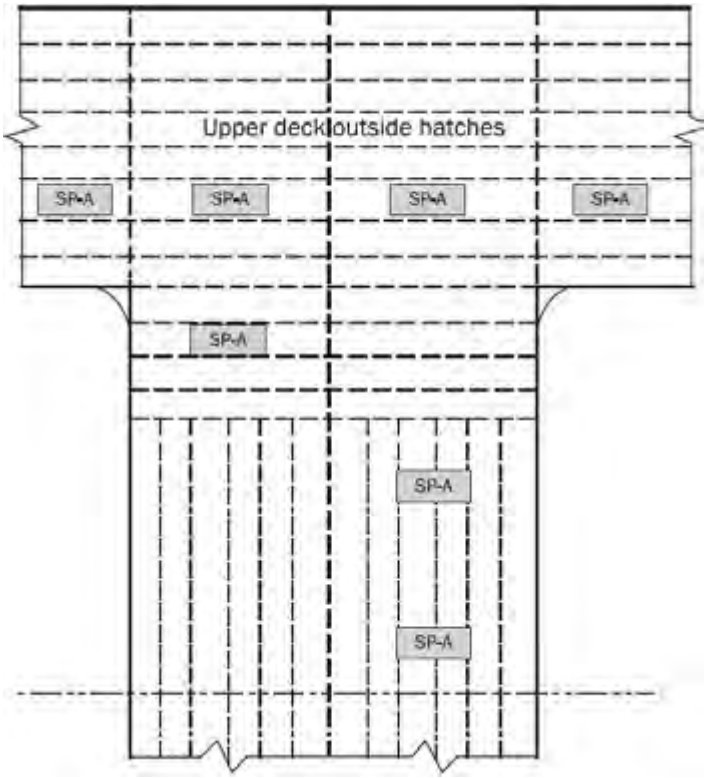


Figure 4.2.1.4: Cross-deck structure

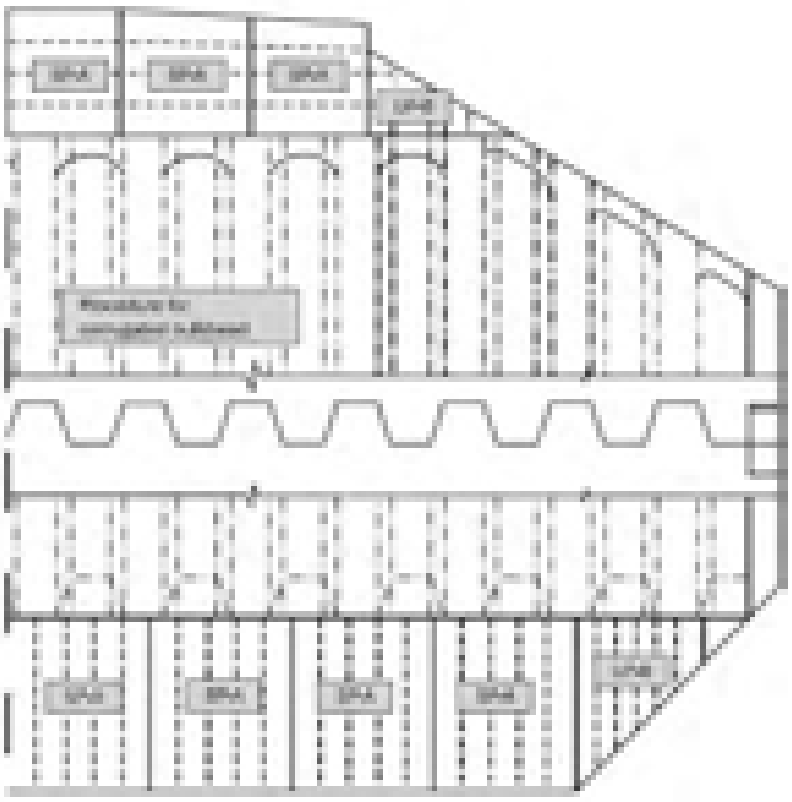


Figure 4.2.1.5: Corrugated bulkhead for bulk carriers

4.2.1.2 Yield stress of the plate panel and stiffener

The panel yield stress $'R_{eH-p}'$ is taken as the minimum value of the specified yield stresses of the elements within the plate panel.

The stiffener yield stress $'R_{eH-s}'$ is taken as the minimum value of the specified yield stresses of the elements within the stiffener.

4.2.1.3 Average thickness of plate panel

Where the plate thickness along a plate panel is not constant, the panel used for the buckling assessment is to be modeled according to Chapter 8 with a weighted average thickness taken as:

$$t_{avr} = \frac{\sum_1^n A_i t_i}{\sum_1^n A_i}$$

where,

A_i = Area of the i-th plate element.

t_i = Net thickness of the i-th plate element.

n = Number of finite elements in the buckling panel.

4.2.2 Stiffened panels

4.2.2.1 To represent the overall buckling behavior, each stiffener with the attached plate is to be modeled as a stiffened panel of the extent defined in Table 4.2.1.

4.2.2.2 If the stiffener properties or stiffener spacing varies within the stiffened panel, the calculations are to be performed separately for all configurations of the panels, i.e., for each stiffener and plate between the stiffeners. Plate thickness, stiffener properties, and stiffener spacing at the considered location need to be assumed for the whole panel.

4.2.2.3 The buckling check of the stiffeners of stiffened panels is only applicable to the stiffeners fitted along the longer side edges of the buckling panel.

4.2.3 Unstiffened panels

4.2.3.1 Irregular plate panel

In way of web frames, stringers, and brackets, the geometry of the panel (i.e., plate bounded by web stiffeners/face plate) may not have a rectangular shape. In this case, for FE analysis, an equivalent rectangular panel is to be defined according to the provisions given in Cl.4.2.3.2 for irregular geometry and Cl.4.2.3.3 for triangular geometry and is to comply with the buckling assessment.

4.2.3.2 Modelling of an unstiffened panel with irregular geometry

Unstiffened panels with irregular geometry are to be idealized into equivalent panels for plate buckling assessment according to the following procedure:

(i) The four corners closest to a right angle, 90 [deg], in the bounding polygon for the plate are identified.

(ii) The distances along the plate-bounding polygon between the corners are calculated, i.e., the sum of all the straight-line segments between the endpoints.

(iii) The pair of opposite edges with the smallest total length is identified, i.e., minimum of $d_1 + d_3$ and $d_2 + d_4$.

(iv) A line joins the middle points of the chosen opposite edges (i.e., a mid-point is defined as the point at half the distance from one end). This line defines the longitudinal direction for the capacity model. The length of the line defines the length of the capacity model 'a' measured from one endpoint.

(v) The length of the shorter side, 'b' in [mm], is to be taken as:

$$b = A/a$$

where,

A = Area of the plate, in [mm^2].

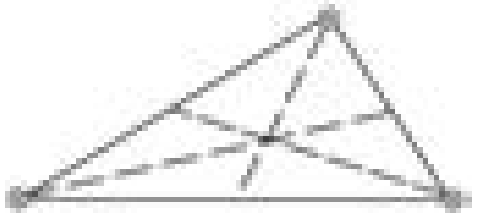
a = Length defined in (iv), in [mm].

(vi) The stresses from the finite element analysis are to be transformed into the local coordinate system of the equivalent rectangular panel. These stresses are to be used for the buckling assessment.

4.2.3.3 Modelling of an unstiffened plate panel with triangular geometry

Unstiffened panels with triangular geometry are to be idealised to equivalent panels for plate buckling assessment according to the following procedure:

(i) Medians are constructed as shown below.



(ii) The longest median is identified. This median the length of which is l_1 in [mm], defines the longitudinal direction for the capacity model.

(iii) The width of the model, l_2, in [mm], is to be taken as:

$$l_2 = A/l_1$$

where,

A = Area of the plate, in [mm^2].

(iv) The lengths of the shorter side, 'b', and of the longer side, 'a', in [mm], of the equivalent rectangular plate panel are to be taken as:

$$b = \frac{l_2}{C_{tr}}$$

$$a = l_1 . C_{tr}$$

where,

$$C_{tr} = 0.4 \frac{l_2}{l_1} + 0.6$$

(v) The stresses from the finite element analysis are to be transformed into the local coordinate system of the equivalent rectangular panel and are to be used for the buckling assessment of the equivalent rectangular panel.

4.2.4 Reference stress

4.2.4.1 The stress distribution is to be taken from the finite element analysis and applied to the buckling model.

4.2.4.2 Stress-based methods for the calculation of reference stresses are defined in Appendix A.

4.2.5 Lateral pressure

4.2.5.1 The lateral pressure applied to the finite element analysis needs to be applied for the buckling assessment.

4.2.5.2 Where the lateral pressure is not constant over a buckling panel defined by a number of finite plate elements, an average lateral pressure, [N/mm^2], is calculated using the following formula:

$$P_{av} = \frac{\sum_{i=1}^{n} A_i \, P_i}{\sum_{i=1}^{n} A_i}$$

where,

A_i = Area of the *i*-th plate element, in [mm^2].

P_i = Lateral pressure of the *i*-th plate element, in [N/mm^2].

n = Number of finite elements in the buckling panel.

4.2.6 Buckling criteria

4.2.6.1 The compressive buckling strength needs to satisfy the following criterion:

$$\eta \leq \eta_{all}$$

where η is the maximum of the relevant utilization factors:

η_{UP-A} = Maximum plate utilisation factor for boundary condition A, Section 5, Cl.5.2.2.

η_{UP-B}= Maximum plate utilisation factor for boundary condition B, Section 5, Cl.5.2.2.

η_{SP-A}= Maximum stiffened panel utilisation factor taken as the maximum of:

- Overall stiffened panel capacity, (See Section 5, Cl.5.2.1).
- Plate capacity calculated as per boundary condition A (See Section 5, Cl. 5.2.2).
- Stiffener buckling strength, (See Section 5, Cl.5.2.3) considering separately the properties (thickness, dimensions), the pressures defined in 4.2.5.2, and the reference stresses of each EPP at both sides of the stiffener.

Note 1: The stiffener buckling capacity check can only be fulfilled when the overall stiffened panel capacity, as defined in Section 5, Cl.5.2.1. is satisfied.

η_{SP-B} = Maximum stiffened panel utilisation factor taken as the maximum of:

- Overall stiffened panel capacity, see Section 5, Cl.5.2.1.
- Plate capacity calculated as per boundary condition B, see Section 5, Cl.5.2.2.
- Stiffener buckling strength, see Section 5, Cl.5.2.3, considering separately the properties (thickness, dimensions), the pressures defined in Cl.4.2.5.2, and reference stresses of each EPP at both sides of the stiffener.

Note 1: The stiffener buckling capacity check can only be fulfilled when the overall stiffened panel capacity, as defined in Section 5, Cl.5.2.1 is satisfied.

4.2.6.2 The web plate of primary supporting members with openings is to satisfy the following criterion:

$$\eta_{\text{opening}} \leq \eta_{\text{all}}$$

where,

η_{opening} = Maximum web plate utilisation factor in way of openings, as defined in Section 5, Cl.5.2.4.

4.3 Corrugated bulkheads

4.3.1 General

Three buckling failure modes are to be assessed on corrugated bulkheads, namely;

- Corrugation overall column buckling.
- Corrugation flange panel buckling.
- Corrugation web panel buckling.

4.3.2 Reference stress

4.3.2.1 Each corrugation flange and web panel are to be assessed.

4.3.2.2 The membrane stresses at the element centroid are to be used.

4.3.2.3 The maximum normal stress parallel to the corrugation, σ_x , is the maximum of the 2 following stresses:

- The normal stress parallel to the corrugation taken at b/2 from the corrugation ends,
- The normal stress parallel to the corrugation within the mid-span of the corrugation.

4.3.2.4 The maximum shear stress is the shear stress which is maximum at the corrugation flange or web at the point b/2 from ends as defined above for the normal stress parallel to the corrugation.

4.3.2.5 In-plane stresses, σ_x and σ_y, and shear stress, τ, are to be taken as element stresses averaged over the width of the considered member (flange or web) at the considered location.

4.3.2.6 When the stress value at b/2 from ends cannot be obtained directly from the finite element, the stress at this location is to be obtained by interpolation on elements in this area. This interpolation needs to be made on elements extending to a distance equal to 3b from the end of the corrugation. The normal stress parallel to the corrugation is to be interpolated as per Chapter 9, Appendix A, i.e., using the 2nd-order polynomial curve. The shear stress at b/2 is obtained by linear interpolation between the elements most close to b/2.

where,

b = width of the considered member of the corrugation, i.e., flange or web.

4.3.2.7 Where more than one plate thickness are used for a flange or web panel, the maximum stress needs to be obtained for each thickness range and is to be checked with the buckling criteria for each thickness.

4.3.3 Overall column buckling

4.3.3.1 The overall buckling failure mode of corrugated bulkheads subjected to axial compression is to be checked for column buckling (e.g., horizontally corrugated bulkheads and vertically corrugated bulkheads subjected to local vertical forces), Refer Table 4.3.3.

Table 4.3.3: Application of overall column buckling for corrugated bulkhead

Bulkhead orientation	Corrugation Orientation	
	Horizontal	Vertical
Longitudinal bulkhead	Required	Required, when subjected to local vertical forces (e.g., crane loads)
Transverse bulkhead	Required	

4.3.3.2 Each corrugation unit within the extension of half flange, web, and half flange, (i.e., single corrugation as shown in grey in Figure 4.3.3), is to satisfy the following criteria:

$$\eta_{overall} \leq \eta_{all}$$

where,

$\eta_{overall}$ = Maximum overall column utilisation factor, as defined in Section 5, Cl. 5.3.1 and Section 5, Cl.5.3.2, considered as a pillar with an unsupported length taken as the length of the corrugation except for vertically corrugated bulkheads where the length, ℓ_{bdg}, is to be applied as per Chapter 3, Section 6, Cl.6.4.

Figure 4.3.3: Single Corrugation

4.3.3.3 End constraint factor, $'f_{end}'$ corresponding to pinned ends is to be applied except for fixed end support to be used in way of stool with the width exceeding two times the depth of the corrugation.

4.3.4 Local buckling

4.3.4.1 The compressive buckling strength of a unit flange and a unit web of corrugation bulkheads is to satisfy the following criterion:

$$\eta_{corr} \leq \eta_{all}$$

where,

η_{corr} = Maximum unit flange or unit web utilisation factor, as defined in Section 5, Cl.5.3.1.

Two stress combinations are to be considered for the application of the above criterion:

- The maximum normal stress parallel to the corrugation, ' σ_x ', combined with the stress perpendicular to the corrugation, ' σ_y ', and with the shear stress, ' τ ', at the location where the maximum normal stress parallel to the corrugation occurs.

- The maximum shear stress, ' τ ', combined with the normal stress parallel to the corrugation, ' σ_x ', and with the stress perpendicular to the corrugation, ' σ_y ', at the location where the maximum shear stress occurs.

The buckling assessment is to be performed for an aspect ratio 'α' equal to 2, and for the thicknesses of the member where the maximum compressive/shear stress occurs (Refer Cl.4.3.2.7).

4.4 Vertically stiffened longitudinal plating in way of the horizontal neutral axis

4.4.1 Buckling criteria

4.4.1.1 Plating

The compressive buckling strength for vertically stiffened longitudinal plating with the shorter edges supported by longitudinal plating/decks and located within $0.7\,z_1$ from the horizontal neutral axis, where 'z_1' is the distance from the horizontal neutral axis to the equivalent deck line or baseline respectively as defined in Chapter 6, Section 2, Cl.2.1.8, may be verified based on the following criterion:

$$\eta_{VSS} \leq \eta_{all}$$

where,

η_{VSS} = Maximum plating utilisation factor calculated according to boundary condition A as defined in Section 5, Cl.5.2.2.1 and considering the following boundary conditions and stress combinations:

(1) The 4 edges are simply supported (Cases 1, 2, and 15 of Section 5, Table 5.2.2.3:

(i) Pure vertical stress:

- The maximum vertical stress of stress elements is used with

$$\alpha = 1 \text{ and } \psi_x = 1.$$

(ii) Maximum vertical stress combined with longitudinal and shear stress:

- The maximum vertical stress in the buckling panel plus the shear and longitudinal stresses at the location where the maximum vertical stress occurs is used with

$$\alpha = 2 \text{ and } \psi_x = \psi_y = 1.$$

- The plate thickness to be considered in the buckling strength check is the one where the maximum vertical stress occurs.

(iii) Maximum shear stress combined with longitudinal and vertical stress:

- The maximum shear stress in the buckling panel plus the longitudinal and vertical stresses at the location where maximum shear stress occurs is used with

$$\alpha = 2 \text{ and } \psi_x = \psi_y = 1.$$

- The plate thickness to be considered in the buckling strength check is the one where the maximum shear stress occurs.

(2) The 2 shorter edges of the plate panel clamped (Cases 11, 12, and 16 of Chapter 9, Section 5, Table 5.2.2.3:

(i) Distributed longitudinal stress associated with vertical and shear stress:

- The actual size of the buckling panel is used to define α.

- The average values for longitudinal, vertical, and shear stresses are to be used.

$$\psi_x = \psi_y = 1.$$

- The plate thickness to be considered in the buckling strength check is the minimum thickness of the buckling panel.

4.4.1.2 Vertical stiffeners

The buckling strength for vertical stiffeners attached to longitudinal plating within $0.7\,z_1$ from the horizontal neutral axis, where $'z_1'$ is the distance from the horizontal neutral axis to the equivalent deck line or baseline respectively as defined in Chapter 6, Section 2, Cl.2.1.8, is to comply with:

$$\eta_{\text{stiffener}} \leq \eta_{\text{all}}$$

where,

$\eta_{\text{stiffener}}$ = Maximum stiffener utilisation factor, as defined in Section 5, Cl.5.2.3.

4.5 Struts, pillars, and cross ties

4.5.1 Buckling criteria

The compressive buckling strength of struts, pillars, and cross ties is to satisfy the following criterion:

$$\eta_{\text{pillar}} \leq \eta_{\text{all}}$$

where,

η_{pillar} = Maximum utilisation factor of struts, pillars, or cross ties, as per Section 5, Cl.5.3.1.

The buckling strength of elementary plate panels of cross ties is to satisfy the following criterion:

$$\eta_{\text{plate}} \leq \eta_{\text{all}}$$

where,

η_{pillar} = Maximum utilisation factor of struts, pillars and cross ties as defined in Section 5, Cl.5.4.

η_{plate} = Maximum plate utilisation factor calculated according to UP-B, as defined in Section 5, Cl.5.2.2.1.

Section 5

Buckling Capacity

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

A_p = Net sectional area of the stiffener attached plating, in [mm^2], taken as:

$$A_p = s\, t_p$$

A_s = Net sectional area of the stiffener without attached plating, in [mm^2].

a = Length of the longer side of the plate panel, in [mm].

b = Length of the shorter side of the plate panel, in [mm].

b_{eff} = Effective width of the attached plating of a stiffener, in [mm], Refer Cl.5.2.3.5.

b_{eff1} = Effective width of the attached plating of a stiffener, in [mm], without the shear lag effect taken as:

- For $\sigma_x > 0$

— For prescriptive assessment:

$$b_{eff1} = \frac{C_{x1} b_1 + C_{x2} b_2}{2}$$

— For FE analysis,

$$b_{eff} = C_x b$$

- For $\sigma_x \leq 0$

$$b_{eff1} = b$$

b_f = Breadth of the stiffener flange, in [mm].

b_1 , b_2 = Width of plate panel on each side of the considered stiffener, in [mm].

C_{x1}, C_{x2} = Reduction factor defined in Table 5.2.2.3 calculated for the EPP1 and EPP2 on each side of the considered stiffener according to Case 1.

d = Length of the side parallel to the axis of the cylinder corresponding to the curved plate panel as shown in Table 5.2.2.4, in [mm].

d_f = Distance for the extension of flange for L2 profiles, in [mm], as defined in Chapter 2, Section 4, Figure 4.6.2.

e_f = Distance from attached plating to the centre of the flange, in [mm], as shown in Figure 1 to be taken as:

$e_f = h_w$ for flat bar profile.

$e_f = h_w - 0.5t_f$ for bulb profile.

$e_f = h_w + 0.5t_f$ for angle, L2, and Tee profiles.

F_{long} = Coefficient defined in Cl.5.2.2.4.

F_{tran} = Coefficient defined in Cl.5.2.2.5.

h_w = Depth of stiffener web, in [mm], as shown in Figure 5.1.1.

l = Span, in [mm], of stiffener equal to the spacing between primary supporting members or span of side frame equal to the distance between the hopper tank and top wing tank ‘h’ as defined in IRS Part 5, Chapter 1, Section 2, Figure 2.5.7.

R = Radius of curved plate panel, in [mm].

S = Partial safety factor and is to be taken as:

= 1.1 for structures that are exposed to local concentrated loads (e.g., container loads on hatch covers, foundations, etc.).

= 1.15 for stiffeners of bulk carriers with L≥ 150[m] and carrying solid bulk cargoes having a density of 1.0[t/m^3] and above, located on:

- hatchway coamings,
- sloping plate of topside and hopper tanks,
- inner bottom,
- inner side,
- top and bottom stools of transverse bulkheads and
- side shell of single side skin construction.

= 1.0 for all other cases.

R_{eH_p} = Specified minimum yield stress of the plate in [N/mm^2].

R_{eH_s} = Specified minimum yield stress of the stiffener in [N/mm^2].

t_p = Net thickness of plate panel, in [mm].

t_w = Net stiffener web thickness, in [mm].

t_f = Net flange thickness, in [mm].

$x-axis$ = Local axis of a rectangular buckling panel parallel to its long edge.

$y-axis$ = Local axis of a rectangular buckling panel perpendicular to its long edge.

α = Aspect ratio of the plate panel, defined in Table 5.2.2.3 to be taken as:

$$\alpha = \frac{a}{b}$$

β = Coefficient taken as:

$$\beta = \frac{1-\psi}{\alpha}$$

ω = Coefficient taken as:

$$\omega = \min(3;\alpha)$$

σ_x = Stress applied on the edge along the x-axis of the buckling panel, in [N/mm^2].

σ_y = Stress applied on the edge along the y-axis of the buckling panel, in [N/mm^2].

σ_1= Maximum stress, in [N/mm^2].

σ_2 = Minimum stress, in [N/mm^2].

τ_c = Buckling strength in shear, in [N/mm^2]. as defined in Section 5, Cl.5.2.2.3:

τ = Applied shear stress, in [N/mm^2].

σ_E = Elastic buckling reference stress, in [N/mm^2], to be taken as:

- For the application of plate limit state according to Section 5, Cl.5.2.2.1:

$$\sigma_E = \frac{\pi^2 E}{12\,(1-\nu^2)}\left(\frac{t_p}{b}\right)^2$$

- For the application of curved plate panels according to Section 5, Cl.5.2.2.6:

$$\sigma_E = \frac{\pi^2 E}{12\,(1-\nu^2)}\left(\frac{t_p}{d}\right)^2$$

Ψ = Edge stress ratio to be taken as:

$$\Psi = \frac{\sigma_1}{\sigma_2}$$

γ = Stress multiplier factor acting on loads. When the factor is such that the loads reach the interaction formulae, $\gamma = \gamma_c$.

γ_c = Stress multiplier factor at failure.

γ_{GEB} = Stress multiplier factor at global elastic buckling capacity.

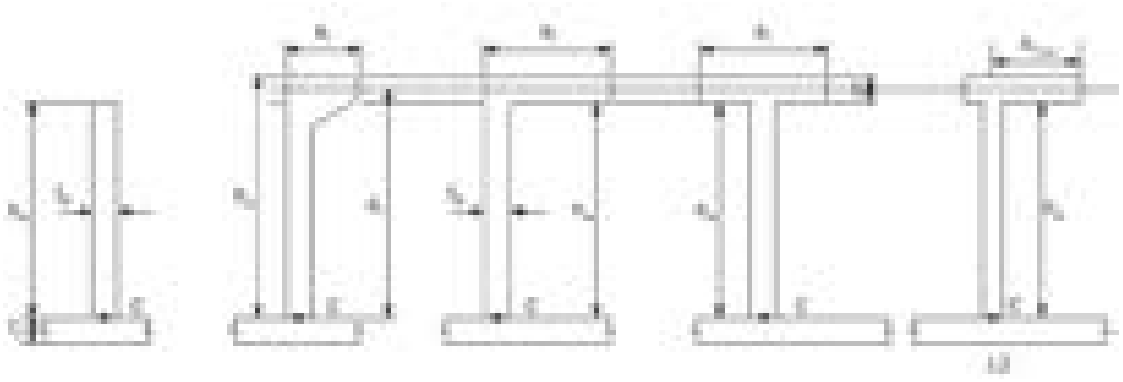


Figure 5.1.1: Stiffener cross-sections

5.1 General

5.1.1 Scope

This section contains the methods for determination of the buckling capacity of plate panels, stiffeners, primary supporting members, struts, pillars, cross ties, and corrugated bulkheads.

5.1.2 For the application of this section, the stresses σ_x, σ_y and τ applied on the structural members are defined in:

- Section 3 for prescriptive requirements.
- Section 4 for FE analysis requirements.

5.1.3 Ultimate buckling capacity

The ultimate buckling capacity is calculated by applying the actual stress combination and then increasing or decreasing the stresses proportionally until the interaction formulae defined in Section 5, Cl.5.2.1, Cl. 5.2.2.1, and Cl.5.2.3.4 are equal to 1.0.

5.1.4 Buckling utilization factor

The buckling utilisation factor of the structural member is equal to the highest utilisation factor obtained for the different buckling modes.

5.1.5 Lateral pressure

The lateral pressure is to be considered a constant in the buckling strength assessment.

5.2 Buckling capacity of plates and stiffeners

5.2.1 Overall stiffened panel capacity

5.2.1.1 The elastic stiffened panel limit state is based on the following interaction formula, which sets a precondition for the buckling check of stiffeners as per Cl.5.2.3.4:

$$\frac{\gamma_c}{\gamma_{GEB}} = 1$$

with the corresponding buckling utilization factor defined as:

$$\eta_{overall} = \frac{1}{\gamma_c}$$

where the stress multiplier factor corresponding to the global elastic buckling capacity, γ_{GEB}, is to be calculated based on the following formulae:

$\gamma_{GEB} = \gamma_{GEB,bi+\tau}$ for $\tau \neq 0$ and ($\sigma_x > 0$ or $\sigma_y > 0$)

$\gamma_{GEB} = \gamma_{GEB,bi}$ for $\tau = 0$ and ($\sigma_x > 0$ or $\sigma_y > 0$)

$\gamma_{GEB} = \gamma_{GEB,\tau}$ for $\tau \neq 0$ and ($\sigma_x \leq 0$ or $\sigma_y \leq 0$)

where $\gamma_{GEB,bi+\tau}$, $\gamma_{GEB,bi}$ and $\gamma_{GEB,\tau}$ are the stress multiplier factors for different load combinations as defined in Section 5, Cl.5.2.1.2, Cl.5.2.1.3, and Cl.5.2.1.4, respectively. For the calculation of $\gamma_{GEB,bi+\tau}$, $\gamma_{GEB,bi}$ and $\gamma_{GEB,\tau}$, neither σ_x nor σ_y is to be taken less than 0.

σ_x, σ_y = Applied normal stresses to the plate panel, in [N/mm^2], to be taken as defined in Cl.5.2.2.7.

τ = Applied shear stress, in [N/mm^2], to be taken as defined in Cl.5.2.2.7.

5.2.1.2 The stress multiplier factor $\gamma_{GEB,bi}$ for the stiffened panel subjected to biaxial loads is taken as:

$$\gamma_{GEB,bi} = \frac{\pi^2}{L_{B1}^2 L_{B2}^2} \frac{[D_{11}L_{B2}^4 + 2(D_{12} + D_{33})n^2 L_{B1}^2 L_{B2}^2 + n^4 D_{22} L_{B1}^4]}{L_{B2}^2 N_x + n^2 L_{B1}^2 N_y}$$

where

N_x = Load per unit length applied on the edge along the x-axis of the stiffened panel, in [N/mm], taken as:

$$N_x = \sigma_{x,av} \frac{(t_p s + t_w h_w + t_f b_f)}{s}$$

For stiffened panels fitted with U-type stiffeners, stiffener spacing 's' is taken as:

$$s = b_1 + b_2$$

where b_1 and b_2 are as defined in Figure 5.2.2.5.

N_y = Load per unit length applied on the edge along the y-axis of the stiffened panel, in [N/mm], taken as:

$$N_y = c\,\sigma_y t_p$$

L_{B1} = Stiffener span, in [mm], equal to the spacing between primary supporting members, i.e., $L_{B1} = l$

For vertically stiffened longitudinal plating with shorter edges supported by longitudinal plating/decks and located within $0.7\,z_1$ from the horizontal neutral axis, where 'z_1' is the distance from the horizontal neutral axis to the

equivalent deck line or baseline respectively as defined in Chapter 6, Section 2, Cl.2.1.8,

$L_{B1} = 0.8\,l$

L_{B2} = Width of the stiffened panel, in [mm], taken as 6 times the stiffener spacing, i.e., $L_{B2} = 6s$.

n = Number of half-waves along the direction perpendicular to the stiffener axis. The factor $\gamma_{GEB,bi}$ is to be minimized with respect to the wave parameter n, i.e., to be taken as the smallest value larger than zero.

c = Factor taking into account the stresses in the attached plating acting perpendicular to the stiffener axis:

$c = 0.5(1 + \psi)$ for $0 \le \psi \le 1$

$c = \frac{1}{2(1-\psi)}$ for $\psi < 0$

ψ = Edge stress ratio for Case 2 according to Table 5.2.2.3

$\sigma_{x,av}$ = Average stress, in [N/mm^2], for both plate and stiffener with Poisson correction, taken as:

$\sigma_{x,av} = \sigma_x - \upsilon\, c\, \sigma_y A_s/(A_p + A_s) \ge 0$ [for $\sigma_x > 0$ and $\sigma_y > 0$]

$\sigma_{x,av} = \sigma_x$ [for $\sigma_x \le 0$ and $\sigma_y \le 0$]

D_{11}, D_{12}, D_{22}, D_{33} = Bending stiffness coefficients, in [N-mm], of the stiffened panel, defined in general as:

$$D_{11} = \frac{E\, I_{eff}\, 10^4}{s}$$

$$D_{12} = \frac{E\, t_p{}^3 \nu}{12\,(1-\nu^2)}$$

$$D_{22} = \frac{E\, t_p{}^3}{12\,(1-\nu^2)}$$

$$D_{33} = \frac{E\, t_p{}^3}{12\,(1+\upsilon)}$$

For stiffened panels fitted with U-type stiffeners, D_{12} and D_{22} are defined as:

$$D_{22} = \frac{E\, t_p{}^3}{12\,(1-\nu^2)} \times \left[1.2 + 4.8 \times Min\left(1.0, \frac{b_1^2}{h_w(b_1+b_2)}\right) \times Min\left(1.0, \left(\frac{t_w}{t_p}\right)^3\right)\right]$$

$$D_{12} = \nu\, D_{22}$$

where

h_w = Breadth of U-type stiffener web as defined in Figure 5.2.2.5.

I_{eff} = Moment of inertia, in [cm^4], of the stiffener including the effective width of attached plating, the same as 'I' defined in Cl.5.2.3.4.

5.2.1.3 The stress multiplier factor $\gamma_{GEB,\tau}$ for the stiffened panel subjected to pure shear load is taken as:

$$\gamma_{GEB,\tau} = \frac{\sqrt[4]{D_{11}^3 D_{22}}}{(L_{B1}/2)^2 N_{xy}}\left[8.125 + 5.64\sqrt{\frac{(D_{12}+D_{33})^2}{D_{11}D_{22}}} - 0.6\frac{(D_{12}+D_{33})^2}{D_{11}D_{22}}\right] \text{ for } D_{11}D_{22} \ge (D_{12}+D_{33})^2$$

$$\gamma_{GEB,\tau} = \frac{\sqrt{2D_{11}(D_{12}+D_{33})}}{(L_{B1}/2)^2 N_{xy}}\left[8.3 + 1.525\frac{D_{11}D_{22}}{(D_{12}+D_{33})^2} - 0.493\frac{D_{11}^2 D_{22}^2}{(D_{12}+D_{33})^4}\right] \text{ for } D_{11}D_{22} < (D_{12}+D_{33})^2$$

where

$N_{xy} = \tau t_p$

5.2.1.4 The stress multiplier factor $\gamma_{GEB,bi+\tau}$ for the stiffened panel subjected to combined loads is taken as:

$$\gamma_{GEB,bi+\tau} = \frac{1}{2}\gamma_{GEB,\tau}^2\left[-\frac{1}{\gamma_{GEB,bi}} + \sqrt{\frac{1}{\gamma_{GEB,bi}^2} + 4\frac{1}{\gamma_{GEB,\tau}^2}}\right]$$

where $\gamma_{GEB,bi}$ and $\gamma_{GEB,\tau}$ are as defined in Cl. 5.2.1.2 and Cl.5.2.1.3, respectively.

5.2.2 Plate Capacity

5.2.2.1 Plate limit state

The plate limit state is based on the following interaction formulae:

$$\left(\frac{\gamma_{c1}\sigma_x S}{\sigma'_{cx}}\right)^{e_o} - B\left(\frac{\gamma_{c1}\sigma_x S}{\sigma'_{cx}}\right)^{e_o/2}\left(\frac{\gamma_{c1}\sigma_y S}{\sigma'_{cy}}\right)^{e_o/2} + \left(\frac{\gamma_{c1}\sigma_y S}{\sigma'_{cy}}\right)^{e_o} + \left(\frac{\gamma_{c1}|\tau|S}{\tau'_c}\right)^{e_o} = 1$$

$$\left(\frac{\gamma_{c2}\sigma_x S}{\sigma'_{cx}}\right)^{2/\beta_p^{0.25}} + \left(\frac{\gamma_{c2}|\tau|S}{\tau'_c}\right)^{2/\beta_p^{0.25}} = 1 \text{ [For } \sigma_x \geq 0]$$

$$\left(\frac{\gamma_{c3}\sigma_y S}{\sigma'_{cy}}\right)^{2/\beta_p^{0.25}} + \left(\frac{\gamma_{c3}|\tau|S}{\tau'_c}\right)^{2/\beta_p^{0.25}} = 1 \text{ [For } \sigma_y \geq 0]$$

$$\frac{\gamma_{c4}|\tau|S}{\tau'_c} = 1$$

with

$$\gamma_c = \min(\gamma_{c1}, \gamma_{c2}, \gamma_{c3}, \gamma_{c4})$$

where,

σ_x, σ_y = Applied normal stress to the plate panel, in [N/mm^2], to be taken as defined in Cl.5.2.2.7.

τ = Applied shear stress to the plate panel, in [N/mm^2].

σ_{cx}' = Ultimate buckling stress, in [N/mm^2], in the direction parallel to the longer edge of the buckling panel as defined in Cl.5.2.2.3.

σ_{cy}' = Ultimate buckling stress, in [N/mm^2], in the direction parallel to the shorter edge of the buckling panel as defined in Cl.5.2.2.3.

$\gamma_{c1}, \gamma_{c2}, \gamma_{c3}, \gamma_{c4}$ = Stress multiplier factors at failure for each of the above different limit states. γ_{c2} and γ_{c3} are only to be considered when $\sigma_x \geq 0$ and $\sigma_y \geq 0$ respectively.

B = Coefficient as given in Table 5.2.2.1.

e_o = Coefficient as given in Table 5.2.2.1

β_p = Plate slenderness parameter to be taken as:

$$\beta_p = \frac{b}{t_p}\sqrt{\frac{R_{eH_P}}{E}}$$

Table 5.2.2.1: Definition of coefficients B and e_o

Applied Stress	B	e_o
$\sigma_x \geq 0$ and $\sigma_y \geq 0$	$0.7 - 0.3\beta_p/\alpha^2$	$2/\beta_p^{0.25}$
$\sigma_x < 0$ or $\sigma_y < 0$	1.0	2.0

5.2.2.2 Reference degree of slenderness

The reference degree of slenderness is to be taken as:

$$\lambda = \sqrt{\frac{R_{eH_p}}{K\,\sigma_E}}$$

where,

K = Buckling factor, as defined in Table 5.2.2.3.and Table 5.2.2.4.

5.2.2.3 Ultimate buckling stresses

The ultimate buckling stresses of plate panels, in [N/mm^2], are to be taken as:

$$\sigma'_{cx} = C_x\, R_{eH_p}$$

$$\sigma'_{cy} = C_y\, R_{eH_p}$$

The ultimate buckling stress of plate panels subject to shear, in [N/mm^2], is to be taken as:

$$\tau'_c = C_\tau \frac{R_{eH_p}}{\sqrt{3}}$$

where,

C_x, C_y, C_τ = Reduction factors, as defined in Table 5.2.2.3.

- For the 1st Equation of Cl.5.2.2.1, when $\sigma_x < 0$ or $\sigma_y < 0$, the reduction factors are to be taken as:

$$C_x = C_y = C_\tau = 1$$

- For the other cases:

— For SP-A and UP-A, C_y is calculated according to Table 5.2.2 (c) by using:

$$c_1 = \left(1 - \frac{1}{\alpha}\right) \geq 0$$

— For SP-B and UP-B, C_y is calculated according to Table 5.2.2.3 by using:

$$c_1 = 1$$

— For vertically stiffened longitudinal plating with shorter edges supported by longitudinal plating/decks and located within $0.7\ z_1$, from the horizontal neutral axis, where $'z_1'$ is the distance from the horizontal neutral axis to the equivalent deck line or baseline respectively as defined in Chapter 6, Section 2, Cl.2.1.8: C_y is calculated according to Table 5.2.2.3 by using:

$$c_1 = \left(1 - \frac{1}{\alpha}\right) \geq 0$$

— For corrugation of corrugated bulkheads, C_y is calculated according to Table 5.2.2.3 by using:

$$c_1 = \left(1 - \frac{1}{\alpha}\right) \geq 0$$

The boundary conditions for plates are to be considered as simply supported, see Cases 1, 2, and 15 of Table 5.2.2.3. If the boundary conditions differ significantly from simple support, a more appropriate boundary condition can be applied according to the different cases of Table 5.2.2.3 subject to the agreement of IRS.

5.2.2.4 Correction factor (F_{long})

The correction factor $'F_{long}'$ depending on the edge stiffener types on the longer side of the buckling panel are defined in Table 5.2.2.2. An average value of F_{long} is to be used for plate panels having different edge stiffeners. For stiffener types other than those mentioned in Table 5.2.2.2, the value of 'c' is to be agreed upon by IRS. In such a case, the value of 'c' higher than those mentioned in Table 5.2.2.2 can be used, provided it is verified by buckling strength check of the panel using non-linear FE analysis and deemed appropriate by IRS.

5.2.2.5 Correction factor (F_{tran})

The correction factor F_{tran} is to be taken as:

• For transversely framed EPP of single side skin bulk carrier, between the hopper and top wing tank with the shorter edges supported by longitudinal plating/decks and located within $0.7z_1$, from the horizontal neutral axis, where z_1, is the distance from the horizontal neutral axis to equivalent deck line or baseline respectively as defined in Chapter 6, Section 2, Cl.2.1.8.

- F_{tran}= 1.25 when the two adjacent frames are supported by one tripping bracket fitted in way of the adjacent plate panels.
- F_{tran} = 1.33 when the two adjacent frames are supported by two tripping brackets each fitted in way of the adjacent plate panels.
- F_{tran} = 1.15 elsewhere.
- For the attached plate of a U-type stiffener fitted on a hatch cover:

$$F_{tran} = \text{Max}(3 - 0.008(F_{tran0} - 6)^2, 1.0) \leq 2.25$$

where

- For EPP b_2

$$F_{tran0} = \text{Min}\left(\frac{b_2}{b_1} + \frac{6b_2^2}{\pi^2 h_w (b_1 + b_2)}\left(\frac{t_w}{t_p}\right)^3, 6.0\right)$$

- For EPP b_1

$$F_{tran0} = \text{Min}\left(\frac{b_1}{b_2} + \frac{6b_1^2}{\pi^2 h_w (b_2 + b_1)}\left(\frac{t_w}{t_p}\right)^3, 6.0\right)$$

with b_1, b_2 and h_w as defined in Figure 5.2.2.5.

Coefficient '*F*', defined in Case 2 of Table 5.2.2.3 is to be replaced by the following formula:

$$F = \left[1 - \left(\frac{K_y}{0.91F_{tran}} - 1\right)/\lambda_p^2\right] c_1 \geq 0$$

• For other cases: F_{tran} = 1

<table>
<tr><th colspan="5">Table 5.2.2.2: Correction factor (F_{long})</th></tr>
<tr><th colspan="3">Structural element types</th><th>F_{long}</th><th>c</th></tr>
<tr><td colspan="3">Unstiffened Panel</td><td>1.0</td><td>N/A</td></tr>
<tr><td rowspan="6">Stiffened Panel</td><td colspan="2">Stiffener not fixed at both ends</td><td>1.0</td><td>N/A</td></tr>
<tr><td rowspan="5">Stiffener fixed at both ends</td><td>Flat bar (1)</td><td rowspan="4">$F_{long} = c + 1$ for $\frac{t_w}{t_p} > 1$
$F_{long} = c\left(\frac{t_w}{t_p}\right)^3 + 1$ for $\frac{t_w}{t_p} \le 1$</td><td>0.10</td></tr>
<tr><td>Bulb profile</td><td>0.30</td></tr>
<tr><td>Angle and L2 profiles</td><td>0.40</td></tr>
<tr><td>T profile</td><td>0.30</td></tr>
<tr><td>Girder of high rigidity (e.g., bottom transverse)</td><td>1.4</td><td>N/A</td></tr>
<tr><td>U-type profile fitted on hatch cover (2)</td><td>• Plate on which the U-type profile is fitted, including EPP $b_1$and EPP b_2:
– For $b_2 < b_1$: $F_{long} = 1$
– For $b_2 \ge b_1$:
$F_{long} = \left(1.55 - 0.55\frac{b_1}{b_2}\right)\left[1 + c\left(\frac{t_w}{t_p}\right)^3\right]$
• Other plates of the U-type profile:
$F_{long} = 1$</td><td>0.20</td></tr>
<tr><td colspan="5">(1) t_w is the net web thickness, in [mm], without the correction defined in Section 5, Cl.5.2.3.2
(2) b_1, b_2 and t_w are defined in Figure 5.2.2.5</td></tr>
</table>

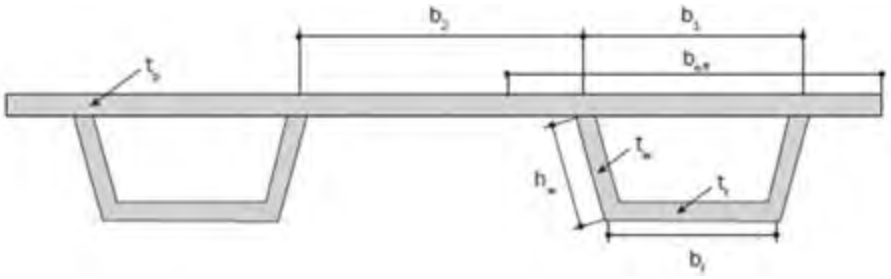


Figure 5.2.2.5: Example of hatch cover fitted with U-type stiffener

5.2.2.6 Curved plate panels

This requirement for the curved plate limit state is applicable when $R/t_p \le 2500$. Otherwise, the requirement for plate limit state given in Section 5, Cl.5.2.2.1 is applicable.

The curved plate limit state is based on the following interaction formula:

$$\left(\frac{\gamma_c\sigma_{ax}S}{C_{ax}R_{eH_P}}\right)^{1.25} - 0.5\left(\frac{\gamma_c\sigma_{ax}S}{C_{ax}R_{eH_P}}\right)\left(\frac{\gamma_c\sigma_{tg}S}{C_{tg}R_{eH_P}}\right) + \left(\frac{\gamma_c\sigma_{tg}S}{C_{tg}R_{eH_P}}\right)^{1.25} + \left(\frac{\gamma_c\tau\sqrt{3}S}{C_\tau R_{eH_P}}\right)^2 = 1$$

where,

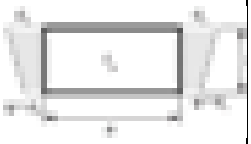
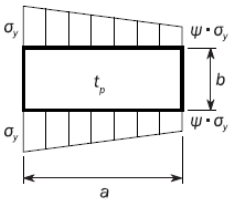
σ_{ax} = Applied axial stress to the cylinder corresponding to the curved plate panel, in [N/mm^2]. In the case of tensile axial stresses, σ_{ax}= 0.

σ_{tg} = Applied tangential stress to the cylinder corresponding to the curved plate panel, in [N/mm^2]. In the case of tensile tangential stresses, σ_{tg}= 0.

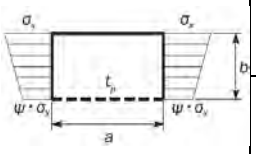
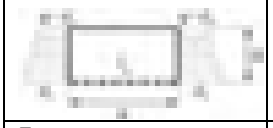
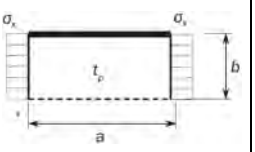
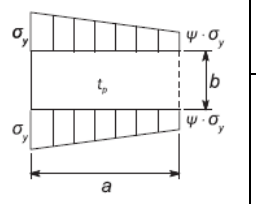
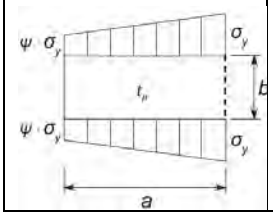
C_{ax}, C_{tg}, C_{τ} = Buckling reduction factor of the curved plate panel, as defined in Table 5.2.2.4.

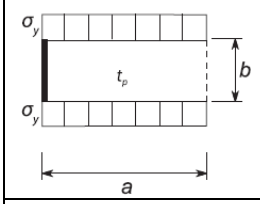
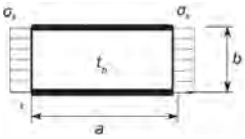
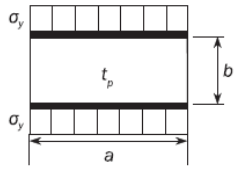
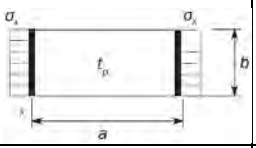
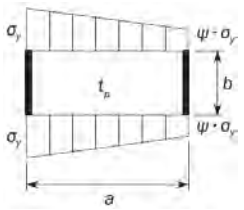
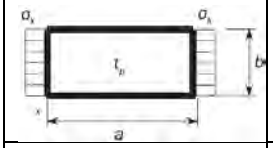
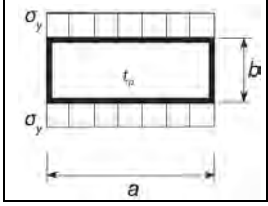
The stress multiplier factor, γ_c, of the curved plate panel need not be taken less than the stress multiplier factor, γ_c, for the expanded plate panel according to Section 5, Cl.5.2.2.1.

Table 5.2.2.3: Buckling factor and reduction factor for plane plate panels

Case	Stress Ratio ψ	Aspect Ratio α	Buckling Factor *K*	Reduction Factor *C*
1. 	$1 \geq \psi \geq 0$		$K_X = F_{long}\dfrac{8.4}{\psi + 1.1}$	When $\sigma_x \leq 0$: $C_X = 1$ When $\sigma_x > 0$: $C_X = 1$ for $\lambda < \lambda_c$ $C_X = c\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda < \lambda_c$ where: $c = (1.25 - 0.12\psi) \leq 1.25$ $\lambda_c = \dfrac{c}{2}\left(1 + \sqrt{1 - \dfrac{0.88}{c}}\right)$
	$0 > \psi > -1$		$K_X = F_{long}[7.63 - \psi(6.26 - 10\psi)]$	
	$\psi \leq -1$		$K_X = F_{long}[5.975(1 - \psi)^2]$	
2. 	$1 \geq \psi \geq 0$		$K_Y = F_{tran}\dfrac{2\left(1 + \frac{1}{\alpha^2}\right)^2}{1 + \psi + \frac{(1-\psi)}{100}\left(\frac{2.4}{\alpha^2} + 6.9 f_1\right)}$	When $\sigma_y \leq 0$: $C_Y = 1$ When $\sigma_y > 0$: $C_Y = c\left(\frac{1}{\lambda} - \frac{R + F^2(H-R)}{\lambda^2}\right)$ where: $c = (1.25 - 0.12\psi) \leq 1.25$ $R = \lambda\left(1 - \dfrac{\lambda}{c}\right)$ for $\lambda < \lambda_c$ $R = 0.22$ for $\lambda \geq \lambda_c$ $\lambda_c = \dfrac{c}{2}\left(1 + \sqrt{1 - \dfrac{0.88}{c}}\right)$ $F = \left[1 - \left(\frac{K}{0.91} - 1\right)/\lambda_p^2\right] c_1 \geq 0$
		$\alpha \leq 6$	$f_1 = (1 - \psi)(\alpha - 1)$	
		$\alpha > 6$	$f_1 = 0.6\left(1 - \dfrac{6\psi}{\alpha}\right)\left(\alpha + \dfrac{14}{\alpha}\right)$ But not greater than: $\left(14.5 - \dfrac{0.35}{\alpha^2}\right)$	
	$0 > \psi \geq (1 - 4\alpha/3)$		$K_Y = \dfrac{200\, F_{tran}(1 + \beta^2)^2}{(1 - f_3)(100 + 2.4\beta^2 + 6.9 f_1 + 23 f_2)}$	
		$\alpha > 6(1 - \psi)$	$f_1 = 0.6\left(\dfrac{1}{\beta} + 14\beta\right)$ But not greater than: $(14.5 - 0.35\beta^2)$ $f_2 = f_3 = 0$	

		$3(1-\psi) \le \alpha \le 6(1-\psi)$	$f_1 = \left(\frac{1}{\beta} - 1\right)$ $f_2 = f_3 = 0$	$\lambda_p^2 = [\lambda^2 - 0.5]$ for $1 \le \lambda_p^2 \le 3$ c_1 as defined in Section 5, Cl.5.2.2.3 $H = \lambda - \frac{2\lambda}{c\left(T + \sqrt{T^2 - 4}\right)} \ge R$ $T = \lambda + \frac{14}{15\lambda} + \frac{1}{3}$
		$1.5(1-\psi) \le \alpha < 3(1-\psi)$	$f_1 = \frac{1}{\beta} - (2 - \omega\beta)^4 - 9(\omega\beta - 1)\left(\frac{2}{3} - \beta\right)$ $f_2 = f_3 = 0$	
		$(1-\psi) \le \alpha < 1.5(1-\psi)$	• For $\alpha > 1.5$: $f_1 = 2\left(\frac{1}{\beta} - 16\left(1 - \frac{\omega}{3}\right)^4\right)\left(\frac{1}{\beta} - 1\right)$ $f_2 = 3\beta - 2$ $f_3 = 0$ • For $\alpha \le 1.5$: $f_1 = 2\left(\frac{1.5}{(1-\psi)} - 1\right)\left(\frac{1}{\beta} - 1\right)$ $f_2 = \frac{\psi\left(1 - 16{f_4}^2\right)}{1-\alpha}$ $f_3 = 0$ $f_4 = \left(1.5 - Min(1.5: \alpha)\right)^2$	
		$0.75(1-\psi) \le \alpha < (1-\psi)$	$f_1 = 0$ $f_2 = 1 + 2.31(\beta - 1) - 48\left(\frac{4}{3} - \beta\right){f_4}^2$ $f_3 = 3f_4(\beta - 1)\left(\frac{f_4}{1.81} - \frac{\alpha - 1}{1.31}\right)$ $f_4 = \left(1.5 - Min(1.5: \alpha)\right)^2$	

	$\psi > \left(1 - \frac{4\alpha}{3}\right)$	$K_Y = 5.972\, F_{tran}\left[\frac{\beta^2}{1 - f_3}\right]$ where: $f_3 = f_5\left(\frac{f_5}{1.81} + \frac{1 + 3\psi}{5.24}\right)$ $f_5 = \frac{9}{16}\left(1 + Max(-1{:}\,\psi)\right)^2$	
3. 	$1 \geq \psi \geq 0$	$K_X = \frac{4\left(0.425 + \frac{1}{\alpha^2}\right)}{3\psi + 1}$	For UP-A: $C_X = 1$ for $\lambda \leq 0.75$ $C_X = \frac{0.75}{\lambda}$ for $\lambda > 0.75$ For UP-B: $C_X = 1$ for $\lambda \leq 0.7$ $C_X = \frac{1}{\lambda^2 + 0.51}$ for $\lambda > 0.7$
	$0 > \psi \geq -1$	$K_X = 4\left(0.425 + \frac{1}{\alpha^2}\right)(1 + \psi) - 5\psi(1 - 3.42\psi)$	
4. 	$1 \geq \psi \geq -1$	$K_X = \left(0.425 + \frac{1}{\alpha^2}\right)\frac{3 - \psi}{2}$	
5. 	–	$\alpha \geq 1.64$: $K_X = 1.28$	
		$\alpha < 1.64$: $K_X = \frac{1}{\alpha^2} + 0.56 + 0.13\alpha^2$	
6. 	$1 \geq \psi \geq 0$	$K_Y = \frac{4\,(0.425 + \alpha^2)}{(3\psi + 1)\alpha^2}$	For UP-A: $C_Y = 1$ for $\lambda \leq 0.75$ $C_Y = \frac{0.75}{\lambda}$ for $\lambda > 0.75$ For UP-B: $C_Y = 1$ for $\lambda \leq 0.7$ $C_Y = \frac{1}{\lambda^2 + 0.51}$ for $\lambda > 0.7$
	$0 > \psi \geq -1$	$K_Y = 4\,(0.425 + \alpha^2)(1 + \psi)\frac{1}{\alpha^2} - 5\psi(1 - 3.42\psi)\frac{1}{\alpha^2}$	
7. 	$1 \geq \psi \geq -1$	$K_Y = (0.425 + \alpha^2)\frac{(3 - \psi)}{2\alpha^2}$	

8. 	–	$K_Y = 1 + \frac{0.56}{\alpha^2} + \frac{0.13}{\alpha^4}$	
9. 	–	$K_X = 6.97$	$C_X = 1$ for $\lambda \leq 0.83$ $C_X = 1.13\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > 0.83$
10. 	–	$K_Y = 4 + \frac{2.07}{\alpha^2} + \frac{0.67}{\alpha^4}$	$C_Y = 1$ for $\lambda \leq 0.83$ $C_Y = 1.13\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > 0.83$
11. 	–	$\alpha \geq 4$: $K_X = 4$ $\alpha < 4$: $K_X = 4 + 2.74\left[\frac{4-\alpha}{3}\right]^4$	$C_X = 1$ for $\lambda \leq 0.83$ $C_X = 1.13\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > 0.83$
12. 	–	$K_Y =$ K_Y determined as per Case 2	For $\alpha < 2$: $C_Y = C_{Y2}$ For $\alpha \geq 2$: $C_Y = \left(1.06 + \frac{1}{10\alpha}\right) C_{Y2}$ where: C_{Y2} : C_Y determined as per Case 2
13. 	–	$\alpha \geq 4$: $K_X = 6.97$ $\alpha < 4$: $K_X = 6.97 + 3.1\left[\frac{4-\alpha}{3}\right]^4$	$C_X = 1$ for $\lambda \leq 0.83$ $C_X = 1.13\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > 0.83$
14. 	–	$K_Y = \frac{6.97}{\alpha^2} + \frac{3.1}{\alpha^2}\left[4 - \frac{1}{\alpha}\right]^4$	$C_Y = 1$ for $\lambda \leq 0.83$ $C_Y = 1.13\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > 0.83$

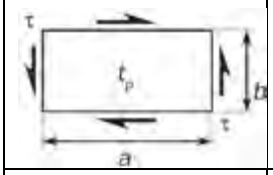
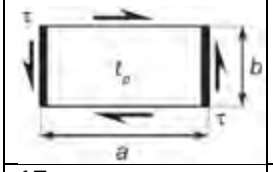
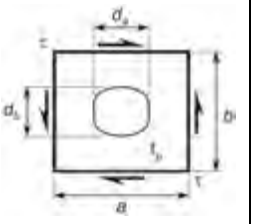
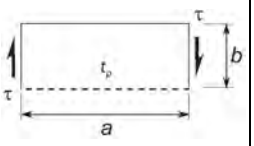
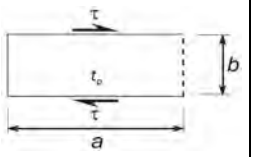
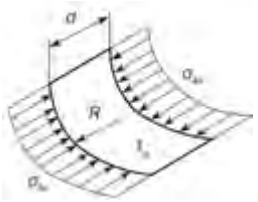
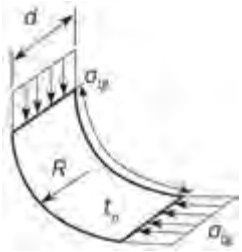
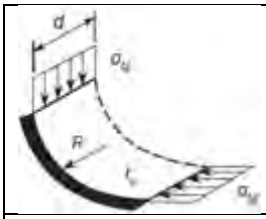
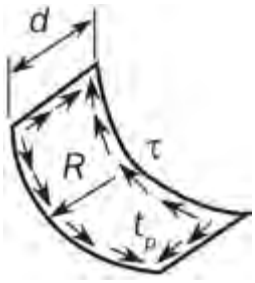
15.  	–	$K_\tau = \sqrt{3}\left[5.34 + \frac{4}{\alpha^2}\right]$	$C_\tau = 1$ for $\lambda \leq 0.84$ $C_\tau = \frac{0.84}{\lambda}$ for $\lambda > 0.84$
16.  	–	$K_\tau = \sqrt{3}\left\{5.34 + Max\left[\frac{4}{\alpha^2}; \frac{7.15}{\alpha^{2.5}}\right]\right\}$	
17.  	–	$K_\tau = K_{\tau_Case15}\, r$ where: K_{τ_Case15} : K_τ according to Case 15 r : Opening reduction factor taken as: $r = \left(1 - \frac{d_a}{a}\right)\left(1 - \frac{d_b}{b}\right)$ With $\frac{d_a}{a} \leq 0.7$ and $\frac{d_b}{b} \leq 0.7$	
18.  	–	$K_\tau = \sqrt{3}\left[0.6 + \frac{4}{\alpha^2}\right]$	$C_\tau = 1$ for $\lambda \leq 0.84$ $C_\tau = \frac{0.84}{\lambda}$ for $\lambda > 0.84$
19.  	–	$K_\tau = 8$	
Edge boundary conditions: – – – – – – – – – Plate edge free Plate edge simply supported Plate edge clamped			
Note 1: Cases listed are general cases. Each stress component (σ_x, σ_y) is to be understood in local coordinates.			

Table 5.2.2.4: Buckling and reduction factor for curved plate panel with $R/t_p \leq 2500$

Case	Aspect Ratio ψ	Buckling Factor K	Reduction Factor C
1. 	$\frac{d}{R} \leq 0.5\sqrt{\frac{R}{t_p}}$	$K = 1 + \frac{2}{3}\frac{d^2}{Rt_p}$	For general application: $C_{ax} = 1$ for $\lambda \leq 0.25$ $C_{ax} = 1.233 - 0.933\lambda$ for $0.25 < \lambda \leq 1$ $C_{ax} = 0.3/\lambda^3$ for $1 < \lambda \leq 1.5$ $C_{ax} = 0.2/\lambda^2$ for $\lambda > 1.5$ For curved single fields, e.g., bilge plating, which is bounded by plane panels as shown in Chapter 7, Section 4, Figure 4.1.3: $C_{ax} = 0.65/\lambda^2 \leq 1.0$
	$\frac{d}{R} > 0.5\sqrt{\frac{R}{t_p}}$	$K = 0.267\frac{d^2}{Rt_p}\left[3 - \frac{d}{R}\sqrt{\frac{t_p}{R}}\right] \geq 0.4\frac{d^2}{Rt_p}$	
2 	$\frac{d}{R} \leq 1.63\sqrt{\frac{R}{t_p}}$	$K = \frac{d}{\sqrt{Rt_p}} + 3\frac{(Rt_p)^{0.175}}{d^{0.35}}$	For general application: $C_{tg} = 1$ for $\lambda \leq 0.4$ $C_{tg} = 1.274 - 0.686\lambda$ for $0.4 < \lambda \leq 1.2$ $C_{tg} = 0.65/\lambda^2$ for $\lambda > 1.2$ For curved single fields, e.g., bilge plating, which is bounded by plane panels as shown in Chapter 7, Section 4, Figure 4.1.3: $C_{tg} = 0.8/\lambda^2 \leq 1.0$
	$\frac{d}{R} > 1.63\sqrt{\frac{R}{t_p}}$	$K = 0.3\frac{d^2}{R^2} + 2.25\left(\frac{R^2}{dt_p}\right)^2$	
3.	$\frac{d}{R} \leq \sqrt{\frac{R}{t_p}}$	$K = \frac{0.6\,d}{\sqrt{Rt_p}} + \frac{\sqrt{Rt_p}}{d} - 0.3\frac{Rt_p}{d^2}$	As in load case 2a.

	$\frac{d}{R} > \sqrt{\frac{R}{t_p}}$	$K = 0.3\frac{d^2}{R^2} + 0.291\left(\frac{R^2}{dt_p}\right)^2$	
4. 	$\frac{d}{R} \leq 8.7\sqrt{\frac{R}{t_p}}$	$K = \sqrt{3}\sqrt{28.3 + \frac{0.67\,d^3}{R^{1.5}\,{t_p}^{1.5}}}$	$C_\tau = 1$ for $\lambda \leq 0.40$ $C_\tau = 1.274 - 0.686\lambda$ for $0.40 < \lambda \leq 1.2$ $C_\tau = 0.65/\lambda^2$ for $\lambda > 1.2$
	$\frac{d}{R} > 8.7\sqrt{\frac{R}{t_p}}$	$K = \sqrt{3}\,\frac{0.28\,d^2}{R\sqrt{Rt_p}}$	
Edge boundary conditions: – – – – – – – – – Plate edge free Plate edge simply supported Plate edge clamped			

5.2.2.7 Applied normal and shear stresses to plate panels

The normal stresses, σ_x and σ_y, in [N/mm^2], to be applied for the overall stiffened panel capacity and plate panel capacity calculation as given in Section 5, Cl.5.2.1.1 and Cl.5.2.2.1 respectively, are to be taken as follows:

- For FE analysis, the reference stresses as defined in Section 4, Cl.2.4.

- For prescriptive assessment of the overall stiffened panel capacity and the plate panel capacity, the axial or transverse compressive stresses calculated according to Section 3, Cl.3.2.2, at load calculation points of the considered stiffener or the considered elementary plate panel, as defined in Chapter 3, Section 7, Cl.7.6. and Cl.7.5, respectively. However, in case of transverse stiffening arrangement, the transverse compressive stress used for the assessment of the overall stiffened panel capacity is to be taken as the compressive stress calculated at load calculation points of the stiffener attached plating, as defined in Chapter 3, Section 7, Cl.7.5.

- For grillage analysis where the stresses are obtained based on beam theory, the stresses taken as:

$$\sigma_x = \frac{\sigma_{xb} + \nu\sigma_{yb}}{1 - \nu^2}$$

$$\sigma_y = \frac{\sigma_{yb} + \nu\sigma_{xb}}{1 - \nu^2}$$

where,

σ_{xb} , σ_{yb} = Stress, in [N/mm^2], from grillage beam analysis respectively along x or y-axis of the plate attached to the PSM web.

The shear stress τ, in [N/mm^2], to be applied for the overall stiffened panel capacity and plate panel capacity calculation as given in Cl.5.2.1.1 and Cl.5.2.2.1 respectively, are to be taken as follows:

- For FE analysis, the reference shear stresses as defined in Section 4, Cl.2.4.

- For prescriptive assessment of the plate panel capacity, the shear stresses calculated according to Section 3, Cl.3.2.2, at load calculation points of the considered elementary plate panel, as defined in Chapter 3, Section 7, Cl.7.5.

- For prescriptive assessment of the overall stiffened panel capacity, the shear stresses are calculated according to Section 3, Cl.3.2.2, at the following load calculation point:

—At the middle of the full span, 'l', of the considered stiffener.

—At the intersection point between the stiffener and its attached plating

• For grillage beam analysis, $\tau = 0$ in the plate attached to the PSM web.

5.2.3 Stiffeners

5.2.3.1 Buckling modes

The following buckling modes are to be checked:

• Stiffener-induced failure (SI).

• Associated plate-induced failure (PI).

5.2.3.2 Web thickness of the flat bar

To account for the decrease of the stiffness due to local lateral deformation, the effective web thickness of flat bar stiffener, in [mm], are to be used in Section 5, Cl.5.2.1.1 and Cl.5.2.3.4 for the calculation of the net sectional area, As, the net section modulus, Z, and the moment of inertia, I, of the stiffener and is taken as:

$$t_{w-red} = t_w \left(1 - \frac{2\pi^2}{3}\left(\frac{h_w}{s}\right)^2 \left(1 - \frac{b_{eff}}{s}\right)\right)$$

5.2.3.3 Idealization of bulb profile

Bulb profiles are to be considered as equivalent angle profiles, as defined in Chapter 3, Section 7, Cl.7.4.

5.2.3.4 Ultimate buckling capacity

When $\sigma_a + \sigma_b + \sigma_w > 0$, the ultimate buckling capacity for stiffeners is to be checked according to the following interaction formula:

$$\frac{\gamma_c \sigma_a + \sigma_b + \sigma_w}{R_{eH}} S = 1$$

where,

σ_a = Effective axial stress, in [N/mm^2], at mid-span of the stiffener, acting on the stiffener with its attached plating.

$$\sigma_a = \left(\frac{s\, t_p + A_s}{b_{eff1}\, t_p + A_s}\right)\sigma_x$$

σ_x = Nominal axial stress, in [N/mm^2], acting on the stiffener with its attached plating.

• For FE analysis, σ_x is the FE-corrected stress as defined in Section 5, Cl.5.2.3.6 in the attached plating in the direction of the stiffener axis.

• For prescriptive assessment, σ_x is the axial stress calculated according to Section 3, Cl.3.2.2 at the load calculation point of the stiffener, as defined in Chapter 3, Section 7, Cl.7.6.

• For grillage beam analysis, σ_x is the stress acting along the x-axis of the attached buckling panel.

R_{eH} = Specified minimum yield stress of the material, in [N/mm^2].

$R_{eH} = R_{eH-s}$ for stiffener-induced failure (SI).

$R_{eH} = R_{eH-p}$ for plate-induced failure (PI).

σ_b = Bending stress in the stiffener, in [N/mm^2].

$$\sigma_b = \frac{M_0 + M_1 + M_2}{1000\, Z}$$

Z = Net section modulus of the stiffener, in [cm^3], including effective width of plating according to Section 5, Cl.5.2.3.5, to be taken as:

• The section modulus calculated at the top of the stiffener flange for stiffener-induced failure (SI).

• The section modulus calculated at the attached plating for plate-induced failure (PI).

C_{PI} = Plate-induced failure pressure coefficient:

C_{PI} = 1 if the lateral pressure is applied on the side opposite to the stiffener.

C_{PI} = -1 if the lateral pressure is applied on the same side as the stiffener.

C_{SI} = Stiffener-induced failure pressure coefficient:

C_{SI} = 1 if the lateral pressure is applied on the side opposite to the stiffener.

C_{SI} = -1 if the lateral pressure is applied on the same side as the stiffener.

M_1 = Bending moment, in [N-mm], due to the lateral load P:

$M_1 = C_i \frac{|P|\, s\, l^2}{24 \times 10^3}$ for continuous stiffener

$M_1 = C_i \frac{|P|\, s\, l^2}{8 \times 10^3}$ for sniped stiffener

$M_1 = C_i \frac{|P|\, s\, l^2}{14.2 \times 10^3}$ for stiffener sniped at one end and continuous at the other end

P = Lateral load, in [kN/m^2].

• For FE analysis, P is the average pressure as defined in Chapter 9, Section 4, Cl.4.2.5.2 in the attached plating.

• For prescriptive assessment, P is the pressure calculated at the load calculation point of the stiffener, as defined in Chapter 3, Section 7, Cl.7.6.

C_i = Pressure coefficient:

$C_i = C_{SI}$ for stiffener-induced failure (SI).

$C_i = C_{PI}$ for plate-induced failure (PI).

M_2 = Bending moment, in [N-mm], due to eccentricity of sniped stiffeners, to be taken as:

$M_2 = 0$ for continuous stiffeners

$M_2 = C_{snip} w_{na} \gamma \sigma_x (A_p + A_s)$ for stiffeners sniped at one or both ends.

C_{snip} = Coefficient to account for the end effect of the stiffener sniped at one or both ends, to be taken as:

$C_{snip} = -1.2$ for stiffener induced failure (SI)

$C_{snip} = 1.2$ for plate induced failure (PI)

M_o = Bending moment, in [N-mm], due to the lateral deformation w of stiffener:

$M_O = F_E C_{sl} \left(\frac{\gamma}{\gamma_{GEB} - \gamma}\right) w_o$ with precondition $(\gamma_{GEB} - \gamma) > 0$

where γ_{GEB} is the stress multiplier factor of global elastic buckling capacity as defined in Section 5, Cl.5.2.1.

F_E = Ideal elastic buckling force of the stiffener, in [N].

$$F_E = \left(\frac{\pi}{l}\right)^2 EI \,.\, 10^4$$

C_{sl} = Deformation reduction factor to account for global slenderness, to be taken as:

$C_{sl} = 1 - \frac{1}{12}\lambda_G^4$ for $\lambda_G \leq 1.56$

$C_{sl} = 3/\lambda_G^4$ for $\lambda_G > 1.56$

λ_G = The reference degree of global slenderness of the stiffened panel, to be taken as:

$$\lambda_G = \sqrt{\frac{\gamma_{ReH}}{\gamma_{GEB}}}$$

where

$$\gamma_{ReH} = \frac{\min(R_{eH_p},\, R_{eH_s})}{\sqrt{\sigma_{x,av}^2 + \sigma_y^2 - \sigma_{x,av}\sigma_y + 3\tau_{xy}^2}}$$

I = Moment of inertia, in [cm^4], of the stiffener including effective width of attached plating according to Section 5, Cl.5.2.3.5 'I' has to comply with the following requirement:

$$I \geq \frac{s\, t_p^3}{12 \times 10^4}$$

t_p = Net thickness of plate, in [mm], to be taken as:

• For prescriptive requirements: the mean thickness of the two attached plating panels,

• For FE analysis: the thickness of the considered EPP on one side of the stiffener.

w_0 = Assumed imperfection, in [mm], to be taken as:

$$w_0 = l/1000$$

w_{na} = Distance in [mm], from the mid-point of attached plating to the neutral axis of the stiffener calculated with the effective width of the attached plating according to Section 5, Cl.5.2.3.5.

σ_W = Stress due to torsional deformation, in [N/mm^2], to be taken as:

- For stiffener induced failure (SI)
  - o For $\sigma_a > 0$

$$\sigma_W = E y_w e_f \Phi_o \left(\frac{m_{tor}\pi}{l_{tor}}\right)^2 \left[\frac{1}{1-\frac{\gamma\sigma_a}{\sigma_{ET}}} - 1\right]$$

with the precondition that $\sigma_{ET} - \gamma\sigma_a > 0$

  - o For $\sigma_a \leq 0$

$$\sigma_W = 0$$

- For plate induced failure (PI)

$$\sigma_W = 0$$

l_{tor} = Stiffener span in [mm], distance equal to the spacing between primary supporting members, i.e., $l_{tor} = l$. When the stiffener is supported by tripping brackets, l_{tor} should be taken as the maximum spacing between the adjacent primary supporting members and fitted tripping brackets.

y_w = Distance, in [mm], from the centroid of the stiffener cross-section to the free edge of the stiffener flange, to be taken as:

$y_w = \frac{t_w}{2}$ For flat bar

$y_w = b_f - \frac{h_w t_w^2 + t_f b_f^2}{2A_s}$ For angle and bulb profile

$y_w = b_{f-out} + 0.5t_w - \frac{h_w t_w^2 + t_f(b_f^2 - 2b_f d_f)}{2A_s}$ For L2 profile

$y_w = \frac{b_f}{2}$ For T-profile

Φ_o = Coefficient to be taken as:

$$\Phi_o = \frac{l_{tor}}{m_{tor} h_w} 10^{-4}$$

σ_{ET} = Reference stress for torsional buckling, in [N/mm^2]:

$$\sigma_{ET} = \frac{E}{I_p}\left[\left(\left(\frac{m_{tor}\pi}{l_{tor}}\right)^2 I_p \cdot 10^2 + \frac{1}{2(1+\nu)} I_T + \left(\frac{l_{tor}}{m_{tor}\pi}\right)^2 \epsilon \cdot 10^{-4}\right)\right]$$

I_p = Net polar moment of inertia of the stiffener, in [cm^4], about point C as shown in Figure 5.1.1, as defined in Table 5.2.3.2.

Table 5.2.3.2: Moments of inertia

Parameter	Flat bars (1)	Bulb, angle, L2, and T profiles
I_p	$\frac{h_w^3 t_w}{3\times 10^4}$	$\left(\frac{A_w(e_f - 0.5t_f)^2}{3} + A_f e_f^2\right) 10^{-4}$
I_t	$\frac{t_w^3 h_w}{3\times 10^4}\left(1 - 0.63\frac{t_w}{h_w}\right)$	$\frac{(e_f - 0.5t_f)t_w^3}{3\times 10^4}\left(1 - 0.63\frac{t_w}{e_f - 0.5t_f}\right) + \frac{b_f t_f^3}{3\times 10^4}\left(1 - 0.63\frac{t_f}{b_f}\right)$
I_w	$\frac{h_w^3\, t_w^3}{36\times 10^6}$	for the bulb, angle and L2 profiles (2) $\frac{A_f^3 + A_w^3}{36\times 10^6} + \frac{e_f^2}{10^6}\left(\frac{A_f b_f^2 + A_w t_w^2}{3} - \frac{(A_f(b_f - 2d_f) + A_w t_w)^2}{4(A_f + A_w)} - A_f d_f(b_f - d_f)\right)$ for T profiles $\frac{b_f^3\, t_f e_f^2}{12\times 10^6}$

(1) t_w is the net web thickness, in [mm]. t_{w-red} as defined in Section 5, Cl.5.2.3.2 is not to be used in this Table.
(2) d_f is to be taken as 0 for bulb and angle profiles.

I_t = Net St. Venant's moment of inertia of the stiffener, in [cm^4], as defined in Table 5.2.3.2.

I_ω = Net sectorial moment of inertia of the stiffener, in [cm^6], about point C as shown in Figure 5.1.1, as defined in Table 5.2.3.2.

m_{tor} = Number of half waves within l_{tor} , taken as a positive integer in order to give the smallest reference stress for torsional buckling.

ε = Degree of fixation in [mm^2], to be taken as:

$$\varepsilon = \left(\frac{3b}{{t_p}^3} + \frac{2h_w}{{t_w}^3}\right)^{-1}$$

For angle, bulb, L2, and T profile.

$$\varepsilon = \frac{{t_p}^3}{3b}$$

For flat bars

A_w = Net web area, in [mm^2].

A_f = Net flange area, in [mm^2].

5.2.3.5 Effective width of attached plating

The effective width of attached plating of stiffeners, b_{eff}, in [mm], is to be taken as:

• For $\sigma_x > 0$:

- For FE analysis,

$$b_{eff} = \min(C_x b, \chi_s s)$$

- For prescriptive assessment,

$$b_{eff} = \min\left(\frac{C_{x1} b_1 + C_{x2} b_2}{2}, \chi_s s\right)$$

• For $\sigma_x \leq 0$:

$$b_{eff} = \chi_s s$$

where,

χ_s = Effective width coefficient to be taken as:

$$\chi_s = min\left[\frac{1.12}{1+\frac{1.75}{\left(\frac{l_{eff}}{s}\right)^{1.6}}}; 1.0\right] \text{ For } \frac{l_{eff}}{s} \geq 1$$

$$\chi_s = 0.407\frac{l_{eff}}{s} \qquad \text{For } \frac{l_{eff}}{s} < 1$$

l_{eff} = Effective length of the stiffener, in [mm], taken as:

$l_{eff} = \frac{l}{\sqrt{3}}$ for stiffener fixed at both ends.

$l_{eff} = 0.75\, l$ for stiffener simply supported at one end and fixed at the other.

$l_{eff} = l$ for stiffener simply supported at both ends.

5.2.3.6 FE corrected stresses for stiffener capacity

When the reference stresses σ_x and σ_y obtained by FE analysis according to Section 4, Cl.2.4 are both compressive, they are to be corrected according to the following formulae:

• If $\sigma_x < \upsilon\sigma_y$:

$$\sigma_{xcor} = 0$$

• If $\sigma_x \geq \upsilon\sigma_y$:

$$\sigma_{xcor} = \sigma_x - \upsilon\sigma_y$$

5.2.4 Primary supporting members

5.2.4.1 Web plate in way of openings

The web plate of primary supporting members with openings is to be assessed for buckling based on the combined axial compressive and shear stresses.

The web plate adjacent to the opening on both sides is to be considered as individual unstiffened plate panels as shown in Table 5.2.4.

The interaction formulae of Section 5, Cl.5.2.2.1 are to be used with:

$\sigma_x = \sigma_{av}$

$\sigma_y = 0$

$\tau = \tau_{av}$

where,

σ_{av} = Weighted average compressive stress, in [N/mm^2], in the area of the web plate being considered, i.e., P1, P2, or P3 as shown in Table 5.2.4.

For the application of Table 5.2.4, the weighted average shear stress is to be taken as:

- Opening modeled in primary supporting members:

τ_{av} = Weighted average shear stress, in [N/mm^2], in the area of the web plate being considered, i.e., P1, P2, or P3 as shown in Table 5.2.4.

- Opening not modeled in primary supporting members:

τ_{av} = Weighted average shear stress, in [N/mm^2], given in Table 5.2.4.

5.2.4.2 Reduction factors of web plate in way of openings

The reduction factors, C_x or C_y in combination with, C_τ of the plate panel(s) of the web adjacent to the opening is to be taken as shown in Table 5.2.4.

5.2.4.3 The equivalent plate panel of the web plate of primary supporting members crossed by perpendicular stiffeners is idealized as shown in Figure 5.2.4.

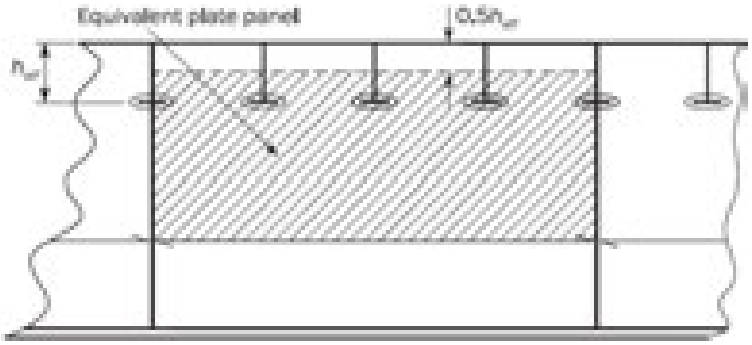
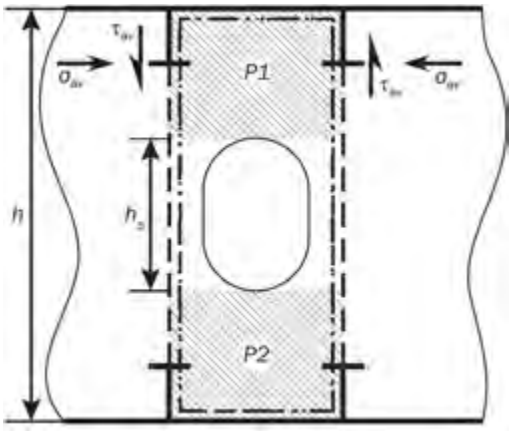
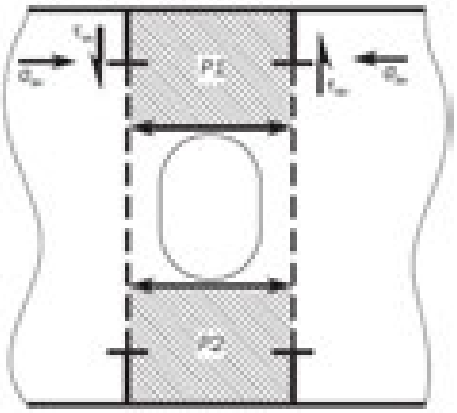
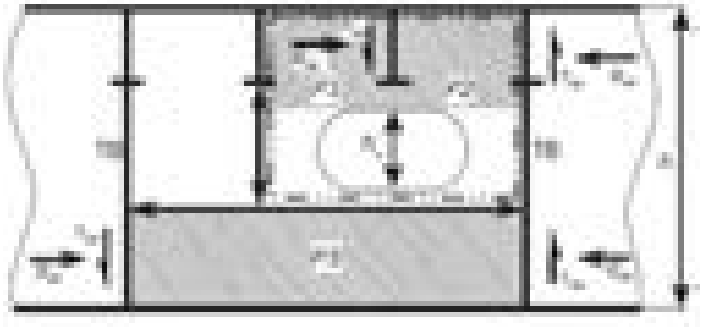


Figure 5.2.4: Web plate idealization

The correction of panel breadth is applicable also for other slot configurations provided that the web or collar plate is attached to at least one side of the passing stiffener

Table 5.2.4: Reduction factors

Configuration	C_x , C_y	C_τ	
		Opening modeled in PSM	**Opening not modeled in PSM**
(a) Without edge reinforcements: 	Separate reduction factors are to be applied to areas P1 and P2 using Case 3 or Case 6 in Table 5.2.2.3, with edge stress ratio: $\Psi = 1.0$	Separate reduction factors are to be applied to areas P1 and P2 using Case 18 or Case 19 in Table 5.2.2.3.	When Case 17 of Table 5.2.2.3 is applicable: A common reduction factor is to be applied to areas P1 and P2 using Case 17 in Table 5.2.2.3 with: $\tau_{av} = \tau_{av(web)}$ When Case 17 of Table 5.2.2.3 is not applicable: Separate reduction factors are to be applied to areas P1 and P2 using Case 18 or Case 19 in Table 5.2.2.3 with: $\tau_{av} = \tau_{av(web)} \frac{h}{h - h_o}$
(b) With edge reinforcements: 	Separate reduction factors are to be applied for areas P1 and P2 using C_x for Case 1 or C_y for Case 2 Table 5.2.2.3 with stress ratio: $\Psi = 1.0$	Separate reduction factors are to be applied for areas P1 and P2 using Case 15 in Table 5.2.2.3.	Separate reduction factors are to be applied to areas P1 and P2 using Case 15 in Table 5.2.2.3 with: $\tau_{av} = \tau_{av(web)} \frac{h}{h - h_o}$
(c) Example of a hole in the web: 		Panels P1 and P2 are to be evaluated as per (a). Panel P3 is to be evaluated as per (b).	

where:
h = Height, in [m], of the web of the primary supporting member in way of the opening.
h_o = Height, in [m], of the opening measured in the depth of the web.
$\tau_{av(web)}$ = Weighted average shear stress, in [N/mm^2] over the web height h of the primary supporting member.
Note 1: Web panels to be considered for buckling in way of openings are shown shaded and numbered P1, P2, etc.

5.3 Corrugated bulkhead

5.3.1 The buckling utilization factor of flange and web of corrugation of corrugated bulkheads is based on the combination of in-plane stresses and shear stress.

The buckling stresses to be used in the interaction formulation in Section 5, Cl.5.2.2.1 are to be determined based on Case 1, 2, and 15, together with the following coefficients:

- $\alpha = 2$
- $\Psi_x = \Psi_y = 1$

5.4 Struts, pillars, and cross ties

5.4.1 Buckling utilization factor

The buckling utilization factor, η, for axially compressed struts, pillars, and cross ties is to be taken as:

$$\eta = \frac{\sigma_{av}}{\sigma_{cr}}$$

where,

σ_{av} = Average axial compressive stress in the member, in [N/mm^2].

σ_{cr} = Minimum critical buckling stress, in [N/mm^2], taken as:

$$\sigma_{cr} = \sigma_E$$

when $\sigma_E \leq 0.5\,\sigma_y$

$$\sigma_{cr} = R_{eH-s}\left(1 - \frac{R_{eH-s}}{4\sigma_E}\right)$$

when $\sigma_E > 0.5\,\sigma_y$

where,

σ_E = Ideal elastic buckling stress, in [N/mm^2], as per Section 5, Cl.5.4.2, Cl.5.4.3 and Cl.5.4.4.

σ_{y-s}= Specified minimum yield stress of the considered member, in [N/mm^2], For built-up members, the lowest specified minimum yield stress is to be used.

5.4.2 Elastic column buckling stress:

The elastic compressive column buckling stress, $'\sigma_{EC}'$, in [N/mm^2], of members subject to axial compression, is to be taken as:

$$\sigma_{EC} = \pi^2 E f_{end} \frac{I}{Al_{pill}{}^2}\, 10^{-4}$$

where:

I = Net moment of inertia about the weakest axis of the cross-section, in [cm^4].

A = Net cross-sectional area of the member, in [cm^2].

l_{pill} = Length of the member, in [m], taken as:

- For pillar and strut: Unsupported length of the member
- For cross tie

- In center tank: distance between the flanges of longitudinal stiffeners on the starboard and port longitudinal bulkheads to which the cross tie's horizontal stringer is attached.

- In wing tank: distance between the flanges of longitudinal stiffeners on the longitudinal bulkhead to which the cross tie's horizontal stringer is attached, and the inner hull plating.

f_{end} = End constraint factor, taken as:

- For pillar and strut

$f_{end} = 1.0$ where both ends are simply supported.

$f_{end} = 2.0$ where one end is simply supported, and the other end is fixed.

$f_{end} = 4.0$ where both ends are fixed.

• For cross tie

$f_{end} = 2.0$

A pillar end may be considered fixed when brackets of adequate size are fitted. Such brackets are to be supported by structural members with greater bending stiffness than the pillar.

5.4.3 Elastic torsional buckling stress:

The elastic torsional column buckling stress, $'\sigma_{ET}'$, in [N/mm^2], with respect to axial compression of members, is to be taken as:

$$\sigma_{ET} = \frac{G\, I_{SV}}{I_{pol}} + \frac{\pi^2\, E\, f_{end}\, c_{warp}}{I_{pol}\, l_{pill}^{\ 2}} 10^{-4}$$

where,

l_{SV}= Net St. Venant's moment of inertia, in [cm^4], Refer Table 5.4.3 for examples of cross sections.

l_{pol}= Net polar moment of inertia about the shear center of cross-section, in [cm^4].

$$l_{pol} = l_y + l_z + A(y_0^2 + z_0^2)$$

c_{warp}= Warping constant, in [cm^6], Refer Table 5.4.3 for examples of cross sections.

l_{pill} = Length of the member, in [m].

y_0 = Transverse position of shear center relative to the cross-sectional centroid, in [cm], Refer Table 5.4.3 for examples of cross sections.

z_0 = Vertical position of shear center relative to the cross-sectional centroid, in [cm], Refer Table 5.4.3 for examples of cross sections.

A = Net cross-sectional area, in [cm^2], as defined in [5.4.2]

I_y= Net moment of inertia about the y-axis, in [cm^4].

I_z= Net moment of inertia about the z-axis, in [cm^4].

5.4.4 Elastic torsional/column buckling stress:

For cross sections where the centroid and the shear center do not coincide, the interaction between the torsional and column buckling mode is to be examined. The elastic torsional/column buckling stress, $'\sigma_{ETF}'$, with respect to axial compression, is to be taken as:

$$\sigma_{ETF} = \frac{1}{2\zeta}\left[(\sigma_{EC} + \sigma_{ET}) - \sqrt{(\sigma_{EC} + \sigma_{ET})^2 - 4\zeta\sigma_{EC}\sigma_{ET}}\right]$$

where,

ζ = Coefficient taken as:

$$\zeta = 1 - \frac{(y_0^2 + z_0^2)A}{I_{pol}}$$

y_0 = Transverse position of shear center relative to the cross-sectional centroid, in [cm], as defined in Cl.5.4.3.

z_0= Vertical position of shear center relative to the cross-sectional centroid, in [cm], as defined in Cl.5.4.3.

A= Net cross-sectional area, in [cm^2], as defined in Cl.5.4.2.

I_{pol}= Net polar moment of inertia about the shear center of cross-section, in [cm^4], as defined in Cl.5.4.3.

σ_{EC}= Elastic column compressive buckling stress, as defined in Cl.5.4.2.

σ_{ET}= Elastic torsional buckling stress, as defined in Cl.5.4.3.

5.5 Stiffened Panels with U-type Stiffeners

5.5.1 Local plate buckling

For stiffened panels with U-type stiffeners, local plate buckling is to be checked for each of the plate panels EPP b_1, b_2, b_f and h_w (Refer Section 5, Figure 5.2.2.5) separately as follows:

- The attached plate panels EPP b_1 and b_2 are to be assessed using SP-A model, where in the calculation of buckling factors K_X as defined in Case 1 of Table 5.2.2.3, the correction factor F_{long} for U-type stiffeners as defined in Table 5.2.2.2 is to be used; and in the calculation of K_y as defined in Case 2 of Table 5.2.2.3, the F_{tran} for U-type stiffeners as defined in Cl.5.2.2.5] is to be used.

- The face plate and web plate panels b_f and h_w are to be assessed using UP-B model with $F_{long} = 1$ and $F_{tran} = 1$.

5.5.2 Overall stiffened panel buckling and stiffener buckling

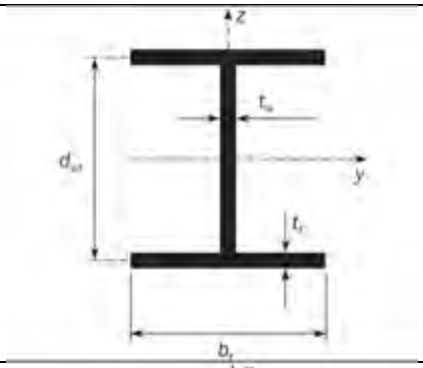
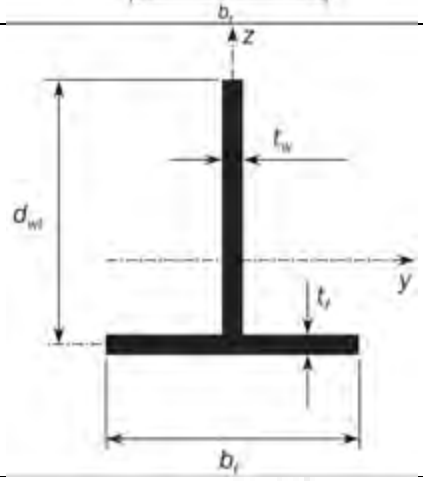
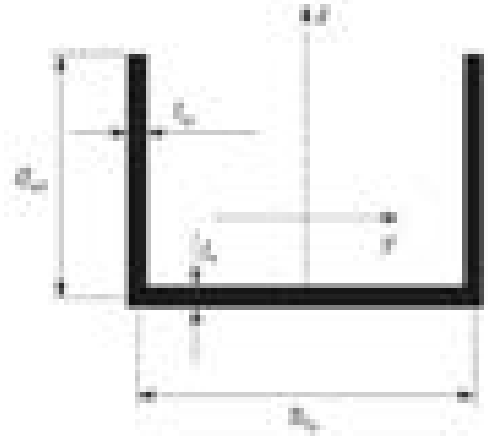
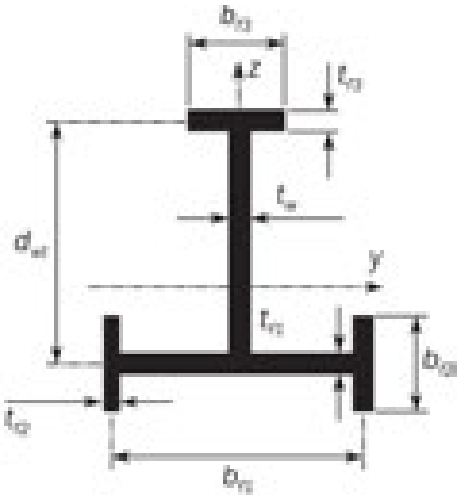
For a stiffened panel with U-type stiffeners, the overall buckling capacity and ultimate capacity of the stiffeners are to be checked with warping stress $\sigma_W = 0$, and with bending moment of inertia including effective width of attached plating being calculated based on the following assumptions:

- The two web panels of a U-type stiffener are to be taken as perpendicular to the attached plate with thickness equal to t_w and height equal to the distance between the attached plate and the face plate of the stiffener.

- Effective width of the attached plating, b_{eff}, taken as the sum of the b_{eff} calculated for the EPP b_1 and b_2 respectively according to SP-A model.

- Effective width of the attached plating of a stiffener without shear lag effect, b_{eff1}, taken as the sum of the b_{eff1} calculated for the EPP b_1 and b_2 respectively.

Table 5.4.3: Cross-sectional properties

	$I_{SV} = \frac{1}{3}(2b_f t_f^3 + d_{wt}\, t_w^3).10^{-4}$	[cm⁴]
	$c_{warp} = \frac{{d_{wt}}^2\, b_f^3\, t_f}{24}\, 10^{-6}$	[cm⁶]
	$I_{SV} = \frac{1}{3}(b_f t_f^3 + d_{wt}\, t_w^3).10^{-4}$	[cm⁴]
	$y_0 = 0$	[cm]
	$z_0 = -\frac{0.5\, {d_{wt}}^2 t_w}{d_{wt} t_w + b_f t_f} 10^{-1}$	[cm]
	$c_{warp} = \frac{b_f^3\, {t_f}^3 + 4\, d_{wt}^3\, {t_w}^3}{144}\, 10^{-6}$	[cm⁶]
	$I_{SV} = \frac{1}{3}(b_{fu}\, t_f^3 + 2\, d_{wt}\, t_w^3).10^{-4}$	[cm⁴]
	$y_0 = 0$	[cm]
	$z_0 = -\frac{{d_{wt}}^2 t_w\, 10^{-1}}{2\, d_{wt} t_w + b_{fu} t_f} - \frac{0.5\, {d_{wt}}^2 t_w 10^{-1}}{d_{wt} t_w + b_{fu}\, t_f/6}$	[cm]
	$c_{warp} = \frac{b_{fu}^2 d_{wt}^3\, t_w (3 d_{wt} t_w + 2 b_{fu} t_f)}{12(6 d_{wt} t_w + b_{fu} t_f)}\, 10^{-6}$	[cm⁶]
	$I_{SV} = \frac{1}{3}(b_{f1}\, t_{f1}^3 + 2\, b_{f2}\, t_{f2}^3 + b_{f3}\, t_{f3}^3 + d_{wt}\, t_w^3).10^{-4}$	[cm⁴]
	$y_0 = 0$	[cm]
	$z_0 = z_s - \frac{(\, b_{f3}\, d_{wt} t_{f3} + 0.5\, {d_{wt}}^2 t_w)}{d_{wt} t_w + b_{f1}\, t_{f1} + 2 b_{f2}\, t_{f2} + b_{f3}\, t_{f3}} 10^{-1}$	[cm]
	$c_{warp} = \left(I_{f1} {z_s}^2 + \frac{I_{f2} {b_{f1}}^2}{200} + I_{f3} \left(\frac{d_{wt}}{10} - z_s \right)^2 \right)$	[cm⁶]
	$I_{f1} = \left(\frac{(b_{f1} - t_{f2})^3\, t_{f1}}{12} + \frac{b_{f2}\, t_{f2} {b_{f1}}^2}{2} \right) 10^{-4}$	[cm⁴]
	$I_{f2} = \frac{t_{f2}\, {b_{f2}}^3}{12}\, 10^{-4}$	[cm⁴]

	$I_{f3} = \frac{t_{f3}\, b_{f3}{}^{3}}{12} 10^{-4}$	[cm^4]
	$z_s = \frac{I_{f3} d_{wt}}{I_{f1}+I_{f3}} 10^{-1}$	[cm]
Note 1: All dimensions are in [mm]. Note 2: Cross-sectional properties are given for typical cross-sections. Properties for other cross-sections are to be determined by direct calculation.		

Appendix A

Stress-based reference stresses

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

A_i = Area, in [mm^2], of the i-th plate element of the buckling panel.

n = Number of plate elements in the buckling panel.

σ_{xi} = Actual stress, in [N/mm^2], at the centroid of the i-th plate element in x direction, applied along the shorter edge of the buckling panel.

σ_{yi} = Actual stress, in [N/mm^2], at the centroid of the i-th plate element in the y-direction, applied along the longer edge of the buckling panel.

τ_i = Actual membrane shear stress, in [N/mm^2], at the centroid of the i-th plate element of the buckling panel.

Ψ = Edge stress ratio as defined in Section 5.

A.1 Stress-based method

A.1.1 Introduction

A.1.1.1 General

This section provides a method to determine stress distribution along edges of the considered buckling panel by 2nd order polynomial curve, by linear distribution using the least square method, or by weighted average approach. This method is called the Stress-based Method.

The reference stress is the stress components at the centre of the plate element transferred into the local system of the considered buckling panel.

A.1.1.2 Definition

A regular panel is a plate panel of rectangular shape. An irregular panel is a plate panel that is not regular, as detailed in Section 4, Cl.4.2.3.

A.1.2 Stress application

A.1.2.1 Regular panel

The reference stresses are to be taken as defined in A.2 for a regular panel when the following conditions are satisfied:

- At least, one plate element center is located in each third part of the long edge ‘a’ of a regular panel and

- This element center is located at a distance in the panel local x-direction not less than ‘a/4’ to at least one of the element centers in the adjacent third part of the panel.

Otherwise, the reference stresses are to be taken as defined in A.2.2 for an irregular panel.

A.1.2.2 Irregular panel and curved panel

The reference stresses of an irregular panel or a curved panel are to be taken as defined in A.2.2.

A.2 Reference stresses

A.2.1 Regular Panel

A.2.1.1 Longitudinal stress

The longitudinal stress σ_x applied on the shorter edge of the buckling panel is to be calculated as follows:

For plate buckling assessment, the distribution of $\sigma_x(x)$ is assumed as 2nd order polynomial curve as:

$$\sigma_x(x) = C.x^2 + D.x + E$$

The best-fitting curve $\sigma_x(x)$ is to be obtained by minimizing the square error Π considering the area of each element as a weighting factor.

$$\Pi = \sum_{i=1}^{n} A_i[\sigma_{ix} - (Cx_i^2 + Dx_i + E)]^2$$

The unknown coefficients C, D, and E must yield zero first derivatives, ∂Π with respect to C, D, and E respectively.

$$\frac{\partial \Pi}{\partial C} = 2\sum_{i=1}^{n} A_i x_i^2[\sigma_{ix} - (Cx_i^2 + Dx_i + E)] = 0$$

$$\frac{\partial \Pi}{\partial D} = 2\sum_{i=1}^{n} A_i x_i[\sigma_{ix} - (Cx_i^2 + Dx_i + E)] = 0$$

$$\frac{\partial \Pi}{\partial E} = 2\sum_{i=1}^{n} A_i[\sigma_{ix} - (Cx_i^2 + Dx_i + E)] = 0$$

The unknown coefficients C, D, and E can be obtained by solving the 3 above equations.

$$\sigma_{x1} = \frac{1}{b}\int_0^b \sigma_x(x)\,dx = \frac{b^2}{3}C + \frac{b}{2}D + E$$

$$\sigma_{x2} = \frac{1}{b}\int_{a-b}^{b} \sigma_x(x)\,dx = \left(a^2 - ab + \frac{b^2}{3}\right)C + \left(a - \frac{b}{2}\right)D + E$$

If -D/2C < b/2 or -D/2C > $\left(a - \frac{b}{2}\right)$, σ_{x3} is to be ignored. Otherwise, σ_{x3} is taken as:

$$\sigma_{x3} = \frac{1}{b}\int_{xmin}^{xmax} \sigma_x(x)\,dx = \frac{b^2}{12}C - \frac{D^2}{4C} + E$$

where,

$$xmin = -\frac{b}{2} - \frac{D}{2C}$$

$$xmax = \frac{b}{2} - \frac{D}{2C}$$

The longitudinal stress is to be taken as:

$$\sigma_x = \max(\sigma_{x1}; \sigma_{x2}; \sigma_{x3})$$

The edge stress ratio is to be taken as:

$$\Psi = 1.0$$

• For overall stiffened panel buckling and stiffener buckling assessments, $\sigma_x(x)$ applied on the shorter edge of the attached plate is to be taken as:

$$\sigma_x = \frac{\sum_1^n A_i \sigma_{xi}}{\sum_1^n A_i}$$

The edge stress ratio Ψ_x for the stress σ_x is equal to 1.0.

A.2.1.2 Transverse stress

The transverse stress σ_y applied along the longer edges of the buckling panel is to be calculated by extrapolation of the transverse stresses of all elements up to the shorter edges of the considered buckling panel.

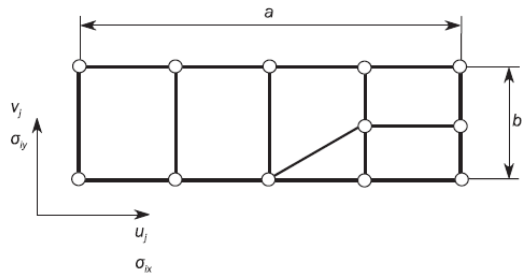


Figure 6.2.1: Buckling panel

The distribution of $\sigma_y(x)$ is assumed as a straight line. Therefore:

$$\sigma_y(x) = A + B.x$$

The best-fitting curve $\sigma_y(x)$ is to be obtained by the least square method minimizing the square error Π considering the area of each element as a weighting factor.

$$\Pi = \sum_{i=1}^{n} A_i\left[\sigma_{iy} - (A + B.x_i)\right]^2$$

The unknown coefficients A and B must yield zero first partial derivatives, ∂Π with respect to A and B, respectively.

$$\frac{\partial \Pi}{\partial A} = 2\sum_{i=1}^{n} A_i\left[\sigma_{iy} - (A + B.x_i)\right] = 0$$

$$\frac{\partial \Pi}{\partial B} = 2\sum_{i=1}^{n} A_i x_i\left[\sigma_{iy} - (A + B.x_i)\right] = 0$$

The unknown coefficients A and B are obtained by solving the 2 above equations and are given as follows:

$$A = \frac{(\sum_{i=1}^n A_i\sigma_{iy})(\sum_{i=1}^n A_i {x_i}^2) - (\sum_{i=1}^n A_i x_i)(\sum_{i=1}^n A_i x_i\,\sigma_{iy})}{(\sum_{i=1}^n A_i)(\sum_{i=1}^n A_i {x_i}^2) - (\sum_{i=1}^n A_i x_i)^2}$$

$$B = \frac{(\sum_{i=1}^n A_i)(\sum_{i=1}^n A_i x_i\,\sigma_{iy}) - (\sum_{i=1}^n A_i x_i)(\sum_{i=1}^n A_i\sigma_{iy})}{(\sum_{i=1}^n A_i)(\sum_{i=1}^n A_i {x_i}^2) - (\sum_{i=1}^n A_i x_i)^2}$$

$$\sigma_y = \max(A, A + Ba)$$

$$\Psi_y = \frac{\min(A, A + Ba)}{\max(A, A + Ba)} \text{ for } \sigma_y \geq 0$$

$$\Psi_y = 1 \text{ for } \sigma_y < 0$$

A.2.1.3 Shear stress

The shear stress τ is to be calculated using a weighted average approach, and is to be taken as:

$$\tau = \frac{\sum_{i=1}^{n} A_i \tau_i}{\sum_{i=1}^{n} A_i}$$

A.2.2 Irregular panel and curved panel

A.2.2.2 Reference stresses

The longitudinal, transverse, and shear stresses are to be calculated using a weighted average approach. They are to be taken as:

$$\sigma_x = \frac{\sum_{i=1}^{n} A_i \sigma_{xi}}{\sum_{i=1}^{n} A_i}$$

$$\sigma_y = \frac{\sum_{i=1}^{n} A_i \sigma_{yi}}{\sum_{i=1}^{n} A_i}$$

$$\tau = \frac{\sum_{i=1}^{n} A_i \tau_i}{\sum_{i=1}^{n} A_i}$$

The edge stress ratios are to be taken as:

$\Psi_x = 1.0$

$\Psi_y = 1.0$

End of Chapter

IRCLASS
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50 YEARS OF EXCELLENCE

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 3
GENERAL HULL REQUIREMENTS
VOLUME II

July 2024

Indian Register of Shipping

Part 3 : General Hull Requirements

This Part (Part 3) of the Rules is arranged in two Volumes (Volume I and Volume II). Chapters 1 to 9 may be referred in Volume I. For Chapter 10 onwards, please refer this Volume.

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Chapter 11	**Special Hull Requirements**	**II**
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Chapter 13	**Rudders**	**II**
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Fatigue

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Appendix A – Detail Design Standard

Section 1

General Principles

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

T_{DF} = Design fatigue life, in year, specified by the designer, but not to be taken less than 25 years**.**

1.1 Application

1.1.1 The requirements of this chapter apply to steel ships with L≥150[m] with unrestricted service and are to be used in conjunction with IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

1.1.2 The additional class notation **FDA** (**F**atigue **D**esign **A**ssessment) will be assigned to vessels where a fatigue design assessment has been carried out in accordance with the relevant requirements of this chapter. The **FDA** notation is optional for vessels of length L≥ 90[m] and mandatory for vessels of length L≥ 150[m]. If fatigue assessment is carried out for vessels for a greater number of years (>25) and/ or restricted service, then the class notation **FDA (N, specific area of operation)**, would be assigned; where 'N' denotes the number of years.

1.1.3 This chapter provides a simplified fatigue analysis procedure. The reference stress for fatigue may be evaluated using Prescriptive (Simplified beam theory) method or using the Finite Element (FE) method.

1.1.4 Fatigue assessment of ships with non-conventional shapes or with restricted service notation will be specially considered.

1.1.5 Fatigue assessment is performed for ship's structural details to prevent the following types of fatigue failure:

• Fatigue cracks initiating from the toe of the weld and propagating into the plate.

• Fatigue cracks initiating from free edge of non-welded details.

1.1.6 The structural details to be checked for fatigue assessment by simplified stress analysis are given in Section 3 and by finite element stress analysis are given in Section 4. Additional specific details may be requested to be checked on a case-by-case basis by IRS.

1.1.7 The fatigue strength assessment indicated in the chapter is applicable for steel material with specified minimum yield stress less than or equal to 390[N/mm^2]. For steel with specified minimum yield stress value higher than 390[N/mm^2] and for steels with improved fatigue performance, the S-N curves are to be specially considered on case to case basis.

1.2 Definitions

1.2.1 Fatigue damage ratio: Ratio of the number of applied stress cycles and the corresponding number of stress cycles to failure at constant stress range.

1.2.2 Fatigue life: Total time corresponding to the number of stress cycles required to cause fatigue failure of the component. The fatigue life is generally expressed in terms of number of years.

1.2.3 Fatigue stress: Fatigue stress is the stress including corrections, for e.g. mean stress, thickness effect, surface finishing and material factor, relevant for fatigue assessment purpose, i.e.:

- Hot spot stress for welded details

- Local stress at free plate edge

1.2.4 Hot spots: Hot spots are locations in the structure where fatigue cracks may initiate due to the combined effect of nominal structural stress fluctuation and stress concentration effects due to the weld geometry.
Hot spots may be located at:

- Weld toe.

- Weld root of partial penetration or fillet weld.

1.2.5 Hot spot stress: Hot spot stress is a local stress at the weld toe taking into account the stress concentration due to structural discontinuities and presence of welded attachments but disregarding the non-linear stress peak caused by the notch at the weld toe. It is the value of stress on the surface at the hot spot also known as geometric stress or structural stress.

- The hot spot stresses to be considered correspond to the two principal stresses on the surface plating at the weld toe. The first principal stress acts within ±45[deg], perpendicular to the weld and the second principal stress acts outside ±45[deg].Refer Figure 1.2.1.

- The hot spot stress is to be obtained by multiplying the nominal stress by a Stress Concentration Factor (SCF), according to Section 3 or directly by a very fine mesh FE analysis, according to Section 4.

1.2.6 Local stress at free edge: Local stress at free edge is the stress at the plate free edge derived using finite element analysis according to Section 4, Cl.4.2.4.

1.2.7 Load case: Load case is the state of the ship subjected to a combination of loads.

1.2.8 Local notch: A local notch is a localized geometric feature, such as the weld toe, which causes stress concentration. The local notch does not modify the structural stress but generates a non-linear stress peak.

1.2.9 Macro-geometric discontinuity: A global discontinuity, the effect of which is usually not taken into account in structural details, such as large openings, a curved part in a beam, a bend in a flange, and eccentricity in lap joints etc.

1.2.10 Macro-geometric effect: A stress raising effect due to macro-geometry in the vicinity of the welded joint, but not due to the welded joint itself.

1.2.11 Membrane stress: Average normal stress across the thickness of a plate or shell.

1.2.12 Nominal stress: Nominal stress is a general stress in a structural component taking into account macro-geometric effect including effective breadth of flanges but disregarding the stress concentration due to structural discontinuities and the presence of welds. Nominal stress is to be obtained either using coarse or fine mesh FE analysis, as required in Section 4 or using analytical calculation based on beam theory with effective breadth of flanges included, as required in Section 3. Refer Figure 1.2.1.

1.2.13 Notch stress: Notch stress is a peak stress in a notch such as the root of a weld or the edge of a cut-out. This peak stress takes into account the stress concentrations due to the presence of structural geometry and notches. Refer Figure 1.2.1.

1.2.14 Palmgren-Miner rule: Method for estimating fatigue life under variable amplitude loading from the constant amplitude S-N curve. Often referred to as Miner's rule. Fatigue failure is expected when the Miner sum reaches unity.

1.2.15 S-N curve: Graphical presentation of the dependence of fatigue life (N) on fatigue strength (S).

1.2.16 Stress cycle: A part of a stress history containing a stress maximum and a stress minimum.

1.2.17 Stress range: The difference between stress maximum and stress minimum in a stress cycle.

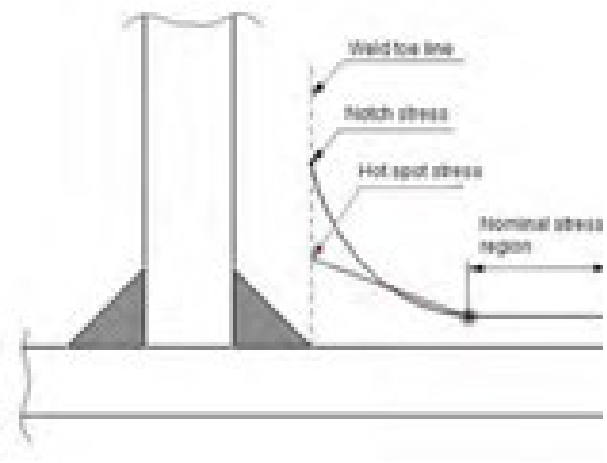


Figure 1.2.1: Stress Definitions

1.3 Assumptions

1.3.1 The following assumptions are made in the fatigue assessment:

- A linear cumulative damage model, i.e. Palmgren-Miner's Rule is used in connection with the design S-N curves with 97.7% probability of survival.

- Design fatigue life, T_{DF}, is taken not less than 25 years.

- Rule quasi-static wave induced loads are based on North Atlantic wave environment. They are determined at 10^{-2} probability level of exceedance.

- Net thickness is used based on net scantling approach according to Chapter 2, Section 4.

- Type of stress used for crack initiating at the weld toe is the hot spot stress. Type of stress used for crack initiating at free edge of non-welded details is local stress at free edge.

- Long term distribution of stress range of a structural detail is assumed to follow a two-parameter Weibull distribution. Weibull shape parameter ξ is equal to 1 and the fatigue stress range $\Delta\sigma_{FS}$ is given at the reference probability level of exceedance equal to 10^{-2}.

- The acceptance criteria for fatigue checking are the total fatigue damage D to be less than or equal to 1 for the design fatigue life.

1.4 Types of environment

1.4.1 Two types of environment are considered for the fatigue detail conditions:

- in air environment

- in corrosive environment without protection

1.5 Corrosion model

1.5.1 The effects of corrosive environment on fatigue life are taken into account by means of:

1.5.1.1 Net scantling approach

The fatigue assessment should be performed based on net scantling approach according to Chapter 2, Section 4.

1.5.1.2 S-N curves.

The basic design S-N curves for corrosive environment are considered to represent the fatigue capacity of material. (Refer Section 2).

1.5.2 Stress correction factor, f_c

1.5.2.1 The global stress is less affected by the corrosion than the local stress due to the probabilistic nature of corrosion, which gives less average corrosion on a global level compared to at a local level. The corrosion allowances are typically higher for local stress calculation than for global stress calculation.

1.5.2.2 For simplified beam model based on net scantlings $t_{n50,}$ the correction factor f_c only applies to the global stresses. When using a global FE model based on net scantlings, the same correction factor applies to both the global and local stresses, i.e. directly on the stress.

The correction factor should be taken as:

- Prescriptive fatigue strength assessment
f_c = 1.0 for local plate and stiffener bending based on t_{n50}

= 0.95 for hull girder stresses based on t_{n50}

- FE based fatigue strength assessment
f_c = 0.95 for stresses based on t_{n50}

1.6 Fatigue failure modes

1.6.1 The following fatigue crack failure modes are generally observed in the ship structure for welded or cut edge details:

- Fatigue cracks initiating from the weld toe into the base material - weld toe failure

- Fatigue cracks initiating from the weld root and propagating into the plate section under the weld - weld root failure

- Fatigue cracks initiating from the weld root and propagating through the weld throat.

- Fatigue cracks initiating from surface irregularity or notch at the free edge into the base material - cut edge failure

1.7 Methodology

1.7.1 Simplified stress analysis

Procedure based on simplified stress analysis, Refer Section 3, is used to determine the hot spot stress at weld toe of longitudinal stiffener end connections. Nominal stresses are calculated by using analytical method based on beam theory according to Section 3, 3.4,3.5 and 3.6. Hot spot stresses are obtained by multiplying nominal stresses by stress concentration factors (SCF) of the considered detail according to Section 3, 3.7.

1.7.2 Finite element stress analysis

Procedure based on finite element stress analysis, Refer Section 4, is used to determine hot spot stress at weld toe of specified structural details, from very fine mesh models. The hot spot stress is generally highly dependent on the finite element model used for representing the structure.

General procedure for the calculation of hot spot stress at weld toe for any welded details except for web stiffened cruciform joints is given in Section 4, 4.5. Procedure for the calculation of hot spot stress at the flange connections for web stiffened cruciform joints is given in Section 4, 4.6. Calculation of local stress for non-welded area is provided in Section 4, Cl.4.5.4.

1.7.3 A simple flowchart of fatigue strength assessment approaches used is shown in Figure 1.7.1. The finite element based fatigue analysis procedure is elaborated in a flowchart given in Figure 1.7.2. Figure 1.7.3 indicates fatigue analysis procedure based on the beam theory approach.

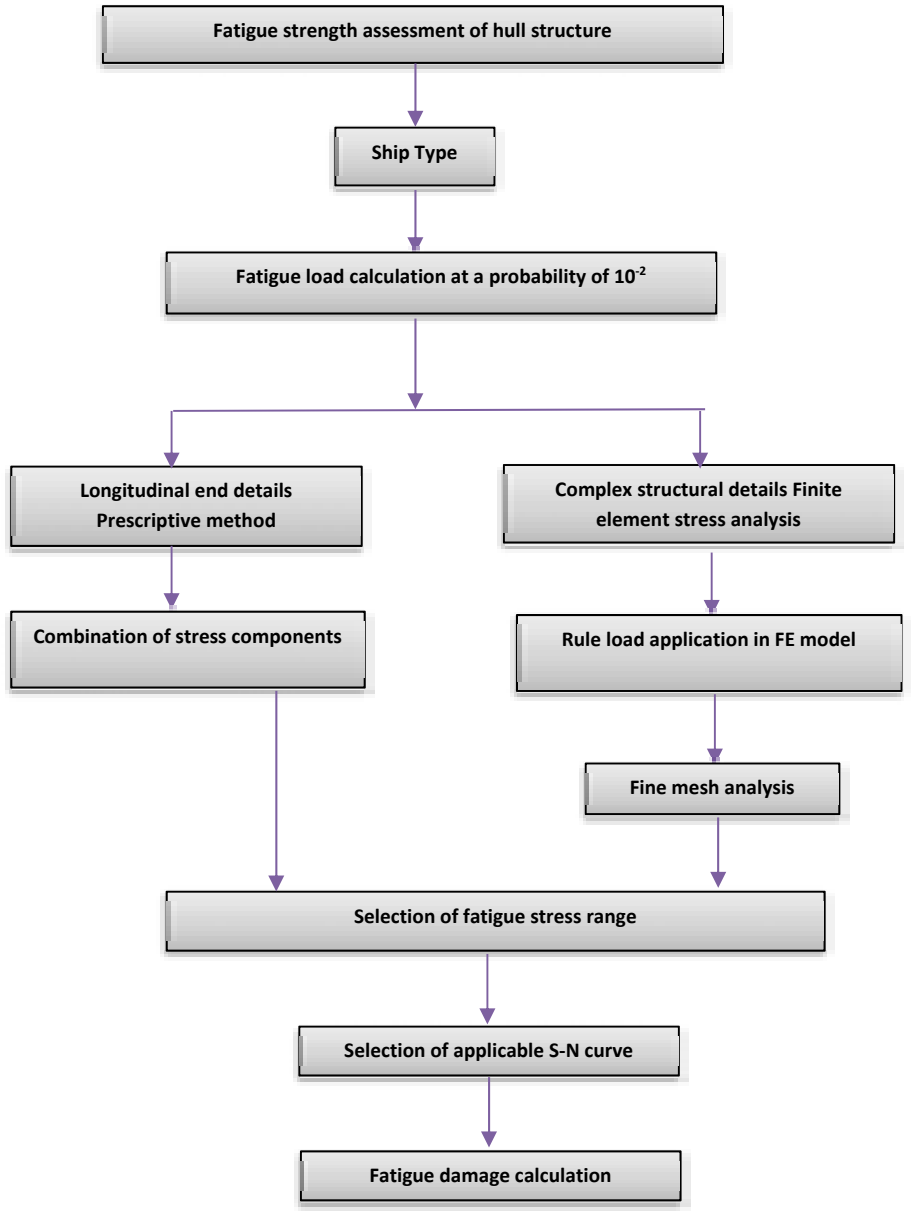


Figure 1.7.1: Flow chart for fatigue strength assessment of hull structure

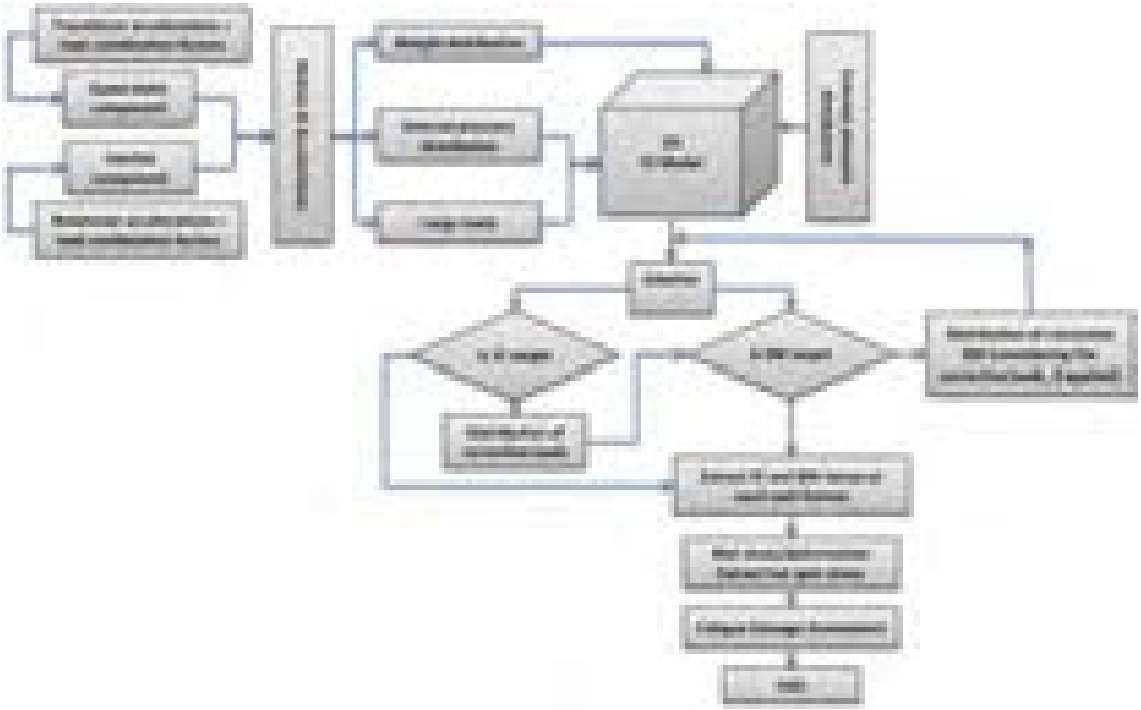


Figure 1.7.2: Procedure for simplified fatigue using FEM

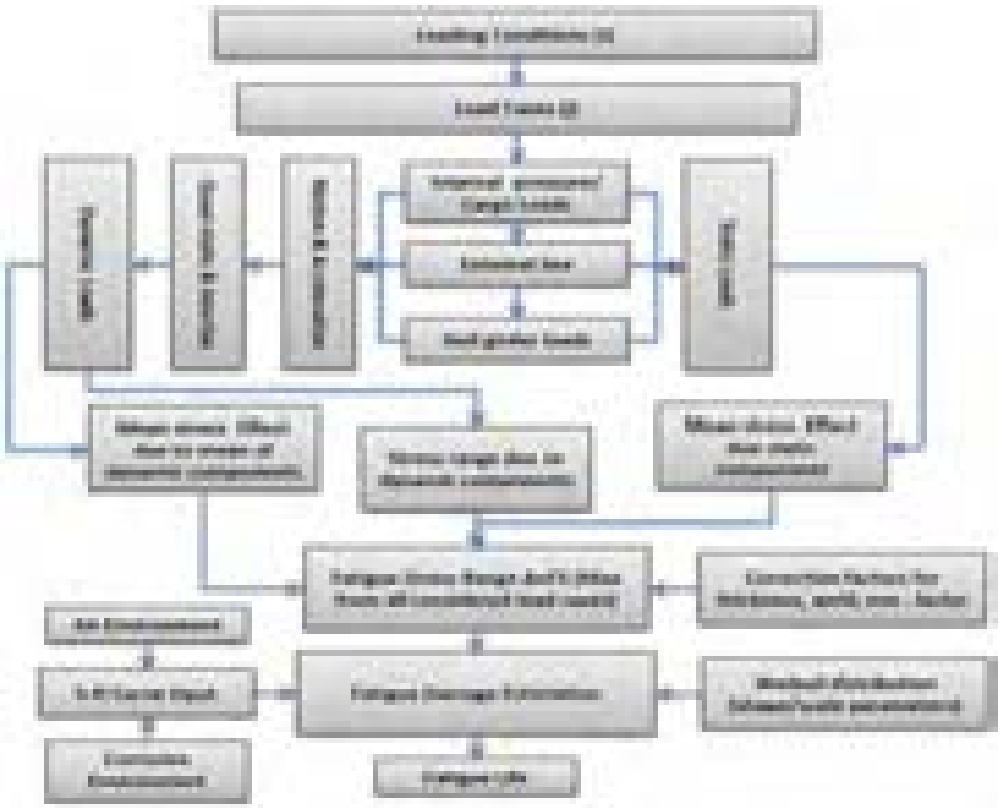


Figure 1.7.3: Procedure for simplified fatigue using beam theory

1.8 Loads

1.8.1 The loads induced in static and dynamic condition are to be considered for fatigue analysis of ship structures which are given below:

- Hull girder loads (bending moments, torsion moment)

- Local loads (wave pressure, inertia and quasi-static loads)

1.8.2 The load cases to be considered for fatigue assessment are given in Chapter 5, Section 2, 2.5. The design load scenario for fatigue assessment is defined in Chapter 5, Section 6, 6.3.

1.8.3 The following loads are not considered for fatigue analysis in these rules:

- dynamic loads resulting from main engine or propeller induced vibratory forces

- dynamic loads due to natural hull girder vibration (springing response)

- dynamic loads due to wave induced hull girder vibrations

- transient loads such as thermal stresses.

- low cycle loads such as cargo variations (cargo loading and unloading) or impact loads such as slamming, sloshing in partially filled tanks

1.9 Loading Conditions

1.9.1 Fatigue analyses are to be carried out for representative loading conditions according to the ship's intended operations.

1.9.2 For novel ship types for which standard loading conditions are generally not defined, the selection of loading conditions is to be performed in consultation with IRS.

1.9.3 Parameters related to a particular loading condition are denoted with a (j) subscript in the rules.

1.9.4 The loading conditions to be considered for some specific ship types and corresponding fraction of time for each loading condition, α(j), are defined in Table 1.9.1. These values are to be used if designers have not specified any values, for corresponding loading conditions.

Table 1.9.1: Standard values for fraction of time in each loading conditions to be used for the fatigue analysis

Vessel type	Full load condition	Normal/partial ballast condition
Tanker	0.5	0.5
Bulk carrier and dry Cargo ship[2)]	0.5	0.5
MPV/ general dry cargo ship[4)]	0.6[3)]	0.4[3)]
Liquefied gas carriers	0.5	0.5
Container ships	0.75[1)]	0.25
Ore Carrier[2)]	0.5	0.5
Passenger Ship[5)]	0.75	0.25
RO-RO[5)]	0.75	0.25
Others [6)]	--	--

1) Two loading conditions, a minimum and maximum hogging condition, is to be checked separately. The lowest fatigue life from the two loading conditions is representative of the calculated fatigue life. For each loading condition, the ballast tank is to be considered as full (to the tank top) 70% and empty 30% of the time, and the fatigue damage is to be calculated as the sum of these two contributions.

2) Applicable for ore carrier, bulk carrier and dry cargo ships except MPV/general dry cargo ships having a long centre cargo hold.

3) For each loading condition, the ballast tank shall be considered as full (to the tank top) 30% and empty 70% of the time, and the fatigue damage shall be calculated as the sum of these two contributions for each loading condition. In the partial condition with empty ballast tanks, bulk cargo pressures should be considered for the hold filled with 70% of the maximum cargo mass and a density of $\rho_c = 1.0$.
4) Applicable for Multi Purpose Vessels (MPV)/general dry cargo ships having a long centre cargo hold.
5) Values used when global FE analysis is required. Prescriptive assessment of longitudinal end connections is not required for passenger vessels.
6) Other ships will be specifically considered.

1.10 Reference Co-ordinate System

1.10.1 The ship geometry, motion, accelerations and loads are defined with respect to right-handed coordinate system are, as shown in Figure 1.10.1.

1.10.2 The sign conventions (Refer Figure 1.10.2) for hull girder loads are as follows:

- The vertical bending moments M_{sw} and M_{wv} are positive when they induce tensile stresses in the strength deck (hogging bending moment) and negative when they induce tensile stresses in the bottom (sagging bending moment).

- The vertical shear forces Q_{sw}, Q_{wv} are positive in the case of downward resulting forces acting aft of the transverse section and upward resulting forces acting forward of the transverse section under consideration.

- The horizontal bending moment M_{wh} is positive when it induces tensile stresses in the starboard side and negative when it induces tensile stresses in the port side.

- The torsional moment M_{wt} is positive in the case of resulting moment acting aft of the transverse section following negative rotation around the X-axis, and of resulting moment acting forward of the transverse section following positive rotation around the X-axis.

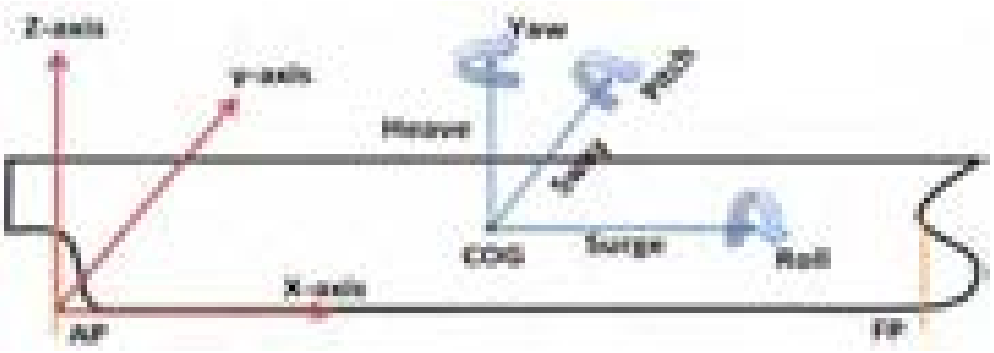


Figure 1.10.1: Reference coordinate system

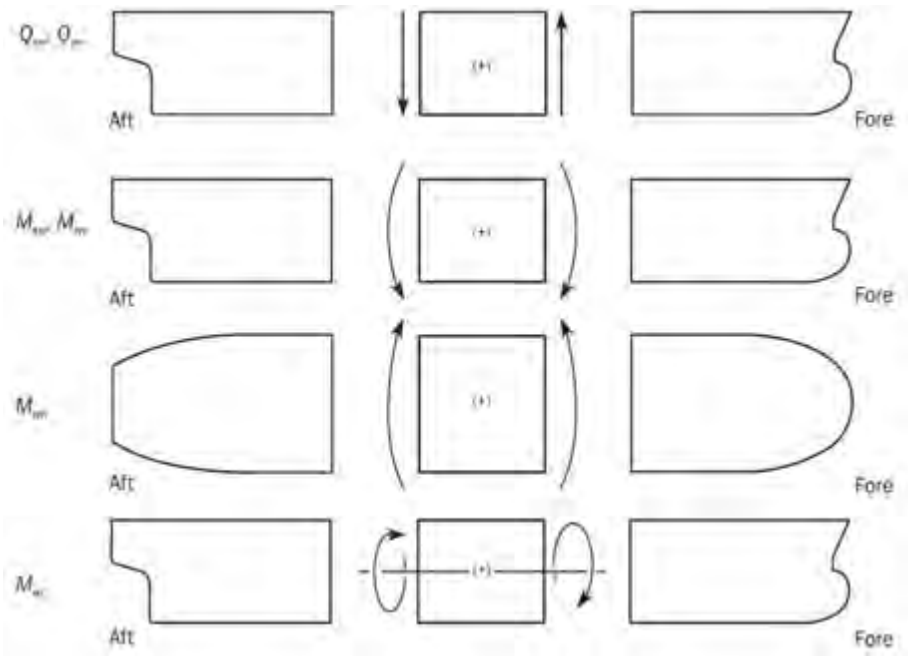


Figure 1.10.2: Sign convention for hull girder loads

Section 2

Fatigue Evaluation

Symbols

For symbols not defined in this section, Refer to Chapter 1 Section 2.

(i) = Suffix which denotes dynamic load case HSM, FSM, BSR-P, BSR-S, BSP-P, BSP-S, OST-P or OST-S specified in Chapter 5.

'i_1' denotes dynamic load case HSM-1, FSM-1, BSR-1P, BSR-1S, BSP-1P, BSP-1S, OST-1P or OST-1S

'i_2' denotes dynamic load case HSM-2, FSM-2, BSR-2P, BSR-2S, BSP-2P, BSP-2S, OST-2P or OST-2S.

(j) = Suffix which denotes loading condition.

T_C = Time in corrosive environment, in years, according to Table 2.5.1.

T_D = Design life, in years, to be taken as 25 years.

T_{DF} = Design fatigue life, in year, specified by the designer, but not to be taken less than 25 years.

T_F = Fatigue life, in year, calculated according to 2.5.

m = Inverse slope of the design S-N curve, as given in Table 2.4.1.1 for in-air environment and in Table 2.4.1.2 for corrosive environment. The inverse slope for S-N curves in-air environment changes from m to m+2 at N = 10^7 cycles.

n_{LC} = Number of applicable loading conditions, as defined in Section 1, Cl.1.9.

f_c = Correction factor as defined in Section 1, Cl.1.5.2.

f_{thick} = Correction factor for plate thickness effect given in Cl.2.3.5.

$f_{mean,i(j)}$ = Correction factor for mean stress effect given in Cl.2.3.4.

2.1 Fatigue analysis methodology

2.1.1 General

In general, assessment of the fatigue strength of structural members includes the following steps:

a) Calculation of stress ranges, according to 2.3.

b) Selection of the design S-N curve, according to 2.4.

c) Calculation of the cumulative damage and the fatigue life calculation, according to 2.5

d) Check the acceptance criteria as per 2.2

2.1.2 Cumulative damage

The fatigue assessment of the structure is based on the application of the Palmgren-Miner rule by estimating the linear cumulative fatigue damage D taken as:

$$D = \sum_{i=1}^{n_{tot}} \frac{n_i}{N_i}$$

where:

n_i = Number of cycles at stress range $\Delta\sigma_i$.

N_i = Number of cycles to failure at stress range $\Delta\sigma_i$.

n_{tot} = Total number of stress range blocks.

i = Stress range block index.

As the long-term stress range distribution of a structural detail in a ship can be described by a two-parameter Weibull distribution, fatigue damage can be obtained by means of a closed form equation, as given in 2.5.

2.2 Acceptance criteria

The design fatigue life, T_{DF}, is to be 25 years, unless a higher value is specified by the designer.

The calculated fatigue life, T_F, is to comply with the following formula:

$$T_F \geq T_{DF}$$

T_{DF} = Design fatigue life, in year, specified by the designer, but not to be taken less than 25 years.

T_F = Fatigue life, in year, calculated according to Cl.2.5.5.

2.3 Reference stresses for fatigue assessment

2.3.1 Fatigue stress

The fatigue stress for each load case of each loading condition is defined in Cl.2.3.2 for welded joints and in Cl.2.3.3 for base material free edge.

The stress range of each loading condition (j) to be considered is the stress range obtained from the predominant load case.

$$\Delta\sigma_{FS,(j)} = \max_i\left(\Delta\sigma_{FS,i(j)}\right)$$

where:

$\Delta\sigma_{FS,i(j)}$ = Fatigue stress range, in [N/mm²], for load case (i) of loading condition (j), as defined in Cl.2.3.2 for welded joints and in Cl.2.3.3 for base material free edge.

2.3.2 Welded joints

For welded joints, the fatigue stress $\Delta\sigma_{FS,i(j)}$, in [N/mm²], corrected for mean stress effect, thickness effect and warping effect, is taken as:

- For simplified stress analysis:

$$\Delta\sigma_{FS,i(j)} = f_{mean,i(j)}.f_{thick}.f_{warp}.\Delta\sigma_{HS,i(j)}$$

- For FE analysis:

2.3.2.1 For web-stiffened cruciform joints:

$$\Delta\sigma_{FS,i(j)} = f_w.f_s.\max\left(\Delta\sigma_{FS1,i(j)}, \Delta\sigma_{FS2,i(j)}\right)$$

2.3.2.2 For other joints:

$$\Delta\sigma_{FS,i(j)} = \max(\text{SideL}, \text{SideR})\left[\max\left(\Delta\sigma_{FS1,i(j)}, \Delta\sigma_{FS2,i(j)}\right)\right]$$

where:

f_w = Correction factor for the effect of stress gradient along weld line = 0.96

f_s = Correction factor for the effect of supporting member = 0.95

$\Delta\sigma_{HS,i(j)}$ = Hot spot stress range, in [N/mm²], due to dynamic loads in load case (i) of loading condition (j) given in Section 3, Cl.3.9.1.

$\Delta\sigma_{FS1,i(j)}$= Fatigue stress range, in [N/mm²], due to the principal hot spot stress range $\Delta\sigma_{HS1,i(j)}$

$$\Delta\sigma_{FS1,i(j)} = f_{mean1,i(j)}.f_{thick}.f_c.\Delta\sigma_{HS1,i(j)}$$

$\Delta\sigma_{FS2,i(j)}$= Fatigue stress range, in [N/mm²], due to the principal hot spot stress range $\Delta\sigma_{HS2,i(j)}$

$$\Delta\sigma_{FS1,i(j)} = 0.9.f_{mean2,i(j)}.f_{thick}.f_c.\Delta\sigma_{HS2,i(j)}$$

SideL, SideR = Left and right side respectively of the line A-A as shown in Section 4, Cl.4.5, Figure 4.5.1 and Figure 4.5.2.

$f_{mean1,i(j)}$ = Correction factor for mean stress effect given in Cl.2.3.4.

$f_{mean2,i(j)}$ = Correction factor for mean stress effect given in Cl.2.3.4.

f_{warp} = Correction factor due to warping effect, taken as:

= 1.07 for the deck longitudinal stiffener of bulk carrier, the closest to the longitudinal hatch coaming in way of the hatch corner as shown in Figure 2.3.1 except f_{warp} = 1.0 when OST is not the dominant load case for all loading conditions.

= 1.04 for following deck longitudinal stiffeners of bulk carrier, except $f_{warp} = 1.0$ when OST is not the dominant load case for all loading conditions.

- The closest stiffener to the longitudinal hatch coaming at one web frame away from the hatch corner, in way of the hatch opening as shown in Figure 2.3.1,

- The second closest stiffener away from the longitudinal hatch coaming in way of the hatch corner as shown in Figure 2.3.1,

= 1.0 for the other cases.

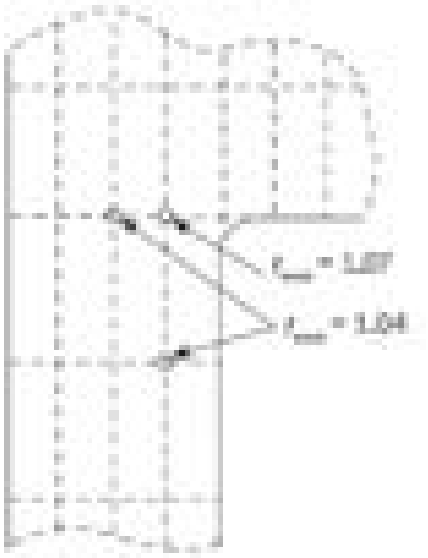


Figure 2.3.1: Warping effect on deck longitudinal stiffeners of bulk carrier

$\Delta\sigma_{HS1,i(j)}$ = Principal hot spot stress ranges, in [N/mm^2], due to dynamic loads for loading condition (j) which acts within ±45° of the perpendicular to the weld toe, determined in Section 4, Cl.4.5.4 and Section 4, Cl. 4.6.2.4 for the two types of shell elements (4-node or 8-node).

$\Delta\sigma_{HS2,i(j)}$ = Principal hot spot stress ranges, in [N/mm^2], due to dynamic loads for loading condition (j) which acts outside ±45° of the perpendicular to the weld toe, determined in Section 4, Cl.4.5.4 and Section 4, Cl.4.6.2.4 for the two types of shell elements (4-node or 8-node).

2.3.3 Base material free edge

For base material free edge, the fatigue stress range, $\Delta\sigma_{FS,(j)}$ in [N/mm^2], is taken as the local stress range at free edge, $\Delta\sigma_{BS,(j)}$, as defined in Section 1, Cl.1.2.6 with correction factors:

$$\Delta\sigma_{FS1,i(j)} = K_{sf}.f_{material}.f_{mean1,i(j)}.f_{thick}.f_c.\Delta\sigma_{BS1,i(j)}$$

where:

K_{sf} = Surface finishing factor for base material given in Cl.2.4.2.2.

$f_{material}$ = Correction factor for material strength, given in Cl.2.3.6.

$\Delta\sigma_{BS,i(j)}$ = Local stress range, in [N/mm^2], due to dynamic loads in load case (i) of loading condition (j) taken as:

$$\Delta\sigma_{BS,i(j)} = \left|\sigma_{BS,i1(j)} - \sigma_{BS,i2(j)}\right|$$

$\sigma_{BS,i1(j)}, \sigma_{BS,i2(j)}$ = Local stress, in [N/mm^2], in load case 'i_1' and 'i_2' of loading condition (j), obtained by very fine mesh FE analysis specified in Section 4.

2.3.4 Mean stress effect

2.3.4.1 Correction factor for mean stress effect

The mean stress correction factor to be considered for each principal hot spot stress range of welded joint, $\Delta\sigma_{HS1,i(j)}$, or for local stress range at free edge, $\sigma_{BS,i1(j)}$, is taken as:

(a) For welded joint:

$$f_{mean,i(j)} = \begin{Bmatrix} \min\left[1.0, 0.9 + 0.2\frac{\sigma_{mcor,i(j)}}{2\Delta\sigma_{HS,i(j)}}\right] \text{ for } \sigma_{m,cor,i(j)} \geq 0 \\ \max\left[0.3, 0.9 + 0.8\frac{\sigma_{mcor,i(j)}}{2\Delta\sigma_{HS,i(j)}}\right] \text{ for } \sigma_{m,cor,i(j)} < 0 \end{Bmatrix}$$

(b) For material:

$$f_{mean,i(j)} = \begin{Bmatrix} \min\left[1.0, 0.8 + 0.4\frac{\sigma_{mcor,i(j)}}{2\Delta\sigma_{BS,i(j)}}\right] \text{ for } \sigma_{m,cor,i(j)} \geq 0 \\ \max\left[0.3, 0.8 + \frac{\sigma_{mcor,i(j)}}{2\Delta\sigma_{BS,i(j)}}\right] \text{ for } \sigma_{m,cor,i(j)} < 0 \end{Bmatrix}$$

where,

$$\sigma_{mcor,i(j)} = \begin{Bmatrix} \sigma_{mean,i(j)} \text{ for } \sigma_{max} \leq R_{eEq} \\ R_{eEq} - \sigma_{max} + \sigma_{mean,i(j)} \text{ for } \sigma_{max} > R_{eEq} \end{Bmatrix}$$

$$\sigma_{max} = \begin{Bmatrix} \text{maxi, (j)}\left(\Delta\sigma_{HS,i(j)} + \sigma_{mean,i(j)}\right) \text{ for welded joint} \\ \text{maxi, (j)}\left(\Delta\sigma_{BS,i(j)} + \sigma_{mean,i(j)}\right) \text{ for base material} \end{Bmatrix}$$

R_{eEq} = max (315,R_{eH})

$\sigma_{mean,i(j)}$ = Fatigue mean stress, in [N/mm^2], for base material calculated according to Cl.2.3.4.2 or for welded joint calculated according to Cl.2.3.4.3 or 2.3.4.4, as applicable.

2.3.4.2 Mean stress for base material free edge
The fatigue mean stress for base material free edge, , $\sigma_{mean,i(j)}$, in [N/mm^2], due to static and dynamic load cases 'i$_1$' and 'i$_2$' of loading condition (j) is calculated by the following formula based on local stress:

$$\sigma_{meani,(j)} = \frac{\sigma_{BS,i1(j)} + \sigma_{BS,i2(j)}}{2}$$

2.3.4.3 Mean stress for simplified stress analysis

The fatigue mean stress to be considered for welded joint assessed by the simplified stress analysis is to be obtained from Section 3, Cl.3.3.5.

The fatigue mean stresses for welded joint due to static and dynamic loads, $\sigma_{meani,(j),pX}$ and $\sigma_{meani,(j),pY}$, in [N/mm^2], for load cases 'i$_1$' and 'i$_2$' of loading condition (j), belonging to the two principal hot spot stress range directions, pX and pY, is calculated by the following formula based on hot spot stress components as defined in Section 4, Cl.4.5.3, 4.6.2.4 and 4.5.5:

θ = Angle between the direction x of the element coordinate system and the principal direction pX of the principal hot spot stress range coordinate system (Section 4, Cl.4.5.3, 4.6.2.4 and 4.5.5). The direction x of the element coordinate system is defined as the normal to the weld toe.

2.3.4.4 Mean stress for FE analysis

$$\sigma_{meani,(j),pX} = \frac{\left(\sigma_{HS,i1(j)}\right)_{xx} + \left(\sigma_{HS,i2(j)}\right)_{xx} + \left(\sigma_{HS,i1(j)}\right)_{yy} + \left(\sigma_{HS,i2(j)}\right)_{yy}}{4} + \frac{\left(\sigma_{HS,i1(j)}\right)_{xx} + \left(\sigma_{HS,i2(j)}\right)_{xx} - \left(\sigma_{HS,i1(j)}\right)_{yy} - \left(\sigma_{HS,i2(j)}\right)_{yy}}{4}.\cos 2\theta + \frac{\left(\sigma_{HS,i1(j)}\right)_{xy} + \left(\sigma_{HS,i2(j)}\right)_{xy}}{2}.\sin 2\theta$$

$$\sigma_{meani,(j),pY} = \frac{\left(\sigma_{HS,i1(j)}\right)_{xx} + \left(\sigma_{HS,i2(j)}\right)_{xx} + \left(\sigma_{HS,i1(j)}\right)_{yy} + \left(\sigma_{HS,i2(j)}\right)_{yy}}{4} - \frac{\left(\sigma_{HS,i1(j)}\right)_{xx} + \left(\sigma_{HS,i2(j)}\right)_{xx} - \left(\sigma_{HS,i1(j)}\right)_{yy} - \left(\sigma_{HS,i2(j)}\right)_{yy}}{4}.\cos 2\theta - \frac{\left(\sigma_{HS,i1(j)}\right)_{xy} + \left(\sigma_{HS,i2(j)}\right)_{xy}}{2}.\sin 2\theta$$

The one of the two mean stresses $\sigma_{meani,(j),pX}$ and $\sigma_{meani,(j),pY}$ which has a principal stress direction with an absolute value less than 45[deg] is defined as $\sigma_{mean1,i(j)}$, belonging to $\sigma_{HS1,i(j)}$. The other mean stress is defined as $\sigma_{mean2,i(j)}$ belonging to $\sigma_{HS2,i(j)}$.

2.3.5 Thickness effect

2.3.5.1 General

Plate thickness primarily influences the fatigue strength of welded joints through the effect of geometry of the weld, and through-thickness stress distribution. The correction factor, f_{thick}, for plate thickness effect is taken as:

$$f_{thick} = 1 \qquad \text{for } t_{eff} \le 22[\text{mm}]$$

$$f_{thick} = \left(\frac{t_{eff}}{22}\right)^{n} \quad \text{for } t_{eff} > 22[\text{mm}]$$

where

n = Thickness exponent given in Table 2.3.5 and Table 2.4.2.2 respectively for welded and non-welded joints. n is to be selected according to the considered stress direction. For this selection, $\Delta\sigma_{HS1}$ and $\Delta\sigma_{HS2}$ are considered perpendicular and parallel to the weld respectively.

t_{eff} = Effective thickness of welds. If values not specified t_{n50} to be used.

t_{n50} = Net thickness, in [mm] of the considered member in way of the hot spot for welded joints or base material free edge.

2.3.5.2 For simplified stress analysis, the net thickness to be considered for stiffeners is as follows:

- Flat bar and Bulb profile: no thickness factor correction,

- Angle bar and T-bar: flange net thickness

2.3.5.3 For FE analysis, the effective thickness to be considered is the thickness of the member where the crack is likely to initiate and propagate.

- For 90[deg] attachments, i.e. cruciform welded joints, transverse T-joints and plates with transverse attachment, the net thickness to be considered is to be taken as:

$$t_{eff} = \min\left(\frac{d}{2}, t_{1n50}\right)$$

where

d = Toe distance in [mm], as shown in Figure 2.3.5, taken as:

$$d = t_{2n50} + 2\ell_{leg}$$

t_{1n50} = Net thickness, in [mm], of the continuous plate as shown in Figure 2.3.5.

t_{2n50} = Net thickness, in [mm], of the transverse attach plate where the hot spot is assessed, as shown in Figure 2.3.5.

ℓ_{leg} = Fillet weld leg length in [mm].

When post-weld treatment methods are applied to improve the fatigue life of considered welded joint, the thickness exponent is provided in 2.6.

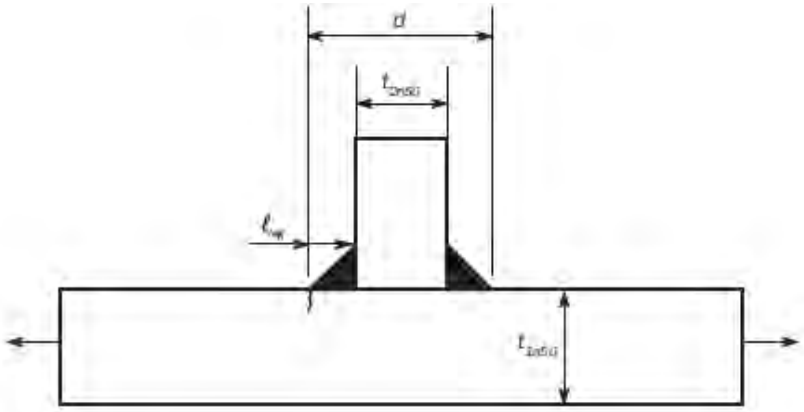
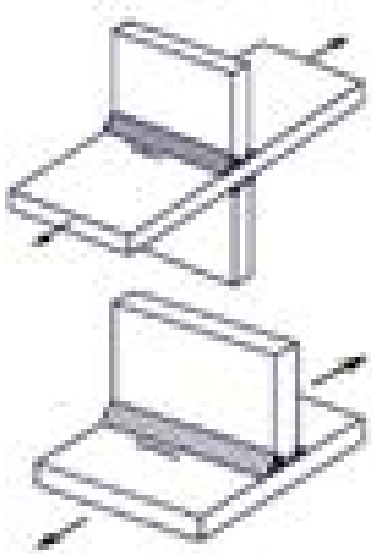
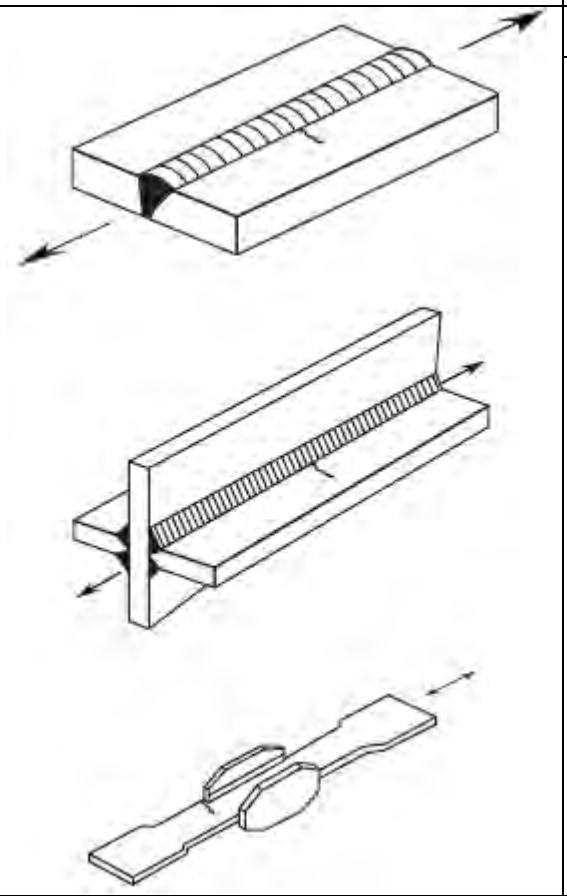
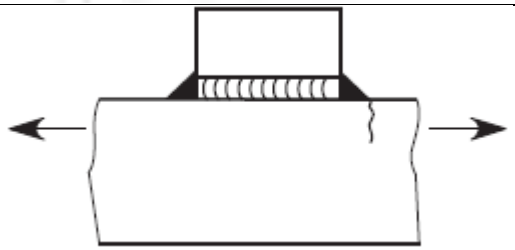
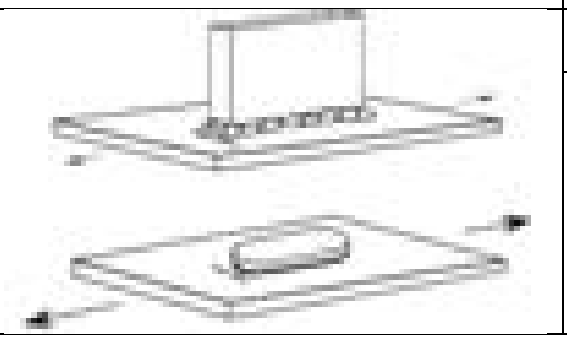
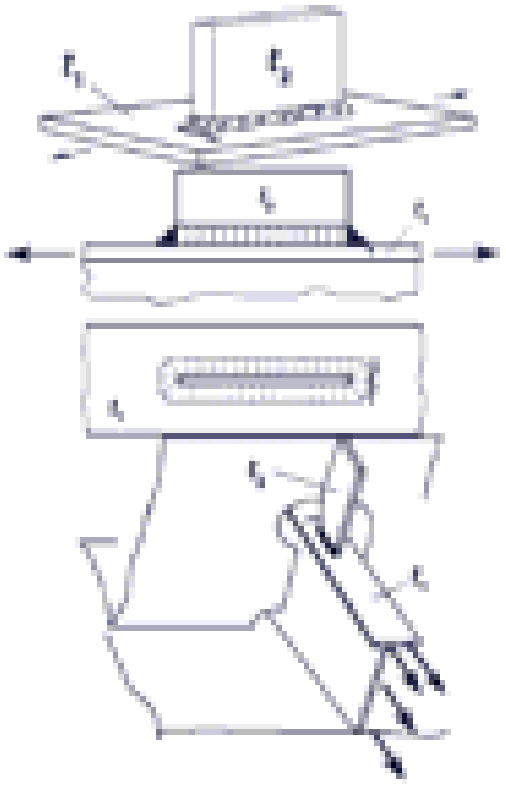


Figure 2.3.5: Toe distance for cruciform welded joints, transverse T-joints and plates with transverse attachment

Table 2.3.5: Welded joints: thickness exponents

Sl. No	Joint category description	Geometry	Condition	n
1	Cruciform joints, transverse T-joints, plates with transverse attachments		As-welded	0.25
			Weld toe treated by post-weld improvement method	0.2

2	Transverse butt welds		As-welded	0.2
			Ground flush or weld toe treated by post-weld improvement method	0.1
3	Longitudinal welds or attachments to plate edges		Any	0.1
			Weld toe treated by post-weld improvement method	0.1
4	Longitudinal attachments on the flat bar or bulb profile		Any	0.0
			Weld toe treated by post-weld improvement method (1)	0.0
5	Longitudinal attachments and doubling plates		As welded	0.2
			Weld toe treated by post-weld improvement method	0.1

6	Longitudinal attachments and doubling plates supported longitudinally		As-welded $t_2 \geq 0.33\ t_1$ else	0.1 0.0
			Weld toe treated by post-weld improvement method (1)	0
(1) No benefit applicable for post-weld treatment of longitudinal end connections				

2.3.5.4 For base material

The thickness effect is not considered for base material and free plate edges assuming good workmanship practice (S-N curve B and C) and when adequate protective measures are taken against wear, tear and corrosion, e.g. from friction from cargo or cargo handling appliances.

2.3.6 Material factor

2.3.6.1 Welded joints

For welded joints, the fatigue strength is independent on the steel yield strength. So, the correction factor for material strength, $f_{material}$, is taken as 1.

2.3.6.2 Non welded details

For non-welded details such as cut plate edges, the fatigue strength increases together with the yield stress of the base material. The fatigue stress range, $\Delta\sigma_{FS,i(j)}$, needs to be corrected for material strength, $f_{material}$, taken as:

$$f_{material} = \frac{1200}{965 + \sigma_y}$$

2.3.7 Environmental factor

The environmental factor is a correction factor representing the ratio of the stress range based on the vertical wave bending moment from a specific trade and the North Atlantic wave environment. For prescriptive loads, the following environmental factors are used

$f_e = 0.8$ for worldwide wave environment

$f_e = 1.0$ for North Atlantic wave environment.

Note:

If hydrodynamic analysis is carried out using a scatter diagram, the environmental factor to be taken as 1.0.

2.4 S-N Curves

2.4.1 Design S-N curves

2.4.1.1 Unless supported by direct measurements, the U.K. HSE (previously DEn) Basic S-N Curves are to be used for determining the fatigue life.

2.4.1.2 The fatigue assessment for welded joints and base material is based on use of S-N curves which are obtained from fatigue tests. The design S-N curves are established at two standard deviations below the mean S-N curves corresponding to 50% of probability of survival for relevant experimental data. Design S-N curves given in Table 2.4.1.1 and Table 2.4.1.2 correspond to a probability of survival of 97.7%.

2.4.1.3 S-N curve scope of application

These S-N curves are applicable to steels with minimum yield strength less than 400[N/mm^2]. For steels with higher yield strength, data obtained from an approved test program are to be used.

2.4.1.4 In-air environment

The basic design curves in-air environment shown in Figure 2.4.1.1 are represented by linear relationships between log (Δσ) and log (N) as follows:

$$\log N = \log(K_2) - m.\log(\Delta\sigma)$$

where

$$\log(K_2) = \log(K_1) - 2.\log(\delta)$$

K_1 = Constant related to mean S-N curve, as given in Table 2.4.1.1.

K_2 = Constant related to design S-N curve, as given in Table 2.4.1.1.

δ = Standard deviation of log (N), as given in Table 2.4.1.1.

$\Delta\sigma_q$ = Stress range at N = 10^7 cycles related to design S-N curve, in [N/mm^2], as given in Table 2.4.1.1.

Table 2.4.1.1: Basic S-N curve data, in-air environment

Class	K_1		m	Standard deviation δ	K_2	Design stress range at 10^7 cycles	Design stress range at $2x10^6$ cycles
	K_1	$\log_{10} K_1$		$\log_{10} \delta$		$\Delta\sigma_q$[N/mm^2]	[N/mm^2]
B	2.343E15	15.3697	4.0	0.1821	1.01E15	100.2	149.9
C	1.082E14	14.0342	3.5	0.2041	4.23E13	78.2	123.9
D	3.988E12	12.6007	3.0	0.2095	1.52E12	53.4	91.3
E	3.289E12	12.5169	3.0	0.2509	1.04E12	47.0	80.4
F	1.726E12	12.2370	3.0	0.2183	0.63E12	39.8	68.1
F2	1.231E12	12.0900	3.0	0.2279	0.43E12	35.1	59.9
G	0.566E12	11.7525	3.0	0.1793	0.25E12	29.2	50.0
W	0.368E12	11.5662	3.0	0.1846	0.16E12	25.1	43.1

2.4.1.5 Corrosive environment

The basic design curves for corrosive environment shown in Fig. 2.4.1.2 are represented by linear relationships between log (Δσ) and log (N) as follows:

$$\log N = \log(K_2) - m.\log(\Delta\sigma)$$

N = Predicted number of cycles to failure under stress range Δσ.

K_2 = Constant related to design S-N curve, as given in Table 2.4.1.2.

Table 2.4.1.2: Basic S-N curve data, corrosive environment			
Class	**m**	**K_2**	**Design stress range at $2x10^6$ Cycles [N/mm²]**
B_{Corr}	4.0	5.05E14	126.1
C_{Corr}	3.5	2.12E13	101.6
D_{Corr}	3.0	7.60E11	72.4
E_{Corr}	3.0	5.18E11	63.7
F_{Corr}	3.0	3.16E11	54.0
$F2_{Corr}$	3.0	2.15E11	47.6
G_{Corr}	3.0	1.24E11	39.6
W_{Corr}	3.0	7.86E11	34.0

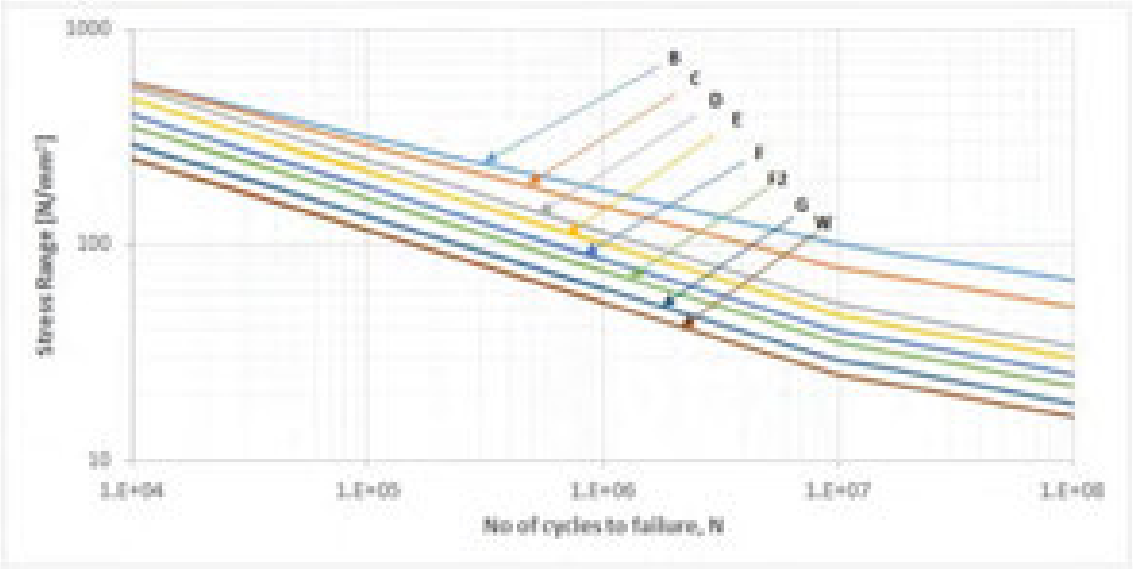


Figure 2.4.1.1: Basic design S-N curves, in-air environment

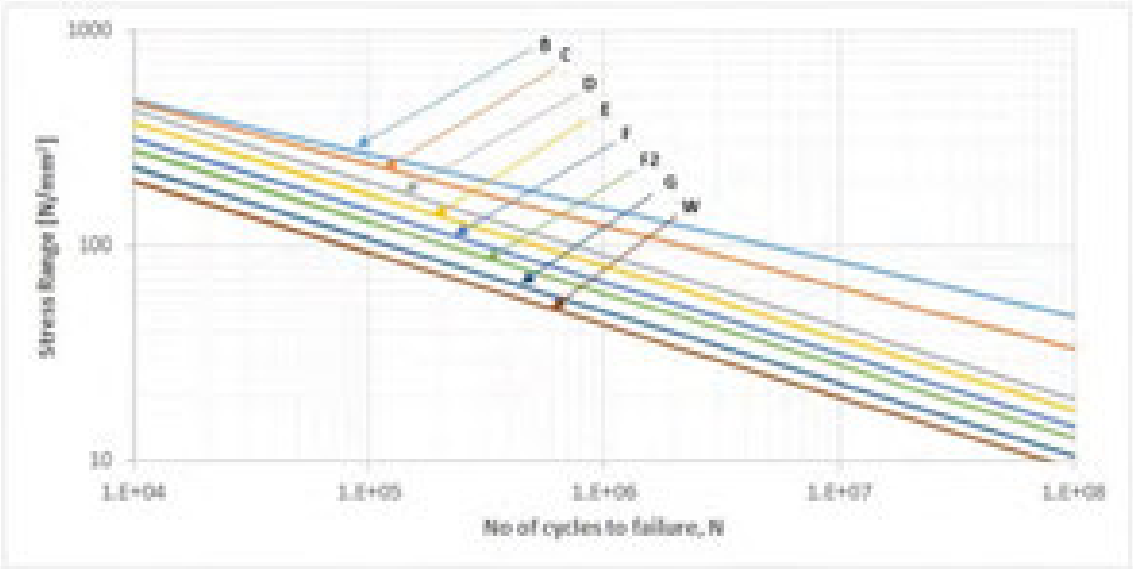


Figure 2.4.1.2: Basic design S-N curves, corrosive environment

2.4.2 Selection of S-N curves

2.4.2.1 General

The HSE S-N Curves for non-tubular joints consist of eight curves, as shown in Table 2.4.1.1 identified by B, C, D, E, F, F2, G and W categories. These curves give the relationship between the nominal stress range and the number of constant amplitude load cycles to failure. Each curve represents a class of welded details depending upon:

- The geometrical arrangement of the detail,

- The direction of the fluctuating stress relative to the detail

- The method of fabrication and inspection of the detail.

The Table 2.4.2.1 provide the summary of details based on above factors. More information can be referred from the UK HSE (1984).

Table 2.4.2.1: Welded Joint Classification for SN curve

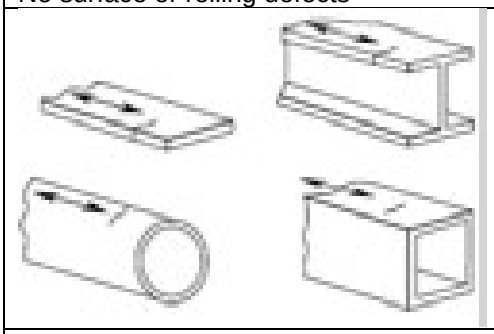
SN curve category for Joint Classification	**Description [1]**
Category B	1) Parent metal in the as-rolled condition with no flame-cut edges or with flame-cut edges ground or machined 2) Full penetration butt welds with the weld cap ground flush with the surface and with the weld proved to be free from defects by NDT.
Category C	1) Parent material in the as-rolled condition with automatic flame-cut edges and ensured to be free from cracks. 2) Butt or fillet welds made by an automatic submerged or open arc process and with no stop-start positions within their length. 3) With the weld cap ground flush with the surface and with the weld proved to be free from significant defects by NDT.
Category D	1) As C(2) but with stop-start positions within the length 2) With the welds made either manually or by an automatic process other than submerged arc and in flat position
Category E	1) Intermittent fillet welds
Category F	1) Welds made on a permanent backing strip between plates of equal width and thickness or tapered with a maximum slope of 1/4. 2) Parent material (of the stressed member) or ends of butt or fillet welded attachments (parallel to the direction of applied stresses) on stressed members: - attachment length l ≤ 150[mm] - edge distance d ≥ 10[mm] 3) Parent metal of cruciform or T Joints made with full penetration welds and with any undercut at the corners of the member ground out 4) Parent metal at the toe of weld connection of web stiffeners to girder flanges: edge distance d ≥ 10[mm]. 5) As E(1) but adjacent to cut-outs.
Category F2	1) Parent material (of the stressed member) or ends of butt or fillet welded attachments (parallel to the direction of applied stresses) on stressed members: - attachment length l > 150[mm] - edge distance d ≥ 10[mm] 2) As F(3) with partial penetration or fillet welds with any undercut at the corners of the member ground out. Parent metal of load-carrying fillet welds transverse to the direction of stresses (member X): edge distance d ≥ 10[mm]

Category G	1) Parent material (of the stressed member) at toes or ends of butt or fillet welded attachments on or within 10[mm] of edges or corners. 2) Parent metal of load-carrying fillet welds transverse to the direction of stresses (member X): edge distance d < 10[mm] 3) Parent metal of load-carrying fillet welds parallel to the direction of stresses, with the weld end on plate edge (member Y). 4) Parent metal at the toe of weld connection of web stiffeners to girder flanges: edge distance d < 10[mm]
[1] For additional guidance also refer, Offshore Installations – Guidance on Design, Construction and Certification: Health & Safety Executive, UK.	

2.4.2.2 Surface finishing factor

The S-N curve C is applicable to most of non-welded locations taking into account the likelihood of some notching from corrosion, wear and tear in service with surface finishing factor as given in Table 2.4.2.2. Higher surface finishing quality may be applied in using S-N curve B as given in Table 2.4.2.2, provided adequate protective measures are taken against wear, tear and corrosion and finite element analysis according to Section 4.3 and 4.4 is carried out.

Table 2.4.2.2: Non-welded joints: thickness exponent and surface finishing factor

Joint configuration, fatigue crack location and stress direction	Edge cutting process	Edge treatment	Surface finishing	n	K_{sf}	S-N Curve
Base material, Rolled or extruded plates and sections as well as seamless pipes. No surface or rolling defects 	N/A	N/A	No surface nor roll defect (1) (2)	0	0.94	B
Cut edges						
	Machine cutting e.g. by a thermal process or sheared edge cutting	Cutting edges chamfered or rounded by means of smooth grinding, groove direction parallel to the loading direction	Smooth surface free of cracks and notches (1) (2)	0.1	1.00	B
		Cutting edges	Smooth	0.1	1.07	B

		broken or rounded	surface free of cracks and notches (1) (2)			
		No edge treatment	Surface free of cracks and severe notches (inspection procedure) (1) (2)	0.1	1.00	C
	Manually thermally cut e.g. by flame cutting	No edge treatment	Surface free of cracks and severe notches (inspection procedure) (1) (2)	0.1	1.24	C
(1) Stress increases due to geometry of cut-outs to be considered. (2) Fine mesh FE analysis according to Section 4.3 and 4.4.						

2.5 Fatigue damage calculation

2.5.1 General

2.5.1.1 The design fatigue life is divided into a number of time periods due to different loading conditions and due to limitation of the corrosion protection.

2.5.1.2 It is assumed that the corrosion protection (i.e. coating system) is only effective for a limited number of years during which the structural details are protected, i.e. in-air environment. During the remaining part of the design life as specified in Table 2.5.1, the structural details are unprotected i.e. exposed to corrosive environment.

The elementary fatigue damage, given in Cl.2.5.2, is the damage accumulated during a specific loading condition (j) associated with a specific environmental condition either protected condition, i.e. in-air environment, or unprotected condition, i.e. corrosive environment.

The combined fatigue damage, given in Cl.2.5.3, is the combination of damage accumulated for a specific loading condition (j) for the in-air and corrosive environment time.

Total fatigue damage, given in Cl.2.5.4, is the sum of the combined fatigue damages obtained for all loading conditions.

2.5.2 Elementary fatigue damage

The elementary fatigue damage for each fatigue loading condition (j) is to be calculated independently for both protected in-air environment and unprotected corrosive environment, based on the fatigue stress range obtained for the predominant load case and calculated based on a closed form formulation as follows:

$$D_{E(j)} = \frac{\alpha_{(j)}.N_D}{K_2} \frac{\Delta\sigma_{FS,(j)}^m}{(\ln N_R)^{m/\xi}} .\mu_{(j)}.\Gamma\left(1+\frac{m}{\xi}\right)$$

where:

N_D = Total number of wave cycles experienced by ship during the design fatigue life, taken as:

$$N_D = \frac{31.557x10^6(f_0 T_D)}{4\log L}$$

f_0 = Factor taking into account time in seagoing operations excluding time in loading and unloading, repairs, etc. taken as 0.85.

$\alpha_{(j)}$ = Fraction of time in each loading condition given in in Section 1.

$\Delta\sigma_{FS,(j)}$ = Fatigue stress range at the reference probability level of exceedance of 10^{-2}, in [N/mm^2].

N_R = Number of cycles corresponding to the reference probability of exceedance of 10^{-2} = 100.

ξ = Weibull shape parameter, = 1.

$\Gamma(x)$ = Complete Gamma function.

K_2 = Constant of the design S-N curve, as given in 2.4 for in-air environment and for corrosive environment.

$\mu_{(j)}$ = Coefficient taking into account the change of inverse slope of the S-N curve, m.

2.5.2.1 For in-air environment:

$$\mu_{(j)} = 1 - \frac{\left\{\gamma\left(1+\frac{m}{\xi},\upsilon_{(j)}\right) - \upsilon_{(j)}^{-\Delta m/\xi}.\gamma\left(1+\left(\frac{m+\Delta m}{\xi}\right),\upsilon_{(j)}\right)\right\}}{\Gamma\left(1+\frac{m}{\xi}\right)}$$

$$\upsilon_{(j)} = \left(\frac{\Delta\sigma_q}{\Delta\sigma_{FS,(j)}}\right)^{\xi} \ln N_R$$

2.5.2.2 For corrosive environment:

$$\mu_{(j)} = 1.0$$

$\gamma(a,x)$ = Incomplete Gamma function.

$\Delta\sigma_q$ = Stress range, in [N/mm^2], corresponding to the intersection of the two segments of design S-N curve at N = 10^7 cycles, as given in 2.4.

Δm = Change in inverse slope of S-N curve at N=10^7 cycles, = 2

2.5.3 Combined fatigue damage

The combined fatigue damage in protected in-air environment and unprotected corrosive environment for each loading condition (j) is to be calculated as follows:

$$D_{(j)} = D_{E,air(j)}.\frac{T_D - T_C}{T_D} + D_{E,corr(j)}.\frac{T_C}{T_D}$$

where:

$D_{E,air(j)}$ = The elementary fatigue damage for in-air environment for loading condition (j) given in Cl.2.5.2.

$D_{E,corr(j)}$ = The elementary fatigue damage for corrosive environment for loading condition (j) as calculated in Cl.2.5.2.

T_C = Time in corrosive environment, in years, according to Table 2.5.1.

T_D = Design life, in years, to be taken as 25 years.

Table 2.5.1: Time in corrosive environment, T_C

Location of weld joint or structural detail	Time in corrosive environment T_C, in years
Water ballast tank	10
Oil cargo tank	
Lower part1) of cargo hold for dry bulk cargo	
Cargo hold for dry bulk cargo except lower part [1)]	5
Void spaces	5
Fuel oil tanks	5
Cofferdam	
Dry cargo holds	
Hull external surfaces	
Other areas	5

1) Lower part including inner bottom and other structure with a height of 1.5[m] above the inner bottom.

2.5.4 Total fatigue damage

The total fatigue damage for all applicable loading conditions is calculated as follows:

$$D = \sum_{j=1}^{n_{LC}} D_{(j)}$$

where:

$D_{(j)}$ = Combined fatigue damage for each applicable loading condition, as given in Cl.2.5.3.

2.5.5 Fatigue life calculation

The fatigue life, T_F, is taken as:

$$T_F = \frac{T_D}{D_{air}} \text{ if } \frac{T_D}{D_{air}} \leq (T_D - T_C)$$

$$T_F = T_D - T_C + \left(\frac{T_D}{D_{air}} - T_D + T_C\right)\frac{D_{air}}{D_{corr}} \text{ otherwise}$$

where:

D_{air} = Total fatigue damage for all loading conditions in-air environment taken as:

$$D_{air} = \sum_{j=1}^{n_{LC}} D_{E,air(j)}$$

D_{corr} = Total fatigue damage for all loading conditions in corrosive environment taken as:

$$D_{corr} = \sum_{j=1}^{n_{LC}} D_{E,corr(j)}$$

2.6 Weld improvement methods

2.6.1 General

Post-weld fatigue strength improvement methods are to be considered as a supplementary means of achieving the required fatigue life, and subjected to quality control procedures. The benefit from post-weld treatment can only be applied for corrosion free condition and may only be considered provided that a protective coating is applied after the post-weld treatment and maintained during the design life time.

2.6.2 Limitation of the benefit of post-weld treatment

For structural details where the benefit of post-weld treatment is applicable, the calculated fatigue life at the design stage for the considered structural detail excluding the post-weld treatment effects, is not to be less than $T_{DF}/1.47$.

2.6.3 The calculated fatigue life at design stage excluding post-weld treatment effects is not to be less than 25 years.

2.6.4 Post-weld treatment at fabrication stage

There is one basic post-weld treatment method considered in these guideline to improve fatigue strength at the fabrication stage, i.e. weld geometry control and defect removal method by burr grinding.

2.6.5 Weld toe

The post-weld improvement method is applied to the weld toe to increase its fatigue life. The possibility of failure initiation at other locations is always to be considered. If the failure is shifted from the weld toe to the root by applying post-weld treatment, there may be no significant improvement in the overall fatigue performance of the joint. Improvements of the weld root cannot be expected from treatment applied to weld toe. A brief description of the method and the degree of improvement which can be achieved is given in Cl.2.6.6.

2.6.6 Weld toe burr grinding

2.6.6.1 The weld may be machined using a burr grinding tool to produce a favorable shape to reduce stress concentrations and remove defects at the weld toe, Refer Fig. 2.6.6.1. In order to eliminate defects, such as intrusions, undercuts and cold laps, the material in way of the weld toe is to be removed. The depth of grinding is to be at least 0.5[mm] below the bottom of any visible undercut. The total depth of the burr grinding is not to be greater than the lesser of 2[mm] and of 7% the local gross thickness of the machined plate. Any undercut not complying with this requirement is to be repaired by an approved method.

2.6.6.2 To avoid introducing a detrimental notch effect due to small radius grooves, the burr diameter is to be scaled to the plate thickness at the weld toe being ground. The diameter is

to be in the 10 to 25[mm] range for application to welded joints with plate thickness from 10 to 50 [mm]. The resulting root radius of the groove is to be no less than $0.25t_{as-built}$. The weld throat thickness and leg length after burr grinding must comply with the rule requirements or any increased weld sizes as indicated on the approved drawings.

The inspection procedure is to include a check of the weld toe radius, the depth of burr grinding, and confirmation that the weld toe undercut has been removed completely.

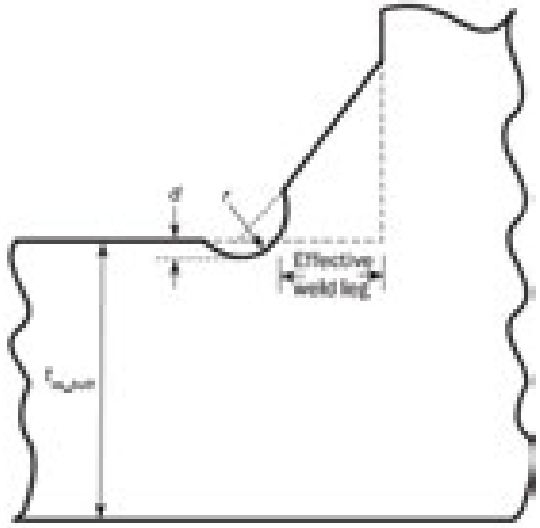


Figure 2.6.6.1: Details of ground weld toe geometry

2.6.7 Fatigue improvement factor

The benefit of burr grinding corresponds to an increase in fatigue strength by a factor of 1.3 (i.e. a reduction of the effective stress range by 1.3), reducing the damage in air to

$$\frac{D_{air}}{2.2}$$

where

D_{air} is the Fatigue damage in air as given in Cl.2.5.5.

2.6.8 Applicability

The application of post-weld improvement and fatigue improvement factor provided is subject to the following:

- The weld type complies with Cl.2.6.9.

- The weld improvement is effective or valid in improving the fatigue strength of structural details under high cycle fatigue conditions and do not apply to low-cycle fatigue conditions, i.e. when $N \leq 5\times10^4$, where N is the number of life cycles to failure.

- Unless otherwise specifically stated, the fatigue improvement factor is to be used for welds, joining steel plates which are between 6 and 50[mm] thick.

- This improvement can only be achieved in a corrosion free condition and may only be considered provided that a suitable protective coating is applied after the post-weld treatment and maintained during the design life time. Hot spots are to be protected from corrosive environment.

- Fatigue improvement factor is to be applied to as-welded transverse butt welds, as-welded T-joint and cruciform welds and as-welded longitudinal attachment welds excluding longitudinal end connections.

- In way of areas prone to mechanical damage, fatigue improvement may occur only if these are adequately protected.

- Treatment of inter-bead toes is required for large multi-pass welds as shown in Figure 2.6.8.

- The yard is to provide the list of details and their locations on the ship for which the post-weld treatment has been applied.

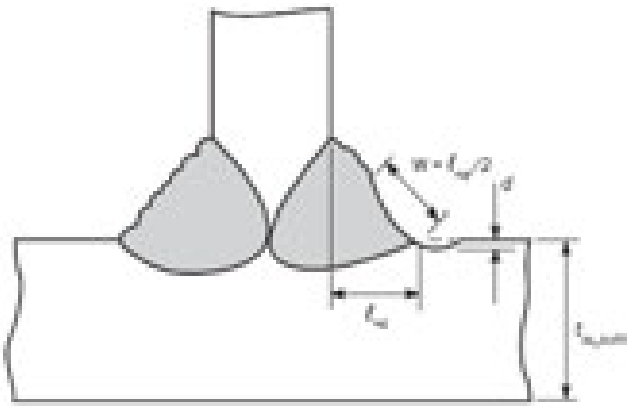


Figure 2.6.8: Extent of weld toe burr grinding to remove inter-bead toes on weld face

(ℓ_{reg}: Weld leg length, W: Width of groove, d: Depth of grinding to be between 0.5 and 1[mm])

2.6.9 Weld type for post-weld treatment

When weld improvements are planned, full or partial penetration welds with a minimum root face according to Chapter 16 are to be used to mitigate or to eliminate the possibility of cracking at the weld root.

Section 3

Simplified Stress Analysis

Symbols

For symbols not defined in this section, Refer to Chapter 1 Section 2.

ℓ_{bdg} = Effective bending span of stiffener, in m, as defined in Chapter 3, Section 7.

I_{y-n50} = Net vertical hull girder moment of inertia in [m^4], at the longitudinal position being considered.

I_{z-n50} = Net horizontal hull girder moment of inertia in [m^4], at the longitudinal position being considered.

Y = Transverse coordinate in [m], of the load calculation point under consideration.

z = Vertical coordinate in [m], of the load calculation point under consideration.

z_n = Distance in [m], from the baseline to the horizontal neutral axis.

K_a = Geometrical stress concentration factor for stress due to axial load given in Cl.3.7.2.

K_b = Geometrical stress concentration factor for stress due to lateral pressure given in Cl.3.7.2.

K_n = Stress concentration factor due to unsymmetrical stiffener geometry, as defined in Cl.3.7.1.

3.1 General

3.1.1 This section provides the procedure for fatigue strength of the longitudinal stiffener end connections. The type of stresses considered in the analysis are explained in 3.3 to 3.6.

3.2 Fatigue approach for longitudinal end connections

3.2.1 The determination of the longitudinal stiffener end connections hot spot stress is based on:

- Nominal stresses from beam theory.

- Stress concentration factors for the different longitudinal end connections as given in 3.7

- Loading conditions specified in Section 1.

3.2.2 The longitudinal stiffener end connection types are described in Cl.3.7.2.

3.3 Axial and bending stress

3.3.1 General

3.3.1.1 This sub-section deals with the assessment of fatigue induced by the axial and bending modes (Figure 3.3.1) in way of longitudinal stiffener connections to primary members.

3.3.1.2 The simplified stress analysis aims at evaluating the hot spot stress at the flange of longitudinal stiffener connections with primary supporting members. The longitudinal stiffener is considered as a beam subjected to:

- axial loads induced by hull girder bending, and

- local bending induced by lateral pressure.

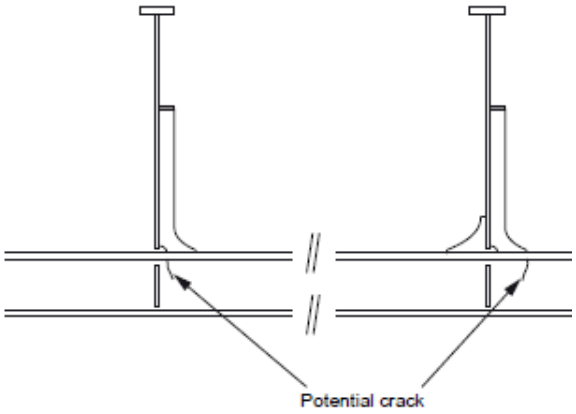


Figure 3.3.1: Areas of potential cracks in longitudinal stiffener web due to bending mode

3.3.2 Effect of primary structure distortion

The hot spot stress ranges and hot spot mean stresses in way of each end connection of longitudinal stiffener, as shown in Figure 3.5.1 are to be evaluated at the flange at following locations:

3.3.2.1 Transverse webs or floors other than those located at transverse bulkhead including swash bulkhead of cargo hold or in way of stool, such that additional hot spot stress due to the relative displacement is not to be considered.

3.3.2.2 Transverse webs or floors located at transverse bulkhead including swash bulkhead of cargo hold or in way of stool, such that additional hot spot stress due to the relative displacement are to be considered.

3.3.3 Stress concentration factors

Stress concentration factors due to unsymmetrical stiffener geometry according to Cl.3.7.1 and due to the stiffener end connection geometry at two potential hot spots (point 'A' and 'B') according to Cl.3.7.2 are to be applied.

For each hot spot of each connection, two geometric stress concentration factors (K_a and K_b) are defined, for both the axial stress and the bending stress respectively. The end connections given in Cl.3.7.2 are based on typical joint geometry under axial and lateral loadings. When a structural detail is different from those shown in Table 3.7.2, a finite element analysis is to be used to demonstrate the adequacy of the detail in terms of fatigue strength, according to Cl.3.7.3.

3.3.4 Hot spot stress range

The hot spot stress range, in [N/mm^2], due to dynamic loads for load case (i) of loading condition (j) is obtained from the following formula:

$$\Delta\sigma_{HS.i(j)} = \left|\left(\sigma_{GD,i1(j)} + \sigma_{LD,i1(j)} + \sigma_{dD,i1(j)}\right) - \left(\sigma_{GD,i2(j)} + \sigma_{LD,i2(j)} + \sigma_{dD,i2(j)}\right)\right|$$

where:

$\sigma_{GD,i1(j)}, \sigma_{GD,i2(j)}$ = Stresses due to global hull girder wave bending moments, in [N/mm^2], as defined in Cl.3.4.1.1.

$\sigma_{LD,i1(j)}, \sigma_{LD,i2(j)}$ = Stresses due to local dynamic pressure, in [N/mm^2], as defined in Cl.3.5.1.1.

$\sigma_{dD,i1(j)}, \sigma_{dD,i2(j)}$ = Stresses due to relative displacement in wave, in [N/mm^2], as defined in Cl.3.5.5.

3.3.5 Hot spot mean stress

The hot spot mean stress, in [N/mm^2], due to static and dynamic loads for load case (i) of loading condition (j) is obtained from the following formula:

$$\sigma_{mean,i(j)} = \sigma_{GS,(j)} + \sigma_{LS,(j)} + \sigma_{dS,(j)} + \sigma_{mLD,i(j)} + \sigma_{mGD,i(j)} + \sigma_{mdD,i(j)}$$

where for the load case (i) of loading condition (j):

$\sigma_{GS,(j)}$ = Stress due to still water hull girder bending moment, in [N/mm^2], as defined in Cl.3.4.1.2.

$\sigma_{LS,(j)}$ = Stress due to local static pressure, in [N/mm^2], as defined in Cl.3.5.1.2.

$\sigma_{dS,(j)}$ = Stress due to relative displacement in still water, in [N/mm^2], as defined in Cl.3.5.6.

$\sigma_{mLD,(j)}$ = Mean stress due to local dynamic pressure, in [N/mm^2], defined as:

$$\sigma_{mLD,i(j)} = \frac{\sigma_{LD,i1(j)} + \sigma_{LD,i2(j)}}{2}$$

$\sigma_{mGD,i(j)}$ = Mean stress due to global wave bending moment, in [N/mm^2], defined as:

$$\sigma_{mGD,i(j)} = \frac{\sigma_{GD,i1(j)} + \sigma_{GD,i2(j)}}{2}$$

$\sigma_{mdD,i(j)}$ = Mean stress due to relative displacement in wave, in [N/mm²], defined as:

$$\sigma_{mdD,i(j)} = \frac{\sigma_{dD,i1(j)} + \sigma_{dD,i2(j)}}{2}$$

3.4 Global Axial Stress

3.4.1 Hull girder bending stress

3.4.1.1 Stress due to hull girder wave bending moments

The hull girder hot spot stress, in [N/mm²], for load cases i_1 and i_2 of considered loading condition (j) is obtained from the following formula:

$$\sigma_{GD,ik(j)} = f_c . K_a . \left(\frac{M_{wv-LC,ik}}{I_{y-n50}} (z - z_n) - \frac{M_{wh-LC,ik}}{I_{z-n50}} . y \right) 10^{-3}$$

where:

$M_{wv-LC,ik}$ = Vertical wave bending moment, in [kN-m], of the considered dynamic load case, at the hull girder load calculation point of the considered longitudinal position for the loading condition (j) for i_k being equal to i_1 and i_2.

$M_{wh-LC,ik}$ = Horizontal wave bending moment, in [kN-m], of the considered dynamic load case, at the hull girder load calculation point of the considered longitudinal position for the loading condition (j) for i_k being equal to i_1 and i_2.

3.4.1.2 Stress due to still water hull girder bending moment

The hull girder hot spot stress due to still water bending moment [N/mm²], in loading condition (j) is obtained from the following formula:

$$\sigma_{GS,(j)} = \frac{f_c . K_a . \beta_{(j)} . M_{sw} . (z - z_n)}{I_{y-n50}} 10^{-3}$$

where:

M_{sw} = Permissible still water vertical bending moment [kN-m], at the hull girder load calculation point of the considered longitudinal position.

$\beta_{(j)}$ = Fraction of permissible still water vertical bending moment, for loading condition (j) as given below in Table 3.4.1.

Table 3.4.1: Fraction of permissible still water vertical bending moments, $\beta_{(j)}$

Vessel type	Full load condition	Normal/partial ballast condition
Tanker	0.6 in sagging condition	0.8 in hogging condition
Bulk carrier and dry Cargo ship[1)]	0.4 in sagging condition	0.8 in hogging condition [2)]
MPV/ general dry cargo ship[3)]	1.0 in sagging condition	1.0 in hogging condition
Liquefied gas carriers	0.7 in sagging condition	0.8 in hogging condition
Container ships	0.90 in hogging condition	0.8 in hogging condition
Ore Carrier[1)]	1.0 in sagging condition	1.0 in hogging condition
Passenger Ship[4)]	--	1.0 in hogging condition
RO-RO[4)]	--	1.0 in hogging condition
Others [5)]	--	

1) Applicable for ore carrier, bulk carrier and dry cargo ships except MPV/general dry cargo ships having a long centre cargo hold.

2) In alternate condition it is taken as 0.75.

3) Applicable for MPV/general dry cargo ships having a long centre cargo hold.

4) Values used when global FE analysis is required. Prescriptive assessment of longitudinal end connections is not required for passenger vessels.

5) Other ships shall be specifically considered.

3.5 Local bending stress

3.5.1 Stress due to stiffener bending

3.5.1.1 Stress due to dynamic pressure

The hot spot stress, in [N/mm^2], due to local dynamic pressure in load case i_1 and i_2 for loading condition (j) is obtained from the following formula:

$$\sigma_{LD,ik(j)} = \frac{K_b . K_n . s . \ell_{bdg}{}^2 . \left(\eta_W . f_{NL} . P_{W,ik(j)} + \eta_{ld} . P_{ld,ik(j)} + \eta_{bd} . P_{bd,ik(j)}\right)\left(1 - \frac{6X_e}{\ell_{bdg}} + \frac{6X_e^2}{\ell_{bdg}{}^2}\right)}{12Z_{eff-n50}}$$

where:

$P_{W,ik(j)}$ = Dynamic wave pressure, at the mid span, in [kN/m^2], in load case i_1 and i_2 for loading condition (j) as given in Chapter 5.

$P_{ld,ik(j)}$ = Dynamic liquid tank pressure, at the mid span, in [kN/m^2]], in load case i_1 and i_2 for loading condition (j) as given in Chapter 5.

Pressure acting on both sides of the stiffener, i.e. applied on the attached plate on stiffener side or on opposite side to the stiffener, could be simultaneously considered if relevant in the loading condition.

For the deck longitudinal stiffeners of bulk carriers, no internal pressure from the topside tank is considered.

$P_{bd,(j)}$ = Dynamic dry bulk cargo pressure at the mid span [kN/m^2], in load case i_1 and i_2 for loading condition (j) as given in Chapter 5.

$\eta_W, \eta_{ld}, \eta_{bd}$ = Pressure normal coefficients, taken as:

= 1 when the considered pressure is applied on the stiffener side,

= −1 otherwise.

f_{NL} = Correction factor for the non-linearity of the wave pressure taken as:

$$f_{NL} = 1 \text{ for } z > T_{LC} + 2h_w$$

$$f_{NL} = 2.5\,\frac{z - T_{LC}}{h_w} - 4 \text{ for } T_{LC} + 1.8h_w < z \leq T_{LC} + 2h_w$$

$$f_{NL} = 0.5\,\frac{z - T_{LC}}{h_w} - 0.4 \text{ for } T_{LC} + 1.fh_w < z \leq T_{LC} + 1.8h_w$$

$$f_{NL} = 0.4 \text{ for } T_{LC} + 1.2h_w < z \leq T_{LC} + 1.6h_w$$

$$f_{NL} = 0.7 - 0.25\,\frac{z - T_{LC}}{h_w} \text{ for } T_{LC} + 0.6h_w < z \leq T_{LC} + 1.2h_w$$

$$f_{NL} = 1 - 0.75\,\frac{z - T_{LC}}{h_w} \text{ for } T_{LC} - 0.2h_w < z \leq T_{LC} + 0.6h_w$$

$$f_{NL} = 0.1875\,\frac{z - T_{LC}}{h_w} + 1.1875 \text{ for } T_{LC} - h_w < z \leq T_{LC} - 0.2h_w$$

$$f_{NL} = 1 \text{ for } z \leq T_{LC} - h_w$$

h_w = Water head equivalent to the pressure at waterline, in [m].

X_e = Distance, in [m], to the hot spot from the closest end of the span ℓ_{bdg}, as defined in Figure 3.5.1.

$Z_{eff-n50}$= Net section modulus [cm^3], of the considered stiffener calculated considering an effective breadth b_{eff} of attached plating and taken at the stiffener flange.

b_{eff} = Effective breadth [mm], of attached plating specified at the ends of the span and in way of end brackets and supports, taken as:

$$b_{eff} = s . \min\left(\frac{\frac{1.04}{3}}{1 + \frac{3}{\left(\frac{\ell_{bdg}}{s}\left(1 - \frac{1}{\sqrt{3}}\right).10^3\right)^{1.35}}}; 1.0\right)$$

for $\frac{\ell_{bdg}}{s}\left(1 - \frac{1}{\sqrt{3}}\right).10^3 \geq 1$

$$b_{eff} = 0.26\ell_{bdg}\left(1 - \frac{1}{\sqrt{3}}\right).10^3$$

for $\frac{\ell_{bdg}}{s}\left(1 - \frac{1}{\sqrt{3}}\right).10^3 < 1$

s = Stiffener spacing in [m].

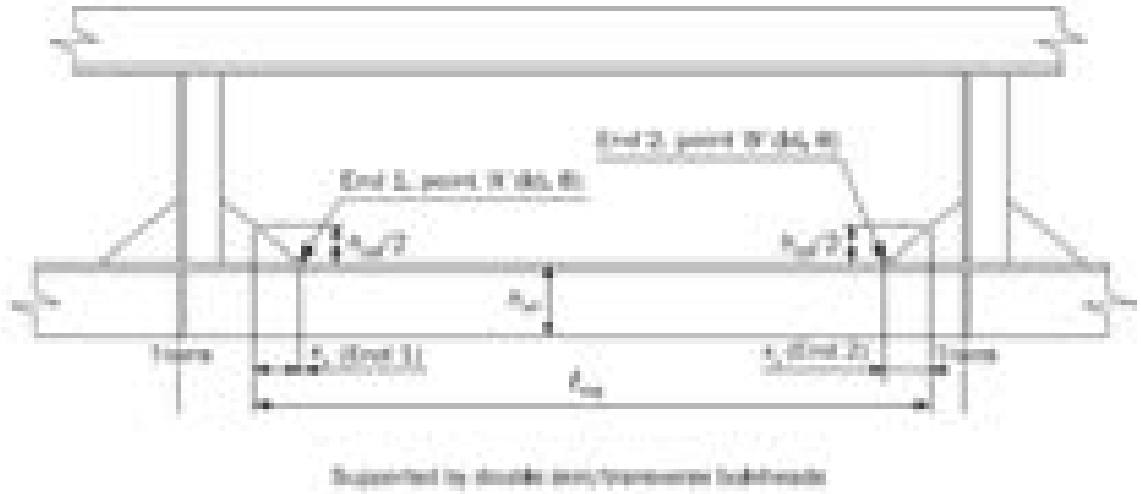


Figure 3.5.1: Definition of effective span and X_e for hot spot

3.5.1.2 Stress due to static pressure

The hot spot stress due to local static pressure [N/mm^2], for loading condition (j) is obtained from the following formula:

$$\sigma_{LS,(j)} = \frac{K_b . K_n . s . \ell_{bdg}{}^2 . \left(\eta_S . P_{S(j)} + \eta_{ls} . P_{ls(j)} + \eta_{bs} . P_{bs(j)}\right)\left(1 - \frac{6X_e}{\ell_{bdg}} + \frac{6X_e^2}{\ell_{bdg}{}^2}\right)}{12Z_{eff-n50}}$$

where:

$P_{S(j)}$ = Static external pressure [kN/m^2], in loading condition (j) as given in Chapter 5.

$P_{ls(j)}$ = Static liquid tank pressure [kN/m^2], in loading condition (j) as given in Chapter 5.

Pressure acting on both sides could be simultaneously considered if relevant in the loading condition.

$P_{bs(j)}$ = Static dry bulk cargo pressure [kN/m^2], in loading condition (j) as given in Chapter 5.

$\eta_S, \eta_{ls}, \eta_{bs}$ = Pressure normal coefficients, taken as:

= 1 when the considered pressure is applied on the stiffener side,

= −1 otherwise.

3.6 Stress due to relative displacements of primary supporting members

3.6.1 The additional hot spot stress due to the relative displacement of primary supporting members is to be considered for longitudinal stiffener end connections fitted on transverse web frames or floors located:

- in way of plane transverse bulkheads (including swash bulkheads in cargo holds), or

- in way of transverse stools and cofferdam bulkheads.

3.6.2 The relative displacement is defined as the lateral displacement of the longitudinal measured at the frame forward (Fwd) or afterward (Aft) relative to the displacement at the transverse bulkhead (or restrained frame).

3.6.3 Displacement of primary supporting members

The displacement of primary supporting members is to be evaluated using a FE model. The displacement of a primary supporting member is the displacement of the point located at the intersection of the stiffener flange, the stiffener web and the supporting member web. This displacement is measured along the line defined by the intersection between the stiffener web and the primary supporting member web in the non-deformed state of the finite element model.

3.6.4 Sign convention

Where the stress at the hot spot location, i.e. at the flange of longitudinal at the transverse

bulkhead (or restrained frame), due to relative displacement is in tension, the sign of the relative displacement is positive.

3.6.5 Stress due to relative displacement derived using FE method

The relative displacement stress can be calculated based on relative displacement derived from FE analysis. If the relative displacement is derived from FE analysis, the local rotation at the position of the hot spot needs to be accounted for.

The stress due to relative deflection, in [N/mm2], for load case i_1 and i_2 of loading condition (j) for both locations 'a' and 'f' is to be calculated directly using the following expression:

$$\sigma_{dD,ik(j)} = \begin{pmatrix} K_b.\sigma_{dFwd-a,ik(j)} + K_b.\sigma_{dAft-a,ik(j)} \text{ for location "a"} \\ K_b.\sigma_{dFwd-f,ik(j)} + K_b.\sigma_{dAft-f,ik(j)} \text{ for location "f"} \end{pmatrix}$$

(k=1,2)

$$\sigma_{dFwd-a,ik(j)} = \frac{3.9\,\delta_{Fwd,ik(j)}.EI_{Aft-n50}.I_{Fwd-n50}}{Z_{Aft-n50}.\ell_{Fwd}(\ell_{Aft}I_{Fwd-n50} + \ell_{Fwd}I_{Aft-n50})}\left(1 - 1.15\frac{|X_{eAft}|}{\ell_{Aft}}\right)10^{-5}$$

$$\sigma_{dAft-a,ik(j)} = \left[\frac{3.9\,\delta_{Aft,ik(j)}.EI_{Aft-n50}.I_{Fwd-n50}}{Z_{Aft-n50}.l_{Aft}(l_{Aft}I_{Fwd-n50} + l_{Fwd}I_{Aft-n50})}\left(1 - 1.15\frac{|X_{eAft}|}{l_{Aft}}\right) - \frac{0.9\delta_{Aft,ik(j)}.EI_{Aft-n50}|X_{eAft}|}{Z_{Aft-n50}\ell_{Aft}^3}\right]10^{-5}$$

$$\sigma_{dFwd-f,ik(j)} = \left[\frac{3.9\,\delta_{Fwd,ik(j)}.EI_{Aft-n50}.I_{Fwd-n50}}{Z_{Fwd-n50}.l_{Fwd}(l_{Aft}I_{Fwd-n50} + l_{Fwd}I_{Aft-n50})}\left(1 - 1.15\frac{|X_{eFwd}|}{l_{Fwd}}\right) - \frac{0.9\delta_{Fwd,ik(j)}.EI_{Fwd-n50}|X_{eFwd}|}{Z_{Fwd-n50}\ell_{Fwd}^3}\right]10^{-5}$$

$$\sigma_{dAft-f,ik(j)} = \frac{3.9\,\delta_{Aft,ik(j)}.EI_{Aft-n50}.I_{Fwd-n50}}{Z_{Fwd-n50}.\ell_{Aft}(\ell_{Aft}I_{Fwd-n50} + \ell_{Fwd}I_{Aft-n50})}\left(1 - 1.15\frac{|X_{eFwd}|}{\ell_{Fwd}}\right)10^{-5}$$

where:

a, f = Suffix which denotes the location as indicated in Figure 3.6.5.

Aft, Fwd = Suffix which denotes the direction, afterward (Aft) or forward (Fwd), from the transverse bulkhead as shown in Figure 3.6.5.

K_b = Stress concentration factor due to bending for the location 'a' or 'f' which may correspond to points 'A' or 'B' as defined in Table 3.7.2.

$\sigma_{dFwd-a,ik(j)}, \sigma_{dAft-a,ik(j)}, \sigma_{dFwd-f,ik(j)}, \sigma_{dAft-f,ik(j)}$ = Additional stress at location 'a' and 'f', in [N/mm^2], due to the relative displacement between the transverse bulkhead including swash bulkhead or floors in way of stool and the forward (Fwd) and afterward (Aft) transverse web or floor respectively for load case i_1 and i_2 of loading condition (j), taken as:

$I_{Aft-n50}, I_{Fwd-n50}$ = Net moment of inertia, in [cm^4], of forward (Fwd) and afterward (Aft) longitudinal.

$Z_{Aft-n50}, Z_{Fwd-n50}$ = Net section modulus in [cm^3], of forward (Fwd) and afterward (Aft) stiffener.

ℓ_{Aft}, ℓ_{Fwd} = Span, in [m], of forward (Fwd) and afterward (Aft) longitudinal, as shown in Figure 3.6.5.

X_{eAft}, X_{eFwd} = Distance, in [m], as shown in Figure 3.5.1, to the hot spot in location 'a' or 'f' from the closest end of ℓ_{Fwd} and ℓ_{Aft} respectively.

$\delta_{Aft,ik(j)}, \delta_{Fwd,ik(j)}$ = Relative displacement in the direction perpendicular to the attached plate, in [mm], between the transverse bulkhead (including swash bulkhead or floor in way of stools) and the forward (Fwd) or afterward (Aft) transverse web (or floor) as shown in Figure 3.6.5.

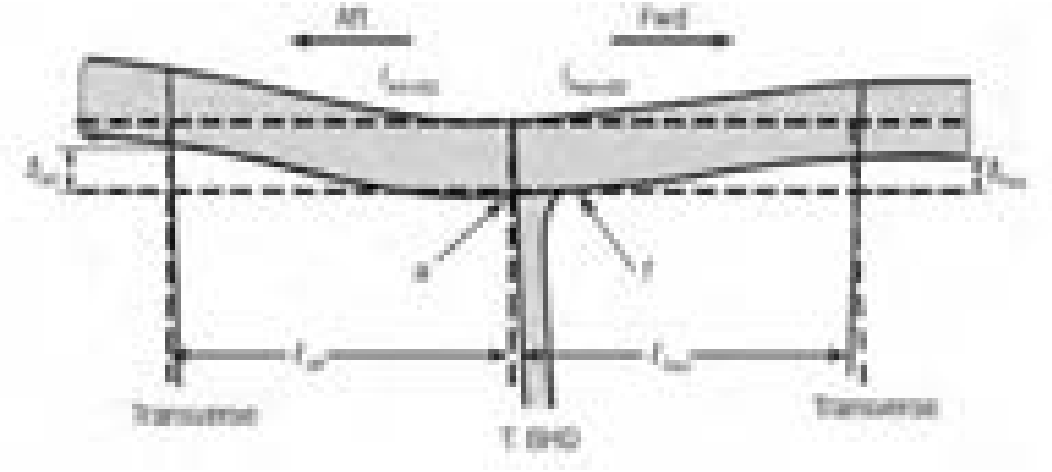


Figure 3.6.5: Definition of the relative displacement (example of the side longitudinal)

3.6.6 Stress due to relative displacement in still water

The additional hot spot stress, in [N/mm^2], in still water, due to the relative displacement is to be obtained according to Cl.3.6.5, replacing dynamic local stress σ_{LD} and dynamic pressure with static local stress σ_{LS} and static pressure.

3.7 Stress concentration factor

3.7.1 Unsymmetrical stiffener

3.7.1.1 The stress concentration factor K_n for unsymmetrical flange of built-up and rolled angle stiffeners under lateral load, calculated at the web's mid-thickness position, as shown in Figure 3.7.1.1, is to be taken as:

$$K_n = \frac{1 + \lambda\,\beta^2}{1 + \lambda\,\beta^2\,\Psi_z}$$

where,

$$\lambda = \frac{3\left(1 + \frac{\eta}{280}\right)}{1 + \frac{\eta}{40}}$$

$$\eta = \frac{\ell_{bdg}^4 . 10^{12}}{b_{f-n50}^3 . t_{f-n50} . {h_{stf-n50}}^2 . \left(\frac{4 . h_{stf-n50}}{t_{w-n50}^3} + \frac{s}{t_{p-n50}^3}\right)}$$

$$\beta = 1 - \frac{2b_{g-n50}}{b_{f-n50}} \text{ for built} - \text{up profiles}$$

$$\beta = 1 - \frac{t_{w-n50}}{b_{f-n50}} \text{ for rolled angle profiles}$$

b_{g-n50} = Eccentricity of the stiffener equal to the distance from flange's edge to web's centreline, in [mm], as shown in Figure 3.7.1.2.

b_{f-n50} = Net breadth of flange, in [mm], as shown in Figure 3.7.1.2.

t_{f-n50} = Net flange thickness, in [mm], as shown in Figure 3.7.1.2.

$h_{stf-n50}$ = Net stiffener height, including face plate, in [mm], as shown in Figure 3.7.1.2.

t_{w-n50} = Net web thickness, in [mm], as shown in Figure 3.7.1.2.

h_{w-n50} = Net web's height stiffener, in [mm], as shown in Figure 3.7.1.2.

t_{p-n50} = Net thickness of attached plating, in [mm], as shown in Figure 3.7.1.2.

Ψ_z = Coefficient given as:

$$\Psi_z = \frac{h_{w-n50}^2 t_{w-n50}}{4Z_{n50}} 10^{-3}$$

Z_{n50} = Net section modulus, in [cm^3], of stiffener with an attached plating breadth equal to the stiffener spacing.

Figure 3.7.1.1: Bending stress in stiffener with symmetrical and unsymmetrical flange

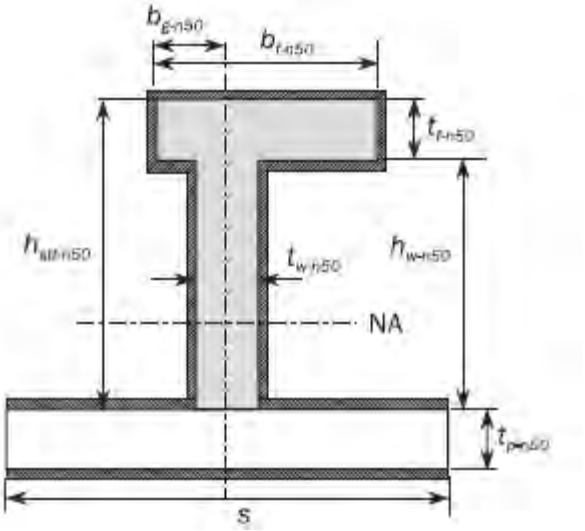


Figure 3.7.1.2: Stiffener - Net scantling

3.7.1.2 Bulb profiles

For bulb profiles K_n factor is to be calculated using the equivalent built-up profile as shown in Figure 3.7.1.3. The flange of the equivalent built-up profile is to have the same properties as the bulb flange, i.e. same cross sectional area and moment of inertia about the vertical axis and neutral axis position. For HP bulb profiles, examples of the equivalent built up profile dimensions are listed in Table 3.7.1.

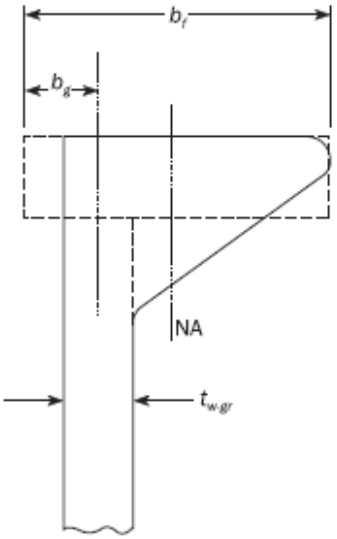


Figure 3.7.1.3: Bulb profile and equivalent built-up profile

Table 3.7.1: HP equivalent built-up profile dimensions				
HP-bulb		**Equivalent built-up flange in gross thickness**		
Height (mm)	**Gross web thickness, t_{w-gr} (mm)**	**b_f (mm)**	**t_{f-gr} (mm)**	**b_0 (mm)**
200	9 – 13	$t_{w-gr} + 24.5$	22.9	$(t_{w-gr} + 0.9)/2$
220	9 – 13	$t_{w-gr} + 27.6$	25.4	$(t_{w-gr} + 1.0)/2$
240	10 – 14	$t_{w-gr} + 30.3$	28.0	$(t_{w-gr} + 1.1)/2$
260	10 – 14	$t_{w-gr} + 33.0$	30.6	$(t_{w-gr} + 1.3)/2$
280	10 – 14	$t_{w-gr} + 35.4$	33.3	$(t_{w-gr} + 1.4)/2$
300	11 – 16	$t_{w-gr} + 38.4$	35.9	$(t_{w-gr} + 1.5)/2$
320	11 – 16	$t_{w-gr} + 41.0$	38.5	$(t_{w-gr} + 1.6)/2$
340	12-17	$t_{w-gr} + 43.3$	41.3	$(t_{w-gr} + 1.7)/2$
370	13-19	$t_{w-gr} + 47.5$	45.2	$(t_{w-gr} + 1.9)/2$
400	14-19	$t_{w-gr} + 51.7$	49.1	$(t_{w-gr} + 2.1)/2$
430	15-21	$t_{w-gr} + 55.8$	53.1	$(t_{w-gr} + 2.3)/2$

3.7.2 Longitudinal stiffener end connections

3.7.2.1 The stress concentration factors K_a and K_b are given in Table 3.7.2 for end connection of stiffeners subjected to axial and lateral loads. The values given in Table 3.7.2 for soft toe are valid provided that toe geometry complies with the requirements given in Cl.3.7.2.5. The stress concentration factor K_b given for lateral loads are to be used also for stress due to relative displacements.

3.7.2.2 Other connection types

When connection types other than those given in Table 3.7.2 are proposed, the fatigue strength for the proposed connection type is to be assessed either by performing a very fine mesh FE analysis as described in Section 4 to obtain directly the hot spot stress, or by calculating the stress concentration factor using FE analysis according to Cl.3.7.3.

3.7.2.3 Overlapped connection

Overlapped connection types for longitudinal stiffeners, i.e. attachments welded to the web of the longitudinals, are to be avoided in areas where fatigue might be critical e.g. in way of cargo hold region. If overlap is used an additional factor of 1.15 should be used.

3.7.2.4 End stiffener without connection to web stiffener: Where the web stiffener is omitted or not connected to the longitudinal flange in way of:

- Side shell below $1.1T_{sc}$

- Bottom.

- Inner hull longitudinal bulkhead below $1.1T_{sc}$

- Hopper.

- Topside tank sloping plating below $1.1T_{sc}$

- Inner bottom.

the following is required:

-A complete collar as defined in Figure 3.7.2.1 (i.e. connection type ID 31 of Table 3.7.2.) or,

-A detail design for cut-outs as described in **Appendix A**.

Equivalence to cut-outs given in **Appendix A**, A.2.1 may be accepted provided it is assessed for fatigue by using comparative FE analysis which is based on hot spot stress around the cut-out in the web plate of the primary supporting member inclusive of the collar, as given in **Appendix A**, A.2.2.

3.7.2.5 Soft toe of web stiffener and backing bracket

The toe geometry end connection of web stiffener and backing bracket is to comply with the following:

$$\theta \leq 20$$

$$h_{toe} \leq \max(t_{bkt-gr}, 15)$$

where:

θ = Angle of the toe in [deg], as shown in Figure 3.7.2.2.

h_{toe} = Height of the toe in [mm], as shown in Figure 3.7.2.2.

t_{bkt-gr} = Gross thickness of the bracket in [mm].

Recommended detail designs for longitudinal end connections with soft toes and backing brackets are given in Figure 3.7.2.2.

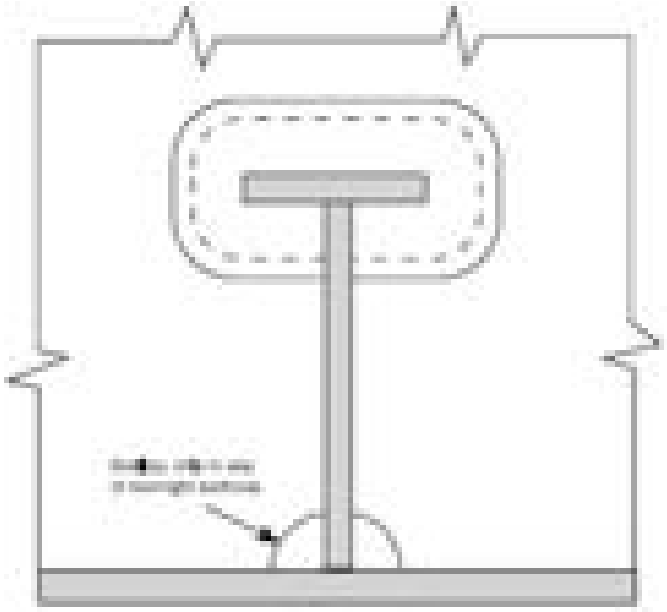
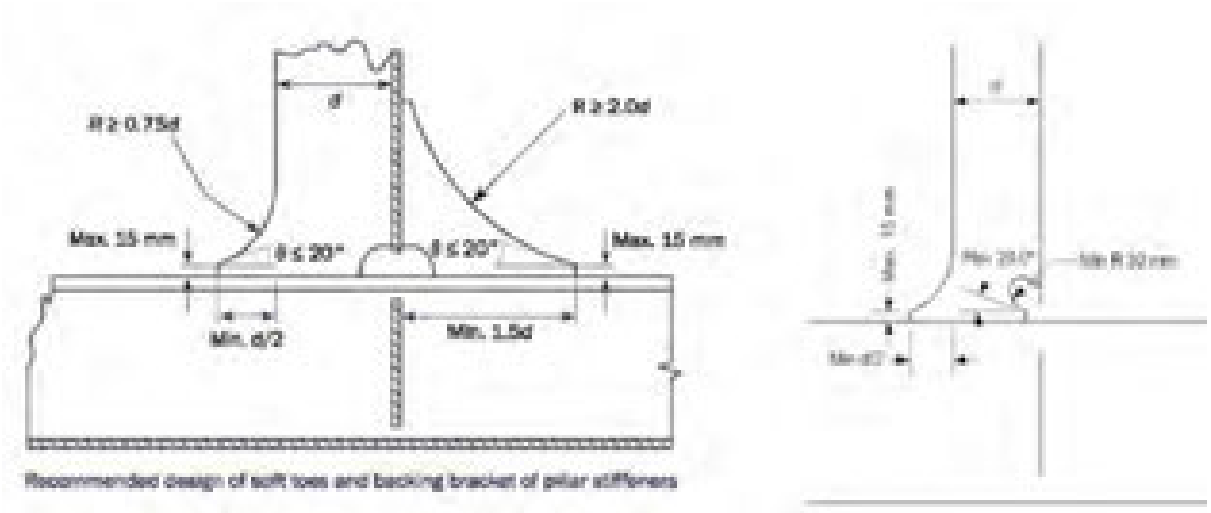


Figure 3.7.2.1: Complete collar fitted before the stiffener is pulled through the web frame



Recommended design of soft toes and backing bracket of pillar stiffeners

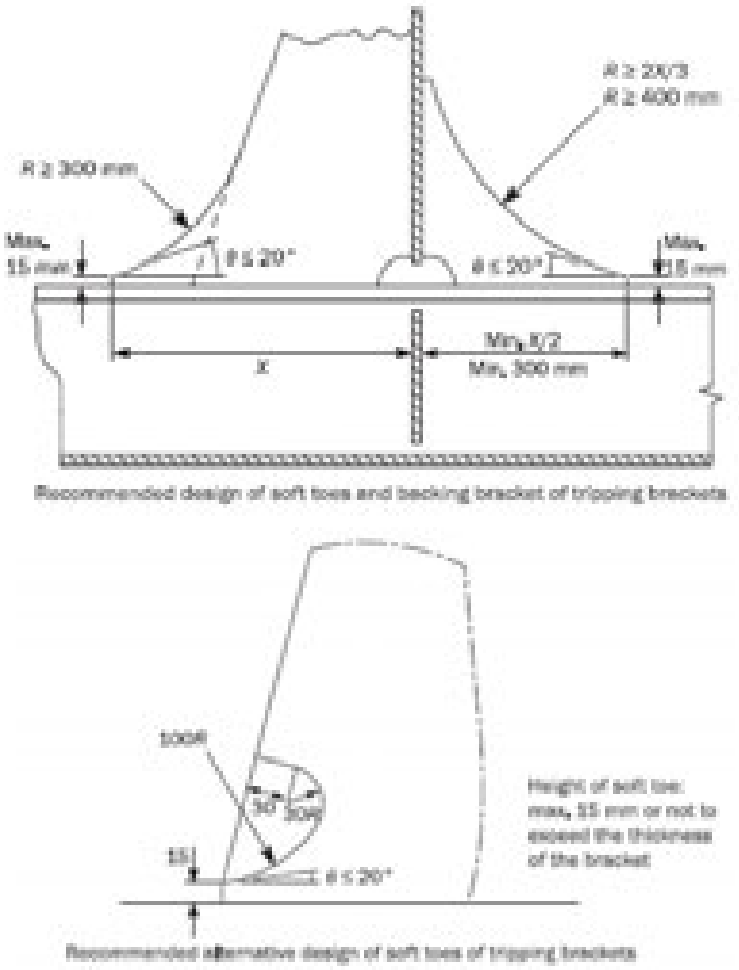
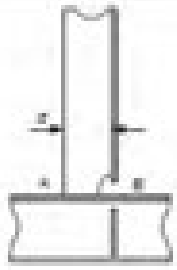
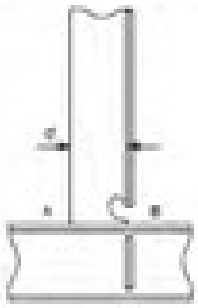
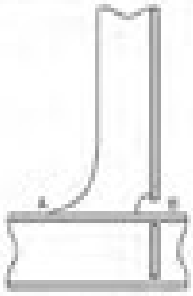
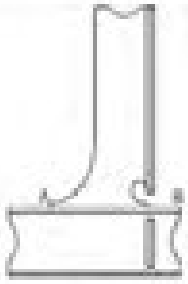
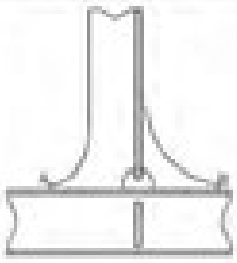
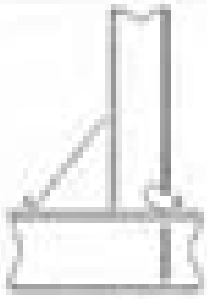
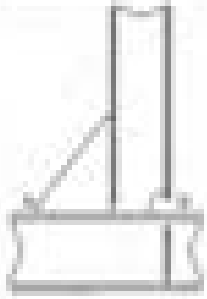
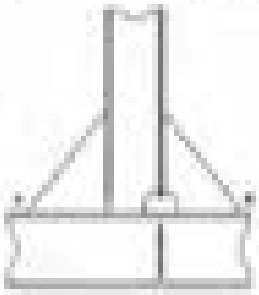
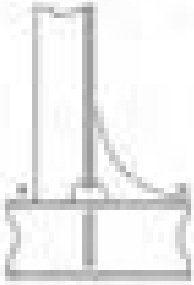
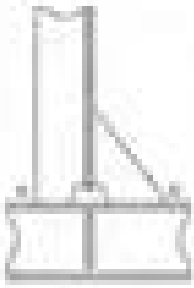
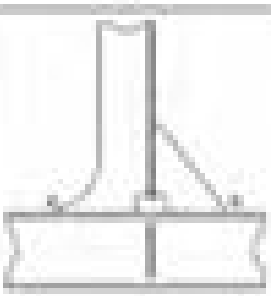
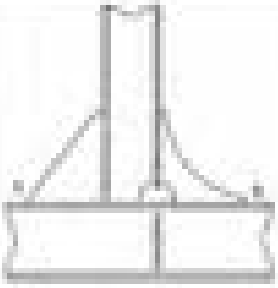
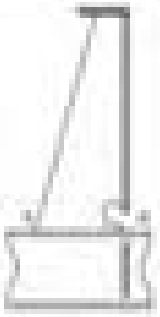
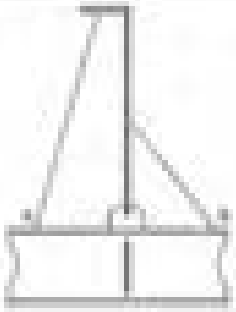


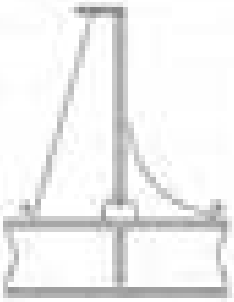
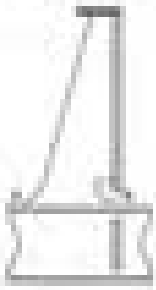
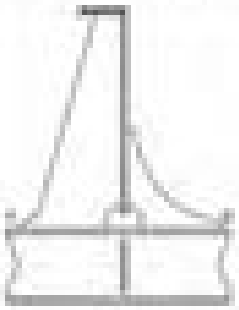
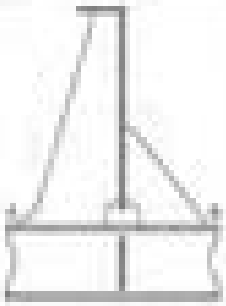
Figure 3.7.2.2: Detail design for soft toes and backing brackets

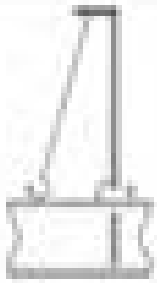
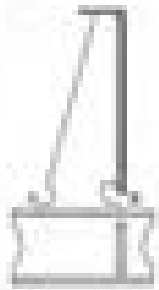
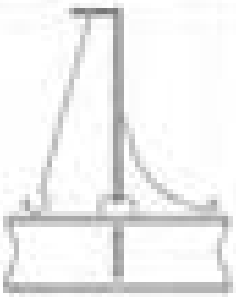
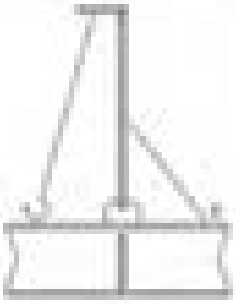
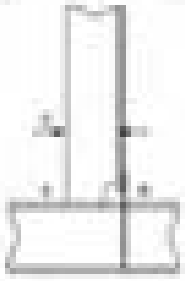
Table 3.7.2: Stress concentration factors

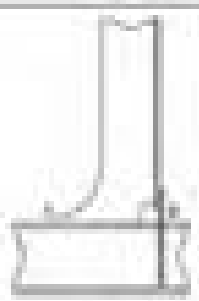
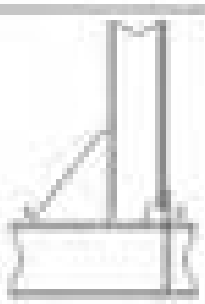
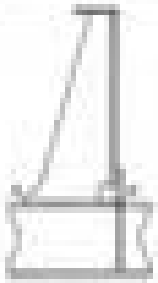
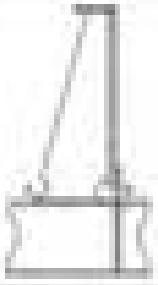
ID	Connection type (Note)	Point 'A'		Point 'B'	
		K_a	K_a	K_a	K_a
1 (Note)		1.28 for $d \le 150$ 1.36 for $150 < d \le 250$ 1.45 for $d > 250$	1.40 for $d \le 150$ 1.50 for $150 < d \le 250$ 1.60 for $d > 250$	1.28 for $d \le 150$ 1.36 for $150 < d \le 250$ 1.45 for $d > 250$	1.60
2 (Note)		1.28 for $d \le 150$ 1.36 for $150 < d \le 250$ 1.45 for $d > 250$	1.40 for $d \le 150$ 1.50 for $150 < d \le 250$ 1.60 for $d > 250$	1.14 for $d \le 150$ 1.24 for $150 < d \le 250$ 1.34 for $d > 250$	1.27
3		1.28	1.34	1.52	1.67
4		1.28	1.34	1.34	1.34
5		1.28	1.34	1.28	1.34

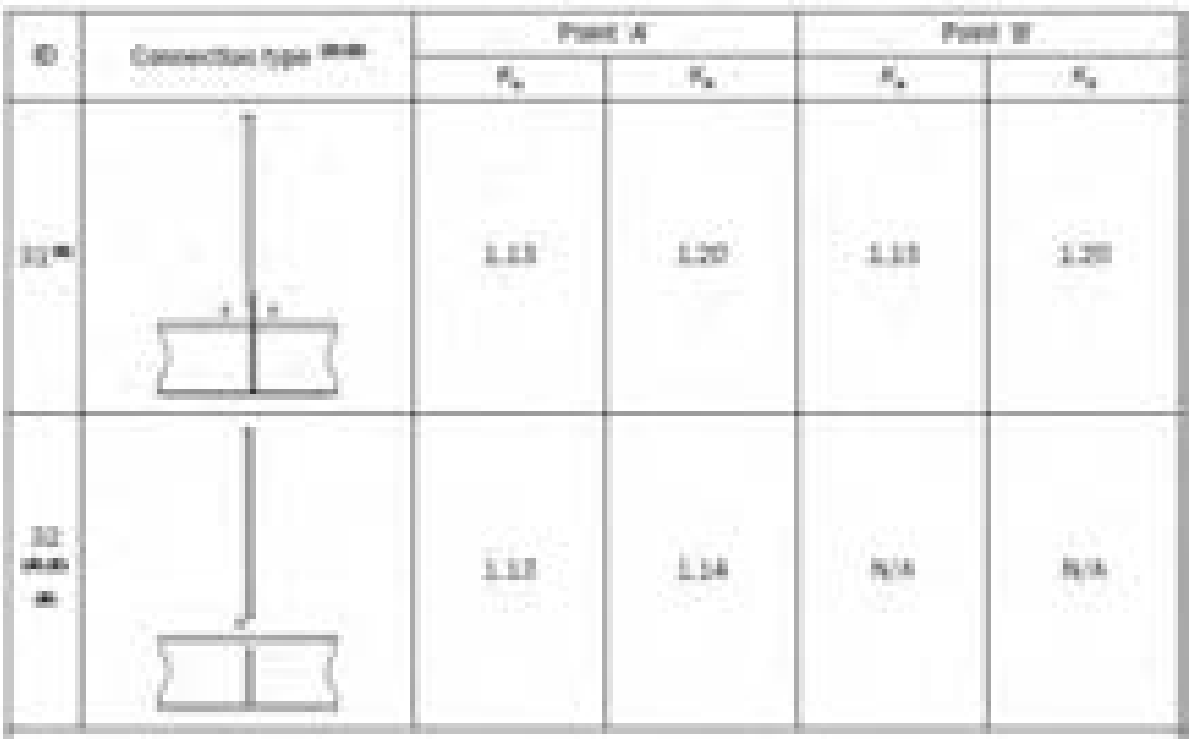
ID	Connection type (R/M)	Point 'A'		Point 'B'	
		K_a	K_a	K_a	K_a
6		1.52	1.67	1.34	1.34
7		1.52	1.67	1.52	1.67
8		1.52	1.67	1.52	1.67
9		1.52	1.67	1.26	1.34
10		1.52	1.67	1.52	1.67

ID	Connection type (With	Point 'A'		Point 'B'	
		K_a	K_a	K_b	K_b
11		1.28	1.34	1.53	1.67
12		1.53	1.67	1.28	1.34
13		1.53	1.67	1.53	1.67
14		1.53	1.67	1.34	1.34
15		1.53	1.67	1.53	1.67

ID	Connection type [Note]	Point 'A'		Point 'B'	
		K_x	K_y	K_x	K_y
16		1.52	1.87	1.28	1.34
17		1.28	1.34	1.52	1.87
18		1.28	1.34	1.34	1.34
19		1.28	1.34	1.28	1.34
20		1.28	1.34	1.52	1.87

ID	Connection type [(a)-(k)]	Point A		Point B	
		F_A	F_A	F_A	F_A
21		1.26	1.34	1.52	1.67
22		1.26	1.34	1.34	1.34
23		1.26	1.34	1.26	1.34
24		1.26	1.34	1.52	1.67
25- (k)		1.26 for $d \le 150$ 1.36 for $150 < d \le 250$ 1.45 for $d > 250$	1.40 for $d \le 150$ 1.50 for $150 < d \le 250$ 1.60 for $d > 250$	1.14 for $d \le 150$ 1.24 for $150 < d \le 250$ 1.34 for $d > 250$	1.20 for $d \le 150$ 1.36 for $150 < d \le 250$ 1.47 for $d > 250$

ID	Connection type [illegible]	Point 'A'		Point 'B'	
		F_{sA}	F_{sA}	F_{sB}	F_{sB}
26		1.28	1.34	1.34	1.47
27		1.52	1.67	1.34	1.47
28		1.52	1.67	1.34	1.47
29		1.28	1.34	1.34	1.47
30		1.28	1.34	1.34	1.47



(1) The attachment length d [mm], is defined as the length of the welded attachment on the longitudinal stiffener flange without deduction of scallop.

(2) Where the longitudinal stiffener is a flat bar and there is a web stiffener/bracket welded to the flat bar stiffener, the stress concentration factor listed in the table (excluding detail 31 and 32) is to be multiplied by a factor of 1.12 when the thickness of attachment is thicker than the 0.7 times thickness of flat bar stiffener. This also applies to unsymmetrical profiles where there is less than 8[mm] clearance between the edge of the stiffener flange and the attachment, e.g. bulb or angle profiles where the clearance of 8[mm] cannot be achieved.

(3) Designs with overlapped connection / attachments, see Cl.3.7.2.3.

(4) ID. 31 and 32 refer to details where web stiffeners are omitted or not connected to the longitudinal stiffener flange. See Cl.3.7.2.4 For detail 31, however, the collar is assumed welded to the stiffener flange.

(5) For connection type ID. 32 with no collar and/or web plate welded to the flange, the stress concentration factors provided in this table are to be used irrespective of slot configuration.

(6) The fatigue assessment point ‘A’ is located at the connection between the stiffener web and the transverse web frame or lug plate. The stress concentration factor represents the hot spot related to the longitudinal stress.

3.7.3 Alternative design

3.7.3.1 Derivation of alternative stress concentration factors

The stress concentration factors for alternative designs are to be calculated by a very fine mesh FE analysis according to the requirements given in Section 4. Additional requirements for derivation of geometrical stress concentration factors for stiffener end connections using very fine mesh FE analysis are given below:

3.7.3.2 FE model extent: the FE model, as shown in Figure 3.7.3, is to cover at least four web frame spacings in the longitudinal stiffener direction with the detail to be considered located at the middle frame. The same type of end connection is to be modelled at all the web frames. In the transverse direction, the model may be limited to one stiffener spacing.

3.7.3.3 Load application: in general, two loading cases are to be considered:

- Axial loading by enforced displacement applied to the model ends and

- Lateral loading by unit pressure load applied to the shell plating.

3.7.3.4 Boundary conditions applied in the FE model considered:

- Symmetry conditions are applied along the longitudinal cut of the plate flange, along transverse and vertical cuts on web frames and on top of the web stiffener.

- For lateral pressure loading: the model is to be fixed in all degrees of freedom at both forward and aft ends.

- For axial loading: the model is to be fixed for displacement in the longitudinal direction at the aft end of the model while enforced axial displacement is applied at the forward end, or vice versa.

3.7.3.5 FE mesh density: At the location of the hot spots under consideration, the element size is to be in the order of the thickness of the stiffener flange or 10[mm] depending on the type of stiffener. In the remaining part of the model, the element size is to be in the order of s/10, where s is the stiffener spacing.

3.7.3.6 The geometric stress concentration factor is defined as the ratio of the hot spot stress derived from FE analysis up on nominal stress calculated according to analytical stress analysis for longitudinal stiffener connections.

3.7.3.7 Two geometric stress concentration factors K_a and K_b are to be determined, for 2 loading cases specified above, and are determined as follows:

- For the axial loading case:

$$K_a = \frac{\sigma_{HSAx}}{\sigma_{NomAx}}$$

- For the bending loading case:

$$K_b = \frac{\sigma_{HSBd}}{\sigma_{NomBd}}$$

σ_{HSAx} = Hot spot stress, [N/mm^2], determined at the stiffener flange for the axial load.

σ_{NomAx} = Nominal axial stress, [N/mm^2], calculated at the stiffener flange according to Cl.3.4.1 for the axial load applied for the FE calculation.

σ_{HSBd} = Hot spot stress, [N/mm^2], determined at the stiffener flange for the unit pressure load.

σ_{NomBd} = Nominal bending stress, [N/mm^2], calculated at the stiffener flange according to Cl.3.5.1 in way of the hot spot for the unit pressure load applied for the FE calculation.

The derivation of geometrical stress concentration factors for alternative designs is to be documented and provided to IRS.

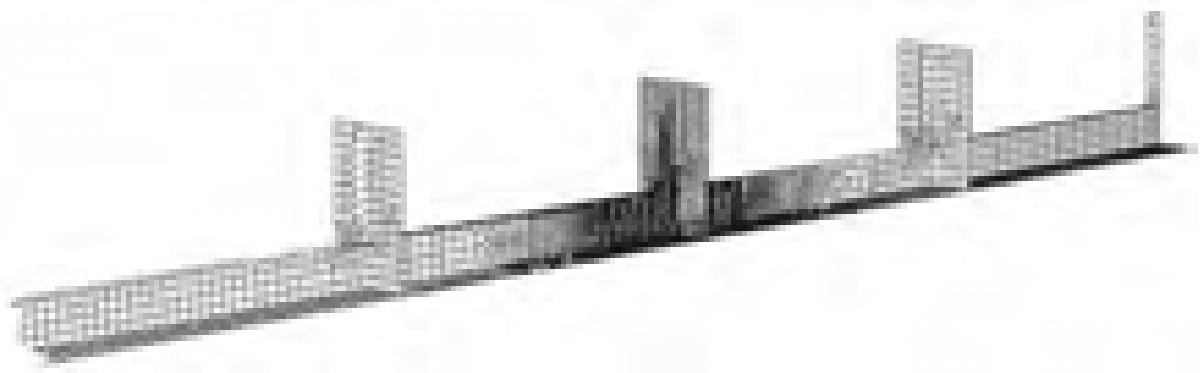


Figure 3.7.3: Fine mesh finite element model for derivation of geometrical stress concentration factor (example of stiffener with flange)

Section 4

Finite Element Stress Analysis

4.1 General

4.1.1 Application

4.1.1.1 This section applies to fatigue assessment by finite element stress analysis. The methods are based on the hot spot stress approach and requirements are given for both plated welded joints and for non-welded hot spots. The hot spot stress takes into account structural discontinuities due to the structural detail of the welded joint, but not taking into account the notch effect at the weld toe.

4.1.1.2 The local nominal stress for non-welded area at cut plate edge requires a specific finite element model and a well specified method for the calculation, Refer Cl.4.2.4.

4.1.1.3 The stress at hot spot for welded joints is generally highly dependent on the finite element model used for the representation of the structure. Where other methods than those described in this Section are adopted for calculation of FEA based hot spot stress, these methods are to be submitted to IRS for review.

4.1.1.4 Limitations of the hot spot stress methodology for simple connections is given in 4.7.

4.1.1.5 Some typical structural details to be assessed for fatigue by very fine mesh analysis according to 4.2 to 4.6 are given below, irrespective of their compliance with the design standard given in **Appendix A**.

- Welded/radiused hopper knuckle (lower/upper) connection (intersection of hopper sloping plate, knuckled inner bottom plate, longitudinal girder, floor and transverse web, inner hull longitudinal bulkhead, transverse web and side stringer) at the most critical frame location.

- Connections of bulkhead (transverse/ corrugated) to the adjoining structures.

- Bracket at the heel of horizontal stringer.

- Large openings

- Upper side frame bracket toe in case of flat bottom of top wing

- Deck plating and longitudinal hatch coaming end bracket toe.

- Side frame end brackets and lower hopper knuckle for single side bulk carrier

The selection of critical locations on the structural members and the fine mesh structural models is to be in accordance with the requirements given in IRS *Guidelines on Structural Assessment of Ships based on Finite Element Method.* Additional specific details may be requested to be checked on a case-by-case basis by IRS.

4.2 Type of details

4.2.1 General

4.2.1.1 The method for calculation of hot spot stress at weld toe for any welded details is given in 4.5 and for web stiffened cruciform joints given in 4.6. The method for calculation of local stress for non-welded area is given in Cl.4.5.4.

4.2.1.2 Two types of hot spots denoted ‘a’ and ‘b’ are described in Table 4.2.1 at weld toe of ordinary welded joints. These are defined depending location on the plate and their orientation to the weld toe as illustrated in Figure 4.2.1.

Table 4.2.1: Types of hot spots

Type	Description
a	Hot spot at the weld toe on plate surface
b	Hot spot at the weld toe around the plate edge

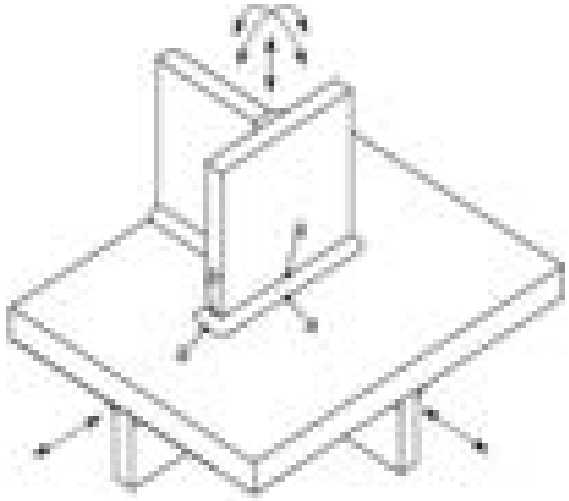


Figure 4.2.1: Types of hot spots

4.2.2 Web-stiffened cruciform joints

A standard cruciform joint corresponds to the intersection of two plates (Figure 4.2.2). A web-stiffened cruciform joint corresponds to a cruciform joint where the two plates are stiffened by another plate perpendicular to the first ones and acting as a web. The method for calculation of the hot spot stress at the weld toes of web-stiffened cruciform joints is given in 4.6.

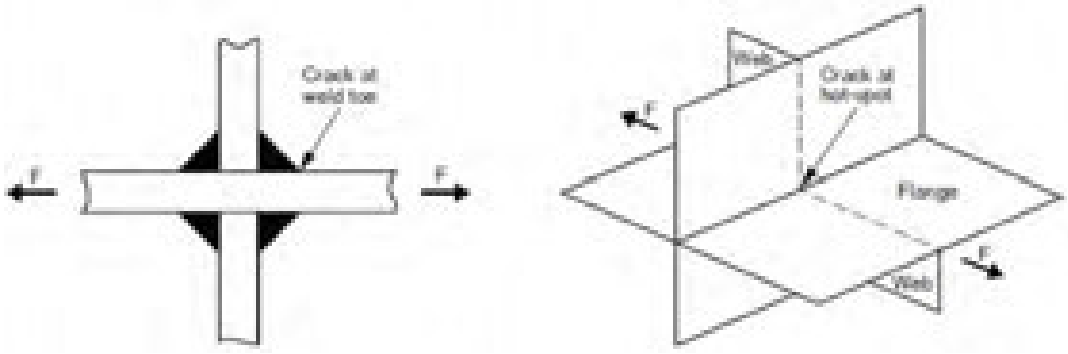


Figure 4.2.2: Standard cruciform joint (left hand side) and web-stiffened cruciform joint (right hand side)

4.2.3 Bent hopper knuckle details

Methods for the calculation of the hot spot stresses for bent hopper knuckles are given in Cl.4.5.5.

4.2.4 Non welded details

Methods for the calculation of the local nominal stress for non-welded area such as cut plate edges, cut-outs for stiffeners connections at web frames, hatch corners, etc. are given in Cl.4.5.4.

4.3 FE model

4.3.1 Three types of FE models are generally used:

- Full ship model

- Partial ship model, like cargo hold model for ships with cargo holds

- Local models, like hot spot models for welded details and local stress models for free plate edges

Evaluation of hot spot stresses for fatigue assessment requires FE models with very fine finite element meshes in way of areas of high stress concentration. These very fine mesh zones are either incorporated into the global FE model (Figure 4.3.1) or by use of sub-modelling technique. The coarse mesh model of cargo holds is to be made according to IRS *Guidelines on Structural Assessment of Ships based on Finite Element Method*. In sub-modelling technique, a separate local finite element model with very fine mesh zones in conjunction with the boundary conditions obtained from a coarse mesh global model is used. Typical examples of local finite element models with very fine mesh are shown in Figure's.4.3.2, 4.3.3 and 4.3.4.

4.3.2 Corrosion model

The thickness to be used in FE analysis is net thickness, t_{n50} together with stress correction factor, f_c, as given in Section 1.

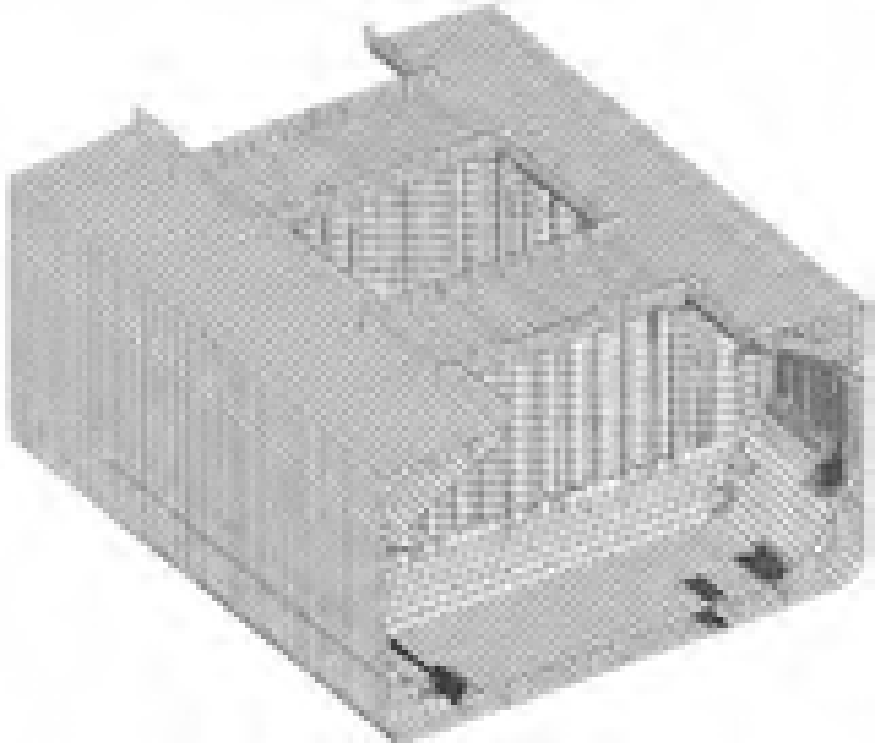


Figure 4.3.1: Very fine mesh areas incorporated directly into the FE model

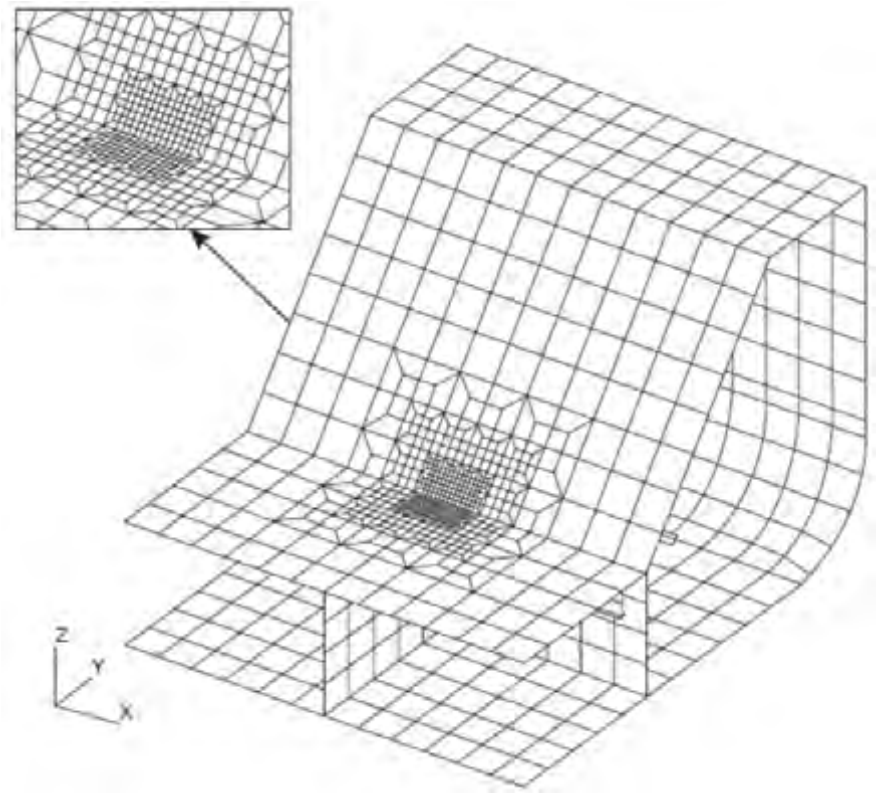


Figure 4.3.2: Local very fine mesh model of hopper knuckle connection between inner bottom and hopper plate

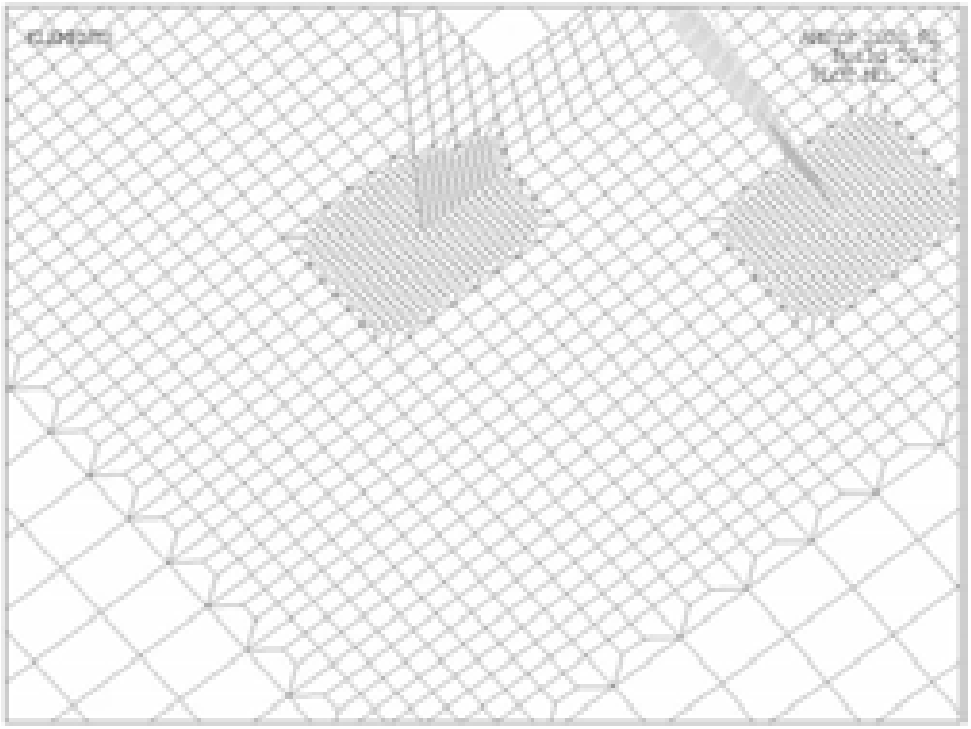


Figure 4.3.3: Local very fine mesh model in LPG carriers

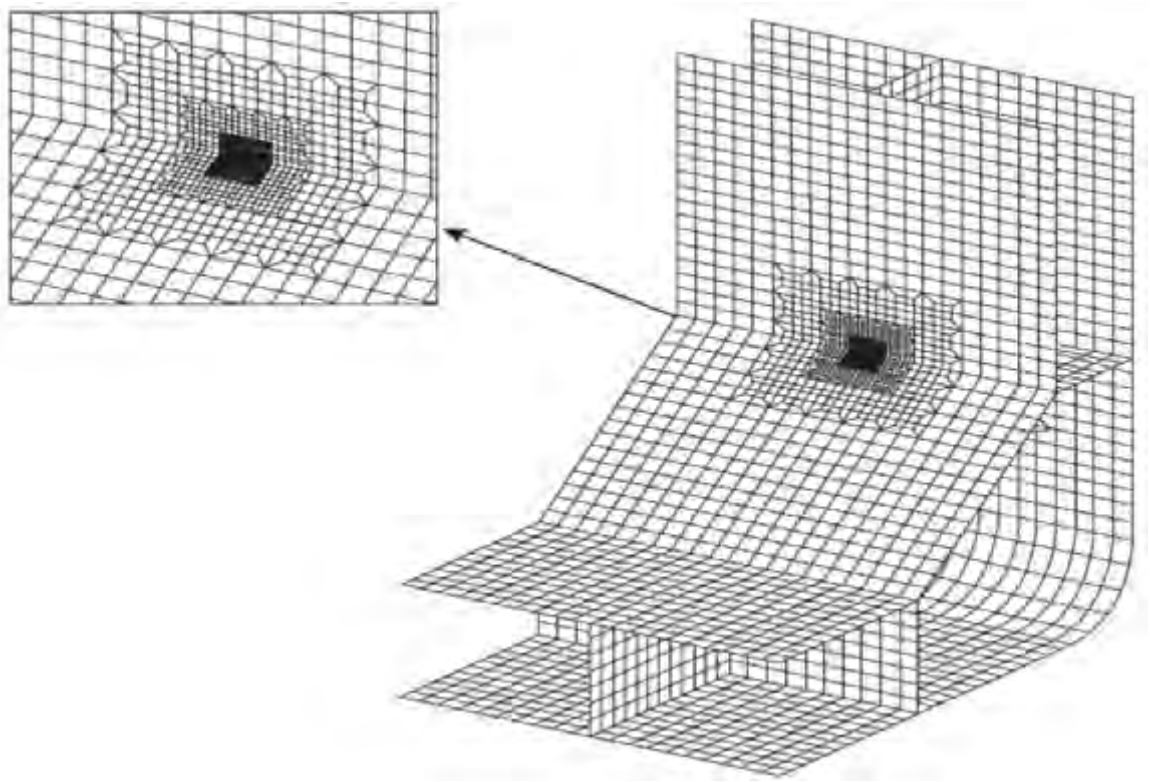


Figure 4.3.4: Local very fine mesh model of upper hopper knuckle connection between inner side shell and hopper plate in Bulk carriers

4.4 FE modelling for hot spot stress calculation

4.4.1 Separate local FE model (Sub-modelling technique)

4.4.1.1 Where a separate local finite element model is used for evaluating the hot spot stress range, the extent of the local model is to be such that the calculated stresses are not significantly affected by the imposed boundary conditions and the applied loads are far enough from the very fine mesh zones.

4.4.1.2 Where a separate local finite element model is used for evaluating the hot spot stress range, the boundary conditions and application of loads are to be in accordance with IRS *Guidelines on Structural Assessment of Ships based on Finite Element Method*.

4.4.1.3 The boundary of the fine mesh model is to be taken at adjacent primary supporting members such as girders, stringers and floors in the cargo hold model as far as practicable. Transverse web frames, stringer plates and girders at the boundaries of the local model need not be represented in the local model.

4.4.2 Mesh size

The evaluation of hot spot stress for 'a' type hot spot is to be based on shell element of mesh size $t_{n50} x\, t_{n50}$, where t_{n50} is the net thickness of the plate in way of the considered hot spot. The evaluation of hot spot stress for a 'b' type hot spot is to be based on shell element of mesh size 10×10[mm]. The aforementioned mesh size is to be maintained within the very fine mesh zone, extending over at least 10 elements in all directions from the fatigue hot spot position. The transition of element size between the coarser mesh and the very fine mesh zone is to be done gradually and an acceptable mesh quality is to be maintained. This transition mesh is to be such that a uniform mesh with regular shape gradually transitions from smaller elements to larger ones. An example of the mesh transition in way of the side frame bracket toe is shown in Figure 4.4.1. Mesh transitions should not be arranged close to the hot spot location.

For efficient read out of element stresses a mesh density in the order of size $t_{n50} x\, t_{n50}$ is generally preferred at the hot spot region.

4.4.3 Four node shell elements with adequate bending and membrane properties are to be used inside the very fine mesh zone. The four node element is to have a complete linear field of in-plane stresses and hence pure in-plane bending of the element can be exactly represented. In case of steep stress gradients, 8 node thin shell elements may be used, if deemed practical. The shell elements are to represent the mid plane of the plating. For practical purposes, adjoining plates of different thickness may be assumed to be median line aligned, i.e. no staggering in way of thickness change is required. The geometry of the weld and construction misalignment is not required to be modelled. The shell elements are to represent the mid plane of the plating as represented in Figure 4.5.1.

4.4.4 All structure in close proximity to the very fine mesh zones is to be modelled explicitly with shell elements. Quadrilateral elements are to have 90[deg] angles as much as possible or angles between 60[deg] and 120[deg]. The aspect ratio is to be close to 1. Triangular elements are to be avoided where possible. Use of extreme aspect ratio (e.g. aspect ratio greater than 3) and distorted elements (e.g. element's corner angle less than 60[deg] or greater than 120[deg]) are to be avoided.

4.4.5 Where stresses are to be evaluated on a free edge, such as cut-outs for stiffener connections at web frames, edge of plating and hatch corners, beam elements having the same depth as the adjoining plate thickness and negligible width is to be used to obtain the required local edge stress values.

Figure 4.4.1: Transition area between coarse and very fine mesh

4.4.6 Hatch corners and hatch coaming end bracket

4.4.6.1 In addition to the general requirements in Cl.4.4.1 to Cl.4.4.4, the modelling requirements in this subsection are applicable to the modelling of hatch corners/ hatch coaming end bracket. The selection of hatch corners/ hatch coaming end bracket for fatigue analysis is to be determined based on the level of stresses obtained from the cargo hold FE analysis.

4.4.6.2 Where separate local finite element models are used, the model extents are to be according to the following:

- Transversely, over the half-breadth of the ship,

- Longitudinally, from the midpoint of the cargo hold in which the concerned hatch corners/hatch coaming end bracket is located to the adjacent cargo hold up to and including the full width of the cross deck nearest to the concerned hatch corners/hatch coaming end bracket.

- Vertically, from the top plate of coaming to the intersection of the topside tank sloping plate with the side or inner side shell.

4.4.6.3 The primary supporting members and coaming stays are to be represented by shell finite elements having both membrane and bending properties. Figure 4.4.6.1 shows a typical FE model of the toe connection of a longitudinal hatch coaming end bracket to the deck plating with the very fine mesh zone having $t_{n50} x\, t_{n50}$ mesh size.

4.4.6.4 The hatch corner area is to be meshed using elements with a sufficiently small size to capture the local stress on the edge. In general, a minimum of 15 elements in 90[deg] arc are to be used to describe the curvature of the hatchway radius plating for a rounded corner (Figure 4.4.6.2). For an elliptical or parabolic corner, a minimum of 15 elements are to be used from the inboard radius end to a point on the edge located at half the longitudinal distance of the semi- major axis. A total of 20 elements are to be used at the elliptical edge of the hatch corner (Figure 4.4.6.3). However, the element edge dimensions along the free edge of the radius need not be less than the thickness of the plating being represented and also not to be greater than 5 times the thickness of the plating being represented. Mesh transitions should not be arranged close to bracket toes.

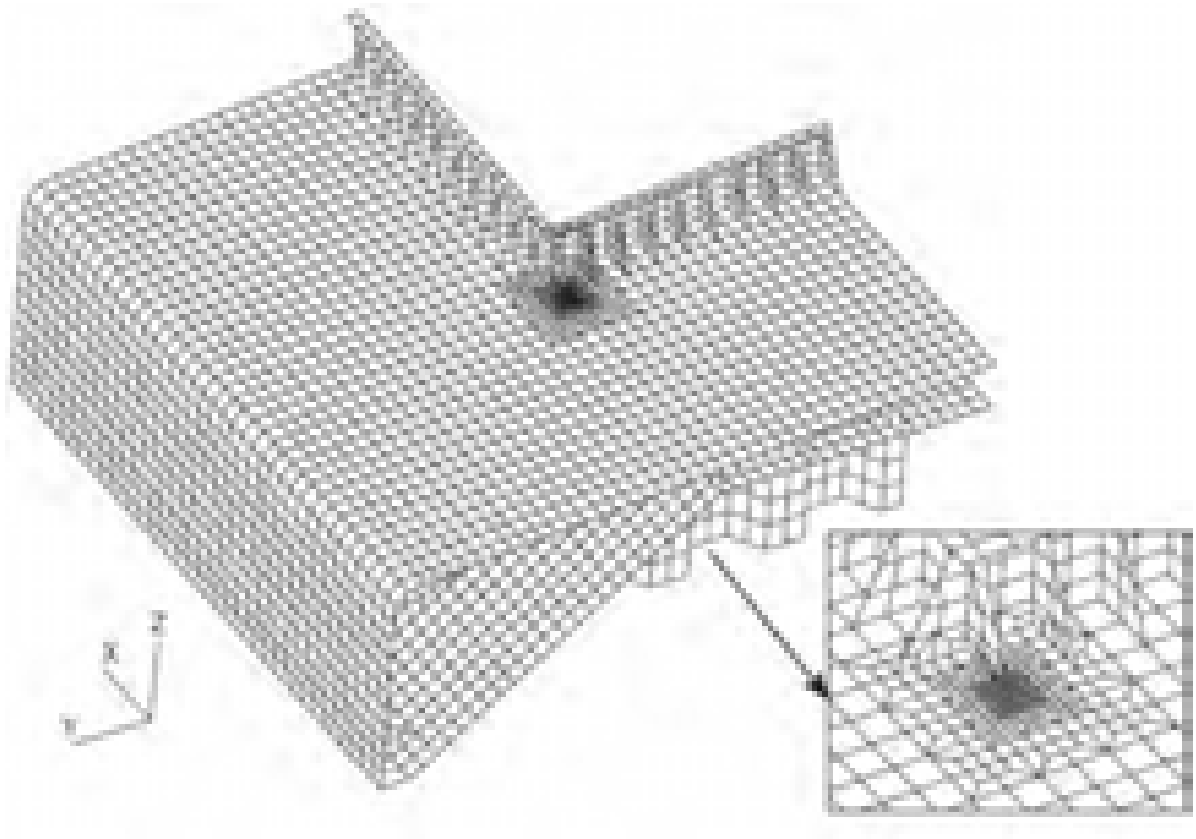


Figure 4.4.6.1: Local FE model of longitudinal hatch coaming end bracket to the deck plating with very fine mesh zone, $t_{n50} \times t_{n50}$ mesh

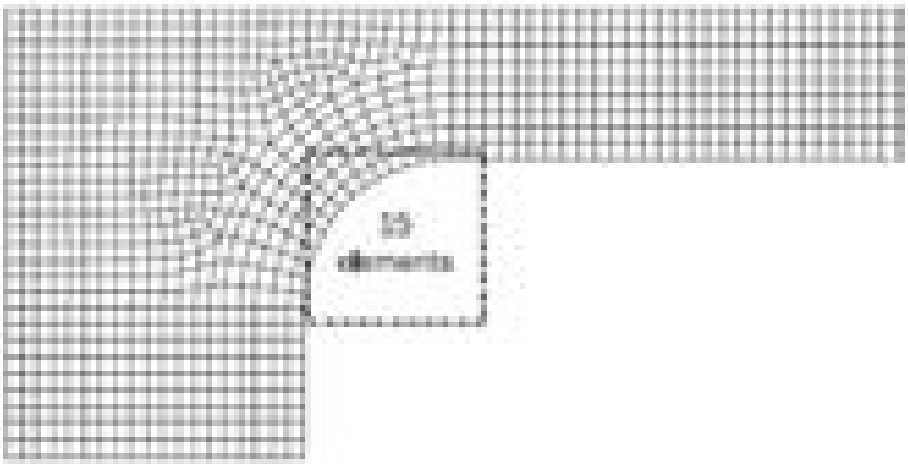


Figure 4.4.6.2: Mesh density for rounded hatch corner

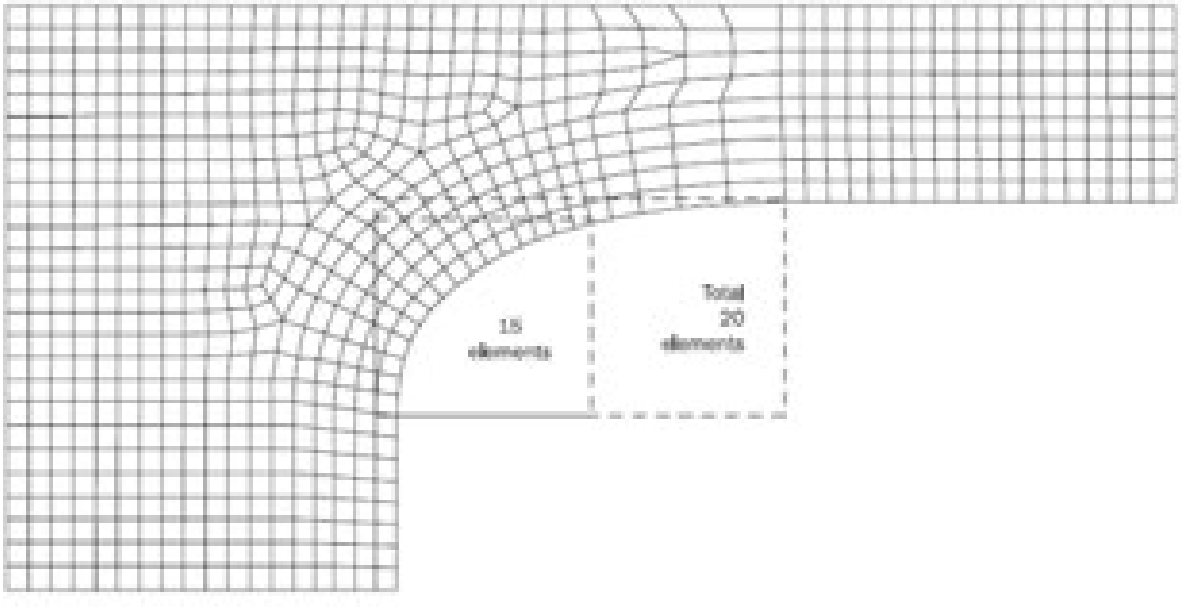


Figure 4.4.6.3: Mesh density for elliptical hatch corner

4.5 Hot spot stress for ordinary details different from web-stiffened cruciform joints

4.5.1 Types of welded details

This section applies to ordinary welded details: butt welds, fillet welds of T joints and ordinary cruciform joints. They do not apply to web-stiffened cruciform joints and bent hopper knuckles.

4.5.2 Welded details

For hot spot type 'a', the structural hot spot stress, σ_{HS}, is calculated from a finite element analysis with $t_{n50} x\, t_{n50}$ mesh density and is obtained by the following formula:

$$\sigma_{HS} = 1.12\sigma$$

where:

σ = Surface principal stress, in [N/mm^2], read out at a distance $t_{n50}/2$ away from the intersection line.

t_{n50} = Plate net thickness, in [mm], in way of the weld toe.

At structural details where the hot spot type 'a' is classified as a web-stiffened cruciform joint, the stress read out procedure of 4.6 is to be applied.

For hot spot type 'b', the stress distribution is not dependent on the plate thickness; the structural hot spot stress, σ_{HS} is derived from a finite element analysis with mesh density 10×10[mm] and is obtained by the following formula:

$$\sigma_{HS} = 1.12\sigma$$

where:

σ = Surface principal stress, [N/mm^2], read out at a distance from the intersection line of 5[mm]. The maximum of the surface principal stress from the two surfaces (edges) should be used.

4.5.3 Stress read out methods

4.5.3.1 The stress evaluation points are located at distances t/2 from the hot spot, where 't' is the plate thickness at the weld toe. This location is denoted as the stress read out point. For hot spot 'b' it would be 5[mm] from the hot spot. For modelling with shell elements without any weld the hot spot stress is taken as the principal stress within an angle ± 45[deg] to the normal to the weld and at the read-out point t/2 away from the intersection line.

4.5.3.2 Depending on the element type, one of the following stress read out methods are to be used:

- With 4-node shell element: Element surface stress (surface where the weld is located)

components at the centre points are linearly extrapolated to the line A-A as shown in Figure 4.5.1 to determine the stress components for load case 'i_1' and 'i_2' at the stress read out point located at a distance $t_{n50}/2$ from the intersection line for type 'a' hot spot. Two principal hot spot stresses are determined at the stress read out point the stress components tensor differences (between load case 'i_1' and 'i_2' in case of EDW approach) calculated from each side (side L, side R) of line A-A. The average from both sides of line A-A is taken. The angle θ between the direction x of the element co-ordinate system and the principal direction pX of the principal hot spot stress range coordinate system has to be determined.

- With 8-node shell element: With a $t_{n50} x\, t_{n50}$ element mesh using 8-node element type, the element mid-side node is located on the line A-A at a distance $t_{n50}/2$ for type 'a' hot spot. This node coincides with the stress read out point. The element surface stress components can be used directly without extrapolation within each adjacent element located on each side (side L, side R) of the line A-A as illustrated in Figure 4.5.2. Two principal hot spot stresses are determined at the stress read out point from the stress components tensor difference (between load case 'i_1' and 'i_2' in case of EDW approach) calculated from each side (side L, side R) of line A-A. The average from both sides of line A-A is taken. The angle θ between the direction x of the element coordinate system and the principal direction pX of the principal hot spot stress range coordinate system has to be determined.

4.5.3.3 For fatigue assessment of type 'b' hot spots, a beam element is to be used to obtain the fatigue stress range at the read out point 5[mm] from the hot spot. The stress range is to be based on axial and bending stress in the beam element. The beam element is to have the same depth as the connecting plate thickness while the in-plane width is negligible, i.e. it should not contribute to the strength and it should be compatible with the shell elements and read out the bending stress correctly.

4.5.3.4 The above read out procedure is based on element surface stresses. Generally, in FE software the element stresses are calculated at the Gaussian integration points located inside the element. Depending on the element type implemented in the FE software, it may be necessary to perform several interpolations in order to determine the actual stress at the considered stress read out point at the surface of the element mid-point or element edge.

4.5.4 Base material free plate edges

For fatigue assessment at a free plate edge, a beam element is to be used to obtain the fatigue stress range. The beam element is to have the same depth as the connecting plate thickness while the in-plane width should be negligible.

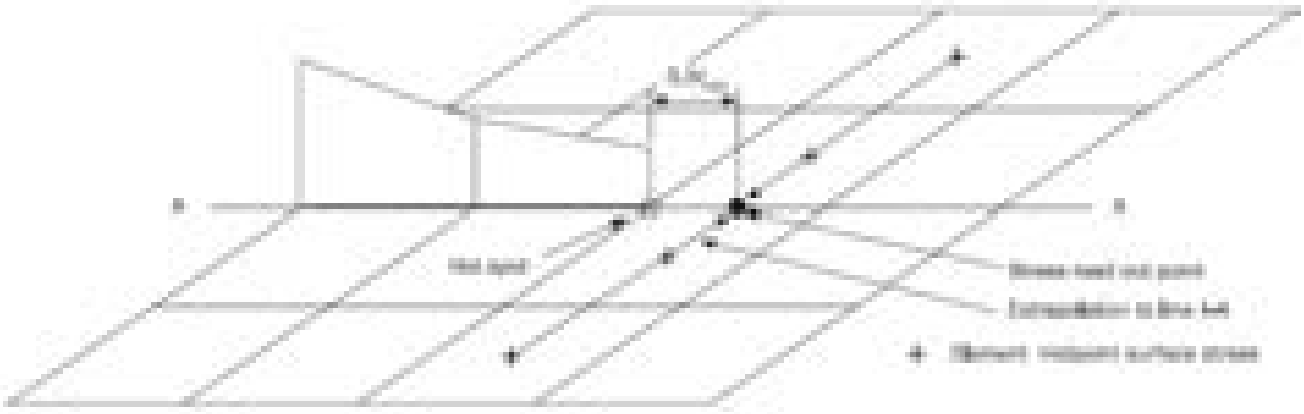


Figure 4.5.1: Determination of stress read out points and hot spot stress for 4-node element

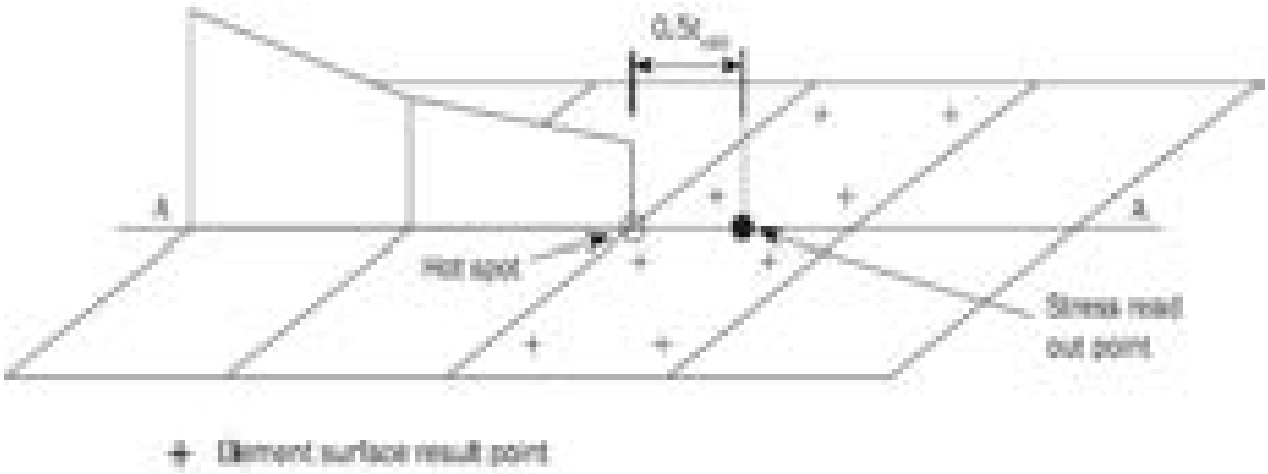


Figure 4.5.2: Determination of stress read out points and hot spot stress for 8-node element

4.5.5 Hot spot stress for bent hopper knuckle details

4.5.5.1 The hot spot stress at the inner bottom/hopper sloping plate in transverse and longitudinal directions of a bent hopper knuckle is to be taken as the surface principal stress read out from a point shifted away from the intersection line between the considered member and abutting member by the weld leg length.

The hot spot stress, in [N/mm^2], is to be obtained as:

$$\sigma_{HS} = \sigma_{shift}$$

where:

σ_{shift} = Surface principal stress, [N/mm^2], at the shifted read out position as defined in Cl.4.6.2.2 and taken as:

$$\sigma_{shift} = \sigma_{membrane}(x_{shift}) + \sigma_{bending}(x_{shift})$$

where,

$\sigma_{membrane}(x_{shift})$ = Membrane stress, in [N/mm^2], at position x_{shift}

$\sigma_{bending}(x_{shift})$ = Bending stress, in [N/mm^2], at position x_{shift}.

The procedure for calculation of hot spot stress at flange such as inner bottom /hopper sloping plate is the same that for web-stiffened cruciform joints as described in Cl.4.6.2.2. The procedure that applies for hot spots on the ballast tank side of the inner bottom/hopper plate in way of a bent hopper knuckle is in principle the same as that applied on the cargo tank side of the inner bottom plate for welded knuckle in Figure 4.6.2.1 and Figure 4.6.2.2. The intersection line is taken at the mid-thickness of the joint assuming median alignment. The plate angle correction factor and the reduction of bending stress as applied for a web-stiffened cruciform joint in Cl.4.6.2.3 are not to be applied for the bent hopper knuckle type.

4.5.5.2 The stress at hot spots located in way of the web (such as transverse web and side girder) of a bent hopper knuckle type is to be derived as described for web-stiffened cruciform joints in Cl.4.6.3.

4.6 Hot spot stress for web-stiffened cruciform joint

4.6.1 Application

4.6.1.1 Concerning the web-stiffened cruciform joint, at the intersection of plates corresponding to the hot spot, there is a high local stress concentration with large gradients. The following structural details are typically considered as a web-stiffened cruciform joint:

- Welded hopper knuckle connection

- Heel of horizontal stringer

- Lower stool – inner bottom connection.

Two kinds of hot spots relative to the web-stiffened cruciform joints are to be assessed:

- Hot spots at the flange of web-stiffened cruciform joint,

- Hot spots in way of the web of web-stiffened cruciform joint.

4.6.1.2 The procedure for calculating hot spot stress at flange of web-stiffened cruciform joint is given in Cl.4.6.2.

4.6.1.3 The procedure for calculating hot spot stress in way of the web of the web-stiffened cruciform joint is given in Cl.4.6.3.

4.6.1.4 Different examples of web stiffened cruciform joints are illustrated in Figure 4.6.1.

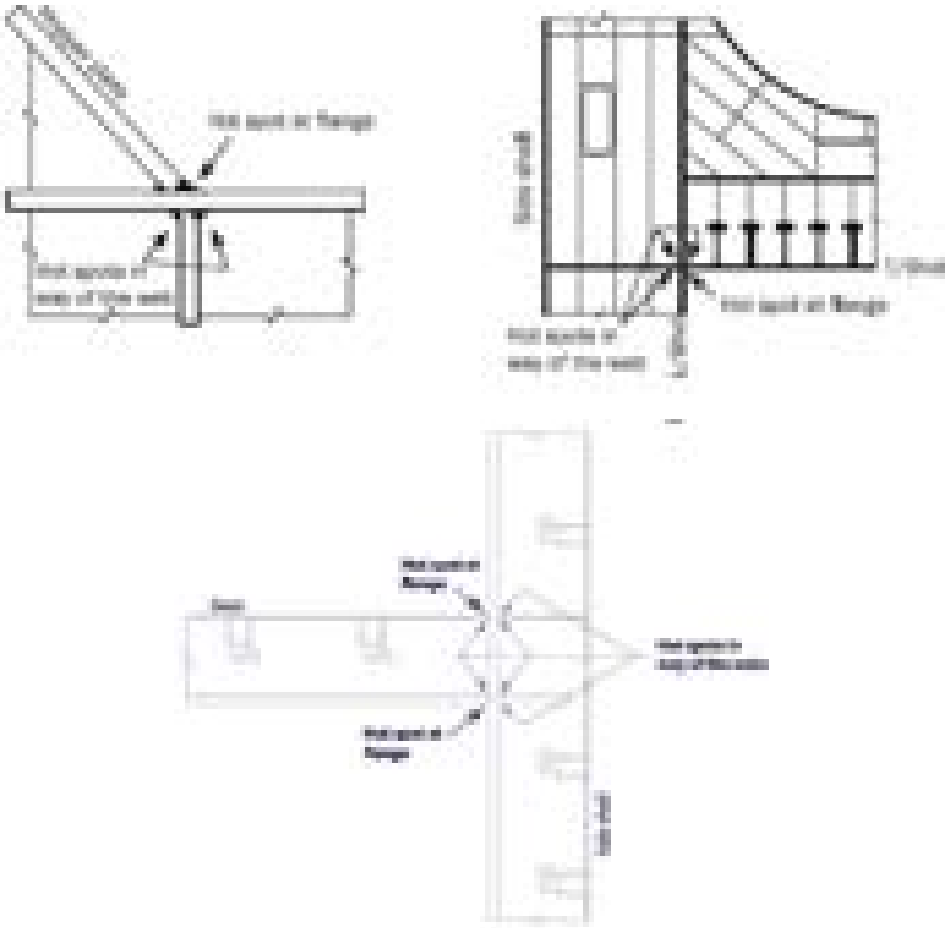


Figure 4.6.1: Web-stiffened cruciform joints

4.6.2 Calculation of hot spot stress at the flange

4.6.2.1 For hot spot at the flange of web-stiffened cruciform joints, the surface principal stress is to be read out from a point shifted away from the intersection line between the considered member and abutting member to the position of the actual weld toe and multiplied by 1.12. The intersection line is taken at the mid-thickness of the cruciform joint assuming a median alignment.

4.6.2.2 The hot spot stress, in [N/mm^2], is to be obtained as:

$$\sigma_{HS} = 1.12\sigma_{shift}$$

where:

σ_{shift} = Surface principal stress, [N/mm^2], at shifted stress read out position.

The stress read out point shifted away from the intersection line is obtained as:

$$x_{shift} = \frac{t_{1-n50}}{2} + x_{wt}$$

where:

t_{1-n50} = Net plate thickness of the plate number 1, in [mm], as shown in Figure 4.6.2.1 associated with hot spot on plate number 1. For hot spot and read out position on plate number 2, t_{1-n50} should be replaced with t_{2-n50} in estimating x_{shift}.

x_{wt} = Extended fillet weld leg length, in [mm], as defined in Figure 4.6.2.1, not taken larger than t_{1-n50} for plate number 1, or t_{2-n50} for plate number 2.

4.6.2.3 The stress at the shifted position is derived according to the following formula and illustrated in Figure 4.6.2.2:

$$\sigma_{shift} = \left(\sigma_{membrane}(x_{shift}) + 0.6\,\sigma_{bending}(x_{shift})\right).\beta$$

where:

$\sigma_{bending}(x_{shift})$ = Bending stress, [N/mm^2], at the shifted position taken as:

$$\sigma_{bending}(x_{shift}) = \sigma_{surfce}x_{shift} - \sigma_{membrane}x_{shift}$$

$\sigma_{surfce}x_{shift}$ = Total surface stress at x_{shift} position (including membrane stress and bending stress) in [N/mm^2].

$\sigma_{membrane}x_{shift}$ = Membrane stress at x_{shift} position in [N/mm^2].

β = Plate angle hot spot stress correction factor, taken as:

For $\alpha = 135[deg]$

$$\beta = 0.96 - 0.13\frac{x_{wt}}{t_{1-n50}} + 0.20\left(\frac{x_{wt}}{t_{1-n50}}\right)^2$$

For $\alpha = 120[deg]$

$$\beta = 0.97 - 0.14\frac{x_{wt}}{t_{1-n50}} + 0.32\left(\frac{x_{wt}}{t_{1-n50}}\right)^2$$

For $\alpha = 90[deg]$

$$\beta = 0.96 - 0.031\frac{x_{wt}}{t_{1-n50}} + 0.24\left(\frac{x_{wt}}{t_{1-n50}}\right)^2$$

α = Angle, in [deg], between the plates forming a web-stiffened cruciform joint as shown in Figure 4.6.2.2.

The procedure is valid for $0 \le x_{wt}/t_{1-n50} \le 1.0$ and correction factors for connections with plate angles intermediate to those given should be derived based on a linear interpolation of the above values. The calculated hot spot stress is to be used in conjunction with the hot spot S-N curve for weld toe connections according to Section 2.

4.6.2.4 Surface principal stresses at the centre point of the two first elements (independent of 4- or 8-node elements) on left and right side of the line A-A are averaged and taken as the surface principal stresses in way of the web position (line A-A). The surface principal stresses for load case 'i_1' and 'i_2' are linearly interpolated along the line A-A in order to determine hot spot principal stresses at the stress read out point located at the x_{shift} position as shown in Figure 4.6.2.3. The two principal hot spot stress ranges are determined at the stress read out point between load case 'i_1' and 'i_2'.

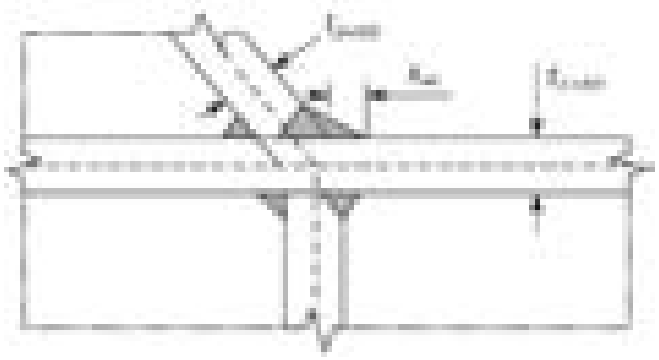


Figure 4.6.2.1: Geometrical parameters of web-stiffened cruciform connections

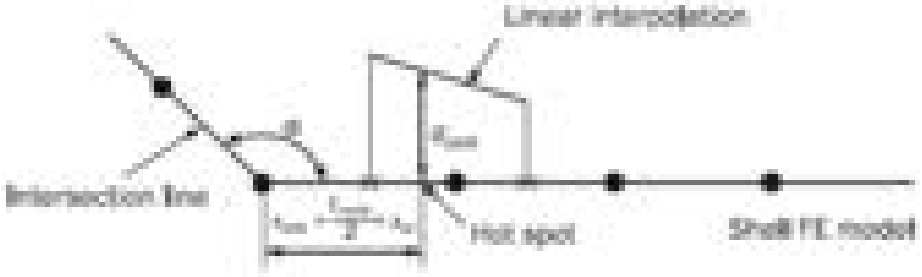


Figure 4.6.2.2: Procedure for calculation of hot spot stress at web-stiffened cruciform connections

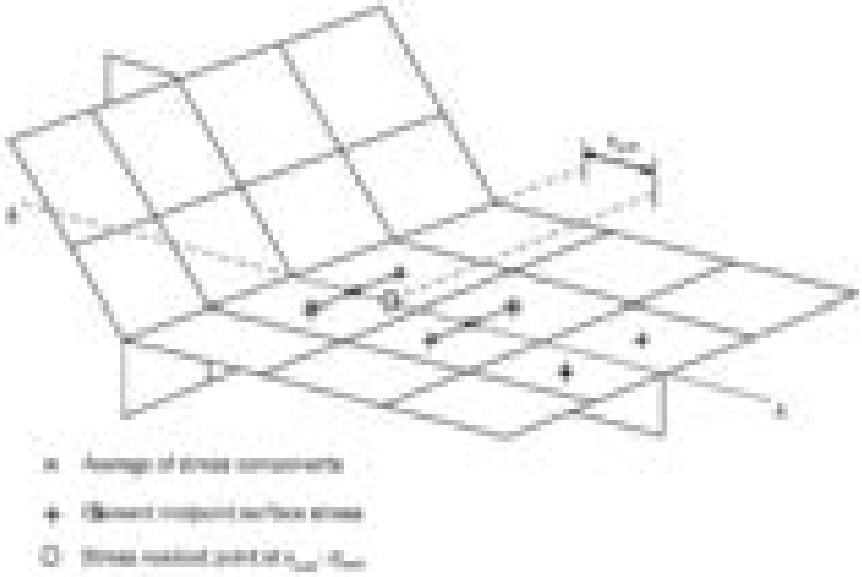


Figure 4.6.2.3: Determination of stress read out points for the flange of web-stiffened cruciform connections

4.6.3 Calculation of hot spot stress in the web

4.6.3.1 Hot spots located in way of the web as indicated in Figure 4.6.3 are to be checked with the hot spot stress defined from the maximum principal surface stress at the intersection offset by the distance x_{shift} from the vertical and horizontal element intersection lines as illustrated in Figure 4.6.3. The intersection line is taken at the mid thickness of the cruciform joint assuming a median alignment.

4.6.3.2 The hot spot stress in [N/mm^2], is to be obtained as:

$$\sigma_{HS} = \sigma_{shift}$$

where:

σ_{shift} = Maximum surface principal stress in [N/mm^2], at the intersection offset by the distance x_{shift}.

The stress read out point at the intersection offset is obtained as:

$$x_{shift} = \frac{t_3}{2} + x_{wt}$$

where:

t_3 = Net plate thickness of the web, in [mm], as shown in Figure 4.6.3.

x_{wt} = Extended fillet weld leg length in [mm], taken as

$$x_{wt} = \min(\ell_{leg1}, \ell_{leg2})$$

ℓ_{leg1}, ℓ_{leg2} = Leg length in [mm], of the vertical and horizontal weld lines as shown in Figure 4.6.3.

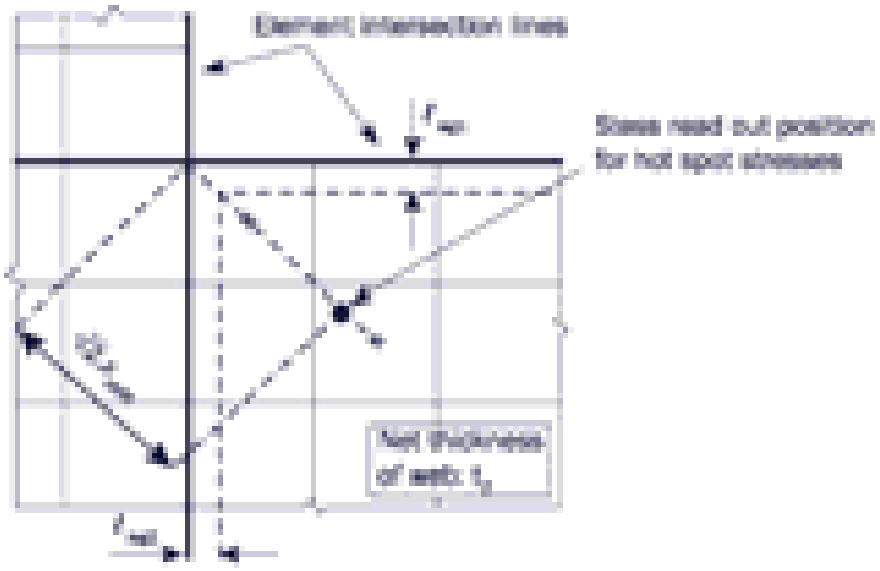


Figure 4.6.3: Hot spots in way of web

4.7 Limitations of hot spot stress approach

4.7.1 Scope of application of hot spot stress approach

The hot spot stress approach given in Section 1, Cl.1.2.5 is not applicable for simple cruciform joints and simple T-joints or simple butt joints when the stress flow in direction I as shown in Figure 4.7.1 is considered. Analysing such connections with shell elements will result in a hot spot stress equal to the nominal stress. The hot spot stress approach is therefore not directly applicable when the stress flow in direction I, is considered. For stresses in the direction normal to the weld at hot spot location “c” (direction I) there is no stress flow into the transverse plating as it is represented only by one plane in the shell model. However, it attracts stresses for in-plane direction (direction II) at hot spot location “a”.

In situations where a bracket (or web) is fitted behind the transverse plate as shown in Figure 4.7.1, acting with stiffness in the direction normal to the transverse plate, stresses flow also into the transverse plate and the hot spot methodology is considered applicable.

4.7.2 Correction of hot spot stress approach for simple cruciform joints

The hot spot stress at position ‘c’ for simple cruciform joints and simple T-joints is to be determined by the stress read out procedure given in 4.5 multiplied by a geometrical stress concentration factor of 1.3 and is taken as:

$$\sigma_{HS} = 1.3 \times 1.12\sigma$$

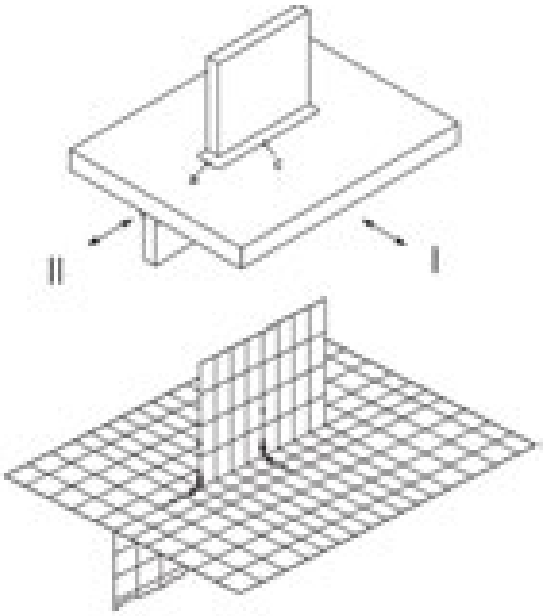


Figure 4.7.1: Illustration of check points in way of a welded attachment under orthogonal applied in plane loads

4.8 Method for hot spot stress extraction

4.8.1 Hot spot extraction method used to obtain hot spot stress from the stress readout points is by linearly extrapolating from two or three readout points to the structural intersection line. (IIW-1823-07) recommendations for fatigue design of welded joints is utilized for this method of hot spot stress extraction.

4.8.2 In case of biaxial stress state at the plate surface, the principal stress which acts approximately in line with the perpendicular to weld toe (within ±60[deg]) are to be used. The other principal stress may need to be analyzed, if necessary, using the fatigue class in the nominal stress approach for welds parallel to the stress (IIW-1823-07).

4.8.3 While using this method of stress extraction, meshing size and the limitation to weld configurations are to be referred from the IIW recommendation

.

End of Chapter

Appendix A – Detail Design Standard

A.1 General

A.1.1 Purpose

A.1.1.1 This detail design standard is provided to ensure improved fatigue performance of structural details. Design standard provides fatigue resistant detail design at an early stage in the structural design process by giving considering the following:

- Application of fatigue design principles.

- Construction tolerances and other practical considerations.

- In-service experience and fatigue performance.

A.1.1.2 The design standard is to be applied to the design of ship structural details in following steps:

- Highlighting potential critical areas within the ship structure.

- Identification of the fatigue hot spot locations for each of the critical structural details.

- Provision of a set of alternative improved configurations from which a suitable solution can be selected.

- Requirements on geometrical configurations, scantlings, welding requirements and construction tolerances.

- Post fabrication method of improving fatigue life, such as weld toe grinding.

A.1.1.3 The detail design standard may provide requirements in order to prevent the following types of fatigue cracks initiating from the:

- Weld toe and propagating into the base material.

- Weld root and propagating into the plate section under the weld.

- Weld root and propagating through the weld throat.

- Surface irregularity or notch at the free plate edge and propagating into the base material

A.1.2 Application

The structural details described in this Appendix are to be designed according to the given design standard. Alternative detail design configurations may be accepted by IRS, subject to demonstration of satisfactory fatigue performance.

A.2 Stiffener-frame connections

A.2.1 Design standard A

A.2.1.1 Designs for cut outs in cases where web stiffeners are omitted or not connected to the longitudinals are required to adopt tight collar or the improved design standard “A” as shown in Table A.2.1 or equivalent, for the following members:

- Side shell below $1.1T_{sc}$

- Bottom.

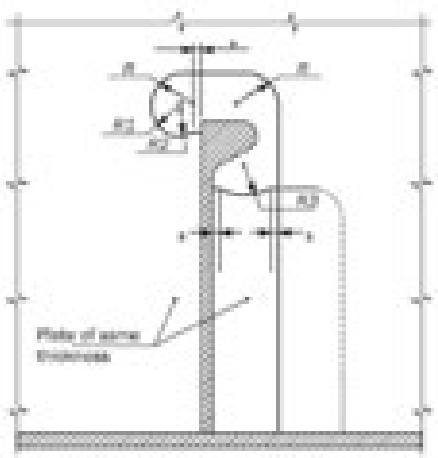
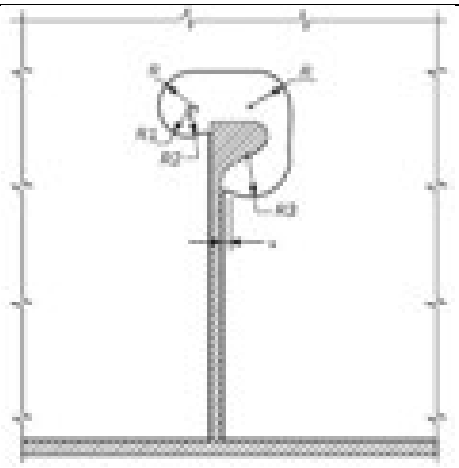
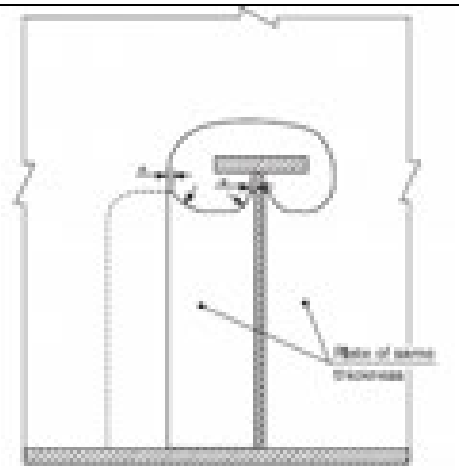
- Inner hull longitudinal bulkhead below $1.1T_{sc}$

- Topside tank sloping plating below $1.1T_{sc}$

- Hopper.

- Inner bottom.

A.2.1.2 Designs that are different from those shown in Table A.2.1 are acceptable subject to demonstration of satisfactory fatigue performance, e.g. by using comparative finite element analysis. The comparative FE analysis is to be performed following the modelling guidance given in Figure A.2.1.

Table A.2.1: Design standard A – stiffener-frame connection	
Cut outs for longitudinals in transverse webs where web stiffeners are omitted or not connected to the longitudinal flange	
Design standard A	
1	2
	
3	4
	
Note 1: Soft toes marked "*" are to be dimensioned to suit the weld leg length such that smooth transition from the weld to the curved part can be achieved. Maximum 15[mm] or thickness of transverse web/collar plates/lug plates whichever is the greater. Note 2: Configurations 1 and 4 indicate acceptable lapped lug plate connections.	
Critical location	Locations around cut-out with high stress concentration and locations in way of weld terminations.
Detail design standard	Improved slot shape to avoid high stress concentrations in transverse webs due to shear loads and local pressure loads transmitted via welded joints.
Building tolerances	Ensure alignment of all connecting members and accurate dimensional control of cut-outs according to IACS Recommendation No.47.
Welding requirements	A wraparound weld, free of undercut or notches, around the transverse web connection to longitudinal stiffener web.

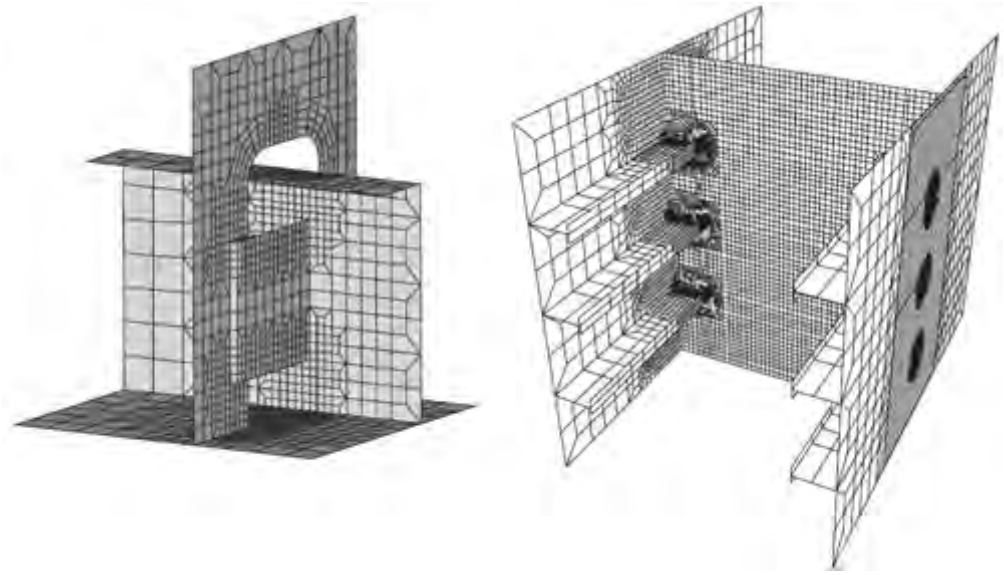


Figure A.2.1: Finite element model for verification of equivalent design

A.2.2 Equivalent design of stiffener-frame connections

(a) If the required designs for stiffener-frame connections A.2.1 are not followed, the alternative design is to be verified to have equivalent fatigue strength to the design standard “A” or to be verified to have satisfactory fatigue performance. The alternative design is to be verified according to the procedure given in A.2.2(b) to A.2.2(e) and documentation of results is to be submitted to IRS.

(b) The procedure in A.2.2(c) and A.2.2(d) is provided to verify the alternative design to have equivalent fatigue strength with respect to any position in the transverse ring, i.e. double bottom and double side. The hot spot stress as well as the stress at the free plate edge of the alternative design and that of the required design is to be compared to the critical hot spots in way of the cut-out. The critical hot spots depend on the detail design and are to be selected in agreement with *IRS.* The hot spot stress is to be derived according to Section 4, Cl.4.5.2 and Cl.4.5.3. It is to be noted that welded hot spots at the free edge are classified as hot spot type “b”.

(c) The very fine mesh finite element models are made to analyze the behavior in way of double side or double bottom. The models should have an extent of 3 stiffeners in cross section, i.e. 4 stiffener spacings, and the longitudinal extent is to be one half frame spacing in both forward and aft direction. A typical model is shown in Figure A.2.1. No cut-outs for access openings are to be included in the models. Connection between the lug or the web-frame to the longitudinal stiffener web, connections of the lug to the web-frame and free edges on lugs and cut-outs in web-frame are to be modelled with elements of net plate thickness size ($t_{n50} x\, t_{n50}$). The mesh with net plate thickness size should extend at least five elements in all directions. Outside this area, the mesh size may gradually be increased in accordance with the requirements in Section 4, 4.3 and 4.4. The eccentricity of the lapped lug plates is to be included in the model. Transverse web and lug plates are to be connected by eccentricity elements (transverse plate elements). The height of eccentricity element is to be the distance between mid-layers of transverse web and lug plates having a thickness equal to 2 times the net thickness of web-frame plate t_{w-n50}. Eccentricity elements representing fillet welds are shown in Figure A.2.2(a).

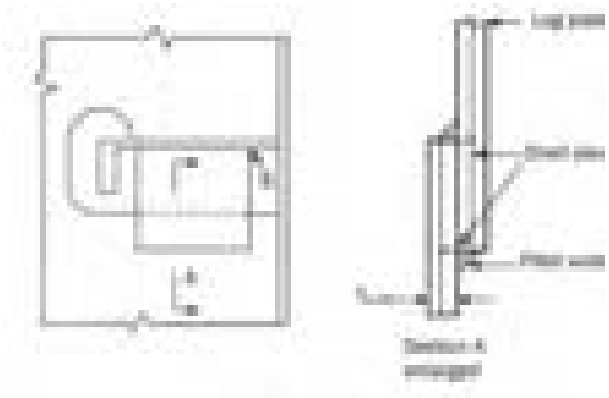


Figure A.2.2(a): Modelling of eccentric lug plate by shell elements

(d) Three load cases are to be applied to the models of the design standard and alternative designs:

- External pressure of unit value, fixed boundary conditions at top and bottom of model.

- Shear stress by prescribed unit displacement at the model top and fixed boundary conditions at the model bottom.

- Axial load by prescribed unit displacement at the model top and fixed boundary conditions at the model bottom.

The forward and aft part of the model is to have symmetry condition describing the behavior in a double hull structure. Load application and boundary conditions are provided in Figure A.2.2(b).

(e) The alternative design may also be verified to have satisfactory fatigue performance using sub-modelling technique where a very fine mesh model of the alternative design located at the actual position of the stiffener-frame connection is analyzed. The alternative design is considered acceptable if the fatigue acceptance criterion given in Section 2 is achieved. The fatigue acceptance criterion is checked by applying the methodology described in Sections 1, 2 and 4. The alternative design is considered acceptable only for the particular position where it is analyzed.

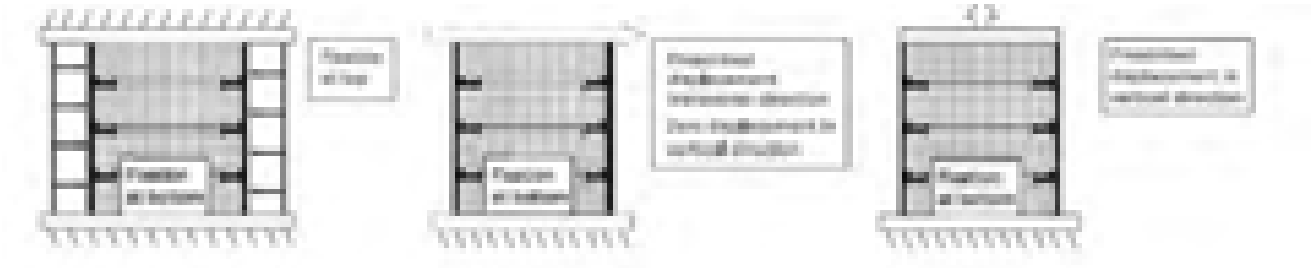


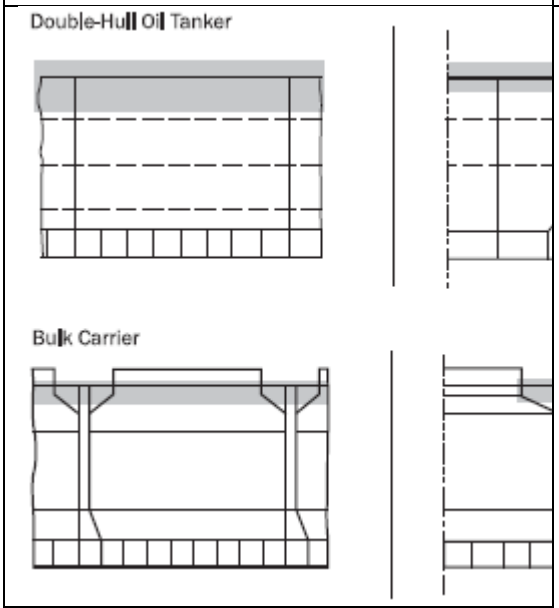
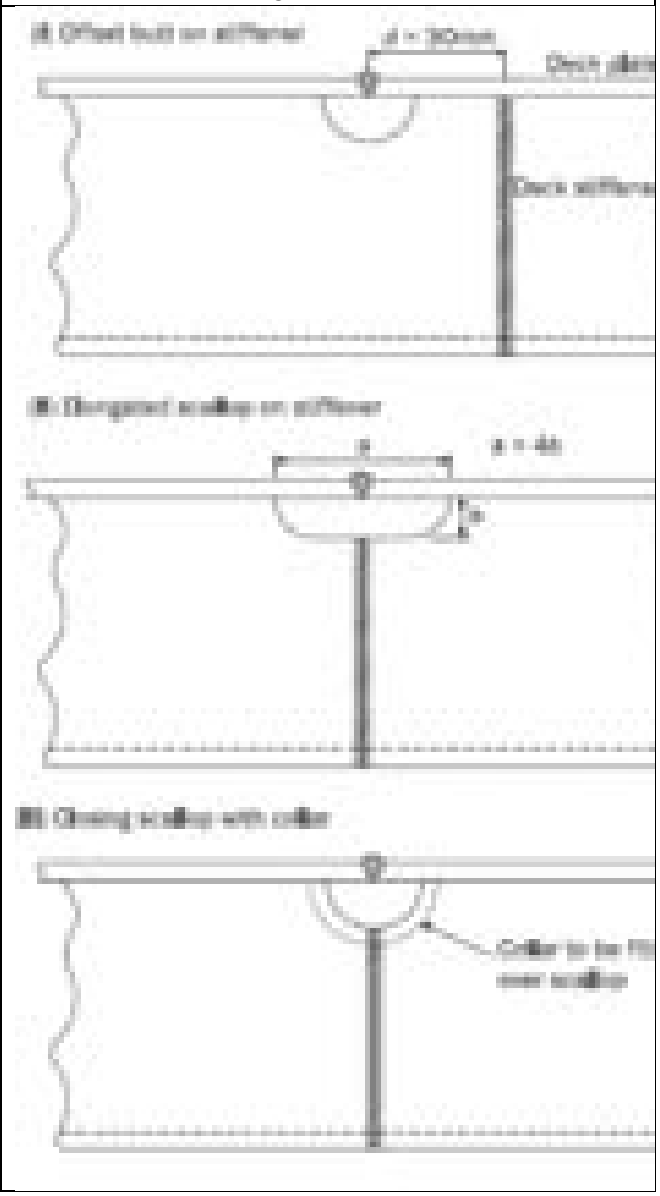
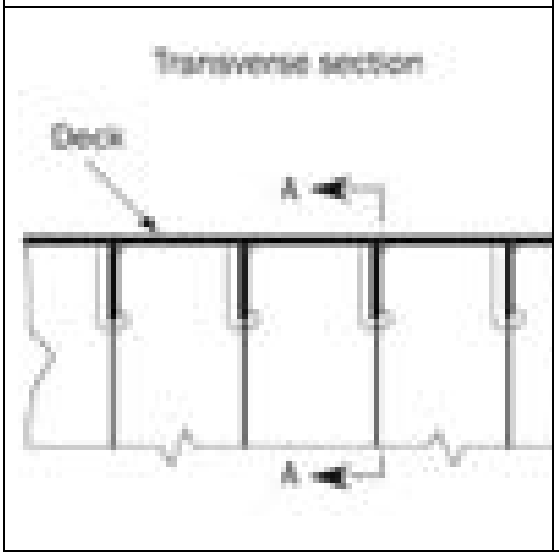
Figure A.2.2(b): Load application and boundary conditions – FE model for verification of alternative design

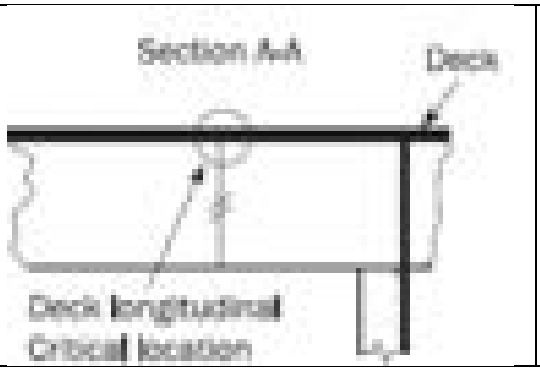
A.3 Scallops in way of block joints

A.3.1 Design standard B

Scallops in way of block joints in the cargo tank/hold region, located on the stiffeners fitted on strength deck, and side above 0.9D from the baseline, are required to be designed according to the design standard B as shown in Table A.3.1.

Table A.3.1: Design standard B – scallops in way of block joints

Welding of deck stiffeners in way of block joints	
Critical areas	Design standard B
	
Critical locations	
	

<table>
<tr><td> 
</td><td></td></tr>
<tr><td colspan="2">Note 1: Alternative scallop geometry to that shown in option II may be accepted subject to demonstration of satisfactory fatigue life based on hull girder loads taking into account additional stress concentration factor in way of weld</td></tr>
<tr><td>Critical location</td><td>Welding of deck stiffeners in way of block joints in cargo tank region, the strength deck and side above 0.9D from the baseline.</td></tr>
<tr><td>Detail design standard</td><td>All scallops are to be fitted according to detail design standard B.</td></tr>
<tr><td>Building tolerances</td><td>Ensure alignment of all structural members according to IACS Recommendation No.47.</td></tr>
<tr><td>Welding requirements</td><td>Full penetration butt weld, free of undercut or notches, around the web and flange of the longitudinal stiffener at block joints, particularly in way of the weld termination at the scallop for option II.</td></tr>
</table>

End Of Appendix A

Chapter 11

Special Hull Requirements

Contents

Section

1 *General*
2 *Bow Flare Impact*
3 *Bottom Slamming*
4 *Stern Slamming*
5 *Wheel Loading*
6 *Special Hull Structures*
7 *Tanks Subject to Sloshing*

Section 1

General

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

1.1 Application

1.1.1 For ships with various Ice Class notations, additional requirements given in Part 5, Chapter 21 are to be complied with.

1.2 Materials

1.2.1 All steel plates and sections, castings and forgings used in the constructions are to be tested and approved in accordance with the requirements of Part 2, Chapter 3, Chapter 4 and Chapter 5 respectively. Material grades for plates and sections are to be selected as defined in Part 3, Chapter 2.

1.2.2 Bar keels and stems may either be steel castings or steel forgings.

1.2.3 Stern frames, rudder horns and shaft brackets may be constructed of cast or forged steel or may be fabricated from plates.

Section 2

Bow Flare Impact

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

f_{bdg} = Bending moment factor.

$$f_{bdg} = 8\left(1 + \frac{n_s}{2}\right)$$

n_s = End fixation factor taken as:

For both ends with low-end fixity (simply supported), $n_s = 0$

For one end fixed and one end simply supported $n_s = 1$

For continuous members or members with bracketed fitted at both ends, $n_s = 2$

λ_w = Wave length in [m].

ζ_a = Average wave amplitude in [m].

x_g = The distance in [m], of the station under consideration from the LCG of the ship.

X = The distance of a section from AP in [m].

θ = Extreme pitching motion in [deg], i.e., Pitch angle as per Chapter 5, Section 2.

$$\theta' = \theta.\frac{\pi}{180}$$

H_s = Significant wave height in [m] $= C.C_w$

where, C is as defined in Cl.2.2.1.

For Coastal vessels, H_s need not be taken more than 5[m].

F_n = Froude number $= V/\sqrt{gL}$

V = Design speed of vessel in [m/s].

V_{th}= Threshold bow flare slamming speed in [m/s] = $0.29\sqrt{L}$

AC-III = Acceptance criteria set as defined in Chapter 1, Section 1.

2.1 Application

2.1.1 Bow flare slamming requirements given in this section are applicable to vessels of L≥ 60[m] on unrestricted or coastal service.

2.1.2 Bow flare slamming impact pressure defined in Cl.2.2.1 is applicable when $V \geq V_{th}$ and $F_n \leq 0.41$. For vessels, when $V \geq V_{th}$ and $F_n > 0.41$, bow flare slamming impact pressure defined in Cl.2.2.1 is not applicable and needs to be specially considered.

2.1.3 The side structure in the region forward of 0.25L abaft of F.P. and vertically above the summer load waterline, is to be strengthened for bow impact pressure 'P_{BOW}'. Refer Figure 2.1.1.

2.1.4 Outside the strengthening area the requirements are to be gradually tapered to the ordinary requirements to maintain adequate continuity of strength in both longitudinal and transverse directions.

2.1.5 The requirements given below are to be applied in addition to applicable scantling requirements in Chapter 7. Local scantling increases due to impact loads are to be made with due consideration given to details and avoidance of hard spots, notches and other harmful stress concentrations.

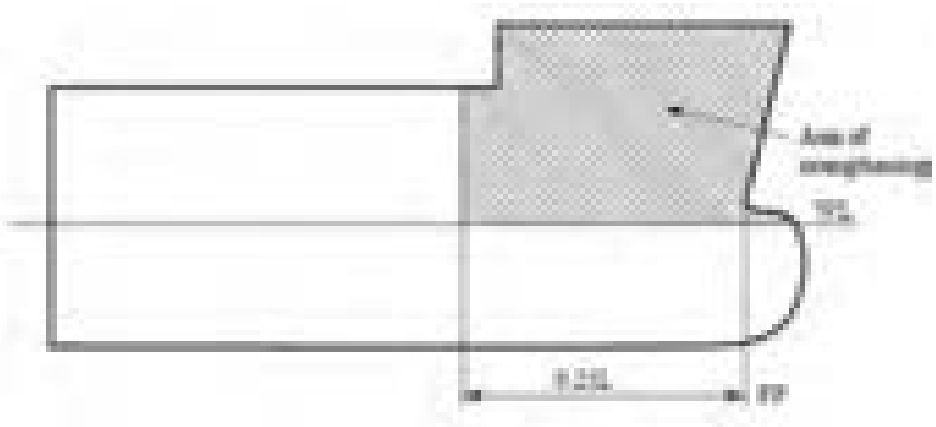


Figure 2.1.1: Extent of strengthening against bow impact

2.2 Design bow impact pressure

2.2.1 The design bow impact pressure, P_{BOW} in [kN/m^2], is to be taken as:

$$P_{BOW} = \frac{1}{2}\rho\, k_b\, U_i^2$$

where,

$k_b = 0.028\alpha^2 - 1.68\alpha + 30 \quad \text{for } \alpha \leq 30°$

$k_b = 7.2 - 0.08\alpha \quad \text{for } 30° < \alpha \leq 75°$

$k_b = 1.2 \quad \text{for } \alpha > 75°$

Impact velocity in [m/s] for finding out Bow flare pressure is the resultant of the normal velocity (U_n) and the tangential velocity (U_t) as given below:

$$U_i = \sqrt{U_n^2 + U_t^2}$$

where,

$$U_n = 2\sqrt{abs(-\omega_{ext}\{z_{ext}\cos(\theta') + x_g\theta' + +\zeta_{ext}\sin(\frac{\left(1.25 - 2.5(1 - \frac{X}{L}\right)\pi}{2})\cos(\theta')\})}$$

$U_t = U/\cos(\beta) \quad \text{for } \beta \leq 60[\text{deg}]$

$U_t = 2.U \text{ for } \beta > 60[\text{deg}]$

$U = (-134F_n^2 + 31F_n + 3)\omega_{ext}\,\zeta_a \text{ for } 0 \leq F_n < 0.16$

$U = (-105F_n^2 + 65F_n - 3)\omega_{ext}\,\zeta_a \text{ for } 0.16 \leq F_n \leq 0.41$

ω_{ext} = Encounter wave frequency, rad/s

$\omega_{ext} = \omega(1 + \frac{\omega V}{g})$

ω = Wave frequency in $[\text{rad/s}]$

$\omega = \sqrt{\frac{2\pi g}{\lambda_w}}$

z_{ext} = Extreme heave motion = $7.2 - 0.012L$

ζ_a =Average wave amplitude in [m].

$\zeta_a = \frac{1}{1.41}(H_s/2)$

ζ_{ext} = Extreme wave amplitude = $0.95(C.C_w)$

$C = \min\left[\sqrt{\frac{L + \lambda_w - 25}{L}}\ ,1.3\right]$

$\lambda_w = 1.05L \text{ for } 0 \leq F_n < 0.16$

$\lambda_w = 1.57L \text{ for } 0.16 \leq F_n < 0.41$

C_w is the wave coefficient as defined in Chapter 5, Section 2

α = Flare angle in [deg] between a tangential plane of side plating and a vertical line, measured at the point considered. (Refer Figure 2.2.1).

β = Local waterline angle, i.e., the angle in degrees made by tangent at a point under consideration with the centreline in the x-y plane (water plane). (Refer Figure 2.2.1).

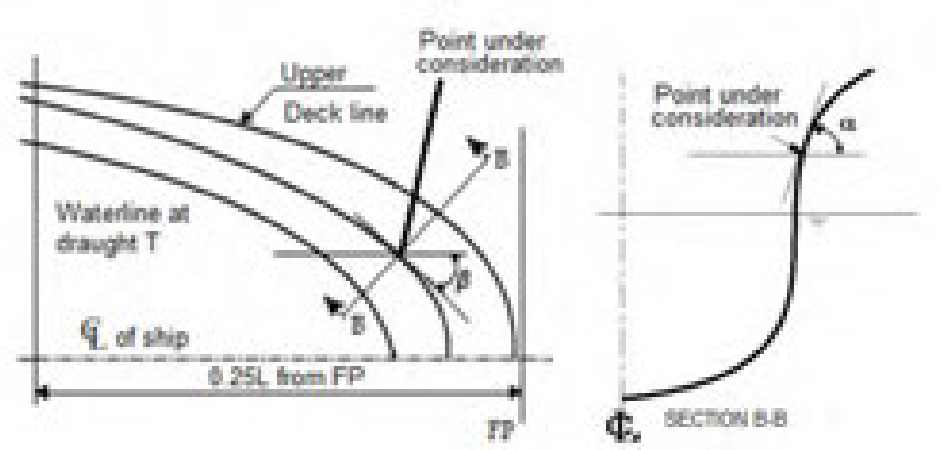


Figure 2.2.1: Local waterline angle and Local body plan angle

2.3 Scantling requirements

2.3.1 Side shell plating

2.3.1.1 The net thickness of the side shell plating, in [mm], subjected to impact pressures from bow flare slamming is not to be less than:

$$t = \frac{0.0158.\alpha_p.b}{C_d}\sqrt{\frac{P_{BOW}}{R_{eH}}}$$

where:

C_d = Plate capacity correction coefficient = 1.3

α_p = correction factor for the panel aspect ratio to be taken as below but not to be greater than 1.0:

$$\alpha_p = 1.2 - \frac{b}{2.1\,a}$$

a,b = Length and breadth of plate panel, in [mm], as defined in Chapter 3 Section 7, Cl.7.5.1.

2.3.2 Side shell stiffeners

2.3.2.1 The design of end connections of stiffeners in the bow impact region is to ensure end fixity, either by making the stiffeners continuous through supports or by providing end brackets complying with Chapter 3, Section 6, Cl.6.2.2.

Where it is not practical to comply with this requirement, the net plastic section modulus, Z_{pl_alt}, in [cm^3], for alternative end fixity arrangements is not to be less than:

$$Z_{pl_alt} = \frac{16\,Z_{pl}}{f_{bdg}}$$

where:

Z_{pl} **=** Effective net plastic section modulus, in [cm^3], required by Cl.2.3.2.3.

2.3.2.2 Scantlings and arrangements of primary supporting members, including decks and bulkheads, in way of the stiffeners, are to comply with Cl.2.5.3. In areas of the greatest bow impact load, the web stiffeners arranged perpendicular to the hull envelope plating and the double-sided lug connections are to be provided.

2.3.2.3 The side shell stiffeners within the strengthening area defined in Cl.2.1.3 are to comply with the following criteria:

(a) The effective net plastic section modulus, Z_{pl}, in [cm^3], calculated in accordance with Chapter 3, Section 7, Cl.7.4.6, is not to be less than:

$$\mathbf{Z_{pl} = \frac{P_{BOW}.s.\ell_{bdg}^2}{f_{bdg}.C_s.R_{eH}}}$$

where:

C_s = Permissible bending stress coefficient taken as:

= 0.9 for acceptance criteria set AC-III.

(b) The net web thickness, $\mathbf{t_w}$, in [mm], is not to be less than:

$$\mathbf{t_w = \frac{\sqrt{3}.P_{BOW}.s.\ell_{shr}}{2.d_{shr}.C_t.R_{eH}}}$$

where:

C_t = Permissible shear stress coefficient taken as:

= 1.0 for acceptance criteria set AC-III.

$\boldsymbol{\ell_{shr}}$ = Effective shear span of the stiffener, in [m], as defined in Chapter 3, Section 7, Cl.7.1.3.

$\mathbf{d_{shr}}$ = Effective shear depth of stiffener, in [mm], as defined in Chapter 3, Section 7, Cl.7.4.3.

2.3.2.4 The slenderness ratio is to comply with Chapter 9, Section 2.

2.3.2.5 Stiffeners are to be connected at ends. For side shell stiffeners, located in the strengthening area as defined in Cl.2.1.3, with the length exceeding 4 times that of the stiffener spacing, tripping brackets are to be fitted for stiffeners with web angle less than 75[deg].

2.3.3 Connection area

Shell stiffeners are to be connected to supports, e.g. stringers, web frames, decks or bulkheads. The connection area is generally obtained through fitting supporting members such as collar plates, lugs, end brackets or web stiffeners. The connection area of stiffeners to PSM and the welding size at the connection are to comply with Chapter 7, Section 7, 7.2, based on design pressure taken equal to P_{BOW}.

2.4 Bow impact load area for primary supporting members

2.4.1 The scantlings of primary supporting members according to Cl.2.5.3 are based on the application of the bow impact pressure, as defined in Cl.2.2.1, to an idealised bow impact load area of hull envelope plating, A_{BL} in [m^2], given by:

$$A_{BL} = 1.1.L.B.C_b.10^{-3}$$

2.5 Primary supporting members, decks and transverse bulkheads

2.5.1 Application

The requirements given in Cl.2.5.2 to Cl.2.5.4 apply for ships as defined in 2.1 and for other ships the evaluation of primary supporting members as described in Cl.2.5.2 to Cl.2.5.4 can be omitted provided the net web thickness of primary support members and decks or bulkheads fitted in lieu of a stringer or a web frame, in [mm], is not less than:

$$t_w = \frac{b}{75}\sqrt{\frac{R_{eH}}{235}}$$

2.5.2 General structural requirements

2.5.2.1 The section modulus of the primary supporting member is to apply along the bending span clear of end brackets. Requirements of the cross-sectional areas of the primary supporting member are to be applied at the ends/supports and may be gradually reduced along the span and clear of the ends/supports following the distribution of f_{dist} indicated in Figure 2.4.3.1.

2.5.2.2 Primary supporting members in the bow impact strengthening area are to be configured to provide effective continuity of strength and the avoidance of hard spots.

2.5.2.3 Web frames supporting side longitudinals in the bow impact region are to be connected to deck transverses in way of decks and platforms.

2.5.2.4 End brackets of primary supporting members are to be suitably stiffened along their edge. Consideration is to be given to the design of bracket toes to minimize abrupt changes of cross-section.

2.5.2.5 Tripping arrangements are to comply with Chapter 9, Section 2. In addition, tripping brackets are to be fitted at the toes of end brackets and at locations where the primary supporting member flange is knuckled or curved.

2.5.3 Scantlings of Primary supporting members

2.5.3.1 Section Modulus of the primary supporting member

The net section modulus of each primary supporting member (stringer and web frames), Z in [cm^3], calculated in accordance with Chapter 3, Section 7, Cl.7.4.7, is not to be less than:

$$\mathbf{Z = \frac{f_{CB}.P_{BOW}.b_{BI}.\ell_{bdg}^2.f_{BL}}{f_{bdg}.C_s.R_{eH}}.10^3}$$

where,

f_{CB} = Correction factor for the bending moment at the ends and considering the patch load taken as:

$$\mathbf{f_{CB} = 3f_{BL}{}^3 - 8\, f_{BL}{}^2 + 6f_{BL}}$$

f_{BL} = Patch load modification factor is taken as:

$$\mathbf{f_{BL} = \frac{\ell_{BL}}{\ell_{bdg}}}$$

$\boldsymbol{\ell_{BL}}$ = Extent of bow impact load area along the span, in [m], taken as:

$\boldsymbol{\ell_{BL}} = \sqrt{\mathbf{A_{BL}}}$ but not to be greater than $\boldsymbol{\ell_{bdg}}$

b_{BI} = Breadth of the impact load area, in [m], supported by the primary supporting member, to be taken as the spacing between primary supporting members, S as defined in Chapter 3, Section 7, but not to be taken as greater than ℓ_{BL}

$\mathbf{A_{BL}}$ = Load area as defined in Cl.2.4.1.

$\mathbf{f_{bdg}}$ = Bending moment factor taken as:

= 12 for primary supporting members with end fixed continuous flange or where brackets at both ends are fitted in accordance with Chapter 3, Section 6, Cl.6.3.5.

C_s = Permissible bending stress coefficient is taken as:

= 0.9 for acceptance criteria set AC-III.

2.5.3.2 Web thickness of the primary supporting member

The net web thickness, t_w in [mm], of primary supporting members including decks/bulkheads in way of the side shell is not to be less than:

$$\mathbf{t_w = \frac{P_{BOW} \cdot b_{BI}}{\sin \varphi_w\, \sigma_{crb}}}$$

where:

$\boldsymbol{\varphi_w}$ = Angle, in [deg], between the primary supporting member web and the shell plate, refer to Figure 2.4.3.2.

σ_{crb} = Critical buckling stress, in [N/mm^2], in compression of the web of the primary supporting member or deck/bulkhead panel in way of the applied load given in Chapter 9. In the calculation, both σ_x and σ_y as given in Chapter 9, Section 5 are to be considered and UP-B is to be applied.

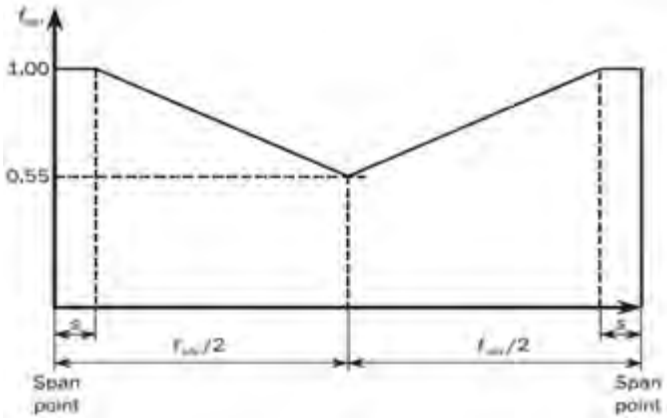


Figure 2.4.3.1: Distribution of $\mathbf{f_{dist}}$ along the span of simple primary supporting members

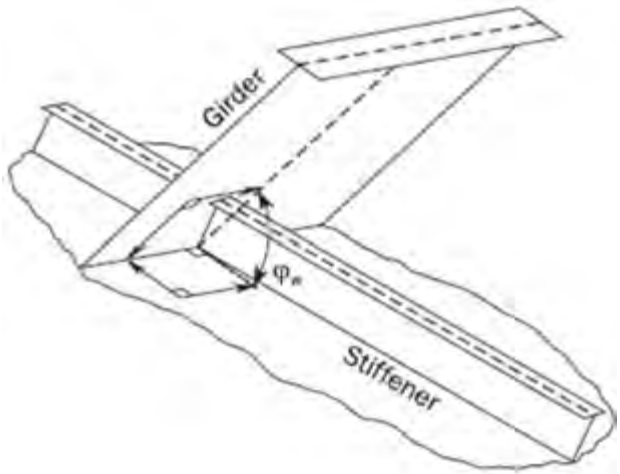


Figure 2.4.3.2: Angle between shell primary member and shell plate

2.5.3.3 Net shear area

The net shear area of the web, $\mathbf{A_{shr-n50}}$, in [cm^2], of each primary supporting member at the support /toe of end brackets, calculated in accordance with Chapter 3, Section 7, Cl.7.4.7, is not to be less than:

$$\mathbf{A_{shr-n50}} = \frac{\mathbf{5}.\sqrt{\mathbf{3}}.\mathbf{P_{BOW}}.\mathbf{b_{BI}}.\boldsymbol{\ell}_{\mathbf{shr}}.\mathbf{f_{BL}}}{\mathbf{C_t}.\mathbf{R_{eH}}}$$

where,

f_{BL} = Patch load modification factor taken as:

$$\mathbf{f_{BL}} = \frac{\boldsymbol{\ell}_{\mathbf{BL}}}{\boldsymbol{\ell}_{\mathbf{shr}}}$$

$\boldsymbol{\ell}_{\mathbf{BL}}$ = Extent of bow impact load area along the span, in [m], taken as:

$\boldsymbol{\ell}_{\mathbf{BL}} = \sqrt{\mathbf{A_{BL}}}$ but not to be greater than $\boldsymbol{\ell}_{\mathbf{shr}}$

C_t = Permissible shear stress coefficient taken as:

= 0.75 for acceptance criteria set AC-III.

2.5.4 Direct calculations

2.5.4.1 Alternative to the formulae in CL.2.2.1, IRS may accept the evaluation of the effects of bow flare impact from direct calculations also, when justified based on the ship's characteristics and intended service. In such cases, the calculations are to be submitted to IRS for approval.

2.5.4.2 Alternative to the requirements given in Cl.2.5.3, for complex arrangements of primary supporting members (grillage system) or for primary supporting members having variable cross sections, the scantling requirements/calculations can be based on direct strength analysis. The procedure for finite element modeling and load application is given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

2.5.4.3 In FE analysis, the equivalent stress, σ_{vm}, as defined in Chapter 8, Section 3, is not to exceed the yield stress, R_{eH}. In the case of beam assessment, the nominal bending stress is not to exceed $0.9R_{eH}$ and the nominal shear stress is not to exceed $0.9\,\tau_{eH}$. Permissible buckling utilization factor for AC-III according to Chapter 9 Section 1, Cl.1.3.3 is to be applied.

Section 3

Bottom Slamming

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

P_{thr} = Theoretical slamming pressure, [kN/m^2] as defined in Cl.3.3.1.

P_{max} = Peak slamming pressure, [kN/m^2] as defined in Cl.3.3.1.

T_{BF} = Draft of the vessel in [m], at forward perpendicular (FP) in ballast condition.

b' = Full breadth in [m], at 1/10th of local draft of a section at 0.25L from FP.

V_{th}= Threshold slamming speed as defined in Section 2.

x_g = The distance in [m], of the station under consideration from the LCG of the ship.

X = The distance of a section from AP in [m].

AC-III = Acceptance criteria set as defined in Chapter 1, Section 1.

3.1 Application

3.1.1 Bottom slamming requirements given in this section are applicable to vessels of L≥ 60[m] on unrestricted or coastal service.

For 'Restricted Service' or 'Sheltered water service', the requirements given in this section are not applicable.

3.1. Bottom slamming pressure defined in Cl.3.3.1 is applicable when V≥ V_{th} and F_n≤ 0.41. For vessels, when V ≥ V_{th} and F_n> 0.41, bottom slamming pressure defined in Cl.3.3.1 are not applicable and needs to be specially considered.

3.1.3 Where the minimum design ballast draught at F.P. is more than 0.045L[m], additional strengthening to resist bottom slamming pressures is not required.

3.1.4 The bottom slamming pressure is significant along the flat bottom forward of approximately 0.3L measured from FP and vertically up to 0.1T_{BF}. This region is to be strengthened for bottom impact pressure 'P_{SL}'. Refer Figure 3.1.1.

The scantlings of bottom plating, stiffeners and girders up to (0.1T_{BF}) above the baseline (up to bilge curvature in case of ships with rise of floor) are to be based on the design slamming pressure P_{SL} given in Cl.3.3.1.

3.1.5 Outside the strengthening area the requirements are to be gradually tapered to the ordinary requirements to maintain adequate continuity of strength in both longitudinal and transverse directions.

3.1.6 The requirements given below are to be applied in addition to applicable scantling requirements in Chapter 7. Local scantling increases due to impact loads are to be made with due consideration given to details and avoidance of hard spots, notches and other harmful stress concentrations.

3.1.7 The draughts for which the bottom has been strengthened are to be indicated on the shell expansion plan and loading guidance information, as required in Chapter 1, Section 4.

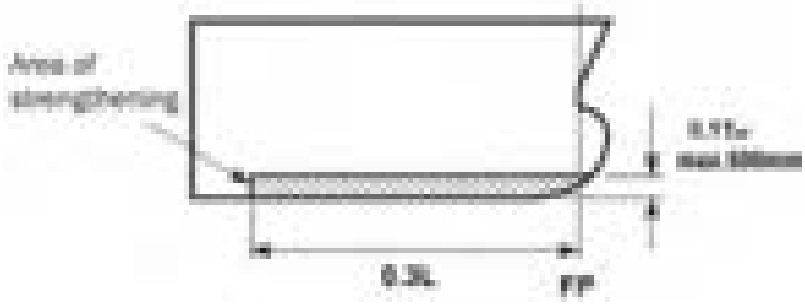


Figure 3.1.1: **Extent of strengthening against bottom slamming**

3.2 Arrangement of floors and girders

3.2.1 The floors and girders in the region forward of 0.3L abaft of F.P. are to be spaced as given below.

3.2.2 Where transverse framing is adopted, floors are to be fitted at every frame. The spacing of longitudinal girders is not to exceed 2.5[m]. In addition, intermediate half height girders or equivalent longitudinals are to be fitted.

3.2.3 Where a longitudinal framing system is adopted, the plate floors are to be fitted at alternate frames. The spacing of longitudinal girders is not to exceed 3 times the longitudinal frame spacing. In the case of longitudinally stiffened single bottoms, the arrangements, and scantlings will be specially considered.

3.3 Design bottom slamming pressure

3.3.1 The design bottom slamming pressure at any point on the keel, P_{SL} in [kN/m^2], is to be taken as:

$$P_{SL} = \mathrm{Min}(P_{thr}, P_{max})$$

where,

$$P_{thr} = \frac{1}{2}\rho\, k_H\, U_r^2$$

$$P_{max} = C_l\, \rho\, g\, h_{max}$$

k_H = Hull shape coefficient

$k_H = 0.028\beta_{DA}{}^2 - 1.68\beta_{DA} + 30$ for $\beta_{DA} \leq 30^0$;

$k_H = 4.8$ for $\beta_{DA} > 30^0$

β_{DA} = The angle made by the line joining the bottommost point, i.e., the intersection of the centreline and the section under consideration) with 60% of the girth at 1/10th of the local draft. Refer to Figure 3.3.1.1.

Relative velocity in [m/s] can be determined as shown below:

$$U_r = 2\sqrt{\mathrm{abs}(-\omega_{ext}\{z_{ext}\cos(\theta') + x_g\theta' + \zeta_{ext}\sin(\frac{\left(1.25 - 2.5\left(1 - \frac{X}{L}\right)\right)\pi}{2})\cos(\theta')\}) + V_{th}\sin(\theta_{sig})}$$

ζ_{ext} = Extreme wave amplitude = $0.95(C.C_w)$

z_{ext} = Extreme heave motion =T_{BF}

ω_{ext} = Encounter wave frequency in [rad/s] as defined in Section 2, Cl.2.2.1

θ_{sig} = Significant pitch angle = $0.33\theta'$

$$h_{max} = \left(\frac{k'}{2\sqrt{g}}\right)\left(\frac{U}{\sqrt{L}}\right)F_n L$$

$$k' = k_1 k_2 k_3$$

$k_1 = -14.6\ln\left(\frac{T_{BF}}{L}\right) - 40.5$ for $0.01 \leq T_{BF}/L \leq 0.045$

$k_1 = 28$ for $T_{BF}/L < 0.01$

$k_1 = 4.8$ for $T_{BF}/L > 0.045$

$k_2 = \frac{10b'}{L}$

Limiting range for k_2: $0.5 \leq k_2 \leq 1.3$

$k_3 = 1.57\exp(-0.01L^{2/3})$

Limiting range for k_3: $0.84 \leq k_3 \leq 1.25$

$U = (-134F_n^2 + 31F_n + 3)\omega_{ext}\zeta_a$

for $0 \leq F_n < 0.16$; [m/s]

$U = (-105F_n^2 + 65F_n - 3)\omega_{ext}\zeta_a$

for $0.16 \leq F_n \leq 0.41$ [m/s]

ζ_a =Average wave amplitude in [m].

$$\zeta_a = \frac{1}{1.41}(H_s/2)$$

C, C_w = Coefficients defined in Section 2, Cl. 2.2.1.

C_l is the coefficient of variation of slamming pressure along the length of the vessel as given in Figure 3.3.1.2 and Figure 3.3.1.3.

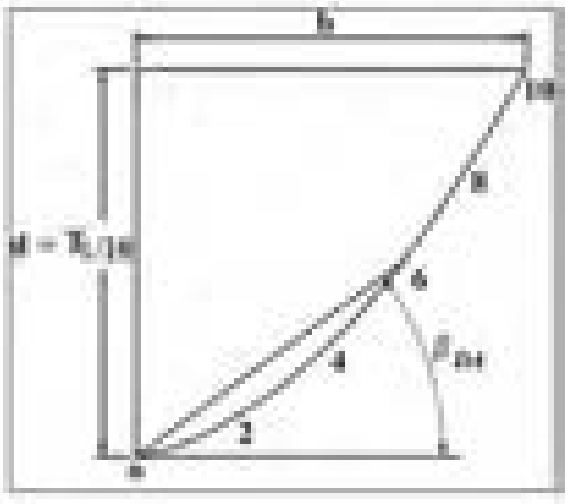


Figure 3.3.1.1: Angle β_{DA} (Cl.3.3.1)

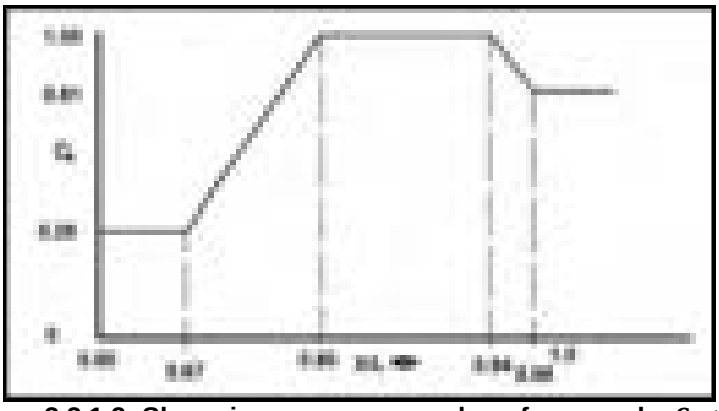


Figure 3.3.1.2: Slamming pressure envelope for vessels, C_b<0.75

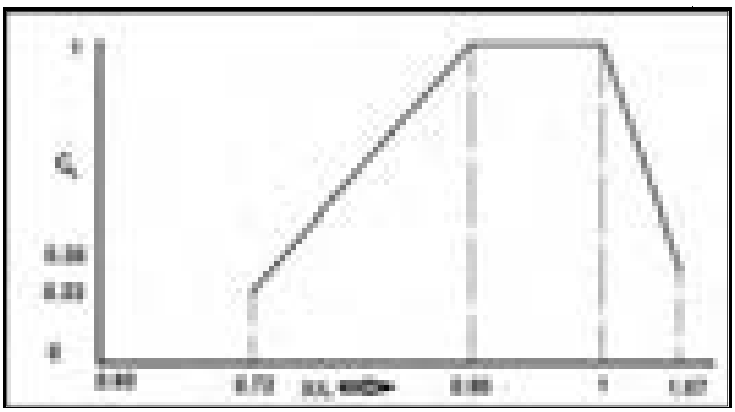


Figure 3.3.1.3: Slamming pressure envelope for vessels, $C_b \geq 0.75$

If the minimum design ballast draught T_{BF} is achieved by means of full ballast tanks in the forebody resulting in internal loads on bottom shell panels, the design slamming pressure P_{SL} on the corresponding panels may be reduced by (14h) [kN/m^2], where, 'h' is the height of the ballast tank [m].

For strength assessment of double bottom floors and girders, h is not to be taken greater than the double bottom height.

3.4 Scantling requirements

3.4.1 Shell plating

The net thickness of the hull envelope plating 't' in [mm], subjected to impact pressures from bottom slamming is not to be less than:

$$t = \frac{0.0158.\alpha_p.b}{C_d}\sqrt{\frac{P_{SL}}{R_{eH}}}$$

where:

C_d = Plate capacity correction coefficient = 1.3

C_a = Permissible bending stress coefficient taken as:

= 1.0 for acceptance criteria set AC-III

α_p = correction factor for the panel aspect ratio to be taken as below but not to be greater than 1.0:

$$\alpha_p = 1.2 - \frac{b}{2.1\,a}$$

a,b = Length and breadth of plate panel, in [mm], as defined in Chapter 3 Section 7, Cl.7.5.1.

3.4.2 Stiffeners

3.4.2.1 The design of end connections of stiffeners in the bottom slamming region is to ensure end fixity, either by making the stiffeners continuous through supports or by providing end brackets complying with Chapter 3, Section 6, Cl.6.2.2. Where it is not practical to comply with this requirement, the net plastic section modulus, Z_{pl_alt}, in [cm^3], for alternative end fixity arrangements is not to be less than:

$$\mathbf{Z_{pl_alt} = \frac{16\,Z_{pl}}{f_{bdg}}}$$

where:

Z_{pl} **=** Effective net plastic section modulus, in [cm^3], required by 3.4.2.2.

Scantlings and arrangements of primary supporting members, including bulkheads in way of stiffeners, are to comply with 3.6.

3.4.2.2 The stiffeners within the strengthening area defined in 3.1 are to comply with the following criteria:

(a) The effective net plastic section modulus, $\mathbf{Z_{pl}}$, in [cm^3], calculated in accordance with Chapter 3, Section 7, Cl.7.4.6, is not to be less than:

$$\mathbf{Z_{pl} = \frac{P_{SL}.s.\ell_{bdg}^2}{f_{bdg}.C_s.R_{eH}}}$$

where:

C_s = Permissible bending stress coefficient taken as:

= 0.9 for acceptance criteria set AC-III.

(b) The net web thickness, t_w, in [mm], is not to be less than:

$$\mathbf{t_w = \frac{\sqrt{3}.P_{SL}.s.\ell_{shr}}{2.d_{shr}.C_t.R_{eH}}}$$

where:

C_t = Permissible shear stress coefficient taken as:

= 1.0 for acceptance criteria set AC-III.

$\boldsymbol{\ell_{shr}}$ = Effective shear span of the stiffener, in [m], as defined in Chapter 3, Section 7, Cl.7.1.3.

$\mathbf{d_{shr}}$ = Effective shear depth of stiffener, in [mm], as defined in Chapter 3, Section 7, Cl.7.4.3.

3.4.2.3 The slenderness ratio is to comply with Chapter 9, Section 2.

3.4.3 Connection area

Bottom stiffeners are to be connected to supports, e.g., floors, girders or bulkheads. The connection area is generally obtained through fitting supporting members such as collar plate, end bracket or web stiffener. The requirements for the net connection area of the supporting members and its welding size are to be according to Section 2, Cl.2.3.3 using $\mathbf{P_{SL}}$.

3.5 Bottom slamming load area for primary supporting members

3.5.1 The scantlings of primary supporting members according to 3.6 are based on the application of slamming pressure, as defined in Cl.3.3.1, to an idealised slamming load area of hull envelope plating, A_{SL} in [m^2], given by:

$$A_{SL} = 1.1.L.B.C_b.10^{-3}$$

3.6 Primary supporting members

3.6.1 Scantlings of Primary supporting members

The size and number of openings in web plating of the floors and girders are to be minimised considering the required shear area as given below in Cl.3.6.1.1:

3.6.1.1 Net shear area

The net shear area $A_{shr-n50}$, in [cm^2], of each primary supporting member web at any position along its span is not to be less than:

$$\mathbf{A_{shr-n50} = \frac{10.\sqrt{3}.Q_{SL}}{C_t.R_{eH}}}$$

where:

$\mathbf{Q_{SL}}$ = The greatest shear force due to slamming for the position being considered, in [kN], based on the application of a patch load, $\mathbf{Q_{SL}}$ to the most onerous location, as determined in accordance with Cl.3.6.1.2 and Cl.3.6.1.3.

C_t = Permissible shear stress coefficient taken as:

= 0.9 for acceptance criteria set AC-III.

3.6.1.2 Simplified calculation of slamming shear force

For simple arrangements of primary supporting members, where the grillage effect may be ignored the shear force, Q_{SL}, in [kN], is given by:

$$Q_{SL} = f_{pt}.f_{dist}.F_{SL}$$

where,

f_{pt} = Correction factor for the proportion of patch load acting on a single primary supporting member, taken as

$$f_{pt} = 0.5({f_{SL}}^3 - 2{f_{SL}}^2 + 2)$$

f_{SL}= Patch load modification factor taken as:

$$\mathbf{f_{SL} = 0.5\frac{b_{SL}}{S}}$$

f_{dist} = Factor for the greatest shear force distribution along the span, according to Figure 3.6.1.

F_{SL} = Patch load, in [kN], taken as:

$$F_{SL} = P_{SL}.\ell_{SL}.b_{SL}$$

$\boldsymbol{\ell_{SL}}$ = Extent of slamming load area along the span, in [m], taken as:

$\boldsymbol{\ell_{SL} = \sqrt{A_{SL}}}$ but not to be greater than $\mathbf{0.5\ell_{shr}}$

$\mathbf{b_{SL}}$ = Breadth of the impact load area, in [m], supported by the primary supporting member, taken as:

$\mathbf{b_{SL} = \sqrt{A_{SL}}}$ but not to be greater than S.

$\mathbf{A_{SL}}$ = Load area as defined in Cl.3.5.1.

3.6.1.3 Direct calculation method for slamming shear force

For complex arrangements of primary supporting members, the greatest shear force, Q_{SL}, at any location along the span of each primary supporting member is to be derived by direct calculation in accordance with Table 3.6.1.

3.6.1.4 Web thickness of the primary supporting member

The net web thickness, t_w, in [mm], of primary supporting members adjacent to the shell, is not to be less than:

$$t_w = \frac{S_w}{100}\sqrt{\frac{R_{eH}}{235}}$$

where,

S_w = Plate breadth, in [mm], taken as the spacing between the web stiffening.

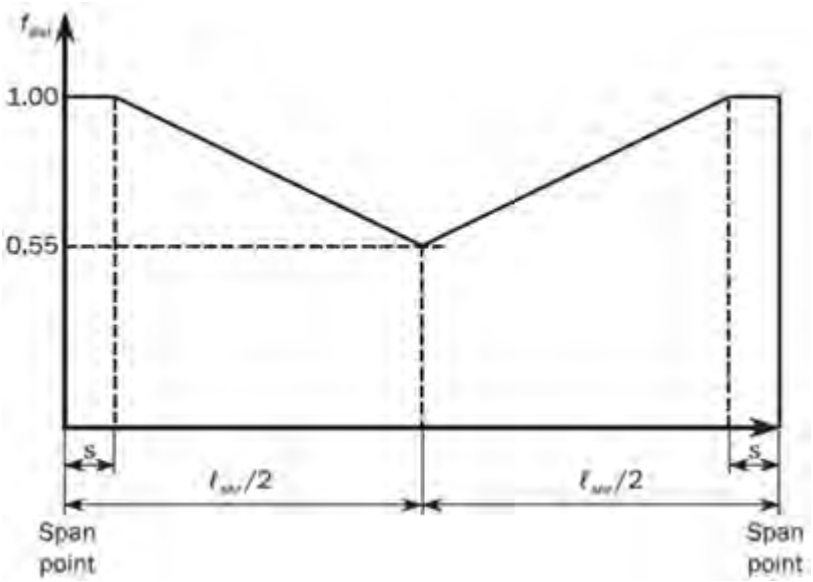


Figure 3.6.1: Distribution of f_{dist} along the span of simple primary supporting members

Table 3.6.1: Direct calculation methods for the derivation of Q_{SL}		
Type of analysis	**Model extent**	**Assumed end fixity of floors**
Beam theory	Overall span of member between effective bending supports.	Fixed at ends
Double bottom grillage	Longitudinal extent to be one cargo tank length. Transverse extent to be between inner hopper knuckle and centreline.	Floors and girders to be fixed at the boundaries of the model.
Note 1: The envelope of greatest shear force along each primary supporting member is to be derived by applying the load patch on a square area as defined in Cl.3.5.1, to several locations along the span. Note 2: A more extensive model in length and breadth can be considered. The procedure for finite element modeling and load application is given in IRS Guidelines on Structural Assessment of Ships based on Finite Element Method. In FE analysis, the equivalent stress, σ_{vm}, as defined in Chapter 8, Section 3, is not to exceed the yield stress, R_{eH}. In the case of beam assessment, the nominal bending stress is not to exceed $0.9R_{eH}$ and the nominal shear stress is not to exceed $0.9\ \tau_{eH}$.		

3.6.2 Direct calculations

Alternative to the formulae in Cl.3.3.1, IRS may accept the evaluation of the effects of bottom slamming from direct calculations also, when justified on the basis of the ship's characteristics and intended service. In such cases, the calculations are to be submitted to IRS for approval.

Section 4

Stern Slamming

4.1 Application

4.1.1 Where the shape of the after sections is such that there are large flat areas of the bottom shell, additional stiffening and/or increased shell plate thickness may be required.

Section 5

Wheel Loading

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

AC-I = Acceptance criteria set as defined in Chapter 1, Section 1 for harbour condition.

AC-II = Acceptance criteria set as defined in Chapter 1, Section 1 for sea going condition.

5.1 General

5.1.1 Application

5.1.1.1 Where it is proposed either to stow wheeled vehicles on the deck or to use wheeled vehicles for cargo handling, the requirements of this section are to be complied with in addition to those given in the relevant chapters.

5.1.1.2 The requirements given below assume that the considered element (Deck plating and/or stiffener) is subjected to one load area only and that the element is continuous over several evenly spaced supports.

5.1.1.3 A "load area" is the tyre print area of individual wheels; for closely spaced wheels it may be taken as the enveloped area of the wheel group as given in Table 5.2.1.

5.1.1.4 The details of wheel loadings are to be forwarded by the shipbuilder. These details are to include the proposed arrangement and dimensions of tyre prints, type of tyre, axle and wheel spacings, maximum axle load and tyre pressure.

5.1.1.5 Signboards stating the maximum permissible axle load, the maximum tyre pressure of pneumatic tyres, wheel arrangement on axles, and specially approved vehicles are to be stated in the Loading Manual and displayed in a notice on each deck fitted in suitable positions.

For decks covered with protection plates, such as wooden plating, an increased footprint area may be considered for the scantling requirements for plates and stiffeners, considering the load spreading through the cover plate.

5.1.1.6 The deck and supporting structure are to be designed based on the maximum loading to which they may be subjected in service. Where applicable, the hatch covers are to be similarly designed. In no case, however, are the scantlings to be less than would be required as a weather or cargo deck or hatch cover, as applicable.

5.2 Wheel loads

5.2.1 The design pressure 'P_{wl}', in [kN/m^2], from the wheels of individual vehicles on deck with specified arrangement and dimension of footprints is to be taken as:

5.2.1.1 In harbour condition:

$$P_{wl-h} = \frac{W}{n.a.b}.\left(g + \frac{3}{\sqrt{W}}\right)10^6$$

5.2.1.2 In sea going condition:

$$P_{wl-s} = \frac{W}{n.a.b}.(g + a_v)10^6$$

where,

W = Maximum axle load in [t]. For fork lift trucks, the total weight is to be taken as the axle load.

n = Number of "load areas" per axle

a = Extent in [mm], of the load area parallel to the stiffeners (Refer Table 5.2.1).

b = Extent in [mm], of the load area perpendicular to the stiffeners (Refer Table 5.2.1).

a_v = Combined vertical acceleration in [m/s^2] as defined in Chapter 5 Section 2, Cl.2.2.3.3 using y = B/2.

5.2.2 If the arrangement and dimensions of footprints are not available for vehicles with pneumatic tyres, the design pressure, in [kN/m^2], may normally be taken as:

5.2.2.1 In harbour condition

$$P_{wl-h} = \frac{P_T}{n_0}.\left(1 + \frac{3}{g\sqrt{W}}\right)$$

5.2.2.2 In sea going condition

$$P_{wl-s} = \frac{P_T}{n_0}.\left(1 + \frac{a_v}{g}\right)$$

where:

P_T = Tyre pressure for pneumatic tyres, in [kN/m^2]. Refer Cl.5.1.1.4.

n_0 = Number of wheels on the plate panel, taken equal to:

= 1 in general

= 1.20 when double wheels are specified

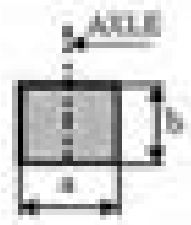
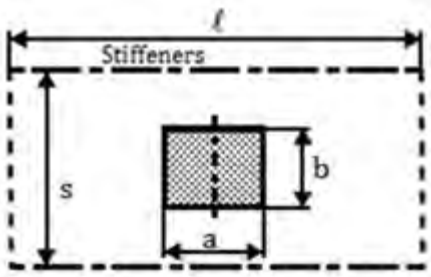
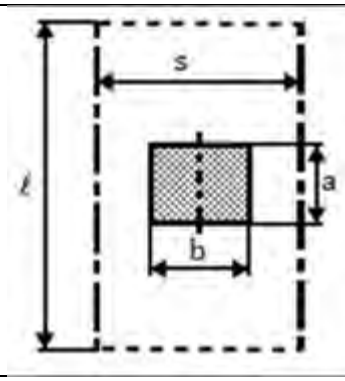
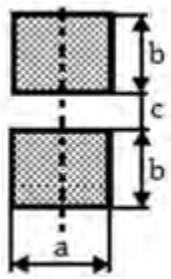
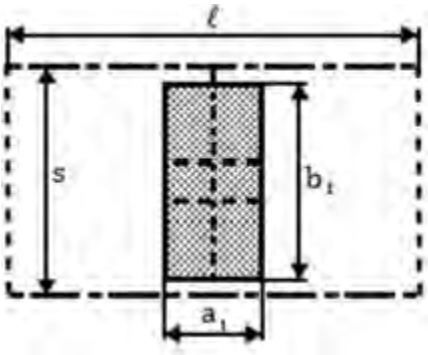
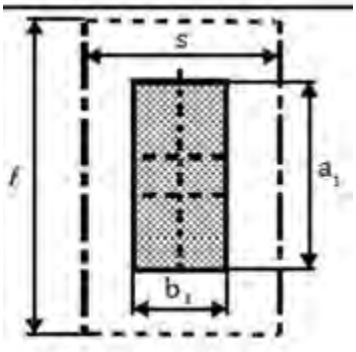
= 1.27 when triple wheels are specified

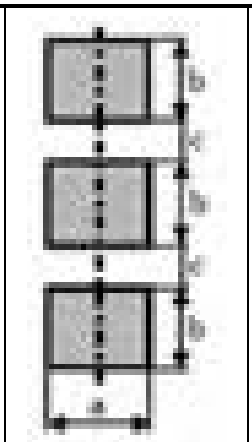
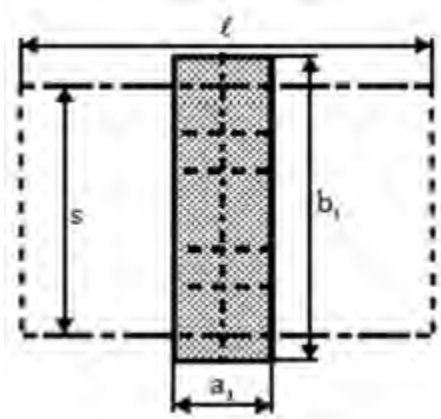
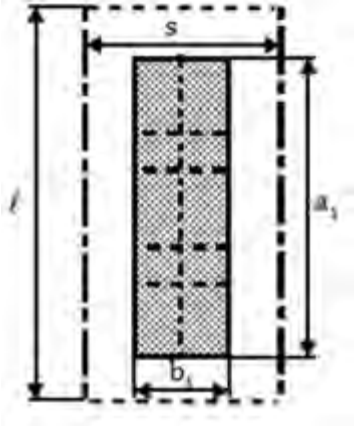
The load area as indicated in Table 5.2.1 is defined as:

- the footprint area of individual wheels or

- the rectangular enveloped area of footprints of a wheel group.

Table 5.2.1: Load area definition

Number of wheels in a group	Wheel print dimensions (Imprint area between tyre and deck)	Design load area Axle perpendicular to stiffeners	Design load area Axle parallel to stiffeners
Single wheel			
Two wheels			

Triple wheels			

NOTE: In general the scantlings are to be checked according to both individual wheel area and wheel group area. If, however, the distance 'c' between individual footprints is less than the breadth 'b' of the prints, the load area may be calculated for the group of wheels only.

5.3 Tyre print area

5.3.1 When the tyre print area (arrangement and dimensions of footprints) is not available for vehicles with pneumatic tyres, it may be taken equal to:

$$A_t\,[m^2] = \frac{9.81 n_0 . W}{n . P_T}$$

where,

P_T = Tyre pressure for pneumatic tyres, in [kN/m^2].

n_0 = Coefficient taken equal to:

= 1 in the case of a single wheel

= 1.20 when double wheels are specified

= 1.27 when triple wheels are specified

n = Number of wheels for the axle considered

= 2, otherwise specified.

5.3.2 The load area dimensions, in [mm], are in general to be taken as:

$$a_1 = \sqrt{k A_t} . 10^3$$

$$b_1 = \sqrt{A_t / k} . 10^3$$

where,

$k = k_1$ in general

$= k_2$ for plating when $k_2 < k_1$ and

$$\frac{n_0 W}{n\, b^2} 10^6 \geq 100$$

k_1 = 2.0 for a single wheel

= 2.0 for multiple wheels with axle parallel to stiffener

= 0.8 for double wheels with axle perpendicular to stiffeners

= 0.5 for triple wheels with axle perpendicular to stiffeners

$$k_2 = \frac{\sqrt{A_t}}{2b} 10^3$$

A_t, n_0, n = As defined in Cl.5.3.1.

5.3.3 For heavy vehicles with axle load, W, exceeding 15t, the footprint data is to be provided.

5.3.4 For heavy vehicles where the stowing and lashing arrangement may significantly affect the load distribution at sea, the design pressure for individual load areas will be specially considered.

5.3.5 Deck areas for wheel loads from cargo handling vehicles, which are frequently operating in all directions, shall be checked for design loads with axle parallel and perpendicular to stiffeners.

5.4 Scantling requirements

5.4.1 Deck plating

The net thickness, 't' in [mm], of deck plating subjected to wheel loadings, is not to be less than;

$$t = 77.4.f_a\sqrt{\frac{c_2.b.c.P_{wl}}{m.c_a.R_{eH}}}10^{-3}$$

where,

$f_a = (1.2 - s/2.1\ell)$, maximum 1.0

c_2 = coefficient taken as:

c_2 = Maximum 1.0 for $a_1/s \geq 1.94$

$$c_2 = 1.3 - \frac{4.2}{(a_1/s + 1.8)^2} \text{ for } a_1/s < 1.94$$

c = coefficient taken as:

= b_1 for $b_1 \leq s$

= s for $b_1 > s$

a_1, b_1 = Extents, in [mm], of the load area as defined in Cl.5.2.1.

P_{wl} = Design pressure in [kN/m^2] as defined in Cl.5.2.1 or Cl.5.2.2, considering the design load set defined in Chapter 7, Section 2.

m = Bending moment factor taken as:

$$m = \frac{38}{(b_1/s)^2 - 4.7(b_1/s) + 6.5} \text{ for } b_1 \leq s$$

$m = 13.57$ for $b_1 > s$

c_a = Permissible bending stress coefficient for plate taken as:

$$c_a = \beta_a - \alpha_a\frac{|\sigma_{hg}|}{R_{eH}}$$

not to be taken greater than c_{a-max}

σ_{hg} = Hull girder longitudinal stress, in [N/mm^2], as defined in Chapter 6, Section 3.

c_{a-max}= Maximum permissible bending stress coefficient for AC-I and AC-II as defined in Table 5.4.1.

α_a, β_a = Coefficient as defined in Table 5.4.1.

Table 5.4.1: Definition of β_a, α_a, and $C_{a\text{-}max}$

Acceptance criteria	Structural member		β_a	α_a	$C_{a\text{-}max}$
AC-I	Longitudinal member	Longitudinal stiffened plating	1.9	0.5	1.8
		Transverse stiffened plating	1.9	1.0	1.8
	Other member		1.8	0	1.8
AC-II	Longitudinal member	Longitudinal stiffened plating	2.1	0.5	2.0
		Transverse stiffened plating	2.1	1.0	2.0
	Other member		2.0	0	2.0

5.4.2 Deck stiffeners

The net section modulus ' Z ' in [cm^3] of deck beams and longitudinal subjected to wheel loading is not to be less than:

$$Z = \frac{c_3.c.d.\ell_{bdg}.P_{wl}}{m.R_{eH}.C_s}10^{-3}$$

where,

P_{wl} = Design pressure in [kN/m^2] as defined in Cl.5.2.1 or Cl.5.2.2, considering the design load set defined in Chapter 7, Section 2.

$c_3 = 1.0$ for $b_1/s \leq 0.6$

$= (1.15 - 0.25\,b_1/s)$ for $0.6 < b_1/s \leq 1.0$

$= (1.15 - 0.25\,b_1/s)\,b_1/s$ for $1.0 < b_1/s < 3.4$

$= 1.0$ for $b_1/s \geq 3.4$

a_1, b_1 = Extents, in [mm], of the load area as defined in Cl.5.2.1.

c = coefficient defined in Cl.5.4.1.

d = coefficient taken as:

$= a_1$ for $a_1 \leq \ell$

$= \ell$ for $a_1 > \ell$

ℓ_{bdg} = Effective bending span, in [m] as defined in Chapter 3.

m = Bending moment factor

$$m = \frac{r}{\left(\frac{a_1}{1000\ell}\right)^2 - 4.7\left(\frac{a_1}{1000\ell}\right) + 6.5}$$

for $\frac{a_1}{1000\ell} \leq 1.0$

$$m = \frac{87}{\left(\frac{a_1}{1000\ell}\right)^2\left[\left(\frac{a_1}{1000\ell}\right)^2 - 6.3\left(\frac{a_1}{1000\ell}\right) + 10.9\right]}$$

for $1.2 < \frac{a_1}{1000\ell} \leq 2.5$

$m = 12$ for $\frac{a_1}{1000\ell} \geq 3.5$

r = Factor depending on the rigidity of girders supporting continuous stiffeners

= 29 for continuous stiffeners supported at girders.

= 38 when the continuous stiffeners can be considered as rigidly supported at girders against rotation.

C_s = Permissible bending stress coefficient for stiffeners taken as:

$$C_s = \beta_s - \alpha_s \frac{|\sigma_{hg}|}{R_{eH}}$$

$C_{s-max.}$= Maximum permissible bending stress coefficient as defined in Table 5.4.2.

β_s = Coefficient as defined in Table 5.4.2.

α_s = Coefficient as defined in Table 5.4.2.

Table 5.4.2: Definition of α_s, β_s, $C_{s-max.}$

Acceptance criteria set	Structural member	β_s	α_s	$C_{s-max.}$
AC-I	Longitudinal strength member	0.85	1.00	0.75
	Transverse or vertical member	0.75	0.00	0.75
AC-II	Longitudinal strength member	1.00	1.00	0.90
	Transverse or vertical member	0.90	0.00	0.90

5.4.3 Web plating

5.4.3.1 The net web thickness, in [mm], of stiffeners subjected to wheel loading, is not to be taken less than:

$$t_w = \frac{\sqrt{3}.f_{shr}.c_3.c.d.P_{wl}}{d_{shr}.c_t.R_{eH}}10^{-3}$$

where:

f_{shr} = Shear force distribution factor taken as:

$$f_{shr} = 1 - 0.5\frac{a_1}{1000\ell} \text{ for } a_1 \leq 1000\ell$$

$$f_{shr} = 0.5 - 0.25\left(\frac{a_1}{1000\ell} - 1\right) \text{ for } a_1 > 1000\ell, \text{Minimum } 0.25$$

P_{wl} = Design pressure in [kN/m^2] as defined in Cl.5.2.1 or Cl.5.2.2 considering the design load set being defined in Chapter 7, Section 2.

c_3 = Coefficient as defined in Cl.5.4.2.

c = Coefficient as defined in Cl.5.4.2.

d = Coefficient as defined in Cl.5.4.2.

d_{shr} = Effective web depth of stiffener, in [mm] as defined in Chapter 3, Section 7.

c_t = Permissible shear stress coefficient as defined in Chapter 7, Section 5 for the design load set being considered.

5.4.3.2 If more than one load area can be positioned simultaneously on the same stiffener span or adjacent spans, the section modulus will be specially considered, based on finite element analysis.

5.4.4 Deck girders

5.4.4.1 Vehicle loads

The scantlings of girders will be specially considered based on the most severe condition of moving or stowed vehicles. Also, see Cl.5.1.1.4. Allowable stresses to be taken as given in Chapter 7, Section 6,6.2. The envelope vertical acceleration, a_{z-env}, in combination with maximum hull girder stress, σ_{hg} are to be considered in the calculation of design concentrated force calculation.

5.4.4.2 The girder structure is to be based on Chapter 5, Section 4, Cl.4.2.3.1 corresponding to the most severe of the uniform deck load, UDL (evenly and unevenly distributed), and the vehicle axle load (cargo handling vehicle and vehicles to be carried). The position of the vehicles is to be taken as the most unfavorable for the girder strength.

5.4.4.3 The scantlings of girders are also to be considered based on the most severe condition of cargo handling or stowed vehicles. Unless otherwise specified, the girder system is to be designed for a condition where the axles of several trailers side by side are acting on the same transverse girder also.

Section 6

Special Hull Structures

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

6.1 Stems, Keel, Breasthooks, Diaphragm plating, and Bulbous bow

6.1.1 Plate stem thickness

6.1.1.1 The net thickness, t_{Stm} in [mm], is not to be less than:

$$t_{Stm} = (0.6 + 0.4S_B)(0.08L + 2.7)\sqrt{k}$$

but need not be greater than $22\sqrt{k} - 1$

where:

S_B = Spacing, in [m], between horizontal stringers (partial or not), breasthooks, or equivalent horizontal stiffening members.

Starting from 0.6[m] above the summer load waterline up to $T_{SC} + C_w$, the net thickness may gradually be reduced to $0.8t_{Stm}$.

The extension in [mm] of the stem plate from its leading edge and aftwards shall not be smaller than determined by the following formula:

$$\ell = 70\sqrt{L}$$

6.1.1.2 Arrangement

The expanded width of the stem is not to be less than the rule breadth of the plate keel, defined in Chapter 3, Section 5, Cl.5.3.7.

The plating forming the stems is to be supported by horizontal diaphragms spaced not more than 1.0[m] apart and connected, as far as practicable, to the adjacent frames and side stringers. The webs are to be extended aft to at least beyond the butts of stem plates and shell plating.

If considered necessary, and particularly where the stem radius is large, a centreline stiffener or web of suitable scantlings is to be fitted.

Continuity of strength is to be ensured at the connection of stems to the surrounding structure.

6.1.2 Bar stem

6.1.2.1 Bar stem area

The cross-sectional area, A_{Stm}, in [cm^2], of a bar stem below the summer load waterline, is not to be less than:

$$A_{Stm} = 1.25L$$

Above the summer load waterline, the cross-sectional area of the bar stem may be gradually reduced to $0.75A_{Stm}$ at the stem head.

6.1.2.2 Arrangement

The lower part of the stem may be constructed of cast steel subject to the examination by IRS; where necessary, a vertical web is to be fitted for welding of the centre keelson.

The bar stem is to be welded to the bar keel generally with butt welding. The shell plating is also to be welded directly to the bar stem with butt welding.

6.1.3 Bar keel

6.1.3.1 The depth and thickness of bar keel, in [mm] are not to be less than:

$$D_{keel} = 100 + 1.5L$$

$$T_{keel} = 10 + 0.6L$$

The scantlings of bar keels at the aft and fore-end are to be the same as in the midship region as given above.

6.1.3.2 Minor deviations from the above values may be accepted provided the required sectional area is maintained.

6.1.4 Breasthooks and diaphragm plating

The net thickness of breasthooks/diaphragm plates, t_w, in [mm], is not to be less than:

$$t_w = \frac{s}{70}\sqrt{\frac{R_{eH}}{235}}$$

where:

s = Spacing of stiffeners on the web, as defined in Chapter 1, Section 2, Table 2.6.5, in [mm]. Where no stiffeners are fitted, s is to be taken as the depth of the web.

6.1.5 Bulbous Bow

6.1.5.1 General

Where a bulbous bow is fitted, the structural arrangements are to be such that the bulb is adequately supported and integrated into the fore peak structure.

6.1.5.2 Diaphragm plates

At the forward end of the bulb, the structure is generally to be supported by horizontal diaphragm plates spaced not more than 1.0[m] apart in conjunction with a deep centreline web. In general, vertical transverse diaphragm plates are to be arranged in way of the transition from the peak framing to the bulb framing.

6.1.5.3 Special bulbous bow designs

Additional strengthening in the form of a centreline wash bulkhead is generally to be fitted in way of wide bulbs. Similarly, transverse wash bulkheads or substantial web frames spaced approximately 5 frame spaces apart are to be fitted in way of long bulbs.

6.1.5.4 Strengthening for anchor and chain cable contact

The shell plating thickness at the fore end of the bulb and in other areas likely to be damaged by the anchors and chain cables during anchor handling is not to be less than that required for plate stems as defined in Section 6, Cl.6.1.1.

6.2 Stern Frames and appendages

6.2.1 General

6.2.1.1 Stern frames, rudder horns, shaft brackets, etc. are to be designed such that they are effectively integrated into the ship's structure.

6.2.1.2 In castings, sudden changes of section or possible constrictions to the flow of metal during castings are to be avoided. All fillets are to have adequate radii, which in general should not be less than 50[mm], depending on the size of the casting.

6.2.1.3 Cast steel and fabricated stern frames are to be strengthened by adequately spaced plates with gross thickness not less than 80% of the required thickness for stern frames as defined in Cl.6.2.3. In way of the upper part of the stern frame arch, these webs are to line up with the floors.

Abrupt changes of the section are to be avoided in castings; all sections are to have an adequate tapering radius.

6.2.1.4 Stern frames may be fabricated from steel plates or made of cast steel with a hollow section. For applicable material specifications and steel grades, Refer Chapter 2, Section 2. Stern frames of other materials or construction will be specially considered.

6.2.1.5 Plating materials for stern frames supporting the rudder and propeller boss, rudders, rudder horns and shaft brackets are not to be of lower grades than those corresponding to the material classes given in Chapter 2, Section 2.

6.2.1.6 It is recommended that the after body of the ship be so shaped as to ensure adequate flow of water to the propeller to prevent the uneven formation of eddies, as far as possible. For a moderately cavitating propeller, the following minimum values of propeller hull clearances may be used as guidance. (Refer Figure 6.2.1)

- For single screw ships:

$$a \geq 0.2R \text{ in [m]}$$

$$b \geq (0.7 - 0.04N_b)R \text{ in [m]}$$

$$c \geq (0.48 - 0.02N_b)R \text{ in [m]}$$

$$d \geq 0.07R \text{ in [m]}$$

- For twin screw ships:

$$b \geq (0.5 - 0.03N_b)R \text{ in [m]}$$

$$c \geq (0.6 - 0.02N_b)R \text{ in [m]}$$

where,

R = Propeller radius, in [m].

N_b = Number of propeller blades.

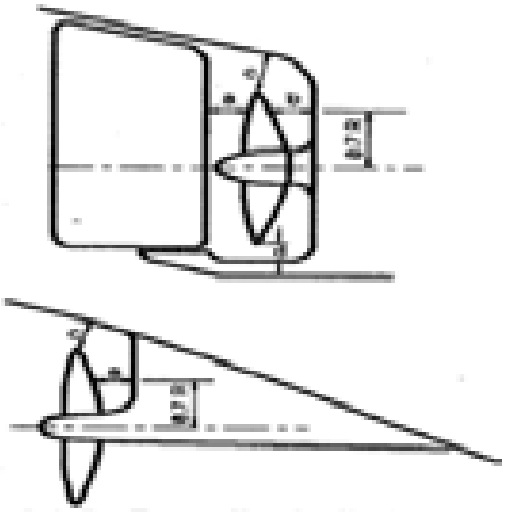


Figure 6.2.1: Propeller hull clearances

6.2.2 Connections

6.2.2.1 Connections with hull structure

Stern frames are to be effectively attached to the aft structure and the lower part of the stern frame is to be extended forward of the propeller post to a length not less than 1500 + 6L[mm], in order to provide an effective connection with the keel. However, the stern frame need not extend beyond the aft peak bulkhead.

The net thickness of shell plating connected with the stern frame is not to be less than that obtained, in [mm], from the following formula:

$$t = 0.094(L_2 - 43) + 0.009b$$

In way of the boss and heel plate, the net thickness of shell plating, in [mm], shall not be less than:

$$t = 0.105(L_2 - 47) + 0.011b$$

where:

b = Breadth of plate panel, in [mm].

L_2 = Rule length but need not be taken greater than 300[m].

The net thickness of the aft peak bulkhead plating in way of the stern tube penetration is to be at least 1.6 times the required thickness for the bulkhead plating.

6.2.2.2 Connection with keel plate

The thickness of the lower part of the stern frames is to be gradually tapered to that of the solid bar keel or keel plate. Where a keel plate is fitted, the lower part of the stern frame is to be designed to ensure an effective connection with the keel.

6.2.2.3 Connection with transom floors

Rudder posts and, in the case of ships greater than 90[m] in length, propeller posts are to be connected with transom floors having height not less than that of the double bottom height and a net thickness not less than that obtained, in [mm], from the following formula:

$$t = 9 + 0.023Lk^{1/2}$$

where:

L = Rule length but need not be taken greater than 250[m]

6.2.2.4 Connection with centre girder

Where the stern frame is made of cast steel, the lower part of the stern frame is to be fitted, as far as practicable, with a longitudinal web for connection with the centre girder.

6.2.3 Propeller posts

6.2.3.1 Gross scantlings of propeller posts

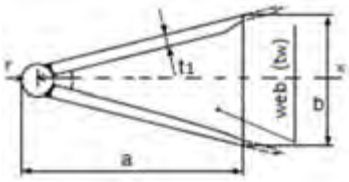
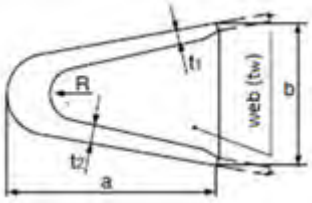
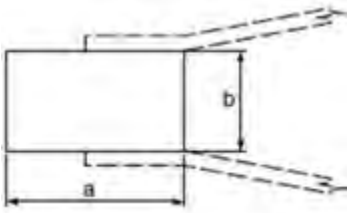
The gross scantlings of propeller posts are not to be less than those obtained from the formulae in Table 6.2.3 for single screw ships.

Where the scantlings and proportions of the propeller post differ from Table 6.2.3, the section modulus of the propeller post about the longitudinal axis is not to be less than that calculated with the propeller post scantlings in Table 6.2.3.

Fabricated and cast propeller posts and rudder posts of twin-screw ships are to be strengthened at intervals by webs. In way of the upper part of the stern frame arch, these webs are to line up with the floors.

Rudder posts and propeller posts are to be connected to floors.

Table 6.2.3: Single screw ships - Gross scantlings of propeller posts

Gross scantlings of Propeller posts, in [mm]	Fabricated propeller post	Cast propeller post	Bar propeller post, cast or forged, having rectangular section
			
a	$50L_1^{1/2}$	$33L_1^{1/2}$	$10\sqrt{2.5L + 25}$ for $L \leq 60$ $10\sqrt{7.2L - 256}$ for $L > 60$
b	$35L_1^{1/2}$	$23L_1^{1/2}$	$10\sqrt{1.6L + 16}$ for $L \leq 60$ $10\sqrt{4.6L - 164}$ for $L > 60$
t_1	$2.5L_1^{1/2}$	$3.2L_1^{1/2}$ and to be taken not less than 19[mm]	NA
t_2	NA	$4.4L_1^{1/2}$ and to be taken not less than 19[mm]	NA
t_w	$1.3L_1^{\frac{1}{2}}$	$2.0L_1^{1/2}$	NA
NOTE: L_1 is not to be taken more than 250[m].			

6.2.3.2 Section modulus below the propeller shaft bossing

In the case of a propeller post without a sole piece, the section modulus of the propeller post may be gradually reduced below the propeller shaft bossing down to 85% of the value calculated with the scantlings in Table 6.2.3.

In any case, the thicknesses of the propeller posts are not to be less than those obtained from the formulae in Table 6.2.3.

The scantlings below the boss of stern frames with sole pieces are to be gradually increased to provide strength and stiffness in proportion to those of sole pieces.

6.2.3.3 Scantlings above the propeller boss

In the upper part of the propeller aperture, where the hull form is full and centreline supports are provided, the thickness of stern frames and internal members may be reduced

to 80% of the applicable requirement of propeller posts given in Table 6.2.3.

6.2.4 Propeller boss

6.2.4.1 The gross wall thickness of the boss 't_b', in [mm], in the propeller post is not to be less than:

$$t_b = 0.16d_{ts} + 30 \text{ for } d_{ts} \leq 300[\text{mm}]$$

$$t_b = 5\sqrt{d_{ts} - 60} \text{ for } d_{ts} > 300[\text{mm}]$$

where,

d_{ts} = Rule diameter of the tail shaft in [mm].

6.2.4.2 In fabricated stern frames the connection of the propeller post to the boss is to be by full penetration welds.

6.2.4.3 In single screw ships, the thickness of the propeller shaft bossing, included in the propeller post, is not to be less than 60% of the dimension 'b' required in Cl.6.2.3 for bar propeller posts with a rectangular section.

6.3 Rudder post and rudder shaft

6.3.1 The section modulus of the rudder post 'Z_T', in [cm^3] about the longitudinal axis is not to be less than:

$$Z_T = \frac{F_r \ell}{1000}$$

where,

F_r = Rudder force in [N] as defined in Part 3, Chapter 13, Section 3.

ℓ = Unsupported span of the rudder post in [m]. (Refer Figure 6.3.1).

Where, due to low rigidity of the sole piece in the transverse direction, the bottom support for the rudder post cannot be regarded as efficient, direct calculations are to be carried out and the section modulus of the rudder post is increased to limit the bending stress to 85[N/mm^2].

6.3.2 The rudder shaft diameter, d_ℓ, in [mm], at the lower bearing is not to be less than:

$$d_\ell = 3.9\sqrt[3]{\left(\frac{F_r . c . k(\ell - c)}{\ell}\right)}$$

where,

F_r = Rudder force in [N] as defined in Part 3, Chapter 13, Section 3.

ℓ, a, b = Refer Figure 6.3.1

$$c = \frac{a + b}{2}$$

6.3.3 The diameter 'd_ℓ' may have to be increased on account of the low rigidity of the sole piece, as described for the rudder post in Cl.6.3.1. The diameter d_f below the coupling, the flange is to be 10% greater than d_ℓ unless the rudder axle is protected by a corrosion-resistant composition above the upper bearing. Where the corrosion-resistant composition is not provided then d_f may be equal to d_ℓ.

6.3.4 The details of the taper, nut, etc. at the lower end of the rudder axle are to be similar to those of rudder stock cone couplings given in Part 3, Chapter 13.

6.3.5 The scantlings and arrangements of vertical coupling at the upper end of the rudder axle are to be similar to those of horizontal bolted couplings of the rudder given in Part 3, Chapter 13, Section 6. In the formula the axle diameter 'd_ℓ' is to be used in place of rudder stock diameter 'd_s'.

6.3.6 The bearing depth l_2 is not to be less than $1.2\ d_\ell$ [mm].

6.4 Sole piece

6.4.1 Requirements for structural strength of sole pieces are given in Part 3, Chapter 13.

6.4.2 The sole piece is to extend at least two frame spaces forward of the forward edge of the propeller boss and beyond this, the cross-section of the extension is to be gradually reduced to that necessary for an efficient connection to the keel plate. Fabricated sole pieces are to have adequate internal stiffening.

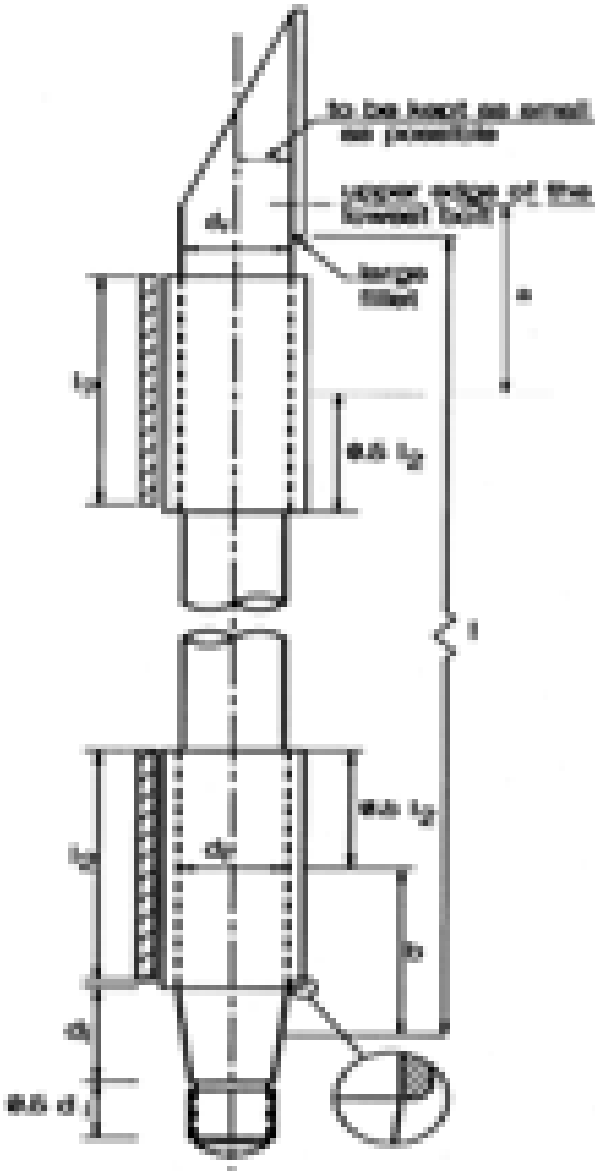


Figure 6.3.1: Rudder shaft details

6.5 Propeller shaft brackets

Propeller shafting is either enclosed in bossing or independent of the main hull and supported by shaft brackets.

6.5.1 Single-arm propeller shaft brackets

6.5.1.1 General

The following requirements are applicable to single arm propeller shaft brackets.

6.5.1.2 Scantling requirement

The moment in a single arm bracket, in [kN-m], is to be obtained from the following formula:

$$M = 0.75\, d_2 F_c \, L/l$$

where,

d_2 = Length of the arm, in [m], measured between the propeller shaft axis and the hull.

L = Length in [m], defined in Figure 6.5.1.

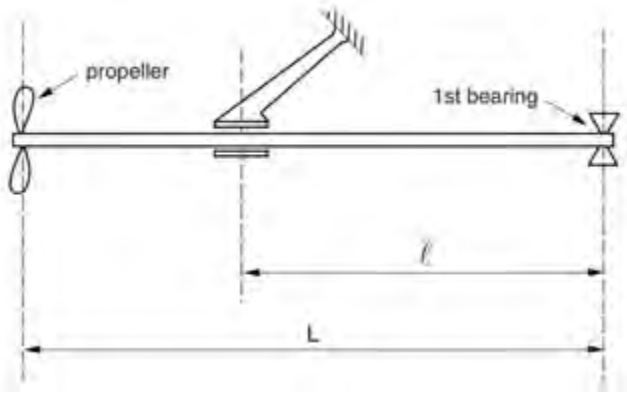


Figure 6.5.1: Schematic showing lengths L and l

F_c = Force, in [kN],

$$= \left(\frac{2\pi N}{60}\right)^2 R_p P$$

N = Number of revolutions per minute of the propeller.

R_p = Distance, in [m], of the centre of gravity of a blade in relation to the rotation axis of the propeller.

P = Mass of a propeller blade, in [t].

6.5.1.3 Checking criteria

The bending stress σ_F and the shear stress τ, is to comply with the following formula:

$$\sqrt{\sigma_F{}^2 + 3\tau^2} \leq \sigma_{all}$$

where,

σ_F = Bending stress to be taken as:

$$\sigma_F = 1000 \frac{M}{z_a}$$

τ = Shear stress

$$\tau = \frac{10.\, F_c.\, L}{A_s l}$$

z_a = Section modulus of the arm, in [cm^3], at the level of the connection to the hull with respect to a longitudinal axis.

A_s = Shear sectional area of the arm, in [cm^2].

σ_{all} = Allowable stress, taken as 70/k [N/mm^2]; where 'k' is the material factor

6.5.2 Double arm propeller shaft brackets

6.5.2.1 General

The following requirements are applicable to propeller shaft brackets having two struts to support the propeller tail shaft boss. The struts are to be of solid or fabricated type. The angle between the struts is not to be less than 50[deg]. It is recommended that the angle included between the arms differs from the angle included between the propeller blades.

6.5.2.2 Arrangement

Solid struts are to be carried continuously through the shell plating and are to be given satisfactory support by the internal ship structure. Fabricated struts may be welded to the shell plating. The shell plating is to be reinforced, and internal brackets in line with strut plating are to be fitted. If the struts are built with a longitudinal centre plate, this plate is to be carried continuously through the shell plating. The struts are to be well-rounded at the fore and aft end at the transition to the hull.

The connection of the arms to the shaft bracket boss is to be by full penetration welding. The strut structure inside the shell needs to terminate within a compartment of limited volume in order to reduce the effect of flooding in the case of damage.

6.5.2.3 Scantling requirement

The scantlings of solid or fabricated shaft brackets are to comply with the following:

$$h \geq 0.4 d_{ts} \text{ [mm]}$$

$$A \geq 0.4 d_{ts}^2 \text{ [mm}^2\text{]}$$

$$Z_T \geq 0.12 d_{ts}^3 \text{ [mm}^3\text{]}$$

where,

d_{ts} = Propeller tail shaft diameter, in [mm].

d_{ts} should be taken as the provided value with a specified minimum tensile strength of 450[N/mm^2] if it is more than the actual value.

h = Thickness of the strut section, in [mm] as given in Figure 6.5.2.

A = Gross area of the strut section, in [mm^2].

Z_T = Gross section modulus of the strut against transverse bending, in [mm^3]. Z_T is to be calculated with reference to the neutral axis YY as indicated in Figure 6.5.2.

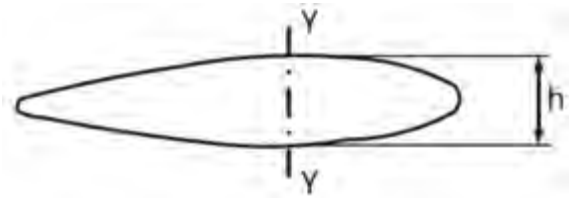


Figure 6.5.2: Sectional view of the strut

6.6 Propeller shaft bossing

6.6.1 Where the propeller shafting is enclosed in bossings extending back to the bearings supporting the propellers, the aft end of the bossings and the bearings are to be adequately supported. End supports are to be designed to transmit the loads from the shaft efficiently to the ship's internal structure.

6.6.2 The boss plating is generally to be radiused into the shell plating and supported at the aft end by diaphragms at every frame. These diaphragms are to be suitably stiffened and connected to floors of increased scantlings or a suitable arrangement of main and deep web frames. At the fore end, the main frames may be shaped to fit the bossing, but deep webs are generally to be fitted not more than four frame spaces apart.

6.6.3 The region where the shafting enters the ship, and the bearing in way, are to be adequately supported by floors or deep webs.

6.7 Shaft Bracket Boss

6.7.1 The length of the shaft bracket boss, ℓ_b, is to be sufficient to support the length of the required bearing and to allow the proper connection of the shaft bracket. Recommendations from the bearing manufacturer are also to be taken into consideration. In general, ℓ_b is recommended

to be not less than $4d_{ts}$, where d_{ts} is the propeller tail shaft diameter, in [mm] and in no case to be less than $2d_{ts}$.

6.7.2 When the shaft and the shaft bracket boss are made of the same material the thickness,t_b [mm] of the shaft bracket boss is to be not less than $(d_{ts}/4)$, where d_{ts} is the propeller tail shaft diameter, in [mm].

6.7.3 Where the shaft and the shaft bracket boss are of dissimilar materials, the thickness of the boss, t_b in [mm] is to be not less than:

$t_b = 0.75\, d_{ts}\, (u^{0.33} - 0.667)$ and in no case is t_b to be less than 12[mm].

where,

d_{ts} = Propeller tail shaft diameter, in [mm] in way of the boss, in the appropriate screw shaft material.

u = Coefficient taken as below but not less than 0.825

$$u = \frac{f_{u-s}}{f_{u-b}}$$

f_{u-s} = Ultimate tensile strength of the shaft material, in [N/mm^2].

f_{u-b} = Ultimate tensile strength of the boss material, in [N/mm^2].

6.8 Rudder horns

6.8.1 Requirements for the structural strength of rudder horns are given in Part 3, Chapter 13.

6.8.2 Rudder horns supporting semi-spade type rudders are to be efficiently integrated into the main hull structure, and additional web frames or side transverses may be required in the 'tween deck above.

6.8.3 The shell plating is to be increased in thickness in way of the horn, and the radius at the shell connection is not to be less than (150 +0.8L)[mm].

6.9 Propeller nozzles

6.9.1 General

6.9.1.1 The requirements for scantlings and arrangements of propeller nozzles given in this section apply, in general, to nozzles of numeral N_N, not greater than 200. Nozzles of numeral greater than 200 will be specially considered.

Nozzle numeral, $N_N = 0.01\ P.D$

where,

P = Power transmitted to the propeller [kW].

D = Diameter of the propeller, in [m].

6.9.1.2 Additional requirements for scantlings and arrangements applicable to steerable nozzles and fixed nozzles are given in Cl.6.9.3 and Cl.6.9.4 respectively.

6.9.1.3 For additional requirements applicable to Azimuth Thrusters see Part 4, Chapter 4, Section 9.

6.9.1.4 All materials used in the construction of the nozzle-body, stocks, pintles, keys, and bolts are to be similar to rudders as per Part 3, Chapter 13, Section 1.

6.9.2 Nozzle scantlings and construction details

6.9.2.1 Nozzle scantlings are to be as per Table 6.9.2. (Refer Figure 6.9.2).

6.9.2.2 If the ship has ice-class notation, nozzle scantlings will be specially considered; however, the part of the outer shell of the nozzle which is situated within the ice belt is to have a plate thickness not less than that of the shell plating in the way.

6.9.2.3 The shroud plating in way of the propeller tips is to be carried well forward and aft of the propeller tips, due to allowance being made on steering nozzles for the rotation of the nozzle in relation to the propeller, however, shroud plate width is not to be less than (L_N/4), where L_N is the length of the nozzle (Refer Figure 6.9.2).

6.9.2.4 The transverse strength of the nozzle is to be maintained by fitting the adequate number of ring webs. At least two ring webs are to be fitted in way of shroud plating, spaced approximately 400[mm]. Fore and aft of these webs the spacing of ring webs are generally not to exceed 600[mm]. Ring webs are to be increased in thickness in way of the headbox and pintle support structure in accordance with Table 6.9.2 and the increased thickness is to be maintained to the adjacent fore and aft web.

Table 6.9.2: Nozzle scantling requirements		
Sl.No	**Item**	**Requirement**
1)	Shroud plating in way of propeller blade tips	• For $N_N \leq 63$ $t_s = (11 + 0.1N_N)$ in [mm] • For $N_N > 63$ $t_s = (14 + 0.052N_N)$ in [mm]
2)	Shroud plating clear of blade tips, flare and cone plating, wall thickness of leading and trailing edge members	$t_p = (t_s - 7)$ in [mm] but not less than 8 [mm].
3)	Webs and ring webs	• In way of the head box and pintle support $t_w = (t_s + 4)$ in [mm] • Elsewhere As for item (2)
Symbols		
N_N = Nozzle numeral as given in Cl.6.9.1. t_s = Thickness of shroud plating in way of propeller tips, in [mm]. t_p = Thickness of plating, in [mm]. t_w = Thickness of webs and ring webs in way of the headbox and pintle support, in [mm].		
NOTE: The thickness given is for mild steel. When corrosion-resistant material is used, the plate thicknesses may be reduced by 15%, however, thickness is not to be less than the minimum thickness as applicable.		

6.9.2.5 Local stiffening is to be fitted in way of the top and bottom supports which are to be integrated with the webs and ring webs. Continuity of bending strength is to be maintained in these regions.

6.9.2.6 Weld details of nozzle plating, webs, coupling flanges, etc. are to be as given in Chapter 13, Section 4 and also as per Chapter 16 'Welding'.

6.9.2.7 Leak testing of the nozzle is to be as required for rudders as per Chapter 17.

6.9.2.8 Arrangement is to be provided to drain the nozzle completely. Drain plugs are to be provided with efficient packing.

6.9.2.9 Internal surfaces of nozzles are to be efficiently coated for corrosion protection after completion of fabrication and testing. Where it is intended to fill the nozzle with plastic foam, details of the foam material are to be submitted.

6.9.2.10 The web plates and shell plates of the support structure are to be in line with the web plates in the nozzle.

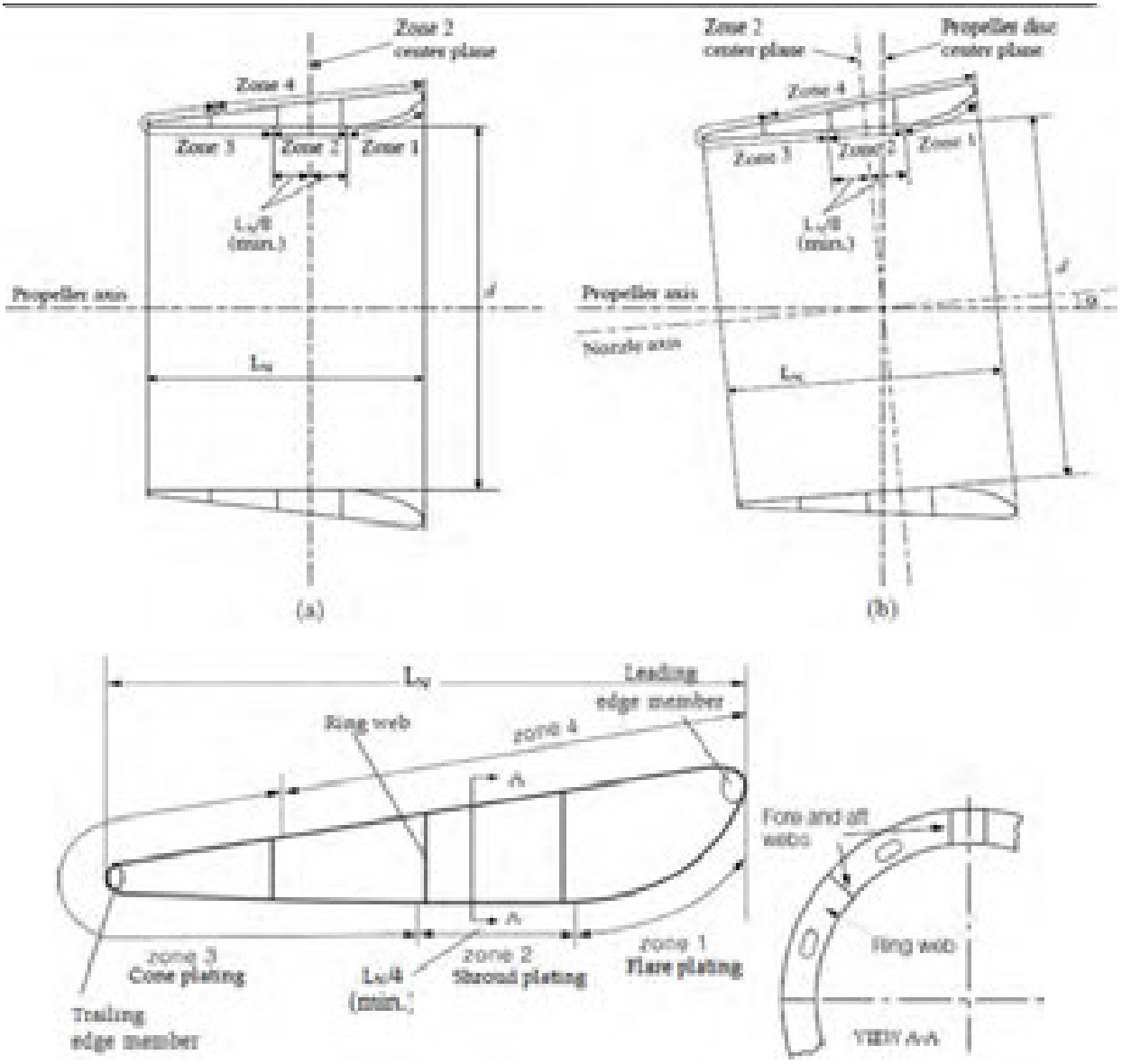


Figure 6.9.2: Propeller Nozzle Section View

where,

L_N = Nozzle length

d = Nozzle inner diameter

Zone 1 = Zone of nozzle inner skin from nozzle leading edge to the fore end of Zone 2.

Zone 2 = Zone of nozzle inner skin in way of propeller tips with two ring webs within the zone, = $L_N/4$ (min)

The minimum length on each side of the Zone 2 centre plane is to be:

= $L_N/8$, where Zone 2 centre plane and propeller disc centre plane coincide as shown in Figure 6.9.2.

= $L_N/8 \cos\alpha + d/2 \tan\alpha$, where Zone 2 centre plane and propeller disc centre plane does not coincide as shown in Figure 6.9.2.

Zone 3 = Zone of nozzle inner and outer skin covering the tail vicinity, from the aft end of Zones 2 to the aft end of Zone 4.

Zone 4 = Zone of nozzle outer skin from the leading edge to the fore end of Zone 3.

6.9.3 Steerable nozzles

6.9.3.1 The nozzle stock, couplings, pintles and bearings are to be designed analogously with the requirements given in Chapter 13, Sections 5 and 6 using the design values of lateral force, F_r, in [N] and turning moment Q_r, in [N-m], as given below:

$$F_r = 376\ A.V^2$$

$$Q_r = F_r.r$$

where,

A = Projected area of the nozzle, in [m^2].

V = Maximum service speed in [knots] with the ship on summer load waterline. When the speed is less than 10[knots], V is to be replaced by the expression

$$V_{min} = \frac{(V + 20)}{3}$$

r = The horizontal distance from the centre line of stock to the centre of pressure of the nozzle, but not to be taken less than 10% of the chord length of the nozzle.

6.9.3.2 Effective means are to be provided for supporting the weight of the nozzle. Where the support is provided by a carrier bearing attached to the nozzle head, the structure in way of the bearing is to be adequately strengthened. The shell plating in way of the nozzle head is to be increased in thickness by 50%.

6.9.3.3 All nozzle bearings are to be accessible for measuring wear without lifting or unshipping the nozzle.

6.9.3.4 Satisfactory arrangement is to be provided to prevent water from entering the steering gear compartment and lubricant from being washed away from the nozzle carrier. A seal or stuffing box is to be fitted above the deepest load water line for this purpose, however, two separate seals or stuffing boxes are to be provided when the nozzle carrier is below the deepest load water line.

6.9.4 Fixed nozzles

6.9.4.1 The nozzle is to be adequately supported preferably by closed box structures of the necessary enclosed area to withstand the lateral force F_r and Torque Q_r given in Cl.6.9.4 and its own weight.

The resultant stresses are not to exceed the following:

Bending stress, $\sigma_b = 75/k\ \text{in}\ [N/mm^2]$

Torsional shear stress, $\tau_t = 50/k\ \text{in}\ [N/mm^2]$

Shear stress, $\tau = 50/k\ \text{in}\ [N/mm^2]$

Equivalent stress,

$$\sigma_{e1} = \sqrt{\sigma^2 + 3\tau^2} = 100/k\ [N/mm^2]$$

$$\sigma_{e2} = \sqrt{\sigma + 3\tau^2} = 90/k\ [N/mm^2]$$

6.9.4.2 The nozzle support box structure is to be well integrated to the webs and ring webs in the nozzle and to the hull structure. Structural continuity is to be maintained between the webs in the nozzle, the top box structure and local girders/floors inside the hull.

6.10 Sea chest

6.10.1 Sea chest plating

In the case where a sea chest is provided in the shell plating for sea suction or discharge, the thickness of the sea chest is not to be less than that obtained from the following formula and to be suitably stiffened to provide sufficient rigidity as necessary. The thickness, however, is not to be less than the thickness of shell plating where the sea chest is installed.

$$t_{sc} = \sqrt{L} + 1[mm]$$

Stiffeners attached to the shell in way of sea chest openings are to be connected at their ends.

The sea water inlet openings in the shell are to be protected by gratings.

6.11 Thruster tunnel

6.11.1 Following requirements apply to transverse thrusters for maneuvering, which are integrated into the ship structure and which can produce transverse thrust at very slow ship speeds. Retractable rudder propellers are not transverse thrusters in the context of these requirements.

6.11.2 Transverse thruster tunnels are to be completely integrated into the ship structure and welded to it.

6.11.3 Where the outboard ends of the tunnel are provided with bars or grids, the bars or grids are to be effectively secured.

6.11.4 The thruster units are to be framed to the same standard as the surrounding shell plating and are to be adequately supported and stiffened.

6.11.5 Thrust element housing structures holding fixtures for propulsion units are to be effectively connected to the tunnel structure.

6.11.6 Welding of structural elements which are part of the watertight integrity of the ship hull is generally to be carried out as welds with full penetration welding. For structural elements where the stress level is low, partial penetration welding may be accepted.

6.11.7 If suction or draining ducts are arranged in the ship's bottom the bottom slamming loads according to Section 3 are to be considered.

6.11.8 Tunnel plate thickness

6.11.8.1 In general, the wall thickness of the thrust (bow and stern) unit is to be in accordance with the manufacturer's recommendation, if any, and is to be not less than 1.1 times more than the thickness of the surrounding shell plating.

6.11.8.2 In addition,t_{tun} is not to be taken less than:

$$t_{tun} = 0.008d_{tun} + 1.8$$

where:

d_{tun} = Inside diameter of the tunnel, in [mm], but not to be taken less than 970[mm].

6.11.8.3 In addition, for ships with L≤ 200 [m], t_{tun} is not to be taken less than:

$$t_{tun} = \sqrt{L.k} + 2.5$$

6.12 Machinery foundations

6.12.1 General

6.12.1.1 Main engines and thrust bearings are to be effectively secured to the hull structure by foundations of strength that can adequately resist the various gravitational, thrust, torque, dynamic, and vibratory forces which may be imposed on them. The recommendations given by the engine manufacturer are also to be considered.

6.12.1.2 In the case of higher power oil engines or turbine installations, the seating should generally be integral with the double bottom structure. The tank top plating in way of the engine foundation plate or the turbine gear case and the thrust bearing is to be substantially increased in thickness, Refer Type 1 of Figure 6.12.1.1.

6.12.1.3 For the main machinery supported on foundations of Type 2, as shown in Figure 6.12.1.2, the forces from the engine into the adjacent structure are to be distributed as uniformly as possible. Longitudinal members supporting the foundation are to be aligned with girders in the double bottom, and transverse stiffening is to be arranged in line with the floors, Refer Type 2 of Figure 6.12.1.2.

They are to be so arranged as to ensure sufficient accessibility for the welding of joints and for surveys and maintenance.

6.12.1.4 For ships with open floors in the machinery space, the foundations are generally to be arranged above the level of the top of the floors and securely bracketed.

6.12.1.5 Lightening holes in engine foundations are to be kept as small as practicable and the edges are to be suitably reinforced.

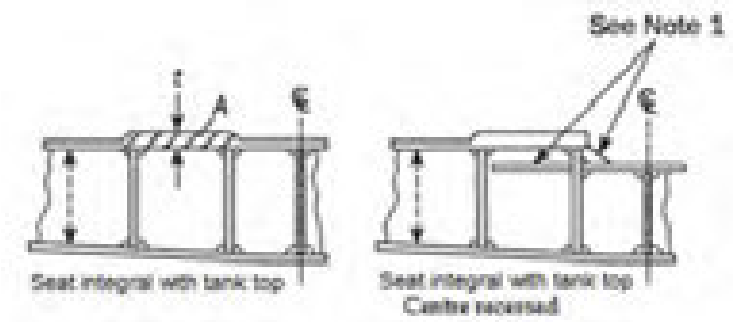


Figure 6.12.1.1: Machinery foundations Type 1-Seat integral with tank top

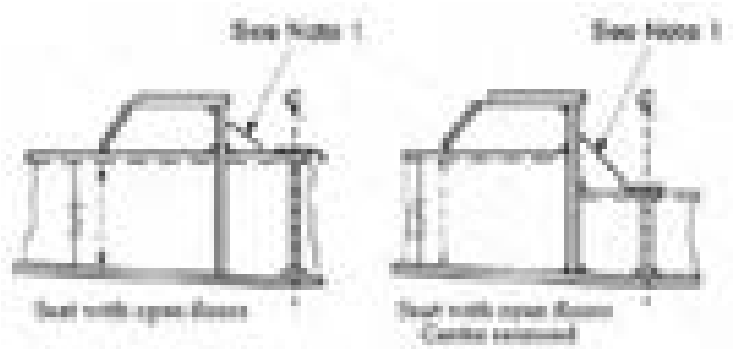


Figure 6.12.1.2: Machinery foundations Type 2-Built up seat

Note 1: Brackets are to be as large as possible. Brackets may be omitted to avoid interference with the girders of the engine foundation, in accordance with the recommendations of the engine manufacturer.

6.12.2 Foundations for internal combustion engines and thrust bearings

6.12.2.1 In determining the scantlings of foundations for internal combustion engines and thrust bearings, consideration is to be given to the general rigidity of the engine and its design characteristics with regard to the out-of-balance forces.

6.12.2.2 Thrust blocks are to be bolted to efficient foundations extending well beyond the thrust blocks and arranged to distribute the loads effectively into the adjacent structure. Extra intercostal girders, effectively attached, are to be fitted in way of the foundations, as may be required.

6.12.3 Auxiliary foundations

6.12.3.1 Auxiliary machinery is to be secured on foundations that are of suitable size and arrangement to distribute the loads from the machinery evenly into the supporting structure.

6.12.4 Boiler foundations

6.12.4.1 Boiler bearers are to be of substantial construction and efficiently supported by transverse and horizontal brackets. These should generally be arranged in line with plate floors and girders in a double bottom or with suitable deep beams or transverses and girders at boiler flats. Suitable allowance is to be made in the design of the supporting structure for the variation in loading due to thermal expansion effects.

6.12.5 Heavy equipment

6.12.5.1 Where heavy equipment is mounted directly on the inner bottom, the thickness of the floors and girders are to be suitably increased.

6.12.6 Recommended scantlings

6.12.6.1 Hold-down bolts are to be arranged as close as possible to floors and longitudinal girders. Plating thickness and the arrangements of hold-down bolts should also consider the manufacturer's recommendations.

6.12.6.2 The scantlings of seating of main engines and thrust bearings are to be adequate in relation to the weight and power of engines and the static and dynamic forces transmitted by the propulsive installation.

6.12.6.3 In general, at least two longitudinal members are to be fitted in way of seating of main engines.

6.12.6.4 One longitudinal member may be fitted only where the following three conditions are complied with:

$$L < 150[m]$$

$$P < 7100[kW]$$

$$P < 2.3\ R\ell$$

6.12.6.5 In general, the scantlings of engine girder face plate, web and floors in way of engine seating may be calculated as given below.

- The minimum top plate area, A in [cm^2], for one side of the seat:

$$A = 50 + 85\left(\frac{P}{Rl}\right) \text{for} \left(\frac{P}{Rl}\right) \geq 2.0$$

$$A = 90 + 65\left(\frac{P}{Rl}\right) \text{ for} \left(\frac{P}{Rl}\right) < 2.0$$

Where twin engines are fitted, a continuous top plate is to be arranged in general if the engines are coupled to one propeller shaft. For elastically mounted engines the sectional area may be adequately reduced.

- The minimum top plate thickness, t_p in [mm],

Where two girders are fitted on each side

$$t_p = \left(300 + 220\left(\frac{P}{Rl}\right)\right)^{0.5}$$

Where one girder is fitted on each side

$$t_p = \left(300 + 220\left(\frac{P}{Rl}\right)\right)^{0.5} + 6$$

where,

P = Maximum power of the engine, in [kW].

R = Rpm of the engine at maximum power.

l = Effective length of engine foundation plate, in [m], required for bolting the engine to the seating.

- The minimum girder web thickness, t_g in [mm],

The sum of the girder web thicknesses (t_{g2}), where two girders are fitted is to be not less than:

$$t_{g2} = \left(400 + 270\left(\frac{P}{Rl}\right)\right)^{0.5}$$

Where a single girder is fitted on each side, the thickness t_{g1}, is to be not less than:

$$t_{g1} = 0.55 t_{g2}$$

- The minimum thickness of floor plates $t_{f,}$ in [mm], below the bed plate

$$t_f = \left(65 + 52\left(\frac{P}{Rl}\right)\right)^{0.5}$$

6.12.6.6 For high-speed engines of less than 1500[kW] and medium speed engines of less than 750[kW] the following may be used instead of the values specified in Cl.6.12.6.5 above.

Top plate area, $A = 20 + 120\frac{P}{R}$ in $[cm^2]$

Top plate thickness, $t_p = 0.1A + 14$ in $[mm]$

Girder web thickness, $t_{g1} = 0.043A + 7$ in $[mm]$

Floor web thickness, $t_f = 0.02A + 6$ in $[mm]$

6.13 Bilge keel

6.13.1 Material

6.13.1.1 The bilge keel and ground bar material are to be of the same yield stress as the material to which they are attached.

6.13.1.2 In addition, when the bilge keel extends over a length more than 0.15L, the material of the bilge keel and ground bar is to be of the same grade as the bilge strake.

6.13.2 Design

6.13.2.1 The design of single web bilge keels is to be such that failure to the web occurs before failure of the ground bar. This may be achieved by ensuring the web thickness of the bilge keel does not exceed that of the ground bar.

Bilge keels are often damaged or ripped off in the ice. The connection of bilge keels to the hull is to be so designed that the risk of damage to the hull, in case a bilge keel is ripped off, is minimized. To limit damage when a bilge keel is partly ripped off, it is recommended that bilge keels are cut up into several shorter independent lengths.

6.13.2.2 Bilge keels of a different design, from that shown in Figure 6.13.2.1, are specially considered. Alternative arrangement as shown in Figure 6.13.2.2 may also be accepted.

6.13.3 Ground bars

6.13.3.1 Bilge keels are not to be welded directly to the shell plating. A ground bar, or doubler, is to be fitted on the shell plating as shown in Figure 6.13.2.1 and Figure 6.13.3. In general, both the bilge keel and the ground bar are to be continuous.

6.13.3.2 It is recommended that bilge keels should not be fitted in the forward 0.3L region on ships intended to navigate in severe ice conditions.

6.13.3.3 The gross thickness of the ground bar is not to be less than the gross thickness of the bilge strake or 14[mm], whichever is the lesser.

6.13.4 End details

6.13.4.1 The ground bar and bilge keel ends are to be tapered or rounded and are to terminate on an internal stiffening member.

6.13.4.2 Tapering is to be gradual with a minimum ratio of 3:1, Refer items (a), (b), (d) and (e) in Figure 6.13.3 and Figure 6.13.4. Rounded ends are to be as shown in item (c) of Figure 6.13.3. Cut-outs on the bilge keel web, within zone 'A' (Refer item (b) of Figure 6.13.3 and item (e) of Figure 6.13.4) are not permitted.

6.13.4.3 The end of the bilge keel web is to be not less than 50[mm] and not greater than 100[mm] from the end of the ground bar, Refer item (a) of Figure 6.13.3 and item (d) of Figure 6.13.4.

6.13.4.4 Ends of the bilge keel and ground bar are to be supported by either transverse or longitudinal members inside the hull, as indicated below:

- Transverse support member is to be fitted at mid length between the end of the bilge keel web and the end of the ground bar, Refer items (a), (b) and (c) of Figure 6.13.3.

- Longitudinal stiffener is to be fitted in line with the bilge keel web, it is to extend to at least the nearest transverse member forward and aft of zone 'A' (Refer item (b) of Figure 6.13.3 and item (e) of Figure 6.13.3). In this case, the requirement of Cl.6.13.4.3 does not apply.

6.13.4.5 Alternative end arrangements may be accepted, provided that they are considered equivalent.

6.13.5 Welding

6.13.5.1 Welding of bilge keel and intermediate plate connections is to be in accordance with Chapter 16, Table 3.2.1.

6.13.5.2 The connections of the bilge keel to the ground bar and the bilge ground bar to the side shell, are to be by double continuous fillet welds

6.13.5.3 Butt welds in shell plating, ground bar and bilge keels are to be staggered. Butt welds in the bilge keel and doubler are to be full penetration.

6.13.5.4 Butt welds, in the bilge keel and ground bar, are to be well clear of each other and of butts in the shell plating. In general, shell butts are to be flush in way of the ground bar and ground bar butts are to be flush in way of the bilge keel.

6.13.5.5 Direct connection between ground bar butt welds and shell plating is not permitted. This may be obtained by the use of removable backing.

6.13.5.6 In general, scallops and cut-outs are not to be used.

6.13.5.7 For ships over 65[m] in length, holes are to be drilled in the bilge keel butt welds. The diameter of the hole is to be greater than the width W of the butt weld and is to be a minimum of 25[mm] (Refer Figure 6.13.2.1.

6.13.5.8 Where the butt weld has been subject to non-destructive examination, the crack arresting hole may be omitted.

6.13.5.9 Within zone 'B', welds at the end of the ground bar and bilge plating, and at the end of the bilge keel web and ground bar, are to have weld factors of 0.44 and 0.34 respectively and they are to be grounded and blended smoothly with the base materials. (Refer Figure 6.13.3 and Figure 6.13.4).

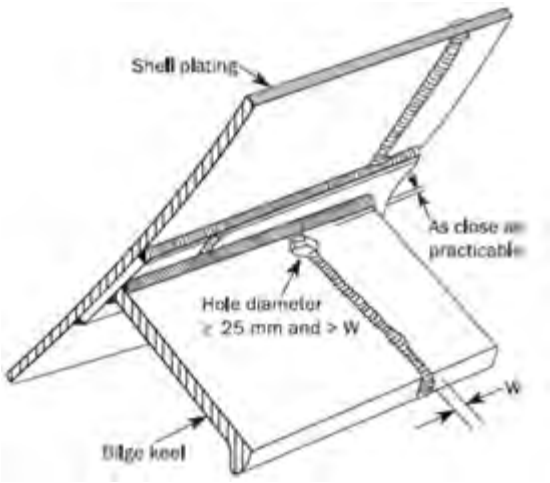


Figure 6.13.2.1: Bilge keel arrangement

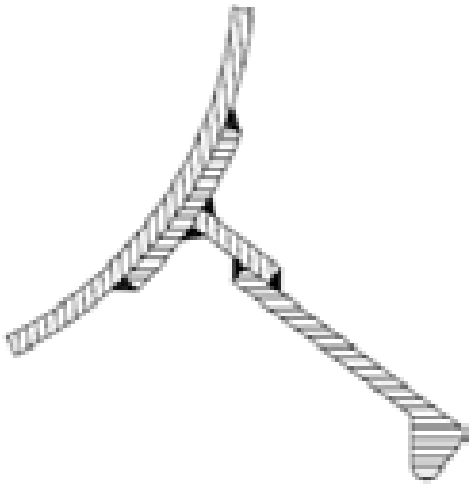


Figure 6.13.2.2: Bilge keel alternative arrangement

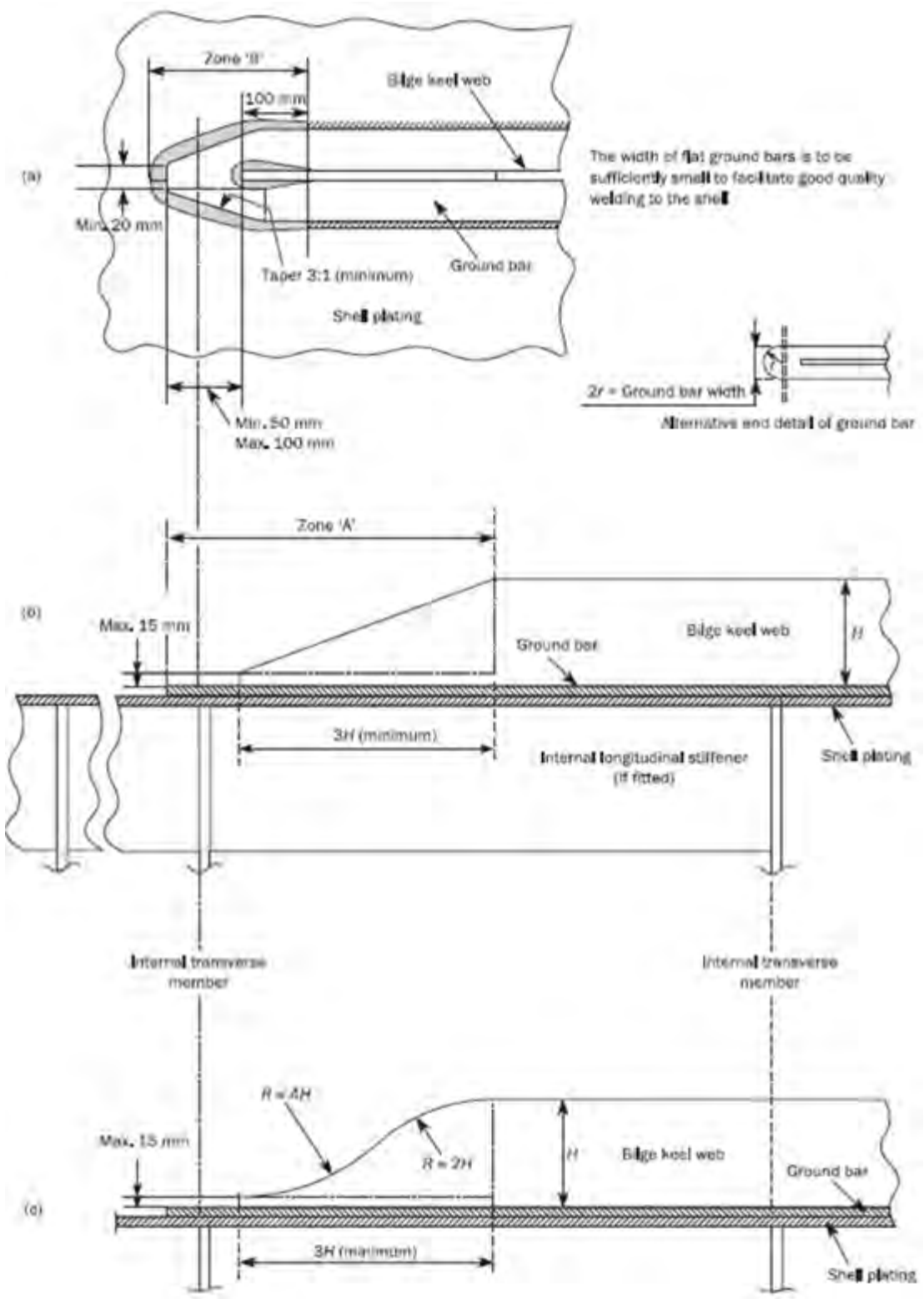


Figure 6.13.3: Bilge keel end design

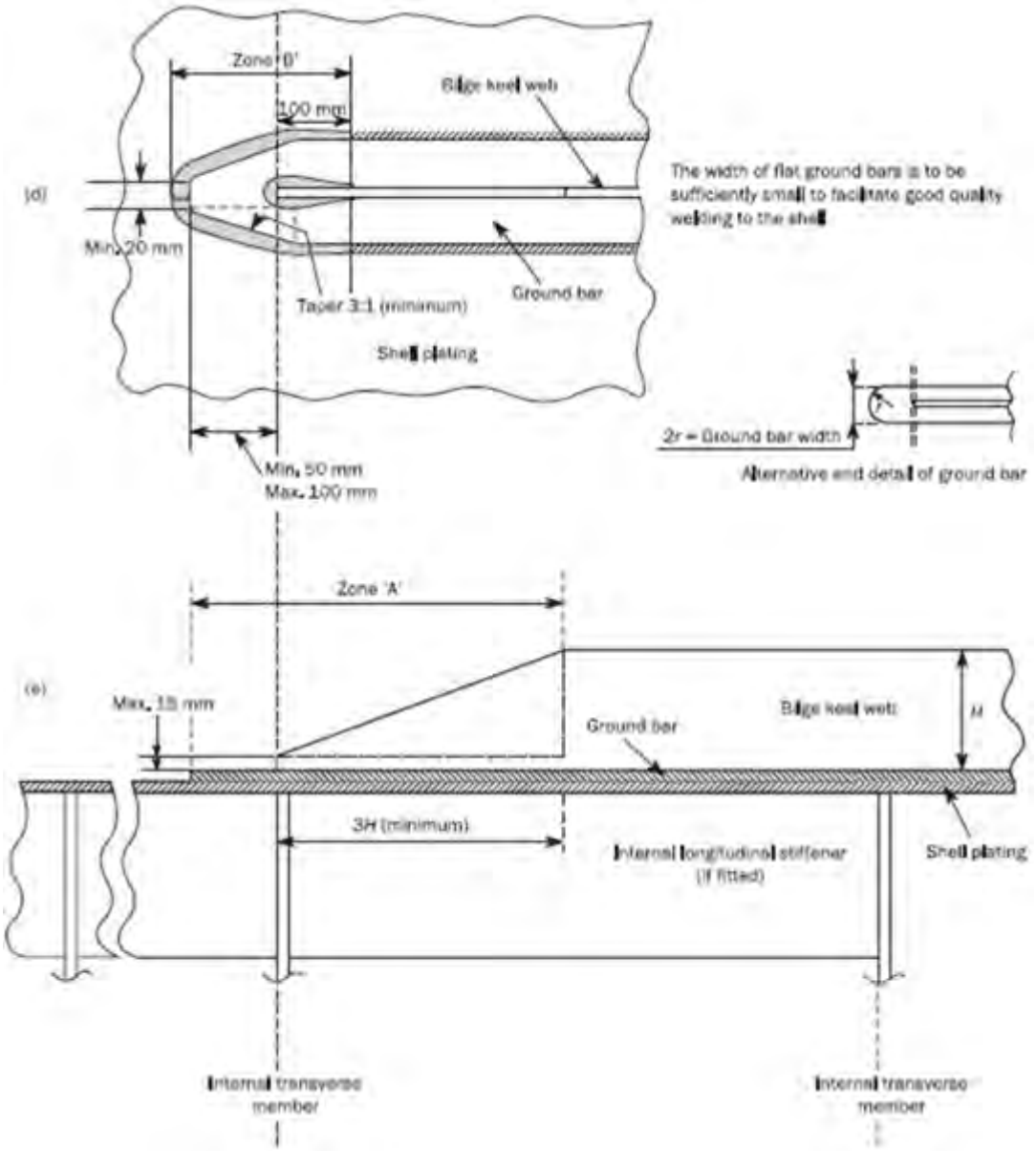


Figure 6.13.4: Bilge keel end design

Section 7

Tanks Subject to Sloshing

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

α_p = Correction factor for the panel aspect ratio to be taken as:

$$\alpha_p = 1.2 - \frac{b}{2.1a}$$

but not to be taken as greater than 1.0

a = Length of plate panel, in [mm], as defined in Chapter 3, Section 7.

b = Breadth of plate panel, in [mm], as defined in Chapter 3, Section 7.

ℓ_{slo} = Effective sloshing length, in [m], as defined in Cl.7.4.2.

P_{slo-l} = Sloshing pressure in [kN/m^2] due to longitudinal liquid motion as defined in Cl.7.4.2.

P_{slo-t} = Sloshing pressure in [kN/m^2] due to transverse liquid motion as defined in Cl.7.4.3.

b_{slo} = Effective sloshing breadth, in [m], as defined in Cl.7.4.3.

ρ_L = Density of liquid in the tank and ballast hold, in [t/m^3], but not less than:

For strength assessment:

= 1.025 for all liquids including oil and product cargoes. If a tank filled at 98% is intended to carry heavier liquid cargoes than 1.025 (i.e.$\rho_{max-LM} > 1.025$), then $\rho_L = \rho_{max-LM}$

For fatigue assessment*:*

= 0.9 for oil and oil product cargoes

= 1.025 for ballast tanks

= ρ_{max-LM} for the cargo tank intended to carry liquid other than oil products for cargo tanks intended to carry oil products with a density exceeding 1.025 [t/m^3].

ρ_{max-LM} = Maximum liquid cargo density in [t/m^3], associated with a full tank at 98%, from any loading condition in the ship's loading manual or the value specified by the designer

ρ_{part} = Maximum permissible high liquid cargo density, in [t/m^3], associated with a partially filled cargo tank but not taken less than ρ_L considered for strength assessment

ρ_{slo} = Liquid density, in [t/m^3], to be used for sloshing assessment, taken as:

= ρ_{part} For heavy liquid cargo density associated with partial filling of cargo tank

= ρ_L For all other cases

h_{max} = Maximum permissible filling level, in [m], taken as:

For ballast tanks: Maximum tank height,

For cargo tanks with cargo density equal to ρ_L: Maximum tank height

For cargo tanks with heavy liquid cargo density equal to ρ_{part} associated with a partially filled cargo tank: h_{part} as defined in Cl.7.2.1.

f_{CD} = Factor for joint probability of occurrence of liquid cargo density and maximum sea state in 25 years design life, to be taken as:

For strength assessment with FE analysis of cargo tanks filled with oil or oil products cargo

$$f_{CD} = 0.88 \text{ for } \rho_L \leq 1.025[t/m3]$$

$$f_{CD} = 1.0 \text{ for } \rho_L > 1.025[t/m3]$$

For other cases: $f_{CD} = 1.0$

7.1 Application

7.1.1 The requirements of this section cover the strengthening requirements for localized sloshing loads that occur in tanks carrying liquid.

7.1.2 The requirements of this section apply to all liquid cargo, ballast tanks, and other tanks with volume exceeding 100[m^3], but does not

apply to the water ballast cargo hold of bulk carriers.

7.1.3 Where the effective sloshing length, ℓ_{slo} is less than 0.03L, calculations involving P_{slo-l} are not required and where the effective sloshing breadth b_{slo} is less than 0.32B, calculations involving P_{slo-t} are not required.

7.1.4 The minimum sloshing pressure, $P_{slo-min}$ for small tanks with a volume less than 100[m^3] and for tanks of cellular construction, i.e., double hull construction with internal structures restricting the fluid motion, is to be taken equal to 12 [kN/m^2]. This pressure is only applicable to plates, stiffeners and tripping brackets at all internal surfaces such as web frames, girders and stringers. No other sloshing requirements are applicable to such tanks.

7.1.5 The minimum sloshing pressure, $P_{slo-min}$ for tanks other than those defined in Cl.7.1.4, is to be taken equal to 20[kN/m^2]. This pressure is only applicable to plates, stiffeners and tripping brackets at all internal surfaces such as web frames, girders, stringers and wash bulkheads.

7.1.6 For larger tanks with an effective sloshing breadth, b_{slo} as defined in Cl.7.4.3, greater than 0.56B or an effective sloshing length, ℓ_{slo}, as defined in Cl.7.4.2, greater than 0.13L, sloshing is to be specially considered.

7.1.7 Dry bulk cargo holds intended for the carriage of ballast water are assumed either full or empty in seagoing condition and are not required to be assessed for sloshing.

7.2 General requirements

7.2.1 Filling heights of tanks

The scantlings of all tanks are to comply with the sloshing requirements given in this section for the following cases:

- Unrestricted filling height for all tanks, except for cargo tanks,

- Unrestricted filling height for cargo tanks with cargo density equal to ρ_L.

- All filling levels up to h_{part} for cargo tanks with cargo density equal to ρ_{part} taken as:

$$h_{part} = \frac{h_{tk} . \rho_L . f_{CD}}{\rho_{part}}$$

where:

h_{part} = Maximum permissible filling height, in [m], associated with a partial filling of the considered cargo tank with a high liquid density equal to ρ_{part}.

h_{tk} = Maximum tank height, in [m].

7.2.2 Structural details

Local scantling increases due to sloshing loads are to be made with due consideration given to details and avoidance of hard spots, notches and other harmful stress concentrations.

7.3 Application of sloshing pressure

7.3.1 General

The structural members of the following tanks are to be assessed for the design sloshing pressures P_{slo-l} and P_{slo-t} in accordance with Cl.7.3.3 and Cl.7.3.4.

- Cargo and slop tanks.

- Fore peak and aft peak ballast tanks.

- Other tanks which allow free movement of liquid, e.g. ballast tanks, fuel oil bunkering tanks and freshwater tanks, etc.

7.3.2 Structural members to be assessed

The following structural members are to be assessed:

- Plates and stiffeners forming boundaries of tanks.

- Plates and stiffeners on wash bulkheads.

- Web plates and web stiffeners of primary supporting members located in tanks.

- Tripping brackets supporting primary supporting members in tanks.

7.3.3 Application of design sloshing pressure due to longitudinal liquid motion

7.3.3.1 The design sloshing pressure due to longitudinal liquid motion, P_{slo-l} as defined in Cl.7.4.2 is to be applied to the following members as shown in Figure 7.3.3.

- Transverse tight bulkheads.

- Transverse wash bulkheads.

- Stringers on transverse tight and wash bulkheads.

- Plating and stiffeners on the longitudinal bulkheads, deck and inner hull within a distance from the transverse bulkhead taken as:

- $0.25\ell_{slo}$,

- The distance between the transverse bulkhead and the first web frame if located inside the tank at the considered level, whichever is less.

7.3.3.2 In addition, the first web frame next to a transverse tight or wash bulkhead if the web frame is located within $0.25\ell_{slo}$ from the bulkhead, as shown in Figure 7.3.3, is to be assessed for the web frame reflected sloshing pressure, P_{slo-wf} as defined in Cl.7.4.2.

7.3.3.3 The minimum sloshing pressure, $P_{slo-min}$ as defined in 7.1 is to be applied to all other members.

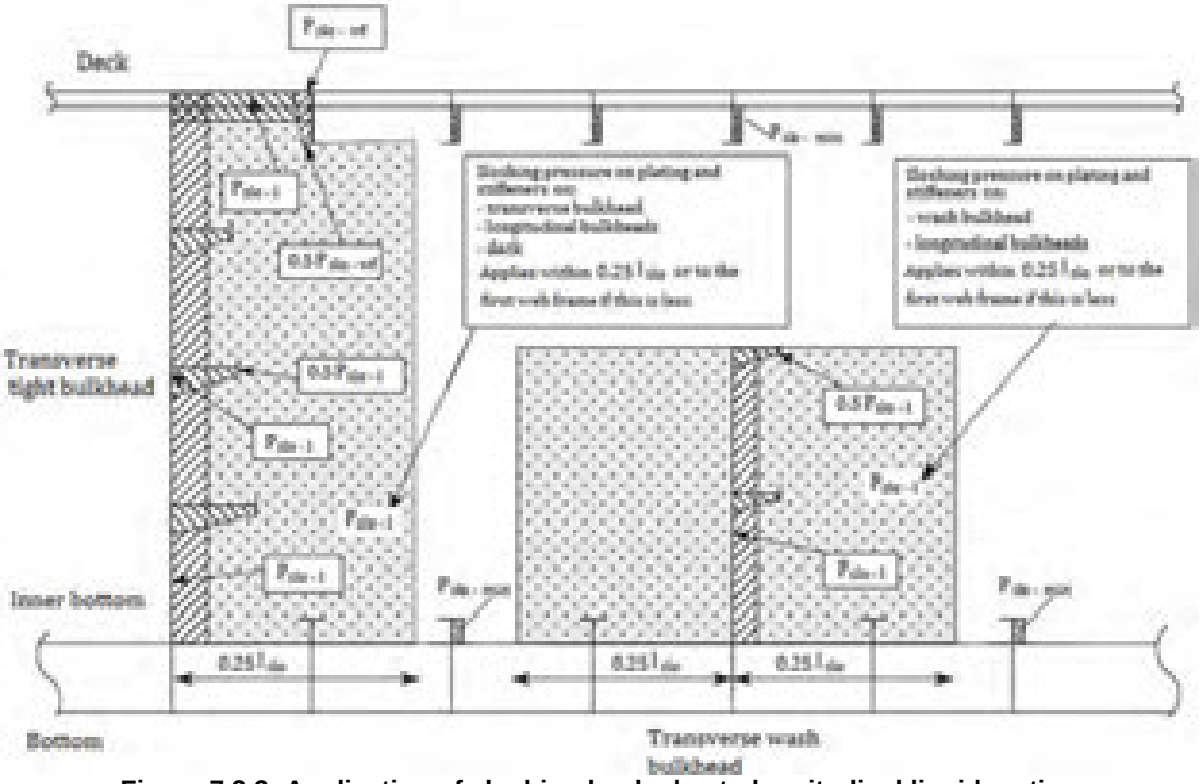


Figure 7.3.3: Application of sloshing loads due to longitudinal liquid motion

7.3.4 Application of design sloshing pressure due to transverse liquid motion

7.3.4.1 The design sloshing pressure due to transverse liquid motion, P_{slo-t} as defined in Cl.8.4.3, is to be applied to the following members as shown in Figure 7.3.4.

- Longitudinal tight bulkhead.

- Longitudinal wash bulkhead and longitudinal girders.

- Horizontal stringers on longitudinal tight and wash bulkheads, and vertical webs on transverse tight and wash bulkheads.

- Plating and stiffeners on the transverse tight bulkheads including stringers and deck within a distance from the longitudinal bulkhead taken as:

- $0.25b_{slo}$,

- The distance between the longitudinal bulkhead and the first girder if located inside the tank at the considered level, whichever is less.

7.3.4.2 In addition, the first girder next to a longitudinal tight or wash bulkhead if the girder is located within $0.25b_{slo}$ from the longitudinal bulkhead, as shown in Figure 7.3.4, is to be assessed for the reflected sloshing pressure, $P_{slo-grd}$ as defined in Cl.7.4.3.

7.3.4.3 The minimum sloshing pressure, $P_{slo-min}$ as defined in 7.1 is to be applied to all other members.

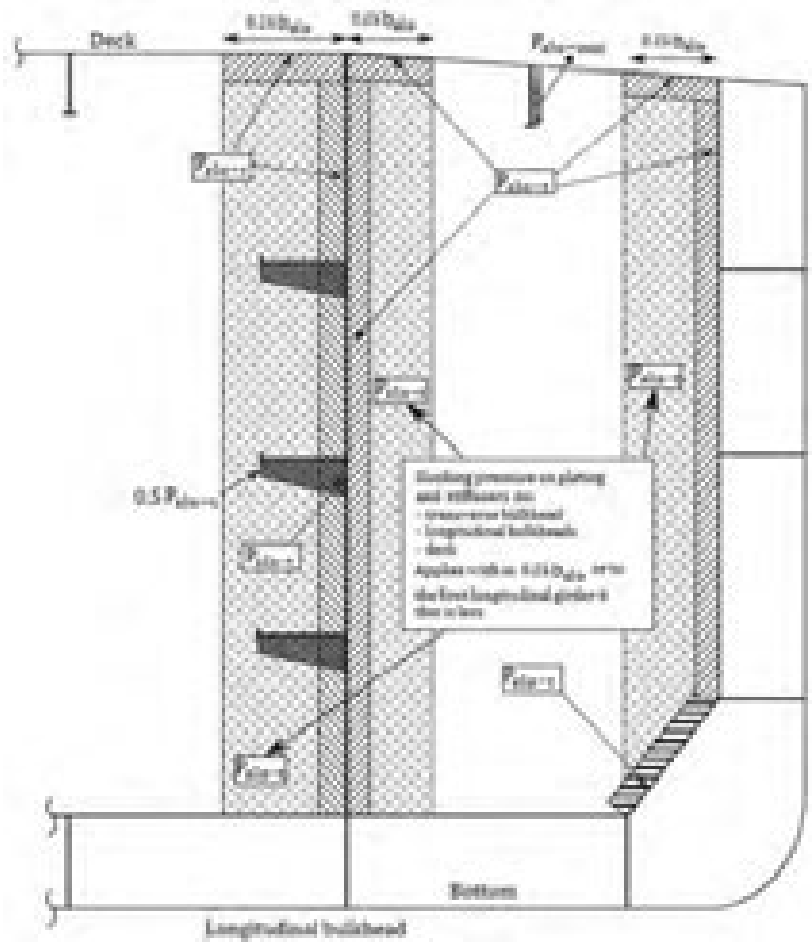


Figure 7.3.4: Application of sloshing loads due to transverse liquid motion

7.3.5 Combination of transverse and longitudinal fluid motion

The sloshing pressures due to transverse and longitudinal fluid motion are assumed to act independently. Structural members are therefore to be evaluated based on the greatest sloshing pressure due to longitudinal and transverse fluid motion.

7.4 Sloshing loads

7.4.1 Sloshing pressure on tank boundaries and internal divisions

The sloshing pressure due to liquid motions in a tank P_{slo} acting on any load point of a tank boundary or internal divisions, in [kN/m^2], for the sloshing design load scenario, given in Cl.7.6.1, is to be taken as follows, but not less than $P_{slo-min}$ as defined in 7.1

$P_{slo} = P_{slo-l}$ for transverse bulkheads, as defined in Cl.7.4.2.

$P_{slo} = P_{slo-wf}$ for web frames and transverse stringers, as defined in Cl.7.4.2.

$P_{slo} = P_{slo-t}$ for longitudinal bulkheads, as defined in Cl.7.4.3.

$P_{slo} = P_{slo-grd}$ for longitudinal girders and stringers, see Cl.7.4.3.

7.4.2 Sloshing pressure due to longitudinal liquid motion

7.4.2.1 Application

The sloshing pressure due to longitudinal liquid motion P_{slo-l} is to be taken as a constant value over the full tank depth and is to be taken as the greater of the sloshing pressures calculated for filling levels from $0.05h_{max}$ to $0.95h_{max}$, and/or $0.05h_{max}$ to h_{part}, typically in 0.05 h_{max} increments.

7.4.2.2 Effective sloshing length

The effective sloshing length, ℓ_{slo} in [m], is to be taken as defined in Table 7.4.2.

Table 7.4.2: Effective sloshing length [ℓ_{slo}]

Type of transverse bulkhead	ℓ_{slo}
Transverse tight bulkheads	$\ell_{slo} = \frac{(1 + n_{WT}\alpha_{WT})(1 + f_{wf}\alpha_{wf})\ell_{tk-h}}{(1 + n_{WT})(1 + f_{wf})}$
Transverse wash bulkheads	$\ell_{slo} = \frac{[1 + (n_{WT} - 1)\alpha_{WT}](1 + f_{wf}\alpha_{wf})\ell_{tk-h}}{(1 + n_{WT})(1 + f_{wf})}$

where:

n_{WT} = Number of transverse wash bulkheads in the tank.

α_{WT} = Transverse wash bulkhead coefficient, to be taken as (Refer Figure 7.4.2.1):

$$\alpha_{WT} = \frac{A_{OWT}}{A_{tk-t-h}}$$

α_{wf} = Transverse web frame coefficient, to be taken as (see Figure 7.4.2.2):

$$\alpha_{wf} = \frac{A_{O-wf-h}}{A_{tk-t-h}}$$

For tanks with changing shape along the length and/or with web frames of different shape the transverse web frame coefficient,α_{wf}, may be taken as the weighted average of all web frame locations in the tank given as:

$$\alpha_{wf} = \frac{\sum_{i=1}^{n_{wf}} \frac{A_{O-wf-h}}{A_{tk-t-h}}}{n_{wf}}$$

A_{OWT} = Total area of openings, in [m^2], in the transverse section in way of the wash bulkhead below the considered filling height.

A_{tk-t-h} = Total transverse cross sectional area, in [m^2], of the tank below the considered filling height.

A_{O-wf-h} = Total area of openings, in [m^2], in the transverse section in way of the web frame below the considered filling height.

f_{wf} = Factor to account for the number of transverse web frames and transverse wash bulkheads in the tank, to be taken as:

$$f_{wf} = \frac{n_{wf}}{1 + n_{WT}}$$

n_{wf} = Number of transverse web frames, excluding wash bulkheads, in the tank

ℓ_{tk-h} = Length of the tank, in [m], at considered filling height.

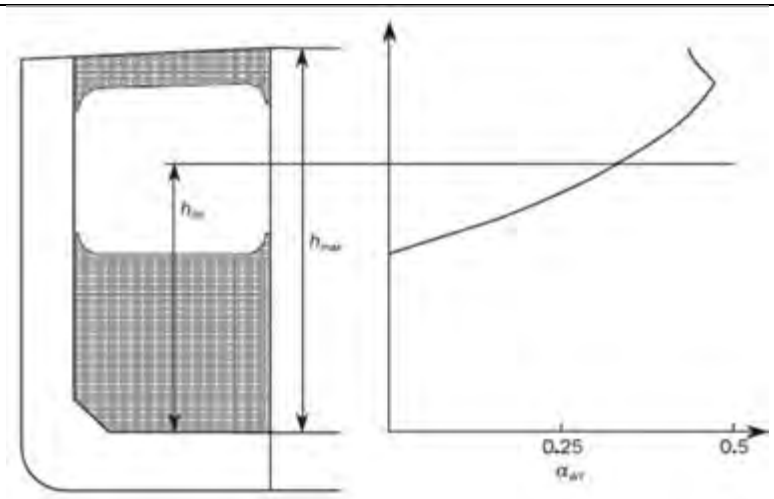


Figure 7.4.2.1: Transverse wash bulkhead coefficient

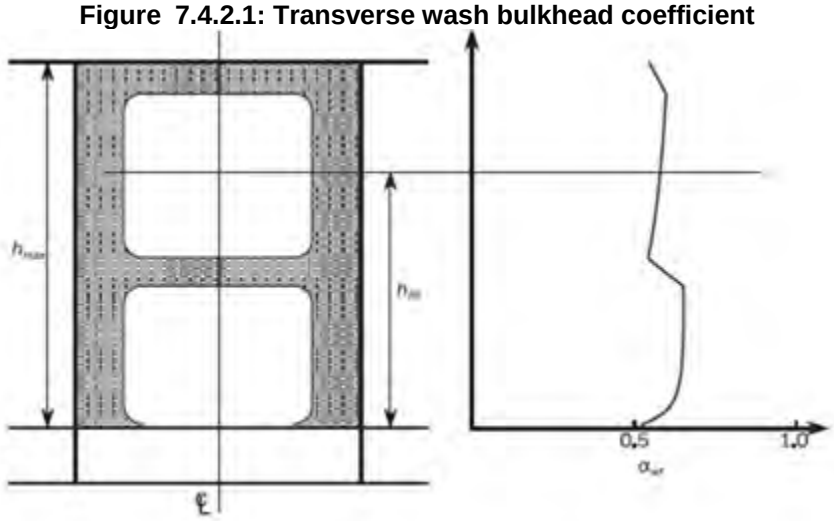


Figure 7.4.2.2: Transverse web frame coefficient

7.4.2.3 Sloshing pressure in way of transverse bulkheads

The sloshing pressure in way of transverse bulkheads including wash bulkheads due to longitudinal liquid motion, P_{slo-l} in [kN/m^2], for a particular filling level, is to be taken as:

$$P_{slo-l} = \rho_{slo} . g . \ell_{slo} . f_{slo} \left[0.4 - \left(0.39 - \frac{1.7 \ell_{slo}}{L} \right) \frac{L}{350} \right]$$

where:

f_{slo} = Coefficient taken as:

$$f_{slo} = 1 - 2 \left(0.7 - \frac{h_{fill}}{h_{max}} \right)^2$$

h_{fill} = Filling height, in [m], measured from the tank bottom, Refer Figure 7.4.2.1.

7.4.2.4 Sloshing pressure on internal web frames or transverse stringers adjacent to a transverse bulkhead

For tanks with internal web frames, the sloshing pressure acting on a web frame or transverse stringer adjacent to transverse bulkheads or transverse wash bulkheads due to longitudinal liquid motion, P_{slo-wf} , in [kN/m^2], provided it is located within $0.25\ell_{slo}$ from the bulkhead, is to be taken as:

$$P_{slo-wf} = P_{slo-l} \left(1 - \frac{S_{wf}}{\ell_{slo}} \right)^2$$

where:

S_{wf} = Distance from transverse bulkhead to web frame under consideration, in [m].

ℓ_{slo} = Effective sloshing length, in [m], as defined in Cl.7.4.2.2.

P_{slo-l} = Sloshing pressure due to longitudinal liquid motion acting on the transverse bulkhead, as defined in Cl.7.4.2.3.

The distribution of pressure across web frames and transverse stringers is given in Figure 7.4.2.3.

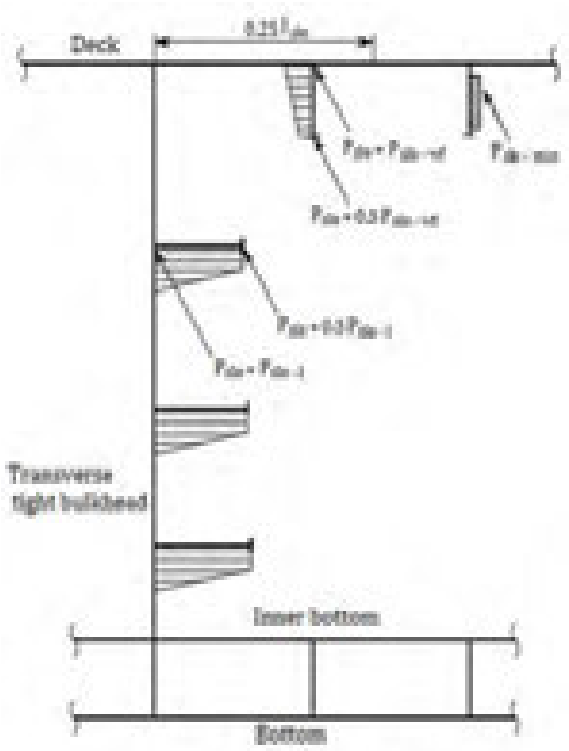


Figure 7.4.2.3: Sloshing pressure distribution on transverse stringers and web frames

7.4.3 Sloshing pressure due to transverse liquid motion

7.4.3.1 Application

The sloshing pressure due to transverse liquid motion, P_{slo-t}, is to be taken as a constant value over the full tank depth and is to be taken as the greater of the sloshing pressures calculated for filling levels from $0.05h_{max}$ to $0.95h_{max}$, and/or $0.05h_{max}$ to h_{part}, typically in $0.05h_{max}$ increments.

7.4.3.2 Effective sloshing breadth

The effective sloshing breadth, b_{slo} in [m], is to be taken as in Table 7.4.3, but not less than 0.3B.

Table 7.4.3: Effective sloshing breadth $[b_{slo}]$

Type of longitudinal bulkhead	b_{slo}
Longitudinal tight bulkheads	$b_{slo} = \frac{(1 + n_{WL}\alpha_{WL})(1 + f_{grd}\alpha_{grd})b_{tk-h}}{(1 + n_{WL})(1 + f_{grd})}$
Longitudinal wash bulkheads	$b_{slo} = \frac{[1 + (n_{WL} - 1)\alpha_{WL}](1 + f_{grd}\alpha_{grd})b_{tk-h}}{(1 + n_{WL})(1 + f_{grd})}$

where:

n_{WL} = Number of longitudinal wash bulkheads in the tank

α_{WL} = Longitudinal wash bulkhead coefficient

$$\alpha_{WL} = \frac{A_{OWL}}{A_{tk-L-h}}$$

α_{grd} = Girder coefficient, to be taken as

$$\alpha_{grd} = \frac{A_{O-grd-h}}{A_{tk-L-h}}$$

A_{OWL} = Total area of openings, in [m^2], in the longitudinal section in way of the wash bulkhead below the considered filling height.

A_{tk-L-h} = Total longitudinal cross-sectional area, in [m^2], of the tank below the considered filling height.

$A_{O-grd-h}$ = Total area of openings, in [m^2], in the longitudinal section in way of the web frame below the considered filling height.

f_{grd} = Factor to account for the number of longitudinal girders and longitudinal wash bulkheads in the tank, to be taken as:

$$f_{grd} = \frac{n_{grd}}{1 + n_{WL}}$$

n_{grd} = Number of longitudinal girders, excluding longitudinal wash bulkheads, in the tank.

b_{tk-h} = Breadth of the tank, in [m], at considered filling height.

7.4.3.3 Sloshing pressure in way of longitudinal bulkheads

The sloshing pressure in way of longitudinal bulkheads including wash bulkheads due to transverse liquid motion, P_{slo-t} in [kN/m^2], for a particular filling level, is to be taken as:

$$P_{slo-t} = 7.\rho_{slo}.g.f_{slo}\left(\frac{b_{slo}}{B} - 0.3\right)GM^{0.75}$$

where:

b_{slo} = Effective sloshing breadth defined in Cl.7.4.3.2.

GM = Metacentric height, in [m], in the considered loading condition.

For the calculation of sloshing pressure in ballast tanks the ‘normal ballast condition’ is to be used. For the calculations of sloshing pressure in cargo tanks, the ‘partial load condition’ is to be used.

f_{slo} = Coefficient as defined in Cl.7.4.3.3.

7.4.3.4 Sloshing pressure on internal girders or longitudinal stringers adjacent to longitudinal bulkheads

For tanks with internal girders or stringers, the sloshing pressure acting on the girder/web frame adjacent to longitudinal bulkheads and longitudinal wash bulkheads, $P_{slo-grd}$ in [kN/m^2], provided it is located within $0.25b_{slo}$ from the bulkhead, is to be taken as:

$$P_{slo-grd} = P_{slo-t}\left(1 - \frac{S_{grd}}{b_{slo}}\right)^2$$

where:

S_{grd} = Distance from longitudinal bulkhead to girder under consideration, in [m].

b_{slo} = Effective sloshing breadth, in [m], as defined in Cl.7.4.3.2.

P_{slo-t} = Sloshing pressure due to transverse liquid motion acting on the longitudinal bulkhead, as defined in Cl.7.4.3.3.

The distribution of pressure across stringers is given in Figure 7.4.3. The distribution of pressure across longitudinal girders is similar to the deck web frame shown in Figure 7.4.2.3.

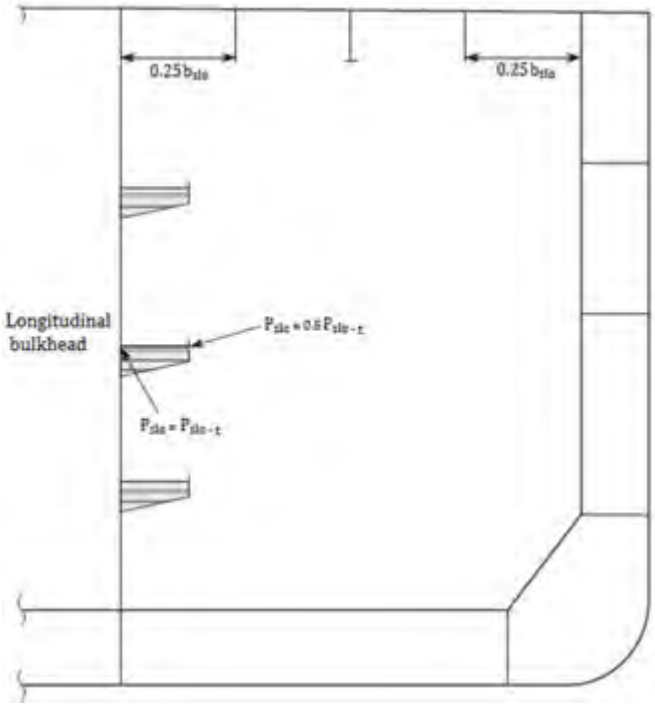


Figure 7.4.3: Sloshing pressure distribution on longitudinal stringers and girders

7.5 Scantling requirements

7.5.1 Plating

The net thickness of the plating, 't' in [mm], subjected to sloshing pressures is not to be less than:

$$t = 0.0158.\alpha_p.b\sqrt{\frac{P_{slo}}{c_a.R_{eH}}}$$

where:

$\mathbf{c_a}$ = Permissible bending stress coefficient for acceptance criteria AC-I as given in Table 7.5.1 but not to be taken greater than c_{a-max}.

$$c_a = \beta_a - \alpha_a \frac{|\sigma_{hg}|}{R_{eH}}$$

Table 7.5.1: Definition of β_a, α_a and $C_{a\text{-}max}$

Acceptance criteria	Structural member		β_a	α_a	$\mathbf{c_{a-max}}$
AC-I	Longitudinal member	Longitudinal stiffened plating	0.9	0.5	0.8
		Transverse stiffened plating	0.9	1.0	0.8
	Other members		0.8	0	0.8

σ_{hg} = Hull girder bending stress, in [N/mm^2], corresponding to the greatest of the sagging and hogging bending moment in absolute value.

P_{slo} = The greater of P_{slo-l}, P_{slo-t} as specified in Cl.7.4.2 and Cl.7.4.3.

For internal members like wash bulkheads the sloshing pressure is not to be taken less than $P_{slo-min}$ as given in 7.1.

7.5.2 Stiffeners

The net section modulus, Z in [cm^3], of stiffeners subjected to sloshing pressures is not to be less than:

$$Z = \frac{P_{slo}.s.\ell_{bdg}^2}{f_{bdg}.C_s.R_{eH}}$$

where:

f_{bdg} = Bending moment factor taken as:

For all continuous horizontal and vertical stiffeners, f_{bdg} is to be taken from Table 7.5.2.1.

For all non-continuous stiffeners, f_{bdg} is taken as:

= 8 For simply supported ends

= 10 For fixed at both ends

C_s = Permissible bending stress coefficient to be taken as:

For members subject to hull girder stress: coefficient for acceptance criteria AC-I as defined in Table 7.5.2.1.

For other cases, Cs = Cs-max.

β_s = Coefficient as defined in Table 7.5.2.2.

α_s = Coefficient as defined in Table 7.5.2.2.

$C_{s-max.}$= Maximum permissible bending stress coefficient as defined in Table 7.5.2.2.

Table 7.5.2.1: Permissible bending stress coefficient C_s

Sign of hull girder bending stress, σ_{hg} (1)	Lateral pressure acting on (2)	Stiffener boundary condition (3)	f_{bdg}	Coefficient C_s
Tension (positive)	Stiffener side	F - F	12	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$
		F - S	8	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$
		S - S	8	$c_s = c_{s-max}$
	Plate side	F - F	24	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than c_{s-max}
		F - S	14	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than c_{s-max}
		S - S	8	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$
Compression (negative)	Stiffener side	F - F	24	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$
		F - S	14	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$
		S - S	8	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$
	Plate side	F - F	12	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than $c_{s-max.}$

		F - S	8	$c_s = \beta_s - \alpha_s \frac{\lvert\sigma_{hg}\rvert}{R_{eH}}$ but not to be taken greater than c_{s-max}.
		S - S	8	$c_s = c_{s-max}$
(1) σ_{hg} is to be considered for hogging and sagging situations. (2) For primary supporting members located inside the considered tank and for wash bulkheads, the sloshing pressure is to be applied both on the stiffener and plate sides. (3) F - F stands for both ends of the stiffener fixed against rotation F - S stands for one end of the stiffener fixed and the other not fixed against rotation S - S stands for both ends of the stiffener not fixed against rotation				

Table 7.5.2.2: Definition of β_s, α_a and $C_{a\text{-}max}$

Acceptance criteria	Structural member		β_s	α_s	$C_{s\text{-}max}$
AC-I	Longitudinal member	Longitudinal stiffeners	0.9	0.5	0.8
		Transverse stiffeners	0.9	1.0	0.8
	Other members		0.8	0	0.8

7.5.3 Primary supporting members

7.5.3.1 Web plating

The net web plating thickness of primary supporting members, 't' in [mm], is not to be less than:

$$t = 0.0158.\alpha_p.b\sqrt{\frac{P_{slo}}{c_a.R_{eH}}}$$

where:

c_a = Permissible plate bending stress coefficient, as given in Cl.7.5.1.

P_{slo} = The greater of P_{slo-l}, P_{slo-t}, P_{slo-wf}, $P_{slo-grd}$ as specified in 8.4 and $P_{slo-min}$ as given in 7.1.

The pressure is to be calculated at the load application point, defined in Chapter 3, Section 7, 7.7, considering the distribution over the height of the member, as shown in Figure 7.3.3 and Figure 7.3.4.

7.5.3.2 Stiffeners on web plating

The net section modulus, Z in [cm^3], of each individual stiffener on the web plating of primary supporting members subjected to sloshing pressures is not to be less than:

$$Z = \frac{P_{slo}.s.\ell_{bdg}^2}{f_{bdg}.C_s.R_{eH}}$$

where:

P_{slo} = The greater of P_{slo-l}, P_{slo-t}, P_{slo-wf}, $P_{slo-grd}$ as specified in 7.4 and $P_{slo-min}$ as given in 7.1.

The pressure is to be calculated at the load application point, defined in Chapter 3, Section 7, 7.6, considering the distribution over the height of the member, as shown in Figure 7.3.3 and Figure 7.3.4.

c_s = Permissible plate bending stress coefficient, as given in Cl.7.5.2.

f_{bdg} = Bending moment factor as given in Cl.7.5.2.

7.5.4 Tripping brackets supporting primary supporting members

The net section modulus, Z in [cm^3], in way of the base within the effective length, d, of tripping brackets and net shear area, A_{shr} in [cm^2], after deduction of cut-outs and slots, of tripping brackets supporting primary supporting members is not to be less than:

$$Z = \frac{1000\, P_{slo}.\, s_{trip}.\, h^2}{2.\, C_s.\, R_{eH}}$$

$$A_{shr} = \frac{\sqrt{3}.\, 10.\, P_{slo}.\, s_{trip}.\, h}{C_t.\, R_{eH}}$$

where:

P_{slo} = The greater of P_{slo-l} , P_{slo-t} , P_{slo-wf} , $P_{slo-grd}$ as specified in 7.4. The average pressure may be calculated at the midpoint of the tripping bracket taking into account the distribution as shown in Figure 7.3.3 and Figure 7.3.4. The sloshing pressure is not being taken less than $P_{slo-min}$ as given in 7.1.

$\mathbf{s_{trip}}$ = Mean spacing, between tripping brackets or other primary supporting members or bulkheads, in [m].

h = Height of tripping bracket, in [m]. Refer Figure 7.5.4

C_s = Permissible bending stress coefficient for tripping brackets to be taken as 0.75.

C_t = Permissible shear stress coefficient for tripping brackets to be taken as 0.75.

The effective breadth of the attached plate to be used for calculating the section modulus of the tripping bracket is to be taken as h/3.

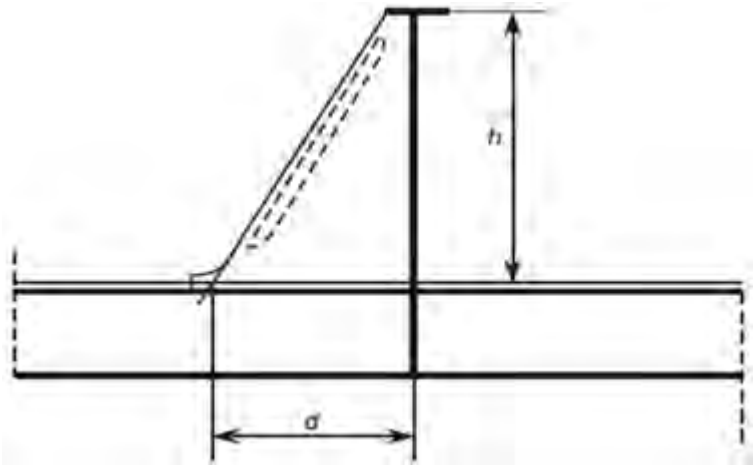


Figure 7.5.4: Effective length and height of tripping bracket

End of Chapter

Chapter 12

Opening and Closing Appliances

Contents

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Section 1

General

1.1 Application

1.1.1 The requirement of this Chapter are applicable to all ship types in general. Additional requirements pertaining to special ship types are given in Part 5 of the Rules.

1.1.2 The requirements of this Chapter apply to the arrangement of openings and closing appliances in hull, deckhouse and superstructure. In general, the closing appliances are to have the same strength or scantling corresponding to the part of the hull in which they are fitted.

1.1.3 The requirements conform, where applicable, to those of the International Convention on Load Lines, 1966 (ICLL, 1966) as amended by the 1988 protocol. Reference should also be made to the additional requirements of the National Authority of the country in which the ship is to be registered.

1.1.4 The requirements given in 2.3 apply to small hatch covers fitted on exposed decks.

1.1.5 The requirements given in 2.4 apply to small hatch covers fitted on the exposed fore deck over the forward 0.25L, for ships equal to or greater than 80[m] in length, where the height of the exposed deck in way of the hatch is less than 0.1L or 22[m] above the summer load waterline, whichever is the lesser.

1.1.6 The requirements given in 2.5 apply to small hatch covers fitted on non-exposed decks.

1.2 Definition

1.2.1 Superstructure deck is a deck forming the upper boundary of a superstructure. Where the summer freeboard is increased such that the resulting draught is not more than that corresponding to the minimum summer freeboard for the same ship from an assumed freeboard deck located at a distance equal to or greater than the standard superstructure height below the actual freeboard deck. The actual freeboard deck may be treated as the superstructure deck, while considering the requirements given in this chapter.

1.2.2 The requirements of hatches, doors and ventilators on the weather decks are given with respect to Position 1 and Position 2 as defined below in Cl.1.2.4.

1.2.3 Small hatches are hatches designed for access to spaces below the deck and are capable of being closed weathertight or watertight, as applicable. Their opening is generally equal to or less than 2.5[m^2]. (Refer Section 2).

1.2.4 For the purpose of Load Line assignment, two positions of hatchways, doorways and ventilators are defined as follows (Refer Figure 1.2.1 for Position 1 and Position 2):

(a) Position 1 includes:

- exposed freeboard and raised quarter decks for their entire length

- exposed superstructure decks situated forward of 0.25 L_L from FP_{LL}.

(b) Position 2 includes:

- exposed superstructure decks situated aft of $0.25L_L$ from FP_{LL} and located atleast one standard height of superstructure above the freeboard deck.

- exposed superstructure decks situated forward of $0.25L_L$ from FP_{LL} and located at least two standard heights of superstructure above the freeboard deck.

Note: The upper limit of the vertical extent of Position 2 need not be considered to exceed two standard superstructure heights at locations abaft the $0.25L_L$ from FP_{LL} and three standard superstructure heights forward of $0.25L_L$ from the FP_{LL}.

1.2.5 For the application of Rule requirements, the following specific ship types are categorized as:

(a) Type-1 ship: includes all ships except bulk carriers, self-unloading bulk carriers, ore carriers and combination carriers, as defined in Part 5, Chapter 1.

(b) Type-2 ship: includes all bulk carriers, self-unloading bulk carriers, ore carriers and combination carriers, as defined in Part 5, Chapter 1.

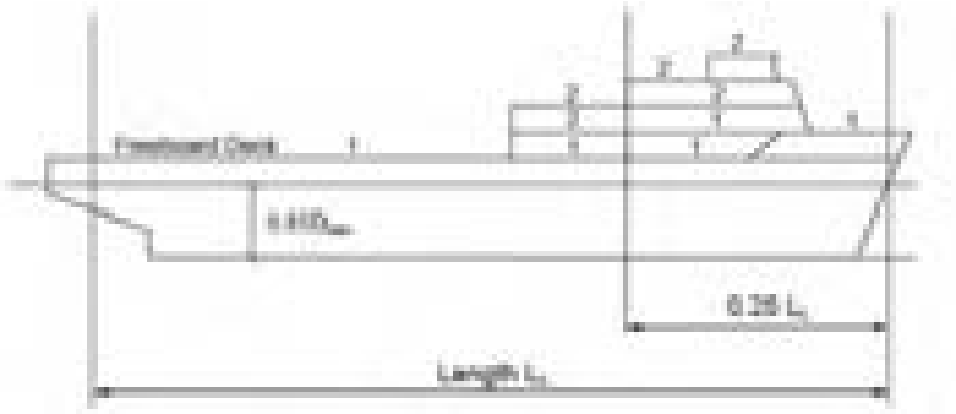


Figure 1.2.1: Position 1 and 2 for general ships

Section 2

Access openings in Freeboard deck and other Exposed decks

2.1 Access openings in Freeboard and Superstructure decks

2.1.1 Manholes and flush scuttles

Manholes and flush scuttles on the decks fitted in position 1 or 2 or within superstructures other than enclosed superstructures are to be closed by substantial covers capable of closing them watertight. Unless secured by closely spaced bolts at a pitch not exceeding 5d, where d is the diameter of bolts, the covers are to be permanently attached.

2.1.2 Companion ways, doors and accesses on weather decks

2.1.2.1 Openings in freeboard decks other than hatchways, machinery space openings, manholes and flush scuttles are to be protected by an enclosed superstructure, or by a deckhouse or companionway of equivalent strength and weather tightness to an enclosed superstructure, effectively secured to deck.

Any such openings:

a) in an exposed superstructure deck, or

b) on top of a deckhouse on the freeboard deck and giving access to space below freeboard deck, or

c) in an exposed position on top of a deckhouse on the enclosed superstructure deck and giving access to space within that enclosed superstructure, is to be protected by an efficient deckhouse or companionway as above.

2.1.2.2 Doorways in such companionways or deckhouses that lead or give access to stairways leading below, need to be fitted with doors complying with the requirements given in Cl.2.2.1.1. Alternatively, if stairways within a deckhouse are enclosed within properly constructed companionways fitted with doors complying with requirements given in Cl.2.2.1.1, the external door need not be weathertight.

2.1.2.3 Regarding the requirement to protect openings in superstructures as described in Cl.2.2.1.1, it is considered that openings in the top of a deckhouse on a raised quarterdeck, or on the deck of a superstructure or on the deck of a deckhouse of less than standard height, having a height equal to or greater than the standard quarterdeck height are to be provided with an acceptable means of closing but need not be protected by an efficient deckhouse or companionway provided the height of the deckhouse is at least the height of superstructure.

The standard height of superstructures is mentioned in Chapter 1 section 2, Table 2.2.5.

2.1.3 Access hatches within enclosed superstructure and tween decks

The requirements of 2.3 are to be complied with as and where applicable.

2.2 Weathertight doors

2.2.1 General

2.2.1.1 All access openings in bulkheads at ends of enclosed superstructures is to be fitted with doors of steel or other equivalent material, permanently and strongly attached to the bulkhead, and framed, stiffened and fitted so that the whole structure is of equivalent strength to the unpierced bulkhead and weathertight when closed. The means for securing these doors weathertight consists of gaskets and clamping devices or other equivalent means and to be permanently attached to the bulkhead or to the doors themselves, and the doors are to be capable of being operated and secured from both sides of the bulkhead. Any fixed lights in the doors are to be provided with hinged steel deadlights fitted internally for maintaining weathertightness.

2.2.1.2 Except pilot doors, which open inwards, weather tight doors are to generally open outwards to provide additional security against the impact of the sea. Doors which open inwards are to be specially approved.

2.2.1.3 Access hatches within a superstructure or deckhouse in Position 1 or 2 need not be provided with means for closing if all openings in the surrounding bulkheads have weather tight closing appliances.

2.2.1.4 Weathertight doors as specified above are to be fitted in all access openings in:

— bulkheads at ends of superstructures

— bulkheads of deckhouses on superstructure deck protecting openings in the superstructure deck

— bulkheads of deckhouses/companionways on freeboard deck protecting openings leading to spaces below the freeboard deck

— bulkheads of deckhouses/companionways protecting openings which give access to spaces within enclosed superstructures.

— companionways and bulkheads of deckhouse upon another deckhouse protecting an opening leading to spaces below the freeboard deck.

— companionways and bulkheads of deckhouse upon another deckhouse which give access to spaces within enclosed superstructures.

2.2.1.5 Weathertight doors in position 1 and 2 are in general to be in accordance with ISO Standards or equivalent.

2.2.2 Sill heights and arrangement

2.2.2.1 The height of doorway sills in companion ways above the deck sheathing, if fitted, is to be at least

— 600[mm] in Position 1

— 380[mm] in Position 2.

2.2.2.2 In the application of Cl.2.1.2.1 and Cl.2.1.2.2 it is understood that:

(i) Where access is provided to the deckhouse, bridge or a poop deck from the deck above as an alternative to the access from the freeboard deck, the height of sill into the deckhouse, bridge or poop can be 380[mm].

(ii) Where access is not provided from above, the height of the sills to doorways in a poop bridge or deckhouse on the freeboard deck should be 600[mm].

(iii) When the closing appliances of openings in superstructures and deckhouses do not comply with the requirements of Cl.2.2.1.1, interior deck openings are to be treated as exposed to the weather deck.

2.2.2.3 The height of sills to doorways in a forecastle if protecting a companionway is to be 600[mm], regardless of whether access is provided from above. If not protecting a companionway the sill height may be 380[mm].

2.2.2.4 Portable sills to be avoided. However, in order to facilitate the loading/unloading of heavy spare parts or similar, portable sills may be fitted on the following conditions:

— They must be installed before the ship leaves port

— Sills are to be gasketed and fastened by closely spaced through bolts.

— Whenever the sills are replaced after removal, the weathertightness of the sills and the related doors must be verified by hose testing. The dates of removal, replacing and hose testing to be recorded in the ship's log book.

2.2.2.5 The height of door sills may be required to be increased on ships of Type 'A', Type 'B-100' or Type 'B-60' where this is shown to be necessary by the floatability calculations required by the ICLL 1966 as amended by the Protocol of 1988. The engine casing in such ships, when not protected by an outer structure will require two weathertight doors in series, with the sill height of the outer and inner door being not less than 600[mm] and 230[mm] respectively.

2.2.2.6 In locations other than those mentioned in 2.2.1.4, doors may be of 50[mm] thick hardwood or equivalent material and strength subject to any requirement of fire protection.

2.2.2.7 Where permitted by the National Authority, reduced companionway sill heights may be accepted on ships engaged on sheltered water service or restricted water service.

2.3 Small hatches on exposed or weather decks

2.3.1 The number and size of access openings are to be kept to the minimum, consistent with the satisfactory operation of the ship.

2.3.2 The height of the coamings for hatches closed by weathertight covers is to be not less than

– 600[mm] in Position 1

– 450[mm] in Position 2

Coaming heights are to be measured from the upper surface of the deck or top of the deck sheathing if fitted.

2.3.3 Lower heights may be considered in relation to operational requirements and the nature of the spaces to which access is given. The height, given in Cl.2.3.2, of hatch coamings closed by steel covers provided with gaskets and securing devices may be reduced or the coamings may be omitted entirely, on condition that the flag administration is satisfied that the safety of the ship is not thereby impaired in any sea conditions. In such cases the scantlings of the covers, their gasketing, their securing arrangements and the drainage of recesses in the deck are considered on a case-by-case basis.

2.3.4 The minimum thickness of the coamings is to be not less than the rule minimum thickness for the deck inside line of openings for that position, or 11[mm], whichever is the lesser. Stiffening of the coaming is to be appropriate to its length and height.

2.3.5 Hatch covers are to be of steel, weathertight and hinged in general. Weathertightness is to be maintained in all sea conditions. Hinges are not to be used as securing devices.

Where toggles are fitted, their diameter and spacing are to be in accordance with or equivalent to, recognized national or international standards.

2.3.6 The thickness of the covers is to be not less than the minimum thickness of the deck, inside the line of opening at that location or (0.01s) where 's' is the spacing of the stiffeners on hatch covers in [mm]. The thickness is not to be less than 6[mm]. The hatch covers are to be adequately stiffened. Stiffeners is to fulfill the requirements in Table 2.4.2.

2.3.7 For hatch covers constructed of materials other than steel, the scantlings are to provide equivalent strength.

2.3.8 Small hatches, including escape hatches, are to be situated clear of cargo containment areas, particularly in the case of offshore supply vessels.

2.3.9 Rope hatches may be accepted with reduced coamings, but generally not less than 380[mm], provided they are well secured and closed before the ship leaves port. A suitable notice is to be displayed at the hatch stating that it is to be closed whilst the ship is at sea.

2.3.10 Where portable plates are required in decks for unshipping machinery, they may be accepted provided they are of equivalent strength to the unpierced deck and are secured by gaskets and closely spaced bolts at a pitch not exceeding 5d, where d is the diameter of bolts. Such portable plates need not be provided with coaming.

2.3.11 Where permitted by the National Authority, access hatch coaming heights may be reduced on ships engaged in sheltered water service or restricted water service. Coaming heights are to be as high as practicable with a minimum height of 230[mm].

2.3.12 Emergency hatch opening

To facilitate a swift and safe means of escape to the lifeboat and life raft embarkation deck, the following provisions apply to overhead hatches fitted along the escape routes addressed by SOLAS Regulation 13 - Means of escape:

– escape hatches and their securing devices are to be capable of being operated and opened from either sides.

– the maximum force needed to open the hatch cover should not exceed 150[N]; and

– the use of a spring equalizing, counterbalance or other suitable device on the hinge side to reduce the force needed for opening is acceptable.

2.3.13 In addition to the requirements, in Cl.2.3.1 to Cl.2.3.12, small hatches on the exposed foredeck of vessels with L≥80[m] are to meet the requirements of 2.4.

2.4 Small hatches fitted on the exposed fore deck

2.4.1 Application

2.4.1.1 These requirements apply to small hatches fitted on the exposed fore deck over the forward 0.25L on ships of L≥80[m], where the height of exposed deck above the summer load waterline in way of the hatch is less than 0.1L or 22[m], whichever is less.

2.4.1.2 For definition of small hatches, Refer Section 1, Cl.1.2.3.

2.4.1.3 Hatches designed for emergency escape need not comply with the requirements given in Cl.2.4.4.2 and Cl.2.4.5.

2.4.1.4 Securing devices of hatches designed for emergency escape are to be of a quick-acting type (e.g. one action wheel handles are provided as central locking devices for latching/unlatching of hatch cover) operable from both sides of the hatch cover. Butterfly nuts and quick acting cleats are not to be used for these hatches.

2.4.1.5 In general, requirements of this subsection do not apply to small hatches on container ships giving access to a cargo hold, which comply with Pt 5, Ch.5, 3.2 for non-weathertight hatch covers. However, the requirements in 2.4.2 for strength and 2.4.3 for primary securing devices, excluding weathertightness, will apply to such small hatch covers. Such hatch covers are considered non-weathertight regardless of whether they are weathertight or not. Requirements for strength in Pt.5, Ch.5, 3.2.7 need not be applied for such small hatch covers.

2.4.2 Strength

2.4.2.1 For small rectangular steel hatch covers, the gross plate thickness, stiffener arrangement and scantlings in [mm], are to be in accordance with Table 2.4.2 and Figure 2.4.2.1. Stiffeners, where fitted, are to be aligned with the metal-to-metal contact points, required in Cl.2.4.4.1 and shown in Figure 2.4.2.1.

2.4.2.2 The primary stiffeners (shown by thick lines) are to be continuous. All stiffeners are to be welded to the inner edge stiffener indicated in Figure 2.4.2.2.

2.4.2.3 The upper edge of the hatchway coaming is to be suitably reinforced by a horizontal section, normally not more than 170 to 190[mm] from the upper edge of the coamings.

2.4.2.4 For small hatch covers of circular or similar shape, the cover plate thickness and reinforcement are to provide strength and stiffness equivalent to the requirements for small rectangular hatches.

2.4.2.5 For small hatch covers constructed of materials other than steel, the required scantlings are to provide equivalent strength and stiffness.

2.4.3 Primary securing devices

2.4.3.1 Small hatches located on exposed fore deck are to be fitted with primary securing devices such that their hatch covers can be secured in place and weathertight by means of a mechanism employing any one of the following methods:

(i) Butterfly nuts tightening onto forks (clamps),

(ii) Quick acting cleats, or

(iii) Central locking device.

2.4.3.2 Dogs (twist tightening handles) with wedges are not acceptable.

2.4.4 Requirement to primary securing

2.4.4.1 The hatch cover is to be fitted with a gasket of elastic material. This is to be designed to allow a metal-to-metal contact at a designed compression and to prevent over compression of the gasket by green sea forces that may cause the securing devices to be loosened or dislodged. The metal-to-metal contacts are to be arranged close to each securing device in accordance with Figure 2.4.2.1 and of sufficient capacity to withstand the bearing force.

2.4.4.2 For a primary securing method using butterfly nuts, the forks (clamps) are to be of robust design. They are to be designed to minimize the risk of butterfly nuts being dislodged while in use; by means of curving the forks upward, a raised surface on the free end, or a similar method. The plate thickness of unstiffened steel forks is not to be less than

16[mm]. An example arrangement is shown in Figure 2.4.2.2.

2.4.4.3 The primary securing method is to be designed and manufactured such that the designed compression pressure is achieved by one person without the need of any tools.

2.4.4.4 For small hatch covers located on the exposed deck forward of the fore-most cargo hatch, the hinges are to be fitted such that the predominant direction of green sea will cause the cover to close, which means that the hinges are normally to be located on the fore edge.

2.4.4.5 On small hatches located between the main hatches, for example between Nos. 1 and 2, the hinges are to be placed on the fore edge or outboard edge, whichever is practicable for protection from green water in beam sea and bow quartering conditions.

2.4.5 Secondary securing device

Small hatches on the fore deck except those designed for emergency escapes are to be fitted with an independent secondary securing device e.g. by means of a sliding bolt, a hasp or a backing bar of slack fit, which is capable of keeping the hatch cover in place, even in the event that the primary securing device became loosened or dislodged. It is to be fitted on the side opposite to the hatch cover hinges.

Table 2.4.2: Gross scantlings for small steel hatch covers on the fore deck

Nominal size [mm x mm]	**Cover plate thickness [mm]**	**Primary stiffeners**	**Secondary stiffeners**
		Flat bar [mm x mm]; number	
630 x 630	8	--	--
630 x 830	8	100 x 8; 1	--
830 x 630	8	100 x 8; 1	--
830 x 830	8	100 x 10; 1	--
1030 x 1030	8	120 x 12; 1	80 x 8; 2
1330 x 1330	8	150 x 12; 2	100 x10; 2

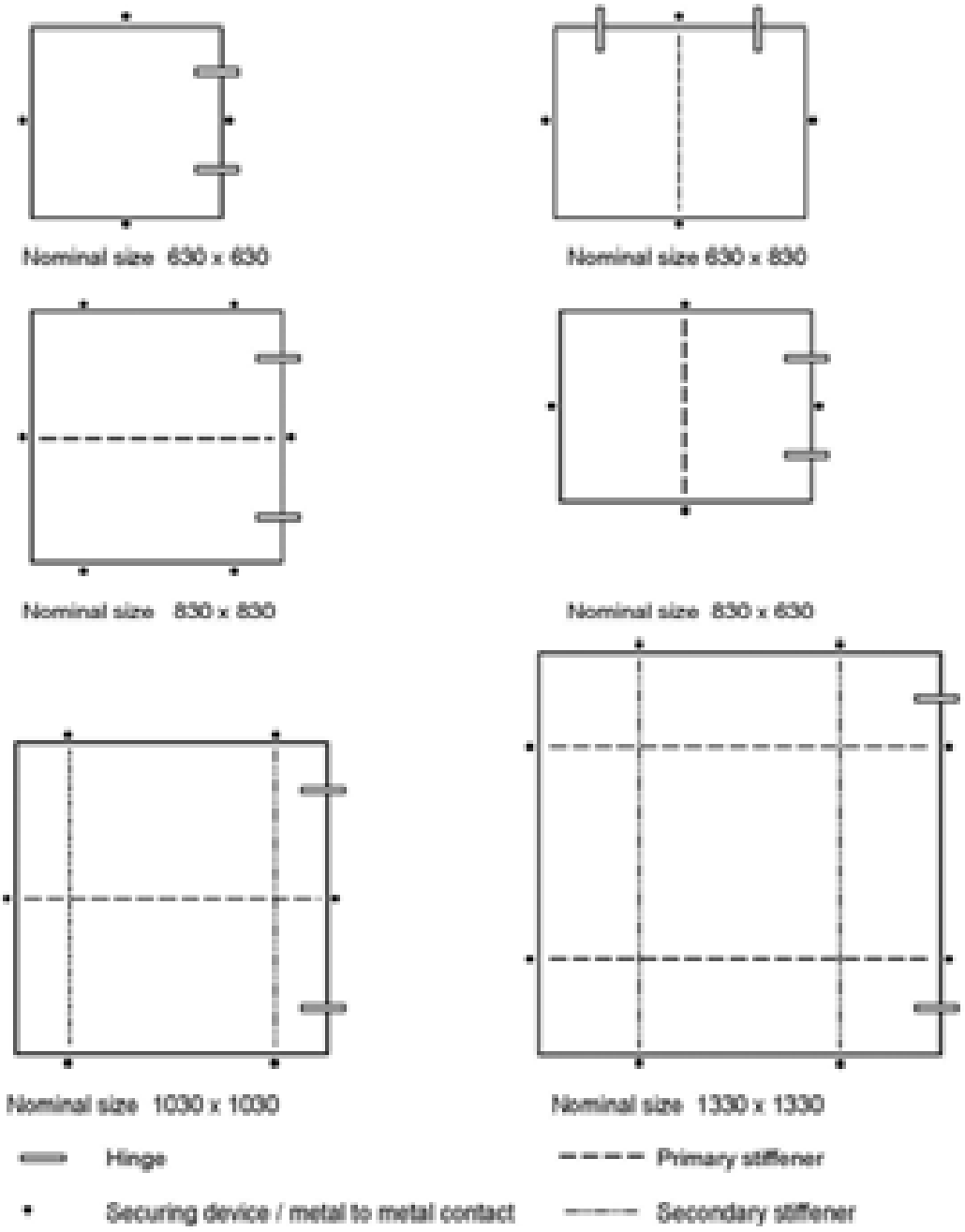


Figure 2.4.2.1: Arrangement of stiffeners

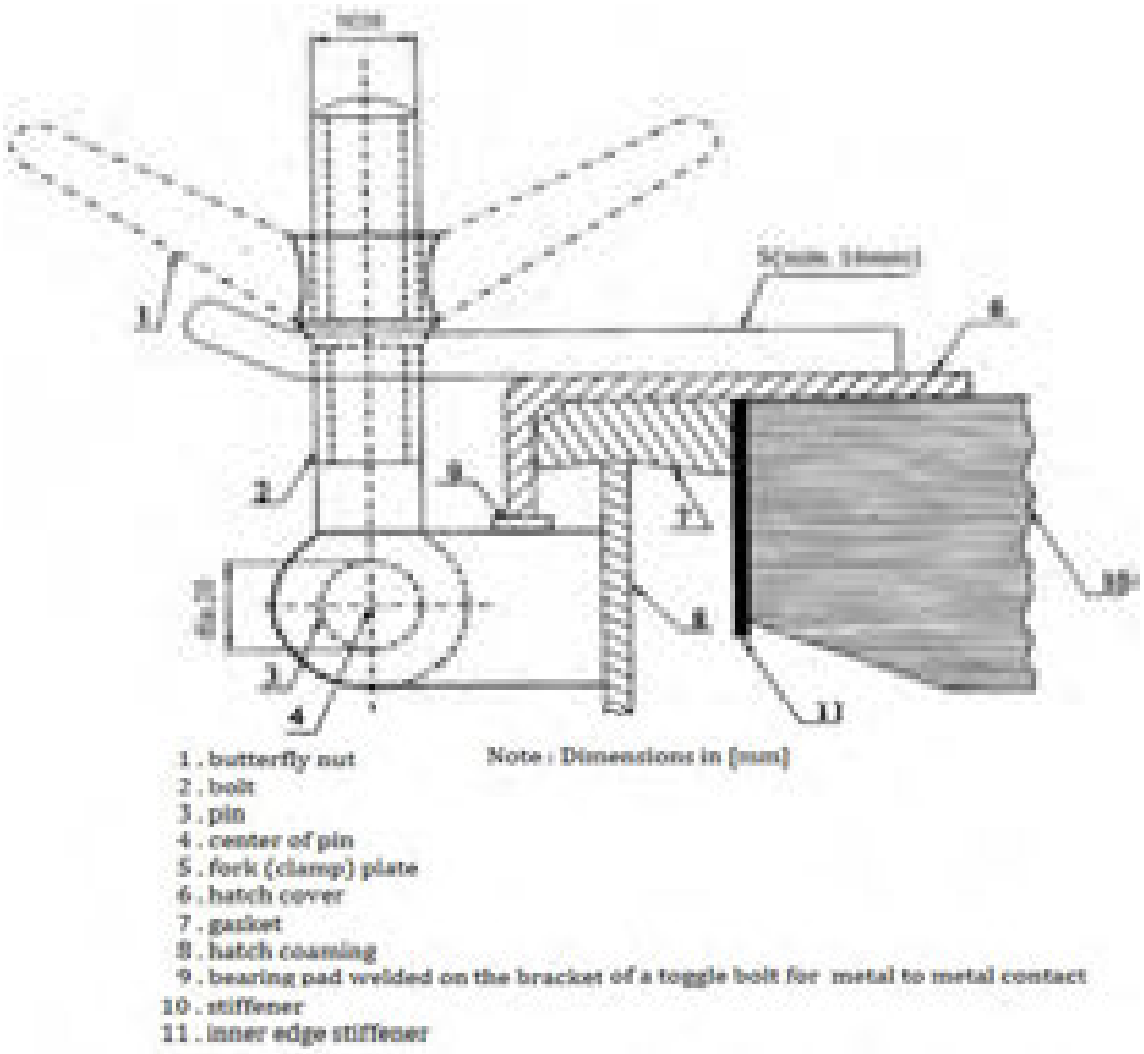


Figure 2.4.2.2: Example or primary securing device

2.5 Small hatch covers fitted on non-exposed decks

2.5.1 Hatch covers on non-exposed decks need not be weathertight.

2.5.2 Small hatch covers fitted on non-exposed decks are to have strength equivalent to that required for the adjacent deck plating.

2.5.3 Small hatch covers fitted on non-exposed decks are to have a level of tightness equivalent to that required for adjacent compartment(s).

2.5.4 If the hatch cover is weathertight or watertight and non-bolted, metal-to-metal contacts are to be provided in order to prevent over compression of the gasket. They are to be designed in order to withstand the bearing force induced by any lateral load on the hatch cover.

Section 3

Machinery Space Openings

3.1 Arrangements

3.1.1 Openings

3.1.1.1 Machinery space openings in position 1 or 2 are to be properly framed and efficiently enclosed by steel casings of ample strength, and where the casings are not protected by other structures their strength are to be specially considered. Access openings in such casings to be fitted with doors complying with the requirements of Section 2.2, Cl.2.2.1.1, the sills to be at least 600[mm] above the deck in position 1, and at least 380[mm] above the deck in position 2. Other openings in such casings are to be fitted with equivalent covers, permanently attached in their proper positions.

3.1.1.2 Machinery space ventilator coamings in an exposed position are to be in accordance with Section 13, Cl.13.5.

3.1.1.3 Machinery space openings in Position 1 or 2 are to have efficient closing appliances. The openings and coamings for fiddley, funnel and machinery space ventilators in the casing in those positions are to be provided with strong covers of steel or other equivalent material permanently attached in their proper positions and capable of being secured weathertight.

3.1.1.4 Skylights are to be of substantial construction and secured firmly to the deck. For skylights in position 1 or 2 the coaming height is not to be less than that of the hatch coaming. Efficient means are to be provided for closing and securing the hinged scuttles, if any. The thickness of glasses in fixed or opening skylights is to be appropriate to their position and size as required for side scuttles. Glasses in any position are to be protected against mechanical damage, and where fitted in Positions 1 and 2 are to be fitted with deadlights or storm covers permanently attached.

3.1.1.5 Side scuttles/skylights in the engine casings are to be provided with fireproof glass.

3.1.2 Special requirements for type 'A' ships

3.1.2.1 Machinery casings on type 'A' ships are to be protected by one of the following arrangements:

- an enclosed poop or bridge of at least standard height; or

- a deckhouse of equal height and equivalent strength.

3.1.2.2 Machinery casings may, however, be exposed if there are no openings giving direct access from the freeboard deck to the machinery space. A door complying with the requirements of Section 2, 2.2 is acceptable in the machinery casing, provided that it leads to a space or passageway which is as strongly constructed as the casing and is separated from the stairway to the engine-room by a second weathertight door or steel or other equivalent material.

Section 4

Windows, Side scuttles and Skylights

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

4.1 Application and general requirements

4.1.1 The requirements given in this section apply to side scuttles, windows and skylights located on the shell, external bulkheads of superstructures and deckhouse and weather decks of ship structures which are exposed to the action of sea and/or rough weather.

4.1.2 Side scuttles and windows together with their glasses, deadlights and storm covers, if fitted, are to be of an approved design and substantial construction in accordance with, or equivalent to, recognized national or international standards. Non-metallic frames are not acceptable.

The minimum design pressures to which windows and side scuttles are to be designed are to be in accordance with applicable requirements of Chapter 7, Section 8 (Superstructure, Deckhouse and Companionways).

4.1.3 'Deadlights', in accordance with recognized standards, are fitted to the inside of windows and side scuttles while 'storm covers', of comparable specifications to deadlights, are fitted to the outside of windows, where accessible, and may be hinged or portable.

4.1.4 The number of openings in the shell plating is to be reduced to the minimum compatible with the design and proper working of the ship.

4.1.5 A plan showing the location of side scuttles and windows is to be submitted.

4.2 Definitions

4.2.1 Side scuttles are round or oval openings with an area not exceeding 0.16[m^2]. Round or oval openings having areas exceeding 0.16[m^2] are to be treated as windows.

4.2.2 Windows are rectangular openings generally, having a radius at each corner depending on the window size in accordance with recognized national or international standards, and also those associated with round or oval openings with an area exceeding 0.16[m^2].

4.3 Arrangement and positioning

4.3.1 Side scuttles to the following spaces are to be fitted with efficient hinged inside deadlights:

– spaces below freeboard deck

– spaces within the first tier of enclosed superstructures

– first tier deckhouses on the freeboard deck protecting openings leading below

– first tier deckhouses on the freeboard deck considered buoyant in stability calculations.

4.3.2 The deadlights are to be capable of being effectively closed and secured watertight if fitted below the freeboard deck and weathertight if fitted above.

4.3.3 Side scuttles are not to be fitted in such a position that their sills are below a line drawn parallel to the freeboard deck at side and having its lowest point 0.025B or 500[mm], whichever is the greatest distance, above the summer load line (or timber summer load line if assigned).

4.3.4 The sills of all side scuttles which are below the bulkhead deck of passenger ships and the freeboard deck of cargo ships, as permitted in Cl.4.3.3, they are to be of such construction as to prevent effectively any person opening them without the consent of the master of the ship.

4.3.5 Efficient hinged inside deadlights, so arranged that they can be easily and effectively closed and secured watertight, are to be fitted to all side scuttles except that, abaft one eighth

of the ship's length from the forward perpendicular and above a line drawn parallel to the bulkhead deck at side and having its lowest point at a height of (3.7+0.025B)[m] above the deepest subdivision summer load line, the deadlights may be portable in passenger accommodation, unless the deadlights are required by the International Convention on Load Lines in force to be permanently attached in their proper positions. Such portable deadlights are to be stowed adjacent to the side scuttles they serve.

4.3.6 No side scuttles are to be fitted in any spaces which are appropriated exclusively to the carriage of cargo.

4.3.7 Side scuttles may, however, be fitted in spaces appropriated alternatively to the carriage of cargo or passengers, but they are to be of such construction as to prevent effectively any person opening them without the consent of the master of the ship.

4.3.8 Side scuttles are to be of the non-opening type in ships subject to damage stability regulations, if the calculations indicate that they would become immersed by any intermediate stage of flooding or the final equilibrium waterplane in any required damage case.

4.3.9 Windows are not to be fitted in the following locations:

- below the freeboard deck

- in the first-tier end bulkheads or sides of enclosed superstructures

- in first tier deckhouses that are considered buoyant in the stability calculations or protecting openings leading below.

4.3.10 Side scuttles and windows at the side shell in the second tier, protecting direct access below or considered buoyant in the stability calculations, are to be provided with efficient hinged inside deadlights capable of being effectively closed and secured weathertight.

4.3.11 Side scuttles and windows in side bulkheads set inboard from the side shell in the second tier, protecting direct access below to spaces listed in Cl.4.3.1, are to be provided with either efficient hinged inside deadlights or, where they are accessible, permanently attached external storm covers of approved design and of substantial construction and capable of being effectively closed and secured weathertight.

4.3.12 Cabin bulkheads and doors in the second tier and above separating side scuttles and windows from a direct access leading below or the second tier considered buoyant in the stability calculations, may be accepted in place of deadlights or storm covers fitted to the side scuttles and windows.

IR4.3.12 Where an opening in a superstructure deck or in the top of a deckhouse on the freeboard deck which gives access to a space below the freeboard deck or to a space within an enclosed superstructure is protected by a deckhouse, then it is considered that only those side scuttles fitted in spaces which give direct access to an open stairway need be fitted with deadlights. A cabin is considered to provide adequate protection against the minimal account of water which will enter through a broken side scuttle glass fitted on the second tier.

4.3.13 Deckhouses situated on a raised quarter deck or on the deck of a superstructure of less than standard height may be regarded as being in the second tier as far as the provision of deadlights is concerned, provided the height of the raised quarter deck or superstructure is equal to, or greater than, the standard quarter deck height.

4.3.14 Side scuttles and windows facing open or enclosed lifeboat and liferaft embarkation areas and below such areas in such a position that their failure during a fire would impede the launching of or embarkation into lifeboats or liferafts are to have adequate fire integrity.

4.3.15 Fixed or opening skylights are to have glass thickness appropriate to their size and position as required for side scuttles and windows. Skylight glasses in any position are to be protected from mechanical damage and where fitted in positions 1 or 2, are to be provided with permanently attached deadlights or storm covers.

4.3.16 Windows are not to be fitted in machinery space boundaries. However, this does not preclude the use of glass in control rooms within the machinery space.

4.3.17 Where windows are permitted in an exposed bulkhead on the weather deck in the forward $0.25L_L$, strong external storm covers

which may be portable and stored adjacent are to be provided.

4.3.18 If necessary, for practical considerations, the storm covers may be in two parts.

Section 5

Hatch Covers and Hatch Coamings

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

x = Longitudinal co-ordinate of midpoint of assessed structural member measured from aft end of length L or L_L, as applicable.

5.1 Application

5.1.1 The requirement of this section applies to all ship types in general and is for all cargo hatch covers and coamings on exposed decks. Additional requirements pertaining to special ship types are given in Part 5.

5.1.2 The requirements conform, where applicable, to those of the International Convention on Load Lines, 1966 (ILLC, 1966) as amended by the 1988 protocol, as amended in 2003. Reference should also be made to the additional requirements of the National Authority of the country in which the ship is to be registered.

5.1.3 The strength requirements are applicable to hatch covers and hatch coamings of stiffened plate construction and their closing arrangements.

5.1.4 This section is applicable to hatch covers and hatch coamings made of steel. Use of alternative materials will be specially considered.

5.1.5 This section does not apply to portable covers secured weathertight by tarpaulins and battening devices, or pontoon covers, as defined in ICLL Regulation 15.

5.2 Hatch cover definitions

5.2.1 Single skin cover

A hatch cover made of steel or equivalent material that is designed to comply with ICLL Regulation 16. The cover has continuous top and side plating, but is open underneath with the stiffening structure exposed. The cover is weathertight and fitted with gaskets and clamping devices unless such fittings are specifically excluded.

5.2.2 Double skin cover

A hatch cover as above but with continuous bottom plating such that all the stiffening structure and internals are protected from the environment.

5.3 Hatch cover types

5.3.1 The type of hatch covers a) to e) as described below may be used as per Table 5.3.1 on the weathertight decks of various ship types mentioned in ICLL 1966 as amended by the 1988 protocol, as amended in 2003.

Table 5.3.1: Applicability of hatch cover type					
Types of hatch covers	**Types of ships as per ICLL, 1966**				
	'A'	**'B- 100'**	**'B-60'**	**'B'**	**'B+'**
a	No	Yes	Yes	Yes	Yes
b	No	No	No	Yes	Yes
c	No	No	No	No	Yes
d	Yes	Yes	Yes	Not applicable	
e	Yes	Yes	Yes	Yes	Yes

Type a: Steel plated cargo hatch covers stiffened by webs or stiffeners and secured by clamping devices. Weather tightness is to be ensured by means of gaskets. Hatch covers used for holds containing liquid cargoes are also included in this category.

Type b: Steel plated pontoon type cargo hatch covers with internal webs and stiffeners extending over the full width of the hatchway. Weather tightness is to be achieved by tarpaulins. Refer Section 10.

Note:
Modern hatch cover designs of lift-away-covers are in many cases called pontoon covers. This definition does not fit to the definition above. Modern lift-away hatch cover designs should belong to one of the two categories-single skin covers or double skin cover.

Type c: Wood or steel hatch covers used in conjunction with the portable beams. Weather tightness to be obtained by tarpaulins. Refer Section 10.

Type d: Access hatch covers for cargo oil tanks and adjacent spaces. The hatch covers are to be of steel and gasketed.

Type e: Access hatch covers other than 'd'. For Types 'A', 'B-100' and 'B-60' ships, the covers are to be of steel, and weathertightness is to be ensured by means of gaskets.

5.3.2 Tween deck hatch covers may be any of the types defined in Cl.5.3.1, but need not be weathertight unless fitted to deep tank or water ballast holds or compartments, in which case the covers are to be of type (a) and oil tight or watertight as appropriate.

5.4 Material

5.4.1 Materials for hatch covers and coamings are to be in accordance with the requirements of Part 3, Chapter 2. Material class I is to be applied for top plate, bottom plate and primary supporting members.

5.4.2 For hatch covers, the application of steel with a minimum yield stress of more than 355[N/mm^2], will be specially considered.

5.5 Structural requirements

5.5.1 Primary supporting members and stiffeners of hatch covers are to be continuous over the breadth and length of hatch covers, as far as practical. When this is impractical, sniped end connections are not to be used and appropriate arrangements are to be adopted to provide sufficient load carrying capacity.

5.5.2 The spacing of primary supporting members parallel to the direction of secondary stiffeners is not to exceed 1/3 of the span of primary supporting members. When strength calculation is carried out by FE analysis, this requirement can be waived.

5.5.3 Stiffeners of hatch coamings are to be continuous over the breadth and length of hatch coamings.

5.5.4 Load bearing connections between the hatch cover panels are to be fitted for the purpose of restricting the relative vertical displacements.

5.6 Net Scantling Approach

5.6.1 Unless otherwise noted, the thicknesses ‘t’ of the following sections is net thicknesses.

5.6.2 The net thicknesses, t_{net}, are the member thicknesses necessary to obtain the minimum net scantlings required by Section 7 (Hatch cover strength criteria) and Section 9 (Hatch coaming strength criteria).

5.6.3 The required gross thicknesses are obtained by adding corrosion additions, t_c in [mm], given in Table 5.6.3 to t_{net}.

5.6.4 Strength calculations using FEM are to be performed with net scantlings.

Table 5.6.3: Corrosion additions t_c for hatch covers and hatch coamings

Application	Structure	t_c [mm]
Weather deck cargo hatches of container ships, car carriers, paper carriers, passenger vessels	Hatch covers	1.0
	Hatch coamings	1.0
Weather deck cargo hatches of all other ship types	Hatch covers in general	2.0
	Weather exposed plating and bottom plating of double skin hatch covers	1.5
	Internal structure of double skin hatch covers and closed box girders	1.0
	Hatch coamings not part of the longitudinal hull structure	1.5
	Hatch coamings part of the longitudinal hull structure	1.0
	Coaming stays and stiffeners	1.5
Weather deck cargo hatches of Type-2 ships	Hatch covers in general	2.0
	Top and bottom plating of double skin hatch covers	2.0
	Internal structure of double skin hatch covers	1.5
	Hatch coamings and coaming stays	1.5
Tween deck hatches	Hatch covers in general	1.0

5.7 Steel renewal

5.7.1 Steel renewal is required where the gauged thickness is less than $t_{net} + 0.5$[mm] for:

- Single skin hatch covers
- The plating of double skin hatch covers, and
- Coaming structures for which corrosion additions t_c is provided in Table 5.6.3

5.7.2 Where the gauged thickness is within the range $t_{net} + 0.5$[mm] and $t_{net} + 1.0$[mm], coating (applied in accordance with the coating manufacturer's requirements) or annual gauging may be adopted as an alternative to steel renewal. Coating is to be maintained in GOOD condition.

5.7.3 For the internal structure of double skin hatch covers, thickness gauging is required when hatch cover top or bottom plating renewal is to be carried out or when this is deemed necessary, at the discretion of the individual class society's surveyor, on the basis of the plating corrosion or deformation condition. In these cases, steel renewal for the internal structures is required where the gauged thickness is less than t_{net}.

5.7.4 For corrosion addition $t_c = 1.0$[mm] the thickness for steel renewal is t_{net} and the thickness for coating or annual gauging is when gauged thickness is between t_{net} and $t_{net} + 0.5$ [mm]. For coaming structures, the corrosion additions t_c of which are not provided in Table 5.6.3, steel renewal and coating or annual gauging are to be specially considered.

Section 6

Hatch Covers and Coaming Load Model

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

x = Longitudinal co-ordinate of midpoint of assessed structural member measured from aft end of length L or L_L, as applicable.

D_{min} = the least moulded depth in [m] as defined in Chapter 1.

h_N = Standard superstructure height in [m].

= $1.05+0.01L_L$, $1.8 \leq h_N \leq 2.3$

6.1 Application

6.1.1 Structural assessment of hatch covers and hatch coamings is to be carried out using the design loads, defined in this section.

6.2 Design Loads on hatch cover and coaming

6.2.1 Vertical weather design load

6.2.1.1 The pressure, P_H in [N/mm²], on the hatch cover panels is given in Table 6.2.1. The vertical design weather load need not be combined with cargo loads given in Cl.6.2.3 and Cl.6.2.4. In Figure 6.2.1.1, the positions 1 and 2 for the purpose of design weather loads are illustrated for an example ship.

6.2.1.2 Where an increased freeboard is assigned, the design load for hatch covers according to Table 6.2.1 on the actual freeboard deck may be as required for a superstructure deck, provided the summer freeboard is such that the resulting draught will not be greater than that corresponding to the minimum freeboard calculated from an assumed freeboard deck situated at a distance at least equal to the standard superstructure height h_N below the actual freeboard deck. (Refer Figure 6.2.1.2).

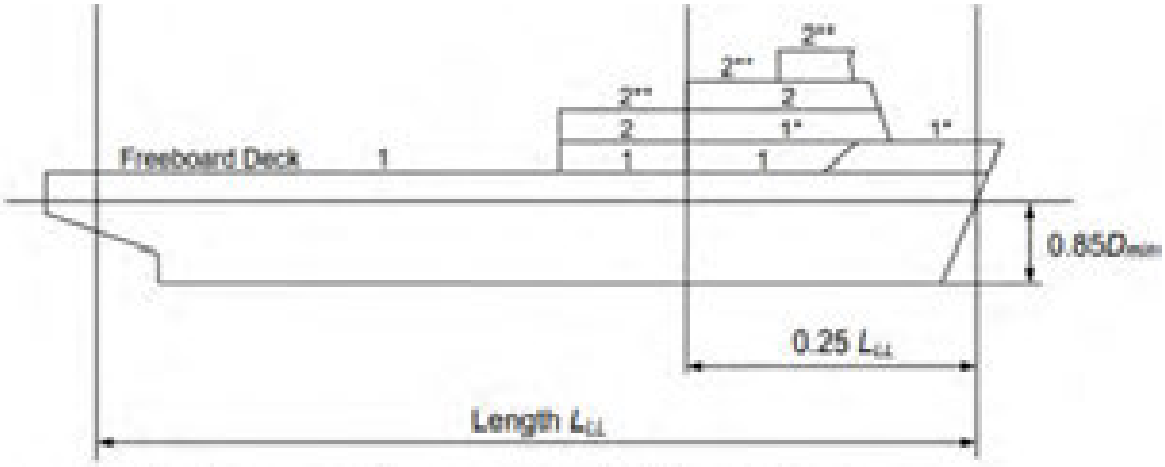
6.2.1.3 Where two or more panels are connected by hinges, each individual panel is to be considered separately.

6.2.1.4 For hatch covers of cargo holds designed for carriage of ballast or liquid cargo, the internal lateral pressures are also to be considered.

Table 6.2.1: Design load P_H of weather deck hatches

Position	Design load P_H [N/mm²]	
1	$\frac{x}{L_L} \leq 0.75$	$0.75 < \frac{x}{L_L} \leq 1.0$
	For $24[m] \leq L_L \leq 100[m]$	
	$\frac{9.81 \times 10^{-3}}{76}(1.5\,L_L + 116)$	On freeboard deck $\frac{9.81 \times 10^{-3}}{76}\left[(4.28\,L_L + 28)\frac{x}{L_L} - 1.71\,L_L + 95\right]$
		Upon exposed superstructure decks located at least one superstructure standard height above the freeboard deck $\frac{9.81 \times 10^{-3}}{76}(1.5\,L_L + 116)$

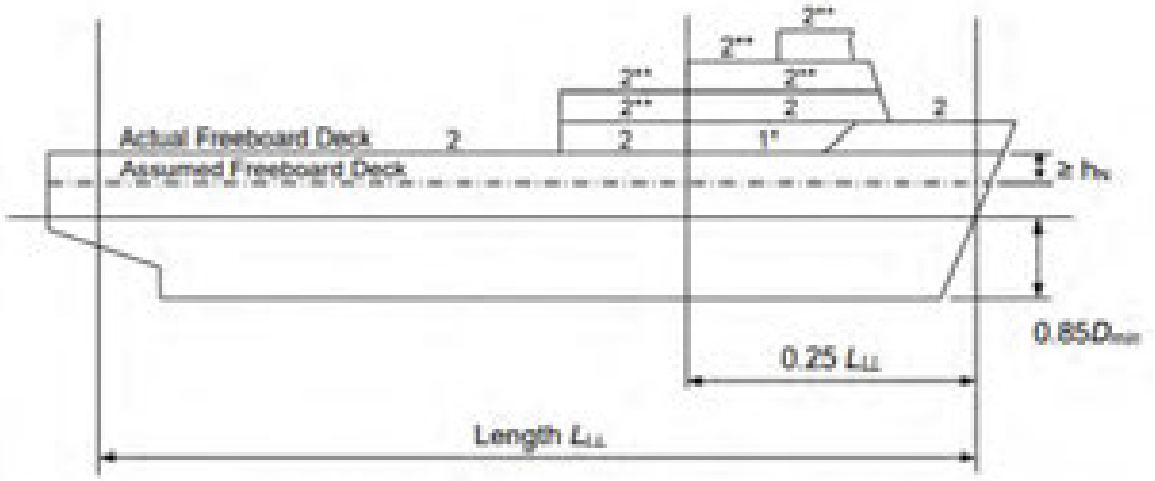
<table>
<tr><td rowspan="4"></td><td colspan="2">For $L_L > 100[m]$</td></tr>
<tr><td rowspan="3">$9.81 \times 3.5 \times 10^{-3}$</td><td>On freeboard deck for type B ships according to ICLL
$9.81 \times 10^{-3} \left[(0.0296L_1 + 3.04)\frac{x}{L_L} - 0.0222L_1 + 1.22\right]$</td></tr>
<tr><td>On freeboard deck for ships with less freeboard than type B according to ICLL
$9.81 \times 10^{-3} \left[(0.1452L_1 - 8.52)\frac{x}{L_L} - 0.1089L_1 + 9.89\right]$
$L_1 = L_L$ but not more than $340[m]$</td></tr>
<tr><td>Upon exposed superstructure decks located at least one superstructure standard height above the freeboard deck
$9.81 \times 3.5 \times 10^{-3}$</td></tr>
<tr><td rowspan="5">2</td><td colspan="2">For $24[m] \leq L_L \leq 100[m]$</td></tr>
<tr><td colspan="2">$\frac{9.81 \times 10^{-3}}{76}(1.1\,L_L + 87.6)$</td></tr>
<tr><td colspan="2">For $L_L > 100[m]$</td></tr>
<tr><td colspan="2">$9.81 \times 2.6 \times 10^{-3}$</td></tr>
<tr><td colspan="2">Upon exposed superstructure decks located at least one superstructure standard height above the lowest Position 2 deck
$9.81 \times 2.1 \times 10^{-3}$</td></tr>
</table>



* reduced load upon exposed superstructure decks located at least one superstructure standard height above the freeboard deck

** reduced load upon exposed superstructure decks of vessels with $L_{LL} > 100$ m located at least one superstructure standard height above the lowest Position 2 deck

Figure 6.2.1.1: Positions 1 and 2 for design weather loads



* reduced load upon exposed superstructure decks located at least one superstructure standard height above the freeboard deck

** reduced load upon exposed superstructure decks of vessels with $L_{LL} > 100$ m located at least one superstructure standard height above the lowest Position 2 deck

Figure 6.2.1.2: Positions 1 and 2 for design weather loads for an increased freeboard

6.2.2 Horizontal weather design load

6.2.2.1 General horizontal weather design load

The horizontal design weather load, P_A in [N/mm²], for determining the scantlings of outer edge girders (skirt plates) of weather deck hatch covers and of hatch coamings is not to be taken less than the greater of:

– P_A and

– P_{A-min}, minimum values given in Table 6.2.2

$$P_A = f_n . f_c . (f_b . c_L . C_w - z) 10^{-3}$$

where

f_n = for unprotected front coamings and hatch cover skirt plates

$$= 20 + \frac{L_1}{12}$$

f_n = for unprotected front coamings and hatch cover skirt plates, where the distance from the actual freeboard deck to the summer load line exceeds the minimum non-corrected tabular freeboard according to ICLL by at least one standard superstructure height h_N

$$= 10 + \frac{L_1}{12}$$

f_n = for side and protected front coamings and hatch cover skirt plates

$$= 5 + \frac{L_1}{15}$$

f_n = for aft end of coamings and aft hatch cover skirt plates abaft amidships

$$= 7 + \frac{L_1}{100} - 8.\frac{x'}{L}$$

f_n = for aft end of coamings and aft hatch cover skirt plates forward of amidships

$$= 5 + \frac{L_1}{100} - 4.\frac{x'}{L}$$

L_1 = L, need not be taken greater than 300[m].

f_b = coefficient given by

$$f_b = 1.0 + \left[\frac{\frac{x'}{L} - 0.45}{C_b + 0.2}\right]^2 \text{ for } \frac{x'}{L} < 0.45$$

$$f_b = 1.0 + 1.5\left[\frac{\frac{x'}{L} - 0.45}{C_b + 0.2}\right]^2 \text{ for } \frac{x'}{L} \geq 0.45$$

$0.6 \leq C_b \leq 0.8$, when determining scantlings of aft ends of coamings and aft hatch cover skirt plates forward of amidships, C_b need not be taken less than 0.8.

x' = distance in [m], between the transverse coaming or hatch cover skirt plate considered and aft end of the length L. When determining side coamings or side hatch cover skirt plates, the side is to be subdivided into parts of approximately equal length, not exceeding 0.15L each, and x' is to be taken as the distance between aft end of the length L and the centre of each part considered.

z = Vertical distance in [m] from the summer load line to the midpoint of stiffener span, or to the middle of the plate field.

$$f_c = 0.3 + 0.7\frac{b'}{B'}$$

b' = Breadth of coaming in [m] at the position considered.

B' = Actual maximum breadth of ship in [m] on the exposed weather deck at the position considered.

b'/B' is not to be taken less than 0.25.

C_w = coefficient given by

$$= 4.1 + \frac{L}{25} \qquad \text{for } L < 90[m]$$

$$= 10.75 - \left(\frac{300 - L}{100}\right)^{1.5} \qquad \text{for } 90 \leq L < 300[m]$$

$$= 10.75 \qquad \text{for } 300 \leq L < 350[m]$$

$$= 10.75 - \left(\frac{L - 350}{150}\right)^{1.5} \qquad \text{for } 350 \leq L \leq 500[m]$$

$$c_L = \sqrt{\frac{L}{90}} \quad \text{for } L < 90[m]$$

$$= 1 \qquad \text{for } L \geq 90[m]$$

The minimum values of design load P_{A-min}, in [N/mm²] is given in Table 6.2.2.

Table 6.2.2: Minimum design load, P_{A-min}

L[m]	P_{A-min} in [N/mm²]	
	For unprotected fronts	Elsewhere
$L \leq 50$	30.10^{-3}	15.10^{-3}
$L > 50$	$\left(25 + \frac{L}{10}\right)10^{-3}$	$\left(12.5 + \frac{L}{20}\right)10^{-3}$
$L < 250$		
$L \geq 250$	50.10^{-3}	25.10^{-3}

6.2.2.2 Horizontal weather design load applicable to coamings of Type-2 ships

The pressure P_{Coam}, in [N/mm²], on the No. 1 forward transverse hatch coaming is given by:

$P_{Coam} = 220x10^{-3}$, when a forecastle is fitted in accordance with Part 5, Chapter 1, Cl.1.7.

$= 290x10^{-3}$, in the other cases

The pressure P_{Coam}, in [N/mm²], on the other coamings is given by:

$P_{Coam} = 220x10^{-3}$

Note: The horizontal design weather loads, P_A and P_{Coam} need not be included in the direct strength calculation of the hatch cover, unless it is utilized for the design of substructures of horizontal support according to Cl.9.12.3.

6.2.3 Cargo loads

6.2.3.1 Distributed loads

The load on hatch covers due to distributed cargo loads P_L in [N/mm²], resulting from heave and pitch (i.e. ship in upright condition) is to be determined according to the following formula:

$$P_L = P_{Cargo}(1 + a_v)$$

where:

P_{Cargo} = Uniform cargo load in [N/mm²].

a_v = Vertical acceleration addition as follows:

$$a_v = F.m$$

$$F = 0.11 \times \frac{v_0}{\sqrt{L}}$$

$$m = m_0 - 5(m_0 - 1)\frac{x}{L} \qquad \text{for } 0 \leq \frac{x}{L} \leq 0.2$$

$$m = 1.0 \qquad \text{for } 0.2 < \frac{x}{L} \leq 0.7$$

$$m = 1.0 + \frac{m_0 + 1}{0.3}\left[\frac{x}{L} - 0.7\right] \text{ for } 0.7 < \frac{x}{L} \leq 1.0$$

$$m_0 = 1.5 + F$$

v_0 = Maximum speed in [knots], at summer load line draught, v_0 is not to be taken less than $\sqrt{L}$.

6.2.3.2 Point loads

The load P in [kN], due to a concentrated force P_S in [kN], except for container load, resulting from heave and pitch (i.e. ship in upright condition) is to be determined as follows:

$$P = P_S(1 + a_v)$$

where:

a_v = Vertical acceleration addition according to Cl.6.2.3.1.

6.2.4 Container loads

6.2.4.1 The loads defined in Cl.6.2.4.2 to Cl.6.2.4.4 are to be applied where containers are stowed on the hatch cover.

6.2.4.2 The load P in [kN], applied at each corner of a container stack, and resulting from heave and pitch (i.e. ship in upright condition) is to be determined as follows:

$$F = 9.81\frac{M}{4}(1 + a_v)$$

where:

a_v = Vertical acceleration addition according to Cl.6.2.3.1.

M = Maximum designed mass of container stack in [t].

6.2.4.3 The loads in [kN], applied at each corner of a container stack, and resulting from heave, pitch, and the ship's rolling motion (i.e. ship in heel condition) are to be determined as follows, (Refer Figure 6.2.4):

$$A_z = 9.81\frac{M}{2}(1 + a_v)\left(0.45 - 0.42\frac{h_m}{b}\right)$$

$$B_z = 9.81\frac{M}{2}(1 + a_v)\left(0.45 + 0.42\frac{h_m}{b}\right)$$

$$B_y = 2.4M$$

where:

a_v = Acceleration addition according to Cl.6.2.3.1.

M = Maximum designed mass of container stack, in [t].

$$M = \sum W_i$$

h_m = Designed height of centre of gravity of stack above hatch cover top in [m], may be calculated as weighted mean value of the stack, where the centre of gravity of each tier is assumed to be located at the centre of each container,

$$= \sum (z_i W_i)/M$$

z_i = Distance in [m], from hatch cover top to the centre of i^{th} container.

W_i = Weight of i^{th} container in [t].

b = Distance in [m], between midpoint of foot points.

A_z, B_z = Support forces in [kN] in z-direction at the forward and aft stack corners.

B_y = Support force in [kN] in y-direction at the forward and aft stack corners.

Values of A_z and B_z applied for the assessment of hatch cover strength are to be shown in the drawings of the hatch covers.

Note:
It is recommended that container loads as calculated above are considered as limit for foot point loads of container stacks in the calculations of cargo securing (container lashing).

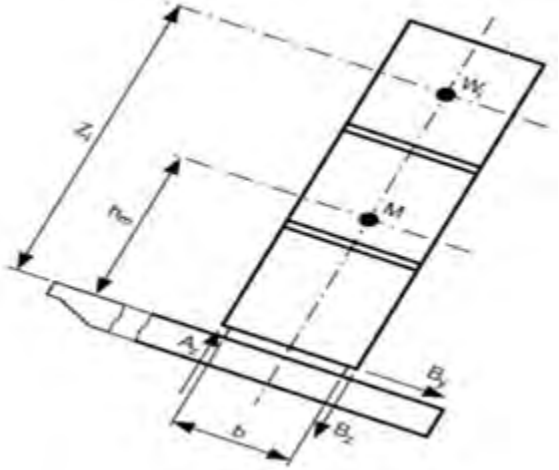


Figure 6.2.4: Forces due to container loads

6.2.4.4 Load cases with partial loading

The load cases defined in Cl.6.2.4.2 and Cl.6.2.4.3 are also to be considered for partial non homogeneous loading which may occur in practice, e.g. where specified container stack places are empty. For each hatch cover, the heel directions, as shown in Table 6.2.4, are to be considered.

The load case *partial loading of container hatch covers* can be evaluated using a simplified approach, where the hatch cover is loaded without the outermost stacks, that are located completely on the hatch cover. If there are additional stacks that are supported partially by the hatch cover and partially by container stanchions then the loads from these stacks are also to be neglected, refer Table 6.2.4, “Partial loading of container hatch covers”. In addition, the case where only the stack places supported partially by the hatch cover and partially by container stanchions are left empty is to be assessed in order to consider the maximum loads in the vertical hatch cover supports. It may be necessary to also consider partial load cases where more or different container stack places are left empty.

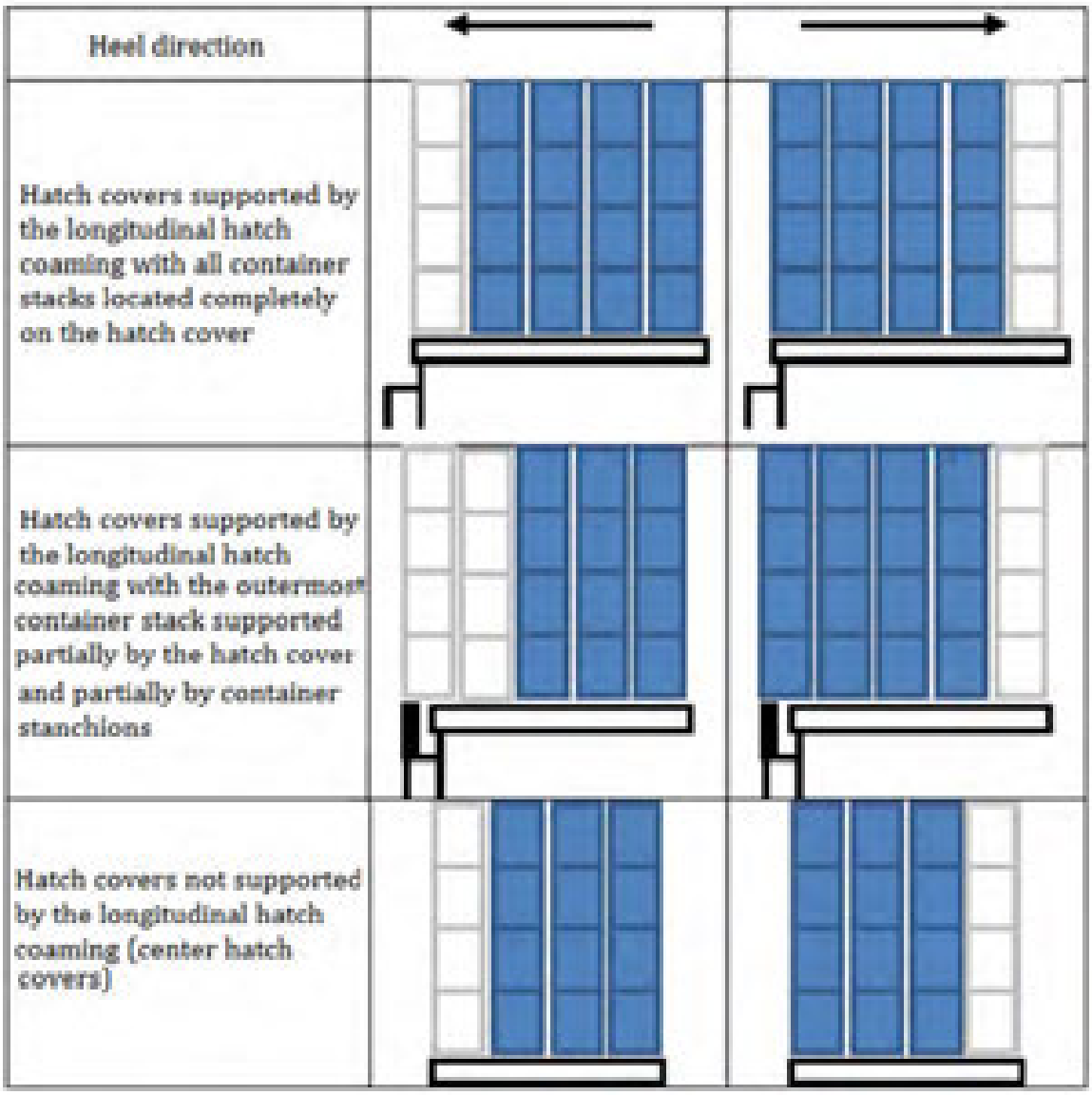
6.2.4.5 Mixed stowage of 20[ft] and 40[ft] containers on hatch covers

In the case of mixed stowage (20[ft] +40[ft] container combined stack), the foot point forces at the fore and aft end of the hatch cover are not to be higher than resulting from the design stack weight for 40[ft] containers, and the foot point forces at the middle of the cover are not to be higher than resulting from the design stack weight for 20[ft] containers.

6.2.5 Loads due to elastic deformations of the ship's hull

Hatch covers, which in addition to the loads according to Cl.6.2.1 to Cl.6.2.4 are loaded in the ship's transverse direction by forces due to elastic deformations of the ship's hull, are to be designed such that the sum of stresses does not exceed the permissible values given in Cl.7.2.1.

Table 6.2.4: Partial loading of container hatch covers



Section 7

Hatch Cover Strength Criteria

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2 and Chapter 9.

7.1 General strength criteria

7.1.1 The scantlings of the hatch cover plating, secondary stiffeners and primary supporting members are to be determined as per 7.3, 7.4 and 7.5 respectively using the design loads given in 6.2.

7.1.2 The scantlings specified in the following sections are applicable to covers of mild steel or higher strength steel. Where other materials are used, it is specially considered. The scantlings apply basically to hatch covers of rectangular shapes, assumed to be simply supported with the stiffening members arranged primarily in one direction and carrying a uniformly distributed load.

7.1.3 Hatch covers of special construction and arrangement e.g. covers designed and constructed as a grillage, covers supported along more than two opposite edges and covers supporting other covers, may require submission of direct strength calculation taking into account the arrangement of stiffeners and the supporting members.

7.1.4 Container loads and other point loads are to be additionally taken in to account, in the calculations.

7.2 Permissible stresses and deflections

7.2.1 Yield strength

All hatch cover structural members are to comply with the following formulae:

$\sigma_{vm} \leq \sigma_a$ for shell elements in general.

$\sigma_{axial} \leq \sigma_a$ for rod or beam elements in general.

where

σ_a = Allowable stress, in [N/mm^2] as defined in Table 7.2.1.

σ_{axial} = Axial stress, in [N/mm^2] in rod or beam elements.

R_{eH} = Specified minimum yield stress of the material in [N/mm^2].

σ_{vm} = Von Mises stress, in [N/mm^2] to be taken as follows:

$$\sigma_{vm} = \sqrt{\sigma_x^2 - \sigma_x.\sigma_y + \sigma_y^2 + 3\tau_{xy}{}^2}$$

where:

σ_x = Normal stress in [N/mm^2] in x-direction.

σ_y = Normal stress in [N/mm^2] in y-direction.

τ_{xy} = Shear stress in [N/mm^2] in the x-y plane.

Indices x and y are coordinates of a two-dimensional Cartesian system in the plane of the considered structural element.

In case of finite element (FE) calculations using shell (or plate) elements, the stresses are to be read from the centre of the individual element. It is to be observed that, in particular, at flanges of unsymmetrical girders, the evaluation of stress from element centre may lead to nonconservative results. Thus, a sufficiently fine mesh is to be applied in these cases or, the stress at the element edges is not exceed the allowable stress. Where shell elements are used, the stresses are to be evaluated at the mid plane of the element.

Stress concentrations are to be assessed to the satisfaction of IRS.

For steels with a minimum yield stress of more than 355[N/mm^2], the value of R_{eH} to be applied throughout this requirement is subject to the individual classification society but is not to be more than the minimum yield stress of the material.

Table 7.2.1: Allowable stresses in [N/mm²]		
Members of	**Subject to**	σ_a
Hatch cover structure	External pressure, as defined in 6.2.1	$0.8R_{eH}$
	Other loads, as defined in 6.2.2 to 6.2.5	$0.9R_{eH}$ for Static (S)+Dynamic (D) load case $0.72R_{eH}$ for Static (S) load case

7.2.2 Deflection

The vertical deflection of primary supporting members in [m], due to the vertical design weather load according to Cl.6.2.1 is to be not more than $(\mu.\ell_g)$

where:

μ = Coefficient taken equal to:

= 0.0056 for weathertight hatch covers.

ℓ_g = Greatest span of primary supporting members in [m].

Note:
Where hatch covers are arranged for carrying containers and mixed stowage is allowed, i.e., a 40[ft] container stowed on top of two 20[ft] containers, particular attention is to be paid to the deflections of hatch covers. Further the possible contact of deflected hatch covers within cargo hold has to be observed.

For curved plating, the stiffener spacing, s or the primary supporting member spacing, 'S' is to be measured on the mean chord between members.

7.3 Local net plate thickness

7.3.1 Hatch cover top plating

The local net plate thickness 't' in [mm], of the hatch cover top plating is not to be less than:

$$t = 0.0158\ F_p s \sqrt{\frac{P \times 10^3}{0.95\ R_{eH}}}$$

and to be not less than 1% of the spacing of the stiffener or 6[mm] if that be greater.

F_p = Factor for combined membrane and bending response

= 1.5, in general

$$= 1.9.\frac{\sigma}{\sigma_a} \text{ for } \frac{\sigma}{\sigma_a} \geq 0.8$$

for the attached plate flange of primary supporting members

where:

s = Stiffener spacing in [mm].

P = Pressure P_H and P_L in [N/mm²], as defined in 6.2.

σ = Maximum normal stress in [N/mm²], of hatch cover top plating, determined according to Figure 7.3.1.

σ_a = As defined in Table 7.2.1.

For flange plates under compression sufficient buckling strength according to 7.7 is to be demonstrated.

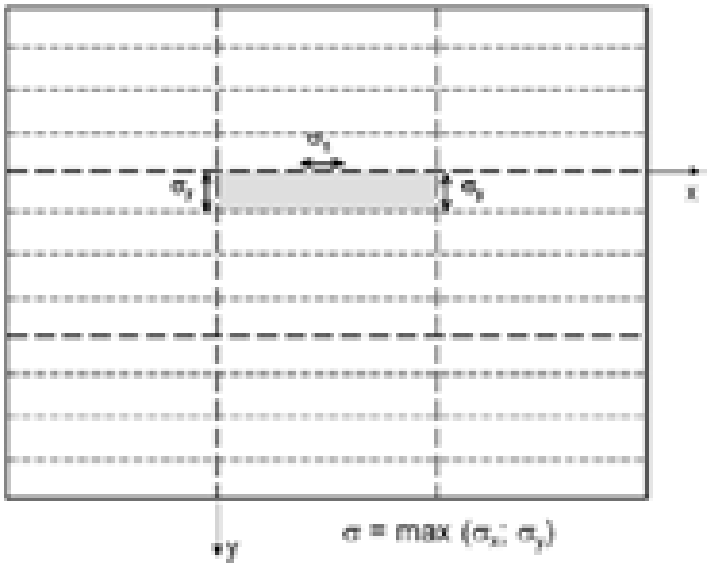


Figure 7.3.1: Determination of normal stress of the hatch cover plating

7.3.2 Local net plate thickness of hatch covers for wheel loading

The local net plate thickness of hatch covers for wheel loading is to be according to Chapter 11, Section 5.

7.3.3 Lower plating of double skin hatch covers and box girders

The thickness to fulfill the strength requirements is to be obtained from the calculation according to 7.6 under consideration of permissible stresses as per Cl.7.2.1. When the lower plating is taken into account as a strength member of the hatch cover, the net thickness in [mm], of lower plating is to be taken not less than 5[mm].

When project cargo is intended to be carried on a hatch cover, the net thickness must not be less than:

$$t = 6.5\, s \times 10^{-3}\ [mm]$$

where,

s = Stiffener spacing in [mm].

Note:
Project cargo means especially large or bulky cargo lashed to the hatch cover. Examples are parts of cranes or wind power stations, turbines, etc. Cargoes that can be considered as uniformly distributed over that hatch cover, e.g., timber, pipes or steel coils need not to be considered as project cargo.

When the lower plating is not considered as a strength member of the hatch cover, the thickness of the lower plating will be specially considered.

7.4 Net scantling of secondary stiffeners

7.4.1 The net section modulus 'Z' in [cm^3] and net shear area A_{shr} in [cm^2] of uniformly loaded hatch cover stiffeners constrained at both ends must not be less than:

$$Z = \frac{P\,.\,s.\ell^2.10^3}{f_{bc}.\sigma_a}$$

$$A_{shr} = \frac{8.7P.\,s.\ell}{\sigma_a}$$

where:

ℓ = Stiffener span in [m], to be taken as the spacing, in [m], of primary supporting members or the distance between a primary supporting member and the edge support, as applicable. When brackets are fitted at both ends of all stiffener spans, the secondary stiffener span may be reduced by an amount equal to 2/3 of the minimum brackets arm length, but not greater than 10% of the unsupported span, for each bracket

s = Stiffener spacing in [mm].

P = Pressure P_H and P_L in [N/mm^2], as defined in 6.2.

f_{bc} = Boundary coefficient of stiffener, taken equal to:

= 8, in the case of stiffener simply supported at both end or simply supported at one end and clamped at the other end

= 12, in the case of stiffener clamped at both ends.

σ_a = Allowable stress as defined in Table 7.2.1.

7.4.2 For stiffeners of lower plating of double skin hatch covers, requirements mentioned above are not applied due to the absence of lateral loads. For double skin hatch covers of holds designed for ballast or liquid cargo, the stiffeners on lower plating are to be strengthened according to Chapter 7, Section 5.

7.4.3 The net thickness, in [mm], of the stiffener (except u-beams/trapeze stiffeners) web is to be taken not less than 4[mm].

7.4.4 The net section modulus of the stiffeners is to be determined based on an attached plate width assumed equal to the stiffener spacing.

7.4.5 Stiffeners parallel to primary supporting members must be continuous at crossing primary supporting member and may be considered for calculating the cross-sectional properties of primary supporting members. It is to be verified that the combined stress of those stiffeners induced by the bending of primary supporting members and lateral pressures does not exceed the permissible stresses according to Cl.7.2.1. The requirements of this paragraph are not applied to stiffeners of lower plating of double skin hatch covers if the lower plating is not considered as strength member.

7.4.6 For hatch cover stiffeners under compression sufficient safety against lateral and torsional buckling according Cl.7.7.3 is to be verified.

7.4.7 For hatch covers subject to wheel loading or point loads stiffener scantlings are to be determined under consideration of the

permissible stresses according to Cl.7.2.1 or are to be determined according to Chapter 11, Section 5.

7.5 Net scantling of primary supporting members

7.5.1 Primary supporting members

7.5.1.1 Scantlings of primary supporting members are obtained from calculations according to 7.6 under consideration of permissible stresses according to Cl.7.2.1.

7.5.1.2 For all components of primary supporting members sufficient safety against buckling must be verified according to 7.7.

7.5.1.3 The net thickness, in [mm], of webs of primary supporting members is not be less than:

$$t = 6.5\, s \times 10^{-3}$$

$$t_{min} = 5.0$$

where,

s = Stiffener spacing [mm]

7.5.2 Edge girders (Skirt plates)

7.5.2.1 Scantlings of edge girders are obtained from the calculations according to 7.6 under consideration of permissible stresses as per Cl.7.2.1.

7.5.2.2 The net thickness, 't' in [mm], of the outer edge girders exposed to wash of sea is not be less than:

$$t = \max\left(\begin{matrix} 0.0158\, s\sqrt{\dfrac{P_A \times 10^3}{0.95 R_{eH}}}; \\ 8.5\, s \times 10^{-3}\,;\ 5.0 \end{matrix}\right)$$

where,

s = Stiffener spacing, in [mm].

p = Horizontal pressure as defined in Cl.6.2.2.

7.5.2.3 The stiffness of edge girders is to be sufficient to maintain adequate sealing pressure between securing devices. The moment of inertia, in [cm^4], of edge girders is not to be less than:

$$I = 6.\, q.\, s_{SD}^4$$

where,

q = Packing line pressure, in [N/mm], minimum 5[N/mm].

s_{SD} = Spacing of securing devices in [m], not to be taken less than 2[m].

7.6 Strength calculations

7.6.1 The stresses in hatch covers are to be determined by FE analysis.

The stress calculation model in this section is to be used for both yielding and buckling strength assessments in accordance with 7.2 and 7.7, respectively.

The net scantlings as defined in Section 5, 5.6 are to be used.

7.6.2 General requirements for FEM calculations

7.6.2.1 For the strength assessments of hatch covers by means of FE analysis, the hatch cover geometry shall be idealized as realistically as possible. In no case element width is to be larger than stiffener spacing. In way of force transfer points and cutouts the mesh has to be refined where applicable. The ratio of element length to width is not exceed 3.

7.6.2.2 The element size along the height of webs of primary supporting member is not to exceed one-third of the web height. Stiffeners, which support plates subjected to lateral pressure loads, are to be included in the FE model idealization. Stiffeners may be modelled by using beam elements, or shell/plate elements. Buckling stiffeners may be disregarded for the stress calculation.

7.6.2.3 Hatch covers fitted with U-type stiffeners as shown in Figure 7.6.2, are to be assessed by means of FE analysis. The geometry of the U-type stiffeners is to be accurately modelled using shell/plate elements. Nodal points are to be properly placed on the intersections between the webs of a U-type stiffener and the hatch cover plate, and between the webs and flange of the U-type stiffener.

7.6.2.4 Wherever applicable the following boundary conditions are to be applied to the FE model:

- Boundary nodes in way of a bearing pad on the hatch coamings are to be fixed against displacement in the direction perpendicular to the pad.
- Lifting stoppers are to be fixed against displacements in the direction determined by the stoppers.
- For a folding type hatch cover, the FE nodes connected through a hinge are to have the same translational displacement in the direction perpendicular to the hatch cover top plating.

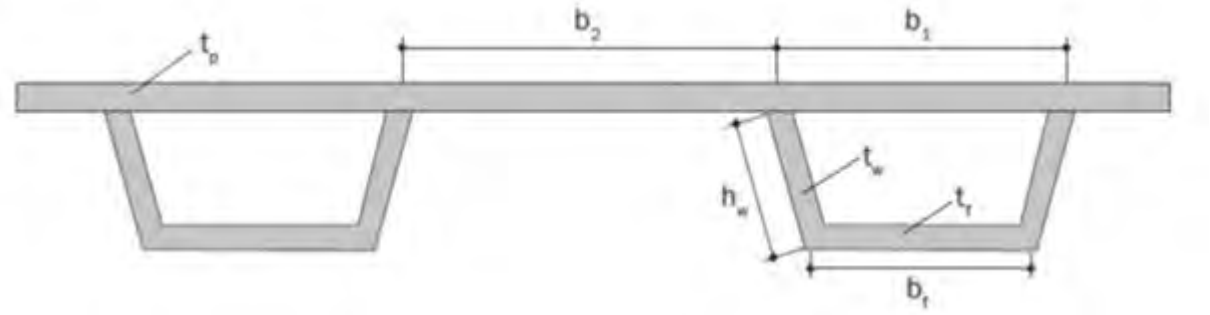


Figure 7.6.2: Example of hatch cover fitted with U-type stiffeners

7.7 Buckling strength of hatch cover structures

7.7.1 General

Buckling strength of all hatch cover structures is to be checked. Buckling assessments are to be done according to Chapter 9 for the conditions specified in 7.7.2 and 7.7.3.

The net scantlings as defined in Section 5, 5.6 are to be used for buckling check.

7.7.2 Slenderness requirements

7.7.2.1 The slenderness requirements are to be in accordance with Chapter 9, Section 2. The slenderness requirements need not be applied to the lower boundary of double skin hatch covers unless the cargo hold is designed for carriage of ballast or liquid cargo.

7.7.2.2 The breadth of the primary supporting member flange is to be not less than 40% of their depth for laterally unsupported spans greater than 3.0[m]. Tripping brackets attached to the flange may be considered as a lateral support for primary supporting members.

7.7.3 Buckling requirements

7.7.3.1 Application

These requirements apply to the buckling assessment of hatch cover structures subjected to compressive and shear stresses and lateral pressures.

The buckling assessment is to be performed for the following structural elements:

- Stiffened and unstiffened panels, including curved panels and panels stiffened with U type stiffeners.
- Web panels of primary supporting members in way of openings.

The buckling strength assessment of coaming parts is to be done according to Chapter 9.

For rule application, the panel types and assessment methods, the applied lateral pressure and stresses, safety factors and buckling check criteria are defined in Cl.7.7.3.2, Cl.7.7.3.3, Cl.7.7.3.4 and Cl.7.7.3.5, respectively.

The procedure and detailed requirements for buckling assessment including idealization of irregular plate panels, definition of reference stresses and buckling criteria, are given in Chapter 9, Section 4.

7.7.3.2 Panel types and assessment methods

The plate panel of a hatch cover structure is to be modelled as stiffened panel (SP) or unstiffened panel (UP) as defined in Chapter 9, Section 4.

Assessment Method A (-A) and Method B (-B) as defined in Chapter 9, Section 1, Cl.1.3.1 are to be used in accordance with Table 7.7.3.1, Figure 7.7.3.1 and Figure 7.7.3.2.

For a web panel with opening, the procedure for opening should be used for its buckling assessment.

For a hatch cover fitted with U-type stiffeners, the additional buckling assessment requirements specific for panels with U-type stiffeners in Chapter 9, Section 5, 5.5 are also to be applied.

Table 7.7.3.1: Structural members and assessment methods

Structural elements	Assessment method (1)(2)	Normal panel definition
Hatch cover top/bottom plating structures, Refer Figure 7.7.3.1		
Hatch cover top/bottom plating	SP-A	Length: between transverse girders Width: between longitudinal girders
Irregularly stiffened panels	UP-B	Plate between local stiffeners/PSM
Hatch cover web panels of primary supporting members, Refer Figure 7.7.3.2		
Web of transverse/longitudinal girder (single skin type)	UP-B	Plate between local stiffeners/face plate/PSM
Web of transverse/longitudinal girder (double skin type)	SP-B (3)	Length: between PSM Width: full web depth
Web panel with opening	Procedure for opening	Plate between local stiffeners/face plate/PSM
Irregularly stiffened panels	UP-B	Plate between local stiffeners/face plate/PSM

Note 1: SP and UP stand for stiffened and unstiffened panel respectively.
Note 2: A and B stand for Method A and Method B respectively.
Note 3: In case that the buckling carlings/brackets are irregularly arranged in the web of transverse/longitudinal girder, UP-B method may be used

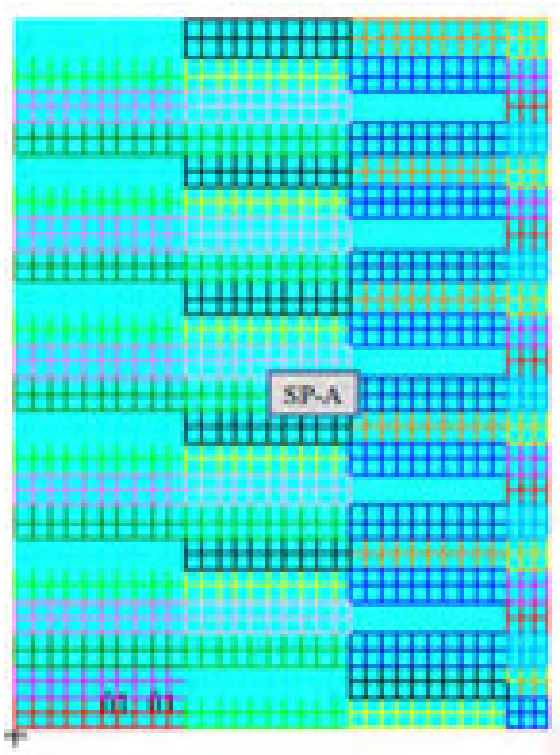


Figure 7.7.3.1: Hatch cover top/bottom plating structures

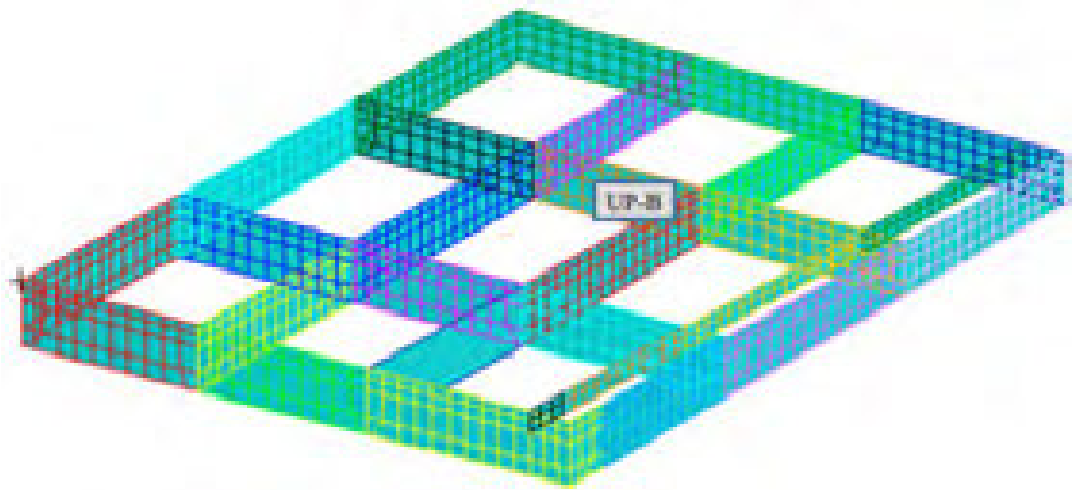


Figure 7.7.3.2: Hatch cover webs of primary supporting members

7.7.3.3 Applied lateral pressure and stresses

The buckling assessment of hatch covers is based on the lateral pressure as defined in 6.2.1 and 6.2.2, and stresses obtained from FE analysis, Refer to 7.6.

7.7.3.4 Safety factors

For all hatch cover structural members, safety factor S=1.0 is to be applied to both of the plating and stiffener buckling capacity formulas as defined in Chapter 9, Section 5, 5.2.

7.7.3.5 Buckling acceptance criteria

A structural member is considered to have an acceptable buckling strength if it satisfies the following criterion

$$\eta_{act} \leq \eta_{all}$$

where,

η_{act} = Buckling utilisation factor based on the applied stress as per Chapter 9, Section 1, Cl.1.3.2, and Chapter 9, Section 4, and calculated per Chapter 9, Section 5.

η_{all} = Allowable buckling utilisation factor as defined in Table 7.7.3.5.

Table 7.7.3.5: Allowable buckling utilisation factor (η_{all})

Structural component	Subject to	Design Load set	η_{all}
Plates and stiffeners	External pressure, as defined in 6.2.1		0.80
Web of PSM	Other loads, as defined in 6.2.2 to 6.2.5	S	0.72
		S + D	0.90

7.8 Hatch covers with variable cross-section

Hatch covers with Z and I gradually reducing towards the ends of the span are to be designed so that the maximum bending stresses and deflections are not increased.

7.8.1 When the section shape is reduced towards the ends (Refer Figure 7.8.1) the requirements of the section modulus Z and the moment of inertia I of the section in the middle uniform part are to be increased by the factors C1 and C2 respectively where C1 and C2 are given below.

7.8.2 The net section modulus *Z*, in [cm^3], of stiffeners and primary supporting members with a variable cross section is to be taken not less than the greater of the values given by the following formulae:

$$Z = Z_{cs}$$

$$Z = \left(1 + \frac{3.2\alpha - \beta - 0.8}{7\beta + 0.4}\right) Z_{cs}$$

where:

Z_{cs} = Net section modulus in [cm^3], for a constant cross section, complying with the checking criteria in Cl.7.4.1.

α = Coefficient taken equal to:

$$\alpha = \frac{\ell_1}{\ell_0}$$

β = Coefficient taken equal to:

$$\beta = \frac{Z_1}{Z_0}$$

ℓ_1 = Length of the variable section part in [m], as shown in Figure 7.8.1.

ℓ_0 = Span measured in [m], between end supports, as shown in Figure 7.8.1.

Z_1 = Net section modulus at end in [cm^3], as shown in Figure 7.8.1.

Z_0 = Net section modulus at mid-span in [cm^3], as shown in Figure 7.8.1.

7.8.3 The net moment of inertia in [cm^4], of stiffeners and primary supporting members with a variable cross section is to be taken not less than the greater of the values given by the following formulae:

$$I = I_{cs}$$

$$I = \left(1 + 8\alpha^3 \left(\frac{1 - \delta}{0.2 + 3\sqrt{\delta}}\right)\right) I_{cs}$$

where:

I_{cs} = Net moment of inertia with a constant cross-section in [cm^4], calculated with wave pressure, as given in Cl.6.2.1 or distributed load as given in Cl.6.2.3.1. It is to be such that the deflection does not exceed (μ α), with μ as defined in Cl.7.2.2.

Δ = Coefficient taken equal to:

$$\delta = \frac{I_1}{I_0}$$

I_1 = Net moment of inertia at end in [cm^4], (Refer Figure 7.8.1)

I_0 = Net moment of inertia at mid-span in [cm^4], (Refer Figure 7.8.1).

The use of these formulae is limited to the determination of the strength of stiffeners and primary supporting members in which abrupt changes in the cross section do not occur along their length.

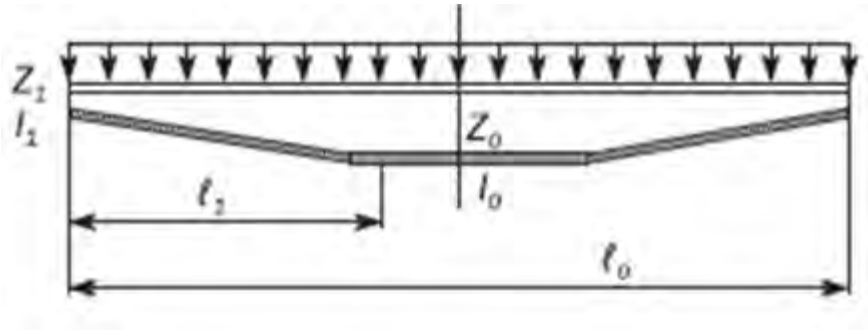


Figure 7.8.1: Variable cross-section stiffener

7.8.4 The net web area, 'A' in [cm^2] of the primary supporting members at any point is not to be less than:

$$A = 10\frac{\left(0.5 - \frac{x}{\ell}\right)}{\tau_a} P.s.\ell$$

where:

τ_a = $0.46R_{eH}$ [Allowable shear stress in [N/mm^2].

x is the distance [m] from the end of the span to the section considered. The value of x is not to be taken more than 0.25ℓ.

Section 8

Details of Hatch Covers

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

8.1 Container foundations on hatch covers

8.1.1 Container foundations and substructures are to be designed for cargo and container loads according to Section 6, Cl.6.2.3 and Cl.6.2.4, applying the permissible stresses according to Section 7, 7.2.

8.2 Weather tightness

8.2.1 General

8.2.1.1 Where the hatchway is exposed, the weather tightness is to be ensured by gaskets and clamping devices sufficient in number and quality. Weather tightness may also be ensured by means of tarpaulins.

8.2.1.2 In general, a minimum of two securing devices or equivalent is to be provided on each side of the hatch cover.

8.2.1.3 The weight of covers and any cargo stowed thereon, together with inertial forces generated by ship motions, are to be transmitted to the ship structure through suitable contact, such as continuous steel to steel contact of the cover skirt plate with the ship's structure or by means of defined bearing pads.

8.2.2 Weathertight Hatch Covers

8.2.2.1The arrangement of weathertight hatch covers is to be such that weather tightness can be maintained in all sea conditions.

8.2.2.2 Weathertight sealing is to be achieved by a continuous gasket of relatively soft elastic material compressed to provide the necessary weather tightness. Similar sealing is to be arranged between cross-joint elements. Where fitted, compression flat bars or angles are to be well rounded where in contact with the gasket and are to be made of a corrosion-resistant material.

8.2.3 Tightness Testing/Operation and maintenance of Weathertight Hatch Covers

8.2.3.1 The tightness testing is to be carried out as per the requirements of Chapter 17, Section 3, Cl.3.4.4.3.

Upon completion of installation of hatch covers, a hose test is to be carried out with a water pressure in the hose nozzle not less than 2 bar. The nozzle is to have a minimum inside diameter of 12[mm] and nozzle is to be held at a perpendicular distance from the hatch cover to be tested not exceeding 1.5[m]. Water pressure is to be sufficient for a free height of water with the stream directed upwards maximum of 10[m]. During frost periods equivalent tightness tests may be carried out to the satisfaction of the surveyor. Alternative methods of tightness testing will be specially considered.

8.2.3.2 All hatch covers are to be tested for proper functioning and operation in presence of the surveyor.

8.2.3.3 It is recommended that ships with steel hatch covers are supplied with an operation and maintenance manual including:

- opening and closing instructions;
- maintenance requirements and specifications for packings, securing devices and operating items;
- cleaning instructions for the drainage system;
- corrosion prevention instructions;
- list of spare parts.

8.2.4 Gaskets

8.2.4.1 Packing material

- For weather deck hatch covers packings are to be provided, for exceptions Refer Cl.8.2.5.
- The gasket material is to be of a quality suitable for all environmental conditions likely to be experienced by the ship, and is to be compatible with the cargoes carried. The material and form of gasket selected is

to be considered in conjunction with the type of cover, dimensions, the securing arrangement and the expected relative movement between cover and ship structure. The gasket is to be effectively secured to the cover. Forces are to be carried by the steel structure only.

- The packings are to be compressed so as to give the necessary tightness effect for all expected operating conditions. Special consideration is to be given to the packing arrangement in ships with large relative movements between hatch covers and coamings or between hatch cover sections. If necessary, suitable devices are to be fitted to limit such movements. The specification or grade of the packing material is to be indicated on the drawings.

- The hatch coamings and steel parts of hatch covers in contact with gaskets are to have no sharp edges and are to be well rounded where necessary.

- Metallic contact is required for an earthing connection between the hatch cover and the hull structures. If necessary, this is to be achieved by means of a special connection for the purpose.

- The gasket material is to be of satisfactory air, seawater and if necessary oil resistant quality. It is to be effectively secured along the edges of the cover in a manner as to ensure that the forces from the hatch covers or cargo stowed on top of the hatch covers are transferred to the coaming or to the deck by direct contact. The sealing is to be achieved by relatively soft packing.

8.2.5 Dispensation of weather tight gaskets

8.2.5.1 For hatch covers of cargo holds solely for the transport of containers, upon request by the owners and subject to compliance with the following conditions the fitting of weather tight gaskets according to Cl.8.2.4 may be dispensed with:

- The hatchway coamings is not to be less than 600[mm] in height;

- The exposed deck on which the hatch covers are located is situated above a depth $H(x)$ in [m]. H(x) is to be shown to comply with the following criteria:

$$H(x) \geq T_{fb} + f_b + h$$

T_{fb} = Draught in [m] corresponding to the assigned summer load line.

f_b = Minimum required freeboard in [m], determined in accordance with ICLL Regulation 28 as modified by further regulations as applicable.

h = Distance, in [m], taken equal to:

$$h = 4.6 \text{ for } x/L_L \leq 0.75$$

$$h = 6.9 \text{ for } x/L_L > 0.75$$

8.2.5.2 Labyrinths, gutter bars or equivalents are to be fitted proximate to the edges of each panel in way of the coamings to minimize the amount of water that can enter the container hold. The clear profile of these opening is to be kept as small as possible.

8.2.5.3 Where a hatch is covered by several hatch cover panels the clear openings of the gap in between the panels is not to be wider than 50[mm].

8.2.5.4 The labyrinths and gaps between hatch cover panels are to be considered as unprotected openings with respect to the requirements of intact and damage stability calculations.

8.2.5.5 With regard to drainage of cargo holds and the necessary fire-fighting system, reference is made to applicable requirements in Piping Systems, Valves and Pumps and Fire Protection and Fire Extinguishing Equipment of Part 4, Chapter 3, Part 6 and Part 5, Chapter 5, Section 2.2 of the Rules.

8.2.5.6 Bilge alarms are to be provided in each hold fitted with non-weathertight covers.

8.2.5.7 Furthermore, Chapter 3 of IMO MSC/Circ. 1087 is to be referred to concerning the stowage and segregation of containers containing dangerous goods.

8.3 Drainage arrangement at the coaming

8.3.1 Drain openings are to be arranged at the ends of drain channels and are to be provided with effective means for preventing ingress of water from outside, such as non-return valves or equivalent. It is unacceptable to connect fire hoses to the drain openings for this purpose.

8.3.2 If drain channels are provided inside the line of gasket by means of a gutter bar or vertical extension of the hatch side and end coaming, drain openings are to be provided at appropriate positions of the drain channels.

8.3.3 Cross-joints of multi-panel covers are to be arranged with drainage of water from the space above the gasket and a drainage channel below the gasket.

8.3.4 If a continuous outer steel contact between cover and ship structure is arranged, drainage from the space between the steel contact and the gasket is also to be provided.

8.3.5 Drain openings in hatch coamings are to be arranged with sufficient distance to areas of stress concentration (e.g. hatch corners, transitions to crane posts).

Section 9

Hatch Coaming Strength Criteria

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

P_A = Horizontal weather design load, in [N/mm^2], as defined in Section 6, Cl.6.2.2.1.

P_{Coam} = Horizontal weather design load applicable to coamings of Type-2 ships, as defined in Section 6, Cl.6.2.2.2.

9.1 Coaming heights

9.1.1 The height of the coamings for hatchways closed by weathertight covers is to be not less than

— 600[mm] in Position 1

— 450[mm] in Position 2

Coaming heights are to be measured from the upper surface of the deck or top of the deck sheathing, if fitted.

9.1.2 The height of coamings less than that required as per Cl.9.1.1 may be accepted on special consideration provided the construction of the hatch covers is such as to ensure that the safety of the ship is not impaired in any sea condition. Such reduced coaming heights will also have to be approved by the concerned National Authorities.

9.1.3 Height of coamings may be required to be increased on certain ships e.g., ships of Type 'B-100' or Type 'B-60' where it is found necessary to satisfy the requirements of floatability stipulated in the ICLL 1966 as amended by the Protocol of 1988.

9.1.4 Coamings are not required for hatchways below the freeboard deck or within weathertight closed superstructures unless they are required for strength purposes.

9.2 Hatch coaming construction

9.2.1 Hatch side coamings are to extend to the lower edge of the deck beams. Side coamings not forming a part of continuous girders, are to extend two frame spaces beyond the hatch ends below the deck.

9.2.2 Hatch end coamings when not in line with the deck transverses are to extend below the deck, at least three longitudinal frame spaces beyond the side coaming.

9.2.3 Continuous hatchway coamings or coamings forming an effective part of the deck girder system are to be made from steel of same tensile strength as that of the deck plating.

9.2.4 If the junction of hatch coamings forms a sharp corner, the side and end coamings are to be extended in the form of tapered brackets in longitudinal and transverse directions respectively. Alternatively tapered brackets may be omitted, provided that the hatch coamings at the corners of the hatchways are well rounded and are joined to the deck by full penetration welds. Suitable supporting brackets are to be provided if the deck overhang at the corner is excessive.

9.2.5 Extension brackets or rails arranged approximately in line with the cargo hatch side coamings and intended for the stowage of steel hatch covers are not to be welded to deckhouse, mast house or to each other unless

they form a part of the longitudinal strength members. The ends of supporting structures of hatch cover stowage rails are not to end abruptly and are to be tapered by suitable end brackets.

9.2.6 Design of local details not covered in this section are to be suitable for the purpose of transferring the loads on the hatch covers to the hatch coamings and through them, to the deck structures below. Hatch coamings and supporting structures are to be adequately stiffened to accommodate the loading from hatch covers, in longitudinal, transverse and vertical directions.

9.2.7 The stresses in the under-deck structures induced by the load transmitted by the stays are not to be greater than the allowable stresses for coaming stays indicated in Cl.7.2.1.

9.2.8 The coamings are to be satisfactorily stiffened against buckling.

9.2.9 Unless otherwise stated, weld connections and materials for hatch coaming are to be dimensioned and selected in accordance with the requirements of Part 3, Chapter 16 .

9.2.10 Coamings are to be stiffened on their upper edges with a stiffener suitably shaped to fit the hatch cover closing appliances.

9.2.11 Coamings which are 600[mm] or more in height is to be stiffened by a horizontal stiffener. Where the unsupported height of a coaming exceeds 1.2[m], additional stiffeners is to be arranged.

9.2.12 When two hatches are close to each other, under deck stiffeners are to be fitted to connect the longitudinal coamings in order to maintaining the continuity of their strength.

9.2.13 Where watertight metallic hatch covers are fitted, other arrangements of equivalent strength may be adopted.

9.2.14 Cut outs in the top of hatch coamings are to be avoided. They are to be circular or elliptical in shape where it is necessary for the securing devices. Also, any local reinforcements should be given a tapered transition in the longitudinal direction with a taper the rate of which should not exceed 1 in 3.

9.2.15 Cut-outs and drain holes are to be avoided in the hatch side coaming continuation brackets. Where these are necessary the size, shape and position will be specially considered.

9.2.16 On ships carrying deck cargo such as timber, coal or coke, coaming stays are to be spaced not more than 1.5[m] apart.

9.3 Local net plate thickness of coamings

9.3.1 The net thickness of weather deck hatch coamings, in [mm], is not be less than the greater of the following values:

(1) For Type-1 ships:

$$t = 0.0142.s\sqrt{\frac{P_A.10^3}{0.95.R_{eH}}} \text{ with } t \geq t_{min}$$

$$t_{min} = 6 + \frac{L_1}{100}$$

(2) For Type-2 ships:

$$t = 0.016.s\sqrt{\frac{P_{Coam}.10^3}{0.95.R_{eH}}} \text{ with } t \geq t_{min}$$

$$t_{min} = 9.5$$

where

s = Stiffener spacing, in [mm].

L_1 = L need not be taken greater than 300[m].

Longitudinal strength aspects are to be verified according to Chapter 6.

9.4 Net scantling of stiffeners of coamings

9.4.1 The stiffeners of coamings must be continuous at the coaming stays. For stiffeners with both ends constraint, the elastic net section modulus Z [cm^3] and net shear area A_{shr} [cm^2], calculated on the basis of net thickness, is not be less than:

(1) For Type-1 ships:

$$Z = \frac{P_A.s.\ell^2.10^3}{f_{bc}.R_{eH}}$$

$$A_{shr} = \frac{10.P_A.s.\ell}{R_{eH}}$$

where

s = Stiffener spacing, in [mm].

ℓ = Stiffener span, in [m], to be taken as the spacing of coaming stays.

f_{bc} = 12, in general.

= 8, for the end spans of stiffeners sniped at the coaming corners.

(2) For Type-2 ships:

$$Z = \frac{1.21\, P_{Coam}.\, s.\, \ell^2.\, 10^3}{f_{bc}.\, c_p.\, R_{eH}}$$

where:

f_{bc} = 16 in general

= 12 for the end spans of stiffeners sniped at the coaming corners

s = Stiffener spacing, in [mm].

ℓ = Stiffener span, in [m].

c_p = Ratio of the plastic section modulus to the elastic section modulus of the stiffeners with an attached plate breadth, in [mm], equal to 40 t, where 't' is the plate net thickness.

= 1.16 in the absence of more precise evaluation

9.4.2 For sniped stiffeners of coaming at hatch corners, section modulus and shear area at the fixed support have to be increased by 35%. The gross thickness of the coaming plate, in [mm], at the sniped stiffener end is not be less than:

$$t_{gr} = 19.6\sqrt{\frac{P_A.\, s(\ell - 0.0005s)}{R_{eH}}}$$

9.4.3 Horizontal stiffeners on hatch coamings, which are part of the longitudinal hull structure, are to be designed according to Chapter 7, Section 5.

9.5 Coaming stays

Coaming stays are to be designed for the loads transmitted through them and permissible stresses according to Cl.7.2.1.

9.5.1 Coaming stay section modulus and web thickness

At the connection with deck, the net section modulus Z [cm^3] and the net thickness t_w [mm], of the coaming stays designed as beams with flange (examples 1 and 2 are shown in Figure 9.5.1) are to be taken not less than:

$$Z = \frac{P.\, s_c.\, H_c^2.\, 10^3}{1.9\, R_{eH}}$$

$$t_w = \frac{2.\, P.\, s_c.\, H_c}{R_{eH}.\, h_w}$$

where:

s_c = Spacing of coaming stays in [mm].

H_c = Height of coaming stays in [m].

h_w = Depth of coaming stay in [mm], at the connection with the deck.

P = Pressure on coaming, in [N/mm^2], taken as P_A defined in 6.2.2.1 in general and as P_{Coam} defined in 6.2.2.2 for Type-2 ships.

For other designs of coaming stays, such as those shown in Figure 9.5.1, examples 3 and 4, the stresses are to be determined through FEM. The calculated stresses are to comply with the permissible stresses according to Cl.7.2.1.

Coaming stays are to be supported by appropriate substructures. For calculating the section modulus of coaming stays, their face plate area is to be taken into account only when it is welded with full penetration welds to the deck plating and adequate underdeck structure is fitted to support the stresses transmitted by it.

Stay webs are to be connected to the deck plating by fillet welds on both sides with a throat thickness of $a = 0.44 t_w [mm]$ or with a leg length of $t_{leg} = 0.62 t_w [mm]$.

For Type-2 ships, toes of stay webs are to be connected to the deck plating with full or partial penetration double bevel welds extending over a distance not less than 15% of the stay width for $t_w > 10 [mm]$.

9.5.2 Coaming stays under friction load

For coaming stays, which transfer friction forces at hatch cover supports, fatigue strength is to be considered. (Also Refer Cl.9.11.2).

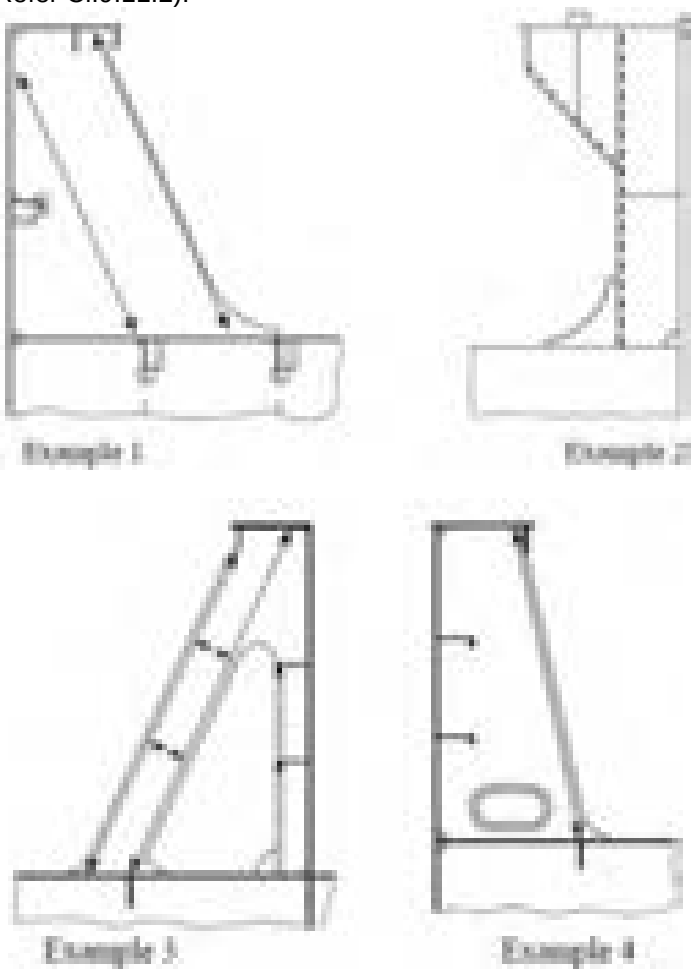


Figure 9.5.1: Examples of coaming stays

9.6 Further requirements for hatch coamings

9.6.1 Longitudinal strength

9.6.1.1 Hatch coamings which are part of the longitudinal hull structure are to be designed according to the requirements of Chapter 6.

9.6.1.2 For structural members welded to coamings and for cutouts in the top of coamings sufficient fatigue strength is to be verified.

9.6.1.3 Longitudinal hatch coamings with a length exceeding 0.1L[m] are to be provided with tapered brackets or equivalent transitions and a corresponding substructure at both ends. At the end of the brackets they are to be connected to the deck by full penetration welds of minimum 300[mm] in length.

9.7 Extend of coaming plates

9.7.1 Coaming plates are to extend to the lower edge of the deck, else hatch side girders are to be fitted that extend to the lower edge of the deck beams. Extended coaming plates and hatch side girders are to be flanged or fitted with face bars or half-round bars. Figure 9.7.1 gives an example.

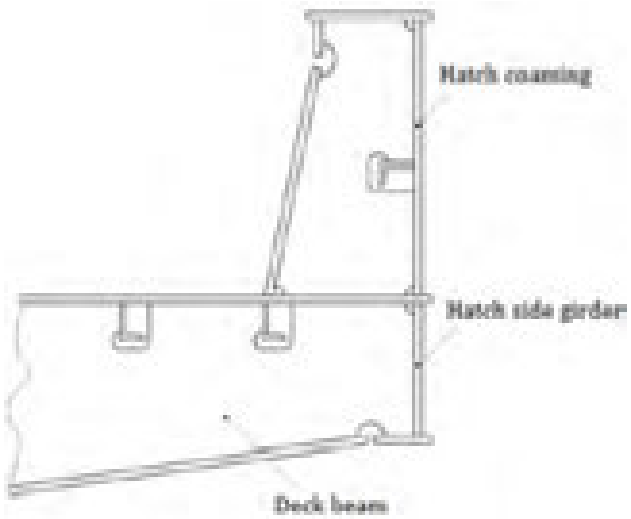


Figure 9.7.1: Example for a hatch side girder

9.8 Closing arrangement

9.8.1 Securing devices

9.8.1.1 Securing devices between cover and coaming and at cross-joints are to be installed to provide weather tightness. Sufficient packing line pressure is to be maintained. Where the packing line pressures is greater than 5[N/mm], it is to be specified in the drawings.

9.8.1.2 Securing devices must be appropriate to bridge displacements between cover and coaming due to hull deformations.

9.8.1.3 Securing devices are to be of reliable construction and effectively attached to the hatchway coamings, decks or covers. Individual securing devices on each cover are to have approximately the same stiffness characteristics.

9.8.1.4 Sufficient number of securing devices is to be provided at each side of the hatch cover considering the requirements of Section 7, Cl.7.5.2. This applies also to hatch covers consisting of several parts.

9.8.1.5 Panel hatch covers are to be secured by appropriate devices (bolts, wedges or similar) suitably spaced alongside the coamings and between cover elements.

9.8.1.6 Hatch covers provided with special sealing devices, insulated hatch covers, flush hatch covers and those having coamings of a reduced height (Refer Cl.9.1.1) are specially considered.

9.8.1.7 In the case of hatch covers carrying containers, the scantlings of the closing devices are to take into account the possible upward vertical forces transmitted by the containers.

9.8.1.8 Securing arrangements with reduced scantlings may be accepted provided it can be demonstrated that the possibility of water reaching the deck is negligible.

9.8.2 Arrangements

9.8.2.1 Securing arrangements together with suitable gasketing material are to ensure weather tightness of the covers to the satisfaction of the surveyors. The arrangements are to be such as to ensure that the weather tightness can be maintained in any sea condition.

9.8.2.2 At cross-joints of multi-panel covers, (male/female) vertical guides are to be fitted to prevent excessive relative vertical deflections between loaded/unloaded panels.

9.8.2.3 The securing and stopping devices are to be arranged so as to ensure sufficient compression on gaskets between hatch covers and coamings and between adjacent hatch covers.

9.8.2.4 The spacing of the securing arrangements is not to be greater than 6[m].

9.8.2.5 Arrangement and spacing are to be determined with due attention to the effectiveness for weather-tightness, depending upon the type and the size of the hatch cover, as well as on the stiffness of the cover edges between the securing devices.

9.8.3 Rod cleats

Where rod cleats are fitted, resilient washers or cushions are to be incorporated.

9.8.4 Hydraulic cleats

In case where hydraulic arrangement is provided for securing the cleats, the system is to remain mechanically locked in closed position in the event of a failure of the hydraulic system.

9.8.5 Materials

Materials used in the manufacturer of stoppers or securing devices including welding electrodes are to be in accordance with relevant requirements given in Part 2. Specifications of the materials are to be shown in the drawings of the hatch covers.

9.9 Cross sectional area of the securing devices

9.9.1 The gross cross-sectional area, in [cm^2] of the securing devices is not to be less than:

$$A = 0.28.q.S_{SD}.k_{\ell}$$

where:

q = Packing line pressure, in [N/mm], not to be less than 5[N/mm].

S_{SD} = Spacing between securing devices in [m], not to be taken less than 2[m].

k_{ℓ} = Material factor, to be taken as:

$$= \left(\frac{235}{R_{eH}}\right)^{e}$$

R_{eH} = The minimum yield strength of the material [N/mm^2], but is not to be taken greater than $0.7\sigma_t$ where σ_t is the tensile strength of the material [N/mm^2].

e = Coefficient taken equal to:

= 0.75 for $R_{eH} > 235$

= 1.00 for $R_{eH} \leq 235$

9.9.2 Where the packing line pressure exceeds 5[N/mm], the cross-sectional area of the securing devices is to be increased in direct proportion.

9.9.3 Rods or bolts are to have a gross diameter not less than 19[mm] for hatchways exceeding 5[m^2] in area.

9.9.4 Securing devices of special design in which significant bending or shear stresses occur may be designed as anti-lifting devices according to 9.10. As load, the packing line pressure 'q' multiplied by the spacing between securing devices 'S_{SD}' is to be applied.

9.10 Anti-lifting devices

9.10.1 The securing devices of hatch covers, on which cargo is to be lashed, are to be designed for the lifting forces resulting from loads according to Cl.6.2.4, refer Figure 9.10.1. Unsymmetrical loadings, which may occur in practice, are to be considered. Under these loadings the equivalent stress, in [N/mm^2], in the securing devices is not to exceed:

$$\sigma_{vm} = \frac{150}{k_{\ell}}$$

Omission of anti-lifting devices will be specially considered.

Note:
The partial load cases given in Table 6.2.4 may not cover all unsymmetrical loadings, critical for hatch cover lifting.

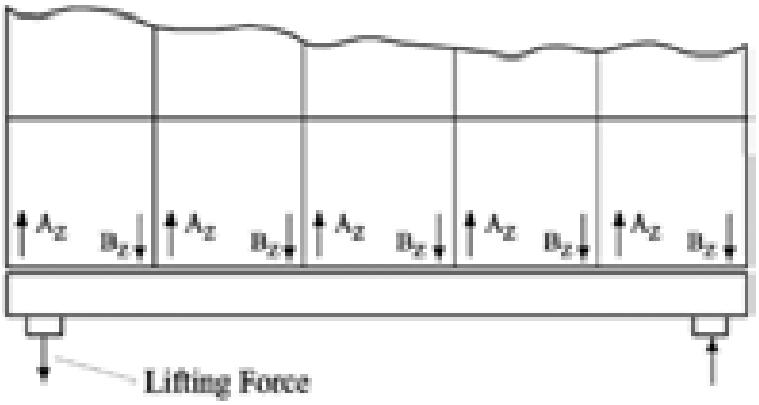


Figure 9.10.1: Lifting forces at a hatch cover

9.11 Hatch cover supports, stoppers and supporting structures

9.11.1 Horizontal mass forces

For the design of hatch cover supports the horizontal mass forces, in [kN], are to be obtained from the following formula.

$$F_h = m.a$$

where:

a = Acceleration, in [m/s^2], to be taken as:

a_x = 0.2g in longitudinal direction

a_y = 0.5g in transverse direction

m = Sum of mass, in [t], of cargo lashed on the hatch cover and of the hatch cover.

The acceleration in longitudinal direction and in transverse direction do not need to be considered as acting simultaneously.

9.11.2 Hatch cover supports

9.11.2.1 For the transmission of the support forces resulting from the load cases specified in Section 6, 6.2 and of the horizontal mass forces specified in Cl.9.11.1, supports are to be provided which are to be designed such that the nominal surface pressures in general do not exceed the following maximum values, P_{n-max} in [N/mm^2]:

In general: $P_{n-max} = d.P_n$

For metallic supporting surfaces not subjected to relative displacements: $P_{n-max} = 3.P_n$

where,

d = Parameter taken equal to:

= 3.75-0.015L, to be taken not greater than, $d_{max} = 3.0$, and not less than d_{min}

d_{min} = Minimum factor, to be taken as:

= 1.0 in general

= 2.0 for partial loading conditions (Refer Cl.6.2.4.4)

P_n = Permissible nominal surface pressure, in [N/mm^2], as defined in Table 9.11.2.

Table 9.11.2: Permissible nominal surface pressure, P_n

Support material	P_n [N/mm²] when loaded by	
	Vertical force	Horizontal force (on stoppers)
Hull structural steel	25	40
Hardened steel	35	50
Plastic materials on steel	50	--

Note:
When the maker of vertical hatch cover support material can provide proof that the material is sufficient for the increased surface pressure, not only statically but under dynamic conditions including relative motion for adequate number of cycles, permissible nominal surface pressure may be relaxed at the discretion of IRS. However, realistic long-term distribution of spectra for vertical loads and relative horizontal motion should be assumed and agreed with IRS.

9.11.2.2 Drawings of the supports are to be submitted. In the drawings of supports the permitted maximum pressure given by the material manufacturer must be specified.

9.11.2.3 Where large relative displacements of the supporting surfaces are to be expected, the use of material having low wear and frictional properties is recommended.

9.11.2.4 The substructures of the supports must be of such a design, that a uniform pressure distribution is achieved.

9.11.2.5 Irrespective of the arrangement of stoppers, the supports must be able to transmit the following force P_h in [kN], in the longitudinal and transverse direction:

$$P_h = \mu . \frac{P_v}{\sqrt{d}}$$

where:

μ =Frictional coefficient, to be taken as:

= 0.50 in general

P_v = Vertical supporting force, in [kN].

9.11.2.6 For non-metallic, low-friction support materials on steel, the friction coefficient may be reduced but not to be less than 0.35 and to the satisfaction of IRS.

9.11.2.7 Supports as well as the adjacent structures and substructures are to be designed such that the permissible stresses according to Cl.7.2.1 are not exceeded.

9.11.2.8 For substructures and adjacent structures of supports subjected to horizontal forces P_h, a fatigue strength analysis is to be carried out.

9.11.3 Hatch cover stoppers

9.11.3.1 Hatch covers are to be sufficiently secured against horizontal shifting. Stoppers are to be provided for hatch covers on which cargo is carried.

9.11.3.2 The greater of the loads resulting from Cl.6.2.2 and Cl.9.11.1 is to be applied for the dimensioning of the stoppers and their substructures.

9.11.3.3 The permissible stress in stoppers and their substructures, in the cover, and of the coamings is to be determined according to Cl.7.2.1. In addition, the provisions in Cl.9.11.2 are to be observed.

9.11.3.4 Specifically for Type-2 ships, the following additional requirements are to be complied with:

- Hatch covers are to be effectively secured, by means of stoppers, against the transverse forces arising from a pressure of 175 [kN/m^2].

- With the exclusion of No.1 hatch cover, hatch covers are to be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 175[kN/m^2].

- No. 1 hatch cover is to be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 230[kN/m^2].

- This pressure may be reduced to 175 [kN/m^2] when a forecastle is fitted in accordance with Part 5, Chapter 1, 1.7.

- The equivalent stress is not to exceed the allowable value of $0.8R_{eH}$:

  — in stoppers and their supporting structures, and

  — calculated in the throat of the stopper welds.

Section 10

Hatchways Closed by Portable Covers and Secured Weathertight by Tarpaulins and Battening Devices

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

10.1 Pontoon Covers

10.1.1 Where steel pontoon covers are used in place of portable beams and covers, the maximum allowable stress and deflection under the minimum design pressures given in Table 10.1.1.1, and the minimum required top plate thickness are as follows:

a) Maximum allowable stress $0.68\ R_{eH}$ [N/mm^2].

b) Maximum allowable deflection 0.0044 times the span of primary supporting member in [m].

c) Top cover plate thickness 0.01s, but not less than 6[mm]

where

s = Stiffener spacing in [mm].

Table 10.1.1.1: Pontoon cover minimum design pressures in [kN/m^2]

1) For ships of 100[m] in length and above: (a) Position 1 hatch covers located in the forward quarter of the ships length is to be designed for wave pressures at the forward perpendicular, calculated from the following equation: $$p = 49.05 + 9.81(L_L - 100)a$$ Where, L_L is L for ships of not more than 340 m but not less than 100 m in length and equal to 340 m for ships of more than 340 m in length: a = 0.0074 for Type B freeboard ships = 0.0363 Ships assigned reduced freeboard The pressure, p, is to be reduced linearly to 34.3[kN/m^2] at the end of the forward quarter's length, as shown in Table 10.1.1.2 Summary of pontoon cover minimum design pressures. The design pressure used for each hatch cover panel is to be determined at its mid-point location. (b) All other position 1 hatch covers is to be designed to 34.3 [kN/m^2]. (c) Position 2 hatch covers is to be designed to 25.51[kN/m^2]. (d) For a position 1 hatchway is located at least one superstructure standard height higher than the freeboard deck, it is to be designed to 34.3[kN/m^2].
2) For ships 24[m] in length: (a) Position 1 hatch covers located in the forward quarter of the ship's length is to be designed for wave pressures of 23.84[kN/m^2] at the forward perpendicular and reduced linearly to 19.6[kN/m^2] at the end of the forward quarter's length, as shown in Table 10.1.1.2 Summary of pontoon cover minimum design pressures. The design pressure used for each hatch cover panel is to be determined at its mid-point location. (b) All other position 1 hatch covers shall be designed to 19.6[kN/m^2].

(c) Position 2 hatch covers shall be designed to 14.7[kN/m^2].

(d) Where a position 1 hatchway is located at least one superstructure standard height higher than the freeboard deck, it may be designed to 19.6[kN/m^2].

3) For ships between 24[m] and 100[m] in length, and for positions between FP and 0,25L, wave pressures is to be obtained by linear interpolation of the values shown in Table 10.1.1.2 Summary of pontoon cover minimum design pressures.

Table 10.1.1.2: Summary of pontoon cover minimum design pressures in [kN/m^2]

Deck Location	**Longitudinal Position**		
	FP	**0.25L**	**Aft of 0.25L**
L_L>100[m]			
Freeboard deck	Equation given in Table 10.1.1.1	34.3	34.3
Superstructure deck	34.3		25.51
L_L=100[m]			
Freeboard deck	49.05	34.3	34.3
Superstructure deck	34.3		25.51
L_L=24[m]			
Freeboard deck	23.84	19.6	19.6
Superstructure deck	19.6		14.7

10.2 Cleats

10.2.1 Cleats are to be of an approved pattern, at least 65[mm] wide, with edges so rounded as to minimize damage to the wedges, and are to be spaced not more than 600[mm] from centre to centre; the first and last cleats along each side or end being not more than 150[mm] from the hatch corners. Cleats are to be so set as to fit the taper of the wedges.

10.3 Wedges

10.3.1 Battens and wedges are to be efficient and in good condition. Wedges are to be of tough wood, generally not more than 200[mm] in length and 50[mm] in width. They should have a taper of not more than 1 in 6 and should not be less than 13[mm] thick at the point.

10.4 Battens and locking bars

10.4.1 For all hatchways in Positions 1 and 2, battens or transverse bars in steel or other equivalent means are to be provided in order to efficiently secure each section of the portable hatch covers and independently after the tarpaulins are battened down. Portable covers of more than 1.5[m] in length are to be secured by at least two such securing devices. Where hatchway covers extend over intermediate supports, steel bars or their equivalent are to be fitted at each end of each section of covers. At all other hatchways in exposed positions on weather decks, ring bolts or other fittings suitable for lashings are to be provided.

10.4.2 Acceptable equivalent means to steel bars are to consist of devices and materials which will provide strength and stiffness equivalent to that of steel. Steel wire ropes cannot be regarded as satisfactory equivalent means. Care is to be taken that tarpaulins are adequately protected from the possibility of damage arising from the usage of securing devices which do not provide a flat bearing surface.

10.5 Tarpaulins

10.5.1 At least two layers of tarpaulin in good condition are to be provided for each hatchway in Positions 1 and 2. Tarpaulins are to be free from jute and are to be waterproof and of ample strength.

The minimum mass of the material before treatment is to be:

— 0.65[kg/m^2] if the material is to be tarred

— 0.60[kg/m^2] if to be chemically dressed

— 0.55[kg/m^2] if to be dressed with black oil

A certificate to this effect is to be supplied by the makers of the tarpaulins. Special consideration will be given to the use of synthetic materials for tarpaulins.

10.6 Wooden hatch covers

10.6.1 Wooden hatch covers are to have a finished thickness of not less than 60[mm] in association with an unsupported span of 1.5 [m] and of not less than 82[mm] with 2[m] unsupported span. The thickness for the intermediate span may be obtained by linear interpolation. Where the tween deck height exceeds 2.6 [m], the thickness of the wood hatch cover is to be increased at the rate of 16.5% per metre of excess tween deck height.

10.6.2 The ends of all wooden hatch covers are to be protected by encircling with galvanized steel bands, about 65[mm] wide and 3[mm] thick, efficiently secured.

10.7 Portable hatch beams

10.7.1 The section modulus and the moment of inertia of the portable hatch beams stiffened at their upper and lower edges by continuous flat bars are to satisfy the requirements of 10.1 for pontoon hatch covers.

10.7.2 Where the cross section of portable beams is not constant along the span, Section 7,7.8 may be used to determine required scantlings.

10.7.3 The ends of the web plates are to be doubled, or inserts fitted for at least 180[mm] along the length of the web.

10.7.4 The beams which carry the ends of wood or steel hatch covers, a 50[mm] vertical flat is to be arranged on the upper face plate. The width of bearing surface for hatch covers is to be not less than 65[mm].

10.7.5 Sliding hatch beams are to be provided with an efficient device for locking them in their correct fore and aft positions when the hatchway is closed.

10.8 Carriers or Sockets

10.8.1 Carriers or sockets, or other suitable arrangements are to be provided as means of the efficient fitting and securing of portable hatch beams. The arrangements are to be of suitable construction and the width of the bearing surface measured along the axis of the beam is not to be less than 75[mm] unless the carriers are of an interlocking type with the beam ends.

10.8.2 Carriers for beams are to extend to the deck level or the coamings are to be fitted with stiffeners or external brackets in way of each beam.

Section 11

Side Shell and Stern Doors

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

11.1 Application

11.1.1 These rule gives requirements for the arrangement, strength and securing of cargo and service doors in the ship side shell, abaft the collision bulkhead and stern doors leading into enclosed spaces, below the freeboard deck and in enclosed superstructures.

11.2 Arrangement

11.2.1 Stern doors for passenger vessels are to be situated above the freeboard deck. Stern doors for Ro-Ro cargo ships and side shell doors may be either below or above the freeboard deck.

11.2.2 Side shell doors and stern doors are to be so fitted as to ensure tightness and structural integrity commensurate with their location and the surrounding structure.

11.2.3 The sill of any side shell door is, in general, not to be below a line drawn parallel to the freeboard deck at side, with its lowest point 230[mm] above the uppermost loadline.

Special consideration shall be given to preventing the spread of any leakage water over the deck. A flat bar welded to the deck and provision of scuppers would be an acceptable arrangement.

11.2.4 Where the sill of any side shell door is below the line specified in Cl.11.2.3 satisfactory additional features are to be fitted to maintain the watertight integrity. Fitting of a second door of equivalent strength and water tightness is one acceptable arrangement. In that case leakage detection device should be provided in the compartment between the two doors. Further, drainage of this compartment to the bilges controlled by an easily accessible screw down valve, should be arranged.

11.2.5 Doors should preferably open outwards.

11.3 Definitions

11.3.1 *Securing device*: a device used to keep the door closed by preventing it from rotating about its hinges or about pivoted attachments to the ship.

11.3.2 *Supporting devices*: a device used to transmit external or internal loads from the door to a securing device and from the securing device to the ship's structure, or a device other than a securing device, such as a hinge, stopper or other fixed device, that transmits loads from the door to the ship's structure.

11.3.3 *Locking device*: a device that locks a securing device in the closed position.

11.4 Design loads

11.4.1 The design forces, in [N], considered for the scantlings of primary members, securing and supporting devices of side shell doors and stern doors are to be not less than:

11.4.1.1 Design forces for securing or supporting devices of doors opening inwards:

External force: $F_e = AP_e 10^6 + F_p$

Internal force: $F_i = F_0 + 10W$

11.4.1.2 Design forces for securing or supporting devices of doors opening outwards:

External force: $F_e = AP_e 10^6$

Internal force: $F_i = F_0 + 10W + F_p$

11.4.1.3 Design forces for primary members:

External force: $F_e = AP_e 10^6$

Internal force: $F_i = F_0 + 10W$

whichever is the greater,

where,

A = Area, in [m^2], of the door opening.

W = Mass of the door, in [kg].

F_p = Total packing force in [N]. Packing line pressure is normally not to be taken less than 5[N/mm].

F_0 = The greater of F_c and 5000 A [N].

F_c = Accidental force, in [N], due to loose cargo etc., to be uniformly distributed over the area A and not to be taken less than 300000[N]. For small doors such as bunker doors and pilot doors, the value of F_c may be appropriately reduced. However, the value of F_c may be taken as zero, provided an additional structure such as an inner ramp is fitted, which is capable of protecting the door from accidental forces due to loose cargoes.

P_e = External design pressure, in [N/mm^2], determined at the centre of gravity of the door opening and not taken less than:

$$P_e = 0.01(T_1 - Z_G) + 0.025 \text{ for } Z_G < T_1$$

$$P_e = 0.025 \text{ for } Z_G > T_1$$

where,

Z_G = Height of the centre of area of the door in [m], above the baseline.

T_1 = Draught in [m], at the highest subdivision load line.

Moreover, for stern doors of ships fitted with bow doors, P_e in [N/mm^2], is not to be taken less than:

$$P_e = 0.6\lambda C_H\left(0.8 + 0.6\sqrt{L}\right)^2 10^{-3}$$

where,

λ = Coefficient depending on the area where the ship is intended to be operated:

= 1 for sea going ships,

= 0.8 for ships operated in coastal waters,

= 0.5 for ships operated in sheltered waters.

C_H = 0.0125L for L < 80[m]

= 1 for L ≥ 80[m]

L = Ship's length in [m], but need not be taken greater than 200[m].

11.5 Strength Criteria

11.5.1 Scantlings of the primary members, securing and supporting devices of side shell doors and stern doors are to be determined considering simply supported end connections and the design loads defined in 11.4. The following permissible stresses, in [N/mm^2], are not to be exceeded:

$$\text{Shear stress, } \tau = \frac{80}{k}$$

$$\text{Bending stress, } \sigma = \frac{120}{k}$$

$$\text{Equivalent stress, } \sigma_c = \sqrt{\sigma^2 + 3\tau^2} = \frac{150}{k}$$

where k, the material factor, is not to be taken less than 0.72 unless a direct strength analysis with regard to relevant modes of failures is carried out.

11.5.2 The buckling strength of primary members is to be verified as per Chapter 9.

For doors acting as ramps in harbour condition, additional buckling check according to Chapter 9 Section 4, is required.

For steel-to-steel bearings in securing and supporting devices, the nominal bearing pressure, σ_b in [N/mm^2], to follow the formulae below:

$$\sigma_b \leq 0.8\,\sigma_y$$

where,

$$\sigma_b = F/A_b$$

with:

F = Design force, in [N], defined in 11.4.

A_b = Projected bearing area, in [mm^2].

σ_y = Yield stress of the bearing material, in [N/mm^2].

For other bearing materials, the permissible bearing pressure is to be determined according to the manufacturer's specification.

11.5.3 The arrangement of securing and supporting devices is to be such that threaded bolts do not carry support forces. The maximum tension in way of threads of bolts not carrying

support forces is not to exceed 125/k [N/mm^2], where k is defined in Cl.11.5.1.

11.6 Scantlings

11.6.1 General

11.6.1.1 The strength of side shell doors and stern doors is to be commensurate with that of the surrounding structure.

11.6.1.2 Side shell doors and stern doors are to be adequately stiffened and means are to be provided to prevent any lateral or vertical movement of the doors when closed. Adequate strength is to be provided in the connections of the lifting/maneuvering arms and hinges to the door structure and to the ship's structure.

11.6.1.3 Where doors also serve as vehicle ramps, the design of the hinges should take into account the ship's angle of trim and heel which may result in uneven loading on the hinges.

11.6.1.4 Shell door openings are to have well rounded corners and adequate compensation is to be arranged with web frames at sides and stringers or equivalent above and below.

11.6.1.5 Arrangements are to be such that the sea pressure is transferred directly to the hull coamings. For doors with opening area A < 12[m^2], securing bolts or similar devices may be accepted as carriers of sea pressure to the hull coamings.

11.6.1.6 If a door is divided into separate sections, each section is to have full strength independent of the other sections.

11.6.2 Plating and secondary stiffeners

11.6.2.1 The gross thickness of the door plating is not to be less than the required thickness for the side shell plating, using the door stiffener spacing, and not less than the minimum required thickness of shell plating as per Chapter 7, Section 3.

Where doors serve as vehicle ramps, the plating thickness is to be not less than required for vehicle decks.

11.6.2.2 The gross section modulus of secondary horizontal or vertical stiffeners is not to be less than that required for side framing as per Chapter 7, Section 5 for the side, using the door stiffener spacing.

Where doors serve as vehicle ramps, the stiffener scantlings are not to be less than required for vehicle decks. Consideration is to be given, where necessary, to differences in fixity between ship's frames and door stiffeners.

11.6.3 Primary Structure

11.6.3.1 The secondary stiffeners are to be supported by primary members constituting the main stiffening of the door.

11.6.3.2 The primary members and the hull structure in way are to have sufficient stiffness to ensure structural integrity of the boundary of the door.

11.6.3.3 Scantlings of the primary members are to be based on direct strength calculations in association with the design forces given in 11.4 and permissible stresses given in 11.5. In general, formulae for simple beam theory may be used to calculate the loads and stresses in primary supporting members. Members are to be considered to have simply supported end connections.

11.6.3.4 The webs of primary members (girders and stringers) are to be adequately stiffened, preferably in a direction perpendicular to the shell plating.

11.6.3.5 Sufficient stiffness in way of door edges and hull structure are provided, to ensure weathertight integrity. Edge stiffeners/girders are to be adequately stiffened against rotation and are to have a moment of inertia, in [cm^4], not less than:

$$I = 8.P_{\ell}.d^4$$

where:

d = Distance between closing devices in [m].

P_{ℓ} = Packing line pressure along edges in [N/mm], not to be taken less than 5[N/mm].

For edge girders supporting main door girders between securing devices, the moment of inertia is to be increased in relation to the additional force.

11.7 Securing and supporting of doors

11.7.1 General

11.7.1.1 Side shell doors and stern doors are to be fitted with adequate means of securing and

supporting so as to be commensurate with the strength and stiffness of the surrounding structure. The hull supporting structure in way of the doors is to be suitable for the same design loads and design stresses as the securing and supporting devices.

Where packing is required, the packing material is to be of a comparatively soft type and the supporting forces are to be carried by the steel structure only. Other types of packing may be specially considered.

11.7.1.2 Maximum design clearance between securing and supporting devices is not generally to exceed 3[mm]. A means is to be provided for mechanically fixing the door in the open position, taking into account the self-weight of the door and a minimum wind pressure of 1.5[kN/m^2] acting on the maximum projected area in the open position.

11.7.1.3 Only the active supporting and securing devices having an effective stiffness in the relevant direction are to be included and considered to calculate the reaction forces acting on the devices.

11.7.1.4 Small and/or flexible devices such as cleats intended to provide local compression of the packing material are not generally to be included in the calculations in Cl.11.7.2.2. The number of securing and supporting devices are generally to be the minimum practical whilst taking into account the requirement for redundant provision given in Cl.11.7.2.3 and the available space for adequate support in the hull structure.

11.7.1.5 Securing devices are to be equipped with positive locking arrangements. Arrangements are to be such that the securing devices are retained in the closed position within design limits of inclination, vibration and other motion-induced loads and in the event of loss of any actuating power supply.

11.7.2 Scantlings

11.7.2.1 Securing and supporting devices are to be adequately designed so that they can withstand the reaction forces within the permissible stresses given in Cl.11.5.1.

11.7.2.2 The distribution of the reaction forces acting on the securing devices and supporting devices may require direct calculations taking into account the flexibility of the hull structure and the actual position of the supports.

11.7.2.3 The arrangement of securing devices and supporting devices in way of these securing devices is to be designed with redundancy so that in the event of failure of any single securing or supporting device the remaining devices are capable to withstand the reaction forces without exceeding by more than 20% of the permissible stresses as given in Cl.11.5.1.

11.7.2.4 All load transmitting elements in the design load path, from the door through securing and supporting devices into the ship's structure, including pins, support brackets, back-up brackets and welded connections, are to be to the same strength standard as required for the securing and supporting devices.

11.8 Securing and locking arrangement

11.8.1 Systems for operation

11.8.1.1 Securing devices are to be simple to operate and easily accessible. Securing devices are to be equipped with mechanical locking arrangement (self-locking or separate arrangement), or are to be of the gravity type. The opening and closing systems as well as securing and locking devices are to be interlocked in such a way that they can only operate in the proper sequence.

Where hinges are used as closing devices they shall be well integrated into the door structure.

11.8.1.2 Doors which are located partly or totally below the freeboard deck with a clear opening area greater than 6[m^2] are to be provided with an arrangement for remote control, from a position above the freeboard deck of:

— the closing and opening of the doors,

— associated securing and locking devices.

For doors which are required to be equipped with a remote control arrangement, indication of the open/closed position of the door and the securing and locking device is to be provided at the remote control stations. The operating panels for operation of doors are to be inaccessible to unauthorized persons. A notice plate, giving instructions to the effect that all securing devices are to be closed and locked before leaving harbour, is to be placed at each operating panel and is to be supplemented by warning indicator lights.

Alternative means of securing are to be provided for emergency use in case of failure of the power systems.

11.8.1.3 Where hydraulic securing devices are applied, the system is to be mechanically lockable in closed position. This means that in the event of loss of the hydraulic fluid, the securing devices remain locked. The hydraulic system for securing and locking devices is to be isolated from other hydraulic circuits, when in closed position.

11.8.2 Systems for indication/monitoring

11.8.2.1 The requirements in Cl.11.8.2.2 to Cl.11.8.2.5 apply to doors in the boundary of special category spaces or ro-ro spaces, as defined in Part 6, Chapter 1, Section 3, through which such spaces may be flooded.

For cargo ships, where no part of the door is below the uppermost waterline and the area of the door opening is not greater than 6[m^2], the requirements of Cl.11.8.2.2 to Cl.11.8.2.5, need not be applied.

11.8.2.2 Separate indicator lights and audible alarms are to be provided on the navigation bridge and on each operating panel to indicate that the doors are closed and that their securing and locking devices are properly positioned.

The indication panel is to be provided with a lamp test function. It shall not be possible to turn off the indicator light.

11.8.2.3 The indicator system is to be designed on the fail safe principle and is to indicate by visual alarms if the door is not fully closed and not fully locked and by audible alarms if securing devices become open or locking devices become unsecured. The power supply for the indicator and alarm system is to be independent of the power supply for operating and closing arrangements of the doors and is to be provided with a backup power supply from the emergency source of power or other secure power supply such as a UPS with a minimum capacity of 30 minutes.

The sensors of the indicator system are to be protected from water, ice formation and mechanical damages.

The requirements for fail safe design principle are to be similar to those for bow doors as indicated in Cl.12.11.2.4.

11.8.2.4 The indication panel on the navigation bridge is to be equipped with a mode selection function "harbour/sea voyage", so arranged that audible alarm is given if the vessel leaves harbour with side shell or stern doors not closed or with any of the securing devices not in the correct position.

11.8.2.5 For passenger ships, a water leakage detection system with audible alarm and CCTV indication to the navigation bridge and to the engine control room of any leakage through the doors.

For cargo ships, a water leakage detection system with audible alarm is to be arranged to provide an indication to the navigation bridge.

11.8.2.6 For ro-ro passenger ships, on international voyages, the special category spaces and ro-ro spaces are to be continuously patrolled or monitored by effective means, such as CCTV surveillance, so that any movement of vehicles in adverse weather conditions and unauthorized access by passengers thereto, can be detected whilst the ship is underway.

11.9 Operating and maintenance manual

11.9.1 An Operating and Maintenance Manual for the side shell and stern doors is to be provided onboard and contain necessary information on:

11.9.1.1 Main particulars and design drawings

— special safety precautions

— details of vessel

— equipment and design loading (for ramps)

— key plan of equipment (doors and ramps)

— manufacturer’s recommended testing for equipment

— description of equipment for

- bow doors
- inner bow doors
- bow ramp/doors
- side doors

- stern doors
- central power pack
- bridge panel
- engine control room panel

11.9.1.2 Service conditions

— limiting heel and trim of ship for loading/unloading

— limiting heel and trim for door operations

— doors/ramps operating instructions

— doors/ramps emergency operating instructions

11.9.1.3 Maintenance and function testing

— schedule and extent of maintenance

— trouble shooting and acceptable clearances

— manufacturer’s maintenance procedures

11.9.1.4 Register of inspections, including inspection of locking, securing and supporting devices, repairs and renewals.

11.9.2 Documented operating procedures for closing and securing side shell and stern doors are to be kept on board and posted at the appropriate places.

11.9.3 This Manual has to be submitted for approval to verify that the above-mentioned items are contained in the OMM and that the maintenance part includes the necessary information with regard to inspections, troubleshooting and acceptance / rejection criteria.

Note: It is recommended that recorded inspections of the door supporting and securing devices be carried out by the ship's personnel at monthly intervals or following incidents that could result in damage, including heavy weather or contact in the region of side shell and stern doors. A record is to be kept and any damage recorded during such inspections is to be reported to IRS.

Section 12

Bow Doors and Inner Doors

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

12.1 Application

12.1.1 These requirements cover the arrangement, strength and securing of bow doors and inner doors leading to a complete or long forward enclosed superstructure or to a long non-enclosed superstructure, where fitted to attain minimum bow height equivalence.

12.1.2 Hull supporting structure in way of the bow doors is to be able to withstand the loads imposed by the bow door securing and supporting devices without exceeding the permissible stresses for the devices given in this section.

12.2 Type of Bow doors

12.2.1 Two types of bow doors have been considered:

12.2.1.1 Visor doors opened by rotating upwards and outwards about a horizontal axis through two or more hinges located near the top of the door and connected to the primary structure of the door by longitudinally arranged lifting arms;

12.2.1.2 Side-opening doors opened either by rotating outwards about a vertical axis through two or more hinges located near the outboard edges or by horizontal translation by means of linking arms arranged with pivoted attachments to the door and the ship. It is anticipated that side-opening bow doors are arranged in pairs.

Other types of bow door will be specially considered in association with the applicable requirements of this section.

12.3 Arrangement

12.3.1 Bow doors are to be situated above the freeboard deck. A watertight recess in the freeboard deck located forward of the collision bulkhead and above the deepest waterline fitted for arrangement of ramps or other related mechanical devices may be regarded as a part of the freeboard deck for the purpose of this requirement.

12.3.2 An inner door is to be fitted. The inner door is to be part of the collision bulkhead. The inner door need not be fitted directly above the bulkhead below, provided it is located within the limits specified for the position of the collision bulkhead. A vehicle ramp may be arranged for this purpose, provided its position is within the limits. (Refer Chapter 3, Section 2, Cl.2.2.1.3. If this is not possible, a separate inner weathertight door is to be installed, as far as practicable within the limits specified for the position of the collision bulkhead.

12.3.3 Bow doors are to be so fitted as to ensure tightness consistent with operational conditions and to give effective protection to inner doors. Inner doors forming part of the collision bulkhead are to be weathertight over the full height of the cargo space and arranged with fixed sealing supports on the aft side of the doors.

12.3.4 Bow doors and inner doors are to be arranged so as to preclude the possibility of the bow door causing structural damage to the inner door or to the collision bulkhead in the case of damage to or detachment of the bow door. If this is not possible, a separate inner weathertight door is to be installed, as indicated in Cl.12.3.2.

12.3.5 The requirements for inner doors are based on the assumption that vehicles are effectively lashed and secured against movement in stowed position.

12.4 Definitions

12.4.1 Securing device: a device used to keep the door closed by preventing it from rotating about its hinges or about pivotted attachments to the ship.

12.4.2 Supporting devices: a device used to transmit external or internal loads from the door to a securing device and from the securing device to the ship's structure, or a device other than a securing device, such as a hinge, stopper or other fixed device, that transmits loads from the door to the ship's structure.

12.4.3 Locking device: a device that locks a securing device in the closed position.

12.5 Bow door design loads

12.5.1 External pressure

12.5.1.1 The external sea pressure, in [N/mm^2], to be considered for the scantlings of primary members, securing and supporting devices of bow doors are to be taken as the greater of the following:

$$P_e = P_{ex} \cdot 10^{-3}$$

$$P_e = 2.75\lambda C_H(0.22 + 0.15\tan\alpha)\left(0.4V\sin\beta + 0.6\sqrt{L}\right)^2 10^{-3}$$

where,

P_{ex} = External sea pressure as given in Chapter 5, Section 4, Cl.4.1.1 to Cl.4.1.3 in [kN/m^2].

V = Maximum service speed, in [knots].

λ = Coefficient depending on the area where the ship is intended to be operated:

= 1 for sea going ships,

= 0.8 for ships operated in coastal waters,

= 0.55 for ships operated in sheltered waters.

C_H = 0.0125L for L < 80[m]

= 1 for L ≥ 80[m]

L = Ship's length in [m] but need not be taken greater than 200[m].

α = Flare angle [deg] at the point to be considered, defined as the angle between a vertical line and the tangent to the side shell plating, measured in a vertical plane normal to the horizontal tangent to the shell plating

β = Entry angle [deg] at the point to be considered, defined as the angle between a longitudinal line parallel to the centerline and the tangent to the shell plating in a horizontal plane. (Refer also Figure 12.5.1 for α and β).

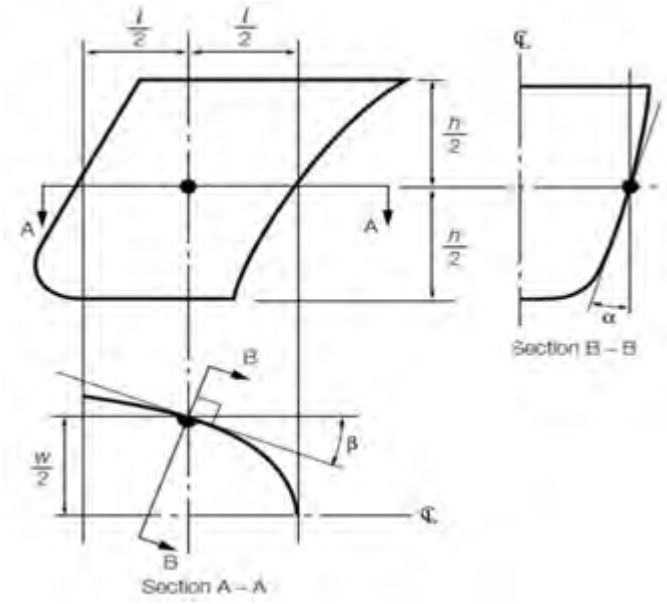


Figure 12.5.1: Definition of α and β angles

12.5.2 External force

12.5.2.1 The design external forces in [kN], considered for the scantlings of securing and supporting devices of bow doors are not to be less than:

$$\text{Longitudinal force, } F_x = P_e A_x 10^3$$

$$\text{Transverse force, } F_y = P_e A_y 10^3$$

$$\text{Vertical force, } F_z = P_e A_z 10^3$$

where,

A_x = Area in [m^2], of the transverse-vertical projection (i.e. in y-z plane) of the door between the levels of the bottom of the door and the top of the upper deck bulwark, or between the bottom of the door and the top of the door, including the bulwark, where it is part of the door, whichever is lesser.

Where the flare angle of the bulwark is at least 15° less than the flare angle of the adjacent shell plating, the height from the bottom of the door may be measured to the upper deck or to the top of the door, whichever is lesser.

In determining the height from the bottom of the door to the upper deck or to the top of the door, the bulwark is to be excluded from the projected area.

A_y = Area in [m^2], of the longitudinal-vertical projection (i.e. in (x-z plane) of the door between the levels of the bottom of the door and the top of the upper deck bulwark, or between the bottom of the door and the top of the door, including the bulwark, where it is part of the door, whichever is lesser.

Where the flare angle of the bulwark is at least 15° less than the flare angle of the adjacent shell plating the height from the bottom of the door may be measured to the upper deck or to the top of the door, whichever is lesser.

A_z = Area in [m^2], of the horizontal projection (i.e. in x-y plane) of the door between the bottom of the door and the top of the upper deck bulwark, or between the bottom of the door and the top of the door, including the bulwark, where it is part of the door, whichever is the lesser.

Where the flare angle of the bulwark is at least 15° less than the flare angle of the adjacent shell plating, the height from the bottom of the door may be measured to the upper deck or to the top of the door, whichever is lesser.

12.5.2.2 If the bulwark is flared inwards such that the horizontal projected area to the top of the bulwark is less than that to the upper deck, then the horizontal projected area to the upper deck is to be considered for the determination of A_z.

h = Height, [m], of the door between the levels of the bottom of the door and the upper deck or between the bottom of the door and the top of the door, whichever is the lesser.

ℓ = Length in [m], of the door at a height h/2 above the bottom of the door.

w = Breadth in [m], of the door at a height h/2 above the bottom of the door.

P_e = External pressure as given in Cl.12.5.1.1 with flare angle and entry angle β measured at a location on the shell h/2 above the bottom of the door and ℓ/2 aft of the intersection of the door with the stem.

12.5.2.3 For bow doors, including bulwark, of unusual form or proportions, e.g. ships with a rounded nose and large stem angles, the areas and angles used for determination of the design values of external forces may require to be specially considered.

12.5.3 Visor door Forces, Moments and Load Cases

12.5.3.1 Closing Moment

For visor doors the closing moment M_y under external loads in [kN-m], is to be taken as:

$$M_y = F_x.a + 10.W.c - F_z.b$$

For visor doors opening outwards, the pivot arrangement are generally to be such that the visor is self-closing under external loads, that is $M_y > 0$. Moreover, the closing moment M_y as given above is to be not less than M_{yo} in [kN-m]:

$$M_{yo} = 10.W.c + 0.1\left(\sqrt{a^2 + b^2}\right)\left(\sqrt{F_x^2 + F_z^2}\right)$$

where:

W = Mass of the visor door in [t].

a = Vertical distance in [m], from visor pivot to the centroid of the transverse vertical projected area of the visor door, as shown in Figure 12.5.3.

b = Horizontal distance in [m], from visor pivot to the centroid of the horizontal projected area of the visor door, as shown in Figure 12.5.3.

c = Horizontal distance in [m], from visor pivot to the centre of gravity of visor mass, as shown in Figure 12.5.3.

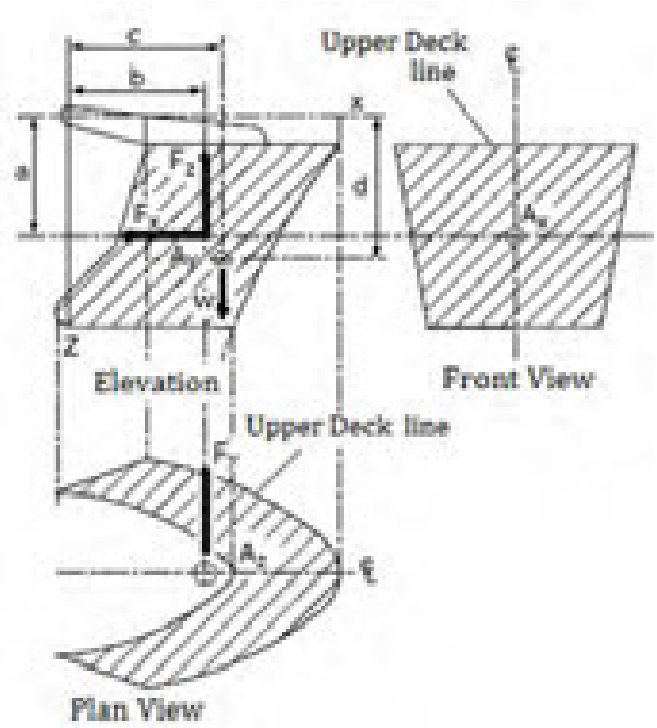


Figure 12.5.3: Bow Door of Visor Type

12.5.3.2 Opening Moment

For visor doors, two securing devices are to be provided at the lower part of the door, each capable of providing the full reaction force required to prevent opening of the door within the permissible stresses given in Cl.12.7.1. The opening moment, M_o in [kN-m], to be balanced by this reaction force, is not to be taken less than:

$$M_o = 5A_x.a + 10.W.d$$

where,

d = Vertical distance [m], from the hinge axis to the centre of gravity of the door, as shown in Figure 12.5.3.

12.5.3.3 Vertical Design Force

For visor doors, the securing and supporting devices excluding the hinges should be capable of resisting the vertical design force F_v in [kN], within the permissible stresses given in Cl.12.7.1 is given below:

$$F_v = F_z - 10W$$

12.5.3.4 Reaction Forces

For visor doors, the reaction forces applied on the effective securing and supporting devices assuming the door as a rigid body are determined for the following combination of external loads acting simultaneously together with the self-weight of the door:

Case 1 : F_x and F_z

Case 2 : $0.7F_y$ acting on each side separately together with $0.7F_x$ and $0.7F_z$

where F_x, F_y and F_z are to be calculated as indicated in Cl.12.5.2 and applied at the centroid of projected areas.

12.5.3.5 Forces acting on the lifting arms

The design of lifting arms of a visor door and its supports is to take into account the static and dynamic forces applied during the lifting and lowering operations, and a minimum wind pressure of 0.0015[N/mm^2].

12.5.4 Side-opening door Reaction forces and Load Cases

12.5.4.1 For side-opening doors, securing devices are to be provided such that in the event of a failure of any single securing device the remaining are capable of providing the full reaction force required to prevent the opening of the door within the permissible stresses given in Cl.12.7.1.

The opening moment about the hinges to be balanced by this reaction force is not to be less than that calculated when the following loads are applied:

— An internal pressure of 5[kN/m^2].

— A force of 10W[kN] acting forward at the centroid of mass.

12.5.4.2 For side-opening doors, the reaction forces applied on the effective securing and supporting devices assuming the door as a rigid body are determined for the following combination of external loads acting simultaneously together with the self-weight of the door:

Case 1 : F_x, F_y and F_z acting on both doors

Case 2 : $0.7F_x$ and $0.7F_z$ acting on both doors and $0.7F_y$ acting on each door separately

where F_x, F_y and F_z are to be calculated as indicated in Cl.12.5.2 and applied at the centroid of projected areas.

12.6 Inner door design loads

12.6.1 External pressure

The design external pressure, P_e in [N/mm^2], considered for the scantlings of primary members, securing and supporting devices and surrounding structure of inner doors is to be taken as the greater of the following:

$$P_e = 0.45.L \times 10^{-3}$$

$$P_h = 0.01h$$

where,

h = The distance in [m], from the load point to the top of the cargo space.

L = Ship's length in [m], but need not be taken greater than 200[m].

12.6.2 Internal pressure

The design internal pressure, P_i in [N/mm^2], considered for the scantlings of securing devices of inner doors is not to be less than:

$$P_i = 0.025$$

12.7 Strength Criteria

12.7.1 Scantlings of the primary members, securing and supporting devices of bow doors and inner doors are to be determined to withstand the design loads defined in 12.5 and 12.6, using the following permissible stresses, in [N/mm^2]:

$$\text{Shear stress}, \tau = \frac{80}{k}$$

$$\text{Bending stress}, \sigma = \frac{120}{k}$$

$$\text{Equivalent stress}, \sigma_c = \sqrt{\sigma^2 + 3\tau^2} = \frac{150}{k}$$

where k, the material factor, is not to be taken less than 0.72 unless a direct fatigue analysis is carried out.

12.7.2 The buckling strength of primary members is to be verified as per Chapter 9.

12.7.3 For steel-to-steel bearings in securing and supporting devices, the nominal bearing pressure, σ_b in [N/mm^2], to follow the formulae below:

$$\sigma_b \leq 0.8\,\sigma_y$$

where,

$$\sigma_b = \frac{F}{A_b}$$

with:

F = Design force in [N], defined in Cl.12.5.2

A_b = Projected bearing area, in [mm^2].

σ_y = Yield stress of the bearing material, in [N/mm^2].

For other bearing materials, the permissible bearing pressure is to be determined according to the manufacturer's recommended value.

12.7.4 The arrangement of securing and supporting devices is to be such that threaded bolts do not carry support forces. The maximum tension in way of threads of bolts not carrying support forces is not to exceed 125/k [N/mm^2], where k is defined in Cl.12.7.1.

12.8 Scantlings of bow doors

12.8.1 General

12.8.1.1 The strength of bow doors is to be commensurate with that of the surrounding structure.

12.8.1.2 Bow doors are to be adequately stiffened and means are to be provided to prevent any lateral or vertical movement of the doors when closed. For visor doors, adequate strength for the opening and closing operations is to be provided in the connections of the lifting arms to the door structure and to the ship's structure.

12.8.2 Plating and secondary stiffeners

12.8.2.1 The gross thickness of the bow door plating is not to be less than the required thickness for the side shell plating, using the door stiffener spacing as per Chapter 7, Section 4, and also not less than the minimum required thickness of fore end shell plating as per Chapter 7, Section 3.

12.8.2.2 The gross section modulus of secondary horizontal or vertical stiffeners is not to be less than that required for end framing as per Chapter 7, Section 5 for the fore end, using the bow door stiffener spacing. Consideration is to be given, where necessary, to differences in fixity between ship's frames and door stiffeners.

12.8.2.3 The net sectional area in [cm^2], of stiffener webs is not to be less than:

$$A = 0.05 P_e . s . \ell . k$$

12.8.3 Primary Structure

12.8.3.1 The bow door secondary stiffeners are to be supported by primary members constituting the main stiffening of the door.

12.8.3.2 The primary members of the bow structure and the hull structure in way are to have sufficient stiffness to ensure structural integrity of the boundary support of the door.

12.8.3.3 Scantlings of the primary members are to be based on direct strength calculations in association with the external pressure given in Cl.12.5.1.1.and permissible stresses given in Cl.12.7.1. In general, formulae for simple beam theory may be applied to determine the bending stresses. Members are to be considered to have simply supported end connections.

12.8.3.4 The webs of primary members (girders and stringers) are to be adequately stiffened, preferably in a direction perpendicular to the shell plating.

12.9 Scantlings of inner doors

12.9.1 Scantlings of the primary members of inner doors are to be based on the external pressure given in Cl.12.6.1 and permissible stresses given in Cl.12.7.1. Supporting calculations for resultant stresses are to be submitted.

12.9.2 Where inner doors serve as vehicle ramps, scantlings are not to be less than those required for vehicle decks.

12.9.3 Calculations to determine the distribution of forces acting on the securing and supporting devices are to take into account the flexibility of the structure and the actual position and stiffness of the supports and are to be submitted for approval.

12.10 Securing and supporting of bow doors

12.10.1 General

12.10.1.1 Bow doors are to be fitted with adequate means of securing and supporting so as to be commensurate with the strength and stiffness of the surrounding structure. The hull supporting structure in way of the bow doors is to be strengthened for the loads and stresses arising from the securing and supporting devices.

Where packing is required, the packing material is to be of a comparatively soft type, and the supporting forces are to be carried by the steel structure only. Other types of packing may be considered. Maximum design clearance between securing and supporting devices is not generally to exceed 3[mm].

12.10.1.2 Only the active supporting and securing devices having an effective stiffness in the relevant direction are to be included and considered while calculating the reaction forces acting on the devices. Small and/or flexible devices such as cleats intended to provide load compression of the packing material are not generally to be included in the calculations in Cl.12.10.2.3. The number of securing and supporting devices is generally to be the minimum practical whilst taking into account the requirements for redundancy given in Cl.12.5.3.2 and Cl.12.10.2.4 and the space

available for adequate support in the hull structure.

12.10.2 Scantlings

12.10.2.1 Scantlings of securing and supporting devices are to be adequately designed so that they can withstand the reaction forces calculated in Cl.12.5.3.4 and Cl.12.5.4.2, within the permissible stresses given in Cl.12.7.1.

12.10.2.2 The support forces as determined according to Cl.12.5.2.1, Case 1 in Cl.12.5.3.4 and Case 1 in Cl.12.5.4.2 are to generally give rise to a zero moment about the transverse axis through the centroid of the area A_X.

For visor doors, longitudinal reaction forces of pin and/or wedge supports at the door base contributing to this moment are not to be in the forward direction.

12.10.2.3 The distribution of the reaction forces acting on the securing and supporting devices may require to be supported by direct calculations taking into account the flexibility of the hull structure and the actual position and stiffness of the supports.

12.10.2.4 The arrangement of securing and supporting devices is to be designed with redundancy so that in the event of failure of any single securing or supporting device the remaining devices are capable to withstand the reaction forces without exceeding the permissible stresses given in Cl.12.7.1, by more than 20%.

12.10.2.5 For side-opening doors, thrust bearing has to be provided in way of girder ends at the closing of the two leaves to prevent one leaf to shift towards the other one under effect of unsymmetrical pressure (Refer Figure 12.10.2). Each part of the thrust bearing has to be kept secured on the other part by means of securing devices. Any other arrangement serving the same purpose may be proposed.

12.10.2.6 All load transmitting elements in the design load path, from door through securing and supporting devices into the ship structure, including pins, supporting brackets, back-up brackets and welded connections, are to be to the same strength standard as required for the securing end supporting devices.

12.10.2.7 Inner doors are to be gasketed and weathertight.

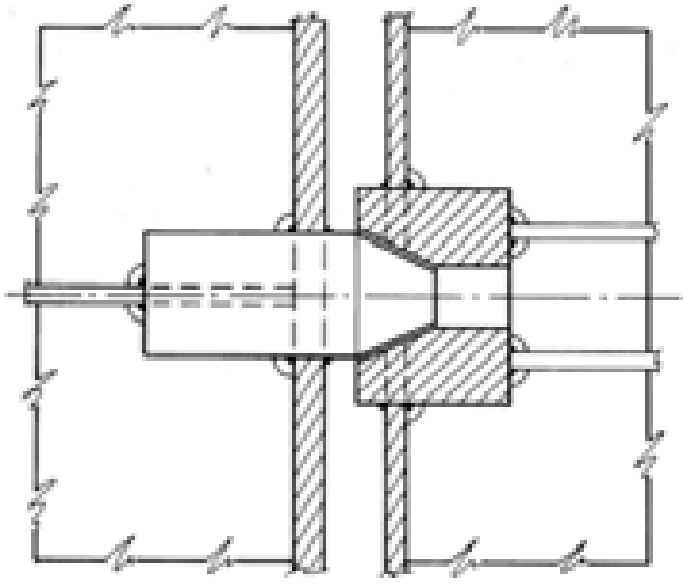


Figure 12.10.2: Thrust bearing

12.11 Securing and locking arrangement

12.11.1 Systems for operation

12.11.1.1 Securing devices are to be simple to operate and easily accessible.

Securing devices are to be equipped with mechanical locking arrangement (self-locking or separate arrangement), or to be of the gravity type.

The opening and closing systems as well as securing and locking devices are to be

interlocked in such a way that they can only operate in the proper sequence.

12.11.1.2 Bow doors and inner doors giving access to vehicle decks are to be provided with an arrangement for remote control of the following from a position above the freeboard deck:

— the closing and opening of the doors, and

— associated securing and locking devices for every door.

Indication of the open/closed position of every door and every securing and locking device is to be available at the remote-control stations.

The operating panels for operation of doors are to be inaccessible to unauthorized persons.

A notice plate, giving instructions to the effect that all securing devices are to be closed and locked before leaving harbour, is to be placed at each operating panel and is to be supplemented by warning indicator lights.

Alternative means of securing are to be provided for emergency use in case of failure of the power systems.

12.11.1.3 Where hydraulic securing devices are applied, the system is to be mechanically lockable in closed position (i.e. in the event of loss of the hydraulic fluid, the securing devices are to remain locked).

The hydraulic system for securing and locking devices is to be isolated from other hydraulic circuits, when in closed position.

12.11.1.4 Securing devices are to be equipped with positive locking arrangements such that the devices are retained in the closed position within design limits of inclination, vibration and other motion-induced or dynamic loads and in the event of loss of any actuating power supply.

12.11.1.5 Means are to be provided to enable the bow doors to be mechanically fixed in the open position taking into account the self-weight of the door and a minimum wind pressure of 1.5[kN/m^2] acting on the maximum projected area in the open position.

12.11.2 Systems for indication/monitoring

12.11.2.1 Separate indicator lights and audible alarms are to be provided on the navigation bridge and on the operating panel to show that the bow door and inner door are closed and that their securing and locking devices are properly positioned.

The indication panel is to be provided with a lamp test function. It shall not be possible to turn off the indicator light.

12.11.2.2 The indicator system is to be designed on the fail-safe principle and is to show by visual alarms if the door is not fully closed and not fully locked and by audible alarms if securing devices become open or locking devices become unsecured.

The power supply for the indicator system is to be independent of the power supply for operating the doors and provided with a back-up power supply from the emergency source of power or other secure power supply such as a UPS.

The sensors of the indicator system are to be protected from water, ice formation and mechanical damages.

12.11.2.3 The indication panel on the navigation bridge is to be equipped with a mode selection function "harbour/sea voyage", so arranged that audible alarm is given on the navigating bridge if vessel leaves harbour with the bow door or inner door not closed and with any of the securing devices not in the correct position.

12.11.2.4 The indicator system is considered designed on the fail-safe principle when the following are satisfied:

• The indication panel is provided with:

— a power failure alarm

— an earth failure alarm

— a lamp test

— separate indication for door closed, door locked, door not closed and door not locked.

• Limit switches electrically closed when the door is closed (when more than one limit switch is provided they may be connected in series).

- Limit switches electrically closed when securing arrangements are in place (when more than one limit switch is provided they may be connected in series).

- Two electrical circuits (may be provided in one multicore cable), one for the indication of door closed/not closed and the other for door locked / not locked.

- In case of dislocation of limit switches, indication to show: not closed / not locked / securing arrangement not in place – as appropriate.

12.11.2.5 A water leakage detection system with audible alarm and CCTV surveillance is to be arranged to provide an indication to the navigation bridge and to the engine control room of leakage through the inner door. The indicator system is considered designed on the fail – safe principle when the requirements of Cl.12.11.2.4 are satisfied.

12.11.2.6 Between the bow door and the inner door a CCTV surveillance system is to be fitted with a monitor on the navigation bridge and in the engine control room.

The system is to monitor the position of doors and a sufficient number of their securing devices.

Special consideration is to be given for lighting and contrasting colour of objects under surveillance. The indicator system is considered designed on the fail – safe principle when the requirements of Cl.12.11.2.4 are satisfied.

12.11.2.7 A drainage system is to be arranged in the area between bow door and ramp, or where no ramp is fitted, between the bow door and inner door.

The system is to be equipped with an audible alarm function to the navigation bridge being set off when the water levels in these areas exceed 0.5[m] above the car deck level or the high water level alarm, whichever is lesser.

12.11.2.8 For ro-ro passenger ships on international voyages, the special category spaces and ro-ro spaces are to be continuously patrolled or monitored by effective means, such as CCTV surveillance, so that any movement of vehicles in adverse weather conditions or unauthorized access by passengers thereto, can be detected whilst the ship is underway.

12.12 Operating and maintenance manual

12.12.1 An Operating and Maintenance Manual for the bow and inner doors is to be provided onboard and contain necessary information on:

12.12.1.1 Main particulars and design drawings

— special safety precautions

— details of vessel

— equipment and design loading (for ramps)

— key plan of equipment (doors and ramps)

— manufacturer's recommended testing for equipment

— description of equipment for

- bow doors
- inner bow doors
- bow ramp/doors
- side doors
- stern doors
- central power pack
- bridge panel
- engine control room panel

12.12.1.2 Service conditions

— limiting heel and trim of ship for loading/unloading

— limiting heel and trim for door operations

— doors/ramps operating instructions

— doors/ramps emergency operating instructions

12.12.1.3 Maintenance and function testing

— schedule and extent of maintenance

— trouble shooting and acceptable clearances

— manufacturer's maintenance procedures

12.12.1.4 Register of inspections, including inspection of locking, securing and supporting devices, repairs and renewals.

12.12.2 Documented operating procedures for closing and securing bow door and inner doors are to be kept on board and posted at the appropriate places.

12.12.3 This Manual has to be submitted for approval to verify that the above-mentioned items are contained in the OMM and that the maintenance part includes the necessary information with regard to inspections, troubleshooting and acceptance / rejection criteria.

Note: It is recommended that recorded inspections of the door supporting and securing devices be carried out by the ship's personnel at monthly intervals or following incidents that could result in damage, including heavy weather or contact in the region of shell doors. A record is to be kept and any damage recorded during such inspections is to be reported to IRS.

Section 13

Ventilators, Air Pipes and Discharges

Symbols

For symbols not defined in this section, Refer to Chapter 1, Section 2.

13.1 General

13.1.1 This section applies to all ships and provides requirements for ventilators, air and sounding pipes and overboard discharges.

13.1.2 The requirements conform, where relevant, to those of the ICLL 1966 as amended by the Protocol of 1988. Reference should also be made to any additional requirements of the National Authority of the country in which the ship is to be registered.

13.2 Definitions

13.2.1 The requirements of ventilators are given with respect to two basic positions – Position 1 and Position 2 as defined in Section 1, Cl.1.2.5.

13.3 Protection

13.3.1 In all cargo spaces and other areas where mechanical damage is likely, all air and sounding pipes, ventilators, scuppers and discharges, including their valves, controls and indicators, are to be well protected. This protection is to be of steel or other equivalent material.

13.4 Strength requirements for fore deck air pipes, ventilator pipes and their closing devices

13.4.1 General

13.4.1.1 The air pipes, ventilators and their closing devices fitted on ships of L ≥ 80[m] on exposed decks in the forward 0.25L are to satisfy the requirements given in Cl.13.4.2 and Cl.13.4.3 if the height of the exposed deck in way of the item is less than 0.1L or 22[m] above the summer load waterline, whichever is the lesser.

However, these requirements need not be applied to cargo tank venting systems and inert gas systems of oil tankers, chemical carriers and liquefied gas carriers.

13.4.2 Design load

13.4.2.1 The green sea pressure 'p' in [kN/m^2] acting on air pipes, ventilators and their closing appliances is to be taken as:

$$p = 0.5\rho V^2 C_d C_s C_p$$

where:

ρ = Density of sea water (1.025 t/m^3)

V = Velocity of water in [m/sec], over the fore deck

= 13.5 for $d \leq 0.5d_1$

= $13.5\sqrt{2\left(1-(d/d_1)\right)}$ for $0.5d_1 < d \leq d_1$

d = Distance in [m], from summer load waterline to exposed deck

d_1 = 0.1L or 22[m] whichever is the lesser

C_d = Shape coefficient

= 0.5 for pipes

= 1.3 for air pipe or ventilator heads in general

= 0.8 for an air pipe or ventilator head of cylindrical form with its axis in the vertical direction.

C_s = Slamming coefficient = 3.2

C_p = Protection coefficient

= 0.7 for pipes and ventilator heads located immediately behind a breakwater or forecastle

= 1.0 elsewhere and immediately behind a bulwark.

13.4.2.2 Forces acting in the horizontal direction on the pipe and its closing device are to be calculated using pressure 'p' and the largest projected area of each component.

13.4.3 Strength criteria

13.4.3.1 Bending moment and stresses: Bending moments and stresses in air pipes and ventilator pipes are to be calculated at critical positions, such as at penetration pieces, weld or flange connections, and toes of supporting brackets. Bending stresses considering the net section (after deduction of 2.0[mm] from the pipes thickness), irrespective of the corrosion protection provided are not to exceed $0.8\sigma_y$, where σ_y is the specified minimum upper yield stress or 0.2% proof stress of the steel at room temperature.

13.4.3.2 Ventilators: For standard ventilators of 900[mm] height closed by heads of not more than the tabulated projected area, the applicable pipe thicknesses and bracket heights are specified in Table 13.4.3.1.

13.4.3.3 For ventilators of height greater than 900[mm], brackets or alternative means of support are to be fitted. Pipe gross thickness is not be taken less than as required in Table 13.4.3.1.

13.4.3.4 For standard air pipes of height 760[mm] closed by heads of not more than the tabulated projected area, the applicable pipe thicknesses and bracket heights are specified in Table 13.4.3.2.

13.4.3.5 For other configurations of air pipes and ventilators, loads according to Cl.13.4.2 are to be applied and means of support determined in order to comply with the requirements of Cl.13.4.3.1.

13.4.3.6 Where brackets are specified for supporting the air pipe or ventilator, three or more radial brackets are to be fitted. These brackets are to have gross thickness of at least 8[mm], minimum length 100[mm] and height according to Table 13.4.3.1 or Table 13.4.3.2 as applicable but need not extend over the joint flange for the head. Bracket toes at the deck are to be suitably supported.

13.4.3.7 Components and Connections: All component parts and connections of the air pipe or ventilator are to be capable of withstanding the green sea pressure 'p' given in Cl.13.4.2.

13.4.3.8 Rotary Heads: Rotating type mushroom ventilator heads are unsuitable for application in the location defined in Cl.13.4.1.1. Screw down type mushroom ventilators may be used in such locations.

Table 13.4.3.1: 900[mm][(a)] ventilator pipe gross thickness and bracket standards			
Standard ventilator details			**Required Height[(b)] of brackets [mm]**
Nominal pipe diameter [mm]	**Minimum fitted gross thickness, [mm]**	**Maximum projected area of head [cm^2]**	
80A	6.3	-	460
100A	7.0	-	380
150A	8.5	-	300
200A	8.5	550	-
250A	8.5	880	-
300A	8.5	1200	-
350A	8.5	2000	-
400A	8.5	2700	-
450A	8.5	3300	-
500A	8.5	4000	-
Notes: (a) For other ventilator heights, the relevant requirements of Cl.13.4.2 and Cl.13.4.3 are to be applied. (b) Brackets need not extend over the joint flange for the head.			

Table 13.4.3.2: 760[mm][(1)] air pipe thickness and bracket standards			
Standard air pipe details			**Required Height[(2)] of brackets [mm]**
Nominal pipe diameter [mm]	**Minimum fitted gross thickness, [mm]**	**Maximum projected area of head [cm^2]**	
65A	6.0	-	480
80A	6.3	-	460
100A	7.0	-	380
125A	7.8	-	300
150A	8.5	-	300
175A	8.5	-	300
200A	8.5[(3)]	1900	300[(3)]
250A	8.5[(3)]	2500	300[(3)]
300A	8.5[(3)]	3200	300[(3)]
350A	8.5[(3)]	3800	300[(3)]
400A	8.5[(3)]	4500	300[(3)]
Notes: (1) For other pipe heights, the relevant requirements of Cl.13.4.2 and Cl.13.4.3 are to be applied. (2) Brackets need not extend over the joint flange for the head. (3) Brackets need not be provided if the as fitted gross thickness of air pipes is not less than 10.5[mm].			

13.5 Ventilators

13.5.1 General

13.5.1.1 Ventilators located on the exposed deck over the forward 0.25L of the rule length, of ships of length 80[m] or more, where the height of the exposed deck in way of the item is less than 0.1L or 22[m] above the summer load waterline, whichever is the lesser, are to comply with the requirements of 13.4. All other ventilators are to comply with the following requirements.

13.5.1.2 Ventilators in position 1 or 2 to spaces below freeboard deck or decks of enclosed superstructures are to have coamings of steel or other equivalent material, substantially constructed and efficiently connected to the deck. Where the coaming of any ventilator exceeds 900[mm] in height it is to be specially supported.

13.5.1.3 Ventilators passing through superstructures other than enclosed superstructures are to have substantially constructed coamings of steel or other equivalent material at the freeboard deck.

13.5.1.4 Special care is to be taken in the design and positioning of ventilator openings and coamings, particularly in the region of the forward end of superstructures and other points of high stress. The deck plating in way of the coamings is to be efficiently stiffened.

13.5.1.5 Ventilators from deep tanks and tunnels passing through 'tween decks are to have scantlings suitable for withstanding the pressures to which they may be subjected, and are to be made watertight.

13.5.1.6 For the requirements of fire precautions on cargo and passenger ships, Refer Part 6, Chapter 2, Section 2.

13.5.2 Coamings

13.5.2.1 The height of ventilator coamings exposed to the weather is to be not less than:

— 900[mm] in position 1

— 760[mm] in position 2

these heights being measured above deck sheathing where fitted.

13.5.2.2 The height of ventilator coamings may be required to be increased on ships of Type 'A', Type 'B-100' and Type 'B-60' where this is shown to be necessary by the floatability criterion required by the ICLL 1966 as amended by the Protocol of 1988.

13.5.2.3 The minimum wall gross thickness of ventilator coamings in position 1 and 2 leading to spaces below the freeboard deck or to spaces within enclosed superstructures, are not to be less than those obtained from Table 13.5.2.

Table 13.5.2: Minimum wall gross thickness of ventilator coamings

Location	External diameter[mm]	Gross thickness[mm]
Position 1 and 2	≤ 80	6.0
	≥ 165	8.5
Above position 2	≤ 155	4.5
	≥ 230	6.0
Note: The thickness for intermediate diameters may be obtained by interpolation.		

13.5.2.4 Where the height of the ventilator exceeds that required by 13.5.2.1, the thickness given in Table 13.5.2 may be gradually reduced, above the required minimum height, to a minimum of 6.0[mm]. The ventilator is to be adequately stayed.

13.5.2.5 For gooseneck ventilators, the coaming height is to be measured to the underside of the bend, this being the lowest point through which water on deck could pass freely to spaces below.

13.5.2.6 Where wall vents are fitted with an internal baffle which rises above the lower edge of the exterior opening, the coaming height is measured to the top of the baffle.

13.5.3 Closing appliances

13.5.3.1 All ventilator openings are to be provided with efficient weathertight closing appliances except where:-

— the height of the coaming is greater than 4.5[m] in position 1, or

— the height of the coaming is greater than 2.3[m] in position 2

in which case closing appliances need not be provided unless unusual features of design make it necessary.

13.5.3.2 The weathertight closing appliance for all ventilators in positions 1 and 2 are to be of steel or other equivalent materials. Wood plugs and canvas covers are not acceptable in these positions.

13.5.3.3 In general, ventilators which are necessary to continuously supply the machinery space and on demand, immediately supply the emergency generator room, should have coamings with sufficient heights specified in Cl.13.5.3.1, without having to fit weathertight closing appliances.

However, where due to vessel size and arrangement this is not practicable, lesser heights for machinery space and emergency generator room ventilator coamings may be considered with provision of weathertight closing appliances in combination with other suitable arrangements to ensure an uninterrupted, adequate supply of ventilation to these spaces.

13.5.3.4 In ships where the load line length, L_L, is not more than 100[m], the closing appliances are to be permanently attached to the ventilator coaming. Whereas in other ships, they may be conveniently stowed near the ventilator to which they are to be fitted.

13.5.3.5 Where, in ferries, ventilators are proposed to be led overboard in an enclosed 'tween deck' the closing arrangements are to be submitted for approval. If such ventilators are led overboard more than 4.5[m] above the main vehicle deck, closing appliances may be omitted, provided that satisfactory baffles and drainage arrangements are provided, as in the case of air intakes or exhaust openings for machinery spaces, which may be arranged in the sides of the ship.

13.5.3.6 On offshore supply ships, to ensure satisfactory operation in all weather conditions, machinery space ventilation inlets and outlets are to be located in such positions that closing appliances will not be necessary.

13.5.3.7 The height of ventilators may be required to be increased depending on where the flooding points in relation to the damaged waterline in the stability calculation.

13.5.3.8 Watertight ventilators and trunks is to be carried at least up to the bulkhead deck in passenger ships and up to the freeboard deck in cargo ships.

13.5.3.9 Where permitted by the National Authority, ventilator coaming heights may be reduced on ships engaged on sheltered water service or restricted water service.

13.6 Air and Sounding pipes

13.6.1 General

13.6.1.1 Air and sounding pipes are to comply with the requirements of Part 4, Chapter 2.

13.6.1.2 Striking plates of suitable thickness, or their equivalent, are to be fitted under all sounding pipes.

13.6.1.3 On offshore supply ships, air pipes are to be situated clear of the cargo containment areas.

13.6.1.4 The venting systems of cargo tanks are to be entirely separated from the air pipes of the other compartments of the ship.

13.6.1.5 All air pipes in cargo spaces are to be well protected.

13.6.2 Height of air pipes

13.6.2.1 The height of air pipes from the upper surface of decks exposed to the weather, to the point where water may have access below is not normally to be less than:

- 760 [mm] on freeboard deck

- 450 [mm] on superstructure decks

these heights being measured above deck sheathing, where fitted. Air pipes with height exceeding 900[mm] are to be additionally supported.

13.6.2.2 Lower heights may be approved in cases where these are essential for the working of the ship, provided that the design and arrangements are otherwise satisfactory. In such cases, efficient, permanently attached closing appliances of an approved automatic type will generally be required.

13.6.2.3 The height of air pipes may be required to be increased on ships of Type 'A', Type 'B-100' and Type 'B-60' where this is shown to be necessary by the floatability calculations required by the ICLL 1966 as amended by the Protocol of 1988.

13.6.2.4 Air pipes are generally to be led to an exposed deck, but alternative arrangements will be considered in the case of ferries where such an arrangement is not practicable.

13.6.2.5 Where air pipes are led through the side of superstructures, the opening is to be at least 2.3[m] above the summer load waterline. Automatic vent heads of approved design are to be provided.

13.6.2.6 The minimum wall gross thickness of air pipe in positions defined in Cl.13.6.2.1 is to be:

• 6.0[mm] for pipes of 80[mm] external diameter or smaller;

• 8.5[mm] for pipes of 165[mm] external diameter or greater.

Intermediate minimum thicknesses are to be determined by linear interpolation.

For other positions not mentioned in Cl.13.6.2.1:

• 4.5[mm] for pipes of 155[mm] external diameter or smaller;

• 6.0[mm] for pipes of 230[mm] external diameter or greater.

Intermediate minimum thicknesses are to be determined by linear interpolation.

13.6.2.7 Where permitted by the National Authority, air pipe coaming heights may be reduced on ships engaged on sheltered water service or restricted water service.

13.6.3 Closing appliances

13.6.3.1 All openings of air and sounding pipes are to be provided with permanently attached, satisfactory means of closing to prevent the free entry of water.

13.6.3.2 Air pipe closing devices are to be of an approved type and so designed that they withstand both ambient and working conditions and be suitable for use at inclinations up to and including 40^{o}.

13.6.3.3 Wooden plugs and trailing canvas hoses are not accepted in Freeboard deck and Superstructure deck.

13.6.3.4 Closing devices shall be automatic if, while the vessel is at its draught corresponding to summer load line, the openings of air pipes to which these closures are fitted submerge at angles up to 40° or up to a lesser angle which may be agreed on the basis of stability requirements. Pressure-vacuum valves (P-V valves) may however, be accepted on tankers.

13.6.3.5 An automatic closing device is to:

• Prevent the free entry of water into the tanks,

• Allow the passage of air or liquid to prevent excessive pressure or vacuum developing in the tank.

13.6.3.6 Air pipe closing devices are to be so constructed as to allow inspection of the closure and the inside of the casing as well as changing of the seals.

13.6.3.7 Efficient ball or float seating arrangements are to be provided for the closures, where fitted. Bars, cage or other devices are to be provided to prevent the ball or float from contacting the inner chamber in its normal state and made in such a way that the ball or float is not damaged when subjected to water impact due to a tank being overfilled.

13.6.3.8 The clear area through an air pipe closing device is to be at least equal to the area of the inlet. Air pipe closing devices are to be self draining.

13.6.3.9 In the case of air pipe closing devices of the float type, suitable guides are to be provided to ensure unobstructed operation. The maximum allowable tolerances for wall thickness of ball floats are not to exceed 10% of the nominal thickness.

13.6.3.10 Casings of air pipe closing devices are to be of approved metallic materials adequately protected against corrosion. The inner and the outer chambers of an automatic air pipe head is to be of a minimum thickness of 6[mm]. Where side covers are provided and their function is integral to providing functions of the closing device as outlined in Cl.13.6.3.5, they are to have a minimum wall thickness of

6[mm]. If the air pipe head can meet the type approval tightness test requirement without the side covers attached, then the side covers are not considered to be integral to the closing device, in which case a wall thickness less than 6[mm] can be considered acceptable for side covers.

13.6.3.11 For galvanized steel air pipe heads, the zinc coating is to be applied by the hot method and the coating thickness is to be 70 to 100 microns. For areas of the head susceptible to erosion (e.g. those parts directly subjected to ballast water impact when the tank is being pressed up, for example the inner chamber area above the air pipe, plus an overlap of 10 ^{o} or more on either side) an additional harder coating such as aluminium bearing epoxy coating or equivalent, is to be applied over the zinc coating.

13.6.3.12 Closures and seats made of non-metallic materials are to be compatible with the media intended to be carried in the tank at temperatures between -25^{o}C and 85^{o}C and sea water.

13.6.3.13 For ships assigned timber freeboards, the air pipes should be provided with automatic closing appliances.

13.7 Scuppers, Inlets and Sanitary Discharges

13.7.1 General

13.7.1.1 Scuppers sufficient in number and size to provide effective drainage are to be fitted in all decks.

13.7.1.2 Scuppers draining weather decks and spaces within superstructures or deckhouses not fitted with efficient weathertight doors are to be led overboard.

13.7.1.3 Scuppers and discharges which drain spaces below the freeboard deck, or spaces within intact superstructures or deckhouses on the freeboard deck fitted with efficient weathertight doors, may be led to the bilges in the case of scuppers.

Alternatively, they may be led overboard provided that the spaces drained are above the load waterline, and the pipes are fitted with efficient and accessible means of preventing water from passing inboard.

13.7.1.4 The drainage arrangements of enclosed 'tween deck spaces in ships with small freeboards are to be specially considered.

13.7.1.5 In the enclosed 'tween deck space on the main vehicle deck of a ferry, which is provided with a drencher fire extinguishing system, scuppers of not less than 150[mm] diameter are to be provided port and starboard, spaced about 9.0[m] apart. The capacity of the scuppers may be required to be increased, dependent on the discharge rate of the drencher system. The mouth of the scupper is to be protected by bars.

13.7.1.6 If a 'tween deck space on the main vehicle deck of a ferry is not totally enclosed, scuppers or freeing ports are to be provided consistent with the requirements of Chapter 7.

13.7.1.7 The following will apply to drainage of enclosed cargo spaces on the bulkhead deck of passenger ships or on the freeboard deck of cargo ships:

• Where the freeboard to the bulkhead deck or the freeboard deck, respectively, is such that the deck edge is immersed when the ship heels more than 5[deg.] the drainage shall be by means of a sufficient number of scuppers of suitable size discharging directly overboard.

• Where the freeboard is such that the edge of the bulkhead deck or the edge of the freeboard deck, respectively, is immersed when the ship heels 5[deg.] or less, the drainage shall be led to a suitable space of adequate capacity, having a high water level alarm and provided with suitable arrangements for discharge overboard.

13.7.1.8 For drainage arrangements of closed vehicle and ro-ro spaces and special category spaces, refer to Part 6, Chapter 7, Cl.3.6.1.4 and Cl.3.6.1.5.

13.7.1.9 Scuppers and discharge pipes should not normally pass through fuel oil or cargo oil tanks. Where scuppers and discharge pipes pass, unavoidably, through fuel oil or cargo oil tanks, and are led through the shell within the tanks, the thickness of the piping should be at least the same thickness as Rule shell plating in way, derived from the appropriate Chapters, but need not exceed 19[mm]. Piping within tanks is to be tested in accordance with Part 4, Chapter 2 and Chapter 3.

13.7.1.10 All piping is to be adequately supported.

13.7.1.11 Where scuppers are carried through watertight bulkheads, arrangements are to be made to ensure the watertight integrity of the bulkheads.

13.7.1.12 For the use of non-metallic pipe, Refer Part 4, Chapter 2, Section 5.

13.7.2 Closing appliances

13.7.2.1 Discharges led through the shell either from spaces below the freeboard deck or from within superstructures and deckhouses on the freeboard deck fitted with doors complying with the requirements given in Section 2, Cl.2.2.1.1, is to be fitted with efficient and accessible means for preventing water from passing inboard. Normally each separate discharge shall have one automatic non-return valve with a positive means of closing it from a position above the freeboard deck. The means for operating the positive action valve shall be readily accessible and provided with an indicator showing whether the valve is open or closed.

It is considered that an acceptable equivalent to an automatic non-return valve with a positive means of closing from a position above the freeboard deck would be one automatic non-return valve and one sluice valve controlled from above the freeboard deck.

Where sanitary discharges and scuppers lead overboard through the shell in way of machinery spaces, the fitting to shell of a locally operated closing valve, together with non-return valve inboard, is considered to provide protection equivalent to above. The controls of the valves is to be in an easily accessible position.

13.7.2.2 Where the vertical distance from the summer load waterline to the inboard end of the discharge pipe exceeds $0.01L_L$, the discharge may be fitted with two automatic non-return valves without positive means of closing, instead of automatic non-return valve with a positive means of closing, provided that the inboard valve is always accessible for examination under service conditions i.e. the inboard valve should be above the level of the tropical load waterline.

If this is not practicable, then, provided a locally controlled sluice valve is interposed between the two automatic non-return valves, the inboard valve need not be fitted above the LWL.

13.7.2.3 Where the vertical distance from the summer load waterline to the inboard end of the discharge pipe exceeds $0.02L_L$, a single automatic non-return valve without positive means of closing may be fitted. Refer Figure 13.7.2.

13.7.2.4 It is considered that the requirements for non-return valves are applicable only to those discharges which remain open during the normal operation of the vessel. For discharges which must necessarily be closed at sea, such as gravity drains from topside ballast tanks, a single screw down valve operated from the deck is sufficient.

The inboard end of a gravity discharge which leads overboard from an enclosed superstructure or space is to be located above the waterline formed by a 5 degree heel, to port or starboard, at a draft corresponding to be assigned summer freeboard.

13.7.2.5 In a ship to which timber freeboards are assigned, the summer load waterline is to be regarded as that corresponding to the timber summer freeboard and the position of the inboard end of discharges is to be regarded as that of timber summer load waterline.

13.7.2.6 Where in an enclosed tween deck space on the main vehicle deck of a ferry, a drencher fire extinguishing system is provided, discharge valve controls are to be operable from spaces above the main vehicle deck and to be protected from mechanical damage.

13.7.2.7 Scuppers and discharge pipes originating at any level which penetrate the shell either more than 450[mm] below the freeboard deck or less than 600[mm] above the summer load waterline, are to be fitted with an automatic non-return valve at the shell. This valve, unless required by Cl.13.7.1.3, may be omitted provided that the pipe is of substantial thickness. Refer Part 4, Chapter 3, Section 3.5.2. However, the thickness need not exceed 12.5[mm], unless greater thickness is required by Cl.13.7.1.8.

13.7.2.8 If, in association with Cl.13.7.1.8, a valve is required by Cl.13.7.1.3, this valve should preferably be fitted as close as possible to the point of entry of the pipe into the tank. If fitted below the freeboard deck, the valve is to be capable of being controlled from an easily

accessible position above the freeboard deck. Local control is also to be arranged, unless the valve is inaccessible. An indicator is to be fitted at the control position showing whether the valve is open or closed

13.7.2.9 Discharges with inboard opening located lower than the ship's uppermost load line may be accepted when a loop of the pipe is arranged between the inboard opening and the outlet in hull. The top of the loop is regarded as the position of the inboard opening, and the pipeline is provided with valves according to Figure 13.7.2.

Refer Figure 13.7.2 for the acceptable arrangements of scuppers, inlets and discharges.

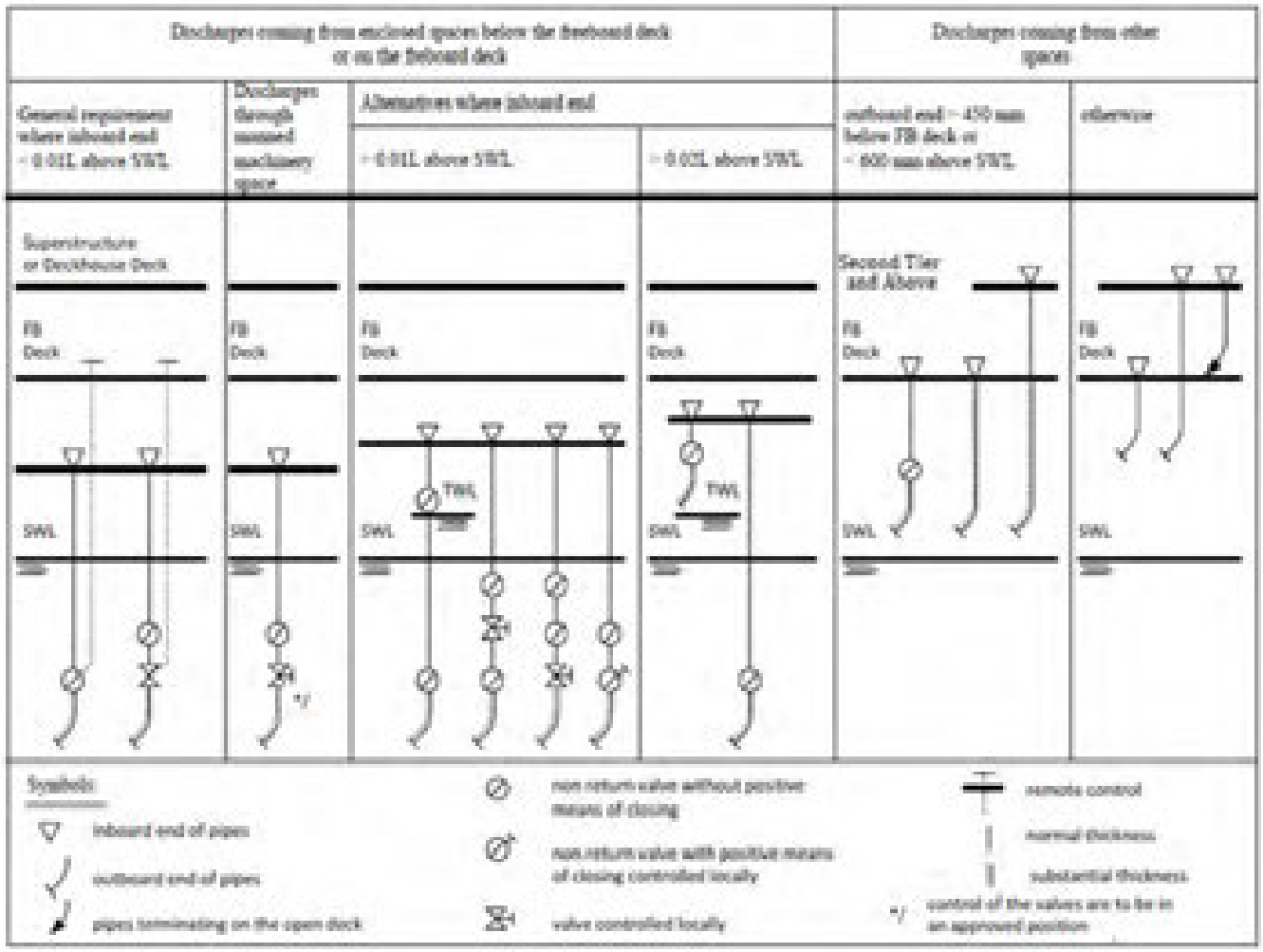


Figure 13.7.2: Discharge Valve Requirements

13.8 Materials for valves, fittings and pipes

13.8.1 All shell fittings, pipes and the valves required by Cl.13.7.2 are to be of steel, bronze or other approved ductile material. Valves of ordinary cast iron or similar material are not acceptable. Materials are to satisfy the requirements of Part 2.

13.8.2 All these items, if made of steel or other approved material with low corrosion resistance, are to be suitably protected against wastage.

13.8.3 The lengths of pipe attached to the shell fittings, elbow pieces or valves are to be of galvanized steel or other equivalent approved material.

13.8.4 Where plastic pipes are used for sanitary discharges and scuppers, they are also subject to the requirement of Figure 13.7.2, and the valve at the shell is to be operated from outside the space in which the valve is located.

Where such plastic pipes are located below the summer waterline, the valve is to be operated

from a position above freeboard deck. The portion of discharge line from the shell to the first valve as well as shell fittings and valves shall be of steel; bronze or other approved ductile material.

The approval of plastic piping in any location will be subject to the consideration of strength and fire hazards involved with special reference to penetrations through bulkheads, decks or other significant compartment boundaries. Attention must be paid to valid fire technical regulations.

13.9 Garbage chutes

13.9.1 General

13.9.1.1 For garbage chutes an acceptable equivalent to the non-return valve with a positive means of closing from a position above the freeboard deck would be two gate valves controlled from the working deck of the chute.

The lowest gate valve should, in addition, be controlled from a position above the freeboard deck. An interlock system is to be arranged between the two valves. The distance between the two gate valves is to be adequate to allow the smooth operation of the interlock system.

13.9.1.2 It is recommended that the inboard end of the garbage chute be located above the waterline formed by an 8.5 deg heel, to port or starboard, at a draft corresponding to the assigned summer freeboard, but not less than 1000[mm] above the summer waterline.

13.9.1.3 Where the inboard end of the garbage chute exceeds $0.01L_L$ above the summer waterline, valve control from the freeboard deck is not required, provided the inboard gate valve is always accessible under service conditions.

13.9.1.4 Alternatively, the upper and lower gate valves may be replaced by a hinged weathertight cover at the inboard end of the chute together with a discharge flap. The cover and the flap are to be arranged with an interlock so that the discharge flap cannot be operated until the hopper cover is closed.

13.9.1.5 The entire chute, including the cover is to be constructed of material of substantial thickness (Refer Part 4, Chapter 3, Section 3.5).

13.9.1.6 The gate valve controls and/or hinged cover are to be clearly marked: “Keep closed when not in use”.

13.9.1.7 Where the inboard end of the chute is below the freeboard deck of a passenger ship or the equilibrium waterlines of a cargo ship to which damage stability requirements apply, then:

— the inboard end hinged cover/valve is to be watertight

— the valve is to be a screw-down non-return valve fitted in an easily accessible position above the deepest load line; and

— the screw-down non-return valve is to be controlled from a position above the bulkhead deck and provided with open/closed indicators. The valve control shall be clearly marked: "Keep closed when not in use".

End of Chapter

Chapter 13

Rudders

Contents

Section	
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2	*Arrangement and Details*
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4	*Rudder Blades*
5	*Rudder Stock and Pintles*
6	*Rudder Couplings*
7	*Sole Pieces and Rudder Horns*
8	*Guidelines for Evaluation of Bending Moments and Shear Force Distribution*

Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter apply to arrangement and scantlings of normal streamlined or plate rudders and their supporting structure. Rudders fitted with special features e.g. special profiles, fins, flaps, steering propellers etc. to increase the lift force will be specially considered.

1.1.2 IRS may accept alternatives to requirements given in this Chapter, provided they are deemed to be equivalent.

1.1.3 Direct analyses adopted to justify an alternative design are to take into consideration all relevant modes of failure, on a case by case basis. These failure modes may include, amongst others: yielding, fatigue, buckling and fracture. Possible damages caused by cavitation are also to be considered.

1.1.4 Lab tests, or full scale tests may be requested to validate the alternative design approach, if deemed necessary by IRS,

1.1.5 Additional requirements for various Ice Class notations are given in Pt.5, Ch.21.

1.2 Material

1.2.1 All materials used in the construction of the rudder are to be tested and approved in accordance with Pt.2.

1.2.2 Material grades for plates and sections for the rudder blade are to be selected as per Pt.3, Ch.2.

1.2.3 For rudder stocks, pintles, keys and bolts, the specified minimum yield stress is not to be less than 200 [N/mm^2]. For materials having a specified yield stress differing from 235 [N/mm^2], material factor 'k' as determined below, is to be applied.

$$k = \left(\frac{235}{R_{eH}}\right)^e$$

with e = 0.75 for R_{eH}> 235 [N/mm^2]

= 1.0 for $R_{eH} \leq 235$ [N/mm^2]

R_{eH} = specified minimum yield stress of the material used, but not to be taken greater than 0.7 σ_u or 450 [N/mm^2] whichever is smaller;

where σ_u is the tensile strength [N/mm^2] of the material.

Where high strength steels are used for rudder stock, evaluation of stock deformations may be required, as large deformations can cause excessive edge pressures in way of bearings.

1.2.4 Bearing materials for bushings are to be stainless steel, bronze, white metal, synthetic material or lignum vitae. If stainless steel is proposed to be used for liners or bushes for the rudder stocks and pintles, the chemical composition is to be submitted for approval. Hardness of the material of the bushing is to be at least 65 Brinell lower than that of the liner or the rudder stock or pintle.

Synthetic bush materials are to be of approved type. Arrangement is to be provided for adequate supply of sea-water to these bearings.

1.3 Testing

1.3.1 Bodies of the rudders are to be tested in accordance with the requirements given in Ch.18.

Section 2

Arrangement and Details

2.1 General

2.1.1 Various types of rudder arrangement are shown in Fig.2.1.1, other combinations of couplings and bearings may however be proposed.

2.1.2 Effective means are to be provided for supporting the weight of the rudder. Where the support is provided by a carrier bearing attached to the rudder head, the structure in way of the bearing is to be adequately strengthened. The plating under all rudder head bearings or rudder carriers is to be increased in thickness.

2.1.3 All rudder bearings are to be accessible for measuring wear without lifting or unshipping the rudder.

2.1.4 Satisfactory arrangement is to be provided to prevent water from entering the steering gear compartment and lubricant from being washed away from the rudder carrier. A seal or stuffing box is to be fitted above the water line at scantling draught (without trim), for this purpose, however two separate watertight seals or stuffing boxes are to be provided when the top of the rudder trunk is below the water line at scantling draught (without trim).

2.1.5 Suitable arrangement is to be provided to prevent the rudder from lifting and accidental unshipping.

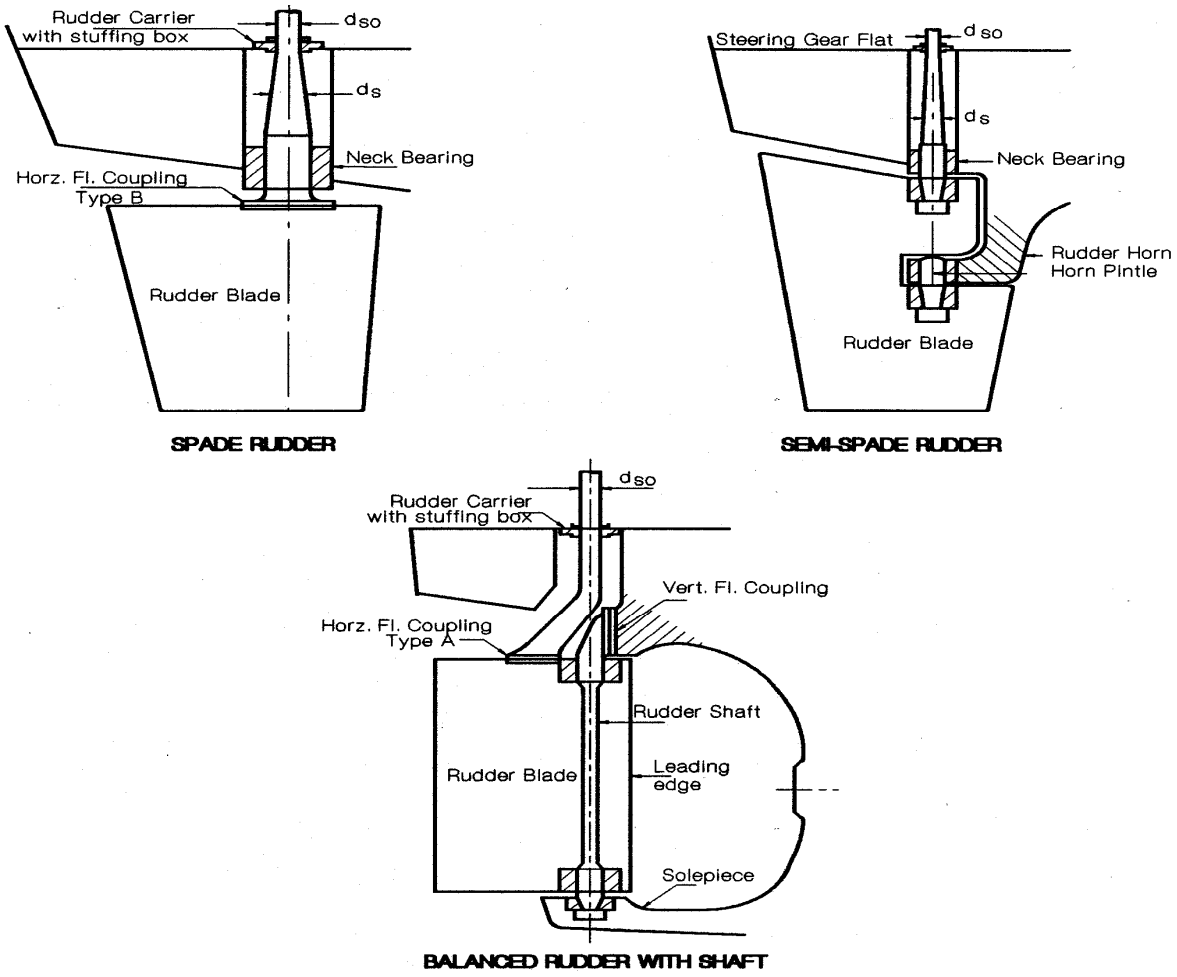


Fig.2.1.1 : Types of rudders

Section 3

Design Loads

3.1 Rudder force

3.1.1 The rudder force upon which rudder scantlings are to be based is to be determined from the following formula:

$F_r = 132 \cdot K_1 \cdot K_2 \cdot K_3 \cdot A \cdot V^2$ [N]

where,

F_r = rudder force [N]

A = area of rudder blade [m^2]

V = Design speed (knots), with vessel running ahead at maximum continuous rated shaft rpm and at the summer load waterline. Special consideration will be given to ships designed to operate at higher speed at a lower draft, and with full immersion of rudder. When the speed is less than 10 knots, V is to be replaced by the expression.

$V_{min} = (V + 20)/3$

For astern condition, the maximum astern speed as defined in SOLAS Regulation II-1/3.15 is to be used, but in no case taken less than:

$V_{astern} = 0.5V$

K_1 = factor based on the aspect ratio, λ of the rudder area;

$K_1 = (\lambda + 2)/3$; with λ not to be taken greater than 2.

$\lambda = b^2/A_t$; where b is the mean height of the rudder area [m] and At, the sum of rudder blade area and area of rudder post or rudder horn, if any, within the height b [m^2]

Mean breadth C [m] and mean height b [m] of rudder are calculated according to the co-ordinate system in Fig.3.1.1.

K_2 = Factor depending on type of the rudder and the rudder profile as per Table 3.1.1.

K_3 = 0.80 for rudders outside the propeller jet

= 1.15 for rudders behind a fixed propeller nozzle

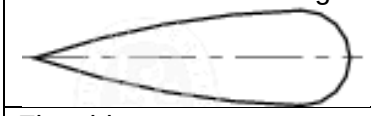
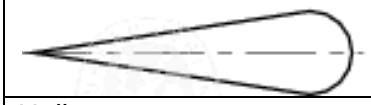
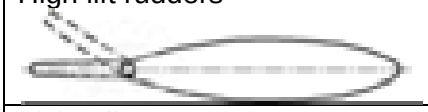
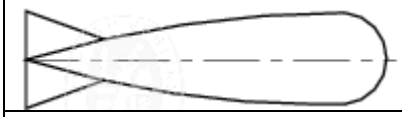
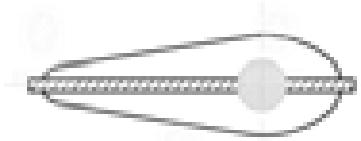
= 1.0 otherwise.

As per Fig, 3.1.1,

$C = \frac{x_2+x_3-x_1}{2}$; Mean Breadth of Rudder

$b = \frac{z_3+z_4-z_2}{2}$; Mean Height of Rudder

Table 3.1.1

Profile Type	K_2	
	Ahead condition	**Astern condition**
NACA-00 series Göttingen 	1.10	0.80
Flat side 	1.10	0.90
Hollow	1.35	0.90
High lift rudders 	1.70*	1.30
Fish tail 	1.40	0.80
Single plate 	1.00	1.00
Mixed profiles (e.g. HSVA)	1.21	0.90

Note :
***** In the case of flap rudder, K_2 values determined by direct calculation methods taking into account geometry of rudder and flap and their angles of turn may be accepted as an alternative to the value indicated in the Table.

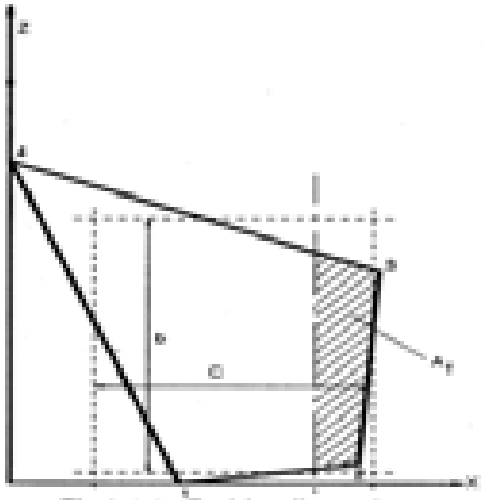


Fig.3.1.1 : Rudder dimensions

3.2 Rudder torque

3.2.1 The rudder torque on regular shaped rudders in both the ahead and astern conditions of travel is to be calculated as follows:

$$Q_r = F_r \cdot r \text{ [N-m]};$$

where,

$r = x_c - f$ [m];

for ahead condition, r is not to be taken less than 0.1C

x_c = the distance of the point of application of the design force F_r from the leading edge

= 0.33 C in ahead condition

= 0.66 C in astern condition.

C = Mean breadth of rudder area [m] See Fig.3.1.1.

$f = C \cdot A_f/A$ where A_f is the portion of the rudder blade area situated ahead of the centre line of the rudder stock.

3.2.2 In case of rudder blades with stepped contours / semi-spade rudders, the total rudder torque is to be obtained as follows:

$$Q_r = \Sigma Q_{ri} \text{ for } i = 1,2,3,.....$$

where,

$Q_{ri} = F_{ri}.r_i$; individual torque component from each part A_i of the total rudder area. See Fig.3.2.2.

The rudder area may be divided into two rectangular or trapezoidal parts with areas A_1 and A_2, so that $A = A_1 + A_2$

A_{1a} = portion of A_1 situated aft of the centre line of the rudder stock.
A_{1f} = portion of A_1 situated ahead of the centre line of the rudder stock.
A_{2a} = portion of A_2 situated aft of the centre line of the rudder stock.
A_{2f} = portion of A_2 situated ahead of the centre line of the rudder stock.

$F_{ri} = F_r . A_i/A$

$r_i = x_{ci} - f_i$; but not to be taken less than $0.1 C_i$.

where,

$f_i = C_i . A_{if}/A_i$

x_{ci}, f_i , A_{if} and C_i are to be taken as x_c, f, A_f and C as in 3.2.1 for each discrete part except that for those rudder parts immediately aft of rudder horn x_{ci} is to be taken as $0.25C_i$ and $0.55C_i$ in ahead and astern conditions respectively.

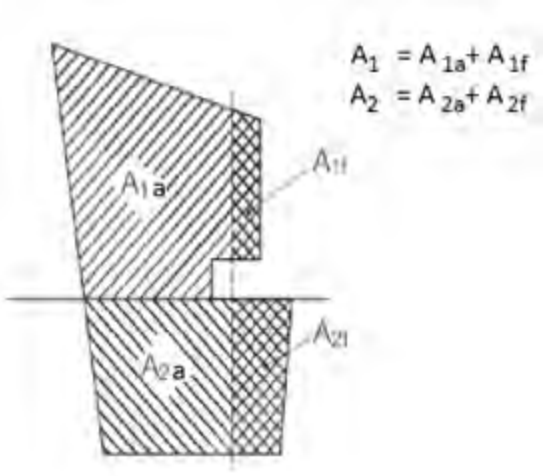


Fig. 3.2.2. Rudder area distribution

3.3 Rudder Strength Calculations

3.3.1 The rudder force and resulting rudder torque as given in 3.1 and 3.2 causes bending

moments and shear forces in the rudder body, bending moments and torques in the rudder stock, supporting forces in pintle bearings and rudder stock bearings and bending moments, shear forces and torques in rudder horns and heel pieces. The rudder body is to be stiffened by horizontal and vertical webs enabling it to act as a bending girder.

3.3.2 The bending moment (BM) and shear force (SF) distributions along the entire height of the rudder blade and rudder stock as well as the bearing reactions may be obtained by direct calculation. The rudder is to be assumed as simply supported at the centres of the upper bearing and the neck bearing. In case of rudders supported by the sole piece or rudder horn the flexibility of the sole piece or rudder horn, and rudder and rudder stock is to be taken into consideration. Guidelines for calculation of bending moment and shear force distribution for various types of rudders are given in Section 8.

Section 4

Rudder Blades

4.1 Construction details

4.1.1 Care is to be taken to avoid notch effects and to maintain continuity of strength around cut-outs and openings in the side plating. The plating thickness is to be increased suitably and corners are to be well rounded and ground smooth.

4.1.2 Side plating and vertical webs transmitting the torque are to be welded to the coupling flange by full penetration welds. Welds in the rudder side plating subjected to significant stresses from rudder bending and welds between plates and heavy pieces (solid parts in forged or cast steel or very thick plating) are to be made as full penetration welds. In way of highly stressed areas e.g. cut-out of semi-spade rudder and upper part of spade rudder, cast or welding on ribs is to be arranged. Two sided full penetration welding is normally to be arranged. Where back welding is impossible, welding is to be performed against ceramic backing bars or equivalent. Steel backing bars may be used and are to be fitted with continuous weld on one side to the bevelled edge, see Figure 4.1.2. The bevel angle is to be at least 15° for one sided welding.

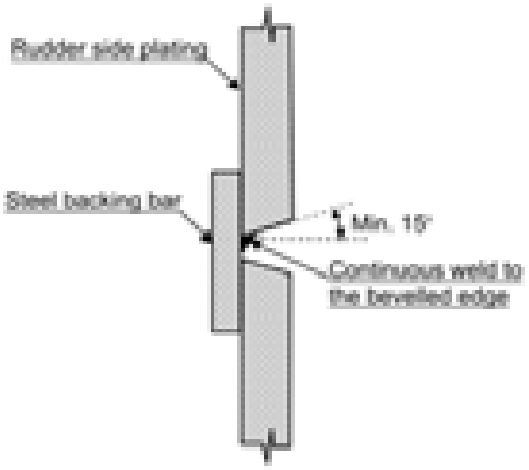


Fig 4.1.2: Use of steel backing bar in way of full penetration welding of rudder side plating

4.1.3 Webs are to be connected to the side plating in accordance with Ch.17. Where fillet welding is not practicable side plating is to be connected by means of slot welding to flat bars welded to the webs. Slot welding is to be limited as far as possible. Slot welding is not to be used in areas with large in-plane stresses transversely to the slots or in way of cut-out areas of semi-spade rudders. Normally slots of

length 75 [mm], breadth at least twice the side plating thickness and spaced 200 [mm] centre to centre will be accepted. The ends of the slots are to be well rounded. The slots are to be fillet welded around the edges and filled with a suitable compound, e.g. epoxy putty. Slots are not to be filled with weld.

4.1.4 Continuous slot welds may be used in lieu of slot welds. When continuous slot welding is used, the root gap is to be between 6-10 [mm] and the bevel angle is to be atleast 15º.

4.1.5 In way of the rudder horn recess of semi-spade rudders, the radii in the rudder plating except in way of solid part in cast steel are not to be less than 5 times the plate thickness, but in no case less than 100 [mm]. Welding in side plate is to be avoided in or at the end of the radii. Edges of side plate and weld adjacent to radii are to be ground smooth.

4.1.6 Requirements for welding and design details of rudder trunks are indicated in 7.3.

4.1.7 Requirements for welding and design details when the rudder stock is connected to the rudder by horizontal flange coupling are indicated in 6.1.6.

4.1.8 Requirements for welding and design details of rudder horns are indicated in 7.2.3.

4.1.9 Arrangement is to be provided to drain the rudders completely. Drain plugs are to be provided with efficient packing.

4.1.10 Internal surfaces of rudders are to be efficiently coated for corrosion resistance after completion of fabrication and testing. Where it is intended to fill the rudder with plastic foam, details of the foam material are to be submitted.

4.2 Double plated rudders

4.2.1 Thickness 't' of the rudder side, top and bottom plating is not to be less than:

$$t = 5.5\, s\, f_a \sqrt{k} \sqrt{\left(T_{SC} + \frac{F_r 10^{-4}}{A}\right)}\, 10^{-3} + 2.5 \ [mm]$$

where,

$$f_a = \sqrt{1.1 - 0.5(s/1000/l)^2};\ \max .1.00$$

s = the smaller of the distances between the horizontal or the vertical web plates/ smallest unsupported width of plating [mm].

l = the larger of the distances between the horizontal or the vertical web plates/ greatest unsupported width of plating [m].

T_{SC} = scantling draught [m]

k = Material factor for the rudder plating.

The thickness 't' is however not to be less than the minimum side shell thickness as per Pt.3, Ch.8.

For nose plates the thickness is to be increased to 1.25 t.

4.2.2 The thickness of the vertical and horizontal webs is not to be less than 70 per cent of the requirement given in 4.2.1 with a minimum of 8 [mm].

4.2.3 Overall strength of the rudder blade (except in way of rudder recess sections where d) applies) based on bending moments (BM) and shear forces (SF) obtained as per Sec.3 is to satisfy the following criteria:

a) bending stress

$$\sigma = \frac{\mathrm{BM}}{\mathrm{Z}_l} \le 110/\mathrm{k}\ [\mathrm{N/mm^2}]$$

b) shear stress $\tau = \dfrac{\mathrm{SF}}{\mathrm{A_w}} \le 50/\mathrm{k}\ [\mathrm{N/mm^2}]$

c) equivalent stresses

$$\sigma_e = \sqrt{\sigma^2 + 3\tau^2} \le 120/\mathrm{k}\ [\mathrm{N/mm^2}]$$

where,

Z_l = Section modulus of the blade cross section against lateral bending obtained considering side plating and web plates within the effective width of the rudder profile. The effective width is to be generally taken as 40 per cent of the net width of the rudder profile; and

A_w = Total Sectional area of the vertical webs.

k = Material factor for the rudder plating

d) In way of cut-outs or openings in the side plating for access to cone coupling or pintle nut:

Bending stress, $\sigma \leq 75$ [N/mm^2]

Torsional stress, $\tau_t \leq 50$ [N/mm^2]

Equivalent stress $\sigma_e = \sqrt{\sigma_b^2 + 3\tau_t^2} \leq 100$ [N/mm^2]

where,

τ_t is torsional stress based on torque arising from the partial rudder force acting on the area below the section in consideration.

Note : The stresses in d) apply equally to high tensile and ordinary steels.

e) In way of the coupling the stresses are to be limited as in d); however for calculation of the stresses only the items within the width of the coupling are to be considered.

4.3 Single plated rudders

4.3.1 Mainpiece diameter

The mainpiece diameter is calculated according to 5.1.1 and 5.1.4 respectively. The diameter of the main piece at top end is not to be less than that of the lower rudder stock, and it may be gradually reduced towards lower end provided stresses are limited as per 4.2.3. For spade rudders the lower third may taper down to 0.75 times stock diameter.

4.3.2 Blade thickness

The blade thickness is not to be less than:

$$t_b = 1.5\, s\, V\sqrt{k} + 2.5 \quad \text{[mm]}$$

where:

s = spacing of stiffening arms, [m], not to exceed 1 m;

V = speed in knots, see 3.1.1.

k = material factor for the rudder plating.

4.3.3 Arms

The thickness of the arms is not to be less than the blade thickness

$t_a = t_b$ [mm]

The section modulus is not to be less than:

$Z_a = 0.5\ s\ C_1^2\ V^2\ k$ [cm^3];

C_1 = horizontal distance from the aft edge of the rudder to the centreline of the rudder stock, [m]

k = material factor

4.3.4 Rudder blade is to be stiffened by horizontal arms spaced not more than 1000 [mm] apart. The arms are to be efficiently attached to the main piece.

4.4 Connection of rudder blade structure with solid parts

4.4.1 Solid parts in forged or cast steel, which house the rudder stock or the pintle, are to be provided with protrusions, except where not required as indicated below.

These protrusions are not required when the web plate thickness is less than:

- 10 [mm] for web plates welded to the solid part on which the lower pintle of a semi-spade rudder is housed and for vertical web plates welded to the solid part of the rudder stock coupling of spade rudders.

- 20 [mm] for other web plates.

4.4.2 The solid parts are in general to be connected to the rudder structure by means of two horizontal web plates and two vertical web plates.

4.4.3 Minimum section modulus of the connection with the rudder stock housing.

The section modulus of the cross-section of the structure of the rudder blade, in [cm^3], formed by vertical web plates and rudder plating, which is connected with the solid part where the rudder stock is housed is to be not less than:

$$Z_s = c_s\, d_c^3 \left(\frac{H_E - H_X}{H_E}\right) \frac{k}{k_s}\ 10^{-4} \quad \text{[cm}^3\text{]}$$

where:

c_s = coefficient, to be taken equal to:

= 1.0 if there is no opening in the rudder plating or if such

openings are closed by a full penetration welded plate

= 1.5 if there is an opening in the considered cross-section of the rudder

d_c = rudder stock diameter, [mm]

H_E = vertical distance between the lower edge of the rudder blade and the upper edge of the solid part, [m]

H_X = vertical distance between the considered cross-section and the upper edge of the solid part, [m]

k = material factor for the rudder blade plating.

k_s = material factor for the rudder stock.

The actual section modulus of the cross-section of the structure of the rudder blade is to be calculated with respect to the symmetrical axis of the rudder.

The breadth of the rudder plating, [m], to be considered for the calculation of section modulus is to be not greater than:

$$b = s_V + 2 H_x / 3 \quad [m]$$

where:

s_V = spacing between the two vertical webs, [m] (see Fig.4.4.3)

Where openings for access to the rudder stock nut are not closed by a full penetration welded plate, they are to be deducted.

4.4.4 The thickness of the horizontal web plates connected to the solid parts, in [mm], as well as that of the rudder blade plating between these webs, is to be not less than the greater of the following values:

$$t_H = 1.2\ t \quad [mm]$$

$$t_H = 0.045\ d_s^2 / s_H \quad [mm]$$

where:

t = defined in 4.2.1

d_S = diameter, [mm], to be taken equal to:

= d_c, as per 5.1.4, for the solid part housing the rudder stock

= d_p, as per 5.2.1, for the solid part housing the pintle

s_H = spacing between the two horizontal web plates, [mm]

The increased thickness of the horizontal webs is to extend fore and aft of the solid part at least to the next vertical web.

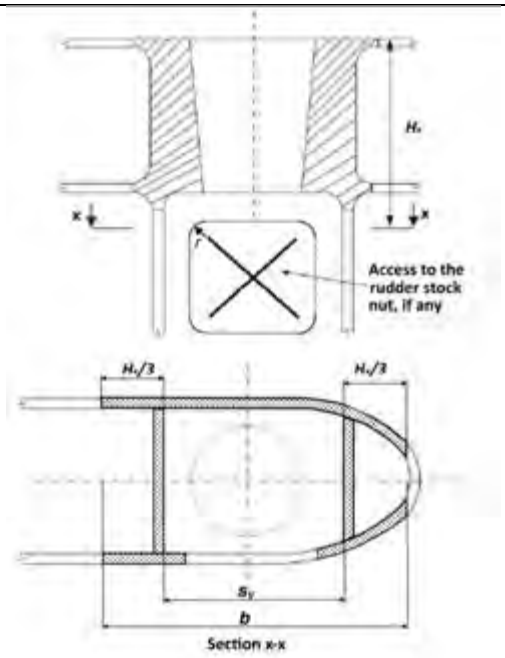


Fig.4.4.3 : Cross-section of the connection between rudder blade structure and rudder stock housing, example with opening in only one side shown

4.4.5 The thickness of the vertical web plates welded to the solid part where the rudder stock is housed as well as the thickness of the rudder side plating under this solid part is to be not less than the values obtained, in [mm], from Table 4.4.5. The increased thickness is to extend below the solid piece at least to the next horizontal web.

Table 4.4.5 : Thickness of side plating and vertical web plates

Type of rudder	Thickness of vertical web plates, [mm]		Thickness of rudder plating, [mm]	
	Rudder blade without opening	Rudder blade with opening	Rudder blade without opening	Area with opening
Rudder supported by sole piece	1.2 t	1.6 t	1.2 t	1.4 t
Semi-spade and spade rudders	1.4 t	2.0 t	1.3 t	1.6 t
t = thickness of the rudder plating, [mm] as defined in 4.2				

Section 5

Rudder Stock and Pintles

5.1 Rudder stock

5.1.1 The rudder stock diameter required for the transmission of the rudder torque is to be dimensioned such that the torsional stress is not exceeding the following value:

$$\tau_T = 68 / k \text{ [N/mm}^2\text{]}$$

The rudder stock diameter for the transmission of the rudder torque is therefore not to be less than:

$$d_t = 4.2\sqrt[3]{Q_r k} \text{ [mm]}$$

Q_r = Total rudder torque [Nm] as calculated in 3.2.

k = Material factor of the rudder stock as given in 1.2.3

5.1.2 If the rudder stock is subjected to combined torque and bending, the Equivalent stress σ_e in the rudder stock is not to exceed 118/k [N/mm^2] i.e.

k = Material factor of the rudder stock as given in 1.2.3

$$\sigma_e = \sqrt{\sigma^2 + 3\tau_t^2} \leq 118 / k \text{ [N/mm}^2\text{]}$$

where,

σ is the bending stress in [N/mm^2],

$\sigma = 10.2 \times 10^3 \, BM / d_c^3$

τ_t is the torsional shear stress in [N/mm^2].

$\tau_t = 5.1 \times 10^3 \, Q_r / d_c^3$

This requirement is regardless of the liners; and both ahead and astern conditions are to be considered.

5.1.3 The diameter of the rudder stock at and above rudder carrier is given by

$$d_t = 4.2\sqrt[3]{(Q_r . k)} \text{ [mm]}$$

5.1.4 The diameter of rudder stock at any other cross section is given by

$$d_c = d_t . \sqrt[6]{\left[1 + \frac{4}{3} . \frac{BM^2}{Q_r^2}\right]} \text{ [mm]}$$

where BM is the bending moment at the cross section under consideration obtained as per 3.3.

For a spade rudder with trunk extending inside the rudder, the rudder stock scantlings are to be checked against the two cases defined in 8.3.

5.1.5 The diameter of the rudder stock at neck bearing is to be maintained to a point as far as practicable above the top of the neck bearing and may subsequently be tapered to that required at the rudder carrier. The length of the taper is to be at least three times the reduction in diameter. Particular care is to be taken to avoid the formation of a notch at the upper end of the taper.

5.1.6 Sudden changes of section or sharp corners in way of the rudder coupling, jumping collars and shoulders for rudder carriers are to be avoided. Jumping collars are not to be welded to the rudder stock. Keyways in the rudder stock are to have rounded ends and the corners at the base of the keyway are to be adequately radiused.

5.1.7 Significant reductions in rudder stock diameter due to the application of steels with specified minimum yield stress exceeding 235 [N/mm^2] are to be granted only after evaluation of rudder stock deformations. Large deformations of the rudder stock are to be avoided in order to avoid excessive pressures in way of bearings.

5.2 Pintles

5.2.1 Scantlings

The diameter d_p of the pintles, is not to be less than

$$d_p = 0.35\sqrt{(B\,k_p)}$$

where,

B = relevant bearing force, [N]

k_p = material factor for pintle as given in 1.2.3

5.2.2 Couplings

Pintles are to have a conical attachment to the gudgeons with a taper on diameter not greater than:
1:8 - 1:12 for keyed and other manually assembled pintles applying locking by slugging nut,

1:12 - 1:20 on diameter for pintles mounted with oil injection and hydraulic nut.

5.2.3 Push-up pressure for pintle

The required push-up pressure for pintle in case of dry fitting, in [N/mm²], is to be determined by *preq1* as given below.

The required push-up pressure for pintle in case of oil injection fitting, in [N/mm²], is to be determined by the maximum pressure of p_{req1} and p_{req2} as given below:

$$p_{req1} = 0.4\,\frac{B d_0}{d_m^2\, l} \quad [N/mm^2]$$

$$p_{req2} = \frac{6\,M_{bp}}{l^2 d_m}\,10^3 \quad [N/mm^2]$$

where:

B = Supporting force in the pintle, [N] e.g., B_1 as defined in Fig 8.5.2 for semi-spade rudder.

d_0, d_m, l = Pintle dimensions, see Fig. 6.3.1

M_{bp} = bending moment in the pintle cone coupling to be determined by:

$$M_{bp} = B l_a\ [Nm]$$

ℓa = length between middle of pintle-bearing and top of contact surface between cone coupling and pintle in m, see Figure 5.2.3.

The required push-up length, $\Delta\ell_1$, is to be calculated similarly as in 6.4.3, using the required push-up pressure as defined above, and properties for the pintle.

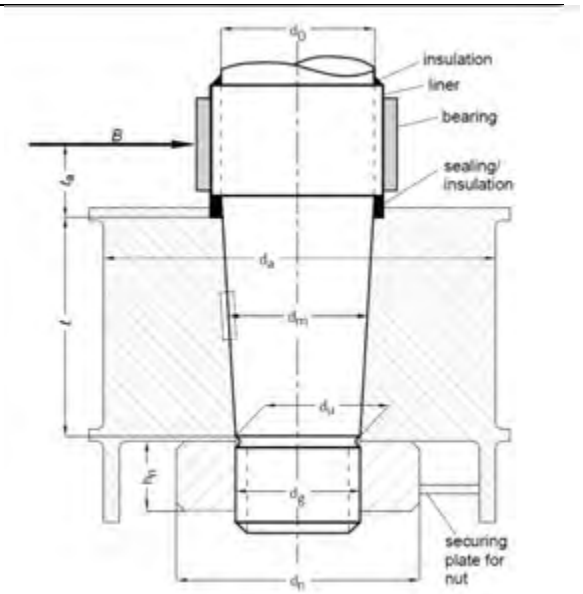


Figure 5.2.3 Pintle cone coupling indicating ℓa

5.2.4 The minimum dimensions of threads and nuts are to be determined according to Cl. 6.3.2. The push-up length is to be calculated similarly as in Cl. 6.4.3, using required push-up pressure and properties for the pintle. An effective sealing against sea water is to be provided at both ends of the cone.

5.2.5 Pintle housing

The length of pintle housing in the gudgeon is not to be less than the pintle diameter d_p. The thickness of the pintle housing is not to be less than 0.25 d_p.

5.2.6 Where liners are fitted to pintles, they are to be shrunk on or otherwise efficiently secured. If liners are to be shrunk on, the shrinkage allowance is to be indicated on the plans. Where liners are formed by stainless steel weld deposit, the pintles are to be of weldable quality steel, and details of the procedure are to be submitted. Bushing is to be effectively secured against movement.

5.3 Bearings of rudder stock, rudder shaft and pintle

5.3.1 Liners and Bushes

5.3.1.1 Rudder stock bearing

Liners and bushes are to be fitted in way of bearings. For rudder stocks and pintles having diameter less than 200 [mm], liners in way of bushes may be provided optionally. The

minimum thickness of liners and bushes is to be equal to:

- t_{min} = 8 [mm] for metallic materials and synthetic material
- t_{min} = 22 [mm] for lignum material

5.3.1.2 Pintle bearing

The thickness of any liner or bush, [mm], is neither to be less than:

$$t = 0.01\sqrt{B}$$

where:

B = relevant bearing force, [N]

nor less than the minimum thickness defined in 5.3.1.1

5.3.2 Minimum Bearing Surface

5.3.2.1 An adequate lubrication is to be provided. The bearing surface A_b (defined as the projected area: length x outer diameter of liner) is not to be less than:

$$A_b = {}^{P}/_{q_a}$$

where:

P = reaction force [N] in bearing as determined in 3.3;

q_a = allowable surface pressure according to the Table 5.3.4.

5.3.3 Bearing clearances are normally to be as given in Table 5.3.3.

Attention is to be paid to the manufacturer's recommendations particularly where bush material requires pre-soaking.

5.3.4 The maximum surface pressure p, for the various combinations is not to exceed the values given in Table 5.3.4. Higher values than given in table may be taken on verification by tests.For the purpose of this calculation, the bearing length is not to be taken greater than 1.2 times the rudder stock or pintle diameter measured outside of liners, if fitted. The bearing length L_p of the pintle is to be such that

$$D_p \leq L_p \leq 1.2\, D_p$$

where,

D_p = Actual pintle diameter measured on the outside of the liners.

Table 5.3.3 : Bearing clearances

- For metal bearing material	Not less than 0.001 d_p + 1.0 [mm]
- For synthetic bearing material	To be specially determined considering the swelling and thermal expansion properties of the material, but not less than 1.5 [mm]#.

A smaller clearance may be accepted if supported by the manufacturer’s recommendation and documented evidence of satisfactory service history with reduced clearance.

Table 5.3.4 : Allowable surface pressure q_a

Bearing material	q_a [N/mm^2]
Lignum vitae	2.5
White metal, oil lubricated	4.5
Synthetic material with hardness greater than 60 shore D(1)	5.5(2)
Steel,(3) bronze and hot pressed bronze- graphite materials	7.0

Note:

(1) Indentation hardness test at 23°C and with 50% moisture, according to a recognised standard. Synthetic bearing materials to be of approved type.

(2)Surface pressures exceeding 5.5 [N/mm^2] may be accepted in accordance with the bearing manufacturer's specification and tests, but in no case more than 10 [N/mm^2]

(3) Stainless and wear resistant steel in an approved combination with stock liner.

Section 6

Rudder Couplings

6.1 Horizontal flange couplings

6.1.1 The diameter of the coupling bolts is not to be less than:

$$d_b = 0.62\left[\frac{d_s^3 . k_b}{n.e_m.k_s}\right]^{1/2} \text{[mm]}$$

where,

d_s = Stock diameter [mm] in way of the coupling flange; equal to the greater of the diameters d_t or d_c according to 5.1.3 and 5.1.4

k_s = material factor for the rudder stock material;

k_b = material factor for the bolt material;

n = total number of bolts;

e_m = mean distance of the bolt axes from the centre of the bolt system [mm].

6.1.2 Coupling bolts are to be fitted bolts and a minimum of six (6) bolts are to be provided. Their nuts are to be effectively locked.

6.1.3 Mean distance e_m from the centre of the bolts to the centre of the bolt system is not to be less than 0.9 d_s [mm]. In addition, where the coupling is subjected to bending stress the mean athwartship distance from the centre of bolts to the longitudinal centreline of the coupling is not to be less than 0.6 d_s [mm].

6.1.4 The thickness of coupling flanges is not to be less than that obtained by the following formula:

$$t_f = d_b\sqrt{\left(\frac{k_f}{k_b}\right)} \text{[mm]}$$

t_f is to be also not less than 0.9 d_b;

d_b = bolt diameter [mm] calculated for a number of bolts not exceeding 8.

k_f = material factor for the flange;

k_b = material factor for the bolt material.

6.1.5 The width of material outside the bolt holes is not to be less than 0.67 d_b [mm].

6.1.6 The welded joint between the rudder stock and the flange is to be made in accordance with Fig.6.1.6 or equivalent.

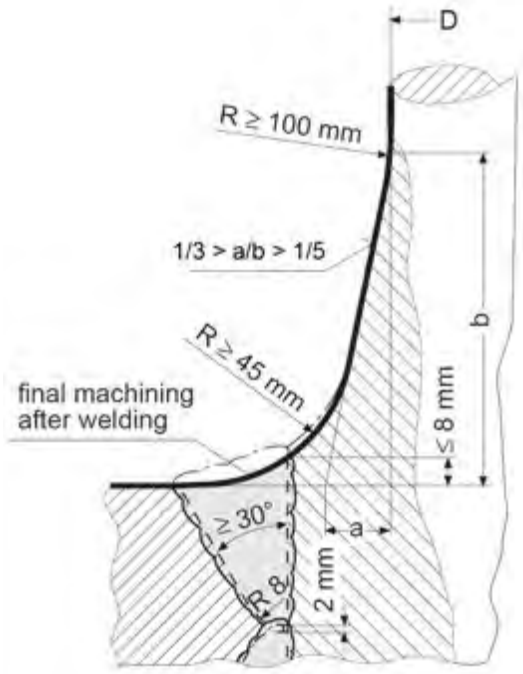


Fig. 6.1.6 : Welded joint between rudder stock and coupling flange

6.2 Vertical flange couplings

6.2.1 The diameter of the coupling bolts is not to be less than:

$$d_b = 0.81\left[\frac{d_s^2 . k_b}{n . k_s}\right]^{1/2} \quad [mm]$$

where,

d_s = Stock diameter [mm] in way of the coupling flange

n = total number of bolts, but not less than 8.

k_b = material factor for bolts.

k_s = material factor for rudder stock.

6.2.2 The first moment of area of the bolts about the centre of the coupling to be not less than:

$$m = 0.00043\ d_s^3\ [cm^3]$$

6.2.3 The thickness of the coupling flanges is to be at least equal to the bolt diameter; and the width of the flange material between the perimeter of the bolt holes and the perimeter of the flange is to be greater than or equal to 0.67 d_b.

6.2.4 Coupling bolts are to be fitted bolts and their nuts are to be locked effectively.

6.3 Cone couplings with key/ Cone couplings without hydraulic arrangement for mounting and dismounting

6.3.1 Cone couplings without hydraulic arrangements for mounting and dismantling the coupling should have a taper 'c' on diameter generally ranging between 1:8 to 1:12.

$c = \frac{d_o - d_u}{l_c}$ according to Fig. 6.3.1 and 6.3.1b)

The diameters d_o in mm, and d_u in mm, are defined in Fig. 6.3.1 and the cone length l_c, in mm, is defined in Fig. 6.3.1 b)

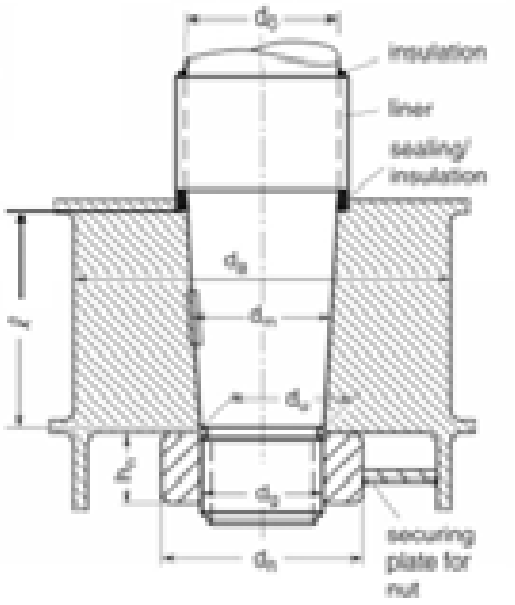


Fig.6.3.1 : Cone coupling with key

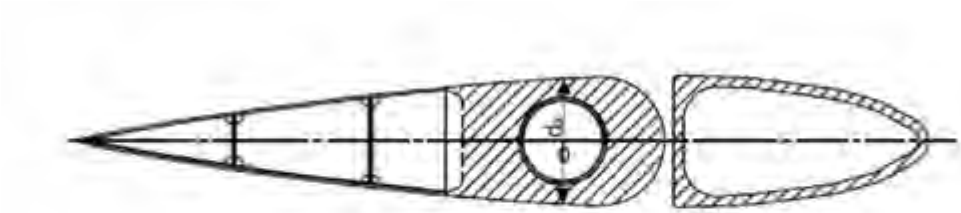


Fig. 6.3.1 a) Gudgeon outer diameter(d_a) measurement

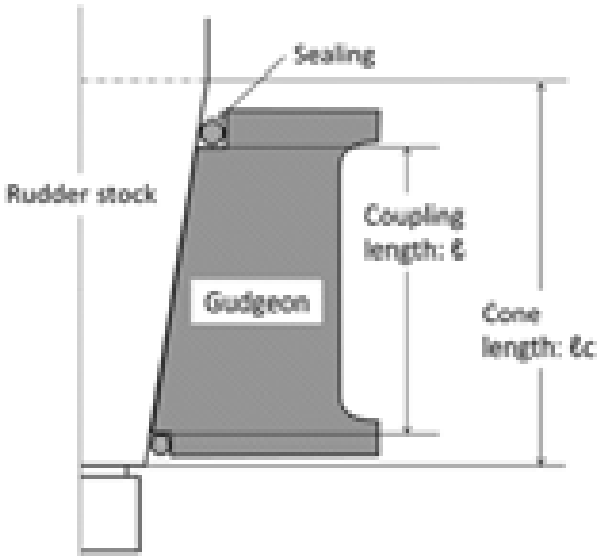


Fig. 6.3.1 b) Cone length and coupling length

The coupling length *l* in mm, is in general not to be less than 1.5 times the rudder stock diameter d_o at the top of rudder.

The cone shape is to be very exact and the slugging nut is to be efficiently secured.

6.3.2 The dimensions of the slugging nut [mm], are not to be less than (See Fig 6.3.1) :

External thread of nut, $d_g \geq 0.65\ d_o$;

Outer diameter of nut, $d_n \geq 1.2\ d_u$ or $1.5\ d_g$, whichever is the greater;

Height of nut, $h_n \geq 0.6\ d_g$

6.3.3 A key is to be fitted in the tapered connection, as close to the nut as possible and ample fillets are to be provided at corners of the keyway.

The shear area of the key is not to be less than:

$$a_s = \frac{17.55\ Q_F}{d_k\ R_{eH1}}$$

where,

Q_F = design yield moment of rudder stock, in [Nm]

$$Q_F = 0.02664\ \frac{d_t^3}{k}$$

Where the actual diameter d_{ta} is greater than the calculated diameter d_t, the diameter d_{ta} is to be used. However, d_{ta} applied to the above formula need not be taken greater than 1.145 d_t.

d_t = stock diameter, [mm], according to 5.1.3.

k = material factor for stock as given in 1.2.3

d_k = mean diameter of the conical part of the rudder stock, [mm], at the key

R_{eH1} = specified minimum yield stress of the key material, [N/mm^2]

The effective surface area, [cm^2], of the key (without rounded edges) between key and rudder stock or cone coupling is not to be less than:

$$a_k = \frac{5\,Q_F}{d_k R_{eH2}}$$

where:

R_{eH2} = specified minimum yield stress of the key, stock or coupling material, [N/mm^2], whichever is less.

6.3.4 Push up

It is to be proved that 50% of the design yield moment is solely transmitted by friction in the cone couplings. This can be done by calculating the required push-up pressure and push-up length according to 6.4.2 and 6.4.3 for a torsional moment $Q'_F = 0.5\ Q_F$.

6.3.5 Notwithstanding the requirements in 6.3.3 and 6.3.4, where a key is fitted to the coupling between stock and rudder and it is considered that the entire rudder torque is transmitted by the key at the couplings, the scantlings of the key as well as the push-up force and push-up length are to be specially considered.

6.4 Cone couplings with hydraulic arrangement for mounting and dismounting (with oil injection and hydraulic nut)

6.4.1 Where the stock diameter exceeds 200 [mm], the press fit is recommended to be achieved by a hydraulic pressure connection. In such cases, the cone diameter is to be, c, generally ranging between 1:12 to 1:20.

In case of hydraulic pressure connections the nut is to be effectively secured against the rudder stock or the pintle.

For the safe transmission of the torsional moment by the coupling between rudder stock and rudder body the push-up pressure and the push-up length are to be determined according to 6.4.2 and 6.4.3 respectively.

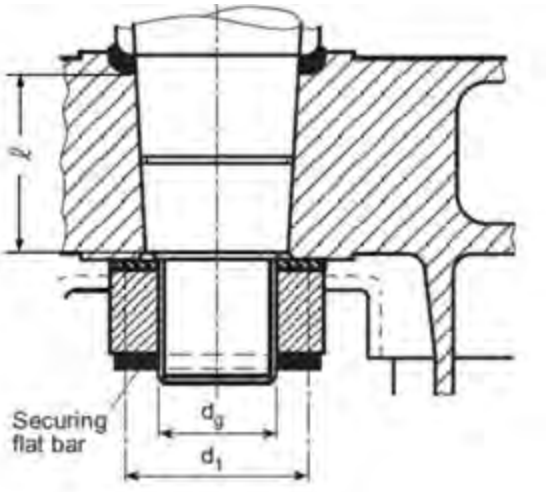


Fig.6.4.1 : Cone coupling without key

6.4.2 Push-up pressure

The push-up pressure is not to be less than the greater of the two following values:

$$p_{req1} = \frac{2Q_F}{d_m^2 \ell \pi \mu_0} 10^3 \quad [\text{N/mm}^2]$$

$$P_{req2} = \frac{6M_C}{l^2 d_m} 10^3 \ [N/mm^2]$$

where:

Q_F = design yield moment of rudder stock, as defined in 6.3.3, [Nm]

d_m = mean cone diameter [mm], see Figure 6.3.1

ℓ = coupling length [mm]

μ_0 = frictional coefficient, equal to 0.15

M_C = bending moment in rudder stock at the top of the cone coupling (e.g. in case of spade rudders), [Nm]

For spade rudder with trunk extending inside the rudder, the coupling is to be checked against the two cases defined in 8.3.

It has to be proved by the designer that the push-up pressure does not exceed the permissible surface pressure in the cone. The permissible surface pressure, [N/mm²], is to be determined by the following formula:

$$p_{perm} = \frac{0.95R_{eH}(1-\alpha^2)}{\sqrt{3+\alpha^2}} - p_b \ [N/mm^2]$$

where:

$$p_b = \frac{3.5M_C}{d_m l^2} 10^3$$

R_{eH} = specified minimum yield stress of the material of the gudgeon [N/mm²]

α = d_m / d_a

d_m = diameter, [mm], see Fig. 6.3.1.

d_a = outer diameter of the gudgeon, [mm], see Fig.6.3.1 and Fig. 6.3.1 a). The least diameter is to be considered.

The outer diameter of the gudgeon in mm is not to be less than $1.25d_o$, with d_o defined in Fig 6.3.1

6.4.3 Push-up length

The push-up length $\Delta\ell$, [mm], $\Delta\ell$ is to comply with the following formula:

$$\Delta\ell_1 \le \Delta\ell \le \Delta\ell_2$$

where:

$$\Delta\ell_1 = \frac{p_{req} d_m}{E\left(\frac{1-\alpha^2}{2}\right)c} + \frac{0.8R_{tm}}{c} \quad \text{[mm]}$$

$$\Delta l_2 = \frac{p_{perm} d_m}{E\left(\frac{1-\alpha^2}{2}\right)c} + \frac{0.8R_{tm}}{c} \ [mm]$$

R_{tm} = mean roughness, [mm] taken equal to 0.01

c = taper on diameter defined in 6.3.1.

Note: In case of hydraulic pressure connections the required push-up force P_e, [N], for the cone may be determined by the following formula:

$$P_e = p_{req} d_m \pi \ell \left(\frac{c}{2} + 0.02\right)$$

The value 0.02 is a reference for the friction coefficient using oil pressure. It varies and depends on the mechanical treatment and roughness of the details to be fixed. Where due to the fitting procedure a partial push-up effect caused by the rudder weight is given, this may be taken into account when fixing the required push-up length, subject to approval by IRS.

Section 7

Sole Pieces and Rudder Horns

7.1 Sole piece

The section modulus around the vertical (z)-axis is not to be less than:

$Z_z = M_b\, k / 80$ [cm^3]

The section modulus around the transverse (y)-axis is not to be less than:

$Z_y = 0.5\, Z_z$

The sectional area is not to be less than:

$A_s = B_1\, k / 48$ [mm^2]

k = material factor as given in 1.2.3

7.1.1 Equivalent stress

At no section within the length ℓ_{50} is the equivalent stress to exceed 115 / k [N/mm^2]. The equivalent stress is to be determined by the following formula (See Fig.8.4.2):

$$\sigma_e = \sqrt{\sigma_b^2 + 3\tau^2} \quad [N/mm^2];$$

where:

$\sigma_b = M_b / Z_z(x)$ [N/mm^2];

$\tau = B_1 / A_s$ [N/mm^2];

M_b = bending moment at the section considered [Nm];

$M_b = B_1\, x$ [Nm];

$M_{bmax} = B_1\, \ell_{50}$ [Nm];

B_1 = supporting force in the pintle bearing [N] (normally $B_1 = F_r / 2$).

k = material factor as given in 1.2.3

7.2 Rudder horn

When the connection between the rudder horn and the hull structure is designed as a curved transition into the hull plating, special consideration is to be given to the effectiveness of the rudder horn plate in bending and to the stresses in the transverse web plates.
The bending moments and shear forces are to be determined by a direct calculation or in line with the guidelines given in Section 8 for semi spade rudder with one elastic support and semi spade rudder with 2-conjugate elastic support respectively.

The section modulus around the horizontal x-axis is not to be less than:

$Z_x = M_b\, k / 67$ [cm^3].

M_b = bending moment at the section considered, [Nm];

The shear stress is not to be larger than:

τ = 48 / k [N/mm^2].

k = material factor as given in 1.2.3

7.2.1 Equivalent stress

At no section within the height of the rudder horn is the equivalent stress to exceed 120 / k N/mm^2. The equivalent stress is to be calculated by the following formula:

$$\sigma_e = \sqrt{\sigma_b^2 + 3\left(\tau^2 + \tau_T^2\right)} \quad [N/mm^2];$$

$\sigma_b = M_b / Z_x$ [N/mm^2];

$\tau = B_1 / A_h$ [N/mm^2];

B_1 = supporting force in the pintle bearing [N];

A_h = effective shear area of rudder horn in y-direction [mm^2];

$\tau_T = M_T\, 10^3 / 2\, A_T\, t_h$ [N/mm^2];

M_T = torsional moment [Nm];

A_T = area in the horizontal section enclosed by the rudder horn [mm^2];

t_h = plate thickness of rudder horn [mm];

k = material factor as given in 1.2.3

7.2.2 Rudder horn plating

The thickness of the rudder horn side plating is not to be less than:

$$t = 2.4\sqrt{Lk} \quad \text{[mm]}$$

where:

L = Rule length as defined in Chapter 1, Cl. 2.1.3;

k = material factor as given in 1.2.3.

7.2.3 Welding and connection to hull structure

.1 The rudder horn plating is to be effectively connected to the aft ship structure, e.g. by connecting the plating to side shell and transverse/ longitudinal girders, in order to achieve a proper transmission of forces, see Figure 7.2.3.

.2 Brackets or stringer are to be fitted internally in horn, in line with outside shell plate, as shown in Figure 7.2.3.

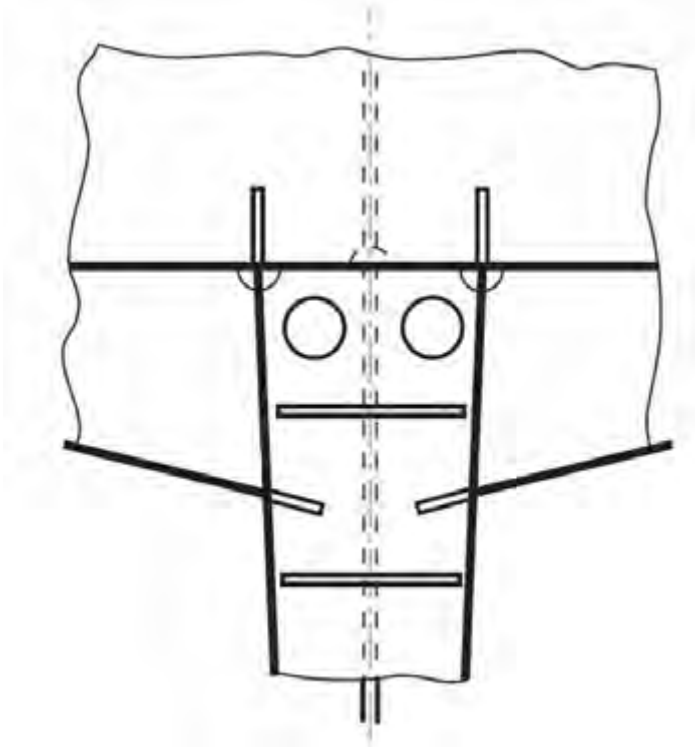


Fig.7.2.3 : Connection of Rudder Horn to aft Ship Structure

.3 Transverse webs of the rudder horn are to be led into the hull up to the next deck in a sufficient number.

.4 Strengthened plate floors are to be fitted in line with the transverse webs in order to achieve a sufficient connection with the hull.

.5 The centre line bulkhead (wash-bulkhead) in the after peak is to be connected to the rudder horn.

.6 Scallops are to be avoided in way of the connection between transverse webs and shell plating.

.7 The weld at the connection between the rudder horn plating and the side shell is to be full penetration. The welding radius is to be as large as practicable and may be obtained by grinding.

7.3 Rudder trunk

The requirements in this sub-section apply to trunk configurations which are extended below stern frame and arranged in such a way that the trunk is stressed by forces due to rudder action.

7.3.1 Materials, welding and connection to hull

The steel used for the rudder trunk is to be of weldable quality, with a carbon content not exceeding 0.23% on ladle analysis or a carbon equivalent C_{EQ} not exceeding 0.41%.
Plating materials for rudder trunks are in general not to be of lower grades than corresponding to Class II as defined in Part 2 of the Rules.

The weld at the connection between the rudder trunk and the shell or the bottom of the skeg is to be full penetration.

For rudder trunks extending below shell or skeg,the fillet shoulder radius r, [mm] (see Figure 7.2.4) is to be as large as practicable and to comply with the following formulae:

$r = 0.1d_c/k$

without being less than

$r = 60$ [mm] when $\sigma \geq 40/k$ [N/mm^2]

$r = 30$ [mm] when $\sigma < 40/k$ [N/mm^2]

where:

d_c = rudder stock diameter axis defined in 5.1.

σ = bending stress in the rudder trunk, [N/mm^2].

k = material factor for the rudder trunk as given in 1.2.3

The radius may be obtained by grinding. If disk grinding is carried out, score marks are to be avoided in the direction of the weld. The radius is to be checked with a template for accuracy. Four profiles at least are to be checked. A report is to be submitted to the attending Surveyor.

Rudder trunks comprising of materials other than steel would be specially considered.

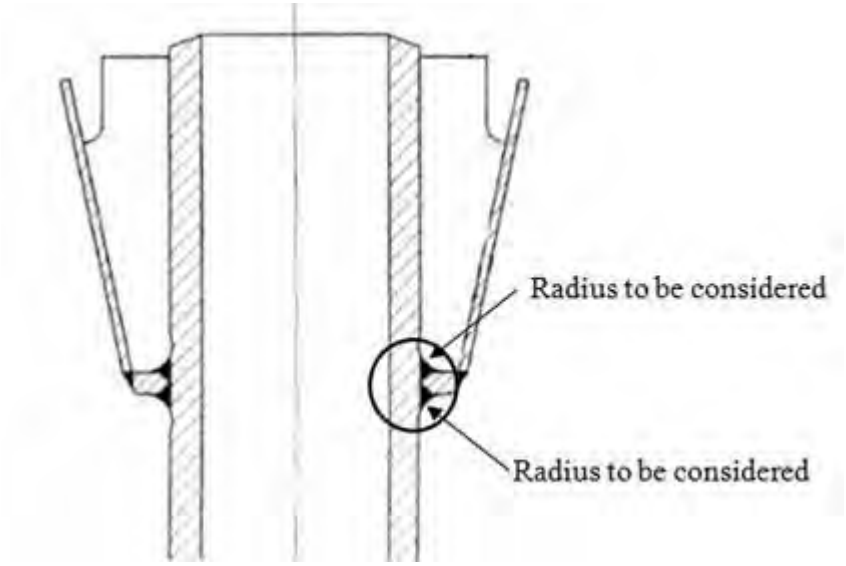


Fig.7.2.4 : Fillet Shoulder Radius

7.3.2 Scantlings

The scantlings of the trunk are to be such that:

- the equivalent stress due to bending and shear does not exceed 0.35 R_{eH},
- the bending stress on welded rudder trunk is to be in compliance with the following formula: $\sigma \leq 80/k$ [N/mm^2]

with:

σ = bending stress in the rudder trunk, as defined in 7.3.1.

k = material factor for the rudder trunk as given in 1.2.3, not to be taken less than 0.7

R_{eH} = specified minimum yield stress [N/mm^2] of the material used.

For calculation of bending stress, the span to be considered is the distance between the mid-height of the lower rudder stock bearing and the point where the trunk is clamped into the shell or the bottom of the skeg.

Section 8

Guidelines for Evaluation of Bending Moment and Shear Force Distribution

8.1 The evaluation of bending moments, shear forces and support forces for the system rudder – rudder stock may be carried out for some basic rudder types as outlined in the succeeding paragraphs:

8.2 Spade rudder

8.2.1 Data for the analysis

ℓ_{10} - ℓ_{30} = Lengths of the individual girders of the system, [m]

I_{10} – I_{30} = Moments of inertia of these girders, [cm^4]

Load of rudder body:

$$P_R = F_r / (\ell_{10}\, 10^3) \quad [kN/m]$$

8.2.2 The moments and forces may be determined by the following formulae:

$$M_b = F_r (\ell_{20} + (\ell_{10} (2\, c_1 + c_2) / 3 (c_1 + c_2))) \quad [Nm]$$

$$B_3 = M_b / \ell_{30} \quad [N]$$

$$B_2 = F_r + B_3 \quad [N].$$

The maximum moment, M_C, in top of the cone coupling as shown in Fig 8.2.2 is applicable for the connection between the rudder and the rudder stock.

8.3 Spade rudder with trunk

8.3.1 Data for the analysis

ℓ_{10} - ℓ_{30} = Lengths of the individual girders of the system, [m]

I_{10} – I_{30} = Moments of inertia of these girders, [cm^4]

Load of rudder body:

$$P_R = Fr / ((\ell_{10} + \ell_{20})10^3) \quad [kN/m]$$

Moments and forces

For a spade rudder with trunk extending inside the rudder, the strength is to be checked against the following two cases:

a) pressure applied on the entire rudder area

b) pressure applied only on rudder area below the middle of neck bearing.

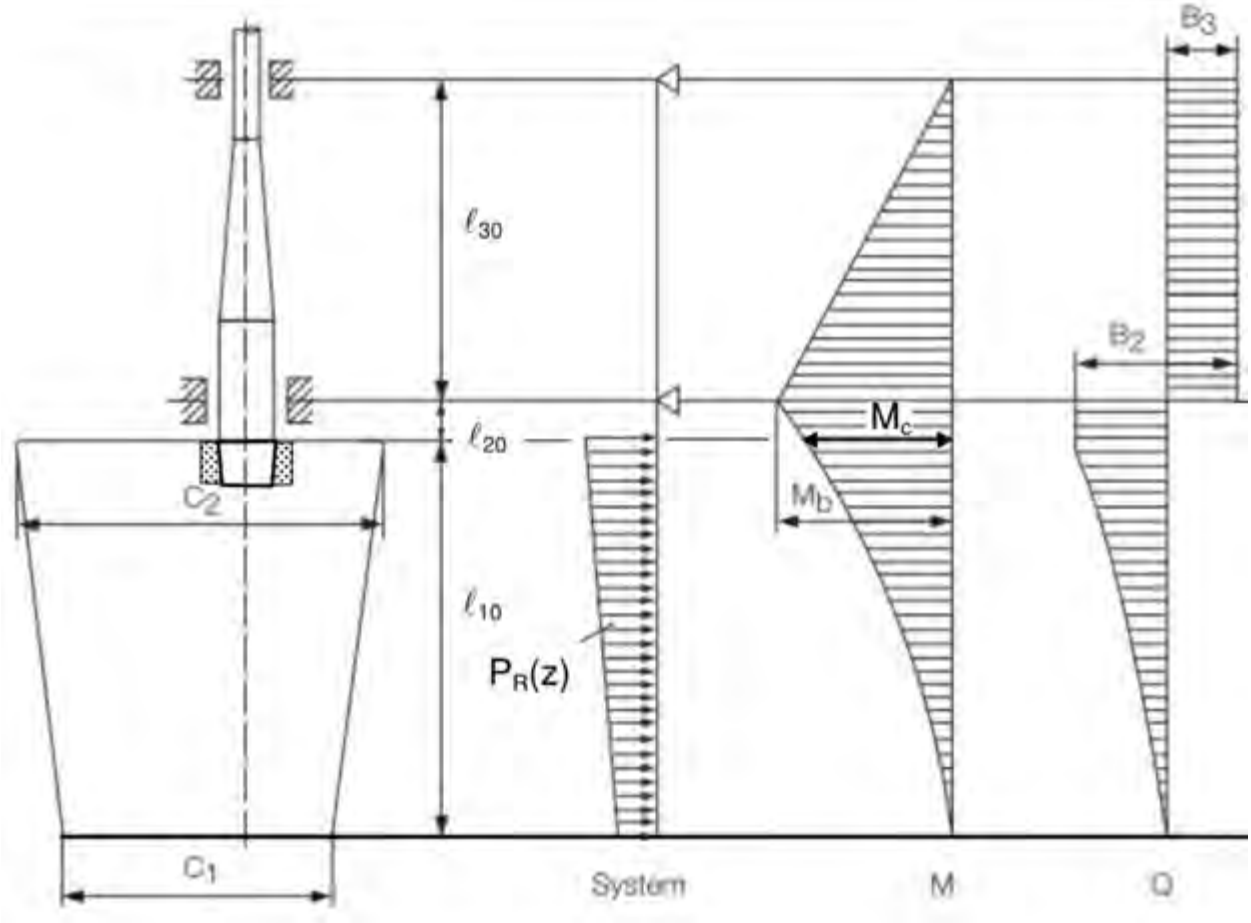


Fig.8.2.2 : Strength Calculations – Spade Rudder

8.3.2 The moments and forces for the two cases defined above may be determined according to Fig 8.3.2 a) and b) respectively.

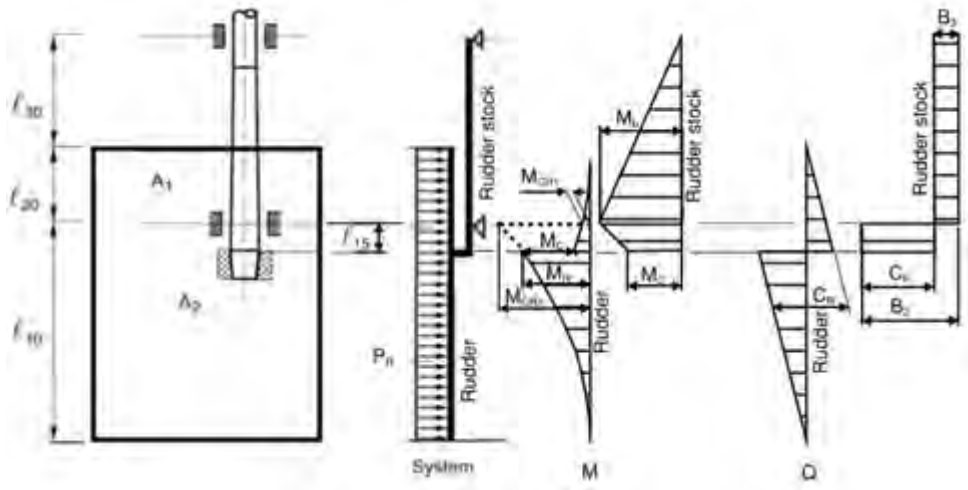


Fig. 8.3.2 a) Full rudder force *CR* = *CR1*+*CR2* and total rudder torque *QR* = *QR1* + *QR2* with rudders stock bending moment *Mb* = *MCR2* - *MCR1*

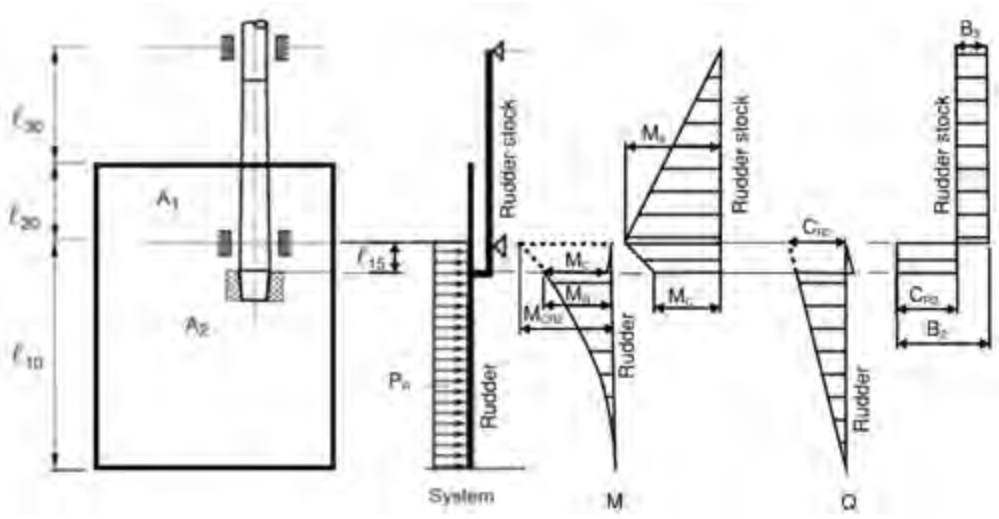


Fig. 8.3.2 b) Rudder force *CR2* corresponding to rudder torque *QR2* acting at rudder blade area *A2* with rudders stock bending moment *Mb = MCR2*

$M_{Fr2} = F_{r2} (\ell_{10} - CG_{2Z})$

$M_{Fr1} = F_{r1} (CG_{1Z} - \ell_{10})$

where:

F_{r1} : Rudder force over the rudder blade area A_1

F_{r2} : Rudder force over the rudder blade area A_2

CG_{1Z} : Vertical position of the centre of gravity of the rudder blade area A_1

CG_{2Z} : Vertical position of the centre of gravity of the rudder blade area A_2

$F_r = F_{r1} + F_{r2}$

$B_3 = (M_{Fr2} - M_{Fr1}) / (\ell_{20} + \ell_{30})$

$B_2 = F_r + B_3$

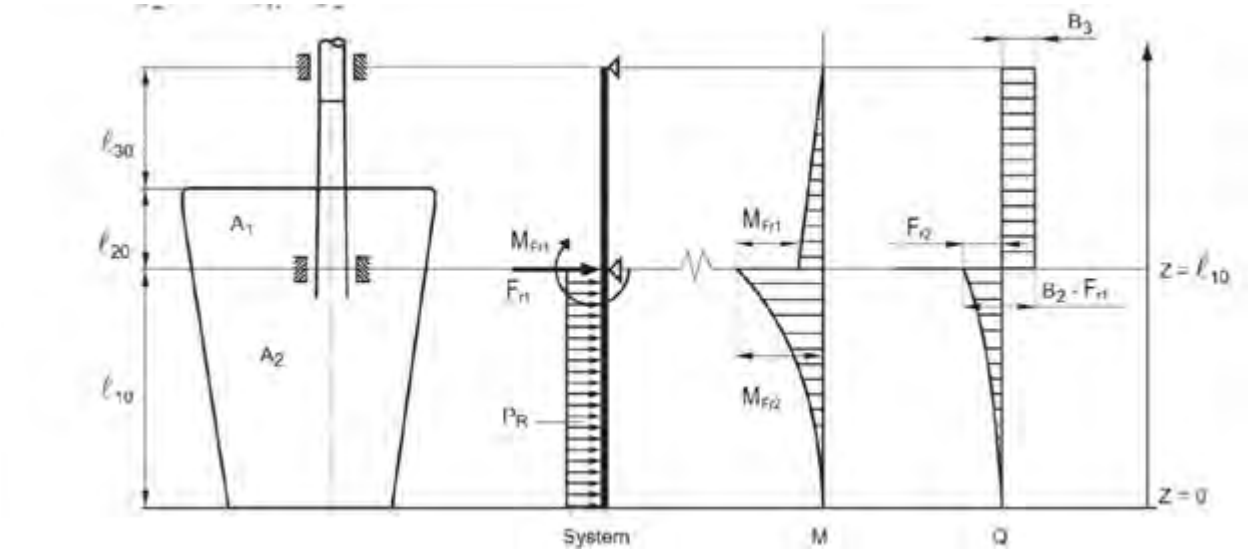


Fig.8.3.2 : Strength Calculations – Spade Rudder with Trunk

8.4 Rudder supported by sole piece

8.4.1 Data for the Analysis

ℓ_{10} - ℓ_{50} = Lengths of the individual girders of the system, [m]

I_{10} – I_{50} = Moments of inertia of these girders, [cm^4]

For rudders supported by a sole piece the length ℓ_{20} is the distance between lower edge of rudder body and centre of sole piece and I_{20} the moment of inertia of the pintle in the sole piece.

I_{50} = moment of inertia of sole piece around the z-axis, [cm^4];

ℓ_{50} = effective length of sole piece, [m];

Load of rudder body:

P_R = F_r / (ℓ_{10} 10^3) [kN/m]

Z = spring constant of support in the sole piece

Z = 6.18 x I_{50} / ℓ_{50}^3 [kN/m]

8.4.2 The moments and shear forces are indicated in Figure 8.4.2.

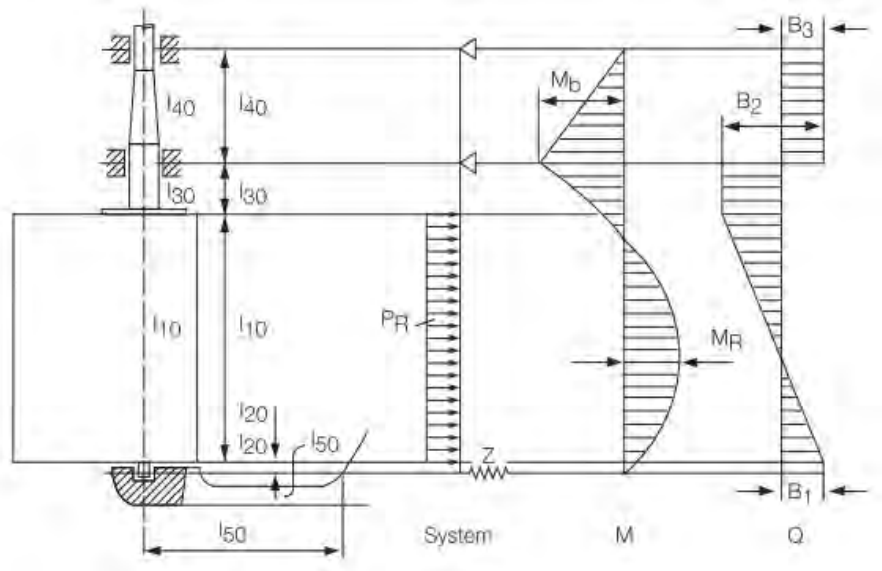


Fig.8.4.2 : Strength Calculations – Rudder supported by Sole Piece

8.5 Semi-spade rudder with one elastic support

8.5.1 Data for the analysis

ℓ_{10} - ℓ_{50} = Lengths of the individual girders of the system, [m];

I_{10} – I_{50} = Moments of inertia of these girders, [cm^4];

Z = spring constant of support in the rudder horn;

Z = 1 / (f_b + f_t) [kN/m] for the support in the rudder horn (Figure 8.5.2);

f_b = unit displacement of rudder horn in [m] due to a unit force of 1 kN acting in the centre of support;

f_b = 1.3 d^3 / (6.18 I_n) [m/kN] (guidance value);

I_n = moment of inertia of rudder horn around the x-axis, [cm^4] (see also Figure 8.5.2);

f_t = unit displacement due to torsion;

$f_t = d\, e^2 \sum {u_i}/{t_i} / [3.14 \times 10^8 F_T^2]$ [m/KN];

F_T = mean sectional area of rudder horn, [m^2];

u_i = breadth, [mm] of the individual plates forming the mean horn sectional area;

t_i = thickness within the individual breadth u_i, [mm];

d = Height of the rudder horn, [m], defined in Figure 8.5.2. This value is measured downwards from the upper rudder horn end, at the point of curvature transition, to the mid-line of the lower rudder horn pintle;

e = distance as defined in Figure 8.5.3

Load of rudder body:

P_{R10} = F_{r2} / (ℓ_{10} x 10^3) [kN/m];

P_{R20} = F_{r1} / (ℓ_{20} x 10^3) [kN/m];

for F_r, F_{r1}, F_{r2}, method indicated in 3.2.2 for estimation may be followed.

8.5.2 Moments and shear forces are indicated in Figure 8.5.2

8.5.3 Rudder horn

The loads on the rudder horn are as follows:

M_b = bending moment = B_1 z [Nm],

M_{bmax} = B_1 d [Nm]

q = shear force = B_1 [N]

$M_T(z)$ = torsional moment = B_1 e (z) [Nm]

An estimate for B_1 is:

B_1 = F_r b / (ℓ_{20} + ℓ_{30}) [N]

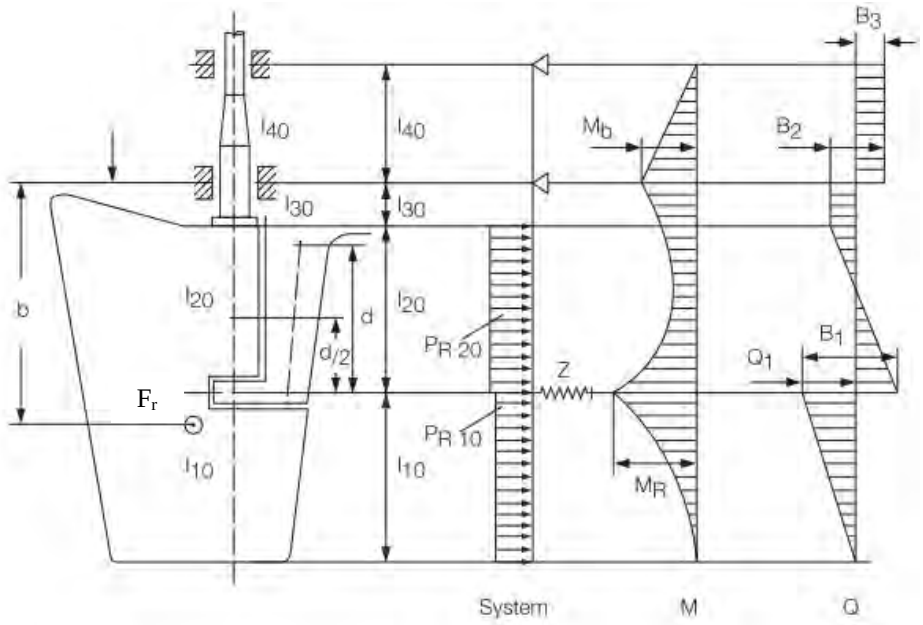


Fig.8.5.2 : Strength Calculations – Semi Spade Rudder with one Elastic Support

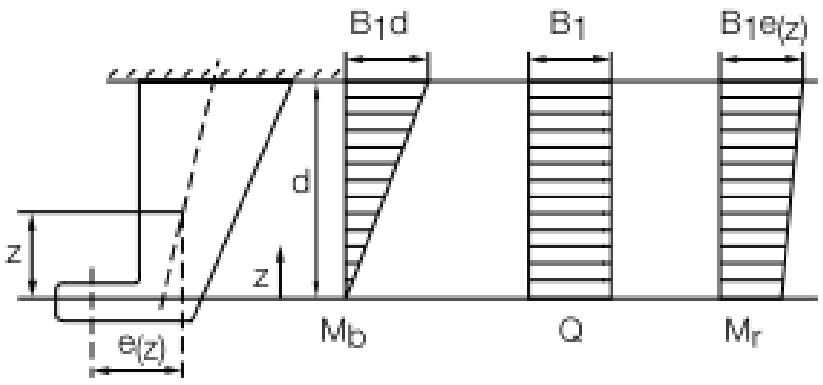


Fig 8.5.3

8.6 Semi-spade rudder with 2-conjugate elastic support

8.6.1 Data for the Analysis

K_{11}, K_{12}, K_{22} : Rudder horn compliance constants calculated for rudder horn with 2-conjugate elastic supports (Figure 8.6.2).The 2-conjugate elastic supports are defined in terms of horizontal displacements, y_i, by the following equations:

at the lower rudder horn bearing:

$$y_1 = - K_{12} B_2 - K_{22} B_1$$

at the upper rudder horn bearing:

$$y_2 = - K_{11} B_2 - K_{12} B_1$$

where:

y_1, y_2 : Horizontal displacements,[m], at the lower and upper rudder horn bearings, respectively.

B_1, B_2 : Horizontal support forces,[kN], at the lower and upper rudder horn bearings, respectively.

K_{11}, K_{12}, K_{22} : Obtained, [m/kN], from the following formulae:

$$K_{11} = 1.3\frac{\lambda^3}{3EJ_{1h}} + \frac{e^2\lambda}{GJ_{th}}$$

$$K_{12} = 1.3\left[\frac{\lambda^3}{3EJ_{1h}} + \frac{\lambda^2(d-\lambda)}{2EJ_{1h}}\right] + \frac{e^2\lambda}{GJ_{th}}$$

$$K_{22} = 1.3\left[\frac{\lambda^3}{3EJ_{1h}} + \frac{\lambda^2(d-\lambda)}{EJ_{1h}} + \frac{\lambda(d-\lambda)^2}{EJ_{1h}} + \frac{(d-\lambda)^3}{3EJ_{2h}}\right] + \frac{e^2 d}{GJ_{th}}$$

d : Height of the rudder horn, [m], defined in Figure 8.6.2. This value is measured downwards from the upper rudder horn end, at the point of curvature transition, to the mid-line of the lower rudder horn pintle.

λ : Length, [m], as defined in Figure 8.6.2. This length is measured downwards from the upper rudder horn end, at the point of curvature transition, to the mid-line of the upper rudder horn bearing. For λ = 0, the above formulae converge to those of spring constant Z for a rudder horn with 1-elastic support, and assuming a hollow cross section for this part.

e : Rudder-horn torsion lever, [m], as defined in Figure 8.6.2 (value taken at z = d/2).

J_{1h} : Moment of inertia of rudder horn about the x axis,[m^4], for the region above the upper rudder horn bearing. Note that J_{1h} is an average value over the length λ (see Figure 8.6.2).

J_{2h} : Moment of inertia of rudder horn about the x axis,[m^4], for the region between the upper and lower rudder horn bearings. Note that J_{2h} is an average value over the length d - λ (see Figure 8.6.2).

J_{th} : Torsional stiffness factor of the rudder horn, [m^4].

For any thin wall closed section:

$$J_{th} = \frac{4F_T^2}{\sum_i \frac{u_i}{t_i}}$$

F_T : Mean of areas enclosed by outer and inner boundaries of the thin walled section of rudder horn, [m^2].

u_i : Length,[mm], of the individual plates forming the mean horn sectional area.

t_i : Thickness,[mm], of the individual plates mentioned above.

The J_{th} value is taken as an average value, valid over the rudder horn height.

Load of rudder body:

$$P_{R10} = F_{r2} / (\ell_{10} \times 10^3) \text{ [kN/m]};$$

$$P_{R20} = F_{r1} / (\ell_{20} \times 10^3) \text{ [kN/m]};$$

for F_r, F_{r1}, F_{r2}, see 3.2.2

8.6.2 The moments and shear forces are indicated in Fig.8.6.2.

8.6.3 Rudder horn bending moment

The bending moment acting on the generic section of the rudder horn is to be obtained, [Nm], from the following formulae:

- between the lower and upper supports provided by the rudder horn:

$M_H = F_{A1}\, z$

- above the rudder horn upper-support:

$M_H = F_{A1}\, z + F_{A2}\, (z - d_{lu})$

where:

F_{A1} : Support force at the rudder horn lower-support, [N], to be obtained according to Figure 8.6.2, and taken equal to B_1.

F_{A2} : Support force at the rudder horn upper-support, [N], to be obtained according to Figure 8.6.2, and taken equal to B_2.

z : Distance, [m], defined in Figure 8.6.3, to be taken less than the distance d, in [m], defined in the same figure.

d_{lu} : Distance, [m], between the rudder-horn lower and upper bearings (according to Figure 8.6.2, $d_{lu} = d - \lambda$).

8.6.4 Rudder horn shear force

The shear force Q_H, [N], acting on the generic section of the rudder horn is to be obtained, from the following formulae:

- between the lower and upper rudder horn bearings:

$Q_H = F_{A1}$

- above the rudder horn upper-bearing:
$Q_H = F_{A1} + F_{A2}$

where:

F_{A1}, F_{A2} : Support forces, in [N].

8.6.5 The torque, [Nm], acting on the generic section of the rudder horn is to be obtained from the following formulae:

- between the lower and upper rudder horn bearings:

$M_T = F_{A1}\, e_{(z)}$

- above the rudder horn upper-bearing:

$M_T = F_{A1}\, e_{(z)} + F_{A2}\, e_{(z)}$

where:

F_{A1}, F_{A2} : Support forces, in [N]

$e_{(z)}$: Torsion lever, [m], defined in Fig.8.6.3.

8.6.6 Rudder horn shear stress calculation

For a generic section of the rudder horn, located between its lower and upper bearings, the following stresses are to be calculated:

T_S: Shear stress, [N/mm^2], to be obtained from the following formula:

$$\tau_S = \frac{F_{A1}}{A_H}$$

T_T : Torsional stress, [N/mm^2], to be obtained for hollow rudder horn from the following formula:

$$T_T = \frac{M_T 10^{-3}}{2F_T t_H}$$

For solid rudder horn, T_T is to be specially considered.

For a generic section of the rudder horn, located in the region above its upper bearing, the following stresses are to be calculated:

T_S: Shear stress, [N/mm^2], to be obtained from the following formula:

$$\tau_S = \frac{F_{A1} + F_{A2}}{A_H}$$

T_T : Torsional stress, [N/mm^2], to be obtained for hollow rudder horn from the following formula:

$$T_T = \frac{M_T 10^{-3}}{2F_T t_H}$$

For solid rudder horn, T_T is to be specially considered, where:

F_{A1}, F_{A2} : Support forces, [N];

A_H : Effective shear sectional area of the rudder horn, [mm^2], in y-direction;

M_T : Torque, [Nm];

F_T : Mean of areas enclosed by outer and inner boundaries of the thin walled section of rudder horn, [m^2];

t_H : Plate thickness of rudder horn,[mm]. For a given cross section of the rudder horn, the maximum value of T_T is obtained at the minimum value of t_H.

8.6.7 Rudder horn bending stress calculation

For the generic section of the rudder horn within the length *d*, the following stresses are to be calculated:

σ_B : Bending stress, [N/mm^2], to be obtained from the following formula:

$$\sigma_B = \frac{M_H}{W_X}$$

where:

M_H : Bending moment, [Nm] at the section considered.

W_X : Section modulus, [cm^3], around the X-axis (see Fig.8.6.3).

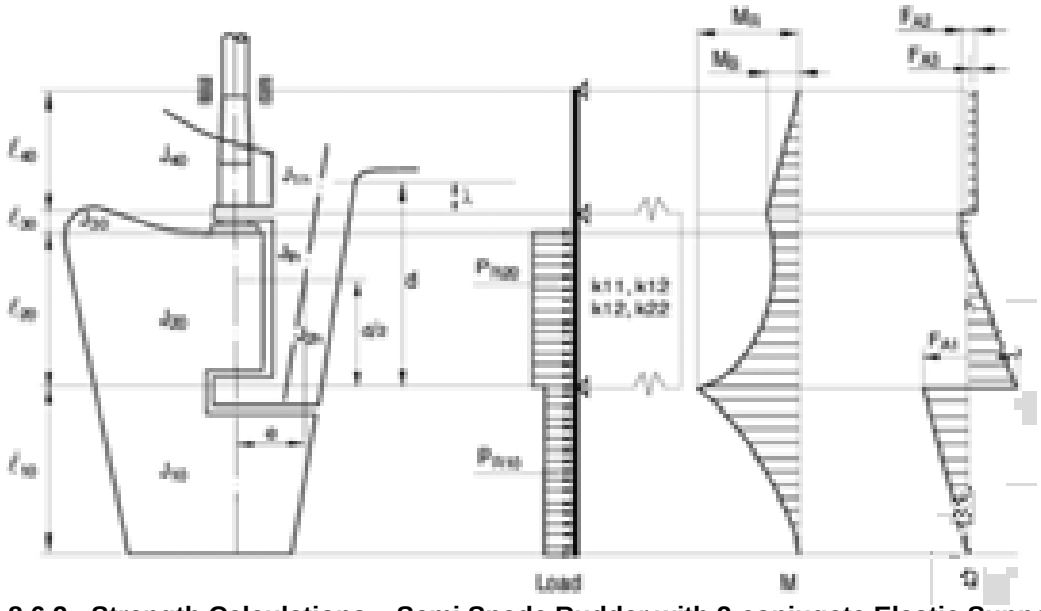


Fig.8.6.2 : Strength Calculations – Semi Spade Rudder with 2-conjugate Elastic Support

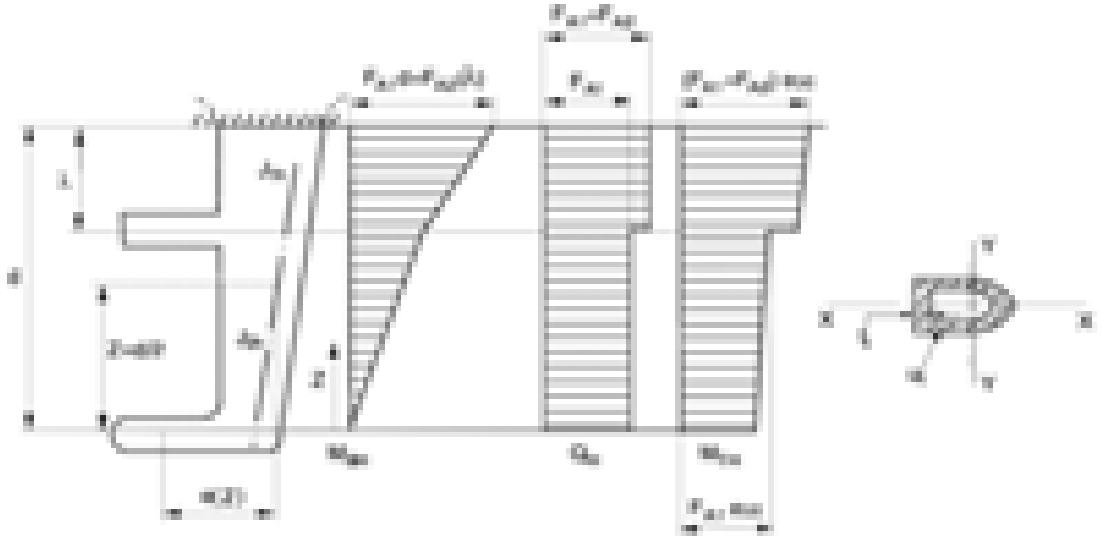


Fig.8.6.3

End of Chapter

Chapter 14

Anchoring and Mooring Equipment

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Section 1

General

1.1 Introduction

1.1.1 To entitle a vessel to the letter 'L' in her character of classification, anchoring and mooring equipment is to be provided in accordance with the requirements of this Chapter.

1.1.2 Except in case of supply vessels engaged in towing operations and tugs, towlines are not subject of classification and the details given in the equipment table are for guidance purpose only. Requirements for towlines for supply vessels engaged in towing operations and tugs are given in 3.2.4. However, for tugs intended to be classed for operations in harbours only, requirement of towline may be waived with written concurrence from the owner.

1.1.3 Attention is drawn to any relevant requirements of the National Authorities of the country in which the ship is to be registered.

1.2 Design of the anchoring equipment

1.2.1 The anchoring equipment required herewith is intended for temporary mooring of a ship within a harbour or sheltered area when the ship is awaiting berth, tide, etc.

1.2.2 Anchoring equipment requirements for ships, designed for anchoring in deep and unsheltered waters, and assigned additional class notation ‘DWA,’ are indicated in Section 8.

1.2.3 The equipment is therefore not designed to hold a ship off fully exposed coasts in rough weather or to stop a ship which is moving or drifting. In this condition the loads on the anchoring equipment increase to such a degree that its components may be damaged or lost owing to the high energy forces generated, particularly in large ships.

1.2.4 The anchoring equipment required herewith is designed to hold a ship in good holding ground in conditions such as to avoid dragging of the anchor. In poor holding ground the holding power of the anchors is significantly reduced.

1.2.5 The Equipment Number (EN) formula for anchoring equipment required here under is based on an assumed current speed of 2.5 [m/s], maximum wind speed of 25 [m/s] and a minimum scope of chain cable of 6, the scope being the ratio between length of chain paid out and water depth. For ships with equipment

length, greater than 135 [m], alternatively the required anchoring equipment can be considered applicable to a maximum current speed of 1.54 [m/s], a maximum wind speed of 11 [m/s] and waves with maximum significant wave height of 2[m].

1.2.6 It is assumed that under normal circumstances a ship uses only one bow anchor and chain cable at a time.

1.2.7 In addition to planned anchoring for normal operations, anchoring equipment is also important for ship safety in emergency situations such as loss of manoeuvrability, unscheduled repairs and other unexpected situations.

1.2.8 The requirements of anchoring equipment in this Chapter are applicable to all self-propelled vessels over 100GT.

1.3 Documentation

1.3.1 The arrangements of anchoring and mooring and Equipment Number calculations are to be submitted for approval.

For vessels of 500 GT and above, towing arrangement plan is to be submitted for approval.

1.3.2 Following details of the proposed equipment are to be submitted for approval:-

1) Number, weight, type and design of anchors.

2) Length, diameter, grade and type of chain cables.

3) Type and breaking load of steel and fibre ropes.

1.4 Symbols and Definitions

1.4.1 L,B,T,C_b and k as defined in Ch.1, Sec.2.

1.4.2 The equipment length of the ship is the length between perpendiculars but is not to be less than 96% nor greater than 97% of the extreme length on the Summer Load waterline (measured from the forward end of the waterline).

Section 2

Structural Arrangement for Anchoring Equipment

2.1 General

2.1.1 The anchors are normally to be housed in hawse pipes and anchor pockets of adequate size, scantlings and suitable form to prevent movement of anchor and chain due to wave action. To hold the anchor tight in against the hull or the anchor pocket, respectively, it is recommended to fit anchor lashings. Anchor lashings are to be designed to resist a load at least corresponding to twice the anchor mass plus 10 [m] of cable without exceeding 40% of the yield strength of the material.

The arrangements are to provide an easy lead of chain cable from windlass to the anchors. Upon release of the brake, the anchors are to immediately start falling by their own weight. Substantial chafing lips are to be provided at shell and deck. These are to have sufficiently large, radiused faces to minimise the probability of cable links being subjected to large bending stresses. Alternatively, roller fairleads of suitable design may be fitted.

Alternative arrangements for housing of anchors will be specially considered.

2.1.2 The shell plating and framing in way of the hawse pipes are to be reinforced as necessary.

2.1.3 On ships provided with a bulbous bow, and where it is not possible to obtain ample clearance between shell plating and anchors during anchor handling, adequate local reinforcements on bulbous bow are to be provided.

2.1.4 The chain locker is to have adequate capacity and depth to provide an easy direct lead for the cable into the spurling pipes, when the cable is fully stowed. The spurling pipes are to be of suitable size and provided with chafing lips. The port and starboard cables are to have separate spaces.

The chain lockers boundaries and spurling pipes are to be watertight upto the weather deck. Bulkheads which form common boundary of chain lockers need not be watertight.

Where means of access to spurling pipes or cable lockers is provided, it is to be closed by a substantial cover and secured by closely spaced bolts.

Where a means of access to spurling pipes or cable lockers is located below the weather deck, the access cover and its securing arrangements are to be in accordance with recognized standards (such as ISO 5894:2018, IS 15876-2009, etc.) or equivalent for watertight manhole covers. Butterfly nuts and/or hinged bolts are not acceptable as the securing mechanism for the access cover.

Provisions are to be made to minimize the ingress of water to the chain locker in bad weather. The spurling pipes are to be provided with permanently attached closing appliances such as steel plates with cutouts for chain links or canvas hoods with lashing arrangements that maintains the cover in the secured position. For requirements regarding drainage of chain lockers, See Pt.4.

Provisions are to be made for securing the inboard ends of the chains to the structure. The strength of this attachment should be between 15 per cent to 30 per cent of the breaking strength of the chain cable. It is recommended that suitable arrangements be provided so that in an emergency the inboard end of the chain (bitter end) can be readily made to slip from an accessible position outside the chain locker.

2.1.5 The windlass and chain stoppers are to be efficiently bedded and secured to deck. The thickness of deck plating is to be increased in way of the windlass and chain stoppers and adequate stiffening underneath is to be provided.

Section 3

Equipment Specification

3.1 Equipment number

3.1.1 The equipment number, EN, on which the requirements of equipment are based is to be calculated as follows :-

$$EN = K.ENc$$

where,

$ENc = \Delta^{2/3} + 2(BH + S_{fun}) + 0.1A$

Δ = moulded displacement, [t], to the summer load water line

B = moulded breadth, [m]

H = effective height, [m], from the summer load waterline to the top of the uppermost deckhouse, to be measured as follows:

$H = a + \Sigma h_i$

a = vertical distance at hull side [m], from summer load waterline amidships to the upper deck

h_i = height [m] on the centreline of each tier of houses having a breadth greater than B/4. For the lowest tier, h_1 is to be measured at centre line from upper deck, or from a notional deck line where there is a local discontinuity in the upper deck. See Fig.3.1.1(a).

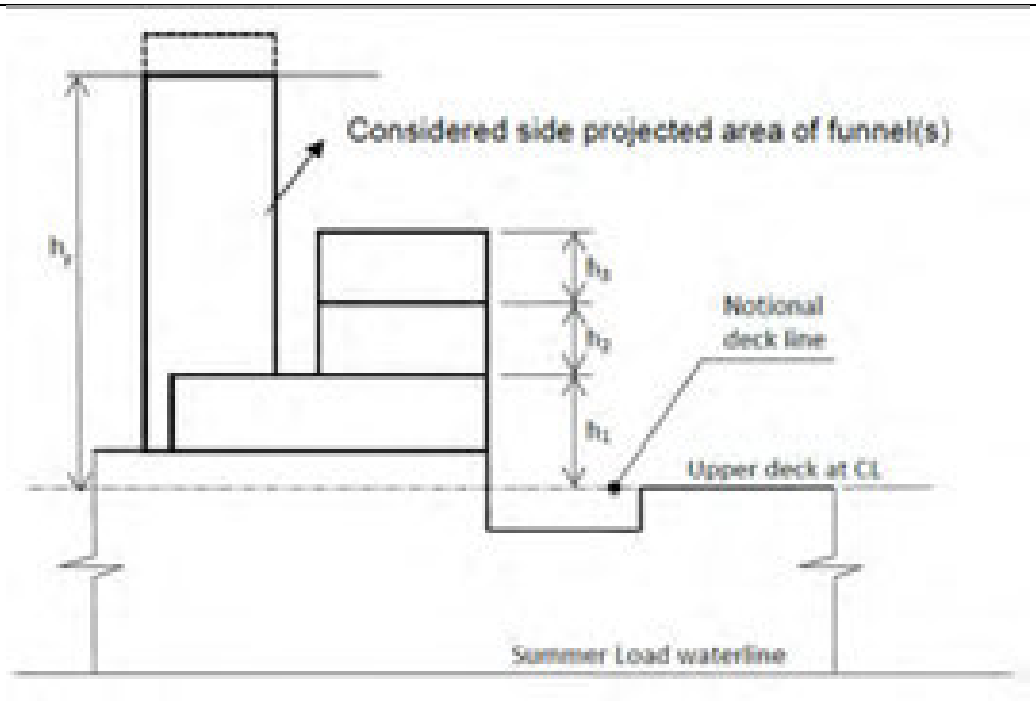


Fig. 3.1.1 (a)

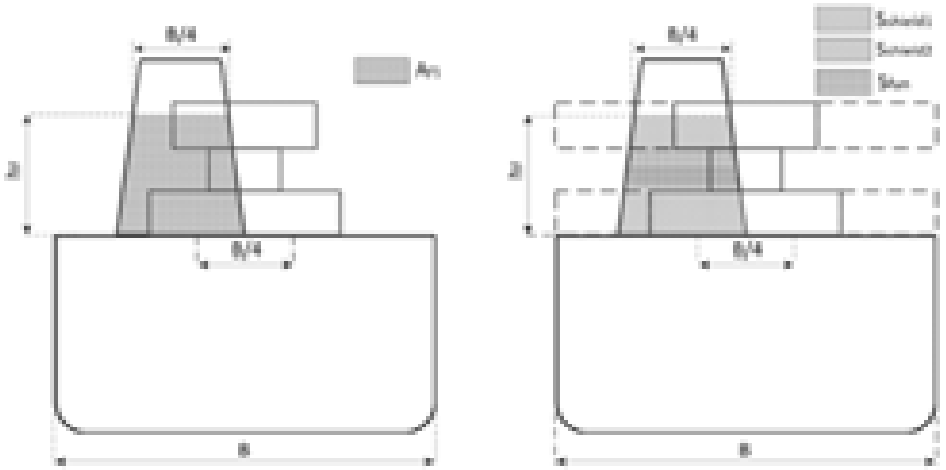


Fig. 3.1.1 (b)

S_{fun} = effective front projected area [m^2] of the funnel, defined as:

$$S_{fun} = A_{FS} - S_{shield}$$

A_{FS} = front projected area of the funnel, [m^2], calculated between the upper deck at centreline, or notional deck line where there is local discontinuity in the upper deck, and the effective height h_F. A_{FS} is taken equal to zero if the funnel breadth is less than or equal to B/4 at all elevations along the funnel height.

h_F = effective height of the funnel, [m], measured from the upper deck at centreline, or notional deck line where there is local discontinuity in the upper deck, and the top of the funnel. The top of the funnel may be taken at the level where the funnel breadth reaches B/4.

S_{shield} = the section of front projected area A_{FS}, [m^2], which is shielded by all deck houses having breadth greater than B/4 . If there are more than one shielded section, the individual shielded sections i.e $S_{shield1}$, $S_{shield2}$ etc as shown in Fig.3.1.1(b) are to be added together. To determine S_{shield}, the deckhouse breadth is assumed as B for all deck houses having breadth greater than B/4 as shown for $S_{shield1}$, $S_{shield2}$ in Fig.3.1.1(b).

A = side projected area [m^2] of the hull, superstructures and houses and funnels above the summer load waterline, which is within the equipment length of the ship and also have a breadth greater than B/4. The side projected area of the funnel is considered in A when A_{FS} is greater than zero. In this case, the side projected area of the funnel should be calculated between the upper deck, or notional deck line where there is local discontinuity in the upper deck, and the effective height h_F.
In the calculation of H , sheer and trim are to be ignored. i.e. H is the sum of freeboard amidships plus the height (at centreline) of each tier of houses having a breadth greater than B/4.

If a deckhouse having a breadth greater than B/4 is above a deckhouse with a breadth of B/4 or less, then the wide deckhouse is to be included but the narrow deckhouse ignored.

Windscreens or bulwarks which are more than 1.5[m] in height are to be regarded as parts of deckhouses when determining H and A. The height of the hatch coamings and that of any deck cargo, such as containers, may be disregarded when determining H and A. With regard to determining A, when a bulwark is more than 1.5 [m] high, the area shown in Fig.3.1.1(c) as A_2 is to be included in A.

When several funnels are fitted on the ship, the above parameters are taken as follows:

H_F: effective height of the funnel, [m], measured from the upper deck, or notional deck line where there is local discontinuity in the upper deck, and the top of the highest funnel. The top of the highest funnel may be taken at the level where the sum of each funnel breadth reaches B/4.

A_{FS}: sum of the front projected area of each funnel, [m^2], calculated between the upper deck, or notional deck line where there is local discontinuity in the upper deck, and the effective height H_F. A_{FS} is to be taken equal to zero if the sum of each funnel breadth is less than or equal to B/4 at all elevations along the funnels height.

A: Side projected area, [m^2], of the hull, superstructures, houses and funnels above the Summer Load waterline which are within the equipment length of the ship. The total side projected area of the funnels is to be considered in the side projected area of the ship, A, when A_{FS} is greater than zero. The shielding effect of funnels in transverse direction may be considered in the total side projected area, i.e., when the side projected areas of two or more funnels fully or partially overlap, the overlapped area needs only to be counted once.

'K' is a factor depending upon the type of vessel and service notation as given below:

For fishing vessels,

K = 1.00

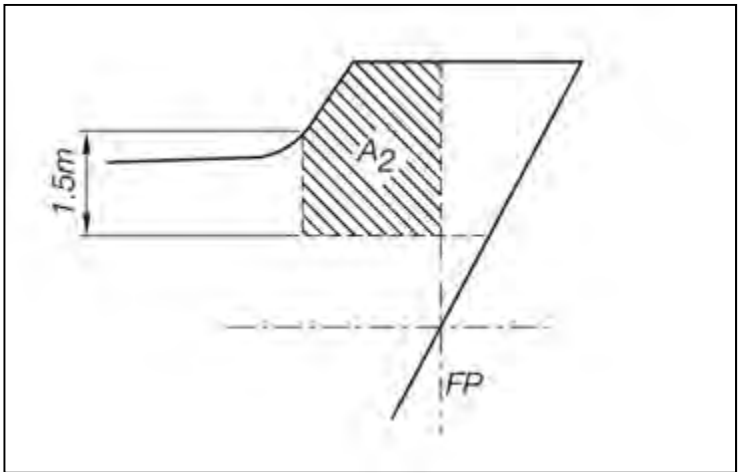


Fig. 3.1.1 (c)

For other vessels,

K = 1.00 for vessels of Unrestricted Service.
K = 0.85 for vessels of Coastal Service
K = 0.775 for vessels of Restricted Water Service.
K = 0.50 for vessels of Sheltered Water Service.

3.1.2 For a barge rigidly connected to a push tug the equipment number is to be calculated for the combination regarded as one unit.

3.1.3 For tugs, while determining the equipment number EN, the term '2BH' in Sec.3.1.1 may be substituted by $2(aB + \Sigma h_i b_i)$ where b_i is the breadth, [m], of the widest superstructure or deckhouse of each tier having a breadth greater than B/4. For tugs under 45 [m] in length intended for towing service only, one anchor may be used onboard provided that the second anchor and its relevant chain cable holds readily available to be installed. In case of loss of anchor, the tug is to remain in port until replace anchor equipment is installed onboard.

3.1.4 Dredgers with unusual design of the underwater part of the hull are not covered by alternative methodology using direct force calculation for anchoring equipment described in 3.3.

3.2 Equipment

3.2.1 For fishing vessels the equipment is to be in accordance with the requirements given in Table 3.2.1 using EN as calculated in 3.1.

3.2.2 For other vessels the equipment is to be in accordance with the requirements given in Table 3.2.2 using EN as calculated in 3.1.

3.2.3 The two bower anchors in Table 3.2.1 and Table 3.2.2 are to be connected to their cables and positioned onboard ready for use. The total length of chain is to be divided in approximately equal parts between the two bower anchors.

3.2.4 For tugs the particulars of towlines are to be based upon the maximum bollard pull and the intended duty of the vessel. However, in no case the breaking strength of the towline is to be less than twice the maximum bollard pull.

3.2.5 For unmanned barges and pontoons of length "L" less than 30 [m] no anchor need be provided.

Where length "L" is greater than or equal to 30 [m] only one anchor of the tabular weight need be provided. The attached chain cable is to be of the tabular diameter and 2 L [m] in length.

Two mooring lines of tabular breaking strength are to be provided where the length 'L' is less than 65 [m], otherwise three mooring lines are to be provided. The length of each mooring line is not be less than the greater of 2L [m] or 80 [m].

For unmanned barges and pontoons where the letter 'L' is not intended to be included in the character of classification only mooring lines of adequate length and strength need be provided.

3.2.6 For Fishing Vessels:

3.2.6.1 For fishing vessels below 40 m in length the anchor chain may be replaced with wire ropes of equal strength of the tabular anchor cables of Grade CC1. Wire ropes of trawl winches complying with this sub-section may be used as anchor chain cables.

3.2.6.2 When wire ropes are substituted for anchor chain cables then:

(i) the length of the ropes is to be equal to 1.5 times the corresponding tabular length of chain cable (Table 3.2.1),

(ii) a short length of chain cable is to be fitted between the wire rope and anchor having a length of 12.5 [m] or the distance between anchor in stowed position and winch, whichever is less,

(iii) all surfaces being in contact with the wire are to be rounded with a radius of not less than 10 times the wire rope diameter (including stem).

(iv) Steel wire is to be selected to fit for purpose based on the manufacturer recommendation and is to be provided with guidance for maintenance and inspection.

3.2.6.3 High holding power anchors of approved design may be used as bower anchors. The mass of each such anchor may be 75% of the tabular mass for ordinary stockless bower anchors.

3.2.6.4 The tabular anchor equipment may be increased for ships fishing and intended for anchoring in very rough waters.

3.2.7 Vessels other than fishing vessels:

3.2.7.1 Wire rope may be used in place of chain cable on ships:

(a) with less than 90 m in length and which will need an anchor for emergency purposes, i.e., not intended to use their anchor in normal temporary anchoring operation, or

(b) with the anchoring equipment used for positioning with a minimum of 4 points anchoring, e.g., for cable or pipe laying.

3.2.7.2 When wire ropes are substituted for anchor chains the following are to be complied with :-

i) The length of rope is to be at least 1.5 times the corresponding tabular length of chain cable and their strength is to be equal to that of chain cable of Grade CC 1.

ii) The anchor weight is to be increased by 25 % compared to anchor associated with chain cable.

iii) A short length of chain is to be fitted between the anchor and the wire rope. The length of the cable need be the smaller of 12.5 [m] and the distance between the anchor in stowed position and the winch.

iv) All surfaces being in contact with the wire need to be rounded with a radius of not less than 10 times the wire rope diameter (including stem).

v) Steel wire is to be selected to fit for purpose based on the manufacturer recommendation and is to be provided with guidance for maintenance and inspection.

3.2.8 For ships of length less than 90m, alternative methodology using direct force calculation for anchoring equipment described in 3.3 may be used.

3.3 Direct force calculation for anchoring equipment

3.3.1 General

3.3.1.1 As an alternative to the prescriptive approach described above, direct force calculation may be performed to determine the necessary anchoring equipment for monohull ships with length less than 90 m.

3.3.2 Total force F_{EN}

3.3.2.1 The total force (static + dynamic) F_{EN} , in [kN], induced by wind and current acting on monohull in anchoring condition as defined in 1.2 may be calculated as follows:

$$F_{EN} = 2\,(F_{SLPH} + F_{SH} + F_{SS})$$

where:

F_{SLPH}: Static force on wetted part of the hull due to current, as defined in 3.3.2.2

F_{SH}: Static force on hull due to wind, as defined in 3.3.2.3

F_{SS}: Static force on superstructures due to wind, as defined 3.3.2.4.

3.3.2.2 Static force on wetted part of hull F_{SLPH}

The theoretical static force induced by current applied on the wetted part of the hull, in [kN], is defined according to the following formula:

$$F_{SLPH} = \frac{1}{2}\rho C_f S_m V_c^2 10^{-3}$$

where:

ρ : Water density, equal to 1025 [kg/m^3]

C_f : Coefficient equal to:

$$C_f = (1+k)\frac{0.075}{(\log R_e - 2)^2}$$

With R_e, Reynolds number:

$$R_e = \frac{(V_c L_{WL})}{1.054 \cdot 10^{-6}}$$

k : Coefficient equal to:

$$k = 0.017 + 20\frac{C_{bWL}}{L_{WL}^2 T^{-0.5} B_{WL}^{-1.5}}$$

With C_{bWL}, block coefficient at waterline:

$$C_{bWL} = \frac{\Delta}{1.025 L_{WL} B_{WL} T}$$

Δ :Moulded displacement at waterline, in [m^3]

S_m :Total wetted surface of the part of the hull below waterline, in [m^2]
The value of S_m is to be given by the designer. When this value is not available, S_m may be taken equal to $6.\Delta^{2/3}$

V_C :Speed of the current, in [m/s], as defined in 1.2.

3.3.2.3 Static force on hull F_{SH}

The theoretical static force induced by wind applied on the upper part of the hull, in [kN], is defined according to the following formula:

$$F_{SH} = \frac{1}{2}\rho\left(C_{hfr}S_{hfr} + 0.02S_{hlat}\right)V_W^2 10^{-3}$$

where:

ρ : Air density, equal to 122 [kg/m^3]

V_W : Speed of the wind, in [m/s], as defined in 1.2

S_{hfr} : Front surface of hull and bulwark if any, in [m^2], projected on a vertical plane of the ship situated aft of the aft end of the ship and perpendicular to the longitudinal axis of the ship

S_{hlat} : Partial lateral surface of one single side of the hull and bulwark if any, in [m^2], through the overall length of the ship, projected on a vertical plane parallel to the longitudinal axis of the ship and delimited according to Figure 3.3.2.2

C_{hfr} :0.8 $\sin\alpha$, with α defined in Figure 3.3.2.2.

In Figure 3.3.2.2, B is the breadth of the hull, in [m].

The upper part of the hull is the part extending from side to side to the uppermost continuous deck extending over the ship length.

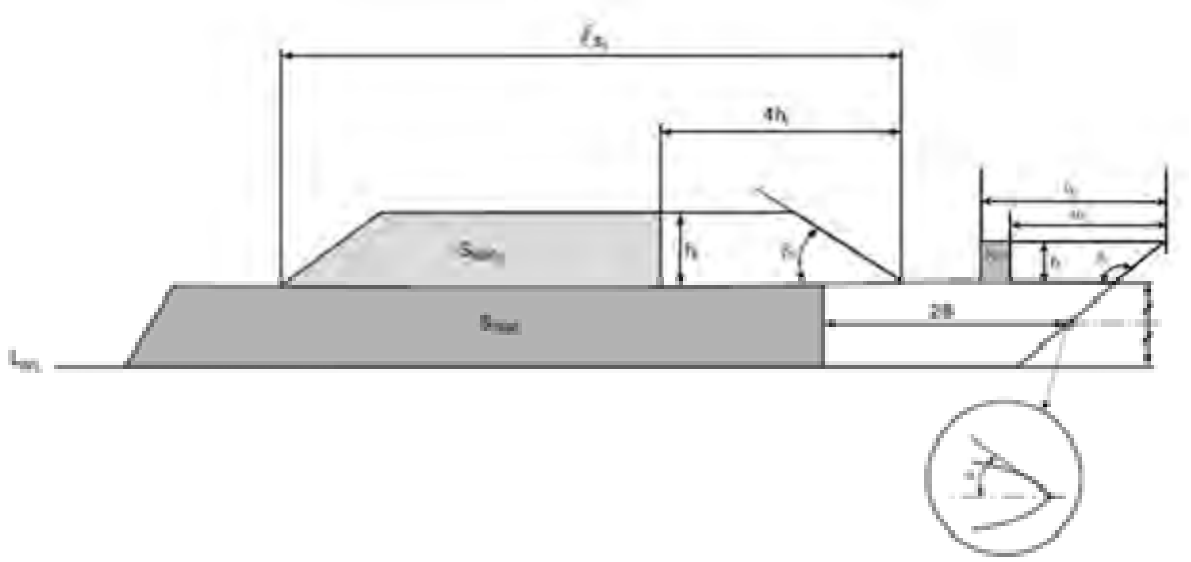


Figure 3.3.2.2

3.3.2.4 Static forces F_{SS} on superstructures and deckhouses

3.3.2.4.1 General case

The theoretical static force induced by wind applied on the superstructures and deckhouses, in [kN], is defined as the sum of the forces applied to each superstructure and deckhouse tier according to the following formula:

$$F_{SS} = \frac{1}{2}\rho\sum\left(C_{sfr_i}S_{sfr_i} + 0.08S_{slat_i}\right)V_W^2\ 10^{-3}$$

where:

ρ :Air density, equal to 1.22 [kg/m^3]

V_W :Speed of the wind, in [m/s], as defined in 3.3.2.3.

S_{sfri}: Front surface of tier i (superstructure or deckhouse, including bulwark if any), in [m^2], projected on a vertical plane of the ship situated aft of the aft end of the ship and perpendicular to the longitudinal axis of the ship

S_{slati} :Partial lateral surface of one single side of tier I (superstructure or deckhouse, including bulwark if any), in [m^2], projected on a vertical plane parallel to the longitudinal axis of the ship and delimited according to Figure 3.3.2.2

When $4h_i \geq l_{si}$, S_{lati} is to be taken equal to 0

C_{sfri} :0.8 sin β_i, with β_i defined in Figure 3.3.2.2 without being greater than 90°

3.3.2.4.2 Superstructures in the forward part of the ship

When superstructures are located in the front of the hull with front and side walls of superstructures in the continuity of the side shell, the static force induced by wind applied on these superstructures, in [kN], is defined as the sum of the forces applied to each superstructure tier according to the following formula:

$$F_{SS} = \frac{1}{2}\rho \sum \left(C_{hfr_i} S_{hfr_i} + 0.08 S_{slat_i}\right) V_W^2 \, 10^{-3}$$

where:

S_{hfri} : Front surface of tier i of the superstructure, in [m^2], projected on a vertical plane of the ship situated aft of the aft end of the ship and perpendicular to the longitudinal axis of the ship

C_{hfri} : $C_{hfri} = 0.8 \sin \alpha_s$, with α_s as defined for α in Figure 3.3.2.2 and measured at mid height of the superstructure tier located in the front of the hull.

ρ, V_W, S_{slati}: As defined in 3.3.2.4.1.

The static force is to be added to the static force calculated for the other superstructures and deckhouses according to 3.3.2.4.1.

3.3.3 Anchor weight

3.3.3.1 The individual mass of anchor, in kg, is to be at least equal to:

for ordinary anchor : $P = (F_{EN} / 7) * 10^2$

for high holding power anchor: $P = (F_{EN} / 10) * 10^2$

for very high holding power: $P = (F_{EN} / 15) * 10^2$

3.3.4 Chain cable

3.3.4.1 Stud link chain cable scantling

Chain cable diameters are to be selected from Pt.2, Ch.10, Table 1.4.1, based on the minimum breaking load BL and proof load PL of steel grades, in [kN], calculated according to the following formulae:

- for steel Grade CC 1:
  BL = 6 F_{EN}
  PL = 0.7 BL
- for steel Grade CC 2:
  BL = 6.8 F_{EN}
  PL = 0.7 BL
- for steel Grade CC 3:
  BL = 7.5 F_{EN}
  PL = 0.7 BL

The chain cable scantling is to be consistent with the mass of the associated anchor. In case the anchor on board is heavier by more than 7% from the mass calculated in 3.3.3, the value of F_{EN} to take into account for the calculation of BL and PL is to be deduced from the actual mass of the anchor according to the formulae in 3.3.3.

3.3.4.2 Length of individual chain cable

The length of chain cable L_{cc}, in [m], linked to each anchor is to be at least equal to:

- When P<180
  L_{cc} = 30 ln(P) - 42
- When P≥180
  L_{cc} to be selected according to Table 3.2.2

where:

P : Anchor weight, in [kg], defined in 3.3.3 for an ordinary anchor according to the considered case.

Table 3.2.1 : Equipment for fishing vessels

EN	Equipment letter	Stockless Bower Anchors		Stud-Link Chain Cables			Mooring Lines (Recommendation)		
					Diameter and chain grade		Steel or natural fibre ropes		
		Number	Mass per anchor [kg]	Total length [m]	CC1 [mm]	CC2 [mm]	Number	Length [m]	Ship design minimum breaking load[kN]
30 & ≤ 40	A_of1	2	80	165	11		2	50	29
> 40 & ≤ 50	A_of2	2	100	192.5	11		2	60	29
> 50 & ≤ 60	Af1	2	120	192.5	12.5		2	60	29
> 60 & ≤ 70	Af2	2	140	192.5	12.5		2	80	29
> 70 & ≤ 80	Bf1	2	160	220	14	12.5	2	100	34
> 80 & ≤ 90	Bf2	2	180	220	14	12.5	2	100	36.8
> 90 & ≤ 100	Cf1	2	210	220	16	14	2	110	36.8
> 100 & ≤ 110	Cf2	2	240	220	16	14	2	110	39
> 110 & ≤ 120	Df1	2	270	247.5	17.5	16	2	110	39
> 120 & ≤ 130	Df2	2	300	247.5	17.5	16	2	110	44
> 130 & ≤ 140	Ef1	2	340	275	19	17.5	2	120	44
> 140 & ≤ 150	Ef2	2	390	275	19	17.5	2	120	49
> 150 & ≤ 175	Ff	2	480	275	22	19	2	120	54
>175 & ≤ 205	Gf	2	570	302.5	24	20.5	2	120	59
> 205 & ≤ 240	Hf	2	660	302.5	26	22	2	120	64
> 240 & ≤ 280	If	2	780	330	28	24	3	120	71
> 280 & ≤ 320	Jf	2	900	357.5	30	26	3	140	78
> 320 & ≤ 360	Kf	2	1020	357.5	32	28	3	140	85.8
> 360 & ≤ 400	Lf	2	1140	385	34	30	3	140	93
> 400 & ≤ 450	Mf	2	1290	385	36	32	3	140	101
> 450 & ≤ 500	Nf	2	1440	412.5	38	34	3	140	108
> 500 & ≤ 550	Of	2	1590	412.5	40	34	4	160	113
> 550 & ≤ 600	Pf	2	1740	440	42	36	4	160	118
> 600 & ≤ 660	QF	2	1920	440	44	38	4	160	123
> 660 & ≤ 720	Rf	2	2100	440	46	40	4	160	127

Table 3.2.2 : Equipment for vessels excluding fishing vessels

EN	Equipment letter	Stockless bower anchors		Stud-link chain cables for bower anchors				Towline (Recommendation)	Mooring lines [1),2),4)]			
					Diameter and grade			Steel or natural fibre ropes				
		Number	Mass per anchor [kg]	Total length [m]	CC1 [mm]	CC2 [mm]	CC3 [mm]	Min. length [m]	Ship design minimum breaking load [kN]	Number	Length of each [m]	Ship design minimum breaking load [kN]
50 & ≤ 70	A	2	180	220	14	12.5		180	98	3	80	37
> 70 & ≤ 90	B	2	240	220	16	14		180	98	3	100	40
> 90 & ≤ 110	C	2	300	247.5	17.5	16		180	98	3	110	42
> 110 & ≤ 130	D	2	360	247.5	19	17.5		180	98	3	110	48
> 130 & ≤ 150	E	2	420	275	20.5	17.5		180	98	3	120	53
> 150 & ≤ 175	F	2	480	275	22	19		180	98	3	120	59
> 175 & ≤ 205	G	2	570	302.5	24	20.5		180	112	3	120	64
> 205 & ≤ 240	H	2	660	302.5	26	22	20.5	180	129	4	120	69
> 240 & ≤ 280	I	2	780	330	28	24	22	180	150	4	120	75
> 280 & ≤ 320	J	2	900	357.5	30	26	24	180	174	4	140	80
> 320 & ≤ 360	K	2	1020	357.5	32	28	24	180	207	4	140	85
> 360 & ≤ 400	L	2	1140	385	34	30	26	180	224	4	140	96
> 400 & ≤ 450	M	2	1290	385	36	32	28	180	250	4	140	107
> 450 & ≤ 500	N	2	1440	412.5	38	34	30	180	277	4	140	117
> 500 & ≤ 550	O	2	1590	412.5	40	34	30	190	306	4	160	134
> 550 & ≤ 600	P	2	1740	440	42	36	32	190	338	4	160	143
> 600 & ≤ 660	Q	2	1920	440	44	38	34	190	371	4	160	160
> 660 & ≤ 720	R	2	2100	440	46	40	36	190	406	4	160	171
> 720 & ≤ 780	S	2	2280	467.5	48	42	36	190	441	4	170	187
> 780 & ≤ 840	T	2	2460	467.5	50	44	38	190	480	4	170	202
> 840 & ≤ 910	U	2	2640	467.5	52	46	40	190	518	4	170	218
> 910 & ≤ 980	V	2	2850	495	54	48	42	190	559	4	170	235
> 980 & ≤ 1060	W	2	3060	495	56	50	44	200	603	4	180	250
> 1060 & ≤ 1140	X	2	3300	495	58	50	46	200	647	4	180	272
> 1140 & ≤ 1220	Y	2	3540	522.5	60	52	46	200	691	4	180	293
> 1220 & ≤ 1300	Z	2	3780	522.5	62	54	48	200	738	4	180	309
> 1300 & ≤ 1390	A+	2	4050	522.5	64	56	50	200	786	4	180	336
> 1390 & ≤ 1480	B+	2	4320	550	66	58	50	200	836	4	180	352
> 1480 & ≤ 1570	C+	2	4590	550	68	60	52	220	888	5	190	352
> 1570 & ≤ 1670	D+	2	4890	550	70	62	54	220	941	5	190	362
> 1670 & ≤ 1790	E+	2	5250	577.5	73	64	56	220	1024	5	190	384
> 1790 & ≤ 1930	F+	2	5610	577.5	76	66	58	220	1109	5	190	411
> 1930 & ≤ 2080	G+	2	6000	577.5	78	68	60	220	1168	5[3]	190[3]	437[3]
> 2080 & ≤ 2230	H+	2	6450	605	81	70	62	240	1259			
> 2230 & ≤ 2380	I+	2	6900	605	84	73	64	240	1356			
> 2380 & ≤ 2530	J+	2	7350	605	87	76	66	240	1453			
> 2530 & ≤ 2700	K+	2	7800	632.5	90	78	68	260	1471			
> 2700 & ≤ 2870	L+	2	8300	632.5	92	81	70	260	1471			

Table 3.2.2 (Contd.)

EN	Equipment letter	Stockless bower anchors		Stud-link chain cables for bower anchors				Towline (Recommendation)		Mooring lines [1),2),4)]		
					Diameter and grade			Steel or natural fibre ropes				
		Number	Mass per anchor [kg]	Total length [m]	CC1 [mm]	CC2 [mm]	CC3 [mm]	Min. length [m]	Ship design minimum breaking load [kN]	Number	Length of each [m]	Ship design minimum breaking load [kN]
> 2870 & ≤ 3040	M+	2	8700	632.5	95	84	73	260	1471			
> 3040 & ≤ 3210	N+	2	9300	660	97	84	76	280	1471			
> 3210 & ≤ 3400	O+	2	9900	660	100	87	78	280	1471			
> 3400 & ≤ 3600	P+	2	10500	660	102	90	78	280	1471			
> 3600 & ≤ 3800	Q+	2	11100	687.5	105	92	81	300	1471			
> 3800 & ≤ 4000	R+	2	11700	687.5	107	95	84	300	1471			
> 4000 & ≤ 4200	S+	2	12300	687.5	111	97	87	300	1471			
> 4200 & ≤ 4400	T+	2	12900	715	114	100	87	300	1471			
> 4400 & ≤ 4600	U+	2	13500	715	117	102	90	300	1471			
> 4600 & ≤ 4800	V+	2	14100	715	120	105	92	300	1471			
> 4800 & ≤ 5000	W+	2	14700	742.5	122	107	95	300	1471			
> 5000 & ≤ 5200	X+	2	15400	742.5	124	111	97	300	1471			
> 5200 & ≤ 5500	Y+	2	16100	742.5	127	111	97	300	1471			
> 5500 & ≤ 5800	Z+	2	16900	742.5	130	114	100	300	1471			
> 5800 & ≤ 6100	A*	2	17800	742.5	132	117	102	300	1471			
> 6100 & ≤ 6500	B*	2	18800	742.5	137	120	107	300	1471			
> 6500 & ≤ 6900	C*	2	20000	770		124	111	300	1471			
> 6900 & ≤ 7400	D*	2	21500	770		127	114	300	1471			
> 7400 & ≤ 7900	E*	2	23000	770		132	117	300	1471			
> 7900 & ≤ 8400	F*	2	24500	770		137	122	300	1471			
> 8400 & ≤ 8900	G*	2	26000	770		142	127	300	1471			
> 8900 & ≤ 9400	H*	2	27500	770		147	132	300	1471			
> 9400 & ≤ 10000	I*	2	29000	770		152	132	300	1471			
> 10000 & ≤ 10700	J*	2	31000	770			137					
> 10700 & ≤ 11500	K*	2	33000	770			142					
> 11500 & ≤ 12400	L*	2	35500	770			147					
> 12400 & ≤ 13400	M*	2	38500	770			152					
> 13400 & ≤ 14600	N*	2	42000	770			157					
> 14600 & ≤ 16000	O*	2	46000	770			162					

1) For individual mooring lines with required breaking force above 490 [kN] according to the table, the required strength may be reduced by the corresponding increase of the number of mooring lines and vice versa, provided that the total of all mooring lines on board is not less than the Rule value. However, the number of mooring lines is not to be less than 6, and no line is to have a breaking force less than 490 [kN]

2) The minimum recommended breaking strength and number of mooring lines for ships with an Equipment Number **EN ≤ 2000** are given in the Table.

3) Value is applicable for ships with equipment number 1930 < EN ≤ 2000.

4) For ships with EN > 2000, ship design minimum breaking load and number of mooring lines is to be provided in accordance with the requirements in Section 6, Cl. 6.3.3.

Section 4

Anchors

4.1 General

4.1.1 Anchors are to be of an approved design.

4.1.2 The mass of each bower anchor as required in Sec.3 is for anchors of equal mass. The masses of individual anchors may vary by ±7 per cent of the tabular masses, provided that the total mass of the anchors is not less than that required for anchors of equal mass.

4.1.3 The mass of the head, including pins and fittings, of an ordinary stockless anchor is not to be less than 60 per cent of the total mass of the anchor.

4.1.4 Stocked bower anchors may be used for ships having 50 ≤ EN < 205. The mass of stocked bower anchors, when used or stream anchors, excluding the stock should be 80 per cent of the tabular mass of ordinary stockless bower anchors. The mass of the stock should be 20 per cent of the total mass of the anchor including the shackle etc. but excluding the stock.

4.1.5 When anchors of a design approved for the designation 'High Holding Power' or 'Super High Holding Power' are used as bower anchors, the mass of each such anchor may be 75 percent or 50 percent respectively of the tabular mass of ordinary stockless bower anchors.

4.1.6 Anchor shackles are to be of a design and material suitable to the service for which the anchor is intended.

4.2 High Holding Power (HHP) anchors

4.2.1 A ‘High Holding Power’ (HHP) anchor is an anchor with a holding power of at least twice that of an ordinary stockless anchor of the same mass. Designs of H.H.P. anchors are to be approved for the designation.

H.H.P. anchors are to be designed for effective hold of the sea bed irrespective of the angle or position at which they first settle on the sea bottom after dropping from a normal type of hawse pipe. In case of doubt a demonstration of these abilities may be required. The designs are to be tested on sea bottom at full scale to show that they have a holding down power per unit of mass at least twice that of approved ordinary stockless anchor.

4.2.2 If approval is sought for a range of anchor sizes, at least two sizes are to be tested. The mass of the larger anchor to be tested is not to be less than one-tenth of that of the largest anchor of which approval is sought. The mass of the maximum size approved is not to be more than 10 times the mass of the largest size tested.

Each test is to comprise of a comparison between the H.H.P. anchor and at least one approved ordinary stockless anchor of nearly equal mass. Where an ordinary stockless anchor is not available, for testing of HHP anchors; a previously approved HHP anchor may be used in place of an ordinary stockless anchor.

Anchors are to be tested in association with the size of chain cable appropriate to this mass.

4.2.3 Full scale tests are to be conducted on various types of bottom, which should normally be soft mud or silt, sand or gravel, and hard clay or similarly compacted material. Three tests are to be taken for each anchor and each type of bottom. The stability of the anchor and ease of breaking out are to be noted where possible.

The tests should normally be carried out from a tug, but alternatively shore tests may be accepted. The pull is to be measured by dynamometer or alternatively derived from recently verified curves of tug rev/min against bollard pull.

The length of the cable with each anchor is to be such that the pull on the shank remains horizontal. For this purpose, a scope of 10 is considered normal but a scope of not less than 6 may be accepted.

4.2.4 The holding power test load is not to exceed the proof load of the anchor.

4.3 Super High Holding Power (SHHP) anchors

4.3.1 A super high holding power anchor is an anchor with a holding power of at least four times that of an ordinary stockless anchor of the same mass. The use of SHHP anchors is to be limited to ships with service restrictions given in Pt.1, Ch.1, Sec. 2.7.3. These anchors are not to require prior adjustment or special placement on the sea bottom.

The SHHP anchor mass is generally not to exceed 1500 [kg].

4.3.2 For approval and/or acceptance as a SHHP anchor satisfactory full scale tests are to be made confirming that the anchor has a holding power of at least four times that of an ordinary stockless anchor or at least two times that of a previously approved HHP anchor, of the same mass. The tests are also to verify that the anchor withstands the test without permanent deformation.

4.3.3 The full scale tests are to be carried out at sea as specified in 4.2.3 for HHP anchors, on various types of bottom. The tests are to be applied to anchors of mass which are as far as possible representative of the full range of sizes proposed.

For a definite group within the range, the two anchors selected for testing (ordinary stockless and SHHP anchors) should be approximately the same mass and should be tested in association with the size of chain required for the anchor mass and anchor type. Where an ordinary stockless anchor is not available, a previously approved HHP anchor may be used in its place.

Tests in comparison with a previously approved SHHP anchor may be also accepted as a basis for approval.

If approval is sought for a range of anchor sizes then at least three anchor sizes are to be tested, indicative of the bottom middle and top of the mass range.

4.3.4 The holding power test load is not to exceed the proof load of the anchor.

4.4 Manufacture and testing

4.4.1 Anchors and anchor shackles are to be manufactured and tested in accordance with the requirements of Pt.2, Ch.10.

Section 5

Anchor Chain Cables

5.1 General

5.1.1 Anchor chain cables and steel wire ropes (where proposed in lieu of chain cables) are to be as required by Sec.3.

5.1.2 Grade CC3 chain cable is to be used only when diameter is 20.5 [mm] or more.

5.1.3 Grade CC1 chain cable having material tensile strength of less than 400 [N/mm^2] is not to be used in association with high holding power or super high holding power anchors.

5.1.4 When the vessel may anchor where the current exceeds 2.5 m/s, a length of heavier chain cable locally between the anchor and the rest of the chain to enhance anchor bedding is recommended.

5.2 Manufacture and testing

5.2.1 Chain cables, steel wire ropes and shackles are to be manufactured and tested in accordance with the requirements of Pt.2, Ch.10.

Section 6

Towlines, Mooring Lines and associated Shipboard Fittings

6.1 General

6.1.1 For the purpose of this Section, the following is defined:

- *Conventional ships* means new displacement-type ships of 500 GT and above, excluding high speed craft, special purpose ships (as defined in MSC. 266(84)), and offshore units of all types.

- *Shipboard fittings* mean those components limited to the following: Bollards and bitts, fairleads, stand rollers, chocks used for normal mooring of the ship and the similar components used for normal or other towing of the ship. Other components such as capstans, winches, etc. are not covered by this Section. Any weld or bolt or equivalent device connecting the shipboard fitting to the supporting structure is part of the shipboard fitting and if selected from an industry standard subject to that standard.

- *Supporting hull structures* means that part of the ship structure on/in which the shipboard fitting is placed and which is directly submitted to the forces exerted on the shipboard fitting. The supporting hull structure of capstans, winches, etc. used for normal or other towing and mooring operations mentioned above is also subject to the requirements of this Section.

- *Industry standard* means international standards (ISO, etc.) or standards issued by national association such as DIN or JMSA, etc. which are recognized in the country where the ship is built.

- *The nominal capacity condition* is defined as the theoretical condition where the maximum possible deck cargoes are included in the ship arrangement in their respective positions. For container ships the nominal capacity condition represents the theoretical condition where the maximum possible number of containers is included in the ship arrangement in their respective positions.

- *Ship Design Minimum Breaking Load* (MBL_{SD}) means the minimum breaking load of new, dry mooring lines or tow line for which shipboard fittings and supporting hull structures are designed in order to meet mooring restraint requirements or the towing requirements of other towing service.

- *Line Design Break Force* (LDBF) means the minimum force that a new, dry, spliced, mooring line will break at. This is for all synthetic cordage materials.

6.1.2 Conventional ships are to be provided with arrangements, equipment and fittings of sufficient safe working load to enable the safe conduct of all towing and mooring operations associated with the normal operations of the ship.

6.1.3 The requirements of this Section are intended to apply to design and construction of shipboard fittings and supporting structures used for normal towing and mooring operations. Normal towing means towing operations necessary for maneuvering in ports and sheltered waters associated with normal operations of the ship.

6.1.4 For ships, not subject to SOLAS Regulation II-1/3-4 Para 1, but intended to be fitted with equipment for towing by another ship or a tug, e.g. such as to assist the ship in case of emergency as given in SOLAS Regulation II-1/3-4 Para 2, the requirements designated as 'other towing' in this Section are to be applied to design and construction of those shipboard fittings and supporting hull structures.

6.1.5 This Section is not applicable to design and construction of shipboard fittings and supporting hull structures used for special towing services defined as:

— *Canal transit towing*: Towing service for ships transiting canals, e.g. the Panama Canal. Local canal transit requirements are to be referred.

— *Emergency towing for tankers*: Towing service to assist tankers in case of emergency. For emergency towing arrangements, ships subject to SOLAS regulation II-1/3-4 Para 1 are to comply with that regulation and resolution MSC.35 (63) as may be amended.

— *Escort towing*: Towing service, in particular, for laden oil tankers or LNG carriers, required in specific estuaries. Local escort requirements and guidance given by organizations such as the Oil Companies International Marine Forum (OCIMF), may be referred.

6.2 Manufacture and Testing

6.2.1 Towlines and mooring lines may be of steel wire, natural fibre or synthetic fibre and are to be made by an approved manufacturer. For synthetic fibre ropes, it is recommended to use lines with reduced risk of recoil (snap-back) to mitigate the risk of injuries or fatalities in the case of breaking mooring lines.

6.2.2 Steel wire ropes are to be manufactured and tested in accordance with the requirements of Pt.2, Ch.10.

6.2.3 Fibre ropes will be specially considered in each case.

6.2.4 The number, length and breaking strength of towlines and mooring lines are to be as required by Table 3.2.2. Also see 1.1.2 and 6.3.

6.3 Mooring Lines

6.3.1 Requirements for mooring lines for ships with EN ≤ 2000 are given in Table 3.2.2. For other ships the mooring line requirements are given in 6.3.3. Deck cargoes at the ship nominal capacity condition is to be included for the determination of the side projected area A. The nominal capacity condition is defined in 6.1.1. Minimum number and minimum strength of mooring lines is specified in 6.3.2 and 6.3.3. As an alternative to 6.3.2 and 6.3.3, the minimum requirements for mooring lines may be determined by direct mooring analysis in line with the procedure given in 6.15. The designer should consider verifying the adequacy of mooring lines based on assessments carried out for the individual mooring arrangement, expected shore-side mooring facilities and design environmental conditions for the berth. The definition of line design break force (LDBF) is given in 6.1.1. This value is declared by the manufacturer on each mooring line certificate and is stated on a manufacturer's line data sheet. LDBF of a line is to be 100%-105% of the ship design minimum breaking load defined in 6.3.3.5. The LDBF for nylon mooring lines is to be specified as break tested wet, because nylon lines change strength characteristics once exposed to water and generally do not fully dry to their original construction state.

6.3.2 Mooring lines for ships with EN ≤ 2000

6.3.2.1 The minimum recommended mooring lines for ships having an EN ≤ 2000 are given in Table 3.2.2.

For ships having the ratio A/EN > 0.9, the following number of lines are to be added to the number of mooring lines as given by Table 3.2.2:

One line where $0.9 < \frac{A}{EN} \leq 1.1,$

Two lines where $1.1 < \frac{A}{EN} \leq 1.2,$

Three lines where $1.2 < \frac{A}{EN}$

6.3.3 Mooring lines for ships with EN > 2000

6.3.3.1 The minimum recommended strength and number of mooring lines for ships with an EN > 2000 are given in 6.3.3.5 and 6.3.3.6 respectively. The length of mooring lines is given by 6.3.4.

6.3.3.2 The strength of mooring lines and the number of head, stern, and breast lines (see Note) for ships with an Equipment Number EN > 2000 are based on the side-projected area A_1. Side projected area A_1 is to be calculated similar to the side-projected area A according to Sec. 3, Cl. 3.1 but considering the following conditions:

- The ballast draft is to be considered for the calculation of the side-projected area A_1. For ship types having small variation in the draft, like e.g. passenger and RO/RO vessels, the side projected area A_1 may be calculated using the summer load waterline.

- Wind shielding of the pier can be considered for the calculation of the side-projected area A_1 unless the ship is intended to be regularly moored to jetty type piers. A height of the pier surface of 3 m over waterline may be assumed, i.e. the lower part of the side-projected area with a height of 3 m above the waterline for the considered loading condition may be disregarded for the calculation of the side-projected area A_1.

- Deck cargoes at the ship nominal capacity condition is to be included for the determination of side-projected area A_1. For the condition with cargo on deck, the summer load waterline may be considered. Deck cargoes may not need to be considered if ballast draft condition generates a larger side-projected area A_1 than the full load condition with cargoes on deck. The larger of both side-projected areas is to be chosen as side-projected area A_1. The nominal capacity condition is defined in 6.1.1.

6.3.3.3 The mooring lines as given here under are based on a maximum current speed of 1.0 [m/s] and the following maximum wind speed v_w, in [m/s]:

V_w = 25.0 – 0.002 (A_1 – 2000) for passenger ships, ferries and car carriers with 2000 [m^2] < A_1 ≤ 4000 [m^2]

= 21.0 for passenger ships, and car carriers with A_1 > 4000 [m^2]

= 25.0 for other ships.

The wind speed is considered representative of a 30 second mean speed from any direction and at a height of 10 [m] above the ground. The current speed is considered representative of the maximum current speed acting on bow or stern (±10°) and at a depth of one-half of the mean draft. Furthermore, it is considered that ships are moored to solid piers that provide shielding against cross current.

Special consideration is to be given to additional loads caused by e.g., higher wind or current speeds, cross currents, additional wave loads, or reduced shielding from non-solid piers for designing the mooring arrangements. It may be noted that unbeneficial mooring layouts can considerably increase the loads on single mooring lines.

6.3.3.4 The following is defined with respect to the purpose of mooring lines (Fig. 6.3.3.4):

.1 *Breast Line*: A mooring line that is deployed perpendicular to the ship, restraining the ship in the off-berth direction.

.2 *Spring line*: A mooring line that is deployed almost parallel to the ship, restraining the ship in fore or aft direction.

.3 *Head/Stern line*: A mooring line that is oriented between longitudinal and transverse direction, restraining the ship in the off-berth and in fore or aft direction. The amount of restraint in fore or aft and off-berth direction depends on the line angle relative to these directions.

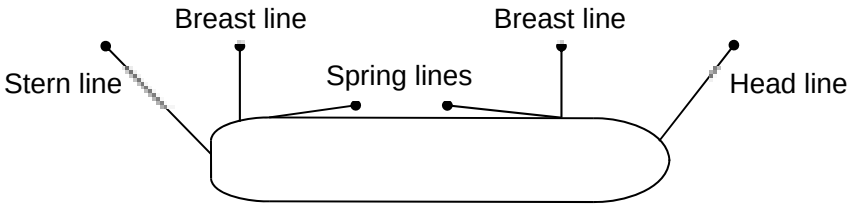


Fig 6.3.3.4 : Mooring Lines

6.3.3.5 Ship design minimum breaking load

.1 The ship design minimum breaking load(MBL_{SD}), in kN, of the mooring lines is to be taken as:

$$MBL_{SD} = 0.1 * A1 + 350$$

The ship design minimum breaking load may be limited to 1275 [kN]. However, in this case the moorings are to be considered as not sufficient for environmental conditions given by 6.3.3. For these ships, the acceptable wind speed vw* [m/s], can be estimated as follows:

$$v_w^* = v_w \cdot \sqrt{MBL_{SD}^* / MBL_{SD}}$$

where v_w is the wind speed as per 6.3.3.3, MBL_{SD}* the ship design minimum breaking load of the mooring lines intended to be supplied and MBL_{SD} the ship design minimum breaking load as recommended according to the above formula.
However, the ship design minimum breaking load is not to be taken less than corresponding to an acceptable wind speed of 21 [m/s]:

$$MBL_{SD}^* \geq \left(\frac{21}{v_w}\right)^2 . \ MBL_{SD}$$

If lines are intended to be supplied for an acceptable wind speed v_w* higher than v_w as per 6.3.3.3, the ship design minimum breaking load is to be taken as:

$$MBL_{SD}^* = \left(\frac{v_w^*}{v_w}\right)^2 . \ MBL_{SD}$$

6.3.3.6 Number of Mooring Lines

.1 The total number of head, stern and breast lines is to be taken as:

$$n = 8.3 * 10^{-4} * A_1 + 6$$

.2 For oil tankers, chemical tankers, bulk carriers, and ore carriers the total number of head, stern and breast lines is to be taken as:

$$n = 8.3 * 10^{-4} * A_1 + 4$$

The total number of head, stern and breast lines is to be rounded to the nearest whole number.

.3 The number of head, stern and breast lines may be increased or decreased in conjunction with an adjustment to the ship design minimum breaking load of the lines. The adjusted ship design minimum breaking load, MBL_{SD}**, is to be taken as:

- for increased number of lines

$$MBL_{SD}^{**} = 1.2 \, MBL_{SD} \times {}^{n}/_{n^{**}} \leq MBL_{SD}$$

- for reduced number of lines

$$MBL_{SD}^{**} = MBL_{SD} \times {}^{n}/_{n^{**}}$$

where MBL_{SD} is MBL_{SD} or MBL_{SD}* specified in 6.3.3.5.1, as appropriate, n** is the increased or decreased total number of head, stern and breast lines and n the number of lines for the considered ship type as calculated by the above formulas without rounding.

Vice versa, the ship design minimum breaking load of head, stern and breast lines may be increased or decreased in conjunction with an adjustment to the number of lines.

.4 The total number of spring lines is to be taken not less than:

Two lines where EN < 5000,

Four lines where EN ≥ 5000.

The ship design minimum breaking load of spring lines is to be the same as that of the head, stern and breast lines. If the number of head, stern and breast lines is increased in conjunction with an adjustment to the ship design minimum breaking load of the lines, the number of spring lines is also to be taken as follows, but rounded up to the nearest even number.

$n_s^* = MBL_{SD}/MBL_{SD}^{**} \; n_s$

where MBL_{SD} is MBL_{SD} or MBL_{SD}^* specified in 6.3.3.5.1, as appropriate, n_s is the number of spring lines as given above and n_s^* the increased number of spring lines.

6.3.4 Length of mooring Lines

6.3.4.1 The length of mooring lines for ships with EN ≤ 2000 may be taken from Table 3.2.2. For ships with EN > 2000 the length of mooring lines may be taken as 200 [m].

6.3.4.2 The lengths of individual mooring lines may be reduced by up to 7% of the above given lengths, but the total length of mooring lines should not be less than would have resulted had all lines been of equal length.

6.4 Mooring and Towing Arrangements

6.4.1 Mooring Arrangements

6.4.1.1 Mooring lines in the same service (e.g. breast lines, as defined in 6.3.3.4) are to be of the same characteristic in terms of strength and elasticity.

6.4.1.2 As far as possible, sufficient number of mooring winches are to be fitted to allow for all mooring lines to be belayed on winches. This allows for an efficient distribution of the load to all mooring lines in the same service and for the mooring lines to shed load before they break. If the mooring arrangement is designed such that mooring lines are partly to be belayed on bitts or bollards, it is to be considered that these lines may not be as effective as the mooring lines belayed on winches.

6.4.1.3 Mooring lines are to have as straight a lead as is practicable from the mooring drum to the fairlead. At points of change in direction sufficiently large radii of the contact surface of a rope on a fitting are to be provided to minimize the wear experienced by mooring lines and as recommended by the rope manufacturer for the rope type intended to be used.

6.4.2 Mooring Winches

6.4.2.1 Each mooring winch is to be fitted with brakes, the holding capacity of which is sufficient to prevent unreeling of the mooring line when the rope tension is equal to 80 % of the ship design minimum breaking load of the rope as fitted on the first layer on the winch drum. The winch is to be fitted with brakes that will allow for the reliable setting of the brake rendering load.

6.4.2.2 For powered winches the maximum hauling tension which can be applied to the mooring line (the reeled first layer) is not to be less than 1/4.5 times, nor be more than 1/3 times the rope's ship design minimum breaking load. For automatic winches these figures apply when the winch is set to the maximum power with automatic control.

6.4.2.3 For powered winches on automatic control, the rendering tension which the winch can exert on the mooring line (the reeled first layer) is not to exceed 1.5 times, nor be less than 1.05 times the hauling tension for that particular power setting of the winch. The winch is to be marked with the range of rope strength for which it is designed.

6.4.3 Towing Arrangements

6.4.3.1 Towing lines are to be led through a closed chock. The use of open fairleads with rollers or closed roller fairleads is to be avoided.

6.4.3.2 For the purpose of towing, bitts or bollards serving a chock are to be located slightly offset and in a distance of at least 2 [m] away from the chock, see Fig. 6.4.3.2:

6.4.3.3 Warping drums are to preferably be positioned not more than 20 [m] away from the chock, measured along the path of the line.

6.4.3.4 Attention is to be given to the arrangement of the equipment for towing and mooring operations in order to prevent interference of mooring and towing lines as far as practicable. It is beneficial to provide dedicated towing arrangements separate from the mooring equipment.

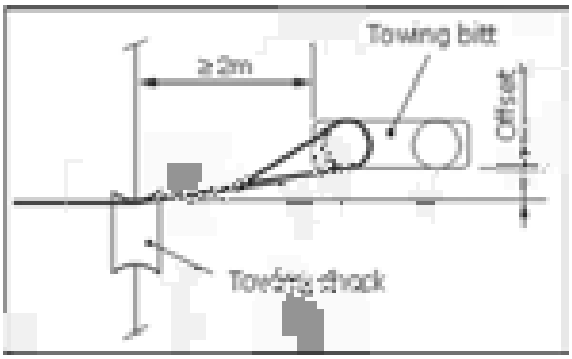


Fig 6.4.3.2

6.5 Shipboard fittings

6.5.1 Means are to be provided to enable towing lines and mooring lines to be adequately secured on board a ship. Bollards and bitts, fairleads, chocks, cleats, etc. are to be so designed and installed as to protect ropes against excessive wear.

6.5.2 The shipboard fittings associated with towing and/or mooring, may be selected from an accepted industry standard, such as. ISO 13795 Ships and Marine Technology – Ship's mooring and towing fittings – welded steel bollards for sea-going vessels and at least based on the following loads:

(a) For normal towing operations, the intended towing load (e.g. static bollard pull) as indicated on the towing and mooring arrangements plan,

(b) For other towing service, the ship design minimum breaking load of the tow line according to Table 3.2.2. Also see 6.7.1 b) and note to 6.7.1.

(c) For fittings intended to be used for, both, normal and other towing operations, the greater of the loads according to (a) and (b).

(d) For mooring operations, the ship design minimum breaking load of the mooring line according to 6.3.2 and 6.3.3, as applicable.

6.5.3 Towing bitts (double bollards) may be chosen for the towing line attached with eye splice if the industry standard distinguishes between different methods to attach the line, i.e. figure-of-eight or eye splice attachment.

6.5.4 Mooring bitts (double bollards) are to be chosen for the mooring line attached in figure-of-eight fashion, if the industry standard distinguishes between different methods to attach the line, i.e. figure-of-eight or eye splice attachment.

6.5.5 When the shipboard fitting is not selected from an accepted industry standard, the strength of the fitting and of its attachment to the ship is to be in accordance with 6.7 and 6.8. Towing bitts (double bollards) are required to resist the loads caused by the towing line attached with eye splice. Mooring bitts (double bollards) are required to resist the loads caused by the mooring line attached in figure--eight fashion, see Note.

Note: With the line attached to a mooring bitt in the usual way (figure-of-eight fashion), either of the two posts of the mooring bitt can be subjected to a force twice as large as that acting on the mooring line. Disregarding this effect, depending on the applied industry standard and fitting size, overload may occur.

6.5.6 For strength assessment beam theory or finite element analysis using net scantlings is to be applied, as appropriate. Corrosion additions are to be as defined in 6.11. A wear down allowance is to be included as defined in 6.12.

6.6 Supporting Hull Structures

6.6.1 Shipboard fittings, winches and capstans are to be suitably located on stiffeners and/or girders, which are part of the deck construction so as to facilitate efficient distribution of the towing and/ or mooring loads. Other

arrangements for mooring such as chocks in bulwarks, may be accepted provided the strength is confirmed adequate for the service.

6.6.2 The supporting structure of capstans and winches used for normal or other towing and mooring operations are also to meet the strength requirements.

6.6.3 The net scantlings of the underdeck supporting structure are to be adequate to meet the design loads specified in 6.7, considering the possible locations for transfer of the load and any horizontal or vertical variation in direction of the applied load. Proper alignment of fittings and supporting hull structure is to be ensured (see sample arrangement in Fig. 6.6.3).

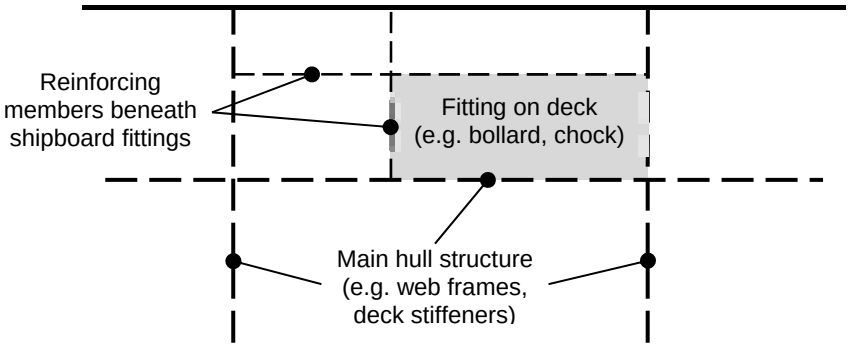


Fig. 6.6.3 : Sample Arrangement

6.6.4 The acting point of the force on the shipboard fittings is to be taken at the attachment point of the towing/mooring line or at a change in its direction. For bollards and bitts the attachment point of the towing/ mooring line is to be taken not less than 4/5 of the tube height above the base, see Fig. 6.6.4(a) and Fig. 6.6.4(b) (i) for towing and mooring respectively. However, if fins are attached to the bollard tubes to keep the mooring lines as low as possible, the attachment point of the mooring line may be taken at the location of the fins, see Fig. 6.6.4 (b) (ii).

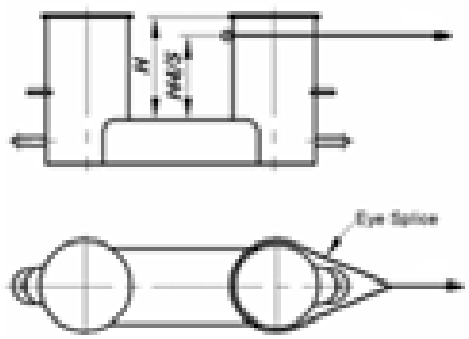
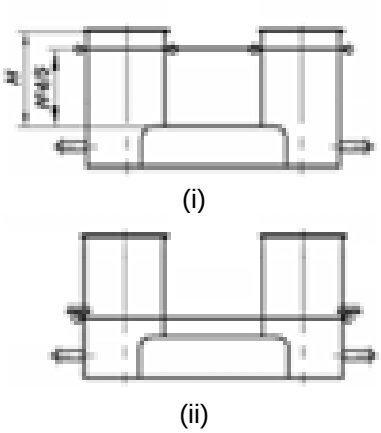


Fig. 6.6.4(a) : Attachment point of the towing line



(i)

(ii)

Fig. 6.6.4(b) : Attachment point of the mooring line

6.6.5 The arrangement of the ship's structure in way of the shipboard fittings and their seats and in way of capstans is to be such that abrupt changes of shape or section are to be avoided in order to minimize stress concentrations. Sharp corners and notches are to be avoided, especially in highly stressed areas.

6.7 Design Loads

6.7.1 Unless the designer specifies a greater safe working load (SWL) or safe towing load (TOW) for the shipboard fitting, the minimum design load is to be as follows:

a) For normal towing operations, 1.25 times the intended maximum towing load (e.g. static bollard pull) as indicated on the towing and mooring arrangement plan.

b) For other towing operations, the ship design minimum breaking load as required in Table 3.2.2 corresponding to the equipment number of the vessel as per 3.1.1. Side projected area including that of deck cargoes as given by the ship nominal capacity condition is to be taken into account for selection of the towing lines and the loads applied to shipboard fittings and supporting hull structure.

c) For fittings intended to be used for both normal and other towing operations, the greater of the design loads according to a) and b).

d) For mooring operations, the minimum design load is to be 1.15 times the ship design minimum breaking load according to 6.3.2 and 6.3.3, as applicable. Side projected area including that of deck cargoes as given by the ship nominal capacity condition is to be taken into account for selection of the towing lines and the loads applied to shipboard fittings and supporting hull structure.

Note: The increase of the line design break force for synthetic ropes according to 6.2.6 need not be taken into account for the loads applied to shipboard fittings and supporting hull structures.

6.7.2 The minimum design load applied to supporting hull structures for winches is to be 1.25 times the intended maximum brake holding load, where the maximum brake holding load is to be assumed not less than 80% of the ship design minimum breaking load according to 6.3.2 and 6.3.3. For supporting hull structures of capstans, 1.25 times the maximum hauling-in force is to be taken as the minimum design load.

6.7.3 The design load is to be applied to fittings in all directions that may occur by taking into account the arrangement shown on the towing and mooring arrangements plan. Where the towing line and mooring line takes a turn at a fitting the total design load applied to the fitting is equal to the resultant of the design loads acting on the line, (see Fig. 6.7.3). However, in no case does the design load applied to the fitting need to be greater than twice the design load on the line

6.7.4 When a safe towing load TOW greater than that determined according to 6.9 is requested by the owner/ designer, then the

design load is to be increased in accordance with the appropriate TOW/ design load relationship given by 6.5 and 6.9.

6.7.5 When a safe working load SWL greater than that determined according to 6.9 is requested by the owner/ designer, then the design load is to be increased in accordance with the appropriate SWL/ design load relationship given by 6.5 and 6.9

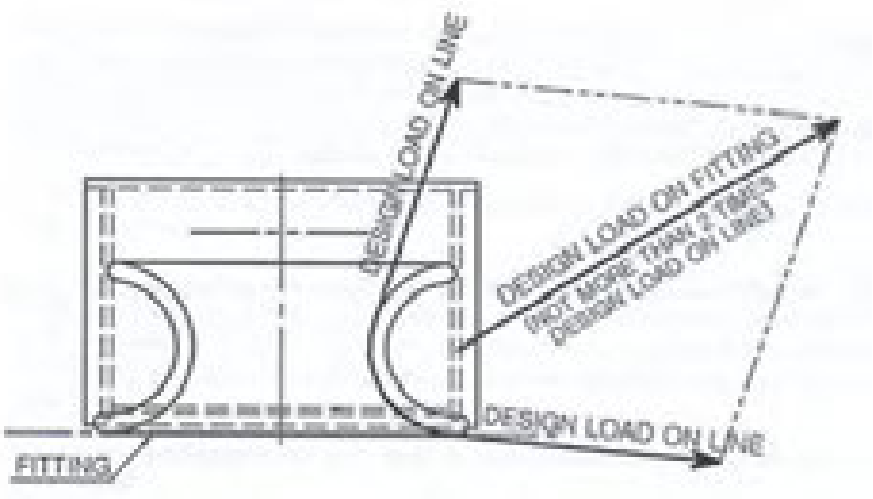


Fig. 6.7.3 : Design Load on Fitting

6.8 Acceptance Criteria

6.8.1 Allowable stresses in the shipboard fittings, supporting structure and welds are not to be exceeded, as given below:

6.8.2 For strength assessment by means of beam theory or grillage analysis:

Normal stress = 1.0 R_{eH}

Shear stress, $\tau = 0.6R_{eH}$

Where R_{eH}= specified minimum yield stress of the material [N/mm^2];

Normal stress is the sum of the bending stress and axial stress. No stress concentration factors being taken into account.

6.8.3 For strength assessment by means of finite element analysis:

Von Mises stress = $1.0R_{eH}$

For strength assessment by means of finite element analysis the mesh is to be fine enough to represent the geometry as realistically as possible. The aspect ratios of elements are not to exceed 3. Girders are to be modelled using shell or plane stress elements. Symmetric girder flanges may be modelled by beam or truss elements. The element height of girder webs must not exceed one-third of the web height. In way of small openings in girder webs the web thickness may be reduced to a mean thickness over the web height. Large openings are to be modelled. Stiffeners may be modelled by using shell, plane stress, or beam elements. The mesh size of stiffeners is to be fine enough to obtain proper bending stress. If flat bars are modeled using shell or plane stress elements, dummy rod elements are to be modelled at the free edge of the flat bars and the stresses of the dummy elements are to be evaluated. Stresses are to be read from the centre of the individual element. For shell elements the stresses are to be evaluated at the mid plane of the element.

6.8.4 Stress concentration factors are not to be taken into account in the above allowable stresses.

6.9 Safe Towing Load (TOW) and Safe Working Load (SWL)

6.9.1 The safe towing load (TOW) is the safe load limit of shipboard fittings used for towing purpose. The following requirements for TOW apply for use with no more than one line. If not otherwise chosen, for towing bitts (double bollards) TOW is the load limit for a towing line attached with eye-splice.

.1 TOW used for normal towing operations is not to exceed 80% of the design load according to 6.7.1 (a)

.2 TOW used for other towing operations is not to exceed 80% of the design load according to 6.7.1 (b)

.3 For fittings used for both normal and other towing operations, the greater of the safe towing loads according to 6.9.1.1. and 6.9.1.2 is to be used

.4 TOW, in [t], of each shipboard fitting is to be marked (by weld bead or equivalent) on the deck fittings used for towing. For fittings intended to be used for, both, towing and mooring, SWL, in [t], according to 6.9.2 is to be marked in addition to TOW.

.5 The towing and mooring arrangements plan as indicated in 6.10 is to define the method of use of towing lines.

6.9.2 The safe working load (SWL) is the safe load limit of shipboard fittings used for mooring purpose. The following requirements for SWL apply for the use with no more than one mooring line:

.1 Unless a greater SWL is requested by the applicant according to 6.7.4, the SWL is not to exceed the ship design minimum breaking load according to 6.3.2 and 6.3.3.

.2 The SWL, in [t], of each shipboard fitting is to be marked (by weld bead or equivalent) on the deck fittings used for mooring. For fittings intended to be used for, both, mooring and towing, TOW, in [t], according to 6.9.1 is to be marked in addition to SWL.

.3 The towing and mooring arrangements plan as indicated in 6.10 is to define the method of use of mooring lines.

6.10 Towing and mooring arrangement plan

.1 The SWL and TOW for the intended use for each shipboard fitting is to be noted in the towing and mooring arrangement plan available on board for the guidance of the Master. It is to be noted that TOW is the load limit for towing purpose and SWL that for mooring purpose. If not otherwise chosen, for towing bitts, it is to be noted that TOW is the load limit for a towing line attached with eye-splice.

.2 Information provided on the plan is to include in respect of each shipboard fitting:

a) location on the ship;
b) fitting type;
c) SWL/ TOW;
d) Purpose (mooring / harbour towing / other towing); and
e) Manner of applying towing or mooring line load including limiting fleet angle i.e. angle of change in direction of an line at the fitting.

Item c) with respect to items d) and e) is subject to approval by IRS.

The following information is also to be clearly indicated on the plan:

a) the arrangement of mooring lines showing number of lines (N), together with

b) the ship design minimum breaking load (MBL_{SD}).

c) the acceptable environmental conditions as given in 6.3.3.3 and 6.3.3.5 for the recommended ship design minimum breaking load for ships with EN > 2000:

• 30 second mean wind speed from any direction. (v_w or v_w^* according to 6.3.3.3 and 6.3.3.5)

• Maximum current speed acting on bow or stern (±10°).

.3 The information as given in 6.10.2 is to be incorporated into the pilot card in order to provide the pilot proper information on harbor and other towing operations.

6.11 Corrosion Addition

6.11.1 The total corrosion addition, t_c, is not to be less than the following values:

- For the supporting hull structure, according to the rules for the surrounding structure (e.g. deck structures, bulwark structures)

- For pedestals and foundations on deck which are not part of a fitting according to an accepted industry standard, 2.0 [mm]

- For shipboard fittings not selected from an accepted industry standard, 2.0 [mm].

6.12 Wear Allowance

6.12.1 In addition to the corrosion allowance specified in 6.11, a wear allowance, t_w, not less than 1.0 [mm] is to be added to surfaces which are intended to regularly contact the line.

6.13 Surveys after construction

6.13.1 The condition of deck fittings, their pedestals or foundations and the hull structures in the vicinity of the fittings are to be surveyed in accordance with the requirements of Pt.1, Ch.2.

6.14 Emergency towing procedure

6.14.1 A ship specific emergency towing procedure is to be provided according to SOLAS Ch.II-1/Reg.3-4 on ships as indicated in the following:

a) all passenger ships, not later than 01 January 2010

b) cargo ships constructed on or after 01 January 2010

c) Cargo ships constructed before 01 January 2010, not later than 01 January 2012.

In the case of oil tankers, chemical carriers and liquefied gas carriers, the requirement for emergency towing arrangements given in Pt.5, Ch.2, Sec.2.8, Pt.5, Ch.3, IR2.8 and Pt.5, Ch.4, IR3.9 respectively are also to be complied with.

6.14.2 The emergency towing procedure is to be based on existing arrangements and equipment available on board the ship

6.14.3 The emergency towing procedure is to include:

a) drawings of fore and aft deck showing possible emergency towing arrangements;

b) inventory of equipment on board that can be used for emergency towing;

c) means and methods of communication; and

d) sample procedures to facilitate the preparation for and conducting of emergency towing operations.

(Refer to the IMO guidelines for owners / operators on preparing emergency towing procedures (MSC.1/Circ.1255)).

6.15 Direct Mooring Analyses

6.15.1 General

6.15.1.1 As an alternative to the prescriptive approach, direct mooring analysis may be performed to determine the necessary mooring restraint, i.e. number and strength of mooring lines. Direct analyses allow to optimize mooring equipment and arrangement for the individual ship and the port mooring facilities typical for the considered ship type and size.

6.15.2 Documentation

6.15.2.1 The calculations are to be documented in a report. The report is to include all assumptions made in calculations for the finally chosen mooring equipment, including lines, and its arrangement, reflected in the mooring arrangement plan as required in this section.

6.15.3 Analysis methodology

6.15.3.1 Three dimensional quasi-static calculations are to be performed to determine the acting mooring line forces. As a minimum, loads from wind and current are to be accounted for in the analysis. Geometrical and material nonlinearities of mooring lines and fenders or breasting dolphins are to be considered. An iterative calculation procedure is to be applied to arrive at a converged solution with forces acting on mooring lines and on fenders or breasting dolphins being in equilibrium with forces and moments applied to the ship.

6.15.4 Environmental conditions

6.15.4.1 Mooring line forces are to be calculated for environmental conditions given in 6.3.3. Additional loads, e.g. wave loads or cross currents, or increased wind and current loads may be considered for certain ship types or for specific ports intended to be regularly called.

6.15.5 Steps to be taken in a direct mooring analysis

6.15.5.1 Direct assessment of mooring forces and determination of the necessary number and strength of mooring lines comprise the following steps:

a) Determine port mooring facilities representative for the considered ship type and size
b) Determine shipboard mooring equipment and arrangement
c) Determine mooring line type(s) to be used
d) Determine mooring layout(s) to be assessed
e) Determine ship loading condition(s) to be assessed
f) Select or determine wind and current drag coefficients
g) Determine wind and current forces and moments
h) Compute forces acting on all mooring line
i) Determine necessary strength of mooring lines
j) If strength of mooring lines should be altered, modify steps b), c) and/or d) with or without changing the number of mooring lines and repeat steps h) and i)

6.15.5.2 Port mooring facilities

.1 Characteristics of port mooring facilities have strong influence on the resulting mooring line forces. Mooring analysis is to be performed for port mooring facilities representative for the considered ship type and size, i.e. type of berth, type and arrangement of hooks/bollards, type and arrangement of fenders or breasting dolphins and height of pier above waterline.

.2 Fenders or breasting dolphins in many cases may not affect the critical mooring line loads. Hence, initially, generic fender or dolphin arrangements and infinitely stiff load-deformation characteristics may be considered. If no fender or dolphin loads occur for load cases yielding the critical mooring line loads, more specific fender or dolphin arrangements and characteristics may be omitted.

.3 If there are substantially different port mooring facilities typically encountered by the considered ship type, additional calculations are to be performed to consider these variations.

6.15.5.3 Shipboard mooring equipment and arrangement

.1 The mooring equipment and arrangement need to be chosen for the mooring analysis, i.e. location of mooring decks and location of mooring winches and fairleads. As a starting point, mooring equipment for the number of lines as determined by the prescriptive approach may be chosen, see 6.3.3.6.

6.15.5.4 Mooring lines

.1 The mooring analysis is to apply the mooring line type(s) intended to be supplied with the vessel. The geometrical and material nonlinearities of the mooring lines are to be considered by the mooring analysis. Load-deflection characteristics of mooring lines can be taken from data sheets of rope manufacturers. If given, characteristics of the broken-in ropes are to be applied.

.2 To achieve a good distribution of mooring line forces, mooring line type and characteristics are to be at least same for lines in the same service, e.g. for head and stern lines, breast lines and spring lines. For very stiff mooring lines, e.g. made of steel or high modulus synthetic fibers, the use of elastic tails is to be considered to enhance the elasticity in the mooring system and taken into account for the mooring analysis.

6.15.5.5 Mooring layout

.1 For the assessment of forces acting on mooring lines, a realistic mooring layout needs to be assumed, i.e. for each mooring line it needs to be determined from which bollard or winch, along which path, through which fairlead it is led and to which shoreside hook or bollard it is connected. Inboard parts of the mooring lines (between fairlead and shipboard fixation point) contribute to the elongation behavior of the line and should be included in the analysis.

.2 The maximum number of lines connected to one shore mooring point needs to be limited to not load the shore side mooring points unrealistically high. For multi-purpose piers the number of lines per shore bollard should be limited to three. For other types of berths, the number mooring lines per shore mooring point is also limited, e.g., by the available number of hooks. Reasonable assumptions are to be made based on typical berth types encountered by the considered ship type.

.3 Alternative mooring layouts are also to be assessed, considering possible and reasonable options to moor the ship to the assumed port mooring facilities. Also, a different position of the ship relative to the shoreside mooring bollards/hooks is to be assessed to find the critical mooring line loads for the normal operation of the ship. Exemptions may be given to e.g. tankers, LNG carriers or ferries if typically moored in the same position relative to the shoreside mooring facilities.

6.15.5.6 Loading conditions

.1 Mooring line forces are to be calculated for loading conditions given in 6.3.3.

6.15.5.7 Wind and current drag coefficients

.1 To calculate the wind and current forces and moments acting on the ship, wind and current drag coefficients are needed for the considered ship type, size and loading condition. Drag coefficients are to be as specific as possible for the considered ship and loading conditions.

.2 There are different sources for drag coefficients. Some Industry Guidelines provide drag coefficients for tankers and LNG carriers which can be applied. Due to the similarity of hull forms and superstructures, these coefficients may also be used for bulk carriers and ore carriers. For other ship types drag coefficients may be taken from the literature, if available, or can be determined by CFD calculations or model tests. CFD calculations are to be justified with suitable validation and sensitivity studies.

.3 There are some effects that can influence the drag coefficients, i.e. blockage (limited under keel clearance, solid quay walls), ship draft and wind shielding by solid quays and buildings or cargo stored on quays (e.g. container stacks). Effects from blockage and ship draft can only be accounted for by appropriate coefficients. Drag coefficient is to be chosen or determined for realistic water depth to draft ratios and for the considered ship draft(s). Some Industry Guidelines provide current drag coefficients for ballast and loaded draft conditions and for different water depth to draft ratios. Wind shielding effects are typically not considered by the wind drag coefficients. The effect of wind shielding of solid quays may be considered by an equivalent reduction of the lateral wind area of the ship. Shielding by buildings or cargo stored on quays should not be considered as their presence is imponderable.

6.15.5.8 Calculation of wind and current forces and moments

.1 Wind and current forces and moments can be calculated for the given environmental conditions with the geometrical particulars of the considered ship and the selected drag coefficients. Usually, the forces in longitudinal and transversal directions as well as the moment about the vertical ship axis (yaw) are calculated.

.2 Wind forces and moments are to be calculated for all directions in intervals of preferably 15°, but not more than 30°. Current forces and moments are to be calculated for selected directions as per 6.3.3. For ships regularly moored to non-solid piers or jetties, cross current may need to be considered in addition.

6.15.5.9 Calculation of mooring line forces

.1 For all considered scenarios and all combinations of applied environmental conditions, the maximum mooring line force is to be determined for groups of lines in the same service.

.2 In case of all lines are intended to be attached to winches, brake rendering can be considered to better distribute line loads among all lines in a group of lines in the same service. Then, the average mooring line force of a group of lines may be determined and taken as mooring line force used to determine the necessary strength of the mooring lines according to 6.15.5.10.

6.15.5.10 Strength of mooring lines

.1 The necessary strength of mooring lines, i.e., the Ship Design Minimum Breaking Load (MBL_{SD}), results from the calculated maximum mooring line force ($F_{L, max}$) divided by the Work Load Limit (WLL) factor of mooring lines. The WLL factor and the resulting MBL_{SD} for different mooring line materials are shown in the below table.

Mooring line material	**WLL factor**	**MBL_{SD}**
Steel wire	0.55	1.82 * $F_{L, max}$
Synthetic fibers	0.5	2.0 * $F_{L, max}$

Preferably, all lines supplied to the ship are to have the same characteristics and strength to avoid confusion of lines. However, for significantly different maximum calculated line loads, lines in different service may also have different strength and characteristics, e.g. for head and stern lines other than for spring lines.

Section 7

Windlass and Chain Stoppers

7.1 General

7.1.1 Windlass of sufficient power and suitable for the size of chain cable is to be fitted. Where wire ropes are proposed and approved in lieu of chain cables, suitable winches capable of controlling wire rope at all times are to be fitted.

7.1.2 Windlass is to have one cable lifter for each anchor required to be kept ready for use. The cable lifter is normally to be connected to the driving shaft by release coupling and provided with brake.

7.1.3 For each chain cable, a chain stopper is normally to be arranged between the windlass and hawse pipe. The chain cables are to reach the hawse pipe through the cable lifter only.

7.1.4 Electrically driven windlasses are to have a torque limiting device. Electric motors are to comply with the requirements of Pt.4, Ch.8.

7.2 Compliance Standards

7.2.1 The design, construction and testing of windlasses are to conform to an acceptable standard such as the following:

- *SNAME T&R Bulletin 3-15:2018 – Guide to the Design and Testing of Anchor Windlasses for Merchant Ships*
- *ISO 7825:2017 – Deck Machinery General Requirements*
- *ISO 4568:2006 – Shipbuilding – Sea going Vessels – Windlasses and Anchor Capstans*
- *JIS F6714:1995 – Windlasses*

7.3 Plans and Documentation

7.3.1 The following plans indicating the design specifications, compliance standards, engineering analyses and construction details; as applicable, are to be submitted:

- Windlass design specifications, anchor and chain cable particulars, anchorage depth, performance criteria, compliance standards
- Windlass arrangement plan showing all of the components of the anchoring/ mooring system such as the prime mover shafting, cable lifter, chain stopper (if fitted), anchors and chain cables; mooring winches, wires and fairleads, if they form part of the windlass machinery, brakes, controls, etc.
- Dimensions, materials, welding details, as applicable, of all torque-transmitting (shafts, gears, clutches, couplings, coupling bolts, etc.) and all load bearing (shaft bearings, cable lifter, sheaves, drums, bed-frames, etc.) components of the windlass and of the winch, where applicable, including brakes, chain stopper (if fitted) and foundation.
- Hydraulic system to include:
  - Piping diagram along with system design pressure
  - Safety valve arrangement and settings
  - Material specifications for pipes and equipment
  - Typical pipe joints, as applicable
  - Technical data and details for hydraulic motors
- Electric one-line diagram along with cable specification and size, motor controller, protective device rating or setting; as applicable
- Control, monitoring and instrumentation arrangements
- Engineering analyses for torque-transmitting and load-bearing components demonstrating their compliance with recognized standards
- Gear analyses in accordance with a recognized standard
- Plans and data for windlass electric motors including associated gears rated 100 [kw] and over

- Calculations demonstrating that the windlass prime mover is capable of attaining the hoisting speed, the required continuous duty pull, and the overload capacity are to be submitted; if the 'load testing' including 'overload capacity' of the entire windlass unit is not carried out at the shop (See 7.6.3)

- Operation and maintenance procedures for the anchor windlass are to be incorporated in the vessel operations manual.

7.4 Materials and Fabrication

7.4.1 Materials used in the construction of torque-transmitting and load-bearing parts of windlasses are to comply with the requirements of Pt. 2 of the Rules. The proposed materials are to be indicated in the construction plans and are to be approved in connection with the design. All such materials are to be certified by the material manufacturers and are to be traceable to the manufacturers' certificates.

7.4.2 Weld joint designs are to be shown in the construction plans and are to be approved in association with the approval of the windlass design. Welding procedures and welders are to be qualified in accordance with the requirements of IRS. Welding consumables are to be approved by IRS in accordance with the requirements of Pt. 2, Ch. 11, as applicable. In case the type and grade of the welding consumables fall outside the scope of Pt. 2, Ch. 11, then the consumables are to comply with suitable national or international standards. The degree of non-destructive examination of welds and post-weld heat treatment, if any, are to be specified and submitted for consideration.

7.5 Design

7.5.1 Notwithstanding the requirements of the chosen standard of compliance, the following requirements are also to be complied with. In lieu of conducting engineering analyses and submitting them for review, approval of the windlass mechanical design may be based on a type test, in which case the testing procedure is to be submitted for consideration.

7.5.2 Mechanical Design

7.5.2.1 Design Loads

(a) Holding Loads

Calculations are to be made to show that, in the holding condition (single anchor, brake fully applied and chain cable lifter declutched), and under a load equal to 80% of the specified minimum breaking strength of the chain cable, the maximum stress in each load bearing component will not exceed yield strength (or 0.2% proof stress) of the material. For installations fitted with a chain cable stopper, 45% of the specified minimum breaking strength of the chain cable may instead be used for the calculation.

(b) Inertia Loads

The design of the drive train, including prime mover, reduction gears, bearings, clutches, shafts, cable lifter and bolting is to consider the dynamic effects of sudden stopping and starting of the prime mover or chain cable so as to limit inertial load.

7.5.2.2 Continuous Duty Pull

The windlass prime mover is to be capable of exerting, for a period of 30 minutes, + a continuous duty pull (e.g. 30-minute short time rating corresponding to S2-30 min. of IEC 60034-1), Z_{cont1}, corresponding to the grade of chain cable, as follows :-

Table 7.5.2.2

Grade of Chain	Z_{cont1}	
	N	Kgf
CC1	37.5 d^2	3.82 d^2
CC2	42.5 d^2	4.33 d^2
CC3	47.5 d^2	4.84 d^2

Where d is the chain diameter [mm]

The values in Table 7.5.2.2 are applicable when using ordinary stockless anchors for anchorage depth down to 82.5 [m].

For anchorage depth deeper than 82.5 [m], a continuous duty pull Z_{cont2} is as follows:

$$Z_{cont2} = Z_{cont1} + (D - 82.5) * 0.27\ d^2$$

Where Z_{cont1} and Z_{cont2} are in [N]

Or

$$Z_{cont2} = Z_{cont1} + (D - 82.5) * 0.0275\ d^2$$

Where Z_{cont1} and Z_{cont2} are in [kgf]

D is the anchor depth in [m]

The anchor masses are assumed to be the masses as given in Pt. 2, Ch. 10. Also, the value of Z_{cont} is based on the hoisting of one anchor at a time, and that the effects of buoyancy and hawse pipe efficiency (assumed to be 70%) have been accounted for. In general, stresses in each torque-transmitting component are not to exceed 40% of yield strength (or 0.2% proof stress) of the material under these loading conditions.

7.5.2.3 Overload Capability

The windlass prime mover is to be able to provide the necessary temporary overload capacity for breaking out the anchor. This temporary overload capacity or "short term pull" is to be at least 1.5 times the continuous duty pull applied for at least 2 minutes. The speed in this period may be lower than normal.

The above criteria do not require both anchors to be raised or lowered simultaneously on windlass fitted with two cable lifters.

7.5.2.4 Hoisting Speed

The mean speed of the chain cable during hoisting of the anchor and cable is to be at least 0.15 [m/sec]. For testing purposes, the speed is to be measured over two shots of chain cable and initially with at least three shots of chain (82.5 [m] or 45 fathoms in length) and the anchor submerged and hanging free.

7.5.2.5 Brake Capacity

.1 The capacity of the windlass brake is to be sufficient for safe stopping of anchor and chain cable when paying out.

.2 The windlass with brakes engaged and release coupling disengaged is to be able to withstand static pull of 45 per cent of the tabular breaking strength of the chain without any permanent deformation of the stressed parts and without brake slip.

.3 If a chain stopper is not fitted, the windlass is to be able to withstand a static pull of 80 per cent of the tabular breaking strength of the chain without any permanent deformation of the stressed parts and without brake slip.

.4 The chain stoppers and their attachments are to withstand a pull of 80 per cent of the tabular breaking strength of the chain without any permanent deformation of the stressed parts. The chain stoppers are to be so designed that additional bending of the individual link does not occur and the links are evenly supported.

.5 Attention is to be paid to stress concentrations in keyways and other stress raisers and also to dynamic effects due to sudden starting or stopping of the prime mover or anchor chain.

7.5.2.6 Hull Supporting Structures of Anchor Windlass and Chain Stopper

The hull supporting structure of anchor windlass and chain stopper is to be sufficient to accommodate the design and sea loads.

7.5.2.6.1 Design Loads

.1 The design loads are to be taken not less than

- for chain stoppers, 80% of the chain cable breaking load
- for windlasses, where no chain stopper is fitted or the chain stopper is attached to the windlass, 80% of the chain cable breaking load
- for windlasses, where chain stoppers are fitted but not attached to the windlass, 45% of the chain cable breaking load

.2 The design loads are to be taken in the direction of the chain cable.

7.5.2.6.2 Sea Loads

.1 In case of ships of length > 80 [m] the windlasses and mooring winches integral with the windlasses fitted on exposed decks in the forward 0.25L are to be adequately secured against the green sea loads given in 7.5.2.6.2.1, if the height of the exposed decks in way of the windlasses is less than 0.1L or 22 [m] above summer LWL. The strength criteria given in 7.5.2.6.2.1 is to be complied with.

.2 Design loads

The following green sea pressures are to be considered:

- Pressure 'p_x' equal to 200 [kN/m^2], normal to the shaft axis as shown in Fig.7.5.2.6.2.1 and acting over the projected area in this direction.

- Pressure 'p_y' equal to 150 [kN/m^2], parallel to the shaft axis as shown in Fig.7.5.2.6.2.1 acting both inboard and outboard separately, over the multiple of 'f' times the projected area in this direction.

where 'f' is defined as:

f = 1+B/H, but need not be taken greater than 2.5

where,

B = width of windlass measured parallel to the shaft axis.

H = overall height of windlass.

.3 For calculation of forces in the bolts securing the windlass to the deck, the bolts may be considered to be divided into 'N' bolt groups each containing one or more bolts. (See Fig.7.5.2.6.2.1).

The axial force R_i in bolt group (or bolt) i, positive in tensile, may be calculated from:

$$R_{xi} = P_x\, hx_i\, A_i / I_x$$
$$R_yi = P_y\, hy_i\, A_i / I_y$$

and $R_i = R_{xi} + R_{yi} – R_{si}$

where,

P_x = force [kN] acting normal to the shaft axis

P_y = force [kN] acting parallel to the shaft axis, either inboard or outboard whichever gives the greater force in bolt group 'i'

h = shaft height above the windlass mounting [cm]

x_i, y_i = x and y coordinates of bolt group 'i' from the centroid of all N bolt groups, positive in the direction opposite to that of the applied force [cm]

A_i = cross sectional area of all bolts in group 'i' [cm^2]

$I_x = \Sigma\, A_i\, x_i^2$ for N bolt groups
$I_y = \Sigma\, A_i\, y_i^2$ for N bolts groups

R_{si} = static reaction at bolt group 'i', due to weight of windlass.

Shear forces F_{xi}, F_{yi} applied to the bolt group 'i', and the resultant combined force F_i may be calculated from:

$$F_{xi} = (P_x - \alpha gM) / N$$
$$F_{yi} = (P_y - \alpha gM) / N$$

and

$$F_i = (F_{xi}^2 + F_{yi}^2)^{0.5}$$

where,

α = coefficient of friction = 0.5
M = mass of windlass (tonnes)
g = gravity acceleration = 9.81 m/sec^2
N = number of bolt groups.

Axial tensile and compressive forces and lateral forces are also to be considered in the design of the supporting structure.

.4 Strength criteria for windlass mounts

Tensile axial stresses in the individual bolts in each bolt group 'i' are to calculated. The horizontal forces F_{xi} and F_{yi} are normally to be reacted by shear chocks. Where "fitted" bolts are designed to support these shear forces in one or both directions, the von Mises equivalent stresses in the individual bolts are to be calculated and compared to the stress under proof load. Where pourable resins are incorporated in the holding down arrangements, due account is to be taken in the calculations.

The safety factor against bolt proof strength is to be not less than 2.0.

.5 Permissible stresses for supporting hull structures of anchor windlass and chain stopper

The permissible stresses acting on the hull structures of windlass and chain stopper, based on net thickness obtained by deducting corrosion addition tc, are not to be greater that the following permissible values:

a) For strength assessment by means of beam theory or grillage analysis:

Normal stress = 1.0 R_{eH}

Shear stress, $\tau = 0.6R_{eH}$

Where R_{eH}= specified minimum upper yield stress [N/mm^2];

The normal stress is the sum of bending stress and axial stress. The shear stress to be considered corresponds to the shear stress acting perpendicular to the normal stress. No stress concentration factors are to be taken into account.

b) For strength assessment with finite element analysis:

Von Mises stress = $1.0R_{eH}$

For strength assessment by means of finite element analysis the mesh is to be fine enough to represent the geometry as realistically as possible. The aspect ratios of elements are not to exceed 3. Girders are to be modelled using shell or plane stress elements. Symmetric girder flanges may be modelled by beam or truss elements. The element height of girder webs must not exceed one-third of the web height. In way of small openings in girder webs, the web thickness may be reduced to a mean thickness over the web height. Large openings are to be modelled. Stiffeners may be modelled using shell, plane stress, or beam elements. The mesh size of stiffeners is to be fine enough to obtain proper bending stress. If flat bars are modeled using shell or plane stress elements, dummy rod elements are to be modelled at the free edge of the flat bars and the stresses of the dummy elements are to be evaluated. Stresses are to be read from the centre of the individual element. For shell elements the stresses are to be evaluated at the mid plane of the element.

7.5.3 Hydraulic Systems

Hydraulic systems where employed for driving windlasses are to comply with the provisions of Pt. 4.

7.5.4 Electrical Systems

7.5.4.1 Electric Motors

Electric motors are to meet the requirements of Pt.4, Ch. 8 and those rated 100 [kW] and over are to be certified. Motors exposed to weather are to have enclosures suitable for their location as provided for in Pt. 4, Ch. 8. Where gears are fitted, they are to meet the requirements of Pt. 4, Ch. 4 and those rated 100 [kW] and over are to be certified.

7.5.4.2 Electrical Circuits

Motor branch circuits are to be protected and cable sizing is to in carried out in accordance with the provisions of Pt. 4, Ch. 8. Electrical cables installed in locations subjected to the sea are to be provided with effective mechanical protection.

7.5.5 Protection of Mechanical Components

To protect mechanical parts including component housings, a suitable protection system is to be fitted to limit the speed and torque at the prime mover. Consideration is to be given to a means to contain debris consequent to a severe damage of the prime mover due to over-speed in the event of uncontrolled rendering of the cable, particularly when an axial piston type hydraulic motor forms the prime mover.

7.5.6 Couplings

Windlasses are to be fitted with couplings which are capable of disengaging between the cable lifter and the drive shaft. Hydraulically or electrically operated couplings are to be capable of being disengaged manually.

7.6 Shop Inspection and Testing

7.6.1 Windlasses are to be inspected during fabrication at the manufacturers' facilities by a Surveyor for conformance with the approved plans. Acceptance tests, as specified in the specified standard of compliance, are to be witnessed by the Surveyor and include the following tests, as a minimum:

.1 *No-load test.* The windlass is to be run without load at nominal speed in each direction for a total of 30 minutes. If the windlass is provided with a gear change, additional run in each direction for 5 minutes at each gear change is required.

.2 *Load test.* The windlass is to be tested to verify that the continuous duty pull, overload capacity and hoisting speed as specified in 7.5.2 can be attained.

Where the manufacturing works does not have adequate facilities, these tests, including the adjustment of the overload protection, can be carried out on board ship. In these cases, functional testing in the manufacturer's works is to be performed under no-load conditions.

.3 *Brake capacity test.* The holding power of the brake is to be verified either through testing or by calculation.

7.7 On-board Testing

7.7.1 Each windlass is to be tested under working conditions, after installation on board, to demonstrate satisfactory working.

7.7.2 Each unit is to be independently tested for braking, clutch functioning, lowering and hoisting of chain cable and anchor, proper riding of the chain over the cable lifter, proper transit of the chain through the hawse pipe and the chain pipe, and effecting proper stowage of the chain and the anchor. It is to be confirmed that anchors properly seat in the stored position and that chain stoppers function as designed if fitted. The mean hoisting speed, as specified in 7.5.2.4, is to be measured and verified. The braking capacity is to be tested by intermittently paying out and holding the chain cable by means of the application of the brake. Where the available water depth is insufficient, the proposed test method will be specially considered.

7.8 Marking

7.8.1 Windlasses are to be permanently marked with the following information:

(a) Nominal size of the windlass (e.g. 100/3/45 is the size designation of a windlass for 100 mm diameter chain cable of Grade CC3, with a holding load of 45% of the breaking load of the chain cable)

(b) Maximum anchorage depth in metres.

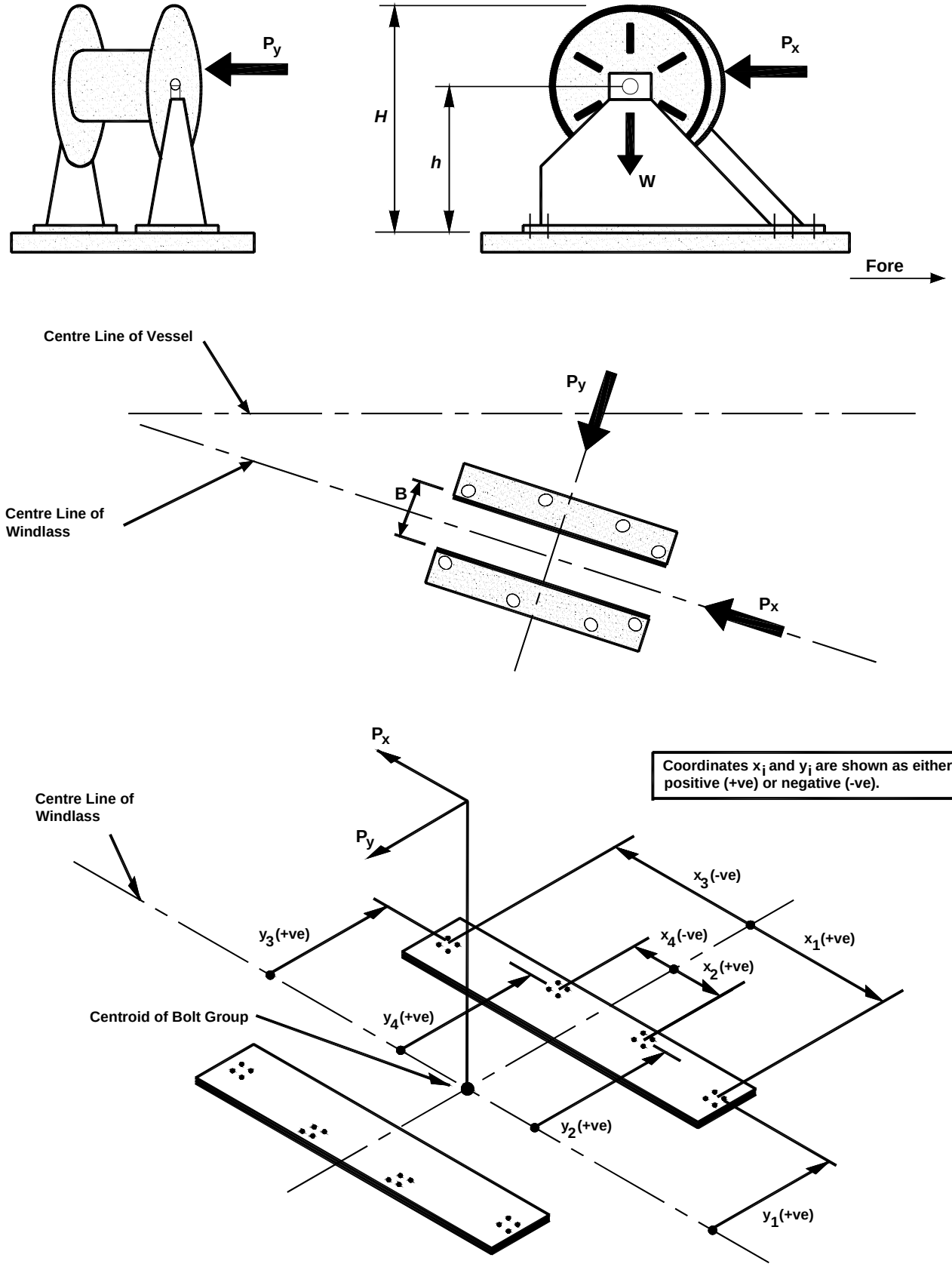


Fig.7.5.2.6.2.1 : Direction of forces and weight

Section 8

Deep Water Anchoring

8.1 General

8.1.1 The requirements in this Section are applicable to ships being assigned additional class notation '**DWA**', These requirements are intended for anchoring equipment of ships anchoring in deep and unsheltered waters.

8.2 Application

8.2.1 The additional class notation, **DWA** indicates that the ship is designed to anchor in deep and unsheltered waters with environmental conditions, as follows:

- Water depth up to 120 [m]
- Current up to 1.54 [m/s]
- Wind speed up to 14 [m/s]
- Waves with significant wave height up to 3[m]

Furthermore, these requirements are applicable to ships with equipment length not less than 135 [m].

8.3 Documentation

8.3.1 The anchoring arrangement plan and Equipment Number calculations are to be submitted for approval.

8.4 Equipment number for deep and unsheltered waters

8.4.1 Anchors and chain cables are to be provided in accordance with the requirements in Table 8.4.1 and based on the Equipment Number, EN_D obtained from the following:

$$EN_D = 0.628 \left[a \left(\frac{EN}{0.628} \right)^{\frac{1}{2.3}} + b((1-a)) \right]^{2.3}$$

Where,

$a = 1.83*10^{-9} * L_E^3 + 2.09 * 10^{-6} * L_E^2 - 6.21 * 10^{-4} * L_E + 0.0866$

$b = 0.156 * L_E + 8.372$

L_E = Equipment Length

EN = Equipment Number as calculated in accordance with Section 3, Cl. 3.1

8.5 Anchors

8.5.1 The bower anchors are to be connected to their chain cables and positioned on board ready for use.

8.5.2 Anchors are to be of the stockless High Holding Power (HHP) type.

8.5.3 The mass of the head of a stockless anchor, including pins and fittings, is not to be less than 60% of the total mass of the anchor. Requirements for HHP anchors are indicated in Section 4.

8.5.4 The mass per anchor, of bower anchors given in Table 8.4.1 is for anchors of equal mass. The mass of individual anchors may vary up to 7% of the tabular mass, but the total mass of anchors is not to be less than that recommended for anchors of equal mass.

8.5.5 Suitable arrangements are to be provided for securing the anchors when stowed, as indicated in Sec. 2.

8.5.6 The manufacture and proof-testing of anchors is to be carried out in accordance with the requirements of Pt. 2, Ch. 10.

Table 8.4.1 Anchoring equipment for ships with DWA notation					
Equipment Number EN_D	High holding power stockless bower anchors		Stud link chain cable for bower anchors		
	Number	Mass per anchor	Length	Min. diameter(d)	
				(CC2)	(CC3 3)
		(kg)	(m)	(mm)	(mm)
< 1790	2	14150	1017.5	105	84
≥ 1790 < 1930	2	14400	990	105	84
≥ 1930 < 2080	2	14800	990	105	84
≥ 2080 < 2230	2	15200	990	105	84
≥ 2230 < 2380	2	15600	990	105	84
≥ 2380 < 2530	2	16000	990	105	84
≥ 2530 < 2700	2	16300	990	105	84
≥ 2700 < 2870	2	16700	990	105	84
≥ 2870 < 3040	2	17000	990	105	84
≥ 3040 < 3210	2	17600	990	105	84
≥ 3210 < 3400	2	18000	990	105	84
≥ 3400 < 3600	2	18300	990	106	84
≥ 3600 < 3800	2	19000	990	107	85
≥ 3800 < 4000	2	19700	962.5	108	87
≥ 4000 < 4200	2	20300	962.5	111	90
≥ 4200 < 4400	2	21100	962.5	114	92
≥ 4400 < 4600	2	22000	962.5	117	95
≥ 4600 < 4800	2	22900	962.5	119	97
≥ 4800 < 5000	2	23500	962.5	122	99
≥ 5000 < 5200	2	24000	935	125	102
≥ 5200 < 5500	2	24500	907.5	130	105
≥ 5500 < 5800	2	25000	907.5	133	107
≥ 5800 < 6100	2	25500	880	137	111
≥ 6100 < 6500	2	25700	880	140	113
≥ 6500 < 6900	2	26000	852.5	143	115
≥ 6900 < 7400	2	26500	852.5	147	118
≥ 7400 < 7900	2	27000	825	152	121
≥ 7900 < 8400	2	27500	825	154	123
≥8400 < 8900	2	28000	797.5	158	127
≥ 8900 < 9400	2	28900	770	162	132
≥ 9400 < 10000	2	29400	770	-	135
≥ 10000 < 10700	2	29900	770	-	139
≥ 10700 < 11500	2	30600	770	-	143
≥ 11500 < 12400	2	31500	770	-	147
≥ 12400 < 13400	2	33200	770	-	152
≥ 13400 < 14600	2	35000	770	-	157
≥ 14600	2	38000	770	-	162

8.6 Chain cables for bower anchors

8.6.1 Bower anchors are to be associated with stud link chain cables of Grade CC2 or Grade CC3 quality. The total length of chain cable, as given in Table 8.4.1 is to be reasonably divided between the two bower anchors. For the proof and breaking loads of stud link chain cables reference is made to Table 2.9.1 (a) and Table 2.9.1 (b) of Pt. 2, Ch. 10.

8.6.2 Manufacture of anchor chain cables is to be carried out in accordance with the requirements of Pt.2, Ch. 10.

8.6.3 Installation of chain cables is to be carried out in accordance with the requirements of Sec.2.

8.7 Anchor windlass and chain stopper

8.7.1 The design and testing of anchor windlass and design of chain stopper is to be carried out in accordance with the requirements of Sec. 7.

8.7.2 In addition to the requirements in Sec. 7, the windlass prime mover is to be able to supply for at least 30 minutes a continuous duty pull Z_{cont} [N] given by:

$$Z_{cont} = 35d^2 + 13.4\, m_A$$

Where:

d = chain diameter [mm] as per Table 8.4.1

m_A = mass of HHP anchor [kg] as per Table 8.4.1

8.7.3 In addition to the requirements of Sec, 7; as far as practicable; for testing purpose the speed of the chain cable during hoisting of the anchor and cable is to be measured over 37.5 [m] of chain cable and initially with at least 120 [m] of chain and the anchor submerged and hanging free. The mean speed of the chain cable during hoisting of the anchor from a depth of 120 [m] to the depth of 82.5 [m] is to be at least 4.5 [m/min].

8.7.4 The hull supporting structure of anchor windlass and chain stopper is to be provided in accordance with the requirements in Sec. 7.

End of Chapter

Chapter 15

Masts and Rigging

Contents

Section

1 *General*

2 *Masts*

3 *Rigging*

Section 1

General

1.1 Application

1.1.1 Masts, derrick posts, crane pedestals etc. and their supporting hull structure are to be in accordance with the requirements of this Chapter. The derricks and the cargo handling gear, are not subject to approval unless required by additional class notations.

1.2 Documentation

1.2.1 The following plans and information are to be submitted:

i) Arrangement plan showing location of mast or derrick post, standing rigging and cargo handling gear alongwith the information about the operation of the derrick booms, their safe working load and working positions.

ii) Plan showing proposed scantlings of mast, derrick post and standing rigging.

iii) Plan showing supporting structures and strengthening of hull in way of mast, post and standing rigging fastenings,

iv) Specification of the steel wire ropes intended to be used for standing rigging, indicating rope construction, scantlings and minimum breaking strength.

1.2.2 Calculations considering the derrick booms at the operating position which results in maximum stresses on the mast, are to be submitted. For masts supporting derricks, angles of heel and trim of the ship in this condition of less than 5° and 2° respectively, may be ignored. Where these angles are exceeded, and in all cases where the mast supports derrick cranes or derricks of special design, the actual angles are to be taken into account in calculating the stresses in the mast. In case of stayed masts, the calculations are to take into account the elongation of the stays under tension.

1.2.3 For masts such as portal, bipod, lattice etc. and those under significant forces other than resulting from cargo gear; detailed stress calculations are to be submitted, taking into account:

i) All horizontal, vertical and torsional forces,
ii) Deflection of the structure,
iii) Variations in the moment of inertia of the parts of the structure,
iv) The effects of outriggers and similar structures,
v) Elasticity and sag in stays, where fitted.

1.3 Materials

1.3.1 Steel for masts, derrick posts, crane pedestals and associated items are to be tested and approved in accordance with the requirements of Pt.2, Ch.3. Material grades of steel are to be as follows:

Thickness [mm]	t ≤ 20	20 ≤ t < 25	25 ≤ t < 40	t > 40
Grade	A/AH	B/AH	D/DH	E/EH

1.3.2 Wire ropes used for stays are to be in accordance with the requirements of Pt.2, Ch.10.

Section 2

Masts

2.1 Scantlings

2.1.1 The outer diameter 'd' of the masts without derricks at the level of upper supporting deck is not to be less than:

d = 37 *l* [mm] for unstayed masts

= 33 *l* [mm] for stayed masts

where,

l = length of the mast [m] measured above upper supporting deck to hounds.

The diameter is to be maintained upto the level of the gooseneck fitting where this is entirely supported by the mast. In case of masts without derricks, the diameter may be reduced to 0.75 d at top.

2.1.2 For masts without derricks, the thickness 't' of the mast plating is not to be less than

t = 0.01d + 2.5 [mm] or 5 [mm] whichever is greater.

2.1.3 The scantlings of masts with derricks are to be based on calculations mentioned in 1.2.2 and 1.2.3. The total stress σ_t at any location, is not to exceed the allowable stress values given below:-

Allowable stresses in masts		
SWL	**Stayed mast**	**Unstayed mast**
SWL ≤ 10t	0.50 σ_y	0.55 σ_y
SWL ≥ 60t	0.625 σ_y	0.675 σ_y
10 < SWL < 60	By linear interpolation	

where,

$$\sigma_t = \sqrt{\left[(\sigma_b + \sigma_c)^2 + 3\tau^2\right]}\ [\mathrm{N/mm^2}]$$

σ_b = bending stress at that location.

σ_c = direct compressive stress at that location due to the vertical components of force. In general, the weight of the mast and cross trees may be ignored in this calculation.

τ = The shear stress due to torque in the mast. The effect of torque need only be considered where cross trees are fitted.

σ_y = Upper yield stress of the material [N/mm^2].

For the above Table, SWL corresponding to the largest derrick operating on the mast is to be taken.

The thickness of the mast plating is to be adequate to prevent buckling and in no case is to be less than 7.5 [mm].

2.2 Construction details

2.2.1 Masts are to be adequately supported by atleast two decks and are to be efficiently scarphed into the main hull structure. The hull structure is to be suitably reinforced. A deckhouse may be considered as a support provided adequate strength is ensured by additional stiffening and increased plate thickness.

2.2.2 In general, mast scantlings are not to be reduced inside the deckhouse and are to be maintained upto the level of the gooseneck fitting where it is entirely supported by the mast.

2.2.3 Cross trees, outriggers, brackets on bridge fronts and similar structures are to be of such design that the stresses on them resulting from the cargo gear and any other significant forces do not exceed the values given in 2.1.3 for unstayed masts. The design is also to be such as to minimize the moments acting on the mast. Attachment to the mast is

to be such as to avoid distortion of the mast under the load.

2.2.4 Special attention is to be paid to structural continuity, and abrupt changes of section are to be avoided. Manholes, lightening holes and other cut outs are to be avoided in way of concentrated loads and areas of high shear. Openings, where unavoidable, are to be well rounded, suitably framed and stiffened.

2.2.5 Care should be taken in the design of masts and fittings to reduce the likelihood of water collecting in inaccessible parts of the structure. Adequate means of drainage should be provided to remove any water which might otherwise accumulate.

2.2.6 Masts are to be strengthened either by addition of doubling plates or equivalent strengthening, or by an increase of the thickness of the plating at the following locations:

i) In way of the fastening of the derrick heels,

ii) In way of the fastening of the derrick spans,

iii) In way of the fastening of all metal fittings.

Suitable arrangements are to be made to avoid notch effects.

2.2.7 Welding and weld details are to comply with Ch.17.

2.2.8 Where a mast supports a derrick with a SWL > 25 [t], all welded joints of the masts below a level 3.0 [m] above the uppermost supporting deck or that of the derrick head, whichever is higher, are to be examined by non-destructive crack or flaw detection method.

2.2.9 Where higher tensile steel is used, preheating or other heat treatments may be required at the Surveyors discretion. Such treatments will normally be required for all ring seams on masts supporting derricks of SWL > 60 [t]. Non-destructive examination may be required in areas of high stresses.

2.2.10 Lightning conductors are to be fitted to masts having wood, aluminium or plastic top masts or where a break in electrical conductivity occurs in other arrangements.

Section 3

Rigging

3.1 General

3.1.1 The positioning of the rigging (shrouds, stays etc.) is to be so designed as to ensure proper working of the cargo gear and that they do not foul with the running rigging or derrick booms under the expected service conditions.

3.1.2 Wire rope stays are to be in one length and are to be set up with an initial tension of 30 [N/mm^2].

3.2 Stay construction

3.2.1 The scantlings of the stay are to be such as to provide the necessary tensile force and elongation for the stayed mast. The breaking load of the stay is to be not less than 3.5 times the maximum calculated force on that stay.

3.2.2 The connection of the stay to a deck, bulwark, house or mast is to be such as to allow rotation at the point of attachment and is to be designed so that the stay cannot become disconnected while the derrick system is in use.

End of Chapter

Chapter 16

Welding

Contents

Section

1 *General*

2 *Welding*

3 *Welded Connections*

Section 1

General

1.1 Scope

1.1.1 Welding in steel hull construction of all types of ships is to comply with the requirements of this Chapter.

Welding in aluminium structures will be specially considered.

1.2 Documentation

1.2.1 Connection details of the welded structural members, including type and size of welds are to be clearly indicated on the plans submitted for approval. An explanation of all symbols or abbreviations used in detailing the weld connections should be included on the plans.

Details of proposed welding procedure is to be submitted indicating preheating temperature and any post-welding heat treatment, if employed. Extent to which automatic welding, including deep penetration welding, is to be employed should also be indicated.

Section 2

Welding

2.1 Welders and supervision

2.1.1 Welders are to be proficient in the type of work on which they are to be engaged. The records of their tests and qualifications are to be kept by the builders and made available to the Surveyors. A sufficient number of skilled supervisors are to be employed to ensure effective control at all stages of assembly and welding operations.

2.2 Welding electrodes

2.2.1 Electrodes and welding consumables approved by IRS in accordance with the requirements of Pt.2, Ch.11 and suitable for the type of joint and grade of steel, are to be used.

2.2.2 For the connection of two different grades of steel of the same tensile strength properties, electrodes suitable for the lower grade will be generally acceptable except at structural discontinuities or other points of stress concentration.

2.2.3 For the connection of steel of different tensile strengths, the electrodes are to be suitable for the tensile strength of the component, on the basis of which the weld fillet size has been determined in Sec.3.

2.3 Preparation for welding

2.3.1 The parts to be welded are to be fitted in accordance with the approved joint detail. The edge preparation is to be accurate and uniform. Means are to be provided for maintaining the parts to be welded, in correct position during the welding operations. Excessive force is not to be employed in aligning the parts before welding and the means employed in maintaining the alignment are to be so arranged as to allow for expansion and contraction during the welding operation. All methods employed in correcting improper alignment are to be to the satisfaction of the Surveyor.

2.3.2 All surfaces to be welded are to be clean, dry and free from rust, scale and grease. The surface and boundaries of each run of deposit are to be thoroughly cleaned and freed from slag before the next run is applied. Before a manual sealing run is applied to the back of a weld, the original root material is to be gouged out to sound metal.

2.3.3 Tack welding is to be kept to a minimum, and where used, should be equal in quality to that of the finished welds. Any defective tack weld is to be cut out before completing the finished welds. Care is to be taken in removing the tack welds to ensure that the structure is not damaged in doing so.

2.4 Welding procedure

2.4.1 Only approved welding procedures are to be used, See 2.5.

2.4.2 Structural arrangements are to be such as to allow adequate access for satisfactory completion of all welding operations. Welded joints are to be so arranged so as to facilitate downhand welding wherever possible.

2.4.3 The sequence of welding is to be so planned that any restraint during welding operations is reduced to a minimum. The ends of the frames and stiffeners should be left unattached to the plating at the subassembly stage until connecting welds are made, in the intersecting systems of plating, framing and stiffeners, at the erection stage.

In general, where a butt meets a seam, the welding of the seam should be interrupted well clear of the junction and not be continued until the butt is completed. Welding of the butt should continue past the open seam and the weld be chipped out for the seam to be welded straight through. Alternative weld sequences may be accepted provided weld distortions are in accordance with recognized fabrication standards such as IACS Recommendation 47 "*Shipbuilding and Repair Quality Standard*".

2.4.4 Adequate precautions are to be taken to ensure that the welding site is protected from the deleterious effects of high moisture, severe wind and extreme cold.

2.5 Approval of procedures

2.5.1 Unless previously approved, welding procedures are to be established by the yard and forwarded to IRS for approval. The welding procedure specifications are to include detailed description of the base material, primer, plate thickness range, joint/groove design, welding consumable, welding position, welding techniques, welding parameters, preheating/ interpass temperature and post heat treatment if any.

The welding for procedure qualification and subsequent testing, are to be witnessed by the IRS Surveyor.

2.6 Inspection of welds

2.6.1 Effective arrangements are to be provided for the inspection of finished welds to ensure that all welding has been satisfactorily completed.

2.6.2 All finished welds are to be visually inspected and are to be sound, uniform and substantially free from slag inclusions, porosity, undercutting or other defects. Welds and adjacent base metal are to be free from injurious arc strikes.

2.6.3 For the examination of important structural welds, visual inspection is to be supplemented by radiography or other acceptable non- destructive crack or flaw detection methods. The extent of such examination is to be to the Surveyors' satisfaction, but particular attention is to be given to the following locations:

a) Junction and crossings of seams and butts in strength deck, sheer strake, side and bottom shell within 0.4L amidships.

b) Butts of keel plating and rounded sheerstrake within 0.4L amidships.

c) Insert plates in way of hatch openings on the strength deck.

d) Butts of longitudinal framing and longitudinal bulkhead stiffeners within 0.4L amidships.

2.6.4 Defective sections of welds as found by visual or non- destructive examination or leakages under hydrostatic tests, are to be gouged out as necessary and carefully rewelded.

Section 3

Welded Connections

3.1 Butt welds

3.1.1 Plates of equal thickness may be manually butt welded as per Fig.3.1.1. For automatic welding procedures and special welding techniques, the welding procedure will be specially considered.

3.1.2 For joints of plates with difference in thickness of more than 4 [mm], the thicker plate is to be tapered. The taper is not to exceed 1:3. Edge preparation after the tapering is to be as indicated in 3.1.1 above.

3.1.3 All manual butt welds are normally to be welded from both sides. Where a back ceiling run is not practicable or in certain cases when the stress level in the members is very low, welding on one side may be permitted provided the welding process is found satisfactory.

3.1.4 Where stiffening members, attached by continuous fillet welds, cross the finished butt or seam welds, these welds are to be made flush in way of the faying surface. Similarly for butt welds in webs of stiffening members, the butt weld is to be first completed and made flush with the stiffening member before the stiffener is connected to the plating by fillet weld. The ends of the flush portion are to run out smoothly without notches or any sudden change of section. Where such conditions can not be complied with, a scallop is to be arranged in the web of the stiffening member. Scallops are to be of such size and in such a position, that a satisfactory weld can be made.

3.2 'T' connections

3.2.1 The throat thickness (See Fig.3.2.1) of the fillet welds is given by:

throat thickness = t_p . weld factor . d/s

where,

t_p = thickness [mm], of the thinner of the two parts being connected.

d = distance [mm], between the successive weld fillets.

s = length [mm], of the correctly proportioned weld fillets, clear of end craters is not to be less than 75 [mm].

The weld factors for various connections are generally to be as given in Table 3.2.1.

Where an approved automatic deep penetration procedure is used, the weld factors may be reduced by 15 per cent.

3.2.2 The throat thickness is not to be less than 3.0 [mm] for t_p upto 8.0 [mm] and 0.21 t_p or 3.25 [mm], whichever is greater, for t_p over 8 [mm]. The throat thickness is also generally not to be greater than 0.44 t_p for double continuous welds and the greater of 0.44 t_p or 4.5 [mm] for intermittent welds.

3.2.3 The leg length is not to be less than $\sqrt{2}$ times the specified throat thickness.

3.2.4 Where the connection is highly stressed, deep penetration or full penetration welding may be required. Where full penetration welding is required, the abutting plate may require to be beveled. (See Fig.3.2.4).

3.2.5 Continuous welding is to be adopted in the following locations and may be used elsewhere if desired.

a) Boundaries of weathertight decks and erections, including hatch coamings, companionways and other openings.

b) Boundaries of tanks and watertight compartments.

c) All structures in the afterpeak and the afterpeak bulkhead stiffeners.

d) All welding inside tanks intended for chemicals or edible liquid cargoes.

e) All lap welds in tanks.

f) Floors and girders to bottom shell in the 0.25L forward.

g) Primary and secondary members to plating in way of end connection and end brackets to plating in the case of lap connection.

h) Where the thickness of the abutting member (i.e. stiffener web) is greater than 15 [mm] and also exceeds the thickness of the table member (e.g. plating).

i) Where loading is mainly of dynamic nature (e.g. bottom and side plating forward subjected to slamming and bow impact).

j) Other connections as given in Table - 3.2.1.

3.2.6 Where intermittent welding is used, the welding is to be made continuous around the ends of brackets, lugs, scallops and at other orthogonal connections with other members. Where intermittent welding is permitted as per Table 3.2.1, only scalloped welding is to be used in tanks for water ballast, cargo oil or freshwater.

Chain or staggered intermittent welding may be used in dry spaces or tanks arranged for fuel oil only.

3.2.7 Where structural members pass through the boundary of a tank, and leakage into the adjacent space could be hazardous or undesirable, full penetration welding is to be adopted for the members for at least 150 [mm] on each side of the boundary. Alternatively, a small scallop of suitable shape may be cut in the member close to the boundary outside the compartment, and carefully welded all round.

3.3 Lap connections

3.3.1 Overlaps are not to be used to connect plates which may be subjected to high strength or compressive loading. However, where they are adopted, the width of overlap is to be adequate to ensure a good weld, the surfaces are to be in close contact and the joints should be closed all round by continuous fillet weld.

3.4 Slot weld

3.4.1 For the connection of plating to internal webs, where access for welding is not practicable, the closing plating is to be attached by continuous full penetration or slot welds to flat bars fitted to the webs. Slots are to be well rounded at ends, to have a minimum length of 75 [mm] and in general, a minimum width of twice the plating thickness. The distance between the slots is not to exceed 150 [mm]. Complete filling of the slots is normally not permitted.

3.5 End connection

3.5.1 In way of the end connections of girders double continuous welding is to be used all around. The weld area is not to be less than the cross-sectional area of the member, and the throat thickness not less than that given by Table 3.2.1 for girder ends.

3.5.2 Where stiffeners have bracketed end connections, bracket arms are to be welded all around and the throat thickness is not to be less than 0.35 times the thickness of bracket.

3.5.3 Where stiffeners are continuous at girder, they are to be connected to the webs, either directly and/or by means of lugs. The weld area is to be such that the shear stress does not exceed 80/k [N/mm^2]. Where the shear forces are high, a double sided connection to the web and/or a web stiffener welded on top of the continuous stiffener may be required.

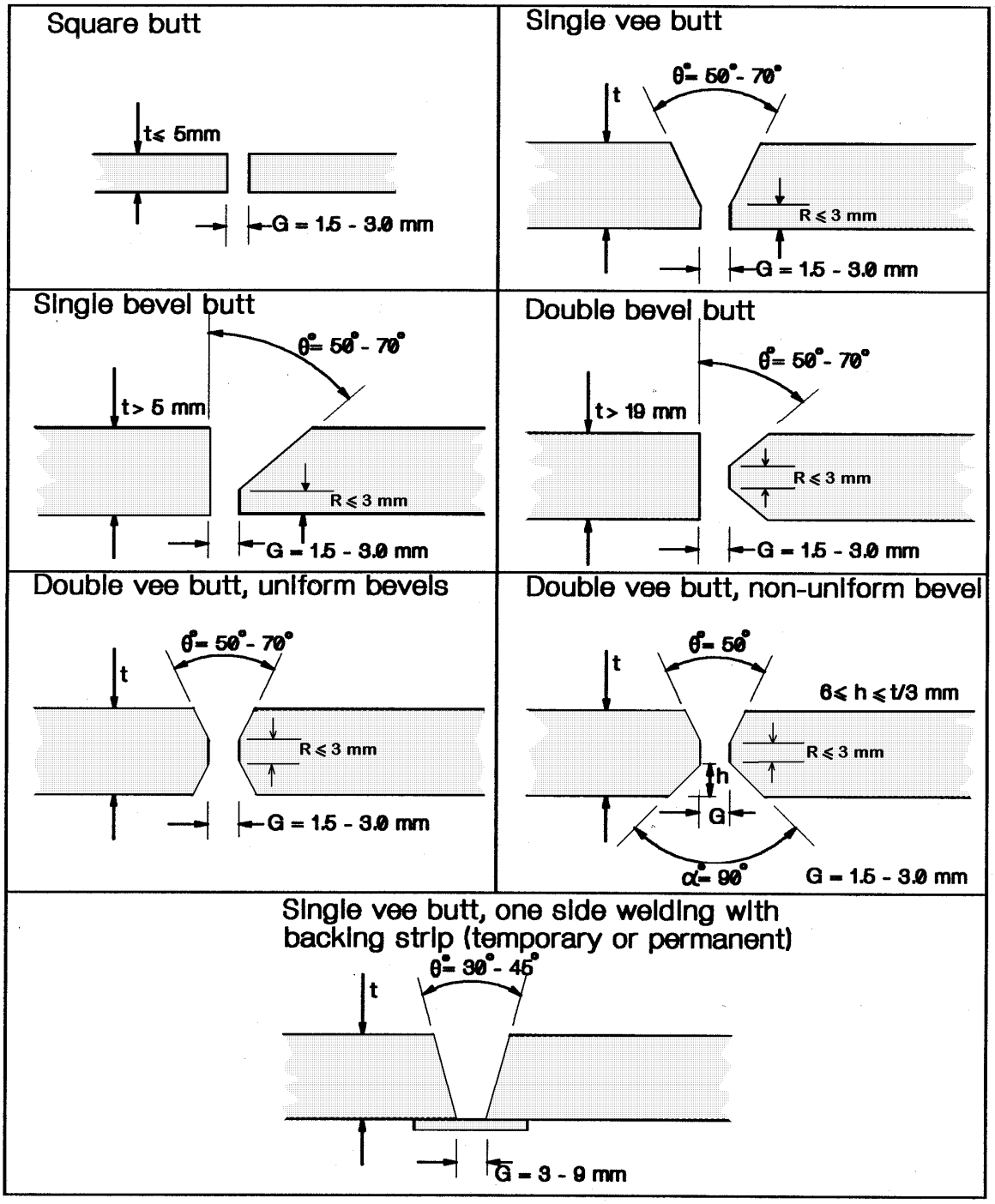


Fig.3.1.1 : Typical manually welded butt joints

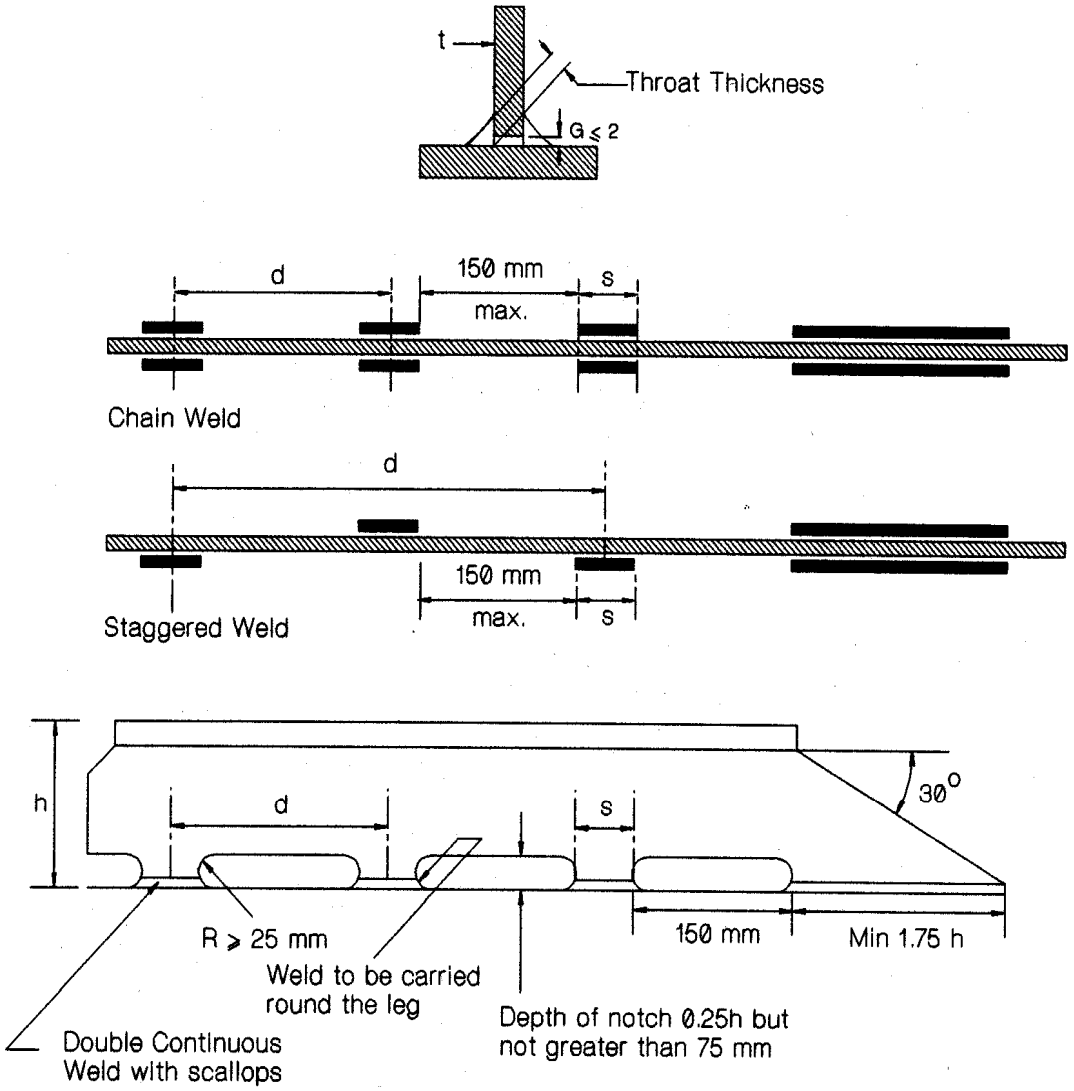


Fig.3.2.1 : Intermittent fillet welds

Table 3.2.1 : Weld factors for fillet welds					
	Structural Items	**Weld Factors**	**Permitted type of weld**		**Remarks**
			Double cont.	**Intermi-ttent**	
Single Bottom					
Centre girder	To keel plate or bar keel	0.3	*		
	To face plate	0.15		*	
Side girder	To bottom shell	0.15		*	
	To face plate	0.13		*	
	To floors	0.20		*	
Floors	To keel plate	0.15	*		
	To shell plating	0.15		*	
	To centre girder	0.35	*		
	To longitudinal bulkheads	0.35	*		
	To face plate	0.15		*	
	Stern-tube covering	0.15	*		
Bottom longitudinal	To shell plating	0.13		*	
Double Bottom (See Note 1)					
Centre girder or duct keel	To keel plate	0.3	*		
	To inner bottom	0.25		*	
Side girder	To bottom shell	0.15		*	
	To inner bottom	0.15		*	
	To floors	0.15		*	
Floors	To shell plating	0.15		*	
	To inner bottom/margin plate	0.15		*	
	To centre girder/keel plate	0.20		*	
Margin plate	To shell plating	0.4	*		
	To inner bottom	0.4	*		
Inner bottom	To side shell	0.4	*		
Tank side brackets	To shell plating	0.3		*	
	To margin plate	0.3		*	
Bracket floor	To inner bottom/bottom shell	0.15		*	
	To centre girder	0.25		*	
	To side shell/margin plate	0.25		*	
Bottom frames	To shell plating	0.13		*	
Reverse frames	To inner bottom	0.13		*	
Longitudinals	To shell plating	0.13		*	
	To inner bottom	0.13		*	
Tank boundaries and bilge wells		0.40	*		
Stiffeners	To floors and girders	0.13		*	

Table 3.2.1 : Weld factors for fillet welds (Contd.)

	Structural Items	Weld Factors	Permitted type of weld		Remarks
			Double cont.	Intermi-ttent	
Structure in Machinery Space					
Floors and girders	To shell and inner bottom	0.3	*		
	To face plate	0.2		*	
Transverse and longitudinal frames	To shell plating	0.15		*	
Floors	To centre girder in way of engine, thrust blocks and boiler seatings				
	- in single bottom	0.50	*		
	- in double bottom	0.30	*		
Main engine foundation girders	To top plate	0.5	*		See Note 2
	To hull structure	0.4	*		
Floors	To engine girder	0.4	*		
Brackets etc.	To engine girders	0.3	*		
Side Structure					
Transverse frames	To side shell				
	- in tanks	0.13		*	
	- elsewhere	0.11		*	For bulk carriers see Pt.5, Ch.1, Sec.2.5
Side longitudinals	To shell plating	0.13		*	
Web frames and side stringers	To shell plating				
	- within 0.2 x span from ends	0.35	*		
	- elsewhere	0.20		*	
	To face plate and tripping bracket	0.15		*	
Web frames	To side stringers	0.3	*		
Deck Structure					
Strength deck	To shell	F.P.			See Note 3
Other decks	To shell and bulkheads	0.3	*		Generally
Deck beams	To deck plating				
	- in tanks	0.13		*	
	- elsewhere	0.11		*	
Deck longitudinals	To decks	0.13		*	
Deck girders	To deck plating				
	- within 0.2 x span from ends	0.35	*		
	- elsewhere	0.20		*	
	To face plating and tripping brackets	0.15		*	
Cantilever webs	To shell, decks, face plates and longitudinal girders at ends	0.35	*		
Pillars	To deck, inner bottom and pillar brackets	0.40	*		

Table 3.2.1 : Weld factors for fillet welds (Contd.)

	Structural Items	Weld Factors	Permitted type of weld		Remarks
			Double cont.	Intermi-ttent	
Construction in 0.25L from F.P.					
Floors and girders	To shell	0.25	*		
	To inner bottom	0.25		*	
Bottom longitudinals	To shell	0.15		*	
Shell	To transverse and longitudinal	0.15		*	
Panting stringers	To shell and frames	0.30	*		
All internal structure	In fore peak (unless a higher factor is specified)	0.13		*	
Aft Peak Construction					
All internal structure	On bottom, side shell and aft peak bulkhead	0.3	*		See 3.2.5
Bulkheads and Partitions					
Boundaries of	Watertight, oiltight and wash bulkheads and shaft tunnels	0.4	*		To be specially considered for chemical cargo tanks
Stiffeners	On tank and wash bulkheads	0.13		*	
	On pillar bulkheads	0.13		*	
	On ordinary bulkheads	0.11		*	
Vertical and horizontal girders in tanks and wash bulkheads	To bulkhead plating				
	- within 0.2 x span from ends	0.40	*		
	- elsewhere	0.30		*	
	- to faceplace	0.30		*	
	- to tripping brackets	0.15		*	
Vertical and horizontal girders elsewhere	To bulkhead plating				
	- within 0.2 x span from ends	0.35	*		
	- elsewhere	0.20		*	
	To faceplate and tripping brackets	0.15		*	
Primary Structures in Cargo Tanks					
Webs	To shell, deck and bulkheads				
	- within 0.2 x span from ends	0.4	*		
	- elsewhere	0.3	*	*	See 3.2.6
Webs	To face plates	0.3	*	*	"
Webs	To webs of other primary members	0.3	*		
Boundaries of	Tripping brackets	0.15	*	*	"
Superstructures and Deck Houses					
External bulkheads	To deck				
	- on 1st and 2nd tiers	0.40	*		
	- elsewhere	0.25	*		
Internal bulkheads	Boundaries	0.13		*	
stiffeners	To external bulkheads	0.10		*	See Note 4

Table 3.2.1 : Weld factors for fillet welds (Contd.)

	Structural Items	Weld Factors	Permitted type of weld		Remarks
			Double cont.	Intermi-ttent	
Hatchways and Closing Appliances					
Hatch coaming	To deck at corners	0.5	*		
	To deck elsewhere	0.4	*		
	To face plate	0.4	*		
	To hatch cover rest bar	0.16	*		
Hatch cover	To stiffeners	0.12		*	
Rudders and Nozzles					
Rudders					See Note 5
Main piece members	To coupling flange	F.P.	*		
	To each other	0.44	*		
Rudder plating	To rudder webs, elsewhere	0.20	*		
Nozzles	Generally as for rudders				
Miscellaneous Fittings and Equipment					
Framing ring for manhole type covers	To deck and bulkhead	0.4	*		
Framing around ports and W.T./oiltight doors	To plating	0.4	*		
Sea-chest boundary welds	Exposed to sea	0.5	*		
	Elsewhere	0.4	*		
Ventilators, air pipes etc.	To deck	0.4	*		
Bulwark stays	To deck	0.4	*		
	To bulwark plating	0.2		*	
Bilge keel	To ground bars	0.2	*	*	
Bilge keel ground bar	To side shell	0.35	*		Continuous fillet weld of minimum 4 [mm] throat thickness
Fabricated anchors		F.P.			
Masts, derrick posts, crane pedestals, deck machinery and mooring equipment seating - to deck etc.		To be considered in each individual case			

Intermittent welding means chain intermittent, staggered intermittent or scalloped welding with rounded ends.
For permitted use see 3.2.6.

F.P. means Full Penetrant weld

Note 1 For tank boundaries see 3.2.5. Where intermittent fillet welding of internals is proposed, see 3.2.6.

Note 2 Preferably to be deep penetration or full penetration weld depending on the thickness of the engine girders.

Note 3 Generally full penetration, but alternative proposals may be considered depending on t_p.

Note 4 For end connections see Chapter 11, 3.1.4.

Note 5 See Chapter 14, Section 4.1.

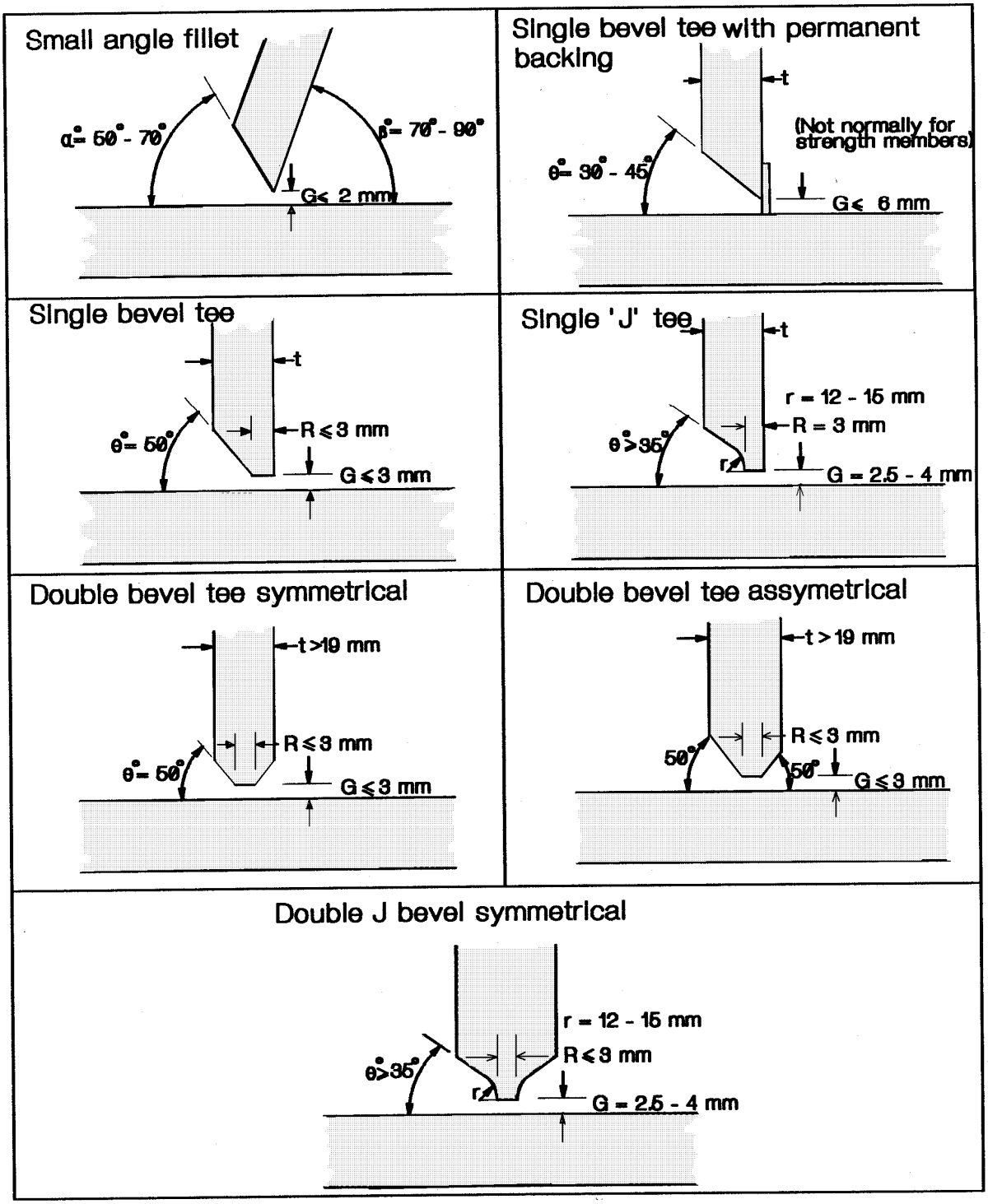


Fig.3.2.4 : Typical edge preparations for manually welded 'T' or cross joints

End of Chapter

Chapter 17

Hull Inspection, Workmanship and Testing

Contents

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1	*Hull Inspection*
2	*Workmanship*
3	*Testing of Tanks and Tight Boundaries for SOLAS Ships*
4	*Testing of Tanks and Tight Boundaries for SOLAS Exempt / Equivalent Ships*
5	*Testing of Tanks and Tight Boundaries for Non-SOLAS Ships*
6	*Construction Monitoring*

Section 1

Hull Inspection

1.1 Approval of works

1.1.1 The builders, intending to class vessels to be built at their yard with IRS, are to demonstrate their capability to carry out the fabrication to acceptable quality standards before the commencement of the fabrication. Similar approval procedure shall apply to subcontractor's works also. Previous experience in the building and repair of relevant structures and equipment can be considered favourably in this regard.

1.2 Inspection facilities

1.2.1 Adequate facilities are to be provided to enable the Surveyor to carry out a satisfactory inspection of all components during each stage of prefabrication and construction.

Section 2

Workmanship

2.1 General

2.1.1 All workmanship is to be of good quality and in accordance with good shipbuilding practice. Any defect is to be rectified to the satisfaction of the Surveyor before being covered with paint, cement or other composition.

2.1.2 The assembly sequence and welding sequence are to be agreed prior to construction and are to be to the satisfaction of the Surveyor.

2.2 Plate edges and cut-outs

2.2.1 Openings, holes and other cut-outs in the main structural components are to be rounded off by adequately large radii. The free edges of cut-outs, hatch corners etc. are to be properly prepared and are to be free from notches. All edges should be faired.

2.3 Cold forming

2.3.1 Flanging and bending of plates while cold forming are not to have an average bending radius less than three times the plating thickness. The minimum radius is not to be less than twice the plating thickness.

2.3.2 During joggling of plates and profiles, the depth of joggle is not to be less than four times and the bending radius not less than twice the web thickness.

2.4 Hammering, bending and straightening

2.4.1 Steel being worked on when hot, is not to be overheated, and it is to be hammered and bent in the appropriate heat condition. Steel which is burnt, is not to be used.

2.4.2 Flame heating may be employed to straighten buckled plating when the buckling is not severe.

Section 3

Testing of Tanks and Tight Boundaries for SOLAS Ships

3.1 General

3.1.1 Testing procedures of watertight compartments for SOLAS Ships are to be carried out in accordance with this section, unless the shipyard provides documentary evidence of the shipowner's agreement to a request to the Flag Administration:

a) for an exemption from the application of SOLAS Chapter II-1, Regulation 11, or

b) for agreeing to an equivalency, that the requirements of Section 4 is equivalent to SOLAS Chapter II-1, Regulation 11;

c) and, the above-mentioned exemption/equivalency has been granted by the Flag Administration

3.1.2 These test procedures are to confirm the watertightness of tanks, watertight boundaries and the structural adequacy of tanks which form part of the watertight subdivisions of ships. These procedures may also be applied to verify the weathertightness of structures and shipboard outfitting. The tightness of all tanks and watertight boundaries of ships during new construction and those relevant to major conversions or major repairs (repairs affecting structural integrity) is to be confirmed by these test procedures prior to the delivery of the ship.

Note: "Watertight Subdivision" means the transverse and longitudinal subdivisions of the ship required to satisfy the subdivision requirements of SOLAS Chapter II-1.

3.2 Application

3.2.1 All gravity tanks and other boundaries required to be watertight or weathertight are to be tested in accordance with this section and proven to be tight and structurally adequate as follows:

- Gravity tanks for their tightness and structural adequacy.
- Watertight boundaries other than tank boundaries for their watertightness, and
- Weathertight boundaries for their weathertightness.

Note : Gravity tank means a tank that is subject to vapour pressure not greater than 0.07 [MPa].

"Watertight" means having scantlings and arrangements capable of preventing the passage of water in any direction under the head of water likely to occur in intact and damaged conditions. In the damaged condition, the head of water to be considered in the worst situation at equilibrium, including intermediate stage of flooding.

"Weathertight" means that in any sea conditions, water will not penetrate into the ship. Weathertight Closing appliances for hatches, doorways and other access openings, ventilators, air-pipes etc are to be such as to prevent the passage of water into the ship in all sea conditions.

3.2.2 The testing of the cargo containment systems of liquefied gas carriers is to be in accordance with testing requirements in Pt.5, Ch.4, Sec.4.21 to 4.26 and Table 3.4.2.

3.2.3 The testing of structures not listed in Table 3.4.1 or 3.4.2 is to be specially considered.

3.3 Test Types and Definitions

3.3.1 The following two types of tests are specified in this requirement:

.1) Structural Test : A test to verify the structural adequacy of tank construction. This may be a hydrostatic test or, where the situation warrants, a hydropneumatic test.

.2) Leak Test : A test to verify the tightness of a boundary. Unless a specific test is indicated, this may be a hydrostatic / hydropneumatic test or an air test. A hose test may be considered an acceptable form of leak test for certain boundaries, as indicated by note 3 of Table 3.4.1.

3.3.2 The definition of each test type is as follows:

Hydrostatic Test: (Leak and Structural)	A test wherein a space is filled with a liquid to a specified head
Hydropneumatic Test: (Leak and Structural)	A test combining a hydrostatic test and an air test, wherein a space is partially filled with a liquid and pressurized with air.
Hose Test (Leak)	A test to verify the tightness of a joint by a jet of water with the joint visible from the opposite side.
Air Tests: (Leak)	A test to verify tightness by means of air pressure differential and leak indicating solution. It includes tank air test and joint air tests, such as compressed air fillet weld tests and vacuum box tests.
Compressed Air Fillet Weld Test: (Leak)	An air test of fillet welded tee joints wherein leak indicating solution is applied on fillet welds
Vacuum Box Test: (Leak)	A box over a joint with leak indicating solution applied on the welds. A vacuum is created inside the box to detect any leaks
Ultrasonic Test: (Leak)	A test to verify the tightness of the sealing of closing devices such as hatch covers by means of ultrasonic design techniques.
Penetration Test: (Leak)	A test to verify that no visual dye penetrant indications of potential continuous leakages exist in the boundaries of a compartment by means of low surface tension liquids (i.e. dye penetrant test).

3.3.3 The ‘top of the overflow’ is defined as being the top of any overflow system which is used to prevent overfilling of a tank. Such a system can be an overflow pipe, airpipe or, an intermediate tank. For gravity tanks (i.e. sewage, grey water and similar tanks, not filled by pumps) the top of the overflow is to be taken as the highest point of the filling line.

Note: Gauging devices are not considered equivalent to an overflow system with the exception of fuel oil overflow tanks not intended to hold fuel which have been fitted with a level alarm.

Where a tank is fitted with multiple means of preventing overfilling, the decision on which overflow system is to be used to determine the test head is to be based on the highest point to which the liquid may rise in service.

3.4 Test procedures

3.4.1 General

3.4.1.1 Tests are to be carried out in the presence of a Surveyor at a stage sufficiently close to the completion of work with all hatches, doors, windows, etc., installed and all penetrations including pipe connections fitted and before any ceiling and cement work is applied over the joints. Specific test requirements are given in 3.4.4 and Table 3.4.1. For the timing of the application of coating and the provision of safe access to joints, see 3.4.5, 3.4.6 and Table 3.4.3.

3.4.2 Structural test procedures

3.4.2.1 Type and time of test

Where a structural test is specified in Table 3.4.1 or Table 3.4.2, a hydrostatic test in accordance with 3.4.4.1 will be acceptable. Where practical limitations, (strength of building berth, light density of liquid, etc.) prevent the performance of a hydrostatic test, a hydropneumatic test in accordance with 3.4.4.2 may be accepted instead

A hydrostatic test or hydropneumatic test for the confirmation of structural adequacy may be carried out while the vessel is afloat provided the results of a leak test are confirmed to be satisfactory before the vessel is afloat.

Alternative equivalent tank testing procedures may be considered for tanks which are constructed from composite materials such as glass reinforced plastic (GRP) and fibre reinforced plastic (FRP) based on the recommendations of the composite manufacturer.

3.4.2.2 Testing Schedule for New Construction or Major Structural Conversion

3.4.2.2.1 Tanks which are intended to hold liquids, and which form part of the watertight subdivision of the ship are to be tested for tightness and structural strength as indicated in Table 3.4.1 and 3.4.2.

3.4.2.2.2 The tank boundaries are to be tested from at least one side. The tanks for structural test are to be selected so that all representative structural members are tested for the expected tension and compression.

3.4.2.2.3 The watertight boundaries of spaces other than tanks for structural testing may be exempted, provided that the water-tightness of boundaries of exempted spaces is verified by leak tests and inspections. Structural testing may not be exempted and the requirements for structural testing of tanks in 3.4.2.2.1 to 3.4.2.2.2 are to be applied, for ballast holds, chain lockers and a representative cargo hold if intended for in-port ballasting.

3.4.2.2.4 Tanks which do not form part of the watertight subdivision of the ship may be exempted from structural testing provided that the water-tightness of boundaries of exempted spaces is verified by leak tests and inspections.

3.4.3 Leak test procedures

3.4.3.1 For the leak tests specified in Table 3.4.1, tank air tests, compressed air fillet weld tests, vacuum box tests in accordance with 3.4.4.4 through 3.4.4.6, or their combination, will be acceptable. Hydrostatic or hydropneumatic tests may also be accepted as leak tests provided that 3.4.5, 3.4.6 and 3.4.7 are complied with. Hose tests will also be acceptable for such locations as specified in Table 3.4.1 with Note 3, in accordance with 3.4.4.3. The application of the leak test for each type of welded joint is specified in Table 3.4.3.

3.4.3.2 Air tests of joints may be carried out in the block stage provided that all work on the block that may affect the tightness of a joint is completed before the test. See also 3.4.5.1 for the application of final coatings and 3.4.6 for the safe access to joints and the summary in Table 3.4.3.

3.4.4 Test Methods

3.4.4.1 Hydrostatic Test

3.4.4.1.1 Unless another liquid is approved, hydrostatic tests are to consist of filling the space with fresh water or sea water, whichever is appropriate for testing, to the level specified in Table 3.4.1 or Table 3.4.2. (See also 3.4.7).

3.4.4.1.2 In cases where a tank is designed for cargo densities greater than sea water and testing is with fresh water or sea water, the testing pressure height is to simulate the actual loading for those greater cargo densities as far as practicable, but the test pressure is not to exceed the maximum design internal pressure at the top of tank.

3.4.4.1.3 All external surfaces of the tested space are to be examined for structural distortion, bulging and buckling, other related damage and leaks.

3.4.4.2 Hydropneumatic test

3.4.4.2.1 Hydropneumatic tests, where approved, are to be such that the test condition, in conjunction with the approved liquid level and supplemental air pressure, will simulate the actual loading. The requirements and recommendations for tank air tests in 3.4.4.4 will also apply to hydropneumatic tests. (See also 3.4.7).

3.4.4.2.2 All external surfaces of the tested space are to be examined for structural distortion, bulging and buckling, other related damage and leaks.

3.4.4.3 Hose test

3.4.4.3.1 Hose tests are to be carried out with the pressure in the hose nozzle maintained at least at 0.2 [MPa] i.e. (2 bar) during the test. The nozzle is to have a minimum inside diameter of 12 [mm] and be at a perpendicular

distance from the joint not exceeding 1.5 [m]. The water jet is to impinge directly upon the weld.

3.4.4.3.2 Where a hose test is not practical because of possible damage to machinery, electrical equipment insulation or outfitting items, it may be replaced by a careful visual examination of welded connections, supported where necessary by means such as a dye penetrant test or ultrasonic leak test or the equivalent.

3.4.4.4 Tank air test

3.4.4.4.1 All boundary welds, erection joints and penetrations, including pipe connections, are to be examined in accordance with approved procedure and under a stabilized pressure differential above atmospheric pressure not less than 0.015 [MPa], with a leak indicating solution such as soapy water/detergent or a proprietary brand applied.

3.4.4.4.2 A U-tube with a height sufficient to hold a head of water corresponding to the required test pressure is to be arranged. The cross sectional area of the U-tube is not to be less than that of the pipe supplying air to the tank. Arrangements involving the use of two calibrated pressure gauges to verify the required test pressure may be accepted taking into account the following:

a) Where there is no protection by suitable barriers / screens,

    i. verification and confirmation that the test procedure is followed,
    ii. Verification and confirmation that test equipment is in good condition,
    iii. Verification and confirmation that the isolating valves are tight,
    iv. Verification and confirmation that the test pressure gauges (at least 2 pcs) are calibrated and are of suitable range as per test pressure,
    v. Confirmation that the pressure gauges are not isolated or bypassed, and that releasing mechanisms or safety valves are of adequate size/rating and correctly set.
    vi. Confirmation that the Pipe work, especially flexible piping, is free from damage or leaking joints.
    vii. Confirmation that the Inlet, outlet and release arrangement for the test medium to or from the item being tested are properly fabricated and supported, including their appropriate dimensions (e.g. Smaller diameter for inlets and bigger diameter for outlets, as well as releasing means).

b) If a releasing mechanism is used in lieu of a U-tube,

    i. It is to be evaluated, whether the design of the mechanism is similar to the U-tube principle (i.e. suitable to be lifted by the predetermined excessive pressure, and not subject to any kind of spring or restricting device).
    ii. The safety precautions adopted for the test are to be carefully reviewed and the risks after taking into account necessary precautions are to be evaluated

3.4.4.4.3 A double inspection is to be made of tested welds. The first is to be immediately upon applying the leak indication solution; the second is to be after approximately four or five minutes in order to detect those smaller leaks which may take time to appear.

3.4.4.5 Compressed air fillet weld test

3.4.4.5.1 In this air test, compressed air is injected from one end of a fillet welded joint and the pressure verified at the other end of the joint by a pressure gauge. Pressure gauges are to be arranged so that an air pressure of at least 0.015 [MPa] can be verified at each end of all passages within the portion being tested.

Note : Where a leak test is required for fabrication involving partial penetration welds, a compressed air test is also to be applied in the same manner as to fillet weld where the root face is large, i.e., 6-8 [mm].

3.4.4.6 Vacuum box test

3.4.4.6.1 A box (vacuum testing box) with air connections, gauges and an inspection window is placed over the joint with a leak indicating solution applied to the weld cap vicinity. The air within the box is removed by an ejector to create a vacuum of 0.02 - 0.026 [MPa] inside the box.

3.4.4.7 Ultrasonic test

3.4.4.7.1 An ultrasonic echo transmitter is to be arranged inside of a compartment and a receiver is to be arranged on the outside. The watertight / weathertight boundaries of the compartment are scanned with the receiver in order to detect an ultrasonic leak indication. A location where sound is detectable by the receiver indicates a leakage in the sealing of the compartment.

3.4.4.8 Dye penetration test

3.4.4.8.1 A test of butt welds or other weld joints uses the application of a low surface tension liquid at one side of a compartment boundary or structural arrangement. If no liquid is detected on the opposite sides of the boundaries after the expiration of a defined period of time, this indicates tightness of the boundaries. In certain cases, a developer solution may be painted or sprayed on the other side of the weld to aid leak detection.

3.4.4.9 Other tests

3.4.4.9.1 Other methods of testing may be considered by IRS upon submission of full particulars prior to the commencement of testing.

3.4.5 Application of coating

3.4.5.1 Final coating

3.4.5.1.1 For butt joints welded by an automatic process, the final coating may be applied any time before the completion of a leak test of spaces bounded by the joints, provided that the welds have been carefully inspected visually to the satisfaction of the Surveyor.

3.4.5.1.2 However, the Surveyor reserves the right to require a leak test prior to the application of the final coating over automatic erection butt welds.

3.4.5.1.3 For all other joints, the final coating is to be applied after the completion of the leak test of the joint. See also Table 3.4.3.

3.4.5.2 Temporary coating

3.4.5.2.1 Any temporary coating which may conceal defects or leaks is to be applied at the time as specified for the final coating (See 3.4.5.1). This requirement does not apply to shop primer.

3.4.6 Safe access to joints

3.4.6.1 For leak tests, a safe access to all joints under examination is to be provided. See also Table 3.4.3.

3.4.7 Hydrostatic or hydropneumatic tightness test

3.4.7.1 In cases where the hydrostatic or hydropneumatic tests are applied instead of a specific leak test, examined boundaries must be dew-free, otherwise small leaks are not visible.

Table 3.4.1 : Test requirements for Tanks and Boundaries				
Sr. No.	**Tank or boundary to be tested**	**Test type**	**Test head or pressure**	**Remarks**
1	Double bottom tanks[4]	Leak and structural[1]	The greater of - top of the overflow[12] - to 2.4 [m] above top of tank[2], or - to bulkhead deck	
2	Double bottom voids[5]	Leak	See 3.4.4.4 through 3.4.4.6, as applicable	Including pump room double bottom and bunker tank protection double hull required by MARPOL Annex I
3	Double side tanks	Leak and structural[1]	The greater of - top of the overflow[12], - to 2.4 [m] above top of tank[2,8], or - to bulkhead deck	
4	Double side voids	Leak	See 3.4.4.4 through 3.4.4.6, as applicable	
5	Deep tanks other than those listed elsewhere in this table	Leak and structural[1]	The greater of - top of the overflow[12], - to 2.4 [m] above top of tank[2,8]	
6	Cargo oil tanks	Leak and structural[1]	The greater of - top of the overflow[12], - to 2.4 [m] above top of tank[2,8], or - top of tank[2] plus the design vapour pressure	
7	Ballast hold of bulk carriers	Leak and structural[1]	- Top of cargo hatch coaming	
8	Peak tanks	Leak and structual[1]	The greater of - top of the overflow[12], or - to 2.4 [m] above top of tank[2,8]	After peak to be tested after installation of stern tube
9	a. Fore peak spaces with equipment	Leak	See 3.4.4.3 through 3.4.4.6, as applicable	
	b. Fore peak voids	Leak	See 3.4.4.4 through 3.4.4.6, as applicable	
	c. Aft peak spaces with equipment	Leak	See 3.4.4.3 through 3.4.4.6, as applicable	
	d. Aft peak voids	Leak	See 3.4.4.4 through 3.4.4.6, as applicable	After peak to be tested after installation of stern tube
10	Cofferdams	Leak	See 3.4.4.4 through 3.4.4.6, as applicable	
11	a. Watertight bulkheads	Leak[10]	See 3.4.4.3 through 3.4.4.6, as applicable[7]	
	b. Superstructure end bulkheads	Leak	See 3.4.4.3 through 3.4.4.6, as applicable	
12	Watertight doors below freeboard or bulkhead deck	Leak [6, 7]	See 3.4.4.3 through 3.4.4.6, as applicable	
13	Double plate rudder blades	Leak	See 3.4.4.4 through 3.4.4.6, as applicable	
14	Shaft tunnels clear of deep tanks	Leak[3]	See 3.4.4.3 through 3.4.4.6, as applicable	
15	Shell doors	Leak[3, 9]	See 3.4.4.3 through 3.4.4.6, as applicable	
16	Weathertight hatch covers and closing appliances	Leak[3,7]	See 3.4.4.3 through 3.4.4.6, as applicable	Hatch covers closed by tarpaulins and battens excluded

Table 3.4.1 : (Contd.)

Sr. No.	Tank or boundary to be tested	Test type	Test head or pressure	Remarks
17	Dual purpose tanks / dry cargo hatch covers	Leak[3,7]	See 3.4.4.3 through 3.4.4.6, as applicable	In addition to structural test in item 6 or 7
18	Chain lockers	Leak and structural[1]	Top of chain pipe	
19	L.O. sump tanks and other similar tanks/spaces under main engines	Leak	- See 3.4.4.3 through 3.4.4.6 as applicable	
20	Ballast ducts	Leak and structural[1]	The greater of - Ballast pump maximum pressure, or - Setting of any pressure relief valve	
21	Fuel Oil Tanks	Leak and structural[1]	The greater of - Top of the overflow[12], - To 2.4 [m] above top of tank[2,8] or - To top of tank[2] plus the design vapour pressure, or - To bulkhead deck	
22	Fuel oil overflow tanks not intended to hold fuel	Leak and structural[1]	The greater of - Top of the overflow[12], - To 2.4 [m] above top of tank[2] or - To bulkhead deck	

Notes :

1. Refer to section 3.4.2.2
2. The top of a tank is the deck forming the top of the tank excluding any hatchways.
3. Hose Test may also be considered as a method of the test, see 3.3.2.
4. Including tanks arranged in accordance with the provisions of SOLAS regulation II-1/9.4.
5. Including duct keels and dry compartments arranged in accordance with the provisions of SOLAS regulation II-1/11.2 and II-1/9.4 respectively, and/or oil fuel tank protection and pump room bottom protection arranged in accordance with the provisions of MARPOL Annex I, Chapter 3, Part A regulation 12A and Chapter 4, Part A, regulation 22 respectively.
6. Where water tightness of a watertight door has not been confirmed by prototype test, testing by filling watertight spaces with water is to be carried out. See SOLAS regulation II-1/16.2 and MSC.1/Circ.1572/Rev.1.
7. An alternative to the hose testing, other testing methods listed in 3.4.4.7 through 3.4.4.9 may be applicable subject to adequacy of such testing methods being verified. See SOLAS regulation II-1/11.1. For watertight bulkheads (item 11 a) alternatives to the hose testing may only be used where a hose test is not practicable.
8. For vessels of L < 90 [m], the head of water above highest point of tank may be (0.02L + 0.6) [m], but not less than 1.0 [m]. However, in mechanically propelled cargo ships of 500 GT and above and passenger ships, for tanks forming part of the watertight subdivision (See Pt.3, Ch.10, 4.2.1), head of water above the highest point of tank is to be 2.4 [m].
9. Watertight shell doors are to be prototype tested prior to installation on board.
10. A “Leak and Structural test” (see 3.4.2.2) is to be carried out for a representative cargo hold if intended for in-port ballasting. The filling level requirement for testing cargo holds intended for in-port ballasting is to be the maximum loading that will occur in-port as indicated in the loading manual.
11. Where L.O sump tanks and other similar spaces under main engines intended to hold liquid form part of the watertight subdivision of the ship, they are to be tested as per the requirements of item 5, Deep tanks other than those listed elsewhere in this table.
12. Refer to section 3.3.3

Table 3.4.2 : Additional Test requirements for Listed Types of Ships / Tanks

Sr. No.	Type of Ship / Tank	Structures to be tested	Type of Test	Test Head of Pressure	Remarks
1	Liquefied gas carriers	Integral tanks	Leak and structural	Refer Pt 5, Ch 4, Sec 4	
		Hull structure supporting membrane or semi-membrane tanks	Refer to Pt.5, Ch.4, Sec 4	Refer Pt 5, Ch 4, Sec 4	
		Independent tanks type A	Refer to Pt.5, Ch.4, Sec 4	Refer Pt 5, Ch 4, Sec 4	
		Independent tanks type B	Refer to Pt.5, Ch.4, Sec 4	Refer Pt 5, Ch 4, Sec 4	
		Independent tanks type C	Refer to Pt.5, Ch.4, Sec4	Refer Pt 5, Ch 4, Sec 4	
2	Edible liquid tanks	Independent tanks	Leak and structural[1]	The greater of - top of the overflow[3], or - to 0.9 [m] above top of tank[2]	
3	Chemical carriers	Integral or independent cargo tanks	Leak and structural[1]	The greater of[4] - to 2.4 [m] above top of tank[2], or - to top of tank[2] plus the design vapour pressure	

Note :

1 Refer to section 3.4.2.2

2. Top of tank is deck forming the top of the tank excluding any hatchways

3. Refer to section 3.3.3

4. Where a cargo tank is designed for the carriage of cargoes with specific gravities greater than seawater, refer to section 3.4.4.1.2.

Table 3.4.3 : Application of leak test, coating and provision of safe access for type of welded joints

Type of welded joints		Leak Test	Coating[1]		Safe Access[2]	
			Before leak test	After leak test but before structural test	Leak test	Structural test
Butt	Automatic	Not required	Allowed[3]	N/A	Not required	Not required
	Manual or semi-automatic[4]	Required	Not allowed	Allowed	Required	Not required
Fillet	Boundary including penetrations	Required	Not allowed	Allowed	Required	Not required

Notes :

1. Coating refers to internal (tank / hold coating), where applied and external (shell/deck) painting. It does not refer to shop primer.
2. Temporary means of access for verification of the leak test.
3. The condition applies provided that the welds have been carefully inspected visually to the satisfaction of the Surveyor
4. Flux Core Arc Welding (FCAW) semiautomatic butt welds need not be tested provided that careful visual inspection show continuous uniform weld profile shape, free from repairs, and the results of NDE testing show no significant defects.

Section 4

Testing of Tanks and Tight Boundaries for SOLAS Exempt/ Equivalent Ships

4.1 General

4.1.1 These test procedures are to confirm the watertightness of tanks, watertight boundaries and the structural adequacy of tanks which form part of the watertight subdivisions of ships. These procedures may also be applied to verify the weathertightness of structures and shipboard outfitting. The tightness of all tanks and watertight boundaries of ships during new construction and those relevant to major conversions or major repairs is to be confirmed by these test procedures prior to the delivery of the ship.

Note: *Major repair means a repair affecting structural integrity*

4.1.2 Testing procedures of watertight compartments are to be carried out in accordance with this section for those SOLAS ships for which the shipyard provides documentary evidence of the shipowner's agreement to a request to the Flag Administration:

a) for an exemption from the application of SOLAS Chapter II-1, Regulation 11, or

b) for agreeing to an equivalency, that the requirements of this section is equivalent to SOLAS Chapter II-1, Regulation 11;

c) and, the above-mentioned exemption/equivalency has been granted by the Flag Administration.

4.2 Application

4.2.1 Testing procedures are to be carried out in accordance with the requirements of Section 3 in association with the following alternative procedures for 3.4.2.2 of Section 3 “Testing Schedule for New Construction or Major Structural Conversion”.

4.2.2 The tank boundaries are to be tested from at least one side. The tanks for structural test are to be selected so that all representative structural members are tested for the expected tension and compression.

4.2.3 Structural tests are to be carried out for at least one tank of a group of tanks having structural similarity (i.e. same design conditions, alike structural configurations with only minor localised differences determined to be acceptable by the attending Surveyor) on each vessel provided all other tanks are tested for leaks by an air test. The acceptance of leak testing using an air test instead of a structural test does not apply to cargo space boundaries adjacent to other compartments in tankers and combination carriers or to the boundaries of tanks for segregated cargoes or pollutant cargoes in other types of ships.

4.2.4 Additional tanks may require structural testing if found necessary after the structural testing of the first tank.

4.2.5 For tanks which are less than 2 [m^3] in volume, structural testing may be replaced by leak testing.

4.2.6 Where the structural adequacy of the tanks and spaces of a vessel was verified by the structural testing required by Section 3 or Section 4.2.3, subsequent vessels in the series (i.e. sister ships built from the same plans at the same shipyard) may be exempted from structural testing of tanks, provided that:

1.water-tightness of boundaries of all tanks and spaces are verified by leak tests and thorough inspections are carried out.

2.structural testing is carried out for at least one tank or space of each type among all tanks/ spaces of each sister vessel.

3.additional tanks and spaces may require structural testing if found necessary after the structural testing of the first tank or if deemed necessary by the attending Surveyor.

For cargo space boundaries adjacent to other compartments in tankers and combination carriers or boundaries of tanks for segregated cargoes or pollutant cargoes in other types of ships, structural tests are to be carried out for at least one tank of a group of tanks having structural similarity (i.e. same design conditions, alike structural configurations with only minor localised differences determined to be acceptable by the attending Surveyor) on each vessel provided all other tanks are tested for leaks by an air test.

4.2.7 Sister ships built (i.e. keel laid) two years or more after the delivery of the last ship of the series, may be tested in accordance with 4.2.6 at the discretion of IRS, provided that:

1.general workmanship has been maintained (i.e. there has been no discontinuity of shipbuilding or significant changes in the construction methodology or technology at the yard, shipyard personnel are appropriately qualified and demonstrate an adequate level of workmanship as determined by IRS); and

2. an NDT plan is implemented and evaluated by IRS for the tanks not subject to structural tests. Shipbuilding quality standards for the hull structure during new construction are to be reviewed and agreed during the kick-off meeting. Structural fabrication is to be carried out in accordance with a recognised fabrication standard (such as, IACS Recommendation 47, “Shipbuilding and Repair Quality Standard”) which has been accepted by IRS prior to the commencement of fabrication/construction. The work is to be carried out in accordance with the Rules and under survey of IRS.

Section 5

Testing of Tanks and Tight Boundaries for Non-SOLAS Ships

5.1 General

5.1.1 These test procedures are to confirm the watertightness of tanks, watertight boundaries and the structural adequacy of tanks which form part of the watertight subdivisions of ships. These procedures may also be applied to verify the weathertightness of structures and shipboard outfitting. The tightness of all tanks and watertight boundaries of ships during new construction and those relevant to major conversions or major repairs is to be confirmed by these test procedures prior to the delivery of the ship.

Note:

Watertight subdivision means the main transverse and longitudinal subdivisions of the ship.

Major repair means a repair affecting structural integrity

5.1.2 For the purpose of application of this section, Non-SOLAS Ships are those ships not engaged on international voyages or cargo ships which are less than 500 GT.

5.1.3 Testing procedures of watertight compartments are to be carried out in accordance with this Section, see SOLAS Chapter I, Regulation 1 and Regulation 3.

5.2 Application

5.2.1 Testing procedures are to be carried out in accordance with the requirements of Section 3 in association with the following alternative procedures for 3.4.2.2 of Section 3 “Testing Schedule for New Construction or Major Structural Conversion”.

5.2.2 The tank boundaries are to be tested from at least one side. The tanks for structural test are to be selected so that all representative structural members are tested for the expected tension and compression.

5.2.3 The requirements given in Table 3.4.1 of Section 3 to structurally test tanks to 2.4 metres above the top of the tank do not apply. Instead, the minimum test pressure for structural testing is to be taken as 0.3D + 0.76 metres above the top of the tank where the top of the tank is the deck forming the top of the tank, excluding any hatchways and D is the depth of the ship. The minimum test pressure need not be taken greater than 2.4 metres above the top of the tank.

5.2.4 Structural tests are to be carried out for at least one tank of a group of tanks having structural similarity (i.e. same design conditions, alike structural configurations with only minor localised differences determined to be acceptable by the attending Surveyor) on each vessel provided all other tanks are tested for leaks by an air test. The acceptance of leak testing using an air test instead of a structural test does not apply to cargo space boundaries adjacent to other compartments in tankers and combination carriers or the boundaries of tanks for segregated cargoes or pollutant cargoes in other types of ships.

5.2.5 Additional tanks may require structural testing if found necessary after the structural testing of the first tank.

5.2.6 For tanks which are less than 2 [m^3] in volume, structural testing may be replaced by leak testing.

5.2.7 Where the structural adequacy of the tanks and spaces of a vessel were verified by the structural testing required by either Section 3 or Section 5.2.4, subsequent vessels in the series (i.e. sister ships built from the same plans at the same shipyard) may be exempted from structural testing of tanks, provided that:

1. water-tightness of boundaries of all tanks and spaces are verified by leak tests and thorough inspections are carried out.

2. structural testing is carried out for at least one tank or space among all tanks/spaces of each sister vessel.

3. additional tanks and spaces may require structural testing if found necessary after the structural testing of the first tank or if deemed necessary by the attending Surveyor.

For cargo space boundaries adjacent to other compartments in tankers and combination carriers or boundaries of tanks for segregated cargoes or pollutant cargoes in other types of ships, structural tests are to be carried out for at least one tank of a group of tanks having structural similarity (i.e. same design conditions, alike structural configurations with only minor localised differences determined to be acceptable by the attending Surveyor) on each vessel provided all other tanks are tested for leaks by an air test.

5.2.8 Sister ships built (i.e. keel laid) two years or more after the delivery of the last ship of the series, may be tested in accordance with 5.2.7 at the discretion of IRS, provided that:

1. general workmanship has been maintained (i.e. there has been no discontinuity of shipbuilding or significant changes in the construction methodology or technology at the yard, shipyard personnel are appropriately qualified and demonstrate an adequate level of workmanship as determined by IRS); and

2. an NDT plan is implemented and evaluated by IRS for the tanks not subject to structural tests. Shipbuilding quality standards for the hull structure during new construction are to be reviewed and agreed during the kick-off meeting. Structural fabrication is to be carried out in accordance with a recognised fabrication standard (such as, IACS Recommendation 47, “Shipbuilding and Repair Quality Standard”) which has been accepted by IRS prior to the commencement of fabrication/construction. The work is to be carried out in accordance with the Rules and under survey of IRS.

Section 6

Construction Monitoring

6.1 Scope

6.1.1 This section provides requirements for new construction ships to be assigned **CM** notation, i.e. Construction Monitoring.

6.1.2 The aim of the **CM** notation is to ensure an acceptable quality of construction at critical locations of the ship thereby ensuring better structural integrity of the hull.

6.1.3 Critical locations are defined as those locations which are identified to be important for structural integrity of the hull (identified from finite element analysis, fatigue assessment), those locations where the fabrication is difficult in practice due to complexity of the details or the finesse of the tolerances to be achieved. Those locations which have been previously reported on ships classed with IRS or the global fleet as being particularly susceptible to failure or cracking are also to be included within the list of critical locations.

6.1.4 A construction monitoring plan is to be submitted to IRS for approval. The construction monitoring plan is to identify and consider all the critical locations, as applicable.

6.1.5 The builder’s shipbuilding quality standard and the details of the builder and subcontractors’ (if any) facilities are also to be submitted for IRS review. These standards are not to be less stringent than IACS Recommendation 47. If deemed necessary, IRS may visit the builder’s facilities including those of the sub-contractors (if any) to assess capability of the builder to fabricate and construct the critical locations.

6.1.6 The construction monitoring plan is to provide details of the following, for each critical location:

- Construction tolerances to be achieved. These may be in terms of dimensional tolerances, maximum values for gaps, misalignments, etc.
- Fabrication procedures/ construction procedures including sequences of welding
- Non-destructive testing plan and acceptance criteria in accordance with IRS Classification Note: *Requirements for Non-Destructive Testing*
- Defect rectification plan.

6.1.7 All critical locations are to be inspected by the Surveyor to ascertain and record the actual construction tolerances achieved.

6.1.8 Records of non-destructive testing for the critical locations are to be made available to the Surveyor.

End of Chapter

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RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 4
MAIN AND AUXILIARY MACHINERY

July 2024

Indian Register of Shipping

Part 4

Main and Auxiliary Machinery

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Chapter 1

General Requirements for the Design and Construction of Machinery

Contents

Section

1	*General*
2	*Machinery Room Arrangements*
3	*Trials*
4	*Certification of Machinery and Components Based on Alternative Certification Scheme*

Section 1

General

1.1 Scope

1.1.1 The Chapters of this Part cover the construction and installation of main propulsion and auxiliary machinery systems, together with their associated equipment, boilers, pressure vessels and pumping and piping arrangements fitted in vessels classed or intended to be classed with IRS.

1.2 Machinery to be constructed under survey

1.2.1 In ships intended to be built under Special Survey, all important units of equipment are to be surveyed at the manufacturer's works. The workmanship is to be to the Surveyor's satisfaction and the Surveyor is to be satisfied that the components are suitable for the intended purpose and duty. Examples of such units are:

- Main propulsion engines, including their associated gearing, flexible couplings, scavenge blowers and superchargers;

- Boilers supplying steam for propulsion or for services essential for the safety or the operation of the ship at sea, including superheaters, economisers, desuper-heaters, steam receivers. All other boilers having working pressures exceeding 0.34 [N/mm^2], and having heating surfaces greater than 4.65 [m^2];

- Auxiliary engines which are the source of power for services essential for safety or for the operation of the ship at sea;

- Steering machinery;

- Athwartship thrust units, their prime movers and control mechanisms;

- All pumps necessary for the operation of vessel, main propulsion and essential machinery, e.g. boiler feed, cooling water circulating, condensate extraction, oil fuel, lubricating oil, etc.;

- All heat exchangers necessary for the operation of main propulsion and essential machinery, e.g. air, water and lubricating oil coolers, oil fuel and water heaters, de-aerators and condensers, evaporators and distiller units;

- Air compressors, air receivers and other pressure vessels necessary for the operation of main propulsion and essential machinery. Any other unfired pressure vessels for which plans are required to be submitted for approval in accordance with the requirements of Ch.5;

- Alarm and control equipment as detailed in Ch.7;

- Electrical equipment and electrical propelling machinery as detailed in Ch.8.

- All pumps essential for safety of the ship, e.g. fire, bilge and ballast pumps; and

- Valves and other components detailed in Ch.2.

1.3 Extent of survey

1.3.1 The Surveyors are to examine and test the materials and workmanship from the commencement of work until the final test of the machinery under full power working conditions. Any defects, etc., are to be indicated as early as possible.

1.3.2 Where items of machinery are manufactured as individual or series produced units, IRS would be prepared to give consideration to the adoption of a survey procedure based upon quality assurance concepts as detailed in Sec.4.

1.4 Departures from the rules

1.4.1 Where it is proposed to depart from the requirements of the Rules, IRS will be prepared to give due consideration to the circumstances of any special case.

1.4.2 Any novelty in the construction of the machinery, boilers or pressure vessels is to be reported to IRS.

1.5 Plans and particulars

1.5.1 Before the work is commenced, plans in triplicate of all machinery items, as detailed in relevant Chapters of this Part giving the requirements for individual systems, are to be submitted for approval. The particulars of the machinery, including power ratings and design calculations, where applicable, necessary to verify the design, are also to be submitted. Any subsequent modifications are subject to approval before being put in to operation.

1.5.2 The strength requirements for rotating parts of the machinery, as specified in relevant Chapters of this Part, are based upon strength consideration only and their application does not relieve the manufacturer from the responsibility for the presence of dangerous vibrations in the installation at speeds within the operating range.

1.6 Availability of machinery for operation

1.6.1 Ship's machinery is to be so arranged that it can be brought in to operation from the "dead ship" condition using only the facilities available on board. "Dead ship" condition is understood to mean a condition under which the main propulsion plant, boilers and auxiliaries are not in operation and in restoring the propulsion, no stored energy for starting and operating the propulsion plant, the main source of electrical power and other essential auxiliaries is assumed to be available. In order to restore operation from the "dead ship" condition, an emergency generator complying with the requirements of Pt.4, Ch.8, Cl.2.8 may be used provided that the emergency power supply from it is available at all times. It is assumed that means are available to start the emergency generator at all times.

1.6.2 "Blackout" is a sudden loss of electric power in the main distribution system. All means of starting by stored energy are available.

1.7 Environmental conditions

1.7.1 All components and systems covered by the Rules are to be designed to operate under the following environmental conditions if not otherwise specified in the detailed requirements for the component or system:

- ambient air temperature in the machinery space between 0°C and 55°C;

- relative humidity of air in the machinery space up to 96 per cent;

- sea water temperature up to 32°C; and

- list, rolling, trim and pitch in accordance with Table 1.7.1.

- ship accelerations and motions in accordance with 1.7.5.

1.7.2 Where the Rules have requirements for capacity or effect of machinery, these are to be based on a total barometric pressure of 1 bar, an engine room ambient temperature or suction air temperature of 45°C, a relative humidity of 60 per cent and a sea water temperature or where applicable the temperature of the charge air coolant at inlet of

32°C. The engine manufacturer is not expected to provide simulated ambient reference conditions at test bed.

1.7.3 In the case of ships to be classed for restricted service, the rating is to be suitable for the temperature conditions associated with the geographical limits of the restricted service.

1.7.4 Where electrical equipment is installed within environmentally controlled spaces the ambient temperature for which the equipment is suitable may be reduced from 45°C to a value not less than 35°C provided:

- The equipment is not for use for emergency services.

Table 1.7.1 : Machinery inclinations

Installations/Components	Angle of inclination [1]			
	Athwartships		Fore and aft	
	Static	Dynamic	Static	Dynamic
Main and auxiliary machinery	15°	22.5°	5°[4]	7.5°
Safety equipment e.g. emergency power installations, emergency fire pumps and their devices Switch gear, electric and electronic appliances[2] and remote control systems	22.5°[3]	22.5°[3]	10°	10°

Notes

1 Athwartships and fore and aft inclinations occur simultaneously.

2 Switches and controls are to remain in their last set position. No undesired switching operations or operational changes are to occur.

3 In ships for the carriage of liquefied gases and of chemicals the emergency power supply must also remain operable with the ship flooded to a final athwartships inclination upto maximum of 30°.

4 For ships where Rule length (L) exceeds 100 [m], the fore and aft static angle of inclination can be taken as 500/L degrees.

5 Any deviations from these angles of inclination taking into consideration the size and service conditions of the ship will be specially considered.

- Temperature control is achieved by at least two cooling units so arranged that in the event of loss of one cooling unit, for any reason, the remaining unit(s) is capable of satisfactorily maintaining the design temperature.

- The equipment is able to be initially set to work safely within a 45°C ambient temperature until such a time that the lesser ambient temperature may be achieved. The cooling equipment is to be rated for a 45°C ambient temperature.

- Audible and visual alarms are provided, at a continually manned control station, to indicate any malfunction of the cooling units.

- It is to be ensured that electrical cables for their entire length are adequately rated for the maximum ambient temperature to which they are exposed along their length.

- The equipment used for cooling and maintaining the lesser ambient temperature is to be classified as a secondary essential service, as defined in Ch.8, sec.1.

1.7.5 Shipboard accelerations

1.7.5.1 General

.1 Main propulsion and steering machinery and auxiliary machinery that is essential to the propulsion and steering, and the safety of the

ship is to be capable of operation under the effects of accelerations and motions.

.2 The requirements in 1.7.5.2 to 1.7.5.4 apply where documented evidence of equipment suitability is specifically required by the Rules.

1.7.5.2 Documentation

.1 The ship builders are to identify and document the ship accelerations and motions periods to which machinery and equipment might be subjected to. The expected accelerations and ship motions periods are to be within machinery and equipment manufacturers requirements. The estimations are to consider vessel type, machinery or equipment location and expected service conditions.

1.7.5.3 Evaluation of equipment suitability

.1 Machinery and equipment manufacturers are to submit evidence of their machinery or equipment, that can operate under the required static and dynamic conditions stated in Table 1.7.1 and at least at the levels of shipboard accelerations as stated in 1.7.5.2 and/or specified in the Rules. Documentation of satisfactory performance is to take the form of:

- Report of testing under representative conditions; or
- Report of theoretical verification using recognized computational techniques accompanied by detailed and relevant validation data: or
- Historical data which provides relevant demonstration of satisfactory experience in service.

1.7.5.4 Installation and operation

.1 Machinery and equipment manufacturers are to submit details of the requirements /recommendations for installation of the machinery and equipment onboard to ensure satisfactory operation in service under the required static and dynamic conditions as described in Table 1.7.1 and at least at the levels of shipboard accelerations as stated in 1.7.5.2 and/or specified in the Rules.

Note: Consideration should be given for positioning machinery in order to minimize the dynamic load on bearings due to ship motion.

.2 Shipbuilders are to submit details demonstrating that the installation of the machinery and equipment onboard is in accordance with manufacturer's requirements /recommendations.

1.8 Power ratings

1.8.1 In the following Chapters, where the dimensions of any particular component are determined from shaft power, P in kW, and revolutions per minute, R, the values to be used are to be derived from the following:

- For main propulsion machinery, the maximum shaft power and corresponding revolutions per minute giving the maximum torque for which the machinery is to be classed; and
- For auxiliary machinery, the maximum continuous shaft power and corresponding revolutions per minute which will be used in service.

1.9 Units

1.9.1 Units and formulae included in the Rules are shown in SI units.

1.9.2 Where the metric version of shaft power, i.e. (shp), appears in the Rules, 1 shp is equivalent to 75 [kgf] [m/sec] or 0.735 [kW].

1.9.3 Pressure gauges may be calibrated in bar

where,

1 bar = 0.1 [N/mm^2] = 1.02 [kgf/cm^2].

1.10 Astern power

1.10.1 In order to maintain sufficient maneuverability and secure control of the ship in all normal conditions, the main propulsion machinery is to be capable of reversing the direction of thrust so as to bring the ship to rest from the maximum service speed. The main propulsion machinery is to be capable of maintaining in free route astern at least 70% of the ahead revolutions.

1.10.2 Where steam turbines are used for main propulsion, they are to be capable of maintaining in free route astern at least 70% of the ahead revolutions for a period of at least 15 minutes. The astern trial is to be limited to 30 minutes or in accordance with

manufacturer's recommendation to avoid overheating of the turbine due to the effects of "windage" and friction.

1.10.3 For main propulsion systems with reversing gears, controllable pitch propeller or electric propeller drive, running astern is not to lead to overload of the propulsion machinery.

1.10.4 Main propulsion systems are to undergo tests to demonstrate the astern response characteristics. The tests are to be carried out at least over the maneuvering range of the propulsion system and from all control positions. A test plan is to be provided by the yard and accepted by the surveyor. If specific operational characteristics have been defined by the manufacturer, these are to be included in the test plan.

1.10.5 The ahead revolutions as mentioned above are understood as those corresponding to the maximum continuous ahead power for which the vessel is classed.

1.10.6 The reversing characteristics of the propulsion plant, including the blade pitch control system of controllable pitch propellers, are to be demonstrated, tested and recorded as specified in Sec.3, 3.2.8 during trials.

1.11 Fuel

1.11.1 The flash point (closed cup test) of oil fuel for use in ships classed for unrestricted service, in general, is not to be less than 60°C.

1.11.2 For emergency generator engines, fuel having a flash point (closed cup test) of not less than 43°C may be used.

1.11.3 Fuels with flash points lower than 60°C, but not less than 43°C unless specially approved, may be used in ships intended for service restricted to geographical limits where it can be ensured that the temperature of the machinery and boiler spaces will always be 10°C below the flash point of fuel. In such cases, safety precautions and the arrangements for storage and pumping will be specially considered. (Also refer Pt.6, Ch.2, 1.2.1).

1.11.4 The use of fuels having a lower flash point than specified in 1.11.1 to 1.11.3 as applicable may be permitted provided that such fuel is not stored in any machinery space and arrangements for the complete installation are specially approved.

1.11.5 For engines operating on "boil-off" vapours from the cargo, the arrangements will be specially considered.

1.12 Materials

1.12.1 The materials used in the construction are to be manufactured and tested in accordance with the requirements of Pt.2 of the Rules. Materials for which provision is not made therein may be accepted, provided that they comply with an approved specification and such tests as may be considered necessary.

1.12.2 Installation of material which contain asbestos is prohibited.

1.13 Alternative design and arrangements

1.13.1 The Rules generally cover the requirements of SOLAS Ch.II-1, Parts C, D and E. Design and arrangements alternative to these requirements may be accepted based on SOLAS Reg. II-1/55 and MSC.1/Circ. 1212, Rev.1.

Section 2

Machinery Room Arrangements

2.1 General

2.1.1 The machinery is to be so designed, installed and protected that risks of fire, explosions, accidental pollution, leakages and accidents thereof, and accidents to personnel working in machinery spaces will be minimised.

2.1.2 The design and arrangement of machinery foundations, shaft connections, piping and ducting is to take into account the effects of thermal expansion, vibrations, mis-alignment and hull interaction to ensure operation within safe limits.

Bolts and nuts exposed to dynamic forces and vibrations are to be properly secured.

2.2 Accessibility

2.2.1 Accessibility, for attendance and maintenance purposes, is to be provided for machinery plants.

2.3 Fire protection

2.3.1 All surfaces of machinery where the surface temperature may exceed 220°C, e.g., steam and exhaust gas lines, silencers, boilers, exhaust gas boilers, turbo blowers, etc. are to be effectively shielded to prevent ignition of combustible materials coming in to contact with them. Where insulation covering these surfaces is oil absorbing or may permit penetration of oil, the insulation is to be encased in steel or its equivalent.

2.3.2 The insulation is to be of a type and so supported that it will not crack or deteriorate when subjected to vibration.

2.3.3 Oil spillages coming into contact with hot surfaces, electrical installations or other sources of ignition.

a) Precautions (e.g. shielding) are to be taken to prevent oil that may escape under pressure from any pump, filter or heater or piping from coming into contact with sources of ignition.

b) Oil tanks, pipes, filters, heaters, etc. are not to be located immediately above or near units of high temperature including boilers, steam pipe lines, exhaust manifolds, silencers or other equipment required to be insulated and electrical equipment, as far as practicable. In particular, oil fuel filters under pressure for diesel engines are to be located such that in the event of leakage, oil cannot be sprayed on to the exhaust manifold.

c) Oil filters fitted in parallel for the purpose of enabling cleaning without disturbing oil supply to engines (e.g. duplex filters) are to be provided with arrangements that will minimise the possibility of a filter under pressure being opened by mistake. Filter/filter chambers are to be provided with suitable means for :

   - venting when put in to operation; and
   - depressurizing before being opened.

   Valves or cocks with drain pipes led to a safe location are to be used for this purpose.

d) Hydraulic units with working pressure above 15 bar are preferably to be placed in separate spaces. If it is impracticable to locate such units in separate space, adequate shielding is to be provided.

2.3.4 Flammable or oil absorbing materials are not to be used in floors, gratings, etc. in boiler and engine rooms, shaft tunnels or in compartments where settling tanks are installed.

2.3.5 Segregation of oil fuel purifiers

a) Oil fuel purifiers for heated oil are to be in separate room enclosed by steel bulkheads extending from deck to deck and provided with self-closing steel doors.

b) The room is to be provided with :

   - independent mechanical ventilation or a ventilation arrangement which can

be isolated from the machinery space ventilation;

- fire detecting system; and

- fixed fire extinguishing installation.

The extinguishing installation is to be capable of being activated from outside the room. Closing of ventilation openings is to be effected from a position close to where the extinguishing system is activated. The extinguishing system is to be separate for the room, but may be part of the main fire extinguishing system for the machinery spaces.

c) If it is impracticable to locate the oil fuel purifiers in a separate space, IRS will give special consideration with regard to location, containment of possible leakages, shielding and ventilation. A local fixed fire extinguishing system is to be provided which is to be capable of being activated manually from the machinery control position or from other suitable location. If automatic release is provided, additional manual release is to be arranged.

2.4 Ventilation

2.4.1 All spaces, including engine and cargo pump spaces, where flammable or toxic gases or vapours may accumulate, are to be provided with adequate ventilation under all conditions.

2.4.2 Machinery spaces of Category A are be adequately ventilated so that a sufficient supply of air is maintained to the spaces for the safety and comfort of personnel and for operation of the machinery, when machinery or boilers are operating at full power in all weather conditions including heavy weather. Any other machinery space is to be adequately ventilated appropriate for the purpose of that machinery space.

IR2.4.2 The openings on deck through which the air is supplied to machinery space should be suitably protected (taking into account the requirements of Pt.3, Ch.13, Sec.2.2 and Sec.2.3).

The machinery spaces of Category A and other machinery spaces are as defined in Pt.6, Ch.1, Clauses 3.31 and 3.30 respectively.

2.5 Communications

2.5.1 At least two independent means of communication are to be provided between the bridge and engine room control station from which the engines are normally controlled.

One of these means is to visually indicate the order and response, both at the engine room control station and on the bridge.

2.5.2 At least one means of communication is to be provided between the bridge and any other control position(s) from which the propulsion machinery may be controlled.

2.5.3 Propulsion machinery orders from the navigation bridge are to be indicated in the main machinery control room and at the manoeuvring platform.

2.6 Machinery controls

2.6.1 Main and auxiliary machinery essential for the propulsion, control and safety of ship are to be provided with effective means for its operation and control. All control systems essential for the propulsion, control and safety of the ship are to be independent or designed such that failure of one system will not affect the performance of the other system.

2.6.2 Where remote control of propulsion machinery from the navigating bridge is provided, the control is to be performed by a single control device for each independent propeller, with automatic control of all associated services, including, where necessary, means of preventing overload of the propulsion machinery.

Section 3

Trials

3.1 General

3.1.1 Tests of components and trials of machinery, as detailed in the Chapters giving the requirements for individual systems are to be carried out to the satisfaction of the Surveyors.

3.2 Sea trials

3.2.1 For all types of installations, the sea trials are to be of sufficient duration, and carried out under normal maneuvering conditions, to prove the machinery under rated power. The trials are also to demonstrate that any vibration which may occur within the operating speed range is acceptable.

3.2.2 The trials are to include demonstrations of the following :

- The adequacy of the starting arrangements to provide the required number of starts of the main engines;

- The ability of the machinery to reverse the direction of thrust of the propeller/propulsion device in sufficient time, under normal maneuvering conditions, and so bring the ship to rest from maximum ahead rated speed; and

- In turbine installations, the ability to permit astern running at 70 per cent of the full power ahead revolutions without adverse effects. The astern trial need not be of more than 30 minutes duration.

3.2.3 Where controllable pitch propellers are fitted, the free route astern trial is to be carried out with the propeller blades set in full pitch astern position. Where emergency manual pitch setting facilities are provided, their operation is to be demonstrated to the satisfaction of the Surveyors.

3.2.4 In geared installations, prior to full power sea trials, the gear teeth are to be suitably coated to demonstrate the contact markings, and on conclusion of sea trials all gears are to be opened up sufficiently to permit the Surveyors to make an inspection of the teeth. The marking is to indicate freedom from hard bearing, particularly towards the end of the teeth, including both ends of each helix where applicable.

3.2.5 The stopping times, ship headings and distances recorded on trials, together with the results of trials to determine the ability of ships having multiple propellers or propulsion/steering arrangements to navigate and maneuver with one or more such devices inoperative, are to be available on board for the use of the master or designated personnel.

3.2.6 Where the ship is provided with supplementary means for maneuvering or stopping, the effectiveness of such means is to be demonstrated and recorded as referred to in 3.2.5.

3.2.7 All trials are to be to Surveyor's satisfaction.

3.2.8 Testing of the control system of controllable pitch propellers

3.2.8.1 General

.1 The purpose of the tests is to ascertain that the pitch control system of controllable pitch propellers for main propulsion is working correctly.

3.2.8.2 Application

.1 The requirements in this sub-section apply to all new buildings and to all replacements, modifications, repairs, or re-adjustments that may affect the pitch control or response characteristics for main propulsion.

3.2.8.3 Scope of the tests

3.2.8.3.1 Pitch response test

.1 A full range of tests is to be carried out to get the pitch response and verify that it coincides with the combinator curve of the propeller (refer Note1). The tests are to be carried out for at least three positions of the control lever in ahead and astern directions (e.g., dead slow ahead / astern, half ahead / astern, full ahead / astern).

.2 The tests are to be carried out in normal and emergency operating conditions.

.3 Tests that are not affected by the control position may be carried out from one control position only.

Note1: The combinator curve is the relationship between the propeller pitch setting and the propeller speed.

3.2.8.3.2 Test of the fail-to-safe characteristics

.1 A test of the fail-to-safe characteristics of the propeller pitch control system is to be carried out to demonstrate that failures in the pitch command and control or feedback signals are alarmed and do not cause any change of thrust. Such failures are to be clearly identified and included in the test procedure.

3.2.8.3.3 Test procedure

.1 Test procedure is to be prepared and proposed by the pitch control system manufacturer or integrator and agreed with IRS.

3.2.8.4 Parameters to be recorded

.1 The list of the parameters to be recorded during the pitch response test is to be established by the pitch control system manufacturer or integrator and agreed with IRS. This is to at least include the following parameters:

- Position of the control handle,
- Actual pitch indication (local indication, remote indications),
- Rotational speed of the propeller,
- Response time between the pitch change order (modification of the lever position) and the instant when the pitch and propeller speed have reached their final position,
- Propelling thrust variation during the transfer of the control from one location to another one.

3.2.8.5 Tests results

.1 Tests are to demonstrate:

- that the propelling thrust is not significantly altered when transferring control from one location to another and in case of failures in the pitch command and control or feedback signals.
- that the pitch response times measured during the test do not exceed the maximum value to be defined by the pitch control system manufacturer or integrator.

Section 4

Certification of Machinery and Components Based on Alternative Certification Scheme

4.1 General

4.1.1 Machinery and components may be certified by Alternative Certification Scheme (ACS) for products including machinery and components as described in Part 1, Chapter 1, Section 4 of the Rules.

4.2 Certification of diesel engines

4.2.1 Certification of diesel engines based on Alternative Certification Scheme described in Part1, Chapter 1, Section 4 of the Rules can be considered only for diesel engines which have been type approved in accordance with IRS type approval scheme.

4.2.2 Certification of individual components of diesel engines is to be undertaken in accordance with the *Classification Note: "Approval of I.C Engines".*

End of Chapter

Chapter 2

Piping Design Requirements

Contents

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Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter apply to the design and construction of piping systems, including pipe fittings and valves forming parts of such systems for the following services:

Air, vapour, gas (excluding liquefied gas cargo and process piping), water, lubricating oil, fuel oil, hydraulic fluid systems for steering gear, toxic gas and liquids, cargo oil and tank cleaning piping and open ended lines such as drains, overflows, vents and boiler escape pipes.

1.1.2 The requirements of this Chapter do not cover the following:

Exhaust pipes from internal combustion engines and gas turbines, and pipes forming integral part of a boiler.

1.1.3 Hydraulic fluid systems other than those for steering gear will be specially considered.

1.1.4 For requirements pertaining to piping systems for liquefied gases (cargo and process) and cargo piping systems of ships carrying chemicals in bulk reference should be made to applicable Chapters of Pt.5 of the Rules.

1.2 Classes of pipes

1.2.1 For the purpose of testing, type of joints to be adopted, heat treatment and welding procedure, piping systems are divided into three classes, as given in Table 1.2.1.

1.3 Design pressure

1.3.1 The design pressure, P, is the maximum permissible working pressure and is to be not less than the highest set pressure of the safety valve or relief valve. For oil fuel pipes the design pressure is to be taken in accordance with Table 1.3.1. For gaseous fuel, the design pressure will be specially considered.

1.3.2 In water tube boiler installations, the design pressure for steam piping between the boiler and integral superheater outlet is to be taken as the design pressure of the boiler, i.e. not less than the highest set pressure of any safety valve on the boiler drum. For piping leading from the superheater outlet, the design pressure is to be taken as the highest set pressure of the superheater safety valves.

1.3.3 The design pressure of feed piping and other piping on the discharge from pumps is to be taken as the pump pressure at full rated speed against a shut valve. Where a safety valve or other protective device is fitted to restrict the pressure to a lower value than the shut valve load, the design pressure is to be the highest set pressure of the protective device.

Table 1.2.1 : Classes of piping systems [3,4]

Piping system	Class I	Class II	Class III
Toxic or corrosive media	Without special safeguards	Not to be used	Not to be used
a) Flammable media heated above flash point[2] b) Flammable media having flash point below 60°C[2] c) Liquefied gas	Without special safeguards	With special safeguards[1]	Not to be used
Steam	P > 16 or T > 300	16 ≥ P > 7 and 300 ≥ T > 170	P ≤ 7 and T ≤ 170
Thermal Oil	P > 16 or T > 300	16 ≥ P > 7 and 300 ≥ T > 150	P ≤ 7 and T ≤ 150
Fuel oil + Lubricating oil + Flammable hydraulic oil	P > 16 or T > 150	16 ≥ P > 7 and 150 ≥ T > 60	P ≤ 7 and T ≤ 60
Other media[5] including water, air, gases, non-flammable hydraulic oil	P > 40 or T > 300	40 ≥ P > 16 and 300 ≥ T > 200	P ≤ 16 and T ≤ 200

Notes:

1 Safeguards for reducing leakage possibility and limiting its consequences will be specially considered e.g. leading pipes in positions where leakage of internal fluids will not cause a potential hazard or damage to surrounding areas or by the usage of pipe ducts, shielding, screening etc.
2 Cargo oil pipes belong to Class III piping systems.
3 P = Design pressure in bar as defined in 1.3
4 T = Design temperature in °C as defined in 1.4.
5 For open ended pipes (drains, overflow, vents, exhaust gas lines, boiler escape lines, etc.) irrespective of the temperature, Class III pipes may be used.

Table 1.3.1 : Definition of the design pressure for fuel oil systems

Working temperature / **Working Pressure (WP)**	**T ≤ 60°C**	**T > 60°C**
WP ≤ **7 bar**	3 bar or max. working pressure, whichever is greater	3 bar or max. working pressure, whichever is greater
WP > 7 bar	Max. working pressure	14 bar or max. working pressure, whichever is greater

1.4 Design temperature

1.4.1 The design temperature is to be taken as the maximum temperature of the internal fluid, but in no case is it to be less than 50°C. For special cases the design temperature will be specially considered.

1.4.2 In the case of pipes for superheated steam, the temperature is to be taken as the designed operating steam temperature for the pipeline, provided that the temperature at the superheater outlet is closely controlled. Where temperature fluctuations exceeding 15°C above the designed temperature are to be expected in normal service, the steam temperature to be used for determining the allowable stress is to be increased by the amount of this excess.

1.5 Design symbols

1.5.1 The symbols used in this Chapter are defined as follows:

a = percentage negative manufacturing tolerance on thickness;

b = bending allowance [mm];

c = corrosion allowance [mm];

D = outside diameter of pipe [mm] (see 1.5.2);

d = inside diameter of pipe [mm] (see 1.5.3);

e = weld efficiency factor (see 1.5.4);

P = design pressure, in bar;

P_t = hydraulic test pressure, in bar;

R = radius of curvature of a pipe bend at the centreline of the pipe [mm];

T = design temperature, in °C;

t = the minimum thickness of a straight pipe [mm], including corrosion allowance and negative tolerance, where applicable;

t_b = the minimum thickness of a straight pipe to be used for a pipe bend [mm]; including bending allowance, corrosion allowance and negative tolerance, where applicable;

K = maximum permissible design stress [N/mm^2].

1.5.2 The outside diameter, D, is subject to manufacturing tolerance, but these are not to be used in the evaluation of formulae.

1.5.3 The inside diameter, d, is not to be confused with nominal size, which is an accepted designation associated with outside diameters of standard rolling sizes.

1.5.4 The weld efficiency factor, e, is to be taken as 1.0 for seamless and electric resistance and induction welded steel pipes delivered by manufacturers approved for making welded pipes which are considered to be equivalent to seamless pipes. Where other methods of pipe manufacture are proposed, the value of e will be specially considered.

1.6 Heat treatment

1.6.1 Method of heat treatment and means of temperature control and recording are to be to the satisfaction of Surveyors.

1.7 Minimum thickness of pipes and bends

1.7.1 The minimum thickness, t, of straight pipes is to be determined by the following formula when ratio of outside-diameter to inside-diameter does not exceed the value 1.7:-

$$t = \left(\frac{PD}{20Ke + P} + c \right) \frac{100}{100 - a} \text{ [mm]}$$

where,

P, D, e and a are defined in 1.5.1.

The maximum permissible design stress, K, is to be taken as the lowest of the following values:-

$$K = \frac{E_t}{1.6} \text{ or } \frac{R_{20}}{2.7} \text{ or } \frac{S_R}{1.6} \text{ or } \sigma_{1/100,000} / 1$$

where,

$\sigma_{1/100,000}/1$ = average stress [N/mm^2] to produce 1% creep in 100,000 hrs at the design temperature.

E_t = specified minimum lower yield or 0.2 per cent proof stress at the design temperature

R_{20} = specified minimum tensile strength at ambient temperature

S_R = average stress to produce rupture in 100,000 hours at the design temperature.

Value of K may be obtained from Sec.2, Table 2.1.1 and Table 2.1.2 for carbon and low alloy steel pipes and fittings or from Sec.3, Table 3.2.1 for copper and copper alloys pipes and fittings.
c is obtained from Sec.2, Table 2.2.1 for carbon and low alloy steel pipes and fittings or from Sec.3, Table 3.2.3 for copper and copper alloys pipes and fittings.

1.7.2 Where it is proposed to use, for high temperature service, alloy steels other than those detailed in Sec.2, particulars of the tube sizes, design conditions and appropriate national or proprietary material specifications are to be submitted for consideration.

1.7.3 The minimum thickness, t_b, of a straight steel pipe to be used for a pipe bend is to be determined by the following formula, except where it can be demonstrated that the use of a thickness less than t_b would not reduce the thickness below 't' at any point after bending:-

$$t_b = \left(\frac{PD}{20Ke + P} + b + c \right) \frac{100}{100 - a} \text{ [mm]}$$

where,

P, D, R, e, b and a are defined in 1.5.1;

K and c are defined in 1.7.1;

$$b = \frac{D}{2.5R} \left(\frac{PD}{20Ke + P} \right) \text{ [mm]}$$

In general, R, is to be not less than 3D.

1.7.4 Notwithstanding the requirements of 1.7.1 and 1.7.3, the minimum thickness of pipes is not to be less than that indicated in Sec.2 or 3, as applicable. For threaded pipes, where permitted, the minimum thickness is to be measured at the bottom of the thread.

1.8 Flanges

1.8.1 The dimensions of flanges and related bolts, gaskets are to be chosen in accordance with recognised national/international standards. For special applications, the dimensions of the flanges and related bolts, will be specially considered.

1.8.2 Gaskets are to be suitable for the medium being conveyed under design pressure and temperature conditions and their dimensions and configurations are to be in accordance with a recognized standard and for the intended service.

1.9 Materials

1.9.1 Materials for ferrous castings and forgings of Class I and Class II piping systems are to be produced at Works approved by IRS and are in general to be tested in accordance with the requirements of Pt.2 of the Rules.

1.9.2 The manufacturer's test certificate for materials of valves and fittings will be accepted in lieu of the IRS certificate provided the maximum conditions are less than given in Table 1.9.1.

1.9.3 Pipes and bodies of valves, intended to be fitted on ship's side and bottom or on collision bulkhead, are to comply with the requirements of Class II piping systems.

1.9.4 Materials for Class III piping systems may be manufactured and tested in accordance with the requirements of an acceptable national/ international standard.

1.9.5 Stainless steel grades 304, 304L, 316 and 316L are generally not considered suitable for use in seawater piping systems due to their limited corrosion resistance in contact with sea water.

Table 1.9.1 : Maximum conditions for valves and fittings for which manufacturer's test certificate is acceptable

Material	Working temp.°C	DN = Nominal Diameter [mm] P_w = Working Pressure in bar
Carbon and low alloy steel Spheroidal or nodular cast iron	< 300	DN < 50 or P_w x DN < 250
Copper alloy	< 200	DN < 50 or P_w x DN < 150

Section 2

Carbon and Low Alloy Steel Pipes and Fittings

2.1 Materials

2.1.1 Pipes having forge butt welded longitudinal seams are not to be used for oil fuel systems, for heating coils in oil tanks, or for pressures exceeding 4.0 bar.

2.1.2 Steel pipes, valves and fittings may be used within the temperature limits indicated in Table 2.1.1 and Table 2.1.2. Where rimming steel is used for pipes manufactured by electric resistance or induction welding processes, the design temperature is limited to 400°C.

2.2 Minimum thickness of steel pipes and bends

2.2.1 The minimum thickness of steel pipes and bends is to be higher of that calculated by Sec.1, Cl.1.7 and that indicated in Table 2.2.2, Table 2.2.3 or Table 2.2.4 as applicable.

2.2.2 The value of 'c' to be used in Sec.1, Cl.1.7 is to be in accordance with Table 2.2.1.

Table 2.1.1 : Carbon and carbon-manganese steel pipes

Specified min. tensile strength [N/mm^2]	Maximum permissible stress [N/mm^2]												
	Maximum design temperature °C												
	50	100	150	200	250	300	350	400	410	420	430	440	450
320	107	105	99	92	78	62	57	55	55	54	54	54	49
360	120	117	110	103	91	76	69	68	68	68	64	56	49
410	136	131	124	117	106	93	86	84	79	71	64	56	49
460	151	146	139	132	122	111	101	99	98	85	73	62	53
490	160	156	148	141	131	121	111	109	98	85	73	62	53

Table 2.1.2 : Alloy steel pipes

Type of steel	Specified min. tensile strength [N/mm^2]	Minimum permissible stress [N/mm^2]									
		Maximum design temperature °C									
		50	**100**	**200**	**300**	**350**	**400**	**440**	**450**	**460**	**470**
1 Cr 1/2 Mo	440	159	150	137	114	106	102	101	101	100	99
2 1/4 Cr 1 Mo annealed	410	76	67	57	50	47	45	44	43	43	42
2 1/4 Cr 1 Mo normalised and tempered (Note 1)	490	167	163	153	144	140	136	130	128	127	116
2 1/4 Cr 1 Mo normalised and tempered (Note 2)	490	167	163	153	144	140	136	130	122	114	105
1/2 Cr 1/2 Mo 1/4 V	460	166	162	147	120	115	111	106	105	103	102
		Maximum design temperature °C									
		480	**490**	**500**	**510**	**520**	**530**	**540**	**550**	**560**	**570**
1 Cr 1/2 Mo	440	98	97	91	76	62	51	42	34	27	22
2 1/4 Cr 1 Mo annealed	410	42	42	41	41	41	40	40	40	37	32
2 1/4 Cr 1 Mo normalised and tempered (Note 1)	490	106	96	86	76	67	58	49	43	37	32
2 1/4 Cr 1 Mo normalised and tempered (Note 2)	490	96	88	79	72	64	56	49	43	37	32
1/2 Cr 1/2 Mo 1/4 V	460	101	99	97	94	82	72	62	53	45	37

Notes:

1 Maximum permissible stress values applicable when the tempering temperature does not exceed 750°C.

2 Maximum permissible stress values applicable when the tempering temperature exceeds 750°C.

Table 2.2.1 : Values of 'c' for steel pipes

Piping service	**c [mm] (See Notes 1,2, 3)**
Superheated steam systems	0.3
Saturated steam systems	0.8
Steam coil systems in cargo tanks	2.0
Feed water for boilers in open circuit systems	1.5
Feed water for boilers in closed circuit systems	0.5
Blow down (for boilers) systems	1.5
Compressed air systems	1.0
Hydraulic/lubricating oil systems	0.3
Fuel oil systems	1.0
Cargo oil systems	2.0
Refrigerating plants	0.3
Fresh water systems	0.8
Sea water systems, in general	3.0

Table 2.2.1 (Contd.)

Notes:

1 For pipes passing through tanks an additional corrosion allowance is to be considered according to the figures given in Table and depending upon the external medium in order to account for the external corrosion.
2 The corrosion allowance may be reduced where pipes and any integral pipe joints are protected against corrosion by means of coating, lining etc., however, the reduction is not to be more than 50 percent in any case.
3 In the case of use of special alloy steel with sufficient corrosion resistance, the corrosion allowance may be reduced to zero.

Table 2.2.2 : Minimum wall thickness for steel pipes [11] [All dimension in mm]

Nominal size	External diameter D	Pipes in general [3, 4, 6, 8, 9, 10]	Venting overflow and sounding pipes for structural tanks [1-4, 6, 8]	Bilge ballast and general sea water pipes [1, 3-8]	Bilge, air, overflow and sounding pipes passing through ballast and fuel tanks, ballast lines through fuel tanks and fuel lines through ballast tanks [1-8]
6	10.2	1.6			
	12	1.6			
8	13.5	1.8			
10	17.2	1.8			
	19.3	1.8			
	20	2			
15	21.3	2		3.2	
	25	2		3.2	
20	26.9	2		3.2	
25	33.7	2		3.2	
	38	2	4.5	3.6	6.3
32	42.4	2	4.5	3.6	6.3
	44.5	2	4.5	3.6	6.3
40	48.3	2.3	4.5	3.6	6.3
	51	2.3	4.5	4	6.3
50	60.3	2.3	4.5	4	6.3
	63.5	2.3	4.5	4	6.3
	70	2.6	4.5	4	6.3
65	76.1	2.6	4.5	4.5	6.3
	82.5	2.6	4.5	4.5	6.3
80	88.9	2.9	4.5	4.5	7.1
90	101.6	2.9	4.5	4.5	7.1
	108	2.9	4.5	4.5	7.1
100	114.3	3.2	4.5	4.5	8
	127	3.2	4.5	4.5	8
	133	3.6	4.5	4.5	8
125	139.7	3.6	4.5	4.5	8
	152.4	4	4.5	4.5	8.8
150	168.3	4	4.5	4.5	8.8
	177.8	4.5	5	5	8.8
175	193.7	4.5	5.4	5.4	8.8
200	219.1	4.5	5.9	5.9	8.8
225	244.5	5	6.3	6.3	8.8

Table 2.2.2 (Contd.)

Nominal size	External diameter D	Pipes in general [3, 4, 6, 8, 9, 10]	Venting overflow and sounding pipes for structural tanks [1-4, 6, 8]	Bilge ballast and general sea water pipes [1, 3-8]	Bilge, air, overflow and sounding pipes passing through ballast and fuel tanks, ballast lines through fuel tanks and fuel lines through ballast tanks [1-8]
250	273	5	6.3	6.3	8.8
	298.5	5.6	6.3	6.3	8.8
300	323.9	5.6	6.3	6.3	8.8
350	355.6	5.6	6.3	6.3	8.8
	368	5.6	6.3	6.3	8.8
400	406.4	6.3	6.3	6.3	8.8
450	457.2	6.3	6.3	6.3	8.8

Notes:

1 IRS may accept reduction in thickness of pipes by an amount not exceeding 1 mm for pipes efficiently protected against corrosion.

2 For sounding pipes, except for those for cargo tanks with cargo having a flash point less than 60°C, the minimum wall thickness is to apply to the part outside the tank.

3 For threaded pipes, where permitted, the minimum wall thickness is to measured at the bottom of the thread.

4 The external diameters and thicknesses have been selected from ISO 4200 for welded and seamless steel pipes. For pipes covered by other standards, slightly less thicknesses may be accepted.

5 The minimum wall thickness for bilge lines and ballast lines through deep tanks will be specially considered. See Pt.5 for thickness of ballast lines passing through oil cargo tanks.

6 The minimum wall thickness for pipes larger than 450 [mm] nominal size is to be in accordance with a national or international standard and in any case not less than the minimum wall thickness indicated for 450 [mm] pipe size.

7 The minimum internal diameter for bilge, sounding, air and overflow pipes shall be 50 [mm], 32 [mm] and 50 [mm] respectively.

8 In general the minimum thickness given in this table is the nominal wall thickness and no allowance need be made for negative tolerance and reduction in thickness due to bending.

9 Minimum wall thickness of exhaust gas pipes will be specially considered.

10 Minimum wall thickness for cargo oil lines will be specially considered.

11 This table is not applicable to air pipes above weather deck, scuppers and discharges which are covered by loadline convention and also to pipes covered by Table 2.2.3.

Table 2.2.3 : Minimum wall thickness for steel pipes for CO_2 fire extinguishing

External diameter D [mm]	From bottles to distribution station	From distribution station to nozzles
21.3 - 26.9	3.2	2.6
30 - 48.3	4	3.2
51 - 60.3	4.5	3.6
63.5 - 76.1	5	3.6
82.5 - 88.9	5.6	4
101.6	6.3	4
108 - 114.3	7.1	4.5
127	8	4.5
133 - 139.7	8	5
152.4 - 168.3	8.8	5.6

Table 2.2.3 (Contd.)

Notes:

1 Pipes are to be galvanised at least internally, except those fitted in the engine room where galvanising may be dispensed with.
2 For threaded pipes, where allowed, the minimum wall thickness is to be measured at the bottom of the thread.
3 The external diameters and thicknesses have been selected from ISO recommendations R 336 for smooth welded and seamless steel pipes. For pipes covered by other standards, slightly less thickness may be accepted.
4 For larger diameters, the minimum wall thickness will be specially considered.
5 In general, the minimum thickness is the nominal wall thickness and no allowance need be made for negative tolerance and reduction in wall thickness due to bending.

Table 2.2.4 : Minimum wall thickness for austenitic stainless steel pipes

External diameter D [mm]	Minimum wall thickness [mm]	External diameter D [mm]	Minimum wall thickness [mm]
10.2 to 17.2	1.0	219.1	2.6
21.3 to 48.3	1.6	273.0	2.9
60.3 to 88.9	2.0	323.9 to 406.4	3.6
114.3 to 168.3	2.3	Over 406.4	4.0

Note : Diameters and thicknesses according to national or international standard may be accepted.

2.3 Types of connection

2.3.1 Pipe lengths may be joined by

- screwed-on or welded on bolted flanges
- Butt welds between pipes or between pipes and valve chests or other fittings
- Socket weld joints
- Threaded sleeve joints
- Mechanical joints.

and are to be to a recognized standard or of a design proven to be suitable for the intended purpose.

2.4 Flange connections

2.4.1 Flange attachments are to be in accordance with recognized national or international standards applicable to piping system keeping in view the boundary fluids, design pressure and temperature conditions, external or cyclic loading and location.

2.4.2 Flanges may be cut from plates or may be forged or cast. The material is to be suitable for the design temperature. Flanges may be attached to the branches by screwing and expanding or by welding. Alternative methods of flange attachment may be accepted provided details are submitted for consideration.

2.4.3 Examples of accepted flanged connections and their uses are given in Fig.2.4.1 and Table 2.4.1 respectively.

2.4.4 Where flanges are secured by screwing, as indicated in Fig.2.4.1, the pipe and flange are to be screwed with a vanishing thread and the diameter of the screwed position of pipe over the thread is not to be appreciably less than the outside diameters of the unscrewed pipe. After the flange has been screwed hard home, the pipe is to be expanded into the flange.

The vanishing thread on a pipe is to be not less than three pitches in length, and the diameter at the root of the thread is to increase uniformly from the standard root diameter to the diameter at the top of the thread. This may be produced by suitably grinding the dies, and the flange should be tapered out to the same formation.

Table 2.4.1 : Type of flange connections

Class of piping	t°C	Steam	Lub. and fuel oil	Other media
		Typical flange application	Typical flange application	Typical flange application
I [1)]	> 400	A	A - B	A
	≤ 400	A - B[2]		A - B
II	> 250	A - B - C	A - B - C	A - B - C
	≤ 250	A - B - C - D - E		A - B - C - D - E
III		A - B - C - D - E	A - B -C - E	A - B - C - D - E

1 For Class I only one condition is sufficient.

2 Type B for outer diameter < 150 [mm] only.

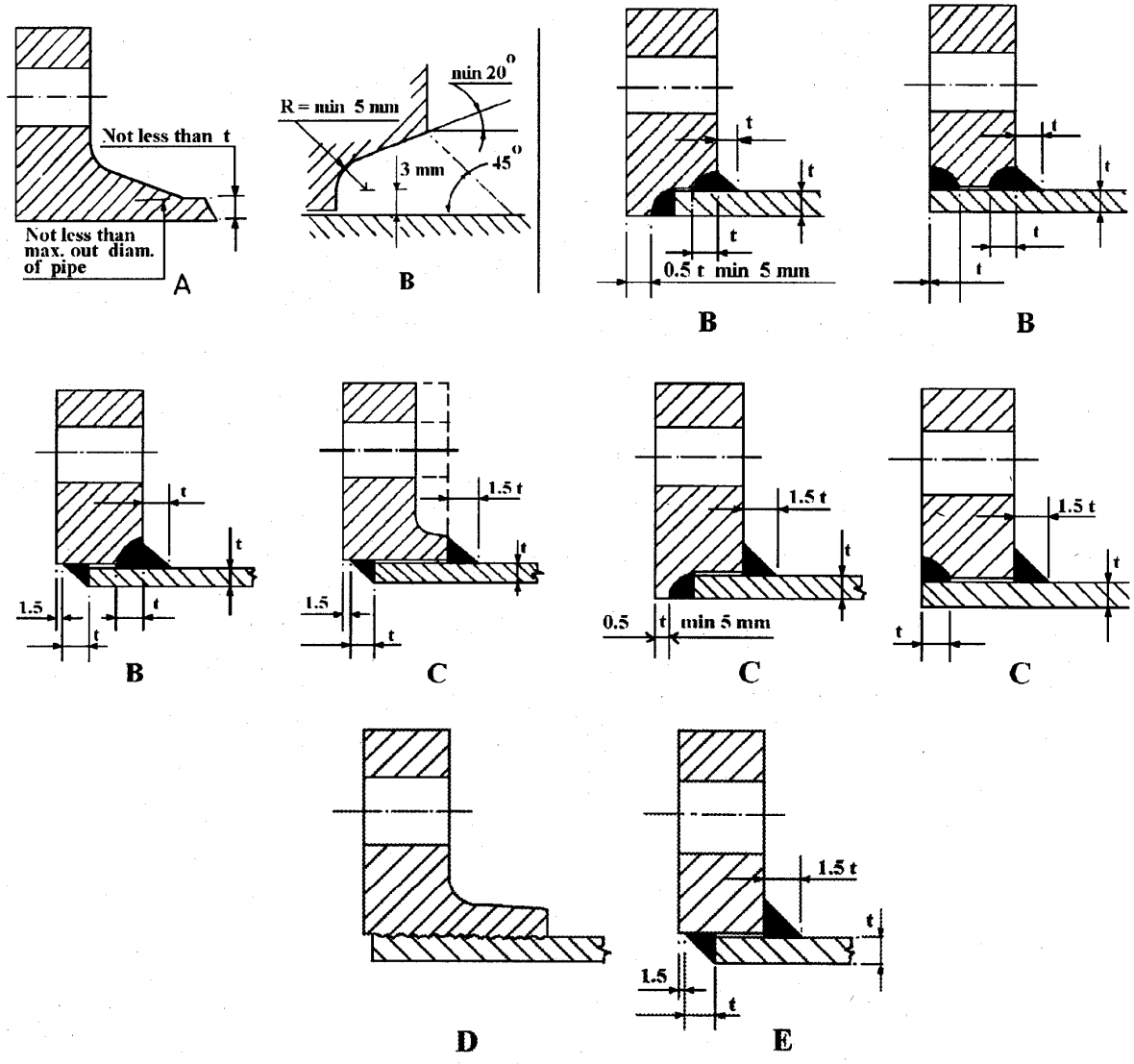

Fig. 2.4.1 : Types of flange connections

2.5 Threaded sleeve joints

2.5.1 Threaded sleeve joints requiring pressure-tight joints, having parallel or tapered threads in accordance with national or other established standards, may be used with carbon steel pipes within the limits given in Table 2.5.1 and for services other than pipe systems conveying combustible or toxic fluids or services where fatigue, severe erosion or crevice corrosion is expected to occur.

Threaded joints in CO_2 systems shall be allowed only inside protected spaces and in CO_2 cylinder rooms.

Threaded joints for direct connections of pipe lengths with tapered threads are to be allowed for

a) Class I piping having outside diameter not more than 33.7 [mm].

b) Class II and Class III piping having outside diameter not more than 60.3 [mm].

Threaded joints with parallel threads are to be allowed for Class III piping having outside diameter not more than 60.3 [mm].

In particular cases, sizes in excess of those mentioned above may be accepted by IRS if in compliance with a recognized national and/or international standard.

Table 2.5.1 : Limiting design conditions for threaded sleeve joints

Nominal bore [mm]	Maximum pressure in bar	Maximum temp.°C
≤ 25	12	260
$> 25 \leq 40$	10	260
$> 40 \leq 80$	8.5	260
$> 80 \leq 100$	7	260

2.6 Welding, non-destructive examination and post-weld heat treatment of welded pipes

2.6.1 Requirements regarding welding procedures, non-destructive examination of welds and post-weld heat treatment are given in Ch.10.

2.7 Mechanical joints

2.7.1 Mechanical joints means devices intended for direct connection of pipe lengths other than by flanges, threaded joints or welding as described in 2.4, 2.5 and 2.6.

2.7.2 The requirements given here are applicable to pipe unions, compression couplings, slip-on joints as shown in Fig.2.7. Similar joints complying with these requirements may be acceptable.

2.7.3 Mechanical joints including pipe unions, compression couplings, slip-on joints and similar joints are to be of approved type for the pressure ratings, service conditions and the intended application. The construction and type are to conform to the examples shown in Fig.2.7. Application of mechanical joints and their acceptable use for each service is indicated in Table 2.7.1. The relevant requirements of the flag Administration are also to be complied with. In case exposure time (t_T) for fire endurance test is greater than 30 minutes, the dry-wet test conditions are 8 minutes dry and, accordingly, the wet period t_T-8 minutes. Application of mechanical joints depending upon the class of piping and pipe dimensions is indicated in Table 2.7.2. (For approval refer Classification Notes “Type Approval of Mechanical Joints used in Piping”).

2.7.4 Where the application of mechanical joints results in reduction in pipe wall thickness due to the use of bite type rings or other structural elements, this is to be taken into account in determining the minimum wall thickness of the pipe to withstand the design pressure.

The materials used in construction of mechanical joints is to be compatible with the piping material and internal and external media. Where appropriate, the mechanical joints are to be of fire resistant type as required by Table 2.7.1.

2.7.5 Mechanical joints, which in the event of damage could cause fire or flooding, are not to

be used in piping sections directly connected to the ship's side below the bulkhead deck of passenger ships and freeboard deck of cargo ships or tanks containing flammable fluids.

2.7.6 The number of mechanical joints in flammable fluid systems is to be kept to a minimum. In general, flanged joints conforming to recognised standards are to be used.

2.7.7 Piping in which a mechanical joint is fitted is to be adequately adjusted, aligned and supported Supports or hangers are not to be used to force alignment of piping at the point of connection.

2.7.8 Slip-on joints are normally not to be used in pipelines in cargo holds, tanks and other spaces which are not easily accessible (refer to MSC/ Circ. 734), except that these joints may be permitted in tanks that contain the same media.

2.7.9 Usage of slip type slip-on joints as the main means of pipe connection is not permitted except for cases where compensation of axial pipe deformation is necessary.

2.7.10 In particular cases, sizes in excess of those mentioned in Table 2.7.2 may be accepted if in compliance with a recognised national and/or international standard.

2.7.11 Mechanical joints are to be subjected to the following tests:

.1 leakage test
.2 vacuum test
.3 vibration (fatigue) test
.4 fire endurance test
.5 burst pressure test at 4 times the design pressure (for design pressures above 200 bar, the burst pressure will be specially considered by IRS)
.6 pressure pulsation test
.7 assembly test
.8 pull out test.

NOTE : For details of tests refer classification notes, "Type Approval of Mechanical Joints used in Piping".

2.7.12 The installation of mechanical joints is to be in accordance with the manufacturer's assembly instructions and using special tools and gauges as required.

Table 2.7.1 : Application of mechanical joints

Systems		Kind of connections			Classification of pipe system	Fire endurance test condition[7]
		Pipe unions	Compression couplings	Slip-on joints		
Flammable Fluids (Flash point ≤ 60°C)						
1	Cargo oil lines[1]	+	+	+	dry	30 min dry (*)
2	Crude oil washing lines[1]	+	+	+	dry	
3	Vent lines[3]	+	+	+	dry	
Inert Gas						
4	Water seal effluent lines	+	+	+	wet	30 min wet (*)
5	Scrubber effluent lines	+	+	+	wet	30 min wet (*)
6	Main lines[1&2]	+	+	+	dry	30 min dry (*)
7	Distributions lines[1]	+	+	+	dry	30 min dry (*)
Flammable Fluids (Flash point > 60°C)						
8	Cargo oil lines[1]	+	+	+	dry	30 min dry (*)
9	Fuel oil lines[2&3]	+	+	+	wet	30 min wet (*)
10	Lubricating oil lines[2&3]	+	+	+	wet	
11	Hydraulic oil[2&3]	+	+	+	wet	
12	Thermal oil[2&3]	+	+	+	wet	
Sea Water						
13	Bilge lines[4]	+	+	+	dry/ wet	8 min dry +22 min wet (*)
14	Permanent water filled fire extinguishing systems, e.g. sprinkler systems[3]	+	+	+	wet	30 min wet (*)
15	Non-permanent water filled fire extinguishing systems, e.g. foam, drencher systems and fire main[3]	+	+	+	dry/ wet	8 min dry + 22 min wet (*) For foam systems, Part 6 Chapter 8. Section 6 is to be complied with.
16	Ballast System[4]	+	+	+	wet	30 min wet (*)
17	Cooling water system[4]	+	+	+	wet	30 min wet (*)
18	Tank cleaning services	+	+	+	dry	Fire endurance test not required
19	Non-essential systems	+	+	+	dry dry/ wet wet	Fire endurance test not required
Fresh Water						
20	Cooling water system[4]	+	+	+	wet	30 min wet (*)
21	Condensate return[4]	+	+	+	wet	30 min wet (*)
22	Non-essential system	+	+	+	dry dry/ wet wet	Fire endurance test not required

Table 2.7.1 (Contd.)

Systems		Kind of connections			Classification of pipe system	Fire endurance test condition[7]
		Pipe unions	Compression couplings	Slip-on joints		
Sanitary/ Drains/ Scuppers						
23	Deck drains (internal)[5]	+	+	+	dry	Fire endurance test not required
24	Sanitary drains	+	+	+	dry	
25	Scuppers and discharge (overboard)	+	+	-	dry	
Sounding/ Vent						
26	Water tanks/ Dry spaces	+	+	+	dry, wet	Fire endurance test not required
27	Oil tanks (f.p.>60°C)[2&3]	+	+	+	dry	
Miscellaneous						
28	Starting/ Control air[4]	+	+	+	dry	30 min dry (*)
29	Service air (non-essential)	+	+	+	dry	Fire endurance test not required
30	Brine	+	+	+	wet	
31	CO_2 system (outside protected space)	+	+	+	dry	30 min dry (*)
32	CO_2 system (inside protected space)	+	+	+	dry	Mechanical joints are to be constructed of materials with melting point above 925°C. Ref. to Part 6, Chapter 8, Section 5.
33	Steam	+	+	+[6]	wet	Fire endurance test not required

Abbreviations

\+ Application is allowed
\- Application is not allowed
***** Fire endurance test as specified in IRS Classification Note: *Type Approval of Mechanical Joints used in Piping.*

Footnotes: - Fire resistance capability

If mechanical joints include any components which readily deteriorate in case of fire, the following footnotes are to be observed:

1) Fire endurance test is to be applied when mechanical joints are installed in pump rooms and open decks.
2) Slip-on joints are not acceptable inside machinery spaces of category A or accommodation spaces. May be accepted in other machinery spaces provided the joints are located in easily visible and accessible positions (Refer to MSC/Circ. 734).
3) Approved fire-resistant types except in cases where such mechanical joints are installed on open decks, as defined in Pt. 6, Ch. 3, Cl. 3.2.3.3.2.2(10) and not used for fuel lines.
4) Fire endurance test is to be applied when mechanical joints are installed inside machinery spaces of category A.

Footnotes – General (Table 2.7.1)

5) Only above bulkhead deck of passenger ships and freeboard deck of cargo ships.
6) Slip type slip-on joints as shown in Fig. 2.7 may be used for pipes on deck with a design pressure of 10 bar or less.
7) If a connection has passed the "30 min dry" test, it is considered suitable also for applications for which the "8 min dry+22 min wet" and/or "30 min wet " tests are required. If a connection has passed the "8 min dry+22 min wet" test, it is considered suitable also for applications for which the "30 min wet" test is required.

Table 2.7.2 : Application of mechanical joints depending upon the class of piping

Types of joints	Classes of piping systems		
	Class I	**Class II**	**Class III**
Pipe Unions			
Welded and brazed type	+ (OD ≤ 60.3 mm)	+ (OD ≤ 60.3 mm)	+
Compression Couplings			
Swage type	+	+	+
Bite type	+ (OD ≤ 60.3 mm)	+ (OD ≤ 60.3 mm)	+
Typical compression type	+ (OD ≤ 60.3 mm)	+ (OD ≤ 60.3 mm)	+
Flared type	+ (OD ≤ 60.3 mm)	+ (OD ≤ 60.3 mm)	+
Press type	-	-	+
Slip-on joints			
Machine grooved type	+	+	+
Grip type	-	+	+
Slip type	-	+	+

Abbreviations:

\+ Application is allowed
\- Application is not allowed

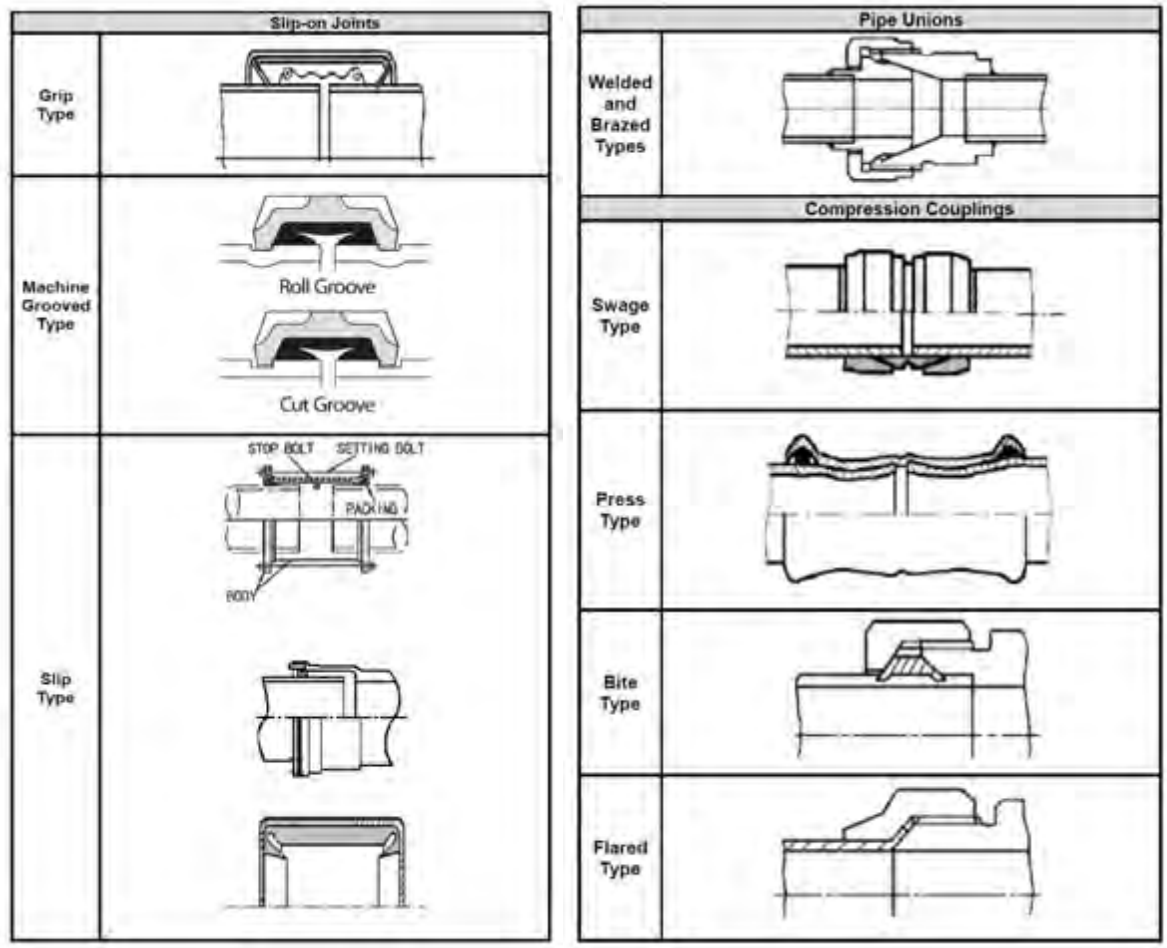

Fig. 2.7 : Examples of Mechanical Joints

Section 3

Copper and Copper Alloy Pipes and Fittings

3.1 Materials

3.1.1 Pipes are to be seamless and branches are to be provided by cast or stamped fittings, pipe pressings or other approved fabrications. Copper pipes for class I and II are to be seamless.

3.1.2 Brazing and welding materials are to be suitable for the operating temperature and for the medium being carried. All brazing and welding are to be carried out to the satisfaction of the Surveyors.

3.1.3 In general, the maximum permissible service temperature of copper and copper alloy pipes, valves and fittings is not to exceed 200°C for copper and aluminium brass, and 300°C for copper nickel. Cast bronze valves and fittings complying with the requirements of Pt.2 Ch.8 may be accepted up to 260°C.

3.2 Minimum thickness of pipes

3.2.1 The minimum thickness of copper and copper alloy pipes and bends is to be higher of that calculated by Sec.1, Cl.1.7 and that indicated in Table 3.2.2.

3.2.2 The value of 'c' to be used in Sec.1, Cl.1.7 is to be in accordance with Table 3.2.3.

3.3 Heat treatment

3.3.1 Pipes which have been hardened by cold bending are to be suitably heat treated on completion of fabrication and prior to being tested by hydraulic pressure. Copper pipes are to be annealed and copper alloy pipes are to be either annealed or stress relief heat treated.

Table 3.2.1 : Copper and copper alloy pipes

Pipe material	Condition of supply	Specified min. tensile strength [N/mm^2]	Permissible stress [N/mm^2]					
			Maximum design temperature °C					
			50	75	100	125	150	175
Copper	Annealed	220	41.2	41.2	40.2	40.2	34.3	27.5
Aluminium brass	Annealed	320	78.5	78.5	78.5	78.5	78.5	78.5
95/5 90/10 copper nickel-iron	Annealed	270	68.6	68.6	67.7	65.7	63.7	61.8
70/30 copper nickel	Annealed	360	81.4	79.4	77.5	75.5	73.5	71.6
			Maximum design temperature °C					
			200	225	250	275	300	
Copper	Annealed	220	18.6	-	-	-	-	
Aluminium brass	Annealed	320	24.5	-	-	-	-	
95/5 90/10 copper nickel-iron	Annealed	270	58.8	55.9	52.0	48.1	44.1	
70/30 copper nickel	Annealed	360	69.6	67.7	65.7	63.7	61.8	

Notes :

1. Intermediate values may be determined by linear interpolation.
2. For materials not included in the Table, the permissible stress shall be specially considered by IRS.

Table 3.2.2 : Minimum thickness for copper and copper alloy pipes

Standard pipe sizes (outside diameter[mm])	Minimum overriding nominal thickness [mm]	
	Copper	Copper alloy
8 to 10	1.0	0.8
12 to 20	1.2	1.0
25 to 44.5	1.5	1.2
50 to 76.1	2.0	1.5
88.9 to 108	2.5	2.0
133 to 159	3.0	2.5
193.7 to 267	3.5	3.0
273 to 457.2	4.0	3.5
508	4.5	4.0

Table 3.2.3 : Corrosion allowance 'c' for copper and copper alloy pipes	
Piping Material	**c [mm]**
Copper, brass and similar alloys, copper-tin alloys except those with lead contents	0.8
Copper-nickel alloys (with Ni ≥ 10%)	0.5
Note: For media without corrosive action in respect of the material employed and in case of special alloys with sufficient corrosion resistance, the corrosion allowance may be reduced to zero.	

Section 4

Cast Iron Pipes and Fittings

4.1 Spheroidal or nodular graphite cast iron

4.1.1 Spheroidal or nodular graphite iron castings for pipes, valves and fittings in Class II and III piping systems are to be made in a grade having a specified minimum elongation not less than 12 per cent on gauge length of $5.65\sqrt{S_o}$, where S_o is the actual cross-sectional area of the test piece.

4.1.2 Proposals for the use of this material in Class I piping systems will be specially considered, but in no case is the material to be used in systems where the design temperature exceeds 350°C.

4.1.3 Where the elongation is less than the minimum required by 4.1.1, the material is, in general, to be subject to the same limitations as grey cast iron.

4.2 Grey cast iron

4.2.1 Grey cast iron pipes, valves and fittings will, in general, be accepted in Class III piping systems except as stated in 4.2.2. Grey cast iron valves and fittings may be accepted in the Class II steam systems, but the design pressure or temperature is not to exceed 13 bar or 220°C respectively.

4.2.2 Grey cast iron is not to be used for the following:

a) Pipes for steam systems and fire extinguishing systems;

b) Pipes, valves and fittings for boiler blow down systems and other piping systems subject to shock or vibration;

c) Ship-side valves and fittings;

d) Valves fitted on collision bulkhead;

e) Clean ballast lines through cargo oil tanks to forward ballast tanks;

f) Bilge lines in tanks;

g) Outlet valves of fuel and lubricating oil tanks with static head.

Section 5

Plastic Pipes and Fittings

5.1 General

5.1.1 Proposals to use plastics material in shipboard piping systems will be considered in relation to the properties of the materials, the operating conditions of temperature and pressure, and the intended service. Any proposed service for plastics pipes not mentioned in these Rules is to be submitted for special consideration.

5.1.2 Attention is also drawn to the Guidelines for the Application of Plastic Pipes on Ships contained in IMO Resolution A.753(18), (as amended by Res MSC.313(88) and 399(95)).

5.1.3 Plastic pipes and fittings will, in general, be accepted in Class III piping systems. Proposals for the use of plastics in Class I and Class II piping systems will be specially considered.

5.1.4 These requirements are applicable to piping systems on ships including pipe joints and fittings, made predominantly of material other than metal. The use of mechanical joints approved for the use in metallic piping systems only are not permitted.

5.1.5 Piping systems intended for non-essential services are to meet only the requirements of recognized standards and that of 5.3.3 (b), 5.7.1.2, 5.8.2 to 5.8.7 and 5.9 of this Section. However, the safety aspects due to fire will require to be considered.

5.1.6 The use of plastics may be restricted by statutory requirements of the National Authority of the country in which the ship is to be registered.

5.1.7 The specification of piping is to be in accordance with a recognized national or international standard acceptable to IRS. In addition, requirements given in 5.3 to 5.6 are to be complied with.

5.2 Terms and definitions

5.2.1 "Plastic(s)" means both thermoplastic and thermosetting plastic materials with or without reinforcement, such as PVC and fibre reinforced plastics - FRP. Plastic includes synthetic rubber and materials of similar thermo/ mechanical properties.

5.2.2 "Pipes/piping systems" means those made of plastic(s) and include the pipes, fittings, system joints, method of joining and any internal or external liners, coverings and coatings required to comply with the performance criteria.

5.2.3 "Joint" means the location at which two pieces of pipe or a pipe and a fitting are connected together. The joint may be made by adhesive bonding, laminating, welding, flanges and mechanical joints according to Fig 2.7.

5.2.4 "Fittings" means bends, elbows, fabricated branch pieces etc. of plastic materials.

5.2.5 "Nominal pressure" means the maximum permissible working pressure which should be determined in accordance with the requirements in 5.3.3.

5.2.6 "Fire endurance" means the capability of piping to maintain its strength and integrity (i.e. capable of performing its intended function) for some predetermined period of time while exposed to fire.

5.2.7 “Essential to the safety of ship” means all piping systems that in event of failure will pose a threat to personnel and the ship (Examples for piping systems essential to the safety are provided by Table 5.7.1).

5.2.8 “Essential services” are those services essential for propulsion and steering and safety of the ship as specified in Pt.4, Ch.8, 1.5.

5.3 Strength

5.3.1 The strength of the pipes is to be determined by hydrostatic test failure pressure of a pipe specimen under the standard conditions: atmospheric pressure equal to 100 kPa, relative humidity 30%, environmental and carried fluid temperature 298 K (25°C).

5.3.2 The strength of fittings and joints is to be not less than that of the pipes.

5.3.3 The nominal pressure is to be determined from the following conditions :

a) Internal Pressure

For an internal pressure the nominal pressure, Pn, is to be lesser of the following :

$$\text{Pn Int} \leq \frac{\text{Psth}}{4}$$

or

$$\text{Pn Int} \leq \frac{\text{Plth}}{2.5}$$

where,

Psth = short-term hydrostatic test pipe failure pressure;

Plth = long-term hydrostatic test pipe failure pressure (> 100,000 h)

b) External pressure (for any installation which may be subject to vacuum conditions inside the pipe or a head of liquid acting on the outside of the pipe; and for any pipe installation required to remain operational in case of flooding damage, as per SOLAS II-1/8-1 or for any pipes that would allow progressive flooding to other compartments through damaged piping or through open ended pipes in the compartments).

For an external pressure the nominal pressure, Pn, is to be as under :

$$\text{Pn ext} \leq \frac{\text{Pcol}}{3}$$

where,

Pcol = pipe collapse pressure.

In no case is the pipe collapse pressure to be less than 3 bar.

The maximum working external pressure is a sum of the vacuum inside the pipe and a head of liquid acting on the outside of the pipe.

5.3.4 Notwithstanding the requirements of 5.3.3 (a) or (b) above as applicable, the pipe or pipe layer minimum wall thickness is to follow recognized standards. In the absence of standards for pipes not subject to external pressure, the requirements of 5.3.3 (b) above are to be met.

5.3.5 The maximum permissible working pressure is to be specified with due regard for maximum possible working temperatures in accordance with Manufacturer's recommendations.

5.4 Axial strength

5.4.1 The sum of the longitudinal stresses due to pressure, weight and other loads is not to exceed the allowable stress in the longitudinal direction.

5.4.2 In the case of fibre reinforced plastic pipes, the sum of the longitudinal stresses is not to exceed half of the nominal circumferential stress derived from the nominal internal pressure condition (See 5.3).

5.5 Impact resistance

5.5.1 Plastic pipes and joints are to have a minimum resistance to impact in accordance with recognised national or international standards such as ASTM D.2444-21 or equivalent.

5.5.2 After the test the specimen is to be subjected to hydrostatic pressure equal to 2.5 times the design pressure for at least 1 hour.

5.6 Temperature

5.6.1 The permissible working temperature depending on the working pressure is to be in accordance with Manufacturer's recommendations, but in each case it is to be at least 20°C lower than the minimum heat distortion/ deflection temperature of the pipe material, determined according to ISO 75-2:2013 method A, or equivalent, such as ASTM D648-18.

5.6.2 The minimum heat distortion/deflection temperature is to be not less than 80°C.

5.7 Requirements for pipes/piping systems depending on service and/or locations

5.7.1 Fire endurance

5.7.1.1 Pipes and their associated joints and fittings whose integrity is essential to the safety of ships including plastic piping required by SOLAS II-2/21.4(safe return to port) to remain operational after a fire casualty, are required to meet the minimum fire endurance requirements of Appendix 1 or 2, as

applicable, of IMO Resolution A.753(18), (as amended by Resolutions MSC.313(88) and MSC. 399(95)).

5.7.1.2 Unless instructed otherwise by the Flag Administration, fire endurance tests are to be carried out with specimen representative for pipes, joints and fittings Note 1:

(i) Pipes:

- for sizes with outer diameter < 200 mm the minimum outer diameter and wall thickness Note 2

- for sizes with outer diameter ≥ 200 mm one test specimen for each category of t/d (D = outer diameter, t = structural wall thickness). A scattering of ±10% for t/D is regarded as the same group. Minimum size approved is equal to the diameter of specimen successfully tested.

(ii) Joints

- Each type of joint applicable for applied fire endurance level tested on pipe to pipe specimen

Notes:

1 A test specimen incorporating several components of a piping system may be tested in a single test.

2 Test conditions are most demanding for minimum wall thickness and thus larger wall thickness is covered. A key factor determining the fire performance of a pipe component variant is the thickness-to-diameter (t/D) ratio and whether it is larger or smaller than that of the variant which has been fire-tested. If fire-protective coatings or layers are included in the variant used in the fire test, only variants with the same or greater thickness of protection, regardless of the (t/D) ratio, is to be qualified by the fire test.

5.7.1.3 Means are to be provided to ensure a constant media pressure inside the test specimen during the fire test as specified in Appendix 1 or 2 of the IMO Res.A.753(18), as amended by IMO Resolutions MSC.313(88) and MSC.399(95). During the test it is not permitted to replace media drained by fresh water or nitrogen.

5.7.1.4 Depending on the capability of a piping system to maintain its strength and integrity, there exist three different levels of fire endurance for piping systems.

a) Level 1. Piping having passed the fire endurance test specified in Appendix 1 of IMO Resolution A.753(18) (as amended by IMO Resolutions MSC.313(88) and MSC. 399(95)) for a duration of a minimum of one hour without loss of integrity in the dry condition is considered to meet level 1 fire endurance standard (L1). Level 1W – Piping systems similar to Level 1 systems except these systems do not carry flammable fluid or any gas and a maximum 5% flow loss in the system after exposure is acceptable (L1W).

b) Level 2. Piping having passed the fire endurance test specified in Appendix 1 of IMO Resolution A.753(18) (as amended by IMO Resolutions MSC.313(88) and MSC. 399(95)) for a duration of a minimum of 30 minutes in the dry condition is considered to meet level 2 fire endurance standard (L2). Level 2W – Piping systems similar to Level 2 systems except a maximum 5% flow loss in the system after exposure is acceptable (L2W).

c) Level 3. Piping having passed the fire endurance test specified in Appendix 2 of IMO Resolution A.753(18) (as amended by IMO Resolutions MSC.313(88) and MSC. 399(95)) for a duration of a minimum of 30 minutes in the wet condition is considered to meet level 3 fire endurance standard (L3).

5.7.1.5 Permitted use of piping depending on fire endurance, location and piping system is given in Table 5.7.1 "Fire Endurance Requirements Matrix".

5.7.1.6 For Safe Return to Port purposes (SOLAS II-2/21.4), plastic piping can be considered to remain operational after a fire casualty if the plastic pipes and fittings have been tested to L1 standard.

5.7.2 Flame spread

5.7.2.1 All pipes, except those fitted on open decks and within tanks, cofferdams, pipe tunnels and ducts separated from accommodation, permanent manned areas and escape ways by means of an A class bulkhead, are to have low surface flame spread characteristics not exceeding average values listed in Appendix 3 of IMO Resolution

A. 753(18), as amended by IMO Resolutions MSC. 313(88) and MSC. 399(95).

5.7.2.2 Surface flame spread characteristics are to be determined using the procedure given in the 2010 FTP Code, Annex 1, Part 5 with regard to the modifications due to the curvilinear pipe surfaces as also listed in Appendix 3 of IMO Resolution A.753(18) (as amended by IMO Resolutions MSC.313(88) and MSC.399(95)).

5.7.2.3 Surface flame spread characteristics may also be determined using the test procedures given in ASTM D635-18, or in other equivalent standards. Under the procedure of ASTM D635-18 a maximum burning rate of 60 mm/min applies.

5.7.3 Fire protection coatings

5.7.3.1 Where a fire protective coating of pipes and fittings is necessary for achieving the fire endurance level required, it is to meet the following requirements:

a) The pipes are generally to be delivered from the manufacturer with the protective coating on.

b) The fire protection properties of the coating are not to be diminished when exposed to salt water, oil or bilge slops. It is to be demonstrated that the coating is resistant to products likely to come into contact with the piping.

c) In considering fire protection coatings, such characteristics as thermal expansion, resistance against vibrations and elasticity are to be taken into account.

d) The fire protection coatings are to have sufficient resistance to impact to retain their integrity.

5.7.4 Electrical conductivity

Where electrical conductivity is to be ensured, the resistance of the pipes and fittings is not to exceed 1×10^5 Ohm/m.

Table 5.7.1 : Fire endurance requirements matrix

Sr. No.	Piping Systems	Location[13]										
		A	B	C	D	E	F	G	H	I	J	K
		Machinery spaces of Category A	Other machinery spaces & pump rooms	Cargo pump rooms	Ro/Ro cargo holds	Other dry cargo holds	Cargo tanks	Fuel oil tanks	Ballast water tanks	Cofferdams void spaces pipe tunnel and ducts	Accommoda-tion service and control spaces	Open decks
1	2	3	4	5	6	7	8	9	10	11	12	13
CARGO (FLAMMABLE CARGOES f.p. ≤ 60°C												
1.	Cargo lines	NA	NA	L1	NA	NA	O	NA	O[10]	O	NA	L1[2]
2.	Crude oil washing lines	NA	NA	L1	NA	NA	O	NA	O[10]	O	NA	L1[2]
3.	Vent lines	NA	NA	NA	NA	NA	O	NA	O[10]	O	NA	X
INERT GAS												
4.	Water seal effluent line	NA	NA	O[1]	NA	NA	O[1]	O[1]	O[1]	O[1]	NA	O
5.	Scrubber effluent line	O[1]	O[1]	NA	NA	NA	NA	NA	O[1]	O[1]	NA	O
6.	Main line	O	O	L1	NA	NA	NA	NA	NA	O	NA	L1[6]
7.	Distribution lines	NA	NA	L1	NA	NA	O	NA	NA	O	NA	L1[2]
FLAMMABLE LIQUIDS (f.p. > 60°C)												
8.	Cargo lines	X	X	L1	X	X	NA[3]	O	O[10]	O	NA	L1
9.	Fuel oil	X	X	L1	X	X	NA[3]	O	O	O	L1	L1
10.	Lubricating	X	X	L1	X	X	NA	NA	NA	O	L1	L1
11.	Hydraulic oil	X	X	L1	X	X	O	O	O	O	L1	L1
SEAWATER[1]												
12.	Bilge main & branches	L1[7]	L1[7]	L1	X	X	NA	O	O	O	NA	L1
13.	Fire main & water spray	L1	L1	L1	X	NA	NA	NA	O	O	X	L1
14.	Foam system	L1W	L1W	L1W	NA	NA	NA	NA	NA	O	L1W	L1W
15.	Sprinkler system	L1W	L1W	L3	X	NA	NA	NA	O	O	L3	L3
16.	Ballast	L3	L1	L3	L3	X	O[10]	O	O	O	L2W	L2W
17.	Cooling water, essential services	L3	L3	NA	NA	NA	NA	NA	O	O	NA	L2W
18.	Tank cleaning services fixed machines	NA	NA	L3	NA	NA	O	NA	O	O	NA	L3[2]
19.	Non-essential systems	O	O	O	O	O	NA	O	O	O	O	O
FRESH WATER												
20.	Cooling water essential	L3	L3	NA	NA	NA	NA	O	O	O	L3	L3
21.	Condensate return	L3	L3	L3	O	O	NA	NA	NA	O	O	O
22.	Non-essential systems	O	O	O	O	O	NA	O	O	O	O	O

Table 5.7.1 : Fire endurance requirements matrix (Contd.)

Sr. No.	Piping Systems	Location[13]										
		A	B	C	D	E	F	G	H	I	J	K
		Machinery spaces of Category A	Other machinery spaces & pump rooms	Cargo pump rooms	Ro/Ro cargo holds	Other dry cargo holds	Cargo tanks	Fuel oil tanks	Ballast water tanks	Cofferdams void spaces pipe tunnel and ducts	Accommodation service and control spaces	Open decks
1	2	3	4	5	6	7	8	9	10	11	12	13
SANITARY/DRAINS/SCUPPERS												
23.	Deck drains (internal)	L1W[4]	L1W[4]	NA	L1W[4]	O	NA	O	O	O	O	O
24.	Sanitary drains (internal)	O	O	NA	O	O	NA	O	O	O	O	O
25.	Scuppers and discharges (overboard)	O[1,8]	O[1,8]	O[1,8]	O[1,8]	O[1,8]	O	O	O	O	O[1,8]	O
SOUNDING/AIR												
26.	Water tanks/dry spaces	O	O	O	O	O	O[10]	O	O	O	O	O
27.	Oil tanks (f.p. > 60°C)	X	X	X	X	X	X[3]	O	O[10]	O	X	X
MISCELLANEOUS												
28.	Control air	L1[5]	L1[5]	L1[5]	L1[5]	L1[5]	NA	O	O	O	L1[5]	L1[5]
29.	Service air (non-essential)	O	O	O	O	O	NA	O	O	O	O	O
30.	Brine	O	O	NA	O	O	NA	NA	NA	O	O	O
31.	Auxiliary low pressure steam (≤ 7 bar)	L2W	L2W	O[9]	O[9]	O[9]	O	O	O	O	O[9]	O[9]
32.	Central vacuum cleaners	NA	NA	NA	0	NA	NA	NA	NA	0	0	0
33.	Exhaust gas Cleaning system Effluent Line	L3[1]	L3[1]	NA	NA	NA	NA	NA	NA	0	L3[1,11] NA	0
34.	Urea Transfer/ Supply System (SCR installations)	L1[12]	L1[12]	NA	NA	NA	NA	NA	NA	0	L3[11] NA	0

Abbreviations:

L1: Fire endurance test (appendix 1 of IMO Resolution A.753(18), as amended by IMO Resolutions MSC. 313(88) and MSC. 399(95)) in dry conditions, 60 min
L1W Fire endurance test (Cl. 5.7.1.2)
L2: Fire endurance test (appendix 1 of IMO Resolution A.753(18), as amended by IMO Resolutions MSC. 313(88) and MSC. 399(95)) in dry conditions, 30 min
L2W Fire endurance test (Cl. 5.7.1.2)
L3: Fire endurance test (appendix 2 of IMO Resolution A.753(18), as amended by IMO Resolutions MSC. 313(88) and MSC. 399(95)) in wet conditions, 30 min
O : No fire endurance test required
NA : Not applicable
X : Plastic piping is not allowed. Metallic materials having a melting point greater than 925°C are permitted.

Footnotes:

1 Where plastic piping is used, remotely controlled valves to be provided at ship's side (valve is to be controlled from outside space).

2 Remote closing valves to be provided at the cargo tanks.

3 When cargo tanks contain flammable liquids with f.p. > 60°C, "O" may replace "NA" or "X".

4 For drains serving only the space concerned, "O" may replace "L1W".

5 When controlling functions are not required by statutory requirements or guidelines, "O" may replace "L1".

6 For pipe between machinery space and deck water seal, "O" may replace "L1".

7 For passenger vessels, "X" is to replace "L1".

8 Scuppers serving open decks in positions 1 and 2, as defined in Regulation 13 of Protocol of 1988 relating to the International Convention on Load Lines, 1966 should be "X" throughout unless fitted at the upper end with the means of closing capable of being operated from a position above the freeboard deck in order to prevent downflooding.

9 For essential services, such as fuel oil tank heating and ship's whistle, "X" is to replace "O".

10 For tankers where compliance with paragraph 6 of Regulation 19 of MARPOL Annex 1, as amended by IMO Resolutions up to MEPC. 314(74) is required, "NA" is to replace "0".

11 L3 in service spaces, NA in accommodation and control spaces.

12 Type Approved plastic piping without fire endurance test (0) is acceptable downstream of the tank valve, provided this valve is metal seated and arranged as fail-to-closed or with quick closing from a safe position outside the space in the event of fire.

13 For Passenger Ships subject to SOLAS II-2/21.4 (Safe return to Port), plastic pipes for services required to remain operative in the part of the ship not affected by the casualty thresholds, such as systems intended to support safe areas, are to be considered essential services. In accordance with MSC.1/Circ.1369, Interpretation 12, for Safe Return to Port purposes, plastic piping can be considered to remain operational after a fire casualty if the plastic pipes and fittings have been tested to L1 standard.

Location Definitions:

A	Machinery spaces of Category A	Machinery spaces of Category A as defined in SOLAS II-2/3.31*.
B	Other machinery spaces and pump rooms	Spaces, other than Category A machinery spaces and cargo pump rooms, containing propulsion machinery, boilers, fuel oil units, steam and internal combustion engines, generators and major electrical machinery, pumps, oil filling stations, refrigerating, stabilising, ventilation and air-conditioning machinery and similar spaces and trunks to such spaces.
C	Cargo pump rooms	Spaces containing cargo pumps and entrances and trunks to such spaces.
D	Ro-ro cargo holds	Ro-ro cargo holds are ro-ro cargo spaces and special category spaces as defined in SOLAS II-2/3.46*.
E	Other dry cargo holds	All spaces other than ro-ro cargo holds used for non-liquid cargo and trunks to such spaces.

F	Cargo tanks	All spaces used for liquid cargo and trunks to such spaces.
G	Fuel oil tanks	All spaces used for fuel oil (excluding cargo tanks) and trunks to such spaces.
H	Ballast water tanks	All spaces used for ballast water and trunks to such spaces.
I	Cofferdams, voids, etc.	Cofferdams and voids are those empty spaces between two bulkheads separating two adjacent compartments.
J	Accommodation, service	Accommodation spaces, service spaces and control stations as defined in SOLAS Chapter II-2/3.1, SOLAS *.
K	Open decks	Open deck spaces as defined in SOLAS II-2/9.2.2.3.2(5)*.
*	SOLAS Chapter II-2	

5.8 Material approval and quality control during manufacture

5.8.1 Except as required in 5.1.5, prototypes of pipes and fittings are to be tested to determine short-term and long-term design strength, fire endurance and low surface flame spread characteristics, electrical resistance (for electrically conductive pipes), impact resistance in accordance with this Section.

5.8.2 For prototype testing representative samples of pipes and fittings are to be selected to the satisfaction of the Classification Society.

5.8.3 The Manufacturer is to have quality system that meets ISO 9000 series standards or equivalent. The quality system is to consist of elements necessary to ensure that pipes and fittings are produced with consistent and uniform mechanical and physical properties.

5.8.4 Each pipe and fitting made by hand lay up technique, is to be tested by the Manufacturer at a hydrostatic pressure not less than 1.5 times the nominal pressure. For other pipes and fittings, manufactured in accordance with the recognized standards by machined processes (e.g. ‘Continuous extruding’, ‘Filament wound and centrifugally cast’) the hydrostatic pressure test may be carried out in accordance with requirements stipulated in the relevant standard to which the pipe or fittings are manufactured, provided that there is an effective quality system in place at manufacturer’s works.

5.8.5 Piping and fittings are to be permanently marked with identification. Identification is to include pressure ratings, the design standards that the pipe or fitting is manufactured in accordance with and the material of which the pipe or fitting is made.

5.8.6 In case the Manufacturer does not have an approved quality system complying with ISO 9000 series or equivalent, pipes and fittings are to be tested in accordance with this Section to the satisfaction of the attending surveyors for every batch of pipes.

5.8.7 Depending upon the intended application, IRS may require, pressure testing of each pipe and/or fitting.

5.9 Installation

5.9.1 Supports

5.9.1.1 Selection and spacing of pipe supports in shipboard systems are to be determined as a function of allowable stresses and maximum deflection criteria. Support spacing is not to be greater than the pipe Manufacturer's recommen-ded spacing. The selection and spacing of pipe supports are to take into account pipe dimensions, length of the piping, mechanical and physical properties of the pipe material, mass of pipe and contained fluid, external pressure, operating temperature, thermal expansion effects, loads due to external forces, thrust forces, water hammer, vibrations, maximum accelerations to which the system may be subjected. Combination of loads is to be considered.

5.9.1.2 Each support is to evenly distribute the load of the pipe and its contents over the full width of the support. Measures are to be taken to minimise wear of the pipes where they contact the supports.

5.9.1.3 Heavy components in the piping system such as valves and expansion joints are to be independently supported.

5.9.2 Expansion

5.9.2.1 Suitable provision is to be made in each pipeline to allow for relative movement between pipes made of plastic and the steel structure, having due regard to:

i) the difference in the coefficients of thermal expansion;
ii) deformations of the ship's hull and its structure.

5.9.2.2 When calculating the thermal expansions, account is to be taken of the system working temperature and the temperature at which assembly is performed.

5.9.3 External loads

5.9.3.1 When installing the piping, allowance is to be made for temporary point loads, where applicable. Such allowances are to include at least the force exerted by a load (person) of 100 kg at mid-span on any pipe of more than 100 mm nominal outside diameter.

5.9.3.2 Besides for providing adequate robustness for all piping including open-ended piping a minimum wall thickness, complying with 5.3.7 may require to be increased taking into account the conditions encountered during service on board ships.

5.9.3.3 Pipes are to be protected from mechanical damage where necessary.

5.9.4 Strength of connections

5.9.4.1 The strength of connections is to be not less than that of the piping system in which they are installed.

5.9.4.2 Pipes may be assembled using adhesive-bonded, welded, flanged or other joints.

5.9.4.3 Adhesive, when used for joint assembly, are to be suitable for providing a permanent seal between the pipes and fittings throughout the temperature and pressure range of the intended application.

5.9.4.4 Tightening of joints is to be performed in accordance with Manufacturers' instructions.

5.9.5 Installation of conductive pipes

5.9.5.1 In piping systems for fluids with conductivity less than 1000 pico siemens per metre (pS/m) such as refined products and distillates use is to be made of conductive pipes.

5.9.5.2 Regardless of the fluid being conveyed, plastic piping is to be electrically conductive if the piping passes through a hazardous area. The resistance to earth from any point in the piping system is not to exceed 1×10^6 Ohm. It is preferred that pipes and fittings be homogeneously conductive. Pipes and fittings having conductive layers are to be protected against a possibility of spark damage to the pipe wall. Satisfactory earthing is to be provided.

5.9.5.3 After completion of the installation, the resistance to earth is to be verified. Earthing wires are to be accessible for inspection.

5.9.6 Application of fire protection coatings

5.9.6.1 Fire protection coatings are to be applied on the joints, where necessary for meeting the required fire endurance for 5.7.3, after performing hydrostatic pressure tests of the piping system.

5.9.6.2 The fire protection coatings are to be applied in accordance with Manufacturer's recommendations, using a procedure approved in each particular case.

5.9.7 Penetration of divisions

5.9.7.1 Where plastic pipes pass through "A" or "B" class divisions, arrangements are to be made to ensure that the fire endurance is not impaired. These arrangements are to be tested in accordance with Recommendations for fire test procedures for "A", "B" and "F" bulkheads specified in Part 3 of Annex 1 to the 2010 FTP Code .

5.9.7.2 When plastic pipes pass through watertight bulkheads or decks, the watertight integrity of the bulkhead or deck is to be maintained. For pipes not able to satisfy the requirements in 5.3.3 (b), a metallic shut-off valve operable from above the freeboard deck is to be fitted at the bulkhead or deck.

5.9.7.3 If the bulkhead or deck is also a fire division and destruction by fire of plastic pipes may cause the inflow of liquid from tanks, a metallic shut-off valve operable from above the freeboard deck should be fitted at the bulkhead or deck.

5.9.8 Control during installation

5.9.8.1 Installation is to be in accordance with the Manufacturer's guidelines.

5.9.8.2 Prior to commencing the work, joining techniques are to be approved by the Classification Society.

5.9.8.3 The tests and examinations specified in this Section are to be completed before shipboard piping installation commences.

5.9.8.4 The personnel performing this work are to be properly qualified and certified to the satisfaction of the Classification Society.

5.9.8.5 The procedure of making bonds is to include:

i) materials used,
ii) tools and fixtures,
iii) joint preparation requirements,
iv) cure temperature,
v) dimensional requirements and tolerances, and
vi) tests acceptance criteria upon completion of the assembly.

5.9.8.6 Any change in the bonding procedure which will affect the physical and mechanical properties of the joint is to require the procedure to be requalified.

5.9.9 Bonding procedure quality testing

5.9.9.1 A test assembly is to be fabricated in accordance with the procedure to be qualified and it is to consist of at least one pipe-to-pipe joint and one pipe-to-fitting joint.

5.9.9.2 When the test assembly has been cured, it is to be subjected to a hydrostatic test pressure at a safety factor 2.5 times the design pressure of the test assembly, for not less than one hour. No leakage or separation of joints is allowed. The test is to be conducted so that the joint is loaded in both longitudinal and circumferential directions.

5.9.9.3 Selection of the pipes used for test assembly, is to be in accordance with the following:

a) When the largest size to be joined is 200 mm nominal outside diameter, or smaller, the test assembly is to be the largest piping size to be joined.

b) When the largest size to be joined is greater than 200 mm nominal outside diameter, the size of the test assembly is to be either 200 mm or 25% of the largest piping size to be joined, whichever is greater.

5.9.9.4 When conducting performance qualifications, each bonder and each bonding operator are to make up test assemblies, the size and number of which are to be as required above.

5.9.10 Testing after installation on board

5.9.10.1 Piping systems for essential services are to be subjected to a test pressure not less than 1.5 times the design pressure or 4 bar whichever is greater. Notwithstanding the requirement above, the requirement in 5.9.10.2 for checking leakage under operational conditions may be applied to open ended pipes (drains, effluent, etc.).

5.9.10.2 Piping systems for non-essential services are to be checked for leakage under operational conditions.

5.9.10.3 For piping required to be electrically conductive, earthing is to be checked and random resistance testing is to be conducted.

5.10 Test specification for plastic pipes

5.10.1 Scope

Section 5.10 contains requirements for the Type Approval of plastic pipes. It is applicable to piping systems, including pipe joints and fittings, made predominantly of other material than metal.

5.10.2 Documentation

The following information for the plastic pipes, fittings and joints is to be submitted for consideration and approval:

I. General information

1) Pipe and fitting dimensions
2) Maximum internal and external working pressure
3) Working temperature range
4) Intended services and installation locations
5) The level of fire endurance
6) If electrically conductive

7) Fluids Intended to be carried
8) Limits on flow rates
9) Serviceable life
10) Installation instructions
11) Details of marking.

II. Drawings and supporting documentation:

1) Certificates and reports for relevant tests previously carried out.
2) Details of relevant standards.
3) All relevant design drawings, catalogues, data sheets, calculations and functional descriptions.
4) Fully detailed section assembly drawings showing pipe, fittings and pipe connections.

III. Materials (as applicable)

1) The resin used for the construction of plastic pipes and fittings.
2) Catalyst and accelerator types and concentration employed in the case of reinforced polyester resin pipes or hardeners where epoxide resins are employed.
3) A statement of all reinforcements employed where the reference number does not identify the mass per unit area or the text number of a roving used in a filament winding process, these are to be detailed.
4) Full information regarding the type of gel-coat or thermoplastic liner employed during construction, as appropriate.
5) Cure/post-cure conditions. The cure and post-cure temperatures and time required and resin/reinforcement ratio.
6) Winding angle and orientation.
7) Joint bonding procedures and qualification results, see Cl. 5.9.8.5.

5.10.3 Testing

Testing is to demonstrate that pipes, fittings and joints for which Type Approval is sought comply with requirements of Section 5.

The standards to be used for testing of pipes, joints and fittings are given in Table 5.10.1 and Table 5.10.2.

Table 5.10.1 : Typical standards for general requirements for all systems			
	Test	**Rule clause no. / Typical Standard**	**Remarks**
1	Internal pressure (1)	5.3.3a) ASTM D 1599 ASTM D 2992 ISO 15493 or equivalent	Top, Middle, Bottom (of range) Tests are to be carried out on pipe spools made of different pipe sizes, fittings and pipe connections
2	External pressure (1)	5.3.3b) ISO 15493 or equivalent	As above, for straight pipes only
3	Axial strength	5.4	As above
4	Load deformation	ASTM D2412 or equivalent	Top, Middle Bottom (of each pressure range)
5	Temperature limitations	ISO 75 Method A GRP piping system: HDT test on each type of resin acc. To ISO 75 method A Thermoplastic piping systems: ISO 75 Method A; ISO 306 Plastics – Thermoplastic materials – Determination of Vicat softening temperature (VST) VICAT test according to ISO 2507 Polyesters with an HDT below 80°C should not be used.	Each type of resin
6	Impact resistance	ISO 9854, ISO 9653 ISO 15493 ASTM D2444 or equivalent	Representative sample of each type of construction
7	Ageing	Manufacturer's standard ISO 9142	Each type of construction
8	Fatigue	Manufacturer's standard or service experience	Each type of construction
9	Fluid absorption	ISO 8361	
10	Material compatibility (2)	ASTM C581 Manufacturer's standard	

Notes:
(1) Test to be witnessed by IRS Surveyor.
(2) If applicable.

Table 5.10.2 : Typical standards for additional requirements depending on service and/or locations of piping

	Test	Rule clause no. / Typical Standard	Remarks
1	Fire endurance (1) (2)	IMO Res. A.753(18) (as amended), Appendix 1,2	Representative samples of each type of construction and type of pipe connection
2	Flame spread (1) (2)	IMO Res. A.753(18) (as amended), Appendix 3	Representative samples of each type of construction
3	Smoke generation (2)	IMO Res. A.753(18) (as amended), Appendix 3	Representative samples of each type of construction
4	Toxicity (2)	IMO Res. A.753(18) (as amended), Appendix 3	Representative samples of each type of construction
5	Electrical conductivity (1) (2)	ASTM F1173-95 or ASTM D257, NS6126 para 11.2 or equivalent	Representative samples of each type of construction

Notes:
(1) Test to be witnessed by IRS Surveyor.
(2) If applicable.

Tests of above table are optional. However, if not carried out, the range of approved applications for the pipes will be limited accordingly (See 5.7).

Section 6

Flexible Hoses

6.1 Definition

6.1.1 Flexible hose assembly is defined as short length of integral metallic or non-metallic hose with prefabricated end fittings ready for installation.

Note: Flexible hose assemblies for essential services or containing either flammable or toxic media are not to exceed 1.5 [m] in length.

6.2 Scope

6.2.1 The requirements of 6.3 to 6.6 apply to flexible hoses of metallic or non-metallic material intended for a permanent connection between a fixed piping system and items of machinery where necessary to accommodate relative movement between the machinery and the fixed piping system. The requirements may also be applied to temporarily connected flexible hoses or hoses of portable equipment.

6.2.2 Flexible hose assemblies as defined in 6.1.1 may be accepted for use in oil fuel, lubricating, hydraulic and thermal oil systems, fresh water and sea water cooling systems, compressed air systems, bilge and ballast systems and Class III steam systems where they comply with 6.3 to 6.6. Flexible hoses are not accepted in high pressure fuel oil injection systems.

6.2.3 These requirements for flexible hose assemblies are not applicable to hoses intended to be used in fixed fire extinguishing systems.

6.3 Design and construction

6.3.1 Flexible hoses are to be designed and constructed in accordance with recognized National or International Standards. Flexible hoses constructed of rubber or plastics materials and intended for use in bilge, ballast, compressed air, oil fuel, lubricating, hydraulic

and thermal oil systems are to incorporate a single double or more closely woven integral wire braid or other suitable material for reinforcement.

Flexible hoses of plastics materials such as Teflon or Nylon, (which are used for the purposes mentioned above), where reinforcement is not possible by incorporating closely woven integral wire braid are to have suitable reinforcing material as far as practicable.

Where rubber or plastic material hoses are to be used in oil supply lines to burners, the hoses are to have external wire braid protection in addition to the reinforcement mentioned above which is provided during manufacturing. Flexible hoses for use in steam systems are to be of metallic construction.

6.3.2 Flexible hoses are to be complete with approved end fittings in accordance with manufacturer's specification. The end connections that do not have a flange are to comply with 2.7 as applicable and each type of hose/fitting combination is to be subject to prototype testing to the same standard as that required by the hose with particular reference to pressure and impulse tests. Refer Pt.4, Ch.4, 4.6.5.2.

6.3.3 The use of hose clamps and similar types of end attachments is not acceptable for flexible hoses in piping systems for steam, flammable media, starting air systems or for sea water systems where failure may result in flooding. In other piping systems, the use of hose clamps may be accepted where the working pressure is less than 5 bar and provided there are double clamps at each end connection.

6.3.4 Flexible hose assemblies intended for installation in piping systems where pressure pulses and/or high levels of vibration are expected to occur in service, are to be designed for the maximum expected impulse peak pressure and forces due to vibration. The tests required by 6.5 are to take into consideration the maximum anticipated in-service pressures, vibration frequencies and forces due to installation.

6.3.5 Flexible hose assemblies constructed of non-metallic materials intended for installation in piping systems for flammable media and sea water systems where failure may result in flooding, are to be of fire-resistant type except in cases where such hoses are installed on open decks, as defined in Pt. 6, Ch. 3, 3.2.3.3.2.2(10) and not used for fuel lines. Fire resistance is to be demonstrated by testing to ISO 15540:2016 and ISO 15541:2016.

6.3.6 Flexible hose assemblies are to be selected for the intended location and application taking into consideration ambient conditions, compatibility with fluids under working pressure and temperature conditions consistent with the manufacturer's instructions and any requirements given elsewhere in the Rules. (Refer Pt.4, Ch.3, 4.3.5; Pt.4, Ch.6, 3.4; Pt.5, Ch.26, 3.2.4.10; Pt.5, Ch.29, 2.1.2).

6.4 Installation

6.4.1 In general, flexible hoses are to be limited to a length necessary to provide for relative movement between fixed and flexibly mounted items of machinery / equipment or systems.

6.4.2 Flexible hose assemblies are not to be installed where they may be subjected to torsional deformation (twisting) under normal operating conditions.

6.4.3 The number of flexible hoses, in piping systems mentioned in 6.2.2 is to be kept to minimum and to be limited for the purpose stated in 6.2.1.

6.4.4 Where flexible hoses are intended to be used in piping systems conveying flammable fluids that are in close proximity of heated surfaces the risk of ignition due to failure of the hose assembly and subsequent release of fluids is to be mitigated as far as practicable by the use of screens or other similar protection to the satisfaction of the Surveyor.

6.4.5 Flexible hoses are to be installed in clearly visible and readily accessible locations.

6.4.6 The installation of flexible hose assemblies is to be in accordance with the manufacturer's instructions and use limitations with particular attention to the following, if any:

- Orientation
- End connection support (where necessary)
- Avoidance of hose contact that could cause rubbing and abrasion
- Minimum bend radii.

6.5 Tests

6.5.1 Acceptance of flexible hose assemblies is subject to satisfactory prototype testing. Prototype test programmes for flexible hose assemblies are to be submitted to IRS by the manufacturer and are to be sufficiently detailed to demonstrate performance in accordance with the specified standards given in 6.5.2 and 6.5.3.

6.5.2 The tests are, as applicable, to be carried out on different nominal diameters of hose type complete with end fittings for pressure, burst, impulse resistance and fire resistance in accordance with the requirements of the relevant standard. The following standards are to be used as applicable:

- ISO 6802:2018 – Rubber and plastics hoses and hose assemblies with wire reinforcements– Hydraulic impulse test with flexing.

- ISO 6803:2017 – Rubber or plastics hoses and hose assemblies – Hydraulic-pressure impulse test without flexing.

- ISO 15540:2016 – Ships and marine technology – Fire resistance of hose assemblies – Test methods.

- ISO 15541:2016 – Ships and marine technology – Fire resistance of hose assemblies – Requirements for test bench.

- ISO 10380:2012 – Pipe work – Corrugated metal hoses and hose assemblies. Other standards may be accepted where agreed by IRS.

Note:

Prototype tests are to be carried out for each size of hose assembly. However, for ranges with more than 3 different diameters, the prototype tests are to be carried out for at least:

  - the smallest diameter,

  - the largest diameter,

  - Intermediate diameters selected based on the principle that prototype tests carried out for a hose assembly with a diameter D are considered valid only for the diameters ranging between 0.5 D and 2 D.

- For fire resistance tests the specimens are to be selected in accordance with ISO 15540:2016.

6.5.3 All flexible hose assemblies are to be satisfactorily prototype burst tested to an international standard***** to demonstrate they are able to withstand a pressure not less than four times its design pressure without indication of failure or leakage.

Note : ***** The international standards, e.g. EN or SAE for burst testing of non-metallic hoses, require the pressure to be increased until burst without any holding period at 4 x MWP.

6.6 Marking

6.6.1 Flexible hoses are to be permanently marked by the manufacturer with the following details:

- Hose manufacturer’s name or trademark.
- Date of manufacture (month/year).
- Designation type reference.
- Nominal diameter.
- Pressure rating.
- Temperature rating.

Where a flexible hose assembly is made up of items from different manufacturers, the components are to be clearly identified and traceable to evidence of prototype testing.

Section 7

Hydraulic Tests on Pipes and Fittings

7.1 Hydraulic tests before installation on board

7.1.1 All Class I and II pipes and their associated fittings are to be tested by hydraulic pressure to the Surveyor's satisfaction. Further, all steam, feed, compressed air and fuel oil pipes, together with their fittings, are to be similarly tested where the design pressure is greater than 3.5 bar. The test is to be carried out after completion of manufacture and before installation on board and, where applicable, before insulating and coating.

7.1.2 Where the design temperature does not exceed 300°C, the test pressure is to be 1.5 times the design pressure.

7.1.3 For steel pipes and integral fittings for use in systems where the design temperature exceeds 300°C, the test pressure is to be as follows but need not exceed twice the design pressure:

$$P_t = 1.5P\frac{K_{100}}{K_t}\text{bar}$$

where,

P_t and P are defined in Sec.1, Cl.1.5.1

K_{100} = allowable stress for 100°C [N/mm^2]

K_t = allowable stress for the design temperature [N/mm^2].

7.1.4 In no case is the membrane stress to exceed 90 per cent of the yield stress at the testing temperature.

7.1.5 All valves and fittings are to be tested in accordance with recognized standards, but not less than 1.5 times the nominal pressure rating at ambient temperature. Valves, cocks and distance pieces intended to be fitted on the ship side below the load waterline are to be hydraulically tested to a pressure not less than 5 bar.

7.1.6 When, for technical reasons, it is not possible to carry out complete hydrotesting before assembly on board for all sections of piping, proposals for testing the remaining/ closing lengths of piping, particularly in respect of the closing seams, are to be submitted for approval.

7.1.7 Pressure testing of small bore pipes (less than 15 mm) may be waived at the discretion of IRS depending on the application.

7.2 Testing after assembly on board

7.2.1 Heating coils in tanks and liquid or gas fuel oil piping are to be tested by hydraulic pressure, after installation on board, to 1.5 times the design pressure but in no case to less than 4 bar.

7.2.2 Where pipes specified in 7.1.1 are butt welded together during assembly on board, they are to be tested by hydraulic pressure in accordance with the requirements of 7.1 after welding. The pipe lengths may be insulated, except in way of the joints made during installation and before hydraulic test is carried out.

7.2.3 The hydraulic test required by 7.2.2 may be omitted provided non- destructive tests by ultrasonic or radiographic methods are carried out on the entire circumference of all butt welds with satisfactory results. Where ultrasonic test has been carried out, the manufacturer is to provide the Surveyor with a signed statement confirming that ultrasonic examination has been carried out by an approved operator and that there were no indications of defects which could be expected to have prejudicial effect on the service performance of the piping.

7.2.4 Where bilge pipes are accepted in way of double bottom tanks or deep tanks, the pipes after fitting are to be tested by hydraulic pressure to the same pressure as the tanks through which they pass.

End of Chapter

Chapter 3

Pumping and Piping

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Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter are applicable to all ships but may be modified for ships classed for restricted services or for special services.

1.1.2 Whilst these requirements are considered to satisfy the relevant requirements of the International Convention for the Safety of Life at Sea, 1974, and applicable amendments, attention should be given to any relevant statutory requirements of the National Authority of the country in which the ship is to be registered.

1.1.3 Consideration will be given to special cases or to arrangements which are considered equivalent to those required by Rules.

1.2 Plans

1.2.1 The following plans in diagrammatic form are to be submitted for consideration before proceeding with the work:

a) General arrangement of pumps and piping systems;

b) Oil-fuel filling, transfer and service piping systems;

c) Boiler feed piping systems;

d) Steam piping and condensate piping systems;

e) Bilge and ballast piping systems;

f) Lubricating-oil piping systems;

g) Liquid cargo pumping systems;

h) Hydraulic power piping systems for essential services;

i) Compressed air piping systems;

j) Steering gear piping systems;

k) Sea water and fresh water service piping systems;

l) Vents, sounding and overflow piping systems;

m) Sanitary piping systems;

n) Fire main and fire extinguishing piping systems.

1.2.2 The plans are to include the information like size, wall thickness, maximum working pressure, temperature and material of all pipes, and type, size and material of the valves and fittings.

1.3 Materials

1.3.1 The materials to be used in piping systems are to be suitable for the service intended. In general, except where otherwise stated, pipes, valves and fittings are to be made of steel, cast iron, copper, copper alloy or other approved material. See Ch.2 also.

Materials for pipes carrying corrosive fluids are to be compatible with the fluids being handled and are to be specially considered. See Ch.2, 1.9.5.

1.3.2 Materials sensitive to heat such as aluminium, lead or plastics, are not to be used in systems essential to the safe operation of the ship.

1.4 Valves and cocks

1.4.1 All the valves and cocks are to be so designed and constructed so that the valve covers or glands will not slacken up when the valves are operated.

1.4.2 All the valves are to be designed to close with right hand (clockwise when facing the end of the stem) motion of the wheel of the valve.

1.4.3 All the valves and cocks are to be fitted in places where they are easily accessible at all times and are to be fitted with legible nameplates indicating their function in the system and their installation is to be such that it can be readily observed that they are open or closed.

1.4.4 All the valves and cocks fitted with remote control are to be provided with local manual control independent of the remote operating mechanism. The operation of the local control is not to render the remote control system inoperable.

1.4.5 The valves, cocks and other fittings which are attached directly to plating, which is required to be of watertight construction, are to be secured to the plating by means of studs screwed into the plating and not by bolts passing through clearance holes. Alternatively the studs may be welded to the plating.

1.5 Shipside fittings (other than sanitary discharges and scuppers)

1.5.1 All sea inlet and overboard discharge valves are to be fitted in either of the following ways :

a) directly on the shell plating;

b) to the plating of fabricated steel water boxes of rigid construction integral with the ship's plating;

c) to short, rigid distance pieces welded to the shell plating.

1.5.2 Valves for ship-side applications are to be installed such that the section of piping immediately inboard of the valve can be removed without affecting the watertight integrity of the hull.

1.5.3 All valves and cocks fitted directly to the shell plating are to have spigots extending through the plating. These spigot on valves may however be omitted, if valves are fitted on pads which themselves form spigots in way of plating.

1.5.4 Boiler blow-down valves or cocks are to have a protection ring fitted on the outside of the shell plating. The spigot of the valve or the cock is to be flush with the outer surface of the protection ring.

1.5.5 Valves and cocks are to be attached to the shell plating by bolts tapped into the plating and fitted with countersunk heads, or by studs screwed into heavy steel pads fitted to the plating. The stud holes are not to penetrate the pad plating.

1.5.6 Ship side valves and fittings, if made of steel or material with low corrosion resistance, are to be suitably protected against wastage.

1.5.7 Gratings are to be fitted at all openings in ship's side for inlet of seawater. The net area through the gratings is to be at least twice the area of the valves connected to the opening. Provision is to be made to clear the gratings by low pressure steam or compressed air.

1.5.8 The scantlings of valves and valve stools fitted with steam, or compressed air clearing connections are to be suitable for the

maximum pressure to which the valves and stools may be subjected.

1.6 Piping installation

1.6.1 Heavy pipes and valves are to be so supported that their weight is not taken up by connected pumps and fittings.

1.6.2 Support of the pipes is to be such that detrimental vibrations do not arise in the system.

1.6.3 Where pipes are carried through watertight bulkheads or tank tops, means are to be made to ensure the integrity of the watertightness of the compartment.

1.6.4 As far as possible, installation of pipes for water, oil, exhaust gases from oil engines or steam, is to be avoided near electric switchboards. If this is impracticable, all the joints in pipe line and valves are to be at a safe distance from the switchboards and shielded to prevent damage to switchboard.

1.6.5 Provision is to be made to take care of expansion or contraction stresses in pipes due to temperature stresses or working of the hull.

1.6.6 Expansion pieces of approved type, made of oil resistant re-inforced rubber or other approved material may be used in circulating water systems in machinery spaces.

1.6.7 All piping systems, where a pressure greater than the designed pressure could be developed, are to be protected by suitable relief valves.

1.6.8 Seawater pipes in cargo holds for dry cargoes, including cargo spaces of container ships and ro-ro ships, are to be protected from impact of cargo where they are liable to be damaged.

1.6.9 All pipes which pass through chambers intended for the carriage or storage of refrigerated cargo are to be well insulated. In case the temperature of the chamber is below 0°C the pipes are to be insulated from the ship's structure also, except at positions where the temperature of the ship's structure is always above 0°C and is controlled by outside temperature.

Air refreshing pipes leading to and from refrigerated chambers need not be insulated from the ship's structure.

1.6.10 Deck wash pipes and discharge pipes from the pumps to domestic water tanks are not to pass through cargo holds.

Section 2

Bilge and Ballast Piping Systems

2.1 General

2.1.1 All ships are to be provided with necessary pumps, suction and discharge piping and means of drainage so arranged that any compartment can be pumped out effectively, when the ship is on an even keel and/or designed trim and is either upright or has a list of not more than 5 degrees, through at least one suction, except for machinery spaces where at least two suctions are required, one of which is to be a branch bilge suction and the other is to be a direct bilge suction. Wing suctions will, generally, be necessary for this purpose, except for short narrow compartments, where a single suction may be sufficient.

2.1.2 All passenger ships are to be provided with an efficient bilge pumping plant capable of pumping from and draining any watertight compartment which is neither a permanent oil compartment nor a permanent water compartment under all practicable conditions, after a casualty, whether the ship is upright or listed.

2.2 Drainage of cargo holds

2.2.1 In ships without a double bottom one bilge suction is normally to be provided on each side of the cargo hold, at the after ends. However, if the rise of floor exceeds 5 degrees, one bilge suction at the after end situated as near the centerline as practicable may be accepted.

In ships where the bilge water cannot be entirely drained by suctions as provided above due to the fitting of watertight floors or keelsons, additional suctions may be required. Recesses and cementing/composition are to be so arranged that water in any part of the hold may find its way to the suction pipes easily.

2.2.2 In the holds of ships with double bottom, bilge suctions are to be provided on each side at the after end of the holds. Where the tanktop extends to the shipside, bilge suctions are to be led from bilge wells of at least 0.15 [m^3] capacity each, situated at the after end at each side of the hold. Where the tanktop has inverse camber, an additional bilge suction from a well situated near the centerline is also to be provided. Bilge wells of smaller size may be considered depending on the size of the vessel and the size of the compartment.

If the tanktop is of a design having discontinuities or depressions, the number and position of bilge suctions will be given special consideration.

2.2.3 Ships with one cargo hold, are to have suctions in the fore and after ends of the hold. In ships having two or more holds, bilge suctions are to be fitted in suitable positions in the fore and after ends of holds over 30 [m] in length.

2.3 Drainage from refrigerated cargo spaces

2.3.1 Provision is to be made for the continuous drainage of the inside of all insulated chambers and cooler trays.

2.3.2 Drains which are led from lower holds and cooler trays situated on the tank top are to be fitted with liquid sealed traps and non-return bilge valves.

2.3.3 Drains from 'tween deck chambers and from cooler trays which are situated well above the tank top are to be fitted with liquid sealed traps.

2.3.4 Where drains from separate chambers join a common main, the branch pipes are each to be provided with a liquid sealed trap.

2.3.5 The liquid sealed traps are to be of adequate depth and are to have a pressure head of at least 100 [mm] when connected to air ducts and 50 [mm] otherwise and arrangements are to be made for ready access to the traps for cleaning and refilling with brine.

2.3.6 Sluices, scuppers or drain pipes, which would permit drainage from compartments outside the insulated chambers into the bilges of the latter, are not to be fitted.

2.3.7 Screwed plugs or other means for blanking off scuppers draining insulating chambers and cooler trays are not to be fitted. If, however, it is especially desired to provide means for temporarily closing these scuppers, they may be fitted with shutoff valves controlled from readily accessible positions on a deck above the load waterline. An indicator is to be provided to indicate whether the valve is open or shut.

2.4 Drainage from fore and aft peaks

2.4.1 Where the peaks are used as tanks, a power pump suction is to be led to each tank, except in case of small tanks (generally not exceeding 2 [m^3] used for the carriage of domestic fresh water where hand pumps may be used.

2.4.2 The peaks may be drained by hand pumps provided the peaks are not used as tanks and they are not connected to bilge main. The suction lift is to be well within the capacity of the hand pumps and is not to exceed 7.3 [m].

In case of vessels of length 24 [m] or less and fishing vessels, after peak may be drained by means of a self closing cock situated in a well lighted and accessible position, and draining into engine room or tunnel.

2.4.3 Except as provided in 2.4.5 below, the collision bulkhead may be pierced below the bulkhead deck of passenger ships and the freeboard deck of cargo ships by not more than one pipe for dealing with fluid in the forepeak tank, provided that the pipe is fitted with a screwdown valve capable of being operated from above the bulkhead deck of passenger ships and the freeboard deck of cargo ships, the valve being located inside the forepeak at the collision bulkhead. The Administration may, however, authorize the fitting of this valve on the after side of the collision bulkhead provided that the valve is readily accessible under all service conditions and the space in which it is located is not a cargo space. Alternatively, for cargo ships, the pipe may be fitted with a butterfly valve suitably supported by a seat or flanges and

capable of being operated from above the freeboard deck. All valves are to be of steel, bronze or other approved ductile material. Valves of ordinary cast iron or similar material are not acceptable.

2.4.4 For ships constructed on or after 1 January 2024, except as provided in paragraph 2.4.5, the collision bulkhead may be pierced below the bulkhead deck of passenger ships and the freeboard deck of cargo ships by not more than one pipe for dealing with fluid in the forepeak tank, provided that the pipe is fitted with a remotely controlled valve capable of being operated from above the bulkhead deck of passenger ships and the freeboard deck of cargo ships. The valve is to be normally closed. If the remote control system should fail during operation of the valve, the valve is to close automatically or be capable of being closed manually from a position above the bulkhead deck of passenger ships and the freeboard deck of cargo ships. The valve is to be located at the collision bulkhead on either the forward or aft side, provided the space on the aft side is not a cargo space. The valve is to be of steel, bronze or other approved ductile material. Valves of ordinary cast iron or similar material are not acceptable.

2.4.5 If the forepeak is divided to hold two different kinds of liquids the Administration may allow the collision bulkhead to be pierced below the bulkhead deck of passenger ships and the freeboard deck of cargo ships by two pipes, each of which is fitted as required by 2.4.3, provided the Administration is satisfied that there is no practical alternative to the fitting of such a second pipe and that, having regard to the additional subdivision provided in the forepeak, the safety of the ship is maintained.

2.5 Drainage from tanks, cofferdams and void spaces

2.5.1 All the tanks except self-draining tanks, whether for water ballast, oil fuel, liquid cargoes etc. are to be provided with suction pipes led to suitable power pumps. The pumping plant is to be so arranged that any water or liquid within any compartment of the ship can be pumped out through atleast one suction, when the ship is on an even keel and/or designed trim and is either upright or has a list of not more than 5 degrees.

2.5.2 Where the length of the ballast tank exceeds 30 [m], an additional suction is to be provided at the forward end of the tank. Where the width of the tank is unusually large, suctions near the centerline in addition to wing suctions may be required.

2.5.3 Suction pipes from the cofferdams and void spaces are to be led to the main bilge line.

2.5.4 In ships where deep tanks may be used for either water ballast or dry cargo, provision is to be made for blanking the water ballast suction and filling when the tank is being used for carrying cargo and for blanking the bilge line when the tank is being used for carriage of water ballast.

2.6 Drainage from spaces above fore and after peaks, chain lockers and above machinery spaces

2.6.1 Provision is to be made for the drainage of chain locker and watertight compartments above the fore peak tank by hand or power pump suctions.

2.6.2 Steering gear compartments or other small enclosed spaces situated above the after peak tank are to be provided with suitable means of drainage, either by hand or power bilge suctions.

2.6.3 If the compartments referred to in 2.6.2 are adequately isolated from the adjacent 'tween decks, they may be drained by scuppers of not less than 38 [mm] bore, discharging into the tunnel (or machinery spaces in case of ships with machinery aft) and fitted with self-closing cocks situated in well lighted and visible positions. These arrangements are not applicable to passenger ships unless they are specially approved in relation to subdivision considerations.

2.7 Drainage from machinery spaces

2.7.1 Machinery spaces with double bottom:

a) Where the inner bottom extends the full length of the engine room forming bilges at the wings, one branch bilge suction and one direct bilge suction will be required at each side. Where the inner bottom extends the full length and breadth of machinery space, one branch and one direct bilge suctions are to be led to each of the two bilge wells, situated one at each side;

b) The capacity of the bilge wells is to be at least 0.15 [m^3];

c) If the inner bottom has inverse camber or discontinuity or depressions formed due to crankpins or other recesses, additional suctions are to be provided.

2.7.2 Machinery spaces without double bottom:

In ships without inner bottom, where the rise of floor is 5 degrees or more, one branch and one direct bilge suctions are to be led to positions as near the center-line as possible. Bilge suctions at wings will be required in all ships having rise of floor less than 5 degrees.

2.7.3 General requirements for machinery spaces:

a) In ships where machinery is situated at the after end, it will generally be necessary to provide suctions at the forward wings in addition to the suctions at the after end of the machinery space;

b) Special means are to be provided to prevent accumulation of bilge water under main propulsion generators and motors;

c) Where the machinery space is divided into watertight compartments, the drainage system for all compartments except for main engine room is to be same as for cargo holds except that one direct bilge suction from each watertight compartment would also be required;

d) The direct bilge suctions in machinery space(s) are to be led to independent power pump(s), and the arrangements are to be such that these direct suctions can be used independently of the main bilge line suctions.

2.7.4 Emergency bilge suction :

a) An emergency bilge suction is to be fitted to the largest independent power pump, the capacity of which is at least equal to the bilge pump. The suction is to be led to a suitable low position in the machinery space fitted with a screw-down non-return valve having the spindle so extended that the hand wheel is at least 450 [mm] above the engine room floor plates. The valve name plate is to be clearly marked - 'EMERGENCY BILGE SUCTION';

b) In motorships, the emergency bilge suction is to be led to the main cooling water pump. The suction is to be the same size as that of the cooling water pump;

c) In steamships, the main sea water circulating pump is to be fitted with an emergency bilge suction. This suction is to have diameter of at least two third of the pump suction;

d) If two or more cooling water pumps are provided for main engine(s), each capable of supplying sufficient cooling water for normal power, only one pump need be fitted with an emergency bilge suction;

e) Where the main circulating pump is not suitable for emergency bilge suction, the emergency bilge suction is to be led to the largest available independent power pump, which is not fitted with a direct bilge suction and is situated in the same compartment;

f) If the pump to which emergency bilge is connected is of self priming type, then the direct bilge suction on the same side as the emergency bilge suction may be omitted except in passenger ships.

2.8 Drainage from tunnel

2.8.1 A bilge suction, connected to the main bilge line is to be provided for draining the tunnel well. This tunnel well may extend to the outer bottom. If the shaft tunnel slopes from aft to forward end, an additional bilge suction at forward end is to be provided.

2.8.2 Internal diameter of the tunnel bilge suction pipe is not to be less than 65 [mm] except for vessels of length less than 60 [m], where tunnel bilge suction pipes of 50 [mm] bore are acceptable.

2.8.3 Where the length of the tunnel is more than 30 [m], bilge suctions are to be provided at both the forward and aft ends.

2.9 Sizes of bilge suctions

2.9.1 The internal diameter of the bilge pipes is not to be less than that found by the following formula to the nearest 5 [mm] commercial size available:

$$dm = 1.68\sqrt{L\,(B + D)} + 25\ [mm]$$

$$db = 2.15\sqrt{C\,(B + D)} + 25\ [mm]$$

where,

dm = internal diameter of bilge main [mm];

db = internal diameter of branch bilge [mm];

L = Rule length of ship [m];

B = Moulded breadth of ship [m];

C = Length of the compartment [m];

D = Moulded depth to bulkhead deck [m].

2.9.2 In any case, bilge main suction line and branch bilge suction line diameters are not to be less than 50 [mm] and the diameter of the main bilge line is not be less than that of the branch bilge line.

2.9.3 The internal diameter of the direct bilge suction is not to be less than the main bilge line.

2.9.4 In oil tankers and similar ships, where the engine room pumps do not deal with bilge drainage outside the machinery spaces, the rule diameter of the bilge main may be reduced provided the proposed cross-sectional area of the bilge main is not less than twice that required for the branch bilge suction in machinery spaces.

2.9.5 The area of each branch pipe connecting the bilge main to a distribution chest is to be not less than the sum of the areas required by the rules for the two largest branch bilge suction pipes connected to that chest, but need not be greater than that required for the main bilge line.

2.10 Bilge pumps and ejectors

2.10.1 All ships, other than passenger ships, are to be provided with at least two independent power bilge pumps. For ships of length 91.5 [m] and below, one of these pumps may be main engine driven. See 2.13 for requirements regarding passenger ships.

2.10.2 The capacity of each bilge pump is to be sufficient enough to give the water a speed of at least 122 [metres/minute] through the rule size of the main bilge line under normal working conditions.

The capacity of the bilge pump may be found by the following formula :

$Q = 5.75 \times 10^{-3} \times d^2$ [m^3/hour]

where,

Q = Capacity of pump [m^3/hour];

d = rule diameter of bilge main [mm].

2.10.3 In ships, other than passenger ships, where one bilge pump is of slightly less than rule capacity, the deficiency may be made good by an excess capacity of the other pump. In general this deficiency is to be limited to 30 per cent.

2.10.4 An ejector in conjunction with a sea water pump may be accepted as a substitute for independent power bilge pump. This, however, is not acceptable on passenger ships.

2.10.5 Pumps required for essential services are not to be connected to a common suction or discharge chest or pipe unless the arrangements are such that the working of any pumps so connected is unaffected by the other pumps being in operation at the same time.

2.10.6 General service pumps and ballast pumps may be accepted as independent power bilge pumps, provided :

a) Their capacity is adequate and in accordance with 2.10.2.

b) These pumps, together with the pipelines to which they are connected, are fitted with necessary devices to ensure that there is no risk of entry of water or oil fuel in the holds or machinery spaces.

c) The arrangements are such that at least one of these pumps is immediately available for bilge duty, when required, even when the remaining pumps are in use for other essential duties such as fire fighting.

2.10.7 For bilge pump capacity for ships having closed vehicle and ro-ro spaces and special category spaces, refer Pt.6, Ch.7, Cl. 3.6.1.4 and 3.6.1.5.

2.11 Pump types

2.11.1 The bilge pumps required by the rules are to be of self-priming type, unless an approved priming system is provided for these ships.

2.12 Bilge piping arrangements and fittings

2.12.1 Bilge pipes are not, as far as possible, to pass through double bottom tanks. If unavoidable, such bilge pipes are to be provided with welded joints or heavy flanged joints and are to be tested after fitting to the same pressure as the tanks through which they pass.

2.12.2 The parts of bilge pipes passing through deep tanks, intended to carry water ballast, fresh water, liquid cargo or fuel oil are normally to be contained in a pipe tunnel, but where this is not done, the pipes are to be of heavy gauge with welded or heavy flanged joints. The open ends of such pipes are to be fitted with non-return valves. The pipes are to be tested, after fitting, to a pressure of not less than the maximum head to which the tanks may be subjected.

2.12.3 Expansion bends, not glands, are to be fitted to pipes passing through double bottom tanks or deep tanks.

2.12.4 The intactness of the machinery spaces, bulkheads and of tunnel plating is not to be impaired by fitting of scuppers discharging into machinery spaces or tunnel from adjacent compartments which are situated below the bulkhead deck. These scuppers may, however, be led into a strongly built scupper drain tank situated in the machinery space or tunnel but closed to these spaces and drained by means of a suction of appropriate size led from the main bilge line through a screw-down non-return valve.

a) The scupper tank air pipe is to be led above the bulkhead deck and provision is to be made for ascertaining the level of the water in the tank;

b) Where one tank is used for the drainage of several watertight compartments, the scupper pipes are to be provided with screw-down non- return valves.

2.12.5 No drain valve or cock is to be fitted to the collision bulkhead. Drain valves or cocks are not to be fitted to other watertight bulkheads if alternative means of drainage are practicable. These arrangements are not permissible in passenger ships.

2.12.6 Where drain valves or cocks are fitted to bulkheads other than collision bulkhead, as permitted by 2.12.5, the drain valves or cocks are to be at all times readily accessible and are to be capable of being shut off from positions above the bulkhead deck. Indicators are to be provided to show whether the drains are open or shut.

2.12.7 Bilge pipes which are required for draining cargo or machinery spaces are to be entirely distinct from sea inlet pipes or from pipes which may be used for filling or emptying spaces where water or oil is carried. This does not, however, exclude a bilge ejection connection, a connecting pipe from a pump to its suction valve chest, or a deep tank suction pipe suitably connected through a change-over device to bilge, ballast or oil line.

2.12.8 The arrangement of pumps, valves and piping is to be such that any pump could be opened up for overhaul and repairs without affecting the operation of the other pumps.

2.12.9 The arrangement of valves, pumps, cocks and their pipe connections is to be such as to prevent the possibility of placing one watertight compartment in communication with another, or of cargo spaces, machinery spaces or other dry spaces coming in communication with the sea or the tanks. For this purpose the bilge suction pipe of any pump also having sea suction is to be fitted with a non-return valve which cannot permit communication between the bilges and the sea or the compartments in use as tanks.

2.12.10 Screw-down non-return valves are to be provided in the following fittings :

a) Bilge distribution chest valves;

b) Direct bilge suction and bilge pump connection to main line;

c) Bilge suction hose connections on the pumps or on the main line;

d) Emergency bilge suctions.

2.12.11 Bilge suction pipes from machinery spaces and shaft tunnel, except emergency bilge suction, are to be led from easily accessible mud boxes fitted with straight tail pipes to the bilges. The open ends of the tail pipes are not to be fitted with strum boxes. The mud boxes are to be provided with covers which can be easily opened and closed for cleaning purposes.

2.12.12 Strum boxes are to be fitted to the open ends of bilge suction pipes from the cargo holds. The diameter of holes from these

strum boxes is not to be more than 10 [mm] and the total area of the holes is not to be less than twice the area of the pipes.

2.12.13 Where access manholes to bilge wells are necessary, they are to be fitted as near to the suction strums as practicable.

2.12.14 Adequate distance is to be provided between the open ends of suction pipes and bilge well bottom to permit adequate and easy flow of water and to facilitate cleaning.

2.12.15 All the valves, cocks and mud boxes are to be located in easily accessible positions above or at the same level as the floor plates. Where this is unavoidable, they may be fitted immediately below the floor plates provided the floor plates are capable of being opened and closed easily and suitable name plates are fitted indicating the fittings below.

2.12.16 Where relief valves are fitted to pumps having sea connections, these valves are to be fitted in readily visible positions above the platform. The arrangement is to be such that any discharge from the relief valves will also be readily visible.

2.12.17 Where non-return valves are fitted to the open ends of bilge suction pipes in cargo holds in order to decrease the risk of flooding, they are to be of an approved type which does not offer undue obstruction to the flow of water.

2.13 Additional requirements for passenger ships

2.13.1 All passenger ships are to have at least three power bilge pumps connected to the bilge main, one of which may be attached to the propelling unit. Where the bilge pump numeral is 30 or more, one additional independent power pump is to be provided. Sanitary, ballast and general service pumps may be accepted as independent power bilge pumps if fitted with necessary connections to the bilge pumping system.

The bilge pump numeral is to be calculated as follows:

When P_1 is greater than P : bilge pump numeral

$$= 72\left[\frac{M + 2P_1}{V + P_1 - P}\right]$$

in other cases: bilge pump numeral

$$= 72\left[\frac{M + 2P}{V}\right]$$

where,

L_L = the load line length of the ship (metres), as defined in Pt.3, Ch.1, Cl.2.1.9;

M = the volume of the machinery space (cubic metres), as defined in Pt.6, Ch.1, Sec.3 that is below the bulkhead deck; with the addition thereto of the volume of any permanent oil fuel bunkers which may be situated above the inner bottom and forward of, or abaft, the machinery space;

P = the whole volume of the passenger and crew spaces below the bulkhead deck (cubic metres) which are provided for the accommodation and use of passengers and crew, excluding baggage, store, and provision rooms;

V = the whole volume of the ship below the bulkhead deck (cubic metres);

P_1 = KN,

where,

N = the number of passengers for which the ship is to be certified, and

$K = 0.056L_L$

However, where the value of KN is greater than the sum of P and the whole volume of the actual passenger spaces above the bulkhead deck, P_1 is to be taken as the above sum (i.e. the sum of P and the whole volume of the actual passenger spaces above the bulkhead deck) or two-thirds KN, whichever is the greater.

2.13.2 Where practicable, the power bilge pumps are to be placed in watertight compartments so arranged or situated that these compartments will not readily be flooded by the same damage. If the engines and boilers are in two or more watertight compartments, the pumps available for bilge service are to be distributed throughout these compartments as far as is possible.

2.13.3 On ships 91.5 [m] or more in length or having a bilge pump numeral of 30 or more, the arrangements are to be such that at least

one power pump is available for use in all ordinary circumstances in which a ship may be flooded at sea and for ships, subject to the provisions of SOLAS Regulation II-1/1.1.1.1, (i.e. for which the building contract is placed on or after 01 January 2020) in all flooding conditions derived from the consideration of minor damages as specified in SOLAS Regulation II-1/8. These requirements will be satisfied if:

a) One of the required pumps is an emergency pump of reliable submersible type having a source of power situated above the bulkhead deck; or

b) The pumps and their sources of power are so disposed throughout the length of the ship that under any condition of flooding which the ship is required to withstand, at least one pump in undamaged compartment will be available.

2.13.4 In passenger ships, independent power bilge pumps situated in machinery spaces are to have direct suctions from these spaces, except that not more than two such suctions are required in any one space, where two or more such suctions are provided there is to be at least one on the port side and one on the starboard side.

2.13.5 Provision is to be made to prevent the compartment served by any bilge suction pipe being flooded in the event of the pipe being severed, or otherwise damaged by collision or grounding in any other compartment. For this purpose where the pipe is at any part situated nearer the side of the ship than one-fifth the breadth of the ship (measured at right angles to the centerline at the level of the deepest subdivision loadline), or in a duct keel, a non-return valve is to be fitted to the pipe in the compartment containing the open end. For ships subject to the provisions of SOLAS Regulation II-1/1.1.1.1, (i.e. for which the building contract is placed on or after 01 January 2020) the deepest subdivision load line is to be taken as the deepest subdivision draught.

For the purpose of 2.13.5 and 2.13.6, 'breadth' means the maximum moulded breadth of the ship at or below the deepest sub-division loadline.

2.13.6 All the distribution boxes, cocks and valves in connection with the bilge pumping arrangements are to be in positions which are accessible at all times under ordinary circumstances. They are to be so arranged that, in the event of flooding, one of the bilge pumps may be operative on any compartment, in addition, damage to a pump or its pipe connecting to the bilge main outboard of a line drawn at one-fifth of the breadth of the ship (measured as per 2.13.5) is not to put the bilge system out of action. If there is only one system of pipes common to all the pumps, the necessary cocks or valves for controlling the bilge suctions must be capable of being operated from above the bulkhead deck. Where in addition to the main bilge pumping system an emergency bilge pumping system is provided, it is to be independent of the main system and so arranged that a pump is capable of operating on any compartment under flooding conditions, in that case only the cocks and valves necessary for the operation of the emergency system need be capable of being operated from above the bulkhead deck.

2.13.7 All valves and cocks mentioned in 2.13.6 which can be operated from above the bulkhead deck shall have their controls at their place of operation clearly marked and provided with means to indicate whether they are open or closed.

2.13.8 Where it is necessary to correct large angles of heel, in case of damage, the means adopted, are where practicable, to be self acting, but in any case where controls to cross flooding fittings are provided, they are to be operable from above the bulkhead deck.

2.13.9 Bilge wells constructed in the double bottom in connection with drainage arrangements of holds, etc. are not to extend downwards more than necessary. The depth of the well is in no case to be more than the depth less 457 [mm] (18 inches) of the double bottom at the center line nor is the well to extend below the horizontal plane passing through the point of intersection with the frame line amidships of a transverse diagonal line inclined at 25 degrees to the base line and cutting it at a point one half the ship's moulded breadth from the middle line. A well extending to the outer bottom is, however, permitted at the after end of the shaft tunnel.

2.14 Drainage arrangements on vessels not fitted with propelling machinery

2.14.1 Vessels not fitted with propelling machinery are to be provided effective hand pumps which can be operated from above the bulkhead deck or the highest convenient level which is always available. There is to be one

pump for each compartment or two pumps connected to a bilge main having at least one branch to each compartment.

2.15 Ballast system

2.15.1 Provision is to be made for ballasting and deballasting all the ballast tanks by pipe lines which are entirely separate and distinct from pipe lines used for bilging.

2.15.2 Where the length of the ballast tanks exceeds 30 [m], an additional suction is to be provided at the forward end of the tanks. Where the width of the tank is unusually large, suction near the centerline in addition to wing suctions may be required.

Section 3

Vents, Sounding, Overflow and Scuppers and Discharges Piping Systems

3.1 General

3.1.1 Reference to oil in this Section is to be taken to mean oil which has a flash point of 60°C or above (closed cup test).

3.1.2 The portions of vent, overflow and sounding pipes fitted above the weather deck are to be of steel.

3.1.3 Name plates are to be affixed to the upper ends of all vent and sounding pipes.

3.1.4 For vent pipes and sounding pipes passing through refrigerated spaces, see Pt.5.

3.1.5 For design and test heads of tanks see Pt.3.

3.2 Vent pipes

3.2.1 Vent pipes are to be fitted to all tanks, cofferdams, tunnels, sea chests and other compartments which are not fitted with alternative ventilation arrangements.

3.2.2 The vent pipes are to be fitted at the opposite end of the tank to which the filling pipes are placed and/or at the highest part of the tank. Where the tank top is of unusual or irregular profile, special consideration will be given to the number and positions of the vent pipes.

3.2.3 Tanks provided with anodes for cathodic protection, are to be provided with vent pipes at forward and aft ends.

3.2.4 Vent pipes to double bottom tanks, deep tanks extending to the shell plating or tanks which can be run up from the sea and sea chests are to be led above the bulkhead deck.

3.2.5 Air pipes for sea chests are to be fitted with shut off valve directly on sea chest.

3.2.6 Vent pipes to oil fuel and cargo oil tanks, cofferdams, all tanks which can be pumped up, shaft tunnels and pipe tunnels are to be led above the bulkhead deck and to open air.

3.2.7 Vent pipes from lubricating oil storage tanks may terminate in the machinery spaces, provided that the open ends are so situated that issuing oil cannot come into contact with electrical equipment or heated surfaces.

3.2.8 The open ends of vent pipes to oil fuel and cargo oil tanks are to be situated where no danger will be incurred from issuing oil or vapour when the tank is being filled.

3.2.9 For details regarding height and closing devices for vent pipes see Pt.3.

3.2.10 The open ends of vent pipes to oil fuel, cargo oil and ballast tanks fitted with anodes for cathodic protection, are to be fitted with a wire gauze diaphragm of incorrodible material which can be readily removed for cleaning. The clear area through the wire gauze is to be at least equal to the area of the vent pipe. Location and arrangement of vent pipes for fuel oil service, settling and lubricating oil tanks is to be such that a broken vent pipe will not directly lead to the risk of ingress of seawater splashes or rainwater.

3.2.11 In the case of all tanks which can be pumped up either by ship's pumps or by shore

pumps through a filling main, the total cross-sectional area of the vent pipes to each tank, or of the overflow pipes where an overflow system is provided, is to be not less than 25 per cent greater than the effective area of the respective filling pipes.

3.2.12 Where tanks are fitted with cross flooding connections, the vent pipes are to be of adequate area for these connections.

3.2.13 Vent pipes are not to be less than 50 [mm] bore.

3.2.14 Shaft tunnels and pipe tunnels are to be fitted with vent pipes with an internal diameter of not less than 75 [mm].

3.2.15 Vent pipes are to be self-draining under normal conditions of trim.

3.2.16 The thickness of the exposed portion of the vent pipes are to be as required in Pt.3, Ch.13, 3.2.5.

3.3 Overflow pipes

3.3.1 All tanks which can be pumped up are to be fitted with overflow pipes when the pressure head corresponding to the height of vent pipe is greater than that for which the tanks are suitable or when the sectional area of the vent pipes is less than that required by 3.2.11.

3.3.2 Oil fuel and lubricating oil tanks which can be pumped up and which have openings for fittings, for example - for a float sounding system, are to be fitted with overflow pipes. The tank openings, for such fittings, are to be situated above the highest point of the overflow piping.

3.3.3 In the case of oil fuel and lubricating oil tanks, the overflow pipe is to be led to an overflow tank of adequate capacity or to a storage tank having space reserved for overflow purposes and be in readily visible position. A sight glass is to be provided in the overflow pipe to indicate when the tanks are overflowing. Such sight glasses are to be placed only in the vertical portion of the pipes and be located in a readily visible position. Alternatively, an alarm device is to be provided to give warning either when the tanks are overflowing or when the oil reaches a predetermined level in the tanks.

3.3.4 Overflow pipes from tanks, other than oil fuel, cargo oil and lubricating oil tanks, are to be led to the open or to suitable overflow tanks arranged as in 3.3.3.

3.3.5 Where overflow from tanks, which are used for the alternative carriage of oil and water ballast, are connected to an overflow system, arrangements are to be made to prevent water ballast overflowing in to tanks containing oil.

3.3.6 The arrangement of the vent and/or overflow system is to be such that in the event of any one of the tanks being bilged, tanks situated in other watertight compartments of the ship cannot be flooded through combined vent pipes or the overflow main. In the case of vessels of 24 [m] length and less, fishing vessels and trawlers the arrangement is to be such that in the event of any one of the tanks being bilged, the other tanks cannot be flooded from the sea through combined vent pipes or the overflow main.

3.3.7 Overflow pipes are to be self-draining under normal conditions of trim.

3.3.8 For size of overflow pipes, see 3.2.11.

3.4 Sounding arrangements (Also refer Pt.6, Ch.2, 1.2.for F.O., L.O. and other flammable oils)

3.4.1 All tanks, cofferdams and pipe tunnels are to be provided with sounding pipes or other approved means for ascertaining the level of liquid in the tanks. Means of ascertaining the level of liquid in oil fuel tanks is to be of safe type. Level switches may be used provided they are contained in steel enclosure or other enclosures not destroyed by fire. Bilges of compartments which are not, at all times, readily accessible are to be provided with sounding pipes. The soundings are to be taken as near the suction pipes as practicable.

3.4.2 Oil level gauges may be used for tanks containing lubricating oil, oil fuel or other flammable liquid in place of sounding pipes, subject to the following conditions:

- In passenger ships such gauges should not require penetration below the top of the tank and their failure or overfilling of the tanks will not allow release of their contents; and

- In cargo ships, the failure of level gauges or overfilling of the tank will not allow release of their contents into the space.

The use of cylindrical gauge glasses is not acceptable. Flat type glasses may be accepted provided they are of heat resisting quality, adequately supported, protected from mechanical damage and fitted with self closing valves between the gauges and the tanks.

3.4.3 Except as permitted by 3.4.4 sounding pipes are to be led to positions above the bulkhead deck which are at all times accessible. In the case of oil fuel tanks, cargo oil tanks and lubricating oil tanks, the sounding pipes are to be led to safe positions on the open deck. The sounding pipes should not terminate in any space where the risk of ignition from spillage from sounding pipe exists. Sounding pipes are not to terminate in passenger or crew spaces.

3.4.4 Where it is impracticable to comply with 3.4.3 above, short sounding pipes may be fitted in readily accessible positions as indicated in the following:

a) Passenger ships:

- Only to double bottom tanks and cofferdams in machinery spaces.
- All short sounding pipes are to be provided with self-closing cocks as described in Pt.6, Ch.2, 1.2.2.3.5.1.

b) Cargo ships:

- Only to tanks and cofferdams in machinery spaces and shaft tunnels.
- Short sounding pipes to oil fuel, lubricating oil and other flammable oil tanks are to be provided as per Pt.6, Ch.2, 1.2.2.3.51 and 1.2.3.1(a) and (b). Short sounding pipes to other types of tanks and cofferdams may be fitted with screw caps attached by chain to the pipe or with shut-off cocks.

3.4.5 Elbow sounding pipes are not to be used for deep tanks unless the elbows and pipes are situated within closed cofferdams or within tanks containing similar liquids. They may, however, be fitted to other tanks and may be used for sounding bilges, provided that it is not practicable to lead them direct to the tanks or compartments.

a) The elbows are to be of heavy construction and adequately supported;
b) In passenger ships, elbow sounding pipes are not permissible.

3.4.6 Striking plates of adequate thickness and size are to be fitted under open ended sounding pipes. Where slotted pipes having closed ends are employed, the closing plugs are to be of substantial construction.

3.4.7 All sounding pipes are to be not less than 32 [mm] bore. All sounding pipes, whether for compartments or tanks, which pass through refrigerated spaces or the insulation thereof, in which the temperatures contemplated are 0°C or below, are to be not less than 65 [mm] bore.

3.4.8 The upper ends of all sounding pipes are to be provided with efficient closing devices. The sounding pipes are to be arranged as straight as practicable, and if curved, the curvature is to be large enough to permit easy passage of sounding rod/chain.

3.5 Scuppers and discharges (Also refer Pt.3, Ch.13)

3.5.1 Scupper and discharge pipes which are not required to be of substantial thickness in accordance with Pt.3 are to be at least 4.5 [mm] thick for pipes of 155 [mm] external diameter and less; and 6.0 [mm] for pipes of external diameter of 230 [mm] and more, intermediate sizes are to be determined by linear interpolation.

3.5.2 Scupper and discharge pipes which are required to be of substantial thickness in accordance with Pt.3 are to be at least of following thickness:

- 7 [mm] thick for pipes of external diameter of equal to or less than 80 [mm];
- 10 [mm] thick for pipes of external diameter equal to 180 [mm];
- 12.5 [mm] thick for pipes of external diameter equal to or more than 220 [mm];

Intermediate sizes are to be determined by linear interpolation.

3.5.3 Overboard discharges are to be so located as to prevent any discharge of water into the life boats.

3.6 Water level detectors on single hold cargo ships other than bulk carriers

3.6.1 Ships of load line length L_{LL} (see Pt.3, Ch.1 for definition of L_{LL}) less than 80 [m] and having a single cargo hold below the freeboard deck or cargo holds below the freeboard deck

which are not separated by at least one full watertight bulkhead, are to be fitted with water level detectors in such cargo hold/s, unless exempted by 3.6.3.

3.6.2 The water level detectors are to:

i) give an audible and visual alarm at the navigation bridge when the water level above the inner bottom in the cargo hold reaches a height of not less than 0.3 [m] and another when such level reaches not more than 15% of the mean depth of the cargo hold; and

ii) be fitted at the aft end of the hold, or above its lowest part where the inner bottom is not parallel to the design waterline. Where webs or partial watertight bulkheads are fitted above the inner bottom, fitting of additional detectors may be required.

3.6.3 The water level detectors required in 3.6.1 need not be fitted in ships having watertight side compartments each side of the cargo hold length extending vertically at least from inner bottom to freeboard deck.

Note : The water level detectors are to be of approved type. For further details regarding performance tests, sensor locations, installation and other testing requirements refer classification notes: “Type approval, installation and testing of water level detectors on ships subject to SOLAS Reg. II-1/25, II-1/25-1 and XII/12”.

3.7 Water level detectors on multiple hold cargo ships other than bulk carriers and tankers

3.7.1 Multiple hold cargo ships other than bulk carriers and tankers constructed on or after 1 January 2024 (Refer MSC.1/Circ.1673) are to be fitted with water level detectors (Refer MSC.188(79)/Rev.1) in each cargo hold intended for dry cargoes. Water level detectors are not required for cargo holds located entirely above the freeboard deck.

3.7.2 The water level detectors required by 3.7.1 are to:

.1 give audible and visual alarms at the navigation bridge, one when the water level above the bottom of the cargo hold reaches a height of not less than 0.3 m, and another at a height not less than 15% of the depth of the cargo hold but not more than 2 m; and

.2 be fitted at the aft end of the cargo holds. For cargo holds which are occasionally used for water ballast, an alarm overriding device may be installed. The audible and visual alarms are to clearly discriminate between the two different water levels detected in each hold.

3.7.3 As an alternative to the water level detector at a height of not less than 0.3 m as per 3.7.2.1, a bilge level sensor (Refer MSC.188(79)/Rev.1) serving the bilge pumping arrangements required by SOLAS Regulation 35-1 (Refer Pt.3, Ch.13, Sec 4) and installed in the cargo hold bilge wells or other suitable location is considered acceptable, subject to:

.1 the fitting of the bilge level sensor at a height of not less than 0.3 m at the aft end of the cargo hold; and
.2 the bilge level sensor giving audible and visual alarm at the navigation bridge which is clearly distinctive from the alarm given by the other water level detector fitted in the cargo hold.

Note: The water level detectors are to be of approved type. For further details regarding performance tests, sensor locations, installation and other testing requirements refer classification notes: “Type approval, installation and testing of water level detectors on ships subject to SOLAS Reg. II-1/25, II-1/25-1 and XII/12”.

Section 4

Fuel Oil Systems

4.1 General

4.1.1 Reference to oil in this Section is to be taken to mean oil which has a flash point of 60°C (closed cup test) or above.

4.2 Oil fuel tanks

4.2.1 Oil fuel tanks are to be separated from fresh water and lubricating oil tanks by means of cofferdams. For requirements in respect of protective location of fuel oil tanks, see Pt.3, Ch.1, Sec.1.6.

4.2.2 The clearance spaces between the boilers and tops of the double bottom tanks, and between the boilers and the sides of the storage tanks in which oil fuel and cargo oil is carried, are to be adequate for the free circulation of the air necessary to keep the temperature of the stored oil sufficiently below it's flash point except in the case of tanks complying with arrangement in 4.5.3 Where water tube boilers are installed, there is to be a space of at least 760 [mm] between the tanktop and the underside of the pans forming the bottom of the combustion spaces. The boilers are to be adequately lagged.

4.2.3 Oil fuel tanks are not to be located directly above the boilers or other highly heated surfaces.

4.2.4 Minimum two fuel oil service tanks for each type of fuel used on board necessary for propulsion and vital systems or equivalent arrangements (See Fig. 4.2.4 for equivalent arrangement) are to be provided with a capacity of at least 8 hours at maximum continuous rating of the propulsion plant and normal operating load of the generating plant.

Fig.4.2.4 : 'Equivalent arrangement' for F.O. service tanks (Refer Clause 4.2.4)

A) Main and auxiliary engines and boilers operating on heavy fuel (one fuel ship)

Rule requirements

HFO Serv Tank Capacity for at least 8.h each Main Eng + Aux. Eng + Aux Boiler	HFO Serv Tank Capacity for at least 8.h each Main Eng + Aux. Eng + Aux Boiler	MDO Tank For initial cold starting or repair work of engines/Boiler

Equivalent arrangements (Any fuel oil which requires post service tank heating to achieve the required injection viscosity is not regarded in this context as MDO).

HFO Serv Tank Capacity for at least 8.h each Main Eng + Aux. Eng + Aux Boiler	MDO Serv Tank Capacity for at least 8.h each Main Eng + Aux. Eng + Aux Boiler

The above arrangement only applies where main and auxiliary engines can operate with heavy fuel oil under all load conditions and, in the case of main engines, during manoeuvring.

For pilot burners of auxiliary boilers if provided, an additional MDO tank for 8 h operation may be necessary.

B) Main engines and boilers on HFO and aux. Engines on MDO (two fuel ship)

Rule requirement

HFO Serv Tank Capacity for at least 8.h each Main Eng + Aux Boiler	HFO Serv Tank Capacity for at least 8.h each Main Eng + Aux Boiler	MDO Serv Tank Capacity for at least 8.h each Aux. Eng	MDO Serv Tank Capacity for at least 8.h each Aux. Eng

Equivalent arrangement (Any fuel oil which requires post service tank heating to achieve the required injection viscosity is not regarded in this context as MDO).

HFO Serv Tank Capacity for at least 8.h each Main Eng + Aux Boiler	MDO Serv Tank Capacity for at least the highest of : 8h each Aux. Eng + Aux. Boiler or 4h each Main Eng + Aux. Eng + Aux Boiler	MDO Serv Tank Capacity for at least the highest of : 8h each Aux. Eng + Aux Boiler or 4h each Main Eng + Aux. Eng + Aux Boiler

Fig.4.2.4 : (Contd.)

Notes:

1. The arrangements in A) and B) above apply, provided the propulsion and vital systems which use two types of fuel, support rapid fuel change over and are capable of operating in all normal operating conditions at sea with both types of fuel (MDO and HFO)

2. Service tank is a fuel oil tank which contains only fuel of a quality ready for use, i.e. fuel of a grade and quality that meets the specification required by the equipment manufacturer. A service tank is to be declared as such and not to be used for any other purpose.

3. Use of a settling tank with or without purifiers alone and one service tank is not acceptable as an “equivalent arrangement” to two service tanks.

4. The equivalent arrangements described above are the interpretations followed by IRS, unless the flag administration has provided a different interpretation.

4.3 Oil fuel piping

4.3.1 Oil fuel pressure pipes are to be led, where practicable, remote from heated surfaces and electrical appliances, but where this is impracticable the pipes are to have a minimum number of joints and are to be led in well lighted and readily visible positions.

4.3.2 Pipes conveying oil heated above 60°C are to be of seamless steel or other approved material having flanged or welded joints.

4.3.3 The flanges are to be machined, and the jointing material, which is to be impervious to oil heated to 150°C, is to be the thinnest possible, so that the flanges are practically metal to metal. The scantlings of the pipes and their flanges are to be suitable for a pressure of at least 14 bar or for the design pressure, whichever is the greater.

4.3.4 The short joining lengths of pipes to the burners from the control valves at the boiler may have cone unions, provided these are of specially robust construction.

4.3.5 Flexible hoses of approved material and design may be used for the burner pipes, provided that spare length complete with couplings, are carried on board.

4.3.6 Transfer, suction and other low pressure oil pipes and all pipes passing through oil storage tanks are to be made of cast iron or steel, having flanged joints suitable for a working pressure of not less than 7 bar. The flanges are to be machined and the jointing material is to be impervious to oil. Where the pipes are of 25 [mm] bore or less, they may be seamless copper or copper alloy, except those which pass through storage tanks.

4.3.7 Pipes for tanks storing fresh water are to be separate and distinct from any pipes which may be used for oil or oily water, and are not to be led through tanks which contain oil, nor are oil pipes to be led through fresh water tanks.

4.3.8 Pipes conveying vegetable oils or similar cargo oils are not to be led through oil fuel tanks, nor are oil fuel pipes to be led through tanks containing such cargoes.

4.3.9 In passenger ships, provision is to be made for the transfer of oil fuel from any oil fuel storage or settling tank to any other oil fuel storage tank.

4.3.10 Where IRS may permit the conveying of oil and combustible liquids through accommodation and service spaces, the pipes conveying oil or combustible liquids are to be of a material approved by IRS having regard to the risk of fire.

4.3.11 Fuel Oil Sampling

.1 Requirements for sampling points are applicable to all ships of 400 GT or above, and, at the flag Administration's discretion, to ships of less than 400 GT.

.2 Sampling points are to be provided within the fuel oil system for measurement of sulphur content in fuel. The design and location of sampling points is to be such as to enable fuel oil samples to be taken easily, when the machinery is running and in a safe manner.

.3 The position of a sampling point is to be such that sample of the fuel oil is representative of the fuel oil quality passing that location within the system, by ensuring steady state flow past the sampling point.

.4 Sampling points are to be

- downstream of the fuel oil service tank, and
- before fuel enters the oil fueled machinery

.5 The sampling points are to be located in positions as far removed as possible from any heated surface or electrical equipment so as to preclude impingement of fuel oil onto such surfaces on equipment under all operating conditions.

.6 Sampling points are to comply with the requirements of MEPC.1/Circ. 864/ Rev.1 *'Guidelines for on-board sampling and verification of the sulphur content of the fuel oil used onboard ships'*.

4.4 Arrangement of valves, cocks, pumps and fittings

4.4.1 The oil fuel pumping arrangements are to be distinct from other pumping systems as far as practicable and the means provided for preventing dangerous interconnection in service are to be thoroughly effective.

4.4.2 All valves and cocks forming part of the oil fuel installation are to be capable of being controlled from readily accessible positions which, in the engine room and boiler spaces are to be above the working platform.

4.4.3 Every oil fuel suction pipe from a double bottom tank is to be fitted with a valve or a cock.

4.4.4 For oil fuel tanks which are situated above the double bottom tanks, the inlet and outlet, pipes which are connected to the tank at a point lower than the outlet of the overflow pipe or below the top of the tanks without an overflow pipe, are to be fitted with shut off valves located on the tank itself.

4.4.5 In the engine and boiler spaces, valves, mentioned in 4.4.4, are to be capable of being closed locally and from positions outside these spaces which will always be accessible in the event of fire occurring in these spaces. Instructions for closing the valves are to be indicated at the valves and at the remote control positions. The controls for remote operation of the quick closing valve for the emergency generator fuel tank are to be in a separate location from other quick closing valves for tanks in the engine room.

4.4.6 Every oil fuel suction pipe which is led into the engine and boiler spaces, from a tank situated above the double bottom, outside these spaces (including shaft tunnel, pipe

tunnel or similar spaces), is to be fitted in the machinery space with a valve controlled as in 4.4.5, except where the valve on the tank is already capable of being closed from an accessible position above the bulkhead deck.

4.4.7 Settling tanks are to be provided with means for draining water from the bottom of the tanks. If the settling tanks are not provided, the oil fuel bunkers or daily service tanks are to be fitted with water drains.

Open drains for removing water from oil tanks are to be fitted with valves or cocks of self-closing type, and suitable provision is to be made for collecting the oily discharge.

4.4.8 Where a power driven pump is necessary for transferring oil fuel, a stand by pump is to be provided and connected ready for use, or, alternatively, emergency connections may be made to another suitable power driven pump.

4.4.9 All pumps which are capable of developing a pressure exceeding the design pressure of the system are to be provided with relief valves. Each relief valve is to be in close circuit, i.e. arranged to discharge back to the suction side of the pump and to effectively limit the pump discharge pressure to the design pressure of the system. Pressure relief valves need not be fitted when the system is secured only by centrifugal pumps so designed that the pressure delivered cannot exceed that for which the piping is designed.

4.4.10 Valves or cocks are to be interposed between the pumps on the suction and discharge pipes in order that any pump may be shut off for opening up and overhaul.

4.4.11 Relief valves are to be fitted on the oil side of the heaters and are to be adjusted to operate at a pressure of 3.5 bar above that of the supply pump relief valves. The discharge from the relief valves is to be led to a safe position.

4.4.12 Drip trays are to be fitted at location where frequent leakage is expected, oil burners at furnace mouths, drain and valves under daily service tanks, filter and all other oil fuel appliances which are required to be opened up frequently for cleaning or adjustment. Where drain pipes are provided from collected leakages, they shall be led to a suitable oil drain tank not forming part of an overflow system.

4.4.13 Where MARPOL Regulation I/12A for protective location of fuel oil tanks applies (see Pt.3, Ch.1, Sec.1.6), the fuel oil lines, suction wells and valves are to satisfy the requirements of paragraph 9 and 10 of the above MARPOL Regulation.
Valves for such fuel oil tanks are to be located above the bottom at distances not less than that required for suction wells` in paragraph 10 of MARPOL Regulation I/12A. Plan showing the location of tanks, suction wells and valves, indicating the distances are to be submitted for approval.

The requirements for locations of oil fuel lines do not apply to fuel oil air escapes and overflow pipes.

4.5 Heating arrangements

4.5.1 Steam heating installation

4.5.1.1 Where steam is used for heating oil fuel, cargo oil or lubricating oil, in bunkers, tanks, heaters or separators, the exhaust drains are to discharge the condensate into an observation tank in a well lighted and accessible position where it can be readily seen whether or not it is free from oil.

4.5.2 Thermal oil installations

4.5.2.1 The inlet and outlet valves of oil-fired thermal oil heaters and exhaust-fired thermal oil heaters is to be controllable from outside the compartment where they are situated. Alternatively an arrangement for quick gravity drainage of the thermal oil contained in the system into a collecting tank is acceptable.

4.5.2.2 Heating of liquid cargoes with flash points below 60°C shall be arranged by means of a separate secondary system, located completely within the cargo area.

A single circuit system is acceptable on the following conditions:

- system is so arranged that a positive pressure in the coil shall be at least 3 [m] water column above the static head of the cargo when circulating pump is not in operation,

- the thermal oil system expansion tank shall be fitted with high and low level alarms,

- means to be provided in the thermal oil system expansion tank for detection of

flammable cargo vapours. Portable equipment is accepted,

- valves for the individual heating coils is to be provided with locking arrangement to ensure that the coils are under static pressure at all times.

4.5.2.3 The thermal oil circulating pumps is to be arranged for emergency stopping from a position outside the space where they are situated.

4.5.2.4 Vents from expansion tanks and thermal oil storage tanks of thermal oil heating plants is to be led to open deck.

4.5.2.5 Exhaust-fired thermal oil heaters

- The heater is to be so designed and installed that all tubes are easily and readily be inspected for signs of corrosion and leakage.

- Visual inspection and tightness testing of the heater tubes is to be not less than the working pressure and is to be carried out annually. Hydraulic testing is to be carried out bi-annually.

- The heater is to be fitted with temperature sensor(s) and an alarm for fire detection.

- A fixed fire extinguishing and cooling system is to be fitted. A drenching system providing copious amounts of water is acceptable. The exhaust ducting below the exhaust boiler is to be arranged for adequate collection and drainage, to prevent water flowing into the diesel engine. The drain is to be led to a suitable location.

4.5.3 Oil fuel in storage tanks is not to be heated to a temperature within 10°C below the flash point of the oil. Where oil fuel in service tanks, settling tanks and any other tanks in the supply system is heated following arrangements are to be provided:

- The length of the vent pipes from such tanks and/or a cooling device is sufficient for cooling the vapours to below 60°C; otherwise the outlet of the vent pipes is to be located at least 3 m away from a source of ignition;

- There are no openings from the vapour space of the fuel tanks in to machinery spaces (bolted manholes are acceptable);

- Any enclosed spaces are not to be located directly above such oil fuel tanks, except for vented cofferdams; and

- Electrical equipment is not to be fitted in the vapour space of the tanks, unless they are certified intrinsically safe.

4.6 Temperature indication

4.6.1 Tanks and heaters in which oil is heated are to be provided with suitable means for ascertaining the temperature of the oil.

4.7 Filling arrangements

4.7.1 The bunkering of the ship is to be carried out through a permanently fitted pipeline, provided with the required fittings and ensuring fuel delivery to all storage tanks. The open end of the filling pipe is to be led to the tank bottom.

In passenger ships fuel bunkering stations are to be isolated from other spaces and are to be efficiently drained and ventilated.

4.7.2 Provision is to be made against over-pressure in the filling pipes, served by pumps on board, and any relief valve fitted for this purpose is to discharge into an overflow tank or other safe position.

4.8 Alternate carriage of oil fuel and water ballast

4.8.1 Where it is intended to carry oil fuel and water ballast in the same tanks alternatively, the valves or cocks connecting the suction pipes of these tanks with the ballast pump and those connecting them with the oil fuel transfer pump are to be so arranged that the oil may be pumped from one tank by the oil fuel pump at the same time as the ballast pump is being used on any other tank.

Where settling or service tanks are fitted, each having a capacity sufficient to permit 12 hour normal service without replenishment, the above requirement may be dispensed with.

4.8.2 Attention is drawn to the statutory requirements issued by the National Authorities in connection with *International Convention for the Prevention of Pollution of the Sea by Oil, 1973/78.*

4.9 Deep tanks for the alternative carriage of oil, water ballast or dry cargo

4.9.1 In the case of deep tanks which can be used for the carriage of oil fuel, cargo oil, water ballast or dry cargo, provision is to be made for blank flanging the oil and water ballast filling and suction pipes, also the steam coils if retained in place, when the tank is used for dry cargo, and for blank flanging the bilge suction pipes when the tanks are used for oil or water ballast.

4.9.2 If the deep tanks are connected to an overflow system, the arrangements are to be such that liquid or vapour from other tanks cannot enter the deep tanks when dry cargo is carried in them.

4.10 Oil fuel burning arrangements

4.10.1 For boilers

4.10.1.1 Where steam is required for the main propelling engines, for auxiliary machinery for essential services, or for heating of heavy fuel oil and is generated by burning oil fuel under pressure, there are to be not less than two oil burning units, each unit comprising a pressure pump, suction and discharge filters and a heater. For auxiliary boilers, a single oil burning unit may be accepted, provided that alternative means, such as an exhaust gas boiler or composite boiler, are available for supplying steam for essential services.

4.10.1.2 In two unit installations, each unit is to be capable of supplying fuel for generating all the steam required for essential services. In installations of three or more units, the capacities and arrangements of the units are to be such that all the steam required for essential services can be maintained with any one unit out of action.

4.10.1.3 In systems where oil is fed to the burners by gravity, duplex filters are to be fitted in the supply pipeline to the burners and so arranged that one filter can be opened up when the other is in use.

4.10.1.4 Where steam is required to bring the boiler plant into operation, starting up oil fuel unit, including an auxiliary heater and hand pump, or other suitable starting up device, which does not require power from shore, is to be provided.

4.10.1.5 Where burners are provided with steam purging and/or atomizing connections, the arrangements are to be such that oil fuel cannot find its way into steam system in the event of valve leakage.

4.10.1.6 The burner arrangements are to be such that a burner cannot be withdrawn unless the oil fuel supply to that burner is shut off and that oil cannot be turned on unless the burner has been correctly coupled to the supply line.

4.10.1.7 A quick-closing master valve is to be fitted to the oil supply to each boiler manifold, suitably located so that the valve can be readily operated in an emergency, either directly or by means of remote control, having regard to the machinery arrangements and location of controls.

4.10.1.8 In the case of top-fired boilers, means are to be provided so that, in the event of flame failure, oil fuel supply to the burners is shut off automatically, and audible and visual warnings are given.

4.10.1.9 Provision is to be made, by suitable non-return arrangements, to prevent oil from spill systems being returned to the burners when oil supply to these burners has been shut off.

4.10.1.10 For alternatively fired furnaces of the boilers using exhaust gases and oil fuel, the exhaust gas inlet pipe is to be provided with an isolating device and interlocking arrangements whereby oil fuel can only be supplied to the burners when the isolating device is closed to the boiler.

4.10.2 For internal combustion engines

4.10.2.1 Filters are to be fitted in the supply lines to the main and auxiliary machinery. For non-redundant units for essential services, it must be possible to clean the filters without stopping the unit or reducing the supply of filtered oil to the unit.

For auxiliary engines one single oil fuel filter for each engine may be accepted.

4.10.2.2 Where an oil fuel booster pump is fitted, which is essential to the operation of the main engine(s), a standby pump is to be provided. The standby pump is to be

connected ready for immediate use but where two or more main engines are fitted, each with its own pump, a complete spare pump may be accepted provided that it is readily accessible and can be easily installed.

4.10.2.3 Where pumps are provided for fuel valve cooling, the arrangements are to be as in 4.10.2.2.

4.10.2.4 In multi-engine installations which are supplied from the same fuel source, means of isolating the fuel supply and spill piping to individual engines, shall be provided. The means of isolation shall not affect the operation of the other engines and shall be operable from a position not rendered inaccessible by a fire on any of the engines.

4.10.3 Oil fired galleys

4.10.3.1 The fuel supply to the burners is to be controlled from a position which will always be accessible in the event of a fire occurring in the galley.

4.10.3.2 The galley is to be well ventilated.

4.10.4 Special requirements for low sulphur Oil fuel burning arrangements:

For ships intending to use Heavy Fuel Oil (HFO) or Marine Diesel Oil (MDO) in non-restricted areas and marine fuels with a sulphur content not exceeding 0,1 % m/m and minimum viscosity of 2 cSt in emission control areas, the oil fuel burning arrangements essential for the normal operation of propulsion machinery are to comply with the following:

4.10.4.1 In non-restricted areas, ships are provided with two (2) fuel oil pumps that can each supply the fuel primarily used by the ship (i.e. HFO or MDO) in the required capacity for normal operation of the propulsion machinery.

4.10.4.2 In emission control areas one of the following configurations are to be provided:

a) Fuel oil pumps as in 4.10.4.1, provided these are each suitable for marine fuels with a sulphur content not exceeding 0.1 % m/m and minimum viscosity of 2 cSt operation at the required capacity for normal operation of propulsion machinery, or,

b) When the fuel oil pumps in 4.10.4.1 are suitable to operate on marine fuels with a sulphur content not exceeding 0.1 % m/m and minimum viscosity of 2 cSt but one pump alone is not capable of delivering marine fuels with a sulphur content not exceeding 0.1 % m/m and minimum viscosity of 2 cSt at the required capacity, then both pumps may operate in parallel to achieve the required capacity for normal operation of propulsion machinery. In this case, one additional fuel oil pump is to be provided. The additional pump is to, when operating in parallel with one of the pumps in 4.10.4.1, be suitable for and capable of delivering marine fuels with a sulphur content not exceeding 0,1 % m/m and minimum viscosity of 2 cSt at the required capacity for normal operation of the propulsion machinery, or,

c) In addition to 4.10.4.1, two separate fuel oil pumps are provided, each capable of and suitable for supplying marine fuels with a sulphur content not exceeding 0,1 % m/m and minimum viscosity of 2 cSt at the required capacity for normal operation of propulsion machinery.
Notes:

1. If a marine distillate grade fuel with a different maximum sulphur content is specified by regulation for the area of operation of the ship (e.g., ECA, specific ports or local areas, etc.) then that maximum is to be applied.

2. In the case of vessels with class notation SYJ, requirement in Pt. 5, Ch. 22 for activating of alarm when pumps are automatically started is applicable regardless of the pump arrangement.

3. Where electrical power is required for the operation of propulsion machinery, the requirements are also applicable for machinery for power generation when such machinery is supplied by common fuel supply pumps.

4. The minimum viscosity of 2cSt specified above is the value corresponding to the standard reference temperature.

4.11 Remote stop of oil fuel pumps and fans

4.11.1 Emergency stop of power supply to the following pumps and fans is to be arranged from a central place outside the engine and boiler room (Also refer Ch.8, Cl. 2.15.2) and Pt.6, Ch.2, Sec.2:

- oil fuel transfer pump;

- oil fuel booster pump;
- nozzle cooling pumps when oil fuel is used as coolant;
- oil fuel purifiers;
- pumps for oil-burning installations;
- fans for forced draught to boilers;
- fans for ventilation of engine and boiler rooms.

4.12 Fuel oil treatment system

4.12.1 Vessel fitted with a fuel oil treatment system complying with the requirements of IRS Classification Note "*Fuel Oil Treatment Systems*" will be assigned additional class notation **FT**.

Section 5

Steam Piping and Condensate Piping Systems

5.1 Expansion and drainage

5.1.1 In all steam piping systems, ample provision is to be made for expansion and contraction to take place without unduly straining the pipes.

5.1.2 The slope of the pipes and the number and position of the drain valves or cocks are to be such that water can be efficiently drained from any portion of the steam piping system, when the ship is on even keel and/or designed trim and is either upright or has a list of not more than 5 degrees.

5.1.3 Drain valves or cocks are to be readily accessible.

5.2 Steam pipes in way of holds, shaft and pipe tunnels

5.2.1 In general, steam pipes are not to be led through spaces which may be used for carrying cargo. But where it is impracticable to avoid steam pipes passing through cargo spaces, the pipes are to be efficiently insulated, secured and protected against mechanical damage. Plans for such arrangements are to be submitted for consideration.

5.2.2 Where the steam pipes pass through shaft tunnels and pipe tunnels, these are to be efficiently secured and insulated in such a way that the lagging surface temperature does not exceed 60°C.

5.3 Relief valves

5.3.1 Auxiliary steam lines not designed to withstand boiler pressure are to be fitted with relief valves, installed on the low pressure side, immediately after pressure reducing valve, having sufficient discharge capacity to protect the piping against excessive pressure.

5.4 Steam supply to auxiliaries

5.4.1 The steam supply to steering gear and essential auxiliaries is to be so arranged that it is not affected in the event of main steam supply to propulsion machinery or cargo oil pumps being shut off.

5.5 Steam for fire extinguishing in cargo holds

5.5.1 Where steam is used for fire extinguishing in cargo holds, provision is to be made to prevent damage to cargo by leakage of steam or drip. Details of proposed precautionary measu-res are to be submitted for approval.

5.6 Condensate pumps

5.6.1 Two or more extraction pumps are to be provided for dealing with the condensate from the main and auxiliary condensers, at least one of which is to be independently driven. Where one of the independent feed pumps is fitted with direct suctions from the condensers and a discharge to the feed tank, it may be accepted as an independently driven extraction pump.

5.6.2 Condensate pumps are to be provided with valves or cocks, interposed between the pumps and the suctions and the discharge pipes, so that any pump may be opened up for overhaul while the others continue in operation.

Section 6

Boiler Feed Piping Systems

6.1 Feed water piping

6.1.1 Two separate means of feed are to be provided for all main and auxiliary boilers, which are required for essential services, with the exception of boilers in which steam is generated exclusively by exhaust gases or steam, where one means of feed will be accepted provided an alternative steam supply is available.
IR6.1.1 Where a steam generating system consists of two or more adequately sized boilers, one independent feed water supply pipe to each boiler from two feed pumps is considered acceptable.

6.2 Feed pumps

6.2.1 Two or more feed pumps are to be provided of sufficient capacity to supply the boilers under full load conditions with any one pump out of action. At least one of these pumps is to be independently driven.

6.2.2 In multiple screw ships in which there is only one independently driven feed pump, each main engine is to be provided with a feed pump.

6.2.3 Where only one independent feed pump has been provided, a harbour feed pump or an injector is to be fitted to provide second means of feed to the boilers which are in use when the main engines are not working.

6.2.4 The harbour feed pump, as required by 6.2.3, may be used as general service pump, provided that it is not connected to tanks containing oil, or to tanks, cofferdams and bilges which may contain oily water.

6.3 Valves, cocks and fittings

6.3.1 Independent feed pumps, required for feeding the main boilers, are to be fitted with automatic regulators for controlling their output.

6.3.2 The valves on the suction pipes from hotwell or condenser and the feed drain tanks or filter are to be of the non-return type.

6.3.3 Feed pumps are to be provided with valves or cocks, interposed between the pumps and the suction and the discharge pipes, so that any pump may be opened up for overhaul while the others continue in operation.

6.4 Fresh-water connections

6.4.1 Feed pumps for water-tube boilers are to have fresh-water connections only.

6.5 Feed water tanks

6.5.1 All ships fitted with boilers are to be provided with storage space for reserve feed water. Capacity of these tanks is to be at least twice the hourly evaporation rate of the boilers.

6.5.2 Feed water tanks are to be separated from oil tanks by cofferdam.

6.6 Evaporators

6.6.1 For main boilers, one or more evaporators, of adequate capacity, are to be provided.

Section 7

Engine Cooling Water Systems

7.1 General

7.1.1 Centrifugal cooling water pumps are to be installed as low as possible in the ship.

7.2 Cooling water main supply

7.2.1 Provision is to be made for an adequate supply of cooling water to the main propelling machinery and essential auxiliary engines, also to lubricating oil and fresh water coolers and air coolers for electric propelling machinery, where these coolers are fitted. The cooling water pump(s) may be worked from the engines or be driven independently.

7.3 Cooling water standby supply

7.3.1 Provision is also to be made for a separate supply of cooling water from a suitable independent pump of adequate capacity.

7.3.2 The following arrangements are acceptable, depending on the purpose for which the cooling water is intended:

a) Where only one main engine is fitted, the standby pump is to be connected ready for immediate use;

b) Where more than one main engine is fitted, each with its own pump, a complete spare pump of each type may be accepted;

c) Where a sea inlet scoop arrangement is fitted, and there is only one independent condenser circulating pump, a further pump, or a connection to the largest available pump suitable for circulation duties, is to be fitted to provide the second means of circulation when the ship is maneuvering. The pump is to be connected ready for immediate use;

d) Where fresh water cooling is employed for main/or auxiliary engines, a standby means of cooling need not be fitted if there are suitable emergency connections from a salt water system;

e) Where each auxiliary is fitted with a cooling water pump, standby means of cooling need not be provided for auxiliaries. Where, however a group of auxiliaries is supplied with cooling water from a common system, a standby cooling water pump is to be provided for this system. This pump is to be connected ready for immediate use and may be a suitable general service pump.

7.3.3 When selecting a pump for standby purposes, consideration is to be given to the maximum pressure which it can develop if the overboard discharge valve is partly or fully closed and, when necessary, condenser doors, water boxes, etc. are to be protected by an approved device against inadvertent over pressure.

7.4 Relief valves on cooling water pumps

7.4.1 Where cooling water pumps can develop a pressure head greater than the design pressure of the system, they are to be provided with relief valves on the pump discharge to effectively limit the pump discharge pressure to the design pressure of the system.

7.5 Sea inlets for cooling water pumps

7.5.1 Sea-water cooling systems for main and auxiliary machinery are to be connected to at least two cooling water inlets, preferably on opposite sides of the ship.

7.5.2 Where sea water is used for the direct cooling of main engines and auxiliaries, the sea water suction pipes are to be provided with strainers which can be cleaned without interrupting the cooling water supply.

Section 8

Lubricating Oil Piping Systems

8.1 General

8.1.1 The lubricating systems are to be so arranged that they will remain efficient with the ship inclined from the normal to any angle up to 15° transversely and when pitching 10° longitudinally and when rolling up to 22.5° from the vertical.

8.1.2 Lubricating oil systems are to be entirely separated from other systems. This requirement, however, does not apply to hydraulic governing and maneuvering systems for main and auxiliary engines.

8.1.3 Lubricating oil tanks are to be separated from other tanks containing water, fuel oil or cargo oil, by means of cofferdams.

8.1.4 For fire protection requirements also refer Pt.6, Ch.2, 1.2.3.

8.2 Pumps

8.2.1 Where lubricating oil for the main engine(s) is circulated under pressure, a standby lubricating oil pump is to be provided where the following conditions apply:

a) The lubricating oil pump is independently driven and the total output of the main engine(s) exceeds 370 [kW];

b) One main engine with its own pump is fitted and the output of the engine exceeds 370 [kW];

c) More than one main engine each with its own lubricating oil pump is fitted and the output of each engine exceeds 370 [kW].

8.2.2 The standby pump is to be of sufficient capacity to maintain the supply of oil for normal conditions with any one pump out of action. The pump is to be fitted and connected ready for immediate use, except that where the conditions referred to in 8.2.1(c) apply, a complete spare pump may be accepted. In all cases satisfactory lubrication of the engines is to be ensured while starting and maneuvering.

8.2.3 Similar provisions to those of 8.2.1 and 8.2.2 are to be made where separate lubricating oil systems are employed for piston cooling, reduction gearing, oil operated couplings and controllable pitch propellers, unless approved alternative arrangements are provided. Where the oil glands for stern tubes are provided with oil circulating pump, and the continuous running of this pump is necessary during normal operation, then a standby pump for this purpose is to be provided.

8.2.4 Independently driven rotary type pumps are to be fitted with non- return valves on the discharge side of the pumps.

8.2.5 A relief valve in close circuit is to be fitted on the pump discharge if the pump is capable of developing a pressure exceeding the design pressure of the system, the relief valve is to effectively limit the pump discharge pressure to the design pressure of the system.

8.3 Control of pumps and alarms

8.3.1 The power supply, to all independently driven lubricating oil service pumps is to be capable of being stopped from a position outside the space which will always be accessible in the event of fire occurring in the compartment in which they are situated, as well as from the compartment itself.

8.3.2 All main and auxiliary engines and turbines intended for essential services are to be provided with means of indicating the lubricating oil pressure supply to them. Alarms for reduction in pressure of the lubricating oil supply are to be actuated from the outlet side of any restrictions such as filters, coolers etc. For main and auxiliary internal combustion engines, above 37 [kW] an alarm device with audible and visual signal for failure of lubricating system is to be provided.

8.4 Emergency supply for propulsion turbines and propulsion turbogenerators

8.4.1 A suitable emergency supply of lubricating oil is to be arranged to come automatically into use in the event of low oil pressure.

The emergency source of supply is to be independent of power from main switchboard.

8.4.2 The emergency supply may be obtained from a gravity tank containing sufficient oil to maintain adequate lubrication for not less than 6 minutes, and, in case of propulsion turbo-generators, until the unloaded turbine comes to rest from its maximum rated running speed.

8.5 Filters

8.5.1 In systems, where lubricating oil is circulated under pressure, provision is to be made for efficient filtration of the oil. For non-redundant units, for essential services, it must be possible to clean the filters without stopping the unit or reducing the supply of filtered oil to the units.

8.5.2 In the case of propulsion turbines and their gears, arrangements are to be made for lubricating oil to pass through magnetic strainers and fine filters.

8.6 Valves and cocks on lubricating oil tanks

8.6.1 The provisions of 4.4.4, 4.4.5 and 4.4.6 are also to apply to lubricating oil tanks except those having a capacity less than 500 litres, storage tanks on which valves are closed during the normal operation mode of the ship, or where it is determined that an unintended operation of a quick closing valve on the oil lubricating tank would endanger the safe operation of the main propulsion and essential auxiliary machinery.

8.6.2 Where an engine lubricating oil drain tank extends to the bottom shell plating in ships that are required to be provided with a double bottom, a shutoff valve is to be fitted in the drain pipe between the engine crank case and the double bottom lubricating oil tank. This valve is to be capable of being closed from an accessible position above the level of the lower platform (floor plates).

End of Chapter

Chapter 4

Prime Movers and Propulsion Shafting Systems

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Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter are applicable to all ships but may be modified for ships classed for restricted services or for special services.

1.1.2 Prime movers of less than 100 [kW] capacity may be accepted based upon manufacturer's certificate provided the prime movers are of an approved type.

1.1.3 Attention is drawn to any relevant statutory requirements of the country in which the ship is to be registered.

1.1.4 Prime movers and power transmission systems not specified in this Chapter will be specially considered by IRS.

1.2 Materials

1.2.1 Materials intended for the main parts of the prime movers and power transmission systems are to be manufactured and tested in accordance with the requirements of Pt.2.

1.3 Turning gear

1.3.1 Arrangements are to be provided to turn the prime movers of main propulsion systems, reduction gears and auxiliary drives. Suitable interlocks are to be provided.

1.4 Power ratings

1.4.1 Where requirements to dimensions in this Chapter are based on power and revolutions per minute, the values to be applied are maximum continuous power measured on engine output shaft and corresponding revolutions per minute.

1.4.2 The maximum continuous power for which the engine is to be approved is defined in ISO 3046/I for oil engines and ISO 3977 for gas turbines.

1.4.3 For plants where overload is used frequently, the scantling criteria may have to be based on over load due to accumulated fatigue.

1.5 Overload capacity

1.5.1 Prime movers of electric generators are to be capable of developing 10 per cent overload for a short period of time (15 minutes). The overload power refers to the power of the generator set.

1.6 Certification of machinery

1.6.1 The certification of machinery is to be in accordance with requirements given in Chapter 1, Section 4 of this part.

1.7 Type approval program

1.7.1 Products that can be consistently manufactured to the same design and specification may be type approved in accordance with "IRS certification scheme for type approval of products".

1.7.2 For non mass-produced engines the requirements given in subsection 4.12 of this chapter are applicable for type approval.

1.7.3 For mass-produced machinery, "Type approval certification scheme for machinery manufactured by mass production system" is to be referred to.

Section 2

Steam Turbines

2.1 Scope

2.1.1 The requirements of this Section are applicable to steam turbines for main propulsion and to those for essential auxiliary services.

2.2 Plan and particulars

2.2.1 The following plans are to be submitted for consideration, together with particulars of materials, maximum shaft powers and revolutions per minute. The pressures and temperatures applicable at maximum shaft power and under the emergency conditions are to be stated or indicated on the plans.

- General technical specifications;
- General arrangement;
- Sections showing rotors, bearings and casings;
- Sections showing typical blades and blade fastenings;
- Integrated steam and drain piping;
- Foundation and fastening;
- Block diagram with functional description of control and monitoring system;
- Lubricating oil system;
- Operating instructions manual; and
- Test programmes.

If required, the following information is also to be submitted:

- Vibration calculation of rotors and/or blades;
- Performance characteristics with functional description;
- Specification of reliability and availability; and
- Analysis proving the specified reliability and availability.

2.2.2 Full particulars of the means proposed for emergency propulsion are to be submitted.

2.2.3 Where rotors and castings are of welded construction, details of the welded joints are also to be submitted for consideration.

2.3 Materials

2.3.1 In the selection of materials, consideration is to be given to their creep strength, corrosion resistance and scaling properties at working temperatures to ensure satisfactory performance and long life under service condition.

2.3.2 Grey cast iron is not to be used for temperatures exceeding 260°C.

2.3.3 Turbine rotors and discs are to be of forged steel. For carbon and carbon-manganese steel forgings, the specified minimum tensile strength is to be selected within the limits of 400 and 600 [N/mm^2]. For alloy steel rotor forgings, the specified minimum tensile strength is to be selected within the limits of 500 and 800 [N/mm^2]. For discs and other alloy steel forgings, the specified minimum tensile strength is to be selected within the limits of 500 and 1000 [N/mm^2].

2.3.4 For alloy steels, details of the proposed chemical composition, heat treatment and mechanical properties are to be submitted for approval.

2.3.5 When it is proposed to use material of higher tensile strength, full details are to be submitted for approval.

2.4 Design and construction

2.4.1 In the design and arrangement of turbine machinery, adequate provision is to be made for the relative thermal expansion of the various turbine parts, and special attention is to be given to minimizing casing and rotor distortion under all operating conditions.

2.4.2 Turbine bearings are to be so disposed and supported that lubrication is not adversely affected by heat flow from adjacent hot parts of the turbine. Effective means are to be provided for intercepting oil leakage and preventing oil from reaching high temperature glands and casings and steam pipes pockets are to be sufficiently large to prevent excessive accumulation and leakage of oil.

2.4.3 The design analysis, if required, is to include a full vibration analysis with measurements as necessary, covering vibration of the rotor as well as blade vibrations.

2.4.4 Turbine rotors, cylinders and associated components fabricated by means of welding will be considered for acceptance if constructed by firms whose works are properly equipped to undertake welding to equivalent standards, for rotors and cylinders respectively, to those required by the Rules for Class 1 welded pressure vessels, See Ch.10.

2.4.5 Before work is commenced, manufacturers are to submit for consideration details of proposed welding procedures and their proposals for routine examination of joints by non-destructive means.

2.4.6 Where it is proposed to construct rotors from two or more forged components joined by welding, full details of the chemical composition, mechanical properties and heat treatment of the materials, together with particulars of the welding consumables, an outline of the welding procedure, method of fabrication and heat treatment, are to be submitted for consideration.

2.4.7 Joints in rotors and major joints in cylinders are to be designed as full-strength welds and for complete fusion of the joint.

2.4.8 Adequate preheating is to be employed for mild steel cylinders and components and where the metal thickness exceeds 44 [mm], and for all low alloy steel cylinders and components and for any part where necessitated by joint restraint.

2.4.9 Stress relief heat treatment is to be applied to all cylinders and associated components on completion of the welding of all joints and attached structures. For details of stress relief procedures, temperature and duration, See Ch.10.

2.4.10 Surveyors are to be satisfied that the desired quality of welding is attainable with the proposed welding equipment and procedure, and for this purpose test specimens representative of the welded joints are to be provided for radiographic examination and mechanical tests.

2.4.11 Smooth fillets are to be provided at abrupt changes of section of rotors, spindles, discs, blade roots and tensions. The rivet holes in blade shrouds are to be rounded and radiused on top and bottom surfaces, and tenons are to be radiused at their junction with blade tips. Balancing holes in discs are to be well rounded and polished.

2.4.12 Surveyors are to be satisfied as to the workmanship and riveting of blades to shroud bands, and that the blade tenons are free from cracks, particularly with high strength blade material. Test samples are to be sectioned and examined, and pull-off tests made if considered necessary by Surveyors.

2.4.13 Main turbine rotor discs fitted by shrinking are to be secured with keys, dowels or other approved means.

2.5 Vibration

2.5.1 Care is to be taken in the design and manufacture of turbine rotors, rotor discs and blades to ensure freedom from undue vibration within the operating speed range. Consideration of blade vibration should include the effect of centrifugal force, blade root fixing, metal temperature and disc flexibility where appropriate.

2.6 External influences

2.6.1 Pipes and ducts connected to turbine casings are to be so designed that no excessive thrust loads or moments are applied by them to the turbines. Grating and any fittings in way of sliding feet or flexible-plate supports are to be so arranged that casing expansion is not restricted. Where main turbine seatings incorporate a tank structure, consideration is to be given to the temperature variation of the tanks in service to ensure that turbine alignment will not be adversely affected.

2.7 Steam supply and water system

2.7.1 In the arrangement of the gland sealing system, the pipes are to be made self-draining and every precaution is to be taken against the possibility of condensed steam entering the glands and turbines. The steam supply to the gland sealing system is to be fitted with an effective drain trap. In the air ejector re-circulating water system, the connection to the condenser is to be so located that water cannot impinge on the L.P. rotor or casing.

2.8 Turning gear

2.8.1 The turning gear for all propulsion turbines is to be power-driven and if electric, is to be continuously rated.

2.9 Safety arrangement

2.9.1 Overspeed protective devices

a) An overspeed protective device is to be provided for main and auxiliary turbines to shut off the steam automatically and prevent the maximum designed speed being exceeded by more than 15 per cent.

b) Where two or more turbines of a compound main turbine installation are separately coupled to the same main gear wheel, and one over-speed protective device is provided, this is to be fitted to the L.P. ahead turbine. Hand trip gear for shutting off the steam in an emergency is to be provided at the maneuvering platform.

c) Where exhaust steam from auxiliary systems is led to the main turbine, activation of the over-speed protective device is to cut off this steam supply to the main turbine.

2.9.2 Speed governors

a) Where a turbine installation incorporates a reverse gear, electric transmission or reversible propeller, a speed governor in addition to, or in combination with, the overspeed protective device is to be fitted, and is to be capable of controlling the speed of the unloaded turbine without bringing the overspeed protective device into action.

b) Auxiliary turbines intended for driving electric generators are to be fitted with speed governors which, with fixed setting, are to control the speed within 10 per cent momentary variation and five per cent permanent variation when full load is suddenly taken off or put on. The permanent speed variations of alternating current machines intended for parallel operations are to equal within a tolerance of $\pm$ 0.5 per cent.

2.9.3 Low vacuum and over pressure protective device

a) Sentinel relief valves are to be provided at the exhaust ends of all turbines, to provide a warning to personnel of allowable pressure being exceeded. The valve discharge outlets are to be visible and suitably guarded if necessary to avoid injury to personnel. Where a low vacuum cut-out device is provided, the sentinel relief valve at the L.P. exhaust may be omitted.

b) Sentinel relief valves are to be provided at the exhaust ends of all auxiliary turbines and the valve discharge outlets are to be visible and suitably guarded if necessary. Low vacuum or over pressure cut-out devices, as appropriate, are also to be provided for auxiliary turbines not installed with their own condensers.

c) When for auxiliary turbines, the inlet steam pressure exceeds the pressure for which the exhaust casing and associated piping

up to exhaust valve are designed, means to relieve the pressure are to be provided.

2.10 Bled steam connections

2.10.1 Non-return or other means, which will prevent steam and water returning to the turbines, are to be fitted in bled steam connections.

2.11 Steam strainers

2.11.1 Efficient steam strainers are to be provided close to the inlets to ahead and astern high pressure turbines, or alternatively at the inlets to the maneuvering valves.

2.12 Emergency arrangements

2.12.1 Lubricating oil failure

a) Arrangements are to be made for the steam to the ahead propulsion turbines to be automatically shut off in the event of failure of the lubricating oil pressure; however, steam is to be made available at the astern turbine for braking purposes in such an emergency.

b) Auxiliary turbine arrangements are to be such that steam supply is automatically shut off in the event of failure of the lubricating oil pressure.

c) Main turbines are to be provided with a satisfactory emergency supply of lubricating oil which will come into use automatically in case of lubricating oil pressure falling below a predetermined value. The emergency supply may be obtained from a gravity tank containing sufficient oil to maintain adequate lubrication until the turbine is brought to rest or by equivalent means. If emergency pumps are used these are to be so arranged that their operation is not affected by failure of the power supply. Suitable arrangements for cooling the bearings after stopping may also be required.

2.12.2 Single screw ships

a) In single screw ships fitted with cross compound steam turbine installations in which two or more turbines are separately coupled to the same main gear wheel, the arrangements are to be such as to enable safe navigation, when the steam supply to any one of the turbines is required to be isolated. For this emergency operation purpose, the steam may be led direct to the L.P. turbine and either the H.P. or I.P. turbine can exhaust direct to the condenser. Adequate arrangements and controls are to be provided for these operating conditions to ensure that the pressure and temperature of the steam will not exceed those which the turbines and condenser can safely withstand.

b) The necessary pipes and valves for these arrangements are to be readily available and properly marked. A fit up test of all combinations of pipes and valves is to be performed prior to the first sea trials.

c) The permissible power/speeds when operating without one of the turbines (all combinations) is to be specified and information provided on board.

d) The operation of the turbines under emergency conditions is to be assessed for the potential influence on shaft alignment and gear teeth loading conditions.

2.12.3 Single main boiler

a) Ships intended for unrestricted service, fitted with steam turbines and having a single main boiler, are to be provided with means to ensure emergency propulsion in the event of failure of the main boiler.

2.13 Tests and equipment

2.13.1 Stability testing of turbine rotors

a) All solid forged H.P. turbine rotors intended for main propulsion service where the inlet steam temperature exceeds 400°C are to be subjected to at least one thermal stability test. This requirement is also applicable to rotors constructed from two or more forged components joined by welding. The test may be carried out at the forge or turbine builders' works (a) after heat treatment and rough machining of the forging or (b) after final machining, or (c) after final machining and blading of the rotor. The stabilizing test temperature is to be not less than 28°C above the maximum steam temperature to which the rotor will be exposed and not more than the tempering temperature of the rotor material.

b) Where main turbine rotors are subjected to thermal stability tests at both forge and

turbine builders' works the foregoing requirements are applicable to both tests. It is not required that auxiliary turbine rotors be tested for thermal stability, but if such tests are carried out, the requirements for main turbine rotors will be generally applicable.

2.14 Balancing

2.14.1 All rotors as finished-bladed and complete with half-coupling are to be dynamically balanced to the Surveyor's satisfaction, in a machine of sensitivity appropriate to the size of rotor.

2.15 Hydraulic tests

2.15.1 Maneuvering valves are to be tested to twice the working pressure. The nozzle boxes of impulse turbines are to be tested to 1.5 times the working pressure.

2.15.2 The cylinders of all turbines are to be tested to 1.5 times the working pressure in the casing, or to 2.0 bar whichever is greater.

2.15.3 For test purposes, the cylinders may be subdivided with temporary diaphragms for distribution of test pressures.

2.15.4 Condensers are to be tested in the steam space to 1.0 bar. The water space is to be tested to the maximum pressure which the pump can develop at ship's full draught with the discharge valve closed plus 0.7 bar with a minimum test pressure of 2.0 bar. Where the operating conditions are not known, the test pressure is to be not less than 3.4 bar.

2.16 Indicators for movement

2.16.1 Indicators for determining the axial position of rotors relative to their casings, and for showing the longitudinal expansion of casings at the sliding feet, if fitted, are to be provided for main turbines. The latter indicators should be fitted at both sides and be readily visible.

2.17 Weardown gauges

2.17.1 Main and auxiliary turbines are to be provided with bridge weardown gauges for testing the alignment of the rotors.

Section 3

Gas Turbines

3.1 Scope

3.1.1 The requirements of this Section are applicable to gas turbines for main propulsion and to those for essential auxiliary services. These requirements do not apply to exhaust gas turbo-blowers.

3.1.2 The spare gear for gas turbines has not been indicated in view of the variation in turbine design and service conditions. In the circumstances, a list of proposed spare gear is to be submitted for consideration.

3.2 Plans and particulars

3.2.1 Following data is to be submitted:

- Rated power;
- Rated speed;
- Maximum turbine entry temperature at which rated power can be achieved;
- Mass and velocity of rotating elements;
- Balancing data;
- Type test schedule, measurements and data;
- Manufacturer’s shop test schedule.

3.2.2 The following plans are to be submitted for consideration together with particulars of materials.

- General technical specifications including effect of inlet temperature on power;
- General arrangement;
- Sectional assembly;
- Casings;
- Combustion chambers and heat exchangers;
- Gasifiers;

- Regenerators or recuperators;

- Rotors (turbine and compressor), including discs, blades, bearings couplings, and clutches;

- Sections showing blades, blade fastenings and sealings;

- Oil fuel and lubricating oil systems including controls and safety devices;

- Governor arrangements;

- Enclosure configurations (where applicable);

- Starting arrangements;

- Air intake and exhaust system;

- Control systems, safety systems and devices and associated failure mode and effect analysis;

- Details of proposed automatic safety devices to safeguard against hazardous conditions arising in the event of malfunctions in the gas turbine installation together with failure mode and effect analysis(FMEA);

- Vibration monitoring systems; and

- All other critical systems such as fire fighting arrangements for enclosures, ventilation, lighting, smoke detector etc.

3.2.3 Where applicable, plans are to indicate the most onerous pressures and temperatures to which each component is subject.

3.2.4 Details are to be submitted, where applicable, of high temperature characteristics of the materials, including (at the working temperatures) the associated creep rate and rupture strength for the designed service life, fatigue strength, corrosion resistance and scaling properties. Particulars of heat treatment, including stress relief, where applicable, are to be submitted.

3.3 Materials

3.3.1 Rotors and discs are to be of forged steel. For carbon and carbon- manganese steel forgings, the specified minimum tensile strength is to be selected within the limits of 400 and 600 [N/mm^2]. For alloy steel rotor forgings, the specified minimum tensile strength is to be selected within the limits of 500 and 800 [N/mm^2]. For discs and other alloy steel forgings, the specified minimum tensile strength is to be selected within the limits of 500 to 1000 [N/mm^2].

3.3.2 For alloy steels, specifications giving the proposed chemical composition and heat treatment are to be submitted for approval.

3.3.3 When it is proposed to use a material of higher tensile strength, full details are to be submitted for approval.

3.4 Design and construction

3.4.1 All parts of turbines, compressors, etc., are to have clearances and fits consistent with adequate provision for the relative thermal expansion of the various components. Special attention is to be given to minimizing casing and rotor distortion under all operating conditions.

3.4.2 Turbine bearings are to be so disposed and supported that lubrication is not adversely affected by heat flow from adjacent hot parts. Effective means are to be provided for intercepting oil leakage and preventing oil from reaching high temperature glands and casings.

3.4.3 Design basis

a) Calculations of the steady stresses, including the effect of stress raisers, etc., in the turbine and compressor rotors and blading at the maximum speed and temperature in service, are to be submitted for consideration. Such calculations should indicate the designed service life and be accompanied, where possible, by test results substantiating the limiting criteria.

b) Details of calculations and tests to establish the service life of other stressed parts, including gearing (where applicable), bearings, seals, etc., are to be submitted. All calculations and tests should take account of all relevant environmental factors including particular type of service and fuels intended to be used.

3.5 Welded components

3.5.1 Components fabricated by means of welding will be considered for acceptance if constructed by firms whose works are properly equipped to undertake welding of the standards

appropriate to the components according to the relevant requirements of Ch.10.

3.5.2 Before work is commenced, manufacturers are to submit for consideration details of proposed welding procedures and their proposals for routine examination of joints by non-destructive means.

3.5.3 Major joints are to be designed as full-strength welds and for complete fusion of the joint.

3.5.4 Stress relief treatment is to be applied to all cylinders, rotors and associated components on completion of the welding of all joints and attached structures. For details of stress relief procedure, temperature and duration, See Ch.10.

3.5.5 Surveyors are to be satisfied that the desired quality of welding is attainable with the proposed welding equipment and procedure, and for this purpose test specimens representative of the welded joints are to be provided for radiographic examination and mechanical tests as required for steam turbines.

3.6 Vibration

3.6.1 Care is to be taken in the design and manufacture of turbine and compressor rotors, rotor discs and rotor blades to ensure freedom from undue vibration within the operating speed range. Calculations of the critical speeds giving full details of the basic assumptions are to be submitted for consideration. Where critical speeds are found by calculation to occur within the operating speed range, vibration tests may be requested in order to verify the calculations.

3.7 Containment

3.7.1 Consideration should be given to the need for containment, with a view to minimizing and localizing damage, in the event of rotor blade failure.

3.8 Inlet and exhaust systems

3.8.1 The air-inlet system is to be designed to minimize the entrance of harmful foreign matter.

3.8.2 The arrangement of the turbine exhaust system is to be such as to prevent exhaust gases being drawn into the compressors.

3.9 Fuel and salt deposits

3.9.1 When it is intended to burn non-distillate fuels forming harmful deposits, adequate provision should be made for periodic removal of the deposits.

3.9.2 Means for preventing the accumulation of salt deposits in the compressors and turbines are to be provided.

3.9.3 Flow path cleaning or washing system is to be provided, where required by the gas turbine manufacturer. Provision is to be made for periodical cleaning / washing by inhibiting liquids, steam or any other cleaning medium. The details of the arrangements and the associated system are to be submitted for approval.

3.10 Indication of temperature

3.10.1 Means are to be provided for indicating the temperature of the power turbine exhaust gases.

3.11 External influences

3.11.1 Pipes and ducting connected to casings are to be so designed that no excessive thrust loads or moments are applied by them to the compressors and turbines.

3.11.2 Platform gratings and fittings in way of the supports are to be so arranged that casing expansion is not restricted.

3.11.3 Where main turbine seatings incorporating a tank structure are proposed, consideration is to be given to the temperature variation of the tank in service to ensure that turbine alignment will not be adversely affected.

3.11.4 Where the turbine is provided with an acoustic enclosure, this is to act as shield to prevent contamination of machinery spaces from external hazards and contain the heat emission from the gas generator. The enclosure is to be structurally sound and is to be a self-contained unit with arrangements for ventilation, lighting and noise attenuation. The drainage arrangements in the enclosure are to be provided to prevent build up of liquids. A self contained and independent fire fighting system is to be provided for such enclosure. One or more observation window/s is/are to be provided for visual inspection of critical components such as auxiliary gearbox and burner manifold.

3.12 Overhaul life

3.12.1 The overhaul schedule recommended by the manufacturer is to be submitted for consideration.

3.13 Safety arrangements

3.13.1 Overspeed protective devices

a) An overspeed protective device is to be provided for each shaft of main and auxiliary turbines to shut off the fuel automatically, near the burners, to prevent a dangerous overspeed condition of the shaft, except where it can be established that such a condition cannot arise.

b) The overspeed device should normally be set to operate when the speed of the line exceeds the rated maximum speed by 15 per cent.

3.13.2 Speed governors

a) Where a main propulsion installation incorporates a reverse gear, electric transmission or controllable (reversible) pitch propeller, a speed governor, independent of the overspeed protective device, is to be fitted and is to be capable of controlling the speed of the unloaded power turbine without bringing the overspeed protective device into action.

b) Where an auxiliary turbine is intended for driving an electric generator, a speed governor, independent of the overspeed protective device, is to be fitted which, with fixed setting, is to control the speed within 10 per cent momentary variation and 5 per cent permanent variation when full load is suddenly taken off or put on. The permanent speed variation of a.c. machines intended for parallel operations are to be equal within a tolerance of 0.5 per cent.

3.13.3 The following turbine services are to be provided with automatic temperature controls so as to maintain steady state conditions throughout the normal operating range of the main turbine:

- lubricating oil system;
- oil fuel supply (or automatic control of oil fuel viscosity as an alternative); and
- exhaust gas.

3.13.4 Alarms and automatic shutdown devices are to be provided in accordance with Table 3.13.1. Although in principle alarming devices listed in Table 3.13.1 are to be provided, they can be added or omitted, taking into account the result of FMEA specified in item 3.2.2.

3.13.5 Hand trip gear for shutting off the fuel in an emergency is to be provided locally at the turbine control platform and where applicable, at the centralized control station.

3.13.6 All non-mobile gas turbines are to be provided with independent, self contained suitable foam or CO_2 or water mist fire fighting system with detection ability to give warning automatically at local and remote control positions when activated. This system is to be distinctly separate from the fire fighting arrangements in the machinery space. The arrangements together with calculations showing the capacities of the cylinders is to be submitted for approval.

3.13.7 Inlet and exhaust silencers are to be fitted to limit the sound power level at one meter from the gas turbine system to 110 dB for unmanned machinery spaces or to 90 dB for manned machinery spaces.

3.14 Starting arrangements

3.14.1 Means are to be provided, preferably automatic or interlocked, to clear all parts of the gas turbine of the accumulation of liquid fuel, or for purging gaseous fuel, before ignition commences on starting, or recommences after failure to start.

3.14.2 The starting arrangements can be pneumatic, hydraulic or electric. The starting air pipes system, safety fittings and number of starts are to comply with the requirements for oil engines as given in Section 4.11.

3.14.3 Starting devices are to be so arranged that firing operation is discontinued and main fuel valve is closed within pre-determined time in case of ignition failure.

Table 3.13.1 : List of alarms and shutdown

Monitoring parameter	Alarm	Automatic Shutdown
Turbines speed	High value	Yes
Lubricating oil pressure	Low value*	Yes
Lubricating oil pressure of reduction gear	Low value*	Yes
Differential pressure across lubricating oil filter	High value	
Lubricating oil temperature	High value	
Oil fuel supply pressure	Low value	
Oil fuel temperature	High value	
Cooling medium temperature	High value	
Bearing temperature	High value	
Flame and ignition failure	Activated	Yes
Automatic starting failure	Activated	
Vibration	High value*	Yes
Axial displacement of rotor	High value	Yes
Exhaust gas temperature	High value*	Yes
Vacuum pressure at the compressor inlet	High value*	Yes
Loss of control system	Activated	

Notes:

1. Alarms marked with "*" are to be activated at the suitable setting points prior to arriving the critical condition for the activation of shutdown devices.

3.14.4 Where the gas turbine is arranged for air starting, the total air receiver capacity is to be sufficient to provide, without replenishment, not less than six consecutive starts. At least two air receivers of equal capacity are to be provided. For details of air receivers, see Chapter 5. For multi engine installations three consecutive starts per engine are required.

3.15 Tests

3.15.1 Balancing

a) The rotors as finished-bladed and complete with half-coupling are to be dynamically balanced to the Surveyor's satisfaction, in a machine of sensitivity appropriate to the size of rotor.

3.15.2 Hydraulic tests

a) All casings are to be tested to a hydraulic pressure equal to 1.5 times the highest pressure in the casing during normal operation or 1.5 times the pressure during starting, whichever is the higher. For test purposes if necessary, the casings may be subdivided with temporary diaphragms for distribution of test pressure.

b) Where hydraulic tests cannot be carried out, the manufacturers are to submit, for approval, alternative proposals for determining the soundness of the component.

c) Inter coolers and heat exchangers are to be tested to 1.5 times the maximum working pressure on each side separately.

3.16 Shop trials

3.16.1 Upon completion of fabrication and assembly, each gas turbine is to be subjected to a shop trial in accordance with the manufacturer's test schedule, which is to be submitted for review before the trial. During the trial, the turbine is to be brought up to its over speed limit to enable testing of the overspeed protective device.

3.17 Electric starting

3.17.1 Where main turbines are fitted with electric starters, two batteries are to be fitted. Each battery is to be capable of starting the turbines when cold and the combined capacity is to be sufficient without recharging to provide the number of starts of the main turbines as required by 3.14.4.

3.17.2 Electric starting arrangements for auxiliary turbines are to have two separate batteries or be supplied by separate circuits from the main turbine batteries when such are provided. Where one of the auxiliary turbines only is fitted with an electric starter one battery will be acceptable.

3.17.3 The combined capacity of the batteries for starting the auxiliary turbines is to be sufficient for at least three starts for each turbine.

3.17.4 The requirements for battery installations are given in Pt.4, Ch.8.

3.18 Type testing of gas turbine

3.18.1 Type tests are to be carried out in accordance with the agreed test programme on a representative machine comprising gas generator and power turbine and are to be witnessed by IRS surveyor. It may be carried out at the manufacturer's works or at any establishment having suitable facilities acceptable to IRS.

3.18.2 Where the gas turbine unit can be separated conveniently into a gas generator and a power turbine, the type test is to include an initial test of the gas generator to ascertain that the fuel consumption and gas horse power are within ±2.5% of design maximum conditions at rated speed.

3.18.3 After completion of the initial test of the gas generator, the gas generator and power turbine are to be assembled together and prepared for the gas turbine type test.

3.18.4 On the over speed test, a propulsion gas generator is to be run at not less than 10% in excess of gas generator speed corresponding to specified full power of the gas turbine engine for not less than 10 minutes continuously and the power turbine disconnected from the dynamometer is to be run at not less than 15% in excess of the maximum designed speed for not less than 10 minutes continuously.

3.18.5 Where it is impracticable to overspeed the complete installation, each rotor, completely bladed and with all relevant parts such as half-couplings, shall be overspeed tested individually at the appropriate speed.

3.18.6 During the type tests at specified full power the pressure and temperature of the gas at outlet from the gas generator is to be recorded and the gas mass flow is to be assessed from air flow and fuel flow measurements.

3.18.7 During the type tests the following features are to be demonstrated:

i) Freedom from air and gas leaks.
ii) Operation of starting arrangements.
iii) Effectiveness of interlocks, if fitted.
iv) Satisfactory lubricating oil and fuel oil system characteristics.
v) Wet and dry motoring.
vi Engine start following a failure to start.
vii) Operation of engine cleaning system.
viii) Reliability of operation at minimum idling speed.

3.18.8 Starting tests are to be carried out to determine:

i) Time to light from cold.
ii) Time to reach idling speed from cold.
iii) Time to reach full output from idling speed.
iv) That the required number of starts can be obtained from the specified starting air bottle capacity.
v) The minimum pressure to ensure a cold start is to be established.

- A cold start is defined as a start after 2 hours or more of natural cooling with the engine at rest.

- A warm start is defined as a start within 15 minutes of shutting down after the engine has been running for not less than two hours.

- An intentional failure to start is defined as the performance of a normal starting sequence with the ignition system deliberately rendered inoperative.

Note : i), ii), iii) and iv) are to be average of four readings.

Section 4

Internal Combustion Engines

4.1 Scope

4.1.1 The requirements of this Section apply to conventional type of internal combustion engines used as prime movers for main propulsion and for essential auxiliary machinery.

4.1.2 The approval of IC Engines is covered in detail in the IRS *Classification Note "Approval of I.C Engines"*. The document flow between the engine designer, IRS Plan Approval Centre, engine builder/ licensee and IRS Surveyors is indicated in the Classification Note. The Classification Note also covers the procedure to be followed for certification of engine components. The requirements with regard to design approval, type testing and certification of turbochargers and their matching on engines is also indicated in the Classification Note (Also refer Cl. 4.11.4).

4.1.3 Engines below 100[kW] including gear boxes used for propulsion and for essential auxiliary machinery may be accepted based on certificate from approved manufacturers. Engines of 100 [kW] and over including gear boxes used for propulsion and for essential auxiliary machinery, are to be type approved and undergo unit certification. Unit certification may be exempted for those engines approved in accordance with the Alternative Certification Scheme of IRS (Refer Pt. 1, Ch. 1, Sec. 4).

4.2 Materials and material testing

4.2.1 Materials used for the construction of oil engines are to comply with the requirements of Pt.2 in so far as applicable. Other materials will be specially considered.

4.2.2 The specified minimum tensile strength of castings and forgings for crankshafts is to be selected within the following general limits:

a) Carbon and carbon-manganese steel castings 400 - 550 [N/mm^2];

b) Carbon and carbon manganese steel forgings (normalized and tempered) : 400 - 600 [N/mm^2];

c) Carbon and carbon-manganese steel forgings (quenched and tempered) : not exceeding 700 [N/mm^2];

d) Alloy steel castings : not exceeding 700 [N/mm^2];

e) Alloy steel-forgings : not exceeding 1000 [N/mm^2];

f) Spheroidal or nodular graphite iron castings: 370 - 800 [N/mm^2].

4.2.3 Where it is proposed to use alloy steel castings, forgings or iron castings for crankshafts, details of the chemical composition, heat treatment and mechanical properties are to be submitted for approval.

4.2.4 Documentation required to be submitted for certification of engine components including their materials are indicated in the IRS *Classification Note "Approval of I.C Engines"*.

4.3 Crankshaft design

4.3.1 Detailed analysis to prove the design of the crankshaft is to be submitted for consideration. Classification Notes on design of crankshafts may be followed in this regard. Manufacturers may also be required to submit detailed analysis for special design features, e.g. serrations in connecting rods and bearing caps.

4.4 Design and construction

4.4.1 The design and construction is to be such as to enable the oil engine to meet the general requirements with regard to environmental conditions, functional capability and reliability.

4.4.2 Cylinders, cylinder liners, cylinder covers, pistons and other parts subject to high temperature or pressure are to be of material suitable for the stress and temperature to which they are exposed.

4.4.3 The engine bedplate or crankcase is to be of rigid and oil-tight construction. The bedplate is to be provided with sufficient number of holding down bolts to effectively secure it to the engine seatings.

4.4.4 Transverse girders, cylinder blocks and columns, when of fabricated construction, are normally to be stress-relieved after welding.

4.4.5 Crankcases and their doors are to be of robust construction and the doors are to be securely fastened so that they will not be readily displaced by an explosion taking into account the installation of explosion relief valve as required by 4.6.

4.4.6 Piston rod glands, which are subjected to pressure from scavenging air, are to be of a design ensuring that air will not leak into the crankcase.

4.5 Crankcase ventilation

4.5.1 Ventilation of crankcase and any arrangement which could produce a flow of external air within the crankcase, is in general not permitted except for dual fuel engines where crankcase ventilation is to be provided in accordance with 4.13.1.3. If forced extraction of the gases from crankcase is provided (e.g. for smoke-detection purposes), the vacuum in the crankcase is not to exceed 25 [mm] of water column.

4.5.2 Crankcase ventilation pipes where provided are to be as small as practicable to minimize the in rush of air after crankcase explosion.

4.5.3 To avoid interconnection between crankcases and the possible spread of fire following an explosion, crankcase ventilation pipes and oil drain pipes for each engine are to be independent of each other.

4.6 Safety devices

4.6.1 For crankcase:

4.6.1.1 Crankcases are to be provided with light weight spring-loaded valves or other quick-acting and self closing devices, of an approved type, to relieve the crankcases of pressure in the event of an internal explosion and to prevent any inrush of air thereafter. The valves are to be designed to open at a pressure not greater than 0.2 bar.

4.6.1.2 The valve lids are to be made of ductile material capable of withstanding the shock of contact with stoppers at the full open position. The discharge from the valves is to be shielded by flame guard or flame trap to minimise the possibility of danger and damage arising from the emission of flame.

4.6.1.3 Crankcase explosion relief valves are to be type tested in a configuration that represents the installation arrangements that will be used on an engine, in accordance with IRS Classification Notes on “Guidance for Type Testing of Crankcase Explosion Relief Valves”.

4.6.1.4 Where crankcase relief valves are provided with arrangements for shielding emissions from the valve following an explosion, the valve is to be type tested to demonstrate that the shielding does not adversely affect the operational effectiveness of the valve.

4.6.1.5 Crankcase explosion relief valves are to be provided with a copy of manufacturer’s installation and maintenance manual that is specific to the size and type of valve being supplied for installation on a particular engine. The manual is to contain the following information:

- Description of valve with details of function and design limits.
- Copy of type test certificate.
- Installation instructions.
- Instructions for testing and renewal of any sealing arrangements.
- Actions required after a crankcase explosion.

4.6.1.6 A copy of the installation and maintenance manual required by 4.6.1.5 is to be provided on board ship.

4.6.1.7 Plans / documentation showing details and arrangements of crankcase explosion relief valves are to be submitted for approval, as indicated in the IRS *Classification Note “Approval of I.C Engines”*.

4.6.1.8 Valves are to be provided with suitable markings that include the following information:

- Name and address of manufacturer
- Designation and size
- Month / Year of manufacture
- Approved installation orientation.

4.6.1.9 In engines having cylinders not exceeding 200 [mm] bore and having a crankcase gross volume not exceeding 0.6 [m^3], relief valves may be omitted.

4.6.1.10 In engines having cylinders exceeding 200 [mm] bore, but not exceeding 250 [mm] bore, at least two relief valves are to be fitted, each valve is to be located at or near the ends of the crankcase. Where the engine has more than 8 crank throws an additional valve is to be fitted near the center of the engine.

4.6.1.11 Engines, with cylinder bores exceeding 250 [mm], but not exceeding 300 [mm], are to have at least one relief valve in way of each alternate crank throw, with a minimum of two relief valves.

4.6.1.12 In engines having cylinders exceeding 300 [mm] bore, at least one relief valve is to be fitted in way of each main crank throw.

4.6.1.13 Additional relief valves are to be fitted for separate spaces on the crankcase, such as gear or chain cases for camshaft or similar drives, when the gross volume of such spaces exceeds 0.6 [m^3].

4.6.1.14 The free area of each crankcase relief valve is not to be less than 45 [cm^2]. The combined free area of the relief valves fitted on the engine is not to be less than 115 [cm^2/m^3] of the crankcase volume.

The free area of the relief valve is the minimum flow area at any section through the valve, when the valve is fully open.

When calculating the gross volume of the crankcase, the volume of stationary parts within the crankcase may be deducted.

4.6.1.15 A warning notice is to be fitted in a prominent position, preferably on a crankcase door on each side of the engine, or alternatively at the engine room control location. This warning notice is to specify that whenever overheating is suspected in the crankcase, the crankcase doors or sight holes are not to be opened until a reasonable time has elapsed after stopping the engine, sufficient to permit adequate cooling within the crankcase.

4.6.1.16 Where crankcase oil mist detection arrangements are to be fitted to engines (as per Pt.4, Ch.7 and Pt.5, Ch.22) they are to be of a type approved by IRS and tested in accordance with Note 1) and comply with 4.6.1.17 to 4.6.1.28. Engine bearing temperature monitors or equivalent devices used as safety devices are to be type approved by IRS for such purposes.

Note 1) : Classification Notes – “Type approval of crankcase oil mist detection and alarm equipment”.

4.6.1.17 The oil mist detection system and arrangements are to be installed in accordance with the manufacturer’s and engine designer’s instructions / recommendations. The following particulars are to be included in the instructions:

- Schematic layout of engine oil mist detection and alarm system showing location of engine crankcase sample points and piping or cable arrangements together with dimensions of the pipe to the detector.

- Evidence of study to justify the selected location of sample points and sample extraction rate (if applicable) in consideration of the crankcase arrangements and geometry and the predicted, crankcase atmosphere where oil mist can accumulate.

- The manufacturer’s maintenance and test manual.

- Information relating to type or in-service testing of the engine with engine protection system test arrangements having approved types of oil mist detection equipment.

4.6.1.18 A copy of the oil mist detection equipment maintenance and test manual required by 4.6.1.17 is to be provided on board ship.

4.6.1.19 Oil mist detection and alarm information is to be capable of being read from a safe location away from the engine.

4.6.1.20 Each engine is to be provided with its own independent oil mist detection arrangement and a dedicated alarm.

4.6.1.21 Oil mist detection and alarm systems are to be capable of being tested on the test bed and onboard under engine at standstill and engine running at normal operating conditions in accordance with test procedures that are acceptable to IRS.

4.6.1.22 Alarms and shutdowns for the oil mist detection system are to be in accordance with Pt.5, Ch.22, Sec.3 and the system arrangements are to comply with Pt.5, Ch.22, Sec.2.

4.6.1.23 The oil mist detection arrangements are to provide an alarm indication in the event of a foreseeable functional failure in the equipment and installation arrangements.

4.6.1.24 The oil mist detection system is to provide an indication when any lenses fitted in the equipment and used in determination of the oil mist level have been partially obscured to a degree that will affect the reliability of the information and alarm indication.

4.6.1.25 Where oil mist detection equipment includes the use of programmable electronic systems, the arrangements are to be in accordance with the rule requirements for such systems.

4.6.1.26 Plans of showing details and arrangements of oil mist detection and alarm arrangements are to be submitted for approval in accordance with Pt.4, Ch.7.

4.6.1.27 The equipment together with detectors is to be tested when installed on the test bed and on board ship to demonstrate that the detection and alarm system functionally operates. The testing arrangements are to be to the satisfaction of IRS.

4.6.1.28 Where sequential oil mist detection arrangements are provided the sampling frequency and time is to be as short as reasonably practicable.

4.6.1.29 Where alternative methods are provided for the prevention of the build-up of oil mist that may lead to a potentially explosive condition within the crankcase details are to be submitted for consideration. The following information is to be included in the details:

- Engine particulars – type, power, speed, stroke, bore and crankcase volume.

- Details of arrangements to prevent the build up of potentially explosive conditions within the crankcase, e.g. bearing temperature monitoring, oil splash temperature, crankcase pressure monitoring, recirculation arrangements.

- Evidence to demonstrate that the arrangements are effective in preventing the build up of potentially explosive conditions together with details of in-service experience.

- Operating instructions and the maintenance and test instructions.

4.6.1.30 Where it is proposed to use the introduction of inert gas into crankcase to minimize a potential crankcase explosion, details of the arrangements are to be submitted to IRS for consideration.

4.6.2 Scavenge spaces:

4.6.2.1 Scavenge spaces in open connection to the cylinders are to be fitted with relief valves. The valves are to open quickly in case of an explosion.

4.6.2.2 For crosshead type engines, scavenge spaces in open connection to the cylinders are to be provided with an approved fire extinguishing system, which is to be entirely separate from the fire extinguishing system of the engine room.

4.6.3 Cylinders:

4.6.3.1 Cylinder relief valves are to be fitted to engines having cylinders over 230 [mm] bore. The valves are to open at a pressure not exceeding 40 per cent above the designed maximum pressure and are to discharge where no damage can occur. Consideration will be given to any other alternative relief arrangement.

4.6.3.2 In the case of auxiliary engines, consideration will be given to the replacement of the relief valve by an efficient warning device of over pressure in the cylinders.

4.6.4 Lubrication system:

4.6.4.1 The lubricating oil drain pipes from the engine sump to drain tank are to be taken down as low as possible, to ensure that their outlet ends are submerged at all times. Where two or more engines are installed, vent pipes, if fitted, and lubricating oil drain pipes are to be installed independently of each other to prevent intercommunication between crankcases.

4.6.5 Fuel systems:

4.6.5.1 All external high-pressure fuel delivery lines between the high-pressure fuel pumps and fuel injectors are to be protected with a jacketed piping system capable of containing fuel from a high-pressure line failure. A

jacketed pipe incorporates an outer pipe into which the high-pressure fuel pipe is placed, forming a permanent assembly. The jacketed piping system is to include means for collection of leakages and arrangements provided for an alarm to be given of a fuel line failure. However, this requirement need not be applied for engines intended to be fitted on board non-passenger vessels of less than 500 GT.

4.6.5.2 Components of a diesel engine fuel system is to be designed considering the maximum peak pressure which will be experienced in service, including any high pressure pulses which are generated and transmitted back into the fuel supply and spill lines by the action of fuel injection pumps. Connections within the fuel supply and spill lines are to be constructed having regard to their ability to prevent pressurized oil fuel leaks while in service and after maintenance.

4.6.6 Starting air:

4.6.6.1 In order to protect air main against explosion arising from improper functioning of starting valves, the following are to be fitted:

a) an isolating non-return valve or equivalent at the starting air supply connection to each engine;

b) a bursting disc or flame arrestor in way of the starting valve of each cylinder for direct reversing engines having a main starting manifold or at the supply inlet to the starting air manifold for non-reversing engines.

4.6.6.2 Bursting disc or flame arrestor may be omitted for engines having a bore not exceeding 230 [mm], or where the power of the engine is less than 1100 [kW].

4.6.7 Overspeed protective devices:

4.6.7.1 Each main engine developing 220 [kW] or over, which can be declutched or which drives a controllable pitch propeller, also each auxiliary engine developing 220 [kW] and over for driving an electric generator, is to be fitted with an approved overspeed protective device.

4.6.7.2 The overspeed protective device, including its driving mechanism is to be independent of the governor required by 4.7 and is to be so adjusted that the speed does not exceed that for which the engine and its driven machinery are to be classed by more than 20 per cent for main engines and 15 per cent for auxiliary engines.

4.7 Governing

4.7.1 For main propulsion engines:

An efficient governor is to be fitted to each main engine and so adjusted that the speed of the engine does not exceed that for which the engine is to be classed by more than 15 per cent.
When electronic speed governors of main engines form part of a remote control system, they are to comply with the following:

a) if lack of power to the governor may cause major and sudden changes in the preset speed and direction of the thrust of the propeller, back up power supply is to be provided;

b) local control of engines is always to be possible;

c) electronic speed governor and its actuators are to be type tested as per *IRS Classification Notes "Type Approval of Electrical Equipment used for Control, Monitoring, Alarm and Protection Systems for use in Ships; and*

d) Also refer requirements of *IRS Classification Notes "Approval of I.C. Engines".*

4.7.2 For auxiliary engines:

Governors on diesel engines driving main or emergency electric generators are to be capable of automatically maintaining the speeds and frequency within following limits:

a) Transient frequency variations within 10% of the rated frequency with a recovery time to steady state conditions not exceeding 5 seconds, when the maximum electrical step load is switched on or off;

b) Momentary variations of 10 per cent of the maximum rated speed when the rated load of the generator is suddenly thrown off, provided this does not cause the intervention of the overspeed device;

c) Momentary variations of 10 per cent of the maximum rated speed when 50 per cent of the rated load is suddenly thrown on followed by the remaining 50 per cent load after an interval sufficient to restore the speed to steady state. Steady state conditions should be achieved in not more than 5 seconds;

Note : Steady state conditions are those at which the envelope of speed variation does not exceed +1% of the declared speed at the new power.

d) At all loads between no load and rated load the permanent speed variation is not to be more than 5 per cent of the maximum rated speed;

e) For generating sets operating in parallel, the governing characteristics of the engine are to be such that within the limits of 20 per cent and 100 per cent of total load, the load on any generating set does not normally differ from its proportionate share of the total load by more than

   i) 15 per cent of the rated output of the largest machine; or
   ii) 25 per cent of the rated output of the individual machine in question, whichever is less;

f) For generating sets intended to operate in parallel, it is to be possible to adjust the governor so that the load is kept within 5 per cent of the rated load at normal frequency;

g) For alternating current installations, the permanent speed variation of the machines intended for parallel operations are to be equal within a tolerance of $\pm$ 0.5 per cent.

h) Application of electrical load in more than 2 load steps (See Fig.4.7.2 for guidance on maximum possible sudden power increase in 4-stroke diesel engines) can only be permitted on the condition that

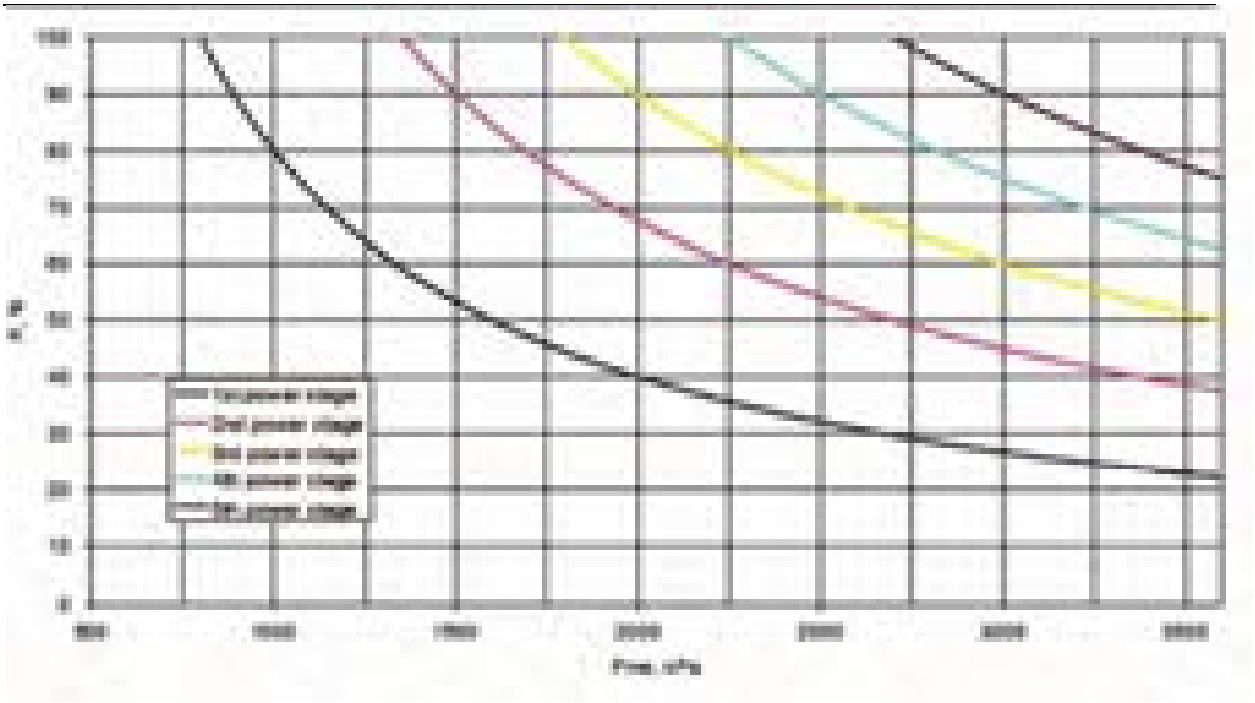
   - the design of the ship's electrical system enables the use of such generator sets
   - such a provision is made in the design stage and same is approved while scrutinising the drawings
   - the safety of the ship's electrical system in the event of parallel operation and failure of a generator is to be demonstrated at ship's trials.

4.7.3 Emergency generator sets must satisfy the above governor conditions even when :

a) their total consumer load is applied suddenly, or

b) their total consumer load is applied in steps, subject to following:

   - the total load is supplied within 45 seconds since power failure of the main switchboard
   - the maximum step load is declared and demonstrated
   - the power distribution system is designed such that the declared maximum step loading is not exceeded
   - the compliance of time delays and loading sequence with the above is to be demonstrated at the time of ship’s trials.



Legend :

Pme : Declared mean effective pressure
P : Power increase referred to declared power at site conditions
1 : First power stage
2 : Second power stage
3 : Third power stage
4 : Fourth power stage
5 : Fifth power stage

Fig. 4.7.2: Reference values for maximum possible sudden power increases as a function of brake mean effective pressure, Pme, at declared power (four-stroke diesel engines)

4.8 Oil fuel, lubricating oil, cooling water and exhaust gas systems

4.8.1 Oil fuel, lubricating oil and cooling water piping systems are to comply with the requirements of Ch.3 in so far as these are applicable.

4.8.2 Exhaust pipes which are led overboard near the waterline are to be protected against the possibility of water finding its way inboard. Where the exhaust is cooled by water spray, the exhaust pipes are to be self-draining overboard.

4.8.3 Exhaust pipes of two or more engines are not to be connected together, but are to be led separately to the atmosphere unless arranged to prevent the return of gases to an idle engine.

4.9 Engine starting arrangements

4.9.1 First start arrangement:

4.9.1.1 Equipments for starting the main and auxiliary engines are to be provided so that the necessary initial charge of starting air or initial electric power can be developed on board ship without external aid. If for this purpose an emergency air compressor or electric generator is required, these units are to be power driven by hand starting oil engines or steam engines, except in the case of small installations where a hand operated compressor of approved capacity may be accepted. Alternatively, other devices of approved type may be accepted as a means of providing the initial start.

4.9.2 Compressed air starting systems:

4.9.2.1 Two or more starting and maneuvering air compressors of sufficient total capacity for the requirements of the main engines are to be fitted. At least one of the compressors is to be driven independent of the main propulsion unit. The capacity of one of the said independently driven compressors or the combined capacity

of independently driven compressors are not to be less than 50 per cent of the total required capacity. The total capacity of air compressors is to be sufficient to supply within one hour the quantity of air needed to satisfy the requirement of 4.9.2.9 by charging the receivers from atmospheric pressure. The main air compressors are to be of approximately the same size.

4.9.2.2 The compressors are to be so designed that the temperature of the air discharged to the starting air receivers will not substantially exceed 93°C in service. A small fusible plug or an alarm device operating at 121°C is to be provided on each compressor to give warning of excessive air temperature. The emergency air compressor need not comply with these requirements.

4.9.2.3 Each compressor is to be fitted with a safety valve so proportioned and adjusted that accumulation of pressure, with the outlet valve closed, will not exceed 10 per cent of the maximum working pressure.

4.9.2.4 The castings of the cooling water spaces are to be fitted with a safety valve or bursting disc so that ample relief will be provided in the event of the bursting of an air cooler tube.

4.9.2.5 Air compressor inlets are to be located in an atmosphere reasonably free from oil vapour or, alternatively, an air duct from outside the machinery space is to be led to the compressors.

4.9.2.6 The air discharge pipe from the compressors is to be led direct to the starting air receivers. Provision is to be made for intercepting and draining oil and water in the air discharge for which purpose a separator or filter is to be fitted in the discharge pipe between compressors and receivers.

4.9.2.7 The starting air pipe system from air receivers to main and auxiliary engines is to be entirely separate from the compressor discharge system. Stop valves on the air receivers are to permit slow opening to avoid sudden pressure rises in the piping system. Valve chests and fittings in the piping systems are to be of ductile material.

4.9.2.8 Drain valves for removing accumulations of oil and water are to be fitted on compressors, separators, filters and air receivers. In the case of any low-level pipelines, drain valves are to be fitted to suitably located drain pots or separators.

4.9.2.9 The total air receiver capacity is to be sufficient to provide, without replenishment, number of starts as per Table 4.9.1.

a) If starting system serves two or more of the above specified purposes, the capacity of the system is to be the sum of the capacity required.

b) At least two air receivers of about equal capacity are to be provided.

Table 4.9.1 : Number of starts of engines

Duty of engine	No. of starts
Propulsion engines – reversible	12 consecutive starts alternating between Ahead and Astern of each engine
Propulsion engines - non-reversible, connected to a controllable pitch propeller or other device enabling the start without opposite torque	6 starts for each engine
Engines for driving electric generators and emergency generators and engines for other purposes	3 starts for each engine

4.9.3 Electric starting

4.9.3.1 Where main engines are fitted with electric starters, two batteries are to be fitted. The arrangement is to be such that batteries cannot be connected in parallel.

4.9.3.2 Each battery is to be capable of starting the engines when in cold and ready to start conditions. The combined capacity is to be sufficient without recharging to provide within 30 minutes the number of starts of the main engines as required by 4.9.2.9.

4.9.3.3 The electric starting arrangements for auxiliary engines are to have two separate batteries or may be supplied by separate circuits from main engine batteries when such are provided. In case of a single auxiliary engine, only one battery is required.

4.9.3.4 The starting batteries are to be used for starting and engine's own monitoring purposes only. Provision is to be made to maintain continuously the stored energy at all times.

4.10 Type testing of oil engines

4.10.1 General

4.10.1.1 Requirements for Type Testing of Oil Engines are specified in *Classification Note "Approval of I.C Engines"*. The documentation required and document flow between the engine designer, IRS Plan Approval Centre, engine builder/ licensee and IRS Surveyors is also indicated in the Classification Note.

4.10.1.2 Upon finalization of the engine design for production of every new engine type intended for the installation on board ships, one engine is to be presented for type testing, as detailed in the IRS *Classification Note "Approval of I.C Engines"*.

4.11 Factory Acceptance Test

4.11.1 Safety Precautions

4.11.1.1 Before any test run is carried out, all relevant equipment for the safety of attending personnel is to be made available by the manufacturer / shipyard and is to be operational.

4.11.1.2 This applies especially to crankcase explosive conditions protection, but also to over-speed protection and any other shut down function.

4.11.1.3 The overspeed protective device is to be set to a value, which is not higher than the overspeed value that was demonstrated during the type test for that engine. This set point will be verified by the surveyor.

4.11.2 General

4.11.2.1 Before any official testing, the engines are to be run-in as prescribed by the engine manufacturer.

4.11.2.2 Adequate test bed facilities for loads as required in 4.11.3.7 are to be provided. All fluids used for testing purposes such as fuel, lubrication oil and cooling water are to be suitable for the purpose intended, e.g. they are to be clean, preheated if necessary and cause no harm to engine parts. This applies to all fluids used temporarily or repeatedly for testing purposes only.

4.11.2.3 The testing consists of workshop and shipboard (quay and sea trial) testing.

4.11.2.4 Engines are to be inspected for:

- Jacketing of high-pressure fuel oil lines including the system used for the detection of leakage.
- Screening of pipe connections in piping containing flammable liquids.
- Insulation of hot surfaces by taking random temperature readings that are to be compared with corresponding readings obtained during the type test. This is to be done while running at the rated power of engine. Use of contact thermometers may be accepted at the discretion of the attending Surveyor. If the insulation is modified subsequently to the Type Approval Test, IRS may request temperature measurements as required by *Classification Note "Approval of I.C Engines" 2.8.9.*

4.11.2.5 These inspections are normally to be made during the works trials by the manufacturer and the attending surveyor, but at the discretion of IRS parts of these inspections may be postponed to the shipboard testing.

4.11.3 Works trials

4.11.3.1 The purpose of the works trials is to verify design premises such as power, safety against fire, adherence to approved limits (e.g.

maximum pressure), and functionality and to establish reference values or base lines for later reference in the operational phase.

4.11.3.2 The following environmental test conditions are to be recorded:

- Ambient air temperature
- Ambient air pressure
- Atmospheric humidity

4.11.3.3 For each required load point, the following parameters are normally to be recorded:

- Power and speed
- Fuel index (or equivalent reading)
- Maximum combustion pressures (only when the cylinder heads installed are designed for such measurement).
- Exhaust gas temperature before turbine and from each cylinder (to the extent that monitoring is required in *Classification Note "Approval of I.C Engines"* Sec 4 and Pt 5, Ch 22)
- Charge air temperature
- Charge air pressure
- Turbocharger speed (to the extent that monitoring is required in *Classification Note "Approval of I.C Engines" Sec 4*)

4.11.3.4 Calibration records for the instrumentation are, upon request, to be presented to the attending Surveyor.

4.11.3.5 For all stages at which the engine is to be tested, the pertaining operational values are to be measured and recorded by the engine manufacturer. All results are to be compiled in an acceptance protocol to be issued by the engine manufacturer. This also includes crankshaft deflections if considered necessary by the engine designer.

4.11.3.6 In each case, all measurements conducted at the various load points are to be carried out at steady state operating conditions. However, for all load points provision should be made for time needed by the Surveyor to carry out visual inspections. The readings for MCR, i.e. 100% power (rated maximum continuous power at corresponding rpm) are to be taken at least twice at an interval of normally 30 minutes.

4.11.3.7 Test loads for various engine applications are given in Tables below. In addition, the scope of the trials may be expanded depending on the engine application, service experience, or other relevant reasons.

Note:
Alternatives to the detailed tests may be agreed between the manufacturer and IRS when the overall scope of tests is found to be equivalent.

Table 4.11.3.7 (a) : Propulsion engines driving propeller or impeller only

A)	100% power (MCR) at corresponding speed n_0 :	At least 60 [min]
B)	110% power at engine speed $1.032n_0$ Note: Only required once for each different engine/turbocharger configuration.	Records to be taken after 15 [min] or after steady conditions have been reached, whichever is shorter
C)	Approved intermittent overload (if applicable)	Testing for duration as agreed with the manufacturer
D)	90% (or normal continuous cruise power), 75%, 50%, and 25% power in accordance with the nominal propeller curve, the sequence to be selected by the engine manufacturer.	
E)	Reversing manoeuvres (if applicable)	

Note:
After running on the test bed, the fuel delivery system is to be so adjusted that overload power cannot be given in service, unless intermittent overload power is approved by IRS. In that case, the fuel delivery system is to be blocked to that power.

Table 4.11.3.7 (b) : Engines driving generators for electric propulsion and Engine driving generators for auxiliary purposes		
A)	100% power (MCR) at corresponding speed n_0 :	At least 60 [min]
B)	110% power at engine speed n_0	15 [min] - after having reached steady conditions
C)	Governor tests for compliance with 4.7 are to be carried out.	
D)	75%, 50%, and 25% power and idle, the sequence to be selected by the engine manufacturer.	

Note:
After running on the test bed, the fuel delivery system is to be adjusted so that full power plus a 10% margin for transient regulation can be given in service after installation onboard.
The transient overload capability is required so that the required transient governing characteristics are achieved also at 100% loading of the engine, and also so that the protection system utilised in the electric distribution system can be activated before the engine stalls.

Table 4.11.3.7 (c) : Propulsion engines also driving power take off (PTO) generator		
A)	100% power (MCR) at corresponding speed n_0 :	At least 60 [min]
B)	110% power at engine speed n_0	15 [min] - after having reached steady conditions
C)	Approved intermittent overload (if applicable)	Testing for duration as agreed with the manufacturer.
D)	90% (or normal continuous cruise power), 75%, 50% and 25% power in accordance with the nominal propeller curve or at constant speed n_0, the sequence to be selected by the engine manufacturer.	

Note:
After running on the test bed, the fuel delivery system is to be adjusted so that full power plus a margin for transient regulation can be given in service after installation onboard. The transient overload capability is required so that the electrical protection of downstream system components is activated before the engine stalls. This margin may be 10% of the engine power but at least 10% of the PTO power.

Table 4.11.3.7 (d) : Engines driving auxiliaries		
A)	100% power (MCR) at corresponding speed n_0 :	At least 30 [min]
B)	110% power at engine speed n_0	15 [min] - after having reached steady conditions
C)	Approved intermittent overload (if applicable)	Testing for duration as agreed with the manufacturer.
D)	For variable speed engines, 75%, 50% and 25% power in accordance with the nominal power consumption curve, the sequence to be selected by the engine manufacturer.	

Note:
After running on the test bed, the fuel delivery system is normally to be so adjusted that overload power cannot be delivered in service, unless intermittent overload power is approved. In that case, the fuel delivery system is to be blocked to that power.

4.11.4 Turbocharger matching with engine

4.11.4.1 Compressor chart

.1 Turbochargers are to have a compressor characteristic that allows the engine, for which it is intended, to operate without surging during all operating conditions and also after extended periods in operation.

.2 For abnormal, but permissible, operation conditions, such as misfiring and sudden load reduction, no continuous surging is to occur.

.3 In this section, *surging* and *continuous surging* are defined as follows:

Surging means the phenomenon, which results in a high pitch vibration of an audible level or explosion-like noise from the scavenger area of the engine.

Continuous surging means that surging happens repeatedly and not only once.

4.11.4.2 Surge margin verification

.1 Category C turbochargers used on propulsion engines are to be checked for surge margins during the engine workshop testing as specified below. These tests may be waived if successfully tested earlier on an identical configuration of engine and turbocharger (including same nozzle rings).

.2 For 4 stroke engines, following are to be performed without indication of surging:

- With maximum continuous power and speed (=100%), the speed is to be reduced with constant torque (fuel index) down to 90% power.

- With 50% power at 80% speed (= propeller characteristic for fixed pitch), the speed is to be reduced to 72% while keeping constant torque (fuel index).

.3 For 2 stroke engines, the surge margin is to be demonstrated by at least one of the following methods:

- The engine working characteristic established at workshop testing of the engine is to be plotted into the compressor chart of the turbocharger (established in a test rig). There is to be at least 10% surge margin in the full load range, i.e. working flow is to be 10% above the theoretical (mass) flow at surge limit (at no pressure fluctuations).

- Sudden fuel cut-off to at least one cylinder is not to result in continuous surging and the turbocharger is to be stabilised at the new load within 20 seconds. For applications with more than one turbocharger the fuel is to be cut-off to the cylinders closest upstream to each turbocharger. This test is to be performed at two different engine loads:

    - The maximum power permitted for one cylinder misfiring.

    - The engine load corresponding to a charge air pressure of about 0.6 [bar] (but without auxiliary blowers running).

- No continuous surging and the turbocharger is to be stabilised at the new load within 20 seconds when the power is abruptly reduced from 100% to 50% of the maximum continuous power.

4.11.5 Integration tests

4.11.5.1 For electronically controlled engines, integration tests are to be made to verify that the response of the complete mechanical, hydraulic and electronic system is as predicted for all intended operational modes and the tests considered as a system are to be carried out at the works. If such tests are technically unfeasible at the works, however, these tests may be conducted during sea trial. The scope of these tests is to be agreed with IRS for selected cases based on the FMEA required in *Classification Note "Approval of I.C Engines" Sec 1.*

4.11.6 Component inspections

4.11.6.1 Random checks of components will be carried out by the attending surveyor.

4.12 Shipboard trials

4.12.1 Safety Precautions

4.12.1.1 Refer to 4.11.1

4.12.2 Objectives

4.12.2.1 The purpose of the shipboard testing is to verify compatibility with power transmission and driven machinery in the system, control systems and auxiliary systems necessary for the engine and integration of engine / shipboard control systems, as well as other items that had not been dealt with in the FAT (Factory Acceptance Testing).

4.12.3 Starting capacity

4.12.3.1 Starting manoeuvres are to be carried out in order to verify that the capacity of the starting media satisfies the required number of start attempts.

4.12.4 Monitoring and alarm system

4.12.4.1 The monitoring and alarm systems are to be checked to the full extent for all engines, except items already verified during the works trials.

4.12.5 Test loads

4.12.5.1 Test loads for various engine applications are given in tables below. In addition, the scope of the trials may be expanded depending on the engine application, service experience, or other relevant reasons.

4.12.5.2 The suitability of the engine to operate on fuels intended for use is to be demonstrated.

Note:
Tests other than those listed below may be required by statutory instruments (e.g. EEDI verification).

Table 4.12.5.1 (a) : Propulsion engines driving fixed pitch propeller or impeller		
A)	At rated engine speed n_0 :	At least 4 [Hrs]
B)	At engine speed $1.032n_0$ (if engine adjustment permits , see 4.11.3.7)	30 [min]
C)	At approved intermittent overload (if applicable)	Testing for duration as agreed with the manufacturer
D)	Minimum engine speed to be determined	
E)	The ability of reversible engines to be operated in reverse direction is to be demonstrated.	
	Note: *During stopping tests according to Resolution MSC.137 (76), see 4.12.6.1 for additional requirements in the case of a barred speed range.*	

Table 4.12.5.1 (b) : Propulsion engines driving controllable fixed pitch propellers		
A)	At rated engine speed n_0 with a propeller pitch leading to rated engine power (or to the maximum achievable power if 100% cannot be reached):	At least 4 [Hrs]
B)	At approved intermittent overload (if applicable)	Testing for duration as agreed with the manufacturer
C)	With reverse pitch suitable for manoeuvring, see 4.12.6.1 for additional requirements in the case of a barred speed range.	

Table 4.12.5.1 (c) : Engine(s) driving generator(s) for electrical propulsion and/or main power supply		
A)	At 100% power (rated electrical power of generator):	At least 60 [Min]
B)	At 110% power (rated electrical power of generator):	At least 10 [Min]
	Note: *Each engine is to be tested 100% electrical power for at least 60 min and 110% of rated electrical power of the generator for at least 10 min. This may, if possible, be done during the electrical propulsion plant test, which is required to be tested with 100% propulsion power (i.e. total electric motor capacity for propulsion) by distributing the power on as few generators as possible. The duration of this test is to be sufficient to reach stable operating temperatures of all rotating machines or for at least 4 hours. When some of the gen. set(s) cannot be tested due to insufficient time during the propulsion system test mentioned above, those required tests are to be carried out separately.*	
C)	Demonstration of the generator prime movers' and governors' ability to handle load steps as described in 4.11.3.2 and 4.11.3.3.	

Table 4.12.5.1 (d) : Propulsion engines also driving power take off (PTO) generator		
A)	100% power (MCR) at corresponding speed n_0 :	At least 4 [Hrs]
B)	100% propeller branch power at engine speed n_0 (unless already covered in A):	2 [Hrs]
C)	100% PTO branch power at engine speed n_0:	At least 1 [Hr]

Table 4.12.5.1 (e) : Engines driving auxiliaries		
A)	100% power (MCR) at corresponding speed n_0 :	At least 30 [Min]
B)	Approved intermittent overload:	Testing for duration as approved

4.12.6 Torsional vibrations

4.12.6.1 Barred speed range

.1 Where a barred speed range (BSR) is required, passages through this BSR, both accelerating and decelerating, are to be demonstrated. The times taken are to be recorded and are to be equal to or below those times stipulated in the approved documentation, if any. This also includes when passing through the BSR in reverse rotational direction, especially during the stopping test.

Note: Applies both for manual and automatic passing-through systems.

.2 The ship's draft and speed during all these demonstrations is to be recorded. In the case of a controllable pitch propeller, the pitch is also to be recorded.

.3 The engine is to be checked for stable running (steady fuel index) at both upper and lower borders of the BSR. Steady fuel index means an oscillation range less than 5% of the effective stroke (idle to full index).

4.13 Reciprocating Internal combustion engines fuelled by natural gas

4.13.1 Application

.1 The following requirements are applicable to marine reciprocating internal combustion engines supplied with natural gas as fuel. Requirements of Pt. 5, Ch. 4 for gas carriers and Pt. 5, Ch. 35 for ships burning low flashpoint fuels are also to be considered as indicated and applicable.

.2 The scope of this section is intended for natural gas fuelled engines. It may also be referred for engines using similar fuels with main component methane such as bio-methane or synthetic methane.

.3 It is to be ensured by the gas supply system that the gas supplied to the engine is always in

gaseous state. This section does not cover requirements for liquid or cryogenic gas.

.4 The engines can be dual fuel engines (hereinafter referred to as DF engines), gas fuel only engines (hereinafter referred to as GF engines), or any variations thereof including fuel sharing capability. DF engines and GF engines may not be permitted for emergency applications.

.5 Requirements of this subsection are to be applied in association with other relevant requirements for internal combustion engines, as far as found applicable to the specific natural gas burning engine design.

.6 The requirements of IGC Code (Pt.5, Ch.4) and IGF Code (Pt.5, Ch.35) are also to be considered, as applicable.

.7 Specific requirements of the IGF Code(Pt.5, Ch.35) as referenced in this subsection are to be applied to engine types covered by this subsection installed on any ship, regardless of type, size and trading area, as long as the IGC Code(Pt.5, Ch.4) is not referenced or explicitly specified otherwise.

4.13.2 Definitions (relevant to this sub-section)

.1 *Certified safe equipment* is equipment certified by an independent national test institution or competent body to be in accordance with a recognised standard for electrical apparatus in hazardous areas.
Note: Refer to IEC 60079 series, Explosive atmospheres and IEC 60092-502:1999 Electrical Installations in Ships – Tankers – Special Features

.2 *Double block and bleed valves* means the set of valves referred to in:

- Pt.5, Ch. 4, Sec. 16, 16.4.5
- Pt. 5, Ch. 35, Sec. 2, 2.2.1.9 and Sec. 9, 9.4.4 to 9.4.6

.3 *Dual fuel engine* (“DF engine”) means an engine that can burn natural gas as fuel simultaneously with liquid fuel, either as pilot oil or bigger amount of liquid fuel (gas mode), and also has the capability of running on liquid diesel fuel oil only (Diesel mode).

.4 Explosion relief device means a device to protect personnel and component against a determined overpressure in the event of a gas explosion. The device may be a valve, a rupture disc or other, as applicable.

.5 *Gas* means natural gas used as fuel consisting primarily of methane.
Note: Gas may also be bio-methane or synthetic methane etc. with methane as main component.

.6 *Gas admission valve* is a valve or injector on the engine, which controls gas supply to the cylinder(s) according to the engine’s gas demand.

.7 *Gas engine* means a DF engine, a GF engine or any variations thereof.

.8 *Gas fuel only engine* (“GF engine”) means an engine capable of operating on gas fuel only and not able to switch over to oil fuel operation.

.9 *Gas piping* means piping containing gas or air / gas mixtures.

.10 *High pressure gas* means gas with a maximum working pressure greater than 10 bar gauge.

.11 *Low pressure gas* means gas with a maximum working pressure lower or equal to 10 bar gauge.

.12 *Lower Heating Value (“LHV”)* means the amount of heat produced from the complete combustion of a specific amount of fuel, excluding latent heat of vaporization of water.

.13 *Methane Number* is a measure of resistance of a gas fuel to knock, which is assigned to a test fuel based upon operation in knock testing unit at the same standard knock intensity.
Note: Pure methane is used as the knock resistant reference fuel, that is, methane number of pure methane is 100, and pure hydrogen is used as the knock sensitive reference fuel, methane number of pure hydrogen is 0. When the antiknock quality of the gas fuel being measured is identical to that of the standard methane-hydrogen mixture fuel mixed by a certain ratio, the value of the volume percentage of the methane contained in such standard fuel is the methane value of the gas fuel.

.14 *Pilot fuel* means the fuel oil that is injected into the cylinder to ignite the main gas-air mixture on DF engines.
Note: The dual-fuel engines use “pilot fuel” to start the combustion process. A small amount of pilot fuel is injected into the cylinder, where it is ignited by the high temperature of the gas air mixture at the end of the compression cycle. Typically, the amount of pilot fuel oil is below 1% of the energy used by the engine.

.15 *Pre-mixed engine* means an engine where gas is supplied in a mixture with air through a common manifold for all cylinders, e.g. mixed before or after the turbocharger.

.16 *Recognized standards* mean applicable international or national standards acceptable to IRS or standards laid down and maintained by an organization which complies with the standards adopted by IMO and which are recognized by IRS.

.17 *Safety Concept* is a document describing the safety philosophy with regard to gas as fuel. It describes how risks associated with this type of fuel are controlled under reasonably foreseeable abnormal conditions as well as possible failure scenarios and their control measures. The results of the risk analysis, (see Section 5, of the Classification Note "*Approval of I.C Engines*")are to be reflected in the safety concept.

Note: A detailed evaluation regarding the hazard potential of injury from a possible explosion is to be carried out and reflected in the safety concept of the engine.

4.13.3 Documents and/ or drawings to be submitted, risk analysis and type approval

The documents and drawings to be submitted, risk analysis document and requirements for type approval of DF and GF engines are specified in Section 5 of the Classification Notes "*Approval of I.C. Engines*".

4.13.4 Design Requirements

4.13.4.1 General Principles

.1 The manufacturer is to declare the allowable gas composition limits for the engine and the minimum and (if applicable) maximum methane number.

.2 Components containing or likely to contain gas are to be designed to:

a) minimize the risk of fire and explosion so as to demonstrate an appropriate level of safety commensurate with that of an oil-fueled engine;

b) mitigate the consequences of a possible explosion to a level providing a tolerable degree of residual risk, due to the strength of the component(s) or the fitting of suitable explosion relief devices of an approved type.

The strength of the component(s) of arrangement of explosion relief devices is to be documented (e.g., as part of risk analysis) or otherwise demonstrated to be sufficient for a worst-case explosion.

Also refer to the Pt.5, Ch.35, Sec.10, 10.2 and 10.3.

.3. Discharge from explosion relief devices are to prevent the passage of flame to the machinery space and be arranged such that the discharge does not endanger personnel or damage other engine components or systems"

.4. Explosion relief devices are to be fitted with a flame arrester.

4.13.4.2 Gas Piping

4.13.4.2.1 General

.1 These requirements are applicable to engine mounted gas piping. The piping is to be designed in accordance with the criteria for gas piping (design pressure, wall thickness, materials, piping fabrication and joining details etc.) as given in Pt.5, Ch. 35, Section 7, or requirements of Pt. 5, Ch. 4, Sections 5.1 to 5.9 and Sec.16 as applicable.

.2 Other connections as mentioned in Pt. 5, Ch. 4, Sec. 5, 5.8 may be accepted subject to type approval in accordance with the requirements of Pt.4, Ch. 2 and Classification Notes "*Type approval of mechanical joints used in piping*".

.3 All single walled or high-pressure gas pipes are to be considered as Class I. Low pressure double walled gas pipes are to be considered as Class II. All secondary enclosures for gas pipes are to be considered as Class II.

.4 Single walled gas vent pipes, if permitted, are to be considered as Class I, except it is justified that the maximum built up pressure is less than 5 bar gauge, in which case they are to be considered as Class II. Gas vent pipes protected by a secondary enclosure are to be considered as Class II. Secondary enclosure for vent pipes are to be considered as Class III.

.5 For design condition of gas pipes refer Table 4.13.4.2.1.5

.6 Flexible bellows used in the fuel gas system on the engine is to be approved based on the requirements of Pt.5, Ch.35, Sec.16, 16.7.2 and Pt.5, Ch.4, Sec.5, 5.13.1.2, as applicable.

.7 The number of cycles, pressure, temperature, axial movement, rotational movement and transverse movement which the bellow will encounter in actual service on the engine should be specified by the engine designer.

.8 Endurance against high cycle fatigue due to vibration loads are to be verified by testing or alternatively be documented by the Expansion Joint Manufacturers Association, Inc. (EJMA) calculation or equivalent (i.e., more than 10^7 cycles).

Note: The fatigue test due to ship deformations in Pt.5, Ch.35, Sec.16, 16.7.2.4 is considered not relevant for bellows which are an integral part of the engine.

Table 4.13.4.2.1.5 : Design pressure for gas pipes

	Design pressure	
	IGF code vessels	**IGC code vessels**
Gas pipe, low pressure	Refer Pt.5, Ch.35, Sec.7, 7.3.3.1	Refer Pt.5, Ch.4, Sec.5, 5.4.1
Gas pipe, high pressure	Refer Pt.5, Ch.35, Sec.7, 7.3.3.1	Refer Pt.5, Ch.4, Sec.5, 5.4.1
outer pipe, low pressure	Refer Pt.5, Ch.35, Sec.9, 9.8.1	Refer Pt.5, Ch.4, Sec.5, 5.4.4
outer pipe, high pressure	Refer Pt.5, Ch.35, Sec.9, 9.8.2	Refer Pt.5, Ch.4, Sec.5, 5.4.4
Open ended pipes	Refer Pt.5, Ch.35, Sec.7, 7.3.3.2	Refer Pt.5, Ch.4, Sec.5, 5.4.1

4.13.4.3 Arrangement of the gas piping system on the engine

.1 Pipes and equipment containing fuel gas are defined as hazardous area Zone 0 (refer to Pt.5, Ch. 35, Sec. 12, 12.5.1).

.2 The space between the gas fuel piping and the wall of the outer pipe or duct is defined as hazardous area Zone 1 (refer to Pt.5, Ch. 35, Sec. 12, 12.5.2.6)

4.13.4.3.1 Normal “double wall” arrangement

.1 The gas piping system on the engine is to be arranged according to the principles and requirements of the Pt.5, Ch. 35, Sec. 9, 9.6. In the case of gas carriers, requirements of Pt.5, Ch. 4, Sec. 16, 16.4.3 would be applicable.

.2 The design criteria for the double pipe or duct are given in Pt. 5, Ch. 35, Sec 9.8 and Sec. 7.4.1.4.

.3 In case of a ventilated double wall, the ventilation inlet is to be located in accordance with the provisions of Pt.5, Ch.35, Sec. 13, 13.8.3. For gas carriers, Pt.5, Ch.4, Sec. 16, 16.4.3.2 applies.

.4 The pipe or duct is to be pressure tested at 1.5 times design pressure to ensure gas tight integrity and to show that it can withstand the expected maximum pressure at gas pipe rupture.

4.13.4.3.2 Alternative arrangement

.1 Single walled gas piping is acceptable in following cases:

a) for engines supplied with low pressure gas installed in ESD protected machinery spaces, as defined in Pt.5, Ch.35, Sec. 5, 5.4.1.2 and in compliance with other relevant parts of Pt.5, Ch. 35 (e.g. 5.6);

b) in the case as per note to 9.6.2 of Pt.5, Ch.35, Sec. 9.

.2 For gas carriers, the requirements of Pt.5, Ch.4 would be applicable.

.3 In case of gas leakage in an ESD-protected machinery space, which would result in the shut-down of the engine(s) in that space, a sufficient propulsion and manoeuvring capability including essential and safety systems is to be maintained.

Therefore, the safety concept of the engine is to clearly indicate application of the “double wall” or “alternative” arrangement.

Note: The minimum power to be maintained is to be assessed on a case-by-case basis from the operational characteristics of the ship.

4.13.4.4 Charge air system and exhaust gas system on the engine

.1 The charge air system and the exhaust gas system on the engine are to be designed in accordance with 4.13.4.1.2. In case of a single engine installation, the engine is to be capable of operating at sufficient load to maintain power to essential consumers after opening of the explosion relief devices caused by an explosion event. Sufficient power for propulsion capability is to be maintained.

Note: Load reduction is to be considered on a case by case basis, depending on engine configuration (single or multiple) and relief mechanism (self-closing valve or rupture disk).

4.13.4.5 Exhaust system on the engine

.1 Continuous relief of exhaust gas (through open rupture disc) into the engine room or other enclosed spaces is not acceptable.

.2 Suitable explosion relief system for air inlet manifolds, scavenge spaces and exhaust system are to be provided unless designed to accommodate the worst-case overpressure due to ignited gas leaks or justified by the safety concept of the engine. A detailed evaluation regarding the hazard potential of overpressure in air inlet manifolds, scavenge spaces and exhaust system are to be carried out and reflected in the safety concept of the engine.

.3 Explosion relief devices for air inlet and exhaust manifold are to be type approved according to Classification Note "*Type Testing Procedure of Explosion Relief Devices for Combustion Air Inlet and Exhaust Gas Manifolds of I.C. Engines using Gas as Fuel*"

.4 The necessary total relief area and the arrangement of the explosion relief devices are to be determined taking into account:

- The worst-case explosion pressure depending on initial pressure and gas concentration,
- the volume and geometry of the component, and
- the strength of the component.

.5 The arrangement is to be determined in the risk analysis (refer *Classification Notes: Approval of I.C. Engines, 5.3.4.1 (g)*) and reflected in the safety concept.

4.13.4.6 Engine crankcase

.1 Crankcase explosion relief valves

Crankcase explosion relief valves are to be installed in accordance with 4.6.1. Also refer Pt.5, Ch.35, Sec.10, 10.3.1.2. For engines not covered by 4.6.1, the detailed evaluation as required in *Classification Notes: Approval of I.C. Engines, 5.3.4.1 (i*), is to determine if crankcase explosion relief valves are necessary.

.2 Inerting

For maintenance purposes, a connection, or other means, are to be provided for crankcase inerting and ventilating and gas concentration measuring.

.3 Crankcase ventilation

Ventilation of crankcase (either supply or extraction), if arranged, is to comply with 4.6. Relevant evidence is to be documented in Safety Concept. The ventilation systems for crankcase, sump and other similar engine spaces are to be independent from the systems on the other engines.

4.13.4.7 Gas ignition in cylinder

.1 Requirements of Pt.5, Ch.35, Sec.10, 10.3 are applicable. For gas carriers, Pt.5, Ch.4, Sec.16, 16.7 applies.

4.13.4.8 Control, monitoring, alarm and safety systems

.1 The engine control system is to be independent and separate from the safety system.

.2 The gas admission valves are to be controlled by the engine control system or by the engine gas demand.

.3 Combustion is to be monitored on an individual cylinder basis.

.4 In the event that poor combustion is detected on an individual cylinder, gas operation may be allowed in the conditions specified in Pt.5, Ch.35, Sec.10, 10.3.1.6.

.5 If monitoring of combustion for each individual cylinder is not practicable due to engine size and design, common combustion monitoring may be accepted.

.6 Unless the risk analysis (required by Section 5, 5.3 of the Classification note: "*Approval of I.C. Engines*") proves otherwise, the monitoring and safety system functions for DF or GF engines are to be provided in accordance with Table 4.13.4.8 in addition to the general monitoring and safety system functions given in Chapter 7.

Note: For DF engines, Table 4.13.4.8 applies only to the gas mode.

4.13.4.9 Gas admission valves

.1 Electrically operated gas admission valves are to be certified safe as follows:

a) The inside of the valve contains gas and shall therefore be certified for Zone 0;

b) When the valve is located within a pipe or duct in accordance with 4.13.4.3.1, the outside of the valve is to be certified for Zone 1;

c) When the valve is arranged without enclosure in accordance with the "ESD-protected machinery space" (see 4.13.4.3.2) concept, no certification is required for the outside of the valve, provided that the valve is de-energized upon gas detection in the space.

However, if they are not rated for the zone they are intended for, it is to be documented that they are suitable for that zone. Documentation and analysis are to be based on IEC 60079-10-1:2015 or IEC 60092-502:1999.

.2 Hydraulically operated gas admission valves are to be provided with sealing arrangement to prevent gas from entering the hydraulic oil system.

4.13.5 Specific Design Requirements

4.13.5.1 DF Engines

4.13.5.1.1 General

.1 The maximum continuous power that a DF engine can develop in gas mode may be lower than the approved MCR of the engine (i.e. in oil fuel mode), depending in particular on the gas composition and its quality or the engine system.

.2 This maximum continuous power available in gas mode and the corresponding conditions is to be stated by the engine manufacturer.

Requirements for type testing of DF and GF engines are given in Classification notes: *Approval of I.C. Engines*.

4.13.5.1.2 Starting, changeover and stopping

.1 DF engines are to be arranged to be started using either oil fuel or gas fuel with pilot oil fuel for ignition. The engines are to be arranged for rapid changeover from gas use to fuel oil use. In the case of changeover to either fuel supply, the engines are to be capable of continuous operation using the alternative fuel supply without interruption to the power supply.

.2 Changeover to gas fuel operation is to be only possible at a power level and under conditions where it can be done with acceptable reliability and safety as demonstrated through testing.

.3 Changeover from gas fuel operation mode to oil fuel operation mode is to be possible at all situations and power levels.

.4 The changeover process itself from and to gas operation is to be automatic but manual interruption is to be possible in all cases.

.5 If the power level or other conditions do not allow safe and reliable gas operation, changeover to oil fuel mode is to be automatically performed.

.6 In case of shut-off of the gas supply, the engines are to be capable of continuous operation by oil fuel only.

4.13.5.1.3 Pilot Injection

.1 Gas supply to the combustion chamber is not to be possible without operation of the pilot oil injection.

Note: Pilot injection is to be monitored for example by fuel oil pressure and combustion parameters.

Table 4.13.4.8 : Monitoring and Safety System Functions for DF and GF Engines

Parameter	Alarm	Automatic activation of the double block-and-bleed valves	Automatic switching over to oil fuel mode [1)]	Engine shutdown
Abnormal pressures in the gas fuel supply line	X	X	X	X [5)]
Gas fuel supply systems - malfunction	X	X	X	X [5)]
Pilot fuel injection or spark ignition systems - malfunction	X	X [2)]	X	X [2),5)]
Exhaust gas temperature after each cylinder - high	X	X [2)]	X	X [2),5)]
Exhaust gas temperature after each cylinder, deviation from average – low [3)]	X	X [2)]	X	X [2),5)]
Cylinder pressure or ignition - failure, including misfiring, knocking and unstable combustion	X	X [2)4)]	X [4)]	X [2),4),5)]
Oil mist concentration in crankcase or bearing temperature [6)] - high	X	X	-	X[9)]
Pressure in the crankcase – high [8)]	X	X	X	-
Engine stops - any cause	X	X	-	-
Failure of the control-actuating medium of the block and bleed valves	X	X	X	-
Failure of crankcase ventilation system, if applicable	X	X[7)]	X[7)]	

Notes:

1) DF engine only, when running in gas mode.

2) For GF engines, the double block-and-bleed valves and the engine shutdown may not be activated in case of specific failures affecting only one cylinder, provided that the concerned cylinder can be individually shutoff and the safe operation of the engine in such conditions is demonstrated by the risk analysis (Refer Section 5, 5.3 of the Classification Note: "*Approval of I.C. Engines"* for risk analysis requirement).

3) Required only if necessary for the detection of misfiring.

4) In the case where the failure can be corrected by an automatic mitigation action, only the alarm may be activated. If the failure persists after a given time, the safety actions are to be activated.

5) GF Engine only.

6) Where required by 4.6

7) Automatic safety actions to be activated as specified by the engine manufacturer, refer 4.6.

8) Only for trunk piston engines. This pressure sensor cannot replace or substitute a gas detector.

9) Only for trunk piston engines. For crosshead engines slow down is to apply (refer Pt.5, Ch.22, Sec.3, Table 3.1.1)

4.13.5.2 GF Engines

4.13.5.2.1 Spark ignition system

.1 In case of failure of the spark ignition, the engine is to be shut down except if this failure is limited to one cylinder, subject to immediate shut off of the cylinder gas supply and provided that the safe operation of the engine is substantiated by the risk analysis and by tests.

4.13.5.3 Pre-mixed Engines

4.13.5.3.1 Charge air system

.1 Inlet manifold, turbo-charger, charge air cooler, etc. are to be regarded as parts of the fuel gas supply system. Failures of those components likely to result in a gas leakage are to be considered in the risk analysis (see Section 5, 5.3 of the Classification Note: "*Approval of I.C. Engines"*).

.2 Flame arresters are to be installed before each cylinder head, unless otherwise justified in the risk analysis, considering design parameters of the engine such as the gas concentration in the charge air system, the path length of the gas-air mixture in the charge air system, etc.

4.13.5.4 Two-stroke engines

4.13.5.4.1 Scavenge air system

.1 The risk analysis required in IRS *Classification Notes: Approval of I.C. Engines,* is to cover the possible gas accumulation in a scavenge space.

4.13.5.4.2 Crankcase

.1 The risk analysis required in IRS *Classification Notes: Approval of I.C. Engines,* is to cover the possible failure of a piston rod stuffing box.

4.13.6 Type Testing, Factory Acceptance Tests and Shipboard Trials

4.13.6.1 Type Testing

The type approval of DF and GF engines is to be carried out in accordance with the Classification Note : *Approval of I.C. Engines*.

4.13.6.2 Factory Acceptance Test

.1 General.

Factory acceptance tests of DF and GF engines are to be carried out in accordance with 4.11, taking into account the additional requirements given below:

(a) For DF engines, the load tests referred to in 4.11.3.7 are to be carried out in diesel mode and in gas mode at the different percentages of the engine's MCR. Maximum continuous power available in gas mode is to be demonstrated (see 4.13.5.1.1).

.2 Safety precautions

In addition to the safety precautions mentioned in 4.11.1, measures to verify that gas fuel piping on engine is gas tight are to be carried out prior to start-up of the engine.

.3 Records

In addition to the records required in 4.11.3.2, the following engine data are to be recorded:

- Fuel index, both gas and diesel as applicable (or equivalent reading)
- Gas pressure and temperature
- Pilot fuel temperature and pressure (supply or common rail as appropriate)

.4 Test loads

Test loads for various engine applications are given in 4.11.3.7. DF engines are to be tested in both diesel and gas mode as found applicable. In addition, the scope of the trials may be expanded depending on the engine application, service experience, or other relevant reasons.

.5 Integration Tests

a) GF and DF engines are to undergo integration tests to verify that the response of the complete mechanical, hydraulic and electronic system is as predicted for all intended operational modes.

The scope of these tests is to be agreed with IRS for selected cases based on the risk analysis required in Section 5, 5.3 of the Classification Note: "*Approval of I.C. Engines"* and is to at least include the following incidents:

- Failure of ignition (spark ignition or pilot injection systems), for one-cylinder unit
- Failure of a gas admission valve
- Failure of the combustion (to be detected by e.g. misfiring, knocking, exhaust temperature deviation, etc.)
- Abnormal gas pressure
- Abnormal gas temperature

The above tests may be carried out using simulation or other alternative methods, subject to special consideration.

4.13.6.3 Shipboard Trials

.1 A leak test is to be carried out for the gas piping system (Pt.5, Ch.35, Sec.16, 16.7.3.3) after assembly on board.

.2 Shipboard trials are to be carried out in accordance with the provisions of 4.12, considering the additional requirements below.

.3 For DF engines, the test loads required in 4.12.5 are to be carried out in all operating modes (gas mode, diesel mode, etc.) as applicable (refer 4.13.5). The maximum continuous power available in gas mode is to be demonstrated.

Note: If a test load is performed in all applicable operation modes without interruption (direct changeover at same power and speed), the duration as required in 4.1.2.5 may be considered as the total duration demonstrated in all fuel modes. However, demonstration at each mode is not to be less than one hour.

.4 The starting maneuvers required in 4.12.3 are to be carried out in diesel mode and gas mode, if applicable.

.5 For DF engines, automatic switching over to oil fuel mode is to be tested. Further, manual change over from diesel to gas mode and vice versa is to be tested.

.6 The efficiency of the ventilation arrangement, or other approved methods of safety (such as inert gas systems), of the double walled gas piping system is to be verified.

4.13.7 Certification of Engine Components

4.13.7.1 The principles, definitions, and general requirements of IRS *Classification Notes: Approval of I.C. Engines* apply. In addition to those engine components specified in 3.2.2 of the Classification Notes, the engine components listed in Table 3.2.3 of the Classification Notes are to be documented as listed in the table.

4.14 Storage and Use of Selective Catalytic Reduction (SCR) Reductants

4.14.1 General

4.14.1.1 The NOx Technical Code, in 2.2.5 and elsewhere, provides for the use of NOx Reducing Devices of which Selective Catalytic Reduction (SCR) is one option. SCR requires the use of a reductant which may be a urea/water solution or, in exceptional cases, aqueous ammonia or even anhydrous ammonia. These requirements apply to the arrangements for the storage and use of SCR reductants which are typically carried on board in bulk quantities.

4.14.2 Reductant using Urea based Ammonia (for e.g. 40% / 60% urea/ water solution)

4.14.2.1 Where urea based ammonia (e.g. AUS 40 – aqueous urea solution specified in ISO 18611-1:2014) is introduced, the storage tank is to be arranged so that any leakage will be contained and prevented from making contact with heated surfaces. All pipes or other tank penetrations are to be provided with manual closing valves attached to the tank. Tank and piping arrangements are to be approved.

4.14.2.2 Urea storage tanks are to be arranged so that they can be emptied of urea, purged and vented.

4.14.2.3 The storage tank may be located within the engine room.

4.14.2.4 The storage tank is to be protected from excessively high or low temperatures applicable to the particular concentration of the solution. Depending on the operational area of the ship, this may necessitate the fitting of heating and/ or cooling systems. The physical conditions recommended by applicable recognized standards (such as ISO 18611-3:2014) are to be taken into account to ensure that the contents of the aqueous urea tank are maintained to avoid any impairment of the urea solution during storage.

4.14.2.5 If a urea storage tank is installed in a closed compartment, the area is to be served by an effective mechanical ventilation system of extraction type providing not less than 6 air changes per hour which is independent from the ventilation system of accommodation, service spaces, or control stations. The ventilation system is to be capable of being controlled from outside the compartment. A warning notice requiring the use of such ventilation before entering the compartment is to be provided outside the compartment adjacent to each point of entry.

Alternatively, where a urea storage tank is located within an engine room a separate ventilation system is not required when the general ventilation system for the space is arranged so as to provide an effective movement of air in the vicinity of the storage tank and is to be maintained in operation continuously except when the storage tank is empty and has been thoroughly ventilated. The following are to be complied with in addition:

- The urea tank location is to be served by an extraction ventilation opening served by an exhaust fan/ blower of the engine room.

- This exhaust fan/ blower is to be provided with an alarm in the ER control panel and at the location of the urea tank, when the fan/ blower is stopping/ tripping off.

- A warning notice is to be placed at the location of the urea tank that the location is ventilated by the E.R. exhaust fan/ blower no.___.

4.14.2.6 Each urea storage tank is to be provided with temperature and level monitoring arrangements. High and low level alarms together with high and low temperature alarms are also to be provided.

4.14.2.7 Where urea based ammonia solution is stored in integral tanks, the following are to be considered during the design and construction:

a) These tanks may be designed and constructed as integral part of the hull, (e.g. double bottom, wing tanks).

b) These tanks are to be coated with appropriate anti-corrosion coating and cannot be located adjacent to any fuel oil and fresh water tank.

c) These tanks are to be designed and constructed as per the structural requirements applicable to hull and primary support members for a deep tank construction.

d) These tanks are to be included in the ship's stability calculations.

4.14.2.8 The requirements specified in 4.14.2.5 also apply to closed compartments normally entered by persons:

- when they are adjacent to the urea integral tanks and there are possible leak points(e.g. manhole, fittings) from these tanks; or
- when the urea piping systems pass through these compartments, unless the piping system is made of steel or other equivalent material with melting point above 925 degrees C and with fully welded joints.

4.14.2.9 The reductant piping and venting systems are to be independent of other ship service piping and/or systems. Reductant piping systems are not to be located in accommodation, service spaces, or control stations. The vent pipes of the storage tank are to terminate in a safe location on the weather deck and the tank venting system is to be arranged to prevent entrance of water into the urea tank.

4.14.2.10 Reductant tanks are to be of steel or other equivalent material with a melting point above 925 degrees C. Pipes/ piping systems

are to be of steel or other equivalent material with melting point above 925 degrees C, except downstream of the tank valve, provided this valve is metal seated and arranged as fail-to-closed or with quick closing from a safe position outside the space in the event of fire; in such case, type approved plastic piping may be accepted even if it has not passed a fire endurance test. Reductant tanks and pipes/ piping systems are to be made with a material compatible with reductant or coated with appropriate anti-corrosion coating.

4.14.2.11 The ship is to have on board suitable personnel protective equipment, for the protection of crew members, Eyewash is to be provided, the location and number of these eyewash stations are to be derived from the detailed installation arrangements.

4.14.2.12 Urea storage tanks are to be arranged so that they can be emptied of urea, and ventilated by means of portable or permanent systems.

4.14.3 Reductant using aqueous ammonia (28% or less concentration of ammonia)

4.14.3.1 Aqueous ammonia is not to be used as a reductant in a SCR except where it can be demonstrated that it is not practicable to use a urea based reductant. Where an application is made to use aqueous ammonia as the reductant then the arrangements for its loading, carriage and use are to be derived from a risk based analysis.

4.14.4 Reductant using anhydrous ammonia (99.5% or greater concentration of ammonia by weight)

4.14.4.1 Anhydrous ammonia is not to be used as a reductant in a SCR except where it can be demonstrated that it is not practicable to use a urea based reductant and where the Flag Administration agrees to its use. Where it is not practicable to use a urea reductant then it is also to be demonstrated that it is not practicable to use aqueous ammonia. Where an application is made to use anhydrous ammonia as the reductant then the arrangements for its loading, carriage and use are to be derived from a risk based analysis.

4.15 Safety Measures for chemical treatment fluids used in Exhaust Gas Cleaning Systems and the Residues which have Hazardous Properties

4.15.1 General

4.15.1.1 With regard to the sulphur content limits of fuel oil specified in MARPOL Annex VI Reg. 14, alternative methods of compliance may be used as allowed by MARPOL Annex VI, Reg. 4.

4.15.1.2 As some types of Exhaust Gas Cleaning Systems (EGCS) to be approved by the Administration as “alternative compliance method” consume chemicals which are typically carried on board in bulk quantities, the safety measures against chemical treatment fluids contained in this subsection apply to EGCSs using such fluids. In this context, the term “chemical treatment fluid” means the aqueous solution of sodium hydroxide (NaOH) or calcium hydroxide ($Ca(OH)_2$) that has corrosive properties or are considered to represent a hazard to personnel.

4.15.1.3 For EGCSs using chemical treatment fluids other than the above, safety measures are to be taken according to the result of a risk assessment to be conducted to analyze the risks, in order to eliminate or mitigate the hazards to personnel brought by the use of such exhaust gas cleaning systems, to an extent equivalent to systems complying with 4.15.2.

4.15.2 Requirements for EGCSs using aqueous solution of NaOH or $Ca(OH)_2$ for chemical treatment fluid

4.15.2.1 The storage tanks are to have sufficient strength to withstand a pressure corresponding to the maximum height of a fluid column in the overflow pipe, with a minimum of 2.4 m above the top plate taking into consideration the specific density of the treatment fluid. Overflow pipes are to be led to appropriate tanks, as specified for drain pipes for drip trays in 4.15.2.12.

4.15.2.2 Where chemical treatment fluid is stored in integral tanks, the following are to be considered during the design and construction:

.1 These tanks may be designed and constructed as integral part of the hull, (e.g. double bottom, wing tanks).

.2 These tanks are to be coated with appropriate anti-corrosion coating and are to be segregated by cofferdams, void spaces, pump rooms, empty tanks or other

similar spaces so as to not be located adjacent to accommodation, cargo spaces containing cargoes which react with chemical treatment fluids in a hazardous manner as well as any food stores, oil tanks and fresh water tanks.

.3 These tanks are to be designed and constructed as per the structural requirements applicable to hull and primary support members for a deep tank construction.

.4 These tanks are to be included in the ship's stability calculation.

4.15.2.3 The storage tank is to be protected from excessively high or low temperatures applicable to the particular concentration chemical treatment fluids. Depending on the operational area of the ship, this may necessitate the fitting of heating and/or cooling systems.

4.15.2.4 Each storage tank for chemical treatment fluids is to be provided with level monitoring arrangements and high/ low level alarms. In cases where heating and/ or cooling systems are provided, high and/ or low temperature alarms or temperature monitoring are also to be provided accordingly.

4.15.2.5 The storage tank may be located within the engine room. In this case, the requirements of 4.15.2.6 are to be complied with, except that a separate ventilation system is not required when the general ventilation system for the space is arranged so as to provide an effective movement of air in the vicinity of the storage tank and is maintained in operation continuously.

4.15.2.6 If a storage tank for chemical treatment fluids is installed in a closed compartment, the area is to be served by an effective mechanical ventilation system of extraction type providing not less than 6 air changes per hour which is independent from the ventilation system of other spaces. The ventilation system is to be capable of being controlled from outside the compartment. A warning notice is to be provided outside the compartment, adjacent to each point of entry, indicating that the ventilation systems to be operated for at least 15 minutes before entry into such compartments.

4.15.2.7 The requirements specified in 4.15.2.6 also apply to closed compartments normally entered by persons:

.1 when they are adjacent to the integral storage tank for chemical treatment fluids and there are possible leak points (e.g. manhole, fittings) from these tanks; or

.2 when the treatment fluid piping systems pass through these compartments, unless the piping system is made of steel or other equivalent material with melting point above 925 degrees C and with fully welded joints.

4.15.2.8 The storage tank for chemical treatment fluids is to be arranged so that any leakage will be contained and prevented from making contact with heated surfaces. All pipes or other tank penetrations are to be provided with manual closing valves attached to the tank. In cases where such valves are provided below top of tank, they are to be arranged with quick acting shutoff valves which are to be capable of being remotely operated from a position accessible even in the event of chemical treatment fluid leakages. Tank and piping arrangements are to be approved by IRS.

4.15.2.9 The chemical treatment fluid piping and venting systems are to be independent of other ship service piping and/ or systems. The chemical treatment fluid piping systems are not to be located in accommodation, service spaces, or control stations. The vent pipes of the storage tank are to terminate in a safe location on the weather deck and the tank venting system is to be arranged to prevent entrance of water into the tank for chemical treatment fluids.

4.15.2.10 Storage tanks and pipes/piping systems and drip trays for chemical treatment fluids are to be of steel or other equivalent material (with a melting point above 925 degrees C), which is compatible with the chemical treatment fluids or coated with appropriate anticorrosion coating.
Note: Several metals are incompatible with the chemical treatment fluids, e.g. NaOH is incompatible with zinc, aluminum, etc.

4.15.2.11 Regardless of design pressure and temperature, piping systems containing chemical treatment fluids only are to comply with the requirements applicable to Class I piping systems. As far as practicable, e.g. except for the flange connections that connect to tank valves, the piping systems are to be joined by welding.

4.15.2.12 The following connections are to be screened and fitted with drip trays to prevent

the spread of any spillage where they are installed:

.1 Detachable connections between pipes (flanged connections and mechanical joints, etc.);

.2 Detachable connections between pipes and equipment such as pumps, strainers, heaters, valves; and

.3 Detachable connections between equipment mentioned in the above subparagraph.

The drip trays are to be fitted with drain pipes of compatible material and size which lead to appropriate tanks of adequate capacities, such as residue tanks, which are fitted with high level alarm, or are to be fitted with alarms for leak detection. In cases where such tank is an integral tank, 4.15.2.2.1 and 4.15.2.2.2 are to be applied to the tank.

4.15.2.13 For the protection of crew members, the ship is to have on board suitable personnel protective equipment. The number of personnel protective equipment carried onboard is to be appropriate for the number of personnel engaged in regular handling operations or that may be exposed in the event of a failure; but in no case is there to be less than two sets available onboard.

4.15.2.14 Personnel protective equipment is to consist of protective clothing, boots, gloves and tight-fitting goggles. Eyewash and safety showers are to be provided, the location and number of these eyewash stations and safety showers are to be derived from the detailed installation arrangements. As a minimum, the following stations are to be provided:

.1 In the vicinity of transfer or treatment pump locations. If there are multiple transfer or treatment pump locations on the same deck then one eyewash and safety shower station may be considered for acceptance provided that the station is easily accessible from all such pump locations on the same deck.

.2 An eyewash station and safety shower are to be provided in the vicinity of a chemical bunkering station on-deck. If the bunkering connections are located on both port and starboard sides, then consideration is to be given to providing two eyewash stations and safety showers, one for each side.

.3 An eyewash station and safety shower are to be provided in the vicinity of any part of the system where a spillage/drainage may occur and in the vicinity of system connections/components that require periodic maintenance.

4.15.2.15 Storage tanks for chemical treatment fluids are to be arranged so that they can be safely emptied of the fluids and ventilated by means of portable or permanent systems.

4.15.3 Requirement for Exhaust Gas Cleaning Systems discharge water pipeline

4.15.3.1 Overboard discharges from exhaust gas cleaning system (EGCS) are not to be interconnected to other systems.

4.15.3.2 Due consideration is to be given to the location of overboard discharges with respect to vessel propulsion features, such as thrusters, propellers or to prevent any discharge water onto survival craft during abandonment.

4.15.3.3 The piping material for the EGCS discharge water pipeline system is to be selected based on the corrosive nature of the liquid media.

4.15.3.4 Special attention is to be paid to the corrosion resistivity of EGCS overboard discharge piping. Where applicable, adequate arrangements are to be provided to prevent galvanic corrosion due to the use of dissimilar metals.

4.15.3.5 In case distance piece is fitted between the outboard discharge valve and the shell plating, it to be made of corrosion resistant material steel or be coated with an anti-corrosive material suitable for the operating environment. The thickness of the distance piece is to be at least the minimum values specified in a) and b) below; otherwise Sch.160 thickness specified in piping standards, as far as practicable, is to be used.

a) 12 mm in cases where complete pipe is made of corrosion resistant material steel.

b) 15 mm for mild steel in cases where the inside of the pipe is treated with an anticorrosive coating or fitted with a sleeve of corrosion resistant material.

4.15.4 Tanks for Residues Generated from Exhaust Gas Cleaning process

4.15.4.1 Tanks for residues generated from the exhaust gas cleaning process are to satisfy the following requirements:

.1 The tanks are to be independent from other tanks, except in cases where these tanks are also used as the over flow tanks for chemical treatment fluids storage tank.

.2 Tank capacities are to be decided in consideration of the number and kinds of installed exhaust gas cleaning systems as well as the maximum number of days between ports where residue can be discharged ashore. In the absence of precise data, a figure of 30 days is to be used.

.3 Where residue tanks used in closed loop chemical treatment systems are also used as the overflow tanks for chemical treatment fluids storage tank, the requirements for storage tanks apply.

4.16 AC Generating Sets

4.16.1 General

4.16.1.1 This sub-section provides requirements for AC Generating sets (i.e. reciprocating internal combustion engines, alternators and couplings), in addition to those stated in this Chapter and Ch.8.

4.16.1.2 The requirements are applicable to AC generating sets driven by reciprocating internal combustion engines irrespective of their types (i.e. diesel engine, dual fuel engine, gas-fuel engine), except for those sets consisting of a propulsion engine which also drives power take off (PTO) generator(s).

4.16.2 Requirements

4.16.2.1 The generating set is to show torsional vibration levels which are compatible with the allowable limits for the alternator, shafts, coupling and damper.

4.16.2.2 The coupling selection for the generating set is to take into account the stresses and torques imposed on it by the torsional vibration of the system. The torsional vibration calculations are to be submitted to IRS for approval when the engine power is 110 [kW] or above.

4.16.2.3 The rated power is to be appropriate for the actual use of the generator set.

4.16.2.4 The firm responsible for assembling the generating set is to install a rating plate marked with at least the following information:

(i) the generating set manufacturer's name or mark;
(ii) serial number of the set;
(iii) date of manufacture (month/year) of the set;
(iv) the rated power (both in kW and KVA) with one of the prefixes COP, PRP (or, only for emergency Generating sets, LTP) as defined in ISO 8528-1:2018;
(v) the rated power factor;
(vi) the set rated frequency [Hz];
(vii) the set rated voltage [V];
(viii) the set rated current [A];and
(ix) the mass [kg].

4.17 Capacity and availability of compressed air for essential services onboard ships other than the supply of compressed air for engine starting

4.17.1 The arrangements for the supply of compressed air to essential services are to ensure that sufficient compressed air to satisfy the total demand of the essential services is available at all times during normal operation, during maintenance, and in the event of a failure of the compressed air system.

4.17.2 Where compressed air is supplied from the engine starting air system, either continuously in normal operation, or periodically during maintenance or in the event of a failure of the compressed air system, the required compressed air demand is not to reduce the capacity and availability of the engine starting air required by 4.9.2.

Section 5

Gearing

5.1 Scope

5.1.1 The requirements of this Section cover the construction, material and inspection of reduction gears for main propelling purposes and for driving electric generators.

5.1.2 Design of bevel gears will be specially considered.

5.1.3 For torsional vibration requirements, See Sec.8.

5.1.4 Rated power of gear is the maximum transmitted power at which the gear is designed to operate continuously at its rated speed.

5.1.5 The rated torque is defined by the rated power and speed and is the torque used in the gear rating calculations.

5.1.6 The gear rating is the rating for which the gear is designed to transmit it's rated torque.

5.2 Plans and particulars

5.2.1 The following plans, in triplicate, of the reduction gearing are to be submitted:

For Approval

a) Pinion(s) and wheel(s);

b) Shafts;

c) Hub(s);

d) Other power transmitting parts;

For information only

e) Longitudinal and transverse sections of the gear box;

f) Clutch(es) and/or coupling(s);

g) Gear casing including propeller thrust bearing housing, if applicable.

5.2.2 The plans are to show clearly all dimensions, details of all fillets and stress raisers, and material of all the parts.

5.2.3 At least the following particulars of the gearing are to be submitted along with the plans:

a) Shaft power and revolutions for each pinion;

b) Number of teeth in each gear;

c) Generating pitch diameters;

d) Helix angles at generating pitch diameters;

e) Normal pitches of teeth at generating pitch diameters;

f) Tip diameters;

g) Root diameters;

h) Face widths and gaps, where applicable;

i) Pressure angles of teeth (normal or transverse) at generating pitch diameters;

j) Minimum backlash;

k) Centre distance;

l) Basic rack tooth form;

m) Details of tooth flank corrections, if adopted;

n) Details of post hobbing process, if any;

o) Case depth for surface-hardened teeth;

p) Shrinkage allowance for shrunk-on rims and hubs;

q) Type of coupling proposed for oil engine applications.

5.2.4 Gears with Multiple Prime Mover Inputs

5.2.4.1 For single helical gears with arrangements utilizing multiple prime mover inputs, and single or multiple outputs, the following analyses for all operating modes are to be conducted:

- All bearing reactions
- Tooth modifications
- Load distributions on the gear teeth
- Contact and tooth root bending stresses

A summary of the results of these analyses for each operating mode is to be submitted for review.

5.3 Materials

5.3.1 Specifications for materials of pinions, pinion sleeves, wheel rims, gear wheels and shafting giving chemical composition, heat treatment and mechanical properties are to be submitted for approval with the plans of gearing and are to be in accordance with Pt.2.

5.3.2 Where the teeth of a pinion or gear wheel are to be surface hardened, the proposed specification and details of the procedure are to be submitted for approval.

5.3.3 In the selection of materials for pinions and wheels consideration should be given to their compatibility in operation. In general, for gears of through hardened steels, except in the case of low reduction ratios, provision should also be made for a hardness differential between pinion teeth and wheel teeth. For this purpose the specified minimum tensile strength of the wheel materials should not be more than 85 per cent of that of the pinion.

5.3.4 Subject to 5.3.3, the specified minimum tensile strength of steel gear forgings is to be selected within the following limits:

- Pinions and pinion sleeves - 550-1050 [N/mm^2];

- Gear wheels and rims - 400-850 [N/mm^2].

A tensile strength range is also to be specified and is not to exceed 120 [N/mm^2] when the specified minimum tensile strength is 600 [N/mm^2] or less. For higher strength steels, the range is not to exceed 150 [N/mm^2].

5.3.5 Unless otherwise agreed, the specified minimum tensile strength of the core is to be 800 [N/mm^2] for induction-hardened or nitrided gearing and 750 [N/mm^2] for carburized gearing.

5.4 Design and construction

5.4.1 The design and construction is to be such as to enable the gearing to meet the general requirements with respect to functional capability and reliability.

5.4.2 Detailed analysis to prove the design of the gearing is to be submitted for consideration. In this regard, Classification Notes on the design of gearing may be followed.

5.4.3 Where castings are used for wheel centres, any radial slots in the periphery are to be fitted with permanent chocks before shrinking-on the rim.

5.4.4 Where bolts are used to secure side plates to rim and hub, the bolts should be tight fit with the holes and the nuts should be suitably locked by means other than welding.

5.4.5 When welding is employed in the construction of wheels, the welding procedure is to be approved by the Surveyors before work is commenced. For this purpose, welding procedure approval tests are to be carried out with satisfactory results. Such tests are to be representative of the joint configuration and materials. Wheels are to be stress relieved after welding. All welds are to have a satisfactory surface finish and contour. Magnetic particle or liquid penetrant examination of all important welding joints is to be carried out to the satisfaction of the Surveyors.

5.4.6 In general arrangements are to be made so that the interior structure of the wheel may be examined. Alternative proposals will be specially considered.

5.5 Accuracy of gear cutting and alignment

5.5.1 Gears are to be cut only on machines which are maintained at a high standard of accuracy. Hobbing machines used in the production of large gears are to be operated under conditions of temperature control with a total temperature variations not exceeding 2°C for the finishing cut. The blank should be allowed sufficient time to stabilize to the machine temperature before cutting commences.

5.5.2 The accuracy of gear-cutting of pinions and wheels is to be demonstrated to the satisfaction of the Surveyors. For this purpose, records of measurements of pitch error, undulations, axial pitch errors, tooth thickness and backlash should be available for review by Surveyors on request.

5.5.3 The alignment is to be demonstrated in the workshop by meshing in the gearbox

without oil clearance in the bearings, or in the meshing frame without oil clearance in the bearings or on rollers. Meshing is to be carried out with the gears locating in their load positions, and a load sufficient to overcome pinion weight and axial movement is to be imposed.

5.5.4 A permanent record is to be made of the meshing contact for the purpose of re-checking the alignment when installed on board ship. The meshing contact on each helix is not to be less than following:

a) For through-hardened gears,

- 40 per cent of the working depth for 35 per cent of the length; and
- 20 per cent of the working depth for further 35 per cent of the length;

b) For through-hardened gears and for all surface hardened gears,

- 40 per cent of the working depth for 50 per cent of length; and
- 20 per cent of the working depth for a further 40 per cent of the length.

5.6 Balancing of gear pinions and wheels

5.6.1 All pinions, gear wheels and flexible couplings or sleeves whose maximum designed speed of rotation exceeds 1000 revolutions per minute are to be dynamically balanced, where the speed of rotation is 1000 revolutions per minute or less, these components are to be statically or, alternatively, dynamically balanced. Parts of couplings, etc., which are to be fixed to the gear in service are normally to be attached before balancing.

5.6.2 For static balancing the final out of balance of each assembly at the balancing planes is not to exceed 2200 [N mm/tonne] of gears whose maximum design speed is less than 300 revolutions per minute and 680 [N mm/tonnes] for gears whose maximum design speed is between 300 and 1000 revolutions per minute. For dynamic balancing the final out of balance is not to exceed $190,000/N_1$ [N mm/tonne], where N_1 = revolutions per minute appropriate to the assembly.

5.6.3 Balancing may, however, be omitted for turbine secondary pinions and for oil engines gearing, provided that the rotating components are of solid forged construction or have a solid forged centre with shrunk-on rim and in both cases are machined to give a concentric and uniform cross-section.

5.7 Gearcases

5.7.1 Gearcases and their supports are to be designed sufficiently stiff such that misalignment at the mesh due to movements of the external foundations and the thermal effects under all conditions of service do not disturb the overall tooth contact. If welding is employed in their construction they are to be stress-relieved on completion.

5.7.2 Inspection openings should be provided at the peripherals of gearcases to enable the teeth of pinions and wheels to be readily examined. When the construction of gearcases is such that sections of the structure cannot readily be moved for inspection purposes, access openings of adequate size are to be also provided at the ends of the gearcases to permit examination of the structure of the wheels. Their attachment to the shafts should be capable of being examined by removal of bearing caps or by equivalent means.

5.8 Type tests and sea trials

5.8.1 Upon completion of fabrication and assembly, reduction gear unit is to be subjected to type testing in accordance with the agreed test programme which is to be submitted by the manufacturer for approval. Type tests are to be witnessed by the Surveyor.

5.8.2 Reduction gearing units, in general, are to be type tested at maker’s works and test load and duration of testing is to be agreed upon in each case.

5.8.3 Tests are to include demonstration of satisfactory operation of clutches and couplings, if fitted.

5.8.4 After completion of type tests, all gear elements are to be examined for tooth contact marking and alignment. All gear tooth contact is to conform to the requirements given in Cl. 5.5.5.

5.8.5 The gear lub.oil system is to be provided with means of indicating the lub.oil pressure supply to them. Alarms are to be provided to indicate low lub.oil pressure, low clutch oil/air pressure and high lub.oil temperature in accordance with Part 4, Chapter 7, Table 1.9.1.

5.9 Alignment and weardown gauges

5.9.1 Reduction gears with sleeve bearings for main and auxiliary turbines and oil engines, are to be provided with weardown gauges or micrometers for testing the internal alignment of the various elements in the gearcases. In certain gears, e.g. gears of the dual tandem type, the direction of loading on the bearings of a gear element may be such that an accurate indication of its alignment under operating conditions cannot be obtained using weardown gauges. In these instances, suitable alternatives such as crown thickness micrometer are to be provided.

5.9.2 Approved means are to be provided by the gear manufacturer to enable the Surveyors to verify that no distortion of the gearcase has taken place, when chocked and secured to its seating on board ship.

5.10 Trials

5.10.1 The sea trials should be of sufficient duration to prove the gears. After these trials, the marking revealed by inspection should indicate freedom from hard bearings, particularly towards both ends of each helix.

5.10.2 In the case of through-hardened gears, not less than 70 per cent contact across effective face width should be indicated. When the teeth of such gears are finished by an approved post hobbing process or profile ground, not less than 90 per cent contact across the effective face width is to be indicated.

5.10.3 For surface hardened gears, the contact across the effective face width is also to be not less than 90 per cent.

Section 6

Main Propulsion Shafting

6.1 Scope

6.1.1 The requirements of this Section apply to shafting for main propulsion of straight forged design and which are driven by rotating machines such as diesel engines, turbines or electric motors. The requirements for couplings, coupling bolts, keys, keyways, sternbushes and associated components are also included. The diameter of shafting as calculated may require to be modified as a result of alignment considerations and vibration characteristics (See Sec.8) or the inclusion of stress raisers, other than those contained in this section.

The requirements given in this section do not cover shafts intended for following application.

- gearing shafts
- electric motor shafts
- generator rotor shafts
- turbine rotor shafts.

6.1.2 The scantlings of shafts that are integral to equipment, such as for gear boxes, podded drives, electrical motors and/or generators, thrusters, turbines and which in general incorporate particular design features are to be determined taking into account appropriate additional criteria including that for stiffness, high temperature application etc. The requirements given in this section may be applied if such shafts are subjected mainly to torsion and are having traditional design features.

6.1.3 For additional strengthening for shafts in ships classed for navigation in waters with ice condition, refer Pt.5, Ch.21 and for diesel engine crankshaft refer IRS Classification Note “Crankshaft for Internal Combustion Engines”.

6.2 Alternative calculation methods

6.2.1 Alternative calculation methods will be considered provided these calculations take into account all relevant loads in the complete dynamic shafting system under all permissible operating conditions giving due consideration to the dimensions and arrangements of all shafting connections. The alternative calculation method is to also take into account design criteria for continuous and transient operating loads for dimensional adequacy for fatigue strength and peak operating loads for yield strength. Refer 6.2.2 to 6.2.5 as guidance for alternative calculations.

6.2.2 The two important considerations that are essential for the design of propulsion shafting are:

a) Fatigue
b) Stress concentration and notch sensitivity.

***Fatigue** : The deterioration of the properties of material which takes place under conditions involving fluctuating stresses. Fatigue failures generally occur at loads, which if applied statically would be below the elastic limit. The fatigue limit of a material is the stress which will not produce failure, even if many fluctuations of it are imposed.*

***Stress concentration in shafts** : Basic stress analysis calculations assume that the components are smooth, have a uniform section and have no irregularities. In practice virtually all engineering components have to have changes in section and/or shape. Common examples are shoulders on shafts, oil holes, key ways and screw threads. Any discontinuity changes the stress distribution in the vicinity of the discontinuity, so that the basic stress analysis equations no longer apply. Such 'discontinuities' or 'stress raisers' cause local increase of stress referred to as 'stress concentration'.*

For static loading the theoretical or geometric stress concentration factors K_t or K_{ts} relate to actual maximum stress at the discontinuity to the nominal stress as follows:

K_t = maximum direct stress/nominal direct stress
K_{ts} = maximum shear stress/nominal shear stress.

The subscript 't' denotes that the stress concentration value is a theoretical calculation based only on the geometry of the component and discontinuity.

Some materials are not as sensitive to notches as implied by the theoretical stress concentration factor. For these materials a reduced value of stress concentration factor K_f may be used where the maximum stress = K_f x nominal stress. The value of K_f which depends on notch sensitivity 'q' is to be determined using the following equation:

$q = (K_f - 1) / (K_t - 1)$ where q is between 0 and 1.

If q = 0, then $K_f = 1$ as the material has no sensitivity to notches. If q = 1, then $K_f = K_t$ and the material is fully notch sensitive.

When designing, usual practice is to first find K_t from the geometry of the component, then specify the material and determine the notch sensitivity, q from the chart for the notch radius.

For cyclic loading, the theoretical stress concentration factor is to be defined as

$$K_{t1} = \frac{\text{endurance limit without stress concentration}}{\text{endurance limit with stress concentration}}$$

Using the notch sensitivity factor 'q' in cyclic loading of shaft, fatigue stress concentration factor K_f is calculated as :

$$K_f = 1 + q\,(K_{t1} - 1)$$

6.2.3 The alternative calculation methods are to take into account following fatigue related issues as have been considered in rule formulations:

a) Low cycle fatigue criterion (typically < 10^4), i.e. the primary cycles represented by zero to full load and back to zero, including reversing torque if applicable.

b) High cycle fatigue criterion (typically > 10^7), i.e. torsional vibration stresses permitted for continuous operation as well as reverse bending stresses.
The limits for torsional vibration stresses are given in 8.4.2.

c) The accumulated fatigue due to torsional vibration when passing through a barred speed range or any other transient condition with associated stresses beyond those permitted for continuous operation is addressed by the criterion for transient stresses in 8.4.2.

6.2.4 The factors k (for low cycle fatigue) and c_k (for high cycle fatigue) as given in Table 6.5.1 and Table 8.4.2 respectively take into account the influence of:

- The stress concentration factors (scf) relative to the stress concentration for a flange with fillet radius of $0.08d_o$ (geometric stress concentration of approximately 1.45).

$$c_k \approx \frac{1.45}{scf} \quad \text{and} \quad k \approx \left[\frac{scf}{1.45}\right]^x$$

where the exponent x considers low cycle notch sensitivity.

- The chosen values of the notch sensitivity are mainly representative for soft steels (σ_B < 600), while the influence of steep stress gradient s in combination with high strength steels may be underestimated.

- The size factor C_D being a function of diameter only does not purely represent a statistical size influence, but rather a combination of this statistical influence and the notch sensitivity.

6.2.5 The stress concentration factor (scf) at the end of slots can be determined by means of the following empirical formulae:

$$scf = \alpha_{t(hole)} + 0.8 \cdot \frac{(l - e)/d}{\sqrt{\left(1 - \frac{d_i}{d}\right).\frac{e}{d}}}$$

where,

e = slot width
l = length of the slot
d_o = outer diameter
d_i = inner diameter (Refer figure given under Table 6.5.1).

This formula applies to:

- slots at 120 or 180 or 360 degrees apart.

- slots with semicircular ends. (Though multi-radii slot end can reduce the local stresses, this is not included in this empirical formula).

- slots with no edge rounding (except chamfering), as any edge rounding increases the scf slightly.

$\alpha_{t(hole)}$ = factor representing the stress concentration of radial hole of diameter e:

$$= 2.3 - 3.\frac{e}{d} + 15.\left(\frac{e}{d}\right)^2 + 10.\left(\frac{e}{d}\right)^2.\left(\frac{d_i}{d}\right)^2$$

6.3 Plans and particulars

6.3.1 The following plans, in triplicate, together with the necessary particulars of the machinery, including the maximum power and revolutions per minute, are to be submitted for approval before the work is commenced.

- Power take-off arrangement if any (e.g. shaft generator), propulsion boosters or similar equipment rated for 100 [kW] and above;
- Final gear shaft;
- Thrust shaft;
- Intermediate shafting, shaft bearings;
- Tube shaft, where applicable;
- Tail shaft;
- Stern tube, shaft seals and Stern bush;
- Stern tube lubrication system;
- Couplings (integral, demountable, keyed or shrink-fit), coupling bolts and keys;
- Flexible coupling including constructional details, static and dynamic torsional stiffness, damping characteristic, rated power, torque and RPM, allowable vibratory torque for continuous and transient operation and allowable misalignment for continuous operations;
- Cardan shafts, if fitted.

6.3.2 The specified minimum tensile strength of each shaft is to be stated.

6.3.3 A shafting arrangement plan indicating the relative position of the main engines, flywheel, flexible coupling, gearing, thrust block, line shafting and bearings, stern tube, 'A' brackets and propeller, as applicable, is to be submitted for information.

6.4 Materials for shafting

6.4.1 The materials are to comply with the relevant requirements of Pt.2, Ch.5. The specified minimum tensile strength of forgings is to be selected within the following general limits:

a) Carbon and carbon-manganese steel 400-760 [N/mm^2];

b) Alloy steels - 400-800 [N/mm^2];

Where shafts may experience vibratory stresses close to the permissible stresses for transient operation, the materials are to have a specified minimum tensile strength of 500 [N/mm^2].

6.4.2 If materials with greater specified or actual tensile strength than the limitations given in 6.4.1 are used, no consideration will be given

for reduction of shaft diameter; when derived from the formulae given in this chapter. Reduction in shaft diameter may be allowed for intermediate shaft of alloy steel, when it is verified that the material used exhibit fatigue life as conventional steel when tested as per 6.20 of this chapter.

6.4.3 Ultrasonic tests are required on shaft forgings where the diameter is 250 [mm] or greater.

6.5 Intermediate and thrust shafts

6.5.1 The diameter, d, of the shaft is to be not less than determined by the following formula:

$$d = Fk \sqrt[3]{\frac{P}{n}\left(\frac{560}{\sigma B + 160}\right)} \text{ [mm]}$$

where,

F = 95 for turbine installations, electric propulsion installations and oil engine installations with slip type couplings;

= 100 for other oil engine installations;

k = shaft design factors as given in Table 6.5.1;

σB = specified minimum tensile strength of the material [N/mm^2] as per 6.4.1.

P = maximum shaft power, in [kW];

n = Revolutions per minutes corresponding to maximum shaft power giving maximum torque.

6.5.2 For shafts with design features other than stated in Table 6.5.1, the value of k will be specially considered.

6.6 Tailshafts and stern tube shafts

6.6.1 The diameter, d_p, of the tailshaft immediately forward of the forward face of the propeller boss or, if applicable, the forward face of the tailshaft flange, is to be not less than determined by the following formula:

$$d_p = 100\,k \sqrt[3]{\frac{P}{n}\left(\frac{560}{\sigma B + 160}\right)} \text{ [mm]}$$

where,

k = shaft design factors as given in Table 6.6.1;

σB = specified minimum tensile strength of the material [N/mm^2]. For calculation purposes, this value is not to be taken greater than 600 [N/mm^2] (for carbon, carbon manganese and alloy steels);

P and n are defined in 6.5.

6.6.2 The diameter, d_p of the tailshaft determined in accordance with the formula in 6.6.1 is to extend over a length not less than that to the forward edge of the bearing immediately forward of the propeller or 2.5 d_p whichever is the greater.

6.6.3 The diameter of the portion of the tailshaft and tubeshaft forward of the length required by 6.6.2 to the forward end of the forward sterntube seal is to be determined in accordance with the formula in 6.6.1 except that:

k = 1.15, where k = 1.22 or 1.26 as required by 6.6.1 (shafts fitted with continuous liner or oil lubricated);

k = 1.18, where k = 1.25 or 1.29 as required by 6.5.1 (shafts with water lubricated bearings).

The change of diameter from that required by 6.6.1 to that required by this clause should be gradual.

6.6.4 Tailshafts which run in sterntubes and tube shafts may have the diameter forward of the forward stern tube seal gradually reduced to the diameter of the intermediate shaft. Abrupt changes in shaft section at the tailshaft/tubeshaft to intermediate shaft couplings are to be avoided.

Table 6.5.1 : Shaft design factors k for line shafts and thrust shafts

Factor	Intermediate shaft with					Thrust shafts external to engine		
	Integral coupling flange [1]	Shrink fit coupling [2]	Key-ways cylindrical and tapered [3], [4]	Radial holes, transverse holes [5]	Longitudinal slots [6]	On both sides of thrust collars [1]	In way of axial bearings where a roller bearing is used for taking up thrust	Straight sections
k	1.0	1.0	1.1	1.1	1.2	1.1	1.1	1.0

Notes:

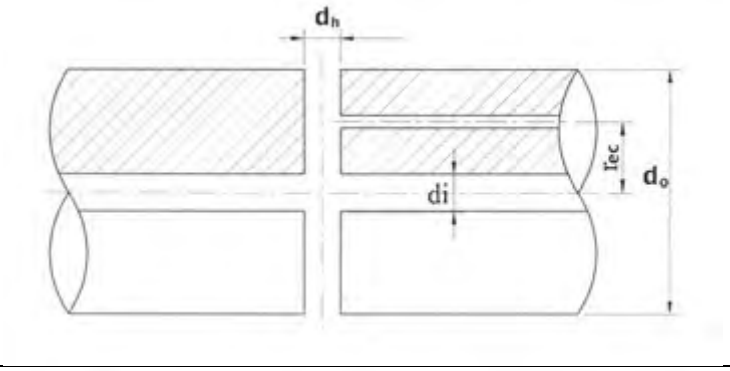
1) Fillet radius is not to be less than 0.08d.

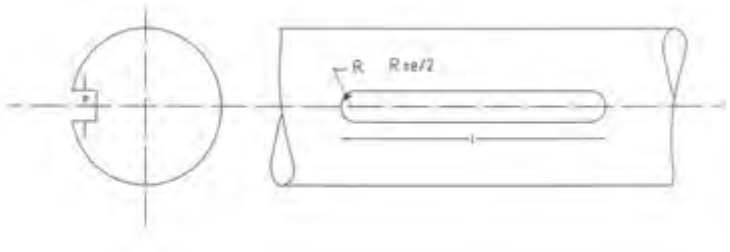
2) k refers to the plain shaft section only. Where shafts may experience vibratory stresses close to the permissible stresses for continuous operation, an increase in diameter to the shrink fit diameter is to be provided, e.g. a diameter increase of 1 to 2% and a blending radius nearly equal to the change in diameter are to be provided.

3) At a distance of not less than 0.2d from the end of the keyway the shaft diameter may be reduced to the diameter calculated with k = 1.0. Fillet radii in the transverse section of the bottom of the key way are not to be less than 0.0125d.

4) It is recommended that keyways are in general not to be used in installations with a slow speed crosshead or 2-stroke engines with a barred speed range.

5) Diameter of radial bore (d_h) not to exceed $0.3d_o$.
The intersection between a radial and an eccentric (r_{ec}) axial bore (see below) the value will be determined on the basis of data submitted in each case.





6) Subject to following limitations being complied with -

i) slot length (l)/outside diameter < 0.8 and
ii) inner diameter (d_i)/outside diameter d_o < 0.7 and
iii) slot width (e)/outside diameter d_o > 0.15.
The end rounding of the slot is not to be less than e/2. An edge rounding should preferably be avoided as this increases the stress concentration slightly. The k value is valid for 1, 2 and 3 slots and they are to be arranged 360, 180 or 120 degrees apart from each other respectively.

Table 6.6.1 : Shaft design factors k for tailshafts [1] and stern tube shafts						
Factor	**Propulsion drive**	**Stern tube configuration**	**Tailshafts : propeller attachment method**			**Stern tube shafts**
			Keyed	**Key less attachment by shrink fit**	**Flanged**	
k	All	Oil lubricating bearings and provided with an approved type of oil seating gland	1.26	1.22	1.22	1.15
	All	Water lubricated bearings : continuous shaft liners / shafts of corrosion resistant material or equivalent	1.26	1.22	1.22	1.15
	All	Water lubricated bearings : non-continuous shaft liners	1.29	1.25	1.25	1.18
Note : 1) Fillet radii in the transverse section at the bottom of the keyway are not to be less than $0.0125d_p$						

6.7 Hollow shafts

6.7.1 For hollow shafts where the bore exceeds 40 per cent of the outside diameter the minimum shaft diameter is not to be less than that given by the following equation:

$$d_o = d \sqrt[3]{\frac{1}{1-\left(\frac{d_i}{d_o}\right)^4}} \quad [mm]$$

where,

d_o = outside diameter [mm];

d = Rule size diameter of shaft [mm], calculated in accordance with 6.5 or 6.6;

d_i = diameter of central hole [mm].

6.7.2 Where the diameter of central hole does not exceed 0.4 times the outside diameter, no increase over Rule size need be provided.

6.8 Integral couplings

6.8.1 The thickness of coupling flanges is not to be less than the minimum required diameter of the coupling bolts calculated as in para 6.9, with tensile strength of the bolt material being taken as the tensile strength of the corresponding shaft material or 0.2 times the rule diameter of the shaft under consideration, whichever is greater. Special consideration will be given by IRS for flanges having non-parallel faces; however, in such cases the thickness of the flanges is not be less than the coupling bolt diameter.

6.8.2 The fillet radius at the base of the coupling flange is to be not less than 0.08 of the diameter of the shaft at the coupling. The fillets are to have a smooth finish and are not to be recessed in way of nuts and bolt heads. The fillet may be formed of multiradii in such a way that the stress concentration factor will not be greater than for a circular fillet with radius 0.08 times the actual shaft diameter.

6.8.3 Where the propeller is attached by means of a flange, the thickness of the flange is to be not less than 0.25 times the actual diameter of the adjacent part of the tailshaft. The fillet radius at the base of the coupling flange is to be not less than 0.125 times the diameter of the shaft at the coupling.

6.9 Demountable couplings

6.9.1 Couplings are to be made of steel or other approved ductile material. The strength of demountable couplings and keys is to be equivalent to that of the shaft. Couplings are to be accurately fitted to the shaft.

6.9.2 Hydraulic and other shrink fit couplings will be specially considered upon submittal of detailed preloading and stress calculations and fitting instructions. In general, the torsional holding capacity is to be at least 2.8 times the transmitted torque and preload stress is not to exceed 70 per cent of the yield strength.

6.9.3 Provision is to be made to resist astern pull.

6.10 Tooth couplings

6.10.1 The contact stress, σ_c, at the flanks of mating teeth of a gear coupling is not to exceed that given in Table below, where:

$$\sigma_c = \frac{24x10^6 P}{nd_p bhz} \ [N/mm^2]$$

P and n are defined in 6.4.1.

d_p = pitch circle diameter of coupling teeth [mm]

b = tooth facewidth [mm]

h = tooth height [mm]

z = number of teeth (per coupling half).

Allowable σ_c values	
Tooth material surface treatment	**Allowable σ_c value [N/mm^2]**
Surface hardened teeth	19
Through hardened teeth	11

6.10.2 Where experience has shown that under similar operating and alignment conditions, a higher tooth loading can be accommodated full details are to be submitted for consideration.

6.11 Flexible couplings

6.11.1 Details of flexible couplings are to be submitted together with the manufacturers' rating capacity, for the designed operating conditions including short term high power operation. Verification of coupling characteristics will be required.

6.11.2 In determining the allowable mean, maximum and vibratory torque ratings, consideration of the mechanical properties of the selected elastic element type in compression, shear and fatigue loading together with heat absorption/generation is to be given.

6.11.3 In determining the allowable torque ratings of the steel spring couplings, consideration of the material mechanical properties to withstand fatigue loading and overheating is to be given.

6.12 Interference fit assemblies

6.12.1 The interference fit assembly is to have a capacity to transmit a torque Q without slippage.

$$Q = 9550 \frac{P}{n} (1+C) . S \ [N-m]$$

Where,

P = power transmitted, kW

n = RPM of the shaft

C = coefficient as per Table 6.12.1

S = 2.0 for assemblies accessible from within the vessel

= 2.5 for assemblies not accessible from within the vessel.

Table 6.12.1 : 'C' values for guidance purposes	
Coupling Location	**C**
High Speed Shafting - IC engine driven	0.3
High Speed Shafting - Electric Motor or Turbine driven	0.1
Low Speed Shafting - Main or PTO stage gearing	0.1

6.12.2 The effect of any axial load acting on the assembly is to be considered.

6.12.3 The resulting equivalent von Mises stress in the assembly is not to be greater than the yield strength of the component material.

6.12.4 Reference marks are to be provided on the adjacent surfaces of parts secured by shrinkage alone.

6.13 Coupling bolts

6.13.1 The diameter of the fitted bolts at the joining faces of the coupling is to be not less than that given by the following formula:

$$d_b = 0.65 \sqrt{\frac{d^3 (T + 160)}{iD T_b}}$$

where,

d_b = diameter of the fitted coupling bolts [mm];

d = required diameter [mm], of the intermediate shaft taking into account ice strengthening requirements, if applicable;

T = specified minimum tensile strength of the shaft material [N/mm^2];

T_b = specified minimum tensile strength of the bolt material [N/mm^2];

= and also $T \leq T_b \leq 1.7\ T$, but not higher than 1000 [N/mm^2]

i = number of bolts in the coupling;

D = pitch circle diameter of bolt holes [mm].

6.13.2 The diameter of the non-fitted bolts will be specially considered upon the submittal of detailed preloading and stress calculations and fitting instructions.

6.14 Tailshaft liners

6.14.1 The thickness, t, of bronze or gunmetal liners fitted on tailshafts, in way of bearings, is not to be less than given by following formula:

$$t = \frac{168 + d_p}{28} \ [mm]$$

where,

t = thickness of liner [mm];

d_p = diameter of tailshaft under the liner [mm].

6.14.2 The thickness of the continuous liner between the bearings is not be less than 0.75t.

6.14.3 Continuous liners are preferably to be cast in one length. If made of several lengths, the joining of the separate pieces is to be made by welding through the whole thickness of liner before shrinking. In general, the lead content of the gunmetal of each length forming a butt welded liner is not to exceed 0.5 per cent. The composition of the electrode or filler rods is to be substantially lead free.

6.14.4 The liners are to withstand a hydraulic pressure of 2.0 bar after rough machining.

6.14.5 The liners are to be carefully shrunk or forced upon the shaft by hydraulic pressure, and they are not to be secured by pins.

6.14.6 Effective means are to be provided for preventing water from reaching the shaft at the part between the after end of the liner and the propeller boss.

6.14.7 If the liner does not fit the shaft tightly between the bearing portions in the stern tube, the space between the shaft and the liner is to be filled with a plastic insoluble non-corrosive compound.

6.15 Keys and keyways for propeller connections

6.15.1 Round ended or sled-runner ended keys are to be used, and the key ways in the propeller boss and cone of the tailshaft are to be provided with a smooth fillet at the bottom of the keyways. The radius of the fillet is to be at least 0.0125 of the diameter of the tailshaft at the top of the cone. The sharp edges at the top of the keyways are to be removed.

6.15.2 Two screwed pins are to be provided for securing the key in the keyway, and the forward pin is to be placed at least one-third of the length of the key from the end. The depth of the tapped holes for the screwed pins is not to exceed the pin diameter and the edges of the holes are to be slightly beveled.

6.15.3 The distance between the top of the cone and the forward end of the keyway is to be not less than 0.2 of the diameter of the tailshaft at the top of the cone.

6.15.4 The effective sectional area of the key in shear, is to be not less than $\frac{d^3}{2.6 d_1}$ [mm^2]

where,

d = diameter [mm], required for the intermediate shaft determined in accordance with 6.4, based on material having a specified minimum tensile strength of 400 [N/mm^2] and k = 1;

d_1 = diameter of shaft at mid-length of the key [mm].

6.16 Stern tube and bearings

6.16.1 The length of the bearing in the sternbush next to and supporting the propeller is to be as follows:

a) For water lubricated bearings; the length is to be not less than 4 times the rule diameter for the tailshaft under the liner;

   i. For a bearing of synthetic material, consideration may be given to a bearing length not less than twice the rule diameter for the tail shaft, provided the bearing design and material is substantiated by experiments to the satisfaction of IRS.

   ii. Synthetic materials for application as water lubricated stern tube bearings are to be type approved.

b) For bearings which are white-metal lined, oil lubricated and provided with an approved type of oil sealing gland;

   i. the length of the bearing is to be not less than twice the rule diameter for the tailshaft.

   ii. the length of the bearing may be less provided the nominal bearing pressure will not exceed 0.8 [N/mm^2] as determined by static bearing reaction calculation taking into account shaft and propeller weight which is deemed to be exerted solely on the aft bearing divided by the projected area of the shaft. However, the length of the bearing is to be not less than 1.5 times its actual diameter;

c) For bearings of cast iron, bronze which are oil lubricated and fitted with an approved oil sealing gland; the length of the bearing is, in general, to be not less than 4 times the rule diameter for tailshaft;

d) For bearings which are grease lubricated, the length of bearing is to be not less than 4 times the rule diameter for the tailshaft;

e) For water lubricated bearings lined with two or more circumferentially spaced sectors of an approved plastics material, in which it can be shown that the sectors operate on hydrodynamic principles, the length of the bearing is to be such that the nominal bearing pressure will not exceed 0.55 [N/mm^2]. The length of the bearing is not to be less than twice its rule diameter;

f) For approved oil lubricated bearings of synthetic rubber, reinforced resin or plastic materials, the length of the bearing is to be not less than 2.0 times the rule diameter of the shaft in way of the bearing. The length of the bearing may be reduced provided the nominal pressure is not more than 6 bar as determined by static bearing reaction calculation taking into account shaft and propeller weight which is deemed to be exerted solely on the aft bearing divided by the projected area of the shaft. In any case the length is not to be less than 1.5 times the actual diameter. Where the material has proven satisfactory testing and operating experience, consideration may be given to an increased bearing pressure. Synthetic materials for application as oil lubricated stern tube bearings are to be type approved.

6.16.2 Forced water lubrication is to be provided for all bearings lined with rubber or plastics where the shaft diameter is 380 [mm] or over. The supply water may come from a circulating pump or other pressure source. The water grooves in the bearings are to be of ample section and of a shape which will be little affected by weardown, particularly for bearings of the plastic type.

6.16.3 The shut off valve or cock controlling the supply of water is to be fitted direct to the after peak bulkhead, or to the sterntube where the water supply enters the sterntube forward of the bulkhead.

6.16.4 Where a tank supplying lubricating oil to the sterntube is fitted, it is to be located above the load water line and is to be provided with a low level alarm device in the engine room.

6.16.5 For vessels with TAILSHAFT CONDITION MONITORING (TCM) notation, the following arrangements are to be provided/ requirements are to be met:

(a) Oil Lubricated Bearings:

(i) Where sternbush bearings are oil lubricated, provision is to be made for cooling the oil by maintaining water in the after peak tank above the level of the sterntube or by other approved means. Means of ascertaining the temperature of the oil in the sternbush are also to be provided;

(ii) At least two independent temperature sensors or other approved arrangements are to be provided for measuring the aft bearing temperature;

(iii) The oil sealing glands used for sterntube bearings are to be of approved type, which will allow for their replacement without shaft withdrawal and removal of the propeller;

(iv) An arrangement for readily obtaining accurate oil samples is to be provided. The sampling point is to be taken from the lowest point in the lub.oil system as far as practicable. Also the arrangements are to be such as to permit the effective removal of the contaminants from the oil lubricating system. The oil sampling points are to be arranged with a test cock and fitted with a signboard;

(v) Possible water content in stern tube lubricating oil is to be monitored by either a test kit provided onboard or by an accredited laboratory;

(vi) An arrangement for bearing wear down measurements is to be provided. Hand operated gauges (like poker gauges) are considered to be acceptable.

(b) Water Lubricated Bearings

(i) An arrangement for bearing wear down measurements is to be provided. Hand operated gauges (like poker gauges) are acceptable. Where used, methods to remotely monitor the aft propeller shaft bearing performance and wear are to be approved and provided with redundancy. Sensors used for wear down monitoring are to be of approved type. Where a single sensor is installed, spare sensor is to be kept onboard the ship. Documentation for trending of rate of bearing weardown obtained from performance monitoring devices is to be in place onboard;

(ii) The lubricant sampling points are to be arranged with a test cock and fitted with a signboard. The open end of the sample pipe is to be located such that the condition of the lubricant and shafting arrangements are most accurately reflected. The sample is not to be taken downstream of filters fitted in the system;

(iii) Filters in the system are to be provided with sufficient redundancy to ensure safe and uninterrupted operation of the propulsion system. Automatic start of pumps is to be arranged on failure of circulation of the lubricant below acceptable limits;

(iv) Provision for alternative means of lubricant supply is to be arranged in order to maintain lubrication in case of loss of system integrity;

(v) Monitoring of lubricant temperature, flow and pressure is to be provided on the lubricant supply piping to the stern tube with means of warning;

(vi) Means of warning to detect contamination by sea water are to be provided (for closed loop system)

(vii) High / low level alarm on the lubricant header tanks are to be provided;

(viii) For closed loop systems, suitable provisions (such as relief valves, vents) for relieving the pressure are to be provided, if the system is capable of building up pressure above the maximum pressure, that the sealing devices are designed to.

(ix) The sealing glands used for sterntube bearings are to be of approved type, which will allow for their replacement without shaft withdrawal and removal of the propeller.

6.16.6 Stern seals are to be of the axially direct face type. Soft packing glands are to be used only if specified by Owners.

6.16.7 Where bulkhead glands are fitted, a watertight sealing arrangement is to be provided. Bulkhead seals is not to be formed by a bulkhead mounted plummer bearing.

6.16.8 Plummer bearings are to be either bulkhead mounted or of pedestal type.

6.17 Roller element bearings

6.17.1 Roller element bearings are to have design life, L_{10h} not less than:

- 40,000 hours for propeller thrust bearing;

- 30,000 hours for other bearings.

Where L_{10h} is the basic rating life in hours which 90% of a sufficiently large group of apparently identical bearings is expected to attain.

6.18 Shaft bearing materials

6.18.1 Shaft bearing fitted in stern bushes and shaft bossings in "A" and "P" brackets are to be constructed from an approved material and effectively secured to prevent rotational and axial movement in the stern tube(s) and stern bush(es)

6.19 Glass Reinforced Plastic coating

6.19.1 The tail shaft may be protected by a fiberglass reinforced plastic coating between liners in accordance with the following procedure, which effectively prevents sea water from contacting the steel shaft. In such cases, the tailshaft survey interval would be 5 years as applicable for the tailshaft survey notation TS(CL). The procedure is to be approved in each case:

a) Coatings are to consist of at least 4 plies of cross-woven glass tape impregnated with resin or equivalent process.

b) The shaft is to be cleaned with solvent or grit blasted.

c) Shaft is to be examined and 1st coat is to be given in presence of Surveyors.

d) Shaft is to be subjected to spark test after coating. There should be freedom from porosity.

e) Effective means are to be provided to prevent water gaining access to the metallic region of the shaft.

f) It is to be ensured that provision is made for overlapping and adequate bonding of the coating.

g) The end of the liner is to stepped and tapered as required to protect the end of the wrapping.

6.20 Special approval of alloy steel used for intermediate shaft material

6.20.1 This sub-section applies to the approval of alloy steel which has a minimum specified tensile strength greater than 800 [N/mm^2], but less than 950 [N/mm^2] intended for use as intermediate shaft material.

6.20.2 Torsional fatigue test

6.20.2.1 A torsional fatigue test is to be performed to verify that the material exhibits similar fatigue life as conventional steels. The torsional fatigue strength of said material is to be equal to or greater than the permissible torsional vibration stress given by the formulae in 8.4.2.

6.20.2.2 The test is to be carried out with notched and unnotched specimens respectively. For calculation of the stress concentration factor of the notched specimen, fatigue strength reduction factor β should be evaluated in consideration of the severest torsional stress concentration in the design criteria.

6.20.2.3 Test Conditions

.1 Test conditions are to be in accordance with Table 6.20.2.3. Mean surface roughness is to be less than 0.2μm Ra with the absence of localized machining marks verified by visual examination at low magnification (x20) as required by section 8.4 of ISO 1352:2021.

.2 Test procedures are to be in accordance with section 10 of ISO 1352:2021.

Table 6.20.2.3: Test Condition

Loading Type	Torsion
Stress ratio	R = -1
Load waveform	Constant-amplitude sinusoidal
Evaluation	S-N Curve
Number of cycles for test termination	1 x 10^7 cycles

6.20.2.4 Acceptance criteria

.1 Measured high-cycle fatigue strength τ_{C1} and low cycle torsional fatigue strength τ_{C2} are to be equal to or greater than the values given by the following formulae:

$$\tau_{C1} \geq \tau_{C,\lambda=0} = \frac{\sigma_B + 160}{6} C_K C_D$$

$$\tau_{C2} \geq 1.7 \frac{1}{\sqrt{c_k}} \tau_{C1}$$

Where

C_k = factor for the particular shaft design features, see 6.2.4

scf = stress concentration factor , see 6.2.4

C_D = Size factor, see 8.4.2

σ_B = specified minimum tensile strength in [N/mm^2] of the shaft material.

6.20.3 Cleanliness requirements

6.20.3.1 The steels are to have a degree of cleanliness as shown in Table 6.20.3.1 when tested according to ISO 4967:2013 method A. Representative samples are to be obtained from each heat of forged or rolled products.

6.20.3.2 The steels are generally to comply with the minimum requirements of Pt.2, Ch.5 Sec.2, Table 2.2.2, with particular attention given to minimizing the concentrations of sulphur, phosphorus and oxygen in order to achieve the cleanliness requirements. The specific steel composition is required to be approved by IRS.

Table 6.20.3.1 : Cleanliness Requirements

Inclusion Group	Series	Limiting chart diagram index
Type A	Fine	1
	Thick	1
Type B	Fine	1.5
	Thick	1
Type C	Fine	1
	Thick	1
Type D	Fine	1
	Thick	1
Type DS	-	1

6.20.4 Inspection

6.20.4.1 The ultrasonic testing required by Pt 2 Ch5 is to be carried out prior to acceptance. The acceptance criteria are to be in accordance with *IRS Classification Note "Guidelines for Non-Destructive Testing of Hull and machinery Steel Forgings*" or a recognized national or international standard.

Section 7

Propellers

7.1 Scope

7.1.1 The requirements of this Section cover the construction, materials and inspection of propellers.

7.2 Plans and particulars

7.2.1 A plan, in triplicate, of the propeller is to be submitted for approval, together with the following particulars:

a) Maximum shaft power, P, in [kW];

b) Revolutions per minute of the propeller at maximum power, R;

c) Propeller diameter, D [m];

d) Pitch at 25 per cent radius (for solid propellers only), $P_{0.25}$ [m];

e) Pitch at 35 per cent radius (for controllable pitch propellers only), $P_{0.35}$ [m];

f) Pitch at 70 per cent radius, $P_{0.7}$ [m];

g) Length of blade section of the expanded cylindrical section at 25 per cent radius (for solid propeller only), $L_{0.25}$ [mm];

h) Length of blade section of expanded cylindrical section at 35 per cent radius (for controllable pitch propellers only) $L_{0.35}$ [mm];

i) Rake at blade tip measured at shaft axis (backward rake positive, forward rake negative), K [mm];

j) Number of blades, N;

k) Developed area ratio, a.

7.2.2 In case of controllable pitch propeller following additional plans / data is to be submitted:

- Hub and it's attachments to tailshaft flange with connecting bolts
- Propeller blade flange and it's attachment bolts
- Internal control mechanisms
- Hydraulic piping control systems
- Instrumentation and alarm systems
- Strength calculations for internal components

In case of highly skewed propellers with skew angle greater than 50 degree and controllable pitch propeller skew angle greater than 25 degrees, propeller load and stress analysis proving adequacy of blade strength are to be submitted.

7.3 Materials

7.3.1 Castings for propellers and propeller blades are to comply with the requirement of Pt.2, Ch.8. The specified minimum tensile strength is to be not less than stated in Table 7.4.1.

7.3.2 When it is proposed to use materials which are not included in Table 7.4.1, details of the chemical composition, mechanical properties and density are to be submitted for approval.

7.4 Design

7.4.1 Minimum blade thickness

7.4.1.1 Where the propeller blades are of conventional design, the thickness, t, of the propeller blades at 25 per cent radius for solid propellers, at 35 per cent for controllable pitch propellers, neglecting any increase due to fillets, is to be not less than:

Table 7.4.1 : Material constants			
Materials	**Specified min. UTS [N/mm²]**	**f**	**w**
Manganese bronze Grade Cu 1	440	20.6	8.3
Ni-Manganese bronze Grade Cu 2	440	20.9	8.0
Ni-Aluminium bronze Grade Cu 3	590	25.7	7.5
Mn-Aluminium bronze Grade Cu 4	630	23.25	7.5
Cast iron	250	11.77	7.2
Carbon and low alloy steels	400	14.0	7.9
Note : The value of f may be increased by 10 percent for twin screw and outboard propellers of triple screw ships.			

a) For fixed propellers

$$t_{0.25} = 1055\sqrt{\frac{AP}{C_n CRN}} + \frac{2.5BKC_s}{CC_n}$$

b) For controllable pitch propellers

$$t_{0.35} = 847\sqrt{\frac{AP}{C_n CRN}} + \frac{1.6BKC_s}{CC_n}$$

where,

$t_{0.25}$ = minimum blade thickness required at 25 per cent radius;

$t_{0.35}$ = minimum blade thickness required at 35 per radius;

C_n = section modulus coefficient at 25 per cent radius or 35 per cent radius as applicable;

$= \frac{l_o}{U_f L T^2}$ and is not to be taken greater than 0.10;

l_o = moment of inertia of the expanded cylindrical section at 25 per cent radius or 35 per cent radius, as applicable, about a straight line passing through the center of gravity parallel to the pitch line or to the nose-tail line [mm⁴];

U_f = maximum normal distance from the moment of inertia axis to points on the face boundary (tension side) of the section at 25 per cent radius or 35 per cent radius, as applicable [mm];

L = length of the blade section of the expanded cylindrical section at 25 per cent radius or 35 per cent radius, as applicable [mm];

T = maximum thickness of the expanded cylindrical section as approved at 25 per cent or 35 per cent radius, as applicable [mm];

C_s = section area coefficient at 25 per cent radius or 35 per cent radius as applicable;

$$= \frac{a_s}{LT}$$

a_s = area of the expanded cylindrical section at 25 per cent radius or 35 per cent radius, as applicable [mm²];

f = material constant as per Table 7.4.1;

w = material constant as per Table 7.4.1;

a) For fixed-pitch propellers

$$A = 1.0 + \frac{6.0D}{P_{0.7}} + \frac{4.3P_{0.25}}{D}$$

$$B = \left(\frac{4300\,w\,a}{N}\right)\left(\frac{R}{100}\right)^2\left(\frac{D}{20}\right)^3$$

$$C = \left(1 + \frac{1.5P_{0.25}}{D}\right)\left(L_{0.25}\,f - B\right)$$

b) For controllable pitch propellers

$$A = 1.0 + \frac{6.0D}{P_{0.7}} + \frac{3.0P_{0.35}}{D}$$

$$B = \left(\frac{4900\,w\,a}{N}\right)\left(\frac{R}{100}\right)^2\left(\frac{D}{20}\right)^3$$

$$C = \left(1 + \frac{0.6P_{0.25}}{D}\right)\left(L_{0.35}\,f - B\right)$$

7.4.1.2 Propellers of unusual design or application will be subject to special consideration upon submittal of detailed stress calculations.

7.4.1.3 Fillets at the root of the blades are not to be considered in the determination of blade thickness.

7.4.2 Controllable pitch propellers

7.4.2.1 In the case of controllable - pitch propellers, means are to be provided to lock the blades in ahead position in case of the failure of the pitch operating mechanism.

7.4.2.2 A propeller pitch indicator is to be fitted at each location from which it is possible to control the pitch of the propeller.

7.5 Fitting of propellers

7.5.1 The propeller boss is to be a good fit on the tailshaft cone. The forward edge of the bore of the propeller boss is to be rounded to about 6 [mm] radius. In general the contact area between propeller hub and tailshaft taper is to be not less than 70 per cent of the theoretical contact area. On completion of final pull-up propeller is to be secured.

7.5.2 The exposed part of the tailshaft is to be protected from the action of water by filling all spaces between propeller hub, cap and shaft with a suitable filling material. The propeller assembly is to be sealed at the forward end with a well-fitted soft rubber packing ring. When the rubber ring is fitted in an external gland, the hub counterbore is to be filled with suitable material, and clearances between shaft liner and hub counterbore are to be kept to a minimum. When the rubber ring is fitted internally, ample clearance is to be provided between liner and hub and the ring is to be sufficiently sized to squeeze in to the clearance space when the propeller is driven up on the shaft, and, where necessary, a filler piece is to be fitted in the propeller - hub keyway to provide a flat unbroken seating for the ring. The recess formed at the small end of the taper by the over hanging propeller hub is to be packed with red lead putty or rust-preventing compound before the propeller nut is put on.

7.5.3 Effective means are to be provided to prevent the slackening of the propeller nut.

7.6 Keyless fittings of propellers without ice strengthening

7.6.1 Design criteria

7.6.1.1 The formulae, etc., given herein are not applicable for propellers where a sleeve is introduced between shaft and boss.

7.6.1.2 The taper of the propeller shaft cone is not to exceed 1/15.

7.6.1.3 Prior to final pull-up, the contact area between the mating surfaces is to be checked and is not to be less than 70% of the theoretical contact area (100%). Non-contact bands extending circumferentially around the boss or over the full length of the boss are not acceptable.

7.6.1.4 After final pull-up, the propeller is to be secured by a nut on the propeller shaft. The nut should be secured to the shaft.

7.6.1.5 The factor of safety against friction slip at 35°C is not to be less than 2.8 under the action of rated torque (based on rated power r.p.m.) plus torque due to torsionals.

7.6.1.6 For the oil injection method the coefficient of friction is to be should be 0.13 for bosses made in copper-based alloy and steel.

7.6.1.7 The maximum equivalent uniaxial stress in the boss at 0°C based on the Mises-Hencky criterion(σ_E) is not to exceed 70% of the yield point or 0.2% proof-stress (0.2% offset yield strength) for the propeller material based on the test piece value. For cast iron the value is not to exceed 30% of the nominal tensile strength.

7.6.1.8 Stress calculations and fitting instructions are to be submitted and are to include at least the following:

i)Theoretical contact surface area

ii)The maximum permissible pull-up length at 0°C as limited by the maximum permissible uniaxial stress specified above

iii)The minimum pull-up length and contact pressure at 35°C to attain a safety factor against slip of 2.8

iv)The proposed pull-up length and contact pressure at fitting temperature

v) The rated propeller ahead thrust

7.6.2 Material Constants

Modulus of elasticity	
Cast and forged steel	20.58 x 10^4 N/mm^2
Cast iron	9.8 x 10^4 N/mm^2
Copper based alloys, Cu 1 and Cu 2	10.78 x 10^4 N/mm^2
Copper based alloys, Cu 3 and Cu 4	11.76 x 10^4 N/mm^2

Possion's ratio	
Cast and forged steel	0.29
Cast iron	0.26
All copper-based alloys	0.33

Coefficient of linear expansion	
Cast and forged steel and cast iron	12 x 10^{-6} mm/mm ˚C
All copper-based alloys	17.5 x 10^{-6} mm/mm ˚C

7.6.3 Formulae used

7.6.3.1 The formulae given below, for the ahead condition, will also give sufficient safety in the astern condition.

7.6.3.2 The formulae are applicable for solid shafts only.

7.6.3.3 Minimum required surface pressure at 35°C is to be:

$$P_{35} = \frac{ST}{AB}\left[-S\theta + \sqrt{\mu^2 + B\left(\frac{F_v}{T}\right)^2}\right]$$

Where

$$B = \mu^2 - S^2\theta^2$$

P_{35} = surface pressure (N/mm^2) between mating surfaces at 35°C

A = 100% theoretical contact area (mm^2) between boss and shaft, as read from drawings and disregarding oil grooves

S = factor of safety against friction slip at 35°C

θ = half taper of propeller shaft, e.g. taper = 1/15. θ = 1/30

μ = coefficient of friction between mating surfaces

T = rated thrust (N)

Corresponding minimum pull-up length at temperature 35°C

$$\delta_{35} = P_{35}\frac{D_s}{2\theta}\left[\frac{1}{E_b}\left(\frac{K^2+1}{K^2-1} + v_b\right) + \frac{1}{E_s}(1 - v_s)\right]$$

δ_{35} = pull-up length (mm) at temperature 35°C

K = D_b/D_s

E_s = modulus of elasticity (N/mm2) of shaft material

E_b = modulus of elasticity (N/mm2) of boss material

D_s = diameter (mm) of propeller shaft at the midpoint of the taper in the axial direction

D_b = mean outer diameter (mm) of propeller boss at the axial position corresponding to D_s

vs = Poisson's ratio for shaft material

vb = Poisson's ratio for boss material

7.6.3.4 Minimum pull-up length at temperature t (where t<35°C)

$$\delta_t = \delta_{35} + \frac{D_s}{2\theta}(\alpha_b - \alpha_s)(35 - t)$$

Corresponding minimum surface pressure at temperature t

$$P_t = P_{35}\frac{\delta_t}{\delta_{35}}$$

Minimum push-up load at temperature t

$$W_t = AP_t(\mu + \theta)$$

δ_t = pull-up length (mm) at temperature *t*°C

*D*s = diameter (mm) of propeller shaft at the midpoint of the taper in the axial direction

α_s = coefficient of linear expansion (mm/mm°C) of shaft material

α_b = coefficient of linear expansion (mm/mm°C) of boss material

*P*t = surface pressure (N/mm^2) between mating surfaces at temperature *t*°C

W_t = push-up load (N) at temperature *t*°C

7.6.3.5 Maximum permissible surface pressure at 0°C

$$P_{max} = \frac{0.7\sigma_y(K^2-1)}{\sqrt{3K^4+1}}$$

Corresponding maximum permissible pull-up length at 0°C

$$\delta_{max} = \frac{P_{max}}{P_{35}}\delta_{35}$$

P_{max} = maximum allowable surface pressure (N/mm^2) at 0°C

σ_y = yield point or 0.2% proof stress (0.2% offset yield strength) of propeller material (N/mm^2)

P_{35} = surface pressure (N/mm^2) between mating surfaces at 35°C

δ_{max} = maximum allowable pull-up length (mm) at temperature 0°C

7.6.3.6 Shear force at interface

$$F_v = \frac{2cQ}{D_s}$$

*F*v = shear force at interface (N)

c = constant,

c = 1.0 for turbines, geared diesel drives, electric drives and for direct diesel drives with a hydraulic or an electromagnetic or high elasticity coupling

c = 1.2 for a direct diesel drive.
IRS reserves the right to increase the *c* constant if the shrinkage has to absorb an extreme high pulsating torque.

Q = rated torque (N mm) transmitted according to rated horsepower, *H*, and speed of propeller shaft

7.6.3.7 Rated thrust developed for free running vessels (if not given)

$$T = 9.8 \times 132\frac{H}{V_s}$$

Or

$$T = 9.8 \times 4.3 \,.\, 10^6 \,\frac{H}{PN}$$

T = rated thrust (N)

H = rated brake horsepower (PS)

*V*s = ship speed (knots) at rated horsepower

P = mean propeller pitch (mm)

N = propeller speed (r.p.m.) at rated brake horsepower

Section 8

Vibrations and Alignment

8.1 Scope

8.1.1 The requirements of this Section are applicable to the following systems:

a) Main oil engine propulsion systems, except in the case of ships classed for smooth water service, when fitted with engines having powers less than 200 [kW].

b) Auxiliary oil engine machinery systems used for essential services, where the power developed by auxiliary engines is 200 [kW] and over.

c) Main propulsion systems formed by turbines or electric motors geared to the shafting and situated aft.

8.1.2 Unless otherwise advised, it is the responsibility of the Shipbuilder as the main contractor to ensure, in co-operation with the Engine builders, that the information required by this Section is prepared and submitted.

8.2 Basic system requirements

8.2.1 The systems are to be free from excessive torsional, axial and lateral vibration, and are to be aligned in accordance with tolerances agreed with the respective manufacturers.

8.2.2 Where changes are subsequently made to a dynamic system which has been approved, revised calculations are to be submitted for consideration.

8.3 Resilient mountings

8.3.1 Where the machinery is installed on resilient mountings, linear vibration (steady state and transient) is not to exceed the limiting values agreed with the manufacturers of the machinery nor those of the resilient mountings.

8.3.2 Misalignment arising from such vibration is not to impose excessive loading on machinery components within the system.

8.4 Torsional vibration

8.4.1 General

8.4.1.1 Torsional vibration calculations, including an analysis of the vibratory torques and stresses for the dynamic systems formed by the oil engines, turbines, motors, generators, flexible couplings, gearing, shafting and propeller, where applicable, including all branches, are to be submitted for approval together with the associated plans.

8.4.1.2 Particulars of the division of power developed throughout the speed range for turbines or from all intended combinations of operation in oil engine installations having more than one engine and/or with power take-off systems are to be submitted.

8.4.1.3 Any special speed requirements for prolonged periods in service are to be indicated, e.g., range of trawling revolutions per minute, range of operation revolutions per minute with a controllable pitch propeller, idling speed, etc.

8.4.1.4 The calculations and/or measurements carried out on oil engine installations containing transmission items sensitive to vibratory torque, e.g. gearing, flexible couplings, or generator rotors and their drives, are to take into account the effects of engine malfunction commonly experienced in service, such as cylinder(s) not firing.

8.4.1.5 Restricted speed ranges will be imposed in regions of speed where stresses are considered to be excessive for continuous running. Similar restrictions will be imposed, or other protective measures required to be taken, where vibratory torques are considered to be excessive for particular machinery items.

8.4.1.6 Where calculations indicate the possibility of excessive torsional vibration within the range of working speeds, torsional vibration measurements, using the appropriate recognized techniques, may be required to be taken from the machinery installation for the purpose of determining the need for restricted speed ranges.

8.4.2 Permissible limits of stresses due to torsional vibrations

8.4.2.1 Alternating torsional vibration stresses are stresses resulting from the alternating torque which is superimposed on the mean torque.

8.4.2.2 In no part of the propulsion system may the alternating torsional vibration stresses exceed the values of $\tau 1$ for continuous operation and $\tau 2$ for transient running.

8.4.2.3 For continuous operation the permissible stresses due to alternating torsional vibrations are not to exceed the following values:

$$\tau_1 = \pm \frac{\sigma B + 160}{18} C_k \, C_D \, (3 - 2\lambda^2) \text{ for } \lambda < 0.9$$

$$\tau_1 = \pm \frac{\sigma B + 160}{18} 1.38 \, C_k \, C_D \text{ for } 0.9 \le \lambda < 1.05$$

where,

τ_1 = permissible stress due to torsional vibrations for continuous operation [N/mm^2];

σB = tensile strength of shaft material [N/mm^2];

For calculation purposes, this value is not to be taken greater than:

- 600 [N/mm^2] for carbon and carbon manganese steels; and
- 800 [N/mm^2] for alloy steels;

C_k = factor for different shaft design features as given in Table 8.4.2;

C_D = size factor = $0.35 + 0.93d^{-0.2}$;

d_o = shaft outside diameter under consideration [mm];

λ = speed ratio = n/n_o;

n = speed in rpm under consideration at rated power;

n_o = rated speed in rpm.

8.4.2.4 Where a vessel, because of its type of employment, is operated predominantly in the lower speed range, special consideration may be given to the permissible stresses for continuous operation.

8.4.2.5 Where the stresses exceed the limiting values of τ_1 for continuous operation, restricted speed ranges are to be imposed which are only allowed to be passed through rapidly. Cl.4.12.6.1 may also be referred for barred speed range requirements.

8.4.2.6 Restricted speed ranges are not acceptable, in the speed range between 0.8 to 1.05 of the rated speed. The limits of the barred speed range are to be calculated in accordance with the following formula unless proved to be otherwise.

$$\frac{16\mathrm{N}_c}{18-\lambda} \le n \le \frac{(18-\lambda)\mathrm{N}_c}{16}$$

where,

n_c = critical speed in [rpm].

Table 8.4.2 : c_k factors for different design features

Intermediate shafts with						Thrust shafts external to engines		Tailshafts		
Integral coupling flange and straight section	shrink fit couplings	Keyway, tapered connection	Keyway, cylindrical connection	Radial hole	Longitudinal slot[1]	On both sides of thrust collar	In way of bearing when a roller bearing is used	Flange mounted or keyless taper fitted propellers	Key fitted propellers	Between forward end of aft most bearing and forward stern tube seal
C_k = 1.0	1.0	0.60	0.45	0.50	0.30 [2)]	0.85	0.85	0.55	0.55	0.80

Note:

1) The c_k value is valid for 1, 2 and 3 slots and they are to be arranged 360, 180 or 120 degrees apart from each other respectively.
2) C_k = 0.3 is an approximation within the limitations given in Note No.6 under Table 6.5.1. More accurate estimate of the stress concentration factor (scf) may be determined from 6.2.3 c) or by direct application of FEM calculation. In which case:

   C_k = 1.45/scf

   The scf is defined as the ratio between the maximum local principal stress and √3 times the nominal torsional stress (determined for the bored shaft without slots).

8.4.2.7 For transient running the permissible stresses due to the alternating torsional vibrations are not, in any case, to exceed the values given by the following formula:

$$\tau_2 = \frac{1.7\,\tau_1}{\sqrt{C_k}} \text{ for } \lambda \le 0.8$$

where,

τ_2 = permissible stress due to torsional vibrations for transient running.

8.4.2.8 If materials with greater specified or actual tensile strength than the limitations given in 8.4.2.3 are used higher permissible vibration stresses are not acceptable when derived from formulae in 8.4.2.3. Higher permissible stress may be allowed for intermediate shaft using alloy steel, when it is verified that the material used exhibit fatigue life as conventional steel when tested as per 6.20 of this chapter.

8.5 Axial vibrations

8.5.1 For all main propulsion shafting systems, the Shipbuilders are to ensure that amplitudes due to axial vibrations are satisfactory throughout the speed range, so far as practicable. Where appropriate, amplitudes may be reduced by the use of suitable vibration dampers or phasing of propeller and engine, etc.

8.5.2 Unless previous experience of similar installation shows it to be unnecessary, calculations of the shafting system are to be carried out. These calculations are to include the effect of the thrust block seating and the surrounding hull structure taking part in the vibration. The result of these calculations or the evidence of previous experience is to be submitted for consideration.

8.5.3 Where calculations indicate the possibility of excessive axial vibration amplitudes within the range of working speeds, measurements using an appropriate recognized technique may be required to be taken from the shafting systems for the purpose of determining the need for restricted speed ranges.

8.6 Lateral vibrations

8.6.1 For all main propulsion shafting systems, the Shipbuilders are to ensure that amplitudes due to lateral vibrations are satisfactory throughout the speed range.

8.6.2 Unless previous experience of similar installations shows it to be unnecessary, calculations of lateral, or bending, vibration characteristics of the shafting system are to be carried out. These calculations, taking account of dynamic bearing stiffnesses, are to cover the frequencies giving rise to all critical speeds which may result in significant amplitudes within

the speed range, and are to indicate relative deflections and bending moments throughout the shafting system.

8.6.3 The results of these calculations, or the evidence of previous experience, is to be submitted for consideration.

8.6.4 Where calculations indicate the possibility of excessive lateral vibration amplitudes within the range of working speeds, measurements using an appropriate recognized technique may be required to be taken from the shafting system for the purpose of determining the need for restricted speed ranges.

8.7 Shaft alignment

8.7.1 For main propulsion installations, the shafting is to be aligned to give reasonable bearing reactions, and bending moments, taking into consideration following factors :

- Forces which may affect the reliability of the propulsion shafting system including weight of the propeller and shafts,
- Hydrodynamic forces acting on the propeller,
- Number of propeller blades in relation to diesel engine cylinders,
- Misalignment forces,
- Thermal expansion,
- Flexibility of engine and thrust bearing foundations,
- Engine induced vibrations, gear tooth loadings, flexible couplings,
- Effect of power take-off,
- Effect of hull deformations at all conditions of ship loading and operation.

Consideration is also to be given to any limits of vibrations and loadings specified by the equipment manufacturer. The Shipbuilder is to position the bearings and construct the bearing seatings to minimize the effects of movements under all operating conditions.

8.7.2 For geared installations, where two or more pinions are driving the final reduction wheel, calculations are to be submitted to verify that shaft alignment is such that proper bearing reactions are maintained under all operating conditions.

8.7.3 Shaft alignment calculations are to be submitted for the following alignment-sensitive types of installations for review:

i) Propulsion shafting with power take-off or with booster power arrangements.
ii) Propulsion shafting for which the tail shaft bearings are to be slope bored.
iii) Propulsion shaft having diameter 300mm. and above in way of after most stern tube bearing
iv) Propulsion shafting arrangement requiring long shaft line.

The alignment calculations are to include bearing reactions, shear forces and bending moments along the shafting.

The alignment calculations are to be performed for the following conditions, as applicable:

- Theoretically aligned cold and hot conditions of the shaft with specified alignment tolerances.
- Deviation from the theoretical aligned conditions due to the forces exerted by power take-off or booster power.

Calculations are to be performed for the maximum allowable alignment tolerances and are to show that:

- Bearing loads under all operating conditions are within the acceptable limits specified by the bearing manufacturer.
- Bearing reactions are always positive (i.e. supporting the shaft).
- Shear forces and bending moments on the shaft are within acceptable limits in association with other stresses in the shaft.
- Forces and moments on propulsion equipment are within the limits specified by the machinery manufacturers.

8.7.4 Shaft alignment is to be verified by measurement.

Section 9

Thrusters

9.1 Scope

9.1.1 The requirements of this section are applicable to:

- fixed thruster for propulsion
- steerable thruster units (azimuth thrusters) for propulsion and steering
- tunnel thruster for transverse propulsion aid to manoeuvring.

9.2 Plans and particulars

The following plans and information in triplicate are to be submitted.

9.2.1 Fixed/azimuth propulsion thrusters

a) A general arrangement sectional assembly plan showing all the connections of the torque transmitting components from the prime mover to the propeller, together with the azimuthing mechanism and if a nozzle is provided, the nozzle ring structure and nozzle support struts.

b) Detailed and dimensional plans of the individual torque transmitting components.

c) Schematic plans for lubricating and hydraulic systems, together with pipe material, relief valves and working pressures.

9.2.2 Tunnel thrusters

In addition to applicable requirements of 9.2.1, structural assembly plan including connections to tunnel.

9.2.3 Calculations and specifications

a) Thruster prime mover type and operational power/speed envelop.

b) Rating and type of motor for the azimuthing mechanism (e.g. type hydraulic or electric).

c) Gearing calculations for the azimuthing and propulsion mechanism which is to be designed in accordance with classification notes on the design of gearing. Calculation for bevel gears is to be on the basis of a conversion to equivalent helical gear.

d) Bearing specifications.

e) Details of control engineering aspects in accordance with Chapter 7.

f) Calculations indicating suitability of components for short term high power operation, where applicable.

g) Where design is carried out using fatigue analysis, a fatigue strength analysis of components indicating a factor of safety of 1.5 at the design loads based on a suitable fatigue failure criteria is required to be submitted for review.

9.3 Materials

9.3.1 The materials used on the construction are to be manufactured and tested in accordance with Part 2.

9.3.2 The grades for various components to be analogous to relevant section of the Rules, Pt.4, Ch.4, Sec.7 for propellers, Pt.4, Ch.4, Sec.6 for shafting and Pt.4, Ch.4, Sec.5 for gearing.

9.4 Design and construction

9.4.1 General

9.4.1.1 The arrangement of all types of thrusters is to be such that the craft can be manoeuvred in accordance with the design specifications.

9.4.1.2 The requirements associated with the structural and watertight integrity and the installation arrangement are to be in accordance with Pt.3.

9.4.1.3 In addition to the requirements of this section reference is to be made to:

- Main transmission gearing (Pt.4, Ch.4, Sec.5). Bevel gears will be specially considered on the basis of a conversion to equivalent helical gears.

- Main transmission shafting (Pt.4, Ch.4, Sec.6).

- Propeller (Pt.4, Sec.7).

- Torsional vibration (Pt.4, Ch.4, Sec.8 of the 'Rules and Regulations for the Construction and Classification of Steel Ships').

- Lateral vibration for shafting systems which include cardan shafts (Pt.4, Ch.4, Sec.8 of the 'Rules and Regulations for the Construction and Classification of Steel Ships').

9.4.2 Azimuth thrusters

9.4.2.1 The following requirements are to be complied with:

a) The azimuthing mechanism is to be capable of a maximum rotational speed of not less than 1.5 rev/min.

b) Gearing for the azimuthing mechanism is to be designed in accordance with classification notes on the design of gearing. Bevel gears will be specially considered on the basis of a conversion to equivalent helical gears.

i) Under dynamic operating conditions, the gear is to be considered for:

- Design maximum dynamic duty steering torque.

- Variable loading, where applicable. A spectrum (duty) factor may be used. The load spectrum value is to be derived using load measurements of similar units, where possible.

ii) Under a static duty (< 10^3 load cycles) steering torque. (for azimuth thruster with nozzle see 9.4.3.1).

iii) The following minimum factor of safety values are to be achieved:

Surface stress S_{Hmin} = 1.2

Bending Stress S_{Fmin} = 1.55

c) For hydraulic pressure retaining parts and load bearing (See Chapter 5).

9.4.3 Azimuth thrusters with a nozzle

9.4.3.1 For steerable rudder propellers contained within a nozzle, the design lateral force Fr and the turning moment Q_r is to be calculated as follows:

$$F_r = 376 \cdot A \cdot V^2 \quad [N]$$

$$Q_r = F_r \cdot r \quad [N\text{-}m]$$

where,

A = projected area of nozzle [M^2]

V = Maximum service speed [knots] with the ship on summer load waterline. When the speed is less than 10 knots, V is to be replaced by the expression $V_{min} = (V+20)/3$

r = the horizontal distance from centre line of stock to centre of pressure of nozzle, but not to be taken less than 10% of the chord length of the nozzle.

The nozzle stock diameter in way of tiller, is to be not less than:

$$d_u = 4.2\sqrt[3]{Q_r.K} \quad [mm]$$

where,

k = material factor for nozzle stock.

9.4.3.2 The scantlings of the nozzle stock or steering tube are to be such that the equivalent stress, σ,does not exceed 118/k [N/mm^2] i.e.

$$\sigma_e = \sqrt{\sigma^2 + 3\tau_t^2} \leq 118/k \ [N/mm^2]$$

where,

σ is the bending stress [N/mm^2] calculated using the maximum bending moment, BM, on the nozzle stock or steering tube.

For nozzles without bottom support :

$$BM = \sqrt{F_r^2 + T^2} \ x\ a \ [Nm]$$

T = Maximum thrust developed by the thruster [N]

F_r = design lateral force [N/mm^2] as per 9.4.3.1

a = Vertical distance from centre line of nozzle to the section under consideration [m] (See Fig.9.4.3.2).

τ_t is the torsional shear stress [N/mm^2] calculated using the torque Qr as per 9.4.3.1.

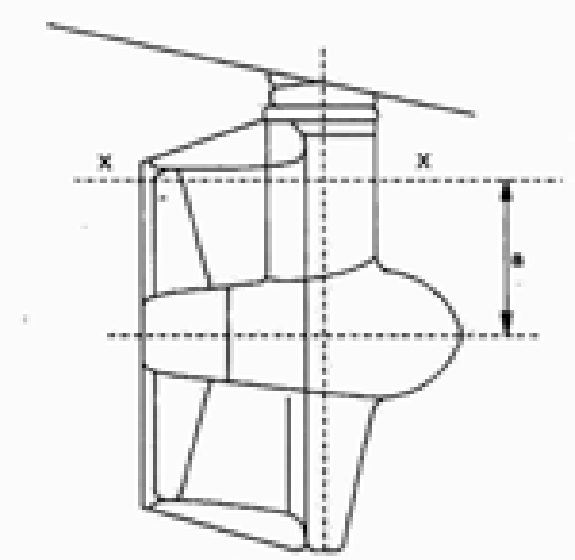


Fig.9.4.3.2 : Azimuth thrusters

9.5 Piping systems

9.5.1 General

The piping system for azimuth thrusters is to comply with the general design requirements given in Pt.4, Ch.2.

The specific requirements for lubricating/ hydraulic oil systems and standby arrangements are given in Pt.4, Ch.3.

9.5.2 Azimuth thruster

The hydraulic power operating systems for each azimuth thruster are to be provided with the following:

a) Arrangements to maintain the cleanliness of the hydraulic fluid, taking into consideration the type and design of the hydraulic system.

b) A fixed storage tank having sufficient capacity to recharge at least one azimuth power actuating system including the reservoir. The piping from the storage tank is to be permanent and arranged in such a manner as to allow recharging from within the thruster space.

c) Where the lubricating oil for the azimuth thrusters is circulated under pressure, provision is to be made for the efficient filtration of the oil. The filters are to be capable of being cleaned without stopping the thruster or reducing the supply of filtered oil.

9.6 Control and monitoring

9.6.1 General

9.6.1.1 Except where indicated in this section the control engineering systems are to be in accordance with Pt.4, Ch.7.

9.6.1.2 Azimuthing control for azimuth thruster(s) and propeller pitch control for azimuth and/or tunnel thruster(s) are to be provided from the navigating bridge, the main machinery control location and locally.

9.6.1.3 Means are to be provided at the remote control location(s) to stop each azimuth or tunnel thruster unit.

9.6.2 Monitoring and alarms

9.6.2.1 Alarms and monitoring requirements are indicated in 9.6.2.2, 9.6.2.3 and Table 9.6.2.

Table 9.6.2		
Item	**Alarm**	**Note**
Thruster, azimuth or tunnel		Indicators, see 9.6.2.2
Azimuthing motor	Power failure, single phase	Also running indication on bridge and at machinery control location
Propeller pitch motor	Power failure	In case of failure the propeller pitch should be locked in full ahead position. Also running indication on bridge and at machinery control location
Electric propulsion motor	Overload, power failure	Also running indication on bridge and at machinery control location
Control system	Failure	
Hydraulic oil supply tank level	Low	
Hydraulic oil system pressure	Low	
Hydraulic oil system temperature	High	Where oil cooler is fitted
Hydraulic oil filters differential pressure	High	Where oil filters are fitted
Lubricating oil supply pressure	Low	If separate forced lubrication

9.6.2.2 An indication of the angular position of the azimuth thruster(s) and the propeller pitch position for azimuth and/or tunnel thruster(s) are to be provided at each location from which it is possible to control the direction of thrust or the pitch.

9.6.2.3 All alarms associated with thruster unit faults are to be indicated individually on the navigating bridge and in accordance with the alarm system specified by Chapter 7.

9.7 Electrical systems

9.7.1 General

The electrical installation is to be designed, constructed and installed in accordance with the requirements of Pt.4, Ch.8.

9.7.2 Emergency power for steering systems and drives

The arrangement to comply with Pt.4, Ch.6, Sec.6, Cl.6.1.1

9.7.3 Circuits

Azimuth thruster auxiliaries and controls are to be served by individual circuits. Services that are duplicated are to be separated throughout their length as widely as is practicable and without the use of common feeders, transformers, converters, protective devices or control circuits.

9.8 Tests

9.8.1 Azimuth thrusters

9.8.1.1 The performance specified for the craft is to be demonstrated.

9.8.1.2 The actual values of steering torque are to be verified during sea trials to confirm that the design maximum dynamic duty torque has not been exceeded.

9.8.2 Tunnel thrusters

It is to be demonstrated that the thruster unit meets the specified performance.

End of Chapter

Chapter 5

Boilers and Pressure Vessels

Contents

Section

1 *General*

2 *Design Requirements*

3 *Fittings and Mountings*

4 *Hydraulic Tests*

Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter are applicable to pressure vessels of seamless and fusion welded construction, and their mountings and fittings, for the following uses :

a) Fired boilers;

b) Exhaust gas heated boilers;

c) Economizers, superheaters, reheaters and steam receivers for, and associated with (a) and(b);

d) Steam heated steam generators;

e) Other pressure vessels containing:

- toxic fluids;
- fluids with pressure above 40 bar; and
- vapours or gases, e.g. air receivers, hydrophore or similar vessels, CO_2 containers, etc. and having $pV \geq 1.5$, where p is the design pressure in bar and V is the volume of the vessel [m^3].

1.1.2 Consideration will be given to arrangements or details of boilers, pressure vessels and equipment which can be shown to comply with other recognized standards, provided they are not less effective.

1.2 Design pressure

1.2.1 The design pressure is the maximum permissible working pressure.

1.2.2 The calculations made to determine the scantlings of the pressure parts are to be based on the design pressure, adjusted where necessary to take account of pressure variations corresponding to the most severe operational conditions.

1.2.3 It is desirable that there should be a margin between the normal pressure at which the boiler or pressure vessel operates and the lowest pressure at which any safety valve is set to lift, to prevent unnecessary lifting of the safety valve.

1.3 Metal temperature

1.3.1 The metal temperature, T, used to evaluate the allowable stress, σ, is to be taken as the actual metal temperature expected under operating conditions for the pressure part concerned, and is to be stated by the manufacturer when plans of the pressure parts are submitted for consideration.

1.3.2 For boilers, the design metal temperature is not to be taken less than the following values, unless justified by an exact calculation of the temperature drop and is in no case to be taken less than 250°C:

a) For steam heated steam generators, secondary drums of double evaporation boilers, steam receivers and pressure parts of fired pressure vessels not heated by hot gases and adequately protected by insulation, the metal temperature, T is to be taken as the maximum temperature of the internal fluid;

b) For pressure parts heated by hot gases, T is to be taken as not less than 25°C in excess of the maximum temperature of the internal fluid;

c) For combustion chambers of the type used in horizontal wet-back boilers, T is to be taken as not less than 50°C in excess of the maximum temperature of the internal fluid;

d) For furnaces, fire boxes, rear-tube plates of dry-back boilers and pressure parts subject to similar rates of heat transfer, T is to be taken as not less than 90°C in excess of the maximum temperature of the internal fluid;

e) For boiler, superheater, reheater and economizer tubes, the design temperature is to be taken as under :

   i) For boiler tubes the design temperature is to be taken as not less than saturated steam temperature plus 25°C for tubes mainly subject to convection heat, or plus 50°C for tubes mainly subject to radiant heat;

   ii) For superheater and reheater tubes, the design temperature is to be taken as not less than steam temperature expected in the part being considered, plus 35°C for tubes mainly subject to convection heat. For tubes mainly subject to radiant heat the design temperature is to be taken as not less than the steam temperature expected in the part being considered, plus 50°C, but the actual metal temperature expected is to be stated when submitting plans;

   iii) The design temperature for economizer tubes is to be taken as not less than 35°C in excess of the maximum temperature of the internal fluid.

1.3.3 In general, any part of boiler drums or headers not protected by tubes, and exposed to radiation from the fire or to the impact of hot gases, is to be protected by a shield of good refractory material or by other approved means.

1.3.4 Drums and headers of thickness greater than 30 [mm] are not to be exposed to combustion gases having an anticipated temperature in excess of 650°C unless they are efficiently cooled by closely arranged tubes.

1.4 Plans and particulars

1.4.1 The following plans, in triplicate, for boiler and pressure vessels are to be submitted for approval, in so far as applicable:

a) General arrangement, including arrangement of valves and fittings;

b) Sectional assembly;

c) Seating arrangements;

d) Steam, water drum and header details;

e) Water wall details;

f) Steam and superheater tubing, including the tube support arrangements;

g) Economizer details;

h) Casing arrangement;

i) Reheat section;

j) Fuel oil burning arrangement;

k) Forced draft system;

l) Boiler mountings including steam stop valves, safety valves and their relieving capacities, feed water connections, blow-off arrangements, watergauges, test cocks, etc.

1.4.2 The plans are to include the following particulars, in so far as applicable :

a) Scantlings;

b) Materials;

c) Weld details;

d) Design pressures and temperatures;

e) Heating surface areas of boilers and superheaters;

f) Estimated pressure drop through superheater;

g) Estimated evaporation rate;

h) Proposed setting pressure of safety valves on steam drum and superheater;

i) Pressure vessel class;

j) Details of heat treatment and testing of welds;

k) Calculations of thicknesses, when required;

l) Test pressures.

1.5 Classification of pressure vessels

1.5.1 For Rule purposes, boilers and pressure vessels are graded as shown in Table 1.5.1.

1.5.2 Pressure vessels which are constructed in accordance with the requirements of Class 2 or Class 3 will, if manufactured in accordance with the requirements of a superior class, be approved with the scantlings appropriate to that class.

1.5.3 In special circumstances relating to service conditions, materials, operating temperature, the carriage of dangerous gases and liquids, etc., it may be required that certain pressure vessels be manufactured in accordance with the requirements of a superior class.

1.6 Definition of symbols

1.6.1 The symbols used in the various formulae in this Chapter are defined as follows and are applicable to the specific part of the pressure under consideration :

d = diameter of hole or opening [mm];

D_i = inside diameter [mm];

D_o = outside diameter [mm];

J = joint factor applicable to the welded seams or ligament efficiency between tube holes, expressed as a fraction;

p = design pressure, in bar;

R_i = inside radius [mm];

R_o = outside radius [mm];

r_o = outside knuckle radius [mm];

r_i = inside knuckle radius [mm];

s = pitch [mm];

T = design temperature, °C;

t = minimum thickness [mm];

σ = allowable stress [N/mm^2].

1.7 Materials

1.7.1 Materials used in the construction of Class 1 & 2 pressure vessels are to be manufactured and tested in accordance with the requirements of Pt.2 under the supervision of IRS Surveyors.

Materials for Class 3 pressure vessels will be accepted with manufacturer's certificate.

Table 1.5.1 : Grading of pressure vessels			
	Boilers	**Steam-heated steam generators**	**Other pressure vessels**
Class 1	p > 3.5	$p > \frac{15000}{D_i + 1000}$	p > 50 or t > 38
Class 2	p ≤ 3.5	$p \leq \frac{15000}{D_i + 1000}$	$\frac{20000}{D_i + 1000}$ < p ≤ 50 or 16 < t ≤ 38 or material temperature > 150°C
Class 3			Pressure vessels not included in Class 1 and 2
Note: p = design pressure, in bar; D_i = internal diameter [mm]; t = shell thickness [mm]			

Valves and fittings for pressure vessels are to be manufactured and tested under the supervision of IRS Surveyors in the following cases:

- cast steel and nodular cast iron valves and fittings - when D > 50 [mm] and PD > 2000
- cast steel valves and fittings - when D > 50 [mm] and T > 400°C or T < -10°C
- copper alloy valve and fittings - when D >50 [mm] and pD > 1000
- Where p is the design pressure in bar and D is the nominal diameter in [mm].

1.7.2 The specified minimum tensile strength of carbon and carbon manganese steel plates, pipes, forgings and castings is to be within the following general limits :

a) For seamless and fusion welded pressure vessels - 340 - 520 [N/mm^2];

b) For boiler furnaces, combustion chambers and flanged plates - 400 - 520 [N/mm^2].

1.7.3 The specified minimum tensile strength of low alloy steel plates, pipes, forgings and castings is to be within the general limits of 400 - 500 [N/mm^2], and pressure vessels made in these steels are to be either seamless or Class 1 fusion welded.

1.7.4 The specified minimum tensile strength of boiler and superheater tubes is to be within the following general limits :

a) Carbon and carbon-manganese steels - 320 - 460 [N/mm^2];

b) Low alloy steels - 400 - 500 [N/mm^2].

1.7.5 Semi-killed, fully killed and fine grain, normal strength structural steels complying with the requirements of Pt.2, Ch.3 will be accepted for following Class 3 pressure vessels when:

$$p < \frac{15000}{D_i + 2000}$$

1.7.6 Where it is proposed to use materials other than those specified in Pt.2, details of the chemical compositions, heat treatment and mechanical properties are to be submitted for approval. In such cases the values of the mechanical properties used for deriving the allowable stress are to be subject to agreement by IRS.

1.7.7 Where a fusion welded pressure vessel is to be made of alloy steel and approval of the scantlings is required on the basis of the high temperature properties of the material, particulars of the welding consumables to be used, including typical mechanical properties and chemical composition of the deposited weld metal, are to be submitted for approval.

1.8 Allowable stress

1.8.1 The term "allowable stress",σ, is the stress to be used in the formulae for calculation of scantlings of pressure parts.

1.8.2 The allowable stress, σ, is to be the lowest of the following values:

$$= \frac{E_t}{1.6} \text{ or } \frac{E_{20}}{1.8} \text{ or } \frac{R_{20}}{2.7} \text{ or } \frac{S_R}{1.6}$$

where,

E_t = specified minimum lower yield strength or 0.2 per cent proof stress at temperature T;

E_{20} = specified minimum lower yield strength or 0.2 per cent proof stress at ambient temperature;

R_{20} = specified minimum tensile strength at room temperature;

S_R = average stress to produce rupture in 100,000 hours at temperature T;

T = metal temperature (see 1.6).

1.8.3 The allowable stress for steel castings is to be taken as 80 per cent of the value determined by the method indicated in 1.8.2, using the appropriate values for cast steel.

1.8.4 Where steel castings, which have been tested in accordance with Pt.2, are also subjected to non-destructive tests, consideration will be given to increasing the allowable stress using a factor up to 90 per cent in lieu of the 80 per cent referred to in 1.8.3. Particulars of the non-destructive test proposals are to be submitted for consideration.

1.9 Joint factors

1.9.1 The following joint factors are to be used in equations in this chapter, where applicable. Fusion welded pressure parts are to be made in accordance with Ch.10 :

Pressure vessel	Joint factor
Class 1	1.0
Class 2	0.80
Class 3	0.60

1.9.2 The longitudinal joints for all classes of vessels are to be butt joints. Circumferential joints for Classes 1 and 2 vessels are also to be butt joints.

1.10 Pressure parts of irregular shape

1.10.1 Where pressure parts are of such irregular shape that it is impracticable to design their scantlings by the application of formulae given in this Chapter, the suitability of their construction is to be determined by hydraulic proof test of a prototype or by an agreed alternative method.

1.11 Adverse working conditions

1.11.1 Where working conditions are adverse, special consideration may be required to be given to increasing the scantlings derived from the formulae, e.g. by increasing the corrosion or other allowance at present shown in the formulae, or by adopting a design pressure higher than defined in 1.2, to offset the possible reduction of life in service caused by the adverse conditions. In this connection, where necessary, account should also be taken of any excess of loading resulting from :

a) impact loads, including rapidly fluctuating pressures;

b) weight of the vessel and normal contents under operating and test conditions;

c) superimposed loads such as other pressure vessels, operating equipment, insulation, corrosion-resistant or erosion-resistant linings and piping;

d) reactions of supporting lugs, rings, saddles or other types of supports; or

e) the effect of temperature gradients on maximum stress.

1.12 Minimum shell thickness

1.12.1 Only plus tolerances are allowed on the design shell thickness.

1.12.2 The thickness after forming of any shell or head is not to be less than $3 + \frac{D_i}{1500}$ [mm] for carbon, carbon-manganese and low-alloy steels or 3 [mm] for stainless steel and non-ferrous materials. For pressure vessels, where the cylindrical part is made of a pipe, a smaller minimum thickness may be approved. See also Sec.2 for minimum thicknesses of plates in case of boilers.

1.13 Heat treatment, non-destructive examination and routine tests

1.13.1 Details regarding heat treatment, non-destructive examination and routine tests are to be in accordance with the requirements of Ch.10.

Section 2

Design Requirements

2.1 Cylindrical shells and drums subject to internal pressure

2.1.1 Minimum thickness

2.1.1.1 The minimum thickness, t, of a cylindrical shell is to be determined by the following formula:

$$t = \frac{p R_i}{10 \sigma J - 0.5p} + 0.75 \text{ [mm]}$$

where,

t, p, R_i and σ are defined in Sec.1;

J = efficiency of ligaments between tube holes or other openings in the shell or the joint factor of the longitudinal joints (expressed as a fraction), as defined in Sec.1, whichever applies. In the case of seamless shells clear of tube holes or other openings, J = 1.0.

2.1.1.2 The formula in 2.1.1.1 is applicable only where the resulting thickness does not exceed half the internal radius, i.e. where R_o is not greater than 1.5 R_i.

2.1.1.3 Irrespective of the thickness determined by 2.1.1.1, t is not to be less than:

For Boilers (Fired and exhaust gas heated), economizers, superheaters, reheaters, steam receivers, steam-heated steam generators and similar vessels	6.0 [mm] for cylindrical shell plates. For tube plates, such thickness as will give a minimum parallel seat of 9.5 [mm], or such greater width as may be necessary to ensure tube tightness.

2.1.2 Efficiency of ligaments between tube holes

2.1.2.1 Where tube holes are drilled in a cylindrical shell in a line or lines parallel to its axis, the efficiency, J, of the ligaments is to be determined as follows:

a) for regular drilling (See Fig.2.1.2.1)

$$J = \frac{s - d}{s}$$

b) for irregular drilling (See Fig.2.1.2.2)

$$J = \frac{S_1 + S_2 - 2d}{S_1 + S_2}$$

where,

d = the mean effective diameter of the tube holes, [mm], after allowing for any serrations, counterboring of recessing, or the compensating effect of tube stub, see 2.1.3 and 2.1.4;

s = pitch of tube holes [mm];

s_1 = the shorter of any two adjacent pitches [mm];

s_2 = the longer of any two adjacent pitches [mm].

2.1.2.2 When applying the formula in 2.1.2.1, the double pitch ($s_1 + s_2$) chosen is to be that which makes J a minimum, and in no case is s_2 to be taken as greater than twice s_1.

2.1.2.3 Where the circumferential pitch between tube holes measured on the mean of the external and internal drum or header diameters is such that the ligament efficiency determined by the formula in 2.1.2.1 is less than one- half of the ligament efficiency of the longitudinal axis, J is to be taken as twice the circumferential efficiency.

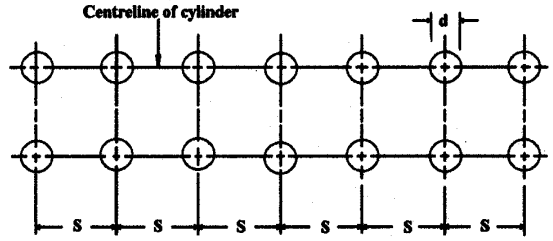


Fig. 2.1.2.1 : Regular drilling

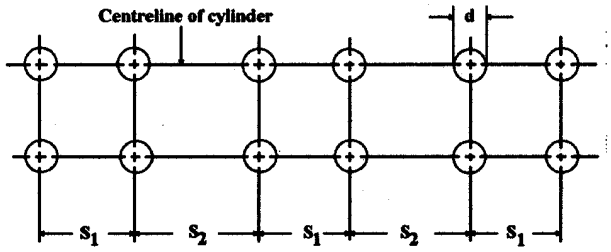


Fig. 2.1.2.2 : Irregular drilling

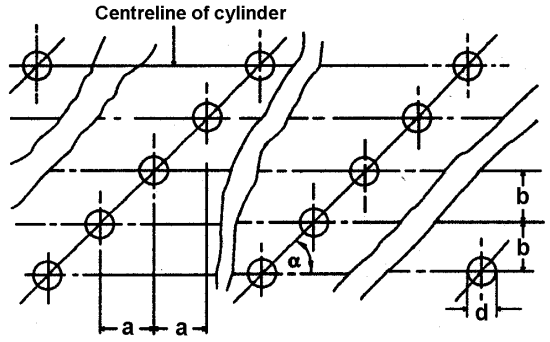


Fig. 2.1.2.3 : Spacing of holes on a diagonal line

2.1.2.4 Where tube holes are drilled in a cylindrical shell along a diagonal line with respect to the longitudinal axis, the efficiency, J, of the ligament is to be determined as in 2.1.2.5 to 2.1.2.8.

2.1.2.5 For spacing of tube holes on a diagonal line as shown in Fig.2.1.2.3, or in a regular saw-tooth pattern as shown in Fig.2.1.2.4, J is to be obtained from the series of curves given in Fig 2.1.2.6 where a and b, as shown in Fig.2.1.2.3 and Fig.2.1.2.4, are measured, in millimeters, on the median line of the plate and d is as defined in 2.1.2.1.

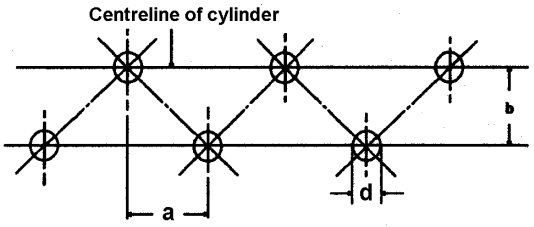


Fig. 2.1.2.4 : Regular saw-tooth pattern of holes

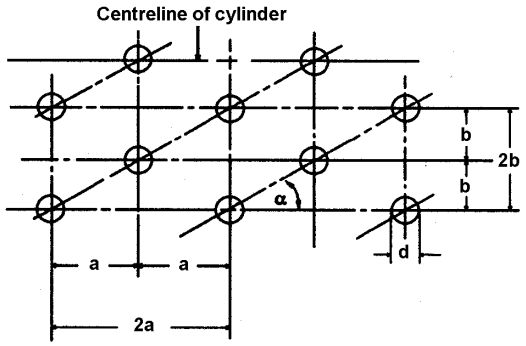


Fig. 2.1.2..5 : Regular staggering of holes

2.1.2.6 The data for Fig.2.1.2.6 is based on the following:

$$J = \frac{2}{A + B + \sqrt{(A - B)^2 + 4C^2}}$$

where,

$$A = \frac{a\,(\mathrm{Cos}^2\,\alpha + 1)}{2\,(a - d\,Cos\alpha)}$$

$$B = 0.5\left(1 - \frac{d\,\mathrm{Cos}\alpha}{a}\right)\left(\mathrm{Sin}^2\alpha + 1\right)$$

$$C = \frac{a\,\mathrm{Sin}\alpha\,Cos\alpha}{2\,(a - d\,Cos\alpha)}$$

$$\mathrm{Cos}\alpha = \frac{a}{\sqrt{a^2 + b^2}}$$

$$\mathrm{Sin}\alpha = \frac{b}{\sqrt{a^2 + b^2}}$$

α = angle between centerline of cylinder and centerline of diagonal holes.

2.1.2.7 For regularly staggered spacing of tube holes as shown in Fig.2.1.2.5, the smallest value of the efficiency, J, of all ligaments (longitudinal, circumferential and diagonal) is obtained from Fig.2.1.2.7, where a and b, as shown in Fig.2.1.2.5 are measured, in millimeters, on the median line of the plate, and d as defined in 2.1.2.1.

2.1.2.8 For irregularly spaced tube holes whose centers do not lie on a straight line, the formula 'b' in 2.1.2.1 is to apply, except that an equivalent longitudinal width is to be used. An equivalent longitudinal width is that which gives, using the formula 'a' in 2.1.2.1 the same efficiency as would be obtained using Fig.2.1.2.6 for the diagonal ligament in question.

2.1.2.9 The spacing of tube holes is to be such that the minimum width, in [mm], of any ligament between tube holes is not less than:

$$0.125\ d + 12.5\ [mm]$$

where,

d = diameter of tube holes [mm].

2.1.3 Compensating effect of tube stubs

2.1.3.1 Where a drum or header is drilled for tube stubs fitted by strength welding, either in line or in staggered formation, the effective diameter of holes is to be taken as follows:

$$d_e = d_a - \frac{A}{t}$$

where,

d_e = the equivalent diameter of the hole [mm];

d_a = the actual diameter of the hole [mm];

A =the compensating area provided by each tube stub and its welding fillets [mm^2];

t = thickness of the shell [mm].

2.1.3.2 The compensating area, A, is to be measured in a plane through the axis of the tube stub parallel to the longitudinal axis of the drum or header and is to be calculated as follows (See Fig.2.1.3.1 and Fig.2.1.3.2):

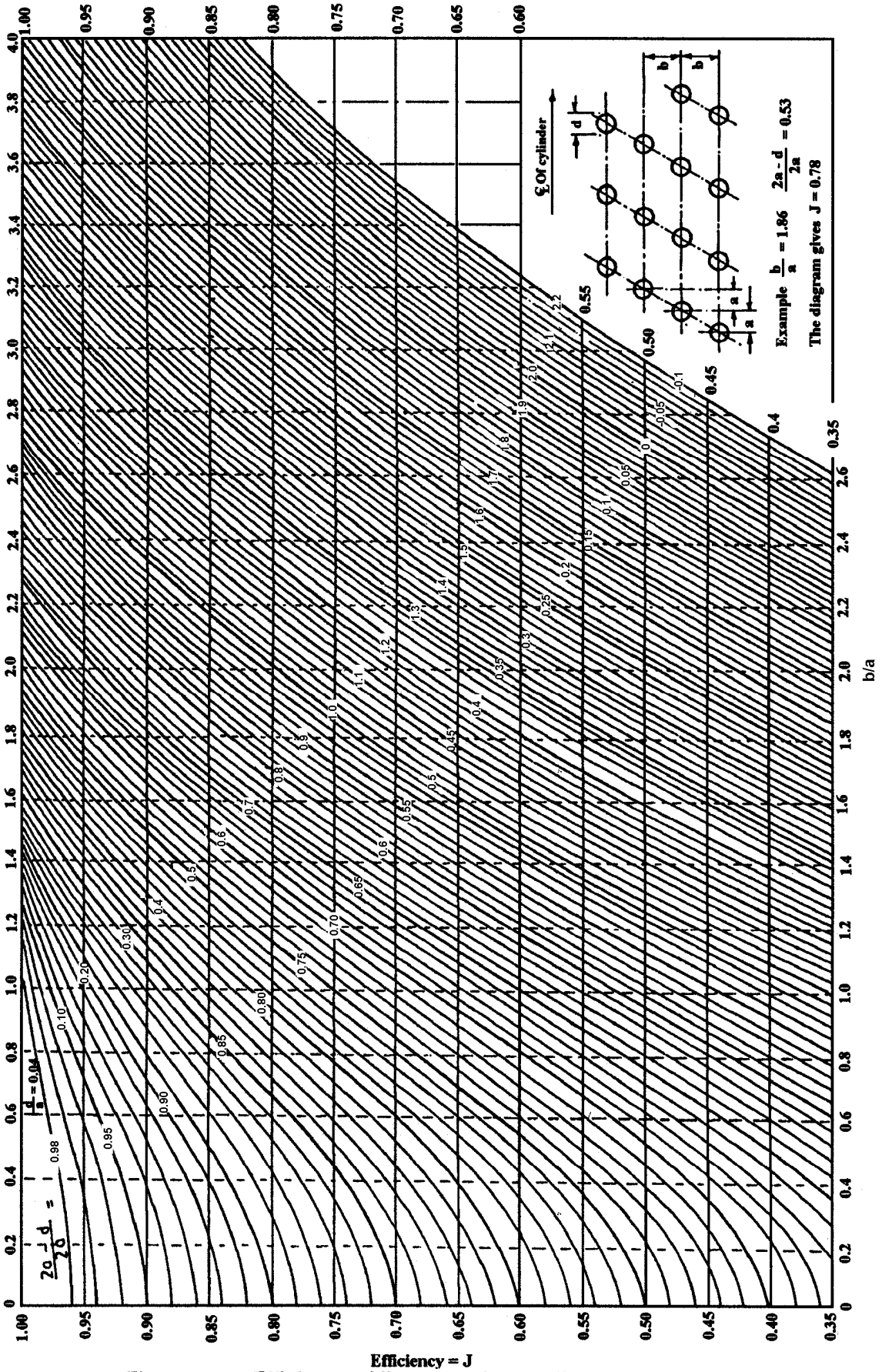


Fig. 2.1.2.6 : Efficiency of ligament along a diagonal line

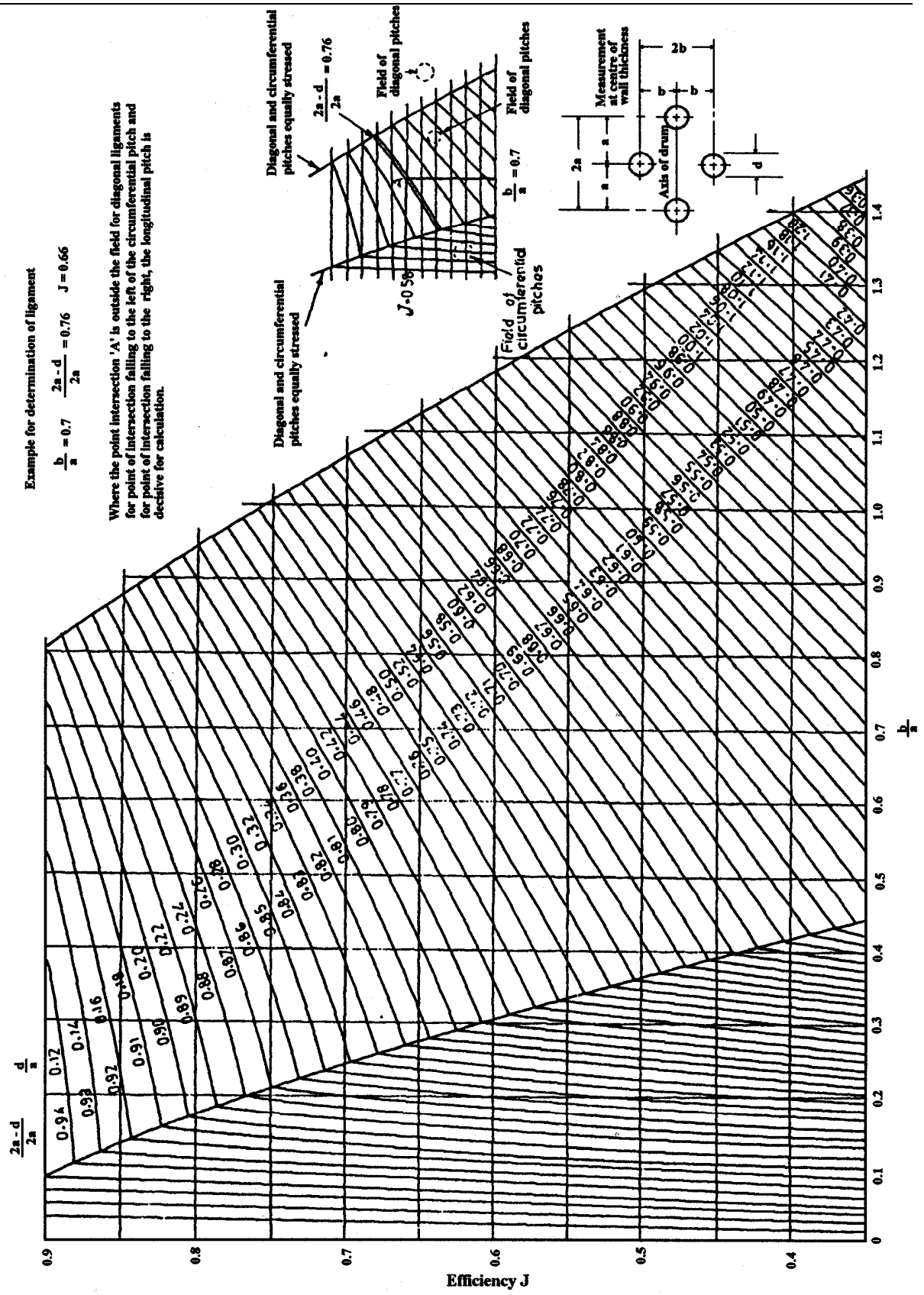


Fig. 2.1.2.7 : Efficiency of ligaments between holes

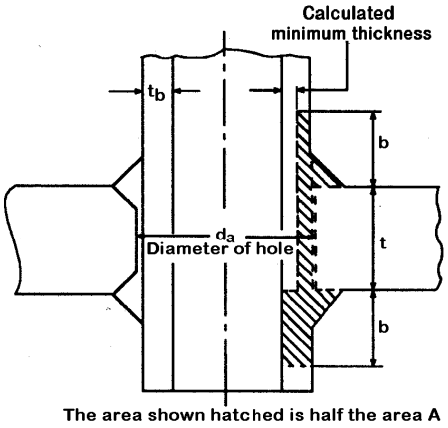


Fig. 2.1.3.1 : Compensation of welded tube stubs

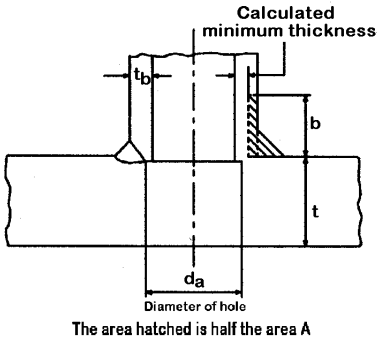


Fig. 2.1.3.2 : Compensation of welded tube stubs

- The cross-sectional area of the stub, in excess of that required by 2.7.1 for the minimum tube thickness, from the interior surface of the shell up to a distance b from the outer surface of the shell;

- plus the cross-sectional area of the stub projecting inside the shell within a distance b from the inner surface of the shell;

- plus the cross-sectional area of the shell welding fillets inside and outside the shell;

where,

$$b = \sqrt{d_a\ t_b}$$

t_b = actual thickness of tube stub [mm].

2.1.3.3 Where the material of the tube stub has an allowable stress lower than that of the shell, the compensating cross-sectional area of the stub is to be multiplied by the ratio:-

$$\frac{\text{allowable stress of stub at design metal temperature}}{\text{allowable stress of shell at desgn metal temperature}}$$

2.1.3.4 No credit will be given for the additional strength of tube stub material having an allowable stress greater than that of the shell.

2.1.4 Unreinforced openings

2.1.4.1 Openings in a definite pattern such as tube holes, may be designed in accordance with the rules for ligaments in 2.1.2, provided that the diameter of the largest hole in group does not exceed that permitted by 2.1.4.2.

2.1.4.2 The maximum diameter, d, of any isolated unreinforced openings is to be obtained from the curves in Fig.2.1.4.1 and Fig.2.1.4.2. The value of K to be used is calculated from the following formula:

$$K = \frac{p\,D_o}{18.2\,\sigma\,t}$$

where,

p, D_o and σ are defined in Sec.1;

t = actual thickness of shell [mm].

2.1.4.3 For elliptical or oval holes, d refers to the major axis when this lies longitudinally or to the mean of the major and minor axes when minor axis lies longitudinally.

2.1.4.4 No unreinforced opening is to exceed 200 [mm] in diameter.

2.1.4.5 Openings may be considered isolated if the center distance between two openings on the longitudinal axis of the cylindrical shell is not less than:

$d+1.1\sqrt{Dt}$ with a minimum of 5d

where,

d = diameter of openings in shell (mean diameter if dissimilarly sized openings are involved)

D = mean diameter of shell

t = actual thickness of shell

Where the center distance is less than so derived, the openings are to be fully compensated.

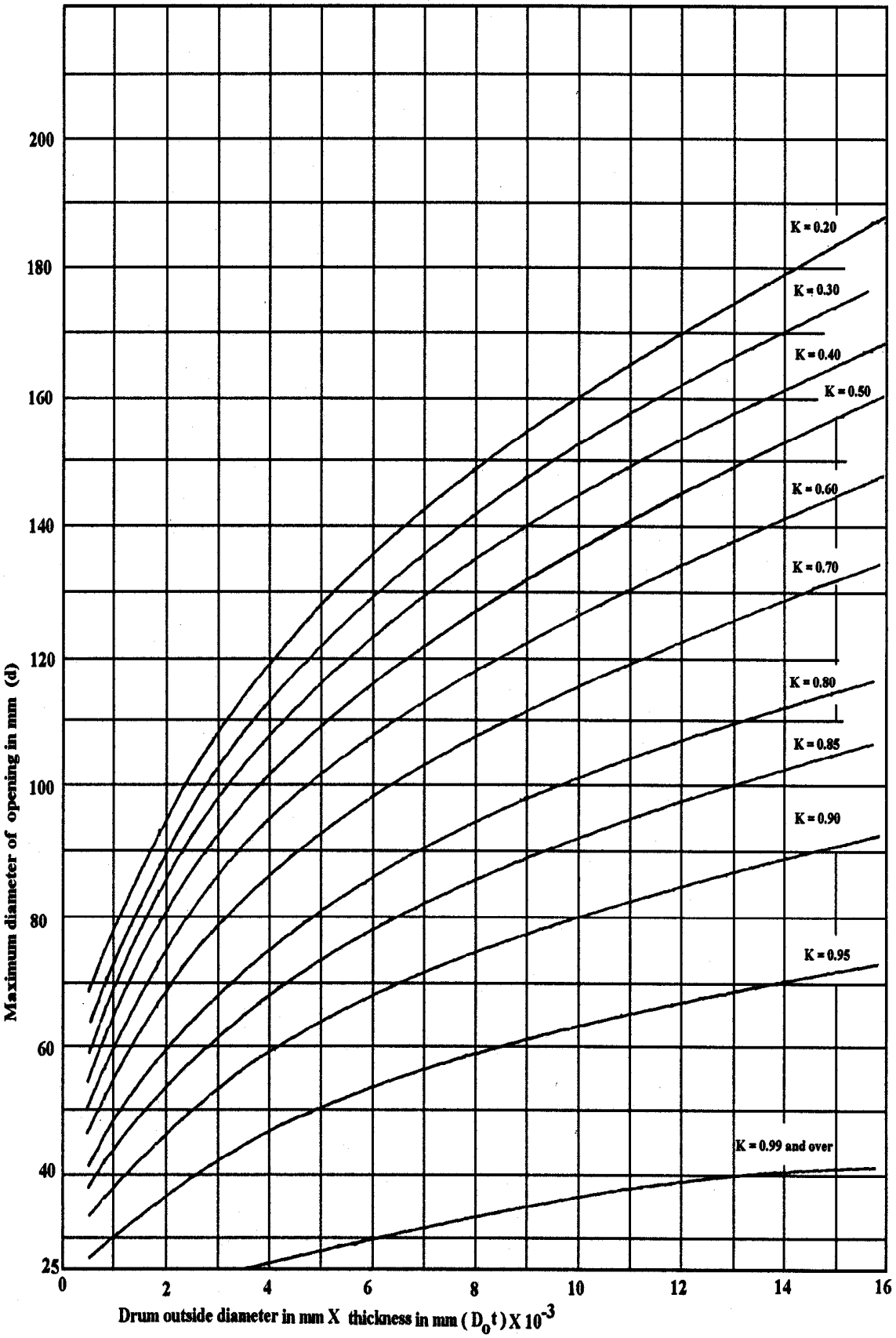


Fig. 2.1.4.1 : Maximum diameter of unreinforced openings

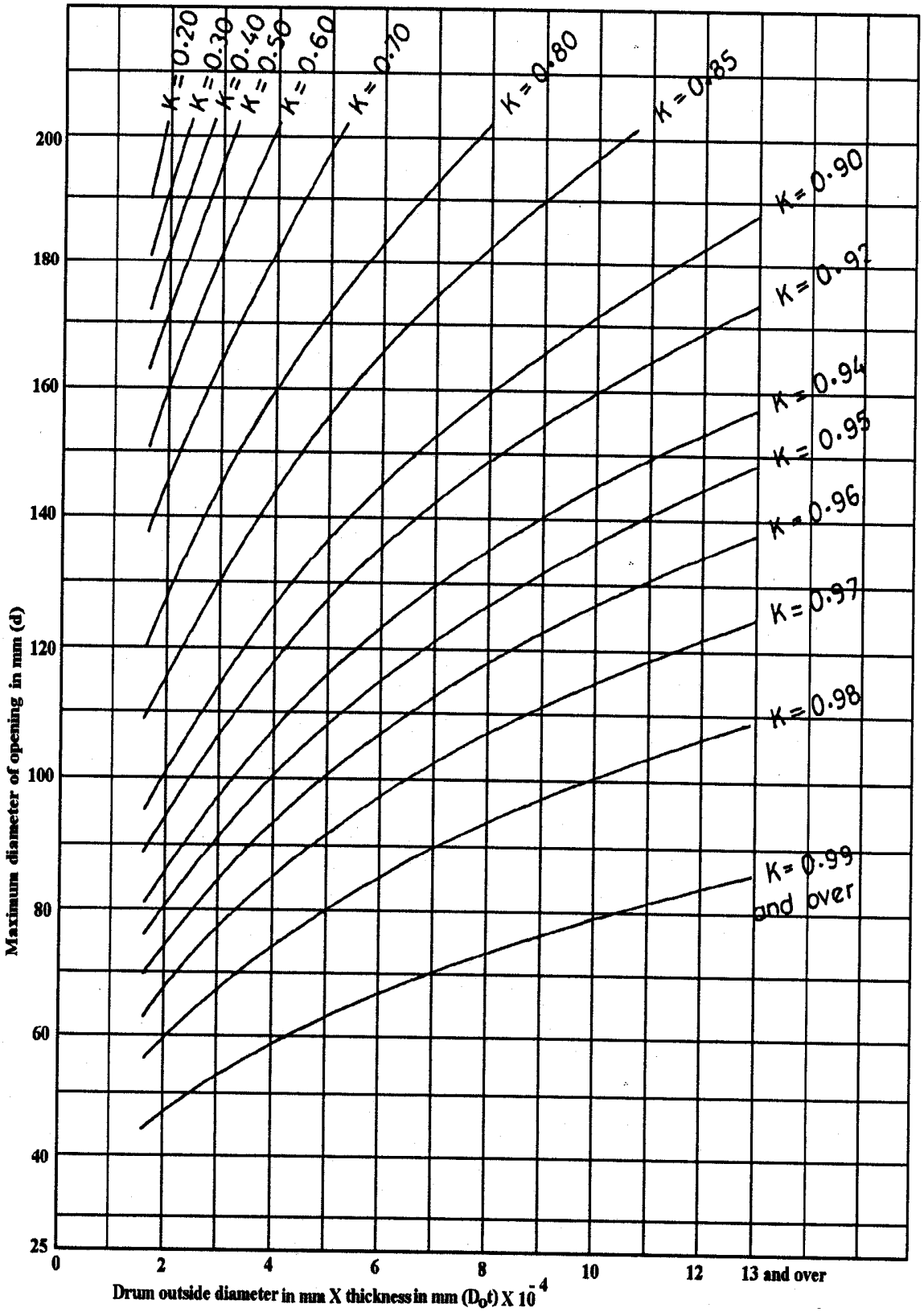


Fig. 2.1.4.2 : Maximum diameter of unreinforced openings

Where two openings are offset on a diagonal line, the diagonal efficiency from Fig.2.1.2.7 may be used to derive an equivalent longitudinal center distance for the purpose of this paragraph.

2.1.5 Reinforced openings

2.1.5.1 The following notations are used in Fig.2.1.5.1(a),(b) and (c):

A = calculated thickness of shell without joint or opening;

B = thickness calculated in accordance with 2.7.1;

$C = 0.8\sqrt{d_i\, t_b}$

$D = \sqrt{D_i\, t_a}$ and is not to exceed 0.5 d_i;

t_a = actual thickness of shell plate [mm];

t_b = actual thickness of standpipe stem or branch [mm];

t_r = thickness of added reinforcement [mm] (t_r will be zero when there is no compensating plate on the side of the shell under consideration);

d_i = internal diameter of standpipe or branch [mm];

D_i = internal diameter of cylindrical shell [mm].

2.1.5.2 Openings larger than those permitted by 2.1.4 are to be reinforced by the method shown in Fig.2.1.5.1. Compensation will be considered adequate when:

$$Y + Z\frac{\sigma_s}{\sigma} \geq X$$

where,

X = the area to be compensated and is indicated by X in Fig.2.1.5.1;

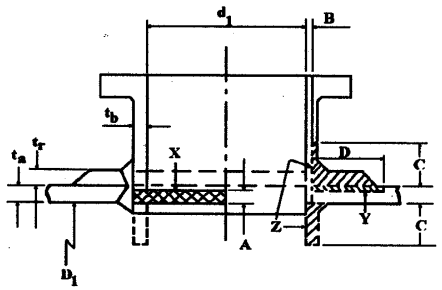
Y = the compensating area available in the shell material and is indicated by Y in Fig.2.1.5.1;

Z = the compensating area available in the standpipe material and is indicated by Z in Fig.2.1.5.1;

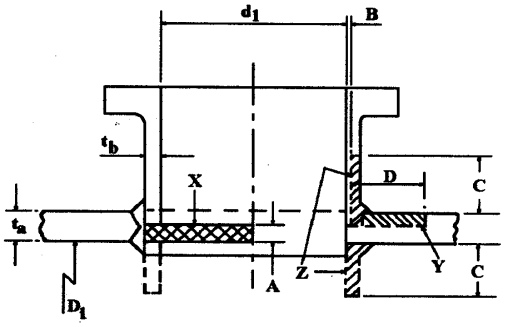
σ_s = the allowable stress of the standpipe material at design temperature;

σ = the allowable stress of shell material at design temperature;

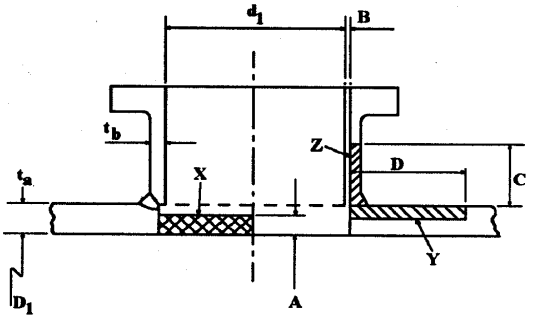
$\frac{\sigma_s}{\sigma}$ is to be taken as not greater than 1.



(a) Welded set-through nozzle with compensating plate



(b) Welded set-through nozzle



(c) Welded set-on nozzle

Fig. 2.1.5.1 : Compensation for welded

2.1.5.3 Area X is to be such that the reinforcement is provided on all planes through the center of the opening and normal to the shell surface, and is to be calculated as the product of the radius of the hole cut in the shell and the thickness, A, that would be required for an equivalent seamless unpierced shell.

2.1.5.4 Area Y is to be measured in the same plane as area X, and is to be calculated as the product of the difference between the actual shell thickness and the equivalent unpierced shell thickness, A, and the dimension from the edge of the opening in shell to limit D.

2.1.5.5 Area Z is to be measured in the same plane as area X, and is to be calculated as follows:

- For that part of standpipe which projects outside the shell, calculate the full cross-sectional area of the stem up to a distance C from the actual outer surface of the shell plate, and deduct from it the cross-sectional area which the stem would have if its thickness was as calculated in accordance with 2.7.1;

- **plus**, in the case of set- through nozzles (see Fig.2.1.5.1 (a) and (b), the full cross-sectional area of that part of the stem which projects inside the shell up to a distance of C, from the inside surface of the shell;

- **plus**, the cross-sectional area of all

- appropriate fillet welds;

- **plus**, if additional reinforcement is fitted as illustrated in Fig.2.1.5.1(b), the cross-sectional area of the reinforcement and the sectional area of its fillet welds.

2.1.5.6 The welds attaching standpipes and reinforcing plates to the shell are to be of sufficient size to transmit the full strength of the reinforcing areas and all other loadings to which they may be subjected.

2.2 Spherical shells subject to pressure on the concave side

2.2.1 Minimum thickness

2.2.1.1 The minimum thickness of a spherical shell is to be determined by the following formula:

$$t = \frac{p R_i}{20 \sigma J - 0.5p} + 0.75 \text{ [mm]}$$

where,

p, R_i, σ, J and t are defined in Sec.1.

2.2.1.2 The formula in 2.2.1.1 is applicable only where the resulting thickness does not exceed half the internal radius.

2.2.2 Openings

2.2.2.1 Openings in spherical shells are to comply with the relevant requirements of 2.3.

2.3 Dished ends subject to pressure on concave side

2.3.1 Minimum thickness

2.3.1.1 The thickness, t, of semi-ellipsoidal, torispherical and hemispherical unstayed ends, dished from plate, having pressure on the concave side and satisfying the conditions listed below, is to be determined by the following formula:

$$t = \frac{p D_o K}{20 \sigma J} + 0.75 \text{ [mm]}$$

where,

p, D_o, σ and J are defined in Sec.1;

K = a shape factor in accordance with 2.3.2 and Fig.2.3.1.1.

2.3.1.2 For semi-ellipsoidal ends:

the external height, H ≥ 0.18 Do, where D_o = the external diameter of the parallel portion of the end [mm].

2.3.1.3 For torispherical ends:

the internal radius, $R_i \leq D_o$;

the internal knuckle radius, $r_i \geq 0.1\ D_o$;

the internal knuckle radius, $r_i \geq 3t$;

the external height, H ≥ 0.18 D_o, and is to be determined as follows:

$$H = R_o - \sqrt{(R_o - 0.5 D_o)(R_o + 0.5 D_o - 2 r_o)}$$

2.3.1.4 In addition to the formula in 2.3.1.1, the thickness, t, is to be not less than that determined by 2.2.

2.3.1.5 In all cases, H is to be measured from the commencement of curvature, see Fig.2.3.1.2(a), (b), (c) and (d).

2.3.1.6 For boilers (fired and exhaust gas heated), economisers, superheaters, reheaters, steam receivers, steam heated steam generators and similar vessels, the minimum thickness of the head, t, is to be not less than 9.5 [mm]. In special cases where it is proposed to use less than 9.5 [mm] thickness, the proposals will be subject to special consideration.

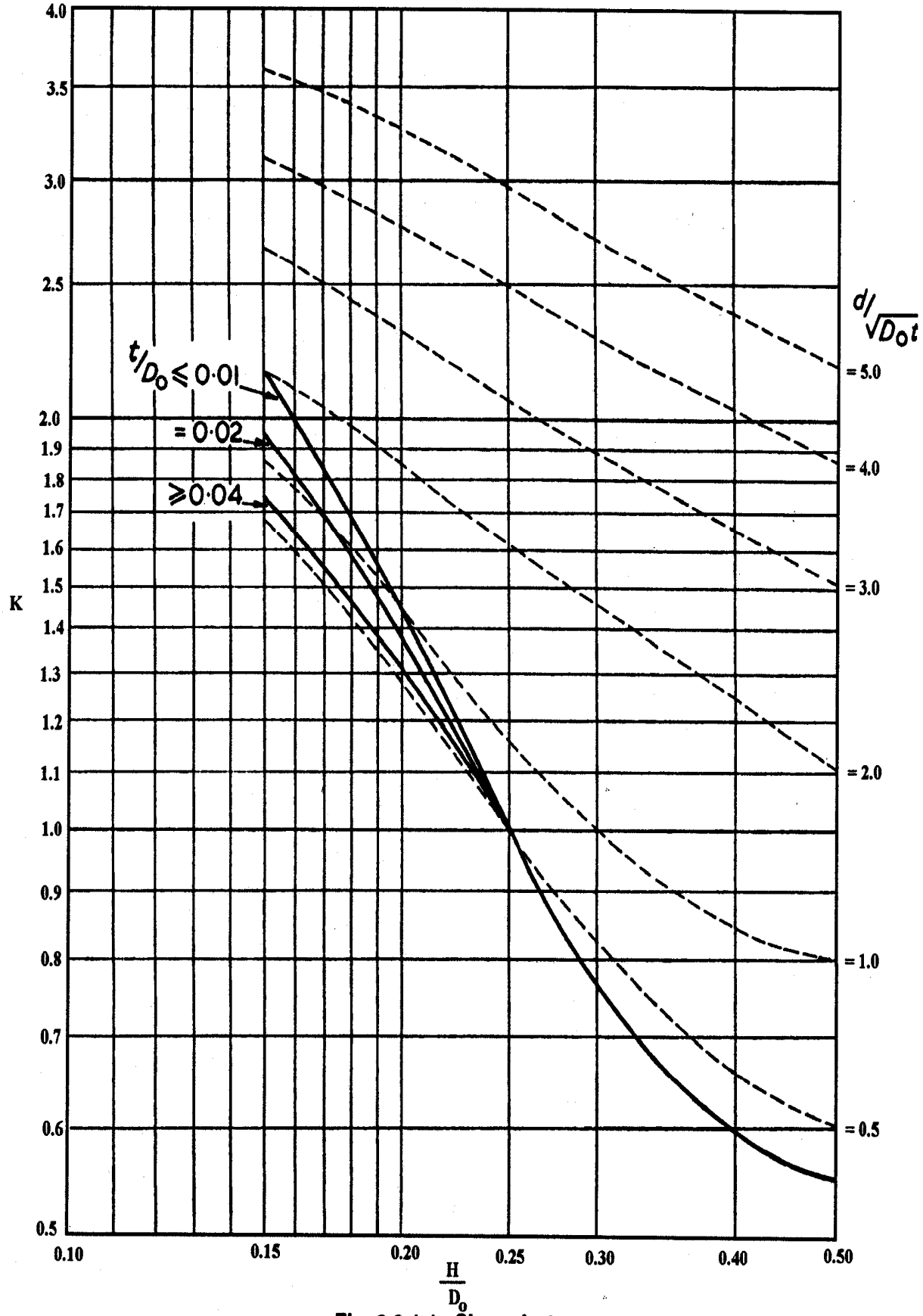


Fig. 2.3.1.1 : Shape factor

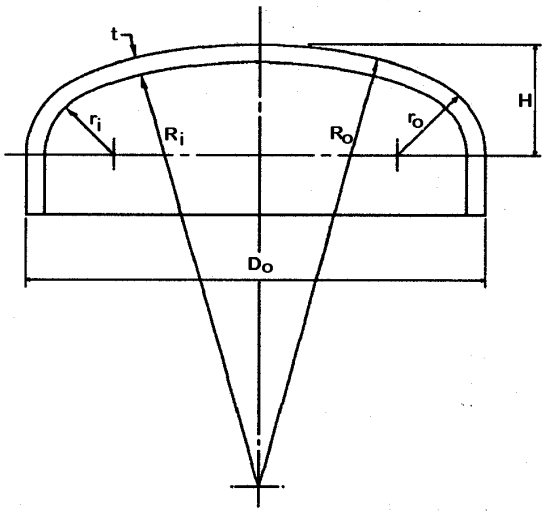


Fig. 2.3.1.2 (a) : Torispherical end

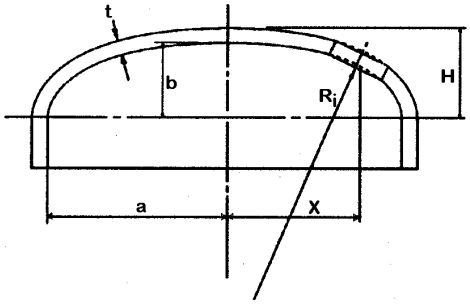


Fig. 2.3.1.2 (b) : Semi-ellipsoidal end

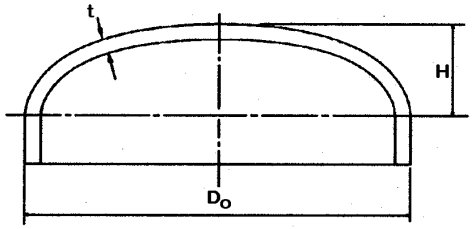


Fig. 2.3.1.2 (c) : Semi-ellipsoidal end (Radius of curvature at opening)

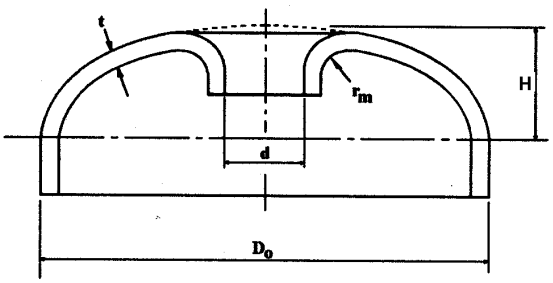
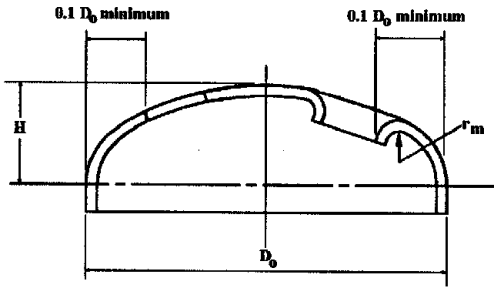


Fig. 2.3.1.2 (d) : End with manhole (Semi-ellipsoidal or torispherical)



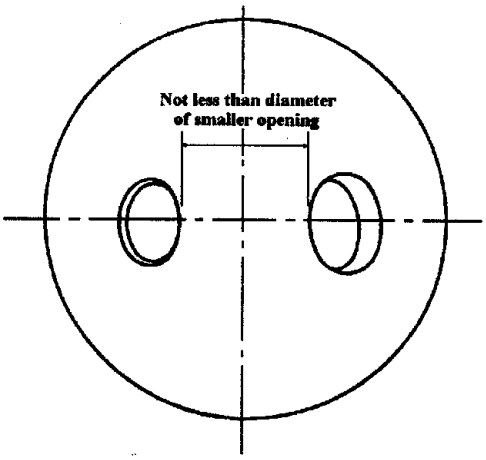


Fig. 2.3.1.3 : Openings in dished ends

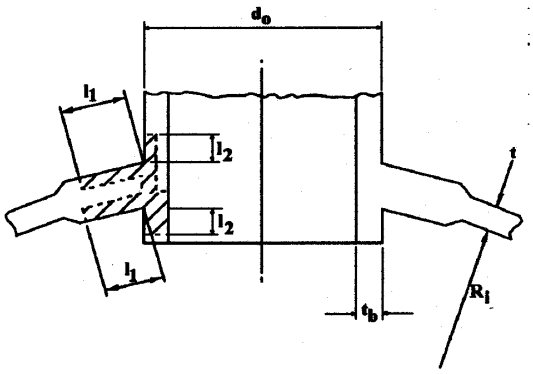


Fig. 2.3.1.4 : Limits of reinforcement

2.3.1.7 For ends which are butt welded to the drum shell, the thickness of the edge of the flange for connection to the shell is to be not less than the thickness of an unpierced seamless or welded shell, whichever is applicable, of the same diameter and material and determined by 2.1.

2.3.2 Shape factors for dished ends

2.3.2.1 The shape factor, K, to be used in 2.3.1.1 is to be obtained from the curves in

Fig.2.3.1.1, and depends upon the ratio of height to diameter H/D_o.

2.3.2.2 The lowest curve in the series provides the factor K for plain ends without openings or with small openings not requiring reinforcement. For lower values of H/D_o, the shape factor, K, depends on the ratio of thickness to diameter, t/D_o, as well as the ratio H/D_o and a trial calculation may be necessary to arrive at the correct value of K.

2.3.3 Dished ends with unreinforced openings

2.3.3.1 The openings in dished ends may be circular or approximately elliptical.

2.3.3.2 The upper curves in Fig.2.3.1.1 provide values of K to be used in 2.3.1.1 for ends with unreinforced openings. The selection of the correct curve depends on the value $\frac{d}{\sqrt{D_o t}}$ and a trial calculation is necessary to select the correct curve.

where,

d = the diameter of the largest opening in the end plate [mm] (in the case of an elliptical opening, the larger axis of the ellipse);

t = minimum thickness, after dishing [mm];

D_o = outside diameter of dished end [mm].

2.3.3.3 The following requirements must in any case be satisfied:

$$\frac{t}{D_o} \leq 0.1$$

$$\frac{d}{D_o} \leq 0.7$$

2.3.3.4 From Fig.2.3.1.1 for any selected ratio H/D_o, the curve for unpierced ends gives a value for $\frac{d}{\sqrt{D_o t}}$ as well as for K. Openings giving a value of $\frac{d}{\sqrt{D_o t}}$ not greater than the value so obtained may thus be pierced through an end designed as unpierced without any increase in thickness.

2.3.4 Flanged openings in dished ends

2.3.4.1 The requirements in 2.3.3 apply equally to flanged openings and to unflanged openings cut in the plate of an end. No reduction may be made in the end plate thickness on account of flanging.

2.3.4.2 Where openings are flanged, the radius, r_m, of the flanging is to be not less than 25 [mm], See Fig.2.3.1.2(d). The thickness of the flanged portion may be less than the calculated thickness.

2.3.4.3 Unreinforced and flanged openings in dished ends are to be so arranged that the distance from the edge of the hole to the outside edge of the plate and the distance between openings are not less than those shown in Fig.2.3.1.3.

2.3.5 Dished ends with reinforced openings

2.3.5.1 Where it is desired to use a large opening on a dished end of less thickness than would be required by 2.3.3, the end is to be reinforced. This reinforcement may consist of a ring or standpipe welded into the hole, or of reinforcing plates welded to the outside and/or inside of the end in the vicinity of the hole, or a combination of both methods, see Fig.2.3.1.4. Forged reinforcement may be used.

2.3.5.2 Reinforcing material within the following limits may be taken as effective reinforcement:

a) The effective width l_1, of reinforcement is not to exceed $\sqrt{2 R_i t}$ or 0.5 d_o whichever is the lesser;

b) The effective length l_2 of a reinforcing ring is not exceed $\sqrt{d_o t_b}$;

where,

R_i = The internal radius of the spherical part of a torispherical end [mm]; or

= the internal radius of the meridian of the ellipse at the center of the opening, of a semi-ellipsoidal end, [mm], and is given by the following formula:

$$\frac{[a^4 - x^2(a^2 - b^2)]^{1.5}}{a^4 b}$$

where,

a, b and x are shown in Fig.2.3.1.2(c);
t_b = actual thickness of ring or standpipe [mm];
d_o = external diameter of ring or standpipe [mm];
l_1 and l_2 are shown in Fig.2.3.1.4.

2.3.5.3 The shape factor, K, for a dished end having a reinforced opening can be read from Fig.2.3.1.1 using the value obtained from:

$$\frac{d_o - A/t}{\sqrt{D_o\, t}} \text{ instead of from } \frac{d}{\sqrt{(D_o\, t)}}$$

where,

A = the effective cross-sectional area of reinforcement and is to be twice the area shown shaded on Fig.2.3.1.4.

A trial calculation is necessary to select the correct curve.

2.3.5.4 The area shown in Fig.2.3.1.4 is to be obtained as follows:-

- calculate the cross-sectional area of reinforcement both inside and outside the end plate within the length l_1;
- **plus** the full cross-sectional area of that part of the ring or standpipe which projects inside the end plate up to the distance l_2;
- **plus** the full cross-sectional area of that part of the ring or standpipe which projects outside the internal surface of the end plate up to a distance l_2;
- and **deduct** the sectional area which the ring or standpipe would have if its thickness were as calculated in accordance with 2.7.1.

2.3.5.5 If the material of the ring or the reinforcing plates has an allowable stress value lower than that of the end plate, then the effective cross-sectional area, A, is to be multiplied by the ratio:-

$$\frac{\text{allowable stress of reinforcing plate at design temperature}}{\text{allowable stress of end plate at design temperature}}$$

2.4 Conical ends subject to internal pressure

2.4.1 Minimum thickness

2.4.1.1 The minimum thickness, t, of the cylinder, knuckle and conical section at the junction and within the distance L from the junction is to be determined by the following formula, but in no case is to be less than the thickness determined by 2.4.1.3:-

$$t = \frac{p\, D_o\, K}{20\, \sigma\, J} + 0.75 \text{ [mm]}$$

where,

t, p, σ and J are defined in Sec.1;

D_o = outside diameter, in [mm], of the conical section or end, see Fig.2.4.1.1(a),(b),(c) and (d);

K = a factor, taking in to account the stress in the knuckle, see Table 2.4.1.1.

2.4.1.2 If the distance of a circumferential seam from the knuckle or junction is not less than L, then J is to be taken as 1.0, otherwise J is to be taken as the weld joint factor appropriate to the circumferential seam;

where,

L = distance, in [mm], from knuckle or junction within which meridional stresses determine the required thickness, see Fig.2.4.1.1;

$$= 0.5\sqrt{\frac{D_o\, t}{\text{Cos}\,\psi}}$$

r_i = inside radius of transition knuckle, in [mm], which is to be taken as 0.01 Dc in case of conical sections without knuckle transition.

2.4.1.3 The minimum thickness, t, of those parts of conical sections not less than a distance L from the junction with a cylinder or other conical section is to be determined by the following formula:-

$$t = \frac{p\, D_c}{(20\, \sigma\, J - p)\, Cos\alpha} + 0.75 \text{ [mm]}$$

where,

D_c = inside diameter, in [mm], of conical section or end at the position under consideration, see Fig.2.4.1.1;

$\alpha 1$, $\alpha 2$, $\alpha 3$ = angle of slope of conical section (at the point under consideration) to the vessel axis, see Fig.2.4.1.1.

2.4.1.4 Conical ends may be constructed of several ring sections of decreasing thickness, as determined by the corresponding decreasing diameter.

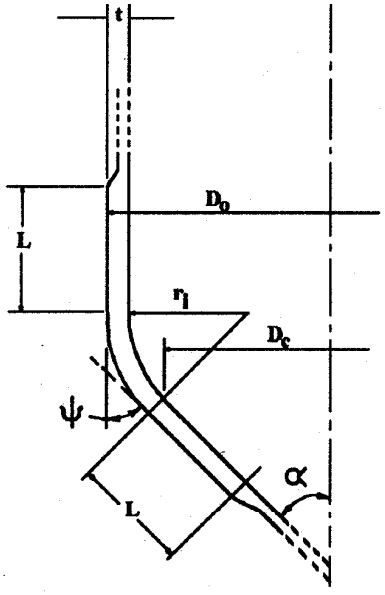


Fig. 2.4.1.1 (a) : Cone/cylinder with knuckle

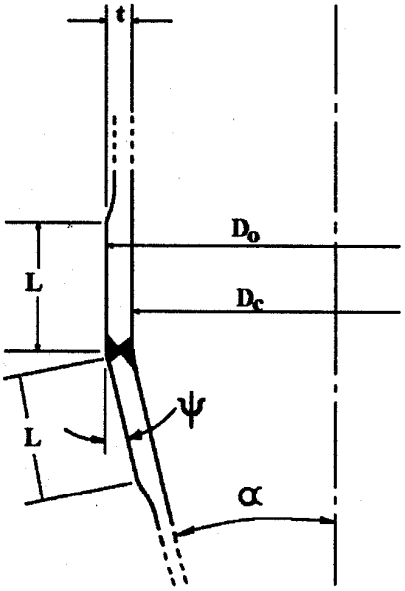


Fig. 2.4.1.1 (b) : Cone/cylinder without knuckle

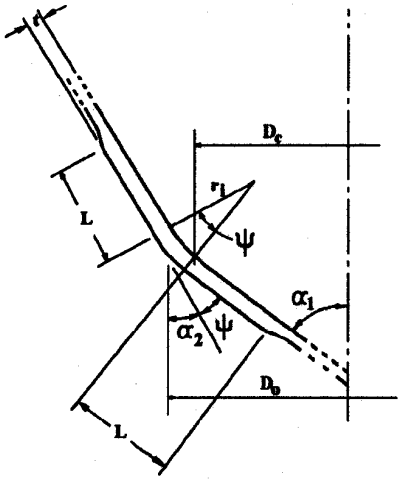


Fig. 2.4.1.1 (c) : Cone/cone with knuckle

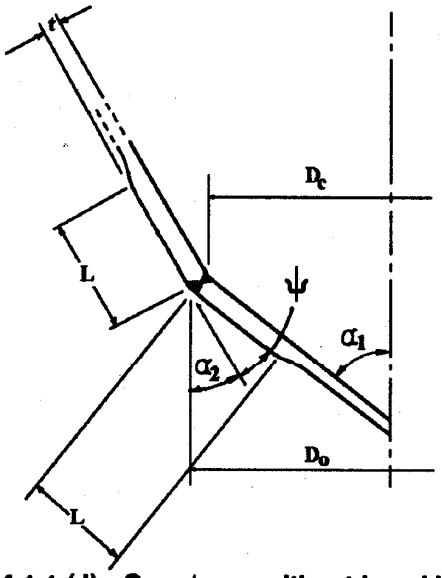


Fig. 2.4.1.1 (d) : Cone/cone without knuckle

Table 2.4.1.1 : Values of K as a function of ψ and r_i/D_o

	Values of K for r_i/D_o ratios											
ψ	**0.01**	**0.02**	**0.03**	**0.04**	**0.06**	**0.08**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
10°	0.70	0.65	0.60	0.60	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
20°	1.00	0.90	0.85	0.80	0.70	0.65	0.60	0.55	0.55	0.55	0.55	0.55
30°	1.35	1.20	1.10	1.00	0.90	0.85	0.80	0.70	0.65	0.55	0.55	0.55
45°	2.05	1.85	1.65	1.50	1.30	1.20	1.10	0.95	0.90	0.70	0.55	0.55
60°	3.20	2.85	2.55	2.35	2.00	1.75	1.60	1.40	1.25	1.00	0.70	0.55
75°	6.80	5.85	5.35	4.75	3.85	3.50	3.15	2.70	2.40	1.55	1.00	0.55

2.4.1.5 The thickness of conical sections having an angle of inclination to the vessel axis of more than 75 degrees is to be determined as for a flat plate.

2.5 Unstayed flat end plates

2.5.1 Minimum thickness

2.5.1.1 The thickness of an unstayed flat end plate is to be determined in accordance with 2.7.3.3 or 2.8.1.7 or 2.8.1.8 as applicable.

2.6 Standpipe and branches

2.6.1 Minimum thickness

2.6.1.1 The minimum wall thickness of standpipes and branches is to be not less than that determined by 2.7.1. In determining the wall thickness of branches, internal pressure in addition to loads by connected piping and vibrations are to be taken into account. The thickness, however, is not to be less than:

$t = 0.04\ D_o + 2.5$ [mm]

where,

t and D_o are as defined in Sec.1.

2.6.1.2 In no case need the wall thickness exceed that of the shell.

2.6.1.3 Where standpipe or branch is connected by screwing, the thickness is to be measured at the root of the threads.

2.7 Particular design requirements for boilers, superheaters, economizers, steam receivers and similar vessels

2.7.1 Boiler tubes subject to internal pressure

2.7.1.1 The minimum wall thickness of straight tubes subject to internal pressure is to be determined by the following formula:

$$t = \frac{p\,D_o}{20\sigma + p}\ [mm]$$

where,

t, p, D_o and σ are as defined in Sec.1. The thickness is in no case to be less than that shown in Table 2.7.1.

2.7.1.2 If the tube is ordered with a minus tolerance, the minimum thickness according to the formula in 2.7.1.1 is to be increased by the necessary amount. Where tubes are bent, the thickness of the thinnest part of the tube is not to be less than the calculated thickness, unless it can be demonstrated that the method of bending results in no decrease in strength at the bend as compared with the straight tube. In connection with any new method of bending, the manufacturer is to prove that this condition is satisfied.

2.7.1.3 For boiler, superheater and economizer tubes, the wall thickness required for the drum or header connection or tube stub is to be calculated as part of the tube.

2.7.1.4 The minimum thickness of downcomer tubes and pipes which form an integral part of the boiler and which are not exposed to combustion gases is to comply with the requirements for steam pipes.

Table 2.7.1 : Minimum thickness of tubes

Nominal outside diameter of tube [mm]	Minimum thickness [mm]	Minimum thickness [mm], See Note
≤ 38	1.75	2.95
> 38 ≤ 50	2.16	2.95
> 50 ≤ 70	2.40	2.95
> 70 ≤ 75	2.67	3.28
> 75 ≤ 95	3.05	3.28
> 95 ≤ 100	3.28	3.28
> 100 ≤ 125	3.50	3.50

Note : Applicable to tubes subject to internal pressure and fitted in cylindrical boilers and also for the tubes of low pressure water tube boilers having a design pressure of 1.72 [N/mm^2] and under with open feed system.

2.7.2 Boiler tubes subject to external pressure

2.7.2.1 The wall thickness of tubes with outside diameter 100 [mm] and less is not to be less than:

$$t = \frac{p D_o}{157 \sigma} \text{ [mm]}$$

where,

t, p, D_o and σ are as defined in Sec.1;

The nominal thickness of plain boiler tubes is, however, in no case to be less than given in Table 2.7.2.

Table 2.7.2 : Thickness of plain tubes under external pressure

Outside diameter of tube [mm]	Nominal thickness of tube [mm]				
	2.9	**3.2**	**3.6**	**4.0**	**4.5**
Design pressure [N/mm^2]					
51	1.1	1.5	2.1		
57	1.0	1.35	1.8	2.2	
63.5	0.9	1.25	1.6	2.1	
70	0.8	1.1	1.5	1.95	
76.1		1.0	1.35	1.75	2.1
82.5		0.9	1.25	1.6	1.95
88.9		0.85	1.15	1.5	1.8

2.7.2.2 Regarding minus tolerance and reduction in thickness due to bending, see 2.7.1.2.

2.7.2.3 Plain tubes may be seal welded at both ends, seal welded at the inlet end and expanded at the outlet end or expanded at both the ends. Where the tubes are seal welded, the tubes are to be expanded into the tube plates in addition to welding.

2.7.2.4 Where plain tubes are expanded only, the process is to be carried out with roller expanders, and the expanded portion of the tube is to be parallel through the full thickness of the tube plate. In addition to expanding, tubes may be bellmouthed or beaded at the inlet end.

2.7.2.5 Where the total number of tubes are arranged in one nest and no stay tubes are fitted, the ends of all tubes are to be welded or expanded and beaded at the inlet end, and welded or expanded at the outlet end.

2.7.2.6 The spacing of the tube holes is to be such that the minimum width, in [mm], of any ligament between the tube holes is not less than:

$$0.125\, d + 12.5 \text{ [mm]}$$

where,

d = diameter of tube hole [mm].

2.7.3 Headers

2.7.3.1 Circular headers

2.7.3.1.1 The minimum thickness of circular section headers is to be calculated in accordance with 2.1.

2.7.3.2 Rectangular section headers

2.7.3.2.1 The thickness of flat surfaces of rectangular solid forged headers is to be not less than (t + 0.75) [mm], where t = the greatest basic thickness, in [mm], derived by the use of Fig.2.7.3.1.

2.7.3.2.2 Fig.2.7.3.1 shows values of t/B corresponding to the values of term K, for parameters of A/B;

where,

A = the distance, in [mm], between centerline of the openings and the limit of the effective width, B, of the header. Where there is more than one row of holes, A is the distance to the row showing the lowest efficiency;

B = the effective width, in [mm], of the pierced surface under consideration measured between the supporting sides of the headers, minus one corner radius. The effective width is not to be taken as less than 0.9 of the full distance between the sides;

$$K = \frac{10\sigma J}{p}$$

where,

σ and p are as defined in Sec.1;

J = the efficiency of the ligaments as calculated in 2.1.2. Where there are several rows of tube holes, the lowest calculated efficiency is to be used.

2.7.3.2.3 Investigations of two stresses are necessary:

- the stress at the corner of the header, where A/B = 0 and J = 1, see Fig.2.7.3.1;
- and the stress in the ligaments between tube holes or other openings piercing the flat face of the header.

2.7.3.2.4 The corner radius is not to be less than 6.5 [mm].

2.7.3.2.5 Where the header surfaces are machined locally at hand holes, the total thickness may be reduced by a maximum of 4 [mm].

2.7.3.2.6 Except for small areas not exceeding 325 [mm^2], where a reduction of designed thickness up to 50 per cent may be permitted, the thickness derived from the use of Fig.2.7.3.1 is to be the minimum. Such minimum is in no case to be less than 7.5 [mm] or where tube holes are drilled, to be less than:-

$$t = 0.5\sqrt{d} + 6.35 \text{ [mm]}$$

where,

d = the diameter of the tube holes [mm].

2.7.3.3 Header ends

2.7.3.3.1 The shape and thickness of ends forged integrally with the bodies of headers are to be the subject of special consideration. Where sufficient experience of previous satisfactory service of headers with integrally forged ends cannot be shown, the suitability of a proposed form of end is to be proved in accordance with 1.10.

2.7.3.3.2 Ends attached by welding are to be designed as follows:-

a) Dished Ends: these are to be in accordance with 2.3

b) Flat Ends: the minimum thickness of flat end plates is to be determined by the following formula:-

$$t = d_i\sqrt{\frac{pC}{10\sigma}}$$

where,

p and σ are as defined in Sec.1;

t = minimum thickness of end plate [mm];

d_i = internal diameter of circular header or least width between walls of rectangular header [mm];

C = a constant depending on method of end attachment, see Fig.2.7.3.2

For end plates welded as shown in Fig.2.7.3.2 (a):

C = 0.19 for circular headers;

= 0.32 for rectangular headers;

For end plates welded as shown in Fig.2.7.3.2 (b) and (c):

C = 0.28 for circular headers;

= 0.40 for rectangular headers.

2.7.3.3.3 Where flat end plates are bolted to flanges attached to the ends of headers, the flanges and end plates are to be in accordance with recognized pipe flange standards.

2.7.4 Flat surfaces and flat tube plates

2.7.4.1 Stayed flat surfaces

2.7.4.1.1 Where flat end plates are flanged for connection to the shell, the inside radius of flanging is to be not less than 1.75 times the thickness of the plate, with a minimum of 38 [mm].

2.7.4.1.2 Where combustion chamber of firebox plates are flanged for connection to the wrapper plate, the inside radius of flanging is to be equal to the thickness of the plate, with a minimum of 25 [mm].

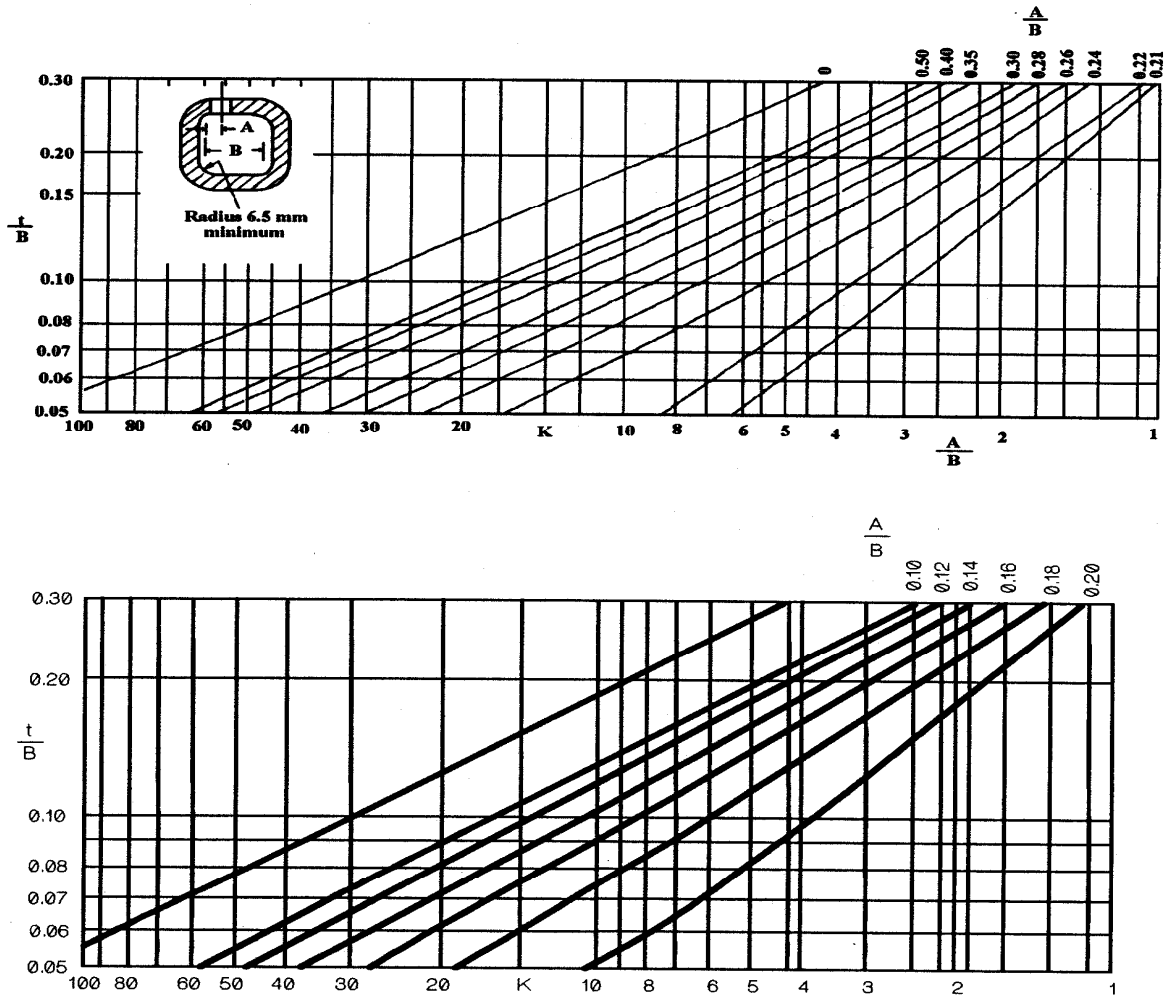


Fig. 2.7.3.1 : Rectangular section headers

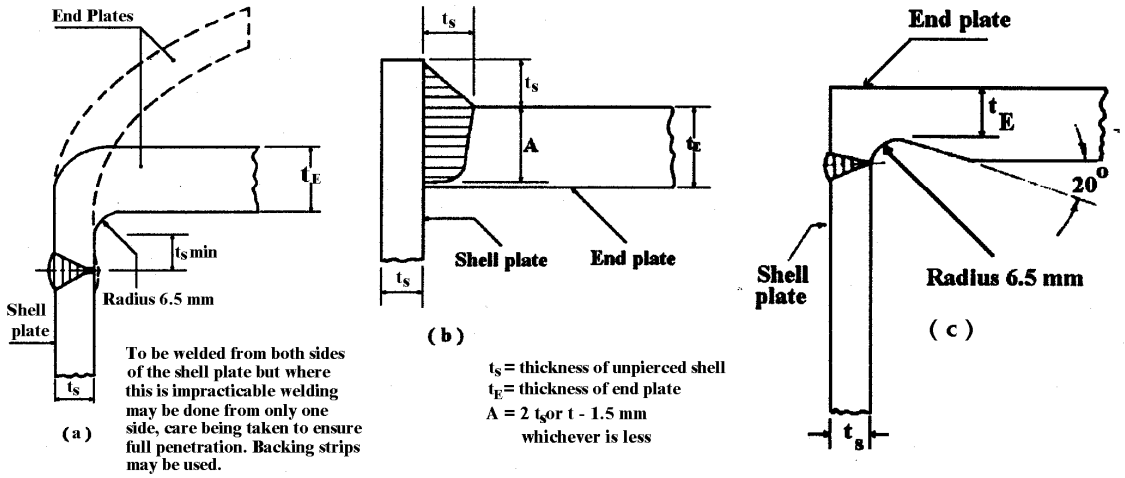


Fig. 2.7.3.2 : Typical methods of attachment of headers and closures

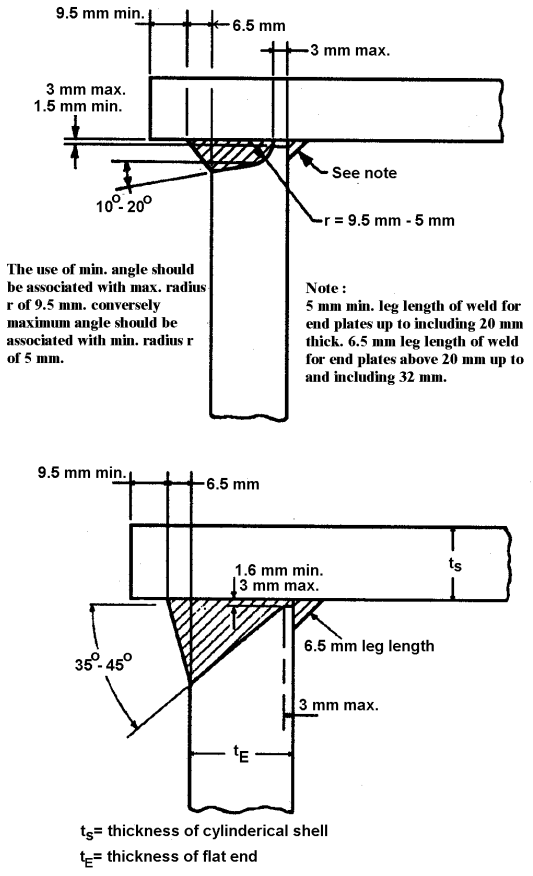


Fig.2.7.4.1 : Attachment of unflanged end plate to shell

2.7.4.1.3 Where unflanged flat plates are connected to the shell by welding, the methods of attachment are to be as shown in Fig.2.7.4.1. Similar forms of attachment may be used where unflanged combustion chamber or firebox plates are connected to the wrapper plate by welding.

2.7.4.1.4 Where the flange curvature is a point of support, this is to be taken at the commencement of curvature, or at a line distant 3.5 times the thickness of the plate from the outside of the plate, whichever is nearer to the flange.

2.7.4.1.5 Where a flat plate is welded directly to a shell or wrapper plate, the point of support is to be taken at the inside of the shell or wrapper plate.

2.7.4.1.6 The thickness, t, of those portions of flat plates supported by stays is to be determined by the following formula:

$$t = C\,d\sqrt{\frac{p}{10\sigma_1}} + 0.75 \ \text{[mm]}$$

where,

t and p are as defined in Sec.1;

σ_1 = 0.85 σ, where σ is as defined in Sec.1;

d = $\sqrt{A^2 + B^2}$ where the stays are regularly pitched

A = horizontal pitch of the stays

B = vertical pitch of the stays

where the tubes are irregularly pitched

d = diameter of the largest circle which can be drawn through three points of support without enclosing another point of support. Only two points of support may lie on one side of any diameter of the circle.

C = a constant, dependent on the method of support as detailed in 2.7.4.1.7. Where various forms of support are used, C is to be the mean of the values for the respective methods adopted.

2.7.4.1.7 All constants given in this paragraph relate to plates which are stress relieved and not exposed to flame. Where the plates are exposed to flame the thickness of the plate is to be increased by 10 per cent. The values of C in the formula in 2.7.4.1.6 is to be as follows:-

a) Where plain bar stays are strength welded in to the plates as shown in Fig.2.7.4.2.

   C = 0.39

b) Where plain bar stays pass through holes in the plates and are fitted on the outside with washers as shown in Fig.2.7.4.3.

   C = 0.35 where the diameter of the washer is 3.5 times the diameter of the stay.

   C = 0.33 where the diameter of the washer is 0.67 times the pitch of the stays.

c) Where the flat plate is flanged for attachment to the shell, flue, furnace or wrapper or, alternatively, is welded directly to the shell, flue, furnace or wrapper.

C = 0.33

d) where the support is a gusset or link stay

C = 0.39

2.7.4.1.8 Alternative methods of support will be specially considered.

2.7.4.1.9 Where a flat plate has a manhole or sighthole and the opening is strengthened by flanging, the total depth, H, of the flange, measured from the outer surface of the plate, is to be not less than:

$$H = \sqrt{t\,W}$$

where,

H = depth of flange [mm];

t = thickness of plate [mm];

W = minor axis of manhole or sight hole [mm].

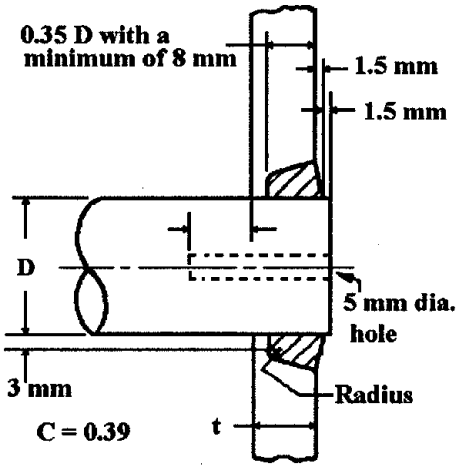


Fig. 2.7.4.2 : Attachment of firebox, combustion chamber and bar stays

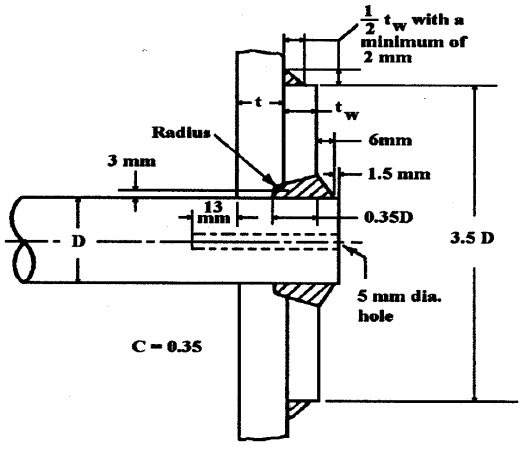


Fig. 2.7.4.3 : Attachment of bar stays

2.7.4.1.10 Where the flat top plates of combustion chambers are supported by welded-on girders, the equation in 2.7.4.1.6 is to apply as follows:

a) In the case of welded-on girders provided with waterways;

C = 0.42;

$$d = \sqrt{X^2 + Y^2}$$

where,

X = width of waterway in the girder plus the thickness of the girder [mm];

Y = pitch of girder [mm];

b) In the case of the continuously welded-on girders;

C = 0.51;

d = D;

where,

D = distance between inside faces of girders [mm].

2.7.4.2 Flat tube plates within tube nests

2.7.4.2.1 Stay tubes are to be fitted:

a) Where the total number of tubes are arranged in one nest, the area of which exceeds 0.65 [m^2] in the case of direct fired boilers, and 2.0 [m^2] in the case of waste heat boilers;

b) In all cases where the total number is arranged in more than one nest.

2.7.4.2.2 Where stay tubes are not fitted, the ends of all tubes are to be welded or expanded and beaded at the inlet end, and welded or expanded at the outlet end. For details of seal welding of plain tubes, see Fig.2.7.4.4.

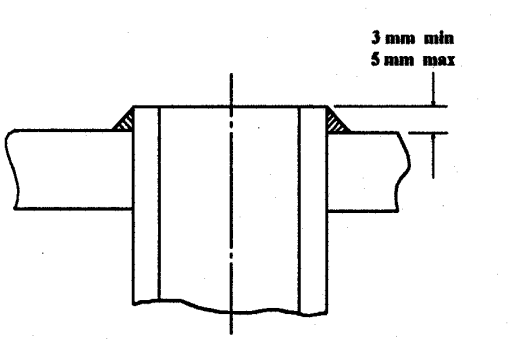


Fig. 2.7.4.4 : Detail of seal weld for plain tube

2.7.4.2.3 Where stay tubes are required to be fitted, the thickness of those parts of the tube plates within the tube nests is to be determined by the following formula:

$$t = C\,M\sqrt{\frac{p}{10\sigma_1}} + 0.75 \ \ [mm]$$

where,

t = thickness of tube plate [mm];

p = design pressure [N/mm^2];

σ_1 = 0.85 σ, where σ is as defined in Sec.1;

M = mean pitch, in [mm], of the stay tubes supporting any positions of the plate (being the sum of the four sides of any quadrilateral divided by 4);

C = 0.42 for plates not exposed to flame with stay tube secured as shown in Fig.2.7.4.5;

C = 0.46 for plates exposed to flame.

2.7.4.2.4 Where the area of the tube nest does not exceed 0.65 [m^2] in the case of direct fired boilers, or 2.0 [m^2] in the case of waste heat boilers, and stay tubes are not fitted, the thickness of the plate is to be determined by the formula in 2.7.4.2.3;

where,

M = Four times the mean pitch [mm], of the plain tubes in the nest;

C = 0.45 for plates not exposed to flame;

= 0.49 for plates exposed to flame.

2.7.4.2.5 The thickness, t, of any tube plate in the tube area is to be not less than:

a) 12.5 [mm], where the diameter of the tube hole does not exceed 50 [mm]; and

b) 14 [mm], where the diameter of the tube hole is greater than 50 [mm].

2.7.4.3 Flat tube plates between wide water spaces and around tube nests

2.7.4.3.1 The thickness of the tube plate in the wide water space between tube nests is to be determined by the following formula:

$$t = C\,d\sqrt{\frac{p}{10\sigma_1}} + 0.75 \ \ [mm]$$

where,

t = thickness of the tube plate [mm];

p = design pressure [N/mm^2];

σ_1 = 0.85 σ, where σ is as defined in Sec.1;

$$d = \sqrt{A^2 + B^2}$$

A = width [mm], of the wide water space between the tube nests (measured from the centerline of the stay tubes);

B = pitch [mm], of the stay tubes in boundary rows of wide water space;

C = 0.42, if the plates are not exposed to flame;

= 0.46, if the plates are exposed to flame.

2.7.4.3.2 The values of C and the method of securing the stay tubes are as indicated in 2.7.4.2.3.

2.7.4.3.3 Where stay tubes are irregularly pitched, d is to be taken as the diameter of the largest circle which can be drawn through any three points of support without enclosing another point of support. Where various forms of supports are used, the value of C is to be the mean of the values for the respective methods adopted.

2.7.4.3.4 For the portions of the end plates between the top rows of tubes and steam

space stays, the formula in 2.7.4.3.1 is to apply, B being taken as the distance between the centerline of the top row of tubes and the center of the bar stays or other point of support, and A being taken as $\frac{A_1 + A_2}{2}$ where A_1 is the horizontal distance between the centers of bar stays or other methods of support, and A_2 is the horizontal distance between the center of one stay tube and the center of next stay tube in the top boundary row. Where no stay tubes are fitted, A_2 is to be taken as equal to four times the horizontal pitch of the plain tubes.

2.7.4.3.5 Where no stay tubes are fitted, the support afforded by the plain tubes are not to be taken to extend beyond the line enclosing the outer surfaces of the tubes except that, between the outside of the wing row of tubes and the attachment of the end plate to shell, there may be an unsupported width equal to the flat plate margin, as given by the formula in 2.7.4.6.1.

2.7.4.4 Combustion chamber tube plates under compression

2.7.4.4.1 The thickness of combustion chamber tube plates under compression due to the pressure on the top plate, based on a compressive stress not exceeding 960 bar is to be determined by the following formula:

$$t = \frac{pWs}{1930(s - d)} \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

W = internal width of combustion chamber [mm], measured from tube plate to back chamber plate;

s = pitch of tubes [mm], measured horizontally where tubes are chain pitched, or diagonally where the tubes are staggered pitched and the diagonal pitch is less than the horizontal pitch;

d = internal diameter of plain tubes.

2.7.4.5 Girders for combustion chamber top plates

2.7.4.5.1 The formula in 2.7.4.5.2 is applicable to plate girders welded continuously to the top combustion chamber plates by means of a full penetration weld.

2.7.4.5.2 The proportion of steel plate girders supporting the tops of combustion chambers is to be determined by the following formula:

$$t = \frac{0.32\, p\, l^2\, s}{d^2\, R_{20}} \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

d = depth of girder [mm];

l = length of girder [mm], measured internally from tube plate to back chamber plate;

s = distance apart of the girders [mm];

R_{20} = specified minimum tensile strength of the girder plate [N/mm^2].

2.7.4.6 Flat plate margins

2.7.4.6.1 The width of margin, b, of a flat plate which may be regarded as being supported by the shell, furnaces or flues to which the flat plate is attached is not to exceed that determined by the following formula:

$$b = \frac{C(t - 0.75)}{\sqrt{p}} \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

b = width of margin [mm];

C = 31.0 for plates not exposed to flame;

= 28.8 for plates exposed to flame.

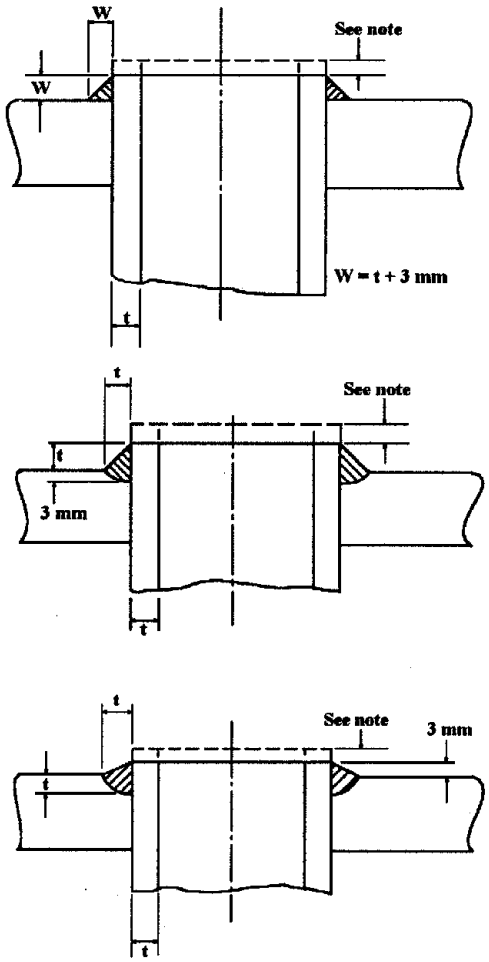
2.7.4.6.2 Where an unflanged flat plate is welded directly to shell, furnaces or flues and it is not practicable to effect the full penetration weld from both sides of the flat plate, the constant C used in formula in 2.7.4.6.1 is to be:

C = 23.6 for plates not exposed to flame;

= 22.3 for plates exposed to flame.

2.7.4.6.3 In the case of plates which are flanged, the margin is to be measured from the commencement of curvature of flanging, or from a line 3.5 times the thickness of the plate, measured from the outside of the plate, whichever is nearer to the flange.

2.7.4.6.4 Where the flat plate is not flanged for attachment to the shell, furnaces or flues, the margin is to be measured from the inside of the shell or the outside of the furnaces or flues, whichever is applicable.



Note : The ends of the tubes are to be dressed flush with the welds and then exposed to flame or comparitively high temperature. When not exposed, the ends of the tubes may extend a maximum of 9.5 [mm] beyond the weld.

Fig. 2.7.4.5 : Attachment of stay tubes

2.7.4.6.5 In no case is the diameter, D, in [mm], of the circle forming the boundary of the margin supported by the uptake of a vertical boiler to be greater than determined by the following formula:

$$D = \sqrt{\frac{345}{p} + d^2} \ [mm]$$

where,

A = cross-sectional area of the uptake tube [mm^2];

p = design pressure in bar;

d = external diameter of uptake [mm].

2.7.5 Flat plates and ends of vertical boilers

2.7.5.1 Tube plates of vertical boilers

2.7.5.1.1 Where vertical boilers have a nest or nests of horizontal tubes, so that there is direct

tension on the tube plates due to the vertical load on the boiler ends or to their acting as horizontal ties across the shell, the thickness of the tube plates in way of the outer rows of tubes are to be determined by the following formula:

$$t = \frac{pD}{5JR_{20}} + 0.75 \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

D = twice the radial distance of the center of the outer row of tube holes from the axis of the shell [mm];

R_{20} = specified minimum tensile strength of tube plate [N/mm^2];

J = efficiency of ligaments between tube holes in the outer vertical rows (expressed as a fraction);

$$= \frac{s - d}{s}$$

where,

s = vertical pitch of tubes [mm];

d = diameter of tube holes [mm].

2.7.5.1.2 Each alternate tube in the outer vertical rows of tubes is to be a stay tube. Further, the arrangement of stay tubes in the nests is to be such that the thickness of the tube plates meets the requirements of 2.7.4.2. and 2.7.4.3.

2.7.5.1.3 Where the vertical height of the tube plates between the top and bottom shelves exceeds 0.65 times the internal diameter of the boiler, the staying of the tube plates, and the scantlings of the tube plates and shell plates to which the sides of the tube plates are connected will require to be specially considered. It is recommended, however, that for this type of boiler the vertical height of the tube plates between the top and bottom shelves should not exceed 1.25 times the internal diameter of the boiler.

2.7.5.2 Horizontal shelves of tube plates forming part of the shell

2.7.5.2.1 For vertical boilers of the type referred to in2.7.5.1, in order to withstand the vertical load due to pressure on the boiler ends, the horizontal shelves of the tube plates are to be supported by gussets in accordance with the following formula:

$$C = \frac{A D_i p}{t}$$

where,

A = maximum horizontal dimension of the shelf from the inside of the shell plate to the outside of the tube plate [mm];

D_i = inside diameter of the boiler [mm];

p = design pressure, in bar;

t = thickness of the tube plate [mm].

2.7.5.2.2 For the combustion chamber tube plate the minimum number of gussets are to be:

- 1 gusset where C exceeds 255 000
- 2 gussets where C exceeds 350 000
- 3 gussets where C exceeds 420 000

2.7.5.2.3 For the smoke box tube plate the minimum number of gussets are to be:

- 1 gusset where C exceeds 255 000
- 2 gussets where C exceeds 420 000

2.7.5.2.4 The shell plates to which the sides of the tube plates are connected are to be not less than 1.5 [mm] thicker than is required by the formula applicable to shell plates with continuous circularity; and where gussets or other stays are not fitted to the shelves, the strength of the parts of the circumferential seams at the top and bottom of these plates from the outside of one tube plate to the outside of the other, is to be sufficient to withstand the whole load on the boiler end with a factor of safety of not less than 4.5 related to R_{20} (where R_{20} is the specified minimum tensile strength
of the shell plates [N/mm^2].

2.7.5.3 Dished and flanged ends for vertical boilers

2.7.5.3.1 The minimum thickness, t, of dished and flanged ends for vertical boilers which are subject to pressure on the concave side and are supported by central uptakes is to be determined by the following formula:

$$t = \frac{P R_i}{10\sigma_2} + 0.75 \text{ [mm]}$$

where,

t, p and R_i are as defined in Sec.1;

$\sigma_2 = 1.3\ \sigma$, where σ is defined in 1.6.

2.7.5.3.2 The inside radius of curvature, R_i, of the end plate is to be not greater than the external diameter of the cylinder to which it is attached.

2.7.5.3.3 The inside knuckle radius, r_i, of the arc joining the cylindrical flange to the spherical surface of the end is to be not less than four times the thickness of the end plate, and in no case less than 65 [mm].

2.7.5.3.4 The inside radius of curvature of flange to uptake is to be not less than twice the thickness of the end plate, and in no case less than 25 [mm].

2.7.5.3.5 If the dished end has a manhole, the opening is to be strengthened by flanging. The total depth, H, of the flange, measured from the outer surface of plate on the minor axis, is to be not less than:

$$H = \sqrt{t\,W}$$

where,

H = depth of flange [mm];

t = thickness of the plate [mm];

W = minor axis of manhole [mm].

2.7.5.4 Flat crowns of vertical boilers

2.7.5.4.1 The minimum thickness of flat crown plates of vertical boilers is to be determined as in 2.7.4; d and c are defined as follows:

a) Where the crown is supported by an uptake only;

   d = diameter [mm], of the largest circle which can be drawn between the connections to the shell or fire box and uptake (see 2.7.4.1.1 to 2.7.4.1.5);

   c = 0.47, if the plates are not exposed to flame;

   = 0.51, if the plates are exposed to flame;

b) Where bar stays are fitted in accordance with 2.7.4.1.6 and 2.7.4.1.7;

   d = diameter, in [mm], of the largest circle which can be drawn through three points of support without enclosing another point of support;

   c = the mean of the values for the respective points of support through which the circle passes.

2.7.5.5 Dished and flanges ends for supported vertical boiler furnaces

2.7.5.5.1 The minimum thickness, t, of dished and flanged ends for vertical boiler furnaces that are subject to pressure on the convex side and are supported by central uptake, is to be determined by the following formula:

$$t = \frac{p R_o}{10\sigma} + 0.75 \text{ [mm]}$$

where,

t, p, R_o and σ are as defined in Sec.1.

2.7.5.5.2 The inside radius of dishing and flanging are to be as required by 2.7.5.3.

2.7.5.6 Dished and flanged ends for unsupported vertical boiler furnaces:

2.7.5.6.1 The minimum thickness, t, of dished and flanged ends for vertical boiler furnaces that are subject to pressure on the convex side and are without support from stays of any kind, is to be determined by the following formula, but is in no case to be less than the thickness of the firebox:

$$t = \frac{C\,p\,R_o}{660} + 0.75 \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

R_o = outside radius of the crown plate, in [mm], (in no case Ro/t to exceed 88);

C = $\frac{2x}{x+\sigma}$ or 0.85 whichever is greater

x = specified minimum 0.2 per cent proof stress in [N/mm^2], at a temperature 90°C above the saturated steam temperature corresponding to the design pressure for carbon and carbon-manganese steel with a specified minimum tensile strength of 400 [N/mm^2];

σ = specified minimum 0.2 per cent proof stress in [N/mm^2], at a temperature 90°C above the saturated steam temperature corresponding to the design pressure for the steel actually used.

2.7.5.6.2 The inside radius of curvature, R_i, see Fig.2.3.1.2 (a), of the end plate is to be not greater than the external diameter of the cylinder to which it is attached.

2.7.5.6.3 The inside knuckle radius, r_i, see Fig.2.3.1.2 (a), of the arc joining the cylindrical flange to the spherical surface of the end is to be not less than four times the thickness of the end plate and in no case less than 65 [mm].

2.7.6 Cylindrical furnaces subject to external pressure

2.7.6.1 Maximum thickness

2.7.6.1.1 Furnaces, plain or corrugated, are not to exceed 22.5 [mm] in thickness.

2.7.6.2 Corrugated furnaces

2.7.6.2.1 The minimum thickness, t, of corrugated furnaces is to be determined by the following formula:

$$t = \frac{P D_o}{C} + 0.75 \text{ [mm]}$$

where,

p is as defined in Sec.1;

D_o = external diameter of the furnace measured at the bottom of the corrugations [mm];

t = thickness, of the furnace plate measured at the bottom of the corrugations [mm];

C = 1060 for Fox, Morision and Deighton corrugations;

= 1130 for suspension and bulb corrugations.

2.7.6.3 Plain furnaces, flue sections and combustion chamber bottoms

2.7.6.3.1 The minimum thickness, t, of plain furnaces or furnaces strengthened by stiffening rings, of flue sections and of the cylindrical bottoms of combustion chambers is to be determined by the following formulae, the greater of two thicknesses being taken:

$$t = \sqrt{\frac{p D_o (L+160)}{102400}} + 0.75 \text{ [mm]}$$

$$t = \frac{CpD_o}{1100} + \frac{L}{320} + 0.75 \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

D_o = external diameter of the furnace, flue or combustion chamber [mm];

L = length of section between the center of points of substantial support [mm];

$$C = \frac{2x}{x+\sigma}$$

where,

x and σ are as defined in 2.7.5.6.

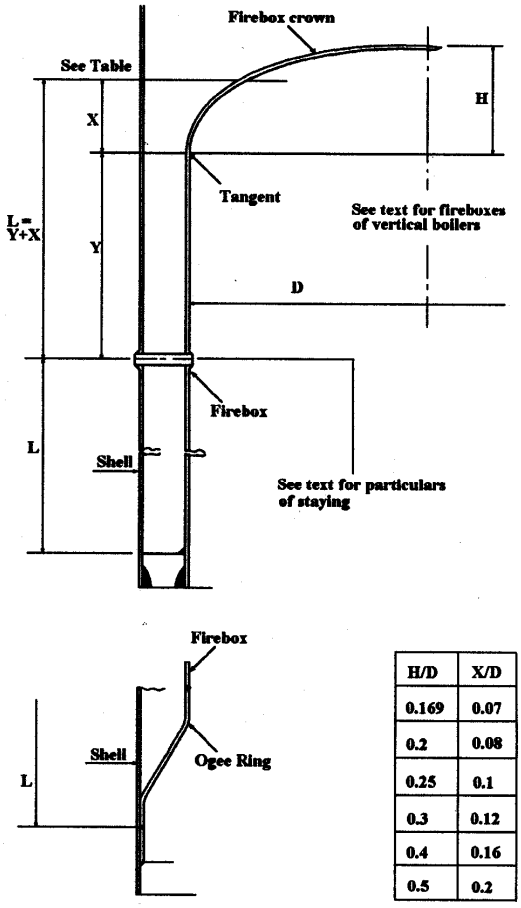
2.7.6.4 Plain furnaces of vertical boilers

2.7.6.4.1 The thickness of plain furnaces not exceeding 1700 [mm] in external diameter is to be determined by the formulae given in 2.7.6.3.1, the greater of the two thicknesses being taken;

where,

D = external diameter of the furnace, in [mm]; where the furnace is tapered, the diameter is to be the mean of that at the top and that at the bottom where it meets substantial support from flange, ring or row of stays;

L = effective length, in [mm], of the furnace between the points of substantial support as indicated in Fig.2.7.6.4.1.



H/D	X/D
0.169	0.07
0.2	0.08
0.25	0.1
0.3	0.12
0.4	0.16
0.5	0.2

Fig. 2.7.6.4.1 : Effective length, L

2.7.6.4.2 For furnaces under 760 [mm] in external diameter, the thickness is to be not less than 8 mm, and for furnaces 760 [mm] in external diameter and over, the thickness is to be not less than 9.5 [mm]. Furnaces exceeding 1700 [mm] in external diameter will be subject of special consideration.

2.7.6.4.3 A circumferential row of stays connecting the furnace to the shell will be considered to provide substantial support to the furnace, provided that:

a) the diameter of the stay is not less than 22.5 [mm] or twice the thickness of the furnace, whichever is the greater. In case of screwed stays the diameter is to be measured over the threads;

b) the pitch of the stays at the furnace does not exceed 14 times the thickness of the furnace.

2.7.6.5 Hemispherical furnaces

2.7.6.5.1 The minimum thickness, t, of unsupported hemispherical furnaces subject to pressure on convex surface is to be determined by the following formula:

$$t = \frac{CpR_o}{608} + 0.75 \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

R_o = outer radius of curvature of furnace [mm];

$C = \frac{2x}{x + \sigma}$ or 0.85, whichever is greater

where,

x and σ are as defined in 2.7.5.6.

2.7.6.5.2 In no case is the minimum thickness to exceed 22.5 mm, or the ratio $\frac{R_o}{(t - 0.75)}$ to exceed 100.

2.7.6.6 Ogee rings

2.7.6.6.1 The minimum thickness, t, of the ogee ring which connects the bottom of the furnace to the shell of a vertical boiler and sustains the whole vertical load on the furnace is to be determined by the following formula:

$$t = \sqrt{\frac{p D_i (D_i - D_o)}{9000}} + 0.75 \text{ [mm]}$$

where,

t and p are as defined in Sec.1;

D_i = inside diameter of boiler shell [mm];

D_o = outside diameter of lower part of the furnace where it joins the ogee ring [mm].

2.7.6.7 Uptakes of vertical boiler

2.7.6.7.1 The minimum thickness, t, of internal uptakes of vertical boiler is to be determined by the following formulae, the greater of the two thicknesses being taken:

$$t = \sqrt{\frac{p\,D_o\,(L + 610)}{102400}} + 4 \ [mm]$$

$$t = \frac{pD_o}{1100} + \frac{L}{320} + 4 \ [mm]$$

where,

t and p are as defined in Sec.1;

D_o = external diameter of uptake [mm];

L = length of uptake between the centers of points of substantial support [mm].

2.7.7 Stay tubes and bar stays for cylindrical boilers

2.7.7.1 Stay tubes

2.7.7.1.1 Each stay tube is to be designed to carry its due proportion of the load on the plates which it supports. No stay tube is to be less than 5 [mm] thick at its thinnest part.

2.7.7.1.2 The thickness of stay tubes welded to the tube plates is to be such that the maximum stress on the thinnest part of the tube does not exceed 69 [N/mm^2].

2.7.7.1.3 Welded-in stay tubes are to be expanded into tube plate in addition to welding.

2.7.7.1.4 Stay tubes may be welded into the boiler after stress relief, provided they are not adjacent in the same tube nest.

2.7.7.2 Combustion chamber and longitudinal bar stays.

2.7.7.2.1 The permissible stress in combustion chamber and other similar bar stays, calculated on minimum sectional area, is not to exceed 62 [N/mm^2].

2.7.7.2.2 The diameter of any stay is to be not less than 19 [mm].

2.7.7.2.3 The permissible stress in longitudinal stays, calculated on the minimum cross-sectional area, is not to exceed:

$$\frac{\text{minimum specified tensile strength [N/mm}^2\text{]}}{5.3}$$

2.7.7.2.4 In no case is the diameter of the longitudinal stays at any section to be less than 25 [mm].

2.7.7.3 Loads on stay tubes and bar stays

2.7.7.3.1 Stay tubes and bar stays are to be designed to carry the whole load due to pressure on the area to be supported.

2.7.7.3.2 For a stay tube or bar stay, the net area to be supported is to be the area [mm^2], enclosed by the lines bisecting at right angles the lines joining the stay and the adjacent points of support, less the area of any tubes or stays embraced. In the case of a stay tube or bar stay in the boundary rows, the support afforded by the flat plate margin, where applicable, should be taken into account.

2.7.7.3.3 Where there are no stay tubes in the tube nest, the area to be supported by a bar stay is to extend to the tangential boundary of the tube nest.

2.8 Access arrangements

2.8.1 General

2.8.1.1 All pressure vessels are to be so made that the internal surfaces may be examined. Wherever practicable, the openings for this purpose are to be sufficiently large for access and for cleaning the inner surfaces.

2.8.1.2 Manholes in cylindrical shells should preferably have their shorter axes arranged longitudinally, and are to be located clear of the welded joints in the shell.

2.8.1.3 Doors for manholes and sightholes are to be formed from steel plate or of other approved construction, and all jointing surfaces are to be machined.

2.8.1.4 Doors of the internal type are to be provided with spigots which have a clearance of not more than 1.5 [mm] all round, i.e. the axes of the opening are not to exceed those of the door by more than 3 [mm].

2.8.1.5 Doors of the internal type for openings not larger than 230 x 180 [mm] need be fitted with only one stud, which may be forged integral with the door. Larger doors are to be

provided with two studs screwed through the door and fitted with nuts on the inside. Alternatively, bolts may be used, screwed through the door with the heads inside. Other methods of attachment may be accepted, provided that details are submitted for consideration.

2.8.1.6 The crossbars or dogs for doors are to be of steel.

2.8.1.7 Circular flat cover plates may be fitted to raised circular manhole frames not exceeding 400 [mm] diameter, and for an approved design pressure not exceeding 18 bar. Thickness of the frames are to be not less than 19 [mm] in all parts. The circular cover plates and joint flanges for such frames are to be not less than:-

a) 25 [mm] thick for an approved design pressure: $\leq$ 8.6 bar;

b) 29 [mm] thick for an approved design pressure: > 8.6 bar $\leq$ 14.5 bar;

c) 32 [mm] thick for an approved design pressure: > 14.5 bar $\leq$ 18 bar.

2.8.1.8 The cover plates are to be provided with spigots and are to be secured by at least 16 bolts not less than 25 [mm] diameter, so that the stress in the bolts at the root of the thread due to pressure does not exceed:

- 34 [N/mm^2] for 25 [mm] diameter bolts
- 41 [N/mm^2] for 28 [mm] diameter bolts
- 45 [N/mm^2] for 32 [mm] diameter bolts

2.8.1.9 For the purpose of calculation, the pressure may be assumed to act on the whole area within the pitch circle of the bolts.

2.8.1.10 For smaller circular openings in headers and similar fittings, an approved type of plug may be used.

2.8.2 Additional requirements for boilers, superheaters, economizers, and similar vessels

2.8.2.1 In water tube boilers, manholes are to be provided in all drums of sufficient size to allow access for internal examinations and cleaning and for fitting and expanding the tubes. In the case of headers for water walls, superheaters or economizers, and of drums which are too small to permit entry, sight holes or mudholes sufficiently large and numerous for these purposes are to be provided.

2.8.2.2 Cylindrical boilers are to be provided, where possible, with means for ingress to permit examination and cleaning of the inner surfaces of plates and tubes exposed to flame. Where the boilers are too small to permit this there are to be sightholes and mudholes sufficiently large and numerous to allow the inside to be satisfactorily cleaned.

2.8.2.3 Where the cross tubes of vertical boilers are large, there is to be a sight hole in the shell opposite to one end of each tube sufficiently large to allow the tube to be examined and cleaned. These sight holes are to be in positions accessible for that purpose.

2.9 Shell type exhaust gas heated economizers that may be isolated from the steam plant system

2.9.1 Scope

2.9.1.1 This requirement is applicable to shell type exhaust gas heated economizers that are intended to be operated in a flooded condition and that may be isolated from the steam plant system and intended to be fitted on board ships contracted for construction on or after 1 January 2007.

2.9.2 Design and construction

2.9.2.1 Design and construction of shell type economizers are to pay particular attention to the welding, heat treatment and inspection arrangements at the tube plate connection to the shell.

2.9.2.2 Every shell type economizer is to be provided with a means of indicating the internal pressure and to be located so that the pressure can be easily read from a position from which the pressure may be controlled.

2.9.2.3 Every shell type economizer is to be provided with removable lagging at the circumference of the tube end plates to enable ultrasonic examination of the tube plate to shell connection.

2.9.3 Feed water

2.9.3.1 Every economizer is to be provided with arrangements for pre-heating and de-aeration, addition of water treatment or

combination thereof to control the quality of feed water to within the manufacturer's recommendations.

2.9.4 Operating instructions

2.9.4.1 The manufacturer is to provide operating instructions for each economizer and is to include reference to:

- Feed water treatment and sampling arrangements.
- Operating temperatures – exhaust gas and feed water temperatures.
- Operating pressure.
- Inspection and cleaning procedures.
- Records of maintenance and inspection.
- The need to maintain adequate water flow through the economizer under all operating conditions.
- Periodical operational checks of the safety devices to be carried out by the operating personnel and to be documented accordingly.
- Procedures for using the exhaust gas economizer in the dry condition.
- Procedures for maintenance and overhaul of safety valves.

Section 3

Fittings and Mountings

3.1 General

3.1.1 All valves over 38 [mm] diameter are to be fitted with outside screws, and the covers are to be secured by bolts or studs. All valves are to be arranged to shut with a right-hand (clockwise rotation) motion of the wheels.

3.1.2 Construction and arrangement of valves and cocks are to be such that it can be seen without difficulty whether they are open or shut, preferably by fitting a suitable indicator.

3.1.3 Where boiler mountings are secured by studs, the studs are to have a full thread holding in the plate for a length of at least one diameter. If the stud hole penetrates the whole thickness of the plate, the stud is to be screwed right through the plate and is to be fitted with a nut inside the boiler. Where bolts are used for securing mountings, they are to be screwed right through the plate with their heads inside the boiler.

3.1.4 Safety valve chests and other boiler and superheater mountings subject to pressures exceeding 10.3 bar or to steam temperatures exceeding 218°C, and boiler blow-down fittings, are to be made of steel or other approved material.

3.1.5 Adequate arrangements are to be provided for draining and venting the separate parts of each pressure vessel.

3.2 Safety valves

3.2.1 For boilers and steam generators

3.2.1.1 Boilers and steam generators are to be fitted with not less than two safety valves, each having a minimum internal diameter of 25 [mm], but those having a total heating surface of less than 50 [m^2] may have one valve not less than 50 [mm] diameter.

3.2.1.2 The valves, spindles, springs and compression screws are to be so encased and locked or sealed that the safety valves and pilot valves after setting to the working pressure, cannot be tampered with or overloaded in service. The spring casing of superheater safety valves should be ventilated, or other arrangements provided to protect the springs from excessive temperature.

3.2.1.3 Valves are to be so designed that in the event of fracture of springs they cannot lift out of their seats.

3.2.1.4 Easing gear is to be provided for lifting the safety valves and is to be operable by mechanical means at a safe position from the boiler or engine room platforms.

3.2.1.5 Safety valves are to be made with working parts having adequate clearances to ensure complete freedom of movement.

3.2.1.6 Valve seats are to be effectively secured in position. Any adjusting devices which control discharge capacity are to be positively secured so that the adjustment will not be affected when the safety valves are dismantled at surveys.

3.2.1.7 All the safety valves of each boiler and steam generator may be fitted in one chest, which is to be separate from any other valve chest and is to be connected directly to the shell by a strong and stiff neck, the passage through which is to be of cross-sectional area not less than the aggregate area of the safety valves in the chest in case of full lift valves, and one half of that area in case of other valves.

3.2.1.8 Each safety valve chest is to be drained by a pipe fitted to the lowest part and led with a downward gradient to the bilges or to a tank, clear of the boilers. No valves or cocks are to be fitted to these drain pipes. It is recommended that the bore of the drain pipes be not less than 19 [mm].

3.2.1.9

a) Where a shell type economizer is capable of being isolated from the steam plant system, it is to be provided with at least one safety valve and when it has a total heating surface of 50 [m^2] or more, it is to be provided with at least two safety valves.

b) To avoid the accumulation of condensate on the outlet side of safety valves, the discharge pipes and/or safety valve housings are to be fitted with drainage arrangements from the lowest part and led with a downward gradient to the bilges or to a tank, clear of the economizer, where it will not pose threat to either personnel or machinery.

3.2.1.10 Full details of the proposed arrangements to satisfy 3.2.1.9(a) to 3.2.1.9(b) are to be submitted for approval.

3.2.1.11 In case of watertube boilers, each saturated steam drum and each superheater are to be provided with at least one safety valve.

3.2.1.12 When a boiler is fitted with an integral superheater without any intervening stop valve, the safety valve(s) on the superheater may be considered as boiler safety valve(s). The safety valves are to be so proportioned and positioned that when relieving, sufficient steam is forced through the superheater to prevent damage to the heater.

Where a superheater, reheater or economizer is fitted with a valve between one of these and the boiler, the unit is to have appropriate safety valves. Such safety valves are not to be regarded as safety valves for the boiler.

3.2.1.13 Where it is impracticable to attach safety valves directly to the superheater, the valves are to be located as near as possible thereto and fitted to a branch piece connected to the superheater outlet pipe.

3.2.1.14 In high temperature installations the drains from safety valves are to be led to a tank or other place where high temperature steam can be safely discharged.

3.2.1.15 The designed discharge capacities of the safety valves on each boiler and steam generators are to be found from the following formulae :

Saturated steam safety valves

$$E = \frac{AC(p + 1.03)}{98.1}$$

Superheated steam safety valves

$$E = \frac{AC(p + 1.03)}{98.1}\sqrt{\frac{v_s}{v_h}}$$

where,

E = the maker's specified peak load evaporation, in [kg/hour] (including all evaporation from waterwalls, integral, or steaming economizers and other heating surfaces in direct communication with the boiler). In no case is the designed evaporation to be based on less than 29 [kg/m^2] hour of heating surface for fired boiler and 14.5 [kg/m^2] hour for exhaust gas heated boilers;

A = for ordinary, high lift or improved high lift safety valves, the aggregate area, in [mm^2], of

the orifices through the seatings of the valves, neglecting the area of guides and other obstructions;

= for full lift safety valves, the net aggregate area, in [mm^2], through the seats after deducting the area of the guides or other obstructions when fully lifted;

C = 4.8 for valves of ordinary type, having a minimum lift of D/24;

= 7.2 for valves of high lift type, having a minimum lift of D/16;

= 9.6 for valves of improved high lift type having a minimum lift of D/12;

= 19.2 for valves of full lift type having a minimum lift of D/4;

D = bore of valve seat [mm];

p = design pressure, in bar;

= specific volume of saturated steam [m^3/kg];

V_h = specific volume of superheated steam [m^3/kg].

3.2.1.16 When the discharge capacity of a safety valve of approved design has been established by type tests, carried out in the presence of the Surveyors or by an independent authority recognized by IRS, on valves representative of the range of sizes and pressures intended for marine application, consideration will be given to the use of a constant higher than C=19.2, based on 90 per cent of the measured capacity up to a maximum of C = 45 for full lift safety valves.

3.2.1.17 Where boilers are not fitted with superheater, the safety valves are to be set to open at a pressure of not more than 3 per cent above the approved design pressure, and in no case at a pressure higher than :

a) the design pressure of the steam piping; or

b) the least sum of the design pressure of the machinery connected to the boiler and the pressure drop in the piping between this machinery and the boiler.

3.2.1.18 Where boilers are fitted with superheaters, the safety valves on the superheaters are to be set to a pressure not higher than :

a) the design pressure of the steam piping; or

b) the least sum of the design pressure of the machinery connected to the boiler and the pressure drop in the piping between this machinery and the boiler.

The safety valves on the boiler drum are to be set to a pressure not less than the superheater valve setting plus 0.35 bar plus the pressure drop through the superheater, when the boiler stop valves are closed and the superheater safety valves are relieving at their rated capacity. In no case, however, are the safety valves to be set to a pressure higher than 3 per cent above the design pressure of the boiler.

3.2.1.19 Tests for accumulation of pressure are to be carried out. The boiler pressure is not to rise more than 10 per cent above the design pressure, when the boiler stop valves are closed under full firing conditions. The duration of the accumulation test is to be 15 minutes for smoke-tube boilers and 7 minutes for water-tube boilers. During this test, no more feed water is to be supplied than is necessary to maintain a safe working water level.

3.2.1.20 For ordinary, high lift and improved high lift type valves, the cross-sectional area of the waste steam pipe and passages leading to it is to be at least 10 per cent greater than the aggregate area of the safety valves as used in the formulae in 3.2.1.15. For full lift and other approved valves of high discharge capacity, the cross-sectional area of the waste steam pipe and passages is to be not less than 0.1C times the aggregate valve area.

3.2.1.21 The cross-sectional area of the main waste steam pipe is to be not less than the combined cross-sectional areas of the branch waste steam pipes leading thereto from the boiler safety valves. In case of water tube boilers, each boiler should have a separate waste steam system to atmosphere.

3.2.1.22 Waste steam pipes are to be led to the atmosphere and are to be adequately supported and provided with suitable expansion joints, bends or other means to relieve the safety valve chests of undue loading.

3.2.1.23 The scantlings of waste steam pipes and silencers are to be suitable for the maximum pressure to which the pipes may be

subjected in service, and in any case not less than 10.0 bar.

3.2.1.24 Silencers fitted to waste steam pipes are to be so designed that the clear area through the baffle plates is not less than that required for pipes.

3.2.1.25 The safety valves of each exhaust gas heated economizer/and each exhaust gas heated boiler which may be used as an economizer are to be provided with entirely separate waste steam pipes.

3.2.1.26 External drains and exhaust steam vents to atmosphere are not to be led to waste steam pipes.

3.2.1.27 It is recommended that a scale trap and means for cleaning be provided at the base of each waste steam pipe.

3.2.1.28 In installations operating with a high degree of superheat, consideration is to be given to the high temperatures which waste steam pipes, silencers and surrounding spaces will attain when the superheater safety valves are blowing during accumulation tests and in service. Adequate protection against heat effects is to be provided to Surveyor's satisfaction.

3.2.1.29 Waste steam pipes are to be led well clear of electric cables and any parts or structures sensitive to heat or likely to distort, the pipes are to be insulated where necessary.

3.2.2 For pressure vessels other than boilers and steam generators

3.2.2.1 Each pressure vessel system or each pressure vessel which can be isolated, is to be provided with arrangements to prevent overpressure.

3.2.2.2 Pressure vessels intended to operate completely filled with liquid are to have a liquid relief valve unless otherwise protected against overpressure.

3.2.2.3 Where a pressure vessel is fitted with heating coils, and fracture in the coils may increase the normal pressure of fluid in the pressure vessel, the relieving capacity of the safety device is to be sufficient to take into account the fracture of one tube.

3.2.2.4 The total capacity of the safety valves, fitted to any pressure vessel or system of pressure vessels, is to be sufficient to discharge the maximum quantity of fluid (liquid or gaseous) that can be generated or supplied without causing a rise in the pressure of more than 10 per cent above the design pressure.

3.2.2.5 The safety valves are to be set to open at a pressure of not more than 3 per cent above the design pressure.

3.2.2.6 The use of bursting discs or a combination of bursting discs and safety valves instead of safety valves is subject to special consideration in each case.

3.2.2.7 The discharge from safety valves is to be led to a place where no hazard will be created by the contents being discharged.

3.2.2.8 Where a gas is used for fire extinguishing in the machinery spaces, the discharge from safety valves or fusion plugs on air receivers is to be led to open deck outside machinery spaces, alternatively, the quantity of gas for fire fighting purposes is to be increased to such an extent that the efficiency of the fire extinguishing installation is not affected.

3.3 Stop valves

3.3.1 Boilers and steam generators

3.3.1.1 One main stop valve is to be fitted to each boiler and located as near the boiler as practicable. There are to be as few auxiliary stop valves as possible so as to avoid piercing the boiler shell than is absolutely necessary.

3.3.1.2 Where two or more boilers are connected together :

a) the steam connections for each boiler are to be provided with two stop valves with a drain connection between them. The valve nearest to the boiler is to be a non-return valve;

b) essential services are to be capable of being supplied from at least two boilers.

3.3.2 Pressure vessels other than boilers and steam generators

3.3.2.1 Each pressure vessel or system is to be fitted with a stop valve situated as close as possible to the shell.

3.4 Water level indicators

3.4.1 Every boiler and steam-heated steam generator is to be fitted with at least two independent means of indicating the water level in it, one of which is to be a gauge glass. The other means is to be either an additional gauge glass or an approved equivalent device.

3.4.2 Where a steam and water drum exceeding 4 [m] in length is fitted athwartships, two gauge glasses are to be fitted in suitable positions, one near each end of the drum.

3.4.3 The water gauges are to be readily accessible and placed so that the water level is clearly visible. The lowest visible part of the glass of the water gauge is to be situated at the lowest safe working water level.

a) In the case of water-tube boilers, the water gauge glasses are to be located so that water is just showing when the water level in the steam drum is just above the top row of tubes when the boiler is cold (generally about 25 [mm] above).

b) In water-tube boilers, the tubes of which are not entirely drowned when cold, the water gauge glasses are to be placed, to the Surveyor's satisfaction, in the positions which have been found by experience to indicate satisfactorily that the water content is sufficient for safety when the boiler is worked under all service conditions.

c) The level of the highest part of the effective heating surfaces, e.g. combustion chamber top of a horizontal boiler and the furnace crown of a vertical boiler, is to be clearly marked in a position adjacent to the water gauge glass.

3.4.4 The length of the water gauge glasses is to be sufficient for verifying the water levels in case of alarm and oil supply cut-off.

3.4.5 The cocks of all water gauges are to be operable from positions free from danger in the event of the glass breaking.

3.4.6 If the water gauges are not fitted directly to the shell of the boiler, but to stand pillars or columns, it is desirable that these pillars or columns should be bolted directly to the shell of the boiler. If they are connected to the boiler by means of pipes, the pipes are to be fitted with terminal cocks, not valves, secured direct to the boiler shell.

a) For boilers exceeding 3 [m] in diameter, the pillars are to be not less than 63 mm, and the connecting pipes not less than 38 [mm] internal diameter. For boilers exceeding 2.3 [m] but not exceeding 3 [m] in diameter, the pillars are to be not less than 50 [mm] and the pipes not less than 32 [mm] internal diameter.

b) The upper ends of the connecting pipes are to be so arranged that there is no pocket or bend where an accumulation of water from the condensation of the steam can lodge. They should not pass through the uptake if they can be otherwise arranged. If, however, this condition cannot be complied with, they may pass through it by means of a passage at least 50 [mm] clear of the pipe all round, open for ventilation.

3.5 Low water level fuel shut-off and alarm

3.5.1 Each fired boiler is to be fitted with a system of water level detection which is to be independent of any other mounting on the boiler and which will operate audible and visible alarms and shut-off automatically the fuel supply to the burners when the water level falls to a predetermined low level.

3.6 Feed check valves

3.6.1 Each main boiler or auxiliary boiler for essential services and each steam-heated steam generator is to have two independent feed pipes; each feed pipe is to be provided with two feed check valves. Boiler heated exclusively by exhaust gases may have only one feed pipe and with one feed check valve.

3.6.2 The feed check valves are to be attached, wherever practicable, direct to the boiler, but where the arrangements necessitate the use of standpipes between the boiler and the check valves, these pipes are to be of steel or other approved material. For boiler feed water systems, See Ch.3, Sec.6.

3.6.3 In boilers, where economizer forms an integral part of the boiler, and where the arrangements necessitate the use of a common inlet pipe on the economizer for both main and auxiliary feed systems, this pipe is to be as short as practicable, and the arrangement of check valves is to be such that either feed line can be effectively isolated without interruption of the feed water supply to the boiler.

3.6.4 The feed check valves are to be fitted with efficient gearing, whereby they can be satisfactorily worked from the stokehold floor, or other convenient position.

3.6.5 Standpipes on boiler, for feed inlet, are to be designed with an internal pipe to prevent direct contact between the feed pipe and the boiler shell or end plates with the object of minimizing thermal stresses in these plates. Similar arrangements are to be provided for desuperheater and other connections, where significant temperature difference occur in service.

3.7 Pressure gauges

3.7.1 Each boiler and superheater is to be provided with a steam-pressure gauge.

3.7.2 The pressure gauges are to be placed where they are easily read. The highest permissible working pressure is to be marked off on pressure gauge in red.

3.8 Blow-down and scum valves

3.8.1 Each boiler is to be fitted with at least one blow-down valve secured to the lower part of the boiler.

3.8.2 Where it is not practicable to attach the blow-down valve direct to water-tube boilers, the valve may be placed immediately outside the boiler casing with a steel pipe of substantial thickness fitted between the boiler and valve. The pipe and valve are to be suitably supported and any pipe which may be exposed to direct heat from the furnace is to be adequately protected.

3.8.3 The blow-down valve and its connections to the sea need not be more than 38 [mm], and is to be not less than 19 [mm] internal diameter. For cylindrical boilers the size of the valve may be generally 0.0085 times the diameter of the boiler.

3.8.4 Blow-down valves and scum valves (where the latter are fitted) of two or more boiler may be connected to one common discharge, but where thus arranged there are to be screw-down non- return valves fitted for each boiler to prevent the possibility of the contents of one boiler passing to another.

3.8.5 The blow-down cock or valve on the ship's side is to be fitted in a readily accessible position, above the level of the floor plates, and is to be arranged such that it can be readily seen whether it is open or shut. The cock handle is not to be removable unless the cock is shut, and if a valve is fitted, the wheel is to be fixed to the spindle.

3.9 Salinometer valve or cock

3.9.1 Each boiler is to be provided with a salinometer valve or cock secured direct to the boiler in a convenient position. The valve or cock is not to be on the water gauge standpipe.

Section 4

Hydraulic Tests

4.1 Boilers (fired and exhaust gas heated), economizers, superheaters, reheaters, steam receivers, steam heated generators and similar vessels

4.1.1 General

4.1.1.1 Boilers and pressure vessels, together with their components are to withstand the following hydraulic tests without any sign of weakness or defect.

4.1.1.2 Having regard to the variation in the types and design of boilers, the hydraulic test may be carried out by either of the methods indicated below:

a) Boilers are to be tested on completion to a pressure 1.5 times the approved design pressure; or

b) Where construction permits, all components of the boiler are to be tested on completion of the work including heat treatment to 1.5 times the design pressure. In the case of components such as drums or headers, which are to be drilled for tube holes, the test may be

made before drilling the tube holes, but is to be after the attachment of standpipes, stubs and similar fittings and also after heat treatment has been carried out. Where all the components have been tested as above, each completed boiler after assembly is to be tested to 1.25 times the design pressure.

4.1.2 Mountings

4.1.2.1 All boiler mountings are to be subjected to a hydraulic test of twice the approved design pressure with the exception of feed check valves and other mountings connected to the main feed system which are to be tested to 2.5 times the approved boiler design pressure, or twice the maximum pressure which can be developed in the feed line in normal service, whichever is greater.

4.2 Other pressure vessels

4.2.1 General

4.2.1.1 Pressure vessels are to be tested on completion to a pressure, p_t, determined by the following formula, without showing signs of weakness or defect :-

$$p_t = 1.3 \frac{\sigma_{50}}{\sigma T} \frac{t}{t - 0.75} p$$

but in no case is to exceed 1.5 $\frac{t}{t - 0.75} p$

where,

p_t = test pressure, in bar

p = design pressure, in bar

t = nominal thickness of shell as indicated on the plan [mm]

σ_{50} = allowable stress at 50°C [N/mm^2]

σT = allowable stress at design temperature [N/mm^2].

4.2.2 Mountings

4.2.2.1 Mountings are to be subjected to a hydraulic test of twice the approved design pressure.

End of Chapter

Chapter 6

Steering Gear

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Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter apply to the design and construction of steering gear.

1.1.2 Whilst the requirements of this Chapter are considered to meet the relevant regulations of the Safety of Life at Sea 1974 and applicable amendments, attention should be given to any relevant statutory requirements of the National Authority of the country in which the ship is to be registered.

1.1.3 Consideration will be given to other cases, or to other arrangements which are equivalent to those required by the Rules.

1.2 Definitions

1.2.1 **Steering System** means ship's directional control system, including main steering gear, auxiliary steering gear, steering gear control system and rudder, if any.

1.2.2 **Steering gear control systems** means the equipment by which orders are transmitted from the navigating bridge to the steering gear power units. Steering gear control systems comprise transmitters, receivers, hydraulic control pumps and their associated motors, motor controllers, piping and cables. Steering gear control system is also understood to cover "the equipment required to control the steering gear power actuating system".

1.2.3 **Main steering gear** means the machinery, rudder actuator(s), the steering gear power units, if any, and ancillary equipment and the means of applying torque to the rudder stock (e.g. tiller or quadrant) necessary for effecting movement of the rudder for the purpose of steering the ship under normal service conditions.

1.2.4 **Steering gear power unit** means :

a) in the case of electric steering gear, an electric motor and its associated electrical equipment;

b) in the case of electrohydraulic steering gear, an electric motor and its associated electrical equipment and connected pump;

c) in the case of other hydraulic steering gear, a driving engine and connected pump.

IR1.2.4 For the purposes of non-traditional steering arrangements such as azimuthing propellers, waterjets, etc, the above definitions may be applied as appropriate. The steering gear power unit relates to the equipment for changing the direction of thrust and does not include those for generating the thrust.

1.2.5 **Auxiliary steering gear** means the equipment other than any part of the main steering gear necessary to steer the ship in the event of failure of the main steering gear but not including the tiller, quadrant or components serving the same purpose.

1.2.6 **Power actuating system** means the hydraulic equipment provided for supplying power to turn the rudder stock, comprising a steering gear power unit or units, together with associated pipes and fittings, and a rudder actuator. The power actuating systems may share common mechanical components, i.e. tiller, quadrant and rudder stock, or components serving the same purpose.

1.2.7 **Maximum ahead service speed** means the greatest speed which the ship is designed to maintain in service at sea at her deepest seagoing draught.

1.2.8 **Rudder actuator** means the compo-nent(s) which converts directly hydraulic pressure into mechanical action to move the rudder.

1.2.9 **Maximum working pressure** means the expected pressure in the system when steering gear is operated to comply with 2.1.2(b).

1.2.10 **Declared steering angle limits** in the case of non-traditional steering arrangements such as azimuthing propellers, waterjets, etc. are the operational limits in terms of maximum steering angle, or equivalent, according to manufacturers guidelines for safe operation, also taking into account the vessel’s speed or propeller torque/speed or other limitation. The "declared steering angle limits" are to be declared by the directional control system manufacturer for each ship specific non-traditional steering mean. Ship's manoeuvrability tests, such as those in the standards for ship maneuverability (Resolution MSC.137(76)) are to be carried out with steering angles not exceeding the declared steering angle limits.

1.2.11 **Hydraulic locking** means all situations where two hydraulic systems (usually identical) oppose each other in such a way that it may lead to loss of steering. It can either be caused by pressure in the two hydraulic systems working against each other or by hydraulic “by-pass” meaning that the systems puncture each other and cause pressure drop on both sides or make it impossible to build up pressure.

1.3 Installation

1.3.1 The steering gear is to be secured to the seating by fitted bolts, and suitable chocking arrangements are to be provided. The seating is to be of substantial construction.

1.4 Steering gear compartment

1.4.1 The steering gear compartment is to be :

a) Readily accessible and, as far as practicable separated from machinery spaces;

b) Provided with suitable arrangements to ensure working access to steering gear machinery and controls. These arrangements are to include handrails and gratings or other non-slip surfaces to ensure suitable working conditions in the event of hydraulic fluid leakage.

1.5 Plans

1.5.1 Before starting construction, all relevant plans and specifications are to be submitted for approval in triplicate.

1.5.2 These plans should give details of scantlings and materials of the steering gear together with proposed rated torque and all relief valve settings.

1.6 Materials

1.6.1 All the steering gear components the rudder stock and/or components of other steering arrangements for directional control are to be of sound and reliable construction to the Surveyor's satisfaction of IRS.

1.6.2 All components transmitting mechanical forces to the rudder stock are to be tested according to the requirements of Pt.2.

1.6.3 Ram; cylinders; pressure housing of rotary vane type actuators; hydraulic power piping; valves; flanges and fittings; and all steering gear components transmitting mechanical forces to the rudder stock (such as tillers, quadrants, or similar components) are to be of steel or other approved ductile material, duly tested in accordance with the requirements of Pt.2. In general, such material is not to have an elongation of less than 12 per cent nor a tensile strength in excess of 650 [N/mm^2]. The use of ductile (nodular) iron castings will be acceptable provided the material has an elongation of not less than 12 per cent.

1.7 Rudder, rudder stock, vanes, tiller and quadrant

1.7.1 For the requirements regarding rudder, rudder stock, See Pt.3, Ch.14.

1.7.2 All components transmitting mechanical forces to the rudder stock are to have a strength of at least equivalent to the rudder stock in way of the tiller. The resultant equivalent stress, σ_e, caused by the transmission of rudder torque, Qr, in tillers, vanes and other power transmitting components is not to exceed 118/k [N/mm^2], i.e.

$$\sigma_e = \sqrt{\sigma^2 + 3\tau^2} \leq 118/k \ \ [\mathrm{N/mm^2}]$$

where,

σ_e = The combined equivalent stress, [N/mm^2]

σ = The bending stress, [N/mm^2]

τ= The torsional shear stress, [N/mm^2]

k = Material factor for the component under consideration as defined in Pt.3, Ch.14, Sec.1;

Q_r = The rudder torque calculated as per Pt.3, Ch.14, Sec.3.2 [N-m].

1.7.3 The section modulus 'Z' [cm^3] and the sectional area 'A' [cm^2] of the tiller arms is not to be less than the following:

$$Z = 0.012\, Q_r \left(1 - \frac{x}{R}\right) k \ [\mathrm{cm^3}]$$

$$A = 2.0 \frac{Q_r}{R} \ k \times 10^{-4} \ [\mathrm{cm^2}]$$

where,

R = The distance [m] from the point of application of the effort on the tiller to the centre of rudder stock; and

x = The distance [m] from the section under consideration to the centre of the rudder stock.

1.7.4 The boss may be fitted on the rudder stock by shrinking with/without key or may be of the split type. The ratio between the mean of outer and inner diameters of the boss is to be not less than 1.75 and the height of the boss is not to be less than the inner diameter of the boss.

1.7.5 Co-efficient of friction for shrink fitting is to be taken as specified in Pt.3, Ch.14, Sec.6.3.

1.7.6 In case of split type boss, the total number of joining bolts is to be at least 4. The distance of the centre of the bolts from the centre of the rudder stock is generally to be 1.15du and the thickness of the coupling flange is to be at least 1.1 times the required bolt diameter. The thickness of shim to be fitted between two halves before machining is to be 0.0015du. The diameter of the coupling bolt, d_b is to be not less than :

$$d_b = 0.60 \frac{du\sqrt{k}}{\sqrt{(nks)}} \ [\mathrm{mm}]$$

where,

du = The rudder stock diameter [mm] in way of the tiller calculated in accordance with Pt.3, Ch.14, Sec.5;

n = Total number of joining bolts.

k = Material factor for the bolts.

ks = Material factor for the rudder stock material.

1.7.7 The dimensions of the key are to comply with the requirements of Pt.3, Ch.14, Sec.6.2.

1.7.8 Where higher tensile bolts are used on bolted tillers and quadrants, the yield and ultimate tensile stresses of the bolt material are to be stated on the plans submitted for approval, together with full details of the methods to be adopted to obtain the required setting-up stress. Where patent nuts or systems are used, the manufacturer's instructions for assembly should be adhered to.

1.7.9 In bow rudders having a vertical locking pin operated from the deck above, positive means are to be provided to ensure that the pin can be lowered only when the rudder is exactly central. In addition, an indicator is to be fitted at the deck to show when the rudder is exactly central.

1.8 Mechanical steering gear

1.8.1 Steel-wire rope, chain and other mechanical systems, when these are used for rudder stock diameters of 120 [mm] and less but excluding allowance for strengthening in ice, will be specially considered. In general, the breaking strength of rods/chains etc. is not to be less than:

$$\text{Breaking strength} = 6\frac{Qr}{R}\ \text{N}$$

Where R is defined in 1.7.3.

Section 2

Performance

2.1 Requirements for traditional type of steering gears

2.1.1 Unless the main steering gear comprises of two or more identical power units, in accordance with 2.1.4 or 8.1.1, every ship is to be provided with a main steering gear and an auxiliary steering gear in accordance with the requirements of the Rules. The main steering gear and the auxiliary steering gear are to be so arranged that the failure of one of them will not render the other one inoperative.

The rudder stock diameters mentioned in 2.1.2c), 2.1.3c) and 6.1.1 are to be taken as having been calculated for mild steel with yield strength of 235 [N/mm^2].

2.1.2 The main steering gear and rudder stock are to be :

a) Of adequate strength and capable of steering the ship at maximum ahead speed which is to be demonstrated in accordance with 7.2;

b) Capable of putting the rudder over from 35° on one side to 35° on the other side with the ship at its deepest seagoing draught and running ahead at maximum ahead service speed and under the same conditions, from 35° on either side to 30° on the other side in not more than 28 seconds;
where it is impractical to demonstrate compliance with this requirement during sea trials with the ship at its deepest seagoing draught and running ahead at the speed corresponding to the number of maximum continuous revolutions of the main engine and maximum design pitch, ships regardless of date of construction may demonstrate compliance with this requirement by one of the following methods:

1 during sea trials the ship is at even keel and the rudder fully submerged whilst running ahead at the speed corresponding to the number of maximum continuous revolutions of the main engine and maximum design pitch or

2 where full rudder immersion during sea trials cannot be achieved, an appropriate ahead speed to be calculated using the submerged rudder blade area in the proposed sea trial loading condition. The calculated ahead speed is to result in a force and torque applied to the main steering gear which is at least as great as if it was being tested with the ship at its deepest seagoing draught and running ahead at the speed corresponding to the number of maximum continuous revolutions of the main engine and maximum design pitch or

3 the rudder force and torque at the sea trial loading condition have been reliably predicted and extrapolated to the full load condition. The speed of the ship is to correspond to the number of maximum continuous revolutions of the main engine and maximum design pitch of the propeller.

c) Operated by power where necessary to meet the requirements of (b) and in any case when the Rules, excluding strengthening for navigation in ice, require a rudder stock over 120 [mm] diameter in way of the tiller (See 2.1.1); and

d) So designed that they will not be damaged at maximum astern speed; however, this design requirement need not be proved by trials at maximum astern speed and maximum rudder angle.

IR 2.1.2.1 In case of 2.1.2 b).2, it is to be ensured that the equivalent stress in the rudder stock due to bending and torque is not to exceed the acceptance limits given in Pt 3, Ch 14, Sec 5.

2.1.3 The auxiliary steering gear is to be :

a) Of adequate strength and capable of steering the ship at navigable speed and of being brought as fast as possible into action in an emergency:

b) Capable of putting the rudder over from 15 on one side to 15 on the other side in not more than 60 seconds with the ship at its deepest seagoing draught and running ahead at one half of the maximum ahead service speed or 7 knots, whichever is the greater;
Where it is impractical to demonstrate compliance with this requirement during sea trials with the ship at its deepest seagoing draught and running ahead at one half of the speed corresponding to the number of maximum continuous revolutions of the main engine and maximum design pitch or 7 [knots], whichever is greater, ships regardless of date of construction, including those constructed before 1 January 2009, may demonstrate compliance with this requirement by one of the following methods:

1 during sea trials the ship is at even keel and the rudder fully submerged whilst running ahead at one half of the speed corresponding to the number of maximum continuous revolutions of the main engine and maximum design pitch or 7 [knots], whichever is greater; or

2 where full rudder immersion during sea trials cannot be achieved, an appropriate ahead speed is to be calculated using the submerged rudder blade area in the proposed sea trial loading condition. The calculated ahead speed is to result in a force and torque applied to the auxiliary steering gear which is at least as great as if it was being tested with the ship at its deepest seagoing draught and running ahead at one half of the speed corresponding to the number of maximum continuous revolutions of the main engine and maximum design pitch or 7 [knots], whichever is greater; or

3 the rudder force and torque at the sea trial loading condition have been reliably predicted and extrapolated to the full load condition; and

c) Operated by power where necessary to meet the requirements of (b) and in any case when the Rules, excluding strengthening for navigation in ice, require a rudder stock over 230 [mm] diameter in way of the tiller. (See 2.1.1).

IR 2.1.3.1 In case of 2.1.3 b).2, it is to be ensured that the equivalent stress in the rudder stock due to bending and torque is not to exceed the acceptance limits given in Pt 3, Ch 14, Sec 5.

2.1.4 Where the main steering gear comprises two or more identical power units, an auxiliary steering gear need not be fitted, provided that :

a) In a passenger ship, the main steering gear is capable of operating the rudder as required by 2.1.2 (b) while any one of the power units is out of operation;

b) In a cargo ship, the main steering gear is capable of operating the rudder as required by 2.1.2 (b) while operating with all power units;

c) The main steering gear is arranged so that after a single failure in its piping system or in one of the power units the defect can be isolated so that steering capability is regained.

2.1.5 Main and auxiliary steering gear power units are to be:

a) Arranged to re-start automatically when power is restored after a power failure;

b) Capable of being brought into operation from a position on the navigating bridge. In the event of a power failure to any one of the steering gear power units, an audible and visual alarm is to be given on the navigating bridge;

c) Arranged so that transfer between units can be readily effected.

2.1.6 Steering gear, other than of the hydraulic type, will be accepted provided the standards are considered equivalent to the requirements of this Section.

2.1.7 Manually operated gears are only acceptable when the operation does not require an effort exceeding 16 [kgf] under normal conditions.

2.2 Requirements for non-traditional (azimuthing propellers, waterjets, etc.) type of steering gears:

2.2.1 For a ship fitted with a single steering-propulsion unit, the requirement in 2.1.1 is considered satisfied if the steering gear is provided with two or more identical power actuating systems and is in compliance with 2.2.4. A detailed risk assessment is to be submitted in order to demonstrate that in the case of any single failure in the steering gear, control system and power supply is maintained.

For a ship fitted with multiple steering - propulsion units , such as but not limited to azimuthing propulsors or water jet propulsion systems each of the steering-propulsion units is to be provided with a main steering gear and an auxiliary steering gear or with two or more identical power actuating systems in compliance with 2.2.5. The main steering gear and the auxiliary steering gear are to be so arranged that the failure of one of them will not render the other one inoperative.

2.2.2 The main steering arrangements for ship directional control are to be:

a) of adequate strength and capable of steering the ship at maximum ahead speed which is to be demonstrated in accordance with 7.2 (Also refer Ch.4, Sec.9);

b) capable of changing direction of the steering- propulsion unit from one side to the other at declared steering angle limits at an average turning speed of not less than 2.3°/s with the ship running ahead at maximum ahead service speed;

c) for all ships, operated by power; and

d) so designed that they will not be damaged at maximum astern speed (Also refer Ch.4, Sec.9), this design requirement need not be proved by trials at maximum astern speed and declared steering angle limits.

Ship maneuverability tests, such as according to Resolution MSC.137(76) on Standards for ship maneuverability, are to be carried out with steering angles not exceeding the declared steering angle limits.

2.2.3 The auxiliary steering arrangements for ship directional control are to be:

a) of adequate strength and capable of steering the ship at navigable speed and of being brought speedily into action in an emergency:

b) capable of changing direction of the ship's directional control system from one side to the other at declared steering angle limits at an average turning speed, of not less than 0.5°/s; with the ship running ahead at one half of the maximum ahead service speed or 7 knots, whichever is the greater; and

c) for all ships, operated by power where necessary to meet the requirements of 2.2.3.b and in any ship having propulsive power of more than 2,500 kW per thruster unit.

Ship maneuverability tests, such as according to Resolution MSC.137(76) on Standards for ship maneuverability, are to be carried out with steering angles not exceeding the declared steering angle limits.

2.2.4 For a ship fitted with a single steering-propulsion unit where the main steering gear comprises two or more identical power units and two or more identical steering actuators, an auxiliary steering gear need not be fitted, provided that the steering gear:

a) in a passenger ship, is capable of satisfying the requirements in 2.2.2 while any one of the power units is out of operation;

b) in a cargo ship, is capable of satisfying the requirements in 2.2.2 while operating with all power units; and

c) is arranged so that after a single failure in its piping system or in one of the power units, steering capability can be maintained or speedily regained.

2.2.5 For a ship fitted with multiple steering-propulsion units, where each main steering system comprises two or more identical power actuating systems, an auxiliary steering gear need not be fitted, provided that each steering gear :

a) in a passenger ship, is capable of satisfying the requirements in 2.2.2 while any one of the steering gear power actuating systems is out of operation;

b) in a cargo ship, is capable of satisfying the requirements in 2.2.2 while operating with all steering gear power actuating systems;

c) is arranged so that after a single failure in its piping or in one of the power actuating systems, steering capability can be maintained or speedily regained

The above capacity requirements apply regardless whether the steering systems are arranged with dedicated power units.

2.2.6 Main and auxiliary steering gear power units are to be:

a) Arranged to re-start automatically when power is restored after a power failure;

b) Capable of being brought into operation from a position on the navigating bridge. In the event of a power failure to any one of the steering gear power units, an audible and visual alarm is to be given on the navigating bridge;

c) Arranged so that transfer between units can be readily effected.

2.3 Other requirements

2.3.1 Where the steering gear is so arranged that more than one power system can be simultaneously operated, the risk of hydraulic locking caused by a single failure is to be considered.

2.3.2 A means of communication is to be provided between the navigating bridge and the steering gear compartment.

2.3.3 Power-operated steering gears are to be provided with positive arrangements, such as limit switches, for stopping the gear before the rudder stops are reached. These arrangements are to be synchronized with the gear itself and not with the steering gear control.

Section 3

Construction and Design

3.1 General

3.1.1 Rudder actuators other than those covered by 8.3 and Sec.9 are to be designed in accordance with the relevant requirement of Ch.5 for Class 1 pressure vessels (notwithstanding any exemptions for hydraulic cylinders).

3.1.2 Accumulators, if fitted, are to comply with the requirements of Ch.5.

3.1.3 The welding details and welding procedures are to be approved. All welded joints within the pressure boundary of a rudder actuator or connecting parts transmitting mechanical loads are to be full penetration type or of equivalent strength.

3.1.4 The construction is to be such as to minimize local concentration of stress.

3.1.5 The design pressure for calculations to determine the scantlings of piping and other steering gear components subjected to internal hydraulic pressure is to be at least 1.25 times the maximum working pressure to be expected under the operational conditions specified in 2.1.2 (b) taking into account any pressure which may exist in the lower pressure side of the system. Fatigue criteria may be applied for the design of piping and components, taking into account pulsating pressures due to dynamic loads (See Sec.9).

3.1.6 The permissible primary general membrane stress is not to exceed the lower of the following values

$$\frac{\sigma B}{A} \text{ or } \frac{\sigma y}{B}$$

where,

σB = specified minimum tensile strength of material at ambient temperature

σY = specified minimum yield stress or 0.2 per cent proof stress of the material, at ambient temperature

A and B are given by the following table :

	Wrought Steel	**Cast Steel**	**Nodular Cast Iron**
A	3.5	4	5
B	1.7	2	3

3.2 Components

3.2.1 Special consideration is to be given to the suitability of any essential component which is not duplicated. Any such essential component is to, where appropriate, utilize anti-friction bearings such as ball bearings, roller bearings or sleeve bearings which are to be permanently lubricated or provided with lubrication fittings.

3.2.2 All steering gear components transmitting mechanical forces to the rudder stock, which are not protected against overload by structural rudder stops or mechanical buffers, are to have a strength at least equivalent to that of the rudder stock in way of the tiller.

3.2.3 Oil seals between non-moving parts, forming part of the external pressure boundary, should be of the metal type or of an equivalent type.

3.2.4 Oil seals between moving parts, forming part of the external pressure boundary, are to be duplicated, so that the failure of one seal does not render the actuator inoperative. Alternative arrangements providing equivalent protection against leakage may be accepted.

3.2.5 Piping, joints, valves, flanges and other fittings are to comply with the requirements of Ch.2 for Class I piping systems components. The design pressure is to be in accordance with 3.1.5.

3.2.6 Hydraulic power operated steering gears are to be provided with the following :-

a) Arrangements to maintain the cleanliness of the hydraulic fluid taking into consideration the type and design of the hydraulic system;

b) A fixed storage tank having sufficient capacity to recharge at least one power actuating system including the reservoir, where the main steering gear is required to be power operated. The storage tank is to be permanently connected by piping in such a manner that the hydraulic systems can be readily recharged from a position within the steering gear compartment and provided with a contents gauge.

3.3 Valve and relief valve arrangement

3.3.1 For all vessels with non-duplicated actuators, isolating valves are to be fitted at the connection of pipes to the actuators, and are to be directly fitted to the actuator. Arrangements for bleeding air from the hydraulic system are to be provided, where necessary.

3.3.2 Relief valves are to be fitted to any part of the hydraulic system which can be isolated and in which pressure can be generated from the power source or from external forces. The setting of the relief valves is not to exceed the design pressure. The valves are to be of adequate size and so arranged as to avoid an undue rise in pressure above the design pressure.

3.3.3 Relief valves, for protecting any part of the hydraulic system which can be isolated, required by 3.3.2 are to comply with the following:

a) The setting pressure is not to be less than 1.25 times the maximum working pressure.

b) The minimum discharge capacity of the relief valve(s) is not to be less than 110 per cent of the total capacity of the pumps which can deliver through it (them). Under such conditions the rise in pressure is not to exceed 10 per cent of the setting pressure. In this regard, due consideration is to be given to extreme foreseen ambient conditions in respect of oil viscosity.

3.4 Flexible hoses

3.4.1 Hose assemblies approved by IRS may be installed between two points where flexibility is required but are not subjected to torsional deflection (twisting) under normal operating conditions. In general, the hose should be limited to the length necessary to provide for flexibility and for proper operation of machinery. (See also Ch.2).

3.4.2 Hoses should be high pressure hydraulic hoses according to recognized standards and suitable for the fluids, pressures, temperatures and ambient conditions in question.

3.4.3 Burst pressure of hoses is not to be less than four times the design pressure.

Section 4

Steering Control Systems

4.1 General

4.1.1 Steering gear control is to be provided:

a) For the main steering gear, both on the navigating bridge and in the steering gear compartment;

b) Where the main steering gear is arranged according to 2.1.4, by two independent control systems, both operable from the navigating bridge. This does not require duplication of the steering wheel or steering lever. Where the control system consists of a hydraulic telemotor, a second independent system need not be fitted, except in a tanker, chemical tanker or gas carrier of 10,000 tons gross and upwards;

c) For the auxiliary steering gear, in the steering gear compartment and, if power operated, it is to be also operable from the navigating bridge and is to be independent of the control system for the main steering gear;

d) Where the steering gear is so arranged that more than one control system can be simultaneously operated, the risk of hydraulic locking caused by a single failure is to be considered.

4.1.2 Any main and auxiliary steering gear control system operable from the navigating bridge is to comply with the following:

a) Means are to be provided in the steering gear compartment for disconnecting any

control system operable from the navigating bridge from the steering gear it serves;

b) The system is to be capable of being brought into operation from a position on the navigating bridge.

4.1.3 The angular position of the rudder is to :

a) If the main steering gear is power operated, be indicated on the navigating bridge. The rudder angle indication is to be independent of the steering gear control system.

b) Be recognizable in the steering gear compartment.

4.1.4 Appropriate operating instructions with a block diagram showing the change-over procedures for steering gear control systems and steering gear actuating systems are to be permanently displayed in the wheelhouse and in the steering gear compartment.

4.1.5 Where the system failure alarms for hydraulic lock (See Ch.7) are provided, appropriate instructions are to be placed on the navigating bridge to shut down the system at fault.

IR4.2 Mechanical, hydraulic and electrical independency and failure detection and response of steering control systems

IR4.2.1 Two independent steering gear control systems are to be provided and so arranged that a mechanical, hydraulic or electrical failure in one of them will not render the other one inoperative.

IR4.2.2 The term “steering gear control system” as defined in 1.2.2 is to be understood as to cover “the equipment required to control the steering gear power actuating system”.

IR4.2.3 Separation of control systems and components

IR4.2.3.1 General

Wires, terminals and the components for duplicated steering gear control systems installed in units, control boxes, switchboards or bridge consoles are to be separated as far as practicable. Where physical separation is not practicable, separation may be achieved by means of a fire retardant plate.

IR4.2.3.2 Steering wheel or steering lever

All electric components of the steering gear control systems are to be duplicated. This does not require duplication of the steering wheel or steering lever.

IR4.2.3.3 Steering mode selector switch

If a joint steering mode selector switch (uniaxial switch) is employed for both steering gear control systems, the connections for the circuits of the control systems are to be divided accordingly and separated from each other by an isolating plate or by air gap.

IR4.2.3.4 Follow-up amplifier

In the case of double follow-up control, the amplifiers are to be designed and fed so as to be electrically and mechanically separated. In the case of non follow-up control and follow-up control, it is to be ensured that the follow-up amplifiers are protected selectively

IR4.2.3.5 Additional control systems

Control circuits for additional control systems, e.g. steering lever or autopilot are to be designed for all pole disconnection.

IR4.2.3.6 Feed-back units and limit switches

The feed-back units and limit switches, if any, for the steering gear control systems are to be separated electrically and mechanically connected to the rudder stock or actuator separately.

IR4.2.3.7 Hydraulic control components

Hydraulic system components in the power actuating or hydraulic servo systems controlling the power systems of the steering gear (e.g. solenoid valves, magnetic valves) are to be considered as part of the steering gear control system and are to be duplicated and separated.

Hydraulic system components in the steering gear control system that are part of a power unit may be regarded as being duplicated and separated when there are two or more separate power units provided and the piping to each power unit can be isolated.

4.2.4 Failure detection and response of all types of steering gear control systems

4.2.4.1 Failure detection

4.2.4.1.1 The most probable failures that may cause reduced or erroneous system performance are to be automatically detected and at least the following failure scenarios are to be considered:

i) Power supply failure

ii) Loop failures in closed loop systems, both command and feedback loops (normally short circuit, broken connections and earth faults)

iii) Earth fault on AC and DC circuits

iv) Data Communication Errors

v) Programmable system failures (hardware and software failures)

vi) Deviation between rudder order and feedback*****.

Note ***** - Deviation alarm is to be initiated if the rudder's actual position does not reach the set point within acceptable time limits for the closed loop control systems (e.g. follow up control and autopilot). Deviation alarm may be caused by mechanical, hydraulic or electrical failures.

All failures detected are to initiate an audible and visual alarm on the navigation bridge.

4.2.4.1.2 System response upon failure

.1 The failures (as defined but not limited to those in 4.2.4.1.1) likely to cause uncontrolled movements of rudder are to be clearly identified. In the event of detection of such failure, the rudder is to stop in the current position without manual intervention or, is to return to the mid ship/ neutral position. For mechanical failures such as sticking valves and failure of static components (pipes, cylinders), the system response without manual intervention is not mandatory, and the operator can follow instructions on the signboard in case of such failures, in accordance with 4.2.4.1.2.2.

Note: For hydraulic locking failure the following are also to be complied with:

(i) Where hydraulic locking, caused by a single failure, may lead to loss of steering, an audible and visual alarm, which identifies the failed system, is to be provided on the navigating bridge.

(ii) This alarm is to be activated whenever:

- position of the variable displacement pump control system does not correspond with the given order; or

- incorrect position of 3-way full flow valve or similar in constant delivery pump system is detected.

.2 Where applicable, following standard signboard is to be fitted at a suitable place on steering control post on the bridge or incorporated into operating instruction on board:

CAUTION

IN SOME CIRCUMSTANCES WHEN 2 POWER UNITS ARE RUNNING SIMULTANEOUSLY THE RUDDER MAY NOT RESPOND TO HELM. IF THIS HAPPENS STOP EACH PUMP IN TURN UNTIL CONTROL IS REGAINED.

The above signboard is related to steering gears provided with two identical power units intended for simultaneous operation, and normally provided with either their own control systems or two separate (partly or mutually) control systems which are/ may be operated simultaneously.

Section 5

Electric Power Circuits, Electric Control Circuits, Monitoring and Alarms

5.1 Electric power circuits

5.1.1 Short circuit protection, an overload alarm and in the case of polyphase circuits, an alarm to indicate single phasing is to be provided for each main and auxiliary motor circuit. Protective devices are to operate at not less than twice the full load current of the motor or circuit protected and are to allow excess current to pass during the normal accelerating period of the motors.

5.1.2 Circuits obtaining their power supply via an electronic converter, e.g. for speed control and which are limited to full load current for continuous rating are exempt from the requirement to provide protection against excess current of magnitude as given in 5.1.1. The overload alarm is to be set to a value for which the electronic converter is designed considering the most severe condition of operation.

5.1.3 Indicators for running indication of each main and auxiliary motor are to be installed on the navigating bridge and at a suitable main machinery control position.

5.1.4 Each electric or electrohydraulic steering gear comprising one or more power units is to be served by at least two exclusive circuits fed directly from the main switchboard; however, one of the circuits may be supplied through the emergency switchboard. An auxiliary electric or electrohydraulic steering gear may be connected to one of the circuits supplying the main steering gear. The circuits supplying an electric or electrohydraulic steering gear are to have adequate rating for supplying all motors which can be simultaneously connected to them and may be required to operate simultaneously.

IR5.1.4 For a ship fitted with multiple steering systems, the above requirements are to be applied to each of the steering systems.

5.1.5 The circuits, required by 5.1.4, are to be separated throughout their length as widely as is practicable.

5.1.6 In ships of less than 1600 tons gross, if an auxiliary steering gear is not electrically powered or is powered by an electric motor primarily intended for other services, the main steering gear may be fed by one circuit from the main switchboard. Consideration would be given to other protective arrangements than described in 5.1.1, for such a motor primarily intended for other services.

5.2 Electric control circuits

5.2.1 Electric control systems are to be independent and separated as far as is practicable throughout their length.

5.2.2 Each main and auxiliary electric control system which is operable from the navigating bridge is to comply with the following:-

a) It is to be served with electric power by a separate circuit supplied from the associated steering gear power circuit, from a point within the steering gear compartment, or directly from the same section of the switchboard busbars, main or emergency, to which the associated steering gear power circuit is connected.

b) Each separate circuit is to be provided with short circuit protection only.

5.3 Monitoring and alarms

5.3.1 All alarms associated with steering gear faults are to be indicated on the navigating bridge and in accordance with the alarm system specified in Ch.7, Table 1.9.1. The alarms are to be both audible and visual.

Section 6

Emergency Power

6.1 General

6.1.1 Where the rudder stock is required to be over 230 [mm] diameter in way of the tiller (See 2.1.1), excluding strengthening for navigation in ice, an alternative power supply, sufficient at least to supply the steering gear power unit which complies with the requirements of 2.1.3 and also its associated control system and the rudder angle indicator, is to be provided automatically, within 45 seconds, either from the emergency source of electrical power or from an independent source of power located in the steering gear compartment. This independent source of power is to be used only for this purpose.

IR 6.1.1 In the case of non-traditional steering arrangements such as azimuthing propellers, waterjets, etc where the propulsion power exceeds 2,500kW per thruster unit, an alternative power supply, sufficient at least to supply the steering arrangements which complies with the requirements of clause 2.1.3.b and also its associated control system and the steering gear response indicator, are to be provided automatically, within 45s, either from the emergency source of electrical power or from an independent source of power located in the steering gear compartment. This independent source of power is to be used only for this purpose.
The above requirement is relevant for the steering-propulsion units having a certain proven steering capability attained due to vessel speed regardless of availability of propulsion power.

6.1.2 In every ship of 10,000 tons gross and upwards, the alternative power supply is to have a capacity for at least 30 minutes of continuous operation and in any other ship for at least 10 minutes.

6.1.3 Where the alternative power source is a generator, or an engine driven pump, starting arrangements are to comply with the requirements relating to the starting arrangements of the emergency generators.

Section 7

Testing and Trials

7.1 Testing

7.1.1 The requirements of the Rules relating to the testing of Class 1 pressure vessels, piping, and related fittings including hydraulic testing apply.

7.1.2 After installation on board the vessel the steering gear is to be subjected to the required hydrostatic and running tests.

7.1.3 Each type of power unit pump is to be subjected to a type test. The type test is to be for a duration of not less than 100 hours, the test arrangements are to be such that the pump may run in idling conditions, and at maximum delivery capacity at maximum working pressure. During the test, idling periods are to be alternated with periods at maximum delivery capacity at maximum working pressure. The passage from one condition to another should occur at least as quickly as on board. During the whole test no abnormal heating, excessive vibration or other irregularities are permitted. After the test, the pump is to be opened out and inspected. Type tests may be waived for a power unit which has been proven to be reliable in marine service.

7.2 Trials

7.2.1 The steering gear is to be tried out on the trial trip in order to demonstrate to the Surveyor's satisfaction that the requirements of the Rules have been met. The trial is to include the operation of the following :-

i) The steering gear; including demonstration of the performances required by 2.1.2 (b) and 2.1.3 (b). In order to demonstrate this ability, the trials may be conducted in accordance with Section 6.1.5.1 of ISO 19019:2005 “Sea-going

vessels and marine technology – Instructions for planning, carrying out and reporting sea trials".

The above is to be to the satisfaction of IRS.

ii) The steering gear power units, including transfer between steering gear power units;

iii) The isolation of one power actuating system, checking the time for regaining steering capability;

iv) The hydraulic fluid recharging system;

v) The emergency power supply required by 6.1.1;

vi) The steering gear controls, including transfer of control and local control;

vii) The means of communication between the steering gear compartment and wheelhouse, also the engine room, if applicable;

viii) The alarm and indicators;

ix) Where the steering gear is designed to avoid hydraulic locking, this feature is to be demonstrated.

Test of items (iv), (vii), (viii) and (ix) may be effected at the dockside.

7.2.2 On all occasions when steering gear trials are conducted with the vessel not at the summer load waterline the loading condition can be accepted on the conditions that either:

i) the rudder is fully submerged (at zero speed waterline) and the vessel is in an acceptable trim condition.

or

ii) the rudder torque at the trial loading condition have been reliably predicted (based on the system pressure measurement) and extrapolated to the summer load waterline condition using the following method to predict the equivalent torque and actuator pressure at the summer load waterline:

$$Q_F = Q_T\ \alpha$$

$$\alpha = 1.25\ \left(\frac{A_F}{A_T}\right)\left(\frac{V_F}{V_T}\right)^2$$

Where:

α = extrapolation factor

Q_F = rudder stock moment for the deepest service draught and design speed condition

Q_T = rudder stock moment for the trial condition

A_F = total immersed projected area of the moveable part of the rudder at summer load waterline

A_T = total immersed projected area of the moveable part of the rudder in the trail condition

V_F = contractual design speed of the vessel corresponding to the maximum continuous revolutions of the [main engine] at the. summer load waterline (also see 7.2.3)

V_T = measured speed of the vessel (considering current) in the trial condition.

Where the rudder actuator system pressure is shown to have a linear relationship to the rudder stock torque the above equation can be taken as:

$$P_F = P_T\ \alpha$$

Where:

P_F = estimated steering actuator hydraulic pressure at summer load waterline.

P_T = maximum measured actuator hydraulic pressure in the trial condition.

Where constant volume fixed displacement pumps are utilized then the regulations can be deemed satisfied if the estimated steering actuator hydraulic pressure at the deepest draught is less than the specified maximum working pressure of the rudder actuator. Where a variable delivery pump is utilized, pump data should be supplied and interpreted to estimate the delivered flow rate corresponding to the summer load waterline in order to calculate the

steering time and allow it to be compared to the required time.

Where A_T is greater than $0.95A_F$ there is no need for extrapolation methods to be applied.

iii) Alternatively the designer or builder may use computational fluid dynamic (CFD) studies or experimental investigations to predict the rudder stock moment at the summer load waterline condition and design speed. These calculations or experimental investigations are to be to the satisfaction of IRS.

7.2.3 In any case for the main steering gear trial, the speed of the ship corresponding to the number of maximum continuous revolution of main engine and maximum design pitch applies.

Section 8

Additional Requirements

8.1 For tankers, chemical tankers or gas carriers of 10,000 tons gross and upwards and every ship of 70,000 tons gross and upwards

8.1.1 The main steering gear is to comprise two or more identical power units complying with the provisions of 2.1.4.

8.2 For tankers, chemical tankers or gas carriers of 10,000 tons gross and upwards

8.2.1 Subject to 8.3, the following are to be complied with :

a) The main steering gear is to be so arranged that in the event of loss of steering capability due to a single failure in any part of one of the power actuating systems of the main steering gear, excluding the tiller, quadrant or components serving the same purpose, or seizure of rudder actuators, steering capability is to be regained in not more than 45 seconds after the loss of one power actuating system.

b) The main steering gear is to comprise either;

   i) two independent and separate power actuating systems, each capable of meeting the requirements of 2.1.2 (b); or

   ii) at least two identical power actuating systems which, acting simultaneously in normal operation are capable of meeting the requirements of 2.1.2(b). Where necessary to comply with these requirements, inter- connection of hydraulic power actuating systems is to be provided. Loss of hydraulic fluid from one system is to be capable of being detected and the defective system automatically isolated so that the other actuating system or systems remain fully operational.

c) Steering gears other than of the hydraulic type are to achieve equivalent standards.

8.3 For tankers, chemical tankers or gas carriers of 10,000 tons gross and upwards but of less than 100,000 tonnes deadweight

8.3.1 Solutions other than those set out in 8.2.1 which need not apply the single failure criterion to the rudder actuator or actuators, may be permitted provided that an equivalent safety standard is achieved and that:

a) Following loss of steering capability due to a single failure of any part of the piping system or in one of the power units, steering capability is regained within 45 seconds; and

b) Where the steering gear includes only a single rudder actuator special consideration is given to stress analysis for the design including fatigue analysis and fracture mechanics analysis, as appropriate, the material used, the installation of sealing arrangements and the testing and inspection and provision of effective maintenance. In consideration of the foregoing, regard will be given to the "GUIDELINES" in Sec.9.

8.3.2 Manufacturers of steering gear who intend their product to, comply with the requirements of the Sec.9 are to submit full details when plans are forwarded for approval.

Section 9

Guidelines for the Acceptance of Non-duplicated Rudder Actuators for Tanker, Chemical Tankers or Gas Carriers of 10,000 tons gross and upwards but of less than 100,000 tonnes Deadweight

9.1 Materials

9.1.1 Parts subject to internal hydraulic pressure or transmitting mechanical forces to the rudder-stock are to be made of duly tested ductile materials complying with recognized standards. Materials for pressure retaining components are to be in accordance with recognized pressure vessel standards. These materials are not to have an elongation less than 12 per cent nor a tensile strength in excess of 650 [N/mm^2].

9.2 Design

9.2.1 **Design pressure** The design pressure should be assumed to be at least equal to the greater of the following :-

a) 1.25 times the maximum working pressure to be expected under the operating conditions required in 2.1.2(b).

b) The relief valves(s) setting.

9.2.2 **Analysis** In order to analyse the design the following are required :

a) The manufacturers of rudder actuators should submit detailed calculations showing the suitability of the design for the intended service.

b) A detailed stress analysis of the pressure retaining parts of the actuator should be carried out to determine the stresses at the design pressure.

c) Where considered necessary because of the design complexity or manufacturing procedures, a fatigue analysis and fracture mechanics analysis may be required. In connection with these analyses, all foreseen dynamic loads should be taken into account. Experimental stress analysis may be required in addition to, or in lieu of, theoretical calculations depending upon the complexity of the design.

9.2.3 **Dynamic loads for fatigue and fracture mechanics analysis** The assumptions for dynamic loading for fatigue and fracture mechanics analysis where required by 3.1.5, 8.3 and 9.2.2 are to be submitted for appraisal. Both the cases of high cycle and cumulative fatigue are to be considered.

9.2.4 **Allowable stresses** For the purpose of determining the general scantlings of parts of rudder actuators subject to internal hydraulic pressure the allowable stresses should not exceed:

$$\sigma_m \leq f$$

$$\sigma_p \leq 1.5\, f$$

$$\sigma_b \leq 1.5\, f$$

$$\sigma_p + \sigma_b \leq 1.5\, f$$

$$\sigma_m + \sigma_b \leq 1.5\, f$$

where,

σ_m = equivalent primary general membrane stress

σ_p = equivalent primary local membrane stress

σ_b = equivalent primary bending stress

f = the lesser of $\frac{\sigma B}{A}$ or $\frac{\sigma Y}{B}$

σB = specified minimum tensile strength of material at ambient temperature

σY = specified minimum yield stress or 0.2 percent proof stress of material at ambient temperature

A and B are as follows :

	Wrought Steel	**Cast Steel**	**Nodular Cast Iron**
A	4	4.6	5.8
B	2	2.3	3.5

9.2.5 **Burst test** Pressure retaining parts not requiring fatigue analysis and fracture mechanics analysis may be accepted on the basis of a certified burst test and the detailed stress analysis required by 9.2.2 need not be provided. The minimum bursting pressure should be calculated as follows :-

$$P_b = P\,A\,\frac{\sigma B_a}{\sigma B}$$

where,

P_b = minimum bursting pressure

P = design pressure as defined in 9.2.1

A = as defined in 9.2.4

σB_a = actual tensile strength

σB = tensile strength as defined in 9.2.4.

9.3 Construction details

9.3.1 General The construction should be such as to minimize local concentration of stress.

9.3.2 Welds

a) The welding details and welding procedures should be approved.

b) All welded joints within the pressure boundary of a rudder actuator or connection parts transmitting mechanical loads should be full penetration type or of equivalent strength.

9.3.3 Oil seals Oil seals forming part of the external boundary are to comply with 3.2.3 and 3.2.4.

9.3.4 Isolating valves Isolating valves are to be fitted at the connection of the pipes to the actuator, and should be directly mounted on the actuator.

9.3.5 Relief valves Relief valves for protecting the rudder actuator against over-pressure as required in 3.3.2 are to comply with the following:

a) The setting pressure is not to be less than 1.25 times the maximum working pressure expected under operating conditions required by 2.1.2(b).

b) The minimum discharge capacity of the relief valve(s) is to be not less than 110 per cent of the total capacity of all pumps which provide power for the actuator. Under such conditions the rise in pressure should not exceed 10 per cent of the setting pressure. In this regard due consideration should be given to extreme foreseen ambient conditions in respect of oil viscosity.

9.4 Non-destructive testing

9.4.1 The rudder actuator should be subjected to suitable and complete non-destructive testing to detect both surface flaws and volumetric flaws. The procedure and acceptance criteria for non-destructive testing should be in accordance with requirements of recognized standards. If found necessary, fracture mechanics analysis may be used for determining maximum allowable flaw size.

9.5 Testing

9.5.1 Tests, including hydrostatic test, of all pressure parts at 1.5 times the design pressure should be carried out.

9.5.2 When installed on board the ship, the rudder actuator should be subjected to a hydrostatic test and a running test.

End of Chapter

Chapter 7

Control Engineering

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Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter are in general to be complied with for all machinery plants where instrumentation is required according to other Chapters of the Rules. In addition, the requirements apply to the following, when installed :-

a) remote control systems for propulsion machinery, controllable pitch propeller, and steering gear;

b) safety systems for propulsion plants and electric power generating plants;

c) instrumentation equipment of boiler plants.

d) Safety systems for emergency diesel engines.

1.1.2 For vessels intended to operate with unattended machinery spaces, refer to Pt.5.

1.1.3 Alarm systems are to satisfy the environmental requirements of IRS Classification Notes "Type Approval of Electrical Equipment used for Control, Protection, Safety and Internal Communication in Marine Environment".

1.2 Remote control systems capability

1.2.1 When it is desired to fit a centralized control system having one or more control locations and embodying various degrees of automatic and remote control of the propulsion plant or associated ship's service systems, the control and monitoring systems are to be designed, assembled and installed in accordance with the requirements given in Section 5, to assure operation as effective as could be obtained with the same systems arranged for manual control and monitoring by watchkeeping personnel.

1.3 Plans and particulars

1.3.1 Plans and specifications for the control systems, are to be submitted, for approval and are to at least include the following information:

a) machinery arrangement plans showing locations control in relation to controlled units;

b) arrangements and details of control consoles including front views, installation arrangements together with schematic diagrams for all power, control and

monitoring systems including their functions;

c) kind and size of all electrical cables and wiring associated with the control systems including voltage rating, service voltage and currents together with overload and short circuit protection;

d) functional block diagrams of all control and monitoring systems;

e) schematic plans of hydraulic and pneumatic control systems together with all inter-connections, piping sizes and materials including working pressures, relief valve settings, reducing valves, filters, compressors and their type and manufacturers;

f) description of all alarm and emergency tripping arrangements, functional sketches or description of all special valves, actuators, sensors and relays;

g) schematic plans and supporting data of fire-protection and extinguishing systems, including fire-detection and alarm systems, bilge high level alarms;

h) control system fault analysis, test program for testing controls, monitoring and alarm systems;

i) Where computer based systems are used for essential services and are integrated with other computer based systems, the failure analysis is to be carried out and should include networks and associated network components (routers, switches etc.)

j) A network plan clearly identifying all the network components for each on board operational technology (OT) network, including location of each network device, details of IT network where the IT and OT Networks are interconnected;

k) documents/plans indicated at Section 7.2 for integrated systems.

1.4 Operational guidance manual

1.4.1 An operational guidance manual is to be provided on board the vessel for reference and is to contain the necessary system technical information and give operating instructions for normal and emergency operations.

1.5 Location of controls

1.5.1 Ship motion and anticipated structural vibrations are to be considered when selecting the locations of main and associated secondary control locations.

1.5.2 The control locations are to be well ventilated and air conditioning is to be provided for the control consoles when required by the operational characteristics of the components within the consoles. There is to be an alarm for failure of the console integral air conditioning.

1.5.3 Enclosed main control location located within the machinery spaces are to have two means of access located as remote from each other as practicable.

1.5.4 The leading of pipes in the vicinity of control console is to be avoided as far as possible. When such leads are necessary, care is to be taken to fit no flanges or joints over or near the console unless provision is made to prevent any leakage from injuring the equipment.

1.5.5 Glass in a control room located within or adjacent to the machinery space is to be of the shatter-resistant type.

1.5.6 When the main control for the propulsion plant is to be situated in a location remote from the propulsion machinery space, details of special arrangements are to be submitted to show that the operation of the propulsion plant would be as effective as with the main control located within or adjacent to the machinery space.

1.6 Main control location

1.6.1 Controls

1.6.1.1 The main control location is to provide control of the propulsion plant and associated ship's service system including starting, stopping, and for change over of vital pumps and motors, and is to provide effective means for monitoring temperatures, pressures, the electrical system, fluid and gas flows, liquid levels and other variables which are essential for the propulsion plant operation.

1.6.1.2 Effective means are to be provided for monitoring and controlling direction of rotation or propeller pitch and speed of propeller for the safe operation of the plant from standby

condition at departure through normal operation to "finished with engines" at the end of the voyage.

1.6.1.3 The main control location is to be fitted with alarms and emergency trips as required together with indicators and is to have means for the assessment of the operational status of all machinery and systems vital to the propulsion of the vessel. Control functions from the console may be designed for either 'remote manual' or 'automatic control'.

1.6.2 Independent manual control

1.6.2.1 Means are to be provided for independent manual control, at or near the machinery concerned, in the event of failure of a particular control in the centralized control system. Necessary instrumentation is to be provided so that satisfactory operation of the machinery under independent manual control can be exercised for lengthy periods. Independent manual control of propulsion machinery is to be demonstrated during the tests/trials to the satisfaction of the Surveyors. This is to include demonstration of independent manual control through the full maneuvering range and transfer from automatic control.

1.6.3 Order of control-location command

1.6.3.1 When the propulsion machinery is arranged to be controlled from two or more locations, the main control location is to have means for transferring control from a secondary location to the main location at all times and blocking any unauthorized control from any secondary location. Consideration will be given to special cases where it may be necessary for a secondary control location to have command over control transfers between control locations.

1.6.4 Control transfer

1.6.4.1 Transfer of control from one location to another, except as required by 1.6.3.1, is to be possible only with acknowledgement by the receiving locations. The main control location and the secondary control location are to have an indicator showing which location is in control.

1.7 Secondary control locations

1.7.1 In general, secondary control locations for the control of propulsion machinery from the bridge or other locations on board are to be kept as simple as possible and provided with only those indicators and controls necessary for the effective control of speed and direction of the propulsion engines, and of the controllable-pitch propeller where fitted, for normal operation.

1.7.2 Precautions are to be taken to ensure that the engine is not normally operated in a barred speed range.

1.7.3 If the control system automatically shuts down the main propulsion engine for any reason, this is to be alarmed at the main control location and each secondary control location. Restoration of normal operating conditions is to be possible only after manual reset. Automatic restarting is not to be possible.

1.8 Control console construction

1.8.1 Control consoles are to be preferably self-supported with the sides and backs suitably protected. Where necessary, protection is to be provided for consoles which might be subject to damage by leaks or falling objects.

1.9 Instrumentation

1.9.1 Main control location

1.9.1.1 Instrumentation and alarms at the main control location are to provide all information necessary for monitoring the operation of the propulsion, electrical and emergency systems, generally in accordance with Table 1.9.1.

1.9.2 Secondary control location

1.9.2.1 Instrumentation and alarms for the bridge control locations and other secondary control locations are to be generally in accordance with Table 1.9.2.

1.10 Displays, indications and alarms

1.10.1 The alarms and displays required by Table 1.9.1 and Table 1.9.2 for essential services are to be readily distinguishable from other alarms (e.g. fire alarm) and are to be grouped functionally in so far as practicable. Alarms are to have individual visual presentation with preferably a common audible signal.

1.10.2 In general, alarms and displays not required by Table 1.9.1 and Table 1.9.2 or not concerned with the management of the

propulsion machinery (non- essential alarms) are to be grouped separately from the essential alarms. Audible signals for non-essential alarms, if provided, are to be of a character distinct from that for essential alarms.

1.10.3 The number of colours used to indicate status of plant operation is to be kept to a minimum. A uniform code is to be used so that like colours indicate like functions or status of operation in the controlled plant or system.

1.10.4 Alarm systems are to be designed so that they cannot remain deactivated when the monitoring system has returned to the normal running condition. Alarm systems are to provide both audible and visual alarm upon each fault condition. Alarms, due to fault condition in machinery, are to remain active until acknowledged and if arrangements are made to mute audible alarms they are not to extinguish visual alarms. Acknowledgement of visual alarms is to be clearly indicated. Alarms are to be of the self-monitoring type so that a circuit failure will cause an alarm condition and are to have provisions for testing all audible and visual alarms and indicating lamps.

1.10.5 Machinery, safety and control system faults are to be indicated at the relevant control locations to advise duty personnel of a fault condition. The presence of unrectified faults is to be clearly indicated at all times.

1.10.6 Where the alarms are displayed as group alarms provision is to be made to identify individual alarms at the main control location (if fitted) or alternatively at subsidiary control locations.

1.10.7 Acknowledgement of alarms at positions outside a machinery space is not to silence the audible alarm or extinguish the visual alarm in that machinery space.

1.10.8 If an alarm has been acknowledged and a second fault occurs prior to the first being rectified, audible and visual alarms are again to operate. Where alarms are displayed at a local panel adjacent to the machinery and with arrangements to provide a group or common fault alarm at the main control room alarm display then the occurrence of a second fault prior to the first alarm being rectified need only be displayed at the local panel, however, the group alarm is to be re-initiated. Unacknowledged alarms on monitors are to be distinguished by either flashing text or a flashing marker adjacent to the text. A change of colour will not in itself be sufficient to distinguish between acknowledged and unacknowledged alarms.

1.10.9 For the detection of transient faults which are subsequently self-correcting, alarms are required to lock in until accepted.

1.10.10 The alarm system is to be arranged with automatic changeover to a standby power supply in the event of a failure of the normal power supply. Where an alarm system could be adversely affected by an interruption in power supply, changeover to the standby power supply is to be achieved without a break.

1.10.11 Failure of any power supply to the alarm system is to operate an audible and visual alarm at the main control location.

1.10.12 The alarm system is to be capable of being tested during normal machinery operation.

1.10.13 The alarm system is to be designed as far as practicable to function independently of control and safety systems such that a failure or malfunction in these systems will not prevent the alarm system from operating.

1.10.14 Disconnection or manual overriding of any part of the alarm system should be clearly indicated.

1.10.15 When alarm systems are provided with means to adjust their set point, the arrangements are to be such that the final settings can be readily identified.

1.10.16 Where monitors are provided at the location in control and if fitted, in the duty engineer's accommodation, they are to provide immediate display of new alarm information regardless of the information display page currently selected. This may be achieved by provision of a dedicated alarm monitor, a dedicated area of screen for alarms or other suitable means.

1.10.17 Alarms are to be displayed in the order in which they occur. Alarms requiring shutdown or slowdown action are to be given visual prominence.

1.11 Safety systems

1.11.1 Where safety systems are provided, the requirements of Pt.5, Ch.22, 2.5.2 to 2.5.8 are to be complied with.

1.11.2 Where arrangements are provided for over-riding a safety system, they are to be such that in-advertent operation is prevented. Visual indication is to be given at the relevant control location(s) when a safety over-ride is operated. The consequences of over-riding a safety system are to be established and documented.

1.11.3 Monitoring and safety systems for DF or GF engines are to be provided in accordance with the requirements of Ch. 4, Sec. 4, 4.14.4.8 in addition to those given in this Section.

Table 1.9.1 : Displays, indications and alarms for main control location				
	System	**Remote Display**	**Alarm activation**	**Remarks**
Main boilers				
1	Main steam superheater outlet	Temperature Pressure	High	-
2	Fuel oil to burners	Pressure Viscosity (temperature)	Low High (low)	-
3	Master fuel oil cut-off valve	-	Closed	-
4	Atomizing medium	Pressure	Outside limits	-
5	Forced draft	Pressure (water column)	Failure	-
6	Water level	Level	High, low Low-low	-
7	Feed pump discharge	Pressure		-
8	Feed water salinity	Salinity	-	-
Main turbine				
1	Steam, ahead chest	Pressure	-	-
2	Steam, astern chest	Pressure	-	-
3	Steam, gland steam	Pressure	-	-
4	L.O. to turbines and reduction gears	Temperature Pressure	High Low	-
5	L.O. sump level	-	Low	-
6	L.O. gravity tank	Level	Low	-
7	Bearings, turbine, thrust and reduction gear (individual)	Temperature	High	-
8	Main condensate pump	Running	Failure	-
9	Main condenser	Vacuum	Low	-
10	Main circulator	Running	Failure	-
11	Condensate level	-	High	-
12	Deaerator feed tank	Level	Low	-
13	Astern guardian valve	Position	Failure to open	-
14	Vibration	-	Excessive	-
15	Rotor axial displacement	-	Excessive	-
16	Shaft roll over	-	Stopped	-
17	Scoop valve	Position	-	-
18	Main circulator sea suction valve	Position	-	-
Turbogenerator				
1	L.O. to turbine and reduction gear	Pressure Temperature	Low High	-
2	Bearings	Temperature	High	-
3	Aux. Condenser	Vacuum	Low	-
4	Aux. Condensate pump	Running	-	-
5	Aux. Circulating pump	running	-	-

Table 1.9.1 : (Contd.)

	System	Remote Display	Alarm activation	Remarks
Main diesel engines				
1	L.O. to engine	Pressure Temperature	Low High	See Note 1
2	Differential pressure across L.O. discharge filter	Pressure	High	-
3	L.O. to reduction gear	Pressure	Low	-
4	L.O. at reduction gear	Temperature	High	-
5	Oil mist concentration in crankcase or temperature of bearings or equivalent devices	-	High	See Note 16
6	L.O. drain tank level	-	Low-high	See Note 2
7	Cylinder lubrication	-	Failure	See Note 1
8	Cooling water to cylinders	Pressure or flow	Low	See Note 9
9	Cylinder cooling water outlet	Temperature	High	See Note 3
10	Level of cylinder cooling water in expansion tank	-	Low	-
11	Piston coolant	Pressure	Low	See Note 4
12	Flow of piston coolant through each piston and temperature in manifold (or temperature of coolant at each piston outlet)	Flow, temperature (or temperature)	Low-high (high)	See Note 4
13	Level of piston coolant in expansion tank	-	Low	See Note 4
14	Fuel valve coolant	Pressure Temperature	Low High	-
15	Level of fuel valve coolant in expansion tank	-	Low	-
16	Sea water coolant	Pressure	Low	See Note 9
17	F.O. to injection pumps	Pressure	Low	-
18	F.O. to engine	Temperature or viscosity	Low-high (Low-high)	See Note 7
19	Level of F.O. in daily service tank	-	Low-high	See Note 5
20	Charging air	Temperature	Low-high	See Note 6
21	Deviation of each cylinder from average and exhaust gas temperature of each engine	Temperature	High	See Note 8
22	Fire in scavenging belt, for two stoke engines	-	Activated	See Note 10
23	Starting air	Pressure	Low	-
24	Overspeed	-	Activated	-

Table 1.9.1 : (Contd.)

	System	Remote Display	Alarm activation	Remarks
Diesel generators				
1	F.O. to engines	Pressure	Failure	-
2	L.O. to engines	Pressure Temperature	Low High	-
3	Starting air	Pressure	Low	-
4	Exhaust gas	Temperature	-	-
5	Cooling medium, outlet	Temperature Pressure	High Low	See Note 9
6	Level in cooling water expansion tank, if not connected to main system	-	Low	-
7	overspeed	-	Activated	-
Steering gear				
1	Rudder position	Indication	-	See Note 12
2	Steering gear power units	-	Failure	-
3	Steering gear motors	Running and stopped indication	Overload and single phase	See Note 13
4	Control system power	-	Failure	-
5	Steering gear hydraulic oil tank level	-	Low	See Note 14
6	Auto pilot indication	Running indication	Failure	-
7	Hydraulic lock	-	Fault	See Note 15
Addition services as applicable				
1	Control electric power	available	Failure	-
2	Control air supply	Pressure	Low	-
3	Hydraulic control system	Pressure	Low	-
4	Control location in operation	In command indication	-	-
5	Propeller pitch indicator	Pitch	-	-
6	Propeller speed, direction	RPM, direction	-	-
7	Stern tube L.O. tank level	-	Low	-
8	F.O. service or settling tank	-	Low, high	-
9	F.O. settling tank temperature	-	High	-
10	Generators, ship service	Volts amperes Frequency	Off limits High off limits	-
11	Fuel oil service pump	Running	-	-
12	Console air conditioning	-	Failure	-
13	Automatic shut down	-	Shut-down	-
14	Turning gear	Engaged	-	-
15	Fire main	Pressure	-	-
16	Bilge pump	Running	-	-
17	Bilge level	-	High	-
18	Clutch air or oil pressure	-	Low	-

Table 1.9.1 : (Contd.)

Notes:

1 Individual alarms are required where separate lubricating oil systems (e.g. for camshaft, rocker arms, turbocharger, etc.) are installed. For turbochargers with an integrated self-contained oil lubrication system such alarm is not required.

2 Where separate lubricating oil systems are installed, individual level alarms are required for the tanks.

3 Where one common cooling space without individual stop valves is employed for all cylinder jackets, one temperature alarm for the common cooling water outlet will be approved.

4 Only applicable for cross-head type engines.

5 High-level alarm is required only if suitable overflow arrangements are not provided.

6 Both the displays and alarms are not required if automatic temperature control is not necessary. Low temperature alarm may be replaced by an alarm for water detection in the charging air duct.

7 Not required for distillate fuels.

8 Alarm for slow speed engines (under 300 RPM) and medium speed engines above 4000 bhp.

9 Not required for attached cooling pumps.

10 Not required for engines with bore 300 [mm], or under.

11 Alarms required for steering gear need to be fitted only on the bridge.

12 The rudder angle indication is to be on the navigation bridge and independent of the steering gear control system.

13 Steering gear motors running indicators are to be fitted both on the bridge and main control location.

14 Each tank is to be monitored.

15 Where more than one system (either power or control) can be operated simultaneously each system is to be monitored and the system at fault is to be identified.

16 If engine power is $\geq$ 2250 [kW] or cylinder bore > 300 [mm], following is to be provided:

- Oil mist detection arrangements or engine bearing temperature monitors or equivalent devices.
- Slow down with alarm for low speed diesel engines
- Shut down with alarm for medium and high speed engines.
- IRS may permit overriding of auto shut off or slow down arrangements, provided consequences of overriding auto shut off or slow down are established and documented.
- For the purpose of this requirement :-
    - Low speed engines means diesel engines having a rated speed of less than 300 rpm
    - Medium speed engines means diesel engines having a rated speed of 300 rpm and above, but less than 1400 rpm.
    - High Speed engines means diesel engines having a rated speed of 1400 rpm and above

Table 1.9.2 : Displays, indications and alarms for secondary control location

	System	Display	Alarm	Remarks
1	Control location	In command	-	-
2	Propeller speed and direction	Ahead, astern, RPM	-	-
3	Propeller pitch	Pitch	-	-
4	Starting air pressure	-	Low	-
5	Propulsion unit	-	Abnormal	-
6	Remote control power	-	Failure	-
7	Shaft roll over	-	Stopped	-
8	Bilge level	-	High	-

Section 2

Control - System Characteristics

2.1 Power supply

2.1.1 The control systems are to be served by two feeders from the main switchboard. One of the feeders may be from the emergency switchboard, if provided. Transfer may be by manual or automatic switch installed in or adjacent to the control console. The feeder from the transfer switch to the control console may be through a single cable. These feeders are not to supply power for any other machinery.

2.1.2 The feeders supplying power to the control console are to be provided with short-circuit protection at the main switchboard and emergency switchboards, if provided. Where circuits within the control consoles are protected by fuses or circuit breakers, the control-system protection is to be subdivided and arranged so that failure of one set of fuses or circuit breakers will not cause maloperation or failure of another circuit or system and it is to be possible to isolate the defective system.

2.1.3 The power for monitoring, alarms and emergency action is to be supplied automatically from an emergency source upon failure of ship's service power supply. An audio / visual alarm is to be provided at the main control location to indicate failure of power supply.

2.1.4 Cables and console wiring for control and monitoring are to be of the flame-retarding type and are to be stranded except that solid conductors may be used in low energy circuits, where they are properly supported and not subject to undue vibration or movements.

2.1.5 Conductors for monitoring circuits which carry low-level or information-level signals are to be installed in such a manner or provided with shields so as to minimize the introduction of spurious signals from outside sources. Wiring within consoles is to be arranged to provide maximum accessibility and protection from steam, water or oil piping.

2.2 Electrical and electronic devices

2.2.1 All electrical and electronic devices are to be suitable for use in marine atmosphere, resistant to corrosion, not affected by shipboard vibration and motion and are to be capable of performing their intended functions at compartment ambient temperature. Electrical and electronic devices are to be applied on the basis of 50°C ambient when located in machinery spaces or 40°C ambient when located in other spaces, including those cases when the component is located in an airconditioned console.

2.2.2 Semiconductor devices are to be selected on the basis of expected shipboard ambient air-temperature ranges of 0°C-50°C for interior compartments and 0°C-60°C inside of consoles. Silicon and selenium semiconductor devices are to be used in

preference to germanium which may be used, where its characteristics are favoured for a circuit provided care is taken to ensure satisfactory operation under shipboard conditions.

2.2.3 Steady-state voltage variations of 10 per cent and frequency variations of ± 5 per cent from the nominal console feeder rating are not to affect the intended functioning of the electrically operated control and monitoring devices. Where close tolerances on voltage and frequency are required, special regulated supplies are to be provided. Voltage transients are not to cause any dangerous malfunctioning or damage to the control and monitoring devices and the control equipment is to be fitted with transient- voltage suppressors.

2.2.4 The design and arrangement of all devices is to provide ready accessibility to parts requiring inspection, adjustment or periodic replacement. Where devices are parts of sub-circuits assembled in physically-identical modular units for easy mounting on and removal from the console, suitable arrangements such as matched plug-in modules with coded plugs are to be provided to facilitate correct replacement of modules in the console.

2.2.5 Built-in circuitry is to be provided for use in the testing of module functions.

2.2.6 Consideration is to be given to minimize, as far as practicable, the probability that failure of any one component or device in the control circuitry will cause unsafe operation of the plant.

2.2.7 Control levers or wheels are to be readily identifiable as to function and position and are to be arranged for a logical sequence of operations. Suitable interlocks are to be provided to prevent incorrect operation.

2.2.8 When logic circuits are used for sequential start-up or for operating individual plant, components, indicators are to be provided at the control console to show the successful completion of the sequence of operations by the logic circuit and start-up and operation of the component. If some particular step is not carried out during the sequence, the sequence is to stop at this point. Manual override is to be fitted in vital functions to permit control in case of failure of a logical circuit.

2.2.9 Electrically-powered actuators for the execution of control commands are to be suitable for shipboard use, and have working and other parts which would not be damaged or rendered ineffective by corrosion. The windings are to be treated to resist oil and water and the enclosures are to be suitable for the location. The ratings of coils is to be based on ambient air temperature of 50°C when located in the machinery spaces and 40°C when located outside such spaces. The power supply for electrical actuators is to be from the same source as the power to the control systems.

2.3 Hydraulic controls

2.3.1 Hydraulic pumps, actuators, motors and accessories are to be suitable for the intended duty, compatible with the working fluid and are to be designed to operate safely at full-power conditions. In general, the hydraulic fluid is to be non-flammable or have a flash point above 157°C.

2.3.2 All control piping is to be readily accessible and supported so as to protect the piping and associated accessories from mechanical damage, vibration and shock. The control piping is to be suitably marked to indicate the character of its service.

2.3.3 The hydraulic pumps are to be fitted in duplicate and have pressure relief protection on the discharge side. The pump suctions are to be from a reservoir of sufficient capacity to contain all the fluid when drained from the system, maintain the fluid level at an effective working height and allow air and foreign matter to separate out. The pump suctions are to be sized and positioned to prevent cavitation or starvation of a pump. A duplex filter which can be cleaned without interrupting the oil supply is to be fitted on the discharge side of the pumps.

2.4 Pneumatic controls

2.4.1 Air compressors, actuators, motors and accessories are to be suitable for the intended duty and have working and other parts which will not be damaged or rendered ineffective by corrosion.

2.4.2 All control piping is to be readily accessible and supported so as to protect the piping and associated accessories from mechanical damage, vibration and shock. The

control piping is to be suitably marked to indicate the character of its service.

2.4.3 Compressed air for pneumatic control is to be available from at least two air compressors. The starting air system may be used as a source of control air. The air pressure to the pneumatic control system is to be automatically maintained at the level required for the operation of the installation and low air pressure is to set off an alarm at the main control location. Means are to be provided in the delivery from the compressors to assure clean, dry and oil-free air to the pneumatic controls.

2.5 Ship motion effects

2.5.1 All control, actuating, monitoring and alarm devices are to be able to operate successfully when inclined at an angle of 30° in any direction from the vertical and when subjected to vibratory frequencies of 2 to 80 Hz, in conjunction with peak to peak amplitudes of 2 [mm] for frequencies 2 to 13.2 Hz and an acceleration of 0.7 g for frequencies of 13.2 to 80 Hz. Care is to be taken to ensure that mounting arrangements for the components will not amplify shipboard vibrations.

2.6 Bridge control for main propulsion machinery

2.6.1 Where a bridge control system for main propulsion machinery is to be fitted, the requirements of 2.6.2 to 2.6.8 are to be complied with.

2.6.2 Means are to be provided to ensure satisfactory control of propulsion from the bridge in both the ahead and astern directions.

2.6.3 Instrumentation to indicate the following is to be fitted on the bridge:

a) Propeller speed.
b) Direction of rotation of propeller for a fixed pitch, propeller or pitch position for controllable pitch propeller.
c) Clutch position where applicable.
d) Shaft brake position where applicable.
e) Engine speed.

2.6.4 The propeller speed, direction of rotation and, if applicable, the propeller pitch are to be controlled from the bridge under all normal sea going and manoeuvring conditions.

2.6.5 Remote control of the propulsion machinery is to be from one control location at any one time. See also 1.6.3. Main propulsion control units on the navigating bridge may be interconnected. Means are to be provided at the main machinery control location to ensure smooth transfer of control between the bridge and machinery control locations.

2.6.6 Means of control, independent of the bridge control system, are to be provided on the bridge to enable the propulsion machinery to be stopped in an emergency.

2.6.7 Audible and visual alarms are to operate on the bridge and in the locations required by Pt.5, Ch.22 if any power supply to the bridge control system fails. Where practicable the preset speed and direction of thrust are to be maintained until corrective action is taken.

2.6.8 At least two means of communication are to be provided between the bridge and the main control location in the machinery space. One of these means may be the bridge control system; the other is to be independent of the main electrical power supply. See also Pt.4, Ch.1, Sec.2, Cl. 2.5.

2.7 Valve control system

2.7.1 Where bilge, ballast, oil fuel transfer and sea valves for engine services are operated by remote or automatic control, the requirements of 2.7.2 to 2.7.5 are to be satisfied.

2.7.2 Failure of actuator power is not to permit a valve to move to an unsafe condition.

2.7.3 Positive indication is to be provided at the remote control location for the service to show the actual valve position or alternatively that the valve is fully open or closed.

2.7.4 Equipment located in places which may be flooded is to be capable of operating when submerged.

2.7.5 A secondary means of operating the valves, which may be by local manual control, is to be provided.

2.7.6 For requirements applicable to closing appliances on scuppers and sanitary discharges, see Pt.3, Ch.13.

Section 3

Requirements for various Machinery Installations

3.1 Boilers

3.1.1 General

3.1.1.1 Where a boiler-control system is fitted for the purpose of automatically controlling the combustion, feed water and firing functions, the arrangements are to be capable of automatically and safely satisfying the steam requirements demanded from the boiler under normal evaporation between minimum and maximum firing rates and be able to maintain complete and stable combustion at the minimum rate of firing or during any sudden change in steam demand.

3.1.1.2 To prevent a build-up of excessive boiler steam which might occur when all burners are in service and burners are at the minimum firing rate, one of the following arrangements or equivalent is to be provided:-

a) Burners sequencing, which may require automatic control of one or more, but not necessarily all, burners in the boiler;

b) an automatic steam dump system, unloading to a condenser of adequate size.
For long-term operation at low loads, the excess burner capacity may be secured.

3.1.2 Fuel oil system

3.1.2.1 A master fuel oil cut-off valve is to be provided in the fuel system to each boiler. The valve is to be automatically closed upon flame failure of all burners in the boiler or upon action of the low-water alarm or upon loss of forced draft. The closing of the master fuel oil valve is to be alarmed at the main control location. Before any attempt at restarting after a complete flame out, manual intervention is to be initiated to determine the cause of flame failure and to reset the master fuel-oil valve.

3.1.2.2 The oil burners are to be so arranged that they cannot be withdrawn unless oil supply to the burners is cut off.

3.1.3 Combustion-control safety

3.1.3.1 Each burner is to be fitted with a flame scanner designed to automatically shut off the fuel oil supply to the burner in the event of flame failure. The flame failure shut off is to be capable of shutting off the oil supply to the burner within 6 seconds following flame failure. In the case of the failure of the flame scanner, the fuel to the burner is to be shut off automatically. On flame failure of all burners in boilers fitted with an automatic ignition system, the control system is to go to the automatic boiler purge condition.

3.1.3.2 Where boilers are fitted with an automatic ignition system, a timed boiler purge with all air registers open is required prior to ignition of the initial burner. The boiler purge may be initiated manually or automatically. The purge time is to be based on a minimum of four air changes of the combustion chamber and furnace passes. It is to be proven that the forced draft fan is operating and the air registers and dampers are open before the purge time commences.

3.1.3.3 Means provided to by-pass the flame-scanner control system temporarily during a trial- for ignition period is to be limited to 15 seconds from the time the fuel reaches the burners. Except for this trial- for-ignition period there is to be no means provided to by-pass one or more of the burner flame scanner systems unless the boiler is being manually fired.

3.1.3.4 Where boilers are fitted with an automatic ignition system, and where residual fuel oil is used, means are to be provided for lighting off the burners with igniters lighting properly heated residual fuel oil. Alternatively, the burners may be lighted off with a light oil used as a pilot to ignite residual fuel oil. If all burners experience a flame failure, the initial burner is to be brought back in to automatic service only in the low-firing position. To avoid the possibility of false indication due to the failure of the flame scanner in the 'flame-on' mode, the initial light-off burner is to be fitted with dual scanners or a scanner of the self checking type.

3.1.3.5 Off-limit conditions of burner primary-air pressure or atomizing- steam pressure are to be alarmed at the main control location.

3.1.4 Boiler alarms and indications

3.1.4.1 Action of the automatic fuel oil or burner shut-off arrangement is to be indicated by audible and visual alarms. Means are to be provided to silence the audible alarms without canceling the visual indicator. Alarm systems are to be designed so that the alarm cannot remain silenced when the monitored system has returned to normal running condition.

3.1.4.2 Each boiler is to be fitted with one high-water level alarm and two independent low water-level sensors and alarms. The low-low water-level alarm is to close the master fuel oil cut- off for the boiler, and is to be set to operate when the water level falls to a minimum safe level, but at a level not lower than that visible in the gauge glass. Water-level sensors are to be so located as to minimize the effect of roll and pitch or they are to be provided with a short- time delay (approximately 5 seconds) to prevent trip-out due to transients or to the vessel's motion.

3.1.4.3 The alarms for forced-draft failure and for low-water level are to be installed in association with limiting controls which are to prevent start-up and cause shut-down when unsafe firing conditions exist. Manual resetting of the control system is to be required before the boiler can be restarted.

3.1.4.4 The operation status of the boiler is to be indicated at the boiler by conventional instruments, gauges, lights or other devices to show the functional condition of fuel, air, feed-water and steam circuits of boiler. Monitoring instruments are also to be provided at the main control location.

3.1.5 Logic sequence for boiler controls

3.1.5.1 Where automatically started boilers are installed, they are to be provided with a programmed control to assure a safe cycle of operation upon initial starting and cycling between temperature and pressure limits.

3.1.5.2 The programmed control is to be designed to cycle the boiler in accordance with a predetermined sequence and, in addition to the automatic boiler purge required by 3.1.3.2, is to include the following:-

a) Ignition (spark coming on) is to precede the opening of the fuel valve;

b) Where it is necessary to cut burners in and out to handle the load on the boiler and the controls are provided to modulate the air-fuel ratio, the automatic boiler purge period is to start with the modulating control in the high firing position and the ignition is not to be turned on until the modulating control has returned to the low-firing position.

3.1.5.3 Means are to be provided for independent manual control at the boiler in the event of failure of the automatic control and sufficient conventional instrumentation is to be provided so that satisfactory operation of the boiler under independent manual control can be exercised for lengthy periods of time.

3.2 Propulsion machinery

3.2.1 General

3.2.1.1 The fitting of centralized automatic controls for propulsion engines and turbines is not to impair the effectiveness of the safety and speed control devices required by the applicable parts of Ch.4.

3.2.1.2 Plant or engine response to throttle control demands is to be tested during trials and after final adjustment to demonstrate that no part of the plant or engine is jeopardized by the rate at which the throttle is moved from one extreme position to the other. Overriding of the remote throttle control is to be possible only from the main control location.

3.2.1.3 The emergency tripping of propulsion machinery is to be possible at the main control location and at the machine site.

3.2.2 Steam turbines

3.2.2.1 The astern guardian valve is to open automatically as a result of a throttle trip or a maneuvering signal, such as the actuation of a specific switch or by movement of the throttle control in the maneuvering range. An alarm is to indicate any failure of the guardian valve to open.

3.2.2.2 If scoop circulation is provided for the main condenser, the main circulating pump is to be automatically started as required for satisfactory operation of the propulsion machinery.

3.2.2.3 In the event of low lubricating-oil pressure there is to be an automatic changeover to the standby lubricating-oil pump(s). The governor is to be arranged to shut off the steam to the ahead turbines upon failure of the lubricating-oil system.

3.2.2.4 Alarms at the main control location are to indicate low vacuum in the condenser, excessive bearing temperature and low lubricating-oil pressure. Remote throttle control-system power failure indication is to be provided at the main control and secondary control locations. An alarm is to be provided at the main control and secondary control locations to indicate that the propeller shaft has been stopped too long on a standby or stop maneuver.

3.2.3 Gas turbines

3.2.3.1 Automatic control arrangements for gas turbine modules are to be provided to regulate the quantity of fuel flowing to the burners during starting, over the operating power range and when stopping the gas turbine, in a controlled manner and with in the safe operating envelope of the gas turbine. The arrangement is to have provision to collect, condition and transmit both primary and secondary surveillance signals to the local and remote operating locations.

3.2.3.2 The control system is to be provided with control functions at the component level to allow direct manual control of the fuel flow and engine shut down in the event of failure of the electrical power supplies or a critical failure of the engine control system.

3.2.3.3 Each gas turbine module is to be provided control unit comprising of minimum instrumentation necessary for monitoring, control and surveillance for safe operation both local and remote positions with facility for transfer of control authority through a suitable selector switch.

The alarms and automatic shut down devices to be provided are as indicated in Pt.4, Ch.4, Table 3.13.1.

3.2.4 Diesel engines

3.2.4.1 In the event of the lubricating oil pressure dropping to a preset level, there is to be an automatic changeover to a standby lubricating oil pump or a reduction in main engine speed to a predetermined level. In the event of a loss of lubricating oil pressure engines are to stop automatically.

3.2.4.2 Engine alarms are to be in accordance with Table 1.9.1 and Table 1.9.2.

3.2.4.3 Precautions are to be taken to ensure that it is not possible to continuously run the main engine in a barred speed range.

3.3 Emergency internal combustion engines

3.3.1 These requirements apply to I.C. engines, which use distillate marine fuels covered by ISO 8217:2017, required to be immediately available in an emergency and capable of being controlled remotely or automatically operated. Information is to be submitted alongwith instructions to test alarm and safety shutdown systems demonstrating compliance with the requirements.

3.3.2 Alarms and safeguards are to be fitted in accordance with Table 3.3.2.

3.3.3 The safety shutdown and alarm systems are to be designed to ‘fail safe’. The characteristics of the ‘fail safe’ operation are to be evaluated on the basis not only of the system and its associated machinery, but also the complete installation, as well as the ship.

3.3.4 Regardless of the engine output, if shutdowns additional to those specified in Table 3.3.2 are provided except for the overspeed shutdown, they are to be automatically overridden when the engine is in automatic or remote control mode during navigation.

3.3.5 The alarm system is to function in accordance with Pt.5, Ch.22, with additional requirements that grouped alarms are to be arranged on the bridge. (See Pt.5, Ch.22, Cl.2.4.14).

3.3.6 In addition to the fuel oil control from outside the space, a local means of engine shutdown is to be provided.

3.3.7 Local indications of at least those parameters listed in Table 3.3.2 are to be provided within the same space as the I.C. engines and are to remain operational in the event of failure of the alarm and safety shutdown systems.

Table 3.3.2 : Alarms and safeguards

Parameter	Alarm activation(Y/N)	Shutdown with alarm(Y/N)
Fuel oil leakage from high pressure pipes (fuel injection pipes and common rails)	Y	N
Lubricating oil temperature[1]	Y, Alarm for high value	N
Lubricating oil pressure	Y, Alarm for low value	N
Activation of oil mist detection arrangements (or activation of the temperature monitoring systems or equivalent devices of : - the engine main and crank bearing oil outlet; or - the engine main and crank bearing)[2]	Y	N
Pressure or flow of cooling water[1]	Y, Alarm for low value	N
Temperature of cooling water (or cooling air)	Y, Alarm for high value	N
Overspeed activated[1]		Y

Note
1 : For engines having a power of or more than 220 kW.

2 : For engines having a power of more than 2250 kW or a cylinder bore of more than 300 [mm].

Section 4

Tests and Trials

4.1 Individual components

4.1.1 The manufacturers of control systems are to certify that the mechanical, electrical, and solid-state components utilized have been satisfactorily tested individually or by acceptable lot sampling to establish their suitability for the intended service including their exposure to the following conditions :-

a) shipboard ambient conditions, See Ch.1, Sec.1;

b) ship motion and vibration as per 2.5;

c) temperature range as per 2.2;

d) voltage and frequency tolerances as per 2.2;

e) pressure tests as per 4.1.2 and 4.2.

4.1.2 Hydraulic and pneumatic piping is to be subjected to pressure tests at 1.5 times the relief-device settings using the service fluid in hydraulic systems and dry air or dry inert gas for pneumatic systems as the testing media. Means are to be provided for the automatic removal of moisture from the piping before delivery to the pneumatic control system.

4.1.3 All electrical control and monitoring circuits are to be checked and proven free of unintentional grounds and short circuits, defective electrical or electronic elements.

4.2 Shop testing

4.2.1 Upon completion of the control consoles or sections or component assemblies of such consoles, the manufacturer is to carry out tests in the presence of the Surveyor to

demonstrate the satisfactory performance of all controls, instruments and alarms. The tests are to simulate all control and alarm functions given in the operational manual as far as practicable. The tests are to demonstrate satisfactory performance under the following conditions :

a) tests to be carried out at rated power supply at prevailing ambient air conditions;

b) loss of power to be simulated;

c) the voltage of the electrical supply to be varied plus or minus 10 per cent and where alternating current is used, the frequency is to be varied simultaneously to plus or minus 5 per cent. The controls are to function satisfactorily during these variations without faults. Where the components selected have been tested individually and are certified to function satisfactorily during these power variations, retesting of final assemblies will not be required;

d) all electrical circuits supplying power to servo-motors and actuators rated over 100 volts are to be subjected to dielectric-strength tests as required by Ch.8;

e) vibration tests required by 2.5 may be conducted in shops other than the plant of manufacture. When equipment is not energized during the test it is to perform satisfactorily after the vibration test;

f) variation of hydraulic or pneumatic system pressure $\pm$ 20 per cent of normal operating pressure, except that relief valve pressure need not be exceeded. The system is to operate satisfactorily at the relief valve pressure setting.

4.3 Trials on board

4.3.1 Upon completion of the installation, complete performance tests of all systems are to be carried out during dock trials and sea trials to demonstrate that the system will perform successfully during standby, maneuvering, steady conditions, and during transfer of controls.

4.3.2 A copy of the record of the trials is to be submitted for information. It is recommended that a copy of the record of the trials be kept on board for reference purposes for future adjustments/testing etc. The record should include details of all adjustments, calibrations, method of simulating failures and expected results due to these simulations. The record should be kept up to date at all times.

Section 5

Machinery Operated from a Centralised Control Station – ‘CCS’ Notation

5.1 General requirements

5.1.1 Where it is proposed to install control, alarm and safety systems to the following equipment, applicable features contained in Section 2 and Section 3 are to be incorporated in the system design.

- Propulsion machinery including essential auxiliaries;
- Electric power generating plant;
- Controllable pitch propellers;
- Oil fuel transfer and storage systems;
- Air compressors;
- Steam generating plant (boilers and their auxiliary equipment).

5.1.2 The arrangements are to be such that corrective actions can be taken at the control location in the event of machinery faults, e.g. stopping of machinery, starting of standby machinery, adjustments of operating parameters, etc. These actions may be effected by either remote manual or automatic control.

5.1.3 The controls, alarms and safeguards required by this chapter and Pt.5, Ch.22, as applicable, are to be provided at the control location. A fire detection system satisfying the requirements of Pt.5, Ch.22, Sec.2.13 is, also to be provided.

5.1.4 Additional requirements for controls, alarms and safeguards are given in 5.2.

5.2 Centralised control station for machinery

5.2.1 A centralized control station is to be provided at some suitable location, which satisfies the requirements of 5.2.2 to 5.2.7.

5.2.2 A system of alarm displays and controls is to be provided which readily ensures identification of faults in the machinery and satisfactory supervision and related equipment. The alarm and control systems are to satisfy the requirements of Section 1 and Section 2 as applicable.

5.2.3 Indication of all essential parameters necessary for the safe and effective operation of the machinery is to be provided, e.g. temperatures, pressures, tank levels, speeds, powers etc.

5.2.4 Indication of the operational status of running and standby machinery is to be provided.

5.2.5 At the centralized control station, means of communication with the bridge area, the accommodation for engineering personnel and if necessary, the machinery space are to be provided.

5.2.6 In addition to the communication required by 5.2.5, a second means of communication is to be provided between the bridge and the centralized control station. One of these means is to be independent of the main electrical power supply.

5.2.7 Arrangements are to be provided in the centralised control station so that the normal supply of electrical power may be restored in the event of failure.

Section 6

Computer Based Systems

6.1 Scope

6.1.1 These requirements apply to design, construction, commissioning and maintenance of computer-based systems where they depend on software for the proper achievement of their functions. These requirements apply to systems which provide control, alarm, monitoring, safety, or internal vessel communication functions that are subject to classification.

6.1.2 The requirements given in this section do not apply to navigation, radio communication systems and loading instruments/ stability computer.

6.1.3 Vessels complying with the requirements in IRS “Guidelines on Maritime Cyber Safety” will be assigned the following optional additional Class Notations:

a) **CyS-I** – Informed Cyber Safety. This notation is assigned where the organisation assesses the issues related to cyber safety, cyber risks and have documented policies. The organization implements minimum cyber safety practices.

b) **CyS-II** – Advanced Cyber Safety. This notation is assigned where the cyber safety policies and procedures are reviewed periodically for continual improvement. The organisation implements advanced cyber safety practices.

6.1.4 Ships and units complying with the requirements in Section 7 will be assigned additional class notation **CyS-E** (i.e. Essential Cyber Safety).

6.2 Definitions and abbreviations (for the purpose of this Section)

6.2.1 Stakeholders

6.2.1.1 **Owner**: The organization or person which orders the vessel in the construction phase or the organization which owns or manages the vessel in service.

6.2.1.2 **Systems Integrator**: A single organization or a person coordinating interaction between suppliers of systems and sub-systems on all stages of life cycle of computer-based systems in order to integrate them into a verified vessel-wide system of systems and to provide proper operation and maintenance of the computer-based systems. During the design and delivery phase, the Shipyard is the default systems integrator, during operations phase, the Owner is the default systems integrator.

6.2.1.3 **Supplier**: A generic term used for any organization or person that is a contracted or a subcontracted provider of services, system components, or software.

6.2.1.4 **System supplier:** An organization or person that is contracted or a subcontracted provider of system components or software under the coordination of the Systems integrator.

6.2.1.5 **Service supplier:** A person or company, not employed by IRS, who at the request of an equipment manufacturer, shipyard, vessel's owner or other client acts in connection with inspection work and provides services for a ship or an offshore unit such as measurements, tests or maintenance of safety systems and equipment, the results of which are used by Surveyors in making decisions affecting classification or statutory certification and services

6.2.2 Objects

6.2.2.1 The hierarchy and relationships of a typical computer based system are shown in Fig. 6.2.2.1.

6.2.2.2 **Vessel** means a ship or offshore unit where the computer-based system is to be installed.

6.2.2.3 **System** means a combination of components, equipment and logic which has a defined purpose, functionality and performance. In the context of this Section, a specific system is delivered by one system supplier.

6.2.2.4 **Sub-system** means an identifiable part of a system, which may perform a specific function or set of functions.

6.2.2.5 **Programmable Device** means a physical component where software is installed.

6.2.2.6 **Software component**means a module is a standalone piece of code that provides specific and closely coupled functionality.

6.2.2.7 **Computer-based system (CBS):** A programmable electronic device, or interoperable set of programmable electronic devices, organized to achieve one or more specified purposes such as collection, processing, maintenance, use, sharing, dissemination, or disposition of information. CBSs onboard include Information Technology (IT) and Operational Technology (OT) systems. A CBS may be a combination of subsystems connected via network. Onboard CBSs may be connected directly or via public means of communications (e.g. Internet) to ashore CBSs, other vessels' CBSs and/or other facilities.

6.2.2.8 **Software master files:** The computer-files that constitute the original source of the software. For custom-made software this may be readable source-code files, and for Commercial off-the-shelf (COTS) software it may be different forms of binary files.

6.2.2.9 **Simulation test:** Monitoring, control, or safety system testing where the equipment under control is partly or fully replaced with simulation tools, or where parts of the communication network and lines are replaced with simulation tools.

6.2.2.10 **Parameterization:** To configure and tune system and software functionality by changing parameters. It does not usually require computer programming and is normally done by the system supplier or a service provider, not the operator or end-user.

6.2.2.11 **Failure mode description:** A document describing the effects due to failures in the system, not failures in the equipment supported by the system. The following aspects are to be covered:

- list of failures which are subject to assessment, with
- description of the system response to each of the above failures
- comments to the consequence of each of these failures

6.2.2.12 **Black-box test methods:** Verification of the functionality, performance, and robustness of a system, sub-system or component by only manipulating the inputs and observing the outputs. This does not require any knowledge of the system's inner workings and focuses only on the observable

behaviour of the system/ component under test in order to achieve the desired level of verification.

6.2.3 Abbreviations

6.2.3.1 Abbreviations used in this Section:

Cat I, Cat II and Cat III are category one, category two and category three systems respectively, as defined in Table 6.3.1

COTS : Commercial off-the-shelf
FAT : Factory acceptance test
FMEA : Failure mode and effect analysis
IT : Information technology
OT : Operational technology
PMS : Planned maintenance system
SAT : System acceptance test
SOST : System of systems test
SSLS : Ship software logging system

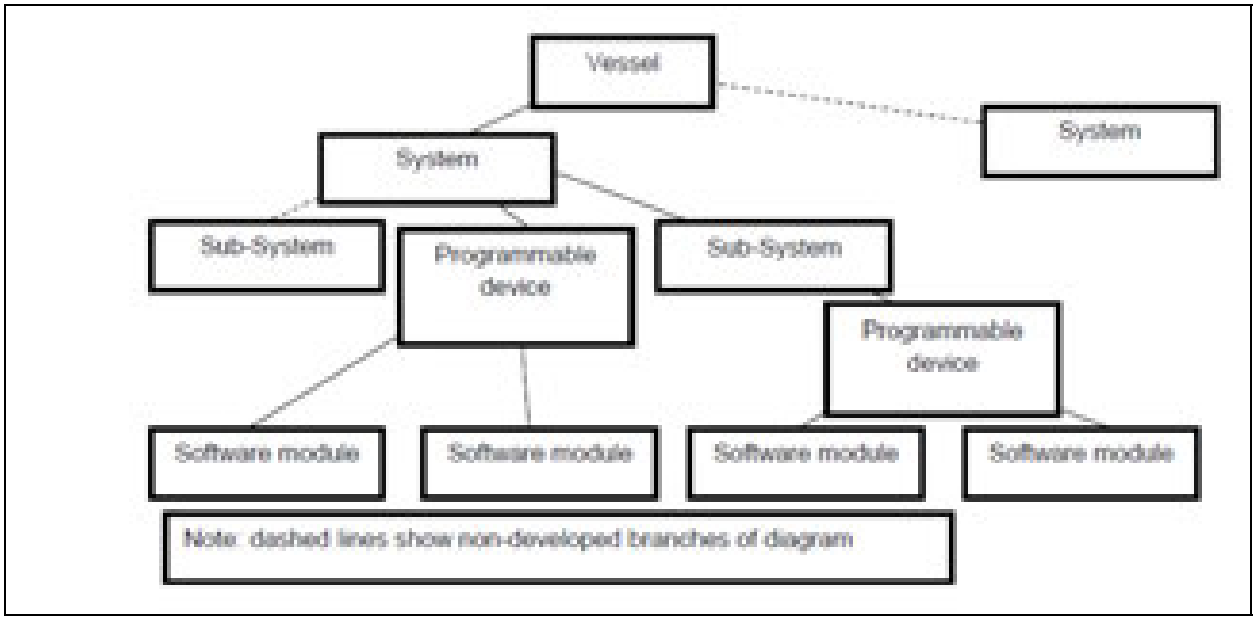


Fig 6.2.2.1 : Illustrative System Hierarchy

6.3 System Categories

6.3.1 The categorization of a system in the context of this Section is based on the potential severity of consequences if the system serving the function fails as shown in Table 6.3.1

Table 6.3.1 : System Categories

Category	Failure Effects	Typical System Functionality
I	Those systems, failure of which will not lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.	- Monitoring, informational and administrative functions
II	Those systems, failure of which could eventually lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment	- Vessel alarm, monitoring and control functions which are necessary to maintain the vessel in its normal operational and habitable conditions
III	Those systems, failure of which could immediately lead to dangerous or catastrophic for human safety, safety of the vessel and/or threat to the environment.	- Control functions for maintaining the vessel's propulsion and steering - Safety functions

6.3.2 Scope of Classification

6.3.2.1 Category I systems are normally not subject to verification by IRS, as failure of these systems are not to lead to dangerous situations. However, information pertinent to category I systems is to be provided upon request to determine the correct category or ensure that they do not influence the operation of systems in category II and category III.

6.3.3 System category examples

6.3.3.1 The category of a system is to be evaluated in the context of the specific vessel in question; thus, the categorization of a system may vary from one vessel to the next. The examples of categories given below are for guidance only. For determining the categorization of systems for a specific vessel, 6.4.3.2 may be referred.

6.3.3.2 Examples of category I systems: Fuel monitoring system, maintenance support system, diagnostics and troubleshooting system, closed circuit television, cabin security, entertainment system, fish detection system.

6.3.3.3 Examples of category II systems: Fuel oil treatment system, alarm and monitoring and safety systems for propulsion and auxiliary machinery, Inert gas system, control, monitoring and safety system for cargo containment system.

6.3.3.4 Examples of category III systems: Propulsion control system, steering gear control system, electric power system (including power management system), dynamic positioning system (IMO classes 2 and 3).

Note: The examples listed above are not exhaustive.

6.4 Requirements for development and certification of computer-based systems

6.4.1 General requirements

6.4.1.1 Life cycle approach with appropriate standards

.1 A global top to bottom approach is to be undertaken in the design and development of both hardware and software the integration in sub-systems, systems, and spanning the complete system lifecycle. This approach is to be accomplished according to standards as listed in this sub-section or other national/ international standards. This requirement will be verified as a part of the quality management system verification described in 6.4.1.2.

6.4.1.2 Quality management System

.1 Systems integrators and system suppliers are to undertake development of computer-based systems for category II and category III, complying with a recognized quality standard such as ISO 9001; also incorporating principles of ISO 90003. The quality management system may be verified by two alternative means:

- IRS confirming that the quality management system is certified as compliant to the recognized standard by an organization with accreditation under a national accreditation scheme.

- IRS confirming compliance to a standard through a specific assessment of quality management system. The documentation requirements will be defined per case.

.2 The quality management system is to have a minimum of the following topics, applicable for both category II and category III systems, as shown in Table 6.4.1.2

Table 6.4.1.2: Quality Management System			
Area		**Role**	
Sr. No.	**Topic**	**System supplier**	**Systems integrator**
1	Responsibilities and competency of the staff	X	X
2	The complete lifecycle of delivered software and of associated hardware	X	X
3	Specific procedure for unique identification of a computer-based system, it's components and versions.	X	
4	Creation and update of the vessel's system architecture		X
5	Organization set in place for acquisition of software and related hardware from suppliers	X	X
6	Organization set in place for software code writing and verification	X	
7	Organization set in place for system validation before integration in the vessel	X	
8	Specific procedure for conducting and approving of systems at FAT and SAT	X	X
9	Creation and update of system documentation	X	
10	Specific procedure for software modification and installation on board the vessel, including interactions with shipyard and owner	X	X
11	Specific procedures for verification of software code	X	
12	Procedures for integrating systems with other systems and testing of the system of systems for the vessel	X	X
13	Procedures for managing changes to software and configurations before FAT	X	
14	Procedures for managing and documenting changes to software and configurations after FAT	X	X
15	Checkpoints for the organization's own follow-up of adherence to the quality management system	X	X

6.4.2 **Requirements for the system supplier:**

6.4.2.1 Define and follow a quality plan

.1 The system supplier is to document that the quality management system is applied for the design, construction, delivery, and maintenance of the specific system to be delivered. All applicable items described in 6.4.1.2 (for the system supplier role) are to be demonstrated to exist and being followed, as relevant and applicable.

.2 For verification of the above requirements for category II and III, a quality plan is to be available during survey (FAT) or submitted for information upon request. For category I items, no documentation is required.

6.4.2.2 Unique identification of systems and software

.1 A method for unique identification of a system, its different software components and different revisions of the same software component is to be applied. The method is to be applied throughout the lifecycle of the system and the software.

.2 The system is to provide means to identify its name, version, identifier, and manufacturer. It is recommended that the system can automatically report the status of its software to a ship software logging system (SSLS) as specified in the international standard ISO 24060.The documentation of the method is typically a part of the quality management system as indicated in 6.4.1.2.

.3 Application of the identification system will be verified by the Surveyor as a part of the FAT (see 6.4.2.7) and SAT (see 6.4.3.5) for category II and III. This verification is not required for category I.

6.4.2.3 System description

.1 The system's specification and design are to be documented in a system description. In addition to serving as a specification for the detailed design and implementation, the purpose of the system description is to document that the entire system-delivery is according to the specifications and in compliance with applicable rule requirements. The system description is to contain information of the following:

- Purpose and main functions, including any safety aspects
- System category as defined
- Key performance characteristics
- Compliance with the technical requirements and Rules
- User interfaces/mimics
- Communication and Interface aspects
  - Identification and description of interfaces to other vessel systems
- Hardware-arrangement related aspects:
  - Network-architecture/topology, including all network components like switches, routers, gateways, firewalls etc.
  - Internal structure with regards to all interfaces and hardware nodes in the system (e.g., operator stations, displays, computers, programmable devices, sensors, actuators, I/O modules etc.)
  - I/O allocation (mapping of field devices to channel, communication link, hardware unit, logic function)
  - Power supply arrangement
  - Failure mode description

Note: The information listed above is collectively referred to as 'system description' in this Section. It may however be divided into several different documents and models.

.2 For verification of the above requirements for category I, the system description documentation is to be submitted for information, upon request. For category II and III, the system description documentation is to be submitted for approval.

6.4.2.4 Environmental compliance of hardware components

.1 Evidence of environmental type testing according to IRS *Classification Note "Type Approval of Electrical Equipment used for Control, Monitoring, Alarm and Protection Systems for Use in Ships"* regarding hardware elements included in the system and sub-systems are to be submitted to IRS.

.2 The above verification is not mandatory for category I systems. Reference to type approval certificate or other evidence of type testing is to be submitted for information, upon request. For category II and III, reference to type approval certificate or other evidence of type testing is to be submitted for information.

6.4.2.5 Software code creation, parameterization, and testing

.1 The software created, modified, or configured for a particular project is to have the quality assurance activities assessed according to the selected standard(s) as indicated in 6.4.2.1. The quality assurance activities are to be performed on several levels of the software structure and include both custom-made software and configured components (e.g., software libraries) as appropriate. The following aspects of the software are to be verified, as a minimum, based on black-box methods:

- Correctness, completeness and consistency of any parameterization and configuration of software components
- Intended functionality
- Intended robustness

.2 For components in systems of category II and III, the scope, purpose, and results of all performed reviews, analyses, tests, and verification activities are to be documented in test reports.

Note: Some of the methods utilized in this activity are referred to as "software unit test" or "developer test" and may also include verification methods like code-reviews and static- or dynamic code analysis.

.3 Software test reports for category II and III systems, are to be submitted for information, upon request. No documentation is required for category I systems.

6.4.2.6 Internal system testing before FAT

.1 The system is to be tested as far as practicable before the FAT. The main purpose of the system test is for the system supplier to verify that the entire system delivery is according to the specifications, approved documentation and in compliance with applicable requirements. Further, the system supplier can verify that the system is completed and ready for the FAT. The testing is to verify the following aspects of the system:

- Functionality
- Effect of faults and failures (including diagnostic functions, detection, alerts response)
- Performance
- Integration between software and hardware components
- Human-machine interfaces
- Interface to other systems

.2 Faults are to be simulated as realistically as possible to demonstrate appropriate system fault detection and system response. Some of the testing may be performed by utilizing simulators and replica hardware. The test environment is to be documented, including a description of any simulators, emulators, test-stubs (programs that simulate behaviour of software components or modules), test-management tools, or other tools affecting the test environment and its limitations. Test cases and test results are to be documented in test programs and test reports respectively.

.3 Internal system test report is to be made available during FAT or submitted for information, upon request for category II and III systems. No documentation is required for category I systems.

6.4.2.7 Factory acceptance testing (FAT) before installation on board

.1 The main purpose of the FAT is to demonstrate that the system is completed and compliant with applicable Rule requirements. The FAT test program is to cover test items from the internal system test indicated in 6.3.1.3.6.

.2 Network testing is to be carried out for category II and III systems to verify the network resilience requirements, as follows:

- A single failure in data link is not to cause loss of vessel functions of category III. Any effect of such failures is to meet the principle of fail-to-safe for the vessel function(s) being served.
- For vessel-functions of category II and III, any loss of functionality in the remote-control system is to be compensated by local/ manual means.
- The data link is to have means to prevent or cope with excessive communication rates.
- Data links are to be self-checking, detecting failures or performance issues on the link itself and data communication failures on nodes connected to the link.
- Detected failures are to initiate an alarm.

If agreed by all stakeholders, the network testing may be performed as part of the system test onboard the vessel.

.3 The FATs are to be performed with the project specific software operating on the actual hardware components to be installed on board, with necessary means for simulation of functions and failure responses, however other solutions such as replica hardware or simulated hardware (emulators) may be agreed with IRS. For each test case, it is to be noted if the test passed or failed, and the test results are to be documented in a test report. The test report is also to contain a list of the software (including software versions) that were installed in the system when the test was executed.

Note: For complex systems there may be a large difference in scope between the “Internal system testing before FAT” activity and the FAT, while for some systems the scope may be identical.

.4 FATs are not required for category I systems. For category II and III systems, the FAT program is to be approved by IRS before the test is executed. The FAT execution is to be witnessed by the Surveyor and the report is to be submitted for information. Additional FAT documentation including user manuals and internal system report is to be made available during FAT or submitted upon request for information.

6.4.2.8 Secure and controlled software installation on the vessel

.1 The initial installation and subsequent updates of the software components of the system are to be carried out according to a management of change procedure which has

been agreed between the system supplier and the systems integrator. The management of change procedure is to comply with the requirements in 6.7.

.2 The management of change procedure is to be submitted for Category II and III systems, upon request, for information. Cyber security measures are to be incorporated as specified in Sections 7 & 8.

6.4.2.9 Summary of documentation to be submitted

.1 The summary of documentation to be submitted by the system supplier is listed in Table 6.4.2.9

Table 6 4.2.9: Documentation to be submitted by the system supplier

Item		System category		
Clause reference	**Document**	**Cat I**	**Cat II**	**Cat III**
6.4.2.1	Quality plan	-	FI on req.	FI on req.
6.4.2.3	System description	FI on req.	AP	AP
6.4.2.4	Environmental compliance	FI on req.	FI	FI
6.4.2.5	Software test reports	-	FI on req.	FI on req.
6.4.2.6	System test reports	-	FI on req.	FI on req.
6.4.2.7	FAT program	-	AP	AP
6.4.2.7	FAT report	-	FI	FI
6.4.2.7	Additional FAT docs. (e.g., user manual, etc.)	-	FI on req.	FI on req.
6.4.2.8	Management of change procedure	-	FI on req.	FI on req.

Note:

AP: Approval
FI: For Information
"- ": No requirement
On req.: Upon request by IRS

6.4.3 **Requirements for the systems integrator**

For the purpose of this Section, the Shipyard is considered as the 'systems integrator' in the development and delivery phase unless another organization or person is appointed by the Shipyard.

6.4.3.1 Quality plan

.1 The systems integrator is to document that a quality management system is applied for the installation, integration, completion, and maintenance of the systems to be installed on board. All applicable items indicated in 6.4.1.2 (for the systems integrator role) are to be demonstrated to exist and being followed, as relevant.

.2 The quality plan is to be made available during survey (at SAT/ SOST) or upon request, submitted for information, for category II and III systems.

6.4.3.2 Categorization of the system

.1 For each system being delivered to a particular vessel, it is to be decided which category the system falls under based on the failure effects of the system as specified in 6.3. The category for a specific system is to be conveyed to the relevant system supplier. IRS may decide that a risk-assessment is needed to verify the proper system category.

.2 The categorization for the different systems (category I, II and III) is to be documented and submitted for approval, upon request.

6.4.3.3 Risk assessment of the system

.1 When requested by IRS, risk assessment of a specific system in the context of a specific vessel is to be performed and documented in order to determine the applicable category for the system.

Note: Standards such as *IEC/ ISO 31010 "Risk Management – Risk Assessment techniques"* may be applied in order to determine method of risk assessment.

.2 The risk assessment report is to be submitted for approval for Category I, II and III systems, upon request.

6.4.3.4 Vessel's system-architecture

.1 The system of systems (SoS) is to be specified and documented. This architecture specification provides the basis for categorization and development of different integrated systems by allocating functionality to the individual systems and by identifying the main interfaces between the systems. The SoS is also to serve as a basis for the testing of the integrated systems on the vessel level as indicated in 6.4.3.6. The vessel's system architecture, is to at least to contain a description of the following:

- Overview of the total systems architecture (the system of systems)
- Each system's purpose and main functionality
- Communication and interface aspects between different systems

.2 The vessel's system architecture is to be submitted for information, upon request, for category I, II and III systems.

Note: Sec. 7.6.2.1 may be referred for inclusion of security zones and conduits, as applicable, in the architecture diagram.

6.4.3.5 System acceptance test (SAT) onboard the vessel

.1 The main purpose of the system acceptance test (SAT) is to verify the system functionality, after installation and integration with the applicable machinery/electrical/process systems on board including possible interfaces with other control and monitoring systems.

.2 For each test-case it is to be noted if the test passed or failed, and the test-results are to be documented in a test report. The test report is also to contain a list of the software (including software versions) that were installed in the system when the test was executed.

.3 For Category II and III systems, the SAT program is to be submitted for approval before the test is executed. The SAT execution is to be witnessed by the Surveyor and the report is to be submitted for information.

6.4.3.6 Testing of integrated systems (System of systems test - SOST)

.1 Integration tests are to be conducted after installation and integration of the different systems in its final environment on board. The purpose of the tests is to verify the functionality of the complete installation (system of systems) including all interfaces and inter-dependencies in compliance with requirements and specifications.
The testing is to at least verify the following aspects of the system of systems:

- The overall functionality of the interacting systems as a whole
- Failure response between systems
- Performance
- Human-machine interfaces
- Interfaces between the different systems

Note: For complex systems there may be a large difference in scope between the "System acceptance test (SAT) onboard the vessel" activity and the SOST, while for some systems the scope may be overlapping or identical. It is possible to combine the two activities into one when the test scope is similar.

.2 For Category II and III systems, the SOST program is to be submitted for approval before the test is executed. The SOST execution is to be witnessed by the Surveyor and SOST report is to be submitted for information.

6.4.3.7 Change management

.1 The systems integrator is to follow procedures for management of change to the system as indicated in 6.7.

.2 The management of change procedure is to be submitted for information, upon request, for category II and III systems.

6.4.3.8 Summary of documentation to be submitted

.1 The summary of documentation to be submitted by the system integrator is listed in Table 6.4.3.8.

Table 6.4.3.8: Documentation to be submitted by the system integrator

Item		System category		
Clause reference	**Document**	**Cat I**	**Cat II**	**Cat III**
6.4.3.1	Quality plan	-	FI on req.	FI on req.
6.4.3.2	List of system categorizations	AP on req.	AP on req.	AP on req.
6.4.3.3	Risk assessment report	AP on req.	AP on req.	AP on req.
6.4.3.4	Vessel's system architecture	FI on req.	FI on req.	FI on req.
6.4.3.5	SAT program	-	AP	AP
6.4.3.5	SAT report	-	FI	FI
6.4.3.6	SOST program	-	AP	AP
6.4.3.6	SOST report	-	FI	FI
6.4.3.7	Change management procedure for software	-	FI on req.	FI on req.

Note:

AP: Approval
FI: For Information
"- ": No requirement
On req.: Upon request by IRS

6.4.4 Summary of activities

6.4.4.1 The summarized list of activities such as test witnessing and survey related to computer-based systems is given in Table 6.4.4.

Table 6.4.4 : Summary of test witnessing and survey					
Item		**Responsible role**	**System category**		
Clause reference	**Activity**		**Cat I**	**Cat II**	**Cat III**
6.4.2.7	FAT witnessing	System supplier	-	X	X
6.4.3.5	SAT witnessing	System integrator	-	X	X
6.4.3.6	SOST witnessing	System integrator	-	X	X
6.7.12	Verification of changes	System integrator	-	X	X
Note: **"X":** Witnessing required. **"-":** Witnessing Not required					

6.5 Approval of system and components

6.5.1 System certification

6.5.1.1 Computer-based systems that are essential to accomplish vessel functions of category II or category III as defined in 6.3 are to be delivered with a vessel-specific certificate. The objective of the vessel-specific system certification is to confirm that design and manufacturing of the system has been completed and that the system complies with applicable Rules. Vessel-specific system certification consist of two main activities:

- Assessment of vessel-specific documentation indicated in 6.4.2 and 6.7.
- Survey and testing of the system to be delivered to the vessel mentioned in 6.4.2.7.

Alternative Certification Scheme (ACS) may be accepted provided that the requirements are met, and that the system is provided with a vessel-specific certificate.

6.5.2 Type approval of computer-based systems

6.5.2.1 Computer-based systems that are routinely manufactured and include standardized software functions may be type approved in accordance with IRS guidelines on *Certification of Software for Computer Based Control Systems*. Hardware is to be documented according to the requirements indicated in 6.4.2.4.

The type approval consists of two main verification activities:

- Assessment of type-specific documentation
- Survey and testing of the standardized functions

Type approval will normally not exempt vessel specific system certification since vessel-specific functions, parameter configurations and installation elements require vessel specific verification and certification.

6.6 Maintenance of computer-based systems

6.6.1 Requirements for the Vessel Owner

6.6.1.1 Responsibilities

.1 In the context of this Section, the vessel Owner is considered to be the systems integrator in the operations phase unless another organization or person is appointed by the Owner.

.2 IRS is to be informed by the Owner, in a timely manner about the appointed systems integrator, who is responsible for implementing any changes to the systems in conjunction with the system supplier(s).

6.6.2 Requirements for the Systems Integrator

6.6.2.1 Change management

.1 The systems integrator is to ensure that necessary procedures for software and

hardware change management exist on board, and that any software modifications/ upgrades are performed according to the procedure(s) specified in 6.7.

.2 Changes to computer-based systems in the operational phase are to be recorded. The records are to contain information about the relevant software versions and other relevant information as indicated in 6.7.11.

.3 The above would be verified by IRS for Category II and III systems, in accordance with the requirements specified in 6.7.12.

6.6.3 Requirements for the System Supplier

6.6.3.1 Change management

.1 The system supplier is to follow procedures for maintenance of the system including procedures for management of change as specified in 6.7.

.2 The above would be verified by IRS for Category II and III systems, in accordance with the requirements specified in 6.7.12.

6.6.3.2 Testing of changes before installation onboard

.1 The system supplier is to make sure that the planned changes to a system have passed relevant in-house tests before the change is made to systems on board.

.2 The above would be verified by IRS for Category II and III systems, in accordance with the requirements specified in 6.7.12.

6.7 Management of change

6.7.1 General

.1 Requirements for the management of change throughout the lifecycle of a computer-based system are specified in this sub-section. Different procedures for the management of change may be defined for specific phases in a system's lifecycle as the different phases typically involve different stakeholders. Requirements for verification by IRS are specified in 6.7.12.

6.7.2 Documented change management procedures

.1 The organization in question is to have defined and documented change management procedures applicable for the computer-based system in question covering both hardware and software. After FAT, the system supplier is to manage all changes to the system in accordance with the procedure. Examples could be qualification of new versions of acquired software, new hardware, modified control logic, changes to configurable parameters, etc.

.2 The procedure(s) are to at least describe the activities listed in 6.7.3 through 6.7.11. The outcome of the impact analysis in 6.7.8 will determine to what extent the activities in 6.7.3 to 6.7.12 are to be performed. Change records (6.7.11) are always to be produced.

6.7.3 Agreement between relevant stakeholders

.1 The management of change process is to be coordinated and agreed between the relevant stakeholders along the different stages of the lifecycle of the computer-based system.

Note:
Typically, the management of change addresses at least three different stages:

- Development and internal verification before FAT; involving the system supplier and sub-suppliers.
- From FAT to handover of the vessel to the owner; involving the system supplier, the systems integrator, IRS, and the owner.
- In operation; involving the system supplier, service suppliers, the owner, and IRS.

6.7.4 Change Management of Approved Software

.1 If changes are required to a system after it has been approved by applicable stakeholders (typically the systems integrator and IRS at FAT) the modifications are to follow defined change management procedures.

6.7.5 Unique identification of system and software versions

.1 The system supplier is to make sure that each system and software version is uniquely identifiable as indicated in 6.4.2.2.

6.7.6 Handling of software master files

.1 There are to be defined mechanisms for handling of the files that constitute the master-

files for a software component. Personnel authorities are to be clearly defined along with the tools and mechanisms used to ensure the integrity of the master files.

6.7.7 Backup and restoration of onboard software

.1 It is to be clearly defined how to perform backup and restoration of the software components of a computer-based system onboard the vessel.

6.7.8 Impact analysis

.1 Before a change to the system is made, an impact analysis is to be performed in order to:

- Determine the criticality of the change.
- Determine the impact on existing documentation.
- Determine the needed verification and test activities.
- Determine the need to inform other stakeholders about the change.
- Determine the need to obtain approval from other stakeholders (e.g., IRS and or Owner) before the change is made.

6.7.9 Roll-back in case of failed software changes

.1 When maintenance includes installation of new versions of the software in the system, it is to be possible to perform a rollback of the software to the previously installed version with the purpose of returning the system to a known, stable state. Roll-backs are to be documented and analyzed to find and eliminate the root cause.

6.7.10 Verification and validation of system changes

.1 Modifications are to be verified before being installed onboard.

.2 After installation, the modification(s) are to be verified onboard according to a documented verification program containing:

- Verification that the new functionalities and/or improvements have had the intended effect.
- Regression test to verify that the modification has not had any negative effects on functionality or capabilities that was not expected to be affected

6.7.11 Change records

.1 Changes to systems and software are to be documented in change records to allow for visibility and traceability of the changes. The change records are to contain at least the following items:

- The purpose for a change
- A description of the changes and modifications
- The main conclusions from the impact analysis, as in 6.7.8.
- The identity and version of any new system or software version(s), as in 6.7.5.
- Test reports or tests summaries, as in 6.7.10.

.2 Documentation of the changes to software may be recorded in the planned maintenance system (PMS), in a software registry or equivalent.

6.7.12 Verification of change management

6.7.12.1 In operation (vessel in service) phase

.1 The verification regarding the management of change in operation will be generally performed during the annual survey of the vessel. Procedures for management of change and relevant change records, as in 6.7.11 are to be made available at the time of survey for verification by the Surveyor.

.2 In cases where the change requires prior approval of IRS, the relevant procedures and documentation for the subject change may be verified at that time.

6.7.12.2 During new construction

.1 The verification of management of change in the new construction phase is to be divided into two phases; Procedures are to be verified as a part of the verification of the quality management system (6.4.1.2), while project specific implementation of the procedures is to be verified during FAT (6.4.2.7), SAT (6.4.3.5) and SOST (6.4.3.6).

6.8 Technical requirements for computer-based systems

6.8.1. Compliance with the requirements in this sub-section is to be documented in the design documentation, as indicated in 6.4.2.3 and verified through the verification activities as detailed .

6.8.2 Reporting of system and software identification and version

6.8.2.1 System identification

The system is to provide means to identify its name, version, and manufacturer. It is recommended that the system should automatically report the status of its software to a ship software logging system (SSLS) as specified in the international standard ISO 24060.

6.8.3 Data links

6.8.3.1 General Requirements for Category II and III systems

.1 Loss of a data link is to be specifically addressed in risk assessment analysis/ Failure Mode and Effect Analysis (FMEA).

..2 A single failure in data link is not to cause loss of vessel-functions of category III. Any effect of such failure is to comply with the principle of fail-to-safe for the vessel-function(s) being served.

.3 For vessel-functions of category II and III, any loss of functionality in the remote-control system is to be compensated for by local/ manual means.

.4 The data link is to have means to prevent or cope with excessive communication rates..

.5 Data link is to be self-checking, detecting failures on the link itself and data communication failures on nodes connected to the link. Detected failures are to initiate an alarm.

.6 The network cables are to be installed to prevent EMI effects from power cables.

.7 Network cables are to be designed and tested as per National/ International Standards.

.8 Network components are to be suitable for the intended area of operation onboard.

Guidance note: The network devices (Hubs, routers, switches etc.) are to be installed in protective cases unless the ingress protection of the device meets the requirements. The protective cases are to be designed for the ambient environment and ease of operation and maintenance of the device.

6.8.3.2 Specific requirements for wireless data links

.1 Category III systems are not to use wireless data links unless specifically considered by IRS on the basis of an engineering analysis carried out in accordance with an International or National Standard acceptable to IRS.

.2 Other categories of systems may use wireless data links with the following requirements

a) Recognized international wireless communication system protocols are to be employed, incorporating:

- *Message integrity*. Fault prevention, detection, diagnosis, and correction so that the received message is not corrupted or altered when compared to the transmitted message.

- *Configuration and device authentication*. Shall only permit connection of devices that are included in the system design.

- *Message encryption*. Protection of the confidentiality and or criticality of the data content.

- *Security management*. Protection of network assets, prevention of unauthorized access to network assets.

b) The internal wireless system within the vessel is to comply with the radio frequency and power level requirements of International Telecommunication Union and flag state requirements. Consideration should be given to system operation in the event of port state and local regulations that pertain to the use of radio-frequency transmission prohibiting the operation of a wireless data communication link due to frequency and power level restrictions.

c) For wireless data communication equipment, tests during harbour and sea trials are to be conducted to demonstrate that radio-

frequency transmission does not cause failure of any equipment and does not itself-fail as a result of electromagnetic interference during expected operating conditions.

6.8.4 Verification of technical requirements

6.8.4.1 The implementation of the technical requirements specified in 6.8 will be verified as part of documentation review of the system description in 6.4.2.3, by the Surveyor during FAT in 6.4.2.7 and SAT in 6.4.3.5

6.9 Performance Management Systems

6.9.1 The requirements in this sub-section are applicable to vessels, which may be installed with additional sensors, and/ or computer based systems, for performance monitoring and analysis.

6.9.2 A performance management system is a monitoring system for a function that is supplementary or additional to those required by the Rules and, is installed to provide operational or decision support. The support provided by the performance management system may either be guidance or recommendations. Examples of such systems could be fuel optimization, vessel performance monitoring systems, condition monitoring, and similar other systems.

6.9.3 In general, performance management systems are arranged as follows:

a) with dedicated operator stations and servers, in which case the loss of hardware will not affect the functions required to be operable as per Rules; or

b) without dedicated input/ output signals, but utilizing data through interface with communication links.

6.9.4 The hardware and associated software are to meet relevant requirements specified in the Rules.

6.9.5 When the performance management system is integrated with multiple control systems, they are to be arranged to meet the requirements of Integrated Systems (Refer Section 7) and are to have sufficient redundancy and/ or segregation, so as to prevent loss of the control of essential functions upon a single failure.

6.9.6 A failure in a performance management system, including the interfaces and communication network, is not to propagate to the interfaced control, monitoring and safety systems.

6.9.7 Cyber risk is to be evaluated and performance management systems are to be protected against cyber incidents.

6.9.8 Plans for the proposed systems including the effect of loss of such systems when they are interfaced with control, monitoring and safety systems are to be submitted for review.

6.9.9 The scope of approval and certification requirements are to be agreed for each vessel for all such additional systems, including testing at manufacturer works and on-board.

Section 7

Cyber Resilience of Ships

7.1 Scope

7.1.1 This Section provides a minimum set of requirements for cyber resilience of ships.

7.1.2 The requirements in this Section are applicable to the following:

- Passenger ships (including passenger high-speed craft) engaged in international voyages
- Cargo ships of 500 GT and upwards engaged in international voyages
- High speed craft of 500 GT and upwards engaged in international voyages
- Mobile offshore drilling units of 500 GT and upwards
- Self-propelled mobile offshore units engaged in construction (i.e. wind turbine installation maintenance and repair, crane units, drilling tenders, accommodation, etc.)

7.1.3 The requirements in this Section may be used as guidance for the following:

- Ships of war and troopships
- Cargo ships less than 500 GT
- Vessels not propelled by mechanical means
- Wooden ships of primitive build
- Passenger yachts (passengers not more than 12)
- Pleasure yachts not engaged in trade
- Fishing vessels
- Site specific offshore installations (i.e. FPSOs, FSUs, etc.)

7.1.4 Ships and units complying with the requirements in this Section will be assigned additional class notation **CyS-E.**

7.1.5 Minimum requirements for cyber resilience of on-board systems and equipment are specified in Section 8.

7.2 Application

Requirements of this Section are applicable to:

a) Operational Technology (OT) systems onboard ships, i.e. those Computer Based Systems (CBSs) using data to control or monitor physical processes that can be vulnerable to cyber incidents and, if compromised, could lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.

In particular, the CBSs used for the operation of the following ship functions and systems, if present onboard, are to be considered:

- Propulsion
- Steering
- Anchoring and mooring
- Electrical power generation and distribution
- Fire detection and extinguishing systems
- Bilge and ballast systems, loading computer
- Watertight integrity and flooding detection
- Lighting (e.g. emergency lighting, low locations, navigation lights, etc.)
- Any other safety system(s) whose disruption or functional impairment may pose risks to ship operations (e.g. emergency shutdown system, cargo safety system, pressure vessel safety system, gas detection system, etc.)

In addition, the requirements are also applicable to the following systems:

- Navigational systems required by Statutory Regulations

- Internal and external communication systems required by the Rules and Statutory Regulations

For navigation and radiocommunication systems, the application of IEC 61162-460 or other equivalent standards in lieu of the required security capabilities in section 8.4 may be accepted by IRS, on the condition that requirements in section 6 are complied with.

b) Any Internet Protocol (IP)-based communication interface from CBSs to other systems. Examples of such systems are, including but not limited to, the following:

- passenger or visitor servicing and management systems
- passenger-facing networks
- administrative networks
- crew welfare systems
- any other systems connected to OT systems, either permanently or temporarily (e.g. during maintenance).

The cyber incidents considered in this Section are events resulting from any offensive manoeuvers that target OT systems onboard ships as defined in section 7.4.

7.3 System Category

System categories are defined in Section 6 based on the consequences of a system failure to human safety, safety of the vessel and/or threat to the environment.

7.4. Definitions

7.4.1 Definition and abbreviations (for the purpose of this Section):

Attack Surface: The set of all possible points where an unauthorized user can access a system, cause an effect on or extract data from. The attack surface comprises two categories: digital and physical. The digital attack surface encompasses all the hardware and software that connect to an organization's network. These include applications, code, ports, servers and websites. The physical attack surface comprises all endpoint devices that an attacker can gain physical access to, such as desktop computers, hard drives, laptops, mobile phones, removable drives and carelessly discarded hardware.

Authentication: Provision of assurance that a claimed characteristic of an entity is correct.

Compensating countermeasure: An alternate solution to a countermeasure employed in lieu of or in addition to inherent security capabilities to satisfy one or more security requirements.

Computer Based System (CBS): A programmable electronic device, or interoperable set of programmable electronic devices, organized to achieve one or more specified purposes such as collection, processing, maintenance, use, sharing, dissemination, or disposition of information. CBSs onboard include IT and OT systems. A CBS may be a combination of sub-systems connected via network. Onboard CBSs may be connected directly or via public means of communications (e.g. Internet) to ashore CBSs, other vessels' CBSs and/or other facilities.

Cyber incident: An event resulting from any offensive manoeuver, either intentional or unintentional, that targets or affects one or more CBS onboard, which actually or potentially results in adverse consequences to an onboard system, network and computer or the information that they process, store or transmit, and which may require a response action to mitigate the consequences. Cyber incidents include unauthorized access, misuse, modification, destruction or improper disclosure of the information generated, archived or used in onboard CBS or transported in the networks connecting such systems. Cyber incidents do not include system failures.

Cyber resilience: The capability to reduce the occurrence and mitigating the effects of cyber incidents arising from the disruption or impairment of operational technology (OT) used for the safe operation of a ship, which potentially lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.

Essential services: Services for propulsion and steering, and safety of the ship. Essential services comprise "Primary Essential Services" and "Secondary Essential Services": Primary Essential Services are those services which need to be in continuous operation to

maintain propulsion and steering; Secondary Essential Services are those services which need not necessarily be in continuous operation to maintain propulsion and steering but which are necessary for maintaining the vessel's safety.

Information Technology (IT): Devices, software and associated networking focusing on the use of data as information, as opposed to Operational Technology (OT).

Integrated system: A system combining a number of interacting sub-systems and/or equipment organized to achieve one or more specified purposes.

Logical network segment: The same as "Network segment", but where two or more logical network segments share the same physical components.

Network: A connection between two or more computers for the purpose of communicating data electronically by means of agreed communication protocols.

Network segment: In the context of this UR, a network segment is an OSI layer-2 Ethernet segment (a broadcast domain).

Note: Network address plan is prefixed by IP addresses and the network mask. Communication between network segments is only possible by the use of routing service at network layer (OSI Layer 3).

Operational Technology (OT): Devices, sensors, software and associated networking that monitor and control onboard systems. Operational technology systems may be thought of as focusing on the use of data to control or monitor physical processes.

Physical network segment: The same as "Network segment", but where physical components are not shared by other network segments.

Protocol: A common set of rules and signals that computers on the network use to communicate. Protocols allow data communication, network management and security. Onboard networks usually implement protocols based on TCP/IP stacks or various fieldbuses.

Security zone: A collection of CBSs in the scope of applicability of this Section that meet the same security requirements. Each zone consists of a single interface or a group of interfaces, to which an access control policy is applied.

Untrusted network: Any network outside the scope of applicability of this Section.

7.4.2 In the context of this Section, the roles and responsibilities of the following are further elaborated as given below:

Shipowner/ Company: The owner of the ship or any other organization or person, such as the manager, agent or bareboat charterer, who has assumed responsibility for operation of the ship from the shipowner and who on assuming such responsibilities has agreed to take over all the attendant duties and responsibilities. The shipowner could be the Shipyard or systems integrator during initial construction. After vessel delivery, the shipowner may delegate some responsibilities to the vessel management company.

Supplier: A manufacturer or provider of hardware and/or software products, system components or equipment (hardware or software) comprising of the application, embedded devices, network devices, host devices etc. working together as system or a subsystem. The supplier is responsible for providing programmable devices, sub-systems or systems to the systems integrator.

Systems Integrator: The specific person or organization responsible for the integration of systems and products provided by suppliers into the system invoked by the requirements in the ship specifications and for providing the integrated system. The systems integrator may also be responsible for integration of systems in the ship. Until vessel delivery, this role is to be performed by the Shipyard unless an alternative organization is specifically contracted/assigned this responsibility.

7.5. Goals and Functional Elements

7.5.1 Primary goal

The primary goal is to support safe and secure shipping, which is operationally resilient to cyber risks. To support safe and secure shipping resilient to cyber risk, the following sub-goals for the management of cyber risk are defined in the five functional elements listed in section 7.5.2 below.

7.5.2 Sub-goals per functional element

.1 *Identify*: Develop an organizational understanding to manage cybersecurity risk to onboard systems, people, assets, data, and capabilities.

.2 *Protect*: Develop and implement appropriate safeguards to protect the ship against cyber incidents and maximize continuity of shipping operations.

.3 *Detect*: Develop and implement appropriate measures to detect and identify the occurrence of a cyber incident onboard.

.4 *Respond*: Develop and implement appropriate measures and activities to take action regarding a detected cyber incident onboard.

.5 *Recover*: Develop and implement appropriate measures and activities to restore any capabilities or services necessary for shipping operations that were impaired due to a cyber incident.

These sub-goals and relevant functional elements are to be concurrent and considered as parts of a single comprehensive risk management framework.

7.5.3 Organization of requirements

The requirements in this Section are organized according to a goal-based approach. Functional/ technical requirements are given for the achievement of specific sub-goals of each functional element (as indicated in 7.5.2). A summary of actions to be carried out and documentation to be made available is also indicated for each phase of the ship's life and relevant stakeholders participating in such phase.

7.6 Requirements

This sub-section contains the requirements to be satisfied in order to achieve the primary goal defined in 7.5.1, organized according to the five functional elements identified in 7.5.2.

The requirements are to be complied with by stakeholders involved in the design, building and operation of the ship. Among them, the following stakeholders are identified (see also section 7.4 for definitions):

- Shipowner/ Company
- Systems integrator
- Supplier
- IRS

Whilst the above requirements may be fulfilled by any of these stakeholders, for the purposes of this Section, responsibility to fulfil them will lie with the stakeholder who has contracted with IRS.

7.6.1 Identify

The requirements for the 'Identify' functional element are aimed at identifying on one side, the CBSs onboard, their interdependencies and the relevant information flows; on the other side, the key resources involved in their management, operation and governance, their roles, and responsibilities.

7.6.1.1 Vessel asset inventory

.1 Requirement

An inventory of hardware and software (including application programs, operating systems, if any, firmware and other software components) of the CBSs in the scope of applicability of this Section and of the networks connecting such systems to each other and to other CBSs onboard or ashore are to be provided and kept up to date during the entire life of the ship.

.2 Requirement details

The vessel asset inventory is to include at least the CBSs indicated in section 7.2, if present onboard.

The inventory is to be kept updated during the entire life of the ship. Software and hardware modifications potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems are to be recorded in the inventory.

If confidential information is included in the inventory (e.g. IP addresses, protocols, port numbers), special measures are to be adopted to limit access to such information only to authorized people.

a) Hardware

For all hardware devices in the scope of applicability of this Section, the vessel asset

inventory is to include at least the information in section 8.3.1.1. In addition, the vessel asset inventory may specify system category and security zone associated with the CBS.

b) Software

For all software in the scope of applicability of this Section (e.g., application program, operating system, firmware), the vessel asset inventory is to include at least the information in section 8.3.1.1.

The software of the CBSs in the scope of applicability of this Section is to be maintained and updated in accordance with the shipowner's process for management of software maintenance and update policy in the Ship cyber security and resilience program.

.3 Demonstration of compliance

a) Design phase

The systems integrator is to submit vessel asset inventory to IRS (ref. section 7.7.1.3).

The vessel asset inventory is to incorporate the asset inventories of all individual CBSs falling under the scope of this Section. Any equipment in the scope of this Section delivered by the systems integrator is also to be included in the vessel asset inventory.

b) Construction phase

The systems integrator is to keep the vessel asset inventory updated.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS that:

- Vessel asset inventory is updated and completed at delivery
- CBSs in the scope of applicability of this Section are correctly represented in the vessel asset inventory
- Software of the CBSs in the scope of applicability of this Section have been kept updated, e.g. by vulnerability scanning or by checking the software versions of CBSs while switched on.

d) Operation phase

The shipowner is to (in the Ship cyber security and resilience program) describe the process of management of change (MoC) for the CBSs in the scope of applicability of this Section, addressing at least the following requirements:

- Management of change (section 7.7.2.3)
- Hardware and software modifications (section 7.6.1.1.2)

The shipowner is to (in the Ship cyber security and resilience program) also describe the management of software updates, addressing at least the following requirements:

- Vulnerabilities and cyber risks (section 7.6.1.1.2)
- Security patching (section 7.6.2.6.2 a))

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.2 Protect

The requirements for the ‘Protect’ functional element are aimed at the development and implementation of appropriate safeguards supporting the ability to limit or contain the impact of a potential incident.

7.6.2.1 Security Zones and Network Segmentation

.1 Requirement

All CBSs in the scope of applicability of this Section are to be grouped into security zones with well-defined security policies and security capabilities. Security zones are either to be isolated (i.e. air gapped) or connected to other security zones or networks by means providing control of data communicated between the zones (e.g. firewalls/routers, simplex serial links, TCP/IP diodes, dry contacts, etc.)

Only explicitly allowed traffic is to traverse a security zone boundary.

.2 Requirement details

A security zone may contain multiple CBSs and networks, all of which are to comply with applicable security requirements given in this Section and Section 8.

The network(s) of a security zone are to be logically or physically segmented from other zones or networks. See also 7.6.2.6.2.

CBSs providing required safety functions are to be grouped into separate security zones and are to be physically segmented from other security zones.

Navigational and communication systems are not to be in same security zone as machinery or cargo systems. If navigation and/or radiocommunication systems are approved in accordance with other equivalent standard(s) (see section 7.2), these systems should be in a dedicated security zone.

Wireless devices are to be in dedicated security zones. See also 7.6.2.5.

Systems, networks or CBSs outside the scope of applicability of this Section are considered untrusted networks and are to be physically segmented from security zones required by this Section. Alternatively, it is accepted that such systems are part of a security zone if these OT-systems meet the same requirements as demanded by the zone.

It is to be possible to isolate a security zone without affecting the primary functionality of the CBSs in the zone, see also section 7.6.4.3.

.3 Demonstration of compliance

a) Design phase

The systems integrator is to submit Zones and conduit diagram and the Cyber security design description (see 7.7.1.1 and 7.7.1.2).

The Zones and conduit diagram is to illustrate the CBSs in the scope of applicability of this Section, how they are grouped into security zones, and include the following information:

- Clear indication of the security zones
- Simplified illustration of each CBS in scope of applicability of this Section, and indication of the security zone in which the CBS is allocated, and indication of physical location of the CBS/equipment.
- Reference to the approved version of the CBS system topology diagrams provided by the suppliers (section 8.3.1.2)
- Illustration of network communication between systems in a security zone
- Illustration of any network communication between systems in different security zones (conduits).
- Illustration of any communication between systems in a security zone and untrusted networks (conduits).

The systems integrator is to include the following information in the Cyber security design description:

- A short description of the CBSs allocated to the security zone. It is to be possible to identify each CBS in the Zones and conduit diagram.
- Network communication between CBSs in the same security zone. The description is to include the purpose and characteristics (i.e. protocols and data flows) of the communication.
- Network communication between CBSs in different security zones. The description is to include the purpose and characteristics (i.e. protocols and data flows) of the communication. The description is also to include zone boundary devices and specify the traffic that is permitted to traverse the zone boundary (e.g. firewall rules).
- Any communication between CBSs in security zones and untrusted networks. The description is to include discrete signals, serial communication, and the purpose and characteristics (i.e. protocols and data flows) of IP-based network communication. The description is also to include zone boundary devices and specify the traffic that is permitted to traverse the zone boundary (e.g. firewall rules).

b) Construction phase

The systems integrator is to keep the Zones and conduit diagram updated.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS that:

- the security zones on board are implemented in accordance with the approved documents (i.e. zones and conduit diagram, cyber security design description, asset inventory, and relevant documents provided by the supplier). This may be done by e.g., inspection of the physical installation, network scanning and/or other methods providing the Surveyor assurance that the installed equipment is grouped in security zones according to the approved design.

- security zone boundaries allow only the traffic that has been documented in the approved Cyber security description. This may be done by e.g., evaluation of firewall rules or port scanning.

d) Operation phase

The shipowner is to (in the Ship cyber security and resilience program) describe the management of security zone boundary devices (e.g., firewalls), addressing at least the following requirements:

- Principle of Least Functionality (section 7.6.2.2.1)

- Explicitly allowed traffic (section 7.6.2.1.1)

- Protection against denial of service (DoS) events (section 7.6.2.2.1)

- Inspection of security audit records (section 7.6.3.1.2)

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.2.2 Network protection safeguards

.1 Requirements

Security zones are to be protected by firewalls or equivalent means as specified in section 7.6.2.1.

The networks are also to be protected against the occurrence of excessive data flow rate and other events which could impair the quality of service of network resources.

The CBSs in scope of this Section are to be implemented in accordance with the principle of Least Functionality, i.e. configured to provide only essential capabilities and to prohibit or restrict the use of non-essential functions, where unnecessary functions, ports, protocols and services are disabled or otherwise prohibited.

.2 Requirement details

The design of the network is to include means to meet the intended data flow through the network and minimize the risk of denial of service (DoS) and network storm/high rate of traffic. Estimation of data flow rate is to at least consider the capacity of network, data speed requirement for intended application and data format.

.3 Demonstration of compliance

a) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate the following to IRS:

- Test denial of service (DoS) attacks targeting zone boundary protection devices, as applicable.

- Test denial of service (DoS) to ensure protection against excessive data flow rate, originating from inside each network segment. Such denial of service (DoS) tests are to cover flooding of network (i.e., attempt to consume the available capacity on the network segment), and application layer attack (i.e., attempt to consume the processing capacity of selected endpoints in the network)

- Test e.g. by analytic evaluation and port scanning that unnecessary functions, ports, protocols and services in the CBSs have been removed or prohibited in accordance with hardening guidelines provided by the suppliers. See section 8.5.7 and 8.6.3.4.7.

The second and third tests may be omitted if performed during the certification of CBSs in accordance with section 7.7.2.1.

b) Operation phase

Requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.2.3 Antivirus, antimalware, antispam and other protections from malicious code

.1 Requirements

CBSs in the scope of applicability of this Section are to be protected against malicious code such as viruses, worms, trojan horses, spyware, etc.

.2 Requirement details

Malware protection is to be implemented on CBSs in the scope of applicability of this Section. On CBSs having an operating system for which industrial-standard anti-virus and anti-malware software is available and maintained up-to-date, anti-virus and/or anti-malware software are to be installed, maintained and regularly updated, unless the installation of such software impairs the ability of CBS to provide the functionality and level of service required (e.g. for Cat.II and Cat.III CBSs performing real-time tasks).
On CBSs where anti-virus and anti-malware software cannot be installed, malware protection is to be implemented in the form of operational procedures, physical safeguards, or according to manufacturer's recommendations.

.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- For each CBS, summary of the approved mechanisms provided by the supplier for protection against malicious code or unauthorized software.

- For CBSs with anti-malware software, information about how to keep the software updated.

- Any operational conditions or necessary physical safeguards to be implemented in the shipowner's management system.

b) Construction phase

The systems integrator is to ensure that malware protection is kept updated during the construction phase.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate the following to IRS:

Approved anti-malware software or other compensating countermeasures are effective (test e.g., with a trustworthy anti-malware test file).

The above tests may be omitted if performed during the certification of CBSs in accordance with section 7.7.2.1.

d) Operation phase

The shipowner is to (in the Ship cyber security and resilience program) describe the management of malware protection, addressing at least the following requirements:

- Maintenance/update (section 7.6.2.3.2)

- Operational procedures, physical safeguards (section 7.6.2.3.2)

- Use of mobile, portable, removable media (section 7.6.2.4.2.4 and 7.6.2.7.2)

- Access control (section 7.6.2.4)

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.2.4 Access control

.1 Requirements

CBSs and networks in the scope of applicability of this Section are to provide physical and/ or logical/ digital measures to selectively limit the ability and means to communicate with or otherwise interact with the system itself, to use system resources to handle information, to gain knowledge of the information the system contains or to control system components and functions. Such measures are to be such as not to hamper the ability of authorized personnel to access CBS for their level of access according to the least privilege principle.

.2 Requirement details

Access to CBSs and networks in the scope of applicability of this Section and all information stored on such systems is only to be allowed to authorized personnel, based on their need to access the information as a part of their responsibilities or their intended functionality.

.1 Physical access control

CBSs of Cat.II and Cat.III are to generally be located in rooms that can normally be locked or in controlled space to prevent unauthorized access or are to be installed in lockable cabinets or consoles. Such locations or lockable cabinets/ consoles are to be however easy to access for the crew and various stakeholders who need to access CBSs for installation, integration, operation, maintenance, repair, replacement, disposal etc. so as not to hamper effective and efficient operation of the ship.

.2 Physical access control for visitors

Visitors such as authorities, technicians, agents, port and terminal officials, and shipowner representatives are to be restricted regarding access to CBSs onboard whilst on board, e.g. by allowing access under supervision.

.3 Physical access control of network access points

Access points to onboard networks connecting Cat.II and/or Cat.III CBSs are to be physically and/or logically blocked except when connection occurs under supervision or according to documented procedures, e.g. for maintenance.

Independent computers isolated from all onboard networks, or other networks, such as dedicated guest access networks, or networks dedicated to passenger recreational activities, are to be used in case of occasional connection requested by a visitor (e.g. for printing documents).

.4 Removable media controls

A policy for the use of removable media devices is to be established, with procedures to check removable media for malware and/or validate legitimate software by digital signatures and watermarks and scan prior to permitting the uploading of files onto a ship's system or downloading data from the ship's system. See also 7.6.2.7.

.5 Management of credentials

CBSs and relevant information are to be protected with file system, network, application, or database specific Access Control Lists (ACL). Accounts for onboard and onshore personnel are to be left active only for a limited period according to the role and responsibility of the account holder and are to be removed when no longer needed.

Note: CBSs are to identify and authenticate human users as per item No.1 in Table 8.4.1 of Section 8. In other words, it is not necessary to "uniquely" identify and authenticate all human users.

Onboard CBSs are to be provided with appropriate access control that complies with the policy of their Security Zone but does not adversely affect their primary purpose. CBSs which require strong access control may need to be secured using a strong encryption key or multi-factor authentication.

Administrator privileges are to be managed in accordance with the policy for access control, allowing only authorized and appropriately trained personnel full access to the CBS, who as part of their role in the company or onboard need to log on to systems using these privileges.

.6 Least privilege principle

Any human user allowed to access CBS and networks in the scope of applicability of this Section is to have only the bare minimum privileges necessary to perform its function.

The default configuration for all new account privileges is to be set as low as possible. Wherever possible, raised privileges are to be restricted only to moments when they are needed, e.g. using only expiring privileges and one-time-use credentials. Accumulation of privileges over time are to be avoided, e.g. by regular auditing of user accounts.

7.6.2.4.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- Location and physical access controls for the CBSs. Devices providing Human Machine Interface (HMI) for operators needing immediate access need not enforce user identification and authentication provided they are located in an area with physical access control. Such devices are to be specified.

b) Construction phase

The systems integrator is to prevent unauthorized access to the CBSs during the construction phase.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate the following to IRS:

- Components of the CBSs are located in areas or enclosures where physical access can be controlled to authorized personnel.

- User accounts are configured according to the principles of segregation of duties and least privilege and that temporary accounts have been removed (may be omitted based on certification of CBSs in accordance with section 7.7.2.1)

d) Operation phase

The shipowner is to (in the Ship cyber security and resilience program) describe the management of logical and physical access, addressing at least the following requirements:

- Physical access control (section 7.6.2.4.2.1)

- Physical access control for visitors (section 7.6.2.4.2.2)

- Physical access control of network access points (section 7.6.2.4.2.3)

- Management of credentials (section 7.6.2.4.2.5)

- Least privilege policy (section 7.6.2.4.2.6)

The shipowner is also to (in the Ship cyber security and resilience program) describe the management of confidential information, addressing at least the following requirements:

- Confidential information (see section 7.6.1.1.2)

- Information allowed to authorized personnel (see section 7.6.2.4.2)

- Information transmitted on the wireless network (see section 7.6.2.5.2)

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.2.5 Wireless communication

.1 Requirements

Wireless communication networks in the scope of this Section are to be designed, implemented and maintained to ensure that:

- Cyber incidents will not propagate to other control systems

- Only authorized human users will gain access to the wireless network

- Only authorized processes and devices will be allowed to communicate on the wireless network

- Information in transit on the wireless network cannot be manipulated or disclosed

.2 Requirement details

Cryptographic mechanisms such as encryption algorithms and key lengths in accordance with industry standards and best practices are to be applied to ensure integrity and confidentiality of the information transmitted on the wireless network.

Devices on the wireless network are to only communicate on the wireless network (i.e. they are not to be “dual-homed”)

Wireless networks are to be designed as separate segments in accordance with 7.6.2.1 and protected as per 7.6.2.2.

Wireless access points and other devices in the network are to be installed and configured such that access to the network can be controlled.

The network device or system utilizing wireless communication is to provide capability to identify and authenticate all users (humans, software processes or devices) engaged in that communication.

7.6.2.5.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- Description of wireless networks in the scope of applicability of this Section and how these are implemented as separate security zones.
- The description is to include zone boundary devices and specify the traffic that is permitted to traverse the zone boundary (e.g. firewall rules)

b) Construction phase

The systems integrator is to prevent unauthorized access to the wireless networks during the construction phase.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate the following to IRS:

- Only authorized devices can access the wireless network.
- Secure wireless communication protocol is used as per approved documentation by the respective supplier (demonstrate e.g. by use of a network protocol analyzer tool).

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

d) Operation phase

Requirements for periodical surveys are covered in Part 1, Chapter 2 of the Rules.

7.6.2.6 Remote access control and communication with untrusted networks

.1 Requirements

CBSs in scope of this Section are to be protected against unauthorized access and other cyber threats from untrusted networks.

.2 Requirement details

User manual is to be submitted for control of remote access to onboard IT and OT systems. Clear guidelines are to identify roles and permissions with functions.

For CBSs in the scope of applicability of this Section, no IP address is to be exposed to untrusted networks.

Communication with or via untrusted networks requires secure connections (e.g. tunnels) with endpoint authentication, protection of integrity and authentication and encryption at network or transport layer. Confidentiality is to be ensured for information that is subject to read authorization.

a) Design phase

CBSs in the scope of applicability of this Section are to:

- have the capability to terminate a connection from the onboard connection endpoint. Any remote access is not to be possible until explicitly accepted by a responsible role on board.
- be capable of managing interruptions during remote sessions so as not to compromise the safe functionality of OT systems or the integrity and availability of data used by OT systems.

- provide a logging function to record all remote access events and retain for a period of time sufficient for offline review of remote connections, e.g. after detection of a cyber incident.

b) Additional requirements for remote maintenance

When remote access is used for maintenance, the following requirements are to be complied with in addition to those in 7.6.2.6.2 a):

- Documentation is to be provided to show how they connect and integrate with the shore side.

- Security patches and software updates are to be tested and evaluated before they are installed to ensure they are effective and do not result in side effects or cyber events that cannot be tolerated. A confirmation report from the software supplier towards above is to be obtained, prior to undertaking remote update.

- Suppliers are to provide plans for and make security updates available to the shipowner, see Sections 8.5.2, 8.5.3 and 8.5.4.

- At any time, during remote maintenance activities, authorized personnel are to have the possibility to interrupt and abort the activity and roll back to a previous safe configuration of the CBS and systems involved.

- Multi-factor authentication is required for any access by human users to CBS's in scope from an untrusted network.

- After a configurable number of failed remote access attempts, the next attempt is to be blocked for a predetermined length of time.

- If the connection to the remote maintenance location is disrupted for some reason, access to the system is to be terminated by an automatic logout function.

7.6.2.6.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- Identification of each CBS in the scope of applicability of this Section that can be remotely accessed or that otherwise communicates through the security zone boundary with untrusted networks.

- For each CBS, a description of compliance with requirements in 7.9.6.2, as applicable

b) Construction phase

The systems integrator is to ensure that any communication with untrusted networks is only temporarily enabled and used in accordance with the requirements of this section.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate the following to IRS:

- Communication with untrusted networks is secured in accordance with 8.4.2 and that the communication protocols cannot be negotiated to a less secure version (demonstrate e.g., by use of a network protocol analyzer tool).

- Remote access requires multifactor authentication of the remote user.

- A limit of unsuccessful login attempts is implemented, and that a notification message is provided for the remote user before session is established.

- Remote connections must be explicitly accepted by responsible personnel on board.

- Remote sessions can be manually terminated by personnel on board or that the session will automatically terminate after a period of inactivity.

- Remote sessions are logged (see section 8.4.1 item 13).

- Instructions or procedures are provided by the respective product suppliers (see section 8.3.1.3).

d) Operation phase

The shipowner is to describe the management of remote access and communication with/via untrusted networks, in the Ship cyber security and resilience program addressing at least the following requirements:

- User's manual (section 7.6.2.6.2)
- Roles and permissions (section 7.6.2.6.2)
- Patches and updates (section 7.6.2.6.2 b))
- Confirmation prior to undertaking remote software update (section 7.6.2.6.2 b))
- Interrupt, abort, roll back (section 7.6.2.6.2 b))

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.2.7 Use of Mobile and Portable Devices

.1 Requirements

The use of mobile and portable devices in CBSs in the scope of applicability of this Section are to be limited to only necessary activities and be controlled in accordance with section 8.4.1 item 10. For any CBS that cannot fully meet these requirements, the interface ports are to be physically blocked.

.2 Requirement details

Mobile and portable devices are to only be used by authorized personnel. Only authorized devices may be connected to the CBSs. All use of such devices is to be in accordance with the shipowner's policy for use of mobile and portable devices, taking into account the risk of introducing malware in the CBS.

7.6.2.7.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- Any CBSs in the scope of applicability that do not meet the requirements in section 8.4.1 item 10, are to have protection of interface ports by physical means such as port blockers.

b) Construction phase

The systems integrator is to ensure that use of physical interface ports in the CBSs is controlled in accordance with section 8.4.1 item 10, and that any use of such devices follows procedures to prevent malware from being introduced in the CBS.

c) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS that capabilities to control use of mobile and portable devices are implemented correctly. The following countermeasures are to be demonstrated as relevant:

- Use of mobile and portable devices is restricted to authorized users
- Interface ports can only be used by specific device types
- Files cannot be transferred to the system from such devices
- Files on such devices will not be automatically executed (by disabling autorun)
- Network access is limited to specific MAC or IP addresses
- Unused interface ports are disabled
- Unused interface ports are physically blocked

d) Operation phase

The shipowner is to describe the management of mobile and portable devices, in the Ship cyber security and resilience program addressing at least the following requirements:

- Policy and procedures (section 7.6.2.4.2.4)
- Physical block of interface ports (section 7.6.2.7.1)
- Use by authorized personnel (section 7.6.2.7.2)

- Connect only authorized devices (section 7.6.2.7.2)

- Consider risk of introducing malware (section 7.6.2.7.2)

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.3 Detect

The requirements for the Detect functional element are aimed at the development and implementation of appropriate means supporting the ability to reveal and recognize anomalous activity on CBSs and networks onboard and identify cyber incidents.

7.6.3.1 Network operation monitoring

.1 Requirements

Networks in scope of this Section are to be continuously monitored, and alarms are to be generated if malfunctions or reduced/degraded capacity occurs.

.2 Requirement details

Measures to monitor networks in the scope of applicability of this Section are to have the following capabilities:

- Monitoring and protection against excessive traffic

- Monitoring of network connections

- Monitoring and recording of device management activities

- Protection against connection of unauthorized devices

- Generate alarm if utilization of the network's bandwidth exceeds a threshold specified as abnormal by the supplier. See section 6.8.3.1.

Intrusion detection systems (IDS) may be implemented, subject to the following:

- The IDS are to be qualified by the supplier of the respective CBS

- The IDS are to be passive and not activate protection functions that may affect the performance of the CBS

- Relevant personnel should be trained and qualified for using the IDS

7.6.3.1.3 Demonstration of compliance

a) Commissioning phase

The systems integrator is to specify in the Ship cyber resilience test procedure and demonstrate to IRS the network monitoring and protection mechanisms in the CBSs.

- Test that disconnected network connections will activate alarm and that the event is recorded.

- Test that abnormally high network traffic is detected, and that alarm and audit record is generated. This test may be carried on together with the test in section 7.6.4.4.3 b).

- Demonstrate that the CBS will respond in a safe manner to network storm scenarios, considering both unicast and broadcast messages (see also 7.6.4.2.3)

- Demonstrate generation of audit records (logging of security-related events)

- If Intrusion detection systems are implemented, demonstrate that this is passive and will not activate protection functions that may affect intended operation of the CBSs.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

Any Intrusion detection systems in the CBSs in scope of applicability to be implemented will be subject to verification by IRS. Relevant documentation is to be submitted for approval, and survey/tests are to be carried out on board.

b) Operation phase

The shipowner is to describe the management activities to detect anomalies in the CBSs and networks, in the Ship cyber security and resilience program addressing at least the following requirements:

- Reveal and recognize anomalous activity (section 7.6.3)

- Inspection of security audit records (section 7.6.3.1.2)

- Instructions or procedures to detect incidents (section 7.6.4.1.1)

The above activities may be addressed together with incident response in section 7.6.4.1.

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.3.2 Verification and diagnostic functions of CBS and networks

.1 Requirements

CBSs and networks in the scope of applicability of this Section are to be capable of checking performance and functionality of security functions required by this Section. Diagnostic functions are to provide adequate information on CBSs integrity and status for the use of the intended user and means for maintaining their functionality for a safe operation of the ship.

.2 Requirement details

CBSs and networks' diagnostics functionality are to be available to verify the intended operation of all required security functions during test and maintenance phases of the ship.

7.6.3.2.3 Demonstration of compliance

a) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS the effectiveness of the procedures for verification of security functions provided by the suppliers.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

b) Operation phase

The shipowner is to describe the management activities to verify correct operation of the security functions in the CBSs and networks, in the Ship cyber security and resilience program, addressing at least the following requirements:

- Test and maintenance periods (section 7.6.3.2.2)

- Periodic maintenance

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.4 Respond

The requirements for the Respond functional element are aimed at the development and implementation of appropriate means supporting the ability to minimize the impact of cyber incidents, containing the extension of possible impairment of CBSs and networks onboard.

7.6.4.1 Incident response plan

.1 Requirements

An incident response plan is to be developed by the shipowner covering relevant contingencies and specifying how to react to cyber security incidents. The Incident response plan is to contain documentation of a predetermined set of instructions or procedures to detect, respond to, and limit consequences of incidents against CBSs in the scope of applicability of this Section.

.2 Requirement details

The various stakeholders involved in the design and construction phases of the ship are to provide information to the shipowner for the preparation of the Incident Response Plan to be placed onboard at the first annual Survey. The Incident Response Plan is to be kept up-to-date (e.g. upon maintenance) during the operational life of the ship.

The Incident response plan is to provide procedures to respond to detected cyber incidents on networks by notifying the proper authority, reporting needed evidence of the incidents and taking timely corrective actions, to limit the cyber incident impact to the network segment of origin.

The incident response plan is to, as a minimum, include the following information:

- Breakpoints for the isolation of compromised systems

- A description of alarms and indicators signalling detected ongoing cyber

events or abnormal symptoms caused by cyber events

- A description of expected major consequences related to cyber incidents

- Response options, prioritizing those which do not rely on either shut down or transfer to independent or local control, if any

- Independent and local control information for operating independently from the system that failed due to the cyber incident, as applicable

The incident response plan is to be kept in hard copy in the event of complete loss of electronic devices enabling access to it.

7.6.4.1.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the cyber security design description:

- References to information provided by the suppliers (see section 8.3.1.8) that may be applied by the shipowner to establish plans for incident response.

b) Operation phase

The shipowner is to describe incident response plans to in the Ship cyber security and resilience program. The plans are to cover the CBSs in scope of applicability of this Section and are to address at least the following requirements:

- Description of who, when and how to respond to cyber incidents in accordance with requirements of section 7.6.4.1

- Procedures or instructions for local/manual control in accordance with requirements in section 7.6.4.2

- Procedures or instructions for isolation of security zones in accordance with requirements in section 7.6.4.3

- Description of expected behaviour of the CBSs in the event of cyber incidents in accordance with requirements in section 7.6.4.4.

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.4.2 Local, independent and/or manual operation

.1 Requirements

Any CBS needed for local backup control as required by SOLAS II-1 Regulation 31 are to be independent of the primary control system. This also includes the necessary Human Machine Interface (HMI) for effective local operation.

.2 Requirement details

The CBS for local control and monitoring are to be self-contained and not depend on communication with other CBS for its intended operation.

If communication to the remote-control system or other CBS's is arranged by networks, segmentation and protection safeguards as described in 7.6.2.1 and 7.6.2.2 are to be implemented. This implies that the local control and monitoring system is to be considered a separate security zone. Notwithstanding the above, special considerations can be given to CBSs with different concepts on case-by-case basis

The CBS for local control and monitoring are to otherwise comply with requirements in this Section.

7.6.4.2.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- Description of how the local controls specified in SOLAS II-1 Reg.31 are protected from cyber incidents in any connected remote or automatic control systems.

b) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1)

and demonstrate to IRS that the required local controls in the scope of applicability of this Section needed for safety of the ship can be operated independently of any remote or automatic control systems. The tests are to be carried out by disconnecting all networks from the local control system to other systems/devices.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

c) Operation phase

Requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.4.3 Network isolation

.1 Requirements

It is to be possible to terminate network-based communication to or from a security zone.

.2 Requirement details

Where the Incident Response Plan indicates network isolation as an action to be carried out, it is to be possible to isolate security zones according to the indicated procedure, e.g. by operating a physical On/Off switch on the network device or similar actions such as disconnecting a cable to the router/firewall. There are to be available instructions and clear marking on the device that allows the personnel to isolate the network in an efficient manner.

Individual system's data dependencies that may affect function and correct operation, including safety, are to be identified, clearly showing where systems must have compensations for data or functional inputs if isolated during a contingency.

7.6.4.3.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- specification of how to isolate each security zone from other zones or networks. The effects of such isolation are also to be described, demonstrating that the CBSs in a security zone do not rely on data transmitted by IP-networks from other zones or networks.

b) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS by disconnecting all networks traversing security zone boundaries, that the CBSs in the security zone will maintain adequate operational functionality without network communication with other security zones or networks.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

c) Operation phase

Requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.4.4 Fallback to a minimal risk condition

.1 Requirements

In the event of a cyber incident impairing the ability of a CBS or network in the scope of applicability of this Section to provide its intended service, the affected system or network is to fall back to a minimal risk condition, i.e. bring itself in a stable, stopped condition to reduce the risk of possible safety issues.

.2 Requirement details

As soon as a cyber incident affecting the CBS or network is detected, compromising the system's ability to provide the intended service as required, the system is to fall back to a condition in which a reasonably safe state can be achieved. Fall-back actions may include:

- bringing the system to a complete stop or other safe state;
- disengaging the system;
- transferring control to another system or human operator;
- other compensating actions.

Fall-back to minimum risk conditions is to occur in a time frame adequate to keep the ship in a safe condition.

The ability of a system to fall back to a minimal risk condition is to be considered from the design phase by the supplier and the systems integrator.

7.6.4.4.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- Specification of safe state for the control functions in the CBSs in the scope of applicability of this Section.

b) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS that CBSs in the scope of applicability of this section respond to cyber incidents in a safe manner (as per section 7.6.4.4.3 a), e.g. by maintaining its outputs to essential services and allowing operators to carry out control and monitoring functions by alternative means. The tests are to at least include denial of service (DoS) attacks and may be done together with related test in section 7.6.3.1.3 a).

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

c) Operation phase

Requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.5 Recover

The requirements for the Recover functional element are aimed at the development and implementation of appropriate means supporting the ability to restore CBSs and networks onboard affected by cyber incidents.

7.6.5.1 Recovery plan

.1 Requirements

A recovery plan is to be made by the shipowner to support restoring CBSs under the scope of applicability of this Section to an operational state after a disruption or failure caused by a cyber incident. Details of where assistance is available and by whom are to be part of the recovery plan.

.2 Requirement details

The various stakeholders involved in the design and construction phases of the ship are to provide information to the ship owner for the preparation of the recovery plan to be placed onboard at the first annual Survey. The recovery plan is to be kept up-to-date (e.g. upon maintenance) during the operational life of the ship.

Recovery plans are to be easily understandable by the crew and external personnel and include essential instructions and procedures to ensure the recovery of a failed system and how to get external assistance if the support from ashore is necessary. In addition, software recovery medium or tools essential for recovery on board are to be available.

When developing recovery plans, the various systems and subsystems involved are to be specified.

The following recovery objectives are also to be specified:

- System recovery: methods and procedures to recover communication capabilities are to be specified in terms of Recovery Time Objective (RTO). This is defined as the time required to recover the required communication links and processing capabilities.

- Data recovery: methods and procedures to recover data necessary to restore safe state of OT systems and safe ship operation are to be specified in terms of Recovery Point Objective (RPO). This is defined as the longest period of time for which an absence of data can be tolerated.

Once the recovery objectives are defined, a list of potential cyber incidents is to be created, and the recovery procedure developed and described. Recovery plans are to include, or refer to the following information;

- Instructions and procedures for restoring the failed system without disrupting the operation from the redundant, independent or local operation.

- Processes and procedures for the backup and secure storage of information.

- Complete and up-to-date logical network diagram.

- The list of personnel responsible for restoring the failed system.

- Communication procedure and list of personnel to contact for external technical support including system support vendors, network administrators, etc.

- Current configuration information for all components.

The operation and navigation of the ship is to be prioritized in the plan in order to help ensure the safety of onboard personnel.

Recovery plans in hard copy onboard and ashore are to be available to personnel responsible for cyber security and who are tasked with assisting in cyber incidents.

7.6.5.1.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- references to information provided by the suppliers (see section 8.3.1.8) that may be applied by the shipowner to establish plans to recover from cyber incidents.

b) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS the effectiveness of the procedures and instructions provided by the suppliers to respond to cyber incidents as specified in section 7.6.5.2 and 7.6.5.3.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

c) Operation phase

The shipowner is to describe incident recovery plans in the Ship cyber security and resilience program. The plans are to cover the CBSs in scope of applicability of this Section and are to address at least the following requirements:

- Description of who, when and how to restore and recover from cyber incidents in accordance with requirements in section 7.6.5.1

- Policy for backup addressing frequency, maintenance and testing of the backups, considering acceptable downtime, availability of alternative means for control, vendor support arrangements and criticality of the CBSs in accordance with requirements in section 7.6.5.2.

- Reference to user manuals or procedures for backup, shutdown, reset, restore and restart of the CBSs in accordance with requirements in section 7.6.5.2 and 7.6.5.3.

Other requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.5.2 Backup and restore capability

.1 Requirements

CBSs and networks in the scope of applicability of this Section are to have the capability to support back-up and restore in a timely, complete and safe manner. Backups are to be regularly maintained and tested.

.2 Requirement details

a) Restore capability

CBSs in the scope of applicability of this Section are to have backup and restore capabilities to enable the ship to safely regain navigational and operational state after a cyber incident.

Data is to be restorable from a secure copy or image.

Information and backup facilities are to be sufficient to recover from a cyber incident.

b) Backup

CBSs and networks in the scope of applicability of this Section are to provide backup for data. The use of offline backups is also to be considered to improve tolerance

against ransomware and worms affecting online backup appliances.

Backup plans are to be developed, including scope, mode and frequency, storage medium and retention period.

7.6.5.2.3 Demonstration of compliance

a) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS the procedures and instructions for backup and restore provided by the suppliers for CBSs in the scope of applicability of this Section.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

b) Operation phase

Requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.6.5.3 Controlled shutdown, reset, roll-back and restart

.1 Requirements

CBS and networks in the scope of applicability of this Section are to be capable of controlled shutdown, reset to an initial state, roll-back to a safe state and restart from a power-off condition in such state, in order to allow fast and safe recovery from a possible impairment due to a cyber incident.

Suitable documentation on how to execute the above-mentioned operations is to be available to onboard personnel.

.2 Requirement details

CBS and networks in the scope of applicability of this Section are to be capable of:

- controlled shutdown allowing other connected systems to commit/rollback pending transactions, terminating processes, closing connections, etc. leaving the entire integrated system in a safe, consistent and known state.

- resetting themselves, instructing the system to go through the process of shutting down, clear memory and reset devices to their initialized state.

- rolling back to a previous configuration and/or state, to restore system integrity and consistency.

- restarting and reloading a fresh image of all the software and data (e.g. after a rollback operation) from a read-only source. Restart time is to be compatible with the system's intended service and are to not bring other connected systems, or the integrated system it is part of, to an inconsistent or unsafe state.

Documentation is to be available to onboard personnel on how to execute the above-mentioned operations in case of a system affected by a cyber incident.

7.6.5.3.3 Demonstration of compliance

a) Design phase

The systems integrator is to include the following information in the Cyber security design description:

- references to product manuals or procedures describing how to safely shut down, reset, restore and restart the CBSs in the scope of applicability of this Section.

b) Commissioning phase

The systems integrator is to submit Ship cyber resilience test procedure (ref. section 7.7.2.1) and demonstrate to IRS that manuals or procedures are established for shutdown, reset and restore of the CBSs in the scope of applicability of this Section. These manuals/procedures are to be provided to the shipowner.

The above tests may be omitted if performed during the certification of CBSs as per section 7.7.2.1.

c) Operation phase

Requirements for periodical surveys are indicated in Part 1, Chapter 2, Section 25 of the Rules.

7.7 Compliance with the Requirements

Evaluation of compliance with requirements in this Section is to be carried out by the Surveyor by assessment of documentation and survey in the relevant phases as specified in the following subsections.

Documentation to be submitted by suppliers to IRS is specified in Section 8. The approved versions of this documentation are also to be provided by the suppliers to the systems integrator as specified in section 8.6.2

Documents to be provided by the systems integrator are listed in section 7.7.1 and 7.7.2.

Documents to be provided by the shipowner are listed in section 7.7.3.

Upon delivery of the ship, the systems integrator is to provide the following documentation to the shipowner:

- Documentation of the CBSs provided by the suppliers (see section 8.6.2)

- Documentation produced by the systems integrator (see sections 7.7.1 and 7.7.2)

Appendix I and Appendix II may be referred for a summary of the documents.

7.7.1 During design and construction phases

The supplier is to demonstrate compliance to IRS by following the certification process specified in section 8.6.

During the design and construction phases, modifications to the design are to be carried out in accordance with the management of change (MoC) requirements in Section 6.
The systems integrator is to demonstrate compliance by submitting documents, as specified in the following subsections to IRS for assessment:

.1 Zones and conduit diagram

The content of this document is specified in section 7.6.2.1.3 a).

.2 Cyber security design description (CSDD)

The content of this document is specified in subsections “Design phase” for each requirement in section 7.6.

.3 Vessel asset inventory

The content of this document is specified in section 7.6.1.1.

.4 Risk assessment for the exclusion of CBSs

The content of this document is specified in section 7.8.

.5 Description of compensating countermeasures

If any CBS in the scope of applicability of this Section has been approved with compensating countermeasures in lieu of a requirement in Section 8, this document is to specify the respective CBS, the security capability that is lacking, as well as provide a detailed description of the compensating countermeasures. See also Section 8.3.1.3 requiring that the supplier describes such compensating countermeasures in the system documentation.

7.7.2 Upon ship commissioning

Before final commissioning of the ship, the systems integrator is to:

a) Submit updated design documentation to IRS (as-built versions of the documents in section 7.7.1)

b) Submit Ship cyber resilience test procedure to IRS describing how to demonstrate compliance with this Section by testing and/or analytic evaluation.

c) Carry out testing, witnessed by the Surveyor, in accordance with the approved Ship cyber resilience test procedure.

7.7.2.1 Ship cyber resilience test procedure

The content of this document is specified for the commissioning phase in each subsection “Demonstration of compliance” in section 7.6.

For each CBS, the required inherent security capabilities and configuration thereof are verified and tested in the certification process of each CBS (see Section 8). Testing of such security functions may be omitted if specified in the respective subsection "Commissioning phase", on the condition that these security

functions have been successfully tested during the certification of the CBS as per Section 8.

Nevertheless, all tests are to be included in the Ship cyber resilience test procedure and the decision to omit tests will be taken by IRS. Tests may generally not be omitted if findings/comments are carried over from the certification process to the commissioning phase, if the respective requirements have been met by compensating countermeasures, or due to other reasons such as modifications of the CBS after the certification process.

The Ship cyber resilience test procedure is also to specify how to test any compensating countermeasures described in Section 7.7.1.5.

The Ship cyber resilience test procedure is to include means to update status and record findings during the testing, and specify the following information:

- Necessary test setup (i.e. to ensure the test can be repeated with the same expected result)
- Test equipment
- Initial condition(s)
- Test methodology, detailed test steps
- Expected results and acceptance criteria

Before submitting the Ship cyber resilience test procedure to IRS, the systems integrator is to verify that the information is updated and placed under change management; that it is aligned with the latest configurations of CBSs and networks connecting such systems together onboard the ship and to other CBSs not onboard (e.g., ashore); and that the tests documented are sufficiently detailed as to allow verification of the installation and operation of measures adopted for the fulfilment of relevant requirements on the final configuration of CBSs and networks onboard.
The systems integrator is to document verification tests or assessments of security controls and measures in the fully integrated ship, maintaining change management for configurations, and noting in the documented test results where safety conditions may be affected by specific circumstances or failures addressed in the Ship cyber resilience test procedure.

The testing is to be carried out on board in accordance with the approved Ship cyber resilience test procedure after other commissioning activities for the CBSs are completed. IRS may request execution of additional tests.

7.7.3 During the operational life of the ship

After the ship has been delivered to the shipowner, the shipowner is to manage technical and organizational security countermeasures by establishing and implementing processes as specified in this Section.

Modifications to the CBSs in scope of applicability of this Section are to be carried out in accordance with the management of change (MoC) requirements in Section 6. This includes keeping documentation of the CBSs up to date.

The shipowner, with the support of suppliers, is to keep the Ship cyber resilience test procedure up to date and aligned with the CBSs onboard the ship and the networks connecting such systems to each other and to other CBSs not onboard (e.g. ashore). The shipowner is to update the Ship cyber resilience test procedure considering the changes occurred on CBSs and networks onboard, possible emerging risks related to such changes, new threats, new vulnerabilities and other possible changes in the ship's operational environment.

The shipowner is to prepare and implement operational procedures, provide periodic training and carry out drills for the onboard personnel and other concerned personnel ashore to familiarize them with the CBSs onboard the ship and the networks connecting such systems to each other and to other CBSs not onboard (e.g. ashore), and to properly manage the measures adopted for the fulfilment of requirements.

The shipowner, with the support of supplier, is to keep the measures adopted for the fulfilment of requirements up to date, e.g. by periodic maintenance of hardware and software of CBSs onboard the ship and the networks connecting such systems.

The shipowner is to retain onboard a copy of results of execution of tests and an updated Ship cyber resilience test procedure and make them available to the Surveyor.

The shipowner is to submit a Ship cyber security and resilience program to IRS for approval, documenting management of cyber security and cyber resilience of the CBSs, as relevant and applicable, well in advance of the first annual survey,

The Ship cyber security and resilience program is to include policies, procedures, plans and/or other information documenting the processes/ activities specified in subsections “demonstration of compliance” in this Section.

The shipowner is to demonstrate compliance by presenting records or other documented evidence of implementation of the processes described in the approved Ship cyber security and resilience program.

Change of vessel management company will require a new verification of the Ship cyber security and resilience program.

Other requirements for periodical surveys are specified in Part 1, Chapter 2, Section 25 of the Rules.

7.8 Risk assessment for exclusion of CBS from the application of requirements

7.8.1 Requirements

A risk assessment is to be carried out in case any of the CBSs falling under the scope of applicability of this Section is excluded from the application of relevant requirements. The risk assessment is to provide evidence of the acceptable risk level associated with the excluded CBSs.

7.8.2 Rationale

Exclusion of a CBS falling under the scope of applicability of this Section from the application of relevant requirements needs to be duly justified and documented. Such exclusion will be accepted by IRS only if evidence is given that the risk level associated to the operation of the CBS is under an acceptable threshold by means of specific risk assessment.

The risk assessment is to be based on available knowledge bases and experience on similar designs, if any, considering the CBS category, connectivity and the functional requirements and specifications of the ship and of the CBS. Cyber threat information from internal and external sources may be used to gain a better understanding of the likelihood and impact of cybersecurity events.

7.8.3 Requirement details

Risk assessment document is to be made and kept up to date by the System integrator during the design and building phase considering possible variations of the original design and newly discovered threats and/or vulnerabilities not known from the beginning.

During the operational life of the ship, the shipowner is to update the risk assessment considering the constant changes in the cyber scenario and new weaknesses identified in CBS onboard in a process of continuous improvement. Should new risks be identified, the shipowner is to update existing or implement new risk mitigation measures.

Should the changes in the cyber scenario be such as to elevate the risk level associated to the CBS under examination above the acceptable risk threshold, the shipowner is to inform IRS and submit the updated risk assessment for evaluation.

The envisaged operational environments for the CBS under examination are to be analyzed in the risk assessment to discern the likelihood of cyber incidents and the impact they could have on the human safety, the safety of the vessel or the marine environment, taking into account the category of the CBS. The attack surface is to be analyzed, taking into account the connectivity of the CBS, possible interfaces for portable devices, logical access restrictions, etc.

Emerging risks related to the specific configuration of the CBS under examination are also to be identified. In the risk assessment, the following elements are to be considered:

- Asset vulnerabilities;
- Threats, both internal and external;
- Potential impacts of cyber incidents affecting the asset on human safety, safety of the vessel and/or threat to the environment;
- Possible effects related to integration of systems, or interfaces among systems, including systems not onboard (e.g. if remote access to onboard systems is provided).

7.8.4 Acceptance criteria

Exclusion of a CBS falling under the scope of applicability of this Section from the application of relevant requirements will be accepted by IRS only if assurance is given that the operation of the CBS has no impact on the safety of operations regarding cyber risk. The said exclusion may be accepted for a CBS which does not fully meet the additional criteria listed below but is provided with a rational explanation together with evidence and is found satisfactory by IRS. IRS may also require submittal of additional documents to consider the said exclusion.

The following criteria are to be met to exclude a system from the scope of applicability of this Section:

a) The CBS is to be isolated (i.e, have no IP-network connections to other systems or networks)

b) The CBS is to have no accessible physical interface ports. Unused interfaces are to be logically disabled. It is not to be possible to connect unauthorized devices to the CBS

c) The CBS is to be located in areas to which physical access is controlled

d) The CBS is not to be an integrated control system serving multiple ship functions as specified in the scope of applicability of this Section (see section 7.1).

The following additional criteria should be considered for the evaluation of risk level acceptability:

a. The CBS should not serve ship functions of category III

b. Known vulnerabilities, threats, potential impacts deriving from a cyber incident affecting the CBS have been duly considered in the risk assessment

c. The attack surface for the CBS is minimized, having considered its complexity, connectivity, physical and logical access points, including wireless access points

Appendix I – Summary of actions and documents

Legend:

Submit	The stakeholder is to submit the document to IRS for verification and approval of compliance with requirements in this Section
Maintain	The stakeholder is to keep the document updated in accordance with procedure for management of change (MoC). Updated document and change management records are to be submitted to IRS as per Section 6.
Demonstrate	The stakeholder is to demonstrate compliance to IRS in accordance with the approved document.
1st AS	First annual survey
AS	Annual survey
SS	Special survey

Document (Ref. Section 7)	Systems integrator			Owner/Company			
	Design	**Construction**	**Commissioning**	**Operation**	**1st AS**	**AS**	**SS**
Approved supplier documentation [7.7]		Maintain	Maintain	Maintain			
Zones and conduit diagram [7.7.1.1]	Submit	Maintain	Maintain	Maintain			
Cyber security design description [7.7.1.2]	Submit	Maintain	Maintain	Maintain			
Vessel asset inventory [7.7.1.3]	Submit	Maintain	Maintain	Maintain			
Risk assessment for the exclusion of CBSs [7.7.1.4] Note 1	Submit	Maintain	Maintain	Maintain			
Description of compensating countermeasures [7.7.1.5]Note 1	Submit	Maintain	Maintain	Maintain			
Ship cyber resilience test procedure [7.7.2.1]		Submit	Demonstrate	Maintain			Demonstrate
Ship cyber security and resilience program [7.7.3.1] - Management of change (MoC) [7.6.1.1.3 d)] - Management of software updates [7.6.1.1.3 d)] - Management of firewalls [7.6.2.1.3 d)] - Management of malware protection [7.6.2.3.3 d)] - Management of access control [7.6.2.4.3 d)] - Management of confidential information [7.6.2.4.3 d)] - Management of remote access [7.6.2.6.3 d)] - Management of mobile and portable devices [7.6.2.7.3 d)]				Maintain	Submit	Demonstrate	

- Detection of security anomalies [7.6.3.1.3 b)] - Verification of security functions [7.6.3.2.3 b)] - Incident response plans [7.6.4.1.3 b)] - Recovery plans [7.6.5.1.3 c)]							
Note 1: If applicable							

Appendix II – Summary of requirements and documents

Vessel asset inventory (section 7.6.1.1)		
CBS security capabilities	Provide documentation of product security updates Provide documentation of dependent component security updates Provide security updates	(sec. 8.5.2) (sec. 8.5.3) (sec. 8.5.4)
CBS documentation	CBS asset inventory Management of change plan	(sec. 8.3.1.1) (sec. 8.3.1.9)
Vessel design documentation	Vessel asset inventory	(7.6.1.1.3 a))
Ship cyber security and resilience program	Management of change	(7.6.1.1.3 d))
	Management of software updates	(7.6.1.1.3 d))

Security zones and network segmentation (section 7.6.2.1)		
CBS security capabilities	-	
CBS documentation	Topology diagrams	(sec. 8.3.1.2)
Vessel design documentation	Zones and conduit diagram Design description Ship cyber resilience test procedure	(7.6.2.1.3 a) (7.6.2.1.3 a)) (7.6.2.1.3 c))
Ship cyber security and resilience program	Management of security zone boundary devices (e.g., firewalls)	(7.6.2.1.3 d))

Network protection safeguards (section 7.6.2.2)		
CBS security capabilities	Denial of service (DoS) protection (# 24) Deterministic output (# 20)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Ship cyber resilience test procedure	(7.6.2.2.3 a))

Ship cyber security and resilience program	-	

Antivirus, antimalware, antispam and other protections from malicious code (section 7.6.2.3)		
CBS security capabilities	Malicious code protection (#18)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.2.3.3 a)) (7.6.2.3.3 c))
Ship cyber security and resilience program	Management of malware protection	(7.6.2.3.3 d))

Access control (section 7.6.2.4)		
CBS security capabilities	Human user id. and auth. (#1) Account management (#2) Identifier management (#3) Authenticator management (#4) Authorization enforcement (#8)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.2.4.3 a)) (7.6.2.4.3 c))
Ship cyber security and resilience program	Management of confidential information	(7.6.2.4.3 d))
	Management of logical and physical access	(7.6.2.4.3 d))

Wireless communication (section 7.6.2.5)		
CBS security capabilities	Wireless access management (#5) Wireless use control (#9)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.2.5.3 a)) (7.6.2.5.3 c))
Ship cyber security and resilience program	-	

Remote access control and communication with untrusted networks (section 7.6.2.6)		
CBS security capabilities	Multifactor authentication (#31) Process / device id. and auth. (#32) Unsuccessful login attempts (#33) System use notification (#34) Access via untrusted networks (#35) Explicit access request approval (#36) Remote session termination (#37) Cryptographic integrity protection (#38) Input validation (#39) Session integrity (#40) Invalidation of session ID (#41)	(sec. 8.4.2)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.2.6.3 a)) (7.6.2.6.3 c))
Ship cyber security and resilience program	Management of remote access and communication with/via untrusted networks	(7.6.2.6.3 d))

Use of mobile and portable devices (section 7.6.2.7)		
CBS security capabilities	Use control for portable devices (#10)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.2.7.3 a)) (7.6.2.7.3 c))
Ship cyber security and resilience program	Management of mobile and portable devices	(7.6.2.7.3 d))

Network operation monitoring (section 7.6.3.1)		
CBS security capabilities	Use control for portable devices (#10) Auditable events (#13) Denial of service (DoS) protection (#24)	(sec. 8.4.1)
	Alarm excessive bandwidth use	(sec. 6.8.3.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities	(sec. 8.3.1.3) (sec. 8.3.1.4)
Vessel design documentation	Ship cyber resilience test procedure	(7.6.3.1.3 a))
Ship cyber security and resilience program	Incident response plans	(7.6.3.1.3 d))

Verification and diagnostic functions of CBS and networks (section 7.6.3.2)		
CBS security capabilities	Security function verification (#19)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities Plans for maintenance and verification	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.7)
Vessel design documentation	Ship cyber resilience test procedure	(7.6.3.2.3 a))
Ship cyber security and resilience program	Verification of security functions	(7.6.3.2.3 b))

Incident response plan (section 7.6.4.1)		
CBS security capabilities	-	
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.4.1.3 a)) (7.6.4.1.3)
Ship cyber security and resilience program	Incident response plans	(7.6.4.1.3 b))

Local, independent and/or manual operation (section 7.6.4.2)		
CBS security capabilities	-	
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.4.2.3 a)) (7.6.4.2.3 b))
Ship cyber security and resilience program	Incident response plans	(7.6.4.2.3 c))

Network isolation (section 7.6.4.3)		
CBS security capabilities	-	
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.4.3.3 a)) (7.6.4.3.3 b))
Ship cyber security and resilience program	Incident response plans	(7.6.4.3.3 c))

Fallback to a minimal risk condition (section 7.6.4.4)		
CBS security capabilities	Deterministic output (#20)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.4.4.3 a) (7.6.4.4.3 b))
Ship cyber security and resilience program	Incident response plans	(7.6.4.1.3 b))

Recovery plan (section 7.6.5.1)		
CBS security capabilities	-	
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.5.1.3 a)) (7.6.5.1.3 b))
Ship cyber security and resilience program	Recovery plans	(7.6.5.1.3 c))

Backup and restore capability (section 7.6.5.2)		
CBS security capabilities	System backup (#26) System recovery and reconstitution (#27)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Ship cyber resilience test procedure	(7.6.5.2.3 a))
Ship cyber security and resilience program	Recovery plan	(7.6.5.1.3 c))

Controlled shutdown, reset, restore and restart (section 7.6.5.3)		
CBS security capabilities	System recovery and reconstitution (#27)	(sec. 8.4.1)
CBS documentation	Description of security capabilities Test procedure for security capabilities Information supporting incident response and recovery plans	(sec. 8.3.1.3) (sec. 8.3.1.4) (sec. 8.3.1.8)
Vessel design documentation	Design description Ship cyber resilience test procedure	(7.6.5.3.3 a)) (7.6.5.3.3 b))
Ship cyber security and resilience program	Recovery plans	(7.6.5.1.3 c))

Risk assessment for exclusion of CBS from the application of requirements (section 7.8)		
CBS security capabilities	-	
CBS documentation	-	
Vessel design documentation	Risk assessment for the exclusion of CBSs	(sec. 7.7.1.4)
Ship cyber security and resilience program	-	

Section 8

Cyber resilience of on-board systems and equipment

8.1 General

8.1.1 Introduction

.1 This Section specifies minimum requirements for cyber resilience of on-board systems and equipment.

.2 This Section does not cover environmental performance for the system hardware and the functionality of the software.

.3 In addition to the requirements of this Section, the following are to be applied:

- Classification Note: *Type Approval of Electrical Equipment used for Control, Monitoring, Alarm and Protection Systems for Use in Ships*, for environmental performance for the system hardware.

- Section 6 for safety of equipment for the functionality of the software.

.4 For navigation and radiocommunication systems, the application of IEC 61162-460 or other equivalent standards in lieu of the required security capabilities in Section 8.4 may be accepted by IRS, on the condition that requirements in Section 6 are complied with.

8.1.2 Scope of application

8.1.2.1 The requirements in this Section are applicable to the following:

- Passenger ships (including passenger high-speed craft) engaged in international voyages

- Cargo ships of 500 GT and upwards engaged in international voyages

- High speed craft of 500 GT and upwards engaged in international voyages

- Mobile offshore drilling units of 500 GT and upwards

- Self-propelled mobile offshore units engaged in construction (i.e. wind turbine installation maintenance and repair, crane units, drilling tenders, accommodation, etc.)

8.1.2.2 The requirements in this Section may be used as guidance for the following:

- Ships of war and troopships

- Cargo ships less than 500 GT

- Vessels not propelled by mechanical means

- Wooden ships of primitive build

- Passenger yachts (passengers not more than 12)

- Pleasure yachts not engaged in trade

- Fishing vessels

- Site specific offshore installations (i.e. FPSOs, FSUs, etc.)

8.1.3 Definitions & Abbreviations

Definition and abbreviations (for the purpose of this Section):

Attack surface: The set of all possible points where an unauthorized user can access a system, cause an effect on or extract data from. The attack surface comprises two categories: digital and physical. The digital attack surface encompasses all the hardware and software that connect to an organization’s network. These include applications, code, ports, servers and websites. The physical attack surface comprises all endpoint devices that an attacker can gain physical access to, such as desktop computers, hard drives, laptops, mobile phones, removable drives and carelessly discarded hardware.

Authentication: Provision of assurance that a claimed characteristic of an identity is correct.

Compensating countermeasure: An alternate solution to a countermeasure employed in lieu of or in addition to inherent

security capabilities to satisfy one or more security requirements.

Computer Based System (CBS): A programmable electronic device, or interoperable set of programmable electronic devices, organized to achieve one or more specified purposes such as collection, processing, maintenance, use, sharing, dissemination, or disposition of information. CBS on-board include IT and OT systems. A CBS may be a combination of subsystems connected via network. On-board CBS may be connected directly or via public means of communications (e.g. Internet) to ashore CBSs, other vessels' CBS and/or other facilities.

Computer Network: A connection between two or more computers for the purpose of communicating data electronically by means of agreed communication protocols.

Control: Means of managing risk, including policies, procedures, guidelines, practices or organizational structures, which can be administrative, technical, management, or legal in nature.

Cyber incident: An event resulting from any offensive cyber manoeuvre, either intentional or unintentional, that targets or affects one or more CBS onboard, which actually or potentially results in adverse consequences to an onboard system, network and computer or the information that they process, store or transmit, and which may require a response action to mitigate the consequences. Cyber incidents include unauthorized access, misuse, modification, destruction or improper disclosure of the information generated, archived or used in onboard CBS or transported into the networks connecting such systems. Cyber incidents do not include system failures.

Cyber resilience: The capability to reduce the occurrence and mitigating the effects of incidents arising from the disruption or impairment of operational technology (OT) used for the safe operation of a ship, which potentially lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.

Defence in depth: Information Security strategy integrating people, technology, and operations capabilities to establish variable barriers across multiple layers and missions of the organization.

Essential Systems: Computer Based System contributing to the provision of services essential for propulsion and steering, and safety of the ship. Essential services comprise "Primary Essential Services" and "Secondary Essential Services": Primary Essential Services are those services which need to be in continuous operation to maintain propulsion and steering; Secondary Essential Services are those services which need not necessarily be in continuous operation to maintain propulsion and steering but which are necessary for maintaining the vessel's safety.

Firewall: A logical or physical barrier that monitors and controls incoming and outgoing network traffic controlled via predefined rules.

Firmware: Software embedded in electronic devices that provide control, monitoring and data manipulation of engineered products and systems. These are normally self-contained and not accessible to user manipulation.

Hardening: Hardening is the practice of reducing a system's vulnerability by reducing its attack surface.

Information Technology (IT): Devices, software and associated networking focusing on the use of data as information, as opposed to Operational Technology (OT).

Integrated system: A system combining a number of interacting sub-systems and/or equipment organized to achieve one or more specified purposes.

Network switch (Switch): A device that connects devices together on a computer network, by using packet switching to receive, process and forward data to the destination device.

Offensive cyber manoeuvre: Actions that result in denial, degradation, disruption, destruction, or manipulation of OT or IT systems.

Operational technology (OT): Devices, sensors, software and associated networking that monitor and control onboard systems. Operational technology systems may be thought of as focusing on the use of data to control or monitor physical processes.

OT system: Computer based systems, which provide control, alarm, monitoring, safety or internal communication functions**.**

Patches: Software designed to update installed software or supporting data to address security vulnerabilities and other bugs or improve operating systems or applications.

Protocols: A common set of rules and signals that computers on the network use to communicate. Protocols allow to perform data communication, network management and security. Onboard networks usually implement protocols based on TCP/IP stacks or various field buses.

Recovery: Develop and implement the appropriate activities to maintain plans for resilience and to restore any capabilities or services that were impaired due to a cyber security event. The Recovery function support s timely return to normal operations to reduce the impact from a cyber security event.

Supplier: A manufacturer or provider of hardware and/or software products, system components or equipment (hardware or software) comprising of the application, embedded devices, network devices, host devices etc. working together as system or a subsystem. The Supplier is responsible for providing programmable devices, sub-systems or systems to the System Integrator.

System: Combination of interacting programmable devices and/or sub-systems organized to achieve one or more specified purposes.

System Categories (I, II, III): System categories based on their effects on system functionality, which are defined in Section 6.

System Integrator: The specific person or organization responsible for the integration of systems and products provided by suppliers into the system invoked by the requirements in the ship specifications and for providing the integrated system. The system integrator may also be responsible for integration of systems in the ship. Until vessel delivery, this role shall be taken by the Shipyard unless an alternative organization is specifically contracted/assigned this responsibility.

Untrusted network: Any network outside the scope of applicability of this Section.

8.2 Security Philosophy

8.2.1 Systems and Equipment

8.2.1.1 A System can consist of group of hardware and software enabling safe, secure and reliable operation of a process. Typical example could be Engine control system, DP system, etc.

8.2.1.2 Equipment may be one of the following:

- Network devices (i.e. routers, managed switches)
- Security devices (i.e. firewall, Intrusion Detection System)
- Computers (i.e. workstation, servers)
- Automation devices (i.e. Programmable Logic Controllers)
- Virtual machine cloud-hosted

8.2.2 Cyber Resilience

The cyber resilience requirements in section 8.4 will be applicable for all systems in scope of Section 7 as applicable. Additional requirements related to interface with untrusted networks will only apply for systems where such connectivity is designed.

8.2.3 Essential Systems Availability

8.2.3.1 Security measures for Essential systems are not to adversely affect the availability of systems.

8.2.3.2 Implementation of security measures is not to cause loss of safety functions, loss of control functions, loss of monitoring functions or loss of other functions which could result in health, safety and environmental consequences.

8.2.3.3 The system is to be adequately designed to allow the ship to continue its mission critical operations in a manner that ensures the confidentiality, integrity, and availability of the data necessary for safety of the vessel, its systems, personnel and cargo.

8.2.4 Compensating Countermeasures

8.2.4.1 Compensating countermeasures may be employed in lieu of or in addition to inherent security capabilities to satisfy one or more security requirements.

Compensating countermeasure(s) are to meet the intent and rigor of the original stated requirement considering the referenced standards as well as the differences between

each requirement and the related items in the standards, and follow the principles specified in section 8.3.1.3.

8.3 Documentation

8.3.1 CBS Documentation

The following documents are to be submitted to IRS for review and approval in accordance with the requirements in this Section. See also section 8.6.2.

8.3.1.1 CBS asset inventory

The CBS asset inventory is to include the information as follows:

- List of hardware components (e.g., host devices, embedded devices, network devices)Name
- Brand/manufacturer
- Model/type
- Short description of functionality/purpose
- Physical interfaces (e.g., network, serial)
- Name/type of system software (e.g., operating system, firmware)
- Version and patch level of system software
- Supported communication protocols

List of software components (e.g., application software, utility software)

- The hardware component where it is installed
- Brand/manufacturer
- Model/type
- Short description of functionality/purpose
- Version of software

8.3.1.2 Topology diagrams

The physical topology diagram is to illustrate the physical architecture of the system. It is to be possible to identify the hardware components in the CBS asset inventory. The diagram is to illustrate the following:

- All endpoints and network devices, including identification of redundant units
- Communication cables (networks, serial links), including communication with I/O units
- Communication cables to other networks or systems

The logical topology diagram is to illustrate the data flow between components in the system. The diagram is to illustrate the following:

- Communication endpoints (e.g. workstations, controllers, servers)
- Network devices (switches, routers, firewalls)
- Physical and virtual computers
- Physical and virtual communication paths
- Communication protocols

One combined topology diagram may be acceptable if all requested information can be clearly illustrated.

8.3.1.3 Description of security capabilities

This document is to describe how the CBS with its hardware and software components meets the required security capabilities in section 8.4.1.

Any network interfaces to other CBSs in the scope of applicability of Section 7 are to be described. The description is to include destination CBS, data flows, and communication protocols. If the System integrator has allocated the destination CBS to another security zone, components providing protection of the security zone boundary (see Section 7.6.2.2.1) are to be described in detail if delivered as part of the CBS.

Any network interfaces to other systems or networks outside the scope of applicability of Section 7 (untrusted networks) are to be described. The description is to specify compliance with the additional security capabilities in section 8.4.2, and include relevant procedures or instructions for the crew. Components providing protection of the security zone boundary (see section 7.6.2.2.1) are to be described in detail if delivered as part of the CBS.

A separate chapter is to be designated for each requirement. All hardware and software components in the system are to be addressed in the description, as relevant.

If any requirement is not fully met, this is to be specified in the description, and compensating countermeasures are to be proposed. The compensating countermeasures should:

- Protect against the same threats as the original requirement
- Provide an equal level of protection as the original requirement
- Not be a security control that is required by other requirements in this Section
- Not introduce higher security risk

Any supporting documents (e.g. OEM information) necessary to verify compliance with the requirements are to be referenced in the description and submitted.

8.3.1.4 Test procedure of security capabilities

This document is to describe how to demonstrate by testing that the system complies with the requirements in section 8.4.1 and 8.4.2, including any compensating countermeasures. Demonstration of compliance by analytic evaluation may be specially considered. The procedure is to include a separate chapter for each applicable requirement and describe:

- Necessary test setup (i.e. to ensure the test can be repeated with the same expected result)
- Test equipment
- Initial condition(s)
- Test methodology, detailed test steps
- Expected results and acceptance criteria

The procedure is also to include means to update test results and record findings during the testing.

8.3.1.5 Security configuration guidelines

This document is to describe recommended configuration settings of the security capabilities and specify default values. The objective is to ensure the security capabilities are implemented in accordance with Section 7 and any specifications by the System integrator (e.g. user accounts, authorization, password policies, safe state of machinery, firewall rules, etc.)

The document is to serve as basis for verification of item no. 29 in section 8.4.1.

8.3.1.6 Secure development lifecycle documents

This documentation is to be submitted to IRS upon request and is to describe the supplier's processes and controls in accordance with requirements for secure development lifecycle in section 8.5. Software updates and patching are to be described. The document is a preparatory document for survey as per section 8.6.3.4.

8.3.1.7 Plans for maintenance and verification of the CBS

This document is to be submitted to IRS upon request and is to include procedures for security-related maintenance and testing of the system. The document is to include instructions for how the user can verify correct operation of the system's security functions as required by item no.19 in section 8.4.1.

8.3.1.8 Information supporting the owner’s incident response and recovery plan

This document is to be submitted to IRS upon request and is to include procedures or instructions allowing the user to accomplish the following:

- Local independent control (see sec. 7.6.4.2)
- Network isolation (see sec. 7.6.4.3)
- Forensics by use of audit records (see sec. 8.4.1 item no.13)
- Deterministic output (see sec. 7.6.4.4 and sec. 8.4.1 item no. 20)
- Backup (see sec. 8.4.1 item no. 26)
- Restore (see sec. 8.4.1 item no. 27)
- Controlled shutdown, reset, roll-back and restart (see sec. 7.6.5.3)

8.3.1.9 Management of change plan

This document is to be submitted to IRS upon request. This procedure is not specific for cyber security and is also required by Section 6.

8.3.1.10 Test reports

CBSs with Type approval certificate covering the security capabilities of this Section may be

exempted from survey by IRS. However, test reports signed by the supplier are to be submitted to IRS, demonstrating that the supplier has completed design, construction, testing, configuration, and hardening as would otherwise be verified by IRS in survey (section 8.6.3).

8.4 System Requirements

This subsection specifies the required security capabilities for CBSs in the scope specified in section 8.1.1.

The requirements in this subsection are based on certain selected requirements in IEC 62443-3-3. To determine the full content, rationale and relevant guidance for each requirement, the subject standard may be referred.

8.4.1 Required security capabilities

Security capabilities required for CBSs in the scope specified in Section 8.1.1 are indicated in Table 8.4.1

Table 8.4.1 : Required security capabilities for all CBSs

Item No	Objective	Requirements
Protect against casual or coincidental access by unauthenticated entities		
1	Human user identification and authentication	The CBS is to identify and authenticate all human users who can access the system directly or through interfaces (Ref. IEC 62443-3-3/SR 1.1)
2	Account management	The CBS is to provide the capability to support the management of all accounts by authorized users, including adding, activating, modifying, disabling and removing account (Ref. IEC 62443-3-3/SR 1.3)
3	Identifier management	The CBS is to provide the capability to support the management of identifiers by user, group and role. (Ref. IEC 62443-3-3/SR 1.4)
4	Authenticator management	The CBS is to provide the capability to: - Initialize authenticator content - Change all default authenticators upon control system installation - Change/refresh all authenticators - Protect all authenticators from unauthorized disclosure and modification when stored and transmitted. (Ref. IEC 62443-3-3/SR 1.5)
5	Wireless access management	The CBS is to provide the capability to identify and authenticate all users (humans, software processes or devices) engaged in wireless communication (Ref. IEC 62443-3-3/SR 1.6)
6	Strength of password-based authentication	The CBS is to provide the capability to enforce configurable password strength based on minimum length and variety of character types. (Ref. IEC 62443-3-3/SR 1.7)
7	Authenticator feedback	The CBS is to obscure feedback during the authentication process. (Ref. IEC 62443-3-3/SR 1.10)
Protect against casual or coincidental misuse		
8	Authorization enforcement	On all interfaces, human users are to be assigned authorizations in accordance with the principles of segregation of duties and least privilege. (Ref. IEC 62443-3-3/SR 2.1)
9	Wireless use control	The CBS is to provide the capability to authorize, monitor and enforce usage restrictions for wireless connectivity to the system according to commonly accepted security industry practices (Ref. IEC 62443-3-3/SR 2.2)

10	Use control for portable and mobile devices	When the CBS supports use of portable and mobile devices, the system is to include the capability to a) Limit the use of portable and mobile devices only to those permitted by design b) Restrict code and data transfer to/from portable and mobile devices Note: Port limits / blockers (and silicone) could be accepted for a specific system (Ref. IEC 62443-3-3/SR 2.3)
11	Mobile code	The CBS is to control the use of mobile code such as java scripts, ActiveX and PDF. (Ref. IEC 62443-3-3/SR 2.4)
12	Session lock	The CBS is to be able to prevent further access after a configurable time of inactivity or following activation of manual session lock. (Ref. IEC 62443-3-3/SR 2.5)
13	Auditable events	The CBS is to generate audit records relevant to security for at least the following events: access control, operating system events, backup and restore events, configuration changes, loss of communication. (Ref. IEC 62443-3-3/SR 2.8)
14	Audit storage capacity	The CBS is to provide the capability to allocate audit record storage capacity according to commonly recognized recommendations for log management. Auditing mechanisms shall be implemented to reduce the likelihood of such capacity being exceeded. (Ref. IEC 62443-3-3/SR 2.9)
15	Response to audit processing failures	The CBS is to provide the capability to prevent loss of essential services and functions in the event of an audit processing failure. (Ref. IEC 62443-3-3/SR 2.10)
16	Timestamps	The CBS is to timestamp audit records. (Ref. IEC 62443-3-3/SR 2.11)
Protect the integrity of the CBS against casual or coincidental manipulation		
17	Communication integrity	The CBS is to protect the integrity of transmitted information. Note: Cryptographic mechanisms are to be employed for wireless networks. (Ref. IEC 62443-3-3/SR 3.1)
18	Malicious code protection	The CBS is to provide capability to implement suitable protection measures to prevent, detect and mitigate the effects due to malicious code or unauthorized software. It is to have the feature for updating the protection mechanisms (Ref. IEC 62443-3-3/SR 3.2)
19	Security functionality verification	The CBS is to provide the capability to support verification of the intended operation of security functions and report when anomalies occur during maintenance (Ref. IEC 62443-3-3/SR 3.3)
20	Deterministic output	The CBS is to provide the capability to set outputs to a predetermined state if normal operation cannot be maintained as a result of an attack. The predetermined state could be: - Unpowered state, - Last-known value, or - Fixed value (Ref. IEC 62443-3-3/SR 3.6)
Prevent the unauthorized disclosure of information via eavesdropping or casual exposure		
21	Information confidentiality	The CBS is to provide the capability to protect the confidentiality of information for which explicit read authorization is supported, whether at rest or in transit. Note: For wireless network, cryptographic mechanisms shall be employed to protect confidentiality of all information in transit. (Ref. IEC 62443-3-3/SR 4.1)

22	Use of cryptography	If cryptography is used, the CBS is to use cryptographic algorithms, key sizes and mechanisms according to commonly accepted security industry practices and recommendations. (Ref. IEC 62443-3-3/SR 4.3)
Monitor the operation of the CBS and respond to incidents		
23	Audit log accessibility	The CBS is to provide the capability for accessing audit logs on read only basis by authorized humans and/or tools. (Ref. IEC 62443-3-3/SR 6.1)
Ensure that the control system operates reliably under normal production conditions		
24	Denial of service protection	The CBS is to provide the minimum capability to maintain essential functions during DoS events. Note: It is acceptable that the CBS may operate in a degraded mode upon DoS events, but it is not to fail in a manner which may cause hazardous situations. Overload-based DoS events should be considered, i.e. where the networks capacity is attempted flooded, and where the resources of a computer is attempted consumed. (Ref. IEC 62443-3-3/SR 7.1)
25	Resource management	The CBS is to provide the capability to limit the use of resources by security functions to prevent resource exhaustion. (Ref. IEC 62443-3-3/SR 7.2)
26	System backup	The identity and location of critical files and the ability to conduct backups of user-level and system-level information (including system state information) is to be supported by the CBS without affecting normal operations (Ref. IEC 62443-3-3/SR 7.3)
27	System recovery and reconstitution	The CBS is to provide the capability to be recovered and reconstituted to a known secure state after a disruption or failure. (Ref. IEC 62443-3-3/SR 7.4)
28	Alternative power source	The CBS is to provide the capability to switch to and from an alternative power source without affecting the existing security state or a documented degraded mode. (Ref. IEC 62443-3-3/SR 7.5)
29	Network and security configuration settings	The CBS traffic is to provide the capability to be configured according to recommended network and security configurations as described in guidelines provided by the supplier. The CBS is to provide an interface to the currently deployed network and security configuration settings. (Ref. IEC 62443-3-3/SR 7.6)
30	Least Functionality	The installation, the availability and the access rights of the following are to be limited to the strict needs of the functions provided by the CBS: - operating systems software components, processes and services - network services, ports, protocols, routes and hosts accesses and any software (Ref. IEC 62443-3-3/SR 7.7)

8.4.2 Additional security capabilities

The following additional security capabilities are required for CBSs with network communication to untrusted networks (i.e. interface to any networks outside the scope of Section 7).

CBSs with communication traversing the boundaries of security zones are also to meet requirements for network segmentation and zone boundary protection in section 7.6.2.1 and 7.6.2.2.

Table 8.4.2: Additional security capabilities for CBSs with network communication to untrusted networks		
Item No	**Objective**	**Requirements**
31	Multifactor authentication for human users	Multifactor authentication is required for human users when accessing the CBS from or via an untrusted network. (Ref. IEC 62443-3-3/SR 1.1, RE 2)
32	Software process and device identification and authentication	The CBS is to identify and authenticate software processes and devices (Ref. IEC 62443-3-3/SR 1.2)
33	Unsuccessful login attempts	The CBS is to enforce a limit of consecutive invalid login attempts from untrusted networks during a specified time period. (Ref. IEC 62443-3-3/SR 1.11)
34	System use notification	The CBS is to provide the capability to display a system use notification message before authenticating. The system use notification message is to be configurable by authorized personnel. (Ref. IEC 62443-3-3/SR 1.12)
35	Access via Untrusted Networks	Any access to the CBS from or via untrusted networks is to be monitored and controlled. (Ref. IEC 62443-3-3/SR 1.13)
36	Explicit access request approval	The CBS is to deny access from or via untrusted networks unless explicitly approved by authorized personnel on board. (Ref. IEC 62443-3-3/SR 1.13, RE1)
37	Remote session termination	The CBS is to provide the capability to terminate a remote session either automatically after a configurable time period of inactivity or manually by the user who initiated the session. (Ref. IEC 62443-3-3/SR 2.6)
38	Cryptographic integrity protection	The CBS is to employ cryptographic mechanisms to recognize changes to information during communication with or via untrusted networks. (Ref. IEC 62443-3-3/SR 3.1, RE1)
39	Input validation	The CBS is to validate the syntax, length and content of any input data via untrusted networks that is used as process control input or input that directly impacts the action of the CBS. (Ref. IEC 62443-3-3/SR 3.5)
40	Session integrity	The CBS is to protect the integrity of sessions. Invalid session IDs are to be rejected. (Ref. IEC 62443-3-3/SR 3.8)
41	Invalidation of session IDs after session termination	The system is to invalidate session IDs upon user logout or other session termination (including browser sessions). (Ref. IEC 62443-3-3/SR 3.8, RE1)

8.5 Secure Development Lifecycle Requirements

A Secure Development Lifecycle (SDLC) broadly addressing security aspects in following stages is to be followed for the development of systems or equipment

- Requirement analysis phase
- Design phase
- Implementation phase
- Verification phase
- Release phase
- Maintenance Phase
- End of life phase

A document is to be developed that records how the security aspects have been addressed in above phases and is to at a minimum integrate controlled processes as set out in 8.5.1 to 8.5.7. The subject document is required to be submitted to IRS for review and approval.

8.5.1 (Ref. IEC 62443-4-1/SM-8) The manufacturer is to have procedural and technical controls in place to protect private keys used for code signing, if applicable, from unauthorized access or modification.

8.5.2 (Ref. IEC 62443-4-1/SUM-2) A process is to be employed to ensure that documentation about product security updates is made available to users (which could be through establishing a cyber security point of contact or periodic publication which can be accessed by the user) that includes but is not limited to:

a) The product version number(s) to which the security patch applies;

b) Instructions on how to apply approved patches manually and via an automated process;

c) Description of any impacts that applying the patch to the product can have, including reboot;

d) Instructions on how to verify that an approved patch has been applied; and

e) Risks of not applying the patch and mediations that can be used for patches that are not approved or deployed by the asset owner.

8.5.3 (Ref. IEC 62443-4-1/SUM-3) A process is to be employed to ensure that documentation about dependent component or operating system security updates is available to users that includes but is not limited to:

a) Stating whether the product is compatible with the dependent component or operating system security update;

8.5.4 (Ref. IEC 62443-4-1/SUM-4) A process is to be employed to ensure that security updates for all supported products and product versions are made available to product users in a manner that facilitates verification that the security patch is authentic.

Note: The manufacturer is to have QA process to test the updates before releasing.

8.5.5 (Ref. IEC 62443-4-1/SG-1) A process is to exist to create product documentation that describes the security defence in depth strategy for the product to support installation, operation and maintenance that includes:

a) Security capabilities implemented by the product and their role in the defence in depth strategy;

b) Threats addressed by the defence in depth strategy; and

c) Product user mitigation strategies for known security risks associated with the product, including risks associated with legacy code.

8.5.6 (Ref. IEC 62443-4-1/SG-2) A process is to be employed to create product user documentation that describes the security defence in depth measures expected to be provided by the external environment in which the product is to be used.

8.5.7 (Ref. IEC 62443-4-1/SG-3) A process is to be employed to create product user documentation that includes guidelines for hardening the product when installing and maintaining the product. The guidelines shall include, but are not limited to, instructions, rationale and recommendations for the following:

a) Integration of the product, including third-party components, with its product security context

b) Integration of the product's application programming interfaces/protocols with user applications;

c) Applying and maintaining the product's defence in depth strategy

d) Configuration and use of security options/capabilities in support of local security policies, and for each security option/capability:

   i. its contribution to the product's defence in depth strategy

   ii. descriptions of configurable and default values that include how each affects security along with any

potential impact each has on work practices; and

iii. setting/changing/deleting its value;

e) Instructions and recommendations for the use of all security-related tools and utilities that support administration, monitoring, incident handling and evaluation of the security of the product;

f) Instructions and recommendations for periodic security maintenance activities;

g) Instructions for reporting security incidents for the product to the supplier;

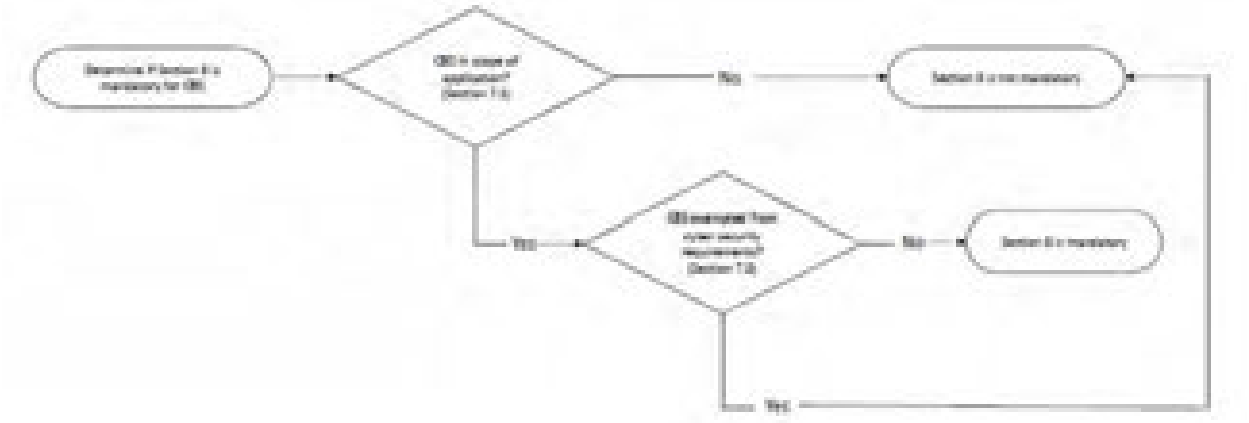
h) Description of the security best practices for maintenance and administration of the product.

8.6 Demonstration of compliance

8.6.1 Introduction

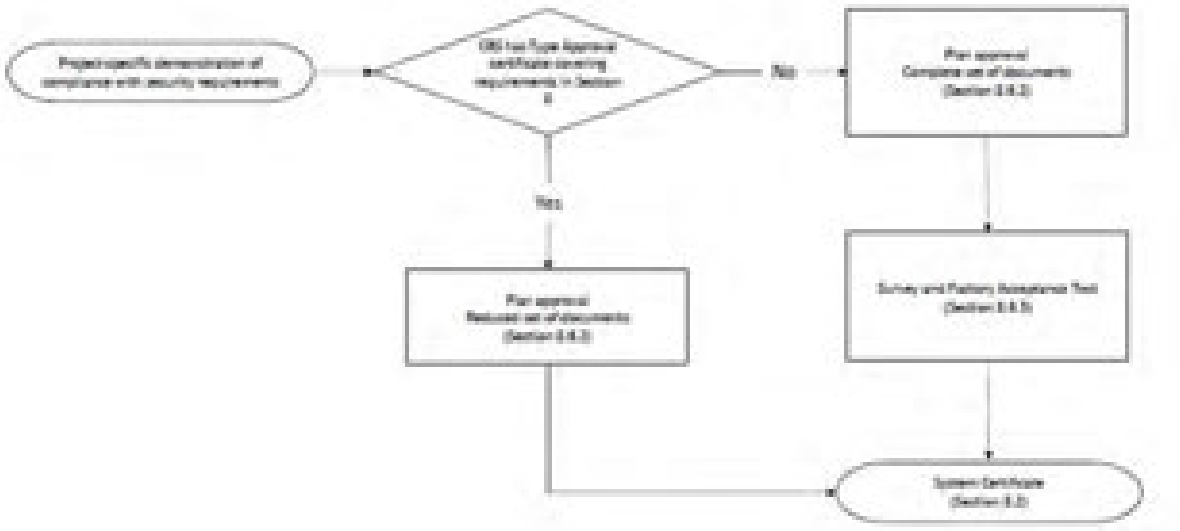
Suppliers are to determine if this section is mandatory for the CBS in cooperation with the System integrator.

Figure 1



Compliance with security requirements is to be demonstrated as indicated in Figure 2. This classification process is ship-specific and shall result in a System certificate.

Figure 2



Type approval is voluntary and applies for CBSs that are standard and routinely manufactured. See Section 6 for definition of System certification and Type approval. In cases where the manufacturer is desirous of type approval, the requirements in IRS Classification Notes: *Type Approval of Cyber Secured Control System Components* are to be complied with.

The process in Figure 1 and Figure 2 applies also if other equivalent standards are applied for navigation and radiocommunication equipment (see section 1.3). In such case:

- the process in Figure 1 illustrates if the equivalent standard is mandatory (in lieu of Section 8)
- the process in Figure 2 illustrates that the certification process is lessened if the CBS has been type approved in accordance with the equivalent standard.

8.6.2 Plan approval

Plan approval is assessment of documents of a CBS intended for a specific vessel. The documents in section 8.3 are required to be submitted by the supplier.

If the CBS holds a valid Type approval certificate covering the requirements of this Section, subject to approval by IRS, the supplier may submit a reduced set of vessel-specific documents to IRS (see Appendix I).

The approved version of the documents is to be included in the delivery of the CBS to the system integrator.

8.6.3 Survey and factory acceptance test

Survey and factory acceptance testing (FAT) is a vessel-specific verification activity required for CBSs that do not hold a valid Type approval certificate covering the requirements of this Section.

The objective of the survey and FAT is to demonstrate by testing and/or analytic evaluation that the CBS complies with applicable requirements in this Section. The survey and FAT is to be carried out at the supplier's premises or at other works having the adequate apparatus for testing and inspection.

After completed plan approval and survey/FAT, IRS will issue a System certificate that is to accompany the CBS upon delivery to the system integrator.

The following subsections specify the survey and FAT activities.

8.6.3.1 General survey items

The supplier is to demonstrate that design, construction, and internal testing has been completed.

It is also to be demonstrated that the system to be delivered is correctly represented by the approved documentation. This is to be done by inspecting the system and comparing the components and arrangement/architecture with the asset inventory (section 8.3.1.1) and the topology diagrams (section 8.3.1.2).

8.6.3.2 Test of security capabilities

The supplier is to test the required security capabilities on the system to be delivered. The tests are to be carried out in accordance with the approved test procedure in section 8.3.1.4 and be witnessed/accepted by the Surveyor.

The tests are to provide the Surveyor with reasonable assurance that all requirements are met. This implies that testing of identical components is normally not required.

8.6.3.3 Correct configuration of security capabilities

The supplier is to test/ demonstrate for the Surveyor that security settings in the system's components have been configured in accordance with the configuration guidelines in section 8.3.1.5. This demonstration may be carried out in conjunction with testing of the security capabilities.

The security settings are to be documented in a report, e.g. a ship-specific instance of the configuration guidelines.

8.6.3.4 Secure development lifecycle

The supplier is to, in accordance with documentation in section 8.3.1.6, demonstrate compliance with requirements for secure development lifecycle in section 8.5.

8.6.3.4.1 Controls for private keys (Ref. IEC 62443-4-1/SM-8)

This requirement applies if the system includes software that is digitally signed for the purpose of enabling the user to verify its authenticity.

The supplier is to present management system documentation substantiating that policies, procedures and technical controls are in place to protect generation, storage and use of private keys used for code signing from unauthorized access.

The policies and procedures are to address roles, responsibilities and work processes. The technical controls are to include e.g. physical access restrictions and cryptographic hardware (e.g. Hardware security module) for storage of the private key.

8.6.3.4.2 Security update documentation (Ref. IEC 62443-4-1/SUM-2)

The supplier is to present management system documentation substantiating that a process is established in the organization to ensure security updates are informed to the users. The information to the users is to include the items listed in section 8.5.2.

8.6.3.4.3 Dependent component security update documentation (Ref. IEC 62443-4-1/SUM-3)

The supplier is to present management system documentation, as required by section 8.5.3, substantiating that a process is established in the organization to ensure users are informed whether the system is compatible with updated versions of acquired software in the system (new versions/patches of operating system or firmware). The information is to address how to manage risks related to not applying the updated acquired software.

8.6.3.4.4 Security update delivery (Ref. IEC 62443-4-1/SUM-4)

The supplier is to present management system documentation, as required by section 8.5.4, substantiating that a process is established in the organization ensuring that system security updates are made available to users, and describing how the user may verify the authenticity of the updated software.

8.6.3.4.5 Product defence in depth (Ref. IEC 62443-4-1/SG-1)

The supplier is to present management system documentation, as required by section 8.5.5, substantiating that a process is established in the organization to document a strategy for defence-in-depth measures to mitigate security threats to software in the CBS during installation, maintenance and operation.

Examples of threats could be installation of unauthorized software, weaknesses in the patching process, tampering with software in the operational phase of the ship.

8.6.3.4.6 Defence in depth measures expected in the environment (Ref. IEC 62443-4-1/SG-2)

The supplier is to present management system documentation, as required by section 8.5.6, substantiating that a process is established in the organization to document defence-in-depth measures expected to be provided by the external environment, such as physical arrangement, policies and procedures.

8.6.3.4.7 Security hardening guidelines (Ref. IEC 62443-4-1/SG-3)

The supplier is to present management system documentation, as required by section 8.5.7, substantiating that a process is established in the organization to ensure that hardening guidelines are produced for the system.

The guidelines are to specify how to reduce vulnerabilities in the system by removal/prohibiting /disabling of unnecessary software, accounts, services, etc.

Appendix I

The following table summarizes documents to be submitted by the supplier to IRS.

Document	Requirements	IRS
CBS asset inventory (sec. 8.3.1.1)	To be incorporated in vessel asset inventory (E26 sec.7.6.1.1)	Approve [1) 2)]
Topology diagrams (sec. 8.3.1.2)	Enabling System integrator to design security zones and conduits (E26 sec.7.6.2.1)	Approve [1) 2)]
Description of security capabilities (sec. 8.3.1.3)	Required security capabilities (sec. 8.4.1)	Approve [1)]
	Additional security capabilities, if applicable (sec. 8.4.2)	
Test procedure for security capabilities (sec. 8.3.1.4)	Required security capabilities (sec. 8.4.1)	Approve [1)]
	Additional security capabilities, if applicable (sec. 8.4.2)	
Security configuration guidelines (sec. 8.3.1.5)	Network and security configuration settings (sec. 8.4.1 item 29)	Info [1)]
Secure development lifecycle (sec. 8.3.1.6)	SDLC requirements (sec. 8.5)	Approve [1)]
Plans for maintenance and verification (sec. 8.3.1.7)	Security functionality verification (sec. 8.4.1 item 19)	Info [1)]
Information supporting incident response and recovery plans (sec. 8.3.1.8)	Auditable events (sec. 8.4.1 item 13)	Info [1)]
	Deterministic output (sec. 8.4.1 item 20)	Info [1)]
	System backup (sec. 8.4.1 item 26)	Info [1)]
	System recovery and reconstitution (sec. 8.4.1 item 27)	Info [1)]
Management of change plan (sec. 8.3.1.9)	Management of change process (sec. 6)	Info [1)]
Test reports (sec. 8.3.1.10)	Configuration of security capabilities and hardening (sec. 8.3.1.5 and sec. 8.5.7)	Info [2)]
Note 1: Required for CBS without type approved security capabilities Note 2: Required for CBS with type approved security capabilities		

End of Chapter

Chapter 8

Electrical Installations

Contents

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Section 1

General Requirements

1.1 Scope

1.1.1 The requirements of this Chapter are applicable to both direct current and alternating current installations unless specifically stated otherwise.

1.1.2 The requirements of this Chapter apply to all ships. Consideration will be given to arrangements or details of equipment which comply with other National or International standards in so far as they are not less effective and reliable.

The arrangements for vessels intended for restricted services or for special services will be specially considered.

1.1.3 Whilst these requirements are considered to meet those of the "International Convention for the Safety of Life at Sea, 1974, including amendments", attention should also be given to any relevant statutory requirements of the National Authority of the country in which the ship is to be registered.

1.1.4 In addition to the requirements of this Chapter, vessels using batteries as the main and/ or additional source of power for propulsion are also to be in accordance with IRS '*Guidelines on Battery Powered Vessels*'.

Requirements for approval of Lithium-ion battery systems to be used in battery powered vessels or hybrid vessels are specified in IRS Classification Note "*Approval of Lithium-ion Battery Systems"*.

1.2 Definitions

1.2.1 **Normal operational and habitable condition** is condition under which the ship as

a whole, the machinery, services, means and aids ensuring propulsion, ability to steer, safe navigation, fire and flooding safety, internal and external communications and signals, means of escape and emergency boat winches, as well as designed comfortable conditions of habitability are in working order and functioning normally.

1.2.2 **Emergency condition** is a condition under which any services needed for normal operational and habitable conditions are not in a working order due to the failure of the main source of electrical power.

1.2.3 **Main source of electrical power** is a source intended to supply electrical power to the main switchboard for distribution to all services necessary for maintaining the ship in normal operational and habitable conditions.

1.2.4 **Main generating station** is the space in which the main source of electrical power is situated.

1.2.5 **Main switchboard** is a switchboard which is directly supplied by the main source of electrical power and is intended to distribute electrical energy to ship's services.

1.2.6 **Emergency switchboard** is a switchboard which in the event of failure of the main electrical power supply system is directly supplied by the emergency source of electrical power or transitional source of emergency power and is intended to distribute electrical energy to the emergency services.

1.2.7 **Emergency source of electrical power** is a source of electrical power, intended to supply the emergency switchboard in the event of failure of the supply from the main source of electrical power.

1.3 Surveys

1.3.1 Electrical propelling machinery and associated equipment together with auxiliary services for the safety of the ship are to be installed in accordance with the relevant requirements of this Chapter, surveyed and have tests witnessed by the Surveyors.

1.3.2 All generators, motors and transformers of 100 [kW] and over, intended for essential services, are to be surveyed during test and if appropriate also during manufacturing.

Where an alternative survey scheme has been approved by IRS as per Ch.1, Sec.4, the attendance of the Surveyor during manufacturing may not be necessary.

1.3.3 All electric propelling machinery including switchgear, control gear, converting equipment, cables, main and auxiliary generators, motors and exciters are to be surveyed by the Surveyors during manufacture and testing.

1.3.4 For refrigerated cargo installations having HY class notation, motors are to be tested and certificates furnished by the manufacturers. Motors of 100 [kW] or over are to be surveyed by the Surveyors during manufacture and testing.

1.4 Plans

1.4.1 The following plans and details for electrical installations including propulsion equipment, if fitted, and auxiliary machinery are to be submitted in triplicate for approval:

a) Schematic diagram of the electrical system including information such as cable sizes, type of insulation, normal working currents, line drop of voltages and details of protection devices together with rating and interrupting capacity of circuit breakers and fuses;

b) Calculations of prospective short circuit current of main busbars and secondary side of transformers (Additionally load schedule is to be submitted for information);

c) Construction details, sectional views, type, rating, make of rotating machines rated at 100 [kW] and above, where welding is to be applied to shafts of machines and plans showing the construction;

d) The arrangement plans and circuit diagrams of switchgear and controlgear giving details of isolating, protective and control devices, details of bus bars, cable sizes, type of insulation and normal working currents in the circuits. Additional details such as mechanical stresses on the bus bars by short circuit, bus bar supporting arrangements, verification of protective devices;

e) General diagrams of emergency circuits, control and alarms for essential services and internal communications together with details of sources of supply, cable sizes and types of insulation and any other

information required for assessment of such circuits;

f) Plans for generators, motors and seatings together with diagrams of control gears, cables and circuits for electric propulsion equipment, if fitted. A harmonic distortion calculation report, indicating distortion levels at MSB and power distribution centers is to be submitted for review for all vessels with electric propulsion and for vessels fitted with harmonic filters. Key diagram and explanation of the system are to be submitted;

g) Plans of electric slip couplings, if fitted, showing details of construction and scantlings together with diagrams of control gear and other electrical components;

h) For tankers and similar vessels, a general arrangement of the ship showing hazardous spaces and the location of the electrical equipment in such spaces. A schedule of safe type electrical equipment located in hazardous spaces giving details of the type of equipment fitted, the certifying authority and copies of the certificates. Where the prescriptive requirements within SOLAS and related codes (IBC, IGC) and the standards published by the International Electro-technical Commission, IEC, such as but not limited to IEC 60092-502, are not aligned, the prescriptive requirements in SOLAS and Codes take precedence and are to be applied.

1.4.2 Additional information may be required for vessels engaged in special services or of a novel design.

1.5 Essential and other services

1.5.1 Essential services

1.5.1.1 Essential services are divided into following types of services:

a) "Primary Essential Services" and

b) "Secondary Essential Services".

Definitions and examples of such services are given in 1.5.1.2 and 1.5.1.3.

1.5.1.2 Primary Essential Services

Primary Essential Services are those services which need to be in continuous operation to maintain propulsion and steering. Examples of equipment for primary essential services are as follows:

- Steering gears
- Pumps for controllable pitch propellers
- Scavenging air blower, fuel oil supply pumps, fuel valve cooling pumps, lubricating oil pumps and cooling water pumps for main and auxiliary engines and turbines necessary for propulsion
- Forced draught fans, feed water pumps, water circulating pumps, vacuum pumps and condensate pumps for steam plants on steam turbine ships, and also for auxiliary boilers on ships where steam is used for equipment supplying primary essential services
- Oil burning installations for steam plants on steam turbine ships and for auxiliary boilers where steam is used for equipment supplying primary essential services
- Azimuth thrusters which are the sole means for propulsion/steering and associated lubricating oil pumps, cooling water pumps
- Electrical equipment for electric propulsion plant with lubricating oil pumps and cooling water pumps
- Electric generators and associated power sources supplying the above mentioned equipment
- Hydraulic pumps supplying the above mentioned equipment
- Viscosity control equipment for heavy fuel oil
- Control, monitoring and safety devices/systems of equipment for primary essential services.

1.5.1.3 Secondary Essential Services

Secondary Essential Services are those services which need not necessarily be in continuous operation to maintain propulsion and steering but which are necessary for maintaining the vessel's safety. Example of equipment for secondary essential services are as follows:

- Windlass
- Fuel oil transfer pumps and fuel oil treatment equipment
- Lubricating oil transfer pumps and lubrication oil treatment equipment
- Pre-heaters for heavy fuel oil
- Starting air and control air compressors
- Bilge, ballast and heeling pumps
- Fire pumps and other fire extinguishing medium pumps

- Ventilating fans for engine and boiler rooms
- Services considered necessary to maintain dangerous spaces in a safe condition
- Navigation lights, aids and signals
- Internal safety communication equipment
- Fire detection and alarm system
- Lighting system
- Electrical equipment for watertight closing appliances
- Electric generators and associated power sources supplying the above equipment
- Hydraulic pumps supplying the above equipment
- Control, monitoring and safety systems for cargo containment systems
- Control, monitoring and safety devices / systems of equipment for secondary essential services.
- Equipment used for cooling and maintaining lesser ambient temperature (refer Ch.1, 1.7.4)

1.5.2 Services such as following are considered necessary for minimum comfortable conditions of habitability :

- cooking;
- heating;
- domestic refrigeration;
- mechanical ventilation;
- sanitary and fresh water;
- electric generators and associated power sources supplying the above mentioned equipment.

1.5.3 Services such as following, which are in addition to 1.5.1 and 1.5.2, are considered necessary to maintain the ship in a normal seagoing operational and habitable condition :

- cargo handling and cargo gear equipment;
- hotel services, other than those required for habitable conditions;
- thrusters, other than those for dynamic positioning.

1.6 Ambient reference conditions

1.6.1 For details regarding ambient reference conditions, See Ch.1.

1.7 Vibrations, accelerations

1.7.1 Electrical equipment is to be constructed to withstand when subjected to following :

vibrations within the frequency range 5 - 50 Hz and with vibration velocity amplitude 20 [mm/sec].

peak accelerations caused by the ship's movements in waves - $\pm$ 0.6 g for vessels of length exceeding 90 [m] and $\pm$ 1 g for other vessels.

1.7.2 All connections for current-carrying parts and earthing connections are to be fixed so that they cannot loosen by vibrations. This also applies to fixing of mechanical parts when found necessary.

1.8 Voltage and frequency variations

1.8.1 All electrical appliances supplied from the main or emergency systems are to be so designed and manufactured that they are capable of operating satisfactorily under the normally occurring variations in voltage and frequency. Unless otherwise stated in national or international standards, provided the variations comply with as stated below, all equipment are to operate satisfactorily with the variations from its rated value as shown in the following Tables 1.8.1 to 1.8.3.

a) For alternating current components; voltage and frequency variations shown in Table 1.8.1 are to be assumed.

b) For direct current components supplied by d.c. generators or converted by rectifiers, voltage variations shown in Table 1.8.2 are to be assumed.

c) For direct current components supplied by electrical batteries, voltage variations shown in Table 1.8.3 are to be assumed.

Table 1.8.1 : Voltage and frequency variations for a.c. distribution systems		
Quantity in operation	**Permanent variations**	**Transient variations**
Frequency	± 5%	± 10% (recovery time 5 seconds)
Voltage	+ 6% - 10%	± 20% (recovery time 1.5 seconds)

Table 1.8.2 : Voltage variations for d.c. distribution systems	
Parameters	**Variations**
Voltage tolerance (continuous)	±10%
Voltage cyclic variation deviation	5%
Voltage ripple (a.c., r.m.s., over steady d.c. voltage)	10%

Table 1.8.3 : Voltage variations for battery systems	
Systems	**Variations**
Components connected to the battery during charging (see Note)	+30%, -25%
Components not connected to the battery during charging	+20%, -25%
Note : Different voltage variations as determined by the charging / discharging characteristics, including ripple voltage from the charging device, may be considered.	

1.8.2 Any special system, e.g. electronic circuits, whose function cannot operate satisfactorily within the limits given in 1.8.1 are not be supplied directly from the system but by alternative means, e.g. through stabilized supply.

1.8.3 Contactors and similar electro-magnetic devices are not to drop out at or above 85 per cent rated voltage.

1.9 Location and construction

1.9.1 Electrical equipment is to be placed in accessible and adequately lighted spaces clear of flammable material and heat sources such as boilers, heated tanks etc. The spaces should be well ventilated, free from accumulations of flammable dusts, vapours or gases and acid fumes. The equipment should not be exposed to risk of mechanical injury or damage from water, excessive moisture, steam, oil or any other dangerous fluid. Where necessarily exposed to such hazards, the equipment is to be suitably constructed or enclosed.

1.9.2 Live parts are to be efficiently shielded from any accidental contact.

1.9.3 All electrical apparatus and equipment is to be constructed and installed so as to avoid injury or electrical shock when handled or touched in the course of normal operation.

1.9.4 Insulating materials and insulated windings are to be resistant to tracking, moisture, sea air, oil vapour and any other similar conditions, unless special precautions are taken to protect them.

1.9.5 Equipment is not to remain alive through the control circuits and/or pilot lamps when switched off by the control switch. This does not apply to synchronising switches and/or plugs.

1.9.6 The operation of all electrical equipment and the lubrication arrangements are to be efficient under such conditions of vibration and shock as arise in normal practice.

1.9.7 All nuts and bolts/screws used to connect or secure current- carrying parts and working

parts are to be effectively locked, to prevent them from working loose due to vibration.

1.9.8 Generators, motors, conductors and generally any electrical equipment producing an external magnetic field are to be placed at such distance from the magnetic compass or any other equipment likely to be affected by such magnetic field or are to be so screened that the interfering external magnetic field is negligible (deviation less than 0.5°), even when circuits are switched on and off.

1.9.9 Where electric power is used for propulsion, the equipment is to be so arranged that it will operate satisfactorily in the event of partial flooding by bilge water above the tank top up to the floor plate level. See also Ch.3, Sec.2.

1.10 Design and construction

1.10.1 Design, construction and installation of electrical propelling machinery and associated equipment together with auxiliary services essential for the safety of ship are to be done in accordance with the relevant requirements of this Chapter and are to be surveyed and tested by the Surveyors.

1.10.2 Design and installation of other equipment is to be such as to ensure minimising the risk of fire due to its failure. As a minimum, it should comply with a National or International Standard, revised where necessary for ambient conditions.

1.10.3 Subject to inspection and testing by the Surveyors, compliance with the recommendations of the International Electrical Commission (IEC) publication 92 : "Electro-technical Installation in Ships" or equivalent national standards, may be accepted as meeting the requirements of this Chapter.

1.10.4 Where the Rules require electrical equipment to be of a 'safe type', such equipment is to be certified for the gases/vapours involved. The construction and type testing is to be in accordance with IEC publication 79 : "Electrical Apparatus for Explosive Gas Atmospheres", or an equivalent national standard.

1.11 Voltages

1.11.1 Standard voltages as recommended by IEC for shipboard installations, will in general be accepted subject to the following limits :

a) For power and the heating equipment, permanently installed and connected to fixed wiring, except for space heaters (radiators) in accommodation spaces - 1000V;

b) For lighting (including signal lamps), space heaters in accommodation spaces, socket-outlets, portable appliances and other consumers supplied by flexible cables and for communication and instrumentation equipment - 250V.

1.11.2 The requirements of 1.11.1 may be waived for the following applications :

a) For permanently installed power and heating equipment, where connection by flexible cable is essential because of the application (e.g. for movable cranes and other hoisting gear), a maximum of 1000V is acceptable,

b) For portable appliances, which are not hand-held during operation (e.g. welding transformers, refrigerated containers), a maximum of 1000V with connection by flexible cable and socket outlet, which is interlocked with a switch. At each such socket outlet a precautionary warning is to be permanently and prominently displayed indicating the nature and voltage of the electric supply and the specific purpose for which the socket outlet is intended. Such arrangements may be installed for applications in 1.11.2(a) also,

c) For instrumentation and control equipment which are part of power and heating installations (e.g. pressure or temperature switches for start/stop of motors), a maximum of 1000V, provided that all components are constructed for this voltage. The wider choice available of such components suitable for a voltage upto 250V is not to be overlooked,

d) Internal circuits of radio, television and navigation equipment or the external transmission circuits of radio and radar equipment. These may be however subject to statutory regulations.

1.11.3 Installations with higher voltages than specified above may be accepted for special purposes (e.g. cold cathode discharge lamps, for large auxiliary machinery and for propulsion machinery) after special consideration, in each case, and subject to the provisions of Sec.13 of this Chapter.

1.12 Electrical equipment for use in explosive gas atmospheres

1.12.1 Where the Rules require electrical equipment to be of a "safe type", such equipment is to be certified for the gases/vapours involved. The equipment should conform to IEC publication 79, "Electrical Apparatus for Explosive Gas Atmospheres", or an equivalent national standard.

1.12.2 Certified safe type equipment includes the following types of protection :

Intrinsically safe Ex "i"

Increased safety Ex "e"

Flameproof Ex "d"

Pressurized enclosure Ex "p"

1.12.3 Further, lighting fittings of the air driven type with pressurized enclosures are considered to be a "safe type" of lighting fitting.

1.12.4 When "safe type" equipment is permitted in hazardous zones or spaces all switches and protective devices are to interrupt all lines or phases and, where practicable, are to be located in a non-hazardous zone or space unless specifically permitted otherwise. Appropriate labels of non-inflammable material are to be permanently affixed to such equipment, switches and protective devices for identification purposes.

1.13 Enclosures

1.13.1 The enclosure types given in Table 1.13.1 are required as a minimum.

Table 1.13.1 : Enclosure types in relation to location
(N : Normally not accepted for installation in this location)

Location	Switchboards	Control gear and motor starters	Rotating machines	Transformers and rectifiers	Lighting fittings	Heating appliances	Socket outlets	Accessories such as switches connection boxes
Engine and boiler rooms								
above the floor	IP 22	IP 22	IP 22	IP 22	IP 22	IP 22	IP 44	IP 44
dry control rooms	IP 22	IP 22	IP 22	IP 22	IP 22	IP 22	IP 22	IP 22
below the floor	N	N	IP 44	N	IP 44	IP 44	N	IP 44
closed compartments for fuel oil and lub.oil separators	IP 44	IP 44	IP 44	IP 44	IP 44	IP 44	N	IP 44
Fuel oil tanks	N	N	N	N	N	N	N	N
Ballast and other water tanks, bilge wells	N	N	IP 68	N	N	IP 68	N	N
Ventilation ducts	N	N	IP 44	N	N	N	N	N
Deckhouses, forecastle spaces, steering gear compartments and similar spaces	IP 22	IP 22	IP 22	IP 22	IP 22	IP 22	IP 44	IP 44
Ballast pump rooms and similar rooms below the loadline	IP 44	IP 44	IP 44	IP 44	IP 44	IP 44	IP 56	IP 56
Cargo holds	N	N	IP 44	N	IP 55	N	IP 56	IP 56
Open deck, keel ducts	IP 56	IP 56	IP 56	IP 56	IP 55	IP 56	IP 56	IP 56
Battery rooms, lamp rooms, paint stores, stores for welding gas bottles	N	N	EX	N	EX	EX	N	EX
Dry accommodation spaces	IP 20	IP 20	IP 20	IP 20	IP 20	IP 20	IP 20	IP 20
Batch rooms and showers	N	N	N	N	IP 44	IP 44	N	IP 56
Galleys, laundries and similar rooms	IP 44	IP 44	IP 44	IP 44	IP 44	IP 44	IP 44	IP 44

1.13.2 *Installation of electrical and electronic equipment in engine rooms protected by fixed water-based local application fire-fighting systems (FWBLAFFS)*

Definitions:

Protected space:

- A machinery space where a FWBLAFFS is installed.

Protected areas:

- Areas within a protected space which is required to be protected by FWBLAFFS.

Adjacent areas:

- Areas, other than protected areas, exposed to direct spray.

- Areas, other than those defined above, where water may extend. See also Fig.1.13.2.

Electrical and electronic equipment enclosures located within areas protected by FWBLAFFS and those within adjacent areas exposed to direct spray are to have a degree of protection not less than IP44, except where evidence of suitability is submitted to and approved by IRS.

The electrical and electronic equipment within adjacent areas not exposed to direct spray may have a lower degree of protection provided evidence of suitability for use in these areas is submitted taking into account :

a) the design and equipment layout,
b) position of inlet ventilation openings and
c) cooling airflow for the equipment.

Note :

1. Additional precautions may be required to be taken in respect of:

    a) tracking as the result of water entering the equipment.

    b) Potential damage as the result of residual salts from sea water systems.

    c) High voltage installations.

    d) Personnel protection against electric shock

2. Also refer to requirements indicated in Pt.6, Ch.3, Sec.4, Cl. 4.5.6.

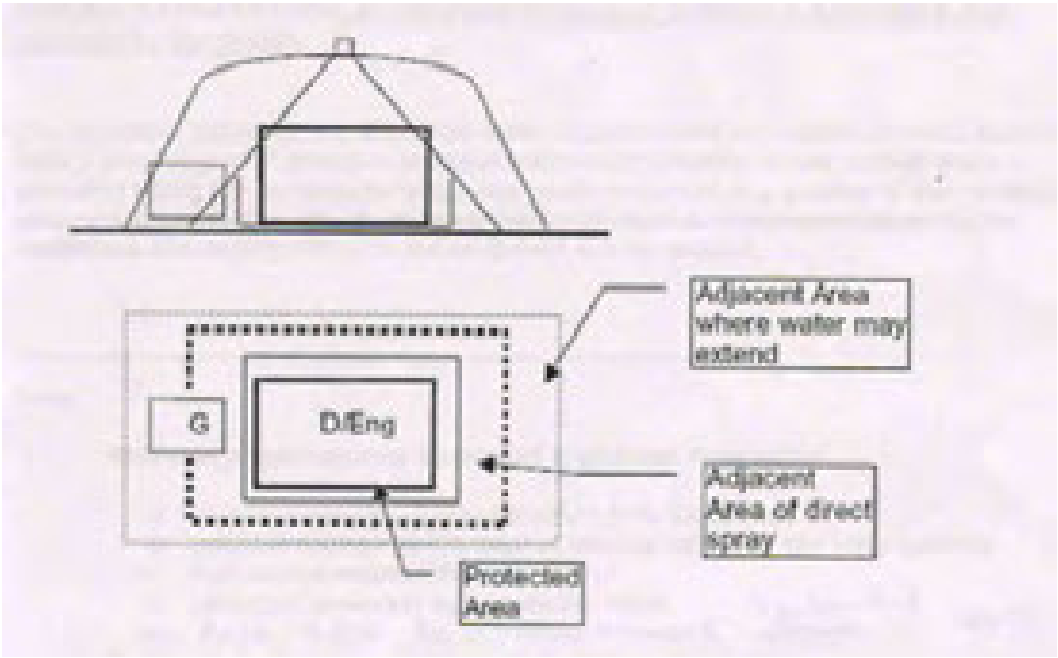


Fig. 1.13.2

1.14 Creepage and clearance

1.14.1 Distance between live parts and between live parts and earthed metal, whether across surfaces or in air, are to be adequate for the working voltages considering the nature of the insulating material and the transient over voltages developed by switch and fault conditions.

1.14.2 For bare busbars the minimum clearance distances in Table 1.14.1 are to be observed. Where necessary these distances are to be increased to allow for the electromagnetic forces involved.

Table 1.14.1 : Minimum clearances between bare busbars

Voltage between phases or poles	Minimum clearance to earth		Minimum clearance between phases or poles	
	In air [mm]	In oil [mm]	In air [mm]	In oil [mm]
660 or less	16	-	19	-
2200	38	-	38	-
3300	51	13	51	10
6600	63	19	89	25

1.15 Earthing of non-current carrying parts

1.15.1 Except where exempted by 1.15.2, all non-current carrying exposed metal parts of electrical equipment and cables are to be earthed.

1.15.2 The following parts may be exempted from the requirements of 1.15.1:

a) lamp-caps, where suitably shrouded;

b) shades, reflectors and guards supported on lamp holders or light fittings constructed of, or shrouded in, non-conducting material;

c) metal parts on, or screws in or through, non-conducting materials, which are separated by such material from current-carrying parts and from earthed non-current carrying parts in such a way that in normal use they cannot become live or come into contact with earthed parts;

d) apparatus which is constructed in accordance with the principle of double insulation;

e) bearing housings which are insulated in order to prevent circulation of current in the bearings;

f) clips for fluorescent lamps;

g) cable clips and short lengths of pipes for cable protection;

h) apparatus supplied at a voltage not exceeding 55 V direct current or 55 V, root mean square, between conductors, or between any conductor and earth in a circuit isolated from the supply. Auto transformers are not to be used for the purpose of achieving the alternating current voltage;

i) apparatus or parts of apparatus which although not shrouded in insulating material is nevertheless otherwise so guarded that it cannot be touched and cannot come in contact with exposed metal.

1.15.3 Metal coverings of cables are to be effectively earthed at both ends of the cable. In final sub-circuits, other than those installed in hazardous zones or spaces, earthing at the supply end only will be considered adequate. Single point earthing may be accepted for instrumentation cables, or cables for intrinsically safe circuits, if desirable for technical reasons.

1.15.4 The electrical continuity of all metal coverings of cables throughout the length of the cable, particularly at joints and trappings, is to be ensured.

1.15.5 Metal parts of portable appliances, other than current-carrying parts and parts exempted by 1.15.2 are to be earthed by means of an earth continuity conductor in the flexible cable or cord through the associated plug and socket outlet.

1.15.6 Earthing conductors are to be of copper or other corrosion-resistant material and be securely installed and protected where necessary against damage and also, where necessary, against electrolytic corrosion. Connections are to be so secured that they cannot work loose under vibration.

1.15.7 The nominal cross-section areas of copper earthing conductors are, in general to be equal to the cross-section of the current-carrying conductor upto 16 [mm^2]. Above this figure they are to be equal to at least half the cross-section of the current-carrying conductor with a minimum of 16 [mm^2]. Every other earthing conductor is to have a conductance not less than that specified for an equivalent copper earthing conductor.

1.15.8 The connection of the earthing conductors to the hull of the ship are to be made in accessible locations to permit their ready examination and to enable their disconnection for testing of insulation. They are to be secured by a screw or stud of diameter not less than 6 [mm] which is to be used for this purpose only. Bright metallic surfaces at the contact areas are to be ensured immediately before the nut or screw is tightened and where necessary, the joint is to be protected against electrolytic corrosion. The connection is to remain unpainted.

1.16 Bonding for the control of static electricity

1.16.1 Bonding straps for the control of static electricity are required for cargo tanks, process plant and piping systems, for flammable products and solids liable to release flammable gas and/or combustible dust, which are not permanently connected to the hull of the ship either directly or via their bolted or welded supports and where the resistance between them and the hull exceeds 1MΩ.

1.16.2 Where bonding straps are required for the control of static electricity, they are to be robust, that is, having a cross-sectional area of about 10 [mm^2] and are to comply with 1.15.6 and 1.15.8.

1.17 Operation under fire conditions

1.17.1 Electrical services required to be operable under fire conditions are as follows:

- Control and power systems to power-operated fire doors and their status indication.
- Control and power systems to power-operated watertight doors and their status indication.
- Emergency fire pump.
- Emergency lighting.
- Fire and general alarms.
- Fire detection systems.
- Fire-extinguishing systems and fire-extinguishing media release alarms.
- Low location lighting, see Pt.6, Ch.4, Sec.2.
- Public address systems.
- Remote emergency stop/shutdown arrangements for systems which may support the propagation of fire and/or explosion.

1.17.2 Where cables for services as described in 1.17.1 including their power supplies, pass through high fire risk areas and in addition for passenger ships, main vertical fire zones, other than those which they serve, they are to be so arranged that a fire in any of these areas or zones does not affect the operation of the service in any other area or zone (refer 1.17.3 for requirements for emergency fire pump). This may be achieved by either of the following measures:

a) Cables of a fire resistant type and complying with

i) IEC 60331-1:2018 for overall diameter greater than 20 [mm] and

ii) IEC 60331-21:1999 and AMD 1:2009 or IEC 60331-2:2018 for overall diameter less than or equal to 20 [mm]

are installed and run continuously within the high fire risk area to keep the fire integrity. See Fig.1.17.2.

b) By running cables in two loops or distributing cables radially as widely apart as practicable and so arranged that in the event of damage by fire atleast one of loops or radial distributions remains operational.

Systems that are self monitoring, fail safe or duplicated with cable runs as widely separated as is practicable may be exempted.

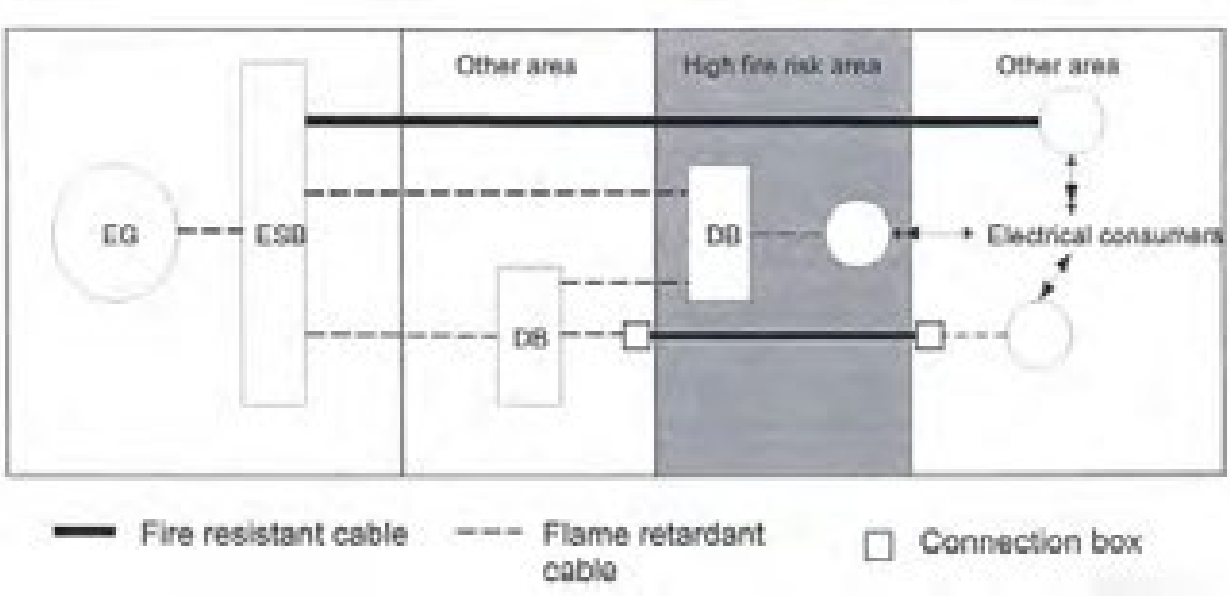


Fig.1.17.2 : The running of fire resistant type cables

1.17.3 The electrical cables to the emergency fire pump are not to pass through the machinery spaces containing the main fire pumps and their source(s) of power and prime mover(s). They are to be of a fire resistant type, in accordance with 1.17.2 (a), where they pass through other high fire risk areas.

1.17.4 “High fire risk areas” are defined as follows:

i) Machinery spaces as defined by Pt.6, Ch.1, 3.30, except spaces having little or no fire risk as defined in Pt 6, Ch 3, 3.2.2.3.2.2 (10) (including interpretations of table 9.3 to 9.8 given in MSC/Circ.1120 as amended by MSC.1/ Circ. 1436 and MSC.1/ Circ. 1510).

ii) Spaces containing fuel treatment equipment and other highly flammable substances.

iii) Galley and pantries containing cooking appliances.

iv) Laundry with drying equipment.

v) Spaces as defined by paragraphs (8), (12) and (14) of Part 6, Chapter 3, Cl.3.2.2.3.2 for ships carrying more than 36 passengers.

1.18 Additions and alterations

1.18.1 Additions or alterations (temporary or permanent) to the approved load of the existing electrical installations are not to be made unless it has been ascertained that the current carrying capacity and condition of the existing installation are adequate for the proposed modification.

1.18.2 Plans for the proposed modifications are to be submitted for approval and the additions or alterations are to be carried out under the inspection and to the satisfaction of the Surveyors.

Section 2

System Design

2.1 General

2.1.1 Electrical installations are to be such that:

- all electrical auxiliary services necessary for maintaining the ship in normal operational and habitable conditions are ensured without recourse to the emergency source of electrical power;

- electrical services essential for safety are ensured under various emergency conditions; and

- the safety of passengers, crew and ship from electrical hazards is ensured.

2.2 Supply and distribution systems

2.2.1 The following systems of generation and distribution as detailed in Table 2.2.1 are acceptable.

2.2.2 Systems of generation and distribution, other than those specified above, will, upon application, be given special consideration.

2.3 Voltage and frequency variations

2.3.1 For installations supplied by generators, the voltage on the main switchboard's bus-bars is to be kept between 97.5 and 102.5 per cent of the installation's nominal voltage under all steady load conditions.

2.3.2 For D.C. installations supplied by batteries, consideration is to be given to the supply voltage variation between the battery's full-charged and minimum-charged voltages. For battery installations with "float charge", the maximum charging voltage is to be considered.

2.3.3 The transient voltage variations on the main switchboard's bus-bars by the maximum power and current variations which can occur (except under fault conditions), are not to exceed - 15% and + 20% of the nominal voltage.

2.3.4 The voltage drop in each consumer circuit, by full load on this circuit, is not to exceed 6% of the nominal voltage, measured from the main switchboard's bus-bars to the consumer. For battery circuits with supply voltage less than 50 Volts, 10% voltage drop may be accepted.

2.3.5 The frequency variations on AC installations are to be kept within the following limits :

- 95 - 105 per cent of rated frequency under steady load conditions; and

- 90 - 110 per cent of rated frequency under transient load conditions.

Table 2.2.1 : Systems of generation and distribution

Description	Tankers intended for the carriage in bulk of oil, liquefied gases and other hazardous liquids having a flashpoint not exceeding 60°C (closed cup test)	Other vessels
d.c. two wire insulated system (See Note 1)	Acceptable	Acceptable
a.c., single-phase, two wire insulated system (See Note 1)	Acceptable	Acceptable
a.c., three-phase, three wire insulated system (See Note 2)	Acceptable	Acceptable
a.c. or d.c. earthed systems	Normally not acceptable (See Note 3)	Acceptable
a.c. three-phase, four wire system with neutral earthed but without hull return	Not acceptable	Acceptable upto 1000V
Hull return system of distribution (a.c. or d.c.)	Normally not acceptable (See Notes 4 and 5)	Normally not acceptable (See Notes 4 and 5)

Note 1 : None of the poles/phases is earthed (see also para 2.4).

Note 2 : Neutral is not earthed.

Note 3 : This may be acceptable for -

a) Power supplied control circuits and instrumentation circuits, where technical or safety reasons require connection to earth, provided the current in the hull is limited to not more than 5 amps in both normal and fault conditions.

b) Earthed intrinsically safe circuits.

c) Limited and locally earthed systems, provided that any possible resulting current does not flow directly through any of the dangerous spaces.

d) Alternating current power networks of 1000 Volts root mean square (line to line) and over, provided that any possible resulting current does not flow directly through any of the dangerous spaces.

e) The electrical systems shall be specially considered for existing vessels built before 1st October, 1994.

Note 4 : This may be acceptable for -

a) Impressed current cathodic protection systems.

b) Limited and locally earthed systems, such as starting and ignition systems of internal combustion engines, provided that any possible resulting current does not flow directly through any of the dangerous spaces.

c) Insulation level monitoring devices, provided the circulation current does not exceed 30 mA under the most unfavourable conditions.

Note 5 : All final sub-circuits, i.e. all circuits fitted after the last protective device are to be of two insulated wires the hull return being achieved by connecting to the hull, one of the busbars of the distribution board from which wires originate.

2.4 Earth indication

2.4.1 Every insulated distribution system is to be provided with a device or devices to continuously monitor the values of electrical insulation to earth and to give an audible or visual indication in case of abnormally low insulated values.

2.5 Number, arrangement and capacity of generators

2.5.1 The number and capacity of generators are to be such that with any one generator not in operation, the capacity of the remaining generators is to be sufficient for :

- supplying all essential and other services mentioned under 1.5.1 and 1.5.2 of this chapter which can be expected to be in use simultaneously, at full power on the propulsion plant, and including the maximum load which can be expected to occur on the lighting installation; and

- to have sufficient reserve capacity for starting the largest electric motor on board, without the transient voltage and frequency variations exceeding the values specified in 2.3.

2.5.2 For a duplicated essential or important consumer with only one unit driven electrically, it is not expected that electrically driven auxiliary would be used with one generator out of action. Similarly the loads due to side/bow thrusters are not to be taken in to account while calculating the electric load with one generator out of action unless necessitated by additional class notation regarding Dynamic Positioning.

2.5.3 The arrangement of the generators is to be such that the requirements of 2.5.1 would be met regardless of the speed and direction of the propulsion machinery or shafting.

2.5.4 In addition, the generating sets are to be such as to ensure that with any one generator or its primary source of power out of operation, the remaining generating sets are to be capable of providing the electrical services necessary to start the main propulsion plant from a dead ship condition. The emergency source of power may be used for the purpose of starting from a dead ship condition if its capability alone or combined with that of any other source of electrical power is sufficient to provide at the same time those services required to be supplied by 2.8.

2.5.5 Where the main source of electrical power is essential for the propulsion and steering of the ship, the system is to be arranged such that the electrical supply to equipment necessary for propulsion, steering and safety of the ship is maintained or immediately restored in case of loss of any one generator in service.

2.5.6 Requirement in 2.5.5 to be fulfilled by the following measures:

a) Where the electrical power is normally supplied by more than one generator set simultaneously in parallel operation, provision of protection, including automatic disconnection of sufficient non-essential services and if necessary services those provided for habitability, should be made to ensure that, in case of loss of any of these generating sets, the remaining ones are kept in operation to permit propulsion and steering and to ensure safety.

b) Where the electrical power is normally supplied by one generator provision shall be made, upon loss of power, for automatic starting and connecting to the main switchboard of stand-by generator(s) of sufficient capacity with automatic restarting of the essential auxiliaries, in sequential operation if required. Starting and connection to the main switchboard of the stand-by generator is to be preferably within 30 seconds, but in any case not more than 45 seconds, after loss of power. Where prime movers with longer starting time are to be used, the starting and connection time are subject to special approval.

c) Load shedding or other equivalent arrangements should be provided to protect the generators against sustained overload.

i) The load shedding is to be automatic.

ii) Primary essential services mentioned in 1.5.1.2 are not to be included in any load shedding.

iii) The secondary essential services mentioned in 1.5.1.3 may be included in load shedding provided:

a) Following systems, required for safety, are not disrupted:
- lighting systems,
- navigation lights, aids and signals,

- internal safety communication equipment,
- fire detection and alarm systems
- electrical equipment for watertight closing appliances.

b) Immediate availability of following services, required for safety, is not affected when the power supply is restored to normal operating conditions:
- fire pumps and other extinguishing medium pumps,
- bilge pumps,
- ventilating fans for engine and boiler rooms.

iv) The services for habitable conditions mentioned in 1.5.2 may be included in load shedding.

2.5.7 For vessels with electric propulsion system, under normal sea-going conditions, the number and capacity of generators is to be such that with any one generator not in operation, the capacity of remaining generator/s is to be sufficient for supplying all essential and other services mentioned at 1.5.1 and 1.5.2 of this Chapter, which can be expected to be used simultaneously, at speed of 7 knots or half the design speed, whichever is lesser.

2.6 Prime movers for generators

2.6.1 Each generator required according to 2.5 is normally to be driven by a separate auxiliary prime mover. If such prime mover for a generator also is used for driving other auxiliary machinery, it is to have sufficient capacity for the total load, or the machinery arrangement is to be such that the generator and the other auxiliary machinery can not be used simultaneously.

If the other auxiliary machinery is connected to the prime mover in such a way that it is possible to overload the prime mover, an interlocking or other effective means for preventing such overloading is to be arranged.

2.6.2 When generators driven by reciprocating steam engines or steam turbines are used, and the operation of the boiler(s) depends on electric power supply, then there is to be at least one generator driven by an auxiliary oil engine or gas turbine to enable the boiler plant to be started.

2.6.3 Generators and generator systems, having the ship's main propulsion machinery as their prime mover, may be accepted as part of the ships main source of electrical power, provided:

a) They are capable of operating under all weather conditions during sailing and during manoeuvring, also when the vessel is stopped, within the limits of voltage and frequency variations as specified in 2.3;

b) Their rated capacity is safeguarded during all operations given above and is such that in the event of any other one of the generators failing, the services given in 1.5.1 and 1.5.2 are maintained;

c) The short circuit current of the generator / generator system is sufficient to trip the generator / generator system circuit-breaker taking into account the selectivity of the protective devices for the distribution system.

d) Protection is to be arranged in order to safeguard the generator / generator system in case of a short-circuit in the main bus bar. The generator / generator system is to be suitable for further use after fault clearance.

e) It is to be possible to disconnect the generator from the propulsion system by a separate clutch.

f) Starting of stand-by generating sets is to be in accordance with 2.5.6b).

2.6.4 Generators and generator systems, having the ship's propulsion machinery as their prime mover but not forming part of the ship's main source of electrical power as specified in 2.6.3a) may be accepted whilst the ship is at sea to supply electrical services required for normal operational and habitable conditions provided :

a) There are sufficient and adequately rated additional generators fitted, which constitute the main source of electrical power, meeting the requirements as specified in 2.5.3.

b) Means are provided to automatically start one or more of the generators, constituting the main source of electrical power as per 2.5.3 or 2.5.6b), upon the frequency

variations exceeding ± 10 of the limits as specified in 2.3 and the voltage variations exceeding the limits as specified in 2.3.

c) The short circuit current of the generator and/or generator system is sufficient to trip the generator/generator system circuit-breaker taking into account the selectivity of the protective devices for the distribution system.

d) Where considered appropriate, load shedding arrangements are fitted to meet the requirements of 2.5.6c).

e) On ships having remote control of the ship's propulsion machinery from the navigating bridge means are provided, or procedures be in place, so as to ensure that supplies to essential services are maintained during manoeuvring conditions in order to avoid a blackout situation.

2.7 Number and capacity of transformers or power convertors

2.7.1 Where essential and other services mentioned under 1.5.1 and 1.5.2 of this chapter are supplied, by transformers or power convertors, the number and rating of transformers or power convertors are to be sufficient to ensure the operation of essential services and important consumers even when any one transformer or power convertor is out of action.

2.7.2 With any one transformer or power convertor out of action, the remaining transformer(s) or power convertor(s) should be adequate for starting the largest essential and important electric motor, without the transient voltage and frequency variations exceeding the limits specified in 2.3.

2.7.3 Each transformer or power convertor fitted in accordance with these requirement is to be installed as a separate unit, with a separate enclosure.

2.7.4 Each transformer is to be served by separate circuits on the primary and secondary sides. Auto transformers used for motor starting are excluded.

2.8 Emergency source of power

2.8.1 A self-contained emergency source of electrical power is to be provided in all ships.

2.8.2 The emergency source of electrical power and associated transforming equipment, if any, transitional source of emergency power, emergency switchboard and emergency lighting switchboard are to be located above the uppermost continuous deck and are to be readily accessible from the open deck. They are not to be located forward of the collision bulkhead.

2.8.3 The location of the emergency source of electrical power and associated transforming equipment, if any, the transitional source of emergency power, the emergency switchboard and the emergency electric lighting switchboards in relation to the main source of electrical power, associated transforming equipment, if any, and the main switchboard are to be such as to ensure to the satisfaction of IRS that a fire or other casualty in spaces containing the main source of electrical power, associated transforming equipment, if any, and the main switchboard or in any machinery space of category A will not interfere with the supply, control and distribution of emergency electrical power. As far as practicable, the space containing the emergency source of electrical power, associated transforming equipment, if any, the transitional source of emergency electrical power and the emergency switchboard is not to be continuous to the boundaries of machinery spaces of category A or those spaces containing the main source of electrical power, associated transforming equipment, if any, or the main switchboard. Requirements for ventilation of emergency generator rooms are given in 2.8.10.

2.8.4 Provided that suitable measures are taken for safeguarding independent emergency operation under all circumstances, the emergency generator may be used exceptionally, and for short periods, to supply non-emergency circuits.

Exceptionally, whilst the vessel is at sea, is understood to mean conditions such as:

- blackout situation
- dead-ship situation
- routine use for testing
- short-term parallel operation with the main source of electrical power for the purpose of load transfer.

Unless instructed otherwise by the Administration the emergency generator may be used during lay time in port for the supply of the ship mains, provided the requirements of 2.8.4.1 and 2.8.4.2 below are complied with.

2.8.4.1 Requirements

a) To prevent the generator or its prime mover from getting overloaded when used in port, arrangements are to be provided to shed sufficient non-emergency loads to ensure its continued safe operation.

b) The prime mover is to be arranged with fuel oil filters and lubrication oil filters, monitoring equipment and protection devices as required for the prime mover for main power generation and for unattended operation.

c) The fuel oil supply tank to the prime mover is to be provided with a low level alarm, arranged at a level ensuring sufficient fuel oil capacity for the emergency services for the period of time as required by 2.8.8 and 2.8.9 of this section.

d) The prime mover is to be designed and built for continuous operation and should be subjected to a planned maintenance scheme ensuring that it is always available and capable of fulfilling its role in the event of an emergency at sea.

e) Fire detectors are to be installed in the location where the emergency generator set and emergency switchboard are installed.

f) Means are to be provided to readily change over to emergency mode from harbour mode.

g) Control, monitoring and supply circuits, for the purpose of the use of the emergency generator in port are to be so arranged and protected that any electrical fault will not influence the operation of the main and emergency services.

h) When necessary for safe operation, the emergency switchboard is to be fitted with switches to isolate the circuits.

2.8.4.2 Operation

Instructions* are to be provided on board to ensure that when the vessel is under way all control devices (e.g. valves, switches) are in a correct position for the independent emergency operation of the emergency generator set and emergency switchboard.

* These instructions are also to contain information on required fuel oil tank level, position of harbour/sea mode switch if fitted, ventilation openings etc.

2.8.5 The emergency generator and other means needed to restore the propulsion are to have a capacity such that the necessary propulsion starting energy is available within 30 minutes of blackout/dead ship condition. For the purpose of this requirement "Blackout" is to be understood to mean a "dead ship" condition. Emergency generator stored starting energy is not to be directly used for starting the propulsion plant, the main source of electrical power and/or other essential auxiliaries (emergency generator excluded). The requirement of propulsion starting energy being available within 30 minutes would remain applicable even when there is no emergency generator installed.

For steam ships, the 30 minutes time limit is the time from blackout / dead ship condition to lighting of the first boiler.

2.8.6 Use of emergency generator during lay time in port for the supply of the ship mains as stated in para 2.8.4 will be specially considered on submission of details of safeguards to ensure its continued availability for its intended emergency duties.

2.8.7 Starting arrangements for emergency generating sets

2.8.7.1 Emergency generating sets are to be capable of being readily started in their cold condition at a temperature of 0°C. If this is impracticable, or if lower temperatures are likely to be encountered, provision acceptable to IRS is to be made for the maintenance of heating arrangements, to ensure ready starting of the generating sets.

2.8.7.2 Each emergency generating set arranged to be automatically started is to be equipped with starting devices approved by IRS with a stored energy capability of at least three

consecutive starts. A second source of energy is to be provided for an additional three starts within 30 minutes unless manual starting can be demonstrated to be effective.

2.8.7.3 The stored energy is to be maintained at all time, as follows:

a) electrical and hydraulic starting systems are to be maintained from the emergency switchboard;

b) compressed air starting systems may be maintained by the main or auxiliary compressed air receivers through a suitable non-return valve or by an emergency air compressor which, if electrically driven, is supplied from the emergency switchboard;

c) all of these starting, charging and energy storing devices are to be located in the emergency generator space; these devices are not to be used for any purpose other than the operation of the emergency generating set. This does not preclude the supply to the air receiver of the emergency generating set from the main or auxiliary compressed air system through the non-return valve fitted in the emergency generator space.

2.8.7.4 Where automatic starting is not required, manual starting is permissible, such as manual cranking, inertia starters, manually charged hydraulic accumulators, or powder charge cartridges, where they can be demonstrated as being effective.

2.8.7.5 When manual starting is not practicable, the requirements of 2.8.7.2 and 2.8.7.3 are to be complied with except that starting may be manually initiated.

2.8.8 Passenger ships

2.8.8.1 The electrical power available is to be sufficient to supply all those services that are essential for safety in an emergency, due regard being paid to such services as may have to be operated simultaneously. The emergency source of electrical power is to be capable, having regard to starting currents and the transitory nature of certain loads, of supplying simultaneously at least the following services for the periods specified herein, if they depend upon an electrical source for their operation:

a) For a period of 36 hours, emergency lighting:

i) at every embarkation station on deck and over sides;

ii) in alleyways, stairways and exits giving access to the muster and embarkation stations;

iii) in all service and accommodation alleyways, stairways and exits, personnel lift cars;

iv) in the machinery spaces and main generating stations including their control positions;

v) in all control stations, machinery control rooms, and at each main and emergency switchboard;

vi) at all stowage positions for firemen's outfits;

vii) at the steering gear; and

viii) at the fire pump, the sprinkler pump and the emergency bilge pump referred to in 2.8.8.1(d) and at the starting position of their motors.

b) For a period of 36 hours

i) the navigation lights and other lights required by the International Regulations for Preventing Collisions at Sea in force; and

ii) on ships constructed on or after 1 February 1995, the VHF radio installation; and if applicable

- the MF radio installation;
- the ship earth station; and
- the MF/HF radio installations.

c) For a period of 36 hours; unless such services have an independent supply for the period of 36 hours from an accumulator battery suitably located for use in an emergency:

i) all internal communication equipment required in an emergency which is generally;

- The means of communication which is provided between the navigating

bridge and the steering gear compartment.

- The means of communication which is provided between the navigating bridge and the position in the machinery space or control room from which the engines are normally controlled.

- The means of communication which is provided between the bridge and the radio telegraph or radio telephone stations.

- The means of communication which is provided between the officer of the watch and the person responsible for closing any watertight door which is not capable of being closed from a central control station.

- The public address system or other effective means of communication which is provided throughout the accommodation, public and service spaces.

- The means of communication which is provided between the navigating bridge and the main fire control station.

ii) the shipborne navigational equipment, where such provision is unreasonable or impracticable IRS may waive this requirement for ships of less than 5,000 tons gross tonnage;

iii) the fire detection and fire alarm system, and the fire door holding and release system; and

iv) for intermittent operation of the daylight signaling lamp, the ship's whistle, the manual fire alarms and all internal signals that are required in an emergency.

d) For a period of 36 hours:

i) one of the fire pumps required by Pt.6, Ch.3, Sec.4, Cl.4.2.2.2 and 4.2.2.3;

ii) the automatic sprinkler pump, if any; and

iii) the emergency bilge pump and all the equipment essential for the operation of electrically powered remote controlled bilge valves.

e) For the period of time required by Ch.6, Sec.6, the steering gear if required to be so supplied.

f) For a period of half an hour:

i) any watertight doors which are required to be power operated together with their indicators and warning signals. Sequential operation of the doors may be permitted provided all doors can be closed in 60 seconds;

ii) the emergency arrangements to bring the lift cars to deck level for the escape of persons. The passenger lift cars may be brought to deck level sequentially in an emergency.

g) In a ship engaged regularly on voyages of short duration, IRS if satisfied that an adequate standard of safety would be attained may accept a lesser period than the 36 hour period specified in 2.8.8.1(a) to 2.8.8.1(e) but not less than 12 hours.

2.8.8.2 The emergency source of electrical power may be either a generator or an accumulator battery, which is to comply with the following:

a) Where the emergency source of electrical power is a generator, it is to be:

i) driven by a suitable prime-mover with an independent supply of fuel having a flashpoint (closed cup test) of not less than 43°C;

ii) started automatically upon failure of the electrical supply from the main source of electrical power and is to be automatically connected to the emergency switchboard; those services referred to in 2.8.8.3 are then to be transferred automatically to the emergency generating set. The automatic starting system and the characteristic of the prime-mover are to be such as to permit the emergency generator to carry its full rated load as quickly as is safe and practicable, subject to a maximum of 45 seconds; unless a second independent means of starting the emergency generating set is provided, the single source of stored

energy is to be protected to preclude its complete depletion by the automatic starting system; and

iii) provided with a transitional source of emergency electrical power according to 2.8.8.3.

b) Where the emergency source of electrical power is an accumulator battery, it is to be capable of:

i) carrying the emergency electrical load without recharging while maintaining the voltage of the battery throughout the discharge period within 12 per cent above or below its nominal voltage;

ii) automatically connecting to the emergency switchboard in the event of failure of the main source of electrical power; and

iii) immediately supplying at least those services specified in 2.8.8.3.

2.8.8.3 The transitional source of emergency electrical power required by 2.8.8.2(a)(iii) is to consist of an accumulator battery suitably located for use in an emergency which is to operate without recharging while maintaining the voltage of the battery throughout the discharge period within 12 per cent above or below its nominal voltage and be of sufficient capacity and so arranged as to supply automatically in the event of failure of either the main or emergency source of electrical power at least the following services, if they depend upon an electrical source for their operation:

a) For half an hour:

i) the lighting required by paragraphs 2.8.8.1(a) and 2.8.8.1(b)(i);

ii) Services required by paragraphs 2.8.8.1(c) (i, iii and iv) unless such services have an independent supply for the period specified from an accumulator battery suitably located for use in an emergency.

b) Power to close the watertight doors but not necessarily all of them simultaneously, unless an independent temporary source of stored energy is provided. Power to the control, indication and alarm circuits as required for half an hour.

2.8.8.4 Where the emergency and/or transitional emergency loads are supplied from a battery via an electronic converter or inverter the maximum permitted voltage variations are to be taken as those on the load side of the converter or inverter.

The above mentioned voltage variants are not to exceed the following limits, as applicable:

a) Where the system on the load side of the converter or inverter is d.c. : limits as given in 2.8.8.3.

b) Where the system on the load side of the converter or inverter is a.c. : limits as given in 1.8.1.

2.8.8.5 The emergency switchboard is to be installed as near as is practicable to the emergency source of electrical power.

2.8.8.6 Where the emergency source of electrical power is a generator, the emergency switchboard is to be located in the same space unless the operation of the emergency switchboard would thereby be impaired.

2.8.8.7 No accumulator battery fitted in accordance with this Section is to be installed in the same space as the emergency switchboard. An indicator is to be mounted in a suitable place on the main switchboard or in the machinery control room to indicate when the batteries constituting either the emergency source of electrical power or the transitional source of emergency electrical power referred to in paragraph 2.8.8.2(b) or 2.8.8.3 are being discharged.

2.8.8.8 The emergency switchboard is to be supplied during normal operation from the main switchboard by an interconnector feeder which is to be adequately protected at the main switchboard against overload and short circuit and which is to be disconnected automatically at the emergency switchboard upon failure of the main source of electrical power. Where the system is arranged for feedback operation, the interconnector feeder is also to be protected at the emergency switchboard at least against short circuit.

2.8.8.9 In order to ensure ready availability of the emergency source of electrical power, arrangements are to be made where necessary to disconnect automatically non-emergency circuits from the emergency switchboard to ensure that power will be available to the emergency circuits.

2.8.8.10 The emergency generator and its prime-mover and any emergency accumulator battery is to be so designed and arranged as to ensure that they will function at full rated power when the ship is upright and when inclined at any angle of list up to 22.5° or when inclined up to 10° either in the fore or aft direction, or is in any combination of angles within those limits.

2.8.8.11 Provision is to be made for the periodic testing of the complete emergency system and is to include the testing of automatic starting arrangements.

2.8.9 Cargo ships

2.8.9.1 The electrical power available is to be sufficient to supply all those services that are essential for safety in an emergency, due regard being paid to such services as may have to be operated simultaneously. The emergency source of electrical power is to be capable, having regard to starting currents and the transitory nature of certain loads, of supplying simultaneously at least the following services for the periods specified herein, if they depend upon an electrical source for their operation:

a) For a period of 3 hours, emergency lighting at every muster and embarkation station and over sides;

b) For a period of 18 hours, emergency lighting:

i) in all service and accommodation alleyways, stairways and exits, personnel lift cars and personnel lift trunks;

ii) in the machinery spaces and main generating stations including their control positions;

iii) in all control stations, machinery control rooms, and at each main and emergency switchboard;

iv) at all stowage positions for firemen's outfits;

v) at the steering gear;

vi) at the fire pump referred to in 2.8.9.2, at the sprinkler pump, if any, and at the emergency bilge pump, if any, and at the starting positions of their motors; and

vii) in all cargo pump-rooms of tankers.

c) For a period of 18 hours

i) the navigation lights and other lights required by the International Regulations for Preventing Collisions at Sea in force; and

ii) on ships constructed on or after 1 February 1995, the VHF radio installation; and if applicable

- the MF radio installation;
- the ship earth station; and
- the MF/HF radio installations.

d) For a period of 18 hours; unless such services have an independent supply for the period of 18 hours from an accumulator battery suitably located for use in an emergency:

i) all internal communication equipment as required in an emergency which is generally as indicated in 2.8.8.1 c) i);

ii) the shipborne navigational equipment, where such provision is unreasonable or impracticable IRS may waive this requirement for ships of less than 5,000 tons gross tonnage;

iii) the fire detection and fire alarm system; and

iv) intermittent operation of the daylight signaling lamp, the ship's whistle, the manual fire alarms, and all internal signals that are required in an emergency.

2.8.9.2 For period of 18 hours one of the fire pumps required by Pt.6, Ch.3, if dependent upon the emergency generator for its source of power.

2.8.9.3 For the period of time required by Ch.6, Sec.6, the steering gear where it is required to be so supplied.

2.8.9.4 In a ship engaged regularly in voyages of short duration, IRS if satisfied that an adequate standard of safety would be attained may accept a lesser period than the 18 hour period specified in paragraphs 2.8.9.1(b) to 2.8.9.1(d) and 2.8.9.2 but not less than 12 hours.

2.8.9.5 The emergency source of electrical power may be either a generator or an accumulator battery, which is to comply with the following:

a) Where the emergency source of electrical power is a generator, it is to be:

i) driven by a suitable prime-mover with an independent supply of fuel, having a flashpoint (closed cup test) of not less than 43°C;

ii) started automatically upon failure of the main source of electrical power supply unless a transitional source of emergency electrical power in accordance with 2.8.9.5(a)(iii) is provided; where the emergency generator is automatically started, it is to be automatically connected to the emergency switchboard; those services referred to in 2.8.9.6 are then to be connected automatically to the emergency generator; and unless a second independent means of starting the emergency generator is provided the single source of stored energy is to be protected to preclude its complete depletion by the automatic starting system; and

iii) provided with a transitional source of emergency electrical power as specified in 2.8.9.6 unless an emergency generator is provided capable both of supplying the services mentioned in 2.8.9.6 and of being automatically started and supplying the required load as quickly as is safe and practicable subject to a maximum of 45 seconds.

b) Where the emergency source of electrical power is an accumulator battery it is to be capable of:

i) carrying the emergency electrical load without recharging while maintaining the voltage of the battery throughout the discharge period within 12 per cent above or below its nominal voltage;

ii) automatically connecting to the emergency switchboard in the event of failure of the main source of electrical power; and

iii) immediately supplying at least those services specified in 2.8.9.6.

2.8.9.6 The transitional source of emergency electrical power where required by 2.8.9.5(a) is to consist of an accumulator battery suitably located for use in an emergency which is to operate without recharging while maintaining the voltage of the battery throughout the discharge period within 12 per cent above or below its nominal voltage and be of sufficient capacity and is to be so arranged as to supply automatically in the event of failure of either the main or the emergency source of electrical power for half an hour at least the following services if they depend upon an electrical source for their operation:

a) the lighting required by 2.8.9.1(a) to 2.8.9.1(c). For this transitional phase, the required emergency electric lighting, in respect of the machinery space and accommodation and service spaces may be provided by permanently fixed, individual, automatically charged, relay operated accumulator lamps; and

b) all services required by 2.8.9.1(d) (i, iii and iv) unless such services have an independent supply for the period specified from an accumulator battery suitably located for use in an emergency.

2.8.9.7 Where the emergency and/or transitional emergency loads are supplied from a battery via an electronic converter or inverter the maximum permitted voltage variations are to be taken as those on the load side of the converter or inverter.

The above mentioned voltage variants are not to exceed the following limits, as applicable:

a) Where the system on the load side of the converter or inverter is d.c. : limits as given in 2.8.9.6.

Where the system on the load side of the converter or inverter is a.c. : limits as given in 1.8.1.

2.8.9.8 The emergency switchboard is to be installed as near as is practicable to the emergency source of electrical power.

2.8.9.9 Where the emergency source of electrical power is a generator, the emergency switchboard is to be located in the same space unless the operation of the emergency switchboard would thereby be impaired.

2.8.9.10 No accumulator battery fitted in accordance with this Section is to be installed in the same space as the emergency switchboard. An indicator is to be mounted in a suitable place on the main switchboard or in the machinery control room to indicate when the batteries constituting either the emergency source of electrical power or the transitional source of electrical power referred to in paragraph 2.8.9.5(b) or 2.8.9.6 are being discharged.

2.8.9.11 The emergency switchboard is to be supplied during normal operation from the main switchboard by an interconnector feeder which is to be adequately protected at the main switchboard against overload and short circuit and which is to be disconnected automatically at the emergency switchboard upon failure of the main source of electrical power. Where the system is arranged for feedback operation, the interconnector feeder is also to be protected at the emergency switchboard at least against short circuit.

2.8.9.12 In order to ensure ready availability of the emergency source of electrical power, arrangements are to be made where necessary to disconnect automatically non-emergency circuits from the emergency switchboard to ensure that electrical power will be available automatically to the emergency circuits.

2.8.9.13 The emergency generator and its prime-mover and any emergency accumulator battery is to be so designed and arranged as to ensure that they will function at full rated power when the ship is upright and when inclined at any angle of list up to 22.5° or when inclined up to 10° either in the fore or aft direction, or is in any combination of angles within those limits. Also refer to Part 4, Chapter 1, Table 1.7.1.

2.8.9.14 Provision is to be made for the periodic testing of the complete emergency system and is to include the testing of automatic starting arrangements.

2.8.10 Ventilation of emergency generator rooms

2.8.10.1 The following requirements apply to closable ventilation louvers and ventilator closing appliances serving emergency generator rooms, where fitted:

a) Ventilation louvers and closing appliances may either be hand-operated or power-operated (hydraulic / pneumatic / electric) and are to be operable under a fire condition;

b) Hand-operated ventilation louvers and closing appliances are to be kept open during normal operation of the vessel. Corresponding instruction plates are to be provided at the location where hand-operation is provided;

c) Power-operated ventilation louvers and closing appliances are to be of a fail-to-open type. Closed power operated ventilation louvers and closing appliances are acceptable during normal operation of the vessel. Power-operated ventilation louvers and closing appliances are to open automatically whenever the emergency generator is starting / in operation;

d) It is to be possible to close ventilation openings by a manual operation from a clearly marked safe position outside the space where the closing operation can be easily confirmed. The louver status (open / closed) is to be indicated at this position. Such closing is not to be possible from any other remote position.

2.9 Requirements for uninterrupted power system (UPS) units as alternative and/or transitional source of power

2.9.1 Scope

These requirements to UPS units, as defined in IEC 62040-3:2011, apply when providing an alternative power supply or transitional power supply to services as defined in 2.8.8 and 2.8.9.

A UPS unit complying with these requirements may provide an alternative power supply as an accumulator battery in terms of being an independent power supply for services defined in 2.8.8.1c) and 2.8.9.1d).

2.9.2 Definitions

Uninterruptible Power System (UPS) – combination of converters, switches and energy storage means, for example batteries, constituting a power system for maintaining continuity of load power in case of input power failure (IEC 62040-3:2011).

Off-line UPS unit – a UPS unit where under normal operation the output load is powered from the bypass line (mains) and only transferred to the inverter if the bypass supply

fails or goes outside preset limits. This transition will invariably result in a brief (typically 2 to 10 milli-seconds) break in the load supply.

Line interactive UPS unit – an off-line UPS unit where the bypass line switches to stored energy power when the input power goes outside the preset voltage and frequency limits.

On-line UPS unit – a UPS unit where under normal operation the output load is powered from the inverter and will therefore continue to operate without break in the event of the supply input failing or going outside preset limits.

2.9.3 Design and construction

a) UPS units are to be constructed in accordance with IEC 62040-1:2017, IEC 62040-2:2016, IEC 62040-3:2011, IEC 62040-4:2013 and/or IEC 62040-5:2016, as applicable, or an acceptable and relevant national or international standard.

b) The operation of the UPS is not to depend upon external services.

c) The type of UPS unit employed, whether off-line, line interactive or on-line, is to be appropriate to the power supply requirements of the connected load equipment.

d) An external bypass is to be provided.

e) The UPS unit is to be monitored and audible and visual alarm is to be given in a normally attended location for –

- Power supply failure (voltage and frequency) to the connected load,
- Earth fault,
- Operation of battery protective device,
- When the battery is being discharged, and
- When the bypass is in operation for on-line UPS units.

2.9.4 Location

a) The UPS unit is to be suitably located for use in an emergency.

b) UPS units utilizing valve regulated sealed batteries may be located in compartments with normal electrical equipment, provided the ventilation arrangements are in accordance with the requirements to IEC 62040-1:2017, IEC 62040-2:2016, IEC 62040-3:2011, IEC 62040-4:2013 and/or IEC 62040-5:2016, as applicable, or an acceptable and relevant national or international standard.

2.9.5 Performance

a) The output power is to be maintained for the duration required for the connected equipment as stated in 2.8.8 and 2.8.9.

b) No additional circuits are to be connected to the UPS unit without verification that the UPS unit has adequate capacity. The UPS battery capacity is, at all times, to be capable of supplying the designated loads for the time specified in the regulations.

c) On restoration of the input power, the rating of the charge unit shall be sufficient to recharge the batteries while maintaining the output supply to the load equipment.

2.9.6 Testing and survey

a) UPS units of 50 kVA and over are to be surveyed during manufacturing and testing.

b) Appropriate testing is to be carried out to demonstrate that the UPS unit is suitable for its intended environment. This is expected to include as a minimum the following tests:

- Functionality, including operation of alarms;
- Temperature rise;
- Ventilation rate
- Battery capacity.

c) Where the supply is to be maintained without a break following a power input failure, this is to be verified after installation by an appropriate test.

2.10 Essential services

2.10.1 Where essential services are duplicated, they are to be served by individual circuits separated throughout their length as widely as is practicable and without the use of common feeders, protective devices or control circuits.

2.11 Diversity factor

2.11.1 Circuits supplying two or more final sub-circuits are to be rated in accordance with the total connected load subject, where justified, to the application of a diversity factor. Where spare ways (feeders) are provided on a section

or distribution board, an allowance for future increase of load is to be added to the total connected load before application of any diversity factor.

2.11.2 The diversity factor may be applied when calculating cable size and when calculating the rating of switchgear and fusegear.

2.11.3 The Diversity factors are not applicable to supply cables to distribution switchboards for lighting and heating.

2.11.4 The calculation of the diversity factor is to be submitted along with all relevant data.

2.12 Lighting circuits

2.12.1 Lighting circuits are to be supplied by final sub-circuits, which are separate from those for heating and power. This provision need not be applied to cabin fans and small wardrobe heaters.

2.12.2 A final sub-circuit of rating exceeding 15 A is not to supply more than one point.

2.12.3 A final sub-circuit of rating 15 A or less is not to supply more than the following number of lighting points:-

10 for ……. 24 - 55 V circuits

14 for ……. 110 - 127 V circuits

24 for ……. 220 - 250 V circuits.

This provision is not applicable to final sub-circuits for cornice lighting, panel lighting and electric signs where lampholders are closely grouped; in such cases, the number of points is unrestricted provided the maximum operating current in the sub-circuit does not exceed 10 A.

2.12.4 Lighting for machinery spaces, control stations, work spaces, public spaces, corridors and stairways leading to boat decks should be supplied from at least two final sub-circuits in such a way that failure of any one of the circuits does not leave the spaces in darkness.

2.12.5 For lighting in hazardous areas, switches are to be of the double-pole type and wherever practicable, located in a non-hazardous area. If fitted in hazardous areas, these switches are to be of flameproof type.

2.12.6 Lighting of unattended spaces, such as cargo spaces and coal bunkers, is to be controlled by multi-pole linked switches located outside such spaces. Provision is to be made for the complete isolation of these circuits and locking in the "OFF" position of the means of control.

2.12.7 The arrangement of the main electric lighting system is to be such that a fire or other casualty in spaces containing the main source of electrical power, associated transforming equipment, if any, the main switchboard and the main lighting switchboard, will not render the emergency lighting system inoperative.

2.13 Motor circuits

2.13.1 A separate final sub-circuit is to be provided for every motor required for essential services and for every motor of 1 [kW] or more.

2.13.2 Motors of permanently installed submersible bilge pumps are to be connected to the emergency switchboard, (if fitted).

2.13.3 Cables and their connections to such pumps are to be capable of operating under a head of water equal to their distance below the bulkhead deck. The cables are to be suitable for operation in permanently wet situations, and installed in continuous lengths from above the bulkhead deck to the motor terminals.

2.13.4 Under all circumstances it is to be possible to start the motor of a permanently installed submersible bilge pump from a position above the bulkhead deck.

If an additional starter is provided at the motor, it is to be so arranged as to permit, in all cases, remote starting.

2.14 Motor controls

2.14.1 Every electric motor is to be provided with an efficient means of starting and stopping so placed as to be easily accessible to the person controlling the motor.

2.14.2 Every motor required for essential services and every motor of 0.5 [kW] or more is to be provided with the control apparatus as mentioned in 2.14.4 to 2.14.8.

2.14.3 When motor control gear is being selected, the maximum current of the motor is to be taken as its rated full load current.

2.14.4 Efficient means of isolation are to be provided so that all voltage may be cut off from the motor and any associated apparatus including any automatic circuit breaker.

2.14.5 Where the primary means of isolation (viz. that provided at the switchboard, section board or distribution fuse board) is remote from a motor, one of the following provisions is to be made :

a) An additional means of isolation fitted adjacent to the motor; or

b) Provision made for locking the primary means of isolation in the OFF position; or

c) Provision made so that the fuses in each line can be readily removed and retained by authorized personnel.

2.14.6 Means to prevent the undesired restarting after a stoppage due to low volts or complete loss of volts are to be provided. This does not apply to motors where a dangerous condition might result from the failure to restart automatically e.g. steering gear motor. It is, however, to be ensured that the total starting current of motors having automatic restart will not cause excessive voltage drop or overcurrent on the installation.

2.14.7 Means for automatic disconnection of the supply in the event of excess current due to mechanical overloading of the motor are to be provided. (This does not apply to steering gear motors and thruster motors. Overload alarms are to be provided for these motors. For requirements regarding steering gear motors, See Ch.6).

2.14.8 Where fuses are installed to protect polyphase motor circuits, means are to be provided to protect the motor against unacceptable overload in the case of single phasing.

2.15 Remote stops for ventilation fans and pumps

2.15.1 Ventilating fans for machinery spaces are to be provided with means for stopping them from easily accessible control stations located outside such spaces. (Also refer Pt.6, Ch.2, 2.2.1.2).

2.15.2 Motors driving forced and induced draught fans, independently driven pumps delivering oil to main propulsion machinery for bearing lubrication and piston cooling, oil fuel transfer pumps, oil fuel unit pumps and other similar fuel pumps, fuel and lubricating oil purifiers and their attached pumps are to be fitted with remote controls situated outside the space concerned so that the electrical supply thereto can be disconnected in the event of fire arising in the space in which they are located. (Also refer Ch.3, Cl. 4.11.1) and Pt.6, Ch.2, Sec.2.

2.15.3 Where overboard discharges from cooling water or similar pumps are likely to jeopardize the safe launching of lifeboats or life rafts, under any condition of loading of the ship, then the motors driving these pumps are to be provided with suitably located remote shut-off arrangements.

2.16 Fire detection and extinguishing systems

2.16.1 Electrical equipment used in operating fire detecting systems is to be served by two circuits, one fed from the main power supply system and the other from the emergency power supply system. These feeders are to be connected to an automatic change-over switch installed in the vicinity of the fire detection panel.

2.16.2 Where an electrically driven fire pump is supplied power from the emergency switchboard, the supply cable to such pump is not to pass through the main machinery spaces.

2.17 Lift circuits

2.17.1 Lifts or hoists to be used by passengers and/or crew are to be supplied from the main switchboard, either directly or through a section board, by circuits which do not supply any other appliance.

2.17.2 Cable or cables supplying current to motors of lifts or hoists are not to be incorporated within the flexible trailing cable used in connection with the control and safety devices.

2.18 Heating and cooking equipment

2.18.1 Every heating or cooking appliance is to be controlled as a complete unit by a multi-pole linked switch mounted in the vicinity of the appliance.

2.18.2 In the case of small heaters, for individual cabins or similar small dry accommodation spaces where the floor coverings, bulkheads and ceiling linings are of insulating materials, a single pole switch is acceptable.

2.18.3 Heating arrangements of the exposed element type are not to be used in any location.

2.18.4 Means of cutting off power to the galley except lighting circuits, in the event of a fire, are to be provided outside the galley exits, at positions which will not readily be rendered inaccessible by such a fire.

2.19 Temporary external supply/shore connection

2.19.1 Where arrangements are provided for the supply of electric power from a source on shore or elsewhere, a connection box is to be installed in an easily accessible location in a manner suitable for the convenient reception of flexible cables from the external source. This box should contain a circuit-breaker or isolating switch and fuses and terminals of ample size and suitable shape to facilitate a satisfactory connection. The mechanical stress of the portable cable is to be conveyed directly to the metallic framework and not to electrical connectors. Suitable cables, permanently fixed, are to be provided, connecting the circuit breaker/isolating switch in the connection box to a linked switch and/or circuit breaker at the main switchboard.

2.19.2 For alternating current systems an earthed terminal is to be provided for the reception of three-phase external supplies with earthed neutrals.

2.19.3 The external connection is to be provided with an indicator at the main switchboard in order to show when the cable is energized.

2.19.4 Means are to be provided for checking the polarity (for direct current) or the phase sequence (for three-phase alternating current) of the incoming supply. This device should be connected between the incoming connectors and the interrupting device in the connection box.

2.19.5 A notice is to be provided at the connection box giving complete information on the system of supply and the normal voltage (and frequency for alternating current) of the ship's installed system. Full details of the procedure for effecting the connection are to be given on this notice.

2.19.6 Alternate arrangements for providing a temporary external supply will be specially considered.

2.20 Permanent external supply

2.20.1 Details are to be submitted for consideration in each specific case.

2.21 Protection

2.21.1 General

2.21.1.1 All installations are to be protected against accidental over-currents including short circuits and other electrical faults. The choice, location and characteristics of the protective device are to provide complete and co-ordinated protection to ensure:

a) Elimination of the fault to reduce damage to the system and hazard of fire.

b) Continuity of service so as to maintain, through the discriminative action of the protective devices, the supply to essential and all other services not directly affected by the fault.

2.21.2 Protection against overload

2.21.2.1 Protection against overloads may be provided by circuit-breakers, automatic switches or fuses. The tripping characteristics of these devices are to be appropriate to the system. Fuses rated above 320 A are not to be used for protection against overload, but may be used for short-circuit protection.

2.21.2.2 The rating or appropriate setting of the overload protective device for each circuit should be permanently indicated at the location of the protective device.

2.21.2.3 The overload releases of circuit-breakers for generators and the setting of preferential trip relays are to be adjustable or, if of the non-adjustable type, are to be readily replaceable by others of different values.

2.21.3 Protection against short-circuit

2.21.3.1 Protection against short-circuit currents is to be provided by circuit-breakers or fuses.

2.21.3.2 The breaking capacity of every protective device is to be not less than the maximum value of the short-circuit current which can flow at the point of installation at the instant of contact separation.

2.21.3.3 The making capacity of every circuit-breaker or switch intended to be capable of being closed, if necessary, on short circuit, is to be not less than the maximum value of the short-circuit current at the point of installation. On alternating current this maximum value corresponds to the peak value allowing for maximum asymmetry.

2.21.3.4 Every protective device or contactor not intended for short circuit interruption is to be adequate for the maximum short-circuit current which can occur at the point of installation having regard to the time required for the short circuit to be removed.

2.21.3.5 In the absence of precise data of rotating machines the following short-circuit currents at the machine terminals are to be assumed. The short circuit current is to be the sum of short circuit currents of generators and that of motors;

a) Direct current systems

- Ten times full load current for generators normally connected (including spare),

- Six times full load current for motors simultaneously in service;

b) Alternating current systems.

- Ten times full load current for generators normally connected (including spare) - symmetrical RMS,

- Three times full load current for motors simultaneously in service.

2.21.4 Combined circuit-breakers and fuses

2.21.4.1 The use of a circuit-breaker of breaking capacity less than the prospective short-circuit current at the point of installation is permitted, provided that it is preceded on the generator side by fuses, or by a circuit-breaker having at least the necessary breaking capacity. The generator breakers are not to be used for this purpose.

2.21.4.2 Fused circuit-breakers with fuses connected to the load side may be used where operation of the circuit-breaker and fuses is co-ordinated.

2.21.4.3 The characteristics of the arrangement are to be such that:

a) When the short-circuit current is broken, the circuit-breaker on the load side is not to be damaged and is to be capable of further service,

b) When the circuit-breaker is closed on the short-circuit current, the remainder of the installation is not to be damaged. However, it is admissible that the circuit-breaker on the load side may require servicing after the fault has been cleared.

2.21.5 Protection of circuits

2.21.5.1 Short circuit protection is to be provided in each live pole of a direct current system and in each phase of an alternating current system.

2.21.5.2 Protection against overloads is to be provided as follows:

a) Two-wire direct current or single-phase alternating current system - at least one line or phase,

b) Insulated three-phase alternating current system - at least two phases,

c) Earthed three-phase alternating current system - all three phases.

2.21.5.3 No fuse, non-linked switch or non-linked circuit-breaker is to be inserted in an earthed conductor. Any switch or circuit-breaker fitted is to operate simultaneously in the earthed conductor and the insulated conductors.

2.21.5.4 These requirements do not preclude the provision (for test purposes) of an isolating link to be used only when the other conductors are isolated.

2.21.6 Protection of generators

2.21.6.1 In addition to over-current protection, the provisions of 2.21.6.2 to 2.21.6.7 are to be adhered to as a minimum.

2.21.6.2 For generators not arranged to run in parallel :

- A multi-pole circuit-breaker arranged to open simultaneously all insulated poles or in the case of generators rated at less than 50 [kW] a multi-pole linked switch with a fuse in each insulated pole on the generator side. The fuse rating in such

cases is to be maximum 125 per cent of the generator rated current.

2.21.6.3 For generators arranged to run in parallel:

- A circuit-breaker arranged to open simultaneously all insulated poles. This circuit-breaker is to be provided with :

a) For direct current generators, instantaneous reverse-current protection operating at not more than 15 percent rated current,

b) For alternating current generators :

i) A reverse-power protection, in accordance with characteristics of prime movers is to be set as shown in Table 2.21.6.3

Table 2.21.6.3 Reverse power protection

Prime Mover	Protection setting (as % of full load of generator)	Time delay
Diesel Generator	Between 2 to15%	Between 3 to 10 sec
Turbo Generator	Between 2 to 6 %	

ii) A device for protection against the effects of parallel connection in opposite phase.

2.21.6.4 The reverse-current protection is to be adequate to deal with the reverse-current conditions emanating from the network, e.g. from winches, the reverse-power protection specified for alternating current generators may be replaced by other devices ensuring adequate protection of the prime movers.

2.21.6.5 A fall of 50 per cent in the applied voltage should not render the reverse-current mechanism inoperative, although it may alter the amount of reverse-current required to open the breaker.

2.21.6.6 The following protective gear is also to be provided for direct current generators arranged to operate in parallel :

a) Where an equaliser connection is in use the reverse-current protection is to be provided in the pole opposite to that in which the series winding is connected,

b) For compound generators an equaliser switch for each generator so interlocked that it closes before and opens after the main contacts of the circuit-breaker with which it is associated, or a three-pole circuit-breaker with all poles acting simultaneously.

2.21.6.7 Generator circuit-breakers are normally to be provided with under voltage release.

2.21.7 Protection of essential services

2.21.7.1 Where generators are operated in parallel and essential services are electrically operated, arrangements are to be made to disconnect automatically the excess non-essential load when the generators are overloaded.

2.21.7.2 If required, this load shedding may be carried out in one or more stages according to the overload ability of the generating sets, taking into consideration the relative importance of the loads being thus disconnected.

2.21.7.3 In cargo ships, circuits for cargo refrigeration machinery are to be included in the last group of services to be disconnected.

2.21.8 Protection of feeder circuits

2.21.8.1 Isolation and protection of each main distribution circuit is to be ensured by a multi-pole circuit-breaker, multi-pole fused circuit-breakers or multi-pole switch and fuses. The provisions of 2.21.2, 2.21.3 and 2.21.5 are to be complied with. The protective devices are to allow excess current to pass during the normal accelerating period of motors. Where multi-pole switch and fuses are used, the fuses are generally to be installed between the busbars and the switch.

2.21.8.2 Circuits which supply motors fitted with overload protection may be provided with short-circuit protection only.

2.21.8.3 Motors of rating exceeding 0.5 [kW] and all motors for essential services are to be protected individually against overload and short- circuit. The short-circuit protection can be provided by the same protective device for the

motor and its supply cable. For essential motors which are duplicated, the overload protection may be replaced by an overload alarm, if desired.

2.21.8.4 For motors intended to provide uninterrupted service the protective gear is to have a delay characteristic to enable the motor to start, and to operate on overload before the windings reach an un-acceptably high temperature. The current which the protective device will allow to pass indefinitely is not to exceed 125 per cent of the rated current.

2.21.8.5 For motors for intermittent service the current setting and the delay are to be chosen in relation to the load factor of the motor.

2.21.8.6 Where fuses are used to protect polyphase motor circuits, means are to be provided to protect the motor against unacceptable overloads in the case of single phasing.

2.21.9 Protection of power transformers

2.21.9.1 The primary circuits of power transformers are to be protected against short-circuit by circuit-breakers or fuses. The rating of fuses or the setting for overcurrent releases of circuit breakers is not to exceed 125 per cent of rated primary current.

2.21.9.2 Each of the secondary circuits is to be provided with a multiple isolating switch as shown in Fig.2.21.9.2.

2.21.9.3 When transformers are arranged to operate in parallel, an automatic protective device is to be provided on the secondary circuits. Switches and circuit-breakers are to be capable of withstanding surge currents.

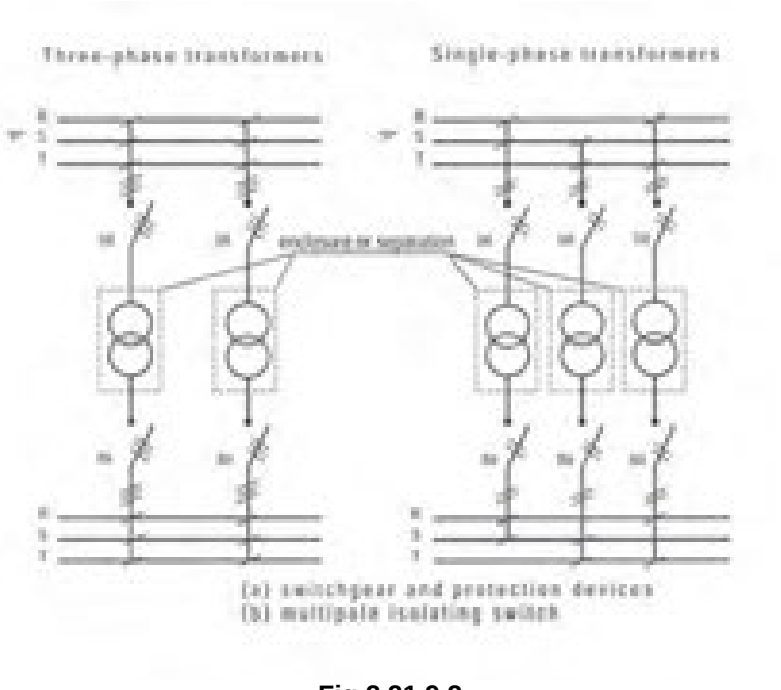


Fig.2.21.9.2

2.21.10 Protection of lighting circuits

2.21.10.1 Lighting circuits are to be provided with overload and short- circuit protection.

2.21.11 Protection of meters, pilot lamps, capacitors, control transformers and control circuits

2.21.11.1 Short circuit protection is to be provided for voltmeters, voltage coils of measuring instruments, earth indicating devices and pilot lamps, control transformers and control circuits together with their connecting leads by means of protective devices fitted to each insulated pole or phase.

2.21.11.2 A pilot lamp installed as an integral part of another item of equipment need not be individually protected, provided it is fitted in the same enclosure. Where a fault in a pilot lamp would jeopardise the supply to essential equipment such lamps are to be individually protected.

2.21.11.3 Where capacitors for suppression of radio interference are fitted to busbars, generators or steering gear, fuses of appropriate size are to be connected in the capacitor circuit.

2.21.12 Protection of batteries

2.21.12.1 Accumulator batteries other than engine starting batteries are to be protected against overload and short circuit by devices, in each insulated pole, placed at a position adjacent to the battery compartment.

2.21.13 Protection of communication circuits

2.21.13.1 Communication circuits other than those supplied from primary batteries are to be protected against overload and short-circuit.

2.21.14 Harmonics

2.21.14.1 **Scope**. The requirements of 2.21.14 to 2.21.18 are applicable to ships where harmonic filters are installed on main bus bars of electrical distribution system, other than those installed for single application frequency drives such as pump motors.

2.21.14.2 The total harmonic distortion (THD) in the voltage waveform of the AC distribution systems is not to exceed 8% and any single order harmonics is not to exceed 5%.

2.21.14.3 Higher values other than those specified in 2.21.14.2 may be accepted provided the distribution equipment and consumers are designed to operate at the higher limits. A harmonic distortion calculation report, indicating distortion levels at MSB and power distribution centers is to be submitted for review for all vessels with electric propulsion and for vessels fitted with harmonic filters. This report is to document the ability of the system and its components to sustain the actual distortion levels. This report is to be made available onboard as a reference for the surveyor at each periodical survey.

2.21.14.4 Total Harmonic Distortion (THD) is a ratio of the rms value of the harmonic content to the rms value of the fundamental frequency, expressed in percent and may be calculated by the expression:

$$THD = \frac{\sqrt{\mathrm{V}_2^2 + \mathrm{V}_3^2 + \mathrm{V}_4^2 + \cdots \mathrm{V}_\mathrm{n}^2}}{\mathrm{V}_1}$$

Where, V_n is the rms voltage of the n^{th} harmonic and n = 1 is the fundamental frequency

2.21.14.5 Equipment producing transient voltage, frequency and current variations are not to cause malfunction of other onboard equipment.

2.21.15 Harmonic Filters

2.21.15.1 If harmonic filter circuits are used in association with semiconductor converter assemblies to reduce the harmonics and transients in the system, they are to be suitably protected against short circuit and overload.

2.21.15.2 The requirements for monitoring harmonic distortion, mitigation of effects and protection arrangements for harmonic filters given in 2.21.16 to 2.21.18 are applicable to ships contracted for construction, on or after 01 July 2017. Existing ships retrofitted with harmonic filters, on or after 01 July 2017 are also to comply with the requirements of 2.21.16 to 2.21.18.

2.21.16 Monitoring of harmonic distortion levels for a ship fitted with harmonic filters

2.21.16.1 The ships are to be fitted with facilities to continuously monitor the levels of harmonic distortion experienced on the main busbar as well as alerting the crew should the level of harmonic distortion exceed the acceptable limits.

2.21.16.2 Where the engine room is provided with automation systems, this reading is to be logged electronically, otherwise it is to be recorded in the engine log book for future inspection by the surveyor.

2.21.17 Mitigation of the effects of harmonic filter failure on a ship's operation

2.21.17.1 Where the electrical distribution system on board a ship includes harmonic filters the system integrator of the distribution system is to show, by calculation, the effect of a failure of a harmonic filter on the level of harmonic distortion experienced.

2.21.17.2 The system integrator of the distribution system is to provide the ship owner with guidance documenting permitted modes of operation of the electrical distribution system while maintaining harmonic distortion levels within acceptable limits during normal operation as well as following the failure of any combination of harmonic filters.

2.21.17.3 The calculation results and validity of the guidance provided are to be verified by the surveyor during sea trials.

2.21.18 Protection arrangements for harmonic filters

2.21.18.1 Arrangements are to be provided to alert the crew in the event of activation of the protection of a harmonic filter circuit.

2.21.18.2 A harmonic filter is to be arranged as a three phase unit with individual protection of each phase. The activation of the protection arrangement in a single phase is to result in automatic disconnection of the complete filter. In addition, a current imbalance detection system is to be installed, independent of the overcurrent protection to alert the crew in case of current imbalance.

2.21.18.3 Individual capacitor elements are to be additionally protected by means such as relief valve or overpressure disconnector in order to protect against damage from rupturing. The additional protection arrangements are to be designed taking into account the type of capacitors used.

Section 3

Cables and Busbar Trunking Systems

3.1 General

3.1.1 The requirements of this Section are applicable to fixed cables on permanent installations. For flexible cables, the requirements apply only as far as applicable.

3.1.2 Cables are to be of a type approved by IRS.

3.1.3 Electric cables for fixed wiring are to be designed, manufactured and tested in accordance with the relevant IEC Publication stated in Table 3.1.1 or an acceptable relevant standard. Cables manufactured and tested to standards other than those specified in Table 3.1.1 may be accepted provided they are in accordance with an acceptable and relevant international or national standard and are of an equivalent or higher safety level than those listed in Table 3.1.1.

Table 3.1.1 : Electric cables

Application	IEC Standard
General constructional and testing requirements	60092-350:2020
Choice and installation of electrical cables	60092-352:2005
Fixed power and control circuits	60092-353:2016
Insulating and sheathing materials for power, control, telecommunication and instrumentation	60092-360:2014
Guidance for selection of cables for telecommunication and data transfer including radio frequency cables.	60092-370:2019
Cables for control and instrumentation circuits 150/250 V (300V)	60092-376:2017

3.1.4 The use of flexible cables on permanent installations is to be limited to applications where flexibility is necessary, and the lengths of such flexible cables are to be kept as short as practicable. Additional requirements may be specified for flexible cable, depending on the applications.

3.1.5 Electric cables complying with IEC Publication 60502 are to have stranded conductors.

3.2 Conductors

3.2.1 High conductivity annealed copper only is to be used.

3.2.2 Where the conductor insulation is composed of vulcanised rubber or of a synthetic rubber liable to contain sulphides, the copper wires are to be tinned or alloy coated and the surface is to be bright. Similarly the copper wires are to be tinned or alloy coated when the insulation is composed of a cambric impregnated with a product other than bitumen, unless an efficient separator is inserted between conductor and insulation.

3.2.3 Conductor composition and stranding is to be selected so that adequate flexibility of the finished cable is assured. Conductors of nominal cross-section 2.5 [mm^2] and less need not be stranded. This requirement does not apply to mineral-insulated cables which have solid conductors.

3.2.4 Cores of multi-core cables are to be readily identifiable.

3.3 Insulating materials

3.3.1 Permitted insulating materials with maximum rated conductor temperatures are given in Table 3.3.1.

3.3.2 Where a rubber or rubber-like material with maximum conductor temperature greater than 60°C is used, it is to be readily identifiable.

3.3.3 Consideration will be given to other insulating materials.

Table 3.3.1 : Insulating materials & maximum rated conductor temperature

Type of insulating compound	Maximum rated conductor temp. °C
Thermoplastics	
Based upon polyvinyl chloride or co-polymer of vinyl chloride and vinyl acetate	60
Based upon polyethylene	60
Elastomeric or thermosettings	
Based upon ethylene propylene rubber or similar (EPM or EPDM)	85
Based upon chemically crosslinked polyethylene	85
Based upon silicon rubber	95
Mineral	95

3.4 Construction

3.4.1 Electric cables are to be at least of a flame-retardant type. Compliance with testing in accordance with IEC 60332-1-2:2004+AMD1:2015 or a test procedure equivalent thereto, will be acceptable.

3.4.2 Exemption from the requirements of 3.4.1 for applications such as radio frequency or digital communication systems, which require the use of particular types of cable, will be subject to special consideration.

3.4.3 Where electric cables are required to be of a 'fire resistant type', they are to comply with the requirements of IEC 60331-1:2018 for cables of greater than 20 [mm] overall diameter, otherwise 60331-21:1999 and AMD 1:2009 or IEC 60331-2:2018.

Notes:

a) Installation methods specified in 3.11 are to be applied.

b) Cables complying with alternative national standards and suitable for use in a marine environment may be considered.

c) Fire resistant type cables are to be easily distinguishable, for e.g. by way of colour code.

d) Following standards may be used for special cables such as electrical data and optical fibre cables:

IEC60331-23:1999 : Electric data cables
IEC60331-25:1999 : Optical fiber cables.

3.4.4 Where electric cables are installed in locations exposed to the weather, in damp and in wet situations, in machinery compartments, refrigerated spaces or exposed to harmful vaporous including oil vapour they are to have the conductor insulating materials enclosed in an impervious sheath of material appropriate to the expected ambient conditions.

3.4.5 Electric cables where it is required that their construction includes metallic sheaths, armouring or braids are to be provided with an overall impervious sheath or other means to protect the metallic elements against corrosion.

3.4.6 Where single core electric cables are used in circuits rated in excess of 20 A and are armoured the armour is to be of a non-magnetic material.

3.4.7 Electric cables are to be constructed such that they are capable of withstanding the mechanical and thermal effects of the maximum short circuit current which can flow in any part of the circuit in which they are installed, taking into consideration not only the time/current characteristics of the circuit protective device but also the peak value of the prospective short circuit current. Where electric cables are to be used in circuits with a maximum short circuit current in excess of 70 kA, evidence is to be submitted for consideration when required demonstrating that the cable construction can withstand the effects of the short circuit current.

3.4.8 All high voltage electric cables are to be readily identified by suitable marking.

3.5 Testing

3.5.1 Routine tests are to include at least

- measurement of electrical resistance of conductors;
- high voltage test;
- insulation resistance measurement; and
- partial discharge tests in case of high voltage cables.

Evidence of successful completion of routine tests is to be provided by the manufacturer for cables supplied to a project.

Note: Routine tests are to be carried out in the presence of the Surveyor for cables intended to be used for electrical propulsion systems, and as required elsewhere by the Rules.

3.5.2 Particular, special and type tests are to be made, when required, in accordance with the requirements of relevant publication or national standard.

3.6 Voltage rating

3.6.1 The rated voltage of any cable is not to be lower than the nominal voltage of the circuit for which it is used. Cables exposed to voltage surges associated with highly inductive circuits, e.g. contactor operating circuits for winches etc., are to be given special consideration.

3.6.2 Electric cables used in unearthed systems are to be suitably rated to withstand the additional stresses imposed on the insulation due to an earth fault.

3.7 Choice of insulating material

3.7.1 The rated operating temperature of the insulating material is to be at least 10°C higher than the maximum ambient temperature liable to be produced in the space where the cable is installed.

3.8 Choice of protective covering

3.8.1 Cables fitted in the following locations are to have an impervious sheath. In permanently wet situations, metallic sheaths are to be used for cables with hydroscopic insulation.

- Decks exposed to weather;
- Bathrooms;
- Refrigerated spaces;
- Machinery spaces; and

- Any other location where water condensation or harmful vapour (e.g. oil vapour) may be present.

3.8.2 All cables are to be of flame-retardant type or fire-resisting type, except that non flame-retardant cables may be accepted for final circuits only in the following cases :

a) Where cables are installed in metallic conduits having internal diameter not exceeding 25 [mm] and provided the conduits are electrically and mechanically continuous.

b) Bare lead sheathed cable having conductor sections not exceeding 4.5 [mm^2].

3.9 Current rating

3.9.1 The highest continuous load carried by a cable is not to exceed its current rating. The diversity factor of the individual loads and the duration of the maximum demand may be allowed for in estimating the maximum continuous load and is to be shown on the plans submitted for approval.

3.9.2 The cross sectional area of the conductors is to be sufficient to ensure that, under short circuit conditions, the maximum rated conductor temperature for short circuit operation is not exceeded, taking in to account the time current characteristics of the circuit protective device and the peak value of the prospective short circuit current.

3.9.3 In assessing the current rating of lighting circuits, every lampholder is to be assessed at the maximum load likely to be connected to it, with a minimum of 60 W, unless the fitting is so connected as to take only a lamp rated at less than 60 W.

Table 3.9.1 : Electric cable current rating, normal operation, based on ambient temp. of 45°C

Nominal cross-section in [mm^2]	Continuous current rating in amperes (based on ambient temp. 45°C)								
	Thermoplastic, PVC, PE			EP rubber and cross linked PE			Silicon rubber or mineral		
	Single core	2 core	3 or 4 core	Single core	2 core	3 or 4 core	Single core	2 core	3 or 4 core
1	8	7	6	16	14	11	20	17	14
1.5	12	10	8	20	17	14	24	20	17
2.5	17	14	12	28	24	20	32	27	22
4	22	19	15	38	32	27	42	36	29
6	29	25	20	48	41	34	55	47	39
10	40	34	28	67	57	47	75	64	53
16	54	46	38	90	77	63	100	85	70
25	71	60	50	120	102	84	135	115	95
35	87	74	61	145	123	102	165	140	116
50	105	89	74	180	153	126	200	175	140
70	135	115	95	225	191	158	255	217	179
95	165	140	116	275	234	193	310	264	217
120	190	162	133	320	272	224	360	306	252
150	220	187	154	365	310	256	410	349	287
185	250	213	175	415	353	291	470	400	329
240	290	247	203	490	417	343	570	485	400
300	335	285	235	560	476	392	660	560	460

3.9.4 Cables supplying winches, cranes, windlasses and capstans are to be suitably rated for their duty. Unless the duty is such as to require a longer time rating, cables for winch or crane motors may be half hour rated on the basis of the half hour [kW] rating of the motors. Cables for windlass and capstan motors are to be not less than one hour rated on the basis of

the one hour [kW] rating of the motor. In all cases the rating is to be subject to the voltage drop being within the specified limits.

3.9.5 The current ratings given in Table 3.9.1 are based on the maximum operating conductor temperatures, given in Table 3.3.1. Where a more precise evaluation of current rating has been carried out based on experimental or calculated data, details may be submitted for approval.

3.10 Correction factors for current rating

3.10.1 Bunching of cables : Where more than six cables belonging to the same circuit are bunched together a correction factor of 0.85 is to be applied.

3.10.2 Ambient temperature : The current ratings in Table 3.9.1 are based on an ambient temperature of 45°C. For other values of ambient temperature the correction factors shown in Table 3.10.1 are to be applied.

3.10.3 Intermittent Service : Where the load is intermittent, the correction factors in Table 3.10.2 may be applied for half hour and one hour ratings. In no case is a shorter rating than one half hour rating to be used, whatever the degree of intermittency.

Table 3.10.1 : Correction factors for temperature

Insulation material	Correction factor for ambient air temperature of °C										
	35	40	45	50	55	60	65	70	75	80	85
PVC, Polyethelene	1.29	1.15	1.00	0.82	-	-	-	-	-	-	-
EPR, XLPE	1.12	1.06	1.00	0.94	0.87	0.79	0.71	0.61	0.5	-	-
Mineral, Silicon rubber	1.10	1.05	1.00	0.95	0.89	0.84	0.77	0.63	0.55	0.45	-

Table 3.10.2 : Correction factors for intermittent rating

Correction factor	Half-hour rating		One-hour rating	
	With metallic sheath [mm^2]	Without metallic sheath [mm^2]	With metallic sheath [mm^2]	Without metallic sheath [mm^2]
1.00	Upto 20	Upto 75	upto 67	upto 230
1.10	21 - 40	76 – 125	68 - 170	231 - 400
1.15	41 - 65	126 – 180	171 - 290	401 - 600
1.20	66 - 95	181 – 250	291 - 430	-
1.25	96 - 130	251 – 320	431 - 600	-
1.30	131 - 170	321 – 400	-	-
1.35	171 - 220	401 – 500	-	-
1.40	221 - 270	-	-	-

3.11 Installation of cables

3.11.1 Cable runs are to be, as far as practicable, straight and accessible and as high as possible above bilges.

3.11.2 The installation of cables across expansion joints in any structure is to be avoided. Where such installation is unavoidable, a loop of cable of length proportional to the expansion of the joint is to be provided. The internal radius of the loop is to be at least twelve times the external diameter of the cable.

3.11.3 Where a duplicate supply is required and provided for any particular service, the two cables are to follow different routes which are separated throughout their length as widely as is practicable, to minimise the probability of simultaneous damage to the two circuits. The provision is also applicable to control circuits.

3.11.4 Generator cables are as far as practicable to be divided between two or more cable runs. These cable runs are to be separated as far apart as practicable.

3.11.5 Cables supplying essential or important consumers are generally not to be installed in rooms where there is an excessive fire hazard such as paint stores, galleys, etc. purifiers, welding-gas bottles etc.

3.11.6 Cables having insulating materials with different maximum-rated conductor temperatures are not to be bunched together, or, where this is not practicable, the cables are to be operated so that no cable reaches temperature higher than that permitted for the lowest temperature-rated cable in the group.

3.11.7 Cables having a protective covering which may damage the covering of other cables are not to be bunched with those of other cables.

3.11.8 The minimum internal radius of cable bends, which are not subjected to movements by expansion when installed, is to be generally in accordance with Table 3.11.1.

3.11.9 Electric cables are not to be coated or painted with materials which may adversely affect their sheath or their fire protection.

3.11.10 Where electric cables are installed in refrigerated spaces they are not to be covered with thermal insulation but may be placed directly on the face of the refrigeration chamber, provided that precautions are taken to prevent the electric cables being used as casual means of suspension.

3.11.11 Cable runs are normally not to include joints. However, if a joint is necessary it is to be carried out with prior approval and with due consideration to methods of splicing that retain the original mechanical and electrical properties of the cable and which ensure that all conductors are adequately secured, insulated and protected from atmospheric action. Terminals and busbars are to be of dimensions adequate for the cable rating.

3.11.12 Where electric cables are installed in bunches, provision is to be made to limit the propagation of fire, which may be achieved by either of the following:

a) Cables which have been tested in accordance with IEC 60332-3-22:2018 Category A or a test procedure for cables installed in bunches equivalent thereto.

b) (Refer to Fig.3.11.12a, b, c and d)

i) Fire stops having at least B0 penetrations fitted as follows:

- Cable entries at the main and emergency switchboard,
- Where cable enter engine control rooms,
- Cable entries at centralized control panels for propulsion machinery and essential auxiliaries,
- At each end of totally enclosed cable trunks, and

ii) In enclosed and semi enclosed spaces, cable runs are to comply with the following:

- to have fire protection coating applied
- to at least 1 metre in every 14 metres
- to entire length of vertical runs or
- Fitted with fire stops having at least B0 penetrations every second deck or approximately 6 metres for vertical runs and at every 14 metres for horizontal runs.

Table 3.11.1 : Minimum internal radii of bends in cables for fixed wiring

Cable construction		Overall diameter of cable	Minimum internal radius of bend (times overall diameter of cable)
Insulation	**Outer covering**		
Thermoplastic and elastomeric 600/1000 V and below	Metal sheathed Armoured and braided	Any	6D
	Other finishes	≤ 25 [mm] > 25 [mm]	4D 6D
Mineral	Hard metal sheathed	Any	6D
Thermoplastic and elastomeric above 600/1000 V			
- single core	Any	Any	20D
- multicore	Any	Any	15D

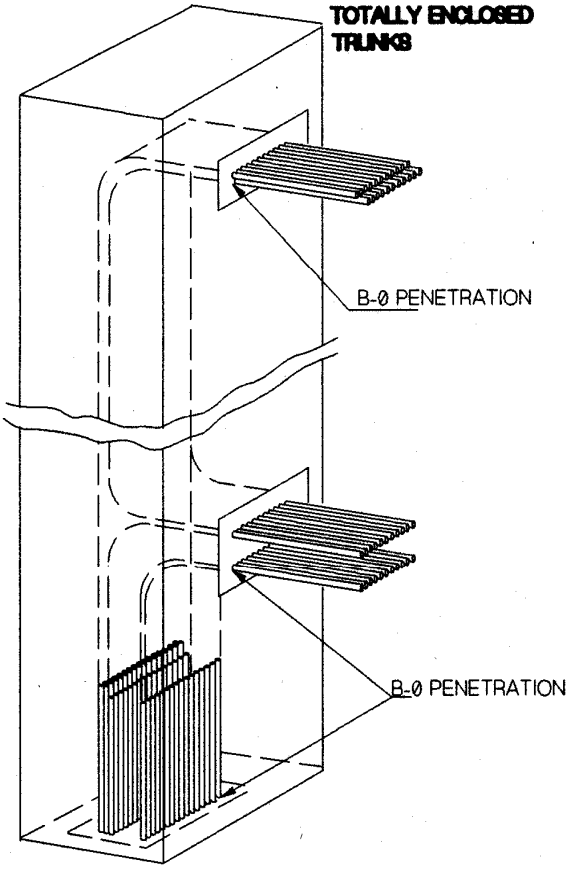


Fig.3.11.12a)

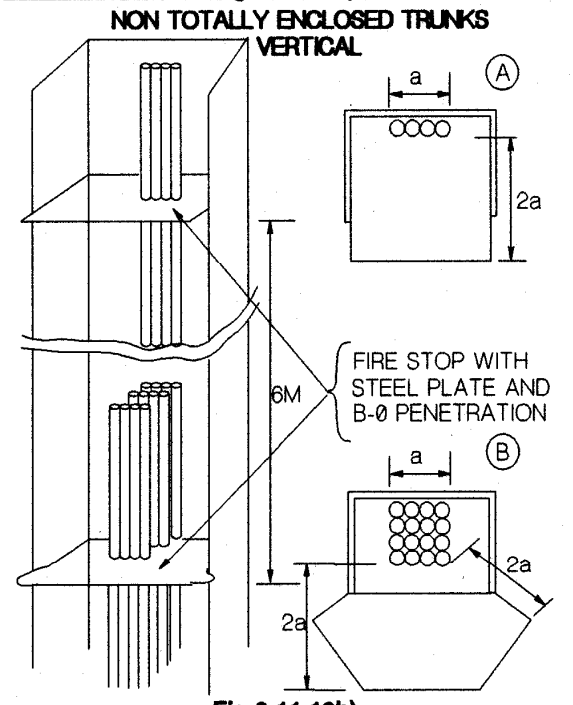
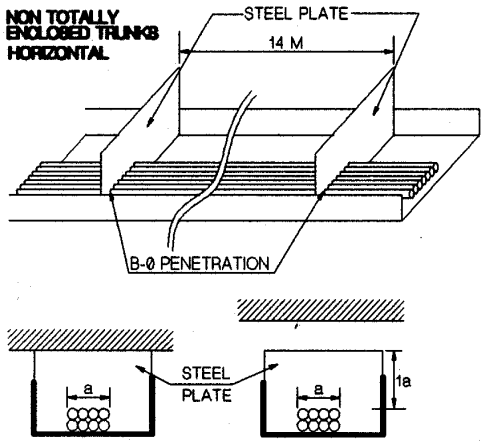


Fig.3.11.12b)



REMARK :
THE LENGHT (a) OF THE FIRE STOPS FOR HORIZONTAL CABLE RUNS SHOULD BE AT LEAST ONCE OF THE LARGER DIMENSION OF THE CABLE BUNCH OR UP TO THE DECK

Fig.3.11.12c)

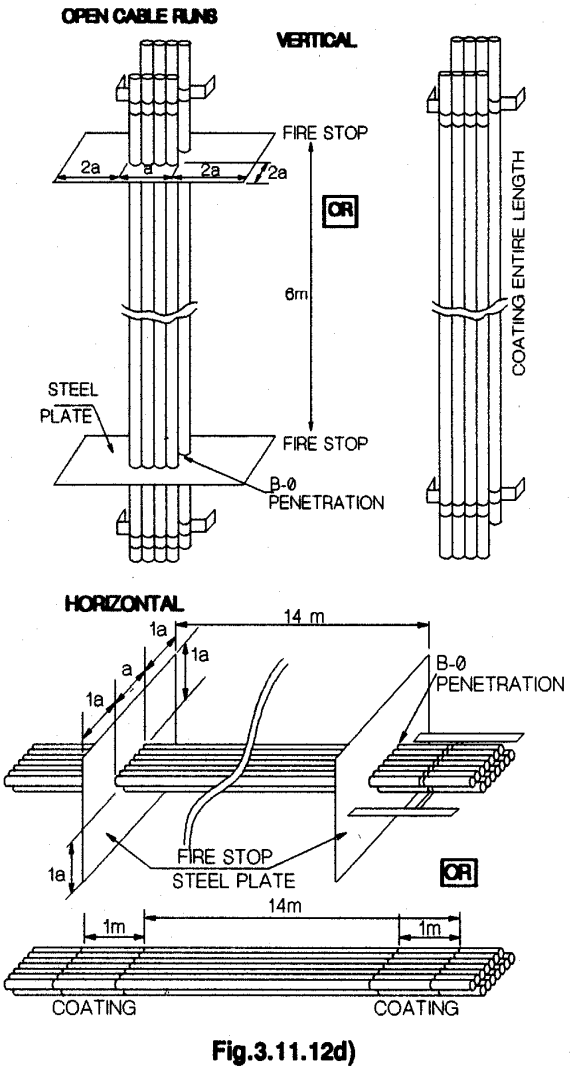


Fig.3.11.12d)

Note : (All fire stops steel plates 3 mm thk)

Fig.3.11.12a) to d) : Installation of cables

3.12 Mechanical protection of cables

3.12.1 Cables exposed to risk of mechanical damage are to be protected by metal channels or casing or enclosed in steel conduit unless the protective covering (e.g. armour or sheath) is adequate to withstand the possible damage.

3.12.2 Cables, in spaces where there is exceptional risk of mechanical damage (e.g. on weather decks, in cargo hold areas and inside the cargo holds) and also below the floor in engine and boiler rooms, are to be suitably protected, even if armoured, unless the steel structure affords adequate protection.

3.12.3 Metal casings for mechanical protection of cables are to be efficiently protected against corrosion.

3.12.4 Non metallic protective casings and fixings are to be flame retardant in accordance with the requirements of IEC Publication 92-101.

3.12.5 If cable trays/protective casings are made of plastics materials, then they are to comply with the requirements in 3.12.5.1 to 3.12.5.4.

3.12.5.1 They are to be of approved type and are to be supplemented by metallic fixing and straps such that in the event of a fire they, and the cables affixed, are prevented from falling and causing an injury to personnel and/or an obstruction to any escape route. The spacing of their metallic fixing and straps is generally not to exceed 2.0 m.

3.12.5.2 When used on open deck, they are additionally to be protected against ultra-violet light.

3.12.5.3 The load on the plastics cable trays/protective casings is to be within the Safe Working Load (SWL) and the spacing of support, in general, is not to exceed 2 metres.

3.12.5.4 The sum of the total cross-sectional area of the cables, based on their external diameter, is not to exceed 40% of the internal cross-sectional area of the protective casing. This does not apply to a single cable in a protective casing.

Note : Cable trays/protective casings made of plastic materials are to be approved in accordance with the Classification Notes issued by IRS.

3.13 Earthing of metal coverings

3.13.1 Metal coverings of cables are to be effectively earthed at both ends of the cable, except its final sub-circuits, other than those installed in hazardous zones or spaces, earthing at the supply end only will be considered adequate. This does not necessarily apply to instrumentation cables where single point earthing may be desirable for technical reasons.

3.13.2 The electrical continuity of all metal coverings of cables throughout the length of the cable, particularly at joints and tappings, is to be ensured.

3.13.3 The lead sheath of lead-sheathed cables is not to be used as the sole means of earthing the non-current carrying parts of items of equipment.

3.14 Securing of cables

3.14.1 Cables, other than those attached to portable appliances and those installed in pipes, conduits or special casing are to be effectively supported and secured in a manner that prevents damage to their coverings.

3.14.2 Supports and accessories are to be robust and are to be of corrosion-resistant material or suitably corrosion inhibited before erection.

3.14.3 The distance between supports, for horizontal as well as vertical runs of cables, is to be chosen according to the type/size of cable, but generally in accordance with Table 3.14.1.

3.14.4 When electric cables are fixed by means of clips or straps manufactured from a material other than metal the material is to be flame retardant and the fixings are to be supplemented by suitable metal clips or straps at regular intervals, each not exceeding 2 [m].

3.14.5 Single core cables are to be firmly fixed, using supports of strength adequate to withstand forces corresponding to the values of the peak prospective short circuit current.

Table 3.14.1 : Distance between supports			
External diameter of cable			
Exceeding [mm]	**Not exceeding [mm]**	**Non-armoured cables [mm]**	**Armoured cables [mm]**
-	8	200	250
8	13	250	300
13	20	300	350
20	30	350	400
30	-	400	450

3.15 Penetration of bulkheads and decks by cables

3.15.1 Penetration of watertight bulkheads or decks is to be carried out with either individual watertight glands or with packed watertight boxes carrying several cables. In either case, the watertight integrity and strength of the bulkheads and decks are to be maintained. Where cables with polyvinyl chloride insulation are being installed, particular care is to be taken to avoid damage to the sheathing during the fitting of watertight bulkhead glands.

3.15.2 Where fire-resisting or fire-retarding bulkheads or decks are drilled for the passage of cables, all arrangements are to be such that this fire resisting or fire retarding property and strength are not diminished.

3.15.3 Where cables pass through gastight bulkheads or decks separating hazardous zones or spaces, arrangements are to be such that the gastight integrity and strength of the bulkhead or deck are not impaired.

3.15.4 Where cables pass through non-watertight bulkheads or structural steel, the holes are to be bushed, in order to protect the cables, with lead or other approved material which will prevent damage to the cables by abrasion. If the steel is 6 [mm] thick, adequately rounded edges may be accepted as the equivalent of bushing.

3.15.5 Cables passing through decks are to be protected by deck tubes or ducts.

3.15.6 Materials used for glands and bushings are to be such that there is no risk of corrosion.

3.15.7 Where rectangular holes are cut in bulkheads or structural steel the corners are to be adequately rounded.

3.15.8 The distance from cable penetrations to flanges of steam pipes or hot oil pipes is to be not less than 500 [mm].

3.16 Installation of cables in pipes and conduits

3.16.1 Installation of cables in pipes and conduits is to be carried out in such a manner that there is no damage to the cable covering.

3.16.2 Metal conduit systems are to be earthed and are to be mechanically and electrically continuous across joints. Individual short lengths of conduit need not be earthed.

3.16.3 The internal radius of bend of pipes and conduit is to be not less than that laid down for cables, provided that for pipes exceeding 64 [mm] diameter the internal radius of bend is not less than twice the diameter of the pipe.

3.16.4 The drawing-in factor (ratio of the sum of the cross-sectional areas of the cables, based on their external diameter, to the internal cross-section area of the pipe) is not to exceed 0.4.

3.16.5 Expansion joints are to be provided where necessary.

3.16.6 Cable pipes and conduits are to be adequately and effectively protected against corrosion. Where necessary, openings are to be provided at the highest and lowest points to permit air circulation and to prevent accumulation of water.

3.16.7 Cable pipes are to be effectively supported, particularly in areas where they are likely to be subject to heavy vibrations.

3.16.8 Cables in a conduit should belong to the same temperature class.

3.16.9 Where cables are laid in trunks, the trunks are to be so constructed as not to afford passage for fire from one deck or compartment to another.

3.16.10 Cables used for cold cathode luminous discharge lamps are not to be installed in metal conduit unless protected by metal sheath or screen.

3.16.11 Non-metallic ducting or conduit is to be of flame-retardant material. PVC conduit is not to be used in refrigerated spaces or on open decks, unless specially approved.

3.17 Installation of cables in refrigerated spaces

3.17.1 The installation of cables in refrigerated spaces is to be avoided as far as possible.

3.17.2 PVC insulated cables are not to be installed in refrigerated spaces.

3.17.3 Cables installed in refrigerated spaces are to have a watertight or impervious sheath and are to be protected against mechanical damage. If an armoured cable is used, the armour, unless galvanised or of non-corrosive material, is to be protected against corrosion by an additional moisture-resisting covering.

3.17.4 Cables are not to be embedded in or covered by the thermal insulation. They may be fixed to galvanised perforated plates with a space left between the back of the plate and the wall of the room.

3.17.5 Where cables entering a refrigerated space have to pass through the thermal insulation, they are to be installed at right angle to such insulation and are to be protected by a pipe, sealed at each end. Alternatively, the cables may be passed through solid door frames, the necessary holes being sealed at each end.

3.17.6 Precautions are to be taken to prevent the placing of hooks around the cable as a casual means of suspension.

3.17.7 Supporting strips, plating or hangers used for securing the cables are to be galvanised or otherwise protected against corrosion.

3.18 Cables for alternating current

3.18.1 Generally, multi-core cables are to be used in A.C. installations. Where it is necessary to use single-core cables for alternating current circuits rated in excess of 20 A the requirements of 3.18.2 to 3.18.9 are to be complied with.

3.18.2 Cables are to be either non-armoured or armoured with non- magnetic material.

3.18.3 If installed in pipe or conduit, cables belonging to the same circuit are to be installed in the same conduit, unless the conduit or pipe is of non-magnetic material.

3.18.4 Cable clips are to include cables of all phases of a circuit unless the clips are of non-magnetic material.

3.18.5 When installing two, three or four single-core cables forming respectively single-phase circuits, three-phase circuits or three-phase and neutral circuits, the cables are to be in contact with one another, as far as possible. In any case, the distance between the external covering of two adjacent cables is not to be greater than one diameter.

3.18.6 In the case of circuits using two or more parallel connected cables per phase, all cables are to have the same length and cross sectional area.

3.18.7 Where single core cables of rating exceeding 50 A and over 30 [m] in length are used, the phases are to be transposed at regular intervals of approximately 15 [m] in order to obtain the same degree of impedance of circuits.

3.18.8 Where single core cables of rating exceeding 50 A are used, magnetic material is not to be placed between single-core cables of a group. If these cables pass through steel plates, all cables of the same circuit are to pass through the plate or gland so constructed that there is no magnetic material between the cables, and suitable clearance is provided between the cable core and magnetic material. This clearance, wherever practicable, is not to be less than 75 [mm] when the current exceeds 300 A. For currents between 50 A and 300 A the clearance may be proportionately reduced.

3.18.9 If single-core cables of current rating greater than 250 A are run along a steel bulkhead, wherever practicable the cables should be spaced away from the steel.

3.19 Cable ends

3.19.1 The ends of all conductors of cross-sectional area greater than 4 [mm^2] are to be fitted with soldering sockets, compression type sockets or mechanical clamps. Corrosive fluxes are not to be used.

3.19.2 Cables having hygroscopic insulation (e.g. mineral insulated) are to have their ends sealed against ingress of moisture.

3.19.3 Cables with a supplementary insulating belt beneath the protective sheath are to have additional insulation at those points where the insulation of each core makes or may make contact with earthed metal.

3.20 Busbar trunking systems

3.20.1 Where busbar trunking systems are used in place of electric cables, they are to comply with the requirements of 3.20.2 to 3.20.6, in addition to the applicable requirements in Section 3.

3.20.2 The busbar trunking, or enclosure system, is to have a minimum ingress protection of IP54, according to IEC 60529 Degrees of protection provided by enclosures (IP Code).

3.20.3 The internal and external arrangements of the busbar trunking or enclosure system are to ensure that the fire and/or watertight integrity of any structure through which it passes is not impaired.

3.20.4 Where the busbar trunking system is employed for circuits on and below the bulkhead deck, arrangements are to be made to ensure that circuits on other decks are not affected in the event of partial flooding under the normal angles of inclination given in Chapter 1, Table 1.7.1.

3.20.5 Supports and accessories are to be robust and are to be of corrosion-resistant material or suitably corrosion inhibited before erection. The support system is to effectively secure the busbar trunking system to the ship's structure.

3.20.6 When accessories are fixed to the busbar system by means of clips or straps manufactured from a material other than metal, the fixings are to be supplemented by suitable metal clips or straps, such that, in the event of a fire or failure, the accessories are prevented from failing and causing injury to personnel and/or an obstruction to any escape route. Alternatively, the busbar system may be routed away from such areas.

3.21 Watertight Cable transits

3.21.1 Watertight cable transits are to be installed and maintained in accordance with the manufacturer's requirements and in accordance with the requirements of the relevant type approval certification.

3.21.2 A Cable Transit Seal Systems Register is to be provided by the shipbuilder for all watertight cable transits fitted on-board the vessel. For an example of a register see Table 3.21.2. The Register can be in either a hard copy or digitized media. It is to include a marking / identification system, documentation referencing manufacturer manual(s) for each type of cable transit installed, the Type Approval certification for each type of transit system, applicable installation drawings, and a recording of each installed transit documenting the as built condition after final inspection in the shipyard. It is to include sections to record any inspection, modification, repair and maintenance.

3.21.3 During new construction the Register is to be reviewed by the attending Surveyor to confirm it contains a list of the watertight cable transits, applicable cable transit information and sections to maintain in-service maintenance and survey records.

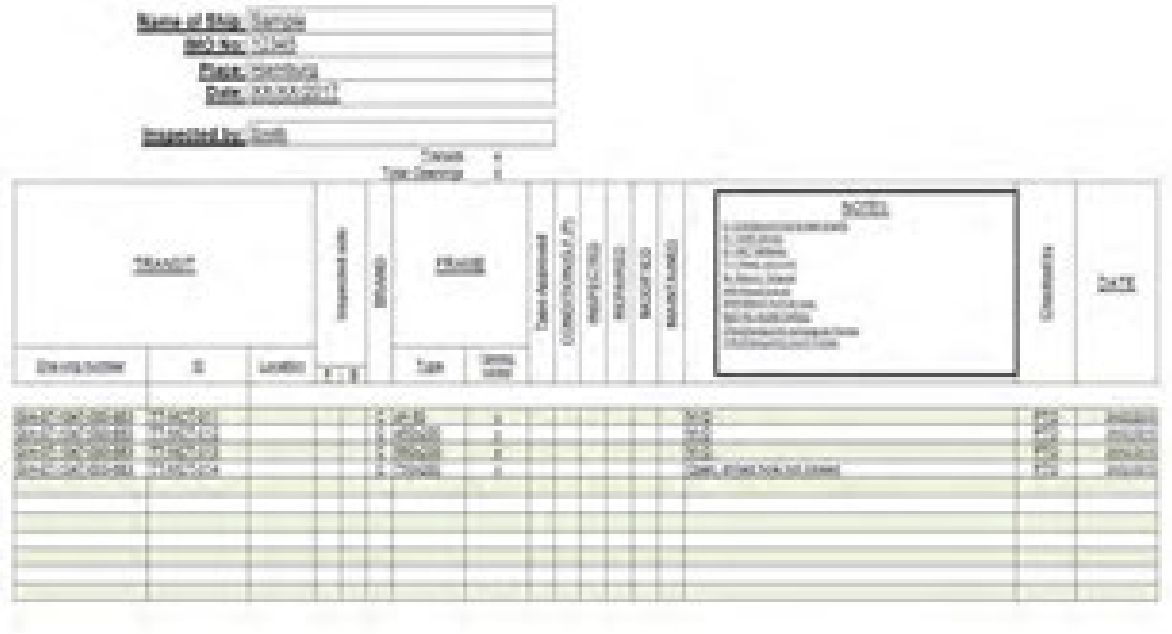
3.21.4 The Register is to be held onboard the vessel.

3.21.5 The owner is to maintain the Register to record any disruption (repair, modification or opening out and closing) to a cable transit or to record the installation of a new cable transit.

3.21.6 At new construction it is to be confirmed by the surveyor that:

- Cable transits have been installed, and where disrupted have been reinstated, in accordance with the manufacturer's requirements and in accordance with the requirements of Type Approval.
- Where specified, appropriate specialized tools have been used.

Table 3.21.2 : Recommended format for Cable Transit Seal System Register



Section 4

Switchgear and Control Gear Assemblies

4.1 Main and emergency switchboards

4.1.1 Location and installation

4.1.1.1 Switchboards are to be installed in accessible and well ventilated dry spaces free from flammable gases and acid fumes.

4.1.1.2 Switchboards are to be secured to a solid foundation and protected against shocks and damage due to leaks and falling objects. They are to be self-supported, or be braced to the bulkhead or the deck above. In case the latter method is used, the means of bracing is to allow normal deflections of the deck without buckling the control cell or assembly structure.

4.1.1.3 Pipes should not be installed directly above or in front of or behind switchboards. If such piping is unavoidable, suitable protection is to be provided in these positions.

4.1.1.4 An adequate, unobstructed working space is to be left in front of switchboards. At the rear, a clearance of at least 0.6 m is to be maintained except that this may be reduced to 0.5 m in way of stiffeners or frames. If switchboards are enclosed at the rear and are fully serviceable from the front, clearance at the rear will not be required unless necessary for cooling.

4.1.1.5 The main switchboard is to be so placed relative to one main generating station that, as far as practicable, the integrity of the normal electrical supply may be affected only by a fire or other casualty in one space. The main switchboard is to be located as close as practicable to the main generating station, within the same machinery space and the same vertical and horizontal A60 fire boundaries. An environmental enclosure for main switchboard, such as may be provided by a machinery

control room situated within the main boundaries of the space, is not to be considered as separating the switchboards from the generators.

4.1.1.6 Where essential services for steering and propulsion are supplied from section boards these and any transformers, converters and similar appliances constituting an essential part of electrical supply system are also to satisfy 4.1.1.5.

4.1.2 Construction of switchboards

4.1.2.1 Switchboards are to have roof with degree of protection IP 22 and are to be of dead front type.

On systems with voltages above 50 V up to and including 1000 V, front and rear dead type switchboards are to be used.

4.1.2.2 All main switchboards are to be guarded by hand rails either made of hardwood or insulated. Where access is provided behind a main switchboard, the handrails on the rear are to be horizontal, and so placed that one cannot accidentally fall into the switchboard. Further, insulated mats or gratings are to be laid on the floor of passage-ways in front of and to the rear of switchboards. Instruments and handles or push buttons for switchgear are to be placed on the front of the switchboard (except for isolating switches, if used). All other parts which require operation, are to be accessible and so placed that the risk of accidental touching of current carrying parts, or accidental making of short-circuits and earthings, is reduced as far as practicable.

4.1.2.3 Section boards (sub-switchboards) and distribution boards are to be enclosed unless they are installed in a cupboard or compartment to which only authorised personnel have access, in which case the cupboard may serve as an enclosure.

4.1.2.4 Framework, panels and doors of switchboards are generally to be of steel or aluminium alloy, and are to be of rigid construction.

4.1.2.5 All parts of the main switchboard are to be accessible for maintenance work.

4.1.2.6 Equipment for each generator and for different distribution systems are to be placed in separate cubicles (panels) or are to be separated from each other by partitions clearly marked with the actual voltages.

4.1.2.7 Doors, behind which equipment requiring operation is placed, are to be hinged. Arrangement is also to be provided to keep the hinged doors open.

4.1.2.8 Cable entrances are generally to be from below or from the side. Cable entries from the top may be accepted provided watertight glands are used.

4.1.2.9 Where the main source of electrical power is essential for propulsion of the ship, the main busbars are to be subdivided into at least two parts which are normally to be connected by circuit breakers (with or without tripping mechanism), disconnecting links or switch by which busbars can be split and reconnected easily and safely. Bolted links (e.g. bolted bus bar sections) are not acceptable. So far is practicable, the connection of generating sets and any other duplicated equipment is to be equally divided among the busbar divisions. Equivalent arrangements to the satisfaction of IRS may be accepted.

4.1.3 Marking and labels

4.1.3.1 All measuring instruments and all apparatus controlling circuits are to be clearly and indelibly marked for identification purposes. An indelible label is to be permanently secured to, or adjacent to, every fuse and every circuit breaker, and marked with particulars of the full load current of the generator or cable which the fuse or circuit-breaker protects. Where inverse time limit and/or reverse current devices are provided in connection with a circuit breaker, the appropriate settings of these devices are to be stated on the lable. The distribution voltage is to be indicated. Labels are to be of flame-retardant material and markings are to correspond with the designations used in the wiring diagrams.

4.1.4 Busbars

4.1.4.1 Busbars and their connections are to be of copper, all connections being so made as to prevent deterioration of the joint by corrosion or oxidation.

4.1.4.2 The sizes of busbars and their connections are to be calculated to ensure that their mean temperature rise does not exceed by more than 45°C from the ambient temperature, when running continuously at the normal rating.

4.1.4.3 Busbars, together with their connections and supports, are to be capable of

withstanding, without detrimental effect, the mechanical stresses which will arise during short-circuits. Further, provision is to be made to allow the busbars to expand without causing any abnormal stress on their supports.

4.1.4.4 Busbars and other bare conductors are to be mounted on non- deteriorating and non-hygroscopic insulating material, maintaining adequate clearance and creepage distance, in accordance with Section 1, to ensure that there is no risk of flash-over under normal service conditions. Where necessary these distances may have to be increased to allow for the electro-magnetic forces involved.

4.1.4.5 Connections from bus-bars and from generator circuit terminals to all circuit breakers and fuses are to be installed "short-circuit proof", i.e. either bare conductors, or insulated conductors or single-core cable without metallic sheath/armour /braid are used and these are mounted on supports of insulating material and with adequate distance between the different poles (phases) and to earthed parts.

4.1.4.6 Horizontally installed bus-bars and bare conductors are to be protected by screens, if they are placed so low that it could be a risk, e.g. by tools falling on them.

4.1.5 Equalizer connections

4.1.5.1 The current rating of equalizer connections and equalizer switches is to be not less than half the rated full load current of the generator.

4.1.5.2 The current rating of equalizer busbars is to be not less than half the rated full load current of the largest generator in the group.

4.1.6 Instruments for direct current generators

4.1.6.1 For generators not arranged to run in parallel, at least one voltmeter and one ammeter are to be provided for each generator.

4.1.6.2 For generators arranged to run in parallel at least one ammeter for each generator and two voltmeters are to be provided. One of these voltmeters is to be permanently connected to the busbars and the other is to be provided with a change-over switch to enable it to be connected to any one generator.

4.1.6.3 For compound wound generators fitted with equalizer connections, the ammeter is to be connected to the pole opposite to that connected to the series winding of the generator.

4.1.7 Instruments for alternating current generators

4.1.7.1 For alternating current generators not arranged to run in parallel, each generator is to be provided with at least one voltmeter, one frequency meter, and one ammeter with an ammeter switch to enable the current in each phase to be read or an ammeter in each phase. Generators above 50 kVA are to be provided with a wattmeter.

4.1.7.2 For alternating current generators arranged to run in parallel, each generator is to be provided with a wattmeter, and an ammeter in each phase conductor or an ammeter with a selector switch to enable measurement of current in each phase.

4.1.7.3 For paralleling of the generators, two voltmeters, two frequency meters and a synchronising aid comprising either a synchroscope and lamps, or an equivalent arrangement, are to be provided. One voltmeter and one frequency meter are to be connected permanently to the busbars, the other voltmeter and frequency meter are to be provided with arrangements to enable the voltage and frequency of any generator to be measured.

4.1.8 Instrument scales

4.1.8.1 In general main switchboard instruments are to be of accuracy class 1.5 and other switchboard instruments are to be of accuracy class 2.5.

4.1.8.2 The upper limit of the scale of every voltmeter is to be approximately 120 per cent of the normal voltage of the circuit, and the normal voltage is to be clearly indicated.

4.1.8.3 The upper limit of the scale of every ammeter is to be approximately 130 per cent of the normal rating of the circuit in which it is installed. Normal full load is to be clearly indicated.

4.1.8.4 Ammeters for use with direct current generators, and wattmeters for use with alternating current generators, which may be operated in parallel, are to be capable of indicating 15 per cent reverse-current or reverse-power respectively.

4.1.8.5 The upper limit of the scale of every wattmeter is to be approximately 130 per cent of the rated full load of the circuit in which it is installed. Rated full load is to be clearly indicated.

4.1.8.6 Frequency meters are to be capable of indicating a variation in the frequency from minus 8 per cent to plus 8 per cent of the nominal frequency of the installation.

4.1.8.7 Instruments are to have effective screening, for example, by metal enclosures, in order to diminish faulty readings caused by induction from adjacent current-carrying parts.

4.1.9 Instrument transformers

4.1.9.1 The secondary windings of instrument transformers are to be earthed.

4.1.10 Circuit-breakers

4.1.10.1 Circuit-breakers are to comply with IEC Publication 947-1 and 947-2, "Low Voltage Distribution Switchgear" or an equivalent national standard, amended where necessary for ambient temperature.

4.1.10.2 Test reports, based on the requirements of IEC Publication 947-1 and 947-2 or an equivalent national standard, are to be submitted for consideration when required.

4.1.10.3 Circuit-breakers are to be of the trip-free type i.e. the breaking action initiated by short-circuit and overcurrent relays, or by undervoltage coil, when fitted, is to be fulfilled independently of the position or operation of manual handle or of other closing devices. Further the arrangement is to be such that automatic repeated breakings/makings by short-circuits and overcurrents cannot occur.

4.1.10.4 Each circuit opening device is to be so arranged that accidental closing and opening does not occur.

4.1.10.5 Handles and operating mechanisms are to be so arranged that the hand of the operator cannot accidentally touch live metal or be injured through an arc arising from the switch or circuit-breaker, or the rupturing of a fuse. If switches are enclosed their handles are not to operate through unprotected slots.

4.1.11 Fuses

4.1.11.1 Fuses are to comply with IEC Publication 269 "Low Voltage Fuse with High Breaking Capacity" or an equivalent national standard, amended where necessary for ambient temperature.

4.1.11.2 A report, giving details of test performance, fusing characteristics, temperature and insulation tests and details of the specification to which the fuse has been tested is to be submitted for consideration when required.

4.1.11.3 Fuse-links and fuse-bases are to be marked with particulars of rated current and rated voltage. Each fuse position is to be permanently and indelibly labeled with the current-carrying capacity of the circuit protected by it and with the appropriate approved size of fuse.

4.1.12 Earthing

4.1.12.1 A good, reliable earth connection of all metallic non-current carrying parts is to be insured. Normally, a continuous earth bar will be required. If other solutions are chosen, earth-continuity test may be required.

4.1.12.2 If components such as switches, instruments, signal lamps, etc. with voltage exceeding 30V AC and 50V DC are mounted on hinged doors of switchboards or other enclosures, the doors are to be connected to the switchboards or enclosures by a separate flexible earth conductor.

4.1.12.3 Earth conductors are to be so installed that they are not likely to suffer mechanical damage or corrosion. The earth conductor is to be carried to a separate earth screw inside the enclosure of the equipment which is earthed, except when separately installed earth conductors are used.

4.1.12.4 Earth conductors inside switchboards and other enclosures are to be insulated, except when the arrangement is such that the earth conductor cannot come into contact with current-carrying parts.

4.1.13 Wiring

4.1.13.1 Insulated cables / wires connecting components are to be stranded, flame retardant and manufactured in accordance with a relevant and acceptable National Standard or equivalent International Standard.

4.2 Distribution switchboards

4.2.1 Large distribution switchboards are to be constructed in accordance with 4.1.

4.2.2 All parts which require operation in normal use are to be placed on the front or easily accessible from behind front doors.

When such parts are placed behind front doors, the interior front is to comply with enclosure type IP 20, except that fuses with accessible current-carrying parts may be permitted, provided that the door is lockable.

4.2.3 Switchboards, supplied from different supply circuits, are not to be placed in the same enclosure unless these are separated by partitions of flame retardant material.

4.2.4 Switchboards, which are provided with two or more supply circuits arranged for automatic standby connection, are to be provided with positive means to show the circuit feeding the switchboard. Such switchboards are to be provided with warning notice that all the supply circuits are to be disconnected before maintenance work is undertaken.

4.2.5 If distribution switchboards are placed in accommodation area then they are also to comply with the requirements given in Pt.6, Ch.3, Cl.3.2.2.3.2 IR.6.

4.3 Navigation lights

4.3.1 Navigation lights are to be connected separately to a distribution board reserved for this purposes only, and connected directly or through transformers to the main or emergency switchboard. The distribution board is to be accessible to the officer of the watch on the bridge. Provision is to be made on the bridge for such navigation lights to be transferred to an alternate circuit.

4.3.2 Each navigation light is to be controlled and protected in each insulated pole by a switch and fuse or circuit breaker mounted on the distribution board.

4.3.3 Where the navigation panel is situated in the midships house, the midships sub-switchboard is regarded as an extension to the main switchboard provided that it is supplied from the main switchboard by two cables each capable of carrying the full load.

4.3.4 Each navigation light is to be provided with an automatic indicator giving audio and visual indication of failure of the light. If a visual signal is used connected in series with the navigation light, means are to be provided to prevent extinction of the navigation light due to failure of the signal.

4.3.5 Transferring of power supply to an alternate circuit should not render the audio/visual alarms inoperative. Means should be provided to test the audio/visual alarms.

4.3.6 Any statutory requirements of the country of registration are to be complied with and, on application to IRS, may be accepted as an alternative to the above.

4.3.7 The voltage drop between the power supply terminals and load terminals for navigation lights is to be not more than 2.5 per cent of the rated voltage to ensure required output and colour.

4.4 Control gear

4.4.1 Control gear is to comply with IEC 60947 “Low-voltage switchgear and control gear” or an equivalent national standard, amended where necessary for ambient temperature.

4.4.2 Control gear, including isolating and reversing switches, is to be so arranged that shunt field circuits are not disconnected without adequate discharging path being provided.

4.4.3 Control gear for essential and important motors are to be separated from each other and from other current carrying parts by screens. The arrangement is to be such that maintenance work can be carried out on each unit without danger when isolated.

4.4.4 When installed in main switchboards motor control gear is to be placed in separate cubicles separated from all other parts of the switchboard by partitions of flame retardant material. The arrangements are to be such that arcs occurring by short-circuit in one cubicle cannot spread to bus-bars.

4.5 Testing

4.5.1 Switchgear and control gear assemblies with supply voltage of 60 V and above are to be tested as follows:

- High voltage test with 1000 V plus twice the rated voltage with a minimum of 2000 V. The test voltage is to be supplied for 1 minute at any frequency between 25 and 100 Hz.

- Insulation resistance measuring.

4.5.2 Switchgear and control gear assemblies with supply voltage less than 60 V are to be tested in accordance with 4.5.1 except that the test voltage is to be 500 V.

Section 5

Rotating Machines - Construction and Testing

5.1 General

5.1.1 Rotating machines are to comply with the relevant part of IEC 60092-301:1980/ AMD2:1995, or an acceptable and relevant national standard, and the requirements of this Section.

5.1.2 For all the rotating machines for essential services, manufacturer's test records are to be provided (See also Sec.1). For other machines they are to be available upon request.

5.1.3 Shaft materials for electric propulsion motors and for main engine driven generators where the shaft is part of the propulsion shafting, is to comply with applicable requirements of Pt.2, and to be certified by IRS. Shaft material for other machines is to be in accordance with either Pt.2 or recognised international or national standard.

5.1.4 The rotating parts are to be so balanced that when running at any speed in the normal working range the vibration level does not exceed the levels specified in IEC 60034.

5.1.5 The lubrication arrangement for bearings are to be effective under all operating conditions including the maximum ship inclinations specified in Ch.1 and there are to be effective means to ensure that lubricant does not reach the machine windings or other conductors and insulators.

5.1.6 Steps are to be taken to prevent the ill effects of flow of currents circulating between the shaft and bearings.

5.1.7 Where welding is proposed to be applied to shafts of machines for securing armature arms or spiders, stress relieving is to be carried out after welding. The proposal is to be submitted for scrutiny and approval.

5.1.8 The construction of alternating machines, are to be capable of withstanding a sudden short circuit at their terminals under any operating condition.

5.1.9 Effective means are to be provided to prevent the accumulation of moisture and condensation within the machines when they are stopped with a provision that such means are switched on at stand-still and switched off at starting.

5.1.10 Coolers : Water-air heat exchangers of rotating machines are to be of the double tube type. In a normally attended position a visual and audible alarm is to be given to monitor water cooler leakage.

5.2 Rating

5.2.1 Ship's service generators including their exciters, and continuously rated motors are to be suitable for continuous duty at their full rated output at maximum cooling air or water temperature for an unlimited period, without the limits of temperature rise in 5.3 being exceeded. Other generators and motors are to be rated in accordance with the duty which they are to perform, and when tested under rated load conditions the temperature rise is not to exceed the values in 5.3.

5.3 Temperature rise

5.3.1 The limits of temperature rise specified in Table 5.3.1 are based on a cooling air temperature of 45°C and a cooling water temperature of 30°C.

5.3.2 If the temperature of the cooling medium is known to exceed the value given in 5.3.1, the permissible temperature rise is to be reduced by an amount equal to the excess temperature of the cooling medium. These temperature rises

are, if necessary to be reduced to satisfy the requirements of flame-proof equipment.

5.3.3 If the temperature of the cooling medium is known to be permanently less than the value given in 5.3.1, the permissible temperature rise may be increased by an amount equal to the difference between the declared temperature and that given in 5.3.1 upto a maximum of 15°C.

Table 5.3.1 : Limits of temperature rise in °C

Item		Part of machines	Method of measurement of temp.	Temperature rise °C					
				Air cooled machines insulation class			Water cooled machines insulation class		
				A	E	B	A	E	B
1	(a)	a.c. windings of turbine type machines having output of 5000 KVA or more	ETD or R	50	60	70	70	80	90
	(b)	a.c. windings of salient-pole and of induction machines having output of 5000 KVA or more, or having a core length of one metre or more	ETD or R	50	60	70	70	80	90
2	(a)	a.c. windings of machines smaller than in item 1	R	50	65	70	70	85	90
			T	40	55	60	60	75	80
	(b)	Field windings of a.c. and d.c. machines having d.c. excitation other than those in items 3 and 4	R	50	65	70	70	85	90
			T	40	55	60	60	75	80
	(c)	Windings of armatures having commutators	R	50	65	70	70	85	90
			T	40	55	60	60	75	80
3		Field windings of turbine-type machines having d.c. excitation	R	-	-	80	-	-	100
4	(a)	Low-resistance field windings of more than one layer and compensating windings	T,R	50	65	70	70	85	90
	(b)	Single-layer windings with exposed bare surfaces	T,R	55	70	80	75	90	100
5		Permanently short-circuited insulated windings	T	50	65	70	70	85	90
6		Permanently short-circuited windings, uninsulated	T	The temperature rise of these parts shall in no case reach such a value that there is a risk of injury to any insulating or other material on adjacent parts					
7		Iron core and other parts not in contact with windings	-	The temperature rise of these parts shall in no case reach such a value that there is a risk of injury to any insulating or other material on adjacent parts					
8		Iron core and other parts in contact with windings	T	50	65	70	70	85	90
9		Commutators and slip-rings, open or enclosed	T	50	60	70	70	80	90

Notes:-

1 T = Thermometer method
R = Resistance method
ETD = Embedded Temperature Detector

2 When the commutators, sliprings or bearings of machines provided with water coolers are not in the enclosed air circuits-cooled by the water cooler, but are cooled by the ambient cooling air, the permissible temperature rise above the ambient cooling air should be the same as for ventilated machines.

3 When Class F or Class H insulation is employed, the permitted temperature rises are respectively 20°C and 40°C higher than the values given for Class B insulation.

4 Classes of insulation are to be in accordance with IEC Publication 85 (1957) - "Recommendations for the classification of material for the insulation of electrical machinery and apparatus in relation to their thermal stability in service".

5.3.4 The limits of temperature rise of electric slip couplings are to be in accordance with Table 5.3.1, except that when a squirrel cage element is used the temperature of this element is not to reach an injurious value. The temperature of the field windings is not to exceed these limits at all speeds of operation. Arrangements for reducing the excitation of self ventilated couplings at low operational speeds are permissible.

5.3.5 Alternating current machines of 5000 kVA output and above and propulsion motors having a total axial core length of 1m or more (including the ventilating duct), are to have at least three embedded temperature detectors. With multicore machines the total length is to be taken as the sum of the individual core lengths.

5.4 Generator control

5.4.1 Each alternating current generator, unless of the self-regulating type, is to be provided with automatic means of voltage regulation; voltage build-up is not to require an external source of power.

5.4.2 The voltage regulation of any alternating current generator with its regulating equipment is to be such that at all loads, from zero to full load at rated power factor, the rated voltage is maintained within $\pm$2.5 per cent under steady conditions. There is to be provision at the voltage regulator to adjust the generator no load voltage.

5.4.3 Generators, and their excitation systems, when operating at rated speed and voltage on no-load are to be capable of absorbing the suddenly switched, balanced, current demand of the largest motor or load at a power factor not greater than 0.4 with a transient voltage dip which does not exceed 15 per cent of rated voltage. The voltage is to recover to rated voltage within a time not exceeding 1.5 seconds.

5.4.4 The transient voltage rise at the terminals of a generator is not to exceed 20 per cent of rated voltage when rated KVA at a power factor not greater than 0.8 is thrown off.

5.4.5 Generators are to be capable of maintaining under steady state short circuit conditions a current of at least three times the full load rated current for a duration of at least two seconds or where precise data is available for the duration of any longer time delay which may be provided by a tripping device for discrimination purposes.

5.4.6 Generators required to run in parallel are to be stable from no load [kW] up to the total combined full load [kW] of the group, and load sharing is to be such that the load on any generator does not normally differ from its proportionate share of the total load by more than 15 per cent of the rated output [kW] of the largest machine or 25 per cent of the rated output [kW] of the individual machine, whichever is less.

5.4.7 When generators are operated in parallel, the kVA loads of the individual generating sets are not to differ from the proportionate share of the total kVA load by more than 5 per cent of the rated kVA output of the largest machines.

5.4.8 Generators running in parallel may have a common neutral connection to earth provided they are suitably designed to avoid excessive circulating currents.
This is particularly important if the generators are of different size and make.
Generators in which the third harmonic content of the waveform does not exceed 5 per cent may be considered adequate.

Note : This would mostly occur with a neutral bus with a single grounding resistor with the associated neutral switching. Where individual resistors are used, circulation of the third harmonic currents between paralleled generators is minimized.

5.5 Overloads

5.5.1 Machines are generally to be capable of withstanding, on test, without injury, the following overload conditions :

a) D.C. generators - an excess current of 50 per cent for 15 seconds after attaining the temperature rise corresponding to rated load, the terminal voltage being maintained as near the rated value as possible. This requirement does not apply to the overload torque capacity of the prime mover.

b) A.C. generators - an excess current of 50 per cent for 2 minutes, at 0.6 power factor, after attaining the temperature rise corresponding to rated load, the terminal voltage and frequency being maintained as near the rated values as possible. This requirement does not apply to the overload torque capacity of the prime mover.

c) Motors - At rated speed or in the case of a range of speeds, at the highest and lowest speeds, under gradual increase of torque, the voltage and frequency being maintained as near to their rated value as possible, the appropriate excess torque given below. Synchronous motors and synchronous induction motors are required to withstand the excess torque without falling out of synchronism and without adjustment of the excitation current preset at the value corresponding to rated load.

d.c. motors	50 per cent for 15 seconds
polyphase a.c. synchronous motors	50 per cent for 15 seconds
polyphase a.c. synchronous induction motors	35 per cent for 15 seconds
polyphase a.c. induction motors	60 per cent for 15 seconds

d) Propulsion machines - Overload tests for propulsion machines will be given special consideration for each installation.

5.6 Brushgear

5.6.1 The final running position of brushgear is to be clearly and permanently marked.

5.6.2 Direct current motors and generators are to operate with fixed brush setting from no load to the momentary overload specified without injurious sparking, or damage to the commutator or brushes. The commutation is to be checked with an excess current of 20 per cent and for a period of time sufficient to judge its quality.

5.6.3 Alternating current commutator motors are to operate over the specified range of load and speed without injurious sparking.

5.7 Inspection and testing

5.7.1 On all machines intended for essential services, tests specified in Table 5.7.1 are to be carried out. See also Section 1 'Surveys', subsection 1.3. Any other relevant tests required by the national standards are also to be carried out.

Type tests are to be carried out on a prototype machine or on the first of a batch of machines and routine tests carried out on subsequent machines.

5.7.2 The high voltage test is to be carried out at 1000 plus twice the rated voltage with a minimum of 2000 V on new machines, preferably at the conclusion of the temperature rise test. The test is to be applied between the windings and the frame with the core connected to the frame and to any windings or sections of windings not under test. Where both ends of each phase are brought out to accessible separate terminals, each phase is to be tested separately. The test is to be made with alternating voltage at any convenient frequency between 25 and 100 Hz of approximately sine wave form. The test is to be commenced at a voltage of not more than one half of the full-test voltage and is to be increased progressively to full value, the time allowed for the increase of the voltage from half to full value being not less than 10 seconds. The full test voltage is then to be maintained for one minute and then reduced to one half full value before switching off.

5.7.3 When additional high voltage tests are required on a machine which has already passed its tests or on machines after repair, the voltage of such further tests is to be 75 per cent of the value given in 5.7.2.

5.7.4 Immediately after the high voltage tests the insulation resistances are to be measured using a direct current insulation tester between:

a) all current carrying parts connected together and earth,

b) all current carrying parts of different polarity or phase, where both ends of each polarity or phase are individually accessible.

The minimum values of test voltages and corresponding insulation resistances are given in Table 5.7.4. The insulation resistance is to be measured close to the operating temperature, or an appropriate method of calculation is to be used.

Table 5.7.1 : Tests to be carried out on electrical rotating machines

Sr. No.	Tests	a.c. Generators		Motors	
		Type test (1)	Routine test (2)	Type test (1)	Routine test (2)
1.	Examination of the technical documentation, as appropriate and visual inspection	X	X	X	X
2.	Insulation resistance measurement	X	X	X	X
3.	Winding resistance measurement	X	X	X	X
4.	Verification of the voltage regulation system	X	X(6)		
5.	Rated load test and temperature rise measurement	X		X	
6.	Overload test	X	X(4)	X	X(4)
7.	Verification of steady short-circuit conditions (3)	X			
8.	Overspeed test	X	X	X(5)	X(5)
9.	Dielectric strength test	X	X	X	X
10.	No load test	X	X	X	X
11.	Verification of degree of protection	X		X	
12.	Verification of bearings	X	X	X	X

(1) For machines of less than 100 kW, type tests on prototype machine is acceptable. For 100 kW or more, tests on at least the first of each batch of machines are to be carried out.
(2) The report of machines routine tested is to contain the manufacturer's serial number of the machine which has been type tested and the test results.
(3) Verification of steady short-circuit condition applies to synchronous machines only.
(4) Only applicable for machines of essential services rated above 100 kW. For routine tests, overcurrent test may be carried out in lieu of overload test.
(5) Not applicable for squirrel cage motors.
(6) Only functional test of voltage regulator system.

Table 5.7.4 : Minimum values of test voltages and corresponding insulation resistances

Related Voltage V_R (V)	Minimum Test Voltage (V)	Minimum Test Insulation Resistance (MΩ)
$V_R \leq 250$	$2 \times V_R$	1
$250 < V_R \leq 1000$	500	1
$1000 < V_R \leq 7200$	1000	$(V_R / 1000) + 1$
$7200 < V_R \leq 15000$	5000	$(V_R / 1000) + 1$

Section 6

Converting Equipment

6.1 Transformers

6.1.1 The following requirements apply to transformers of 5 kVA and above.

6.1.2 Transformers are to comply with the requirements of IEC Publication 76 or an acceptable and relevant national standard amended where necessary for ambient temperature.

6.1.3 Transformers may be of the dry type, encapsulated or liquid filled type.

6.1.4 Transformers are to be placed in easily accessible well ventilated spaces free from any gaseous or acid fumes. They are to be clear of non-protected ignitable materials, and so arranged as to be protected against shocks and any damage resulting from water, oil, liquid fuel, steam etc.

6.1.5 The live parts of transformers are to be provided with means of protection against accidental contact.

6.1.6 Transformers are to be double wound except those for motor starting.

6.1.7 Windings of air cooled transformers are to be treated to efficiently resist moisture, sea air and oil vapours.

Effective means are to be provided to prevent the accumulation of moisture and condensation within the transformers, when de-energized.

6.1.8 Each transformer is to be provided with a nameplate of corrosion- resistant metal giving information on make, type, serial number, insulation class and any other technical data necessary for the application of the transformer.

6.1.9 Liquid fillings for transformers are to be non-toxic and of a type which does not readily support combustion. Liquid filled transformers are to have a pressure relief-device with an alarm and there is to be a suitable means provided to contain any liquid which may escape from the transformer.

6.1.10 When forced cooling is employed, whether air or liquid, there is to be monitoring of the cooling medium and transformer winding temperature with an alarm should these exceed preset limits. There are to be arrangements so that the load may be reduced to a level commensurate with the cooling available.

6.1.11 The inherent regulation of transformers at their rated output is to be such that voltage drop to any point in the installation does not exceed that allowed by 2.3.

6.1.12 All transformers are to be constructed to withstand, without damage, the thermal and mechanical effects of a short-circuit at the terminals of any winding for 2 seconds with rated primary voltage and frequency without damage. This may be required to be verified by type test or random test.

6.1.13 The temperature rise of windings of transformers above the ambient temperatures given in Ch.1, when measured by resistance, during continuous operation at the maximum rating, is not to exceed :

a) For dry type transformers, air cooled

- 50°C for Class A;
- 60°C for Class E;
- 70°C for Class B;
- 90°C for Class F; and
- 110°C for Class H.

b) For liquid filled transformers

- 50°C - where air provides cooling of the fluid;
- 65°C - where water provides cooling of the fluid.

6.1.14 The tests given hereunder are to be carried out on all transformers at the manufacturer's works and a certificate of tests issued by the manufacturer.

a) High voltage test : The test voltage is to be applied, preferably after the temperature rise test, to each winding in turn, between the winding under test and the remaining

windings, core, frame and tank or casing connected together and to earth. The test is to be made with 1 kV a.c. plus twice the highest voltage between lines with a minimum of 2.5 KV at any frequency between 25 and 100 Hz and maintained for 1 minute without failure.

b) When additional high voltage tests are carried out and for transformers that have been rewound or subject to extensive repair, the test voltage may be limited to a value which is 75 per cent of that stated in (a).

c) Induced high voltage test : To test between turns, coils and terminals, an a.c. voltage is to be applied between the above parts corresponding to twice the voltage appearing between these parts when rated voltage is applied to the terminals. The duration of the test is to be 1 minute for any test frequency up to and including twice the rated frequency.

d) Insulation resistance : The insulation resistance of each winding in turn to all the other windings, core, frame and tank or casing connected together and to earth is to be measured after the high voltage test and recorded together with the temperature of the transformer at the time of the test.

e) Temperature rise : One transformer of each size and type is to be given a temperature rise test. For transformers of rating 100 KVA and above, it will be accepted that the temperature rise test is made on one of several identical transformers manufactured and tested at the same time.

6.2 Semiconductor equipment

6.2.1 The requirements of 6.2.2 to 6.2.16 apply to semiconductor equipment rated for 5 [kW] upwards.

6.2.2 Semiconductor equipment is to comply with the requirements of IEC 146: Semiconductor convertors, or an acceptable and relevant national standard amended where necessary for ambient temperature.

6.2.3 Semiconductor static power converter equipment is to be rated for the required duty having regard to peak loads, system transients and overvoltage.

6.2.4 Converter equipment may be air or liquid cooled and is to be so arranged that it cannot remain loaded unless effective cooling is maintained. Alternatively the load may be automatically reduced to a level commensurate with the cooling available.

6.2.5 Liquid cooled converter equipment is to be provided with leakage alarms and there is to be a suitable means provided to contain any liquid which may leak from the system. Where the semiconductors and other current carrying parts are in direct contact with the cooling liquid, provision is to be made for monitoring the liquid to ensure it has a satisfactory resistivity.

6.2.6 Where forced cooling is used there is to be temperature monitoring of the heated cooling medium with an alarm and shutdown when the temperature exceeds a preset value.

6.2.7 Cooling fluids are to be non-toxic and of low flammability.

6.2.8 Converter equipment is to be so arranged that the semiconductor devices, fuses, control and firing circuit boards may be readily removed from the equipment for repair or replacement.

6.2.9 Test and monitoring facilities are to be provided to permit identification of control circuit faults and faulty components.

6.2.10 Protection devices fitted for converter equipment protection are to ensure that, under fault conditions, the protective action of circuit breakers, fuses or control systems is such that there is no further damage to the convertor or the installation.

6.2.11 Converter equipment, including any associated transformers, reactors, capacitors and filters, if provided, is to be so arranged that the harmonic distortion, and voltage spikes, introduced in to the ships electrical system are restricted to a level necessary to ensure that it causes no malfunction of equipment connected to the electrical installation.

6.2.12 Overvoltage spikes or oscillations caused by commutation or other phenomena, are not to result in the supply voltage waveform deviating from the superimposed equivalent sine wave by more than 10 percent of the maximum value of the equivalent sine wave.

6.2.13 When converter equipment is operated in parallel, load sharing is to be such that under normal operating conditions overloading of any unit does not occur and the combination of

parallel equipment is stable throughout the operating range.

6.2.14 When converter equipment has parallel circuits there is to be provision to ensure that the load is disturbed uniformly between the parallel paths.

6.2.15 Transformers, reactors, and other circuit devices associated with convertor equipment are to be suitable for the distorted voltage and current waveforms to which they may be subjected.

6.2.16 Tests at the manufacturer's works are to include

- high voltage test for one minute applied between terminals and earthed parts at a frequency between 25 - 100 Hz. For system voltage up to 60 V the test voltage is to be 600 V; for system voltage between 60 - 90 V the test voltage is to be 900 V; for system voltage over 90 V the test voltage is to be twice the system voltage plus 1000 V;
- functional tests;
- temperature rise test; and
- such other agreed tests as are necessary to demonstrate the suitability of the equipment for its intended duty. Details of tests are to be submitted for consideration when required.

Section 7

Miscellaneous Equipment

7.1 Accumulator batteries

7.1.1 General

7.1.1.1 The requirements of this Section apply to permanently installed secondary batteries of the vented type.

7.1.1.2 A vented type battery is one in which the electrolyte can be replaced and which may release gas whilst operating on charge and overcharge.

7.1.1.3 Proposals for the use of other types of secondary batteries may be submitted for consideration.

7.1.2 Construction

7.1.2.1 The cells of all vented type batteries are to be so constructed and secured as to prevent spilling of the electrolyte due to inclinations of 40 degrees from normal. The filling plugs are to be so constructed as to prevent spilling of electrolyte due to ship's movements, e.g. rolling and pitching.

7.1.2.2 All batteries are to be provided with durable labels of flame retardant material, giving information on the application for which the battery is intended, voltage and capacity. Instructions are to be fitted either at the battery or at the charging device giving information on maintenance and charging.

7.1.3 Location

7.1.3.1 Alkaline batteries and lead acid batteries of the vented type are not to be installed in the same compartment.

7.1.3.2 Batteries are to be located in such a manner that they are not exposed to excessive heat, extreme cold, spray, steam or other conditions which would impair performance or accelerate deterioration.

7.1.3.3 Batteries are not to be located in an accommodation space.

7.1.3.4 Large batteries are to be installed in a compartment assigned to them only. A box on deck would meet this requirement if adequately ventilated and provided with means to prevent ingress of water, and provided the intended service is assured in cold conditions.

7.1.3.5 Engine starting batteries are to be located as close as practicable to the engine(s) served. If such batteries cannot be accommodated in the battery compartment, they are to be installed so that adequate ventilation is ensured.

7.1.4 Installation

7.1.4.1 Batteries should be so arranged that each cell or crate of cells is accessible from the top and at least one side.

7.1.4.2 Cells or crates are to be carried on non-absorbent insulating supports. Similar insulators are to be fitted to prevent any movement of cells arising from the motion of the vessel. Adequate space for circulation of air is to be ensured.

7.1.4.3 Where acid is used as the electrolyte for a vented type battery, a tray of acid resisting material is to be provided below the cells unless the deck below is similarly protected.

7.1.4.4 The interiors of all compartments for vented type batteries, including the shelves, are to be painted with corrosion resistant paint with consideration given to the type of electrolyte actually used.

7.1.4.5 A permanent notice is to be fitted to all compartments for vented type batteries prohibiting naked lights and smoking in the compartment.

7.1.5 Ventilation

7.1.5.1 Battery compartments, lockers and boxes are to be adequately ventilated by an independent ventilating system to avoid accumulation of inflammable gases. Particular attention should be given to the fact that these gases are lighter than air and tend to accumulate at the top of the spaces.

7.1.5.2 The battery compartment is to be provided with an effective air inlet near the floor level.

7.1.5.3 Natural ventilation may be employed if ducts of adequate size can be run directly from the top of the compartment to the open air with no part of the duct more than 45 degrees from the vertical. If natural ventilation is impracticable, mechanical ventilation providing at least 30 changes of air per hour is to be installed. Interior surfaces of ducts and fans are to be painted with corrosion-resistant paint. Fan motors are not to be located in the air stream.

7.1.5.4 Necessary precautions are to be taken to prevent sparking due to possible contact by the ventilation fan blades with fixed parts.

7.1.5.5 All openings through the battery compartment bulkheads or decks, other than ventilation openings, are to be effectively sealed. (Also refer Pt.6, Ch.2, Cl.2.2.1).

7.1.6 Electrical equipment

7.1.6.1 Switches, fuses and other electrical equipment liable to cause an arc should not normally be installed in battery compartments. Where such equipment is necessary for operational reasons, the equipment should be certified for Group II C gases and temperature Class T1 in accordance with IEC 79, "Electrical Apparatus for Explosive Gas Atmospheres" or an equivalent national standard.

7.1.6.2 Cables, with the exception of those pertaining to the battery or the local lighting, are not to be installed in battery compartments except where installation in other locations is impracticable.

7.1.7 Number and rating of batteries

7.1.7.1 Where batteries are used for starting main engines, at least two batteries are to be fitted of such combined size that requirements of Ch.4 are complied with.

7.1.8 Charging facilities

7.1.8.1 All batteries are to be provided with charging devices, suitable for the actual service. Generally, automatically regulated charging devices are to be used, operating as floating service with the battery and giving trickle-charging when the battery is fully charged, or with change-over arrangement for full(quick)-charging and automatic trickle-charging. Other forms for charging arrangements, e.g. charging through resistors from a DC net with higher voltage, will be specially considered.

7.1.8.2 In direct current systems means are to be provided to isolate the batteries from the low voltage systems when being charged from a higher voltage system.

7.1.8.3 Charging devices are to be provided with suitable switch and fuse gear for protection against faults such as short circuit, overload and connection failures.

7.1.9 Maintenance of records for batteries

a) Where batteries are fitted for use for essential and emergency services a schedule of such batteries is to be compiled and maintained. The schedule, which is to be submitted for review during new building

survey and subsequently verified during annual surveys, is to include at least the following information regarding the battery(ies):

- Type and manufacturer's type designation
- Voltage and ampere-hour rating
- Location
- Equipment and/or system(s) served
- Maintenance / replacement cycle dates
- Date(s) of last maintenance and/or replacement
- For replacement batteries in storage, the date of manufacture and shelf life.[1)]

b) Procedures are to be put in place to ensure that when batteries are replaced, they are of an equivalent type.

c) Where vented[2)] type batteries replace valve-regulated sealed[3)] types, it is to be ensured that there is adequate ventilation[4)] and that IRS requirements relevant to the location and installation of vented types batteries are complied with.

d) Details of the schedule and of the procedures are to be included in the ship's safety management system and be integrated into the ship's operational maintenance routine.

Notes:

1) Shelf life is the duration of storage under specified conditions at the end of which a battery retains the ability to give a specified performance.

2) A vented battery is one in which the cells have a cover provided with an opening through which products of electrolysis and evaporation are allowed to escape freely from the cells to atmosphere.

3) A valve-regulated battery is one in which cells are closed but have an arrangement (valve) which allows the escape of gas if the internal pressure exceeds a predetermined value.

4) The ventilation arrangements for installation of vented type batteries which have charging power higher than 2kW are to be such that the quantity of air expelled is at least equal to:

$Q = 110\ I\ n$

where,

n = number of cells in series

I = maximum current delivered by the charging equipment during gas formation, but not less than 25 percent of the maximum obtainable charging current in amperes.

Q = quantity of air expelled in litres/hr.

The ventilation rate for compartments containing valve-regulated batteries may be reduced to 25 per cent of that given above.

7.2 Luminaries - construction and testing

7.2.1 General

7.2.1.1 Lighting which is essential for safety and working is to comply with the following provisions.

7.2.1.2 Lighting fittings installed in engine rooms or similar spaces where they are exposed to the risk of mechanical damage are to be provided with suitable grilled mechanical guards to protect their lamps and glass globes against such damage.

7.2.1.3 Precautions are to be taken so that a lamp for one voltage cannot be inserted in a lampholder for another voltage.

7.2.2 Incandescent lighting

7.2.2.1 The voltage of tungsten filament lampholders is not to exceed:

a) Bayonet fitting

Normal	B22 250 V
Small (single contact)	B15 s 130 V
Small (double contact)	B15 d 130 V

b) Screw fitting

Goliath	E40 250 V
Medium	E27 250 V
Small	E14 250 V
Miniature	E10 24 V

7.2.2.2 Lamps are to be in accordance with the following :

B22	upto 200 W
E27	upto 200 W
E40	upto 3000 W

7.2.2.3 Lampholders are to be constructed of flame-retarding and non- hygroscopic material. All metal parts are to be of robust construction. Goliath lampholders are to be provided with means for locking the lamp in the holder. The temperature of cable connections is not to exceed the maximum conductor temperature permitted for the cable as given in Table 3.3.1.

7.2.3 Fluorescent lighting

7.2.3.1 The ratings of tubular fluorescent lamps are not to exceed 250 V and 80 W.

7.2.3.2 Fittings, reactors, capacitors and other auxiliaries are not to be mounted on surfaces which are subject to high temperatures.

7.2.3.3 Capacitors of 0.5mF and above are to be provided with a means of prompt discharge on disconnection of the supply.

7.2.4 Discharge lighting

7.2.4.1 Where cold cathode luminous discharge lamps of normal operating voltage above 250 V are used, a warning notice calling attention to the voltage is to be displayed at points of access to the lamps and where otherwise necessary.

7.2.4.2 Inductance and high reactance transformers are to be installed as close as practicable to the associated discharge lamp.

7.2.4.3 Every capacitor of 0.5mF or more is to be provided with means for reducing the voltage of capacitor to less than 55V within one minute after disconnections from supply source.

7.2.4.4 All live parts of discharge lamp luminaries are to be so designed, placed and installed that they cannot be touched accidentally or inadvertently. The creepage distance along the surface of the glass tube is to be taken into considerations.

7.2.4.5 All parts of searchlights or arc lamps to be handled for their operations or adjustment while in use, are to be so arranged that there is no risk of shock to the operator. Disconnection of every searchlight or arc lamp is to be by a multi-pole disconnecting switches.

If a series resistor is used with an arc lamp, the disconnecting switch is to be so placed in the supply circuit that both the series resistor and arc lamp are disconnected when the switch is in the "off" position.

7.3 Accessories - Construction and testing

7.3.1 Enclosures

7.3.1.1 Enclosures are to be of metal or of flame-retardant insulating materials.

7.3.2 Inspection and draw boxes

7.3.2.1 If metal conduit systems are used, inspection and draw boxes are to be of metal and are to be in rigid electrical and mechanical connection with the conduits.

7.3.3 Socket outlets and plugs

7.3.3.1 The temperature rise on the live parts of socket outlet and plugs is not to exceed 30°C.

7.3.3.2 Socket outlets and plugs are to be so constructed that they cannot be readily short-circuited whether the plug is in or out, and so that a pin of the plug cannot be made to earth either pole of the socket outlet.

7.3.3.3 All socket outlets of current rating 16 A or more are to be provided with a switch so interlocked that the plug cannot be inserted or withdrawn with the switch in the "ON" position.

7.3.3.4 Where it is necessary to earth the non-current-carrying parts of portable or transportable equipment, an effective means of earthing is to be provided at the socket outlet.

7.3.3.5 Where distribution systems with different voltages are in use, the socket outlets and plugs are to be of such design that a plug for one voltage cannot be inserted into the socket outlet for a different voltage.

7.3.3.6 On weather decks, galleys, laundries, machinery spaces and all wet situations socket outlets and plugs are to be effectively shielded against rain and spray and are to be provided with means for maintaining this quality after removal of the plug.

7.4 Heating and cooking equipment

7.4.1 Heaters are to be so constructed, installed and protected that clothing, bedding and other inflammable material cannot come in contact with them in such a manner as to cause risk of fire. There is to be no excessive heating of adjacent bulkheads or decks.

7.4.2 Parts of heaters or cooking appliances which must necessarily be handled should not become heated to a temperature exceeding 55°C for metallic parts and 65°C for nonmetallic parts; for other parts the temperatures obtained in service are to be such that the various materials are not subjected in the course of normal working to any excessive thermal stress.

7.4.3 Each separate element rated more than 15 A is considered as a separate consumer, for which a separate circuit from a switchboard or distribution board is required.

7.5 Lightning conductors

7.5.1 General

7.5.1.1 Lightning conductors are to be fitted to each mast of all wood, composite, and steel ships having masts or topmasts of wood or other non-conductive material. They need not be fitted to steel ships having steel masts, unless the mast is partly or completely insulated from the ship's hull.

7.5.2 Construction

7.5.2.1 In wooden and composite ships fitted with wooden masts, the lightning conductors are to be composed of continuous copper tape and/or rope, having a Section not under 100 [mm^2]. These are to be riveted with copper rivets or fastened with copper clamps to an appropriate copper spike not less than 13 [mm] in diameter and projecting at least 150 [mm] above the top of the mast. If tape is used the lower end of the tape is to terminate at the point at which the shrouds leave the mast, and is to be securely clamped to a copper rope of not less than 13 [mm] diameter. This copper rope is to be led down the shrouds and is to be securely clamped to a copper plate having an area of at least 0.2 [m^2]. This copper plate is to be fixed to the ship's hull well below the light load waterline in such a manner that it is to be immersed under all conditions of heel.

7.5.2.2 In wooden and composite ships fitted with steel masts, each mast is to be connected to a copper plate in accordance with 7.5.2.1. The copper rope is to be securely attached to and in good electrical contact with the mast at or above the point at which the shrouds leave the mast.

7.5.2.3 In steel ships fitted with wooden masts, the lightning conductors are to be composed of copper tape or rope terminating in a spike in accordance with 7.5.2.1. At the lower end this copper tape or rope is to be securely clamped to the nearest metal forming part of hull of the ship.

7.5.2.4 Lightning conductors are to be run as straight as possible, and sharp bends in the conductors are to be avoided. All clamps used are to be of brass or copper, preferably of the serrated contact type, and efficiently locked. Soldered connections are not acceptable.

7.5.2.5 The resistance of the lightning conductors, measured between the mast head and the position on the earth plate or hull to which the lightning conductor is earthed, is not to exceed 0.02 ohms.

7.5.2.6 Suitable means should be provided to enable ships when in drydock or on a slipway to have their lightning conductors or steel hulls connected to an efficient earth on shore. When the ships are in floating docks, suitable means should be provided for earthing these lightning conductors to the sea.

Section 8

Electric Propulsion Systems

8.1 General

8.1.1 All electric propelling machinery including switchgear, control gear, cables, main and auxiliary generators, motors, exciters and electro-magnetic coupling, is to comply with the relevant Chapter/Sections of the Rules and is to be constructed under Special Survey.

8.1.2 Prime movers are to comply with the relevant Section(s) of Ch.4.

8.1.3 Armature shafts and other important steel forgings and castings are to comply with the requirements of Pt.2.

8.1.4 The torsional vibration characteristics of the propulsion system are to be submitted as required by Ch.4, as applicable.

8.1.5 Cooling water and lubricating oil systems are to comply with Ch.3, where applicable.

8.1.6 Where the arrangements permit a propulsion motor to be connected to a generating plant having a continuous rating greater than the motor rating, means are to be provided to limit the continuous full load torque for which the motor and shafts are approved.

8.1.7 The torque available for maneuvering is to be greater than the trailing torque of the propeller and of sufficient magnitude to be capable of stopping the propeller and reversing its direction of rotation (within a reasonable time) when the vessel is underway at the maximum service speed. In alternating current installations the power torque is to be sufficiently high to ensure that there is no risk of the motor pulling out of step in bad weather or when turning in the case of vessels having several lines of shafting.

8.1.8 Motors and generators of 400 [kW] or over are to be provided with means of heating the windings to prevent condensation when idle. If steam pipes are used for this purpose the joints are not to be within the machine.

8.2 System Design

8.2.1 The power supply for the propulsion systems may be provided from dedicated generator/s or from the vessel's main power distribution centre.

8.2.2 The prime movers for generators supplying dedicated power to electrical propulsion motors are to be controlled and monitored as follows:

- The control, alarm and monitoring systems are to be as per requirements specified for main engines in Ch. 7 or Pt.5, Ch.22 as relevant and applicable.
- Safety (shutdowns) and testing is to be as per relevant and applicable provisions for auxiliary engines in Ch. 7 or Pt. 5, Ch. 22.

8.2.3 Prime movers which supply power to both propulsion motor and other consumers on board are to be controlled and monitored as follows:

- The control, alarm and monitoring systems, safety and testing are to be as per requirements specified for auxiliary engines in Ch. 7 or Pt.5, Ch.22 as relevant and applicable.

8.2.4 A power management system is to be provided. Power management system failure is to be alarmed at normally manned control location. The power management system is to prevent overloading of generator and ensure a safe distribution of power to ship services and propulsion by tripping of non-essential services and /or reduction in propulsion power.

8.2.5 Single failure in propulsion systems (i.e. electrical propulsion motor and drives) is not to result in total loss of vessel propulsion.

8.3 Excitation

8.3.1 Systems dependent on the auxiliary generators for excitation are to be capable of maneuvering and of maintaining power at all times with a fall of 10 per cent excitation voltage at the busbars.

8.3.2 Where motor driven exciters, boosters, balancers or rectifiers are provided for excitation purposes, provision for an alternative

supply of excitation is to be made. Where two machines are used, each of at least 50 per cent of the required power, it will be sufficient to provide one spare machine.

8.3.3 Negative boosters are to be provided with overspeed protection where necessary.

8.3.4 In direct current constant pressure systems, arrangements for generator and motor excitation are to be such that if the motor excitation circuit is opened by a switch or contactor, the generator excitation is simultaneously interrupted, or the generator voltage is immediately reduced to zero.

8.4 Controls

8.4.1 In addition to the requirements of Ch.7, the following provisions are to be complied with.

8.4.2 Where bridge or deck control is employed, alternative control in the engine room is to be provided.

8.4.3 Suitable interlocks, operating preferably by mechanical means, are to be provided to prevent damage to the plant as a result of incorrect switching, such as the opening of switches or contactors not intended to be operated while carrying current or such as the simultaneous closing of the ahead and astern circuits.

8.4.4 Provision is to be made for the manual operation, without undue manual effort, of all maneuvering contactors, switches, field regulators and controllers. Where electric, pneumatic or hydraulic aid is used for normal operation, failure of such aid is not to result in interruption of power to the propeller shaft and any such device is to be capable of purely manual operation without delay. This latter requirement does not apply to bridge control equipment.

8.4.5 In propulsion installations with two or more generators or two or more motors on one line of shafting the propulsion circuit is to be so arranged that any of these machines can be cut out of the circuit without preventing the others from working.

8.4.6 Where steam and oil gauges are mounted on the main control station, provisions are to be made so that in case of leakage, steam or oil may not come into contact with the energized parts.

8.4.7 Alternative arrangements will be specially considered.

8.4.8 Emergency stop independent of normal control system is to be provided at each control station.

8.4.9 Means to identify the cause of propulsion motor power limitation or automatic reduction are to be provided at manned control location.

8.5 Cables

8.5.1 Conductors in circuits essential for maneuvering or maintenance of propelling power are to be stranded, having not less than seven strands, and are to have a nominal cross-sectional area of not less than 2.5 [mm^2].

8.5.2 Cables which are connected to the slip rings of synchronous motors are to be suitably insulated for the voltage to which they are subjected during maneuvering.

8.5.3 Cable ends are to be fitted with connectors or connecting sockets of appropriate size and in such a manner as to inhibit corrosion. They are to be arranged and supported in a manner suitable for withstanding the electro-mechanical forces due to a short circuit.

8.6 Overload and short circuit protection

8.6.1 Provision is to be made for protection against severe overloads, and electrical faults likely to result in damage to the plant.

8.6.2 The overload protection in excitation circuits should not lead to opening of the circuit.

8.7 Earth leakage detection

8.7.1 The main propulsion circuit is to be provided with means for detecting earth faults. For direct current equipments exceeding 500 V and for all alternating current equipments, aural and visual alarms are to be automatically operated on the occurrence of an earth fault, but the operation of such devices is not to interrupt the power supply. A switch may be provided to switch off the aural devices, but in such cases the visual alarm is to remain switched on to indicate that the aural device is switched off. Alternative arrangements will be specially considered.

8.7.2 If an earth connection is used for operating the detector arrangements, then in direct current systems the earth circuit is to be automatically opened in order to stop the circulation of fault current. In alternating current systems, the fault current is to be interrupted or limited to a safe value.

8.7.3 Earth leakage devices are to be arranged to function for all earth faults exceeding 5 A. In three-phase star connected alternating current generators and motors with neutral points earthed, the earth leakage device is to operate on the occurrence of an earth fault in the windings of the machine, provided that 5 per cent of the coils at the neutral end of each phase may be left unprotected by the device. In high voltage alternating current systems, where the capacitive leakage current is high, consideration will be given to increasing this figure of five per cent.

8.7.4 Excitation circuits are to be provided with lamps, voltmeters or other means to indicate continuously the state of the insulation of the excitation circuits under running conditions.

8.8 Discharge protection

8.8.1 For the protection of field windings and cables, means are to be provided for limiting the induced voltage when the field circuits are opened or, alternatively, the induced voltage, when the field circuits are opened, is to be taken as the nominal design voltage.

8.8.2 Where excitation is obtained from the auxiliary busbars, means are to be provided to limit the voltage induced at the busbars when the auxiliary circuit-breaker or the distribution circuit-breaker opens.

8.8.3 Shunt resistors which are connected across the field circuit of synchronous propulsion motors, when they are functioning as asynchronous motors, are to be suitably insulated for the voltage induced when reversing, and are to be amply rated to allow for inadvertent delay during the reversing operation.

8.9 Safety devices

8.9.1 Where separately driven direct current generators are connected electrically in series, means are to be provided to prevent reversal of the direction of rotation of any of them on the failure of the prime movers.

8.9.2 Where, on stopping or reversing the propeller, the regenerated energy transmitted by the propulsion motor is such as to cause a dangerous increase of speed in the prime mover, means are to be provided for suitably absorbing or limiting such energy.

8.9.3 Contactors and switches used for reversing the rotation of the propulsion motors are to be provided with means for forcibly opening them if they should in advertently remain closed.

8.10 Electro-magnetic couplings

8.10.1 Propulsion arrangements incorporating electro-magnetic couplings will receive special consideration.

8.11 Alarms

8.11.1 Where machines have enclosed ventilation systems, an aural alarm device is to be provided and arranged to operate if the temperature of the heated air exceeds the predetermined safe value.

8.12 Identification

8.12.1 All important circuits, instruments and apparatus are to be clearly labeled for identification.

Section 9

Crew and Passenger Emergency Safety Systems

9.1 Emergency lighting

9.1.1 For the purpose of this section emergency lighting, transitional emergency lighting and supplementary emergency lighting are hereafter referred to under the generic name 'emergency lighting'.

9.1.2 Emergency lighting is to be arranged so that a fire or other casualty in the spaces containing the emergency source of electrical power, associated transforming equipment and the emergency lighting switchboard does not render the main lighting system inoperative.

9.1.3 The level of illumination provided by the emergency lighting is to be adequate to permit safe evacuation in an emergency, having regard to the possible presence of smoke.

9.1.4 The exit(s) from every main compartment occupied by passengers or crew is to be continuously illuminated by an emergency lighting fitting.

9.1.5 Switches are not to be installed in the final sub-circuits to emergency light fittings unless the light fittings are serving normally unmanned spaces, i.e. storage-rooms, cold rooms, etc or where they are normally required to be extinguished for operational reasons (e.g. for night visibility from the navigating bridge). Where switches are fitted they are to be accessible only to ships crew with provision made to ensure that the emergency lighting is energised when such spaces are manned and/or during emergency conditions.

9.1.6 Where emergency lighting fittings are connected to dimmers, provision is to be made, upon the loss of the main lighting, to automatically restore them to their normal level of illumination.

9.1.7 Fittings are to be specially marked to indicate that they form part of the emergency lighting system.

9.2 General emergency alarm system

9.2.1 An electrically operated bell or klaxon or other equivalent warning system installed in addition to the ship's whistle or siren, for sounding the general emergency alarm signal is to comply with the requirement of this Section.

9.2.2 The system is to be capable of operation from at least the navigating bridge and except for the ship's whistle, also from other strategic points.

9.2.3 The alarm system is to be fed by exclusive circuits, one from the main source of electrical power and one from an emergency source of electrical power with automatic change-over facilities located in, or adjacent to the main alarm signal distribution panel.

9.2.4 The alarm system is to be audible throughout all the accommodation and normal crew working spaces with all doors and accesses closed and is to have a sound pressure level, in the 1/3-octave band above the fundamental, of not less than 75dB(A) and at least 10 dB(A) above normal ambient noise levels, with the ship underway in moderate weather, when measured at the sleeping positions in the cabins and one metre from the source. An audible alarm level of 120 dB(A) is not to be exceeded in any space.

9.2.5 With the exception of bells, the alarm is to have a signal frequency between 200 Hz and 2.5 kHz.

9.2.6 Where the special alarm fitted to summon the crew from the navigation bridge, of fire control station, forms part of the ship's general alarm system, it is to be capable of being sounded independently of the alarm to the passenger spaces.

9.3 Public address system

9.3.1 Public address systems installed in accordance with Pt.6, Ch.4 are to comply with requirements of this Section.

9.3.2 Where the public address system forms part of the internal communication equipment required in an emergency it is to be fed by exclusive circuits, one from the main source of electrical power and one from an emergency source of electrical power with automatic change-over facilities located adjacent to the public address system.

9.3.3 Amplifiers are to be continuously rated for the maximum power that they are required to deliver into the system for audio and, where alarms are to be sounded through the public address system, for tone signals.

9.3.4 Loudspeakers are to be continuously rated for their proportionate share of amplifier output.

9.3.5 Amplifiers and loudspeakers are to be selected and arranged to prevent feedback and other interference.

9.3.6 Where the public address system does not form part of the internal communication equipment required in an emergency, provision is to be made, at a position adjacent to the emergency system control panel, to silence the public address system.

9.3.7 The public address system may be used for sounding the general emergency alarm and the fire alarm provided that in addition to the requirements of 9.2.

a) the emergency system is given automatic priority over any other system input;

b) there are means to automatically override any volume controls so as to ensure the specified sound pressure levels are met;

c) there are multiple amplifiers having their power supplies so arranged that a single fault will not cause the loss of more than one amplifier;

d) there are segregated cable routes to public rooms, alleyways, stairways, and control stations so arranged that any single electrical fault or a fire in any one main vertical fire zone as defined by Pt.6, Ch.3, other than the zone in which the public address control station is located, will not interfere with the sounding of the emergency alarm through the remaining system;

e) more than one device is provided for generating the sound signals for the emergency alarms;

f) short circuit protection is fitted for individual loudspeakers.

9.3.8 Where more than one alarm is to be sounded through the public address system they are to have recognizably different characteristics and additionally be arranged so that any single electrical failure which prevents the sounding of any one alarm will not affect the sounding of the remaining alarms.

Section 10

Ship Safety Systems

10.1 Watertight doors

10.1.1 The electrical power required for power-operated sliding watertight doors is to be separate from any other power circuit and supplied from the emergency switchboard either directly or by a dedicated distribution board situated above the bulkhead deck. The associated control, indication and alarm circuits are to be supplied from the emergency switchboard either directly or by a dedicated distribution board situated above the bulkhead deck and for passenger ships be capable of being automatically supplied by the transitional source of emergency electrical power in the event of failure of either the main or emergency source of electrical power.

10.1.2 For passenger ships, where the source for opening and closing the watertight doors have electric motors, unless an independent temporary source of stored energy is provided, the electric motors are to be capable of being automatically supplied from the transitional source of emergency electrical power.

10.1.3 A single electrical failure in the power operating or control system of power-operated sliding watertight doors is not to result in a closed door opening or prevent the hand operation of any door.

10.1.4 Availability of the power supply is to be continuously monitored at a point in the electrical circuit adjacent to the door operating equipment. Loss of any such power supply is to activate an audible and visual alarm at the central operating console at the navigating bridge.

10.1.5 Electrical power, control, indication and alarm circuits are to be protected against fault

in such a way that a failure in one door circuit will not cause a failure in any other door circuit. Short circuits or other faults in the alarm or indicator circuits of a door are not to result in a loss of power operation of the door. Arrangements are to be such that leakage of water into the electrical equipment located below the bulkhead deck will not cause the door to open.

10.1.6 The enclosures of electrical components necessarily situated below the bulkhead deck are to provide suitable protection against the ingress of water with ratings as defined in IEC Publication 529 or an acceptable and relevant national standard, as follows :

a) Electrical motors, associated circuits and control components, protected to IPX7 standard.

b) Door position indicators and associated circuit components protected to IPX8 standard, where the water pressure testing of the enclosures is to be based on the pressure that may occur at the location of the component during flooding for a period of 36 hours.

c) Door movement warning signals, protected to IPX6 standard.

10.1.7 Watertight door electrical controls including their electric cables are to be kept as close as is practicable to the bulkhead in which the doors are fitted and so arranged that the likelihood of them being involved in any damage which the ship may sustain is minimized.

10.1.8 An audible alarm, distinct from any other alarm in the area, is to sound whenever the door is closed remotely by power and sound for at least five seconds but no more than ten seconds before the door begins to move and is to continue sounding until the door is completely closed. The audible alarm is to be supplemented by an intermittent visual signal at the door in passenger areas and areas where the noise level exceeds 85 dB(A).

10.1.9 A central operating console is to be fitted on the navigating bridge and is to be provided with a 'master-mode' switch having :

a) a 'local control' mode for normal use which is to allow any door to be locally opened and locally closed after use without automatic closure, and;

b) a 'doors closed' mode for emergency use which is to allow any door that is opened to be automatically closed whilst still permitting any doors to be locally opened but with automatic reclosure upon release of the local control mechanism.

10.1.10 The 'master mode' switch is to be arranged to be normally in the 'local control' mode position; be clearly marked as to its emergency function.

10.1.11 The central operating console at the navigating bridge is to be provided with a diagram showing the location of each door, with visual indicators to show whether each door is open or closed. A red light is to indicate a door is fully open and a green light, a door fully closed. When the door is closed remotely a red light is to indicate the intermediate position by flashing. The indicating circuit is to be independent of the control circuit for each door.

10.1.12 The arrangements are to be such that it is not possible to remotely open any door from the central operating console.

10.2 Shell doors, loading doors and other closing appliances

10.2.1 Where it is required that indicators be provided for shell doors, loading doors and other closing appliances, which are intended to ensure the watertight integrity of the ships structure in which they are located, the indicator system is to be designed on the fail-safe principle. The system is to indicate if any of the doors or closing appliances are open or are not fully closed or secured.

10.2.2 Where such doors and appliances are to be operated at sea, the requirements of 10.1 are to be complied with as far as is practicable.

10.2.3 The electrical power supply for the indicator system is to be independent of any electrical power supply for operating and securing the doors.

10.2.4 For the requirements of side shell doors, stern doors and bow doors, also refer to Pt.3, Ch.12, Sections 5 and 6.

Section 11

Electrical Equipment for use in Explosive Gas/Combustible Dust Atmospheres

11.1 General

11.1.1 The installation of electrical equipment in areas containing flammable gas or vapour and/or combustible dust, is to be minimized as far as is consistent with operational necessity and the provision of lighting, monitoring, alarm or control facilities enhancing the overall safety of the ship.

11.1.2 Combustible dust is defined as finely divided solid particles, 500 μm or less in nominal size, which may be suspended in air, may settle out of the atmosphere under their own weight, may burn or glow in air and may form explosive mixtures with air at atmospheric pressure and normal temperatures. (See IEC 60079-0).

Note : Examples of cargoes that can generate combustible dust with explosive hazard as per the IMSBC Code are:

Brown coal briquettes, sulphur UN 1350, iron oxide-spent or iron sponge-spent UN1376, wood pellets, peat moss, pitch prill etc.

11.2 Selection of equipment

11.2.1 When apparatus is to be installed in areas where an explosive gas atmosphere may be present, unless permitted otherwise by 11.2.2, it is to be of a 'safe type', as listed below, certified or approved by a competent authority for the gases encountered. The construction and type testing is to be in accordance with IEC Publication 79: Electrical Equipment for Explosive Gas Atmospheres or an acceptable and relevant national standard.

Intrinsically safe	Ex 'i'
Increased safety	Ex 'e'
Flameproof	Ex 'd'
Pressurized enclosure	Ex 'p'
Power filled	Ex 'q'
Encapsulated	Ex 'm'.

11.2.2 Consideration may be given to the use of equipment of the following types:

a) equipment such as control panels, protected by purging and pressurisation and capable of being verified by inspection as meeting the requirements of IEC 60079-2;

b) simple non-energy-storing apparatus having negligible surface temperature rise in normal operation, such as limit switches, strain gauges, etc., incorporated in intrinsically-safe circuits;

c) radio aerials having robust construction, meeting the relevant requirements of IEC 60079-15;

d) electrical apparatus with type of protection 'n' or 'N' provided it is in a well ventilated area on open deck and not within 3 [m] of any flammable gas or vapour outlet.

11.2.3 Where apparatus is to be installed in areas where combustible dusts may be present in quantities sufficient to create an explosive atmosphere, it is, when practicable, to be of a type certified or approved by a competent authority for the dusts and additionally any explosive gases encountered.

11.2.4 Electrical equipment for use in combustible dust atmospheres is to be so designed and installed as to minimize the accumulation of dust which may interfere with the safe dissipation of heat from the enclosure.

11.2.5 Where equipment certified for combustible dust, is not available, consideration will be given to the use of apparatus complying as a minimum, with the following requirements provided no explosive gases will be present:

a) the enclosure is to be at least dust protected (IP5X) having, when type tested, an ingress of fine dust within the enclosure not exceeding 10 g per [m^3] of free air space, and

b) the surface temperature of the apparatus, under the most onerous combination of normal operating conditions, but in the absence of a dust layer, is to be at least 10°C below the auto-ignition temperature of the dusts encountered, or

c) the equipment is to be certified intrinsically-safe having a temperature classification ensuring compliance with (b), or

d) pressurized and operated in accordance with procedures ensuring, prior to its reenergisation, the absence of dust within the enclosure following loss of pressurisation and consequent shutdown, and having surface temperature complying with (b), or

e) simple apparatus included in intrinsically-safe circuits or radio aerials, complying with 11.2.2 (b) or (c) respectively.

11.3 Installation of electrical equipment and cabling

11.3.1 The method of installation and application of safe type equipment is to be in accordance with IEC 79-14, or the national code of practice relevant to the standard to which the equipment has been certified. Any special requirements laid down by the equipment certification documentation are also to be observed. The ambient temperature range for which the apparatus is certified, is to be taken to be -20°C to 40°C, unless otherwise stated, and account is to be taken of this when assessing the suitability of the equipment for the auto-ignition temperature of the gases and dusts encountered.

11.3.2 All switches and protective devices from which equipment located in dangerous zones or spaces is supplied are to interrupt all poles or phases and, where practicable are to be located in non-hazardous zone or space. Such equipment, switches and protective devices are to be suitably labeled for identification purposes.

11.3.3 Metal coverings of cables installed in dangerous zones or spaces are to be effectively earthed at both ends.

11.3.4 Cables associated with intrinsically-safe circuits are to be used only for such circuits. They are to be physically separated from cables associated with non-intrinsically-safe circuits, e.g. neither installed in the same protective casing nor secured by the same fixing clip.

11.3.5 Where there is risk of intermittent contact between armour and exposed metalwork, non-metallic impervious sheath is to be applied over metallic armour of cable.

11.4 Dangerous zones and spaces

11.4.1 Dangerous zones or spaces and sources of hazard for ships intended for the carriage in bulk of oil, liquefied gases and other hazardous liquids, and the requirements for ships carrying vehicles with fuel in their tanks, are defined in 11.9 to 11.12. The following principles are to apply in general, and where any specific arrangement does not fall into any of the categories covered by 11.9 to 11.12.

11.4.2 A dangerous zone or space may arise from the presence of any of the following:

a) spaces or tanks containing either

   i) flammable liquid having a flashpoint (closed-cup test), not exceeding 60°C;

   ii) flammable liquid having a flashpoint exceeding 60°C, heated or raised by ambient conditions to a temperature within 15°C of its flashpoint;

   iii) flammable gas.

b) piping systems or equipment containing fluid defined by (a) and having flanged joints or glands or other openings through which leakage of fluid may occur under normal operating conditions;

c) spaces containing solids, such as coal or grain, liable to release flammable gas and/or combustible dust;

d) piping systems or equipment associated with processes (such as battery charging or electrochlorination) generating flammable gas as a by-product and having openings from which the gas may escape under normal operating conditions;

e) piping system or equivalent containing flammable liquids not defined by (a), having flanged joints, glands or other openings through which leakage of fluid in the form of a mist or fine spray may occur under normal operating conditions.

11.4.3 The following zones or spaces are regarded as dangerous:

a) the interiors of those spaces or tanks defined by 11.4.2(a) and (c);

b) spaces separated by a single bulkhead or deck from a cargo defined by 11.4.2(a);

c) enclosed or semi-enclosed spaces containing pipework or equipment defined by 11.4.2(b) and (d);

d) enclosed or semi-enclosed spaces with direct opening into a dangerous space;

e) zones within a 3 [m] radius of ventilation inlets or outlets, hatches or doorways or other openings into dangerous spaces, and within 3 [m] of the ventilation outlets of spaces regarded by 11.5 as open areas and which contain the pipework or equipment defined by 11.4.2(b); space where the hazard results from flammable gas or vapour having a density relative to that of air of more than 0.5, the dangerous zone is considered to extend vertically downward to solid deck, or for a distance of 9 [m], whichever is the lesser;

f) zones within a 3 [m] radius of flanged joints, or glands or other openings defined by 11.4.2(b); in the case of gas or vapour having a relative density of more than 0.5, the dangerous zone is considered to extend vertically downwards as described under (e);

g) zones within a 1.5 [m] radius of the ventilation outlets of spaces regarded as open areas containing items defined under 11.4.2(d);

h) zones within a 1.5 [m] radius of flanged joints, or glands or other openings defined by 11.4.2(d) and (e);

i) zones within a 3 [m] radius of bunds or barriers intended to contain spillage of liquids defined by 11.4.2(a).

11.5 Semi-enclosed spaces

11.5.1 Semi-enclosed spaces are considered to be spaces limited by decks and/or bulkheads in such a manner that the natural conditions of ventilation are sensibly different from those obtained on open deck.

11.6 Ventilation

11.6.1 Where an enclosed or semi-enclosed space is provided with mechanical ventilation ensuring at least 12 air changes/hour, and leaving no areas of stagnant air, it may be regarded in consideration of dangerous zones, as an open area.

11.6.2 Where the rate of ventilation air flow, in relation to the maximum rate of release of flammable substances reasonably to be expected under normal conditions, is sufficient to prevent the concentration of flammable substances approaching their lower explosive limit, consideration may be given to regarding as non-dangerous, the space, ventilation and other openings into it, and the zone around the equipment contained within.

11.7 Pressurisation

11.7.1 A space having access to a dangerous space or zone as defined under 11.4.3(c) to (i) may be regarded as non-dangerous if fulfilling all the following conditions:

a) access is by means of an air-lock, having gas-tight steel doors, the inner of which as a minimum, is self-closing without any hold-back arrangement;

b) it is maintained at an overpressure relative to the external hazardous area by ventilation from a non-dangerous area;

c) the relative air pressure within the space is continuously monitored and, so arranged, that in the event of loss of overpressure an alarm is given and the electrical supply to all equipment not of a safe type is automatically disconnected. Where the shut-down of equipment could introduce a hazard, an alarm may be given, in lieu of shutdown, upon loss of overpressure, and a means of disconnection of non-safe type electrical equipment, capable of being controlled from a manned station, provided in conjunction with an agreed operational procedure; where the means of disconnection is located within the space then it is to be effected by equipment of a safe type;

d) any electrical equipment required to operate upon loss of overpressure, lighting fittings and equipment within the air-lock, is to be of a safe type;

e) means are to be provided to prevent electrical equipment other than of a safe type, being energized until the atmosphere within the space is made safe, by air renewal of at least 10 times the capacity of the space.

11.8 Cable and cable installation

11.8.1 Electric cables are not to be installed in dangerous zones or spaces, except where specifically permitted by 11.9. to 11.11 or when associated with intrinsically-safe circuits.

11.8.2 In addition to other requirements in this chapter, cables in dangerous zones or spaces, or which may be exposed to cargo oil, oil vapour or gas, are to be either:

a) mineral insulated with copper sheath, or

b) armoured or braided (for mechanical protection and earth detection) with non-metallic impervious sheath.

11.9 Requirements for tankers intended for the carriage in bulk of oil cargoes having a flash point not exceeding 60°C (closed cup test)

11.9.1 The following requirements define the electrical equipment permitted within dangerous spaces and zones and are in addition to the requirements of 11.1 to 11.8.

11.9.2 The requirements for cargo tanks also apply to cargo slops tanks.

11.9.3 The relevant gas group and temperature class for safe type equipment in the defined locations are IIA T3.

11.9.4 Where intrinsically-safe equipment is required, consideration will be given to the use of simple apparatus incorporated in intrinsically-safe circuits, as defined in 11.2.2(b).

11.9.5 Cargo tanks: intrinsically-safe equipment of category 'ia':

11.9.6 Cofferdams adjoining cargo tanks

a) intrinsically-safe equipment of category 'ia';

b) electric depth-sounding devices hermetically enclosed, located clear of the cargo tank bulkhead, with cables installed in heavy gauge steel pipes with gastight joints up to the main deck;

c) cables for impressed current cathodic protection systems (for external hull protection only) installed in heavy gauge steel pipes with gas tight joints up to the upper deck;

d) through runs of cables, installed in heavy gauge steel pipes with gas tight joints.

11.9.7 Cargo pump rooms:

a) intrinsically-safe equipment;

b) electrical equipment as defined in 11.9.6(b) and (c);

c) flameproof lighting fittings (symbol d);

d) pressurised lighting fittings (symbol p) of either the air driven type, of pressurised from an external source of protective gas and arranged to be de-energised automatically on loss of pressurization;

e) gas detector heads having sinter-type flametrap protection, included within an intrinsically-safe circuit, the gas detector system is to be certified;

f) general alarm sounders of flameproof type, without internal sparking contacts;

g) through runs of cables, confined to pump room entrances only, installed in heavy gauge steel pipes with gas tight joints.

11.9.8 Spaces under cargo tanks (e.g. duct keels): Electrical equipment as defined in 11.9.6(a) and (b) and 11.9.7(c) to (f).

11.9.9 Enclosed or semi-enclosed spaces immediately above cargo tanks or having bulkheads immediately above and in line with cargo tank bulkheads, compartments for cargo hoses, spaces other than cofferdams adjoining and below the top of a cargo tanks, e.g. trunk, passageways and holds:

a) intrinsically-safe equipment, this is to be of category 'ia' where the spaces or compartments are not mechanically ventilated;

b) safe type lighting fittings;

c) through runs of cable;

d) general alarm sounders as defined by 11.9.7(f).

11.9.10 Electrical equipment or cables are not normally to be installed in hazardous areas. Where essential for operational purposes, electrical equipment may be installed in accordance with IEC 60092-502:1999 as per the following area classification:

11.9.10.1 Areas on open deck adjoining cargo openings (refer Part 5, Chapter 2, Clause 8.1.4.a) permitting flow of small volumes of vapour or gas mixtures:

a) Zone 1- Areas on open deck, or semi-enclosed spaces on open deck within 3 [m] of any cargo oil tank outlet or vapour outlet (e.g. cargo tank or cofferdam hatch; sight port; tank cleaning opening; ullage opening; sounding pipe cargo pump room entrance and ventilation intakes and exhausts) which permit the flow of small volumes of vapour or gas mixtures caused by thermal variation as given in IEC 60092-502:1999 para 4.2.2.7.

b) Zone 2 - Areas within 2 [m] beyond the zone 1 above.(note: 1.5 [m] given in IEC 60092-502:1999 para 4.2.3.1).

11.9.10.2 Areas on open deck adjoining cargo openings (refer Part 5, Chapter 2, Clause 8.1.4.b) permitting flow of large volumes of vapour or gas mixtures:

Zone 1- Areas on open deck, or semi-enclosed spaces on open deck, within a vertical cylinder of unlimited height and 6m radius centered upon the centre of the outlet, and within a hemisphere of 6m radius below the outlet which permit the flow of large volumes of vapour or gas mixtures during loading/discharging/ballasting as given in IEC 60092-502 para 4.2.2.8. (e.g. pressure/vacuum valve, high velocity vents, mast riser).

a) Zone 2- Areas within 4m beyond the zone 1 above as given in IEC 60092-502 para 4.2.3.2.

11.9.10.3 Areas on open deck over all cargo tanks (including all ballast tanks within the cargo tank area) to the full width of the vessel, plus 3 [m] forward and 3 [m] aft of the cargo tank area or any spillage barrier installed aft of the cargo tanks area, up to a height of 2.4 [m] above the deck

a) Zone 1- Where structures are restricting the natural ventilation as given in IEC 60092-502 para 4.2.2.11.

b) Zone 2- Where unrestricted natural ventilation is guaranteed as given in IEC 60092-502 para 4.2.3.5.

11.9.11 Spaces below the level of, and having direct openings onto, the main deck, but outside the dangerous zone previously described:

a) safe type equipment;

b) through runs of cable.

11.9.12 Mechanically ventilated or pressurized spaces:

a) where a space of the type defined by 11.9.13 is provided with a self closing door for the opening onto the main deck and has mechanical ventilation, the air intake for which is remote from any dangerous space or zone, non-safe equipment is permitted within the space;

b) where a space opening into a dangerous into a dangerous zone or space is provided with an airlock, is separated from the cargo by at least two gastight bulkhead, and is pressurized in accordance with 11.7, non-safe type equipment is permitted within the space.

11.9.13 Electrical installations in enclosed or semi-enclosed spaces having a direct opening into any dangerous space or zone are to comply with the requirements for the space or zone to which the opening leads.

11.10 Requirements for ships for the carriage of liquefied gases in bulk

11.10.1 See Pt.5 for relevant Rules for Ships for Liquefied Gases.

11.11 Requirements for ships intended for the carriage in bulk of other flammable liquid cargoes

11.11.1 See Pt.5 for relevant Rules for Ships for Liquid Chemicals.

11.12 Special requirements for ships with spaces for carrying vehicles with fuel in their tanks, for their own propulsion

11.12.1 Passenger ships with special category spaces above the bulkhead deck for carrying vehicles:

a) electrical equipment fitted within a height of 45 [cm] above the vehicle deck, or any platform on which vehicles are carried, or within the exhaust ventilation trunking of the special category space, is to be of a safe type;

b) electrical equipment situated elsewhere within the space is to have an enclosure of ingress protection rating of at least IP55. (See IEC Publication 529: Classification of Degrees of Protection Provided by Enclosures), if not of a safe type.

11.12.2 Passenger ships with special category spaces below the bulkhead deck for carrying vehicles: electrical equipment fitted within the space and within the space's exhaust ventilation trunking, is to be of certified safe type.

11.12.3 Passenger ship with cargo spaces, other than special category spaces, for carrying vehicles:

a) electrical equipment within such a cargo space, or within the space's exhaust ventilation trunking, is to be of a safe type;

b) all electrical circuits terminating in the cargo space are to be provided with multi-pole linked isolating switches located outside the cargo hold. Provision is to be made for locking in the off position. This does not apply to safety circuits such as those for fire, smoke or gas detection.

11.12.4 Cargo ships with spaces for carrying vehicles:

a) electrical equipment fitted within a height of 45 [cm] above the vehicle deck, or within a height of 45 [cm] above the vehicle deck, or within the cargo space's exhaust ventilation trunking, and any equipment associated with safety circuits such as those for fire, smoke or gas detection, located within the cargo space, is to be of a safe type;

b) the requirements of 11.12.1(b) apply to other equipment within the space;

c) the requirements of 11.12.3(b) apply to circuits terminating in the space.

11.13 Requirements of electrical equipment allowed in paint stores and in the enclosed spaces leading to paint stores

11.13.1 Electrical equipment is to be installed in paint stores and in ventilation ducts serving such spaces only when it is essential for operational services.

Certified safe type equipment of the following type is acceptable:

a) intrinsically safe Ex 'i'
b) flameproof Ex 'd'
c) pressurised Ex 'p'
d) increased safety Ex 'e'
e) special protection Ex 's'.

Cables (through-runs or terminating cables) of armoured type or installed in metallic conduits are to be used.

11.13.2 The minimum requirements for the certified safe type equipment are as follows:

- explosion group II B

- temperature class T3.

11.13.3 Switches, protective devices, motor control gear of electrical equipment installed in a paint store are to interrupt all poles or phases and preferably are to be located in non-hazardous space.

11.13.4 In the areas on open deck within 1 m of inlet and exhaust ventilation openings or within 3 m of exhaust mechanical ventilation outlets, the following electrical equipment may be installed:

- electrical equipment with the type of protection as permitted in paint stores;
- equipment of protection class Ex 'n';
- appliances which do not generate arcs in service and whose surface does not reach unacceptably high temperature;
- appliances with simplified pressurised enclosures or vapour-proof enclosures (minimum class of protection IP55) whose surface does not reach unacceptably high temperature; or
- cables as specified in 11.13.1.

11.13.5 The enclosed spaces giving access to the paint store may be considered as non-hazardous, provided that:

- the door to the paint store is a gastight door with self-closing devices without holding back arrangements,
- the paint store is provided with an acceptable, independent, natural ventilation system ventilated from a safe area, and
- warning notices are fitted adjacent to the paint store entrance stating that the store contains flammable liquids.

Note: The paint stores and inlet and exhaust ventilation ducts under 11.13.1 are classified as Zone-1 and areas on open deck under 11.13.4 as Zone 2, as defined in IEC 60092-502:1999.

A watertight door may be considered as being gastight.

Section 12

High Voltage Systems

12.1 Scope

12.1.1 The requirements of this Section apply to a.c. three-phase systems with nominal voltage above 1 kV and upto 15 kV.

12.1.2 The nominal voltage is the voltage between phases.

12.1.3 Where necessary for special application, higher voltages may be accepted by IRS.

12.1.4 Vessels complying with the '*Guidelines on High Voltage Shore Connection Systems for Ships'* may be assigned additional class notation **HVSCS**.

12.2 High voltage and low-voltage segregation

12.2.1 Equipment with voltage above 1 kV is not to be installed in the same enclosure as low voltage equipment, unless segregation or other suitable measures are taken to ensure that access to low voltage equipment is obtained without danger.

12.3 System design

12.3.1 It is to be possible to split the main switchboard into atleast two independent sections, by means of at least one circuit breaker or other suitable disconnecting devices, each supplied by at least one generator. If two separate switchboards are provided and interconnected with cables, a circuit breaker is to be provided at each end of the cable.

12.3.2 Services which are duplicated are to be divided between the sections.

12.3.3 In a neutral earthed system, in case of earth fault the current is not to be greater than full load current of the largest generator on the switchboard or relevant switchboard section and not less than three times the minimum current required to operate any device against earth fault. It is to be assured that at least one source neutral to ground connection is available whenever the system is in the energized mode.

12.3.4 Electrical equipment in directly earthed neutral or other neutral earthed systems is to withstand the current due to a single phase fault against earth for time necessary to trip the protection device.

12.3.5 Means of disconnection are to be fitted in the neutral earthing connection of each generator so that the generator may be disconnected for maintenance and for insulation resistance measurement.

12.3.6 All earthing impedances are to be connected to the hull. The connection to the hull is to be so arranged that any circulating currents in the earth connections do not interfere with radio, radar, communication and control equipment circuits.

12.3.7 In the divided systems with neutral earthed, connection of the neutral to the hull is to be provided for each section.

12.4 Degrees of protection

12.4.1 Each part of the electrical installation is to be provided with a degree of protection appropriate to the location, and as a minimum the requirements of IEC Publication 60092-201:2019 are to be followed.

12.4.2 The degree of protection of enclosures of rotating electrical machines is to be at least IP23. The degree of protection of terminals is to be at least IP44.

12.4.3 For motors installed in spaces accessible to unqualified personnel, a degree of protection against approaching or contact with live or moving parts of at least IP4X is required.

12.4.4 The degree of protection of enclosures of transformers is to be at least IP23.

12.4.5 For Transformers installed in spaces accessible to unqualified personnel a degree of protection of at least IP4X is required. For transformers not contained in enclosures, see clause 12.14.1.

12.4.6 The degree of protection of metal enclosed switchgear, controlgear assemblies and static converters is to be atleast IP32. For switchgear, control gear assemblies and static converters installed in spaces accessible to unqualified personnel, a degree of protection of at least IP4X is required.

12.5 Air clearances

12.5.1 In general, for Non Type Tested equipment phase-to-phase air clearances and phase-to-earth air clearances between non-insulated parts are to be as specified in Table 12.5.1.

Table 12.5.1

Nominal Voltage [kV]	Minimum air clearance [mm]
3 - 3.3	55
6 - 6.6	90
10 - 11	120
15	160

Notes
1 Intermediate values may be accepted for nominal voltages provided that the next higher air clearance is observed.

2 In the case of smaller distances, appropriate voltage impulse test must be applied.

12.6 Creepage distances

12.6.1 Creepage distance between live parts and between live parts and earthed metal parts are to be in accordance with IEC 60092-503:2007 for the nominal voltage of the system the nature of the insulation material and the transient overvoltage developed by switch and fault conditions.

12.7 Protection

12.7.1 **Faults on the generator side of circuit breaker** : Protective devices are to be provided against phase-to-phase faults in the cables connecting the generators to the main switchboard and against interwinding faults within the generators. The protective devices are to trip the generator circuit breaker and to automatically de-excite the generator. In distribution systems with a neutral earthed, phase to earth faults are also to be treated as above.

12.7.2 **Faults to earth** : Any earth fault in the system is to be indicated by means of a visual and audible alarm. In low impedance or direct earthed systems, provision is to be made to automatically disconnect the faulty circuits. In high impedance earthed system, where outgoing feeders will not be isolated in case of an earth fault, insulation of the equipment is to be designed for the phase to phase voltage.

Note: Earthing factor is defined as the ratio between the phase to earth voltage of the healthy phase and the phase to phase voltage. This factor may very between 1/√3 and 1.

A system is defined effectively earthed (low impedance) when this factor is lower than 0.8.

A system is defined non-effectively earthed (high impedance) when this factor is higher than 0.8.

12.7.3 **Power transformers** : Power transformers are to be provided with overload and short circuit protections. When transformers are connected in parallel, tripping of the protective devices at the primary side has to automatically trip the switch connected at the secondary side.

12.7.4 **Voltage transformers for control and instrumentation** : Voltage transformers are to be provided with overload and short circuit protections on the secondary side.

12.7.5 **Fuses** : Fuses are not to be used for overload protection.

12.7.6 **Low voltage systems** : Low voltage systems supplied through transformers from high voltage systems are to be protected against overvoltage. This may be achieved by:

i) direct earthing of the low voltage system;
ii) appropriate neutral voltage limiters;
iii) earthed screen between the primary and secondary windings of transformers.

12.8 Rotating machinery

12.8.1 **Stator windings of generators** : Generators stator windings are to have all phase ends brought out for the installation of the differential protection.

12.8.2 **Temperature detectors** : Rotating machinery is to be provided with temperature detectors in their stator windings to actuate audiovisual alarm in a normally attended position whenever the temperature exceeds the permissible limit. If embedded temperature detectors are used, means are to be provided to protect the circuit against overvoltage.

12.8.3 Tests : In addition to the tests normally required for rotating machinery, a high frequency high voltage test in accordance with IEC Publication 60034-15:2009 is to be carried out on the individual coils in order to demonstrate a satisfactory withstand level of the inter-turn insulation to steep fronted switching surges.

12.9 Power Transformers

12.9.1 Dry and Liquid cooled transformers are to comply with relevant parts of IEC 60076. Oil immersed transformers are to be provided with the following alarms and protections:

- liquid level (Low) - alarm
- liquid temperature (High) - alarm
- liquid level (Low) - trip or load reduction
- liquid level (High) - trip or load reduction
- gas pressure relay (High) - trip.

12.10 Cables

12.10.1 Cables are to be constructed in accordance with the IEC Publication 60092-353: 2016 and 60092-354:2020 or other equivalent Standard.

12.11 Switchgear and controlgear assemblies

12.11.1 Switchgear and controlgear assemblies are to be constructed according to the IEC Publication 62271-200:2011 and the following additional requirements.

12.11.2 Mechanical construction : Switchgear is to be of metal - enclosed type in accordance with IEC Publication 62271-200:2011 or of the insulation - enclosed type in accordance with the IEC Publication 62271-201:2014.

12.11.3 Locking facilities : Withdrawable circuit breakers and switches are to be provided with mechanical locking facilities in both service and disconnected positions. For maintenance purpose, key locking of withdrawable circuit breakers and switches and fixed disconnectors is to be possible. Withdrawable circuit breakers are to be located in the service position so that there is no relative motion between fixed and moving portions.

12.11.4 Shutters : The fixed contacts of withdrawable circuit breakers and switches are to be so arranged that in the withdrawable position the live contacts are automatically covered. Shutters are to be clearly marked for incoming and outgoing circuits. This may be achieved with the use of colours or labels.

12.11.5 Earthing and short-circuiting : For maintenance purposes an adequate number of earthing and short-circuiting devices are to be provided to enable circuits to be worked upon with safety.

12.11.6 Internal Arc Classification (IAC) : Switchgear and control gear assemblies are to be internal arc classified (IAC). Accessibility

Type A is sufficient (IEC 62271-200:2011; Annex AA; AA 2.2), where switchgear and control gear are accessible by authorized personnel only Accessibility Type B is required if accessible by non-authorised personnel.
Installation and location of the switchgear and control gear is to correspond with its internal arc classification and classified sides (F, L and R).

12.12 Auxiliary systems

12.12.1 Source and capacity of supply : If electrical energy and/or physical energy is required for the operation of circuit breakers and switches, a store supply of such energy is to be provided for at least two operations of all the components.
However, the tripping due to overload or short-circuit, and under-voltage is to be independent of any stored electrical energy sources. This does not preclude shunt tripping provided that alarms are activated upon lack of continuity in the release circuits and power supply failures.

12.12.2 Number of external supply sources : When external source of supply is necessary for auxiliary circuits, at least two external sources of supply are to be provided and so arranged that a failure or loss of one source will not cause the loss of more than one generator set and/or set of essential services.

Where necessary one source of supply is to be from the emergency source of electrical power for the start up from dead ship condition.

12.12.3 High voltage test : A power-frequency voltage test is to be carried out on any switchgear and controlgear assemblies.

The test procedure and voltages are to be according to the IEC Publication 62271-200:2011, Section 7 / Routine Test.

12.13 Installation

12.13.1 **Electrical equipment** : Where equipment is not contained in an enclosure but a room forms the enclosure of the equipment, the access doors are to be so interlocked that they cannot be opened until the supply is isolated and the equipment earthed down. At the entrance of the spaces where high-voltage electrical equipment is installed, a suitable marking is to be placed which indicates danger of high voltage. As regard the high-voltage electrical equipment installed out-side above mentioned spaces, the similar marking is to be provided. An adequate, unobstructed working space is to be left in the vicinity of high voltage equipment for preventing potential severe injuries to personnel performing maintenance activities. In addition, the clearance between the switchboard and the ceiling/ deckhead above is to meet the requirements of the Internal Arc Classification according to IEC 62271-200:2011 (see 12.11.6).

12.13.2 Cables

12.13.2.1 In accommodation spaces, high voltage cables are to be run in enclosed cable transit systems.

12.13.2.2 High voltage cables are to be segregated from cables operating at different voltage ratings and from each other, in particular, they are not to be run in the same cable bunch, nor in the same ducts or pipes, or, in the same box. Where high voltage cables of different voltage ratings are installed on the same cable tray, the air clearance between cables is not be less than the minimum air clearance for the higher voltage side in 12.6. However, high voltage cables are not to be installed on the same cable tray for the cables operating at the nominal system voltage of 1 kV and less.

12.13.2.3 High voltage cables, in general, are to be installed on carrier plating when they are provided with a continuous metallic sheath or armour which is effectively bonded to earth; otherwise they are to be installed for their entire length in metallic casings effectively bonded to earth.

12.13.2.4 Terminations in all conductors of high voltage cables are to be, as far as practicable, effectively covered with suitable insulating material. In terminal boxes, if conductors are not insulated, phases are to be separated from earth and from each other by substantial barriers of suitable insulating materials.

High voltage cables of the radial field type, i.e. having a conductive layer to control the electric field within the insulation, are to have terminations which provide electric stress control.
Terminations are to be of a type compatible with the insulation and jacket material of the cable and are to be provided with means to ground all metallic shielding components (i.e. tapes, wires etc.)

12.13.2.5 High voltage cables are to be readily identifiable by suitable marking.

12.13.2.6 Before a new high voltage cable installation or an addition to an existing installation is put into service, a voltage withstand test is to be satisfactorily carried out on each completed cable and its accessories.

The test is to be carried out after an insulation resistance test.

For cables with rated voltage (U_0/U) above 1.8/3 kV (U_m=3.6 kV) an a.c. voltage withstand test may be carried out upon advice from high voltage cable manufacturer. One of the following test methods are to be used:

a) test for 5 min with the phase-to-phase voltage of the system applied between the conductor and the metallic screen/sheath;

b) test for 24 h with the normal operating voltage of the system.

Alternatively, a d.c. test voltage equal to 4 U_0 may be applied for 15 minutes:

For cables with rated voltage (U_0/U) up to 1.8/3 kV (Um=3.6 kV) a d.c. voltage equal to 4 U_0 is to be applied for 15 minutes.

After completion of the test, the conductors are to be connected to earth for a sufficient period in order to remove any trapped electric charge.

An insulation resistance test is then to be repeated.

Section 13

Trials

13.1 General

13.1.1 Before a new installation, or any alteration or addition to an existing installation, is put into service the tests and trials specified in this Section are to be carried out. These tests and trials are intended to demonstrate the general condition of the installation at the time of completion. They are in addition to any acceptance tests which may have been carried out at the manufacturer's works.

13.2 Insulation resistance measurement

13.2.1 Insulation resistance is to be measured using a self-contained instrument such as a direct reading ohm-meter of the generator type applying a voltage of at least 500 V. Where a circuit incorporates capacitors of more than 2mF total capacitance, a constant-voltage type instrument is to be used to ensure accurate test readings.

13.2.2 Power and light circuits : The insulation resistance between all insulated poles and earth and, where practicable, between poles, is to be at least 1MΩ. The installation may be subdivided and appliances may be disconnected if initial tests produce results less than this figure.

13.2.3 Internal communication circuits : Circuits operating at 55 V and above are to have an insulation resistance between conductors and between each conductor and earth of at least 1 MΩ. Circuits operating at less than 55 V are to have an insulation resistance of at least 0.33 MΩ.

13.2.4 Switchboards, Section boards and distribution boards : The insulation resistance is to be at least 1 MΩ when measured between each busbar and earth and between busbars. This test may be made with all circuit-breakers and switches open, all fuse links for pilot lamps, earth fault-indicating lamps, voltmeters, etc., removed and voltage coils temporarily disconnected, where otherwise damage may result.

13.2.5 Generators and motors : The insulation resistance of generators and motors, in normal working condition and with all parts in place, is to be measured and recorded. The test should be carried out with the machine hot, if possible. The insulation resistance of generator and motor cables, field windings and control gear is to be at least 1 MΩ.

13.3 Earth continuity

13.3.1 Tests are to be made to verify that all earth continuity conductors are effective and that the bonding and earthing of metallic conduit and/or sheathing of cables is effective.

13.4 Performance

13.4.1 It is to be established that the provisions of the Rules have been complied with respect to the criteria mentioned in this sub-section.

13.4.2 Temperatures of joints, connections, circuit-breakers and fuses.

13.4.3 The operation of engine governors, synchronising aids and devices, overspeed trips, reverse-current, reverse-power, over-current and under-voltage trips and other safety devices.

13.4.4 Efficient lubrication, absence of vibration or abnormal noises and satisfactory alignment/ holding down arrangements.

13.4.5 Satisfactory commutation, excitation and performance of each generator throughout a run at full rated load.

13.4.6 Voltage regulation of every generator when full rated load is suddenly thrown off.

13.4.7 For alternating current and direct current generators, satisfactory parallel operation and [kW] load sharing of all generators capable of being operated in parallel at all loads up to normal working load. For alternating current generators satisfactory parallel operation and kVA load sharing of all generators capable of being operated in parallel at all loads up to normal working load.

13.4.8 All essential motors and other important equipment are to be operated under service conditions, though not necessarily at full load or simultaneously, for a sufficient length of time to demonstrate that they are satisfactory.

13.4.9 Normal and emergency lighting, heating and various appliances (galley, pantry etc.) and corresponding electrical circuits are to be tested, as far as practicable, under normal working conditions.

13.4.10 Internal communication systems are to be checked to demonstrate their suitable and specified functioning. Particular attention is to be given to engine order telegraphs, docking telegraphs, helm indicators, fire detection and alarm systems, emergency alarms, morse signal lamp, navigation light indicator panel and telephones.

13.4.11 Operation of the emergency source of power is to be checked. The working of any apparatus intended to be fed by this source is to be checked as well as automatic change-over switches, if any.

13.4.12 Propulsion equipment is to be tested under working conditions and operated in the presence of the Surveyors and to their satisfaction. The equipment is to have sufficient power for going astern to secure proper control of the unit in all normal circumstances. In passenger ships the ability of the machinery to reverse the direction of thrust of the propeller in sufficient time, under normal maneuvering conditions, and so bring the ship to rest from maximum ahead service speed, is to be demonstrated at the sea trial.

13.5 Voltage drop

13.5.1 Voltage drop is to be measured, where necessary, to verify that this is not excessive.

End of Chapter

Chapter 10

Requirements for Fusion Welding

Contents

Section

1 *General*
2 *Fusion Welded Pressure Vessels*
3 *Welded Pressure Pipes*

Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter apply to the welding of pressure vessels, pressure pipes and the welded structures of machinery.

1.1.2 The term 'fusion weld', for the purpose of these requirements, is applicable to welded joints made by manual, semi-automatic or automatic electric arc welding processes. Special consideration will be given to the proposed use of other fusion welding processes.

1.2 General requirements for welding plant and welding quality

1.2.1 The welding plant and equipment are to be installed under cover and so arranged that all welding is carried out in positions free from draughts and adverse weather conditions.

1.2.2 In the first instance, and before work is commenced, the Surveyors are to be satisfied that the required quality of welding is attainable with the proposed welding plant, equipment and procedures.

1.2.3 The welding plant and equipment are to be maintained in an efficient working condition.

1.2.4 The procedures are to include the regular systematic supervision of all welding, and the welders are to be subjected by the works' supervisors to periodic tests for quality of workmanship. Records of these tests are to be kept and are to be available for inspection by the Surveyors.

1.2.5 All welding is to be to the satisfaction of the Surveyors.

Section 2

Fusion Welded Pressure Vessels

2.1 Manufacture of Class 1 and Class 2 fusion welded pressure vessels

2.1.1 Fusion welded pressure vessels constructed to Class 1 and Class 2 requirements will be accepted only if manufactured by firms equipped and competent to undertake high quality welding. In order that firms may be approved for this purpose, it will be necessary for the Surveyors to visit the works for the purpose of inspecting the welding plant, equipment and procedures and to arrange for carrying out preliminary tests as stated in 2.3.

2.1.2 In the case of Class 1 approval, arrangements are to be made for the survey

during construction and testing of a full size welded pressure vessel.

2.1.3 The welding plant and equipment are to be suitable for undertaking work of the standard required for Class 1 and Class 2.

2.1.4 The works are to have an efficient testing laboratory, suitably equipped to carry out tensile, bend and impact tests, the x-ray examination of pressure vessels, and the metallographic examination of welds. The works are also to be equipped with a suitable heat treating furnace with satisfactory means of temperature control.

2.1.5 Alternative arrangements which, in the opinion of the Surveyors, ensure an equally high standard of quality control may be submitted for consideration.

2.1.6 On completion of inspection and tests, the Surveyor's report, including the results of the preliminary tests and also, for Class 1 approval, the results of the tests on the pressure vessel, as required by 2.1.2, is to be submitted for the consideration of IRS. The report should also include the radiographs and particulars of any fusion welded pressure vessels previously constructed at these works.

2.2 Manufacture of Class 3 fusion welded pressure vessels

2.2.1 Class 3 pressure vessels will be accepted if constructed by firms whose works are equipped to undertake the welding of pressure vessels of this Class. Preliminary tests would require to be carried out as stated in 2.4.

2.3 Preliminary tests for Class 1 and Class 2 fusion welded pressure vessels

2.3.1 Preliminary tests to demonstrate the quality of welding are to be carried out by the firm under supervision of the Surveyors. The test requirements will be based on the types of steels and on the welding process to be used. For approval purposes, the types of rolled steel plates specified in Pt.2, Ch.3 are grouped as follows :

Group 1 : Carbon and carbon-manganese steels - specified minimum tensile strength not exceeding 500 [N/mm^2];

Group 2 : Carbon and carbon-manganese steels - Specified minimum tensile strength 500-520 [N/mm^2];

Group 3 : Alloy steels.

2.3.2 The maximum plate thickness which will be approved in pressure vessel construction will depend on the thickness of the test plates used in the preliminary tests; the test plates are, however, to be at least 20 [mm] thick.

2.3.3 The test plates and the pressure vessel, mentioned in 2.1.2, are to be representative as regards materials and approximate shell thickness of the production vessels for which approval is desired.

2.3.4 The welded seams of the test plates are to be radiographed, and the Surveyors are to select portions of the test plates containing the welded joint from which specimens are to be provided for the following tests:

a) Tensile;

b) Bend;

c) Hardness;

d) Impact - For Class 1 application and for steels in Groups 2 and 3;

e) Fatigue - For Class 1 application and for steels in Groups 2 and 3;

f) Micrographs, at 100 and 300 magnifications, of weld center, fusion zone and parent plate, for Class 1 application and for steels in Group 2 and 3;

g) Macrograph of full section weld;

h) Chemical analysis of deposited weld metal;

i) Chemical analysis of test plates.

2.3.5 Where the welding is carried out by an established and approved process, the fatigue tests and micrographs in 2.3.4(e) and (f) will not, in general, be required. Furthermore, as an alternative to 2.3.4(i), a guaranteed analysis obtained from the steelmaker will be accepted.

2.3.6 If a firm intends to manufacture pressure vessels either of a different group of steel or by means of a different welding process from that used in the preliminary tests on which the original approval was based, further tests will

be required to cover the proposed welding procedure. In such cases full details of the material, plate thickness and welding process proposed are to be submitted for consideration, when the requirements for further preliminary tests will be indicated.

2.4 Preliminary tests for Class 3 fusion welded pressure vessels

2.4.1 Preliminary tests to demonstrate the quality of welding are to be carried out by the firm under the supervision of the Surveyors. The test requirements will be based on the welding process used, but are to be similar to those described in 2.4.2 to 2.4.6.

2.4.2 Two test assemblies, of sufficient dimensions to prepare test specimens, as required by 2.4.3, are to be prepared using plates of thickness representative of the thickness of the pressure vessels to be manufactured. Alternatively, one test assembly may be prepared to provide all the test specimens required by 2.4.3 and for retest purposes. Test assemblies need not be prepared for circumferential seams, except in cases where the vessel has only circumferential seams or where the process for welding the circumferential joints is significantly different from that used for longitudinal joints.

2.4.3 Test specimens, as shown in Fig.2.4.1, are to be tested as follows :

a) Specimen No. 1 :- Tensile test for joint. The shape and preparation of the specimen are to conform to the requirements of 2.5.12. The tensile strength obtained is to be not less than the minimum specified tensile strength for the plate material;

b) Specimen No. 2 :- Transverse bend test. The shape and preparation of the specimen and procedure for testing are to comply with the requirements of 2.5.11;

c) Specimen No. 3 :- Nicked bend test. The specimen is to have a slot cut into each side on the centreline of the weld and perpendicular to the plate surface. The specimen is then to be broken in the weld, and the fracture is to reveal a sound homogeneous weld, substantially free from slag inclusions, porosity and coarse crystalline form.

2.4.4 If any of the tests fail, further two re-test specimens are to be prepared and tested. Should any of these two further re-tests also fail, the reasons for the failure are to be investigated and the results are to be submitted to IRS for review.

2.4.5 Where two test assemblies have been prepared, one re-test is to be taken from each test assembly.

2.4.6 If both the re-test values are satisfactory, the re-test values may be accepted.

2.5 Routine tests for Class 1 and Class 2 fusion welded pressure vessels

2.5.1 Two test plates, each of sufficient dimensions to provide one complete set of specimens required by 2.5.9 are to be prepared for each pressure vessel. They are to be attached to the shell plate in such a manner that the edges to be welded are a

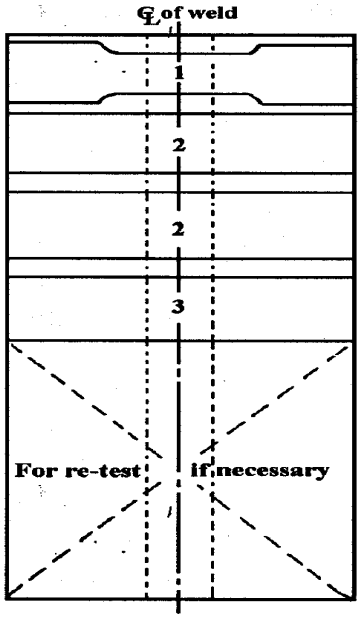


Fig. 2.4.1 : Test sample

continuation and simulation of the corresponding edges of the longitudinal joint. The welding process, procedure and technique are to be the same as employed in the welding of the longitudinal joint.

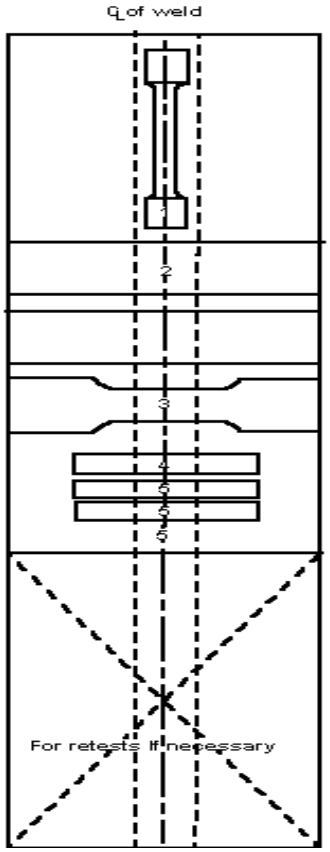


Fig.2.5.1 : Test specimen

2.5.2 As an alternative to 2.5.1, one test plate may be prepared to provide all the test specimens required by 2.5.9 and for re-test purposes.

2.5.3 Test plates need not be prepared for the circumferential seams, except in cases where a pressure vessel has circumferential seams only, or where the process for welding the circumferential joints is significantly different from that used for the longitudinal joints; in which case one test plate is to be prepared having a welded joint which so far as possible is a simulation of the circumferential seams. The test plate is to provide all the test specimens required by 2.5.9 and for re-test purposes. Where a number of similar vessels are made at the same time, it will suffice if, test plates are provided for each 30 [m] of circumferential welded seam.

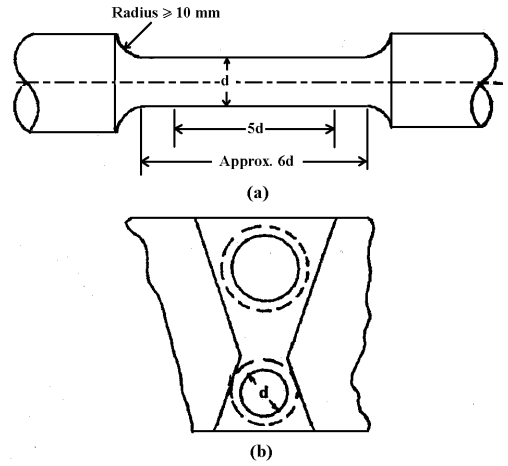


Fig. 2.5.2 : All weld test specimen

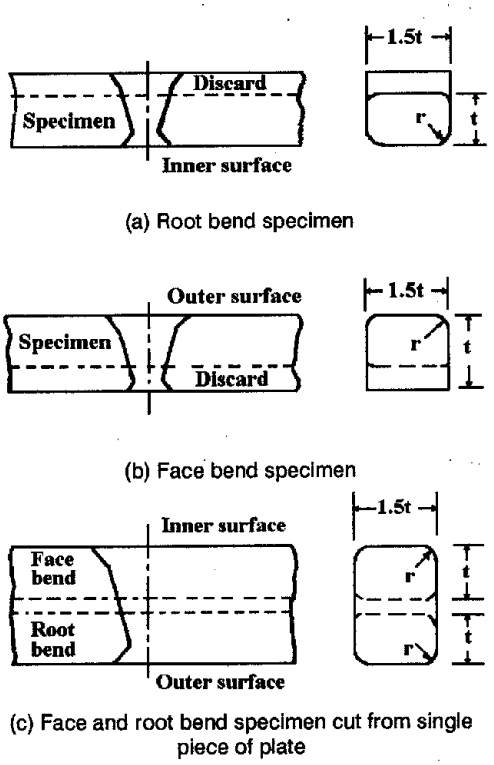


Fig. 2.5.3 : Bend test specimen

2.5.4 The test plates are to be cut from the shell plate or plates forming the appropriate seam, and before being detached are to be stamped by the Surveyor.

2.5.5 Where there is insufficient material available on the shell plates for the provision of test plates, acceptance may be given to test plates cut from another plate, provided that this plate is from the same cast and in the same heat treatment condition.

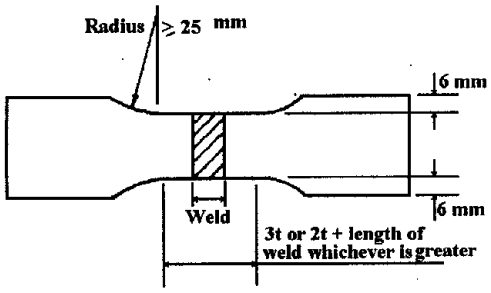


Fig. 2.5.4 : Tensile test for joint

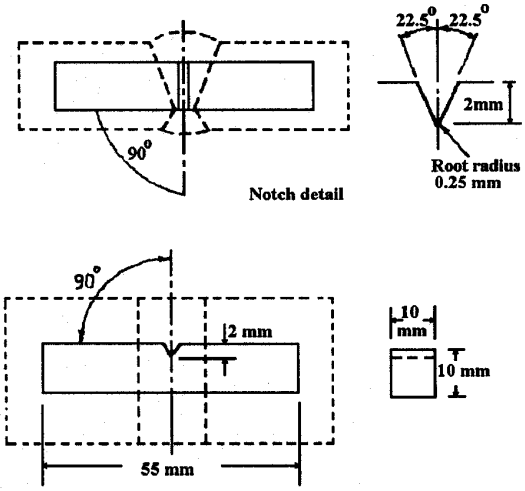


Fig. 2.5.5 : Impact test specimens

2.5.6 The thickness of test plates is to be the same as that of the pressure vessel.

2.5.7 Test plates are to be so supported, during welding, that warping is reduced to a minimum.

2.5.8 The test plates are to be straightened before being subjected to heat treatment, and for this purpose the test plates may be heated to a temperature below that required for the final heat treatment.

2.5.9 One set of test specimens is to be cut from the test plates as shown in Fig.2.5.1. The results of the tests are to comply with the requirements detailed in 2.5.10 to 2.5.14.

2.5.10 Specimen no. 1 : Tensile test for the weld metal - The following conditions are to be satisfied :

a) One weld metal tensile specimen is to be taken from Class 1 pressure vessels having a shell thickness not exceeding 70 [mm], and for all Class 2 pressure vessels. Where the shell thickness of Class 1 pressure vessels exceeds 70 [mm], two such specimens are to be taken, one above the other;

b) The diameter of the all weld metal test specimen at the reduced parallel position is to be not less than 14 [mm], except in the case of thin plates, where the largest practicable diameter should be used. The gauge length of the specimen is to be 5 times the diameter;

c) The dimensions of the all weld metal test specimen are shown in Fig.2.5.2 (a), and their location when two specimens are used, in Fig.2.5.2 (b);

d) The tensile strength of the weld metal is not to be less than the minimum specified for the plate material and not more than 145 [N/mm2] above this value;

e) The percentage elongation, A, is to be not less than that given by :

$$A = \frac{980 - R}{21.6}$$

where R is the tensile strength [N/mm^2]. In addition, this elongation is to be not less than 80 per cent of the equivalent minimum elongation specified for the plate.

2.5.11 Specimen no. 2 :- Transverse bend test. The following conditions are to be satisfied :

a) Two bend test specimens of rectangular section are to be cut from the test plate transversely to the weld, one to be bent with the outer surface of the weld in tension, and the other with the inner surface in tension;

b) The specimens are to have a width equal to 1.5 times the thickness of the specimen, and the mid-portion is to coincide with the centreline of the weld. The edges are to be rounded to a radius, r, not exceeding 10 per cent of the thickness;

c) Where the plate thickness does not exceed 25 [mm], the thickness of the specimens is to be the full thickness of the plate. Where the plate thickness exceeds 25 [mm], the specimens, in all cases, are to have a thickness of 25 [mm] and are to be prepared by discarding metal from the

surfaces which will be in compression when the test is applied. See Fig.2.5.3 (a) and (b);

d) Where the thickness of the plate permits, the bend specimens may be prepared as shown in Fig.2.5.3 (c);

e) For each specimen the weld reinforcement is to be removed by grinding or machining so that the outer and inner surfaces of the weld are flush with the surface of the plate;

f) The specimen is to be mounted on roller supports with the centre of the weld midway between the supports. A former, with its axis perpendicular to the specimen, is to bend the specimen by pushing it through the clear space between the supports. The diameter of the former and the clear space between the supports will depend upon thickness of the specimens, and these dimensions are shown in the following table in terms of the thickness, t, of the specimen;

Minimum specified tensile strength of plate [N/mm^2]	**Diameter of former**	**Clear space between supports**
< 460	2t	4.2t
≥ 460 < 510	3t	5.2t
≥ 510 < 620	4t	6.2t

g) After bending, there is to be no crack or defect exceeding 1.5 [mm] measured across the specimen or 3 [mm] measured along the specimen. Premature failure at the edges of the specimen will not lead to rejection.

2.5.12 Specimen no. 3 : Tensile test for joint. The following conditions are to be satisfied :

a) One reduced section tensile test specimen is to be cut transversely to the weld or, in the thick plate, as many tensile test specimens as may be necessary to investigate the tensile strength throughout the whole thickness of the joint;
The weld reinforcement is to be removed by grinding or machining so that the outer and inner surfaces of the weld are flush with the surface of the plate. The dimensions of the reduced section tensile test specimens are shown in Fig.2.5.4. The width, B, at the reduced section is to be at least 25 [mm];

b) Where the plate thickness exceeds 30 [mm], the tensile test may be effected on several reduced section specimens, each with a thickness of at least 30 [mm] and a width at the effective cross-section of at least 25 [mm];

c) The tensile strength obtained is to be not less than the minimum specified tensile strength for the plate material.

2.5.13 Specimen no. 4 : Macro-specimen. The following conditions are to be satisfied :

a) Macrograph of a complete cross-section of the weld, including the heat affected zone, is to show a satisfactory penetration and fusion, and an absence of significant inclusions or other defects;

b) Should there be any doubt as to the condition of the weld as shown by macro-etching, the area concerned is to be examined by a micrograph.

2.5.14 Specimen no. 5 : Charpy V-notch impact test - Class 1 pressure vessels only

The following conditions are to be satisfied :

a) Three Charpy V-notch impact test specimens are to be cut transversely to the weld, parallel to the plate surface and at mid-plate thickness. The notch is to be cut at approximately the centre of the weld, and the axis of the notch is to be perpendicular to the surface of the plate. See Fig.2.5.5;

b) The dimensions and tolerances of the specimens are to be in accordance with Pt.2 Ch.2;

c) The average energy value obtained from the Charpy V-notch test specimen is to be not less than 27 J when the temperature of the specimen at the time of the test does not exceed 50°C;

d) Where it is proposed to use impact tests other than the Charpy V-notch type, details are to be submitted for consideration;

e) The foregoing impact test are not applicable to pressure vessels operating at low metal temperatures, and for such cases the results of the weld metal impact tests will be specially considered.

2.5.15 The test assemblies are to be heat treated in accordance with 2.10.

2.5.16 If any of the tests fail, further two re-test specimens are to be prepared and tested. Should any of these two further re-test also fail, the reasons for the failure are to be investigated and the results are to be submitted to IRS for review.

2.5.17 Where two test assemblies have been prepared, one re-test is to be taken from each test assembly.

2.5.18 If both the re-test values are satisfactory, the re-test values may be accepted.

2.6 Routine tests for Class 3 fusion welded pressure vessels

2.6.1 Routine weld tests are not required for Class 3 pressure vessels, but occasional check tests on the quality of the welding may be carried out at the discretion of the Surveyors.

2.7 Construction and workmanship

2.7.1 Cutting of plates

2.7.1.1 Plates are to be cut to size and shaped by machine flame cutting and/or machining.

2.7.1.2 Where the plate thickness does not exceed 25 [mm], cold shearing may be used provided that the sheared edge is cut back by machining or chipping for a distance of one-quarter of the plate thickness but in no case by less than 3 [mm].

2.7.1.3 All plate edges, after being cut and before further work is carried out upon them, are to be examined for laminations, and also to ensure that any sheared edges are free from cracks. Visual methods may be supplemented by other techniques at the discretion of the Surveyors.

2.7.2 Welded joints

2.7.2.1 Only full penetration butt welds are acceptable for longitudinal and circumferential main joints.

2.7.2.2 The joints are to be welded from both sides of the plates unless otherwise approved.

2.7.2.3 Details of the welded joints between dished ends and shells, flat ends and shells or rectangular headers and methods of attaching standpipes, branches, compensating plates and pads are generally to be in accordance with Fig.2.7.2.1, Fig.2.7.2.2 and Fig.2.7.2.3 respectively.

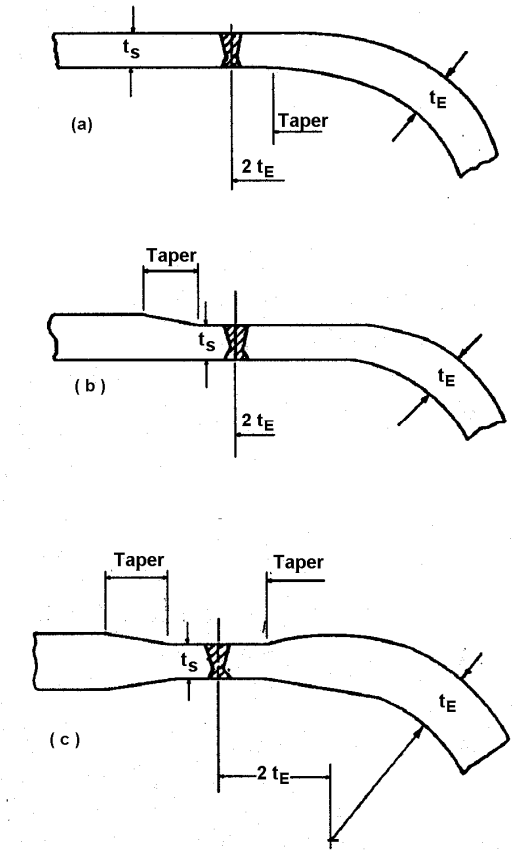


Fig.2.7.2.1 : Typical attachments of dished ends to cylindrical shells

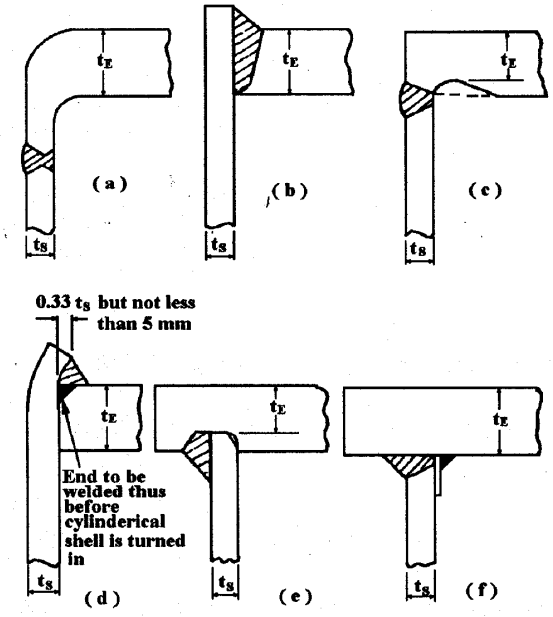
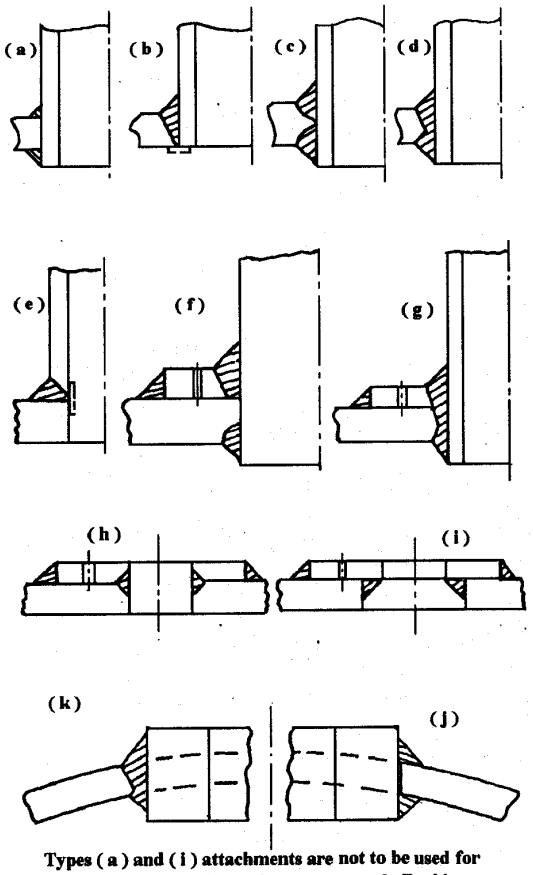


Fig. 2.7.2.2 : Typical attachments of flat ends



Types (a) and (i) attachments are not to be used for openings which require to be compensated . Backing rings may be used with types (b) and (e).

Fig. 2.7.2.3 : Typical acceptable methods of attaching branches and pads

2.7.2.4 Before welding is commenced, it is to be ascertained that the plate edges are in alignment within the following limits :

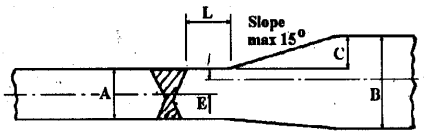
a) 10 per cent of the plate thickness with a maximum of 3 [mm] for longitudinal joints;

b) 10 per cent of the plate thickness plus 1 [mm] with a maximum of 4 [mm] for circumferential joints.

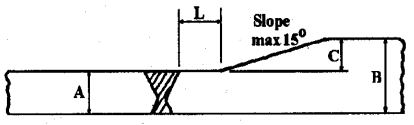
2.7.2.5 If the plates are of unequal thickness, and the difference between the surfaces exceeds that given in 2.7.2.4 for the thicker plate, the thicker plate is to have a smooth taper with a slope not exceeding 15 degrees. See Fig.2.7.2.4.

2.7.2.6 Wherever practicable, the welding is to be carried out in the downhand position. In the case of circumferential joints in cylindrical shells, means are to be adopted to ensure compliance with this requirement.

2.7.2.7 Steel backing strips, if used, are to be of same nominal composition as the plates to be welded and, where practicable, are to be removed and the surface dressed smooth by grinding prior to radiography.



Longitudinal Joint Where C Exceeds 10 % of B or 3 mm



Longitudinal Joint Where C Exceeds 10 % of B or 3 mm

Fig. 2.7.2.4 : Attachment of plates of unequal thickness

2.7.2.8 Fillet welds are to be so made as to ensure proper fusion and penetration of the weld metal at the root of the fillet.

2.7.2.9 The arc is to be struck only on those parts of the parent metal where the weld metal is to be applied, or on the weld metal already deposited. Accidental arc strikes are to be removed by grinding followed by magnetic particle or dye penetrant testing to the satisfaction of the Surveyors.

2.7.2.10 Preheating and maintenance of minimum inter-pass temperature are to be employed where necessitated by the joint restraint, thickness of the plate or composition of the material to be welded.

2.7.3 Welding consumables

2.7.3.1 All consumables intended for use in the welding of pressure vessels are to be stored in a dry place and are to be treated in accordance with the manufacturer's instructions.

2.7.3.2 In order to ensure that the quality of welding consumables is being consistently maintained, they are to be subjected to a regular system of periodic testing and inspection.

2.7.3.3 Where routine tests are frequently carried out in respect of pressure vessels made in the normal course of production, such tests may be regarded as meeting the requirements of 2.7.3.2.

2.7.4 Forming shell sections and end plates

2.7.4.1 Plates for shell sections and end plates are to be formed to the required shape by any process that will not impair the quality of the material. Tests to demonstrate the suitability of a process may be required at the discretion of the Surveyors.

2.7.4.2 Shell plates are to be formed to the correct contour up to the extreme edges of the plates.

2.7.4.3 So far as possible, hot and cold forming is to be carried out by machine. Forming by hammering, with or without local heating, is not to be employed.

2.7.4.4 All plates which have been hot formed or locally heated for forming are to be normalized on completion of this operation. If, however, hot forming is carried out entirely at a temperature within the normalizing range, subsequent heat treatment will not be required for carbon and carbon- manganese steels. In both instances alloy steels may, in addition, be required to be tempered.

2.7.4.5 All plates which have been cold formed to an internal radius less than 10 times the plate thickness are to be given an appropriate heat treatment.

2.7.4.6 Seams in plates may be welded prior to forming provided that, on completion of forming and subsequent heat treatment, they meet the specified mechanical test requirements and that they are examined radiographically throughout their length after forming. After forming, the surfaces of such seams in alloy steel parts, also carbon steel parts over 25 [mm] in thickness are to be ground smooth and inspected for cracks by magnetic particle crack detection, dye penetration or other means at the discretion of the Surveyors.

2.7.5 Fitting of tubes to drums and headers

2.7.5.1 The tube holes in drums or headers are to be formed in such a way that tubes can be effectively tightened in them. Where the tube ends are not normal to the tube plates, there is to be a neck or belt of parallel seating of at least 13 [mm] in depth, measured in a plane through the axis of the tube at the holes. Where the tubes are practically normal to their plates, this parallel seating is to be not less than 10 [mm] in depth.

2.7.5.2 Tubes are to be carefully fitted in the tube holes and secured by means of welding, expanding and bevelling or by other approved methods. The tubes are to project through the neck or belt of parallel seating by at least 6 [mm] and, where they are secured from drawing out by means of bellmouthing only, the included angle of belling is to be not less than 30 degrees.

2.7.6 Attachments and fittings

2.7.6.1 All lugs, brackets, branches, manhole frames and reinforcements around openings and other members are to conform to the shape of the surface to which they are attached.

2.7.6.2 Doubling plates with well rounded corners are to be fitted in way of attachments such as lifting lugs, supporting brackets and feet, to minimise load concentrations on pressure shells and ends. Compensating plates, pads, brackets, and supporting feet are to bedded closely to the surface before being welded, and are to be provided with a 'tell-tale' hole not greater than 9.5 [mm] in diameter, open to the atmosphere to provide for the release of entrapped air during heat treatment of the vessel, or as a means of indicating any leakage during hydraulic testing and in service

2.7.6.3 The attachment by welding of such fittings to the main pressure shell after post-weld heat treatment is not permitted, except where the material involved is mild steel, when welding will be permitted only if it is necessitated by the method of construction employed. Prior approval of the Surveyor must be obtained before any welding is carried out. In no circumstances is any welding to be carried out after heat treatment on vessels made of carbon or carbon- manganese steels with tensile strength exceeding 500[N/mm^2], or of alloy steel.

2.7.6.4 Where the fittings referred to in 2.7.6.3, together with flats and other attachments for supporting internal and external components, are welded to the main pressure shell, the welding is to be of comparable standard to that required for the vessel, and the material used is to be of compatible composition.

2.7.6.5 The finish of all welds attaching pressure parts and non-pressure parts to the main pressure shell is to be such as to allow satisfactory examination of the welds. In the case of Class 1 pressure vessels, these welds are to be ground smooth, if necessary, to provide a suitable finish for crack detection tests, which are to be carried out to the Surveyor's satisfaction on completion of the hydraulic test.

2.7.7 Tolerances for cylindrical shells

2.7.7.1 Measurements are to be made to the surface of the parent plate, not to a weld, fitting or other raised part.

2.7.7.2 In assessing the out-of-roundness of pressure vessels, the difference between the maximum and minimum internal diameter measured at one cross-section is not to exceed the amount given in Table 2.7.7.1.

2.7.7.3 The profile, measured on the inside or outside of the shell by means of a gauge of designed form of the shell, and having a length equal to one-quarter of the internal diameter of the vessel, is not to depart from the designed form by more than the amount given in Table 2.7.7.1 This amount corresponds to, x, in Fig.2.7.7.1.

Table 2.7.7.1 : Permitted out of roundness		
Nominal internal diameter of vessel [mm]	**Difference between maximum and minimum diameters**	**Maximum departure from designed form**
≤ 300	1.0 per cent of internal diameter	1.2 [mm]
> 300 ≤ 460		1.6 [mm]
> 460 ≤ 600		2.4 [mm]
> 600 ≤ 900		3.2 [mm]
> 900 ≤ 1220		4.0 [mm]
> 1220 ≤ 1520		4.8 [mm]
> 1520 ≤ 1900		5.6 [mm]
> 1900 ≤ 2300	19 [mm]	6.4 [mm]
> 2300 ≤ 2670		7.2 [mm]
> 2670 ≤ 3950		8.0 [mm]
> 3950 ≤ 4650	19 [mm]	0.2 per cent of internal diameter
> 4650	0.4 per cent of internal diameter	0.2 per cent of internal diameter

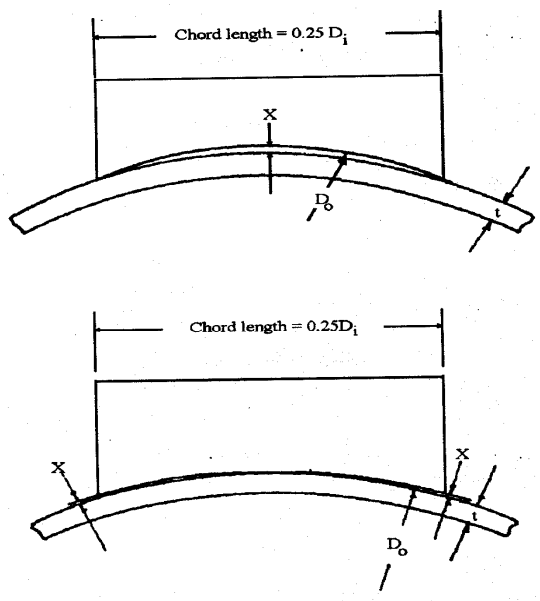


Fig. 2.7.7.1 : Tolerances for cylinderical shells

Table 2.7.7.2 : Circumferential tolerance	
Outside diameter (nominal inside diameter plus twice actual plate thickness) [mm]	**Circumferential tolerance**
300 to 600 inclusive	± 5 [mm]
Greater than 600	± 0.25 per cent

2.7.7.4 Shell sections are to be measured for out-of-roundness, either when laid on their sides or when set up on end. When the shell sections are checked while lying on their sides, each measurement for diameter is to be repeated after turning the shell through 90 degrees about its longitudinal axis. The two measurements for each diameter are to be

averaged, and the amount of out-of-roundness calculated from the average values so determined.

2.7.7.5 There are to be no flats or peaks at welded seams, and any local departure from circularity is to be gradual.

2.7.7.6 The external circumference of the completed shell is not to depart from the calculated circumference (based upon nominal inside diameter and actual plate thickness) by more than the amount given in Table 2.7.7.2.

2.8 Non-destructive examination of fusion welded pressure vessels

2.8.1 Radiographic examination

2.8.1.1 The extent of the radiographic examination of welded seams of Class 1 and Class 2 pressure vessels is to be as follows :

a) Class 1 pressure vessels - All butt welded seams in drums, shells, headers and pipes together with the test plate or plates are to be subjected to 100 per cent radiographic examination. For circumferential butt welds in extruded connections, pipes, tubes, headers and other tubular parts of 170 [mm] outside diameter or less, 10 per cent of the total number of welds are to be radiographed;

b) Class 2 pressure vessels - Spot radiographs are to be taken at selected regions of each main seam. The test plate or plates are to be fully radiographed and at least 10 per cent of the length of each main seam is to be so examined.

2.8.1.2 Butt welds in furnaces, combustion chambers and other pressure parts for fired pressure vessels under external pressure, are to be subjected to spot radiographic examination.

2.8.1.3 Where the surface finish of any weld which has to be radiographed is such that it will prevent accurate radiographic examination, the surface is to be machined or ground to provide a smooth contour to the Surveyor's satisfaction.

2.8.1.4 Lead type is to be fixed to the plate adjacent to the weld so that each radiograph is marked in such a way that the corresponding portion of the welded seam can be readily and accurately identified.

2.8.1.5 The length of weld covered by each radiograph is to be such that the metal thickness along the incidental beam at the extremity of the exposure is not to exceed the actual thickness by more than 10 per cent.

2.8.1.6 Image quality indicators (penetrameters) of an approved type are to be placed at either end of each radiograph and on the surface of the plate facing the source of radiation.

2.8.1.7 Image quality indicators of the step-hole type are to be placed alongside the welded seam, parallel to its length, and are to have a hole in each step of a diameter corresponding to thickness at that step, or are to have some similar device whereby the step thickness can be identified when the radiographic film is examined.

2.8.1.8 The radiographic technique employed is to be such that the smallest diameter hole visible in the radiograph is not to exceed 3 per cent of the weld thickness for welds not exceeding 50 [mm] thick, or 2.5 per cent for welds exceeding 50 [mm] thick. The steps are to bear these proportions to the weld thickness radiographed, and the radiographic technique is to be capable of revealing changes of metal thickness of these percentages.

2.8.1.9 Image quality indicators of the wire type are to be placed across the weld, and the smallest diameter wire which can be seen in the radiograph is to have a diameter not greater than 1.5 per cent of the weld thickness if the weld thickness is between 10 [mm] and 50 [mm], and not greater than 1.25 per cent of the weld metal thickness if the thickness is between 50 and 200 [mm].

2.8.1.10 The use of gamma-rays may be permitted in certain circumstances, and details should be submitted for consideration and approval.

2.8.1.11 Radiographs are to be examined by the Surveyors on the original films using a viewing device of suitable illuminating power.

2.8.2 Ultrasonic examination

2.8.2.1 In Class 1 pressure vessels where plate thickness exceeds 50 [mm], ultrasonic examination may be accepted as an alternative to radiographic examination. Such examination is to be effected by an approved operator using an approved technique and an

approved recording system. Supplementary examination by radiography may be required at selected locations.

2.8.3 Magnetic particle testing

2.8.3.1 In Class 1 and Class 2 pressure vessels, the welds on standpipes, compensating plates, stubs and branches, etc. of ferritic steel which have not been radiographed are to be magnetic particle tested at the rate of 10 per cent of such welds. This may be increased or decreased, if not practicable, at the discretion of the Surveyors. For non-magnetic materials, dye penetrant examination will be accepted.

2.9 Repairs to welded seams on fusion welded pressure vessels

2.9.1 Class 1

2.9.1.1 In the case of Class 1 pressure vessels where non-destructive tests show unacceptable defects in the welded seams, the defects are to be repaired and are to be shown by further non-destructive tests to have been eliminated to the Surveyor's satisfaction.

2.9.2 Class 2

2.9.2.1 In the case of Class 2 pressure vessels, when a spot radiograph reveals unacceptable defects in the welded seam, at least two further radiographs are to be made in the length of weld represented by the first radiograph, in locations selected by the Surveyor. If these reveal no further unacceptable defects, the defects revealed by the first radiograph are to be repaired and re-radiographed. If the check radiographs reveal unacceptable defects, then either :

a) the whole length of weld represented is to be cut out and rewelded, then subjected to spot radiography as if it were a new weld, and the original test plates associated with the weld are to be similarly treated; or

b) the whole length of the weld represented is to be radiographed. Unacceptable defects are to be repaired and are to be shown by radiography to have been eliminated.

2.10 Post-weld heat treatment of pressure vessels

2.10.1 General

2.10.1.1 Dependent upon the grade of steel and plate thickness, Class 1 and Class 2 pressure vessels, where indicated in Table 2.10.1, are to be efficiently heat treated on completion of the welding of the seams and of all attachments to the shell and ends, and before the hydraulic test is carried out.

Table 2.10.1 : Post-weld heat treatment

Type of steel	Plate thicknesses [mm], above which post-weld heat treatment is required	
	Steam raising plant	Other pressure vessels
Carbon and carbon manganese steels with minimum specified tensile strengths between 360 to 510 [N/mm^2] without low temperature impact values	20	30
Carbon and carbon manganese steels with minimum specified tensile strengths between 360 to 510 [N/mm^2] with low temperature impact values of 27 J at -20°C or 10°C below design temperature whichever is lower	Not applicable	40
Alloy steels	All thicknesses to be heat treated	

2.10.2 Basic requirements

2.10.2.1 Heat treatment is to be carried out in a properly constructed furnace which is efficiently maintained and has adequate means of temperature control, and is fitted with pyrometers which will measure and record the temperature of the furnace charge.

2.10.2.2 The heat treatment is to consist of heating the vessel slowly and uniformly to a suitable stress relieving temperature, soaking for a suitable period, followed by cooling slowly and uniformly in the furnace to a temperature not exceeding 400°C and subsequently cooling in a still atmosphere.

2.10.2.3 The temperature and soaking periods are to be so selected as to relieve residual stresses without undue deterioration of the properties of the material.

2.10.2.4 Recommended soaking temperatures and periods are given in Table 2.10.2.

Table 2.10.2 : Recommended soaking temperatures and periods

Type of steel	Soaking temp. in °C	Time at recommended temperatures, per 25 [mm] of thickness
Carbon and carbon manganese	580 - 620	1 hour, minimum period 1 hour
1 Cr 1/2 Mo	620 - 660	1 hour, minimum period 2 hour
2 1/4 Cr 1 Mo	660 - 690	2 hours, minimum period 2 hours

2.10.2.5 Where materials other than those detailed in Table 2.10.2 are used for pressure vessel construction, full details of the proposed heat treatment are to be submitted for consideration.

2.10.2.6 Where pressure vessels are of such dimensions that the whole length cannot be accommodated in the furnace at one time, the pressure vessels may be heated in sections, provided that sufficient overlap is allowed to ensure the heat treatment of the entire length of the longitudinal seam.

2.10.2.7 Where it is proposed to adopt special methods of heat treatment, full particulars are to be submitted for consideration. In such cases it may be necessary to carry out tests to show the effect of the proposed heat treatment.

2.10.3 Test assemblies

2.10.3.1 Test assemblies are to be heat treated in the same furnace and at the same time as the pressure vessels which they represent.

2.10.3.2 Alternatively, it may be permissible to heat treat the test plates separately from the pressure vessels, provided that the Surveyor is satisfied with the means adopted to ensure that the following factors will be the same for the pressure vessels as for their respective test assemblies:

a) Rate of heating;

b) Maximum temperature;

c) Time held at maximum temperature; and

d) Condition of cooling.

Section 3

Welded Pressure Pipes

3.1 General

3.1.1 Manual or semi-automatic electric arc welding is to be used for butt and socket welded joints in pipes, for branch pieces and for the attachment of flanges. Oxy-acetylene welding may also be used, but, in general, is suitable only for butt joints in pipes not exceeding 100 [mm] diameter or 9.5 [mm] thickness.

3.1.2 Where pressure pipelines are assembled and butt welded in place, the piping is to be arranged well clear of adjacent structures to allow sufficient access for preheating welding, heat-treatment and examination of the joints.

Joint preparation, tolerances and edge preparation, are to be appropriate to the welding process and in accordance with recognized standards and/or approved drawings. The preparation of edges is to be carried out by mechanical means and where flame cutting is used, oxide scales, notches are to be removed by grinding or chipping back to sound metal.

3.1.3 Butt welded joints shall be full penetration type with details according to the following:

a) For class I piping:

- Double vee groove joint or
- with a backing strip or equivalent

b) For class II and III piping:

- a single vee groove joint or equivalent

The tolerances on the alignment of pipes to be welded are to be in accordance with Table 3.1.1. Higher tolerances for class III pipes may be specially considered.

3.1.4 Slip-on sleeve and socket welded joints

Slip-on sleeve and socket welded joints are to have sleeves, sockets and weldments of adequate dimensions conforming to IRS Rules or recognized standard.

Slip-on sleeve and socket welded joints may be used in Class III piping systems, of any outside diameter.

In particular cases, slip-on sleeve and socket welded joints may be allowed by IRS for piping systems of Class I and Class II having outside diameter less than or equal to 88.9 [mm] except for piping systems conveying toxic media or services where fatigue, severe erosion or crevice corrosion is expected to occur.

3.1.5 Acceptable methods of flange attachment are illustrated in Fig.2.3.1 in Ch.2.

3.1.6 Welding consumables meeting the rule requirements and, where used, fusible root inserts, are to be suitable for the materials being joined.

3.1.7 All welds in high pressure and high temperature pipelines are to have a smooth surface finish and even contour, if necessary they are to be made smooth by grinding.

3.1.8 Preheating under temperature control is to be employed in accordance with Table 3.1.2, when necessitated by the dimensions and composition of the materials to be welded and is to be effected by a method which ensures uniformity of temperature at the joint. The method of heating and the means adopted for temperature control are to be to the satisfaction of the Surveyors.

Table 3.1.1 : Tolerances on alignment of pipes

Diameter and thickness of pipes	Tolerances [mm]
All diameter and thickness with permanently fitted backing ring	0.5
Pipes welded without backing ring	
- Inside diameter < 150 mm and thickness upto 6.0 mm included	1.0
- Inside diameter < 300 mm and thickness upto 9.5 mm included	1.5
- Inside diameter 300 mm and above and thickness > 9.5 mm	2.0

Table 3.1.2 : Preheating temperatures

Type of steel	Thickness of thicker part [mm]	Minimum preheating temperature (°C)
C and C-Mn steels $C + \frac{Mn}{6} \leq 0.40$	≥ 20 (2)	50
$C + \frac{Mn}{6} \leq 0.40$	≥ 20 (2)	100
0.3 Mo	> 13 (2)	100
1 Cr – 0.5 Mo	< 13 ≥ 13	100 150
2.25 Cr – 1 Mo and 0.5 Cr – 0.5 Mo – 0.25 V (1)	< 13 ≥ 13	150 200

Notes :

1. For these materials, preheating may be omitted for thicknesses upto 6 [mm] if the results of hardness tests carried out on welding procedure qualification are considered acceptable by IRS.

2. For welding in ambient temperature below 0°C, the minimum preheating temperature is required regardless of the thickness unless specifically approved by IRS.

The values given in the Table 3.1.2 are based on the use of low hydrogen processes. Higher preheating temperatures may be employed when low hydrogen processes are not used.

3.1.9 The welding procedure proposed for the attachment of flanges, valve chests and other fittings to pipes, pipes-to-pipes and the fabrication of branch pieces, whether in carbon or alloy steel, is to be approved by the Surveyors in the first instance before work is commenced. For this purpose representative specimens of such parts will be required for examination and testing.

Tack welds forming part of the finished weld are to be made using approved, welding procedure.

3.1.10 The assembly for the weld procedure test and the welding technique should simulate the conditions under which the work is to be done on the installation. Test welds are to be examined for defects by the appropriate method specified in 3.2.2 to 3.2.4. The test welds are then to be sectioned at positions selected by the Surveyor, one surface of each section being prepared, etched and examined for defects in the weld and heat effected zones.

3.1.11 In the case of pipes of branch pieces of alloy steel, mechanical tests and tests to destruction may also be required to demonstrate that the joints are of adequate strength.

3.1.12 Check tests of the quality of the welding are to be carried out periodically at the discretion of the Surveyors.

3.2 Non-destructive examination of welded pipes

3.2.1 In addition to visual examination of pipes welds by the Surveyors, non-destructive examination of butt and fillet welds is to be carried out in accordance with 3.2.2 to 3.2.5 to the satisfaction of the Surveyors.

Radiographic and ultrasonic examination is to be carried out with an appropriate technique by trained operators using an approved procedure depending on the criticality of the joint.

3.2.2 Butt welds are to be subjected to radiographic examination in accordance with Table 3.2.1. For radiographic examinations and required standards of sensitivity, See 2.8.

3.2.3 The use of ultrasonic examination in lieu of radiographic examination as required by 3.2.2 will be specially considered.

Table 3.2.1 : Radiographic examination of butt welds

Class of pipe	Outside diameter of pipe	Extent of examination
I	≤ 76.1 [mm]	Selected welds at Surveyor's discretion
	> 76.1 [mm]	100 percent of welds
II	> 101.6 [mm]	Selected welds at Surveyor's discretion Atleast 10% 1)

Note 1) Surveyor may apply more stringent requirement depending upon kind of materials, welding procedure and controls exercised during fabrication.

Table 3.2.2 : Magnetic particle or liquid penetrant flaw detection testing of fillet welds

Class of pipe	Outside diameter of pipe	Extent of examination
I	≤ 76.1 [mm]	Selected welds at Surveyor's discretion
	> 76.1 [mm]	100 percent of welds
II	> 101.6 [mm]	Selected welds at Surveyor's discretion

3.2.4 Fillet welds are to be subjected to magnetic particle or liquid penetrant flaw detection testing in accordance with Table 3.2.2.

The equipment is to be suitable for flaw detection and may require calibration against standard samples. The acceptance criteria for welds is to be generally in accordance with the national or international standard.

3.2.5 Defects in welds are to be rectified and re-examined by the appropriate test method, all to the satisfaction of the Surveyors.

3.2.6 Ultrasonic examination may be required in special cases in addition to above non-destructive testing.

3.3 Post-weld heat treatment

3.3.1 Carbon and carbon-manganese steel pipes and fabricated branch pieces having a thickness exceeding 30 [mm] are to be heat treated on completion of fusion welding. All thicknesses of alloy steel pipes and branch pieces are to be heat treated on completion of fusion welding.

3.3.2 Recommended soaking temperatures and periods for post-weld heat treatment are given in Table 3.3.1.

3.3.3 Any proposal to use oxy-acetylene welding for the fabrication of pipes and branch pieces will be the subject of special consideration. Due consideration should be given to the need for normalizing and tempering after such welding.

3.4 Heat treatment after forming of pipes

3.4.1 Heat treatment should preferably be carried out in a suitable furnace provided with temperature recording equipment without impairing the specified properties of the materials. Tests may be carried out if necessary. Localized heat treatment of welded joints extended sufficiently along the pipe length on either side of the joint, carried out with approved procedure, can be also accepted.

Table 3.3.1 : Post-weld heat treatment

Type of steel	Heat treatment of temperature °C
Carbon and carbon-manganese	Normalizing 880 to 940
0.3 Mo	Normalizing 900 to 940
1 Cr 1/2 Mo	Normalizing 900 to 960 tempering 640 to 720
2 1/4 Cr 1 Mo	Normalizing 900 to 960 Tempering 650 to 780
0.5 Cr – 0.5 Mo – 0.25V	Normalizing 930 to 980 Tempering 670 to 720
Other alloys steels	Subject to special consideration

Table 3.3.2 : Stress relieving temperatures

Type of steel	Thickness of thicker part [mm]	Stress relief heat treatment temperature (°C)
C and C-Mn	≥ 15 (1) (3)	550 to 620
0.3 Mo	≥ 15 (1)	580 to 640
1 Cr – 0.5 Mo	> 8	620 to 680
2.25 Cr – 1 Mo and 0.5 Cr – 0.5 Mo – 0.25 V	Any (2)	650 to 720

Notes:

1. When steels with specified Charpy V notch impact properties at low temperature are used, the thickness above which postweld heat treatment shall be applied may be increased by special agreement with IRS.

2. Heat treatment may be omitted for pipes having thickness ≤ 8 [mm], diameter ≤ 100 [mm] and minimum service temperature 450°C.

3. For C and C-Mn steels, stress relieving heat treatment may be omitted upto 30 [mm] thickness by special agreement with IRS.

3.4.2 Hot forming should generally be carried out within the normalizing temperature range of 850 – 1000°C for all grades. When carried out within this temperature range no subsequent heat treatment is required for carbon, carbon-manganese and carbon-molybdenum steels. For Chrome-molybdenum and Carbon-molybdenum-vanadium steels, a subsequent stress relieving heat treatment is to be carried out in accordance with Table 3.3.2. When the hot forming is carried out outside the normalizing temperature range, a subsequent heat treatment in accordance with Table 3.3.1 is required.

3.4.3 After cold forming, when bent to a radius measured at the centerline of the pipe of less than four times the outside diameter, heat treatment in accordance with Table 3.3.1 is required.

In any case, a stress relieving heat treatment is to be carried out in accordance with Table 3.3.2 for all grades other than carbon and carbon-manganese steels having UTS 320, 360 and 410 [N/mm^2].

3.4.4 Stress relieving is to be carried out after welding for other than oxy-acetylene welding process in accordance with Table 3.3.2.

The stress relieving heat treatment is to consist in heating the piping slowly and uniformly to a temperature within the range indicated in the table, soaking at this temperature for a suitable period, in general one hour per 25 [mm] of thickness with minimum half an hour, cooling slowly and uniformly in the furnace to a temperature not exceeding 400°C and subsequently cooling in a still atmosphere.

3.4.5 For oxy-acetylene welding, heat treatment is to be carried out depending upon the type of steel in accordance with Table 3.3.1.

End of Chapter

Chapter 11

Spare Gear

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1 *General Requirements*

2 *Spare Parts Recommended for Main and Auxiliary Machinery Installations*

Section 1

General Requirements

1.1 General

1.1.1 Adequate spare parts for propelling and essential auxiliary machinery together with the necessary tools for maintenance and repair are to be readily available for use.

1.1.2 The spare parts to be supplied and their location is the responsibility of the Owners but must take into account the design and arrangements of the machinery and the intended service and operation of the ship. Account should also be taken of the recommendations of the manufacturers and any applicable statutory requirements of the country of registration of the ship.

Section 2

Spare Parts Recommended for Main and Auxiliary Machinery Installations

2.1 List of minimum required spare parts

2.1.1 The spare parts recommended for main and auxiliary machinery installations are shown in the following Tables:-

Table 2.1.1 : Spare parts for main internal combustion engines;

Table 2.1.2 : Spare parts for main steam turbines;

Table 2.1.3 : Spare parts for auxiliary steam turbines;

Table 2.1.4 : Spare parts for auxiliary internal combustion engines driving electric generators for essential services;

Table 2.1.5 : Spare parts for main and auxiliary boilers;

Table 2.1.6 : Spare parts for auxiliary machinery.

2.1.2 In ships with multi-engine installations, spare parts need only be carried for one main engine.

2.1.3 Where additional units of adequate capacity are fitted, for auxiliary machinery of each type required for essential services, no spare parts are required.

Table 2.1.1 : Spare parts for main internal combustion engines

Sr. No.	Item	Spare parts	Number required	
			Ships for unrestricted service	Ships for restricted service
1	Main bearings	Main bearings or shells for one bearing of each size and type fitted, complete with shims, bolts and nuts	1	-
2	Main thrust block	Pads for one face of Michell type thrust block, or	1 set	1 set
		Complete white metal thrust shoe of solid ring type, or	1	1
		Inner and outer race with rollers, where roller thrust bearings are fitted	1	1
3	Connecting rod bearings	Bottom end bearings or shells of each size and type fitted, complete with shims, bolts and nuts, for one cylinder	1 set	-
		Top end bearings or shells of each size and type fitted, complete with shims, bolts and nuts, for one cylinder	1 set	-
4	Cylinder liner	Cylinder liner, complete with joint rings and gaskets	1	-
5	Cylinder cover	Cylinder cover, complete with valves, joint rings and gaskets. For engines without covers the respective valves for one cylinder unit	1	-
		Cylinder cover bolts and nuts for one cylinder	Half set	-
6	Cylinder valves	Exhaust valves, complete with casings, seats, springs and other fittings for one cylinder	2 sets	1 set
		Air inlet valves, complete with casings, seats, springs and other fittings for one cylinder	1 set	1 set
		Starting air valve, complete with casing, seat, springs and other fittings	1	1
		Cylinder overpressure sentinel valve, complete	1	1
		Fuel valves of each size and type fitted complete with all fittings, for one engine	1 set see foot note 1	1/4 set
7	Pistons	Crosshead type: Piston of each type fitted, complete with piston rod, stuffing box, skirt, rings, studs and nuts	1	-
		Trunk piston type: Piston of each type fitted, complete with skirt, rings, studs, nuts, gudgeon pin and connecting rod	1	-
8	Piston rings	Piston rings, for one cylinder	1 set	-
9	Piston cooling	Telescopic cooling pipes and fittings or their equivalent, for one cylinder unit	1 set	-
10	Cylinder lubricators	Lubricator complete, of the largest size, with its chain drive or gear wheels	1	-
11	Fuel injection pumps	Fuel pump complete, or when replacement at sea is practicable, a complete set of working parts for one pump (plunger, sleeve, valves springs etc).	1	-
12	Fuel injection piping	High pressure fuel pipe of each size and shape fitted, complete with couplings	1	-
13	Scavenge blowers (including turbochargers)	Rotors, rotor shafts, bearings, nozzle rings and gear wheels or equivalent working parts of other types	1 set see foot note 2	-
14	Scavenging system	Suction and delivery valves for one pump of each type fitted	1 set	-

Table 2.1.1 : Spare parts for main internal combustion engines (Contd.)

Sr. No.	Item	Spare parts	Number required	
			Ships for unrestricted service	Ships for restricted service
15	Reduction and/or reverse gear	Complete bearing bush, of each size fitted in the gear case assembly	1 set	-
		Roller or ball race, of each size fitted in the gear case assembly	1 set	-
16	Main engine driven air compressors	Piston rings of each size fitted	1 set	-
		Suction and delivery valves complete of each size fitted	Half set	-
17	Special gaskets and packing of each size and type fitted for cylinder cover and cylinder liner for one cylinder		1 set	-

Footnotes :

1 (a) Engines with one or two fuel valves per cylinder: one set of valves, complete.

(b) Engines with 3 or more fuel valves per cylinder : two fuel valves complete per cylinder and a sufficient number of valve parts, including the body, to form with those fitted in the complete valves, a full engine set.

2 The spare parts may be omitted where it has been demonstrated, at the builder's test bench for one engine of the type concerned that the engine can be maneuvered satisfactorily with one blower out of action.

The requisite blanking and blocking arrangements for running with the blower out of action are to be available on board.

Notes

1 The availability of other spare parts, such as gears and chains for camshaft drive, should be specially considered and decided upon by the Owners.

2 When the spare parts are utilized, new spare parts are to be supplied as soon as possible.

Table 2.1.2 : Spare parts for steam turbines for main propulsion

Sr. No.	Item	Spare parts	Number required	
			Ships for unrestricted service	Ships for restricted service
1	Main bearing	Bearing bushes, of each size and type fitted for the rotor, pinion and gear wheel shafts, for one engine	1	-
2	Turbine thrust	Pads of each size for one face of Michell type thrust or rings for turbine adjusting block, or each size fitted for one engine. Assorted liners for 1 block where fitted	1 set	1 set
3	Main thrust block	Pads for one face of Michell type thrust block, or	1 set	1 set
		Complete white metal thrust shoe of solid ring type, or	1	1
		Inner and outer race with rollers where roller thrust bearings are fitted	1	1
4	Turbine shaft	Carbon sealing rings, where fitted, with springs for each size of sealing rings and type of gland	1 set	1 set
5	Oil filters	Strainer baskets or inserts for filters of special design of each type and size	1 set	1 set

Notes

1 The availability of other spare parts should be specially considered and decided upon by the Owners.

2 When the spare parts are utilized, new spare parts are to be supplied as soon as possible.

Table 2.1.3 : Spare parts for auxiliary steam turbines

Sr. No.	Item	Spare parts	Number required	
			Ships for unrestricted service	Ships for restricted service
1	Main bearings	Bearing bushes or roller bearings of each size and type fitted for the shafts of the turbine rotor and of the reduction gearing, if any, for one engine	1	1
2	Turbine thrust	Pads of each size for one face of Michell type thrust or rings for turbine adjusting block, or each size fitted for one engine. Assorted liners for 1 block where fitted	1 set	1 set
3	Turbine shaft	Carbon sealing rings, where fitted, with springs, for each size of sealing rings and type of gland, for one engine	1 set	1 set
4	Oil filters	Strainer baskets or inserts, for filters of special design, of each type and size	1 set	1 set

Notes

1 The availability of other spare parts should be specially considered and decided upon by the Owners.

2 When the spare parts are utilized, new spare parts are to be supplied as soon as possible.

Table 2.1.4 : Spare parts for auxiliary internal combustion engines driving electric generators for essential services

Sr. No.	Item	Spare parts	Number required	
			Ships for unrestricted service	Ships for restricted service
1	Main bearings	Main bearings or shells for one bearing of each size and type fitted, complete with shims, bolts and nuts	1	-
2	Cylinder valves	Exhaust valves, complete with casings, seats, springs and other fittings for one cylinder	2 sets	-
		Air inlet valves, complete with casings, seats, springs and other fittings for one cylinder	1 set	-
		Starting air valve, complete with casing, seat, springs and other fittings	1	-
		Cylinder overpressure sentinel valve, complete	1	-
		Fuel valves of each size and type fitted complete with all fittings, for one engine	Half set	-
3	Connecting rod bearings	Bottom end bearings or shells of each size and type fitted, complete with shims, bolts and nuts, for one cylinder	1 set	-
		Top end bearings or shells of each size and type fitted, complete with shims, bolts and nuts, for one cylinder	1 set	-
		Trunk piston type : gudgeon pin with bush for one cylinder	1 set	-
4	Piston rings	Piston rings, for one cylinder	1 set	-
5	Piston cooling	Telescopic cooling pipes and fittings or their equivalent, for one cylinder unit	1 set	-
6	Fuel injection pumps	Fuel pump complete, or when replacement at sea is practicable, a complete set of working parts for one pump (plunger, sleeve, valves springs etc).	1	-
7	Fuel injection piping	High pressure fuel pipe of each size and shape fitted, complete with couplings	1	-
8	Gaskets and packings	Special gaskets and packings of each size and type fitted, for cylinder covers and cylinder	1 set	-

Table 2.1.5 : Spare parts for boilers (main and auxiliary)				
Sr. No.	**Item**	**Spare parts**	**Number required**	
			Ships for unrestricted service	**Ships for restricted service**
1	Tube stoppers or plugs	Tube stoppers or plugs, of each size used, for boiler superheater and economiser tubes	20	10
2	Fire bars	Fire bars for one boiler, where coal fired	1 set	Half set
3	Oil fuel burners	Oil fuel burners complete, for one boiler	1 set	1 set
4	Gauge glasses	Gauge glasses of round type	2 sets per boiler	2 sets per boiler
		Gauge glasses of flat type	1 set for every two boilers	1 set for every two boilers

Table 2.1.6 : Spare parts for auxiliary machinery				
Sr. No.	**Item**	**Spare parts**	**Number required**	
			Ships for unrestricted service	**Ships for restricted service**
Pumps				
1	**Reciprocating pumps**			
1.1	Valves	Valve with seats and springs, each size fitted	1 set	1 set
1.2	Piston rings	Piston rings, each type and size for one piston	1 set	1 set
2	**Centrifugal pumps**			
2.1	Bearings	Bearing of each type and size	1	1
2.2	Rotor	Rotor sealing of each type and size	1	1
3	**Gear type pumps**			
3.1	Bearing	Bearing of each type and size	1	1
3.2	Rotor	Rotor sealing of each type and size	1	1
Air compressor				
1	Piston rings	Rings of each size fitted for one piston	1 set	1 set
2	Valves	Suction and delivery valves, complete, of each size fitted	Half set	Half set

Notes

1 The availability of other spare parts should be specially considered and decided upon by the Owners.

2 When the spare parts are utilized, new spare parts are to be supplied as soon as possible.

3 When a sufficiently rated standby pump is available, spare part for the pump may be dispensed with.

End of Chapter

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50 YEARS OF EXCELLENCE

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 5
SPECIAL SHIP TYPES – VOLUME I

July 2024

Indian Register of Shipping

Part 5 : Special Ship Types

This Part (Part 5) of the Rules is arranged in two Volumes (Volume I and Volume II) . Chapters 1 to 14 are covered in this Volume. Chapters 15 to 20 are intentionally left blank. For Chapter 21 onwards, please refer Volume II)

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Chapter 1

Dry Bulk Cargo Carriers

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Section 1

General

For symbols not defined in this section, Refer to Pt.3, Ch.1, Sec.2.

1.1 Application, definitions, class notations and documentation

1.1.1 Application

1.1.1.1 The requirements of this chapter apply in general to all bulk carriers as defined in Cl.1.1.2.1 except for hull structures of those vessels mentioned in Cl.1.1.1.2 below. The requirements are supplementary to those given for the assignment of main character of class.

1.1.1.2 The IACS common structural rules (CSR), for bulk carriers which are covered in IRS “Rules for Bulk Carriers and Oil Tankers” are to be applied to hull structures of seagoing self-propelled bulk carriers of unrestricted service and having a length L≥90[m], which are constructed generally with single deck, double bottom, hopper side tanks and topside tanks and with single or double skin construction in cargo area, excluding ore and combination carriers. The CSR is also applicable to hybrid bulk carriers, when at least one cargo hold is constructed with hopper tank and top side tank.

Accordingly, the requirements of this chapter will not apply to CSR bulk carriers.

1.1.1.3 The requirements of SOLAS Chapter XII - Additional safety measures for bulk carriers, apply to bulk carriers of "single side skin construction" and of “double side skin construction” as defined in Cl.1.1.2.3.

1.1.1.4 The applicable requirements of International Maritime Solid Bulk Cargoes (IMSBC) Code are also to be complied with.

1.1.2 Definitions

1.1.2.1 Bulk carrier means a ship which is intended primarily to carry dry cargo in bulk, including such types as ore carriers and combination carriers.

1.1.2.2 “Bulk carrier of single side skin construction” means a bulk carrier in which:

a) any part of a cargo hold is bounded by the side shell; or

b) where one or more cargo holds are bounded by a double side skin, the width of which is less than 760[mm] in bulk carriers constructed before 1 January, 2000 and less than 1000[mm] in bulk carriers constructed on or after 1 January, 2000 but before 1 July, 2006, the distance being measured perpendicular to the side shell. Such ships include combination carriers in which any part of a cargo hold is bounded by the side shell.

1.1.2.3 “Bulk carriers of double side skin construction” means a bulk carrier in which all cargo holds are bounded by a double-side skin, other than as defined in b) above.

1.1.2.4 “Double side skin” means a configuration where each ship side is constructed by the side shell and longitudinal bulkhead connecting the double bottom and the deck. Hopper side tanks and top-side tanks may, where fitted, be integral parts of the double side skin construction. Double side skin construction is to comply with the requirements of 1.10.

1.1.3 Class notations

1.1.3.1 When the IACS Common Structural Rules (CSR) are applied as mentioned in Cl.1.1.1.2 above, the vessel will be eligible to be assigned class notation ‘**CSR**’ and additional notations as per the Common Structural Rules, as applicable.

1.1.3.2 In general, bulk carriers other than those to which CSR is applied built in compliance with the above requirements, as applicable, will be eligible to be assigned one of the following class notations:

BULK CARRIER, ESP
BULK CARRIER, "Strengthened for heavy Cargoes", ESP
BULK CARRIER, "Strengthened for heavy Cargoes, hold(s) ... (to be specified) **... may be empty", ESP.**

1.1.3.3 The notation "BULK CARRIER" implies that the vessel is largely designed for homogeneous loading in all holds and any hold may be filled up to the top of hatch coaming with bulk cargo of density at least up to 0.8[t/m^3] when draught in way of the hold is 0.6T_L or more. It is also implied that no hold will be empty when the draught in way of the hold is greater than 0.6T_L.

1.1.3.4 The notation "BULK CARRIER", "Strengthened for heavy cargoes" implies that the vessel is designed for heavier bulk cargo loadings due to uneven distributions among the cargo holds and any hold may be filled up to the top of hatch coaming with bulk cargo of density at least up to 1.0[t/m^3] when draught in way of the hold is 0.8T_L or more. It is also implied that no hold will be empty when the draught in way of the hold is greater than 0.8T_L.

1.1.3.5 The notation "BULK CARRIER", "Strengthened for heavy cargoes, hold(s) ... (to be specified) ... may be empty" implies that the vessel is also designed for heavy bulk cargo loading in non-homogeneous loading conditions with approved arrangement of empty holds at the fully loaded draught T_L. Where found appropriate, the 'specification' of empty hold(s) may be suitably worded to include the requested combinations of empty holds. e.g. "(Any hold may be empty)" or "(Holds 2, 4, 6 or 3, 5, 7 may be empty)".

1.1.3.6 Assignment of class notation **ESP** (Enhanced Survey Program) is mandatory for the following:

a) sea going self-propelled bulk carriers of single or double skin construction in cargo length area, with single deck, double bottom, hopper side tanks and topside tanks and intended primarily to carry dry cargoes in bulk. Typical cross sections of such bulk carriers are given in Figure 1.1.3.6(a). Bulk Carriers as described above will be eligible to be assigned class notation “**BULK CARRIER, ESP” Class** notation ‘**ESP**’ may be assigned to other types of bulk carriers, when required by the Administration.

b) sea going self-propelled ore carriers with single deck, two longitudinal bulkheads and a double bottom throughout the cargo length area and intended primarily to carry ore cargoes in the centre holds only. Typical midship section of such ore carriers is given in Figure 1.1.3.6(b). Ore carriers as described above will be eligible to be assigned class notation "**ORE CARRIER, ESP**"

c) sea going self-propelled self-unloading bulk carriers with single deck, double bottom, hopper side tanks and topside tanks and with single or double side skin construction in cargo length area and intended to carry and self-unload dry cargoes in bulk. Typical midship sections are given in Figure 1.1.3.6(c). Self-unloading bulk carriers as described above will be eligible to be assigned class notation “**SELF-UNLOADER, ESP**”.

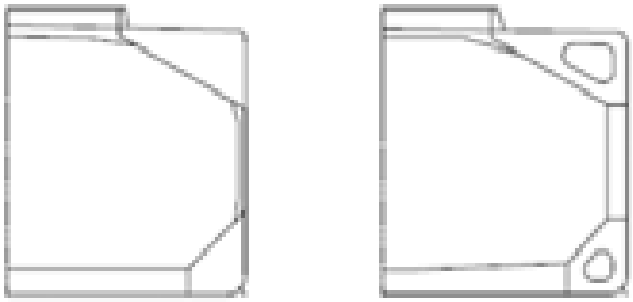


Figure 1.1.3.6(a): Bulk Carriers

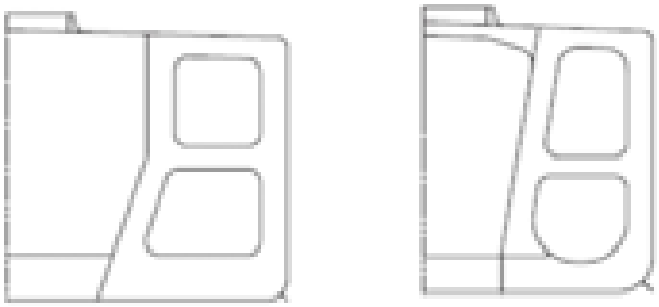


Figure 1.1.3.6(b): Ore Carriers

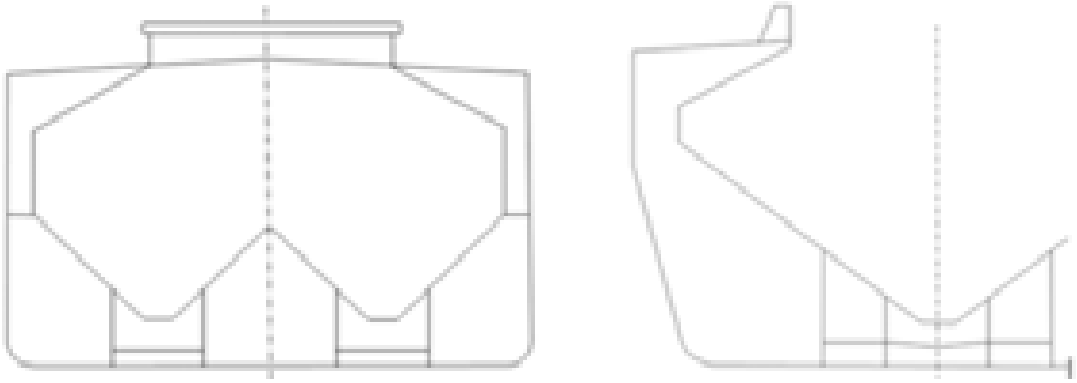


Figure 1.1.3.6(c): Self-Unloading Bulk Carriers

1.1.3.7 Bulk carriers which are in accordance with the definition of SOLAS Ch XII and its construction features not being as per the definition of SOLAS Ch IX, Reg 1.6. (i.e. not fitted with hopper and topside tanks) will be eligible for class notation **BULK CARRIER (BC-XII)** (Note: Vessels with **BULK CARRIER, ESP** notation will meet the definition in SOLAS Ch.XII as well as Ch IX).

1.1.4 Documentation

1.1.4.1 The following additional documents are to be submitted for approval, as applicable:

a) Plan showing the cross sections taken at mid-length of each hold and the hold capacities upto the top of hatch coaming.

b) Maximum allowable and minimum required mass of cargo and double bottom contents of each hold as a function of the draught at mid-hold position.

c) Design values of maximum density of bulk cargo to be carried in each hold when the cargo is filled upto top of the hatch coaming. Corresponding angles of repose to be indicated.

d) Details of all (mandatory and other envisaged) homogeneous and/or non-homogeneous loading conditions giving various combinations of filling levels in holds, combinations of empty and loaded holds including possible part loading/unloading conditions in the homogeneous mode and loading conditions where two adjacent holds are loaded fully with adjacent holds empty (block loading). In ballast conditions where water ballast is intended to be carried in holds, details of filling levels are to be included. Calculations of still water bending moments and shear forces in these conditions are to be submitted along with the limiting values to be included in the loading manual and the loading instrument.

e) Where applicable, calculations of still water bending moments and shear forces in flooded conditions, as detailed in Cl.1.3.3.2.

f) Typical cargo loading/unloading sequences and sequences for change of ballast at sea as indicated in Cl.1.2.3(g) and (h) respectively.

1.2 Loading guidance information - additional requirements

1.2.1 Bulk carriers, ore carriers and combination carriers of length L≥150[m], are to comply with the following in addition to the general requirements given in Pt.3, Ch.1, Sec.4

i) The loading manual is to contain the additional information as per Cl.1.2.2 and also include additional loading conditions given in Cl.1.2.3; and

ii) The computer-based loading instrument is to have additional capabilities as per Cl.1.2.4 and its approval is to subjected to the additional conditions of approval given in Cl.1.2.5.

1.2.2 Loading manual shall contain complete information with respect to:

a) The loading conditions on which the design of the ship has been based, including permissible limits of still water bending moments and shear forces;

b) The results of the calculations of still water bending moments, shear forces and where applicable, limitations due to torsional loads;

c) Where the longitudinal strength in flooded condition is required to be checked according to Sec.2, Cl.2.2.2, envelope results and permissible limits of still water bending moments and shear forces in the hold flooded condition;

d) The cargo hold(s) or combination of cargo holds that might be empty at full draught. If no cargo hold is allowed to be empty at full draught, this is to be clearly stated in the loading manual;

e) Maximum allowable and minimum required mass of cargo and double bottom contents of each hold as a function of the draught at mid-hold position.

f) Maximum allowable and minimum required mass of cargo and double bottom contents of any two adjacent holds as a function of the mean draught in way of these holds. This mean draught may be calculated by averaging the draught of the two mid-hold positions.

g) Maximum allowable tank top loading together with specifications of the nature of the cargo for cargoes other than bulk cargoes;

h) Maximum allowable load on deck and hatch covers. If the vessel is not approved to carry load on the deck or hatch covers, this is to be clearly stated in the loading manual;

i) The maximum rate of ballast change together with the advice that a load plan is to be agreed with the terminal on the basis of the achievable rates of change of ballast.

1.2.3 The following additional conditions, subdivided into departure and arrival conditions as appropriate, are to be included in the Loading Manual:

a) Alternate light and heavy cargo loading conditions at maximum draught, where applicable;

b) Homogeneous light and heavy cargo loading conditions at maximum draught;

c) Ballast conditions. For vessels having ballast holds adjacent to topside wing, hopper and double bottom tanks, it shall be strengthwise acceptable that the ballast holds are filled when the topside wing, hopper and double bottom tanks are empty. Partial filling of the peak tanks is not acceptable in the design ballast conditions unless effective means are provided to prevent accidental overfilling;

d) Short voyage conditions where the vessel is to be loaded to maximum draught but with limited amount of bunkers;

e) Multiple port loading/unloading conditions;

f) Deck cargo conditions, where applicable;

g) Typical loading/unloading sequences as detailed in Cl.1.2.4;

h) Typical sequences for change of ballast at sea, where applicable.

1.2.4 Typical loading/unloading sequences are to be prepared, considering the various intermediate stages from commencement to completion of the loading/unloading operation relevant to all envisaged loading conditions including the following:

- alternate light and heavy cargo load condition,
- homogeneous light and heavy cargo load condition,
- short voyage condition where the ship is loaded to maximum draught but with limited bunkers,
- multiple port loading/unloading condition,
- deck cargo condition,
- block loading.

The loading/unloading sequences may be port specific or typical.

The sequence is to be built up step by step from commencement of cargo loading to reaching full deadweight capacity. Each time the loading equipment changes position to a new hold defines a step. Each step is to be documented and submitted. In addition to longitudinal strength, the local strength of each hold is to be considered.

For each loading condition a summary of all steps is to be included. This summary is to highlight the essential information for each step such as:

- How much cargo is filled in each hold during the step.

- How much ballast is discharged from each ballast tank during the step.

- The maximum still water bending moment and shear at the end of the step.

- The ship's trim and draught at the end of the step.

1.2.5 The digital loading instrument shall also be capable of ascertaining, as applicable, the following:

- The mass of cargo and double bottom contents in way of each hold as a function of the draught at mid-hold position;

- The mass of cargo and double bottom contents of any two adjacent holds as a function of the mean draught in way of these holds;

- Where the longitudinal strength in flooded condition is required to be checked according to Cl.2.2.2, the still water bending moment and shear forces in the hold flooded conditions are within permissible values

1.2.6 The approval of loading instrument is to be based on the following additional conditions, as applicable:

a) Acceptance of hull girder bending moment limits for all read-out points.

b) Acceptance of hull girder shear force limits for all read-out points.

c) Acceptance of limits for mass of cargo and double bottom contents of each hold as a function of draught.

Acceptance of limits for mass of cargo and double bottom contents in any two adjacent holds as a function of draught.

1.3 Design loads in cargo holds

Requirements of this subsection are in lieu of Pt.3, Ch.5, 5.4.

1.3.1 Hold Definitions and Dry cargo loading parameters

For symbols not defined in this section, Refer to Pt.3, Ch.1, Sec.2.

M_{sw-f} = Permissible vertical still water bending moment in flooded condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Cl.1.3.3.2(a).

Q_{sw-f} = Positive and negative permissible vertical still water shear force in flooded condition, in [kN], at the hull transverse section being considered, as defined in Cl.1.3.3.2(a).

M_{wv-f} = Permissible vertical wave bending moment in flooded condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Cl.1.3.3.2(b).

Q_{wv-f} = Positive and negative permissible vertical wave shear force in flooded condition, in [kN], at the hull transverse section being considered, as defined in Cl.1.3.3.2(b).

K_{C-f} = Coefficient taken as:

$$K_{C-f} = \tan^2\left(45 - \frac{\Psi}{2}\right)$$

Ψ = Assumed angle of repose of bulk cargo (considered drained and removed) in [deg], as given in Table 1.3.3.2.

S_c = Half pitch of the corrugation flange, in [mm] as defined in Pt.3, Ch.7 Sec.4, Figure 4.1.2.

α = Angle, in [deg], between panel considered and the horizontal plane. The angle is to be measured inward and upward from horizontal plane outside cargo hold, between 0 and 180 [deg], as shown in Figure 1.3.1.

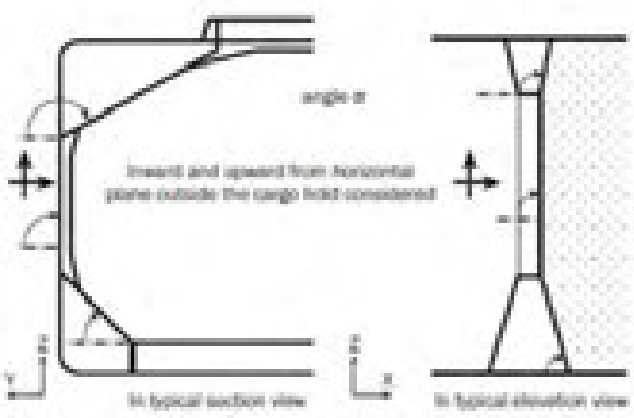


Figure 1.3.1: Measurement of angle α

1.3.1.1 Hold Definitions

(a) Geometrical characteristics

The main geometrical elements of dry bulk cargo hold with different arrangements of bulkheads are given in Figure 1.3.1.1(a) to Figure 1.3.1.1(c).

(b) Fully and partially filled cargo holds

The definitions of a fully and partially filled dry bulk cargo holds are as follows:

(i) Fully filled hold:

The dry bulk cargo density is such that the cargo hold is filled up to the top of the hatch coaming, as shown in:

- cargo hold with hopper and top wing tank: Figure 1.3.1.2(a1)
- box shaped cargo hold: Figure 1.3.1.2(a2)
- ore carrier with slanted longitudinal bulkhea d: Figure 1.3.1.2(a3).

The upper surface of the cargo and its effective height in the hold h_c are to be determined in accordance with Cl.1.3.1.2(a).

(ii) Partially filled hold:

The cargo density is such that the cargo hold is not filled up to the top of the hatch coaming, as shown in:

- cargo hold with hopper and top wing tank: Figure 1.3.1.2(b1) or Figure 1.3.1.2(b2)
- box shaped cargo hold: Figure 1.3.1.2(b3)
- ore carrier with slanted longitudinal bulkhead: Figure 1.3.1.2(b4).

The upper surface of the cargo and its effective height in the hold h_c are to be determined in accordance with Cl.1.3.1.2(b).

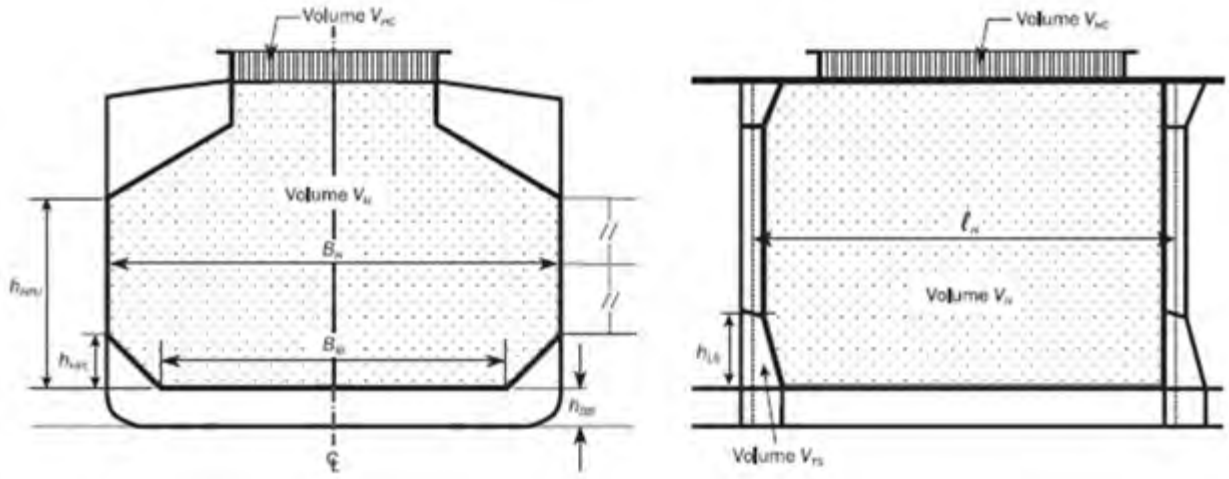


Figure 1.3.1.1(a): Definition of cargo hold parameters for bulk carrier (Cargo hold with hopper tank and top wing tank)

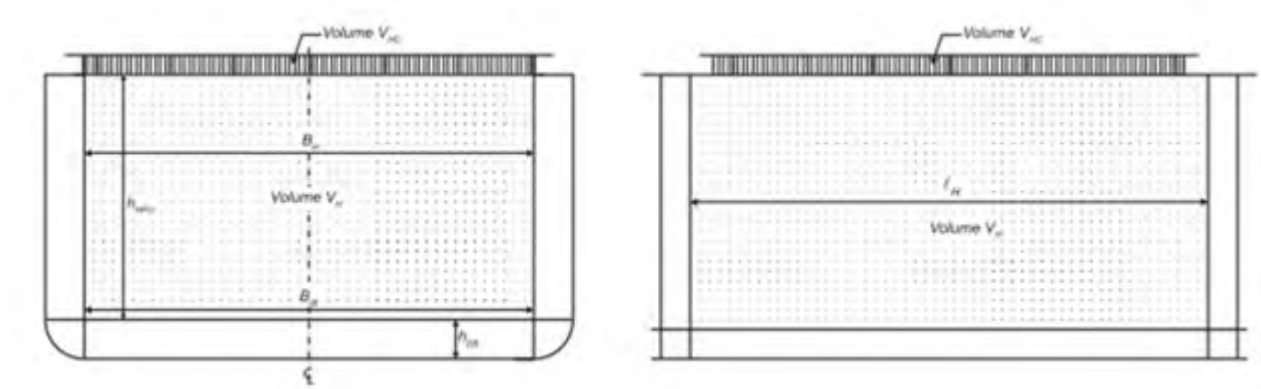


Figure 1.3.1.1(b): Definition of cargo hold parameters (Box shaped cargo hold)

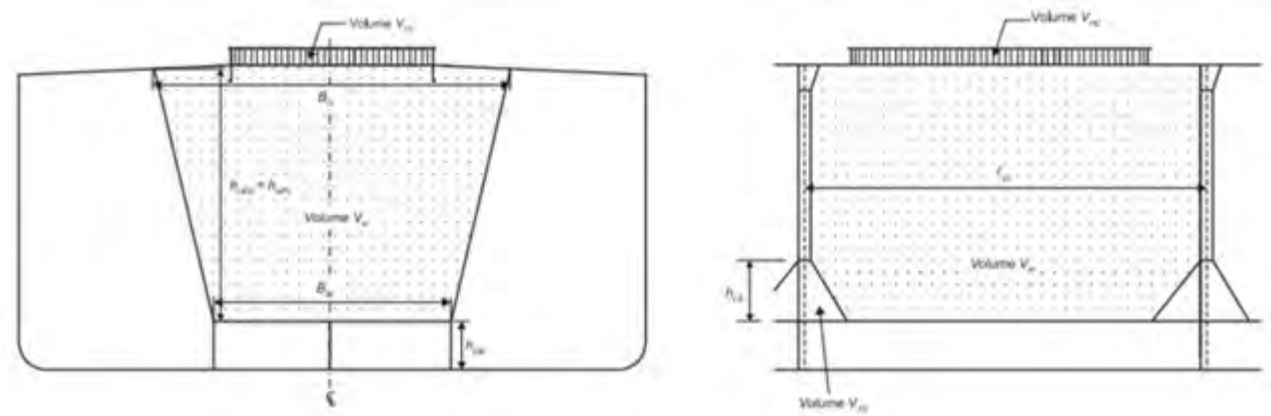


Figure 1.3.1.1(c): Definition of cargo hold parameters (Ore carrier with slanted longitudinal bulkhead)

1.3.1.2 Dry cargo loading parameters

1.3.1.2(a) Definition of the upper surface of dry bulk cargo for full cargo holds

For a fully filled cargo hold as defined in Cl.1.3.1.1(b), including non-prismatic holds, the effective upper surface of the cargo is an equivalent horizontal surface at h_c, in [m], above inner bottom at centreline as shown in Figure 1.3.1.2(a1) to Figure 1.3.1.2(a3).

The value of h_c is to be calculated at mid length of the cargo hold at the midship, is to be kept constant over the cargo hold region area and is determined as follows:

$$h_c = h_{HPU} + h_O$$

where,

$$h_O = \frac{S_A}{B_H}$$

$$S_A = S_O + \frac{V_{HC}}{\ell_H}$$

S_O = Shaded area, in [m^2], given below:

Figure 1.3.1.2(a1): S_O = shaded area above the lower intersection of top side tank and side shell or inner side, as the case may be, and up to the level of the intersection of the main deck with the hatch coaming, determined for the cargo hold at the midship.

Figure 1.3.1.2(a2): $S_O = 0$

Figure 1.3.1.2(a3): S_O = shaded area above the intersection of longitudinal bulkhead and main deck and up to the level of the intersection of the main deck with the hatch coaming, determined for the cargo hold at the midship

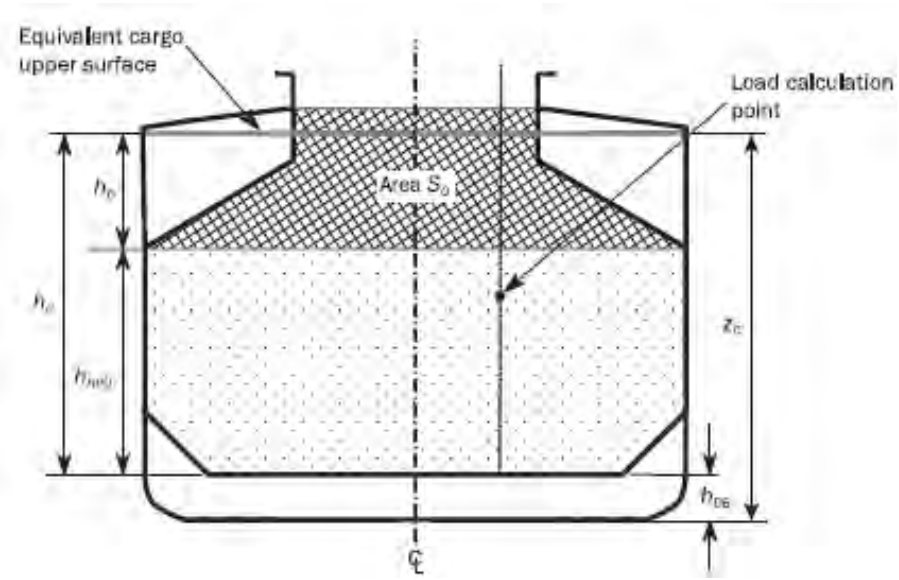


Figure 1.3.1.2(a1): Definition of effective upper surface of cargo for a full cargo hold (Cargo hold with hopper tank and top wing tank)

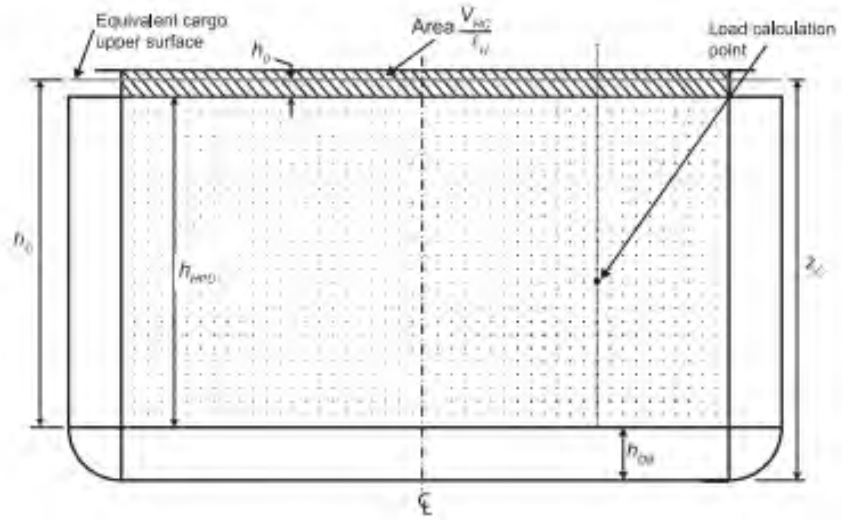


Figure 1.3.1.2(a2): Definition of effective upper surface of cargo for a full cargo hold (Box shaped cargo hold)

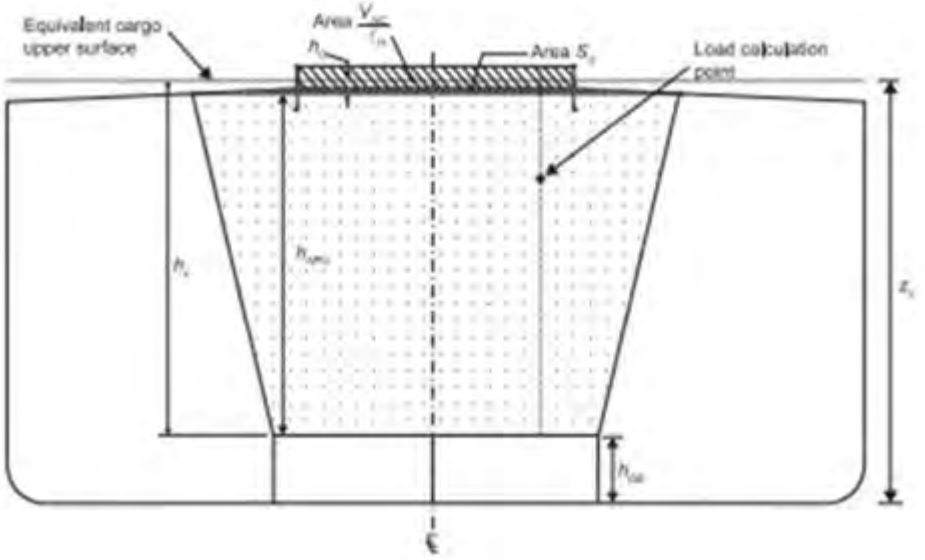


Figure 1.3.1.2(a3): Definition of effective upper surface of cargo for a full cargo hold (Ore carrier with slanted longitudinal bulkhead)

1.3.1.2(b) Definition of upper surface of dry bulk cargo for partially filled cargo holds

For any partially filled cargo hold, as defined in Cl.1.3.1.1(b), including non-prismatic holds, the effective upper surface of the cargo is to be made of three parts:

- One central horizontal surface of breadth $0.5B_H$, in [m], at a height h_{C-CL}, in [m], above the inner bottom
- A sloped surface at each side with an angle ψ/2, in [deg], between the central horizontal surface, and the side shell or inner hull, as shown in Figure 1.3.1.2(b1), Figure 1.3.1.2(b3) and Figure 1.3.1.2(b4) or the hopper plating, as shown in Figure 1.3.1.2(b2), as the case may be.

The height of cargo surface h_c, in [m], is to be calculated at mid length of the considered cargo hold and is to be taken as constant over the length of the hold as follows:

For $|y| \leq 0.25B_H$:

$$h_c = h_{C-CL}$$

For $0.25B_H < |y| \leq 0.5B_2$:

$$h_c = h_{C-CL} - (|y| - 0.25B_H)\tan\frac{\psi}{2}$$

For $|y| > 0.5B_2$:

$$h_c = 0$$

h_1 = Height, in [m], to be taken as:

$$h_1 = \frac{M}{\rho_c . B_H . \ell_H} - \left(\frac{B_H + B_{IB}}{2B_H}\right) h_{HPL} - \frac{3}{16} B_H \tan\frac{\psi}{2} + \frac{V_{TS}}{B_H . \ell_H}$$

- For $h_1 \geq 0$ as shown in Figure 1.3.1.2(b1) and Figure 1.3.1.2(b3):

$$h_{C-CL} = h_{HPL} + h_1 + h_2$$

$$h_2 = 0.25 B_H \tan\frac{\psi}{2}$$

$$B_2 = B_H$$

- For $h_1 < 0$ as shown in Figure 1.3.1.2(b2) and Figure 1.3.1.2(b4):

$$h_{C-CL} = h_{11} + h_{22}$$

$$h_{11} = h_{HPL}\left(\frac{B_2 - B_{IB}}{B_H - B_{IB}}\right)$$

$$h_{22} = \left(\frac{B_2}{2} - \frac{B_H}{4}\right)\tan\frac{\psi}{2}$$

$$B_2 = \frac{\left(\frac{M}{\rho_c} + V_{TS}\right)\frac{1}{\ell_H} + 0.5\left(\frac{h_{HPL} . B_{IB}^2}{B_H - B_{IB}}\right) + \left(\frac{B_H^2}{16}\tan\frac{\psi}{2}\right)}{0.5\left[\left(\frac{h_{HPL}}{B_H - B_{IB}}\right) + 0.5\tan\frac{\psi}{2}\right]}$$

where,

h_{C-CL}= Height, in [m], of the cargo surface at the centreline, as shown in Figure 1.3.1.2(b1) to Figure 1.3.1.2(b4).

M = Mass, in [t], of the bulk cargo being considered.

B_2 = Maximum breadth of the cargo, in [m], as shown in Figure 1.3.1.2(b2) and Figure 1.3.1.2(b4).

ρ_c = Density of bulk cargo, in [t/m^3] as defined in Cl.1.3.1.3.

B_H = Breadth of the cargo hold, as defined below:

For holds with hopper tank and top wing tank:
B_H= Breadth of the cargo hold in [m], measured at mid-length of the cargo hold and at the mid-height between the top of hopper tank and the bottom of topside tank as shown in Figure 1.3.1.1(a).

For holds with vertical inner side connected to inner bottom:
$B_H = B_{IB}$, as shown in Figure 1.3.1.1(b).

For holds with slanted longitudinal bulkhead connected to inner bottom:
B_H = Breadth of the cargo hold, in [m], measured at mid-length of the cargo hold and at the intersection of longitudinal bulkhead and main deck, as shown in Figure1.3.1.1(c).

B_{IB}= breadth of inner bottom, in [m], measured at mid-length of the cargo hold, Refer Figure 1.3.1.1(a) to Figure 1.3.1.1(c).

h_{HPL} = As defined below:

For holds with vertical inner side connected to inner bottom:
h_{HPL} = 0

For holds with slanted longitudinal bulkhead connected to inner bottom:
$h_{HPL} = h_{HPU}$

For holds with hopper tank:
h_{HPL} = Vertical distance, in [m], from the inner bottom at centreline to the upper intersection of hopper tank and side shell or inner side for double side ships, determined at mid length of the considered cargo hold, as shown in Figure 1.3.1.1(a).

h_{HPU} = As defined below:
For cargo holds with no top wing tank:
h_{HPU} = Vertical distance, in [m], from the inner bottom at centreline to the intersection of longitudinal bulkhead and main deck, determined at mid length of the cargo hold at midship, as shown in Figure 1.3.1.1(b) and Figure 1.3.1.1(c).

For cargo holds with top wing tank:
h_{HPU} = Vertical distance, in [m], from the inner bottom at centreline to the lower intersection of topside tank and side shell or inner side for double side ships, determined at mid length of the cargo hold at midship, as shown in Figure 1.3.1.1(a).

V_{TS} = Total volume, in [m^3], of the portion of the lower bulkhead stools within the cargo hold length ℓ_H and inboard of the hopper tanks.

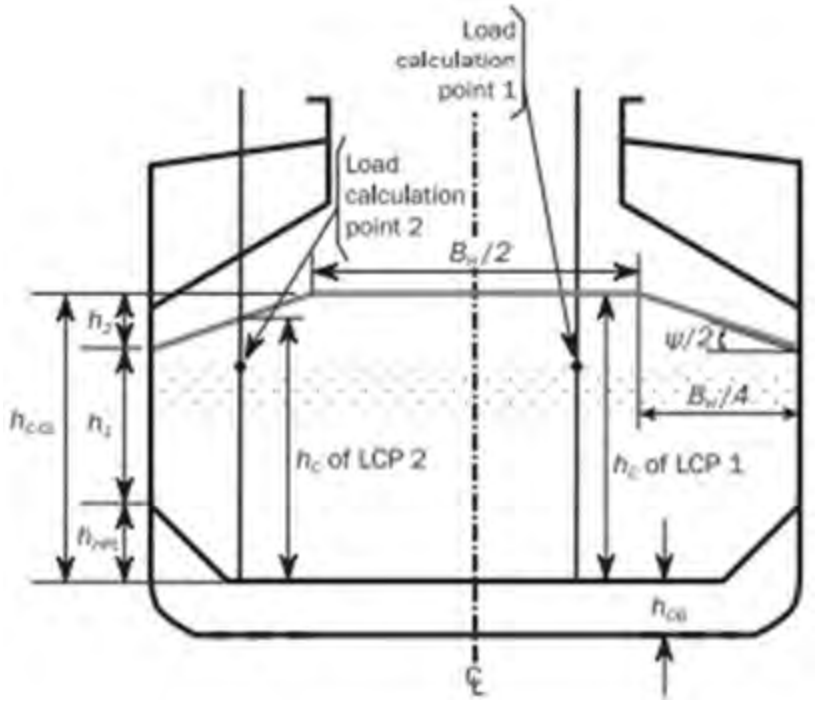


Figure 1.3.1.2(b1): Definition of effective upper surface of cargo for a partially filled cargo hold, when $\mathbf{h_1 \geq 0}$ (Cargo hold with hopper tank and top wing tank)

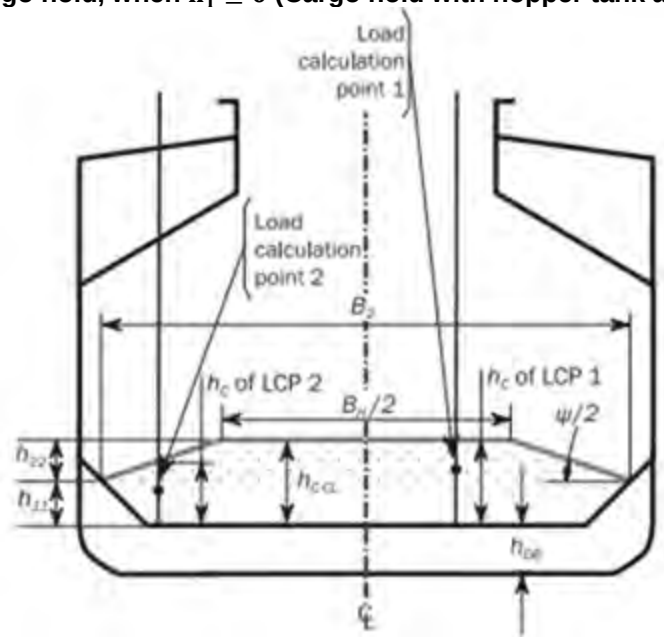


Figure 1.3.1.2(b2): Definition of effective upper surface of cargo for a partially filled cargo hold, when $\mathbf{h_1 < 0}$ (Cargo hold with hopper tank and top wing tank)

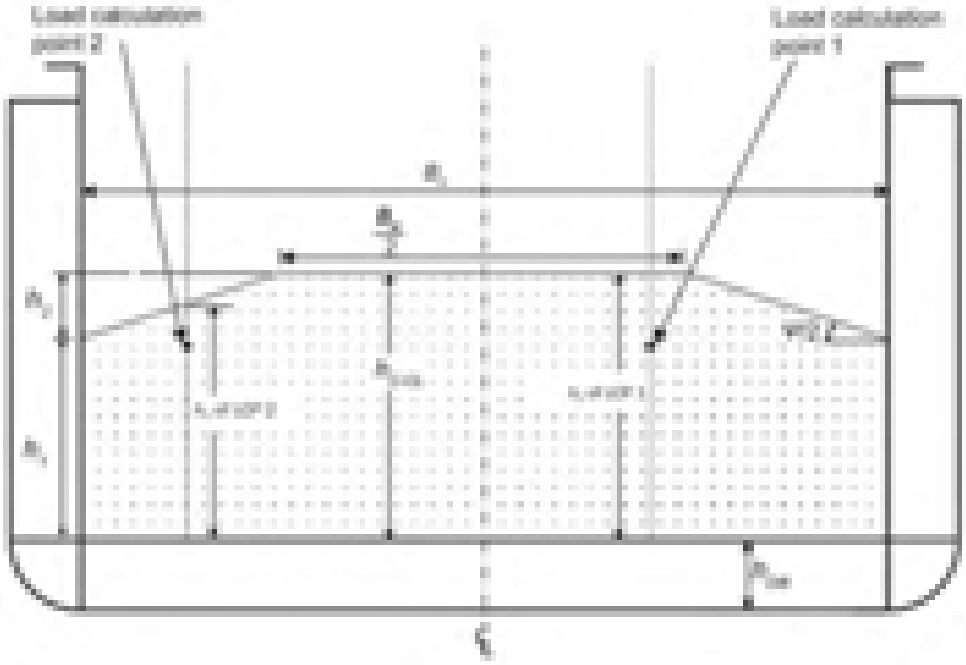


Figure 1.3.1.2(b3): Definition of effective upper surface of cargo for a partially filled cargo hold (Box shaped cargo hold)

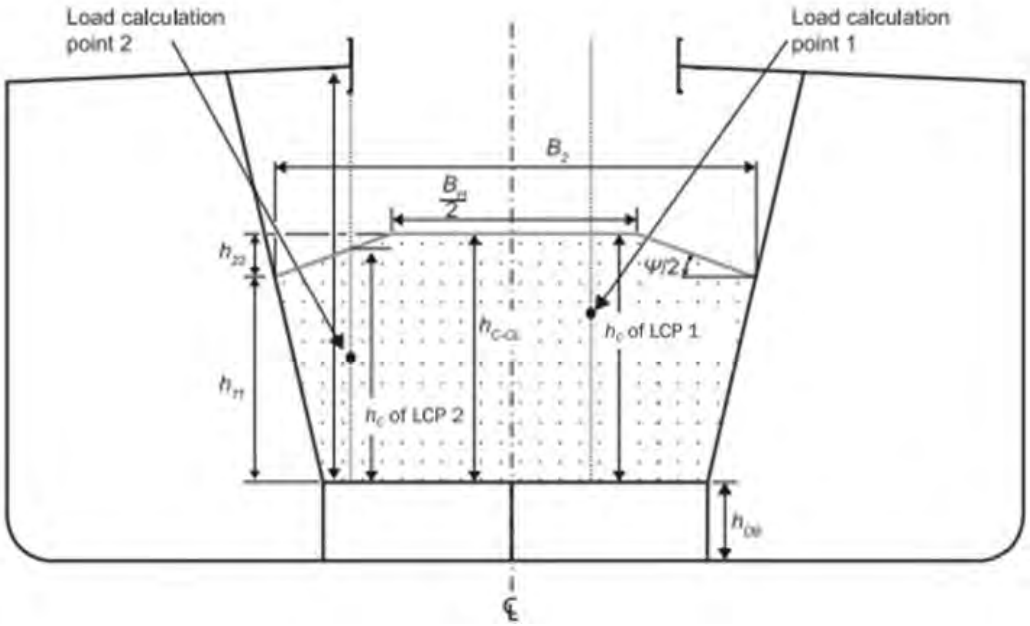


Figure 1.3.1.2(b4): Definition of effective upper surface of cargo for a partially filled cargo hold (Ore carrier with slanted longitudinal bulkhead)

1.3.1.3 Mass and density

The dry cargo mass and the density of the cargo are to be taken as given in Table 1.3.1.3 for strength and fatigue assessment.

Table 1.3.1.3: Dry bulk cargo mass and density for strength and fatigue assessment

Ship type	Cargo mass Cargo density (3)	Homogeneous loading condition		Alternate loading condition (2)	
		Fully filled hold	Partially filled hold	Fully filled hold	Partially filled hold
Strength assessment	M	$M = M_{Full}$	$M = M_H$	$M = M_{HD} + 0.1M_H$	$M = M_{HD} + 0.1M_H$
	ρ_c	$\rho_c = \frac{M_{Full}}{V_{Full}}$ but not less than 1.0	$\rho_c = 3.0$(1)	$\rho_c = \frac{M_{HD} + 0.1M_H}{V_{Full}}$	$\rho_c = 3.0$(1)
Strength assessment in flooded conditions	M	$M = M_H$	$M = M_H$	$M = M_{HD}$	$M = M_{HD}$
	ρ_c	$\rho_c = \frac{M_H}{V_{Full}}$	$\rho_c = 3.0$(1)	$\rho_c = \frac{M_{HD}}{V_{Full}}$	$\rho_c = 3.0$(1)
Fatigue assessment	M	$M = M_H$	NA	NA	$M = M_{HD}$
	ρ_c	$\rho_c = \frac{M_H}{V_{Full}}$			$\rho_c = 3.0$(1)

(1) To be taken as 3.0[t/m^3] unless an alternative maximum cargo density is specified in the loading manual.

(2) Hold loaded with $\rho_c \leq 1.78$[t/m^3], in flooded condition following values to be taken:
$M = M_{HD}$[t]
$\rho_c = 1.78$[t/m^3]

(3) Definitions:
M_{Full} = Cargo mass, in [t], in a cargo hold corresponding to the volume up to the top of the hatch coaming with a density of the greater of M_H/V_{Full} or 1.0[t/m^3].

$M_{Full} = 1.0V_{Full}$ but not less than M_H.

M_H = Cargo mass, in [t], in a cargo hold that corresponds to the homogeneously loaded condition at maximum draught with 50% consumables.

M_{HD} = Maximum allowable cargo mass, in [t], in a cargo hold according to design loading conditions with specified holds empty at maximum draught with 50% consumables and all ballast water tanks in cargo hold region empty.

V_{Full} = Volume, in [m^3], of cargo hold up to top of the hatch coaming:

$$V_{Full} = V_H + V_{HC}$$

V_H = Volume, in [m^3], of cargo hold up to level of the intersection of the main deck with the hatch coaming excluding the volume enclosed by hatch coaming, Refer Figure 1.3.1.1(a) to Figure 1.3.1.1(c).

V_{HC} = Volume, in [m^3], of the hatch coaming, from the level of the intersection of the main deck with the hatch side coaming to the top of the hatch coaming, determined for the cargo hold at midship, as shown in Figure 1.3.1.1(a) to Figure 1.3.1.1(c).

1.3.1.4 FE application

The following process is to be applied for the bulk cargo pressure loads used in FE analysis:

a) Determine h_c according to Cl.1.3.1.2(a) for fully filled cargo hold or Cl.1.3.1.2(b) for partially filled cargo hold.

b) Determine the corresponding static pressure as defined in Cl.1.3.2.2 and static shear pressure as defined in Cl.1.3.2.4(b) using ρ_c and apply them in the FE model.

c) Calculate the actual mass of cargo, M_{actual}, in [t].

d) Determine the effective cargo density, in [t/m^3]:

$$\rho_{eff} = \frac{M}{M_{actual}} . \rho_c$$

e) Calculate the final pressure distribution and shear load using ρ_{eff} instead of ρ_c.

1.3.2 Pressures and Forces due to dry bulk cargo in intact conditions

1.3.2.1 Total pressure

The total pressure due to dry bulk cargo acting on any load point of a cargo hold boundary, in [kN/m^2], is to be:

$P_{in} = P_{bs}$ for strength assessment of intact conditions for static (S) design load scenarios as given in Pt.3, Ch.5, Sec.6.

$P_{in} = P_{bs} + P_{bd}$ for strength assessment of intact conditions and fatigue assessment for static plus dynamic (S+D) design load scenarios as given in Pt.3, Ch.5, Sec.6.

where:

P_{bs}= Static pressure due to dry bulk cargo in [kN/m^2], as defined in Cl.1.3.2.2.

P_{bd} = Dynamic inertial pressure due to dry bulk cargo for each dynamic load case in [kN/m^2], as defined in Cl.1.3.2.3.

1.3.2.2 Static pressure

The dry bulk cargo static pressure P_{bs}, in [kN/m^2], is to be taken as:

$$P_{bs} = \rho_c \, g \, K_c \, (z_c - z) \text{ but not less than 0.}$$

1.3.2.3 Dynamic dry bulk cargo pressure

The dynamic dry bulk cargo pressure P_{bd}, in [kN/m^2], for each load case is to be taken as:

$$P_{bd} = f_\beta . \rho_c \left[0.25 a_x (x_G - x) + 0.25 a_y (y_G - y) + a_z \, K_c \, f_{dc} \, (z_c - z)\right] \text{ for } z \le z_c$$
$$P_{bd} = 0 \text{ for } z > z_c$$

Static and dynamic pressures as defined in Cl.1.3.2.2 and Cl.1.3.2.3 for FE analysis are to be determined using ρ_{eff} instead of ρ_c.

where,

a_X, a_Y, a_Z = Longitudinal, transverse and vertical accelerations, in [m/s^2], at x_G, y_G, z_G, as defined in Pt.3, Ch.5, Sec.2.

x_G, y_G, z_G = X, Y and Z coordinates, in [m], of the volumetric centre of gravity of the tank, considered with respect to the reference coordinate system defined in Pt.3, Ch.5, Sec.1.

In case of partially filled cargo hold, x_G, y_G, z_G to be taken as:

x_G, y_G = Volumetric centre of gravity of the cargo hold.
$z_G = h_{DB} + 0.5 h_{C-CL}$

f_β = heading correction factor as defined in Pt.3, Ch.5, Sec.2.

z_C = Height of the upper surface of the cargo above the baseline in way of the load point, in m, to be taken as:

$$z_C = h_{DB} + h_C$$

h_{DB} = Height of the double bottom at the centreline, measured at mid-length of the cargo hold, in [m], Refer Figure 1.3.1.1(a) to Figure 1.3.1.1(c).

h_c = Height of bulk cargo, in [m], from the inner bottom to the upper surface of bulk cargo as defined in Cl.1.3.1.2(a) and Cl.1.3.1.2(b).

ρ_c = Density of bulk cargo, in [t/m^3] as defined in Cl.1.3.1.3.

f_{dc} = Dry cargo factor:
= 1.0 for strength assessment
= 0.5 for fatigue assessment

K_c = Coefficient taken as defined in Pt.3, Ch.5, Sec.5, Cl.5.4.

1.3.2.4 Shear load in FE assessment

(a) Application

For FE strength assessment and FE fatigue assessment, the following shear load pressures are to be considered in addition to the dry bulk cargo pressures defined in Cl.1.3.2.1 to Cl.1.3.2.3, when the load point elevation, z, is lower or equal to z_c:

- For static (S) design load scenarios: Static shear load, P_{bs-s}, due to gravitational forces acting on hopper tanks and lower stools plating, as defined in Cl.1.3.2.4(b).

- For static plus dynamic (S+D) design load scenarios: The following dynamic shear load pressures:

  - $P_{bs-s} + P_{bs-d}$ for the hopper tank and the lower stool plating, as defined in Cl.1.3.2.4(c).

  - P_{bs-dx} for the inner bottom plating in the longitudinal direction, as defined in Cl.1.3.2.4(d).

  - P_{bs-dy} for the inner bottom plating in the transverse direction, as defined in Cl.1.3.2.4(d).

  Shear loads as defined in Cl.1.3.2.4(b) to Cl.1.3.2.4(d) for FE analysis are to be determined using ρ_{eff} instead of ρ_c.

(b) Static shear load on the hopper tank and lower stool plating

The static shear load pressure, P_{bs-s} (positive downward to the plating) due to dry bulk cargo gravitational forces acting on hopper tank and lower stool plating, in [kN/m^2], is to be taken as:

$$P_{bs-s} = \rho_c \, g \, \frac{(1 - K_c) \; (z_c - z)}{\tan \alpha}$$

(c) Dynamic shear load on the hopper tank and lower stool plating

The dynamic shear load pressure, P_{bs-d} (positive downward to the plating) due to dry bulk cargo forces on the hopper tank and lower stool plating, in [kN/m^2], for each dynamic load case is to be taken as:

$$P_{bs-d} = f_\beta . \rho_c . a_z \, \frac{(1 - K_c) \; (z_c - z)}{\tan \alpha}$$

(d) Dynamic shear load along the inner bottom plating for FE analyses

The dynamic shear load pressures, P_{bs-dx} in the longitudinal direction (positive to bow) due to dry bulk cargo forces acting along the inner bottom plating, in [kN/m^2], for each dynamic load case is to be taken respectively as:

$$P_{bs-dx} = -0.75 \, f_\beta . \rho_c . a_x . h_c$$

The dynamic shear load pressures, P_{bs-dy} in the transverse direction (positive to port) due to dry bulk cargo forces acting along the inner bottom plating, in[kN/m^2], for each dynamic load case is to be taken respectively as:

$$P_{bs-dy} = -0.75 \, f_\beta . \rho_c . a_y . h_c$$

The dynamic shear load pressures P_{bs-dx} and P_{bs-dy} are only used for FE strength assessment.

1.3.3 Pressures and Forces due to dry cargoes in Flooded Conditions

1.3.3.1 Application

(i) These requirements apply to ships intended for carriage of heavy dry bulk cargo which are required to be considered flooded for the hull girder longitudinal strength investigation in flooded conditions, as per Sec.2, Cl.2.2.2 and Sec.3, Cl.3.6.1.

(ii) This also applies to self-unloading bulk carriers where the unloading system maintains the water-tightness during seagoing operations. In self-unloading bulk carriers with unloading systems that do not maintain water-tightness, the longitudinal strength in the flooded conditions is to be considered using the extent to which the flooding may occur.

(iii) Only cargo holds in way of the double side space which do not meet the criteria given above is to be considered flooded.

(iv) This sub-section describes enhanced flooded requirements for dry cargo ships in addition to the requirements described in Pt.3, including:

- Hull girder loads in flooded conditions, Refer Cl.1.3.3.2.
- Local loads on vertically corrugated transverse bulkheads in flooded conditions, Refer Cl.1.3.3.3.
- Double bottom in cargo hold region in flooded conditions, Refer Cl.1.4.

1.3.3.2 Hull girder loads in flooded conditions

(a) Vertical still water hull girder loads

The designer is to provide the permissible still water bending moment M_{sw-f} and shear force

Q_{sw-f} in flooded condition. Each cargo hold shall be considered individually flooded to the equilibrium waterline.

The permissible vertical still water bending moment and shear force in flooded condition at sea at any longitudinal position shall envelope the most severe flooded seagoing loading conditions defined in the loading manual, i.e. all seagoing loaded and ballast conditions. Flooding check of harbour conditions, docking condition afloat, loading/unloading sequences and Intermediate conditions of loading encountered during ballast water exchange need not be considered.

(b) Wave hull girder loads

The wave loads in flooded conditions are taken as follows:
Vertical wave bending moment, M_{wv-f} and Vertical wave shear force, Q_{wv-f}, in flooded conditions, at the hull transverse section under consideration, is to be taken equal to 80% of the most probable maximum lifetime values given in Pt.3, Ch.5, Sec.3, Cl.3.6.

(c) Flooding assumptions

Calculation of the quantity of ingressed water due to flooding of the hold is to be based on the following assumptions:
— the permeability of empty cargo spaces and volume left in loaded cargo spaces above any cargo is to be 0.95

— appropriate values of permeability and bulk density are to be used for the cargo carried, Refer Table 1.3.3.2. In this respect, permeability for solid bulk cargo means the ratio of the floodable volume between the particles, granules or any larger pieces of the cargo, to the gross volume of the bulk cargo (that is the ratio of the volume of the voids within the cargo mass to the volume occupied by the cargo). For packed cargo conditions (such as steel mill products), the actual density of the cargo is to be used with a permeability of zero.

Table 1.3.3.2: Characteristics of select bulk cargoes

Bulk cargo	Angle of repose, ψ [deg]	Minimum Permeability 'μ'	Bulk density ρ_c [t/m^3]
Iron ore	35	0.3	3.0
Cement	25	0.3	1.3
Coal	20	0.3	0.9
Grain	20	0.5	0.8

1.3.3.3 Local loads on vertically corrugated transverse bulkheads in flooded conditions

(a) General

(i) The pressure defined in this section applies to vertically corrugated transverse watertight bulkheads of the cargo holds of ships intended for the carriage of dry cargoes in bulk for flooded condition assessment.

(ii) Each cargo hold is to be considered individually flooded, Refer Figure 1.3.3.3(a), Figure 1.3.3.3(b) and Figure 1.3.3.3(c).

(iii) The loads to be considered as acting on the bulkheads are those given by the combination of the cargo loads with those induced by the flooding of one hold adjacent to the bulkhead under examination. In any case, the pressure due to the flooding water alone is also to be considered.

(iv) The most severe combinations of cargo induced loads and flooding loads are to be used for the check of the scantlings of each bulkhead, depending on the loading conditions included in the loading manual considering the individual flooded condition of both loaded and empty holds:
- homogeneous loading conditions
- non-homogeneous loading conditions

(v) In this context, the following items are defined as:
- Design load limits: The specified design load limits for the cargo holds are to be represented by loading conditions defined by the designer in the loading manual.

- Maximum cargo mass to consider: Unless the ship is intended to carry, in non-homogeneous conditions, only iron ore or cargo having bulk density equal to or greater than 1.78[t/m3], the maximum mass of cargo which may be carried in the hold is also to be considered to fill that hold up to the upper deck level at centreline.

- Homogeneous loading conditions: Homogeneous loading condition means a loading condition in which the ratio between the highest and the lowest filling level, evaluated for each hold and corrected for differences in cargo densities, if any, does not exceed 1.20.

- Packed cargoes: Holds carrying packed cargoes (such as steel mill products) are to be considered as empty.

- Unconsidered loading conditions: Non homogeneous part loading conditions associated with multiport loading and unloading operations for homogeneous loading conditions need not be considered.

(b) Flooded level

The flooded level z_F is the distance, in [m], measured vertically from the baseline with the ship in the upright position, and obtained from Table 1.3.3.3.

Table 1.3.3.3: Flooded level z_F, for vertically corrugated transverse bulkheads

Ship Type	Loading condition	Vertically corrugated transverse bulkhead position	
		Foremost	Others
Ships less than 50,000 [t] deadweight with Type B freeboard	Non-homogeneous loading conditions with cargo density less than 1.78 [t/m^3]	$z_F = 0.9D_1$	$z_F = 0.8D_1$
	Other cases	$z_F = 0.95D_1$	$z_F = 0.85D_1$
Other ships	Non-homogeneous loading conditions with cargo density less than 1.78 [t/m^3]	$z_F = 0.95D_1$	$z_F = 0.85D_1$
	Other cases	$z_F = D_1$	$z_F = 0.9D_1$
D_1 = Distance from the base line to the freeboard deck at side amidships, in [m].			

(c) Flooded patterns

Three different flooded patterns are to be considered:

- The flooded level is below the upper surface of the cargo, (Refer Figure 1.3.3.3(a), $z_C > z_F$)

- The flooded level is above the upper surface of the cargo, (Refer Figure 1.3.3.3(b), $z_C < z_F$)

- The flooded hold is empty, (Refer Figure 1.3.3.3(c), $z_C = h_{DB}$)

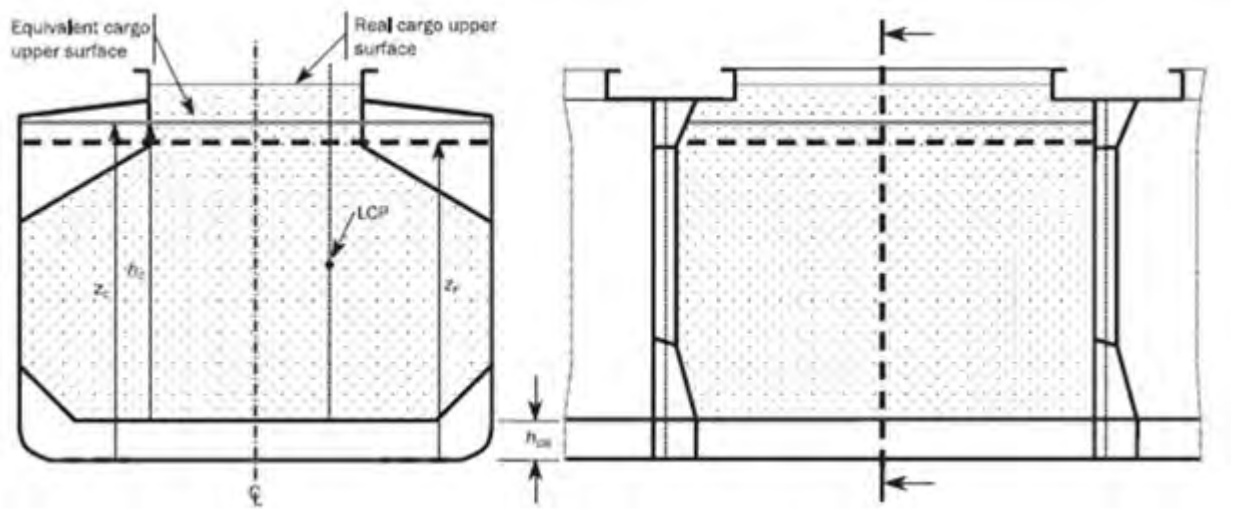


Figure 1.3.3.3(a): Flooded level below upper surface of bulk cargo

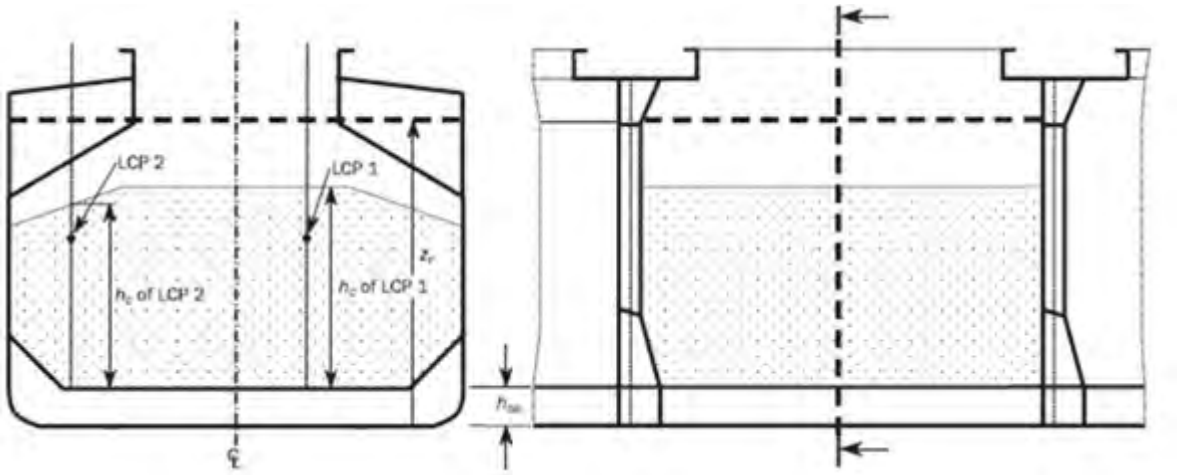


Figure 1.3.3.3(b): Flooded level above upper surface of bulk cargo

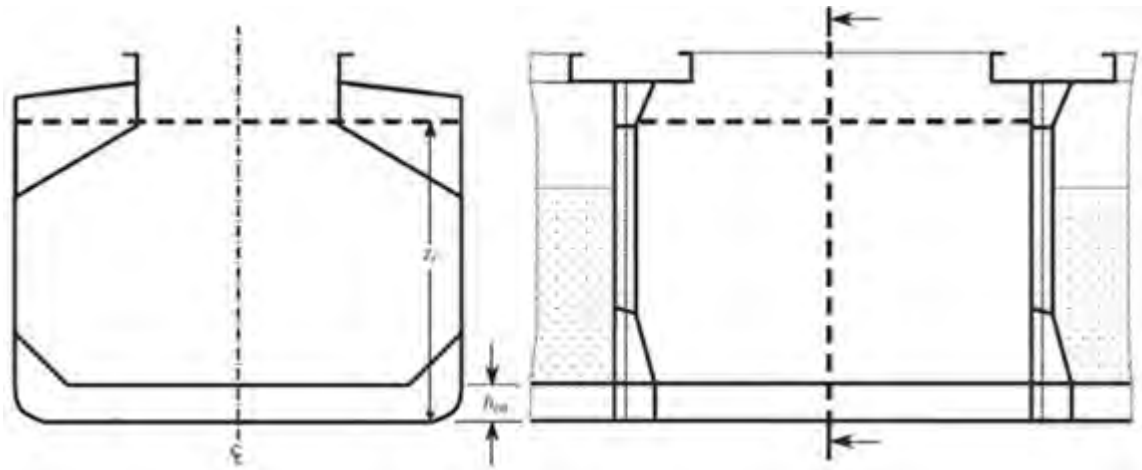


Figure 1.3.3.3(c): Flooded cargo hold without cargo

(d) Pressures and forces on vertically corrugated transverse bulkheads of flooded cargo holds

The static pressure P_{bf-s}, in [kN/m^2], at any point of the vertically corrugated transverse bulkhead located at a level z from the baseline is given in Table 1.3.3.3(a) for each flooded pattern defined in Cl.1.3.3.3(c).

The force F_{bf-s}, in [kN], acting on a corrugation of a transverse bulkhead is given by Table 1.3.3.3(b) for each flooded pattern defined in Cl.1.3.2.3(c).
where:

$P_{bf-s-LE}$ = Static pressure calculated according to Table 1.3.3.3(a) for z = $h_{LS} + h_{DB}$.

Table 1.3.3.3(a): Static pressure on vertically corrugated transverse bulkhead of a flooded cargo hold P_{bf-s}

Flooded case	Load calculation point	Pressure, P_{bf-s} [kN/m^2]
$z_C > z_F$	$z > z_c$	$P_{bf-s} = 0$
	$z_C \geq z \geq z_F$	$P_{bf-s} = \rho_c . g(z_C - z)K_{C-f}$
	$z_F > z \geq h_{DB}$	$P_{bf-s} = \rho . g(z_F - z) + [\rho_c(z_C - z) - \rho(1-\mu)(z_F - z)]gK_{C-f}$
$h_{DB} \leq z_C \leq z_F$	$z > z_F$	$P_{bf-s} = 0$
	$z_F \geq z \geq z_C$	$P_{bf-s} = \rho . g(z_F - z)$
	$z_C > z \geq h_{DB}$	$P_{bf-s} = \rho . g(z_F - z) + [\rho_c - \rho(1-\mu)]g(z_C - z)K_{C-f}$

Table 1.3.3.3(b): Force acting on a corrugation in the flooded cargo holds F_{bf-s}	
Flooded case	**Force, F_{bf-s} [kN]**
$z_C > z_F$	$F_{bf-s} = S_c \left\{ 0.5\rho_c . g . K_{C-f}(z_C - z_F)^2 + (z_F - h_{DB} - h_{LS}) \left[\frac{\rho_c . g . K_{C-f}(z_C - z_F) + P_{bf-s-LE}}{2} \right] \right\}$
$z_F \geq z_C$	$F_{bf-s} = S_c \left\{ 0.5\rho . g(z_F - z_C)^2 + (z_C - h_{DB} - h_{LS}) \left[\frac{\rho . g(z_F - z_C) + P_{bf-s-LE}}{2} \right] \right\}$

(e) Pressures and forces on vertically corrugated transverse bulkheads of non-flooded cargo holds

The static pressure P_{bs}, in [kN/m^2], at a point of the vertically corrugated transverse bulkhead located, located at the level z from the baseline, due to dry bulk cargo of a non-flooded cargo hold acting on the intact side of the transverse bulkhead which is flooded on the other side is to be taken as:

$$P_{bs} = \rho_c . g(z_C - z) K_{C-f} \text{ but not less than } 0.$$

The resultant force F_{bs}, in [kN], acting on a corrugation is to be taken as:

$$F_{bs} = 0.5\rho_c . g . S_c . K_{C-f} . (z_C - h_{DB} - h_{LS})^2$$

(f) Resultant pressures and forces on vertically corrugated transverse bulkheads of flooded cargo holds

The resultant pressure P_R, in [kN/m^2], at each point of the bulkhead, and the resultant force F_R, in [kN], acting on a corrugation, given in Table 1.3.3.3(c), are to be considered for the assessment in flooded conditions of vertically corrugated transverse bulkhead structures, where:

P_{bf-s} = Pressure in the flooded cargo holds, in [kN/m^2], as defined in Cl.1.3.3.3(d).

F_{bf-s} = Force acting on a corrugation in the flooded cargo holds, in [kN], as defined in Cl.1.3.3.3(d).

P_{bs} = Pressure in the non-flooded cargo holds, in [kN/m^2], as defined in Cl.1.3.3.3(e).

F_{bs} = Force acting on a corrugation in the non-flooded cargo holds, in [kN], as defined in Cl.1.3.3.3(e).

Table 1.3.3.3(c): Resultant pressure and resultant force on vertically corrugated transverse bulkhead in flooded condition			
Loading condition	**Resultant pressure P_R, in [kN/m^2]**	**Resultant force F_R, in [kN]**	**Application**
Homogeneous	$P_R = P_{bf-s} - 0.8\, P_{bs}$	$F_R = F_{bf-s} - 0.8\, F_{bs}$	In general [1)]
Non-homogeneous	$P_R = P_{bf-s}$	$F_R = F_{bf-s}$	Ships with criteria as given in Cl.1.3.3.1
Empty hold, flooded	$P_R = \rho . g(z_F - z)$	$P_R = 0.5\rho . g . S_c (z_F - h_{DB} - h_{LS})^2$	In general
1) Loading conditions in which the ratio between h_C , evaluated for each hold, does not exceed 1.2.			

1.4 Double bottom in cargo hold region in flooded conditions

1.4.1 General

These requirements apply to ships intended for carriage of heavy dry bulk cargo which are required to be considered flooded for the hull girder longitudinal strength investigation in flooded conditions, as per Sec.2, Cl.2.2.2 and Sec.3, Cl.3.6.1.

The loads to be considered as acting on the double bottom are those given by the external sea pressures and the combination of the cargo

loads with those induced by the flooding of the hold to which the double bottom belongs.

The most severe combinations of cargo induced loads and flooding loads are to be used, depending on the loading conditions included in the loading manual:

- homogeneous loading conditions;
- non homogeneous loading conditions;
- packed cargo conditions (such as steel mill products).

For each loading condition, the maximum bulk cargo density to be carried is to be considered in calculating the allowable hold loading limit.

1.4.2 Flooded level

The flooded level z_F is the distance, in [m], measured vertically from the baseline with the ship in the upright position, and obtained from Table 1.4.1.

Table 1.4.1: Flooded level z_F, for double bottom in cargo hold region

Ship Type	Cargo hold	
	Foremost	Others
Ships less than 50,000 [t] deadweight with Type B freeboard	$z_F = 0.95D_1$	$z_F = 0.85D_1$
Other ships	$z_F = D_1$	$z_F = 0.9D_1$
D_1 = Distance from the base line to the freeboard deck at side amidships, in [m].		

1.4.3 Allowable hold loading in flooded conditions

1.4.3.1 Evaluation of the double bottom Shear capacity

The shear capacity of the double bottom is defined as the sum of the shear strength at each end of:

- all floors adjacent to both hoppers, less one half of the strength of the two floors adjacent to each stool (or transverse bulkhead if no stool is fitted), Refer Figure 1.4.1. The shear strength of floors is to be calculated according to Cl.1.4.3.2.
- all double bottom girders adjacent to both stools (or transverse bulkheads if no stool is fitted). The shear strength of girders shall be calculated according to Cl.1.4.3.3.

The floors and girders to be considered when calculating the shear capacity of the double bottom are those inside the hold boundaries formed by the hopper tanks and stools or transverse bulkheads if no stool is fitted.

Where both ends of girders or floors are not directly connected to the hold boundaries, or where, in the end holds, girders or floors run out and are not directly attached to the boundary stool or hopper girder, their strength is to be evaluated for the connected end only.

The hopper side girders and the floors directly below the connection of the bulkhead stools (or transverse bulkheads if no stool is fitted) to the inner bottom are not to be included.

When the geometry and/or the structural arrangement of the double bottom are such as to make the above assumptions inadequate, the shear capacity of double bottom will be specially considered.

In calculating the shear strength, the net thickness of floors and girders is to be used, where the net thickness, t_{net} in [mm] is given by:

$$t_{net} = t - 2.5[\text{mm}]$$

where,

t = As-built thickness of floors and girders in [mm].

1.4.3.2 Floor shear strength

The floor shear strength, in [kN], is to be taken as given in the following formulae:

- In way of the floor panel adjacent to the hopper tank:

$$S_{f1} = A_f \frac{\tau_a}{\eta_1} 10^{-3}$$

- In way of the openings in the outermost bay (i.e., that bay which is closer to the hopper tank):

$$S_{f2} = A_{f,h} \frac{\tau_a}{\eta_2} 10^{-3}$$

where:

A_f = Net sectional area, in [mm^2], of the floor panel adjacent to the hopper tank.

$A_{f,h}$ = Net sectional area, in [mm^2], of the floor panels in way of the openings in the outermost

bay (i.e., the bay which is closer to the hopper tank).

τ_a = Allowable shear stress, in [N/mm^2], to be taken as the lesser of:

$$\tau_a = 0.645.R_{eH}^{0.6}.\left(\frac{t_{net}}{s}\right)^{0.8}$$

$$\tau_a = \frac{R_{eH}}{\sqrt{3}}$$

For floors adjacent to the stools or transverse bulkheads, τ_a is taken as:

$$\tau_a = \frac{R_{eH}}{\sqrt{3}}$$

t_{net} = Floor web net thickness, in [mm].

s = Spacing of stiffening members of the panel considered, in [m].

η_1 = Coefficient to be taken equal to 1.10.

η_2 = Coefficient to be taken equal to 1.20. It may be reduced to 1.1 where adequate reinforcements are fitted in way of the openings in the floors.

1.4.3.3 Girder shear strength

The girder shear strength, in [kN], is to be taken as given in the following formulae:

- In way of the girder panel adjacent to the stool or transverse bulkhead, if no stool is fitted:

$$S_{g1} = A_g \frac{\tau_a}{\eta_1} 10^{-3}$$

- In way of the largest opening in the outermost bay (i.e., that bay which is closer to the stool) or transverse bulk-head, if no stool is fitted:

$$S_{g2} = A_{g,h} \frac{\tau_a}{\eta_2} 10^{-3}$$

A_g = Net sectional area in [mm^2], of the girder panel adjacent to the stool (or transverse bulkheads, if no stool is fitted).

$A_{g,h}$ = Net sectional area in [mm^2], of the girder panel in way of the largest opening in the outermost bay (i.e. that bay which is closer to the stool) or transverse bulkhead, if no stool is fitted.

τ_a = Allowable shear stress in [N/mm^2], as defined in Cl.1.4.3.2, where 't_{net}' is the girder web net thickness.

η_1 = Coefficient to be taken equal to 1.10.

η_2 = Coefficient to be taken equal to 1.15. It may be reduced to 1.10 where adequate reinforcements are fitted in way of the openings in girders.

1.4.3.4 Allowable hold loading

The allowable hold loading W in [t], is to be taken as:

$$W = \rho_c.V.\frac{1}{F}$$

where,

F = Coefficient to be taken as:

F = 1.1 in general

F = 1.05 for steel mill products

ρ_c = Maximum applicable value of cargo density, in [t/m^3]; for bulk cargoes, Refer Table 1.3.3.2 and for steel products, ρ_c is to be taken as the density of steel.

V = Volume in [m^3], occupied by the cargo up to the level h_B.

h_B = Level of cargo, in [m], to be taken as:

$$h_B = \frac{P}{\rho_c.g}$$

P = Pressure, in [kN/m^2], to be taken as:

- For dry bulk cargoes, the lesser of:

$$P = \frac{Z + \rho.g(z_f - 0.1D_1 - h_F)}{1 + \frac{\rho}{\rho_c}(\mu - 1)}$$

$$P = Z + \rho.g(z_f - 0.1D_1 - h_F.\mu)$$

- For steel mill products:

$$P = \frac{Z + \rho.g(z_f - 0.1D_1 - h_F)}{1 - \frac{\rho}{\rho_{st}}}$$

D_1 = Distance, in [m], from the baseline to the freeboard deck at side amidships.

h_F = Inner bottom flooded height, in [m], measured vertically with the ship in the upright position, from the inner bottom to the flooded level z_F.

z_F = Flooded level, in [m], as defined in Cl.1.4.2.

μ = Permeability of cargo as defined in Table 1.3.3.2, however need not be taken greater than 0.3. For steel mill products, $\mu = 0$.

ρ_{st} = Density of steel, in [t/m^3], to be taken as 7.85.

ρ = Sea water density, in [t/m^3]

g = 9.81 [m/s^2], gravity acceleration

Z = Pressure, in [kN/m^2], to be taken as the lesser of:

$$Z = \frac{C_H}{A_{DB,H}}$$

$$Z = \frac{C_E}{A_{DB,E}}$$

C_H = Shear capacity of the double bottom, in [kN], to be calculated according to Cl.1.4.3.1, considering, for each floor, the lesser of the shear strengths S_{f1} and S_{f2} as defined in Cl.1.4.3.2 and, for each girder, the lesser of the shear strengths S_{g1} and S_{g2} as defined in Cl.1.4.3.3.

$A_{DB,H}$ = Area, in [m^2], taken as:

$$A_{DB,H} = \sum_{i=1}^{n} S_i \, B_{DB,i}$$

C_E = Shear capacity of the double bottom, in [kN], to be calculated according to Cl.1.4.3.1, considering, for each floor, the shear strength S_{f1} as defined in Cl.1.4.3.2 and, for each girder, the lesser of the shear strengths S_{g1} and S_{g2} as defined in Cl.1.4.3.3.

$A_{DB,E}$ = Area, in [m^2], taken as:

$$A_{DB,E} = \sum_{i=1}^{n} S_i (B_{DB} - S)$$

n = Number of floors between stools or (transverse bulkheads, if no stool is fitted).

S_i = Space of i-th floor, in [m].

$B_{DB,i}$ = Length, in [m], to be taken equal to:

$B_{DB,i} = B_{DB} - S$ for floors for which $S_{f1} < S_{f2}$.

$B_{DB,i} = B_{DB,h}$ for floors for which $S_{f1} \geq S_{f2}$.

B_{DB} = Breadth, in [m], of double bottom between the hopper tanks as shown in Figure 1.4.2.

$B_{DB,h}$ = Distance, in [m], between the two openings considered as shown in Figure 1.4.2.

s = Spacing, in [m], of inner bottom longitudinal ordinary stiffeners adjacent to the hopper tanks.

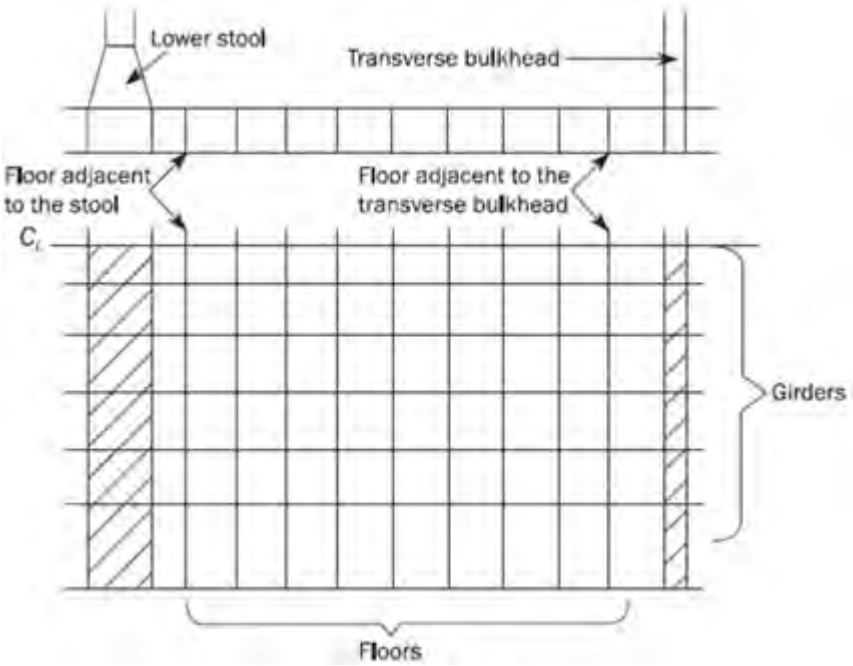


Figure 1.4.1: Double bottom structure

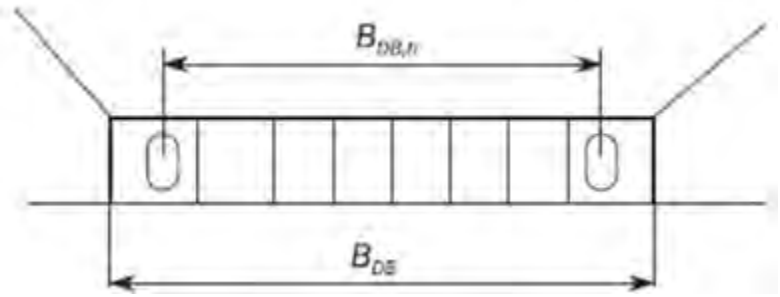


Figure 1.4.2: Dimensions B_{DB} and $B_{DB,h}$

1.5 Transverse vertically corrugated watertight bulkheads separating cargo holds in flooded condition

1.5.1 Application

These requirements apply to vertically corrugated transverse watertight bulkheads of the cargo holds of ships intended for the carriage of dry cargoes in bulk which are required to be considered flooded for the hull girder longitudinal strength investigation in flooded conditions, as per Sec.2, Cl.2.2.2 and Sec.3, Cl.3.6.1.

1.5.2 Structural arrangement

1.5.2.1 Lower stool

(a) The lower stool, where fitted, is generally to have a height not less than:

- 3 times the corrugation depth for dry cargo hold bulkheads.
- one corrugation depth for tank and water ballast cargo hold bulkheads.

(b) The stool side vertical stiffeners are to be bracketed at the top and bottom ends. Lower stool side vertical stiffeners and their brackets in the stool are to be aligned with the inner bottom structures such as longitudinals or similar. The sloping stool side plating is to align with the corrugation flanges. Lower stool side plating is not to be knuckled anywhere between the inner bottom plating and the stool top plate.

(c) The distance of the edge of the stool top plate to the surface of the corrugation flange is to be in accordance with Figure 1.5.1.

(d) The stool bottom is to be arranged in line with the double bottom floors or girders as the case may be and should have a width not less than

- 2.5 times the corrugation depth for dry cargo hold bulkheads.
- one corrugation depth for tank and water ballast cargo hold bulkheads.

The net thickness of these supporting floors is not be less than the net required thickness stool side plating (excluding the application of GRAB requirements) connected to the inner bottom and is to be of at least the same material yield strength. If material of different yield strength is used, the required thickness is to be adjusted by the ratio of the two material factors ‘k’, as defined in Pt.3, Ch.2, Sec.1. Cut-outs for the connection of inner bottom longitudinals to these double bottom floors are to be closed by collar plates.

(e) The stool is to be fitted with diaphragms in line with longitudinal double bottom girders or floors for effective support of the bulkhead. Scallops in the brackets and diaphragms in way of the top and bottom connections to the top plate and the double bottom floors/girders are to be avoided.

(f) The stool side plating is to be connected to the stool top plate and the inner bottom plating by either full penetration or partial penetration welds. The supporting floors are to be connected to the inner bottom by either full penetration or partial penetration welds.

(g) The net thickness of the stool top plate is not to be less than that required for the attached corrugated bulkhead and is to be of at least the same material yield strength as the attached corrugation. The extension of the top plate beyond the corrugation is not to be less than the as-built flange thickness of the corrugation.

(h) The net thickness of the stool side plate, within the region of the corrugation depth from the stool top plate, is not to be less than the corrugated bulkhead flange net required thickness at the lower end and is to be of at least the same material yield strength. The net thickness may be reduced to 90% of corrugation flange thickness if continuity is provided between the corrugation web and supporting brackets inside the stool as defined in Cl.1.5.2.1(i).

(i) Continuity between corrugation web and lower stool supporting brackets is to be maintained inside the stool. Alternatively, lower stool supporting brackets inside the stool is to be aligned with every knuckle point of corrugation web.

(j) The net thickness of supporting bracket is not to be less than 80% of the required net thickness of the corrugation webs and is to be of at least the same material yield strength.

(k) Particular attention is to be given to the through-thickness properties, and arrangements for continuity of strength, at the connection of the bulkhead stool to the inner bottom, where a lower stool is fitted.

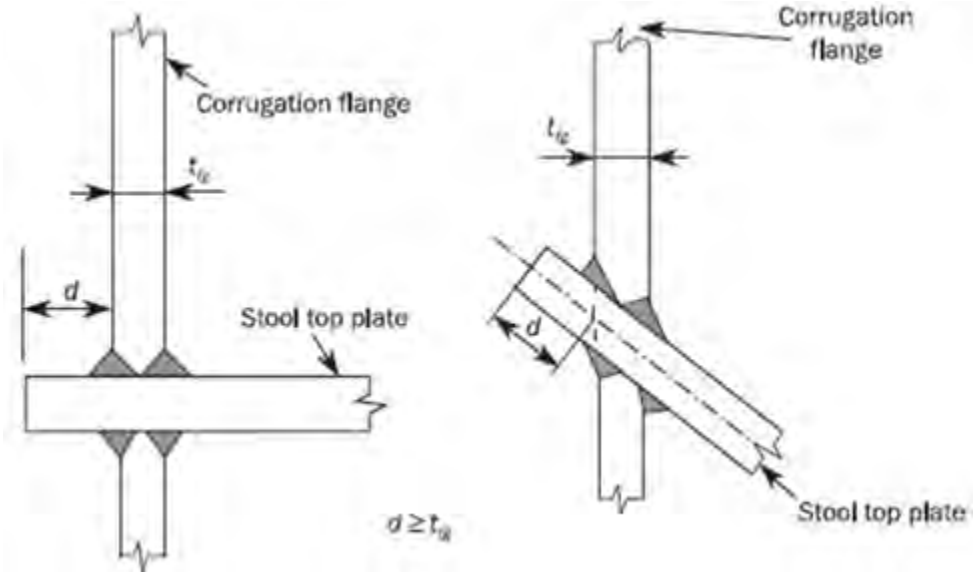


Figure 1.5.1: Permitted distance, d, from edge of stool top plate to surface of the corrugation flange

1.5.2.2 Upper stool

(a) The upper stool, where fitted, is generally to have a height, measured at the hatch side girder as below:

- Not less than 2 times the corrugation depth for dry cargo hold bulkheads.
- At least one corrugation depth for tank and water ballast cargo hold bulkheads.

Rectangular stools are generally to have a height of 2 times the depth of corrugation, measured from the deck level and at the hatch side girder or at the inner hull as applicable. Brackets or deep webs are to be fitted to connect the upper stool to the deck transverse or hatch end beams.

(b) The upper stool of a transverse bulkhead is to be properly supported by deck girders or deep brackets between the adjacent hatch end beams. The width of the upper stool bottom plate is generally to be the same as that of the lower stool top plate. The stool top of non-rectangular stools is to have a width not less than 2 times the depth of corrugations. The ends of stool side ordinary stiffeners when fitted in a vertical plane, are to be attached to brackets at the upper and lower end of the stool.

(c) The stool is to be fitted with diaphragms in line with and effectively attached to longitudinal deck girders extending to the hatch end coaming girders or transverse deck primary supporting members. Scallops in the brackets and diaphragms in way of the connection to the stool bottom plate are to be avoided.

(d) The net thickness of the stool bottom plate is not be less than that required for the attached corrugated bulkhead and is to be of at least the same material yield strength as the attached corrugation. The extension of the top plate beyond the corrugation is not to be less than the as-built flange thickness of the corrugation.

(e) The net thickness of the lower portion of the stool side plating is to be not less than 80% of the upper part of the bulkhead plating as required by Cl.1.5.3, where same material is used. If material of different yield strength is used, the required thickness is to be adjusted by the ratio of the two material factors ‘k’, as defined in Pt.3, Ch.2, Sec.1.

(f) Local supporting structure in way of corrugated bulkheads without a lower stool

- Where no stool is fitted at bottom, supporting floors are to be provided in line with corrugation flanges. Where no stool is fitted at deck, two transverse reinforced beams are to be fitted in line with the corrugation flanges.

- The net thickness of the supporting floors in way of a corrugated bulkhead are not to be less than the required net thickness of the corrugation flanges and are to be of at least the same material yield strength. The inner bottom and hopper tank in way of the corrugation is to be of at least the same material yield strength as the attached corrugation.
- Brackets/carlings arranged in line with the corrugation web are to have a depth of not

less than 0.5 times the corrugation depth and a net thickness not less than 80% of the net thickness of the corrugation webs and are to be of at least the same material yield strength. Where support is provided by gussets with shedder plates instead of brackets/carlings, the height of the gusset plate, refer h_G in Sec.1, Cl.1.5.3, Figure 1.5.3.1, is to be at least equal to the corrugation depth. The gusset plates are to be fitted in line with and between the corrugation flanges. The net thickness of the gusset and shedder plates is not to be less than 100% and 80%, respectively, of the net thickness of the corrugation flange and is to be of at least the same material yield strength.

- The plating of supporting floors shall be connected to the inner bottom by either full penetration or partial penetration weld.

1.5.2.3 Bulkheads acting as pillars

(a) Each vertical stiffener, in association with a width of plating equal to:

- 35 times the plating net thickness
- 1/12 of stiffener length, or
- the stiffener spacing, whichever is the smaller,

is to comply with the applicable requirements in Pt.3, Ch.7, for the load being supported.

(b) Non-tight bulkheads acting as pillars are to be provided with bulkhead stiffeners with a maximum spacing equal to:

- Two frame spacings, when the frame spacing does not exceed 750[mm].
- One frame spacing, when the frame spacing is greater than 750[mm].

(c) Where non-tight bulkheads are corrugated, the depth of the corrugation is not to be less than 100[mm].

1.5.3 Scantlings

1.5.3.1 Net thickness of corrugation

(a) Cold and hot formed corrugations

The net plate thickness t_{net}, in mm, of transverse vertically corrugated watertight bulkheads separating cargo holds is not to be taken less than:

$$t_{net} = 0.0149\, s_{cw} \sqrt{\frac{1.05\, P_R}{R_{eH}}}$$

where:

s_{cw} = Plate width, in [mm], taken as the width of the corrugation flange a or the web c, whichever is greater as defined in Pt.3, Ch.3, Sec.6, Figure 6.4.1.

P_R = Resultant pressure, in [kN/m^2], as defined in Section 1, Cl.1.3.3.3(f), Table 1.3.3.3(c).

The net thicknesses is also to comply with the requirements given in Pt.3, Ch.7, Sec 4, Cl.4.1.2.1.

(b) Built-up corrugations

Where the thicknesses of the flange and web of built-up corrugations of transverse vertically corrugated watertight bulkheads separating cargo holds are different, the net plate thicknesses are not to be taken less than that obtained from the following formula.

The net thickness t_N, in [mm], of the narrower plating is not to be taken less than:

$$t_N = 0.0149\, s_N \sqrt{\frac{1.05\, P_R}{R_{eH}}}$$

where:

s_N = Plate width, in [mm], taken as the width of the corrugation flange a or the web c, whichever is smaller as defined in Pt.3, Ch.3, Sec.6, Figure 6.4.1.

The net thickness t_W, in [mm], of the wider plating is not to be taken less than the greater of the following formulae:

$$t_W = 0.0149\, s_{cw} \sqrt{\frac{1.05\, P_R}{R_{eH}}}$$

$$t_W = \sqrt{\frac{4.62\, s_{cw}^2 . P_R}{R_{eH} . 10^4} - t_{NO}^2}$$

where:

t_{NO} = Net offered thickness of the narrower plating, in [mm], not to be taken greater than:

$$t_{NO} = 0.0149\, s_{cw} \sqrt{\frac{1.05\, P_R}{R_{eH}}}$$

The net thicknesses is also to comply with the requirements given in Pt.3, Ch.7, Sec.4, Cl.4.1.2.2.

(c) Lower part of corrugation

The net thickness of the lower part of corrugations is to be maintained for a distance of not less than $0.15\ell_c$ measured from the top of the lower stool, or from the inner bottom where no lower stool is fitted. The span of the corrugations ℓ_c, in [m], is to be taken as given in Pt.3, Ch.3, Sec.6, Cl.6.4.3.

(d) Middle part of corrugation

The net thickness of the middle part of corrugations is to be maintained for a distance not greater than $0.3\ell_c$ from the bottom of the upper stool, or from the deck if no upper stool is fitted.

1.5.3.2 Net section modulus at the lower end of the corrugations

(a) Effective flange width

The net section modulus of the corrugations is to be calculated with the compression flange having an effective flange width b_{eff} not larger than the following formula:

$$b_{eff} = C_E . a$$

where:

C_E = Coefficient to be taken equal to:

$$C_E = \frac{2.25}{\beta} - \frac{1.25}{\beta^2} \text{ for } \beta > 1.25$$

$$C_E = 1.0 \text{ for } \beta \leq 1.25$$

β = Coefficient to be taken equal to:

$$\beta = \frac{a}{t_f} . \sqrt{\frac{R_{eH}}{E}}$$

c = Width, in [mm], of the corrugation flange as shown in Pt.3, Ch.3, Sec.6, Figure 6.4.1.
t_f = Net flange thickness, in [mm].

(b) Webs not supported by local brackets

Unless welded to a sloping stool top plate as defined in Cl.1.5.3.2(e), if the corrugation webs are not supported by local brackets below the stool top plate (or below the inner bottom) in the lower part, the section modulus of the corrugations is to be calculated considering the corrugation webs 30% effective.

(c) Effective shedder plates

Provided that effective shedder plates are fitted as shown in Figure 1.5.3, when calculating the section modulus at the lower end of the corrugations (Sections '1' in Figure 1.5.3), the net area, in [cm^2], of flange plates may be increased by I_{SH} to be taken as:

$$I_{SH} = 2.5.10^{-3}.a\sqrt{t_f.t_{SH}} \text{ without being taken greater than } 2.5.a.t_f.10^{-3}$$

where:

a = Width, in [mm], of the corrugation flange as shown in Pt.3, Ch.3, Sec.6, Figure 6.4.1.
t_{SH} = Net shedder plate thickness, in [mm].
t_f = Net flange thickness, in [mm].

Effective shedder plates are those which:
- are not knuckled;
- are welded to the corrugations and the top of the lower stool by one side penetration welds or equivalent;
- are fitted with a minimum slope of 45[deg] and their lower edge is in line with the lower stool side plating;
- have net thickness not less than 75% of the net required for the corrugation flanges,
- have material properties are at least equal to that of the corrugation flanges.

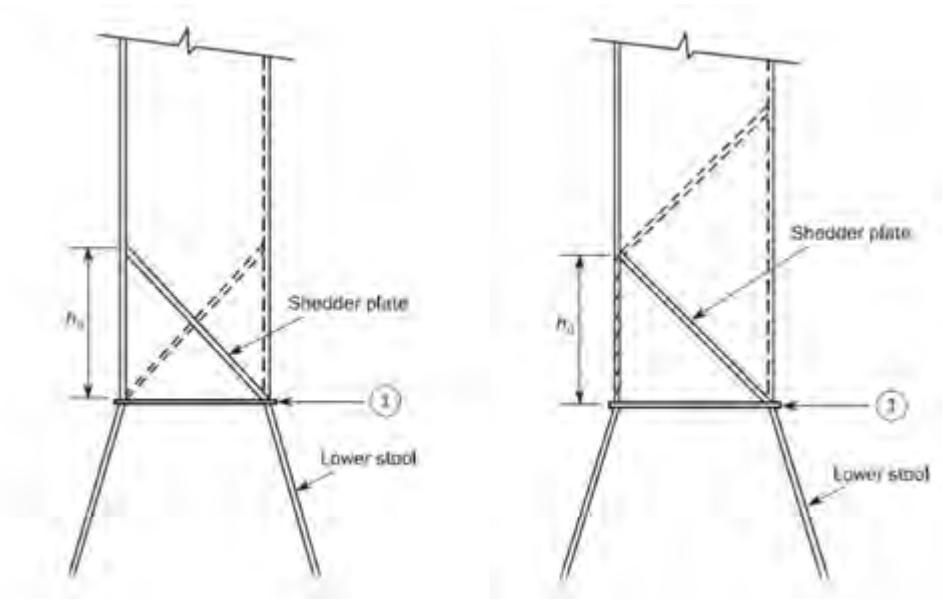


Figure 1.5.3: Symmetrical and unsymmetrical shedder plates

(d) Effective gusset plates

Provided that effective gusset plates are fitted, when calculating the section modulus at the lower end of the corrugations (Sections '1' in Figure 1.5.3(a) and Figure 1.5.3(b)), the net area, in [cm^2], of flange plates may be increased by I_G to be taken as:

$$I_G = 7.h_G.t_{GU}$$

where:

h_G = Height, in [m], of gusset plates as shown in Figure 1.5.3(a) and Figure 1.5.3(b) but not to be taken greater than:

$$\frac{10.S_{GU}}{7}$$

S_{GU} = Width, in [m], of the gusset plates.

t_{GU} = Net gusset plate thickness, in [mm], not to be taken greater than t_f.

t_f = Net flange thickness, in [mm].

Effective gusset plates are those which:

- are in combination with shedder plates having thickness, material properties and welded connections as requested for shedder plates in Cl.1.5.3.2(c);
- have a height not less than half of the flange width,
- are fitted in line with the lower stool side plating,
- are generally welded to the top of the lower stool by full penetration welds, and to the corrugations and shedder plates by one side penetration welds or equivalent
- have net thickness and material properties not less than those net required for the flanges.

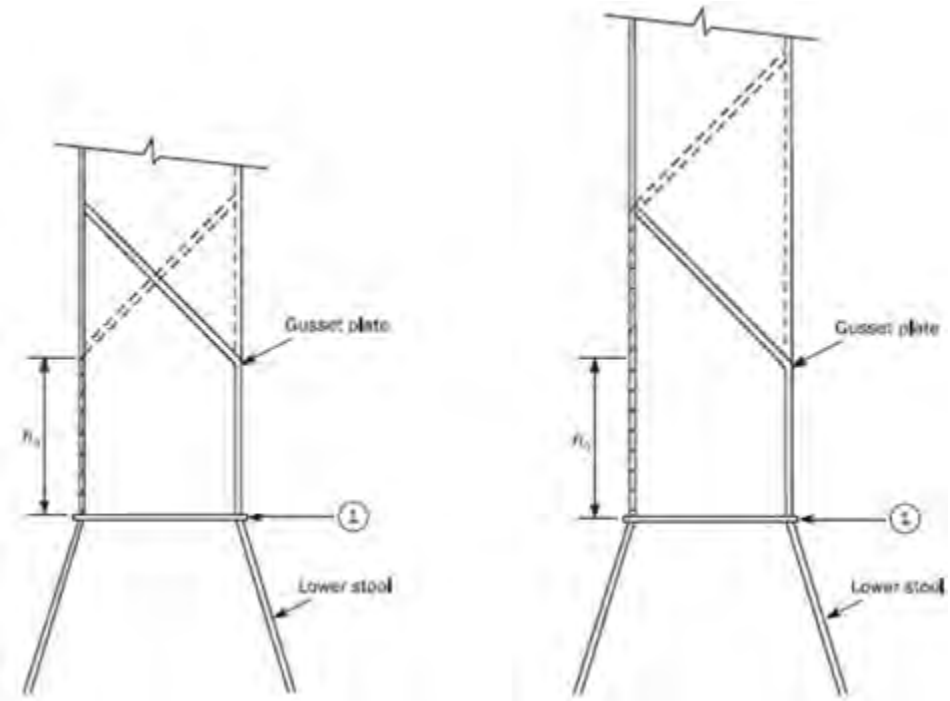


Figure 1.5.3(a): Symmetrical and unsymmetrical gusset / shedder plates

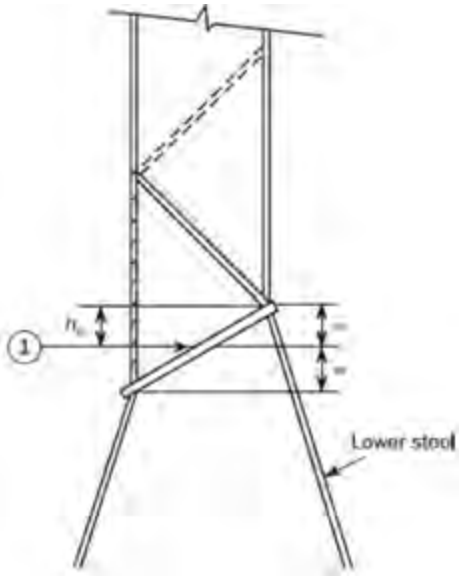


Figure 1.5.3(b): Asymmetrical gusset / shedder plates

(e) Corrugation web efficiency in way of sloping stool top plate

Where the corrugation webs are welded to a sloping stool top plate which has an angle not less than 45[deg] with the horizontal plane, the section modulus at the lower end of the corrugations may be calculated considering the corrugation webs fully effective. For angles less than 45[deg], the effectiveness of the web may be obtained by linear interpolation between 30% efficient for 0[deg] and 100% efficient for 45[deg].

Where effective gusset plates are fitted, when calculating the net section modulus of corrugations, the net area of flange plates may be increased as specified in Cl.1.5.3.2(d) above. No credit can be given to shedder plates only.

(f) Section modulus of corrugations at cross-sections other than the lower end

The section modulus of corrugations at cross sections other than the lower end is to be calculated with the corrugation webs

considered effective and the compression flange having an effective flange width, b_{eff}, as defined in Cl.1.5.3.2(a).

(g) Corrosion addition

The total corrosion addition, in [mm], is to comply with the requirements given in Pt.3, Ch.2, Sec.5, but not less than the minimum total corrosion addition value given below for both sides of the structural member:

$$t_c = 3.5$$

(h) Local details

- The design of local details is to be adequate for the purpose of transferring the corrugated bulkhead forces and moments to the boundary structures, in particular, the double bottom and cross-deck structures.
- The thickness and stiffening of effective gusset and shedder plates, are to be based on loads as per Cl.1.3.3.3 as applicable.
- Where the corrugation angle is less than 50[deg], a horizontal row of staggered shedder plates is to be fitted at approximately mid depth of the corrugations to help preserve dimensional stability of the bulkhead under flooding loads. The shedder plates are to be welded to the corrugations by double continuous welding, but they are not to be welded to the side shell. (Refer Figure 1.5.3(c))

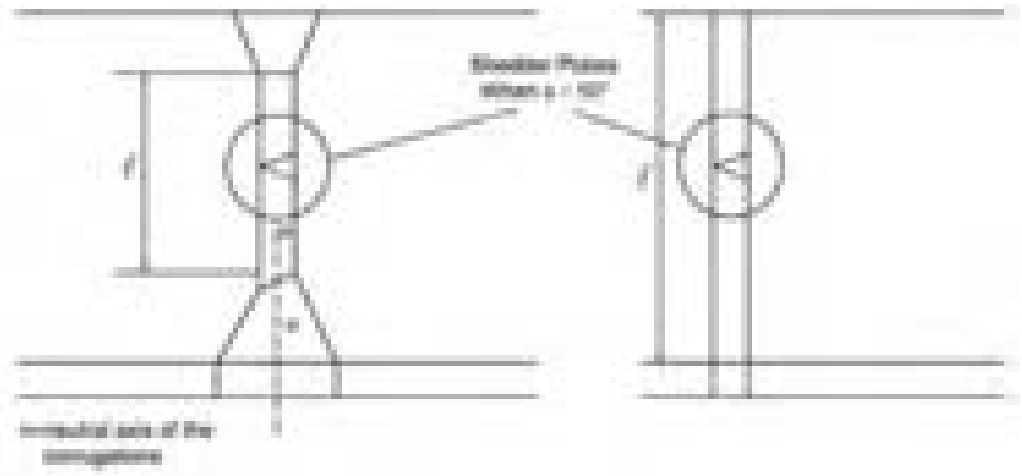


Figure 1.5.3(c): Staggered shedder plates

1.5.3.3 Bending, shear and buckling check

(a) General

The bending moment M and the shear force Q in the bulkhead corrugations are obtained using the formulae given in Cl.1.5.3.3(b). The M and Q values are to be used for the checks in Cl.1.5.3.3(c).

(b) Bending moment and shear Force

- The design bending moment M, in [kN-m], for the bulkhead corrugations is given by:

$$M = \frac{F_R . \ell_c}{8}$$

- The shear force Q, in [kN], at the lower end of the bulkhead corrugations is given by:

$$Q = 0.8F_R$$

where,

F_R = Resultant force, in [kN], as defined in Section 1, Cl.1.3.3.3(f), Table 1.3.3.3(c).

ℓ_c = Span of the corrugations, in [m], as given in Pt.3, Ch.3, Sec.6, Cl.6.4.3.

(c) Bending and shear capacity

The bending and the shear capacity of the corrugations of transverse watertight corrugated bulkheads separating cargo holds are to comply with the following formulae:

$$0.5W_{LE} + W_M \geq \frac{M}{0.95R_{eH}} 10^3$$

$$\tau \leq 0.5R_{eH}$$

where,

W_{LE} = Net section modulus, in [cm^3], of one-half pitch corrugation, to be calculated at the lower end of the corrugations according to Cl.1.5.3.2, not to be taken greater than:

$$W_{LE,M} = W_G + \frac{Q.h_G.10^3 - 0.5.h_G^2.S_C.P_R}{R_{eH}}$$

W_G = Net section modulus, in [cm^3], of one-half pitch corrugation, to be calculated in way of the upper end of shedder or gusset plates, as applicable, according to Cl.1.5.3.2.

W_M = Net section modulus, in [cm^3], of one-half pitch corrugation, to be calculated at the mid-span of corrugations according to according to Cl.1.5.3.2, without being taken greater than $1.15W_{LE}$.

S_C = Half pitch, in [mm], of the corrugation flange as defined in Pt.3, Ch.3, Sec.6, Cl.6.4.1, Figure 6.4.1.

h_G = Height, in [m], of shedders or gusset plates as shown in Figure 1.5.3, Figure 1.5.3(a) and Figure 1.5.3(b).

P_R = Resultant pressure, in [kN/m^2], to be calculated in way of the middle of the shedders or gusset plates, as applicable, according to Section 1, Cl.1.3.3.3(f), Table 1.3.3.3(c).

τ = Shear stress, in [N/mm^2], in the corrugation to be taken as:

$$\tau = \frac{10.Q}{A_{shr}}$$

A_{shr} = Net shear area, in [cm^2], of one-half pitch corrugation. The calculated net shear area is to consider possible reduced shear efficiency due to non-straight angles between the corrugation webs and flanges. In general, the reduced shear area may be obtained by multiplying the web sectional area by $\sin\varphi$.

φ = Angle between the web and the flange, Refer Pt.3, Ch.3, Sec.6, Cl.6.4.1, Figure 6.4.1.

The net section modulus of the corrugations in the upper part of the bulkhead, as defined in Figure 1.5.3(d), is not to be less than 75% of that of the middle part complying with this requirement, corrected for different minimum yield stresses.

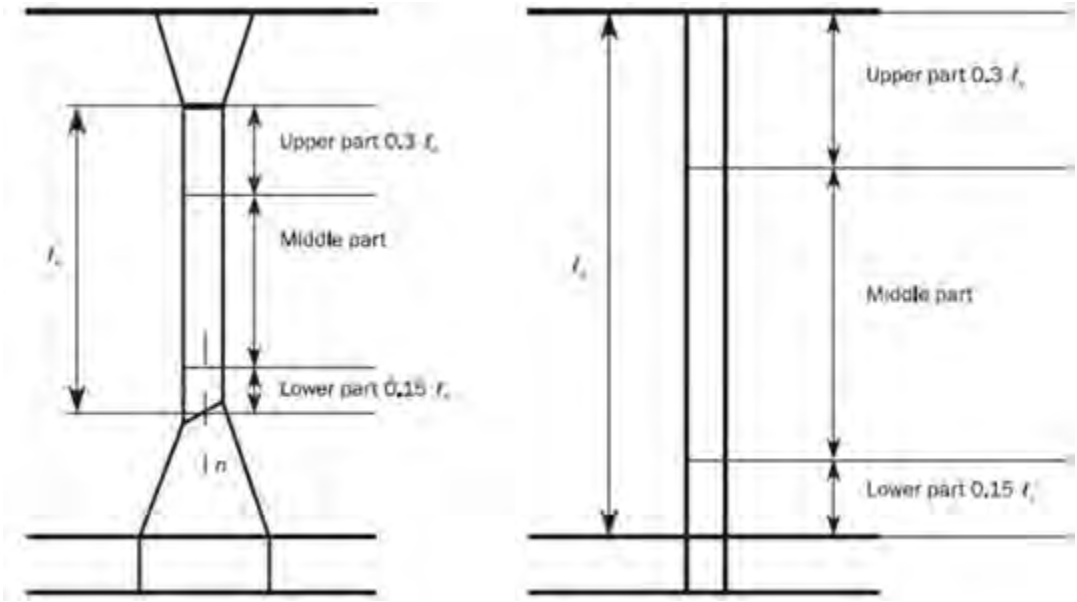


Figure 1.5.3(d): Parts of corrugation

(d) Shear buckling check of the bulkhead corrugation webs

The buckling check is to be performed for the web plates at the corrugation ends.

The shear stress τ, calculated according to Cl.1.5.3.3(c), is to comply with the following formula:

$$\tau \leq \tau_c$$

where:

τ_c = Critical shear buckling stress, in [N/mm^2], to be taken as:

$$\tau_r = \tau_E \qquad \text{for } \tau_E \leq 0.5\,\tau_y$$

$$\tau_{cr} = \tau_y\left(1 - \frac{\tau_y}{4\tau_E}\right) \text{ for } \tau_E > 0.5\,\tau_y$$

τ_E = Euler shear buckling stress, in [N/mm^2], to be taken as:

$$\tau_E = 0.9\, K_t.\, E.\left(\frac{t_w}{c}\right)^2$$

K_t = Coefficient, to be taken equal to 6.34.

t_w = Net thickness, in [mm], of the corrugation webs.

c = Width, in [mm], of the corrugation webs as shown in Pt.3, Ch.3, Sec.6, Cl.6.4.1, Figure 6.4.1.

τ_y = Yield stress in shear of the material, in [N/mm^2].

$$\tau_y = \frac{R_{eH}}{\sqrt{3}}$$

1.5.3.4 Permissible Vertical hull girder bending and shear strength in flooded conditions

(a)Permissible still water bending moments for flooded conditions

The permissible still water bending moments, M_{sw-f} in [kN-m], for flooded conditions in hogging and sagging is to comply with the following criteria:

$$|M_{sw-f}| \leq |M_{sw} + 0.2M_{wv}|$$

The bending moment M_{wv} used above is to have the same sign as the considered bending moment M_{sw-f}.

(b) Permissible still water shear force in flooded conditions

The positive and negative permissible vertical still water shear force, in [kN], for flooded condition is to comply with the following criteria:

$$|Q_{sw-f}| \leq |Q_{sw} + 0.2Q_{wv}|$$

The shear force Q_{wv} used above is to be taken with the same sign as the considered shear force Q_{sw-f}.

The vertical still water shear forces, in [kN], for all loading conditions in flooded condition is to comply with the following criteria, except for Ore carrier:

$$|Q_{sw-lcd-f} - \Delta Q_s| \leq |Q_{sw-f}|$$

For Ore Carrier:

$$|Q_{sw-lcd-f}| \leq |Q_{sw-f}|$$

The shear force Q_{sw-f} used above is to be taken with the same sign as the considered shear force $Q_{sw-lcd-f}$.

where:

M_{sw} = Permissible vertical still-water bending moment in [kN-m], for hogging and sagging, in seagoing condition at the hull transverse section being considered as defined in Pt.3, Ch.5, Sec.3, Cl.3.4.2.

M_{wv} = Vertical wave bending moment in [kN-m], for hogging and sagging, in seagoing condition at the hull transverse section being considered as defined in Pt.3, Ch.5, Sec.3, Cl.3.7.1.

Q_{sw} = Permissible positive or negative still water shear force for seagoing operation, in [kN], at the hull transverse section considered, as defined in Pt.3, Ch.5, Sec.3, Cl.3.6.2.

Q_{wv} = Positive and negative vertical wave shear force, in [kN], for seagoing condition at the hull transverse section being considered, as defined in Pt.3, Ch.5, Sec.3, Cl.3.7.2.

ΔQ_s = Shear force correction at the transverse section considered, in [kN], if applicable. as defined in Pt.3, Ch.6, Sec.2, Cl.2.2.5.

1.5.3.5 Hull girder ultimate strength check

(a) In addition to the hull girder ultimate strength check requirements given in Pt.3, Ch.6, Sec.4 for intact conditions, the same hull girder ultimate strength requirement applies to flooded conditions using hull girder ultimate bending loads in accordance with Cl.1.5.3.5(b).

(b) Hull girder ultimate bending loads

The vertical hull girder bending moment in hogging and sagging conditions, in [kN-m], to be considered in the ultimate strength check is to be taken as:

$$M = \gamma_S M_{sw-f} + 0.8\gamma_W M_{wv}$$

where:

γ_S = Partial safety factor for the still water bending moment, taken in accordance with Pt.3, Ch.6, Sec.4.

γ_W = Partial safety factor for the vertical wave bending moment, taken in accordance with Pt.3, Ch.6, Sec.4.

1.6 Design load scenarios and Design Load sets

The design load scenarios given in Pt.3, Ch.5, Sec.6 is to be complied with, in addition to the additional principal design load scenarios for dry cargo ships given in Cl.1.6.1.

1.6.1 Additional principal design load scenarios for strength assessment

The additional principal design load scenarios for strength assessment of dry cargo ships are given in Table 1.6.1.

Table 1.6.1: Additional principal design load scenarios for strength assessment

		Design Load scenarios	
		1	**5**
Load Component		**Normal operation at sea**	**Flooding requirements (1)**
		Static + Dynamic (S+D)	Accidental (A)
Hull Girder Loads (3)	VBM		$M_{sw-f} + M_{wv-f}$
	VSF		$Q_{sw-f} + Q_{wv-f}$
Local Loads - P_{in} (4)	Boundaries of bulk cargo holds	$P_{bs} + P_{bd}$	P_{bf-s} (2)

(1) The application of design load scenario 5 is limited to the following:
- allowable hold loading in flooded conditions: Sec.1, Cl.1.4.3.
- transverse vertically corrugated watertight bulkheads separating cargo holds in flooded condition: Sec.1, Cl.1.5.
- vertical hull girder bending and shear strength in flooded conditions: Sec.1, Cl.1.5.3.4.

(2) Static pressure P_{bf-s} is to be applied to vertically corrugated transverse bulkheads only.

(3) Hull girder loads:
M_{sw-f} = Permissible vertical still water bending moment in flooded condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Cl.1.3.3.2(a).
Q_{sw-f} = Positive and negative permissible vertical still water shear force in flooded condition, in [kN], at the hull transverse section being considered, as defined in Cl.1.3.3.2(a).
M_{wv-f} = Permissible vertical wave bending moment in flooded condition, in [kN-m], at the hull transverse section being considered in hogging and sagging, as defined in Cl.1.3.3.2(b).
Q_{wv-f} = Positive and negative permissible vertical wave shear force in flooded condition, in [kN], at the hull transverse section being considered, as defined in Cl.1.3.3.2(b).

(4) Local loads:
P_{bs} = Static dry bulk cargo pressure as given in Cl.1.3.2.
P_{bd} = Dynamic dry bulk cargo pressure as given in Cl.1.3.2.
P_{bf-s} = Static pressure on vertically corrugated transverse bulkhead of a flooded cargo hold as given in Sec.1, Cl.1.5.

1.6.2 Design load sets for ships intended to carry dry bulk cargo

(a) Application

The design load sets given in Cl.1.6.2(c) and Cl.1.6.2(d) apply to the cargo hold region of dry cargo ships, in addition to the design loads sets given in Pt.3, Ch.7, Sec.2, for the following structural members:

- additional design load sets for plating and stiffeners, in Table 1.6.2(a).
- additional design load sets for primary supporting members, in Table 1.6.2(b).

(b) Load components

The static and dynamic load components is to be determined in accordance with the principal design load scenarios given in Cl.1.6.1.

Radius of gyration, k_r, and metacentric height, GM, is to be in accordance with Pt.3, Ch.5, Sec.2, Table 2.1.2.

(c) Additional design load sets for plating and stiffeners of dry cargo ships

Additional design load sets for plating and stiffeners of dry cargo ships are given in Table 1.6.2(a).

(d) Additional design load sets for primary supporting members of dry cargo ships

Additional design load sets for primary supporting members of dry cargo ships are given in Table 1.6.2(b). The severest loading conditions from the loading manual or otherwise specified by the designer is to be considered for the calculation of $P_{bs} + P_{bd}$ and P_{bs} in design load sets BC-9 to BC-10.

For Primary supporting member in bilge hopper tanks and topside tanks, applicable design load sets in Pt.3, Ch.7, Sec.2 are to be considered.

(e) PSM's in holds intended to carry dry bulk cargo

The strength check of primary supporting members in cargo hold structures, subjected to lateral pressure is to comply with Rules for Bulk Carriers and Oil Tankers, Volume 2: Common Structural Rules, Part 2, Chapter 1, Section 4, 4.3 to 4.7.

In the calculation, following are to be used:
The design load set to be considered for Design Pressure, P in [kN/m^2], is to be according to Pt.3, Ch.7, Sec.2 and Table 1.6.2(b),

The reference coordinate system used for the evaluation point is to be taken as defined in Pt.3, Ch.1, Sec.2, Cl.2.5.1.

Effective bending span, in [m], of primary supporting members, measured between the supporting members is to be taken as defined in Pt.3, Ch.3, Sec.7, Cl.7.1.7.

Effective shear span, in [m], of primary supporting members, measured between the supporting members is to be taken as defined in Pt.3, Ch.3, Sec.7, Cl.7.1.8.

C_{t-pr} = Permissible shear stress coefficient for primary supporting members taken equal to:
C_{t-pr} = 0.70 for AC-I.
C_{t-pr} = 0.85 for AC-II.

C_{s-pr} = Permissible bending stress coefficient for primary supporting members taken equal to:
C_{s-pr} = 0.70 for AC-I.
C_{s-pr} = 0.85 for AC-II.

Table 1.6.2(a): Additional design load sets for plating and stiffeners of dry cargo ships

Structural member	Design load set	Design load scenario	Load component[1)]	Draught	Acceptance criteria	Loading condition
Boundaries of bulk cargo hold	BC-1	1	$P_{bs} + P_{bd}$	T_{SC}	AC-II	Homogeneous loading, fully filled
	BC-2	2	P_{bs}		AC-I	
	BC-3	1	$P_{bs} + P_{bd}$	T_{SC}	AC-II	Homogeneous loading, Partially filled
	BC-4	2	P_{bs}		AC-I	
	BC-5	1	$P_{bs} + P_{bd}$	T_{SC}	AC-II	Alternate light cargo, fully filled
	BC-6	2	P_{bs}		AC-I	
	BC-7	1	$P_{bs} + P_{bd}$	T_{SC}	AC-II	Alternate heavy cargo, partially filled
	BC-8	2	P_{bs}		AC-I	

1) Local loads:

P_{bs} = Static dry bulk cargo pressure as given in Cl.1.3.2.

P_{bd} = Dynamic dry bulk cargo pressure as given in Cl.1.3.2.

Table 1.6.2(b): Additional design load sets for primary supporting members of dry cargo ships

Structural member	Design load set	Design load scenario	Load component[1)]	Draught	Acceptance criteria	Loading condition
Double bottom floors and girders Stringers and transverse web frames in double side structures	WB-1(1)	1	$P_{in} - P_{ex}$	T_{BH}	AC-II	Heavy ballast condition
	WB-2(1)	2	P_{in}		AC-I	Harbour and Tank Testing condition
	BC-9	1	$(P_{bs} + P_{bd}) - (P_S + P_W)$	T_{SC}	AC-II	Full load condition
	BC-10	2	$P_{bs} - P_S$	T_{SC}	AC-I	
	FD-2(2)	5	P_{bf-s}		(3)	Flooded condition

(1) Only applicable to holds assigned as ballast hold.

(2) Not applicable to holds assigned as ballast hold.

(3) Applicable only to vertically corrugated bulkheads, specific criteria as given in Sec.1, Cl.1.5.

P_{bf-s} = Static pressure on vertically corrugated transverse bulkhead of a flooded cargo hold as given in Sec.1, Cl.1.5.

All possible flooding scenarios are to be investigated in the flooded condition which includes below:

- Empty hold flooded and opposite holds empty
- Homogeneous, flooded and opposite holds fully filled
- Homogeneous heavy cargo, flooded and opposite holds partially filled
- Alternate light, flooded hold fully filled, opposite hold empty
- Alternate heavy, flooded hold partially filled, opposite hold empty

1.7 Requirements for the fitting of a forecastle for bulk carriers, ore carriers and combination carriers

1.7.1 All vessels with any of the following class notations are to be fitted with an enclosed forecastle on the freeboard deck as detailed in Cl.1.7.2:

'BULK CARRIER'
BULK CARRIER "Strengthened for heavy cargoes"
BULK CARRIER "Strengthened for heavy cargoes, hold(s) …. may be empty"
'ORE CARRIER', 'ORE OR OIL CARRIER', 'OIL OR BULK CARRIER'.

1.7.2 The forecastle is to be located on the freeboard deck with its aft bulkhead fitted in way or aft of the collision bulkhead as shown in Figure 1.7.2.

However, if this requirement hinders hatch cover operation, the aft bulkhead of the forecastle may be fitted forward of the forward bulkhead of the foremost cargo hold provided the forecastle length is not less than 7% of ship length abaft the forward perpendicular where the ship length and forward perpendicular are defined in the International Convention on Load Line 1966 and its Protocol 1988.

The forecastle height H_F above the main deck is to be not less than:

- the standard height of a superstructure as specified in the International Convention on Load Line 1966 and its Protocol of 1988, or

- $H_C + 0.5$[m], where H_C is the height of the forward transverse hatch coaming of the foremost cargo hold.

whichever is the greater.

1.7.3 Reduced horizontal loading on the forward transverse hatch coaming and hatch cover of the foremost cargo hold as per Pt.3, Ch.12, Sec.6, Cl.6.2.2 and Sec.9, Cl.9.11.3 due to the shielding effect provided by the forecastle, can be considered only for all points located at a distance less than or equal to ℓ_F from the aft edge of the forecastle deck, taken as:

$$\ell_F = 5\sqrt{H_F - H_C}$$

where,

ℓ_F, H_F and H_C are as shown in Figure 1.7.2.

1.7.4 A breakwater is not to be fitted on the forecastle deck with the purpose of protecting the hatch coaming or hatch covers. If fitted for other purposes, it is to be located such that its upper edge at centre line is not less than H_B/tan 20° forward of the aft edge of the forecastle deck, where H_B is the height of the breakwater above the forecastle (Refer Figure 1.7.2).

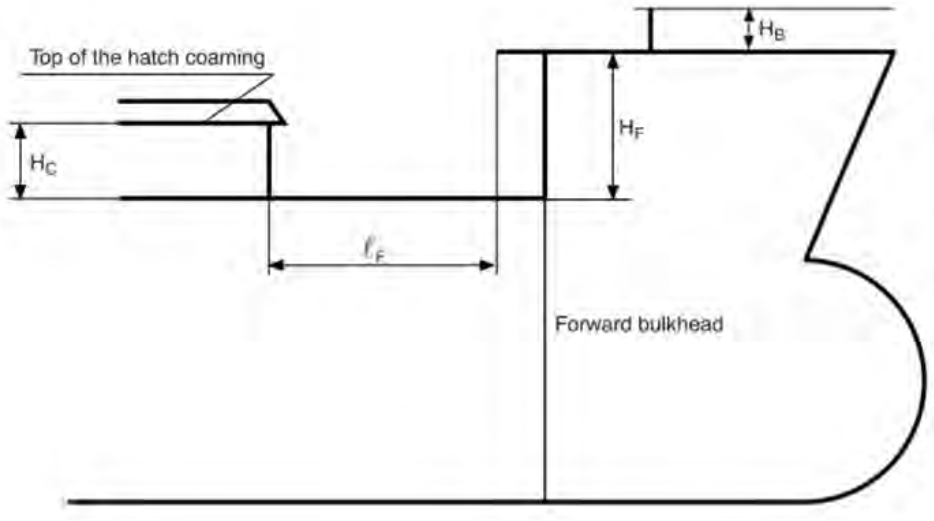


Figure 1.7.2: Forecastle arrangement

1.8 Arrangements for access in the cargo area and forward spaces

1.8.1 The requirements in 1.8 and 1.9 apply to bulk carriers of single or double side skin construction, with double bottom, hopper side tanks and top side tanks fitted below the upper deck, ore carriers and combinations carriers of 20,000 gross tonnage and over.

1.8.2 Each space is to be provided with a means of access to enable, throughout the life of the ship, overall and close-up inspections and thickness measurements of the ship's structures. Such means of access are to comply with the requirements of the Technical provisions for means of access for inspections, specified in 1.9

Where a permanent means of access may be susceptible to damage during normal cargo loading and unloading operations or where it is impracticable to fit permanent means of access, the provision of movable or portable means of access, as specified in 1.9 may be considered, provided that the means of attaching, rigging, suspending or supporting the portable means of access forms a permanent part of the ship's structure. All portable equipment may be capable of being readily erected or deployed by ship's personnel.

Each space for which close-up inspection is not required such as fuel oil tanks and void spaces forward of cargo area, may be provided with a means of a access necessary for overall survey intended to report on the overall conditions of the hull structure.

1.8.3 The construction and materials of all means of access and their attachment to the ship's structure are to be approved by IRS.

1.8.4 Access to cargo holds, cofferdams, ballast tanks and other spaces in the cargo area is to be direct from the open deck and such as to ensure their complete inspection. Access to double bottom spaces or to forward ballast tanks may be from a pipe tunnel, cargo hold, double hull space or similar compartment not intended for the carriage of hazardous cargoes. Access to a double side skin spaces may be from a topside tank or double bottom tank.

1.8.5 Tanks and subdivisions of tanks, having a length of 35[m] or more are to be fitted with at least two access hatchways and ladders, as far apart as practicable. Tanks less than 35[m] in length is to be served by at least one access hatchway and ladder. When a tank is subdivided by one or more swash bulkheads or similar obstructions which do not allow ready means of access to the other parts of the tank, at least two hatchways and ladders are to be fitted.

1.8.6 Each cargo hold is to be provided with at least two means of access as far apart as practicable. In general, these accesses should be arranged diagonally, for example one access near the forward bulkhead on the port side, the other one near the aft bulkhead on the starboard side.

1.8.7 For access through horizontal openings, hatches or manholes, the dimensions are to be sufficient to allow a person wearing a self-contained air-breathing apparatus and protective equipment to ascend or descend any ladder without obstruction and also provide a clear opening to facilitate the hoisting of an injured person from the bottom of the space.

1.8.8 A ship's means of access to carry out overall and close-up inspections and thickness measurements is to be described in a Ship Structure Access Manual which is to consist of two parts.

The first part should include the following for each space:

a) plans showing the means of access to the space, with appropriate technical specifications and dimensions;

b) plans showing the means of access within each space to enable an overall inspection to be carried out, with appropriate technical specifications and dimensions. The plans are to indicate from where each area in the space can be inspected;

c) plans showing the means of access within the space to enable close-up inspection to be carried out, with appropriate technical specifications and dimensions. The plans are to indicate the positions of critical structural areas, whether the means of access is permanent or portable and from where each area can be inspected. Critical structural areas are to be identified by advanced calculation techniques for structural strength and fatigue performance, and service history of similar ships;

d) instructions for regularly inspecting and maintaining the structural strength of all means of access and means of attachment,

taking into account any corrosive atmosphere that may be within the space;

e) instructions for safety guidance when rafting is used for close-up inspections and thickness measurements;

f) instructions for the rigging and use of any portable means of access in a safe manner;

g) an inventory of all portable means of access; and

The second part of the Ship Structure Access Manual is to contain a form of record of periodical inspections and maintenance, and change of inventory of portable equipment due to additions or replacement after construction. The format of this part is to be approved at the time of construction of the ship. The manual is to include a re-approval procedure for any changes to the permanent, portable or movable means of access.

The Ship Structure Access Manual is to be approved by IRS and an updated copy including all revisions / re-approvals is to be kept onboard.

1.9 Technical provisions for means of access for inspections

1.9.1 **Definitions**: For the purpose of these technical provisions, the following definitions apply:

a) **Rung** means the step of vertical ladder or step on the vertical surface.

b) **Tread** means the step of inclined ladder, or step for the vertical access opening.

c) **Flight of an inclined ladder** means the actual stringer length of an inclined ladder. For vertical ladders, it is the distance between the platforms.

d) **Stringer** means:

i) the frame of a ladder; or

ii) the stiffened horizontal plating structure fitted on side shell, transverse bulkheads and/or longitudinal bulkheads in the space. For the purpose of ballast tanks of less than 5[m] width forming double side spaces, the horizontal plating structure is credited as a stringer and a longitudinal permanent means of access, if it provides a continuous passage of 600[mm] or more in width past frames or stiffeners on the side shell or longitudinal bulkhead. Openings in stringer plating utilized as permanent means of access are to be arranged with guard rails or grid covers to provide safe passage on the stringer or safe access to each transverse web.

e) **Vertical ladder** means a ladder of which the inclined angle is 70° and over up to 90°. A vertical ladder is not to be skewed by more than 2°.

f) **Overhead obstructions** mean the deck or stringer structure including stiffeners above the means of access.

g) **Distance below deck head** means the distance below the plating.

h) **Cross deck** means the transverse area which is located inboard of the line of hatch openings of the main deck and between adjacent transverse hatch coamings.

1.9.2 Structural members subject to the close-up inspections and thickness measurements of the ship's structure referred to in Pt.1, Ch.2, except those in double bottom spaces, are to be provided with a permanent means of access to the extent as specified in Table 1.9.2(a) and Table 1.9.2(b), as applicable. For wing ballast tanks of ore carriers, approved alternative methods such as rafting may be used in combination with the fitted permanent means of access, provided that the structure allows for its safe and effective use.

1.9.3 Permanent means of access should as far as possible be integral to the structure of the ships, thus ensuring that they are robust and at the same time contributing to the overall strength of the structure of the ship.

Table 1.9.2(a): Means of access to ballast tanks in bulk carriers*

Top side tanks

1.1 For each topside tank of which the height is 6[m] and over, one longitudinal continuous permanent means of access shall be provided along the side shell webs and installed at a minimum of 1.6[m] to a maximum of 3[m] below deck with a vertical access ladder in the vicinity of each access to that tank.

1.2 If no access holes are provided through the transverse webs within 600[mm] of the tank base and the web frame rings have a web height greater than 1[m] in way of side shell and sloping plating, then stop rings/grab rails shall be provided to allow safe access over each transverse web frame ring.

1.3 Three permanent means of access, fitted at the end bay and middle bay of each tank, shall be provided spanning from tank base up to the intersection of the sloping plate with the hatch side girder. The existing longitudinal structure, if fitted on the sloping plate in the space may be used as part of this means of access. If the longitudinal structures on the sloping plate are fitted outside of the tank a means of access is to be provided.

1.4 For topside tanks of which the height is less than 6[m], alternative means as defined in Cl.1.9.10 of the technical provisions or a portable means may be utilized in lieu of the permanent means of access.

Bilge hopper tanks

1.5 For each bilge hopper tank of which the height is 6[m] and over, one longitudinal continuous permanent means of access shall be provided along the side shell webs and installed at a minimum of 1.2[m] below the top of the clear opening of the web ring with a vertical access ladder in the vicinity of each access to the tank.

Note: The height of a bilge hopper tank located outside of the parallel part of vessel is to be taken as the maximum of the clear vertical distance measured from the bottom plating to the hopper plating of the tank.

1.5.1 An access ladder between the longitudinal continuous permanent means of access and the bottom of the space shall be provided at each end of the tank.

1.5.2 Alternatively, the longitudinal continuous permanent means of access can be located through the upper web plating above the clear opening of the web ring, at a minimum of 1.6[m] below the deck head, when this arrangement facilitates more suitable inspection of identified structurally critical areas. An enlarged longitudinal frame of at least 600[mm] clear width can be used for the purpose of the walkway.

The foremost and aftermost bilge hopper ballast tanks with raised bottom of which the height is 6[m] and over, a combination of transverse and vertical means of access for access to the sloping plate of hopper tank connection with side shell plating for each transverse web can be accepted in place of the longitudinal permanent means of access.

Table 1.9.2a (Contd.)

1.5.3 For double side skin bulk carriers the longitudinal continuous permanent means of access may be installed within 6[m] from the knuckle point of the bilge, if used in combination with alternative methods to gain access to the knuckle point.

1.6 If no access holes are provided through the transverse ring webs within 600[mm] of the tank base and the web frame rings have a web height greater than 1[m] in way of side shell and tank base, then step rungs/grab rails are to be provided to allow safe access over each transverse web frame ring.

1.7 For bilge hopper tanks of which the height is less than 6[m], alternative means as defined in Cl.1.9.10 or a portable means may be utilized in lieu of the permanent means of access. It is to be demonstrated that such means of access can be deployed and made readily available in the areas where needed.

Double skin side tanks

1.8 Permanent means of access is to be provided in accordance with the applicable sections of Pt.5, Ch.2, Table 2.4.2.

Fore peak tanks

1.9 For fore peak tanks with a depth of 6[m] or more at the centre line of the collision bulkhead, a suitable means of access shall be provided for access to critical areas such as the underdeck structure, stringers, collision bulkhead and side shell structure.

1.9.1 Stringers of less than 6[m] in vertical distance from the deck head or a stringer immediately above are considered to provide suitable access in combination with portable means of access.

1.9.2 In case the vertical distance between the deck head and stringers, stringers or the lowest stringer and the tank bottom is 6[m] or more, alternative means of access as defined in Cl.1.9.10 is to be provided.

Table 1.9.2(b): Means of access to cargo holds of bulk carriers*

Access to underdeck structure

1.1 Permanent means of access is to be fitted to provide access to the overhead structure at both sides of the cross deck and in the vicinity of the centerline. Each means of access is to be accessible from the cargo hold access mentioned in Cl.1.8.6 or directly from the main deck and installed at a minimum of 1.6[m] to a maximum of 3[m] below the deck. Means of access is to be provided to the cross-deck structures of the foremost and aftermost part of the cargo hold.

1.2 An athwartship permanent means of access fitted on the transverse bulkhead at a minimum 1.6[m] to a maximum 3[m] below the cross-deck head is accepted as equivalent to 1.1

1.3 Access to the permanent means of access to overhead structure of the cross deck may also be via the upper stool.

1.4 Ships having transverse bulkheads with full upper stools (extending between top side tanks and upto the hatch end beam) with access from the main deck which allows monitoring of all framing and plates from inside, do not require permanent means of access of the cross deck.

1.5 Special attention is to be paid to the structural strength where any access opening is provided in the main deck or cross deck. The requirements for access to cross deck structure are also applicable to ore carriers.

1.6 Alternatively, movable means of access may be utilized for access to the overhead structure of cross deck if its vertical distance is 17[m] or less above the tank top. This movable means of access need not be necessarily be carried on board the vessel, but is to be made available when required.

Access to vertical structures

1.7 Permanent means of vertical access shall be provided in all cargo holds and built into the structure to allow for an inspection of a minimum of 25% of the total number of hold frames port and starboard equally distributed throughout the hold including at each end in way of transverse bulkheads. But in no circumstance shall this arrangement be less than 3 permanent means of vertical access fitted to each side (fore and aft ends of hold and mid-span). Permanent means of vertical access fitted between two adjacent hold frames is counted for an access for the inspection of both hold frames. A means of portable access may be used to gain access over the sloping plating of lower hopper ballast tanks.

1.8 In addition, portable or movable means of access are to be utilized for access to the remaining hold frames upto their upper brackets and transverse bulkheads.

1.9 Portable or movable means of access may be utilized for access to hold frames upto their upper bracket in place of the permanent means required in 1.6. These means of access are to be carried on board the ship and able to be transported to location in cargo hold and safely erected by ship's staff.

1.10 The width of vertical ladders for access to hold frames is to be at least 300[mm], measured between stringers.

1.11 A single vertical ladder over 6[m] in length is acceptable for the inspection of the hold side frames in a single skin construction.

1.12 For double side skin construction no vertical ladders for the inspection of the cargo hold surfaces are required. Inspection of this structure should be provided from within the double hull space.

1.9.4 Elevated passageways forming sections of a permanent means of access, where fitted, is to have a minimum clear width of 600[mm], except for going around vertical webs where the minimum clear width may be reduced to 450[mm] and have guard rails over the open side/ sides of their entire length. Sloping structures that are sloped 5[deg] or more from horizontal plane, providing part of the access is to be of a non-skid construction. Non-skid construction is to be such that the surface on which the personnel walks provides sufficient friction to the sole of boots even when the surface is wet and covered with thin sediment.

Guard rails are to be atleast 1000[mm] in height and consist of a rail and an intermediate rail. Guard rails are to be fitted on the open side. For stand-alone passageways guard rails are to be fitted on both sides of these structures. Guardrail stanchions are to be attached to the permanent means of access. The distance between the passageway and the intermediate rail and the distance between intermediate rail and the top rail shall not be more than 500[mm]. They are to be of substantial construction ensuring adequate design strength as well as residual strength during service life. Stanchions are to be not more than 3[m] apart. Discontinuous handrails are allowed provided the gap does not exceed 50[mm]. (Refer Figure 1.9.4). The distance between adjacent stanchions across the handrail gaps is to be not more than 350[mm].

Durability of passage ways and guard rails are to be ensured by corrosion protection and inspection and maintenance during services. Fire resistant materials are to be used for all means of access.

1.9.5 Access to permanent means of access and vertical openings from the ship's bottom is to be provided by means of easily accessible passageways, ladders or treads. Treads are to be provided with lateral support for the foot. Where the rungs of ladders are fitted against a vertical surface, the distance from the centre of the rungs to the surface is to be at least 150[mm]. Where vertical manholes are fitted higher than 600[mm] above the walking level, access is to be facilitated by means of treads and hand grips with platform landings on both sides. In such cases, it is to be demonstrated that an injured person can be easily evacuated.

* For ore carriers, permanent means of access shall be provided in accordance with the applicable sections of Table 1.9.2 and Pt.5, Ch.2, Table 2.4.2(a) and Table 2.4.2(b).

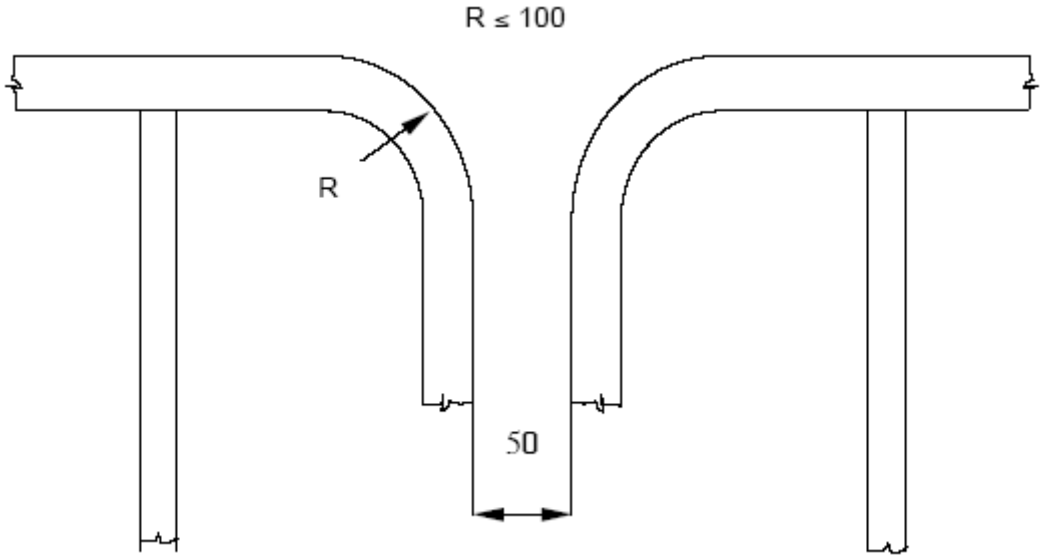


Figure 1.9.4: Arrangement of discontinuous handrails

1.9.6 Permanent inclined ladders are to be inclined at an angle of less than 70[deg]. There are to be no obstructions within 750[mm] of the face of the inclined ladder, except that in way of an opening this clearance may be reduced to 600[mm]. The clearance distances are to be measured perpendicular to the face of the ladder. Resting platforms of adequate dimensions are to be provided normally at a maximum of 6[m] vertical height. Requirements for resting platforms are to be similar to that of elevated passageways.

Ladders and handrails are to be constructed of steel or equivalent material of adequate strength and stiffness and securely attached to the structure by stays. The method of support and length of stay are to be such that vibration is reduced to a practical minimum. In cargo holds, ladders are to be designed and arranged so that cargo handling difficulties are not increased and the risk of damage from cargo handling gear is minimized.

1.9.7 The width of inclined ladders between stringers are not to be less than 400[mm] in general. However, the width of inclined ladders for access to a cargo hold is to be at least 450[mm]. The treads are to be equally spaced at a distance apart, measured vertically, of between 200[mm] and 300[mm]. When steel is used, the treads are to be formed of two square bars of not less than 22[mm] by 22[mm] in section, fitted to form a horizontal step with the edges pointing upward. The treads are to be carried through the side stringers and attached thereto by double continuous welding. All inclined ladders are to be provided with two course handrails of substantial construction on both sides, fitted at a convenient distance above the treads. Vertical height of handrails is not to be less than 890[mm] from the center of the step and two course handrails need only be provided where the gap between stringer and top handrail is greater than 500[mm].

1.9.8 For vertical ladders or spiral ladders, the width and construction should be in accordance with accepted international or national standards.
The width of vertical ladders is to be not less than 350[mm] and the vertical distance between the rungs is to be between 250[mm] and 350[mm]. The minimum climbing clearance in width is to be 600[mm] other than for ladders between hold frames.

The vertical ladders are to be secured at intervals not exceeding 2.5[m] apart to prevent vibration.

1.9.9 No free-standing portable ladder is to be more than 5[m] long. Mechanical devices such as hooks for securing the upper end of a ladder is considered as an appropriate securing device if movement fore/ aft and sideways can be prevented at the upper end of the ladder.

1.9.10 Alternative means of access include, but are not limited to such devices as:

a) hydraulic arm fitted with a stable base;
b) wire lift platform;
c) staging;
d) rafting;
e) robot arm or remotely operated vehicle (ROV);
f) portable ladders (ladders more than 5[m] long shall only be utilized if fitted with a mechanical device to secure the upper end of the ladder);
g) other means of access, approved by and acceptable to IRS.

(Refer to Classification Notes: “Guidelines for approval / acceptance of alternative means of access to spaces in oil tankers, bulk carriers, ore carriers and combination carriers”).

Means for safe operation and rigging of such equipment to and from and within these spaces are to be clearly described in the Ship Structure Access Manual.

1.9.11 For access through horizontal openings, hatches or manholes, the minimum clear opening is not to be less than 600[mm] x 600[mm], which may have corner radii of not more than 100[mm]. Where a larger corner section is desired to reduce the stress level around the opening, the size of the opening is to be suitably increased to ensure the required size of clear opening e.g. 600[mm] x 800[mm] opening with 300[mm] corner radius is acceptable.

When access to a cargo hold is arranged through the cargo hatch, the top of the ladder is to be placed as close as possible to the hatch coaming.

Access hatch coamings having a height greater than 900[mm] shall also have steps on the outside in conjunction with the ladder.

1.9.12 For access through vertical openings, or manholes, in swash bulkheads, floors, girders and web frames providing passage through the length and breadth of the space, the minimum opening is to be not less than 600[mm] x 800[mm] at a height of not more than 600[mm]

from the passage unless gratings or other foot holds are provided.

The opening of 600[mm] x 800[mm] may have corner radii of 300[mm]. An opening of 600[mm] in height x 800[mm] in width may be accepted as access opening in vertical structures where it is not desirable to make large opening considering structural strength aspects, such as in girders and floors in double bottom tanks.

Subject to verification of easy evacuation of injured person on a stretcher, a vertical opening 850[mm] x 620[mm] as shown in Figure 1.9.12 is considered as an acceptable alternative to the opening of 600[mm] width x 800[mm] height with corner radii of 300[mm].

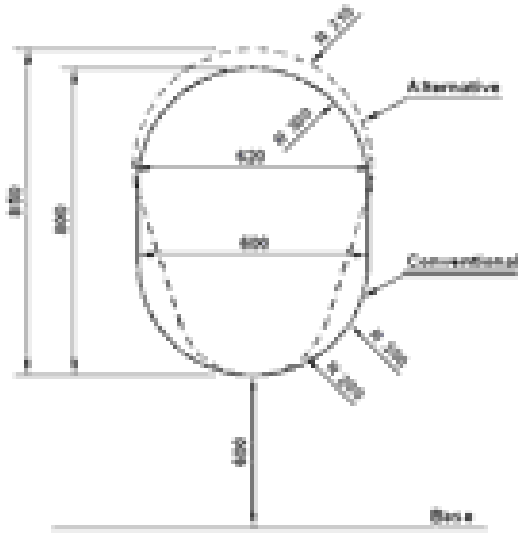


Figure 1.9.12: Access opening in vertical structure

1.9.13 Access ladders to cargo holds and other spaces are to be as follows:

a) Where the vertical distance between the upper surface of adjacent decks or between deck and the bottom of the cargo space is not more than 6[m], either a vertical ladder or an inclined ladder or a combination of both.

b) Where the vertical distance between the upper surface of adjacent decks or between deck and the bottom of the cargo space is more than 6[m], an inclined ladder or series of inclined ladders at one end of the cargo hold, except the uppermost 2.5[m] of a cargo space measured clear of overhead obstructions and the lowest 6 [m] may have vertical ladders, provided that the vertical extent of the inclined ladder or ladders connecting the vertical ladders is not less than 2.5[m].

   The second means of access at the other end of the cargo hold may be formed of a series of staggered vertical ladders, which should comprise of one or more ladders linking platforms spaced not more than 6 [m] apart vertically and displaced to one side of the ladder. Adjacent sections of ladder should be laterally offset from each other by at least the width of the ladder. The uppermost entrance section of the ladder directly exposed to a cargo hold should be vertical for a distance of 2.5[m] measured clear of overhead obstructions and connected to a ladder-linking platform.

c) A vertical ladder may be used as a means of access to topside tanks, where the vertical distance is 6[m] or less between the deck and the longitudinal means of access in the tank or the stringer or the bottom of the space immediately below the entrance. The uppermost entrance section from deck of the vertical ladder of the tank should be vertical for a distance of 2.5[m] measured clear of the overhead obstructions and comprises a ladder linking platform unless landing on the longitudinal means of access, the stringer or the bottom within the vertical distance. The platform is to be displaced to one side of a vertical ladder.

d) Unless allowed in c) above, an inclined ladder or combination of ladders should be used for access to a tank or a space where the vertical distance is greater than 6[m] between the deck and a stringer immediately below the entrance, between stringers, or between the deck or a stringer and the bottom of the space immediately below the entrance.

   In such cases the uppermost entrance section from deck of the ladder should be vertical for a distance of 2.5[m] clear of the overhead obstructions and connected to a landing platform and continued with an inclined ladder. The flights of inclined ladders are not to be more than 9[m] in actual length and the vertical height is not normally be more than 6[m]. The lowermost section of the ladders may be vertical for a vertical distance of not less than 2.5[m].

e) In spaces of less than 2.5[m] width, the access to the space may be by means of vertical ladders that comprise of one or

more ladder linking platforms spaced not more than 6[m] apart vertically and displaced to one side of the ladder. Adjacent sections of ladder should be laterally offset from each other by at least the width of the ladder.

f) A spiral ladder is considered acceptable as an alternative for inclined ladders. In this case the spiral ladder may be continued in the uppermost 2.5[m] and the change over to vertical ladder is not required.

1.9.14 The uppermost entrance section from weather deck of the vertical ladder providing access to a tank is to be vertical for a distance of 2.5[m] measured clear of the overhead obstructions and comprise a ladder linking platform, displaced to one side of a vertical ladder. The vertical ladder can be between 1.6[m] and 3[m] below weather deck structure if it lands on a longitudinal or athwartship permanent means of access fitted within that range.

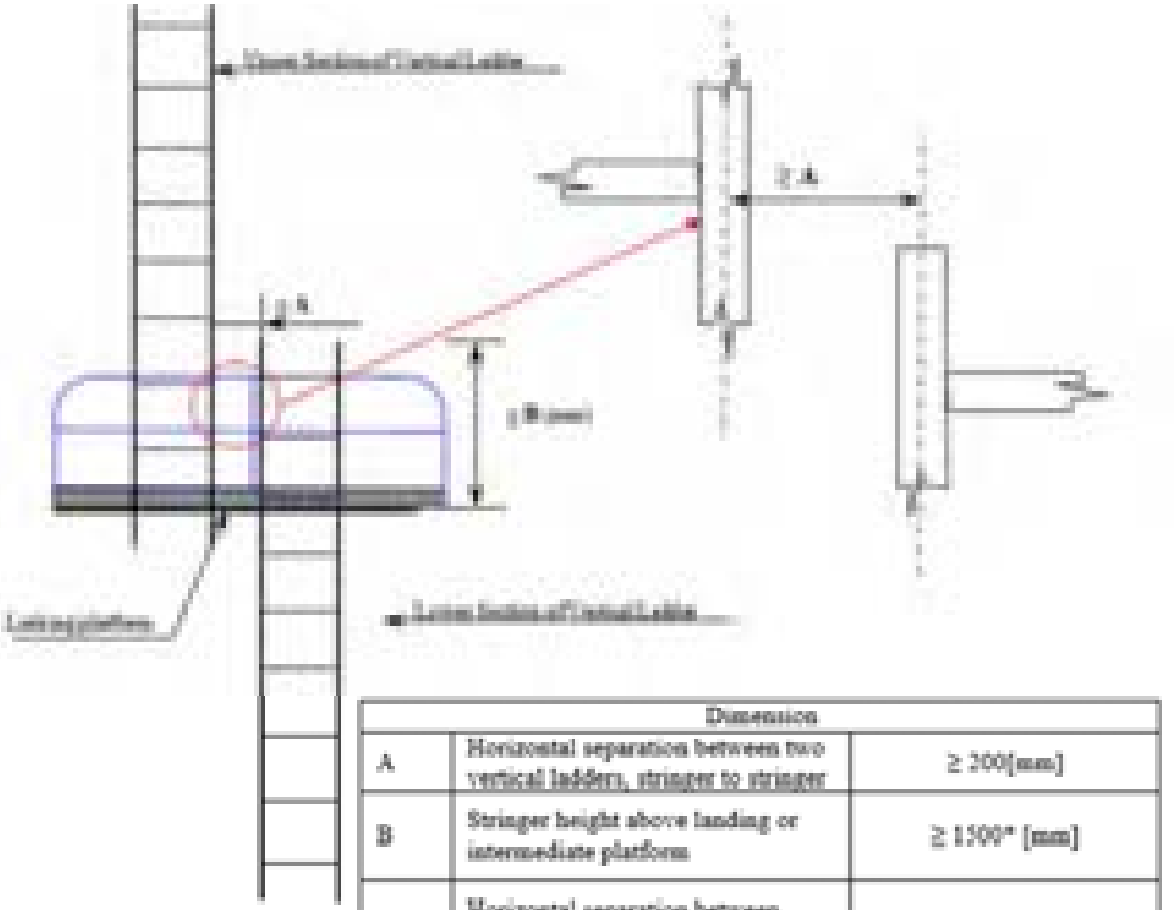
1.9.15 With respect to Cl.1.9.13 b) and e), adjecent sections of vertical ladder need to be installed so that the following provisions are complied with,(Refer Figure “A” and Figure “B”)

a) The minimum “lateral offset” , between two adjacent sections of vertical ladder, is the distance between the sections, upper and lower, so that the sdjecent stringers are spaced of at least 200mm, measured from half thickness of each stringer.

b) Adjacent section is vertically overlapped, in respect to the lower end of the upper section, to a height of 1500[mm] in order to permit a safe transfer between ladders.

c) no section of the access ladder is to be terminated directly or partly above an access opening.

Figure "A"
Vertical Ladder – Ladder through the linking platform



Dimension		
A	Horizontal separation between two vertical ladders, stringer to stringer	≥ 300[mm]
B	Stringer height above landing or intermediate platform	≥ 1500* [mm]
C	Horizontal separation between ladder and platform	100[mm] ≤ C< 300[mm]
*Note: the minimum height of the handrail of resting platform is of 1000[mm] (Technical Provision, resolution MSC.158(78), paragraph 3.3)		

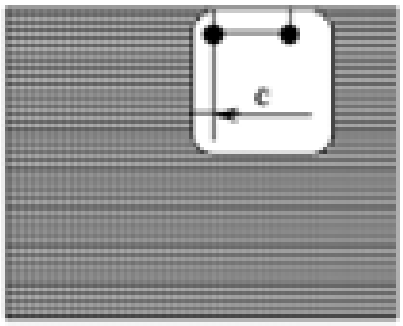
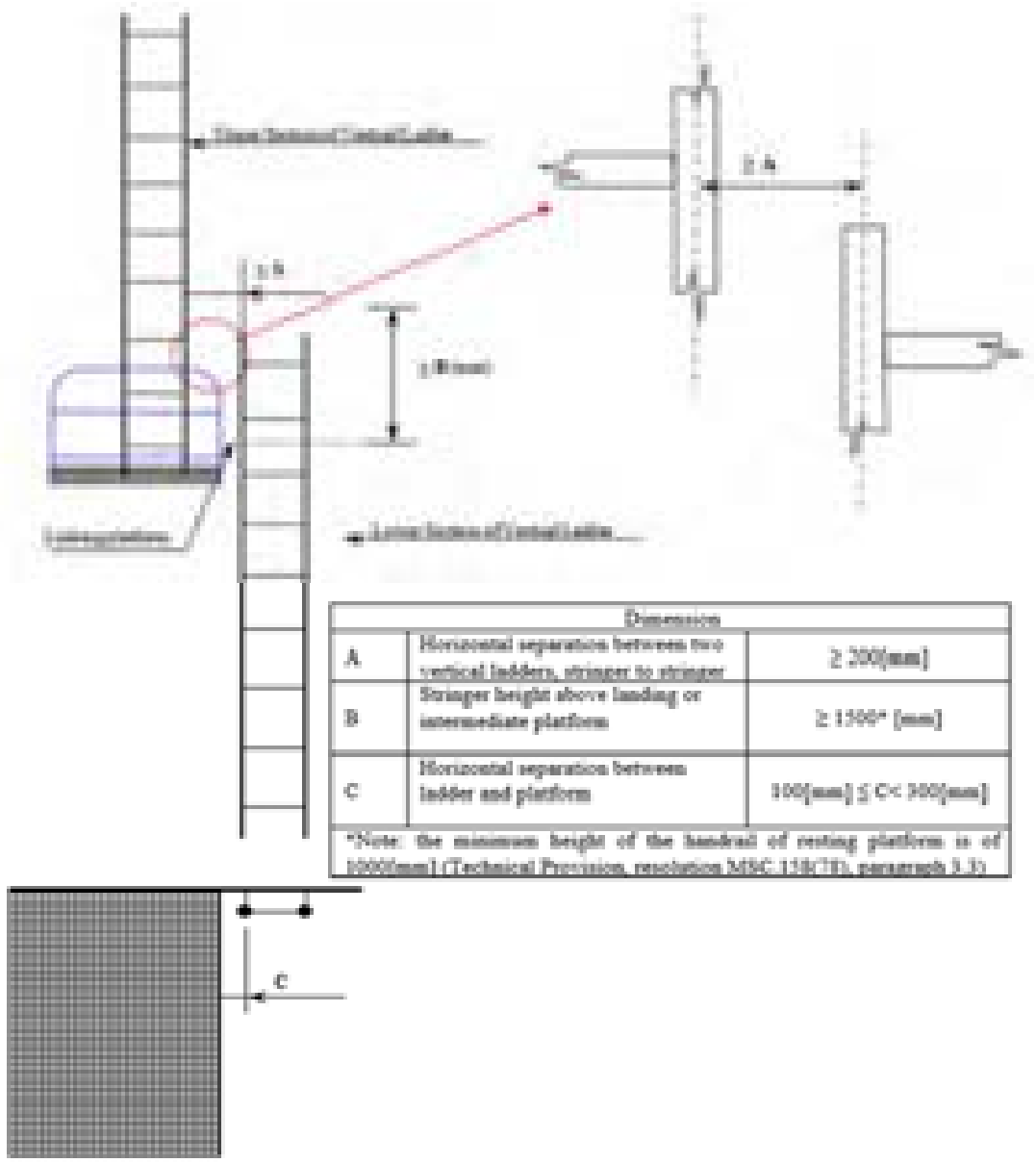


Figure "B"
Vertical Ladder – Side mount



Dimension		
A	Horizontal separation between two vertical ladders, stringer to stringer	≥ 200[mm]
B	Stringer height above landing or intermediate platform	≥ 1500* [mm]
C	Horizontal separation between ladder and platform	100[mm] ≤ C< 300[mm]

*Note: the minimum height of the handrail of resting platform is of 1000[mm] (Technical Provision, resolution MSC.133(76), paragraph 3.3)

1.10 General requirements for double side skin construction

1.10.1 Bulk carriers of Length L≥150[m] are to comply with the following in all areas of double side skin construction.

a) Primary stiffening structures of the double side skin shall not be placed inside the cargo hold space.

b) Subject to the provisions below, the distance between the outer shell and the inner shell at any transverse section shall not be less than 1000[mm]. This distance is to be measured perpendicular to the outer shell from the top of the double bottom to the main deck, as indicated in Figure 1.10.1. The distance is to be maintained throughout the whole double-side skin construction.

The double side skin construction shall be such as to allow access for inspection as provided in regulation 1.9 and 1.10:

i) The clearances below need not be maintained in way of cross ties, upper and lower end brackets of transverse framing or end brackets of longitudinal framing.

ii) The minimum width of the clear passage through the double side skin space in way of obstructions such as piping or vertical ladders shall not be less than 600[mm].

iii) Where the inner and/or outer skins are transversely framed, the minimum clearance between the inner surfaces of the frames shall not be less than 600[mm].

iv) Where the inner and outer skins are longitudinally framed, the minimum clearance between the inner surfaces of the frames shall not be less than 800[mm]. Outside the parallel part of the cargo hold length, this clearance may be reduced where necessitated by the structural configuration but in no case shall be less than 600[mm].

v) The minimum clearance referred to above shall be the shortest distance measured between assumed lines connecting the inner surfaces of the frames on the inner and outer skins.

c) The double-side skin spaces, with the exception of top-side wing tanks, if fitted, shall not be used for the carriage of cargo.

1.11 Buckling - Single failure of cargo hold structures

1.11.1 Requirements on lateral buckling for ordinary stiffeners in Pt.3 Ch.3 are to be complied with.

1.11.2 For bulk carriers of length L≥150[m], carrying solid bulk cargoes having a density of 1.0[t/m^3] and above, the safety factor with respect to lateral buckling of longitudinal and transverse ordinary stiffeners is to be increased by a factor at least of 1.15 (allowable utilization factor to be reduced by at least 1/1.15 =0.87) for the following cases:

a) Hatchway coaming
b) Inner bottom
c) Sloped stiffened panel of topside tanks and hopper tanks (if any)
d) Inner side (if any)
e) Top stool and bottom stool of transverse bulkhead (if any)
f) Stiffened transverse bulkhead (if any)
g) Side shell (if directly bounding the cargo hold).

1.12 Protection against loading/discharge equipment

For bulk carriers of length L≥150[m], carrying solid bulk cargoes having a density of 1.0[t/m^3] and above, the following requirement is applicable, excluding ships with inner bottom construction adapted for self-unloading,

- Wire rope grooving in way of cargo holds openings is to be prevented by fitting suitable protection such as half-round bar on the hatch side girders (i.e. upper portion of top side tank plates)/hatch end beams in cargo hold and upper portion of hatch coamings.

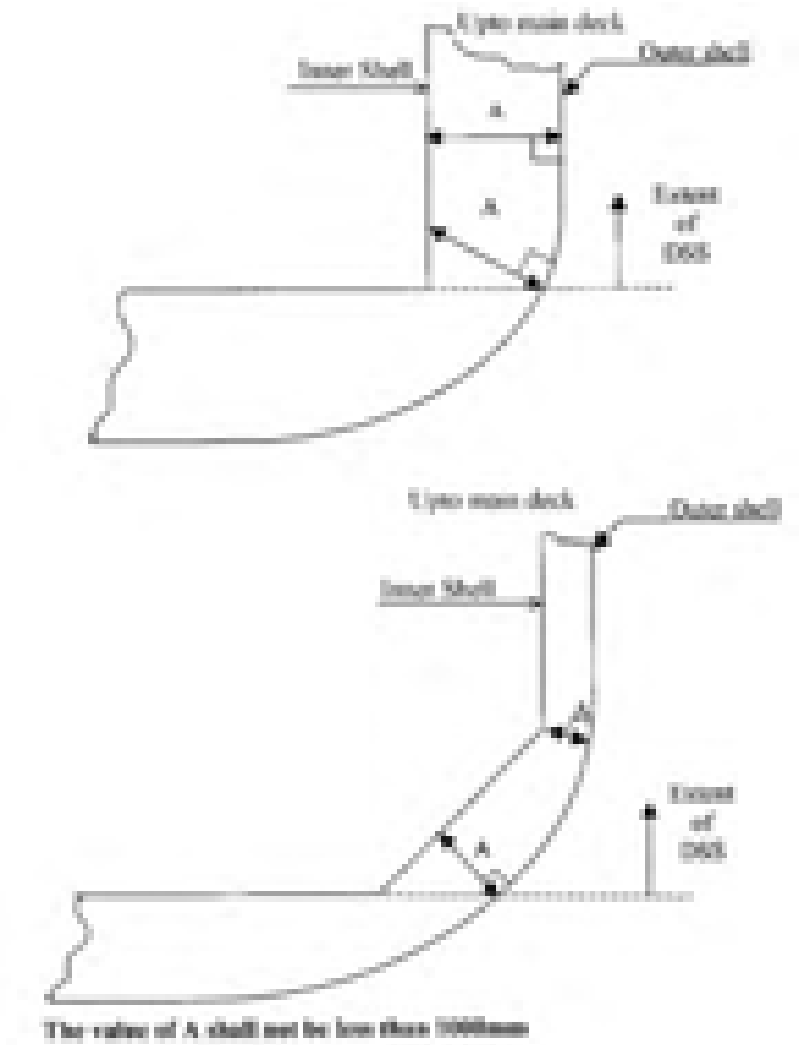


Figure 1.10.1: Distance between inner and outer shell in way of double-side skin

Section 2

Bulk Carriers

For symbols not defined in this section, Refer to Pt.3, Ch.1, Sec.2.

2.1 Hull arrangement

2.1.1 Within the cargo region, longitudinal stiffening system, in general, is to be adopted for the double bottom, strength deck outside line of openings and in the hopper and topside tanks. Proposals for transverse framing on sloping bulkheads and side shell within the hopper side tanks and top side tanks will be specially considered.

2.1.2 In the hopper side tanks and top side tanks, tank end bulkheads or wash bulkheads are to be provided in line with the main hold bulkheads.

2.2 Longitudinal strength

2.2.1 The longitudinal strength in intact condition is to be in accordance with the requirements given in Pt.3 Ch.6.

2.2.2 Bulk carriers of the following configurations are to have sufficient strength to withstand flooding of each cargo hold in all cargo loading and ballast conditions:

a) Single side skin bulk carriers of length L≥150[m] and designed to carry bulk cargoes having a density of 1.0[t/m^3] and above.

b) Double side skin bulk carriers of length L≥150[m] in which any part of the longitudinal bulkhead is located within B/5 or 11.5[m] whichever is less, inboard from the ship's side at right angle to the centre line at the assigned summer load line, designed to carry bulk cargoes having a density of 1.0[t/m^3] and above.

2.2.3 The hull girder section modulus and shear strength requirements for the flooded conditions at any location are to be calculated in a way similar to that for intact conditions (Refer Pt.3, Ch.6) considering the following:

- Maximum Stillwater bending moment and Shear force values in flooded conditions, at the section under consideration. Refer Cl.1.3.3.2(a)
- The wave bending moment and wave shear force values for the flooded condition as given in Section 1, Cl.1.3.3.2(b).
- The structure is assumed to remain fully effective in resisting the applied loading.

2.2.4 For intact and flooded conditions featuring uneven distribution of cargo loading (e.g. loading conditions with empty holds or ballast conditions with ballast in cargo holds), shear force correction ΔQ_S as per Pt.3, Ch.6, Sec.2, Cl.2.2.5 is to be applied.

2.2.5 The buckling strength of the structure participating in longitudinal strength is to comply the requirements given in Pt.3, Ch.9.

2.3 Bottom structure

2.3.1 The scantlings and arrangements are, in general, to be as per Pt.3, Ch.3 (Arrangements) and Ch.7(Scantlings), except as given below.

2.3.2 For bulk carriers with notation "Strengthened for heavy cargo" (with or without the additional empty hold notation) and for all ships in way of cargo hold(s) designated for ballast, the spacing of plate floors in the double bottom is generally not to exceed 2.5[m].

2.3.3 The inner bottom plating is to be based on the design loads given in Section 1, Cl.1.3.2 for intact conditions and Section 1, Cl.1.3.3 for hold flooded conditions, considering the design load scenarios and load sets as given in Pt.3 (as applicable) and Section 1.6, Cl.1.6.1 and Cl.1.6.2.

The permissible stress for Intact conditions is to be as per Pt.3, Ch.7, Sec.4 based on still water bending moment and wave bending moment given in Pt.3, Ch.5, Sec.3.

The permissible stress for hold flooded conditions is to be as per Pt.3, Ch.7, Sec.4 based on still water bending moment and wave bending moment in flooded conditions as given in Section 1, Cl.1.3.3.2.

For bulk carriers with notation "Strengthened for heavy cargo" (with or without the additional empty

hold notation) or the thickness of inner bottom plating between the hopper or side tanks is not to be less than:

$$t = 5.5 + 0.03L\sqrt{K} + t_c \text{ [mm]}$$

where,

L need not be taken as greater than 200[m].

2.3.4 The section modulus of inner bottom longitudinals is to be based on the design loads given in Section 1, Cl.1.3.2 for intact conditions and Section 1, Cl.1.3.3 for hold flooded conditions, considering the design load scenarios and load sets as given in Pt.3 (as applicable) and Section 1.6, Cl.1.6.1 and Cl.1.6.2.

The permissible stress for Intact conditions is to be as per Pt.3, Ch.7, Sec.5 based on still water bending moment and wave bending moment given in Pt.3, Ch.5, Sec.3.

The permissible stress for hold flooded conditions is to be as per Pt.3, Ch.7, Sec.5 based on still water bending moment and wave bending moment in flooded conditions as given in Section 1, Cl. 1.3.3.2.

2.3.5 Structural details in way of double bottom tank and hopper tank knuckle is to be given special attention during design/fabrication.

In all dry holds where the double bottom tank and hopper tank knuckle is of radiused construction and the floor spacing is 2.5[m] or greater, brackets as shown in Figure 2.3.5 are to be arranged mid-length between floors in way of the intersection. The brackets are to be attached to the adjacent inner bottom and hopper longitudinals. The thickness of the brackets is to be in accordance with Pt.3, Ch.3, Sec.5, Cl.5.3.11.2 but need not exceed 15[mm].

In way of floodable holds, two intermediate brackets as shown in Figure 2.3.5, are to be provided in all cases where the hopper to double bottom knuckle is radiused and one such intermediate bracket is to be provided where the double bottom tank and hopper tank knuckle is of welded construction.

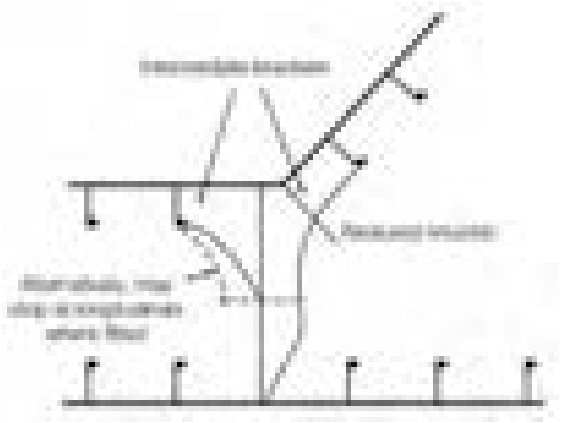


Figure 2.3.5: Intermediate brackets at hopper/Inner bottom interface

2.3.6 The connections at the knuckle are to be as follows:

a) Where of welded construction, the corner scallops in floors and transverses are to be omitted, or closed by welded collars where arranged for purposes of construction. In such cases, to ensure satisfactory welding of the collars, the radius of the scallops should not be less than 150[mm]. Refer Figure 2.3.6(a). Alternatively, the scallop may be retained on the hopper tank side, provided gusset plates are arranged in line with the inner bottom plating. Refer Figure 2.3.6(b)

b) Where of radiused construction, the corner scallops are to be omitted, and full penetration welding arranged locally for the connection to the inner bottom plating. The centre of the flange at the knuckle is not to be greater than 70[mm] from the side girder, Refer Figure 2.3.6(c).

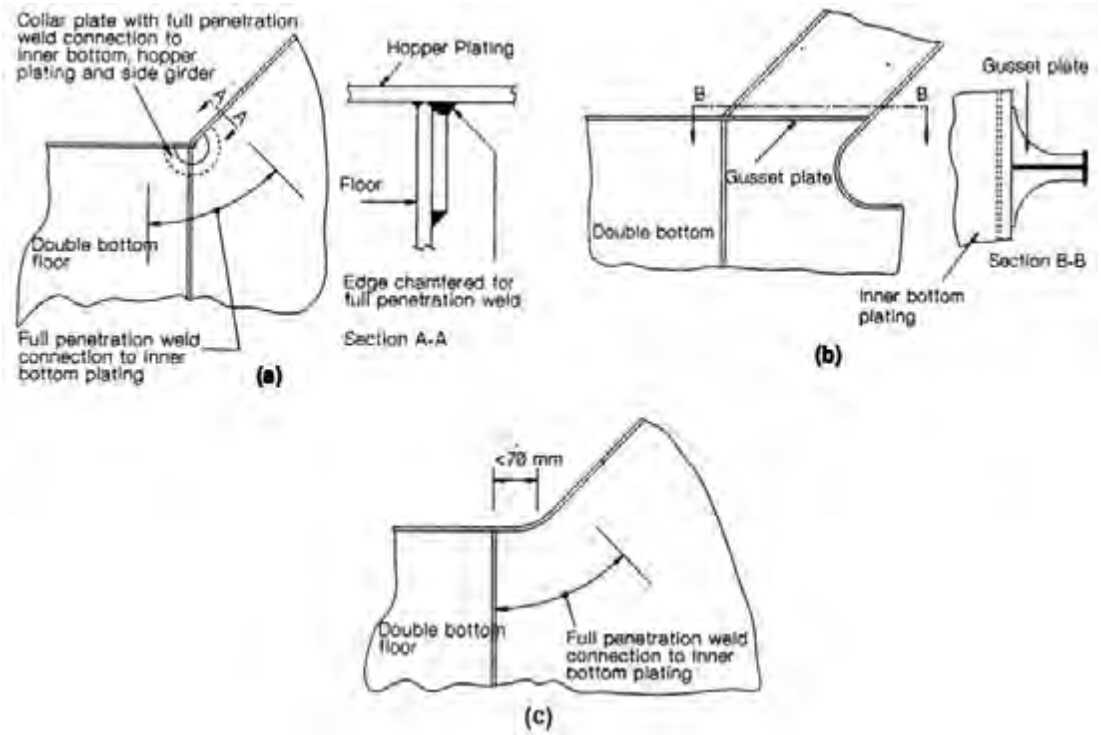


Figure 2.3.6: Connection at intersection of double bottom and hopper

2.4 Double side structure

2.4.1 General

(a) A ship has a double side configuration, if each ship side for all cargo holds is constructed by the side shell and a longitudinal bulkhead connecting the double bottom and the strength deck. Hopper side tanks and top wing tanks may, where fitted, be integral parts of the double side configuration.

(b) Side shell and inner hull bulkheads are generally to be longitudinally framed. Alternative framing arrangements will be specially considered.

2.4.2 Primary supporting members

(a) The spacing of transverse side primary supporting members is to be not greater than 3 frame spaces.

(b) Transverse side primary supporting members are to be fitted in line with web frames in double bottom or in hopper tanks and topside tanks, or aligned with large brackets in the spaces.

(c) Transverse primary supporting members is to be fitted in way of hatch end beams or similar large deck opening supporting transverse structure. Horizontal side stringers or scarfing brackets are to be fitted aft of the collision bulkhead in line with fore peak stringers, and forward of engine room bulkhead in line with platform decks in machinery spaces.

2.4.3 Transverse stiffeners

Transverse stiffeners on side shell and inner side, where fitted, are to be continuous within the height of the double side or fitted with bracket end connections. The transverse stiffeners are to be effectively connected to stringers. At their upper and lower ends, shell and inner side transverse stiffeners are to be connected by brackets to supporting stringer plates.

2.4.4 Longitudinal stiffeners

(a) Longitudinal stiffeners on side shell and inner side, where fitted, are to be continuous within the length of the parallel part of the cargo hold region. They are to be fitted with soft toe brackets in way of transverse bulkheads aligned with cargo hold bulkheads and are to be effectively connected to transverse web frames of the double side structure.

(b) Longitudinal framing of the side shell is to extend outside the cargo hold region as far forward as practicable.

2.4.5 Plating connections

(a) Inner hull plating and hopper tank structures, where fitted, shall be supported at forward and aft ends, e.g.by scarfing brackets in way of the collision bulkhead and the engine room bulkhead.

(b) Connection between the inner hull plating and the inner bottom plating is to be designed such that stress concentration is minimized.

(c) Connections of hopper tank plating with inner hull and with inner bottom are to be supported by a longitudinal girder. When a hopper tank is not fitted, the inner hull plating is to be supported by a longitudinal girder below the inner bottom plating and the inner bottom plating is to be supported by scarfing brackets.

2.5 Single side structure

2.5.1 Application

2.5.1.1 The requirements given in this section apply to the single side structure within the cargo hold region of dry cargo ships intended primarily to carry dry cargo in bulk with transverse framing, which are contracted for construction on or after 1 July 1998.

2.5.2 Minimum thickness requirements

a) Side shell plating

The thickness of the side shell plating located between the hopper and topside tanks is to be not less than:

$$t_{pmin} = \sqrt{L}[mm]$$

b) Web frames

The thickness of web of main frames situated within cargo and ballast holds, except in the foremost hold, is not to be less than:

$$t_{wmin} = (7 + 0.03L)[mm]$$

where,

L need not be taken as greater than 200[m].

In foremost hold, the frame web thickness is to be at least 15% greater than that given above.

2.5.3 Net section modulus and net shear sectional area

2.5.3.1 The net section modulus Z, in [cm^3], and the net shear sectional area A_{shr}, in [cm^2], in the mid-span area of side frames subjected to lateral pressure are not to be taken less than:

$$Z = 1.125\alpha_m \frac{P.s.\ell_{SF}^2}{f_{bdg}.C_s.R_{eH}}$$

$$A_{shr} = 5\alpha_s \frac{P.s.\ell_{SF}}{C_t.\tau_{eH}}\left(\frac{\ell_{SF} - 2\ell_B}{\ell_{SF}}\right)10^{-3}$$

where:

α_m = Coefficient taken as:
α_m = 0.42 for side frames of holds that may be empty in alternate conditions.
α_m = 0.36 for other ships.
f_{bdg} = Bending coefficient taken as 10.
C_s = Permissible bending stress coefficient for the design load set being considered taken as:
C_s = 0.75 for acceptance criteria set AC-I.
C_s = 0.90 for acceptance criteria set AC-II.
α_s = Coefficient taken as:
α_s = 1.1 for side frames of empty holds in alternate conditions.
α_s = 1.0 for other side frames.
ℓ_B = Lower bracket length, in [m], as defined in Figure 2.5.3.
P = Design pressures, in [kN/m^2], for design load sets as defined in Sec.1, 1.6.
C_t = Permissible shear stress coefficient for the design load set being considered, taken as:
C_t = 0.75 for acceptance criteria set AC-I.
C_t = 0.90 for acceptance criteria set AC-II.

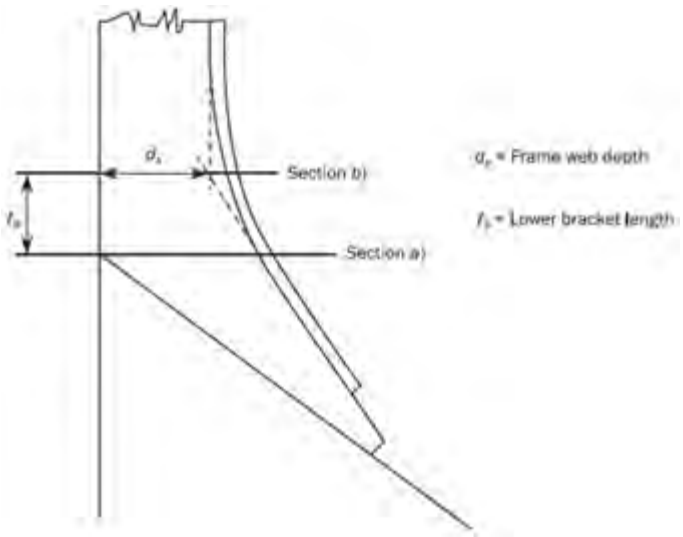


Figure 2.5.3: Side frame lower bracket length

2.5.3.2 Side frames in ballast holds

In addition to Cl. 2.5.3.1, for side frames in cargo holds designed to carry ballast water in heavy ballast condition, the net section modulus Z, in [cm^3], and the net web thickness, t_w, in [mm], all along the span are to be in accordance with Pt.3, Ch.7, Sec.5 where the span of the side frame is 'ℓ' as defined in Pt.3, Ch.7, Sec.7, Cl.7.1 with consideration to brackets at ends.

2.5.4 Additional strength requirements

The scantlings of hold side frames immediately adjacent to the collision bulkhead are to be suitably increased in order to prevent excessive deformation imposed on the shell plating.
The net moment of inertia I, in [cm^4], of the three side frames located immediately abaft the collision bulkhead is not to be taken less than:

$$I = 0.18 \frac{P \ell_{SF}^4}{n}$$

where:

n = Frame number of considered side frame counted from the collision bulkhead to the frame in question, taken equal to 1, 2 or 3.

As an alternative, supporting structures, such as horizontal stringers, are to be fitted between the collision bulkhead and a side frame which is in line with transverse webs fitted in both the topside tank and hopper tank, maintaining the continuity of the forepeak stringers within the foremost hold. Refer Figure 2.5.4.

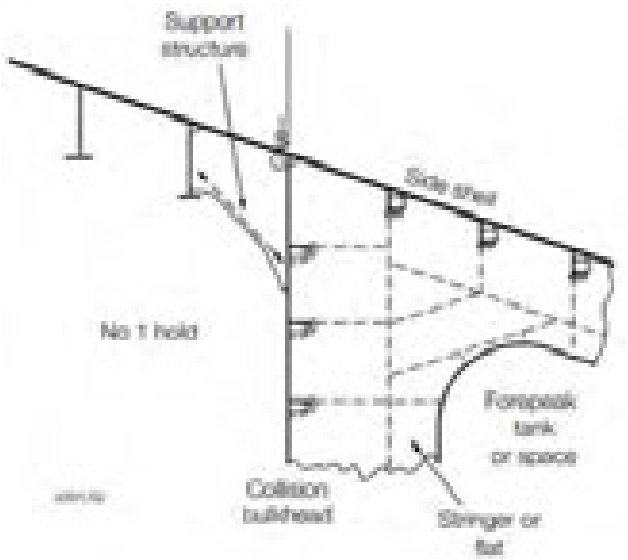


Figure 2.5.4: Hold frame supporting structures aft of collision bulkhead

2.5.5 Side Frames

Frames are to be fabricated symmetrical sections with integral upper and lower brackets and are to be arranged with soft toes.

The frame face plate is to be curved (not knuckled) at the connection with the end brackets. The radius of curvature, 'r', in [mm] is not to be less than:

$$r = \frac{0.4 b_f^2}{t_f + t_c}$$

where:

t_c = Corrosion addition, in [mm], as given in Pt.3, Ch.2, Sec.5.

b_f, t_f = Flange width and net thickness of the curved flange, in [mm]. The end of the flange is to be sniped.

In ships of length less than 190[m], frames when made of ordinary mild steel, may be asymmetric and fitted with separate brackets. The face plate or flange of the bracket is to be sniped at both ends. Brackets are to be arranged with soft toes. The dimensions of side frames are defined in Figure 2.5.5.

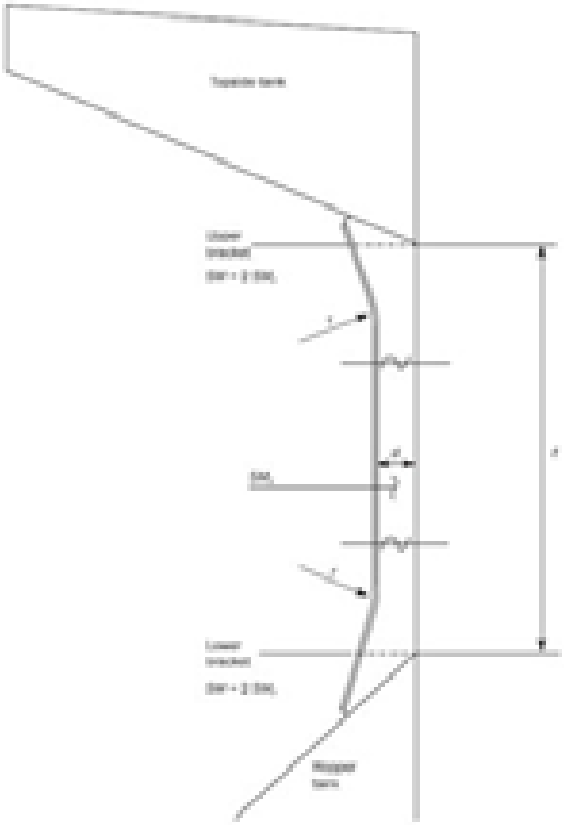


Figure 2.5.5: Dimensions of side frames

2.5.6 Lower and upper brackets of side frame

2.5.6.1 The face plates or flange of the brackets is to be sniped at both ends. Brackets are to be arranged with soft toes. The as-built thickness of the brackets is not to be less than the as-built thickness of the side frame webs to which they are connected.

2.5.6.2 The dimensions (in particular the height and length) of the lower brackets and upper brackets are not to be less than those shown in Figure 2.5.6.

2.5.6.3 Lower bracket of side frame

(a) At the level of the lower bracket as shown in Figure 2.5.5, the net section modulus of the frame and bracket, or integral bracket, with associated shell plating, is not to be taken less than twice the required net section modulus Z, in [cm^3], for the frame mid-span area obtained from Cl.2.5.3.1.

(b) For holds intended to carry ballast water in heavy ballast condition, the net section modulus Z, in [cm^3], at the level of the lower bracket is not to be taken less than twice the greater of the required net section moduli given in Cl.2.5.3.1 and Cl.2.5.3.2.

(c) The net thickness t_{LB}, in [mm], of the lower bracket is not to be taken less than:

$$t_{LB} = t_w + 1.5$$

where t_w is the net thickness of the side frame web, in [mm].

(d) The net thickness t_{LB} of the lower bracket is to comply with the following formula:

- For symmetrically flanged frames:

$$\frac{h_{LB}}{t_{LB}} \leq 87\sqrt{k}$$

- For asymmetrically flanged frames:

$$\frac{h_{LB}}{t_{LB}} \leq 73\sqrt{k}$$

The web depth h_{LB} of lower bracket is to be measured from the intersection between the hopper tank sloping plating and the side shell plate, perpendicularly to the face plate of the lower bracket as shown in Figure 2.5.8.

For the three side frames located immediately abaft the collision bulkhead, where the frames are strengthened in accordance with Cl.2.5.4 and the offered t_{LB} is greater than $1.73t_w$, the t_{LB} applied in Cl.2.5.6.3(d) may be taken as t'_{LB} given by:

$$t'_{LB} = (t^2_{LB}.t_w)^{1/3}$$

where t_w is the net thickness of the side frame web, in [mm], corresponding to A_{shr} determined in accordance to Cl.2.5.3.1.

2.5.6.4 Upper bracket of side frame

(a) At the level of the upper bracket as shown in Figure 2.5.5, the net section modulus of the frame and bracket, or integral bracket, with associated shell plating, is not to be taken less than twice the net section modulus Z required for the frame mid-span area obtained from Cl.2.5.3.1.

(b) For holds intended to carry ballast water in heavy ballast condition, the net section modulus Z, in [cm^3], at the level of the upper bracket is not to be taken less than twice the greater of the required net sections modulus obtained from Cl.2.5.3.1 and Cl.2.5.3.2.

The net thickness t_{UB}, in [mm], of the upper bracket is not to be less than the net thickness of the side frame web.

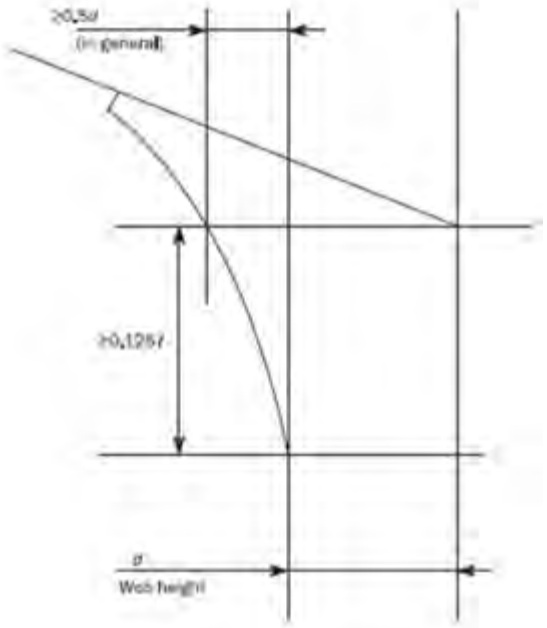


Figure 2.5.6: Dimensions of lower and upper brackets

2.5.7 Tripping brackets

In way of the foremost hold, side frames of asymmetrical section are to be effectively supported by tripping brackets at every two frames, as shown in Figure 2.5.7.

The as-built thickness of the tripping brackets is not to be less than the as-built thickness of the side frame webs to which they are connected. Double continuous welding is to be adopted for the connections of tripping brackets with side shell frames and plating.

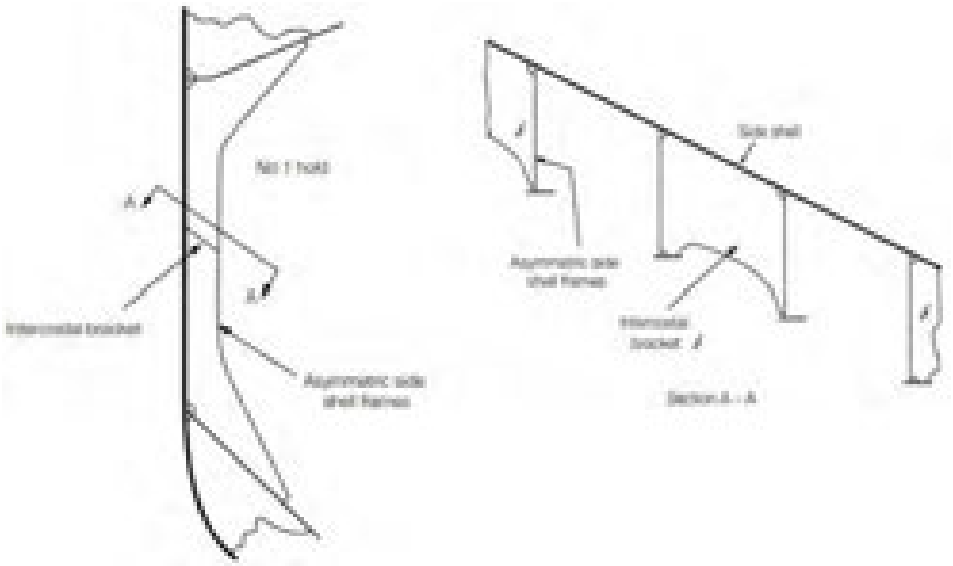


Figure 2.5.7 Typical arrangement of inter-coastal brackets supporting asymmetric side shell frames in foremost hold

2.5.8 Support structure

(a) At upper and lower end of frames, supporting brackets are to be arranged inside hopper and topside tanks, in line with the main frame brackets to provide adequate resistance to the rotation and displacement of the joint.

(b) Where the toe of the hold frame bracket is situated on or in close proximity of the first longitudinal on the hopper or topside tank sloping bulkhead from the side shell, the supporting brackets are to be extended to the next longitudinal. This extension is to be obtained by enlarging the supporting bracket or by fitting an intercostal flat bar stiffener of same depth as that of the longitudinal.

(c) Structural continuity with the lower and upper end connections of side frames is to be ensured within hopper and topside tanks by connecting brackets as shown in Figure 2.5.8. The thickness of the supporting brackets is generally to be the same as that of the main frame bracket.

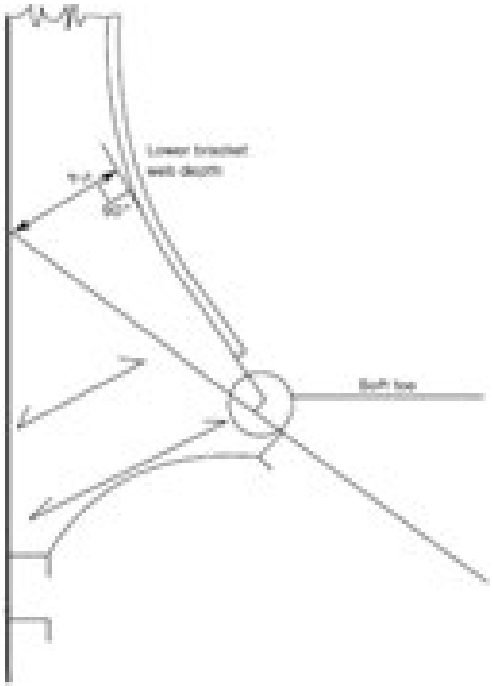


Figure 2.5.8: Connection with bracket within hopper and topside tanks

2.5.9 Provided support at upper and lower connections of side frames

2.5.9.1 Net section modulus

The net section modulus of the:
- Side shell and hopper tank longitudinals supporting the lower connecting brackets.
- Side shell and topside tank longitudinals supporting the upper connecting brackets.

is to comply with the following formula:

$$\sum_{n} Z_{pli} d_i \geq \alpha_T \frac{P.\ell_{SF}^2.\ell_1^2}{16.R_{eH}}$$

where:

n = Number of the longitudinal stiffeners on the side shell and hopper/topside tank supporting the lower/upper end connecting bracket of the side frame, as applicable.

Z_{pli} = Net plastic section modulus, in [cm^3], of the ith longitudinal stiffener on the side shell or hopper/topside tank supporting the lower/upper end connecting bracket of the side frame, as applicable.

d_i = Distance, in [m], of the above ith longitudinal stiffener from the intersection point of the side shell and hopper/topside tank.

ℓ_1 = Spacing, in [m], of transverse supporting webs in hopper/topside tank, as applicable.

R_{eH} = Lowest value of specified yield stress, in [N/mm^2], among the materials of the longitudinal stiffeners of side shell and hopper/topside tanks that support the lower/upper end connecting bracket of the side frame.

α_T = Coefficient taken as:

α_T = 150 for the longitudinal stiffeners supporting the lower connecting brackets.

α_T = 75 for the longitudinal stiffeners supporting the upper connecting brackets.

ℓ_{SF} = Side frame span ℓ, in [m], as defined in Figure 2.5.5, not to be taken less than 0.25D.

P = Design pressure in [kN/m^2], for the design load set being considered according to Pt.3, Ch.7, Sec.2, and calculated at the load calculation point defined in Pt.3, Ch.3, Sec.7.

2.5.9.2 Net connection area of brackets

The net connection area of the lower or upper connecting bracket to the supporting longitudinal stiffener is to comply with the following formula:

$$\sum_{i} A_i.d_i.R_{eH,bkt-i} \geq 0.02\alpha_T P.\ell_{SF}^2.10^{-3}$$

A_i = The offered net connection area of the bracket connecting with the ith longitudinal stiffener, in [cm^2].

α_T = As defined in Cl. 2.5.9.1.

$R_{eH,bkt-i}$ = The specified minimum yield stress of the bracket connecting with the i^{th} longitudinal stiffener, in [N/mm^2].

s = The space of the side frame, in [mm].

2.5.10 Double continuous welding is to be adopted for the connections of frames and brackets to side shell, hopper and upper wing tank plating and web to face plates.

Refer Figure 2.5.10, for extent of zone 'a' and 'b'.

Where the hull form is such that an effective fillet weld cannot be made, edge preparation of the web of frame and bracket may be required, in order to ensure the required efficiency of the weld connection.

For zones 'a' and 'b' as shown in Figure 2.5.10, double continuous fillet welding should be used with leg lengths of $0.62t_{as-built}$ and $0.57t_{as-built}$ respectively, where $t_{as-built}$ is the as-built thickness of the thinner of two connected members, in [mm].

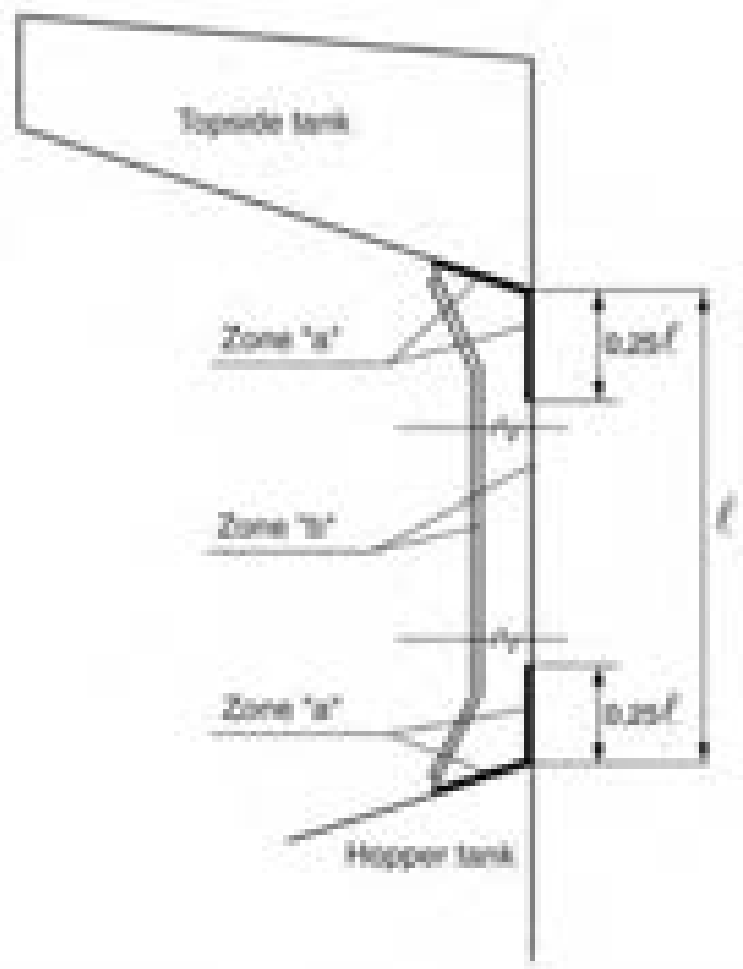


Figure 2.5.10: Zones "a" and "b" of weld connections of side frames

2.6 Hopper side tank structure

2.6.1 Scantlings of sloped bulkhead plating and stiffeners are to be as per Pt.3, Ch.7 considering the design loads given in Sec.1.3 and design load sets given in Sec.1.6.

2.6.2 Transverses supporting longitudinals are to be arranged in line with double bottom floors.

2.7 Topside tank structure

2.7.1 Scantlings of sloped bulkhead plating and stiffeners are to be as per Pt.3, Ch.7 considering the design loads given in Sec.1.3 and design load sets given in Sec.1.6

2.7.2 Where the sloping bulkhead stiffeners are fitted on the hold side of the bulkhead, suitable arrangement to prevent tripping are to be provided.

2.7.3 Topside tank structures, where fitted, are to extend as far as possible within the machinery space and supported at forward and aft ends, e.g. by scarfing brackets in way of the collision bulkhead and the engine room bulkhead.

2.8 Deck structure

2.8.1 Deck between hatches

(a) The deck within the line of hatchway openings is preferably to be stiffened transversely or alternatively the arrangements are to be such as to provide adequate buckling strength to resist athwartships forces acting on ship's sides.

(b) In case of large bulk carriers with narrow deck strips between hatchways and for vessels of class notation “Strengthened for heavy cargoes”, with empty hold notation, the cross-deck scantlings will be specially considered. Refer Pt.3, Ch.3, Sec 5, Cl.5.5.1.

(c) Connection of the strength deck at side with the deck between hatches is to be ensured by a plate of intermediate thickness.

2.8.2 Connection of hatch end beams with deck structures

Inside the line of openings, where a transversely framed structure is adopted for the cross-deck structures, hatch end beams and cross deck beams are to be adequately supported by girders and extended outward to the second longitudinal from the hatch side girders towards the deck side. Where the extension of girders outward is impracticable, intercostal stiffeners are to be fitted between the hatch side girder and the second longitudinal and checks of the structure are to be performed in compliance with the requirements in Pt.3, Ch.8.

The transverse primary members supporting the cross deck are to be supported by side or topside tank primary supporting members.

2.8.3 Web frame spacing in topside tanks

The spacing of web frames in topside tanks is to be not greater than 6 frame spaces. Other arrangements will be specially considered.

2.8.4 Large openings and hatchways

For openings and hatchways located within and outside of the cargo area, the requirements of Pt.3, Ch.3, Sec.6, Cl.6.5.4.3 are to be applied.

2.9 Corrugated bulkheads

2.9.1 General

2.9.1.1 Transverse vertically corrugated watertight bulkheads are to be fitted with a lower stool, and generally with an upper stool below deck in the following cases:

- bulkheads normally subjected to liquid pressure, e.g. tank bulkheads and ballast hold bulkheads on ships of D≥16[m],
- any bulkhead on ships of D≥18[m], and
- any bulkhead on ships of length L≥190[m].

Otherwise, corrugations may extend from inner bottom to deck. However, where a lower and/or upper stool is fitted in smaller ships, the requirements of Section 1, Cl.1.5.2.1 and/or Cl.1.5.2.2 are to be complied with, as applicable.

2.9.1.2 For corrugated bulkhead construction, arrangement, end supports and scantlings, Refer to Pt.3, Ch.3, Sec.6, Cl.6.4 and the requirements given in Cl.1.5, as applicable.

2.10 Plane bulkheads

Floors are to be fitted in the double bottom in line with the plane transverse bulkhead.

2.11 Protective coatings for cargo hold spaces

2.11.1 All internal and external surfaces of hatch coamings and hatch covers and all internal surfaces of the cargo holds, excluding the flat tank top areas and the hopper tank sloping plating extending upwards upto approximately 300[mm] below the side shell frame and brackets, are to have an efficient protective coating (epoxy or equivalent) applied in accordance with the manufacturers recommendations.

2.11.2 In the selection of coating, due consideration is to be given by the Owner to intended cargo conditions expected in service.

2.12 Damage stability requirements

2.12.1 Bulk carriers of single side skin construction as defined in Cl.1.1.2.2 and of

length L≥150[m], designed to carry solid bulk cargoes having a density of 1.0[t/m^3] and above, the keels of which are laid or are at a similar stage of construction on or after 1 July, 1999 shall, when loaded to the summer load line, be able to withstand flooding of any one cargo hold in all loading conditions and remain afloat in a satisfactory condition of equilibrium, as specified in Cl.2.12.4.

2.12.2 Bulk carriers of single side skin construction as defined in Cl.1.1.4, in the foremost cargo hold and of length L≥150[m], carrying solid bulk cargoes having a density of 1.78[t/m^3] and above, the keels of which are laid or are at a similar stage of construction before 1 July 1999 shall, when loaded to the summer loadline, be able to withstand flooding of the foremost cargo hold in all loading conditions and remain afloat in a satisfactory condition of equilibrium, as specified in Cl.2.12.4. This requirement shall be complied with in accordance with the schedule given in Pt.1, Ch.2, Sec.2.4.12 for compliance with IMO standards for scantlings of transverse bulkhead between the two foremost cargo holds and allowable hold loading for the foremost cargo hold.

2.12.3 Bulk carriers of double side skin construction as defined in Cl.1.1.2.3 and of length L≥150[m] in which any part of the longitudinal bulkhead is located within B/5 or 11.5[m], whichever is less, inboard from the ship's side at right angle to the centerline at the assigned summer load line, designed to carry solid bulk cargoes having a density of 1.0[t/m^3] and above, shall be able to withstand when loaded to the summer load line, flooding of any one cargo hold in all loading conditions and remain afloat in a satisfactory condition of equilibrium as specified in Cl.2.12.4.

2.12.4 Subject to the provisions of Cl.2.12.7 below, the condition of equilibrium after flooding shall satisfy the condition of equilibrium laid down in the annex to resolution A.320(IX) - Regulation equivalent to Regulation 27 of the International Convention on Load Lines, 1966, as amended by Resolution A.514(13). The assumed flooding need only take into account flooding of the cargo hold space. The permeability of a loaded hold shall be assumed as 0.9 and the permeability of an empty hold shall be assumed as 0.95, unless a permeability relevant to a particular cargo is assumed for the volume of a flooded hold occupied by cargo and a permeability of 0.95 is assumed for the remaining empty volume of the hold.

2.12.5 Bulk carriers contracted before 1 July, 1998 which have been assigned a reduced freeboard in compliance with regulation 27(7) of the International Convention on Load Lines, 1966, may be considered as complying with the requirements of Cl.2.12.2.

2.12.6 Bulk carriers which have been assigned a reduced freeboard in compliance with the provisions of paragraph (8) of the regulation equivalent to regulation 27 of the International Convention on Load Lines, 1966, adopted by resolution A.320(IX), as amended by resolution A.514(13), may be considered as complying with the requirements of Cl.2.12.1, Cl.2.12.2 or Cl.2.12.3, as appropriate.

2.12.7 On bulk carriers which have been assigned reduced freeboard in compliance with the provisions of regulation 27(8) set out in Annex B of the Protocol of 1988 relating to the International Convention on Load Lines, 1966, the condition of equilibrium after flooding shall satisfy the relevant provisions of that Protocol.

2.13 Hold, ballast and dry space water level detectors

2.13.1 All bulk carriers are to be fitted with water level detectors as indicated in a), b) and c) below:

a) In each cargo hold, giving audible and visual alarms, one when the water level above the inner bottom in any hold reaches a height of 0.5[m] and another at a height not less than 15% of the depth of the cargo hold but not more than 2[m].

   If bulk carriers are exempted by the Administration from the compliance of SOLAS Chapter XII Reg.4.2 and 6 according to the provisions of SOLAS Chapter XII Reg.9, then only one alarm at a water level of not less than 15% of the depth of the cargo hold but not more than 2[m] needs to be provided.

   The water level detectors are to be fitted in the aft end of the cargo holds.

   For cargo holds which are used for water ballast, an alarm overriding device is to be installed.

The visual alarms are to clearly discriminate between the two different water levels detected in each hold.

b) In any ballast tank forward of the collision bulkhead giving an audible and visual alarm when the liquid in the tank reaches a level not exceeding 10% of the tank capacity. An alarm overriding device is to be installed for activation when the tank is in use; and

c) In any dry or void space other than a chain cable locker, any part of which extends forward of the foremost cargo hold, giving an audible and visual alarm at a water level of 0.1[m] above the deck. Such alarms need not be provided in enclosed spaces if the volume of the space does not exceed 0.1% of the ship's maximum displacement volume.

The audible and visual alarms described in a), b) and c) above are to be located on the navigation bridge.

Note: The water level detectors are to be of approved type. For further details regarding performance tests, sensor locations, installation and other testing requirements refer classification notes: "Type approval, installation and testing of water level detectors on ships subject to SOLAS Reg. II-1/25, II-1/25-1 and XII/12".

2.14 De-watering of forward spaces

2.14.1 On bulk carriers, the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold, is to be capable of being brought into operation from a readily accessible enclosed space, the location of which is accessible from the navigation bridge or propulsion machinery control position without traversing exposed freeboard or superstructure decks. Where pipes serving tanks or bilges pierce the collision bulkhead, valve operation by means of remotely operated actuators may be accepted, as an alternative to the valve control specified in Pt.4, Ch.3, 2.4.3, provided that the location of such valve controls complies with the above.

IR2.14.1.1 Where the piping arrangements for dewatering closed dry spaces are connected to the piping arrangements for the drainage of water ballast tanks, two non-return valves located in accessible positions, are to be provided to prevent the ingress of water into dry spaces from those intended for the carriage of water ballast. One of these non-return valves is to be fitted with shut-off isolation arrangement capable of being controlled from the same readily accessible enclosed space which is readily accessible from the navigation bridge or the propulsion machinery control position without travelling exposed freeboard or superstructure decks. In this context, a position which is accessible via an under-deck passage, a pipe trunk or other similar means of access is not to be taken as being in the "readily accessible enclosed space".

IR 2.14.1.2

.1 the valve specified in Pt.4, Ch.3, 2.4.3.and 2.4.4 is to be capable of being controlled from the navigation bridge, the propulsion machinery control position or enclosed space which is readily accessible from the navigation bridge or the propulsion machinery control position without travelling exposed freeboard or superstructure decks. In this context, a position which is accessible via an under-deck passage, a pipe trunk or other similar means of access is not to be taken as being in the "readily accessible enclosed space";

.2 the valve is not to move from the demanded position in the case of failure of the control system power or actuator power;

.3 positive indication is to be provided at the remote-control station to show that the valve is fully open or closed; and

.4 local hand-powered valve operation from above the freeboard deck, as permitted in Pt.4, Ch.3, 2.4.3, is required. An acceptable alternative to such arrangement may be remotely operated actuators as specified in Cl.2.14.1, on the condition that all of the provisions of Cl.2.14.1 are met.

IR2.14.1.3 The dewatering arrangements are to be such that any accumulated water can be drained directly by a pump or educator.

IR2.14.1.4 When the dewatering arrangements are in operation, other systems essential for the safety of the ship including fire-fighting and bilge systems are to remain available and ready for immediate use. The systems for normal operation of electric power supplies, propulsion and steering should not be affected by the operation of the dewatering systems. It must also be possible to immediately start fire pumps

and have a ready available supply of fire-fighting water and to be able to configure and use bilge system for any compartment when the dewatering system is in operation.

IR2.14.1.5 Bilge wells are to be provided with gratings or strainers that will prevent blockage of the dewatering system with debris.

IR2.14.1.6 The enclosures of electrical equipment for the dewatering system installed in any of the forward dry spaces are to provide protection to IPX8 standard as defined in IEC 60529:1989/AMD:2013/COR1:2019 for a water head equal to the height of the space in which the electrical equipment is installed for a time duration of at least 24 hours.

IR2.14.1.7 The spaces where availability of pumping systems is required in accordance with Cl.2.14.1 are to be the same watertight spaces where water level detectors are required in accordance with Cl.2.13.1 c). This means that requirements given in 2.14.1 and IR2.14.1.1 to IR 2.14.1.6 are not applicable to the enclosed spaces whose volume does not exceed 0.1% of the ship's maximum displacement volume and to the chain locker.

2.15 Dewatering capacity

2.15.1 The dewatering system for ballast tanks located forward of the collision bulkhead and for bilges of dry spaces any part of which extends forward of the foremost cargo hold is to be designed to remove water from the forward spaces at a rate of not less than 320A[m^3/h], where A is the cross-sectional area in [m^2] of the largest air pipe or ventilator pipe connected from the exposed deck to a closed forward space that is required to be dewatered by these arrangements.

Section 3

Ore Carriers

For symbols not defined in this section, Refer to Pt.3, Ch.1, Sec.2.

3.1 Hull arrangement

3.1.1 The vessel is to have two effective longitudinal bulkheads and a double bottom in way of the cargo holds. It is assumed that only spaces between the longitudinal bulkheads are used as cargo holds. A double bottom is to be fitted in way of the cargo holds.

3.1.2 The bottom and deck outside the line of openings are to be longitudinally framed. The side shell and longitudinal bulkheads also, in general, are to be longitudinally framed. Inside line of openings, the deck is generally to be transversely stiffened.

3.2 Longitudinal strength

3.2.1 The longitudinal strength requirements as detailed in Pt.3, Ch.6 are to be complied with.

3.3 Bottom structure

3.3.1 The scantlings and arrangements are in general, to be as per Pt.3, Ch.3 (Arrangements) and Ch.7 (Scantlings) except as given below.

3.3.2 Spacing of floors in the double bottom in centre hold is not to exceed 2.5[m] or 0.01L whichever is greater. Additional side girders are to be provided below the centre hold so that the spacing of longitudinal girders generally does not exceed 3.6[m].

3.3.3 The spacing of bottom transverses in the wing tanks is not to exceed the greater of 0.02L or 3.6[m].

3.3.4 The thickness of inner bottom plating in cargo holds is to be based on the design loads given in Section 1.3, however not to be less than:

$$t = 5.5 + 0.03L\sqrt{K} + t_c \text{ [mm]}$$

3.4 Wing tank structure

3.4.1 In wing tanks, primary bottom structure is to be so arranged as to maintain structural continuity of the hold double bottom structure in the transverse direction.

3.4.2 The inner bottom plating is to be extended into the wing tank in the form of a horizontal gusset plate. The gusset plates are to be of sufficient width to provide effective scarfing of the inner bottom into the wing tank structure.

3.4.3 At locations where bottom transverses are not provided in line with plate floors in the hold double bottom, substantial brackets are to be arranged in line with such plate floors. These brackets are to extend transversely over at least three longitudinal spaces and vertically well above the inner bottom level.

3.4.4 In the wing tanks, bulkheads are to be arranged in line with the centre hold bulkheads so that continuity of transverse strength is maintained.

3.4.5 All watertight and non-watertight bulkheads in wing tanks are to be suitably reinforced in way of double bottom scarfing arrangements and also at the ends of centre hold deck transverses. Openings in wing tank bulkheads are to be kept clear of these areas.

3.4.6 In general, the spacing of vertical primary supporting members is to be not greater than 6 times the frame spacing.

3.5 Deck structure

3.5.1 The arrangement and scantlings of deck plating inside line of ore hatchways are to be in accordance with the requirements for bulk carriers given in Section 2.8.

3.5.2 When the hatch coamings are situated inboard of the longitudinal bulkhead, the portion of the deck between the two is to be suitably supported by longitudinals.

3.5.3 The connection of hatch end beams with deck structures is to be properly ensured by fitting inside the wing tanks additional web frames or brackets.

3.6 Additional requirements for ore carriers with narrow wing tanks

3.6.1 For ore carriers of 150[m] in length and above, in which any part of longitudinal bulkhead is located within B/5 or 11.5[m], whichever less, inboard from ship side at right angle to the centerline at the assigned summer load line, are to comply with the following additional requirements given in Cl.3.6.2 to Cl.3.6.6.

3.6.2 The longitudinal strength in the hold flooded condition is to comply with the requirements given in Section 1, Cl.1.3.3 and Section 2, Cl.2.2.3 to Cl.2.2.5.

3.6.3 The thickness of the inner bottom plating and the section modulus of inner bottom longitudinals are to comply with the requirements given in Cl.2.3.3 and Cl.2.3.4 respectively.

3.6.4 The shear capacity of the double bottom is to comply with the requirements given in Section 1, Cl.1.4.3.

3.6.5 The thickness of longitudinal bulkhead plating and stiffeners are to be in accordance with Pt.3, Ch.7 using the design loads in hold flooded conditions given in Section 1, Cl.1.3.3.

3.6.6 The construction and strength of transverse corrugated bulkheads is to be as per Section 1, 1.5.

3.7 Buckling

The buckling strength of the structure participating in longitudinal strength is to comply the requirements given in Pt.3, Ch.9, applying the additional design load sets given in Section 1, 1.6.

Section 4

Direct Strength Calculations

4.1 General

4.1.1 Direct strength calculations are required in cases where simplified formulations are not able to take into account special stress distributions, boundary conditions or structural arrangements with sufficient accuracy.

4.1.2 The scantlings of the double bottom and transverse bulkhead structures in cargo region and top wing tank/deck structures over long holds, obtained from simplified formulae may have to be increased based on the results of the direct strength calculations.

4.1.3 The computer programs used are to take into account the effects of bending, shear, axial and torsional deformations.

4.1.4 For deep girders, bulkhead panels, bracket zones etc. FEM or equivalent methods are to be applied. For systems consisting of slender girders, calculations may be based on beam theory.

4.1.5 For ships having Length, L≥150[m], Cargo hold finite element analysis is to be carried out for midship region according to the requirements, methods and procedures as given in Pt.3, Ch.8, Sec.3 and in the IRS Guidelines on Structural Assessment of Ships based on Finite Element Method.

However, cargo hold FEA may be optionally carried out for ships of Length, L<150[m] applying the requirements in Pt.3, Ch.8, Sec. 3.

4.1.6 FE models for cargo hold FE analyses, local mesh FE analysis, are to be based on the net scantling approach, as defined in Pt.3, Ch.2, Sec.4.

The calculations are to be carried out using net thicknesses obtained after deduction of applicable corrosion additions specified Pt.3, Ch.2, Sec.5.

4.2 Load cases and design loads

4.2.1 The calculations are to be carried out for realistic intact and flooded conditions which cause most severe loading on double bottom, bulkhead and top side structures.

The ship and individual hold loading conditions specified in 1.2 are also to be considered for the evaluation of strength.

4.2.2 The following cases are generally to be included:

a) Ballast in ballast hold with adjacent holds empty, at minimum ballast draught T_b - with respect to double bottom, transverse bulkhead and top wing tank/ship side strength.

b) Ballast in top wing tank - with respect to top wing tank strength in the upright and heeled conditions, where angle of heel is to be taken equal to half the angle of roll as given in Pt.3, Ch.5.

c) Heavy ore cargo in hold, hold considered flooded with adjacent hold empty - with respect to double bottom of the loaded and adjacent empty holds and transverse bulkhead in between.

d) Maximum cargo in the hold filled upto the top of hatch coaming, hold considered flooded - with respect of transverse bulkhead structure.

e) Specified cargo on deck and external sea pressure on deck (in particular forward holds) with respect to deck and top wing tank structure.

4.2.3 Normally, harbour conditions need not be considered provided the minimum draught in harbour with cargo hold filled, is not less than two thirds of the draught in the associated approved seagoing condition.

4.2.4 The internal pressures on transverse bulkheads and inner bottom (Bulk pressures and shear loads) to be applied to the FE model in accordance with Section 1, Cl.1.3. The external sea pressures are to be taken as per Pt.3, Ch.5, Sec.4.

4.3 Analysis Criteria

4.3.1 Coarse mesh permissible yield and buckling utilization factors

The coarse mesh permissible yield utilization factors, λ_{yperm}, for cargo hold analysis is given in Pt.3, Ch.8, Sec.3, Table 3.4.2.3.

The allowable buckling utilization factors, η_{all} is given in Pt.3, Ch.9, Sec.1, Table 1.3.3.

End of Chapter

Chapter 2

Oil Tankers

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Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to sea going self-propelled ships which are constructed generally with integral tanks and intended primarily to carry crude oil or petroleum products in bulk, in liquid state at atmospheric pressure and ambient temperature (or thus maintained by heating) and whose Reid vapour pressure is below the atmospheric pressure. These requirements are supplementary to those given for the assignment of main characters of class. The list of oils and petroleum products which can be carried in such vessels is given in Appendix A. A notation **OIL TANKER**, may be assigned to such ships.

1.1.2 Other cargoes which may also be carried in these vessels are listed in Appendix B.

1.1.3 For general requirements of fire protection and fire safety of oil tankers intended to carry petroleum products with a flashpoint exceeding 60[°C] (closed cup test), refer to Pt.6, Ch.1, 1.5.4.

1.1.4 Typical midship sections of oil tankers are shown in Fig.1.1.4.

1.1.5 The additional requirements given in Sec.12 apply to ships intended primarily to carry either oil in bulk with flash point below 60°C (closed cup test) or alternatively dry bulk cargo. Typical midship sections of such combination carriers are shown in Figure 1.1.5(a) (Ore or Oil Carriers) and Figure 1.1.5(b) (Oil or Bulk Carriers).

1.1.6. The following additional notations may be assigned to Oil tankers

1.1.6.1 IRS Rules for Bulk Carriers and Oil Tankers are to be applied to double hull oil tankers of 150[m] length and above for unrestricted service. The additional notation **CSR** is assigned to such vessels.

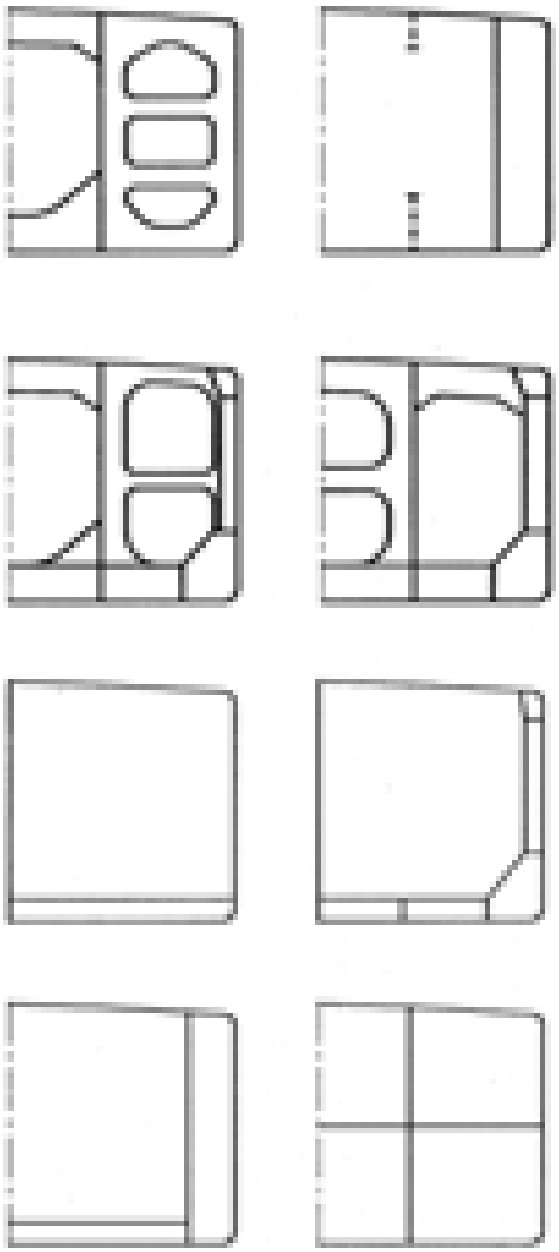

Figure 1.1.4: Typical midship sections of oil tankers

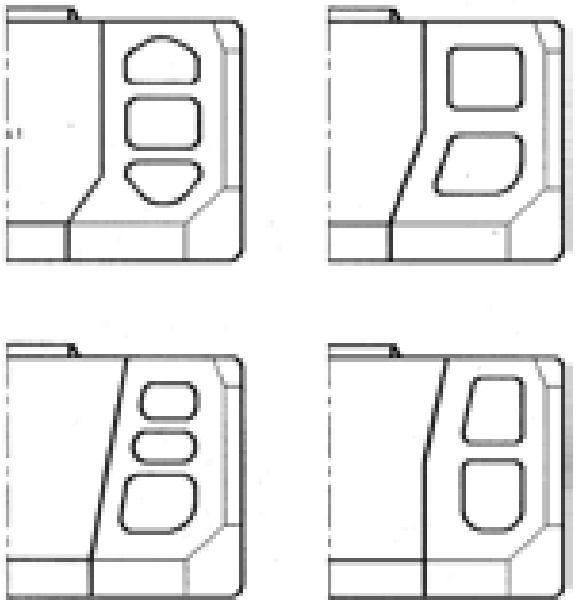

Figure 1.1.5(a): Typical midship sections of Ore or Oil Carriers

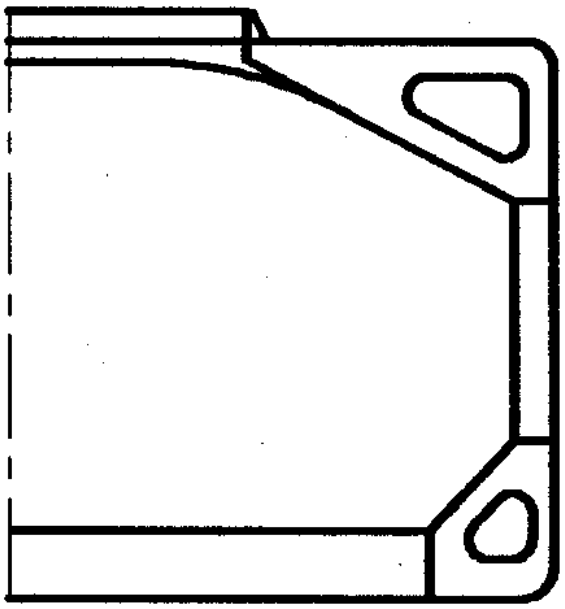

Figure 1.1.5(b): Typical midship sections of Oil or Bulk Carriers

1.1.6.2 **FOR CARRIAGE OF……(to be specified),** where the ship is intended to carry liquid cargoes as per Cl.1.1.2.

1.1.6.3 **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C**, as applicable, where the ship is intended to carry only such type of products, under certain conditions. The above notation is applicable to ships intended to carry unheated cargoes having F.P exceeding 60[°C] and cargoes heated to a temperature below and not within 15[°C] of their flashpoint.

1.1.6.4 **ASPHALT CARRIER (INTEGRAL TANK, MAX. CARGO TEMP ≤ XXX℃),ESP** where the ship is intended to carry such type of products. The additional requirements in Section 14 are applicable to these ships.

1.1.6.5 For ships which are intended to carry only Asphalt in independent (non-integral) cargo tanks, the notation **ASPHALT CARRIER (INDEPENDENT TANKS, MAX. CARGO TEMP ≤ XXX℃)**, may be assigned. The notation ESP need not be assigned to such ships. The additional requirements of Section 14 are to be complied with.

1.1.7 For ships with integral cargo tanks, the notation **OIL TANKER** is always appended with the additional notation **ESP** (i.e. **OIL TANKER ESP**), which means that these ships are subjected to the Enhanced Survey Program as laid down in Pt. 1, Ch.2.

1.1.8 Combination carriers as described in 1.1.5 will be eligible to be assigned class notation **"ORE OR OIL CARRIER, ESP" or "OIL OR BULK CARRIER, ESP".** Vessels also complying with appropriate additional requirements given in Pt.5, Ch.1 may be assigned additional class notations **"Strengthened for heavy cargoes"** or **"Strengthened for heavy cargoes, hold(s) ....** (to be specified) **... may be empty".**

1.1.9 The requirements of the following statutory regulations (as amended) are to be complied with in so far as they are applicable:

a) SOLAS 1974; Chapter II-1
   - for general safety measures (construction, subdivision and stability, machinery and electrical installations);

b) SOLAS 1974; Chapter II-2
   - for fire safety measures;

c) MARPOL; Annex (I)
   - for ship arrangement (Refer Cl.2.2.2) and pollution prevention;

d) MARPOL; Annex II
   - for equipment when carrying noxious substance indicated in Cl.1.1.2 above.

1.1.10 Where the prescriptive requirements within SOLAS and related Codes (IBC, IGC) and the standards published by the International Electrotechnical Commission, such as but not limited to IEC 60092-502, are not aligned, the prescriptive requirements in SOLAS and Codes take precedence and are to be applied. This is applicable to hazardous area classification in respect of selection of electrical equipment, cables and wiring and positioning of openings and air intakes.

1.2 Documentation

1.2.1 Additional documents containing details of the items mentioned below are to be submitted for approval, as applicable.

- General Arrangement
  - cargo hatches, tank cleaning hatches and all other openings to cargo tank
  - doors, hatches and all other openings to gas dangerous zones
  - ventilation pipes and openings for cargo hatches, pump rooms and other gas dangerous spaces
  - doors, air locks, hatches, vent pipes and openings, openable hinged scuttles and other openings to gas-safe spaces adjacent to cargo area including spaces in and below forecastle deck
  - cargo oil pipes over the deck with shore connection
  - extent and location of gas-dangerous zones

- Capacity Plan
  - details of cargo tanks and segregated ballast tanks

- Access Plan
  - means of access to all spaces in cargo region and ballast spaces to meet the requirements of close-up survey

- Pumping and Piping Arrangements
  - cargo piping system including expansion element and flange connections
  - hydraulic system for cargo pump

  - bilge piping systems in pump rooms, cofferdams, pipe tunnels and other dry spaces in cargo area
  - permanent ballast piping in cargo area and arrangements in the forward and aft ends of the ship

- Electrical Installation

  - all electrical equipment in cargo area
  - area classification drawings
  - location of all electrical equipment in gas dangerous zone
  - single line diagrams for intrinsically safe circuits
  - list of explosion proof equipment with drawings and certificates

- Documents on

  - inert gas plant and piping system together with details of inert gas control and monitoring devices
  - crude oil washing system
  - slop tanks
  - cargo tank vents indicating the type and position of vent outlets from superstructure, erection, air intakes etc.
  - ventilation arrangement of cargo and/or ballast pump rooms and other enclosed spaces which contain cargo handling equipment
  - tank level gauges details
  - emergency towing arrangements.

1.2.2 Following documents are to be submitted for reference:

- A diagrammatic plan indicating compliance with the requirements of MARPOL Annex I, Regulation 19 regarding the dimensions of double bottom and wing tank spaces. When alternatives to double bottom or double hull as per the above regulations are envisaged, details are to be submitted accordingly.

- Compliance with intact and damage stability requirements of the statutory authorities.

1.3 Materials and material protection

1.3.1 Materials used for the construction of hull, piping and fittings are to be compatible with the liquids and their vapours. Hatch packing materials should be resistant to the liquids and their vapours. Synthetic materials for components and piping are to be approved in each separate case.

1.3.2 Materials used in the inert gas system are to be suitable for their intended purpose.

1.3.3 Impressed current systems and magnesium or magnesium alloy anodes are not permitted in oil cargo tanks and in tanks adjacent to oil cargo tanks. Where aluminium anodes are fitted in oil cargo tanks and in tanks adjacent to oil cargo tanks, the installations are to satisfy the requirements of Pt.3, Ch.2, Sec.3.4.

1.3.4 Aluminium coatings containing greater than 10% aluminium by weight in the dry film are not to be applied in cargo tanks, cargo tank deck area, in pump rooms, cofferdams or any other spaces where flammable cargo vapour may accumulate.

1.3.5 Aluminised pipes may be permitted in ballast tanks, in inerted cargo tanks and, provided the pipes are protected from accidental impact, in hazardous areas on open deck.

1.4 Definitions

1.4.1 *Cargo area* is that part of the vessel that contains cargo tanks, slop tanks and cargo pump-rooms including, cofferdams, ballast and void spaces adjacent to cargo tanks and also deck areas throughout the entire length and breadth of the part of the ship over the above mentioned spaces.

1.4.2 *Slop tank* means a tank specifically designated for the collection of tank drainings, tank washings and other oily mixtures.

1.4.3 *Cargo tank block* is the part of the ship extending from the aft bulkhead of the aftmost cargo or slop tank to the forward bulkhead of the forward most cargo or slop tank, extending to the full depth and beam of the ship, but not including the area above the deck of the cargo or slop tank.

1.5 Intact stability of tankers during liquid transfer operations

1.5.1 All oil tankers of 5000 tonnes deadweight and above are to satisfy the requirements of intact stability during liquid transfer operations given in MARPOL, Annex I, Reg.27, where applicable. (For conditions of applicability Refer Reg.27).

All other tankers to which the MARPOL, Annex I, Reg.27 does not apply are to comply with the requirements given in Cl.1.5.2 during liquid transfer operations which include cargo loading and unloading, lightering, ballasting and deballasting, ballast water exchange, and tank cleaning operations. Alternatively, requirements of MARPOL Annex I, Reg.27 could be applied as a matter of equivalence.

1.5.2 For any operating draught reflecting actual, partial or full load conditions, including the intermediate stages of liquid transfer operations the following intact stability criteria is to be complied with:

a) In port (see note below), the initial metacentric height GM_0 is not to be less than 0.15[m]. Positive intact stability is to extend from the initial equilibrium position at which GM_θ is calculated over a range of at least 20[deg] to port and to starboard.

b) At sea, the intact stability criteria contained in paragraphs Chapter 3 of IMO Resolution A.749(18), the Intact Stability Code, or the criteria contained in the national requirements of the flag administration if the national stability requirements provide at least an equivalent degree of safety are to be complied with.

Note: At some port locations where the environmental conditions are similar to those at sea, the requirements given in para (b) are to be applied.

For all loading conditions in port and at sea, including intermediate stages of liquid transfer operations, the initial metacentric height and the righting lever curve are to be corrected for the effect of free surfaces of liquids in tanks.

1.5.3 For tankers to which MARPOL Annex I, Reg.27 has not been applied, the intact stability criteria specified in Cl.1.5.2 is to be met preferably by design of the ship i.e. the design should allow for maximum free surface effects of all relevant slack tanks during the liquid transfer operations.

Where the intact stability criteria specified in Cl.1.5.2 is not met through the design of the ship alone, the master is to be provided with clear instructions covering the operational restrictions and methods necessary to ensure compliance with the intact stability criteria. These instructions should be simple, clear and concise and shall provide pre-planned requirements for cargo/ballast transfer operations which satisfy the intact stability criteria.

Additionally, adequate instructions to enable pre-planning of other sequences of cargo/ballast transfer operations, which comply with the intact stability criteria throughout the operations, are to be provided. These instructions are to also provide for corrective actions to be taken by the officer-in-charge in case of unexpected technical difficulties in adhering to the pre-planned transfer sequences or in case of emergency situations.

These instructions are to be prominently displayed in the approved trim and stability booklet and at the cargo/ballast transfer control station. They are also to be included in any computer software available on board which is used for calculating and monitoring the intact stability or hull strength.

Section 2

Ship Arrangement

2.1 Location and separation of spaces

2.1.1 Location and separation of cargo spaces from machinery, accommodation, service spaces, control stations etc. are to be in accordance with Pt.6, Ch.2, Sec.1.5.

Vessels assigned the additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C** need not comply with the requirements of Pt.6, Ch.2, Cl.1.5.2 for location and separation of cargo spaces. However, the access doors, air inlets and openings to accommodation spaces, service spaces and control stations are not to face the cargo area.

2.1.2 All dry spaces and tanks intended for water ballast which can remain empty in loaded condition are to be so arranged that they cannot be used for any other purpose.

2.1.3 Slop tanks are to be designed for efficient decantation. Positions of inlets, outlets, baffles and weirs where fitted, are to be located to ensure minimum turbulence and entrainment of oil or emulsion with water.

2.1.4 Cargo tanks are to be segregated from machinery spaces, accommodation spaces and other spaces of electrical hazard by means of cofferdams at least 760[mm] in length and covering the whole area of the end bulkheads of cargo tanks. A pump room, oil fuel bunker or water-ballast tank will be accepted in lieu of a cofferdam. Oil engines or electrical equipment of potential fire hazard are not to be sited in these pump rooms or cofferdams.

2.1.5 Where a corner-to-corner situation occurs between a safe space and a cargo tank, the safe space is to be protected by a cofferdam. This protection may however be obtained by an angle bar or a diagonal plate across the corner. Such cofferdams, if accessible, is to be ventilated and if not accessible, is to be filled with a suitable and compatible compound.

2.1.6 A cofferdam between the forward cargo tank and the forepeak may be dispensed with, if:

- direct access is provided to the forepeak from the open deck,
- the air and sounding pipes to the forepeak space are led to the open deck, and
- portable means are provided for gas detection and inerting the forepeak compartment.

2.1.7 Vessels assigned the additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C** need not comply with the requirements of Cl.2.1.4, Cl.2.1.5 & Cl.2.1.6 above. However, the following provisions are to be complied with:

a) Tanks containing cargo or cargo residues are to be segregated from accommodation, service and machinery spaces by means of a cofferdam or similar space.

2.2 Tank arrangement

2.2.1 The disposition of transverse bulkheads should in general comply with the requirements of Pt.3, Ch.3, Sec.2 as applicable to ships with machinery aft.

2.2.2 The arrangement of the spaces within the cargo region with respect to the following features (Refer Table 2.2.2) are to be in accordance with the MARPOL 73/78 (as amended), Annex I.

Table 2.2.2 Applicable regulations (ref MARPOL Annex I) for arrangement of spaces within cargo region		
	Feature	**Regulation**
a)	Protection of cargo tank region with double bottom and wing ballast tank/spaces	19
b)	Segregated ballast tank (SBT)	18
c)	Protective location of SBT	18
d)	Crude oil washing (COW) (Crude carriers only),	33
e)	Segregation of fuel oil/ballast water	16
f)	Slop tanks and oil/water interface detectors	29
g)	Sludge tank for fuel oil	12
h)	Minimization of retention of oil on board	30(4) & (5)
i)	Tank size limitation / Accidental oil outflow performance	23
j)	Subdivision and damage stability	Reg.19, Reg.24 and Reg.28
k)	Intact stability	27
l)	Pump room bottom protection	22
Note : The requirements of regulation 29, are not applicable to asphalt carriers.		

2.2.3 The aggregate capacity of wing tanks, double bottom tanks, forepeak tanks and aft peak tanks intended to carry water ballast is not to be less than the capacity of segregated ballast tanks necessary to meet the requirements of MARPOL Annex I, Regulation 18.

2.2.4 Small tankers with single hull arrangements will be specially considered.

2.2.5 Location of fuel tanks in cargo area

2.2.5.1 On oil and chemical tankers carrying liquid cargoes having a flashpoint not exceeding 60[deg] and/or toxic liquid cargoes, fuel tanks located with a common boundary to cargo or slop tanks are not to be situated within nor extend partly into the cargo tank block (see note below). Such tanks may, however, be situated at the aft and/ or forward of the cargo tank block. They may be accepted when located as independent tanks on open deck in the cargo area subject to spill and fire safety considerations.

Note: Toxic liquid cargoes include those for which toxic vapour detection is specified in column 'k' of Table 17.1 in Part 5, Ch. 3, Sec. 17.

2.2.5.2 The arrangement of independent fuel tanks and associated fuel piping systems, including the pumps, can be as for fuel tanks and associated fuel piping systems located in the machinery spaces. For electrical equipment, requirements to hazardous area classification must however be met.

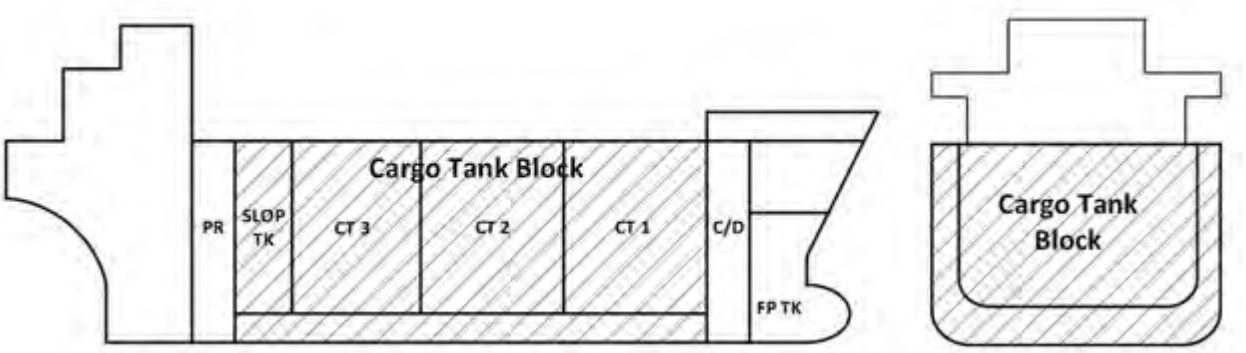

Fig 2.2.5.1 Cargo Tank Block

2.3 Arrangement for access in the cargo area and forward spaces

2.3.1 Each space is to be provided with a means of access to enable, throughout the life of a ship, overall and close-up inspections and thickness measurements of the ship's structures. Such means of access are to comply with the requirements of the technical provisions for means of access for inspections, specified in 2.4.

The technical provisions do not apply to cargo tanks of combined chemical/ oil tankers complying with Part 5, Ch.3. However, they are to be applied to ballast tanks in such vessels.

Each space for which close-up inspection is not required such as fuel oil tanks and void spaces forward of cargo area, may be provided with a means of a access necessary for overall survey intended to report on the overall conditions of the hull structure.

2.3.2 Where it is impracticable to fit permanent means of access, the provision of movable or portable means of access, as specified in 2.4 may be considered, provided that the means of attaching, rigging, suspending or supporting the portable means of access forms a permanent part of the ship's structure. All portable equipment is to be capable of being readily erected or deployed by ship's personnel.

2.3.3 The construction and materials of all means of access and their attachment to the ship's structure are to be approved by IRS.

2.3.4 Access to cofferdams, ballast tanks, cargo tanks and other spaces in the cargo area are to be direct from the open deck and such as to ensure their complete inspection. (For typical arrangement Refer Figure 2.3.4(a) and Figure 2.3.4(b)). Access to double bottom or to forward ballast spaces may be through a pump room, deep cofferdam, pipe tunnel, double hull space or similar compartment not intended for the carriage of oil or hazardous cargoes.

Every double bottom space is to be provided with separate access without having to pass through other neighbouring double bottom space.

The wording "not intended for the carriage of oil or hazardous cargoes" applies only to "similar compartments" i.e. safe access can be through a pump-room, deep cofferdam, pipe tunnel, cargo hold or double hull space.

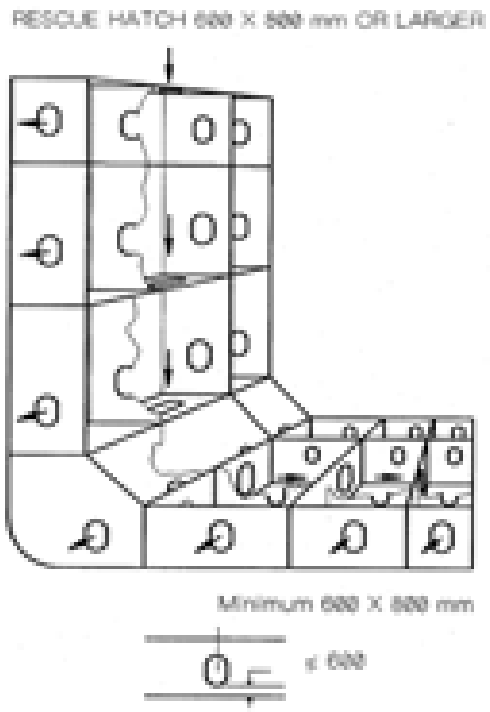

Figure 2.3.4(a): Access through side ballast space and double bottom

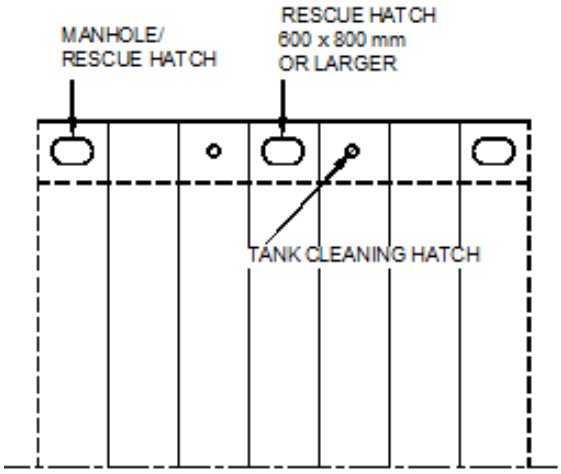

Figure 2.3.4(b): Access arrangement through deck into wing ballast space

2.3.5 Where a duct keel or pipe tunnel is fitted and access is normally required for operational purposes, access is to be provided at each end and at least one other location at approximately mid-length. Access is to be directly from the exposed deck. There is to be no connection to engine room. Where an after access is to be provided from the pump room to the duct keel, requirements of Pt.6, Ch.2,

Cl.1.5.2.4 are to be complied with. Mechanical ventilation is to be provided and such spaces are to be adequately ventilated prior to entry. A suitable notice is to be posted at the entry stipulating a sufficient period of ventilation prior to any attempt made to enter the space. In addition, the atmosphere in the tunnel is to be sampled by a reliable gas monitor, and when an inert gas system is fitted in the cargo tanks, an oxygen monitor is to be provided.

2.3.6 Tanks and subdivisions of tanks, having a length of 35[m] or more are to be fitted with at least two access hatchways and ladders, as far apart as practicable. Tanks less than 35[m] in length are to be served by at least one access hatchway and ladder. When a tank is subdivided by one or more swash bulkheads or similar obstructions which do not allow ready means of access to the other parts of the tank, at least two hatchways and ladders are to be fitted.

2.3.7 For access through horizontal openings, hatches or manholes, the dimensions shall be sufficient to allow a person wearing a self-contained, air-breathing apparatus and protective equipment to ascend or descend any ladder without obstruction and also to provide a clear opening to facilitate the hoisting of an injured person from the bottom of the space.

2.3.8 A ship's means of access to carry out overall and close-up inspections and thickness measurements is to be described in a Ship Structure Access Manual which is to consist of two parts.

The first part is to include the following for each space:

a) plans showing the means of access to the space, with appropriate technical specifications and dimensions;

b) plans showing the means of access within each space to enable an overall inspection to be carried out, with appropriate technical specifications and dimensions. The plans are to indicate from where each area in the space can be inspected;

c) plans showing the means of access within the space to enable close-up inspections to be carried out, with appropriate technical specifications and dimensions. The plans are to indicate the positions of critical structural areas, whether the means of access is permanent or portable and from where each area can be inspected. Critical structural areas are to be identified by advanced calculation techniques for structural strength and fatigue performance and service history of similar ships;

d) instruction for regularly inspecting and maintaining the structural strength of all means of access and means of attachment, taking into account any corrosive atmosphere that may be within the space;

e) instructions for safety guidance when rafting is used for close-up inspections and thickness measurements;

f) instructions for the rigging and use of any portable means of access in a safe manner;

g) an inventory of all portable means of access; and

The second part of the Ship Structure Access Manual is to contain a form of record of periodical inspections and maintenance, and change of inventory of portable equipment due to additions or replacement after construction. The format of this part is to be approved at the time of construction of the ship. The manual is to include a re-approval procedure for any changes to the permanent, portable or movable means of access.

The Ship Structure Access Manual is to be approved by IRS and an updated copy including all revisions / re-approvals is to be kept onboard.

2.4 Technical provisions for means of access for inspections

2.4.1 **Definitions**: For the purpose of these technical provisions the following definitions apply:

a) **Rung** means the step of a vertical ladder or step on the vertical surface.

b) **Tread** means the step of an inclined ladder or step for the vertical access opening.

c) **Flight of an inclined ladder** means the actual stringer length of an inclined ladder. For vertical ladders, it is the distance between the platforms.

d) **Stringer** means:

i) the frame of a ladder; or

ii) the stiffened horizontal plating structure fitted on the side shell, transverse bulkheads and/or longitudinal bulkheads in the space. For the purpose of ballast tanks of less than 5[m] width forming double side spaces, the horizontal plating structure is credited as a stringer and a longitudinal permanent means of access, if it provides a continuous passage of 600[mm] or more in width past frames or stiffeners on the side shell or longitudinal bulkhead. Openings in stringer plating utilized as permanent means of access are to be arranged with guard rails or grid covers to provide safe passage on the stringer or safe access to each transverse web.

e) **Vertical ladder** means a ladder of which the inclined angle is 70[deg] and over upto 90[deg]. A vertical ladder is not to be skewed by more than 2[deg].

f) **Overhead obstructions** means the deck or stringer structure including stiffeners above the means of access.

g) **Distance below deck head** means the distance below the plating.

2.4.2 Structural members subject to the close-up inspections and thickness measurements of the ship's structure referred to in Pt.1, Ch.2, except those in double bottom spaces are to be provided with a permanent means of access to the extent as specified in Table 2.4.2(a) and Table 2.4.2(b), as applicable. Approved alternative methods such as rafting may be used in combination with the fitted permanent means of access, provided that the structure allows for its safe and effective use.

2.4.3 Permanent means of access should as far as possible be integral to the structure of the ships, thus ensuring that they are robust and at the same time contributing to the overall strength of the structure of the ship.

2.4.4 Elevated passageways forming sections of a permanent means of access, where fitted are to have a minimum clear width of 600[mm], except for going around vertical web where the minimum clear width may be reduced to 450[mm] and have guard rails over the open side of their entire length. Sloping structures providing part of the access that are sloped 5[deg] or more from the horizontal plane are to be of a non-skid construction. Non-skid construction is to be such that the surface on which the personnel walks provides sufficient friction to the sole of boots even when the surface is wet and covered with thin sediment.

Guard rails are to be atleast 1000[mm] in height and consist of a rail and an intermediate rail. Guard rails are to be fitted on the open side. For stand-alone passageways guard rails are to be fitted on both sides of these structures. Guardrail stanchions are to be attached to the permanent means of access. The distance between the passageway and the intermediate rail and the distance between intermediate rail and the top rail shall not be more than 500[mm]. They are to be of substantial construction ensuring adequate design strength as well as residual strength during service life. Stanchions are to be not more than 3[m] apart. Discontinuous handrails are allowed provided the gap does not exceed 50[mm]. (Refer Figure 2.4.4). The distance between adjacent stanchions across the handrail gaps is to be not more than 350[mm]. Durability of passage ways and guard rails are to be ensured by corrosion protection and inspection and maintenance during services. Use of alternative materials such as GRP are to be subject to compatibility with the liquid carried in the tank. Fire resistant materials are to be used for all means of access.

Table 2.4.2(a): Means of access for ballast and cargo tanks of oil tankers
Access to the underdeck and vertical structure

Water ballast wing tanks of less than 5[m] width forming double side spaces and their bilge hopper sections

Double side spaces above hopper side tank

1. **For double side spaces above the upper knuckle point** of the bilge hopper sections, permanent means of access are to be provided in accordance with 1.1 to 1.3:

1.1 where the vertical distance between horizontal uppermost stringer and deck head is 6[m] or more, one continuous longitudinal permanent means of access is to be provided for the full length of the tank with a means to allow passing through transverse webs installed at a minimum of 1.6[m] to a maximum of 3[m] below the deck head with a vertical access ladder at each end of the tank;

1.2 continuous longitudinal permanent means of access, which are integrated in the structure, at a vertical distance not exceeding 6[m] apart; (when the permanent means of access is integral with the structure, the allowable vertical distance may be increased by 10%) and

1.3 plated stringers are as far as possible, be in alignment with horizontal girders of transverse bulkheads.

Notes:

a) 1.1 above is for access to under deck structures, whereas 1.2 is for access for inspection of vertical structures (transverse webs) on longitudinal bulkheads.

b) 1.1 to 1.3 above are also applicable to wing tanks designed as void spaces

c) For a tank where the vertical distance between horizontal upper stringer and deck head varies at different sections item 1.1 is to be applied to such sections which fall under the criteria.

d) The continuous permanent means of access may be a wide longitudinal, which provides access to critical details on the opposite side by means of platforms as necessary on web frames. In case the vertical opening of the web frame is located in way of the open part between the wide longitudinal and the longitudinal on the opposite side, platforms shall be provided on both sides of the web frames to allow safe passage through the web frame.

e) Where two access hatches are required by Cl.2.3.6, access ladders at each end of the tank are to be led to the deck.

Hopper side tanks of height ≥ 6[m]

2. For bilge hopper sections of which the vertical distance from the tank bottom to the upper knuckle point is 6[m] and over, one longitudinal permanent means of access is to be provided for the full length of the tank. It is to be accessible by vertical permanent means of access at each end of the tank.

Table 2.4.2a (Contd.)

Notes:

a) Permanent means of access is to be provided between the longitudinal continuous permanent means of access and the bottom of the space.

b) The height of a bilge hopper tank located outside of the parallel part of vessel is to be taken as the maximum of the clear vertical distance measured from the bottom plating to the hopper plating of the tank.

c) The foremost and aftmost bilge hopper ballast tanks with raised bottom, of which the height is 6[m] and over, a combination of transverse and vertical means of access to the upper knuckle point for each transverse web may be accepted in place of the longitudinal permanent means of access.

2.1 The longitudinal continuous permanent means of access may be installed at a minimum 1.6[m] to maximum 3[m] from the top of the bilge hopper section. In this case, a platform extending the longitudinal continuous permanent means of access in way of the web frame may be used to access the identified structural critical areas.

2.2 Alternatively, the continuous longitudinal permanent means of access may be installed at a minimum of 1.2[m] below the top of the clear opening of the web ring allowing a use of portable means of access to reach identified structural critical areas.

Hopper side tanks of height < 6[m]

3. Where the vertical distance referred to in 2 is less than 6[m], alternative means as defined in Cl.2.4.10 or portable means of access may be utilized in lieu of the permanent means of access. To facilitate the operation of the alternative means of access, in-line openings in horizontal stringers are to be provided. The openings are to be of an adequate diameter and are to have suitable protective railings.

Table 2.4.2(b): Means of access for ballast and cargo tanks of oil tankers Access to the underdeck and vertical structure
Water ballast tanks, except those specified in Table 2.4.2(a) and cargo oil tanks
Tanks of height ≥ 6[m] 1. **For tanks of which the height is 6 [m] and over** containing internal structures, permanent means of access is to be provided in accordance with 1.1 to 1.6: 1.1 continuous athwartship permanent access arranged at each transverse bulkhead on the stiffened surface, at a minimum of 1.6[m] to a maximum of 3[m] below the deck head; 1.2 at least one continuous longitudinal permanent means of access at each side of the tank. One of these accesses is to be at a minimum of 1.6[m] to a maximum of 6[m] below the deck head and the other is to be at a minimum of 1.6[m] to a maximum of 3[m] below the deck head; (1.2 is applicable also when the deck longitudinals and deck transverses are fitted on deck but the supporting brackets are fitted under the deck) 1.3 access between the arrangements specified in 1.1 and 1.2 and from the main deck to either 1.1 or 1.2; 1.4 continuous longitudinal permanent means of access which are integrated in the structural member on the stiffened surface of a longitudinal bulkhead, in alignment, where possible, with horizontal girders of transverse bulkheads, are to be provided for access to the transverse webs unless permanent fittings are installed at the upper most platform for use of alternative means, as defined in Cl.2.4.10 for inspection at intermediate heights; (rafting is not acceptable for this purpose) For water ballast tanks of 5[m] or more in width, such as on an ore or oil carrier, side shell plating shall be considered in the same way as “longitudinal bulkhead”. 1.5 for ships having cross-ties which are 6[m] or more above tank bottom, a transverse permanent means of access on the cross-ties providing inspection of the tie flaring brackets at both sides of the tank, with access from one of the longitudinal permanent means of access in 1.4; 1.6 alternative means as defined in Cl.2.4.10 may be provided for small ships as an alternative to 1.4 for cargo oil tanks of which the height is less than 17[m]. Notes: a) 1.1, 1.2 and 1.3 above are for access to underdeck structure, uppermost sections of transverse webs and connection between these structures b) 1.4, 1.5 and 1.6 above are meant for access to vertical structures only and are linked to the presence of transverse webs on longitudinal bulkheads. c) If there are no underdeck structures (deck longitudinals and deck transverses) but there are vertical structures in the cargo tank supporting transverse and longitudinal bulkheads, access in accordance with 1.1 to 1.6 is to be provided for inspection of the upper parts of vertical structure on transverse and longitudinal bulkheads. d) If there are no stiffening structures inside the cargo tank, 1.1 to 1.6 are not applicable. e) The requirements in 1.1 to 1.6 are also to be applied to void spaces of comparable volume in the cargo area.

Table 2.4.2b (Contd.)

f) The vertical distance below the overhead structure is to be measured from the underside of the main deck plating to the top of the platform of the means of access at a given location

g) The height of the tank is to be measured at each tank. For a tank where the height varies at different bays, the requirements are to be applied to only such bays of a tank that have height 6[m] and over.

Tanks of height < 6 [m]

2. For tanks of which the height is less than 6[m], alternative means as defined in Cl.2.4.10 or portable means may be utilized in lieu of the permanent means of access.

Fore peak tanks

3. For fore peak tanks with a depth of 6[m] or more at the centre line of the collision bulkhead, a suitable means of access is to be provided for access to critical areas such as the underdeck structure, stringers, collision bulkhead and side shell structure.

3.1 Stringers of less than 6[m] in vertical distance from the deck head or a stringer immediately above are considered to provide suitable access in combination with portable means of access.

3.2 In case the vertical distance between the deck head and stringers, stringers or the lowest stringer and the tank bottom is 6[m] or more, alternative means of access as defined in Cl.2.4.10 is to be provided.

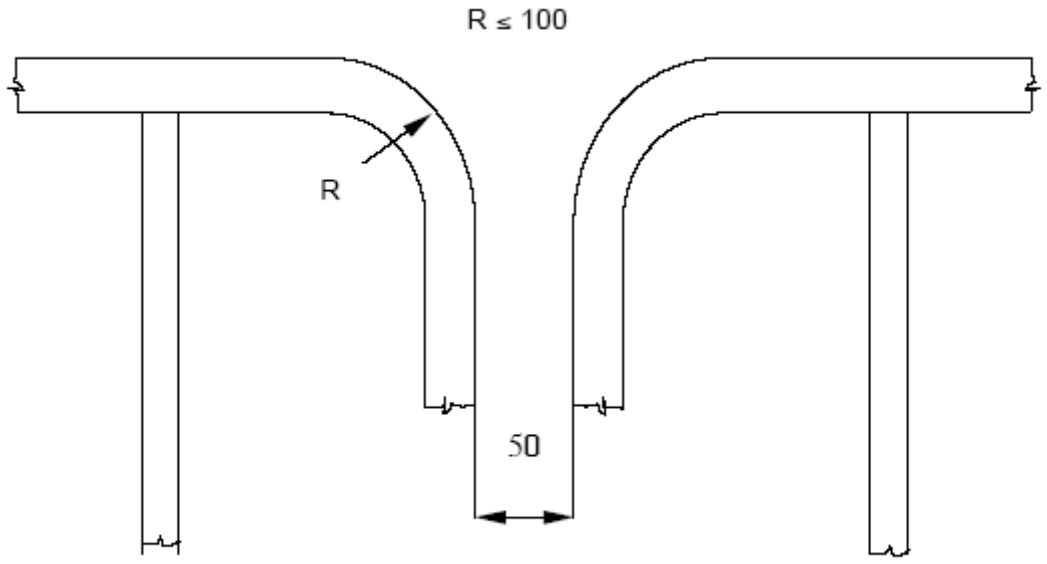

Figure 2.4.4: Discontinuous handrail arrangement

2.4.5 Access to permanent means of access and vertical openings from the ship's bottom is to be provided by means of easily accessible passageways, ladders or treads. Treads are to be provided with lateral support for the foot. Where the rungs of ladders are fitted against a vertical surface, the distance from the centre of the rungs to the surface is to be at least 150[mm]. Where vertical manholes are fitted higher than 600[mm] above the walking level, access are to be facilitated by means of treads and hand grips with platform landings on both sides. In such cases it is to be demonstrated that an injured person can be easily evacuated.

2.4.6 Permanent inclined ladders are to be inclined at an angle of less than 70[deg]. There are to be no obstructions within 750[mm] of the face of the inclined ladder, except that in way of an opening this clearance may be reduced to 600[mm]. Resting platforms of adequate dimensions are to be provided, normally at a maximum of 6[m] vertical height. Ladders and handrails are to be constructed of steel or equivalent material of adequate strength and stiffness and securely attached to the structure by stays. The method of support and length of stay are to be such that vibration is reduced to a practical minimum.

2.4.7 The width of inclined ladders between stringers are not to be less than 400 [mm]. The treads are to be equally spaced at a distance apart, measured vertically of between 200[mm] and 300[mm]. When steel is used, the treads are to be formed of two square bars of not less than 22[mm] by 22[mm] in section fitted to form a horizontal step with the edges pointing upward. The treads are to be carried through the side stringers and attached thereto by double continuous welding. All inclined ladders are to be provided with two course handrails of substantial construction on both sides, fitted at a convenient distance above the treads. Vertical height of handrails is not to be less than 890[mm] from the center of the step and two course handrails need only be provided where the gap between stringer and top handrail is greater than 500[mm].

2.4.8 For vertical ladders or spiral ladders, the width and construction are to be in accordance with international or national standards.

The width of vertical ladders is to be not less than 350[mm] and the vertical distance between the rungs is to be between 250[mm] and 350[mm]. The minimum climbing clearance in width is to be 600[mm].

The vertical ladders are to be secured at intervals not exceeding 2.5[m] apart to prevent vibration.

2.4.9 No free-sanding portable ladder is to be more than 5[m] long. Mechanical devices such as hooks for securing the upper end of a ladder is considered as an appropriate securing device if movement fore/ aft and sideways can be prevented at the upper end of the ladder.

2.4.10 Alternative means of access include, but are not limited to, such devices as:

a) hydraulic arm fitted with a stable base;
b) wire lift platform;
c) staging;
d) rafting;
e) robot arm or remotely operated vehicle (ROV);
f) portable ladders (ladders of more than 5[m] long are to be utilized only if fitted with a mechanical device to secure the upper end of the ladder);
g) other means of access, approved by and acceptable to IRS.

(Refer to Classification Notes: "Guidelines for approval / acceptance of alternative means of access to spaces in oil tankers, bulk carriers, ore carriers and combination carriers").

Means for safe operation and rigging of such equipment to and from and within the spaces are to be clearly described in the Ship Structure Access Manual.

Where an unmanned robot arm or ROVs are proposed to be used for overall and close-up surveys and thickness measurements of deck head structure such as deck transverses and deck longitudinals of cargo oil tanks and ballast tanks, they are to be capable of:

a) safe operation in ullage space in gas free environment; and
b) introduction into the place directly from a deck access.

2.4.11 For access through horizontal openings, hatches or manholes, the minimum clear opening is not to be less than 600[mm] x 600[mm], which may have corner radii of not more than 100[mm]. Where a larger corner sections is desired to reduce the stress level

around the opening, the size of the opening is to be suitably increased to ensure the required size of clear opening e.g. 600[mm] x 800[mm] opening with 300[mm] corner radius is acceptable.

2.4.12 For access through vertical openings, or manholes, in swash bulkheads, floors, girders and web frames providing passage through the length and breadth of the space, the minimum opening is not to be less than 600[mm] x 800[mm] at a height of not more than 600[mm] from the passage unless gratings or other foot holds are provided.

The opening of 600[mm] x 800[mm] may have corner radii of 300[mm]. An opening of 600[mm] in height x 800[mm] in width may be accepted as access opening in vertical structures where it is not desirable to make large opening considering structural strength aspects, such as in girders and floors in double bottom tanks.

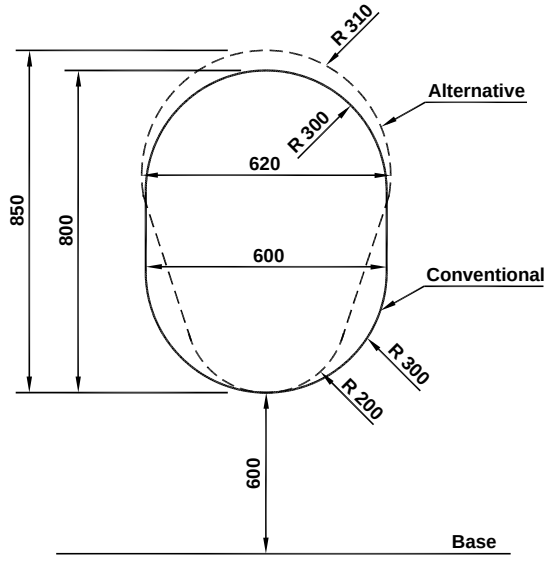

Fig.2.4.12 : Access opening in vertical structure

Subject to verification of easy evacuation of injured person on a stretcher, a vertical opening 850[mm] x 620[mm] as shown in Figure 2.4.12 is considered as an acceptable alternative to the opening of 600[mm] x 800[mm] height with corner radii of 300[mm].

2.4.13 For oil tankers of less than 5,000[t] deadweight, in special circumstances, smaller dimensions may be accepted for the openings referred to in Cl.2.4.11 and Cl.2.4.12, if the ability to traverse such openings or to remove an injured person can be proven.

2.4.14 In tanks and subdivisions of tanks having a length of 35[m] or more requiring two accesses, one access is to consist of inclined ladder or ladders and for the other access a vertical ladder may be used as follows:

a) Where an inclined ladder or combination of ladders is used for access to the space, the uppermost section of the ladder, measured clear of the overhead obstructions in way of the tank entrance, is to be vertical for not less than 2.5[m] but not exceeding 3.0[m] and is to comprise of a landing platform continuing with an inclined ladder. However, the vertical distance of the upper most section of the vertical ladder may be reduced to 1.6[m], if the ladder lands on a longitudinal or athwartship permanent means of access fitted within that range. The flights of the inclined ladders are normally to be not more than 6[m] in vertical height. The lowermost section of the ladders may be vertical for a distance not exceeding 2.5[m].

b) Where vertical ladders are used for access and the vertical distance is more than 6[m], they are to comprise of one or more ladder linking platforms spaced not more than 6[m] apart vertically and displaced to one side of the ladder. The uppermost entrance section from deck of the vertical ladder providing access to a tank is to be vertical for a distance of 2.5[m] but not exceeding 3.0[m] measured clear of overhead obstructions and comprise a ladder linking platform, displaced to one side of a vertical ladder. The vertical distance of the uppermost section of the ladder may be reduced to 1.6[m] if it lands on a longitudinal or athwartship permanent means of access fitted within that range.

2.4.15 In tanks of less than 35[m] in length served by one access hatchway, an inclined ladder or combination of ladders are to be used as specified in Cl.2.4.14 (a) above.

2.4.16 In spaces of less than 2.5[m] width, the access to the space may be by means of vertical ladders as specified in Cl.2.4.14(b) above. Adjacent sections of ladder are to be laterally offset from each other by at least the width of the ladder.

2.4.17 Access from deck to a double bottom space may be by means of vertical ladders through a trunk, with resting platforms spaced not more than 6[m] vertically.

2.5 Equipment in tanks and cofferdams

2.5.1 Anodes, crude oil washing machines and other permanently attached equipment units in tanks and cofferdams are to be firmly secured to the structure. The units and their supports are to be able to withstand sloshing in the tanks and vibratory loads as well as other loads which may be imposed in service. Due consideration should be paid to anti-spark properties of the materials in these permanently attached equipment in tanks and cofferdams. The fixed tank washing machines are to be permanently earthed to the ship.

2.6 Chain locker, anchor windlass and emergency fire pump

2.6.1 The chain locker, anchor windlass and diesel engines for emergency fire pumps are to be installed outside spaces susceptible to the danger of fire and explosion. The distance from the nearest tank hatch or cargo tank opening to windlass and chain pipe is not to be less than 6.0[m]. (Also Refer Cl.8.1.15(c)).

2.6.2 The exhaust pipe of the diesel engine is to have an effective spark arrester and is to be led out to the atmosphere well outside the cargo area and away from all possible sources of fire and explosion.

2.7 Testing of cargo, ballast tanks and other spaces in cargo region

2.7.1 The procedure for testing of cargo tanks and ballast and other spaces in the cargo region are to be generally in accordance with Pt.3, Ch.17.

Due consideration may however be given in cases where the procedures are rendered impracticable due to large size of the tanks.

2.8 Emergency towing arrangements

2.8.1 All tankers of deadweight 20000[t] and above are to be provided with emergency towing arrangements at forward and aft ends. Emergency towing arrangements are to be of adequate strength taking into account the size and deadweight of the ship and the expected forces during bad weather conditions. The design, construction and prototype testing of emergency towing arrangements are to be approved by IRS.

2.8.2 The arrangements, at all times are to be capable of rapid deployment in the absence of main power on the ship to be towed and easy connection to the towing ship. At least one of the emergency towing arrangements are to be pre-rigged, ready for rapid deployment.

2.8.3 Towing pennant, chafing gear, fairleads and strong points on the ship are to have a minimum working strength as follows:

DWT[t]	SWL*[kN]
≥ 20,000 and < 50,000	1000
≥ 50,000	2000
* SWL= 0.5 x Ultimate Strength	

The strength of towing components and their supporting structure is also to be sufficient for all relevant angles of towline, i.e. upto 90[deg] from the ship's centreline to port and starboard and upto 30[deg] vertically downwards from horizontal.

2.8.4 Chafing chain for emergency towing arrangements

a) **Scope**: These requirements apply to the chafing chain for chafing gear of two types of emergency towing arrangement (ETA) with specified safe working load (SWL) of 1000[kN] (ETA1000) and 2000[kN] (ETA2000). Chafing chains other than those specified can be used subject to special approval by IRS.

b) Approval of manufacturing, materials used for the manufacture of the chafing chain, design, manufacture, testing and certification of chafing chain are to be in accordance with Part 2, Chapter 10, Section 2.

c) The arrangement at the end connected to the strong point and the dimensions of the chafing chain are determined by the type of ETA. The other end of the chafing chain is to be fitted with a pear-shaped open link allowing connection to a shackle corresponding to the type of ETA and chain cable grade. A typical arrangement of this chain end is shown in Figure 2.8.4.

d) The common link is to be of stud link type grade 2 or 3.

e) The chafing chain is to be able to withstand a breaking load not less than twice the SWL. For each type of ETA, the nominal diameter of common link for chafing chains is to comply with the value indicated in the following table:

Nominal diameter of common link for chafing chains		
Type of ETA	**Nominal diameter of common link, d min.**	
	Grade CC2	**Grade CC3**
ETA 1000	62[mm]	52[mm]
ETA 2000	90[mm]	76[mm]

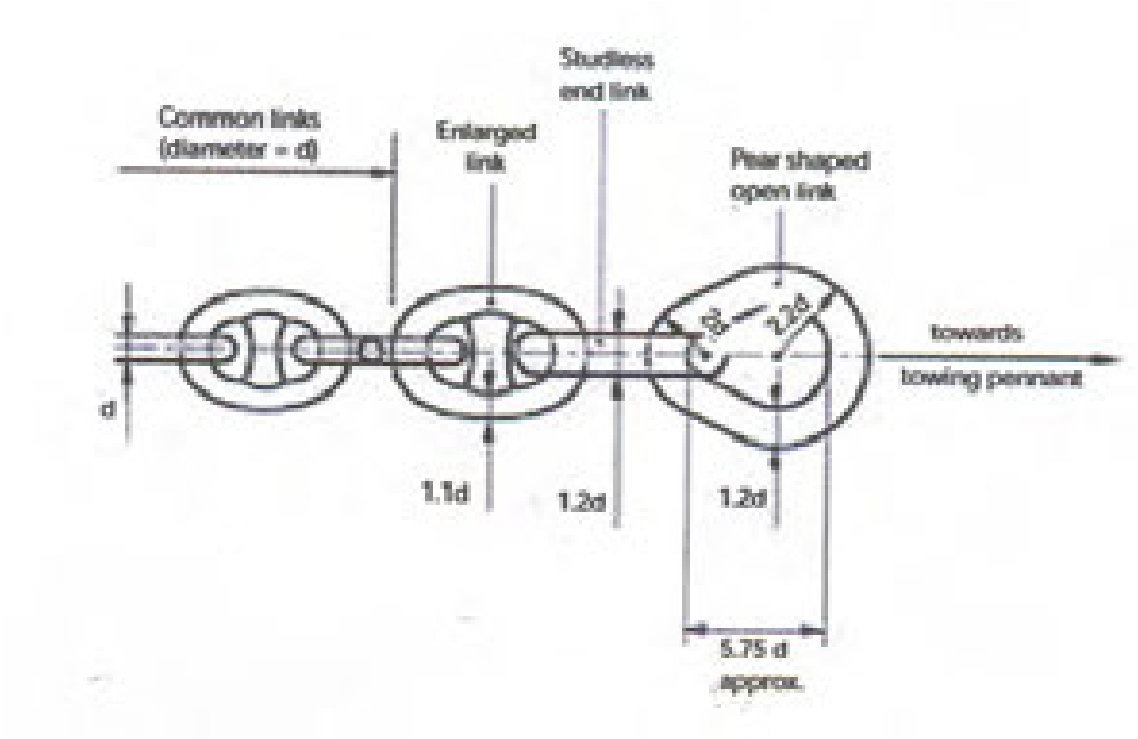

Figure 2.8.4: Typical outboard chafing chain end

2.9 Cargo manifold gutter bars - freeing arrangements and intact stability

2.9.1 Where gutter bars are installed on the weather decks of tankers in way of cargo manifolds and are extended aft as far as the after house front for the purpose of containing cargo spills on deck during loading and discharge operations the free surface effects caused by containment of a cargo spill during liquid transfer operations or of boarding seas while underway require consideration with respect to the vessel's available margin of positive initial stability (GM_0).

2.9.2 Where the gutter bars installed are greater than 300[mm] in height, they are to be treated as bulwarks according to the Load Line Convention, with freeing ports arranged in accordance with Pt.3, Ch.7, Sec.9 Cl.9.2.3 and effective closures provided for use during loading and discharge operations. Attached closures are to be arranged in such a way that jamming cannot occur while at sea, ensuring that the freeing ports will remain fully effective.

2.9.3 On ships without deck camber, or where the height of the installed gutter bars exceeds the camber, and for tankers having cargo tanks exceeding 60% of the vessel's maximum beam at midships regardless of gutter bar height, an assessment of the initial stability (GM_0) for compliance with the relevant intact stability requirement taking into account the free surface effect caused by liquids contained by the gutter bars will be required if gutter bars are provided.

Section 3

Longitudinal Strength

3.1 General

3.1.1 The longitudinal strength is to be determined as per Pt.3, Ch.6.

3.1.2 The read-out points for loading instruments fitted in accordance with Pt.3, Ch.1, Sec.4 are to be positioned at the transverse bulkheads. In general, except when the instrument calculates the maximum values between read-out points, the spacing of read-out points within the cargo tank length is not to exceed five percent of the ship length with intermediate points arranged between bulkheads as necessary.

3.2 Loading conditions

3.2.1 Where a double bottom tank is omitted in accordance with MARPOL Annex I, Reg. 19(4)(1), the minimum operating draught in any loading condition is to be indicated on the midship section plan, loading manual and loading instrument.

3.2.2 Conditions which provide for wing and centre cargo tanks abreast to be filled, with adjacent wing and centre cargo tanks empty, should be avoided as far as possible. Similarly, conditions which provide for differential loading of port and starboard wing cargo tanks with centre cargo tanks empty should also be avoided as far as possible. Where such conditions are contemplated, they will be subject to special consideration.

3.2.3 When loading conditions envisage partially filled tanks, attention is drawn to the need to ensure that the boundary bulkheads are capable of withstanding the loads imposed by the movement of liquid in the tanks.

Section 4

Hull Structure

4.1 General

4.1.1 The bottom shell, inner bottom and deck are generally to be longitudinally framed in the cargo tank region. For ships length L ≥ 150[m], the side shell, inner hull and longitudinal bulkheads are also to be longitudinally framed. The longitudinal bulkheads may however, be horizontally corrugated provided the ship's length does not exceed 200[m].

4.1.2 Inner hull and longitudinal bulkheads are to extend beyond the cargo tank region as far forward and aft as practicable and are to be effectively scarfed into the adjoining structure.

4.1.3 Primary members are to be so arranged as to ensure effective continuity of strength throughout the tank structure. Abrupt changes in depth of sections are to be avoided. Vertical webs on structure are to be arranged in line with the double bottom floors, deck transverses and vertical transverses at the longitudinal bulkheads to ensure continuity of transverse structure.

Longitudinal deck girders are to be supported at transverse bulkheads by vertical webs or equivalent. The depth and scantlings of the continuous girders are to be increased in way to provide effective support.

Where members abutt on both sides of bulkhead or other deeper members, the alignment of webs and faceplates are to be ensured.

4.1.4 The minimum thickness requirement of all structural parts in the cargo tank region is to be as per Pt.3, Ch.7, Sec.3.

4.1.5 The arrangement and scantlings of all tank boundary panels are to be capable of withstanding the dynamic loads from the liquids inside the tanks. Where tanks are intended to be partially filled, the scantling

calculations together with the predicted dynamic loadings are to be submitted.

4.2 Bottom structure

4.2.1 The scantlings and arrangements are, in general, to be as per Pt.3, Ch.3 (Arrangements) and Ch.7 (Scantlings) except as given below.

4.2.2 The strengthening of bottom forward is to be based on the minimum draught forward obtained using segregated ballast tanks only.

4.2.3 Longitudinal girders are to be provided at

- centreline (or duct keel)
- under longitudinal bulkhead (or sloping plates of bulkhead stool in case of vertically corrugated longitudinal bulkheads)
- under sloping plate of hopper side tank where fitted.

4.2.4 In way of vertically corrugated transverse bulkheads supported by stools, additional longitudinal girders are to be arranged extending at least to the first plate floor adjacent to the bulkhead stool sloping plates on each side. These girders are to be spaced not more than 3.6[m] apart.

4.2.5 Plate floors are to be arranged in way of transverse bulkheads and sloping plates of bulkhead stools.

4.2.6 Transverse continuity of inner bottom is to be maintained outboard of inner hull.

4.3 Side structure

4.3.1 Where a hopper side tank is fitted, a transverse is to be arranged in the hopper tank in line with each double bottom plate floor. Scarfing brackets are to be fitted in the hopper in line with the inner bottom on both sides of each transverse to ensure continuity of the inner bottom plating into the hopper side tank. Longitudinal knuckles in the hopper tank plating are to be adequately supported. A horizontal girder is to be arranged at the top of the hopper space and is to be located close to the knuckle between the hopper and the inner hull. Where additional longitudinal girders are provided to satisfy access requirements as per Cl.2.3.7, these are to be arranged in line with horizontal girders on the transverse bulkhead and wing tank cross ties where fitted.

4.3.2 The scantlings of vertical webs and horizontal girders are to be determined by means of direct calculations.

4.3.3 The sectional areas of the cross ties connecting the side transverses to the vertical webs on inner hull or longitudinal bulkheads effectively are not to be less than required in Pt.3, Ch.9.

4.3.4 Brackets are to be provided at the ends of the crossties to connect to the transverses or girders. Transverses and vertical webs are to be fitted with tripping brackets at the junctions with cross ties. Where the width of the face plate of the cross ties exceeds 150[mm] on any one side of the web, additional tripping brackets are to be provided to support the face plate.

4.3.5 End connections of cross-ties are to ensure adequate area of connection and may require additional bracket thickness. Full penetration welding may be required particularly in way of toes of the end brackets.

4.4 Deck structure

4.4.1 Where small diameter threaded openings are arranged on the upper decks for staging wires, these are to be located clear of the other openings and similar areas of stress concentration. Care is to be taken to ensure a gradual transition at the thread ends and the edges of the holes are to be ground smooth. The closing arrangements are to be as per Cl.2.3.8.

4.4.2 The scantlings of deck transverses and girders are to be determined by means of direct calculations.

4.4.3 A trunk deck, if fitted is to extend over the full length of the cargo tanks and is to be effectively scarfed into the main hull structure. The trunk deck and the sides are to be longitudinally framed and the transverse primary members are to be aligned with the outboard deck transverses.

4.4.4 Where external stiffening is carried in way of the trunk deck, appropriate tripping brackets are to be fitted in way of the underdeck supporting structure. The arrangement and details of the external girders will be specially considered.

4.4.5 The thickness of the trunk top and side plating as well as the scantlings of the stiffeners and girders are to be obtained as per Pt.3, Ch.7.

4.5 Tank bulkheads

4.5.1 The arrangement and stiffening of transverse oil tight bulkheads are to efficiently support the lateral liquid pressure as well as the loads transmitted by end connection of inner hull, longitudinal bulkheads, shell and deck longitudinals. Where transverse bulkheads are vertically corrugated, horizontal stringers or equivalent are to be fitted to provide adequate resistance to transverse compressive forces.

4.5.2 The top and bottom strakes of longitudinal corrugated bulkheads are to be plane over width of 0.1D from the deck and bottom. The thickness of this plating is not to be less than 75% of the adjoining deck and inner bottom plating.

4.5.3 Particular attention is to be paid to the through thickness properties at the connection to the deck and inner bottom.

4.5.4 Where longitudinal bulkheads are corrugated horizontally, the corrugations are to be aligned, and stiffening arrangements on plane members are to be arranged to provide adequate support in way of flanges of abutting corrugations. Where both the longitudinal and transverse bulkheads are horizontally corrugated, the ends are to be connected to ensure continuity.

4.5.5 Bulkhead stools in way of vertically corrugated bulkheads where fitted are to be generally as per Pt.5, Ch.1, Sec.2.9. An efficient system of reinforcement is to be arranged in line with the tank transverse bulkheads or bulkhead stools at the intersection with the sloped plating of double bottom hopper tanks and topside tanks.

4.5.6 The arrangement of stools and adjacent structure common to the cargo tank is to be designed to avoid pockets in which gas could collect.

4.5.7 Non-oil tight wash bulkheads are generally to be of plane construction having an area of perforation between 5 to 10 percent of the total area of the bulkhead. Large perforations should not be provided on the top and bottom strakes of longitudinal bulkheads. The stiffening arrangement and the perforations are to be arranged so as not to impair strength of the bulkhead and provide adequate support to the loads from the end connections. Where tanks are intended to be partially filled, the scantlings and arrangement are to be capable of withstanding the dynamic loads.

4.5.8 The plate thickness of wash bulkheads is not to be less than the minimum thickness specified in Pt.3, Ch.7. Where the wash bulkheads support primary members perpendicular to their plane, the plating with stiffeners are to have adequate shear rigidity in way and also satisfy panel stability requirement.

4.5.9 Where horizontal girders (or vertical webs) on the transverse bulkheads do not form part of a ring structure, they are to be arranged with substantial end brackets forming a buttress extending to the adjacent vertical web (or transverse). The shear and combined stresses in the buttress arrangement is to be specially examined.

4.6 Construction details

4.6.1 The members are to have adequate end fixity, lateral support and web stiffening, and the structure is to be arranged to minimize hard spots or other sources of stress concentration. Openings are to have well rounded corners and smooth edges and are to be located having regard to the stress distribution and buckling strength of the plate panel.

4.6.2 To maintain continuity of strength, substantial horizontal and vertical brackets are to be fitted to transverses or stringers at the ends of the cross ties. Horizontal brackets are to be aligned with the cross tie face plates, and vertical end brackets are to be aligned with the cross tie web.

4.6.3 In a ring system where the end bracket is integral with the webs of the members, and the face plate is carried continuously along the edges of the members and the bracket, the full area of the largest face plate is to be maintained upto the mid-point of the bracket and then gradually tapered to the smaller face plates. Butts in face plates are to be kept well clear of the toes of brackets.

4.6.4 The thickness of separate end brackets is generally to be not less than that of the

thicker of the primary member webs being connected, but may be required to be locally increased at the toes. The bracket is to extend to adjacent tripping brackets, stiffeners or other support points. Bracket toes are to be well radiused. Where the bracket is attached to a corrugated bulkhead, the plating at the bracket toe is to be suitably reinforced.

4.6.5 Tripping brackets are generally to be fitted close to the toes of end brackets, in way of cross ties and generally at every fourth stiffener elsewhere. Arrangements should also be made to prevent tripping at the intersection with other primary members. Refer Pt.3, Ch.3, Sec.6 for Tripping bracket arrangement.

4.6.6 In way of cross ties and their end connections lightening holes are not to be cut in side and longitudinal bulkhead stringers. Lightening holes are also to be avoided on vertical webs on longitudinal bulkheads and in wing ballast tanks.

4.6.7 Holes cut in primary longitudinal members within 0.1D of deck and bottom are, in general to be reinforced. Where holes are cut in primary longitudinal members in areas of high stress and where primary members are of higher tensile steel, they are to be elliptical, or equivalent, to minimise stress concentration.

4.6.8 Longitudinals within the range of cargo tanks are not permitted to have closely spaced scallops except in way of ballast pipe suctions. Reinforcement in these areas will be specially considered. Small air and drain holes, cut-outs at erection butts and similar widely spaced openings are, in general not to be less than 200[mm] clear of the toes of end brackets, intersections with primary supporting members and other areas of high stress. All openings are to be well rounded with smooth edges.

4.6.9 Where holes are cut for heating coils, the lower edge of the hole is to be not less than 100[mm] from the inner bottom. Where large notches are cut in the transverses for the passage of longitudinal framing, adjacent to openings for heating coils, the notches for longitudinals are to be collared.

4.7 Design load sets for PSM

Design load sets for primary supporting members within the cargo hold region are given in Table 4.7.1.

Table 4.7.1 Design load sets for PSM						
Structural member	**Design load set (9) (10)**	**Design load scenario**	**Load component**	**Draught**	**Acceptance Criteria (11)**	**Loading condition**
Double bottom floors and girders (2)	SEA-1	1	P_{ex}	T_{SC}	AC-II	Full load condition
	SEA-2	2	P_{ex}	T_{SC}	AC-I	Harbour condition
	LT-3	1	$P_{in} - P_{ex}$ (1)	$0.6T_{SC}$	AC-II	Partial load condition
	LT-2	2	$P_{in} - P_{ex}$ (1)	(3)	AC-I	Harbour condition
Side Transverses (2)	SEA-1	1	P_{ex}	T_{SC}	AC-II	Full load condition
	SEA-2	2	P_{ex}	T_{SC}	AC-I	Harbour condition
	LT-1	1	P_{in}	T_{SC}	AC-II	Full load condition
	LT-3			$0.6T_{SC}$		Partial load condition
	LT-2	2	P_{in}	-	AC-I	Harbour and Tank Testing condition
Deck Transverses	SEA-1	1	P_{ex} (4)	T_{SC}	AC-II	Full load condition
	LT-1	1	P_{in}	T_{SC}	AC-II	Full load condition
	LT-3			$0.6T_{SC}$		Partial load condition
	LT-2	2	P_{in}	-	AC-I	Harbour and Tank Testing condition
Vertical web frames on longitudinal bulkheads (5)	LT-1	1	P_{in}	T_{SC}	AC-II	Full load condition
	LT-3			$0.6T_{SC}$		Partial load condition
	LT-2	2	P_{in}	-	AC-I	Harbour and Tank Testing condition
Horizontal stringers on transverse bulkhead (6)	LT-1	1	P_{in}	T_{SC}	AC-II	Full Load condition
	LT-3			$0.6T_{SC}$		Partial load condition
	LT-2	2	P_{in}	–	AC-I	Harbour and Tank Testing condition
Cross ties in centre tanks (7)	LT-1	1	$0.5(P_{in-pt} + P_{in-stb})$	T_{SC}	AC-II	Full Load condition
	LT-3	1	$0.5(P_{in-pt} + P_{in-stb})$	$0.6T_{SC}$	AC-II	Partial load condition

	LT-2	2	P_{in}	–	AC-I	Harbour and Tank Testing condition
Cross ties in Wing tanks (8)	LT-1	1	$0.5(P_{in} + P_{ex})$	T_{SC}	AC-II	Full Load condition
	LT-3	1	$0.5(P_{in} + P_{ex})$	$0.6T_{SC}$	AC-II	Partial load condition
	LT-2	2	$0.5(P_{in} + P_{ex})$	T_{SC}	AC-I	Harbour and Tank Testing condition

where,

P_{in-pt} = Design pressure from port side wing cargo tank, in [kN/m^2].

P_{in-stb} = Design pressure from starboard side wing cargo tank, in [kN/m^2].

P_{ex} = External pressure for static (S) and static plus dynamic (S+D) design load scenario as defined in Pt.3, Ch.5, Sec.4, Cl.4.1.1.

P_{in} = Internal pressure due to liquid for static (S) and static plus dynamic (S+D) design load scenario as defined in Pt.3, Ch.5, Sec.5, Cl.5.1.1.

(1) (P_{ex}) is considered for external shell only.

(2) Draughts specified for bottom floors, girders and side transverses are based on operational limits specified in Pt.3, Ch.5, Sec.6. Where the optional loading conditions exceed the minimum Rule required loading conditions, the draughts will be subject to special consideration.

(3) For tankers with two oil-tight longitudinal bulkheads, the draught is to be taken as $0.25T_{SC}$. For tankers with a centreline bulkhead, the draught is to be taken as $0.33T_{SC}$.

(4) Green sea pressure only or other loads on deck.

(5) Pressure from one side only. Full cargo tank with adjacent cargo tank empty.

(6) Pressure from one side only. Full cargo tank with adjacent forward or aft cargo tank empty.

(7) Full wing cargo tanks, centre tank empty

(8) Full centre tank, wing cargo tanks empty

(9) For a void or dry space, the pressure component from the void side is to be ignored.

(10) When the ship's configuration cannot be described by the structural members or structural configurations identified above, then the applicable Design Load Sets to determine the scantling requirements of PSM's are to be selected so as to specify all applicable cases from the following:

- A full tank on one side of the member with the tank or space on the other side empty.
- A full tank on one side of the member with the external pressure minimized.
- External pressure maximized with the adjacent tank or space empty.

The boundary is to be evaluated for loading from both sides. Design Load Sets are to be selected based on the tank or space contents and are to maximize the net pressure on the structural boundary, the draught to use is to be taken in accordance with the Design Load Set and this table. Design Load Sets covering the AC-I and AC-II acceptance criteria are to be selected.

(11) See acceptance criteria definition in Pt.3, Ch.1, Sec.1, Cl.1.10.

Section 5

Direct Strength Calculations

5.1 General

5.1.1 Direct strength calculations are required in cases where simplified formulations are not able to take into account special stress distributions, boundary conditions or structural arrangements with sufficient accuracy.

5.1.2 For those girder systems where such direct strength calculations are mentioned in the preceeding sections of this Chapter, scantlings obtained from simplified formulae may have to be increased based on the results obtained.

5.1.3 Wherever required by Cl. 5.1.1 and Cl. 5.1.2, a simplified engineering analysis based on elastic beam theory, two-dimensional grillage or finite element analysis is be carried out to demonstrate that the stresses are acceptable when the structure is loaded as given in 5.2.

Calculations as mentioned above needs to be carried out for:

- transverse, horizontal and vertical girders in cargo tanks
- bulkhead structures
- double bottom structures

other structures as deemed necessary by IRS

5.1.4 For ships having Length, L≥150[m], Cargo hold finite element analysis is to be carried out for midship region according to the requirements given in Pt.3, Ch.8, Sec.3. However, cargo hold FEA may be optionally carried out for ships of Length, L<150[m] applying the requirements in Pt 3, Ch 8, Sec. 3.

5.1.5 Where FEA is carried out, local fine mesh analysis is to be carried out according to the requirements given in Pt.3, Ch.8, Sec.4:

- for laterally loaded local stiffeners and their connected brackets, subject to relative deformation between supports in the midship region
- for a connection of the corrugation and lower supporting structure to inner bottom if no lower stool is fitted.

5.2 Load cases and design loads

5.2.1 The calculations are to be carried out for the realistic condition which cause most severe loading on the various parts of cargo tank structures.

5.2.2 For transverse and longitudinal girders in the cargo region following conditions are to be considered:

- Sea-going conditions:
  - any cargo tank empty with adjacent cargo tanks full and the ship on scantling draught, (T_{SC}).
  - any cargo tank filled with adjacent cargo tanks full and the ship at the minimum relevant sea-going draught.
  - all cargo tanks within a transverse section of the ship filled, with adjoining cargo tanks forward and aft empty and the ship at the minimum relevant sea-going draught.
- Harbour Conditions:
  - any cargo tank filled with adjacent tanks empty and the ship is at a draught of 0.35T
  - all tanks in a section of the ship filled with adjacent tanks empty and the ship floating at a draught of 0.5T.

All load combinations based on the above are to be investigated for various types of longitudinal bulkhead arrangements.

5.2.3 For girders on transverse bulkheads, loading conditions involving alternate loading in centre tank and wing tanks are to be considered in addition to those mentioned in Cl.5.2.2, with minimum sea-going draught.

5.2.4 Ship with two (2) longitudinal bulkheads and with cross ties in the centre tank is to be considered for an asymmetric load condition with one wing tank filled and other tanks empty, where such loading pattern is included in the ship loading manual for seagoing

conditions. If such loading patterns are not considered, an operational restriction describing that the difference in filling level between corresponding port and starboard wing cargo tanks is not to exceed 25% of the filling height in the wing cargo tank, is to be added in the loading manual.

The actual loading pattern, draught, GM and k_y from the loading manual are to be used in the FE analysis. Where the GM and k_y are not given in the ship's loading manual, GM and k_y are to be determined in accordance with Pt.3, Ch 5, Sec 2.

5.2.5 External Sea pressures and Internal tank pressure related to the specified draught and the filling height are to be considered, generally as per Pt.3, Ch.5 and Ch.7.

For seagoing conditions realistic combinations of dynamic components of the external and internal pressures are to be considered. For harbour conditions only static pressures may be considered.

5.3 Analysis Criteria

5.3.1 Coarse mesh permissible yield and buckling utilization factors

The coarse mesh permissible yield utilization factors, λ_{yperm}, for cargo hold analysis is given in Pt.3, Ch.8, Sec.3, Table 3.4.2.3.

The allowable buckling utilization factors, η_{allow} is given in Pt.3, Ch.9, Sec.1, Table 1.3.3.

In addition, for a vertically corrugated bulkhead where lower stool is not fitted, coarse mesh allowable yield and buckling utilization factor are given in Table 5.3.1.

5.3.2 Fine mesh permissible yield utilization factors

Analysis criteria for fine mesh analysis are given in Pt.3, Ch.8, Sec.4, Table 4.4.3.

Where a lower stool is not fitted to a vertically corrugated bulkhead, the permissible stresses given in Pt.3, Ch.8, Sec.4, Cl.4.4.3 is to be reduced by 10% for the areas under investigation by fine mesh analysis.

Table 5.3.1: Permissible yield utilization, λ_{yperm}, and allowable buckling utilization, η_{allow}, factors

Structural Members	Load Components	Acceptance criteria	λ_{yperm}	η_{allow}
Supporting structure in way of lower end of corrugated bulkheads without lower stool [1)]	S	AC-I	0.72	0.72
	S+D	AC-II	0.90	0.90
Corrugation of vertically corrugated bulkheads without lower stool under lateral pressure from liquid loads, for shell elements only.	S	AC-I	0.65	0.65
	S+D	AC-II	0.81	0.81

1) Supporting structure for a transverse corrugated bulkhead refers to the structure in the longitudinal direction within half a web frame space forward and aft of the bulkhead, and within a vertical extent equal to the corrugation depth. Supporting structure for a longitudinal corrugated bulkhead refers to the structure in transverse direction within 3 longitudinal stiffener spacings from each side of the bulkhead, and within a vertical extent equal to the corrugation depth.

Section 6

Pumping and Piping Systems

6.1 Cargo pump rooms

6.1.1 Cargo pump rooms are to be totally enclosed and are to have no direct communication with machinery spaces.

6.1.2 Pump rooms are to be situated within, or adjacent to, the cargo tanks area and are to be provided with ready means of access from the open deck.

6.1.3 In cargo pump rooms any drain pipes from steam or exhaust pipes or from the steam cylinders of the pumps are to terminate well above the level of the bilges.

6.1.4 Cargo pump rooms and other closed spaces which contain cargo handling equipment, and to which regular access is required during cargo handling are to be provided with permanent ventilation systems of the mechanical extraction type.

6.1.5 The ventilation system is to be capable of being operated from outside the compartment being ventilated, and a notice is to be fixed near the entrance stating that no person is to enter the space until the ventilation system has been in operation for at least 15 minutes.

6.1.6 The ventilation system is to be capable of 20 changes/hour. The number of air changes is to be based on the gross volume of the pump room or space.

6.1.7 The ventilation ducting is to be arranged to permit extraction from the vicinity of the pump room bilges, immediately above the transverse floor plates or bottom longitudinals. An emergency intake is also to be provided in the ducting at a height of 2[m] above the pump room lower platform and is to be provided with a damper capable of being opened or closed from the weather deck and lower platform level. An arrangement involving a specific ratio of areas of upper emergency and lower main ventilation openings, which can be shown to result in at least the required number of air changes through the lower inlets, can be accepted without the use of dampers. When the lower inlets are sealed off owing to the flooding of the bilges, then at least 75% of the required number of air changes is to be obtainable through the upper inlets. Means are to be provided to ensure the free flow of gases through the upper platform to the duct intakes.

6.1.8 The arrangements and materials of mechanical ventilator components are to be designed to prevent the risk of incendive sparking. Where non-metallic materials are used they are to have anti-static properties.

6.1.9 Renewable flame screens are to be provided in ventilation ducts, and ventilation intakes are to be so arranged as to minimise the possibility of re-cycling hazardous vapours from any ventilation discharge openings. Vent exits are to be arranged to discharge upwards.

6.1.10 Vent exits from pump rooms are to discharge at least 3[m] above the deck and from the nearest air intakes or openings to accommodation and enclosed working spaces and from possible sources of ignition.

6.2 Piping systems for bilge, ballast, oil fuel etc.

6.2.1 There is to be no connection between piping systems in the cargo area and the piping systems in the remainder of the vessel, unless explicitly specified herein.

6.2.2 The oil fuel bunkering system is to be entirely separate from the cargo handling system.

6.2.3 The pumping arrangements in the machinery space and forward end of the ship are to comply with the requirements for general cargo ships, in so far as they are applicable.

6.2.4 Cargo pump rooms are to have a drainage system connected to pumps or bilge ejectors. The cargo pumps may be used for this purpose provided each bilge suction pipe is fitted with a screw-down non-return valve and an additional valve/cock is fitted to the pipe connection between the pump and the non-return valve. Pump room suctions are not to enter the engine room.

6.2.5 Cofferdams, which are required to be provided at the fore and aft ends of the cargo spaces, are to be fitted with suitable drainage arrangements, generally, in accordance with following:

6.2.5.1 Where deep cofferdams can be filled up with water ballast, a ballast pump in the main engine room may be used for emptying the after cofferdam. Where fitted, a ballast pump in a forward pump room may be used for draining the forward cofferdam. In each case, the suctions are to be led direct to the pump and not to a pipe system.

6.2.5.2 Where intended to be dry compartments, after cofferdam adjacent to the pump room may be drained as provided in 6.2.4. Forward cofferdam may be drained by a bilge/ballast pump in a forward pump room. Alternatively, cofferdams may be drained by bilge ejectors.

6.2.6 Cofferdams are to have no direct connections to cargo tanks or cargo lines.

6.2.7 Ballast tanks and void spaces within the cargo area are not to be connected to cargo pumps or have any connections to the cargo system. A separate ballast/bilge pump is to be provided for dealing with the contents of these spaces. This pump is to be located in the cargo pump room or other suitable space within the cargo area.

Consideration will be given to connecting double bottom and/or wing tanks, which are in the cargo area, to pumps in the machinery spaces where the tanks are completely separated from the cargo tanks by cofferdams, heating ducts, or containment spaces, etc.

The forepeak tank can be ballasted with the system serving ballast tanks within the cargo area, provided:

- the forepeak tank is considered as hazardous area;

- the vent pipe openings are located on open deck at an appropriate distance from sources of ignition. In this respect, the hazardous zones distances are to be defined in accordance to IEC 60092-502: Electrical installations in ships - Tankers – Special features;

- means are provided, on the open deck, to allow measurement of flammable gas concentrations within the forepeak tank by a suitable portable instrument;

- the sounding arrangement to the forepeak tank is direct from open deck;

- the access to the forepeak tank is direct from open deck. Alternatively, indirect access from the open deck to the forepeak tank through an enclosed space may be accepted provided that:

  a) In case the enclosed space is separated from the cargo tanks by cofferdams, the access is through a gas tight bolted manhole located in the enclosed space and a warning sign is to be provided at the manhole stating that the forepeak tank may only be opened after:

     i) it has been proven to be gas free; or
     ii) any electrical equipment which is not certified safe in the enclosed space is isolated.

  b) In case the enclosed space has a common boundary with the cargo tanks and is therefore a hazardous area, the enclosed space can be well ventilated.

The hazardous area classification for above is in accordance to IEC 60092-502: Electrical installations in ships - Tankers - Special features.

6.2.8 Ballast piping is not to pass through cargo tanks as far as possible and is not to be connected to cargo oil piping. Provision may, however, be made for emergency discharge of water ballast by means of a portable spool connection to a cargo oil pump and where this is arranged, a non-return valve is to be fitted in the ballast suction to the cargo oil pump. The portable spool piece is to be mounted in a conspicuous position in the pump room and a permanent notice restricting its use is to be prominently displayed adjacent to it. Shut-off valves shall be provided to shut-off the cargo and ballast lines before the spool piece is removed.

The ballast pump is to be located in the cargo pump room, or a similar space within the cargo area not containing any source of ignition.

Ballast piping passing through cargo tanks and cargo oil pipes passing through segregated ballast tanks, as permitted by Reg.19.3.6 of MARPOL Annex I, are to be of heavy gauge steel of minimum wall thickness according to the table hereunder with welded or heavy flanged joints the number of which is to be kept to a minimum. Only expansion bends (not glands) are permitted in these lines within cargo tanks for serving the ballast tanks and within the ballast tanks for serving the cargo tanks.

Nominal diameter [mm]	Minimum wall thickness of carbon steel pipes [mm]
50	6.3
100	8.6
125	9.5
150	11.0
200 and above	12.5

Thickness for intermediate sizes may be calculated by interpolation.

6.2.9 Deep cofferdams at the fore and aft ends of the cargo spaces and other tanks or cofferdams within the range of the cargo tanks, which are not intended for cargo, are to be provided with air and sounding pipes led to the open deck. The air pipes are to be fitted with wire gauge diaphragms at their outlets.

This requirement is not applicable to ships assigned additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C.**

6.2.10 So far as practicable, the air and sounding pipes required by Cl.6.2.9 are not to pass through cargo tanks. Where this cannot be avoided e.g. on oil tankers of deadweight less than 5000[t], where wing ballast tanks or spaces need not be provided, the sounding and air pipes may pass through the cargo tanks. However, the pipes are to be of steel having a wall thickness of not less than 12.5[mm] and they are to be in continuous lengths or with welded joints.

This requirement is not applicable to ships assigned additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C.**

6.2.11 The requirements for integrated cargo and ballast systems are given in Sec.7.6.

6.3 Steam connection to cargo tanks

6.3.1 Where steaming out and/or fire extinguishing connections are provided for cargo tanks or cargo pipe lines, they are to be fitted with valves of the screw-down non-return type. The main supply to these connections is to be fitted with a master valve placed in a readily accessible position clear of the cargo tanks.

6.4 Equipment in dangerous spaces

6.4.1 Oil engines, or, any other equipment which could constitute a possible source of ignition, are not to be situated within cargo tanks, pump rooms, cofferdams or other spaces liable to contain explosive vapours or in spaces immediately adjacent to cargo oil or slop tanks. The temperature of steam or other fluid, in pipes (or heating coils) in these spaces is not to exceed 220[°C].

6.4.2 Measures are to be taken to prevent explosions in cargo pump rooms. Cargo pumps, ballast pumps and stripping pumps, installed in cargo pump rooms and driven by shafts passing through pump room bulkheads are to be fitted with temperature sensing devices for bulkhead shaft glands, bearing, and pump casings. Alarms are to be initiated in the cargo control room or the pump control station.

The requirements for temperature sensing devices and alarms are not applicable to ships assigned additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C**

6.4.3 Lighting in cargo pump room is to be interlocked with ventilation such that ventilation shall be in operation when switching on the lighting. Failure of the ventilation system is not to cause the lighting to go out. Emergency lighting if fitted is not to be interlocked.

6.4.4 A system for continuously monitoring the concentration of hydrocarbon gases is to be fitted. Sequential sampling is acceptable as long as it is dedicated for the pump room only, including exhaust ducts and the sampling time is reasonably short. Sampling points or detector heads shall be located in suitable positions e.g. exhaust ventilation ducts and lower parts of pump rooms above floor plates, in order that potentially dangerous leakages

are readily detected. The system is to raise an alarm if the concentration of hydrocarbon reaches a pre-set level which shall not be higher than 10% of the lower flammable limit (LFL). Audible and visual signals are to be provided in the pump room, engine control room, cargo control room and on the navigation bridge.

Gas analysing units with non-explosion proof measuring equipment may be located in areas outside cargo areas, e.g. in cargo control room, navigation bridge or engine room when mounted on the forward bulkhead provided the following requirements are observed:

a) Sampling lines are not to run through gas safe spaces, except where permitted under (e).

b) The gas sampling pipes are to be equipped with flame arresters. Sample gas is to be led to the atmosphere with outlets arranged in a safe location.

c) Bulkhead penetrations of sample pipes between safe and dangerous areas are to be of approved type and have same fire integrity as the division penetrated. A manual isolating valve is to be fitted in each of the sampling lines at the bulkhead on the gas safe side.

d) The gas detection equipment including sample piping, sample pumps, solenoids, analysing units etc. are to be located in a reasonably gas tight enclosure (e.g. a fully enclosed steel cabinet with a gasketed door) being monitored by its own sampling point. At gas concentrations above 30 per cent LFL inside the enclosure the entire gas analysing unit is to be automatically shut down.

e) Where the enclosure cannot be arranged directly on the bulkhead, sample pipes are to be of steel or other equivalent material and without detachable connections, except for the connection points for isolating valves at the bulkhead and analysing units and are to be routed on their shortest ways.

The above requirements are also applicable to gas analysing units of the sampling type located outside gas dangerous zones and fitted on board gas carriers or chemical tankers.

6.4.5 All pump rooms are to be provided with bilge level monitoring devices together with audio-visual alarms in the wheel house, pump control station and other appropriate locations.

IR6.4.5 The above requirements Cl.6.4.2 to Cl.6.4.5 are only applicable to the pump-rooms where pumps for cargo, such as cargo pumps, stripping pumps, pumps for slop tanks, pumps for COW or similar pumps are provided. Pump rooms intended solely for ballast transfer and/or fuel oil transfer need not comply with Cl.6.4.2 to Cl.6.4.5 above.

6.5 Non-sparking fans

6.5.1 A fan is considered as non-sparking if in either normal or abnormal conditions it is unlikely to produce sparks.

6.5.2 The air gap between the impeller and the casing is to be not less than 0.1 of the shaft diameter in way of the impeller bearing but not less than 2[mm] and in any case not more than 13[mm].

6.5.3 Protection screens of not more than 13[mm] square mesh are to be fitted in the inlet and outlet ventilation openings on the open deck to prevent the entrance of objects into the fan housing.

6.5.4 The impeller and the housing in way of the impeller are to be made of alloys which are approved as being spark proof by appropriate test.

6.5.5 Electrostatic charges both in the rotating body and the casing are to be prevented by the use of antistatic materials. Furthermore, the installation on board of the ventilation units is to be such as to ensure the safe bonding to the hull of the units themselves.

6.5.6 Tests may not be required for fans having the following combinations:

a) impellers and/or housings of nonmetallic material, due regard being paid to the elimination of static electricity,

b) impellers and housings of non-ferrous materials,

c) impellers of aluminium alloys or magnesium alloys and a ferrous (including austenitic stainless steel) housing on which a ring of suitable thickness on non-ferrous materials is fitted in way of the impeller,

d) any combination of ferrous (including austenitic stainless steel) impellers and housings with not less than 13[mm] tip design clearance.

6.5.7 The following impellers and housings are considered as sparking and are not permitted:

a) impellers of an aluminium alloy or magnesium alloy and a ferrous housing, regardless of tip clearance,

b) housing made of an aluminium alloy or a magnesium alloy and a ferrous impeller, regardless of tip clearance,

c) any combination of ferrous impeller and housing with less than 13[mm] design tip clearance.

6.5.8 Type tests on the finished product are to be carried out in accordance with the requirements of the 'Rules and Regulations for the Construction and Classification of Steel Ships' or an equivalent national or international standard.

6.6 Earthing and bonding of cargo tanks and piping systems for cargo of flash point not exceeding 60°C

6.6.1 Independent cargo tanks and cargo piping systems intended for cargo with flash point not exceeding 60[°C] are to be connected to the hull of the ship so that resistance between them and the hull is less than 10^6 [ohm].

6.6.2 Bonding straps are required for cargo tanks/piping systems which are not permanently connected to the hull of the ship, e.g.

a) Independent cargo tanks;

b) Cargo tanks/piping systems which are electrically separated from the hull of the ship;

c) Pipe connections arranged for the removal of spool pieces.

d) Wafer-style valves with non-conductive (e.g. PTFE) gaskets or seals.

6.6.3 Where bonding straps are required, they are to be:

a) Clearly visible so that any shortcomings can be clearly detected;

b) Designed and sited so that they are protected against mechanical damage and that they are not affected by high resistivity contamination e.g. corrosive products or paint;

c) Easy to install and replace.

Section 7

Cargo Handling Systems

7.1 General

7.1.1 A complete system of piping and pumps is to be fitted for dealing with the cargo.

7.1.2 Standby means for pumping out each cargo tank are to be provided.

7.1.3 Where cargo tanks are provided with single deep well pumps, or submerged pumps, it will be necessary to provide alternative means for emptying the tanks in the event of failure of a pump. Portable submersible pumps may be provided on board for this purpose, but the arrangements are to be such that a portable pump could be safely introduced in to a full or part full tank. Details of the arrangements are to be submitted for approval.

7.1.4 Cargo tank access hatches and all other openings to cargo tanks, such as ullage and tank cleaning openings, sighting ports and restricted sounding devices, are to be located on the weather deck and are not to be arranged in enclosed compartments.

7.2 Cargo pumps

7.2.1 Cargo oil pumps are to be designed so as to minimise the risk of sparking and oil leakage at the stuffing box.

7.2.2 Where cargo pumps are driven by shafting which passes through a pump room bulkhead, shafting is to be provided with flexible couplings and gastight stuffing boxes are to be fitted to the shaft at the pump room bulkhead. The glands are to be efficiently lubricated from outside the pumproom and constructed so as to reduce the risk of overheating. The bulkhead shaft glands, bearings and pump casings are to be provided with temperature sensing devices. Alarms are to be initiated in the cargo control room or the pump control station.
The requirements for temperature sensing devices and alarms are not applicable to ships assigned additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C.**

Parts, which may come in contact, if the seals are misaligned or bearings are damaged, are to be of materials which will not spark. Where bellows are incorporated in the design, these are to be hydraulically tested to 3.4 bar before fitting.

7.2.3 A stop valve and a relief valve of adequate capacity are to be fitted on the delivery side of each pump. Relief valves are to be fitted in close- circuit, i.e. discharging to the suction side of the pumps. Relief valve may be omitted in case of centrifugal pumps, which are so designed that the discharge pressure cannot exceed the design pressure.

7.2.4 A pressure gauge is to be fitted on the delivery side of each pump. Where the pump is driven by a prime mover, which is installed in a space other than the pump room, an additional pressure gauge is to be fitted at a suitable position visible from the controlling position of the prime mover.

7.2.5 Where cargo pumps are driven by hydraulic motors which are located inside cargo tanks, the design is to be such that the contamination of the operating media and cargo oil cannot take place under normal operating conditions.

7.2.6 Means are to be provided for stopping the cargo oil pumps from a position outside the pump rooms, as well as at the pumps.

7.2.7 In general, cargo oil pumps are to be driven by steam where the prime movers are installed in the pump rooms.

7.2.8 Motors of electrically driven cargo pumps may be installed in cargo pump rooms, provided they comply with the applicable requirements for electrical equipment in hazardous areas, as indicated in IEC 60092-502: Tankers – Special Features. Additional requirements of the flag Administration may also be taken cognizance of.

7.2.9 Pumps with a submerged electric motor are not permitted in cargo tanks.

7.3 Cargo piping systems

7.3.1 The complete cargo piping system is to be located within the cargo tank area.

7.3.2 Expansion joints of approved type or bends are to be provided, where necessary, in the cargo pipe lines.

7.3.3 Means for drainage of cargo lines are to be provided. Drain lines may be led to a cargo tank or to a separate tank for this purpose.

7.3.4 The cargo piping system is not to have any connection to permanent ballast tanks.

7.3.5 Filling lines to cargo tanks are to be so arranged that the formation of static electricity is reduced. When cargo is loaded directly into tanks, the loading pipes are to be led as low as practicable in the tank. This requirement is not applicable to ships assigned additional notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60°C**

7.3.6 Terminal pipes, valves and other fittings in the cargo loading and discharge lines to which shore installation hoses are directly connected, are to be of steel or approved ductile material. They are to be strongly supported. A manually operated shut-off valve is to be fitted to each shore loading/discharging connection.

7.3.7 Where a cargo hose connection is arranged outside the cargo area, the pipe leading to such connection is to be provided with means of segregation such as a spectacle flange or removable spool piece or equivalent (see MSC Circ.474) located within cargo area. The space within 3[m] of the cargo hose connection flange is to be considered as a dangerous area with regard to electrical or incendiary equipment.

7.3.8 Isolation of piping systems which serve tanks containing incompatible cargoes are to be made by means of removable pipe lengths and blank flanges. Isolating shut off valves, single or double, or spectacle flanges are not acceptable as equivalent arrangement.

7.3.9 Cargo piping is not to be led through any tank containing a cargo which is incompatible with that contained in the tank served by such piping, unless it is encased in a pipe tunnel.

7.3.10 Where cargo suction and/or filling lines are led through cargo tanks, or through other spaces situated below the weather deck, the connection to each tank is to be provided with a valve situated inside the tank, and capable of being operated from the deck. In the case of cargo tanks which are located adjacent to below-deck pump rooms, or pipe tunnels, the deck operated valve may be located in these spaces at the bulkhead. In any case, not less than two isolating shut off valves are to be provided in the pipelines between the tanks and cargo pumps.

7.4 Remote control of valves

7.4.1 Valves on deck and in pump rooms which are provided with remote control, are, in general, to be arranged for local manual operation independent of the remote operating mechanism.

7.4.2 Where the valves and their actuators are located inside the cargo tanks, two separate suctions are to be provided in each tank, or alternative means of emptying the tank, in the event of a defective actuator, are to be provided.

7.4.3 All actuators are to be of a type which will prevent the valves from opening inadvertently in the event of the loss of pressure in the operating medium. Indication is to be provided at the remote control station showing whether the valve is open or shut.

7.4.4 Compressed air is not to be used for operating actuators inside cargo tanks. The actuator operating medium in hydraulic systems is to have a flash point of 60[°C] or above (Closed Cup Test) and is to be compatible with the intended cargoes.

7.4.5 The design of the actuator is to be such that contamination of the operating medium with cargo liquid cannot take place under normal operating conditions.

7.4.6 Where the operating medium is oil or other fluid, the supply tank is to be located as high as practicable above the level of the top of the cargo tanks, and all actuator supply lines are to enter the tanks through the highest part of the tanks. Furthermore the supply tank is to be of the closed type with an air pipe led to a safe space on the open deck and fitted with a flameproof wire gauge diaphragm at it's open end. This tank is also to be fitted with a high and low level audible and visual alarm. The requirements of this paragraph need not be complied with if the actuators and piping are located external to the cargo tanks.

7.4.7 It is recommended that for remote control valves not arranged for manual operation, emergency means be provided for

operating the valve actuators in the event of damage to the main hydraulic circuits on deck. In the case of valves located inside cargo tanks, this could be achieved by ensuring that the supply lines to actuators are led vertically inside the tank from deck, and that connections with necessary isolating valves, are provided on deck for coupling to a portable pump carried on board.

7.5 Cargo handling controls

7.5.1 Electrical measuring, monitoring control and communication circuits located in dangerous spaces are to be intrinsically safe.

7.5.2 The handling controls and instruments are to be arranged for safe and easy operation. They may be grouped at a number of control stations or at one main control station.

7.5.3 A satisfactory means of communication is to be provided between cargo handling stations, open deck, the bridge and the machinery spaces.

7.5.4 The cargo handling controls and instrumentation are, so far as possible, to be separate from the propulsion and auxiliary machinery controls and instrumentation.

7.6 Integrated cargo and ballast systems on tankers

7.6.1 The following requirements apply to integrated cargo and ballast systems meaning any integrated hydraulic and/or electric system used to drive both cargo and ballast pumps (including active control and safety systems, but excluding passive components, e.g. piping).

7.6.2 Adequate measures are to be taken to prevent cargo and ballast pumps becoming inoperative simultaneously due to a single failure in the integrated cargo and ballast system, including its control and safety systems in order to ensure that these systems are available for use even in emergency condition whilst at sea

7.6.3 The following features are to be ensured:

- The emergency stop circuits of the cargo and ballast systems are to be independent from the circuits for the control systems. A single failure in the control system circuits or the emergency stop circuits are not to render the integrated cargo and ballast system inoperative;

- Manual emergency stops of the cargo pumps are to be arranged in such a way that they do not de-activate the power pack affecting the operation of the ballast pump;

- The control systems are to be provided with backup power supply, which may be served by a duplicate power supply from the main switch board. The failure of any power supply is to initiate audible and visual alarm at each location where the control panel is fitted;

- In the event of failure of the automatic or remote control systems, a secondary means of control is to be made available for the operation of the integrated cargo and ballast system. This is to be achieved by manual overriding and/or redundant arrangements within the control systems.

Section 8

Gas Freeing and Venting of Cargo Tank

8.1 Cargo tank venting

8.1.1 Provision is to be made for the gas freeing of the cargo oil tanks when the cargo has been discharged and for the ventilation and gas freeing of all compartments adjacent to cargo oil tanks.

The venting systems of cargo tanks are to be entirely distinct from the air pipes of the other compartments of the ship. The arrangements and position of openings in the cargo tank deck from which emission of flammable vapours can occur is to be such as to minimize the possibility of flammable vapours being admitted to enclosed spaces containing a source of ignition, or collecting in the vicinity of deck machinery and equipment which may constitute an ignition hazard.

8.1.2 Every oil tanker is to be provided with at least two portable O_2 analysers and two

portable gas detectors capable of measuring flammable vapour concentrations in air (%LEL). Alternatively, at least two gas detectors each capable of measuring both oxygen and flammable vapour concentrations in air (%LEL) are to be provided.

In addition, for tankers fitted with inert gas systems, at least two portable gas detectors are to be capable of measuring concentrations of flammable vapours in inerted atmosphere (% gas by volume).

8.1.3 There is to be no connection between the gas freeing system and the ventilation system for cargo pump room.

8.1.4 Each cargo tank is to be fitted with venting arrangements which will limit the pressure or vacuum in the tanks. Venting arrangements are to be designed to provide for:

a) the flow of the small volumes of vapour, air or inert gas mixtures caused by thermal variations in a cargo tank in all cases through pressure/ vacuum valves;

b) the passage of large volumes of vapour, air or inert gas mixtures during cargo loading and ballasting, or during discharging; and

c) a secondary means of allowing full flow relief of vapour, air or inert gas mixtures to prevent over-pressure or under-pressure in the event of failure of the arrangements provided for in (b) above. In addition, for tankers constructed on or after 01 Jan 2017, the secondary means are to be capable of preventing over-pressure or under-pressure in the event of damage to, or inadvertent closing, of the means of isolation required in 8.1.7. Alternatively, pressure sensors may be fitted in each tank protected by arrangements required by (b) above, with a monitoring system in the ship's cargo control room or the position from which cargo operations are normally carried out. Such monitoring equipment is to also provide an alarm facility which is activated by detection of over-pressure or under-pressure conditions within a tank.

A P/V breaker fitted on the IG main may be accepted as secondary means of relief where the cargo is homogenous or for multiple cargoes where the vapours are compatible and do not require isolation. Where the venting arrangements are of the free flow type and the masthead isolation valve is closed for the unloading condition, the IG system will serve as the primary under-pressure protection with the P/V breaker serving as the secondary means.

In-advertent closure or mechanical failure of the isolation valves required by Cl.8.1.7 and Cl.11.6.15 need not be considered in establishing the secondary means where the cargo is homogenous or for multiple cargoes where the vapours are compatible and do not require isolation since:

- the valves are operated under the control of the ship’s responsible officer and the clear visual indication of the operational status of the valve is to be provided as required by Cl.8.1.7 and

- the possibility of mechanical failure of the valves is remote due to their simplicity.

For ships that apply pressure sensors in each tank as an alternative secondary means of relief as per (c) above, the setting of the over-pressure alarm is to be above the pressure setting of the P/V-valve and the setting of the under-pressure alarm is to be below the vacuum setting of the P/V-valve. The alarm settings are to be within the design pressures of the cargo tanks. The settings are to be fixed and not arranged for blocking or adjustment in operation*.

* An exception is permitted for ships that carry different types of cargo and use P/V-valves with different settings, one setting for each type of cargo. The settings may be adjusted to account for the different types of cargo.

8.1.5 Provision is to be made to ensure that the liquid head in any tank does not exceed the test head of the tank; suitable high level alarms, together with gauging devices and tank filling procedures, may be accepted for this purpose.

8.1.6 The pressure/vacuum system and venting system for the cargo tanks may be separate or combined and may be connected to an inert gas system. Due regard is given to the cargo segregation requirements.

8.1.7 The vent and/or pressure/vacuum valve stand pipes are to be connected to the highest part of each tank and where combined

systems are adopted, a means of isolation is to be provided between each tank and common main. Where stop valves are fitted, these are to be provided with locking arrangements and which shall be under the control of responsible ship's officer. There is to be a clear visual indication of the operational status of the valves or other acceptable means. Where tanks have been isolated, it is to be ensured that relevant isolating valves are opened before cargo loading or ballasting or discharging of those tanks is commenced. The means of isolation are to be such that tank breathing is maintained when the tank is isolated. Any isolation is to continue to permit the flow caused by thermal variations in a cargo tank in accordance with Cl.8.1.4(a).

For tankers constructed on or after 01 Jan 2017, any isolation is to continue to permit the passage of large volumes of vapour, air or inert gas mixtures during cargo loading and ballasting, or during discharging in accordance with Cl.8.1.4(b).

If cargo loading and ballasting or discharging of a cargo tank or cargo tank group is intended, which is isolated from the common venting system, that cargo tank or cargo tank group is to be fitted with a means for over-pressure or under-pressure protection required by Cl.8.1.5.

8.1.8 Pressure/vacuum valves are to be set at a positive pressure of not more than 0.2bar above atmospheric and a negative pressure of not more than 0.07bar below atmospheric unless the tank scantlings are specially considered.

8.1.9 In no case are shut-off valves to be fitted either above or in the pipe leading to a pressure/vacuum valve. However, bypass valves may be fitted or provision may be made to enable the tank pressure/vacuum valves to be held in an open position. The arrangements are to be such that clear indication is given when the bypass valve is open or the pressure/vacuum valve is secured in the open position. Means are to be provided to check the functioning of the pressure/vacuum valves.

8.1.10 The area of the venting system used during cargo loading is to be based on the maximum design loading rate and a gas evolution factor of 1.25.

8.1.11 Suitable drainage arrangements are to be provided in the vapour lines. The venting arrangements are to be self-draining to the cargo tanks under all normal condition of trim and list of the ship. Where it may not be possible to provide self draining lines, permanent arrangements are to be provided to drain the vent lines to a cargo tank.

8.1.12 The venting system is to be provided with devices to prevent the passage of flame into the cargo tanks. The design, testing and locating of these devices are to comply with the requirements approved by the National Statutory Authority*. Ullage openings are not to be used for pressure equalization. They are to be provided with self-closing and tightly sealing covers. Flame arresters and screens are not permitted in these openings.

Ullage openings do not include cargo tank openings that are fitted with standpipe arrangements with its own manually operated shutoff valves.

Examples include the common 25 to 50[mm] diameter standpipe arrangements that are used for sampling, monitoring or measuring of ullage/temperature/interface, oxygen, liquid and hand dipping in the cargo tank.

** Refer to MSC/Circ.677 as amended by MSC / Circ.1009 on Revised standards for the design, testing and locating of devices to prevent the passage of flame into cargo tanks in tankers and to MSC/Circ.450/Rev.1 on Revised factors to be taken into consideration when designing cargo tank venting and gas-freeing arrangements.*

8.1.13 Vent outlets and pressure/vacuum valve outlets, if used during loading, are to be arranged to discharge the vapour in an upward vertical direction. All outlets are to be arranged to prevent the entrance of water into the cargo tanks.

8.1.14 Openings for pressure release required by Cl.8.1.4(a) are to:

a) have as great a height as is practicable above the cargo tank deck to obtain maximum dispersal of flammable vapours but in no case less than 2[m] above the cargo tank deck, and

b) be arranged at the furthest distance practicable but not less than 5[m] from the nearest air intakes and openings to enclosed spaces containing a source of ignition and from deck machinery and equipment which may constitute an ignition hazard.

IR-b) Electrical equipment fitted in compliance with IEC Publication 60092 – "Electrical installations in ships – Part 502 : Tankers – Special features" is not considered a source of ignition or ignition hazard (Refer Pt 4, Ch. 8, Clause 12.9.10.1 for area classification for electrical equipment installation).

For tankers constructed on or after 01 Jan 2017, the openings are to be arranged in accordance with Cl.8.1.15.

The height requirements given in Cl.8.1.14 and Cl.8.1.15 and the requirements for devices to prevent the passage of flame are not applicable to the P/V breaker provided the settings are above those of the venting arrangements given in Cl.8.1.4 (a) and (b).

8.1.15 Vent outlets for cargo loading, discharging and ballasting required by Cl.8.1.4(b) are to:

a) permit the free flow of vapour mixtures or alternatively, permit the throttling of the discharge of the vapour mixtures to achieve a velocity of not less than 30[m/s].

b) be so arranged that the vapour mixture is discharged vertically upwards.

c) where the method is by free flow of vapour mixtures, be such that the outlet is not less than 6[m] above the cargo tank deck or fore and aft gangway if situated within 4[m] of gangway and located not less than 10[m] measured horizontally from the nearest air intakes and openings to enclosed spaces containing a source of ignition and from deck machinery, which may include anchor windlass and chain locker openings and equipment which may constitute an ignition hazard.

IR-c) Electrical equipment fitted in compliance with IEC 60092-502:1999 is not considered a source of ignition or ignition hazard (Refer Pt 4, Ch. 8, Cl.11.9.10 for area classification for electrical equipment installation).

d) where the method is by high velocity discharge, be located at a height not less than 2[m] above the cargo tank deck and not less than 10[m] measured horizontally from the nearest air intakes and openings to enclosed spaces containing a source of ignition and from deck machinery, which may include anchor windlass and chain locker openings and equipment which may constitute an ignition hazard. These outlets are to be provided with high velocity devices of an approved type.

IR d) Electrical equipment fitted in compliance with IEC 60092-502:1999 is not considered a source of ignition or ignition hazard (refer Pt 4, Ch. 8, Cl.11.9.10 for area classification for electrical equipment installation).

e) the Master is to be provided with information regarding the maximum permissible loading rate for each cargo tank and in case of combined venting systems, for each group of cargo tanks.

8.1.16 The arrangements for the venting of all vapours displaced from the cargo tanks loading and ballasting are to comply with this section and are to consist of either one or more mast risers, or a number of high velocity vents. The inert gas supply mains may be used for such venting.

8.2 Cargo tank purging and/or gas freeing

8.2.1 Arrangements for purging/and or gas freeing are to be such as to minimise the hazards due to the dispersal of flammable vapours in the atmosphere and to flammable mixtures in cargo tank.

8.2.2 When the ship is provided with an inert gas system the cargo tanks are first to be purged until the concentration of hydrocarbon vapours has been reduced to less than 2% by volume. Thereafter gas freeing may take place at the cargo tank deck level.

8.2.3 The arrangements for inerting, purging or gas-freeing of empty tanks as required in Cl.8.2.2 are to be to the satisfaction of IRS and are to be such that the accumulation of hydrocarbon vapours in pockets formed by the internal structural members in a tank is minimized and that:

.1 on individual cargo tanks, the gas outlet pipe, if fitted, shall be positioned as far as practicable from the inert gas/air inlet and in accordance with Sec.8.1 and Pt.6, Ch.3, Sec.5.6. The inlet of such outlet pipes may be located either at deck level or at not more than 1[m] above the bottom of the tank;

.2 the cross-sectional area of such gas outlet pipe referred to in Cl.8.2.3.1 is to be such that an exit velocity of at least 20[m/s] can be maintained when any three tanks are being simultaneously supplied with inert gas. Their outlets shall extend not less than 2[m] above deck level; and

.3 each gas outlet referred to in paragraph Cl.8.2.3.2 is to be fitted with suitable blanking arrangements.

.4 if a connection is fitted between the inert gas supply mains and the cargo piping system, arrangements are to be made to ensure an effective isolation having regard to the large pressure difference which may exist between the systems. This is to consist of two shut-off valves with an arrangement to vent the space between the valves in a safe manner or an arrangement consisting of a spool-piece with associated blanks. The valve separating the inert gas supply main from the cargo main and which is on the cargo main side is to be a non-return valve with a positive means of closure.

IR.4 as a guide, the effective isolation required by .4 above may be achieved by the two arrangements shown in Figure 8.2.3.4.

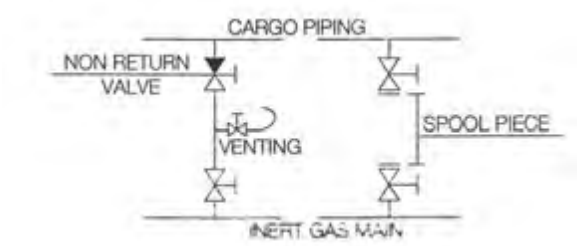

Fig.8.2.3.4

8.2.4 One or more pressure-vacuum breaking devices are to be provided to prevent the cargo tanks from being subject to:

a) a positive pressure in excess of the test pressure of the cargo tank if the cargo were to be loaded at the maximum rated capacity and all other outlets were left shut; and

b) a negative pressure in excess of 700[mm] water gauge if cargo were to be discharged at the maximum rated capacity of the cargo pumps and the inert gas blowers were to fail.

Such devices are to be installed on the inert gas main unless they are installed in the venting system or on individual cargo tanks.

8.2.5 When the ship is not provided with an inert gas system, the operation is to be such that the flammable vapour is initially discharged either:

a) through the vent outlets as specified in Cl.8.1.15; or

b) through outlets at least 2[m] above the cargo tank deck level with a vertical efflux velocity of at least 30[m/s] maintained during gas freeing operation; or

c) through outlets at least 2[m] above the cargo tank deck level with a vertical efflux velocity of at least 20[m/s] and which are protected by suitable devices to prevent the passage of flame.

8.2.6 When the flammable vapour concentration at the outlet has been reduced to 30 per cent of the lower flammable limit, gas freeing may thereafter be continued at the cargo tank deck level.

8.2.7 At least two oxygen sensors are to be positioned at appropriate locations in the space or spaces containing the inert gas system. Also Refer Pt.6, Ch.8, Sec.15, Cl.15.2.2.4.5.4, regarding measurements in inert atmosphere. Suitable means are to be provided for the calibration of such instruments.

8.3 Inerting, ventilation and gas measurement detection of double hull and double bottom spaces

8.3.1 Double hull and double bottom spaces are to be fitted with suitable connections for the supply of air.

8.3.2 On tankers required to be fitted with inert gas systems:

a) double hull spaces are to be fitted with suitable connections for the supply of inert gas;
b) where hull spaces are connected to a permanently fitted inert gas distribution system, means are to be provided to prevent hydrocarbon gases from the cargo tanks entering the double hull spaces through the system;

c) where such spaces are not permanently connected to an inert gas distribution system, appropriate means are to be provided to allow connection to the inert gas main.

8.3.3 At least two suitable portable instruments for measuring oxygen and flammable vapour concentrations are to be provided. In selecting these instruments, due attention shall be given to their use in combination with the fixed gas-sampling-line systems referred to in paragraph Cl.8.3.4.

8.3.4 Where the atmosphere in double hull spaces cannot be reliably measured using flexible gas sampling hoses, such spaces are to be fitted with permanent gas sampling lines. The configuration of such line systems are to be adapted to the design of such spaces.

8.3.5 The materials of construction and the dimensions of gas sampling lines are to be such as to prevent restriction. Where plastic materials are used, they should be electrically conductive.

8.3.6 Arrangements for fixed hydrocarbon gas detection systems in double-hull and double-bottom spaces of oil tankers

8.3.6.1 In addition to the requirements in Cl.8.3.3 to Cl.8.3.5 above, oil tankers of 20,000[t] deadweight and above, constructed on or after 1 January 2012, are to be provided with a fixed hydrocarbon gas detection system complying with Pt.6, Ch.8 for measuring hydrocarbon gas concentrations in all ballast tanks and void spaces of double-hull and double-bottom spaces adjacent to the cargo tanks, including the forepeak tank and any other tanks and spaces under the bulkhead deck adjacent to cargo tanks. The terms used in this requirement are to be interpreted as follows:

a) *"cargo tanks"* in the phrase *"spaces adjacent to the cargo tanks"* includes slop tanks except those arranged for the storage of oily water only.
b) *"spaces"* in the phrase *"spaces under the bulkhead deck adjacent to cargo tanks"* includes dry compartments such as ballast pump-rooms and bow thruster rooms and any tanks such as freshwater tanks, but excludes fuel oil tanks.
c) *"adjacent"* in the phrase *"adjacent to the cargo tanks"* includes ballast tanks, void spaces, other tanks or compartments located below the bulkhead deck located adjacent to cargo tanks and includes any spaces or tanks located below the bulkhead deck which form a cruciform (corner to corner) contact with the cargo tanks.

IR8.3.6.1 Double-hull spaces required to be fitted with suitable connections for the supply of inert gas as per Cl.8.3.2 (a)/ Pt. 6, Ch. 2, Cl.1.5.5.1.4.1 are all ballast tanks and void spaces of double-hull and double-bottom spaces adjacent to the cargo tanks, including the forepeak tank and any other tanks and spaces under the bulkhead deck adjacent to cargo tanks, except cargo pump-rooms and ballast pump rooms.

8.3.6.2 Oil tankers provided with constant operative inerting systems for such spaces need not be equipped with fixed hydrocarbon gas detection equipment.

8.3.6.3 Notwithstanding the above, cargo pump-rooms complying with the provisions of Cl.6.4.2 to Cl.6.4.5 and IR6.4.5 need not comply with the requirements of this paragraph (i.e. Cl.8.3.6).

8.4 The above requirements (i.e. 8.1 to 8.3) are not applicable to ships assigned with the notation **FOR CARRIAGE OF LIQUID HAVING F.P. EXCEEDING 60[°C].** However, the following provisions apply:

a) Spaces located within the cargo area are to be efficiently ventilated. Enclosed spaces within the cargo area are to be provided with efficient means of ventilation. Portable means are permitted for that purpose. Ventilation fans are to be of non-sparking construction.

b) Ventilation inlets and outlets, especially for machinery spaces, are to be situated as far aft as practicable. Due consideration in this regard is to be given when the ship is equipped to load or discharge at the stern. Sources of ignition such as electrical equipment are to be so arranged as to avoid an explosion hazard.

c) Ventilation of the cargo pump room is to comply with Section 7.

d) Tank venting systems are to open to the atmosphere at a height of at least 760[mm] above the weather deck.

e) Tanks may be fitted with venting systems of the open type provided with a flame screen.

Section 9

Cargo Tank Level Measurement

9.1 General

9.1.1 Following types of tank level measurements may be accepted:

a) Open type: A method which makes use of an opening in the tank and directly exposes the operator to the cargo or it's vapours. Examples of this type are ullage openings and gauge hatches, etc. (Also Refer Cl.7.1.4).

b) Restricted type: A device which penetrates the tank and which, when in use, permits a limited quantity of cargo vapour or liquid to be expelled to the atmosphere. Examples of this type are sounding pipe, rotary tube, etc. (Also Refer Cl.7.1.4).

c) Closed type: A device which penetrates the tank, but which is a part of the closed system which keeps the cargo completely sealed off from the atmosphere. The examples of this kind are sight glasses, pressure cells, float-type, radar type systems etc.
(Also refer Pt.6, Ch.2, 1.5.5.3.2).

9.1.2 Each cargo tank is to be fitted with at least one level gauging device. Where only one liquid level measuring device is fitted, it is to be arranged so that any necessary maintenance can be carried out while the cargo tank is in service.

9.1.3 If a closed measuring device is not mounted directly on the tank, it is to be provided with shut off valves situated as close as possible to the tank.

Section 10

Cargo Heating Arrangements

10.1 General

10.1.1 In addition to the relevant requirements of Part 4, Ch.3, following are also to be complied with for vessels fitted with heating arrangements for cargoes.

10.1.2 Spectacle flanges or spool pieces are to be provided in the heating medium supply and return pipes to the cargo heating system, in a suitable position in the cargo area, so that the lines can be blanked off in circumstances where the cargo does not require to be heated or where the heating coils have been removed from the tank. Alternatively, blanking arrangements may be provided for each tank heating circuit.

10.1.3 The heating medium is to be compatible with the cargoes to be heated. Where a combustible liquid is used as the heating medium it is to have a flash point of 60[°C] or above (Closed Cup Test). In general the temperature of the heating medium is not to exceed 220[°C].

10.1.4 The heating medium supply and return lines are not to penetrate the cargo tank plating, other than at the top of the tank, and the main supply lines are to be run above the weather deck.

10.1.5 Isolating shut-off valves or cocks are to be provided at the inlet and outlet connections of each tank, and means are to be provided for regulating the flow.

10.1.6 Where steam or water is employed in the heating circuits, the returns are to be led to

an observation tank which is to be in a well ventilated and well lighted part of the machinery space remote from the boilers.

10.1.7 Where a thermal oil is employed in the heating circuits, the arrangements will be specially considered.

10.1.8 In any heating system, a higher pressure is to be maintained within the heating circuit than the maximum pressure head which can be exerted by the contents of the cargo tank on the circuit. Alternatively, when the heating circuit is not in use, it may be drained and blanked.

10.1.9 Means are to be provided for measuring the cargo temperature.

Section 11

Inert Gas Systems

11.1 General

11.1.1 The applicability of inert gas systems for tankers is given in Pt.6, Ch.2, Cl.1.5.5.

11.1.2 Inert gas systems on tankers are to be provided in accordance with the requirements in Pt.6, Ch.8, Sec.15.

Section 12

Ships for Carriage of Oil or Dry Cargo in Bulk

12.1 General

12.1.1 Ships intended to carry either oil in bulk with flash point not exceeding 60[°C] (closed cup test) or alternatively dry bulk cargo are to comply with the additional requirements of this section.

12.1.2 Relevant requirements for bulk carriers and ore carriers given in Pt.5, Ch.1 are also to be complied with, as applicable.

12.1.3 Dry cargoes and oil (with flash point below 60[°C]) are not be carried simultaneously, except for oil retained in protected slop tanks when the ship is carrying dry cargoes. Prior to employing the ship for the carriage of dry cargoes, the entire cargo area is to be cleaned, gas freed, inerted and isolated in accordance with the requirements of the National or Port Authorities.

12.1.4 Attention is drawn to the requirements of Pt.6, Ch.2, Cl.1.5.4.2 regarding ventilation and gas freeing of enclosed spaces in combination carriers which are to be complied with. In way of cargo holds for oil, enclosed spaces in which explosive gases may accumulate are to be avoided.

12.2 Structural arrangement

12.2.1 Cargo tanks are to facilitate efficient cleaning. Tank boundaries are to have plane surface, corrugated surface or vertical stiffeners as far as practicable. Horizontal primary members are to be avoided inside tanks; where primary structural members are unavoidable, attention is to be paid to the cleaning facilities.

12.2.2 Openings which may be used for cargo operations, for example in the bottom of topside tanks, are not permitted in bulkheads and decks separating oil cargo spaces from other spaces not designed and equipped for the carriage of oil cargoes unless they are equipped with alternative approved means to ensure equivalent integrity.

12.2.3 Where transverse bulkheads in wing tanks have different structural arrangement than in way of the holds, arrangements are to be made to ensure continuity of transverse strength across the longitudinal bulkheads.

12.2.4 Suitable arrangements are to be provided for gas freeing the double bottom, double hull, hopper, side and topside tanks. Similar arrangements are to be provided for cargo oil ducts which are used as pipe tunnels when the ship is carrying dry cargo.

12.3 Hatch covers

12.3.1 The hatch covers and the sealing system are to avoid leakages caused by possible elastic deformation of the hatchways.

12.3.2 The scantlings and arrangement of hatch covers are to be obtained as per Pt.3, Ch.12. Where cargo holds are intended to be partially filled, the hatch covers may require to be additionally strengthened.

12.4 Slop tanks

12.4.1 These requirements are applicable to combination carriers where oil residues are retained in the slop tanks and the ship is otherwise gas free.

12.4.2 The slop tanks are to be surrounded by cofferdams except where the boundaries are the hull, main cargo deck, cargo pump-room bulkhead or oil fuel bunker tank. These cofferdams are not to be open to a double bottom, pipe tunnel, pump room or other enclosed spaces nor they are to be used for cargo or ballast and are not to be connected to piping systems serving oil cargo or ballast. Means are to be provided for filling the cofferdams with water and draining them. Where the boundary of a slop tank is the cargo pump room bulkhead, the pump room should not be open to the double bottom, pipe tunnel or other enclosed spaces except when they are closed gastight by bolted plates.

12.4.3 Means are to be provided for isolating the piping connecting the pump room with the slop tanks. The means of isolation is to consist of a valve followed by a spectacle flange or a spool piece with appropriate blank flanges. This arrangement is to be located adjacent to the slop tanks, but where this is unreasonable or impracticable it may be located within the pump room directly after the piping penetrates the bulkhead. A separate pumping and piping arrangement is to be provided for discharging the contents of the slop tanks directly over the open deck when the ship is in the dry cargo mode. When the transfer system is used for slop transfer in the dry cargo mode, it is to have no connection to other systems. Separation from other systems by means of removal of spool pieces may be accepted.

12.4.4 Hatches and tank cleaning openings to slop tanks are to be provided only on the open deck and are to be fitted with closing arrangements. Except when they are closed watertight by bolted plates, these closing arrangements are to be provided with locking arrangements, under the control of a responsible ship's officer.

12.4.5 Slop tanks are to be provided with an approved independent venting system, generally in accordance with Sec.8.2.

12.4.6 At least two portable instruments are to be available on board for gas detection.

12.4.7 Adequate ventilation is to be provided for spaces surrounding slop tanks.

12.4.8 Warning notices are to be erected at suitable points detailing the precautions to be observed prior to the ship loading or unloading, or when the ship is carrying dry cargo with liquid in the slop tanks.

12.4.9 Attention is drawn to the requirements of certain Flag state/ Port state Authorities whereby an inert gas system is to be provided for blanketting the slop tank contents.

12.5 Cargo piping

12.5.1 Where cargo wing tanks are provided, cargo oil lines below deck are to be installed inside these tanks. However, IRS may permit cargo oil lines to be placed in special ducts provided these are capable of being adequately cleaned and ventilated to the satisfaction of IRS. Where cargo wing tanks are not provided, cargo oil lines below deck are to be placed in special ducts.

Section 13

Requirements Concerning use of Crude Oil or Slops as Fuel for Tanker Boilers

13.1 In tankers crude oil or slops may be used as fuel for main or auxiliary boilers according to the following requirements. For this purpose all arrangement drawings of a crude oil installation with pipeline layout and safety equipment are to be submitted for approval in each case.

13.2 Crude oil or slops may be taken directly from cargo tanks or from slop tanks or from other suitable tanks. These tanks are to be fitted in the cargo tank area and are to be separated from non-gas-dangerous areas by means of cofferdams with gas-tight bulkheads.

13.3 The construction and workmanship of the boilers and burners are to be proved to be satisfactory in operation with crude oil. The whole surface of the boilers shall be gas-tight separated from the engine room. The boilers themselves are to be tested for gas-tightness before being used. The whole system of pumps, strainers, separators and heaters, if any, shall be fitted in the cargo pump room or in another room, to be considered as dangerous and separated from engine and boiler room by gas-tight bulkheads. When crude oil is heated by steam or hot water the outlet of the heating coils should be led to a separate observation tank installed together with above mentioned components. This closed tank is to be fitted with a venting pipe led to the atmosphere in a safe position according to the rules for tankers and with the outlet fitted with a suitable flame proof wire gauze of corrosion resistant material which is to be easily removable for cleaning.

13.4 Electric, internal combustion and steam (when the steam temperature is higher than 220°C) prime movers of pumps, of separators (if any), etc., shall be fitted in the engine room or in another non-dangerous room. Where drive shafts pass through pump room bulkhead or deck plating, gas-tight glands are to be fitted. The glands are to be efficiently lubricated from outside the pump room.

13.5 Pumps shall be fitted with a pressure relief bypass from delivery to suction side and it shall be possible to stop them by a remote control placed in a position near the boiler fronts or machinery control room and from outside the engine room.

13.6 When it is necessary to preheat crude oil or slops, their temperature is to be automatically controlled and a high temperature alarm is to be fitted.

13.7 The piping for crude oil or slops and the draining pipes for the tray defined in 13.9 are to have a thickness as follows:

External diameter of pipes, d_e thickness, t

OD[mm]	Thickness[mm]
82.5	6.3
88.9 - 108	7.1
114.3 > 139.7	8.0
152.4	8.8

Their connections (to be reduced to a minimum) are to be of the heavy flange type. Within the engine room and boiler room these pipes are to be fitted within a metal duct, which is to be gas-tight and tightly connected to the fore bulkhead separating the pump room and to the tray. This duct (and the enclosed piping) is to be fitted at a distance from the ship's side of at least 20% of the vessel's beam amidships and be at an inclination rising towards the boiler so that the oil naturally returns towards the pump room in the case of leakage or failure in delivery pressure. It is to be fitted with inspection openings with gas-tight doors in way of connections of pipes within it, with an automatic closing drain-trap placed on the pump room side, set in such a way as to discharge leakage of crude oil into the pump room. In order to detect leakages, level position indicators with relevant alarms are to be fitted on the drainage tank defined in 13.9. Also a vent pipe is to be fitted at the highest part of the duct and is to be led to the open in a safe position. The outlet is to be fitted with a suitable flame proof wire gauze of corrosion resistant material which is to be easily removable for cleaning.

The duct is to be permanently connected to an approved inert gas system or steam supply in order to make possible:

- injection of inert gas or steam in the duct in case of fire or leakage

- purging of the duct before carrying out work on the piping in case of leakage.

13.8 In way of the bulkhead to which the duct defined in 13.7 is connected, delivery and return oil pipes are to be fitted on the pump room side, with shut-off valves remotely controlled from a position near the boiler fronts or from the machinery control room. The remote control valves should be interlocked with the hood exhaust fans (defined in 13.10) to ensure that whenever crude oil is circulating the fans are running.

13.9 Boilers shall be fitted with a tray or gutter way of acceptable height and be placed in such a way as to collect any possible oil leakage from burners, valves and connections.

Such a tray or gutter way shall be fitted with a suitable flame proof wire gauze, made of corrosion resistant material and easily dismountable for cleaning. Delivery and return oil pipes shall pass through the tray or gutter way by means of a tight penetration and shall then be connected to the oil supply manifolds.

A quick closing master valve is to be fitted on the oil supply to each boiler manifold. The tray or gutter way shall be fitted with a draining pipe discharging into a collecting tank in pump room. This tank is to be fitted with a venting pipe led to the open in a safe position and with the outlet fitted with wire gauze made of corrosion resistant material and easily dismountable for cleaning. The draining pipe is to be fitted with arrangements to prevent the return of gas to the boiler or engine room.

13.10 Boilers shall be fitted with a suitable hood placed in such a way as to enclose as much as possible of the burners, valves and oil pipes, without preventing, on the other side, air inlet to burner register. The hood, if necessary, is to be fitted with suitable doors placed in such a way as to enable inspection of and access to oil pipes and valves placed behind it. It is to be fitted with a duct leading to the open in a safe position, the outlet of which is to be fitted with a suitable flame wire gauze, easily dismountable for cleaning. At least two mechanically driven exhaust fans having spark proof impellers are to be fitted so that the pressure inside the hood is less than that in the boiler room. The exhaust fans are to be connected with automatic change over in case of stoppage or failure of the one in operation. The exhaust fan prime movers shall be placed outside the duct and a gas-tight bulkhead penetration shall be provided for shaft.

Electrical equipment installed in gas dangerous areas or in areas which may become dangerous (i.e. in the hood or duct in which crude-oil piping is placed) is to be of certified safe type.

13.11 When using fuel oil for delivery to and return from boilers fuel oil burning units in accordance with the Rules shall be fitted in the boiler room. Fuel oil delivery to, and returns from, burners shall be effected by means of suitable mechanical interlocking devices so that running on fuel oil automatically excludes running on crude oil or vice versa.

13.12 The boiler compartments are to be fitted with a mechanical ventilation plant and shall be designed in such a way as to avoid the formation of gas pockets.

Ventilation is to be particularly efficient in way of electrical plants and machinery and other plants which may generate sparks. These plants shall be separated from those for service of other compartments.

13.13 A gas detector plant shall be fitted with intakes in the duct defined in 13.7, in the hood duct (downstream of the exhaust fans in way of the boilers) and in all zones where ventilation may be reduced. An optical warning device is to be installed near the boiler fronts and in the machinery control room. An acoustical alarm, audible in the machinery space and control room, is to be provided.

13.14 Means are to be provided for the boiler to be automatically purged before firing.

13.15 Independent of the fire extinguishing plant as required by the Rules, an additional fire extinguishing plant is to be fitted in the engine and boiler rooms in such a way that it is possible for an approved fire extinguishing medium to be directed on to the boiler fronts and on to the tray defined in 13.9. The emission of extinguishing medium should automatically stop the exhaust fan of the boiler hood.

13.16 A warning notice must be fitted in an easily visible position near the boiler front. This notice must specify that when an explosive mixture is signalled by the gas detector plant defined in 13.13 the watch keepers are to immediately shut off the remote controlled valves on the crude oil delivery and return pipes in the pump room, stop the relative pumps, inject inert gas into the duct defined in 13.7 and turn the boilers to normal running on fuel oil.

13.17 One pilot burner in addition to the normal burning control is required.

Section 14

Requirements for Asphalt Carriers

14.1 General

14.1.1 Application

14.1.1.1 The additional requirements in this Section apply to ships carrying melted asphalt (also commonly referred to as “bitumen”) in bulk, hereinafter referred to as “asphalt carriers”.

14.1.1.2 Unless specified otherwise, asphalt carriers are to comply with the requirements for tankers carrying petroleum products having a flash point exceeding 60[℃] (closed cup test). Where the temperature of asphalt being carried is not more than 15[℃] below the flash point, the requirements for tankers carrying petroleum products having a flash point not exceeding 60[℃] (closed cup test) are to be complied with.

14.1.1.3 The requirements of MARPOL Annex I, Reg 29, 31 and 32 do not apply to Asphalt Carriers.

14.1.2 Notations

14.1.2.1 Vessels that comply with the requirements of this Section will be assigned following additional class notations, as applicable:

ASPHALT CARRIER (INTEGRAL TANK, MAX. CARGO TEMP ≤ XXX℃), ESP

ASPHALT CARRIER (INDEPENDENT TANK, MAX. CARGO TEMP ≤ XXX℃)

14.1.3 Definitions

14.1.3.1 **Independent tank** means a cargo containment envelope which is not contiguous with, or part of, the hull structure. An independent tank is built and installed so as to eliminate whenever possible (or in any event to minimize) its stressing as a result of stressing or motion of the adjacent hull structure. An independent tank is not essential to the structural completeness of the ship's hull.

14.1.3.2 **Integral tank** means a cargo containment envelope which forms part of the ship's hull and which may be stressed in the same manner and by the same loads which stress the contiguous hull structure and which is normally essential to the structural completeness of the ship's hull.

14.1.3.3 **Hold space** is the space enclosed by the ship’s structure in which an independent cargo tank is fitted.

14.1.4 Documentation

14.1.4.1 Following plans and documents are to be submitted for approval:

.1 Structural plan of cargo tank;

.2 Arrangement plan of cargo tank;

.3 Arrangement plan of heat-insulating material on cargo tank;

.4 Arrangement of means of access for inspections;

.5 Details of heating system (for example: drawing of thermal oil heating system, arrangement of thermal oil boiler room, diagram of control system of remotely controlled valve of thermal oil system, diagram of fuel oil system of thermal oil boiler etc.)

.6 Drawings and arrangement diagrams of monitoring and alarm systems related to liquid cargo;

.7 Cargo information (type and property of cargoes allowed to be carried, e.g. specific gravity, flash point, temperature of cargo being carried etc.);

.8 Calculations of heat-insulating performance of cargo tank.

14.1.4.2 For asphalt carriers fitted with integral tanks, the following plans and documents are to be additionally submitted:

.1 Calculations for thermal stress analysis of hull structural members (in full and half load conditions), in respect to the temperature of cargo being carried over 80℃.

14.1.4.3. For asphalt carriers fitted with independent tanks, the following plans and documents are to be additionally submitted for approval:

.1 Arrangement plan of longitudinal and transverse limiting devices of cargo tank;

.2 Structural plan of supports and chocks arrangements for cargo tanks;

.3 Drawing and arrangement diagram of high level alarm system for cargo tank bilge well.

.4 Analysis of thermal stress of cargo tank supporting structure and analysis of thermal stress of cargo tank structure for temperature of cargo being carried over 80℃.

.5 Following additional information is to be submitted for support and chocks of independent tanks:

.1 general arrangement on the ship (positioning, total surface)

.2 detailed fabrication and integration in the ship hull structure

.3 mechanical characteristics of the support pads, including compressive and shear strengths, frictional coefficients and elasticity.

.4 Test report of interface temperature of hull support structures in way of pads and pads with chocking compound

14.2 Arrangement

14.2.1 General

14.2.1.1 The ship arrangements are to comply with the requirements in Annex I of MARPOL 1973, as amended.

14.2.2 Cargo Tank Arrangement

14.2.2.1 Asphalt Carriers with Integral Tanks

.1 For asphalt carriers fitted with integral tanks carrying cargo at a temperature over 80℃, tank boundaries are not to be in direct contact with seawater.

14.2.2.2 Asphalt Carriers with Independent tanks

.1, The independent cargo tanks are to be so located that the distance from the moulded line of the bottom shell and side shell is to be not less than the limits as required by Regulation 19 of Annex I of MARPOL 1973, as amended.

14.2.3. Requirements for Access in Asphalt Carriers with Independent Tanks

14.2.3.1 Asphalt carriers with independent tanks need not comply with SOLAS Regulation II-1/3-6, which is applicable to oil tankers having integral tanks for carriage of oil in bulk as contained in the definition of oil in Annex I of MARPOL 1973, as amended. Asphalt carriers with independent tanks, are to comply with the requirements of Part 5, Ch. 4, Sec 3.5.

14.2.3.2 Where fitted, cofferdams are to have sufficient size for easy access to all their parts. The width of the cofferdams is not to be less than 760[mm].

14.2.3.3 Pipe tunnels are to have enough space to permit inspection of pipes. The pipes in pipe tunnels are to as high as possible from the ship’s bottom.

14.2.4 Supports and Chocks of Independent tanks

14.2.4.1 Supports of independent tanks are to be designed to allow, when expected, free

thermal expansion in the three directions (longitudinal, transverse and vertical) to avoid thermal stress in the tank structures and transmission of loads from the tanks to the supports. Generally, the supports are also to be designed so as to limit the transmission of loads relative to hull deflection (global and local) from the ship structure to the tanks. Adequate support for the tanks may be achieved by means of:

- vertical supports
- anti-rolling chocks
- anti-pitching chocks
- anti-flotation chocks
- anti-collision chocks

14.2.4.2 Contact surfaces of supports (vertical supports, anti-roll and anti-pitch chocks, etc.) are usually made of synthetic material such as resin, plywood and adhesive. Such materials are to be type approved.

14.2.5 Cargo Pump Room

14.2.5.1 A separate cargo pump room may not be required in an asphalt carrier.

14.2.5.2 For asphalt carriers fitted with independent tanks, cargo pumps may be located in hold space.

14.2.5.3 For asphalt carriers fitted with integral tanks, if the cargo pump room is adjacent to a cargo tank, means for heat insulation is to be provided. The space where cargo pumps are located is to have good ventilation, normally with 30 to 45 air changes per hour.

14.3 Structure

14.3.1 General Requirements

14.3.1.1 In general, the hull structure is to be in accordance with requirements in Sec 3, 4 & 5.

14.3.1.2 For asphalt carriers whose cargoes are carried at an elevated temperature (i.e. above 80[°C]), the thermal effect is to be considered in the strength assessment of the tank and surrounding structures. A thermal calculation, or evidence showing temperature distribution in the structure considering insulation covering cargo tanks, is to be submitted by the designer to determine the temperature distribution in hull structures.

14.3.1.3 The temperature distribution calculation is to consider at least the following two conditions for ambient temperatures of normal service:

i) 5[°C] for air and 0[°C] for seawater;

ii) 45[°C] for air and 32[°C] for seawater;

For service zones outside of the above conditions, the specified design temperatures are also to be submitted and considered in the design.

14.3.1.4 The hull structure requirements at an elevated temperature greater than 80°C are to be obtained by modifying the relevant basic requirements in previous sections and Part 3 of the Rules. Guidance regarding modified properties (i.e. yield point and modulus of elasticity) due to high temperature is provided in the "*IRS Guidelines on Structural Assessment of Ships carrying Asphalt in Independent Cargo Tanks"*.

14.3.2 Hull Structure of Asphalt Carriers with independent tanks

14.3.2.1 General

.1 Hull Structure, Independent Tanks, Supports and Chocks are to be comply with the requirement of "*IRS Guidelines on Structural Assessment of Ships carrying Asphalt in Independent Cargo Tanks"*

14.3.2.2 Thermal Stresses

.1 For hull structures where the air temperature in the space between the hold and independent cargo tank is not more than 50[°C] higher than the ambient temperature, the thermal stresses in hull structures may be neglected.

.2 For independent cargo tanks, where the design of the tank supports allows free thermal elongation or contraction and temperature distribution in the tank structures is relatively uniform in all designed loading conditions, the thermal stress may also be neglected. In addition, the tanks are to be pre-heated when loading hot cargo in order to minimize the variations in temperature profile. Otherwise, the thermal stresses in hull structures and cargo tanks

are to be considered and analyzed. Additionally, when the piping lines are heated or cooled, they may put additional loads into the ship structures; therefore, sufficient expansion bends are to be provided to reduce this thermal loading.

14.3.2.3 Hull structure in way of supports of independent tanks

.1 Hull structures in way of tank supports (e.g., vertical supports, anti-roll and anti-pitch chocks) are to be of sufficient strength when subjected to reaction loads from independent tanks. These loads are to be transmitted to the hull main supporting members as directly as possible to minimize stress concentrations. Additional web plates and brackets are to be provided to transmit these loads to the adjacent main supporting members if the loads are not in the plane of them. The number of openings in structures in way of tank supports is to be kept to a minimum, compatible with the design and proper working of the vessel. Local reinforcements in way of those openings are to be provided as necessary.

The strength criteria of hull structures in way of supports as specified in 14.3 are to be complied with.

14.3.3 Structure of Independent Tanks

14.3.3.1 General

.1 The structural design of the independent tank should take into account the temperature of carriage and relative density.

14.3.3.2 Hull Girder Stress

.1 In general, the cargo tank supporting system is to be designed so that no hull girder loads are transmitted to the independent cargo tanks. However, if the hull girder loads are not negligible for the cargo tank (due to the tank supports arrangement), the stresses in cargo tanks due to the hull girder loads are to be considered and determined by finite elements analysis of hull and cargo tanks with tank supporting system.

14.3.3.3 Sloshing

.1 In vessels fitted with long tanks, the scantlings of oil tight transverse bulkheads in smooth-sided tanks are to be specially considered when the spacing between tight bulkheads, non-tight bulkheads, or partial bulkheads exceeds 12[m] in the case of corrugated-type construction, or 15[m] in the case of flat-plate type of construction.

.2 Special consideration is to be given to the scantlings of longitudinal oil-tight bulkheads forming the boundaries of wide tanks.

.3 Where the length of the smooth-sided tanks exceeds 0.1L or the breadth exceeds 0.6B, non-tight bulkheads are to be fitted, unless calculations are submitted to prove that no resonance due to sloshing will occur in service, Alternatively, reinforcements to the bulkheads and decks, without non-tight bulkheads, may be determined by an acceptable method of engineering analysis.

14.3.3.4 Non-tight bulkhead

.1 Non-tight bulkheads are to be fitted in line with transverse webs, bulkheads, or other structures with equivalent rigidity. They are to be suitably stiffened.

.2 Openings in the non-tight bulkhead are to have generous radii and their aggregate area is not to exceed 33%, nor be less than 10% of the area of the non-tight bulkhead.

.3 Alternatively, the opening ratio and scantlings may be determined by an acceptable method of engineering analysis.

14.3.3.5 Supports and Chocks

.1 Independent cargo tanks are to be supported by the hull in a manner that will prevent bodily movement of the tanks under static and dynamic loads while allowing contraction and expansion of the tanks under temperature variations and hull deflections without undue stressing of the tanks and of the hull.

.2 The strength of each contact surface is to be verified under the maximum contact force perpendicular to the surface and the associated friction force parallel to the surface. The loads on each tank support are to be obtained from global finite element analysis. The representative value of frictional coefficient in global FEM analysis is to be taken as 0.1, while for evaluating the strength of each individual support or chock, the friction coefficient is to be taken as 0.3 based on the supports or chocks made of synthetic materials such as resin, plywood,

and adhesive. Alternatively, the frictional coefficient may be taken from measurement.

.3 Average compressive and shear stresses in each layer of the synthetic contact surface are to be separately checked against the safe working stresses. The safety factors as specified in "*Guidelines on Structural Assessment of Ships carrying Asphalt in Independent Cargo Tanks"* for vertical supports and chocks under sea-going and accidental load cases are to be complied with.

.4 Attention is to be paid to the temperature of hull structure in way of vertical supports and chocks if the cargo tank insulation is not provided in these areas.

14.4. Machinery and Systems

14.4.1 Specific requirements for Asphalt Carriers (heating of cargo with temperatures above 120[°C]), in addition to those in Section 10.

14.4.1.1 Cargo tanks intended for the carriage of asphalt solutions are to be equipped with a heating system capable of preserving the asphalt solutions in their liquid state. Valves are to be fitted on the heating system inlet and outlet.

14.4.1.2 Redundancy is to be provided for heating plants for asphalt tanks. Redundancy is required for boilers/ thermal oil heaters, heat exchangers and as well as active components (e.g. circulation pumps). The installed heating capacity is not to be reduced by more than 50% due to failure of a redundant component.

14.4.1.3 Heating coils in asphalt tanks are to be separated into at least two independent systems. Emergency cross connections may however be accepted.

14.4.1.4 Cargo pumps, cargo piping and associated fittings are to be provided with arrangements for heating.

14.4.1.5 Cargo piping and associated fittings outside tanks are to be suitably insulated, where necessary.

14.4.1.6 Temperature sensors are to be arranged in each cargo tank enabling the recording of temperatures at bottom, midway between bottom and deck and at deck level in order to prevent overheating of cargo.

14.4.1.7 Heating coils are to be tested according to the non-destructive testing requirements indicated in Pt.4, Ch.2, Sec.7.

Appendix A

List of oils*

Asphalt solutions

Blending stocks
Roofers flux
Straight run residue

Oils

Clarified
Crude oil
Mixtures containing crude oil
Diesel oil
Fuel oil no.4
Fuel oil no.5
Fuel oil no.6
Residual fuel oil
Road oil
Transformer oil
Aromatic oil (excluding vegetable oil)
Lubricating oils and blending stocks
Mineral oil
Motor oil
Penetrating oil
Spindle oil
Turbine oil

Distillates

Straight run
Flashed feed stocks

Gas oil

Cracked

Gasoline blending stocks

Alkylates - fuel
Reformaters
Polymer - fuel

Gasolines

Casinghead (natural)
Automotive
Aviation
Straight run
Fuel oil no.1 (kerosene)
Fuel oil no.1-D
Fuel oil no.2
Fuel oil no.2-D

Jet fuels

JP-1 (kerosene)
JP-3
JP-4
JP-5 (kerosene, heavy)
Turbo fuel
Kerosene
Mineral spirit

Naphtha

Solvent
Petroleum
Heartcut distillate oil

*This list of oils may not necessarily be considered as comprehensive

Appendix B

List of cargoes other than oils which can be carried on oil tankers

Product Name	Pollution Category
Acetone	Z
Alcoholic beverages, n.o.s.	Z
Apple juice	OS
n-Butyl alcohol	Z
sec-Butyl alcohol	Z
Calcium nitrate solutions (50% or less)	Z
Clay slurry	OS
Coal slurry	OS
Diethy lene glycol	Z
Ethyl alcohol	Z
Ethylene carbonate	Z
Glucose solution	OS
Glycerine	Z
Glycerol monooleate	Z
Hexamethylenetetramine solutions	Z
Hexylene glycol	Z
Hydrogenated starch hydrolysate	OS
Isopropyl alcohol	Z
Kaolin slurry	OS
Lecithin	OS
Magnesium hydroxide slurry	Z
Maltitol solution	OS
N-Methylglucamine solution (70% or less)	Z
Methyl propyl ketone	Z
Molasses	OS
Noxious liquid, (11) n.o.s. (trade name …, contains …) Cat.Z	Z
Non-noxious liquid, (12) n.o.s. (trade name …, contains …) Cat OS	OS
Polyaluminium chloride solution	Z
Polyglycerin, sodium salt solution (containing less than 3% sodium hydroxide)	Z
Potassium formate solutions	Z
Propylene carbonate	Z
Propylene glycol	Z
Sodium acetate solutions	Z
Sodium sulphate solutions	Z
Sorbitol solution	OS
Sulphonated polyacrylate solution	Z
Tetraethyl silicate monomer/oligomer (20% in ethanol)	Z
Triethylene glycol	Z
Vegetable protein solution (hydrolysed)	OS
Water	OS
Note 1 : Pollution Category : The letter Z means the Pollution category assigned to each product under Annex II of MARPOL 73/78. OS means the product was evaluated and found to fall outside Categories X, Y or Z.	
Note 2 : Some liquid substances are identified as falling into Pollution Category Z and therefore, subject to certain requirements of Annex II of MARPOL 73/78.	

End of Chapter

Chapter 3

Chemical Carriers

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Introduction

IR 1.0 General

IR 1.1 These Rules have been prepared to ensure that Bulk Chemical Tankers built with a view to classification with Indian Register of Shipping, will also comply with the requirements of the IBC Code, as interpreted by IRS.

IR 1.2 Responsibility for interpretation of the IBC Code requirements for the purpose of issuing an International Certificate of Fitness for Chemical Tankers lies with the Government of the State whose flag the ship is entitled to fly.

IR 1.3 The Rules incorporate the final text of the IBC Code in full. For purpose of classification with Indian Register of Shipping and assignment of the notations provided for in IR 4.0, chemical tankers are required to comply with these Rules and the relevant provisions of IRS Rules and Regulations for the Classification of Ships. The paragraphs which are not included in the IMO code, but have been included in these Rules for the purpose of classification have been prefixed with the letters "IR".

IR 1.4 Ships built prior to the coming-into-force date of these Rules where the provisions of these Rules have not been applied will continue to be dealt with for classification purposes on the basis of the Rules for Chemical Tankers to which these vessels were built. In considering new or additional chemical cargoes for inclusion in the cargo lists of these ships individual consideration, taking account of the ships' arrangements and the nature of the proposed cargoes, will be given.

IR 1.5 For the purpose of classification, the following words in the IMO Code text have been changed as appropriate.

Administration	to	IRS
'Should be'	to	'is to be' or 'are to be'
'Chapter'	to	Section
'Code'	to	Chapter
Section	to	Clause (Cl).

IR 2.0 Ship operational requirements

IR 2.1 The IBC Code contains requirements in respect of operational matters which are not within the scope of classification as defined in the Rules for Ships, but are the responsibility of the National Authority or Administration responsible for issuing the Certificate of Fitness.

IR 2.2 Similarly, operational requirements which appear in the IBC Code will not be dealt with by IRS for classification purposes.

IR 2.3 Those paragraphs which are considered to be operational are indicated in Sec.16.

IR 3.0 Chemicals to which the Code does not apply

IR 3.1 Sec.18 contains a list of chemicals to which the IBC Code does not apply. For classification purposes the carriage of these listed products is therefore not restricted to Chemical Tankers constructed in accordance with these Rules.

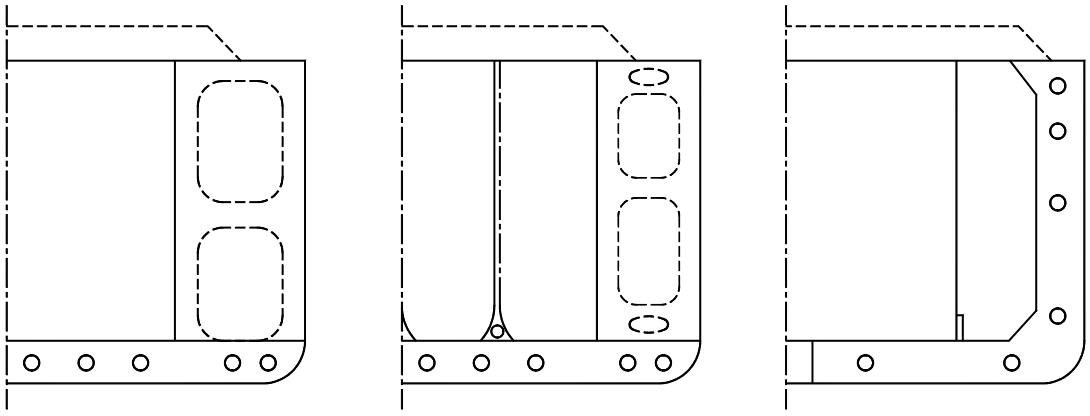

Fig. IR 4.2

IR 4.0 Classification and class notations

IR 4.1 The requirements of this chapter apply to sea going ship's intended primarily to carry chemicals in bulk (i.e. Chemical Tankers) and are supplementary to those given for the assignment of main characters of class.

IR 4.2 Assignment of class notation ESP (Enhanced Survey Program) is mandatory for seagoing self propelled Chemical Tankers which are constructed generally with integral tanks. This notation shall be assigned to chemical tankers of both single or double hull construction, as well as those with alternative structural arrangements. Typical midship sections are given in Fig.IR 4.2.

IR 4.3 Ships complying with the above mentioned requirements as applicable will be eligible to be assigned one of the following class notation:

CHEMICAL TANKER, ESP

in association with a Ship Type notation (Type 1, 2 or 3) and a list of defined cargoes.

IR 4.4 The assignment of Ship Type will not imply that the ship is suitable for all cargoes listed in Sec.17 as requiring that Ship Type. Those cargoes from Sec.17 and 18 for the carriage of which the ship has been approved will be named on a list of Defined Cargoes which will be attached to the Classification Certificate.

IR 4.5 Additional notations

IR 4.5.1 Additional notations may be given for the following features:-

a) Independent tanks, when fitted;

b) Maximum permissible specific gravity for which the scantlings have been approved, where greater than 1.025, e.g. "SG 2.0";

c) Maximum permissible positive pressure/ vacuum relief valve setting for which the scantlings have been approved, where greater than 0.21 bar e.g. "PV + 0.4 bar gauge" ;

d) Tanks constructed of corrosion resistant materials, e.g. stainless steel ("CR (S. Stl)"), or lined with corrosion resistant linings, e.g. rubber lining ("CR (r.l.)");

e) Where temperature control systems are incorporated in compliance with the requirements of Sec.7, the vessel will be eligible for additional notations as follows :

TC (Temperature Control), and where applicable HY(BC) - Refrigerated Machinery (Bulk Chemicals).

IR 4.5.2 Where features given in IR4.5.1 are confined to certain tanks, the tanks concerned will be identified in the notation.

IR 4.5.3 Other notations, as appropriate to the arrangements, scantlings and service, may be assigned.

IR 4.5.4 Chemical tankers will not generally be eligible for a reduction of cargo tank scantlings in association with a corrosion control notation.

IR 5.0 List of defined Cargoes

IR 5.1 The tanks for which each cargo has been approved will be indicated on the list of Defined Cargoes attached to the Classification Certificate as provided for in IR 4.4 together with any specific conditions of carriage.

IR 5.2 Where Chemical Cargoes which are dealt with in these Rules but which are not on the list of Defined Cargoes for classification purposes are to be carried, it will be Owner's responsibility to ensure that particulars are forwarded to IRS for consideration of the classification aspects prior to carriage.

Preamble (To the IBC Code)

1. The purpose of this Code is to provide an international standard for the safe carriage by sea in bulk of dangerous and noxious liquid chemicals listed in Ch.17 of the Code by prescribing the design and construction standards of ships regardless of tonnage, involved in such carriage and the equipment they should carry so as to minimize the risk to the ship, to its crew and to the environment, having regard to the nature of the product involved.

2. The basic philosophy is to assign to each chemical tanker one of the ship types according to the degree of the hazards of the products carried by such ship. Each of the products may have one or more hazard properties which include flammability, toxicity, corrosivity and reactivity, as well as the hazard they may present to the environment if accidentally released.

3. Throughout the development of the Code it was recognized that it must be based upon sound naval architectural and engineering principles and the best understanding available as to the hazards of the various product covered; furthermore that chemical tanker design technology is not only a complex technology but is rapidly evolving and that the Code should not remain static. Therefore, the Organization will periodically review the Code taking into account both experience and technical development.

4. Amendments to the Code involving requirements for new products and their conditions of carriage will be circulated as recommendations, on an interim basis, when adopted by the Maritime Safety Committee and the Marine Environment Protection Committee of the Organization, in accordance with the provisions of article VIII of the International Convention for the Safety of Life at Sea, 1974 (SOLAS 74), and article 16 of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol related thereto (MARPOL 73/78) respectively, pending the entry into force of these amendments.

5. The Code primarily deals with ship design and equipment. In order to ensure the safe transport of the products the total system must, however, be appraised. Other important facts of the safe transport of the products, such as training, operation, traffic control and handling in port, are being or will be examined further by the Organization.

6. The development of the Code has been greatly assisted by relevant work of the International Association of Classification Societies (IACS) and of the International Electrotechnical Commission (IEC).

7. Ch.16 of the Code, dealing with operation requirements of chemical tankers, highlights the regulations in other Chapters that are operational in nature and mentions those other important safety features that are peculiar to chemical tanker operation.

8. The layout of the Code is in line with the International Code for the Construction and Equipment of Ship Carrying Liquefied Gases in Bulk (IGC Code), adopted by the Maritime Safety Committee at its forty-eight session. Gas carriers may also carry in bulk liquid chemicals covered by this Code as prescribed in the IGC Code.

9. The 1998 edition of the Code was based on the original text as adopted by MSC resolution MSC.4(48). In response to resolution 15 of the International Conference on Marine Pollution, 1973, the Marine Environment Protection Committee at its twenty-second session, adopted, by resolution MEPC.19(22), the IBC Code extended to cover the marine pollution-prevention aspects for the implementation of Annex II to the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (MARPOL 73/78).

10. This edition of the Code includes amendments adopted by the following resolutions:

	Resolution	**Adoption**	**Deemed acceptance**	**Entry into force**
1	MSC.10(54)	29 April 1987	29 April 1988	30 October 1988
2	MSC.14(57) MEPC.32(27)	11 April 1989 17 March 1989	12 April 1990 12 April 1990	13 October 1990 13 October 1990
3	MSC.28(61) MEPC.55(33)	11 December 1992 30 October 1992	11 January 1994 11 January 1994	1 July 1994 1 July 1994
4	MSC.50(66) MEPC.69(38)	4 June 1996 10 July 1996	1 January 1998 1 January 1998	1 July 1998 1 July 1998
5	MSC.58(67) MEPC.73(39)	5 December 1996 10 March 1997	1 January 1998 10 January 1998	1 July 1998 10 July 1998
6	MSC.102(73)	5 December 2000	1 January 2002	1 July 2002
7	MSC.176(79) MEPC.119(52)	9 December 2004 15 October 2004	1 July 2006 1 July 2006	1 January 2007 1 January 2007
8	MSC 219(82) MEPC.166(56)	8 December 2006 13 July 2007	1 July 2008 1 July 2008	1 January 2009 1 January 2009
9	MSC.340(91) MEPC.225(64)	30 November 2012 5 October 2012	1 December 2013 1 December 2013	1 June 2014 1 June 2014
10	MSC.369(93) MEPC.250(66)	22 May 2014 4 April 2014	1 July 2015 1 July 2015	1 January 2016 1 January 2016
11	MSC.460(101) MEPC.318(74)	14 June 2019 17 May 2019	1 July 2020 1 July 2020	1 January 2021 1 January 2021

11. As from the date of entry into force of the 1983 amendments to the 1974 SOLAS Convention (i.e. 1 July 1987) and the date of implementation of Annex II of MARPOL 73/78 (i.e. 6 April 1987), this Code became subject to mandatory requirements under these Conventions. Any future amendment to the Code, whether from the point of view of safety or of marine pollution, must be adopted and brought into force in accordance with the procedures laid down in Article VIII of SOLAS 74 and Article 16 of MARPOL 73/78 respectively.

Section 1

General

1.1 Application

IR 1.1.1 The requirements of this Chapter are in addition to the requirements of Pt.1 to 4 of the Rules as applicable.

1.1.1 This Chapter applies to ships regardless of size, including those of less than 500 tons gross tonnage, engaged in the carriage of bulk cargoes of dangerous or noxious liquid chemical substances, other than petroleum or similar flammable products as follows:

.1 products having significant fire hazards in excess of those of petroleum products and similar flammable products;

.2 products having significant hazards in addition to or other than flammability.

1.1.2 Products that have been reviewed and determined not to present safety and pollution hazards to such an extent as to warrant the application of the Code are found in Sec.18.

1.1.3 Liquids covered by the Chapter are those having a vapour pressure not exceeding 0.28 MPa at a temperature of 37.8°C.

1.1.4 For the purpose of the 1974 SOLAS Convention, the Chapter applies to ships which are engaged in the carriage of products included in Sec.17 on the basis of their safety characteristics and identified as such by an entry of S or S/P in column d.

1.1.5 For the purposes of MARPOL 73/78, the Code applies only to NLS tankers, as defined in regulation 1.16.2 of Annex II thereof, which are engaged in the carriage of Noxious Liquid Substances identified as such by an entry of X, Y or Z in column c of Section 17.

1.1.6 For a product proposed for carriage in bulk, but not listed in Sec.17 or 18, the Administration and port Administrations involved in such carriage should prescribe the preliminary suitable conditions for the carriage, having regard to the criteria for hazard evaluation of bulk chemicals. The Organization should be notified of the conditions for consideration for inclusion of the product in the Code. For the evaluation of the pollution hazard of such a product and assignment of its pollution category, the procedure specified in regulation 6.3 of Annex II of MARPOL 73/78 must be followed.

IR 1.1.6 Products covered in 1.1.6 will be specially considered by IRS.

1.1.7 Unless expressly provided otherwise the Chapter applies to ships the keels of which are at a stage at which:

.1 construction identifiable with the ship begins; and

.2 assembly has commenced comprising at least 50 tonnes or 1% of the estimated mass of all structural material, whichever is less;

on or after 1 July 1986.

IR 1.1.7 For classification purposes these Rules may, but need not be, applied to ships for which the midship section is approved prior to 1 July 1986.

1.1.8 A ship, irrespective of the date of construction, which is converted to a chemical tanker on or after 1 July 1986, is to be treated as a chemical tanker constructed on the date on which such conversion commences. This conversion provision does not apply to the modification of a ship referred to in regulation 1(12) of Annex II of MARPOL 73/78.

1.1.9 Where reference is made in the Chapter to a paragraph, all the provisions of the subparagraphs of that designation will apply.

1.2 Hazards

Hazards of products covered by this Chapter include:

1.2.1 Fire hazard defined by flashpoint, boiling point, flammability limits and auto-ignition temperature of the chemical.

1.2.2 Health hazard defined by:

.1 corrosive effects on the skin in the liquid state; or

.2 acute toxic effect, taking into account values of

LD_{50} (oral): a dose which is lethal to 50% of the test subjects when administered orally;

LD_{50} (dermal) : a dose which is lethal to 50% of the test subjects when administered to the skin;

LC_{50} (inhalation): the concentration which is lethal by inhalation to 50% of the test subjects.

.3 other health effects such as carcinogenicity and sensitization.

1.2.3 Reactivity hazard defined by reactivity:

.1 with water;
.2 with air;
.3 with other products; or
.4 of the product itself
(e.g.polymerization).

1.2.4 Marine pollution hazard as defined by:

.1 bioaccumulation;

.2 lack of ready biodegradibility;

.3 acute toxicity to aquatic organisms;

.4 chronic toxicity to aquatic organisms;

.5 long term human health effects; and

.6 physical properties resulting in the product floating or sinking and so adversely affecting marine life.

1.3 Definitions

The following definitions apply unless expressly provided otherwise. (Additional definitions are given in individual chapters).

1.3.1 **Accommodation spaces** are those spaces used for public spaces, corridors, lavatories, cabins, offices, hospitals, cinemas, games and hobbies rooms, barber shops, pantries containing no cooking appliances and similar spaces. Public spaces are those portions of the accommodation spaces which are used for halls, dining rooms, lounges and similar permanently enclosed spaces.

1.3.2 **Administration** means the Government of the State whose flag the ship is entitled to fly.

1.3.3 **Anniversary date** means the day and the month of each year which will correspond to the date of expiry of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk.

1.3.4 **Boiling point** is the temperature at which a product exhibits a vapour pressure equal to the atmospheric pressure.

1.3.5 **Breadth (B**) means the maximum breadth of the ship, measured amidships to the moulded line of the frame in a ship with a metal shell and to the outer surface of the hull in a ship with a shell of any other material. The breadth (B) is to be measured in metres [m].

IR 1.3.1 For the purpose of calculation of scantlings, Pt.3 of the Rules is to be referred for the definition of "Breadth".

1.3.6 **Cargo area** is that part of the ship that contains cargo tanks, slop tanks, cargo pump-rooms including pump-rooms, cofferdams, ballast or void spaces adjacent to cargo tanks or slop tanks and also deck areas throughout the entire length and breadth of the part of the ship over the above-mentioned spaces. Where independent tanks are installed in hold spaces, cofferdams, ballast or void spaces at the after end of the aftermost hold space or at the forward end of the forwardmost hold space are excluded from the cargo area.

1.3.7 **Cargo pump-room** is a space containing pumps and their accessories for the handling of the products covered by this Chapter.

1.3.8 Cargo service spaces are spaces within the cargo area used for workshops, lockers and store-rooms of more than 2 [m^2] in area, used for cargo-handling equipment.

1.3.9 **Cargo tank** is the envelope designed to contain the cargo.

1.3.10 **Chemical tanker** is a cargo ship constructed or adapted and used for the carriage in bulk of any liquid product listed in Sec.17.

1.3.11 **Cofferdam** is the isolating space between two adjacent steel bulkheads or decks. This space may be a void space or a ballast space.

1.3.12 **Control stations** are those spaces in which ship's radio or main navigating equipment or the emergency source of power is located or where the fire-recording or fire-control equipment is centralized. This does not include special fire-control equipment which can be most practically located in the cargo area.

1.3.13 **Dangerous chemicals** means any liquid chemicals designated as presenting a safety hazard, based on the safety criteria for assigning products to chapter 17.

1.3.14 **Density** is the ratio of the mass to the volume of a product, expressed in terms of kilograms per cubic metre. This applies to liquids, gases and vapours.

1.3.15 **Explosive/Flammability limits/range** are the conditions defining the state of fuel-air mixture at which application of an adequately strong external ignition source is only just capable of producing flammability in a given test apparatus.

1.3.16 **Flashpoint** is the temperature in degrees Celsius at which a product will give off enough flammable vapour to be ignited. Values given in this Chapter are "closed cup test" determined by an approved flashpoint apparatus.

1.3.17 **Gas-freeing** means the process where a portable or fixed ventilation system is used to introduce fresh air into a tank in order to reduce the concentration of hazardous gases or vapours to a level safe for tank entry.

1.3.18 **Hold space** is the space enclosed by the ship's structure in which an independent cargo tank is situated.

1.3.19 **Independent** means that a piping or venting system, for example, is in no way connected to another system and that there are no provisions available for the potential connection to other systems.

1.3.20 **Length (L)** means 96% of the total length on a waterline at 85% of the least moulded depth measured from the top of the keel, or the length from the foreside of the stem to the axis of the rudder stock on that waterline, if that be greater. In ships designed with rake of keel, the waterline on which this length is measured is to be parallel to the designed waterline. The length (L) is to measured in metres [m].

IR 1.3.20A For the purpose of calculation of scantlings, Pt.3 of the Rules is to be referred for the definition of "Length".

1.3.21 **Machinery spaces of category A** are those spaces and trunks to such spaces which contain:

.1 internal combustion machinery used for main propulsion; or

.2 internal combustion machinery used for purposes other than main propulsion where such machinery has in the aggregate a total power output of not less than 375 [kW]; or

.3 any oil-fired boiler or oil fuel unit or any oil-fired equipment other than boilers, such as inert gas generators, incinerators, etc.

1.3.22 **Machinery spaces** are all machinery spaces of category A and all other spaces containing propelling machinery, boilers, oil fuel units, steam and internal combustion engines, generators and major electrical machinery, oil filling stations, refrigerating, stabilizing, ventilation and air-conditioning machinery, and similar spaces, and trunks to such spaces.

1.3.23 **MARPOL** means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto and by the Protocol of 1997, as amended.

1.3.24 **Noxious Liquid Substance** means any substance indicated in the Pollution Category column of chapters 17 or 18 of the International Bulk Chemical Code, or the current MEPC.2/Circular or provisionally assessed under the provisions of regulation 6.3 of MARPOL Annex II as falling into categories X, Y or Z.

1.3.25 **Oil fuel unit** is the equipment used for the preparation of oil fuel for delivery to an oil-fired boiler, or equipment used for the preparation for delivery of heated oil to an internal combustion engine, and includes any oil pressure pumps, filters and heaters dealing with oil at a gauge pressure of more than 0.18 [MPa].

1.3.26 **Organization** is the International Maritime Organization (IMO).

1.3.27 **Permeability** of a space means the ratio of the volume within that space which is assumed to be occupied by water to the total volume of that space.

1.3.28 **Port Administration** means the appropriate authority of the country in the port of which the ship is loading or unloading.

1.3.29 **Products** is the collective term used to cover both Noxious Liquid Substances and Dangerous Chemicals.

1.3.30 **Pump-room** is a space, located in the cargo area, containing pumps and their accessories for the handling of ballast and oil fuel.

1.3.31 **Purging** means the introduction of inert gas into a tank which is already in an inert condition with the object of further reducing the oxygen content; and/or reducing the existing hydrocarbon or other flammable vapours content to a level below which combustion cannot be supported if air is subsequently introduced into the tank.

1.3.32 **Recognized organization** is an organization authorized by an Administration in accordance with MARPOL Annex II regulation 8.2.2 and SOLAS regulation XI-1/1.

1.3.33 **Recognized standards** are applicable international or national standards acceptable to the Administration or standards laid down and maintained by an organization which complies with the standards adopted by IMO and which is recognised by the Administration.

1.3.34 **Reference temperature** is the temperature at which the vapour pressure of the cargo corresponds to the set pressure of the pressure-relief valve.

1.3.35 **Separate** means that a cargo piping system or cargo vent system, for example, is not connected to another cargo piping or cargo vent system.

1.3.36 **Service spaces** are those spaces used for galleys, pantries containing appliances, lockers, mail and specie rooms, store-rooms, workshop other than those forming part of the machinery spaces and similar spaces and trunks to such spaces.

1.3.37 **SOLAS** means the International Convention for the Safety of Life at Sea, 1974, as amended.

1.3.38 **Vapour pressure** is the equilibrium pressure of the saturated vapour above a liquid expressed in Pascals [Pa] at a specified temperature.

1.3.39 **Void space** is an enclosed space in the cargo area external to a cargo tank, other than a hold space, ballast space, oil fuel tank, cargo pump-room, pump-room, or any space in normal use by personnel.

1.4 Equivalents

IR 1.4.1 The construction, equipment, etc. which do not comply with the provisions of these Rules but are considered to be equivalent to these Rules would be specially considered by IRS.

1.4.1 Where the Chapter requires that a particular fitting, material, appliance, apparatus, item of equipment or type thereof is to be fitted or carried in a ship, or that any particular provision should be made, or any procedure or arrangement is to be complied with, the Administration may allow any other fitting, material, appliance, apparatus, item of equipment or type thereof to be fitted or carried, or any other provision, procedure or arrangement to be made in that ship, if it is satisfied by trial thereof or otherwise that such fitting, material, appliance, apparatus, item of equipment or type thereof or that any particular provision, procedure or arrangement is at least as effective as that required by the Chapter. However, the Administration may not allow operational methods or procedures to be made an alternative to a particular fitting, material appliance, apparatus, item of equipment, or type thereof, which are prescribed by the Chapter, unless such substitution is specifically allowed by the Chapter.

1.4.2 When the Administration so allows any fitting, material, appliance, apparatus, item of equipment, or type thereof, or provision, procedure, or arrangement, or novel design or application to be substituted thereafter, it should communicate to the Organization the particulars thereof together with a report on the evidence submitted so that the Organization may circulate the same to other Contracting Governments to SOLAS and

parties to MARPOL for the information of their officers.

1.5 Surveys and certification

1.5.1 Survey procedure

IR 1.5.1.1 The survey of ships with regard to the classification of the ship is to be carried out by the Surveyors of IRS.

1.5.1.1 The survey of ships so far as regards the enforcement of the provisions of the regulations and granting of exemptions therefrom, should be carried out by the officers of the Administration. The Administration may, however, entrust the surveys either to Surveyors nominated for the purpose or to organisations recognized by it.

1.5.1.2 The recognized organization referred to in 1.3.32 is to comply with the provisions of SOLAS and MARPOL and with Parts 1 and 2 of the Code for Recognized Organizations (RO Code), as adopted by resolutions MSC.349(92) and MEPC.237(65), as may be amended.

1.5.1.3 When a nominated surveyor or recognized organization determines that the condition of the ship or its equipment does not correspond substantially with the particulars of the certificates or is such that the ship is not fit to proceed to sea without danger to the ship, or persons on board, such surveyor or organization should immediately ensure that corrective action is taken and should in due course notify the Administration. If such corrective action is not taken the relevant certificate should be withdrawn immediately; and, if the ship is in a port of another Contracting Government, the Port State authority concerned should also be notified immediately.

1.5.1.4 In every case, the Administration should guarantee the completeness and efficiency of the survey, and should undertake to ensure the necessary arrangements to satisfy this obligation.

1.5.2 Survey requirements

IR 1.5.2.1 The Classification Regulations for New Construction Surveys, the classification of ships not built under Survey and Periodical Survey Regulations are given in Pt.1 of the Rules. The following requirements are also to be complied with.

1.5.2.1 The structure, equipment, fittings, arrangements and material (other than items in respect of which a Cargo Ship Safety Construction Certificate, Cargo Ship Safety Equipment Certificate and Cargo Ship Safety Radiotelegraphy Certificate or Cargo Ship Safety Radiotelephony Certificate are issued) of a chemical tanker should be subjected to the following surveys:

.1 An initial survey before the ship is put in service or before the International Certificate of Fitness for the Carriage of Dangerous Chemical in Bulk is issued for the first time, which should include a complete examination of its structure, equipment, fittings, arrangements and material in so far as the ship is covered by the Code. This survey should be such as to ensure that the structure, equipment, fittings, arrangements and material fully comply with the applicable provisions of the Code.

.2 A periodical survey at intervals specified by the Administration, but not exceeding 5 years which should be such as to ensure that the structure, equipment, fittings, arrangements and material comply with the applicable provisions of the Code.

.3 A minimum of one intermediate survey within 3 months before or after the second anniversary date or within 3 months before or after the third anniversary date of the certificate, which shall take the place of one of the annual surveys specified in 1.5.2.1.4. Intermediate surveys should be such as to ensure that the safety equipment, and other equipment, and associated pump and piping systems comply with the applicable provisions of the Code and are in good working order. Such surveys should be endorsed on the International Certificate of fitness for the Carriage of Dangerous Chemicals in Bulk.

IR.3 Those items which are additional to the requirements of Annual Survey may be examined between the second or third annual survey.

.4 A mandatory annual survey within 3 months before or after the anniversary of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk which should include a general examination to ensure that the structure, equipment, fittings, arrangements and materials remain in all respects satisfactory

for the service for which the ship is intended. Such a survey should be endorsed in the International Certificate of Fitness for the Carriage of Dangerous Chemicals in bulk.

.5 An additional survey, either general or partial according to the circumstances, should be made when required after an investigation prescribed in 1.5.3.3 or whenever any important repairs or renewals are made. Such a survey should ensure that the necessary repairs or renewals have been effectively made, that the material and workmanship of such repairs or renewals are satisfactory; and that the ship is fit to proceed to sea without danger to the ship or persons on board or without presenting unreasonable threat of harm to the marine environment.

1.5.3 Maintenance of conditions after survey

1.5.3.1 The condition of the ship and its equipment should be maintained to confirm with the provisions of the Code to ensure that the ship will remain fit to proceed to sea without danger to the ship or persons on board or without presenting an unreasonable threat of harm to the marine environment.

1.5.3.2 After any survey of the ship under 1.5.2 has been completed, no change should be made in the structure, equipment, fittings, arrangements and material covered by the survey, without the sanction of the Administration, except by direct replacement.

1.5.3.3 Whenever an accident occurs to a ship or a defect is discovered, either of which affects the safety of the ship or the efficiency or completeness of its life-saving appliances or other equipment, the master or owner of the ship should report at the earliest opportunity to the Administration, the nominated surveyor or recognised organization responsible for issuing the relevant certificate, who should cause investigations to be initiated to determine whether a survey, as required by 1.5.2.1.5 is necessary. If the ship is in a port of another Contracting Government, the master or owner should also report immediately to the port State authority concerned and the nominated surveyor or recognised organization should ascertain that such a report has been made.

1.5.4 Issue of International certificate of fitness

1.5.4.1 An International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk the model form of which is set out in the appendix to the Code, should be issued after an initial or periodical survey to a chemical tanker engaged in international voyages which complies with the relevant requirements of the Code.

1.5.4.2 The certificate issued under provisions of this Section should be available on board for inspection at all times.

1.5.5 Issue or endorsement of International certificate of fitness by another Government

1.5.5.1 A Party to the 1974 SOLAS Convention and to MARPOL 73/78 may, at the request of another Party, cause a ship entitled to fly the flag of the other State to be surveyed and, if satisfied that the requirements of the Code are complied with, issue or authorise the issue of the certificate to the ship, and, where appropriate, endorse or authorise the endorsement of the certificate on board the ship in accordance with the Code. Any certificate so issued should contain a statement to the effect that it has been issued at the request of the Government of the State whose flag the ship is entitled to fly.

1.5.6 Duration and validity of the International certificate of fitness

1.5.6.1 An International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk should be issued for a period specified by the Administration which should not exceed 5 years.

1.5.6.2 No extension of the 5 years period of the certificate should be permitted.

1.5.6.2.1 Notwithstanding the provisions of 1.5.6.1, when the renewal survey is completed within 3 months before the expiry date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding 5 years from the date of expiry of the existing certificate.

1.5.6.2.2 When the renewal survey is completed after the expiry date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding 5 years from the date of expiry of the existing certificate.

1.5.6.2.3 When the renewal survey is completed more than 3 months before the expiry date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding 5 years from the date of completion of the renewal survey.

1.5.6.3 If a certificate is issued for a period of less than 5 years, IRS may extend the validity of the certificate beyond the expiry date to the maximum period specified in 1.5.6.1, provided that the surveys referred to in 1.5.2.1.3 and 1.5.2.1.4, applicable when a certificate is issued for a period of 5 years, are carried out as appropriate.

1.5.6.4 If a renewal survey has been completed and a new certificate cannot be issued or placed on board the ship before the expiry date of the existing certificate, the Surveyor may endorse the existing certificate. Such a certificate shall be accepted as valid for a further period which shall not exceed 5 months from the expiry date.

1.5.6.5 If a ship, at the time when a certificate expires, is not in a port in which it is to be surveyed, IRS may extend the period of validity of the certificate but this extension shall be granted only for the purpose of allowing the ship to complete its voyage to the port in which it is to be surveyed, and then only in cases where it appears proper and reasonable.

1.5.6.6 A certificate, issued to a ship engaged on short voyages which has not been extended under the foregoing provisions of this section, may be extended by the Administration for a period of grace of up to one month from the date of expiry stated on it. When the renewal survey is completed, the new certificate shall be valid to a date not exceeding 5 years from the date of expiry of the existing certificate before the extension was granted.

1.5.6.7 In special circumstances, as determined by the Administration, a new certificate need not be dated from the date of expiry of the existing certificate as required by 1.5.6.2.2, 1.5.6.5 or 1.5.6.6. In these special circumstances, the new certificate shall be valid to a date not exceeding 5 years from the date of completion of the renewal survey.

1.5.6.8 If an annual or intermediate survey is completed before the period specified in 1.5.2, then:

.1 the anniversary date shown on the certificate shall be amended by endorsement to a date which shall not be more than 3 months later than the date on which the survey was completed;

.2 the subsequent annual or intermediate survey required by 1.5.2 shall be completed at the intervals prescribed by that section using the new anniversary date; and

.3 the expiry date may remain unchanged provided one or more annual or intermediate surveys, as appropriate, are carried out so that the maximum intervals between the surveys prescribed by 1.5.2 are not exceeded.

1.5.6.9 The certificate issued under 1.5.4 or 1.5.5 should cease to be valid in any of the following cases:

.1 if the relevant surveys are not completed within the periods specified under 1.5.2;

.2 if the Certificate is not endorsed in accordance with 1.5.2.1.3 or 1.5.2.1.4;

.3 upon transfer of the ship to the flag of another State. A new certificate should only be issued when the Government issuing the new certificate is fully satisfied that the ship is in compliance with the requirements of 1.5.3.1 and 1.5.3.2. In the case of a transfer between Governments that are both a Contacting Government to the 1974 SOLAS Convention and a party to MARPOL 73/78, if requested within 3 months after the transfer has taken place, the Government of the State whose flag the ship was formerly entitled to fly shall, as soon as possible transmit to the Administration copies of the certificates carried by the ship before the transfer and, if available, copies of the relevant survey reports.

IR 1.6 Plans and particulars

IR 1.6.1 A general arrangement is to be submitted for approval giving location of:

- cargo hatches, tank cleaning hatches and any other openings to the cargo tanks.

- doors, hatches and any other openings to pump rooms and other gas-dangerous spaces.

- ventilating pipes and openings for cargo tanks, pump rooms and other gas-dangerous spaces.

- doors, air locks, hatches, ventilating pipes and openings, hinged scuttles which can be opened, and other openings to gas safe spaces adjacent to the cargo area including spaces in and below the forecastle.

- cargo pipes and gas return pipes over the deck with shore-connections including stern pipes for cargo discharge or gas-freeing of cargo tanks.

- deckplan showing location and type of all monitoring equipment for cargo handling such as, level gauging, overflow control, temperature read out etc.

IR 1.6.2 Plans with the following particulars for the tanks are to be submitted for approval:

- drawings of cargo tanks including information on non-destructive testing of welds and strength and tightness testing of tanks.

- drawings of support and staying of independent cargo tanks.

- a complete stress analysis is to be submitted for pressure tanks.

- damage stability calculations.

IR 1.6.3 Plans of the following pumping and piping arrangements with accessories are to be submitted for approval:

- diagrams of the cargo piping system including details such as expansion elements and flange connections.

- bilge piping systems in pump rooms, cofferdams, pipe tunnels and hold spaces.

- arrangement for drainage of cargo pumps and piping in the pumproom.

- drawing of cargo pump.

- arrangements for cargo tank stripping and drainage/stripping of cargo lines.

- diagram of tank washing.

- arrangement and location of underwater discharge outlet(s).

IR 1.6.4 Plans showing the following equipment and systems are to be submitted for approval:

- arrangement and capacity of air ducts, fans and their motors in the cargo area, drawing and material specification of rotating parts and casing of the fans.

- drawings of portable ventilators and drawings showing where and how these are to be fitted.

- arrangement for gas-freeing of cargo tanks and cargo lines. Arrangement of cargo tank venting systems.

- arrangement and specification of all monitoring systems and devices for liquid level indication.

- quick-closing arrangements for cargo valves.

- drawings of gastight bulkhead stuffing boxes.
- arrangement of cargo heating systems.

- arrangement of cargo cooling systems.

- arrangement of possible thermal insulation and specification of insulation materials.

- drawings showing details of design, attachment and location of anodes and other fittings in tanks and cofferdams.

IR 1.6.5 Plans of electrical installations giving the following particulars are to be submitted for approval:

- area classification drawings.

- drawings showing location of all electrical equipment in gas dangerous area.

- single line diagram for intrinsically safe circuits.

- list of explosion protected equipment with reference to drawings together with certificates.

Section 2

Ship Survival Capability and Location of Cargo Tanks

2.1 General

2.1.1 Ships subject to the requirements of this Chapter are to survive the normal effects of flooding following assumed hull damage caused by some external force. In addition, to safeguard the ship and the environment, the cargo tanks of certain types of ships are to be protected from penetration in the case of minor damage to the ship resulting, for example, from contact with a jetty or tug, and given a measure of protection from damage in the case of collision or stranding, by locating them at specified minimum distances inboard from the ship's shell plating. Both the damage to be assumed and the proximity of the cargo tanks to the ship's shell is to be dependent upon the degree of hazard presented by the products to be carried.

2.1.2 Ships subject to the requirements of this Chapter are to be designed to one of the following standards:

.1 **A Type 1 ship** is a chemical tanker intended to transport products listed in Sec.17 with very severe environmental and safety hazards which require maximum preventive measures to preclude an escape of such cargo.

.2 **A Type 2 ship** is a chemical tanker intended to transport products listed in Sec.17 with appreciably severe environmental and safety hazards which require significant preventive measures to preclude an escape of such cargo.

.3 **A Type 3 ship** is a chemical tanker intended to transport products listed in Sec.17 with sufficiently severe environmental and safety hazards which require a moderate degree of containment to increase survival capability in a damaged condition.

Thus a Type 1 ship is a chemical tanker intended for the transportation of products considered to present the greatest overall hazard and Type 2 and Type 3 for products of progressively lesser hazards. Accordingly, a Type 1 ship is to survive the most severe standard of damage and its cargo tanks are to be located at the maximum prescribed distance inboard from the shell plating.

2.1.3 The ship type required for individual products is indicated in column "e" in the Table 17.1.1.

2.1.4 If a ship is intended to carry more than one product listed in Sec.17, the standard of damage is to correspond to that product having the most stringent ship type requirement. The requirements for the location of individual cargo tanks, however, are those for ship types related to the respective products intended to be carried.

2.2 Freeboard and stability

2.2.1 Ships subject to this Chapter may be assigned the minimum freeboard permitted by the International Convention on Load Lines in force. However, the draught associated with the assignment is not to be greater than the maximum draught otherwise permitted by this Chapter.

2.2.2 The stability of the ship in all seagoing conditions is to be to a standard which is acceptable to the Administration.

2.2.3 When calculating the effect of free surfaces of consumable liquids for loading conditions it is to be assumed that, for each type of liquid, at least one transverse pair or a single centre tank has a free surface and the tank or combination of tanks to be taken in to account is to be that where the effect of the free surfaces is the greatest. The free surface effect in undamaged compartments is to be calculated by a method acceptable to the Administration.

2.2.4 Solid ballast is not normally to be used in double bottom spaces in the cargo area. Where, however, because of stability considerations, the fitting of solid ballast in such spaces becomes unavoidable, then its disposition is to be governed by the need to ensure that the impact loads resulting from bottom damage are not directly transmitted to the cargo tank structure.

2.2.5 The master of the ship is to be supplied with a loading and stability information booklet. This booklet is to contain details of typical service and ballast conditions, provisions for evaluating other conditions of loading and a summary of the ship's survival capabilities. In addition, the booklet is to contain sufficient information to enable the master to load and operate the ship in a safe and seaworthy manner.

2.2.6 All ships, subject to the requirements of this Chapter, are to be fitted with a stability instrument, capable of verifying compliance with intact and damage stability requirements, approved by the Administration/ IRS, having regard to the recommended performance standards

(Refer to Part B, Chapter 4, of the International Code on Intact Stability, 2008 (2008 IS Code), as amended; the Guidelines for the Approval of Stability Instruments (MSC.1/Circ.1229), Annex, Section 4, as amended; and the technical standards defined in Part 1 of the Guidelines for verification of damage stability requirements for tankers (MSC.1/Circ.1461)

.1 ships constructed before 1 January 2016 are to comply with this requirement at the first scheduled renewal survey of the ship on or after 1 January 2016 but not later than 1 January 2021;

.2 notwithstanding the requirements of 2.2.6.1, a stability instrument fitted on a ship constructed before 1 January 2016 need not be replaced provided it is capable of verifying compliance with intact and damage stability, to the satisfaction of the Administration/ IRS; and

.3 for the purposes of control under regulation 16 of MARPOL Annex II, the Administration/ IRS may issue a document of approval for the stability instrument.

2.2.7 The Administration may waive the requirements of 2.2.6 for the following ships provided the procedures employed for intact and damage stability verification maintain the same degree of safety, as being loaded in accordance with the approved conditions *(Refer to operational guidance provided in Part 2 of the guidelines for verification of damage stability requirements for tankers (MSC.1/Circ.1461).* Any such waiver would be duly noted on the International Certificate of Fitness referred to in 1.5.4:

.1 ships which are on a dedicated service, with a limited number of permutations of loading such that all anticipated conditions have been approved in the stability information provided to the master in accordance with the requirements of 2.2.5;

.2 ships where stability verification is made remotely by a means approved by the Administration/ IRS;

.3 ships which are loaded within an approved range of loading conditions; or

.4 ships constructed before 1 January 2016 provided with approved limiting KG/ GM curves covering all applicable intact and damage stability requirements.

2.3 Shipside discharges below the freeboard deck

2.3.1 The provision and control of valves fitted to discharges led through the shell from spaces below the freeboard deck or from within the superstructures and deckhouses on the freeboard deck fitted with weathertight doors are to comply with the requirements of the relevant regulation of the International Convention on Load Lines in force, except that the choice of valves is to be limited to:

.1 one automatic nonreturn valve with a positive means of closing from above the freeboard deck; or

.2 where the vertical distance from the summer load waterline to the inboard end of the discharge pipe exceeds 0.01L, two automatic non-return valves without positive means of closing, provided that the inboard valve is always accessible for examination under service conditions.

2.3.2 For the purpose of this Chapter "summer waterline" and "freeboard deck", have the meanings as defined in the International Convention on Load Lines in force.

2.3.3 The automatic nonreturn valves referred to in 2.3.1 are to comply with recognized standards and are to be fully effective in preventing admission of water into the ship, taking into account the sinkage, trim and heel in survival requirements given in 2.9.

2.4 Conditions of loading

2.4.1 Damage survival capability is to be investigated on the basis of the loading information submitted to IRS for all anticipated conditions of loading and variations in draught and trim. Ballast conditions where the chemical tanker is not carrying products covered by this Chapter, or is carrying only residues of such products, need not be considered.

2.5 Damage assumptions

2.5.1 The assumed maximum extent of damage is to be :

Side damage		
Longitudinal extent	$\frac{1}{3}$ $L^{2/3}$ or 14.5 [m], whichever is less	
Transverse extent	B/5 or 11.5 [m], whichever is less (measured inboard from the ship's side at the right angles to the centreline at the level of the summer load line)	
Vertical extent	upwards without limit (from the moulded line of the bottom shell plating at centreline)	
Bottom damage		
	For 0.3L from the forward perpendicular of the ship	Any other part of the ship
Longitudi-nal extent	$\frac{1}{3}$ $L^{2/3}$ or 14.5 [m], whichever is less	$\frac{1}{3}$ $L^{2/3}$ or 5 [m], whichever is less
Transver-se extent	B/6 or 10 [m], whichever is less	B/6 or 5 [m], whichever is less
Vertical extent	B/15 or 6 [m], whichever is less. (measured from the moulded line of the bottom shell plating at centreline (see 2.6.2))	B/15 or 6 [m], whichever is less. (measured from the moulded line of the bottom shell plating at centreline (see 2.6.2))

2.5.2 If any damage of a lesser extent than the maximum damage specified in 2.5.1 would result in a more severe condition, such damage is to be considered.

2.6 Location of cargo tanks

2.6.1 Cargo tanks are to be located at the following distances inboard:

.1 Type 1 ships : from the side shell plating not less than the transverse extent of damage specified in table above titled 'Side damage' and from the moulded line of the bottom shell plating at centreline not less than the vertical extent of damage specified in table above titled 'Bottom damage' and nowhere less than 760 [mm] from the shell plating. This requirement does not apply to the tanks for diluted slops arising from tank washing.

.2 Type 2 ships : from the moulded line of the bottom shell plating at centreline, not less than the vertical extent of damage specified in table above titled 'Bottom damage' and nowhere less than 760 [mm] from the shell plating. This requirement does not apply to the tanks for diluted slops arising from tank washing.

.3 Type 3 ships : no requirement.

2.6.2 Except for Type 1 ships, suction wells installed in cargo tanks may protrude into the vertical extent of bottom damage specified in table above titled 'Bottom damage' provided that such wells are as small as practicable and the protrusion below the inner bottom plating does not exceed 25% of the depth of the double bottom or 350 [mm], whichever is less. Where there is no double bottom, the protrusion of the suction well of independent tanks below the upper limit of bottom damage is not to exceed 350 [mm]. Suction wells installed in accordance with this paragraph may be ignored in determining the compartments affected by damage.

2.7 Flooding assumptions

2.7.1 The requirements of 2.9 are to be confirmed by calculations which take into consideration the design characteristics of the ship; the arrangements, configuration and contents of the damaged compartments; the distribution, relative densities and the free surface effects of liquids; and the draught and trim for all conditions of loading.

2.7.2 The permeabilities of spaces assumed to be damaged are to be as follows:

Spaces	Permeabilities
Appropriated to stores	0.60
Occupied by accommodation	0.95
Occupied by machinery	0.85
Voids	0.95
Intended for consumable liquids	0 to 0.95*
Intended for other liquids	0 to 0.95*
* The permeability of partially filled compartments is to be consistent with the amount of liquid carried in the compartment.	

2.7.3 Wherever damage penetrates a tank containing liquids it is to be assumed that the contents are completely lost from that compartment and replaced by salt water up to the level of the final plane of equilibrium.

2.7.4 Every watertight division within the maximum extent of damage defined in 2.5.1 and considered to have sustained damage in positions given in 2.8.1 is to be assumed to be penetrated. Where damage less than the maximum is being considered in accordance with 2.5.2, only watertight divisions or combinations of watertight divisions within the envelope of such lesser damage is to be assumed to be penetrated.

2.7.5 The ship is to be so designed as to keep unsymmetrical flooding to the minimum consistent with efficient arrangements.

2.7.6 Equalization arrangements requiring mechanical aids such as valves or cross-levelling pipes, if fitted, are to be considered for the purpose of reducing an angle of heel or attaining the minimum range of residual stability to meet the requirements of 2.9 and sufficient residual stability is to be maintained during all stages where equalization is used. Spaces which are linked by ducts of large cross-sectional area may be considered to be common.

2.7.7 If pipes, ducts, trunks or tunnels are situated within the assumed extent of damage penetration, as defined in 2.5, arrangements are to be such that progressive flooding cannot thereby extend to compartments other than those assumed to be flooded for each case of damage.

2.7.8 The buoyancy of any superstructure directly above the side damage is to be disregarded. The unflooded parts of superstructures beyond the extent of damage, however, may be taken into consideration provided that:

.1 they are separated from the damaged space by watertight divisions and the requirements of 2.9.3 in respect of these intact spaces are complied with; and

.2 openings in such divisions are capable of being closed by remotely operated sliding watertight doors and unprotected openings are not immersed within the minimum range of residual stability required in 2.9; however, the immersion of any other openings capable of being closed weathertight may be permitted.

2.8 Standard of damage

2.8.1 Ships are to be capable of surviving the damage indicated in 2.5 with the flooding assumptions in 2.7 to the extent determined by the ship's type according to the following standards:

.1 A Type 1 ship is to be assumed to sustain damage anywhere in its length;

.2 A Type 2 ship of more than 150 [m] in length is to be assumed to sustain damage anywhere in its length;

.3 A Type 2 ship of 150 [m] in length or less is to be assumed to sustain damage anywhere in its length except involving either of the bulkheads bounding a machinery space located aft;

.4 A Type 3 ship of more than 225 [m] in length is to be assumed to sustain damage anywhere in its length;

.5 A Type 3 ship of 125 [m] in length or more but not exceeding 225 m in length is to be assumed to sustain damage anywhere in its length except involving either of the bulkheads bounding a machinery space located aft;

.6 A Type 3 ship below 125 [m] in length is to be assumed to sustain damage anywhere in its length except involving damage to the machinery space when located aft. However, the ability to survive

the flooding of the machinery spaces would be considered by IRS.

2.8.2 In the case of small Type 2 and Type 3 ships which do not comply in all respects with the appropriate requirements of 2.8.1.3 and 2.8.1.6, special dispensation may only be considered by the Administration provided that alternative measures can be taken which maintain the same degree of safety. The nature of the alternative measures are to be approved and clearly stated and be available to the Port Administration. Any such dispensation is to be duly noted on the International Certificate of fitness referred to in 1.5.4.

2.9 Survival requirements

2.9.1 Ships subject to this Chapter are to be capable of surviving the assumed damage specified in 2.5 to the standard provided in 2.8 in a condition of stable equilibrium and are to satisfy the following criteria.

2.9.2 In any stage of flooding:

.1 the waterline, taking into account sinkage, heel and trim, are to be below the lower edge of any opening through which progressive flooding or down-flooding may take place. Such openings are to include air pipes and openings which are closed by means of weathertight doors or hatch covers and may exclude those openings closed by means of :

- watertight manhole covers and watertight flush scuttles,
- small watertight cargo tank hatch covers which maintain the high integrity of the deck,
- remotely operated watertight sliding doors,
- hinged watertight access doors with open/closed indication locally and at the navigation bridge, of the quick-acting or single-action type that are normally closed at sea,
- hinged watertight doors that are permanently closed at sea, and ;
- sidescuttles of the non-opening type;

.2 the maximum angle of heel due to unsymmetrical flooding is not to exceed 25°, except that this angle may be increased to 30° if no deck immersion occurs;

.3 the residual stability during intermediate stages of flooding is to be to the satisfaction of IRS. However, it is never to be significantly less than that required by 2.9.3.

2.9.3 At final equilibrium after flooding:

.1 the righting lever curve is to have a minimum range of 20° beyond the position of equilibrium in association with a maximum residual righting lever (GZ_{max}) of at least 0.1 [m] within the 20° range; the area under the curve within this range is not to be less than 0.0175 [m/rad]. Unprotected openings are not to be immersed within this range unless the space concerned is assumed to be flooded. Within this range, the immersion of any of the openings listed in 2.9.2.1 and other openings capable of being closed weathertight may be permitted; and

IR.1 Other openings capable of being closed weathertight do not include ventilators (complying with ILLC 19(4)) that for operational reasons have to remain open to supply air to the engine room or emergency generator room (if the same is considered buoyant in the stability calculation or protecting openings leading below) for the effective operation of the ship.

.2 the emergency source of power is to be capable of operating.

Section 3

Ship Arrangements

3.1 Cargo segregation

3.1.1 Unless expressly provided otherwise, tanks containing cargo or residues of cargo subject to the requirements of this Chapter are to be segregated from accommodation, service and machinery spaces and from drinking water and stores for human consumption by means of a cofferdam, void space, cargo pump-room, pump-room, empty tank, oil fuel tank or other similar space. Location of fuel tanks in cargo area are to comply with 2.2.5 of Chapter 2 “Oil Tankers”

3.1.2 Cargo piping is not to pass through any accommodation, service or machinery space other than cargo pump-rooms or pump-rooms.

3.1.3 Cargoes, residues of cargoes or mixtures containing cargoes which react in a hazardous manner with other cargoes, residues or mixtures, are to :

.1 be segregated from such other cargoes by means of a cofferdam, void space, cargo pump-room, pump-room, empty tank, or tank containing a mutually compatible cargo;

.2 have separate pumping and piping systems which are not to pass through other cargo tanks containing such cargoes, unless encased in a tunnel; and

.3 have separate tank venting systems.

3.1.4 If cargo piping systems or cargo ventilation systems are to be separated, this separation may be achieved by the use of design or operational methods. Operational methods shall not be used within a cargo tank and shall consist of one of the following types:

.1 removing spool-pieces or valves and blanking the pipe ends;

.2 arrangement of two spectacle flanges in series, with provisions for detecting leakage into the pipe between the two spectacle flanges.

3.1.5 Cargoes subject to the requirements of this Chapter are not to be carried in either the fore or aft peak tank.

3.2 Accommodation, service and machinery spaces and control stations

3.2.1 No accommodation or service spaces or control stations are to be located within the cargo area except over a cargo pump-room recess or pump-room recess that complies with regulation 11-2/56 of the 1983 SOLAS amendments and no cargo or slop tank is to be aft of the forward end of any accommodation.

3.2.2 In order to guard against the danger of hazardous vapours, due consideration is to be given to the location of air intakes and openings into accommodation, service and machinery spaces and control stations in relation to cargo piping and cargo vent systems.

3.2.3 Entrances, air inlets and openings to accommodation, service and machinery spaces and control stations are not to face the cargo area. They are to be located on the end bulkhead not facing the cargo area and/or on the outboard side of the superstructure or deckhouse at a distance of at least 4% of the length of the ship but not less than 3 [m] from the end of the superstructure or deckhouse facing the cargo area. This distance, however, need not exceed 5 [m]. No doors are to be permitted within the limits mentioned above, except that doors to those spaces not having access to accommodation and service spaces and control stations, such as cargo control stations and store-rooms may be fitted. Where such doors are fitted, the boundaries of the space are to be insulated to "A-60" standard. Bolted plates for removal of machinery may be fitted within the limits specified above. Wheelhouse doors and wheelhouse windows may be located within the limits specified above so long as they are so designed that a rapid and efficient gas and vapour tightening of the wheelhouse can be ensured. Windows and sidescuttles facing the cargo area and on

the sides of the superstructures and deckhouses within the limits specified above are to be of the fixed (non-opening) type. Such sidescuttles in the first tier on the main deck are to be fitted with inside covers of steel or equivalent material.

IR3.2.3 Access to forecastle spaces containing sources of ignition may be permitted through doors facing cargo area provided the doors are located outside hazardous areas as defined in IEC Publication 60092-502:1999.

3.3 Cargo pump-rooms

3.3.1 Cargo pump-rooms are to be so arranged as to ensure:

.1 unrestricted passage at all times from any ladder platform and from the floor; and
.2 unrestricted access to all valves necessary for cargo handling for a person wearing the required personnel protective equipment.

3.3.2 Permanent arrangements are to be made for hoisting an injured person with a rescue line while avoiding any projecting obstacles.

3.3.3 Guard railings are to be installed on all ladders and platforms.

3.3.4 Normal access ladders are not to be fitted vertical and are to incorporate platforms at suitable intervals.

3.3.5 Means are to be provided to deal with drainage and any possible leakage from cargo pumps and valves in cargo pump-rooms. The bilge system serving the cargo pump-room is to be operable from outside the cargo pump-room. One or more slop tanks for storage of contaminated bilge water or tank washings are to be provided. A shore-connection with a standard coupling or other facilities are to be provided for transferring contaminated liquids to on-shore reception facilities.

3.3.6 Pump discharge pressure gauges are to be provided outside the cargo pump-room.

3.3.7 Where machinery is driven by shafting passing through a bulkhead or deck, gastight seals with efficient lubrication or other means of ensuring the permanence of the gas seal are to be fitted in way of the bulkhead or deck.

3.4 Access to spaces in the cargo area

3.4.1 Access to cofferdams, ballast tanks, cargo tanks and other spaces in the cargo area is to be direct from the open deck and such as to ensure their complete inspection. Access to double bottom spaces may be through a cargo pump-room, pump-room, deep cofferdam, pipe tunnel or similar compartments, subject to consideration of ventilation aspects.

3.4.2 For access through horizontal openings, hatches or manholes, the dimensions are to be sufficient to allow a person wearing a self-contained air-breathing apparatus and protective equipment to ascend or descend any ladder without obstruction and also to provide a clear opening to facilitate the hoisting of an injured person from the bottom of the space. The minimum clear opening is to be not less than 600 [mm] by 600 [mm].

3.4.3 For access through vertical openings, or manholes providing passage through the length and breadth of the space, the minimum clear opening is to be not less than 600 [mm] by 800 [mm] at a height of not more than 600 [mm] from the bottom shell plating unless gratings or other footholds are provided.

3.4.4 Smaller dimensions may be approved by IRS in special circumstances, if the ability to traverse such openings or to remove an injured person can be proved to the satisfaction of IRS.

3.5 Bilge and ballast arrangements

3.5.1 Pumps, ballast lines, vent lines and other similar equipment serving permanent ballast tanks are to be independent of similar equipment serving cargo tanks and of cargo tanks themselves. Discharge arrangements for permanent ballast tanks sited immediately adjacent to cargo tanks are to be outside machinery spaces and accommodation spaces. Filling arrangements may be in the machinery spaces provided that such arrangements ensure filling from tank deck level and non-return valves are fitted.

3.5.2 Filling of ballast in cargo tanks may be arranged from deck level by pumps serving permanent ballast tanks, provided that the filling line has no permanent connection to cargo tanks or piping and that non-return valves are fitted.

3.5.3 Bilge pumping arrangements for cargo pump-rooms, pump-rooms, void spaces, slop tanks, double bottom tanks and similar spaces

are to be situated entirely within the cargo area except for void spaces, double bottom tanks and ballast tanks where such spaces are separated from tanks containing cargo or residues of cargo by a double bulkhead.

3.6 Pump and pipeline identification

3.6.1 Provisions are to be made for the distinctive marking of pumps, valves and pipelines to identify the service and tanks which they serve.

3.7 Bow or stern loading and unloading arrangements

3.7.1 Cargo piping may be fitted to permit bow or stern loading and unloading. Portable arrangements are not permitted.

3.7.2 Bow or stern loading and unloading lines are not to be used for the transfer of products required to be carried in type 1 ships. Bow and stern loading and unloading lines are not to be used for the transfer of cargoes emitting toxic vapours required to comply with 15.12.1, unless specifically approved by IRS.

3.7.3 In addition to the requirements of 5.1 the following provisions apply :

.1 The piping outside the cargo area is to be fitted at least 760 [mm] inboard on the open deck. Such piping is to be clearly identified and fitted with a shutoff valve at its connection to the cargo piping system within the cargo area. At this location, it is also to be capable of being separated by means of a removable spool piece and blank flanges when not in use.

.2 The shore-connection is to be fitted with a shutoff valve and a blank flange.

.3 The piping is to be full penetration butt welded, and fully radiographed. Flange connections in the piping are only to be permitted within the cargo area and at the shore-connection.

.4 Spray shields are to be provided at the connections specified in .1 as well as collecting trays of sufficient capacity with means for the disposal of drainage.

.5 The piping is to be self-draining to the cargo area and preferably into a cargo tank. Alternative arrangements for draining the piping may be accepted by IRS.

.6 Arrangements are to be made to allow such piping to be purged after use and maintained gas-safe when not in use. The vent pipes connected with the purge are to be located in the cargo area. The relevant connections to the piping are to be provided with shutoff valve and blank flange.

3.7.4 Entrances, air inlets and openings to accommodation, service and machinery spaces and control stations are not to face the cargo shore-connection location of bow or stern loading and unloading arrangements. They are to be located on the outboard side of the superstructure or deckhouse at a distance of at least 4% of the length of the ship but not less than 3 [m] from the end of the house facing the cargo shore-connection location of the bow or stern loading and unloading arrangements. This distance, however, need not exceed 5 [m]. Sidescuttles facing the shore-connection location and on the sides of the superstructure or deckhouse within the distance mentioned above are to be of the fixed (non-opening) type. In addition, during the use of the bow or stern loading and unloading arrangements, all doors, ports and other openings on the corresponding superstructure or deck-house side should be closed. Where, in the case of small ships, compliance with 3.2.3 and this paragraph is not possible, IRS may approve relaxations from the above requirements.

3.7.5 Air pipes and other openings to enclosed spaces not listed in 3.7.4 are to be shielded from any spray which may come from a burst hose or connection.

3.7.6 Escape routes are not to terminate within the coamings required by 3.7.7 or within a distance of 3 [m] beyond the coamings.

3.7.7 Continuous coamings of suitable height are to be fitted to keep any spills on deck and away from the accommodation and service areas.

3.7.8 Electrical equipment within the coamings required by 3.7.6 or within a distance of 3 [m] beyond the coamings is to be in accordance with the requirements of Sec.10.

3.7.9 Fire-fighting arrangements for the bow or stern loading and unloading areas are to be in accordance with 11.3.16.

3.7.10 Means of communication between the cargo control station and the cargo shore-connection location are to be provided and certified safe, if necessary. Provision is to be made for the remote shutdown of cargo pumps from the cargo shore-connection location.

IR3.8 Emergency towing arrangement (ETA)

IR3.8.1 All chemical carriers of deadweight 20,000 tonnes and above are to be provided with emergency towing arrangements at both forward and aft ends. Emergency towing arrangements are to be as per Pt.5, Ch.2, Sec.2.

Section 4

Cargo Containment

4.1 Definitions

4.1.1 **Independent tank** means a cargo containment envelope which is not contiguous with, or part of, the hull structure. An independent tank is built and installed so as to eliminate whenever possible (or in any event to minimize) its stressing as a result of stressing or motion of the adjacent hull structure. An independent tank is not essential to the structural completeness of the ship's hull.

4.1.2 **Integral tank** means a cargo containment envelope which forms part of the ship's hull and which may be stressed in the same manner and by the same loads which stress the contiguous hull structure and which is normally essential to the structural completeness of the ship's hull.

4.1.3 **Gravity tank** means a tank having a design pressure not greater than 0.7 bar gauge at the top of the tank. A gravity tank may be independent or integral. A gravity tank is to be constructed and tested according to recognized standards taking account of the temperature of carriage and relative density of the cargo.

4.1.4 **Pressure tank** means a tank having a design pressure greater than 0.7 bar gauge. A pressure tank is to be an independent tank and is to be of a configuration permitting the application of pressure vessel design criteria according to recognized standards.

4.2 Tank type requirements for individual products

4.2.1 Requirements for both installation and design of tank types for individual products are shown in column "f" in Sec.17, Table 17.1.1.

Section 5

Cargo Transfer

5.1 Piping scantlings

5.1.1 Subject to the requirements of 5.1.4 the wall thickness [t] of pipes is not to be less than:

$$t = \frac{t_o + b + c}{1 - \frac{a}{100}} [mm]$$

where,

t_o = theoretical thickness

$t_o = PD/(2Ke+P)$ [mm]

where,

P = design pressure [MPa] referred to in 5.1.2

D = outside diameter [mm]

K = allowable stress [N/mm^2] referred to in 5.1.5

e = efficiency factor; equal to 1.0 for seamless pipes and for longitudinally or spirally welded pipes delivered by approved manufacturers of

welded pipes which are considered equivalent to seamless pipes when non-destructive testing on welds is carried out in accordance with recognized standards. In other cases, an efficiency factor of less than 1.0, in accordance with recognized standards may be required depending on the manufacturing process.

b = allowance for bending [mm]. The value of b is to be chosen so that the calculated stress in the bend, due to internal pressure only, does not exceed the allowable stress. Where such justification is not given, b is to be not less than:

$$b = \frac{D\, t_o}{2.5\, r} \text{ [mm]}$$

where,

r = mean radius of the bend [mm]

c = corrosion allowance [mm]. If corrosion or erosion is expected, the wall thickness of piping is to be increased over that required by the other design requirements
a = negative manufacturing tolerance for thickness (per cent).

5.1.2 The design pressure P in the formula for to in 5.1.1 is the maximum gauge pressure to which the system may be subjected in service, taking into account the highest set pressure on any relief valve on the system.

5.1.3 Piping and piping system components which are not protected by a relief valve, or which may be isolated from their relief valve, are to be designed for at least the greatest of:

.1 piping systems or components which may contain some liquid, the saturated vapour pressure at 45°C;

.2 the pressure setting of the associated pump discharge relief valve;

.3 the maximum possible total pressure head at the outlet of the associated pumps when a pump discharge relief valve is not installed.

5.1.4 The design pressure is not to be less than 1 [MPa] gauge except for open-ended lines where it is to be not less than 0.5 [MPa] gauge.

5.1.5 For pipes, the allowable stress to be considered in the formula for to in 5.1.1 is the lower of the following values:

$$\frac{R_m}{A} \text{ or } \frac{R_e}{B}$$

where,

R_m = specified minimum tensile strength at ambient temperature [N/mm^2]

R_e = specified minimum yield stress at ambient temperature [N/mm^2]. If the stress-strain curve does not show a defined yield stress, the 0.2% proof stress applies.

A and B are to have values of at least

A = 2.7

B = 1.8.

5.1.6.1 The minimum wall thickness is to be in accordance with the recognized standards.

5.1.6.2 Where necessary for mechanical strength to prevent damage, collapse, excessive sag or buckling of pipes due to weight of pipes and content and to superimposed loads from supports, ship deflection or other causes, the wall thickness is to be increased over that required by 5.1.1 or, if this is impracticable or would cause excessive local stresses, these loads are to be reduced, protected against or eliminated by other design methods.

5.1.6.3 Flanges, valves and other fittings are to be in accordance with recognized standards, taking into account the design pressure defined under 5.1.2.

5.1.6.4 For flanges not complying with a standard the dimensions of flanges and associated bolts are to be to the satisfaction of IRS.

5.2 Piping fabrication and joining details

5.2.1 The requirements of this Section apply to piping inside and outside the cargo tanks. However, relaxations from these requirements may be accepted in accordance with recognized standards for open-ended piping and for piping inside cargo tanks except for cargo piping serving other cargo tanks.

5.2.2 Cargo piping is to be joined by welding except:

.1 for approved connections to shutoff valves and expansion joints; and

.2 for other exceptional cases specifically approved by IRS.

5.2.3 The following direct connections of pipe lengths, without flanges may be considered:

.1 Butt welded joints with complete penetration at the root may be used in all applications.

.2 Slip-on welded joints with sleeves and related welding having dimensions in accordance with recognized standards are only to be used for pipes with an external diameter of 50 [mm] or less. This type of joint is not to be used when crevice corrosion is expected to occur.

.3 Screwed connections in accordance with recognized standards are only to be used for accessory lines and instrumentation lines with external diameters of 25 [mm] or less.

5.2.4 Expansion of piping is normally to be allowed for by the provision of expansion loops or bends in the piping system.

.1 Bellows in accordance with recognized standards may be specially considered.

.2 Slip joints are not to be used.

5.2.5 Welding, post weld heat treatment and non-destructive testing are to be performed in accordance with Part 4 of IRS Rules.

5.3 Flange connections

5.3.1 Flanges are to be of the welded neck, slip-on or socket-welded type. However, socket-welded-type flanges are not to be used in nominal size above 50 [mm].

5.3.2 Flanges are to comply with recognized standards as to their type, manufacture and test.

5.4 Test requirements for piping

5.4.1 The test requirements of this Section apply to piping inside and outside cargo tanks. However, relaxations from these requirements may be accepted in accordance with recognized standards for piping inside tanks and open-ended piping.

5.4.2 After assembly, each cargo piping system is to be subject to a hydrostatic test to at least 1.5 times the design pressure. When piping systems or parts of systems are completely manufactured and equipped with all fittings, the hydrostatic test may be conducted prior to installation aboard the ship. Joints welded on board are to be hydrostatically tested to at least 1.5 times the design pressure.

5.4.3 After assembly on board, each cargo piping system is to be tested for leaks to a pressure depending on the method applied.

5.5 Piping arrangements

5.5.1 Cargo piping is not to be installed under deck between the outboard side of the cargo-containment spaces and the skin of the ship unless clearances required for damage protection (see 2.6) are maintained; but such distances may be reduced where damage to the pipe would not cause release of cargo provided that the clearance required for inspection purposes is maintained.

5.5.2 Cargo piping, located below the main deck, may run from the tank it serves and penetrate tank bulkheads or boundaries common to longitudinally or transversally adjacent cargo tanks, ballast tanks, empty tanks, pump-rooms or cargo pump-rooms provided that inside the tank it serves, it is fitted with a stop valve operable from the weather deck and provided cargo compatibility is assured in the event of piping failure. As an exception, where a cargo tank is adjacent to a cargo pump-room, the stop valve operable from the weather deck may be situated on the tank bulkhead on the cargo pump-room side, provided an additional valve is fitted between the bulkhead valve and the cargo pump. A totally enclosed hydraulically operated valve located outside the cargo tank may, however, be accepted provided that the valve is:

.1 designed to preclude the risk of leakage;

.2 fitted on the bulkhead of the cargo tank which it serves;

.3 suitably protected against mechanical damage;

.4 fitted at a distance from the shell, as required for damage protection; and

.5 operable from the weather deck.

5.5.3 In any cargo pump-room where a pump serves more than one tank, a stop valve is to be fitted in the line to each tank.

5.5.4 Cargo piping installed in pipe tunnels is also to comply with the requirements of 5.5.1 and 5.5.2. Pipe tunnels are to satisfy all tank requirements for construction, location and ventilation and electrical hazard requirements. Cargo compatibility is to be assured in the event of a piping failure. The tunnel is not to have any other openings except to the weather deck and cargo pump-room or pump-room.

5.5.5 Cargo piping passing through bulkheads is to be so arranged as to preclude excessive stresses at the bulkhead and is not to utilize flanges bolted through the bulkhead.

5.6 Cargo transfer control systems

5.6.1 For the purpose of adequately controlling the cargo, cargo transfer systems are to be provided with:

.1 one stop valve capable of being manually operated on each tank filling and discharge line, located near the tank penetration; if an individual deepwell pump is used to discharge the contents of a cargo tank, a stop valve is not required on the discharge line of that tank;

.2 one stop valve at each cargo hose connection;

.3 remote shutdown devices for all cargo pumps and similar equipment.

5.6.2 The controls necessary during transfer or transport of cargoes covered by this Chapter other than in cargo pump-rooms which have been dealt with elsewhere in this Chapter are not to be located below the weather deck.

5.6.3 For certain products additional cargo transfer control requirements are shown in column "o" in Sec.17, Table 17.1.1.

5.7 Ship's cargo hoses

5.7.1 Liquid and vapour hoses used for cargo transfer are to be compatible with the cargo and suitable for the cargo temperature.

5.7.2 Hoses subject to tank pressure or the discharge pressure of pumps are to be designed for a bursting pressure not less than 5 times the maximum pressure the hose will be subjected to during cargo transfer.

5.7.3 Each new type of cargo hose, complete with end fittings, is to be prototype-tested at a normal ambient temperature with 200 pressure cycles from zero to at least twice the specified maximum working pressure. After this cycle pressure test has been carried out, the prototype test is to demonstrate a bursting pressure not less than 5 times its specified maximum working pressure at the intended extreme service temperature. Hoses used for prototype testing are not to be used for cargo service. Thereafter, before being placed in service, each new length of cargo hose produced is to be hydrostatically tested at ambient temperature to a pressure not less than 1.5 times its specified maximum working pressure but not more than two-fifths of its bursting pressure. The hose is to be stenciled or otherwise marked with its specified maximum working pressure and, if used in other than ambient temperature services, its maximum and minimum service temperature, as applicable. The specified maximum working pressure is not to be less than 1 [MPa] gauge.

IR5.8 Integrated cargo and ballast system

IR5.8.1 The requirements given in Pt.5, Ch.2, Sec.7.6 are to be applied.

Section 6

Materials of Construction, Protective Linings and Coatings

6.1 Structural materials used for tank construction, together with associated piping, pumps, valves, vents and their jointing materials, are to be suitable at the temperature and pressure for the cargo to be carried in accordance with recognized standards. Steel is assumed to be the normal material of construction.

6.2 The shipyard is responsible for providing compatibility information to the ship operator and/or master. This must be done in a timely manner before delivery of the ship or on completion of a relevant modification of the material of construction.

6.3 Where applicable the following are to be taken into account in selecting the material of construction:

.1 notch ductility at the operating temperature;

.2 corrosive effect of the cargo;

.3 possibility of hazardous reactions between the cargo and the material of construction; and

.4 suitability of linings.

6.4 The shipper of the cargo is responsible for providing compatibility information to the ship operator and/or master. This must be done in a timely manner before transportation of the product. The cargo shall be compatible with all materials of construction such that:

.1 no damage to the integrity of the materials of construction is incurred; and/or

.2 no hazardous, or potentially hazardous reaction is created.

IR 6.5 For use of aluminium coatings onboard chemical tankers also see 1.3.4 & 1.3.5 of Ch.2.

Section 7

Cargo Temperature Control

7.1 General

7.1.1 When provided, any cargo heating or cooling systems are to be constructed, fitted and tested to the satisfaction of IRS. Materials used in the construction of temperature-control systems are to be suitable for use with the product intended to be carried.

7.1.2 Heating or cooling media is to be of a type approved for use with the specific cargo. Consideration is to be given to the surface temperature of heating coils or ducts to avoid dangerous reactions from localized overheating or overcooling of cargo. (See also 15.13.7).

7.1.3 Heating or cooling systems are to be provided with valves to isolate the system for each tank and to allow manual regulation of flow.

7.1.4 In any heating or cooling system, means are to be provided to ensure that, when in any condition other than empty, a higher pressure can be maintained within the system than the maximum pressure head that could be exerted by the cargo-tank contents on the system.

7.1.5 Means are to be provided for measuring the cargo temperature as under:

.1 The means for measuring the cargo temperature are to be of restricted or closed type, respectively, when a restricted or closed gauging device is required for individual substances as shown in column "j" in Sec.17, Table 17.1.1.

.2 A restricted temperature measuring device is subject to the definition for a restricted gauging device in 13.1.1.2, e.g. a portable thermometer lowered inside a gauge tube of the restricted type.

.3 A closed temperature-measuring device is subject to the definition for closed gauging device in 13.1.1.3, e.g. a remote-reading thermometer of which the sensor is installed in the tank.

.4 When overheating or overcooling could result in a dangerous condition, an alarm system which monitors the cargo temperature is to be provided. (See also operational requirements in 16.6).

7.1.6 When products for which 15.12, 15.12.1 or 15.12.3 are listed in column "o" in the table of Sec.17, Table 17.1.1 are being heated or cooled, the heating or cooling medium is to operate in a circuit:

.1 which is independent of other ship's services, except for another cargo heating or cooling system, and which does not enter the machinery space; or

.2 which is external to the tank carrying toxic products; or

.3 where the medium is sampled to check for the presence of cargo before it is recirculated to other services of the ship or into the machinery space. The sampling equipment is to be located within the cargo area and be capable of detecting the presence of any toxic cargo being heated or cooled. Where this method is used, the coil return is to be tested not only at the commencement of heating or cooling or a toxic product, but also on the first occasion the coil is used subsequent to having carried an unheated or uncooled toxic cargo.

7.2 Additional requirements

7.2.1 For certain products, additional requirements contained in Sec.15 are shown in column "o" in Sec.17, Table 17.1.1.

Section 8

Cargo Tank Venting and Gas-freeing Arrangement

8.1 Application

8.1.1 Unless expressly provided otherwise, this section applies to ship's constructed on or after 1 January 1994.

8.1.2 Ship's constructed before 1 January 1994 should comply with the requirements of Chapter 8 of this Code which were inforce prior to the said date.

8.1.3 For the purpose of this regulation, the term "ship constructed" is as defined in regulation II-1/1.3.1 of the 1974 SOLAS convention as amended.

8.1.4 Ships constructed on or after 1 July 1986 but before 1 January 1994 which fully comply with the requirements of the chapter applicable at that time may be regarded as complying with the requirements of regulation II-2/59 of SOLAS 74.

8.1.5 For ships to which the Chapter applies, the requirements of this Section should apply in lieu of SOLAS regulations II-2/4.5.3, 4.5.6 and 16.3.2.

8.1.6 Ships constructed on or after 1 July 1986, but before 1 July 2002 are to comply with the requirements of paragraph 8.3.3 by the date of the first scheduled dry-docking after 1 July 2002, but not later than 1 July 2005. However, IRS may approve relaxation of paragraph 8.3.3 for ships of less than 500 gross tonnage which were constructed on or after 1 July 1986, but before 1 July 2002.

8.2 Cargo tank venting

8.2.1 All cargo tanks are to be provided with a venting system appropriate to the cargo being carried and these systems are to be independent of the air pipes and venting systems of all other compartments of the ship. Tank venting systems are to be designed so as to minimize the possibility of cargo vapour accumulating about the decks, entering accommodation, service and machinery spaces and control stations and in the case of flammable vapours entering or collecting in spaces or areas containing sources of ignition. Tank venting systems are to be arranged to prevent entrance of water into the cargo tanks and at the same time, vent outlets should direct the vapour discharge upwards in the form of unimpeded jets.

8.2.2 The venting systems are to be connected to the top of each cargo tank and as far as practicable the cargo vent lines are to be self-draining back to the cargo tanks under all normal operational conditions of list and trim. Where it is necessary to drain venting systems above the level of any pressure/vacuum valve, capped or plugged drain cocks are to be provided.

8.2.3 Provision is to be made to ensure that the liquid head in any tank does not exceed the design head of that tank. Suitable high-level alarms, overflow control systems or spill valves, together with gauging and tank filling procedures may be accepted for this purpose. Where the means of limiting cargo tank overpressure includes an automatic closing valve, the valve is to comply with the appropriate provisions 15.18.

8.2.4 Tank venting systems are to be designed and operated so as to ensure that neither pressure nor vacuum created in the cargo tanks during loading or unloading exceeds tank design parameters. The main factors to be considered in the sizing of a tank venting system are as follows:

.1 design loading and unloading rate;

.2 gas evolution during loading; this should be taken account of by multiplying the maximum loading rate by a factor of at least 1.25;

.3 density of the cargo vapour mixture;

.4 pressure less in vent piping and across valves and fittings;

.5 pressure/vacuum settings of relief devices.

8.2.5 Tank vent piping connected to cargo tanks of corrosion-resistant material, or to tanks which are lined or coated to handle special cargoes, as required by this Chapter,

is to be similarly lined or coated, or constructed of corrosion-resistant material.

8.2.6 The master is to be provided with the maximum permissible loading and unloading rates for each tank or group of tanks consistent with design of the venting systems.

8.3 Types of tank venting systems

8.3.1 **An Open tank venting system** is a system which offers no restriction except for friction losses to the free flow of cargo vapours to and from the cargo tanks during normal operations. An open venting system may consist of individual vents from each tank, or such individual vents may be combined into a common header or headers, with due regard to cargo segregation. In no case should shut-off valves be fitted either to the individual vents or to the header.

8.3.2 **A Controlled tank venting system** is a system in which pressure and vacuum relief valves or pressure/vacuum valves are fitted to each tank to limit the pressure or vacuum in the tank. A controlled venting system may consist of individual vents from each tank or such individual vents on the pressure side only as may be combined into a common header or headers with due regard to cargo segregation. In no case should shut-off valves be fitted either above or below pressure or vacuum relief valves or pressure/vacuum valves. Provision may be made for bypassing a pressure or vacuum valve or pressure/vacuum valve under certain operating conditions provided that the requirements of 8.2.5 is maintained and that there is suitable indication to show whether or not the valve is bypassed.

IR8.3.2 By-passing of high velocity valves is not allowed. In other cases by-passing pressure/vacuum valves may be allowed during cargo operations for cargoes which do not require a vapour return system, provided the vent-line outlet is fitted with flame arrestors and is located at a height above deck level as required by 8.3.4.1.

8.3.3 On ships constructed on or after 1 July 2002, controlled tank venting systems should consist of a primary and a secondary means of allowing full flow relief of vapour to prevent over-pressure or under-pressure in the event of failure of one means. Alternatively, the secondary means may consist of pressure sensors fitted in each tank with a monitoring system in the ship's cargo control room or position from which cargo operations are normally carried out. Such monitoring equipment should also provide an alarm facility which is activated by detection of over-pressure or under-pressure conditions within a tank.

8.3.4 The position of vent outlets of a controlled tank venting system is to be arranged:

.1 at a height of not less than 6 [m] above the weather deck or above a raised walkway if fitted within 4 [m] of the raised walkway;

.2 at a distance of at least 10 [m] measured horizontally from the nearest air intake or opening to accommodation, service and machinery spaces and ignition sources.

8.3.5 The vent outlet height referred to in 8.3.4.1 may be reduced to 3 [m] above the deck or a raised walkway, as applicable, provided that high velocity venting valves of an approved type directing the vapour/air mixture upwards in a unimpeded jet with an exit velocity of at least 30 [m/s] are fitted.

8.3.6 Controlled tank venting systems fitted to tanks to be used for cargoes having a flashpoint not exceeding 60°C (closed cup test) are to be provided with devices to prevent the passage of flame into the cargo tanks. The design, testing and locating of the devices should contain at least the standards adopted by the Organisation.

8.3.7 In designing venting systems and in the selection of devices to prevent the passage of flame for incorporation into the tank venting system, due attention is to be paid to the possibility of the blockage of these systems and fittings by, for example, the freezing of cargo vapour, polymer build up, atmospheric dust or icing up in adverse weather conditions. In this context it is to be noted that flame arresters and flame screens are more susceptible to blockage. Provisions are to be made such that the system and fittings may be inspected, operationally checked, cleaned or renewed as applicable.

8.3.8 Reference in 8.3.1 and 8.3.2 to the use of shut-off valves in the venting lines are to be interpreted to extend to all other means of stoppage including spectacle blanks and blank flanges.

8.4 Venting requirements for individual products

8.4.1 Venting requirements for individual products are shown in column "g" and additional requirements in column "o" in Sec.17, Table 17.1.1.

8.5 Cargo Tank Purging

8.5.1 When the application of inert gas is required by 11.1.1, before gas-freeing, the cargo tanks are to be purged with inert gas through outlet pipes with cross-sectional area such that an exit velocity of at least 20 [m/s] can be maintained when any three tanks are being simultaneously supplied with inert gas. The outlets are to extend not less than 2 [m] above the deck level. Purging is to continue until the concentration of hydrocarbon or other flammable vapours in the cargo tanks has been reduced to less than 2% by volume.

8.6 Cargo tank gas-freeing

8.6.1 The arrangements for gas-freeing cargo tanks used for cargoes other than those for which open venting is permitted are to be such as to minimize the hazards due to the dispersal of flammable or toxic vapours in the atmosphere and to flammable or toxic vapour mixtures in a cargo tank. Accordingly, gas-freeing operations are to be carried out such that vapour is initially discharged:

.1 through the vent outlets specified in 8.3.4 and 8.3.5; or

.2 through outlets at least 2 [m] above the cargo tank deck level with a vertical efflux velocity of at least 30 [m/s] maintained during the gas freeing operation; or

.3 through outlets at least 2 [m] above the cargo tank deck level with a vertical efflux velocity of at least 20 [m/s] which are protected by suitable devices to prevent the passage of flame.

When the flammable vapour concentration at the outlets has been reduced to 30% of the lower flammable limit and in the case of a toxic product the vapour concentration does not present a significant health hazard, gas freeing may thereafter be continued at cargo tank deck level.

8.6.2 The outlets referred to in 8.6.1.2 and 8.6.1.3 may be fixed or portable pipes.

8.6.3 In designing a gas-freeing system in conformity with 8.6.1 particularly in order to achieve the required exit velocities of 8.6.1.2 and 8.6.1.3, due consideration are to be given to the following:

.1 materials of construction of system;

.2 time to gas-free;

.3 flow characteristics of fans to be used;

.4 the pressure losses created by ducting, piping, cargo tank inlets and outlets;

.5 the pressure achievable in the fan driving medium (e.g. water or compressed air);

.6 the densities of the cargo vapour/air mixtures for the range of cargoes to be carried.

Section 9

Environmental Control

9.1 General

IR 9.1.1 The requirements of this Section are not requirements for the maintenance of class.

9.1.1 Vapour spaces within cargo tanks and, in some cases, spaces surrounding cargo tanks may require to have specially controlled atmospheres.

9.1.2 There are four different types of control for cargo tanks, as follows:

.1 **Inerting** by filling the cargo tank and associated piping systems and, where specified in Sec.15, the spaces surrounding the cargo tanks, with a gas or vapour which will not support combustion and which will not react with the cargo, and maintaining that condition.

.2 **Padding** by filling the cargo tank and associated piping systems with a liquid, gas or vapour which separates the cargo from the air, and maintaining that condition.

.3 **Drying** by filling the cargo tank and associated piping systems with moisture-free gas or vapour with a dewpoint of -40°C or below at atmospheric pressure, and maintaining that condition.

.4 **Ventilation** forced or natural.

9.1.3 Where inerting or padding of cargo tanks is required by this Chapter in column ‘h’ of Section 17:

.1 An adequate supply of inert gas for use in filling and discharging the cargo tanks is to be carried or is to be manufactured on board unless a shore supply is available. In addition, sufficient inert gas is to be available on the ship to compensate for normal losses during transportation.

.2 The inert gas system on board the ship is to be able to maintain a pressure of at least 0.07 bar gauge within the containment system at all times. In addition, the inert gas system is not to raise the cargo tank pressure to more than the tank's relief valve setting.

.3 Where padding is used, similar arrangements for supply of the padding medium is to be made as required for inert gas in .1 and .2.

.4 Means are to be provided for monitoring ullage spaces containing a gas blanket to ensure that the correct atmosphere is being maintained.

.5 Inerting or padding arrangements or both, where used with flammable cargoes, are to be such as to minimize the creation of static electricity during the admission of the inerting medium.

9.1.4 Where drying is used and dry nitrogen is used as the medium, similar arrangements for supply of the drying agent are to be made to those required in 9.1.3. Where drying agents are used as the drying medium on all air inlets to the tank, sufficient medium is to be carried for the duration of the voyage, taking into consideration the diurnal temperature range and the expected humidity.

9.2 Environmental control requirements for individual products

9.2.1 The required types of environmental control for certain products are shown in column "h" in Sec.17, Table 17.1.1.

Section 10

Electrical Installations

10.1 General

10.1.1 The provisions of this Section are applicable to ships carrying cargoes which are inherently, or due to their reaction with other substances, flammable or corrosive to the electrical equipment and are to be applied in conjunction with applicable electrical requirements of Part D, Chapter II-1 of SOLAS.

10.1.2.1 Electrical installations are to be such as to minimize the risk of fire and explosion from flammable products.

10.1.2.2 Where the specific cargo is liable to damage the materials normally used in electrical apparatus, due consideration is to be given to the particular characteristics of the materials chosen for conductors, insulation, metal parts, etc. As far as necessary, these components are to be protected to prevent contact with gases or vapours liable to be encountered.

10.1.3 IRS should take appropriate steps to ensure uniformity in the implementation and the application of the provisions of this Section in respect of electrical installations.

10.1.4 Electrical equipment, cables and wiring, is not to be installed in the hazardous locations unless it conforms with the standards not inferior to those acceptable to the Organization*. However, for locations not covered by such standards, electrical equipment, cables and wiring which do not conform to the standards may be installed in

hazardous locations based on a risk assessment to the satisfaction of the Administration, to ensure that an equivalent level of safety is assured.

* Reference is made to the recommendations published by the International Electrotechnical Commission, in particular to Publication IEC 60079-1-1 : 2002.

IR 10.1.4 Where the prescriptive requirements within SOLAS and related Codes (IBC, IGC) and the standards published by the International Electrotechnical Commission, such as but not limited to IEC 60092-502, are not aligned, the prescriptive requirements in SOLAS and Codes take precedence and are to be applied.

10.1.5 Where electrical equipment is installed in hazardous locations, as permitted in this Section, it is to be to the satisfaction of IRS and of certified type for operation in the flammable atmosphere concerned, as indicated in column "i" in Sec.17, Table 17.1.1.

10.1.6 For guidance, indication is given if the flashpoint of a substance is in excess of 60°C (closed cup test). In the case of heated cargo, carriage conditions might need to be established and the requirements for cargoes having a flashpoint not exceeding 60°C.

10.2 Bonding

Independent cargo tanks are to be electrically bonded to the hull. All gasketed cargo pipe joints and hose connections are to be electrically bonded.

10.3 Electrical requirements for individual products

Electrical requirements for individual products are shown in column "i" in Sec.17, Table 17.1.1.

Section 11

Fire Protection and Extinction

11.1 Application

11.1.1 The requirements for tankers in chapter II-2 of SOLAS would apply to ships covered by this chapter, irrespective of tonnage, including ships of less than 500 tons gross tonnage, except that:

.1 regulations 10.8 and 10.9 do not apply;

.2 regulation 4.5.1.2, i.e. the requirement for location of the main cargo control station, need not apply;

.3 regulations 10.2, 10.4 and 10.5 as applicable to cargo ships are to apply as they would apply to cargo ships of 2,000 tons gross tonnage and over;

.4 regulation 10.5.6 shall apply to ships of 2000 gross tonnage and over;

.5 the provisions of 11.3 are to apply in lieu of regulation 10.8; and

.6 the provisions of 11.2 are to apply in lieu of regulation 10.9.

.7 regulation 4.5.10 shall apply to ships of 500 gross tonnage and over, replacing “hydrocarbon gases” by “flammable vapours” in the regulation; and

.8 regulations 13.3.4 and 13.4.3 shall apply to ships of 500 gross tonnage and over.

IR11.1.1 The following interpretation is to be applied to 11.1.1.3 and 11.1.1.4 above:

a) SOLAS regulations II-2/10.2 and 10.4 apply to cargo ships of 500 gross tonnage and over, under SOLAS and to chemical carriers, regardless of size, under the IBC Code.

b) SOLAS II-2/10.5, except for sub-paragraph 10.5.6, applies to chemical carriers, regardless of size, constructed on/after 1 July 1986.

c) SOLAS II-2/10.5.6 applies only to chemical carriers constructed on/after 1 July 2002 and of 2,000 gt and above.

11.1.2 Notwithstanding the provisions of 11.1.1, ships engaged solely in the carriage of products which are non-flammable (entry NF in column 'i' of the table of minimum requirements) need not comply with requirements for tankers specified in SOLAS Chapter II-2, the requirements for cargo ships of that chapter, except that regulation 10.7 need not apply to such ships and 11.2 and 11.3 hereunder need not apply.

11.1.3 For ships engaged solely in the carriage of products with flashpoint above 60°C (entry "yes" in column 'i' of the table of minimum requirements) requirements of Chapter II-2 of SOLAS may apply as specified in regulation II-2/ 1.6.4 in lieu of the provisions of this Section.

11.1.4 In lieu of the provisions of SOLAS regulation II-2/1.6.7, the requirements of regulations II-2/4.5.10.1.1 and II-2/4.5.10.1.4 shall apply and a system for continuous monitoring of the concentration of flammable vapours shall be fitted on ships of 500 gross tonnage and over which were constructed before 1 January 2009 by the date of the first scheduled dry-docking after 1 January 2009, but not later than 1 January 2012. Sampling points or detector heads should be located in suitable positions in order that potentially dangerous leakages are readily detected. When the flammable vapour concentration reaches a pre-set level which shall not be higher than 10% of the lower flammable limit, a continuous audible and visual alarm signal shall be automatically effected in the pump-room and cargo control room to alert personnel to the potential hazard. However, existing monitoring systems already fitted having a pre-set level not greater than 30% of the lower flammable limit may be accepted. Notwithstanding the above provisions, the Administration may exempt ships not engaged on international voyages from those requirements.

11.2 Cargo pump-rooms

11.2.1 The cargo pump-room of any ship is to be provided with a fixed carbon dioxide fire-extinguishing system as specified in SOLAS regulation II-2/10.9.1.1. A notice is to be exhibited at the controls stating that the system is only to be used for fire-extinguishing and not for inerting purposes, due to the electrostatic ignition hazard. The alarms referred to in regulation (II-2/19.9.1.1.1) of SOLAS amendments are to be safe for use in a flammable cargo vapour-air mixture. For the purpose of this requirement, an extinguishing system is to be provided which would be suitable for machinery spaces. However, the amount of gas carried is to be sufficient to provide a quantity of free gas equal to 45% of the gross volume of the cargo pump-room in all cases.

11.2.2 Cargo pump-rooms of ships which are dedicated to the carriage of a restricted number of cargoes are to be protected by an appropriate fire- extinguishing system approved by IRS.

11.2.3 If cargoes are to be carried which are not suited to extinguishment by carbon dioxide or equivalent media, the cargo pump-room should be protected by a fire extinguishing system consisting of either a fixed pressure water spray or high expansion foam system. The International Certificate of Fitness for the Carriage of Dangerous Chemical in Bulk will reflect this conditional requirement.

11.3 Cargo area

11.3.1 Every ship is to be provided with a fixed deck foam system in accordance with the requirements of 11.3.2 to 11.3.12.

11.3.2 Only one type of foam concentrate is to be supplied, and it is to be effective for the maximum possible number of cargoes intended to be carried. For other cargoes for which foam is not effective or is incompatible, additional arrangements to the satisfaction of IRS are to be provided. Regular protein foams are not to be used.

11.3.3 The arrangements for providing foam are to be capable of delivering foam to the entire cargo tanks deck area as well as into any cargo tank, the deck of which is assumed to be ruptured.

11.3.4 The deck foam system is to be capable of simple and rapid operation. The main control station for the system is to be suitably located outside of the cargo area, adjacent to the accommodation spaces and readily accessible and operable in the event of fires in the area protected.

11.3.5 The rate of supply of foam solution is to be not less than the greatest of the following:

.1 2 [l/min] per square metre of the cargo tanks deck area, where cargo tanks deck

area means the maximum breadth of the ship times the total longitudinal extent of the cargo tank spaces;

.2 20 [l/min] per square metre of the horizontal sectional area of the single tank having the largest such area;

.3 10 [l/min] per square metre of the area protected by the largest monitor, such area being entirely forward of the monitor, but not less than 1,250 [l/min]. For ships of less than 4,000 tonnes deadweight, the minimum capacity of the monitor is to be to the satisfaction of IRS.

11.3.6 Sufficient foam concentrate is to be supplied to ensure at least 30 min of foam generation when using the highest of the solution rates stipulated in 11.3.5.1, 11.3.5.2 and 11.3.5.3.

11.3.7 Foam from the fixed foam system is to be supplied by means of monitors and foam applicators. At least 50% of the foam rate required in 11.3.5.1 or 11.3.5.2 is to be delivered from each monitor. The capacity of any monitor is to be at least 10 [l/min] of foam solution per square metre of deck area protected by that monitor, such area being entirely forward of the monitor. Such capacity is to be not less than 1,250 [l/min]. For ships of less than 4,000 tonnes deadweight, the minimum capacity of the monitor is to be to the satisfaction of IRS.

11.3.8 The distance from the monitor to the farthest extremity of the protected area forward of that monitor is to be not more than 75% of the monitor throw in still air conditions.

11.3.9 A monitor and hose connection for a foam applicator is to be situated both port and starboard at the poop front or accommodation spaces facing the cargo area.

11.3.10 Applicators are to be provided for flexibility of action during fire-fighting operations and to cover areas screened from the monitors. The capacity of any applicator is not to be less than 400 [l/min] and the applicator throw in still air conditions is to be not less than 15 [m]. The number of foam applicators provided are to be not less than four. The number and disposition of foam main outlets are to be such that foam from at least two applicators can be directed to any part of the cargo tanks deck area.

11.3.11 Valves are to be provided in the foam main, and in the fire main where this is an integral part of the deck foam system, immediately forward of any monitor position to isolate damaged sections of those mains.

11.3.12 Operation of a deck foam system at its required output is to permit the simultaneous use of the minimum required number of jets of water at the required pressure from the fire main.

11.3.13 Ships which are dedicated to the carriage of a restricted number of cargoes are to be protected by alternative provisions to the satisfaction of IRS when they are just as effective for the products concerned as the deck foam system required for the generality of flammable cargoes.

11.3.14 Suitable portable fire-extinguishing equipment for the products to be carried is to be provided and kept in good operating order.

11.3.15 Where flammable cargoes are to be carried all sources of ignition are to be excluded from hazardous locations referred to in 10.2.

11.3.16 Ships fitted with bow or stern loading and unloading arrangements are to be provided with one additional foam monitor meeting the requirements of 11.3.7 and one additional applicator meeting the requirements of 11.3.10. The additional monitor is to be located to protect the bow or stern loading and unloading arrangements. The area of the cargo line forward or aft of the cargo area is to be protected by the above-mentioned applicator.

11.4 Special requirements

Fire-extinguishing media considered to be suitable for certain products are listed for information in column "I" in Sec.17, Table 17.1.1.

Section 12

Mechanical Ventilation in the Cargo Area

12.0 For ships to which the Code applies, the requirements of this Section replace the requirements of SOLAS regulation II-2/4.5.2.6 and 4.5.4. However, for products addressed under paragraphs 11.1.2 and 11.1.3, except acids and products for which paragraph 15.16 apply, SOLAS regulation II-2/4.5.2.6 and 4.5.4 may apply in lieu of the provisions of this Section.

12.1 Spaces normally entered during cargo handling operation

12.1.1 Cargo pump-rooms and other enclosed spaces which contain cargo handling equipment and similar spaces in which work is performed on the cargo are to be fitted with mechanical ventilation systems, capable of being controlled from outside such spaces.

12.1.2 Provision is to be made to ventilate such spaces prior to entering the compartment and operating the equipment and a warning notice requiring the use of such ventilation is to be placed outside the compartment.

12.1.3 Mechanical ventilation inlets and outlets are to be arranged to ensure sufficient air movement through the space to avoid the accumulation of toxic or flammable vapours or both (taking into account their vapour densities) and to ensure sufficient oxygen to provide a safe working environment, but in no case is the ventilation system to have a capacity of less than 30 changes of air per hour based upon the total volume of the space. For certain products, increased ventilation rates for cargo pump-rooms are prescribed in 15.17.

12.1.4 Ventilation systems are to be permanent and are normally to be of the extraction type. Extraction from above and below the floor plates is to be possible. In rooms housing motors driving cargo pumps, the ventilation is to be of the positive pressure type.

12.1.5 Ventilation exhaust ducts from spaces within the cargo area is to discharge upwards in locations at least 10 [m] in the horizontal direction from ventilation intakes and openings to accommodation, service and machinery spaces and control stations and other spaces outside the cargo area.

12.1.6 Ventilation intakes are to be so arranged as to minimize the possibility of recycling hazardous vapours from any ventilation discharge opening.

12.1.7 Ventilation ducts are not to be led through accommodation, service and machinery spaces or other similar spaces.

12.1.8 Electric motors driving fans are to be placed outside the ventilation ducts if the carriage of flammable products is intended. Ventilation fans and fan ducts, in way of fans only, for hazardous locations referred to in Sec.10 are to be of non-sparking construction whose details are given in Pt.5, Ch.2, Sec.6.5 of these Rules.

12.1.9 Sufficient spare parts are to be carried for each type of fan on board, required by this Section.

12.2 Pump-rooms and other enclosed spaces normally entered

Pump-rooms and other enclosed spaces normally entered, which are not covered by 12.1.1, are to be fitted with mechanical ventilation systems, capable of being controlled from outside such spaces and complying with the requirements of 12.1.3, except that the capacity is not to be less than 20 changes of air per hour, based upon the total volume of the space. Provision is to be made to ventilate such spaces prior to entering.

12.3 Spaces not normally entered

Double bottoms, cofferdams, duct keels, pipe tunnels, hold spaces and other spaces where cargo may accumulate, are to be capable of being ventilated to ensure a safe environment when entry into the spaces is necessary. Where a permanent ventilation system is not provided for such spaces, approved means of portable mechanical ventilation are to be provided. Where necessary, owing to the arrangement of spaces, for instance hold spaces, essential ducting for such ventilation

is to be permanently installed. For permanent installations, the capacity of eight air changes per hour is to be provided and for portable systems the capacity of 16 air changes per hour. Fans or blowers are to be clear of personnel access openings, and are to comply with 12.1.8.

Section 13

Instrumentation

13.1 Gauging

13.1.1 Cargo tanks are to be fitted with one of the following types of gauging devices:

.1 Open device which makes use of an opening in the tanks and may expose the gauges to the cargo or its vapour. An example of this is the ullage opening.

.2 Restricted device which penetrates the tank and which, when in use, permits a small quantity of cargo vapour or liquid to be exposed to the atmosphere. When not in use, the device is completely closed. The design is to ensure that no dangerous escape of tank contents (liquid or spray) can take place in opening the device.

IR.2 A restricted device could be a sounding pipe with vapour tight cover.

.3 Closed device which penetrates the tank, but which is part of a closed system and keeps tank contents from being released. Examples are the float-type systems, electronic probe, magnetic probe and protected sight glass. Alternatively, an indirect device which does not penetrate the tank shell and which is independent of the tank may be used. Examples are weighing of cargo, pipe flow meter, etc.

13.1.2 Gauging devices are to be independent of the equipment required under 15.19.

13.1.3 Open gauging and restricted gauging are to be allowed only where:

.1 open venting is allowed by this Chapter; or

.2 means are provided for relieving tank pressure before the gauge is operated.

13.1.4 Types of gauging for individual products are shown in column "j" in Sec.17, Table 17.1.1.

IR13.1.5 Openings for open and restricted devices are to be located on the weather deck and are not to be arranged in enclosed compartments.

13.2 Vapour detection

13.2.1 Ships carrying toxic or flammable products or both are to be equipped with at least two instruments designed and calibrated for testing for the specific vapours in question. If such instruments are not capable of testing for both toxic concentrations and flammable concentrations, then two separate sets of instruments are to be provided.

13.2.2 Vapour detection instruments may be portable or fixed. If a fixed system is installed, at least one portable instrument is to be provided.

The installation requirements for analysing units are given in Pt.5, Ch.2, Sec.6, Cl.6.4.4 of these Rules.

13.2.3 When toxic-vapour detection equipment is not available for some products which require such detection, as indicated in column "l" in Sec.17, Table 17.1.1. IRS may exempt the ship from the requirement, provided an appropriate entry is made in the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk. When granting such an exemption, the Administration should recognize the necessity for additional breathing-air supply and an entry should be made on the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk drawing attention to the provisions of 14.2.4 and 16.4.2.2.

13.2.4 Vapour detection requirements for individual products are shown in column "k" in Sec.17, Table 17.1.1.

Section 14

Personnel Protection

14.1 Protective equipment

IR 14.1.1 The requirements of this Section are not requirements for the maintenance of class.

14.1.1 For the protection of crew members who are engaged in loading and discharging operations, the ship is to have on board suitable protective equipment consisting of large aprons, special gloves with long sleeves, suitable footwear, coveralls of chemical-resistant material, and tight-fitting goggles or face shields or both. The protective clothing and equipment is to cover all skin so that no part of the body is unprotected.

14.1.2 Work clothes and protective equipment are to be kept in easily accessible places and in special lockers. Such equipment is not to be kept within accommodation spaces, with the exception of new, unused equipment and equipment which has not been used since undergoing a thorough cleaning process. IRS may, however, approve storage rooms for such equipment within accommodation spaces if adequately segregated from living spaces such as cabins, passageways, dining rooms, bathrooms, etc.

14.1.3 Protective equipment is to be used in any operation which may entail danger to personnel.

14.2 Safety equipment

14.2.1 Ships carrying cargoes for which 15.12, 15.12.1 or 15.12.3 is listed in column "o" in Sec.17, Table 17.1.1 are to have on board sufficient but not less than three complete sets of safety equipment each permitting personnel to enter a gas-filled compartment and perform work there for at least 20 min. Such equipment is to be in addition to that required by SOLAS regulation II-2/10.10.

14.2.2 One complete set of safety equipment is to consist of:

.1 one self-contained air-breathing apparatus (not using stored oxygen);

.2 protective clothing, boots, gloves and tight-fitting goggles;

.3 fire proof lifeline with belt resistant to the cargoes carried; and

.4 explosion-proof lamp.

14.2.3 For the safety equipment required in 14.2.1, all ships are to carry the following, either:

.1 one set of fully charged spare air bottles for each breathing apparatus;

.2 a special air compressor suitable for the supply of high-pressure air of the required purity;

.3 a charging manifold capable of dealing with sufficient spare breathing apparatus air bottles for the breathing apparatus; or

.4 fully charged spare air bottles with a total free air capacity of at least 6,000 l for each breathing apparatus on board in excess of the requirements of SOLAS Regulation II-2/10.10.

14.2.4 A cargo pump-room on ships carrying cargoes which are subject to the requirements of 15.18 or cargoes for which in column "k" in Sec.17, Table 17.1.1 toxic vapour detection equipment is required but is not available is to have either:

.1 a low-pressure line system with hose connections suitable for use with the breathing apparatus required by 14.2.1. This system is to provide sufficient high-pressure air capacity to supply, through pressure reduction devices, enough low-pressure air to enable two men to work in a gas-dangerous space for at least 1 hour without using the air bottles of the breathing apparatus. Means should be provided for recharging the fixed air bottles and breathing apparatus air bottles from a special air compressor suitable for the supply of high-pressure air of the required purity; or

.2 an equivalent quantity of spare bottled air in lieu of the low-pressure air line.

14.2.5 At least one set of safety equipment as required by 14.2.2 is to be kept in a suitable clearly marked locker in a readily accessible place near the cargo pump-room. The other sets of safety equipment are also to be kept in suitable, clearly marked, easily accessible, places.

14.2.6 The breathing apparatus is to be inspected at least once a month by a responsible officer, and the inspection recorded in the ship's log-book. The equipment is to be inspected and tested by an expert at least once a year.

14.3 Emergency equipment

14.3.1 Ships carrying cargoes, for which 'Yes' is indicated in column 'n' of Table 17.1.1 are to be provided with suitable respiratory and eye protection sufficient for every person on board for emergency escape purposes, subject to the following:

.1 filter-type respiratory protection is unacceptable;

.2 self-contained breathing apparatus is normally to have at least a duration of service of 15 min;

.3 emergency escape respiratory protection is not to be used for fire-fighting or cargo handling purposes and is to be marked to that effect.

Individual cargoes to which the provisions of this paragraph apply are indicated in column "n" in Sec.17, Table 17.1.1.

14.3.2 The ship is to have on board medical first-aid equipment including oxygen resuscitation equipment and antidotes for cargoes carried, based on the guidelines developed by the Organisation.

Reference is made to the Medical First Aid Guide for Use in Accidents Involving Dangerous Goods (MFAG), which provides advice on the treatment of casualties in accordance with the symptoms exhibited as well as equipment and antidotes that may be appropriate for treating the casualty.

14.3.3 A stretcher which is suitable for hoisting an injured person up from spaces such as the cargo pump-room is to be placed in a readily accessible location.

14.3.4 Suitably marked decontamination showers and an eyewash are to be available on deck in convenient locations. The showers and eyewash are to be operation in all ambient conditions.

Section 15

Special Requirements

15.1 General

15.1.1 The provisions of this Section are applicable where specific reference is made in column "o" in Sec.17, Table 17.1.1. These requirements are additional to the general requirements of this Chapter.

15.2 Ammonium nitrate solution (93% or less)

15.2.1 The ammonium nitrate solution is to contain at least 7% by weight of water. The acidity (pH) of the cargo when diluted with ten parts of water to one part of cargo by weight is to be between 5.0 and 7.0. The solution is to not contain more than 10 ppm chloride ions, 10 ppm ferric ions, and is to be free of other contaminants.

15.2.2 Tanks and equipment for ammonium nitrate solution are to independent of tanks and equipment containing other cargoes or combustible products. Equipment which may, in service or when defective, release combustible products into the cargo, e.g. lubricants, is not to be used. Tanks are not to be used for seawater ballast.

15.2.3 Except where expressly approved by IRS, ammonium nitrate solutions are not to be transported in tanks which have previously contained other cargoes unless tanks and associated equipment have been cleaned to the satisfaction of IRS.

15.2.4 The temperature of the heat exchanging medium in the tank heating system is not to exceed 160°C. The heating system is to be provided with a control system to keep the cargo at a bulk mean temperature of 140°C. High-temperature alarms at 145°C and 150°C and a low-temperature alarm at 125°C are to be provided. Where the temperature of the heat exchanging medium exceeds 160°C an alarm is also to be given. Temperature alarms and controls are to be located on the navigating bridge.

15.2.5 If the bulk mean cargo temperature reaches 145°C, a cargo sample is to be diluted with ten parts of distilled or demineralized water to one part of cargo by weight and the acidity (pH) is to be determined by means of a narrow range indicator paper or stick. Acidity (pH) measurements is then to be taken every 24 hours. If the acidity (pH) is found to be below 4.2, ammonia gas is to be injected into the cargo until the acidity (pH) of 5.0 is reached.

15.2.6 A fixed installation is to be provided to inject ammonia gas into the cargo. Controls for this system are to be located on the navigation bridge. For this purpose, 300 kg of ammonia per 1,000 tonnes of ammonium nitrate solution is to be available on board.

15.2.7 Cargo pumps are to be of the centrifugal deepwell type or of the centrifugal type with water flushed seals.

15.2.8 Vent piping is to be fitted with approved weatherhoods to prevent clogging. Such weatherhoods are to be accessible for inspection and cleaning.

15.2.9 Hot work on tanks, piping and equipment which have been in contact with ammonium nitrate solution is only to be done after all traces of ammonium nitrate have been removed, inside as well as outside.

15.3 Carbon disulphide

Carbon disulphide may be carried either under a water pad or under a suitable inert gas pad as specified in the following paragraphs:

Carriage under water pad

15.3.1 Provision is to be made to maintain a water pad in the cargo tank during loading, unloading and transit. In addition, a suitable inert gas pad is to be maintained in the ullage space during transit.

15.3.2 All openings are to be in the top of the tank, above the deck.

15.3.3 Loading lines are to terminate near the bottom of the tank.

15.3.4 A standard ullage opening is to be provided for emergency sounding.

15.3.5 Cargo piping and vent lines are to be independent of piping and vent lines used for other cargo.

15.3.6 Pumps may be used for discharging cargo, provided they are of the deepwell or hydraulically driven submersible types. The means of driving a deepwell pump are not to present a source of ignition for carbon disulphide and are not to employ equipment that may exceed a temperature of 80°C.

15.3.7 If a cargo discharge pump is used, it is to be inserted through a cylindrical well extending from the tank top to a point near the tank bottom. A water pad should be formed in this well before attempting pump removal unless the tank has been certified as gas-free.

15.3.8 Water or inert gas displacement may be used for discharging cargo, provided the cargo system is designed for the expected pressure and temperature.

15.3.9 Safety relief valves are to be of stainless steel construction.

15.3.10 Because of its low ignition temperature and close clearances required to arrest its flame propagation, only intrinsically safe systems and circuits are permitted in the hazardous locations described in Sec.10, Cl.10.2.3.

Carriage under suitable inert gas pad

15.3.11 Carbon disulphide is to be carried in independent tanks with a design pressure of not less than 0.06 MPa gauge

15.3.12 All openings are to be located on the top of the tank, above the deck.

15.3.13 Gaskets used in the containment system are to be of a material which does not react with, or dissolve in, carbon disulphide.

15.3.14 Threaded joints are not permitted in the cargo containment system, including the vapour lines.

15.3.15 Prior to loading, the tank(s) is to be inerted with suitable inert gas until the oxygen level is 2% by volume or lower. Means are to be provided to automatically maintain a positive pressure in the tank using suitable inert gas during loading, transport and discharge. The system is to be able to maintain this positive pressure between 0.01 and 0.02 MPa and is to be remotely monitored and fitted with over/under-pressure alarms.

15.3.16 Hold spaces surrounding an independent tank carrying carbon disulphide is to be inerted by a suitable inert gas until the oxygen level is 2% or less. Means are to be provided to monitor and maintain this condition throughout the voyage. Means is also to be provided to sample these spaces for carbon disulphide vapour.

15.3.17 Carbon disulphide is to be loaded, transported and discharged in such a manner that venting to the atmosphere does not occur. If carbon disulphide vapour is returned to shore during loading or to the ship during discharge, the vapour return system is to be independent of all other containment systems.

15.3.18 Carbon disulphide is to be discharged only by submerged deepwell pumps or by a suitable inert gas displacement. The submerged deepwell pumps are to be operated in a way that prevents heat build-up in the pump. The pump is also to be equipped with a temperature sensor in the pump housing with remote readout and alarm in the cargo control room. The alarm is to be set at 80°C. The pump is also to be fitted with an automatic shut-down device, if the tank pressure falls below atmospheric pressure during the discharge.

15.3.19 Air is not to be allowed to enter the cargo tank, cargo pump or lines while carbon disulphide is contained in the system.

15.3.20 No other cargo handling, tank cleaning or deballasting is to take place concurrent with loading or discharge of carbon disulphide.

15.3.21 A water spray system of sufficient capacity is to be provided to blanket effectively the area surrounding the loading manifold, the exposed deck piping associated with product handling and the tank domes. The arrangement of piping and nozzles is to be such as to give a uniform distribution rate of 10 $l/m^2/min$. Remote manual operation is to be arranged such that remote starting of pumps supplying the water-spray system and remote operation of any normally closed valves in the system can be carried out from a suitable location outside the cargo area adjacent to the accommodation spaces and readily accessible and operable in the event of fire in the areas protected. The water-spray system is to be capable of both local and remote manual

operation and the arrangement is to ensure that any spilled cargo is washed away. Additionally, a water hose with pressure to the nozzle when atmospheric temperature permits is to be connected ready for immediate use during loading and unloading operations.

15.3.22 No cargo tanks are to be more than 98% liquid-full at the reference temperature (R).

15.3.23 The maximum volume (V_L) of cargo to be loaded in a tank is to be:

$$V_L = 0.98\ V\ \frac{\rho_R}{\rho_L}$$

where,

V = volume of the tank

ρ_R = relative density of cargo at the reference temperature (R)

ρ_L = relative density of cargo at the loading temperature

R = reference temperature.

15.3.24 The maximum allowable tank filling limits for each cargo tank is to be indicated for each loading temperature which may be applied and for the applicable maximum reference temperature, on a list approved by the Administration. A copy of the list is to be permanently kept on board by the master.

15.3.25 Zones on open deck, or semi-enclosed spaces on open deck within three metres of a tank outlet, gas or vapour outlet, cargo pipe flange or cargo valve of a tank certified to carry carbon disulphide, are to comply with the electrical equipment requirements specified for carbon disulphide in column "i", of Section 17. Also, within the specified zone, no other heat sources, like steam piping with surface temperatures in excess of 80°C is to be allowed.

15.3.26 Means are to be provided to ullage and sample the cargo without opening the tank or disturbing the positive suitable inert gas blanket.

15.3.27 The product is to be transported only in accordance with a cargo handling plan that has been approved by the Administration. Cargo handling plans are to show the entire cargo piping system. A copy of the approved cargo handling plan is to be made available on board. The International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk is to be endorsed to include reference to the approved cargo handling plan.

15.4 Diethyl ether

15.4.1 Unless inerted, natural ventilation is to be provided for the voids around the cargo tanks while the vessel is under way. If a mechanical ventilation system is installed, all blowers are to be of nonsparking construction. Mechanical ventilation equipment is not to be located in the void spaces surrounding the cargo tanks.

15.4.2 Pressure relief valve settings are not to be less than 0.02 MPa gauge for gravity tanks.

15.4.3 Inert gas displacement may be used for discharging cargo from pressure tanks provided the cargo system is designed for the expected pressure.

15.4.4 In view of the fire hazard, provision is to be made to avoid any ignition source or heat generation or both in the cargo area.

15.4.5 Pumps may be used for discharging cargo, provided that they are of a type designed to avoid liquid pressure against the shaft gland or are of a hydraulically operated submerged type and are suitable for use with the cargo.

15.4.6 Provisions are to be made to maintain the inert-gas pad in the cargo tank during loading, unloading and transit.

15.5 Hydrogen peroxide solutions

15.5.1 Hydrogen peroxide solutions over 60% but not over 70%.

15.5.1.1 Hydrogen peroxide solutions over 60% but not over 70% is to be carried in dedicated ships only and no other cargoes are to be carried.

15.5.1.2 Cargo tanks and associated equipment are to be either pure aluminium (99.5%) or solid stainless steel (304L, 316, 316L or 316Ti), and passivated in accordance with approved procedures. Aluminium is not to be used for piping on deck. All nonmetallic materials of construction for the containment system are neither to be attacked by hydrogen peroxide nor contribute to its decomposition.

15.5.1.3 Pump-rooms are not to be used for cargo transfer operations.

15.5.1.4 Cargo tanks are to be separated by cofferdams from oil fuel tanks or any other space containing flammable or combustible materials.

15.5.1.5 Tanks intended for the carriage of hydrogen peroxide are not to be used for seawater ballast.

15.5.1.6 Temperature sensors are to be installed at the top and bottom of the tank. Remote temperature readouts and continuous monitoring are to be located on the navigating bridge. If the temperature in the tanks rises above 35°C, visible and audible alarms are to be activated on the navigation bridge.

15.5.1.7 Fixed oxygen monitors (or gas-sampling lines) are to be provided in void spaces adjacent to tanks to detect leakage of the cargo into these spaces. Remote readouts, continuous monitoring (if gas sampling lines are used, intermittent sampling is satisfactory) and visible and audible alarms similar to those for the temperature sensors are also to be located on the navigating bridge. The visible and audible alarms are to be activated if the oxygen concentration in these void spaces exceeds 30% by volume. Two portable oxygen monitors are also to be available as back-up systems.

15.5.1.8 As a safeguard against uncontrolled decomposition, a cargo jettisoning system is to be installed to discharge the cargo overboard. The cargo is to be jettisoned if the temperature rise of the cargo exceeds a rate of 2°C per hour over a 5 hour period or when the temperature in the tank exceeds 40°C.

15.5.1.9 Cargo tank venting system is to have pressure/vacuum relief valves for normal controlled venting, and rupture discs or a similar device for emergency venting, if tank pressure rises rapidly as a result of uncontrolled decomposition. Rupture discs are to be sized on the basis of tank design pressure, tank size and anticipated decomposition rate.

15.5.1.10 A fixed water-spray system is to be provided for diluting and washing away any concentrated hydrogen peroxide solution spilled on deck. The areas covered by the water-spray are to include the manifold/hose connections and the tank tops of those tanks designated for carrying hydrogen peroxide solutions. The minimum application rate is to satisfy the following criteria:

.1 The product is to be diluted from the original concentration to 35% by weight within 5 minutes of the spill.

.2 The rate and estimated size of the spill is to be based upon maximum anticipated loading and discharge rates, the time required to stop flow of cargo in the event of tank overfill or a piping/hose failure, and the time necessary to begin application of dilution water with actuation at the cargo control location or on the navigating bridge.

15.5.1.11 Only those hydrogen peroxide solutions which have a maximum decomposition rate of 1% per year at 25°C are to be carried. Certification from the shipper that the product meets this standard is to be presented to the master and kept on board. A technical representative of the manufacturer is to be on board to monitor the transfer operations and have the capability to test the stability of the hydrogen peroxide. He is to certify to the master that the cargo has been loaded in a stable condition.

15.5.1.12 Protective clothing that is resistant to hydrogen peroxide solutions is to be provided for each crew member involved in cargo transfer operations. Protective clothing is to include non flammable coveralls, suitable gloves, boots and eye protection.

15.5.2 Hydrogen peroxide solutions over 8% but not over 60% by mass

15.5.2.1 The ship's shell plating is not to form any boundaries of tanks containing this product.

15.5.2.2 Hydrogen peroxide is to be carried in tanks thoroughly and effectively cleaned of all traces of previous cargoes and their vapours or ballast. Procedures for inspection, cleaning, passivation and loading of tanks are to be in accordance with the MSC/Circ.394. A certificate is to be on board the vessel indicating that the procedures in the circular have been followed. The passivation requirement may be waived by an Administration for domestic shipments of short duration. Particular care in this respect is essential to ensure the safe carriage of hydrogen peroxide.

.1 When hydrogen peroxide is carried no other cargoes are to be carried simultaneously.

.2 Tanks which have contained hydrogen peroxide may be used for other cargoes after cleaning in accordance with the procedures outlined in MSC/Circ.394.

.3 Consideration in design is to provide minimum internal tank structure, free draining, no entrapment and ease of visual inspection.

15.5.2.3 Cargo tanks and associated equipment is to be either pure aluminium (99.5%) or solid stainless steel of types suitable for use with hydrogen peroxide (e.g.304, 304L, 316, 316L, 316Ti). Aluminium is not to be used for piping on deck. All nonmetallic materials of construction for the containment system are neither be attacked by hydrogen peroxide nor contribute to its decomposition.

15.5.2.4 Cargo tanks are to be separated by a cofferdam from fuel oil tanks or any other space containing materials incompatible with hydrogen peroxide.

15.5.2.5 Temperature sensors are to be installed at the top and bottom of the tank. Remote temperature readouts and continuous monitoring are to be located on the navigating bridge. If the temperature in the tank rises above 35°C, visible and audible alarms are to activate on the navigating bridge.

15.5.2.6 Fixed oxygen monitors (or gas sampling lines) are to be provided in void spaces adjacent to tanks to detect leakage of the cargo into these spaces. The enhancement of flammability by oxygen enrichments is to be recognized. Remote readouts, continuous monitoring (if gas sampling lines are used, intermittent sampling is satisfactory) and visible and audible alarms similar to those for the temperature sensors are also to be located on the navigating bridge. The visible and audible alarms are to activate if the oxygen concentration in these void spaces exceeds 30% by volume. Two portable oxygen monitors are also to be available as back-up systems.

15.5.2.7 As a safeguard against uncontrolled decomposition, a cargo jettisoning system is to be installed to discharge the cargo overboard. The cargo is to be jettisoned if the temperature rise of the cargo exceeds a rate of 2°C per hour over a 5 hour period or when the temperature in the tank exceeds 40°C.

15.5.2.8 Cargo-tank venting systems with filtration are to have pressure vacuum-relief valves for normal controlled venting, and a device for emergency venting, if tank pressure rises rapidly as a result of an uncontrolled decomposition rate, as stipulated in 15.5.20. These venting systems are to be designed in such a manner that there is no introduction of seawater into the cargo tank even under heavy sea conditions. Emergency venting is to be sized on the basis of tank design pressure and tank size.

15.5.2.9 A fixed water-spray system is to be provided for diluting and washing away any concentrated solution spilled on deck. The areas covered by the water-spray are to include the manifold/hose connections and the tank tops of those tanks designated for the carriage of hydrogen peroxide solutions. The minimum application rate is to satisfy the following criteria:

.1 The product is to be diluted from the original concentration to 35% by weight within 5 minutes of the spill.

.2 The rate of estimated size of the spill is to be based upon maximum anticipated loading and discharge rates, the time required to stop flow of the cargo in the event of tank overfill or a piping/hose failure, and the time necessary to begin application of dilution water with actuation at the cargo control location or on the navigating bridge.

15.5.2.10 Only those hydrogen peroxide solutions which have maximum decomposition rate of 1% per year at 25°C are to be carried. Certification from the shipper that the product meets this standards is to be presented to the master and kept on board. A technical representative of the manufacturer is to be on board to monitor the transfer operations and have the capability to test the stability of the hydrogen peroxide. He is to certify to the master that the cargo has been loaded in a stable condition.

15.5.2.11 Protective clothing that is resistant to hydrogen peroxide is to be provided for each crew member involved in cargo-transfer operations. Protective clothing is to include

coveralls that are nonflammable, suitable gloves, boots and eye protection.

15.5.2.12 During transfer of hydrogen peroxide the related piping system is to be separated from all other systems. Cargo hoses used for transfer of hydrogen peroxide are to be marked "For Hydrogen Peroxide Transfer only".

15.5.3 Procedures for inspection, cleaning, passivation and loading of tanks for the carriage of hydrogen peroxide solutions over 8% but not over 60% by mass, which have contained other cargoes, or for the carriage of other cargoes after the carriage of hydrogen peroxide

15.5.3.1 Tanks having contained cargoes other than hydrogen peroxide are to be inspected, cleaned and passivated before re-use for the transport of hydrogen peroxide solutions. The procedures for inspection and cleaning, as given in paragraphs 15.5.3.2 to 15.5.3.8 below, apply to both stainless steel and pure aluminium tanks (see paragraph 15.5.3.2). Procedures for passivation are given in paragraph 15.5.3.9 for stainless steel and 15.5.3.10 for aluminium. Unless otherwise specified, all steps apply to the tanks and to all associated equipment having been in contact with the other cargo.

15.5.3.2 After unloading the previous cargo the tank should be rendered safe and inspected for any residues, scale and rust.

15.5.3.3 Tanks and associated equipment shall be washed with clean filtered water. The water to be used shall at least have the quality of potable water with a low chlorine content.

15.5.3.4 Trace residues and vapours of the previous cargo shall be removed by steaming of tank and equipment.

15.5.3.5 Tanks and equipment are washed again with clean water (quality as above) and dried, using filtered, oil-free air.

15.5.3.6 The atmosphere in the tanks shall be sampled and investigated for the presence of organic vapours and oxygen concentration.

15.5.3.7 The tank shall be checked again by visual inspection for residues of the previous cargo, scale and rust as well as for any smell of the previous cargo.

15.5.3.8 If inspection or measurements indicate the presence of residues of the previous cargo or its vapours, actions described in paragraphs 15.5.3.3 to 15.5.3.5 shall be repeated.

15.5.3.9 Tanks and equipment made from stainless steel have contained other cargoes than hydrogen peroxide or which have been under repair shall be cleaned and passivated, regardless of any previous passivation, according to the following procedure:

.1 New welds and other repaired parts shall be cleaned and finished using stainless steel wire brush, chisel, sandpaper or buff. Rough surfaces shall be given a smooth finish. A final polishing is necessary.

.2 Fatty and oily residues shall be removed by the use of appropriate organic solvents or detergent solutions in water. The use of chlorine-containing compounds shall be avoided as they can seriously interfere with passivation.

.3 The residues of the degreasing agent shall be removed, followed by a washing with water.

.4 In the next step, scale and rust shall be removed by the application of acid (e.g. a mixture of nitric and hydrofluoric acids), followed again by a washing with clean water.

.5 All the metal surfaces which can come into contact with hydrogen peroxide shall be passivated by the application of nitric acid of a concentration between 10 and 35% by mass. The nitric acid must be free from heavy metals, other oxidizing agents or hydrogen fluoride. The passivation process shall continue for 8 to 24 h, depending upon the concentration of acid, the ambient temperature and other factors. During this time a continuous contact between the surfaces to be passivated and the nitric acid shall be ensured. In the case of large surfaces this may be achieved by recirculating the acid. Hydrogen gas may be evolved in the passivation process, leading to the presence of an explosive atmosphere

in the tanks. Therefore, appropriate measures must be taken to avoid the build-up or the ignition of such an atmosphere.

.6 After passivation the surfaces shall be thoroughly washed with clean filtered water. The washing process shall be repeated until the effluent water has the same pH value as the incoming water.

.7 Surfaces treated according to the above steps may cause some decomposition when coming into contact with hydrogen peroxide for the first time. This decomposition will cease after a short time (usually within two or three days). Therefore, an additional flushing with hydrogen peroxide for a period of at least two days is recommended.

.8 Only degreasing agents and acid cleaning agents which have been recommended for this purpose by the manufacturer of the hydrogen peroxide shall be used in the process.

15.5.3.10 Tanks and equipment made from aluminium and which have contained cargoes other than hydrogen peroxide, or which have been under repair, shall be cleaned and passivated. The following is an example of a recommended procedure:

.1 The tank shall be washed with a solution of a sulphonated detergent in hot water, followed by a washing with water.

.2 The surface shall then be treated for 15 to 20 min with a solution of sodium hydroxide of a concentration of 7% by mass or treated for a longer period with a less concentrated solution (e.g. for 12 h with 0.4 to 0.5% sodium hydroxide). To prevent excessive corrosion at the bottom of the tank when treating with more concentrated solutions of sodium hydroxide, water shall be added continuously to dilute the sodium hydroxide solution which collects there.

.3 The tank shall be thoroughly washed with clean, filtered water. As soon as possible after washing, the surface shall be passivated by the application of nitric acid of a concentration between 30 and 35% by mass. The passivation process shall continue for 16 to 24 h. During this time a continuous contact between the surfaces to be passivated and the nitric acid shall be ensured.

.4 After passivation the surfaces shall be thoroughly washed with clean, filtered water. The washing process shall be repeated until the effluent water has the same pH value as the incoming water.

.5 A visual inspection shall be made to ensure that all surfaces have been treated. It is recommended that an additional flushing is carried out for a minimum of 24 h with dilute hydrogen peroxide solution of a concentration approximately 3% by mass.

15.5.3.11 The concentration and stability of the hydrogen peroxide solution to be loaded shall be determined.

15.5.3.12 The hydrogen peroxide is loaded under intermittent visual supervision of the interior of the tank from an appropriate opening.

15.5.3.13 If substantial bubbling is observed which does not disappear within 15 min after the completion of loading, the contents of the tank shall be unloaded and disposed of in an environmentally safe manner. The tank and equipment shall then be repassivated as described above.

15.5.3.14 The concentration and stability of the hydrogen peroxide solution shall be determined again. If the same values are obtained within the limits of error as in paragraph 15.5.3.10, the tank is considered to be properly passivated and the cargo ready for shipment.

15.5.3.15 Actions described in paragraphs 15.5.3.2 to 15.5.3.8 shall be carried out under the supervision of the master or shipper. Actions described in paragraphs 15.5.3.9 to 15.5.3.15 shall be carried out under the on-site supervision and responsibility of a representative of the hydrogen peroxide manufacturer or under supervision and responsibility of another person familiar with the safety-relevant properties of hydrogen peroxide.

15.5.3.16 The following procedure shall be applied when tanks having contained hydrogen peroxide solution are to be used for

other products (unless otherwise specified, all steps apply to the tanks and to all associated equipment having been in contact with hydrogen peroxide):

.1 Hydrogen peroxide cargo residue shall be drained as completely as possible from tanks and equipment.

.2 Tanks and equipment shall be rinsed with clean water, and subsequently thoroughly washed with clean water.

.3 The interior of the tanks hall be dried and inspected for any residues.

Steps .1 to .3, in 15.5.3.16 shall be carried out under the supervision of the master or the shipper. Step .3) in paragraph 15.5.3.16 shall be carried out by a person familiar with the safety-relevant properties of the chemical to be transported and of hydrogen peroxide.

SPECIAL CAUTIONS :

1) Hydrogen peroxide decomposition may enrich the atmosphere with oxygen and appropriate precautions shall be observed.

2) Hydrogen gas may be evolved in the passivation processes described in paragraphs 15.5.3.9, 15.5.3.10 .2) and 15.5.3.10 .4), leading to the presence of an explosive atmosphere in the tank. Therefore, appropriate measures must be taken to avoid the build-up or the ignition of such an atmosphere.

15.6 Motor fuel anti-knock compounds (containing lead alkyls)

15.6.1 Tanks used for these cargoes are not to be used for the transportation of any other cargo except those commodities to be used in the manufacture of motor fuel anti-knock compounds containing lead alkyls.

15.6.2 If a cargo pump-room is located on deck level according to 15.17, the ventilation arrangements are to be in compliance with 15.16.

15.6.3 Entry into cargo tanks used for the transportation of these cargoes is not permitted unless approved by IRS.

15.6.4 Air analysis is to be made for lead content to determine if the atmosphere is satisfactory prior to allowing personnel to enter the cargo pump-room or void spaces surrounding the cargo tank.

15.7 Phosphorus, yellow or white

15.7.1 Phosphorus is to, at all times, be loaded, carried and discharged under a water pad of 760 [mm] minimum depth. During discharge operations, arrangements are to be made to ensure that water occupies the volume of phosphorus discharged. Any water discharged from a phosphorus tank is to be returned only to a shore installation.

15.7.2 Tanks are to be designed and tested to a minimum equivalent water head of 2.4 [m] above the top of the tank, under designed loading conditions, taking into account the depth, relative density and method of loading and discharge of the phosphorus.

15.7.3 Tanks are to be so designed as to minimize the interfacial area between the liquid phosphorus and its water pad.

15.7.4 A minimum ullage space of 1% is to be maintained above the water pad. The ullage space is to be filled with inert gas or naturally ventilated by two cowled standpipes terminating at different heights but at least 6 [m] above the deck and at least 2 [m] above the pump house top.

15.7.5 All openings are to be at the top of cargo tanks, and fittings and joints attached thereto are to be materials resistant to phosphorus pentoxide.

15.7.6 Phosphorus is to be loaded at a temperature not exceeding 60°C.

15.7.7 Tank heating arrangements are to be external to tanks and have a suitable method of temperature control to ensure that the temperature of the phosphorus does not exceed 60°C. A high-temperature alarm is to be fitted.

15.7.8 A water drench system acceptable to IRS is to be installed in all void spaces surrounding the tanks. The system is to operate automatically in the event of an escape of phosphorus.

15.7.9 Void spaces referred to in 15.7.8 are to be provided with effective means of mechanical ventilation which is to be capable of being sealed off quickly in an emergency.

15.7.10 Loading and discharge of phosphorus is to be governed by a central system on the ship which, in addition to incorporating high-level alarms, is to ensure that no overflow of tanks is possible and that such operations can be stopped quickly in an emergency from either ship or shore.

15.7.11 During cargo transfer, a water hose on deck is to be connected to a water supply and kept flowing throughout the operation so that any spillage of phosphorus may be washed down with water immediately.

15.7.12 Ship-to-shore loading and discharge connections is to be of a type approved by IRS.

15.8 Propylene oxide and mixtures of ethylene oxide/propylene oxide with an ethylene oxide content of not more than 30% by mass

15.8.1 Products transported under the provisions of this Section are to be acetylene free.

15.8.2 Unless cargo tanks are properly cleaned, these products are not to be carried in tanks which have contained as one of the three previous cargoes any products known to catalyse polymerization, such as:

.1 mineral acids (e.g. sulphuric, hydrochloric, nitric);

.2 carboxylic acids and anhydrides (e.g. formic, acetic);

.3 halogenated carboxylic acids (e.g. chloroacetic);

.4 sulphonic acids (e.g. benzene sulphonic);

.5 caustic alkalies (e.g. sodium hydroxide, potassium hydroxide);

.6 ammonia and ammonia solutions;

.7 amines and amine solutions;

.8 oxidizing substances.

15.8.3 Before loading, tanks are to thoroughly and effectively cleaned, to remove all traces of previous cargoes from tanks and associated pipework, except where the immediately prior cargo has been propylene oxide or ethylene oxide/propylene oxide mixtures. Particular care is to be taken in the case of ammonia in tanks made of steel other than stainless steel.

15.8.4 In all cases, the effectiveness of cleaning procedures for tanks and associated pipework is to checked by suitable testing or inspection, to ascertain that no traces of acidic or alkaline materials remain that might create a hazardous situation in the presence of these products.

15.8.5 Tanks are to be entered and inspected prior to each initial loading of these products to ensure freedom from contamination, heavy rust deposits and visible structural defects. When cargo tanks are in continuous service for these products, such inspections are to be performed at intervals of not more than two years.

15.8.6 Tanks for the carriage of these products are to be of steel or stainless steel construction.

15.8.7 Tanks for the carriage of these products may be used for other cargoes after thorough cleaning of tanks and associated pipework systems by washing or purging.

15.8.8 All valves, flanges, fittings and accessory equipment are to be of a type suitable for use with the products and are to be constructed of steel or stainless steel in accordance with recognized standards. The chemical composition of all material used is to be submitted to IRS for approval prior to fabrication. Discs or disc faces, seats and other wearing parts of valves are to be made of stainless steel containing not less than 11% chromium.

15.8.9 Gaskets are to be constructed of materials which do not react with, dissolve in, or lower the autoignition temperature of, these products and which are fire-resistant and possess adequate mechanical behaviour. The surface presented to the cargo is to be polytetrafluoroethylene (PTFE), or materials giving a similar degree of safety by their inertness. Spirally- wound stainless steel, with a filler of PTFE or similar fluorinated polymer, may be accepted.

15.8.10 Insulation and packing, if used, is to be of a material which does not react with, dissolve in, or lower the autoignition temperature of, these products.

15.8.11 The following materials are generally found unsatisfactory for gaskets, packing and similar uses in containment systems for these products and would require testing before being approved by IRS:

.1 Neoprene or natural rubber, if it comes into contact with the products.

.2 Asbestos, or binders used with asbestos.

.3 Materials containing oxides of magnesium, such as mineral wools.

15.8.12 Threaded joints are not permitted in the cargo liquid and vapour lines.

15.8.13 Filling and discharge piping are to extend to within 100 [mm] of the bottom of the tank or any sump pit.

15.8.14.1 The containment system for a tank containing these products is to have a valved vapour-return connection.

15.8.14.2 The products are to be loaded and discharged in such a manner that venting of the tanks to atmosphere does not occur. If vapour return to shore is used during tank loading, the vapour-return system connected to a containment system for the product is to be independent of all other containment systems.

15.8.14.3 During discharge operations, the pressure in the cargo tank must be maintained above 0.007 MPa gauge.

15.8.15 The cargo may be discharged only by deepwell pumps, hydraulically operated submerged pumps, or inert gas displacement. Each cargo pump is to be arranged to ensure that the product does not heat significantly if the discharge line from the pump is shut off or otherwise blocked.

15.8.16 Tanks carrying these products are to be vented independently of tanks carrying other products. Facilities are to be provided for sampling the tank contents without opening the tank to atmosphere.

15.8.17 Cargo hoses used for transfer of these products are to be marked "FOR ALKYLENE OXIDE TRANSFER ONLY".

15.8.18 Cargo tanks, void spaces and other enclosed spaces, adjacent to an integral gravity cargo tank carrying propylene oxide, are to either contain a compatible cargo (those cargoes specified in 15.8.2 are examples of substances considered incompatible) or be inerted by injection of a suitable inert gas. Any hold space in which an independent cargo tank is located is to be inerted. Such inerted spaces and tanks are to be monitored for these products and oxygen. The oxygen content of these spaces is to be maintained below 2 percent. Portable sampling equipment is satisfactory.

15.8.19 In no case is air to be allowed to enter the cargo pump or piping system while these products are contained within the system.

15.8.20 Prior to disconnecting shore-lines, the pressure in liquid and vapour lines are to be relieved through suitable valves installed at the loading header. Liquid and vapour from these lines are not to be discharged to atmosphere.

15.8.21 Propylene oxide may be carried in pressure tanks or in independent or integral gravity tanks. Ethylene oxide/propylene oxide mixtures are to be carried in independent gravity tanks or pressure tanks. Tanks are to be designed for the maximum pressure expected to be encountered during loading, conveying and discharging cargo.

15.8.22.1 Tanks for the carriage of propylene oxide with a design pressure less than 0.6 bar gauge and tanks for the carriage of ethylene oxide/propylene oxide mixtures with a design pressure less than 0.12 [MPa] gauge are to have a cooling system to maintain the cargo below the reference temperature. (See 1.3.3.1).

15.8.22.2 The refrigeration requirement for tanks with a design pressure less than 0.6 bar gauge may be waived by IRS for ships operating in restricted areas or on voyages of restricted duration, and account may be taken in such cases of any insulation of the tanks. The area and times of year for which such carriage would be permitted would be included in the conditions of carriage of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk.

15.8.23.1 Any cooling system is to maintain the liquid temperature below the boiling temperature at the containment pressure. At least two complete cooling plants automatically regulated by variations within the tanks are to be provided. Each cooling plant is to be complete with the necessary auxiliaries for proper operation. The control system is also to be capable of being manually operated.

An alarm is to be provided to indicate malfunctioning of the temperature controls. The capacity of each cooling system is to be sufficient to maintain the temperature of the liquid cargo below the reference temperature (see 1.3.3.1) of the system.

15.8.23.2 An alternative arrangement may consist of three cooling plants, any two of which are to be sufficient to maintain the liquid temperatures below the reference temperature (see 1.3.3.1).

15.8.23.3 Cooling media which are separated from the products by a single wall only is to be non reactive with the products.

15.8.23.4 Cooling systems requiring compression of the products are not to be used.

15.8.24 Pressure-relief-valve settings are not to be less than 0.02 MPa gauge and for pressure tanks not greater than 0.7 MPa gauge for the carriage of propylene oxide and not greater than 0.53 MPa gauge for the carriage of propylene oxide/ethylene oxide mixtures.

15.8.25.1 The piping system for tanks to be loaded with these products are to be separated (as defined in 3.1.4) from piping systems for all other tanks, including empty tanks. If the piping system for the tanks to be loaded is not independent (as defined in 1.3.19), the required piping separation is to be accomplished by the removal of spool pieces, valves, or other pipe section, and the installation of blank flanges at these locations. The required separation applies to all liquid and vapour piping, liquid and vapour vent lines and any other possible connections, such as common inert gas supply lines.

15.8.25.2 These products may be transported only in accordance with cargo handling plans that have been approved by IRS. Each intended loading arrangement is to be shown on a separate cargo-handling plan. Cargo-handling plans are to show the entire cargo piping system and the locations for installation of blank flanges needed to meet the above piping separation requirements. A copy of each approved cargo-handling plan is to be maintained on board the ship. The International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk is to be endorsed to include reference to the approved cargo- handling plans.

15.8.25.3 Before each initial loading of these products and before every subsequent return to such service, certification verifying that the required piping separation has been achieved is to be obtained from a responsible person acceptable to the Port Administration and carried on board the ship. Each connection between a blank flange and a pipeline flange is to be fitted with a wire and seal by the responsible person to ensure that inadvertent removal of the blank flange is impossible.

15.8.26.1 No cargo tanks are to be more than 98% liquid full at the reference temperature. (see 1.3.3.1).

15.8.26.2 The maximum volume to which a cargo tank is to be loaded is:

$$V_L = 0.98V\frac{\rho R}{\rho L}$$

where,

V_L = maximum volume to which the tank may be loaded

V = volume of the tank

ρR = density of cargo at the reference temperature (see 1.3.3.1)

ρL = density of cargo at the loading temperature and pressure.

15.8.26.3 The maximum allowable tank filling limits for each cargo tank is to be indicated for each loading temperature which may be applied, and for the applicable maximum reference temperature, on a list to be approved by IRS. A copy of the list is to be permanently kept on board by the master.

15.8.27 The cargo is to be carried under a suitable protective padding of nitrogen gas. An automatic nitrogen make-up system is to be installed to prevent the tank pressure falling below 0.07 bar gauge in the event of product temperature fall due to ambient conditions or maloperation of refrigeration systems. Sufficient nitrogen is to be available on board to satisfy the demand of the automatic pressure control. Nitrogen of commercially pure quality (99.9% by volume) is to be used for padding. A battery of nitrogen bottles connected to the cargo tanks through a pressure reduction valve satisfies the intention of the expression "automatic" in this context.

15.8.28 The cargo tank vapour space is to be tested prior to and after loading to ensure that the oxygen content is 2% by volume or less.

15.8.29 A water-spray system of sufficient capacity is to be provided to blanket effectively the area surrounding the loading manifold, the exposed deck piping associated with product handling, and the tank domes. The arrangement of piping and nozzles is to be such as to give a uniform distribution rate of 10 [l/m^2/min]. Remote manual operation should be arranged such that remote starting of pumps supplying the water spray system and remote operation of any normally closed valves in the system can be carried out from a suitable location outside the cargo area, adjacent to the accommodation spaces and readily accessible and operable in the event of fire in the areas protected. The water-spray system is to be capable of both local and remote manual operation, and the arrangement is to ensure that any spilled cargo is washed away. Additionally, a water hose with pressure to the nozzle, when atmospheric temperature permit, is to be connected ready for immediate use during loading and unloading operations.

15.8.30 A remotely operated, controlled closing-rate, shutoff valve is to be provided at each cargo-hose connection used during cargo transfer.

15.9 Sodium chlorate solution (50% or less by mass)

15.9.1 Tanks and associated equipment which have contained this product may be used for other cargoes after thorough cleaning by washing or purging.

15.9.2 In the event of spillage of this product, all spilled liquid is to be thoroughly washed away without delay. To minimize fire risk, spillage is not to be allowed to dry out.

15.10 Sulphur liquid (molten)

15.10.1 Cargo tank ventilation is to be provided to maintain the concentration of hydrogen sulphide below one half of its lower explosive limit throughout the cargo-tank vapour space for all conditions of carriage, i.e. below 1.85% by volume.

15.10.2 Where mechanical ventilation systems are used for maintaining low gas concentrations in cargo tanks, an alarm system is to be provided to give warning if the system fails.

15.10.3 Ventilation systems are to be so designed and arranged as to preclude depositing of sulphur within the system.

15.10.4 Openings to void spaces adjacent to cargo tanks are to be so designed and fitted as to prevent the entry of water, sulphur or cargo vapour.

15.10.5 Connections are to be provided to permit sampling and analysing of vapour in void spaces.

15.10.6 Cargo temperature controls are to be provided to ensure that the temperature of the sulphur does not exceed 155°C.

15.10.7 Sulphur (molten) has a flashpoint above 60°C; however, electrical equipment are to be certified safe for gases evolved.

15.11 Acids

15.11.1 The ship's shell plating is not to form any boundaries of tanks containing mineral acids.

15.11.2 Proposals for lining steel tanks and related piping systems with corrosion-resistant materials may be considered by IRS. The elasticity of the lining is not to be less than that of the supporting boundary plating.

IR15.11.2 “Lining” (refer clause 15.11.2 above) is an acid-resistant material that is applied to the tank or piping system in a solid state i.e. not spray on. The requirement for the elasticity of a lining to be not less than the supporting boundary plating (in 15.11.2 above) is to prevent debonding at the interface between the lining and the lined surface.

15.11.3 Unless constructed wholly of corrosion-resistant materials or fitted with an approved lining, the plating thickness is to take into account the corrosivity of the cargo.

15.11.4 Flanges of the loading and discharge manifold connections are to be provided with shields, which may be portable, to guard against the danger of the cargo being sprayed; and in addition, drip trays are also to be provided to guard against leakage on to the deck.

15.11.5 Because of the danger of evolution of hydrogen when these substances are being carried, the electrical arrangements are to comply with 10.1.4. The certified safe type equipment is to be suitable for use in hydrogen-air mixtures. Other sources of ignition are not to be permitted in such spaces.

15.11.6 Substances subjected to the requirements of this clause are to be segregated from oil fuel tanks, in addition to the segregation requirements in 3.1.1.

15.11.7 Provision is to be made for suitable apparatus to detect leakage of cargo into adjacent spaces.

15.11.8 The cargo pump-room bilge pumping and drainage arrangements are to be of corrosion-resistant materials.

15.12 Toxic products

15.12.1 Exhaust openings of tank vent systems are to be located:

.1 at a height of B/3 or 6 m, whichever is greater, above the weather deck or, in the case of a deck tank, the access gangway;

.2 not less than 6 [m] above the fore-and-aft gangway, if fitted with 6 [m] of the gangway; and

.3 15 [m] from any opening or air intake to any accommodation and service spaces;

.4 the vent height may be reduced to 3 [m] above the deck or fore-and-aft gangway, as applicable, provided high-velocity vent valves of an approved type, directing the vapour-air mixture upwards in an unimpeded jet with and exit velocity of at least 30 [m/s], are fitted.

15.12.2 Tank venting systems are to be provided with a connection for a vapour return line to the shore installation.

IR15.12.2 Tank venting systems are to be provided with a stop valve for vapour return to shore.

15.12.3 Products are:

.1 not be to stowed adjacent to oil fuel tanks;

.2 to have separate piping systems; and

.3 to have tank vent systems separate from tanks containing nontoxic products.

15.12.4 Cargo tank relief valve settings are to be a minimum of 0.02 MPa gauge.

15.13 Cargoes protected by additives

15.13.1 Certain cargoes, with a reference in column "o" in Sec.17, Table 17.1.1, by the nature of their chemical make-up, tend, to under certain conditions of temperature, exposure to air or contact with a catalyst, to undergo polymerization, decomposition, oxidation or other chemical changes. Mitigation of this tendency is carried out by introducing small amounts of chemical inhibitors into the liquid cargo or controlling the cargo-tank environment.

15.13.2 Ships carrying these cargoes are to be so designed as to eliminate from the cargo tanks and cargo-handling system any material of construction or contaminants which could act as a catalyst or destroy the inhibitor.

15.13.3 Care is to be taken to ensure that these cargoes are sufficiently protected to prevent deleterious chemical change at all times during the voyage. Ships carrying such cargoes are to be provided with a certificate of protection from the manufacturer, and kept during the voyage, specifying:

.1 the name and amount of additive present;

.2 whether the additive is oxygen dependent*;

*(*Refer to MSC-MEPC on products requiring oxygen-dependent inhibitors)*

.3 date additive was put in the product and duration of effectiveness;

.4 any temperature limitations qualifying the additives' effective lifetime; and

.5 the action to be taken should the length of the voyage exceed the effective lifetime of the additives.

15.13.4 Ships using the exclusion of air as the method of preventing oxidation of the cargo are to comply with 9.1.3.

15.13.5 When a product containing an oxygen-dependent inhibitor is to be carried:

.1 in a ship for which inerting is required under SOLAS regulation II-2/4.5.5 (Pt.6, Ch.2, 1.5.5), as amended, the application of inert gas not to take place before loading or during the voyage, but is to be applied before commencement of unloading*;

.2 in a ship to which SOLAS regulation II-2/4.5.5 (Pt.6, Ch.2, 1.5.5), as amended, does not apply, the product may be carried without inertion (in tanks of a size not greater than 3000 m^3). If inertion is to be applied on such a ship, then the application of inert gas is not to take place before loading or during the voyage, but is to be applied before commencement of unloading*.

*(*Refer to MSC-MEPC.2/Circ.14 on products requiring oxygen-dependent inhibitors)*

15.13.6 Venting systems are to be of a design that eliminates blockage from polymer build-up. Venting equipment is to be of a type that can be checked periodically for adequacy of operation.

15.13.7 Crystallization or solidification of cargoes normally carried in the molten state can lead to depletion of inhibitor in parts of the tank contents. Subsequent remelting can thus yield pockets of uninhibited liquid, with the accompanying risk of dangerous polymerization. To prevent this, care is to be taken to ensure that at no time are such cargoes allowed to crystallize or solidify, either wholly or partially, in any part of the tank. Any required heating arrangements are to be such as to ensure that in no part of the tank does cargo become overheated to such an extent that any dangerous polymerization can be initiated. If the temperature from steam coils would induce overheating, an indirect low-temperature heating system is to be used.

15.14 Cargoes with a vapour pressure greater than 0.1013 MPa absolute at 37.8°C

15.14.1 For a cargo referenced in column "o" in Sec.17, Table 17.1.1 to this clause, a mechanical refrigeration system is to be provided unless the cargo system is designed to withstand the vapour pressure of the cargo at 45°C. Where the cargo system is designed to withstand the vapour-pressure of the cargo at 45°C, and no refrigeration system is provided, a notation would be made in the conditions of carriage on the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk to indicate the required relief valve setting for the tanks.

15.14.2 A mechanical refrigeration system is to maintain the liquid temperature below the boiling temperature at the cargo-tank design pressure.

15.14.3 When ships operate in restricted areas and at restricted times of the year, or on voyages of limited duration, IRS may agree to waive requirements for a refrigeration system. A notation of any such agreement, listing geographic area restrictions and times of the year, or voyage duration limitations, would be included in the conditions of carriage on the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk.

15.14.4 Connections are to be provided for returning expelled gases to shore during loading.

IR15.14.4 Tank venting systems are to be provided with a stop valve for vapour return to shore.

15.14.5 Each tank is to be provided with a pressure gauge which indicates the pressure in the vapour space above the cargo.

15.14.6 Where the cargo needs to be cooled, thermometers are to be provided at the top and bottom of each tank.

15.14.7.1 No cargo tanks is to be more than 98% liquid-full at the reference temperature (R).

15.14.7.2 The maximum volume (V_L) of cargo to be loaded in a tank is to be:

$$V_L = 0.98V\frac{\rho R}{\rho L}$$

where,

V = volume of the tank

ρR = density of cargo at the reference temperature (R)

ρL = density of cargo at loading temperature

R = reference temperature is the temperature at which the vapour pressure of the cargo corresponds to the set pressure of the pressure-relief valve.

15.14.7.3 The maximum allowable tank filling limits for each cargo tank is to be indicated for each loading temperature which may be applied, and for the applicable maximum reference temperature, on a list approved by IRS. A copy of the list should be permanently kept on board by the master.

15.15 Hydrogen sulphide (H_2S) detection equipment for bulk liquids

Hydrogen sulphide (H_2S) detection equipment is to be provided on board ships carrying bulk liquids prone to H_2S formation. It may be noted that scavengers and biocides, when used, may not be 100% effective in controlling the formation of H_2S. Toxic vapour detection instruments complying with the requirements in 13.2.1 of this Chapter for testing for H_2S may be used to satisfy this requirement."

15.16 Cargo contamination

15.16.1 Deleted

15.16.2 Where column "o" in the Sec.17, Table 17.1.1 refers to this Section, water is not to be allowed to contaminate this cargo. In addition, the following provisions apply:

.1 Air inlets to pressure/vacuum relief valves of tanks containing the cargo are to be situated at least 2 [m] above the weather deck.

.2 Water or steam is not to be used as the heat transfer media in a cargo temperature control system required by Sec.7.

.3 The cargo is not to be carried in cargo tanks adjacent to permanent ballast or water tanks unless the tanks are empty and dry.

.4 The cargo is not to be carried in tanks adjacent to slop tanks or cargo tanks containing ballast or slops or other cargoes containing water which may react in a dangerous manner. Pumps, pipes or vent lines serving such tanks are to be separate from similar equipment serving tanks containing the cargo. Pipelines from slop tanks or ballast lines are not to pass through tanks containing the cargo unless encased in a tunnel.

15.17 Increased ventilation requirements

For certain products, the ventilation system as described in 12.1.3 is to have a minimum capacity of at least 45 changes of air per hour based upon the total volume of space. The ventilation system exhaust ducts are to discharge at least 10 [m] away from openings into accommodation spaces, work areas or other similar spaces, and intakes to ventilation systems, and at least 4 [m] above the tank deck.

15.18 Special cargo pump-room requirements

For certain products, the cargo pump-room is to be located on the deck level or cargo pumps are to be located in the cargo tank. IRS may give special consideration to cargo pump-rooms below deck.

15.19 Overflow control

15.19.1 The provisions of this clause are applicable where specific reference is made in column "o" in Sec.17, Table 17.1.1, and are in addition to the requirements for gauging devices.

15.19.2 In the event of a power failure on any system essential for safe loading, an alarm is to be given to the operators concerned.

15.19.3 Loading operations are to be terminated at once in the event of any system essential for safe loading becoming inoperative.

15.19.4 Level alarms are to be capable of being tested prior to loading.

15.19.5 The high-level alarm system required under 15.19.6 are to be independent of the overflow control system required by 15.19.7 and are to be independent of the equipment required by 13.1.

15.19.6 Cargo tanks are to be fitted with a visual and audible high-level alarm which complies with 15.19.1 to 15.19.5 and which indicates when the liquid level in the cargo tank approaches the normal full condition.

15.19.7 A tank overflow control system required by this Section is to:

.1 come into operation when the normal tank loading procedures fail to stop the tank liquid level exceeding the normal full condition;

.2 give a visual and audible tank-overflow alarm to the ship's operator; and

.3 provide an agreed signal for sequential shutdown of onshore pumps or valves or both and of the ship's valves. The signal, as well as the pump and valve shutdown, may be dependent on operator's intervention. The use of shipboard automatic closing valves would be permitted only when specific approval has been obtained from IRS and the Port State Authority concerned.

15.19.8 The loading rate (LR) of the tank is not to exceed:

$$LR = \frac{3600\,U}{t}\ [m^3/h]$$

where,

U = ullage volume [m^3] at operating signal level;

t = time[s] needed from the initiating signal to fully stopping the cargo flow into the tank, being the sum of times needed for each step in sequential operations such as operator's responses to signals, stopping pumps and closing valves; and is to also take into account the pipeline system design pressure.

15.20 Alkyl (C7-C9) nitrates, all isomers

15.20.1 The carriage temperature of the cargo should be maintained below 100°C to prevent the occurrence of self- sustaining, exothermic decomposition reaction.

15.20.2 The cargo may not be carried in independent pressure vessels permanently affixed to the vessel's deck unless:

.1 the ranks are sufficiently insulated from fire; and

.2 the vessel has a water deluge system for the tanks such that the cargo temperature is maintained below 100°C and the temperature rise in the tanks does not exceed 1.5°C/hr for a fire of 650°C (1200°F).

15.21 Temperature sensors

Temperature sensors should be used to monitor the cargo pump temperature to detect overheating due to pump failures.

Section 16

Operational Requirements

IR 16.1 General

IR 16.1.1 The requirements of this Section are not the conditions or maintenance of class but are for the guidance of the master and the Owners of the vessel.

16.1 Maximum allowable quantity of cargo per tank

16.1.1 The quantity of a cargo required to be carried in a type 1 ship is not to exceed 1,250 m^3 in any one tank.

16.1.2 The quantity of cargo required to be carried in a type 2 ship is not to exceed 3,000 m^3 in any one tank.

16.1.3 Tanks carrying liquids at ambient temperatures are to be so loaded as to avoid the tank becoming liquid-full during the voyage, having due regard to the highest temperature which the cargo may reach.

16.2 Cargo information

16.2.1 A copy of this Chapter is to be available on board every ship covered by this Chapter.

16.2.2 Any cargo offered for bulk shipment is to be indicated in the shipping documents by the product name, under which it is listed in Section 17 and 18 of this chapter or the latest edition of MEPC.2/Circ. or under which it has been provisionally assessed. Where the cargo is a mixture, an analysis indicating the dangerous components contributing significantly to the total hazard of the product are to be provided, or a complete analysis if this is available. Such an analysis is to be certified by the manufacturer or by an independent expert acceptable to the National Authority.

16.2.3 Information is to be on board, and available to all concerned, giving the necessary data for the safe carriage of the cargo. Such information is to include a cargo stowage plan to be kept in an accessible place, indicating all cargo on board, including each dangerous chemical carried:

.1 full description of the physical and chemical properties, including reactivity necessary for the safe containment of the cargo;

.2 action to be taken in the event of spills or leaks;

.3 countermeasures against accidental personal contact;

.4 fire-fighting procedures and fire-fighting media;

.5 procedures for cargo transfer, tank cleaning, gas-freeing and ballasting;

.6 for those cargoes required to be stabilized or inhibited, the cargo is to be refused if the certificate required by these paragraphs is not supplied.

16.2.4 If sufficient information necessary for the safe transportation of the cargo is not available, the cargo is to be refused.

16.2.5 Cargoes which evolve highly toxic imperceptible vapours is not to be transported unless perceptible additives are introduced into the cargo.

16.2.6 Where column "o" in Sec.17, Table 17.1.1 refers to this paragraph, the cargo's viscosity at 20°C is to be specified on shipping document and if the cargo's viscosity exceeds 50 mPa.s at 20°C, the temperature at which the cargo has a viscosity of 25 mPa.s is to be specified in the shipping document.

16.2.7 Where column “o” in Sec. 17, Table 17.1.1 refers to this paragraph, the cargo is subject to the prewash requirements in regulation 13.7.1.4 of Annex II of MARPOL.

16.2.8 Deleted

16.2.9 Where column "o" in Sec.17, Table 17.1.1 refers to this paragraph, the cargo's melting point is to be indicated in the shipping document.

16.3 Personnel training

16.3.1 All personnel are to be adequately trained in the use of protective equipment and have basic training in the procedures appropriate to their duties, necessary under emergency conditions.

16.3.2 Personnel involved in cargo operations are to be adequately trained in handling procedures.

16.3.3 Officers are to be trained in emergency procedure to deal with conditions of leakage, spillage or fire involving the cargo based on the guidelines developed by the Organisation* and a sufficient number of them are to be instructed and trained in essential first aid for cargoes carried.

(Refer to the Medical First Aid Guide for use in Accidents Involving Dangerous Goods (MFAG), which provides advice on the treatment of casualties in accordance with the symptoms exhibited as well as equipment and antidotes that may be appropriate for treating the casualty and to the relevant provisions of the STCW Code, parts A and B.)*

16.4 Opening of and entry into cargo tanks

16.4.1 During handling and carriage of cargoes producing flammable or toxic vapours, or both, or when ballasting after the discharge of such cargo, or when loading or unloading cargo, cargo tank lids are always to be kept closed. With any hazardous cargo, cargo tank lids, ullage and sighting ports and tank washing access covers are to be open only when necessary.

16.4.2 Personnel are not to enter cargo tanks, void spaces around such tanks, cargo handling spaces or other enclosed spaces unless:

.1 the compartment is free of toxic vapours and not deficient in oxygen; or

.2 personnel wear breathing apparatus and other necessary protective equipment, and the entire operation is under the close supervision of a responsible officer.

16.4.3 Personnel are not to enter such spaces when the only hazard is of a purely flammable nature, except under the close supervision of a responsible officer.

16.5 Stowage of cargo samples

16.5.1 Samples which have to be kept on board are to be stowed in a designated spaces situated in the cargo area or, exceptionally, elsewhere, subject to the approval of IRS.

16.5.2 The stowage space is to be:

.1 cell-divided in order to avoid shifting of the bottles at sea;

.2 made of material fully resistant to the different liquids intended to be stowed; and

.3 equipped with adequate ventilation arrangements.

16.5.3 Samples which react with each other dangerously are not to be stowed close to each other.

16.5.4 Samples are not to be retained on board longer than necessary.

16.6 Cargoes not to be exposed to excessive heat

16.6.1 Where the possibility exists of a dangerous reaction of a cargo such as polymerization, decomposition, thermal instability or evolution of gas, resulting from local overheating of the cargo in either the tank or associated pipelines, such cargo is to be loaded and carried adequately segregated from other products whose temperature is sufficiently high to initiate a reaction of such cargo. (see 7.1.5.4).

16.6.2 Heating coils in tanks carrying this product are to be blanked off or secured by equivalent means.

16.6.3 Heat-sensitive products are not to be carried in deck tanks which are not insulated.

16.6.4 In order to avoid elevated temperatures, this cargo should not be carried in deck tanks.

Section 17

Summary of Minimum Requirements

17.1 General

17.1.1 The summary of the requirements of this chapter are given in Table 17.1.1 and the notes applicable to the Table 17.1.1 are given in Table 17.1.

Mixtures of noxious liquid substances presenting pollution hazards only and which are assessed or provisionally, assessed under regulation 6.3 of Annex II of MARPOL may be carried under the requirements of the code applicable to the appropriate position of the entry in this section for "noxious liquid substances not otherwise specified" (n.o.s.).

Table 17.1 : Notes		
Product name	(column a)	The product names are to be used in the shipping document for any cargo offered for bulk shipments. Any additional name may be included in brackets after the product name. In some cases, the product names are not identical with the names given in previous issues of the Code.
UN Number	(column b)	*Deleted*
Pollution category	(column c)	The letter X, Y, Z means the Pollution Category assigned to each product under MARPOL Annex II
Hazards	(column d)	S means that the product is included in the Code because of its safety hazards; P means that the product is included in the Code because of its pollution hazards; and S/P means that the product is included in the Code because of both safety and pollution hazards.
Ship Type	(column e)	1 = Ship Type 1 (2.1.2.1) 2 = Ship Type 2 (2.1.2.2) 3 = Ship Type 3 (2.1.2.3)
Tank type	(column f)	1 = independent tank (4.1.1) 2 = integral tank (4.1.2) G = gravity tank (4.1.3) P = pressure tank (4.1.4)
Tank vents	(column g)	Open: open venting Cont: controlled venting
Tank environmental control	(column h)	Inert: inerting (9.1.2.1) Pad: liquid or gas padding (9.1.2.2) Dry: drying (9.1.2.3) Vent: natural or forced ventilation (9.1.2.4) No: no special requirements under this Code
Electrical equipment	(column i', i'', i''')	Temperature classes (i') T1 to T6 - indicates no requirements blank no information Apparatus group (i'') IIA, IIB or IIC: - indicates no requirements blank no information Flash point (i''') Yes : flashpoint exceeding 60°C (10.1.6) No : flashpoint not exceeding 60°C (10.1.6) NF : nonflammable product (10.1.6)
Gauging	(column j)	O : open gauging (13.1.1.1) R : restricted gauging (13.1.1.2) C : closed gauging (13.1.1.3)
Vapour detection	(column k)	F : flammable vapours T : toxic vapours No: indicates no special requirements under this Code
Fire protection	(column l)	A : alcohol-resistant foam or multi-purpose foam B : regular foam; encompasses all foams that are not of an alcohol-resistant type, including fluoro-protein and aqueous-film - forming foam (AFFF) C : water-spray D : drychemical No : no special requirements under this code
Emergency equipment	(column n)	Yes: see 14.3.1 No: no special requirements under this Code
Specific and operational requirements	(column o)	When specific reference is made to chapters 15 and/or 16, these requirements shall be additional to the requirements in any other column

Table 17.1.1 : Summary of minimum requirements (For details of various columns, see Table 17.1)

a	c	d	e	f	G	h	i'	i''	i'''	j	k	l	n	O
Acetic acid	Z	S/P	3	2G	Cont.	No	T1	IIA	No	C	F	AC	Yes	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.17, 15.19, 16.2.9
Acetic anhydride	Z	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	AC	Yes	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.12.3, 15.12.4, 15.19.6
Acetochlor	X	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.6, 16.2.9
Acetone cyanohydrin	Y	S/P	1	1G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.13, 15.17, 15.19, 16.6.1 to 16.6.3
Acetonitrile	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4 15.19.6
Acetonitrile (Low purity grade)	Y	S/P	3	2G	Cont.	No	T1	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Acid oil mixture from soyabean, corn (maize) and sunflower oil refining	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Acrylamide solution (50% or less)	Y	S/P	3	2G	Cont.	No			NF	C	T	No	No	15.12, 15.13, 15.17, 15.19, 16.2.9, 16.6.1.
Acrylic acid	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.11.2, 15.11.3, 15.11.4, 15.11.6, 15.11.7, 15.11.8, 15.12.3, 15.12.4, 15.13,15.17, 15.19,16.2.9, 16.6.1
Acrylic acid/ethenesulphonic acid copolymer with phosphonate groups, sodium salt solution	Z	P	3	2G	Open	No			Yes	O	No	AB C	No	
Acrylonitrile	Y	S/P	2	2G	Cont.	No	T1	IIB	No	C	F-T	AC	Yes	15.12, 15.13, 15.17, 15.19
Acrylonitrile Styrene copolymer dispersion in polyether polyol	Y	P	3	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6
Adiponitrile	Z	S/P	2	2G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Alachlor technical (90% or more)	X	S/P	2	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6, 16.2.9
Alcohol (C_9-C_{11}) Poly (2-5-9) ethoxylate	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Alcohol (C_6-C_{17})(secondary) Poly (3-6) ethoxylates	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Alcohol (C_6- C_{17}) (secondary) Poly (7-12) ethoxylates	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Alcohol (C10-C18) poly(7) ethoxylate	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Alcohol (C_{12}-C_{16}) Poly (1-6) ethoxylates	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Alcohol (C_{12}-C_{16}) Poly (7-19) ethoxylates	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Alcohol (C_{12}-C_{16}) Poly (20+) ethoxylates	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15,12,3, 15.12.4, 15.19.6,16.2.9

a	c	d	e	f	g	h	i'	i''	i'''	j	k	l	n	o
Alcohols (C_{13}+)	Y	P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.9
Alcohols (C_{12}+), Primary, linear	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Alcohols (C_8-C_{11}), primary, linear and essentially linear	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Alcohols (C_{12}-C_{13}), primary, linear and essentially linear	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Alcohols (C_{14}-C_{18}), primary, linear and essentially linear	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6
Alkanes (C_6-C_9)	X	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19.6
Iso- and cyclo-alkanes(C_{10}-C_{11})	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Iso- and cyclo-alkanes (C_{12}+)	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
n-Alkanes(C_9-C_{11})	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AB C	No	15.19.6
n-Alkanes(C_{10}-C_{20})	Y	P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Alkaryl polyethers (C_9-C_{20})	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.17, 15.19.6, 16.2.6
Alkenoic acid, polyhydroxy ester borated	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Alkenyl (C_{11}+) amide	X	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Alkenyl (C_{16}-C_{20}) succinic anhydride	Z	S/P	3	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.17, 15.19
Alkyl acrylate-vinylpyridine copolymer in toluene	Y	S/P	2	2G	Cont.	No	T1	IIB	No	C	F-T	AB C	No	15.12, 15.17, 15.19.6, 16.2.9
Alkylaryl phosphate mixtures (more than 40% Diphenyl tolyl phosphate, less than 0.02% ortho-isomers)	X	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6
Alkylated (C_4-C_9) hindered phenols	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Alkybenzene, alkylindane, alkylidine mixture (each C_{12}-C_{17})	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Alkyl benzene distillation bottoms	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6
Alkylbenzene mixtures (containing at least 50% of toluene)	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	AB C	No	15.12, 15.17, 15.19.6
Alkylbenzene mixtures (containing napthalene)	X	S/P	2	2G	Cont.	No			Yes	C	T	AB C	No	15.12, 15.17, 15.19.6
Alkyl (C_3 - C_4) benzenes	Y	S/P	2	2G	Cont.	No	T1	IIA	No	R	F-T	AB C	No	15.12.3, 15.12.4, 15.19.6
Alkyl (C_5-C_8) benzenes	X	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6

a	c	d	e	f	g	h	i'	i''	i'''	j	k	l	n	o
Alkyl (C_{9+}) benzenes	Y	S/P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6
Alkyl (C_{11}-C_{17}) benzene sulphonic acid	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Alkylbenzene sulphonic acid, sodium salt solution	Y	S/P	2	2G	Cont.	No	-	-	NF	C	T	No	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Alkyl/cyclo (C_4-C_5) alcohols	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Alkyl (C_{10}-C_{15}, C_{12} rich) phenol poly (4-12) ethoxylate	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Alkyl(C_{12+}) dimethylamine	X	S/P	1	2G	Cont.	No	-	-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Alkyl dithiocarbamate (C_{19}-C_{35})	Y	P	3	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Alkyldithiothiadiazole (C_6-C_{24})	Y	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6, 16.2.6
Alkyl ester copolymer (C_4-C_{20})	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Alkyl (C_7-C_9) nitrates	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 15.20, 16.6.1, 16.6.2, 16.6.3
Alkyl (C_8-C_{10}) / (C_{12}-C_{14}):(40% or less/60% or more) polyglucoside solution (55% or less)	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Alkyl (C_8-C_{10}) / (C_{12}-C_{14}):(60% or more/40% or less) polyglucoside solution (55% or less)	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Alkyl (C_7-C_{11}) phenol poly (4-12) ethoxylate	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Alkyl (C_8-C_{40}) phenol sulphide	Z	S/P	3	2G	Open	No			Yes	O	No	ABC	No	
Alkyl (C_8-C_9) phenylamine in aromatic solvents	Y	S/P	2	2G	Cont.	No	T1	IIB	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Alkyl (C_9-C_{15}) phenyl propoxylate	Z	S/P	3	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Alkyl (C_8-C_{10}) polyglucoside solution (65% or less)	Y	S/P	3	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Alkyl (C_8-C_{10}) / (C_{12}-C_{14}):(50% / 50%) polyglucoside solution (55% or less)	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9, 16.2.6
Alkyl (C_{12}-C_{14}) polyglucoside solution (55% or less)	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Alkyl(C_{12}-C_{16}) propoxyamine ethoxylate	X	S/P	2	2G	Cont	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.6
Alkyl (C_{10}-C_{20}, saturated and unsaturated) phosphite	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Alkyl sulphonic acid ester of phenol	Y	P	3	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Alkyl (C_{18+}) toluenes	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.9

a	c	d	e	f	g	h	i'	i''	i'''	j	k	l	n	o
Alkyl(C_{18}-C_{28}) toluenesulfonic acid	Y	S/P	2	2G	Cont	No	-	-	Yes	C	T	ABC	Yes	15.11.2, 15.11.3, 15.11.4, 15.11.6, 15.11.7, 15.11.8, 15.12, 15.17 15.19, 16.2.6 16.2.9
Alkyl (C_{18}-C_{28}) toluenesulfonic acid,calcium salts, borated	Y	S/P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Alkyl (C_{18}-C_{28}) toluenesulphonic acid, calcium salts, high overbase	Y	S/P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Alkyl (C_{18}-C_{28}) toluenesulphonic acid, calcium salts, low overbase	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Allyl alcohol	Y	S/P	2	2G	Cont.	No	T2	IIB	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Allyl chloride	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19
Aluminium chloride/Hydrogen chloride solution	Y	S/P	2	2G	Cont	No	-	-	NF	C	T	No	Yes	15.11, 15.12, 15.17, 15.19
Aluminium hydroxide, sodium hydroxide, sodium carbonate solution (40% or less)	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19
Aluminium sulphate solution	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19
2-(2-Aminoethoxy) ethanol	Z	S/P	3	2G	Cont.	No			Yes	C	T	AD	Yes	15.12, 15.17, 15.19
Aminoethyldiethanolamine/ Aminoethylethanolamine solution	Z	S/P	3	2G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Aminoethyl ethanolamine	Z	S/P	3	2G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19
N-Aminoethylpiperazine	Z	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
2-Amino-2-methyl-1-propanol	Z	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Ammonia aqueous (28% or less)	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19
Ammonium chloride solution (less than 25%) (*)	Z	S/P	3	2G	Open	No	-	-	NF	O	No	No	No	
Ammonium hydrogen phosphate solution	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Ammonium lignosulphonate solutions	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Ammonium nitrate solution (93% or less) (*)	Z	S/P	2	1G	Cont.	No			NF	R	T	No	No	15.2, 15.11.4, 15.11.6, 15.12.3, 15.12.4, 15.18, 15.19.6, 16.2.9
Ammonium polyphosphate solution	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	
Ammonium sulphate solution	Z	P	3	2G	Open	No			NF	O	No	No	No	
Ammonium sulphide solution (45% or less) (*)	Y	S/P	2	2G	Cont.	Inert	T4	IIB	No	C	F-T	AC	No	15.12, 15.17, 15.19, 16.6.1, 16.6.2, 16.6.3

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Ammonium thiosulphate solution (60% or less)	Z	S/P	3	2G	Open	No			NF	O	No	No	No	
Amyl acetate (all isomers)	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
n-Amyl alcohol	Z	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	ABC	Yes	15.12, 15.17, 15.19
Amyl alcohol, primary	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
sec-Amyl alcohol	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
tert-Amyl alcohol	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
tert-Amyl ethyl ether	Z	P	3	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
tert-Amyl methyl ether	X	S/P	2	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Aniline	Y	S/P	2	2G	Cont.	No	T1	IIA	Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Aryl polyolefins (C_{11}-C_{50})	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Aviation alkylates (C_8 paraffins and iso-paraffines BPT.95- 120°C)	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
Barium long chain (C_{11}-C_{50}) alkaryl sulphonate	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19, 16.2.6, 16.2.9
Benzene and mixtures having 10% benzene or more (i)	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19.6, 16.2.9
Benzene sulphonyl chloride	Y	S/P	3	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.9
Benzenetricarboxylic acid, triotyl ester	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Benzyl acetate	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Benzyl alcohol	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Benzyl chloride	Y	S/P	2	2G	Cont.	No	T1	IIA	No	C	F-T	ABC	Yes	15.12, 15.13, 15.17, 15.19
Bio-fuel blends of Diesel/gas oil and FAME (>25% but <99% by volume)	X	S/P	2	2G	Cont	No	-	-	Yes	C	T	ABC	No	15.12, 15.17, 15.19.6
Bio-fuel blends of Diesel/gas oil and vegetable oil (>25% but <99% by volume)	X	S/P	2	2G	Cont	No	-	-	Yes	C	T	ABC	No	15.12, 15.17, 15.19.6
Bio-fuel blends of Gasoline and Ethyl alcohol (>25% but <99% by volume)	X	S/P	2	2G	Cont	No	T3	IIA	No	R	F-T	AC	No	15.12, 15.17 15.19.6
Bis (2-ethylhexyl) terephthalate	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Brake fluid base mix: Poly (2-8) alkylene (C_2-C_3) glycols/Polyalkylene (C_2-C_{10}) glycols monoalkyl (C_1-C_4) ethers and their borate esters	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	

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Bromochloromethane	Z	P	3	2G	Open	No			NF	O	No	No	No	
Butene oligomer	X	P	2	2G	Cont.	No	T4	IIB	No	R	F	ABC	No	15.19.6
2-Butoxyethanol (58%)/Hyperbranched polyesteramide (42%) (mixture)	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	No	15.12.3, 15.12.4, 15.19
Butyl acetate (all isomers)	Y	P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Butyl acrylate (all isomers)	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F	ABC	No	15.13, 15.19.6, 16.6.1, 16.6.2
tert-Butyl alcohol	Z	P	3	2G	Cont.	No	T1	IIA	No	R	F	AC	No	15.19.6
Butylamine (all isomers)	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Butylbenzene (all isomers)	X	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Butyl benzyl phthalate	X	S/P	2	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
Butyl butyrate (all isomers)	Y	S/P	3	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
Butyl/Decyl/Cetyl/Eicosyl methacrylate mixture	Y	S/P	2	2G	Open	No	T3	IIA	No	R	F	ABC	No	15.13, 15.19.6, 16.6.1, 16.6.2,
Butylene glycol	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	
1,2-Butylene oxide	Y	S/P	3	2G	Cont.	Inert	T2	IIB	No	C	F-T	AC	No	15.8.1 to 15.8.7, 15.8.12, 15.8.13, 15.8.16, 15.8.17, 15.8.18, 15.8.19, 15.8.21, 15.8.25, 15.8.27, 15.8.29, 15.12, 15.17, 15.19.6
n-Butyl ether	Y	S/P	3	2G	Cont.	Inert	T4	IIB	No	R	F	AC	No	15.4.6, 15.19
Butyl methacrylate	Z	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.13, 15.19.6, 16.6.1, 16.6.2
n-Butyl propionate	Y	P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
Butyraldehyde (all isomers)	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Butyric acid	Y	S/P	3	2G	Cont.	No			Yes	O	No	AC	No	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.19.6
gamma-Bytyrolactone	Y	S/P	3	2G	Cont.	No			Yes	C	T	ABC	No	15.12, 15.17, 15.19.6
Calcium alkaryl sulphonate (C_{11}-C_{50})	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	
Calcium alkyl (C_{10}-C_{28}) salicylate	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4 15.19.6, 16.2.9
Calcium hydroxide slurry	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4 15.19.6, 16.2.9
Calcium hypochlorite solution (15% or less)	Y	S/P	2	2G	Cont.	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19.6
Calcium hypochlorite solution (more than 15%)	X	S/P	1	2G	Cont.	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19
Calcium lignosulphonate solutions	Z	P	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Calcium long-chain alkyl (C_5-C_{10}) phenate	Y	P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6
Calcium long-chain alkyl (C_{11}-C_{40}) phenate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6

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Calcium long-chain alkyl phenate sulphide (C_8-C_{40})	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Calcium long-chain alkyl salicylate (C_{13+})	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Calcium long-chain alkyl (C_{18}-C_{28}) salicylate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6 16.2.9
Calcium nitrate/ Magnesium nitrate/ Potassium chloride solution	Z	P	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Calcium nitrate solution (50% or less)	Z	S	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Camelina oil	Y	S/P	2(k)	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7
epsilon-Caprolactam (molten or aqueous solutions)	Z	S/P	3	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Carbolic oil	Y	S/P	2	2G	Cont.	No			Yes	C	F-T	ABC	Yes	15.12, 15.17, 15.19, 16.2.9
Carbon disulphide	Y	S/P	1	1G	Cont.	Pad +Inert	T6	IIC	No	C	F-T	C	Yes	15.3, 15.12, 15.17, 15.18, 15.19
Carbon tetrachloride	Y	S/P	2	2G	Cont.	No			NF	C	T	No	No	15.12, 15.17, 15.19.6
Cashew nut shell oil (untreated)	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.7, 16.2.9
Castor oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Cesium formate solution(*)	Y	S/P	3	2G	Open	No	-	-	NF	O	No	No	No	15.19.6
Cetyl/Eicosyl methacrylate mixture	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.13, 15.19.6, 16.6.1, 16.6.2, 16.2.9
Chlorinated paraffins (C_{10}-C_{13})	X	S/P	1	2G	Cont	No			NF	C	T	No	No	15.12, 15.17, 15.19, 16.2.6
Chlorinated paraffins (C_{14}-C_{17}) (with 50% chlorine or more and less than 1% C_{13} or shorter chains)	X	S/P	1	2G	Cont	No	-	-	Yes	C	T	AC	No	15.12, 15.17, 15.19
Chloroacetic acid (80% or less)	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.11.2, 15.11.3, 15.11.4, 15.11.6, 15.11.7, 15.11.8, 15.12, 15.17, 15.18, 15.19, 16.2.9
Chlorobenzene	Y	S/P	2	2G	Cont.	No	T1	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Chloroform	Y	S/P	3	2G	Cont.	No			NF	C	T	No	No	15.12, 15.17, 15.19.6
Chlorohydrins (crude)	Y	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
4-Chloro-2-methylphenoxyacetic acid, dimethylamine salt solution	Y	S/P	2	2G	Cont	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
o-Chloronitrobenzene	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	No	15.12.3, 15.12.4, 15.19, 16.2.6, 16.2.9
1-(4-Chlorophenyl)-4,4-dimethyl-pentan-3-one	Y	S/P	2	2G	Open	No			Yes	O	No	ABD	No	15.19.6, 16.2.6, 16.2.9

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2-or 3-Chloropropionic acid	Z	S/P	2	2G	Cont	No			Yes	C	T	AC	No	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.12.3, 15.12.4, 15.19, 16.2.9
Chlorosulphonic acid	Y	S/P	1	2G	Cont.	No			NF	C	T	No	Yes	15.11.2 to 15.11.8, 15.12, 15.16.2, 15.17, 15.18, 15.19
m-Chlorotoluene	Y	S/P	2	2G	Cont.	No	T4	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19
o-Chlorotoluene	Y	P	2	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
p-Chlorotoluene	Y	P	2	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6, 16.2.9
Chlorotoluenes (mixed isomers)	Y	P	2	2G	Cont.	No	T4	IIA	No	R	F	ABC	No	15.19.6
Choline chloride solutions	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Citric acid (70% or less)	Z	S/P	3	2G	Cont	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Coal tar	X	S/P	2	2G	Cont.	No	T2	IIA	Yes	C	T	BD	No	15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Coal tar naphtha solvent	Y	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19.6, 16.2.9
Coal tar pitch (molten) (*)	X	S/P	2	1G	Cont.	No	T2	IIA	Yes	C	T	ABC D	No	15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Cocoa butter	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Coconut oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Coconut oil fatty acid	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Coconut oil fatty acid methyl ester	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6
Copper salt of long chain (C_{17+}) alkanoic acid	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Corn Oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Cotton seed Oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Creosote (coal tar)	X	S/P	1	2G	Cont.	No	T2	IIA	Yes	C	T	AD	No	, 15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Cresols (all isomers)	Y	S/P	1	2G	Cont	No	T1	IIA	Yes	C	T	ABC	Yes	15.12, 15.18, 15.19, 16.2.9
Cresol/Phenol/Xylenol mixture	Y	S/P	2	2G	Cont	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Cresylic acid, dephenolized	Y	S/P	2	2G	Cont	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Cresylic acid, sodium salt solution	Y	S/P	2	2G	Cont	No	T4	IIB	No	C	FT	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Crotonaldehyde	X	S/P	1	1G	Cont	No	T3	IIB	No	C	F-T	AC	Yes	15.12, 15.17, 15.18, 15.19
1,5,9-Cyclododecatriene	X	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.13, 15.19.6, 16.6.1, 16.6.2
Cycloheptane	X	S/P	2	2G	Cont.	No	T4	IIA	No	R	F	AC	No	15.19.6
Cyclohexane	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6, 16.2.9

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Cyclohexane-1,2-dicarboxylic acid, diisononyl ester	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Cyclohexane oxidation products, sodium salts solution	Z	P	3	2G	Open	No			NF	O	No	No	No	
Cyclohexanol	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Cyclohexanone	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Cyclohexanone, Cyclohexanol mixture	Y	S/P	3	2G	Cont.	No			Yes	R	F	AC	No	15.19.6
Cyclohexyl acetate	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Cyclohexylamine	Y	S/P	3	2G	Cont.	No	T3	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
1,3-Cyclopentadiene dimer (molten)	Y	S/P	2	2G	Cont.	No	T1	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19, 16.2.6, 16.2.9
Cyclopentane	Y	P	2	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Cyclopentene	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
p-Cymene	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Decahydronaphthalene	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Decanoic acid	X	S/P	2	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Decene	X	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Decyl acrylate	X	S/P	1	2G	Cont	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.13, 15.19, 16.6.1, 16.6.2
Decyl alcohol (all isomers)	Y	P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9 (e)
Decyl/Dodecyl/Tetradecyl alcohol mixture	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Decyloxytetrahydrothiophene dioxide	X	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Diacetone alcohol	Z	S/P	3	2G	Cont.	No	T1	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.9.6
Dialkyl (C_8-C_9) diphenylamines	Z	P	3	2G	Open	No			Yes	O	No	ABC	No	
Dialkyl (C_7-C_{13}) phthalates	X	S/P	2	2G	Cont	No			Yes	C	T	ABC	No	15.12, 15.17, 15.19.6, 16.2.6
Dialkyl (C_9-C_{10}) phthalates	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Dialkyl thiophosphates sodium salts solution	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
2,6-Diaminohexanoic acid phosphonate mixed salts solution	Z	S/P	3	2G	Cont	No			NF	R	No	No	No	15.11, 15.17, 15.19.6
Dibromomethane	Y	S/P	2	2G	Open	No			NF	O	No	No	No	15.19.6
Dibutylamine	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	ABC	Yes	15.12, 15.17, 15.19
Dibutyl hydrogen phosphonate	Y	S/P	2	2G	Cont	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
2,6-Di-tert-butylphenol	X	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.9
Dibutyl phthalate	X	S/P	2	2G	Cont	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6

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Dibutyl terephthalate	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.9
Dichlorobenzenes (all isomers)	X	S/P	2	2G	Cont.	No	T1	IIA	Yes	C	T	ABD	No	15.12, 15.17, 15.19.6
3,4 Dichloro-1-butene	Y	S/P	2	2G	Cont.	No	T1	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
1,1-Dichloroethane	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Dichloroethyl ether	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.18, 15.19
1,6-Dichlorohexane	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6
2,2'-Dichloroisopropyl ether	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19
Dichloromethane	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19.6
2,4-Dichlorophenol	Y	S/P	2	2G	Cont.	Dry			Yes	C	T	AD	Yes	15.12, 15.16.2, 15.17, 15.19, 16.2.6, 16.2.9
2,4-Dichlorophenoxyacetic acid, diethanolamine salt solution	Y	S/P	3	2G	Cont	No			NF	C	T	No	Yes	15.12, 15.17, 15.1916.2.9
2,4-Dichlorophenoxyacetic acid, diethanolamine salt solution (70% or less)	Y	S/P	3	2G	Cont	No			NF	C	T	No	Yes	15.12, 15.17, 15.1916.2.9
2,4-Dichlorophenoxyacetic acid, triisopropanolamine salt solution	Y	S/P	3	2G	Cont	No			NF	C	T	No	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
1,1-Dichloropropane	Y	S/P	2	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
1,2-Dichloropropane	Y	S/P	3	2G	Cont.	No	T1	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
1,3-Dichloropropene	X	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	ABC	Yes	15.12, 15.17, 15.19
Dichloropropene / Dichloropropane mixtures	X	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	ABD	No	15.12, 15.17, 15.19
2,2-Dichloropropionic acid	Y	S/P	2	2G	Cont.	Dry			Yes	C	T	AD	Yes	15.11.2, 15.11.4, 15.11.6, 15.11.7, 15.11.8, 15.12, 15.16.2, 15.17, 15.1916.2.9
Dicyclopentadiene, Resin Grade, 81-89%	Y	S/P	2	2G	Cont	Inert	T2	IIB	No	C	F-T	ABC	Yes	15.12, 15.13, 15.17, 15.19
Diethanolamine	Y	S/P	3	2G	Cont	No	T1	IIA	Yes	C	T	AC	No	15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Diethylamine	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19.6
Diethylaminoethanol	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
2,6-Diethylaniline	Y	S/P	2	2G	Cont	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Diethylbenzene	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Diethylene glycol	Z	S/P	3	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Diethylene glycol dibutyl ether	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	

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Diethylene glycol diethyl ether	Z	S/P	3	2G	Cont	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Diethylene glycol phthalate	Y	S/P	3	2G	Cont	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Diethylenetriamine	Y	S/P	3	2G	Cont	No	-	-	Yes	C	T	ABC	No	15.12, 15.17, 15.19
Diethylenetriaminepentaacetic acid, pentasodium salt solution	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	
Diethyl ether (*)	Z	S/P	2	1G	Cont.	Inert	T4	IIB	No	R	F	AC	No	15.4, 15.14, 15.19
Di-(2-ethylhexyl) adipate	Y	S/P	2	2G	Cont	No			Yes	C	T	ABC	No	15.12, 15.17, 15.19.6
Di-(2-ethylhexyl) phosphoric acid	Y	S/P	2	2G	Cont	No			Yes	R	T	AD	No	15.12.3, 15.12.4, 15.19.6
Diethyl phthalate	Y	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6
Diethyl sulphate	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Diglycidyl ether of bisphenol A	X	S/P	2	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Diglycidyl ether of bisphenol F	Y	S/P	2	2G	Cont	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6, 16.2.6
Diheptyl phthalate	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6
Di-n-hexyl adipate	X	S/P	1	2G	Open	No			Yes	O	No	AC	No	15.19
Dihexyl phthalate	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	No	15.12, 15.17, 15.19.6
Diisobutylamine	Y	S/P	2	2G	Cont.	No	T4	IIB	No	C	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Diisobutylene	Y	P	2	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Diisobutyl ketone	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Diisobutyl phthalate	X	S/P	2	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
Diisononyl adipate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6
Diisooctyl phthalate	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Diisopropanolamine	Z	P	3	2G	Open	No			Yes	O	No	AC	No	16.2.9
Diisopropylamine	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.17, 15.19.6
Diisopropylbenzene (all isomers)	X	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Diisopropylnaphthalene	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6
N,N-Dimethylacetamide	Z	S/P	3	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
N,N-Dimethylacetamide solution (40% or less)	Z	S/P	3	2G	Cont.	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19.6
Dimethyl adipate	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Dimethylamine solution (45% or less)	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19
Dimethylamine solution (greater than 45% but not greater than 55%)	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19
Dimethylamine solution (greater than 55% but not greater than 65%)	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.14, 15.19

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N,N-Dimethylcyclohexylamine	Y	S/P	2	2G	Cont.	No	T3	IIB	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Dimethyl disulphide	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
N-N-Dimethyldodecylamine	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Dimethylethanolamine	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Dimethylformamide	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19.6
Dimethyl glutarate	Y	S/P	3	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Dimethyl hydrogen phosphite	Y	S/P	3	2G	Cont.	No	T4	IIB	No	R	F	AC	No	15.19.6
Dimethyl octanoic acid	Y	S/P	2	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Dimethyl phthalate	Y	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Dimethylpolysiloxane	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6
2,2-Dimethylpropane-1,3-diol (molten or solution)	Z	P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	16.2.9
Dimethyl succinate	Y	P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Dinitrotoluene (molten)	X	S/P	2	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19, 15.21, 16.2.6, 16.2.9, 16.6.4
Dinonyl phthalate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6
Dioctyl phthalate	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6
1,4-Dioxane	Y	S/P	3	2G	Cont.	No	T2	IIB	No	C	F-T	AC	No	15.12, 15.17, 15.19.6, 16.2.9
Dipentene	Y	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Diphenyl	X	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Diphenylamine (molten)	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Diphenylamine, reaction product with 2,2,4-Trimethylpentene	Y	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19, 16.2.6
Diphenylamines, alkylated	Y	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19, 16.2.6, 16.2.9
Diphenyl/Diphenyl ether mixtures	X	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Diphenyl ether	X	P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Diphenyl ether/Diphenyl phenyl ether mixture	X	P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Diphenylmethane diisocyanate	Y	S/P	2	2G	Cont.	Dry	-	-	Yes (a)	C	T (a)	AB (b)D	Yes	15.12, 15.16.2, 15.17, 15.19, 16.2.6, 16.2.9
Diphenylol propane-epichlorohydrin resins	X	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Di-n-propylamine	Y	S/P	2	2G	Cont.	No	T3	IIB	No	C	F-T	AC	Yes	15.12.3, 15.12.4, 15.17, 15.19.6
Dipropylene glycol	Z	P	3	2G	Open	No			Yes	O	No	AC	No	

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Dithiocarbamate ester (C_7-C_{35})	X	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6
Ditridecyl adipate	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Ditridecyl phthalate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6
Diundecyl phthalate	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Dodecane (all isomers)	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
tert-Dodecanethiol	Y	S/P	3	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
1-Dodecene	Y	S/P	3	2G	Open	No			Yes	O	No	ABC	No	15.19.6
Dodecene (all isomers)	X	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Dodecyl alcohol	Y	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
n-dodecyl mercaptan	X	S/P	1	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Dodecylamine/Tetradecyl amine mixture	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.9
Dodecylbenzene	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Dodecyl diphenyl ether disulphonate solution	X	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19, 16.2.6
Dodecyl hydroxypropyl sulphide	X	P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6
Dodecyl methacrylate	Y	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.13, 15.19.6
Dodecyl/Octadecyl methacrylate mixture	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.13, 15.19.6, 16.2.6, 16.6.1, 16.6.2
Dodecyl/Pentadecyl methacrylate mixture	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.13, 15.19.6 16.6.1, 16.6.2
Dodecyl phenol	X	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.6
Dodecyl Xylene	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Drilling brines (containing zinc chloride)	X	S/P	2	2G	Open	No			NF	O	No	No	Yes	15.19.6
Drilling brines (containing calcium bromide)	Z	S/P	3	2G	Open	No			NF	O	No	No	No	15.19.6
Epichlorohydrin	Y	S/P	2	2G	Cont.	No	T2	IIB	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Ethanolamine	Y	S/P	3	2G	Cont.	No	T2	IIA	Yes	C	F-T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
2-Ethoxyethyl acetate	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	FT	AC	No	15.12, 15.17, 15.19.6
Ethoxylated long chain (C_{16+}) alkyloxyalkylamine	Y	S/P	2	2G	Cont.	No	-	-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.9
Ethoxylated tallow amine (> 95%)	X	S/P	2	2G	Cont	Inert	-	-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Ethyl acetate	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
Ethyl acetoacetate	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	
Ethyl acrylate	Y	S/P	2	2G	Cont.	No	T2	IIB	No	C	F-T	AC	No	15.12, 15.13, 15.17, 15.19, 16.6.1, 16.6.2

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Ethylamine (*)	Y	S/P	2	1G	Cont.	No	T2	IIA	No	C	F	AC	No	15.12.3.2, 15.14, 15.19
Ethylamine solutions (72% or less)	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F	AC	No	15.12.3.2, 15.14, 15.19
Ethyl amyl ketone	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Ethylbenzene	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19.6
Ethyl tert-butyl ether	Y	S/P	2	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Ethyl butyrate	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Ethylcyclohexane	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
N-Ethylcyclohexylamine	Y	S/P	2	2G	Cont.	No	T3	IIB	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
S-Ethyl dipropylthiocarbamate	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.9.6, 16.2.9
Ethylene carbonate	Z	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Ethylene chlorohydrin	Y	S/P	1	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.18, 15.19
Ethylene cyanohydrin	Y	S/P	2	2G	Cont.	No		IIB	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Ethylenediamine	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Ethylenediaminetetraacetic acid, tetrasodium salt solution	Y	S/P	3	2G	Cont	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Ethylene dibromide	Y	S/P	2	2G	Cont.	No			NF	C	T	No	No	15.12, 15.17, 15.19, 16.2.9
Ethylene dichloride	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19
Ethylene glycol	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6
Ethylene glycol acetate	Y	S/P	3	2G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Ethylene glycol butyl ether acetate	Y	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6
Ethylene glycol diacetate	Y	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6
Ethylene glycol methyl ether acetate	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
Ethylene glycol monoalkyl ethers	Y	S/P	3	2G	Cont.	No	T2	IIB	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19 16.2.9
Ethylene glycol phenyl ether	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Ethylene glycol phenyl ether/Diethylene glycol phenyl ether mixture	Z	S/P	3	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Ethylene glycol (>75%)/sodium alkyl carboxylates/borax mixture	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
Ethylene glycol (>85%)/sodium alkyl carboxylates mixture	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6

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Ethylene oxide/Propylene oxide mixture with an ethylene oxide content of not more than 30% by mass	Y	S/P	2	1G	Cont.	Inert	T2	IIB	No	C	F-T	AC	Yes	15.8, 15.12, 15.14, 15.17, 15.19
Ethylene-Vinyl acetate copolymer (emulsion)	Y	S/P	3	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Ethyl-3-ethoxypropionate	Y	P	2	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
2-Ethylhexanoic acid	Y	S/P	3	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
2-Ethylhexyl acrylate	Y	S/P	3	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.13, 15.19.6, 16.6.1, 16.6.2
2-Ethylhexylamine	Y	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19.6
2-Ethyl-2 (hydroxymethyl) propane-1,3-diol, C_8-C_{10} ester	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Ethylidene norbornene	Y	S/P	2	2G	Cont.	No	T3	IIB	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Ethyl methacrylate	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.13, 15.19.6, 16.6.1, 16.6.2
N-Ethylmethylallylamine	Y	S/P	2	2G	Cont.	No	T2	IIB	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19
Ethyl propionate	Y	S/P	3	2G	Cont.	No	T1	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
2-Ethyl-3-propylacrolein	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6, 16.2.9
Ethyl toluene	Y	P	2	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
Fatty acid (saturated C_{13+})	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Fatty acid methyl esters (m)	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Fatty acids, C_8-C_{10}	Y	S/P	2	2G	Cont.	No	-	-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Fatty acids, (C_{12}+)	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Fatty acids, (C_{16+})	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Fatty acids, essentially linear, C_6-C_{18}, 2-ethylhexyl ester	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6
Ferric chloride solutions	Y	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.17, 15.19, 16.2.9
Ferric nitrate/nitric acid solution	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.17, 15.19
Fish oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Fish silage protein concentrate (containing 4% or less formic acid)	Y	P	2	2G	Open	No			NF	O	No	No	No	15.19.6, 16.2.6
Fish protein concentrate (containing 4% or less formic acid)	Z	P	3	2G	Open	No			NF	O	No	No	No	
Fluorosilicic acid solution (20-30%)	Y	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.17, 15.19

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Formaldehyde solutions (45% or less)	Y	S/P	3	2G	Cont.	No	T2	IIB	No	C	F-T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Formamide	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6, 16.2.9
Formic acid (85% or less acid)	Y	S/P	3	2G	Cont.	No	-	-	Yes	C	T (g)	AC	Yes	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.12.3, 15.12.4, 15.17, 15.19, 16.2.9
Formic acid (over 85%)	Y	S/P	3	2G	Cont	No	T1	IIA	No	C	F-T(g)	AC	Yes	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.12.3, 15.12.4 15.17, 15.19, 16.2.9
Formic acid mixture (containing up to 18% propionic acid and up to 25%sodium formate)	Z	S/P	3	2G	Cont	No	-	-	Yes	R	T(g)	AC	No	15.11.2, 15.11.3, 15.11.4, 15.11.6, 15.11.7, 15.11.8, 15.12.3, 15.12.4, 15.19.6
Furfural	Y	S/P	3	2G	Cont.	No	T2	IIB	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Furfuryl alcohol	Y	S/P	3	2G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Glucitol/glycerol blend propoxylated (containing less than 10% amines)	Z	S/P	3	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Glucitol/glycerol blend propoxylated (containing 10% or more amines)	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Glutaraldehyde solutions (50% or less)	Y	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19
Glycerine	Z	S	3	2G	Open	No			Yes	O	No	AC	No	16.2.9
Glycerol monooleate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6, 16.2.6, 16.2.9
Glycerol propoxylated	Z	S/P	3	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Glycerol, propoxylated and ethoxylated	Z	P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	
Glycerol/sucrose blend propoxylated and ethoxylated	Z	P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	
Glyceryl triacetate	Z	P	3	2G	Open	No			Yes	O	No	ABC	No	15.19.6
Glycidyl ester of C_{10} trialkylacetic acid	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Glycine, sodium salt solution	Z	S/P	3	2G	Open	No			NF	O	No	No	No	
Glycolic acid solution (70% or less)	Z	S/P	3	2G	Cont.	No	-	-	NF	C	T	No	Yes	15.12.3, 15.12.4, 15.17, 15.19, 16.2.9
Glyoxal solution (40% or less)	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9

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Glyoxylic acid solution (50% or less)	Y	S/P	3	2G	Cont.	No	-	-	Yes	C	T	ACD	Yes	15.11.2, 15.11.3 15.11.4, 15.11.6, 15.11.7, 15.11.8, 15.12, 15.17, 15.19, 16.2.9, 16.6.1, 16.6.2, 16.6.3
Glyphosate solution (not containing surfactant)	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Grapeseed Oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7
Groundnut oil	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Heptane (all isomers)	X	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
n-Heptanoic acid	Z	S/P	3	2G	Cont.	No			Yes	R	No	ABC	No	15.17, 15.19.6
Heptanol (all isomers) (d)	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Heptene (all isomers)	Y	P	2	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
Heptyl acetate	Y	S/P	2	2G	Cont	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
1-Hexadecylnaphthalene / 1,4-bis(hexadecyl)naphthalene mixture	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Hexamethylenediamine (molten)	Y	S/P	3	2G	Cont.	No	-	-	Yes	C	T	A,C	Yes	15.12, 15.17, 15.19, 16.2.9
Hexamethylenediamine adipate (50% in water)	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Hexamethylenediamine solution	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19
Hexamethylene diisocyanate	Y	S/P	2	2G	Cont.	Dry	T1	IIB	Yes	C	T	AC(b) D	Yes	15.12, 15.16.2, 15.17, 15.18,15.19
Hexamethylene glycol	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	
Hexamethyleneimine	Y	S/P	2	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19
Hexamethylenetetramine solutions	Z	S	3	2G	Open	No			Yes	O	No	AC	No	15.19.6
Hexane (all isomers)	Y	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19.6
1,6-Hexanediol, distillation overheads	Y	S/P	3	2G	Cont.	No	-	-	Yes	R	T	AC	No	15,12,3, 15.12.4, 15.19.6, 16.2.9
Hexanoic acid	Y	S/P	3	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Hexanol	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Hexene (all isomers)	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Hexyl acetate	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
Hexylene glycol	Z	S	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Hydrocarbon wax	X	S/P	2	2G	Cont.	No	-	-	Yes	C	T	ABC	No	15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Hydrochloric acid (*)	Z	S/P	3	1G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.17, 15.19

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Hydrogen peroxide solutions (over 8% but not over 60% by mass)	Y	S/P	3	2G	Cont.	No			NF	R	T	No	No	15.5.2, 15.18, 15.12.3, 15.12.4, 15.19.6
Hydrogen peroxide solutions (over 60% but not over 70% by mass)	Y	S/P	2	2G	Cont.	No			NF	R	T	No	No	15.5.1, 15.12.3, 15.12.4, 15.19.6
2-Hydroxyethyl acrylate	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.13, 15.17, 15.19, 16.6.1, 16.6.2
N-(Hydroxyethyl) ethylenediaminetriacetic acid, trisodium salt solution	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
2-Hydroxy-4-(methylthio) butanoic acid	Z	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Illipe oil	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Isoamyl alcohol	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Isobutyl alcohol	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
Isobutyl formate	Z	P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
Isobutyl methacrylate	Z	S/P	3	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.13, , 15.19.6, 16.6.1, 16.6.2
Isophorone	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Isophoronediamine	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Isophorone diisocyanate	Y	S/P	2	2G	Cont.	Dry			Yes	C	T	ABD	Yes	15.12, 15.16.2, 15.17, 15.19
Isoprene	Y	S/P	2	2G	Cont.	No	T3	IIB	No	C	F-T	ABC	No	15.12, 15.13, 15.14, 15.17, 15.19.6, 16.6.1, 16.6.2
Isopropanolamine	Y	S/P	3	2G	Cont.	No	T2	IIA	Yes	R	No	AC	No	15.19.6, 16.2.6, 16.2.9
Ispropyl acetate	Z	P	3	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
Isopropylamine	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12.3.2, 15.14, 15.19
Isopropylamine (70% or less) solution	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12.3.2, 15.19
Isopropylcyclohexane	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6, 16.2.9
Isopropyl ether	Y	S/P	3	2G	Cont.	Inert	T2	IIA	No	R	F	AC	No	15.4.6, 15.13, 15.19.6, 16.6.1, 16.6.2
Jatropha Oil	Y	P	2(k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7
Lactic acid	Z	S/P	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Lactonitrile solution (80% or less)	Y	S/P	1	1G	Cont.	No			NF	C	T	No	Yes	15.12, 15.13, 15.17, 15.18,15.19, 16.6.1, 16.6.2, 16.6.3

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Lard	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Latex, ammonia (1% or less), inhibited	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6, 16.2.6, 16.2.9
Latex: Carboxylated styrene-Butadiene copolymer; Styrene-Butadiene rubber	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Lauric acid	X	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Ligninsulphonic acid, magnesium salt solution	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	
Ligninsulphonic acid, sodium salt solution	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Linseed oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Liquid chemical wastes	X	S/P	2	2G	Cont.	No			No	C	F-T	AC	No	15.12, 15.17, 15.19, 20.5.1, 20.7
Long-chain alkaryl polyether (C_{11}-C_{20})	Y	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Long-chain alkaryl sulphonic acid (C_{16}-C_{60})	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Long-chain alkylphenate/Phenol sulphide mixture	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Long-chain alkylphenol (C14-C18)	Y	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Long-chain alkylphenol (C18-C30)	Y	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
L-Lysine solution (60% or less)	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Magnesium chloride solution	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Magnesuim hydroxide slurry	Z	S	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Magnesium long-chain alkaryl sulphonate (C_{11}-C_{50})	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Magnesium long-chain alkyl salicylate (C_{11+})	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Maleic anhydride	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC (f)	Ye s	15.12, 15.17, 15.19, 16.2.9
Maleic anhydride-sodium allylsulphonate copolymer solution	Z	P	3	2G	Open	No			Yes	O	No	AB C	No	
Mango kernel oil	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16..2.7, 16.2.9
Mercaptobenzothiazol, sodium salt solution	X	S/P	2	2G	Open	No			NF	O	No	No	No	15.19.6, 16.2.9
Mesityl oxide	Z	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6

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Metam sodium solution	X	S/P	2	2G	Cont.	No	-	-	NF	C	T	No	No	15.12.3, 15.12.4, 15.19
Methacrylic acid	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.13, 15.12.3, 15.12.4, 15.19, 16.2.9, 16.6.1
Methacryclic acid – alkoxypoly (alkylene oxide) methacrylate copolymer, sodium salt aqueous solution (45% or less)	Z	S/P	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Methacrylic resin in ethylene dichloride	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19, 16.2.9
Methacrylonitrile	Y	S/P	2	2G	Cont.	No	T1	IIA	No	C	F-T	AC	Yes	15.12, 15.13, 15.17, 15.19
3-Methoxy-1-butanol	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.19.6
3-Methoxybutyl acetate	Y	S/P	3	2G	Open	No			Yes	O	No	ABC	No	15.19.6
N-(2-Methoxy-1-methyl ethyl)-2-ethyl-6-methyl chloroacetanilide	X	S/P	1	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19, 16.2.6
Methyl acetate	Z	P	3	2G	Cont.	No	T1	IIA	No	R	F	AC	No	15.19.6
Methyl acetoacetate	Z	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Methyl acrylate	Y	S/P	3	2G	Cont.	No	T1	IIB	No	C	F-T	AC	No	15.12, 15.13, 15.17, 15.19,
Methyl alcohol (*)	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	AC	No	15.12.1, 15.12.2, 15.12.3.2, 15.12.3.3, 15.12.4, 15.17, 15.19
Methylamine solutions (42% or less)	Y	S/P	2	2G	Cont.	No	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Methylamyl acetate	Y	P	2	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
Methylamyl alcohol	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Methyl amyl ketone	Z	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
N-Methylaniline	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Alpha-methylbenzyl alcohol with acetophenone (15% or less	Y	S/P	2	2G	Cont.	No	-	-	Yes	C	T	ABC	Yes	15.12, 15.17 15.19, 16.2.6 16.2.9
Methylbutenol	Y	S/P	3	2G	Cont.	No	T4	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Methyl tert-butyl ether	Z	P	3	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
Methyl butyl ketone	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19.6
Methylbutynol	Z	S/P	3	2G	Cont.	No	T4	IIB	No	R	F	AC	No	15.19.6
Methyl butyrate	Y	S/P	3	2G	Cont.	No	T4	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6

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Methylcyclohexane	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Methylcyclopentadiene dimer	Y	S/P	2	2G	Cont.	No	T4	IIB	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Methylcyclopentadienyl manganese tricarbonyl	X	S/P	2	2G	Cont.	No	-	-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.18, 15.19, 16.2.9
Methyl diethanolamine	Y	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6,16.2.6
2-Methyl-6-ethyl aniline	Y	S/P	3	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
Methyl ethyl ketone	Z	S/P	3	2G	Cont.	No	T1	IIA	No	R	F	AC	No	15.19.6
2-Methyl-5-ethyl pyridine	Y	S/P	2	2G	Cont.	No		-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Methyl formate	Z	S/P	2	2G	Cont.	No	T1	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.14, 15.19.6
2-Methylglutaronitrile with 2-Ethylsuccinonitrile (12% or less)	Z	S/P	3	2G	Cont	No	-	-	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
2-Methyl-2-hydroxy-3-butyne	Z	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6, 16.2.9
Methyl isobutyl ketone	Z	S/P	3	2G	Cont.	No	T1	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6
Methyl methacrylate	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.13, 15.19.6,
3-Methyl-3-methoxybutanol	Z	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Methylnaphthalene (molten)	X	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
N-Methylglucamine solution (70% or less)	Z	S	3	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
2-Methyl-1,3-propanediol	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	
2-Methylpyridine	Z	S/P	3	2G	Cont.	No	T1	IIA	No	C	F	AC	No	15.12.3.2, 15.19
3-Methylpyridine	Z	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19
4-Methylpyridine	Z	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19, 16.2.9
N-Methyl-2-pyrrolidone	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
Methyl propyl ketone	Z	S	3	2G	Cont.	No	T1	IIA	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.9.6
Methyl salicylate	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
alpha- Methylstyrene	Y	S/P	2	2G	Cont.	No	T1	IIB	No	C	F-T	AD(j)	No	15.12, 15.13, 15.17, 15.19.6, 16.6.1, 16.6.2
3-(methylthio) propionaldehyde	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	ABC	No	15.12, 15.17, 15.19.6
Molybdenum Polysulfide Long Chain Alkyl Dithiocarbamide Complex	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9

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Morpholine	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12.3, 15.12.4, 15.19
Motor fuel anti-knock compounds (containing lead alkyls)	X	S/P	1	1G	Cont.	Inert	T4	IIA	No	C	F-T	AC	Yes	15.6, 15.12, 15.17, 15.18, 15.19
Myrcene	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Naphthalene (molten)	X	S/P	2	2G	Cont.	No	T1	IIA	Yes	C	T	ABC	No	15.12, 15.17, 15.19.6, 16.2.9
Naphthalene crude (molten)	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	No	15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Naphthalenesulphonic acid-Formaldehyde copolymer, sodium salt solution	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Neodecanoic acid	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Nitrating acid (mixture of sulphuric and nitirc acids)	Y	S/P	1	1G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.16.2, 15.17, 15.18, 15.19
Nitric acid (less than 70%)	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.17, 15.19
Nitric acid (70% and over)	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.6.2, 15.17, 15.19
Nitrilotriacetic acid, trisodium salt solution	Y	S/P	3	2G	Cont.	No			Yes	C	T	AC	No	15.12, 15.17, 15.19.6
Nitrobenzene	Y	S/P	2	2G	Cont.	No	-	-	Yes	C	T	ABC	No	15.12, 15.17, 15.19, 16.2.9
Nitroethane	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	ABC (f)	No	15.12.3, 15.12.4, 15.19.6, 16.6.1, 16.6.2, 16.6.4
Nitroethane (80%)/Nitropropane (20%)	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	ABC (f)	No	15.12.3, 15.12.4, 15.19.6, 16.6.1 to 16.6.3
Nitroethane, 1-Nitropropane (each 15% or more) mixture	Y	S/P	3	2G	Cont.	No	T2	IIB	No	R	F-T	ABC (f)	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.6.1, 16.6.2, 16.6.3
o-Nitrophenol (molten)	Y	S/P	2	2G	Cont.	No	T4	IIB	No	R	F	ABC	No	15.19.6, 16.2.6, 16.2.9
1- or 2-Nitropropane	Y	S/P	3	2G	Cont.	No	T2	IIB	No	C	F-T	AC	No	15.12, 15.17, 15.19
Nitropropane (60%)/Nitroethane (40%) mixture	Y	S/P	2	2G	Cont.	No	T2	IIB	No	C	F-T	ABC (f)	No	15.12, 15.17, 15.19.6
o- or p-Nitrotoluenes	Y	S/P	2	2G	Cont.	No		IIB	Yes	C	T	ABC	No	15.12, 15.17, 15.19.6
Nonane (all isomers)	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
Nonanoic acid (all isomers)	Y	S/P	2	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.9

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Non-edible industrial grade palm oil	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.7, 16.2.9
Nonene (all isomers)	Y	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Nonyl alcohol (all isomers)	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Nonyl methacrylate monomer	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.9
Nonylphenol	X	S/P	1	2G	Cont.	No			Yes	C	T	AC	Ye s	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Nonylphenol poly(4+)ethoxylate	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6
Noxious liquid, N.F.(1) n.o.s. (trade name..., contains...) S.T.1, Cat.X	X	P	1	2G	Open	No	-	-	Yes	O	No	AC	No	15.19, 16.2.6
Noxious liquid, F, (2) n.o.s. (trade name..., contains...) S.T.1, Cat.X	X	P	1	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19, 16.2.6
Noxious liquid, N.F.(3) n.o.s. (trade name..., contains...) S.T.2, Cat.X	X	P	2	2G	Open	No	-		Yes	O	No	AC	No	15.19, 16.2.6
Noxious liquid, F, (4) n.o.s. (trade name..., contains...) S.T.2,Cat.X	X	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19, 16.2.6
Noxious liquid, N.F. (5) n.o.s. (trade name..., contains...) S.T.2, Cat.Y	Y	P	2	2G	Open	No	-		Yes	O	No	AC	No	15.19, 16.2.6, 16.2.9 (l)
Noxious liquid, F. (6) n.o.s. (trade name..., contains...) S.T.2, Cat.Y	Y	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19, 16.2.6, 16.2.9 (l)
Noxious liquid, N.F. (7) n.o.s. (trade name..., contains...) S.T.3, Cat.Y	Y	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	15.19, 16.2.6, 16.2.9(l)
Noxious liquid, F. (8) n.o.s. (trade name..., contains...) S.T.3, Cat.Y	Y	P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19, 16.2.6, 16.2.9 (l)
Noxious liquid, N.F. (9) n.o.s. (trade name..., contains...) S.T.3, Cat.Z	Z	P	3	2G	Open	No	-		Yes	O	No	AC	No	
Noxious liquid, F, (10) n.o.s. (trade name..., contains...) S.T.3, Cat.Z	Z	P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Octamethylcyclotetrasiloxane	Y	P	2	2G	Cont	No	T2	IIA	No	R	F	AC	No	15.19.6 16.2.9
Octane (all isomers)	X	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Octanoic acid (all isomers)	Y	S/P	2	2G	Cont.	No	-	-	Yes	C	T	AB C	Ye s	15.12, 15.17, 15.19
Octanol (all isomers)	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Octene (all isomers)	Y	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
n-Octyl acetate	Y	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Octyl aldehydes	Y	S/P	2	2G	Cont.	No	T4	IIB	No	R	F	AC	No	15.19.6, 16.2.9
Octyl decyl adipate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6, 16.2.9

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n-Octyl mercaptan	X	S/P	1	2G	Open	No			Yes	O	No	ABC	No	15.19
Offshore contaminated bulk liquid P (o)	X	P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6
Offshore contaminated bulk liquid S (o)	X	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	AC	Yes	15.12, 15.15, 15.17, 15.19
Olefin-Alkyl ester copolymer (molecular weight 2000+)	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Olefin Mixture (C_7-C_9) C_8 rich, stabilised	X	P	2	2G	Cont	No	T3	IIB	No	R	F	ABC	No	15.13, 15.19.6
Olefin mixtures (C_5-C_7)	Y	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	A	No	15.19.6
Olefin mixtures (C_5-C_{15})	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6
Olefin (C_{13+}, all isomers)	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Alpha-Olefins (C_6-C_{18}) mixtures	X	S/P	2	2G	Cont.	No	T4	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Oleic acid	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Oleum	Y	S/P	2	2G	Cont.	Dry	-	-	NF	C	T	No	Yes	15.11.2 to 15.11.8, 15.12, 15.16.2, 15.17, 15.19, 16.2.6
Oleylamine	X	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Olive oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Oxygenated aliphatic hydrocarbon mixtures	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	ABC	No	
Palm acid oil	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm fatty acid distillate	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm kernel acid oil	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm kernel fatty acid distillate	Y	S/P	2	2G	Cont	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm kernel oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm kernel olein	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm kernel stearin	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm mid fraction	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm oil	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Palm oil fatty acid methyl ester	Y	P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6, 16.2.9
Palm olein	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9

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Palm stearin	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Paraffin wax, highly refined	Y	P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Paraffin wax, semi-refined	X	S/P	2	2G	Cont.	No	-	-	Yes	C	T	AB C	No	15.12, 15.17, 15.19.6, 16.2.6, 16.2.9
Paraldehyde	Z	S/P	3	2G	Cont.	No	T3	IIB	No	R	F	AC	No	15.19.6, 16.2.9
Paraldehyde-ammonia reaction product	Y	S/P	2	2G	Cont.	No	T1	IIB	No	C	F-T	AB C	Yes	15.12, 15.17, 15.19
Pentachloroethane	Y	S/P	2	2G	Cont.	No			NF	C	T	No	No	15.12, 15.17, 15.19.6
1,3-Pentadiene	Y	P	3	2G	Cont.	No	T1	IIA	No	R	F	AB C	No	15.13, 15.19.6, 16.6.1 to 16.6.3
1,3-Pentadiene (greater than 50%),cyclopentene and isomers, mixtures	Y	S/P	2	2G	Cont	Inert	T3	IIB	No	C	F-T	AB C	Yes	15.12, 15.13, 15.17, 15.19
Pentaethylenehexamine	X	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.17, 15.19
Pentane (all isomers)	Y	P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	15.14, 15.19.6
Pentanoic acid	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.17, 15.19
n-Pentanoic acid (64%)/2-Methyl butyric acid (36%) mixture	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.11.2, 15.11.3, 15.11.4, 15.11.6 to 15.11.8, 15.12, 15.17, 15.19
Pentene (all isomers)	Y	P	2	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.14, 15.19.6
n-Pentyl propionate	Y	S/P	3	2G	Cont.	No	T2	IIA	No	R	F-T	AB C	No	15.12.3, 15.12.4, 15.19.6
Perchloroethylene	Y	S/P	2	2G	Cont.	No			NF	C	T	No	No	15.12, 15.17, 15.19.6
Phenol	Y	S/P	2	2G	Cont.	No	T1	IIA	Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
1-Phenyl-1-xylyl ethane	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6
Phosphate esters, alkyl (C_{12}-C_{14}) amine	Y	S/P	2	2G	Cont.	No	T4	IIB	No	R	F-T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Phosphoric acid	Z	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.11.1 to 15.11.4, 15.11.6 to 15.11.8, 15.12, 15.17, 15.19, 16.2.9
Phosphorus, yellow or white (*)	X	S/P	1	1G	Cont.	Pad + (Vent or Inert)			No (c)	C	No	AB C	No	15.7, 15.19, 16.2.9
Phthalic anhydride (molten)	Y	S/P	2	2G	Cont.	No	T1	IIA	Yes	C	T	AB C	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
alpha-Pinene	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	AB C	No	15.19.6

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beta-Pinene	X	S/P	2	2G	Cont.	No	T1	IIB	No	R	F	ABC	No	15.19.6
Pine oil	X	S/P	2	2G	Open	No			Yes	O		ABC	No	15.19.6,16.2.6 16.2.9
Piperazine, 68% solution	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Polyacrylic acid solution (40% or less)	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	
Polyalkyl (C_{18}-C_{22}) acrylate in xylene	Y	S/P	2	2G	Cont.	No	T1	IIB	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6 16.2.9
Polyalkylalkenaminesuccinimide, molybdenum oxy sulphide	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Poly(2-8)alkylene glycol monoalkyl (C_1-C_6) ether	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	
Poly(2-8)alkylene glycol monoalkyl (C_1-C_6) ether acetate	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6
Polyalkyl (C_{10}-C_{20}) methacrylate	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6 16.2.9
Polyalkyl (C_{10}-C_{18}) methacrylate/ethylene-propylene copolymer mixture	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyaluminium chloride solution	Z	S	3	2G	Open	No			NF	O	No	No	No	
Polybutene	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Polybutenyl succinimide	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Poly(2+)cyclic aromatics	X	S/P	1	2G	Cont.	No			Yes	C	T	ABC	No	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Polyether (molecular weight 1350+)	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6
Polyethylene glycol	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Polyethylene glycol dimethyl ether	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	
Poly(ethylene glycol) methylbutenyl ether(MW>1000)	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Polyethylene polyamines	Y	S/P	2	2G	Cont.	No	-	-	Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.6, 16.2.9
Polyethylene polyamines(more than 50% C_5-C_{20} paraffin oil)	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Polyferric sulphate solution	Y	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19

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Poly(imioethylene)-graft-N-poly(ethyleneoxy) solution (90% or less)	Z	S/P	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Polyisobutenamine in aliphatic (C_{10}-C_{14}) solvent	Y	S/P	2	2G	Cont.	No	-	-	Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6
(Polyisobutene) amino products in aliphatic hydrocarbons	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Polyisobutenyl anhydride adduct	Z	S/P	3	2G	Open	No			Yes	O	No	ABC	No	
Poly(4+)isobutylene (MW>224)	X	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyisobutylene (MW≤224)	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Polyglycerin, sodium salt solution (containing less than 3% sodium hydroxide)	Z	S	2	2G	Cont.	No			Yes	C	T	AC	Yes	15,12, 15,17, 15,19, 16.2.9
Polymethylene polyphenyl isocyanate	Y	S/P	3	2G	Cont.	Dry			Yes (a)	C	T (a)	AD	Yes	15.12, 15.16.2, 15.17, 15.19.6, 16.2.9
Polyolefin (molecular weight 300+)	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyolefin amide alkeneamine (C_{17+})	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Polyolefin amide alkeneamine borate (C_{28}-C_{250})	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyolefin amide alkeneamine polyol	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyolefinamine (C_{28}-C_{250})	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Polyolefinamine in alkyl (C_2-C_4) benzenes	Y	S/P	2	2G	Cont.	No	T2	IIB	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Polyolefinamine in aromatic solvent	Y	S/P	2	2G	Cont.	No	T2	IIB	No	R	F-T	ABC	No	15.12.3, 15.12.4, 15.19.6,16.2.6, 16.2.9
Polyolefin aminoester salts (molecular weight 2000+)	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyolefin anhydride	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Polyolefin ester (C_{28}-C_{250})	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyolefin phenolic amine (C_{28}-C_{250})	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Polyolefin phosphorosulphide, barium derivative (C_{28}-C_{250})	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Poly(20)oxyethylene sorbitan monooleate	Y	P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.6, 16.2.9
Poly(5+)propylene	Y	P	3	2G	Open	no	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.9
Polypropylene glycol	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6

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Polysiloxane	Y	P	2	2G	Cont.	No	T2	IIB	No	R	F	ABC	No	15.19.6, 16.2.9
Potassium chloride solution	Z	P	3	2G	Open	No	-	-	NF	O	No	No	No	16.2.9
Potassium hydroxide solution (*)	Y	S/P	3	2G	Open	No			NF	C	No	No	No	15.12.3.2, 15.19
Potassium formate solutions (*)	Z	S	3	2G	Open	No			NF	R	No	No	No	15.19.6
Potassium oleate	Y	S/P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.6, 16.2.9
Potassium thiosulphate (50% or less)	Y	S/P	3	2G	Cont.	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
n-Propanolamine	Y	S/P	3	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 16.2.9, 15.19
2-Propene-1-aminium, N,N-dimethyl-N-2-propenyl-, chloride, homopolymer solution	Y	P	3	2G	Open	No	-	-	NF	O	No	No	No	15.19.6
beta-Propiolactone	Y	S/P	1	2G	Cont.	No		IIA	Yes	C	T	AC	Yes	15.12, 15.17, 15.18, 15.19
Propionaldehyde	Y	S/P	3	2G	Cont.	Inert	T4	IIB	No	R	F	AC	No	15.19.6
Propionic acid	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	AC	Yes	15.11.2 to 15.11.4, 15.11.6 to 15.11.8, 15.12, 15.17, 15.19
Propionic anhydride	Y	S/P	2	2G	Cont.	No	T2	IIA	Yes	C	T	AC	Yes	15.12 15.17, 15.19
Propionitrile	Y	S/P	1	1G	Cont.	No	T1	IIB	No	C	F-T	AC	Yes	15.12, 15.17 to 15.19
n-Propyl acetate	Y	P	3	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6
n-propyl alcohol	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19.6
n-Propylamine	Z	S/P	2	2G	Cont.	Inert	T2	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19
Propylbenzene (all isomers)	Y	P	3	2G	Cont.	No	T2	IIA	No	R	F	ABC	No	15.19.6
Propylene carbonate	Z	S	3	2G	Cont.	No			Yes	C	T	ABC	Yes	15.12, 15.17, 15.19
Propylene glycol methyl ether acetate	Z	P	3	2G	Cont.	No	T2	IIA	No	R	F	AC	No	
Propylene glycol monoalkyl ether	Z	S/P	3	2G	Cont.	No	T3	IIA	No	R	F	AC	No	15.19.6
Propylene glycol phenyl ether	Z	S/P	3	2G	Open	No			Yes	O	No	ABC	No	
Propylene oxide	Y	S/P	2	2G	Cont.	Inert	T2	IIB	No	C	F-T	AC	No	15.8, 15.12.1, 15.14, 15.17, 15.19
Propylene tetramer	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
Propylene trimer	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F	ABC	No	15.19.6
Pyridine	Y	S/P	3	2G	Cont.	No	T1	IIA	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.19.6

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Pyrolysis gasoline (containing benzene)	Y	S/P	2	2G	Cont.	No	T3	IIA	No	C	F-T	ABC	No	15.12, 15.17, 15.19.6
Rapeseed oil	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Rapeseed oil(low erucic acid containing less than 4% free fatty acids)	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Rape seed oil fatty acid methyl esters	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6
Resin oil, distilled	Y	S/P	2	2G	Cont	No	T1	IIA	No	C	FT	ABC	No	15.12,15.17,15.19.6
Rice bran oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Rosin	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Safflower oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Shea butter	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Sodium alkyl (C_{14}-C_{17}) sulphonates (60-65% solution)	Y	S/P	2	2G	Cont.	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Sodium aluminosilicate slurry	Z	P	3	2G	Open	No			NF	O	No	No	No	16.2.9
Sodium benzoate	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	16.2.9
Sodium borohydride (15% or less)/Sodium hydroxide solution (*)	Y	S/P	3	2G	Open	No			NF	C	No	No	No	15.19, 16.2.6, 16.2.9
Sodium bromide solution (less than 50%)(*)	Y	S/P	3	2G	Open	No	-	-	NF	R	No	No	No	15.19.6
Sodium carbonate solution (*)	Z	S/P	3	2G	Open	No			NF	R	No	No	No	15.19.6
Sodium chlorate solution (50% or less) (*)	Z	S/P	3	2G	Open	No			NF	R	No	No	No	15.9, 15.12, 15.19, 16.2.9
Sodium dichromate solution (70% or less)	Y	S/P	1	1G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.18, 15.19
Sodium hydrogen sulphide (6% or less)/Sodium carbonate (3% or less) solution	Z	S/P	3	2G	Open	No			NF	O	No	No	No	15.19.6, 16.2.9
Sodium hydrogen sulphite solution (45% or less)	Z	P	3	2G	Open	No			NF	O	No	No	No	16.2.9
Sodium hydrosulphide/Ammonium sulphide solution (*)	Y	S/P	2	2G	Cont.	No	T4	IIB	No	C	F-T	AC	Yes	15.12, 15.15, 15.17, 15.19, 16.6.1, 16.6.2, 16.6.3
Sodium hydrosulphide solution (45% or less) (*)	Z	S/P	3	2G	Cont.	Vent or pad (gas)			NF	R	T	No	Yes	15.12, 15.15, 15.19.6, 16.2.9

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Sodium hydroxide solution (*)	Y	S/P	3	2G	Open	No			NF	C	No	No	No	15.19, 16.2.6, 16.2.9
Sodium hypochlorite solution (15% or less)	Y	S/P	2	2G	Cont.	No	-	-	NF	R	No	No	No	15.17, 15.19.6
Sodium methylate 21-30% in methanol	Y	S/P	2	2G	Cont	No	T1	IIA	No	C	F-T	AC	Yes	15.12, 15.17, 15.19, 16.2.6(only if >28%), 16.2.9
Sodium nitrite solution	Y	S/P	3	2G	Cont.	No			NF	C	T	No	No	15.12.3, 15.12.4, 15.19, 16.2.6, 16.2.9
Sodium petroleum sulphonate	Y	S/P	2	2G	Cont.	No			Yes	R	T	ABC	Yes	15.12.3, 15.12.4, 15.19.6, 16.2.6
Sodium poly(4+) acrylate solutions	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AC	No	16.2.9
Sodium silicate solution	Y	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19,16.2.9
Sodium sulphate solutions	Z	S	3	2G	Open	No			NF	O	No	No	No	16.2.9
Sodium sulphide solution (15% or less)	Y	S/P	3	2G	Cont.	No			NF	C	T	No	Yes	15.12, 15.17, 15.19, 16.2.9
Sodium sulphite solution (25% or less)	Y	S/P	3	2G	Open	No			NF	O	No	No	No	15.19.6, 16.2.9
Sodium thiocyanate solution (56% or less)	Y	S/P	3	2G	Open	No			NF	O	No	No	No	15.19.6, 16.2.9
Soyabean oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Soybean Oil Fatty Acid Methyl Ester	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.9
Styrene monomer	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	ABC	No	15.12, 15.13, 15.17, 15.19.6, 16.6.1, 16.6.2
Sulphohydrocarbon (C_3-C_{88})	Y	P	2	2G	Open	No	-	-	Yes	O	No	ABC	No	15.19.6, 16.2.6, 16.2.9
Sulpholane	Y	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Sulphur (molten) (*)	Z	S	3	1G	Open	Vent or pad (gas)	T3		Yes	O	F-T	No	No	15.10, 16.2.9
Sulphuric acid	Y	S/P	2	2G	Cont.	No			NF	C	T	No	Yes	15.11, 15.12, 15.16.2, 15.17, 15.19, 16.2.9
Sulphuric acid, spent	Y	S/P	2	2G	Open	No			NF	C	T	No	Yes	15.11, 15.12, 15.16.2, 15.17, 15.19
Sulphurized fat (C_{14}-C_{20})	Z	S/P	3	2G	Open	No			Yes	O	No	ABC	No	
Sulphurized polyolefinamide alkene (C_{28}-C_{250}) amine	Z	P	3	2G	Open	No	-	-	Yes	O	No	AC	No	

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Sunflower seed oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Tall oil, crude	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6
Tall oil, distilled	Y	P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6
Tall oil fatty acid (resin acids less than 20%)	Y	S/P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6
Tall oil pitch	Y	P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Tall oil soap, crude	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.17, 15.19, 16.2.6
Tallow	Y	P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Tallow fatty acid	Y	P	2	2G	Open	No	-	-	Yes	O	No	AC	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Tetrachloroethane	Y	S/P	2	2G	Cont.	No			NF	R	T	No	No	15.12.3, 15.12.4, 15.19
Tetraethylene glycol	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Tetraethylene pentamine	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19
Tetrahydrofuran	Z	S	3	2G	Cont.	No	T3	IIB	No	R	F	AC	No	15.19.6
Tetrahydronaphthalene	Y	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6
Tetramethylbenzene (all isomers)	X	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.9
Titanium dioxide slurry	Z	P	3	2G	Open	No			NF	O	No	No	No	
Toluene	Y	S/P	3	2G	Cont.	No	T1	IIA	No	C	F-T	AC	No	15.12, 15.17, 15.19.6
Toluenediamine	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.17, 15.18, 15.19, 16.2.6, 16.2.9
Toluene diisocyanate	Y	S/P	2	2G	Cont.	Dry			Yes	C	T	AB C(b) D	Yes	15.12, 15.16.2, 15.17, 15.18, 15.19, 16.2.9
o-Toluidine	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	No	15.12, 15.17, 15.19
Tributyl phosphate	Y	S/P	3	2G	Cont.	No			Yes	C	T	AB C	No	15.12.3, 15.12.4, 15.19.6
1,2,3-Trichlorobenzene (molten)	X	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, , 15.19.6, 16.2.9, 16.2.6
1,2,4-Trichlorobenzene	X	S/P	1	2G	Cont.	No			Yes	C	T	AB C	No	15.12, 15.17, 15.19, 16.2.9
1,1,1-Trichloroethane	Y	P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6
1,1,2-Trichloroethane	Y	S/P	3	2G	Open	No			NF	O	No	No	No	15.19.6
Trichloroethylene	Y	S/P	2	2G	Cont.	No			NF	C	T	No	No	15.12, 15.17, 15.19.6

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1,2,3-Trichloropropane	Y	S/P	3	2G	Cont.	No			Yes	C	T	AB C	No	15.12, 15.17, 15.19
1,1,2-Trichloro-1,2,2-Trifluoroethane	Y	P	2	2G	Open	No			NF	O	No	No	No	15.19.6
Tricresyl phosphate (containing less than 1% ortho-isomer)	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	No	15.12, 15.17, 15.19, 16.2.6
Tricresyl phosphate (containing 1% or more ortho-isomer)	Y	S/P	2	2G	Cont.	No	T2	IIA	Yes	C	T	AB C	No	15.12, 15.17, 15.19.6, 16.2.6
Tridecane	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6
Tridecanoic acid	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Tridecyl acetate	Y	S/P	3	2G	Cont.	No	-	-	Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6
Triethanolamine	Z	S/P	3	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Triethylamine	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	F-T	AB C	No	15.12.3, 15.12.4, 15.19
Triethylbenzene	X	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6
Triethylenetetramine	Y	S/P	2	2G	Cont.	No			Yes	C	T	AC	Yes	15.12, 15.17, 15.19, 16.2.9
Triethyl phosphate	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6
Triethyl phosphite	Z	S/P	3	2G	Cont.	No	T3	IIA	No	R	F-T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Triisopropanolamine	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.9
Triisopropylated phenyl phosphates	X	P	2	2G	Open	No			Yes	O	No	AC	No	15.19.6, 16.2.6
Trimethylacetic acid	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.11, 15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
Trimethylamine solution (30% or less)	Z	S/P	2	2G	Cont.	No	T3	IIB	No	R	F-T	AC	No	15.12.3, 15.12.4, 15.14, 15.19.6
Trimethylbenzene (all isomers)	X	S/P	2	2G	Cont.	No	T1	IIA	No	R	F	AB C	No	15.19.6
Trimethylol propane propoxylated	Z	S/P	3	2G	Open	No	-	-	Yes	O	No	AB C	No	
2,2,4-Trimethyl-1,3-pentanediol diisobutyrate	Y	S/P	3	2G	Open	No			Yes	O	No	AB C	No	15.19.6
2,2,4-Trimethyl-1,3-pentanediol-1-isobutyrate	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6
1,3,5-Trioxane	Y	S/P	3	2G	Cont.	No	T2	IIB	No	C	F-T	AC	No	15.12, 15.17, 15.19.6, 16.2.9
Tripropylene glycol	Z	P	3	2G	Open	No			Yes	O	No	AC	No	
Trixylyl phosphate	X	S/P	1	2G	Cont.	No			Yes	C	T	AB C	No	15.12, 15.17, 15.19.6, 16.2.6

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Tung oil	Y	S/P	2 (k)	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Turpentine	X	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	AC	No	15.19.6
Undecanoic acid	Y	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.6, 16.2.9
1-Undecene	X	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6
Undecyl alcohol	X	S/P	2	2G	Cont.	No			Yes	R	T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Urea/Ammonium nitrate solution	Y	S/P	3	2G	Open	No			NF	O	No	No	No	15.19.6
Urea/Ammonium phosphate solution	Y	S/P	2	2G	Cont.	No			Yes	R	T	AC	No	15.12.3, 15.12.4, 15.19.6
Urea solution	Z	S/P	3	2G	Open	No			Yes	O	No	AC	No	16.2.9
Used cooking oil (m)	X	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.9
Used cooking oil (Triglycerides, C16-C18 and C18 unsaturated) (m) (n)	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Valeraldehyde (all isomers)	Y	S/P	3	2G	Cont.	Inert	T3	IIB	No	R	F	AB C	No	15.4.6, 15.13, 15.19.6, 16.6.1, 16.6.2
Vegetable acid oils (m)	Y	S/P	2	2G	Open	No	-	-	-Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Vegetable fatty acid distillates (m)	Y	P	2	2G	Open	No	-	-	Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Vegetable oil mixtures, containing less than 15% free fatty acid (m)	Y	S/P	2	2G	Open	No			Yes	O	No	AB C	No	15.19.6, 16.2.6, 16.2.7, 16.2.9
Vinyl acetate	Y	S/P	3	2G	Cont.	No	T2	IIA	No	C	FT	AB C	No	15.12, 15.13, 15.17, 15.19.6, 16.16.1, 16.6.2
Vinyl ethyl ether	Z	S/P	2	2G	Cont.	Inert	T3	IIB	No	R	F	AB C	No	15.4, 15.13, 15.14, 15.19.6, 16.6.1, 16.6.2
Vinylidene chloride	Y	S/P	2	2G	Cont.	Inert	T2	IIA	No	C	F-T	AB C	No	15.12, 15.13, 15.14, 15.17, 15.19, 16.6.1, 16.6.2
Vinyl neodecanoate	Y	S/P	2	2G	Cont.	No			Yes	C	T	AB C	Yes	15.12, 15.13, 15.17, 15.19, 16.6.1, 16.6.2
Vinyltoluene	Y	S/P	2	2G	Cont	No	T1	IIA	No	C	FT	AB C	No	15.12, 15.13, 15.17, 15.19.6, 16.6.1, 16.6.2
White spirit, low (15-20%) aromatic	Y	S/P	2	2G	Cont.	No	T3	IIA	No	R	F-T	AB C	No	15.12.3, 15.12.4, 15.19.6, 16.2.9
Wood lignin with sodium acetate/oxalate	Z	S/P	3	2G	Open	No	-	-	NF	O	No	No	No	

a	c	d	e	f	g	h	i'	i''	i'''	j	k	l	n	o
Xylenes	Y	P	2	2G	Cont.	No	T1	IIA	No	R	F	ABC	No	15.19.6, 16.2.9 (h)
Xylenes/ethylbenzene (10% or more) mixture	Y	S/P	2	2G	Cont.	No	T2	IIA	No	R	F-T	ABC	No	15.12.,3 15.12.4, 15.19.6
Xylenol	Y	S/P	2	2G	Cont.	No	-	IIA	Yes	C	T	ABC	Yes	15.12, 15.17, 15.19, 16.2.9
Zinc alkaryl dithiophosphate (C_7-C_{16})	Y	P	2	2G	Open	No			Yes	O	No	AB	No	15.19.6, 16.2.6, 16.2.9
Zinc alkenyl carboxamide	Y	S/P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6
Zinc alkyl dithiophosphate (C_3-C_{14})	Y	P	2	2G	Open	No			Yes	O	No	ABC	No	15.19.6, 16.2.6

Notes:

a If the product to be carried contains flammable solvents such that the flashpoint does not exceed 60°C then special electrical systems and a flammable-vapour detector is to be provided.

b Although water is suitable for extinguishing open-air fires involving chemicals to which this footnote applies, water is not to be allowed to contaminate closed tanks containing these chemicals because of the risk of hazardous gas generation.

c Phosphorus, yellow or white is carried above its auto-ignition temperature and therefore flashpoint is not appropriate. Electrical equipment requirements may be similar to those for substances with a flashpoint above 60°C.

d Requirements are based on those isomers having a flashpoint of 60°C, or less; some isomers have a flashpoint greater than 60°C, and therefore the requirements based on flammability would not apply to such isomers.

e Applies to n-decyl alcohol only.

f Dry chemical is not to be used as fire extinguishing media.

g Confined spaces are to be tested for both formic acid vapours and carbon monoxide gas, a decomposition product.

h Applies to p-xylene only.

i For mixtures containing no other components with safety hazards and where the pollution category is Y or less.

j Only certain alcohol-resistant foams are effective.

k Requirements for Ship Type identified in column e might be subject to regulation 4.1.3 of Annex II of MARPOL 73/78.

l Applicable when the melting point is equal to greater than 0°C.

m From vegetable oils, animal fats and fish oils specified in this chapter.

n Confirmation that the product is composed of Triglycerides, C16-C18 and C18 unsaturated would be required in order for the entry to be used. Otherwise, the more generic entry "Used cooking oil (m)" must be used.

o Indicates that the entries are to be used solely for backloading of contaminated bulk liquids from offshore installations used in the search and exploitation of seabed mineral resources.

* indicates that with reference to Chapter 21 of the IBC Code (paragraph 21.1.3), deviations from the normal assignment criteria for some carriage requirements have been implemented.

Section 18

List of Products to which this Chapter does not Apply

18.1 The following are products which have been reviewed for their safety and pollution hazards and determined not to present hazards to such an extent as to warrant application of this Chapter.

18.2 Although the products listed in this Section fall outside the scope of the Chapter, it is to be noted that some safety precautions may be needed for their safe transportation. Accordingly, IRS would prescribe appropriate safety requirements.

18.3 Some chemicals are identified as falling into pollution category Z and therefore, subject to certain operational requirements of Annex II of MARPOL 73/78.

18.4 Liquid mixtures which are assessed or provisionally assessed under regulation 6.3 of Annex II of MARPOL as falling into pollution category Z or OS and which do not present safety hazards, may be carried under the appropriate entry in this Section for “Noxious or Non-Noxious Liquid Substances, not otherwise specified (n.o.s.)”.

18.5 Explanatory notes to Table 18.1.1 are given below:

Product name	The product name is to be used in the shipping document for any cargo offered for bulk shipments. Any additional name may be included in brackets after the product name. In some cases, the product names are not identical with the names given in previous issues of the Code.
Pollution category	The letter Z means the pollution category assigned to each product under Annex II of MARPOL 73/78. OS means the product was evaluated and found to fall outside the categories X, Y or Z.

Table 18.1.1	
Product Name	**Pollution Category**
Acetone	Z
Alcoholic beverages, n.o.s.	Z
Apple juice	OS
n-Butyl alcohol	Z
sec-Butyl alcohol	Z
Calcium Carbonate Slurry	OS
Clay slurry	OS
Coal slurry	OS
Ethyl alcohol	Z
Glucose solution	OS
Glycerol ethoxylated	OS
Hydrogenated starch hydrolysate	OS
Isopropyl alcohol	Z
Kaolin slurry	OS
Lecithin	OS
Maltitol solution	OS
Microsilica Slurry	OS
Molasses	OS
Noxious liquid, (11) n.o.s. (trade name …, contains …) Cat.Z	Z
Non-noxious liquid, (12) n.o.s. (trade name …, contains …) Cat OS	OS
Orange Juice (Concentrated)	OS
Orange Juice (Not Concentrated)	OS
Potassium Chloride Solution (<26%)	OS
Propylene glycol	OS
Sodium acetate solutions	Z
Sodium Bicarbonate Solution (<10%)	OS
Sorbitol solution	OS
Sulphonated polyacrylate solution	Z
Tetraethyl silicate monomer/oligomer (20% in ethanol)	Z
Triethylene glycol	OS
Vegetable protein solution (hydrolysed)	OS
Water	OS

Section 19

Index of Products Carried in Bulk

19.1 General

IR19.1.1 For complete list of products covered by Sec.17 and 18 of this chapter, reference is to be made to the “Index of Products” given in IBC Code Chapter 19. The index of products contains information regarding the Index Name, Product Name and the relevant Chapter Number of the IBC Code. The Index has been developed for information purposes only. None of the Index Names indicated in non-bold lower case in the first column are to be used as the Product Name on the shipping document.

Section 20

Transport of Liquid Chemical Wastes

20.1 Preamble

20.1.1 Maritime transport of liquid chemical wastes could present a threat to human health and to the environment.

20.1.2 Liquid chemical wastes are, therefore, to be transported in accordance with relevant international conventions and recommendations and, in particular, where it concerns maritime transport in bulk, with the requirements of this Chapter.

20.2 Definitions

For the purpose of this section:

20.2.1 "Liquid chemical wastes" are substances, solutions or mixtures, offered for shipment, containing or contaminated with one or more constituents which are subject to the requirements of this Chapter and for which no direct use is envisaged but which are carried for dumping, incineration or other methods of disposal other than at sea.

20.2.2 "Transboundary movement" means maritime transport of wastes from an area under the national jurisdiction of one country to or through an area under the national jurisdiction of another country, or to or through an area not under the national jurisdiction of any country, provided at least two countries are concerned by the movement.

20.3 Applicability

20.3.1 The requirements of this Section are applicable to the transboundary movement of liquid chemical wastes in bulk by seagoing ships and are to be considered in conjunction with all other requirements of this Chapter.

20.3.2 The requirements of this Section do not apply to:

.1 wastes derived from shipboard operations which are covered by the requirements of MARPOL 73/78;

.2 substances, solutions or mixtures containing or contaminated with radioactive materials which are subject to the applicable requirements for radioactive materials.

20.4 Permitted shipments

20.4.1Transboundary movement of wastes is permitted to commence only when:

.1 notification has been sent by the competent authority of the country of origin, or by the generator or exporter through the channel of the competent authority of the country of origin, to the country of final destination; and

.2 the competent authority of the country of origin, having received the written consent of the country of final destination stating that the wastes will be safely incinerated or treated by other methods of disposal, has given authorization to the movement.

20.5 Documentation

20.5.1 In addition to the documentation specified in 16.2 of this Chapter, ships engaged in transboundary movement of liquid chemical wastes are to carry on board a waste movement document issued by the competent authority of the country of origin.

20.6 Classification of liquid chemical wastes

20.6.1 For the purpose of the protection of the marine environment, all liquid chemical wastes transported in bulk are to be treated as Category A noxious liquid substances, irrespective of the actual evaluated category.

20.7 Carriage and handling of liquid chemical wastes

20.7.1 Liquid chemical wastes are to be carried in ships and cargo tanks in accordance with the minimum requirements for liquid chemical wastes specified in Sec.17, unless there are clear grounds indicating that the hazards of the wastes would warrant:

.1 carriage in accordance with the ship type 1 requirements; or

.2 any additional requirements of this Chapter applicable to the substance or, in case of a mixture, its constituent presenting the predominant hazard.

Section 21

Inert Gas Systems

IR21.1 Application

IR21.1.1 Chemical tankers of deadweight in excess of 20,000 tons having individual cargo tank capacity exceeding 3000 m^3 or cargo tanks fitted with washing machine with a nozzle capacity exceeding 17.5 m^3/hr or a total through put per tank of 110 m^3/hr and intended for carriage of chemicals with flash point less than 60°C are to be fitted with an inert gas system complying with requirements given in part 5, Ch.2, Sec 11. (Also refer Pt.6, Ch.2, 1.5.5.2).

IR21.1.2 As an alternative to water seal in the inert gas line on deck, as required by Pt.5, Ch.2, Sec.11.6.5, an arrangement consisting of automatically operated double shut off valves in series with a venting valve in between may be accepted (double block and bleed type), subject to the following conditions:

The operation of the valve is to be automatically executed. Signal(s) for opening / closing is (are) to be taken from the process directly, e.g. inert gas flow or differential pressure.

Alarm for faulty operation of the valves is to be provided, e.g. the operation status of "Blower stop" and "supply valve(s) open" is an alarm condition.

21.2 Operation of inert gas systems

21.2.1 The inert gas systems for all tankers are to be operated in accordance with requirements laid down in Pt. 6, Ch.5, 3.3.3. In applying Pt. 6, Ch.5, 3.3.3, the requirements of Pt.6, Ch. 8, 15.2.2.1.4 are also to be complied with.

End of Chapter

Chapter 4

Liquefied Gas Carriers

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Introduction

IR1.0 General

IR1.1 These Rules have been prepared to ensure that ships for the carriage of liquefied gases in bulk built with a view to classification with IRS, will also comply with the requirements of IMO International Code for the construction and Equipment of ships carrying liquefied gases in bulk as interpreted by IRS.

IR1.2 Responsibility for interpretation of the IGC Code requirements for the purpose of issuing an International Certificate of Fitness for the carriage of Liquefied Gases in Bulk lies with the statutory authority of the country where the ship is to be registered.

IR1.3 These Rules are considered to incorporate the final text of the IGC code in full. The number of various clauses in these Rules are the same as those appearing in the IMO code. The numbering of paragraphs which are not there in the IMO code, but have been included in these Rules for the purpose of classification have been prefixed by the letters 'IR'.

IR1.4 The IGC code contains requirements for operational matters which are not within the scope of classification requirements and IRS does not require these to be investigated for the purpose of classification. However, these matters are the responsibility of the National Authority or Administration responsible for issuing the International Certificate of Fitness.

IR1.5 For the purpose for classification the following words in the construction rules of the IMO code (not in the survey regulations) have been changed as appropriate:

'Organization'	to	'IRS'
'should be'	to	'is to be' or 'are to be'
'chapter'	to	'section'
'code'	to	'chapter'
'Section'	to	'Clause'

IR1.6 Where for the purpose of issuing an International Certificate of Fitness, a National Authority has specifically accepted an equivalent under the terms of the IGC code or has adopted an interpretation different from that agreed to by IRS, individual consideration will be given to the acceptance of the equivalent or interpretation concerned for the purpose of classification.

IR2.0 Classification and class notations

IR2.1 The regulations for classification and the assignment of class notations are given in Pt.1 of the Rules, to which reference is to be made. In general, the class notation to be assigned would be "**SUL Liquefied gas carrier**" where the vessel is designed and constructed primarily for the carriage of liquefied gases in bulk in integral, membrane or independent tanks.

IR2.2 Additional class notation in respect of following items will be assigned as appropriate:

- Ship Type, i.e. 1G, 2G, 2PG or 3G;
- Type of Tanks;
- Name(s) of gas(es);
- Maximum vapour pressure (at sea and in harbor); minimum and (where necessary) maximum cargo temperature;
- Design ambient temperatures (when the ship is suitable for continuous service in high and/or low temperature climatic conditions).

Preamble (To the IGC Code)

1. The purpose of this Code is to provide an international standard for the safe carriage by sea in bulk, of liquefied gases and certain other substances listed that are listed in chapter 19 (Section 19 of this Chapter). Through consideration of the products carried, it prescribes the design and construction standards of the ships involved and the equipment they should carry to minimize the risk to the ship, its crew and the environment.

2. The basic philosophy is one of ship types related to the hazards of the products covered by the Code. Each of the products may have one or more hazard properties which include flammability, toxicity, corrosivity and reactivity. A further possible hazard may arise where products are transported under cryogenic or pressure conditions.

3. Severe collisions or strandings could lead to cargo tank damage and result in uncontrolled release of the product. Such a release could result in evaporation and dispersion of the product and, in some cases, could cause brittle fracture of the ship's hull. The requirements in the Code are intended to minimize this risk as far as practicable, based upon present knowledge and technology.

4. Throughout the development of the Code, it was recognized that it must be based on sound naval architectural and engineering principles and the best understanding available as to the hazards of the various products covered. Gas carrier design technology is not only a complex technology but is rapidly evolving and the Code shall not remain static. The Organization will periodically review the Code, continually taking into account both experience and future development.

5. Requirements for new products and their conditions of carriage will be circulated as recommendations, on an interim basis, when adopted by the Maritime Safety Committee of the Organization, prior to the entry into force of the appropriate amendments, under the terms of article VIII of the International Convention for the Safety of Life at Sea, 1974.

6. The Code primarily deals with ship design and equipment. To ensure the safe transport of the products the total system must, however, be appraised. Other important facets of the safe transport of the products, such as training, operation, traffic control and handling in port, are being or will be examined further by the Organization.

7. The development of the Code has been greatly assisted by a number of organizations in consultative status, such as Society of International Gas Tanker and Terminal Operators Limited (SIGTTO) and other organizations, such as members of the International Association of Classification Societies (IACS)

8. Chapter.18 (Section 18 of this chapter) of the Code dealing with operation of liquefied gas carriers highlights the regulations in other chapters (Sections of this chapter) that are operational in nature and mentions those other important safety features that are peculiar to gas carrier operation.

9. The layout of the Code is in line with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (IBC Code) adopted by the Maritime Safety Committee at its forty-eighth session. Gas carriers may also carry in bulk liquid chemicals covered by IBC Code(Chapter 3), as Prescribed in the IGC Code(this chapter).

10. Floating production, storage and offloading (FPSO) facilities, which are designed to handle liquefied gases in bulk, do not fall under the IGC Code. However, designers of such units may consider using IGC Code to the extent that the Code provides the most appropriate risk mitigation measures for the operations the unit is to perform. Where other more appropriate risk mitigation measures are determined that are contrary to this Code, they shall take precedence over the Code.

Section 1

General

1.0 Goal

To provide an international standard for the safe carriage, by sea in bulk, of liquefied gases by laying down the design and construction standards of ships involved in such carriage and the equipment, they shall carry to minimize the risk to the ship, its crew and to the environment, having regard to the nature of the products including flammability, toxicity, asphyxiation, corrosivity, reactivity and low temperature and vapour pressure.

1.1 Application and implementation

1.1.1 This Chapter applies to ships regardless of their size, including those of less than 500 gross tonnage, engaged in carriage of liquefied gases having a vapour pressure exceeding 2.8 [bar] absolute at a temperature of 37.8°C, and other products as shown in Sec.19, when carried in bulk.

IR 1.1.1 The requirements of this Chapter are in addition to the requirements of Pt.1 to 4 of the Rules.

1.1.2.1 Unless expressly provided otherwise, the Chapter applies to ships whose keels are laid or which are at a similar stage of construction where:

.1 construction identifiable with the ship begins; and

.2 assembly of that ship has commenced, comprising at least 50 tonnes or 1 per cent of the estimated mass of all structural material, whichever is less;

on or after 01 July, 2016.

1.1.2.2 For the purpose of this chapter, the expression “ship constructed” means ships the keels of which are at a similar stage of construction.

1.1.2.3 Unless expressly provided otherwise, for ships constructed on or after 1 July 1986 and before 1 July 2016, the requirements which are applicable under IGC Code, as adopted by resolution MSC.5(48) as amended by resolutions MSC.17(58), MSC.30(61), MSC.32(63), MSC.59(67), MSC.103(73), MSC.177(79) and MSC.220(82), are to be complied with.

1.1.3 A ship, irrespective of the date of construction, which is converted to a gas carrier on or after 1 July, 2016, is to be treated as a gas carrier constructed on the date on which such conversion commences.

IR 1.1.3 For classification, these Rules apply to ships where the midship section is approved on or after the effective date.

1.1.4.1 When cargo tanks contain products for which the Chapter requires a type 1G ship, neither flammable liquids having a flashpoint of 60°C (closed cup test) or less nor flammable products listed in Sec.19 are to be carried in tanks located within the protective zones described in 2.4.1.1.

1.1.4.2 Similarly, when cargo tanks contain products for which the Chapter requires a type 2G/2PG ship, the flammable liquids as described in 1.1.4.1 are not to be carried in tanks located within the protective zones described in 2.4.1.2.

1.1.4.3 In each case, for the cargo tanks loaded with products for which this chapter requires a Type 1G or 2G/2PG ship, the restriction applies to the protective zones within the longitudinal extent of the hold spaces for those tanks.

1.1.4.4 The flammable liquids and products described in 1.1.4.1 may be carried within these protective zones when the quantity of products retained in the cargo tanks, for which the Chapter requires a Type 1G or 2G/2PG ship is solely used for cooling, circulation or fuelling purposes.

1.1.5 Except as provided in 1.1.7.1, when it is intended to carry products covered by this chapter and products covered by the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (IBC Code (Chapter 3)) adopted by resolution MSC.4(48), as may be amended by the Organization, the ship is to comply with the requirements of both Codes appropriate to the products carried.

1.1.6.1 Where it is proposed to carry products that may be considered to come within the scope of the chapter that are not at present designated in Sec.19, the Administration and the port Administrations involved in such carriage should establish Tripartite Agreement based on a provisional assessment and lay down preliminary suitable conditions of carriage based on the principles of the Code.

1.1.6.2 For the evaluation of such products, the manufacturer of the product should submit to the Administration a completed assessment form (see Appendix 1, of IGC Code), which includes the proposed ship type and carriage requirements.

1.1.6.3 When a provisional assessment for a pure or technically pure product has been completed and agreed with the other parties, the Administration will submit the assessment form and a proposal for a new and complete entry in the IGC Code, to the relevant sub-committee of the IMO.

1.1.6.4 After provisional assessment by tripartite agreement and express or tacit agreement has been established, an addendum to the relevant ship's certificate may be issued.

1.1.7.1 The requirements of this chapter are to take precedence when a ship is designed and constructed for the carriage of the following products:

.1 those listed exclusively in Sec.19 of this Chapter; and

.2 one or more of the products that are listed both in this chapter and in the the International Bulk Chemical Code (Chapter 3). These products are marked with an asterisk (*) in column "a" in the Table 19.1.1.

1.1.7.2 When a ship is intended to exclusively carry one or more of the products referred to in 1.1.7.1.2, the requirements of the International Bulk Chemical Code (Chapter 3) as amended will apply.

1.1.8 The ship's compliance with the requirements of the International Gas Carrier Code is to be shown by its International Certificate of Fitness for the carriage of liquefied gases in bulk as described in 1.5. Compliance with the amendments to the Code, as appropriate, is also to be indicated in the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk.

1.1.9 Where reference is made in this chapter to a paragraph, all the provisions of the subparagraph of that designation shall apply.

1.1.10 When a ship is intended to operate for periods at a fixed location in a re-gasification and gas discharge mode or a gas receiving, processing, liquefaction and storage mode, the Administration/ IRS and port Administrations involved in the operation will take appropriate steps to ensure implementation of the provisions of the IGC Code as are applicable to the proposed arrangements. Furthermore, additional requirements will be established based on the principles of the IGC Code as well as recognized standards that address specific risks not envisaged by it. Such risks may include, but not be limited to:

.1 fire and explosion;

.2 evacuation;

.3 extension of hazardous areas;

.4 pressurized gas discharge to shore;

.5 high-pressure gas venting;

.6 process upset conditions;

.7 storage and handling of flammable refrigerants;

.8 continuous presence of liquid and vapour cargo outside the cargo containment system;

.9 tank over-pressure and under-pressure;

.10 ship-to-ship transfer of liquid cargo; and

.11 collision risk during berthing manoeuvres.

1.1.11 Where a risk assessment or study of similar intent is utilized within this chapter, the results are also to include, but not be limited to, the following as evidence of effectiveness:

.1 description of methodology and standards applied;

.2 potential variation in scenario interpretation or sources of error in the study;

.3 validation of the risk assessment process by an independent and suitable third party;

.4 quality system under which the risk assessment was developed;

.5 the source, suitability and validity of data used within the assessment;

.6 the knowledge base of persons involved within the assessment;

.7 system of distribution of results to relevant parties; and

.8 validation of results by an independent and suitable third party.

1.1.12 Although the IGC Code is legally treated as a mandatory instrument under SOLAS Convention, the provisions of Sec 4.28 (4.28 of this chapter) and appendices 1, 3, and 4 of the IGC Code are recommendatory or informative

1.2 Definitions

IR 1.2.1 Following definitions are in addition to the various definitions given in Pt.4 of the Rules.

1.2.1 "**Accommodation spaces**" are those spaces used for public spaces, corridors, lavatories, cabins, offices, hospitals, cinemas, games and hobby rooms, barber shops, pantries without cooking appliances and similar spaces.

1.2.2 "**"A" Class divisions**" are divisions as defined in regulation II-2/3.2 of SOLAS Convention (Rules Pt.6, Ch.1, Cl.3.2).

1.2.3 "**Administration**" means the Government of the State whose flag the ship is entitled to fly. For Administration (Port), see port Administration

1.2.4 **Anniversary date** means the day and the month of each year that will correspond to the date of expiry of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk.

1.2.5 "**Boiling point**" is the temperature at which a product exhibits a vapour pressure equal to the atmospheric barometric pressure.

1.2.6 "**Breadth (B)**" means the maximum breadth of the ship, measured amidships to the moulded line of the frame in a ship with a metal shell and to the outer surface of the hull in a ship with a shell of any other material. The breadth (B) is to measured in metres.

IR 1.2.6 For the determination of scantlings for hull construction, the breadth (B) to be taken is defined in Pt.3 of the Rules.

1.2.7 "**Cargo area**" is that part of the ship which contains the cargo containment system and cargo pump and compressor rooms and includes deck areas over the full length and breadth of the part of the ship over these spaces. Where fitted, the cofferdams, ballast or void spaces at the after end of the aftermost hold space or at the forward end of the foremost hold space are excluded from the cargo area.

1.2.8 "**Cargo containment system**" is the arrangement for containment of cargo including, where fitted, a primary and secondary barrier, associated insulation and any intervening spaces, and adjacent structure, if necessary, for the support of these elements. If the secondary barrier is part of the hull structure, it may be a boundary of the hold space.

1.2.9 "**Cargo control room**" is a space used in the control of cargo handling operations.

1.2.10 "**Cargo machinery spaces"** are the spaces where cargo compressors or pumps, cargo processing units, are located, including those supplying gas fuel to the engine-room.

1.2.11 "**Cargo Pumps"** are pumps used for the transfer of liquid cargo including main pumps, booster pumps, spray pumps, etc.

1.2.12 "**Cargoes**" are products listed in Sec.19 that are carried in bulk by ships subject to this chapter.

1.2.13 "**Cargo service spaces**" are spaces within the cargo area, used for workshops, lockers and store-rooms that are of more than 2 [m^2] in area.

1.2.14 "**Cargo tank**" is the liquid-tight shell designed to be the primary container of the cargo and includes all such containment systems whether or not they are associated with the insulation and/or the secondary barriers.

1.2.15 "**Closed loop sampling"** is a cargo sampling system that minimizes the escape of cargo vapour to the atmosphere by returning product to the cargo tank during sampling.

1.2.16 "**Cofferdam**" is the isolating space between two adjacent steel bulkheads or decks. This space may be a void space or ballast space.

1.2.17 "**Control stations**" are those spaces in which ships' radio, main navigating equipment or the emergency source of power is located or where the fire recording or fire control equipment is centralized. This does not include special fire control equipment which can be most practically located in the cargo area.

1.2.18 "**Flammable products**" are those identified by an "F" in column "f" in Sec.19 (Table 19.1.1).

1.2.19 "**Flammability limits**" are the conditions defining the state of fuel-oxidant mixture at which application of an adequately strong external ignition source is only just capable of producing flammability in a given test apparatus.

1.2.20 FSS Code is the Fire Safety Systems Code meaning the International Code for Fire Safety Systems, adopted by the Maritime Safety Committee of the IMO by resolution MSC.98(73), as amended.

1.2.21 "**Gas carrier**" is a cargo ship constructed or adapted and used for the carriage in bulk of any liquefied gas or other products listed in the table of Sec.19.

1.2.22 "**Gas combustion unit (GCU)**" is a means of disposing excess cargo vapour by thermal oxidation.

1.2.23 "**Gas consumer"** is any unit within the ship using cargo vapour as a fuel.

1.2.24 "**Hazardous area"** is an area in which an explosive gas atmosphere is, or may be expected to be present, in quantities that require special precautions for the construction, installation and use of electrical equipment. When a gas atmosphere is present, the following hazards may also be present: toxicity, asphyxiation, corrosivity, reactivity and low temperature. These hazards shall also be taken into account and additional precautions for the ventilation of spaces and protection of the crew will need to be considered. Examples of hazardous areas include, but are not limited to, the following:

.1 the interiors of cargo containment systems and any pipework of pressure-relief or other venting systems for cargo tanks, pipes and equipment containing the cargo;

.2 interbarrier spaces;

.3 hold spaces where the cargo containment system requires a secondary barrier;

.4 hold spaces where the cargo containment system does not require a secondary barrier;

.5 a space separated from a hold space by a single gastight steel boundary where the cargo containment system requires a secondary barrier;

.6 cargo machinery spaces;

.7 areas on open deck, or semi-enclosed spaces on open deck, within 3 [m] of possible sources of gas release, such as cargo valve, cargo pipe flange, cargo machinery space ventilation outlet, etc.;

.8 areas on open deck, or semi-enclosed spaces on open deck within 1.5 [m] of cargo machinery space entrances, cargo machinery space ventilation inlets;

.9 areas on open deck over the cargo area and 3 [m] forward and aft of the cargo area on the open deck up to a height of 2.4 [m] above the weather deck;

.10 an area within 2.4 [m] of the outer surface of a cargo containment system where such surface is exposed to the weather;

.11 enclosed or semi-enclosed spaces in which pipes containing cargoes are

located, except those where pipes containing cargo products for boil-off gas fuel burning systems are located;

.12 an enclosed or semi-enclosed space having a direct opening into any hazardous area;

.13 void spaces, cofferdams, trunks, passageways and enclosed or semi-enclosed spaces, adjacent to, or immediately above or below, the cargo containment system;

.14 areas on open deck or semi-enclosed spaces on open deck above and in the vicinity of any vent riser outlet, within a vertical cylinder of unlimited height and 6 [m] radius centred upon the centre of the outlet and within a hemisphere of 6 [m] radius below the outlet; and

.15 areas on open deck within spillage containment surrounding cargo manifold valves and 3 [m] beyond these up to a height of 2.4 [m] above deck.

1.2.25 "**Non-hazardous area"** is an area other than a hazardous area.

1.2.26 "**Hold space**" is the space enclosed by the ship's structure in which a cargo containment system is situated.

1.2.27 "**IBC Code"** means the *International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk*, adopted by the Maritime Safety Committee of the IMO by resolution MSC.4(48), as amended.

1.2.28 "**Independent**" means that a piping or venting system, for example, is in no way connected to another system and there are no provisions available for the potential connection to other systems.

1.2.29 "**Insulation space**" is the space, which may or may not be an interbarrier space, occupied wholly or in part by insulation.

1.2.30 "**Interbarrier space**" is the space between a primary and a secondary barrier, whether or not completely or partially occupied by insulation or other material.

1.2.31 "**Length (L)**" is the length as defined in the International Convention on Load Lines in force.

IR 1.2.31 For the determination of scantlings for hull construction, the length (L) to be taken is defined in Pt.3 of the Rules.

1.2.32 "**Machinery spaces of category A**" are those spaces and trunks to those spaces, which contain either:

.1 internal combustion machinery used for main propulsion; or

.2 internal combustion machinery used for purposes other than main propulsion where such machinery has in the aggregate a total power output of not less than 375 [kW]; or

.3 any oil-fired boiler or oil fuel unit or any oil-fired equipment other than boilers, such as inert gas generators, incinerators, etc.

1.2.33 "**Machinery spaces**" are machinery spaces of category A and other spaces containing propelling machinery, boilers, oil fuel units, steam and internal combustion engines, generators and major electrical machinery, oil filling stations, refrigerating, stabilizing, ventilation and air-conditioning machinery, and similar spaces; and the trunks to such spaces.

1.2.34 "**MARVS**" is the maximum allowable relief valve setting of a cargo tank (gauge pressure).

1.2.35 **"Nominated surveyor"** is a surveyor nominated/appointed by an Administration to enforce the provisions of the SOLAS Convention regulations with regard to inspections and surveys and the granting of exemptions therefrom.

1.2.36 "**Oil fuel unit**" is the equipment used for the preparation of oil fuel for delivery to an oil-fired boiler, or equipment used for the preparation for delivery of heated oil to an internal combustion engine, and includes any oil pressure pumps, filters and heaters dealing with oil at a pressure of more than 1.8 bar gauge.

1.2.37 "**Organization**" is the International Maritime Organization (IMO).

1.2.38 "**Permeability of a space**" means the ratio of the volume within that space which is assumed to be occupied by water to the total volume of that space.

1.2.39 **"Port Administration"** means the appropriate authority of the country for the port where the ship is loading or unloading.

1.2.40 "**Primary barrier**" is the inner element designed to contain the cargo when the cargo containment system includes two boundaries.

1.2.41 "**Products"** is the collective term used to cover the list of gases indicated in sec 19 of this chapter.

1.2.42 "**Public spaces"** are those portions of the accommodation that are used for halls, dining rooms, lounges and similar permanently enclosed spaces.

1.2.43 **"Recognized organization"** is an organization authorized by an Administration in accordance with SOLAS regulation XI-1/1.

1.2.44 "**Recognized standards**" are applicable international or national standards acceptable to the Administration or standards laid down and maintained by the recognized organization

1.2.45 "**Relative density**" is the ratio of the mass of a volume of a product to the mass of an equal volume of fresh water.

1.2.46 **"Secondary barrier"** is the liquid-resisting outer element of a cargo containment system, designed to afford temporary containment of any envisaged leakage of liquid cargo through the primary barrier and to prevent the lowering of the temperature of the ship's structure to an unsafe level. Types of secondary barrier are more fully defined in sec 4.

1.2.47 "**Separate systems**" are those cargo piping and vent system that are not permanently connected to each other.

1.2.48 "**Service spaces**" are those used for galleys, pantries containing cooking appliances, lockers, mail and specie rooms, store-rooms, workshops other than those forming part of the machinery spaces and similar spaces and trunks to such spaces.

1.2.49 "**SOLAS Convention**" means the International Convention for the Safety of Life at Sea, 1974, as amended.

1.2.50 "**Tank cover**" is the protective structure intended to either protect the cargo containment system against damage where it protrudes through the weather deck or to ensure the continuity and integrity of the deck structure.

1.2.51 "**Tank dome**" is the upward extension of a portion of a cargo tank. In the case of below-deck cargo containment systems, the tank dome protrudes through the weather deck or through a tank cover.

1.2.52 **"Thermal oxidation method"** means a system where the boil-off vapours are utilized as fuel for shipboard use or as a waste heat system subject to the provisions of sec 16 or a system not using the gas as fuel complying with this Code.

1.2.53 "**Toxic products**" are identified by a "T" in column "f" in Sec.19 (Table 19.1.1).

1.2.54 **"Turret compartments"** are those spaces and trunks that contain equipment and machinery for retrieval and release of the disconnectable turret mooring system, high-pressure hydraulic operating systems, fire protection arrangements and cargo transfer valves.

1.2.55 "**Vapour pressure**" is the equilibrium pressure of the saturated vapour above the liquid, expressed in bars absolute at a specified temperature.

1.2.56 "**Void space**" is the enclosed space in the cargo area external to a cargo containment system, other than a hold space, ballast space, fuel oil tank, cargo pumps or compressor room, or any space in normal use by personnel.

1.3 Equivalents

1.3.1 Where this chapter requires that a particular fitting, material, appliance, apparatus, item of equipment or type thereof is to be fitted or carried in a ship, or that any particular provision is to be made, or any procedure or arrangement is to be complied with, the Administration may allow any other fitting, material, appliance, apparatus, item of equipment or type thereof to be fitted or carried, or any other provision, procedure or arrangement to be made in that ship, if it is satisfied by trial thereof or otherwise that such fitting, material, appliances, apparatus, item of equipment or type thereof or that any particular provision, procedure or arrangement is at least

as effective as that required by this Chapter. However, the Administration may not allow operational methods or procedures to be made as an alternative to a particular fitting, material, appliance, apparatus, item of equipment, or type thereof that is prescribed by the Chapter, unless such a substitution is specifically allowed by the chapter.

IR 1.3.1 Alternative arrangements or fittings which are considered to be equivalent to those specified in these Rules will be accepted. Arrangements or systems incorporating features not provided for in these Rules will be specially considered.

1.3.2 When the Administration so allows any fitting, material, appliance, apparatus, item of equipment, or type thereof, or provision, procedure or arrangement or novel design or application to be substituted, it should communicate to the Organization(IMO) the particulars thereof together with a report on the evidence submitted, so that the Organization(IMO) may circulate the same to other Contracting Governments to the SOLAS Convention for the information of their officers.

1.4 Surveys and certification

1.4.1 Survey procedure

1.4.1.1 The survey of ships, so far as regards the enforcement of the provisions of this chapter and granting of exemptions therefrom, should be carried out by officers of the Administration. The Administration may, however, entrust the surveys either to surveyors nominated for the purpose or to organizations recognized by it.

1.4.1.2 The recognized organization, referred to in 1.2.43 is to comply with the provisions of the SOLAS Convention and with the Code of recognized organizations (RO Code)

1.4.1.3 The Administration nominating surveyors or recognizing organizations to conduct surveys should, as a minimum, empower any nominated surveyor or recognized organization to:

.1 require repairs to a ship; and

.2 carry out surveys if requested by the appropriate authorities of a port State.

The Administration should notify the Organization(IMO) of the specific responsibilities and conditions of the authority delegated to nominated surveyors or recognized organizations for circulation to the Contracting Governments.

1.4.1.4 When a nominated surveyor or recognized organization determines that the condition of a ship or its equipment does not correspond substantially with the particulars of the International certificate of fitness for the carriage of liquefied gases in bulk or is such that the ship is not fit to proceed to sea without danger to the ship, or persons on board, or without presenting unreasonable threat of harm to the marine environment. the surveyor or organization should immediately ensure that corrective action is taken and should in due course notify the Administration. If such corrective action is not taken the certificate should be withdrawn and the Administration should be notified immediately. If the ship is in a port of another Contracting Government, the appropriate authorities of the port State should be notified immediately. When an officer of the Administration, a nominated surveyor or a recognized organization has notified the appropriate authorities of the port State, the Government of the port State concerned should give the officer, surveyor or organization any necessary assistance to carry out their obligations under this paragraph. When applicable, the Government of the port State concerned should take such steps as will ensure that the ship does not sail until it can proceed to sea or leave the port for the purpose of proceeding to the nearest appropriate repair yard available without danger to the ship or persons on board or without presenting an unreasonable threat of harm to the marine environment.

1.4.1.5 In every case, the Administration should guarantee the completeness and efficiency of the survey, and should undertake to ensure the necessary arrangements to satisfy this obligation.

1.4.2 Survey requirements

IR 1.4.2.1 The Classification Regulations for new construction surveys, the classification of ships not built under survey and Periodical Survey regulations are given in Pt.1 of the Rules.

1.4.2.1 The structure, equipment, fittings, arrangements and material (other than items in respect of which a Cargo Ship Safety Construction Certificate, Cargo Ship Safety

Equipment Certificate and Cargo Ship Safety Radio Certificate or Cargo Ship Safety Certificate, required by the SOLAS Convention, are issued) of a gas carrier should be subjected to the following surveys:

.1 An initial survey before the ship is put in service or before the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk is issued for the first time, which should include a complete examination of its structure, equipment, fittings, arrangements and material in so far as the ship is covered by the chapter. This survey should be such as to ensure that the structure, equipment, fittings, arrangements and material fully comply with the applicable provisions of the Chapter.

.2 A renewal survey at intervals specified by the Administration, but not exceeding 5 years, except where 1.4.6.2.1, 1.4.6.5, 1.4.6.6, or 1.4.6.7 is applicable. The renewal survey should be such as to ensure that the structure, equipment, fittings, arrangements and material fully comply with the applicable provisions of the Chapter.

.3 An intermediate survey within 3 months before or after the second anniversary date or within 3 months before or after the third anniversary date of the certificate, which should take the place of one of the annual surveys mentioned in 1.4.2.4. Intermediate surveys should be such as to ensure that the safety equipment, and other equipment and associated pump and piping systems fully comply with the applicable provisions of the Chapter and are in good working order. Such intermediate surveys should be endorsed on the Certificate issued under 1.4.4 or 1.4.5.

.4 An annual survey within 3 months before or after each anniversary date of the Certificate including a general inspection of the structure, equipment, fittings, arrangements and material referred to in 1.4.2.1 to ensure that they remain satisfactory for the service for which the ship is intended. Such annual surveys should be endorsed on the Certificate issued under 1.4.4 or 1.4.5.

.5 An additional survey, either general or partial according to the circumstances, should be made when required after an investigation prescribed in 1.4.3.3, or whenever any important repairs or renewals are made. Such a survey should ensure that the necessary repairs or renewals have been effectively made, that the material and workmanship of such repairs or renewals are satisfactory; and that the ship is fit to proceed to sea without danger to the ship or persons on board or without presenting unreasonable threat of harm to the marine environment.

1.4.3 Maintenance of conditions after survey

1.4.3.1 The condition of the ship and its equipment should be maintained to conform with the provisions of the Chapter to ensure that the ship will remain fit to proceed to sea without danger to the ship or persons on board or without presenting unreasonable threat or harm to the marine environment.

1.4.3.2 After any survey of the ship, as described in 1.4.2 has been completed, no change should be made in the structure, equipment, fittings, arrangements and material covered by the survey, without the sanction of IRS, except by direct replacement.

1.4.3.3 Whenever an accident occurs to a ship or a defect is discovered, either of which affects the safety of the ship or the efficiency or completeness of its life-saving appliances or other equipment covered by the chapter, the master or owner of the ship should report at the earliest opportunity to the Administration/ IRS, who should cause investigations to be initiated to determine whether a survey, as required by 1.4.2.5, is necessary. If the ship is in a port of another Contracting Government, the master or owner should also report immediately to the appropriate authorities of the port State and the nominated surveyor or recognized organization should ascertain that such a report has been made.

1.4.4 Issue and endorsement of an International Certificate of Fitness of Liquefied Gases in Bulk

1.4.4.1 An International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk should be issued after an initial or renewal survey to a gas carrier engaged on international voyages that comply with the relevant provisions of the Code.

1.4.4.2 Such a certificate should be drawn up in the form corresponding to the model given in the Appendix 2 of IGC Code. If the language used is not English, French or Spanish, the text

should include a translation into one of these languages.

1.4.4.3 The certificate issued under the provisions of this clause should be available on board for examination at all times.

1.4.4.4 Notwithstanding any other provisions of the amendments to the Code, adopted by the Maritime Safety Committee by resolution MSC.17(58), any International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk that is current when these amendments enter into force shall remain valid until it expires under the terms of IGC Code prior to the amendments entering into force.

1.4.5 Issue or endorsement of an International Certificate of Fitness of Liquefied Gases in Bulk by another Government

1.4.5.1 A contracting Government to the SOLAS Convention may, at the request of another Contracting Government, cause a ship entitled to fly the flag of the other state to be surveyed and if satisfied that the requirements of the Code are complied with, issue or authorize the issue of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk to the ship and, where appropriate, endorse or authorize the endorsement of the certificate on board the ship in accordance with the Code. Any certificate so issued should contain a statement to the effect that it has been issued at the request of the Government of the State whose flag the ship is entitled to fly.

1.4.6 Duration and validity of an International Certificate of Fitness of Liquefied Gases in Bulk

1.4.6.1 An International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk is to be issued for a period specified by the Administration which is not to exceed 5 years.

1.4.6.2.1 Notwithstanding the provisions of 1.4.6.1, when the renewal survey is completed within 3 months before the expiry date of the existing certificate, the new certificate is to be valid from the date of completion of the renewal survey to a date not exceeding 5 years from the date of expiry of the existing Certificate.

1.4.6.2.2 When the renewal survey is completed after the expiry date of the existing certificate, the new certificate is to be valid from the date of completion of the renewal survey to a date not exceeding 5 years from the date of expiry of the existing certificate.

1.4.6.2.3 When the renewal survey is completed more than 3 months before the expiry date of the existing certificate, the new certificate is to be valid from the date of completion of the renewal survey to a date not exceeding 5 years from the date of completion of the renewal survey.

1.4.6.3 If a certificate is issued for a period of less than 5 years, IRS may extend the validity of the certificate beyond the expiry date to the maximum period specified in 1.4.6.1, provided that the surveys referred to in 1.4.2.3 and 1.4.2.4, applicable when a certificate is issued for a period of 5 years, are carried out as appropriate.

1.4.6.4 If a renewal survey has been completed and a new certificate cannot be issued or placed on board the ship before the expiry date of the existing certificate, IRS may endorse the existing certificate. Such a certificate is to be accepted as valid for a further period which is not to exceed 5 months from the expiry date.

1.4.6.5 If a ship is not in a port in which it is to be surveyed at the time when a certificate expires, IRS may extend the period of validity of the certificate. However, the extension is to be granted only for the purpose of allowing the ship to complete its voyage to the port in which it is to be surveyed and then only in cases where it appears proper and reasonable to do so.

1.4.6.6 A certificate, issued to a ship engaged on short voyages, that has not been extended under the foregoing provisions of this section, may be extended by the Administration for a period of grace of upto one month from the date of expiry stated on it. When the renewal survey is completed, the new certificate shall be valid to a date not exceeding 5 years from the date of expiry of the existing certificate before the extension was granted.

1.4.6.7 In special circumstances, as determined by the Administration, a new certificate need not be dated from the date of expiry of the existing certificate as required by 1.4.6.2.2, 1.4.6.5 or 1.4.6.6. In these special circumstances, the new certificate shall be valid to a date not exceeding 5 years from the date of completion of the renewal survey.

1.4.6.8 If an annual or intermediate survey is completed before the period specified in 1.4.2; then:

.1 the anniversary date shown on the certificate shall be amended by endorsement to a date that is not to be more than 3 months later than the date on which the survey was completed;

.2 the subsequent annual or intermediate survey required by 1.4.2 shall be completed at the intervals prescribed by that section using the new anniversary date;

.3 the expiry date may remain unchanged provided one or more annual or intermediate surveys, as appropriate, are carried out so that the maximum intervals between the surveys prescribed by 1.4.2 are not exceeded.

1.4.6.9 The certificate issued under 1.4.4 or 1.4.5 shall cease to be valid in any of the following cases:

.1 If the relevant surveys are not completed within the periods specified by 1.4.2;

.2 if the certificate is not endorsed in accordance with 1.4.2.3 or 1.4.2.4; and

.2 Upon transfer of the ship to the flag of another State. A new certificate shall only be issued when the Government issuing the new certificate is fully satisfied that the ship is in compliance with the provisions of 1.4.3.1 and 1.4.3.2. In case of transfer between Contracting Governments to the SOLAS Convention, if requested within 3 months after the transfer has taken place, the Government of the State whose flag the ship was formerly entitled to fly shall, as soon as possible, transmit to the Administration copies of the certificates carried by the ship before the transfer and, if available, copies of the relevant survey reports.

IR 1.5 Plans

IR 1.5.1 The plans, to be submitted, are to show clearly that the requirements of this Chapter are fulfilled. Other plans, specifications or information may be required depending on the arrangement and the equipment used in each separate case.

IR 1.5.2 A general arrangement is to be submitted for approval giving location of:

- machinery and boiler spaces, accommodation, service and control station spaces, chain lockers, cofferdams, fuel oil tanks, drinking and domestic water tanks and stores;
- cargo tanks and cargo containment systems;
- cargo pump and compressor rooms;
- cargo control rooms;
- cargo piping with shore connections including stern loading/discharge arrangements and emergency cargo dumping arrangement, if fitted;
- cargo hatches, vent pipes and any other openings to the cargo tanks;
- ventilating pipes, doors and openings to cargo pump rooms, cargo compressor rooms and other gas-dangerous spaces;
- doors, air locks, hatches, ventilating pipes and openings, hinged scuttles which can be opened, and other openings to gas-safe spaces within and adjacent to the cargo area including spaces in and below the forecastle;
- entrances, air inlets and openings to accommodation, service and control station spaces; and
- gas-safe spaces and zones and gas-dangerous spaces and zones to be clearly identified.

IR 1.5.3 Plans of the cargo containment system with the following particulars are to be submitted for approval:

- drawing of cargo tanks including information on non-destructive testing of welds and strength and tightness testing of tanks;
- drawings of support structure of independent tanks;
- drawing of antifloatation arrangement for independent tanks;
- specification of materials in cargo tanks and cargo piping systems;

- specifications of welding procedures for cargo tanks;

- specification of stress relieving procedures for independent tanks Type C (thermal or mechanical);

- specification of design loads and structural analysis of cargo tanks;

- a complete stress analysis is to be submitted for independent tanks, Type B and Type C;

- detailed analytical calculation of hull and tank system for independent tanks, Type B;

- specification of cooling-down procedure for cargo tanks;

- arrangement and specifications of secondary barriers, including method for periodically checking of tightness;

- documentation of model tests of primary and secondary barriers of membrane tanks;

- drawings and specifications of tank insulation; and

- drawing of marking plate for independent tanks.

IR 1.5.4 Plans of the following piping systems are to be submitted for approval:

- drawings and specifications of cargo and process piping including vapour piping and vent lines of safety relief valves or similar piping, and relief valves discharging liquid cargo from the cargo piping system;

- drawings and specifications of offsets, loops, bends and mechanical expansion joints, such as bellows, slip joints (only inside tank) or similar means in the cargo piping;

- drawings and specifications of flanges, valves and other fittings in the cargo piping system. For valves intended for piping systems with a design temperature below -55°C, documentation for leak test and functional test at design temperature (type test) is required;

- complete stress analysis of piping system when design temperature is below -110°C;

- documentation of type tests for expansion components in the cargo piping system;

- specification of materials, welding, post-weld heat treatment and non-destructive testing of cargo piping;

- specification of pressure tests (structural and tightness tests) of cargo and process piping;

- program for functional tests of all piping systems including valves, fittings and associated equipment for handling cargo (liquid or vapour);

- specifications of control system for all quick-closing shut-off valves;

- drawings and specifications and insulation for low temperature piping where such insulation is installed;

- specification of electrical bonding of piping; and

- specification of means for removal of liquid contents from cargo loading and discharging crossover headers and/or cargo hoses prior to disconnecting the shore connection.

IR 1.5.5 The following plans and particulars for the safety relief valves are to be submitted for approval:

- drawings and specifications for safety relief valves and pressure/vacuum relief valves and associated vent piping;

- calculation of required cargo tank relief valve capacity; and

- specification of procedures for changing of set pressures of cargo tank safety relief valves if such arrangements are contemplated.

IR 1.5.6 Plans of the following equipment and systems with particulars are to be submitted:

- construction and specifications of pressure relief systems for hold spaces, interbarrier spaces and cargo piping if such systems are required;

- calculation of hull steel significant temperature when cargo temperature is below -20°C;

- specification of tightness test of hold spaces for membrane tank system;

- arrangement and specifications of means for maintaining the cargo tank vapour pressure below MARVS. (cooling plant, gas burning arrangement, etc.);

- drawings showing location and construction of air locks with alarms equipment;

- drawings of gastight bulkhead stuffing boxes;

- arrangements and specifications of mechanical ventilation systems for spaces in the cargo area, giving capacity and location of fans and their motors. Drawings and material specifications of rotating parts and casings for fans and portable ventilators;

- drawings and specifications of protection of hull steel beneath liquid piping where liquid leakage may be anticipated, such as at shore connections and at pump seals;

- arrangement and specifications of piping system for gas freeing and purging of cargo tanks;

- arrangement of piping for inerting of interbarrier and hold spaces (not required for independent tanks Type C);

- specifications of equipment for provision of dry inert gas (dry air in hold spaces containing independent tanks Type C) for the maintenance of an inert atmosphere in interbarrier and hold spaces;

- specification of instruments for measurement of oxygen content in inert atmospheres;

- arrangement and specifications of all monitoring systems and devices for indicating liquid level, vapour pressure and temperature in the cargo tanks, interbarrier and hold spaces;

- specifications of liquid level alarms;

- arrangement of automatic shut-down of cargo pump and compressors;

- arrangement and specifications of gas detecting equipment;

- location of gas sampling points within cargo tanks;

- bilge and drainage arrangements in cargo pump rooms, cargo compressor rooms, cofferdams, pipe tunnels, hold spaces and interbarrier spaces;

- drawings and specifications of inert gas plants if installed, see Sec. 9; and

- documentation for fire protection, see Sec.11.

IR 1.5.7 Plans of electrical installations giving the following particulars are to be submitted for approval:

- area classification drawing(s);

- drawing(s) showing location of all electrical equipment in gas dangerous area;

- single line diagram for intrinsically safe circuits; and

- list of explosion protected equipment with reference to drawings together with certificates.

Section 2

Ship Survival Capability and Location of Cargo Tanks

2.0 Goal

To ensure that the cargo tanks are in a protective location in the event of minor hull damage, and that the ship can survive the assumed flooding conditions.

2.1 General

2.1.1 Ships subject to this chapter are to survive the hydrostatic effects of flooding following assumed hull damage caused by some external force. In addition, to safeguard the ship and the environment, the cargo tanks are to be protected from penetration in the case of minor damage to the ship resulting, for example, from contact with a jetty or tug, and also given a measure of protection from damage in the case of collision or grounding, by locating them at specified minimum distances inboard from the ship's shell plating. Both the damage to be assumed and the proximity of the tanks to the ship's shell are to be dependent upon the degree of hazard presented by the product to be carried. In addition, the proximity of the cargo tanks to the ship's shell is to be dependent upon the volume of the cargo tank.

2.1.2 Ships subject to this chapter are to be designed to one of the following standards:

- **A Type 1G ship** is a gas carrier intended to transport the products indicated in Sec.19 that require maximum preventive measures to preclude their escape.

- **A Type 2G ship** is a gas carrier intended to transport the products indicated in Sec.19 that require significant preventive measures to preclude their escape.

- **A Type 2PG ship** is a gas carrier of 150 [m] in length or less intended to transport the products indicated in Sec.19 that require significant preventive measures to preclude their escape, and where the products are carried in Type C independent tanks (refer 4.23) designed for a MARVS of at least 7.0 bar gauge and a cargo containment system design temperature of -55°C or above. A ship of this description that is over 150 [m] in length is considered a Type 2G ship;

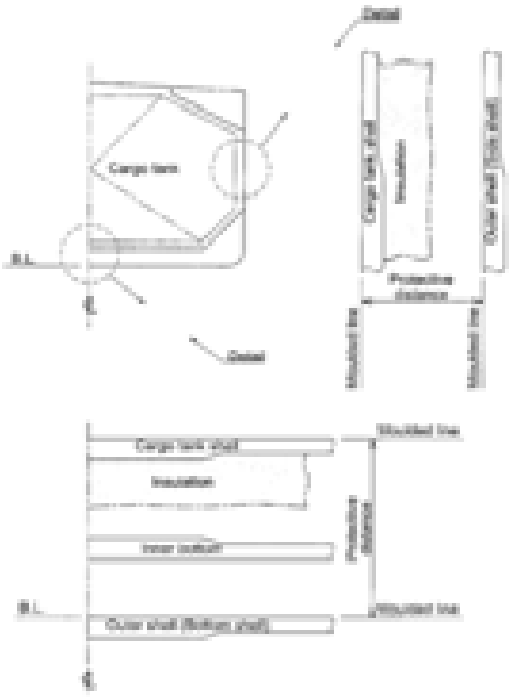
- **A Type 3G ship** is a gas carrier intended to carry products indicated in Sec.19 that require moderate preventive measures to preclude their escape.

Therefore, a type 1G ship is a gas carrier intended for the transportation of products considered to present the greatest overall hazard and types 2G/2PG and type 3G for products of progressively lesser hazards. Accordingly, a type 1G ship is to survive the most severe standard of damage and its cargo tanks are to be located at the maximum prescribed distance inboard from the shell plating.

2.1.3 The ship type required for individual products is indicated in column "c" in Table 19.1.1 of Sec.19.

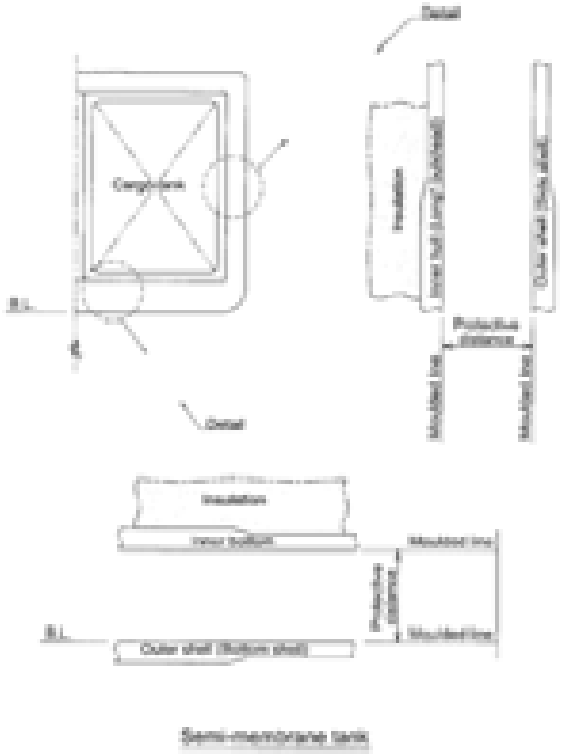
2.1.4 If a ship is intended to carry more than one of the products listed in Sec.19, the standard of damage is to correspond to the product having the most stringent ship type requirements. The requirements for the location of individual cargo tanks, however, are those for ship types related to the respective products intended to be carried.

2.1.5 For the purpose of this chapter, the position of the moulded line for different containment systems is shown in Fig.2.1.5 (a) to (e).



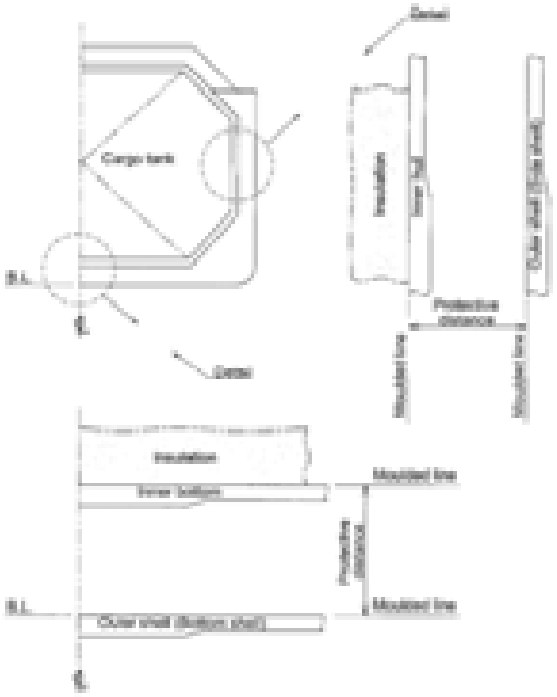
Independent prismatic tank

Fig 2.1.5 (a)



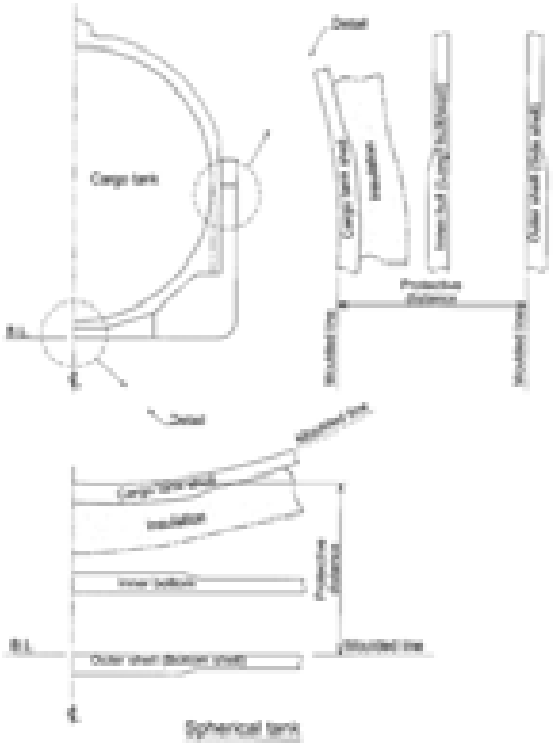
Semi-membrane tank

Fig 2.1.5 (b)



Membrane tank

Fig 2.1.5 (c)



Spherical tank

Fig 2.1.5 (d)

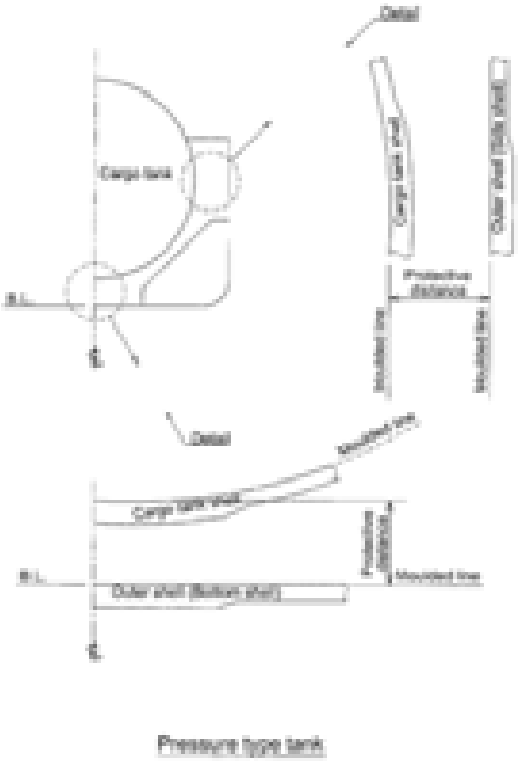


Fig 2.1.5 (e)

2.2 Freeboard and stability

2.2.1 Ships subject to this chapter may be assigned the minimum freeboard permitted by the International Convention on Load Lines, in force. However, the draught associated with the assignment is not to be greater than the maximum draught otherwise permitted by this chapter.

2.2.2 The stability of the ship, in all seagoing conditions and during loading and unloading cargo is to comply with the requirements of the International Code on Intact Stability. This includes partial filling and loading and unloading at sea, when applicable. Stability during ballast water operations is to fulfill stability criteria.

2.2.3 When calculating the effect of free surfaces of consumable liquids for loading conditions it is to be assumed that for each type of liquid, at least one transverse pair or a single centre tank has a free surface. The tank or combination of tanks to be taken into account are to be those where the effect of free surfaces is greatest. The free surface effect in undamaged compartments is to be calculated by a method according to the International Code on Intact Stability.

2.2.4 Solid ballast is not normally to be used in double bottom spaces in the cargo area. Where, however, because of stability considerations the fitting of solid ballast in such spaces becomes unavoidable, its disposition is to be governed by the need to enable access for inspection and to ensure that the impact loads resulting from bottom damage are not directly transmitted on to the cargo tank structure.

2.2.5 The master of the ship is to be supplied with a Loading and Stability Information booklet. This booklet is to contain details of typical service conditions, loading, unloading and ballasting operations, provisions for evaluating other conditions of loading and a summary of the ship's survival capabilities. The booklet is to contain sufficient information to enable the master to load and operate the ship in a safe and seaworthy manner.

2.2.6 All ships, subject to this chapter shall be fitted with a stability instrument, capable of verifying compliance with intact and damage stability requirements, approved by IRS having regard to the performance standards recommended by the Organization.

.1 ships constructed before 1 July 2016 shall comply with this paragraph at the first

scheduled renewal survey of the ship after 1 July 2016 but not later than 1 July 2021;

.2 notwithstanding the requirements of paragraph 2.2.6.1 a stability instrument installed on a ship constructed before 1 July 2016 need not be replaced provided it is capable of verifying compliance with intact and damage stability, to the satisfaction of the Administration; and

.3 for the purposes of control under SOLAS regulation XI-1/4, IRS would issue a document of approval for the stability instrument.

2.2.7 The Administration may waive the requirements of paragraph 2.2.6 for the following ships, provided the procedures employed for intact and damage stability verification maintain the same degree of safety, as being loaded in accordance with the approved conditions. Any such waiver shall be duly noted on the International Certificate of Fitness referred to in paragraph 1.4.4:

.1 ships which are on a dedicated service, with a limited number of permutations of loading such that all anticipated conditions have been approved in the stability information provided to the master in accordance with the requirements of paragraph 2.2.5;

.2 ships where stability verification is made remotely by a means approved by the Administration/ IRS;

.3 ships which are loaded within an approved range of loading conditions; or

.4 ships constructed before 1 July 2016 provided with approved limiting KG/GM curves covering all applicable intact and damage stability requirements

2.2.8 Conditions of loading

Damage survival capability is to be investigated on the basis of loading information submitted to IRS for all anticipated conditions of loading and variations in draught and trim. This is to include ballast and, where applicable, cargo heel.

2.3 Damage assumptions

2.3.1 The assumed maximum extent of damage is to be:

.1 Side damage	
Direction	**Extent Of Damage**
.1.1 Longitudinal extent	$\frac{1}{3}L^{2/3}$ or 14.5 [m], whichever is less
.1.2Transverse extent : measured inboard from the moulded line of the outer shell at right angles to the centreline at the level of the summer waterline	B/5 or 11.5 [m], whichever is less
.1.3 Vertical extent : from the moulded line of the outer shell	upwards without limit

.2 Bottom damage		
Direction	**For 0.3L from the forward perpendicular of the ship**	**Any other part of the ship**
.2.1 Longitudinal extent	$\frac{1}{3}L^{2/3}$ or 14.5 [m], whichever is less	$\frac{1}{3}L^{2/3}$ or 14.5 [m], whichever is less
.2.2 Transverse extent	B/6 or 10 [m], whichever is less	B/6 or 5 [m], whichever is less
.2.3 Vertical extent	B/15 or 2 [m], whichever is less, measured from the moulded line of the bottom shell plating at centreline (see 2.4.3)	B/15 or 2 [m], whichever is less, measured from the moulded line of the bottom shell plating at centreline (see 2.4.3)

2.3.2 Other damage:

2.3.2.1 If any damage of lesser extent than the maximum damage specified in 2.3.1 would result in a more severe condition, such damage is to be assumed.

2.3.2.2 Local damage anywhere in the cargo area extending inboard distance “d” as defined

in 2.4.1 measured normal to the moulded line of the outer shell is to be considered. Bulkheads are to be assumed damaged when the relevant sub-paragraphs of 2.6.1 apply. If a damage of a lesser extent than “d” would result in a more severe condition, such damage is to be assumed.

2.4 Location of cargo tanks

2.4.1 Cargo tanks are to be located at the following distances inboard:

.1 **Type 1G ships**: from the moulded line of the outer shell, not less than the transverse extent of damage specified in 2.3.1.1.2 and, from the moulded line of the bottom shell at centerline, not less than the vertical extent of damage specified in 2.3.1.2.3 and nowhere less than “d” where “d” is as follows:

.1) for Vc below or equal 1000 [m^3], d=0.8 [m];

.2) for 1000 [m^3] < Vc < 5000 [m^3], d=0.75 + Vc x 0.2/4000 [m];

.3) for 5000 [m^3] ≤ Vc < 30000 [m^3], d = 0.8 + Vc/25000 m and

.4) for Vc ≥ 30000 m^3, d=2 [m]

Where:

Vc corresponds to 100% of the gross design volume of the individual cargo tank at 20 [deg C], including domes and appendages (see the Fig.2.4.1 (a) and (b)). For the purpose of cargo tank protective distances, the cargo tank volume is the aggregate volume of all the parts of tank that have a common bulkhead(s); and

“d” is measured at any cross section at a right angle from the moulded line of outer shell.

Tank size limitation may apply to type 1G ship cargoes in accordance with section 17.

.2 **Types 2G/2PG** : from the moulded line of the bottom shell at centreline not less than the vertical extent of damage specified in 2.3.1.2.3 and nowhere less than “d” as indicated in 2.4.1.1 (see Fig.2.4.1 (a) and (c))

.3 **Type 3G Ships**: from the moulded line of the bottom shell at centerline not less than the vertical extent of damage specified in 2.3.1.2.3 and nowhere less than “d”, where “d” = 0.8 [m] from moulded line of outer shell. (see Figs.2.4.1 (a) and (d))

2.4.2 For the purpose of tank location, the vertical extent of bottom damage is to be measured to the inner bottom when membrane or semi-membrane tanks are used, otherwise to the bottom of the cargo tanks. The transverse extent of side damage is to be measured to the longitudinal bulkhead when membrane or semi-membrane tanks are used, otherwise to the side of the cargo tanks. The distance indicated in 2.3 and 2.4 is to be applied as in Figs.2.1.5 (a) to (e). These distances is to be measured plate to plate, from the moulded line to the moulded line, excluding insulation.

2.4.3 Except for Type 1G ships, suction wells installed in cargo tanks may protrude into the vertical extent of bottom damage specified in 2.3.1.2.3 provided that such wells are as small as practicable and the protrusion below the inner bottom plating does not exceed 25 per cent of the depth of the double bottom or 350 [mm], whichever is less. Where there is no double bottom, the protrusion below the upper limit of bottom damage is not to exceed 350 mm. Suction wells installed in accordance with this paragraph may be ignored in determining the compartments affected by damage.

2.4.4 Cargo tanks are not to be located forward of the collision bulkhead.

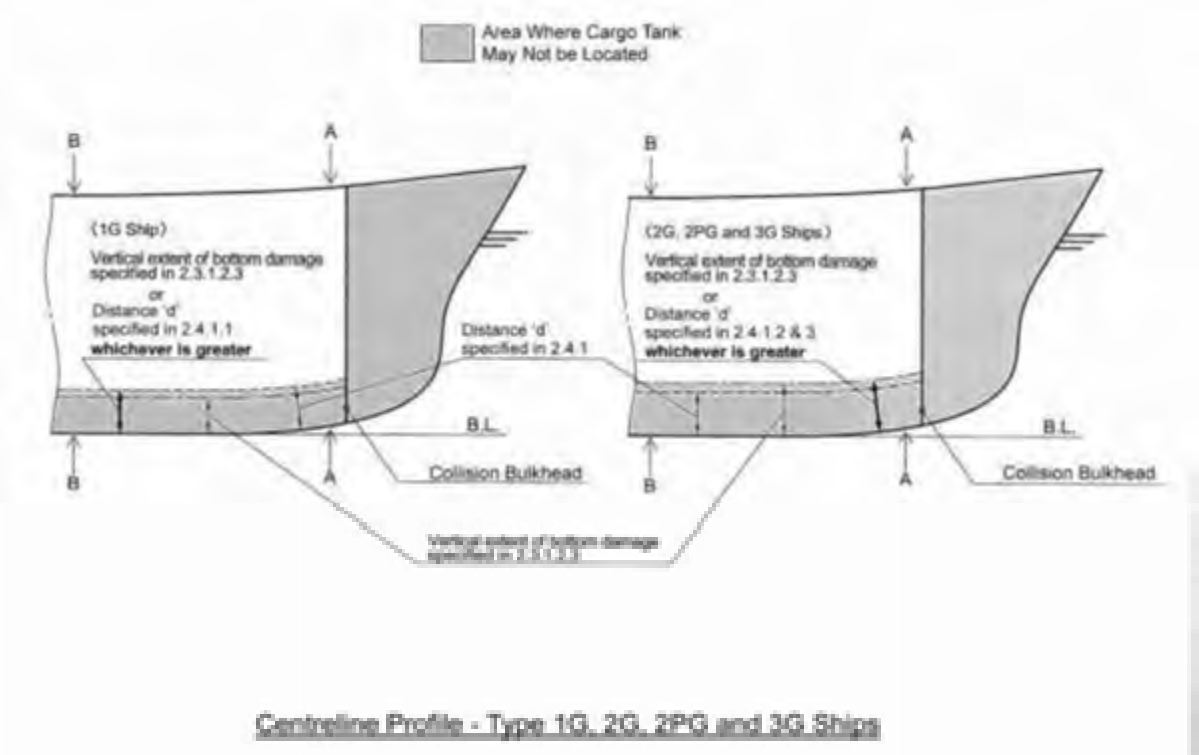


Fig.2.4.1 (a) : Cargo tank location requirements

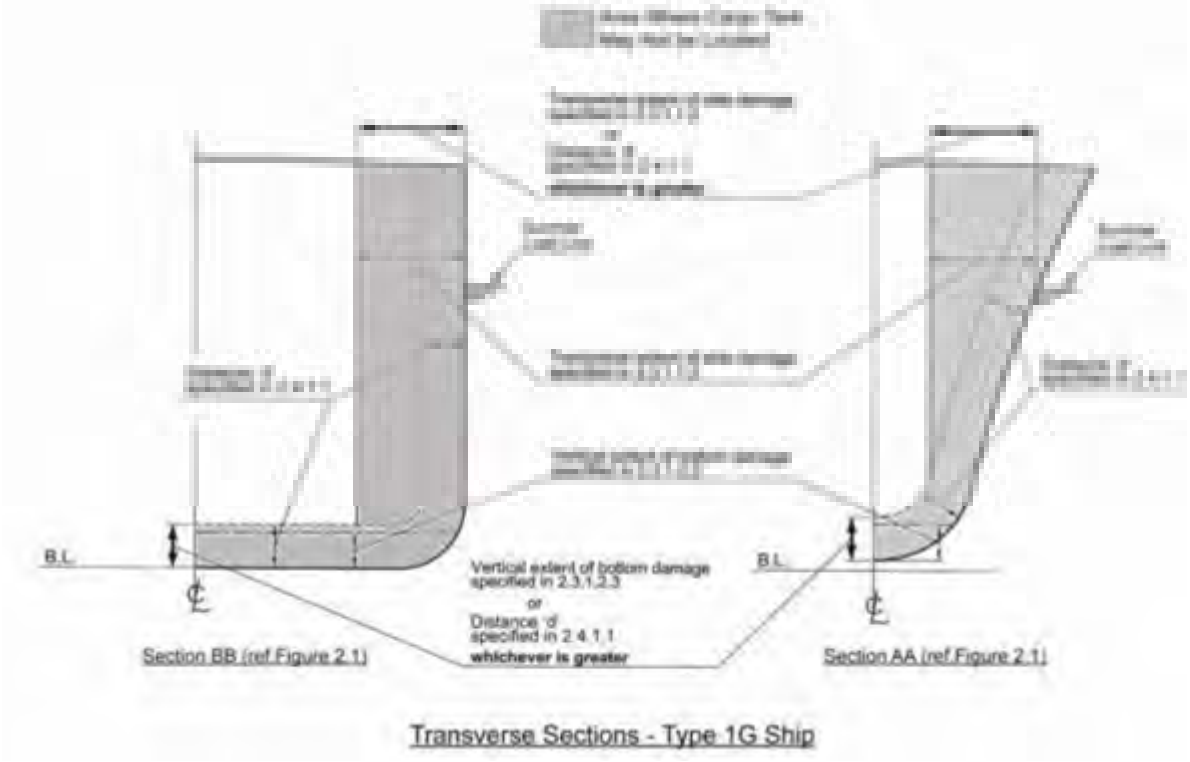


Fig.2.4.1 (b) : Cargo tank location requirements

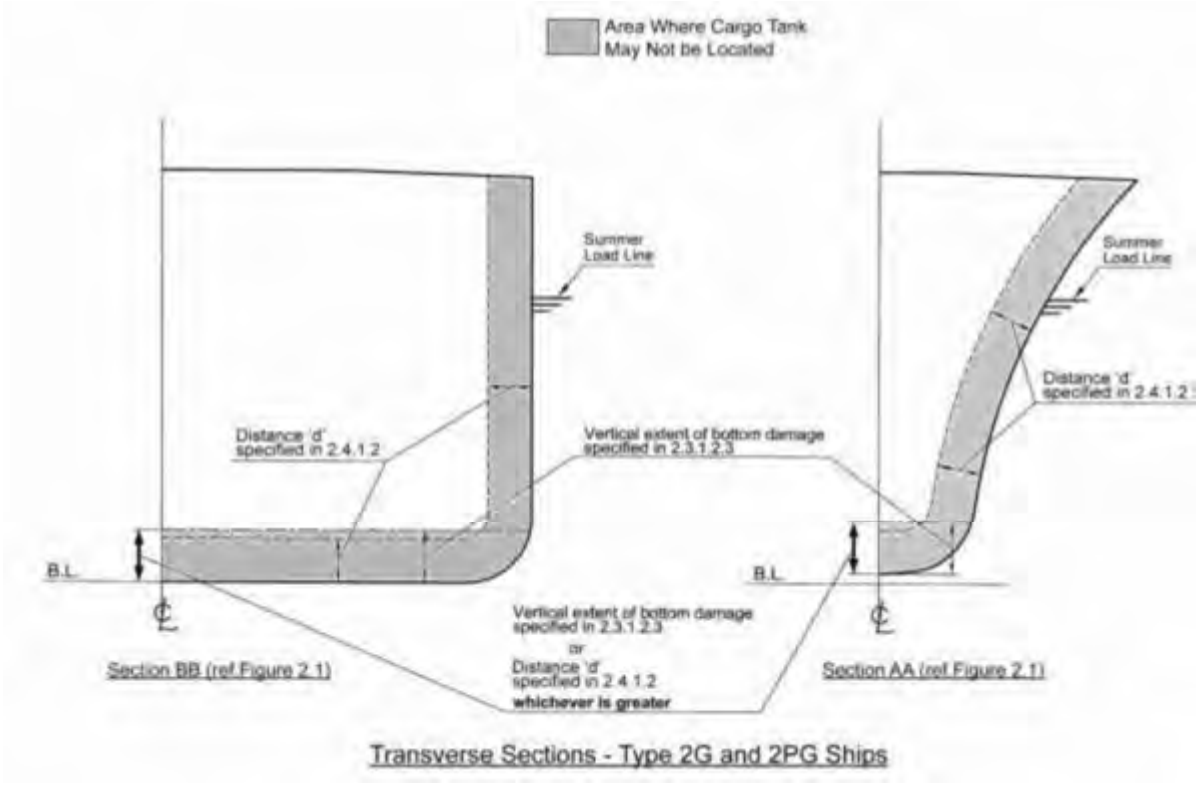


Fig.2.4.1 (c) : Cargo tank location requirements

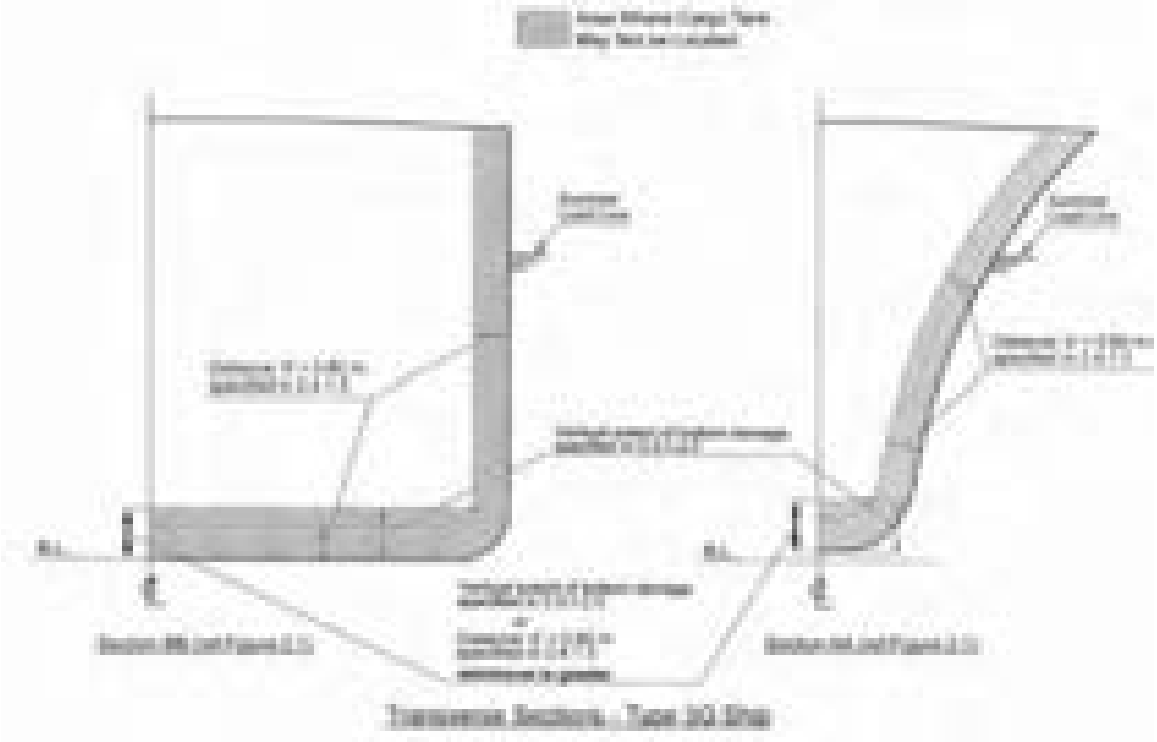


Fig.2.4.1 (d) : Cargo tank location requirements

2.5 Flooding assumptions

2.5.1 The requirements of 2.7 are to be confirmed by calculations that take into consideration the design characteristics of the ship; the arrangements, configuration and contents of the damaged compartments; the distribution, relative densities and the free surface effects of liquids; and the draught and trim for all conditions of loading.

2.5.2 The permeabilities of spaces assumed to be damaged are to be as follows:

Spaces	Permeabilities
stores	0.60
Accommodation	0.95
Machinery	0.85
Voids	0.95
Hold spaces	0.95**
Consumable liquids	0 to 0.95*
Other liquids	0 to 0.95*
* The permeability of partially filled compartments is to be consistent with the amount of liquid carried. ** Other values of permeability can be considered based on the detailed calculations. Interpretations of regulation of part B-1 of SOLAS Chapter II-1 (MSC/Circ.651) are referred.	

2.5.3 Wherever damage penetrates a tank containing liquids, it is to be assumed that the contents are completely lost from that compartment and replaced by salt water up to the level of the final plane of equilibrium.

2.5.4 Where the damage between transverse watertight bulkheads is envisaged as specified in 2.6.1.4, .5, and .6, transverse bulkheads are to be spaced at least at a distance equal to the longitudinal extent of damage specified in 2.3.1.1.1 in order to be considered effective. Where transverse bulkheads are spaced at a lesser distance, one or more of these bulkheads within such extent of damage are to be assumed as non-existent for the purpose of determining flooded compartments. Further, any portion of a transverse bulkhead bounding side compartments or double bottom compartments is to be assumed damaged if the watertight bulkhead boundaries are within the extent of vertical or horizontal penetration required by 2.3. Also, any transverse bulkhead is to be assumed damaged if it contains a step or recess of more than 3 [m] in length located within the extent of penetration of assumed damage. The step formed by the after peak bulkhead and after peak tank top is not to be regarded as a step for the purpose of this paragraph.

2.5.5 The ship is to be so designed as to keep unsymmetrical flooding to the minimum consistent with efficient arrangement.

2.5.6 Equalization arrangements requiring mechanical aids such as valves or cross-leveling pipes, if fitted, are not to be considered for the purpose of reducing an angle of heel or attaining the minimum range of residual stability to meet the requirements of 2.7.1 and sufficient residual stability is to be maintained during all stages where equalization is used. Spaces linked by ducts of large cross-sectional area may be considered to be common.

2.5.7 If pipes, ducts, trunks or tunnels are situated within the assumed extent of damage penetration, as defined in 2.3, arrangements are to be such that progressive flooding cannot thereby extend to compartments other than those assumed to be flooded for each case of damage.

2.5.8 The buoyancy of any superstructure directly above the side damage is to be disregarded. However, the unflooded parts of superstructures beyond the extent of damage, however, may be taken into consideration provided that:

.1 they are separated from the damaged space by watertight divisions and the requirements of 2.7.1.1 in respect of these intact spaces are complied with; and

.2 openings in such divisions are capable of being closed by remotely operated sliding watertight doors and unprotected openings are not immersed within the minimum range of residual stability required in 2.7.2.1. However, the immersion of any other openings capable of being closed weathertight may be permitted.

2.6 Standard of damage

2.6.1 Ships are to be capable of surviving the damage indicated in 2.3 with the flood assumptions in 2.5, to the extent determined by the ship's type according to the following standards:

.1 A Type 1G ship is to be assumed to sustain damage anywhere in its length;

.2 A Type 2G ship of more than 150 [m] in length is to be assumed to sustain damage anywhere in length;

.3 A Type 2G ship of 150 [m] in length or less is to be assumed to sustain damage anywhere in its length except involving either of the bulkheads bounding a machinery space located aft;

.4 A Type 2PG ship it to be assumed to sustain damage anywhere in its length except involving transverse bulkheads spaced further apart than the longitudinal extent of damage as specified in 2.3.1.1.1;

.5 A Type 3G ship of 80 [m] in length or more is to be assumed to sustain damage anywhere in its length except involving transverse bulkheads spaced further apart than the longitudinal extent of damage specified in 2.3.1.1.1;

.6 A Type 3G ship less than 80 [m] in length is to be assumed to sustain damage anywhere in its length except involving transverse bulkheads spaced further apart than the longitudinal extent of damage specified in 2.3.1.1.1 and except damage involving the machinery space when located aft.

2.6.2 In the case of small Type 2G/2PG and 3G ships that do not comply in all respects with the appropriate requirements of 2.6.1.3,.4 and .6, special dispensations may only be considered by the Administration provided that alternative measures can be taken which maintain the same degree of safety. The nature of the alternative measures are to be approved and clearly stated and be available to the port Administration. Any such dispensation will be duly noted on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk referred to in 1.4.4.

2.7 Survival requirements

Ships subject to this Chapter are to be capable of surviving the assumed damage specified in 2.3, to the standard provided in 2.6 in a condition of stable equilibrium and are to satisfy the following criteria.

2.7.1 In any stage of flooding:

.1 the waterline, taking into account sinkage, heel and trim, is to be below the lower edge of any opening through which progressive flooding or downflooding may take place. Such openings are to include air pipes and openings that are closed by means of weathertight doors or hatch covers and may exclude those openings closed by means of watertight manhole covers and watertight flush scuttles, small watertight cargo tank hatch covers that maintain the high integrity of the deck, remotely operated sliding watertight doors, hinged watertight access doors with open/closed indication locally and at the navigation bridge, of the quick-acting or single-action type that are normally closed at sea, hinged watertight doors that are permanently closed at sea, and sidescuttles of the non-opening type;

.2 the maximum angle of heel due to unsymmetrical flooding is not to exceed 30 [deg]; and

.3 the residual stability during intermediate stages of flooding is not to be less than that required by 2.7.2.1.

2.7.2 At final equilibrium after flooding:

.1 the righting lever curve is to have a minimum range of 20 [deg] beyond the position of equilibrium in association with a maximum residual righting lever of at least 0.1 [m] within the 20 [deg] range; the area under the curve within this range is not to be less than 0.0175 [m-radians] The 20 [deg] range may be measured from any angle commencing between the position of equilibrium and the angle of 25 [deg] (or 30 [deg] if no deck immersion occurs) Unprotected openings are not to be immersed within this range unless the space concerned is assumed to be flooded. Within this range, the immersion of any of the openings listed in 2.7.1.1 and other openings capable of being closed weathertight may be permitted; and

IR.1 Other openings capable of being closed weathertight do not include ventilators (complying with ILLC 19(4)) that for operational reasons have to remain open to supply air to the engine room or emergency

generator room (if the same is considered buoyant in the stability calculation or protecting openings leading below) for the effective operation of the ship.

.2 the emergency source of power is to be capable of operating.

Section 3

Ship Arrangements

3.0 Goal

To ensure that the cargo containment and handling system are located such that the consequences of any release of cargo will be minimized, and to provide safe access for operation and inspection.

3.1 Segregation of the cargo area

3.1.1 Hold spaces are to be segregated from machinery and boiler spaces, accommodation spaces, service spaces, control station, chain lockers, domestic water tanks and from stores. Hold spaces are to be located forward of machinery spaces of category 'A'. Alternative arrangements, including locating machinery spaces of category A forward, may be accepted, based on SOLAS regulation II-2/17(Pt 6, Ch 6), after further consideration of involved risks, including that of cargo release and the means of mitigation.

3.1.2 Where cargo is carried in a cargo containment system not requiring a complete or partial secondary barrier, segregation of hold spaces from spaces referred to in 3.1.1 or spaces either below or outboard of the hold spaces may be effected by cofferdams, fuel oil tanks, or a single gastight bulkhead of all welded construction forming an A-60 Class division. A gas-tight A-0 Class division is acceptable if there is no source of ignition or fire hazard in the adjoining spaces.

3.1.3 Where cargo is carried in a cargo containment system requiring a complete or partial secondary barrier, segregation of hold spaces from spaces referred to in 3.1.1 or spaces either below or outboard of the hold spaces that contain a source of ignition or fire hazard is to be effected by cofferdams or fuel oil tanks. A gastight A-0 class division is acceptable if there is no source of ignition or fire hazard in the adjoining spaces.

3.1.4 Turret compartment segregation from spaces referred in 3.1.1, or spaces either below or outboard of the turret compartment that contain a source of ignition or fire hazard, are to be effected by cofferdams or an A-60 class division. A gastight A-0 class division is acceptable if there is no source of ignition or fire hazard in the adjoining spaces.

3.1.5 In addition, the risk of fire propagation from turret compartments to adjacent spaces are to be evaluated by risk analysis (see 1.1.11) and further preventive measures, such as the arrangement of a cofferdam around the turret compartment, is to be provided if need.

3.1.6 When cargo is carried in a cargo containment system requiring a complete or partial secondary barrier:

.1 at temperatures below -10°C, hold spaces are to be segregated from the sea by a double bottom, and

.2 at temperatures below -55°C, the ship is to also have a longitudinal bulkhead forming side tanks.

3.1.7 Arrangements are to be made for sealing the weather decks in way of openings for cargo containment systems.

3.2 Accommodation, service and machinery spaces and control stations

3.2.1 No accommodation space, service space or control station is to be located within the cargo area. The bulkhead of accommodation spaces, service spaces or control stations that face the cargo area is to be so located as to avoid the entry of gas from the hold space to such spaces through a single failure of a deck or bulkhead on a ship having a containment system requiring a secondary barrier.

3.2.2 To guard against the danger of hazardous vapours, due consideration is to be given to the location of air intakes/outlets and openings into accommodation, service and machinery

spaces and control stations in relation to cargo piping, cargo vent systems and machinery space exhausts from gas burning arrangements.

3.2.3 Access through doors, gastight or otherwise, is not to be permitted from a non-hazardous area to a hazardous area, except for access to service spaces forward of the cargo area through air-locks as permitted by 3.6.1 when accommodation spaces are aft.

3.2.4.1 Entrances, air inlets and openings to accommodation spaces, service spaces, machinery spaces and control stations are not to face the cargo area. They are to be located on the end bulkhead not facing the cargo area or on the outboard side of the superstructure or deckhouse or on both at a distance of at least 4 percent of the length of the ship but not less than 3 [m] from the end of the superstructure or deck-house facing the cargo area. This distance, however, need not exceed 5 [m].

3.2.4.2 Windows and side scuttles facing the cargo area and on the sides of the superstructures or deckhouses within the distance mentioned above are to be of the fixed (non-opening) type. Wheelhouse windows may be non-fixed and wheelhouse doors may be located within the above limits so long as they are designed in a manner that a rapid and efficient gas and vapour tightening of the wheelhouse can be ensured.

3.2.4.3 For ships dedicated to the carriage of cargoes that have neither flammable nor toxic hazards, IRS may approve relaxations from the above requirements.

3.2.4.4 Access to forecastle spaces containing sources of ignition may be permitted through a single door facing cargo area, provided the doors are located outside hazardous areas as defined in Sec 10.

3.2.5 Windows and side scuttles facing the cargo area and on the sides of the superstructures and deckhouses within the limits specified in 3.2.4, except wheelhouse windows, are to be constructed to "A-60" class. Side scuttles in the shell below the uppermost continuous deck and in the first tier of the superstructure or deckhouse are to be of the fixed (non-opening) type.

3.2.6 All air intakes, outlets and other openings into the accommodation spaces, service spaces and control stations are to be fitted with closing devices. When carrying toxic products, they are to be capable of being operated from inside the space. The requirement for fitting air intakes and openings with closing devices operated from inside the space for toxic products need not apply to spaces not normally manned, such as deck stores, forecastle stores, workshops. In addition, the requirement does not apply to cargo control rooms located within the cargo area.

IR 3.2.6.1 The closing devices that need not be operable from within the single spaces may be located in centralized positions. Engine room casings, cargo machinery spaces, electric motor rooms and steering gear compartments are generally considered as spaces not covered by 3.2.6 and therefore the requirement for closing devices not to be applied to these spaces.

IR 3.2.6.2 The closing devices should give a reasonable degree of gas tightness. Ordinary steel fire-flaps without gaskets/seals should not be considered satisfactory.

IR 3.2.6.3 Regardless of the above (IR 3.2.6.1 and IR 3.2.6.2), the closing devices are to be operable from outside of the protected space (SOLAS regulation II-2/5.2.1.1)

3.2.7 Control rooms and machinery spaces of turret systems may be located in the cargo area forward or aft of cargo tanks in ships with such installations. Access to such spaces containing sources of ignition may be permitted through doors facing the cargo area, provided the doors are located outside hazardous areas or access is through airlocks.

3.3 Cargo machinery spaces and turret compartments

3.3.1 Cargo machinery spaces are to be situated above the weather deck and located within the cargo area. Cargo machinery spaces and turret compartments are to be treated as cargo pump rooms for the purpose of fire protection according to SOLAS regulation II-2/9.2.4 (Rules Pt.6, Ch.3, Cl.3.2.4), and for the prevention of potential explosion according to SOLAS regulation II-2/4.5.10 (Pt 6, Ch.2, 1.5.10).

3.3.2 When cargo machinery spaces are located at the after end of the aftermost hold space or at the forward end of the foremost hold space, the limits of the cargo area as defined is 1.2.7 are to be extended to include the cargo

machinery spaces for the full breadth and depth of the ship and deck areas above those spaces.

3.3.3 Where the limits of the cargo area are extended by 3.3.2, the bulkhead that separates the cargo machinery spaces from accommodation and service spaces, control stations and machinery spaces of category 'A' are to be located so as to avoid the entry of gas to these spaces through a single failure of a deck or bulkhead.

3.3.4 Cargo compressors and cargo pumps may be driven by electric motors in an adjacent non-hazardous space separated by a bulkhead or deck, if the seal around the bulkhead penetration ensures effective gastight segregation of the two spaces. Alternatively, such equipment may be driven by certified safe electric motors adjacent to them if the electrical installation complies with the requirements of Sec 10.

3.3.5 Arrangements of cargo machinery spaces and turret compartments are to be such as to ensure safe unrestricted access for personnel wearing protective clothing and breathing apparatus, and in the event of injury to allow unconscious personnel to be removed. At least two widely separated escape routes and doors shall be provided in cargo machinery spaces, except that a single escape route may be accepted where the maximum travel distance to the door is 5 [m] or less.

3.3.6 All valves necessary for cargo handling are to be readily accessible to personnel wearing protective clothing. Suitable arrangements are to be made to deal with drainage of pump and compressor rooms.

3.3.7 Turret compartments are to be designed to retain their structural integrity in case of explosion or uncontrolled high-pressure gas release (overpressure and/or brittle fracture), the characteristics of which are to be substantiated on the basis of a risk analysis with due consideration of the capabilities of the pressure relieving devices.

3.4 Cargo control rooms

3.4.1 Any cargo control room is to be above the weather deck and may be located in the cargo area. The cargo control room may be located within the accommodation spaces, service spaces or control station provided the following conditions are complied with:

.1 The cargo control room is a non-hazardous area

.2 If the entrance complies with 3.2.4.1, the control room may have access to the spaces described above; and

.3 If the entrance does not comply with 3.2.4.1, the control room is to have no access to the spaces described above and the boundaries to such spaces are to be insulated to "A-60" class.

3.4.2 If the cargo control room is designed to be a non-hazardous area, instrumentation is, as far as possible, to be by indirect reading systems and is in any case to be designed to prevent any escape of gas into the atmosphere of that space. Location of the gas detection system within the cargo control room will not cause the room to be classified as a hazardous area, if installed in accordance with 13.6.11.

3.4.3 If the cargo control room for ships carrying flammable cargoes is classified as a hazardous area, sources of ignition are to be excluded and any electrical equipment is to be installed in accordance with sec 10.

3.5 Access to spaces in the cargo area

3.5.1 Visual inspection of at least one side of the inner hull structure is to be possible without the removal of any fixed structure or fitting. If such a visual inspection, whether combined with those inspections required in 3.5.2, 4.6.2.4 or 4.20.3.7 or not, is only possible at the outer face of the inner hull, the inner hull is not to be a fuel-oil tank boundary wall.

3.5.2 Inspection of one side of any insulation in hold spaces is to be possible. If the integrity of the insulation system can be verified by inspection of the outside of the hold space boundary when tanks are at service temperature, inspection of one side of the insulation in the hold space need not be required.

3.5.3 Arrangements for hold spaces, void spaces, cargo tanks and other spaces classified as hazardous areas, are to be such as to allow entry and inspection of any such space by personnel wearing protective clothing and breathing apparatus and are to also allow for the evacuation of injured and/or unconscious personnel. Such arrangements are to comply with the following:

.1 Access are to be provided as follows:

.1 access to all cargo tanks. Access is to be direct from the weather deck;

.2 access through horizontal openings, hatches or manholes. The dimensions are to be sufficient to allow a person wearing a breathing apparatus to ascend or descend any ladder without obstruction, and also to provide a clear opening to facilitate the hoisting of an injured person from the bottom of the space. The minimum clear opening is to be not less than 600 [mm] x 600 [mm];

IR.2 The minimum clear opening of 600 [mm] X 600 [mm] may have corner radii up to 100 [mm] maximum. In such a case where as a consequence of structural analysis of a given design the stress is to be reduced around the opening, it is considered appropriate to take measures to reduce the stress such as making the opening larger with increased radii, e.g. 600 [mm] X 800 [mm] with corner radii up to 100 [mm] maximum fits.

.3 access through vertical openings or manholes providing passage through the length and breadth of the space. The minimum clear opening is to be not less than 600 [mm] x 800 [mm] at a height of not more than 600 [mm] from the bottom plating unless gratings or other footholds are provided; and

IR.3.1 The minimum clear opening of not less than 600 [mm] X 800 [mm] may also include an opening with corner radii of 300 [mm]. An opening of 600 [mm] in height and 800 [mm] in width may be accepted as access openings in vertical structures where it is not desirable to make large opening in the structural strength aspects, i.e. girders and floors in double bottom tanks. (Refer to Figure 3.5.3.1)

IR.3.2 Subject to verification of easy evacuation of injured person on a stretcher the vertical opening 850 [mm] X 620 [mm] with wider upper half than 600 [mm], while the lower half may be less than 600 [mm] with the overall height not less than 850 [mm] is considered an acceptable alternative to the traditional opening of 600 [mm] X 800 [mm] with corner radii of 300 [mm]. (Refer to figure 3.5.3.2)

IR.3.3 If a vertical opening is at a height of more than 600 [mm], steps an handgrips are to be provided. In such arrangements it is to be demonstrated that an injured person can be easily evacuated.

.4 circular access openings to type C tanks are to have a diameter of not less than 600 [mm].

.2 The dimensions referred to in 3.5.3.1.2 and 3.5.3.1.3 may be decreased, if the requirements of 3.5.3 can be met to the satisfaction of the IRS.

.3 Where cargo is carried in a containment system requiring a secondary barrier, the requirements of 3.5.3.1.2 and 3.5.3.1.3 do not apply to spaces separated from a hold space by a single gastight steel boundary. Such spaces are to be provided only with direct or indirect access from the weather deck, not including any enclosed non-hazardous area.

.4 Access required for inspection are to be a designated access through structures below and above cargo tanks, which are to have at least the cross-sections as required by 3.5.3.1.3.

.5 For the purpose of 3.5.1 or 3.5.2, the following applies:

.1 where it is required to pass between the surface to be inspected, flat or curved, and structures such as deck beams, stiffeners, frames, girders, etc., the distance between that surface and the free edge of the structural elements is to be at least 380 [mm]. The distance between the surface to be inspected and the surface to which the above structural elements are fitted, e.g. deck, bulkhead or shell, is to be at least 450 [mm] for a curved tank surface (e.g. for a type C tank), or 600 [mm] for a flat tank surface (e.g. for a type A tank) (see Fig.3.1);

.2 where it is not required to pass between the surface to be inspected and any part of the structure, for visibility reasons the distance between the free edge of that structural element and the surface to be inspected is to be at least 50 [mm] or half the breadth of the structure's face plate, whichever is the larger (see Fig.3.2);

.3 if for inspection of a curved surface where it is required to pass between that surface and another surface, flat or curved, to which no structural elements are fitted, the distance between both surfaces are to be at least 380 [mm] (see Fig.3.3). Where it is not required to pass between that curved surface and another surface, a smaller distance than 380 [mm] may be accepted taking into account the shape of the curved surface;

.4 if for inspection of an approximately flat surface where it is required to pass between two approximately flat and approximately parallel surfaces, to which no structural elements are fitted, the distance between those surfaces is to be at least 600 [mm]. Where fixed access ladders are fitted, a clearance of at least 450 [mm] is to be provided for access (see Fig.3.4);

.5 the minimum distances between a cargo tank sump and adjacent double bottom structure in way of a suction well is not to be less than those shown in Fig.3.5 (Fig.3.5 shows that the distance between the plane surfaces of the sump and the well is a minimum of 150 [mm] and that the clearance between the edge between the inner bottom plate, and the vertical side of the well and the knuckle point between the spherical or circular surface and sump of the tank is at least 380 [mm]). If there is no suction well, the distance between the cargo tank sump and the inner bottom is not to be less than 50 [mm];

.6 the distance between a cargo tank dome and deck structures is not to be less than 150 [mm] (see Fig.3.6);

.7 fixed or portable staging is to be installed as necessary for inspection of cargo tanks, cargo tank supports and restraints (e.g. anti-pitching, anti-rolling and anti-flotation chocks), cargo tank insulation etc. This staging are not to impair the clearances specified in 3.5.3.5.1 to 3.5.3.5.4; and

.8 if fixed or portable ventilation ducting are to be fitted in compliance with 12.1.2, such ducting are not to impair the distances required under 3.5.3.5.1 to 3.5.3.5.4.

3.5.4 Access from the open weather deck to non-hazardous areas is to be located outside the hazardous areas as defined in Sec 10,

unless the access is by means of an airlock in accordance with 3.6.

3.5.5 Turret compartments are to be arranged with two independent means of access/egress.

3.5.6 Access from a hazardous area below the weather deck to a non-hazardous area is not permitted.

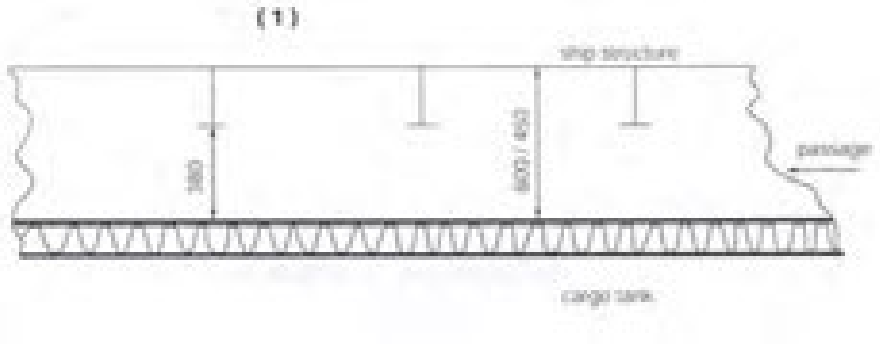


Fig.3.5.2(1)

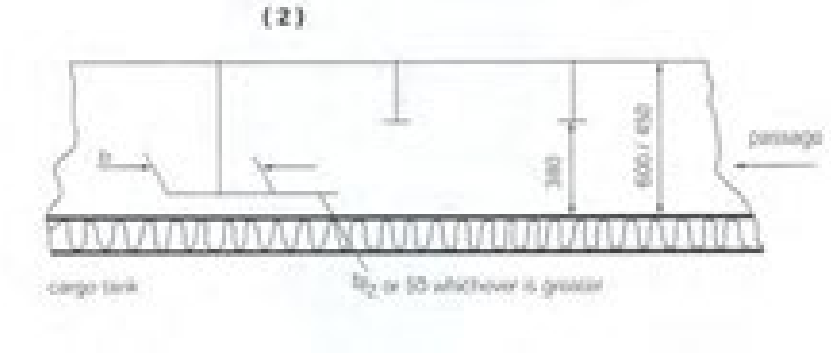


Fig.3.5.2(2)

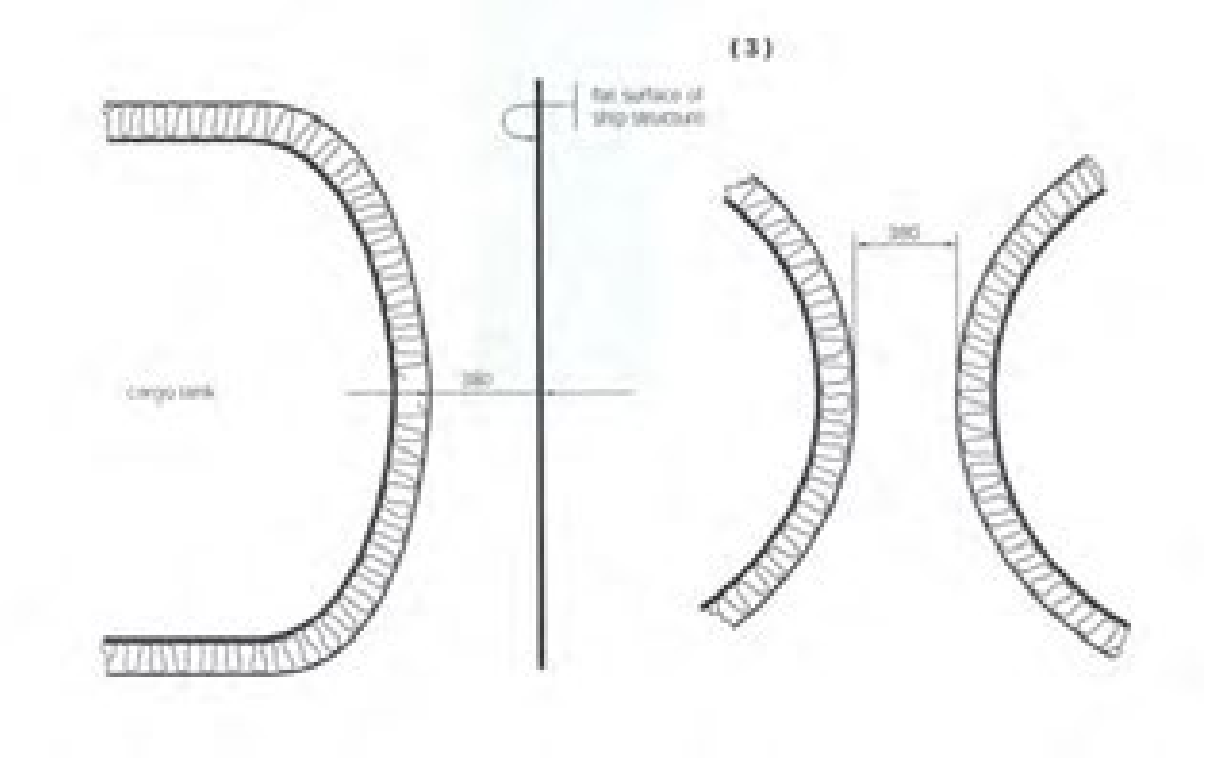


Fig.3.5.2(3)

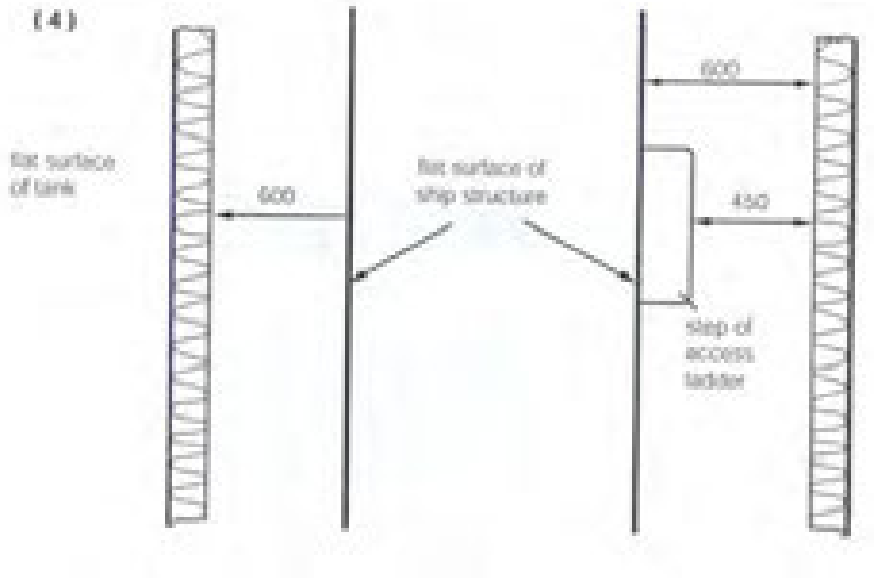


Fig.3.5.2(4)

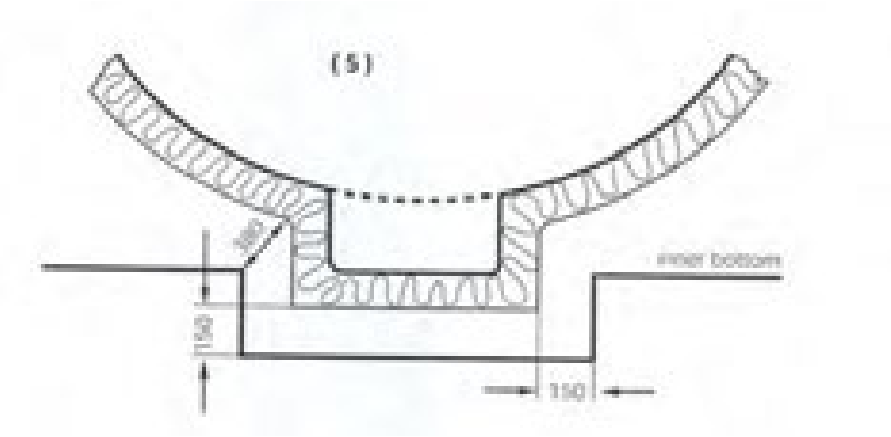


Fig.3.5.2(5)

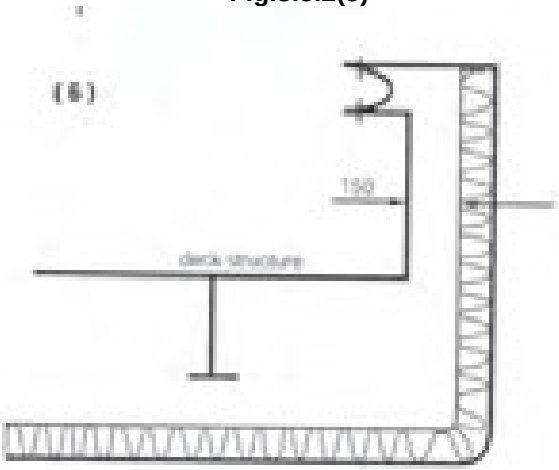


Fig.3.5.2(6)

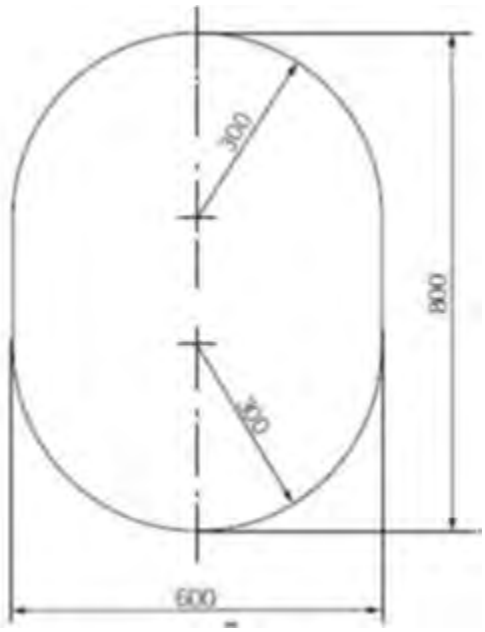


Fig.3.5.3.1

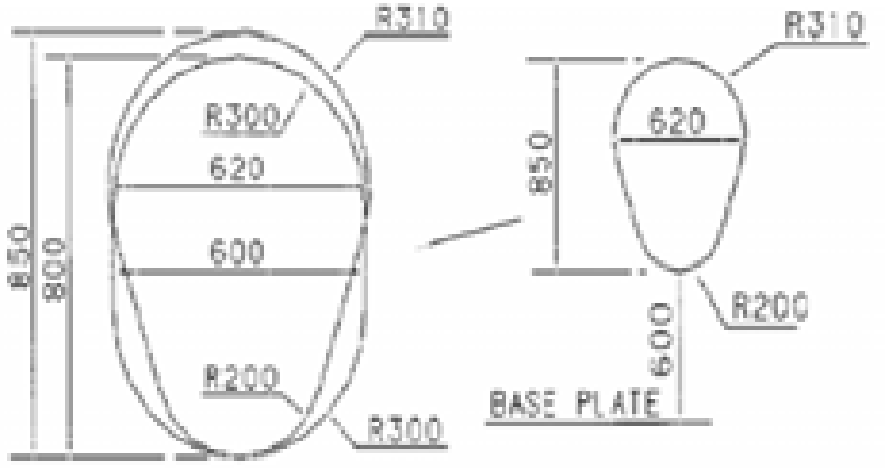


Fig.3.5.3.2

3.6 Air-locks

3.6.1 Access between hazardous area on the open weather deck and non-hazardous spaces is to be by means of an airlock. This is to consist of two self-closing, substantially gastight, steel doors without any holding back arrangements, capable of maintaining the overpressure, at least 1.5 m but no more than 2.5 m apart. The airlock space is to be artificially ventilated from a non-hazardous area and maintained at an overpressure to the hazardous area on the weather deck.

3.6.2 Where spaces are protected by pressurization, the ventilation is to be designed and installed in accordance with IEC 60092-502:1999.

3.6.3 An audible and visual alarm to give a warning on both sides of the air-lock is to be provided. The visible alarm is to indicate if one door is open. The audible alarm is to sound if doors on both sides of the air-lock are moved from the closed positions.

3.6.4 In ships carrying flammable products, electrical equipment that is located in spaces protected by air-locks and not of the certified safe type, are to be de-energized in case of loss of over-pressure in the space

3.6.5 Electrical equipment for manoeuvring, anchoring and mooring, as well as emergency fire pumps that are located in spaces protected by airlocks, are to be of a certified safe type.

3.6.6 The air-lock space is to be monitored for cargo vapour.(see 13.6.2)

3.6.7 Subject to the requirements of the International Convention on Load Lines in force, the door sill is not to be less than 300 [mm] in height.

3.7 Bilge, ballast and oil fuel arrangements

3.7.1 Where cargo is carried in a cargo containment system not requiring a secondary barrier, suitable drainage arrangements for the hold spaces that are not connected with the machinery space to be provided. Means of detecting such leakage are to be provided.

3.7.2 Where there is a secondary barrier, suitable drainage arrangements for dealing with any leakage into the hold or insulation spaces through adjacent ship structure are to be provided. The suction is not to be led to pumps inside the machinery space. Means of detecting such leakage are to be provided.

3.7.3 The hold or interbarrier space of Type A independent tanks are to be provided with a drainage system suitable for handling liquid cargo in the event of cargo tank leakage or rupture. Such arrangements are to provide for the return of any cargo leakage to the liquid cargo piping.

3.7.4 The arrangements referred to in 3.7.3 is to be provided with a removable spool piece.

3.7.5 Ballast spaces including wet duct keels used as ballast piping, fuel oil tanks and non-hazardous spaces, may be connected to pumps in the machinery spaces. Dry duct keels with ballast piping passing through, may be connected to pumps in the machinery spaces, provided the connections are led directly to the pumps and the discharge from the pumps are led directly overboard with no valves or manifolds in either line that could connect the line from the duct keel to the lines serving non-hazardous spaces. Pump vents are not to be open to machinery spaces.

IR 3.7.4 The requirements of “Pump vents are not to be open to machinery spaces” apply only to pumps in the machinery spaces serving dry duct keels through which ballast piping passes.

3.8 Bow or stern loading and unloading arrangements

3.8.1 Subject to the requirements of this Section and Sec 5, cargo piping may be arranged to permit bow or stern loading and unloading.

3.8.2 Bow or stern loading and unloading lines that are led past accommodation spaces, service spaces or control stations are not to be used for the transfer of products requiring a Type 1G ship. Bow or stern loading and unloading lines are not to be used for the transfer of toxic products as specified in 1.2.53, where the design pressure is above 2.5 [Mpa].

3.8.3 Portable arrangements are not permitted.

3.8.4.1 Entrances, air inlets and openings to accommodation spaces, service spaces, machinery spaces and control stations are not to face the cargo shore connection location of bow or stern loading and unloading arrangements. They are to be located on the outboard side of the superstructure or deckhouse at a distance of at least four per cent of the length of the ship but not less than 3 [m] from the end of the superstructure or deckhouse facing the cargo shore connection location of the bow or stern loading and unloading arrangements. This distance need not exceed 5 [m].

3.8.4.2 Windows and side scuttles facing the shore connection location and on the sides of the superstructure or deckhouse within the distance mentioned above are to be of the fixed (non-opening) type.

3.8.4.3 In addition, during the use of the bow or stern loading and unloading arrangements, all doors, ports and other openings on the corresponding superstructure or deckhouse side are to be kept closed.

3.8.4.4 Where, in the case of small ships, compliance with 3.2.4.1 to 3.2.4.4 and 3.8.4.1 to 3.8.4.3 is not possible, IRS may approve relaxations from the above requirements.

3.8.5 Deck openings and air inlets and outlets to spaces within distances of 10 [m] from the cargo shore connection location are to be kept closed during the use of bow or stern loading or unloading arrangements.

3.8.6 Fire-fighting arrangements for the bow or stern loading and unloading areas are to be in accordance with 11.3.1.3 and 11.4.7.

3.8.7 Means of communication between the cargo control station and shore connection location are to be provided and where applicable, certified for use in hazardous areas.

IR 3.9 Emergency towing arrangement (ETA)

IR 3.9.1 All gas carriers of deadweight 20,000 tonnes and above are to be provided with emergency towing arrangements at both forward and aft ends. Emergency towing arrangements are to be as per Pt.5, Ch.2, Sec.2.

IR 3.10 Hull construction

IR 3.10.1 The scantlings and arrangements of the hull structure would be specially considered in each case.

Section 4

Cargo Containment

4.0 Goal

To ensure the safe containment of cargo under all design and operating conditions having regard to the nature of the cargo carried. This will include measures to:

.1 provide strength to withstand defined loads;

.2 maintain the cargo in a liquid state;

.3 design for or protect the hull structure from low temperature exposure; and

.4 prevent the ingress of water or air into the cargo containment system.

4.1 Definitions

4.1.1 *A cold spot* is a part of the hull or thermal insulation surface where a localized temperature decrease occurs with respect to the allowable minimum temperature of the hull or of its adjacent hull structure, or to design capabilities of cargo pressure/temperature control systems required in Sec 7.

4.1.2 *Design vapour pressure* "P_0" is the maximum gauge pressure, at the top of the tank, to be used in the design of the tank.

4.1.3 *Design temperature* for selection of materials is the minimum temperature at which cargo may be loaded or transported in the cargo tanks.

4.1.4 *Independent tanks* are self-supporting tanks. They do not form part of the ship's hull and are not essential to the hull strength. There are three categories of independent tank, which are referred to in 4.21, 4.22 and 4.23.

4.1.5 *Membrane tanks* are non-self-supporting tanks that consist of a thin liquid and gastight layer (membrane) supported through insulation by the adjacent hull structure. Membrane tanks are covered in 4.24.

4.1.6 *Integral tanks* are tanks that form a structural part of the hull and are influenced in the same manner by the loads that stress the adjacent hull structure. Integral tanks are covered in 4.25.

4.1.7 *Semi-membrane tanks* are non-self-supporting tanks in the loaded condition and consist of a layer, parts of which are supported through insulation by the adjacent hull structure. Semi-membrane tanks are covered in 4.26.

4.1.8 In addition to the definitions in 1.2, the definitions given in this chapter shall apply throughout the Chapter

4.2 Application

Unless otherwise specified in part E, the requirements of parts A to D will apply to all types of tanks, including those covered in part F.

Part A
Cargo Containment

4.3 Functional Requirements

4.3.1 The design life of the cargo containment system is not to be less than the design life of the ship.

4.3.2 Cargo containment systems are to be designed for North Atlantic environmental conditions and relevant long-term sea state scatter diagrams for unrestricted navigation. Lesser environmental conditions, consistent with the expected usage, may be accepted by the IRS for cargo containment systems used exclusively for restricted navigation. Greater environmental conditions may be required for cargo containment systems operated in conditions more severe than the North Atlantic environment.

4.3.3 Cargo containment systems are to be designed with suitable safety margins:

.1 to withstand, in the intact condition, the environmental conditions anticipated for the cargo containment system's design life and the loading conditions appropriate for them, which include full homogeneous and partial load conditions, partial filling within defined limits and ballast voyage loads; and

.2 being appropriate for uncertainties in loads, structural modelling, fatigue, corrosion, thermal effects, material variability, ageing and construction tolerances.

IR4.3.3 Guidance: Safety Margins

IR4.3.3.1 The Safety Margins to be used for design are specified in the sections as mentioned below:

Independent Type A Tank:	Section 4.21
Independent Type B Tank:	Section 4.22
Independent Type C Tank	Section 4.23
Membrane Tank:	Section 4.24
Integral Tank	Section 4.25

IR4.3.3.2,Notwithstanding IR4.3.3.1, the safety margin is to be taken not less than 2 for membrane tanks. IRS may specially consider lower safety margins provided they are accompanied with satisfactory technical justification which may include detailed analyses, calculations, experimental and model tests, records of satisfactory service history etc. If the flag Administration specifies a more stringent margin of safety, then the same is to be utilized.

4.3.4 The cargo containment system structural strength is to be assessed against failure modes, including but not limited to plastic deformation, buckling and fatigue. The specific design conditions which is to be considered for the design of each cargo containment system are given in 4.21 to 4.26. There are three main categories of design conditions:

.1 Ultimate design conditions – the cargo containment system structure and its structural components are to withstand loads liable to occur during its construction, testing and anticipated use in service, without loss of structural integrity. The design is to take into account proper combinations of the following loads:

.1 internal pressure;

.2 external pressure;

.3 dynamic loads due to the motion of the ship;

.4 thermal loads;

.5 sloshing loads;

.6 loads corresponding to ship deflections;

.7 tank and cargo weight with the corresponding reaction in way of supports;

.8 insulation weight;

.9 loads in way of towers and other attachments; and

.10 test loads.

.2 Fatigue design conditions – the cargo containment system structure

and its structural components are not to fail under accumulated cyclic loading.

.3 The cargo containment system is to meet the following criteria:

.1 Collision – the cargo containment system is to be protectively located in accordance with 2.4.1 and withstand the collision loads specified in 4.15.1 without deformation of the supports, or the tank structure in way of the supports, likely to endanger the tank structure.

.2 Fire – the cargo containment systems is to sustain, without rupture, the rise in internal pressure specified in 8.4.1 under the fire scenarios envisaged therein.

.3 Flooded compartment causing buoyancy on tank – the anti-flotation arrangements are to sustain the upward force, specified in 4.15.2, and there shall be no endangering plastic deformation to the hull.

4.3.5 Measures are to be applied to ensure that scantlings required meet the structural strength provisions and be maintained throughout the design life. Measures may include, but are not limited to, material selection, coatings, corrosion additions, cathodic protection and inerting. Corrosion allowance need not be required in addition to the thickness resulting from the structural analysis. However, where there is no environmental control, such as inerting around the cargo tank, or where the cargo is of a corrosive nature, IRS may require a suitable corrosion allowance.

IR4.3.5 A corrosion margin of not less than 1 [mm] is to be considered for steel in spaces which do not have any provisions for environmental control.

4.3.6 An inspection/survey plan for the cargo containment system is to be developed and approved by IRS. The inspection/survey plan is to identify areas that need inspection during surveys throughout the cargo containment system's life and, in particular, all necessary in-service survey and maintenance that was assumed when selecting cargo containment system design parameters. Cargo containment systems are to be designed, constructed and equipped to provide adequate means of access to areas that need inspection as specified in the inspection/survey plan. Cargo containment systems, including all associated internal equipment, shall be designed and built to ensure safety during operations, inspection and maintenance (see 3.5).

4.4 Cargo containment safety principles

4.4.1 The containment systems are to be provided with a full secondary liquid-tight barrier capable of safely containing all potential leakages through the primary barrier and, in conjunction with the thermal insulation system, of preventing lowering of the temperature of the ship structure to an unsafe level.

4.4.2 However, the size and configuration or arrangement of the secondary barrier may be reduced where an equivalent level of safety is demonstrated in accordance with the requirements of 4.4.3 to 4.4.5, as applicable.

4.4.3 Cargo containment systems for which the probability for structural failures to develop into a critical state has been determined to be extremely low, but where the possibility of leakages through the primary barrier cannot be excluded, shall be equipped with a partial secondary barrier and small leak protection system capable of safely handling and disposing of the leakages. The arrangements shall comply with the following requirements:

.1 failure developments that can be reliably detected before reaching a critical state (e.g. by gas detection or inspection) shall have a sufficiently long development time for remedial actions to be taken; and

.2 failure developments that cannot be safely detected before reaching a critical state are to have a predicted development time that is much longer than the expected lifetime of the tank.

4.4.4 No secondary barrier is required for cargo containment systems, e.g. type C independent tanks, where the probability for structural failures and leakages through the primary barrier is extremely low and can be neglected.

4.4.5 No secondary barrier is required where the cargo temperature at atmospheric pressure is at or above -10 [°C].

4.5 Secondary barriers in relation to tank types

Secondary barriers in relation to the tank types defined in 4.21 to 4.26 shall be provided in accordance with the following table.

<table>
<tr><th>Cargo temperature at atmospheric pressure</th><th>-10 [°C] and above</th><th>Below -10 [°C] down to -55 [°C]</th><th>Below -55 [°C]</th></tr>
<tr><td>Basic tank type</td><td rowspan="8">No secondary barrier required</td><td>Hull may act as secondary barrier</td><td>Separate secondary barrier where required</td></tr>
<tr><td>Integral</td><td colspan="2">Tank type not normally allowed (1)</td></tr>
<tr><td>Membrane</td><td colspan="2">Complete secondary barrier</td></tr>
<tr><td>Semi-membrane</td><td colspan="2">Complete secondary barrier (2)</td></tr>
<tr><td>Independent</td><td colspan="2"></td></tr>
<tr><td>Type A</td><td colspan="2">Complete secondary barrier</td></tr>
<tr><td>Type B</td><td colspan="2">Partial secondary barrier</td></tr>
<tr><td>Type C</td><td colspan="2">No secondary barrier required</td></tr>
<tr><td colspan="4">Notes:
1 A complete secondary barrier is normally to be required if cargoes with a temperature at atmospheric pressure below -10°C are permitted in accordance with 4.25.1.
2 In the case of semi-membrane tanks that comply in all respects with the requirements applicable to type B independent tanks, except for the manner of support, IRS may, after special consideration, accept a partial secondary barrier.</td></tr>
</table>

4.6 Design of secondary barriers

4.6.1 Where the cargo temperature at atmospheric pressure is not below -55 [°C], the hull structure may act as a secondary barrier based on the following:

.1 the hull material is to be suitable for the cargo temperature at atmospheric pressure as required by 4.19.1.4; and

.2 the design is to be such that this temperature will not result in unacceptable hull stresses.

4.6.2 The design of the secondary barrier is to be such that:

.1 it is capable of containing any envisaged leakage of liquid cargo for a period of 15 days, unless different criteria apply for particular voyages, taking into account the load spectrum referred to in 4.18.2.6;

.2 physical, mechanical, or operational events within the cargo tank that could cause failure of the primary barrier is not to impair the due function of the secondary barrier, or vice versa;

.3 failure of a support or an attachment to the hull structure will not lead to loss of liquid tightness of both the primary and secondary barriers;

.4 it is capable of being periodically checked for its effectiveness by means acceptable to IRS. This may be by means of a visual inspection or a pressure/vacuum test or other suitable means carried out according to a documented procedure agreed with IRS.

.5 the methods required in .4 above are to be approved by IRS and are to include, where applicable to the test procedure:

.1 details on the size of defect acceptable and the location within the secondary barrier, before its liquid-tight effectiveness is compromised;

.2 accuracy and range of values of the proposed method for detecting defects in .1 above;

.3 scaling factors to be used in determining the acceptance criteria, if full scale model testing is not undertaken; and

.4 effects of thermal and mechanical cyclic loading on the effectiveness of the proposed test; and

.6 the secondary barrier is to fulfill its functional requirements at a static angle of heel of 30°.

IR 4.6.2 For containment system with glued secondary barriers:

.1 At the time of construction, a tightness test is to be carried out in accordance with approved system designer's procedures and acceptance criteria before and after initial cool down. Low differential pressures tests are not considered as acceptable test.

.2 If the designer's threshold values are exceeded, an investigation is to be carried out and additional testing such as thermographic or acoustic emissions testing is to be carried out.

.3 The values recorded are to be used as reference for future assessment of secondary barrier tightness.

For containment systems with welded metallic secondary barriers, a tightness test after initial cool down is not required.

4.7 Partial secondary barriers and primary barrier small leak protection system

4.7.1 Partial secondary barriers as permitted in 4.4.3 are to be used with a small leak protection system and meet all the requirements in 4.6.2. The small leak protection system is to include means to detect a leak in the primary barrier, provision such as a spray shield to deflect any liquid cargo down into the partial secondary barrier, and means to dispose of the liquid, which may be by natural evaporation.

4.7.2 The capacity of the partial secondary barrier is to be determined, based on the cargo leakage corresponding to the extent of failure resulting from the load spectrum referred to in 4.18.2.6, after the initial detection of a primary leak. Due account may be taken of liquid evaporation, rate of leakage, pumping capacity and other relevant factors.

4.7.3 The required liquid leakage detection may be by means of liquid sensors, or by an effective use of pressure, temperature or gas detection systems, or any combination thereof.

4.8 Supporting arrangements

4.8.1 The cargo tanks are to be supported by the hull in a manner that prevents bodily movement of the tank under the static and dynamic loads defined in 4.12 to 4.15, where applicable, while allowing contraction and expansion of the tank under temperature variations and hull deflections without undue stressing of the tank and the hull.

4.8.2 Anti-flotation arrangements are to be provided for independent tanks and capable of withstanding the loads defined in 4.15.2 without plastic deformation likely to endanger the hull structure.

4.8.3 Supports and supporting arrangements are to withstand the loads defined in 4.13.9 and 4.15, but these loads need not be combined with each other or with wave-induced loads.

4.9 Associated structure and equipment

4.9.1 Cargo containment systems are to be designed for the loads imposed by associated structure and equipment. This includes pump towers, cargo domes, cargo pumps and piping, stripping pumps and piping, nitrogen piping, access hatches, ladders, piping penetrations, liquid level gauges, independent level alarm gauges, spray nozzles, and instrumentation systems (such as pressure, temperature and strain gauges).

4.10 Thermal insulation

4.10.1 Thermal insulation is to be provided, as required, to protect the hull from temperatures below those allowable (see 4.19.1) and limit the heat flux into the tank to the levels that can be

maintained by the pressure and temperature control system applied in Sec 7.

4.10.2 In determining the insulation performance, due regard is to be given to the amount of the acceptable boil-off in association with the reliquefaction plant on board, main propulsion machinery or other temperature control system.

Part B
Design Loads

4.11 General

This section defines the design loads to be considered with regard to the requirements in 4.16, 4.17 and 4.18. This includes:

.1 load categories (permanent, functional, environmental and accidental) and the description of the loads;

.2 the extent to which these loads are to be considered depending on the type of tank, and is more fully detailed in the following paragraphs; and

.3 tanks, together with their supporting structure and other fixtures, that are to be designed taking into account relevant combinations of the loads described below.

4.12 Permanent loads

4.12.1 *Gravity loads*

The weight of tank, thermal insulation, loads caused by towers and other attachments shall be considered.

4.12.2 *Permanent external loads*

Gravity loads of structures and equipment acting externally on the tank shall be considered.

4.13 Functional loads

4.13.1 Loads arising from the operational use of the tank system are to be classified as functional loads. All functional loads that are essential for ensuring the integrity of the tank system, during all design conditions, are to be considered. As a minimum, the effects from the following criteria, as applicable, are to be considered when establishing functional loads:

.1 internal pressure;

.2 external pressure;

.3 thermally induced loads;

.4 vibration;

.5 interaction loads;

.6 loads associated with construction and installation;

.7 test loads;

.8 static heel loads; and

.9 weight of cargo.

4.13.2 ***Internal pressure***

.1 In all cases, including 4.13.2.2, P_o is not to be less than MARVS.

.2 For cargo tanks, where there is no temperature control and where the pressure of the cargo is dictated only by the ambient temperature, P_o is not to be less than the gauge vapour pressure of the cargo at a temperature of 45 [°C] except as follows:

.1 lower values of ambient temperature may be accepted by IRS for ships operating in restricted areas. Conversely, higher values of ambient temperature may be required; and

.2 for ships on voyages of restricted duration, P_o may be calculated based on the actual pressure rise during the voyage, and account may be taken of any thermal insulation of the tank.

.3 Subject to special consideration by the Administration/ IRS and to the limitations given in 4.21 to 4.26, for the various tank types, a vapour pressure P_h higher than P_o may be accepted for site specific conditions (harbour or other locations), where dynamic loads are reduced. Any relief valve setting resulting from this paragraph is to be recorded in the International Certificate

of Fitness for the Carriage of Liquefied Gases in Bulk.

.4 The internal pressure P_{eq} results from the vapour pressure P_o or P_h plus the maximum associated dynamic liquid pressure P_{gd}, but not including the effects of liquid sloshing loads. Guidance formulae for associated dynamic liquid pressure P_{gd} are given in 4.28.1.

4.13.3 ***External pressure***

External design pressure loads are to be based on the difference between the minimum internal pressure and the maximum external pressure to which any portion of the tank may be simultaneously subjected.

4.13.4 ***Thermally induced loads***

4.13.4.1 Transient thermally induced loads during cooling down periods are to be considered for tanks intended for cargo temperatures below -55 [°C].

4.13.4.2 Stationary thermally induced loads are to be considered for cargo containment systems where the design supporting arrangements or attachments and operating temperature may give rise to significant thermal stresses (see 7.2).

4.13.5 ***Vibration***

The potentially damaging effects of vibration on the cargo containment system are to be considered.

4.13.6 ***Interaction loads***

The static component of loads resulting from interaction between cargo containment system and the hull structure, as well as loads from associated structure and equipment, are to be considered.

4.13.7 ***Loads associated with construction and installation***

Loads or conditions associated with construction and installation, e.g. lifting, are to be considered.

4.13.8 ***Test loads***

Account is to be taken of the loads corresponding to the testing of the cargo containment system referred to in 4.21 to 4.26.

4.13.9 ***Static heel loads***

Loads corresponding to the most unfavourable static heel angle within the range 0° to 30° are to be considered.

4.13.10 ***Other loads***

Any other loads not specifically addressed, which could have an effect on the cargo containment system, are to be taken into account.

4.14 Environmental loads

Environmental loads are defined as those loads on the cargo containment system that are caused by the surrounding environment and that are not otherwise classified as a permanent, functional or accidental load.

4.14.1 ***Loads due to ship motion***

4.14.1.1 The determination of dynamic loads is to be take into account the long-term distribution of ship motion in irregular seas, which the ship will experience during its operating life. Account may be taken of the reduction in dynamic loads due to necessary speed reduction and variation of heading.

4.14.1.2 The ship's motion is to include surge, sway, heave, roll, pitch and yaw. The accelerations acting on tanks are to be estimated at their centre of gravity and include the following components:

.1 vertical acceleration : motion accelerations of heave, pitch and, possibly, roll (normal to the ship base);

.2 transverse acceleration: motion accelerations of sway, yaw and roll and gravity component of roll; and

.3 longitudinal acceleration : motion accelerations of surge and pitch and gravity component of pitch.

4.14.1.3 Methods to predict accelerations due to ship motion are to be proposed and approved by IRS.

IR4.14.1.3 Motions and accelerations are to be determined using computational methods or model tests acceptable to IRS. The North Atlantic environment conditions or the specific site environmental conditions may be considered. If site-specific environmental conditions are used to evaluate the motions and accelerations, this will be indicated in the Class Certificate.

4.14.1.4 Guidance formulae for acceleration components are given in 4.28.2.

4.14.1.5 Ships for restricted service may be given special consideration.

4.14.2 ***Dynamic interaction loads***

Account is to be taken of the dynamic component of loads resulting from interaction between cargo containment systems and the hull structure, including loads from associated structures and equipment.

4.14.3 ***Sloshing loads***

4.14.3.1 The sloshing loads on a cargo containment system and internal components are to be evaluated based on allowable filling levels.

4.14.3.2 When significant sloshing-induced loads are expected to be present, special tests and calculations are to be required covering the full range of intended filling levels.

4.14.4 ***Snow and ice loads***

Snow and icing are to be considered, if relevant.

4.14.5 ***Loads due to navigation in ice***

Loads due to navigation in ice are to be considered for vessels intended for such service.

4.15 Accidental loads

Accidental loads are defined as loads that are imposed on a cargo containment system and its supporting arrangements under abnormal and unplanned conditions.

4.15.1 ***Collision loads***

The collision load is to be determined based on the cargo containment system under fully loaded condition with an inertial force corresponding to 0.5 g in the forward direction and 0.25 g in the aft direction, where "g" is gravitational acceleration.

4.15.2 ***Loads due to flooding on ship***

For independent tanks, loads caused by the buoyancy of an empty tank in a hold space flooded to the summer load draught are to be considered in the design of the anti-flotation chocks and the supporting hull structure.

Part C
Structural Integrity

4.16 General

4.16.1 The structural design is to ensure that tanks have an adequate capacity to sustain all relevant loads with an adequate margin of safety. This is to take into account the possibility of plastic deformation, buckling, fatigue and loss of liquid and gas tightness.

IR4.16.1 Guidance provided in 4.3.3 may be referred.

4.16.2 The structural integrity of cargo containment systems is to be demonstrated by compliance with 4.21 to 4.26, as appropriate, for the cargo containment system type.

4.16.3 The structural integrity of cargo containment system types that are of novel design and differ significantly from those covered by 4.21 to 4.26 is to be demonstrated by compliance with 4.27 to ensure that the overall level of safety provided in this section is maintained.

4.17 Structural analyses

4.17.1 ***Analysis***

4.17.1.1 The design analyses is to be based on accepted principles of statics, dynamics and strength of materials.

4.17.1.2 Simplified methods or simplified analyses may be used to calculate the load effects, provided that they are conservative. Model tests may be used in combination with, or instead of, theoretical calculations. In cases where theoretical methods are inadequate, model or full-scale tests may be required.

4.17.1.3 When determining responses to dynamic loads, the dynamic effect is to be taken

into account where it may affect structural integrity.

IR4.17.1.3 It is recommended that the use of simplified methods or analyses, model/ full scale tests, consideration of dynamic effects be finalized in consultation with IRS.

4.17.2 ***Load scenarios***

4.17.2.1 For each location or part of the cargo containment system to be considered and for each possible mode of failure to be analysed, all relevant combinations of loads that may act simultaneously are to be considered.

4.17.2.2 The most unfavourable scenarios for all relevant phases during construction, handling, testing and in service, and conditions are to be considered.

4.17.3 When the static and dynamic stresses are calculated separately, and unless other methods of calculation are justified, the total stresses are to be calculated according to:

$$\boldsymbol{\sigma_x} = \boldsymbol{\sigma_{x.st}} \pm \sqrt{\sum \left(\boldsymbol{\sigma_{x.dyn}}\right)^2}$$

$$\boldsymbol{\sigma_y} = \boldsymbol{\sigma_{y.st}} \pm \sqrt{\sum \left(\boldsymbol{\sigma_{y.dyn}}\right)^2}$$

$$\boldsymbol{\sigma_z} = \boldsymbol{\sigma_{z.st}} \pm \sqrt{\sum \left(\boldsymbol{\sigma_{z.dyn}}\right)^2}$$

$$\boldsymbol{\tau_{xy}} = \boldsymbol{\tau_{xy.st}} \pm \sqrt{\sum \left(\boldsymbol{\tau_{xy.dyn}}\right)^2}$$

$$\boldsymbol{\tau_{xz}} = \boldsymbol{\tau_{xz.st}} \pm \sqrt{\sum \left(\boldsymbol{\tau_{xz.dyn}}\right)^2}$$

$$\boldsymbol{\tau_{yz}} = \boldsymbol{\tau_{yz.st}} \pm \sqrt{\sum \left(\boldsymbol{\tau_{yz.dyn}}\right)^2}$$

Where:

$\sigma_{x.st}, \sigma_{y.st}, \sigma_{z.st}, \tau_{xy.st}, \tau_{xz.st}, and\ \tau_{yz.st}$ are static stresses and

$\boldsymbol{\sigma_{x.dyn}, \sigma_{y.dyn}, \sigma_{z.dyn}, \tau_{xy.dyn}, \tau_{xz.dyn}\ and\ \tau_{yz.dyn}}$ are dynamic stresses.

Each to be determined separately from acceleration components and hull strain components due to deflection and torsion.

4.18 Design conditions

All relevant failure modes are to be considered in the design for all relevant load scenarios and design conditions. The design conditions are given in the earlier part of this chapter, and the load scenarios are covered by 4.17.2.

4.18.1 ***Ultimate design condition***

Structural capacity may be determined by testing, or by analysis, taking into account both the elastic and plastic material properties, by simplified linear elastic analysis or by the provisions of this chapter.

4.18.1.1 Plastic deformation and buckling are to be considered.

4.18.1.2 Analysis is to be based on characteristic load values as follows:

Permanent loads :	Expected values
Functional loads :	Specified values
Environmental loads:	For wave loads: most probable largest load encountered during 10^8 wave encounters.

4.18.1.3 For the purpose of ultimate strength assessment, the following material parameters apply:

.1.1 R_e = specified minimum yield stress at room temperature (N/mm^2). If the stress-strain curve does not show a defined yield stress, the 0.2% proof stress applies.

.1.2 R_m = specified minimum tensile strength at room temperature (N/mm^2).

For welded connections where under-matched welds, i.e. where the weld metal has lower tensile strength than the parent metal, are unavoidable, such as in some aluminium alloys, the respective R_e and R_m of the welds, after any applied heat treatment, is to be used. In such cases, the transverse weld tensile strength is not to be less than the actual yield strength of the parent metal. If this cannot be achieved, welded structures made from such materials is not to be incorporated in cargo containment systems.

.2 The above properties are to correspond to the minimum specified mechanical properties of the material, including the weld metal in the as-fabricated condition. Subject to special

consideration by IRS, account may be taken of the enhanced yield stress and tensile strength at low temperature. The temperature on which the material properties are based is to be shown on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk required in 1.4.

4.18.1.4 The equivalent stress σ_c (von Mises, Huber) is to be determined by:

$$\boldsymbol{\sigma_c = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - \sigma_x\sigma_y - \sigma_x\sigma_z - \sigma_y\sigma_z + 3\left(\tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2\right)}}$$

Where:

σ_x=total normal stress in x-direction
σ_y=total normal stress in y-direction
σ_z=total normal stress in z-direction
τ_{xy}=total shear stress in x-y plane
τ_{xz}=total shear stress in x-z plane
τ_{yz}=total shear stress in y-z plane

The above values to be calculated as described in 4.17.3

4.18.1.5 Allowable stresses for materials other than those covered by Sec 6 are subject to approval by the IRS in each case.

4.18.1.6 Stresses may be further limited by fatigue analysis, crack propagation analysis and buckling criteria.

4.18.2 ***Fatigue design condition***

4.18.2.1 The fatigue design condition is the design condition with respect to accumulated cyclic loading.

4.18.2.2 Where a fatigue analysis is required, the cumulative effect of the fatigue load shall comply with:

$$\boldsymbol{\sum \frac{n_i}{N_i} + \frac{n_{Loading}}{N_{Loading}} \leq C_w}$$

Where:

n_i=number of stress cycles at each stress level during the life of the tank

N_i=number of cycles to fracture for the respective stress level according to the Wohler (S-N) Curve.

$n_{Loading}$=number of loading and unloading cycles during the life of the tank, not to be less than 1000. Loading and unloading cycles include a complete pressure and thermal cycle;

$N_{Loading}$=number of cycles to fracture for the fatigue loads due to loading and unloading; and

C_w=maximum allowable cumulative fatigue damage ratio.

The fatigue damage is to be based on the design life of the tank but not less than 10^8 wave encounters.

4.18.2.3 Where required, the cargo containment system shall be subject to fatigue analysis, considering all fatigue loads and their appropriate combinations for the expected life of the cargo containment system. Consideration shall be given to various filling conditions.

4.18.2.4.1 Design S-N curves used in the analysis shall be applicable to the materials and weldments, construction details, fabrication procedures and applicable state of the stress envisioned.

4.18.2.4.2 The S-N curves shall be based on a 97.6% probability of survival corresponding to the mean-minus-two-standard-deviation curves of relevant experimental data up to final failure. Use of S-N curves derived in a different way requires adjustments to the acceptable C_w values specified in 4.18.2.7 to 4.18.2.9.

4.18.2.5 Analysis shall be based on characteristic load values as follows:

Permanent loads :	Expected values
Functional loads :	Specified values or specified history
Environmental loads:	Expected load history, but not less than 10^8 cycle.

If simplified dynamic loading spectra are used for the estimation of the fatigue life, they shall be specially considered by the Administration/ IRS.

4.18.2.6.1 Where the size of the secondary barrier is reduced, as is provided for in 4.4.3, fracture mechanics analyses of fatigue crack growth shall be carried out to determine:

.1 crack propagation paths in the structure;

.2 crack growth rate;

.3 the time required for a crack to propagate to cause a leakage from the tank;

.4 the size and shape of through thickness cracks; and

.5 the time required for detectable cracks to reach a critical state.

The fracture mechanics are, in general, based on crack growth data taken as a mean value plus two standard deviations of the test data.

4.18.2.6.2 In analysing crack propagation, the largest initial crack not detectable by the inspection method applied is to be assumed, taking into account the allowable non-destructive testing and visual inspection criterion, as applicable.

4.18.2.6.3 Crack propagation analysis under the condition specified in 4.18.2.7: the simplified load distribution and sequence over a period of 15 days may be used. Such distributions may be obtained as indicated in Fig.4.4. Load distribution and sequence for longer periods, such as in 4.18.2.8 and 4.18.2.9 are to be approved by IRS.

4.18.2.6.4 The arrangements are to comply with 4.18.2.7 to 4.18.2.9, as applicable.

4.18.2.7 For failures that can be reliably detected by means of leakage detection:
C_w is to be less than or equal to 0.5.

Predicted remaining failure development time, from the point of detection of leakage till reaching a critical state, is not to be less than 15 days, unless different requirements apply for ships engaged in particular voyages.

4.18.2.8 For failures that cannot be detected by leakage but that can be reliably detected at the time of in-service inspections:

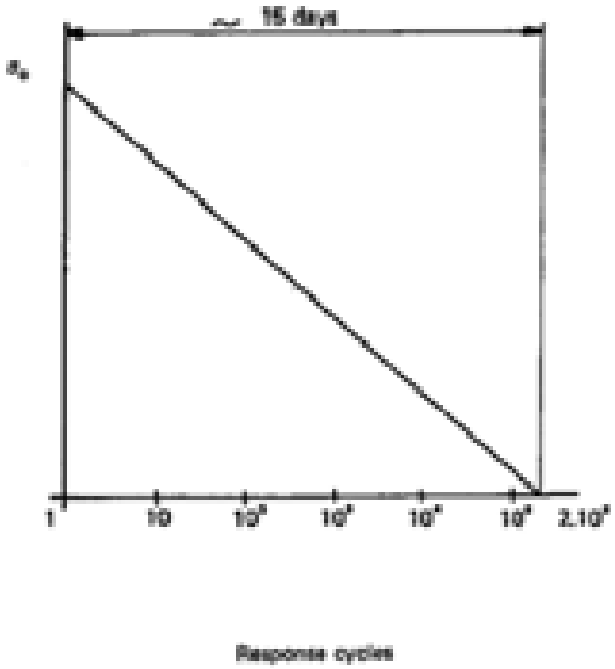
Cw is to be less than or equal to 0.5.

Predicted remaining failure development time, from the largest crack not detectable by in-service inspection methods until reaching a critical state, is not to be less than three times the inspection interval.

4.18.2.9 In particular locations of the tank, where effective defect or crack development detection cannot be assured, the following, more stringent, fatigue acceptance criteria is to be applied as a minimum:

Cw is to be less than or equal to 0.1.

Predicted failure development time, from the assumed initial defect until reaching a critical state, is not to be less than three times the lifetime of the tank.



σ_0 = most probable maximum stress over the life of the ship

Response cycle scale is logarithmic; the value of 2.10^8 is given as an example of estimate.

Fig.4.4 : Simplified load distribution

4.18.3 ***Accident design condition***

4.18.3.1 The accident design condition is a design condition for accidental loads with extremely low probability of occurrence.

4.18.3.2 Analysis is to be based on the characteristic values as follows:

Permanent loads :	Expected values
Functional loads :	Specified values
Environmental loads:	Specified values
Accidental loads	Specified values or expected values

4.18.3.3 Loads mentioned in 4.13.9 and 4.15 need not be combined with each other or with wave-induced loads.

Part D
Materials and construction

4.19 Materials

4.19.0 Goal

To ensure that the cargo containment system, primary and secondary barriers, the thermal insulation, adjacent ship structure and other materials in the cargo containment system are constructed from materials of suitable properties for the conditions they will experience, both in normal service and in the event of failure of the primary barrier, where applicable.

4.19.1 ***Materials forming ship structure***

4.19.1.1 To determine the grade of plate and sections used in the hull structure, a temperature calculation is to be performed for all tank types when the cargo temperature is below -10°C. The following assumptions are to be made in this calculation:

.1 the primary barrier of all tanks is to be assumed to be at the cargo temperature;

.2 in addition to .1, where a complete or partial secondary barrier is required, it is to be assumed to be at the cargo temperature at atmospheric pressure for any one tank only;

.3 for worldwide service, ambient temperatures is to be taken as 5°C for air and 0ºC for seawater. Higher values may be accepted for ships operating in restricted areas and, conversely, lower values may be fixed by the Administration for ships trading to areas where lower temperatures are expected during the winter months;

.4 still air and seawater conditions are to be assumed, i.e. no adjustment for forced convection;

.5 degradation of the thermal insulation properties over the life of the ship due to factors such as thermal and mechanical ageing, compaction, ship motions and tank vibrations, as defined in 4.19.3.6 and 4.19.3.7, is to be assumed;

.6 the cooling effect of the rising boil-off vapour from the leaked cargo is to be taken into account, where applicable;

.7 credit for hull heating may be taken in accordance with 4.19.1.5, provided the heating arrangements are in compliance with 4.19.1.6;

.8 no credit is to be given for any means of heating, except as described in 4.19.1.5; and

.9 for members connecting inner and outer hulls, the mean temperature may be taken for determining the steel grade.

The ambient temperatures used in the design, described in this paragraph, are to be shown on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk required in 1.4.4.

4.19.1.2 The shell and deck plating of the ship and all stiffeners attached thereto are to be in accordance with recognized standards. If the calculated temperature of the material in the design condition is below -5 [°C] due to the influence of the cargo temperature, the material is to be in accordance with table 6.5.

4.19.1.3 The materials of all other hull structures for which the calculated temperature in the design condition is below 0 [°C], due to the influence of cargo temperature and that do not form the secondary barrier, are also to be in accordance with table 6.5. This includes hull structure supporting the cargo tanks, inner bottom plating, longitudinal bulkhead plating, transverse bulkhead plating, floors, webs, stringers and all attached stiffening members.

4.19.1.4 The hull material forming the secondary barrier is to be in accordance with table 6.2. Where the secondary barrier is formed by the deck or side shell plating, the material grade required by table 6.2 is to be carried into the adjacent deck or side shell plating, where applicable, to a suitable extent.

4.19.1.5 Means of heating structural materials may be used to ensure that the material temperature does not fall below the minimum allowed for the grade of material specified in table 6.5. In the calculations required in 4.19.1.1, credit for such heating may be taken in accordance with the following:

.1 for any transverse hull structure;

.2 for longitudinal hull structure referred to in 4.19.1.2 and 4.19.1.3 where colder ambient temperatures are specified, provided the material remains suitable for the ambient temperature conditions of +5°C for air and 0°C for seawater with no credit taken in the calculations for heating; and

.3 as an alternative to .2, for longitudinal bulkhead between cargo tanks, credit may be taken for heating, provided the material remain suitable for a minimum design temperature of -30 [°C], or a temperature 30 [°C] lower than that determined by 4.19.1.1 with the heating considered, whichever is less. In this case, the ship's longitudinal strength shall comply with SOLAS regulation II-1/3-1 for both when those bulkhead(s) are considered effective and not.

4.19.1.6 The means of heating referred to in 4.19.1.5 is to comply with the following requirements:

.1 the heating system is to be arranged so that, in the event of failure in any part of the system, standby heating can be maintained equal to not less than 100% of the theoretical heat requirement;

IR.1.1 Heating system referred to in.1 above is to be such that in case of a single failure of a mechanical or electrical component in any part of the system, heating can be maintained at not less than 100% of the theoretical heat requirement.

IR.1.2 Where the above requirements are met by duplication of the system components, i.e., heaters, glycol circulation pumps, electrical control panel, auxiliary boilers etc., all electrical components of at least one of the systems are to be supplied from the emergency source of electrical power.

IR.1.3 Where duplication of the primary source of heat, e.g., oil-fired boiler is not feasible, alternative proposals can be accepted such as an electric heater capable of providing 100% of the theoretical heat requirement provided and supplied by an individual circuit arranged separately on the emergency switchboard. Other solutions may be considered towards satisfying the requirements of .1 provided a suitable risk assessment is conducted to the satisfaction of IRS. The requirement in para .2 continues to apply to all other electrical components in the system

.2 the heating system is to be considered as an essential auxiliary. All electrical components of at least one of the systems provided in accordance with 4.19.1.5.1 is to be supplied from the emergency source of electrical power; and

.3 the design and construction of the heating system is to be included in the approval of the containment system by the IRS.

4.19.2 Materials of primary and secondary barriers

4.19.2.1 Metallic materials used in the construction of primary and secondary barriers not forming the hull, are to be suitable for the design loads that they may be subjected to, and be in accordance with, table 6.1, 6.2 or 6.3.

4.19.2.2 Materials, either non-metallic or metallic but not covered by tables 6.1, 6.2 and 6.3, used in the primary and secondary barriers may be approved by the IRS, considering the design loads that they may be subjected to, their properties and their intended use.

4.19.2.3 Where non-metallic materials, including composites, are used for, or incorporated in the primary or secondary barriers, they are to be tested for the following properties, as applicable, to ensure that they are adequate for the intended service:

.1 compatibility with the cargoes;

.2 ageing;

.3 mechanical properties;

.4 thermal expansion and contraction;

.5 abrasion;

.6 cohesion;

.7 resistance to vibrations;

.8 resistance to fire and flame spread; and

.9 resistance to fatigue failure and crack propagation.

4.19.2.4 The above properties, where applicable, are to be tested for the range between the expected maximum temperature in service and +5 [°C] below the minimum design temperature, but not lower than -196 [°C].

4.19.2.5.1 Where non-metallic materials, including composites, are used for the primary and secondary barriers, the joining processes are also to be tested as described above.

4.19.2.5.2 Guidance on the use of non-metallic materials in the construction of primary and secondary barriers is provided in appendix 4 of IGC Code.

4.19.2.6 Consideration may be given to the use of materials in the primary and secondary barrier, which are not resistant to fire and flame spread, provided they are protected by a suitable system such as a permanent inert gas environment, or are provided with a fire-retardant barrier.

4.19.3 Thermal insulation and other materials used in cargo containment systems

4.19.3.1 Load-bearing thermal insulation and other materials used in cargo containment systems are to be suitable for the design loads.

4.19.3.2 Thermal insulation and other materials used in cargo containment systems are to have the following properties, as applicable, to ensure that they are adequate for the intended service:

.1 compatibility with the cargoes;

.2 solubility in the cargo;

.3 absorption of the cargo;

.4 shrinkage;

.5 ageing;

.6 closed cell content;

.7 density;

.8 mechanical properties, to the extent that they are subjected to cargo and other loading effects, thermal expansion and contraction;

.9 abrasion;

.10 cohesion;

.11 thermal conductivity;

.12 resistance to vibrations;

.13 resistance to fire and flame spread; and

.14 resistance to fatigue failure and crack propagation.

4.19.3.3 The above properties, where applicable, are to be tested for the range between the expected maximum temperature in service and 5 [°C] below the minimum design temperature, but not lower than -196 [°C].

4.19.3.4 Due to location or environmental conditions, thermal insulation materials are to have suitable properties of resistance to fire and flame spread and are to be adequately protected against penetration of water vapour and mechanical damage. Where the thermal insulation is located on or above the exposed deck, and in way of tank cover penetrations, it is to have suitable fire resistance properties in accordance with recognized standards or be covered with a material having low flame-spread characteristics and forming an efficient approved vapour seal.

4.19.3.5 Thermal insulation that does not meet recognized standards for fire resistance may be used in hold spaces that are not kept permanently inerted, provided its surfaces are covered with material with low flame-spread characteristics and that forms an efficient approved vapour seal.

4.19.3.6 Testing for thermal conductivity of thermal insulation is to be carried out on suitably aged samples.

4.19.3.7 Where powder or granulated thermal insulation is used, measures are to be taken to reduce compaction in service and to maintain the required thermal conductivity and also prevent any undue increase of pressure on the cargo containment system.

4.20 Construction processes

4.20.0 Goal

To define suitable construction processes and test procedures in order to ensure, as far as reasonably practical, that the cargo containment system will perform satisfactorily in service in accordance with the assumptions made at the design stage.

4.20.1 Weld joint design

4.20.1.1 All welded joints of the shells of independent tanks are to be of the in-plane butt weld full penetration type. For dome-to-shell connections only, tee welds of the full penetration type may be used depending on the results of the tests carried out at the approval of the welding procedure. Except for small penetrations on domes, nozzle welds are also to be designed with full penetration.

IR4.20.1.1 Paragraph 4.20.1.1 above is applicable to independent tanks of type A or type B, primarily constructed of plane surfaces. This includes the tank corners which are constructed using bent plating which is aligned with the tank surfaces and connected with in-plane welds. The applicability of the expression “For dome-to-shell connections only” is clarified as follows:

- Welded corners (i.e. corners made of weld metal) are not to be used in the main tank shell construction, i.e. corners between shell side (sloped plane surfaces parallel to hopper

or top side inclusive if any) and bottom or top of the tank, and between tank end transverse bulkheads and bottom, top or shell sides (sloped plane surfaces inclusive if any) of the tank. Instead, tank corners which are constructed using bent plating aligned with the tank surfaces and connected with in-plane welds are to be used.

- Tee welds can be accepted for other localised constructions of the shell such as suction well, sump, dome, etc. where tee welds of full penetration type shall also be used.

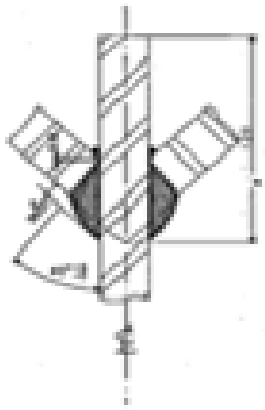
4.20.1.2 Welding joint details for type C independent tanks, and for the liquid-tight primary barriers of type B independent tanks primarily constructed of curved surfaces, are to be as follows:

.1 all longitudinal and circumferential joints are to be of butt welded, full penetration, double vee or single vee type. Full penetration butt welds are to be obtained by double welding or by the use of backing rings. If used, backing rings are to be removed except from very small process pressure vessels. Other edge preparations may be permitted, depending on the results of the tests carried out at the approval of the welding procedure; and

.2 the bevel preparation of the joints between the tank body and domes and between domes and relevant fittings are to be designed according to a standard acceptable to IRS. All welds connecting nozzles, domes or other penetrations of the vessel and all welds connecting flanges to the vessel or nozzles are to be full penetration welds.

IR4.20.1.2 Paragraph 4.20.1.2 above is applicable to type C independent tanks including bi-lobe and multi-lobe tanks, primarily constructed of curved surfaces fitted with longitudinal bulkhead/s. The applicability of the expression “Other edge preparations” is clarified as follows:

- Cruciform full penetration welded joints in a bi-lobe or multi-lobe tank with longitudinal bulkhead/s can be accepted for the tank structure construction at tank longitudinal bulkhead welds with bevel preparation subject to the approval of the Administration/IRS, based on the results of the tests carried out at the approval of the welding procedure. (See below example)



4.20.1.3 Where applicable, all the construction processes and testing, except that specified in 4.20.3, are to be done in accordance with the applicable provisions of Sec 6.

4.20.2 Design for gluing and other joining processes

The design of the joint to be glued (or joined by some other process except welding) is to take account of the strength characteristics of the joining process.

4.20.3 Testing

4.20.3.1 All cargo tanks and process pressure vessels are to be subjected to hydrostatic or hydropneumatic pressure testing in accordance with 4.21 to 4.26, as applicable for the tank type.

4.20.3.2 All tanks are to be subject to a tightness test which may be performed in combination with the pressure test referred to in 4.20.3.1.

4.20.3.3 Requirements with respect to inspection of secondary barriers are to be decided by IRS in each case, taking into account the accessibility of the barrier (see 4.6.2).

4.20.3.4 The Administration/ IRS may require that for ships fitted with novel type B independent tanks, or tanks designed according to 4.27 at least one prototype tank and its supporting structures are to be instrumented with strain gauges or other suitable equipment to confirm stress levels. Similar instrumentation may be required for type C independent tanks, depending on their

configuration and on the arrangement of their supports and attachments.

4.20.3.5 The overall performance of the cargo containment system is to be verified for compliance with the design parameters during the first full loading and discharging of the cargo, in accordance with the survey procedure and requirements in 1.4 and the requirements of IRS. Records of the performance of the components and equipment essential to verify the design parameters, are to be maintained and be available to the Administration/ IRS.

4.20.3.6 Heating arrangements, if fitted in accordance with 4.19.1.5 and 4.19.1.6, are to be tested for required heat output and heat distribution.

4.20.3.7 The cargo containment system is to be inspected for cold spots during, or immediately following, the first loaded voyage. Inspection of the integrity of thermal insulation surfaces that cannot be visually checked is to be carried out in accordance with recognized standards.

IR 4.20.4 Verifications before and after the first loaded voyage for Liquefied Gas Carriers.

IR 4.20.4.1 Following requirements applies to all ships carrying liquefied gases in bulk.

IR 4.20.4.2 The following initial certificates would be "conditionally" issued at delivery subject to satisfactory completion of all required verifications and examinations, as applicable.:

i) Certificate of class

ii) International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk.

The conditions would be stated on the Certificate of Class or issued as a Condition of Class in the ship's Record.

IR 4.20.4.3 Surveyor attendance is required at the first cargo loading and first cargo unloading.

IR 4.20.4.4 Surveyor attendance during new building gas trials can be considered to comply with the below applicable verifications and examinations survey requirements, with the exceptions of the survey requirements marked (**).[1]

IR 4.20.4.5 Verifications and examinations at gas trials or first full cargo loading, as applicable to cargo containment system[2]

- verify the satisfactory functionality of the emergency shutdown system during testing;
- satisfactory operation of gas detection system;
- satisfactory operation of cargo tank pressure monitoring system;
- satisfactory operation of inter barrier space(s) and insulation space(s) pressure monitoring system, as applicable;
- satisfactory operation of cargo tank temperature monitoring system;
- satisfactory operation of cargo tank level indicating system;
- satisfactory operation of inter barrier space(s) and inner hull temperature monitoring system, as applicable;
- inert gas generator, if operating;
- nitrogen generating plant, if operating;
- nitrogen pressure control system for insulation, inter-barrier, and annular spaces, as applicable;
- reliquefaction plant, if fitted;
- equipment fitted for the burning of cargo vapours such as boilers, engines, gas combustion units, etc., if operating;
- examination of on-deck cargo piping systems including expansion and supporting arrangements;
- verification and examination of all piping systems, including valves, fittings and associated equipment for handling cargo or vapours;
- advise Master to carry out cold spot examination of the hull and external insulation during transit voyage to unloading port and record in ship's logbook; and
- advise Master to test high-level alarm(s) with liquid cargo during voyage and record in ship's logbook, when loading condition permits.

IR 4.20.4.6 Verifications and examinations at gas trials or first full cargo unloading, as applicable[3]

- examination of on-deck cargo piping systems including expansion and supporting arrangements;

- review logbook entry of emergency shutdown system testing prior to commencement of unloading;
- (**) review cargo logs and alarm reports for cargo tank pressure, temperature, and level indicating systems;
- satisfactory operation of cargo compressors;
- satisfactory operation of cargo pumps;
- inert gas generator, if operating;
- nitrogen generating plant, if operating;
- nitrogen pressure control system for insulation, inter-barrier, and annular spaces, as applicable;
- (**) review of records for satisfactory operation of the reliquefaction plant, if fitted;
- review of records for equipment fitted for the burning of cargo vapours such as boilers, engines, gas combustion units, etc.;
- (**) on ships fitted with membrane tanks, review records of the cofferdam and inner hull temperature sensors to verify the readings are not below the allowable temperature for the selected grade of steel;
- (**) cofferdam heating system, if in operation;
- (**) review logbook entries for cold spot examination; and
- (**) review logbook entry for testing of high-level alarm(s) with liquid cargo. If cargo conditions did not permit testing, surveyor to require testing at the first occasion where cargo conditions allow for testing. Master to be advised to record testing in ship's logbook which is to be verified no later than the first annual survey.

IR 4.20.4.7 Documentation to be requested to the Master

To demonstrate satisfactory functionality of the verifications, ship's Master is required to arrange and provide to the surveyor print outs or screen shots showing:

- trends of cargo tanks pressure and temperature;
  - trends of pressure and temperature distribution of inter-barrier space(s) and insulation space(s), and temperature distribution of inner hull, as applicable;
  - trends record of performance of cofferdam heating system, when fitted;
  - trends record of consumption of nitrogen gas, and whether any abnormality has been observed;
- list of any gas alarms, if occurred:
  - Cargo Tanks Containment System Cold Spot Inspection Statement; and
  - activation of Cargo Tanks High-Level Alarm and Overfill Protection tests.

Note 1: The symbol (**) indicates survey requirements only feasible to be carried out at the time of first full cargo loading/unloading.

Note 2: When attending at first full cargo loading, priority is to be given to latter stages of loading.

Note 3: When attending at first full cargo unloading, priority is to be given to the commencement of unloading.

Part E
Tank Types

4.21 Type A independent tanks

4.21.1 Design basis

4.21.1.1 Type A independent tanks are tanks primarily designed using classical ship-structural analysis procedures in accordance with recognized standards. Where such tanks are primarily constructed of plane surfaces, the design vapour pressure P_o is to be less than 0.07 [MPa].

4.21.1.2 If the cargo temperature at atmospheric pressure is below -10 [°C], a complete secondary barrier is to be provided as required in 4.5. The secondary barrier is to be designed in accordance with 4.6.

4.21.2 Structural analysis

4.21.2.1 A structural analysis is to be performed taking into account the internal pressure as indicated in 4.13.2, and the interaction loads with the supporting and keying system as well as a reasonable part of the ship's hull.

IR4.21.2.1 A finite element analysis is to be performed. Stresses obtained from direct calculations using the finite element model of the tank along with the supports and the hull are required to be submitted to IRS. The support structures and the hull are to be suitably idealized or modeled so that the structural rigidity of the system is simulated accurately.

4.21.2.2 For parts, such as supporting structures, not otherwise covered by the requirements of the chapter, stresses are to be determined by direct calculations, taking into account the loads referred to in 4.12 to 4.15 as far as applicable, and the ship deflection in way of supporting structures.

4.21.2.3 The tanks with supports are to be designed for the accidental loads specified in 4.15. These loads need not be combined with each other or with environmental loads.

4.21.3 Ultimate design condition

4.21.3.1 For tanks primarily constructed of plane surfaces, the nominal membrane stresses for primary and secondary members (stiffeners, web frames, stringers, girders), when calculated by classical analysis procedures, are not to exceed the lower of R_m /2.66 or R_e /1.33 for nickel steels, carbon-manganese steels, austenitic steels and aluminium alloys, where R_m and R_e are defined in 4.18.1.3. However, if detailed calculations are carried out for the primary members, the equivalent stress σ_c, as defined in 4.18.1.4, may be increased over that indicated above to a stress acceptable to IRS. Calculations are to take into account the effects of bending, shear, axial and torsional deformation as well as the hull/cargo tank interaction forces due to the deflection of the double bottom and cargo tank bottoms.

4.21.3.2 Tank boundary scantlings are to meet at least the requirements of IRS for deep tanks taking into account the internal pressure as indicated in 4.13.2 and any corrosion allowance required by 4.3.5.

IR4.21.3.2 To comply with 4.21.3.2, the tank boundary scantlings are to be determined in accordance with Part 3, Chapter 10, considering these as tank bulkheads.

4.21.3.3 The cargo tank structure is to be reviewed against potential buckling.

4.21.4 Accident design condition

4.21.4.1 The tanks and the tank supports are to be designed for the accidental loads and design conditions specified in 4.3.4.3 and 4.15, as relevant.

4.21.4.2 When subjected to the accidental loads specified in 4.15, the stress is to comply with the acceptance criteria specified in 4.21.3, modified as appropriate, taking into account their lower probability of occurrence.

4.21.5 Testing

All type A independent tanks are to be subjected to a hydrostatic or hydropneumatic test. This test is to be performed such that the stresses approximate, as far as practicable, the design stresses, and that the pressure at the top of the tank corresponds at least to the MARVS. When a hydropneumatic test is performed, the conditions are to simulate, as far as practicable, the design loading of the tank and of its support structure, including dynamic components, while avoiding stress levels that could cause permanent deformation.

4.22 Type B independent tanks

4.22.1 ***Design basis***

4.22.1.1 Type B independent tanks are tanks designed using model tests, refined analytical tools and analysis methods to determine stress levels, fatigue life and crack propagation characteristics. Where such tanks are primarily constructed of plane surfaces (prismatic tanks), the design vapour pressure P_o is to be less than 0.07 [MPa].

4.22.1.2 If the cargo temperature at atmospheric pressure is below -10°C, a partial secondary barrier with a small leak protection system is to be provided as required in 4.5. The small leak protection system shall be designed according to 4.7.

4.22.2 *Structural analysis*

4.22.2.1 The effects of all dynamic and static loads are to be used to determine the suitability of the structure with respect to:

.1 plastic deformation;

.2 buckling;

.3 fatigue failure; and

.4 crack propagation.

Finite element analysis or similar methods and fracture mechanics analysis, or an equivalent approach, are to be carried out.

4.22.2.2 A three-dimensional analysis shall be carried out to evaluate the stress levels, including interaction with the ship's hull. The model for this analysis is to include the cargo tank with its supporting and keying system, as well as a reasonable part of the hull.

4.22.2.3 A complete analysis of the particular ship accelerations and motions in irregular waves, and of the response of the ship and its cargo tanks to these forces and motions are to be performed, unless the data is available from similar ships.

4.22.3 *Ultimate design condition*

4.22.3.1 Plastic deformation

4.22.3.1.1 For type B independent tanks, primarily constructed of bodies of revolution, the allowable stresses are not to exceed:

$$\sigma_m \leq f$$
$$\sigma_L \leq 1.5f$$
$$\sigma_b \leq 1.5F$$
$$\sigma_L + \sigma_b \leq 1.5F$$
$$\sigma_m + \sigma_b \leq 1.5F$$
$$\sigma_m + \sigma_b + \sigma_g \leq 3F$$
$$\sigma_L + \sigma_b + \sigma_g \leq 3F$$

Where:

σ_m = equivalent primary general membrane stress;

σ_L = equivalent primary local membrane stress;

σ_b = equivalent primary bending stress;

σ_g = equivalent secondary stress;

f = the lesser of (R_m/A) or (R_e/B);

F = the lesser of (R_m/C) or (R_e/D)

with R_m and R_e as defined in 4.18.1.3. With regard to the stresses σ_m, σ_L, σ_b and σ_g, the definition of stress categories in 4.28.3 are referred. The values A and B are to be shown on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk and shall have at least the following minimum values:

	Nickel steels and carbon manganese steels	Austenitic steels	Aluminium alloys
A	3	3.5	4
B	2	1.6	1.5
C	3	3	3
D	1.5	1.5	1.5

The above figures may be altered, taking into account the design condition considered in acceptance with the Administration/ IRS.

4.22.3.1.2 For type B independent tanks, primarily constructed of plane surfaces, the allowable membrane equivalent stresses applied for finite element analysis are not to exceed:

.1 for nickel steels and carbon-manganese steels, the lesser of R_m /2 or R_e /1.2;

.2 for austenitic steels, the lesser of R_m /2.5 or R_e /1.2; and

.3 for aluminium alloys, the lesser of R_m /2.5 or R_e /1.2.

The above figures may be amended, taking into account the locality of the stress, stress analysis methods and design condition considered in acceptance with the Administration/ IRS.

4.22.3.1.3 The thickness of the skin plate and the size of the stiffener is not to be less than those required for type A independent tanks.

4.22.3.2 Buckling

Buckling strength analyses of cargo tanks subject to external pressure and other loads causing compressive stresses are to be carried out in accordance with recognized standards. The method is to adequately account for the difference in theoretical and actual buckling stress as a result of plate edge misalignment, lack of straightness or flatness, ovality and deviation from true circular form over a specified arc or chord length, as applicable.

4.22.4 Fatigue design condition

4.22.4.1 Fatigue and crack propagation assessment is to be performed in accordance

with 4.18.2. The acceptance criteria is to comply with 4.18.2.7, 4.18.2.8 or 4.18.2.9, depending on the detectability of the defect.

4.22.4.2 Fatigue analysis is to consider construction tolerances.

4.22.4.3 Where deemed necessary by the IRS, model tests may be required to determine stress concentration factors and fatigue life of structural elements.

4.22.5 Accident design condition

4.22.5.1 The tanks and the tank supports are to be designed for the accidental loads and design conditions specified in 4.3.4.3 and 4.15, as applicable.

4.22.5.2 When subjected to the accidental loads specified in 4.15, the stress is to comply with the acceptance criteria specified in 4.22.3, modified as appropriate, taking into account their lower probability of occurrence.

4.22.6 Testing

Type B independent tanks are to be subjected to a hydrostatic or hydropneumatic test as follows:

.1 the test is to be performed as required in 4.21.5 for type A independent tanks; and

.2 in addition, the maximum primary membrane stress or maximum bending stress in primary members under test conditions are not to exceed 90% of the yield strength of the material (as fabricated) at the test temperature. To ensure that this condition is satisfied, when calculations indicate that this stress exceeds 75% of the yield strength, the prototype test shall be monitored by the use of strain gauges or other suitable equipment.

4.22.7 Marking

Any marking of the pressure vessel is to be achieved by a method that does not cause unacceptable local stress raisers.

4.23 Type C independent tanks

4.23.1 Design basis

4.23.1.1 The design basis for type C independent tanks is based on pressure vessel criteria modified to include fracture mechanics and crack propagation criteria. The minimum design pressure defined in 4.23.1.2 is intended to ensure that the dynamic stress is sufficiently low, so that an initial surface flaw will not propagate more than half the thickness of the shell during the lifetime of the tank.

4.23.1.2 The design vapour pressure is not to be less than:

$$P_o = 0.2 + AC(\rho_r)^{1.5} \quad [MPa]$$

Where:

$$A = 0.00185\left(\frac{\sigma_m}{\Delta\sigma_A}\right)^2$$

With:

σ_m=design primary membrane stress

$\Delta\sigma_A$=allowable dynamic membrane stress (double amplitude at probability level Q=10^{-8}) and equal to :

- 55 [N/mm^2] for ferritic-perlitic, martensitic and austenitic steel;
- 25 [N/mm^2] for aluminium alloy (5083-O)

C = a characteristic tank dimension to be taken as the greatest of the following:

$$h, 0.75b \; or \; 0.45l$$

with:

h = height of tank (dimension in ship's vertical direction) [m]

b = width of tank (dimension in ship's transverse direction) [m]

l = length of tank (dimension in ship's longitudinal direction) [m]

ρ_r=the relative density of the cargo (ρ_r=1 for fresh water) at the design temperature.

When a specified design life of the tank is longer than 10^8 wave encounters, $\Delta\sigma_A$ is to be modified to give equivalent crack propagation corresponding to the design life.

IR4.23.1.2.1 If the carriage of products not covered by the Chapter is intended, it is to be verified that the double amplitude of the primary membrane stress $\Delta\sigma_m$ created by the maximum dynamic pressure differential ΔP does not exceed the allowable double amplitude of the dynamic membrane stress $\Delta\sigma_A$ as specified in 4.23.1.2

$$\Delta\sigma_m \leq \Delta\sigma_A$$

IR4.23.1.2.2 The dynamic pressure differential ΔP is to be calculated as follows:

$$\Delta P = \frac{\rho}{1.02\times10^5}\left(a_{\beta1}Z_{\beta1} - a_{\beta2}Z_{\beta2}\right)\text{[MPa]}$$

Where:

ρ is maximum liquid cargo density in [kg/m^3] at the design temperature

a_β , Z_β are as defined in 4.28.1.2 of this Chapter, see also Figure IR 4.23.1.2.2.

$a_{\beta1}$, $Z_{\beta1}$ are the a_β and Z_β values giving the maximum liquid pressure $\left(P_{gd}\right)_{max}$.

$a_{\beta2}$, $Z_{\beta2}$ are the a_β and Z_β values giving the minimum liquid pressure $\left(P_{gd}\right)_{min}$.

In order to evaluate the maximum pressure differential ΔP, pressure differentials are to be evaluated over the full range of the acceleration ellipse as shown in the Fig.IR 4.23.1.2.2.

The above mentioned procedure is only applicable to products having a relative density exceeding 1.

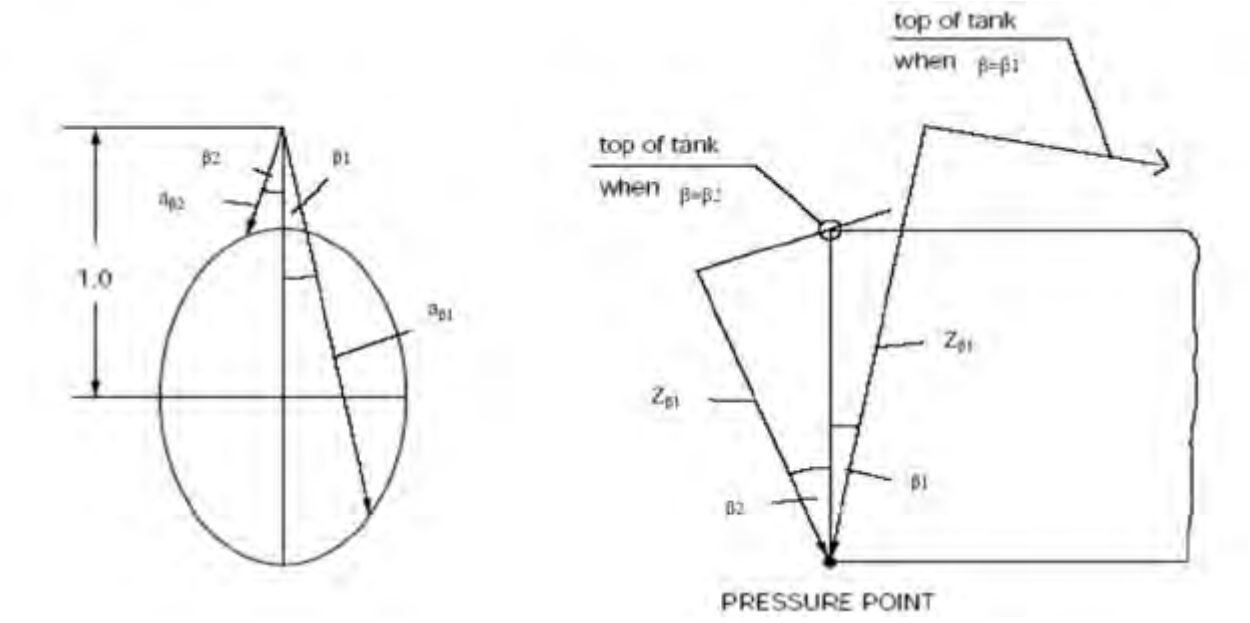


Fig.IR 4.23.1.2.2 : Acceleration Ellipse and Internal Pressure Heads

4.23.1.3 The Administration / IRS may allocate a tank complying with the criteria of type C tank minimum design pressure as in 4.23.1.2, to a type A or type B, dependent on the configuration of the tank and the arrangement of its supports and attachments.

4.23.2 Shell thickness

4.23.2.1 The shell thickness is to be as follows:

.1 For pressure vessels, the thickness calculated according to 4.23.2.4 is to be considered as a minimum thickness after forming, without any negative tolerance.

.2 For pressure vessels, the minimum thickness of shell and heads including corrosion allowance, after forming, is not to be less than 5 [mm] for carbon-manganese steels and nickel steels, 3

[mm] for austenitic steels or 7 [mm] for aluminium alloys.

.3 The welded joint efficiency factor to be used in the calculation according to 4.23.2.4 is to be 0.95 when the inspection and the non-destructive testing referred to in 6.5.6.5 are carried out. This figure may be increased up to 1 when account is taken of other considerations, such as the material used, type of joints, welding procedure and type of loading. For process pressure vessels, IRS may accept partial non-destructive examinations, but not less than those of 6.5.6.5, depending on such factors as the material used, the design temperature, the nil-ductility transition temperature of the material, as fabricated, and the type of joint and welding procedure, but in this case an efficiency factor of not more than 0.85 is to be adopted. For special materials, the above-mentioned factors are to be reduced, depending on the specified mechanical properties of the welded joint.

4.23.2.2 The design liquid pressure defined in 4.13.2 is to be taken into account in the internal pressure calculations.

4.23.2.3 The design external pressure P_e, used for verifying the buckling of the pressure vessels, is not to be less than that given by:

$$P_e = P_1 + P_2 + P_3 + P_4 \quad [MPa]$$

Where:

P_1 = setting value of vacuum relief valves. For vessels not fitted with vacuum relief valves, P_1 is to be specially considered, but, in general, is not to be taken as less than 0.025 [MPa];

P_2 = the set pressure of the pressure relief valves (PRVs) for completely closed spaces containing pressure vessels or parts of pressure vessels; elsewhere P_2=0;

P_3 = compressive actions in or on the shell due to the weight and contraction of thermal insulation, weight of shell including corrosion allowance and other miscellaneous external pressure loads to which the pressure vessel may be subjected. These include, but are not limited to, weight of domes, weight of towers and piping, effect of product in the partially filled condition, accelerations and hull deflection. In addition, the local effect of external or internal pressures or both shall be taken into account; and

P_4 = external pressure due to head of water for pressure vessels or part of pressure vessels on exposed decks; elsewhere P_4 = 0.

4.23.2.4 Scantlings based on internal pressure are to be calculated as follows: the thickness and form of pressure-containing parts of pressure vessels, under internal pressure, as defined in 4.13.2, including flanges, are to be determined. These calculations in all cases are to be based on accepted pressure vessel design theory. Openings in pressure-containing parts of pressure vessels are to be reinforced in accordance with recognized standards.

IR4.23.2.4 The ASME Boiler and Pressure vessel Code is considered as an acceptable recognized standard. Further, the scantlings of the Type C tank including reinforcements of the openings are not to be less than those provided by Part 4, Chapter 5 considering the tank to be a Class 1 pressure vessel.

4.23.2.5 Stress analysis in respect of static and dynamic loads are to be performed as follows:

.1 Pressure vessel scantlings are to be determined in accordance with 4.23.2.1 to 4.23.2.4 and 4.23.3.

.2 Calculations of the loads and stresses in way of the supports and the shell attachment of the support are to be made. Loads referred to in 4.12 to 4.15 are to be used, as applicable. Stresses in way of the supporting structures are to be to a recognized standard acceptable to IRS. In special cases, a fatigue analysis may be required by IRS.

.3 If required by IRS, secondary stresses and thermal stresses are to be specially considered.

4.23.3 *Ultimate design condition*

4.23.3.1 Plastic deformation

4.23.3.1.1 For type C independent tanks, the allowable stresses are not to exceed:

$$\sigma_m \leq f$$
$$\sigma_L \leq 1.5f$$
$$\sigma_b \leq 1.5f$$

$$\sigma_L + \sigma_b \leq 1.5f$$
$$\sigma_m + \sigma_b \leq 1.5f$$
$$\sigma_m + \sigma_b + \sigma_g \leq 3f$$
$$\sigma_L + \sigma_b + \sigma_g \leq 3f$$

Where:

σ_m = equivalent primary general membrane stress;

σ_L = equivalent primary local membrane stress;

σ_b = equivalent primary bending stress;

σ_g = equivalent secondary stress;

f = the lesser of (R_m/A) or (R_e/B);

with R_m and R_e as defined in 4.18.1.3. With regard to the stresses σ_m, σ_L, σ_b and σ_g, the definition of stress categories in 4.28.3 are referred. The values A and B are to be shown on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk and is to have at least the following minimum values:

	Nickel steels and carbon manganese steels	Austenitic steels	Aluminium alloys
A	3	3.5	4
B	1.5	1.5	1.5

IR 4.23.3.1 Permissible stresses in way of supports of type C Cargo tanks

IR 4.23.3.1.1 The circumferential stresses at supports are to be calculated by a procedure acceptable to IRS for a sufficient number of load cases.

IR 4.23.3.1.2 Permissible stresses in stiffening rings

.1 For horizontal cylindrical tanks made of C-Mn steel supported by saddles, the equivalent stress in the stiffening rings are not to exceed the following values if calculated using finite element method:

$$\sigma_e \leq \sigma_{all}$$

Where:

$$\sigma_{all} = min(0.57R_m\,;\, 0.85R_e)$$

$$\sigma_e = \sqrt{(\sigma_n + \sigma_b)^2 + 3\tau^2}$$

σ_e= von Mises equivalent stress in [N/mm^2]

σ_n= normal stress in [N/mm^2] in the circumferential direction of the stiffening ring

σ_b= bending stress in [N/mm^2] in the circumferential direction of the stiffening ring

τ= shear stress in [N/mm^2] in the stiffening ring

R_m and R_e as defined in 4.18.1.3

Equivalent stress value $\sigma_e$is to be calculated over the full extent of the stiffening ring by a procedure acceptable to IRS

IR 4.23.3.1.3 Assumptions to be made for the stiffening rings

.1 The stiffening ring is to be considered as a circumferential beam formed by web, face plate, doubler plate, if any, and associated shell plating. The effective width of the associated plating is to be taken as:

.1 For cylindrical shells:
an effective width [mm] not greater than 0.78 $\sqrt{rt}$ on each side of the web. A doubler plate, if any, may be included within that distance.

where:

r = mean radius of the cylindrical shell [mm]

t = shell thickness [mm]

.2 For longitudinal bulkheads (in the case of lobe tanks):

the effective width is to be determined according to established standards. A value of 20 t_b on each side of the web may be taken as a guidance value.

where:

t_b = bulkhead thickness (mm).

.2 The stiffening ring is to be loaded with circumferential forces, on each side of the ring, due to the shear stress, determined by the bi-dimensional shear flow theory from the shear force of the tank.

IR 4.23.3.1.4 For calculation of reaction forces at the supports, the following factors are to be taken into account:

.1 Elasticity of support material (intermediate layer of wood or similar material).

.2 Change in contact surface between tank and support, and of the relevant reactions, due to:

– thermal shrinkage of tank.

– elastic deformations of tank and support material.

The final distribution of the reaction forces at the supports should not show any tensile forces.

IR 4.23.3.1.5 The buckling strength of the stiffening rings is to be examined.

4.23.3.2 Buckling criteria is to be as follows: the thickness and form of pressure vessels subject to external pressure and other loads causing compressive stresses are to be based on calculations using accepted pressure vessel buckling theory and are to adequately account for the difference in theoretical and actual buckling stress as a result of plate edge misalignment, ovality and deviation from true circular form over a specified arc or chord length.

4.23.4 *Fatigue design condition*

For large type C independent tanks, where the cargo at atmospheric pressure is below -55°C, IRS may require additional verification to check their compliance with 4.23.1.1 regarding static and dynamic stress.

4.23.5 Accident design condition

4.23.5.1 The tanks and the tank supporting structures are to be designed for the accidental loads and design conditions specified in 4.3.4.3 and 4.15, as applicable.

4.23.5.2 When subjected to the accidental loads specified in 4.15, the stress is to comply with the acceptance criteria specified in 4.23.3.1, modified as appropriate taking into account their lower probability of occurrence.

4.23.6 Testing

4.23.6.1 Each pressure vessel is to be subjected to a hydrostatic test at a pressure measured at the top of the tanks, of not less than 1.5 P_o. In no case during the pressure test shall the calculated primary membrane stress at any point exceed 90% of the yield stress of the material. To ensure that this condition is satisfied where calculations indicate that this stress will exceed 0.75 times the yield strength, the prototype test is to be monitored by the use of strain gauges or other suitable equipment in pressure vessels other than simple cylindrical and spherical pressure vessels.

4.23.6.2 The temperature of the water used for the test is to be at least 30 [°C] above the nil-ductility transition temperature of the material, as fabricated.

4.23.6.3 The pressure is to be held for 2 h per 25 mm of thickness, but in no case less than 2 h.

4.23.6.4 Where necessary for cargo pressure vessels, a hydropneumatic test may be carried out under the conditions prescribed in 4.23.6.1 to 4.23.6.3.

4.23.6.5 Special consideration may be given to the testing of tanks in which higher allowable stresses are used, depending on service temperature. However, the requirements of 4.23.6.1 are to be fully complied with.

4.23.6.6 After completion and assembly, each pressure vessel and its related fittings are to be subjected to an adequate tightness test which may be performed in combination with the pressure testing referred to in 4.23.6.1.

4.23.6.7 Pneumatic testing of pressure vessels other than cargo tanks are only to be considered on an individual case basis. Such testing is only to be permitted for those vessels designed or supported such that they cannot be safely filled with water, or for those vessels that cannot be dried and are to be used in a service

where traces of the testing medium cannot be tolerated.

4.23.7 Marking

The required marking of the pressure vessel is to be achieved by a method that does not cause unacceptable local stress raisers.

4.24 Membrane tanks

4.24.1 Design basis

4.24.1.1 The design basis for membrane containment systems is that thermal and other expansion or contraction is compensated for without undue risk of losing the tightness of the membrane.

4.24.1.2 A systematic approach based on analysis and testing is to be used to demonstrate that the system will provide its intended function in consideration of the events identified in service as specified in 4.24.2.1.

4.24.1.3 If the cargo temperature at atmospheric pressure is below -10°C, a complete secondary barrier shall be provided as required in 4.5. The secondary barrier is to be designed according to 4.6.

4.24.1.4 The design vapour pressure P_o is normally not to exceed 0.025 [MPa]. If the hull scantlings are increased accordingly and consideration is given, where appropriate, to the strength of the supporting thermal insulation, P_o may be increased to a higher value, but less than 0.07 [MPa].

4.24.1.5 The definition of membrane tanks does not exclude designs such as those in which non-metallic membranes are used or where membranes are included or incorporated into the thermal insulation.

4.24.1.6 The thickness of the membranes is normally not to exceed 10 [mm].

4.24.1.7 The circulation of inert gas throughout the primary insulation space and the secondary insulation space, in accordance with 9.2.1, is to be sufficient to allow for effective means of gas detection.

IR4.24.1.8 IRS Guidelines on *Hull Design and Construction including Containment System for Liquefied Natural Gas Carriers – Membrane Type Tanks* may also be referred for design of membrane type tanks.

4.24.2 Design considerations

4.24.2.1 Potential incidents that could lead to loss of fluid tightness over the life of the membranes is to be evaluated. These include, but are not limited to:

.1 Ultimate design events:

.1 tensile failure of membranes;

.2 compressive collapse of thermal insulation;

.3 thermal ageing;

.4 loss of attachment between thermal insulation and hull structure;

.5 loss of attachment of membranes to thermal insulation system;

.6 structural integrity of internal structures and their supporting structures; and

.7 failure of the supporting hull structure.

.2 Fatigue design events:

.1 fatigue of membranes including joints and attachments to hull structure;

.2 fatigue cracking of thermal insulation;

.3 fatigue of internal structures and their supporting structures; and

.4 fatigue cracking of inner hull leading to ballast water ingress.

.3 Accident design events:

.1 accidental mechanical damage (such as dropped objects inside the tank while in service);

.2 accidental overpressurization of thermal insulation spaces;

.3 accidental vacuum in the tank; and

.4 water ingress through the inner hull structure.

Designs where a single internal event could cause simultaneous or cascading failure of both membranes are unacceptable.

4.24.2.2 The necessary physical properties (mechanical, thermal, chemical, etc.) of the materials used in the construction of the cargo containment system are to be established during the design development in accordance with 4.24.1.2.

4.24.3 Loads and load combinations

Particular consideration is to be given to the possible loss of tank integrity due to either an overpressure in the interbarrier space, a possible vacuum in the cargo tank, the sloshing effects, hull vibration effects, or any combination of these events.

4.24.4 Structural analyses

4.24.4.1 Structural analyses and/or testing for the purpose of determining the ultimate strength and fatigue assessments of the cargo containment and associated structures, e.g. structures as defined in 4.9, are to be performed. The structural analysis is to provide the data required to assess each failure mode that has been identified as critical for the cargo containment system.

4.24.4.2 Structural analyses of the hull is to take into account the internal pressure as indicated in 4.13.2. Special attention is to be paid to deflections of the hull and their compatibility with the membrane and associated thermal insulation.

4.24.4.3 The analyses referred to in 4.24.4.1 and 4.24.4.2 are to be based on the particular motions, accelerations and response of ships and cargo containment systems.

4.24.5 Ultimate design condition

4.24.5.1 The structural resistance of every critical component, subsystem or assembly is to be established, in accordance with 4.24.1.2, for in-service conditions.

4.24.5.2 The choice of strength acceptance criteria for the failure modes of the cargo containment system, its attachments to the hull structure and internal tank structures, are to reflect the consequences associated with the considered mode of failure.

4.24.5.3 The inner hull scantlings are to meet the requirements for deep tanks, taking into account the internal pressure as indicated in 4.13.2 and the specified appropriate requirements for sloshing load as defined in 4.14.3.

4.24.6 Fatigue design condition

4.24.6.1 Fatigue analysis is to be carried out for structures inside the tank, i.e. pump towers, and for parts of membrane and pump tower attachments, where failure development cannot be reliably detected by continuous monitoring.

4.24.6.2 The fatigue calculations are to be carried out in accordance with 4.18.2, with relevant requirements depending on:

.1 the significance of the structural components with respect to structural integrity; and

.2 availability for inspection.

4.24.6.3 For structural elements for which it can be demonstrated by tests and/or analyses that a crack will not develop to cause simultaneous or cascading failure of both membranes, C_w is to be less than or equal to 0.5.

4.24.6.4 Structural elements subject to periodic inspection, and where an unattended fatigue crack can develop to cause simultaneous or cascading failure of both membranes, are to satisfy the fatigue and fracture mechanics requirements stated in 4.18.2.8.

4.24.6.5 Structural element not accessible for in-service inspection, and where a fatigue crack can develop without warning to cause simultaneous or cascading failure of both membranes, are to satisfy the fatigue and fracture mechanics requirements stated in 4.18.2.9.

4.24.7 Accident design condition

4.24.7.1 The containment system and the supporting hull structure are to be designed for the accidental loads specified in 4.15. These loads need not be combined with each other or with environmental loads.

4.24.7.2 Additional relevant accident scenarios are to be determined based on a risk analysis. Particular attention is to be paid to securing devices inside tanks.

4.24.8 Design development testing

4.24.8.1 The design development testing required in 4.24.1.2 is to include a series of analytical and physical models of both the primary and secondary barriers, including corners and joints, tested to verify that they will withstand the expected combined strains due to static, dynamic and thermal loads. This will culminate in the construction of a prototype-scaled model of the complete cargo containment system. Testing conditions considered in the analytical and physical models are to represent the most extreme service conditions the cargo containment system will be likely to encounter over its life. Proposed acceptance criteria for periodic testing of secondary barriers required in 4.6.2 may be based on the results of testing carried out on the prototype-scaled model.

4.24.8.2 The fatigue performance of the membrane materials and representative welded or bonded joints in the membranes are to be determined by tests. The ultimate strength and fatigue performance of arrangements for securing the thermal insulation system to the hull structure are to be determined by analyses or tests.

4.24.9 Testing

4.24.9.1 In ships fitted with membrane cargo containment systems, all tanks and other spaces that may normally contain liquid and are adjacent to the hull structure supporting the membrane, is to be hydrostatically tested.

4.24.9.2 All hold structures supporting the membrane are to be tested for tightness before installation of the cargo containment system.

4.24.9.3 Pipe tunnels and other compartments that do not normally contain liquid need not be hydrostatically tested.

4.25 Integral tanks

4.25.1 Design basis

Integral tanks that form a structural part of the hull and are affected by the loads that stress the adjacent hull structure are to comply with the following:

.1 the design vapour pressure P_o as defined in 4.1.2 is normally not to exceed 0.025 [MPa]. If the hull scantlings are increased accordingly, P_o may be increased to a higher value, but less than 0.07 [MPa];

.2 integral tanks may be used for products, provided the boiling point of the cargo is not below -10°C. A lower temperature may be accepted by IRS subject to special consideration, but in such cases a complete secondary barrier shall be provided; and

.3 products required by sec 19 to be carried in type 1G ships are not to be carried in integral tanks.

4.25.2 Structural analysis

The structural analysis of integral tanks is to be in accordance with recognized standards.

4.25.3 Ultimate design condition

4.25.3.1 The tank boundary scantlings are to meet the requirements for deep tanks, taking into account the internal pressure as indicated in 4.13.2.

4.25.3.2 For integral tanks, allowable stresses are normally to be those given for hull structure in the requirements of the IRS.

4.25.4 Accident design condition

4.25.4.1 The tanks and the tank supports are to be designed for the accidental loads specified in 4.3.4.3 and 4.15, as relevant.

4.25.4.2 When subjected to the accidental loads specified in 4.15, the stress is to comply with the acceptance criteria specified in 4.25.3, modified as appropriate, taking into account their lower probability of occurrence.

4.25.5 Testing

All integral tanks are to be hydrostatically or hydropneumatically tested. The test is to be performed so that the stresses approximate, as far as practicable, to the design stresses and that the pressure at the top of the tank corresponds at least to the MARVS.

4.26 Semi-membrane tanks

4.26.1 Design basis

4.26.1.1 Semi-membrane tanks are non-self-supporting tanks when in the loaded condition and consist of a layer, parts of which are supported through thermal insulation by the adjacent hull structure, whereas the rounded parts of this layer connecting the above-mentioned supported parts are designed also to accommodate the thermal and other expansion or contraction.

4.26.1.2 The design vapour pressure P_o is normally not to exceed 0.025 [MPa]. If the hull scantlings are increased accordingly, and consideration is given, where appropriate, to the strength of the supporting thermal insulation, P_o may be increased to a higher value, but less than 0.07 [MPa].

4.26.1.3 For semi-membrane tanks the relevant requirements in this section for independent tanks or for membrane tanks are to be applied as appropriate.

4.26.1.4 In the case of semi-membrane tanks that comply in all respects with the requirements applicable to type B independent tanks, except for the manner of support, the Administration/ IRS may, after special consideration, accept a partial secondary barrier.

Part F
Cargo containment systems of novel configuration

4.27 Limit state design for novel concepts

4.27.1 Cargo containment systems that are of a novel configuration that cannot be designed using sections 4.21 to 4.26 are to be designed using this section and parts A and B of this section, and also parts C and D, as applicable. Cargo containment system design according to this section shall be based on the principles of limit state design which is an approach to structural design that can be applied to established design solutions as well as novel designs. This more generic approach maintains a level of safety similar to that achieved for known containment systems as designed using 4.21 to 4.26.

4.27.2.1 The limit state design is a systematic approach where each structural element is evaluated with respect to possible failure modes related to the design conditions identified in 4.3.4. A limit state can be defined as a condition beyond which the structure, or part of a structure, no longer satisfies the requirements.

4.27.2.2 For each failure mode, one or more limit states may be relevant. By consideration of all relevant limit states, the limit load for the structural element is found as the minimum limit load resulting from all the relevant limit states. The limit states are divided into the three following categories:

.1 Ultimate limit states (ULS), which correspond to the maximum load-carrying capacity or, in some cases, to the maximum applicable strain or deformation; under intact (undamaged) conditions.

.2 Fatigue limit states (FLS), which correspond to degradation due to the effect of time varying (cyclic) loading.

.3 Accident limit states (ALS), which concern the ability of the structure to resist accidental situations.

4.27.3 The procedure and relevant design parameters of the limit state design are to comply with the Standards for the Use of limit state methodologies in the design of cargo containment systems of novel configuration (LSD Standard), as set out in appendix 5 of IGC Code.

Part G
Guidance

4.28 Guidance notes for Sec 4

4.28.1 Guidance to detailed calculation of internal pressure for static design purpose

4.28.1.1 This section provides guidance for the calculation of the associated dynamic liquid pressure for the purpose of static design calculations. This pressure may be used for determining the internal pressure referred to in 4.13.2.4, where:

.1 (P_{gd})max is the associated liquid pressure determined using the maximum design accelerations.

.2 $(P_{gd}$ site)max is the associated liquid pressure determined using site specific accelerations.

.3 P_{eq} should be the greater of P_{eq1} and P_{eq2} calculated as follows:

$$P_{eq1} = P_o + (P_{gd})max \quad [MPa]$$

$$P_{eq2} = P_h + (P_{gd}site)max \quad [MPa]$$

4.28.1.2 The internal liquid pressures are those created by the resulting acceleration of the centre of gravity of the cargo due to the motions of the ship referred to in 4.14.1. The value of internal liquid pressure P_{gd} resulting from combined effects of gravity and dynamic accelerations should be calculated as follows:

$$P_{gd} = \alpha_\beta Z_\beta \frac{\rho}{1.02 \times 10^5} \quad [MPa]$$

Where:

α_β = dimensionless acceleration (i.e. relative to the acceleration of gravity), resulting from gravitational and dynamic loads, in an arbitrary direction β (see Fig.4.1).

For large tanks, an acceleration ellipsoid taking account of transverse vertical and longitudinal accelerations, should be used.

Z_β = largest liquid height (m) above the point where the pressure is to be determined measured from the tank shell in the *β* direction (see Fig.4.2).

Tank domes considered to be part of the accepted total tank volume shall be taken into account when determining Z_β, unless the total volume of tank domes V_d does not exceed the following value:

$$V_d = V_t\left(\frac{100 - FL}{FL}\right)$$

With:

V_t = tank volume without any domes; and

FL = filling limit according to Sec 15

ρ = maximum cargo density [kg/m^3] to the design temperature.

The direction that gives the maximum value (P_{gd})max or $(P_{gd}$site$)$max should be considered. The above formula applies only to full tanks.

4.28.1.3 Equivalent calculation procedures may be applied.

4.28.2 Guidance formulae for acceleration components

4.28.2.1 The following formulae are given as guidance for the components of acceleration due to ship's motions corresponding to a probability level of 10^{-8} in the North Atlantic and apply to ships with a length exceeding 50 m and at or near their service speed:

- Vertical acceleration, as defined in 4.14.1:

$$a_z = \pm a_0 \sqrt{1 + \left(5.3 - \frac{45}{L_0}\right)^2 \left(\frac{x}{L_0} + 0.05\right)^2 \left(\frac{0.6}{C_B}\right)^{1.5} + \left(\frac{0.6yK^{1.5}}{B}\right)^2}$$

- Transverse acceleration, as defined in 4.14.1:

$$a_y = \pm a_0 \sqrt{0.6 + 2.5\left(\frac{x}{L_0} + 0.05\right)^2 + K\left(1 + 0.6K\frac{z}{B}\right)^2}$$

- Longitudinal acceleration, as defined in 4.14.1:

$$a_y = \pm a_0 \sqrt{0.06 + A^2 - 0.25A}$$

Where:

$$a_0 = 0.2\frac{V}{\sqrt{L_0}} + \frac{34 - \left(\frac{600}{L_0}\right)}{L_0}$$

L_0 = length of the ship for determination of scantlings as defined in recognized standards (m);

C_B = block coefficient;

B = greatest moulded breadth of the ship (m);

x = longitudinal distance (m) from amidships to the centre of gravity of the tank with contents; x is positive forward of amidships, negative aft of amidships;

y = transverse distance (m) from centreline to the centre of gravity of the tank with contents;

z = vertical distance (m) from the ship's actual waterline to the centre of gravity of tank with contents; z is positive above and negative below the waterline;

K = 1 in general. For particular loading conditions and hull forms, determination of K according to the following formula may be necessary:

$K = 13GM/B$, where $K \geq 1$ and GM = metacentric height (m);

$$A = \left(0.7 - \frac{L_0}{1200} + 5\frac{z}{L_0}\right)\left(\frac{0.6}{C_B}\right)$$; and

V = service speed (knots);

a_x, a_y, a_z = maximum dimensionless accelerations (i.e. relative to the acceleration of gravity) in the respective directions. They are considered as acting separately for calculation purposes, and a_z does not include the component due to the static weight, a_y includes the component due to the static weight in the transverse direction due to rolling and a_x includes the component due to the static weight in the longitudinal direction due to pitching. The accelerations derived from the above formulae are applicable only to ships at or near their service speed, not while at anchor or otherwise near stationary in exposed locations.

4.28.3 Stress categories

4.28.3.1 For the purpose of stress evaluation, stress categories are defined in this section as follows.

4.28.3.2 Normal stress is the component of stress normal to the plane of reference.

4.28.3.3 Membrane stress is the component of normal stress that is uniformly distributed and equal to the average value of the stress across the thickness of the section under consideration.

4.28.3.4 Bending stress is the variable stress across the thickness of the section under consideration, after the subtraction of the membrane stress.

4.28.3.5 Shear stress is the component of the stress acting in the plane of reference.

4.28.3.6 Primary stress is a stress produced by the imposed loading, which is necessary to balance the external forces and moments. The basic characteristic of a primary stress is that it is not self-limiting. Primary stresses that considerably exceed the yield strength will result in failure or at least in gross deformations.

4.28.3.7 Primary general membrane stress is a primary membrane stress that is so distributed in the structure that no redistribution of load occurs as a result of yielding.

4.28.3.8 Primary local membrane stress arises where a membrane stress produced by pressure or other mechanical loading and associated with a primary or a discontinuity

effect produces excessive distortion in the transfer of loads for other portions of the structure. Such a stress is classified as a primary local membrane stress, although it has some characteristics of a secondary stress. A stress region may be considered as local, if:

$$S_1 \leq 0.5\sqrt{Rt}$$

and

$$S_2 \geq 2.5\sqrt{Rt}$$

Where:

S_1 = distance in the meridional direction over which the equivalent stress exceeds 1.1f;

S_2 = distance in the meridional direction to another region where the limits for primary general membrane stress are exceeded;

R = mean radius of the vessel;

t = wall thickness of the vessel at the location where the primary general membrane stress limit is exceeded; and

f = allowable primary general membrane stress.

4.28.3.9 Secondary stress is a normal stress or shear stress developed by constraints of adjacent parts or by self-constraint of a structure. The basic characteristic of a secondary stress is that it is self-limiting. Local yielding and minor distortions can satisfy the conditions that cause the stress to occur.

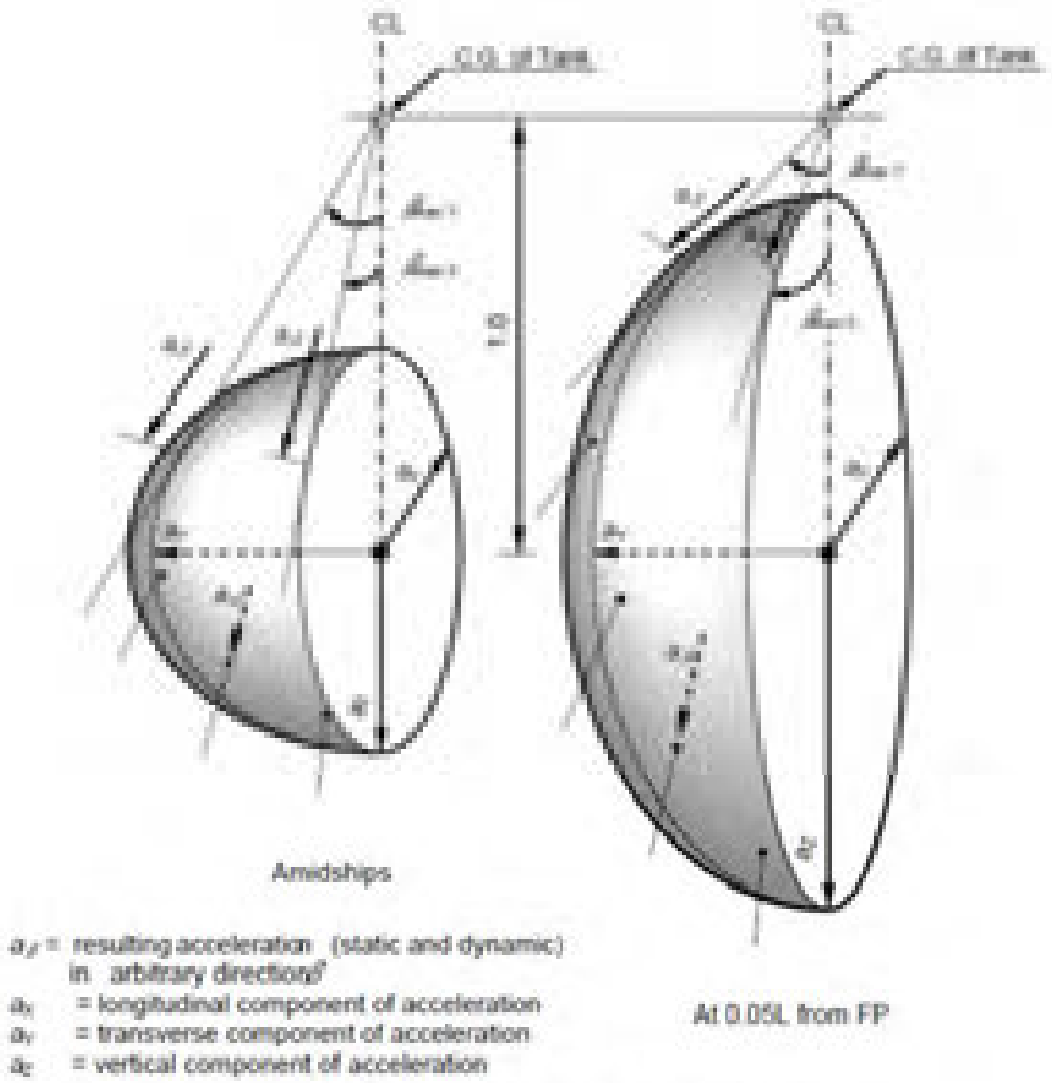


Figure 4.1 – Acceleration ellipsoid

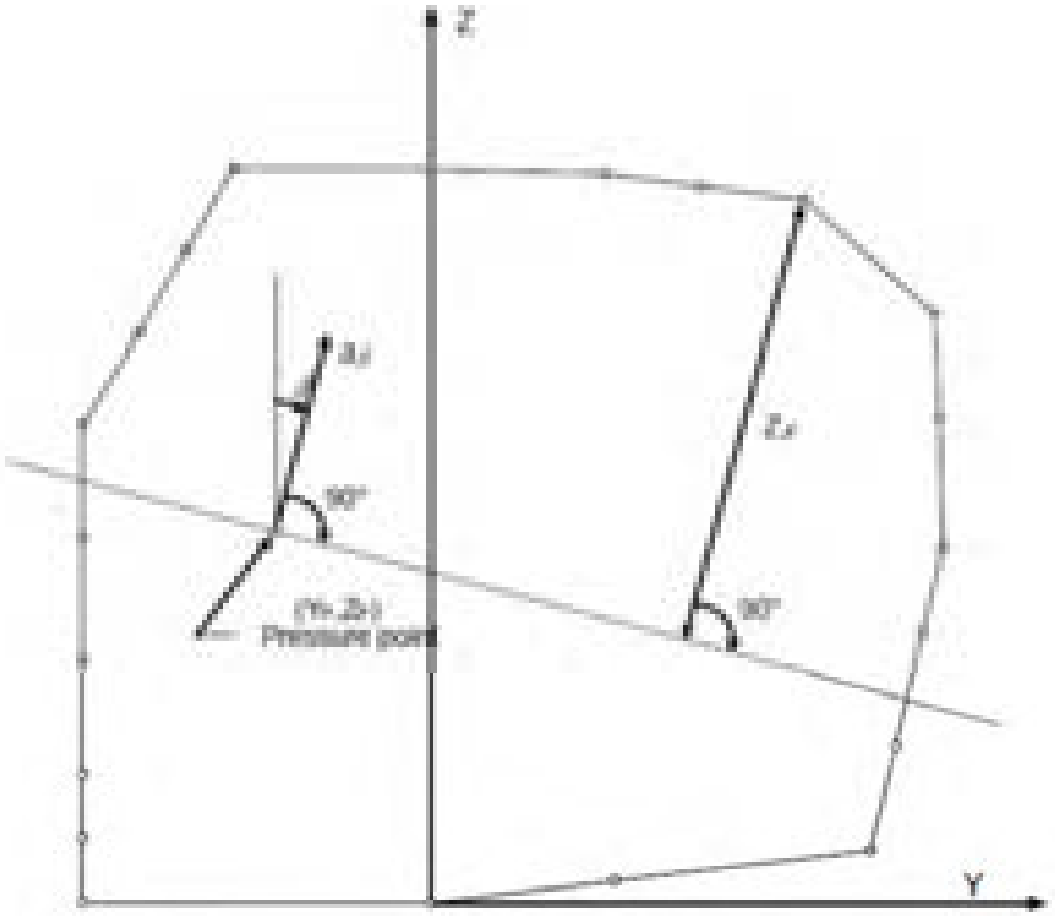


Figure 4.2 – Determination of internal pressure heads

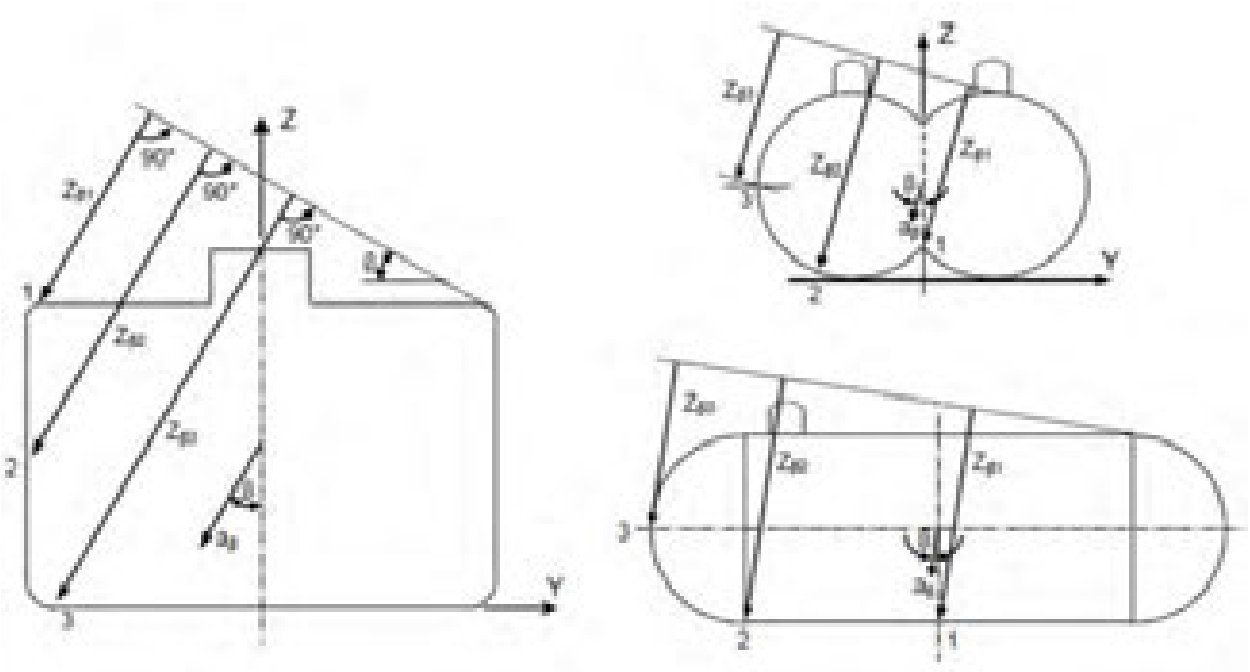


Figure 4.3 – Determination of liquid height Z_p for points 1, 2 and 3

Section 5

Process Pressure Vessels and Liquid, Vapour and Pressure Piping Systems

5.0 Goal

To ensure the safe handling of all cargo and process liquid and vapour, under all operating conditions, to minimize the risk to the ship, crew and to the environment, having regard to the nature of the products involved. This will:

.1 ensure the integrity of process pressure vessels, piping systems and cargo hoses;

.2 prevent the uncontrolled transfer of cargo;

.3 ensure reliable means to fill and empty the containment systems; and

.4 prevent pressure or vacuum excursions of cargo containment systems, beyond design parameters, during cargo transfer operations.

5.1 General

IR 5.1.1 Process pressure vessels and piping systems are to comply with the relevant Chapters of Pt.4.

5.1.1 The requirements of this section apply to products and process piping, including vapour piping, gas fuel piping and vent lines of safety valves or similar piping. Auxiliary piping systems not containing cargo are exempt from the general requirements of this section.

5.1.2 The requirements for Type C independent tanks provided in Sec.4 may also apply to process pressure vessels. If so required, the term "pressure vessels" as used in Sec.4 cover both Type C independent tanks and process pressure vessels.

5.1.3 Process pressure vessels include surge tanks, heat exchangers and accumulators that store or treat liquid or vapour cargo.

5.2 System requirements

5.2.1 The cargo handling and cargo control systems are to be designed taking into account the following:

.1 prevention of an abnormal condition escalating to a release of liquid or vapour cargo;

.2 the safe collection and disposal of cargo fluids released;

.3 prevention of the formation of flammable mixtures;

.4 prevention of ignition of flammable liquids or gases and vapours released; and

.5 limiting the exposure of personnel to fire and other hazards.

5.2.2 Arrangements: general

5.2.2.1 Any piping system that may contain cargo liquid or vapour is to:

.1 be segregated from other piping systems, except where interconnections are required for cargo-related operations such as purging, gas-freeing or inerting. The requirements of 9.4.4 shall be taken into account with regard to preventing back-flow of cargo. In such cases, precautions are to be taken to ensure that cargo or cargo vapour cannot enter other piping systems through the interconnections;

.2 except as provided in sec 16, not pass through any accommodation space, service space or control station or through a machinery space other than a cargo machinery space;

.3 be connected to the cargo containment system directly from the weather decks except where pipes installed in a vertical trunkway or equivalent are used to traverse void spaces above a cargo containment system and except where pipes for drainage, venting or purging traverse cofferdams;

.4 be located in the cargo area above the weather deck except for bow or stern loading and unloading arrangements in accordance with 3.8, emergency cargo jettisoning piping systems in accordance with 5.3.1, turret compartment systems in accordance with 5.3.3 and except in accordance with sec 16; and

.5 be located inboard of the transverse tank location requirements of 2.4.1, except for athwartship shore connection piping not subject to internal pressure at sea or emergency cargo jettisoning piping systems.

5.2.2.2 Suitable means is to be provided to relieve the pressure and remove liquid cargo from loading and discharging crossover headers; likewise, any piping between the outermost manifold valves and loading arms or cargo hoses to the cargo tanks, or other suitable location, prior to disconnection.

5.2.2.3 Piping systems carrying fluids for direct heating or cooling of cargo are not to be led outside the cargo area unless a suitable means is provided to prevent or detect the migration of cargo vapour outside the cargo area (see 13.6.2.6).

5.2.2.4 Relief valves discharging liquid cargo from the piping system are to discharge into the cargo tanks. Alternatively, they may discharge to the cargo vent mast, if means are provided to detect and dispose of any liquid cargo that may flow into the vent system. Where required to prevent overpressure in downstream piping, relief valves on cargo pumps are to discharge to the pump suction.

5.3 Arrangements for cargo piping outside the cargo area

5.3.1 Emergency cargo jettisoning

If fitted, an emergency cargo jettisoning piping system is to comply with 5.2.2, as appropriate, and may be led aft, external to accommodation spaces, service spaces or control stations or machinery spaces, but is not to pass through them. If an emergency cargo jettisoning piping system is permanently installed, a suitable means of isolating the piping system from the cargo piping is to be provided within the cargo area.

5.3.2 Bow and stern loading arrangements

5.3.2.1 Subject to the requirements of 3.8, this section and 5.10.1, cargo piping may be arranged to permit bow or stern loading and unloading.

5.3.2.2 Arrangements are to be made to allow such piping to be purged and gas-freed after use. When not in use, the spool pieces are to be removed and the pipe ends blank-flanged. The vent pipes connected with the purge is to be located in the cargo area.

5.3.3 Turret compartment transfer systems

For the transfer of liquid or vapour cargo through an internal turret arrangement located outside the cargo area, the piping serving this purpose is to comply with 5.2.2, as applicable, 5.10.2 and the following:

.1 piping are to be located above the weather deck, except for the connection to the turret;

.2 portable arrangements are not to be permitted; and

.3 arrangements are to be made to allow such piping to be purged and gas-freed after use. When not in use, the spool pieces for isolation from the cargo piping are to be removed and the pipe ends blank-flanged. The vent pipes connected with the purge are to be located in the cargo area.

5.3.4 Gas fuel piping systems

Gas fuel piping in machinery spaces are to comply with all applicable clauses of this section in addition to the requirements of sec 16.

5.4 Design pressure

5.4.1 The design pressure P_o, used to determine minimum scantlings of piping and piping system components, are not to be less than the maximum gauge pressure to which the system may be subjected in service. The minimum design pressure used is not to be less than 1 [MPa] gauge, except for open-ended lines or pressure relief valve discharge lines, where it is not to be less than the lower of 0.5 [MPa] gauge, or 10 times the relief valve set pressure.

5.4.2 The greater of the following design conditions are to be used for piping, piping systems and components, based on the cargoes being carried:

.1 for vapour piping systems or components that may be separated from their relief valves and which may contain some liquid, the saturated vapour pressure at a design temperature of 45 [°C]. Higher or lower values may be used (see 4.13.2.2); or

.2 for systems or components that may be separated from their relief valves and which contain only vapour at all times, the superheated vapour pressure at 45°C. Higher or lower values may be used (see 4.13.2.2), assuming an initial condition of saturated vapour in the system at the system operating pressure and temperature; or

.3 the MARVS of the cargo tanks and cargo processing systems; or

.4 the pressure setting of the associated pump or compressor discharge relief valve; or

.5 the maximum total discharge or loading head of the cargo piping system considering all possible pumping arrangements or the relief valve setting on a pipeline system.

5.4.3 Those parts of the liquid piping systems that may be subjected to surge pressures are to be designed to withstand this pressure.

5.4.4 The design pressure of the outer pipe or duct of gas fuel systems is not to be less than the maximum working pressure of the inner gas pipe. Alternatively, for gas fuel piping systems with a working pressure greater than 1 [MPa], the design pressure of the outer duct is not to be less than the maximum built-up pressure arising in the annular space considering the local instantaneous peak pressure in way of any rupture and the ventilation arrangements.

IR 5.4.4.1 Refer IR 5.13.2.4.1

IR 5.4.4.2 The expression "design pressure of the outer pipe or duct" in 5.4.4 above is either of the following:

.1 the maximum pressure that can act on the outer pipe or equipment enclosure after the inner pipe rupture as documented by suitable calculations taking into account the venting arrangements; or

.2 for gas fuel systems with inner pipe working pressure greater than 1 MPa, the "maximum built-up pressure arising in the annular space", after the inner pipe rupture, which is to be calculated in accordance with Pt.5, Ch.35, 9.8.2 (para 9.8.2 of the IGF Code as adopted by MSC.391(95)).

5.5 Cargo system valve requirements

5.5.1.1 Every cargo tank and piping system is to be fitted with manually operated valves for isolation purposes as specified in this sub-section.

5.5.1.2 In addition, remotely operated valves are also to be fitted, as appropriate, as part of the emergency shutdown (ESD) system the purpose of which is to stop cargo flow or leakage in the event of an emergency when cargo liquid or vapour transfer is in progress. The ESD system is intended to return the cargo system to a safe static condition so that any remedial action can be taken. Due regard is to be given in the design of the ESD system to avoid the generation of surge pressures within the cargo transfer pipework. The equipment to be shut down on ESD activation includes manifold valves during loading or discharge, any pump or compressor, etc., transferring cargo internally or externally (e.g. to shore or another ship/barge) and cargo tank valves, if the MARVS exceeds 0.07 [MPa].

5.5.2 Cargo tank connections

5.5.2.1 All liquid and vapour connections, except for safety relief valves and liquid level gauging devices, are to have shutoff valves located as close to the tank as practicable. These valves are to provide full closure and are to be capable of local manual operation. They may also be capable of remote operation.

5.5.2.2 For cargo tanks with a MARVS exceeding 0.07 [MPa] gauge, the above connections are also to be equipped with remotely controlled ESD valves. These valves are to be located as close to the tank as practicable. A single valve may be substituted for the two separate valves, provided the valve complies with the requirements of 18.10.2 and provides full closure of the line.

5.5.3 Cargo manifold connections

5.5.3.1 One remotely controlled ESD valve is to be provided at each cargo transfer connection in use to stop liquid and vapour transfer to or from the ship. Transfer connections not in use are to be isolated with suitable blank flanges.

5.5.3.2 If the cargo tank MARVS exceeds 0.07 [MPa], an additional manual valve is to be provided for each transfer connection in use, and may be inboard or outboard of the ESD valve to suit the ship's design.

5.5.4 Excess flow valves may be used in lieu of ESD valves, if the diameter of the protected pipe does not exceed 50 [mm]. Excess flow valves are to close automatically at the rated closing flow of vapour or liquid as specified by the manufacturer. The piping including fittings, valves and appurtenances protected by an excess flow valve is to have a capacity greater than the rated closing flow of the excess flow valve. Excess flow valves may be designed with a bypass not exceeding the area of a 1 [mm] diameter circular opening to allow equalization of pressure after a shutdown activation.

5.5.5 Cargo tank connections for gauging or measuring devices need not be equipped with excess flow valves or ESD valves, provided that the devices are constructed so that the outward flow of tank contents cannot exceed that passed by a 1.5 [mm] diameter circular hole.

5.5.6 All pipelines or components which may be isolated in a liquid full condition are to be protected with relief valves for thermal expansion and evaporation.

5.5.7 All pipelines or components which may be isolated automatically due to a fire with a liquid volume of more than 0.05 [m^3] entrapped are to be provided with PRVs sized for a fire condition.

5.6 Cargo transfer arrangements

5.6.1 Where cargo transfer is by means of cargo pumps that are not accessible for repair with the tanks in service, at least two separate means are to be provided to transfer cargo from each cargo tank, and the design is to be such that failure of one cargo pump or means of transfer will not prevent the cargo transfer by another pump or pumps, or other cargo transfer means.

5.6.2 The procedure for transfer of cargo by gas pressurization is to preclude lifting of the relief

valves during such transfer. Gas pressurization may be accepted as a means of transfer of cargo for those tanks where the design factor of safety is not reduced under the conditions prevailing during the cargo transfer operation. If the cargo tank relief valves or set pressure are changed for this purpose, as it is permitted in accordance with 8.2.7 and 8.2.8, the new set pressure is not to exceed P_h as is defined in 4.13.2.

5.6.3 Vapour return connections

Connections for vapour return to the shore installations are to be provided.

5.6.4 Cargo tank vent piping systems

The pressure relief system is to be connected to a vent piping system designed to minimize the possibility of cargo vapour accumulating on the decks, or entering accommodation spaces, service spaces, control stations and machinery spaces, or other spaces where it may create a dangerous condition.

5.6.5 Cargo sampling connections

5.6.5.1 Connections to cargo piping systems for taking cargo liquid samples are to be clearly marked and are to be designed to minimize the release of cargo vapours. For vessels permitted to carry toxic products, the sampling system is to be of a closed loop design to ensure that cargo liquid and vapour are not vented to atmosphere.

5.6.5.2 Liquid sampling systems are to be provided with two valves on the sample inlet. One of these valves is to be of the multi-turn type to avoid accidental opening, and is to be spaced far enough apart to ensure that they can isolate the line if there is blockage, by ice or hydrates for example.

5.6.5.3 On closed loop systems, the valves on the return pipe is also to comply with 5.6.5.2.

5.6.5.4 The connection to the sample container is to comply with recognized standards and be supported so as to be able to support the weight of a sample container. Threaded connections are to be tack-welded, or otherwise locked, to prevent them being unscrewed during the normal connection and disconnection of sample containers. The sample connection is to be fitted with a closure plug or flange to prevent any leakage when the connection is not in use.

5.6.5.5 Sample connections used only for vapour samples may be fitted with a single valve in accordance with 5.5, 5.8 and 5.13, and is also to be fitted with a closure plug or flange.

5.6.5.6 Sampling operations are to be undertaken as prescribed in 18.9.

IR 5.6.5 Requirements in 5.6.5 above are only applicable if such a sampling system is fitted on board. Connections used for control of atmosphere in cargo tanks during inerting or gassing up are not considered as cargo sampling connections.

5.6.6 Cargo filters

5.6.6.1 The cargo liquid and vapour systems are to be capable of being fitted with filters to protect against damage by extraneous objects. Such filters may be permanent or temporary, and the standards of filtration are to be appropriate to the risk of debris, etc., entering the cargo system. Means are to be provided to indicate that filters are becoming blocked, and to isolate, depressurize and clean the filters safely.

IR 5.6.6.1 Means to indicate that filters are becoming blocked and filter maintenance is required is to be provided for fixed in-line filter arrangement and portable filter installations where dedicated filter housing piping is provided. Where portable filters for fitting to manifold presentation flanges are used without dedicated filter housing, and these can be visually inspected after each loading and discharging operation, no additional arrangements for indicating blockage or facilitating drainage are required.

5.7 Installation requirements

5.7.1 Design for expansion and contraction

Provision is to be made to protect the piping, piping system and components and cargo tanks from excessive stresses due to thermal movement and from movements of the tank and hull structure. The preferred method outside the cargo tanks is by means of offsets, bends or loops, but multi-layer bellows may be used if offsets, bends or loops are not practicable.

5.7.2 Precautions against low temperature

Low temperature piping is to be thermally isolated from the adjacent hull structure, where

necessary, to prevent the temperature of the hull from falling below the design temperature of the hull material. Where liquid piping is dismantled regularly, or where liquid leakage may be anticipated, such as at shore connections and at pump seals, protection for the hull beneath is to be provided.

5.7.3 Water curtain

For cargo temperatures below -110 [°C], a water distribution system is to be fitted in way of the hull under the shore connections to provide a low-pressure water curtain for additional protection of the hull steel and the ship's side structure. This system is in addition to the requirements of 11.3.1.4, and is to be operated when cargo transfer is in progress.

5.7.4 Bonding

Where tanks or cargo piping and piping equipment are separated from the ship's structure by thermal isolation, provision is to be made for electrically bonding both the piping and the tanks. All gasketed pipe joints and hose connections are to be electrically bonded. Except where bonding straps are used, it is to be demonstrated that the electrical resistance of each joint or connection is less than 1MΩ.

5.8 Piping fabrication and joining details

5.8.1 General

The requirements of this sub-section apply to piping inside and outside the cargo tanks. Relaxation from these requirements may be accepted, in accordance with recognized standards for piping inside cargo tanks and open-ended piping.

5.8.2 Direct connections

The following direct connection of pipe lengths, without flanges, may be considered:

.1 butt-welded joints with complete penetration at the root may be used in all applications. For design temperatures colder than -10 [°C], butt welds shall be either double welded or equivalent to a double welded butt joint. This may be accomplished by use of a backing ring, consumable insert or inert gas backup on the first pass. For design pressures in excess of 1 [MPa] and design temperatures of -10 [°C] or colder, backing rings shall be removed;

.2 slip-on welded joints with sleeves and related welding, having dimensions in accordance with recognized standards, are only to be used for instrument lines and open-ended lines with an external diameter of 50 [mm] or less and design temperatures not colder than -55 [°C]; and

.3 screwed couplings complying with recognized standards are only to be used for accessory lines and instrumentation lines with external diameters of 25 [mm] or less.

5.8.3 Flanged connections

5.8.3.1 Flanges in flanged connections are to be of the welded neck, slip-on or socket welded type.

5.8.3.2 Flanges are to comply with recognized standards for their type, manufacture and test. For all piping, except open ended, the following restrictions apply:

.1 for design temperatures colder than -55 [°C], only welded-neck flanges are to be used; and

.2 for design temperatures colder than -10 [°C], slip-on flanges are not to be used in nominal sizes above 100 [mm] and socket welded flanges are not to be used in nominal sizes above 50 [mm].

5.8.4 Expansion joints

Where bellows and expansion joints are provided in accordance with 5.7.1, the following requirements apply:

.1 if necessary, bellows are to be protected against icing; and

.2 slip joints are not to be used except within the cargo tanks.

5.8.5 Other connections

Piping connections are to be joined in accordance with 5.8.2 to 5.8.4, but for other exceptional cases the Administration/ IRS may consider alternative arrangements.

5.9 Welding, post-weld heat treatment and non-destructive testing

5.9.1 General

Welding is to be carried out in accordance with 6.5.

5.9.2 Post-weld heat treatment

Post-weld heat treatment is to be required for all butt welds of pipes made with carbon, carbon-manganese and low alloy steels. IRS may waive the requirements for thermal stress relieving of pipes with wall thickness less than 10 [mm] in relation to the design temperature and pressure of the piping system concerned.

5.9.3 Non-destructive testing

In addition to normal controls before and during the welding, and to the visual inspection of the finished welds, as necessary for proving that the welding has been carried out correctly and according to the requirements of this paragraph, the following tests shall be required:

.1 100% radiographic or ultrasonic inspection of butt-welded joints for piping systems with design temperatures colder than -10 [°C], and with inside diameters of more than 75 [mm], or wall thicknesses greater than 10 [mm];

.2 when such butt-welded joints of piping sections are made by automatic welding procedures approved by IRS, then a progressive reduction in the extent of radiographic or ultrasonic inspection can be agreed, but in no case to less than 10% of each joint. If defects are revealed, the extent of examination is to be increased to 100% and is to include inspection of previously accepted welds. This approval can only be granted if well-documented quality assurance procedures and records are available to assess the ability of the manufacturer to produce satisfactory welds consistently; and

.3 for other butt-welded joints of pipes not covered by 5.9.3.1 and 5.9.3.2, spot radiographic or ultrasonic inspection or other non-destructive tests are to be carried out depending upon service, position and materials. In general, at least 10% of butt-welded joints of pipes are to be subjected to radiographic or ultrasonic inspection.

5.10 Installation requirements for cargo piping outside the cargo area

5.10.1 Bow and stern loading arrangements

The following requirements shall apply to cargo piping and related piping equipment located outside the cargo area:

.1 cargo piping and related piping equipment outside the cargo area are to have only welded connections. The piping outside the cargo area are to run on the weather decks and are to be at least 0.8 [m] inboard, except for athwartships shore connection piping. Such piping is to be clearly identified and fitted with a shutoff valve at its connection to the cargo piping system within the cargo area. At this location, it is also to be capable of being separated by means of a removable spool piece and blank flanges, when not in use; and

.2 the piping are to be full penetration butt-welded and subjected to full radiographic or ultrasonic inspection, regardless of pipe diameter and design temperature. Flange connections in the piping are only to be permitted within the cargo area and at the shore connection.

5.10.2 Turret compartment transfer systems

The following requirements shall apply to liquid and vapour cargo piping where it is run outside the cargo area:

.1 cargo piping and related piping equipment outside the cargo area are to have only welded connections; and

.2 the piping are to be full penetration butt-welded, and subjected to full radiographic or ultrasonic inspection, regardless of pipe diameter and design temperature. Flange connections in the piping are only to be permitted within the cargo area and at connections to cargo hoses and the turret connection.

5.10.3 Gas fuel piping

Gas fuel piping, as far as practicable, are to have welded joints. Those parts of the gas fuel piping that are not enclosed in a ventilated pipe or duct according to 16.4.3, and are on the weather decks outside the cargo area, are to have full penetration butt-welded joints and are to be subjected to full radiographic or ultrasonic inspection.

5.11 Piping system component requirements

5.11.1 Piping scantlings.

Piping systems are to be designed in accordance with recognized standards.

5.11.2.1 The following criteria are to be used for determining pipe wall thickness.

5.11.2.2 The wall thickness of pipes is not to be less than:

$$t = \frac{t_0 + b + c}{1 - \frac{a}{100}} \qquad [mm]$$

Where:

t_0 = theoretical thickness, determined by the following formula:

$$t_0 = \frac{P\ D}{2K\ e + P} \qquad [mm]$$

With:

P = design pressure [MPa] referred to in 5.4;

D = outside diameter [mm];

K = allowable stress [N/mm^2] referred to in 5.11.3;

e = efficiency factor equal to 1 for seamless pipes and for longitudinally or spirally welded pipes, that are considered equivalent to seamless pipes when non-destructive testing on welds is carried out in accordance with recognized standards. In other cases, an efficiency factor of less than 1, in accordance with recognized standards, may be required, depending on the manufacturing process;

b= allowance for bending [mm]. The value of b is to be chosen so that the calculated stress in the bend, due to internal pressure only, does not exceed the allowable stress. Where such justification is not given, b is to be:

$$b = \frac{D\ t_0}{2.5r} \qquad [mm]$$

With:

r = mean radius of the band [mm];

c = corrosion allowance [mm]. If corrosion or erosion is expected, the wall thickness of the piping is to be increased over that required by the other design requirements. This allowance is to be consistent with the expected life of the pricing; and

a = negative manufacturing tolerance for thickness (%).

5.11.2.3 The minimum wall thickness is to be in accordance with recognized standards.

5.11.2.4 Where necessary for mechanical strength to prevent damage, collapse, excessive sag or buckling of pipes due to superimposed loads, the wall thickness shall be increased over that required by 5.11.2.2 or, if this is impracticable or would cause excessive local stresses, these loads may be reduced, protected against or eliminated by other design methods. Such superimposed loads may be due to: supporting structures, ship deflections, liquid pressure surge during transfer operations, the weight of suspended valves, reaction to loading arm connections, or otherwise.

5.11.3 Allowable stress

5.11.3.1 For pipes, the allowable stress K referred to in the formula in 5.11.2 is the lower of the following values:

$$\frac{R_m}{A}\ \ or\ \ \frac{R_e}{B}$$

Where:

R_m = specified minimum tensile strength at room temperature [N/mm^2]; and

R_e = specified minimum yield stress at room temperature [N/mm^2]. If the stress-strain curve does not show a defined yield stress, the 0.2% proof stress applies.

The values of A and B are to be shown on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk required in 1.4.4, and have values of at least A = 2.7 and B = 1.8.

5.11.4 High-pressure gas fuel outer pipes or ducting scantlings

In fuel gas piping systems of design pressure greater than the critical pressure, the tangential membrane stress of a straight section of pipe or ducting shall not exceed the tensile strength divided by 1.5 (R_m /1.5) when subjected to the design pressure specified in 5.4. The pressure ratings of all other piping components are to reflect the same level of strength as straight pipes.

5.11.5 Stress analysis

When the design temperature is -110 [°C] or lower, a complete stress analysis, taking into account all the stresses due to the weight of pipes, including acceleration loads if significant, internal pressure, thermal contraction and loads induced by hog and sag of the ship for each branch of the piping system is to be submitted. For temperatures above -110 [°C], a stress analysis may be required by IRS in relation to such matters as the design or stiffness of the piping system and the choice of materials. In any case, consideration is to be given to thermal stresses even though calculations are not submitted. The analysis may be carried out according to a code of practice acceptable to IRS.

5.11.6 Flanges, valves and fittings

5.11.6.1 Flanges, valves and other fittings are to comply with recognized standards, taking into account the material selected and the design pressure defined in 5.4. For bellows expansion joints used in vapour service, a lower minimum design pressure may be accepted.

5.11.6.2 For flanges not complying with a recognized standard, the dimensions of flanges and related bolts are to be to the satisfaction of IRS.

5.11.6.3 All emergency shutdown valves are to be of the "fail-closed" type (see 5.13.1.1 and 18.10.2).

5.11.6.4 The design and installation of expansion bellows are to be in accordance with recognized standards and be fitted with means to prevent damage due to over-extension or compression.

5.11.7 Ship's cargo hoses

5.11.7.1 Liquid and vapour hoses used for cargo transfer are to be compatible with the cargo and suitable for the cargo temperature.

5.11.7.2 Hoses subject to tank pressure, or the discharge pressure of pumps or vapour compressors, are to be designed for a bursting pressure not less than five times the maximum pressure the hose will be subjected to during cargo transfer.

5.11.7.3 Each new type of cargo hose, complete with end-fittings, is to be prototype-tested at a normal ambient temperature, with 200 pressure cycles from zero to at least twice the specified maximum working pressure. After this cycle pressure test has been carried out, the prototype test is to demonstrate a bursting pressure of at least 5 times its specified maximum working pressure at the upper and lower extreme service temperature. Hoses used for prototype testing is not to be used for cargo service. Thereafter, before being placed in service, each new length of cargo hose produced is to be hydrostatically tested at ambient temperature to a pressure not less than 1.5 times its specified maximum working pressure, but not more than two fifths of its bursting pressure. The hose is to be stencilled, or otherwise marked, with the date of testing, its specified maximum working pressure and, if used in services other than ambient temperature services, its maximum and minimum service temperature, as applicable. The specified maximum working pressure is not to be less than 1 [MPa] gauge.

5.12 Materials

5.12.1 The choice and testing of materials used in piping systems are to comply with the requirements of Sec 6, taking into account the minimum design temperature. However, some relaxation may be permitted in the quality of material of open-ended vent piping, provided that the temperature of the cargo at the pressure relief valve setting is not lower than -55 [°C], and that no liquid discharge to the vent piping can occur. Similar relaxations may be permitted under the same temperature conditions to open-ended piping inside cargo tanks, excluding discharge piping and all piping inside membrane and semi-membrane tanks.

5.12.2 Materials having a melting point below 925 [°C] are not to be used for piping outside the cargo tanks except for short lengths of pipes attached to the cargo tanks, in which case fire-resisting insulation is to be provided.

5.12.3 Cargo piping insulation system

5.12.3.1 Cargo piping systems are to be provided with a thermal insulation system as required to minimize heat leak into the cargo during transfer operations and to protect personnel from direct contact with cold surfaces.

IR 5.12.3.1.1 The expression "*a thermal insulation system as required to minimize heat leak into the cargo during transfer operations*" means that properties of the piping insulation are to be taken into consideration when calculating the heat balance of the containment system and capacity of the pressure/temperature control system.

IR 5.12.3.1.2 The expression "*cargo piping systems shall be provided with a thermal insulation system as required ... to protect personnel from direct contact with cold surfaces*" means that surfaces of cargo piping systems with which personnel is likely to contact under normal conditions shall be protected by a thermal insulation, with the exception of the following examples:

- surfaces of cargo piping systems which are protected by physical screening measures to prevent such direct contact;

- surfaces of manual valves, having extended spindles that protect the operator from the cargo temperature;

- surfaces of cargo piping systems whose design temperature (to be determined from inner fluid temperature) is above minus 10 °C.

5.12.3.2 Where applicable, due to location or environmental conditions, insulation materials are to have suitable properties of resistance to fire and flame spread and is to be adequately protected against penetration of water vapour and mechanical damage.

5.12.4 Where the cargo piping system is of a material susceptible to stress corrosion cracking in the presence of a salt-laden atmosphere, adequate measures to avoid this occurring is to be taken by considering material selection, protection of exposure to salty water and/or readiness for inspection.

5.13 Testing requirements

5.13.1 Testing of piping components

5.13.1.1 Valves

5.13.1.1.1 Prototype testing

Each type of valve intended to be used at a working temperature below -55 [^{o}C] is to be subject to the following type tests:

.1 each size and type of valve is to be subjected to seat tightness testing over the full range of operating pressures for bi-directional flow and temperatures, at intervals, up to the rated design pressure of the valve. Allowable leakage rates are to be to the requirements of IRS. During the testing, satisfactory operation of the valve is to be verified;

.2 the flow or capacity is to be certified to a recognized standard for each size and type of valve;

IR .2 The expression "Each type of valve …… is to be certified to a recognized standard" means that:

IR .2.1 For pressure relief valves (PRVs) that are subject to 8.2.5, the flow or capacity are to be certified by IRS; and

IR .2.2 For other types of valves, the manufacturer is to certify the flow properties of the valves based on tests carried out according to recognized standards.

.3 pressurized components are to be pressure tested to at least 1.5 times the rated pressure; and

.4 for emergency shutdown valves, with materials having melting temperatures lower than 925 [°C], the type testing is to include a fire test to a standard acceptable to the Administration/ IRS.

IR.4 " *Emergency shutdown valves, with materials having melting temperatures lower than 925°C*" do not

include an emergency shutdown valve in which components made of materials having melting temperatures lower than 925°C do not contribute to the shell or seat tightness of the valve

IR.5 for valves other than safety valves, a seat and stem leakage test at a pressure equal 1.1 times the design pressure.

IR 5.13.1.1.2 Unit Production testing

All valves are to be tested at the plant of manufacturer in the presence of the IRS's surveyor. Testing is to include hydrostatic test of the valve body at a pressure equal to 1.5 times the design pressure for all valves, seat and stem leakage test at a pressure equal to 1.1 times the design pressure for valves other than safety valves. In addition, cryogenic testing consisting of valve operation and leakage verification for a minimum of 10 % of each type and size of valve for valves other than safety valves intended to be used at a working temperature below -55 [˚C]. The set pressure of safety valves is to be tested at ambient temperature. For valves used for isolation of instrumentation in piping not greater than 25 [mm], unit production testing need not be witnessed by the surveyor. Records of testing are to be available for review.
As an alternative to the above, if so requested by the relevant Manufacturer, the certification of a valve may be issued subject to the following:

- The valve has been approved as required by 5.13.1.1.1 for valves intended to be used at a working temperature below -55[˚C], and

- The manufacturer has a recognized quality system that has been assessed and certified by the IRS subject to periodic audits, and

- The quality control plan contains a provision to subject each valve to a hydrostatic test of the valve body at a pressure equal to 1.5 times the design pressure for all valves and seat and stem leakage test at a pressure equal to 1.1 times the design pressure for valves other than safety valves. The set pressure of safety valves is to be tested at ambient temperature. The manufacturer is to maintain records of such tests, and

- Cryogenic testing consisting of valve operation and leakage verification for a minimum of 10% of each type and size of valve for valves other than safety valves intended to be used at a working temperature below -55[˚C] in the presence of the IRS's representative.

5.13.1.2 Expansion bellows

The following type tests are to be performed on each type of expansion bellows intended for use on cargo piping outside the cargo tank and where required by IRS, on those installed within the cargo tanks:

.1 elements of the bellows, not pre-compressed, are to be pressure tested at not less than five times the design pressure without bursting. The duration of the test is not to be less than 5 [min];

.2 a pressure test is to be performed on a type expansion joint, complete with all the accessories such as flanges, stays and articulations, at the minimum design temperature and twice the design pressure at the extreme displacement conditions recommended by the manufacturer, without permanent deformation;

.3 a cyclic test (thermal movements) is to be performed on a complete expansion joint, which is to withstand at least as many cycles under the conditions of pressure, temperature, axial movement, rotational movement and transverse movement as it will encounter in actual service. Testing at ambient temperature is permitted when this testing is at least as severe as testing at the service temperature; and

.4 a cyclic fatigue test (ship deformation) is to be performed on a complete expansion joint, without internal pressure, by simulating the bellows movement corresponding to a compensated pipe length, for at least 2,000,000 cycles at a frequency not higher than 5 [Hz]. This test is only required when, due to the piping arrangement, ship deformation loads are actually experienced.

IR5.13.1.3 Cargo pumps

IR5.13.1.3.1 Prototype testing

Each size and type of pump is to be approved through design assessment and prototype testing. Prototype testing is to be witnessed by the Surveyor. Consideration may be given to acceptance of satisfactory in service experience report submitted by the manufacturer, for a class approved existing pump design.

Prototype testing is to include hydrostatic test of the pump body equal to 1.5 times the design pressure and a capacity test. For submerged electric motor driven pumps, the capacity test is to be carried out with the design fluid or with a fluid below the minimum working temperature. For shaft driven deep well pumps, the capacity test may be carried out with water. In addition, for shaft driven deep well pumps, a spin test to demonstrate satisfactory operation of bearing clearances, wear rings and sealing arrangements is to be carried out at the minimum design temperature. The full length of shafting is not required for the spin test, but must be of sufficient length to include at least one bearing and sealing arrangements. After completion of tests, the pump is to be opened out for examination.

IR5.13.1.3.2 Unit production testing

All pumps are to be tested at the plant of manufacturer in the presence of the Surveyor. Testing is to include hydrostatic test of the pump body equal to 1.5 times the design pressure and a capacity test. For submerged electric motor driven pumps, the capacity test is to be carried out with the design fluid or with a fluid below the minimum working temperature. For shaft driven deep well pumps, the capacity test may be carried out with water.

As an alternative to the above, the certification of a pump may be carried out subject to the following:

- The pump has been approved as required by IR5.13.1.3.1, and
- The manufacturer has a recognized quality system that has been assessed and certified by IRS subject to periodic audits, and
- The quality control plan contains a provision to subject each pump to a hydrostatic test of the pump body equal to 1.5 times the design pressure and a capacity test. The manufacturer is to maintain records of such tests.

5.13.2 System testing requirements

5.13.2.1 The requirements of this sub-section shall apply to piping inside and outside the cargo tanks.

5.13.2.2 After assembly, all cargo and process piping are to be subjected to a strength test with a suitable fluid. The test pressure is to be at least 1.5 times the design pressure (1.25 times the design pressure where the test fluid is compressible) for liquid lines and 1.5 times the maximum system working pressure (1.25 times the maximum system working pressure where the test fluid is compressible) for vapour lines. When piping systems or parts of systems are completely manufactured and equipped with all fittings, the test may be conducted prior to installation on board the ship. Joints welded on board are to be tested to at least 1.5 times the design pressure.

5.13.2.3 After assembly on board, each cargo and process piping system are to be subjected to a leak test using air, or other suitable medium, to a pressure depending on the leak detection method applied.

5.13.2.4 In double wall gas-fuel piping systems, the outer pipe or duct is also to be pressure tested to show that it can withstand the expected maximum pressure at gas pipe rupture.

IR 5.13.2.4.1 The expression “duct” in 5.4.4 and 5.13.2.4 above, means to include the equipment enclosure required in 16.4.3.1 and 16.4.3.2 (e.g. Gas Valve Unit enclosure) as well as the structural pipe duct intended to contain any release of gas from inner pipe or equipment. The term “structural pipe duct” means an outer duct forming part of a structure such as hull structure or superstructure or deck house, where permitted other than gas valve unit rooms.

The GVU rooms are to be:

(a) gastight toward other enclosed spaces;

(b) equipped with mechanical exhaust ventilation having a capacity of at least 30 air changes per hour and arranged to maintain a pressure less than the atmospheric pressure; and

able to withstand the maximum built-up pressure arising in the room in case of a gas pipe rupture, as documented by suitable calculations taking into account the ventilation arrangements.

IR5.13.2.4.2 The expression "maximum pressure at gas pipe rupture" in 5.13.2.4 above, is the maximum pressure to which the outer pipe or duct is subjected after the inner pipe rupture and for testing purposes it is the same as the design pressure used in 5.4.4.

5.13.2.5 All piping systems, including valves, fittings and associated equipment for handling cargo or vapours, are to be tested under normal operating conditions not later than at the first loading operation, in accordance with recognized standards.

5.13.3 Emergency shutdown valves

The closing characteristics of emergency shutdown valves used in liquid cargo piping systems are to be tested to demonstrate compliance with 18.10.2.1.3. This testing may be carried out on board after installation.

Section 6

Materials of Construction and Quality Control

6.0 Goal

To identify the required properties, testing standards and stability of metallic and non-metallic materials and fabrication processes used in the construction of cargo containment and piping systems to ensure they serve the functions for which they have been selected, as required in Section 4 and 5.

6.1 Definitions

6.1.1 Where reference is made in this chapter to A, B, D, E, AH, DH, EH and FH hull structural steels, these steel grades are hull structural steels according to recognized standards.

IR 6.1.1 Hull structural steel in accordance with relevant requirements of Pt 2 Ch.3, Sec 2 and 3 will be accepted.

6.1.2 A piece is the rolled product from a single slab or billet or from a single ingot, if this is rolled directly into plates, strips, sections or bars.

6.1.3 A batch is the number of items or pieces to be accepted or rejected together, on the basis of the tests to be carried out on a sampling basis. The size of a batch is given in the recognized standards.

6.1.4 Controlled rolling (CR) is a rolling procedure in which the final deformation is carried out in the normalizing temperature range, resulting in a material condition generally equivalent to that obtained by normalizing.

6.1.5 Thermo-mechanical controlled processing (TMCP) is a procedure that involves strict control of both the steel temperature and the rolling reduction. Unlike CR, the properties conferred by TMCP cannot be reproduced by subsequent normalizing or other heat treatment. The use of accelerated cooling on completion of TMCP may also be accepted, subject to approval by the Administration/ IRS. The same applies for the use of tempering after completion of TMCP.

6.1.6 Accelerated cooling (AcC) is a process that aims to improve mechanical properties by controlled cooling with rates higher than air cooling, immediately after the final TMCP operation. Direct quenching is excluded from accelerated cooling. The material properties conferred by TMCP and AcC cannot be reproduced by subsequent normalizing or other heat treatment.

6.2 Scope and general requirements

6.2.1 This section gives the requirements for metallic and non-metallic materials used in the construction of the cargo system. This includes requirements for joining processes, production process, personnel qualification, NDT and inspection and testing including production testing. The requirements for rolled materials, forgings and castings are given in 6.4 and tables 6.1, to 6.5. The requirements for

weldments are given in 6.5, and the guidance for non-metallic materials is given in appendix 4 of IGC Code. A quality assurance/quality control programme is to be implemented to ensure that the requirements of 6.2 are complied with.

6.2.2 The manufacture, testing, inspection and documentation are to be in accordance with recognized standards and the specific requirements given in this Chapter.

6.2.3 Where post-weld heat treatment is specified or required, the properties of the base material are to be determined in the heat-treated condition, in accordance with the applicable table of this section, and the weld properties are to be determined in the heat treated condition in accordance with 6.5. In cases where a post-weld heat treatment is applied, the test requirements may be modified at the discretion of the Administration/ IRS.

6.3 General test requirements and specifications

6.3.1 Tensile test

6.3.1.1 Tensile testing is to be carried out in accordance with recognized standards.

IR 6.3.1.1 Tensile testing in accordance with Pt 2, Ch 2, Sec 2, will be accepted.

6.3.1.2 Tensile strength, yield stress and elongation are to be to the satisfaction of the Administration/ IRS. For carbon-manganese steel and other materials with definitive yield points, consideration is to be given to the limitation of the yield to tensile ratio.

6.3.2 Toughness test

6.3.2.1 Acceptance tests for metallic materials are to include Charpy V-notch toughness tests, unless otherwise specified by the Administration. The specified Charpy V-notch requirements are minimum average energy values for three full size (10 [mm] × 10 [mm]) specimens and minimum single energy values for individual specimens. Dimensions and tolerances of Charpy V-notch specimens are to be in accordance with recognized standards. The testing and requirements for specimens smaller than 5 [mm] in size are to be in accordance with recognized standards. Minimum average values for subsized specimens are to be:

Charpy V-notch specimen size (mm)	Minimum average energy of three specimens
10 x 10	KV
10 x 7.5	5/6 KV
10 x 5	2/3 KV

Where:

KV = the energy values (J) specified in tables 6.1 to 6.4

Only one individual value may be below the specified average value, provided it is not less than 70% of that value.

IR 6.3.2.1 Dimensions and tolerances of impact test specimens in accordance with the requirements of Pt.2, Ch.2, Sec.3 will be accepted.

6.3.2.2 For base metal, the largest size Charpy V-notch specimens possible for the material thickness shall be machined with the specimens located as near as practicable to a point midway between the surface and the centre of the thickness and the length of the notch perpendicular to the surface as shown in Fig.6.1.

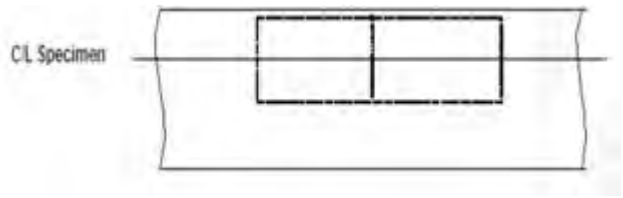


Fig.6.1 : Orientation of base metal test specimen

6.3.2.3 For a weld test specimen, the largest size Charpy V-notch specimens possible for the material thickness is to be machined, with the specimens located as near as practicable to a point midway between the surface and the centre of the thickness. In all cases, the distance from the surface of the material to the edge of the specimen is to be approximately 1 [mm] or greater. In addition, for double-V butt welds, specimens are to be machined closer to the surface of the second welded section. The specimens are to be taken generally at each of the following locations, as shown in Fig.6.2, on the centreline of the welds, the fusion line and 1 [mm], 3 [mm] and 5 [mm]from the fusion line.

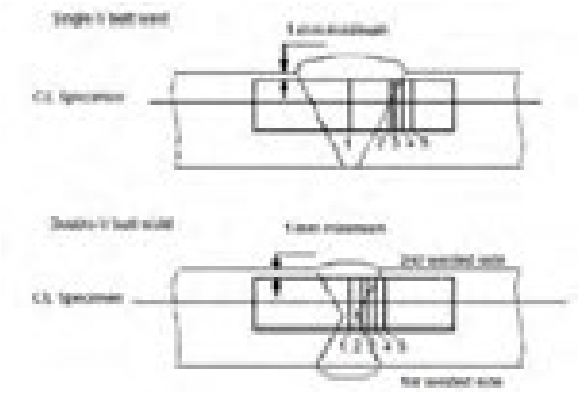


Fig.6.2 : Orientation of weld test specimen

Notch locations in Fig.6.2:

.1 Centreline of the weld.

.2 Fusion line.

.3 In heat-affected zone (HAZ), 1 [mm] from the fusion line.

.4 In HAZ, 3 [mm] from the fusion line.

.5 In HAZ, 5 [mm] from the fusion line.

6.3.2.4 If the average value of the three initial Charpy V-notch specimens fails to meet the stated requirements, or the value for more than one specimen is below the required average value, or when the value for one specimen is below the minimum value permitted for a single specimen, three additional specimens from the same material may be tested and the results be combined with those previously obtained to form a new average. If this new average complies with the requirements and if no more than two individual results are lower than the required average and no more than one result is lower than the required value for a single specimen, the piece or batch may be accepted.

6.3.3 Bend test

6.3.3.1 The bend test may be omitted as a material acceptance test, but is required for weld tests. Where a bend test is performed, this is to be done in accordance with recognized standards.

IR 6.3.3.1 Bend testing in accordance with Pt.2, Ch.11, Sec.1 will be accepted.

6.3.3.2 The bend tests are to be transverse bend tests, which may be face, root or side bends at the discretion of the Administration/ IRS. However, longitudinal bend tests may be required in lieu of transverse bend tests in cases where the base material and weld metal have different strength levels.

6.3.4 Section observation and other testing

Macrosection, microsection observations and hardness tests may also be required by the Administration/ IRS, and they are to be carried out in accordance with recognized standards, where required.

6.4 Requirements for metallic materials

6.4.1 General requirements for metallic materials

6.4.1.1 The requirements for materials of construction are shown in the tables as follows:

.1 Table 6.1: Plates, pipes (seamless and welded), sections and forgings for cargo tanks and process pressure vessels for design temperatures not lower than 0 [°C].

.2 Table 6.2: Plates, sections and forgings for cargo tanks, secondary barriers and process pressure vessels for design temperatures below 0 [°C] and down to -55 [°C].

.3 Table 6.3: Plates, sections and forgings for cargo tanks, secondary barriers and process pressure vessels for design temperatures below -55 [°C] and down to -165 [°C].

.4 Table 6.4: Pipes (seamless and welded), forgings and castings for cargo and process piping for design temperatures below 0 [°C] and down to -165 [°C].

.5 Table 6.5: Plates and sections for hull structures required by 4.19.1.2 and 4.19.1.3.

IR6.4.1.1 High Manganese Austenitic Steel may also be used for cargo and fuel tanks of LNG Carriers. IRS *'Guidelines on Approval of High Manganese Austenitic Steel for Cryogenic Service'* may be referred for manufacturing approval of such steels.

Table : 6.1		
PLATES, PIPES (SEAMLESS AND WELDED)[See note 1 and note 2], SECTIONS AND FORGING OFOR CARGO TANKS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES NOT LOWER THAN 0 ˚C		
CHEMICAL COMPOSITION AND HEAT TREATMENT		
• Carbon-manganese steel		
• Fully killed fine grain steel		
• Small additions of alloying elements by agreement with the Administration/ IRS		
• Composition limits to be approved by the Administration/ IRS		
• Normalized, or quenched and tempered [See note 4]		
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS		
Sampling Frequency		
• Plates	Each “piece” to be tested	
• Sections and forgings	Each “batch” to be tested	
Mechanical properties		
• Tensile properties	Specified minimum yield stress not to exceed 410 [N/mm^2][See note 5]	
Toughness (Charpy V-notch test)		
• Plates	Transverse test pieces. Minimum average energy value (KV) 27J	
• Sections and forgings	Longitudinal test pieces. Minimum average energy (KV) 41J	
• Test temperature	Thickness t (mm)	Test temperature (˚C)
	t < 20	0
	20 < t ≤ 40 [See note 3]	-20

Notes

1 For seamless pipes and fittings normal practice applies. The use of longitudinally and spirally welded pipes is to be specially approved by the Administration or IRS.

2 Charpy V-notch impact tests are not required for pipes.

3 This table is generally applicable for material thicknesses up to 40 mm.

4 A controlled rolling procedure or TMCP may be used as an alternative.

5 Materials with specified minimum yield stress exceeding 410 [N/mm^2] may be approved by the Administration or IRS. For these materials, particular attention is to be given to the hardness of the welded and heat affected zones.

In addition to Table 6.1 above, for design temperature not lower than 0°C, the following applies:

Table 6.1.1		
Plates, pipes (seamless and welded), sections and forgings for cargo tanks, fuel tanks and process pressure vessels for design temperatures not lower than 0°C.		
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS		
TEST TEMPERATURE	Thickness t (mm)	Test temperature (°C)
	40 < t ≤ 50 (1)	-20 (2)
	40 < t ≤ 50 (1)	-30 (3)
NOTES:		
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4. (2) Applies to type C independent tanks and process pressure vessels. In addition, post-weld stress relief heat treatment is to be performed. Exemption to post-weld stress relief heat treatment based on alternative approach (e.g. Engineering Critical Assessment) is to be approved by IRS or is to be in accordance with recognized standards. (3) Applies to tank other than type C.		

Table 6.2					
PLATES, SECTIONS AND FORGINGS See note 1 FOR CARGO TANKS, SECONDARY BARRIERS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES BELOW 0°C AND DOWN TO -55 [°C] MAXIMUM THICKNESS 25 [mm] see note 2					
CHEMICAL COMPOSITION AND HEAT TREATMENT					
• Carbon-manganese steel					
• Fully killed, aluminium treated fine grain steel					
• Chemical composition (ladle analysis)					
C	Mn	Si	S	P	
0.16%max see note 3	0.7 – 1.60 %	0.1 – 0.50 %	0.025% max	0.025% max	
Optional additions: Alloys and grain refining elements may be generally in accordance with the following:					
Ni	Cr	Mo	Cu	Nb	V
0.8% max	0.25% max	0.08% max	0.35% max	0.05% max	0.1% max
Al content total 0.02% min (Acid soluble 0.015% min)					
• Normalized, or quenched and tempered See note 4					
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS					
Sampling frequency					
• Plates	Each “piece” to be tested				
• Sections and forgings	Each “ batch” to be tested				
Mechanical properties					
• Tensile properties	Specified minimum yield stress not to exceed 410 [N/mm^2] See note 5				
Toughness (Charpy V-notch test)					
• Plates	Transverse test pieces. Minimum average energy value (KV) 27J				
• Sections and forgings	Longitudinal test pieces. Minimum average energy (KV) 41J				
• Test temperature	5 [˚C] below the design temperature or -20 [˚C], whichever is lower				

Notes

1 The Charpy V-notch and chemistry requirements for forgings may be specially considered by the Administration/ IRS.

2 For material thickness of more than 25 mm, Charpy V-notch tests is to be conducted as follows:

Material Thickness [mm]	Test temperature [˚C]
25 < t ≤ 30	10 [˚C] below design temperature or -20[˚C], whichever is lower
30 < t ≤ 35	15 [˚C] below design temperature or -20[˚C], whichever is lower
35 < t ≤ 40	20 [˚C] below design temperature
40 < t	Temperature approved by Administration/ IRS

The impact energy value is to be in accordance with the table for the applicable type of test specimen.

Materials for tanks and parts of tanks which are completely thermally stress relieved after welding may be tested at a temperature 5 [°C] below design temperature or -20 [°C], whichever is lower.

For thermally stress relieved reinforcements and other fittings, the test temperature is to be the same as that required for the adjacent tank-shell thickness.

3 By special agreement with the Administration/ IRS, the carbon content may be increased to 0.18% maximum, provided the design temperature is not lower than -40 [°C].

4 A controlled rolling procedure or TMCP may be used as an alternative.

5 Materials with specified minimum yield stress exceeding 410 [N/mm^2] may be approved by Administration/IRS. For these materials, particular attention is to be given to the hardness of the welded and heat affected zones.

Guidance:
For materials exceeding 25 [mm] in thickness for which the test temperature is -60 [°C] or lower, the application of specially treated steels or steels in accordance with Table 6.3 may be necessary.

In addition to Table 6.2 above, the following applies:

Table 6.2.1		
Plates, sections and forgings for cargo tanks, fuel tanks, secondary barriers and process pressure vessels for design temperatures below 0°C and down to -10°C		
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS		
TEST TEMPERATURE	Thickness t (mm)	Test temperature (°C)
	40 < t ≤ 50 (1)	5°C below design temperature or –20°C, whichever is lower (2)
	40 < t ≤ 45 (1)	25 °C below design temperature (3)
	45 < t ≤ 50 (1)	30 °C below design temperature (3)
NOTES:		
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4. (2) Applies to type C independent tanks and process pressure vessels. In addition, post-weld stress relief heat treatment is to be performed. Exemption to post-weld stress relief heat treatment based on alternative approach (e.g. Engineering Critical Assessment) is to be approved by IRS or is to be accordance with recognized standards. (3) Applies to tank other than type C.		

Table 6.2.2		
Plates, sections and forgings for cargo tanks, fuel tanks, secondary barriers and process pressure vessels for design temperatures below -10°C and down to -55°C		
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS		
TEST TEMPERATURE	Thickness t (mm)	Test temperature (°C)
	40 < t ≤ 50 (1)	5°C below design temperature or –20°C, whichever is lower (2)
	40 < t ≤ 45 (1)	25 °C below design temperature (3)
	45 < t ≤ 50 (1)	30 °C below design temperature (3)
NOTES:		
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4. (2) 6.6.2.2 applies with regards to post-weld stress relief heat treatment. Exemption to post-weld stress relief heat treatment based on alternative approach (e.g. Engineering Critical Assessment) is to be approved by IRS or is to be in accordance with recognized standards. (3) Applies to tank other than type C.		

Table 6.3		
PLATES, SECTIONS AND FORGINGS See note 1 FOR CARGO TANKS, SECONDARY BARRIERS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES BELOW -55 [°C] AND DOWN TO -165 [°C] See note 2 MAXIMUM THICKNESS 25 [mm] see note 3 and note 4		
Minimum design temperature [˚C]	**Chemical composition See note 5 and heat treatment**	**Impact test temperature [˚C]**
-60	1.5% nickel steel – normalized or normalized and tempered or quenched and tempered or TMCP See note 6	-65
-65	2.25% nickel steel – normalized or normalized and tempered or quenched and tempered or TMCP See note 6 and 7	-70
-90	3.5% nickel steel – normalized or normalized and tempered or quenched and tempered or TMCP See note 6 and 7	-95
-105	5% nickel steel – normalized or normalized and tempered or quenched and tempered See note 6,7and 8	-110
-165	9% nickel steel – double normalized and tempered or quenched and tempered See note 6	-196
-165	Austenitic steels, such as types 304, 304L, 316, 316L, 321 and 347 solution treated See note 9	-196
-165	Aluminium alloys; such as type 5083 annealed	Not required
-165	Austenitic Fe-Ni alloy (36% nickel). Heat treatment as agreed.	Not required
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS		
Sampling frequency		
• Plates	Each “piece” to be tested	
• Sections and forgings	Each “ batch” to be tested	
Toughness (Charpy V-notch test)		
• Plates	Transverse test pieces. Minimum average energy value (KV) 27J	
• Sections and forgings	Longitudinal test pieces. Minimum average energy (KV) 41J	

Table 6.3 : Notes

1 The impact test required for forgings used in critical application is to be subject to special consideration by the Administration/ IRS.

2 The requirements for design temperatures below -165[°C] are to be specially agreed with the Administration/ IRS.

3 For materials 1.5% Ni, 2.25% Ni, 3.5% Ni and 5% Ni, with thicknesses greater than 25 [mm], the impact tests are to be conducted as follows:

Material Thickness [mm]	Test temperature [˚C]
25 < t ≤ 30	10 [˚C] below design temperature
30 < t ≤ 35	15 [˚C] below design temperature
35 < t ≤ 40	20 [˚C] below design temperature

The energy value is to be in accordance with the table for the applicable type of test specimen. For material thickness of more than 40 [mm], the Charpy V-notch values is to be specially considered.

4 For 9% Ni steels, austenitic stainless steels and aluminium alloys, thickness greater than 25 [mm] may be used.

5 The chemical composition limits are to be in accordance with recognized standards.

6 TMCP nickel steels will be subject to acceptance by the Administration/ IRS.

7 A lower minimum design temperature for quenched and tempered steels may be specially agreed with the Administration/ IRS.

8 A specially heat treated 5% nickel steel, for example triple heat treated 5% nickel steel, may be used down to -165 [°C], provided that the impact tests are carried out at -196 [°C].

9 The impact test may be omitted, subject to agreement with the Administration/ IRS.

In addition to Table 6.3 above, the following applies:

Table 6.3.1	
Plates, sections and forgings for cargo tanks, fuel tanks, secondary barriers and process pressure vessels for design temperatures below -55°C and down to -165°C.	
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS	
40 < t ≤ 45 mm (1)	25°C below design temperature
45 < t ≤ 50 mm (1)	30°C below design temperature
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4.	

Table 6.4			
PIPES(SEAMLESS AND WELDED) See note 1 , FORGINGS See note 2, AND CASTINGS See note 2 FOR CARGO AND PROCESS PIPING FOR DESIGN TEMPERATURES BELOW 0 [˚C] AND DOWN TO -165 [˚C] See note 3 MAXIMUM THICKNESS 25 [mm]			
Minimum design temperature [˚C]	**Chemical composition See note 5 and heat treatment**	**Impact test**	
		Test temp **[˚C]**	Minimum average energy (KV)
-55	Carbon-manganese steel. Fully killed fine grain. Normalized or as agreed See note 6	See note 4	27
-65	2.25% nickel steel. Normalized, normalized and tempered or quenched and tempered See note 6	-70	34
-90	3.5% nickel steel. Normalized, normalized and tempered or quenched and tempered See note 6	-95	34
-165	9% nickel steel See note 7. Double Normalized, and tempered or quenched and tempered	-196	41
	Austenitic steels, such as types 304, 304L, 316, 316L, 321, and 347.Solution treated See note 8	-196	41
	Aluminium alloys; such as type 5083 annealed		Not required
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS			
Sampling frequency			
• Each “batch” to be tested			
Toughness (Charpy V-notch test)			
• Impact test: Longitudinal test pieces			

Notes

1 The use of longitudinally or spirally welded pipes is to be specially approved by the Administration/ IRS.

2 The requirements for forgings and castings may be subject to special consideration by the Administration/ IRS.

3 The requirements for design temperatures below -165 [°C] are to be specially agreed with the Administration/ IRS.

4 The test temperature is to be 5 [°C] below the design temperature or -20 [°C], whichever is lower.
5 The composition limits are to be in accordance with recognized standards.

6 A lower design temperature may be specially agreed with the Administration/ IRS for quenched and tempered materials.

7 This chemical composition is not suitable for castings.

8 Impact tests may be omitted, subject to agreement with the Administration/ IRS.

Table 6.5								
PLATES AND ECTIONS FOR HULL STRUCTURES REQUIRED BY 4.19.1.2 AND 4.19.1.3								
Minimum design temperature of hull structure (˚C)	**Maximum thickness (mm) for steel grades**							
	A	**B**	**D**	**E**	**AH**	**DH**	**EH**	**FH**
0 and above See note 1 **-5 and above** See note 2	**Recognized standards**							
down to -5	**15**	**25**	**30**	**50**	**25**	**45**	**50**	**50**
down to -10	**X**	**20**	**25**	**50**	**20**	**40**	**50**	**50**
down to -20	**X**	**X**	**20**	**50**	**X**	**30**	**50**	**50**
down to -30	**X**	**X**	**X**	**40**	**X**	**20**	**40**	**50**
Below -30	**In accordance with table 6.2, except that the thickness limitation given in table 6.2 and in note 2 of that table does not apply.**							
Notes " X " means steel grade not to used 1 for the purpose of 4.19.1.3 2 for the purpose of 4.19.1.2								

6.5 Welding of metallic materials and non-destructive testing

6.5.1 General

6.5.1.1 This sub-section shall apply to primary and secondary barriers only, including the inner hull where this forms the secondary barrier. Acceptance testing is specified for carbon, carbon-manganese, nickel alloy and stainless steels, but these tests may be adapted for other materials. At the discretion of the Administration/ IRS, impact testing of stainless steel and aluminium alloy weldments may be omitted and other tests may be specially required for any material.

6.5.2 Welding Consumables

6.5.2.1 Consumables intended for welding of cargo tanks are to be in accordance with Recognized Standards. Deposited weld metal tests and butt weld tests would be required for all consumables. The results obtained from tensile and Charpy V-notch impact tests are to be in accordance with the Recognized Standards. The chemical composition of the deposited weld metal is to be recorded for information.

6.5.3 Welding procedure tests for cargo tanks and process pressure vessels

6.5.3.1 Welding Procedure tests for cargo tanks and process pressure vessels are required for all butt welds.

6.5.3.2 The test assemblies are to be representative of:

.1 Each base material

.2 Each type of consumable and welding process and,

.3 Each welding position.

6.5.3.3 For butt welds in plates, the test assemblies are to be so prepared that the rolling direction is parallel to the direction of welding. The range of thickness qualified by each welding procedure test is to be in accordance with Recognized Standards. Radiographic or ultrasonic testing may be performed at the option of the fabricator.

6.5.3.4 The following welding procedure tests for cargo tanks and process pressure vessels are to be carried out in accordance with 6.3, with specimens made from each test assembly:

.1 Cross-weld tensile tests.

.2 Longitudinal all-weld testing, where required by recognized standards.

.3 Transverse bend tests: These bend tests may be face, root or side bends. However, longitudinal bend tests may be required in lieu of transverse bend tests in cases where the base material and weld metal have different strength levels.

.4 One set of three Charpy V-notch impacts, generally at each of the following locations, as shown in Fig.6.2:

.1 centreline of the weld;

.2 fusion line;

.3 1 mm from the fusion line;

.4 3 mm from the fusion line; and

.5 5 mm from the fusion line; and

.5 macrosection, microsection and harness survey may also be required.

6.5.3.5 Each test is to satisfy the following requirements:

.1 **Tensile tests** : cross-weld tensile strength shall not be less than the specified minimum tensile strength for the appropriate parent materials. For materials such as aluminium alloys, reference shall be made to 4.18.1.3 with regard to the requirements for weld metal strength of under-matched welds (where the weld metal has a lower tensile strength than the parent metal). In every case, the position of fracture is to be recorded for information;

.2 **Bend tests** : No fracture is acceptable after a 180° bend over a former of a diameter four times the thickness of the test pieces, and

.3 **Charpy V-notch impact tests**: Charpy v-notch tests are to be conducted at the temperature prescribed for the base material being joined. The results of weld metal impact tests, minimum average energy (KV), are to be no less than 27J. The weld metal requirements for subsize specimens and single energy values are to be in accordance with 6.3.2. The results of fusion line and heat affected zone impact tests are to show a minimum average energy (KV) in accordance with the transverse or longitudinal requirements of the base material, whichever is applicable, and for subsize specimens, the minimum average energy (KV) is to be in accordance with 6.3.2. If the material thickness does not permit machining either full size or standard subsize specimens, the testing procedure and acceptance standards are to be in accordance with Recognized Standards.

6.5.3.6 Procedure tests for fillet welding are to be in accordance with recognized standards. In such cases, consumables are to be so selected that exhibit satisfactory impact properties.

6.5.4 Welding Procedure Tests for Piping

Welding procedure tests for piping are to be carried out and are to be similar to those detailed for cargo tanks in 6.5.3.

6.5.5 Production weld tests

6.5.5.1 For all cargo tanks and process pressure vessels except integral and membrane tanks, production weld tests are generally to be performed for approximately each 50 [m] of butt weld joints and are to be representative of each welding position. For secondary barriers, the same type production tests as required for primary tanks are to be performed except that the number of tests may be reduced subject to agreement with IRS. Tests, other than those specified in 6.5.5.2 to 6.5.5.5, may be required for cargo tanks or secondary barriers.

6.5.5.2 The production tests for type A and type B independent tanks and semi-membrane tanks shall include bend tests and, where required for procedure tests, one set of three Charpy V-notch tests. The tests are to be made for each 50 [m] of weld. The Charpy V-notch tests are to be made with specimens having the notch alternately located in the centre of the weld and in the heat-affected zone (most critical location based on procedure qualification results). For austenitic stainless steel, all notches are to be in the centre of the weld.

6.5.5.3 For type C independent tanks and process pressure vessels, transverse weld tensile tests are required in addition to the tests listed in 6.5.5.2. Tensile tests are to meet the requirements of 6.5.3.5.

6.5.5.4 The quality assurance/quality control programme are to ensure the continued conformity of the production welds as defined in the material manufacturers quality manual.

6.5.5.5 The test requirements for integral and membrane tanks are the same as the applicable test requirements listed in 6.5.3.

6.5.6 Non-destructive testing

6.5.6.1 All test procedures and acceptance standards are to be in accordance with recognized standards, unless the designer specifies a higher standard in order to meet design assumptions. Radiographic testing is to be used, in principle, to detect internal defects. However, an approved ultrasonic test procedure in lieu of radiographic testing may be conducted, but, in addition, supplementary radiographic testing at selected locations is to be carried out to verify the results. Radiographic and ultrasonic testing records are to be retained.

6.5.6.2 For Type A independent tanks and semi-membrane tanks where the design temperature is below -20 [°C] , and for Type B independent tanks, regardless of temperature, all full penetration butt welds of the shell plating of cargo tanks are to be subjected to no-destructive testing suitable to detect internal defects over their full length. Ultrasonic testing in lieu of radiographic testing may be carried out under the same conditions as described in 6.5.6.1.

6.5.6.3 Where the design temperature is higher than -20 [°C], all full penetration butt welds in way of intersections and at least 10 per cent of the remaining full penetration welds of tank structures are to be subjected to radiographic testing or ultrasonic testing under the same condition as described in 6.5.6.1.

6.5.6.4 In each case the remaining tank structure including the welding of stiffeners and other fittings and attachments are to be examined by magnetic particle or dye penetrant methods as considered necessary.

6.5.6.5 For type C independent tanks, the extent of non-destructive testing is to be total or partial according to recognized standards, but the controls to be carried out are not to be less than the following:

.1 Total non-destructive testing referred to in 4.23.2.1.3:

Radiographic testing:

.1 all butt welds over their full length;

Non-destructive testing for surface crack detection:

.2 all welds over 10% of their length;

.3 reinforcement rings around holes, nozzles, etc., over their full length.

As an alternative, ultrasonic testing as described in 6.5.6.1 may be accepted as a partial substitute for the radiographic testing. In addition, the Administration/ IRS may require total ultrasonic testing on welding of reinforcement rings around holes, nozzles, etc.

.2 Partial non-destructive testing referred to in 4.23.2.1.3:

Radiographic testing:

.1 all butt-welded crossing joints and at least 10% of the full length of butt welds at selected positions uniformly distributed;
Non-destructive testing for surface crack detection:

Non-destructive testing for surface crack detection:

.2 reinforcement rings around holes, nozzles, etc., over their full length;

Ultrasonic testing:

.3 as may be required by the Administration/ IRS.

6.5.6.6 The quality assurance/quality control programme is to ensure the continued conformity of the non-destructive testing of welds, as defined in the material manufacturer's quality manual.

6.5.6.7 Inspection of piping is to be carried out in accordance with the requirements of Sec.5.

6.5.6.8 The secondary barrier is to be non-destructive tested for internal defects as considered necessary. Where the outer shell of the hull is part of the secondary barrier, all sheer strake butts and the intersections of all butts and seams in the side shell are to be tested by radiographic testing.

6.6 Other requirements for construction in metallic materials

6.6.1 **General**

6.6.1.1 Inspection and non-destructive testing of welds are to be in accordance with the requirements of 6.5.5 and 6.5.6. Where higher standards or tolerances are assumed in the design, they are also to be satisfied.

6.6.2 **Independent tank**

6.6.2.1 For type C tanks and type B tanks primarily constructed of bodies of revolution, the tolerances relating to manufacture, such as out-of-roundness, local deviations from the true form, welded joints alignment and tapering of plates having different thicknesses, are to comply with recognized standards. The tolerances are also to be related to the buckling analysis referred to in 4.22.3.2 and 4.23.3.2.

6.6.2.2 For type C tanks of carbon and carbon-manganese steel, post-weld heat treatment is to be performed after welding, if the design temperature is below -10 [°C]. Post-weld heat treatment in all other cases and for materials other than those mentioned above are to be to recognized standards. The soaking temperature and holding time are to be to the recognized standards.

6.6.2.3 In the case of type C tanks and large cargo pressure vessels of carbon or carbon-manganese steel, for which it is difficult to perform the heat treatment, mechanical stress relieving by pressurizing may be carried out as an alternative to the heat treatment and subject to the following conditions:

.1 complicated welded pressure vessel parts such as sumps or domes with nozzles, with adjacent shell plates are to be heat treated before they are welded to larger parts of the pressure vessel;

.2 the mechanical stress relieving process is to be preferably be carried out during the hydrostatic pressure test required by 4.23.6, by applying a higher pressure than the test pressure required by 4.23.6.1. The pressurizing medium is to be water;

.3 for the water temperature, 4.23.6.2 applies;

.4 stress relieving is to be performed while the tank is supported by its regular saddles or supporting structure or, when stress relieving cannot be carried out on board, in a manner which will give the same stresses and stress distribution as when supported by its regular saddles or supporting structure;

.5 the maximum stress relieving pressure is to be held for 2 h per 25 [mm] of thickness, but in no case less than 2 h;

.6 the upper limits placed on the calculated stress levels during stress relieving are to be the following:

.1 equivalent general primary membrane stress: 0.9 *Re*;

.2 equivalent stress composed of primary bending stress plus membrane stress: 1.35 *Re*, where *R*e is the specific lower minimum yield stress or 0.2% proof stress at test temperature of the steel used for the tank;

.7 strain measurements will normally be required to prove these limits for at least the first tank of a series of identical tanks built consecutively. The location of strain gauges is to be included in the mechanical stress relieving procedure to be submitted in accordance with 6.6.2.3;

.8 the test procedure is to demonstrate that a linear relationship between pressure and strain is achieved at the end of the stress relieving process when the pressure is raised again up to the design pressure;

.9 high-stress areas in way of geometrical discontinuities such as nozzles and other openings are to be checked for cracks by dye penetrant or magnetic particle inspection after mechanical stress relieving. Particular attention in this respect is to be paid to plates exceeding 30 [mm] in thickness;

.10 steels which have a ratio of yield stress to ultimate tensile strength greater than 0.8 are generally not to be mechanically stress relieved. If, however, the yield stress is raised by a method giving high ductility of the steel, slightly higher rates may be accepted upon consideration in each case;

.11 mechanical stress relieving cannot be substituted for heat treatment of cold formed parts of tanks, if the degree of cold forming exceeds the limit above which heat treatment is required;

.12 the thickness of the shell and heads of the tank are not to exceed 40 [mm]. Higher thicknesses may be accepted for parts which are thermally stress relieved;

.13 local buckling is to be guarded against, particularly when tori-spherical heads are used for tanks and domes; and

.14 the procedure for mechanical stress relieving is to be to a recognized standard.

6.6.3 Secondary barriers

During construction, the requirements for testing and inspection of secondary barriers are to be approved or accepted by IRS (see 4.6.2.5 and 4.6.2.6).

6.6.4 Semi-membrane tanks

For semi-membrane tanks, the relevant requirements in 6.6 for independent tanks or for membrane tanks are to be applied as appropriate.

6.6.5 Membrane tanks

The quality assurance/quality control programme are to ensure the continued conformity of the weld procedure qualification, design details, materials, construction, inspection and production testing of components. These standards and procedures are to be developed during the prototype testing programme.

6.7 Non-metallic materials

6.7.1 General

The information in the attached appendix 4 of IGC Code is given for guidance in the selection and use of these materials, based on the experience to date.

Section 7

Cargo Pressure/Temperature Control

7.0 Goal

To maintain the cargo tank pressure and temperature within design limits of the containment system and/or carriage requirements of the cargo.

7.1 Methods of control

7.1.1 With the exception of tanks designed to withstand full gauge vapour pressure of the cargo under conditions of the upper ambient design temperatures, cargo tanks' pressure and temperature is to be maintained at all times within their design range by either one, or a combination of, the following methods:

.1 reliquefaction of cargo vapours;

.2 thermal oxidation of vapours;

.3 pressure accumulation; and

.4 liquid cargo cooling.

7.1.2 For certain cargoes, where required by Sec 17, the cargo containment system is to be capable of withstanding the full vapour pressure of the cargo under conditions of the upper ambient design temperatures, irrespective of any system provided for dealing with boil-off gas.

7.1.3 Venting of the cargo to maintain cargo tank pressure and temperature is not acceptable except in emergency situations. The Administration may permit certain cargoes to be controlled by venting cargo vapours to the atmosphere at sea. This may also be permitted in port with the authorization of the port Administration.

7.2 Design of systems

For normal service, the upper ambient design temperature is to be:

- sea: 32 [°C]

- air: 45 [°C]

For service in particularly hot or cold zones, these design temperatures are to be increased or decreased, to the satisfaction of the Administration/ IRS. The overall capacity of the system is to be such that it can control the pressure within the design conditions without venting to atmosphere.

7.3 Reliquefaction of cargo vapours

7.3.1 General

The reliquefaction system may be arranged in one of the following ways:

.1 a direct system, where evaporated cargo is compressed, condensed and returned to the cargo tanks;

.2 an indirect system, where cargo or evaporated cargo is cooled or condensed by refrigerant without being compressed;

.3 a combined system, where evaporated cargo is compressed and condensed in a cargo/refrigerant heat exchanger and returned to the cargo tanks; and

.4 if the reliquefaction system produces a waste stream containing methane during pressure control operations within the design conditions, these waste gases, as far as reasonably practicable, are disposed of without venting to atmosphere.

Note:
The requirements of sec 17 and 19 may preclude the use of one or more of these systems or may specify the use of a particular system.

7.3.2 Compatibility

Refrigerants used for reliquefaction are to be compatible with the cargo they may come into contact with. In addition, when several refrigerants are used and may come into contact, they are to be compatible with each other.

7.4 Thermal oxidation of vapours

7.4.1 General

Maintaining the cargo tank pressure and temperature by means of thermal oxidation of cargo vapours, as defined in 1.2.52 and 16.2 are to be permitted only for LNG cargoes. In general:

.1 thermal oxidation systems shall exhibit no externally visible flame and shall maintain the uptake exhaust temperature below 535 [°C];

.2 arrangement of spaces where oxidation systems are located are to comply with 16.3 and supply systems shall comply with 16.4; and

.3 if waste gases coming from any other system are to be burnt, the oxidation system is to be designed to accommodate all anticipated feed gas compositions.

7.4.2 Thermal oxidation systems

Thermal oxidation systems are to comply with the following:

.1 each thermal oxidation system are to have a separate uptake;

.2 each thermal oxidation system are to have a dedicated forced draught system; and

.3 combustion chambers and uptakes of thermal oxidation systems are to be designed to prevent any accumulation of gas.

7.4.3 Burners

Burners are to be designed to maintain stable combustion under all design firing conditions.

7.4.4 Safety

7.4.4.1 Suitable devices are to be installed and arranged to ensure that gas flow to the burner is cut off unless satisfactory ignition has been established and maintained.

7.4.4.2 Each oxidation system is to have provision to manually isolate its gas fuel supply from a safely accessible position.

7.4.4.3 Provision are to be made for automatic purging the gas supply piping to the burners by means of an inert gas, after the extinguishing of these burners.

7.4.4.4 In case of flame failure of all operating burners for gas or oil or for a combination thereof, the combustion chambers of the oxidation system are to be automatically purged before relighting.

7.4.4.5 Arrangements are to be made to enable the combustion chamber to be manually purged.

7.5 Pressure accumulation systems

The containment system insulation, design pressure or both are to be adequate to provide for a suitable margin for the operating time and temperatures involved. No additional pressure and temperature control system is required. Conditions for acceptance are to be recorded in the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk required in 1.4.4.

7.6 Liquid cargo cooling

The bulk cargo liquid may be refrigerated by coolant circulated through coils fitted either inside the cargo tank or onto the external surface of the cargo tank.

7.7 Segregation

Where two or more cargoes that may react chemically in a dangerous manner are carried simultaneously, separate systems as defined in 1.2.47, each complying with availability criteria as specified in 7.8, is to be provided for each cargo. For simultaneous carriage of two or more cargoes that are not reactive to each other but where, due to properties of their vapour, separate systems are necessary, separation may be by means of isolation valves.

7.8 Availability

The availability of the system and its supporting auxiliary services are to be such that:

.1 in case of a single failure of a mechanical non-static component or a component of the control systems, the cargo tanks' pressure and temperature can be maintained within their design

range without affecting other essential services;

.2 redundant piping systems are not required;

.3 heat exchangers that are solely necessary for maintaining the pressure and temperature of the cargo tanks within their design ranges are to have a standby heat exchanger, unless they have a capacity in excess of 25% of the largest required capacity for pressure control and they can be repaired on board without external resources. Where an additional and separate method of cargo tank pressure and temperature control is fitted that is not reliant on the sole heat exchanger, then a standby heat exchanger is not required; and

.4 for any cargo heating or cooling medium, provisions are to be made to detect the leakage of toxic or flammable vapours into an otherwise non-hazardous area or overboard in accordance with 13.6. Any vent outlet from this leak detection arrangement shall be to a safe location and be fitted with a flame screen.

Section 8

Vent Systems for Cargo Containment

8.0 Goal

To protect cargo containment systems from harmful overpressure or underpressure at all times.

8.1 General

8.1.1 All cargo tanks are to be provided with a pressure relief system appropriate to the design of the cargo containment system and the cargo being carried. Hold spaces, interbarrier spaces and cargo piping which may be subject to pressures beyond their design capabilities are also to be provided with a suitable pressure relief system. Pressure control systems specified in Sec.7 are to be independent of the pressure relief systems.

8.2 Pressure relief systems

8.2.1 Cargo tanks, including deck tanks, are to be fitted with a minimum of two pressure relief valves (PRVs), each being of equal size within manufacturer's tolerances and suitably designed and constructed for the prescribed service.

8.2.2 Interbarrier spaces are to be provided with pressure relief devices complying with recognized standards.

8.2.3 The setting of the pressure relief valves are not to be higher than the vapour pressure which has been used in the design of the tank. Where two or more pressure relief valves are fitted, valves comprising not more than 50% of the total relieving capacity may be set at a pressure up to 5% above MARVS to allow sequential lifting, minimizing unnecessary release of vapour.

8.2.4 The following temperature requirements apply to PRVs fitted to pressure relief systems:

.1 PRVs on cargo tanks with a design temperature below 0 [ºC] are to be designed and arranged to prevent their becoming inoperative due to ice formation;

.2 the effects of ice formation due to ambient temperatures are to be considered in the construction and arrangement of PRVs;

.3 PRVs are to be constructed of materials with a melting point above 925 [°C]. Lower melting point materials for internal parts and seals may be accepted, provided that fail-safe operation of the PRV is not compromised; and

.4 sensing and exhaust lines on pilot operated relief valves are to be of suitably robust construction to prevent damage.

8.2.5 Valve testing

8.2.5.1 PRVs are to be type-tested. Type tests are to include:

.1 verification of relieving capacity;

.2 cryogenic testing when operating at design temperatures colder than -55 [°C];

.3 seat tightness testing; and

.4 pressure containing parts are pressure tested to at least 1.5 times the design pressure.

PRVs are to be tested in accordance with recognized standards (such as ISO 21013-1:2008 – Cryogenic vessels – Pressure-relief accessories for cryogenic service – part 1: Recloseable pressure-relief valves; and ISO 4126-1; 2004 Safety devices for protection against excessive pressure – part 1 and part 4: Safety valves.)

8.2.5.2 Each PRV is to be tested to ensure that:

.1 it opens at the prescribed pressure setting, with an allowance not exceeding ± 10% for 0 to 0.15 [MPa], ± 6% for 0.15 to 0.3 [MPa], ± 3% for 0.3 [MPa] and above;

.2 seat tightness is acceptable; and

.3 pressure containing parts will withstand at least 1.5 times the design pressure.

8.2.6 PRVs are to be set and sealed by IRS, and a record of this action, including the valves' set pressure, is to be retained on board the ship.

8.2.7 Cargo tanks may be permitted to have more than one relief valve set pressure in the following cases :

.1 installing two or more properly set and sealed PRVs and providing means as necessary for isolating the valves not in use from the cargo tank; or

.2 installing relief valves whose settings may be changed by the use of a previously approved device not requiring pressure testing to verify the new set pressure. All other valve adjustments are to be sealed.

8.2.8 Changing the set pressure under the provisions of 8.2.7 and the corresponding resetting of the alarms referred to in 13.4.2 are to be carried out under the supervision of the master in accordance with approved procedures and as specified in the ship's operating manual. Changes in set pressure are to be recorded in the ship's log and a sign is to be posted in the cargo control room, if provided, and at each relief valve, stating the set pressure.

8.2.9 In the event of a failure of a cargo tank-installed PRV, a safe means of emergency isolation is to be available:

.1 Procedures are to be provided and included in the cargo operations manual (see 18.2).

.2 The procedures are to allow only one of the cargo tank installed PRVs to be isolated.

.3 Isolation of the PRV is to be carried out under the supervision of the master. This action is to be recorded in the ship's log and a sign posted in the cargo control room, if provided, and at the PRV.

.4 The tank is not to be loaded until the full relieving capacity is restored.

8.2.10 Each PRV installed on a cargo tank is to be connected to a venting system, which is to be:

.1 so constructed that the discharge will be unimpeded and directed vertically upwards at the exit;

.2 arranged to minimize the possibility of water or snow entering the vent system;

.3 arranged such that the height of vent exits are not to be less than B/3 or 6 [m,] whichever is the greater, above the weather deck; and

.4 6 [m] above working areas and walkways.

8.2.11.1 Cargo PRV vent exits are to be arranged at a distance at least equal to B or 25 [m], whichever is less, from the nearest air intake, outlet or opening to accommodation spaces, service spaces and control stations, or other non-hazardous areas. For ships less than

90 [m] in length, smaller distances may be permitted.

8.2.11.2 All other vent outlets connected to the cargo containment system are to be arranged at a distance of at least 10 [m] from the nearest air intake, outlet or opening to accommodation spaces, service spaces and control stations, or other non-hazardous areas.

8.2.12 All other cargo vent exits not dealt with in other Sections are to be arranged in accordance with 8.2.10, 8.2.11.1 and 8.2.11.2. Means are to be provided to prevent liquid overflow from vent mast outlets, due to hydrostatic pressure from spaces to which they are connected.

8.2.13 If cargoes that react in a dangerous manner with each other are carried simultaneously, a separate pressure relief system is to be fitted for each one.

8.2.14 In the vent piping system, means for draining liquid from places where it may accumulate is to be provided. The PRVs and piping are to be arranged so that liquid can, under no circumstances, accumulate in or near the PRVs.

8.2.15 Suitable protection screens of not more than 13 [mm] square mesh are to be fitted on vent outlets to prevent the ingress of extraneous objects without adversely affecting the flow. Other requirements for protection screens apply when carrying specific cargoes (see 17.9 and 17.21).

8.2.16 All vent piping are to be designed and arranged not to be damaged by the temperature variations to which it may be exposed, forces due to flow or the ship's motions.

8.2.17 PRVs are to be connected to the highest part of the cargo tank above deck level. PRVs are to positioned on the cargo tank so that they will remain in the vapour phase at the filling limit (FL) as defined in Section 15, under conditions of 15° list and 0.015L trim, where L is as defined in 1.2.31.

IR 8.2.17 Vapour pockets not in communication with cargo tank vapour/liquid domes

.1 Clause 8.2.17 (IGC Code Chapter 8, 8.2.17) states that the PRVs are to be in the vapour phase under conditions of 15° list and 0.015L trim and presumes that no isolated vapour pockets are formed within this range in principle, this scenario can occur at other trim and list values based upon the filling level of the tank since the ship is designed to survive a damage condition up to 30° of list.

.2 In above condition, there is the potential for liquid build-up in the vapour/liquid domes caused by a pressure differential between the isolated vapour pocket and the vapour/liquid domes resulting in a possible overflow of cargo liquid into the vapour line or into the tank PRVs. Even though the likelihood of this situation occurring may be minimal, the consequences could be quite severe and lead up to the loss of the ship.

.3 Owners/operators of liquefied gas carriers, in consultation with the cargo containment system/cargo handling system designers, are recommended to develop emergency procedures to mitigate the risks to the vessel caused by isolated vapour pockets.

.4 These procedures should identify the condition when isolated vapour pockets can be present and contain measures to reduce or eliminate them and/or mitigate their consequences such as cargo jettisoning, transfer of cargo between tanks, and cargo vapourization/utilization based upon different scenarios following the accident, including, but not limited to, loss of power, limited ability to reduce angle of heel or trim.

.5 These emergency procedures are not a substitute for requirement 15.4.1.1 when determining the increased filling limits.

8.2.18 The adequacy of the vent system fitted on tanks loaded in accordance with 15.5.2 is to be demonstrated, taking into account the recommendation developed by IMO (Guidelines for the evaluation of the adequacy of type C tank vent systems - IMO Res.A.829(19)). A relevant certificate is to be permanently kept on board the ship. For the purposes of this paragraph, vent system means:

a) the tank outlet and the piping to the pressure relief valve;

b) the pressure relief valve;

c) the piping from the pressure relief valve to the location of discharge to the atmosphere and including any interconnections and piping that joins other tanks.

8.3 Vacuum protection systems

8.3.1 Cargo tanks not designed to withstand a maximum external pressure differential 0.025 [MPa] (0.25 [bar]), or tanks that cannot withstand the maximum external pressure differential that can be attained at maximum discharge rates with no vapour return into the cargo tanks, or by operation of a cargo refrigeration system, or by thermal oxidation , are to be fitted with:

.1 two independent pressure switches to sequentially alarm and subsequently stop all suction of cargo liquid or vapour from the cargo tank, and refrigeration equipment if fitted, by suitable means at a pressure sufficiently below the maximum external designed pressure differential of the cargo tank; or

.2 vacuum relief valves with a gas flow capacity at least equal to the maximum cargo discharge rate per cargo tank, set to open at a pressure sufficiently below the external design differential pressure of the cargo tank;

8.3.2 Subject to the requirements of Sec.17, the vacuum relief valves are to admit an inert gas, cargo vapour or air to the cargo tank and are to be arranged to minimize the possibility of the entrance of water or snow. If cargo vapour is admitted, it is to be from a source other than the cargo vapour lines.

8.3.3 The vacuum protection system is to be capable of being tested to ensure that it operates at the prescribed pressure.

8.4 Sizing of pressure relieving system

8.4.1 Sizing of pressure relief valves

PRVs are to have a combined relieving capacity for each cargo tank to discharge the greater of the following with not more than a 20 per cent rise in cargo tank pressure above the MARVS:

8.4.1.1 The maximum capacity of the cargo tank inerting system if the maximum attainable working pressure of the cargo tank inerting system exceeds the MARVS of the cargo tanks; or

8.4.1.2 Vapours generated under fire exposure computed using the following formula:

$$Q = FGA^{0.82}\ [m^3/s]$$

where,

Q = minimum required rate of discharge of air at standard conditions of 273.15 K and 0.1013 [MPa] (1.013 [bar]);

F = fire exposure factor for different cargo tank types;

F = 1.0 for tanks without insulation located on deck;

F = 0.5 for tanks above the deck when insulation is approved by Administration / IRS. (Approval will be based on the use of a fireproofing material, the thermal conductance of insulation, and its stability under fire exposure);

F = 0.5 for uninsulated independent tanks installed in holds;

F = 0.2 for insulated independent tanks in holds (or uninsulated independent tanks in insulated holds);

F = 0.1 for insulated independent tanks in inerted holds (or uninsulated independent tanks in inerted, insulated holds);

F = 0.1 for membrane and semi-membrane tanks.

For independent tanks partly protruding through the open deck, the fire exposure factor is to be determined on the basis of the surface areas above and below deck.

G = Gas factor

$$G = \frac{12.4}{L\,D}\sqrt{\frac{ZT}{M}}$$

with:

T = temperature in kelvins (K) at relieving conditions, i.e. 120 per cent of the pressure at which the pressure relief valve is set;

L = latent heat of the material being vaporized at relieving conditions, in [kJ/kg];

D = constant based on relation of specific heats k, and is calculated as follows:

$$D = \sqrt{k\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}$$

Where:

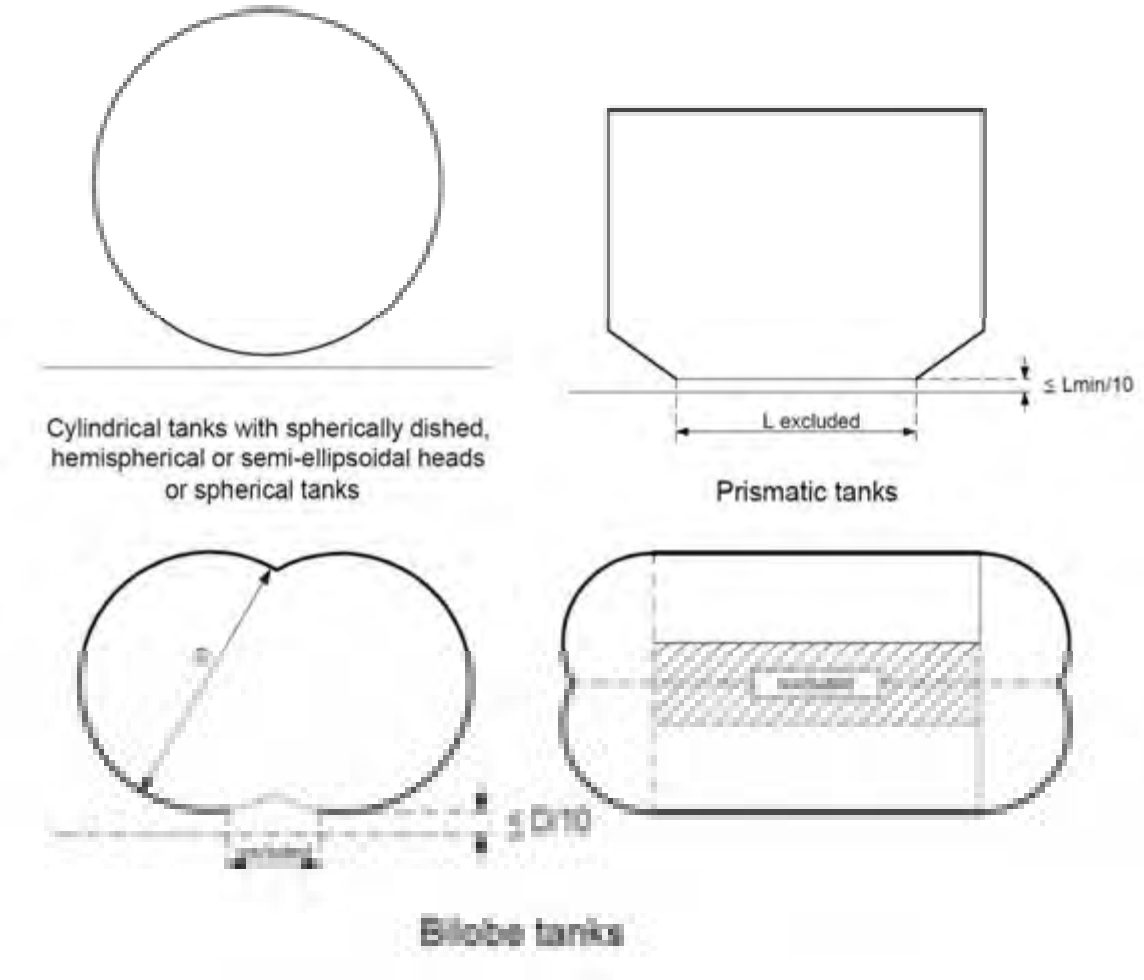
k = ratio of specific heats at relieving conditions and the valve of which is between 1 and 2.2. If k is not known, D=0.606 is to be used.

Z = compressibility factor of the gas at relieving conditions; if not known, Z = 1.0 should be used;

M = molecular mass of the product;

The gas factor of each cargo to be carried is to be determined and the highest value is to be used for PRV sizing.

A = external surface area of the tank [m^2] as defined in 1.2.14, for different tank types, as shown in Fig.8.1



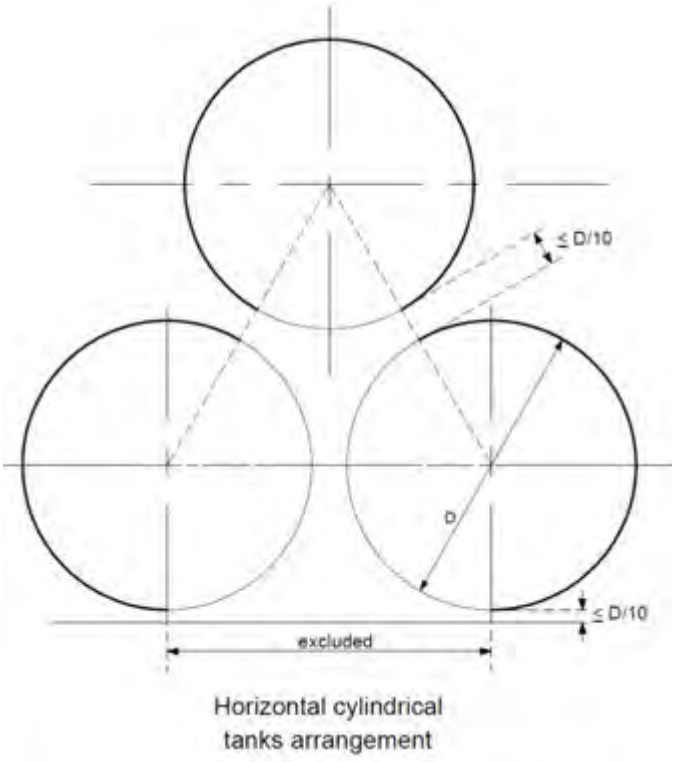


Figure 8.1

IR 8.4.1.2 For Prismatic Tanks:

.1 L_{min} , for non-tapered tanks, is the smaller of the horizontal dimensions of the flat bottom of the tank. For tapered tanks, as would be used for the forward tank, L_{min} is the smaller of the length and the average width.

.2 For prismatic tanks whose distance between the flat bottom of the tank and bottom of the hold space is equal to or less than $L_{min}/10$:

A=external surface area minus flat bottom surface area

.3 For prismatic tanks whose distance between the flat bottom of the tank and bottom of the hold space is greater than $L_{min}/10$:

A= external surface area.

8.4.1.3 The required mass flow of air at relieving conditions is given by the formula:

$$M_{air} = Q\ \rho_{air}\ \text{[Kg/s]},$$

where:

density of air (ρ_{air}) = 1.293 [Kg/m^3] (air at 273.15 [K], 0.1013 [MPa]).

8.4.2 Sizing of vent pipe system

Pressure losses upstream and downstream of the PRVs are to be taken into account when determining their size to ensure the flow capacity required by 8.4.1.

8.4.3 Upstream pressure losses

8.4.3.1 The pressure drop in the vent line from the tank to the PRV inlet is not to exceed 3% of the valve set pressure at the calculated flow rate, in accordance with 8.4.1.

8.4.3.2 Pilot-operated PRVs are to be unaffected by inlet pipe pressure losses when the pilot senses directly from the tank dome.

8.4.3.3 Pressure losses in remotely sensed pilot lines are to be considered for flowing type pilots.

8.4.4 Downstream pressure losses

8.4.4.1 Where common vent headers and vent masts are fitted, calculations are to include flow from all attached PRVs.

8.4.4.2 The built-up back pressure in the vent piping from the PRV outlet to the location of discharge to the atmosphere, and including any vent pipe interconnections that join other tanks, are not to exceed the following values:

.1 for unbalanced PRVs: 10% of MARVS;

.2 for balanced PRVs: 30% of MARVS; and

.3 for pilot operated PRVs: 50% of MARVS.

Alternative values provided by the PRV manufacturer may be accepted.

8.4.5 To ensure stable PRV operation, the blow-down is not to be less than the sum of the inlet pressure loss and 0.02 MARVS at the rated capacity.

IR 8.5 Sizing of pressure relief systems for interbarrier spaces

IR 8.5.1 General

IR 8.5.1.1 The formula for determining the relieving capacity given in IR 8.5.2 is for interbarrier spaces surrounding independent type A cargo tanks, where the thermal insulation is fitted to the cargo tanks.

IR 8.5.1.2 The relieving capacity of pressure relief devices of interbarrier spaces surrounding independent type B cargo tanks may be determined on the basis of the method given in IR 8.5.2, however, the leakage rate is to be determined in accordance with Sec 4, 4.7.2.

IR 8.5.1.3 The relieving capacity of pressure relief devices for interbarrier spaces of membrane and semimembrane tanks is to be evaluated on the basis of specific membrane/semi-membrane tank design.

IR 8.5.1.4 The relieving capacity of pressure relief devices for interbarrier spaces adjacent to integral type cargo tanks may, if applicable, be determined as for type A independent cargo tanks.

IR 8.5.2 Size of pressure relief devices

IR 8.5.2.1 The combined relieving capacity of the pressure relief devices for interbarrier spaces surrounding type A independent cargo tanks where the insulation is fitted to the cargo tanks may be determined by the following formula:

$$Q_{sa} = 3.4\, A_c\, \frac{\rho}{\rho_v}\, \sqrt{h} \ \text{[m}^3\text{/s]}$$

Where:

Q_{sa}=minimum required discharge rate of air at standard conditions of 273 [K] and 1.013 [bar]

A_c = design crack opening area [m^2]

$A_c = \frac{\pi}{4}\delta . l$ [m^2]

δ = max, crack opening width [m]

δ = 0.2 t [m]

t = thickness of tank bottom plating [m]

l = design crack length [m] equal to the diagonal of the largest plate panel of the tank bottom (See Fig IR 8.5.2)

h = max liquid height above tank bottom plus 10 x MARVS [m]

ρ = density of product liquid phase [kg/m^3] at the set pressure of the interbarrier space relief device.

ρ_v = density of product vapour phase [kg/m^3] at the set pressure of the inter barrier space relief device and a temperature of 273 K

MARVS = max allowable relief valve setting of the cargo tank [bar]

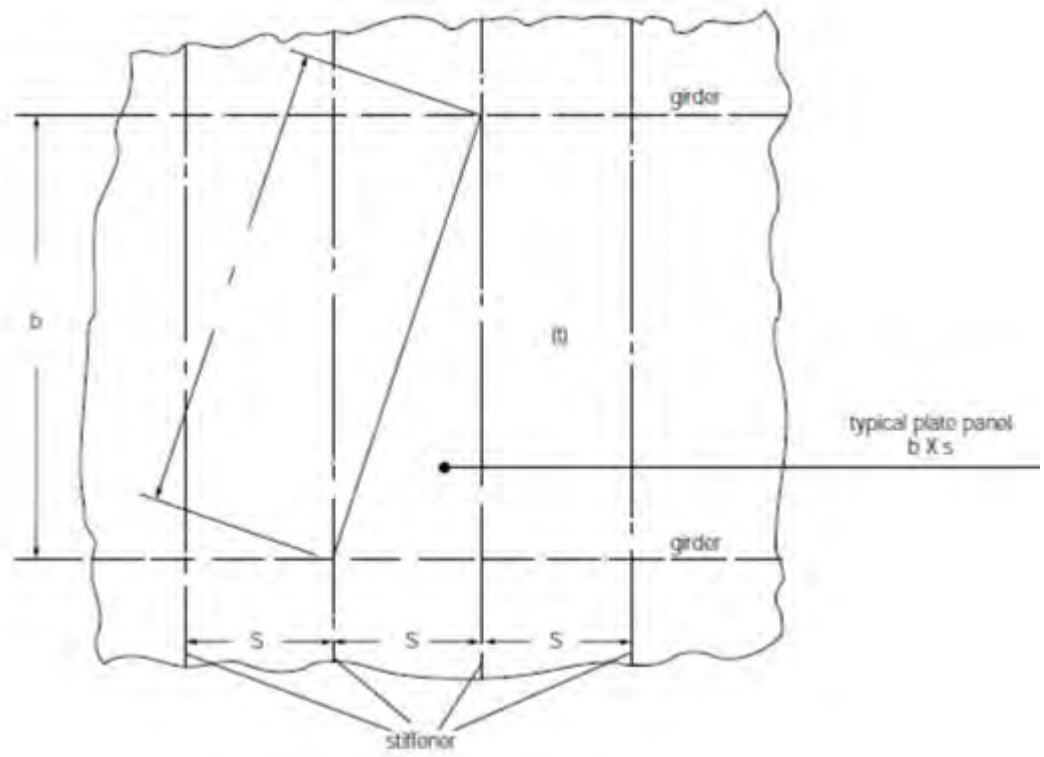


Fig : IR 8.5.2

Section 9

Cargo Containment System Atmosphere Control

9.0 Goal

To enable monitoring of the integrity of the containment system and to ensure that the atmosphere within the system and hold spaces is maintained in a safe condition at all times that the ship is in service.

9.1 Atmosphere control within the cargo containment system

9.1.1 A piping system is to be arranged to enable each cargo tank to be safely gas-freed, and to be safely filled with cargo vapour from a gas-free condition. The system is to be arranged to minimize the possibility of pockets of gas or air remaining after changing the atmosphere.

9.1.2 For flammable cargoes, the system is to be designed to eliminate the possibility of a flammable mixture existing in the cargo tank during any part of the atmosphere change operation by utilizing an inerting medium as an intermediate step.

9.1.3 Piping systems that may contain flammable cargoes is to comply with 9.1.1 and 9.1.2.

9.1.4 A sufficient number of gas sampling points are to be provided for each cargo tank and cargo piping system to adequately monitor the process of atmosphere change. Gas sampling connections are to be fitted with a single valve above the main deck, sealed with a suitable cap or blank. (See also 5.6.5.5)

9.1.5 Inert gas utilized in these procedures may be provided from the shore or from the ship.

9.2 Atmosphere control within the hold spaces (cargo containment systems other than Type C independent tanks)

9.2.1 Interbarrier and hold spaces associated with cargo containment systems for flammable gases requiring full or partial secondary barriers are to be inerted with a suitable dry inert gas and kept inerted with make-up gas provided by a shipboard inert gas generation system, or by shipboard storage, which is to be sufficient for normal consumption for at least thirty days.

9.2.2 Alternatively, subject to the restrictions specified in Sec.17, the spaces referred to in 9.2.1 requiring only a partial secondary barrier may be filled with dry air provided that the ship maintains a stored charge of inert gas or is fitted with an inert gas generation system sufficient to inert the largest of these spaces and provided that the configuration of the spaces and the relevant vapour detection systems, together with the capability of the inerting arrangements, ensure that any leakage from the cargo tanks will be rapidly detected and inerting effected before a dangerous condition can develop. Equipment for the provision of sufficient dry air of suitable quality to satisfy the expected demand is to be provided.

9.2.3 For non-flammable gases, the spaces referred to in 9.2.1 and 9.2.2 may be maintained with a suitable dry air or inert atmosphere.

9.3 Environmental control of spaces surrounding Type C independent tanks

Spaces surrounding cargo tanks that do not have secondary barriers are to be filled with suitable dry inert gas or dry air and be maintained in this condition with make-up inert gas provided by a shipboard inert gas generation system, shipboard storage of inert gas, or with dry air provided by suitable air drying equipment. If the cargo is carried at ambient temperature, the requirement for dry sir or inert gas is not applicable.

9.4 Inerting

9.4.1 Inerting refers to the process of providing a non-combustible environment. Inert gases are to be compatible chemically and operationally, at all temperatures likely to occur within the spaces and the cargo. The dew points of the gases are to be taken into consideration.

9.4.2 Where inert gas is also stored for fire-fighting purposes, it is to be carried in separate containers and is not to be used for cargo services.

9.4.3 Where inert gas is stored at temperature below 0 [°C], either as a liquid or as a vapour, the storage and supply system is to be designed so that the temperature of the ship's structure is not reduced below the limiting values imposed on it.

9.4.4 Arrangements to prevent the back flow of cargo vapour into the inert gas system that are suitable for the cargo carried, are to be provided. If such plants are located in machinery spaces or other spaces outside the cargo area, two non-return valves or equivalent devices and, in addition, a removable spool piece is to be fitted in the inert gas main in the cargo area. When not in use, the inert gas system is to be made separate from the cargo system in the cargo area except for connections to hold spaces or interbarrier spaces.

9.4.5 The arrangements are to be such that each space being inerted can be isolated and the necessary controls and relief valves etc. are to be provided for controlling pressure in these spaces.

9.4.6 Where insulation spaces are continually supplied with an inert gas as part of a leak detection system, means are to be provided to monitor the quantity of gas being supplied to individual spaces.

9.5 Inert gas production on board

9.5.1 The equipment is to be capable of producing inert gas with an oxygen content at no time greater than 5 per cent by volume subject to the Special Requirements of Sec.17. A continuous-reading oxygen content meter is to be fitted to the inert gas supply from the equipment and is to be fitted with an alarm set at a maximum of 5 per cent oxygen content by volume subject to the requirements of Sec.17.

9.5.2 An inert gas system is to have pressure controls and monitoring arrangements appropriate to the cargo containment system.

9.5.3 Spaces containing inert gas generation plants are to have no direct access to accommodation spaces, service spaces or control stations, but may be located in machinery spaces. Inert gas piping is not to pass through accommodation spaces, service spaces or control stations.

9.5.4 Combustion equipment for generating inert gas is not to be located within the cargo area. Special consideration may be given to the location of inert gas generating equipment using the catalytic combustion process.

Section 10

Electrical Installations

10.0 Goal

To ensure that electrical installations are designed such as to minimize the risk of fire and explosion from flammable products, and that electrical generation and distribution systems relating to the safe carriage, handling and conditioning of cargo liquid and vapour are available.

10.1 Definitions

For the purpose of this section, unless expressly provided otherwise, the definitions below shall apply.

10.1.1 *Hazardous area* is an area in which an explosive gas atmosphere is or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of electrical apparatus.

10.1.1.1 *Zone 0 hazardous area* is an area in which an explosive gas atmosphere is present continuously or is present for long periods.

10.1.1.2 *Zone 1 hazardous area* is an area in which an explosive gas atmosphere is likely to occur in normal operation.

10.1.1.3 *Zone 2 hazardous area* is an area in which an explosive gas atmosphere is not likely to occur in normal operation and, if it does occur, is likely to do so infrequently and for a short period only.

10.1.2 *Non-hazardous area* is an area in which an explosive gas atmosphere is not expected to be present in quantities such as to require special precautions for the construction, installation and use of electrical apparatus.

10.2 General requirements

IR 10.2.1 The provisions of this Section are applicable to ships carrying flammable products and are in addition to the requirements of Pt.4, Ch.8 of the Rules.

10.2.1 Electrical installations are to be such as to minimize the risk of fire and explosion from flammable products.

10.2.2 Electrical installations are to in accordance with recognized standards (such as IEC 60092-502:1999).

10.2.3 Electrical equipment and/or wiring is not to be installed in hazardous areas, unless essential for operational purposes or safety enhancement.

10.2.4 Where electrical equipment is installed in hazardous areas as provided in 10.2.3, it is to be selected, installed and maintained in accordance with standards not inferior to those acceptable to the IMO. Equipment for hazardous areas is to be evaluated and certified or listed by an accredited testing authority or notified body recognized by the Administration. Automatic isolation of non-certified equipment on detection of a flammable gas is not to be accepted as an alternative to the use of certified equipment.

IR 10.2.4 Where the prescriptive requirements within SOLAS and IGC Code (this Chapter) and the standards published by the International Electrotechnical Commission, such as but not limited to IEC 60092-502, are not aligned, the prescriptive requirements in SOLAS and IGC Code (this Chapter) take precedence and are to be applied.

10.2.5 To facilitate the selection of appropriate electrical apparatus and the design of suitable electrical installations, hazardous areas are divided into zones in accordance with recognized standards.

10.2.6 Electrical generation and distribution systems, and associated control systems are to be designed such that a single fault will not result in the loss of ability to maintain cargo tank pressures, as required by 7.8.1, and hull structure temperature, as required by 4.19.1.6, within normal operating limits. Failure modes and effects are to be analysed and documented to a standard not inferior to those acceptable to the Administration/ IRS (such as IEC 60812, edition 2.0 2006-01 “Analysis techniques for system reliability – Procedure for failure mode and effects analysis (FMEA)”).

10.2.7 The lighting system in hazardous areas is to be divided between at least two branch circuits. All switches and protective devices are to interrupt all poles or phases and are to be located in a non-hazardous area.

10.2.8 Electrical depth sounding or log devices and impressed current cathodic protection system anodes or electrodes are to be housed in gastight enclosures.

10.2.9 Submerged cargo pump motors and their supply cables may be fitted in cargo containment systems. Arrangements are to be made to automatically shut down the motors in the event of low-liquid level. This may be accomplished by sensing low pump discharge pressure, low motor current or low liquid level. This shutdown is to be alarmed at the cargo control station. Cargo pump motors are to be capable of being isolated from their electrical supply during gas-freeing operations.

Section 11

Fire Protection and Extinction

11.0 Goal

To ensure that suitable systems are provided to protect the ship and crew from fire in the cargo area.

11.1 Fire safety requirements

11.1.1 The requirements for tankers in Chapter II-2 of SOLAS are to apply to ships covered by this Chapter, irrespective of the tonnage including ships of less than 500 tons gross tonnage, except that:

.1 regulation 4.5.1.6 and 4.5.10 (Rules Pt.6, Ch.2, Cl.1.5.1.6 and Cl.1.5.10.1) does not apply;

.2 regulations 10.4 and 10.5 (Rules Pt.6, Ch.3, Cl.4.4 and Cl.4.5) are to apply as they would apply to tankers of 2000 tons gross and over;

.3 regulation 10.5.6 (Rules Pt.6, Ch.3, Cl.4.5.6) is to apply to ships of 2000 gross tonnage and over.

.4 the following regulations of Chapter II-2 of SOLAS related to tankers do not apply and are replaced by clauses/sections of this Chapter as under -

Regulation/(Rules)	Replaced by
10.10 (Pt.6, Cl.3, Cl.4.10)	11.6
4.5.1.1 and 4.5.1.2 (Pt.6, Ch.2, Cl.1.5.1.1 and Cl.1.5.1.2)	Sec.3
4.5.5 and 10.8 (Pt.6, Ch.2, Cl.1.5.5 and Pt.6, Ch.3, Cl.4.8)	Relevant Sections in this chapter 11.3 and 11.4
10.9 (Pt.6, Ch.3, Cl4.9)	11.5
10.2	11.2.1 to 11.2.4

.5 regulations 13.3.4 and 13.4.3 shall apply to ships of 500 gross tonnage and over.

11.1.2 All sources of ignition are to be excluded from spaces where flammable vapour may be present except otherwise provided in Secs. 10 and 16.

11.1.3 The provisions of the Section apply in conjunction with Sec.3.

11.1.4 For the purpose of fire fighting, any open deck areas above cofferdams, ballast or void spaces at the after end of the aftermost hold space or at the forward end of the forward most hold space are to be included in the cargo area. (Also refer to IR 11.3.6)

11.2 Fire mains and hydrants

11.2.1 Irrespective of size, ships carrying products that are subject to this chapter are to comply with requirements of regulations II-2/10.2 (Rules Pt.6, Ch.3, Cl.4.2) of the SOLAS convention, as applicable to the cargo ships, except that the required fire pump capacity and fire main and water service pipe diameter is not to be limited by the provisions of regulations II-2/10.2.2.4.1 and II-2/10.2.1.3 (Rules Pt.6, Ch.3, Cl.4.2.2.4.1 and Cl.4.2.1.3), when a fire pump is used to supply the water spray system as permitted by 11.3.3. The capacity of this fire pump is to be such that these areas can be protected when simultaneously supplying two jets of water from fire hoses with 19 [mm] nozzles at a pressure of at least 0.5 [MPa] gauge.

11.2.2 The arrangements are to be such that at least two jets of water can reach any part of the deck in the cargo area and those portions of the cargo containment system and tank covers that are above the deck. The necessary number of fire hydrants are to be located to satisfy the above arrangements and to comply with the requirements of Regulations II-2/10.2.1.5.1 and II-2/10.2.3.3 (Rules Pt.6, Ch.3, Cl.4.2.3.3) of SOLAS with hose lengths as specified in II-2/10.2.3.1.1 (Rules Pt.6, Ch.3, Cl.4.2.3.1.1). In addition, the requirements of regulation II-2/10.2.1.6 (Rules Pt6, Ch 3, Cl 4.2.1.6) of SOLAS is to be meet at a pressure of at least 0.5 [MPa] gauge.

11.2.3 Stop valves are to be fitted in any cross-over provided and in the fire main(s) in a protected location before entering the cargo area and at intervals ensuring isolation of any damaged single section of the fire main, so that 11.2.2 can be complied with using not more than two lengths of hoses from the nearest fire hydrant. The water supply to the fire main serving the cargo area is to be a ring main supplied by the main fire pumps or a single main supplied by fire pumps positioned fore and aft of the cargo area, one of which is to be independently driven.

11.2.4 Nozzles are to be of an approved dual-purpose type (i.e spray/jet type) incorporating a shutoff.

11.2.5 After installation, the pipes, valves, fittings and assembled system are to be subject to a tightness and function test.

IR11.2 Maximum capacity calculation for emergency pump:

.1 In 11.3.4, the term "fire pumps" where not qualified by the word "emergency" refers to the fire pumps required in accordance with Pt 6, Ch. 3, Cl. 4.2.2.2.

.2 If all the fire pumps mentioned in .1 above supplying the water spray system (for covering the superstructures and deckhouses) are disabled due to a fire in any one compartment; then the emergency fire pump is to be sized to cover:

a) the water spray system for the boundaries of the superstructures and deckhouses, and lifeboats, liferafts and muster areas facing the cargo area, (as per 11.3.4); and

b) two fire hydrants (as per 11.2).

.3 When the ship is also fitted with a total flooding high expansion foam system protecting the engine-room (to comply with Pt. 6, Ch. 3, Cl, 4.4.1.1.2 and 4.5.1.1) and the emergency fire pump is intended to supply sea water to this system, then, the emergency fire pump is also to be sized to cover the foam system for dealing with an engine-room fire, when the main fire pumps are disabled.

.4 On the basis of the principle of dealing with one single fire incident at a time, the emergency fire pump does not need to be sized to cover all three systems in .2 and .3 above (i.e. water spray, hydrants and foam) at the same time and need only be sized to cover the most demanding area and required systems, as follows:

.1 the foam system and two hydrants; or

.2 the water spray system and two hydrants;

whichever is greater.

11.3 Water spray system

11.3.1 On ships carrying flammable and/or toxic products, a water spray system for cooling, fire prevention and crew protection is to be installed to cover:

.1 exposed cargo tank domes, any exposed parts of cargo tanks and any part of cargo tank covers that may be exposed to heat from fires in adjacent equipment containing cargo such as exposed booster pumps/heaters/re-gasification or re-liquefaction plants, hereafter addressed as gas process units, positioned on weather decks;

.2 exposed on-deck storage vessels for flammable or toxic products;

.3 gas process units positioned on deck;

.4 cargo liquid and vapour discharge and loading connections, including the presentation flange and the area where their control valves are situated, which are to be at least equal to the area of the drip trays provided;

.5 all exposed emergency shut-down (ESD) valves in the cargo liquid and vapour pipes, including the master valve for supply to gas consumers;

.6 exposed boundaries facing the cargo area, such as bulkheads of superstructures and deckhouses normally manned, cargo machinery spaces, store-rooms containing high fire-risk items and cargo control rooms. Exposed horizontal boundaries of these areas do not require protection unless detachable cargo piping connections are arranged above or below. Boundaries of unmanned forecastle structures not containing high fire risk items or equipment do not require water spray protection.

.7 exposed lifeboats, liferafts and muster stations facing the cargo area, regardless of distance to cargo area; and

IR.7 With reference to para .7 above, the survival crafts on board including remote survival crafts (ref. SOLAS III/Reg. 31.1.4) facing the cargo area are to be protected by a water-spray system taking into consideration cargo area extension for fire-fighting purposes as stated in 11.1.4. Remote liferafts located in areas covered by water-spray protection as required in para .6 may be considered as adequately protected.

.8 any semi-enclosed cargo machinery spaces and semi-enclosed cargo motor room.

IR11.3.1.7 Water spray protection should be considered for exposed embarkation stations and exposed launching routes from the life rafts stowage location to the ship side unless the life rafts are located and ready for launching at both sides.

Ships intended for operation as listed in 1.1.10 are to be subject to special consideration (see 11.3.3.2).

11.3.2.1 The system is to be capable of covering all areas mentioned in 11.3.1.1 to 11.3.1.8, with a uniformly distributed water application rate of at least 10 [l/m^2] per minute for the largest projected horizontal surfaces and 4 [l/m^2] per minute for vertical surfaces. For structures having no clearly defined horizontal or vertical surfaces, the capacity of the water-spray system is not to be less than the projected horizontal surface multiplied by 10 [l/m^2] per minute.

11.3.2.2 On vertical surfaces, spacing of nozzles protecting lower areas may take account of anticipated rundown from higher areas. Stop valves are to be fitted in the main supply lines(s) in the water spray system, at intervals not exceeding 40 [m], for the purpose of isolating damaged sections. Alternatively, the system may be divided into two or more sections which may be operated independently provided the necessary controls are located together in the readily accessible locations outside the cargo areas. A section protecting any area included in 11.3.1.1 and .2 is to cover at least the entire athwartship tank grouping in that area. Any gas process unit(s) included in 11.3.1.3 may be served by an independent section.

11.3.3 The capacity of the water spray pumps is to be capable of simultaneous protection of the greater of the following:

.1 any two complete athwartship tank groupings, including any gas process units within these areas; or

.2 for ships intended for operation as listed in 1.1.10, necessary protection subject to special consideration under 11.3.1 of any added fire hazard and the adjacent athwartship tank grouping, in addition to the surfaces specified in 11.3.1.4 to 11.3.1.8.

Alternatively, the main fire pumps may be used for this service provided that their total capacity is increased by the amount needed for the water spray system. In either case, a connection, through a stop valve, is to be made between the fire main and water spray system main supply line outside the cargo area.

IR11.3.3 The expression “two complete athwartship tank groupings” in 11.3.3.1 above means any two groups of tanks where one group is defined as tanks located in transverse direction from ship side to ship side. Where there is only one cargo tank occupying a hold space from ship side to ship side, it will be considered as a “grouping” for the purpose of this interpretation. “Any two complete athwartship tank groupings” represents an area equal to the combined area of the two largest tank groupings including any gas process units within these areas

11.3.4 The boundaries of superstructures and deckhouses normally manned, and lifeboats, liferafts and muster areas facing the cargo area, shall also be capable of being served by one of the fire pumps or the emergency fire pump, if a fire in one compartment could disable both fire pumps.

IR11.3.4 Fire pumps used as spray pumps: In cases where the emergency fire pump is used to meet this requirement, its capacity, in addition to being capable of maintaining two jets of water as required by Pt. 6, Ch. 8, Cl. 12.2.2.1.1 is to be increased taking into account the spray application rates stated in 11.3.2.1, but limiting coverage to boundaries of normally manned superstructures and deckhouses, survival crafts and their muster areas.

For the purpose of this interpretation:

.1 the expression "one of the fire pumps or emergency fire pump" is related to fire pumps required by Pt.6, Ch. 3, Cl. 4.2.2, installed outside the space where spray pump(s) are located; and

.2 the expression "fire in one compartment" means a compartment provided with A-Class boundaries in which is located the fire pump(s), or the source of power of the fire pump(s), serving the water-spray system in accordance with 11.3.3.

Note: Also see IR11.2 for interpretation regarding maximum capacity calculation for emergency fire pump.

11.3.5 Water pumps normally used for other services may be arranged to supply the water-spray system main supply line.

11.3.6 All pipes, valves, nozzles and other fittings in the water-spray system are to be resistant to corrosion by seawater. Piping, fittings and related components within the cargo area (except gaskets) are to be designed to withstand 925 [°C]. The water-spray system is to be arranged with in-line filters to prevent blockage of pipes and nozzles. In addition, means are to be provided to back-flush the system with fresh water.

IR 11.3.6 Where 'F.O. tanks' are installed at the after end of the aftermost hold space or at the forward end of the forwardmost hold space instead of cofferdams as allowed in 3.1.2 and 3.1.3, the weather deck area above these tanks are to be regarded as a 'cargo area' for the purpose of applying 11.3.6 above.

11.3.7 Remote starting of pumps supplying the water-spray system and remote operation of any normally closed valves in the system is to be arranged in suitable locations outside the cargo area, adjacent to the accommodation spaces and readily accessible and operable in the event of fire in the protected areas.

11.3.8 After installation, the pipes, valves, fittings and assembled system are to be subject to a tightness and function test.

11.4 Dry chemical powder fire extinguishing systems

11.4.1 Ships in which carriage of flammable products is intended are to be fitted with fixed dry chemical powder fire-extinguishing systems approved by the Administration/ IRS based on MSC .1/Circ 1315/Rev.1 for the purpose of fighting fire on the deck in the cargo area, including any cargo liquid and vapour discharge and loading connections on deck and bow or stern cargo handling areas, as applicable.

11.4.2 The system is to be capable of delivering powder from at least two hand hose lines or combination monitor/hand hose line(s) to any part of the exposed cargo liquid and vapour piping, load/unload connection and exposed gas process units.

11.4.3 The dry chemical powder fire-extinguishing system is to be designed with not less than two independent units. Any part required to be protected by 11.4.2 is to be capable of being reached form not less than two independent units with associated controls, pressurizing median fixed piping, monitors or hand hose lines. For ships with a cargo capacity of less than 1000 [m^3], only one such unit need be fitted. A monitor is to be arranged to protect any load/unload connection area and be capable of actuation and discharge both locally and remotely. The monitor is not required to be remotely aimed if it can deliver the necessary powder to all required areas of coverage from a single position. One hose line is to be provided at both port- and starboard side at the end of the cargo area facing the accommodation and readily available from the accommodation.

11.4.4 The capacity of a monitor is to be not less than 10 [kg/sec]. Hand hose lines are to be non-kinkable and be fitted with a nozzle capable of on/off operation and discharge at a rate not less than 3.5 [kg/sec]. The maximum discharge rate is to be such as to allow operation by one man. The length of a hand hose line is not to exceed 33 [m]. Where fixed piping is provided between the powder container and a hand hose line or monitor, the length of piping is not to exceed that length which is capable of maintaining the powder in a fluidized state during sustained or intermittent use, and which can be purged of powder when the system is shut down. Hand hose lines and nozzles are to be of weather-resistant construction or stored in weather-resistant housing or covers and be readily accessible.

11.4.5 Hand hose lines are to be considered to have a maximum effective distance of coverage equal to the length of hose. Special consideration would be given where areas to be

protected are substantially higher than the monitor or hand hose reel locations.

11.4.6 Ships fitted with bow/stern load/unload connections are to be provided with independent dry powder unit protecting the cargo liquid and vapour piping, aft and forward of cargo area, by hose lines and a monitor covering the bow/stern load/unload complying with the requirements of 11.4.1 to 11.4.5.

11.4.7 Ships intended for operation as listed 1.1.10 are to be subject to special consideration.

11.4.8 After installation, the pipes, valves, fittings and assembled systems are to be subjected to a tightness test and functional testing of the remote and local release stations. The initial testing is also to include a discharge of sufficient amounts of dry chemical powder to verify that the system is in proper working order. All distribution piping is to be blown through with dry air to ensure that the piping is free of obstructions.

IR11.4.8 Testing arrangements are to involve the discharge using dry chemical powder from all monitors and hand hose lines on board, but it is not required that there is a full discharge of the installed quantity of dry powder. This testing can also be used to satisfy the requirement that the piping is free of obstructions, in lieu of blowing through with dry air all the distribution piping. However, after the completion of this testing, the system, including all monitors and hand hose lines, are to be blown through with dry air; but only for the purpose of the system subsequently being clear from any residues of dry chemical powder.

11.5 Enclosed spaces containing cargo handling equipment

11.5.1 Enclosed spaces meeting the criteria of cargo machinery spaces in 1.2.10, and the cargo motor room within the cargo area of any ship, are to be provided with a fixed fire-extinguishing system complying with the provisions of the FSS Code and taking into account the necessary concentrations/application rate required for extinguishing gas fires.

11.5.2 Enclosed spaces meeting the criteria of cargo machinery spaces in Sec 3.3, within the cargo area of ships that are dedicated to the carriage of a restricted number of cargoes, are to be protected by an appropriate fire-extinguishing system for the cargo carried.

11.5.3 Turret compartments of any ship are to be protected by internal water spray, with an application rate of not less than 10 [ℓ/m^2/min] of the largest projected horizontal surface. If the pressure of the gas flow through the turret exceeds 4 [MPa], the application rate is to be increased to 20 [ℓ/m^2/min]. The system is to be designed to protect all internal surfaces.

11.6 Fire Fighter's Outfits

IR11.6.1 Every ship carrying flammable products is to carry firefighter's outfits complying with the requirements of Part 4.

11.6.1 Every ship carrying flammable products is to carry firefighter's outfits complying with the requirements of regulation II-2/10.10 of SOLAS (Rules Pt.6, Ch.3, Cl.4.10) as follows:

Total cargo capacity	Number of outfits
5000 [m^3] and below	4
above 5000 [m^3]	5

11.6.2 Additional requirements for safety equipment are given in Sec.14.

11.6.3 Any breathing apparatus required as part of a fireman's outfit is to be a self-contained compressed air-operated breathing apparatus having a capacity of at least 1,200 litres of free air.

Section 12

Artificial Ventilation in Cargo Area

Note: The requirements of this Chapter replace regulation II-2/4.5.2.6 and II-2/4.5.4.1 of SOLAS (Rules Pt.6, Ch.2, Cl.1.5.2.6 and 1.5.4).

12.0 Goal

To ensure that arrangements are provided for enclosed spaces in the cargo area to control the accumulation of flammable and/or toxic vapours.

12.1 Spaces required to be entered during normal cargo handling operations

12.1.1 Electric motor rooms, cargo compressor and pump-rooms, spaces containing cargo handling equipment and other enclosed spaces where cargo vapours may accumulate are to be fitted with fixed artificial ventilation systems capable of being controlled from outside such spaces. The ventilation is to be run continuously to prevent the accumulation of toxic and/or flammable vapours, with a means of monitoring acceptable to the Administration/ IRS to be provided. A warning notice requiring the use of such ventilation prior to entering is to be placed outside the compartment.

12.1.2 Artificial ventilation inlets and outlets are to be arranged to ensure sufficient air movement through the space to avoid accumulation of flammable, toxic or asphyxiant vapours and to ensure a safe working environment.

12.1.3 The ventilation system is to have a capacity of not less than 30 changes of air per hour, based upon the total volume of the space. As an exception, non-hazardous cargo control rooms may have eight changes of air per hour.

12.1.4 Where a space has an opening into an adjacent more hazardous space or area, it is to be maintained at an overpressure. It may be made into a less hazardous space or non-hazardous space by overpressure protection in accordance with recognized standards.

12.1.5 Ventilation ducts, air intakes and exhaust outlets serving artificial ventilation systems are to be positioned in accordance with IEC 60092-502:1999.

12.1.6 Ventilation ducts serving hazardous areas are not to be led through accommodation, service and machinery spaces or control stations, except as allowed in Sec.16.

12.1.7 Electric motors driving fans are to be placed outside the ventilation ducts that may contain flammable vapours. Ventilation fans are not to produce a source of ignition in either the ventilated space or the ventilation system associated with the space. For hazardous areas, ventilation fans and ducts, adjacent to the fans are to be of non-sparking construction, as defined below:

.1 impellers or housing of non-metallic construction, with due regard being paid to the elimination of static electricity;

.2 impellers and housing of non-ferrous materials;

.3 impellers and housing of austenitic stainless steel; and

.4 ferrous impellers and housing with design tip clearance of not less than 13 [mm].

Any combination of an aluminium or magnesium alloy fixed or rotating component and a ferrous fixed or rotating component, regardless of tip clearance, is considered a sparking hazard and are not to be used in these places.

12.1.8 Where fans are required by this section, full required ventilation capacity for each space is to be available after failure of any single fan, or spare parts are to be provided comprising a motor, starter spares and complete rotating element, including bearings of each type.

12.1.9 Protection screens of not more than 13 [mm] square mesh are to be fitted to outside openings of ventilation ducts.

12.1.10 Where spaces are protected by pressurization, the ventilation shall be designed and installed in accordance with IEC 60092-502:1999.

12.2 Spaces not normally entered

12.2.1 Enclosed spaces where cargo vapours may accumulate are to be capable of being ventilated to ensure a safe environment when entry into them is necessary. This is to be capable of being achieved without the need for prior entry.

12.2.2 For permanent installations, the capacity of 8 air changes per hour is to be provided and for portable systems, the capacity of 16 air changes per hour.

12.2.3 Fans or blowers are to be clear of personnel access openings, and are to comply with 12.1.7.

Section 13

Instrumentation and Automation Systems

13.0 Goal

To ensure that the instrumentation and automation systems provides for the safe carriage, handling and conditioning of cargo liquid and vapour.

13.1 General

13.1.1 Each cargo tank is to be provided with means for indicating level, pressure and temperature of the cargo. Pressure gauges and temperature indicating devices are to be installed in the liquid and vapour piping systems, in cargo refrigeration installations.

13.1.2 If loading and unloading of the ship is performed by means of remotely controlled valves and pumps, all controls and indicators associated with a given cargo tank are to be concentrated in one control position.

13.1.3 Instruments are to be tested to ensure reliability in the working conditions and recalibrated at regular intervals. Test procedures for instruments and the intervals between recalibration are to be in accordance with manufacturer’s recommendations.

13.2 Level indicators for cargo tanks

13.2.1 Each cargo tank is to be fitted with liquid level gauging device(s), arranged to ensure that a level reading is always obtainable whenever the cargo tank is operational. The device(s) are to be designed to operate throughout the design pressure range of the cargo tank and at temperatures within the cargo operating temperature range.

13.2.2 Where only one liquid level gauge is fitted, it is to be arranged so that it can be maintained in an operational condition without the need to empty or gas-free the tank.

IR 13.2.2 In order to assess whether or not only one level gauge is acceptable in relation to the aforesaid sentence, the expression ‘can be maintained’ means that any part of the level gauge other than passive parts can be overhauled while the cargo tank is in service.

Note : passive parts are those parts assumed not subject to failures under normal service conditions.

13.2.3 Cargo tank liquid level gauges may be of the following types, subject to any special requirements for particular cargoes shown in column "g" in the table of Sec.19.

.1 indirect devices, which determine the amount of cargo by means such as weighing or in-line flow metering ;

.2 closed devices, which do not penetrate the cargo tank, such as devices using radioisotopes or ultrasonic devices;

.3 closed devices, which penetrate the cargo tank, but which form part of a closed system and keep the cargo from being released, such as float type systems, electronic probes, magnetic probes and bubble tube indicators. If a closed gauging device is not mounted directly onto the tank it is to be provided with a shut-off valve located as close as possible to the tank; and

.4 restricted devices, which penetrate the tank and when in use permit a small quantity of cargo vapour or liquid to escape to the atmosphere, such as fixed tube and slip tube gauges. When not in use, the devices are to be kept completely closed. The design and installation is to ensure that no dangerous escape of cargo can take place when opening the device. Such gauging devices are to be so designed that the maximum opening does not exceed 1.5 [mm] diameter or equivalent area, unless the device is provided with an excess flow valve.

IR.4 The openings of restricted devices are to be located on the weather deck and not in enclosed compartments.

13.3 Overflow control

13.3.1 Except as provided in 13.3.4, each cargo tank is to be fitted with a high liquid level alarm operating independently of other liquid level indicators and giving an audible and visual warning when activated.

13.3.2 An additional sensor operating independently of the high liquid level alarm is to automatically actuate a shutoff valve in a manner that will both avoid excessive liquid pressure in the loading line and prevent the tank from becoming liquid full.

13.3.3 The emergency shutdown valve referred to in 5.5 and 18.10 may be used for this purpose. If another valve is used for this purpose, the same information as referred to in 18.10.2.1.3 is to be available on board. During loading, whenever the use of these valves may possibly create a potential excess pressure surge in the loading system, alternative arrangements such as limiting the loading rate is to be used.

13.3.4 A high liquid level alarm and automatic shut-off of cargo tank filling need not be required when the cargo tank:

.1 is a pressure tank with a volume of not more than 200 [m^3]; or

.2 is designed to withstand the maximum possible pressure during the loading operation and such pressure is below that of the set pressure of the cargo tank relief valve.

13.3.5 The position of the sensors in the tank is to be capable of being verified before commissioning. At the first occasion of full loading after delivery and after each dry-docking, testing of high-level alarms is to be conducted by raising the cargo liquid level in the cargo tank to the alarm point.

IR13.3.5 The expression “each dry-docking” is considered to be the survey of the outside of the ship’s bottom required for the renewal of the Cargo Ship Safety Construction Certificate and or the Cargo Ship Safety Certificate; and for vessels below 500GT, during renewal of the Class Certificate.

13.3.6 All elements of the level alarms, including the electrical circuit and the sensor(s), of the high, and overfill alarms, are to be capable of being functionally tested. Systems are to be tested prior to cargo operation in accordance with 18.6.2.

13.3.7 Where arrangements are provided for overriding the overflow control system, they are to be such that inadvertent operation is prevented. When this override is operated, continuous visual indication is to be given at the relevant control station(s) and the navigation bridge.

13.4 Pressure monitoring

13.4.1 The vapour space of each cargo tank is to be provided with a direct reading gauge. Additionally, an indirect indication is to be provided at the control position required by 13.1.2. Maximum and minimum allowable pressures are to be clearly indicated.

13.4.2 A high-pressure alarm and, if vacuum protection is required, a low-pressure alarm is to be provided on the navigation bridge and at the control position required by 13.1.2. Alarms are to be activated before the set pressures are reached.

13.4.3 For cargo tanks fitted with PRVs which can be set at more than one set pressure in accordance with 8.2.7, high-pressure alarms are to be provided for each set pressure.

13.4.4 Each cargo-pump discharge line and each liquid and vapour cargo manifold is to be provided with at least one pressure indicator

13.4.5 Local reading manifold indication is to be provided to indicate the pressure between ship's manifold valves and hose connections to the shore.

13.4.6 Hold spaces and interbarrier spaces without open connection to the atmosphere are to be provided with pressure indication

13.4.7 All pressure indications provided are to be capable of indicating throughout the operating pressure range.

13.5 Temperature indicating devices

13.5.1 Each cargo tank is to be provided with at least two devices for indicating cargo temperatures, one placed at the bottom of the cargo tank and the second near the top of the tank, below the highest allowable liquid level. The lowest temperature for which the cargo tank has been designed, as shown on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk required by 1.4.4, is to be clearly indicated by means of a sign on or near the temperature indicating devices

13.5.2 The temperature indicating devices are to be capable of providing temperature indication across the expected cargo operating temperature range of the cargo tanks.

13.5.3 Where thermowells are fitted, they are to be designed to minimize failure due to fatigue in normal service.

13.6 Gas detection

13.6.1 Gas detection equipment is to be installed to monitor the integrity of the cargo containment, cargo handling and ancillary systems, in accordance with this section.

13.6.2 A permanently installed system of gas detection and audible and visual alarms is to be fitted in :

.1 all enclosed cargo and cargo machinery spaces (including turrets compartments) containing gas piping, gas equipment or gas consumers;

.2 other enclosed or semi-enclosed spaces where cargo vapours may accumulate, including interbarrier spaces and hold spaces for independent tanks other than type C tanks;

.3 airlocks;

.4 spaces in gas-fired internal combustion engines, referred to in 16.7.3.3;

.5 ventilation hoods and gas ducts required by Sec 16;

.6 cooling/heating circuits, as required by 7.8.4;

.7 inert gas generator supply headers; and

.8 motor rooms for cargo handling machinery.

13.6.3 Gas detection equipment is to be designed, installed and tested in accordance with IEC 60079-29-1 and is to be suitable for the cargoes to be carried in accordance with column "f" in table of sec 19.

13.6.4 Where indicated by an "A" in column "*f* " in the table of Sec 19 ships certified for carriage of non-flammable products, oxygen deficiency monitoring is to be fitted in cargo machinery spaces and hold spaces for independent tanks other than type C tanks. Furthermore, oxygen deficiency monitoring equipment are to be installed in enclosed or semi-enclosed spaces containing equipment that may cause an oxygen-deficient environment such as nitrogen generators, inert gas generators or nitrogen cycle refrigerant systems.

IR 13.6.4 Two oxygen sensors are to be positioned at appropriate locations in the space or spaces containing the inert gas system, in accordance with Pt.6, Ch.8, 15.2.2.4.5.4, for all gas carriers, irrespective of the carriage of cargo indicated by an "A" in column "f" in the table in Sec 19.

13.6.5 In the case of toxic products or both toxic and flammable products, except when columns 'i' in the table of section 19 refers to 17.5.3, portable equipment can be used for the detection of toxic products as an alternative to a permanently installed system. This equipment is to be used prior to personnel entering the spaces listed in 13.6.2 and 30-minute intervals while they remain in the space.

13.6.6 In the case of gases classified as toxic products, hold spaces and interbarrier spaces are to be provided with a permanently installed piping system for obtaining gas samples from the spaces. Gas from these spaces is to be

sampled and analysed from each sampling head location.

13.6.7 Permanently installed gas detection is to be of the continuous detection type, capable of immediate response. Where not used to activate safety shutdown functions required by 13.6.9 and Sec 16, sampling type detection may be accepted.

13.6.8 When sampling type gas detection equipment is used, the following requirements are to be met:

.1 the gas detection equipment is to be capable of sampling and analysing for each sampling head location sequentially at intervals not exceeding 30 min;

.2 individual sampling lines from sampling heads to the detection equipment is to be fitted; and

.3 pipe runs from sampling heads are not to be led through non-hazardous spaces except as permitted by 13.6.9.

13.6.9 The gas detection equipment may be located in a non-hazardous space, provided that the detection equipment such as sample piping, sample pumps, solenoids and analysing units are located in a fully enclosed steel cabinet with the door sealed by a gasket. The atmosphere within the enclosure is to be continuously monitored. At gas concentrations above 30% lower flammable limit (LFL) inside the enclosure, the gas detection equipment is to be automatically shut down.

13.6.10 Where the enclosure cannot be arranged directly on the forward bulkhead, sample pipes are to be of steel or equivalent material and be routed on their shortest way. Detachable connections, except for the connection points for isolating valves required in 13.6.11 and analysing units, are not permitted.

13.6.11 When gas sampling equipment is located in a non-hazardous space, a flame arrester and a manual isolating valve is to be fitted in each of the gas sampling lines. The isolating valve is to be fitted on the non-hazardous side. Bulkhead penetrations of sample pipes between hazardous and non-hazardous areas are to maintain the integrity of the division penetrated. The exhaust gas is to be discharged to the open air in a safe location.

13.6.12 In every installation, the number and the positions of detection heads are to be determined with due regard to the size and layout of the compartment, the compositions and densities of the products intended to be carried and the dilution from compartment purging or ventilation and stagnant areas.

13.6.13 Any alarms status within a gas detection system required by this section is to initiate an audible and visible alarm:

.1 on the navigation bridge;
.2 at the relevant control station(s) where continuous monitoring of the gas levels is recorded; and

.3 at the gas detector readout location.

13.6.14 In the case of flammable products, the gas detection equipment provided for hold spaces and interbarrier spaces that are required to be inerted are to be capable of measuring gas concentrations of 0% to 100% by volume.

13.6.15 Alarms are to be activated when the vapour concentration by volume reaches the equivalent of 30% LFL in air.

13.6.16 For membrane containment systems, the primary and secondary insulation spaces are to be able to be inerted and their gas content analysed individually (refer to '*Gas Concentration in the insulation Spaces of Membrane LNG Carriers, March 2007*' published by SIGTTO). The alarm in the secondary insulation space is to be set in accordance with 13.6.15, that in the primary space is set at a value approved by IRS.

13.6.17 For other spaces described by 13.6.2, alarms are to be activated when the vapour concentration reaches 30% LFL and safety functions required by Sec 16 are to be activated before the vapour concentration reaches 60% LFL. The crankcases of internal combustion engines that can run on gas are to be arranged to alarm before 100% LFL.

13.6.18 Gas detection equipment is to be so designed that it may readily be tested. Testing and calibration is to be carried out at regular intervals. Suitable equipment for this purpose shall be carried on board and be used in accordance with the manufacturer's recommendations. Permanent connections for such test equipment are to be fitted.

13.6.19 Every ship is to be provided with at least two sets of portable gas detection equipment that meet the requirement of 13.6.3 or an acceptable national or international standard.

13.6.20 A suitable instrument for the measurement of oxygen levels in inert atmospheres is to be provided.

13.7 Additional requirements for containment systems requiring a secondary barrier

13.7.1 Integrity of barriers

Where a secondary barrier is required, permanently installed instrumentation is to be provided to detect when the primary barrier fails to be liquid-tight at any location or when liquid cargo is in contact with the secondary barrier at any location. This instrumentation is to consist of appropriate gas detecting devices according to 13.6. However, the instrumentation need not be capable of locating the area where liquid cargo leaks through the primary barrier or where liquid cargo is in contact with the secondary barrier.

13.7.2 Temperature indication devices

13.7.2.1 The number and position of temperature-indicating devices are to be appropriate to the design of the containment system and cargo operation requirements.

13.7.2.2 When cargo is carried in a cargo containment system with a secondary barrier, at a temperature lower than -55 [°C], temperature-indicating devices are to be provided within the insulation or on the hull structure adjacent to cargo containment systems. The devices are to give readings at regular intervals and, where applicable, alarm of temperatures approaching the lowest for which the hull steel is suitable.

13.7.2.3 If cargo is to be carried at temperatures lower than -55 [°C], the cargo tank boundaries, if appropriate for the design of the cargo containment system, shall be fitted with a sufficient number of temperature-indicating devices to verify that unsatisfactory temperature gradients do not occur.

13.7.2.4 For the purposes of design verification and determining the effectiveness of the initial cooldown procedure on a single or series of similar ships, one tank is to be fitted with devices in excess of those required in 13.7.2.1. These devices may be temporary or permanent and only need to be fitted to the first ship, when a series of similar ships is built.

13.8 Automation systems

13.8.1 The requirements of this Sub-section shall apply where automation systems are used to provide instrumented control, monitoring/alarm or safety functions required by this Chapter.

13.8.2 Automation systems are to be designed, installed and tested in accordance with IEC 60092-504:2001.

13.8.3 Hardware is to be capable of being demonstrated to be suitable for use in the marine environment by type approval or other means.

13.8.4 Software is to be designed and documented for ease of use, including testing, operation and maintenance.

13.8.5 The user interface is to be designed such that the equipment under control can be operated in a safe and effective manner at all times.

13.8.6 Automation systems are to be arranged such that a hardware failure or an error by the operator does not lead to an unsafe condition. Adequate safeguards against incorrect operation are to be provided.

13.8.7 Appropriate segregation is to be maintained between control, monitoring/alarm and safety functions to limit the effect of single failures. This shall be taken to include all parts of the automation systems that are required to provide specified functions, including connected devices and power supplies.

13.8.8 Automation systems are to be arranged such that the software configuration and parameters are protected against unauthorized or unintended change.

13.8.9 A management of change process is to be applied to safeguard against unexpected consequences of modification. Records of configuration changes and approvals are to be maintained on board.

13.8.10 Processes for the development and maintenance of integrated systems are to be in accordance with ISO/IEC 15288:2008 and ISO

17894:2005. These processes are to include appropriate risk identification and management.

13.9 System integration

13.9.1 Essential safety functions are to be designed such that risks of harm to personnel or damage to the installation or the environment are reduced to a level acceptable to the Administration/ IRS, both in normal operation and under fault conditions. Functions are to be designed to fail-safe. Roles and responsibilities for integration of systems are to be clearly defined and agreed by relevant parties.

13.9.2 Functional requirements of each component subsystem are to be clearly defined to ensure that the integrated system meets the functional and specified safety requirements and takes account of any limitations of the equipment under control.

13.9.3 Key hazards of the integrated system are to be identified using appropriate risk-based techniques.

IR13.9.3 The expression “integrated system” means a combination of computer-based systems which are used for the control, monitoring/alarm and safety functions required for the carriage, handling and conditioning of cargo liquid and vapours and are interconnected in order to allow communication between computer-based systems and to allow centralized access to monitoring/alarm and safety information and/or command/control.

13.9.4 The integrated system is to have a suitable means of reversionary control.

13.9.5 Failure of one part of the integrated system is not to affect the functionality of other parts, except for those functions directly dependent on the defective part.

13.9.6 Operation with an integrated system is to be at least as effective as it would be with individual stand-alone equipment or systems.

13.9.7 The integrity of essential machinery or systems, during normal operation and fault conditions, are to be demonstrated.

Section 14

Personnel Protection

14.0 Goal

To ensure that protective equipment is provided for ship staff, considering both routine operations or emergency situations and possible short- or long-term effects of the product being handled.

14.1 Protective equipment

14.1.1 Suitable protective equipment including eye protection to a recognized national or international standard, are to be provided for protection of crew members engaged in normal cargo operations, taking into account the characteristics of the products being carried.

14.1.2 Personal protective and safety equipment required in this section are to be kept in suitable, clearly marked lockers located in readily accessible places.

14.1.3 The compressed air equipment is to be inspected at least once a month by a responsible officer and the inspection logged in the ship's records. This equipment is also to be inspected and tested by a competent person at least once a year.

14.2 First-aid equipment

14.2.1 A stretcher that is suitable for hoisting an injured person from spaces below deck is to be kept in a readily accessible location.

14.2.2 The ship is to have onboard medical first-aid equipment, including oxygen resuscitation equipment, based on the requirements of the Medical First Aid Guide (MFAG) for the cargoes listed on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk shown in appendix 5 of IGC Code.

14.3 Safety Equipment

14.3.1 Sufficient, but not less than three complete sets of safety equipment are to be provided in addition to the firemen's outfits required by 11.6.1. Each set is to provide

adequate personal protection to permit entry and work in a gas-filled space. This equipment is to take into account the nature of the cargoes, listed on the International Certificate of Fitness for the Carriage of Liquified Gases in Bulk shown in appendix 2 of IGC Code.

14.3.2 Each complete set of safety equipment is to consist of:

.1 one self-contained positive pressure air-breathing apparatus incorporating full face mask, not using stored oxygen and having a capacity of at least 1200 [l] of free air. Each set is to be compatible with that required by 11.6.1;

.2 protective clothing, boots, and gloves to a recognized standard ;

.3 steel cored rescue line with belt; and

.4 explosion-proof lamp.

14.3.3 An adequate supply of compressed air is to be provided and is to consist either of :

.1 at least one dfully charged spare air bottles for each breathing apparatus required by 14.3.1;

.2 an air compressor of adequate capacity capable of continuous operation suitable for the supply of high-pressure air of breathable quality; and

.3 a charging manifold capable of dealing with sufficient spare breathing apparatus air bottles for the breathing apparatus required by 14.3.1

14.4 Personnel protection requirement for individual products

14.4.1 Requirements of this sub-section applicable to ships carrying products for which those paragraphs are listed in column "i" in the table 19.1.1.

14.4.2 Suitable respiratory and eye protection for emergency escape purposes are to be provided for every person on board subject to following :

.1 filter type respiratory protection is unacceptable,
.2 self-contained breathing apparatus is to have at least a duration of service of 15 minutes; and

.3 emergency escape respiratory protection is not to be used for fire-fighting or cargo handling purposes and is to be marked to that effect;

14.4.3 One or more suitably marked decontamination showers and an eyewash are to be available on deck, taking into account the size and layout of the ship. The showers and eyewash are to be operable in all ambient conditions.

14.4.4 The protective clothing required under 14.3.2.2 is to be gastight.

Section 15

Filling Limits for Cargo Tanks

15.0 Goal

To determine the maximum quantity of cargo that can be loaded.

15.1 Definitions

15.1.1 *Filling limit (FL)* means the maximum liquid volume in a cargo tank relative to the total tank volume when the liquid cargo has reached the reference temperature.

15.1.2 *Loading limit (LL)* means the maximum allowable liquid volume relative to the tank volume to which the tank may be loaded.

15.1.3 *Reference temperature* means (for the purposes of this section only):

.1 when no cargo vapour pressure/temperature control, as referred to in Sec 7, is provided, the temperature corresponding to the vapour pressure of the cargo at the set pressure of the PRVs; and

.2 when a cargo vapour pressure/temperature control, as referred to in Sec 7, is provided, the temperature of the cargo upon termination of loading, during transport or at unloading, whichever is the greatest.

15.1.4 *Ambient design temperature for unrestricted service* means sea temperature of 32°C and air temperature of 45 [°C]. However, lesser values of these temperatures may be accepted by the Administration/ IRS for ships operating in restricted areas or on voyages of restricted duration, and account may be taken in such cases of any insulation of the tanks. Conversely, higher values of these temperatures may be required for ships permanently operating in areas of high-ambient temperature.

15.2 General requirements

The maximum filling limit of cargo tanks are to be so determined that the vapour space has a minimum volume at reference temperature allowing for:

.1 tolerance of instrumentation such as level and temperature gauges;

.2 volumetric expansion of the cargo between the PRV set pressure and the maximum allowable rise stated in 8.4; and

.3 an operational margin to account for liquid drained back to cargo tanks after completion of loading, operator reaction time and closing time of valves, see 5.5 and 18.10.2.1.4.

15.3 Default filling limit

The default value for the filling limit (*FL*) of cargo tanks is 98% at the reference temperature. Exceptions to this value are to meet the requirements of 15.4.

15.4 Determination of increased filling limit

15.4.1 A filling limit greater than the limit of 98% specified in 15.3 may be permitted under the trim and list conditions specified in 8.2.17, providing:

.1 no isolated vapour pockets are created within the cargo tank;

.2 the PRV inlet arrangement shall remain in the vapour space; and

.3 allowances need to be provided for:

.1 volumetric expansion of the liquid cargo due to the pressure increase from the MARVS to full flow relieving pressure in accordance with 8.4.1;

.2 an operational margin of minimum 0.1% of tank volume; and

.3 tolerances of instrumentation such as level and temperature gauges.

IR 15.4.1.2 Determining PRV inlet remains in vapour space. (Refer 15.4.1.2 above)

.1 The PRV inlet is to remain in the vapour space at a minimum distance of 40% of the diameter of the suction funnel measured at the centre of the funnel above the liquid level under conditions of 15° list and 0.015L trim.

IR 15.4.1.3 Calculation of Allowances (Refer 15.4.1.3 above)

.1 The following method may be used to determine the allowance. IRS may accept other methods to determine the allowance provided the method meets an equivalent level of safety.

The parameters specified under 15.4.1.3 may be expressed by the expansion factors α_1 through α_4 as follows:

α_1 = relative increase in liquid volume due to tolerance of level gauges

α_2 = relative increase in liquid volume due to the tolerance of temperature gauges

α_3 = expansion of cargo volume due to pressure rise when pressure relief valves are relieving at maximum flow rate

α_4 = operational margin of 0.1%

The factors α_1 through α_4 are to be determined as follows:

$$\alpha_1 = \frac{dV}{dh}\frac{\Delta h}{V}\ 100\ (\%)$$

where:

dV/dh = variation of tank volume per metre filling height at the filling height *h* (m^3/m)

h = filling height (m) at the filling limit FL to be investigated (FL > 98%)

V = accepted total tank volume (m^3)

Δh = max. total tolerance of level gauges (m)

$$\alpha_2 = \beta\ \Delta T\ (\%)$$

where:

β = volumetric thermal expansion coefficient at reference temperature (%/ °K)

ΔT = max. tolerance of temperature gauge (°K)

$\alpha_3 = \left(\frac{\rho_{PRV}}{\rho_{PRV\ 1.2}} - 1\right)100$ (%) expansion due to pressure rise when relieving at full capacity

ρ_{PRV} = ρ_R cargo density at reference conditions, i.e. corresponding to the temperature of the cargo at set opening pressure of the pressure relief valve (PRV)

$\rho_{PRV\cdot 1.2}$ = cargo density corresponding to the temperature of the cargo at 1.2 times the set opening pressure of the pressure relief valve (PRV)

α_4= 0.1% operational margin

Based on the factors α_1 through α_4 the following total expansion factor α_t is to be determined

$$\alpha_t = \sqrt{\alpha_1^2 + \alpha_2^2} + \alpha_3 + \alpha_4\ (\%)$$

15.4.2 In no case shall a filling limit exceeding 99.5% at reference temperature be permitted.

15.5 Maximum loading limit

15.5.1 The maximum loading limit (LL) to which a cargo tank is to be determined by following formula :

$$\mathrm{LL} = \mathrm{FL}\frac{\mathrm{d_R}}{\mathrm{d_L}}$$

where,

LL = loading limit defined in 15.1.2, expressed in percentage

FL = filling limit as specified in 15.3 or 15.4 expressed in percentage

d_R = relative density of cargo at the reference temperature; and

d_L = relative density of cargo at the loading temperature.

15.5.2 IRS may allow type C tanks to be loaded according to the formula in 15.5.1 with relative density d_R as defined below, provided that the tank vent system has been approved in accordance with 8.2.18:

d_R = relative density of cargo at the highest temperature that the cargo may reach upon termination of loading, during transport, or at unloading, under the ambient design temperature conditions described in 15.1.4;

This paragraph does not apply to products requiring a type 1G ship.

15.6 Information to be provided to the master

15.6.1 A document is to be provided to the ship, specifying the maximum allowable loading limit for each cargo tank and product, at each applicable loading temperature and maximum reference temperature. The information in this document is to be approved by IRS.

15.6.2 Pressures at which the PRVs have been set is also to be stated in the document.

15.6.3 A copy of the above document is to be permanently kept on board by the master.

Section 16

Use of Cargo as Fuel

16.0 Goal

To ensure the safe use of cargo as fuel.

16.1 General

Except provided in 16.9, methane (LNG) is the only cargo whose vapour or boil-off gas may be utilized in machinery spaces of category A and in these spaces, it may be utilized only in systems such as boilers, inert gas generators, internal combustion engines, gas combustion unit and gas turbines.

16.2 Use of cargo vapour as fuel

This sub-section addresses the use of cargo vapour as fuel in systems such as boilers, inert gas generators, internal combustion engines, gas combustion units and gas turbines.

16.2.1 For vaporized LNG, the fuel supply system is to comply with the requirements of 16.4.1, 16.4.2 and 16.4.3.

16.2.2 For vaporized LNG, gas consumers are to exhibit no visible flame and are to maintain the uptake exhaust temperature below 535 [°C].

16.3 Arrangement of spaces containing gas consumers

16.3.1 Spaces in which gas consumers are located are to be fitted with a mechanical ventilation system that is arranged to avoid areas where gas may accumulate, taking into account the density of the vapour and potential ignition sources. The ventilation system is to be separated from those serving the other spaces.

16.3.2 Gas detectors are to be fitted in these spaces, particularly where air circulation is reduced. The gas detection system is to comply with the requirements of Sec.13.

16.3.3 Electrical equipment located in the double wall pipe or duct specified in 16.4.3 are to comply with the requirements of Sec 10.

16.3.4 All vents and bleed lines that may contain or be contaminated by gas fuel are to be routed to a safe location external to the machinery space and be fitted with a flame screen.

16.4 Gas fuel supply

16.4.1 General

16.4.1.1 The requirements of this sub-section is applicable to gas fuel supply piping outside of the cargo area. Fuel piping is not to pass through accommodation spaces, service spaces, electrical equipment rooms or control stations. The routeing of the pipeline is to take into account potential hazards, due to mechanical damage, in areas such as stores or machinery handling areas.

16.4.1.2 Provision is to be made for inerting and gas-freeing that portion of the gas fuel piping systems located in the machinery space.

16.4.2 Leak detection

Continuous monitoring and alarms are to be provided to indicate a leak in the piping system in enclosed spaces and shut down the relevant gas fuel supply.

16.4.3 Routeing of fuel supply pipes

Fuel piping may pass through or extend into enclosed spaces other than those mentioned in 16.4.1, provided it fulfill one of the following conditions:

.1 it is of a double-wall design with the space between the concentric pipes pressurized with inert gas at a pressure greater than the gas fuel pressure. The master gas fuel valve, as required by 16.4.6, closes automatically upon loss of inert gas pressure; or

.2 it is installed in a pipe or duct equipped with mechanical exhaust ventilation having a capacity of at least 30 air changes per hour and is arranged to maintain a pressure less than the atmospheric pressure. The mechanical ventilation is in accordance with section 12, as applicable. The ventilation is always in operation when there is fuel in the piping and the master gas fuel valve, as

required by 16.4.6, closes automatically if the required air flow is not established and maintained by the exhaust ventilation system. The inlet or the duct may be from a non-hazardous machinery space, and the ventilation outlet is in a safe location.

16.4.4 Requirements for gas fuel with pressure greater than 1 MPa

16.4.4.1 Fuel delivery lines between the high-pressure fuel pumps/compressors and consumers are to be protected with a double-walled piping system capable of containing a high pressure line failure, taking into account the effects of both pressure and low temperature. A single-walled pipe in the cargo area up to the isolating valve(s) required by 16.4.6 is acceptable.

16.4.4.2 The arrangement in 16.4.3.2 may also be acceptable providing the pipe or trunk is capable of containing a high pressure line failure, according to the requirements of 16.4.7 and taking into account the effects of both pressure and possible low temperature and providing both inlet and exhaust of the outer pipe or trunk are in the cargo area.

16.4.5 Gas consumer isolation

The supply piping of each gas consumer unit is to be provided with gas fuel isolation by automatic double block and bleed, vented to a safe location, under both normal and emergency operation. The automatic valves are to be arranged to fail to the closed position on loss of actuating power. In a space containing multiple consumers, the shutdown of one is not to affect the gas supply to the others.

16.4.6 Spaces containing gas consumers

16.4.6.1 It is to be possible to isolate the gas fuel supply to each individual space containing a gas consumer(s) or through which fuel gas supply piping is run, with an individual master valve, which is located within the cargo area. The isolation of gas fuel supply to a space is not to affect the gas supply to other spaces containing gas consumers if they are located in two or more spaces, and it is not to cause loss of propulsion or electrical power.

16.4.6.2 If the double barrier around the gas supply system is not continuous due to air inlets or other openings, or if there is any point where single failure will cause leakage into the space, the individual master valve for the space is to operate under the following circumstances:

.1 automatically by:

.1 gas detection within the space;

.2 leak detection in the annular space of a double-walled pipe;

.3 leak detection in other compartments inside the space, containing single-walled gas piping;

.4 loss of ventilation in the annular space of a double-walled pipe; and

.5 loss of ventilation in other compartments inside the space, containing single-walled gas piping; and

.2 manually from within the space, and at least one remote location.

16.4.6.3 If the double barrier around the gas supply system is continuous, an individual master valve located in the cargo area may be provided for each gas consumer inside the space. The individual master valve is to operate under the following circumstances:

.1 automatically by:

.1 leak detection in the annular space of a double-walled pipe served by that individual master valve;

.2 leak detection in other compartments containing single-walled gas piping that is part of the supply system served by the individual master valve; and

.3 loss of ventilation or loss of pressure in the annular space of a double-walled pipe; and

.2 manually from within the space, and at least one remote location.

16.4.7 Piping and ducting construction

Gas fuel piping in machinery spaces is to comply with 5.1 to 5.9, as applicable. The piping

is to, as far as practicable, have welded joints. Those parts of the gas fuel piping that are not enclosed in a ventilated pipe or duct according to 16.4.3, and are on the weather decks outside the cargo area, is to have full penetration butt-welded joints and is to be fully radiographed.

16.4.8 Gas detection

Gas detection systems provided in accordance with the requirements of this section is to activate the alarm at 30% LFL and shut down the master gas fuel valve required by 16.4.6 at not more than 60% LFL (see 13.6.17).

16.5 Gas fuel plant and related storage tanks

16.5.1 Provision of gas fuel

All equipment (heaters, compressors, filters, etc.) for conditioning the cargo and/or cargo boil off vapour for its use as fuel, and any related storage tanks are to be located in the cargo area. If equipment is in an enclosed space, the space is to be ventilated according to 12.1 and be equipped with a fixed fire-fighting system, according to 11.5 and with a gas detection system according to 13.6, as applicable

16.5.2 Remote stops

16.5.2.1 All rotating equipment utilized for conditioning the cargo for its use as fuel are to be arranged for manual remote stop from the engine-room. Additional remote stops are to be located in areas that are always easily accessible, typically cargo control room, navigation bridge and fire control station.

16.5.2.2 The fuel supply equipment is to be automatically stopped in the case of low suction pressure or fire detection. Unless expressly provided otherwise, the requirements of 18.10 need not apply to gas fuel compressors or pumps when used to supply gas consumers.

16.5.3 Heating and cooling mediums

If the heating or cooling medium for the gas fuel conditioning system is returned to spaces outside the cargo area provisions are to be made to detect and alarm the presence of cargo/cargo vapour in the medium. Any vent outlet is to be in a safe position and fitted with an effective flame screen of an approved type.

16.5.4 Piping and pressure vessels

Piping or pressure vessels fitted in the gas fuel supply system are to comply with Sec.5.

16.6 Special requirements for main boilers

16.6.1 Arrangements

16.6.1.1 Each boiler are to have a separate exhaust uptake.

16.6.1.2 Each boiler is to have a dedicated forced draught system. A crossover between boiler force draught systems may be fitted for emergency use providing that any relevant safety functions are maintained.

16.6.1.3 Combustion chambers and uptakes of boilers are to be designed to prevent any accumulation of gaseous fuel.

16.6.2 Combustion equipment

16.6.2.1 The burner systems are to be of dual type, suitable to burn either oil fuel or gas fuel alone or oil and gas fuel simultaneously.

16.6.2.2 Burners are to be designed to maintain stable combustion under all firing conditions.

16.6.2.3 An automatic system is to be fitted to change over from gas fuel operation to oil fuel operation without interruption of the boiler firing, in the event of loss of gas fuel supply.

16.6.2.4 Gas nozzles and the burner control system are to be configured such that gas fuel can only be ignited by an established oil fuel flame, unless the boiler and combustion equipment is designed and approved by recognized organization to light on gas fuel.

16.6.3 Safety

16.6.3.1 There are to be arrangements to ensure that gas fuel flow to the burner is automatically cut-off, unless satisfactory ignition has been established and maintained.

16.6.3.2 On the pipe of each gas-burner, a manually operated shut-off valve shall be fitted.

16.6.3.3 Provisions are to be made for automatically purging the gas supply piping to the burners, by means of an inert gas, after the extinguishing of these burners.

16.6.3.4 The automatic fuel changeover system required by 16.6.2.3 is to be monitored with alarms to ensure continuous availability.

16.6.3.5 Arrangements are to be made that, in case of flame failure of all operating burners, the combustion chambers of the boilers are automatically purged before relighting.

16.6.3.6 Arrangements are to be made to enable the boilers to be manually purged.

16.7 Special requirements for gas-fired internal combustion engines

Dual fuel engines are those that employ gas fuel (with pilot oil) and oil fuel. Oil fuels may include distillate and residual fuels. Gas only engines are those that employ gas fuel only.

16.7.1 Arrangements

16.7.1.1 When gas is supplied in a mixture with air through a common manifold, flame arrestors shall be installed before each cylinder head.

16.7.1.2 Each engine is to have its own separate exhaust.

16.7.1.3 The exhausts are to be configured to prevent any accumulation of unburnt gaseous fuel.

16.7.1.4 Unless designed with the strength to withstand the worst case overpressure due to ignited gas leaks, air inlet manifolds, scavenge spaces, exhaust system and crank cases are to be fitted with suitable pressure relief systems. Pressure relief systems is to lead to a safe location, away from personnel.

IR 16.7.1.4 A suitable pressure relief system for air inlet manifolds, scavenge spaces and exhaust system is to be provided unless designed to accommodate the worst-case overpressure due to ignited gas leaks or justified by the safety concept of the engine. A detailed evaluation regarding the hazard potential of overpressure in air inlet manifolds, scavenge spaces and exhaust system is to be carried out and reflected in the safety concept of the engine. In the case of crankcases, the explosion relief valves, as required by Regulation 27.4 of SOLAS Chapter II-1 as amended by IMO resolutions up to MSC.436(99), are to be considered suitable for the gas operation of the engine (See Part 4, Chapter 4, Cl. 4.6.1). For engines not covered by said Regulation, a detailed evaluation regarding the hazard potential of fuel gas accumulation in the crankcase is to be carried out.

16.7.1.5 Each engine is to be fitted with vent systems independent of other engines for crankcases, sumps and cooling systems.

16.7.2 Combustion equipment

16.7.2.1 Prior to admission of gas fuel, correct operation of the pilot oil injection system on each unit is to be verified.

16.7.2.2 For a spark ignition engine, if ignition has not been detected by the engine monitoring system within an engine specific time after opening of the gas supply valve, this is to be automatically shut off and the starting sequence terminated. It is to be ensured that any unburnt gas mixture is purged from the exhaust system.

16.7.2.3 For dual-fuel engines fitted with a pilot oil injection system, an automatic system is to be fitted to change over from gas fuel operation to oil fuel operation with minimum fluctuation of the engine power.

16.7.2.4 In the case of unstable operation on engines with the arrangement in 16.7.2.3 when gas firing, the engine is to automatically change to oil fuel mode.

16.7.3 Safety

16.7.3.1 During stopping of the engine, the gas fuel is to be automatically shut off before the ignition source.

16.7.3.2 Arrangements are to be provided to ensure that there is no unburnt gas fuel in the exhaust gas system prior to ignition.

16.7.3.3 Crankcases, sumps, scavenge spaces and cooling system vents are to be provided with gas detection (see 13.6.17).

16.7.3.4 Provision is to be made within the design of the engine to permit continuous monitoring of possible sources of ignition within the crank case. Instrumentation fitted inside the crankcase is to be in accordance with the requirements of Sec 10.

16.7.3.5 A means is to be provided to monitor and detect poor combustion or misfiring that may lead to unburnt gas fuel in the exhaust system during operation. In the event that it is detected, the gas fuel supply is to be shut down. Instrumentation fitted inside the exhaust system shall be in accordance with the requirements of Sec 10.

16.8 Special requirements for gas turbine

16.8.1 Arrangements

16.8.1.1 Each turbine is to have its own separate exhaust.

16.8.1.2 The exhausts are to be appropriately configured to prevent any accumulation of unburnt gas fuel.

16.8.1.3 Unless designed with the strength to withstand the worst case overpressure due to ignited gas leaks, pressure relief systems are to be suitably designed and fitted to the exhaust system, taking into consideration explosions due to gas leaks. Pressure relief systems within the exhaust uptakes are to be lead to a non-hazardous location, away from personnel.

16.8.2 Combustion equipment

An automatic system is to be fitted to change over easily and quickly from gas fuel operation to oil fuel operation with minimum fluctuation of the engine power.

16.8.3 Safety

16.8.3.1 Means are to be provided to monitor and detect poor combustion that may lead to unburnt gas fuel in the exhaust system during operation. In the event that it is detected, the gas fuel supply is to be shut down.

16.8.3.2 Each turbine is to be fitted with an automatic shutdown device for high exhaust temperatures.

16.9 Alternative fuels and technologies

16.9.1 If acceptable to the Administration/ IRS, other cargo gases may be used as fuel, providing that the same level of safety as natural gas in this Chapter is ensured.

16.9.2 The use of cargoes identified as toxic products are not to be permitted.

16.9.3 For cargoes other than LNG, the fuel supply system is to comply with the requirements of 16.4.1, 16.4.2, 16.4.3 and 16.5, as applicable, and is to include means for preventing condensation of vapour in the system.

16.9.4 Liquefied gas fuel supply systems is to comply with 16.4.5.

16.9.5 In addition to the requirements of 16.4.3.2, both ventilation inlet and outlet is to be located outside the machinery space. The inlet is to be in a non-hazardous area and the outlet is to be in a safe location.

Section 17

Special Requirements

17.0 Goal

To set out the additional requirements in respect of specific cargoes.

17.1 General

The requirements of this Section are applicable where reference thereto is made in column "i" in the table of sec 19. These requirements are additional to the general requirements of the Chapter.

17.2 Materials of construction

Materials that may be exposed to cargo during normal operations are to be resistant to the corrosive action of the gases. In addition, the following materials of construction of cargo tanks, and associated pipelines, valves, fittings and other items of equipment normally in direct contact with cargo liquid or vapour are not to be used for certain products as specified in column "i" in the table of sec 19.

.1 mercury, copper and copper-bearing alloys, and zinc;

.2 copper, silver, mercury, magnesium and other acetylide-forming metals;

.3 aluminum and aluminium-bearing alloys;

.4 copper, copper alloys, zinc and galvanized steel;

.5 aluminum, copper and alloys of either;

.6 copper and copper-bearing alloys with greater than 1 per cent copper.

17.3 Independent tanks

17.3.1 Products are to be carried in independent tanks only.

17.3.2 Products are to be carried in Type C independent tanks and the provisions of 7.1.2 apply. The design pressure of the cargo tank is to take into account any padding pressure or vapour discharge unloading pressure.

17.4 Refrigeration systems

17.4.1 Only the indirect system described in 7.3.1.2 is to be used.

17.4.2 For a ship engaged in the carriage of products that readily form dangerous peroxides, recondensed cargo is not to be allowed to form stagnant pockets of uninhibited liquid. This may be achieved either by:

.1 using the indirect system described in 7.3.1.2 with the condenser inside the cargo tank; or

.2 using the direct system or combined system described in 7.3.1.1 and .3 respectively, or the indirect system described in 7.3.1.2 with the condenser outside the cargo tank, and designing the condensate system to avoid any places in which liquid could collect and be retained. Where this is impossible inhibited liquid is to be added upstream of such a place.

17.4.3 If the ship is to consecutively carry products as specified in 17.4.2 with a ballast passage between, all uninhibited liquid is to be removed prior to the ballast voyage. If a second cargo is to be carried between such consecutive cargoes, the reliquefication system is to be thoroughly drained and purged before loading the second cargo. Purging is to be carried out using either inert gas or vapour from the second cargo, if compatible. Practical steps are to be taken to ensure that polymers or peroxides do not accumulate in the cargo system.

17.5 Cargoes requiring type 1G ship

17.5.1 All butt-welded joints in cargo piping exceeding 75 mm in diameter are to be subject to 100% radiography.

17.5.2 Gas sampling lines are not to be led into or through non-hazardous areas. Alarms referred to in 13.6.2 are to be activated when the vapour concentration reaches the threshold limiting value.

17.5.3 The alternative of using portable gas detection equipment in accordance with 13.6.5 shall not be permitted.

17.5.4 Cargo control rooms are to be located in a non-hazardous area and, additionally, all instrumentation shall be of the indirect type.

17.5.5 Personnel are to be protected against the effects of a major cargo release by the provision of a space within the accommodation area that is designed and equipped to the satisfaction of the Administration/ IRS.

17.5.6 Notwithstanding the requirements in 3.2.4.3, access to forecastle spaces are not to be permitted through a door facing the cargo area, unless airlock in accordance with 3.6 is provided.

17.5.7 Notwithstanding the requirements in 3.2.7, access to control rooms and machinery spaces of turret systems are not to be permitted through doors facing the cargo area.

17.6 Exclusion of air from vapour spaces

Air is to be removed from the cargo tanks and associated piping before loading and then subsequently excluded by:

.1 introducing inert gas to maintain a positive pressure. Storage or production capacity of the inert gas is to be sufficient to meet normal operating requirements and relief valve leakage. The oxygen content of inert gas is at no time to be greater than 0.2 per cent by volume; or

.2 control of cargo temperatures such that a positive pressure is maintained at all times.

17.7 Moisture control

For gases which are non-flammable and may become corrosive or react dangerously with water, moisture control is to be provided to ensure that cargo tanks are dry before loading and that during discharge, dry air or cargo vapour is introduced to prevent negative pressures. For the purposes of this paragraph, dry air is air which has a dewpoint of -45°C or below at atmospheric pressure.

17.8 Inhibition

17.8.1 Care is to be taken to ensure that the cargo is sufficiently inhibited to prevent self-reaction (e.g. polymerization or dimerization) at all times during the voyage. Ships are to be provided with a certificate from the manufacturer stating:

.1 name and amount of inhibitor added;

.2 date inhibitor was added and the normally expected duration of its effectiveness;

.3 any temperature limitations affecting the inhibitor;

.4 the action to be taken when the length of the voyage exceeds the effective lifetime of the inhibitors.

17.9 Flame screens on vent outlets

When carrying a cargo referenced in this section, cargo tank vent outlets are to be provided with readily renewable and effective flame screens or safety heads of an approved type. Due attention is to be paid in the design of flame screens and vent heads to the possibility of the blockage of these devices by the freezing of cargo vapour or by icing up in adverse weather conditions. Flame screens are to be removed and replaced by protection screens, in accordance with 8.2.15, when carrying cargoes not referenced in this section.

17.10 Maximum allowable quantity of cargo per tank

When carrying a cargo referenced to this Section, the quantity of the cargo is not be exceed 3000 [m^3] in any one tank.

17.11 Cargo pumps and discharge arrangements

17.11.1 The vapour space of cargo tanks equipped with submerged electric motor pumps is to be inerted to a positive pressure prior to loading, during carriage and during unloading of flammable liquids.

17.11.2 The cargo is to be discharged only by deepwell pumps or by hydraulically operated submerged pumps. These pumps are to be of a type designed to avoid liquid pressure against the shaft gland.

17.11.3 Inert gas displacement may be used for discharging cargo from type C independent tanks, provided the cargo system is designed for the expected pressure.

17.12 Ammonia

17.12.1 Anhydrous ammonia may cause stress corrosion cracking in containment and process

systems made of carbon manganese steel or nickel steel. To minimize the risk of this occurring, measures detailed in 17.12.2 to 17.12.8 are to be taken as appropriate.

17.12.2 Where carbon manganese steel is used, cargo tanks, process pressure vessels and cargo piping are to be made of fine grained steel with a specified minimum yield strength not exceeding 355 [N/mm^2] and with an actual yield strength not exceeding 440 [N/mm^2]. One of the following constructional or operational measures is to also be taken:

.1 lower strength material with a specified minimum tensile strength not exceeding 410 [N/mm^2] is to be used; or

.2 cargo tanks, etc., are to be post weld stress relief heat treated; or

.3 carriage temperature is to be maintained preferably at a temperature close to the product's boiling point of -33°C but in no case at a temperature above -20°C; or

.4 the ammonia is to contain not less than 0.1% w/w water and the master is to be provided with documentation confirming this.

17.12.3 If carbon-manganese steels with higher yield properties are used other than those specified in 17.12.2, the completed cargo tanks, piping, etc. are to be given a post weld stress relief heat treatment.

17.12.4 Process pressure vessels and piping of the condensate part of the refrigeration system is to be given a post-weld stress relief heat treatment when made of materials mentioned in 17.12.1.

17.12.5 The tensile and yield properties of the welding consumables are to exceed those of the tank or piping material by the smallest practical amount.

17.12.6 Nickel steel containing more than 5% nickel and carbon manganese steel, not complying with the requirements of 17.12.2 and 17.12.3 are particularly susceptible to ammonia stress corrosion cracking and are not to be used in containment and piping systems for the carriage of this product.

17.12.7 Nickel steel containing not more than 5% nickel may be used provided the carriage temperature complies with the requirements specified in 17.12.2.3.

17.12.8 To minimize the risk of ammonia stress corrosion cracking, it is advisable to keep the dissolved oxygen content below 2.5 ppm/w/w. This can best be achieved by reducing the average oxygen content in the tanks prior to the introduction of liquid ammonia to less than the values given as a function of the carriage temperature T in the Table below:

T (°C)	O_2 (%v/v)
- 30 and below	0.90
-20	0.50
-10	0.28
0	0.16
10	0.10
20	0.05
30	0.03

Oxygen percentages for intermediate temperatures may be obtained by direct interpolation.

17.13 Chlorine

17.13.1 Cargo containment system

17.13.1.1 The capacity of each tank is not to exceed 600 [m^3] and the total capacity of all cargo tanks is not to exceed 1200 [m^3].

17.13.1.2 The tank design vapour pressure is not to be less than 13.5 bar (see also 7.1.2 and 17.3.2).

17.13.1.3 Parts of tanks protruding above the upper deck are to be provided with protection against thermal radiation, taking into account total engulfment by fire.

17.13.1.4 Each tank is to be provided with two pressure relief valves. A bursting disc of appropriate material is to be installed between the tank and the pressure relief valves. The rupture pressure of the bursting disc is to be 1 bar lower than the opening pressure of the pressure relief valve, which is to be set at the design vapour pressure of the tank but not less than 13.5 bar gauge. The space between the bursting disc and the relief valve is to be connected through an excess flow valve to a pressure gauge and a gas detection system. Provision is to be made to keep this space at or

near the atmospheric pressure during normal operation.

17.13.1.5 Outlets from pressure relief valves are to be arranged in such a way as to minimize the hazards on board the ship as well as to the environment. Leakage from the relief valves is to be led through the absorption plant to reduce the gas concentration as far as possible. The relief valve exhaust line is to be arranged at the forward end of the ship to discharge outboard at deck level with an arrangement to select either port or starboard side, with a mechanical interlock to ensure that one line is always open.

17.13.1.6 The Administration and the Port Administration may require that chlorine is carried in refrigerated state at a specified maximum pressure.

17.13.2 Cargo piping systems

17.13.2.1 Cargo discharge is to be performed by means of compressed chlorine vapour from shore, dry air or another acceptable gas or fully submerged pumps. Cargo discharge compressors on board ships are not to be used for this. The pressure in the vapour space of the tank during discharging is not to exceed 10.5 bar gauge.

17.13.2.2 The design pressure of the cargo piping system is to be not less than 21 bar gauge. The internal diameter of the cargo pipes is not to exceed 100 mm. Only pipe bends will be accepted for compensation of pipeline thermal movement. The use of flanged joints is to be restricted to a minimum, and when used the flanges are to be of the welding neck type with tongue and groove.

17.13.2.3 Relief valves of the cargo piping system are to discharge to the absorption plant, and the flow restriction creted by this unit is to be taken into account when designing the relief valve system (see 8.4.3 and 8.4.4).

17.13.3 Materials

17.13.3.1 The cargo tanks and cargo piping systems are to be made of steel suitable for the cargo and for a temperature of -40°C, even if a higher transport temperature is intended to be used.

17.13.3.2 The tanks are to be thermally stress relieved. Mechanical stress relief is not to be accepted as an equivalent.

17.13.4 Instrumentation - safety devices

17.13.4.1 The ship is to be provided with a chlorine absorbing plant with connections to the cargo piping system and the cargo tanks. The absorbing plant is to be capable of neutralizing at least 2 per cent of the total cargo capacity at a reasonable absorption rate.

17.13.4.2 During the gas-freeing of cargo tanks, vapours are not to be discharged to the atmosphere.

17.13.4.3 A gas detecting system is to be provided that is capable of monitoring chlorine concentrations of at least 1 ppm by volume. Suction points are to be located:

.1 near the bottom of the hold spaces;

.2 in the pipes from the safety relief valves;

.3 at the outlet from the gas absorbing plant;

.4 at the inlet to the ventilation systems for the accommodation, service and machinery spaces and control stations;

.5 on deck at the forward end, in the middle and at the after end of the cargo area. This is only required to be used during cargo handling and gas-freeing operations.

The gas detection system is to be provided with an audible and visual alarm with a set point of 5 ppm.

17.13.4.4 Each cargo tank is to be fitted with a high-pressure alarm giving an audible alarm at a pressure equal to 10.5 bar gauge.

17.13.5 Personnel protection

The enclosed space required by 17.5.5 is to meet the following requirements:

.1 The space is to be easily and quickly accessible from the weather decks and from accommodation spaces by means of air locks and is to be capable of being rapidly closed gastight.

.2 one of the decontamination showers required by 14.4.3 is to be located near waeather deck airlock to the space;

.3 the space is to be designed to accommodate the entire crew of the ship and be provided with a source of

uncontaminated air for a period of not less than 4 h; and

.4 One set of oxygen therapy equipment is to be carried in the space

17.13.6 Filling limits for cargo tanks

17.13.6.1 The requirements of 15.1.3.2 do not apply when it is intended to carry chlorine.

17.13.6.2 The chlorine content of the gas in the vapour space of the cargo tank after loading is to be greater than 80 per cent by volume.

17.14 Ethylene oxide

17.14.1 For the carriage of ethylene oxide the requirements of 17.18 apply, with the additions and modifications as given in this sub-section.

17.14.2 Deck tanks are not to be used for the carriage of ethylene oxide.

17.14.3 Stainless steel types 416 and 442 as well as cast iron are not to be used in ethylene oxide cargo containment and piping systems.

17.14.4 Before loading, tanks are to be thoroughly and effectively cleaned to remove all traces of previous cargoes from tanks and associated pipework, except where the immediate prior cargo has been ethylene oxide, propylene oxide or mixtures of these products. Particular care is to be taken in the case of ammonia in tanks made of steel other than stainless steel.

17.14.5 Ethylene oxide is to be discharged only by deepwell pumps or inert gas displacement. The arrangement of pumps is to comply with 17.18.15.

17.14.6 Ethylene oxide is to be carried refrigerated only and maintained at temperatures of less than 30°C.

17.14.7 Pressure relief valves are to be set at a pressure of not less than 5.5 bar gauge. The maximum set pressure is to be specially approved by Administration/ IRS.

17.14.8 The protective padding of nitrogen gas as required by 17.18.27 is to be such that the nitrogen concentration in the vapour space of the cargo tank will at no time be less than 45 per cent by volume.

17.14.9 Before loading and at all times when the cargo tank contains ethylene oxide liquid or vapour, the cargo tank is to be inerted with nitrogen.

17.14.10 The water-spray system required by paragraph 17.18.29 and that required by 11.3 is to operate automatically in a fire involving the cargo containment system.

17.14.11 A jettisoning arrangement is to be provided to allow the emergency discharge of ethylene oxide in the event of uncontrollable self-reaction.

17.15 Separate piping systems

Separate piping systems, as defined in 1.2.47, are to be provided.

17.16 Methyl acetylene-propadiene mixtures

17.16.1 Methyl acetylene-propadiene mixtures are to be suitably stabilized for transport. Additionally, upper limits of temperature and pressure during the refrigeration are to be specified for the mixtures.

17.16.2 Examples of acceptable, stabilized compositions are:

.1 Composition 1

.1 maximum methyl acetylene to propadiene molar ratio of 3 to 1;

.2 maximum combined concentration of methyl acetylene and propadiene of 65 mol per cent;

.3 minimum combined concentration of propane, butane, and isobutane of 24 mol per cent, of which at least one third (on a molar basis) must be butanes and one third propane; and

.4 maximum combined concentration of propylene and butadiene of 10 mol per cent;

.2 Composition 2

.1 maximum methyl acetylene and propadiene combined concentration of 30 mol per cent;

.2 maximum methyl acetylene concentration of 20 mol per cent;

.3 maximum propadiene concentration of 20 mol per cent;

.4 maximum propylene concentration of 45 mol per cent;

.5 maximum butadiene and butylenes combined concentration of 2 mol per cent;

.6 minimum saturated C4 hydrocarbon concentration of 4 mol per cent; and

.7 minimum propane concentration of 25 mol per cent.

17.16.3 Other compositions may be accepted provided the stability of the mixture is demonstrated to the satisfaction of Administration/ IRS.

17.16.4 If a ship has a, direct vapour compression refrigeration this is to comply with following requirements, subject to pressure and temperature limitations depending on the composition. For the example compositions given in 17.18.2, the following features are to be provided:

.1 A vapour compressor that does not raise the temperature and pressure of the vapour above 60C and 17.5 bar gauge during its operation, and that does not allow vapour to stagnate in the compressor while it continues to run.

.2 Discharge piping from each compressor stage or each cylinder in the same stage of a reciprocating compressor is to have:

.1 two temperature-actuated shutdown switches set to operate at 60C or less;

.2 a pressure-actuated shutdown switch set to operate at 17.5 bar gauge or less; and

.3 a safety relief valve set to relieve at 18.0 bar gauge or less.

.3 The relief valve required by .2.3 is to vent to a mast meeting the requirements of 8.2.10, 8.2.11, and 8.2.15 and is not to relieve into the compressor suction line.

.4 An alarm that sounds in the cargo control position and in the navigation bridge when a high-pressure switch, or a high-temperature switch operates.

17.16.5 The piping system, including the cargo refrigeration system, for tanks to be loaded with methyl acetylene-propadiene mixtures is to be either independent (as defined in 1.2.28) or separate (as defined in 1.2.47) from piping and refrigeration systems for other tanks. This segregation applies to all liquid and vapour vent lines and other possible connections, such as common inert gas supply lines.

17.17 Nitrogen

Materials of construction and ancillary equipment such as insulation are to be resistant to the effects of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo system. Due consideration is to be given to ventilation in such areas where condensation might occur to avoid the stratification of oxygen-enriched atmosphere.

17.18 Propylene oxide and mixtures of ethylene oxide-propylene oxide with ethylene oxide content of not more than 30 per cent by weight

17.18.1 Products transported under the provisions of this Section are to be acetylene-free

17.18.2 Unless cargo tanks are properly cleaned, these products are not to be carried in tanks that have contained as one of the three previous cargoes any product known to catalyse polymerization, such as:

.1 anhydrous ammonia and ammonia solutions;

.2 amines and amine solutions;

.3 oxidizing substances (e.g. chlorine).

17.18.3 Before loading, tanks are to be thoroughly and effectively cleaned to remove all traces of previous cargoes from tanks and associated pipework, except where the immediate prior cargo has been propylene oxide or ethylene oxide-propylene oxide mixtures. Particular care is to be taken in the case of ammonia in tanks made of steel other than stainless steel.

17.18.4 In all cases, the effectiveness of cleaning procedures for tanks and associated pipework is to be checked by suitable testing or inspection to ascertain that no traces of acidic or alkaline materials remain that might create a

hazardous situation in the presence of these products.

17.18.5 Tanks are to be entered and inspected prior to each initial loading of these products to ensure freedom from contamination, heavy rust deposits and any visible structural defects. When cargo tanks are in continuous service for these products, such inspections are to be performed at intervals of not more than 2 years.

17.18.6 Tanks for the carriage of these products are to be of steel or stainless steel construction.

17.18.7 Tanks which have contained these products may be used for other cargoes after thorough cleaning of tanks and associated pipework systems by washing or purging.

17.18.8 All valves, flanges, fittings and accessory equipment are to be of a type suitable for use with these products and are to be constructed of steel or stainless steel in accordance with recognized standards. Discs or disc faces, seats and other wearing parts of valves are to be made of stainless steel containing not less than 11 per cent chromium.

17.18.9 Gaskets are to be constructed of materials which do not react with, dissolve in, or lower the autoignition temperature of these products and which are fire- resistant and possess adequate mechanical behavior. The surface presented to the cargo is to be polytetrafluoroethylene (PTFE) or materials giving a similar degree of safety by their inertness. Spirally-wound stainless steel with a filler of PTFE or similar fluorinated polymer may be accepted, if approved by IRS.

17.18.10 Insulation and packing if used are to be of a material which does not react with, dissolve in, or lower the autoignition temperature of these products.

17.18.11 The following materials are generally found unsatisfactory for gaskets, packing and similar uses in containment systems for these products and would require testing before being approved:

.1 Neoprene or natural rubber if it comes into contact with the products;

.2 Asbestos or binders used with asbestos;

.3 Materials containing oxides of magnesium, such as mineral wools.

17.18.12 Filling and discharge piping are to extend to within 100 [mm] of the bottom of the tank or any sump.

17.18.13 The products are to be loaded and discharged in such a manner that venting of the tanks to atmosphere does not occur. If vapour return to shore is used during tank loading, the vapour return system connected to a containment system for the product is to be independent of all other containment systems.

17.18.14 During discharging operations, the pressure in the cargo tank is to be maintained above 0.07 bar gauge.

17.18.15 The cargo is to be discharged only by deepwell pumps, hydraulically operated submerged pumps, or inert gas displacement. Each cargo pump is to be arranged to ensure that the products does not heat significantly if the discharge line from the pump is shut off or otherwise blocked.

17.18.16 Tanks carrying these products are to be vented independently of tanks carrying other products. Facilities are to be provided for sampling the tank contents without opening the tank to atmosphere.

17.18.17 Cargo hoses used for transfer of these products are to be marked "FOR ALKYLENE OXIDE TRANSFER ONLY".

17.18.18 Hold spaces are to be monitored for these products. Hold spaces surrounding Type A and B independent tanks are also to be inerted and monitored for oxygen. The oxygen content of these spaces is to be maintained below 2 per cent by volume. Portable sampling equipment is satisfactory.

17.18.19 Prior to disconnecting shore-lines, the pressure in liquid and vapour lines is to be relieved through suitable valves installed at the loading header. Liquid and vapour from these lines are not to be discharged to atmosphere.

17.18.20 Tanks are to be designed for the maximum pressure expected to be encountered during loading, carriage or unloading cargo.

17.18.21 Tanks for the carriage of propylene oxide with a design vapour pressure of less than 0.6 bar and tanks for the carriage of ethylene oxide-propylene oxide mixtures with a design vapour pressure of less than 1.2 bar are to have a cooling system to maintain the cargo

below the reference temperature. The reference temperatures are referred to in 15.13.

17.18.22 Pressure relief valve settings is not to be less than 0.2 bar gauge and for Type C independent tanks not greater than 7.0 bar gauge for the carriage of propylene oxide and not greater than 5.3 bar gauge for the carriage of ethylene oxide-propylene oxide mixtures.

17.18.23 The piping system for tanks to be loaded with these products is to be completely separate from piping systems for all other tanks, including empty tanks, and from all cargo compressors. If the piping system for the tanks to be loaded with these products is not independent as defined in 1.2.28 the required piping separation is to be accomplished by the removal of spool pieces, valves, or other pipe sections and the installation of blank flanges at these locations. The required separation applies to all liquid and vapour piping, liquid and vapour vent lines and any other possible connections such as common inert gas supply lines.

17.18.24 The products are to be transported only in accordance with cargo handling plans approved by Administration/ IRS. Each intended loading arrangement is to be shown on a separate cargo handling plan. Cargo handling plans are to show the entire cargo piping system and the locations for installation of blank flanges needed to meet the above piping separation requirements. A copy of each approved cargo handling plan is to be kept on board the ship. The International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk is to be endorsed to include reference to the approved cargo handling plans.

17.18.25 Before each initial loading of the products and before every subsequent return to such service, certification verifying that the required piping separation has been achieved is to be obtained from a responsible person acceptable to the Port Administration and carried on board the ship. Each connection between a blank flange and pipeline flange is to be fitted with a wire and seal by the responsible person to ensure that inadvertent removal of the blank flange is impossible.

17.18.26 The maximum allowable tank loading limits for each cargo tank is to be indicated for each loading temperature that may be applied, in accordance with 15.5.

17.18.27 The cargo is to be carried under a suitable protective padding of nitrogen gas. An automatic nitrogen make-up system is to be installed to prevent the tank pressure falling below 0.07 bar gauge in the event of product temperature fall due to ambient conditions or malfunctioning of refrigeration system. Sufficient nitrogen is to be available on board to satisfy the demand of the automatic pressure control. Nitrogen of commercially pure quality (99,9 per cent by volume) is to be used for padding. A battery of nitrogen bottles connected to the cargo tanks through a pressure reduction valve satisfies the intention of the expression "automatic" in this context.

17.18.28 The cargo tank vapour space is to be tested prior to and after loading to ensure that the oxygen content is 2 per cent by volume or less.

17.18.29 A water-spray system of sufficient capacity is to be provided to blanket effectively the area surrounding the loading manifold, the exposed deck piping associated with product handling and the tank domes. The arrangement of piping and nozzles is to be such as to give a uniform distribution rate of 10 [l/m^2] per minute. The arrangement is to ensure that any spilled cargo is washed away.

17.18.30 The water-spray system is to be capable of local and remote manual operation in case of a fire involving the cargo containment system. Remote manual operation is to be arranged such that the remote starting of pumps supplying the water-spray system and remote operation of any normally closed valves in the system can be carried out from a suitable location outside the cargo area, adjacent to the accommodation spaces and readily accessible and operable in the event of fire in the areas protected.

17.18.31 When ambient temperatures permit, a pressurized water hose ready for immediate use shall be available during loading and unloading operations, in addition to the above water-spray requirements.

17.19 Vinyl chloride

In cases where polymerization of vinyl chloride is prevented by addition of an inhibitor, 17.8 is applicable. In cases where no inhibitor has been added or the inhibitor concentration is in sufficient, any inert gas used for the purposes

of 17.6 is to contain not more oxygen than 0,1 per cent. Before, loading is started, inert gas samples from the tanks and piping are to be analysed. When vinyl chloride is carried, a positive pressure is always to be maintained in the tanks, and during ballast voyages between successive carriages.

17.20 Mixed C4 cargoes

17.20.1 Cargoes that may be carried individually under the requirements of this chapter, notably butane, butylenes and butadiene, may be carried as mixtures subject to the provisions of this section. These cargoes may variously be referred to as "Crude C4", "Crude butadiene", "Crude steam-cracked C4", "Spent steam-cracked C4", "C4 stream", "C4 raffinate", or may be shipped under a different description. In all cases, the material safety data sheets (MSDS) shall be consulted as the butadiene content of the mixture is of prime concern as it is potentially toxic and reactive. While it is recognized that butadiene has a relatively low vapour pressure, if such mixtures contain butadiene they are to be regarded as toxic and the appropriate precautions applied.

17.20.2 If the mixed C4 cargo shipped under the terms of this section contains more than 50% (mole) of butadiene, the inhibitor precautions in 17.8 applies.

17.20.3 Unless specific data on liquid expansion coefficients is given for the specific mixture loaded, the filling limit restrictions of chapter 15 shall be calculated as if the cargo contained 100% concentration of the component with the highest expansion ratio.

17.21 Carbon dioxide: high purity

17.21.1 Uncontrolled pressure loss from the cargo can cause "sublimation" and the cargo will change from the liquid to the solid state. The precise "triple point" temperature of a particular carbon dioxide cargo is to be supplied before loading the cargo, and will depend on the purity of that cargo, and this is to be taken into account when cargo instrumentation is adjusted. The set pressure for the alarms and automatic actions described in this sub-section are to be set to at least 0.5 Bar (0.05 MPa) above the triple point for the specific cargo being carried. The "triple point" for pure carbon dioxide occurs at 0.5 Bar (0.5 MPa) gauge and -54.4°C.

17.21.2 There is a potential for the cargo to solidify in the event that a cargo tank relief valve, fitted in accordance with 8.2, fails in the open position. To avoid this, a means of isolating the cargo tank safety valves are to be provided and the requirements of 8.2.9.2 do not apply when carrying this carbon dioxide. Discharge piping from safety relief valves are to be designed so they remain free from obstructions that could cause clogging. Protective screens are not to be fitted to the outlets of relief valve discharge piping, so the requirements of 8.2.15 do not apply.

17.21.3 Discharge piping from safety relief valves are not required to comply with 8.2.10, but shall be designed so they remain free from obstructions that could cause clogging. Protective screens are not to be fitted to the outlets of relief valve discharge piping, so the requirements of 8.2.15 do not apply.

17.21.4 Cargo tanks are to be continuously monitored for low pressure when a carbon dioxide cargo is carried. An audible and visual alarm is to be given at the cargo control position and on the bridge. If the cargo tank pressure continues to fall to within 0.5 Bar (0.05 MPa) of the "triple point" for the particular cargo, the monitoring system is to automatically close all cargo manifold liquid and vapour valves and stop all cargo compressors and cargo pumps. The emergency shutdown system required by 18.10 may be used for this purpose.

17.21.5 All materials used in cargo tanks and cargo piping system are to be suitable for the lowest temperature that may occur in service, which is defined as the saturation temperature of the carbon dioxide cargo at the set pressure of the automatic safety system described in 17.21.1.

17.21.6 Cargo hold spaces, cargo compressor rooms and other enclosed spaces where carbon dioxide could accumulate are to be fitted with continuous monitoring for carbon dioxide build-up. This fixed gas detection system replaces the requirements of 13.6, and hold spaces are to be monitored permanently even if the ship has type C cargo containment.

17.22 Carbon dioxide: reclaimed quality

17.22.1 The requirements of 17.21 also apply to this cargo. In addition, the materials of construction used in the cargo system are also

to take account of the possibility of corrosion, in case the reclaimed quality carbon dioxide cargo contains impurities such as water, sulphur dioxide, etc., which can cause acidic corrosion or other problems.

Section 18

Operating Requirements

18.0 Goal

To ensure that all ship staff involved in cargo operations have sufficient information about cargo properties and operating the cargo system so they can conduct cargo operations safely.

18.1 General

18.1.1 Those involved in liquefied gas carrier operations are to be made aware of the special requirements associated with, and precautions necessary for, their safe operation.

18.1.2 A copy of the IGC Code, or national regulations incorporating the provisions of the Code, is to be on board every ship covered by the Code.

18.2 Cargo operations manuals

18.2.1 The ship is to be provided with copies of suitably detailed cargo system operation manuals approved by the Administration/ IRS such that trained personnel can safely operate the ship with due regard to the hazards and properties of the cargoes that are permitted to be carried.

18.2.2 The content of the manuals are to include, but not be limited to:

.1 overall operation of the ship from dry-dock to dry-dock, including procedures for cargo tank cooldown and warm-up, transfer (including ship-to-ship transfer), cargo sampling, gas-freeing, ballasting, tank cleaning and changing cargoes;

.2 cargo temperature and pressure control systems;

.3 cargo system limitations, including minimum temperatures (cargo system and inner hull), maximum pressures, transfer rates, filling limits and sloshing limitations;

.4 nitrogen and inert gas systems;

.5 firefighting procedures: operation and maintenance of firefighting systems and use of extinguishing agents;

.6 special equipment needed for the safe handling of the particular cargo;

.7 fixed and portable gas detection;

.8 control, alarm and safety systems;

.9 emergency shutdown systems;

.10 procedures to change cargo tank pressure relief valve set pressures in accordance with 8.2.8 and 4.13.2.3; and

.11 emergency procedures, including cargo tank relief valve isolation, single tank gas-freeing and entry and emergency ship-to-ship transfer operations.

18.3 Cargo information

18.3.1 Information is to be on board and available to all concerned in the form of a cargo information data sheet(s) giving the necessary data for the safe carriage of the cargo. Such information is to include for each product carried:

.1 a full description of the physical and chemical properties necessary for the safe carriage and containment of the cargo;

.2 reactivity with other cargoes that are capable of being carried on board in accordance with the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;

.3 action to be taken in the event of spills or leaks;

.4 counter measures against accidental personal contact;

.5 fire-fighting procedures and fire-fighting media;

.6 special equipment needed for the safe handling of the particular cargo;

.7 emergency procedures.

18.3.2 The physical data supplied to the master, in accordance with 18.3.1.1, is to include information regarding the relative cargo density at various temperatures to enable the calculation of cargo tank filling limits in accordance with the requirements of sec 15.

18.3.3 Contingency plans in accordance with 18.3.1.3, for spillage of cargo carried at ambient temperature, shall take account of potential local temperature reduction such as when the escaped cargo has reduced to atmospheric pressure and the potential effect of this cooling on hull steel.

18.4 Suitability of carriage

18.4.1 The master is to ascertain that the quantity and characteristics of each product to be loaded are within the limits indicated on the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk required by 1.4 and in the loading and stability information booklet required by 2.2.5 and that products are listed in the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk as required under Sec.4 of the certificate.

18.4.2 Care is to be taken to avoid dangerous chemical reactions if cargoes are mixed. This is of particular significance in respect of:

.1 tank cleaning procedures required between successive cargoes in the same tank; and

.2 simultaneous carriage of cargoes that react when mixed. This is to be permitted only if the complete cargo systems including, but not limited to, cargo pipe work, tanks, vent systems and refrigeration systems are separated as defined in 1.2.27.

18.4.3 Where products are required to be inhibited, the certificate required by 17.8 is to be supplied before departure, otherwise the cargo shall not be transported.

18.5 Carriage of cargo at low temperature

When carrying cargoes at low temperatures:

.1 the cooldown procedure laid down for that particular tank, piping and ancillary equipment is to be followed closely;

.2 loading is to be carried out in such a manner as to ensure that design temperature gradients are not exceeded in any cargo tank, piping or other ancillary equipment; and

.3 if provided, the heating arrangements associated with the cargo containment systems are to be operated in such a manner as to ensure that the temperature of the hull structure does not fall below that for which the material is designed.

18.6 Cargo transfer operations

18.6.1 A pre-cargo operations meeting is to take place between ship personnel and the persons responsible at the transfer facility. Information exchanged is to include the details of the intended cargo transfer operations and emergency procedures. A recognized industry checklist is to be completed for the intended cargo transfer and effective communications shall be maintained throughout the operation.

18.6.2 Essential cargo handling controls and alarms shall be checked and tested prior to cargo transfer operations.

18.7 Personnel training

18.7.1 Personnel are to be adequately trained in the operational and safety aspects of liquefied gas carriers as required by the International Convention on Standards of Training, Certification and Watchkeeping for Seafareres, 1978, as amended, the International Safety Management Code and Medical Firat Aid Guide (MFAG). As a minimum:

.1 All personnel are to be adequately trained in the use of protective equipment provided on board and have basic training in the procedures, appropriate to their duties, necessary under emergency conditions.

.2 Officers are to be trained in emergency procedures to deal with conditions of leakage, spillage or fire involving the cargo and a sufficient number of them are to be instructed and trained in essential first aid for the cargoes carried.

18.8 Entry into enclosed spaces

18.8.1 Under normal operational circumstances, personnel are not to enter cargo tanks, hold spaces, void spaces or other enclosed spaces where gas may accumulate, unless the gas content of the atmosphere in such space is determined by means of fixed or portable equipment to ensure oxygen sufficiency and the absence of toxic atmosphere.

18.8.2 If it is necessary to gas-free and aerate a hold space surrounding a type A cargo tank for routine inspection, and flammable cargo is carried in the cargo tank, the inspection is to be conducted when the tank contains only the minimum amount of cargo "heel" to keep the cargo tank cold. The hold is to be re-inerted as soon as the inspection is completed.

18.8.3 Personnel entering any space designated as hazardous area on a ship carrying flammable products are not to introduce any potential source of ignition into the space unless it has been certified gas-free and is maintained in that condition.

18.9 Cargo sampling

18.9.1 Any cargo sampling is to be conducted under the supervision of an officer who is to ensure that protective clothing appropriate to the hazards of the cargo is used by everyone involved in the operation.

18.9.2 When taking liquid cargo samples, the officer shall ensure that the sampling equipment is suitable for the temperatures and pressures involved, including cargo pump discharge pressure, if relevant.

18.9.3 The officer is to ensure that any cargo sample equipment used is connected properly to avoid any cargo leakage.

18.9.4 If the cargo to be sampled is a toxic product, the officer is to ensure that a "closed loop" sampling system as defined in 1.2.15 is used to minimize any cargo release to atmosphere.

18.9.5 After sampling operations are completed, the officer is to ensure that any sample valves used are closed properly and the connections used are correctly blanked.

IR 18.9 Requirements in 18.9 above are only applicable if such a sampling system is fitted on board. Connections used for control of atmosphere in cargo tanks during inerting or gassing up are not considered as cargo sampling connections.

18.10 Cargo emergency shutdown (ESD) system

18.10.1 General

18.10.1.1 A cargo emergency shutdown system is to be fitted to stop cargo flow in the event of an emergency, either internally within the ship, or during cargo transfer to ship or shore. The design of the ESD system is to avoid the potential generation of surge pressures within cargo transfer pipe work (see 18.10.2.1.4).

18.10.1.2 Auxiliary systems for conditioning the cargo that use toxic or flammable liquids or vapours are to be treated as cargo systems for the purposes of ESD. Indirect refrigeration systems using an inert medium, such as nitrogen, need not be included in the ESD function.

18.10.1.3 The ESD system is to be activated by the manual and automatic initiations listed in table 18.1. Any additional initiations are only to be included in the ESD system if it can be shown that their inclusion does not reduce the integrity and reliability of the system overall.

18.10.1.4 Ship's ESD systems are to incorporate a ship-shore link in accordance with ISO 28460:2010.

18.10.1.5 A functional flow chart of the ESD system and related systems shall be provided in the cargo control station and on the navigation bridge.

18.10.2 ESD valve requirements

18.10.2.1 General

18.10.2.1.1 The term *ESD valve* means any valve operated by the ESD system.

18.10.2.1.2 ESD valves are to be remotely operated, be of the fail-closed type (closed on

loss of actuating power), be capable of local manual closure and have positive indication of the actual valve position. As an alternative to the local manual closing of the ESD valve, a manually operated shut-off valve in series with the ESD valve shall be permitted. The manual valve shall be located adjacent to the ESD valve. Provisions are to be made to handle trapped liquid should the ESD valve close while the manual valve is also closed.

18.10.2.1.3 ESD valves in liquid piping systems are to close fully and smoothly within 30 s of actuation. Information about the closure time of the valves and their operating characteristics shall be available on board, and the closing time shall be verifiable and repeatable.

18.10.2.1.4 The closing time of the valve referred to in 13.3.1 to 13.3.3 (i.e. time from shutdown signal initiation to complete valve closure) shall not be greater than:

$$\frac{3600U}{LR}\ Seconds$$

Where:

U = ullage volume at operating signal level (m^3);

LR = maximum loading rate agreed between ship and shore facility (m^3/h).

The loading rate is to be adjusted to limit surge pressure on valve closure to an acceptable level, taking into account the loading hose or arm, the ship and the shore piping systems, where relevant.

18.10.2.2 Ship-shore and ship-ship manifold connections

One ESD valve is to be provided at each manifold connection. Cargo manifold connections not being used for transfer operations are to be blanked with blank flanges rated for the design pressure of the pipeline system.

18.10.2.3 Cargo system valves

If cargo system valves as defined in section 5.5 are also ESD valves within the meaning of 18.10, then the requirements of 18.10 applies.

18.10.2.4 Fail-close action of ESD valve

When ESD valve is actuated by hydraulic or pneumatic system, the following are be complied with.

1. Audible and visible alarm is to be given in the event of loss of pressure that causes activation of fail-close action. The alarm is to be provided in a normally manned control station (e.g. Cargo Control Room and/or the navigation bridge, etc.).

2. The following conditions are also to be complied to ensure the fail-close action:

1. Failure of hydraulic or pneumatic system is not to lead to loss of fail-close functionality (i.e. activated by spring or weight); or

2. Hydraulic or pneumatic system for fail-close action is to be arranged with stored power and separated from normal valve operation.

18.10.3 ESD system controls

18.10.3.1 As a minimum, the ESD system is to be capable of manual operation by a single control on the bridge and either in the control position required by 13.1.2 or the cargo control room, if installed, and no less than two locations in the cargo area.

18.10.3.2 The ESD system is to be automatically activated on detection of a fire on the weather decks of the cargo area and/or cargo machinery spaces. As a minimum, the method of detection used on the weather decks shall cover the liquid and vapour domes of the cargo tanks, the cargo manifolds and areas where liquid piping is dismantled regularly. Detection may be by means of fusible elements designed to melt at temperatures between 98°C and 104°C, or by area fire detection methods.

18.10.3.3 Cargo machinery that is running is to be stopped by activation of the ESD system in accordance with the cause and effect matrix in table 18.1.

18.10.3.4 The ESD control system is to be configured so as to enable the high-level testing required in 13.3.5 to be carried out in a safe and controlled manner. For the purpose of the testing, cargo pumps may be operated while

the overflow control system is overridden. Procedures for level alarm testing and re-setting of the ESD system after completion of the high-level alarm testing shall be included in the operation manual required by 18.2.1.

18.10.4 Additional shutdowns

18.10.4.1 The requirements of 8.3.1.1 to protect the cargo tank from external differential pressure may be fulfilled by using an independent low pressure trip to activate the ESD system, or, as minimum, to stop any cargo pumps or compressors.

18.10.4.2 An input to the ESD system from the overflow control system required by 13.3 may be provided to stop any cargo pumps or compressors' running at the time a high level is detected, as this alarm may be due to inadvertent internal transfer of cargo from tank to tank.

18.10.5 Pre-operations testing

Cargo emergency shutdown and alarm systems involved in cargo transfer is to be checked and tested before cargo handling operations begin.

18.11 Hot work on or near cargo containment systems

18.11.1 Special fire precautions are to be taken in the vicinity of cargo tanks and, particularly, insulation systems that may be flammable or contaminated with hydrocarbons or that may give off toxic fumes as a product of combustion.

18.12 Additional operating requirements

Additional operating requirements will be found in the following paragraphs of this Chapter:

2.2.2, 2.2.5, 2.2.8, 3.8.4, 3.8.5,5.3.2, 5.3.3.3, 5.7.3, 7.1, 8.2.7,8.2.8, 8.2.9, 9.2, 9.3, 9.4.4, 12.1.1, 13.1.3, 13.3.6, 13.6.18, 14.3.3, 15.3, 16.6.3, 17.4.2, 17.6, 17.7, 17.9, 17.10, 17.11, 17.12, 17.13, 17.14, 17.16, 17.18,17.19, 17.21, 17.22.

Table 18.1 : ESD Functional Arrangements								
	Pumps		**Compressor systems**				**Valves**	**Link**
Shut down action Initiation	Cargo pumps/Cargo booster pumps	Spray/stripping pumps	Vapour return compressors	Fuel gas compressors	Reliquification *** plant including condensante return pumps, if fitted	Gas combustion unit	ESD Valve	Signal to ship/shore link***
Emergency push buttons (see 18.10.3.1)	✓	✓	✓	Note 2	✓	✓	✓	✓
Fire detection on deck or in compressor house* (see 18.10.3.2	✓	✓	✓	✓	✓	✓	✓	✓
High level in cargo tank (see 13.3.2 and 13.3.3)	✓	✓	✓	Note 1, Note 2	Note 1, Note 3	Note 1	Note 6	✓
Signal from ship/shore link (see 18.10.1.4)	✓	✓	✓	Note 2	Note 3	n/a	✓	n/a
Loss of motive power to ESD Valves**	✓	✓	✓	Note 2	Note 3	n/a	✓	✓
Main electric power failure (“blackout”)	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	✓	✓
Level alarm override (see 13.3.7)	Note 4	Note 4, Note 5	✓	Note 1	Note 1	Note 1	✓	✓

Note 1: These items of equipment can be omitted from these specific automatic shutdown initiators, provided the equipment inlets are protected against cargo liquid ingress.

Note 2: If the fuel gas compressor is used to return cargo vapour to shore, it shall be included in the ESD system when operating in this mode.

Note 3: If the reliquefaction plant compressors are used for vapour return/shore line clearing, they shall be included in the ESD system when operating in that mode.

Note 4: The override system permitted by 13.3.7 may be used at sea to prevent false alarms or shutdowns. When level alarms are overridden, operation of cargo pumps and the opening of manifold ESD valves shall be inhibited except when high-level alarm testing is carried out in accordance with 13.3.5 (see 18.10.3.4).

IR Note 4: In applying the second sentence of note 4 above, a hardware system such as an electric or mechanical interlocking device is to be provided to prevent inadvertent operation of cargo pumps and inadvertent opening of manifold ESD valves.

Note 5: Cargo spray or stripping pumps used to supply forcing vaporizer may be excluded from the ESD system only when operating in that mode.

Note 6: The sensors referred to in 13.3.2 may be used to close automatically the tank filling valve for the individual tank where the sensors are installed, as an alternative to closing the ESD valve referred to in 18.10.2.2. If this option is adopted, activation of the full ESD system shall be initiated when the high-level sensors in all the tanks to be loaded have been activated.

Note 7: These items of equipment shall be designed not to restart upon recovery of main electric power and without confirmation of safe conditions.

* Fusible plugs, electronic point temperature monitoring or area fire detection may be used for this purpose on deck.

** Failure of hydraulic, electric or pneumatic power for remotely operated ESD valve actuators.
*** Indirect refrigeration systems which form part of the reliquefaction plant do not need to be included in the ESD function if they employ an inert medium such as nitrogen in the refrigeration cycle.

**** Signal need not indicate the event initiating ESD.

✓ Functional requirement.

N/A Not applicable.

Section 19

Summary of Minimum Requirements

19.1 General

19.1.1 The summary of the requirements of this Chapter is given in Table 19.1.1.

19.1.2 Explanatory notes to the summary of minimum requirements :

Product name *(column a)*	The product name is to be used in the shipping document for any cargo offered for bulk shipments. Any additional name may be included in brackets after the product name. In some cases, the product names are not identical with the names given in previous issues of the IGC Code.
(column b)	Deleted
Ship type *(column c)*	1: Ship type 1G (2.1.2.1) 2: Ship type 2G (2.1.2.2) 3: Ship type 2PG (2.1.2.3) 4: Ship type 3G (2.1.2.4)
Independent tank type C required *(column d)*	Type C independent tank (4.23)
Tank environmental control *(column e)*	Inert: Inerting (9.4) Dry: Drying (17.7) - : No special requirements under the Code
Vapour detection *(column f)*	F: Flammable vapour detection T: Toxic vapour detection F+T: Flammable and toxic vapour detection A: Asphixiant
Gauging *(column g)*	I: Indirect or closed (13.2.3.1 and .2) R: Indirect, closed or restricted (13.2.3.1, .2, .3 and .4) C: Indirect or closed (13.2.3.1, .2 and .3)
(column h)	Deleted
Special requirements *(column i)*	When specific reference is made to chapters 14 and/or 17, these requirements shall be additional to the requirements in any other column.
Refrigerant gases	Non-toxic and non-flammable gases

Unless otherwise specified, gas mixtures containing less than 5% total acetylenes may be transported with no further requirements than those provided for the major components.

Table 19.1.1 : Summary of minimum requirements								
a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**
Product name	**-**	**Ship type**	**Independent tank type C required**	**Control of vapour space within cargo tanks**	**Vapour detection**	**Gauging**	**-**	**Special requirements**
Acetaldehyde	-	2G/2PG	-	Inert	F+T	C	-	14.4.3, 14.3.3.1, 17.4.1, 17.6.1
Ammonia, anhydrous	-	2G/2PG	-	-	T	C		14.4, 17.2.1, 17.12
Butadiene (all isomers)	-	2G/2PG	-	-	F+T	R	-	14.4, 17.2.2, 17.4.2, 17.4.3, 17.6, 17.8
Butane (all isomers)	-	2G/2PG	-	-	F	R	-	
Butane/Propane mixtures	-	2G/2PG	-	-	F	R	-	
Butylenes (all isomers)	-	2G/2PG	-	-	F	R	-	
Carbon dioxide (high purity)	-	3G	-	-	A	R	-	17.21
Carbon dioxide (Reclaimed quality)	-	3G	-	-	A	R	-	17.22
Chlorine	-	1G	Yes	Dry	T	I	-	14.4, 17.3.2, 17.4.1, 17.5, 17.7, 17.9, 17.13
Diethyl ether*	-	2G/2PG	-	Inert	F+T	C	-	14.4.2, 14.4.3, 17.2.6, 17.3.1, 17.6.1, 17.9, 17.10, 17.11.2, 17.11.3
Dimethylamine	-	2G/2PG	-	-	F+T	C	-	14.4, 17.2.1
Dimethyl ether	-	2G/2PG	-	-	F+T	C	-	
Ethane	-	2G	-	-	F	R	-	
Ethyl Chloride	-	2G/2PG	-	-	F+T	C	-	
Ethylene	-	2G	-	-	F	R	-	
Ethylene Oxide	-	1G	Yes	Inert	F+T	C	-	14.4, 17.2.2, 17.3.2, 17.4.1, 17.5, 17.6.1, 17.14

a	b	c	d	e	f	g	h	i
Product name	**-**	**Ship type**	**Independent tank type C required**	**Control of vapour space within cargo tanks**	**Vapour detection**	**Gauging**	**-**	**Special requirements**
Ethylene oxide-propylene oxide mixtures with ethylene oxide content of not more than 30% by weight*	-	2G/2PG	-	Inert	F+T	C	-	14.4.3, 17.3.1, 17.4.1, 17.6.1, 17.9, 17.10, 17.18
Isoprene* (all isomers)	-	2G/2PG	-	-	F	R	-	14.4.3, 17.8, 17.9, 17.11.1
Isoprene (Part refined)*	-	2G/2PG	-	-	F	R	-	14.4.3, 17.8, 17.9, 17.11.1
Isopropylamine*	-	2G/2PG	-	-	F+T	C	-	14.4.2, , 17.2.3, 17.3.2, 17.4.1, 17.5
Methane (LNG)	-	2G	-	-	F	C	-	
Methyl Acetylene-Propadiene mixture	-	2G/2PG	-	-	F	R	-	17.16
Methyl Bromide	-	1G	Yes	-	F+T	C	-	14.4, 17.2.3, 17.3.2, 17.4.1, 17.5,
Methyl Chloride	-	2G/2PG	-	-	F+T	C	-	17.2.3
Mixed C4 Cargoes	-	2G/2PG	-	-	F + T	C	-	14.4, 17.2.2, 17.4.2, 17.4.3, 17.6, 17.20
Monoethylamine*	-	2G/2PG	-	-	F+T	C	-	14.4, 17.2.1, 17.3.1, 17.9, 17.10, 17.11.1, 17.15
Nitrogen	-	3G	-	-	A	C	-	17.17
Pentanes (all isomers)*	-	2G/2PG	-	-	F	R	-	17.9, 17.11
Pentenes (all isomers)*	-	2G/2PG	-	-	F	R	-	17.9, 17.11
Propane	-	2G/2PG	-	-	F	R	-	
Propylene	-	2G/2PG	-	-	F	R	-	

a	b	c	d	e	f	g	h	i
Product name	**-**	**Ship type**	**Independent tank type C required**	**Control of vapour space within cargo tanks**	**Vapour detection**	**Gauging**	**-**	**Special requirements**
Propylene oxide*	-	2G/2PG	-	Inert	F+T	C	-	14.4.3, 17.3.1, 17.4.1, 17.6.1, 17.9, 17.10, 17.18
Refrigerant Gases	-	3G	-	-	-	R	-	
Sulphur Dioxide	-	1G	Yes	Dry	T	C	-	14.4, 17.3.2, 17.4.1, 17.5, 17.7
Vinyl Chloride	-	2G/2PG	-	-	F+T	C	-	14.4.2, 17.2.2, 17.2.3, 17.3.1, 17.6, 17.19
Vinyl ethyl ether*	-	2G/2PG	-	Inert	F+T	C	-	14.4.2, 17.2.2, 17.3.1, 17.6.1, 17.8, 17.9, 17.10, 17.11.2, 17.11.3
Vinylidene chloride*	-	2G/2PG	-	Inert	F+T	R	-	14.4.2, 17.2.5, 17.6.1, 17.8, 17.9, 17.10

*** The cargo are covered also by Rules for Chemical Carriers**

APPENDIX 4 of IGC Code- NON-METALLIC MATERIALS

1 General

1.1 The guidance given in this appendix is in addition to the requirements of 4.19, where applicable to non-metallic materials.

1.2 The manufacture, testing, inspection and documentation of non-metallic materials are to in general comply with recognized standards, and with the specific requirements of this chapter, as applicable.

1.3 When selecting a non-metallic material, the designer should ensure that it has properties appropriate to the analysis and specification of the system requirements. A material can be selected to fulfill one or more requirements.

1.4 A wide range of non-metallic materials may be considered. Therefore, the section below on material selection criteria cannot cover every eventuality and should be considered as guidance.

2 Material selection criteria

2.1 Non-metallic materials may be selected for use in various parts of liquefied gas carrier cargo systems based on consideration of the following basic properties:

.1 insulation – the ability to limit heat flow;

.2 load bearing – the ability to contribute to the strength of the containment system;

.3 tightness – the ability to provide liquid and vapour tight barriers;

.4 joining – the ability to be joined (for example by bonding, welding or fastening).

2.2 Additional considerations may apply depending on the specific system design.

3 Properties of materials

3.1 Flexibility of insulating material is the ability of an insulating material to be bent or shaped easily without damage or breakage.

3.2 Loose fill material is a homogeneous solid generally in the form of fine particles, such as a powder or beads, normally used to fill the voids in an inaccessible space to provide an effective insulation.

3.3 Nanomaterial is a material with properties derived from its specific microscopic structure.

3.4 Cellular material is a material type containing cells that are either open, closed or both and which are dispersed throughout its mass.

3.5 Adhesive material is a product that joins or bonds two adjacent surfaces together by an adhesive process.

3.6 Other materials are materials that are not characterized in this section of the Chapter and are to be identified and listed. The relevant tests used to evaluate the suitability of material for use in the cargo system should be identified and documented.

4 Material selection and testing requirements

4.1 **Material specification**

4.1.1 When the initial selection of a material has been made, tests are to be conducted to validate the suitability of this material for the use intended.

4.1.2 The material used is to clearly be identified and the relevant tests should be fully documented.

4.1.3 Materials should be selected according to their intended use. They are :

.1 to be compatible with all the products that may be carried;

.2 not to be contaminated by any cargo nor react with it;

.3 not to have any characteristics or properties affected by the cargo; and

.4 to be capable to withstand thermal shocks within the operating temperature range.

.2 ***Material testing***

The tests required for a particular material depend on the design analysis, specification and intended duty. The list of tests below is for illustration. Any additional tests required, for example in respect of sliding, damping and galvanic insulation, is to be identified clearly and documented. Materials selected according to 4.1 of this appendix is to be tested further according to the following table:

Function	Insulation	Load bearing structural	Tightness	Joining
Mechenical Tests		**X**		**X**
Tightness Tests			**X**	
Thermal tests	**X**			

Thermal shock testing is to subject the material and/or assembly to the most extreme thermal gradient it will experience when in service.

4.2.1 Inherent properties of materials

4.2.1.1 Tests are to be carried out to ensure that the inherent properties of the material selected will not have any negative impact in respect of the use intended.

4.2.1.2 For all selected materials, the following properties are to be evaluated:

1 density; example standard ISO 845; and

.2 linear coefficient of thermal expansion (LCTE); example standard ISO 11359 across the widest specified operating temperature range. However, for loose fill material the volumetric coefficient of thermal expansion (VCTE) should be evaluated, as this is more relevant.

4.2.1.3 Irrespective of its inherent properties and intended duty, all materials selected should be tested for the design service temperature range down to 5°C below the minimum design temperature, but not lower than -196°C.

4.2.1.4 Each property evaluation test is to be performed in accordance with recognized standards. Where there are no such standards, the test procedure proposed is to be fully detailed and submitted to the Administration/ IRS for acceptance. Sampling is to be sufficient to ensure a true representation of the properties of the material selected.

4.2.2 Mechanical tests

4.2.2.1 The mechanical tests are to be performed in accordance with the following table.

Mechanical Tests	Load bearing structural
Tensile	**ISO 527** **ISO 1421** **ISO 3346** **ISO 1926**
Shearing	**ISO 4587** **ISO 3347** **ISO 1922** **ISO 6237**
Compressive	**ISO 604** **ISO 844** **ISO 3132**
Bending	**ISO 3133** **ISO 14679**
Creep	**ISO 7850**

4.2.2.2 If the chosen function for a material relies on particular properties such as tensile, compressive and shear strength, yield stress, modulus or elongation, these properties is to be tested to a recognized standard. If the properties required are assessed by numerical simulation according to a high order behaviour law, the testing is to be performed to the satisfaction of the Administration/ IRS.

4.2.2.3 Creep may be caused by sustained loads, for example cargo pressure or structural loads. Creep testing is to be conducted based on the loads expected to be encountered during the design life of the containment system.

4.2.3 Tightness tests

4.2.3.1 The tightness requirement for the material is to relate to its operational functionality.

4.2.3.2 Tightness tests are to be conducted to give a measurement of the material's permeability in the configuration corresponding to the application envisaged (e.g. thickness and stress conditions) using the fluid to be retained (e.g. cargo, water vapour or trace gas).

4.2.3.3 The tightness tests are to be based on the tests indicated as examples in the following table.

Tightness Test	Tightness
Porosity/Permeability	ISO 15106 ISO 2568 ISO 2782

4.2.4 Thermal conductivity tests

4.2.4.1 Thermal conductivity tests are to be representative of the lifecycle of the insulation material so its properties over the design life of the cargo system can be assessed. If these properties are likely to deteriorate over time, the material is to be aged as best possible in an environment corresponding to its lifecycle, for example operating temperature, light, vapour and installation (e.g. packaging, bags, boxes, etc.).

4.2.4.2 Requirements for the absolute value and acceptable range of thermal conductivity and heat capacity is to be chosen taking into account the effect on the operational efficiency of the cargo containment system. Particular attention should also be paid to the sizing of the associated cargo handling system and components such as safety relief valves plus vapour return and handling equipment.

4.2.4.3 Thermal tests are to be based on the tests indicated as examples in the following table or their equivalents:

Thermal tests	**Insulating**
Thermal conductivity	**ISO 8301** **ISO 8302**
Heat capacity	**x**

4.2.5 Physical tests

4.2.5.1 In addition to the requirements of 4.19.2.3 and 4.19.3.2, the following table provides guidance and information on some of the additional physical tests that may be considered.

Physical tests	Flexible Insulating	Loose Fill	Nano-material	Cellular	Adhesive
Particle size		X			
Closed cells content				ISO 4590	
Absorption/Desorption	ISO 12571	X	X	ISO 2896	
Viscosity			X		ISO 2555 ISO 2431
Open time					ISO 10364
Thixotropic properties					X
Hardness					ISO 868

4.2.5.2 Requirements for loose fill material segregation is to be chosen considering its potential adverse effect on the material properties (density, thermal conductivity) when subjected to environmental variations such as thermal cycling and vibration.

4.2.5.3 Requirements for a material with closed cell structures is to be based on its eventual impact on gas flow and buffering capacity during transient thermal phases.

4.2.5.4 Similarly, adsorption and absorption requirements is to take into account the potential adverse effect an uncontrolled buffering of liquid or gas may have on the system.

5 Quality assurance and quality control (QA/QC)

5.1 ***General***

5.1.1 Once a material has been selected, after testing as outlined in section 4 of this appendix, a detailed quality assurance/quality control (QA/QC) programme is to be applied to ensure the continued conformity of the material during installation and service. This programme is to consider the material starting from the manufacturer's quality manual (QM) and then follow it throughout the construction of the cargo system.

5.1.2 The QA/QC programme is to include the procedure for fabrication, storage, handling and preventive actions to guard against exposure of a material to harmful effects. These may include, for example, the effect of sunlight on some insulation materials or the contamination of material surfaces by contact with personal products such as hand creams. The sampling methods and the frequency of testing in the QA/QC programme is to be specified to ensure the continued conformity of the material selected throughout its production and installation.

5.1.3 Where powder or granulated insulation is produced, arrangements are to be made to prevent compacting of the material due to vibrations.

5.2 ***QA/QC during component manufacture***

The QA/QC programme in respect of component manufacture is to include, as a minimum but not limited to, the following items.

5.2.1 Component identification

5.2.1.1 For each material, the manufacturer is to implement a marking system to clearly identify the production batch. The marking system is not to interfere, in any way, with the properties of the product.

5.2.1.2 The marking system is to ensure complete traceability of the component and is to include:

.1 date of production and potential expiry date;

.2 manufacturer's references;

.3 reference specification;

.4 reference order; and

.5 when necessary, any potential environmental parameters to be maintained during transportation and storage.

5.2.2 Production sampling and audit method

5.2.2.1 Regular sampling is required during production to ensure the quality level and continued conformity of a selected material.

5.2.2.2 The frequency, the method and the tests to be performed are to be defined in QA/QC programme; for example, these tests will usually cover, inter alia, raw materials, process parameters and component checks.

5.2.2.3 Process parameters and results of the production QC tests are to be in strict accordance with those detailed in the QM for the material selected.

5.2.2.4 The objective of the audit method as described in the QM is to control the repeatability of the process and the efficacy of the QA/QC programme.

5.2.2.5 During auditing, auditors are to be provided with free access to all production and QC areas. Audit results are to be in accordance with the values and tolerances as stated in the relevant QM.

6 Bonding and joining process requirement and testing

6.1 ***Bonding procedure qualification***

6.1.1 The bonding procedure specification and qualification test is to be defined in accordance with recognized standards.

6.1.2 The bonding procedures are to be fully documented before work commences to ensure the properties of the bond are acceptable.

6.1.3 The following parameters are to be considered when developing a bonding procedure specification:

.1 surface preparation;

.2 materials storage and handling prior to installation;

.3 covering-time;

.4 open-time;

.5 mixing ratio, deposited quantity;

.6 environmental parameters (temperature, humidity); and

.7 curing pressure, temperature and time.

6.1.4 Additional requirements may be included as necessary to ensure acceptable results.

6.1.5 The bonding procedures specification is to be validated by an appropriate procedure qualification testing programme.

6.2 ***Personnel qualifications***

6.2.1 Personnel involved in bonding processes are to be trained and qualified to recognized standards.

6.2.2 Regular tests are to be made to ensure the continued performance of people carrying out bonding operations to ensure a consistent quality of bonding.

7 Production bonding tests and controls

7.1 ***Destructive testing***

During production, representative samples are to be taken and tested to check that they correspond to the required level of strength as required for the design.

7.2 ***Non-destructive testing***

7.2.1 During production, tests which are not detrimental to bond integrity is to be performed using an appropriate technique such as:

.1 visual examination;

.2 internal defects detection (for example acoustic, ultrasonic or shear test); and

.3 local tightness testing.

7.2.2 If the bonds have to provide tightness as part of their design function, a global tightness test of the cargo containment system is to be completed after the end of the erection in accordance with the designer's and QA/QC programme.

7.2.3 The QA/QC standards are to include acceptance standards for the tightness of the bonded components when built and during the lifecycle of the containment system.

APPENDIX 5 of IGC Code - STANDARD FOR THE USE OF LIMIT STATE METHODOLOGIES IN THE DESIGN OF CARGO CONTAINMENT SYSTEMS OF NOVEL CONFIGURATION

1 General

1.1 The purpose of this standard is to provide procedures and relevant design parameters of limit state design of cargo containment systems of a novel configuration in accordance with section 4.27 of this Chapter.

1.2 Limit state design is a systematic approach where each structural element is evaluated with respect to possible failure modes related to the design conditions identified in section 4.3.4 of this Chapter. A limit state can be defined as a condition beyond which the structure, or part of a structure, no longer satisfies the requirements.

1.3 The limit states are divided into the three following categories:

.1 Ultimate Limit States (ULS), which correspond to the maximum load-carrying capacity or, in some cases, to the maximum applicable strain, deformation or instability in structure resulting from buckling and plastic collapse; under intact (undamaged) conditions;

.2 Fatigue Limit States (FLS), which correspond to degradation due to the effect of cyclic loading; and

.3 Accident Limit States (ALS), which concern the ability of the structure to resist accident situations.

1.4 Part A through part D of Sec 4 of this Chapter is to be complied with as applicable depending on the cargo containment system concept.

2. Design format

2.1 The design format in this standard is based on a Load and Resistance Factor Design format. The fundamental principle of the Load and Resistance Factor Design format is to verify that design load effects, L_d , do not exceed design resistances,R_d , for any of the considered failure modes in any scenario:

$$\boldsymbol{L_d \leq R_d}$$

A design load F_{dk} is obtained by multiplying the characteristic load by a load factor relevant for the given load category:

$$F_{dk} = \gamma_f F_k$$

where:

γ_f is load factor; and

F_k is the characteristic load as specified in part B and part C of Section 4

A design load L_d (e.g. stresses, strains, displacements and vibrations) is the most unfavourable load effect derived from the design loads, and may be expressed by:

$$L_d = q(F_{d1}, F_{d2}, \ldots . F_{dN})$$

Where q denoted the functional relationship between load and load effect determined by structural analyses.

The design resistance R_d is determined as follows:

$$R_d = R_k/(\gamma_R \gamma_C)$$

Where:

R_k is the characteristic resistance. In case of materials covered by section 6, it may be, but not limited to, specified minimum yield stress, specified minimum tensile strength, plastic resistance of cross sections and ultimate buckling strength;

γ_R is the resistance factor defined as $\gamma_R = \gamma_m \gamma_s$

γ_m is the partial resistance factor to take account of the probabilistic distribution of the material properties (material factor);

γ_s is the partial resistance factor to take account of the uncertainties on the capacity of the structure, such as the quality of the construction, method considered for determination of the capacity including accuracy of analysis; and

$\boldsymbol{\gamma_C}$ is the consequence class factor, which accounts for the potential results of failure with

regard to release of cargo and possible human injury.

2.2 Cargo containment design is to take into account potential failure consequences. Consequence classes are defined in table 1, to specify the consequences of failure when the mode of failure is related to the Ultimate Limit State, the Fatigue Limit State, or the Accident Limit State.

Table 1: Consequences classes

Consequence class	Definition
Low	Failure implies minor release of the cargo
Medium	Failure implies release of the cargo and potential for human injury
High	Failure implies significant release of the cargo and high potential for human injury/fatality

3 Required analyses

3.1 Three dimensional finite element analyses is to be carried out as an integrated model of the tank and the ship hull, including supports and keying system as applicable. All the failure modes shall be identified to avoid unexpected failures. Hydrodynamic analyses are to be carried out to determine the particular ship accelerations and motions in irregular waves, and the response of the ship and its cargo containment systems to these forces and motions.

3.2 Buckling strength analyses of cargo tanks subject to external pressure and other loads causing compressive stresses are to be carried out in accordance with recognized standards. The method is to adequately account for the difference in theoretical and actual buckling stress as a result of plate out of flatness, plate edge misalignment, straightness, ovality and deviation from true circular form over a specified arc or chord length, as relevant.

3.3 Fatigue and crack propagation analysis is to be carried out in accordance with paragraph 5.1 of this standard.

4 Ultimate Limit States

4.1 Structural resistance may be established by testing or by complete analysis taking account of both elastic and plastic material properties. Safety margins for ultimate strength are to be introduced by partial factors of safety taking account of the contribution of stochastic nature of loads and resistance (dynamic loads, pressure loads, gravity loads, material strength, and buckling capacities).

4.2 Appropriate combinations of permanent loads, functional loads and environmental loads including sloshing loads are to be considered in the analysis. At least two load combinations with partial load factors as given in table 2 are to be used for the assessment of the ultimate limit states.

Table 2 : Partial load factors

Load Combina-tion	Permanent Loads	Functional Loads	Enviro-mental Loads
‘ a ‘	1.1	1.1	0.7
‘ b ‘	1.0	1.0	1.3

The load factors for permanent and functional loads in load combination '*a*' are relevant for the normally well-controlled and/or specified loads applicable to cargo containment systems such as vapour pressure, cargo weight, system self-weight, etc. Higher load factors may be relevant for permanent and functional loads where the inherent variability and/or uncertainties in the prediction models are higher.

4.3 For sloshing loads, depending on the reliability of the estimation method, a larger load factor may be required by IRS.

4.4 In cases where structural failure of the cargo containment system are considered to imply high potential for human injury and significant release of cargo, the consequence class factor shall be taken as $\gamma_C = 1.2$ This value may be reduced if it is justified through risk analysis and subject to the approval by IRS. The risk analysis is to take account of factors including, but not limited to, provision of full or partial secondary barrier to protect hull structure from the leakage and less hazards associated with intended cargo. Conversely, higher values may be fixed by the IRS, for example, for ships carrying more hazardous or higher pressure cargo. The consequence class factor is to in any case not be less than 1.0.

4.5 The load factors and the resistance factors used are to be such that the level of safety is equivalent to that of the cargo containment systems as described in sections 4.21 to 4.26 . This may be carried out by calibrating the factors against known successful designs.

4.6 The material factor γ_m is to in general reflect the statistical distribution of the mechanical properties of the material, and needs to be interpreted in combination with the specified characteristic mechanical properties. For the materials defined in section 6, the material factor γ_m may be taken as:

1.1 : when the characteristic mechanical properties specified by the recognized organization typically represents the lower 2.5% quantile in the statistical distribution of the mechanical properties; or

1.0 : when the characteristic mechanical properties specified by the recognized organization represents a sufficiently small quantile such that the probability of lower mechanical properties than specified is extremely low and can be neglected.

4.7 The partial resistance factors γ_{si} in general be established based on the uncertainties in the capacity of the structure considering construction tolerances, quality of construction, the accuracy of the analysis method applied, etc.

4.7.1 For design against excessive plastic deformation using the limit state criteria given in paragraph 4.8 of this standard, the partial resistance factors γ_{si} is to be taken as follows:

$$\gamma_{s1} = 0.76\frac{B}{k_1}$$

$$\gamma_{s1} = 0.76\frac{D}{k_1}$$

$$k_1 = Min\left(\frac{R_m}{R_e}\frac{B}{A}\,;1\right)$$

$$k_2 = Min\left(\frac{R_m}{R_e}\frac{D}{C}\,;1\right)$$

Factors A, B, C and D are defined in section 4.22.3.1. R_m and R_e are defined in section 4.18.1.3.

The partial resistance factors given above are the results of calibration to conventional type B independent tanks.

4.8 Design against excessive plastic deformation

4.8.1 Stress acceptance criteria given below refer to elastic stress analyses.

4.8.2 Parts of cargo containment systems where loads are primarily carried by membrane response in the structure are to satisfy the following limit state criteria:

$$\sigma_m \le f$$

$$\sigma_L \le 1.5f$$

$$\sigma_b \le 1.5F$$

$$\sigma_L + \sigma_b \le 1.5F$$

$$\sigma_m + \sigma_b \le 1.5F$$

$$\sigma_m + \sigma_b + \sigma_g \le 3F$$

$$\sigma_L + \sigma_b + \sigma_g \le 3F$$

Where:

σ_m=equivalent primary general membrane stress

σ_L=equivalent primary local membrane stress

σ_b=equivalent primary bending stress

σ_g=equivalent secondary stress

$$f = \frac{R_e}{\gamma_{s1}\,\gamma_m\,\gamma_C}$$

$$F = \frac{R_e}{\gamma_{s2}\,\gamma_m\,\gamma_C}$$

With regard to the stresses σ_m , σ_L , σ_b and σ_g , see also the definition of stress categories in section 4.28.3.

4.8.3 Parts of cargo containment systems where loads are primarily carried by bending of girders, stiffeners and plates, are to satisfy the following limit state criteria:

$$\sigma_{ms} + \sigma_{bp} \le 1.25F \quad (note\ 1\ and\ 2)$$

$$\sigma_{ms} + \sigma_{bp} + \sigma_{bs} \le 1.25F \quad (note\ 2)$$

$$\sigma_{ms} + \sigma_{bp} + \sigma_{bs} + \sigma_{bt} + \sigma_g \le 3F$$

Note 1: The sum of equivalent section membrane stress and equivalent membrane stress in primary structure ($\sigma_{ms} + \sigma_{bp}$) will

normally be directly available from three-dimensional finite element analyses.

Note 2: The coefficient, 1.25, may be modified by IRS considering the design concept, configuration of the structure, and the methodology used for calculation of stresses.

Where:

σ_{ms}= equivalent section membrane stress in primary structure

σ_{bp}= equivalent membrane stress in primary structure and stress in secondary and tertiary structure caused by bending of primary structure

σ_{bs}= section bending stress in secondary structure and stress in tertiary structure caused by bending of secondary structure

σ_{bt}= section bending stress in tertiary structure

σ_g= equivalent secondary stress

$$f = \frac{R_e}{\gamma_{s1}\,\gamma_m\,\gamma_C}$$

$$F = \frac{R_e}{\gamma_{s2}\,\gamma_m\,\gamma_C}$$

The stresses σ_{ms} , σ_{bp} , σ_{bs} and σ_{bt} , are defined in 4.8.4. For a definition of σ_g , see section 4.28.3.

Skin plates are to be designed in accordance with the requirements of IRS. When membrane stress is significant, the effect of the membrane stress on the plate bending capacity is to be appropriately considered in addition.

4.8.4 Section stress categories

Normal stress is the component of stress normal to the plane of reference.

Equivalent section membrane stress is the component of the normal stress that is uniformly distributed and equal to the average value of the stress across the cross section of the structure under consideration. If this is a simple shell section, the section membrane stress is identical to the membrane stress defined in paragraph 4.8.2 of this standard.

Section bending stress is the component of the normal stress that is linearly distributed over a structural section exposed to bending action, as illustrated in Fig.1.

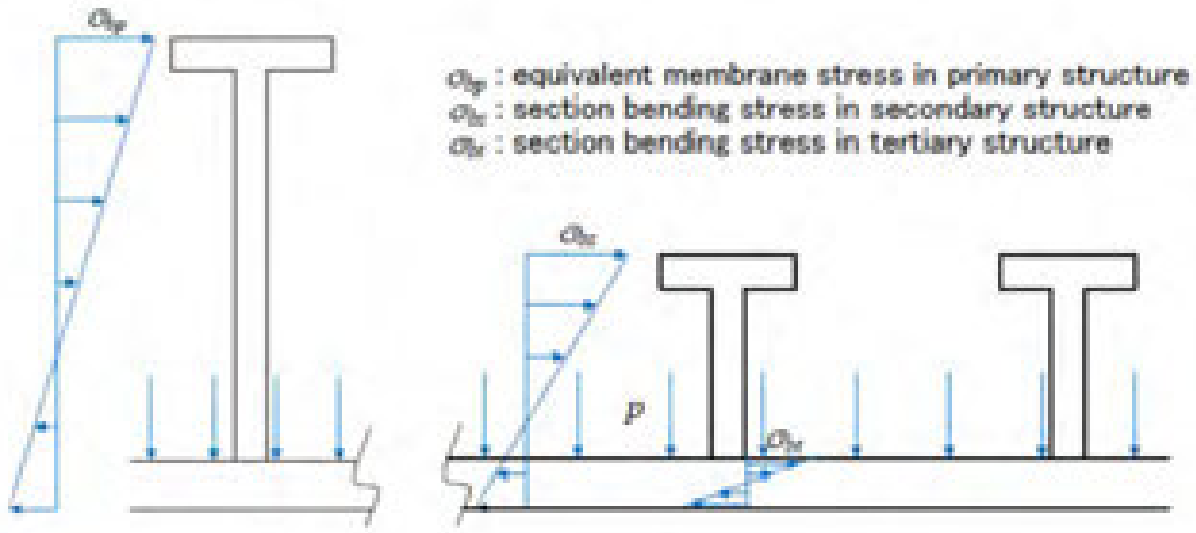


Figure 1: Definition of the three categories of section stress (Stresses σ_{bp} and σ_{bs} are normal to the cross section shown.)

4.9 The same factors γ_C , γ_m , γ_{si} are to be used for design against buckling unless otherwise stated in the applied recognized buckling standard. In any case the overall level of safety is not to be less than given by these factors.

5 Fatigue Limit States

5.1 Fatigue design condition as described in section 4.18.2 is to be complied with as applicable depending on the cargo containment system concept. Fatigue analysis is required for the cargo containment system designed under section 4.27 and this annexure.

5.2 The load factors for FLS shall be taken as 1.0 for all load categories.

5.3 Consequence class factor γ_C and resistance factor γ_R are to be taken as 1.0.

5.4 Fatigue damage is to be calculated as described in sections 4.18.2.2 to 4.18.2.5. The calculated cumulative fatigue damage ratio for the cargo containment systems are to be less than or equal to the values given in Table 3.

Table 3 : Maximum allowable cumulative fatigue damage ratio

	Consequence Class		
C_W	Low	Medium	High
	1	0.5	0.5 (Note 1)

Note 1: Lower value are to be used in accordance with sections 4.18.2.7 to 4.18.2.9 , depending on the detectability of defect or crack, etc.

5.5 Lower values may be fixed by IRS, for example for tank structures where effective detection of defect or crack cannot be assured, and for ships carrying more hazardous cargo.

5.6 Crack propagation analyses are required in accordance with sections 4.18.2.6 to 4.18.2.9. The analysis is to be carried out in accordance with methods laid down in a standard recognized by IRS.

6 Accident Limit States

6.1 Accident design condition as described in section 4.18.3 is to be complied with as applicable, depending on the cargo containment system concept.

6.2 Load and resistance factors may be relaxed compared to the ultimate limit state considering that damages and deformations can be accepted as long as this does not escalate the accident scenario.

6.3 The load factors for ALS are to be taken as 1.0 for permanent loads, functional loads and environmental loads.

6.4 Loads mentioned in section 4.13.9 (Static heel loads) and section 4.15 (Collision and Loads due to flooding on ship) need not be combined with each other or with environmental loads, as defined in section 4.14.

6.5 Resistance factor γ_R is to in general be taken as 1.0.

6.6 Consequence class factors γ_C are to be in general be taken as defined in paragraph 4.4, but may be relaxed considering the nature of the accident scenario.

6.7 The characteristic resistance R_k is to in general be taken as for the ultimate limit state, but may be relaxed considering the nature of the accident scenario.

6.8 Additional relevant accident scenarios are to be determined based on a risk analysis.

7 Testing

7.1 Cargo containment systems designed according to this standard is to be tested to the same extent as described in section 4.20.3, as applicable depending on the cargo containment system concept.

End of Chapter

Chapter 5

Container Ships

Contents

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3	*Ship Arrangement*
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6	*Buckling Capacity*
7	*Ultimate Hull Girder Strength Assessment*
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9	*Container Stowage and Securing Arrangement*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to container ships and are supplementary to those given for the assignment of main characters of class.

1.1.2 Vessels built in compliance with the above requirements will be eligible to be assigned the class notation "**CONTAINER SHIP**".

1.1.3 Container ships are seagoing ships of single or double skin side construction, designed exclusively for the carriage of standard general cargo freight containers on holds and on deck.

1.1.4 When oil or dangerous goods are intended to be carried in limited quantities inside the containers, the safety aspects will be specially considered.

1.1.5 Container Ships provided with approved lashing software will be assigned additional class notation **CLS**. (See Section 9).

1.2 Documentation

1.2.1 In addition to the information and plans required by Pt.3, Ch.1, Sec.3, the following are to be submitted:

- Hull section modulus calculations about vertical and horizontal axes at sufficient number of sections, distributed over the length of the ship so as to include sections where abrupt structural changes are present.
- Combined stress distribution and envelope of still water bending moments in graphical or tabular forms.
- Outline plan showing stowage arrangement of containers.
- Plans showing details of cell guides and supporting structure; and their attachment to main structural members.
- Details of reinforcement to structure in way of container corners.

1.2.2 A copy of the Cargo Securing Manual approved by the National Administration is to be submitted for reference.

1.2.3 *Revised Guidelines for the Preparation of the Cargo Securing Manual* (MSC.1/ Circ. 1353, Rev 2) may be referred whilst preparing the Cargo Securing Manual.

1.3 Materials and protection

1.3.1 Materials and grades of steel are to comply with the requirements of Pt.3, Ch.2. The material grades of steel are generally to be maintained over the entire region of the container holds.

1.3.2 Steels used for container supporting structures and fixed fittings welded to the ship are in general to comply with the requirements of Pt.2. Other steels and materials may also be specially considered.

Section 2

Usage of Extremely Thick Plates of High Strength Steels

2.1 General

2.1.1 The requirements in this section are applicable in case of container ships using high strength steel plates of grade; EH36, EH40, EH47 of thicknesses over 50[mm] and not exceeding 100[mm] for longitudinal structural members.

2.1.2 This section identifies when measures for the prevention of brittle fracture of extremely thick steel plates are required for longitudinal structural members.

2.1.3 This section gives the basic concepts for application of extremely thick steel plates to longitudinal structural members in the upper deck region. The upper deck region means the upper deck plating, hatch side coaming, hatch coaming top plating and their attached longitudinals.

2.1.4 Measures for preventing brittle fracture of longitudinal members are given in 2.3 and Table 2.3.1.

For steel plates with thicknesses exceeding 100[mm] appropriate measures acceptable to IRS are to be taken to prevent brittle crack initiation and propagation.

2.2 Hull Structural Design Requirements for EH47 Grade Steel

2.2.1 The material factor k, (as defined in Pt.3, Ch.2, Sec.1) of EH47 steel for the assessment of hull girder strength, is to be taken as 0.62. The material factors for EH36 and EH40 steel are to be in accordance with Pt.3, Ch.2, Sec.1.

2.2.2 Fatigue assessment is also to be performed on longitudinal structural members to the satisfaction of IRS.

2.2.3 Details of construction of structural members such as connections between outfitting and hull structures are to be specially approved.

2.3 Measures to prevent brittle fracture

2.3.1 The thickness and the yield strength shown in the Table 2.3.1 apply to the hatch coaming top plating and side plating and are the controlling parameters for the application of countermeasures for longitudinal members. These controlling parameters are not applicable to the upper deck.

If the as built thickness of the hatch coaming top plating and side plating is below the values contained in the table, countermeasures are not necessary regardless of the thickness and the yield strength of the upper deck plating.

2.3.2 NDT during construction (Measure No. 1 of Table 2.3.1):

2.3.2.1 NDT other than visual inspection on all target block joints is to be carried out as per 2.3.2. Enhanced NDT as specified in 2.3.4.3 is to be carried out in accordance with an appropriate standard.

2.3.2.2 General

Ultrasonic testing (UT) is to be carried out to the satisfaction of IRS, on all block-to-block butt joints of all the upper flange longitudinal

structural members in the cargo hold region. Upper flange longitudinal structural members include the topmost strakes of the inner hull/bulkhead, the sheer strake, main deck, coaming plate, coaming top plate, and all attached longitudinal stiffeners (See Fig 2.3.2.2).

2.3.2.3 Acceptance Criteria of UT

Acceptance criteria of UT are to be acceptable to IRS.

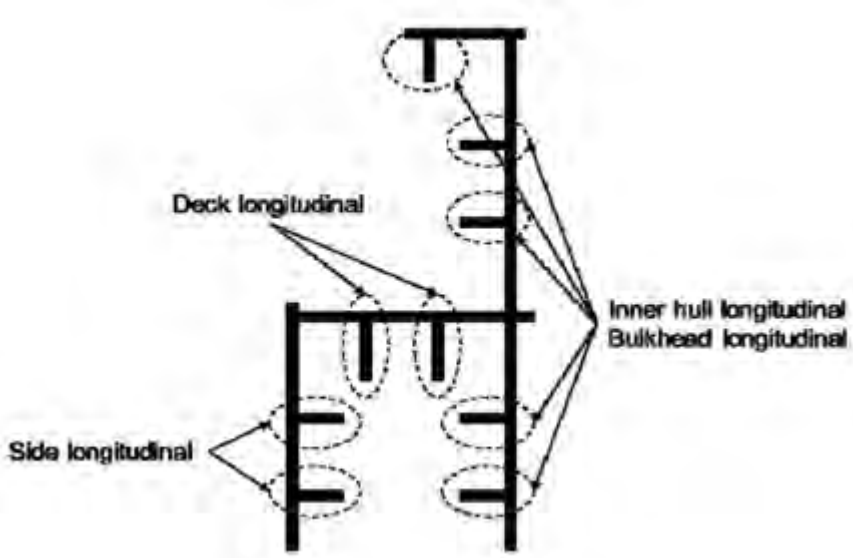


Fig.2.3.2.2 : Upper flange longitudinal structural members

Table 2.3.1 : Measures to prevent Brittle Fracture

Yield Strength [N/mm^2]	Thickness (mm)	Option	Measures			
			1	2	3+4	5
355	50<t≤85		-.	-.	-.	-.
	85<t≤100		X	-.	-.	-.
390	50<t≤85		X	-.	-.	-.
	85<t≤100	A	X	-.	X	X
		B	X*	-.**	-.	X
460 (FCAW)	50<t≤100	A	X	-.	X	X
		B	X*	-.**	-.	X
460 (EGW)	50<t≤100		X	-.	X	X

Measures:

1: NDT other than visual inspection on all target block joints (during construction). See 2.3.2

2: Periodic NDT other than visual inspection on all target block joints (after delivery). See 2.3.3

3: Brittle crack arrest design against straight propagation of brittle crack along the weld line to be taken (during construction). See 2.3.4.3.1 i, ii and iii.

4: Brittle Crack arrest design against deviation of brittle crack from weld line (during construction). See 2.3.4.3.2

5: Brittle crack arrest design against propagation of cracks from other welds areas such as fillets and attachment welds, as defined in 2.3.4.2.1 (b) (during construction). See 2.3.4.3.

Table 2.3.1 (Contd.)

Symbols:
a. "X" means "To be applied".
b. "-." means "Need not be applied"
c. A, B under the column titled "options" means option "A" or "B" may be selected.

Note:
*: See 2.3.4.3.1 iv.
**: May be required at the discretion of IRS.

2.3.3 Periodic NDT after delivery (Measure No.2 of Table 2.3.1):

2.3.3.1 General

The procedure of the NDT is to be acceptable to IRS.

2.3.3.2 Timing of Ultrasonic Testing (UT)

Where UT is carried out, the frequency of survey is to acceptable to IRS

2.3.3.3 Acceptance Criteria of UT

Where UT is carried out, acceptance criteria of UT are to be acceptable to IRS.

2.3.4 Brittle crack Arrest Design (Measure No 3, 4 and 5 of Table 2.3.1)

2.3.4.1 General

2.3.4.1.1 The brittle crack arrest steel method detailed in this sub-section may be used when the measures No.3, 4 and 5 of Table 2.3.1 are applied and the steel grade material of the upper deck is not higher than EH40. Otherwise other means for preventing the crack initiation and propagation are to be agreed with IRS.

2.3.4.1.2 Measures for prevention of brittle crack propagation, are to be taken within the cargo hold region. A brittle crack arrest design means a design using these measures.

2.3.4.1.3 The measures given in this sub-section generally apply to the block-to-block joints but it should be noted that cracks can initiate and propagate away from such joints. Therefore, appropriate measures should also be considered for cases specified in 2.3.4.2.1 b)(I).

2.3.4.1.4 Brittle crack arrest steels are defined as steel plate with the specified brittle crack arrest properties measured by either the brittle crack arrest toughness Kca or Crack Arrest Temperature (CAT). Requirements for brittle crack arrest steels are specified in Part 2, Chapter 3, Section 10 of the Rules

2.3.4.2 Functional Requirements of brittle crack arrest design

2.3.4.2.1 The purpose of the brittle crack arrest design is to arrest propagation of a crack at a proper position and to prevent large scale fracture of the hull girder.

a) The locations of most concern for brittle crack initiation and propagation are the block-to-block butt weld joints either on hatch side coaming or on upper deck plating. Other locations in block fabrication where joints are aligned may also present higher opportunity for crack initiation and propagation along butt weld joints.

b) Both of the following cases are to be considered:

I. where the brittle crack runs straight along the butt joint, and
II. where the brittle crack initiates in the butt joint but deviates away from the weld and into the plate, or where the brittle crack initiates from any other weld and propagates into the plate.

"Other Weld Areas" includes the following (Refer to Fig.2.3.4.2).

1. Fillet weld between hatch side coaming plating,

including top plate and longitudinals.

2. Fillet weld between hatch side coaming plating, including top plating and attachments (e.g. fillet welds between hatch side top plating and hatch cover pad plating).
3. Fillet weld between hatch side coaming top plating and hatch side coaming plating.
4. Fillet weld between hatch side coaming plating and upper deck plating;
5. Fillet weld between upper deck plating and inner hull/bulkheads;
6. Fillet weld between upper deck plating and longitudinal; and
7. Fillet weld between sheer strakes and upper deck plating.

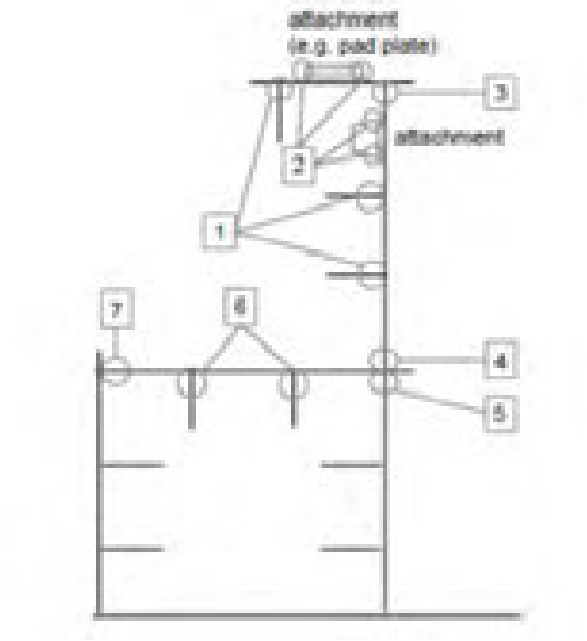


Fig.2.3.4.2 : Other Weld Areas

2.3.4.3 Concept examples of brittle crack arrest design

The following are considered acceptable examples of measures that can be used in a brittle crack arrest-design to prevent brittle crack propagations. The detail design arrangements are to be submitted for approval to IRS. Other concept designs may be considered and accepted for review by IRS.

2.3.4.3.1 Brittle crack arrest design for 2.3.4.2.1 (b)(I)

i. Where the block-to-block butt welds of the hatch side coaming and those of the upper deck are shifted, this shift is to be at least 300 [mm]. Brittle crack arrest steel is to be provided for the hatch side coaming plating.

ii. Where crack arrest holes are provided in way of the block-to-block butt welds at the region where hatch side coaming weld meets the deck weld, the fatigue strength of the lower end of the butt weld is to be assessed. Additional countermeasures are to be taken for the possibility that a running brittle crack may deviate from the weld line into the upper deck or the hatch side coaming. These countermeasures are to include the application of brittle crack arrest steel in the hatch side coaming plating.

iii. Where the Insert plates of brittle crack arrest steel or weld metal inserts with high crack arrest toughness properties

are provided in way of the block-to-block butt welds at the region where the hatch side coaming weld meets the deck weld, additional countermeasures are to be taken for the possibility that a running brittle crack may deviate from the weld line into upper deck or the hatch side coaming. These countermeasures are to include the application of brittle crack arrest steel in hatch side coamings plating.

iv. The application of enhanced NDT particularly Time of Flight Diffraction (TOFD) technique using stricter defect acceptance in lieu of standard UT technique specified in 2.3.2 can be an alternative to (i) to (iii) mentioned above.

Brittle crack arrest design for 2.3.4.2.1 (b)(II):

- Brittle crack arresting steel is to be used for the upper deck plating along the cargo hold region in a way suitable to arrest a brittle crack initiating from the coaming and propagating into the structure below.

Measures to arrest propagation of cracks from other weld areas such as fillets and attachment welds during construction are identical to the ones mentioned above.

2.3.4.4 Selection of brittle crack arrest steels

.1 The brittle crack arrest steels fitted in the upper deck region of container ships are to comply with Table 2.3.4.4 where suffixes BCA1 and BCA2 are, as defined in Part 2, Chapter 3. Section 10 of the Rules.

.2 The brittle crack arrest steel property is to be selected for each individual structural member with thickness above 50 [mm] according to Table 2.3.4.4.

.3 When brittle crack arrest steels as specified in Table 2.3.4.4 are used, the weld joints between the hatch coaming side and the upper deck are to be partial penetration weld details approved by IRS.

In the vicinity of ship block joints, alternative weld details may be used for the deck and hatch coaming side connection provided additional means for preventing the crack propagation are implemented and agreed with IRS in this connection area.

Table 2.3.4.4 : Brittle crack arrest requirement in function of structural members and thickness

Structural Members plating*****	Thickness [mm]	Brittle crack arrest steel requirement
Upper Deck	50 < t ≤ 100	EH 36 or 40 with suffix BCA1
Hatch Coaming	50 < t ≤ 80	EH 40 or 47 with suffix BCA2
	80 < t ≤ 100	EH 40 or 47 with suffix BCA2

*** -** Excluding their attached longitudinals

Section 3

Ship Arrangement

3.1 General

3.1.1 Container ships are in general to be provided with closed hatches with weathertight means of closure.

3.1.2 Proposals for omission of hatch covers altogether will be specially considered, generally in accordance with MSC/Circ.608, "Interim Guidelines for Open-top Containerships" issued by IMO. The proposals will also require to be agreed by the National Statutory Authority in order to obtain an exemption from the requirement of hatch covers for the assignment of loadline.

3.2 Non-weathertight hatch covers above superstructure deck

3.2.1 Non-weathertight hatch covers may be accepted above a level equivalent to second tier superstructure; or above a level equivalent to third tier superstructure in the forward quarter of the ship's length; subject to the approval of the flag Administration and satisfying the requirements given in 3.2.2 to 3.2.10.

3.2.2 These covers would be accepted only on containerships.

3.2.3 They may be fitted to hatchways located on weatherdecks which are at least two standard superstructure heights above an actual freeboard deck or an assumed freeboard deck from which a freeboard can be calculated which will result in a draught not less than that corresponding to the freeboard actually assigned. Where any part of a hatchway is forward of a point located one quarter of the ship's length (0.25L) from the forward perpendicular, that hatchway is to be located on a weatherdeck at least three standard superstructure heights above the actual or assumed freeboard deck. It is to be noted that the assumed freeboard deck is used only for the purpose of measuring the height of the deck on which the hatchways are situated and may be an imaginary, or virtual deck and in this case is not to be used for the actual assignment of freeboard. The vessels freeboard is to be assigned from an actual deck, designated as the freeboard deck, which is to be determined in accordance with the International Loadline Convention.

3.2.4 The hatchway coaming is to be not less than 600 [mm] in height.

3.2.5 The non-weathertight gaps between hatch cover panels are to be considered as unprotected openings with respect to the requirements of intact and damage stability calculations. They are to be as small as possible commensurate with the capacity of the bilge system and expected water ingress and the capacity and operational effectiveness of the fire-fighting system and generally, not to exceed 50 [mm].

3.2.6 Labyrinths, gutter bars, or equivalents are to be fitted proximate to the edges of each panel in way of the gaps to minimise the amount of water that can enter the container hold from the top surface of each panel.

3.2.7 Scantlings of the hatch cover panels as well as details on the securing arrangements to the vessel's support structure and coamings are to be equivalent to those for weathertight covers.

3.2.8 If a fixed gas fire extinguishing system is fitted in the hold, the capacity of the system is to be increased by 10 percent over that for the same vessel with weathertight hatch covers, provided the gap between hatches is not more than 50 [mm]. Alternatively, a fixed water spray system is to be provided.

3.2.9 The bilge system required for cargo holds with non-weathertight hatches is to have sufficient additional capacity for water ingress based on the greater of:

i) a steady rainfall of 100 [mm/hr] applied through the total area of the gaps between panels; or

ii) the capacity of the fire fighting water spray system where fitted.

The size of the bilge main is to be in accordance with the increased bilge pump capacity. Bilge alarms are to be provided in each hold fitted with non-weathertight covers.

3.2.10 Container holds fitted with non-weathertight hatch covers on vessels intended to carry dangerous goods are to be considered in the same manner as open-top container holds according to 3.1.2.

3.3 Ships with large hatch openings

3.3.1 A ship is to be considered to have large hatch openings if any one of the following conditions are met (See Fig.3.3.1):

a) $\frac{b}{B_1} \geq 0.7$

b) $\frac{l_H}{l_{BH}} \geq 0.89$

c) $\frac{b}{B_1} \geq 0.6$ and $\frac{l_H}{l_{BH}} > 0.7$

where,

b = breadth [m], of the opening. Where there are multiple openings abreast, these are regarded as a single opening, and b is to be the sum of individual widths of these openings.

l_{BH} =distance [m], between centres of the deck strip at each end of the opening. Where there is no further opening beyond the one under consideration, the point to which is measured will be specially considered.

l_H = length of the opening [m].

B_1 = extreme breadth of deck including opening, measured at the mid-length of the opening [m].

3.3.2 The structural configuration of these ships are to provide for adequate torsional rigidity in the form of either a double skin side construction or a single skin side shell with efficient torsion box girders or equivalent structures at the top sides.

The double skin or the torsion box is to extend over the entire cargo hold area and efficiently scarphed at the ends.

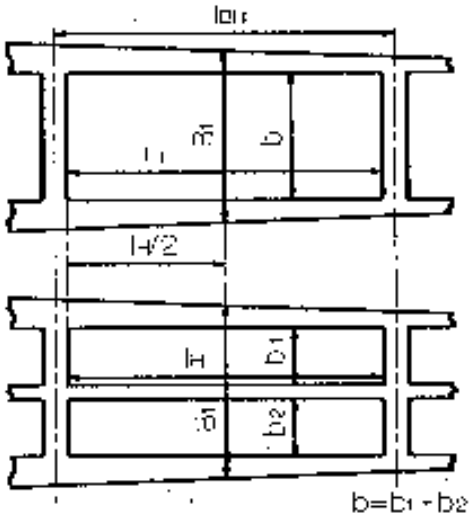


Fig.3.3.1 : Hatch opening dimensions

Section 4

Hull Girder Strength

4.1 General

4.1.1 Ships of L > 90 [m] are to be longitudinally framed at topsides and at bottom. Side shell clear of the torsion box may be longitudinally or transversely framed.

4.2 Application

4.2.1 The requirements in this section are applicable to steel ships with a length L of 90 [m] and greater and operated in unrestricted service:

- Container Ships
- Ships dedicated primarily to carry their load in containers

4.2.2 Wave induced load requirements specified in this section apply to monohull displacement ships in unrestricted service and are limited to ships meeting the following criteria:

.1 length of ship: 90 [m] ≤ L ≤ 500 [m]

.2 Proportion : 5≤L/B≤9, 2≤B/T≤6

.3 Block coefficient at scantling draft : $0.55 \leq C_B \leq 0.9$

4.2.3 For ships that do not meet all of the criteria specified in 4.2.2, applied wave induced loads is to be obtained from direct analysis method.

4.2.4 For ships of length 150 [m] or above, a finite element analysis is to be carried out (Refer to Sec 7) in addition to the requirements of this section.

4.2.5 Longitudinal extent of strength assessment

4.2.5.1 The stiffness, yield strength, buckling strength and hull girder ultimate strength assessment are to be carried out in way of 0.2L to 0.75L with due consideration given to locations where there are significant changes in hull cross section, e.g. changing of framing system and the fore and aft end of the forward bridge block in case of two-island designs.

4.2.5.2 In addition, strength assessments are to be carried, out outside areas mentioned in 4.2.5.1, by the method deemed necessary by IRS. As a minimum assessments are to be carried out at forward end of the foremost cargo hold and the aft end of the aft most cargo hold.

4.3 Corrosion margin and net thickness

4.3.1 Net scantling definitions

4.3.1.1 The strength is to be assessed using the net thickness approach on all scantlings.

4.3.1.2 The net thickness for the plates, webs and flanges is obtained by subtracting the voluntary addition and the factored corrosion addition from the as built thickness, as follows:

$$t_{net} = t_{as_built} - t_{vol_add} - \alpha t_c$$

Where,

t_{net}:net thickness [mm]

t_{as_built}: as built thickness [mm]

t_{vol_add}: voluntary addition [mm]

t_c: corrosion addition, in [mm]

α: corrosion addition factor

Voluntary addition, if being used, is to be clearly indicated on the drawings.

Structural requirement	Property/analysis type	α
Strength assessment	Section properties	0.5
Buckling strength	Section propertied (stress determination)	0.5
	Buckling capacity	1
Hull girder ultimate strength	Section properties	0.5
	Buckling/collapse capacity	0.5

4.3.2 Determination of corrosion addition

4.3.2.1 The corrosion addition for each of the two sides of a structural member, t_{c1} or t_{c2} is specified in table. The total corrosion addition t_c, [mm], for both sides of the structural member is obtained by the following formula:

$$t_c = (t_{c1} + t_{c2}) + t_{res}$$

4.3.2.2 For an internal member within a given compartment, the total corrosion addition, t_c is obtained from the following formula:

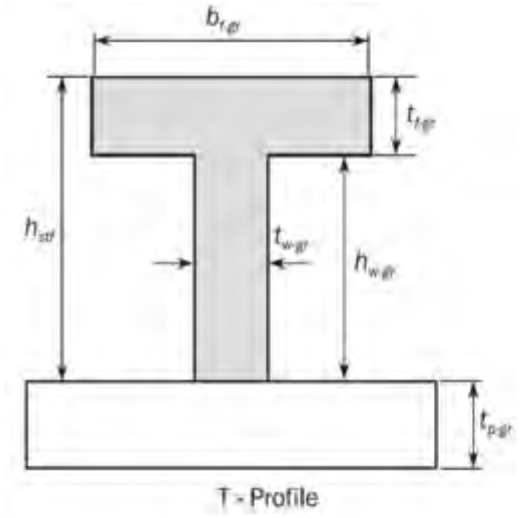
$$t_c = (2t_{c1}) + t_{res}$$

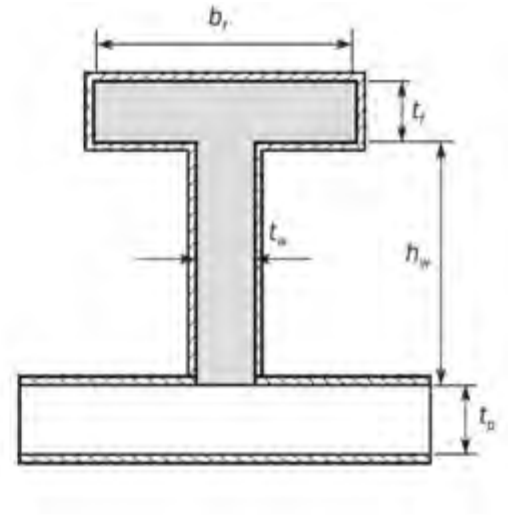
4.3.2.3 The corrosion addition of a stiffener is to be determined according to the location of its connection to the attached plating.

Corrosion addition for one side of the structural member	
Compartment type	**One side corrosion addition t_{c1} or t_{c2} [mm]**
Exposed to sea water	1
Exposed to atmosphere	1
Ballast water tank	1
Void and dry spaces	0.5
Fresh water, fuel oil and lube oil tank	0.5
Accommodation spaces	0
Container holds	1
Compartment types not mentioned above	0.5

4.3.3 Determination of net section properties

4.3.3.1 The net section modulus, moment of inertia and shear area properties of a supporting member are to be calculated using net dimensions of the attached plate, web and flange, as defined in Fig.4.3.3.1. The net cross-sectional area, the moment of inertia about axis parallel to the attached plate and the associated neutral axis position are to be determined through applying a corrosion magnitude of $0.5\alpha t_c$ deducted from the surface of the profile cross-section.





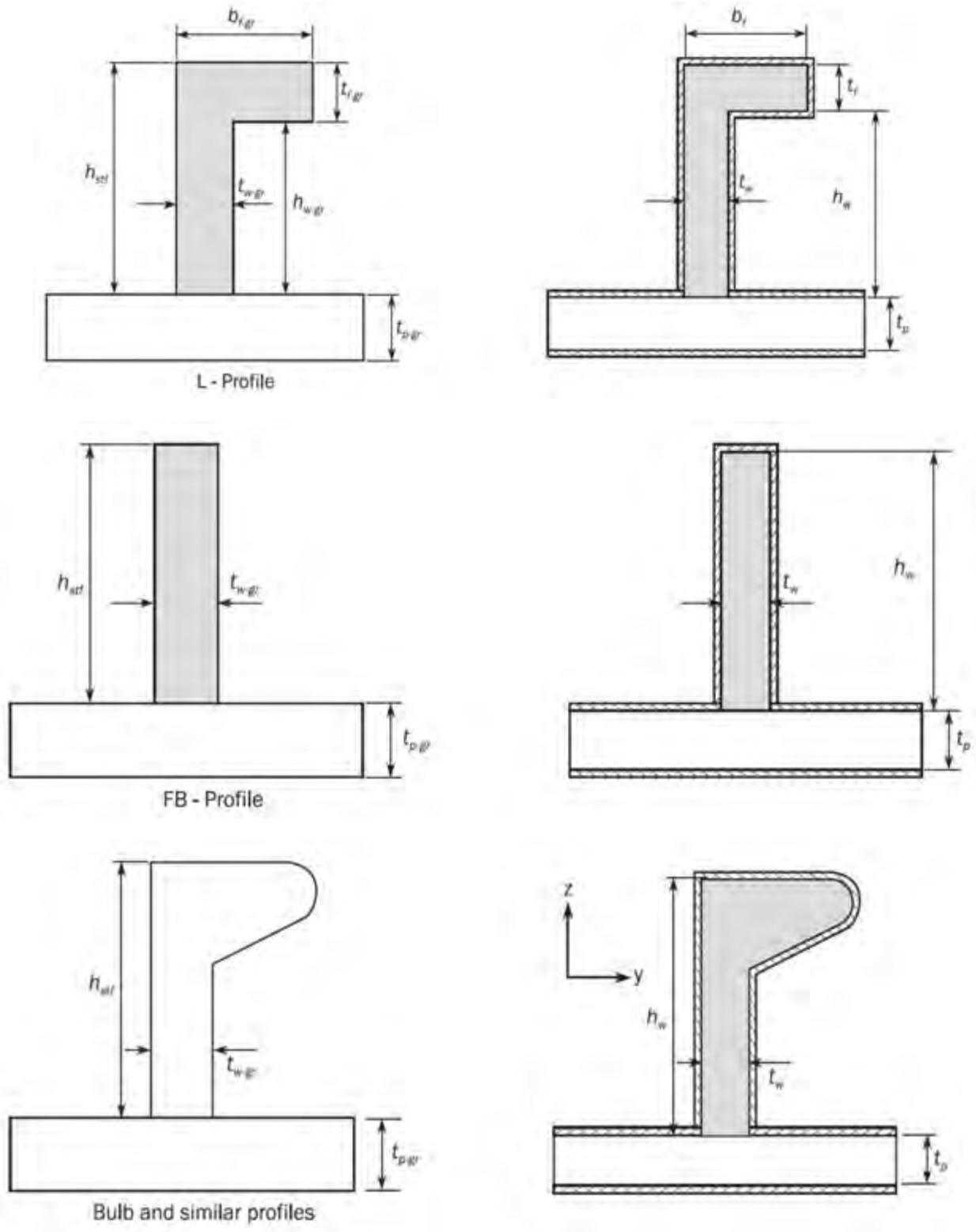


Fig.4.3.3.1 : Net sectional properties of supporting members

4.4 Loads

4.4.1 Still water bending moments, *Ms* [kNm], and shear forces *Fs* [kN] are to be calculated at each section along the ship length for the design loading conditions

4.4.2 In general, the design cargo and ballast loading conditions, based on amount of bunker, fresh water and stores at departure and arrival, are to be considered for the *Ms* and *Fs* calculations.

4.4.3 Where the amount and disposition of consumables at any intermediate stage of the voyage are considered more severe, calculations for such intermediate conditions are to be submitted in addition to those for departure and arrival conditions.

4.4.4 Where any ballasting and/or de-ballasting is intended during voyage, calculations of the intermediate condition just before and just after ballasting and/or de-ballasting any ballast tank are to be submitted and where approved included in the loading manual for guidance.

4.4.5 The permissible vertical still water bending moments M_{Smax} and M_{Smin} and the permissible vertical still water shear forces F_{Smax} and F_{Smin} in seagoing conditions at any longitudinal position are to envelop:

.1 The maximum and minimum still water bending moments and shear forces for the seagoing loading conditions defined in the Loading Manual.

.2 The maximum and minimum still water bending moments and shear forces specified by the designer

4.4.6 The Loading Manual is to include the relevant loading conditions, which envelop the still water hull girder loads for seagoing conditions, including following loading conditions:

.1 The design loading and ballast conditions, subdivided into departure and arrival conditions, and ballast exchange at sea conditions, where applicable, upon which the approval of the hull scantlings is based.

.2 Homogeneous loading conditions at maximum draught

.3 Ballast conditions

.4 Special loading conditions, e.g. container or light load conditions at less than the maximum draught, heavy cargo, empty holds or non-homogeneous cargo conditions deck cargo conditions, etc., where applicable

.5 Short voyage or harbour conditions, where applicable

.6 Docking condition afloat

.7 Loading and unloading transitory conditions, where applicable

4.4.7 The distribution of the vertical wave induced bending moments, M_w [kNm], along the ship length is given in Fig.4.4.7 a) where:

$$M_{W-Hog} = +1.5 f_R L^3 C C_W \left(\frac{B}{L}\right)^{0.8} f_{NL-Hog}$$

$$M_{W-Sag} = -1.5 f_R L^3 C C_W \left(\frac{B}{L}\right)^{0.8} f_{NL-Sag}$$

f_R : Factor related to the operational profile, to be taken as 0.85

f_{NL-Hog} : Non-linear correction for hogging, to be taken as

$$f_{NL-Hog} = 0.3 \frac{C_B}{C_W} \sqrt{T}$$

Not to be taken greater than 1.1

f_{NL-Sag} : Non-linear correction for sagging, to be taken as

$$f_{NL-Sag} = 4.5 \frac{1 + 0.2 f_{Bow}}{C_W \sqrt{C_B}\ L^{0.3}}$$

Not to be taken greater than 1

f_{Bow}: Bow flare shape coefficient, to be taken as:

$$f_{Bow} = \frac{A_{DK} - A_{WL}}{0.2 L Z_f}$$

A_{DK}: Projected area in horizontal plane of uppermost deck [m^2] including the forecastle deck, if any, extending from 0.8L forward (see Fig.4.4.7 b)). Any other structures, e.g. plated bulwark, are to be excluded.

A_{WL}: Waterplane area, in [m^2] at draught T, extending from 0.8L forward.

Z_f: Vertical distance [m], from the waterline at draught T to the uppermost deck (or fore castle deck), measured at fore end (see Fig.4.4.7 b)), Any other structures, e.g plated bulwark, are to be excluded.

C: wave parameter, to be taken as:

$$C = 1 - 1.5\left(1 - \sqrt{\frac{L}{L_{ref}}}\right)^{2.2} \quad for\ L \le L_{ref}$$

$$C = 1 - 0.45\left(\sqrt{\frac{L}{L_{ref}}} - 1\right)^{1.7} \quad for\ L > L_{ref}$$

L_{ref}: Reference length [m], to be taken as:

$$L_{ref} = 315 C_W^{-1.3}$$

C_W: waterplane coefficient at the designed maximum load draught, to be taken as:

$$C_W = \frac{A_W}{L\,B}$$

A_W: waterplane area at the scantling draught

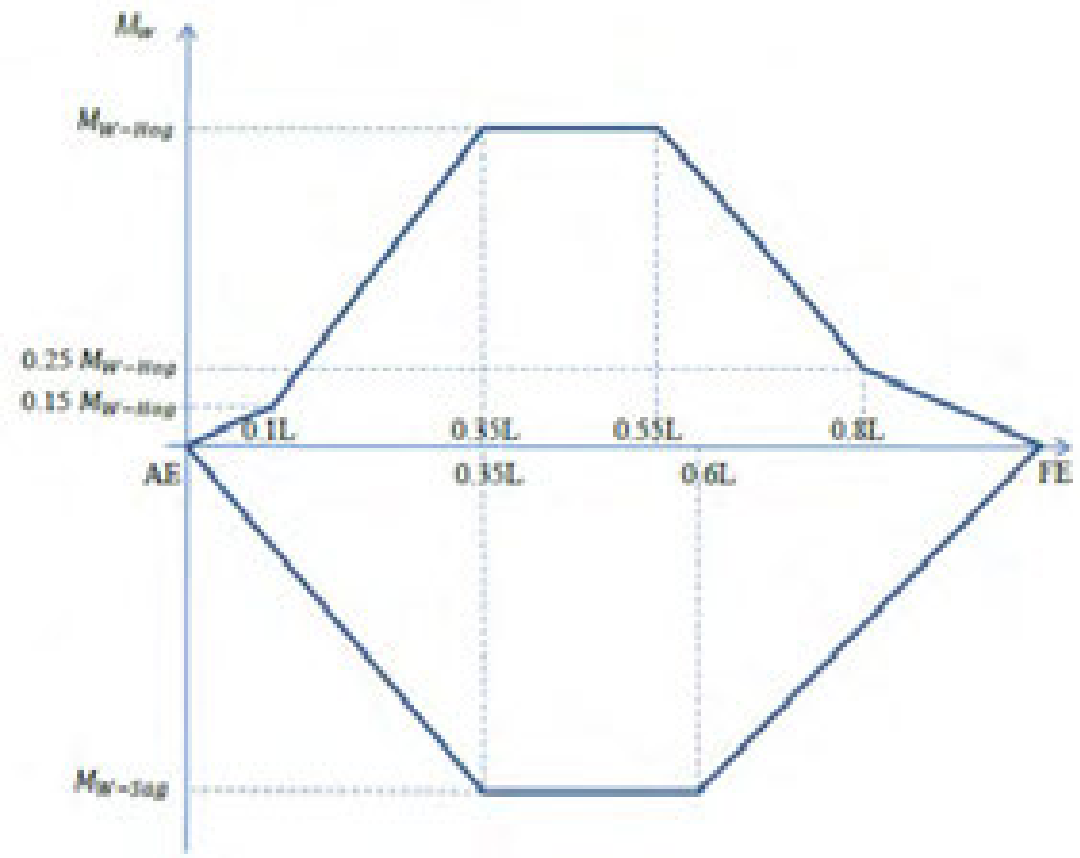


Fig.4.4.7 a) : Distribution of vertical wave bending moment along the ship length

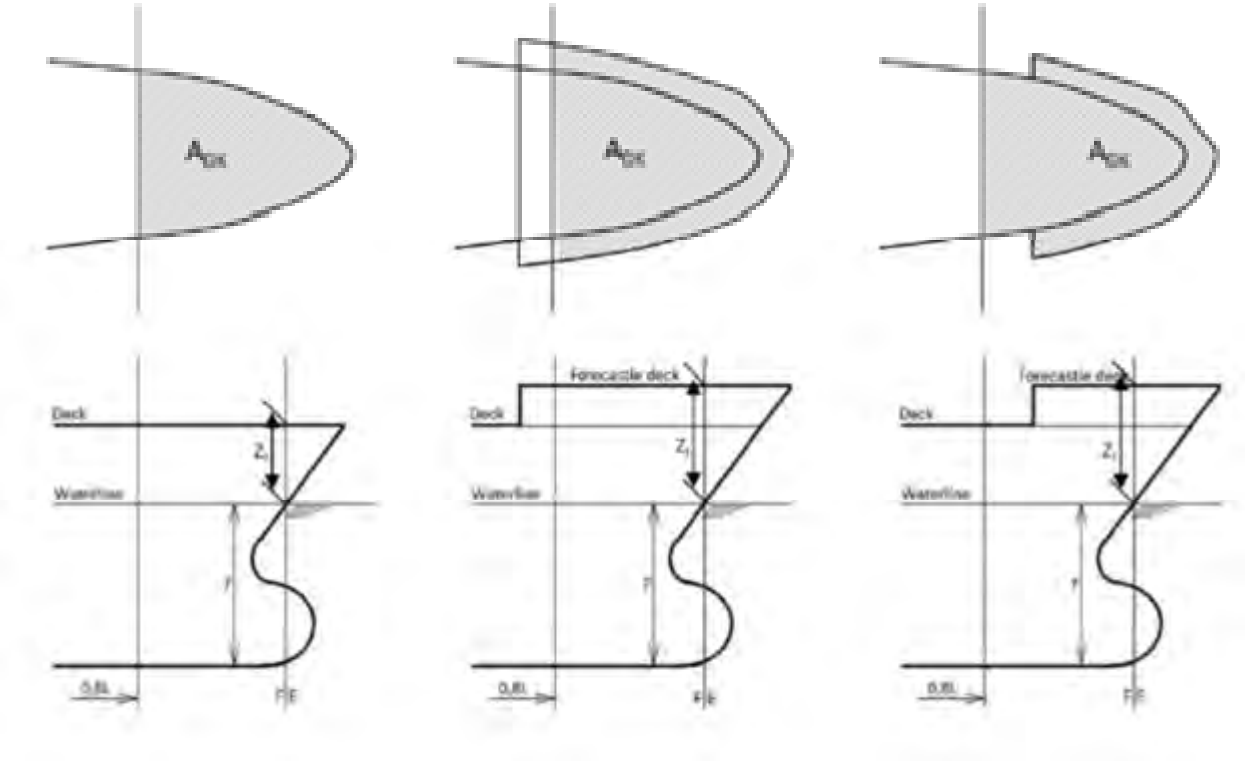


Fig.4.4.7 b) : Projected area A_{DK} and Vertical distance Z_f

4.4.8 The distribution of the vertical wave induced shear forces F_W (kN) along the ship length is given in Fig.4.4.8 are to be obtained using the following formulae:

$$F_{W-Hog-Aft} = +5.2 f_R L^2 C C_W \left(\frac{B}{L}\right)^{0.8} (0.3 + 0.7\, f_{NL-Hog})$$

$$F_{W-Hog-Fore} = -5.2 f_R L^2 C C_W \left(\frac{B}{L}\right)^{0.8} f_{NL-Hog}$$

$$F_{W-Sag-Aft} = -5.2 f_R L^2 C C_W \left(\frac{B}{L}\right)^{0.8} (0.3 + 0.7\, f_{NL-Sag})$$

$$F_{W-Sag-Fore} = +5.7 f_R L^2 C C_W \left(\frac{B}{L}\right)^{0.8} (0.25 + 0.75\, f_{NL-Sag})$$

$$F_{W-Mid} = +4 f_R L^2 C C_W \left(\frac{B}{L}\right)^{0.8}$$

Where,

L_{ref}: Reference length [m], to be taken as:

$$L_{ref} = 330 C_W^{-1.3}$$

Other parameters in the above equations are to be taken in accordance with 4.4.7

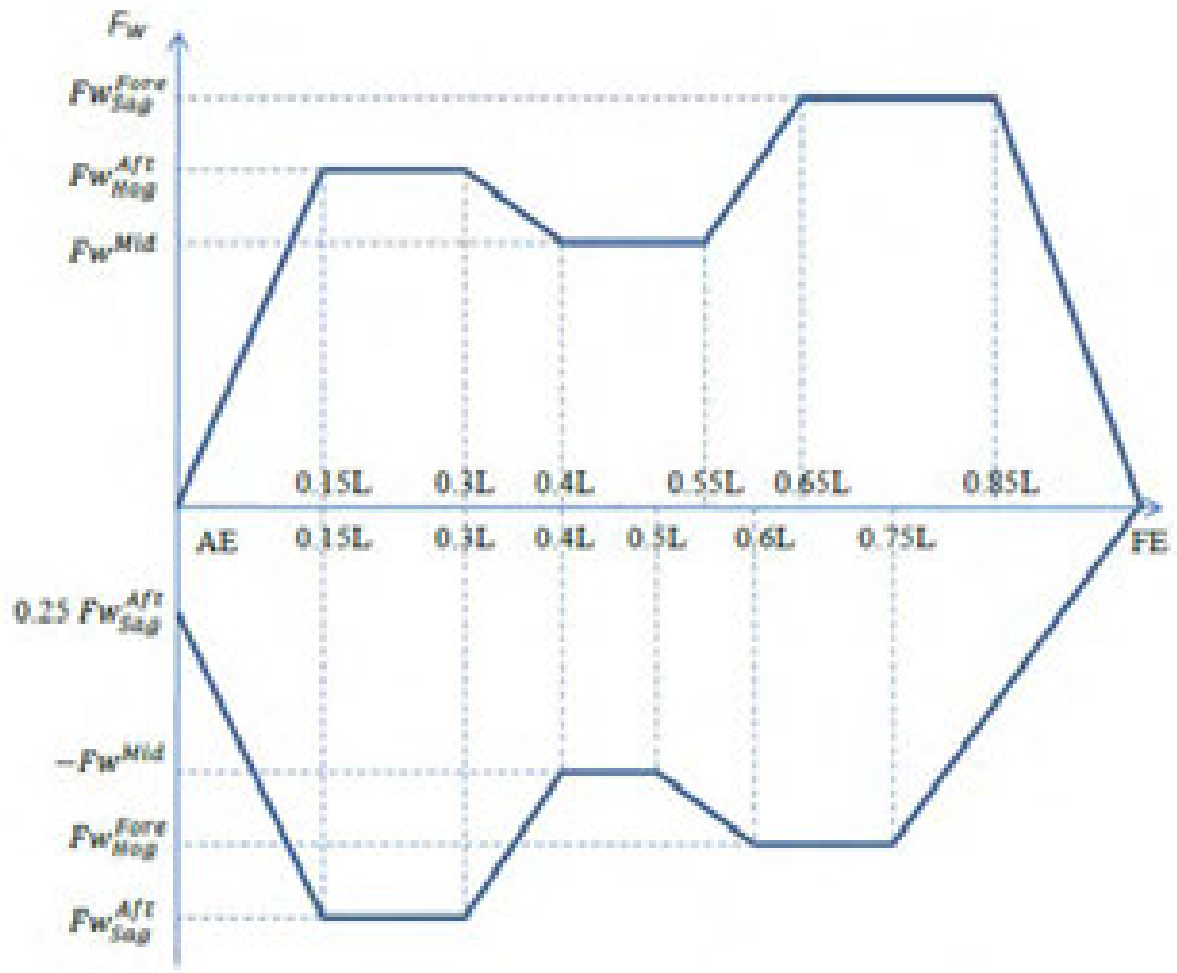


Fig.4.4.8 : Distribution of vertical wave shear force along the ship length

4.4.9 For strength assessments, the maximum hogging and sagging load cases given in Table 4.4.9 are to be checked. For each load case, the still water condition at each section, as defined in 4.4.1 to 4.4.6, is to be combined with the wave condition, as defined in 4.4.7 and 4.4.8.

Table 4.4.9 : Combination of still water bending moment and shear forces

Load Cases	Bending moment		Shear Forces	
	M_S	M_W	F_S	F_W
Hogging	M_{Smax}	M_{WH}	F_{Smax} for x ≤ 0.5L	F_{Wmax} for x ≤ 0.5L
			F_{Smin} for x > 0.5L	F_{Wmin} for x > 0.5L
Sagging	M_{Smin}	M_{WS}	F_{Smin} for x ≤ 0.5L	F_{Wmin} for x ≤ 0.5L
			F_{Smax} for x > 0.5L	F_{Wmax} for x > 0.5L

M_{WH}: Wave bending moment in hogging at the cross section under consideration, to be taken as the positive value of M_W as defined Fig.4.4.7 a)

M_{WS}: Wave bending moment is sagging at the cross section under consideration, to be taken as the negative value of M_W as defined in Fig.4.4.7 a)

F_{Wmax} : Maximum value of the wave shear force at the cross section under consideration, to be taken as the positive value of F_W as defined in Fig.4.4.8

F_{Wmin} : Minimum value of the wave shear force at the cross section under consideration, to be taken as the negative value of F_W as defined in Fig.4.4.8

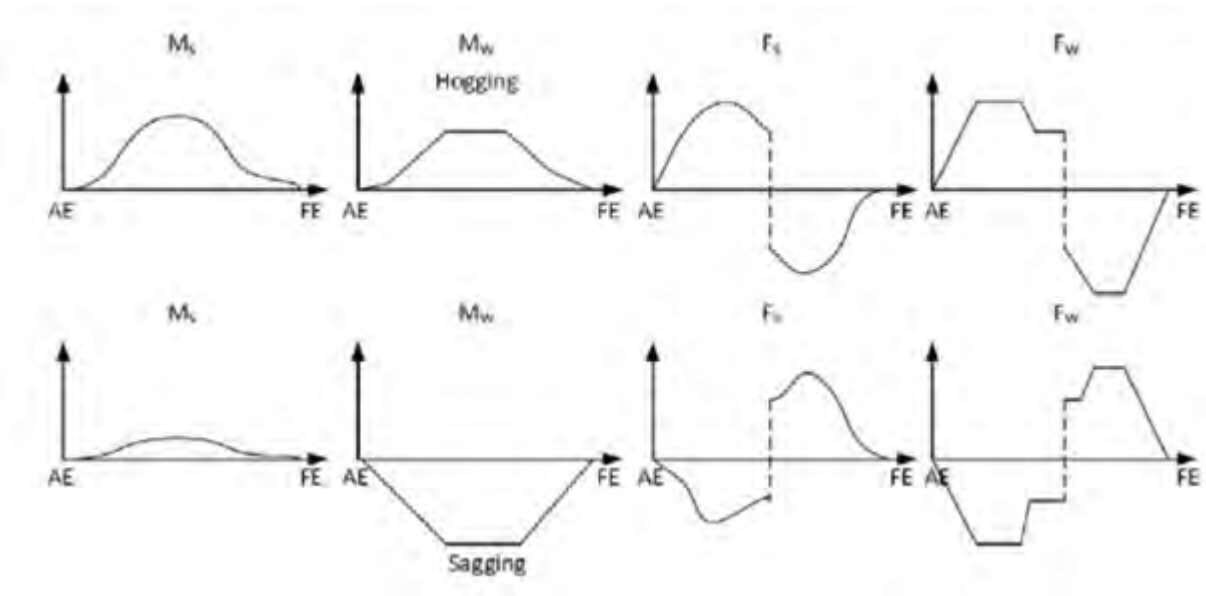


Fig.4.4.9 : Load combination to determine the maximum hogging and sagging load cases as given in Table 4.4.9

4.5 Hull girder stress

4.5.1 The hull girder bending stress and hull girder shear stress are to be determined at the load calculation point under consideration, for the "hogging" and "sagging" load cases as follows:

.1 Hull girder bending stress

$$\sigma_{HG} = \frac{\gamma_S M_S + \gamma_W M_W}{I_{net}}(Z - Z_n)10^{-3}$$

.2 Hull girder shear stress

$$\tau_{HG} = \frac{\gamma_S F_S + \gamma_W F_W}{t_{net}/q_v}10^3$$

γ_S, γ_W: Partial safety factors, to be taken as 1.

4.6 Strength assessment

4.6.1 General

4.6.1.1 Continuity of structure is to be maintained throughout the length of the ship. Where significant changes in structural occur adequate transitional structure is to be provided.

4.6.2 Stiffness assessment

4.6.2.1 The moment of inertia (in [m^4]) is to be in accordance with the formula:

$$I \geq 1.55L|M_S + M_W|10^{-7}$$

4.6.3 Yield strength assessment

4.6.3.1 For each of the load cases "hogging" and "sagging" the equivalent hull girder stress is to be in accordance with the formula :

$$\sigma_{eq} < \sigma_{perm}$$

$$\sigma_{eq} = \sqrt{\sigma_x^2 + 3\tau^2}$$

$$\sigma_{Perm} = \frac{R_{eH}}{\gamma_1 \gamma_2}$$

γ_1: Partial safety factor for material, to taken as

$$\gamma_1 = k\frac{R_{eH}}{235}$$

γ_2: Partial safety factor for load combinations and permissible stress, to be taken as:

γ_2= 1.24, for bending strength assessment according to 4.6.3.2

γ_2= 1.13, for shear stress assessment according to 4.3.3.3

4.6.3.2 The assessment of the bending stresses is to be carried out according to

4.6.3.1 at the following locations of the cross section:

- at bottom
- at deck
- at top of the hatch coaming
- at any point where there is a change of steel yield strength.

The following combination of hull girder stress as defined in 4.5 is to be considered:

$$\sigma_x = \sigma_{HG}$$

$$\tau = 0$$

4.6.3.3 The assessment of shear stress is to be carried out according to 4.6.3.1 for all structural elements that contribute to the shear strength capability. The following combination of hull girder stress as defined in 4.5 is to be considered:

$$\sigma_x = 0$$

$$\tau = \tau_{HG}$$

4.6.4 Buckling assessment

4.6.4.1 General

a) The requirements of 4.6.4 apply to plate panels and longitudinal stiffeners subject to hull girder bending and shear stress.

b) Acceptance criteria for the buckling assessment is defined as follows:

$$\eta_{act} \leq 1$$

c) The utilization factor, η_{act}, is defined as the inverse of the stress multiplication factor at failure γ_c , see Fig.4.6.4.1

$$\eta_{act} = \frac{1}{\gamma_c}$$

d) Failure limit states are defined in Sec 6, where each limit state is defined by an equation and γ_c is to be determined such that it satisfies the equation.

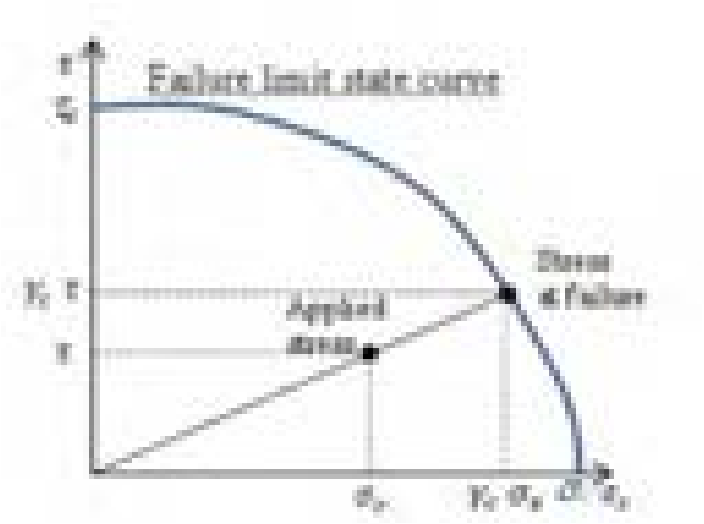


Fig.4.6.4.1 : Example of failure limit state curve and stress multiplication factor at failure

σ_x, τ : Applied stress combination for buckling given in Cl.4.6.4.2

σ_c, τ_c : Critical buckling stresses to be obtained according to Sec 6 for stress combination for buckling σ_x and τ .

4.6.4.2 Stress combination for buckling assessment

4.6.4.2.1 Two stress combination as given in Table 4.6.4.2.1 are to be considered for each of the load cases “hogging” and “sagging”. The stresses are to derived at the load calculation points

Table : 4.6.4.2.1				
		σ_x	σ_y	τ
Longitudinal stiffening	Stress combination 1	σ_{HG}	0	$0.7\tau_{HG}$
	Stress combination 2	$0.7\sigma_{HG}$	0	τ_{HG}
Transverse stiffening	Stress combination 1	0	σ_{HG}	$0.7\tau_{HG}$
	Stress combination 2	0	$0.7\sigma_{HG}$	τ_{HG}

4.6.4.3 Load calculation points

4.6.4.3.1 The hull girder stresses for elementary plate panels (EPP) are to be calculated at the load calculation points defined in Table 4.6.4.3.1

4.6.4.3.2 The hull girder stresses for longitudinal stiffeners are to be calculated at the following load calculation point:

- at the mid length of the considered stiffener
- at the intersection point between the stiffener and its attached plate.

Table 4.6.4.3.1 : Load calculation points (LCP) coordinates for plate buckling assessment			
LCP coordinates	Hull girder bending stress		Hull girder shear stress
	Non horizontal plating	Horizontal plating	
X coordinate	Mid-length of the EPP		
Y coordinate	Both upper and lower ends of the EPP (points A1 and A2 in Fig.4.6.4.3.1)	Out board and inboard ends of the EPP (points A1 and A2 in Fig.4.6.4.3.1)	Mid-point of EPP (point B in Fig.4.6.4.3.1)
Z coordinate	Corresponding to x and y values		

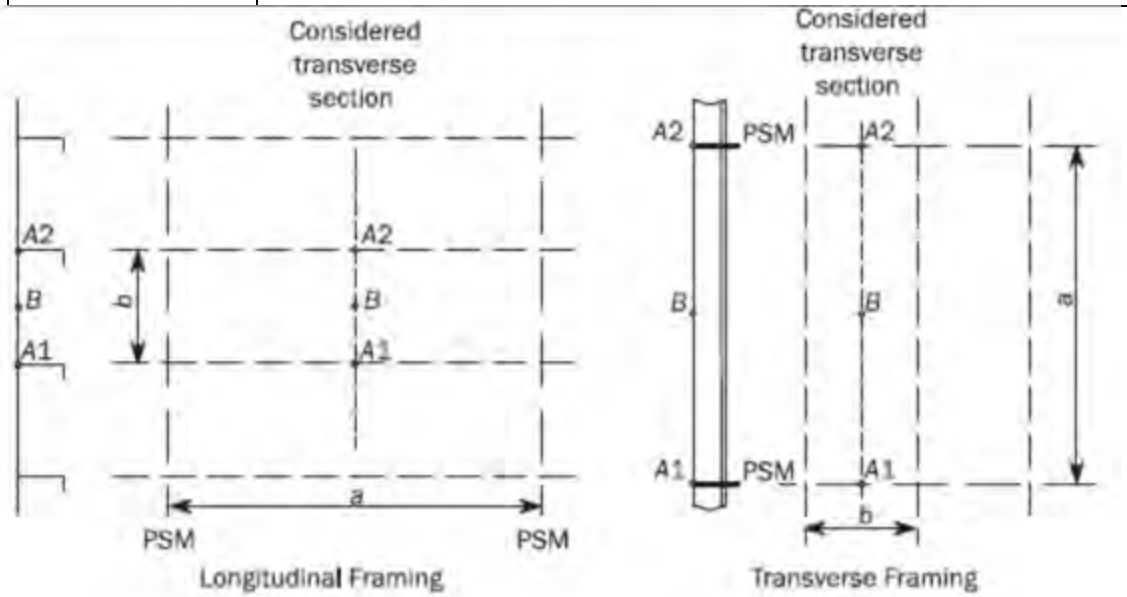


Fig.4.6.4.3.1 : LCP for plate buckling - assessment, PSM stands for primary supporting members

4.7 Hull girder ultimate strength

4.7.1 The hull girder ultimate strength is to be assessed for ships with length L equal or greater than 150 [m]. The method for calculation is given in detail in Sec 7.

4.7.2 Acceptance criteria given in 4.7.7 are applicable to intact ship structures.

4.7.3 Hull girder ultimate bending capacity is to be checked for the load cases “hogging” and “sagging”.

4.7.4 The vertical hull girder bending moment, M in hogging and sagging conditions, to be considered in the ultimate strength check is to be taken as:

$$M = \gamma_S M_S + \gamma_W M_W$$

Where,

M_S = permissible still water bending moment [kNm]

M_W = Vertical wave bending moment [kNm]

γ_S=Partial safety factor for the still water bending moment, to be taken as 1

γ_W= Partial safety factor for the vertical wave bending moment, to be taken as 1.2

4.7.5 The Hull girder ultimate bending moment capacity, M_U is defined as the maximum bending moment capacity of the hull girder beyond which the hull structure collapses. The ultimate bending moment capacities of a hull girder transverse section, in hogging and sagging conditions, are defined as the maximum values of the curves of bending moment *M* versus the curvature *X* of the transverse section considered (M_{UH} for hogging condition and M_{US} for sagging condition). The curvature *X* is positive for hogging condition and negative for sagging condition.

4.7.6 The hull girder ultimate bending moment capacity M_U is to be calculated using the incremental-iterative method as given in Sec 7.2 or using an alternative method as indicated in Sec 7.3.

4.7.7 The hull girder ultimate bending capacity at any hull transverse section is to satisfy the following criteria:

$$M \leq \frac{M_U}{\gamma_M \gamma_{DB}}$$

M_U = Hull girder ultimate bending moment capacity [kNm] to be obtained as specified in 4.7.5 and 4.7.6.

γ_M=Partial safety factor for the hull girder ultimate bending capacity, covering material geometric and strength prediction uncertainties, to be taken as 1.05

γ_{DB}=Partial safety factor for the hull girder ultimate bending moment capacity, covering the effect of double bottom bending, to be taken as:

- For hogging condition

$$\gamma_{DB} = 1.15$$

- For sagging condition

$$\gamma_{DB} = 1$$

4.7.8 For cross sections where the double bottom breadth of the inner bottom is less than that at amidships or where the double bottom structure differs from that at amidships (e.g. engine room sections), the factor γ_{DB} for hogging condition may be reduced based upon agreement with IRS.

4.8 Requirements for large container ships

4.8.1 Application

4.8.1.1 For container ships with a breadth B greater than 32.26 [m], following requirements are applicable in addition to the requirements of 4.2 to 4.7.

4.8.2 Yield and buckling assessment

4.8.2.1 Yield and buckling assessment is to be specially considered, taking into consideration additional hull girder loads:

- Wave torsion
- Wave horizontal bending
- Static cargo torque
- Local loads

4.8.2.2 All in plane stress components (i.e. bi-axial and shear stress) induced by hull girder

loads and local loads are to be considered during assessment.

4.8.3 Whipping

4.8.3.1 Hull girder strength assessment is to take into consideration the whipping contribution to the vertical bending moment.

4.9 Direct calculation of Shear Flow

4.9.1 General

4.9.1.1 This sub-section describes the procedures of direct calculation of shear flow around a ship's cross section due to hull girder vertical shear force. The shear flow q_v at each location in the cross section, is calculated by considering the cross section is subjected to a unit vertical shear force of 1 [N].

The unit shear flow per mm, q_v, [N/mm], is to be taken as:

$$q_v = q_D + q_I$$

where:

q_D : Determinate shear flow, as defined in 4.9.2.

q_I : Indeterminate shear flow which circulates around the closed cells, as defined in 4.9.3.

In the calculation of the unit shear flow, q_v, the longitudinal stiffeners are to be taken into account.

4.9.2 Determinate Shear Flow

4.9.2.1 The determinate shear flow, q_D [N/mm] at each location in the cross section is to be obtained from the following line integration:

$$q_D(s) = -\frac{1}{10^6 I_{y-net}} \int_0^s (z - z_n) t_{net} \, d_s$$

where:

s : Coordinate value of running coordinate along the cross section [m].

$I_{y\text{-}net}$: Net moment of inertia of the cross section [m^4].

t_{net} : Net thickness of plating [mm].

z_n : Z coordinate of horizontal neutral axis from baseline [m].

4.9.2.2 It is assumed that the cross section is composed of line segments as shown in Fig.4.9.2.1: where each line segment has a constant plate net thickness. The determinate shear flow is obtained by the following equation.

$$q_{Dk} = -\frac{t_{net}\ell}{2 \cdot 10^6 I_{Y-net}} (z_k + z_i - 2z_n) + q_{Di}$$

where:

q_{Dk}, q_{Di} : Determinate shear flow at node k and node i respectively [N/mm].

ℓ : Length of line segments [m].

y_k, y_i : Y coordinate of the end points k and i of line segment [m], as defined in Fig.4.9.2.1.

z_k, z_i : Z coordinate of the end points k and i of line segment [m], as defined in Fig.4.9.2.1.

4.9.2.3 Where the cross section includes closed cells, the closed cells are to be cut with virtual slits, as shown in Fig.4.9.2.2: in order to obtain the determinate shear flow. These virtual slits must not be located in walls which form part of another closed cell.

4.9.2.4 Determinate shear flow at bifurcation points is to be calculated by water flow calculations, or similar, as shown in Fig.4.9.2.2.

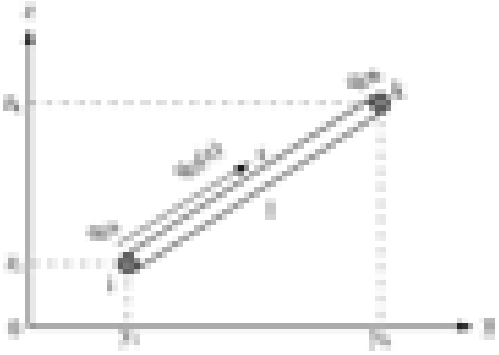


Fig.4.9.2.1 : Definition of line segment

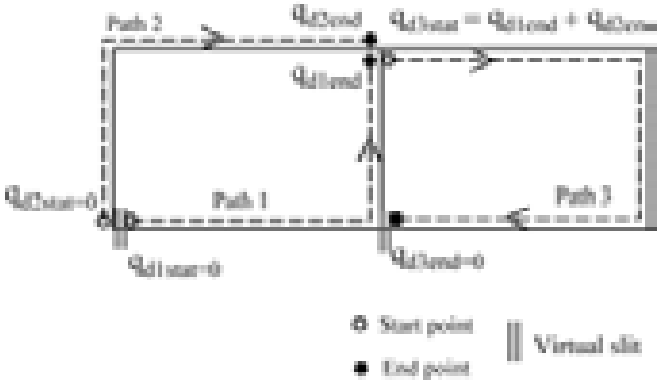


Fig.4.9.2.2 : Placement of virtual slits and calculation of determinate shear flow at bifurcation points

4.9.3 Indeterminate Shear Flow

4.9.3.1 The indeterminate shear flow around closed cells of a cross section is considered as a constant value within the same closed cell. The following system of equation for determination of indeterminate shear flows can be developed. In the equations, contour integrations of several parameters around all closed cells are performed.

$$q_{Ic}\oint_C \frac{1}{t_{net}}ds - \sum_{m=1}^{N_w}\left(q_{\mathrm{Im}}\oint_{c\&m}\frac{1}{t_{net}}ds\right) = -\oint_c \frac{q_D}{t_{net}}ds$$

where:

N_w : Number of common walls shared by cell c and all other cells.

$c\&m$: Common wall shared by cells *c* and *m*

q_{Ic}, q_{Im} : Indeterminate shear flow around the closed cell *c* and *m* respectively [N/mm].

4.9.3.2 Under the assumption of the assembly of line segments shown in Fig.4.9.2.1 and constant plate thickness of each line segment, the above equation can be expressed as follows:

$$q_{Ic}\sum_{j=1}^{N_c}\left(\frac{l}{t_{net}}\right)_j-\sum_{m=1}^{N_w}\left\{q_{Im}\left[\sum_{j=1}^{N_m}\left(\frac{l}{t_{net}}\right)_j\right]_m\right\}=-\sum_{j=1}^{N_c}\emptyset_j$$

$$\emptyset_j=\left[-\frac{l^2}{6\times10^3 I_{Y-net}}(z_k+2z_i-3z_n)+\frac{l}{t_{net}}q_{Di}\right]_j$$

where:

N_c : Number of line segments in cell *c*.

N_m : Number of line segments on the common wall shared by cells *c* and *m*.

q_{Di} : Determinate shear flow [N/mm], calculated according to 4.9.2.

4.9.3.3 The difference in the directions of running coordinates specified in 4.9.2 and in this section has to be considered.

4.9.4 Computation of sectional properties

4.9.4.1 Properties of the cross section are to be obtained by the following formulae where the cross section is assumed as the assembly of line segments:

$$\ell=\sqrt{\left(y_k-y_i\right)^2+\left(z_k-z_i\right)^2}$$

$$a_{net}=10^{-3}\ell t_{net}$$

$$A_{net}=\sum a_{net}$$

$$s_{y-net}=\frac{a_{net}}{2}(z_k+z_i)$$

$$s_{y-net}=\sum s_{y-net}$$

$$i_{y0-net}=\frac{a_{net}}{3}(z_k^2+z_k z_i+z_i^2)$$

$$I_{y0-net}=\sum i_{y0-net}$$

where:

a_{net}, A_{net}: Area of the line segment and the cross section respectively [m^2].

$s_{y\text{-}net}$, $S_{y\text{-}net}$: First moment of the line segment and the cross section about the baseline [m^3].

$i_{y0\text{-}net}$, $I_{y0\text{-}net}$: Moment of inertia of the line segment and the cross section about the baseline [m^4].

4.9.4.2 The height of horizontal neutral axis, z_n, in [m], is to be obtained as follows:

$$z_n=\frac{s_{y-net}}{A_{net}}$$

4.9.4.3 Inertia moment about the horizontal neutral axis [m^4], is to be obtained as follows:

$$I_{y-net}=I_{y0-net}-z_n^2 A_{net}$$

Section 5

Local Structure

5.1 Bottom structure

5.1.1 The bottom structure is to be as per Pt.3, Ch.7, Sec.3 and Sec.4.

5.1.2 Plate floors are to be fitted under bulkheads, midhold supports and quarter length supports and the maximum spacing is not to exceed 3.6[m].

5.1.3 Local stiffening is to be provided under container corner seatings.

5.1.4 Minimum thickness of floors is not to be less than

$$t = 6.0 + 0.03L\sqrt{K} \text{ [mm]}$$

5.1.5 Where mid-hold support or quarter length supports are arranged for the double bottom, these are to take the form of an efficiently stiffened transverse box or open structure.

5.2 Side structure

5.2.1 Thickness of the strake immediately below the sheer strake is generally not to be less than 80 percent of the sheer strake thickness.

5.2.2 Side shell plating at positions of high local loads and also at the termination of double skin structure, may require local increase in thickness.

5.2.3 Side shell primary supporting structure is to be effectively supported at deck level. Scantlings of primary structure may require verification by direct calculations.

5.2.4 Transverses supporting side longitudinals are to be arranged in line with the plate floors in the double bottoms to ensure continuity of transverse strength.

5.2.5 Transverses and horizontal perforated flats inside double skin construction are to have a thickness not less than

$t = 7.5 + 0.015L\sqrt{K}$ [mm] or

= 10 [mm] whichever is lower.

The flats and transverses are to be efficiently stiffened, and the thickness increased locally where necessary, to account for high shear stress.

5.3 Deck structure

5.3.1 The midship scantlings are generally to extend over the region of the container holds.

5.3.2 Decks are to be efficiently scarphed into the machinery space, and the fore end and aft end structure.

5.3.3 The scantlings of lower deck plating and stiffeners are to comply with the requirements of Pt.3, Ch.7 ; but the thickness may be required to be increased when the deck acts as a primary support for the side shell stiffening.

5.3.4 The scantlings of cross deck strips forming top of the bulkheads are to be specially considered in accordance with Pt.3, Ch.3, Sec.5.

The cross deck strips are in general to be transversely framed and the plating thickness is to satisfy the buckling criteria given in Pt.3, Ch.9.

5.3.5 Inserts are required at the intersection of the longitudinal underdeck girders and the cross deck strip. The inserts are to have a thickness not less than that of the girder top and bottom plates, as appropriate.

5.3.6 Longitudinal underdeck girders are, in general, to be fitted at deck level to support the hatch coamings. In case of ships with twin hatch openings abreast, the arrangement at deck centre may consist of one girder on the centreline, or one girder each on port and starboard in the breadth of the hatchway. The girders may take the form of open sections or closed box sections. Special consideration will be given to ships where the girders are omitted.

The top and bottom flanges of these girders are to have a minimum width suitable to accommodate the hatch coamings and the hatch cover securing arrangements, and the

scantlings will be specially considered. They are, in general, to be continuous throughout the container hold area, including the engine room where this is situated between container holds. Special attention is to be given to the integration of these girders into the fore end structure, machinery spaces where situated aft, and the intersection of the girders and the transverse box girders.

5.3.7 Coamings are, in general, to have a thickness, t, not less than the greater of the following:

a) $t = 8\, h_{HC}\, \sqrt{k}$ [mm]

where h_{HC} is the height of coaming [m]

b) t = 11 [mm] where length $L \geq 60$ [m], and not less than 9 [mm] where length $L \leq 30$ [m]. Intermediate values are to be obtained by interpolation.

The coamings are to be efficiently stiffened and stayed, see also Pt.3, Ch.7, Sec.9. Continuous coamings are to be effectively scarphed into the deckhouse structure or gradually tapered at ends, as applicable.

5.3.8 The corners of main hatchway openings are generally be rounded with radii not less than:

i) 350 + 0.525L [mm] outboard radius

ii) 200+0.3L [mm] inboard radius (min.250[mm]).

The radius of the hatch corners of the main hatchway openings adjacent to the engine room is to be made as large as practicable, approximately 40B [mm]. Parabolic or elliptic corners will be given special consideration.

5.3.9 Attention is to be paid to structural continuity, and abrupt change of shape, section and plate thicknesses are to be avoided. Arrangements in way of corners and openings are to be such as to minimize stress concentration.

In general, small openings should not be arranged in the strength deck outside line of main hatchways in the region of container holds with wide hatches.

5.4 Bulkheads

5.4.1 Where fitted, inner skin longitudinal bulkheads may be transversely or longitudinally framed. The continuity of the longitudinal material must be maintained in way of the container holds and also the machinery space where it is situated between container holds.

5.4.2 Where the inner skin bulkheads are stepped to accommodate the container stowage arrangement, the scantlings and stiffening arrangement will be required to be specially considered. The scarphing arrangements in way of the steps are to be sufficient to ensure an efficient overlap of the inner skin bulkheads.

5.4.3 For inner skin bulkheads, in general, the thickness of the strake below the top strake is to be not less than 80 per cent of the top strake thickness.

5.4.4 At transverse watertight bulkheads, in general, a transverse box structure is to be arranged at tank top and deck level, and a system of vertical webs and/or horizontal stringers arranged to support the secondary stiffeners. The webs are to be fitted in line with the longitudinal underdeck girders and the corresponding double bottom side girders. Where the depth to breadth ratio of the bulkhead is large, the scantlings of the primary supporting members are to be determined by direct calculation.

5.4.5 Where non-watertight bulkheads are arranged in conjunction with the double bottom mid-hold support, a transverse box is to be arranged at deck level.

Section 6

Buckling Capacity

The buckling assessment required by 4.6.4 for plates and stiffeners are to be determined using the methods detailed in this section.

6.1 Elementary Plate Panel

6.1.1 General

6.1.1.1 Definitions

6.1.1.1 An Elementary Plate Panel (EPP) is the unstiffened part of the plating between stiffeners and/or primary supporting members.

6.1.1.2 All the edges of the elementary plate panel are forced to remain straight (but free to move in the in-plane directions) due to the surrounding structure/neighbouring plates (usually longitudinal stiffened panels in deck, bottom and inner-bottom plating, shell and longitudinal bulkheads).

6.1.2 Symbols

6.1.2.1 Refer to table 6.1.2, for description of symbols used in this section.

6.1.3 EPP with different thicknesses

6.1.3.1 Longitudinally stiffened EPP with different thicknesses

6.1.3.1.1 In longitudinal stiffening arrangement, when the plate thickness varies over the width, *b* [mm], of a plate panel, the buckling capacity is calculated on an equivalent plate panel width, having a thickness equal to the smaller plate thickness, t_1. The width of this equivalent plate panel, b_{eq} [mm], is defined by the following formula:

$$b_{eq} = \ell_1 + \ell_2 \left(\frac{t_1}{t_2}\right)^{1.5}$$

where:

ℓ_1 : Width of the part of the plate panel with the smaller plate thickness, t_1 [mm], as defined in Fig.6.1.3.1.1.

ℓ_2 : Width of the part of the plate panel with the greater plate thickness, t_2 [mm], as defined in Fig.6.1.3.1.1.

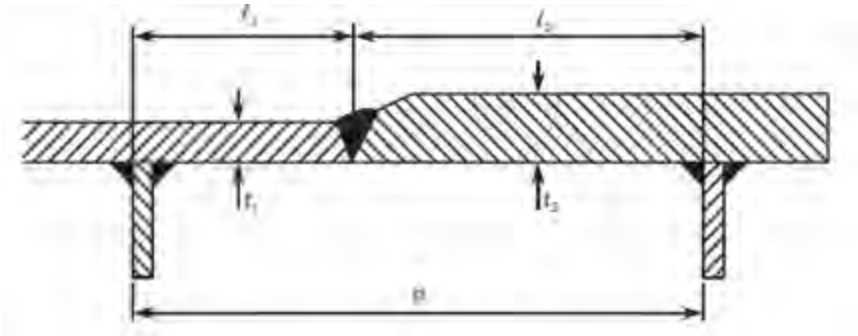


Fig.6.1.3.1.1 : Plate thickness change over the width

Table 6.1.2 : Symbols

x axis	*:* Local axis of a rectangular buckling panel parallel to its long edge.
y axis	*:* Local axis of a rectangular buckling panel perpendicular to its long edge.
σ_x	: Membrane stress applied in *x* direction [N/mm^2]
σ_y	: Membrane stress applied in *y* direction [N/mm^2]
τ	: Membrane shear stress applied in *xy* plane [N/mm^2]
σ_a	: Axial stress in the stiffener [N/mm^2]
σ_b	: Bending stress in the stiffener [N/mm^2]
σ_w	: Warping stress in the stiffener [N/mm^2]
$\sigma_{cx}, \sigma_{cy}, \tau_c$	: Critical stress [N/mm^2], defined in 6.2.1.1 for plates.
R_{eH_S}	: Specified minimum yield stress of the stiffener [N/mm^2]
R_{eH_P}	: Specified minimum yield stress of the plate [N/mm^2]
a	: Length of the longer side of the plate panel as shown in Table 6.2.1.2.1 [mm].
b	: Length of the shorter side of the plate panel as shown in Table6.2.1.2.1 [mm].
d	: Length of the side parallel to the axis of the cylinder corresponding to the curved plate panel as shown in Table 6.2.1.2.2 [mm].
σ_E	: Elastic buckling reference stress [N/mm^2] to be taken as: • For the application of plate limit state according to 6.2.1.2: $\sigma_E = \frac{\pi^2 E}{12(1-\nu^2)}\left(\frac{t_p}{b}\right)^2$ • For the application of curved plate panels according to 6.2.2: $\sigma_E = \frac{\pi^2 E}{12(1-\nu^2)}\left(\frac{t_p}{d}\right)^2$
v	: Poisson's ratio to be taken equal to 0.3
t_p	: Net thickness of plate panel [mm]
t_w	: Net stiffener web thickness [mm]
t_f	: Net flange thickness [mm]
b_f	: Breadth of the stiffener flange [mm]
h_w	: Stiffener web height [mm]
e_f	: Distance from attached plating to centre of flange [mm], to be taken as: $e_f = h_w$ for flat bar profile. $e_f = h_w - 0.5\ t_f$ for bulb profile. $e_f = h_w + 0.5\ t_f$ for angle and Tee profiles.
α	: Aspect ratio of the plate panel, to be taken as $\alpha = \frac{a}{b}$
β	: Coefficient taken as $\beta = \frac{1-\psi}{\alpha}$
ψ	: Edge stress ratio to be taken as $\psi = \frac{\sigma_2}{\sigma_1}$
σ_1	: Maximum stress [N/mm^2]
σ_2	: Minimum stress [N/mm^2]
R	: Radius of curved plate panel [mm]
ℓ	: Span, in mm, of stiffener equal to the spacing between primary supporting members
s	: Spacing of stiffener [mm], to be taken as the mean spacing between the stiffeners of the considered stiffened panel.

6.1.3.2 Transversally stiffened EPP with different thicknesses

6.1.3.2.1 In transverse stiffening arrangement, when an EPP is made of different thicknesses, the buckling check of the plate and stiffeners is to be made for each thickness considered constant on the EPP.

6.2 Buckling capacity of plates

6.2.1 Plate panel

6.2.1.1 Plate limit state

6.2.1.1.1.The plate limit state is based on the following interaction formulae:

a) Longitudinal stiffening arrangement:

$$\left(\frac{\gamma_C \sigma_x}{\sigma_{cx}}\right)^{2/\beta_p^{0.25}} + \left(\frac{\gamma_C |\tau|}{\tau_c}\right)^{2/\beta_p^{0.25}} = 1$$

b) Transverse stiffening arrangement:

$$\left(\frac{\gamma_C \sigma_y}{\sigma_{cy}}\right)^{2/\beta_p^{0.25}} + \left(\frac{\gamma_C |\tau|}{\tau_c}\right)^{2/\beta_p^{0.25}} = 1$$

where:

σ_x, σ_y : Applied normal stress to the plate panel [N/mm²], as defined in *Sec 4.6.3.2*, at load calculation points of the considered elementary plate panel.

τ : Applied shear stress to the plate panel [N/mm²], as defined in *Sec 4.6.3.2*, at load calculation points of the considered elementary plate panel.

σ_{cx} : Ultimate buckling stress [N/mm²] in direction parallel to the longer edge of the buckling panel as defined in 6.2.1.3

σ_{cy} : Ultimate buckling stress [N/mm²] in direction parallel to the shorter edge of the buckling panel as defined in 6.2.1.3

τ_c :Ultimate buckling shear stress [N/mm²] as defined in 6.2.1.3

β_p : Plate slenderness parameter taken as:

$$\beta_p = \frac{b}{t_p}\sqrt{\frac{R_{eH_P}}{E}}$$

R_{eH_P} : Specified minimum yield stress of the stiffener [N/mm^2]

t_p : Net thickness of plate panel [mm]

6.2.1.2 Reference degree of slenderness

6.2.1.2.1 The reference degree of slenderness is to be taken as:

$$\lambda = \sqrt{\frac{R_{eH_P}}{K\sigma_E}}$$

where:

K : Buckling factor, as defined in Table 6.2.1.2.1 and Table 6.2.1.2.2.

σ_E : Elastic buckling reference stress [N/mm^2]

$$\sigma_E = \frac{\pi^2 E}{12(1-\nu^2)}\left(\frac{t_p}{b}\right)^2$$

6.2.1.3 Ultimate buckling stresses

6.2.1.3.1 The ultimate buckling stress of plate panels [N/mm²], is to be taken as:

$$\sigma_{cx} = C_x R_{eH_P}$$

$$\sigma_{cy} = C_y R_{eH_P}$$

6.2.1.3.2 The ultimate buckling stress of plate panels subject to shear [N/mm²], is to be taken as:

$$\tau_c = C_\tau \frac{R_{eH_P}}{\sqrt{3}}$$

where:

C_x, C_y, $C\tau$: Reduction factors, as defined in Table 6.2.1.2.1

6.2.1.3.3 The boundary conditions for plates are to be considered as simply supported (see cases 1, 2 and 15 of Table 6.2.1.2.1. If the boundary conditions differ significantly from simple support, a more appropriate boundary condition can be applied according to the different cases of Table 6.2.1.2.2 subject to the agreement of IRS.

6.2.1.4 Correction Factor F_{long}

6.2.1.4.1 The correction factor F_{long} depending on the edge stiffener types on the longer side of the buckling panel is defined in Table 6.2.1.4. An average value of F_{long} is to be used for plate panels having different edge stiffeners. For stiffener types other than those mentioned in Table 6.2.1.4, the value of *c* is to be agreed by IRS. In such a case, value of *c* higher than those mentioned in Table 6.2.1.4 can be used, provided it is verified by buckling strength check of panel using non-linear FE analysis and deemed appropriate by IRS.

Table 6.2.1.4 : Correction Factor F_{long}

Structural element types			F_{long}	c
Unstiffened Panel			1.0	N/A
Stiffened Panel	Stiffener not fixed at both ends		1.0	N/A
	Stiffener fixed at both ends	Flat bar (1)	$F_{long} = c + 1 \quad for \quad \frac{t_w}{t_p} > 1$	0.10
		Bulb profile		0.30
		Angle profile		0.40
		T profile	$F_{long} = c\left(\frac{t_w}{t_p}\right)^3 + 1 \quad for \quad \frac{t_w}{t_p} \le 1$	0.30
		Girder of high rigidity (e.g. bottom transverse)	1.4	N/A

(1) t_w is the net web thickness, in [mm], without the correction defined in 6.4.3.5

Table 6.2.1.2.1: Buckling Factor and reduction factor for plane plate panels

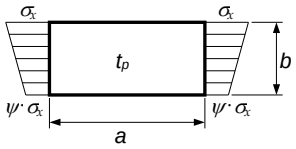
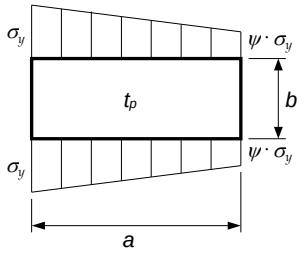
Case	Stress ratio ψ	Aspect ratio α	Buckling factor K	Reduction factor C
1 	1 ≥ ψ ≥ 0		$K_x = F_{long}\frac{8.4}{\psi + 1.1}$	$C_x = 1$ for $\lambda \le \lambda_c$ $C_x = c\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > \lambda_c$ where: $c = (1.25 - 0.12\psi) \le 1.25$ $\lambda_c = \frac{c}{2}\left(1 + \sqrt{1 - \frac{0.88}{c}}\right)$
	0 > ψ > -1		$K_x = F_{long}\left[7.63 - \psi\,(6.26 - 10\,\psi)\right]$	
	ψ ≤ -1		$K_x = F_{long}\left[5.975(1-\psi)^2\right]$	
2 	1 ≥ ψ ≥ 0		$K_y = \frac{2\left(1+\frac{1}{\alpha^2}\right)^2}{1+\psi+\frac{(1-\psi)}{100}\left(\frac{2.4}{\alpha^2}+6.9\,f_1\right)}$	$C_y = c\left(\frac{1}{\lambda} - \frac{R + F^2(H-R)}{\lambda^2}\right)$ where: $c = (1.25 - 0.12\psi) \le 1.25$ $R = \lambda(1 - \lambda / c)$ for $\lambda < \lambda_c$ $R = 0.22$ for $\lambda \ge \lambda_c$ $\lambda_c = 0.5\,c\,\left(1 + \sqrt{1 - 0.88 / c}\right)$ $F = \left(1 - \left(\frac{K}{0.91} - 1\right) / \lambda_p^2\right) c_1 \ge 0$ $\lambda_p^2 = \lambda^2 - 0.5$ for $1 \le \lambda_p^2 \le 3$
		α ≤ 6	$f_1 = (1-\psi)(\alpha - 1)$	
		α > 6	$f_1 = 0.6\left(1 - \frac{6\psi}{\alpha}\right)\left(\alpha + \frac{14}{\alpha}\right)$ But not greater than $14.5 - \frac{0.35}{\alpha^2}$	

Table 6.2.1.2.1: Buckling Factor and reduction factor for plane plate panels

Case	Stress ratio ψ	Aspect ratio α	Buckling factor K	Reduction factor C
	$0 > \psi \geqslant 1-4\alpha/3$		$K_y = \frac{200\left(1+\beta^2\right)^2}{\left(1-f_3\right)\left(100+2.4\beta^2+6.9f_1+23f_2\right)}$	$c_1 = \left(1-\frac{1}{\alpha}\right) \geq 0$ $H = \lambda - \frac{2\lambda}{c(T+\sqrt{T^2-4})} \geq R$ $T = \lambda + \frac{14}{15\lambda} + \frac{1}{3}$
		$\alpha > 6(1-\psi)$	$f_1 = 0.6\left(\frac{1}{\beta}+14\beta\right)$ but not greater than $14.5-0.35\beta^2$ $f_2 = f_3 = 0$	
		$3(1-\psi) \leq \alpha \leq 6(1-\psi)$	$f_1 = \frac{1}{\beta} - 1$ $f_2 = f_3 = 0$	
		$1.5(1-\psi) \leq \alpha < 3(1-\psi)$	$f_1 = \frac{1}{\beta} - (2-\omega\beta)^4 - 9(\omega\beta-1)\left(\frac{2}{3}-\beta\right)$ $f_2 = f_3 = 0$	
		$1-\psi \leq \alpha < 1.5(1-\psi)$	For $\alpha > 1.5$: $f_1 = 2\left(\frac{1}{\beta} - 16\left(1-\frac{\omega}{3}\right)^4\right)\left(\frac{1}{\beta}-1\right)$ $f_2 = 3\beta - 2$ $f_3 = 0$ For $\alpha \leq 1.5$: $f_1 = 2\left(\frac{1.5}{1-\psi}-1\right)\left(\frac{1}{\beta}-1\right)$ $f_2 = \frac{\psi\left(1-16f_4^2\right)}{1-\alpha}$ $f_3 = 0$ $f_4 = (1.5-Min(1.5;\alpha))^2$	
		$0.75(1-\psi) \leq \alpha < 1-\psi$	$f_1 = 0$ $f_2 = 1 + 2.31(\beta-1) - 48\left(\frac{4}{3}-\beta\right)f_4^2$ $f_3 = 3f_4(\beta-1)\left(\frac{f_4}{1.81} - \frac{\alpha-1}{1.31}\right)$ $f_4 = (1.5-Min(1.5;\alpha))^2$	

Table 6.2.1.2.1: Buckling Factor and reduction factor for plane plate panels

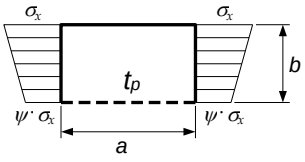
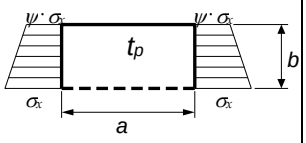
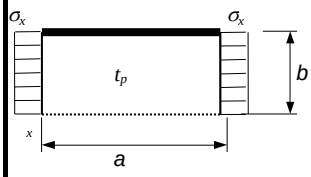
Case	Stress ratio ψ	Aspect ratio α	Buckling factor K	Reduction factor C
	$\psi < 1-4\alpha/3$		$K_y = 5.972\,\frac{\beta^2}{1-f_3}$ where: $f_3 = f_5\left(\frac{f_5}{1.81}+\frac{1+3\psi}{5.24}\right)$ $f_5 = \frac{9}{16}(1+Max(-1;\psi))^2$	
3 	$1 \geq \psi \geq 0$		$K_x = \frac{4(0.425+1/\alpha^2)}{3\psi+1}$	$C_x = 1$ for $\lambda \leq 0.7$ $C_x = \frac{1}{\lambda^2+0.51}$ for $\lambda > 0.7$
	$0 > \psi \geq -1$		$K_x = 4(0.425+1/\alpha^2)(1+\psi) - 5\psi(1-3.42\psi)$	
4 	$1 \geq \psi \geq -1$		$K_x = \left(0.425+\frac{1}{\alpha^2}\right)\frac{3-\psi}{2}$	
5 	—	$\alpha \geq 1.64$	$K_x = 1.28$	
		$\alpha < 1.64$	$K_x = \frac{1}{\alpha^2}+0.56+0.13\alpha^2$	
6	$1 \geq \psi \geq 0$		$K_y = \frac{4(0.425+\alpha^2)}{(3\psi+1)\alpha^2}$	$C_y = 1$ for $\lambda \leq 0.7$ $C_y = \frac{1}{\lambda^2+0.51}$ for $\lambda > 0.7$

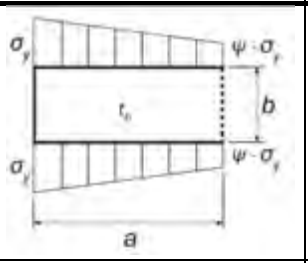
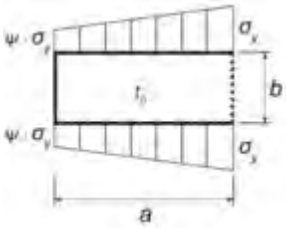
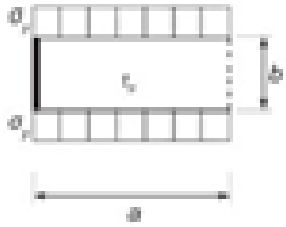
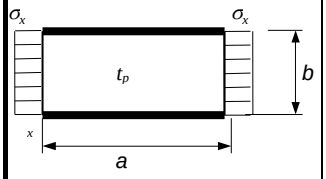
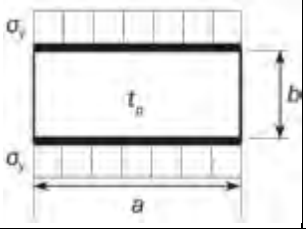
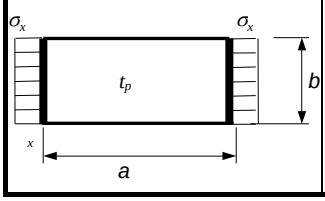
Table 6.2.1.2.1: Buckling Factor and reduction factor for plane plate panels				
Case	**Stress ratio *ψ***	**Aspect ratio *α***	**Buckling factor *K***	**Reduction factor *C***
	$0 > \psi \geq -1$	$K_y = 4(0.425 + \alpha^2)(1 + \psi)\frac{1}{\alpha^2} - 5\psi(1 - 3.42\psi)\frac{1}{\alpha^2}$		
7 	$-1 \geq \psi \geq -1$	$K_y = (0.425 + \alpha^2)\frac{(3 - \psi)}{2\alpha^2}$		
8 	-	$K_y = 1 + \frac{0.56}{\alpha^2} + \frac{0.13}{\alpha^4}$		
9 	-	$K_x = 6.97$		$C_x = 1$ for $\lambda \leq 0.83$ $C_x = 1{,}13\left[\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right]$ for $\lambda > 0.83$
10 	-	$K_y = 4 + \frac{2.07}{\alpha^2} + \frac{0.67}{\alpha^4}$		
11 	-	$\alpha \geq 4$	$K_x = 4$	
		$\alpha < 4$	$K_x = 4 + 2.74\left(\frac{4 - \alpha}{3}\right)^4$	

Table 6.2.1.2.1: Buckling Factor and reduction factor for plane plate panels

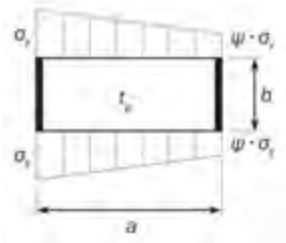
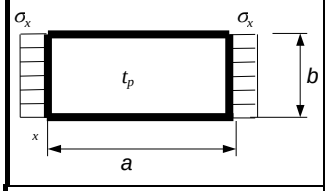
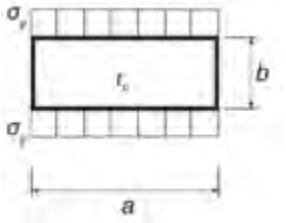
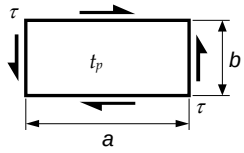
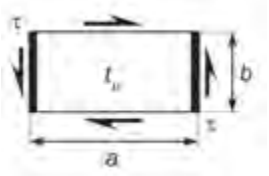
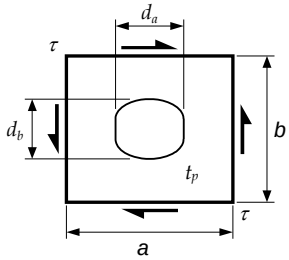
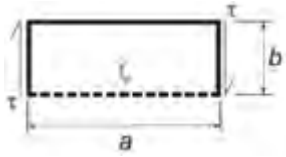
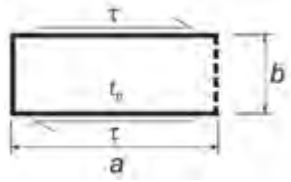
Case	Stress ratio ψ	Aspect ratio α	Buckling factor K	Reduction factor C
12 	-	$K_y = K_y$ determined as per case 2		• For α < 2: $C_y = C_{y2}$ • For α ⩾ 2: $C_y = \left(1.06 + \frac{1}{10\alpha}\right) C_{y2}$ where: $C_{y2} = C_y$ determined as per case 2
13 	-	$\alpha \geq 4$	$K_x = 6.97$	$C_x = 1$ for $\lambda \leq 0.83$ $C_x = 1{,}13\left[\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right]$ for $\lambda > 0.83$
		α < 4	$K_x = 6.97 + 3.1\left(\frac{4-\alpha}{3}\right)^4$	
14 	-	$K_y = \frac{6.97}{\alpha^2} + \frac{3.1}{\alpha^2}\left[\frac{4 - 1/\alpha}{3}\right]^4$		$C_y = 1$ for $\lambda \leq 0.83$ $C_y = 1.13\left(\frac{1}{\lambda} - \frac{0.22}{\lambda^2}\right)$ for $\lambda > 0.83$
15 	-	$K_\tau = \sqrt{3}\left[5.34 + \frac{4}{\alpha^2}\right]$		$C_\tau = 1$ for $\lambda \leq 0.84$ $C_\tau = \frac{0.84}{\lambda}$ for $\lambda > 0.84$
16 	-	$K_\tau = \sqrt{3}\left\{5.34 + Max\left[\frac{4}{\alpha^2}; \frac{7.15}{\alpha^{2.5}}\right]\right\}$		
17 	-	$K = K'r$ K' = K according to case 15. r = opening reduction factor taken as $r = \left(1 - \frac{d_a}{a}\right)\left(1 - \frac{d_b}{b}\right)$ with $\frac{d_a}{a} \leq 0.7$ and $\frac{d_b}{b} \leq 0.7$		

Table 6.2.1.2.1: Buckling Factor and reduction factor for plane plate panels				
Case	**Stress ratio ψ**	**Aspect ratio α**	**Buckling factor *K***	**Reduction factor *C***
18 	-		$K_\tau = 3^{0.5}(0.6 + 4/\alpha^2)$	$C_\tau = 1$ for $\lambda \le 0.84$ $C_\tau = \frac{0.84}{\lambda}$ for $\lambda > 0.84$
19 	-		$K_\tau = 8$	
Edge boundary conditions: - - - - - - - - - Plate edge free. ———— Plate edge simply supported. ▬▬▬▬ Plate edge clamped.				
Notes: 1) Cases listed are general cases. Each stress component (σx, σy) is to be understood in local coordinates.				

6.2.2 Curved plate panels

6.2.2.1 This requirement for curved plate limit state is applicable when $R/t_p \le 2500$. Otherwise, the requirement for plate limit state given in 6.2.1.1 is applicable.

Where,

R : Radius of curved plate panel, in [mm]

6.2.2.2 The curved plate limit state is based on the following interaction formula:

$$\left(\frac{\gamma_c \sigma_{ax}}{C_{ax} R_{eH_P}}\right)^{1.25} + \left(\frac{\gamma_c \tau \sqrt{3}}{C_\tau R_{eH_P}}\right)^2 = 1.0$$

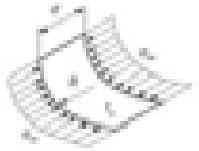
where:

σ_{ax} : Applied axial stress to the cylinder corresponding to the curved plate panel [N/mm²]. In case of tensile axial stresses, σ_{ax}=0.

C_{ax}, C_τ : Buckling reduction factor of the curved plate panel, as defined in Table 6.2.1.2.2.

6.2.2.3 The stress multiplier factor γ_c of the curved plate panel needs not be taken less than the stress multiplier factor γ_c for the expanded plane panel according to 6.2.1.1.

Table 6.2.1.2.2 : Buckling Factor and reduction factor for curved plate panel with $R/t_p \leq 2500$

Case	Aspect ratio	Buckling factor *K*	Reduction factor *C*
1 	$\frac{d}{R} \leq 0.5\sqrt{\frac{R}{t_p}}$	$K = 1 + \frac{2}{3}\frac{d^2}{Rt_p}$	For general application: $C_{ax} = 1$ for $\lambda \leq 0.25$ $C_{ax} = 1.233\text{-}0.933\lambda$ for $0.25 < \lambda \leq 1$ $C_{ax} = 0.3/\lambda^3$ for $1 < \lambda \leq 1.5$ $C_{ax} = 0.2/\lambda^2$ for $\lambda > 1.5$ For curved single fields, e.g. bilge strake, which are bounded by plane panels: $C_{ax} = 0.65/\lambda^2 \leq 1.0$
	$\frac{d}{R} > 0.5\sqrt{\frac{R}{t_p}}$	$K = 0.267\frac{d^2}{Rt_p}\left[3 - \frac{d}{R}\sqrt{\frac{t_p}{R}}\right] \geq 0.4\frac{d^2}{Rt_p}$	
2	$\frac{d}{R} \leq 8.7\sqrt{\frac{R}{t_p}}$	$K = \sqrt{3}\sqrt{28.3 + \frac{0.67d^3}{R^{1.5}t_p^{1.5}}}$	$C_\tau = 1$ for $\lambda \leq 0.4$ $C_\tau = 1.274 - 0.686\lambda$ for $0.4 < \lambda \leq 1.2$ $C_\tau = 0.65/\lambda^2$ for $\lambda > 1.2$
	$\frac{d}{R} > 8.7\sqrt{\frac{R}{t_p}}$	$K = \sqrt{3}\frac{0.28d^2}{R\sqrt{Rt_p}}$	
Explanations for boundary conditions: ———— Plate edge simply supported.			

6.3 Buckling Capacity of overall stiffened panel

6.3.1 The elastic stiffened panel limit state is based on the following interaction formula:

$$\frac{P_z}{c_f} = 1$$

where P_z and c_f are defined in 6.4.4.3.

6.4 Buckling capacity of longitudinal stiffeners

6.4.1 Stiffeners limit states

6.4.1.1 The buckling capacity of longitudinal stiffeners is to be checked for the following limit states:

- Stiffener induced failure (*SI*).
- Associated plate induced failure (*PI*).

6.4.2 Lateral pressure

6.4.2.1 The lateral pressure is to be considered as constant in the buckling strength assessment of longitudinal stiffeners.

6.4.3 Stiffener idealization

6.4.3.1 Effective length of the stiffener ℓ_{eff}

6.4.3.1.1 The effective length of the stiffener ℓ_{eff}, in mm, is to be taken equal to:

$\ell_{eff} = \frac{\ell}{\sqrt{3}}$ for stiffener fixed at both ends.

$\ell_{eff} = 0.75\ell$ for stiffener simply supported at one end and fixed at the other.

$\ell_{eff} = \ell$ for stiffener simply supported at both ends.

6.4.3.2 Effective width of the attached plating b_{eff1}

6.4.3.2.1 The effective width of the attached plating of a stiffener b_{eff1} [mm], without the shear lag effect is to be taken equal to:

$$b_{eff1} = \frac{C_{x1}b_1 + C_{x2}b_2}{2}$$

where:

C_{x1}, C_{x2} : Reduction factor defined in Table 6.2.1.2.1 calculated for the EPP1 and EPP2 on each side of the considered stiffener according to case 1.

b_1, b_2 : Width of plate panel on each side of the considered stiffener [mm].

6.4.3.3 Effective width of attached plating b_{eff}

6.4.3.3.1 The effective width of attached plating of stiffeners, b_{eff} [mm], is to be taken as:

$$b_{eff} = min(b_{eff1}, \chi_s s)$$

where:

χ_s : Effective width coefficient to be taken as:

- $\chi_s = min\left[\frac{1.12}{1+\frac{1.75}{\left(\frac{\ell_{eff}}{s}\right)^{1.6}}}; 1\right]$ for $\frac{\ell_{eff}}{s} \geq 1$
- $\chi_s = 0.407\frac{\ell_{eff}}{s}$ for $\frac{\ell_{eff}}{s} < 1$

6.4.3.4 Net thickness of attached plating t_p

6.4.3.4.1 The net thickness of plate t_p [mm], is to be taken as the mean thickness of the two attached plating panels

6.4.3.5 Effective web thickness of flat bar

6.4.3.5.1 For accounting the decrease of stiffness due to local lateral deformation, the effective web thickness of flat bar stiffener, in mm, is to be used for the calculation of the net sectional area, A_s, the net section modulus, Z, and the moment of inertia, I, of the stiffener and is taken as:

$$t_{w_red} = t_w\left(1 - \frac{2\pi^2}{3}\left(\frac{h_w}{s}\right)^2\left(1 - \frac{b_{eff1}}{s}\right)\right)$$

6.4.3.6 Net section modulus Z of a stiffener

6.4.3.6.1 The net section modulus Z of a stiffener [cm^3], including effective width of plating b_{eff} is to be taken equal to:

- the section modulus calculated at the top of stiffener flange for stiffener induced failure (*SI*).
- the section modulus calculated at the attached plating for plate induced failure (*PI*).

6.4.3.7 Net moment of inertia I of a stiffener

6.4.3.7.1 The net moment of inertia I [cm^4], of a stiffener including effective width of attached plating b_{eff} is to comply with the following requirement:

$$\boldsymbol{I \geq \frac{s\, t_p^3}{12 \cdot 10^4}}$$

6.4.3.8 Idealisation of bulb profile

6.4.3.8.1 Bulb profiles may be considered as equivalent angle profiles. The net dimensions of the equivalent built-up section are to be obtained [mm], from the following formulae.

$$h_w = h_w^{'} - \frac{h_w^{'}}{9.2} + 2$$

$$b_f = \alpha\left(t_w^{'} + \frac{h_w^{'}}{6.7} - 2\right)$$

$$t_f = \frac{h_w^{'}}{9.2} - 2$$

$$t_w = t_w^{'}$$

where:

h'_w, t'_w : Net height and thickness of a bulb section [mm], as shown in Fig.6.4.3.8.

α : Coefficient equal to:

$\alpha = 1.1 + \frac{\left(120 - h_w^{'}\right)^2}{3000}$ for $h'_w \leq 120$

$\alpha = 1.0$ for $h'_w > 120$

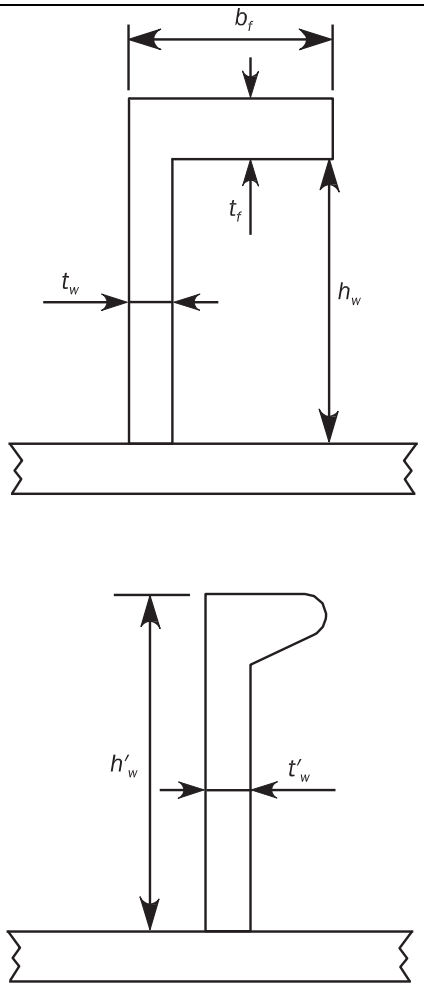


Fig.6.4.3.8 : Idealisation of bulb stiffener

6.4.4 Ultimate buckling capacity

6.4.4.1 Longitudinal stiffener limit state

6.4.4.1.1 When $\sigma_a + \sigma_b + \sigma_w > 0$, the ultimate buckling capacity for stiffeners is to be checked according to the following interaction formula:

$$\frac{\gamma_c \, \sigma_a + \sigma_b + \sigma_w}{R_{eH}} = 1$$

where:

σ_a : Effective axial stress [N/mm²], at mid-span of the stiffener, defined in 6.4.4.2.

σ_b : Bending stress in the stiffener [N/mm²], defined in 6.4.4.3.

σ_w : Stress due to torsional deformation [N/mm²], defined in 6.4.4.4.

R_{eH} : Specified minimum yield stress of the material [N/mm²]:

$R_{eH} = R_{eH-S}$ for stiffener induced failure (*SI*).

$R_{eH} = R_{eH-P}$ for plate induced failure (*PI*).

6.4.4.2 Effective axial stress σ_a

6.4.4.2.1 The effective axial stress σ_a [N/mm²], at mid-span of the stiffener, acting on the stiffener with its attached plating is to be taken equal to:

$$\sigma_a = \sigma_x \frac{s t_p + A_s}{b_{eff1} t_p + A_s}$$

where:

σ_x : Nominal axial stress [N/mm^2], acting on the stiffener with its attached plating, calculated according to Row 1 of Table 4.6.3.2.1 at load calculation point of the stiffener.

A_s : Net sectional area [mm^2], of the considered stiffener.

6.4.4.3 Bending stress σ_b

6.4.4.3.1 The bending stress in the stiffener σ_b [N/mm^2], is to be taken equal to:

$$\sigma_b = \frac{M_0 + M_1}{Z} 10^{-3}$$

where:

M_1 : Bending moment [Nmm], due to the lateral load P:

$M_1 = C_i \frac{|P| s \ell^2}{24} 10^{-3}$ for continuous stiffener

$M_1 = C_i \frac{|P| s \ell^2}{8} 10^{-3}$ for sniped stiffener

P : Lateral load [kN/m^2], to be taken equal to the static pressure at the load calculation point of the stiffener.

C_i : Pressure coefficient:

C_i=C_{SI} for stiffener induced failure (*SI*).

C_i=C_{PI} for plate induced failure (*PI*).

C_{PI} : Plate induced failure pressure coefficient:

C_{PI} = 1 if the lateral pressure is applied on the side opposite to the stiffener.

C_{PI} = -1 if the lateral pressure is applied on the same side as the stiffener.

C_{SI} : Stiffener induced failure pressure coefficient:

C_{SI} = -1 if the lateral pressure is applied on the side opposite to the stiffener.

C_{SI} = 1 if the lateral pressure is applied on the same side as the stiffener.

M_0 : Bending moment [Nm], due to the lateral deformation w of stiffener:

$$M_0 = F_E \left(\frac{P_z w}{c_f - P_z} \right) \quad \text{with} \quad c_f - P_z > 0 .$$

F_E : Ideal elastic buckling force of the stiffener [N].

$$F_E = \left(\frac{\pi}{\ell}\right)^2 EI \, 10^4$$

P_z : Nominal lateral load [N/mm^2], acting on the stiffener due to stresses σ_x and τ, in the attached plating in way of the stiffener mid span:

$$P_z = \frac{t_p}{s} \left(\sigma_{x\ell} \left(\frac{\pi s}{\ell}\right)^2 + \sqrt{2}\tau_1 \right)$$

$\sigma_{x\ell} = \gamma_c \sigma_x \left(1 + \frac{A_s}{s t_p}\right)$ but not but not less than 0

$\tau_1 = \left(\gamma_c |\tau| - t_p \sqrt{R_{eH-p} E \left(\frac{m_1}{a^2} + \frac{m_2}{s^2} \right)} \right) \geq 0$ but not less than 0

m_1, m_2 : Coefficients taken equal to:

m_1 = 1.47, m_2 = 0.49 for $\alpha \geq 2$.

m_1 = 1.96, m_2 = 0.37 for $\alpha < 2$.

w : Deformation of stiffener [mm], taken equal to:

$$w = w_0 + w_1$$

w_0 : Assumed imperfection [mm], taken equal to:

$w = \ell \, 10^{-3}$ in general

$w_0 = -w_{na}$ for stiffeners sniped at both ends, considering stiffener induced failure (*SI*)

$w_0 = w_{na}$ for stiffeners sniped at both ends, considering plate induced failure (*PI*)

w_{na} : Distance [mm], from the mid-point of attached plating to the neutral axis of the stiffener calculated with the

effective width of the attached plating b_{eff}.

w_1 : Deformation [mm] of stiffener at midpoint of stiffener span due to lateral load P. In case of uniformly distributed load, w_1 is to be taken as:

$w_1 = C_i \frac{|P|s\,\ell^4}{384\,EI} 10^{-7}$ in general

$w_1 = C_i \frac{5\,|P|s\,\ell^4}{384\,EI} 10^{-7}$ for stiffener sniped at both ends

c_f : Elastic support provided by the stiffener [N/mm^2], to be taken equal to:

$$c_f = F_E \left(\frac{\pi}{\ell}\right)^2 \left(1 + c_p\right)$$

c_p : Coefficient to be taken as:

$$c_p = \frac{1}{1 + \frac{0.91}{c_{xa}} \left(\frac{12 I 10^4}{s t_p^3} - 1 \right)}$$

c_{xa} : Coefficient to be taken as:

$c_{xa} = \left(\frac{\ell}{2s} + \frac{2s}{\ell}\right)^2$ for $\ell \geq 2s$

$c_{xa} = \left(1 + \left(\frac{\ell}{2s}\right)^2\right)^2$ for $\ell < 2s$

6.4.4.4 Stress due to torsional deformation σ_w

6.4.4.4.1 The stress due to torsional deformation σ_w [N/mm^2], is to be taken equal to:

$\sigma_w = E\, y_w \left(\frac{t_f}{2} + h_w\right) \Phi_0 \left(\frac{\pi}{\ell}\right)^2 \left(\frac{1}{1 - \frac{0.4\,R_{eH} - S}{\sigma_{ET}}} - 1 \right)$

for stiffener induced failure (*SI*).

$\sigma_w = 0$ for plate induced failure (*PI*).

where:

γ_w : Distance [mm], from centroid of stiffener cross-section to the free edge of stiffener flange, to be taken as:

$\gamma_w = \frac{t_w}{2}$ for flat bar.

$y_w = b_f - \frac{h_w t_w^2 + t_f b_f^2}{2A_s}$ for angle and bulb profiles.

$y_w = \frac{b_f}{2}$ for Tee profile.

$\Phi_0 = \frac{\ell}{h_w} 10^{-3}$

σ_{ET} : Reference stress for torsional buckling, in [N/mm^2]:

$$\sigma_{ET} = \frac{E}{I_P} \left(\frac{\varepsilon\, \pi^2\, I_\omega}{\ell^2} 10^2 + 0.385 I_T \right)$$

I_P : Net polar moment of inertia of the stiffener about point C as shown in Fig.6.4.4.4, as defined in Table 6.4.4.4 [cm^4].

I_T : Net St. Venant's moment of inertia of the stiffener, as defined in Table 6.4.4.4 [cm^4].

I_ω : Net sectional moment of inertia of the stiffener about point C as shown in Fig.6.4.4.4, as defined in Table 6.4.4.4 [cm^6].

ε : Degree of fixation.

$$\varepsilon = 1 + \frac{\left(\frac{\ell}{\pi}\right)^2 10^{-3}}{\sqrt{I_\omega \left(\frac{0.75\,s}{t_p^3} + \frac{e_f - 0.5 t_f}{t_w^3} \right)}}$$

Table 6.4.4.4 : Moments of inertia		
	Flat bars	**Bulb, angle and Tee profiles**
I_P	$\frac{h_w^3 t_w}{3 \cdot 10^4}$	$\left(\frac{A_w (e_f - 0.5 t_f)^2}{3} + A_f \, e_f^2 \right) 10^{-4}$
I_T	$\frac{h_w t_w^3}{3 \cdot 10^4} \left(1 - 0.63 \frac{t_w}{h_w} \right)$	$\frac{(e_f - 0.5 t_f) t_w^3}{3 \cdot 10^4} \left(1 - 0.63 \frac{t_w}{e_f - 0.5 t_f} \right) + \frac{b_f t_f^3}{3 \cdot 10^4} \left(1 - 0.63 \frac{t_f}{b_f} \right)$
I_ω	$\frac{h_w^3 t_w^3}{36 \cdot 10^6}$	$\frac{A_f e_f^2 b_f^2}{12 \cdot 10^6} \left(\frac{A_f + 2.6 A_w}{A_f + A_w} \right)$ for bulb and angle profiles. $\frac{b_f^3 t_f e_f^2}{12 \cdot 10^6}$ for Tee profiles.
A_w : Net web area, in [mm^2]. A_f : Net flange area, in [mm^2].		

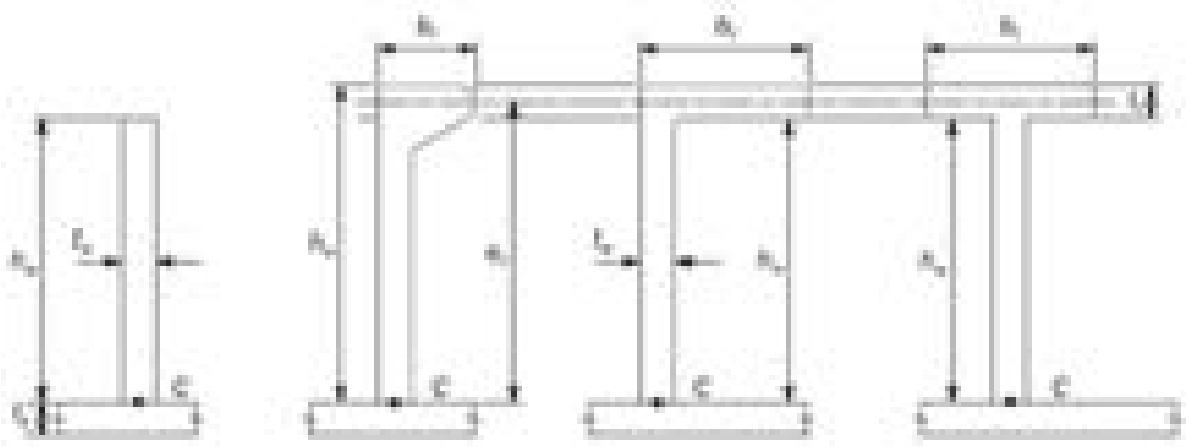


Fig.6.4.4.4 : Stiffener cross sections

Section 7

Ultimate Hull Girder Strength Assessment

This section gives the method of calculation of ultimate hull girder strength as required in 4.7

7.1 General

7.1.1 General Assumptions

7.1.1.1 The method for calculating the ultimate hull girder capacity is to identify the critical failure modes of all main longitudinal structural elements.

7.1.1.2 Structures compressed beyond their buckling limit have reduced load carrying capacity. All relevant failure modes for individual structural elements, such as plate buckling, torsional stiffener buckling, stiffener web buckling, lateral or global stiffener buckling and their interactions, are to be considered in order to identify the weakest inter-frame failure mode.

7.1.2 Symbols

7.1.2.1 Refer to table 7.1.2, for description of symbols used in this section.

Table 7.1.2 : Symbols

$I_{y\text{-}net}$:	Net moment of inertia [m^4], of the hull transverse section around its horizontal neutral axis
$Z_{B\text{-}net}$, $Z_{D\text{-}net}$:	Section moduli [m^3], at bottom and deck, respectively,
R_{eH_S}:	Minimum yield stress [N/mm^2], of the material of the considered stiffener.
R_{eH_P}:	Minimum yield stress [N/mm^2], of the material of the considered plate.
$A_{s\text{-}net}$:	Net sectional area [cm^2], of stiffener, without attached plating.
$A_{p\text{-}net}$:	Net sectional area [cm^2], of attached plating.

7.2 Incremental iterative method

7.2.1 Assumptions

7.2.1.1 In applying the incremental-iterative method, the following assumptions are generally to be made:

- The ultimate strength is calculated at hull transverse sections between two adjacent transverse webs.
- The hull girder transverse section remains plane during each curvature increment.
- The hull material has an elasto-plastic behaviour.
- The hull girder transverse section is divided into a set of elements, see 7.2.2.2, which are considered to act independently.

7.2.1.2 According to the iterative procedure, the bending moment M_i acting on the transverse section at each curvature value χ_i is obtained by summing the contribution given by the stress σ acting on each element. The stress σ corresponding to the element strain, ε is to be obtained for each curvature increment from the non-linear load-end shortening curves σ-ε of the element.

7.2.1.3 These curves are to be calculated, for the failure mechanisms of the element, from the formulae specified in 7.2.3. The stress σ is selected as the lowest among the values obtained from each of the considered load-end shortening curves σ-ε.

7.2.1.4 The procedure is to be repeated until the value of the imposed curvature reaches the value χ_F in m^{-1}, in hogging and sagging condition, obtained from the following formula:

$$X_F = \pm 0.003 \frac{M_y}{EI_{y-net}}$$

where:

M_y: Lesser of the values M_{Y1} and M_{Y2}, in [kNm].

$M_{Y1} = 10^3 R_{eH} Z_{B\text{-}net}$.

$M_{Y2} = 10^3 R_{eH} Z_{D\text{-}net}$.

7.2.1.5 If the value χ_F is not sufficient to evaluate the peaks of the curve $M\text{-}\chi$, the procedure is to be repeated until the value of the imposed curvature permits the calculation of the maximum bending moments of the curve.

7.2.2 Procedure

7.2.2.1 General

7.2.2.1.1 The curve $M\text{-}\chi$ is to be obtained by means of an incremental-iterative approach, summarised in the flow chart in Fig.7.2.2.1.

7.2.2.1.2 In this procedure, the ultimate hull girder bending moment capacity, M_U is defined as the peak value of the curve with vertical bending moment M versus the curvature χ of the ship cross section as shown in Fig.7.2.2.1. The curve is to be obtained through an incremental-iterative approach.

7.2.2.1.3 Each step of the incremental procedure is represented by the calculation of the bending moment M_i which acts on the hull transverse section as the effect of an imposed curvature χ_i.

7.2.2.1.4 For each step, the value χ_i is to be obtained by summing an increment of curvature, $\Delta\chi$ to the value relevant to the previous step χ_{i-1}. This increment of curvature corresponds to an increment of the rotation angle of the hull girder transverse section around its horizontal neutral axis.

7.2.2.1.5 This rotation increment induces axial strains ε in each hull structural element, whose value depends on the position of the element. In hogging condition, the structural elements above the neutral axis are lengthened, while the elements below the neutral axis are shortened, and vice-versa in sagging condition.

7.2.2.1.6 The stress σ induced in each structural element by the strain ε is to be obtained from the load-end shortening curve $\sigma\text{-}\varepsilon$ of the element, which takes into account the behaviour of the element in the non-linear elasto-plastic domain.

7.2.2.1.7 The distribution of the stresses induced in all the elements composing the hull transverse section determines, for each step, a variation of the neutral axis position due to the nonlinear $\sigma\text{-}\varepsilon$, relationship. The new position of the neutral axis relevant to the step considered is to be obtained by means of an iterative process, imposing the equilibrium among the stresses acting in all the hull elements on the transverse section.

7.2.2.1.8 Once the position of the neutral axis is known and the relevant element stress distribution in the section is obtained, the bending moment of the section M_i around the new position of the neutral axis, which corresponds to the curvature χ_i imposed in the step considered, is to be obtained by summing the contribution given by each element stress.

7.2.2.1.9 The main steps of the incremental-iterative approach described above are summarised as follows (see also Fig.7.2.2.1):

a) Step 1: Divide the transverse section of hull into stiffened plate elements.

b) Step 2: Define stress-strain relationships for all elements as shown in Fig.7.2.2.1.

c) Step 3: Initialise curvature χ_1 and neutral axis for the first incremental step with the value of incremental curvature (i.e. curvature that induces a stress equal to 1% of yield strength in strength deck) as:

$$\chi_{1.} = \Delta\chi = 0.01 \frac{R_{eH}}{E} \frac{1}{z_D - z_n}$$

where:

z_D: Z coordinate [m], of strength deck at side.

z_n: Z coordinate [m], of horizontal neutral axis of the hull transverse section

d) Step 4: Calculate for each element the corresponding strain, $\varepsilon_i = \chi(z_i - z_n)$ and the corresponding stress σ_i

e) Step 5: Determine the neutral axis z_{NA_cur} at each incremental step by establishing force equilibrium over the whole transverse section as:

$\Sigma A_{i\text{-}net}\ \sigma_i = \Sigma A_{j\text{-}net}\ \sigma_j$ (*i*-th element is under compression, *j*-th element under tension).

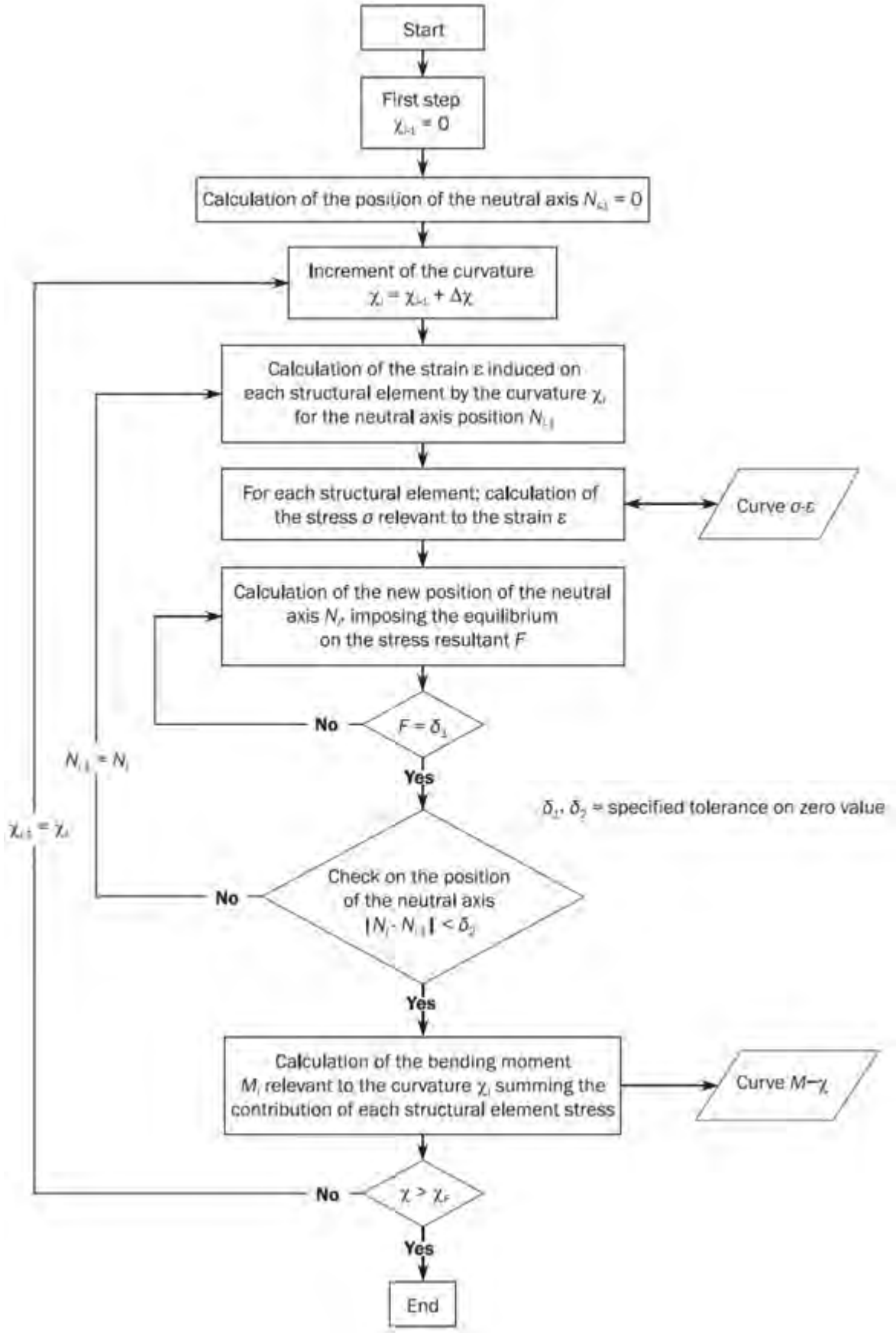


Fig.7.2.2.1 : Flow chart of the procedure for the evaluation of the curve M-χ

f) Step 6: Calculate the corresponding moment by summing the contributions of all elements as:

$$M_U = \sum \sigma_{Ui} A_{i-net} \left| (z_i - z_{NA_cur}) \right|$$

g) Step 7: Compare the moment in the current incremental step with the moment in the previous incremental step. If the slope in M-χ relationship is less than a negative fixed value, terminate the process and define the peak value M_U. Otherwise, increase the curvature by the amount of $\Delta\chi$ and go to Step 4.

7.2.2.2 Modelling of the hull girder cross section

7.2.2.2.1 Hull girder transverse sections are to be considered as being constituted by the members contributing to the hull girder ultimate strength.

7.2.2.2.2 Sniped stiffeners are also to be modelled, taking account that they do not contribute to the hull girder strength.

7.2.2.2.3 The structural members are categorised into a stiffener element, a stiffened plate element or a hard corner element.

7.2.2.2.4 The plate panel including web plate of girder or side stringer is idealised into a stiffened plate element, an attached plate of a stiffener element or a hard corner element.

7.2.2.2.5 The plate panel is categorised into the following two kinds:

- Longitudinally stiffened panel of which the longer side is in ship's longitudinal direction, and
- Transversely stiffened panel of which the longer side is in the perpendicular direction to ship's longitudinal direction.

a) Hard corner element:

Hard corner elements are sturdier elements composing the hull girder transverse section, which collapse mainly according to an elasto-plastic mode of failure (material yielding); they are generally constituted by two plates not lying in the same plane.

The extent of a hard corner element from the point of intersection of the plates is taken equal to 20 t_{net} on a transversely stiffened panel and to 0.5 s on a longitudinally stiffened panel, see Fig.7.2.2.2.

where:

t_{net} : Net thickness of the plate [mm].

s: Spacing of the adjacent longitudinal stiffener [m].

Bilge, sheer strake-deck stringer elements, girder-deck connections and face plate-web connections on large girders are typical hard corners.

b) Stiffener element:

The stiffener constitutes a stiffener element together with the attached plate.

The attached plate width is in principle:

- Equal to the mean spacing of the stiffener when the panels on both sides of the stiffener are longitudinally stiffened, or
- Equal to the width of the longitudinally stiffened panel when the panel on one side of the stiffener is longitudinally stiffened and the other panel is of the transversely stiffened, see Fig.7.2.2.2.

c) Stiffened plate element:

The plate between stiffener elements, between a stiffener element and a hard corner element or between hard corner elements is to be treated as a stiffened plate element, see Fig.7.2.2.2.

The typical examples of modelling of hull girder section are illustrated in Fig.7.2.2.3.

Notwithstanding the foregoing principle, these figures are to be applied to the modelling in the vicinity of upper deck, sheer strake and hatch coaming.

In case of the knuckle point as shown in Fig. 7.2.2.4, the plating area adjacent to knuckles in the plating with an angle greater than 30

degrees is defined as a hard corner. The extent of one side of the corner is taken equal to 20 t_{net} on transversely framed panels and to 0.5 s on longitudinally framed panels from the knuckle point.

- Where the plate members are stiffened by non-continuous longitudinal stiffeners, the non-continuous stiffeners are considered only as dividing a plate into various elementary plate panels.

- Where the opening is provided in the stiffened plate element, the openings are to be specially considered.

- Where attached plating is made of steels having different thicknesses and/or yield stresses, an average thickness and/or average yield stress obtained from the following formula are to be used for the calculation.

$$t_{net} = \frac{t_{1-net}\, s_1 + t_{2-net}\, s_2}{s}$$

$$R_{eH_P} = \frac{R_{eH_P1} t_{1-net} s_1 + R_{eH_P2} t_{2-net} s_2}{t_{net} s}$$

where R_{eH_P1}, R_{eH_P2}, $t_{1\text{-}net}$, $t_{2\text{-}net}$, s_1, s_2 and s are shown in Fig.7.2.2.5

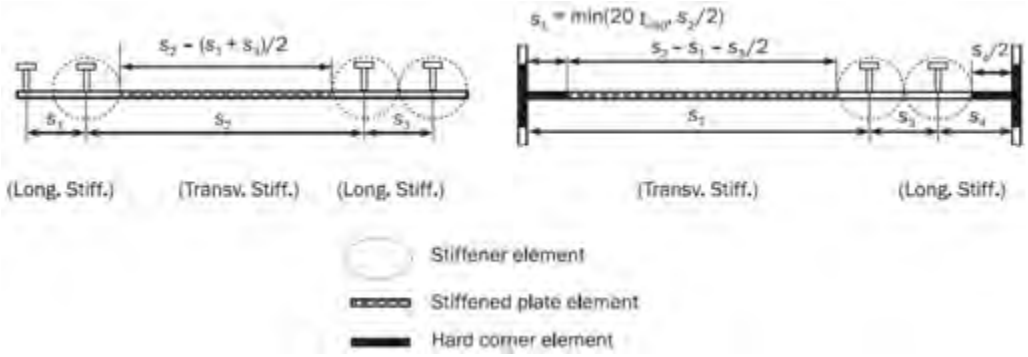


Fig.7.2.2.2 : Extension of the breadth of the attached plating and hard corner element

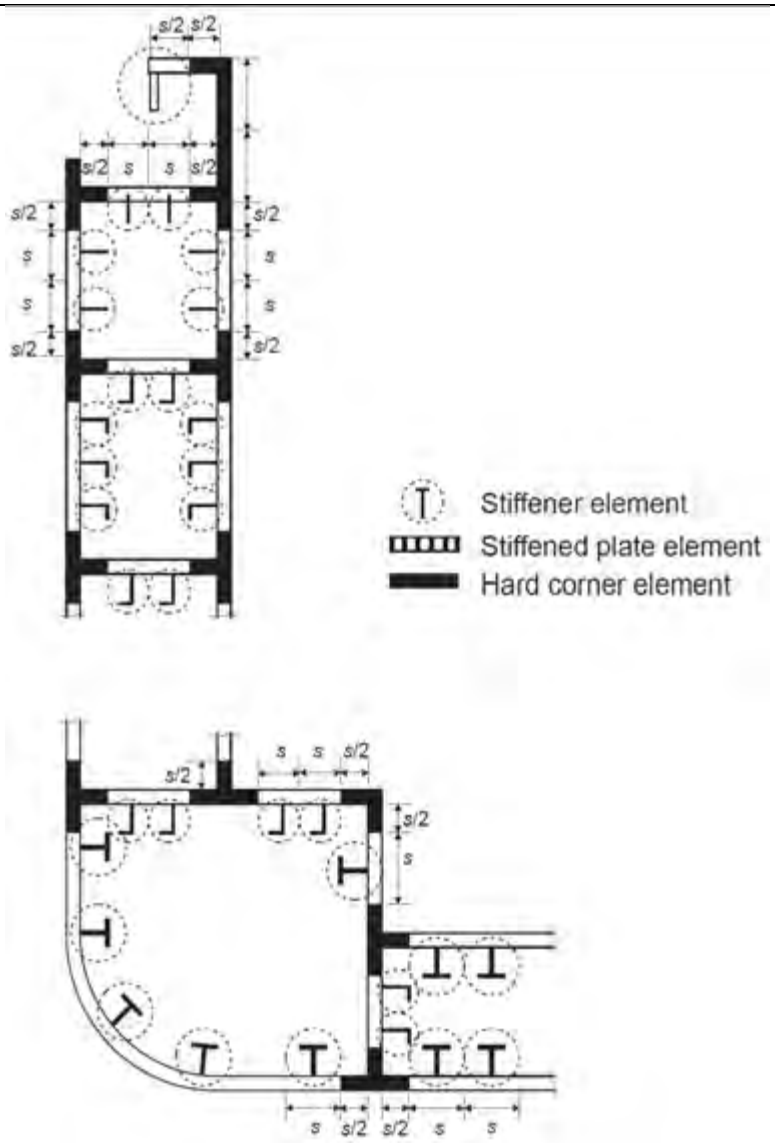


Fig.7.2.2.3 : Examples of the configuration of stiffened plate elements, stiffener elements and hard corner elements on a hull section

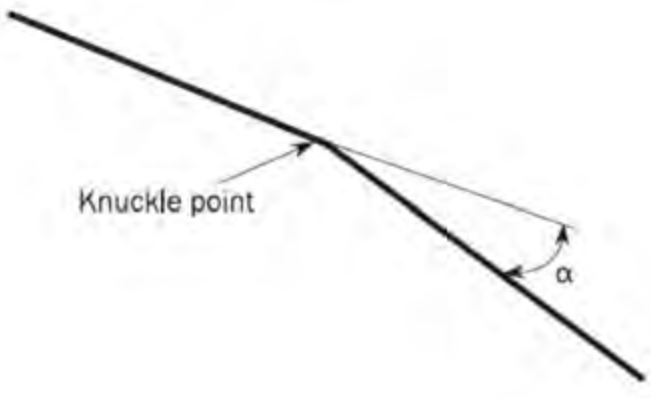


Fig.7.2.2.4 : Plating with knuckle point

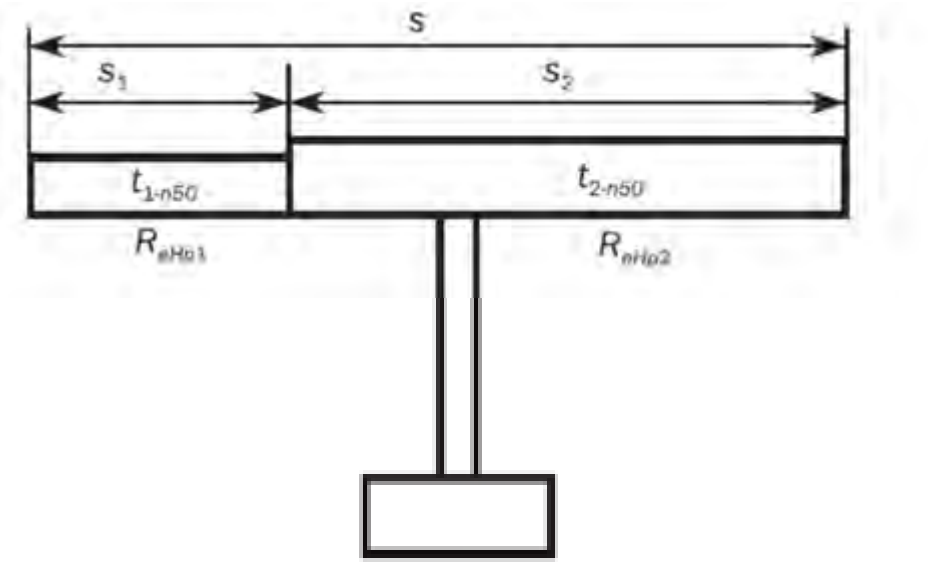


Fig.7.2.2.5 : Element with different thickness and yield strength

7.2.3 Load-end shortening curves

7.2.3.1 Stiffened plate element and stiffener element

7.2.3.1.1 Stiffened plate element and stiffener element composing the hull girder transverse sections may collapse following one of the modes of failure specified in Table 7.2.3.

- Where the plate members are stiffened by non-continuous longitudinal stiffeners, the stress of the element is to be obtained in accordance with 7.2.3.2 to 7.2.3.7, taking into account the non-continuous longitudinal stiffener.

7.2.3.1.2 In calculating the total forces for checking the hull girder ultimate strength, the area of non-continuous longitudinal stiffener is to be assumed as zero.

- Where the opening is provided in the stiffened plate element, the considered area of the stiffened plate element is to be obtained by deducting the opening area from the plating in calculating the total forces for checking the hull girder ultimate strength.

- For stiffened plate element, the effective width of plate for the load shortening portion of the stress-strain curve is to be taken as full plate width, i.e. to the intersection of other plate or longitudinal stiffener – neither from the end of the hard corner element nor from the attached plating of stiffener element, if any. In calculating the total forces for checking the hull girder ultimate strength, the area of the stiffened plate element is to be taken between the hard corner element and the stiffener element or between the hard corner elements, as applicable.

Table 7.2.3 : Modes of failure of stiffened plate element and stiffener element

Element	Mode of failure	Curve σ-ε defined in
Lengthened stiffened plate element or stiffener element	Elasto-plastic collapse	7.2.3.2
Shortened stiffener element	Beam column buckling Torsional buckling Web local buckling of flanged profiles Web local buckling of flat bars	7.2.3.3, 7.2.3.4, 7.2.3.5, 7.2.3.6
Shortened stiffened plate element	Plate buckling	7.2.3.7

7.2.3.2 Elasto-plastic collapse of structural elements (Hard corner element)

7.2.3.2.1 The equation describing the load-end shortening curve σ-ε for the elasto-plastic collapse of structural elements composing the hull girder transverse section is to be obtained from the following formula.

$$\sigma = \Phi R_{eHA}$$

where:

R_{eHA}: Equivalent minimum yield stress [N/mm^2], of the considered element, obtained by the following formula:

$$R_{eHA} = \frac{R_{eH_P} A_{p-net} + R_{eH_S} A_{s-net}}{A_{p-net} + A_{s-net}}$$

Φ: Edge function, equal to:

Φ = -1 for ε < -1
Φ = ε for -1 ≤ ε ≤ 1
Φ = 1 for ε > 1

ε: Relative strain, equal to:

$$\varepsilon = \frac{\varepsilon_E}{\varepsilon_Y}$$

ε_E : Element strain.

ε_Y : Strain at yield stress in the element, equal to:

$$\varepsilon_y = \frac{R_{eHA}}{E}$$

7.2.3.3 Beam column buckling

7.2.3.3.1 The positive strain portion of the average stress – average strain curve σ_{CR1} - ε based on beam column buckling of plate-stiffener combinations is described according to the following:

$$\sigma_{CR1} = \phi \sigma_{C1} \frac{A_{s-net} + A_{pE-net}}{A_{s-net} + A_{p-net}}$$

where:

Φ: Edge function, as defined in 7.2.3.2.

σ_{C1}: Critical stress [N/mm^2], equal to:

$$\sigma_{C1} = \frac{\sigma_{E1}}{\varepsilon} \text{ for } \sigma_{E1} \le \frac{R_{eHB}}{2}\varepsilon$$

$$\sigma_{C1} = R_{eHB}\left(1 - \frac{R_{eHB}\varepsilon}{4\sigma_{E1}}\right) \text{for } \sigma_{E1} > \frac{R_{eHB}}{2}\varepsilon$$

R_{eHB}: Equivalent minimum yield stress [N/mm^2], of the considered element, obtained by the following formula:

$$R_{eHB} = \frac{R_{eH_p} A_{pEI-net} \ell_{pE} + R_{eH_S} A_{s-net} \ell_{sE}}{A_{pEI-net} \ell_{pE} + A_{s-net} \ell_{sE}}$$

$A_{pEI-net}$: Effective area [cm^2], equal to:

$$A_{pEI-net} = 10 b_{E1} t_{net}$$

ℓ_{pE}: Distance [mm], measured from the neutral axis of the stiffener with attached plate of width b_{E1} to the bottom of the attached plate

ℓ_{sE}: Distance [mm], measured from the neutral axis of the stiffener with attached plate of width b_{E1} to the top of the stiffener

ε: Relative strain, as defined in 7.2.3.2

σ_{E1}: Euler column buckling stress [N/mm^2], equal to:

$$\sigma_{E1} = \pi^2 E \frac{I_{E-net}}{A_{E-net}\ell^2} 10^{-4}$$

I_{E-net}: Net moment of inertia of stiffeners [cm^4], with attached plate of width b_{E1}

A_{E-net}: Net area [cm^2], of stiffeners with attached plating of width b_E

b_{E1}: Effective width corrected for relative strain, in [m], of the attached plating, equal to:

$$b_{E1} = \frac{s}{\beta_E} \text{ for } \beta_E > 1.0$$

$$b_{E1} = s \quad \text{for } \beta_E \le 1.0$$

β_E: $$\beta_E = 10^3 \frac{s}{t_{net}} \sqrt{\frac{\varepsilon R_{eH_P}}{E}}$$

$A_{pE\text{-}net}$: Net area [cm^2], of attached plating of width b_E, equal to:

$$A_{pE-net} = 10 b_E t_{net}$$

b_E: Effective width [m], of the attached plating, equal to:

$$b_E = \left(\frac{2.25}{\beta_E} - \frac{1.25}{\beta_E^2} \right) s \text{ for } \beta_E > 1.25$$

$b_E = s$ for $\beta_E \leq 1.25$

7.2.3.4 Torsional buckling

7.2.3.4.1 The load-end shortening curve $\sigma_{CR2\text{-}\varepsilon}$ for the flexural-torsional buckling of stiffeners composing the hull girder transverse section is to be obtained according to the following formula:

$$\sigma_{CR2} = \phi \frac{A_{s-net}\sigma_{C2} + A_{p-net}\sigma_{CP}}{A_{s-net} + A_{p-net}}$$

where:

Φ: Edge function, as defined in 7.2.3.2

σ_{C2}: Critical stress [N/mm^2], equal to:

$$\sigma_{C2} = \frac{\sigma_{E2}}{\varepsilon} \text{ for } \sigma_{E2} \leq \frac{R_{eH_S}}{2}\varepsilon$$

$$\sigma_{C2} = R_{eH_S}\left(1 - \frac{R_{eH_S}\varepsilon}{4\sigma_{E2}}\right) \text{ for } \sigma_{E2} > \frac{R_{eH_S}}{2}\varepsilon$$

ε: Relative strain, as defined in 7.2.3.2

σ_{E2}: Euler column buckling stress [N/mm^2], taken as σ_{ET} defined in 6.4.4.4

σ_{CP}: Buckling stress of the attached plating [N/mm^2], equal to:

$$\sigma_{CP} = \left(\frac{2.25}{\beta_E} - \frac{1.25}{\beta_E^2} \right) R_{eH_P} \text{ for } \beta_E > 1.25$$

$\sigma_{CP} = R_{eH_P}$ for $\beta_E \leq 1.25$

β_E: Coefficient, as defined in 7.2.3.3

7.2.3.5 Web local buckling of stiffeners made of flanged profiles

7.2.3.5.1 The load-end shortening curve $\sigma_{CR3\text{-}\varepsilon}$ for the web local buckling of flanged stiffeners composing the hull girder transverse section is to be obtained from the following formula:

$$\sigma_{CR3} = \phi \frac{10^3 b_E t_{net} R_{eH_P} + (h_{we} t_{w-net} + b_f t_{f-net}) R_{eH_s}}{10^3 s t_{net} + h_w t_{w-net} + b_f t_{f-net}}$$

where:

Φ: Edge function, as defined in 7.2.3.2

b_E: Effective width [m], of the attached plating, as defined in 7.2.3.3

h_{we}: Effective height [mm], of the web, equal to:

$$h_{we} = \left(\frac{2.25}{\beta_w} - \frac{1.25}{\beta_w^2} \right) h_w \text{ for } \beta_w \geq 1.25$$

$h_{we} = h_w$ for $\beta_w < 1.25$

β_w: $$\beta_w = \frac{h_w}{t_{w-net}} \sqrt{\frac{\varepsilon R_{eH_s}}{E}}$$

ε: Relative strain, as defined in 7.2.3.2

7.2.3.6 Web local buckling of stiffeners made of flat bars

7.2.3.6.1 The load-end shortening curve $\sigma_{CR4\text{-}\varepsilon}$ for the web local buckling of flat bar stiffeners composing the hull girder transverse section is to be obtained from the following formula:

$$\sigma_{CR4} = \phi \frac{A_{p-net}\sigma_{CP} + A_{s-net}\sigma_{C4}}{A_{p-net} + A_{s-net}}$$

where:

Φ: Edge function, as defined in 7.2.3.2.

σ_{CP}: Buckling stress of the attached plating, in [N/mm^2], as defined in 7.2.3.4.

σ_{C4}: Critical stress [N/mm^2], equal to:

$$\sigma_{C4} = \frac{\sigma_{E4}}{\varepsilon} \text{ for } \sigma_{E4} \le \frac{R_{eH_S}}{2}\varepsilon$$

$$\sigma_{C4} = R_{eH_S}\left(1 - \frac{R_{eH_S}\varepsilon}{4\sigma_{E4}}\right) \text{ for } \sigma_{E4} > \frac{R_{eH_S}}{2}\varepsilon$$

σ_{E4}: Local Euler buckling stress [N/mm^2], equal to:

$$\sigma_{E4} = 160000\left(\frac{t_{w-net}}{h_w}\right)^2$$

ε: Relative strain, as defined in 7.2.3.2.

7.2.3.7 Plate buckling

7.2.3.7.1 The load-end shortening curve σ_{CR5}-ε for the buckling of transversely stiffened panels composing the hull girder transverse section is to be obtained from the following formula:

$$\sigma_{CR5} = \min\left\{\Phi R_{eH_P}\left[\frac{s}{\ell}\left(\frac{2.25}{\beta_E} - \frac{1.25}{\beta_E^2}\right)^{R_{eH_P}\Phi} + 0.1\left(1 - \frac{s}{\ell}\right)\left(1 + \frac{1}{\beta_E^2}\right)^2\right]\right\}$$

where:

Φ: Edge function, as defined in 7.2.3.2.

β_E: Coefficient as defined in 7.2.3.3.

s: Plate breadth [m], taken as the spacing between the stiffeners.

ℓ: Longer side of the plate [m].

7.3 Alternative Methods

7.3.1 General

7.3.1.1 Application of alternative methods is to be agreed by IRS prior to commencement. Documentation of the analysis methodology and detailed comparison of its results are to be submitted for review and acceptance. The use of such methods may require the partial safety factors to be recalibrated.

7.3.1.2 The bending moment-curvature relationship, M-χ, may be established by alternative methods. Such models are to consider all the relevant effects important to the non-linear response with due considerations of:

a) Non-linear geometrical behaviour.

b) Inelastic material behaviour.

c) Geometrical imperfections and residual stresses (geometrical out-of-flatness of plate and stiffeners).

d) Simultaneously acting loads:

- Bi-axial compression.
- Bi-axial tension.
- Shear and lateral pressure.

e) Boundary conditions.

f) Interactions between buckling modes.

g) Interactions between structural elements such as plates, stiffeners, girders, etc.

h) Post-buckling capacity.

i) Overstressed elements on the compression side of hull girder cross section possibly leading to local permanent sets/buckle damages in plating, stiffeners etc. (double bottom effects or similar).

7.3.2 Non-linear finite element analysis

7.3.2.1 Advanced non-linear finite element analyses models may be used for the assessment of the hull girder ultimate capacity. Such models are to consider the relevant effects important to the non-linear responses with due consideration of the items listed in7.3.1.2.

7.3.2.2 Particular attention is to be given to modelling the shape and size of geometrical imperfections. It is to be ensured that the shape and size of geometrical imperfections trigger the most critical failure modes.

Section 8

Load Cases for Finite Element Analysis

8.1 General

8.1.1 The requirements in this section are functional requirements on load cases to be considered on finite element analysis for the structural strength assessment (yielding and buckling).

8.1.2 The procedure for yielding and buckling assessment is to be agreed upon by IRS.

8.1.3 All in-plane stress components (i.e. bi-axial and shear stresses) induced by hull girder loads and local loads as specified in this section are to be considered.

8.1.4 All aspects and principles not mentioned explicitly in this section are to be considered specially.

8.2 Definitions

8.2.1 Global Analysis

8.2.1.1 A Global Analysis is a finite element analysis, using a full ship model, for assessing the structural strength of global hull girder structure, cross deck structures and hatch corner radii.

8.2.2 Cargo Hold Analysis

8.2.2.1 A Cargo Hold Analysis is a finite element analysis for assessing the structural strength of the cargo hold primary structural members (PSM) in the midship region.

8.2.3 Primary Structural Members (PSM)

8.2.3.1 Primary structural members are members of girder or stringer type which provide the overall structural integrity of the hull envelope and cargo hold boundaries, such as:

(i) double bottom structure (bottom plate, inner bottom plate, girders, floors)

(ii) double side structure (shell plating, inner hull, stringers and web frames)

(iii) bulkhead structure

(iv) deck and cross deck structure

8.3 Analysis

8.3.1 Global Analysis

8.3.1.1 A Global Analysis is to be carried out for ships of length 290 [m] or above. Hull girder loads (including torsional effects) are to be considered. The following methods may be used for Global Analysis:

Method 1: Analysis where hull girder loads only (vertical bending moment, horizontal bending moment and torsional moment) are directly applied to the full ship finite element model

Method 2: Analysis where direct loads transferred from direct load analysis are applied to the full ship finite element model

8.3.2 Cargo Hold Analysis

8.3.2.1 Cargo Hold Analysis is to be carried out for ships of length 150 [m] or above. Local loads such as sea pressure and container loads as well as hull girder loads are to be considered.

8.4 Load principles

8.4.1 Wave environment

8.4.1.1 The ship is to be considered sailing in the North Atlantic wave environment for yielding and buckling assessments. The corresponding vertical wave bending moments are to be in line with Sec 6 and the other hull girder loads are to be taken in consultation/post-agreement with IRS. The corresponding local loads are also to be taken in consultation/post-agreement with IRS.

8.4.2 Ship operating conditions

8.4.2.1 Seagoing conditions are to be considered. Harbour conditions and special conditions such as flooded conditions, tank testing conditions may be considered in consultation with IRS.

8.5 Load components

8.5.1 Global Analysis

8.5.1.1 The load components to be considered in Global Analysis are shown in Table 8.5.1.1a).

Table 8.5.1.1a) : Load components to be considered in Global Analysis

	Static Load	Dynamic Load
Method 1	• Still water vertical bending moment • Still water torsional moment	• Wave induced vertical bending moment • Wave induced horizontal bending moment • Wave induced torsional moment
Method 2	• Static sea pressure • Static container loads • Static loads for ballast and fuel oil • Self-weight of hull structure	• Wave induced sea pressure • Dynamic loads for hull structure, containers, ballst and fuel oil

Table 8.5.1.1b) : Load Components to be considered in Cargo Hold Analysis

	Static Load	Dynamic Load
Hull girder loads	• Still water vertical bending moment	• Wave induced vertical bending moment
Local loads	• Static sea pressure • Static container loads • Static loads for ballast and fuel oil(1) • Self-weight of hull structure	• Wave induced sea pressure • Dynamic loads for hull structure, containers, ballst and fuel oil

(1) For the minimum set of loading conditions specified in Table 3, all ballast and fuel oil tanks in way of the cargo hold model are to be empty. If additional loading conditions other than those given in Table 3 are considered, ballast and fuel oil loads may be taken into consideration at the discretion of IRS.

8.6 Loading conditions

8.6.1 Global Analysis

8.6.1.1 Loading conditions to be considered for the Global Analysis are to be in accordance with the Loading Manual (Refer 4.4.6).

8.6.2 Cargo Hold Analysis

8.6.2.1 The minimum set of loading conditions is specified in Table 8.6.2.1. In addition, loading conditions from the Loading Manual are to be considered in the Cargo Hold Analysis where deemed necessary.

Table 8.6.2.1 : Minimum set of loading conditions for Cargo Hold Analysis

Loading condition	Draught	Container weight	Ballast and fuel oil tanks	Still water hull girder moment
Full load condition	Scantling Draught	Heavy cargo weight (1) (40ft containers)	Empty	Permissible hogging
Full load condition	Scantling Draught	Light cargo weight (2) (40ft containers)	Empty	Permissible hogging
Full load condition	Reduced Draught(3)	Heavy cargo weight (1) (20ft containers)	Empty	Permissible sagging (minimum hogging)
One bay empty condition(4)	Scantling draught	Heavy cargo weight (1) (40ft containers)	Empty	Permissible hogging

(1) Heavy cargo weight of a container unit is to be calculated as the permissible stacking weight divided by the maximum number of tiers planned.

(2) Light cargo weight corresponds to the expected cargo weight when light cargo is loaded in the considered holds.

- Light cargo weight of a container unit in hold is not to be taken more than 55% of its related heavy cargo weight (see (1) above).

- Light cargo weight of a container unit on deck is not to be taken more than 90% of its related heavy cargo weight (see (1) above) or 17 metric tons, whichever is the lesser.

(3) Reduced draught corresponds to the expected draught amidships when heavy cargo is loaded in the considered holds while lighter cargo is loaded in other holds. Reduced draught is not to be taken more than 90% of scantling draught.

(4) For one bay empty condition, if the cargo hold consists of two or more bays, then each bay is to be considered entirely empty in hold and on deck (other bays full) in turn as separate load cases.

8.7 Wave conditions

8.7.1 Global Analysis

8.7.1.1 Wave conditions presumed to lead to the most severe load combinations due to vertical bending moment, horizontal bending moment and torsional moment are to be considered.

8.7.2 Cargo Hold Analysis

8.7.2.1 The following wave conditions are to be considered:

(i) Head sea condition yielding the maximum hogging and sagging vertical bending moments.

(ii) Beam sea condition yielding the maximum roll motion. This condition may be disregarded for some loading conditions defined in Table 8.6.2.1 where deemed not necessary.

Section 9

Container Stowage and Securing Arrangement

9.1 General

9.1.1 Details of stowage and securing arrangement of containers is to be submitted for approval.

9.2 Stowage arrangement

9.2.1 Containers are generally to be stowed in fore and aft direction in holds, tween decks and on top of exposed decks and hatch covers. The arrangement for securing the containers in position are to be designed on the basis of the most severe distribution of loads which may arise in the container stack.

9.2.2 In general containers are to be assumed to be fully loaded. Where, however, specified loading patterns are proposed, the securing arrangements may be considered on the basis of these loading patterns which are to be clearly indicated on the Cargo Securing Manual to be kept on board.

9.2.3 Where containers certified to higher strength standards are to be incorporated in the stowage arrangement, the container stowage plan is to indicate clearly the locations where these containers are stowed.

9.2.4 Containers stowed on the deck are not to extend beyond the ship's side. Adequate supports are to be provided when they overhang hatch coamings or other deck structures.

9.2.5 Where containers are stowed on the hatch covers, the cover and the hatch coamings are to be additionally strengthened. In addition, the hatch covers are to be effectively restrained against sliding by approved type stoppers or equivalent.

9.2.6 All equipment on deck and in holds essential for maintaining the safety of the ship and which are to be accessible at sea, e.g. fire fighting equipment, sounding pipes etc., are not to be made inaccessible by the containers or their securing arrangement.

9.3 Securing systems

9.3.1 Containers are to be secured generally by any of the following systems:

- Corner lashing devices;
- Rod, wire or chain lashings;
- Buttress, shores or equivalent structural restraint;
- Cell guides.

9.3.2 Local reinforcements are to be fitted as necessary in way of the fittings attached to the ship structure.

9.3.3 When cell guide system is provided, the guides are to be constructed to effectively transmit the loads to double bottom, side structure and bulkheads.

The cell guides are not to be connected to the projections of deck plating edges in way of the hatchways. Any flame cutting or welding is to be avoided on the deck plating in the hatchway corners.

9.4 Container Lashing Software

9.4.1 As actual loading conditions of containerships can vary significantly due to varying container carrying arrangements and weights from voyage to voyage, deviations from the sample loading conditions mentioned in the Container Stowage Manual/ plans can exist. In order to efficiently evaluate actual stowage and securing of cargo containers, lashing software can be used by the crew as a supplement to the Stowage and Securing Plans included in the approved Cargo Securing Manual.

9.4.2 Container Ships provided with IRS approved container lashing software will be assigned additional class notation **CLS** (Container Lashing Software).

9.4.3 As a minimum, the lashing software is to provide the following information:

a) The accelerations (also refer Part 3, Chapter 4) which can be expected in various positions on board the ship in adverse sea conditions and with a range of applicable metacentric height values;

b) The forces acting on typical cargo units when subjected to the accelerations referred to in 9.4.3 (a) and angles of roll and metacentric height values above which the forces acting on the cargo units exceed the permissible limit for the specified securing arrangements;

c) Calculations on the number and strength of portable securing devices required to counteract the forces referred to in 9.4.3 (b) as well as safety factors to be used for different types of portable cargo securing devices. Calculations may be carried out according to Annex 13 to the CSS Code or other methods accepted by IRS/ the Administration.

9.4.4 The lashing software and supporting hardware are to comply with the requirements in Part 4, Chapter 7, Section 6.3 of the Rules, as relevant and applicable.

End of Chapter

Chapter 6

Passenger Ships

Contents

Section

1	*General*
2	*Ship Arrangement*
3	*Hull Structure*
4	*Openings and their Closing Appliances*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to sea-going passenger ships intended to carry more than 12 passengers and are supplementary to those given for the main character of class.

1.1.2 The requirements of this chapter also apply to:

a) ferries intended for the carriage of passenger and/or vehicles on a regular scheduled service of short duration; and

b) roll-on roll-off ferries intended to carry passengers, vehicles and cargo in pellet form or in containers and loaded/unloaded by wheeled vehicles.

1.1.3 Ships complying with the requirements of the above will be eligible to be assigned one of the class notations

- **"PASSENGER SHIP";**
- **"FERRY";**
- **"RO-RO FERRY";**
- **“TRAIN FERRY”.**

1.1.4 The applicable requirements of SOLAS 1974 regulations and its amendments including SOLAS 1995 and IMO Resolution MSC 47(66), (as on June, 1996) are to be complied with in their entirety except when waived or relaxed by the National Administration.

1.2 Documentation

1.2.1 In addition to the information and plans required in Pt.3, Ch.1, Sec.3, following details are to be submitted:

Stern or bow ramps;

- Bow, stern and side doors;
- Movable decks, if any, including stowage arrangements for portable components;
- Details of deck loadings including wheeled vehicles and trains, as applicable; as well as the fixed securing points and the lashing arrangements for such loads;
- Arrangement of openings on watertight bulkheads.

1.2.2 Following documents are to be submitted for reference:

- 'Damage Control Plan' see Pt.3, Ch.4, Sec.4.
- Cargo Securing Manual for wheeled vehicles as per Chapter VI, Regulation 5 of SOLAS 74.

Section 2

Ship Arrangement

2.1 Structural configuration

2.1.1 The structural configuration detailed in Chapter II-1, Part B of SOLAS 74, and amendments as applicable to passenger ships, are to be complied with.

2.1.2 Where a reduced number of bulkheads are envisaged above the bulkhead deck, partial bulkheads and appropriate transverse ring structures are to be fitted to provide equivalent transverse strength.

2.2 Machinery, control and electrical arrangement

2.2.1 In addition to complying with the requirements of Pt.4, as applicable to passenger ships, the following are to be complied with.

2.2.2 Machinery & equipments detailed below are not to be located forward of the collision bulkhead:

- bilge pumps except those specially serving the spaces forward of the collision bulkhead;
- fire pumps;
- emergency source of electrical power;
- any fixed fire-extinguishing system required by Pt.6;
- other emergency installations essential for the safety of the ships.

These equipments are also not to be located within the shaft tunnel or the space containing the forward propeller gland.

2.2.3 In passenger ships constructed on or after 01 July 2010, supplementary lighting is to be provided in all cabins to clearly indicate the exit so that occupants will be able to find their way to the door. Such lighting, which may be connected to an emergency source of power or have a self-contained source of electrical power in each cabin, are automatically illuminated when power to the normal cabin lighting is lost and remain on for a minimum of 30 min.

2.2.4 Every passenger ship with ro-ro cargo spaces is to be provided with supplementary emergency lighting system of adequate illumination in all passenger public spaces and alleyways. These should operate for at least three hours under any condition of heel when all other sources of electrical power have failed. The source of power for the supplementary lighting is to consist of accumulator batteries located within the lighting units and are to be continuously charged. All crew space alleyway, recreational space and every working space that is normally occupied are to be provided with portable rechargeable battery operated lamps unless they are fitted with the supplementary emergency lighting mentioned earlier.

2.3 Fire safety measures

2.3.1 The fire safety measures comprising of structural fire protection, detection and extinction are to comply with the requirements of Pt.6.

2.4 Flooding detection system

2.4.1 A flooding detection system for watertight spaces below the bulkhead deck is to be provided on passenger ships carrying 36 or more persons, constructed on or after 01 July 2010. (Refer to MSC.1/Circ. 1291 “Guidelines for flooding detection systems on passenger ships”) and MSC.1/Circ. 1539, Rev. 1 “Unified interpretations of SOLAS Chapters II-1 and safe return to port requirements for flooding detection systems”).

2.5 System capabilities and operational information after a flooding casualty

2.5.1 Application

Clause 2.5 applies to passenger ships constructed on or after 1 July 2010 to which Pt.6, Ch.7, Sec.4 applies.

*2.5.2 Availability of essential systems in case of flooding damage**

A passenger ship is to be designed so that the systems specified in Pt.6, Ch.7, Cl.4.4 remain operational when the ship is subject to flooding of any single watertight compartment.

* Refer to “Interim explanatory notes for the assessment of passenger ship systems’ capabilities after a fire or flooding casualty” MSC.1/Circ.1369 as amended by MSC.1/Circ.1369/Add.1.

2.5.3 Operational information after a flooding casualty

For the purpose of providing operational information to the Master for safe return to port after a flooding casualty, passenger ships constructed on or after 1 January 2014 are to have.

1. Onboard stability computer; or
2. Shore-based support,

based on MSC.1/Circ.1400 “Guidelines on operational information for the masters of passenger ships for safe return to port by own power or under tow”.

Section 3

Hull Structure

3.1 Longitudinal strength

3.1.1 Longitudinal strength calculations are to be carried out in accordance with Pt.3, Ch.6.

For ships with multideck arrangements, the effectiveness of superstructures will be specially considered.

3.2 Bottom structure

3.2.1 All passenger ships are to be provided with a double bottom structure in accordance with SOLAS 74, Chapter II-1, Part B, Regulation 12.

3.2.2 The scantlings of the bottom structure are to comply with Pt.3, Ch.7.

3.2.3 In ships where transfer of loads between deck and bottom is carried out through a series of pillars, the double bottom structure is to be reinforced to avoid high shear and bending stresses due to the concentrated loads transmitted by the pillar system.

3.3 Side structure

3.3.1 The thickness of side shell plating and sides of enclosed superstructures which participate in the longitudinal strength may require to be increased to comply with the buckling requirements given in Pt.3, Ch.9.

3.3.2 Openings for doors and windows in the side shell and superstructure plating, are to be suitably stiffened. The thickness and material grade of plating in way of large openings will be specially considered.

3.3.3 Where ramp openings are provided on the ship-side, the structure in way is to be adequately strengthened.

3.4 Deck structure

3.4.1 Vehicle decks are to comply with the requirements for wheel loading as given in Pt.3, Ch.11, Sec.5. In addition, the vehicle deck structure is to be also capable of withstanding the concentrated loads coming from the fixed securing points.

3.5 Bulkhead structure

3.5.1 The arrangement and disposition of bulkheads is to be such as to ensure compliance with the subdivision and damage stability requirements of SOLAS 74, Chapter II-1, Part B.

3.5.2 In case of ships fitted with long forward superstructure, the weathertight extension of the forepeak or collision bulkhead is to be in accordance with the requirements of SOLAS 74, Chapter II-1, Part B, Regulation 10 (last amendment 1995).

3.6 Superstructures

3.6.1 The scantlings and arrangements of superstructure of multideck passenger ships will be specially considered in light of the length of the superstructure, alignment of the side walls with the side shell, size of openings on the sides and continuity of internal decks.

Section 4

Openings and their Closing Appliances

4.1 Openings on ship sides

4.1.1 The number of openings on the side shell are to be kept to a minimum compatible with the design and working of the ship. The arrangement, scantlings and efficiency of the means for closing any opening in the shell plating are to comply with the requirements of Pt.3, Ch.12, Sec.11.

4.1.2 Details of bow doors and inner doors are to be as per Pt.3, Ch.12, Sec.12.

4.1.3 The inboard openings of ash and rubbish chutes, etc. where fitted, are to be provided with efficient covers. If the inboard opening is situated below the bulkhead deck, the cover is to be watertight and, in addition a screw down non-return valve is to be fitted in the chutes above the deepest subdivision load line. The valve is to be controlled from a position above the bulkhead deck and provided with open/shut indicators.

4.1.4 All side scuttles to spaces on the first tier above the bulkhead deck are to be fitted with efficient inside deadlights, so arranged that they can be easily and effectively closed and secured watertight.

4.1.5 Sidescuttles, gangway, cargo and fuelling ports and other means for closing openings in the shell plating above the bulkhead deck are to be of efficient design and construction and of sufficient strength having regard to the spaces in which they are fitted and their positions relative to the deepest subdivision draught. (Refer resolution A.793(19).)

4.1.6 Openings in the shell plating below the bulkhead deck of passenger ships are to be in accordance with SOLAS Chapter II-1, Part B-2, Regulation 17 and 17-1.

4.2 Openings on the deck

4.2.1 All openings on the exposed weather deck are to have coamings of ample height and strength and are to be provided with efficient means for expeditiously closing them weathertight. Freeing ports, open rails and scuppers are to be fitted as necessary for rapidly clearing the weather deck of water under all weather conditions.

4.2.2 In ro-ro passenger vessels arrangements, of all access openings, leading to spaces below the bulkhead deck are to be in accordance with SOLAS 74 Chapter II-1, Part B-2, Regulation 17 and 17-1.

4.2.2.1 Stern, bow and side doors of large dimensions, when manual devices would not be readily accessible, are to be normally secured by means of power systems. Alternative means of securing is also to be provided for emergency use in case of failure of the power systems.

4.3 Watertight integrity above bulkhead deck

4.3.1 For passenger ships subject to the provisions of SOLAS Reg.II-1/1.1.1.1 and constructed before 1 January 2024, the Administration may require that all reasonable and practicable measures are to be taken to limit the entry and spread of water above the bulkhead deck. Such measures may include partial bulkheads or webs. When partial watertight bulkheads and webs are fitted on the bulkhead deck, above or in the immediate vicinity of watertight bulkheads, they are to have watertight shell and bulkhead deck connections so as to restrict the flow of water along the deck when the ship is in a heeled damaged condition. Where the partial watertight bulkhead does not line up with the bulkhead below, the bulkhead deck between is to be made effectively watertight. Where openings, pipes, scuppers, electric cables, etc. are carried through the partial watertight bulkheads or decks within the immersed part of the bulkhead deck, arrangements are to be made to ensure the watertight integrity of the structure above the bulkhead deck. (Refer MSC/Circ.541, as may be amended).

4.3.2 For ships constructed on or after 01 January 2024, the internal watertight subdivision arrangements to limit the entry and spread of water above the bulkhead deck is to be in accordance with the design arrangements necessary for compliance with the stability requirements in SOLAS parts B-1,

and B-2 if applicable. Where pipes, scuppers, electric cables, etc. are carried through internal watertight boundaries that are immersed at any intermediate or final stage of flooding in damage cases that contribute to the attained subdivision index A, arrangements are to be made to ensure their watertight integrity.

4.3.3 For ships constructed on or after 01 January 2024, doors in internal watertight subdivision arrangements above the bulkhead deck, and also above the worst intermediate or final stage of flooding waterlines, are to be capable of preventing the passage of water when immersed in the required range of positive stability for any damage cases contributing to the attained subdivision index A. These doors may remain open provided they can be remotely closed from the navigation bridge. They shall always be ready to be immediately closed.

4.3.4 Air pipes terminating within a superstructure which are not fitted with watertight means of closure are to be considered as unprotected openings when applying SOLAS regulation II-1/7- 2.6.1.1.

4.3.5 Where the main vehicle deck is enclosed, all companionways and openings in the deck which lead to spaces below are generally to be protected by steel doors or hatch covers. Approved fire doors may be accepted in lieu of steel doors. The sills or coamings are to be not less than 230 [mm] above the main vehicle deck, with the exception of those leading to machinery spaces which are to have sills or coamings not less than 380 [mm]. Where such openings are to be kept closed while at sea and a permanent notice to this effect is put, the height of sills or coamings may be reduced provided sealing arrangements are adequately improved by sufficient number of clamping devices. Such items as portable plates in the main vehicle deck arranged for the removal of machinery parts, etc., may be arranged flush with the deck, provided they are secured by gaskets and closely spaced bolts at a pitch not exceeding five times the bolt diameters.

4.3.6 Provisions are to be made for efficient drainage of enclosed cargo spaces above the bulkhead deck. The space between the bow door/visor and inner door/ramp is to be adequately drained, either by a bilge suction or by scuppers (both port and starboard) of diameter not less than 50[mm]. Details of scuppers and discharges are to be as per Pt.3, Ch.12, Sec.13.

4.4 Watertight integrity below the bulkhead deck

4.4.1 Openings in watertight boundaries below the bulkhead deck

4.4.1.1 The number of openings in watertight boundaries are to be reduced to the minimum compatible with the design and proper working of the ship; satisfactory means are to be provided for closing these openings.

4.4.1.2 Where pipes, scuppers, electric cables, etc., are carried through watertight boundaries, arrangements are to be made to ensure the watertight integrity of the boundaries.

4.4.1.3 Valves not forming part of a piping system are not to be permitted in watertight boundaries.

4.4.1.4 Lead or other heat sensitive materials are not to be used in systems which penetrate watertight boundaries, where deterioration of such systems in the event of fire would impair the watertight integrity of the boundaries.

4.4.1.5 No doors, manholes or access openings are permitted in watertight transverse bulkheads dividing a cargo space from an adjoining cargo space, except as provided in paragraph 4.4.1.22 and in SOLAS Reg. II-1/14.

4.4.1.6 Subject to paragraph 4.4.1.24, not more than one door, apart from the doors to shaft tunnels, may be fitted in each watertight bulkhead within spaces containing the main and auxiliary propulsion machinery including boilers serving the needs of propulsion. Where two or more shafts are fitted, the tunnels are to be connected by an intercommunicating passage. There is to be only one door between the machinery space and the tunnel spaces where two shafts are fitted and only two doors where there are more than two shafts. All these doors are to be of the sliding type and are to be so located as to have their sills as high as practicable. The hand gear for operating these doors from above the bulkhead deck is to be situated outside the spaces containing the machinery.

4.4.1.7 Watertight doors, except as provided in paragraph 4.4.1.22 or SOLAS Reg. II-1/14, are

to be power-operated sliding doors complying with the requirements of paragraph 6.

4.4.1.8 The means of operation whether by power or by hand of any power-operated sliding watertight door is to be capable of closing the door with the ship listed to 15° either way. Consideration is to also be given to the forces which may act on either side of the door as may be experienced when water is flowing through the opening applying a static head equivalent to a water height of at least 1 m above the sill on the centreline of the door.

4.4.1.9 Watertight door controls, including hydraulic piping and electric cables, are to be kept as close as practicable to the bulkhead in which the doors are fitted, in order to minimize the likelihood of them being involved in any damage which the ship may sustain. The positioning of watertight doors and their controls are to be such that if the ship sustains damage within one fifth of the breadth of the ship, as defined in SOLAS II-1/Reg.2, such distance being measured at right angles to the centreline at the level of the deepest subdivision draught, the operation of the watertight doors clear of the damaged portion of the ship is not impaired.

4.4.1.10 Each power-operated sliding watertight door:

.1 is to have a vertical or horizontal motion;

.2 is to, subject to paragraph 4.4.1.24, be normally limited to a maximum clear opening width of 1.2 m. The Administration may permit larger doors only to the extent considered necessary for the effective operation of the ship provided that other safety measures, including the following, are taken into consideration:

.1 special consideration is to be given to the strength of the door and its closing appliances in order to prevent leakages; and

.2 the door is to be located inboard the damage zone *B*/5;

.3 is to be fitted with the necessary equipment to open and close the door using electric power, hydraulic power or any other form of power that is acceptable to the Administration;

.4 is to be provided with an individual hand-operated mechanism. It is to be possible to open and close the door by hand at the door itself from either side, and in addition, close the door from an accessible position above the bulkhead deck with an all-round crank motion or some other movement providing the same degree of safety acceptable to the Administration. Direction of rotation or other movement is to be clearly indicated at all operating positions. The time necessary for the complete closure of the door, when operating by hand gear, is not to exceed 90 s with the ship in the upright position. Visual indicators to show whether the door is open or closed are to be provided at the accessible position above the bulkhead deck;

.5 is to be provided with controls for opening and closing the door by power from both sides of the door and also for closing the door by power from the central operating console(s) required by paragraph 4.4.1.18;

.6 is to be provided with an audible alarm, distinct from any other alarm in the area, which will sound whenever the door is closed remotely by power and which is to sound for at least 5 s but no more than 10 s before the door begins to move and is to continue sounding until the door is completely closed. In the case of remote hand operation it is sufficient for the audible alarm to sound only when the door is moving. Additionally, in passenger areas and areas of high ambient noise the Administration may require the audible alarm to be supplemented by an intermittent visual signal at the door; and

.7 is to have an approximately uniform rate of closure under power. The closure time, from the time the door begins to move to the time it reaches the completely closed position, is to in no case be less than 20 s or more than 40 s with the ship in the upright position.

4.4.1.11 The electrical power required for power-operated sliding watertight doors is to be supplied from the emergency switchboard either directly or by a dedicated distribution board situated above the bulkhead deck. The associated control, indication and alarm circuits are to be supplied from the emergency switchboard either directly or by a dedicated distribution board situated above the bulkhead deck and be capable of being automatically supplied by the transitional source of emergency electrical power required by SOLAS Reg.II-1/42.3.1.3 in the event of failure of either the main or emergency source of electrical power.

4.4.1.12 Power-operated sliding watertight doors are to have either:

.1 a centralized hydraulic system with two independent power sources each consisting of a motor and pump capable of simultaneously closing all doors. In addition, there is to be for the whole installation hydraulic accumulators of sufficient capacity to operate all the doors at least three times, i.e. closed-open-closed, against an adverse list of 15º. This operating cycle is to be capable of being carried out when the accumulator is at the pump cut-in pressure. The fluid used is to be chosen considering the temperatures liable to be encountered by the installation during its service. The power operating system is to be designed to minimize the possibility of having a single failure in the hydraulic piping adversely affect the operation of more than one door. The hydraulic system is to be provided with a low-level alarm for hydraulic fluid reservoirs serving the power-operated system and a low gas pressure alarm or other effective means of monitoring loss of stored energy in hydraulic accumulators. These alarms are to be audible and visual and are to be situated on the central operating console(s) required by paragraph 4.4.1.18; or

.2 an independent hydraulic system for each door with each power source consisting of a motor and pump capable of opening and closing the door. In addition, there is to be a hydraulic accumulator of sufficient capacity to operate the door at least three times, i.e. closed-open-closed, against an adverse list of 15º. This operating cycle is to be capable of being carried out when the accumulator is at the pump cut-in pressure. The fluid used is to be chosen considering the temperatures liable to be encountered by the installation during its service. A low gas pressure group alarm or other effective means of monitoring loss of stored energy in hydraulic accumulators are to be provided at the central operating console(s) required by paragraph 4.4.1.18. Loss of stored energy indication at each local operating position is also to be provided; or

.3 an independent electrical system and motor for each door with each power source consisting of a motor capable of opening and closing the door. The power source is to be capable of being automatically supplied by the transitional source of emergency electrical power as required by SOLAS Reg.II-1/42.4.2 - in the event of failure of either the main or emergency source of electrical power and with sufficient capacity to operate the door at least three times, i.e. closed-open-closed, against an adverse list of 15º.

For the systems specified in paragraphs 4.4.1.12.1, 4.4.1.12.2 and 4.4.1.12.3, provision should be made as follows: Power systems for power-operated sliding watertight doors are to be separate from any other power system. A single failure in the electric or hydraulic power-operated systems excluding the hydraulic actuator is not to prevent the hand operation of any door.

4.4.1.13 Control handles are to be provided at each side of the bulkhead at a minimum height of 1.6 m above the floor and are to be so arranged as to enable persons passing through the doorway to hold both handles in the open position without being able to set the power closing mechanism in operation accidentally. The direction of movement of the handles in opening and closing the door is to be in the direction of door movement and is to be clearly indicated.

4.4.1.14 As far as practicable, electrical equipment and components for watertight doors are to be situated above the bulkhead deck and outside hazardous areas and spaces.

4.4.1.15 The enclosures of electrical components necessarily situated below the bulkhead deck are to provide suitable protection against the ingress of water.*

* Refer to the following publication IEC 60529:2003:
.1 electrical motors, associated circuits and control components; protected to IPX 7 standard;
.2 door position indicators and associated circuit components; protected to IPX 8 standard; and
.3 door movement warning signals; protected to IPX 6 standard.

Other arrangements for the enclosures of electrical components may be fitted provided the Administration is satisfied that an equivalent protection is achieved. The water pressure IPX 8 is to be based on the pressure that may occur at the location of the component during flooding for a period of 36 h.

4.4.1.16 Electric power, control, indication and alarm circuits are to be protected against fault in such a way that a failure in one door circuit

will not cause a failure in any other door circuit. Short circuits or other faults in the alarm or indicator circuits of a door are not to result in a loss of power operation of that door. Arrangements are to be such that leakage of water into the electrical equipment located below the bulkhead deck will not cause the door to open.

4.4.1.17 A single electrical failure in the power operating or control system of a power-operated sliding watertight door is not to result in a closed door opening. Availability of the power supply should be continuously monitored at a point in the electrical circuit as near as practicable to each of the motors required by paragraph 4.4.1.12. Loss of any such power supply should activate an audible and visual alarm at the central operating console(s) required by paragraph 4.4.1.18.

4.4.1.18 A central operating console for all power-operated sliding watertight doors are to be located in the safety centre in accordance with SOLAS Reg. II-2/23. If the safety centre is located in a separate space adjacent to the navigation bridge, a central operating console is to also be located on the navigation bridge. The central operating console(s) are to have a "master mode" switch with two modes of control: a "local control" mode, which is to allow any door to be locally opened and locally closed after use without automatic closure, and a "doors closed" mode, which is to automatically close any door that is open in not more than 60 s with the ship in an upright position. The "doors closed" mode is to permit doors to be opened locally and is to automatically re-close the doors upon release of the local control mechanism. The "master mode" switch is to normally be in the "local control" mode. The "doors closed" mode is to only be used in an emergency or for testing purposes.

4.4.1.19 For ships subject to the provisions of SOLAS Reg. II-1/1.1.1.1 and constructed before 1 January 2024, the central operating console at the navigation bridge is to be provided with a diagram showing the location of each door, with visual indicators to show whether each door is open or closed. A red light is to indicate a door is fully open and a green light is to indicate a door is fully closed. When the door is closed remotely the red light is to indicate the intermediate position by flashing. The indicating circuit is to be independent of the control circuit for each door.

4.4.1.20 For ships constructed on or after 1 January 2024, the central operating console(s) is to be provided with a diagram showing the location of each power-operated sliding watertight door, with visual indicators to show whether each door is open or closed. A red light is to indicate a door is fully open and a green light is to indicate a door is fully closed. When the door is closed remotely the red light is to indicate the intermediate position by flashing. The indicating circuit is to be independent of the control circuit for each door. Indication is also be provided to the onboard stability computer, if installed in accordance with SOLAS Reg. II-1/8-1.3.1.

4.4.1.21 It is not to be possible to remotely open any door from the central operating console.

4.4.1.22 If the Administration is satisfied that such doors are essential, watertight doors of satisfactory construction may be fitted in watertight bulkheads dividing cargo spaces on 'tween decks. Such doors may be hinged, rolling or sliding doors but are not to be remotely controlled. They are to be fitted at the highest level and as far from the shell plating as practicable, but in no case the outboard vertical edges is to be situated at a distance from the shell plating which is less than one fifth of the breadth of the ship, as defined in SOLAS II-1/Reg.2, such distance being measured at right angles to the centreline at the level of the deepest subdivision draught.

4.4.1.23 Should any such doors be accessible during the voyage, they are to be fitted with a device which prevents unauthorized opening. When it is proposed to fit such doors, the number and arrangements are to receive the special consideration of the Administration.

4.4.1.24 Portable plates on bulkheads are not to be permitted except in machinery spaces. The Administration may permit not more than one power-operated sliding watertight door larger than those specified in paragraph 4.4.1.10.2 to be substituted for these portable plates in each watertight bulkhead, provided these doors are intended to remain closed during navigation except in case of urgent necessity at the discretion of the master. These doors need not meet the requirements of paragraph 4.4.1.10.4 regarding complete closure by hand-operated gear in 90 s.

4.4.1.25 Where trunkways or tunnels for access from crew accommodation to the machinery spaces, for piping, or for any other purpose are carried through watertight bulkheads, they are to be watertight and in accordance with the requirements of SOLAS Reg.II-1/16-1. The access to at least one end of each such tunnel or trunkway, if used as a passage at sea, is to be through a trunk extending watertight to a height sufficient to permit access above the bulkhead deck. The access to the other end of the trunkway or tunnel may be through a watertight door. Such trunkways or tunnels are not to extend through the first subdivision bulkhead abaft the collision bulkhead.

4.4.1.26 Where it is proposed to fit tunnels piercing watertight bulkheads, these are to receive the special consideration of the Administration.

4.4.1.27 Where trunkways in connection with refrigerated cargo and ventilation or forced draught trunks are carried through more than one watertight bulkhead, the means of closure at such openings are to be operated by power and be capable of being closed from a central position situated above the bulkhead deck.

End of Chapter

Chapter 7

Tugs

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2	*Hull Arrangement and Strength*
3	*Towing Arrangement*
4	*Stability*
5	*Tests and Trials*
6	*Additional Requirements for Escort Tugs*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to tugs, and are supplementary to those given for the assignment of main characters of class.

1.1.2 Vessels built in compliance with the above requirements will be eligible to be assigned the class notation " **TUG**". Tugs which are intended for escort service (i.e., accompanying another vessel in transit and providing assistance in its manoeuvring), and which comply with the additional requirements in Section 6 will be eligible for assignment of class notation '**ESCORT**'.

1.1.3 The maximum value of bollard pull [tonnes] measured during bollard pull test will be entered in the Register of Ships as " **BPmax** ..... ". In addition, if requested, the continuous bollard pull [tonnes] measured as per the procedure given in Sec.5.2 will be entered in the Register of Ships as "**BPcont** ..... ". In case of escort tugs, the class notation is to be supplemented with values of design maximum steering force (Fs, in kN), design maximum braking force (Fb, in kN), and design maximum escort speed (V, in knots).

1.2 Documentation

1.2.1 The following additional plans and documents are to be submitted for approval, as applicable. The list of additional documents to be submitted for assignment of '**ESCORT**' notation is provided in Section 6.

Towing arrangement

- Maximum and continuous bollard pull and the breaking strength of the tow rope.

Towing hook

- It's attachment and corresponding strengthening of hull structure, slip arrangements.

Towing winch

- Primemover, shafting, coupling details, maximum braking power.

Towing winch seats

- Underdeck strengthening.

Bollard Pull test program

- Items specified in Sec.5.

1.2.2 Additional Certificates of Approval are to be submitted for :-

a) Towing hook with attachments
b) Towline
c) Towing winch.

1.3 Materials

1.3.1 Towing hook including its attachment is to be made of forged steel in accordance with Pt.2, Ch.5 or of special quality carbon and carbon-manganese steel castings in accordance with Pt.2, Ch.4.

Section 2

Hull Arrangement and Strength

2.1 General

2.1.1 The draught T, used for determination of scantlings is not to be taken less than 0.90 D.

2.1.2 The structure in the forebody and afterbody is to be adequately reinforced against forces arising from pushing operations.

2.2 Longitudinal strength

2.2.1 The requirements of Pt.3, Ch.5, Sec.6 regarding loading instruments, are not applicable to tugs.

2.3 Bottom structure

2.3.1 The requirements of Pt.3, Ch.7, Sec.7 regarding strengthening against slamming, are not applicable to tugs.

2.3.2 Structure in way of openings provided for fitment of propulsion units is to be reinforced to ensure the continuity of longitudinal and transverse strength.

2.3.3 Single bottom floors clear of the machinery space may be flanged in lieu of a face plate.

2.4 Side structure

2.4.1 In fore peak space, side stringers supporting vertical peak frames are to be fitted at a spacing not exceeding 2.0 [m].

2.4.2 For harbour tugs engaged in berthing operations, it is recommended to provide a stringer all around the vessel at a suitable height, to provide additional stiffness against contact.

2.5 Deck structure

2.5.1 Foundations of towing winch and towing hook are to be capable of withstanding the breaking strength of the towline without any permanent deformations. The design of structures under these foundations and under heavy duty bollards is to be based on additional loads imposed by the tow line at its breaking strength.

2.6 Machinery casings, emergency exits, scuttles, air pipes, ventilators & bulwark etc.

2.6.1 Exposed machinery casings are generally to be not less than 900 [mm] high above the upper surface of the deck. Proposals of reduced height to facilitate lowering of the towline, will receive individual consideration on the basis of safety against the ingress of water.

The scantlings of the exposed machinery casings are to be 20% more than those required for exposed deckhouses in the same location and at the casing stiffeners are to be connected to beams at both ends.

2.6.2 Emergency exit from the machinery space to the weather deck is to be capable of being used at extreme angles of heel and is to be located on or near the ship's centreline. The coaming height is to be not less than 600 [mm]. The hatch cover is to have hinges arranged athwartships, and is to be capable of being opened and closed weathertight from either side.

2.6.3 Side scuttles are generally not permitted below the main deck except under special consideration when the distance from the lower edge of side scuttles to the maximum waterline is at least 750 [mm], and the scuttles with hinged inside deadlights meeting the requirements of Type A (heavy) scuttles according to ISO Recommendation 1751, are provided. Scuttles on the sides of any superstructures are to be of the same standard.

Fixed lights of skylights on the deck are to have glass thickness appropriate to their location as required for side scuttles, and fitted with hinged deadlight on the weather side.

2.6.4 In the area aft of the tow hook, the air pipes and vent pipes are to be so arranged as to prevent damage from the towline and to provide maximum practicable angle of downflooding. Closing appliances on air pipes on upper deck are to be of automatic type.

2.6.5 The bulwarks are to be sloped inboard to avoid damage due to contact.

2.7 Sternframe, rudder & steering gear

2.7.1 In the case of tugs designed for maximum helm angle more than 35°, the scantlings of the rudder, rudder stock, stern frame and the steering gear will be specially considered.

2.8 Fenders

2.8.1 In addition to the special fendering provided for pushing operations, an efficient fender is to be fitted all around on the ship's side at deck level.

Section 3

Towing Arrangement

3.1 General

3.1.1 The towline is to be in accordance with Pt.3, Ch.15, Sec.3.2.

3.1.2 The position of the towing hook or towing winch is to be carefully selected so as to minimise the heeling moment as well as the risk of girting due to the pull exerted by the tow rope.

3.1.3 For tugs to be assigned with additional notation '**ESCORT**' , the requirements in this Section are to be read in conjunction with the additional requirements provided in Section 6.

3.2 Strength of towing gear and supporting structure

3.2.1 The design and scantlings of the towing hook or towing winch and their supporting structure are to be based on the design load equal to the breaking strength of the largest towline to be used.

3.2.2 The scantlings of all components of towing gear and its supporting structure are to be derived from direct calculations considering all possible directions of the tow rope from port to starboard in horizontal plane as well as in the plane inclined at 45° above the horizontal and the permissible stresses given below:

For towing hook and towing winch

$\sigma_e = 0.9\ \sigma_y$

For towing gear supporting structure

$\sigma_e = 0.85\ \sigma_y$

where,

$$\sigma_e = \sqrt{\sigma^2 + 3\tau^2}$$

σ = Normal stress

τ= shear stress

σ_y = yield stress of the material.

For tugs with conventional propellers, alternatively, for the purpose of calculations the range of tow rope angle in the horizontal plane may be considered to be limited to within 30° port to 30° starboard.

3.3 Quick release devices for tow hooks

3.3.1 Towing hook should be provided with an efficient slip arrangement to facilitate release of the towline regardless of the angle of heel and the direction of the towline. The releasing device is also to be operable from the bridge.

3.3.2 Releasing devices may be mechanical, hydraulic or pneumatic type. A releasing device is to be so designed that unintentional slipping is avoided.

3.3.3 Mechanical releasing devices are to be designed so that the required release force does not exceed 150 [N] at the towing hook and 250 [N] on the bridge when the hook is subjected to the design load.

The release rope is to be guided over rope sheaves. Slipping should be possible by downward drawing, using the whole body weight.

3.3.4 For a pneumatic releasing device, a complete mechanical slip device has to be additionally provided.

3.4 Safety devices for towing winches

3.4.1 Towing winch is to be provided with disconnecting coupling operable from the bridge.

3.4.2 The towing winch brakes are to have sufficient static holding capacity which is to be not less than 1.8 times the maximum continuous bollard pull of the vessel. The holding capability should be calculated for the outermost towline layer on the winch drum at which towing will be performed.

3.4.3 It should be possible to release the tension on the winch drum(s) in an emergency and in all operational modes. After an emergency release the winch brakes should revert to normal function without delay. It should also be possible to carry out the emergency release sequence (emergency release/application of brakes) even during a black-out.

3.4.4 Means should be provided to spool the towline effectively on the drum(s).

3.4.5 Towing winch location is to facilitate safe guiding of the tow rope in accordance with load direction.

3.4.6 Winch is to be safely operated from control stands on the bridge and on deck. In case winch drum is not clearly visible from any of the control stands, suitable rope guide is to be provided.

3.4.7 Operating direction of winch drum control levers is to be the same as the direction of motion of the rope. When released, control levers are to return to the stop position automatically. Arrangement to secure control lever in stop position is to be provided.

3.4.8 Mechanism to disengage winch drum from the drive is to be provided.

3.4.9 Arrangement for quick release of the winch drum brake from all control stands is to be provided. After quick release of brake, the winch driving motor is not to start automatically.

3.5 Testing

3.5.1 At manufacturer's works

3.5.1.1 Towing hooks and towing winches completed in all respect, including the quick release devices are to be subjected to a design load as per para 3.2.1 at an approved testing facility.

3.5.1.2 Components of towing hooks and towing winches, subjected to pressure, are to be pressure tested at 1.5 times admissible working pressure or opening pressure of safety valves.

3.5.2 Testing onboard

3.5.2.1 All towing hooks and towing winches along with the quick release devices are to be subjected to bollard pull test as per Sec.5. After bollard pull test quick release devices are to be subjected to an operational trial at no load.

3.5.3 Periodical testing of towing gears

3.5.3.1 The functional safety of towing gear and quick release devices are to be tested by the ship's staff, at least once a month.

3.5.3.2 Following initial testing onboard, towing gears and quick release devices are to be thoroughly examined every two years and subjected to design load at an approved testing facility.

3.6 Towing winch emergency release systems on ships performing towing operations within close quarters, ports or terminals

3.6.1 Scope

3.6.1.1 This subsection provides minimum safety standards for emergency release systems provided on towing winches that are used on towing ships within close quarters, ports or terminals including those ships normally not intended for towing operation in transverse direction.

3.6.1.2 This subsection is not intended to cover towing winches on board ships used solely for long distance ocean towage, anchor handling or similar offshore activities.

3.6.2 Purpose

3.6.2.1 The purpose of this sub-section is to provide requirements to prevent the capsize of a tug when in the act of towage as a result of the towline force acting transversely to the tug (in beam direction) as a consequence of an unexpected event (could be loss of propulsion/steering or otherwise), whereby the resulting couple generated by offset and opposing transverse forces (towline force is opposed by thrust or hull resistance force) causes the tug to heel and, ultimately, to capsize. This capsize may be referred to as “girting”, “girthing”, “girding” or “tripping”. See Fig. 3.6.2.1 which shows the forces acting during towage operations.

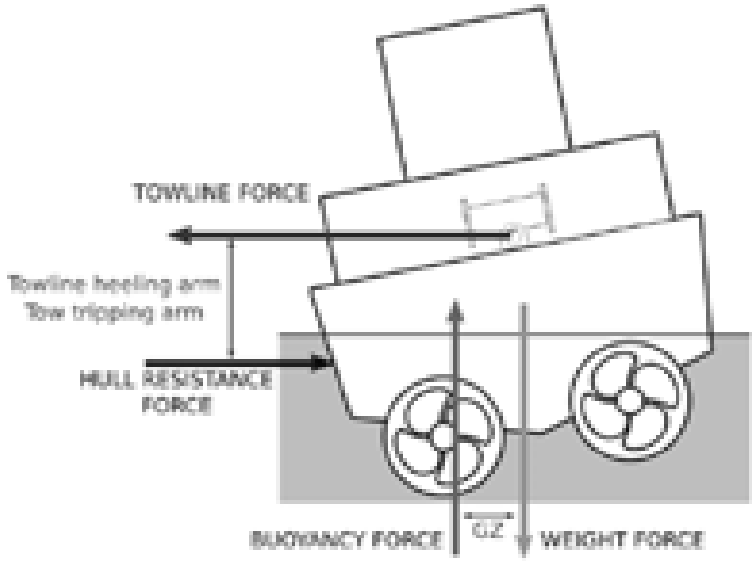

Fig. 3.6.2.1 Forces during towing

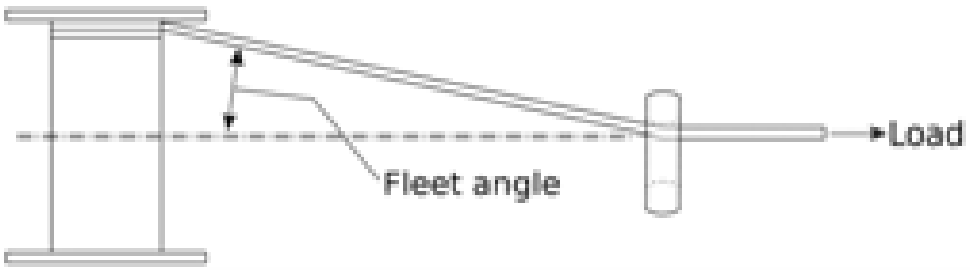

Fig. 3.6.3.3 Towline ‘Fleet Angle’

3.6.3 Definitions

3.6.3.1 The following definitions are applicable, for this sub-section:

.1 '**Emergency release system**' refers to the mechanism and associated control arrangements that are used to release the load on the towline in a controlled manner under both normal and black out conditions.

.2 '**Maximum design load**' is the maximum load that can be held by the winch as defined by the manufacturer (the manufacturer's rating).

.3 '**Fleet angle**' is the angle between the applied load (towline force) and the towline as it is wound onto the winch drum. (see Fig. 3.6.3.3).

3.6.4 General requirements

3.6.4.1 The in-board end of the towline is to be attached to the winch drum with a weak link or similar arrangement that is designed to release the towline at low load.

3.6.4.2 All towing winches are to be fitted with an emergency release system.

3.6.5 Emergency release systems requirements

3.6.5.1 Performance requirements

.1 The emergency release system is to operate across the full range of towline load, fleet angle and ship heel angle under all normal and reasonably foreseeable abnormal conditions (these may include, but are not limited to, the following: vessel electrical failure, variable towline load (for example due to heavy weather), etc.).

.2 The emergency release system is to be capable of operating with towline loads up to at least 100 per cent of the maximum design load.

.3 The emergency release system is to function as quickly as is reasonably practicable and within a maximum of three seconds after activation.

.4 The emergency release system is to allow the winch drum to rotate and the towline to pay out in a controlled manner such that, when the emergency release system is activated, there is sufficient resistance to rotation to avoid uncontrolled unwinding of the towline from the drum. Spinning (free, uncontrolled rotation) of the winch drum is to be avoided, as this could cause the towline to get stuck and disable the release function of the winch.

.5 Once the emergency release is activated, the towline load required to rotate the winch drum is to be no greater than:

(a) the lesser of five tonnes or five per cent of the maximum design load when two layers of towline are on the drum, or

(b) 15 per cent of the maximum design load where it is demonstrated that this resistance to rotation does not exceed 25 per cent of the force that will result in listing sufficient for the immersion of the lowest unprotected opening.

.6 Emergency release of the towline is to be possible in the event of a blackout. For this purpose, where additional sources of energy are required, such sources are to comply with 3.6.5.1.7

.7 The sources of energy required by .6 are to be sufficient to achieve the most onerous of the following conditions (as applicable):

(a) sufficient for at least three attempts to release the towline (i.e. three activations of the emergency release system). Where the system provides energy for more than one winch it is to be sufficient for three activations of the most demanding winch connected to it.

(b) Where the winch design is such that the drum release mechanism requires continuous application of power (e.g. where the brake is applied by spring tension and released using hydraulic or pneumatic power) sufficient power is to be provided to operate the emergency release system (e.g. hold the brake open and allow release of the towline) in the event of a blackout for a minimum of five minutes. This may be reduced to the time required for the full length of the towline to feed off the winch drum at the load specified in 3.6.5.1.5 if this is less than five minutes.

3.6.5.2 Operational requirements

.1 Emergency release operation is to be possible from the bridge and from the winch control station on deck. The winch control station on deck is to be in a safe location. A position in close proximity to the winch is not regarded as “safe location”, unless it is documented that the position is at least protected against towline break or winch failure.

.2 The emergency release control is to be located close to an emergency stop button for winch operation, if provided, and is to be clearly identifiable, clearly visible, easily accessible and positioned to allow safe operability.

.3 The emergency release function is to take priority over any emergency stop function. Activation of the winch emergency stop from any location is not to inhibit operation of the emergency release system from any location.

.4 Emergency release system control buttons are to require positive action to cancel, the positive action may be made at a different control position from the one where the emergency release was activated. It must always be possible to cancel the emergency release from the bridge regardless of the activation location and without manual intervention on the working deck.

.5 Controls for emergency use are to be protected against accidental use.

.6 Indications are to be provided on the bridge for all power supply and/or pressure levels related to the normal operation of the emergency release system. Alarms are to activate automatically if any level falls outside of the limits within which the emergency release system is fully operational.

.7 Wherever practicable, control of the emergency release system is to be provided by a hard-wired system, fully independent of programmable electronic systems.

.8 Computer based systems that operate or may affect the control of emergency release systems are to meet the requirements for Category III systems of Pt.4, Ch.7, Sec.6.

.9 Components critical for the safe operation of the emergency release system are to be identified by the manufacturer.

3.6.6 Testing requirements

3.6.6.1 General

.1 All testing is to be witnessed by a IRS surveyor.

.2 For each emergency release system or type thereof, the performance requirements of 3.6.5.1 are to be verified either at the manufacturer’s works or as part of the commissioning of the towing winch when it is installed on board. Where verification solely through testing is impracticable (e.g. due to health and safety), testing may be combined with inspection, analysis or demonstration in agreement with IRS.

.3 The performance capabilities and operating instructions of the emergency release system are to be documented by the manufacturer and made available on board the ship on which the winch has been installed.

.4 Instructions for surveys of the emergency release system are to be documented by the manufacturer, agreed with IRS and made available on board the ship on which the winch has been installed.

.5 Where necessary for conducting the annual and special surveys of the winch, adequately sized strong points are to be provided on deck.

3.6.6.2 Installation trials

.1 The full functionality of the emergency release system is to be tested as part of the shipboard commissioning trials to the satisfaction of the surveyor. Testing may be conducted either during a bollard pull test or by applying the towline load against a strong point on the deck of the tug that is certified to the appropriate load.

.2 Where the performance of the winch in accordance with 3.6.5.1 has previously been verified, the load applied for the installation trials is to be at least the lesser of 30% of the maximum design load or 80% of vessel bollard pull.

Section 4

Stability

4.1 General

4.1.1 In addition to the general requirements in Pt.3, Ch.1, Sec.1.4, the stability of tugs is to be assessed considering the effect of transverse heeling force caused when the tow rope is not in line with the tug's longitudinal centerline. Compliance with the following criteria is recommended, as a minimum :

(i) The residual area between a righting lever curve and a heeling lever curve developed from 70% of the maximum bollard pull force acting in 90° to the ship length direction is not to be less than 0.09 [m-rad];

The residual areas under the curves are to be calculated from the 1st intercept upto an angle of 40° beyond the first intercept or upto the downflooding angle, whichever is less.

OR

(ii) alternatively, the area under a righting lever curve should not be less than 1.4 times the area under a heeling lever curve developed from 70% of the maximum bollard pull force acting in 90° to ship-length direction.

In this case the areas under the curves are to be calculated from upright condition upto an angle of 40° or the downflooding angle, whichever is less.

The heeling lever curve is to be derived by using the following formula:

$$b_h = 0.7\ TH \cos \varphi / 9.81\Delta$$

b_h = heeling arm [m];
T = maximum bollard pull [kN];
H = vertical distance [m] between the towing hook and the centre of the propeller;
Δ = loading condition displacement [t];
φ = angle of heel.

4.1.2 As an alternative to the requirements in paragraph 4.1.1, the stability criteria for vessels engaged in towing operations, as included in Part B of the 2008 IS Code, will be considered as an acceptable equivalent standard for classification purposes.

4.1.3 In addition to the above, compliance may need to be demonstrated for any alternate/additional standards laid down by the Administration/local authority.

4.1.4 For tugs to be assigned with additional notation '**ESCORT**', compliance with the additional stability requirements included in Section 6 is to be demonstrated.

Section 5

Tests and Trials

5.1 Towing gear

5.1.1 In addition to the tests at the manufacturer's works, the towing gear including the towing hook, winch and their emergency release systems are to be tested after installation.

5.2 Bollard pull test procedure

5.2.1 The proposed test programme is to be submitted prior to the testing.

5.2.2 Test for continuous bollard pull is to be carried out with the main engines running at the maximum attainable engine RPM without exceeding the maximum RPM and torque recommended by the engine builder for continuous operation.

5.2.3 The test is to be carried out with the vessel's own propellers only. All auxiliary machinery which are normally driven from the main engine(s) or propeller shaft(s) while towing, are to be connected during the test.

5.2.4 It is recommended that the water depth at test location be not less than 20 [m] within a radius of 100 [m] from the vessel.

5.2.5 The length of the towline, measured between the vessel's stern and the shore bollard, is not to be less than 300 [m].

5.2.6 The test should be carried out at a wind speed not exceeding 5 [m/s] and a current not exceeding 1 [knot] in any direction.

5.2.7 An approved and calibrated load measuring device giving a continuous read-out is to be fitted between the eye of the towline and the bollard. Fitting of a load cell connected to instruments giving a continuous digital read-out and also a graphical record of the bollard pull against time is recommended.

5.2.8 During the test, efficient communication is to be maintained between the vessel and the shore personnel recording the bollard pull.

5.2.9 The vessel is to maintain a fixed course for at least 10 minutes during which the bollard pull is to be recorded.

5.2.10 The pull maintained uniformly for minimum of 10 minutes without any tendency to decline shall be certified as the vessel's continuous bollard pull, subject to a limit of 50% of the breaking strength of the towline supplied.

5.3 Additional trial requirements for assignment of 'ESCORT' notation

5.3.1 Additional trial requirements applicable to vessels assigned with '**ESCORT**' notation are provided in Section 6.

Section 6

Additional Requirements for Escort Tugs

6.1 General

6.1.1 The requirements in this Section are applicable to tugs which are intended for escort operations.

6.1.2 During escort operations, the tug follows the assisted ship providing control by steering and braking, as necessary, using a combination of propulsive and hydrodynamic forces developed by the tug. The tug is connected by means of a permanent towline to the stern of the escorted vessel.

6.1.3 Escort operations are typically carried out at speeds from 6 to 10 knots. These rules assume that no escort operations will be conducted at speeds greater than 10 knots. Vessels designed to conduct escort at speeds greater than 10 knots will be specially considered.

6.1.4 The '**ESCORT**' notation is to be supplemented by the following additional parameters, which are estimated based on Escort performance simulations and full scale trials:

- design maximum steering force (Fs)
- design maximum braking force (Fb)
- design maximum escort speed (V)

E.g., **+SUL, TUG, ESCORT (Fs=, Fb=, V=)**

In case of Escort Tugs operating within specific areas/limits, the class notation is to be further supplemented by a 'service restriction notation' or 'service area notation', as applicable

6.2 Definitions

6.2.1 *Winch brake holding load (WBL),* [kN]: is the maximum towline force the towing winch can withstand without slipping of the winch brake (engaged), considering the towline at the first inner layer. This load is a reference for the strength assessment and testing of towing winches and associated towing fittings (e.g. fairlead, staple, gob-eye) as well as their supporting structures.

6.2.2 *Direct Towing Mode:* The towing force is generated directly by the escort vessel's propulsion system.

6.2.3 *Indirect Towing Mode*: The towing force is a combination of hydrodynamic lift and drag forces acting on the escort vessel hull. The escort vessel's propulsion system is only used indirectly to maintain the orientation (drift angle) of the escort tug.

6.2.4 *Rated towline force*, [kN]: is the highest anticipated towline force, as obtained from the evaluation of the escort forces for a particular loading condition and escort speed, taking into account the applicable stability and strength criteria;

6.2.5 *Rated steering force*,[kN]: is the highest anticipated steering force, as obtained from the evaluation of the escort forces for a particular loading condition and escort speed, taking into account the applicable stability and strength criteria in this guidance note;

6.2.6 *Rated braking force,*[kN]: is the highest anticipated braking force F_b as obtained from the evaluation of the escort forces for a particular loading condition and escort speed, taking into account the applicable stability and strength criteria.

6.2.7 *The design maximum steering force*, [kN]: is the highest steering force value obtained over the applicable range of loading conditions and escort speeds

6.2.8 *The design maximum braking force*,[kN]: is the highest braking force value obtained over the applicable range of loading conditions and escort speeds.

6.3 Documentation

6.3.1 In addition to the plans required to be submitted in accordance with Section 1, Cl.1.2, the following additional details/ plans are to be submitted for **information**:

- **Design service restriction and limiting environmental conditions** for performing escort operations;

- **Escort Towing arrangement plan**, showing the location and general layout of the towing equipment, the range of anticipated lines of action of the towlines with the associated maximum steady towline forces and the corresponding points of application of the towline forces on the towing equipment;

- **Arrangement drawings** of towing winches, towing hooks and towline guiding fittings (fairleads, staples, gob-eyes, towing pins, stern roller, etc.);

- **Design information** of towing winches, including maximum rated line pull, winch brake holding force, rendering load and specification of emergency quick-release arrangements;

- **Design information** of towing hook, towline guiding fittings, and supporting structures.

- Escort Forces and speed

The steady towline force is to be determined from full-scale ship escort vessel trials. The results of such trials are to be submitted to IRS for information. The results are to include among others, the design maximum steering force and design maximum braking force, calculated using the highest among the steady towline forces estimated at anticipated towing speeds and conditions (defined by the position of escort tug relative to the escorted ship and the drift angle).

A preliminary estimate of these forces may be obtained through computer model simulations, and may be submitted to IRS for information, along with the preliminary stability booklet. The final values of escort forces and speed, as confirmed by full scale ship escort vessel trials are to be appended to the class notation, and are to be used for the calculation of final stability particulars.

- A **table with permissible values of heeling angle and steady towline force as function of loading condition and escort speed** (based on rated steering and braking forces complying with intact stability criteria), as applicable.

- **Escort Tug Operational information booklet** which includes

- Instructions to the master regarding the handling of the escort tug and the associated towing equipment; and regarding indicator/alarms alert the master when the heeling angle and/or steady towline force exceeds the permissible value(s) applicable to the relevant loading condition and escort speed.

- A table with permissible values of heeling angle and steady towline force as function of loading condition and escort speed. This table is also to be displayed in the wheelhouse next to

the control desk or another appropriate location.

In addition to the above, IRS may request the designer to submit design calculations or other supporting documents, for information.

6.4 Calculation of Escort forces

6.4.1 The steady towline force during escorting, which is the towline force associated with the equilibrium in indirect towing mode, for a given loading condition and escort speed V, is applied by the tug on the stern of the escorted ship.

6.4.2 The following angles are defined in relation to escort operations:

• The towline angle α, in deg, is the angle between the towline and the centreline of the escorted ship, and

• The drift angle β, in deg (*see Fig 6.1*), is the angle between the centreline of the tug and the centreline of the escorted ship (also referred to as yaw angle).

6.4.3 The steady towline force, can be resolved into two components, a transverse steering force Fs and a longitudinal braking force Fb.

6.4.4 The designer is to specify the rated steering force and braking force, evaluated for each loading condition and escort speed, taking into account the applicable stability and strength criteria.

6.4.5 The matrix of steady towline forces, steering forces and braking forces is to be initially specified by the designer and is to be verified on the basis of the results of full scale trials.

6.4.6 Full scale trials are to be performed in accordance with the requirements of 6.7.2. The procedure for conduct of full scale trials should be provided to IRS, for approval.

6.4.7 The final values of escort forces and speed, as confirmed by full scale ship escort vessel trials, are to be used for verification of compliance with stability criteria, as indicated in 6.5. The design maximum values of steering and braking forces are to be appended to the class notation.

6.4.8 The propulsion system should have sufficient speed and thrust so that the tug can manoeuvre aft of the escorted vessel.

6.4.9 The sizing of the prime mover and the propulsion chain to develop sufficient thrust for the vessel to perform her duties is beyond the scope of Classification.

6.4.10 In case of loss of propulsion, or breaking/ quick release of tow line, the remaining forces are to be so balanced that the resulting turning moment will turn the escort tug to a safer position with reduced heel.

6.5 Stability

6.5.1 Heeling lever for escort operations

6.5.1.1 For the evaluation of the stability particulars during escort operations the ship is considered to be in an equilibrium position determined by the combined action of the hydrodynamic forces acting on hull and appendages, the thrust force and the towline force as shown in Fig. 6.1.

6.5.1.2 For each equilibrium position the corresponding steering force, braking force, heel angle and heeling lever are to be obtained from the results of a computer simulation program (at the design stage) and confirmed by full scale trials.

6.5.1.3 For each relevant loading condition the evaluation of the equilibrium positions is to be performed over the applicable escort speed range, whereby the speed of the assisted ship through the water is between 6 to 10 knots.

6.5.1.4 For each relevant combination of loading condition and escort speed, the maximum heeling lever is to be used for stability calculations.

6.5.1.5 For the purpose of stability calculations the heeling lever is to be taken as constant.

6.5.1.6 The stability analysis is to include the effects of skegs and other appendages on both the reserve buoyancy and the lateral resistance of the escort vessel.

6.5.1.7 The stability analysis is to take into account the heel and trim caused by the thrust produced by the propulsion system, along with maximum line forces.

6.5.2 Stability criteria

6.5.2.1 In addition to the stability criteria given in Section 4, compliance with the following escort stability criteria is to be demonstrated for assignment of notation '**ESCORT**'.

6.5.2.2 For vessels engaged in escort operations the maximum heeling lever determined in accordance with 6.5.1.2 to 6.5.1.4 is to comply with the following criteria:

.1 Area A ≥ 1.25 × Area B;

.2 Area C ≥ 1.40 × Area D; and

.3 φe ≤ 15 degrees.

where:

Area A = Righting lever curve area measured from the heel angle φe to a heel angle of 20 degrees (see Fig 6.2);

Area B = Heeling lever curve area measured from the heeling angle φe to a heel angle of 20 degrees (see Fig 6.2);

Area C = Righting lever curve area measured from the zero heel (φ = 0) to φd (see Fig 6.2);

Area D = Heeling lever curve area measured from zero heel (φ = 0) to the heeling angle φd (see Fig 6.2);

φe = Equilibrium heel angle corresponding to the first intersection between heeling lever curve and the righting lever curve;

φd = the heel angle corresponding to the second intersection between heeling lever curve and the righting lever curve or the angle of down-flooding or 40 degrees, whichever is less.

6.5.2.3 The results of computer simulations carried out at the design stage, need to be confirmed by way of full scale trials prior to issuance of Class certificate.

6.6 Equipment for escort operations

6.6.1 General

6.6.1.1 Towing equipment used for escort services are normally to be arranged towards the centreline of the vessel, and as low as practicable so as to minimise heeling moment.

6.6.1.2 Materials used in towing equipment are to comply with the relevant requirements in Part 2 of the Rules.

Materials used for winch drums, drum shafts, winch brake components, winch supporting frames and towline guiding fittings, are to be class certified.

6.6.1.3 The strength of towing equipment and their foundations are to comply with the requirements provided in Section 3.

6.6.1.4 Where deemed necessary by IRS, buckling and/or fatigue analysis, performed in accordance with a standard or code of practice recognized by IRS, may be required to be submitted for information.

6.6.2 Design loads

6.6.2.1 The design loads to be considered for the strength assessment of the towing equipment used for escort services and the associated supporting structures are indicated in Table 6.6.2.

Table 6.6.2

Bollard Pull (kN)	Design load (kN)
T<500	3 x maximum rated towline force*
500<=T<=1000	maximum rated towline force* +1000
T>1000	2 x maximum rated towline force*

Note : *The maximum rated towline force, is the highest of the rated towline forces, from among the values calculated for different loading conditions.

6.6.3 Escort winches

6.6.3.1 The winch brake will normally act directly on the drum and is to be operable in case of failure of the primary power supply system (either manually or otherwise arranged).

6.6.3.2 Escort winches are to be equipped with an active pay-out and haul-in system.

This system is to automatically pay-out the towline in a controlled manner when the towline force exceeds a pre-set (adjustable) level equal to 110 per cent of the rated towline force; and actively haul-in the towline, as the

towline force is reduced, to prevent slack-line events and maintain a pre-set or adjustable towline force consistent with the rated towline force.

Pay-out and haul-in speeds and pull capability are to be chosen taking into account the anticipated escort services and the dynamic characteristics of the escort tug.

6.6.3.3 Escort operations performed by escort tugs with the operating area limited to non-exposed waters and calm water conditions, such as in ports and sheltered waters, may be based on the use of the brakes of the winch drum; instead of the active pay-out and haul-in system.

As a minimum, the winch brake holding load (WBL) is to be equal to or greater than two times the maximum steady towline force.

6.6.3.4 Escort winches are to be fitted with equipment to continuously measure the tension in the towline.

The measured data is to be continuously displayed in the wheelhouse next to the control desk or another appropriate location.

6.6.3.5 The escort towing system is to be designed so as to enable the proper spooling of the towline on the winch drum when hauling in.

Where a spooling device is fitted, this device should be designed for the same design load and stress criteria as the towline guiding fittings.

6.6.3.6 Winches are to be provided with an emergency release system as specified in Section 3, Cl. 3.6.

6.6.3.7 The dimensioning of the winch drum is to take into account the rope bending specifications provided by the towline manufacturer.

Due consideration is to be given to preventing the towline to slip over the flanges of the drum.

6.6.3.8 Where deemed necessary, buckling and/or fatigue analysis, performed in accordance with a standard or code of practice recognized by IRS, may be required to be submitted for information.

6.6.4 Towlines

6.6.4.1 The breaking strength of towlines used for escort services is not to be less than the design load, as specified in 6.6.2.

In addition, the breaking strength for towlines used for escort services on an escort winch is not to be less than the winch brake holding load (WBL) of the associated escort winch.

6.6.4.2 The towline is to be protected from being damaged by chafing and abrasion. To this end cargo rails, bulwarks, and other elements supporting the towline are to be adequately rounded.

6.6.4.3 It is recommended that the total length of the towline applied on a towing winch is such that under normal operation, at least half a layer remains on the drum. In no case less than three turns are to remain on the drum in normal operation.

6.6.5 Towline guiding fittings

6.6.5.1 Towline guiding fittings used for escort services, such as fairleads, staples and equivalent components which guide the towline, are to be able to sustain the force exerted by the towline loaded under a tension equal to the design load, as specified in 6.6.2, in the most unfavourable anticipated position of the towline without exceeding the stress level criteria specified in Section 3, 3.2.

6.6.5.2 In addition, towline guiding fittings are to be able to sustain the force exerted by the towline loaded under a tension equal to the brake holding load of the associated winch,, in the most unfavourable anticipated position of the towline without exceeding the stress level criteria specified in Section 3, 3.2.

6.6.5.3 Where a towline guiding fitting has been designed for a specific safe working load SWL, defined as the maximum static working load, the fitting is to be able to sustain a force equal to 2 times the SWL without exceeding the stress level criteria specified in Section 3, 3.2.

6.6.5.4 In case the strength evaluation of the towline guiding fittings is carried out by means of a three dimensional finite element model, the permissible stress levels given above may be increased by 10 per cent (compared to a beam model).

6.6.6 Supporting structures

6.6.6.1 The supporting structures of towing equipment used for escort services are to be able to sustain the load exerted on the supporting structure under the action of the towline loaded under a tension equal to the design load, as specified in 6.6.2, in the most unfavourable anticipated position of the towline, without exceeding the stress level criteria specified in Section 3, 3.2.

6.6.6.2 In addition, supporting structures of towing equipment used for escort services are to be able to sustain the load exerted on the supporting structure under the action of the towline loaded under a tension equal to the brake holding load of the associated winch, in the most unfavourable anticipated position of the towline without exceeding the stress level criteria specified in Section 3, 3.2

6.6.6.3 In case the evaluation of the towing equipment supporting structures is carried out by means of a three dimensional finite element model, the permissible stress levels given above may be increased by 10 per cent (compared to a beam model).

6.7 Simulations, Tests and Trials

6.7.1 Escort performance simulations

6.7.1.1 For the purposes of preliminary stability assessment, the highest anticipated heeling moment as obtained from the results of a computer simulation program, as well as the basic assumptions and theoretical models underlying the software are to be presented in detail. Items to be addressed include:

- Hydrodynamic lift and drag computation (hull and appendices)
- modelling of thrust forces
- Interaction effects between hull, skeg and (steerable) propulsion units
- flow separation effects
- Water pile-up effects
- Effects of waves and/or swell
- Dynamic effects before a steady state is reached (e.g. during initiation and turning manoeuvres) and scaling effects (if any).
- Meshing details
- Model assumptions and relevant limitations

6.7.1.2 A validation report, containing comparisons between simulation results and full scale, is to be presented to IRS, for information.

6.7.1.3 A clear description of the input and output data is to be provided, along with explanations on how the output data are obtained/ calculated by the software.

6.7.1.4 As a minimum, for each relevant loading condition, the following set of results is to be provided in tabular form as function of the escort speed for the rated values of the steering force and the braking force:

- Rated steering force or steering force corresponding to rated braking force, as applicable
- Rated braking force or braking force corresponding to rated steering force, as applicable
- Corresponding towline force
- All corresponding forces acting in transverse direction (hydrodynamic, thrust and towline)
- Corresponding heeling angle
- Corresponding heeling moment
- Corresponding towline angle relative to the escorted ship
- Corresponding drift angle of the escort tug.

Note: The highest anticipated values of the steering force, braking force, towline force and heeling moment do not normally all occur in the same orientation (defined by the position of escort tug relative to the escorted ship and the drift angle), although more than one parameter may have its highest value in a particular orientation.

Hence it is necessary to consider at least two orientations: one for the highest anticipated steering force and one for the highest anticipated braking force. In case the highest anticipated heeling moment and/or towline

force do not occur in either one of these two conditions, the relevant conditions are to be added.

6.7.1.5 It is recommended that the results of the escort performance simulations are presented in the form of diagrams showing the envelope of the (steady state) combinations of steering and braking forces obtained from the simulations.

Such diagrams should cover the applicable escort speed range, with a recommended step of 2 knots.

6.7.2 Escort performance trials

6.7.2.1 Escort performance trials at full scale are to be carried out in order to confirm the results of escort performance simulations.

6.7.2.2 The trials are to cover the applicable range of loading conditions and escort speeds. The confirmatory full scale trials for each loading condition may be undertaken for representative orientations only (defined by the position of escort tug relative to the escorted ship and the drift angle) only, for the validation of results of the computer simulation program; and may not cover the full range of escort tug orientations.

6.7.2.3 The following documents are to be submitted for information prior to testing:

- Relevant loading conditions, defined by draught (or displacement) and trim, for which the tug is designed to perform escort services
- Applicable range of test speeds and orientations (defined by the position of escort tug relative to the escorted ship and the drift angle) of the escorted ship: the speed is defined as the relative speed with respect to the sea, taking into account current effects
- Main propulsion characteristics, in particular power and maximum orientation angle of the rudder(s) (propellers);
- Preliminary calculation of the rated steering force, rated braking force and rated steady towline force as defined in, as well as the corresponding heeling moments and heeling angles, for the range of test speeds
- Preliminary stability calculations for the above mentioned conditions
- Escort towing arrangement plan, including the load cell and specification of the components
- Documentation relevant to the Bollard Pull test.

6.7.2.4 Prior to commencing the escort performance trials the following data are to be recorded:

- Wind speed and direction
- Sea state, including significant wave height and peak period
- Current speed and direction
- Water depth
- Loading condition of the escort tug: draught (or displacement) and trim
- Loading condition of the escorted ship.

6.7.2.5 Testing is to be performed over the applicable range of escort tug orientation. The length of the towline and the angle of the towline with the horizontal plane are to represent typical operating conditions.

6.7.2.6 As a minimum, the following data is to be collected during testing for post-processing and analysis:

- Towline force (tension)
- Towline angle
- Drift angle
- Heeling angle of the escort tug
- Towline length and angle of towline with the horizontal plane.

6.7.2.7 It is also recommended to measure the following data:

- Power setting and orientation angle of rudder(s) (propellers) of the escort tug

• Time needed to make the tug swing from the equilibrium position to its mirror position.

6.7.2.8 For each combination of loading condition and test speed:

• The rated steering force and rated braking force are to be calculated on the basis of the corresponding measured steady towline force and the associated measured towline angle, drift angle and the angle between the towline and the horizontal plane.

• The maximum heeling arm is to be calculated on the basis of the corresponding measured steady towline force, the associated measured heeling angle and the GZ curve applicable to the loading condition considered.

Note: The GZ curve is to be based on the escort tug in upright position before commencing the escort operation.

6.7.2.9 As a minimum, for each tested loading condition the following set of results is to be provided in tabular form as function of the escort speed for the rated values of the steering force and the braking force :

• Rated steering force and braking force corresponding to rated steering force.

• Rated braking force and steering force corresponding to rated braking force.

• Corresponding towline force

• Corresponding heeling angle

• Corresponding heeling moment

• Corresponding towline angle relative to the escorted ship

• Corresponding drift angle of the escort tug.

6.8 Escort equipment testing

6.8.1 Load testing

Escort winches, including the associated emergency release system are normally to be load tested at the design load and the brake holding load. Load testing is to be witnessed by an attending Surveyor.

In case an escort winch is of conventional, proven design, for which load testing has been previously performed and deemed acceptable by the IRS, it is sufficient to perform on board function testing in accordance with 6.8.2. In such a case, the winch manufacturer is to supply records of the load tests performed to IRS for information.

6.8.2 Function testing

In general, the proper functioning of the towing equipment used for escort services is to be verified by on board testing and to be witnessed by the attending Surveyor. Function testing is to be performed both for normal operating conditions and in accordance with the escort towing arrangement plan as well as in emergency conditions (emergency release, failure of main power supply). The safe operation of the escort winch from all control stations is to be demonstrated.

Escort winches are to be function tested on board.

The correct functioning of the winch brake, the load carrying winch components and the associated supporting structure is to be demonstrated at a towline force equal to the design Bollard Pull.

The emergency release systems are to be function tested.

Winch operating modes to be function tested include hauling in and paying out of the towline, braking and the active pay-out and haul-in system when fitted.

Hydraulic and electrical systems are to be function tested on-board in accordance with the requirements for machinery and electrical systems.

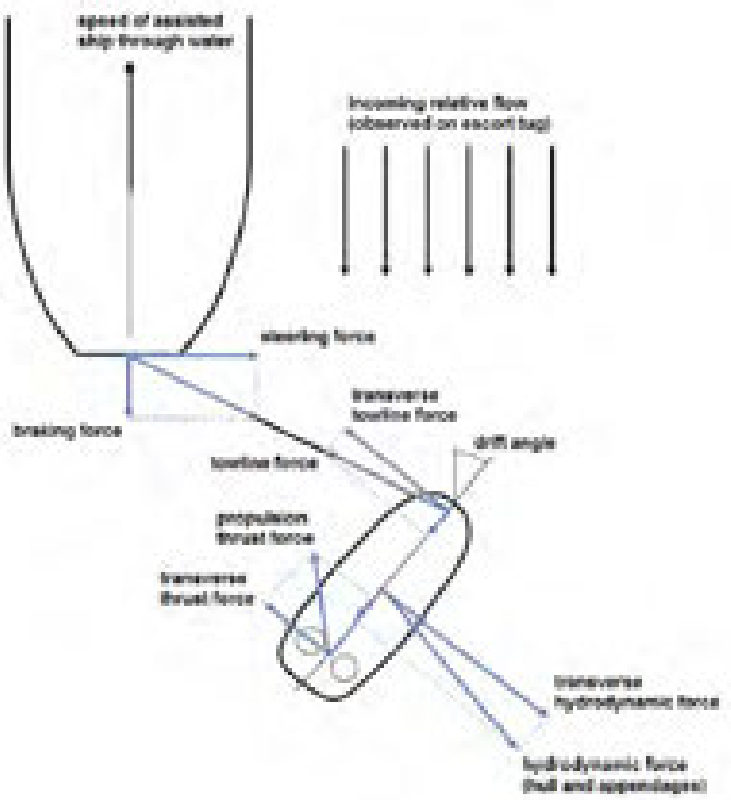

Fig. 6.1: Diagram of Forces – Escort Operations

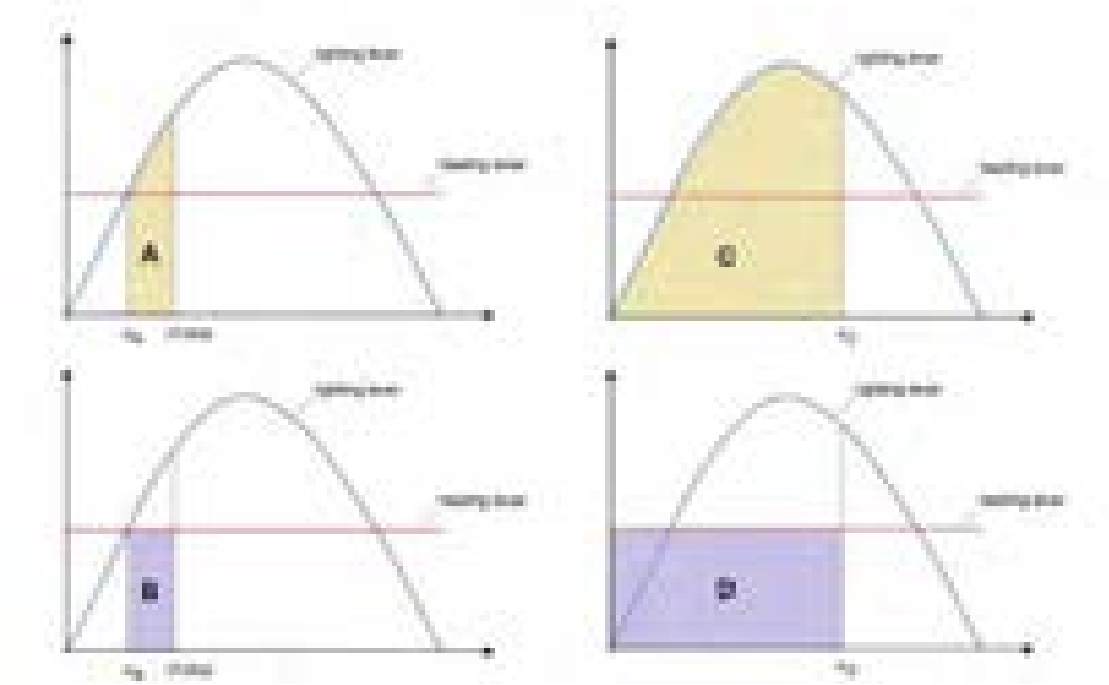

Fig. 6.2 : Stability Criteria for Escort Operations

End of Chapter

Chapter 8

Offshore Support Vessels

Contents

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1	*General*
2	*Offshore Support Vessels (Supply)*
3	*Offshore Support Vessels (Anchor Handling)*
4	*Safety Standby Rescue Vessels*
5	*Wind Farm Service Vessels*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to vessels intended for support services to offshore installations. These requirements are supplementary to those given for the assignment of main characters of class.

1.2 Class Notations

1.2.1 Vessels built in compliance with the requirements of this Chapter will be eligible to be assigned class notation **OFFSHORE SUPPORT VESSEL**.

1.2.2 Vessels constructed and equipped to the applicable requirements for specialized functional services will be assigned additional class notations as listed in Table 1.2.2.

For example:

OFFSHORE SUPPORT VESSEL (SUPPLY).

1.2.3 Vessels intended for several roles/ services related to offshore support will be assigned notation **MULTIPURPOSE OFFSHORE SUPPORT VESSEL**, in combination with the applicable additional class notations listed in Table 1.2.2, provided that the specific requirements for each intended role/ service are complied with and the vessels are equipped and prepared at all times to engage in operations related to the relevant roles/ services.

For example, an Offshore Support Vessel capable of anchor handling, towing and supply would be assigned the class notations:

MULTIPURPOSE OFFSHORE SUPPORT VESSEL (AH, SUPPLY, OCC, TOW).

Table 1.2.2 List of Additional Class Notations for Offshore Support Vessels

Additional Class Notation	Description	Reference to the applicable requirements to be complied with
SUPPLY	Vessels engaged in the transport of stores, materials, equipment and/or personnel (excluding crew boats) to, from and between offshore installations	Section 2
SUPPLY-PLUS	Vessels designed for operations in North Sea[1)] , engaged in the transport of stores, materials, equipment and/or personnel (excluding crew boats) to, from and between offshore installations and complying with the additional requirements given in 2.8.	Section 2, and 2.8
STS	Supply vessels complying with the requirements for Survival Capability (Floatability in damaged condition) of Ch.3 of the IMO Resolution MSC 235(82).	Pt.3, Ch.4
LHNS	Vessels intended to carry limited amounts of hazardous and noxious liquid substances in bulk.	Section 2, 2.7
OCC	Vessels complying with OSV Chemical Code (IMO Resolution A.1122 (30)).	Section 2, 2.7
AGNI 1, AGNI 2, AGNI 3	Vessels equipped for fire-fighting operations.	Pt.5, Ch.25
AH	Vessels equipped for anchor handling operations.	Section 3
TOW	Vessels equipped for towing operations.	Pt.5, Ch.7
DSV	Vessels equipped for providing support during diving operations.	Pt.5, Ch.26
WSV	Vessels equipped for performing well stimulation operations	Pt.5, Ch.33
OIL RECOVERY	Vessels equipped to recover oil spills from the sea surface at a safe distance, handle, store and transport the recovered oil.	Pt.5, Ch.34
SSR	Vessels equipped to carry out rescue and standby services for offshore installations	Section 4
WFSV, WFSV-HSC	Vessels equipped to serve and support wind farm activities	Section 5

Note:

1) The North Sea is situated between the eastern coasts of the British Isles and the western coasts of Europe. The geographical limits of the North Sea may be referred in the International Hydrographic Organization's document: 'Limits of the Oceans and Seas (Special Publication No. 23), 3rd edition, 1953' (as may be amended).

1.3 Selection of additional notations

1.3.1 It is the responsibility of the Owner to select the additional notations suitable for the intended service and ensure that the vessel is operated in a safe environment with respect to the risk of fire and explosion.

1.4 Requirements of Administrations

1.4.1 Requirements additional to those given in this Chapter may be imposed by the Administration with whom the vessel is registered or by the Administration within whose territorial jurisdiction the vessel is intended to operate.

Approval of structural fire protection, fire extinguishing equipment and/or stability of the vessel by an Administration, in accordance with requirements equivalent to those specified by these Rules, may be considered as complying with the class requirements provided such approval can be satisfactorily documented.

Section 2

Offshore Support Vessels (Supply)

For symbols not defined in this section, Refer to Pt.3, Ch.1, Sec.2.

2.1 Application

2.1.1 Requirements in this Section are applicable to vessels which are primarily engaged in the transport of stores, materials, equipment and/or personnel (excluding crew boats) to, from and between offshore installations.

2.2 Class Notation

2.2.1 Vessels complying with the requirements of this Section will be assigned additional class notation **SUPPLY or SUPPLY-PLUS**.

2.2.2 Supply vessels intended to carry limited amounts of hazardous and noxious liquid substances in bulk, meeting the requirements of sub-section 2.8 will be assigned notation **LHNS**.

2.2.3 Supply vessels complying with OSV Chemical Code (IMO Resolution A.1122 (30), as amended) will be assigned additional class notation **OCC**.

2.2.4 Supply vessels complying with the requirements for Survival Capability (Floatability in damaged condition) of Ch.3 of the IMO Resolution MSC 235(82) will be assigned additional class notation **STS**

For Example:

1) An Offshore support vessel, intended to carry limited amounts of hazardous and noxious liquid substances complying with OSV Chemical Code (IMO Resolution A.1122 (30)); and also the requirements for Survival Capability (Floatability in damaged condition) of Ch.3 of the IMO Resolution MSC 235(82), and intended to provide firefighting support, will be assigned following additional notation

MULTIPURPOSE OFFSHORE SUPPORT VESSEL (SUPPLY, OCC, STS , AGNI 1)

2.3 Documentation

2.3.1 In addition to the requirements in Part 3, following plans are to be submitted:

a) Information regarding loading arrangement of deck cargoes, weights and their centres of gravity

b) Structural details and arrangements of structures in way of cargo deck

c) Lashing arrangement of deck cargoes

d) Details of integral liquid cargo tanks including vents and/or overflows height and location

e) Details of independent liquid and/or dry cargo tanks

f) Details of independent tank supports and fastening arrangements

g) Piping diagrams of liquid cargo transfer systems

h) Piping diagrams of dry bulk cargo transfer systems

i) Ventilation diagrams of liquid cargoes

j) Arrangement in dry cargo compartments

k) Stability data and calculations

2.3.2 For notation "**STS**" following details are to be submitted for approval:

- Internal watertight integrity plan including details of any tunnels, ducts or pipes etc., in the damage penetration zone and arrangements to prevent progressive flooding;

- Preliminary damage stability calculations;

- Final damage stability calculations (in case the final lightship data compares favourably with the estimated lightship data, the final damage stability calculations need not be submitted);

2.4 Ship Arrangement and Strength

2.4.1 General

2.4.1.1 It is recommended that supply vessels be provided with deep tanks at the wings to provide protection against accidental flooding of the machinery spaces.

2.4.1.2 Longitudinal fenders are to be fitted on shipside's at freeboard deck and deck above. The fenders are to extend not less than 0.02L forward of the section where the deck has its full breadth. In addition, sloping fenders are to be arranged in between the longitudinal fenders.

2.4.1.3 Carling plates or other means of stiffening are to be provided so that fender loads are effectively distributed to the hull structure.

2.4.1.4 In general, Plates, Stiffeners and Primary supporting members are to be evaluated in accordance with Pt.3, Ch.7. Additional hull local scantling requirements for offshore support vessels are given in the following.

2.4.2 Longitudinal strength

2.4.2.1 The requirements of Pt.3, Ch.1, Sec.4 regarding loading instruments, are not applicable to supply vessels.

2.4.2.2 The longitudinal strength is to comply with the relevant requirements of Pt.3, Ch.6.

2.4.3 Bottom structure

2.4.3.1 The flat of bottom in the stern region, if any, is to be additionally stiffened.

2.4.4 Side structure

2.4.4.1 The net thickness, $t_1$in [mm] of side shell plating including bilge strake, up to deck above freeboard deck is generally not to be less than:

$$t_1 = \max(4.5 + 0.05L\sqrt{k}, 7.0)$$

where,

L = Rule length but need not be taken greater than 90[m].

Requirements given for side plating in Pt.3, Ch.7 Sec.4 also to be complied with, as applicable.

2.4.4.2 The net section modulus, Z_1 in [cm^3] of the transverse stiffeners or side longitudinals in any region is not to be less than:

$$Z_1 = 1.15\ Z$$

where,

Z = Required net section modulus, in [cm^3], as given in Pt.3, Ch.7, Sec.5 and Pt.3, Ch.7, Sec.8.

2.4.4.3 The net section modulus of transverse stiffeners or side longitudinals which are inclined at more than 20[deg] with the vertical, is not be less than 1.25 times the net section modulus required as per Pt.3, Ch.7.

The side frames are not to be scalloped and are to be bracketed at both ends.

All stiffeners up to deck above freeboard deck, and stiffeners forward of 0.2L from F.E. up to forecastle deck, are to have end connections with brackets.

Continuous welds are to be used in connections between stiffeners and shell plating.

2.4.5 Deck structure

2.4.5.1 Deck Arrangement

.1 Safe havens and escape routes for personnel from the cargo deck are to be properly marked and kept clear at all times. A cargo rail (crash barrier) fitted along each side of the deck and/or aft may be the method of providing a safe haven.

.2 The vessel is to be equipped with a sufficient quantity and types of lashing and securing materials for the safe operation of the vessel, and is to comply with the *Code of Safe Practice for Cargo Stowage and Securing* (CSS) Code (IMO resolution A.714(17)) and IMO/ILO/UN/ECE *Guidelines for Packing of Cargo Transport Units* (CTUs),(MSC/Circ.787).

2.4.5.2 Reinforcement against Heavy Cargoes

.1 Where heavy cargo is carried on deck, means such as a steel cradle or steel or wooden dunnage is to be provided so that the cargo weight is uniformly distributed on deck structures.

2.4.5.3 Deck Covering

.1 Knot free wooden sheathings, in the form of planks not attached permanently to the plating, are to be provided on cargo decks to protect steel deck plate from mechanical damage and/or rubbing. Thickness of wooden sheathing is to be at least 50 [mm]. The deck plating under the wooden planks is to be protected by heavy duty paint or equivalent coating. Adequate arrangements allowing drainage from beneath the wooden sheathing are to be provided.

2.4.5.4 Securing deck cargo

.1 Pad eyes for cargo securing and/or moving are to be welded directly to the deck plating (i.e. without use of doublers plating). The pad eyes are to be adequately supported by approved deck reinforcement. Dismountable pad eyes and stanchions are to have firm attachments to the deck sockets or holdings. All pad eyes are to be permanently marked with bead welded SWL values.

2.4.5.5 Cargo rail

.1 Where cargo rails are fitted on cargo deck, they are to have freeing ports allowing quick drainage of water from the deck.

.2 The cargo rails and stanchions may be designed using direct calculations or FE analysis(Refer Pt.3, Ch.8, Sec.4) based on the design loads and load distribution.

2.4.5.6 Weather decks aft

.1 The scantlings of the weather deck intended to carry deck cargo, are to be based on a pressure 'p' given below :

$$p = p_c + c.\, p_e \qquad [N/mm^2]$$

where,

p_c = Static pressure due to deck cargo, in $[N/mm^2]$ to be not less than 1.5 g $[kN/m^2]$.

p_e = External sea pressure, $[kN/m^2]$ as given in Pt.3, Ch.5, Sec.4.1.

$c = 1.28 - 0.32.q$ for $q < 4$ $[t/m^2]$
$= 0$ for $q \geq 4$ $[t/m^2]$

q = deck cargo loading, in $[t/m^2]$, but not to be taken less than $1.5[t/m^2]$.

.2 Additional local loads on deck, due to anchor handling, specialized cargoes, towing winch, as well as from supports of other heavy components, are to be considered and scantlings increased accordingly.

.3 The net thickness of deck plating is not to be less than 7.0[mm].

.4 Stow racks of substantial construction are to be provided for protection of deck cargoes. These are to be efficiently attached to the deck structure and adequately supported.

All hatches, air pipes, ventilators, valve controls, etc. are to be located outboard of the stow racks and adequately protected.

.5 The disposition of freeing ports is to be carefully considered to ensure most effective drainage of water trapped in pipe deck cargoes and in way of recesses at the after end of the forecastle. In way of such recesses, appropriate scuppers with discharge pipes led overboard may be required.

The area of the freeing ports in the side bulwarks on the cargo deck is to meet the requirements of Pt.3, Ch.7, Sec.9, Cl.9.2.3.

.6 Gross thickness of bulwark plating is not to be less than 7[mm]. Bulwark stays are generally not to be spaced more than 1.3[m] apart and the depth of the stay at the lower end is not to be less than 350[mm]. The stays are to be adequately supported under the deck.

.7 Scantlings of flush hatch covers in the cargo deck areas are normally to be based on the same load as the adjacent deck.

2.4.6 Superstructures and deckhouses

2.4.6.1 The thickness of plating forming the unprotected sides and ends of superstructures and deckhouses is to be 1[mm] higher than that required as per Pt.3, Ch.7, Sec.8.

2.4.6.2 The section modulus of stiffeners on these bulkheads is to be 50% higher than that required by Pt.3, Ch.7, Sec.8. The stiffeners on front bulkheads are to be bracketed at both ends. The stiffeners on sides and aft end bulkheads are to be bracketed at upper ends and their webs welded to the deck at lower ends.

2.4.7 Miscellaneous openings and their closing devices

2.4.7.1 Requirements for miscellaneous openings are to be in accordance with Pt.3, Ch.12, Sec.2, as applicable.

2.4.7.2 Direct access from the freeboard deck to machinery and other spaces below the freeboard deck is not permitted. As far as practicable, it is to be arranged within the forecastle or above; other equivalent arrangements may be considered.

2.4.7.3 Requirements for windows and side scuttles are to be determined in accordance with Pt 3, Ch.12, Sec.4.

2.4.8 Primary supporting members

2.4.8.1 Direct strength analysis

Primary supporting members are to be evaluated in accordance with Pt.3, Ch.7, Sec.6 considering the design load sets given in Pt.3, Ch.7, Sec.2.

The strength of primary structural members that form part of a grillage system, such as deck girders, side web frames, pillars, floors and girders in double bottom may be determined by using direct strength analysis. For this purpose, beam analysis as described in Cl.2.4.8.2 are to be used.

2.4.8.2 Beam analysis

Beam analysis is used to evaluate bending and shear stresses in webs and flanges of grillage structure under lateral loads such as decks, double bottom and side structure under cargo or liquid pressure, e.g. sea and tank pressures. The effective plate breadth in bending of the primary strength members is to be calculated according to Pt.3, Ch.3, Sec.7, Cl.7.3.

For allowable stress, Refer Pt.3, Ch.7, Sec.6, Cl.6.2.

2.4.8.3 Buckling check of plate panels based on the results of beam analysis

Plate panels are subjected to buckling check as given in Pt.3, Ch.9. The stresses used for the buckling check are to be obtained from beam analysis (Refer Cl.2.4.8.2) and the stresses corrected as per Pt.3, Ch.9, Cl. 5.2.2.7.

2.5 Cargo Tanks and System

2.5.1 Liquid Cargo Tanks

2.5.1.1 Integral Cargo Tanks

.1 For cargo tanks intended for storage of combustible liquid cargoes such as fuel oil, base oil and oil-based liquid mud having a flashpoint exceeding 60°C, the provisions in Pt.4, Ch.3, Sec.4 are applicable for cargo tank arrangements and protection.

.2 Due consideration is to be given to separation arrangements for tanks containing drinking water and stores for human consumption complying with current regulations of national health authorities.

.3 The tank structure is to comply with the requirements in Pt.3, Ch.3, Sec.5.6, Sec.5.9.5 and Sec.5.11.7(as applicable).

2.5.1.2 Tank Access Arrangements

.1 The access hatch is to have clear dimensions of minimum 380 x 580 [mm] when rectangular or oval, or 460 [mm] diameter, when circular, and the cover equipped with an ullage hole 200 [mm] diameter.

.2 The hatch center is preferably to be aligned with the cargo rail to allow for use of a small block and/or rope, to help lifting up an incapacitated person, in case of injury. Adequate means of access into the tank, by either fixed or removable ladders, are to be provided.

2.5.2 Dry Cargo Tanks

2.5.2.1 Where cargo tanks for dry cement or dry mud (pulverized materials such as barite, bentonite, etc.) are fitted, these cargo tanks are to be separated from Category A machinery spaces, accommodation spaces and service spaces by watertight bulkheads and decks.

2.5.2.2 Where cargo tanks are subjected to pressurized media for cargo loading or discharging, independent tanks are to be used, and tank structures are to comply with the applicable requirements for pressure vessels in Pt.4, Ch.5.

2.5.3 Spill Coaming

2.5.3.1 Deck areas for handling oils and oil-based liquids are to be provided with a coaming around all pumps, transfer flanges, connections and other equipment where leakage may occur. Each coaming is to be adequately sized to contain deck spills and prevent spilled cargoes from entering accommodation, machinery, control and service spaces or passing overboard. The coaming is to have a height of

at least 150 [mm]. Where drains are provided for the coaming, closing devices for these drains are to be permanently attached.

2.5.4 Cargo Piping Systems

2.5.4.1 General

.1 Cargo piping systems are in general to comply with the relevant requirements in Pt.4. Due consideration is to be given to prevent drinking water and stores for human consumption from being contaminated, in accordance with the current national regulations of health authorities.

2.5.4.2 Liquid Cargoes

2.5.4.2.1 Combustible Liquid Cargoes

.1 For cargo fuel oil, base oil and oil-based liquid mud having a flashpoint exceeding 60°C, the provisions in Pt.4, Ch.3, Sec.4 are applicable for, cargo transfer piping, tank level gauging, venting and heating systems.

.2 Cargo pumps are to be of approved type.

.3 An emergency means for pumping out the tanks is to be provided. For this purpose, a portable pump of sufficient capacity, which can be used safely, may be accepted.

.4 A relief valve is to be installed in the discharge of each cargo and stripping pump. The outlet from the relief valve is to be led to the suction side of the pump. This relief valve need not be fitted in the case where centrifugal pumps are installed and the piping is designed to withstand the shut-off head of the pumps.

.5 Remote shutdown devices for all cargo pumps are to be fitted and they are to be capable of being activated from a dedicated cargo control location which is manned at the time of cargo transfer and from at least one other location outside of the cargo area and; at a safe distance from it.

.6 One pressure gauge for each pump is to be located at the pump discharge. Where pumps are operated at cargo control station remote from the cargo pumps, additional pressure gauges are to be installed at the cargo control station.

.7 Where cross-contamination either by cargoes or cargo residues causes safety hazards or marine pollution hazards, separation between cargo piping systems, is to be by means of spectacle flanges, spool pieces or equipment.

2.5.4.2.2 Liquid Mud Cargo Tanks

.1 Liquid mud cargo tanks are to be provided with vent pipes in accordance with Pt.4, Ch.3. In order to prevent overpressure or under-pressure in the event of overflow into the vent pipe or clogging of the flame screen in the case of oil based mud vents for liquid mud tanks are to also be provided with a suitable burst disc(s) rated below the mud tank design pressure. Spare burst discs are to be carried on board so that damage burst disc can be replaced. Suitable means of gauging the mud tanks such as a tank ullage method or level indicating devices may be fitted in lieu of sounding pipe per Pt.4, Ch.3, 3.4.

2.5.4.3 Dry Cargo Piping Systems

2.5.4.3.1 Separation

.1 Dry cargo transfer piping systems are, in general, not to be led through machinery spaces. However, when such design is impractical, piping systems may pass through machinery spaces, provided that all pipe connections located within the machinery space are welded and any detachable connections are arranged outside of the space. Any access door in the bulkhead is to be provided with a notice affixed to each such closing appliance to the effect that it is to be kept closed.

2.5.4.3.2 Piping Scantlings

.1 The minimum requirements for wall thicknesses of steel pipes in cement and dry mud system passing through machinery spaces are to be in accordance with Pt.4 Ch.2, Table 2.2.2.

2.5.4.4 Cargo Transfer Import and Export System

.1 Cargo Transfer Hose

.1 Where fitted, cargo transfer hoses are to be compatible with the cargo and suitable for the cargo temperature and have a burst pressure not less than 5 times the maximum expected service pressure of the cargo transfer system.

.2 For combustible cargoes, cargo hoses are to have electrical continuity over their entire lengths, including couplings and flanges

(except shore connections) and are to be earthed for removal of electrostatic charges.

.3 In the case of transfer operations involving pressures in excess of 5 [MPa], arrangements for emergency depressurizing and disconnection of the transfer hose are to be provided. The controls for activating emergency depressurization and disconnection of the transfer hose are to meet the requirements given in 2.5.4.2.1.5 and 2.5.4.4.2.

.2 Cargo Transfer Hose Emergency Disconnection

.1 Means for emergency disconnection of the cargo hose are to be provided and capable of being activated from the vessel position control station or cargo transfer control station. The connecting coupling is to be of self-sealing type (automatic closing at disconnection).

2.6 Machinery Arrangement

2.6.1 Engine exhaust outlets

2.6.1.1 Engine exhaust outlets are to be located, as high as practicable, above the deck and are to be fitted with spark arresters.

2.6.2 Steering gear

2.6.2.1 The steering gear is to be capable of bringing the rudder from 35 degrees on one side to 30 degrees on the other side in 20 seconds, when the vessel is running ahead at maximum service speed.

2.7 Carriage of Hazardous and Noxious Liquid Substances

2.7.1 Application

2.7.1.1 In addition to the above requirements, the requirements in IMO Resolution A.673 (16), "*Guidelines for the Transport and Handling of Limited Amounts of Hazardous and Noxious Liquid Substances in Bulk in Offshore Support Vessels*" as amended by IMO Resolutions MSC.236(82) and MEPC.158(55), apply to the design, construction and operation of offshore supply vessels intended to carry limited amounts of hazardous and noxious liquid substances in bulk.

2.7.1.2 For the purpose of this subsection, 'limited quantities', would be as defined in IMO Resolution A.673(16). Where the Administration permits the carriage of hazardous and noxious liquid substances in amounts greater than the relevant maximum amount specified above, the vessel is to comply with the survival capability requirements in Pt.5, Ch.3, Sec.2 for chemical cargoes or Pt.5, Ch.4, Sec.2 for liquefied gases.

2.7.1.3 The *International Maritime Dangerous Goods Code* (IMDG) may be referred for provisions regulating the transport of dangerous goods and marine pollutants in packaged form, including transport of dangerous goods in portable tanks.

2.7.1.4 The IMO Resolution A.1122(30), "Code for the Transport and Handling of Hazardous and Noxious Liquid Substances in Bulk on Offshore Support Vessels" (OSV Chemical Code), as applicable**,** may be applied as an alternative to IMO Resolution A.673(16), as amended, if acceptable to the Flag Administration.

2.7.1.5 The above mentioned requirements also apply to offshore support vessels other than those in offshore supply service when, due to their operations, they are designed and constructed to carry limited amounts of hazardous and noxious liquid substances in bulk, and as referred to in other Sections of this Chapter.

2.8 Requirements for vessels assigned with SUPPLY-PLUS notation

2.8.1 Local hull scantlings

2.8.1.1 Requirements given in this sub section are to be applied in lieu of the requirements in Cl. 2.4.7, 2.4.4.2 and 2.4.4.3.

2.8.1.2 Where subjected to heavy loads when handling anchors for offshore floating units drilling rigs, the stern is to be adequately strengthened.

- The gross plate thickness, in [mm], adjacent to the stern roller and shark jaw is not to be taken less than:

$$t = 8 + 0.2L$$

- The deck adjacent to the stern is to be strengthened accordingly.

- The net thickness of the side plating up to forecastle deck is not to be less than as given in Cl.2.4.4.1.

2.8.1.3 The net section modulus, Z_1 in [cm^3] of the transverse stiffeners or side longitudinals up to second deck above the freeboard deck is not to be taken less than:

$$Z_1 = 0.0015.L.\ell_{bdg}.s.k$$

where:

ℓ_{bdg} = Effective bending span of stiffener, in [m], as defined in Pt.3, Ch.3, Sec.7.

s = Stiffener spacing in [mm], as defined in Pt.3, Ch.3, Sec.7.

The net section modulus, Z_{min} in [cm^3] of the transverse stiffeners or side longitudinals in any region, not to be taken less than:

$$Z_{min} = 1.25\,Z$$

where,

Z = Required net section modulus, in [cm^3], as given in Pt.3, Ch.7, Sec.5 and Pt.3, Ch.7, Sec.8.

The requirement of Z_1 given above refers to the ship's sides, which are inclined at less than 20[deg] with the vertical (along the ship's depth). For greater inclinations the requirement given for Z_{min} is to be applied.

2.8.1.4 All stiffeners up to deck above freeboard deck, and stiffeners forward of 0.2L from F.E. up to forecastle deck, are to have end connections with brackets.

2.8.1.5 Continuous welds are to be used in connections between stiffeners and shell plating.

2.8.1.6 Deckhouses and superstructures

The net section modulus of stiffeners, in [cm^3], not contributing to longitudinal strength is not to be taken less than:

$$Z = \frac{|P|.s.\ell_{bdg}^2}{f_{bdg}.\chi.C_s.R_{eH}}$$

P = Design pressure in [kN/m^2] but not less than 10 [kN/m^2] for weather decks.

= P_D for exposed decks

= max (P_W, P_{SI}) for exposed superstructure side

= P_A for exposed superstructure end bulkheads and deckhouse boundaries

C_s = 0.75 for acceptance criteria set AC-II.

2.8.1.7 Loading conditions

2.8.1.7.1 The following loading conditions are to be included in the stability booklet:

a. vessel in fully loaded departure condition with cargo distributed below deck and with deck cargo specified by position and weight, with full stores and fuel, corresponding to the worst service condition in which all stability criteria are met

b. vessel in fully loaded arrival condition with cargo as specified, but with 10% stores and fuel

c. vessel in ballast departure condition, without cargo but with full stores and fuel

d. vessel in ballast arrival condition, without cargo but with 10% stores and fuel

e. vessel in the worst anticipated operating condition

f. if the vessel is equipped with towing gear, vessel in a typical condition ready for towing.

2.8.1.7.2 Assumptions for calculating loading conditions are as follows:

g. if a vessel is fitted with cargo tanks, the fully loaded condition as specified in Cl. 2.8.1.7.1 is to be modified, assuming first the cargo tanks full and then the cargo tanks empty

h. in all cases when deck cargo is carried a realistic stowage weight is to be assumed and stated in the stability information, including the height of the cargo and its centre of gravity

Where pipes are carried on deck, a quantity of trapped water equal to a certain percentage of the net volume of the pipe deck cargo is to be assumed in and around the pipes. The net volume to be taken as the internal volume of the pipes plus the volume between the pipes. This percentage is to be 30 if the freeboard amidships is equal to or less than $0.015L_{LL}$ and 10 if the freeboard amidships is equal to or greater than $0.03L_{LL}$. For intermediate values of the freeboard amidships the percentage may be obtained by linear interpolation.

2.8.2 Windows and side scuttles

2.8.2.1 The requirements given below are to be applied in conjunction with Pt 3, Ch.12, Sec.4.

2.8.2.2 In superstructure and deckhouses, windows may only be fitted in the following locations:

(a) Second tier and higher above the freeboard deck:
(i) in the after-end bulkhead of deckhouses and superstructures,
(ii) in the sides of deckhouses and superstructures which are not part of the shell plating.

(b) Third tier and higher above the freeboard deck:
(i) in the front bulkheads of deckhouse and superstructures, except that in the first tier of the front bulkhead above the weather deck, (forecastle deck) only side scuttles will be accepted.

In locations not specified above, only side scuttles with hinged deadlights will be accepted.

2.8.2.3 Hinged deadlights are to be provided as follows:

(a) Side scuttles:
(i) in the side shell plating,
(ii) in the front bulkheads of superstructures and deckhouses,
(iii) in the sides of deckhouses and superstructures up to and including the third tier above the freeboard deck,
(iv) in the after-end bulkheads of superstructures, deckhouses, casings and companionways in the first and second tiers above the freeboard deck.
(v) in all bulkheads of the first tier on the weather deck.

(b) Windows in locations permitted in Cl.2.8.2.2:
(i) in the sides of deckhouses and superstructures in the second and third tiers above the freeboard deck.
(ii) in the after-end bulkheads of superstructures, deckhouses, casings and companionways in the second tier above the freeboard deck.
(iii) in the front bulkheads of superstructures and deckhouses in the third tier
(iv) in all bulkheads of the first tier on the weather deck.

2.8.2.4 Windows in the front bulkhead for tier four and above, unless it is the first tier above the forward weather deck, the deadlights may be portable if they are stowed adjacent to the windows for quick mounting. At least two of the deadlights are to have the means of providing a clear view.

Deadlights on above mentioned windows and side scuttles are to be of steel and may be hinged externally provided there is easy and safe access for closing.
Other deadlights are to be internally hinged.

Typical arrangements in line with the above requirements are indicated in Fig. 2.8.2.

2.8.2.5 The dimensions of the windows are to be kept to a minimum, compatible with operational requirements.

2.8.2.6 Side scuttles are to comply with ISO 1751, as follows:

- Type A (heavy): In the hull, in the sides of superstructures and in the front bulkheads of superstructures and deckhouses on the weather deck.
- Type B (medium): In the after end of superstructures and in the sides and ends of deckhouses (except front in weather deck tier).

2.8.3 Anchoring, mooring and towing equipment

2.8.3.1 For supply vessels that are intended for operation in the North Sea and not provided with dynamic positioning systems or other means of control by thrusters; the length and the diameter of chain cable is to be based on an equipment letter two steps higher than that corresponding to the EN of the vessel.

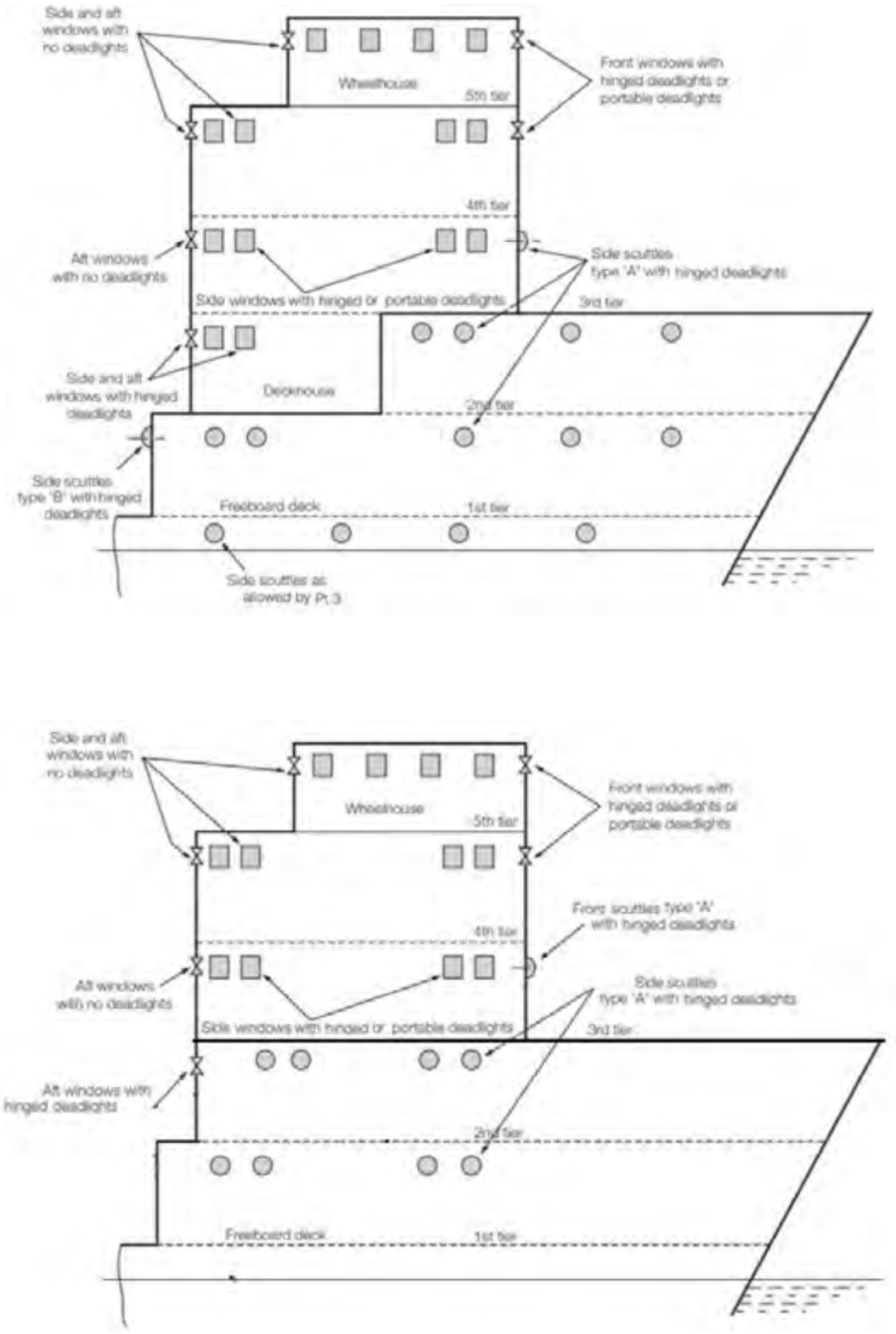

Figure 2.8.2: Location and arrangement of windows and side scuttles

Section 3

Offshore Support Vessels (Anchor Handling)

For symbols not defined in this section, Refer to Pt.3, Ch.1, Sec.2.

3.1 General

3.1.1 Application

3.1.1.1 The requirements in this section are applicable to Offshore Support Vessels equipped for the handling of anchors of offshore installations.

3.1.2 Notation

3.1.2.1 An additional qualifier notation **AH** will be assigned to Offshore Support Vessels equipped for anchor handling and designed and built to the requirements of this section and other relevant sections of this chapter.

Offshore Support Vessel (AH)

3.1.3 Submission of Plans and Data

Following additional plans and particulars are to be submitted:

3.1.3.1 General Arrangement Plan

i) Anchor handling arrangement plan showing all equipment/components of the handling system such as winches, ropes and/or wires, components on anchor handling line path, quick release devices/emergency release systems and control stations as well as design criteria including the expected static bollard pull (BP), other loads, hoisting, brake holding and rendering capacities of the winch, quick release speed, as applicable and the most extreme line directions, etc.

3.1.3.2 Structure Plans

i) Structural details of supporting structures in way of the anchor handling winches

ii) Structural details of stern roller, guide pins, shark jaw and their supporting structure

iii) Details of stow racks, cargo rails, crash rails and supporting structures

iv) Spare chain locker(s) structural details including chutes (if installed)

v) Structural details of supporting structures of A-frames and deck cranes

vi) Detailed drawings showing dimensions, materials, welding details, as applicable, of stern rollers, etc.

vii) Design analysis of structures and deck fittings on the load path

viii) Non-destructive examination (NDE) plans

3.1.3.3 Machinery Plans

i) Anchor handling winch assembly plans including prime mover and brakes

ii) Detailed drawings showing dimensions, materials, welding details, as applicable, of all load bearing components of the winch

iii) Shark jaws and guide pins units assembly plans

iv) Detailed drawings showing dimensions, materials, welding details, as applicable, of shark jaws and guide pins

v) Design analyses for load bearing components. Analyses for gears are to be in accordance with a recognized standard

vi) Non-destructive examination (NDE) plans

vii) Hydraulic piping system diagram along with system design pressure, relief valve setting, bill of materials, typical pipe joints, as applicable

viii) Electric one line diagram along with cable specification and size; motor controller; protective device rating or setting; as applicable

ix) Control, monitoring and instrumentation arrangements

x) Plans and data for winch hydraulic or electric motors including associated gears rated 100 kW and over.

3.1.3.4 Stability Documents

i) Stability calculations and corresponding information for the Master are to be submitted for review and approval. See 3.2 for detailed requirements.

3.1.3.5 Information Documents

i) Locations, weights and centres of gravity of anchor handling and secondary winches (storage reels)

ii) Arrangement and details of communication systems between anchor handling operation control stations and navigation bridge

iii) Design loads and locations of rollers, eyes/hooks A-frame and deck cranes, shark jaws and pins and other components on anchor handling line path as well as quick release devices (if installed)

iv) Anchor launch and recovery unit for deep penetrating anchors (if installed)

v) Pad-eyes for securing and lashing anchors on deck

vi) Type, lengths, diameters and minimum specified breaking strength of ropes and/or wires to be set on the above winches

vii) Laying arrangement and weights of anchors carried as cargo

viii) Estimated static bollard pull, together with the method of prediction. (The estimated value is to be confirmed at trials prior to final certification)

ix) Bollard Pull Test Procedure (to be reviewed by the attending Surveyor in advance of the test)

3.1.4 Definitions

3.1.4.1 Anchor handling winch **-** Winch dedicated to anchor handling operations.

3.1.4.2 Wire **-** A dedicated line (wire rope, synthetic rope or chain cable) used for handling of anchors by means of the anchor handling winch.

3.1.4.3 Design Bollard Pull (BP) **-** The maximum sustained line force the vessel is capable of generating at maximum power (100% MCR) and zero forward speed. This force (BP) is to be initially specified by the designer and is to be verified by a bollard pull test.

3.1.4.4 Safe working load (SWL) **-** Load to be specified for each shipboard fitting. Individual SWLs are to be marked on each concerned fitting.

3.1.4.5 Loads on anchor handling winch

.1 **Rated line pull (RP)**: Maximum rope tension, in [kN], that the winch can haul at the relevant layer, in normal service condition, when the drum rotates at its maximum service speed. The rated line pull is defined for a specific reeled layer which is to be specified by the owner/ designer. The reeled layer may be the first layer (in contact with the drum) or the outer layer.

.2 **Brake holding load (BHL)**: Maximum rope tension, in [kN], with holding brake activated

.3 **Rendering load (RL):** Maximum rope tension, in [kN], at the drum exit when the drum just starts rotating in the opposite direction of the applied driving torque with a first layer of rope wound on the drum.

3.1.4.6 Stern roller **-** A roller or similar equipment arranged at the aft boundary of the ship to launch and recover the anchors. The stern roller is designed to adequately support the working line during the launching/ recovering of anchors.

3.1.4.7 Wire stopper (i.e. shark jaw, karm fork) Wire stopper means a fitting used to secure the anchor wire or chain, thereby relieving the strain on the winch.

3.1.4.8 Guide pin **–** Guide pins means elevating pins mounted aft of the wire stopper(s) which are fitted with guide arms that can be individually controlled to help guide the chain.

3.2 Stability

3.2.1 General

3.2.1.1 During anchor handling, the wire comes from the anchor handling winch and is controlled by the guide pins and connected, over the stern roller, to the lifted item.

3.2.1.2 For vessels used for anchor handling while their towing capacity and/ or the traction power of their winches is/ are already in action, calculations are to be made showing the

maximum acceptable tension to which the vessel is able to be exposed.

3.2.1.3 The compliance with the relevant stability criteria are to be made for each set of towing pins and its associated permissible line tensions, including any physical element or arrangement that can restrict the line movement.

Moreover, the loading conditions defined in 3.2.1.4 below are to meet the stability criteria defined in 3.2.4 when applying the design tension F_d, for the tow pin set nearest to centreline, as a minimum for the lowest α equal to 5 degrees. Allowance is to be made for the anticipated weight of cargo on and below deck, chain in lockers, anticipated type of wire or rope on storage reels and wire on the winches when calculating loading conditions.

3.2.1.4 In addition to the standard loading conditions, the following loading cases are to be included in the trim and stability booklet:

.1 service loading condition at the maximum draught at which anchor handling operations may occur with the heeling levers as defined in 3.2.2 for the line tension the ship is capable of with a minimum of 67% stores and fuel, in which all the relevant stability criteria defined in 3.2.4 are met ;

.2 service loading condition at the minimum draught at which anchor handling operations may occur with the heeling levers as defined in 3.2.2 for the line tension the ship is capable of with 10% stores and fuel, in which all the relevant stability criteria as defined in 3.2.4 are met.

3.2.2 Heeling Lever

3.2.2.1 A heeling lever, HL_{φ}, generated by the action of a heeling moment caused by the vertical and horizontal components of the tension applied to the wire is to be calculated as:

$$HL_{\varphi} = (M_{AH}/\Delta_2)\cos\varphi$$

Where:

$$M_{AH} = F_P \times (h\sin\alpha \times \cos\beta + y \times \sin\beta)$$

Δ_2= displacement in a loading condition, including action of the vertical loads added (F_V), at the centreline at the stern of ship;

$$F_V = F_P \times \sin\beta$$

α =the horizontal angle between the centreline and the vector at which the wire tension is applied to the ship in the upright position, positive outboard;

β =the vertical angle between the waterplane and the vector at which the wire tension is applied to the ship, positive downwards, should be taken at the maximum heeling moment angle as $\tan^{-1}(y/(h \times \sin\alpha))$, but not less than $\cos^{-1}(1.5B_P/(F_P \times \cos\alpha))$ consistent units;

B_P =the Bollard pull that is the documented maximum continuous pull obtained from a static pull test on sea trial, carried out in accordance with Appendix A of MSC/Circ.884 or an equivalent standard acceptable to IRS;

F_P =(Permissible tension) the wire tension which can be applied to the ship as loaded while working through a specified tow pin set, at each *α*, for which all stability criteria can be met. *Fp* should in no circumstance be taken as greater than *Fd*;

F_d =(Design maximum wire tension) the maximum winch wire pull or maximum static winch brake holding force, whichever is greater;

h = the vertical distance (m) from the centre the propulsive force acts on the ship to either:

- the uppermost part at the towing pin, or
- a point on a line defined between the highest point of the winch pay-out and the top of the stern or any physical restriction of the transverse wire movement;

y = the transverse distance (m) from the centreline to the outboard point at which the wire tension is applied to the ship given by:

$y_0 + x\tan\alpha$; but not greater than *B*/2;

B = the moulded breadth (m);

y_0 = the transverse distance (m) between the ship centreline to the inner part of the towing pin or any physical restriction of the transverse wire movement;

x = the longitudinal distance (m) between the stern and the towing pin or any physical restriction of the transverse wire movement.

F_t shows the vector of the applied wire tension.

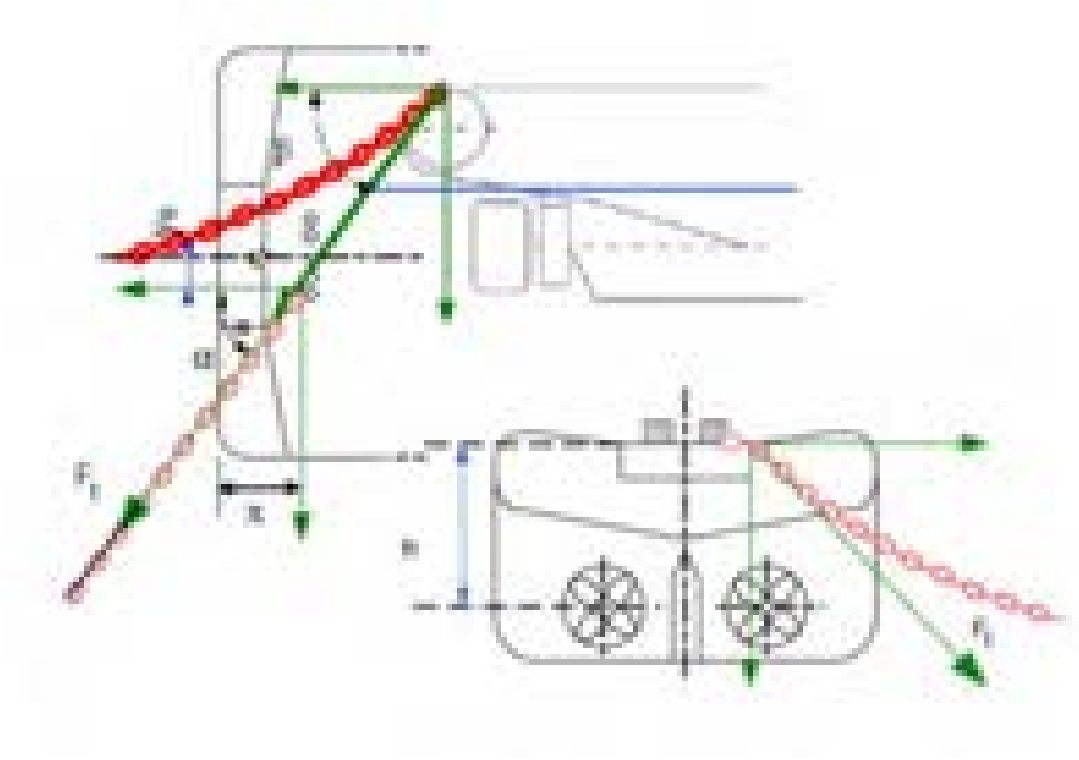

Figure 3.2.2.1: Diagram showing the intended meaning of parameters referred in Cl. 3.2.2.1.

3.2.3 Permissible tension

3.2.3.1 The permissible tension as a function of *α*, defined in 3.2.2, is not to be greater than the tension given by 3.2.3.2 below.

3.2.3.2 Permissible tension as a function of *α* can be calculated by direct stability calculations, provided that the following are met:

.1 the heeling lever is to be taken as defined in 3.2.2 for each *α*;

.2 the stability criteria in 3.2.4, is to be met;

.3 *α* is not to be taken less than 5 degrees, except as permitted by 3.2.3.3; and

.4 Intervals of *α* is not to be more than 5 degrees, except that larger intervals may be accepted, provided that the permissible tension is limited to the higher *α* by forming working sectors.

3.2.3.3 For the case of a planned operation to retrieve a stuck anchor in which the ship is on station above the anchor and the ship has low or no speed, *α* may be taken as less than 5 degrees.

3.2.4 Stability Criteria

3.2.4.1 For the loading conditions intended for anchor handling, but before commencing the operation, the stability criteria given in paragraph 2.2 of part A of IMO IS Code, or where a ship's characteristics render compliance with paragraph 2.2 of part A of IMO IS Code impracticable, the equivalent stability criteria given in paragraph 2.4 of part B of IMO IS Code, will apply. During operation, under the action of the heeling moment, the criteria under 3.2.4.2 to 3.2.4.4 will apply.

3.2.4.2 The residual area between the righting lever curve and the heeling lever curve calculated in accordance with 3.2.2 is not to be less than 0.070 [metre-radians]. The area is determined from the first intersection of the two curves, φ_e, to the angle of the second intersection, φ_c, or the angle of down-flooding, φ_f, whichever is less.

3.2.4.3 The static angle at the first intersection,φ_e, between the righting lever curve and the heeling lever curve calculated in accordance with 3.2.2 is not to be greater than:

.1 the angle at which the righting lever equals 50% of the maximum righting lever;

.2 the deck edge immersion angle; or

.3 15°,

whichever is less.

3.2.4.4 A minimum freeboard at stern, on centreline, of at least 0.005*L* is to be maintained in all operating conditions, with a displacement given by Δ_2, as defined in 3.2.2. In the case of the anchor retrieval operation covered by 3.2.3.3, a lower minimum freeboard may be accepted provided that due consideration has been given to this in the operation plan.

3.2.5 Stability Manual

3.2.5.1 The stability manual is to contain additional information on the following:

.1 maximum bollard pull, winch pull capacity and brake holding force;

.2 details on the anchor handling arrangement such as location of the fastening point of the wire, type and arrangement of towing pins, stern roller, all points or elements where the tension is applied to the ship;

.3 identification of critical down-flooding openings;

.4 guidance on the permissible tensions for each mode of operation and for each set of towing pins, including any physical element or arrangement that can restrict the wire movement, as function of all relevant stability criteria; and

.5 recommendations on the use of roll reduction systems.

3.2.6 Information to be displayed

3.2.6.1 Information stating the maximum allowed tension in the wire, as well as the corresponding angle, in accordance with the calculations performed for each loading condition, are to be communicated to the vessel's crew and displayed next to the control desk or at any location where the navigator on duty can easily see the information from his command post.

3.2.7 Precautions against capsizing

3.2.7.1 A stability instrument may be used for determining the permissible tension and checking compliance with relevant stability criteria. Two types of stability instruments may be used on board:

• either a software checking the intended or actual tension on the basis of the permissible tension curves; or

• a software performing direct stability calculations to check compliance with the relevant criteria, for a given loading condition (before application of the tension force), a given tension and a given wire position (defined by angles α and β).

3.2.7.2 Access to the machinery space, excluding emergency access and removal hatches, should, if possible, be arranged within the forecastle. Any access to the machinery space from the exposed cargo deck is to be provided with two weather-tight closures. Access to spaces below the exposed cargo deck is to be preferably from a position within or above the superstructure deck.

3.2.7.3 The area of freeing ports in the side bulwarks of the cargo deck is to at least meet the requirements of Regulation 24 of the International Convention on Load Lines, 1966 or the Protocol of 1988 relating thereto, as amended, as applicable. The disposition of the freeing ports is to be carefully considered to ensure the most effective drainage of water trapped in working deck and in recesses at the after end of the forecastle. In ships operating in areas where icing is likely to occur, no shutters are to be fitted in the freeing ports.

3.2.7.4 The winch systems are to be provided with means of emergency release.

3.2.7.5 The following anchor handling arrangements are to be considered:

.1 stop pins or other design features meant to impede the movement of the wire further outboard are to be installed; and

.2 the working deck is to be marked with contrasting colours or other identifiers such as guide pins, stop pins or similar easily identifiable points that identify operational zones for the line to aid operator observation.

3.2.7.6 The arrangement of cargo stowed on deck is to be such as to avoid any obstruction of the freeing ports or sudden shift of cargo on deck.

3.2.7.7 Counter-ballasting to correct the list of the ship during anchor handling operations is to be avoided.

3.3 Vessel Design

3.3.1 Design loads

3.3.1.1 The design loads(DL) defined below are to be taken not less than the maximum value of the tension in the wire (or chain cable) during anchor handling resulting from the stability analysis. Local strength of deck structure is to be based on the following design loads:

Deck Structure in way of	Design Load
Anchor handling Winch	max(1.5BP;1.5RP;BHL)
Guide Pins, Wire Stopper, Stern Roller	max (2SWL; 1.5RP; BHL)

3.3.1.2 Strength of anchor handling winch is to be based on the two following design loads DL:

Case 1:

DL = max (1.5 BP; 1.5 RP)

Case 2:

DL = BHL as defined in 3.1.4

3.3.1.3 Strength of deck equipment (guide pins, wire stopper, stern roller) is to be based on the same design loads as those considered for the deck foundations.

3.3.2 Deck

3.3.2.1 Structure

.1 Local reinforcements are to be provided in way of deck areas subjected to concentrated loads.

.2 The strength checks required in 3.4.4.1.1 to 3.4.4.1.3 are to be based on the following criteria:

a) Stress level in the deck structure is not to exceed the following permissible stresses:

- normal stress: $\sigma \leq 0.75R_{eH}$

- shear stress: $\tau \leq 0.47R_{eH}$

- equivalent stress: $\sigma_{VM} \leq 0.85R_{eH}$

where:

R_{eH} = Minimum yield stress, in [N/mm^2], of the material, to be taken equal to 235/k unless otherwise specified.

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2}$$

b) In case the yielding check of the structure is carried out by means of a three-dimensional finite element model, the permissible stress levels may be increased by 10 per cent.

c) Where necessary, buckling strength is to be checked.

.3 Strength of the deck structure is to be checked considering:

a) in way of anchor handling winch, including welds:

- design load as defined in 3.3.1.1, with the following assumptions:
  - wire considered at the most unfavourable layer of the drum, in general
  - direction of wire cable taken in the most unfavourable direction allowed by the anchor handling equipment.

b) in way of anchor handling deck equipment other than winches (i.e. guide pins, wire stopper), including welds:

- design load as defined in 3.3.1.1, exerted in the most unfavourable directions, with the most severe vertical location.

c) in way of stern roller, including welds:

- design load as defined in 3.3.1.1, considering an anchor hanging underwater below the stern roller at a negative angle of 30° with respect to a vertical axis.

d) in way of handling and/or lifting appliances, if any, including welds:

- the maximum dynamic reactions exerted by lifting appliance pedestal, according to the lifting appliance certification.

3.3.2.2 Reinforcement against Impact, Wear and Tear

.1 Plating thickness at the aft portion of the work deck is to be increased to protect the structure against heavy impact loads and wear and tear. It is recommended that minimum plating thickness in this area be not less than 25 [mm].

Alternative arrangements will be considered on case by case basis for re-enforcement against impact and wear and tear. Where heavy anchors and/or chains are carried on deck, suitable means for distributing their weights properly to deck structures are to be provided.

3.3.2.3 Work Deck Protection

.1 The aft deck areas exposed to anchor drags should not be fitted with sheathings or if present, the sheathings are to be suitably protected. In addition, any protrusion above deck such as coamings, manholes, lashing pad eyes, etc. are to be avoided. The deck plating thickness in these areas is to be suitably increased to allow for abrasion and mechanical damage.

3.3.2.4 Anchors and Chains Securing

.1 Pad eyes for securing and/or moving the anchors and/or chains are to be welded directly to the deck plating without doublers. The deck in way of the pad-eyes is to be adequately reinforced. Removable pad eyes are to have firm attachment to the deck sockets or holdings. All pad eyes are to be permanently marked with bead welded SWL values.

3.3.2.5 Cargo Rails

.1 Cargo rails are to be as per 2.4.5.5. The cargo rail may be constructed from stanchions topped with the rails, which are to be provided with line stoppers limiting the line's sweep. The line lay over the rail is to be such as to provide safe and unobstructed passage behind the cargo rails. Cargo rails installed athwartships just abaft of the anchor handling winch, are to protect the winch against damage from work deck operations, as well as to separate and support the wires/chains coming off the winch.

3.3.2.6 Arrangements for Shifting Anchors and Chains

.1 The foundations of tugger winches and/or capstans are to be welded directly to the deck plate and with adequate reinforcement underneath.

3.3.3 Weather Deck Openings

3.3.3.1 Openings in the weather deck leading to spaces below the freeboard or superstructure deck, including emergency exits, are to be protected as required in Pt.3, Ch.12, Sec.2, Cl.2.1.2 with sill height of doors at least as required by Pt.3, Ch.12, Sec.2, Cl.2.2.2.1 for companionways. Access openings, including emergency exits, are to be located clear off the towline sweep area.

3.3.4 Steering Gear

3.3.4.1 General

.1 Anchor handling vessels are to have suitable steering gear and arrangements to ensure course keeping capability under all normal anchor handling operation conditions.

3.3.4.2 Steering Gear

.1 In addition to the applicable requirements in Pt.4, Ch.4, the steering gear is to be capable of turning the rudder from 35° on one side to 30° on the other side within 20 seconds, when the vessel is running ahead at half the service speed. An alternative test procedure may be considered for demonstrating this capability.

3.3.4.3 Rudder

.1 Rudder size and rudder force are to be suitable for the expected anchor handling conditions and speed. The speed of the vessel used for the rudder design is not to be taken less than 10 knots.

3.3.4.4Thrusters

.1 Where azimuthal thrusters or azimuthing pods are fitted as necessary means for manoeuvring of the vessel, the provisions in Pt.4 Ch.4 are to be complied with.

3.3.4.5 Special Arrangements

.1 Steering units, thruster types and combination arrangements not explicitly covered by this sub-section will be considered based on the manufacturer's submittal regarding design and engineering analysis.

3.4 Anchor Handling Gear

3.4.1 General

3.4.1.1 In addition to compliance with applicable IRS Rules and statutory requirements, equipment and systems for anchor handling services are also to comply with the requirements in this sub-section. Alternatively, equipment complying with a recognized standard may be accepted provided the recognized standard is not less effective and the same can be verified from the design documentation submitted. Equipment and systems for anchor handling services are to be approved and inspected by IRS.
Vessels are to be fitted with the following items:

a) Anchor handling winch(es)

b) Stern roller for anchor handling operations

c) Guide pins in way of the stern roller for limiting transverse movement of the line or chain

d) Equipment for temporarily securing an anchor i.e. wire stoppers (e.g., shark jaw, etc.)

3.4.2 Arrangement

3.4.2.1 Work Deck

.1 Safe havens and escape routes for personnel from the work deck are to be properly marked and kept clear at all times. A crash barrier (cargo rail) fitted along each side of the deck and/or aft may be the method of providing a safe haven.

3.4.2.2 Gear

.1 All anchor handling related gears are to be located as low as practicable.

.2 Guide pins and wire stoppers are to be arranged so as to contain the anchor handling line within the design limits of its sweep and slack. Means are to be provided to lead and spool the line on the winch drum in a controlled manner under all foreseeable conditions and direction of the line.

3.4.3 Anchor Handling Lines

3.4.3.1 The anchor handling line may be either steel wire or fiber rope of the appropriate diameter.

3.4.3.2 The minimum breaking strength of the wire is not to be less than the design load defined in 3.3.1.1 for the anchor handling winch.

3.4.3.3 The anchoring handling line is to be protected from being damaged by chafing and abrasion. To this end, cargo rails, bulwarks, and all the elements supporting the towline are to be adequately rounded.

3.4.3.4 Strength of the wire attachment on the winch is to allow breaking in case the line is to be run out. Under normal operation, at least 3 turns of wire on the drum are considered.

3.4.3.5 Cargo rails, bulwarks, stern rail, tailgate and other structural elements on which the anchor handling line may bear during normal operations are to have a suitable bend radius.

3.4.4 Anchor Handling Winch and Accessories

3.4.4.1 Anchor Handling Winch

3.4.4.1.1 Design

.1 The combined stress is not to exceed the following permissible stress depending on the design load, for both cases defined in 3.3.1:

- for case 1: $\sigma_{VM} \leq R_{eH}$
- for case 2: $\sigma_{VM} \leq 0.9R_{eH}$

where:

R_{eH} = Minimum specified yield stress of material, in [N/mm^2]

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2}$$

.2 Evaluation by means of Finite Element Analysis, direct calculation, or equivalent, is to be provided for verifying the maximum stress and deformation of load bearing components of

winch under the required holding capacity or braking capacity respectively. Buckling and fatigue are to be considered according to recognized standard or code of practice.

.3 Each winch is to be provided with a power control braking means such as regenerative, dynamic, counter torque breaking, controlled lowering or a mechanically controlled braking means capable of maintaining controlled lowering speeds. Brakes are to be applied automatically upon loss of power or when the winch lever is returned to neutral.

3.4.4.1.2 Speed Control and Handling

.1 The hoisting or lowering speed of the winch is to be adjustable between stop and maximum speed.

.2 The winch operating controls for hoisting/lowering of the anchor should be designed to pay out the anchor by moving the control lever away from the winch operator and hoist the anchor by pulling the handle towards the operator. All winch operating controls are to be permanently marked with signs indicating their purpose and the operating direction.

.3 The winch operating controls are to be of the "hold-to run" type which will cause the hoisting or lowering motion to stop when released by the operator.

3.4.4.1.3 Tension Control

.1 Anchor handling winches are to be equipped with tension control in order to ensure that the system is not overloaded in cases of the anchor being handled is stuck or entangled or is exposed to similar situations.

3.4.4.1.4 Overload Alarm

.1 Winches are to be provided with continuous load monitors. Such monitors are to be fitted with an audible and visual overload alarm.

3.4.4.1.5 Control Panel

.1 The main control panel is to be placed in a position on the bridge which has a clear view of the deck area. Operators are to be able to visually monitor anchor handling winches and associated equipment and, if the view is obstructed, cameras or video monitoring devices may be used for this purpose.

The anchor handling winch may be controlled from more than one position remote from the winch provided that an arrangement to prevent more than one position from exercising control at any one time is fitted.

Each control station is to be provided with:

.1 means for two-way communication with the main control station;

.2 an arrangement to prevent inadvertent actuation;

.3 adequate protection of personnel; and

.4 sufficient illumination.

3.4.4.1.6 Spooling Devices

.1 Where necessary, anchor handling winches should be equipped with automatically operated automatic spooling devices except in cases where the winch can maintain the fleet angle within manufacturer specified limits without the use of spooling gear.

3.4.4.1.7 Emergency Release

.1 Emergency release of wire is to be provided on the bridge and at local stations. Control handles, buttons, etc. for emergency release are to be protected against unintentional activation.

.2 Emergency release is to be operable in all conditions and be independent of the main source of power.

3.4.4.1.8 Power redundancy

.1 Where the power supply for normal operation of the anchor handling winch is taken from the same source for propulsion, such as shaft generator, shaft power take-off (PTO), an independent (redundant) power supply with sufficient capacity for the winch operation is to be available to ensure the vessel’s maneuvering capability during anchor handling operations is not degraded.

3.4.4.2 Anchor handling deck equipment other than winches

.1 Strength of the deck equipment used for anchor handling, including guide pins and wire stopper, is to be checked under the maximum

load on the wire equal to design load as defined in Cl.3.3.1.1, exerted in the most unfavourable directions, with the most severe vertical location. Stress levels in the deck structure in way of anchor handling deck equipment are not to exceed the following permissible stresses:

- normal stress: $\sigma \leq 0.75R_{eH}$
- shear stress: $\tau \leq 0.47R_{eH}$
- equivalent stress: $\sigma_{VM} \leq 0.85R_{eH}$

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2}$$

In case the yielding check of the structures is carried out by means of a three-dimensional finite element model, these permissible stress levels may be increased by 10 per cent.

Where necessary, buckling strength is to be checked.

3.4.4.3 Stern roller

.1The structure of the stern roller is to be checked under design load as defined in 3.3.1.1, considering an anchor hanging underwater below the stern roller at a negative angle of 30° with respect to a vertical axis.

For strength purposes, design load as defined in 3.3.1.1 is to be considered in any transverse position on the stern roller, as allowed by the actual locations of the guide pins and the anchor handling deck equipment.

The bending, shearing and combined stresses are not to exceed the following permissible stresses:

- normal stress: $\sigma \leq 0.75R_{eH}$
- shear stress: $\tau \leq 0.47R_{eH}$
- equivalent stress: $\sigma_{VM} \leq 0.85R_{eH}$

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2}$$

Where necessary, buckling strength is to be checked.

.2 The length of stern roller (or rollers) is to be kept to a minimum, and sufficient to accommodate the widest anticipated anchor to be served. The minimum external diameter of the stern roller is to be:

Dsr = 17dw mm

where dw, is the nominal anchor handling wire rope diameter in mm

3.4.4.4 Maintenance

.1 For anchor handling winches and associated equipment, the inspection and maintenance is to be in compliance with the manufacturer's recommendations with applicable national standards of the Administration.

.2 Records of the routine inspection and maintenance of anchor handling winches should be maintained on board and made available to the Surveyor during surveys.

3.5 Tests

3.5.1 Load test

3.5.1.1 Load test is to be performed at the manufacturer workshop, for hoisting operation, and witnessed by the Surveyor.

3.5.1.2 Winches are to be tested at design load, as defined in 3.3.1.1. However, in case the winch is not of novel design, it is sufficient to perform load test at the design bollard pull BP. In this case, it may be performed on board during commissioning trials and is to be witnessed by the Surveyor.

3.5.2 Functional test

3.5.2.1 The tests are performed to check the proper:

.1 operation of the equipment within the specified limitations

.2 arrangement of the towline sectors and the towline paths, as shown on the arrangement drawing

.3 functioning under the normal operation modes

.4 functioning under the emergency operation modes, including the emergency release and the dead ship operations. In particular, the emergency quick-release systems are to be function tested at the design bollard pull BP.

3.5.3 Bollard Pull Test

3.5.3.1 The bollard pull test procedure is to be submitted for review by the attending Surveyor in advance of the test.

3.5.3.2 The bollard pull is to be measured with the vessel at the maximum continuous rpm and at or near the maximum towing depth. The bollard pull is the pull that is recorded over the state of equilibrium without any tendency to decline.

3.5.3.3 The depth of water under the keel in the testing area should be at least two times the vessel draft at amidships.

3.5.3.4 For additional test criteria, refer MSC/Circ.884.

Section 4

Safety Standby Rescue Vessels

4.1 General

4.1.1 Application

4.1.1.1 The requirements in this Section are applicable to Offshore Support Vessels intended to carry out rescue and standby services for offshore installations. These vessels are generally known as "*Safety Standby Rescue Vessels*" or "*Emergency Response and Rescue Vessels*". However, for the purpose of this Chapter these vessels will be referred to as "Safety Standby Rescue Vessels (SSRVs)"

4.1.1.2 Other Standards

.1 In order to ensure consistency with other applicable standards (e.g. national authorities, industry bodies), the requirements of this Section may be supplemented or replaced, as necessary, by any of the following recognized standards upon agreement of IRS:

• Canada Atlantic Standby Vessel Guidelines (June 2015 or later)

• Danish Maritime Authority Guidance on standby vessels (July 1997 or later)

• Norwegian Maritime Authority (NMA) Regulation No.853 on standby vessels (December 2014 or later)

• NMA Instructions to Class regarding Standby Vessels (February 2014 or later)

• NOGEPA Industry Guideline No. 6 (V1 - June 2004 or later)

• Oil & Gas UK - Emergency Response & Rescue Vessel - Management Guidelines (Issue 5 - April 2013 or later)

• Oil & Gas UK - Emergency Response & Rescue Vessel - Survey Guidelines (Issue 6 - April 2013 or later)

Standards not listed above would be considered by IRS on a case-by-case basis.

4.1.2 Notation

4.1.2.1 An additional qualifier notation **SSR** would be assigned to Offshore Support Vessels equipped to carry out rescue and standby services for offshore installations and designed and built to the requirements of this Section and other relevant sections of Chapter

Offshore Support Vessel (SSR-A(XX))

or;

Offshore Support Vessel (SSR-B(XX))

or;

Offshore Support Vessel (SSR-C(XX))

Where,

A, **B** and **C** indicates the category of the SSRV as defined in 4.2.1 and **XX** indicates the number of survivors the vessel can accommodate (e.g., **A(320)**, **B(250)**, **C(15)**, etc.).

4.1.3 Submission of Plans

4.1.3.1 Following additional plans are to be submitted for review:

i) An operating booklet is to be provided and sighted by the Surveyor on-board the vessel. It is to include information on the use of safety and rescue equipment and procedures to be

followed to evacuate persons from an offshore structure, from another vessel or for rescue from the sea, as well as information on the arrangements and use of the vessel's radio communication equipment required by these Rules.

ii) General arrangement (including accommodations for survivors)

iii) Rescue and safety equipment

4.2 Definitions

4.2.1 Safety Standby Service

.1 A vessel classed for Safety Standby Service is an unrestricted service vessel which has been adapted and has special features for the evacuation and reception of personnel from an offshore installation and the rescue and care of persons from another vessel or from the sea. Vessels which have been designed and constructed for other purposes, such as offshore supply, firefighting, diving support, etc. may be classed for safety standby service, provided the requirements of these Section are complied with. Depending on the number of survivors the vessel is capable of accommodating, safety standby vessels are categorized in three groups, as indicated in the following table:

Table 4.2.1.1 SSRV Groups based on number of survivors rescued

Category	Number of Survivors
Group A	>300
Group B	300 to 20
Group C	<20

All SSRVs are to have a length of 35 [m] and above except for Group C which is to have a length of at least 30 [m].

4.2.2 Survivor

4.2.2.1 A person evacuated from an offshore installation or from another vessel on account of an emergency or rescued from the sea is referred to as a 'Survivor'.

4.3 General Requirements

4.3.1 Propulsion and Manoeuvrability

4.3.1.1 SSRVs are to be capable in calm conditions of a speed of at least 10 knots and as a minimum, be equipped with any one of the following propulsion configurations:-

.1 A 360 degree azimuth bow thruster unit and either single screw propulsion with reversing gearbox or controllable pitch propeller. The bow thruster unit is to be capable of producing an ahead speed of 4 knots and is to be independent of the main engine for its source of power;

.2 Twin screw propulsion and bow thruster such that the SSRV is capable in calm conditions of at least 4 knots in the ahead direction with one main propulsion unit out of action;

.3 Equivalent main propulsion and manoeuvring systems providing similar levels of redundancy and manoeuvring capabilities.

4.3.1.2 Control of the propulsion machinery and station keeping system is to be possible by one person on the navigation bridge.

4.3.1.3 It is to be possible to turn the SSRV without headway or sternway and to hold it in a desired position and on a set heading while carrying out rescue operations, with sufficient accuracy to avoid hazard to the SSRV, other units or persons in the water, in all but exceptional weather.

4.3.2 Navigation Bridge

4.3.2.1 Visibility

.1 The navigating bridge deck is to be so designed that the person in charge is able to move easily from side to side of the SSRV, have an unobstructed view all round to within approximately 0.3 [m] of the SSRV's sides and, in particular, a clear sight of the Rescue Zones at the deck edge, the Winching Area and the areas in which rescue boats are launched and recovered. The points from which this view is obtained need not be fully enclosed. If it is necessary to move external to the bridge to obtain this viewpoint, means are to be provided to make sure that the observer is not isolated by reason of lack of visual or verbal contact with the occupants of the bridge interior.

.2 In general the navigation bridge is to be not lower than the second tier above the forecastle deck.

.3 Deck lighting is to provide adequate illumination of areas in which rescue operations

are being undertaken but is to be installed to prevent glare on the bridge.

.4 The design of vessel is to ensure full visibility of all operational areas, when operating in the SSRV role and when manoeuvring the vessel.

4.3.2.2 Bridge Windows

.1 Windows are to be in accordance with International or National standards in which the design pressures are given in terms of the windows longitudinal location and height above the design waterline. The thickness of the toughened safety glass is to be finalized in accordance with standard based on design pressure and size of window.

.2 Windows are to be fitted from the outside.

.3 Bridge front windows and those of the forward end of the bridge sides are to be provided with efficient means of being blanked off, internally or externally. Portable storm shutters may be used for this purpose, provided they are stowed in an accessible position, so as to be readily mounted and secured. Sufficient storm shutters are to be carried to cover any three bridge front and side windows. Storm shutters are to be of strength equivalent to the bulkhead and are to provide at least limited visibility from the bridge.

.4 Polarized or tinted windows are not to be fitted.

.5 The stiffening is to be arranged so that mullion stiffeners are continuous between decks, with horizontal stiffeners intercostal between the mullion stiffeners. The front, end and side stiffeners are to be in alignment with the stiffeners of the deckhouse below.

.6 Flat bars are not to be used for mullion stiffeners, where used elsewhere, flat bars are to have a depth to thickness not greater than 10.

4.3.3 Survivor rescue and recovery

4.3.3.1 Rescue Zone

.1 Rescue Zones are to be located clear of the effects of the propellers and thrusters and any fendering systems which might impede survivor recovery from the sea or rescue craft. They are to be created on each side of the SSRV for a total minimum length of 5 [m] and clearly indicated by highly visible markings. Where there are bulwark openings, or stanchions, the opening is to comprise a minimum accumulated unencumbered length of 4 [m]. Climbing aids are not to be less than 0.8 [m] wide and cover a minimum length of 4 [m].

.2 Cargo rails with height 1.83 [m] and less are to be of the openable or removable type. Where cargo rails are more than 1.83 [m] high, they are to be permanently attached and have sufficient openings of at least 9.14 [m^2] in close proximity to rescue zone (not more than 1 [m] on either side of zone) to perform rescue operations.

.3 Arrangements are to be provided for deploying boarding nets and for attachment of lifelines. In general, the rescue zone is to be free of hull fittings and projections and overboard discharges which would impede rescue operations. Electric lights in watertight fixtures are to be provided to illuminate the rescue zones and reception areas. Rescue zones are to be marked so as to be readily identified from the seaward side.

4.3.3.2 Recovery from the Sea

.1 Every SSRV is to be provided with a system to aid personnel climbing the ship's side from the sea or from rescue craft alongside. This system is to provide a rigid, non-slip climbing and grip surface than is provided by traditional rope scramble nets and be constructed from materials which are resistant to the marine environment. The system is to be designed to avoid injury to survivors and be secured to the SSRV in such a way that it hangs clear of the ship's side by at least 0.25 [m] when deployed.

.2 In addition, every SSRV is to be provided with at least one power-assisted method of recovering both able and disabled/ injured persons from the sea.

4.3.4 Freeboard

4.3.4.1 The loading condition for a SSRV is to be such that the highest freeboard mark would not be submerged if the number of survivors for which vessel is certified would be taken aboard. Weight of survivors is to be taken as 75 [kg] each.

4.3.4.2 The freeboard in the rescue zone area is not to exceed 1.75 [m]. A higher freeboard may be allowed for exposed locations if the ability to recover persons from the sea can be proved.

4.3.5 Emergency Power

4.3.5.1 SSRV is to be capable of simultaneous operation of all electrical equipment which may be required during an emergency situation and is to have adequate emergency power to allow rescue services to continue in the event of a main power failure. As a minimum, during the loss of main power, the SSRV is to be capable of continuing to launch and recover one Fast Rescue Craft (FRC), the operation of emergency lighting providing coverage as described in 4.5.6 and the operation of external communications equipment.

4.3.5.2 The emergency power supply to FRC/DC (see 4.5.11) davits is to be sufficient to permit recovery of the FRC/DC fully laden from the water to the stowed position within 5 minutes or less.

4.3.5.3 It is not expected that emergency power sources for rescue and survivor purposes should provide the same level of illumination as main power sources but they are to as sufficient to permit rescue and recovery and survivor treatment activities to continue on a restricted basis for at least 30 minutes.

4.4 Accommodation for Survivors

4.4.1 General Requirements

.1 All survivors are to be accommodated in spaces which afford protection from the elements and are adequately furnished with heating, lighting, ventilation and general conditions satisfactory for the survivors' comfort.

.2 In an emergency, survivors may be accommodated in crew accommodation except for sanitary accommodation, galley, berths for the master and two crew members, the radio room (where provided), the wheelhouse and main access passageways, which should be kept clear.

4.4.2 Seating Arrangements

4.4.2.1 The seating arrangement is be provided in accordance with the requirements given in the table below:

Table 4.4.2.1: Seating Arrangements

SSRV	Minimum Number of seats required[1]
Group A	66
Group B	50
Group C	20

Note:

1 The required minimum number of seats is including those in the Reception Area, available for survivors use, and additional to those provided in crew accommodation, with as many persons as possible provided with seating.

4.4.3 Access to and Means of Escape from Survivor Areas

4.4.3.1 All spaces intended for survivors are to be provided with safe access and means of escape.

4.4.4 Sanitary Facilities

4.4.4.1 In addition to the crew accommodation facilities, sanitary facilities as mentioned in the table below are to be provided.

Table 4.4.4.1: Sanitary Facilities

SSRV	Minimum Number wash hand basins, Showers and WCs
Group A	13 each
Group B	10 each
Group C	4 each

4.4.4.2 These facilities are to be readily accessible from spaces which may accommodate survivors.

4.4.5 Water for the use of Survivors

4.4.5.1 Amount of water to be provided for the use of survivors is to be as per table below.

Table 4.4.5.1: Domestic Water Requirements	
SSRV	**Minimum amount of water for the use of Survivors**
Group A	6 tonnes for washing and showers 6 tonnes for survivors' consumption
Group B	5 tonnes for washing and showers 4 tonnes for survivors' consumption
Group C	2 tonnes for washing and showers 2 tonnes for survivors' consumption

4.4.5.2 Where this water is separate from the crew supplies it is to be changed at least every 6 months.

4.4.5.3 The water heating arrangements are to provide for a constant thermostatically controlled supply of 160 [litres/hour] of water of at least 40 deg. celsius for 2 hours to each shower. In determining the heating input required to maintain the supply the amount of preheated water in any calorifier provided may be taken into account.

4.4.6 Survivor Area

4.4.6.1 The Survivor Areas are to be situated out with the crew accommodation and positioned to provide ready and easy access to/ from the Rescue Zones and Winching Area.

4.4.6.2 These areas are to be arranged to provide an easy flow of survivors including survivors on stretchers.

4.4.6.3 The necessity for vertical transfer is to be avoided to the maximum extent feasible.

4.4.6.4 The Treatment Room, Reception Area and at least 12 of the Recovery Area berths (including the 2 single berths) are to be located on the same deck as the Rescue Zones or the rescue craft disembarkation deck unless special arrangements for casualty transfer are demonstrated and accepted by IRS.

4.4.6.5 The declivity of the walkway or stairwell, and its width, is to be such that the transfer of injured parties on a stretcher can be easily carried out by those crew that would be available in an emergency situation.

4.4.6.6 Where vertical transfer is required either for helicopter winching or moving to recovery area, etc., there are to be practical means of transfer for stretcher cases between recovery area level and heli-winching level.

4.4.6.7 The Treatment Area and Recovery Area are to be segregated from other areas by bulkheads and/or curtains to ensure that survivors undergoing medical treatment or requiring rest and quiet are not unduly disturbed by the passage of other personnel.

4.4.6.8 The Survivor Areas are to consist of the following and be designed to allow an easy flow of survivors. The spaces, clearly marked for the guidance and ease of distribution of survivors, are to consist of:

a) Decontamination Area

b) Reception Area

c) Treatment Area

d) Recovery Area

e) Sanitary Area

4.4.6.9 Decontamination Area

.1 The Decontamination Area is to be equipped with a shower system suitable for cleaning of survivors and crew members before they proceed into the Reception Area.

.2 The arrangement of the Decontamination Area is to provide protection of the occupants from the weather but full enclosure is not required. The means of protection from the weather of the shower occupants need not be permanently installed.

4.4.6.10 Reception Area

.1 The Reception Area is to consist of a well-lit and heated space equipped with a desk and seat, filing and locker arrangements suitable for documenting survivors and additional seats for a minimum of 5 survivors.

.2 Arrangements are to be made for separating those needing treatment from able bodied survivors and providing the latter with changing facilities and access to the accommodation. Where possible, the changing facilities are to be contained within the Reception Area but, if lack of suitable space prevents this, alternative arrangements within reasonable proximity, e.g. within the seating area, may be accepted.

4.4.6.11 Treatment Area

.1 The Treatment Area is to have a deck area of not less than 15 [m^2] and provided with scuppers.

.2 Access and exits are to allow for easy transportation of stretchers.

.3 It is to be well-lit, heated and ventilated and comply with the requirements indicated in 5.18.2.

4.4.6.12 Recovery Area

.1 The Recovery Area, which is to be separated from but near to the Treatment Area to assist monitoring of the injured, is to provide a well-lit and heated space to accommodate survivor berths as mentioned in table below

Table 4.4.6.12.1: Berth Arrangements in Recovery Area

SSRV	Minimum number of berths to be provided
Group A	26
Group B	20
Group C	10

Note:

.1 At least two of the survivor berths are to be single tier.

.2 All survivor berths are to have lee boards, and upper bunks are to be equipped with bunk ladders, all berths are to be accessible by stretcher.

.3 Survivor berths are to be in accordance with IMO MLC Requirements.

4.4.6.13 Sanitary Area

.1 Following are to be provided in the Recovery Area:

Table 4.4.6.13.1: Sanitary Facilities in Recovery Area

SSRV	Minimum number of:		
	WashBasins	Showers	WCs
Group A	5	5	5
Group B	4	4	4
Group C	2	2	2

Note

.1 The above may be included in the total sanitary facilities stipulated in 4.4.4.1.

4.4.6.14 Morgue

.1 Facilities to store corpses in a cool, ventilated and illuminated space, with shelving and the means of securing corpses thereon, are to be provided. Storage capacity is to be as shown in table below:

Table 4.4.6.14.1: Morgue Capacity

SSRV	Minimum capacity of Morgue
Group A	13
Group B	10
Group C	4

.2 The location of the facilities is to permit safe access by stretcher and be screened from survivors.

4.5 Rescue and Safety Equipment

4.5.1 Rescue basket

4.5.1.1 Temporary refuge for survivors is to be supplied by a rescue basket fitted with retro reflective tape visible from the sides and above, and be provided with powered recovery arrangements. The powered recovery arrangements need not be dedicated to the rescue basket alone.

4.5.2 Lifebuoys

4.5.2.1 Vessels of all SSRV Groups are to carry at least twelve lifebuoys.

4.5.2.2 Two of the twelve lifebuoys are to be fitted with self-igniting electric lights and smoke

signals and two with self-igniting electric lights. The remaining eight are to be provided with 30 [m] buoyant lines and conveniently situated for easy access by the crew engaged in recovery of survivors.

4.5.3 Survivor Assistance Aids

4.5.3.1 All SSRVs are to be equipped with two devices, which may be of the same type and are to be capable of deploying a lifeline to survivors at distances of up to 30 [m] from the SSRVs' sides. Such equipment are not part of the statutory requirements for Life Saving Appliances and are to be certified by their manufacturers as providing the required range in still air.

4.5.3.2 In addition, vessels of all SSRV Groups are to be equipped with at least two devices besides the lifebuoys referred to in 4.5.2, to assist survivors who are close to or alongside the SSRV; to reach a position from which they may be recovered on board using the equipment referred to in 4.3.3.2.1 or 4.3.3.2.2. Extended hooks offering no hazard to survivors, e.g. "Dutch Hooks" or similar devices providing an extended reach from the SSRV's sides and capable of arresting and guiding the movement of a survivor, would satisfy this requirement.

4.5.4 Protective Water Spraying System

4.5.4.1 Arrangements are to be provided to protect from heat and fire those engaged in rescue operations on exposed deck areas, the boundaries of control stations and accommodation areas; by a water spraying system. The capacity of the system is to be such as to ensure overall wetting of the subject areas.

4.5.5 Person Overboard Alarms

4.5.5.1 A bridge operated "Person Overboard" alarm is to be installed which is clearly audible throughout the SSRV and is separate and distinct from all other alarms. In areas of high ambient noise levels, audible alarms may be supplemented by visual alarms.

4.5.6 Lighting

4.5.6.1 Electric lighting is to be provided in the following areas from both a main and emergency source of power:-

a) To illuminate Fast Rescue Crafts (FRC) stowage and launching areas and Survivor Reception, Treatment and Recovery Areas;

b) To provide floodlighting of the sea in way of the Rescue Zone and the FRC launch and recovery area;

c) To illuminate the helicopter winching deck area and access routes leading to and from the reception area.

4.5.6.2 In providing any flood lighting system, including emergency systems, the "run up" time from switch on to achieving adequate lighting levels is to be taken into account and, wherever possible, minimised.

Other requirements for the operation of emergency lighting may be referred in 4.3.5

4.5.7 Signalling Equipment

4.5.7.1 A daylight signalling lamp, mains and/or battery powered, plus a portable battery-operated loud hailer is to be provided on the bridge.

4.5.8 Location Aids

4.5.8.1 At least two directional searchlights are to be provided that together provide a capability of illuminating any area of the sea over a 360 degree arc and for a distance of at least 75 [m]. Such lights are to be capable of remote operation from within the bridge and be designed to illuminate as large an area of the sea as possible.

4.5.9 Helicopter Winching

4.5.9.1 All SSRVs are to have an illuminated, designated emergency winching area for the use of helicopters transferring personnel. The intention is to establish a safe emergency operating site offering the least possible hazard. As far as is practicable, it should be free of obstructions, which may cause injury during winching operations or the entanglement of winch wires. The deck surface is to be non-slip. The lighting indicated in 5.8 is to be installed such that it does not dazzle the helicopter crew.

4.5.9.2 Unobstructed stretcher access to the winching area is to be available.

4.5.10 Operations with Totally Enclosed Motor Propelled Survival Craft (TEMPSC) or Life Rafts

4.5.10.1 All SSRV's are to have documented procedures for and the means of:-

a) directly towing life rafts and TEMPSC;

b) securing such craft alongside and

c) transferring people both injured and uninjured from life rafts and TEMPSC.

4.5.11 Fast Rescue Craft

4.5.11.1 Definitions

.1 SOLAS Fast Rescue Boat: A rescue boat certificated as complying with Section 5.1 of the International Life Saving Appliance (LSA) Code adopted by Res. MSC 48 (66). These are sometimes referred to as FRB.

.2 Fast Rescue Craft: A specialised Craft, whether open (FRC) or enclosed (Fast Rescue Daughter Craft, or FRDC) meeting the criteria of this subsection. These craft may also meet the requirements of SOLAS Fast Rescue Boats.

4.5.11.2 General

.1 Irrespective of the number of DC's or FRC's fitted aboard an SSRV, there is to be at least one FRC/ FRB with its own launching and recovery arrangements that comply with the SOLAS requirements of a Rescue Boat, and is not to be of a length exceeding 8.5 metres.

4.5.11.3 Numbers and Capacities

.1 Every Group 'A' SSRV is to be equipped with at least three Fast Rescue Crafts, each capable of carrying 15 persons. Class "A" SSRVs may stow one FRC in a cradle to be launched when required by a crane that complies fully with these subsections.

.2 Every Group 'B' SSRV is to be equipped with a minimum of two Fast Rescue Crafts, both of which should be capable of carrying 15 persons.

.3 Every Group 'C' SSRV is to carry at least two Fast Rescue Craft. The second craft may be of a lesser capacity, capable of carrying only nine persons.

4.5.11.4 FRC Launch and Recovery Systems

.1 Each FRC is to be capable of being launched while the SSRV is making way and provided with its own launching system compliant with this Subsection.

.2 FRCs and/or FRDCs, are to be launched by davits, cranes or ramp launch systems.

.3 The launch and recovery system used to deploy these crafts are to be regarded as lifesaving appliances for load test purposes which is to include a five yearly dynamic overload test.

.4 Unless expressly requested by the SSRV owner, the weight of persons used for load tests is to be 82.5 kg.

.5 Fall wires are to be considered as LSA, have a factor of safety of six and be made from rotation-resistant and corrosion-resistant steel wire rope.

.6 The lowering speed is to be between 0.3 and 1.0 [m/sec] and the hoisting speed of a fully loaded FRC not less than 0.3 [m/sec] under normal power conditions.

.7 The height of the suspension point of any FRC launched from a standard davit should not in general, exceed a height of 7.5 [m] from the water in still water conditions and precautions should be taken to prevent swinging. The height of the suspension point is to be measured from the water level to the suspension point at the davits full out-reach. The suspension point on a davit with a docking head is the point where the davit head meets the boats' docking head. For davits not fitted with a docking head the suspension point is at the centre of the sheave from which the boat is launched.

.8 The lowering or raising of FRC is not to be impeded by side fenders or is to be accomplished well clear of the ship's side when using davit systems designed to control the FRC's movement.

.9 FRC davits are to be installed such that the withdrawal of any one davit from service does not affect the continued, unrestricted, availability of the other(s).

.10 In the case of hydraulic powered FRC launch and recovery systems, two hydraulic pumps are to be provided and the means of isolating hydraulic circuits arranged such that loss of supply to any one davit does not affect the continued operation of the other(s). Where one of the two sources is an emergency back-

up system, this does not require to be of the same capacity as the main source as long as it can achieve safe recovery of the fully laden FRC in an acceptable time. Hydraulic pumps may share the same hydraulic oil reservoir and pipe work up to the point of separate davit supply.

.11 Electrically powered launch and recovery systems are to have their electrical supply arranged such that failure of the motor(s) for any one davit or electrical supply to any one davit, does not affect the continued operation of any of the other davits. Common cable runs from generators and switchboard to the point of electrical isolation of individual davits are acceptable.

.12 For FRC launch and recovery arrangements under emergency power conditions refer to 4.3.5. While observing the above, care should be taken that when the SSRV is of such a size that FRCs are carried in lieu of lifeboats under equivalence clauses (specifically approved by the concerned flag Administration), the LSA Code requirements continue to be observed.

4.5.11.5 FRC Crew Lifejackets

.1 For each FRC, four lifejackets complying with EN 399 or IS 6685 standard or equivalent are to be provided. Inflatable lifejackets should give 275 Newtons of buoyancy and be designed to assure self-righting of the wearer when properly worn.

4.5.11.6 FRC Crew Protective Clothing

.1 Protective clothing appropriate to the working environment, including the possibility of capsize, is to be provided for all FRC crew.

.2 For vessels operating in extreme environmental conditions, consideration should be given to the provision of SOLAS Standard submersible, dry insulated suits.

.3 Clothing should be compatible with the lifejackets provided.

.4 Head protection is to be provided for all FRC crew meeting BS EN397: 1995 or BS 6658:85 or an equivalent standard.

4.5.11.7 FRC Maintenance

.1 Tools and spare parts as recommended by the manufacturer are to be provided to enable FRC and their engines to be maintained whilst the vessel is on standby duty.

.2 FRC engines are to be provided with the facility for running with the FRC in the inboard position and, where practical, should be started and run daily to ensure constant readiness.

.3 Each FRC fitted with petrol outboard engines is to have available in the near vicinity a spare engine of the same horsepower for each FRC. Lifting aids are to be provided to enable the crew to exchange FRC outboard engines in safety.

4.5.11.8 FRC Fuel Supply and Storage

.1 Adequate supplies of FRC fuel should be readily available in suitable safe storage.

4.5.11.9 FRC Equipment

.1 The following equipment are to be provided for each FRC, and properly secured:

a) An efficient radar reflector.

b) VHF radio equipment as set out in Section 6.2

c) A portable searchlight.

d) A compass.

e) A net or cradle system for recovery of persons from the water. The system should be designed to recover the person in a horizontal position with least effort by the crew.

f) A first-aid kit

g) Paddles

h) Boathook

4.5.12 Medical Stores and Equipment in Survivor Spaces

4.5.12.1 Stowage Locations

.1 The medical stores as required by the national authorities are to be stowed, clearly marked, in the appropriate locations. In case of absence of any guidance from national authorities, Appendix A of "*Oil & Gas UK - Emergency Response & Rescue Vessel - Survey Guidelines (Issue 6 - April 2013 or later)*" is to be followed.

.2 Oxygen bottles are to have suitable secure means of stowage both in the Treatment Room and Recovery Area.

4.5.12.2 Treatment Area Outfitting

.1 The Treatment Area should be equipped with:-

a) High luminance illumination with special note taken of lighting directly above the treatment table;

b) Wash hand basin provided with hot and cold water supplied via elbow-operated taps;

c) Treatment table accessible from both sides and the foot;

d) Portable shower head provided with a thermostatically regulated water supply via an extended hose capable of reaching both ends of the treatment table;

e) Fixed frame(s) for holding two stretchers and their occupants securely;

f) Moveable instrument table capable of being secured to the treatment table;

g) Waste bin for soiled clothing and/or dressings;

h) Cupboards enabling the stowage of medical stores in a systematic and readily available format;

i) Desk with file drawers for the stowage of medical documents;

j) Hands free radio equipment capable of direct communications with medical advisors onshore and offshore.

k) A bulkhead-mounted clock with sweep second hand

4.5.12.3 Recovery Area Preparation

.1 The Recovery Area is to be equipped such that is ready to receive survivors with the minimum of preparation. Hooks are to be located at bunks for the suspension of intravenous fluid bags.

4.5.12.4 Medical Inventory

.1 An inventory of all medical equipment is to be maintained on board in a format that permits rapid confirmation of current stock levels and expiry dates (where appropriate). The inventory is to be made available for examination by surveyors.

4.5.12.5 Survivor Showers

.1 Each shower is to be provided with soap, also large size dispensers, containing skin degreasing fluid. Degreasing dispensers may be shared between users on the basis of one dispenser to each two showers depending upon the shower arrangements. The total amount of skin degreasing fluid on board should be 100 litres, 75 litres and 35 litres for Group A, B and C SSRVs respectively

4.6 Navigation and Communication Equipment

4.6.1 General

4.6.1.1 Effective arrangements are to be made for radio communications, on appropriate frequencies, at all times between the SSRV and the Installation which it serves and between the SSRV and the appropriate coastal station. This may include MF, HF, VHF and Satellite Systems. Alternative means of communication between the Installation(s) and the SSRV for use in the event of radio or power failure.

4.6.1.2 Vessels are to be equipped with a voice/fax/data Satcom, UHF telephone patch and a mobile phone.

4.6.1.3 Radio communication equipment used by rescue crafts is subject to particularly adverse operating conditions. Its construction and protection should reflect the environment in which it operates.

4.6.1.4 The radio equipment noted in the following subsections is additional to that required by Flag State, Global Maritime Distress and Safety System (GMDSS), and SOLAS Regulations.

4.6.2 FRC Radio Communications

4.6.2.1 FRC radio communication is to be provided for by either:-

a) A permanently installed VHF radio with hand-microphone and waterproof loudspeaker for use by the coxswain plus a radio helmet complete with headphones, microphone and a portable VHF radio in a weather tight housing for one crewman

or

b) Two radio helmets, fit for purpose, equipped with earphones and microphone connected to a waterproof radio or one housed in a waterproof container.

or

d) Alternative arrangements which can be demonstrated to provide means of radio communication between the coxswain and other craft, vessels or Installations, without hazarding his control of the FRC and also providing back-up in the case of the failure of the coxswain’s radio. At least two VHF are to be used in the FRC to achieve this purpose and the arrangements should minimise the effect of interference from external noise upon communication with the FRC.

4.6.3 SRRV Internal Communications

4.6.3.1 Internal communications on the SSRV are to be sufficient to provide communication between the bridge, Survivor Reception and Treatment areas, the Master's cabin, those spaces occupied by crew members on immediate call and the Winching Area.

4.6.3.2 Where the Survivor Recovery Area is not immediately adjacent to the Treatment Area, this should also be added to the internal communication system and for this purpose inclusion in a talk-back or telephone system or the provision of a dedicated portable radio and charger are acceptable.

4.6.3.3 In addition, a fixed, talk-back intercommunication system is to be provided between the bridge, the rescue zone and the FRC launching stations.

Section 5

Wind Farm Service Vessels

5.1 Scope

5.1.1 The requirements of this Section are applicable to Wind Farm Service Vessels (WFSVs) and are in addition to other relevant and applicable requirements in this Chapter.

5.1.2 Wind Farm Service Vessels are vessels serving offshore wind farm activities. These are primarily engaged in the following activities:

i) Transportation of persons from the shore/ offshore accommodation ships/ units to the wind farm and vice versa.
ii) Transfer of light cargoes (as necessary to support the functions of the wind farm) to and from the wind farm, using the lifting appliances installed onboard for this purpose.

5.1.3 WFSVs (in the scope of this Section) are intended to operate solely on domestic voyages within 200 nautical miles from the nearest onshore refuge, considering the present status of technology of wind farms. However, WFSVs which may operate on short international voyages will be specially considered by IRS.

5.1.4 Vessels/ units performing major work on wind farms such as replacement or repair of damaged blades, towers, foundations, nacelles, wind turbine power generation and other equipment etc. which involve lifting operations of these components will not be classed as WFSVs. For such vessels/ units, reference is made to IRS Classification Note: *Wind Turbine Installation Units.*

5.1.5 The following items are additionally included in the scope of Classification for Wind Farm Service Vessels:

i) permanently installed arrangements for personnel transfer
ii) lifting appliances for transfer of light cargoes
iii) dynamic positioning systems in accordance with the requirements in Chapter 24. WFSVs are to be fitted with DP(2) or DP(3) systems

5.1.6 For vessels carrying industrial personnel, the applicable requirements of SOLAS Chapter XV are to be complied with. Where, in addition to industrial personnel, the vessel carries more than 12 passengers on-board, the vessel is to be classed in accordance with Part 5 Chapter 6 of the Rules as a Passenger vessel. (Note: *Industrial personnel (IP)* means all persons transported or accommodated on board for the purpose of offshore industrial activities performed on board other ships and/or offshore facilities.)

5.2 Class Notations

5.2.1 Vessels complying with the requirements of this Section may be assigned the following additional class notations as applicable:

WFSV – This notation may be assigned to such vessels which are not high speed crafts.

WFSV – HSC – This notation may be assigned to vessels which additionally comply with the requirements applicable to high speed crafts, as given in the IRS *Rules and Regulations for Construction and Classification of High Speed Crafts and Light Crafts*.

5.2.2 The service restrictions in terms of area of operation and environmental conditions are to accompany the class notations for the WFSVs. These would typically be as follows:

e.g. “For Operation in Maximum Significant Wave Height of 2m”.

5.2.3 Operational weather condition limits for transfer of personnel are also to be specified.

5.3 Requirements for Personnel Transfer Arrangements, Systems and Equipment

5.3.1 Designated location(s) and platform(s) are to be provided for personnel transfer. The locations and platforms are to be clearly visible from the bridge and provided with sufficient illumination having a level of at least 200 lux.

5.3.2 The transfer location and platform deck are to have non-skid surfaces.

5.3.3 Handrails and guardrails (permanent or temporary) are to be installed at the transfer location and platform.

5.3.4 The personnel transfer location and platform are to be as far as practicable from the propellers/ thrusters.

5.3.5 Gangways are to be certified according to recognized standards such as ISO 7061:2015. The supporting structure on the vessel for such gangways is to be checked for its structural adequacy.

5.4 Requirements for Fender System

5.4.1 A suitable fender system is to be installed for safe transfer of the berthing forces between the wind turbine structure boat landing area and the vessel.

5.5 Lifting Appliances

5.5.1 Lifting appliances to be used for transfer of personnel baskets or cargoes/ supplies are to be certified in accordance with recognized standards such as EN 13852:2013, API Spec 2C, etc..

5.5.2 Intact stability for vessels engaged in lifting operations is to be assessed in accordance with IMO Intact Stability Code 2008 (as amended), Part B, Chapter 2, 2.9.

End of Chapter

Chapter 9

Trawlers and Fishing Vessels

Contents

Section

1 *General*

2 *Hull Arrangement and Strength*

3 *Fish Holds*

4 *Pumping and Piping Arrangement*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to trawlers and fishing vessels, and are supplementary to those given for the assignment of main characters of class.

For the purpose of this chapter, a fishing vessel is a ship used for fishing operations, but not equipped for trawling.

1.1.2 Vessels built in compliance with the above requirements will be eligible to be assigned one of the following class notations:

- **"TRAWLER"**
- **"STERN TRAWLER"**
- **"FISHING VESSEL"**

1.2 Documentation

1.2.1 Following documents are required to be submitted in addition to those specified in Pt.3, Ch.1, Sec.3.

- Winch and crane foundations along with the winch capacities and wire forces.
- Mast and rigging plans.
- Additional deck loads in way of masts, rigging, heavy machinery and equipment.
- Bilge and drainage arrangement from cargo spaces and working areas.
- Details of insulated compartments.

Section 2

Hull Arrangement and Strength

2.1 General

2.1.1 The draught T, used for determination of scantlings is not to be taken less than 0.9D.

2.1.2 All fishing vessels and trawlers are to have pronounced sheer on the forebody or a forecastle.

2.2 Longitudinal strength

2.2.1 The requirements of Pt.3, Ch.5 regarding the loading instruments are not applicable to trawlers and fishing vessels.

2.3 Bottom and side shell structure

2.3.1 The thickness of bottom and side shell plating is not to be less than :

$$t = 6 + 0.06L \text{ [mm]}$$

2.3.2 In the case of vessels engaged in side trawling, the thickness of sheer strake, side shell and bilge strake are to be increased by 30% over a length of not less than 4 [m] abaft the forward gallow leg.

2.3.3 In case of stern trawlers, the thickness of trawl ramp and ramp sides, stern and side shell plating abaft the point where trawling boards are normally taken on board; is to be increased by 30%.

2.3.4 Bulwark plating between the gallows is to be of the same thickness as the adjacent side shell plating and the bulwark is to be supported by stays at every frame.

2.3.5 Bulwark, side shell and transom plating where excessive wear could occur, are to be suitably protected with rubbing bars.

2.3.6 The section modulus of stiffeners in the trawl ramp is not to be less than:

$$Z = 0.015\, l^2 s \quad [cm^3]$$

where,

l,s are the stiffener span [m] and spacing [mm] respectively.

2.4 Deck structure

2.4.1 The thickness of deck plating under the trawl winch, windlass, mast, bollards and gallows is not to be less than:

$t = 7.5 + 0.05L$ [mm]

2.4.2 Adequate scarphing is to be arranged in way of raised decks.

2.5 Bulkhead structure

2.5.1 A collision bulkhead located at a distance greater than 0.08L from the forward end may be considered, provided it is shown that flooding of the forepeak spaces would not cause a deleterious effect on the ship's trim and stability.

2.5.2 Cargo holds designed to carry fish in bulk are to have sufficient number of divisions to ensure that the catch is adequately secured against cargo shifting which could cause dangerous trim or heel of the vessel.

2.6 Hull openings and their closing appliances

2.6.1 Hull openings and their closing appliances are to be according to Pt.3, Ch.12 except as specified otherwise.

2.6.2 Skylights leading to accommodation and machinery spaces below the freeboard deck are to be of substantial construction and capable of being closed weathertight from both sides and be positioned clear of deck working areas. The coaming height of the skylights should be not less than 900 [mm]; lower coaming height may however be considered depending on the practicability of operations and integrity of the weathertightness. Scuttles may be fitted on skylights for accommodation spaces only and are to be of toughened glass fitted with deadlights.

2.6.3 Hatchways are to be fitted with steel coamings, complete with all necessary fittings and covers to ensure weathertight closure. Covers made of materials other than mild steel are to be of equivalent strength. For vessels of $L < 24$ [m], hatch coaming and/or sill heights of less than 600 [mm] may be accepted depending on the location and size of the opening and the construction details of closing appliances.

2.6.4 Access, loading and discharge hatches on the freeboard deck are to be located along the ship's centreline as far as practicable. Ice scuttles and other small flush type hatches may however, be located away from the centreline, subject to satisfactory means of weathertight closing.

2.6.5 Openings on the deck sides for fishing lines are to be so constructed as to facilitate quick closing of the weathertight cover by one member of the crew. The sill height of such openings is normally not to be less than 1000 [mm]. Side ports, if fitted are to have securing devices with a strength equivalent to the structure to which they are fitted. Suitable notices are to be provided to all such openings stating "TO BE KEPT CLOSED AT SEA WHILE NOT ENGAGED IN FISHING".

2.6.6 Stern trawlers are to be provided with suitable protection such as doors, gates or nets at the top of the stern ramp up to the same height as the adjacent bulwark or guard

rail. Where such protection is not in position, a chain or other means of protection are to be fitted across the ramp.

2.6.7 Inboard openings for garbage chutes from factory decks and galley are not to be less than 0.7 [m] above the load waterline and are to be fitted with weathertight covers. The outboard end is to be fitted with watertight screw down non-return valve operable from 1.5 [m] above the deck.

2.7 Masts and rigging

2.7.1 Masts and rigging are to comply with the requirement of Pt.3, Ch.16. Derricks and posts used in conjunction with side trawling are to be approved by IRS.

2.7.2 Where gallows and/or gantries are fitted, they are to be complete with all necessary leads, cleats, hanging blocks, eyebolts, stopper chains and supporting stays, and arranged to allow safe stowage of the trawl doors.

2.7.3 Warping bollards and leads are to be provided with guards as far as practicable and are to be located clear of the working areas.

Section 3

Fish Holds

3.1 Temporary bulkheads in cargo holds

3.1.1 Generally, for vessels of B $\leq$ 6 [m], one longitudinal bulkhead is to be fitted and for vessels of B > 6 [m] two such bulkheads are to be fitted. Spacing of transverse bulkheads in the cargo holds is normally not to exceed 9 [m].

3.2 Refrigerated fish hold

3.2.1 Refrigerated fish hold spaces are to be comply with the requirements of Pt.5, Ch.23, Sec.3.

Section 4

Pumping and Piping Arrangement

4.1 Piping in way of refrigerated chambers

4.1.1 Piping in way of refrigerated chambers is to comply with the requirements of Pt.4, Ch.3 and Pt.5, Ch.23; as applicable.

4.2 Drainage from refrigerated fish hold

4.2.1 Provision is to be made for adequate and continuous drainage of water, oil and brine from inside of all insulated chambers and cooler trays. The drainage arrangement is to comply with the requirements of Pt.4, Ch.3.

4.3 Bilge arrangements of spaces for carriage of fish in bulk

4.3.1 Each fish hold of length less than 9 [m] is to have a bilge well of adequate size suitably located to ensure good drainage of water, oil and brine from the cargo.

Fish holds more than 9 [m] in length are to have minimum two bilge wells.

4.3.2 Separate branch suction lines are to be led from each of the bilge well to the bilge system valve chest in the engine room. The valve chest collecting branch suction lines from the fish holds is to have no connections to other compartments. All valves in the chest are to be screwdown non-return valves and are to be easily accessible. The valve chest is to be directly connected to the largest bilge pump. The second bilge pump should also be connected to the system.

4.3.3 Suitable arrangement is to be made between the bilge suction and the non-return valve to connect a portable water supply to carry out back flushing of each bilge suction.

4.3.4 Tween deck spaces not exposed to the sea and intended to carry loose fish in bulk may be drained into the engine room bilges provided no processing is carried out in that space requiring supply of water. The drainage pipes are normally not to exceed 50 [mm] in diameter and are to have self closing valves on the engine room side.

4.4 Drainage arrangement for working deck

4.4.1 Vessels with processing facilities on the tween deck requiring supply of water for washing down the remains and vessels with side ports or flush deck hatch covers which remain open while fishing at sea, are to have a substantial bilge system similar to the requirements of Sec.4.3. For working decks of length greater than 18 [m], the number of drainage wells will be specially considered.

4.4.2 The volume of each drainage well is not to be less than 0.15 [m^3] and a loose strain-off grating is to be provided on top having clear opening area not less than 3 times the internal sectional area of the bilge suction pipe. The sizes of the bilge suctions are also to be compatible with the capacity of the bilge pumps.

4.4.3 Where only two bilge wells are provided, additional suctions are to be arranged on each side.

4.4.4 Each drainage well is to be provided with an independent bilge suction. The piping system is to be so arranged as to ensure alternative drainage facility from other bilge suctions when any one of the bilge suctions may be clogged.

4.4.5 In addition to the bilge suctions required in 4.4.3 above, outlets from the bilge wells may be led overboard provided the inboard drainage opening on the deck are situated 0.02L (min. 0.7 [m]) above the load waterline. Each such outlet is to have an approved type automatic non-return valve with manual means of closing, operable from a position 1.5 [m] above the deck. In addition, remote closing arrangement is to be provided from the bridge. The outlet flap opening is to be free from obstructions to prevent clogging by offal that may impair the closability of the flap. Direct overboard discharge may be considered in lieu of the additional suctions required in 4.4.3.

4.4.6 Submerged bilge pumps with rotating knife edge blades may be installed in the bilge wells without the strain-off gratings. The discharge from the submerged pump is to be fitted with screw-down non-return valve capable of operating from 1.5 [m] above the deck. The outboard opening is to be located 0.02L (min. 0.7 [m]) above the load waterline.

4.4.7 The bilge pumping capacity for direct suction from each well is not to be less than 1.25 times the available wash down capacity for the deck wash system, nor less than the requirement of bilge pump capacity as per Pt.4, Ch.3.

4.4.8 The total capacity Q of bilge pumps for drainage of spaces with side ports is not to be less than:

$$Q = 3BA_s \ [m^3/h]$$

where,

A_s = area of the side port [m^2]

The total bilge pump capacity, however, need not exceed

$Q = 0.75\ A_D\ [m^3/h]$

where,

A_D = deck area of the compartment fitted with side ports [m^2].

4.5 Alarms and cutouts

4.5.1 Visual and audible alarms are to be fitted on the bridge for the free water on the working deck.

4.5.2 The pumps for the deckwash system are to be capable of being shut down from the bridge in case of emergency.

End of Chapter

Chapter 10

Dredgers

Contents

Section

1 *General*

2 *Hull Arrangement and Strength*

3 *Miscellaneous*

4 *Self-Unloading Barges*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to self-propelled or non-self-propelled steel vessels engaged in dredging or reclamation operation and are supplementary to those given for the assignment of main characters of class.

1.1.2 Vessels complying with the above requirements will, in general, be eligible to be assigned one of the following class notations :

- **'DREDGER'**, for dredgers designed to operate wholly or generally for the purpose of raising spoil such as mud, silt, gravel, clay, sand etc. from the bed of sea, rivers, lakes, canals or harbours. The dredged material may be placed in suitably designed holds or similar spaces within the ship.

- **'HOPPER DREDGER'**, for dredgers designed to raise spoil and so arranged that the dredged material may be placed in one or more hoppers (with arrangements to discharge the spoil through doors or valves in the bottom of the ship).

- **'RECLAMATION CRAFT'**, for vessels which receive spoil from a dredger and discharge it ashore.

- **'HOPPER BARGE'**, for non-self-propelled vessels which carry spoil or the dredged material in hoppers within the ship.

- **'SPLIT HOPPER BARGE'**, for vessels similar to hopper barges but with arrangements for discharging the spoil through the bottom of the ship by means of the split hull, separated at the hinges using actuating devices.

1.1.3 In addition to the above notations, appropriate description may be entered in the Register Book to indicate the type of dredging or reclamation craft (e.g, 'TRAILING SUCTION DREDGER', 'CUTTER SUCTION DREDGER', 'BUCKET DREDGER', 'GRAB DREDGER' etc.).

1.1.4 Bucket dredgers and non-propelled vessels will, in general, be assigned unrestricted service notation only after special consideration.

1.2 Documentation

1.2.1 In addition to the documents required as per Pt.3, Ch.1, Sec.3, the following details are to be submitted.

- General arrangement plan, showing also the arrangement of the dredging equipment.

- Longitudinal and transverse hopper bulkheads, with information regarding density of the spoil and height of overflow.

- Arrangement and scantlings of substructures attached to or integrated into main structural members, such as gantries, gallows, dredging machinery and pumps, hopper, doors and their operating gear, positioning equipment, other dredging equipment and devices and their seats.

1.3 Statutory requirements

1.3.1 Attention is drawn to the special requirements imposed by National Authorities involving loading limitations and stability requirements. Certain types of dredge material with thixotropic properties can cause shifting of spoil within the spoil spaces, resulting in undesirable changes in trim or angles of heel. This can lead to dangerous situations in ships with closed top spaces.

Section 2

Hull Arrangement and Strength

2.1 General

2.1.1 While determining the principal dimensions of dredgers, local structures and deviations from the principal design dimensions associated with the attachment of the dredging gear, are to be ignored.

2.1.2 Where freeboards less than minimum freeboards are assigned, the length L, draught T and block coefficient C_b are to be determined considering this freeboard.

2.2 Longitudinal strength

2.2.1 For dredgers and other reclamation craft, the longitudinal strength requirements apply as per Pt.3, Ch.6.

2.3 Bottom structure

2.3.1 Where hopper doors are fitted on the vessel's centreline or where there is a centreline well for dredging gear (bucket ladder, suction tube etc.), a plate strake is to be fitted on each side of the well or door opening. The width of this strake is not to be less than 50% of the rule width of the flat keel and thickness not less than that of the rule flat keel.

The same applies where the centreline box keel is located above the base line at such a distance that it cannot serve as a docking keel. In addition, brackets are to be fitted at the longitudinal bulkheads of dredging wells and hopper spaces. These brackets are to extend to the adjacent longitudinals and longitudinal bulkhead stiffeners. Where the spacing of bottom transverses is less than 2.5 [m], one such bracket is to be fitted and for greater spacings, two such brackets are to be fitted. The thickness of the brackets is not to be less than the web thickness of the adjacent bottom transverses. The free edge of the brackets is to be stiffened. Where longitudinal bulkheads and the side shell are framed transversely, the brackets are to be fitted at every frame and are to extend to the bilge.

2.3.2 The corners of hopper door openings and of dredging gear wells generally are to comply with the requirements of Pt.3, Ch.3, Sec.6.5.4 and 6.5.5. The structural details and welded connections in this area are to be designed with particular care.

2.3.3 In transversely framed single bottoms, the height of floors in spaces abreast of hoppers or centreline dredging wells, is to be calculated as per Pt.3, Ch.7, Sec.3.3 using the actual breadth in these spaces (with a minimum of 0.4B) in place of B.

Floors, longitudinal girders etc. below dredging machinery and pump seats are to be adequately designed for the additional loads.

Where floors are additionally stressed by the reactions of the pressure required for closing the hopper doors, their section modulus and their depth are to be increased accordingly.

Floors in line with the hopper lower cross members fitted between hopper doors are to be connected with the hopper side wall by brackets. The free edge of the brackets is to be stiffened.

2.3.4 In longitudinally framed single bottoms, the spacing of bottom transverses generally is not to exceed 3.6 [m]. Section modulus and web cross sectional area are to be increased by 10% and 20% respectively over that required by Pt.3, Ch.7, Sec.6.

The bottom transverses are to be stiffened by means of flatbar stiffeners, at every longitudinal, of depth equal to the depth of the bottom longitudinals; however, it need not exceed 150 [mm].

2.3.5 Double bottoms need not be fitted in way of the spoil spaces.

Where it is intended to unload the spoil by means of grabs and a double bottom is fitted in way of spoil spaces, plate floors are to be fitted at each frame in case of transverse framing system.

2.3.6 On non-self-propelled dredgers and on self-propelled dredgers with restricted service notation, strengthening of the bottom forward in accordance with Pt.3, Ch.11, Sec.3 need not be provided.

2.3.7 The flat bottom plating of swim ends is to have a thickness not less than that of the rule bottom shell plating amidships, up to 500 [mm] above the maximum load waterline.

2.3.8 Unless specified otherwise, non-self-propelled dredging and reclamation crafts are considered to operate while aground.

In such cases, the bottom structure is to be additionally strengthened as follows:

The thickness of the bottom shell plating is to be increased by 20% above the value required as per Pt.3, Ch.7, Sec.4; however, it is not to be less than 8 [mm].

In transversely framed single bottom construction, floors are to be fitted at every frame, with vertical buckling stiffeners spaced not more than 1.2 [m] apart.

Where the single bottom is longitudinally framed, the spacing of the bottom transverses is not to exceed 1.8 [m]. The webs are to be stiffened to withstand buckling. Side girders, spaced not more than 2.2 [m] apart, are to be provided. The section modulus of the bottom longitudinals is to be increased by 50%, over that required by Pt.3, Ch.7, Sec.5.

In double bottom construction, the requirements as specified for single bottoms, are to be applied analogously.

2.4 Hopper and well structure

2.4.1 The net plate thickness, in [mm], of the boundaries of hopper spaces and wells is to be determined in accordance with Pt.3, Ch.7, Sec.4 and not to be less than the minimum value as prescribed in Pt.3, Ch.7, Sec.3.

2.4.2 Section modulus of stiffeners of boundaries of hopper and well spaces are to be as per the requirements of tank bulkheads given in Pt.3, Ch.7, Sec.2 and Sec.5 using the pressure given in Pt.3, Ch.5.

2.4.3 In general, the strength of longitudinal bulkheads of hopper spaces and wells is to be at least equal to that of the ship's sides.

The top and bottom strakes of the longitudinal bulkheads are to be extended through the end bulkheads, or else scarphing brackets are to be fitted in line with the bulkheads in conjunction with strengthenings at deck and bottom.

Where the length of wells does not exceed 0.1 L and where the wells and/or ends of hopper spaces are located beyond 0.6L amidships, special scarphing may not be required.

2.4.4 In vessels fitted with hopper doors, lower cross members are to be fitted between the doors at spacing generally not exceeding 3.6 [m].

The depth of the cross members is not be less than 2.5 times the depth of floors as per Pt.3, Ch.7, Sec.5. The web plate thickness is not to be less than the thickness of the side shell plating. The top and bottom edges of the cross members are to be fitted with face plates. The thickness of the face plates is to be at least 50% greater than the Rule web thickness.

Vertical stiffeners spaced not more than 900 [mm] apart are to be fitted on the webs.

Where the lower cross members are constructed as watertight box girders, the scantlings are not to be less than required by Sec.2.4.1. The top plate is to be increased in thickness by at least 50%.

2.4.5 The transverse bulkheads at the ends of the hoppers are to extend the full breadth of the vessel.

2.4.6 Wash bulkheads or transverse ring structure is to be provided in the wing spaces abreast of the hopper, in line with the lower cross members.

Upper cross members are to be fitted at deck level in line with the ring structure, which are to be supported by means of pillars at the box keel, if fitted. The scantlings of the upper cross members are to based on the actual loads arising from hopper door supporting arrangements. The maximum reactions of hydraulically operated rams, where fitted for hopper door operations, are to be taken as the actual load.

2.4.7 On bucket dredgers, the ladder wells are to be isolated by transverse and longitudinal cofferdams at the bottom, of such size as to prevent the adjacent compartments from being flooded in case of any damage to the shell by dredging equipment and dredged objects.

2.5 Box keel structure

2.5.1 The scantlings of the bottom plating of box keel are to be as follows:

Where the box keel can serve as a docking keel, the requirements for flat plate keels apply.

In other cases, the requirements for bottom shell plating apply.

2.5.2 The remaining plating of the box keel are to be as follows :

- Outside the hopper space, the requirements for bottom shell plating apply.

- Within the hopper space, the requirements for hopper space plating as per 2.4.1 apply. The thickness of the upper portion particularly subjected to damage is to be increased by at least 50%.

2.5.3 Box keel plating is to be adequately stiffened in consonance with the adjoining hopper well structure.

2.5.4 Plate floors are to be fitted within the box keel in line with the web frames as per 2.4.6 to ensure continuity of strength across the vessel.

2.5.5 The ends of the centreline box keel are to be effectively scraphed to the vertical keels or equivalent structure at the ends of the hopper spaces.

Section 3

Miscellaneous

3.1 Equipment

3.1.1 The requirements for equipment of non-self-propelled dredgers are the same as that for barges.

3.1.2 Considering rapid wear and tear, it is recommended that the anchor chain cables, where additionally employed for positioning of the vessel during dredging operations, be increased in size.

3.2 Rudders

3.2.1 In the case of bucket dredgers having bucket ladders at their stern, the ship's rudders are to be kept well clear of the buckets to minimize the possibility of damage to the rudders by large objects that may be dredged up.

3.3 Bulwark and overflow arrangement

3.3.1 Bulwarks are not to be fitted in way of hoppers where the hopper weirs discharge onto the deck instead of into enclosed overflow trunks. Even where overflow trunks are provided, it is recommended not to fit bulwarks. Adequate protection for crew is to be provided by suitable alternate means in such cases.

However, if bulwarks are fitted, freeing ports are to be provided throughout their length which should be of sufficient width to allow undisturbed overboard discharge of any spoil spilling out of the hopper in the event of rolling.

3.3.2 Dredgers without restricted service notation, are to be fitted with overflow trunks on either side, suitably arranged and of sufficient size, to permit safe overboard discharge of excess water during dredging operations.

The construction is to be such as not to require cutouts at the upper edge of the sheer strake. Where overflow trunks are carried through the wing compartments, they are to be arranged such as to pierce the sheer strake at an adequate distance from the deck; and also from

discontinuities such as breaks of superstructure.

3.3.3 Dredgers with restricted service notation may have overflow arrangements which permit discharge of excess water during dredging operations onto the deck.

3.4 Scuppers, sanitary discharges and side scuttles

3.4.1 In all areas where mechanical damage is likely (due to barges coming alongside), all side scuttles, scuppers and discharges including their valves, controls and their indicators are to be well protected. Special attention is to be paid to the possibility of scuppers becoming blocked by sand and other spoil that may spill on to the deck and other areas being drained.

3.5 Ladder wells

3.5.1 Where the ladder well length exceeds 1.5 times the width of deck remaining on each side of the well, cross connections of adequate strength are to be provided in the region of the free ends, unless alternative arrangements are made to avoid high stress concentrations at the inboard ends of the well.

Section 4

Self-Unloading Barges

4.1 General

4.1.1 The requirements of this section apply to self-unloading barges, the port and starboard portions of which are hinged at the hopper end bulkheads to facilitate rotation around the longitudinal axis, when the bottom is to be opened.

4.2 Longitudinal strength

4.2.1 Longitudinal strength requirements given in Pt.3, Ch.6 are to be applied to self-unloading barges in sailing conditions (hull closed).

Additional calculations for the split hull condition (while unloading) are to be carried out as follows:

4.2.2 The combined stress at any point on the cross section of one half hull, is to be determined according to the formula:

$$\sigma = \left(\frac{M_{Y`}}{Z_{Y`}} + \frac{M_{Z`}}{Z_{Z`}} \right) x10^3 \ [N/mm^2]$$

where,

$M_{Y`} = \pm M_Y \text{ Cos } \alpha \pm M_Z \text{ Sin } \alpha$ [kN-m]

$M_{Z`} = \pm M_Y \text{ Sin } \alpha \pm M_Z \text{ Cos } \alpha$ [kN-m]

M_Y, M_Z = Total bending moments (i.e. still water + wave components) on one half of the hull in the vertical and horizontal directions, respectively. For calculation of M_Y and M_Z see Sec.4.2.3 and Sec.4.2.4.

α = the angle between the usual horizontal and vertical axis system (Y-Y and Z-Z) and the principal inertia axis system (Y'-Y' and Z'-Z') See Fig.4.2.2.

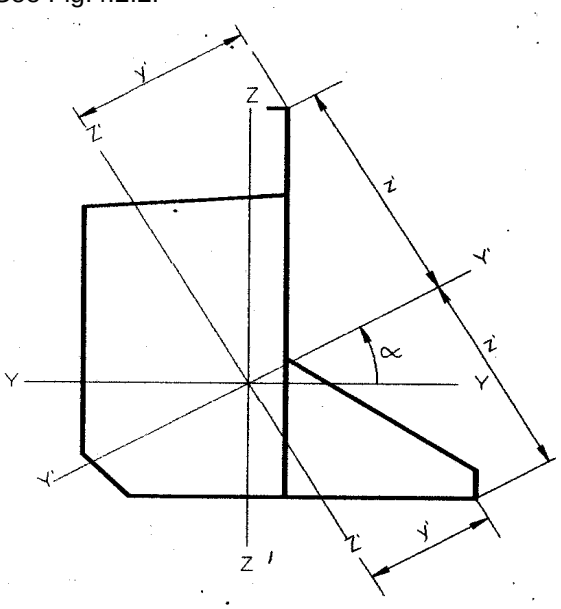

Fig.4.2.2 : Principal inertia axis system.

$$\propto = 0.5 \tan^{-1} \left(\frac{2I_{yz}}{I_{zz} - I_{yy}} \right)$$

I_{YZ} = product of inertia of the cross section with respect to the axis Y-Y and Z-Z [cm^4].

I_{YY}, I_{ZZ} = moment of inertia of the cross section with respect to the axis Y-Y and Z-Z, respectively [cm^4].

$Z_{Y'}$, $Z_{Z'}$ = modulus of section with respect to the principal inertia axes Y'-Y' and Z'-Z' [cm^3].

$= \frac{I_{Y`Y`}}{z`}$ and $\frac{I_{Z`Z`}}{y`}$ respectively. See Fig.4.2.2.

4.2.3 The total vertical bending moment acting on one half of the hull, M_Y is given by:

$$M_Y = M_{Y1} + M_{Y2} \quad [kN\text{-}m]$$

where,

M_{Y1} = 50% of the vertical still water bending moment determined for the whole hull for the most unfavourable distribution of cargo and consumables.

M_{Y2} = 50% of the vertical wave bending moment as per Pt.3, Ch.5, Sec.3.

4.2.4 The total horizontal bending moment acting on one half of the hull, Mz is given by:

$M_Z = M_{Z1} + M_{Z2}$ [kN-m]

where,

M_{Z1} = horizontal still water bending moment due to the horizontal pressure difference between the external hydrostatic pressure on the side shell and the spoil pressure on the hopper bulkhead; which may be taken as:

$M_{Z1} = P \cdot l_{ho} (2l_h - l_{ho})/8$ [kN-m]

where,

l_h = distance [m], between the hinges

l_{ho} = length of hopper [m]

P = net horizontal loading [kN/m] along the length of the hopper which may be approximated as $0.5\, g_o\, (\rho H^2 - 1.025T^2)$

H = height of the spoil [m], from base line to the top of overflow

ρ = design density of the spoil [t/m^3]

M_{Z2} = horizontal wave bending moment which may be taken as:

$M_{Z2} = 0.286\, R_s\, C_w\, (T - 0.143\, C_w)\, l_h^2$ [kN-m]

C_w, R_s = As given in Pt.3, Ch.5, Sec.2.

4.2.5 The hull girder bending stress, in the split hull condition, calculated as per Sec.4.2.2 is not to exceed σ_L as given below:

σ_L = 175/k within 0.4L amidships

= 125/k within 0.1L from ends

Between the specified regions, σ_L is to be obtained by linear interpolation.

4.3 Hinge arrangement

4.3.1 The forces acting on the hinges, actuating mechanisms and locking devices are to be determined by direct calculations, taking into consideration, the static and dynamic forces acting on the hulls and their respective locations.

A typical arrangement of hinges and mechanisms together with the associated static loads is shown in Fig.4.3.1. In general, the aft and forward hinge assemblies are assumed to share the loads equally.

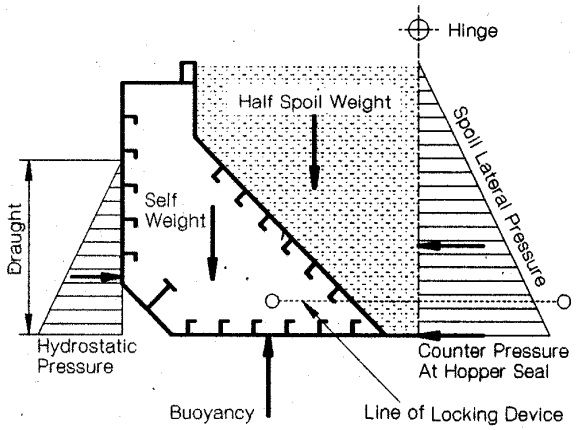

Fig.4.3.1 : Forces acting on split hull.

4.3.2 All elements of the hinge and its associated structure are to be such that the following permissible stress values are not exceeded:

bending stress, $\sigma_b \leq 90/k$ [N/mm^2]

shear stress, $\tau \leq 55/k$ [N/mm^2].

4.3.3 Efficient locking devices of adequate strength are to be provided to prevent the accidental separation of the hulls due to ship motions and vibrations.

4.3.4 Hinge seatings, ram anchorages and locking devices are to be effectively integrated into the local structure and suitably reinforced.

End of Chapter

Chapter 11

Barges and Pontoons

Contents

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2	*Hull Arrangement and Strength*
3	*Pushing, Towing - Devices and Connecting Elements*
4	*Machinery and Electrical Installation*
5	*Requirements for Accommodation Barges/ Pontoons*

Section 1

General

1.1 Application

1.1.1 The requirements of this Chapter apply to manned or unmanned barges and pontoons and are supplementary to those given for the assignment of main characters of class.

Barges are non-self propelled vessels designed and constructed, generally, for carriage of dry cargoes in holds or liquid cargoes in tanks.

Pontoons are non-self propelled vessels designed and constructed, generally, for carriage of non-perishable cargoes on deck.

1.1.2 Vessels built in compliance with the above requirements will be eligible to be assigned class notations " **BARGE** " or " **PONTOON** ", as appropriate.

1.1.3 Barges or Pontoons, which are intended to be manned at sea, whether on a voyage or on-site offshore by crew and/ or special personnel and/ or industrial personnel; will be assigned notation "**MANNED BARGE**" or "**MANNED PONTOON**". In addition, such barges/ pontoons are to comply with the applicable requirements of Section 4.

Note: Industrial Personnel means all persons who are transported or accommodated on board for the purpose of offshore industrial activities performed on board other ships and/ or offshore facilities.

1.1.4 Barges and pontoons for special purpose or for carriage of specialised cargoes are also to comply with the requirements of relevant Chapters of Pt.5, and such vessels would be eligible for assignment of additional or appropriately modified class notations e.g "**OIL BARGE", "SULLAGE BARGE',"SHIPBORNE BARGE", "BARGE FOR LIQUEFIED GASES ", " CRANE PONTOON** " etc.

1.1.5 Barges and pontoons intended to be operated only in agreed areas or conditions, will be individually considered for service restriction notations as per Pt.1, Ch.1, Sec.2. In particular, the 'Shipborne barges' will normally be considered for notation 'Restricted Water Service'.

1.1.6 Sullage barges are those used primarily for carriage of grey water i.e. domestic waste water resulting from baths, sinks, washing machines, etc.

1.1.7 Where position mooring systems are provided, the systems are in general to comply with the requirements of Chapter 10 of *Rules and Regulations for the Construction and Classification of Mobile Offshore Drilling Units*. Such vessels will be assigned additional class

notation '**PM**'. The mooring configuration is to be analyzed for a combination of environmental conditions, as relevant and applicable to the area of operation. The vessel is to be operated in accordance with the approved mooring analysis assessment report which is to be available on-board. Change in mooring configuration and/ or location would necessitate fresh mooring analysis assessment. If the limiting environmental parameters (as indicated in the report) are expected to be exceeded, the vessel is to be moved to a safe location as soon as possible.

1.2 Documentation

1.2.1 The following additional plans and documents are to be submitted for approval, as applicable.

- Towing arrangement and details of towing brackets, bollards and other fittings with under deck stiffening.
- Details of structure and fittings, if any, to which deck cargo securing lashings etc. are attached.
- Details of arrangements for lifting fully loaded shipborne barges.
- In case of pusher tugs or integral tug/ barge systems or combination units comprising many modules, details of the connecting elements, attachments and supporting structures.
- Details of any position mooring systems, equipment and associated foundations for barges/ pontoons.
- In the case of accommodation barges, an operating manual containing appropriate details of the vessel, including at least the following information:
  - Operating restrictions (if any)
  - Limiting environmental parameters such as significant wave height, wind speed, current speed, water depth
  - Mooring configurations and procedures for a combination of environmental parameters
  - Details of watertight and weathertight closures
  - Design deck loading and mooring loads
  - Details of lifesaving appliances and emergency escapes

Section 2

Hull Arrangement and Strength

2.1 General

2.1.1 Where a rudder is not fitted, the Rule length, L, is to be taken as 97% of the length of the load waterline at draught T.

In case of pusher tug/barge units with rigid connections, the Rule length, L, is to be based on the combined length of the tug and barge.

2.2 Longitudinal strength

2.2.1 The requirements of Pt.3, Ch.1, Sec.4 regarding loading instruments is not applicable to barges and pontoons.

2.2.2 The hull section modulus may be 5% less than that required by Pt.3, Ch.6, Sec.2.

2.2.3 For special harbour conditions, (e.g. shipborne barge fully loaded at crane or transient conditions during loadout of offshore jackets), the wave bending moment M_w and wave shear force Q_w may be neglected, provided the longitudinal bending and shear stresses in these loading conditions are limited to 145/k [N/mm^2] and 100/k [N/mm^2] respectively.

2.3 Bottom structure

2.3.1 For barges and pontoons of L < 90 [m] having no rise of floor, the keel plate thickness may be same as adjacent bottom shell.

2.3.2 On hard chine vessels, where a solid round bar is provided at the knuckles, the

diameter of round chine bar is not to be less than three times the bottom plate thickness.

2.4 End construction

2.4.1 Flat bottom areas in way of raked fore ends are to be strengthened according to Pt.3, Ch.11, Sec.3.

2.4.2 In case of vessels having no definite "fore" or "aft", the collision bulkhead and strengthening of bottom forward are to be provided at both ends.

2.5 Truss arrangements

2.5.1 A truss is a system of internal framing members comprising deck and bottom girders in association with regularly spaced stanchions and diagonal bracings inclined at about 45 degrees with the horizontal, in each space between the stanchions.

2.5.2 The scantlings of , stiffeners and girders are not to be less than the general requirements given in Pt.3, Ch.7, except as specified in 2.5.3. below.

2.5.3 The section modulus of bottom girders is not to be less than that required by Pt.3, Ch.7, Sec 6 with the value of the coefficient f_{bdg} taken as 6. The section modulus of deck girders is not to be less than that required by Pt.3, Ch.7 with the value of the coefficient f_{bdg} taken as 8.

2.5.4 The scantlings of stanchions are to be based on the external sea pressure on bottom or the static cargo load on deck, whichever is higher; and the buckling requirements given in Pt.3, Ch.9. Stanchions in tank spaces are also to be checked for tension caused by internal pressure.

2.5.5 The cross-sectional area of diagonals may be approximately 50% of that of the adjacent stanchion.

Section 3

Pushing, Towing - Devices and Connecting Elements

3.1 General

3.1.1 Devices for pushing and towing of linked barges as well as the elements connecting the modular units are to be adequately dimensioned for the acting external forces calculated considering all possible load combinations. Towing gear, brackets and bollards are to be adequately dimensioned for the estimated towing pull considering the displacement and towing speed.

3.1.2 The scantlings of these devices and elements as well as their supporting structures are to be based on the following permissible stresses:

bending and normal stress σ = 100/k [N/mm^2]

shear stress τ = 60/k [N/mm^2]

equivalent stress,

$\sqrt{\sigma^2 + 3\tau^2}$ = 120/k [N/mm^2].

Section 4

Machinery and Electrical Installation

4.1 General

4.1.1 Unless otherwise specified, machinery, boilers, pressure vessels, pumps, piping, materials and electrical installations on-board barges/ pontoons are to be in accordance with the applicable requirements of Pt. 4 of the Rules.

4.2 Internal Combustion Engines

4.2.1 Engines and gearing are to be designed such that they will function properly under the environmental conditions specified for normal operation, including their lubrication and cooling arrangements.

4.2.2 Internal combustion engines having a rated power of 100 [kW] and over are to be of an approved type if they are intended for services considered essential for the maneuvering or safety of the barge/ pontoon.

4.2.3 Internal combustion engines having a rated power of 100 [kW] and over that are intended for services considered not essential for maneuvering and safety need not be of an approved type. This would include engines driving cargo pumps. Industrial engines constructed and equipped in accordance with good commercial practice are acceptable provided they meet the criteria specified in 4.2.1. Further, the engines are to comply with safety features, such as crankcase explosion relief valve, overspeed protection, etc. as provided in Part 4 of the Rules. Crankcase explosion valves are not required for engines with less than 200 [mm] bore. After installation, the engines are to be subject to a satisfactory performance test conducted in the presence of the Surveyor.

4.2.4 Internal combustion engines located on the weather deck are to be provided with ventilated metal hoods or with a well-ventilated metal housing of sufficient size to allow for proper operation and maintenance.

4.2.5 Exhaust manifolds and piping are to be insulated with non-combustible material or are to be water-cooled such that the temperature of exposed parts does not exceed 220°C. Exhaust manifold connections are to be constructed of pipe or pipe fittings with screwed or welded connections or bolted connections with fireproof gaskets.

4.3 Generators

4.3.1 Generators and Motors of 100 [kW] and over and intended for essential services (for maneuvering and safety) are to be of approved type.

4.3.2 Generators and motors of 100 [kW] and over that are intended for non-essential services need not be of an approved type. Generators and motors constructed and equipped in accordance with recognised standards such as IEC etc. are acceptable. After installation, the equipment is to be subjected to a satisfactory performance test conducted in the presence of the Surveyor.

4.4 Ventilation

4.4.1 Machinery spaces including pump rooms are to be provided with adequate ventilation for the operation of machinery and the safety of personnel. All air inlets to spaces where personnel may enter are to be located outside hazardous areas.

4.5 Air Pipes

4.5.1 In general, vent pipes are to comply with the applicable requirements of Pt.3, Ch.12, Sec.13 and Pt.4, Ch.3 of the Rules.

4.5.2 Closed compartments and tanks are to be provided with air pipes. Air pipes may not be required for dedicated dry voids which are permanently preserved and closed, and do not contain piping. Access is to be arranged for detecting possible water ingress and enable use of portable bilge pumps.

4.5.3 Where air pipes of full height may cause difficulties in operation of the vessel, a lower height may be accepted, with the approval of the flag Administration.

4.6 Sounding Arrangements

4.6.1 All tanks, except as indicated below, are to be provided with separate sounding pipes or with approved tank level indicating apparatus. Where a tank-level indicating system is used, a supplementary manual means of sounding is to be provided, where practicable, for tanks which are not always accessible.

4.6.2 Void compartments adjacent to the sea or to tanks containing liquids and void compartments through which piping carrying liquids pass are to be fitted with separate sounding pipes, approved tank liquid level indicating apparatus, or be fitted with other means such as bolted manholes to determine if the void tanks contain liquids.

4.6.3 Unmanned barges in unrestricted service are to be provided with sounding pipes to cargo holds.

4.7 Ballast System

4.7.1 Ballast systems are to comply with applicable requirements of Pt.4, Ch.3 of the Rules.

4.7.2 For barges of 20 [m] in length or greater all ballast tanks are to be connected to at least two power driven ballast pumps. Alternative means of de-ballasting, such as an eductor or a suitable liquid cargo pump with an appropriate temporary connection to the ballast system may be accepted in lieu of a second ballast pump.

4.8 Bilge System

4.8 1 Where barges are fitted with below deck machinery spaces or where fixed piping systems are led through void spaces, a satisfactory means is to be provided capable of pumping from and draining such spaces. This means may be by use of suitable hand pumps through fixed bilge piping arrangements or by means of portable pumps stored onboard the barge.

4.9 Fuel Oil System

4.9.1 The fuel oil pumping arrangements are to be separate from other pumping systems, as far as practicable.

4.9.2 No fuel-oil tank is to be situated where spillage or leakage can constitute a hazard by falling on heated surfaces. Adequate precautions are to be taken to ensure that any oil that may escape under pressure from any pump, filter or heater does not come into contact with heated surfaces.

4.9.3 Tanks not forming part of the structure (independent tanks) are to have suitable drip trays with drains to oil drain containment tanks.

4.9.4 Oil pipes and other pipes when passing through fuel-oil tanks are to be of steel, except that other materials may be considered where it is demonstrated that the material is suitable for intended service.

4.9.5 Fuel-oil transfer pumps and other similar pumps are to be fitted with remote controls so that they may be stopped in the event of a fire. These controls are to be located outside of the spaces in which the pumps are located.

4.10 Electrical Installations

4.10.1 In general, electrical installations, where provided onboard barges/ pontoons, are to be in accordance with Pt.4, Ch.8 of the Rules.

4.10.2 Electrical equipment is to be so placed or protected as to minimize the probability of injury or mechanical damage from the accumulation of dust, oil vapors, steam or dripping liquids.

4.10.3 Equipment liable to arc is to be ventilated or placed in ventilated compartments in which flammable gases, acid fumes and oil vapors cannot accumulate.

4.10.4 Electrical equipment is to be located clear of skylights and ventilators so as to avoid damage to such equipment.

4.10.5 All electrical equipment exposed to the weather or located in spaces where it would be exposed to the sea, splashing or other severe moisture condition is to be of a watertight type or be protected by means of watertight enclosures.

4.10.6 In general, all electrical equipment is to be constructed of durable, flame-retardant, moisture-resistant materials which are resistant to deterioration in the atmosphere and at the temperatures to which they are likely to be exposed.

4.10.7 In general, the frames or cases of all permanently installed generators, motors, controllers, instruments and similar equipment for which the arrangement and method of installation does not assure positive grounding are to be permanently grounded through separate conductors and protected against damage.

4.10.8 All generators, motors and other electrical equipment are to be so arranged that they cannot be damaged by bilge water, and, if necessary, a watertight coaming is to be provided to form a well around the base of such equipment with provision for removing water from the well.

4.11 Additional requirements for Manned Barges/ Pontoons

4.11.1 Additional requirements for manned barges/ pontoons are indicated in this sub-section.

4.11.2 Manned barges are to be provided with sounding pipes to cargo holds.

4.11.3 In general, bilge and ballast piping systems are to be provided in accordance with Pt.4, Ch.3 of the Rules.

4.11.4 Manned barges carrying more than 12 persons, excluding members of the crew, are to be provided with a fixed power operated bilge system capable of pumping from and draining any compartment below the freeboard or bulkhead deck.

4.11.5 Manned barges for unrestricted service are to be equipped with two permanently installed bilge pumps. Manned barges in restricted service may be provided with at least one bilge pump. Ballast pumps may be used as bilge pumps. The bilge pumps are to be of self-priming type. Where only one permanently installed bilge pump is provided, this pump is not to serve as fire pump.

4.11.6 Dry compartments in fore and aft peak may be drained by effective hand pumps.

4.11.7 For manned barges/ pontoons carrying less than 12 persons, excluding crew members, the bilge pumps may be substituted by portable submersible pumps of requisite capacity.

4.11.8 In the absence of specific safety requirements from the flag Administration, a manned barge/ pontoon carrying 12 persons or less is to comply with the cargo ship fire safety requirements of Pt. 6 of the Rules, as relevant.

4.11.9 Portable fire extinguishers are to be provided as per scales indicated in Pt. 6 of the Rules, for cargo ships.

4.11.10 At least two main generator sets are to be provided with combined capacity sufficient to maintain the barge/ pontoon in normal operational conditions, including any special purpose equipment. Additionally, the capacity of these sets is to be sufficient to maintain the barge/ pontoon in normal operational conditions, excluding any special purpose equipment, with any one main generator in reserve.

4.11.11 A self-contained emergency source of power and its associated equipment, located on or above the freeboard deck, is to be provided if a fire or other casualty in any one space will make inoperative any combination of the independent sources of normal electrical power required by 4.11.10.

4.11.12 When an emergency source of power is not fitted, the two independent sources of electrical power required by 4.2.6 are to have their own systems, including power distribution and control systems, completely independent of the system in the other space, such that a fire or other casualty in any one space will not affect the power distribution in the other. In addition, the two independent sources of electrical power are to comply with 4.11.13 to 4.11.19.

4.11.13 The power available is to be sufficient to supply for at least 18 hours all services necessary for safety in an emergency, particular attention being given to the following:

i) Navigation and special purpose lights and warning systems including helicopter landing lights

ii) Emergency lighting for machinery spaces, control stations, alleyways, stairways, exits and
elevators

iii) Emergency lighting for embarkation stations on decks and over sides

iv) General alarm and communications systems

v) Emergency lighting for stowage position(s) for firemen's outfits

vi) Fire detection and alarm systems

vii) Fire extinguishing systems

viii) Emergency lighting for helicopter landing decks (where provided)

ix) Abandonment systems dependent on electrical power

4.11.14 Where the emergency source of electrical power is a generator, the generator is to:

- be driven by a suitable prime mover with an independent supply of fuel, having a flashpoint (closed cup test) of not less than 43 deg C.

- to start automatically upon failure of the electrical supply from the main source of electrical power, unless a transitional source of emergency power in accordance with 4.11.16 is provided; where the emergency generator is automatically started, it is to be automatically connected to the emergency switchboard.

4.11.15 The automatic starting system and the characteristic of the prime mover is to be such as to permit the emergency generator to carry its full rated load as quickly as is safe and practicable.

4.11.16 When the arrangements are such that more than 45 seconds are needed for the generator to supply the required load, a transitional source of emergency power is to be provided in accordance with 4.11.15.

4.11.17 When a transitional source of emergency power is required in accordance with 4.11.15, it is to be installed consisting of storage batteries of sufficient capacity to supply the rated voltage and amperage for at least 30 minutes to the following:

- Emergency lighting
- Fire detection and alarm systems
- General alarm and communications systems

4.11.18 Arrangements are to be such that the transitional source of emergency power will come into operation automatically in the event of failure of the main electrical supply.

4.11.19 Means are to be provided for periodic testing of the emergency source of power and the transitional source of power, if provided, and is to include testing of the automatic arrangements.

Section 5

Requirements for Accommodation Barges/ Pontoons

5.1 General

5.1.1 Non-propelled, barges and pontoons which are designed and constructed for accommodation of industrial personnel, may be assigned class notation '**ACCOMMODATION PONTOON**', when all accommodation/ storage/ auxiliary machinery spaces are provided above main deck. For '**ACCOMMODATION BARGE**', spaces below the deck may also be used for accommodation/ storage/ auxiliary machinery etc. Barges/ Pontoons assigned notation 'ACCOMMODATION BARGE/ ACCOMMODATION PONTOON' need not be assigned 'MANNED BARGE/ MANNED PONTOON' notations.

5.1.2 In addition to the requirements in this Chapter, these vessels are to additionally comply with the requirements applicable to vessels carrying industrial personnel, as specified by the Flag Administration.

5.1.3 Accommodation barges/ pontoons are to be equipped with suitable equipment for station keeping. Accommodation barges/ pontoons equipped with position mooring equipment will be assigned additional class notation **PM** (also refer Section 1.1.7).

Additional class notations for comfort may be provided at the request of the owner.

5.1.4 In general, stability, fire protection and safety measures for accommodation barges/ pontoons are to comply with the applicable requirements of the *Code of Safety for Special Purpose Ships*, as amended or *Code for the Construction and Equipment of Mobile Offshore Drilling Units, 2009, as amended*.

5.1.5 The accommodation facilities provided for persons onboard are to comply with the relevant requirements of the flag Administration, as applicable to such persons.

End of Chapter

Chapter 12

Floating Dry Docks

Contents

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2	*Hull Arrangement and Strength*
3	*Machinery and Electrical Installations*
4	*Fire Safety*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to steel floating docks of the types defined in 1.1.2. Other design configurations would be specially considered.

1.1.2 The requirements apply directly to floating docks of the following types:

a) **Caisson type**, in which the bottom caisson and both wing walls are continuous and inseparable.

b) **Pontoon type**, in which the wing walls are continuous and the bottom is formed of non-continuous pontoons. The pontoons may be either permanently attached to the wing walls or may be detachable.

1.1.3 The docks built in compliance with the above requirements will be eligible to the assigned the class notation “FLOATING DOCK”. For special types and configurations of floating docks not covered in 1.1.2, appropriate additional notations may be assigned, if necessary.

1.2 Documentation

1.2.1 The following plans are to be submitted for approval:

- General arrangement
- Transverse section at mid-length
- Structural plan of wing walls, bottom caisson or pontoons
- Structural plan of top deck, safety deck and bulkheads.
- Stability calculations
- Pumping arrangements
- Machinery and electrical plans
- Fire extinguishing systems

1.2.2 The following information is to be submitted:

a) Required lifting capacity (W).

b) Length of shortest contemplated ship having a docking weight equal to the required lifting capacity.

c) The maximum draught of the vessel when entering the dock. Also maximum associated dock submergence draught.

d) Estimated dock lightweight, its main components and distribution.

e) Depth of rest water.

f) Depth of keel and side blocks.

g) Hydrostatic curves

h) Tank arrangements showing maximum service heads and heights of overflow pipes and air pipes.

i) The hydrostatic head assumed for the design of boundary bulkheads separating adjacent ballast tanks. It is equal to the maximum allowable difference between the filling levels of adjacent ballast tanks. This head should be included in the dock operating instructions.

j) Operating manual including ballasting manual.

k) Details of operating site and method of dock securing

l) Full details of the type and proposed arrangement of deflection monitoring equipment.

m) Particulars of indicating system for tank water level and draughts

n) Details of the mooring attachments to the dock structure.

o) Coating specifications of external structure and tanks.

1.3 Definitions

1.3.1 General

1.3.1.1 The lifting capacity "W" of the dock is equal to the displacement [t] of the heaviest ship that the dock is designed to lift in normal service.

1.3.1.2 The light displacement includes the structural weight of the dock complete with all machinery, crane(s), equipment, fresh water, fuel oil for the use of the dock, compensating ballast water (see 1.3.1.9), and rest-water (see 1.3.1.8).

1.3.1.3 The length, L_D [m], is the length of the bottom caisson or the distance from the aft end of the aftermost pontoon to the fore end of the forward pontoon, but is exclusive of non-integral end platforms or swing bridges.

1.3.1.4 Amidships is to be taken as the middle of the length L_D.

1.3.1.5 The breadth, B_D, is the moulded breadth, [m].

1.3.1.6 The depth, D_D [m], is the vertical distance from the lowest point of the bottom framing to the lower surface of the uppermost deck plating.

1.3.1.7 The top deck is the deck extending over the top of the wing walls.

1.3.1.8 The safety deck is a watertight deck arranged at such a distance below the top deck that when all tanks below the safety deck are flooded, a freeboard of at least 1 [m] is available to the top deck. Alternative arrangements to the fitting of a safety deck will be specially considered.

1.3.1.9 The rest-water is the ballast water in the tanks which the pumps cannot discharge.

1.3.1.10 Compensating ballast water is ballast water which may be used to limit the deflection of the dock by counteracting the ship-induced longitudinal bending moment. Rest-water should not be included in compensating ballast water.

1.4 Freeboard

1.4.1 The freeboard to the pontoon deck at the centerline of the dock when supporting a ship having a displacement equal to the lifting capacity (as defined in 1.3.1.1) is to be not less than 300 [mm]. When the pontoon deck at the inner side walls is lower than at the center, the freeboard to the pontoon deck at the inner side walls is to be not less than 75 [mm] and the freeboard at the centerline is to be not less than 300 [mm].

1.4.2 The above freeboard values are for level trim condition assuming that the crane(s) are positioned suitably to achieve level trim. Movement of cranes to the forward end or to the after end of the dock should not result in submergence of any part of the pontoon deck.

1.4.3 The above freeboards are considered adequate for sheltered water operation. In other areas it is recommended that they are suitably increased.

1.5 Stability

1.5.1 The dock should have a metacentric height (GM) of at least 1.0 [m] for the following conditions of submergence:

a) Full dock submergence upto the minimum freeboard level to top deck.
b) the submergence draught to the top of keel block assuming full weight of the docked ship supported on blocks.

c) The dock in final working condition with the likely most unfavourable docked ship.

1.5.2 Any additional requirements of the Statutory Authorities regarding Stability are to be satisfied by the dock.

1.5.3 For operating sites exposed to wind heeling moments, these effects should be taken into account.
In general, the angle of heel due to wind heeling moment should not result in submergence of any part of the pontoon deck.
The wind heeling moment may be calculated from the following formula:

$0.613 \times 10^{-3} \times V^2AH$ [kN-m]

where,

A = the longitudinal projected area of the exposed surface including exposed areas of docked ship [m^2].

H = ΔH + ½ d [m]

ΔH = vertical distance from the center of A to the water line of the dock [m].

d = draught of the dock [m]

V = wind velocity [m/sec.], the wind velocity is not to be less than 25 [m/s] in general. However, the values of the wind velocity will depend on the service location and the mode of operation of the dock and may be considered more precisely in each case.

1.6 Immersion trials

1.6.1 On completion of the dock, trials are to be carried out to ascertain the following:

a) The freeboard to top deck with the dock flooded

b) The light displacement and the lifting capacity of the dock corresponding to the minimum freeboard

c) The position of center of gravity by an inclining test

d) Any permanent deflection in the initial condition. In the initial condition, all tanks for consumables are completely filled, but all other tanks are empty, only the rest water remaining in ballast tanks. The traveling cranes may be parked in positions giving equal drafts forward and aft.

e) Calibration of the deflection meters are to be carried out, simulating the most severe intended loading condition as far as practicable.

1.7 Ocean towage

1.7.1 The class of a floating dock will be assigned after it has reached its port of operation and been subjected to a satisfactory general examination.

1.7.2 Where it is intended that the dock be towed at sea from its port of construction to port of operation or from one port of operation to another, the aspects specified below will need to be investigated at an initial stage of design development:

a) The longitudinal strength is to be evaluated considering the estimated design voyage wave bending moment. Estimates of design voyage bending moment will be based on draught in towage condition, towage route and season.

   In order to minimize the design voyage wave bending moment, it is recommended that the voyage be undertaken at as close to the light draught as possible.

b) Particular attention should be given to the resistance to buckling of transversely framed bottom structure when the dock is subjected to wave induced hogging moments.

c) In addition the following will be required for issuance of a Towage Certificate:

   i) The aprons/working platforms are to be removed from the ends of a dock prior to an ocean tow.

   ii) Where the traveling crane(s) are to be mounted on the wing walls prior to towage then the proposed sea fastenings should be submitted for approval. The securing arrangements for any items carried on the pontoon deck should be to the Surveyor's satisfaction.

   iii) Emergency anchoring arrange-ments should be provided to the

Surveyor's satisfaction. Consideration should be given to the method by which the dock could be boarded during the tow by the personnel required to release the anchors.

iv) Other information for towage is to be submitted such as:

- details of the tug and the towing gear;
- details of the towing attachments, bridle recovery arrangements, secondary towing gear, etc.

Section 2

Hull Arrangement and Strength

2.1 General

2.1.1 It is recommended that the bottom, side and deck plating are longitudinally stiffened.

2.1.2 In general, the requirements and general principles given in Pt.3, Ch.2 "Materials for Construction and Corrosion Additions" Pt.3, Ch.3 " General Arrangement Design" and Pt.3, Ch.7 "Hull Local Scantlings" are to be applied, as appropriate.

2.2 Longitudinal strength

2.2.1 General

2.2.1.1 The general requirements for longitudinal strength given in Pt.3, Ch.6 is to applied for steel floating docks along with the requirements in Sec.2.2.2 to 2.2.5 of this chapter.
The floating dock is to be considered as a Category I vessel as per Pt.3, Ch.1, Sec.4. Operating and ballasting manual and deflection control system are to be provided in lieu of loading manual and loading instrument.

2.2.2 Still water bending moment

2.2.2.1 The SWBM is to be calculated for the condition when the ship of length L_s is supported on the keel blocks, the center of the ship's length being over the mid-length of the dock and the freeboard at the pontoon deck is at least equal to that given in 1.4.1.

2.2.2.2 Dock buoyancy distribution may be assumed rectangular over the length of the dock L_D.

2.2.2.3 The weight curve of the ship can be taken as a rectangle with a superimposed parabola of half the area of the rectangle, the length of each area being L_s.

2.2.2.4 Considering the distribution of weight and buoyancy indicated in 2.2.2.3 and 2.2.2.4 above, an approximate value of the maximum sagging BM generated during docking of the ship is given by the following expression:

$$M_s = W\left[1.25\, L_D - 1.146\ \ L_s\right] \quad [\text{kN m}]$$

W = Lifting capacity of the dock [t]

L_S = The length between perpendiculars of the shortest ship whose displacement is equal to the lifting capacity of the dock [m]; Normally L_S is not to be taken less than 0.8 L_D.

2.2.2.5 Where compensating ballast water is proposed to be used to reduce the bending moments in the normal operating conditions, lower values of still water bending moments may be approved. The operating manual should include instructions for distribution of ballast in such cases.

2.2.3 Required section modulus

2.2.3.1 The required section modulus of the dock at mid length is to be evaluated based on the still water bending moment given in 2.2.2 and a permissible stress of 140 [N/mm^2]. Reduced values of still water bending moment may be considered where compensating ballast water is used as mentioned in 2.2.2.5.

2.2.3.2 Scantlings of all related longitudinal material of the hull girder are to be continuous over a length of 0.4 L_D in the mid length region of the dock. In the region outside 0.4 L_D the scantlings may be gradually tapered to the requirement at ends.

2.2.3.3 The section modulus required as per 2.2.3.1 above is to be used for the evaluation of permissible stresses for local structure according Part 3 of these Rules as mentioned in Sec 2.5 of this chapter.

2.2.4 Calculation of section modulus

2.2.4.1 The material to be included in the calculation of the section modulus will be, for the caisson type dock, all continuous fore and aft material.

For the pontoon type dock, all continuous fore and aft material of the wing walls structure may be included. The fore and aft material of the parts of the pontoon structure which are continuous or scarphed longitudinally for at least 2 [m] at the end of each pontoon may also be included.

2.2.4.2 The crane rail is not to be included in section modulus calculation.

2.2.5 Deflection control system

2.2.5.1 Two completely independent deflection meters are to be fitted and be capable of outputting deflections over the length of the dock L_D. One of the two deflection monitoring systems required should be of the hydraulic type. In all cases the systems should be arranged so as to measure the longitudinal deflection of a wing wall.

2.2.5.2 The readings of one of the systems installed should be displayed on an indicator board in the control room of the dock.

2.2.5.3 Consideration will be given to acceptance of only one deflection control system for docks designed to operate without using differential ballasting. Such a system could then be of optical type.

2.2.5.4 In all cases the methods of monitoring and limiting the dock deflections in service are to be submitted for approval. These methods are to include arrangements for visual and audible warning and also for ballast pumps to be stopped automatically before the maximum allowable deflection is reached.

2.2.5.5 The maximum allowable deflection is that corresponding to a bending stress of 140 [N/mm^2] when lifting a ship as detailed in 2.2.2.

2.3 Transverse strength

2.3.1 The primary transverse strength members of the dock are to be evaluated for the following load cases considering combined effect of the self weight of the dock, weight of the docked ship, external hydrostatic pressure and weight of ballast water that are acting in way of the transverse member.

a) Load case 1: Docked ship condition with the minimum freeboard at the pontoon deck.

b) Load case 2: Docked ship condition as in load case 1 with no load from the docked ship on the keel blocks at the ends of the dock. The transverse structure of the dock at the ends is to be evaluated for loads due to self-weight, buoyancy and compensating ballast water.

c) Load case .3: Transient condition with dock emerging out of water with ship fully supported on the blocks and water level up to the top of the keel blocks.

2.3.2 In the load cases mentioned in 2.3.1, the load due to docked ship at any section is to be based on the maximum value of ship weight per metre obtained from the weight distribution in 2.2.2.3, which is given by:

$$w_s = \frac{1.167 \quad W}{L_s} \quad [t/m]$$

where,

w_s = ship weight per unit length [t/m],
W = maximum lifting capacity [t],
L_s = length [m] of shortest contemplated ship associated with docking weight W

The whole of the ship weight should be assumed supported by keel blocks.

2.3.3 The wing wall reactions are to be considered acting on the transverse section as required to give equilibrium to the section. These reactions at inner and outer wing walls may be taken as equal.

2.3.4 The compressive and tensile stresses computed for the loading conditions given in 2.3.1 should not exceed 170 [N/mm^2]. The mean shear stress at any section through the web plating of the primary transverse members should not exceed 100 [N/mm^2] and the Von Mises combined stress at any point should not exceed 200 [N/mm^2].

2.3.5 For the purpose of stress evaluation the section modulus of the primary transverse strength members is to comprise the continuous transverse material over the effective width of the primary member.

2.4 Buckling

2.4.1 The structural panels, stiffeners and other members are to satisfy the requirements of resistance to buckling as given in Pt.3, Ch.9.

2.5 Local strength

2.5.1 Shell and tank structure

2.5.1.1 The design pressures for plating, stiffeners and girders of the shell, tank top and tank bulkheads are to be based on the following:

p = applicable design pressure

= 0.01 h [N/mm^2],

where:

a) For ballast tanks, “h” is the greatest of the following distances [m] from the load point of the structural member (see Pt.3, Ch.3, Sec.7 for definition of load point):

i) To a point located at two-thirds of the distance from the top of the tank to the top of the overflow. As an alternative, the maximum differential head in service may be used based on the highest levels to which water will rise on each side of the structure. Where the head is obtained using the maximum differential head in service, data on operating the dock within such limits should be included in the operating manual;

ii) 3.5 [m].

b) For all other tanks, the greatest of the following distances [m] from the load point:

i) To a point located two-thirds of the distance from the top of the tank to the top of the overflow

ii) To the maximum immersion waterline for wing wall and pontoon plating

iii) 3.5 [m]

c) For void spaces and cofferdams, the greater of the following distances [m] from the load point:

i) To the maximum immersion waterline, for wing wall and pontoon plating
ii) 3.5 [m]

2.5.1.2 The plate thickness and scantlings of structural members of bottom shell, side shell and tank bulkheads are to be not less than that calculated using the design pressure given in 2.5.1.1 in the basic formulae given in Pt.3, Ch.7, Sec.4, Sec.5 and Sec.6.

The permissible bending stress coefficient to be used for the above calculations are to be as given in Pt.3, Ch.7, considering acceptance criteria AC-I

2.5.1.3 The minimum required thickness of tank and shell plating is 7.0 [mm].

2.5.2 Cross ties

2.5.2.1 The cross ties connecting transverses in wing walls are to meet the requirements for resistance against buckling using the criteria given in Pt.3, Ch.9. The loads are to be based on the area supported by the cross tie and design pressures in 2.5.1.

2.5.3 Top deck

2.5.3.1 The plate thickness and scantlings of structural members of top deck are to be not less than that calculated using the basic formulae given in Pt.3, Ch.7, Sec.4, Sec.5 and Sec.6 and a design pressure of 0.005 [N/mm^2]. The permissible bending stress coefficient is to be determined as per Pt.3, Ch.7, considering acceptance criteria AC-I

The thickness of top deck plating is to be also not less than 7.0 [mm].

2.5.4 Safety deck

2.5.4.1 The plate thickness and scantlings of structural members in the area of the safety deck forming part of tanks is to be based on

design pressure as given in 2.5.1 and the basic formulae as per Pt.3, Ch.7. However, the minimum head of 3.5 [m] specified in 2.5.1.1 a), b), c) need not be applied to safety deck tank top. The areas of safety deck which are not forming top of tanks may be based on a design pressure of 0.005 [N/mm^2].

The permissible stresses are to be based on Pt.3, Ch.7, considering acceptance criteria AC-I

The thickness of safety deck plating is not to be less than 7.0 [mm].

2.5.5 Keel block and supporting structure

2.5.5.1 The keel block and their supporting structure are to be designed for a uniform load equal to the maximum ship weight per unit length "w_s" given in 2.3.2. and permissible stress of 160/k [N/mm^2].

2.5.6 Cranes

2.5.6.1 Where cranes are fitted, the resulting loads on the dry-dock structure are to be indicated in the plans. The total crane weight including hook load and the arrangement of wheels and rails are to be submitted for approval of the supporting structure. Certification of cranes would be specially considered.

2.6 Ventilation and access

2.6.1 All tanks are to have vent or overflow pipes that terminate well above the water line at the maximum submergence draft of the dock. All compartments are to be provided with manholes for access and openings are to be arranged to provide adequate ventilation and access to all parts of the structure.

2.7 Tank testing

2.7.1 Ballast tanks, oil tanks, freshwater tanks and cofferdams are to be tested by a head of water to the highest point to which the liquid will rise in service. Where the scantlings of a tank boundary are based on the maximum differential head in service, care is to be taken to ensure that the test heads do not exceed the design differential head.

Leak testing may be accepted (See Pt.3, Ch.17, Sec.3) as an alternative, provided selected tanks are subject to hydraulic testing as above.

Section 3

Machinery and Electrical Installations

3.1 Machinery

3.1.1 Machinery such as boilers, pressure vessels, auxiliary engines, compressors, pumps etc., essential for the operation of the floating dock are to be constructed and installed in accordance with the relevant requirements of Part 4 of the Rules.

3.2 Piping systems

3.2.1 The piping systems are to be generally in accordance with the relevant requirements of Pt.4, Ch. 2 and Ch.3.

3.2.2 The dock is to have at least two water ballast pumps. The arrangements for de-ballasting are to be such that in case of failure to one pump an alternative pumping is available for each ballast tank.

3.2.3 Where the valves in ballast system are power operated, arrangements are to be provided for manual operation in the event of power failure. Instructions are to be provided in the operating manual of the dock that the sea inlet, discharge and distribution valves, in the ballast systems are to be immediately closed in the event of power failure during ballasting / de-ballasting.

In this case it is recommended that the ballast distribution valves are fitted with fail safe devices, automatically closing them in case of power failure.

3.3 Electrical equipment

3.3.1 Electrical installations are to be generally in accordance with the requirements of Pt.4, Ch.8 of the Rules.

Section 4

Fire Safety

4.1 General

4.1.1 This section gives the minimum requirements for fire protection, detection and extinction applicable to the floating dock and do not cover equipment fitted for fighting fires that may occur on the docked ship.

However, provision is to be made for connecting the fire fighting system of the dock to the docked vessel fire fighting system as indicated in 4.4.1. Attention should also be given to any relevant statutory requirements of the National Authority of the country in which the dock is to operate.

4.1.2 For definitions and details not covered in this section, reference may be made to Part 6 of the Rules.

4.2 Fire protection

4.2.1 Boundaries of machinery spaces and interior stairways below top deck are to be of steel or equivalent material.

4.2.2 Deckhouses are to be constructed of steel or equivalent material. All internal bulkheads, ceilings and linings are to be of non-combustible materials. Corridor bulkheads are to be constructed of steel or to "B" class standards of SOLAS. Bulkheads of galley, paint stores, storage spaces for flammable materials and other areas of high fire risk are to be of steel or equivalent material.

4.2.3 Exposed surfaces in corridors and stairway enclosures and surfaces in concealed and inaccessible spaces in accommodation, service spaces and control stations are to have low flame spread characteristics.

4.2.4 Paints, varnishes and other finishes used on exposed interior surfaces are not to be capable of producing excessive quantities of smoke and toxic products. Primary deck coverings within accommodation, service spaces and control stations are to be of a type which will not give rise to smoke or toxic or explosive hazards at elevated temperatures.

4.3 Fire detection systems

4.3.1 Where machinery spaces containing essential machinery of 736 kW and above are not continuously manned and a remote centralized control system is installed for the operation of machinery, a fire detection and alarm system is to be provided in those spaces.

4.3.2 There are to be not less than two sources of power supply for the electrical equipment used in the operation of the fire detection and fire alarm system, one of which is to be an emergency source.

4.3.3 The location and spacing of detectors are to be suitable for optimum performance. The activation of any detector is to initiate a visual and audible signal at the continuously manned central control station.

4.4 Fire pumps, hydrants and hoses

4.4.1 The fire pumps, associated piping and fire main are to be so designed that a minimum pressure can be maintained at all hydrants of the dry dock and the largest vessel to be docked, sufficient to produce jet throw of at least 12 [m] through any adjacent hydrants with nozzles of size as required by 4.4.4. A fire main is to be provided on each dock wing. Two separate means of water supply are to be provided for the fire main.

4.4.2 If fire in one compartment could put all the fire pumps out of action, an alternative means through an independently driven emergency fire pump is to be provided. The emergency fire pump is to be located in a space, which will be accessible in case of fire in the space containing the main fire pumps. If shore side supply for fire fighting is readily available, consideration may be given for dispensing with the requirement of emergency fire pump.

4.4.3 The number and positions of the hydrants are to be such that at least two jets of water not emanating from the same hydrant,

one of which is to be from a single length of hose, may reach any part of the dock except the water ballast tanks under any operating conditions.

Two hydrants are to be provided in spaces containing machinery with a total power of 736 kW and over. One hydrant may be accepted in other machinery spaces. Where, fire fighting from within a small compartment is impracticable due to limitations in space, the hydrants may be situated outside and adjacent to the compartment entrance.

4.4.4 For accommodation and service spaces, a nozzle size of 12 [mm] may be adequate. For machinery space and exterior locations, nozzle sizes of 12 [mm], 16[mm] or 19[mm] may be used so as to obtain maximum discharge possible from the fire pumps.

4.4.5 It is recommended that an international shore connection is provided according to the specification given in Pt.6, Ch.8, Sec 2.

4.5 Portable fire extinguishers

4.5.1 Accommodation, service spaces and control stations are to be provided with portable fire extinguishers so placed that at least one extinguisher will be readily accessible from any part of the spaces.

4.5.2 For galleys and spaces containing domestic boilers, one portable fire extinguisher suitable for dealing with oil fires or fires in electric cooking equipment, as appropriate, is to be provided.

4.5.3 In machinery spaces portable fire extinguishers appropriate to the type of machinery are to be provided.

4.6 Fixed gas fire extinguishing systems

4.6.1 Where provision is made for injection of gas into machinery spaces for fire extinguishing purposes, the necessary pipes for conveying the gas are to be provided with control valves so marked as to indicate clearly the spaces to which the pipes are led. The valves are to be so positioned that they are readily accessible and not likely to be cut off by a fire in the protected space. Suitable provision is to be made to prevent inadvertent release of the medium into that space.

End of Chapter

Chapter 13

Pipe Laying and Cable Laying Vessels

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1	*General*
2	*Ship Strength & Arrangement*
3	*Stability*
4	*Station Keeping*
5	*Pipe Laying Equipment and Systems*
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7	*Tests and Trials*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter apply to vessels that are primarily engaged in installation of pipelines and cables. The requirements in this chapter are supplementary to the assignment of main characters of class.

1.1.2 The following aspects are subject of Classification:

- Hull structural details related to laying operations
- Laying equipment and systems
- Equipment for station-keeping/maneuvering during laying
- Supporting structures for equipment
- Anchoring and Mooring Equipment

1.2 Notations

1.2.1 Vessels in compliance with the relevant requirements of Sec 1 to 5 and 7 will be assigned notation:

PIPE LAYING VESSEL

1.2.2 Vessels in compliance with the relevant requirements of Sec 1 to 4, 6 and 7 will be assigned notation:

CABLE LAYING VESSEL.

1.2.3 Non self-propelled vessels complying with the requirements for the class notation **BARGE** (refer Pt.5, Ch.11) and which comply with the relevant requirements mentioned above in 1.2.1 and 1.2.2, may be given the following class notation, as appropriate

PIPE LAYING BARGE

CABLE LAYING BARGE

1.3 Documentation

1.3.1 In addition to the information and plans required by Pt.3, Ch.1, Sec.3, the following are to be submitted as applicable for approval:

1.3.1.1 Plans

i)Hull structural details related to pipe/cable laying. For pipe laying vessels hull structural details related to abandonment and recovery (A&R) are also to be provided.

ii)Plans of equipment and installations and their supporting structure for pipe/cable laying. For pipe laying vessels, plans of equipment and installations and their supporting structure for A&R operations are also to be provided.

iii)Plans of equipment for station-keeping/ maneuvering during pipe/cable laying.

iv)Electrical schematics, load analysis, short circuits analysis and coordination study for the vessel's electrical systems supplying pipe/cable laying. For pipe laying vessel above details are also to be provided for vessel's electrical systems supplying A&R system operations.

v)Piping schematics and calculations for the vessel's piping systems supplying pipe/cable laying systems.

vi)Arrangement and details of communication systems between the vessel and pipe/cable laying systems.

vii)For pipe laying vessels, arrangements and schematic diagrams for the anti-heeling and/or ballast systems, as appropriate.

1.3.1.2 **Calculations**

i) demonstrating the adequacy of the vessel's stability during all pipe/cable laying operations.

ii) demonstrating adequacy of maneuvering power required for the vessel to maintain station during pipe/cable laying operations.

iii) for the strength of the supporting structure in way of all pipe/cable laying and pipe/cable storing equipment.

iv)for the design loads and allowable deflection at each foundation of the pipe/cable laying and pipe/cable storing equipment

1.3.2 The following plans/ documents are to be submitted for information (as applicable) :-

i)Plans showing the locations of all pipe/cable laying equipment and control stations.

ii)Location and support details of the cranes indicating the reaction forces.

iii)Descriptions of equipment for moving skid frames, substructures, including piping and electrical systems, details of mechanical components, including hold-down devices and applicable strength calculations.

iv)Plans for towing and towing support structure including tow forces, angles, supporting structures for winches etc.

v)For Pipe Laying Vessels,

- Pipe Support arrangements (maximum forces expected to be indicated) on the pipe ramp.
- Arrangement of tensioners and supporting structures along with their tension capacities.
- Maximum forces expected on fastening of stinger to hull.
- Supporting structures for pipes stowed on racks in holds and/or deck; maximum weights are to be stated.
- Supporting structures for reels and/or carousels and/or in baskets, in holds and/or deck; maximum weights are to be stated.

vi)For Cable laying vessels,

- support arrangement for equipment used in cable laying operation such as cable drum, cable reels; maximum force are to be stated.
- Supporting structures for cable stowed on racks in holds and/or deck; maximum weights are to be stated.

vii)Any other documentation that IRS may require.

Section 2

Ship Strength & Arrangements

2.1 General

2.1.1 Limiting weather conditions for pipe/ cable laying operations are to be clearly defined. Calculation of the vessel's motions and associated structural loads for the lay equipment and its interface with the ship structure is to be carried out taking into account design parameters like intended operating and abandonment swell heights, together with associated wind, current speeds and pipe pull.

2.1.2 The work deck is to be strengthened for the specified design loads and an allowance for corrosion, wear and tear is recommended. Uniform deck loadings are to be specified.

2.1.3 The working deck, as far as possible, is to be kept clear of engine room intakes and exhausts. Obstructions from tank vents are to be minimized.

2.1.4 Additional requirements for Pipe Laying Vessels

2.1.4.1 Fuel tanks are not to be located directly under the working decks, unless a void space, hold, store, cofferdam or water ballast tank forms a separation space between the working deck and fuel tank.

2.1.4.2 If a pipe laying moon pool is provided, then the same is to be located near the center of gravity of the vessel. Openings are to be at a suitable distance from the deck edge, from cargo hatch covers, from superstructure breaks and from other area of structural discontinuities. The side structure of moon pools are to be designed for impact loads from pipe laying. As a minimum, moon pool side structure is to comply with requirements for side shell plating given in Pt.3. The corner radius of the moon pool opening is not be less than 0.125 times the width of the moon-pool opening but it need not exceed 600 [mm]. Free edges of the moon pool opening are to be suitably rounded. Means are to be provided to prevent personnel from falling into the moon pool.

2.1.4.3 Provision of a clamping system capable of dealing with emergencies, such as flooded pipeline, and an A&R winch system capable of abandoning and recovering the pipeline are to be considered when designing the foundations and their integration into the vessel's structure.

2.2 Cranes

2.2.1 Cranes fitted on the vessel are to be approved by IRS.

2.3 Hull Strength & Design Loads

2.3.1 The hull structural strength is to be as required for the main class taking into account necessary strengthening of supporting structures for equipment applied in the laying, abandonment and recovery operations in pipe laying vessels.

2.3.2 The scantlings of supporting structures for laying equipment and cranes are to be worked out based on the maximum expected operational loads.

2.3.3 Ship structures supporting heavy components of laying equipment are to be designed considering accelerations given in Pt.3, Ch.4.

2.3.4 Acceleration loads need not be combined with normal laying or lifting operation loads of pipe laying equipment and deck cranes. Alternatively, accelerations derived from other recognized standards or direct calculations, model tests considering the most severe environmental conditions the vessel is expected to encounter may be considered.

Section 3

Stability

3.1 General

3.1.1 In addition to the stability requirements mentioned in Pt.3 Ch.1 the stability calculations have to be performed in accordance with 3.1.3. Loading conditions covering departure and arrival in full load as well as ballast conditions, along with anticipated operational or intermediate conditions at site are to be included in the Trim and Stability booklet.

3.1.2 Vessel equipped with lifting devices in addition to the laying devices(s), are to comply with the relevant requirement related to vessels engaged in lifting operation in IS Code 2008, as amended.

3.1.3 In evaluating the loading conditions, the following loads are to be included:

3.1.3.1 General

i)The overturning moments due to environmental and operational loads on the laying devices are to be included. The full range of laying device positions is to be considered (in order to investigate the most critical condition), if the laying device is movable from stowage to operating condition.

ii)The effect of pipe racks/cable drum and reels on the stability is to be assessed for each operating condition.

iii)If the vessel is fitted with or carries open cargo bins on the deck that may accumulate water, either an appropriate free surface correction is to be applied or an effective means to drain water from these spaces is to be provided.

iv) If large and heavy equipment or structures are intended to be stowed on deck, the estimated weight and height of the center of gravity in the worst possible scenario are to be considered in the stability analysis.

v) Where pipes/cable are carried on deck, a quantity of trapped water equal to a certain percentage of the net volume of the pipe/cable deck cargo should be assumed in and around the pipes/cables. The net volume is to be taken as the internal volume of the pipes/cables, plus the volume between the pipes/cables. This percentage is to be 30% if the freeboard amidships is equal to or less than 0.015*L* and 10% if the free board amidships is equal to or greater than 0.03*L*. For intermediate values of the freeboard amidships, the percentage is to be obtained by linear interpolation. In assessing the quantity of trapped water, the positive or negative sheer aft, actual trim and area of operation may be taken into account. Where pipe/cable is stored on decks located above the freeboard deck, the distance from the freeboard deck to the storage deck is to be added to the freeboard in the above calculation to determine the amount of trapped
water.

3.1.3.2 Pipe Laying Vessel

3.1.3.2.1 If the vessel's layout results in the pipeline creating a transverse moment relative to centreline, the vessel is to comply with the stability requirements for vessels engaged in lifting operations (refer 3.1.2), with the transverse heeling moment of the pipeline substituted for the crane heeling moment.

3.1.3.2.2 The following requirements are to be satisfied for the case where the pipe is flooded during the pipe lay operations:

i) The final waterline, at the equilibrium heel angle due to the maximum design holding capacity of the abandonment and recovery (A&R) system, must be below the lower edge of any weathertight opening through which progressive flooding may occur.

ii The righting arm curve must be positive for a range of 20 degrees after the equilibrium heel angle due to the maximum design holding capacity of the A&R system.

iii)The righting arm must be at least 100 [mm] within the 20 degree range.

iv)Each opening submerged in the range of stability must be weathertight.

Section 4

Station keeping

4.1 Station Keeping

4.1.1 Vessels are to be capable of maintaining their positions safely during laying operations. Position keeping may be attained by a mooring system with anchors or a dynamic positioning system.

4.1.2 Anchors, cables and mooring winches when used to maintain the vessel's position during pipe laying operations, are to comply with the requirements given in Chapter 10 of IRS Rules and Regulations for the construction and Classification of Mobile Offshore Drilling Units. Safety precautions are to be considered to prevent damaging seabed equipment and installations by anchor deployment, recovery and station keeping.

4.1.3 Dynamic positioning systems, when used to maintain the vessel's position during pipe laying operations, are to comply with the requirements for the class notation **DP(2)** or **DP(3)**.

4.1.4 For pipe laying vessel, where the DP system has interfaces with the pipe tensioner system, the potential effects of the operations of the pipe tensioner system on the DP station keeping capability are to be addressed in the DP FMEA.

Section 5

Pipe Laying Equipment and Systems

5.1 General

5.1.1 The recognized standard used in design of pipe laying equipment and systems is to be specified by designer and acceptable to IRS. Installations and onboard testing are to be supervised in the aspects of operational safety as to reduce to a minimum any danger to persons on board and marine pollution, due regard to be paid to moving parts, hot surfaces and other hazards. Considerations are to be given to the consequences of the failure of systems and equipment essential to the safety of the vessel by suitable FMEA analysis.

5.1.2 Pipe laying methods

5.1.2.1 S Lay

5.1.2.1.1 In S-lay method of installation, the pipe is laid from a near-horizontal position using a combination of horizontal tensioner and a stinger controlling the curvature at over-bend. The required lay tension is to be determined based on the water depth, the submerged weight of the pipeline, the allowable radius of curvature at over-bend, departure angle and the allowable curvature at the sag-bend. The stinger limitations for minimum and maximum radius of curvature and the pipeline departure angle are to be met.

5.1.2.2 J Lay

5.1.2.2.1 In J-Lay method, the pipe is laid from an elevated tower on a lay vessel using longitudinal tensioner. In this way, over-bend at the sea surface is avoided. In general, J-Lay follows the same procedure as S-Lay, but it is near vertical.

5.1.2.3 Reel Lay

5.1.2.3.1 For reel lay, the pipe is spooled onto a large radius reel aboard a reel lay vessel. The reel-off at location will normally occur under tension and involve pipe straightening through reverse bending on the lay vessel.

5.2 Control Systems & Communication

5.2.1 A central control station is to be provided for controlling and/ or coordinating pipe laying equipment operations. The central control station is to be located at a position that allows the operating personnel to have a clear view of the pipe departure location and to provide an

overview of all systems and activities associated with the pipe laying operations. Tension measurements on the paid-out pipe are to be displayed at the control stations.

5.2.2 The central control station and the vessel station keeping control station are to be linked by both hard-wired communication system and manually operated alarm system. Means of communications are to be provided between the central control station and the local control stations for the pipe laying equipment.

5.3 Pipe Lay Equipment & Systems

5.3.1 Abandonment & Recovery system (A&R)

5.3.1.1 Abandonment and Recovery (A&R) system is to be provided to lay the pipe down on operation completion or the onset of harsh weather and for recovering the pipe after such an event. The system may consist of a hydraulic or electric motor driven traction winch, storage winch, and sheaves.

5.3.1.2 The A&R system is to be designed to provide adequate dynamic and brake holding capacity to control combined loads of the expected maximum tension generated by flooding the heaviest pipe and environmental forces such as waves, currents and tides, etc., exerting on the vessel as well as the laid-off pipes.

5.3.1.3 A&R system is to be provided with a power control braking or a mechanically controlled braking means capable of maintaining controlled lowering speeds. Brakes are to be applied automatically upon loss of power or when the control lever is returned to neutral

5.3.1.4 Detail drawings of the foundation and supporting structure on which the A&R winch are installed are to be submitted in accordance with section 1. The foundations and supporting structures of the A&R winch and accessories are to be designed for the design static and dynamic loading conditions of the A&R winch defined in 5.3.1.2.

5.3.2 Pipe Racks, Reels and Support Structure

5.3.2.1 Pipe reels are to be designed, constructed and installed in accordance with a recognized standard.

5.3.2.2 Pipe racks and reel support structure including the reinforcements for the hull are to be designed to adequately resist the load effects of pipes, risers or reels imposed on the supports in the severe storm, normal operating and transit conditions. Considerations should also be given to the unit in damaged conditions, where the pipe racks and reel support structure are to withstand the load effects caused by the trim and heel of the vessel.

5.3.3 J-lay/S-lay/Reel-lay Tower and Skid Frame

5.3.3.1 J-lay, S-lay, Reel-lay towers and skid frames are to be designed in accordance with a recognized standard.

5.3.3.2 Detail drawings of the foundation and supporting structure on which the pipe-lay towers and skid frames rest or other stowage arrangements are installed are to be submitted for approval.

5.3.3.3 The following are to be considered whilst undertaking strength calculations of supports :-

- the tower's weight at extreme positions - upright and tilted position for J-lay/S-lay towers,
- upright elevation angles for pipe spooling-out and angled for pipe spooling-on and skid fleeting locations for Reel-lay towers, including associated center of gravity variations
- static tension of the pipe and dynamic effects caused by environmental forces.

5.3.4 Pipe Tensioners

5.3.4.1 Reliable tensioners are to be provided for maintaining tension in the pipeline while it is being lowered onto seabed.

5.3.4.2 The tensioner is to be designed to provide adequate dynamic and brake holding capacity to control combined loads of the tension generated by the paid-out pipeline, motions and environmental loads. The required type depends on pipe specifics and configuration of the pipe-lay system.

5.3.4.3 The tensioners are to be provided with power controlled system operable at a remote control station to control and monitor the paid out length with feedback to the lay speed setting or vessel's station keeping system.

5.3.4.4 The foundations and supporting structures of the tensioner and accessories are to be designed for the load effects of pipe tensioner capacity defined in 5.3.4.2.

5.3.4.5 Hang-off clamps are to be provided to hold the pipe when it is not suspended by tensioners or alternative means to hold the pipe such as A&R systems. Means are to be provided to ensure the holding power in case of electrical power black-out.

Section 6

Cable laying Equipment and Systems

6.1 General

6.1.1 The recognized standard used in design of cable laying equipment and systems is to be specified by designer and acceptable to IRS. Installations and onboard testing are to be supervised in the aspects of operational safety as to reduce to a minimum any danger to persons on board and marine pollution, due regard to be paid to moving parts, hot surfaces and other hazards. Considerations are to be given to the consequences of the failure of systems and equipment essential to the safety of the vessel by suitable FMEA analysis.

6.2 Control Systems & Communication

6.2.1 A central control station is to be provided for controlling or coordinating the operations of the cable laying equipment. The central control station is to be located at a position that allows the operating personnel to have a clear view of the cable departure location and to provide an overview of all systems and activities associated with the cable laying operations.

6.2.2 The cable laying central control station and the vessel station keeping control station are to be linked by a hard-wired communication system and a manually operated alarm system. Means of communications are to be provided between the central control station and the local control stations for the cable laying equipment.

6.3 Cable-lay Equipment and Systems

6.3.1 Cable Drums, Reels, Deployment Sheaves and Support Structure

6.3.1.1 Cable drums and reels are to be designed, constructed and installed in accordance with a recognized standard.

6.3.1.2 Cable drum, reel and deployment sheave support structure including the reinforcements for the hull are to be designed to adequately resist the load effects of cables or reels imposed on the supports in the severe storm, normal operating and transit conditions. Considerations should also be given to the unit in damaged conditions, where the cable reel support structure are to withstand the load effects caused by the trim and heel of the vessel.

Section 7

Tests & Trials

7.1 Installation of the laying equipment and systems on pipe/ cable laying vessels is to be to the satisfaction of the attending Surveyor.

7.2 Upon installation, functional integration tests of the laying systems on board to the extent and as per the method agreed are to be carried out to the satisfaction of the attending Surveyor.

7.3 Position mooring equipment is to be tested in accordance with the specifications of the owner and in the presence of a Surveyor.

7.4 Dynamic positioning system is to be tested in accordance with Pt 5, Ch.24 of the Rules.

End of Chapter

Chapter 14

Low Flash Point Fuel Bunkering Vessels

Contents

Section	
1	*General*
2	*Liquefied Natural Gas*

Section 1

General

1.1 Application

1.1.1 The requirements of this Chapter are applicable to vessels carrying low flash point fuel and intended to transfer the fuel to ships using such fuel.

1.1.2 In general, this Chapter is applicable to bunkering and transfer systems of the ship, which are to additionally comply with the applicable requirements of Part 4.

1.2 Notations

1.2.1 Vessels complying with the requirements of this Chapter may be assigned additional class notation **LFPF BUNKERING VESSEL ([fuel name])** Or **LFPF BUNKERING BARGE ([fuel name])**

1.2.2 Vessels complying with the requirements in Section 2 may be assigned the following notations:

LFPF BUNKERING VESSEL (LNG)
LFPF BUNKERING BARGE (LNG)

1.3 Definitions

1.3.1 **Bunkering emergency shut-down system (ESD) -** A system that safely and effectively stops the transfer of fuel (and fuel vapour as applicable) between the receiving vessel and the bunkering vessel in the event of an emergency during the bunkering operation, and puts the system in a safe condition.

1.3.2 **Bunkering connections -** Bunkering connections correspond to the end of the fixed piping to the bunkering vessel (i.e. manifold for a system with flexible hose and before the swivel for a system with transfer arm).

1.3.3 **Emergency release coupling (ERC) -** An ERC is a coupling located on the receiving vessel bunkering manifold or on the fuel transfer system, which separates at a predetermined section, when required, each separated section containing a self-closing shut-off valve, which seals automatically. An emergency release coupling can be activated:

• by maximum allowable forces applied to the predetermined section

• by manual or automatic control, in case of emergency.

1.3.4 **Enclosed space -** Enclosed space means any space within which, in the absence of artificial ventilation, the ventilation will be limited and any explosive atmosphere will not be dispersed naturally.

1.3.5 **Hazardous area -** An area in which an explosive gas atmosphere is or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of electrical apparatus. Hazardous areas are divided into Zone 0, 1 and 2 as defined below and according to the area classification specified in 2.3.4 for LNG:

• Zone 0:
Area in which an explosive gas atmosphere is present continuously or is present for long periods.

• Zone 1:
Area in which an explosive gas atmosphere is likely to occur in normal operation.

• Zone 2:
Area in which an explosive gas atmosphere is not likely to occur in normal operation and, if it does occur, is likely to do so only infrequently and will exist for a short period only.

1.3.6 **Transfer system -** A system used to connect the bunkering vessel and the receiving ship to transfer fuel only or both fuel and fuel vapour. The transfer system includes:

• rigid pipes, hoses, swivels, valves, couplings

• supporting structure

• handling system and its control/monitoring system.

It also includes the compressors or blowers intended for the boil-off gas pressure management, when required and applicable.

1.3.7 **Vapour return lines -** A connection between the bunkering vessel and the receiving vessel to prevent pressure increase in the receiving tank due to liquid transfer and associated boil-off.

1.3.8 **Non-hazardous area -** An area in which an explosive gas atmosphere is not expected to be present in quantities such as to require special precautions for the construction, installation and use of electrical apparatus.

1.3.9 **Open deck -** A deck that is open on both ends, or is open on one end equipped with adequate natural ventilation that is effective over the entire length of the deck through permanent openings distributed in the side panels or in the deck above.

1.3.10 **Quick connect disconnect coupler (QCDC) -** A QCDC is a manual or hydraulic mechanical device used to connect the transfer system to the receiving vessel manifold.

1.3.11 **Receiving vessel -** A receiving vessel is a vessel receiving the bunker fuel.

1.3.12 **Safety zone -** A zone around the bunkering vessel, the bunkering station of the receiving vessel and the transfer system, where the only activities performed are the bunkering operations and related activities and where measures are taken to prevent leakage of fuel or fuel vapour and to control sources of ignition.

1.3.13 **Semi-enclosed space -** A space limited by decks and or bulkheads in such manner that the natural conditions of ventilation are notably different from those obtained on open deck.

1.3.14 **Transfer arm -** Transfer arm refers to any system allowing supporting hoses or rigid pipes during bunkering operations.

1.3.15 **Pendant ESD Unit** – A hand held portable unit for controlling the ESD

1.3.16 **Presentation Flange** - The part of the ship's pipeline onto which the reducer is fitted when connecting the loading/discharging hoses/arms prior to cargo operations.

1.3.17 **Toxic Areas** - Toxic areas are areas where toxicity risk from potential leak sources exist.

1.4 Documentation

1.4.1 Following additional documents are to be submitted for approval, as relevant and applicable:

a) General arrangement of the vessel showing the location of the bunkering station and bunkering control station
b) Details of maximum bunkering flow and maximum pressure
c) Details of transfer system
d) Details of ESD for Bunkering system
e) Instrumentation list
f) Drawing of transfer arm
g) Toxic Area Classification Plan

1.4.2 Following documents are to be submitted for review/ information:

a) Risk analysis of transfer system
b) Safety certificates for electrical equipment, concerning the bunkering, located in hazardous areas, where applicable
c) Bunkering procedure, including inerting and gas freeing
d) BOG management procedure (LNG Fuel)
e) Operating envelop of the bunkering vessel
f) Risk analysis of the bunkering operations

1.4.3 Operation Manual

1.4.3.1 An operation manual is to be provided for crew and it is to contain instructions and guidance for the following:

a) preparation before the bunkering including restriction of activities/operations during gas fuel transfer
b) information to be exchanged between bunker vessel and receiving ship prior to operation
c) hose handling guidelines reflecting specific hose manufacturer instructions for the hose handling
d) procedures for connection including line inerting and tightness test
e) preparations for start of bunkering, i.e. preparedness of fire-fighting, tightness testing, establishing communications, allocation of personnel/responsibilities
f) pre-cooling of transfer connection(s) and transfer procedures
g) draining of the pipeline, purging and disconnection on completion of the transfer
h) operational restrictions to prevent dangerous pressure surge effect in the pipes
i) fire safety during the transfer
j) procedures for raising alarms
k) procedures in case of communication failure
l) suspension of operation during emergencies
m) procedures for authorization of ERS activation
n) emergency procedures for: gas fuel leakage, termination of the bunkering and emergency disconnection, response in case of unintentional disconnection of ERS, etc.

1.4.3.2 Cargo system piping diagram, function flowchart or cause and effect (C&E) diagram for ESD and related systems are to be available in the bunker/cargo control room where such room is established.

Section 2

Liquefied Natural Gas

2.1 General

2.1.1 The requirements of this Section are applicable to vessels carrying LNG fuel and intended to transfer the LNG to ships using LNG as fuel.

2.1.2 Requirements in this Section are in addition to the requirements of those given in Part 5 Chapter 4 (i.e. IGC Code).

2.2 Notation

2.2.1 Following additional qualifiers, mentioned in the individual sub-sections, as applicable, may be assigned:

RE - where the vessel is designed to receive LNG from a gas fuelled vessel for which the LNG fuel tanks have to be emptied(Refer Sub-section 2.8).

ICD - where the vessel is designed for initial cooling down of the gas fuelled vessel's LNG fuel tank(Refer Sub-section 2.8).

IGS - where the vessel is designed to supply inert gas and dry air, to ensure gas freeing and aeration, to a gas fuelled vessel complying with Pt.5, Ch.35, 6.10.4 (Refer Sub-section 2.8).

VR - where the vessel is designed to recover and manage the boil-off gas generated during the bunkering operation (Refer Sub-section 2.8).

2.3 Arrangement

2.3.1 Material Requirements

2.3.1.1 Materials used in LNG transfer systems, piping system for liquefied gas and other systems or components in contact with gas are to be in accordance with Pt. 5, Ch. 4, Sec 6 of the Rules.

2.3.2 Tanks

2.3.2.1 The location of cargo tanks is to be in accordance with Pt. 5, Ch. 4, Sec 2 of the Rules.

2.3.2.2 A bunkering vessel is to be able to abort bunkering operations at any stage in case of emergency. Cargo tanks on bunkering vessel therefore are not to have restrictions on intermediate filling. However, internal transfer between cargo tanks within short period of time to leave dangerous sloshing zone may be specially considered.

2.3.3 LNG Bunkering Station

2.3.3.1 Bunker Manifold Area

2.3.3.1.1 The LNG bunkering station(s) and manifold(s) are to be located in the open deck so that sufficient natural ventilation is provided and are to be sufficiently illuminated. Bunker manifold area and escape routes are to have safe access for crew engaged in operation.

2.3.3.1.2 The bunker manifold and connections are to be clearly visible from the bunkering control station and cargo control station or bridge, as applicable. CCTV arrangements may be accepted as an alternative where direct line of sight from the aforementioned locations is not possible.

2.3.3.1.3 Arrangement of work platforms in areas where LNG spill may occur is to prevent spill accumulation at the platform surface. Gratings used in this location are to be suitable for low temperatures and correspond to boiling point of LNG. Areas under the gratings are to be equipped with drip trays with drainage arrangements suitable for draining the accumulated spill overboard. The drain is to be fitted with a valve.

2.3.3.1.4 Hull in the manifold area is to be effectively protected from low temperature in case of an LNG spill. A water piping system is to be fitted in way of the hull under the bunkering manifold to provide low-pressure water curtain for additional protection of the hull steel and the vessel's structure. This system is to be in operation when transfer system is in progress. Other solutions are acceptable with justification. Where water curtain is used for hull protection, the pumps are to be arranged with redundancy

2.3.3.1.5 The manifold area is to be clear of obstructions which may interfere with quick release.

2.3.3.1.6 The bunkering station is to be physically separated or structurally shielded from accommodation areas and control stations.

2.3.3.1.7 Structural strength calculations and drawings of manifolds are to be submitted and SWL indicated on the manifolds.

2.3.3.2 Hazardous areas created during bunkering operations

2.3.3.2.1 Specific attention is to be paid to the hazardous areas created during the bunkering operations and to restrict access in order to avoid the presence of unauthorized persons in the vicinity of these hazardous areas and the possibility to create source of ignition.

2.3.3.3 Drip trays

2.3.3.3.1 Drip trays are to be fitted below the bunkering connections and where leakage may occur which can cause damage to the vessel structure.

2.3.3.3.2 Thermal sensors of approved type are to be positioned in way of bunkering connections in the drip tray.

2.3.3.3.3 The drip trays are to be made of stainless steel, and are to be able to remotely emptied without risk of damage to the vessel structure and to the receiving vessel.

2.3.3.4 Bow and stern arrangements

2.3.3.4.1 LNG bunkering station may be accepted at the ship bow and stern provided that the relevant requirements of Pt.5, Ch.4, 3.8, are satisfied.

2.3.3.5 Control Stations for Bunkering Operations

2.3.3.5.1 The bunkering control station is to be considered as a control station with regard to the applicable requirements of Part 4, Part 6 and Part 5, Chapter 4 (IGC Code) of the Rules.

2.3.3.5.2 Control of the bunkering operation is to be possible from a safe location with regards to bunkering operations and may be from the cargo control room. Overfilling alarms, automatic and manual shutdown are be indicated at this location.

2.3.3.6 Ventilation

2.3.3.6.1 Ventilation of closed or semi-enclosed spaces is to be of mechanical type

and take place in the lower part of the compartment. A gas detection system is to be fitted in such spaces.

2.3.3.6.2 Ducts used for the ventilation of hazardous areas are to be separated from that used for the ventilation of non-hazardous areas. The ventilation is to be capable of functioning at all temperature conditions the vessel is designed to operate in. Electric fan motors are not to be located in ventilation ducts for hazardous areas unless the motor is certified for the same area classification and operating conditions as the space served.

2.3.3.6.3 Ventilation ducts are to have the same area classification as the ventilated space.

2.3.3.6.4 Ventilation capacity is to be in accordance with Pt.5, Ch.4, Sec.12.

2.3.4 Hazardous areas

2.3.4.1 General

The hazardous areas are to be in accordance with Pt.5, Ch.4.

2.3.4.2 Zone 1

In addition to the areas indicated in Pt.5, Ch.4, this zone includes the following:

- LNG bunkering station
- areas on the open deck within spillage coamings surrounding gas bunkering manifold valves and 3 [m] beyond these, up to a height of 2.4 [m] above the deck
- transfer arm operating amplitude, when applicable.

2.3.4.3 Classification of spaces adjacent to hazardous areas

A space separated by gastight boundaries (with or without opening) from an hazardous area may be classified as zone 1, 2 or considered as hazardous, taking into account the sources of release inside that space, the type and arrangements of openings and the conditions of ventilation, in accordance with IEC 60092-502, paragraph 4.1. A type approved gastight bulkhead penetration device is not considered as a source of release.

2.3.5 Mooring Equipment

2.3.5.1 The bunkering vessel is to be equipped with sufficient number of closed type fairleads for safe mooring to receiving vessel. Mooring fittings and arrangements are to have their safe working load (SWL) corresponding to wind and current forces acting on the bunkering vessel with allowance for dynamic forces occurring due to vessel's relative movement.

2.3.5.2 Steel mooring wire ropes are not to be used unless synthetic tails are fitted at the wire ends.

2.3.5.3 Provisions are to be made for emergency cast-off of mooring lines. Where fixed or portable quick release mooring hooks are used, they are to comply with a recognised standard.

2.3.5.4 Sufficient number of fittings are to be available for primary (parallel body length) and secondary (bow and stern quarters and superstructures) fenders.

2.3.5.5 The fenders are to be sized to provide sufficient energy absorption for the size of the vessel at approach velocities and sea mooring conditions. Number and size of the fenders are to meet recommendations of OCIMF/ SIGTTO/ ICS "*Ship to ship transfer guide for petroleum, chemicals and liquefied gases*".

2.4 Bunker Transfer System

2.4.1 General Requirements

2.4.1.1 The design and installation of the LNG transfer system are to be substantiated by a risk analysis. The risk analysis is to assess the consequences of the following (including, but not limited to):

- o A failure affecting the concerned system
- o LNG leakage
- o Tank over-pressurisation

2.4.1.2 The LNG transfer system is to include a Quick Connect Disconnect Coupler (QCDC), a Break-away coupling or an Emergency Release Coupling (ERC) and insulation flanges.

2.4.1.3 Transfer systems and their associated systems are to be considered as essential services.

2.4.1.4 The transfer system is to be designed to avoid the release of gas or liquid to the atmosphere during bunkering operations.

2.4.1.5 Cargo sampling connections are to be provided in accordance with Pt.5, Ch.4.

2.4.1.6 Possibility to perform tightness test of the bunker connections between bunker vessel and receiving ship prior to operation is to be provided. Such procedure is to be described in the operation manual.

2.4.1.7 Suitable arrangements are to be provided for inerting of the lines before filling them with the bunker vapour and for safe displacement of bunker liquid and vapour from bunker lines prior to disconnection.

2.4.1.8 Sections of the line where liquid may remain after the bunker transfer are to be equipped with drain system leading back to the cargo tanks. Means are to be provided to ensure that section of the transfer line outboard of manifold valve is free of liquid. Where such arrangement is a control valve with drain opened to atmosphere, this control valve bore has to be limited to maximum 1.5 [mm].

2.4.1.9 The section of the line between cross-over valve and presentation flange is to be equipped with pressure gauge visible from the drain valve position.

2.4.1.10 Bunker transfer piping system is to be thermally insulated to minimise heat leaks to transferred gas bunker and protect personnel from direct contact with cold surfaces.

2.4.1.11 Arrangement for control of distance between the bunker ship and receiving ship manifolds in transverse and longitudinal directions is to be fitted. If the permitted working distance is exceeded, then alarm is to be triggered, followed by automatic stop of transfer and closing of the manifold valves.

2.4.1.12 Where quick connect/disconnect couplers (QCDC) are used they are to be equipped with mechanical locking device to prevent inadvertent release. Powered QCDC are to stay in “as is” position at loss of power.

2.4.1.13 In order to prevent cryogenic liquid spills, the design of the transfer system is to be such that the lines can be drained before disconnection and purged after an emergency disconnection.

2.4.2 Manifold

2.4.2.1 The manifold for transfer of bunker is to be fitted with manually operated stop valve and a remotely/ automatically operated valve (ESD valve) fitted in series. The requirement for two valves is applicable irrespective of design pressure of cargo tanks fitted at the bunker vessel to make the transfer connection suitable for operation with different MARVS setting at supply and receiving ships. A combined manually operated and remote valve may also be accepted.

2.4.2.2 The vapour return manifold is to be fitted with a remotely/ automatically operated stop valve. Manual local operation of the valve is to be possible, e.g. by portable means.

2.4.2.3 Safe working load (SWL) of the manifold is to comply with the requirements in “*Recommendations for liquefied gas carrier manifolds (OCIMF, 2018,as may be amended)*” and be capable of accommodating forces occurring due to vessel’s relative motion, ERC self-release force and dynamic forces due to self-release.

2.4.2.4 Information about maximum safe working load (SWL) of bunker connection is to be available on-board and posted at the bunker station.

2.4.2.5 All pipelines, hoses or components which may be isolated in a liquid full condition, or where liquid may accumulate, are to be protected by relief valves. The outlet from these pressure relief valves is to be led to the LNG tanks. Alternatively, they may discharge to the vent mast if means are provided to detect and dispose of any liquid cargo which may flow into the vent system.

2.4.2.6 Strainers are to be fitted to protect control valves and systems.

2.4.2.7 Manifold connections not being used for bunker transfer operations are to be blanked with blind flanges rated for the design pressure of the pipeline system.

2.4.3 Hoses

2.4.3.1 General

2.4.3.1.1 Bunker transfer hoses are to be certified in accordance with requirements for cargo hose in Pt.5 Ch.4 Sec.5, 11.7.

2.4.3.1.2 Hoses are to be protected by relief valves fitted outboard manifold valve.

2.4.3.1.3 Hoses used are to be single length hoses from export manifold to import manifold and provide sufficient allowance for vessel's relative motion (including rolling) within operation envelope. Combination of hoses connected in series may be accepted when effective control of the connections' integrity can be arranged and when such connections will not impose excessive stress on the flexible part of the hoses or change hose radius beyond acceptable limits.

2.4.3.1.4 Arrangements for support of the hoses at critical points such as, hose rails, handrails, hose suspension points, etc. are to be provided to avoid excessive local stress and for maintaining of acceptable bending radius of the hoses. Such arrangements are not to constrain the emergency release function. Hoses are to be arranged with enough slack to allow for all possible movements between the receiving ship and the bunkering ship.

2.4.3.1.5 The arrangement of the hoses when they are connected for bunker transfer is to consider possible direct contact or close proximity with hull structures. The hull in such locations is to be effectively protected from the effect of low temperatures.

2.4.3.1.6 Sufficient space for hose stowage after disconnection from receiving ship is to be provided. The stowage arrangement is also to consider permitted bending radius for the hoses.

2.4.3.1.7 Where the stowed hoses may contain residues of liquid with boiling temperature below the hull steel design temperature, the ship hull structure is to be protected from the low temperature effect in way of the bunker hoses stowage location.

2.4.3.2 Design requirements

2.4.3.2.1 General

The following characteristics are to be defined and submitted by the designer:

- Extreme service temperature
- Maximum working load
- Maximum design pressure
- Minimum bend radius (MBR)
- Maximum allowable applied twist (MAAT).

2.4.3.2.2 Maximum design pressure

The maximum design pressure is not to be less than 1 [MPa] in accordance with Pt.5, Ch.4, 5.11.7.3

2.4.3.2.3 Materials

All materials are to be compatible with each other and with the fluid conveyed (i.e. LNG and LNG vapours).

2.4.3.2.4 End connection and coupling

The end fittings are to be made of stainless steel and be in accordance with IGC Code.

2.4.3.2.3 Approval of bunkering hose

2.4.3.2.3.1 Hoses are to comply with recognised standards such as EN 1474-2.

2.4.3.2.3.2 All hoses are to be tested at the plant of manufacturer in the presence of the Surveyor.

2.4.3.3 Hoses onboard

2.4.3.3.1 Manufacturer's instructions regarding testing, storage and number of temperature and pressure operating cycles before removal from service of transfer hoses, are to be strictly followed.

2.4.3.3.2 The maximum service life of the hose assembly is not to exceed 5 years and hoses are to be inspected periodically during the annual survey of the bunker vessel.

2.4.3.3.3 A document containing the following information is to be kept on board:

- Hose identification number
- Type approval certificate
- Date of initial entry into service
- Initial test and certificates
- Records of all transfer operations.

2.4.3.3.4 Each hose is to be permanently marked with at least the following information:

- Manufacturer's name or logo
- Hose designation and size
- Maximum working pressure
- Maximum and minimum working temperature
- Overall weight of the hose and end fittings assembly
- Date of manufacture

- Society's brand as relevant
- Date of last inspection and testing.

2.4.4 Quick Connect Disconnect Coupler (QCDC)

2.4.4.1 Type approval of QCDC

2.4.4.1.1 QCDCs are to be type approved.

2.4.4.1.2 All QCDCs are to be tested at the plant of manufacturer in the presence of the Surveyor. An alternative survey scheme, may be agreed with IRS.

2.4.4.1.3 Each manufactured QCDC is to be tested as defined in 2.4.4.3.1 (QCDC used for prototype testing are not to be used onboard).

2.4.4.2 Type testing

2.4.4.2.1 The QCDC is to be subjected to a type test to confirm the release performance under ice built up condition.

2.4.4.3 Workshop testing

2.4.4.3.1 The QCDC is to be subjected to a hydraulic pressure test, at ambient temperature, to a pressure not less than 1.5 times the design pressure, to demonstrate that the QCDC is capable of withstanding its pressure without leaking.

2.4.5 Break-away and emergency release coupling (ERC)

2.4.5.1 General

2.4.5.1.1 The bunkering line is to be designed and arranged to withstand the surge pressure that may result from the quick closing of ESD valves and activation of the break-away or the ERC. If not demonstrated to be required at a higher value due to pressure surge considerations, a default time of 5 seconds from the trigger of the alarm to full closure of the ESD valves is to be adjusted, in accordance with Pt.5, Ch.35, 16.7.3.7

2.4.5.1.2 Justifications are to be submitted regarding the compatibility with hoses and the maximum axial and shear forces likely to be exerted on the break-away or the ERC during bunkering operations. Details of the manifold loads are to be submitted to IRS for information.

2.4.5.1.3 ERC is to be designed for:

- Remote and local manual activation
- Automatic activation in case of excessive forces
- Automatic activation in case the safe working envelope of the loading arm is exceeded.

2.4.5.1.4 In the event of activation of the break-away or the ERC, the hoses are to be adequately supported and protected to prevent potential damage, spark or rupture due to mechanical shocks.

2.4.5.1.5 All electrical components of the emergency release coupling actuator are to be of a suitable safe type. When applicable, the availability of hydraulic power is to be monitored. If the power supply of ERC by the hydraulic source is no longer available, bunkering operation is to be stopped.

2.4.5.2 Type approval of break-away and ERC

2.4.5.2.1 Break-away and ERC are to be type approved.

2.4.5.2.2 All break-away and ERC are to be tested at the plant of manufacturer in the presence of the Surveyor. An alternative survey scheme, may be agreed with IRS.

2.4.5.2.3 Each manufactured break-away and ERC are to be tested as indicated in 2.4.5.3.1 (break-away and ERC used for prototype testing are not to be used onboard).

2.4.5.3 Type testing

2.4.5.3.1 The break-away or the ERC are to be subjected to a type test to confirm the values of axial and shear forces at which it automatically separates. The tightness of the self-closing shut-off valves after separation is to be checked.

2.4.5.3.2 The break-away or the ERC are to be subjected to a type test to confirm the release performance under ice built up condition.

2.4.5.3.3 When applicable, the ERC is to be subjected to a type test to confirm the automatic release in case of activation.

2.4.5.4 Workshop testing

2.4.5.4.1 The break-away or the ERC are to be subjected to a hydraulic pressure test, at ambient temperature, to a pressure not less than 1.5 times the design pressure, to demonstrate that the break-away or the ERC are capable of withstanding its pressure without leaking.

2.4.6 Electrical isolation flanges

2.4.6.1 A single insulation (isolation) flange (in accordance with applicable requirements of recognised industry standards such as "International Safety Guide for Oil Tankers and Terminals (ISGOTT), 6th Edition" and "SIGTTO "LNG Ship to Ship Transfer Guide for Petroleum, Chemicals and Liquefied Gases") is to be provided in each transfer arm or hose of the transfer system between the receiving vessel manifold and the bunker pipeline. The earthing point is to conform to local electrical codes.

2.4.6.2 Each insulation flange is to be subjected to a test of electrical resistance in air and the resistance is to be not less than 10000Ω.

2.4.6.3 The resistance of each insulation flange is to be measured after installation in the complete LNG transfer system and the resistance is to be not less than 1000Ω. To protect against the accumulation of static charge, the resistance is to be less than 1MΩ

2.4.6.4 Bonding cables/straps should not be used unless required by the Administration. In case bonding cable/strap is to be used, the circuit continuity is to be made via a "certified-safe" switch (such as one housed inside an inherently-safe enclosure) and the connection on board the vessel is to be in a location remote from the hazardous area. The switch is not to be in "on" position until the bonding cable/strap has been connected and it is to be in "off" position prior to disconnection of the bonding strap. Warning is to be posted stating that the switch is to be in "off" position prior connecting and disconnecting the bonding cable/strap.

2.4.6.5 Insulated section of the bunker hose or rigid transfer arm is to have possibility to be stowed without electrical contact with hull upon disconnection.

2.4.7 Lifting Appliances

2.4.7.1 Deck Cranes

2.4.7.1.1 Bunker vessel is to be fitted with deck crane or cranes for handling of transfer hoses, reducers, spool pieces and other equipment used for transfer, including transfers at sea.

2.4.7.1.2 Where the crane is used for hose support during the transfer, it is to be certified to a recognized standard.

2.4.7.1.3 Arrangements to maintain hose bend radius within limits are to be provided. Where cradles are used for support they have to be able to handle all hoses used for bunkering.

2.4.7.2 Transfer arm

2.4.7.2.1 Rigid transfer arms and transfer systems based on foldable arms intended for use as transfer arrangement will be specially considered. Acceptance criteria may be based on assessment of acceleration forces acting on the transfer arm, permissible manifold loads for the nominal diameter of the transfer arm connection, transfer arm location on bunker vessel and working envelope considering allowance for fenders, transfer arm support arrangements in operational and stowed positions, effect of the hull vibration on the transfer arm, maintenance of the transfer arm and testing program.

2.4.7.2.2 Transfer arms are as a minimum to conform to the requirements of ISO 16904 or EN 1474-3:2016 or the OCIMF's publication "*Design and Construction Specification for Marine Loading Arms"*.

2.4.7.2.3 The maximum allowable operating amplitude for the system is to be defined.

2.4.8 Swivels

2.4.8.1 The pressure parts of a pressure swivel are to be designed and manufactured according to the requirements of Pt.4 Ch.5 or other recognised pressure vessel code. Swivel joints are to comply with ISO 16904:2016 or any other recognised standard.

2.4.8.2 A pressure swivel is to be isolated from the structural loads due to the connection with the receiving ship.

2.4.8.3 Means are to be provided to collect and safely dispose of liquid leaks.

2.4.8.4 Pressure swivels are to be subjected to a pressure resistance static test, according to its design code.

2.4.8.5 Rotation and oscillation test including rest periods are to be performed at design pressure with measurement of starting and running moments. At least two complete rotations, or equivalent, in each direction are to be performed.

2.4.9 Testing of Complete LNG transfer system

2.4.9.1 Following requirements are applicable to complete LNG transfer systems including additional safety devices such as dry break-away coupling/self-sealing quick release, ERC, swivels, etc. (i.e: all parts which are after the bunkering manifold).

2.4.9.2 The LNG transfer system is to be subjected to a hydraulic pressure test, at ambient temperature, to a pressure not less than 1.5 times the nominal pressure, to demonstrate that the hose assembly is capable of withstanding its pressure without leaking.

2.4.9.3 When applicable, the welds of the LNG transfer system with connections systems are to be subjected to a non-destructive examination (NDE) test and all butt welds of the LNG transfer system with connections systems are to be subjected to a 100% radiography examination.

2.4.10 Bunkering transfer rate

2.4.10.1 The bunkering transfer rate is to be kept within the capabilities of the receiving ship.

2.4.10.2 The maximum LNG transfer rate is to be justified, taking into consideration:

- The management of the BOG generated during bunkering operation
- The temperature of the LNG supplied to the ship
- Characteristics of the receiving tank
- The maximum flow permitted by the ERC
- The maximum flow permitted by the hose
- The maximum flow permitted by the QCDC.

2.4.10.3 The LNG velocity in the piping system is not to exceed 10m/s in order to avoid the generation of static electricity and to limit the heat transfer due to friction inside the pipes.

2.4.11 Compatibility between receiving ship and bunkering ship

2.4.11.1 The ship working limits of bunkering are to be checked with regards to at least the following aspects:

- Draught and freeboard difference between the receiving ship and the LNG bunkering ship
- Compatibility of the bunkering arm or hose operating amplitude with the bunkering station location
- Pressure and temperature difference between the LNG tanks of receiving ship and bunkering ship
- Vapour management
- Vapour return line (pressure and temperature)
- Delivery flow rate (maximum and minimum)
- Type and size of hose connections systems.
- Compatibility of the ESD link
- Mooring arrangement.

2.4.12 Inert Gas System

2.4.12.1 The bunker ship is to have onboard source of suitable inert gas for inerting, purging of gas fuel transfer lines and for testing of the bunker connection for tightness prior to transfer.

2.4.12.2 Installed onboard inert gas systems are to comply with requirements of Pt.5 Ch.4 Sec.9, 9.5.

2.4.12.3 The inerting capacity is to be designed according the bunkering operations and it is not to be less than 5 times the volume of the hose and pipes to be purged.

2.4.12.4 The inert gas stored or produced on board for purging of gas fuel transfer lines is to have dewpoint sufficiently low to eliminate risk of water condensate accumulation in the piping system.

2.4.12.5 Bunker ship is to be equipped with nitrogen supply system for gas fuel transfer lines inerting/purging. For fuel gas with warmer boiling point, inert gas produced by combustion of fuel or gas may be used.

2.4.12.6 Suitable arrangement to prevent back-flow of hydrocarbons from cargo system into the inert gas system is to be provided. When the source of inert gas is located outside the cargo area, the line is to be

equipped with removable spool piece located on the open deck of cargo area.

2.4.12.7 When the back flow preventer consists of non-return valves installed in series, the spool piece is to be removed after every inerting/ purging operation. A warning sign is to be posted at the valves location and corresponding instruction included into the cargo plant operation manual.

2.4.12.8 Where a double block and bleed device is fitted to prevent the back-flow, it is to be equipped with alarm for faulty operation.

2.4.12.9 The inert gas supply connections to cargo system are to be equipped with shut off valves and have a branch connection to supply the inert gas to receiving ship through a flexible hose of sufficient length for cases when inert gas source is not available on the other ship.

2.5 Safety, Control and Monitoring Systems

2.5.1 Gas Detection System

2.5.1.1. General

.1 Installed onboard gas detection system is to be capable to measure gas concentration in the manifold connections area in addition to location described in Pt.5 Ch.4 Sec.13, 13.6.2 and have arrangement to provide a remote gas detection point for receiving ship. This additional gas detector may be part of gas detection system required by Pt.5 Ch.4 Sec.13, 13.6 if requirements of 2.5.1.2 are met.

.2 Gas detecting equipment are to provide continuous monitoring and activate alarm when concentration of hydrocarbons reaches 30% of lower flammable limit (LFL).

.3 Audible and visible alarms from the permanently installed gas detection equipment are to be located on the navigation bridge, in the bunkering operation control position and at the gas detector readout location.

2.5.1.2 Gas detection in enclosed spaces

.1 Permanently installed gas detectors are to be fitted in all hazardous areas including bunkering station, bunkering process room and other enclosed spaces containing gas piping or other equipment without ducting.

.2 The number of detectors in each space is to be considered taking into account the size, layout and ventilation of the space. At least two independent gas detectors are required in each hazardous area.

.3 The detection equipment is to be located where gas may accumulate and in the ventilation outlets. Gas dispersion analysis or a physical smoke test is to be used to find the best arrangement.

.4 Audible and visual alarms from the gas detection equipment are to be located on the bridge and in the bunkering control room.

2.5.1.3 Gas detection in open areas

.1 Permanently installed thermal imaging cameras are to be fitted for the following hazardous areas when located in open areas:

• Bunkering station

• Bunkering process systems.

Monitoring of thermal imaging camera is to be possible from a safe location with respect to bunkering operations and with a permanent observation during transfer operations.

2.5.2 Control and Monitoring System

2.5.2.1 The bunkering control and monitoring system may be connected to an integrated control system or be a stand-alone system.

2.5.2.2 A Failure Modes Effects Analysis (FMEA) is to be carried out for the bunkering control system identifying component criticality.

2.5.2.3 The bunkering control and monitoring system design is to meet the single failure criteria (i.e., no single control system component failure or single fault condition is to lead to loss of control of bunkering operations or result in an unsafe situation). For the purpose of the FMEA, any excessive transient surge pressure is to be treated as a single fault.

2.5.2.4 Control of bunkering is to be possible from a safe location in regard to bunkering operations.

2.5.2.5 Automatic control, alarm and safety functions are to be provided to maintain operations within preset parameters for all

LNG bunkering operations. Operation outside the preset parameters or activation of the safety functions is to provide audible and visual alarms in the bunkering control location.

2.5.2.6 The temperatures, pressures, flow rates and functions of the LNG bunkering system are to be controlled as follows:

a) The design of the control system is to be such as to identify faults in the equipment, as well as the process system.

b) All computer-based control systems are to comply with the applicable requirements of Part 4, Ch.7.

2.5.2.7 The tank pressure and tank level are to be monitored at the bunkering control location. Overfill alarm and automatic shutdown are also to be indicated at this location.

2.5.2.8 Local reading manifold pressure gauges and transmitters with isolation valves are to be fitted to indicate the pressure between stop valves and hose connections.

2.5.2.9 The alarms and safety function actions required for the transfer system are given in table below.

Table 2.5.2 : Control and Monitoring			
Monitored Parameter	**Alarm(Visual and Audible) at bunkering control station**	**Activation of the ESD System**	**Automatic activation of the emergency release coupling**
Loss of ventilation in LNG cargo machinery space	**X**		
Loss of ventilation in closed or semi-enclosed bunkering station during bunkering operations	**X**	**X**	
Gas detection in, LNG cargo machinery space above 30% LEL	**X**		
Gas detection in LNG cargo machinery space above 60% LEL	**X**	**X**	
Low oxygen concentration alarm in inert gas storage space	**X**		
High or low liquid level in supply tank	**X**		
High-high liquid level in supply tank	**X**	**X**	
High pressure in supply tank	**X**		
Low pressure in supply tank	**X**	**X**	
High-high pressure in supply tank	**X**	**X**	
High level in the receiving tank	**X**		
High-high level in the receiving tank	**X**	**X**	
High pressure in the receiving tank	**X**		
High-high pressure in the receiving tank	**X**	**X**	
High pressure at bunker manifold	**X**	**X**	
High delivery flow rate	**X**		
High pressure in vapour return line	**X**		
High-high pressure in vapour return line	**X**	**X**	
Gas detection at bunker manifold area above 30% LEL	**X**		
Gas detection at bunker manifold area above 60% LEL	**X**	**X**	
High or low temperature at LNG pump inlet	**X**		
High or low pressure at LNG pump discharge	**X**		
Transfer arm limit switches out of range	**X**		**X**
Activation of the Emergency Release System	**X**	**X**	**X**
Manual ESD shutdown	**X**	**X**	**X**
Manual or automatic ESD signal from LNG receiving vessel or facility	**X**	**X**	
Loss of ESD valve motive power	**X**	**X**	
Fire detection onboard	**X**	**X**	
Sudden pressure drop at the transfer pump discharge	**X**	**X**	
LNG leak detection or vapour detection	**X**	**X**	
Gas detection around the bunkering lines	**X**	**X**	
Safe working envelope of the loading arm exceeded	**X**	**X**	**X**

2.5.3 Emergency shut-down systems (ESD)

2.5.3.1 The design of the ESD systems is to take into account the possible excessive bunkering ship (or receiving ship) motions.

2.5.3.2 Any activation of the ESD systems is to be implemented simultaneously on both bunkering facility and receiving ship. The timing sequence is to ensure that the involved pumps and vapour return compressors (if any) stop before the closure of any manifold valves.

2.5.3.3 At least one local manual activation position for the ESD system is to be at a safe distance from the manifold and is to have a clear view of the manifold area.

2.5.3.4 Any pipeline or component containing liquid, which may be isolated due to the ESD activation, is to be provided with pressure relief valve.

2.5.3.5 An emergency shutdown system (ESD) pendant with manual activation button is to be available for receiving ship. If a bunker ship has capability to connect own ESD system to receiving ship ESD system this arrangement can replace the pendant requirement.

2.5.3.6 In addition to Pt.5 Ch.4 requirements for ESD, the ESD function is to be initiated in following circumstances:

— automatically, if the distance of connection on receiving ship from the connection on bunker vessel exceeds safe operational envelope for transfer arrangement

— by activating manual ESD button on ESD pendant

— automatically at Emergency Release System (ERS) activation

2.5.3.7 Opening of remotely operated main transfer valves is not to be possible after ERS/ERC activation unless the ERS is re-assembled. ERS function of the valves may be utilised to meet this requirement. Main transfer valves are considered to be remotely operated cross over (manifold) valves as described in Pt. 5, Ch.4, 5.5.3.1.

2.5.4 Emergency Release System

2.5.4.1 The bunker ship is to have an arrangement for quick release of bunker connections in case of emergency.

2.5.4.2 Emergency release couplings (ERC) used in bunker connection are to be of “dry-break” type and be capable to self-disconnect upon application of force at any direction of vessel’s relative motion which exceeds design loads and at pressure surge exceeding the coupling design pressure. ERC fitted in lines for transfer of gas fuel at temperature below zero degrees Celsius are to be capable to break-away through the ice accumulated on the coupling during the transfer.

2.5.4.3 In addition to ERC, the bunker connections are to be equipped with powered emergency release system (ERS) operational in all conditions. The actuating ERS power is to have reserve storage of energy sufficient for disconnection of all transfer lines in case the main source of actuating power becomes unavailable (for example in case of cargo area black-out). Where ERS is fitted outboard the insulating flange, the insulation flange is not to be shorted by use of electrically continuous hydraulic hoses.

2.5.4.4 The ERS is not to disconnect the coupling at accidental black-out.

2.5.4.5 Activation of ERS or ERC is not to impose excessive pressure surge effect in the system. Pressure surge effect for different flow rates and closing time of valves in transfer system is to be documented. Operational restrictions caused by closing time of valves on receiving ship is to be reflected in the vessel’s cargo plant operation manual.

2.5.4.6 Where the bunker connections are supported by hose crane, the quick release couplings (ERC/ERS) are to be fitted at the receiving ship end (outward of the hose crane suspension point). Alternatively, the quick release couplings may be fitted at bunker ship end when the hose suspension point has arrangements for automatic release synchronized with the quick release coupling.

2.5.4.7 When ERS is activated it is to release in two steps: first the ESD is to be triggered to stop the transfer and then release the ERS flanges.

2.5.4.8 Testing of ERS circuits is to be possible without disconnecting the ERS coupling.

2.5.4.9 Where hoses are used for transfer, release of ERC or ERS may potentially lead to hard impact of the hoses ends against the ships structures. Arrangements preventing the impact are to be provided. Release of the hoses is not to impose excessive stress to the manifold valves of the vessel.

2.5.4.10 Where rigid transfer arms are used for gas fuel transfer, the part of the transfer arm inboard of the quick release coupling is to retract automatically toward bunker vessel to avoid contact with other part and receiving vessel due to vessels motion.

2.5.4.11 ERS is to be capable of being activated by manual controls situated in at least two remote locations on bunker ship. One of these locations should be the bunker operation control position.

2.5.4.12 Excessive pressure in bunker connections is to activate alarm and automatic stop of transfer. Capability to adjust the activation set point has to be provided, but it is not to exceed design pressure of the transfer lines.

2.5.5 Communication systems

2.5.5.1 The bunker ship is to be equipped with effective means of communication for internal purpose (onboard communications), for communication between the bunker ship and receiving ship and for external communications. A communication system with back-up is to be provided between the bunkering ship and the receiving ship. This may be achieved by electric, fibre-optic or pneumatic links, or a combination of these systems.

2.5.5.2 Communications are to be maintained between the bunkering ship and the receiving ship at all times during the bunkering operation. In the event that communications cannot be maintained, bunkering is to be stopped and not resumed until communications are restored.

2.5.5.3 The components of the communication system located in hazardous and safety zones are to be of a suitable certified safe type.

2.5.5.4 Where portable radios are used for communication with receiving ship, they are to be of approved type suitable for gas-hazardous areas. The bunker ship is to have sufficient set of spare radios with the same operating frequencies table for the receiving ship duty personnel. Minimum two portable radios are to be available for receiving ship. Where the ships have wired communication link the number of portable radios can be reduced to one.

2.5.5.5 To reduce risk of current induction in hose cranes or rigid transfer arms, high power transmitting equipment such as MF/HF radios are to be provided with capability to be earthed manually and equipment with low transmitting power such as VHF radios, AIS transceivers are to have low power operation mode with transmission power not exceeding 1 W.

2.6 Electrical Installations

2.6.1 Application

2.6.1.1 The requirements in this Section are to be complied with, for bunkering vessels, in addition to those pertaining to electrical installations of Pt.5, Ch.4.

2.6.2 System of supply

2.6.2.1 Acceptable systems of supply

The systems of generation and distribution of electrical energy as detailed in Pt.4, Ch.8, Table 2.2.1 for Gas Carriers, are acceptable. In insulated distribution systems, no current carrying part is to be earthed, other than:

a) through an insulation level monitoring device

b) through components used for the suppression of interference in radio circuits.

2.6.2.2 Earthed system with hull return

Earthed systems with hull return are not permitted, with the following exceptions to the satisfaction of IRS:

a) impressed current cathodic protection systems

b) limited and locally earthed systems, such as starting and ignition systems of internal combustion engines, provided that any possible resulting current does not flow directly through any hazardous area

c) insulation level monitoring devices, provided that the circulation current of the device does not exceed 30 [mA] under the most unfavourable conditions.

2.6.2.3 Earthed system without hull return

Earthed systems without hull return are not permitted, with the following exceptions:

a) earthed intrinsically safe circuits and the following other systems to the satisfaction of IRS

b) power supplies, control circuits and instrumentation circuits in non-hazardous areas where technical or safety reasons preclude the use of a system with no connection to earth, provided the current in the hull is limited to not more than 5 A in both normal and fault conditions, or

c) limited and locally earthed systems, such as power distribution systems in galleys and laundries to be fed through isolating transformers with the secondary windings earthed, provided that any possible resulting hull current does not flow directly through any hazardous area, or

d) alternating current power networks of 1000 V root mean square (line to line) and over, provided that any possible resulting current does not flow directly through any hazardous area; to this end, if the distribution system is extended to areas remote from the machinery space, isolating transformers or other adequate means are to be provided.

2.6.3 Earth detection

2.6.3.1 Monitoring of circuits in hazardous areas

2.6.3.1.1 The devices intended to continuously monitor the insulation level of all distribution systems are also to monitor all circuits, other than intrinsically safe circuits, connected to apparatus in hazardous areas or passing through such areas. An audible and visual alarm is to be given, at a manned position, in the event of an abnormally low level of insulation.

2.7 Fire Safety

2.7.1 General

2.7.1.1 This Section provides requirements for fire safety in the safety zone.

2.7.2 Fire protection

2.7.2.1 When applicable, the bunkering station is to be separated by class A-60 insulation towards other spaces, except for spaces such as tanks, voids, auxiliary machinery spaces of no fire risk, sanitary and similar spaces where insulation standard may be reduced to class A-0.

2.7.2.2 Spark arresting arrangements are to be provided at exhaust outlets of internal combustion engines and boilers using oil as fuel and at incinerators exhaust outlets. Spark arrestors fitted in the exhaust outlets are to comply with a recognized standard

2.7.2.3 Use of vapour oxidising equipment not in compliance with 2.8.4.6 is to be restricted during bunker operations.

2.7.3 Fire extinction

2.7.3.1 Water spray systems

2.7.3.1.1 Water spray systems are to comply with the requirements of Pt.5, Ch.4.

2.7.3.1.2 The water spray system is to be fitted to protect the bunkering manifold, associated piping installations and the transfer area.

2.7.3.1.3 The system is to be designed to cover all areas as specified in 2.7.3.1.2 with an application rate of 10 [l/min/m^2] for horizontal projected surfaces and 4 [l/min/m^2] for vertical surfaces.

2.7.3.1.4 For the purpose of isolating damage sections, stop valves are to be fitted at least every 40 [m] or the system may be divided into two or more sections with control valves located in a safe and readily accessible position not likely to be cut-off in case of fire.

2.7.3.1.5 The capacity of the water spray pump is to be sufficient to deliver the required

amount of water to the hydraulically most demanding area as specified in 2.7.3.1.4 in the protected areas.

2.7.3.1.6 Remote start of pumps supplying the water spray system and remote operation of any normally closed valves to the system is to be located in a readily accessible position which is not likely to be cut off in case of fire in the protected areas.

2.7.3.1.7 The nozzles are to be of an approved full bore type and they are to be arranged to ensure an effective distribution of water throughout the space being protected.

2.7.3.1.8 An equivalent system to the water spray system may be fitted provided it has been tested.

2.7.3.2 Dry chemical powder fire-extinguishing system

2.7.3.2.1 Dry chemical powder fire-extinguishing systems are to comply with the requirements of Pt.5, Ch.4.

2.7.3.2.2 A permanent dry chemical powder extinguishing system is to be installed in the bunkering station area to cover all possible leak points. The capacity is to be at least 3.5 [kg/s] for a maximum of 45 s discharges. The system is to be arranged for easy manual release from a safe location outside of the protected area.

2.7.3.2.3 One portable dry powder extinguisher of at least 5 [kg] capacity is to be located near the bunkering station.

2.8 Additional Features

2.8.1 Bunkering ships designed to receive LNG from a gas fuelled ship for which the LNG fuel tanks have to be emptied.

2.8.1.1 The BOG handling system of the LNG bunkering ship is to be sized to handle the extra vapours generated during this operation taking into account the fact that the level in the receiving cargo tanks is increasing.

2.8.1.2 Bunkering procedure for LNG receiving from a gas fuelled ship is to be submitted for information.

2.8.2 Bunkering ship designed to supply inert gas and dry air, to ensure gas freeing and aeration, to a gas fuelled ship complying with Part 5, Ch.35, 6.10.4 (IGF Code, paragraph 6.10.4).

2.8.2.1 The lines used for the inert gas are to be independent from the liquid and vapour lines used for normal operation.

2.8.2.2 Procedure for gas freeing and diagram of the gas freeing system are to be submitted for review.

2.8.3 Bunkering ship designed for initial cooling down of the gas fuelled ship LNG fuel tank.

2.8.3.1 The bunkering ship is to be capable of handling all or part of the boil-off gas from receiving ship, in addition to its own boil-off, generated during the initial cooling down without release to the atmosphere. The boil-off gas handling capacity of the bunkering ship is to be indicated and justified.

2.8.3.2 Procedure of initial cooling down is to submitted for information.

2.8.4 Bunkering ship designed to recover and manage the boil-off gas generated during the bunkering operation.

2.8.4.1 The bunkering ship is to be capable of handling all or part of the boil-off gas from receiving ship, in addition to its own boil-off, generated during the LNG bunkering operation without release to the atmosphere. The boil-off gas handling capacity of the bunkering ship is to be indicated and justified.

2.8.4.2 Different ways to dispose of the BOG may be considered:

- re-liquefaction
- cooling
- utilization by the gas consuming equipment of the ship (e.g. gas or dual-fuel engines or boilers)
- gas combustion unit.

A combination of these means is possible for the BOG and other solutions may be accepted if they are duly justified.

2.8.4.3 Such arrangement and systems are to meet design conditions stated in Pt.5 Ch.4 Sec.7 unless other ambient conditions are specially agreed.

2.8.4.4 Redundancy requirements in Pt.5 Ch.4 Sec.7 are not applicable for this optional system.

2.8.4.5 Process piping for handling excess vapour from receiving ship is to be adequately separated to avoid over-pressurising cargo system on bunker vessel. Risk evaluation of the simultaneous operations of liquid bunker transfer, vapour recovery from receiving ship and pressure/temperature control in own tanks are to be made.

2.8.4.6 If vapour disposal is by means of oxidation, the installation is to comply with Pt.5 Ch.4 Sec.7, 7.4 except that the exhaust gas is to have a maximum exit temperature of 250°C. Exhaust outlet for such system is to be located on the side opposite to receiving vessel and directed outward. As an alternative to temperature limitation a dispersion analysis may be taken for consideration. Such analysis is to contain operational conditions (wind speed and relative direction) at which required dilution is achievable and operation of GCU during BOG vapours recovery may be considered safe.

2.8.4.7 When the system is based on accumulation of compressed vapour on bunker vessel, due considerations are to be made for safe operational margin. Additional pressure control systems with capacity to maintain cargo pressure/temperature within the limits are to be fitted.

2.8.4.8 Compressor installed for vapour extraction from receiving ship are to be fitted with automatic pressure and flow control, and be provided with surge protection and provisions for emergency stop.

2.8.4.9 Procedure for boil off gas management is to be submitted for information.

2.8.4.10 The system is to be operational and capable of accepting the maximum vapour flow rate generated at loading rate defined by the designer. The maximum vapour flow rate is to be indicated and justified.

End of Chapter

IRCLASS
Indian Register of Shipping

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RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 5
SPECIAL SHIP TYPES – VOLUME II

July 2024

Indian Register of Shipping

Part 5 : Special Ship Types

This Part (Part 5) of the Rules is arranged in two Volumes (Volume I and Volume II). Chapters 1 to 14 may be referred in Volume I. Chapters 15 to 20 are intentionally left blank. For Chapter 21 onwards, please refer this Volume.

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Ice Class Requirements

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Section 1

General

1.1 Application

1.1.1 The Rules in this Chapter apply to vessels occasionally or primarily intended for navigation in waters with ice conditions. The requirements are to be regarded as supplementary to those given for the assignment of main class.

1.2 Class notations

1.2.1 Vessels complying with relevant additional requirements of this Chapter will be eligible to be assigned one of the following class notations:

a) **HAT(B)** : This strengthening is for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 1.0 [m].

b) **HT(B)** : This strengthening is for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 0.8 [m].

c) **HM(B)** : This strengthening is for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 0.6 [m].

d) **Ha(B)** : This strengthening is for first year ice conditions in Northern Baltic in winter or areas of similar ice with condition equivalent to unbroken level ice with a maximum thickness of 0.4 [m].

e) **Ha** : This strengthening is for light first year ice conditions in areas other than Northern Baltic. The standard of strengthening is equivalent to that for **Ha(B)** but only the requirements for strengthening the bow region, the rudder and steering arrangements are applicable.

(Note: For definition of first year ice and thin first year ice, refer Pt. 5, Ch. 32 of the Rules).

1.3 Symbols

L, B, D, T, C_b as defined in Pt.3, Ch.1.

Δ = moulded displacement, [t] at summer load waterline.

Δ_f = is the displacement of the ship, [t] determined from the waterline on the UIWL. Where multiple waterlines are used for determining the UIWL, the displacement must be determined from the waterline corresponding to the greatest displacement.

s = stiffener spacing, [mm] measured along the plating between ordinary and/or intermediate stiffeners.

l = stiffener span, [m] measured along the top flange of the member.

S = girder span, [m].

σ_y = minimum yield stress of material, [N/mm^2].

g_o = standard acceleration due to gravity.
= 9.81 [m/s^2].

Section 2

Hull Strengthening for First Year Ice Conditions

2.1 General

2.1.1 The requirements of this Section apply to vessels for service in the Northern Baltic in winter or areas with similar first year ice conditions.

2.1.2 Vessels built in compliance with the following requirements may be given one of the class notations **HAT(B), HT(B), HM(B)** or **Ha(B)** whichever is relevant. For class notation **Ha** the requirements of **Ha(B)** are to be applied as per 1.2.1(d) above.

2.1.3 The formulae given for plating, stiffeners and girders are based on the distribution of ice loads from plating to stiffeners and girders as well as redistribution of loads on stiffeners and girders.

2.1.4 If scantlings derived from these regulations are less than those required for an unstrengthened ship, the latter are to be used.

2.1.5 Assistance from icebreakers is normally assumed when navigating in ice bound waters.

2.2 Definitions

2.2.1 *Upper Ice Waterline (UIWL)* : The upper ice waterline is the envelope of the highest points of the waterlines at which the ship is intended to operate in ice. The line may be a broken line. The maximum permissible ice class draft amidships is the draught on UIWL measured amidships.

2.2.2 *Lower Ice Waterline (LIWL)* : The lower ice waterline is the envelope of the lowest points of the waterlines at which the ship is intended to operate in ice. The line may be a broken line.

2.2.3 *Maximum and Minimum Draught Fore and Aft:* The maximum and minimum ice class draughts at fore and aft perpendiculars are to be determined in accordance with the upper and lower ice waterlines and the draught of the ship at fore and aft perpendiculars, when ice conditions require the ship to be ice-strengthened is always to be between the upper and lower ice waterlines. The maximum and minimum ice class draughts fore, amidships and aft are to be indicated in the class certificate.

Restrictions on draughts when operating in ice are to be documented and kept on board readily available to the master. If the summer load line in fresh water is anywhere located at a higher level than the UIWL, the ship's sides are to be provided with a warning triangle and with an ice class draught mark at the maximum permissible ice class draught amidships. (See Figure 2.2.3).

The draught and trim, limited by the UIWL, is not to be exceeded when the ship is navigating in ice. The salinity of the sea water along the intended route is to be taken into account when loading the ship. The ship is to always be loaded down at least to the draught of LIWL amidships when navigating in ice. Any ballast tank, situated above the LIWL and needed to load down the ship to this water line, is to be equipped with devices to prevent the water from freezing. In determining the LIWL, regard is to be paid to the need to ensure a reasonable degree of ice going capability in ballast. The highest point of the propeller is to be submerged and if possible at a depth of at least h_o below the water surface in all loading conditions. The forward draught is to be at least:

$(2 + 0.00025\ \Delta_f)\ h_o$ [m],

but need not exceed 4 h_o

where,

h_o = level ice thickness as per 2.3.1.

2.2.4 The ice belt is divided into regions as follows : (See Fig.2.2.4).

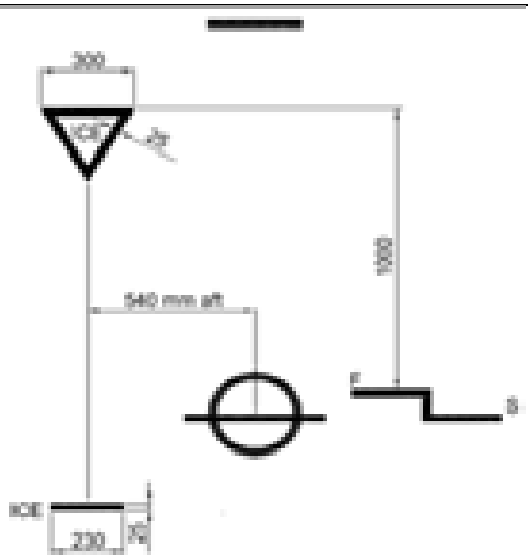


Fig.2.2.3 : Ice Class Draught Marking

1. The upper edge of the warning triangle is to be located vertically above the “ICE” mark, 1000 mm higher than the Summer Load Line in fresh water but in no case higher than the deck line. The sides of the triangle are to be 300 mm in length.
2. The ice class draught mark is to be located 540 mm abaft the centre of the load line ring or 540 mm abaft the vertical line of the timber load line mark, if applicable.
3. The marks and figures are to be cut out of 5 - 8 mm plate and then welded to the ship's side. The marks and figures are to be painted in a red or yellow reflecting colour in order to make the marks and figures plainly visible even in ice conditions.
4. The dimensions of all letters are to be the same as those used in the load line mark.

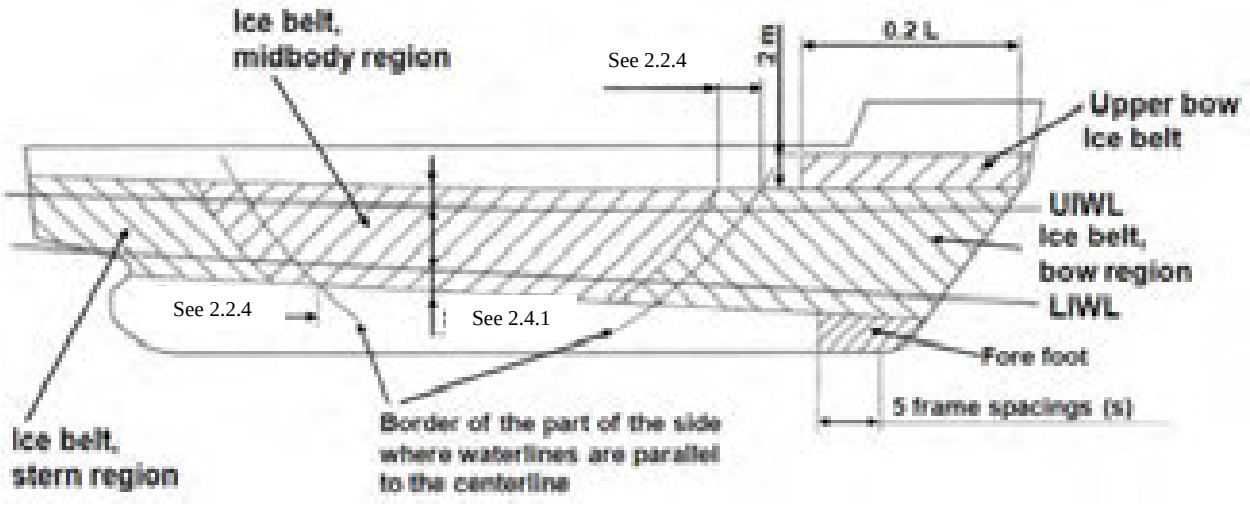


Fig 2.2.4 : Ice Belt Regions

2.2.4.1 Bow region : From the stem to a line parallel to and 0.04L aft of the forward borderline of the part of the hull where the waterlines run parallel to the centreline. For ice classes **HAT(B)** and **HT(B)** the overlap of the borderline need not exceed 6 [m], for ice classes **HM(B)** and **Ha(B)** this overlap need not exceed 5 [m].

2.2.4.2 Midbody region : From the aft boundary of the bow region to a line parallel to and 0.04 L aft of the aft borderline of the part of the hull

where the waterlines run parallel to the centreline. For ice classes **HAT(B)** and **HT(B)** the overlap of the borderline need not exceed 6 [m], for ice classes **HM(B)** and **Ha(B)** this overlap need not exceed 5 [m].

2.2.4.3 Stern region : From the aft boundary of the Midbody region to the stern.

2.2.4.4 LIWL and UIWL are to be indicated on the shell expansion plan together with the lines separating the bow, midbody and stern regions of the ice belt. The displacement Δ_f, and the output of propulsion machinery P, are to be stated on the shell expansion and/or the framing plan.

2.3 Design loads

2.3.1 An ice strengthened ship is assumed to operate in open sea conditions corresponding to a level ice thickness not exceeding h_o. The design ice load height (h) of the area actually under ice pressure at any particular point of time is, however, assumed to be only a fraction of the ice thickness. The values of h_o and h are given in Table 2.3.1.

Table 2.3.1 : Ice thicknesses

Ice Class	h_o [m]	h [m]
HAT(B)	1.0	0.35
HT(B)	0.8	0.30
HM(B)	0.6	0.25
Ha(B)	0.4	0.22

2.3.2 The design ice pressure is determined by the formula:

$p = c_d\, c_p\, c_a\, p_o$ [N/mm^2]

p_0 is nominal ice pressure; the value 5.6 [N/mm^2] to be used.

c_d = a factor which takes account of the influence of the size and engine output of the ship. This factor is not to be taken more than 1.0.

$$c_d = \frac{aK + b}{1000}$$

$$K = \frac{\sqrt{\Delta_f P}}{1000}$$

a and b are given in Table 2.3.2.

Table 2.3.2 : Values of 'a' and 'b' for calculation of c_d

	Region			
	Bow		Midbody and Stern	
	K ≤ 12	K > 12	K ≤ 12	K > 12
a	30	6	8	2
b	230	518	214	286

Δ_f = displacement [t] of the ship at maximum ice class draught as defined in 2.2.1.

P = is the actual continuous engine output of the ship [kW] (See 3.1.1) available when sailing in ice. If additional power sources are available for propulsion power (e.g shaft motors) in addition to the power of the main engines(s), they are also to be included in the total engine output used as the basis for hull scantling calculations. The engine output used for the calculation of the hull scantlings is to be clearly stated on the shell expansion drawing.

c_p = a factor that reflects the magnitude of the load expected in the hull area in question relative to the bow area .

The value of c_p is given in the following Table:

Ice class	Region		
	Bow	Midbody	Stern
HAT(B)	1.0	1.0	0.75
HT(B)	1.0	0.85	0.65
HM(B)	1.0	0.70	0.45
Ha(B)	1.0	0.50	0.25

c_a = a factor which takes account of the probability that the full length of the area under consideration will be under pressure at the same time.

$$c_a = \sqrt{\frac{l_0}{l_a}}$$

however, not to be taken greater than 1.0 or less than 0.35

l_0 =0.6 [m]

l_a is to be taken as follows:

Structure	Type of framing	l_a [m]
Shell	Transverse	Frame Spacing
	Longitudinal	1.7 x Frame Spacing
Frames	Transverse	Frame spacing
	Longitudinal	Span of frame
Ice stringer		Span of stringer
Web frame		2 x spacing of web frames

2.4 Shell plating

2.4.1 The vertical extension of the ice belt is not to be less than the following:

Table 2.4.1 : Vertical extension of the ice belt

Ice class	Hull Region	Above UIWL [m]	Below LIWL [m]
HAT(B)	Bow	0.6	1.2
	Midbody	0.6	1.2
	Stern	0.6	1.0
HT(B)	Bow	0.5	0.9
	Midbody	0.5	0.75
	Stern	0.5	0.75
HM(B) and Ha(B)	Bow	0.4	0.7
	Midbody	0.4	0.6
	Stern	0.4	0.6

2.4.2 In addition, the following areas are to be strengthened:

2.4.2.1 *Fore foot:* For ice class **HAT(B)**, the shell plating below the ice belt from the stem to a position five main frame spaces abaft the point where the bow profile departs from the keel line is to be ice-strengthened in the same way as the bow region .

2.4.2.2 *Upper bow ice belt*: For ice classes **HAT(B)** and **HT(B)** on ships with an open water service speed equal to or exceeding 18 knots, the shell plate from the upper limit of the ice belt to 2 [m] above it and from the stem to a position at least 0.2L abaft the forward perpendicular, is to be ice-strengthened in the same way as the midbody region A similar strengthening of the bow region is also recommended for a ship with a lower service speed when, on the basis of the model tests, for example, it is evident that the ship will have a high bow wave.

2.4.3 Side scuttles are not to be situated in the ice belt. If the weather deck in any part of the ship is situated below the upper limit of the ice belt (e.g. in way of the well of a raised quarter decker), the bulwark is to be given at least the same strength as is required for the shell in the ice belt. The strength of the construction of the freeing ports are to meet the same requirements.

2.4.4 For transverse framing, the thickness of the shell plating is to be determined by the formula:

$$t = 0.667s\sqrt{\frac{x_1 p_{pl}}{\sigma_y}} + t_c \ [mm]$$

For longitudinal framing the thickness of the shell plating is to be determined by the formula:

$$t = 0.667s\sqrt{\frac{p}{x_2 \sigma_y}} + t_c \ [mm]$$

p_{pl} =0.75 p [N/mm^2], where p is as given in 2.3.2.

$$x_1 = 1.3 - \frac{4.2}{(h_1/s + 1.8)^2};$$

but not to be taken greater than 1.

$$x_2 = 0.6 + \frac{0.4}{h_1/s};\ \text{for } (h_1/s) \le 1$$

= 1.4 - 0.4 (h_1/s); for 1 ≤ (h_1/s) < 1.8

h_1 = 1000 h

h = as given in 2.3.1;

t_c = addition for abrasion and corrosion [mm]; normally 2 [mm]. Where special surface coatings, with proven ability to withstand the abrasion of ice is applied and maintained, lower values of t_c may be considered.

2.5 Frames

2.5.1 The vertical extension of the ice strengthening of the framing is to be at least as per Table 2.5.1.

Where an upper bow ice belt is required, the ice strengthened part of the framing is to be extended at least to the top of this ice belt.

Table 2.5.1 : Vertical extension of Ice strengthening of frames			
Ice Class	**Region**	**Above UIWL [m]**	**Below LIWL [m]**
HAT(B)	Bow	1.2	Down to tank top or below top of floors
	Midbody	1.2	2.0
	Stern	1.2	1.6
HT(B) HM(B) Ha(B)	Bow	1.0	1.6
	Midbody	1.0	1.3
	Stern	1.0	1.0

2.5.2 Where the ice strengthening would go beyond a deck, the top or bottom plating of a tank or tank top by not more than 250 [mm], it can be terminated at that deck, top or bottom plating of the tank or tank top.

2.5.3 The section modulus of main or intermediate transverse frame is to be calculated by the formula:

$$Z = \frac{s\,p\,h\,l}{m_t\,\sigma_y}\,10^3\ [cm^3]$$

and the effective shear area is calculated from

$$A = \frac{8.66\ sphx_3}{\sigma_y}\ \ [cm^2]$$

p = ice pressure [N/mm^2], as given in 2.3.2

h = height of load area [m], as given in 2.3.1

$$m_t = \frac{7m_o}{7 - 5\,h/l}$$

m_o takes the boundary conditions into account. The values of m_o are given in the Table 2.5.3

x_3 is a factor which takes into account the maximum shear force versus the load location and shear stress distribution, $x_3 = 1.2$

The boundary conditions are those for the main and the intermediate frames. Possible different conditions for main frames are assumed to be taken care of by interaction between the frames and this is included in the m_o values. Load is applied at mid span.

If the ice belt covers less than half the span of a transverse frame ($0.15 \le b < 0.5\ l$), the following modified formula is to be used for the section modulus:

$$Z = \frac{p\,s\,h\,b\,(l-b)^2}{\sigma_y\,l^2}\,10^3\ [cm^3]$$

b = distance [m], between upper or lower boundary of the ice belt and the nearest deck or stringer within the ice belt.

When $b < 0.15\ l$, main class rule scantlings for ordinary frames may be used.

2.5.4 Upper end of transverse framing

i) The upper end of the strengthened part of a main frame and of an intermediate ice frame is to be attached to a deck, top or bottom plating of a tank or an ice stringer (See 2.6).

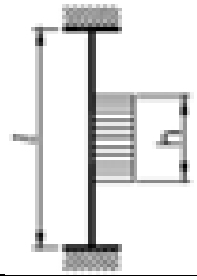
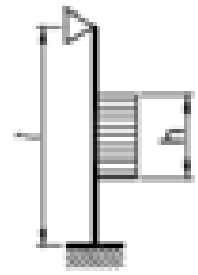
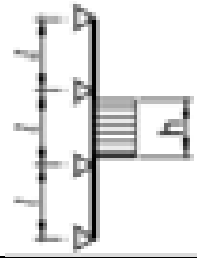
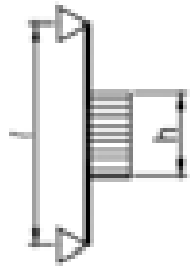
ii) Where a frame terminates above a deck or an ice stringer which is situated at or above the upper limit of the ice belt, the part above the deck or stringer may have the scantlings required for a non-ice strengthened ship and the upper end of an intermediate frame may be connected to the adjacent frames by a horizontal member of the same scantlings as the main frame.

2.5.5 Lower end of transverse framing

i) The lower end of the strengthened part of a main frame and of an intermediate ice frame is to be attached to a deck, top or bottom plating of a tank, tank top or an ice stringer (See 2.6).

ii) Where an intermediate frame terminates below a deck, top or bottom plating of a tank, tanktop or ice stringer which is situated at or below the lower limit of the ice belt, the lower end may be connected to the adjacent main frames by a horizontal member of the same scantlings as the main frame. The main frames below the lower edge of ice belt are to be ice strengthened. (See 2.5.1).

Table 2.5.3 : Factor m_o for various boundary conditions of frames

Boundary condition	m_o	Example
	7	Frames in a bulk carrier with top wing tanks
	6	Frames extending from the tank top to the main deck of a single–decked vessel
	5.7	Continuous frames between several decks or stringers
	5	Frames extending between two decks only

2.5.6 The section modulus of longitudinal frame is to be calculated by the formula:

$$Z = \frac{x_4\ p\ h\ l^2}{m_l \sigma_y}\ \ 10^6\ \ [cm^3]$$

The effective shear area of longitudinals is to be:

$$A = \frac{8.66\ x_4\ x_5\ p\ h\ l}{\sigma_y}\ \ 10^3\ \ [cm^2]$$

In calculating the actual shear area of the frames, the shear area of the brackets is not to be taken into account.

X_4 = factor which takes account of the load distribution over adjacent frames:

X_4 = (1 - 200 h/s)

X_5 = factor which takes into account the maximum shear force versus load location and the shear stress distribution:

X_5 = 2.16

p = ice pressure [N/mm^2], as given in 2.3.2

h = height of load area [m], as given in 2.3.1

m_l = boundary condition factor;

= 13.3 for a continuous beam with brackets. Where the boundary conditions deviate significantly from those of a continuous beam with brackets, e.g. in an end field, a smaller boundary condition factor may be required.

2.5.7 Within the ice strengthened area, all frames are to be effectively attached to all supporting structures. A longitudinal frame is to be attached by brackets to all supporting web frames and bulkheads by brackets. When a transverse frame terminates at a stringer or deck, a bracket or similar construction is to be fitted. Frames crossing supporting structures such as web frames or stringers are to be connected to these structures on both sides (by collar plates or lugs in way of cutouts). When a bracket is fitted, it is to have at least the same thickness as the web plate of the frame and the edge is to be appropriately stiffened against buckling.

2.5.8 Support of frames against instability, in particular tripping

i) Frames are to be attached to the shell by double continuous welds. No scalloping is allowed (except when crossing shell plate butts).

ii) The web thickness of the frames is to be not less than the maximum of the following :

- $\frac{h_w\sqrt{\sigma_y}}{C}$, h_w is the web height and C = 805 for profiles and C = 282 for flat bars
- Half of the net thickness of the shell plating, t-t_c. For the purpose of calculating the minimum web thickness of frames, the required thickness of the shell plating is to be calculated according to 2.4.4 using the yield strength of the frames
- 9 mm

iii) Where there is a deck, top or bottom plating of a deck, tank top or bulkhead in lieu of a frame, the plate thickness of it is to be calculated as in (ii) above, to a depth corresponding to the height of adjacent frames. In such a case, the material properties of the deck, top or bottom plating of the tank, tank top or bulkhead and the frame height h_w of the adjacent frames is to be used in the calculations, and the constant C is to be 805.

iv) Asymmetrical frames and frames which are not at right angles to the shell (web less than 90 degrees to the shell) are to be supported against tripping by brackets, intercoastals, stringers or similar, at a distance not exceeding 1300 [mm]. For frames with spans greater than 4 [m], the extent of anti-tripping supports must be applied to all regions and for all ice classes. For frames with spans less than or equal to 4 [m], the extent of anti-tripping supports must be applied to all regions for HAT(B), to the bow and midbody regions for HT(B), and to the bow region for HM(B) and Ha(B). Direct calculation methods may be applied to demonstrate the equivalent level of support provided by alternative arrangements.

2.6 Ice stringers

2.6.1 *Stringers within the Ice Belt* : The section modulus of a stringer situated within the ice belt is to be calculated by the formula:

$$Z = \frac{1.62\ p\ h\ l^2}{m_l \sigma_y}\ 10^6\ [cm^3]$$

The effective shear area is not to be less than:

$$A = \frac{16.83\ p\ h\ l}{\sigma_y}\ 10^3\ [cm^2]$$

p = ice pressure [N/mm^2], as given in 2.3.2.

h = height of load area [m] as given in 2.3.1.

The product "p h" is not to be taken as less than 0.15

l = span of stringer [m]

m_l = boundary condition factor as given in 2.5.6.

2.6.2 *Stringers outside the Ice Belt* : The section modulus of a stringer, situated outside the ice belt but supporting ice strengthened frames, is to be calculated by the formula:

$$Z = \frac{1.44\ p\ h\ l^2}{m_l \sigma_y}\left(1 - \frac{h_s}{l_s}\right)\ 10^6\ [cm^3]$$

The effective shear area is not to be less than:

$$A = \frac{14.96\ p\ h\ l}{\sigma_y}\left(1 - \frac{h_s}{l_s}\right)\ 10^3\ [cm^2]$$

p, h, l, m_l as in 2.6.1

The product "p h" is not to be taken as less than 0.15.

l_s = distance to the adjacent ice stringer [m]

h_s = distance to the ice belt [m].

2.6.3 *Deck Strips* : Narrow deck strips abreast of hatches and serving as ice stringers are to comply with the section modulus and shear area requirements in 2.6.1 and 2.6.2 respectively. In the case of very long hatches, the lower limit of the product "p h" may be taken as less than 0.15, but in no case as less than 0.10.

2.6.4 The deflection of the ship's sides due to ice pressure in way of very long (more than B/2) hatch openings is to be accounted for, whilst designing weather deck hatchcovers and their fittings.

2.7 Web frames

2.7.1 For transverse web frames supporting longitudinals or ice stringers, the load transferred to the web frame is given by:

$$F = 1.8\ p\ h\ S\ 10^3\ [kN]$$

p = ice pressure [N/mm^2], obtained as per 2.3.2, considering the value of c_a which is to be calculated taking l_a = 2S.

h = height [m] of load area, as given in 2.3.1.

The product "p h" is not to be taken less than 0.15.

S = web frame spacing, [m].

In case the supported stringer is outside the ice belt, the load F may be multiplied by the factor (1 - h_s / l_s).

For h_s, l_s See 2.6.2.

2.7.2 For a web frame the section modulus requirement is given by:

$$Z = \frac{M}{\sigma_y}\sqrt{\frac{1}{1 - [\gamma(A/A_a)]^2}}\ 10^3\ [cm^3]$$

where,

l = span [m] of web frame

γ = as given in Table 2.7.3.

M = maximum calculated bending moment under the ice load F;

M = 0.193 F l

A = required shear area [cm^2], from 2.7.3.

A_a = actual cross sectional area of web frame, [cm^2].

$$A_a = A_f + A_w$$

A_f = cross sectional area of free flange, [cm^2].

A_w = cross sectional area of web plate, [cm^2].

F = load force as given in 2.7.1.

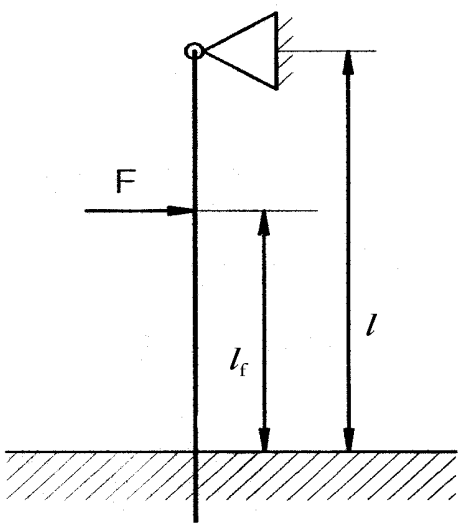


Fig.2.7.2 : Web frame

2.7.3 The shear area of a web frame is given by:

$$A = \frac{19.05\ \alpha\ Q}{\sigma_y} \quad [cm^2]$$

where,

Q = maximum calculated shear force [kN] under the ice load F, as given in 2.7.1

l = as given in 2.7.2.

α = factor given in Table 2.7.3.

2.8 Special arrangements and strengthening forward

2.8.1 The stem is to be made of rolled, cast or forged steel or of shaped steel plates.

2.8.2 The plate thickness of a shaped plate stem and in the case of a blunt bow, any part of the shell where α ≥ 30° and ψ ≥ 75° (see Fig 3.1.1.) is to be calculated according to the formulae in 2.4.4 assuming that:

s = spacing of elements supporting the plate [mm]

l_a = spacing of vertical supporting elements [m].

p_{pl} = p (see 2.3.2)

Table 2.7.3 : Value of A, α , γ

$\frac{A_f}{A_w}$	α	γ
0.0	1.5	0
0.2	1.23	0.44
0.4	1.16	0.62
0.6	1.11	0.71
0.8	1.09	0.76
1.0	1.07	0.80
1.2	1.06	0.83
1.4	1.05	0.85
1.6	1.05	0.87
1.8	1.04	0.88
2.0	1.04	0.89

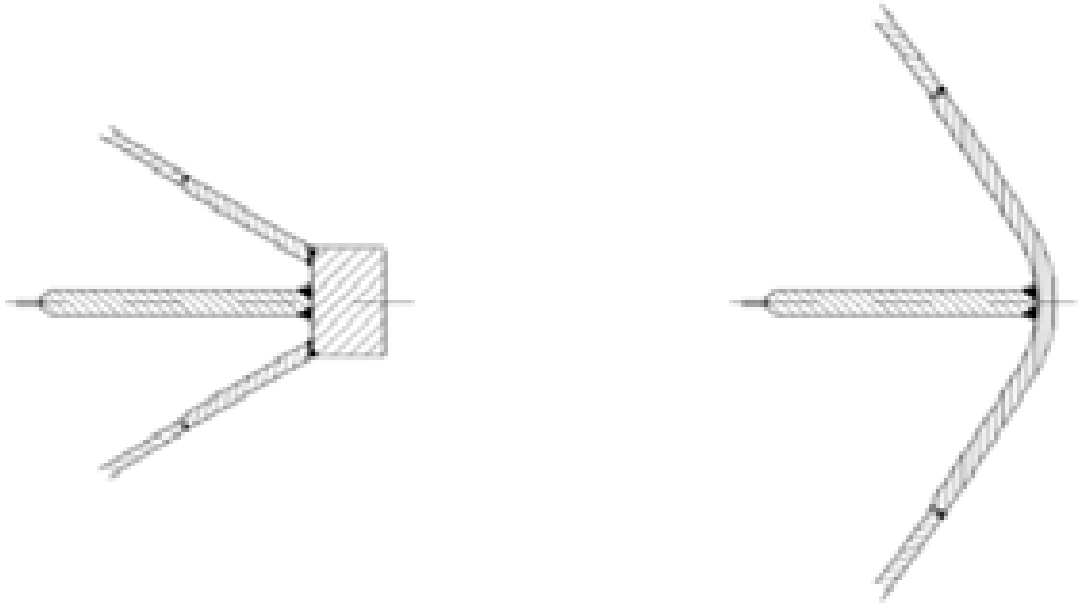


Fig.2.8.1 : Examples of suitable stems

2.8.3 The stem and the part of a blunt bow defined above are to be supported by floors or brackets spaced not more than 0.6 [m] apart and having a thickness not less than half the plate thickness. The reinforcement of the stem is to extend from the keel to a point 0.75 [m] above UIWL or, in case an upper foreward ice belt is required, to the upper limit of this.

2.8.4 A mooring pipe with an opening not less than 250 x 300 [mm], a length of at least 150 [mm] and an inner surface radius of at least 100 [mm] is to be fitted in the bow bulwark at the centre line.

2.8.5 A bitt or other means for securing a towline, dimensioned to withstand the breaking force of the towline of the ship is to be fitted.

2.8.6 On ships with a displacement not exceeding 30,000 tons, the part of the bow which extends to a height of at least 5 [m] above the UIWL and at least 3 [m] aft of the stem, is to be strengthened to take the stresses caused by fork towing. For this purpose, intermediate frames supported by stringers or decks are to be fitted.

2.8.7 The connection of bilge keels to the hull is to be so designed that the risk of damage to the hull, in case the bilge keel is ripped off, is minimized.

2.8.8 To limit damage when a bilge keel is partly ripped off, it is recommended that bilge keels be cut up into several shorter independent lengths.

2.9 Special arrangements and strengthening aft

2.9.1 An extremely narrow clearance between the propeller blade tip and the stern frame is to be avoided, as a small clearance would cause very high loads on the blade tip. In order to avoid very high loads on propeller blade tips, the minimum distance between propeller(s) and hull (including stern frame) should not be less than h_o (see Table 2.3.1).

2.9.2 On twin and triple screw ships, the ice strengthening of the shell and framing is to be extended to the double bottom for 1.5 [m] forward and aft of the side propellers.

2.9.3 Shafting and stern tubes of side propellers are normally to be enclosed within plated bossings. If detached struts are used, their design, strength and attachment to the hull are to be duly considered.

2.10 Rudder and steering arrangements

2.10.1 The scantlings of rudder, rudder post, rudder stock, pintles, steering gear etc. as well as the capacity of the steering gear are to be as per Pt.3, Ch.14. The maximum service speed of the ship in ahead condition, however, is not to be taken less than that stated below:

HAT(B) 20 knots

HT(B) 18 knots

HM(B) 16 knots

Ha(B) 14 knots

In the astern condition, half the speed value is to be used.

The factors K2 and K3 are to be taken as 1.0.

2.10.2 For the ice classes **HAT(B)** and **HT(B),** the rudder stock and the upper part of the rudder is to be protected from direct contact with intact ice by an ice knife that extends below the LIWL, if practicable (or equivalent means). Special consideration is to be given to the design of the rudder and the ice knife for the ships with flap-type rudders.

2.10.3 For ice classes **HAT(B)** and **HT(B),** due regard is to be paid to the large loads that arise when the rudder is forced out of the midship position while going astern in ice or into ice ridges. Suitable arrangements such as rudder stoppers are to be installed to absorb these loads.

2.10.4 The local scantlings of rudders are to be determined assuming that the whole rudder belongs to the ice belt. Further, the rudder plating and frames are to be designed using the ice pressure *p* for the plating and frames in the midbody region.

2.10.5 Relief valves for hydraulic pressure in rudder turning mechanism(s) are to be installed. The components of the steering gear (e.g. rudder stock, rudder coupling, rudder horn etc.) are to be dimensioned to withstand loads causing yield stresses within the required diameter of the rudder stock.

Section 3

Machinery

3.1 Required engine output

3.1.1 Definitions (See Fig.3.1.1)

L_{BP} = length of ship between the perpendiculars [m]

L_{BOW} = length of the bow [m]

L_{PAR} = length of the parallel midship body [m]

B_{max} = maximum breadth of the ship at UIWL [m]

T_{ice} = actual ice class draughts of the ship [m] according to 3.1.2.2.

A_{wf} = area of the waterplane of the bow [m^2]

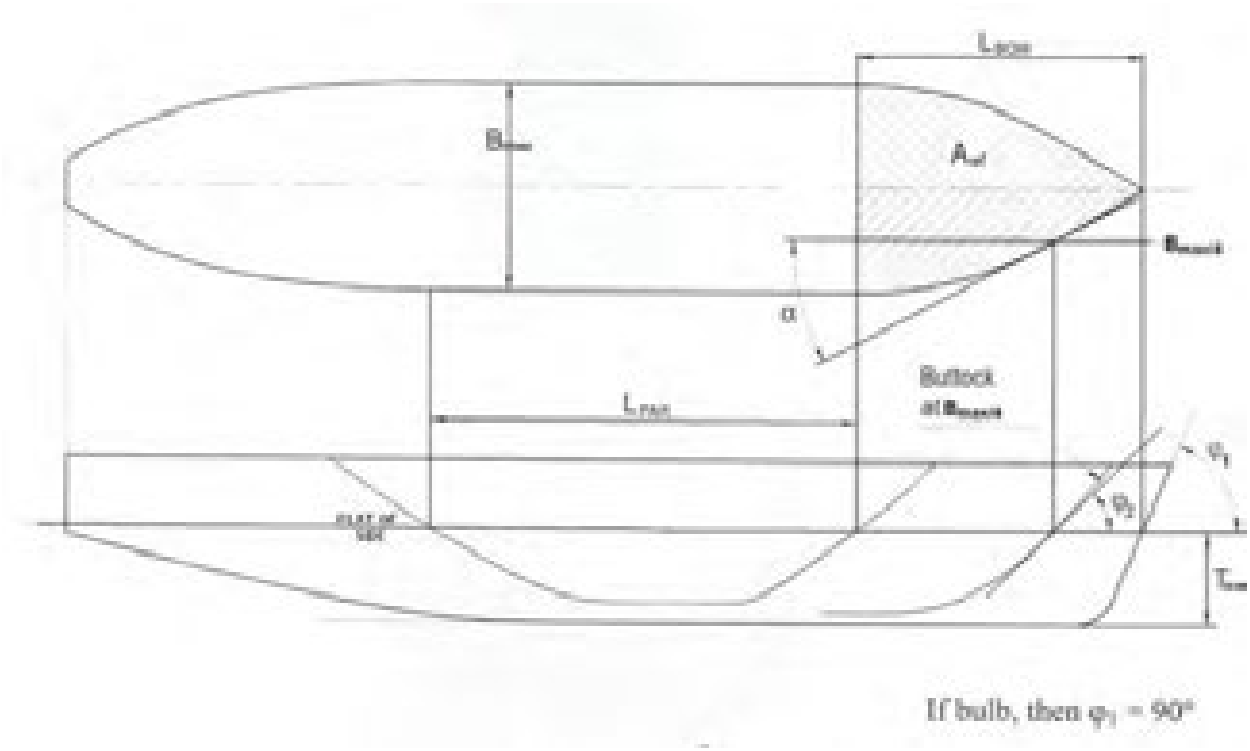


Fig. 3.1.1

α = the angle of the waterline at $B_{max}/4$ [degrees]

φ_1 = the rake of the stem at the centerline [degrees]

= 90° for ship with a bulbous bow

φ_2 = the rake of the bow at $B_{max}/4$ from centerline [degrees]

Ψ = flare angle calculated as $\psi = \arctan(\tan\varphi / \sin\alpha)$ using angles α and φ at each location. For Section 3, flare angle is calculated using $\varphi = \varphi_2$

D_p = diameter of the propeller [m]

H_M = thickness of the brash ice in mid channel [m]

H_F = thickness of the brash ice layer displaced by the bow [m]

H_M = thickness of the brash ice in mid channel [m]

H_F = thickness of the brash ice layer displaced by the bow [m]

P = The engine output 'P' [kW] is the total maximum output the propulsion machinery can continuously deliver to the propeller(s).If the output of machinery is restricted by technical means or by any regulations applicable to the ship, 'P' is to be taken as the restricted output. If additional power sources are available for propulsion power (e.g. shaft motors), in addition to the power of the main engines(s), they are also to be included in the total engine output.

3.1.2 The engine output 'P' is not to be less than the requirements given in 3.1.2.1 for ice class notation Ha and 3.1.2.2 for ice class notations Ha(B), HM(B), HT(B) and HAT(B). The engine output in no case is to be less than 1000 [kW] for ice class notations Ha(B), HM(B) and HT(B) and not less than 2800 [kW] for HAT(B).

3.1.2.1 For vessels with Ice Class "Ha" and Length > 60 [m]

P = 0.72 L B [kW].

Where,

L & B as defined in Pt.3, Ch.1

In the case of smaller vessels, P will be specially considered based on the type and duties of the vessel.

3.1.2.2 For ice class notations HAT(B), HT(B), HM(B) and Ha(B). The engine output 'P' is to be not less than the greater of the values calculated for:

a) Maximum draught amidships ("UIWL"), and

b) Minimum draught ("LIWL")

See 2.2 for definitions of "UIWL" and "LIWL".

The ship's parameters in the calculations are to be determined at the appropriate draught excepting L_{BP} and B_{max}, which are to be determined at UIWL.

$$P = K_e \frac{(R_{CH}/1000)^{\frac{3}{2}}}{D_P} \text{ [kW];}$$

where,

K_e is to be taken as follows:

Number of propellers	Controllable Pitch or electric or hydraulic propulsion machinery	Fixed Pitch propeller
1 propeller	2.03	2.26
2 propellers	1.44	1.60
3 propellers	1.18	1.31

These K_e values apply for conventional propulsion systems. For advanced propulsion systems, other methods may be used for determining the required power. (See 3.1.4).

R_{CH} is the resistance [N] of the ship in a channel with brash ice and a consolidated layer, as applicable, given by:

$$R_{CH} = \Big\{ C_1 + C_2$$

$$+ C_3\, C_\mu (H_F + H_M)^2 (B_{max} + C_\psi H_F)$$

$$+ C_4 L_{PAR} H_F^2 + C_5 \left(\frac{L_{BP} T_{ice}}{B_{max}^2} \right)^3 \frac{A_{WF}}{L_{BP}} \Big\}$$

$\left(\frac{L_{BP} T_{ice}}{B_{max}^2} \right)^3$ is not to be taken as less than 5 and not to be taken as more than 20.

Coefficients C_1 and C_2, which take into account a consolidated upper layer of the brash ice are to be taken as follows:

For ice class notations HT(B), HM(B) and Ha(B): $C_1 = 0$; . $C_2 = 0$

For ice class HAT(B):

$$C_1 = f_1 \frac{B_{max} L_{PAR}}{2\frac{T_{ice}}{B_{max}} + 1} + (1 + 0.021\varphi_1)$$

$$(f_2 B_{max} + f_3 L_{BOW} + f_4 B_{max} L_{BOW})$$

and

$$C_2 = (1 + 0.063\varphi_1)(g_1 + g_2 B_{max}) + g_3\left(1 + 1.2\frac{T_{ice}}{B_{max}}\right)\frac{B_{max}^2}{\sqrt{L_{BP}}}$$

For ship with bulbous bow, φ_1 is to be taken as 90°

where,

f_1 = 23 [N/m^2]

f_2 = 45.8 [N/m]

f_3 = 14.7 [N/m]

f_4 = 29 [N/m^2]

g_1 = 1530 [N]

g_2 = 170 [N/m]

g_3 = 400 [N/m$^{1.5}$]

$C\mu = 0.15 \cos\varphi_2 + \sin\psi \sin\alpha$; $C\mu$ is not to be taken less than 0.45

$C\psi = 0$ if $\psi \leq 45°$

$= (0.047\psi - 2.115)$ if $\psi > 45°$

where $\psi = \arctan\left(\frac{\tan\varphi_2}{\sin\alpha}\right)$

c3 = 845 kg/(m^{2}s^2)

c4 = 42 kg/(m^{2}s^2)

c5 = 825 kg/s^2.

3.1.3 **Existing Ships**. In order to retain ice class **HAT(B)** or **HT(B)** a ship; the keel of which has been laid or which has been at a similar stage of construction before 1 September 2003; is to comply with the requirements of 3.1.2.2, not later than 1 January in the year when twenty years have elapsed since the year the ship was delivered.

If the ship does not comply with the requirements in 3.1.2.2 on the date given above, the highest lower ice class for which the engine output is sufficient will be assigned to the ship.

When, for an existing ship, values for some of the hull form parameters required for the calculation method in 3.1.2.2 are difficult to obtain, the following alternative formulae can be used:

$$R_{CH} = C_1 + C_2 + C_3(H_F + H_M)^2(B_{max} + 0.658H_F) + C_4 L_{BP} H_F^2 + C_5\left(\frac{L_{BP}T_{ice}}{B_{max}^2}\right)^3 \frac{B_{max}}{4}$$

Where for ice class **HT(B)**, C_1=0 and C_2=0

For ice class **HAT(B)**, ship without bulb, C_1 and C_2 are to be calculated as follows:

$$C_1 = f_1\frac{B_{max}L_{BP}}{2\frac{T_{ice}}{B_{max}} + 1} + 1.84(f_2 B_{max} + f_3 L_{BP} + f_4 B_{max} L_{BP})$$

$$C_2 = 3.52(g_1 + g_2 B_{max}) + g_3\left(1 + 1.2\frac{T_{ice}}{B_{max}}\right)\frac{B_{max}^2}{\sqrt{L_{BP}}}$$

For ice class **HAT(B)**, ship with bulb, C_1 and C_2 are to be calculated as follows:

$$C_1 = f_1\frac{B_{max}L_{BP}}{2\frac{T_{ice}}{B_{max}} + 1} + 2.89(f_2 B_{max} + f_3 L_{BP} + f_4 B_{max} L_{BP})$$

$$C_2 = 6.67(g_1 + g_2 B_{max}) + g_3\left(1 + 1.2\frac{T_{ice}}{B_{max}}\right)\frac{B_{max}^2}{\sqrt{L_{BP}}}$$

where,

f_1 = 10.3 [N/m^2]

f_2 = 45.8 [N/m]

f_3 = 2.94 [N/m]

f_4 = 5.8 [N/m^2]

g_1 = 1530 [N]

g_2 = 170 [N/m]

g_3 = 400 [N/m$^{1.5}$]

c_3 = 460 kg/(m^{2}s^2)

c_4 = 18.7 kg/(m^{2}s^2)

c_5 = 825 kg/s^2

$\left(\frac{L_{BP}T_{ice}}{B^2_{max}}\right)^3$ is not to be taken as less than 5 and not to be taken as more than 20.

3.1.4 Other methods of determining K_e or R_{CH}

For an individual ship, in lieu of the K_e or R_{CH} values defined in 3.1.2 and 3.1.3, the use of K_e or R_{CH} values based on more precise calculations or values based on model tests may be approved. For this purpose a minimum speed of 5 knots in the following brash ice channels according to the desired class notation are to be assumed:

HAT(B) H_M= 1.0 m and a 0.1 m thick consolidated layer of ice

HT(B) H_M = 1.0 m

HM(B) H_M = 0.8 m

Ha(B) H_M = 0.6 m

Such approval will be subject to satisfactory performance of the ship in service.

3.1.5 All components of the main propulsion system are to be of steel or other approved ductile material.

3.1.6 For tail shafts in ships intended for the notation 'Ice Class **HAT(B)** or **HT(B)** and where the connection between the propeller and the tail shaft is by means of a key, Charpy impact tests are to be made in accordance with the requirements of Pt.2.

3.1.7 Propellers and propeller blades are to be of cast steel or copper alloys.

3.1.8 For steel propellers, the elongation of the material used is to be not less than 19 per cent for a test piece length of 5d. Charpy impact tests are to be carried out in accordance with the requirements of Pt.2.

3.2 Ice class 'Ha'

3.2.1 Where the notation '**Ha**' is desired, the requirements of this Section, in addition to those for open water service, are to be complied with.

3.2.2 Main engine shafting, gearing and propellers

The diameters of the shafting and propeller blade thickness as required by the Rules for open water service are to be increased by the following percentages.

Tailshaft diameter (as required by Pt.4, Ch.4, Sec.6)	5%
Propeller blade thickness at root and at 60 per cent radius (as required by Pt.5, Ch.4, Sec.7)	8%
Keyless propeller, mean torque	15%

No increase in the diameter of crankshafts, thrustshafts or intermediate shafts is required.

The tailshaft may be tapered at the forward end in accordance with Pt.4, Ch.4, Sec.6.

3.2.3 Minimum propeller blade tip thickness

The tip thickness, t, of the blade at 95 per cent radius is to be not less than that obtained by the following formula:

$$T = 0.14\ (T_p + 57)\ \sqrt[3]{\frac{430}{\sigma_u}}\ \text{[mm]}$$

where,

T_p = blade root thickness required by 3.2.3 [mm]

σ_u = specified minimum tensile strength of material, [N/mm^2].

3.2.4 Blade edge thickness

The edges of the blades are to be suitably thickened for the operating conditions, but are to be not less than 50 per cent of the required tip thickness, t, measured at 1.25 t, from the edge. For controllable pitch propellers, this requirement need only be applied to the leading edges of the blades.

3.3 Propeller and Shafting for Ice Class HAT(B), HT(B), HM(B) and Ha(B)

3.3.1 General

.1 This sub-section applies to propulsion machinery covering open- and ducted-type propellers with a controllable pitch or fixed pitch design for the ice classes HAT(B), HT(B), HM(B) and Ha(B). The given propeller loads are the expected ice loads for the entire ship's service life under normal operational conditions, including loads resulting from the changing rotational direction of FP propellers.

However, these loads do not cover off-design operational conditions, for example when a stopped propeller is dragged through ice. The load models of the section do not include propeller/ice interaction loads when ice enters the propeller of a turned azimuthing thruster from the side (radially).
The sub-section also applies to azimuthing and fixed thrusters for main propulsion, considering loads resulting from propeller/ice interaction and loads on the thruster body/ice interaction. The given azimuthing thruster body loads are the expected ice loads for the ship's service life under normal operational conditions. The local strength of the thruster body is to be sufficient to withstand local ice pressure when the thruster body is designed for extreme loads.
The thruster global vibrations caused by blade order excitation on the propeller may cause significant vibratory loads.

3.3.2 Design Ice Conditions

.1 In estimating the ice loads of the propeller for various ice classes, different types of operation as given in Table 3.3.2.1 a) were taken into account. For the estimation of design ice loads, a maximum ice block size is to be determined. The maximum design ice block entering the propeller is a rectangular ice block with the dimensions H_{ice} X $2H_{ice}$ X $3H_{ice}$. The thickness of the ice block (H_{ice}) is given in Table 3.3.2.1 b).

Table 3.3.2.1 a)	
Ice Class Notation	**Operation of Ship**
HAT(B)	Operation in ice channels and in level ice. The ship may proceed by ramming
HT(B), HM(B), and Ha(B)	Operation in ice channels

Table 3.3.2.1 b)	
Ice Class Notation	**Thickness of the design maximum ice block entering the propeller (H_{ice})**
HAT(B)	1.75 m
HT(B)	1.5 m
HM(B)	1.2 m
Ha(B)	1.0 m

3.3.3 Materials

3.3.3.1 Materials of components exposed to sea water, such as propeller blades, propeller hubs, and thruster body, are to have an elongation of not less than 15% on a test specimen, the gauge length of which is five times the diameter. A Charpy V impact test is to be carried out for materials other than bronze and austenitic steel. An average impact energy value of 20 J taken from three tests is to be obtained at minus 10 ºC. For nodular cast iron, average impact energy of 10J at minus 10 ºC is required accordingly.

3.3.3.2 Materials exposed to sea water temperature are to be of steel or other ductile material. An average impact energy value of 20 J taken from three tests is to be obtained at minus 10 ºC. This requirement applies to the propeller shaft, blade bolts, CP mechanisms, shaft bolts, strut-pod connecting bolts etc. This does not apply to surface hardened components, such as bearings and gear teeth. The nodular cast iron of a ferrite structure type may be used for relevant parts other than bolts. The average impact energy for nodular cast iron is to be minimum of 10 J at minus 10 ºC.

3.3.4 Design Loads

3.3.4.1 The given loads are intended for component strength calculations only and are total loads including ice-induced loads and hydrodynamic loads during propeller/ice interaction. The presented maximum loads are based on a worst-case scenario that occurs once during the service life of the ship. Thus, the load level for a higher number of loads is lower.

3.3.4.2 If the highest point of the propeller is not at a depth of at least h_o below the water surface when the ship is in ballast condition, the propulsion system shall be designed according to ice class HT(B) for ice classes HM(B) and Ha(B).

3.3.4.3 Design Loads on Propeller Blades

3.3.4.3.1 F_b is the maximum force experienced during the lifetime of the ship that bends a propeller blade backwards when the propeller mills an ice block while rotating ahead. F_f is the maximum force experienced during the lifetime of the ship that bends a propeller blade forwards when the propeller mills an ice block while rotating ahead. F_b and F_f originate from different propeller/ice interaction phenomena, not acting simultaneously. Hence they are to be applied to one blade separately.

3.3.4.3.2 Maximum backward blade force F_b for open propellers

$$F_b = 27[nD]^{0.7}\left[\frac{A_e}{Z}\right]^{0.3} D^2\ [kN]\ when\ D \le D_{limit}$$

$$F_b = 23[nD]^{0.7}\left[\frac{A_e}{Z}\right]^{0.3} DH_{ice}^{1.4}\ [kN]\ when\ D > D_{limit}$$

Where

$$D_{limit} = 0.85\ H_{ice}^{1.4}\ [m]$$

n is the nominal rotational speed, in [rev/sec] (at MCR in free running condition) for a controllable pitch propeller and 85% of the nominal rotational speed (at MCR in free running condition) for an fixed pitch propeller.

The direction of the forces, F_b and F_f are perpendicular to 0.7R chord line. These forces are used for strength calculation of the propeller blade, applied separately.

D is propeller diameter in [m]

A_e means expanded blade area ratio.

Z is number of propeller blades

3.3.4.3.3 Maximum forward blade force F_f for open propellers

$$F_f = 250\left[\frac{A_e}{Z}\right] D^2\ [kN]\ when\ D \le D_{limit}$$

$$F_f = 500\left[\frac{A_e}{Z}\right] D\frac{1}{\left(1-\frac{d}{D}\right)} H_{ice}\ [kN]\ when\ D > D_{limit}$$

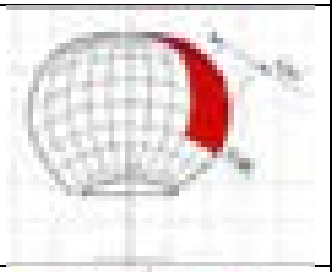
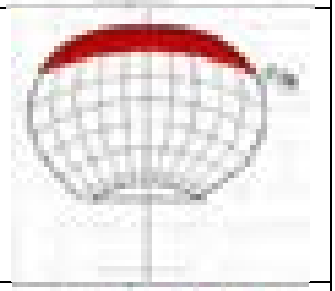
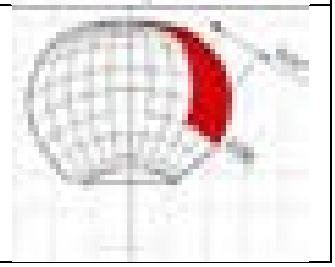
where

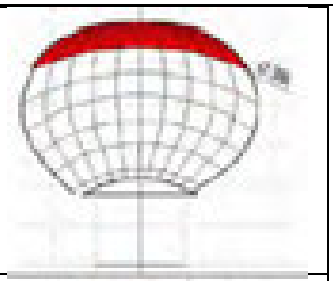
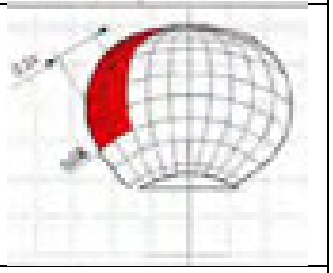
$$D_{limit} = \frac{2}{\left(1-\frac{d}{D}\right)} H_{ice}\ [m]$$

d is external diameter of propeller hub (at propeller plane) in [m]

3.3.4.3.4 Loaded area on the blade for open propellers (refer to Table 3.3.4.3.4).

Table 3.3.4.3.4 : Loaded Area on the blade for open propellers

Load Case No.	Force	Loaded Area	Right –handed propeller blade seen from aft
1	$\boldsymbol{F_b}$	Uniform pressure applied on the back of the blade (suction side) to an area from 0.6R to the tip and from the leading edge to 0.2 times the chord length	
2	50% of $\boldsymbol{F_b}$	Uniform pressure applied on the back of the blade (suction side) on the blade tip area outside 0.9R radius	
3	$\boldsymbol{F_f}$	Uniform pressure applied on the blade face (Pressure side) to an area from 0.6R to the tip and from the leading edge to 0.2 times the chord length.	

4	50% of $\boldsymbol{F_f}$	Uniform pressure applied on the blade face (pressure side) of the blade tip area outside 0.9R radius.	
5	60% of $\boldsymbol{F_f}$ or $\boldsymbol{F_b}$, whichever is greater	Uniform pressure applied on the blade face (pressure side) to an area from 0.6R to the tip and from the trailing edge to 0.2 times the chord length	
Note: Load cases 1-4 have to be covered for CP and FP propellers. In order to obtain blade ice loads for reversing propeller, load case 5 also has to be covered for FP propeller.			

3.3.4.3.5 Maximum backward blade ice force F_b for ducted propellers.

$$F_b = 9.5[nD]^{0.7}\left[\frac{A_e}{Z}\right]^{0.3} D^2\ [kN]\ when\ D \le D_{limit}$$

$$F_b = 66[nD]^{0.7}\left[\frac{A_e}{Z}\right]^{0.3} D^{0.6}H_{ice}^{1.4}\ [kN]\ when\ D > D_{limit}$$

where

$$D_{limit} = 4\ H_{ice}\ [m]$$

n is the nominal rotational speed in [rev/sec] (at MCR in free running condition) of a CP propeller and 85% of the nominal rotational speed (at MCR in free running condition) of an FP propeller.

3.3.4.3.6 Maximum forward blade ice force F_f for ducted propellers

$$F_f = 250\left[\frac{A_e}{Z}\right] D^2\ [kN]\ when\ D \le D_{limit}$$

$$F_b = 500\left[\frac{A_e}{Z}\right] D\frac{1}{\left(1-\frac{d}{D}\right)}H_{ice}\ [kN]\ when\ D > D_{limit}$$

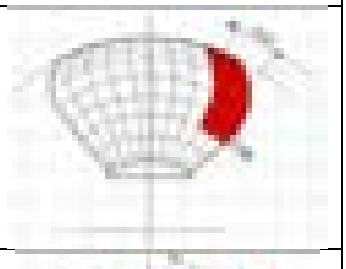
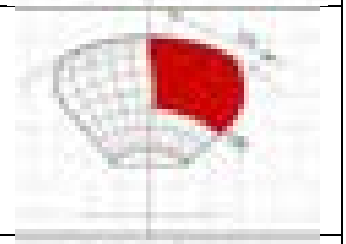
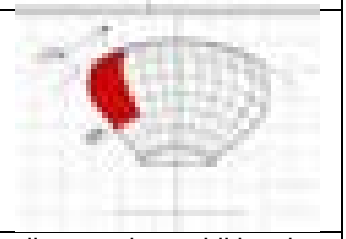
where

$$D_{limit} = \frac{2}{\left(1-\frac{d}{D}\right)}H_{ice}\ \ [m]$$

d as defined in 3.3.4.3.3

3.3.4.3.7 Loaded area on the blade for ducted propellers

Table 3.3.4.3.7 : Loaded area on the blade for ducted propellers

Load Case No.	Force	Loaded Area	Right –handed propeller blade seen from aft
1	$\boldsymbol{F_b}$	Uniform pressure applied on the back of the blade (suction side) to an area from 0.6R to the tip and from the leading edge to 0.2 times the chord length	
3	$\boldsymbol{F_f}$	Uniform pressure applied on the blade face (Pressure side) to an area from 0.6R to the tip and from the leading edge to 0.5 times the chord length.	
5	60% of $\boldsymbol{F_f}$ or $\boldsymbol{F_b}$, whichever is greater	Uniform pressure applied on face (pressure side) to an area from 0.6R to the tip and from the trailing edge to 0.2 times the chord length	
Note: Load cases 1 and 3 have to be covered as given above for all propellers, and an additional load case (load case 5) for FP propeller, to cover ice loads when the propeller is reversed.			

3.3.4.3.8 Maximum blade spindle torque Q_{smax} for open and ducted propellers

.1 The spindle torque Q_{smax} around the axis of the blade fitting is to be determined both for the maximum backward blade force F_b and forward blade force F_f, which are applied as in Table 3.3.4.3.4 and 3.3.4.3.7. The larger of the obtained torques is used as the dimensioning torque. If the above method gives a value which is less than the default value given by the formula below, the default value shall be used

$$Default\ Value\ Q_{max} = 0.25\ F\ c_{0.7}\ [kNm]$$

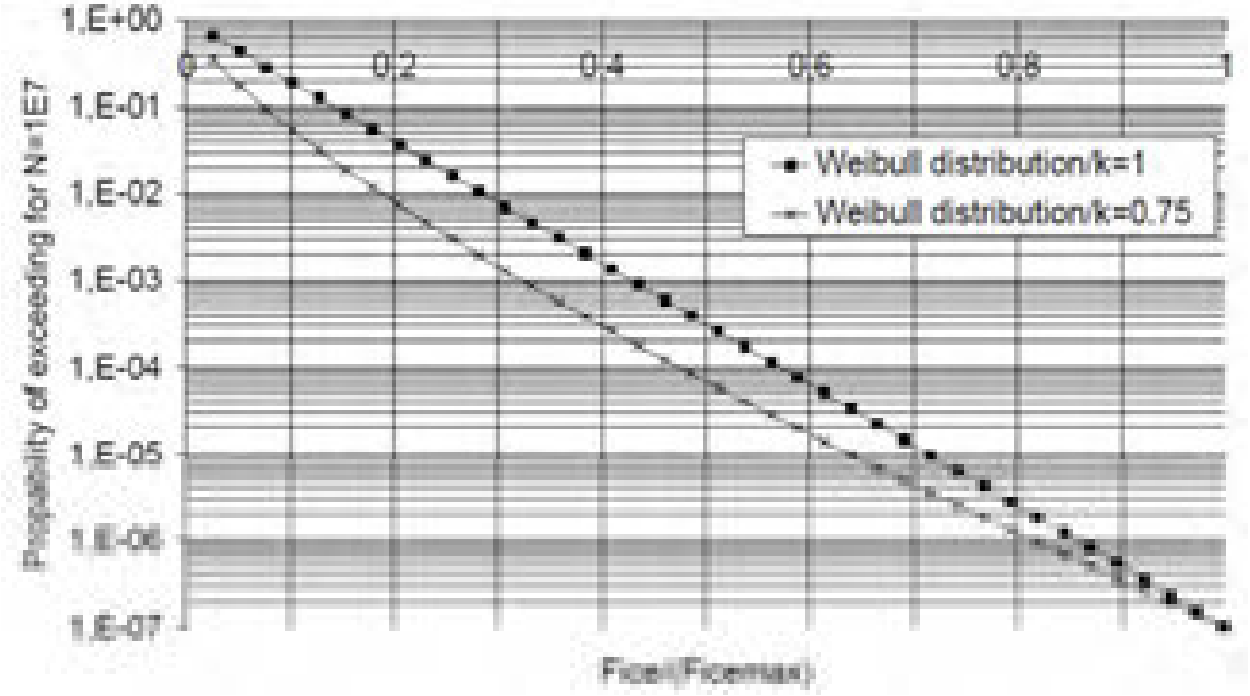
where $c_{0.7}$ is the length of the blade section at 0.7R radius and F is either F_b or F_f, whichever has the greater absolute value.

3.3.4.3.9 Load Distribution for blade Loads

.1 The weibull-type distribution (probability that F_{ice} exceeds $(F_{ice})_{max}$), as given in the graph below, is used for the fatigue design of the blade.

$$P\left(\frac{F_{ice}}{(F_{ice})_{max}} \geq \frac{F}{(F_{ice})_{max}}\right) = e^{\left(-\left(\frac{F}{(F_{ice})_{max}}\right)^{k} \ln(N_{ice})\right)}$$

where k is the shape parameter of the spectrum, N_{ice} is the number of load cycles in the spectrum, and F_{ice} is the random variable for ice loads on the blade, $0 \leq F_{ice} \leq (F_{ice})_{max}$. The shape parameter k=0.75 shall be used for the ice force distribution of an open propeller and the shape parameter k=1.0 for that of a ducted propeller blade.



3.3.4.3.10 Number of ice loads

.1 The number of load cycles per propeller blade in the load spectrum is to be determined according to the formula:

$$N_{ice} = k_1 k_2 k_3 N_{class} n$$

where

Number of loads for ice classes N_{class}	
Class	**Impacts in life/n**
HAT (B)	9 x 10^6
HT(B)	6 x 10^6
HM(B)	3.4 x 10^6
Ha(B)	2.1 x 10^6

Propeller Location factor , k_1	
Location	**k_1**
Centre propeller Bow first operation	1
Wing propeller Bow first operation	2
Pulling propeller (wing and centre) Bow propeller or stern first operation	3

Propulsion Type factor , k_3	
Type	**k_3**
Fixed	1
Azimuthing	1.2

The submersion factor K_2 is determined from the equation

$K_2 = 0.8 - f$ when $f < 0$
$= 0.8 - 0.4f$ when $0 \leq f \leq 1$
$= 0.6 - 0.2f$ when $1 < f \leq 2.5$
$= 0.1$ when $f > 2.5$

where the immersion function f is

$$f = \frac{h_o - H_{ice}}{D/2} - 1$$

where h_o is the depth of the propeller centerline at the lower ice waterline (LIWL) of the ship.

For components that are subject to loads resulting from propeller/ice interaction with all the propeller blades, the number of load cycles (N_{ice}) is to be multiplied by the number of propeller blades (Z).

n = rotational speed of the propeller, rps.

3.3.4.4 Axial design loads for open and ducted propellers

3.3.4.4.1 Maximum ice thrust on propeller T_f and T_b for open and ducted propellers are:

$$T_f = 1.1\, F_f\ [kN]$$
$$T_b = 1.1\, F_b\ [kN]$$

3.3.4.4.2 Design thrust along the propulsion shaft line for open and ducted propellers

.1 The design thrust along the propeller shaft line is to be calculated with the formulae below. The greater value of the forward and backward direction loads are to be taken as the design load for both directions. The factors 2.2 and 1.5 take into account the dynamic magnification resulting from axial vibration.

In a forward direction

$$T_r = T + 2.2T_f \ [kN]$$

In a backward direction

$$T_r = 1.5T_b \ [kN]$$

If the hydrodynamic bollard thrust, T [kN], is not known, T is to be taken as follows:

Propeller Type	T
CP Propeller (Open)	1.25 T_n
CP Propeller (Ducted)	1.1 T_n
FP Propellers driven by turbine or electric motor	T_n
FP propellers driven by diesel engine (open)	0.85 T_n
FP propellers driven by diesel engine (ducted)	0.75T_n

where T_n is the nominal propeller thrust [kN] at MCR in the free running open water condition.

3.3.4.5 Torsional Design Loads

3.3.4.5.1 Design ice torque on propeller Q_{max} for open propellers

.1 Q_{max} is the maximum torque on a propeller resulting from ice/propeller interaction during the service life of the ship.

where $D \leq D_{limit}$

$$Q_{max} = 10.9 \left[1 - \frac{d}{D}\right]\left[\frac{P_{0.7}}{D}\right]^{0.16} (nD)^{0.17} D^3 \ [kNm]$$

where $D > D_{limit}$

$$Q_{max} = 20.7 \left[1 - \frac{d}{D}\right]\left[\frac{P_{0.7}}{D}\right]^{0.16} (nD)^{0.17} D^{1.9} H_{ice}^{1.1} \ [kNm]$$

where,

$$D_{limit} = 1.8 \, H_{ice} \ [m]$$

n is the rotational propeller speed at MCR in bollard condition. If not known, n is to be taken as follows:

Propeller Type	Rotational Speed n
CP Propellers	n_n
FP Propellers driven by turbine or electric motor	n_n
FP propellers driven by diesel engine	0.85n_n

Here n_n [rev/sec] is the nominal rotational speed at MCR in the free running open water condition.

For CP propellers, the propeller pitch, $P_{0.7}$ is to correspond to MCR in bollard condition. If not known, $P_{0.7}$ is to be taken as $0.7P_{0.7n}$, where $P_{0.7n}$ is the propeller pitch at MCR in free running condition.

3.3.4.5.2 Design ice torque on propeller Q_{max} for ducted propellers.

.1 Q_{max} is the maximum torque on a propeller during the service life of the ship resulting from ice/propellers interaction.

Where $D \leq D_{limit}$

$$Q_{max} = 7.7 \left[1 - \frac{d}{D}\right]\left[\frac{P_{0.7}}{D}\right]^{0.16} (nD)^{0.17} D^3 \ [kNm]$$

Where $D > D_{limit}$

$$Q_{max} = 14.6 \left[1 - \frac{d}{D}\right]\left[\frac{P_{0.7}}{D}\right]^{0.16} (nD)^{0.17} D^{1.9} H_{ice}^{1.1} \ [kNm]$$

Where, D_{limit} and n are to be taken as indicated in 3.3.4.5.1

3.3.4.5.3 Design torque for non-resonant shaft line

.1 If there is no relevant first blade order torsional resonance in the operational speed range or in the range 20% above and 20% below the maximum operating speed (bollard condition), the following estimation of the maximum response torque can be used.

Directly coupled two stroke diesel engines without flexible coupling

$$Q_{peak} = Q_{emax} + Q_{vib} + Q_{max}\frac{I_e}{I_t}\ [kNm]$$

And other plants

$$Q_{peak} = Q_{emax} + Q_{max}\frac{I_e}{I_t}\ [kNm]$$

Where I_e is equivalent mass moment of inertia of all parts on engine side of component under consideration and I_t is equivalent mass moment of inertia of the whole propulsion system.

Q_{vib} is vibratory torque taken from frequency domain open water torsional vibration calculations [kNm]

.2 All the torques and the inertia moments shall be reduced to the rotation speed of the component being examined.

.3 If the maximum torque Q_{emax} [kN m] is not known, it shall be taken as follows:

Table 3.3.4.5.3.1 : Default values for prime mover maximum torque Q_{emax}

Propeller Type	**Q_{emax}**
Propeller driven by electric motor	Q_{motor}
CP propellers not driven by electric motor	Q_n
FP propellers driven by turbine	Q_n
FP propellers driven by diesel engine	$0.75Q_n$

Here Q_{motor} [kN m] is the electric motor peak torque.

3.3.4.5.4 Design torque for shaft lines having resonances.

.1 If there is first blade order torsional resonance in the operational speed range or in the range 20% above and 20% below the maximum operating speed (bollard condition), the design torque (Q_{peak}) of the shaft component is to be determined by means of torsional vibration analysis of the propulsion line. There are two alternative ways of performing the dynamic analysis.

a) Time domain calculation for estimated milling sequence excitation

b) Frequency domain calculation for blade orders sinusoidal excitation.

The frequency domain analysis is generally considered conservative compared to the time domain simulation, provided that there is a first blade order resonance in the considered speed range.

.2 Time domain calculation of torsional response.

a) Time domain calculations are to be calculated for the MCR condition, MCR bollard conditions and for blade order resonant rotational speeds so that the resonant vibration responses can be obtained.

b) The load sequence given in this chapter, for a case where a propeller is milling an ice block, is to be used for the strength evaluation of the propulsion line. The given load sequence is not intended for propulsion system stalling analyses.

c) The following load cases are intended to reflect the operational loads on the propulsion system, when the propeller interacts with ice, and the respective reaction of the complete system. The ice impact and system response causes loads in the individual shaft line components. The ice torque Q_{max} may be taken as a constant value in the complete speed range. When considerations at specific shaft speeds are performed, a relevant Q_{max} may be calculated using the relevant speed.

d) Diesel engine plants without an elastic coupling is to be calculated at the least favourable phase angle for ice versus engine excitation, when calculated in the time domain. The engine firing pulses is to be included in the calculations and their standard steady state harmonics can be used.

e) If there is a blade order resonance just above the MCR speed, calculations are to cover rotation-al speeds up to 105% of the MCR speed.

f) The propeller ice torque excitation for shaft line transient dynamic analysis in the time domain is defined as a

sequence of blade impacts which are of half sine shape. The excitation frequency is to follow the propeller rotational speed during the ice interaction sequence. The torque due to a single blade ice impact as a function of the propeller rotation angle is then defined using the formula:

$$Q(\phi) = C_q Q_{max} \sin\left(\phi(180/\alpha_i)\right)$$

When φ rotates from 0 to α_i plus integer revolutions

$$Q(\phi) = 0$$

When φ rotates from α_i to 360 plus integer revolutions

Where

φ is the rotation angle from when the first impact occurs and parameters Cq and α_i are given in Table

α_i is the duration of propeller blade/ice interaction expressed in terms of the propeller rotation angle.

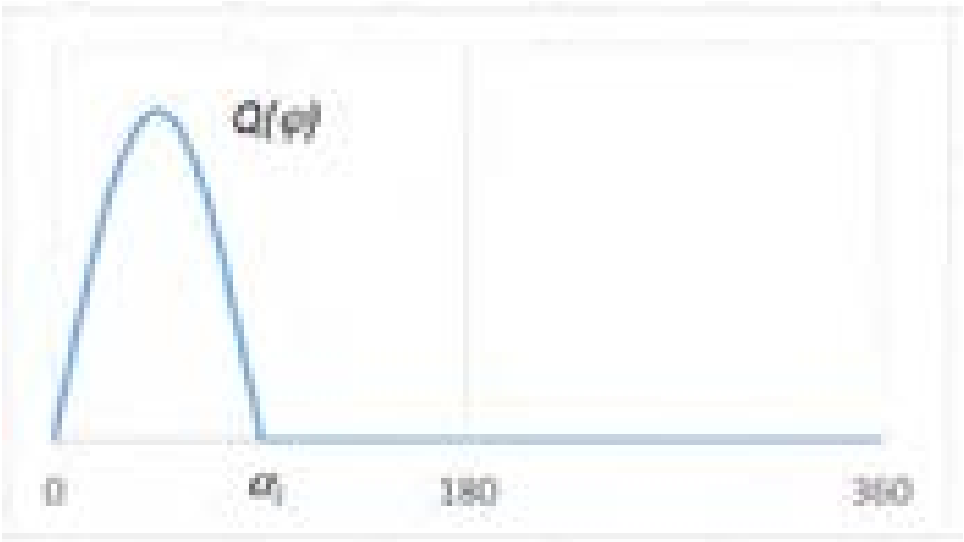


Fig. 3.3.4.5.4 a) : Schematic ice torque due to a single blade ice impact as a function of the propeller rotation angle.

Table 3.3.4.5.4.1 : Ice impact magnification and duration factors for different blade numbers

Torque Excitation	Propeller/ice interaction	Cq	α_i [deg]			
			Z=3	Z=4	Z=5	Z=6
Excitation case 1	Single ice block	0.75	90	90	72	60
Excitation case 2	Single ice block	1.0	135	135	135	135
Excitation case 3	Two ice blocks (phase shift 360/(2.Z)deg.)	0.5	45	45	36	30
Excitation case 4	Single ice block	0.5	45	45	36	30

g) The total ice torque is obtained by summing the torque of single blades, while taking account of the phase shift 360 deg./*Z*. At the beginning and end of the milling sequence (within the calculated duration) linear ramp functions are to be used to increase *Cq* to its maximum value within one

propeller revolution and vice versa to decrease it to zero (see the examples of different Z numbers in Figs. 3.3.4.5.4 b) and 3.3.4.5.4 c)).

h) The number of propeller revolutions during a milling sequence is to be obtained from the formula:

$$N_Q = 2H_{ice}$$

i) The number of impacts is $Z \cdot N_Q$ for blade order excitation. An illustration of all excitation cases for different numbers of blades is given in Fig. 3.3.4.5.4 b) and 3.3.4.5.4 c).

j) A dynamic simulation is to be performed for all excitation cases at the operational rotational speed range. For a fixed pitch propeller propulsion plant, a dynamic simulation must also cover the bollard pull condition with a corresponding rotational speed assuming the maximum possible output of the engine.

k) If a speed drop occurs until the main engine is at a standstill, this indicates that the engine may not be sufficiently powered for the intended service task. For the consideration of loads, the maximum occurring torque during the speed drop process must be used.

l) For the time domain calculation, the simulated response torque typically includes the engine mean torque and the propeller mean torque. If this is not the case, the response torques are to be obtained using the formula:

$$Q_{peak} = Q_{emax} + Q_{rtd}$$

Where Q_{rtd} is the maximum simulated torque obtained from the time domain analysis.

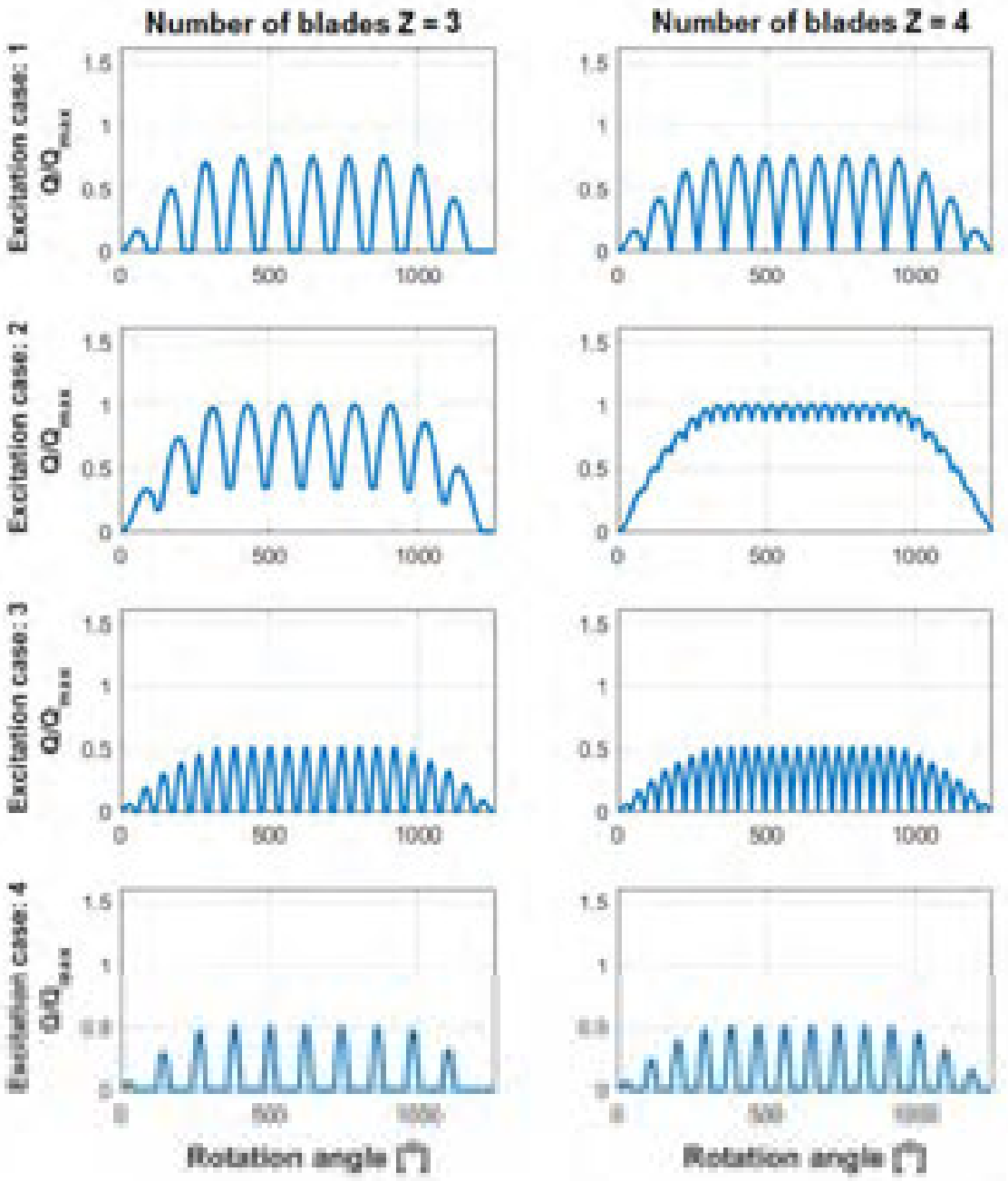


Fig. 3.3.4.5.4 b) : The shape of the propeller ice torque excitation sequences for propellers with 3 or 4 blades.

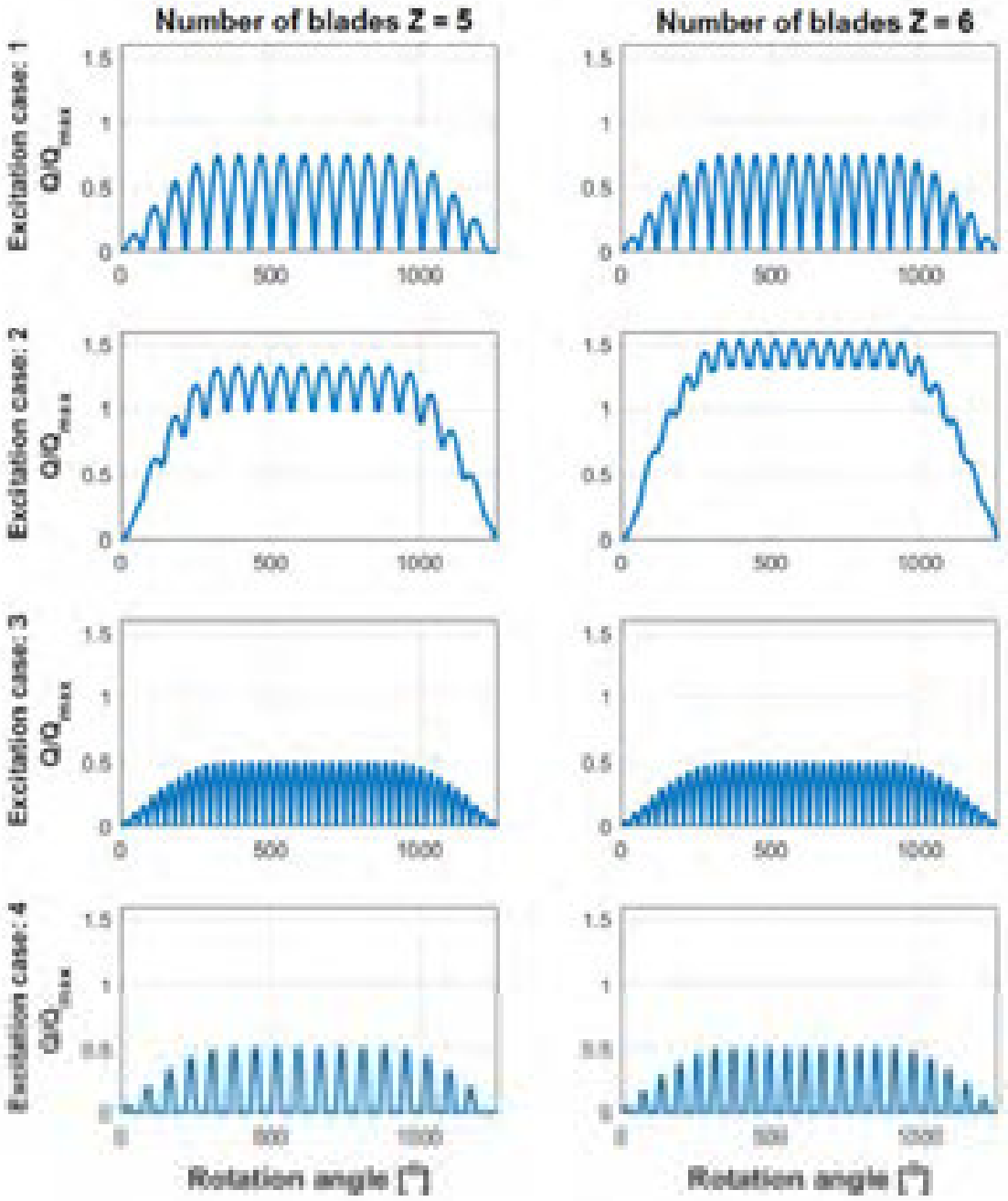


Fig. 3.3.4.5.4 c) : The shape of the propeller ice torque excitation sequences for propellers with 5 or 6 blades.

.3 Frequency domain calculation of torsional response

a) For frequency domain calculations, blade order and twice-the-blade-order excitation may be used. The amplitudes for the blade order and twice-the-blade-order sinusoidal excitation have been derived based on the assumption that the time domain half sine impact sequences were continuous, and the Fourier series components for blade order and twice-the-blade-order components have been derived. The propeller ice torque is then:

$$Q_F(\varphi) = Q_{max}\big(C_{q0} + C_{q1}\sin(ZE_0\varphi + \alpha_1) + C_{q2}\sin(2ZE_0\varphi + \alpha_2)\big)\ [kNm]$$

Where

C_{q0} is mean torque parameter
C_{q1} is the first blade order excitation parameter
C_{q2} is the second blade order excitation parameter
α_1 and α_2 are phase angles of the excitation component.
Φ is the angle of rotation
E_0 is the number of ice blocks in contact
The values of the parameters are given in the table 3.3.4.5.4.2

b) The design torque for the frequency domain excitation case must be obtained using the formula:

$$Q_{peak} = Q_{emax} + Q_{vib} + \left(Q^n_{max}C_{q0}\right)I_e/I_t + Q_{rf1} + Q_{rf2}$$

Where,

$Q^n_{max}$is the maximum propeller ice torque at the operation speed in consideration
C_{q0} is the mean static torque coefficient from Table 3.3.4.5.4.2
Q_{rf1} is the blade order torsional response from the frequency domain analysis
Q_{rf2} is the second order blade torsional response from the frequency domain analysis.

c) If the prime mover maximum torque, Q_{emax}, is not known, it is to be taken as given in Table 3.3.4.5.3.1. All the torque values have to be scaled to the shaft revolutions for the component in question.

Table 3.3.4.5.4.2 : Coefficient values for frequency domain excitation calculation

Torque Excitation Z=3						
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.375	0.36	-90	0	0	1
Excitation case 2	0.7	0.33	-90	0.05	-45	1
Excitation case 3	0.25	0.25	-90	0		2
Excitation case 4	0.2	0.25	0	0.05	-90	1
Torque Excitation Z=4						
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.45	0.36	-90	0.06	-90	1
Excitation case 2	0.9375	0	-90	0.0625	-90	1
Excitation case 3	0.25	0.25	-90	0	0	2
Excitation case 4	0.2	0.25	0	0.05	-90	1
Torque Excitation Z=5						
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.45	0.36	-90	0.06	-90	1
Excitation case 2	1.19	0.17	-90	0.02	-90	1
Excitation case 3	0.3	0.25	-90	0.048	-90	2
Excitation case 4	0.2	0.25	0	0.05	-90	1
Torque Excitation Z=6						
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.45	0.36	-90	0.05	-90	1
Excitation case 2	1.435	0.1	-90	0	0	1
Excitation case 3	0.3	0.25	-90	0.048	-90	2
Excitation case 4	0.2	0.25	0	0.05	-90	1

.4 Guidance for torsional vibration calculation

a) The aim of time domain torsional vibration simulations is to estimate the extreme torsional load for the ship's lifespan. The simulation model can be taken from the normal lumped mass elastic torsional vibration model, including damping. For a time domain analysis, the model should include the ice excitation at the propeller, other relevant excitations and the mean torques provided by the prime mover and hydrodynamic mean torque in the propeller. The calculations should cover variation of phase between the ice excitation and prime mover excitation. This is extremely relevant to propulsion lines with directly driven combustion engines. Time domain calculations shall be calculated for the MCR condition, MCR bollard conditions and for resonant speed, so that the resonant vibration responses can be obtained.

b) For frequency domain calculations, the load should be estimated as a Fourier component analysis of the continuous sequence of half sine load sequences. First and second order blade components should be used for excitation.

c) The calculation should cover the entire relevant rpm range and the simulation

of responses at torsional vibration resonances.

3.3.4.6 Blade failure load

3.3.4.6.1 Bending force, F_{ex}

.1 The ultimate load resulting from blade failure as a result of plastic bending around the blade root is to be calculated from the formula below or alternatively by means of an appropriate stress analysis, reflecting non-linear plastic material behavior of the actual blade. In such a case, the blade failure area may be outside the root section. The ultimate load is assumed to be acting on the blade at the 0.8*R* radius in the weakest direction of the blade. A blade is regarded as having failed if the tip is bent into an offset position by more than 10% of propeller diameter D.

$$F_{ex} = \frac{300\ c\ t^2 \sigma_{ref1}}{0.8D - 2r}\ [kN]$$

where

$$\sigma_{ref1} = 0.6\sigma_{0.2} + 0.4\sigma_u$$

c, t, and r are respectively, the actual chord length, maximum thickness and radius of the cylindrical root section of the blade, which is the weakest section outside the root filet typically located at the point where the fillet terminates at the blade profile.

σ_u = ultimate tensile strength of blade material [N/mm^2]

$\sigma_{0.2}$ = Proof yield strength (at 0.2% offset) of blade material [N/mm^2]

3.3.4.6.2 Spindle torque, Q_{sex}

.1 The maximum spindle torque due to a blade failure load acting at 0.8*R* is to be determined. The force that causes blade failure typically reduces when moving from the propeller centre towards the leading and trailing edges. At a certain distance from the blade centre of rotation, the maximum spindle torque will occur. This maximum spindle torque is to be defined by an appropriate stress analysis or using the equation given below.

$$Q_{sex} = max(C_{LE0.8}; 0.8C_{TE0.8})C_{spex}F_{ex}\ [kNm]$$

Where

$$C_{spex} = C_{sp}C_{fex} = 0.7\left(1 - \left(\frac{4A_e}{Z}\right)^3\right)$$

C_{sp} is a non-dimensional parameter taking account of the spindle arm

C_{fex} is a non-dimensional parameter taking account of the reduction of the blade failure at the location of the maximum spindle torque.
If C_{spex} is below 0.3, a value of 0.3 is to be used for C_{spex}.

$C_{LE0.8}$ is the leading edge portion of the chord length at 0.8R.
$C_{TE0.8}$ is the trailing edge portion of the chord length at 0.8R.

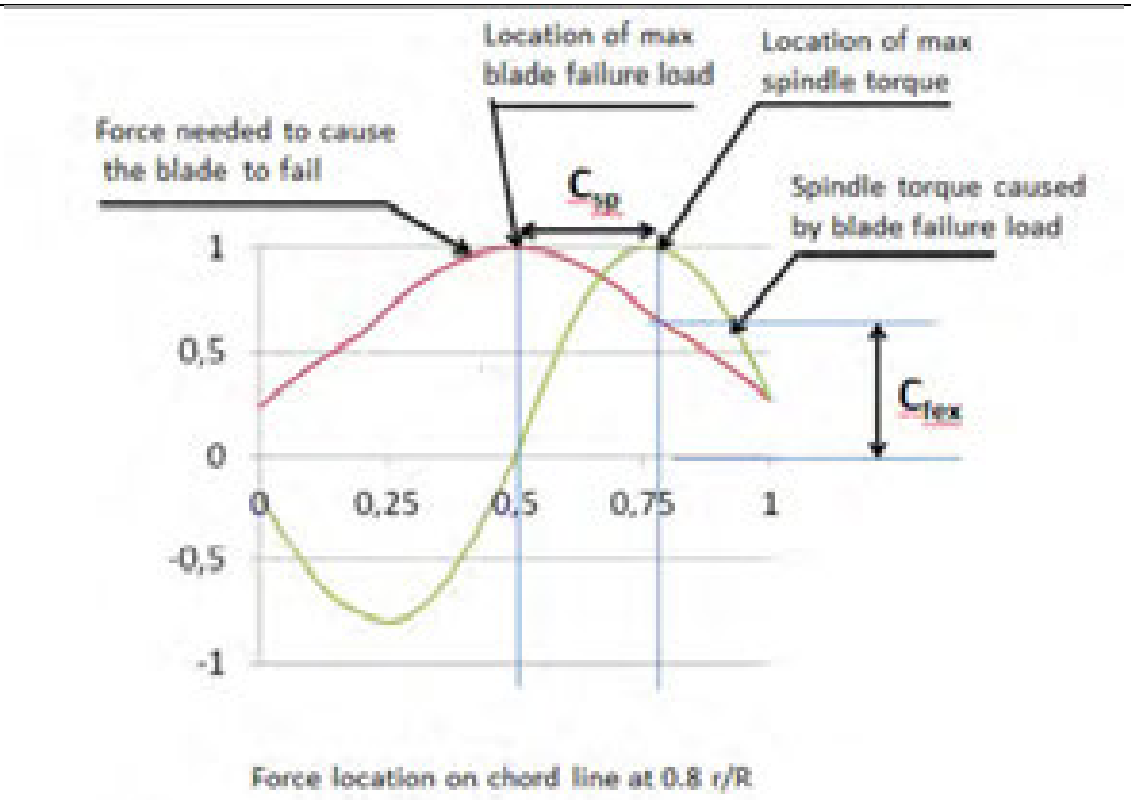


Fig. 3.3.4.6.2 : Schematic figure showing a blade failure load and the related spindle torque when the force acts at a different location on the chord line at radius 0.8R

3.3.5 Design

3.3.5.1 Design Principle

.1 The strength of the propulsion line is to be designed according to the pyramid strength principle. This means that the loss of the propeller blade is not to cause any significant damage to other propeller shaftline components.

3.3.5.2 Propeller Blade

3.3.5.2.1 Calculation of blade stresses

The blade stress is to be calculated for the design loads given in 3.3.4. Finite element analysis is to be used for stress analysis for final approval for all propellers. The following simplified formulae can be used in estimating the blade stresses for all propellers at the root area (r/R<0.5). The root area dimensions based on formula given below can be accepted even if the FEM analysis would show greater stresses at the root area.

$$\sigma_{st} = C_1 \frac{M_{BL}}{100\, ct^2} \quad [N/mm^2]$$

Where

C_1 is the ratio of actual stress to stress obtained with beam equation. If the actual value is not available, C_1 is to be taken as 1.6.

For relative radius r/R <0.5:

$$M_{BL} = (0.75 - r/R)RF$$

F is the force F_b or F_f whichever has greater absolute value.

3.3.5.2.2 Acceptability criterion for blade stresses

The following criterion for calculated blade stresses is to be fulfilled:

$$\frac{\sigma_{ref2}}{\sigma_{st}} \geq 1.3$$

σ_{st} is the calculated stress for the design loads. If FEM analysis is used in estimating the stresses, von Mises stresses are to be used.

σ_{ref2} is the reference stress, defined as:

$$\sigma_{ref2} = 0.7\sigma_u \quad or$$

$$\sigma_{ref2} = 0.6\sigma_{0.2} + 0.4\sigma_u \, , whichever \; is \; less$$

σ_u = ultimate tensile strength of blade material [N/mm^2]

$\sigma_{0.2}$ = Proof yield strength (at 0.2% offset) of blade material [N/mm^2]

3.3.5.2.3 Fatigue Design of Propeller Blade

The fatigue design of the propeller blade is based on an estimated load distribution for the service life of the ship and the S-N curve for the blade material. An equivalent stress that produces the same fatigue damage as the expected load distribution is to be calculated and the acceptability criterion for fatigue is to be fulfilled as given in this sub-section. The equivalent stress is normalized for 10^8 cycles.

For materials with a two-slope SN curve (Fig. 3.3.5.2.3 a), fatigue calculations in accordance with this sub-section are not required if the following criterion is fulfilled.

$$\sigma_{exp} \geq B_1 \sigma_{ref2}^{B_2} \log(N_{ice})^{B_3}$$

where B_1, B_2 and B_3 coefficients for open and ducted propellers are given in the Table 3.3.5.2.3.

Table 3.3.5.2.3 : Values for B1, B2 & B3

	Open Propeller	Ducted Propeller
B_1	0.00328	0.00223
B_2	1.0076	1.0071
B_3	2.101	2.471

For calculation of equivalent stress two types of S-N curves are available.

(a) Two slope S-N curve (slopes 4.5 and 10), Fig 3.3.5.2.3 a)
(b) One slope S-N curve (the slope can be chosen), Fig 3.3.5.2.3 b)

The type of S-N curve is to be selected to correspond to the material properties of the blade. If the S-N-curve is not known, the two slope S-N curve is to be used.

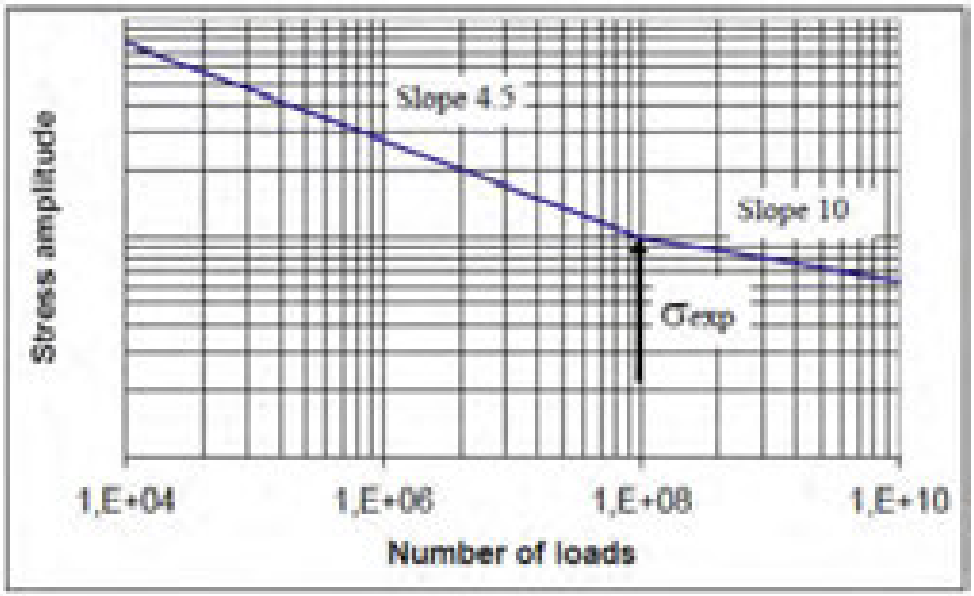


Fig.3.3.5.2.3 a) : Two-slope S-N curve

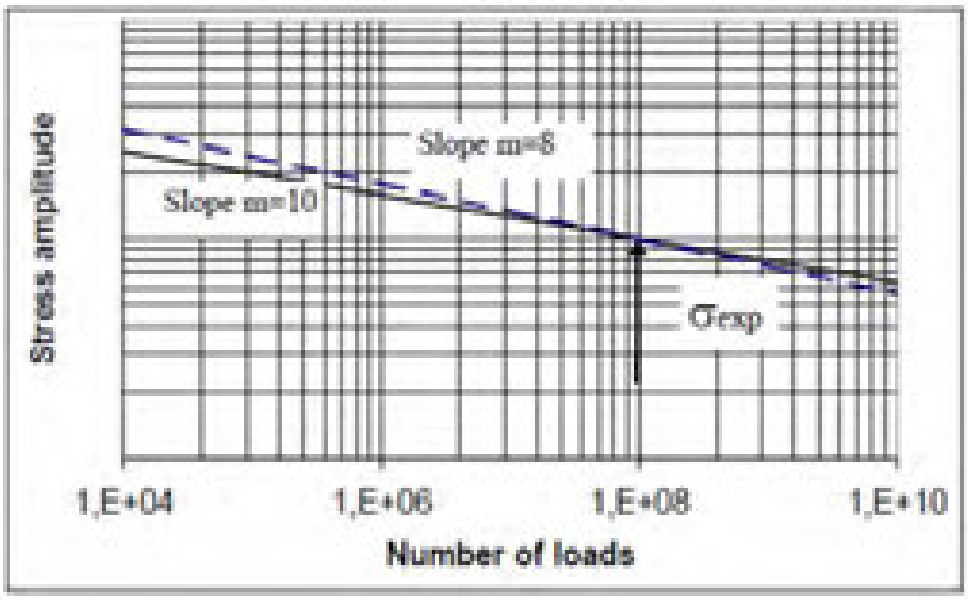


Fig.3.3.5.2.3 b) : Constant slope S-N curve

The equivalent fatigue stress for 10^8 stress cycles which produces the same fatigue damage as the load distribution for the service life of the ship is:

$$\sigma_{fat} = \rho(\sigma_{ice})_{max}$$

where

$$(\sigma_{ice})_{max} = 0.5\big((\sigma_{ice})_{fmax} - (\sigma_{ice})_{bmax}\big)$$

$(\sigma_{ice})_{max}$ is the mean value of the principal stress amplitudes resulting from design forward and backward blade forces at the location being studied.

$(\sigma_{ice})_{fmax}$ is the principal stress resulting from forward load

$(\sigma_{ice})_{bmax}$ is the principal stress resulting from backward load

In calculation of $(\sigma_{ice})_{max}$, case 1 and case 3 (or case 2 and case 4) are considered as a pair for $(\sigma_{ice})_{fmax}$ and $(\sigma_{ice})_{bmax}$ calculations. Case 5 is excluded from the fatigue analysis.

.1 Calculation of Parameter ρ for two slope S-N Curve

The parameter ρ relates the maximum ice load to the distribution of ice loads according to the regression formula.

$$\rho = C_1(\sigma_{ice})_{max}^{C_2}\sigma_{fl}^{C_3}\log(N_{ice})^{C_4}$$

where,

$$\sigma_{fl} = \gamma_{\varepsilon 1}\gamma_{\varepsilon 2}\gamma_v\gamma_m\sigma_{exp},$$

where

$\gamma_{\varepsilon 1}$ is the reduction factor due to scatter (equal to one standard deviation)

$\gamma_{\varepsilon 2}$ is the reduction factor for test specimen size effect

γ_v is the reduction factor for variable amplitude loading

γ_m is the reduction factor for mean stress

σ_{exp} is the mean fatigue strength of the blade material at 10^8 cycles to failure in sea water. The following values are to be used for the reduction factors if actual values are not available:

$$\gamma_\varepsilon = \gamma_{\varepsilon 1}\gamma_{\varepsilon 2} = 0.67$$
$$\gamma_v = 0.75$$
$$\gamma_m = 0.75$$

The applicable range of N_{ice} for calculating ρ is $5 \times 10^6 \le N_{ice} \le 10^8$

	Open Propeller	**Ducted Propeller**
C_1	0.000747	0.000534
C_2	0.0645	0.0533
C_3	-0.0565	-0.0459
C_4	2.22	2.584

.2 Calculation of Parameter ρ for constant slope S-N Curve

For materials with a constant-slope S-N curve, the factor ρ is to be calculated from the following formula:

$$\rho = \left(G\frac{N_{ice}}{N_R}\right)^{1/m} (\ln(N_{ice}))^{-1/k}$$

where

k is the shape parameter of the Weibull distribution k=1.0 for ducted propellers and k=0.75 for open propellers. N_R is the reference number of load cycles (=10^8)

The applicable range of N_{ice} for calculating ρ is $5 \times 10^6 \leq N_{ice} \leq 10^8$

m is the slope of the S-N curve

Values for the parameter 'G', for different m/k ratios are given in Table 3.3.5.2.3.2. Linear interpolation may be used to calculate the value of m/k ratios other than those given in the Table 3.3.5.2.3.2.

Table 3.3.5.2.3.2 : Values for G for different m/k ratios

m/k	G	m/k	G	m/k	G
3	6	5.5	287.9	8	40320
3.5	11.6	6	720	8.5	119292
4	24	6.5	1871	9	362880
4.5	52.3	7	5040	9.5	1.133×10^6
5	120	7.5	14034	10	3.629×10^6
				10.5	11.899×10^6
				11	39.917×10^6
				11.5	136.843×10^6
				12	479.002×10^6

.3 Acceptability criterion for fatigue

The equivalent fatigue stress at all locations on the blade has to fulfill the following acceptability criterion:

$$\frac{\sigma_{fl}}{\sigma_{fat}} \geq 1.5$$

where

$$\sigma_{fl} = \gamma_{\varepsilon 1}\gamma_{\varepsilon 2}\gamma_v\gamma_m\sigma_{exp}$$

where

$\gamma_{\varepsilon 1}$ is the reduction factor due to scatter(equal to one standard deviation)

$\gamma_{\varepsilon 2}$ is the reduction factor for test specimen size effect

γ_v is the reduction factor for variable amplitude loading

γ_m is the reduction factor for mean stress

σ_{exp} is the mean fatigue strength of the blade material at 10^8 cycles to failure in seawater. The following values should be used for the reduction factors if actual values are not available:

$$\gamma_\varepsilon = \gamma_{\varepsilon 1}\gamma_{\varepsilon 2} = 0.67$$
$$\gamma_v = 0.75$$
$$\gamma_m = 0.75$$

3.3.5.3 Propeller bossing and CP mechanism

.1 The blade bolts, the CP mechanism, the propeller boss, and the fitting of the propeller to the propeller shaft are to be designed to withstand the maximum and fatigue design loads, as defined in 3.3.4.3 to 3.3.4.6. The safety factor against yielding is to be greater than 1.3 and that against fatigue greater than 1.5. In addition, the safety factor for loads resulting from loss of the propeller blade through plastic bending as defined in 3.3.4.6 is to be greater than 1.0 against yielding.

3.3.5.4 Propulsion shaft line

.1 The shaft and shafting components, such as the thrust and stern tube bearings, couplings, flanges and sealing, are to be designed to withstand the propeller/ ice interaction loads as given in 3.3.4.3 to 3.3.4.6. The safety factor is to be at least 1.3 against yielding for extreme operational loads, 1.5 for fatigue loads and 1.0 against yielding for the blade failure load.

.2 The ultimate load resulting from total blade failure as defined in 3.3.4.6 is not to cause yielding in shafts and shaft components. The loading is to consist of the combined axial, bending, and torsion loads, wherever this is significant. The minimum safety factor against yielding is to be 1.0 for bending and torsional stress.

3.3.5.5 Azimuthing main propulsors

.1 Design Principles

a) In addition to the above requirements for propeller blade dimensioning, azimuthing thrusters are to be designed for thruster body/ ice interaction loads. Load formulae given, are for estimating once in a lifetime extreme loads on the thruster body, based on the estimated ice condition and ship operational parameters. Examples of load scenarios are illustrated in Fig. 3.3.5.5.1. In addition, blade order thruster body vibration responses may be estimated for propeller excitation. The following load scenario types are to be considered:

   i) Ice block impact on the thruster body or propeller hub

   ii) Thruster penetration into an ice ridge that has a thick consolidated layer

   iii) Vibratory response of the thruster at blade order frequency

b) The steering mechanism, the fitting of the unit, and the body of the thruster is to be designed to withstand the plastic bending of a blade without damage. The loss of a blade is to be taken into account for the propeller blade orientation causing the maximum load on the component being studied. Top-down blade orientation typically places the maximum bending loads on the thruster body.

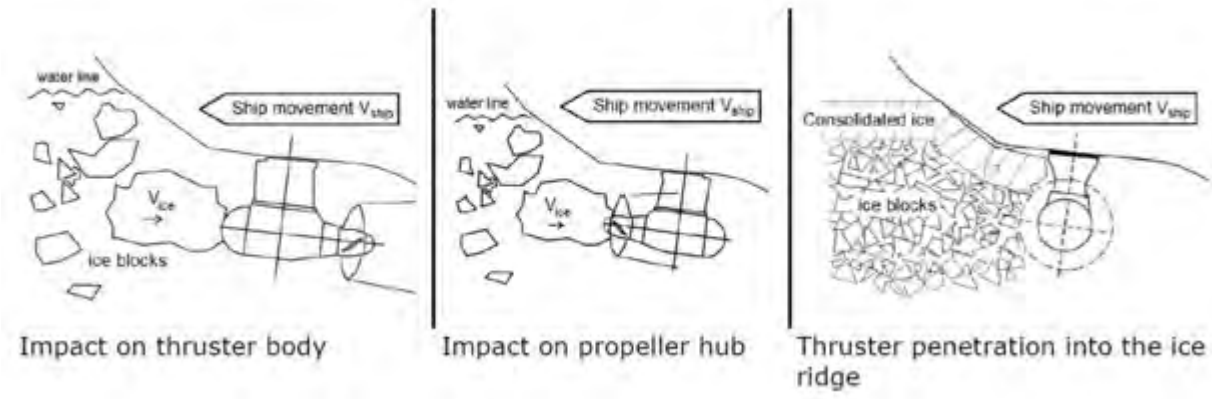


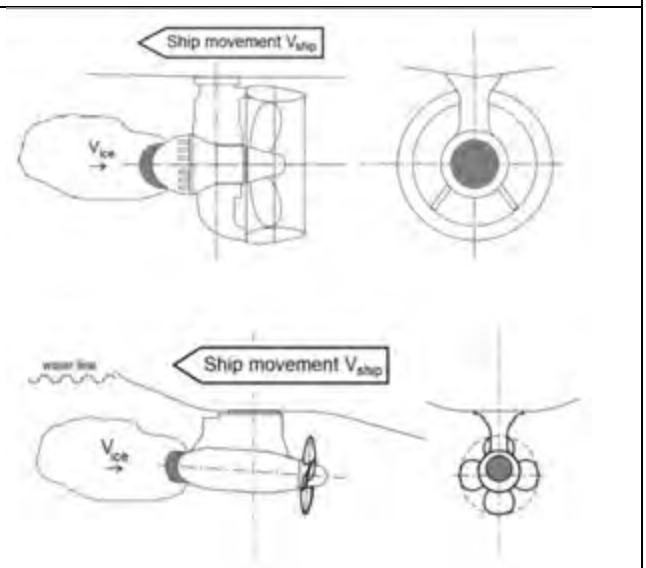
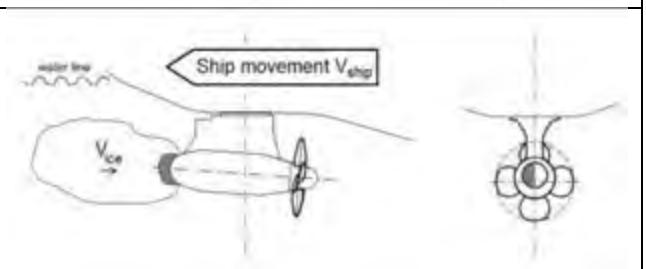
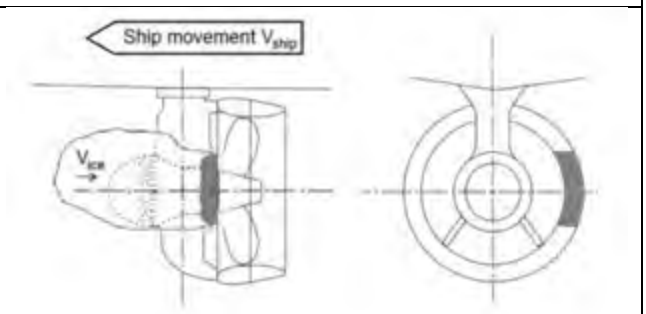
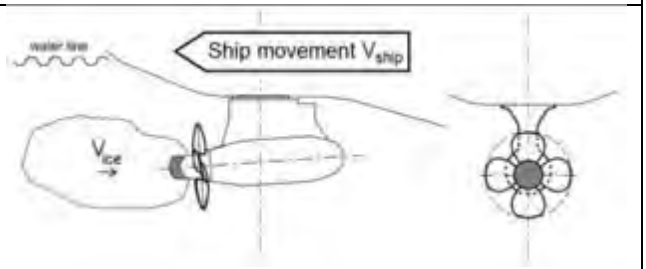
Fig. 3.3.5.5.1 : Examples of load scenario types

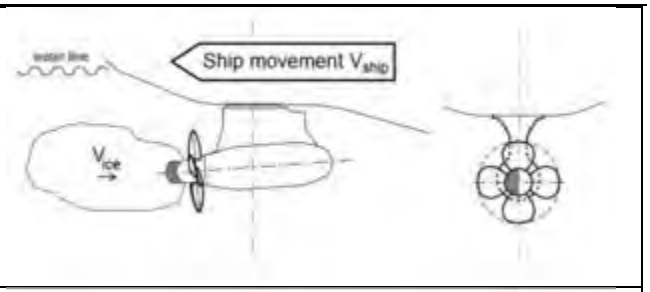
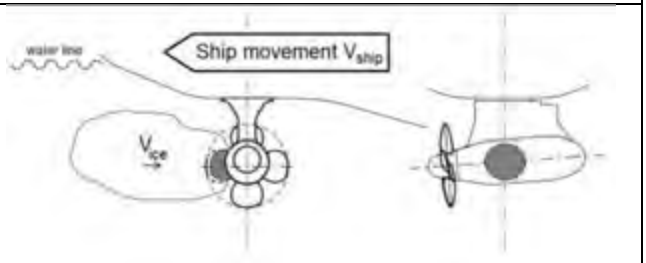
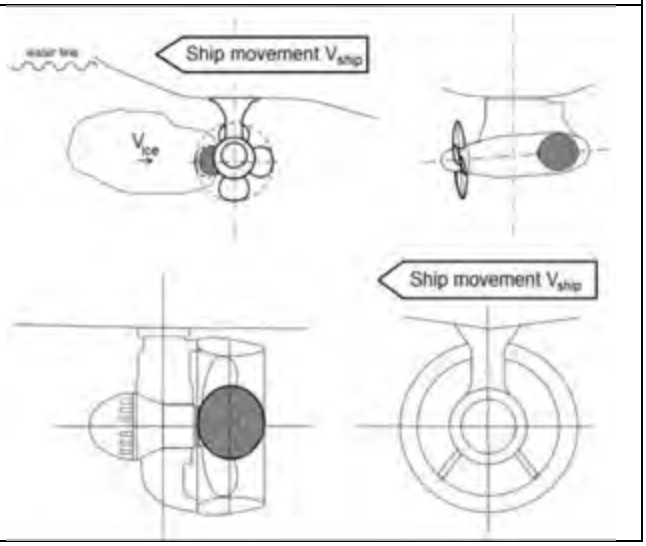
.2 Extreme Ice Impact Loads

a) When the ship is operated in ice conditions, ice blocks formed in channel side walls or from the ridge consolidated layer may impact on the thruster body and the propeller hub. Exposure to ice impact is dependent on the ship size and ship hull design, as well as the location of the thruster. The contact force will grow in terms of thruster/ ice contact until the ice block reaches the ship speed.

b) The thruster is to withstand the loads occurring when the design ice block defined in Table 3.3.2.1 b) impacts on the thruster body when the ship is sailing at a typical ice operating speed. Load cases for impact loads are given in Table 3.3.5.5.1 a). The contact geometry is estimated to be hemi-spherical in shape. If the actual contact geometry differs from the shape of the hemisphere, a sphere radius is to be estimated so that the growth of the contact area as a function of penetration of ice corresponds as closely as possible to the actual geometrical shape penetration.

Table 3.3.5.5.1 a) : Load cases for azimuthing thruster ice impact loads.

	Force	Loaded area	
Load case T1a Symmetric longitudinal ice impact on thruster	F_{ti}	Uniform distributed load or uniform pressure, which are applied symmetrically on the impact area.	
Load case T1b Non-symmetric longitudinal ice impact on thruster	50% of F_{ti}	Uniform distributed load or uniform pressure, which are applied on the other half of the impact area.	
Load case T1c Non-symmetric longitudinal ice impact on nozzle	F_{ti}	Uniform distributed load or uniform pressure, which are applied on the impact area. Contact area is equal to the nozzle thickness (H_{nz})*the contact height (H_{ice}).	
Load case T2a Symmetric longitudinal ice impact on propeller hub	F_{ti}	Uniform distributed load or uniform pressure, which are applied symmetrically on the impact area.	

Load case T2b Non-symmetric longitudinal ice impact on propeller hub	50% of F_{ti}	Uniform distributed load or uniform pressure, which are applied on the other half of the impact area.	
Load case T3a Symmetric lateral ice impact on thruster body	F_{ti}	Uniform distributed load or uniform pressure, which are applied symmetrically on the impact area.	
Load case T3b Non-symmetric lateral ice impact on thruster body or nozzle	F_{ti}	Uniform distributed load or uniform pressure, which are applied on the impact area. Nozzle contact radius *R* to be taken from the nozzle length (L_{nz}).	

c) The ice impact contact load is to be calculated using formula given below. The related parameter values for ice dimensions and dynamic magnification are given in Table 3.3.5.5.1 b). The design operation speed in ice may be derived from Tables 3.3.5.5.1 c) and 3.3.5.5.1 d), or the ship's actual design operation speed in ice can be used. The longitudinal impact speed in Tables 3.3.5.5.1 c) and 3.3.5.5.1 d) refers to the impact in the thruster's main operational direction. For the pulling propeller configuration, the longitudinal impact speed is used for load case T2, impact on hub; and for the pushing propeller unit, the longitudinal impact speed is used for load case T1, impact on thruster end cap. For the opposite direction, the impact speed for transversal impact is applied.

$$F_{ti} = C_{DMI} 34.5 R_c^{0.5} (m_{ice} v_s^2)^{0.333} \text{ [kN]}$$

R_c is the impacting part sphere radius

m_{ice} is the ice block mass [kg]

v_s is the ship speed at the time of contact [m/s]

C_{DMI} is the dynamic magnification factor for impact loads.

C_{DMI} can be taken from Table 3.3.5.5.1 b) if unknown.

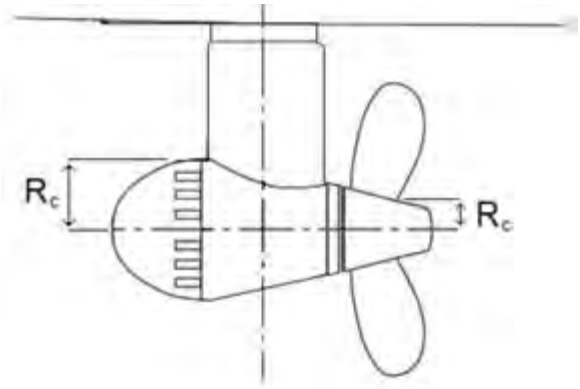


Fig. 3.3.5.5.2 : Dimensions used for Rc

d) For impacts on non-hemispherical areas, such as the impact on the nozzle, the equivalent impact sphere radius is to be estimated using the equation below.

$$R_{ceq} = \sqrt{\frac{A}{\pi}} \ [m]$$

If the 2*R_{ceq} is greater than the ice block thickness, the radius is set to half of the ice block thickness. For the impact on the thruster side, the pod body diameter can be used as a basis for determining the radius. For the impact on the propeller hub, the hub diameter can be used as a basis for the radius.

Table 3.3.5.5.1 b) : Parameter values for ice dimensions and dynamic magnification

	HAT(B)	**HT(B)**	**HM(B)**	**Ha(B)**
Thickness of the design ice block impacting thruster (2/3 of H_{ice})	1.17 m	1.0 m	0.8 m	0.67 m
Extreme ice block mass (m_{ice})	8670 kg	5460 kg	2800 kg	1600 kg
C_{DMI} (if not known)	1.3	1.2	1.1	1

Table 3.3.5.5.1 c) : Impact speeds for aft centerline thruster

	HAT(B)	**HT(B)**	**HM(B)**	**Ha(B)**
Longitudinal impact in main operational direction	6 m/s	5 m/s	5 m/s	5 m/s
Longitudinal impact in reversing direction (pushing unit propeller hub or pulling unit cover end cap impact)	4 m/s	3 m/s	3 m/s	3 m/s
Transverse impact in bow first operation	3 m/s	2 m/s	2 m/s	2 m/s
Transverse impact in stern first operation (double acting ship)	4 m/s	3 m/s	3 m/s	3 m/s

Table 3.3.5.5.1 d) : Impact speeds for aft wing, bow centerline and bow wing thrusters

	HAT(B)	**HT(B)**	**HM(B)**	**Ha(B)**
Longitudinal impact in main operational direction	6 m/s	5 m/s	5 m/s	5 m/s
Longitudinal impact in reversing direction (pushing unit propeller hub or pulling unit cover end cap impact)	4 m/s	3 m/s	3 m/s	3 m/s
Transverse impact	4 m/s	3 m/s	3 m/s	3 m/s

.3 Extreme ice loads on thruster hull when penetrating an ice ridge

a) In icy conditions, ships typically operate in ice channels. When passing other ships, ships may be subject to loads caused by their thrusters penetrating ice channel walls. There is usually a consolidated layer at the ice surface, below which the ice blocks are loose. In addition, the thruster may penetrate ice ridges backing. Such a situation is most likely in the case of **HAT(B)** ships, because they may operate independently in difficult ice conditions. However, the thrusters in ships with lower ice classes may also have to withstand such a situation, but at a remarkably lower ship speed.

b) In this load scenario, the ship is penetrating a ridge in thruster first mode with an initial speed. This situation occurs when a ship with a thruster at the bow moves forward, or a ship with a thruster astern moves in backing mode. The maximum load during such an event is considered the extreme load. An event of this kind typically lasts several seconds, due to which the dynamic magnification is considered negligible and is not taken into account.

c) The load magnitude is to be estimated for the load cases shown in Table 3.3.5.5.2 a), using equation given below. The parameter values for calculations are given in Table 3.3.5.5.2 b) and Table 3.3.5.5.2 c). The loads are to be applied as uniform distributed load or uniform pressure over the thruster surface. The design operation speed in ice may be derived from Table 3.3.5.5.2 b) or Table 3.3.5.5.2 c). Alternatively, the actual design operation speed in ice of the ship may be used.

$$F_{tr} = 32 v_s^{0.66} H_r^{0.9} A_t^{0.74} \ [kN]$$

v_s is ship speed [m/s]

H_r is design ridge thickness (the thickness of the consolidated layer is 18% of the total ridge thickness) [m]
A_t is the projected area of the thruster [m^2]

When calculating the contact area for thruster-ridge interaction, the loaded area in the vertical direction is limited to the ice ridge thickness, as shown in Fig. 3.3.5.5.3**.**

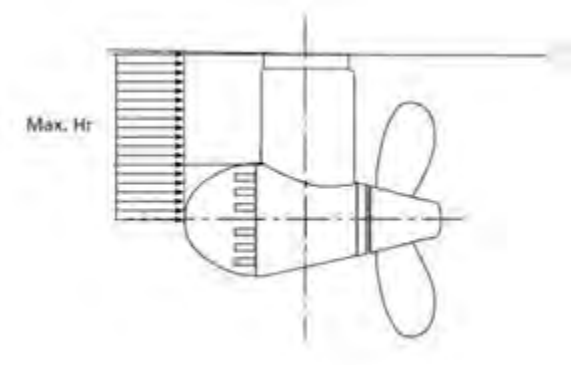
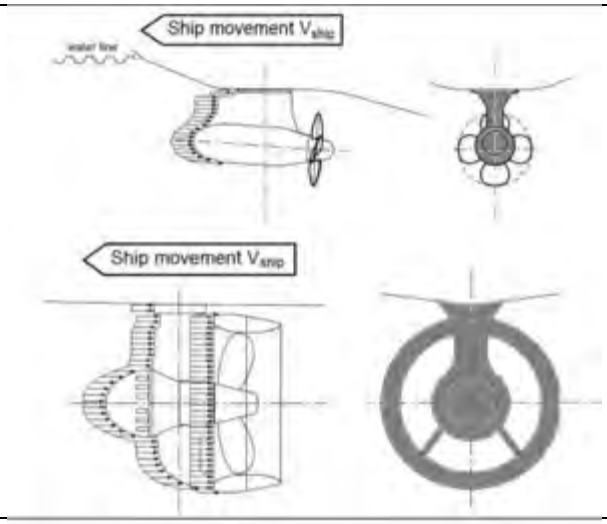
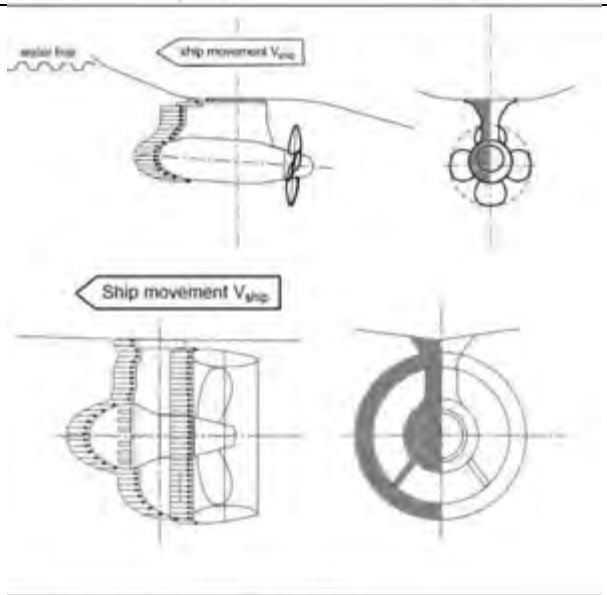
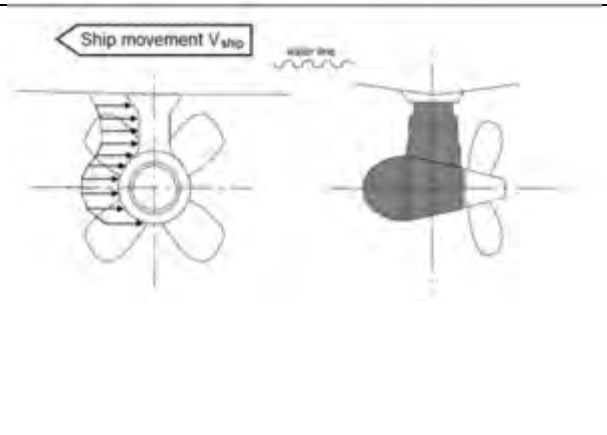


Fig. 3.3.5.5.3 : Schematic figure showing the reduction of the contact area by the maximum ridge thickness

Table 3.3.5.5.2 a) : Load cases for ridge ice loads.			
	Force	Loaded area	
Load case T4a Symmetric longitudinal ridge penetration loads	F_{tr}	Uniform distributed load or uniform pressure, which are applied symmetrically on the impact area.	
Load case T4b Non-Symmetric longitudinal ridge penetration loads	50% of F_{tr}	Uniform distributed load or uniform pressure, which are applied on the other half of the contact area.	
Load case T5a Symmetric lateral ridge penetration loads for ducted azimuthing unit and pushing open propeller unit	F_{tr}	Uniform distributed load or uniform pressure, which are applied symmetrically on the contact area.	

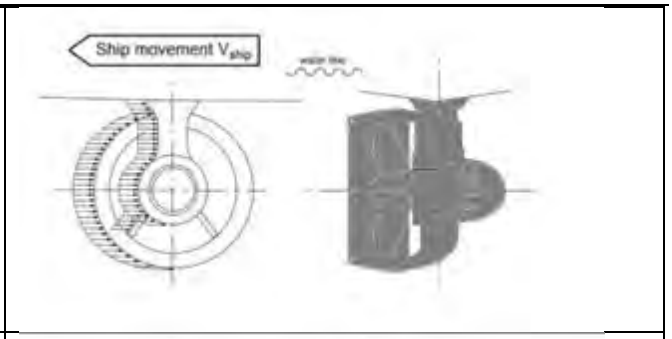
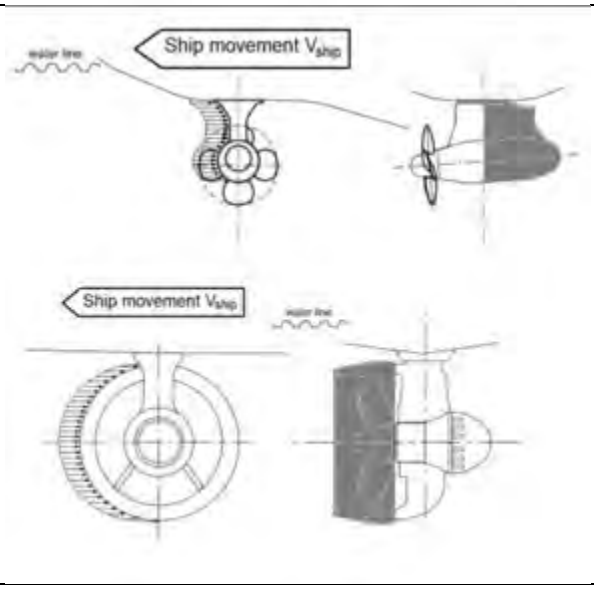
			
Load case T5b Non-symmetric lateral ridge penetration loads for all azimuthing units	50% of F_{tr}	Uniform distributed load or uniform pressure, which are applied on the other half of the contact area.	

Table 3.3.5.5.2 b) : Parameters for calculating maximum loads when the thruster penetrates an ice ridge. Aft thrusters. Bow first operation

	HAT(B)	HT(B)	HM(B)	Ha(B)
Thickness of the design ridge consolidated layer	1.5 m	1.5 m	1.2 m	1.0 m
Total thickness of the design ridge, *Hr*	8 m	8 m	6.5 m	5 m
Initial ridge penetration speed (longitudinal loads)	4 m/s	2 m/s	2 m/s	2 m/s
Initial ridge penetration speed (transversal loads)	2 m/s	1 m/s	1 m/s	1 m/s

Table 3.3.5.5.2 c) : Parameters for calculating maximum loads when the thruster penetrates an ice ridge. Thruster first mode such as double acting ships

	HAT(B)	HT(B)	HM(B)	Ha(B)
Thickness of the design ridge consolidated layer	1.5 m	1.5 m	1.2 m	1.0 m
Total thickness of the design ridge, *Hr*	8 m	8 m	6.5 m	5 m
Initial ridge penetration speed (longitudinal loads)	6 m/s	4 m/s	4 m/s	4 m/s
Initial ridge penetration speed (transversal loads)	3 m/s	2 m/s	2 m/s	2 m/s

.4 Acceptability criterion for static loads

a) The stresses on the thruster are to be calculated for the extreme once-in-a-lifetime loads described in 3.3.5.5. The nominal von Mises stresses on the thruster body are to have a safety margin of 1.3 against the yield strength of the material. At areas of local stress concentrations, stresses are to have a safety margin of 1.0 against yielding. The slewing bearing, bolt connections and other components are to be able to maintain operability without incurring damage that requires repair when subject to the loads given in sections

3.3.5.5.2 and 3.3.5.5.3 multiplied by a safety factor of 1.3.

.5 Thruster body global vibration

a) Evaluating the global vibratory behavior of the thruster body is important, if the first blade order excitations are in the same frequency range with the thruster global modes of vibration, which occur when the propeller rotational speeds are in the high power range of the propulsion line. This evaluation is mandatory and it is to be shown that there is either no global first blade order resonance at high operational propeller speeds (above 50% of maximum power) or that the structure is designed to withstand vibratory loads during resonance above 50% of maximum power.

b) When estimating thruster global natural frequencies in the longitudinal and transverse direction, the damping and added mass due to water are to be taken into account. In addition to this, the effect of ship attachment stiffness is to be modelled.

3.3.6 Alternative design procedure

.1 Scope

a) As an alternative to 3.3.4 and 3.3.5, a comprehensive design study may be performed to the satisfaction of IRS/ the Administration. The study is to be based on the ice conditions given for different ice classes in 3.3.2. It is to include both fatigue and maximum load design calculations and fulfil the pyramid strength principle, as given in 3.3.5.1.

.2 Loading

a) Loads on the propeller blade and propulsion system are to be based on an acceptable estimation of hydrodynamic and ice loads.

.3 Design levels

a) The analysis is to confirm that all components transmitting random (occasional) forces, excluding propeller blade, are not subjected to stress levels in excess of the yield stress of the component material, with a reasonable safety margin.

b) Cumulative fatigue damage calculations are to give a reasonable safety factor. Due account is to be taken of material properties, stress raisers, and fatigue enhancements.

c) A vibration analysis is to be performed and demonstrated that the overall dynamic system is free of the harmful torsional resonances resulting from propeller/ice interaction.

3.4 Miscellaneous Machinery Requirements

3.4.1 For Ice Class HAT(B), HT(B), HM(B) and Ha(B)

3.4.1.1 Starting Arrangement

3.4.1.1.1 The capacity of the air receivers is to be sufficient to provide, without reloading, not less than 12 consecutive starts of the propulsion engine, if this has to be reversed for going astern, or 6 consecutive starts if the propulsion engine does not have to be reversed for going astern.

3.4.1.1.2 If the air receivers serve any other purposes than starting the propulsion engine, they are to have additional capacity sufficient for these purposes.

3.4.1.1.3 The capacity of the air compressors is to be sufficient for charging the air receivers from atmospheric to full pressure in one (1) hour, except for a ship with the ice class HAT(B), if its propulsion engine has to be reversed for going astern, in which case the compressor is to be able to charge the receivers in half an hour.

3.4.1.2 Sea Inlet and Cooling Water Systems

3.4.1.2.1 The cooling water system is to be designed to ensure supply of cooling water when navigating in ice.

3.4.1.2.2 For this purpose at least one cooling water inlet chest is to be arranged as follows:

.1 The sea inlet is to be situated near the centreline of the ship and well aft if possible.

.2 As guidance for design the volume of the chest is to be about one cubic metre for

every 750 [kW] engine output of the ship including the output of auxiliary engines necessary for the ship's service.

.3 The chest is to be sufficiently high to allow ice to accumulate above the inlet pipe.

.4 A pipe for discharge cooling water, allowing full capacity discharge, is to be connected to the chest.

.5 The open area of the strainer plates is not to be less than four (4) times the inlet pipe sectional area.

3.4.1.2.3 If there are difficulties to meet the requirements of 3.4.2.2.2 and 3.4.2.2.3 above, two smaller chests may be arranged for alternating intake and discharge of cooling water. Otherwise the arrangements and other details are to be as above.

3.4.1.2.4 Heating coils may be installed in the upper part of the sea chest.

3.4.1.2.5 Arrangements for using ballast water for cooling purposes may be useful as a reserve in ballast condition but cannot be accepted as a substitute for sea inlet chest as described above.

3.4.2 For Ice Class Ha

3.4.2.1 Ship-side valves

3.4.2.1.1 The sea inlet and overboard discharge valves which are situated at or below the maximum Load Line, are to be provided with low pressure steam or compressed air connection for clearing purposes.

3.4.2.1.2 When steam is not available for clearing, it is recommended that arrangements be made for supplying water for machinery cooling purposes by circulating from ballast tank(s) of adequate capacity, preferably situated in the double bottom. Such tank(s) must be used only for storage of water ballast or fresh water.

3.4.2.2 Cooling water lines

3.4.2.2.1 Connections are to be fitted between the cooling water overboard discharge lines and sea inlets for main and/or auxiliary engine cooling water systems so that warm water may be used to assist in maintaining the suction pipes free from ice.

3.4.2.2.2 Where the cooling water inlet valves are fitted to a common water box, the connections from the cooling water discharge lines may be led to the water box in a position as near as possible to the inlet valves.

3.4.3 Fire pumps in motor ships

3.4.3.1 In motor ships where clearing steam is not available, at least one of the fire pumps are to be provided with suctions from the main cooling water inlet pipe.

End of Chapter

Chapter 22

Vessels with Unattended Machinery Spaces

Contents

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3 *Requirements for Monitoring*

4 *Vessels fitted with Internal Combustion Propulsion Engines of less than 1000 [kW]*

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6 *Testing of the Plant on Board*

Section 1

General

1.1 Scope

1.1.1 The requirements of this Chapter are applicable to machinery spaces intended for unattended operation during normal service at sea and in port.

1.1.2 Cargo handling systems are not covered by the requirements of this Chapter.

1.1.3 The requirements of this Chapter are in addition to the requirements of Pt.4 of the Rules.

1.2 Class notation

1.2.1 Machinery installations complying with the requirements of this Chapter may be given the additional class notation SYJ.

1.3 Assumptions

1.3.1 The requirements of this Chapter are based on the assumptions that :

a) engineering staff can reach the engine room at short notice;

b) systematic maintenance and functional testing of instrumentation are performed and documented; and

c) the test equipment is kept on board. The equipment is to be sufficient for all tests specified in the instructions for systematic maintenance and function testing on board.

1.4 Plans and particulars

1.4.1 In addition to the plans and particulars required by Pt.4, Ch.7, the following plans and particulars are to be submitted for approval :

- location of automatic fire detectors, manual fire alarm controls and fire alarm bells;
- location of fire alarm system central unit, together with location of remote stop controls for fans, pumps, etc.;
- supply of electric power to fire alarm system;
- connection and arrangement of automatic stop system (if fitted) for ventilation fans;
- connection and arrangement of detector sections and warning actuators; and
- location and power supply arrangements of internal communication systems.

1.5 Signboards and instruction manuals

1.5.1 Instruction manuals are to be kept on board. One copy of the manuals is to be submitted for information. The manuals are to contain necessary instructions on :

- operation;
- testing;
- identification of faults;
- repairs;
- emergency operation; and
- systematic maintenance and function testing.

1.5.2 The plan for systematic maintenance and function testing is to show how components and systems are to be tested and what is to be observed during the tests.

1.5.3 On the bridge and the control stand in the engine room, instructions are to be fitted, stating routines to be carried out in connection with transfer to/from unmanned engine room and precautions to be taken at alarm condition.

1.6 Spares

1.6.1 It is advisable to keep on board spare parts for important, replaceable parts/units of the automatic control systems, monitoring systems, remote control systems and programmed control systems.

1.6.2 The quantity of spare parts is, preferably to be as recommended by the manufacturers, and sufficient to cover replacements necessary for at least one year's service. Spare parts should amount to 10 per cent, minimum one of each, of such components as:

- transducers;
- controllers;
- indicating instruments;
- printed circuit cards;
- actuators; and
- relays, fuses, lamps, etc.

1.6.3 Spare parts should preferably consist of complete units. However, solenoids, springs, gaskets and similar parts will be useful as substitute for complete units, if replacement of defective parts within the unit is easily made by the engine staff.

Section 2

System Design

2.1 General

2.1.1 The arrangements provided are to be such as to ensure that the safety of the ship in all conditions, including maneuvering, is equivalent to that of a ship having machinery spaces manned.

2.1.2 The extent of automation is to be sufficient for 24 hours of unattended engine room operation during normal service at sea and normal maneuvers. Normal maneuvers do not include emergency maneuvers, where alarm and safety limits may be exceeded.

2.1.3 Starting of engine plant and transfer to various operating modes may be accepted as manual operations, if the need for such actions will not arise at short notice.

2.2 Control locations

2.2.1 A system of alarm displays and controls is to be provided which readily ensures identification of faults in the machinery and satisfactory supervision of related equipment. This may be provided at a main control location or, alternatively at subsidiary control locations. In the latter case, a master alarm display is to be provided at the main control location showing which of the subsidiary control locations is indicating a fault condition.

2.2.2 Provision is to be made at the main control location, or subsidiary control locations as appropriate, for the operation of engineer's alarm which is to be clearly audible in the engineer's accommodation.

2.2.3 Provision is to be made at the main control location and any other subsidiary control location from which the main propulsion and auxiliary or associated equipment may be controlled to indicate which location is in control.

2.2.4 Remote control of propulsion machinery and associated equipment is to be possible from only one location at a time; at such locations interconnected control positions are permitted. The transfer of control between the navigating bridge and machinery spaces is to be possible only in the main machinery space or in the main machinery control room. Changeover between control locations is to be arranged so that it may be effected with the acceptance of the location taking control. The system is to be provided with interlocks or other suitable means to ensure effective transfer of control. The system is to include means to prevent the propelling thrust from altering significantly when transferring control from one location to another.

2.2.5 Under all sailing conditions, including maneuvering, the speed, direction of thrust and, if applicable, the pitch of the propeller are to be fully controllable from the bridge. Such remote control is to be performed by a single control device for each independent propeller, with automatic performance of all associated services, including, where necessary, means of preventing overload and prolonged running in critical speed ranges of propelling machinery.

2.2.6 Propulsion machinery orders from the navigating bridge are to be indicated in the main machinery control room or at the propulsion machinery control position as appropriate.

2.2.7 It is to be possible for all machinery essential for the safe operation of the ship to be controlled from a local position, even in case of failure in any part of the automatic or remote control systems.

2.2.8 The bridge control system is to be independent from the other transmission system; however, one control lever for both system may be accepted.

2.2.9 Operations following any setting of the bridge control device including reversing from the maximum ahead service speed in case of emergency are to take place in an automatic sequence and with time intervals acceptable to the machinery.

2.2.10 Instrumentation to indicate at least the following is to be fitted on the bridge :

a) Propeller speed;

b) Direction of rotation of propeller for a fixed propeller or pitch position for controllable pitch propeller;

c) Clutch position, where applicable; and

d) Shaft brake position, where applicable.

2.2.11 Means of control, independent of bridge control system, are to be provided on bridge to enable the watchkeeper to stop the propulsion machinery in an emergency.

2.2.12 Remote starting of the propulsion machinery is to be automatically inhibited if conditions exist which may hazard the machinery, e.g. shaft turning gear engaged, drop of lubricating oil pressure.

2.2.13 For steam turbines a slow-turning device should be provided which operates automatically if the turbine is stopped longer than admissible. Discontinuation of this automatic turning from the bridge must be possible.

2.2.14 Audible and visual alarms are to operate on the bridge, if the power supply to the bridge control system or the alarm system fails. Where practical, the preset speed and direction of thrust are to be automatically maintained until corrective action is taken.

2.3 Control systems

2.3.1 Automatic control systems are to keep all parameters within the limits for safe operation under all load conditions.

2.3.2 Expansion/compensating tanks are normally not to be automatically filled. If automatic filling of expansion/compensating tanks is arranged, an alarm is to be activated before such filling is started or at abnormal long or frequent operation.

2.3.3 Protection against flooding

1. Bilge wells in periodically unattended machinery spaces are to be located and monitored in such a way that the

accumulation of liquids is detected at normal angles of trim and heel, and are to be large enough to accommodate easily the normal drainage during unattended period.

2. Where the bilge pumps are capable of being started automatically, means are to be provided to indicate when the influx of liquid is greater than the pump capacity or when the pump is operating more frequently than would normally be expected. In these cases, smaller bilge wells to cover a reasonable period of time may be permitted. Where automatically controlled bilge pumps are provided, special attention is to be given to oil pollution prevention requirements.

3. The location of the controls of any valve serving a sea inlet, a discharge below the waterline or a bilge injection system is to be so sited as to allow adequate time for operation in case of influx of water to the space, having regard to the time likely to be required to reach and operate such controls. If the level to which the space could become flooded with the ship in the fully loaded condition so requires, arrangements are to be provided to operate the controls from a position above such level.

IR 2.3.3.3.1 'Bilge injection system' referred above is same as 'Emergency bilge suction' referred in Pt. 4, Ch. 3, Sec 2.7.4 which is used to discharge overboard large quantities of sea water accumulated in engine room bilges using the main circulating pump or another suitable pump as permitted by Pt. 4, Ch. 3, Sec 2.7.4.

IR 2.3.3.3.2 The requirements for the controls as referred in 2.3.3.3 above are not applicable to valves serving an emergency bilge system provided:

(1) The emergency bilge valve is normally maintained in a closed position,

(2) A non-return device is installed in the emergency bilge piping, and

(Note: A normally closed non-return valve with positive means of closing is considered to satisfy both (1) and (2) above.)

(3) The emergency bilge suction piping is located inboard of a shell valve (overboard valve) that is fitted with the control arrangements required by 2.3.3.3.

2.3.4 Starting air receivers are to be automatically filled, if the air consumption during normal operation at sea, not including maneuvering, can reduce the air pressure by 20 per cent or more in the course of 24 hours.

2.3.5 Control systems should be designed to "fail safe". The characteristics of the "fail safe" operation are to be evaluated on the basis not only of the control system and its associated machinery, but also the complete installation.

2.3.6 The arrangements are to be such that machinery may be operated with the automatic controls out of action. This may be achieved by manual control or redundancy arrangements within the control system. Instrumentation is to be provided at the local manual control locations to ensure effective operation of the machinery.

2.3.7 The design of the remote automatic control system is to be such that in the case of its failure an alarm will be given. Unless IRS considers it impracticable, the preset speed and direction of thrust of the propeller is to be maintained until local control is in operation.

2.3.8 The number of consecutive automatic attempts which fail to produce a start are to be limited to safeguard sufficient starting air pressure. An alarm is to be provided to indicate low starting air pressure set at a level which still permits starting operations of the propulsion machinery.

2.4 Alarm systems

2.4.1 An alarm system is to be provided indicating any fault requiring attention and is to:

a) be capable of sounding an audible alarm in the main machinery control room or at the propulsion machinery control location, and indicate visually each separate alarm function at a suitable position;

b) have a connection to the public rooms and to each of the engineers' cabins through a selector switch, to ensure connection to at least one of those cabins. Equivalent arrangements will be specially considered;

c) activate an audible and visual alarm on the navigating bridge for any situation which requires action by or attention of officer on watch;

d) as far as is practicable be designed on the fail-to-safety principle; and

e) activate an alarm in the engineers accommodation if an alarm function has not received attention locally within a limited time.

f) satisfy the environmental requirements of IRS Classification Notes “Type Approval of Electrical Equipment used for Control, Protection, Safety and Internal Communication in Marine Environment”.

2.4.2 The alarm system is to be continuously powered and is to have an automatic change-over to a stand-by power supply in case of loss of normal power supply.

2.4.3 Failure of the normal power supply of the alarm system is to be indicated by an alarm.

2.4.4 The alarm system is to be able to indicate at the same time more than one fault and the acceptance of any alarm is not to inhibit another alarm.

2.4.5 Acceptance at any of the positions referred to in 2.4.1(b) of any alarm condition is to be indicated at the positions where it was shown. Alarms are to be maintained until they are accepted and the visual indicators of individual alarms are to remain until the fault has been corrected, when the alarm system is to automatically reset to the normal operating condition.

2.4.6 Alarms associated with machinery, safety and control system faults are to be clearly distinguishable from other alarms (e.g. Fire, General alarm).

2.4.7 Where alarms are displayed as group alarms, provision is to be made to identify individual alarms at the main control location (if fitted) or alternatively at subsidiary control locations.

2.4.8 The alarm system should be designed with self-monitoring properties. Insofar as practical, any fault in the alarm system should cause it to fail to the alarm condition.

2.4.9 The alarm system is to be capable of being tested during normal machinery operation.

2.4.10 The alarm system is to be designed as far as practical to function independently of control and safety systems such that a failure or malfunction in these systems will not prevent the alarm from operating.

2.4.11 Disconnection or manual overriding of any part of the alarm system should be clearly indicated as long as the disconnection/manual override is in use.

2.4.12 Any alarm condition in the engine room is to activate an alarm in the watch-keeping engineer officer's cabin and dayroom. Acknowledgement in the cabin is to be indicated on the bridge when the engine room is unattended.

2.4.13 Audible and visual indications of machinery space alarms are to be relayed to the navigating bridge control location in such a way that the navigating officer of the watch is made aware when :

a) a machinery fault has occurred;

b) the machinery fault is being attended to; and

c) the machinery fault has been rectified.

2.4.14 Group alarms may be arranged in the bridge to indicate machinery faults, but alarms associated with faults requiring speed or power reduction or the automatic shut down of propulsion machinery are to be separately identified.

2.5 Safety systems

2.5.1 A safety system is to be provided to ensure that a serious malfunction in machinery or boiler operations, which presents an immediate danger, is to initiate the automatic shutdown of that part of the plant and that an alarm is to be given. Shutdown of the propulsion system is not to be automatically activated except in cases which could lead to serious damage, complete break down, or explosion. Where arrangements for overriding the shutdown of the main propelling machinery are fitted, these are to be such as to preclude inadvertent operation. Visual means are to be provided to indicate when the override has been activated.

2.5.2 The safety system is to be designed as far as practical to operate independently of the control and alarm system, such that a failure or malfunction in these systems will not prevent the safety system from operating.

2.5.3 Safety systems are to operate automatically in case of serious faults endangering the machinery, so that :

a) normal operating conditions are restored, e.g. by the starting of standby machinery; or

b) the operation of the machinery is temporarily adjusted to the prevailing conditions, e.g. by reducing the output of the machinery; or

c) the machinery is protected from critical conditions by shutting off the fuel or power supplies thereby stopping the machinery.

2.5.4 Safety systems for different items of the machinery plant are to be arranged so that failure of the safety system of one part of the plant is not to interfere with the operation of the safety system in another part of the plant.

2.5.5 The safety system should be designed to "fail safe". The characteristics of the "fail safe" operation are to be evaluated on the basis not only of the safety system and it's associated machinery, but also the complete installation.

2.5.6 When a safety system is activated, an audible and visual alarm is to be provided to indicate the cause of the safety action.

2.5.7 The safety system is to be provided with a manual reset.

2.5.8 The safety system is to be arranged with automatic changeover to a standby power supply in the event of a failure of the normal supply. The failure of the power supply to a safety system is to operate an audible and visual alarm.

2.6 Automatic start of pumps

2.6.1 Faults in the mechanical or electrical system of the running pump are not to inhibit automatic start of the standby pump.

2.6.2 Manual start and stop of the pumps are to be possible without activating the alarm for automatic start of the pump.

2.6.3 When a pump is standby, this is to be clearly indicated on the control panel by indicating lamps, etc.

2.7 Automatic start/connection of diesel generator units and essential auxiliaries for propulsion

2.7.1 Where the electrical power can normally be supplied by one generator, suitable load-shedding arrangements are to be provided to ensure the integrity of supplies to services required for propulsion and steering as well as the safety of the ship. In the case of loss of the generator in operation, adequate provision is to be made for automatic starting and connecting to the main switchboard of a stand-by generator of sufficient capacity to permit propulsion and steering and to ensure the safety of the ship with automatic restarting of the essential auxiliaries including, where necessary, sequential operations. IRS may dispense with this requirement for a ship of less than 1,600 tons gross tonnage, if it is considered impracticable.

2.7.2 If the electrical power is normally supplied by more than one generator simultaneously in parallel operation, provision is to be made, for instance by load shedding, to ensure that, in case of loss of one of these generating sets, the remaining ones are kept in operation without overload to permit propulsion and steering, and to ensure the safety of the ship.

2.7.3 Where stand-by machinery are required for other auxiliary machinery essential to propulsion, automatic change over devices are to be provided.

Above is applicable to stand-by machinery are required by the Rules for operation of:

a) oil engines for propulsion purposes,
b) steam turbines for propulsion purposes,
c) gas turbines for propulsion purposes,
d) controllable pitch propellers.

2.7.4 When several generator units have individual systems for automatic start and connection on to the main switchboard, each system is to be selectively fused (separate short circuit protection), and alarm is to be activated for voltage failure.

2.7.5 Automatic start attempts are to be limited in time to restrict consumption of starting energy.

2.7.6 Tachometer feedback to the starting system is to be arranged so that broken mechanical pinions or external electrical connections do not lead to stop of a running generator unit. Neither are such faults to inhibit automatic stop or alarm functions.

2.7.7 Manual start and stop of generator are not to activate alarm.

2.7.8 The generator circuit breaker is to be provided with automatic wind up of the closing spring of the breaker.

2.7.9 Simultaneous connection of generators on to the main switchboard is not to be possible.

2.7.10 If there is no arrangement for automatic synchronization, paralleling and load sharing, connection is only to be possible when auxiliary contacts on the generator circuit breakers show directly that all generators are disconnected from the main switchboard.

2.7.11 When a generator unit is standby, this is to be indicated on the control panel.

2.7.12 The voltage of a generator being connected on to a de-energized switchboard is to be sufficiently high for safe operation of all connected circuits.

2.7.13 No more than one attempt of automatic connection on to a de-energized switchboard is permitted.

2.7.14 Systems with automatic synchronization, connection, and load sharing are to be designed so that deviations in voltage, frequency, and phase at the instant of connection are within adequate safety limits for generator and engine.

2.7.15 Means for manual synchronization and paralleling are to be provided.

2.7.16 When automatic start of the standby unit is caused by reduction of voltage or frequency below the criteria for automatic connection, the running unit is to be stopped when standby unit is ready for connection.

2.7.17 Systems with automatic start of the standby unit at heavy load on running units are to be arranged with adequate delay to prevent false start attempts, e.g. caused by short load peaks.

2.7.18 The battery supplying the diesel engine start system is to have at least twice the capacity required for maximum load for half an hour for all consumers connected to the system.

2.7.19 Batteries are to be continuously charged (trickle charge) from a charge rectifier having sufficient capacity to supply the sum of the trickle charge current, the current according to the fuse rating of the largest consumer, and the normal load current for all other consumers. A lower charge rectifier capacity may be approved, provided the distribution system is monitored for ground faults. The capacity is not to be less than the sum of the trickle charge current plus average normal load during 24 hours for all consumers with a 10 per cent allowance in addition.

2.8 Automatic stop of auxiliary engines

2.8.1 The system for automatic stop and alarm is to be selectively fused (separate short circuit protection). Similarly, automatic stop circuits for individual units are to be selectively fused. An alarm is to be activated for voltage failure.

2.8.2 The safety system is to be arranged so that a single open circuit in wiring between sensors and control unit, or between control unit and actuators, does not cause unintentional stop.

2.9 Automatic stop of propulsion machinery

2.9.1 Propulsion machinery is defined as all machinery which will cause loss of propulsion function if stopped, with the exception of main boilers.

2.9.2 The safety system is to be arranged so that a single open circuit in the wiring between sensors and control unit, or between control unit and actuators, does not cause unintentional stop.

2.9.3 Voltage failure behind the last fuse in the system is to activate an alarm.

2.9.4 Separate automatic stop system, whenever practicable, is to be arranged for each unit in multi-unit propulsion plants. The requirements of 2.9.2 could be waived if the maneuvering ability is maintained after shut-down of one propulsion unit.

2.9.5 All parameters which may cause automatic stop are normally to activate an alarm prior to causing a shut- down.

2.10 Automatic stop of oil fired auxiliary boilers

2.10.1 Connections between sensors and control unit are to be based upon normally closed contacts, so that an open circuit will lead to shut-off of the oil supply.

2.10.2 The parameter causing an automatic stop is to be identified on the control panel.

2.11 Re-establishment of propulsion after blackout

2.11.1 Automatic restart after a blackout is to be arranged for all components which are necessary for re- establishment of propulsion from the bridge.

2.11.2 In multi-unit installations, where stopping of one unit will not cause loss of propulsion, the requirements of 2.11.1 need not be complied with.

2.11.3 If necessary, restart is to be arranged in a sequence ensuring that the power system or other systems are not overloaded, and that most important components are started first.

2.12 Communications

2.12.1 A reliable means of vocal communication is to be provided between the main machinery control room or the propulsion machinery control position as appropriate, the navigating bridge and the engineer officers' accommodation.

2.12.2 It is to be possible at any time (including during blackout) to be able to communicate with the engineer officers, from the bridge and from the engine room.

2.12.3 For ships over 2000 gt., at least 4 simultaneous voice connections are to be possible. Alternatively the connection between the bridge and the engine rooms is to have priority above other connections.

2.13 Fire detection and alarm systems

2.13.1 A fire detector indicator panel is to be located in the navigating bridge area, or in such a position that a fire in the machinery spaces will not render it inoperative. The panel is to indicate the source of the fire in accordance with arranged fire zones by means of a visual signal.

2.13.2 An audible fire alarm is to be provided having a characteristic tone which distinguishes it from any other alarm system required by the Rules. The audible fire alarm is to be audible on all parts of the navigating bridge, the fire control location, the accommodation areas and the machinery space.

2.13.3 Facilities are to be provided in the detecting system to manually initiate the fire alarm from the following locations :

a) positions adjacent to entrances to engine and boiler rooms;

b) navigating bridge;

c) control location in engine room; and

d) fire control location.

2.13.4 The alarm system is to be designed with self monitoring properties. Power or system failures are to initiate an audible alarm distinguishable from the fire alarm. This alarm may be incorporated in the machinery alarm systems.

2.13.5 Detector heads and manual call points are to be of a type approved by IRS.

2.13.6 Detector heads are to be located in the machinery space so that all potential fire outbreak points are guarded. A combination of detector types is recommended in order that the system will react to all possible fire characteristics.

2.13.7 When fire detectors are provided with means to adjust their sensitivity, the arrangements are to be such that the set point can be fixed and readily identified.

2.13.8 When it is intended that a particular section or detector is to be temporarily switched off, this state is to be clearly indicated. Reactivation of the section or detector is to be performed automatically after a preset time.

2.13.9 The fire detector heads are to be of a type which can be tested. Facilities are to be provided on the fire control panel for functional testing of the system.

2.13.10 It is to be demonstrated to the Surveyor's satisfaction that detector heads are so located that air currents will not render the system ineffective at sea and in port.

2.13.11 Also refer to Pt.6, Ch.3, Sec.1.4 for other requirements.

Section 3

Requirements for Monitoring

3.1 Extent of monitoring

3.1.1 The monitoring system is to cover machinery and equipment having functions necessary for the safety and maneuverability of the ship.

3.1.2 When alarm and automatic actions (safety actions) are required for one failure condition, alarm is to be activated first.

3.1.3 The extent of monitoring required for internal combustion engines used for propulsion is given in Table 3.1.1. and 3.1.2. Requirements for steam turbine plants used for propulsion are given in Table 3.1.3.

3.1.4 In case of multi-engine propulsion plants, automatic start of standby pump is not required in auxiliary systems which are separate for each engine or group of engines.

3.1.5 The extent of monitoring required for shafting and propeller plant is given in Table 3.1.4.

3.1.6 The extent of monitoring required for electrical power generating plant is given in Table 3.1.5.

3.1.7 The extent of monitoring required for steam generating plant is given in Table 3.1.6.

3.1.8 The extent of monitoring required for tanks is given in Table 3.1.7.

3.1.9 The extent of monitoring required for miscellaneous items is given in Table 3.1.8.

Table 3.1.1 : Cross-Head Diesel Engines					
	Gr 1 (Common sensor)			Gr 2	Gr 3
Monitored parameters for cross-head diesel engines	**Remote Indication**	**Alarm activation**	**Slow down with alarm**	**Automatic start of standby pump with alarm**	**Shut down with alarm**
1.0 Fuel oil system					
Fuel oil pressure after filter (engine inlet)	x	low		x	
Fuel oil viscosity before injection pumps or Fuel oil temp before injection pumps		high low			
Leakage from high pressure pipes		x			
Level of fuel oil in daily service tank[1]		low			
Common rail fuel oil pressure		low			
2.0 Lubricating oil system					
Lub oil to main bearing and thrust bearing, pressure	x	low	x	x	x
Lub oil to crosshead bearing pressure[2]	x	low	x	x	x
Lub oil to camshaft pressure[2]		low		x	x
Lub oil to camshaft temp[2]		high			
Lub oil inlet temp		high			
Thrust bearing pads temp or bearing outlet temp		high	x		x
Activation of oil mist detection arrangements (or activation of the temperature monitoring systems or equivalent devices of: - the engine main, crank and crosshead bearing oil outlet; or - the engine main, crank and crosshead bearing) [3]		x	x		
Main, crank, crosshead bearing, oil outlet temp or Oil mist concentration in crankcase[3]		high	x		
Flow rate cylinder lubricator. Each apparatus		low	x		
Level in lubricating oil tanks[4]		low			
Common rail servo oil pressure		low			
3.0 Turbocharger system					
Turbocharger lub oil inlet pressure[9]		low			
Turbocharger lub oil outlet temp each bearing[10]		high			
Speed of turbocharger[11]	x	high			

Table 3.1.1 : (Contd.)					
	Gr 1 (Common sensor)			Gr 2	Gr 3
Monitored parameters for cross-head diesel engines	**Remote Indication**	**Alarm activation**	**Slow down with alarm**	**Automatic start of standby pump with alarm**	**Shut down with alarm**
4.0 Piston cooling system					
Piston coolant inlet pressure[5]		low	x	x	
Piston coolant outlet temp each cylinder		high	x		
Piston coolant outlet flow each cylinder[8]		low	x		
Level of piston coolant in expansion tank		low			
5.0 Sea water cooling system					
Sea water pressure		low		x	
6.0 Cylinder fresh cooling water system					
Cylinder water inlet pressure		low	x	x	
Cylinder water outlet temp (from each cylinder) or Cylinder water outlet temp (general)[6]		high	x		
Oily contamination of engine cooling water system[7]		x			
Level of cylinder cooling water in expansion tank		low			
7.0 Starting and control air systems					
Starting air pressure before main shut-off valve	x	low			
Control air pressure		low			
Safety air pressure		low			
8.0 Scavenge air system					
Scavenge air receiver pressure	x				
Scavenge air box temp (fire)		high	x		
Scavenge air receiver water level		high			
9.0 Exhaust gas system					
Exhaust gas temp after each cylinder	x	high	x		
Exhaust gas temp after each cylinder. Deviation from average.		high			
Exhaust gas temp before each T/C	x	high			
Exhaust gas temp after each T/C	x	high			
10.0 Fuel valve coolant					
Pressure of fuel valve coolant		low		x	
Temperature of fuel valve coolant		high			
Level of fuel valve coolant in expansion tank		low			

Table 3.1.1 : (Contd.)						
		Gr 1 (Common sensor)			Gr 2	Gr 3
Monitored parameters for cross-head diesel engines		**Remote Indication**	**Alarm activation**	**Slow down with alarm**	**Automatic start of standby pump with alarm**	**Shut down with alarm**
11.0	**Engine speed/direction of rotation.** **Wrong way**	x	x			
12.0	**Engine overspeed**					x
13.0	**Control-Safety-Alarm system power supply failure**		x			

1 High-level alarm is also required if no suitable overflow arrangement is provided.

2 If separate lub oil systems are installed.

3 Slow down with Alarm is to be provided for Engines of 2250 [KW] and above or having cylinders of more than 300 [mm] bore.

4 Where separate lubricating oil systems are installed (e.g. camshaft, rocker arms, etc.), individual level alarms are required for the tanks.

5 The slow down is not required if the coolant is oil taken from the main cooling system of the engine.

6 Where one common cooling space without individual stop valves is employed for all cylinder jackets.

7 Where main engine cooling water is used in fuel and lubricating oil heat exchangers.

8 Where outlet flow cannot be monitored due to engine design, alternative arrangement may be accepted.

9 Unless provided with a self-contained lubricating oil system integrated with the turbocharger.

10 Where outlet temperature from each bearing cannot be monitored due to the engine / turbocharger design alternative arrangements may be accepted. Continuous monitoring of inlet pressure and inlet temperature in combination with specific intervals for bearing inspection in accordance with the turbocharger manufacturer's instructions may be accepted as an alternative.

11 Only required for turbochargers of categories B and C (see IRS Classification Note "*Approval of IC Engines*")

Table 3.1.2 : Trunk-Piston Engines					
	Gr 1 (Common Sensor)			Gr 2	Gr 3
Monitored parameters for trunk-piston diesel engines	**Remote Indication**	**Alarm activation**	**Slow down with alarm**	**Automatic start of standby pump with alarm**	**Shut down With alarm**
1.0 Fuel oil system					
Fuel oil pressure after filter (engine inlet)	x	low		x	
Fuel oil viscosity before injection pumps or Fuel oil temp before injection pumps[1]		high low			
Leakage from high pressure pipes		x			
Level of fuel oil in daily service tank[2]		low			
Common rail fuel oil pressure		low			
2.0 Lubrication oil system					
Lub oil to main bearing and thrust bearing, pressure	x	low		x	x
Lub oil filter differential pressure	x	high			
Lub oil inlet temp	x	high			
Activation of oil mist detection arrangements (or activation of the temperature monitoring systems or equivalent devices of: - the engine main and crank bearing oil outlet; or - the engine main and crank bearing)[3]		x			x
Flow rate cylinder lubricator. Each apparatus		low	x		
Common rail servo oil pressure		low			
3.0 Turbocharger system					
Turbocharger lub oil inlet pressure[5]	x	low			
Turbocharger lub oil temperature each bearing[8]		high			
Speed of turbocharger[9]	X	high			
4.0 Sea Water cooling system					
Sea Water pressure	x	low		x	
5.0 Cylinder fresh cooling water system					
Cylinder water inlet pressure or flow	x	low	x	x	
Cylinder water outlet temp (general)[6]	x	high	x		
Level of cylinder cooling water in expansion tank		low			

Table 3.1.2 : (Contd.)					
	Gr 1 (Common Sensor)			Gr 2	Gr 3
Monitored parameters for trunk-piston diesel engines	**Remote Indication**	**Alarm activation**	**Slow down with alarm**	**Automatic start of standby pump with alarm**	**Shut down With alarm**
6.0 Starting and control air systems					
Starting air pressure before main shut-off valve	x	low			
Control air pressure	x	low			
7.0 Scavenge air system					
Scavenge air receiver temp		high			
8.0 Exhaust Gas system					
Exhaust gas temp after each cylinder[7]	x	high	x		
Exhaust gas temp after each cylinder. Deviation from average[7]		high			
9.0 Engine speed	x				
10.0 Engine overspeed					x
11.0 Control-Safety-Alarm system power supply failure		x			

1 For heavy fuel oil burning engines only.

2 High-level alarm is also required if no suitable overflow arrangement is provided.

3 Alarm and automatic shut off is to be provided for Engines of 2250 [KW] and above or having cylinders of more than 300 [mm] bore. For each engine, one oil mist detector (or engine bearing temperature monitoring system or equivalent device) having two independent outputs for initiating the alarm and shut-down would satisfy the requirement for independence between alarm and shut-down systems.

4 If necessary for the safe operation of the engine.

5 Unless provided with a self-contained lubricating oil system integrated with the turbocharger.

6 Two separate sensors are required for alarm and slow down.

7 For engine power > 500 kW/cyl.

8 Where outlet temperature from each bearing cannot be monitored due to the engine/ turbocharger design alternative arrangements may be accepted. Continuous monitoring of inlet pressure and inlet temperature in combination with specific intervals for bearing inspection in accordance with the turbocharger manufacturer's instructions may be accepted as an alternative.

9 Only required for turbochargers of categories B and C (see IRS Classification Note "*Approval of IC Engines*")

Table 3.1.3 : Steam Turbine Plants				
System		**Required monitoring (stated by an x)**		**Comments**
		Alarm	**Automatic start of standby pump with alarm**	
Lubricating oil inlet	Temperature, high	x		
	Pressure, low	x		Independent of safety system
	Pressure, low		x	
	Pressure, low	x		Automatic shut down
	Filter differential pressure, high	x		
Bearings, turbine and gear	Temperature, high			
Lubricating oil	Level, low	x		In gravity tank
Gland steam	Pressure, high	x		
	Pressure, low	x		
	Exhaust fan, stopped	x		
Hydraulic system	Pressure, low			If separate system
	Pressure, low		x	If failure results in loss of maneuverability
Turbine slow-turning arrangement	Failure	x		
Rotor	Axial displacement, high	x		Turbine shut down
Vibration	Level, high	x		
Condensate system	Level, high	x		
	Level, high		x	
	Level, high	x		Turbine shut down
	Level, low	x		If non-cavitating condensate pumps
	Vacuum, low	x		
	Vacuum pump, stopped		x	
	Vacuum, low	x		Turbine shut down
	Salinity, high	x		
Cooling water	Flow, low		x	

Table 3.1.4 : Monitoring of Shafting and Propeller Plant				
System		**Required monitoring (stated by an x)**		**Comments**
		Alarm	**Automatic start of standby pump with alarm**	
Shafting	Temperature, thrust bearing, high	x		Automatic or manual reduction, see 3.1.14. Two sensors for diesel engine built in thrust bearing and for steam turbine plants
	Temperature, shaft bearing, high	x		Shaft power ≥ 1500 [kW]. Sensor to be located near the bearing surface
	Temperature, stern tube bearing, high	x		Shaft power ≥ 15000 [kW]. Sensor to be located near the bearing surface at the area of highest load
	Level, stern tube lubricating oil gravity tank, low	x		
Controllable pitch propeller	Pressure, servo oil, low	x		
	Pressure, servo oil, low		x	
	Temperature, servo oil, high	x		Shaft power ≥ 1500 [kW]
Gear	Pressure, lubricating oil inlet, low	x		
	Pressure, lubricating oil inlet, low		x	
	Pressure, lubricating oil inlet, low			Automatic or manual stop, see 3.1.11
	Temperature, lubricating oil, high	x		
Hydraulic coupling	Pressure, oil, low	x		If separate system
	Pressure, oil, low		x	If failure results in loss of maneuverability

Table 3.1.5 : Monitoring of Electric Power Generating Plant				
System		**Required monitoring (stated by an x)**		**Comments**
		Alarm	**Automatic shut down of prime mover with alarm**	
Generators driven by trunk piston IC engines	Lubricating oil pressure, low	x	x	
	Lubricating oil temperature, high	x		
	Fuel oil leakage from high pressure pipes	x		
	Activation of oil mist detection arrangements (or activation of the temperature monitoring systems or equivalent devices of: - the engine main and crank bearing oil outlet; or - the engine main and crank bearing)[1]	x	x	
	Pressure or flow of cooling water, low	x		
	Temperature of cooling water or cooling air, high	x		
	Level in cooling water expansion tank, low	x		If not connected to main system
	Level in fuel oil daily service tank, low	x		
	Starting air pressure, low	x		
	Over speed activated		x	
	Fuel oil viscosity before injection pumps or fuel oil temp before injection pumps, low/ high	x		For heavy fuel oil burning engines only
	Exhaust gas temperature after each cylinder, high	x		For engine power above 500 kW/cyl
	Common rail servo oil pressure, low	x		
	Common rail fuel oil pressure, low	x		
	Speed of turbocharger, high	x		Only required for turbochargers of categories B and C (see IRS Classification Note "*Approval of IC Engines*")
	Voltage, low	x		
	Frequency, low	x		
	Disconnection of non-essential consumers	x		
	1. Alarm and automatic shut off is to be provided for Engines of 2250 [KW] and above or having cylinders of more than 300 [mm] bore. For each engine, one oil mist detector (or engine bearing temperature monitoring system or equivalent device) having two independent outputs for initiating the alarm and shut-down would satisfy the requirement for independence between alarm and shut-down systems.			
Steam turbine	Temperature, lubricating oil, high	x		
	Pressure, lubricating oil inlet, low	x		Independent of safety system
	Pressure, lubricating oil inlet, low		x	
	Pressure, condenser, high	x		
	Pressure, condenser, high		x	
	Pressure, steam inlet line, low	x		
	Axial displacement rotor, large		x	Multistage turbines

Table 3.1.6 : Monitoring of Steam Generating Plant				
System		**Required monitoring (stated by an x)**		**Comments**
		Alarm	**Automatic shut down of boiler with alarm**	
Main Steam and Feed Water Installation				
Boiler water	Level, high	x		
	Level, high	x		Turbine shut down
	Level, low	x		Independent of safety system
	Level, low		x	
Water circulation	Stopped		x	Forced circulation boilers
Combustion air supply	Fan stopped		x	
Uptake gas	Temperature, high	x		For fire detection
Fuel oil	Pressure, low	x		
	Pressure, low			Automatic start of standby pump with alarm
Fuel oil temperature or viscosity	Heavy oil, high	x		
	Heavy oil, low	x		
Oil burner	Pressure of atomizing medium, low	x		
Steam	Pressure, high	x		
	Pressure, low	x		Turbine slow down
	Temperature, high	x		
Ignition/flame	Ignition/flame failure		x	Separate monitoring and fuel shutoff for each burner
Feed water system	Atmospheric tank level, high	x		
	Atmospheric tank level, low	x		
	Deaerator level, high	x		
	Deaerator level, low	x		
	Deaerator pressure, high	x		
	Deaerator pressure, low	x		
Feed water	Temperature, high	x		
	Pressure, low	x		
	Pressure, low			Automatic start of standby pump with alarm
High pressure feed water heaters	Level, high	x		
Fresh water generator	Salinity, high	x		Automatic stop of fresh water generator

Table 3.1.6 : (Contd.)				
System		**Required monitoring (stated by an x)**		**Comments**
		Alarm	**Automatic shut down of boiler with alarm**	
Auxiliary Boiler Plant				
Non-dual pressure boiler	Water level, high	x		
	Water level, low	x		Independent of safety system
	Water level, low		x	
Dual pressure boilers	Water level primary system, low	x		Independent of safety system
	Primary system, low		x	
	Secondary system, high	x		
	Secondary system, low	x		
Water circulation	Stopped		x	Forced circulation boilers
Fuel oil temperature or viscosity	Heavy oil, high	x		
	Heavy oil, low	x		
Steam	Pressure, high	x		
	Pressure, high		x	When the automatic control system does not cover the entire load range from zero load
	Temperature, high	x		For superheated steam ≥ 350°C
Feed water	Salinity, high	x		Boiler pressure ≥ 20 bar
Condenser	Pressure, high	x		
Ignition/flame	Failure		x	Separate monitoring and fuel shut off for each burner

Table 3.1.7 : Monitoring of Tanks		
Item	**Alarm**	**Comments**
Sludge and drain tanks, level high	x	All tanks arranged with automatic filling are to have alarm for both high and low level
Service tanks, level, low	x	
Expansion tanks, level low	x	
Circulating tanks, level low	x	

Table 3.1.8 : Monitoring of Miscellaneous Items		
Item	**Alarm**	**Comments**
Bilges and bilge wells		
Engine room, level high	x	2 independent alarm circuits, minimum 2 detectors
Purifiers		
Lubricating oil inlet, temperature, high	x	
Fuel oil inlet, temperature high	x	
Fuel oil inlet, temperature low	x	For heavy fuel oil
Water seal, loss	x	
Alarm and safety systems, power failure	x	Electric, pneumatic, hydraulic
Remote control system, power failure	x	
Steering gear		
Electric motors, stopped	x	Manual start of standby motor from the bridge, alternatively automatic start with alarm

3.1.10 Proposals for extent of monitoring, different from those specified in Table 3.1.1 to Table 3.1.6, may be accepted provided these are shown to be as effective.

3.1.11 The fault conditions listed below are to effect an alarm requiring manual stop or alternatively automatic stop:

- lubricating oil pressure, inlet engine, low;
- lubricating oil pressure, inlet gear, low;
- thrust bearing temperature, high;
- crankcase explosive conditions for medium and high speed engines (for cylinder dia > 300 [mm] or engine power ≥ 2250 kW).

For transmitters for propulsion engines in Table 3.1.1, refer Note 7) in Table 3.1.1.

The alarm for gear is to have separate transmitter and in addition to the transmitters required by Table 3.1.4.

IRS may permit overriding of auto shut down provided consequences of overriding auto shut down are established and documented.

3.1.12 Automatic stop of single engine plant will be accepted only if the engine can be restarted from the bridge. However, automatic stop by overspeed and low lubricating oil pressure may be accepted without starting possibility from the bridge.

3.1.13 For multi-engine plants, canceling of automatic stop is not required if maneuverability of the vessel is maintained.

3.1.14 The fault conditions listed below are to cause an alarm requiring manual load reduction or alternatively automatic load reduction :

- lubricating oil pressure inlet engine, low;
- lubricating oil pressure inlet gearbox, low;
- cylinder lubrication flow, low;
- thrust bearing temperature, high;
- cooling medium pressure, inlet engine, low;
- cooling medium pressure, inlet piston, low;
- cooling medium flow, outlet pistons, low;
- cooling medium temperature outlet engine, high;
- scavenging belt temperature, high (fire); and
- crankcase explosive conditions for low speed engines (for cylinder dia > 300 [mm] or engine power ≥ 2250 kW).
- exhaust gas temperature outlet of each cylinder, high.

3.2 Electric power supply

3.2.1 Means are to be provided to avoid overloading of supply units.

3.2.2 The ship's main power supply is to be arranged according to one of the following methods :

- In ships where one generator is sufficient to cover normal load at sea there is to be an arrangement for automatic start and connection to the main switchboard of a standby generator. Start and connection are to be completed within 30 seconds;

- In ships where more than one generator is necessary to cover normal load at sea, there is, in order to maintain maneuverability, to be an arrangement for necessary tripping non-essential consumers in case of disconnections of, or failures which will lead to disconnection of a generator.

3.2.3 Standby units are normally to have separate cooling water and lubricating oil pumps. Alternatively, automatic start of standby pumps is to be arranged when they also serve other generating sets.

3.2.4 In the event of automatic start of standby power supply unit, automatic start of essential machinery is to follow in a pre-determined sequence. Units necessary for maneuvering of the ship are regarded as essential machinery. Starting air compressors, bilge, ballast and fire pumps need not be included.

3.2.5 When the maneuverability of the ship is independent of electric power, the requirements of 3.2.3 and 3.2.4 do not apply.

3.2.6 For essential consumers having power supply from the lighting system, precautions against power failure are to be similar to those taken for units having power supply from the main generators, e.g. the following means may be applied:

- adequate automatic emergency lighting for access to standby transformer for the lighting system and operating gear for manual connection;

- automatic connection of standby transformer;

- parallel connection of a sufficient number of transformers, and arrangement for selective disconnection;

- automatic connection of emergency source of power; and
- dividing the system in two or more circuits with automatic switch over.

In this context, essential consumers are units and equipment necessary for maneuvering of the ship, including navigation lights and sufficient lighting (either as part of the normal lighting or as separate emergency lighting) in the engine room, on the bridge, in the chartroom, in all passageways and stairways of the accommodation.

Section 4

Vessels fitted with Internal Combustion Propulsion Engines of less than 1000 [kW]

4.1 General

4.1.1 The requirements of this section are applicable to internal combustion propulsion engines of less than 1000 [kW], in lieu of the requirements specified in Section 3. Monitoring of parameters listed in 4.2 will normally be required. Other combinations of measuring points other than those listed may be accepted, when the chosen monitoring can detect fault conditions in an equivalent satisfactory manner.

4.1.2 The limitations of installations in ships with propulsion machinery of less than 1000 [kW] apply to the extent of monitoring only. The system arrangement is to be in accordance with Sec.2, unless otherwise specially stated.

4.2 Alarm system

4.2.1 Alarm is to be activated for the following fault conditions:

- fire in engine room;
- bilge level, high;
- variable pitch propeller, servo oil pressure, low;
- reduction gearing, lubricating oil pressure, low;
- power failure for alarm and remote control systems;
- main engine lubricating oil pressure, low;
- main engine cooling water (sea and fresh water) pressure, low. Indication of cooling water pressure on the bridge will be accepted as an alternative to alarms;
- main engine cooling water temperature, high;
- main engine cooling water expansion tank level - low; and
- low lubricating oil pressure and high cooling water temperature for electric power generating plants greater than 35 [kW].

4.3 Arrangements on bridge

4.3.1 In the event of fault condition the parameters, required to be monitored as per 4.2, are to activate individual or groupwise alarm signals on the bridge.

4.3.2 The following parameters are to be indicated on the bridge:

- rpm;
- propeller pitch (controllable pitch propeller);
- starting air pressure (reversible engine);
- direction of rotation (reversible engine);
- lubricating oil pressure for main engines and reduction gearing. This indication may be omitted if the pressure is monitored in accordance with the requirements of Sec.3.

4.4 Electric power supply

4.4.1 When the maneuverability of the ship is dependent on electric power, precautions are to be taken to avoid power supply units from being overloaded.

Section 5

Precautions against Fire

5.1 Oil fuel systems

5.1.1 Fuel oil injection pipes on all engines, irrespective of cylinder bore, are to be effectively shielded and secured. Safe drainage is to be provided for the leak- off oil.

5.1.2 Oil purifiers, burner units, and similar equipments, are to be efficiently shielded in order to prevent oil spray and leakages from reaching electric motors, hot surfaces, etc.

5.1.3 Also refer Pt.6. Ch.2, 1.2.5.

5.2 Fire detection

5.2.1 Fire detectors are to be fitted in the machinery spaces. All potential fire outbreak points are to be effectively guarded. The fire detection systems as well as number and location of detectors are subject to approval in each case.
A mixture of temperature and smoke/ionization detectors is advised. Flame detectors should be used to cover large spaces, e.g. over the main engine. The location and minimum number of detectors should preferably be as follows:

a) one detector for each internal combustion engine, boiler front, oil purifier installation, electric power distribution board and for each separate room in machinery spaces;

b) if temperature detectors are used exclusively, detectors should be fitted under all horizontal surfaces exceeding 10 [m^2] with a maximum distance of 5 [m] between detectors;

c) if smoke/ionization detectors are used exclusively, the number of detectors should not be less than -

$$0.03.L.B$$

L = total length of machinery spaces [m];

B = total breadth of machinery spaces [m];

d) the air flow in the machinery spaces under normal ventilation condition should be considered when the location of smoke/ionization detectors is considered.

5.2.2 Heat detectors only are not accepted for fire detection in the engine room.

5.2.3 The fire detectors are to be arranged in sections. There should be at least two sections for ships above 5000 gt.

5.3 Fire detector indicator panel

5.3.1 A fire detector indicator panel is to be located in the navigating bridge area, the fire control location or in such a position that a fire in the machinery spaces will not render it inoperative. The panel is to indicate the source of fire in accordance with arranged fire zones by means of a visual signal.

Section 6

Testing of the Plant on Board

6.1 General

6.1.1 After installation of the plant has been completed, function testing of the plant is to be carried out in the presence of the Surveyors.

6.1.2 Only after all work in connection with adjustments and starting of the various units of the plant has been completed, sea trials are to be carried out. The sea trials are to include at least 4 hours of continuous operation with unattended machinery spaces. Agreement is to be made in advance in each case for the personnel that will be present in the control room.

6.1.3 The builders are to prepare a detailed test programme. The test programme is to be kept onboard, all filled in and signed by the Surveyors.

6.1.4 If required by the Surveyor, vibrations and temperatures of exposed units are to be measured. If it is found that a unit is subjected to strains beyond the limits specified for it, precautions are to be taken to improve the conditions.

6.1.5 Recording of important automatically controlled parameters may be required as part of the testing.

6.2 Monitoring systems

6.2.1 Failure conditions are to be simulated as realistically as possible, preferably by letting the monitored parameters exceed the alarm and safety limits. The alarm and safety limits are to be recorded in the test programme.

6.2.2 Any calibrations of sensors prior to installation onboard are to be approved by the Surveyor.

6.3 Automatic and remote control systems

6.3.1 Automatic control systems are to be tested by varying the parameters having effect upon the controlled process. As far as practicable it is to be verified that all normal control ranges are covered.

6.3.2 All tests included in the test programme for the remote control system are to be carried out without any manual assistance from the engine room, and all systems are to be in operation as normal for unattended periods.

6.4 Blackout test

6.4.1 During bridge control at about half speed ahead, a failure is to be simulated to cause automatic stop of electric power generator.

6.4.1.1 For installations with automatic start and connection of standby units, it is to be ensured that the automatic start of standby power supply unit and machinery necessary for maneuvering of the ship from the bridge is effected without manual intervention in the engine room.

6.4.1.2 For installations with two or more generator units run in parallel at a time, it is to be ensured that the stopped unit is automatically disconnected from the switchboard, and that the capacity of the remaining running unit(s) is adequate for propulsion and steering functions after automatic tripping of non-essential consumers.

End of Chapter

Chapter 23

Refrigerated Cargo Installations

Contents

Section

1	*General*
2	*Refrigerating Plant*
3	*Cargo Chambers*
4	*Tests*
5	*Spare Gear*

Section 1

General

1.1 Class notation

1.1.1 On application from an Owner, refrigerated cargo installations which comply with the requirements of this Chapter will be assigned an appropriate notation. The class notations assigned, together with particulars of the installation, will be entered in the Register Book. The class will be maintained so long as the installation is found, at the prescribed periodical surveys, to be in a fit and efficient condition and in accordance with requirements of the Rules. Any damage or breakdown which could affect the temperature assigned or maintenance of class should be reported to IRS at the earliest practical opportunity.

1.1.2 The class notation generally assigned will be 'HY' followed by the minimum temperature(s) approved by IRS for the installation with the maximum sea temperature stated e.g. 'to maintain temperature - 20°C in the lower and 0°C in the upper chamber with sea temperature of 32°C maximum'.

1.1.3 For refrigerated cargo installations aboard container ships with approved refrigerating plant and arrangements to supply refrigerated air through ducting to insulated containers, the class notation assigned will additionally specify the maximum number and characteristics of the containers for which the plant is approved, e.g., 'to supply refrigerated air at a temperature of -20°C to certified 600 insulated containers with an average thermal transmittance per container of 27 W/C with sea temperature 32°C maximum'.

1.1.4 On application from an Owner, consideration will be given by IRS to an alternative temperature notation being assigned to that appearing in the Register Book. The temperature at which the cargo is to be carried is not the responsibility of IRS.

1.1.5 Installations constructed under Special Survey in accordance with the requirements of this Section will be eligible for the distinguishing mark ++ before class notation. In other cases, the mark will not be assigned.

1.1.6 If the refrigerating plant is found to be capable of cooling down fruit in general or a catch of fish the symbol * may be added after the class notation.

1.1.7 IRS will give consideration to ships engaged on voyages of short duration, to installations of small capacity, or to other special circumstances. In such cases, the class may include a service limitation or other restriction.

1.1.8 The requirements of Pt.1 of the Rules, regarding reclassification and withdrawal of

class respectively, apply also to refrigerated cargo installations.

1.2 Survey during construction

1.2.1 New installations intended for classification are to be constructed and tested under Special Survey in accordance with the requirements of this Chapter.

1.2.2 The materials used in the construction are to be manufactured and tested in accordance with the requirements of Pt. 2. Materials for which provision is not made in Pt.2 may be accepted, provided that they comply with an approved specification and such tests as may be considered necessary.

1.2.3 From the commencement of the construction and installation of the refrigerating plant and of the insulation and fitting out of the cargo chambers, to the testing of the completed installation, the Surveyors are to examine the materials and workmanship and are to indicate at the earliest opportunity, and require the rectification of, any items not in accordance with the Rules or the approved specifications and plans or any material, workmanship or arrangement found to be defective or unsatisfactory.

1.3 Refrigerants

1.3.1 These Rules are applicable to the following primary refrigerants :

- R 717 Ammonia

- R 22 Monochlorodifluoromethane.

1.3.2 Proposals to use other refrigerants will be specially considered on application, but methylchloride will not be accepted. Attention is also drawn to any statutory requirements of the country, in which the ship is to be registered.

1.4 Plans and particulars

1.4.1 The following plans and particulars, and any other information which may be specially requested, are to be submitted, in triplicate, for consideration before construction is commenced:

a) Refrigerating plant - detailed specification;

b) Insulation - detailed specification;

c) General arrangement of insulated chambers in elevation and plan. The plans are to be to a scale adequate for the measurement of external surfaces and the deck and bulkhead edges. Dimensions and spacing of frames, beams and stiffeners, and details of other steel works intruding into the insulation and within the chambers, are to be shown. Oil fuel and liquid cargo tanks adjacent to or below the chambers, and whether heating is provided for such tanks, are to be indicated along with the maximum design temperature to be maintained in those tanks, ventilating and air conditioning trunks and ducts passing through chambers are to be shown. The plans are to include a diagram showing the position of the chambers in relation to other parts of the ship if this is not otherwise apparent;

d) Detailed plans showing the thickness and methods of attachment of the insulation and lining on all surfaces including girders, hatch coamings and pillars; details of insulated doors and hatch access, bilge and manhole plugs and their frames. Methods of attachment of brine or direct expansion grids and meat rails are also to be shown;

e) Arrangements of air ducts (including method of cooling spaces within hatch coamings), air coolers, fans and their motors;

f) Air cooler defrosting arrangements;

g) Arrangement of brine grids and method of construction;

h) Arrangement of chamber thermometers and/or temperature sensoring devices;

i) General arrangement of refrigerating machinery;

j) Brine circuit diagram with particulars of piping;

k) Primary refrigerant gas and liquid circuit diagrams and full particulars of all pressure piping;

l) Sectional arrangement of refrigerant compressors;

m) Detailed dimensional plans of :

- Reciprocating compressor crankshafts, or rotors of screw-type compressors
- Reciprocating compressor crankcases where exposed to the refrigerant pressure or the casing of screw-type compressors
- Condensers
- Evaporators (brine coolers)
- Air coolers
- Oil separators
- Liquid receivers
- Any other pressure vessels.

1.4.2 Where it is proposed to use components such as compressors, condensers, oil separators, etc., that currently have general approval, the type and model number of the component are to be stated. Plans of items that have been so approved need not be re-submitted.

1.4.3 Where refrigerated air circulating fans are electrically driven, plans of the motors are to be submitted in accordance with Pt.4, Ch.8.

1.4.4 Where it is proposed to apply centralized bridge or automatic controls, a description of the scheme and particulars of the spares to be carried are to be submitted.

1.4.5 Any subsequent modifications or additions to particulars, and/or arrangements shown on the approved plans or in the specifications, are also to be submitted for approval.

1.4.6 Where refrigeration is effected by forced air circulation over coolers, the proposed fan output in each chamber will be considered in relation to the heat to be removed, the nature of the cargo, and the service(s) intended.

1.5 Heat balance test

1.5.1 A heat balance test may be required, as prescribed in this Chapter, on a classed installation, or one being considered for reclassification, when extensive repairs or alterations have been carried out, or when the Surveyors consider that an amended temperature notation should be assigned.

1.6 Novel arrangement and design

1.6.1 Where the proposed construction of the refrigerating plant or refrigerated chambers is novel in design or involves the use of unusual materials, special tests may be required, and a suitable class notation may be assigned when IRS considers this necessary.

Section 2

Refrigerating Plant

2.1 Refrigerating unit

2.1.1 A refrigerating unit is considered to comprise a compressor (refrigerating machine), its driving motor and one gas condenser. Where a secondary refrigerant is employed the unit is also to include a brine cooler. It is usual for the compressor and the condenser to be permanently connected to the installation with all necessary piping, fittings and electrical equipment. Alternative arrangements to the above will be given consideration in relation to the number of units in the installation and related factors.

2.1.2 Two or more refrigerating machines driven by a single motor, or having only one condenser or brine cooler, are to be regarded as one unit.

2.1.3 The refrigerating units of a classed installation must be completely independent of any refrigerating machinery associated with air conditioning plant, or any domestic refrigerated installation, or any process plant, unless full details of any proposal have been submitted and approved.

2.2 Requirements regarding refrigeration units

2.2.1 The refrigeration provided is to be sufficient to maintain the temperatures specified in the class notation when operating

24 hours per day with one unit in reserve as stand by. In order to compensate for deterioration of machinery and insulation over the life of the installation, the plant is to be designed to have at least 5 per cent excess capacity over that required for maximum design output for the desired class notation.

2.2.2 The proposals of both machinery and insulating contractors will be evaluated by IRS in determining the theoretical capabilities of the plant to maintain the duty temperatures. IRS will advise the contractors after appraisal of the specification and plans if it is considered that additional refrigeration or insulation effect is required, but the temperature assigned on completion of the heat balance tests will be determined on the actual results of the test.

2.2.3 Where the units are not connected in common to all refrigerated chambers, the plant serving each Section is to comply with 2.2.1.

2.2.4 In the case of installations having a large number of small units arranged to serve individual chambers or groups of chambers, the question of standby capacity will be specially considered.

2.2.5 Where only two refrigerating units are fitted, the working parts are to be interchangeable.

2.2.6 Where a refrigerating plant is provided for sub-cooling the liquid refrigerant of other units, but is not arranged for cooling the cargo chambers independently, it will not be regarded as a unit.

2.2.7 The standby or reserve unit may be considered as an operating unit during the limited cooling down period of a non-precooled fruit cargo.

2.3 Location of refrigerating machinery

2.3.1 Refrigerating machinery is to be located in an efficiently ventilated compartment.

2.3.2 Where ammonia is used as the refrigerant, the machinery compartment is to be isolated by gastight bulkheads and decks from any adjacent accommodation or working spaces and provided with at least two access doors.

2.3.2.1 Ammonia piping is not to pass through accommodation spaces.

2.3.3 Compartments containing ammonia machinery (including process vessels) are to be fitted with:

a) A negative ventilation system independent of ventilation systems serving other ship spaces and having a capacity not less than 30 changes per hour based upon the total volume of the space. Other suitable arrangements which are equally effective, may be considered;

b) A fixed ammonia detector system with alarms inside and outside the compartment;

c) Water screens above all access doors, operable manually from outside the compartment;

d) An independent bilge system;

e) At least two sets of ammonia gas masks with hermetically sealed filters and protective clothing are to be available in a glass door case located immediately outside ammonia machinery rooms.

2.3.4 In case of ammonia plants on fishing vessels under 55 [m] in length or other ammonia plants with a quantity of ammonia not greater than 25 [kgs], the plants may be located in machinery space provided:

The area where the ammonia machinery is installed is served by a hood with negative ventilation system, so as to prevent leakage of ammonia from dissipating into other areas in the space;

A water spray system is provided for the area

Requirements of paras 2.3.2.1, 2.3.3b) and 2.3.3e) are complied with.

2.3.5 Where the machinery compartment is unmanned, arrangements are to be made to raise an alarm if the cooling water ceases to circulate, and to shut down the affected compressor(s).

2.3.6 Machinery using non-toxic and non-flammable refrigerants will not in general, be subject to restriction on location, but proposals to install relatively large plants in other than separate compartments are to conform to statutory requirements and will be given special consideration.

2.4 Prime movers

2.4.1 Where the refrigerating plant is electrically driven, the motors and their control gear are to comply with the requirements of Pt.4, Ch.8. The electrical power is to be available as a source from at least two electrical alternators/ generators, with each alternator/generator being of sufficient capacity to take the entire refrigeration and normal sea load.

2.5 Circulating pumps

2.5.1 At least two separate gas condenser cooling water pumps are to be installed. One of the pumps may be considered as a standby pump and may be used for other purposes, provided that it is of adequate capacity and its use on other services does not interfere with the supply of cooling water to the gas condensers.

2.5.2 Where the primary and/or secondary refrigerants are circulated round the system by pumps, a standby pump(s) capable of operating on all cargo chambers is to be provided.

2.6 Sea connections

2.6.1 Condenser cooling water is to be taken from not less than two sea connections, it being recommended that one of the sea connections be provided on the port and the other on starboard side. Proposals are to be submitted for approval.

2.7 Refrigerating machinery design pressures

2.7.1 The scantlings of various parts of refrigerating plant are to be based on the pressures specified in Table 2.7.1.

Table 2.7.1 : Design pressure of refrigerating plant

Refrigerant	Minimum design pressure (g) in bar	
	HP side of system	LP side of system
R 22	20.6	16.7
R 717	21.2	17.2

2.7.2 If refrigerants other than those specified in Table 2.7.1 are used, the design pressure is subject to approval in each individual case. It may, in general, be assumed to be equal to the vapour saturation pressure of the refrigerant at 55°C and 45°C on the HP and LP sides, respectively.

2.8 Compressors

2.8.1 Compressors are to be of an approved type and are to be equipped with all the accessories and instruments necessary for effective and dependable operation.

2.8.2 Crankcases of compressors having cylinders integral with crankcase are normally to be designed to withstand an internal pressure at least equal to the compressor maximum working pressure. Any crankcase designed for a pressure lower than the compressor discharge maximum pressure, is to be provided with a relief valve set to lift at not more than the designed pressure, and the discharge led to a safe place. Compressor crankcases of large machines where cylinders are separate from the crankcase and which are not exposed to the refrigerant pressure when working, and are isolated when the plant is not in use, may have a lower design pressure according to the setting of the relief valve, but in no case is this to be lower than 2 bar.

2.8.3 In case of rotary displacement compressors, the rotor casing is to be designed for the maximum design pressure for the refrigerant.

2.9 Refrigerant pressure vessels

2.9.1 Welded cylindrical pressure vessels which are of steel construction and exposed to the pressure of refrigerants, are to be constructed in accordance with the requirements of Pt.4, Ch.5, so far as applicable.

2.9.2 Where ammonia is the refrigerant, the vessels are to be constructed to at least Class 2 requirements. Radiographic inspection may, however, not be waived regardless of

pressure. The specified minimum tensile strength of steel used for shells and dished ends is not to exceed 430 [N/mm^2].

2.9.3 Where pressure vessel shells are made from steel tubes, these are to be seamless, electric resistance welded or longitudinally submerged arc welded. Forge butt welded and spirally welded tubes are not acceptable.

2.9.4 All pressure vessels with a design temperature below - 40°C and also vessels with a design temperature below 0°C where the pressure/saturation temperature relationship does not apply, are to be manufactured from steels with mechanical properties, including notch toughness, suitable for the thickness and the lowest design temperature. Details of the proposed specifications for these steels are to be submitted for approval.

2.10 Pressure piping

2.10.1 Steel piping for primary refrigerant systems is to comply with the requirements of Pt,4, Ch.2. The tubes are to be seamless, electric resistance or submerged arc welded. Forge butt welded tubes are not acceptable. See also 2.14.

2.10.2 Where the piping is galvanized in lengths and then joined by butt welding, the galvanizing is to be removed in way of pipe ends before welding commences.

2.10.3 Copper piping is to be in accordance with Pt.4, Ch.2 except in the case of small air coolers having finned pipe of sizes not greater than 19 [mm] outside diameter, and fabricated under shop conditions. The finned pipe may have a minimum wall thickness of 0.5 [mm] when used with R 22 refrigerant.

2.11 Oil separators, filters and refrigerant driers

2.11.1 Suitable oil separators with drains are to be provided to the refrigerant lines. If wire gauze is used in the separator, it is to be sufficiently robust and supported to prevent disintegration.

2.11.2 Suitable filters are to be provided in the gas lines to compressors and in the liquid lines to regulators. Wire gauze in filters is to be sufficiently robust and supported to prevent disintegration. A filter may be combined with the required oil separator.

2.11.3 Driers are to be fitted in R22 refrigerant systems, and the arrangement is to be such that a drier can be by-passed, isolated and opened up without interrupting plant operations.

2.12 Safety devices

2.12.1 A pressure relief valve and or safety disc is to be fitted between each compressor and its gas delivery stop valve, the discharge being led to the suction side of the compressor. Where the motive power for the compressor does not exceed 10 [kW], the pressure relief valve and/or safety disc may be omitted.

2.12.2 All pressure vessels or other components of refrigerant systems which could become filled with liquid refrigerant and isolated are to be provided with safety discs and relief valves in series, or other approved arrangements, the discharge being led to a safe place above deck.

2.12.3 Suitable spring-loaded safety valves are to be provided to the cooling liquid side of condensers and the brine side of evaporators, where the pressure from any pump in the circuit could exceed the design pressure of the piping or any component forming part of the cooling systems.

2.13 Access to plant

2.13.1 In general, the arrangements are to be such that all components of the refrigerating machinery can be readily opened up for inspection or overhaul. Space is to be provided for the withdrawal and renewal of the tubes in 'shell and tube' type evaporators (brine coolers) and condensers. Proposals for alternative arrangements are to be submitted for consideration.

2.14 Pressure tests

2.14.1 Before installation

a) Components intended for use with a primary refrigerant are to be subject to strength and leak tests as detailed in Table 2.14.1, where p is in the design pressure as defined in 2.7.

b) The leak test normally requires an air pressure to be applied to the component while under water. Any alternative methods of leak testing will be considered on their merits.

c) Components for use with brine or cooling water are to be hydraulically tested to 2.0p but not less than 3.5 bar.

Table 2.14.1 : Pressure tests		
Component	**Strength test**	**Leak test**
Pressure vessels	1.5p (hydraulic)	1.0p
Compressors		
Crankcase and cylinders in one casting	1.5p (paraffin or oil)	1.0p
Separate Cylinders	1.5p (paraffin or oil)	1.0p
Crankcase separate from Cylinders	1.2p (paraffin or oil	0.8p
Valves and fittings	2.0p (paraffin or oil)	1.0p
Pressure piping, fabricated headers, air coolers, etc.,	1.5p (hydraulic)	1.0p

2.14.2 After erection on board ship

For pressure piping welded in place, hydraulic or pneumatic strength tests of welds are to be carried out at a test pressure of 1.5p. For pneumatic tests the following precautions are to be observed:

- All pressure vessels connected to the system must have already been strength and leak tested;
- All items such as compressor crankcases, gauges and automatic equipment that may be damaged by the test pressure are to be isolated;
- All spaces containing pipe lines under test are to be clear of personnel during the test, and adequate precautions are to be taken to completely prevent any danger to personnel and minimize risk to property;
- The pressure in the piping is to be observed by a gauge remote from the systems;
- The test pressure is to be built up slowly using an inert gas;
- The test pressure is to be held for at least one hour before being released.

a) After completion of the strength test, a leak test is to be carried out at a pressure equal to the design pressure in the presence of the Surveyor.

2.15 Automation

2.15.1 Where plants are operated by thermostatic refrigerant control, efficient manual controls are also to be provided, and the arrangement is to be such that thermostatic controls can be by-passed and isolated.

2.15.2 As an alternative, duplicate thermostatically operated refrigerant control valves may be fitted, each valve to be capable of the required duty and operable with the other out of action.

2.15.3 Where the system is to operate unattended, it is to conform to the general requirements of Ch.22.

2.16 Other refrigerating plant

2.16.1 Particulars of refrigerating plant using refrigerants other than those listed in this Chapter, are to be submitted for special consideration.

2.17 Cooling appliances in refrigerated chambers

2.17.1 Chambers may be refrigerated by pipe grids on the ceiling and sides or by circulation of air over coolers.

2.17.2 Refrigeration by pipe grids in the chamber is to be effected by the circulation of brine except in the case of installations or chambers of very small capacity when direct expansion of the refrigerant will be considered.

The pipe grids in each chamber are to be arranged in not less than two sections, and each section is to be fitted with valves or cocks so that it can be shut off. A single section may be approved for a chamber of small capacity but details must be submitted for approval.

2.17.3 Either brine or direct expansion of the refrigerant may be employed in the coils of air coolers. The coils are to be arranged in not less than two sections, each of which is to be capable of being readily isolated when necessary. Alternatively, at least two independent air coolers should be fitted. A single cooler with one circuit may be approved for a chamber of small capacity, but details must be submitted for approval.

2.17.4 In order to minimise the dehydration of the cargo and the frosting of the cooling appliances in each chamber aboard bulk refrigerated cargo ships, it is recommended that the installation be designed to maintain the required minimum temperature under the maximum difference of not more than 4.5°C between the cooling medium and the chamber.

2.17.5 Steel piping used to convey primary or secondary refrigerants within the refrigerated cargo chambers, or where embedded in insulation, is to be galvanized externally. For brine piping, an alternative method of protecting the piping against corrosion may be approved. Details are to be submitted for approval.

2.17.6 Where the pipe connections are by screwed couplings or by butt welds, the ungalvanized portion of the piping is to be suitably coated and taped after pressure testing to prevent the ingress of water vapour to the ungalvanized pipe and to reduce the incidence of corrosion. The locations of the joints are to be marked on the outside of the insulation.

2.17.7 Brine tanks and piping are not to be galvanized on the brine side. However, if any parts of the brine system have been galvanized, the brine cooling and return tanks, if closed, are to be provided with a ventilating pipe or pipes led to the atmosphere in a location where no damage will arise from the gas discharged, and the ventilating pipes are to be fitted with wire gauze diaphragms which can readily be renewed. Where the brine tanks are not closed, the compartments in which they are situated are to be efficiently ventilated.

2.17.8 Means are to be provided for effectively defrosting air coolers. Air coolers are to be provided with trays of suitable depth arranged to collect all condensate. The trays are to be provided with drains at the bottom so that the whole of the condensate can be drained away when the chambers are in service.

2.17.9 Where circulation of refrigerated air in a chamber is dependent on a single fan and electric motor, access arrangements are to be such that both the fan and the motor can be readily removed for repair or renewal, when the chamber is loaded with refrigerated cargo. Access for servicing only is required where several fans and motors are installed.

2.17.10 The air circulation system is to be arranged for positive air delivery to, or suction from all parts of the chamber, including any space within hatch coamings.

2.18 Heating arrangements for fruit cargoes

2.18.1 Where it is intended to carry fruit cargoes which may be adversely affected by low temperatures into areas where the ambient temperature may be below the carrying temperature, facilities for heating the chambers are to be provided.

2.19 Air refreshing arrangements

2.19.1 Where chambers are intended for the carriage of refrigerated cargoes requiring controlled ventilations, air refreshing appliances are to be provided. The positions of the air inlets are to be carefully selected to minimize the possibility of contaminated air entering the chambers. Each individual chamber is to be provided with its own separate inlet and discharge vent. Each vent is to have a positive airtight valve capable of closing on to a seat. It is recommended that a distance of at least 3 [m] is maintained between inlet and exhaust vents.

2.20 Drying of the refrigerating plant

2.20.1 After tightness testing and before charging with refrigerant and oil, the completed plant is to be dried carefully.

Guidance :

The following procedure is advised:

The plant is to be evacuated until an absolute pressure of maximum 0.007 bar is reached, which is to be held constant until all water is

evaporated. The plant is then filled with air until atmospheric pressure, and evacuated again until 0.007 bar. Care must be taken that no part of the plant has a temperature lower than 15℃ during evacuation.

2.20.2 Vessels, compressors, etc. are to be dried carefully at their place of manufacture.

2.20.3 After drying, all pipe connections, valves, etc. are to be closed gastight.

2.21 Refrigerating plant on ships not classed with IRS

2.21.1 In the case of the refrigerating installations being constructed under special survey on ships not intended to be classed with IRS, the generator engines and electrical equipment, which supply power to the refrigerating installations, are to be constructed in accordance with requirements of the Classification Society concerned. Such plant is to be examined generally and under working conditions by IRS Surveyors.

Section 3

Cargo Chambers

3.1 Airtightness of chambers

3.1.1 The following Rules are mainly concerned with avoiding infiltration of odours, gases, water vapour and air, which may taint or adversely affect the refrigerated cargo or the insulation, or cause undesirable frosting of cooling equipment.

3.1.2 Each individual chamber is to be of steel construction throughout and hose-tested for tightness. Alternative proposals to test more thoroughly by gas or air under slight pressure will be considered.

3.1.3 Consideration will be given to the construction of divisional bulkheads of materials other than steel, between refrigerated cargo chambers, where the chambers concerned are intended for cargo which will not taint or otherwise adversely affect the cargo in any other chamber; for example, gases given off by fruit of one category may adversely affect fruit of another category by promoting rapid ripening or, alternatively, adverse organic effect.

3.1.4 Hatch closing appliances, access doors, tonnage doors, bilge and manhole plugs forming part of the insulated envelope of independently refrigerated chambers, are to be made air tight. Where hatch covers or plugs are exposed to ambient conditions, they are to be provided with a double seal.

3.1.5 Ventilators, ducts or pipes passing through refrigerated chambers to other compartments are to be made airtight and efficiently insulated. Particular attention is to be given to insulation linings forming surfaces of air ducts. Ventilators to refrigerated spaces, if fitted, are to be provided with airtight closing appliances.

3.1.6 Refrigeration pipes passing through bulkheads or decks of refrigerated cargo chambers are not to be in direct contact with the steel work, and the holes through which they pass are to be true and of suitable finish for effectively sealing by the method intended. It is recommended that holes be trepanned and not burnt out. The airtightness of the bulkheads and decks is to be maintained.

3.1.7 Where cooling pipes pass through watertight plating and bulkheads, the fittings and packing of the glands are to be both fire resisting and watertight. Refrigerating pipelines are to be effectively insulated outside the chambers they serve, except within insulated brine cooler and control rooms.

3.1.8 Insulation lining, bilge limbers and plugs, and chamber and access doors are to be constructed of water-vapour- resisting material, or covered with such material, and where exposed to bilges or external conditions, they are to be sealed.

3.2 Insulation

3.2.1 Steelwork is to be thoroughly cleaned and suitably coated with an approved composition and is to be examined for

airtightness of fittings before insulation is applied.

3.2.2 Insulating material approved by IRS is to be used throughout, and the thickness of insulation over all surfaces and the manner in which it is supported is to be in accordance with the approved specifications and plans.

3.2.3 The insulation is to be efficiently packed and, where it is of slab form, the joints are to be butted closely together and staggered. Unavoidable crevices are to be filled with insulating material, but bitumen is not to be used for filling crevices. When it is intended to use a foam 'in-situ' type of insulant, full details of the process are to be submitted for approval before the work commences. The insulants, linings, sealants and paints are to be free from odour likely to cause taint.

3.2.4 Organic foam insulants are to be of self-extinguishing grade and are to be type approved. Certification by an independent authority is a requirement of type approval. For protection from fire, organic foam insulants are acceptable only when fitted behind a metallic lining or its equivalent. Special foams having exceptional fire resisting qualities may be accepted without a lining, provided that they have been approved for this type of application.

3.2.5 Where the insulation is to support fork lift trucks, the strength of the lining and its supports is to be demonstrated. A sample of the insulation 4 [m] x 4 [m] is to be prepared and tested by a fully loaded fork lift truck being driven and maneuvered over the sample to the satisfaction of the Surveyors.

3.2.6 If the cargo to be loaded on to tank top insulation could cause damage to the lining, then additional protection is to be provided in way of the hatch and 0.6 [m] beyond. The protection may either be of a permanent or temporary nature.

3.2.7 Insulation linings and air screens, together with supports, are to be strong enough to withstand the loads imposed by either refrigerated or general cargo.

3.3 Access plugs and panels

3.3.1 Insulated plugs are to be provided in the insulation where required for easy access to the bilges, bilge suction roses, cooler and chamber drains and tank manhole lids. Removable panels are to be provided for access to tank air and sounding pipes and drains.

3.3.2 It is recommended that where the insulation is covered with reinforced asphalt, a number of small insulated inspection plugs be fitted for leak detection.

3.3.3 Successive coatings of an approved oil impervious composition are to be applied to the surface of oil storage tank tops and bulkheads when insulation is to be fitted in way of tank plating. The total thickness of the coating required will depend on the construction of the tank, the composition used and the method of application.

3.3.4 Tank top insulation in way of manholes and bilge hats is to be provided with a liquidtight steel coaming to prevent seepage into insulation.

3.4 Piping in way of refrigerated chambers

3.4.1 Sounding pipes are to be not less than 32 [mm] bore.

3.4.2 All sounding pipes, whether for compartments or tanks, which pass through refrigerated spaces or the insulation thereof, in which the temperatures contemplated are 0°C or below, are to be not less than 65 [mm] bore.

3.4.3 Sounding pipes to oil compartments are not to terminate within refrigerated chambers or in the fan and battery rooms for these chambers, nor are these pipes to terminate in enclosed spaces from which access is provided to refrigerated cargo chambers or their fan and battery rooms.

3.4.4 All pipes, including scupper pipes, air pipes and sounding pipes that pass through chambers, intended for the carriage or storage of refrigerated produce are to be well insulated.

3.4.5 Where the pipes, referred to in 3.4.4 pass through chambers intended for a temperature of 0°C or below, they are also to be insulated from the steel structure, except in positions where the temperature of the structure is mainly controlled by the external temperature and will normally be above freezing point. Pipes passing through a deckplate within the shipside insulation, where the deck is fully insulated below and has an insulation ribband on top, are to be attached to the deckplating. In the case of pipes adjacent

to shell plating, metallic contact between the pipes and the shell plating or frames is to be arranged so far as practicable.

3.4.6 The air refreshing pipes to and from refrigerated compartments need not, however be insulated from the steel work.

3.5 Drainage from refrigerated cargo spaces

3.5.1 Provision is to be made for the continuous drainage of the inside of all insulated chambers and cooler trays.

3.5.2 Drains which are led from lower holds and cooler trays situated on the tank top are to be fitted with liquid sealed traps and non-return bilge valves.

3.5.3 Drains from tween deck chambers and from cooler trays which are situated well above the tank top are to be fitted with liquid sealed traps.

3.5.4 Where drains from separate chambers join a common main, the branch pipes are each to be provided with a liquid sealed trap.

3.5.5 The liquid sealed traps are to be of adequate depth and are to have a pressure head of at least 100 [mm] when connected to air ducts and 50 [mm] otherwise and arrangements are to be made for ready access to the traps for cleaning and refilling with brine.

3.5.6 Sluices, scuppers or drain pipes which would permit drainage from compartments outside the insulated chambers into the bilges of the latter, are not to be fitted.

3.5.7 Screwed plugs or other means for blanking off scuppers draining insulated chambers and cooler trays are not to be fitted. If, however, it is especially desired to provide means for temporarily closing these scuppers, they may be fitted with shut off valves controlled from ready accessible positions on a deck above the load water line. An indicator is to be provided to indicate whether the valve is open or shut.

3.6 Airtightness of insulation lining and air ducts

3.6.1 Air ducts and insulation linings are to be so constructed and fitted that moving air is prevented from entering the insulation. Special care is necessary where cooling pipes, air refreshing ducts, fan supports, etc. protrude through the lining.

3.7 Cargo battens

3.7.1 Where the carriage of fruit is intended and provision is made for adequate air circulation through the stow, then cargo battens may be omitted.

3.7.2 Where the carriage of frozen cargo is intended, provision is to be made for the circulation of air between the insulation lining and the frozen cargo. Where cargo battens are used to maintain the air passage, they may be integral with the lining, fastened to the lining, or added as dunnage during cargo loading, to all exposed vertical or near vertical, surfaces of the insulation lining. They are to be arranged to suit the air flow.

They need not be fitted to air screens and cooler casings for the purpose of providing an air passage. Where it is intended to carry refrigerated cargo within the hatch trunk, a full flow of air over the cargo and below the hatch covers is considered essential. The air flow may be promoted by use of air trunking or special battening arrangements.

3.8 Galvanizing of fixtures

3.8.1 All steel bolts, nuts, hangers, brackets and fixtures which support or secure cooling appliances, insulation, meat rails, etc. are to be galvanized.

3.9 Temperature measurement

3.9.1 Thermometers are to be of approved type, number and position in each chamber.

3.9.2 Thermometer tubes with their flanges and covers are to be insulated from the deck plating, and on weather decks they are to be so arranged that water will not run down the tubes when temperatures are being taken.

3.9.3 The inside diameter of thermometer tubes is to be not less than 50 [mm], and the tubes are not to be in contact with cold decks.

3.9.4 Where thermometer tubes pass through compartments other than those which they serve, they are to be efficiently insulated.

3.9.5 The thermometers used in thermometer tubes are to have a suitably slow reaction with respect to temperature change.

3.9.6 When in refrigerated cargo chambers only remote thermometers are fitted, at least two instruments are to be provided for each installation. The sensing elements in each chamber are to be divided between the two instruments in an appropriate manner. One of the instruments may be a data logger. The data logger is to comply with the requirements of 3.9.8.

3.9.7 The accuracy of remote thermometer is to be compatible with the sensitivity of the cargo to temperature variations. If no special require-ments are given, the following applies:-

Accuracy of measuring device

Below 0°C	At 0°C and above
± 0.5°C	0.25°C

In the above are included possible errors from resistance variations in cables and instrument reading errors. The combined errors of instrument reading and hysteresis are to be less than 0.1°C.

3.9.8 The scale of the instruments is to cover the whole expected range of chamber air temperatures ± 3°C. The scale deflection for fruit cargo is not to be less than approximately 5 [mm/°C] and for frozen cargo not less than 2.5 [mm/°C]. The temperature reading is to be possible within 0.1°C of both scale instruments and digital panels.

3.9.9 The sensing elements are to be well protected against mechanical damage and moisture. They are to be permanently connected to their instruments, i.e. no plug- in connections are allowed.

3.9.10 Remote thermometers and their installation is to be in accordance with the requirements of Pt.4, Ch.7.

Section 4

Tests

4.1 Thermometers

4.1.1 All thermometers and equipment for the measuring of chamber and air suction and delivery temperatures are to be installed in accordance with the approved plans and the whole installation completed to the satisfaction of the Surveyors.

4.1.2 Each thermometer is to be checked for accuracy by the contractors, and a statement is to be given to the Surveyors giving the readings to the nearest 0.1 of a degree at the freezing point of water. The Surveyors may use their discretion as to whether check tests are necessary. In the case of electric remote reading instruments, a statement of calibration for each instrument indicating the method employed is to be indicated.

4.2 Fan and air circulation systems

4.2.1 Air cooler fans are to be tested by the contractors after the air circulating system has been completed, and a statement of the results for each chamber handed to the Surveyors. The statement is to show the static pressure, the volume of air circulated per minute, the fan speed and the power consumption. The air circulation arrangements in the chambers are to be checked for distribution.

4.3 Refrigeration tests

4.3.1 All the plant is to be tested under working conditions, and when found to be operating satisfactorily, the capability of the installation to perform its maximum duty, as stated in the temperature qualification, is to be determined by the IRS thermal balance test.

4.3.2 The IRS balance test measures the capability of the combined plant and insulated chambers to carry out the duty required by the temperature qualification. It does not separately measure the insulation value or the efficiency of the machinery. Failure of the installation to perform satisfactorily may require the machinery manufacturer and the insulating contractor separately to prove their respective parts of the installation.

4.3.3 The thermal balance test is to be carried out under the supervision of and to the satisfaction of the Surveyors.

4.3.4 The insulated envelope is to be closed up, including air refreshing arrangements and the priming of scuppers. The hatch covers are to be made airtight to the satisfaction of the Surveyors.

4.3.5 During the test, comprehensive logs of machinery working conditions and chamber temperatures are to be made.

4.3.6 External thermometers are to be provided, and the external temperatures are to be logged at frequent intervals during the test.

4.3.7 The electrical energy input to fan and brine pumps is to be measured by wattmeter.

4.3.8 Prior to, or on completion of, the tests, the air coolers are to be defrosted.

4.3.9 If the heat balance test is held at some temperature in chambers above the qualification temperature, then on completion of test, the chambers are to be lowered to the qualification temperature.

4.3.10 The temperature of the chambers should not be reduced more than 1°C below the minimum designed temperature without the express permission of the Owner, the Builder and IRS.

4.3.11 Observations are to be recorded from the beginning of the cooling down to the end of the test. In the early stages, logging may be at intervals of about 4 hours, but external temperatures are to be logged hourly for the final 18 hours of the test, and all other relevant temperatures and pressure, speed and power consumption of compressor and fan motors, power consumption of brine and cooling water pumps, are to be logged hourly for the final 12 hours of the test.

4.3.12 Where an installation is to be completed and tested during a period when extremely low external temperatures are probable and a reasonable difference between the external and internal temperature will not be possible, the case is to be submitted for special consideration.

4.3.13 The air refreshing arrangements when fitted, safety devices (where practicable) and automatic arrangements are to be tested and found satisfactory.

Section 5

Spare Gear

5.1 General

5.1.1 It is recommended that adequate spares, together with the tools necessary for maintenance, or repair, be carried. The spares are to be determined by the Owner according to the design and intended service. The maintenance of the spares is the responsibility of the Owner.

End of Chapter

Chapter 24

Dynamic Positioning Systems

Contents

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1 *General*
2 *Thruster System*
3 *Power Systems*
4 *DP - Control System*
5 *Tests and Trials*

Section 1

General

1.1 Scope, application

1.1.1 The requirements of this Chapter apply to ships with installed dynamic positioning systems and are in addition to those applicable in other Parts of these Rules.

1.1.2 A ship provided with a dynamic positioning system in accordance with these Rules will be eligible for an appropriate class notation which will be recorded in the Register Book.

1.1.3 Requirements, additional to these Rules may be imposed by the National Authority with whom the ship is registered and/or by the administration within whose territorial jurisdiction it is intended to operate.

1.1.4 These rules specify design criteria, necessary equipment, tests and documentation requirements for dynamic positioning systems to reduce risk to personnel, the vessel, other vessels or structures and the environment while performing operations under dynamic positioning control.

1.1.5 For the purpose of these Rules, the area of operation is the specified allowable position deviation from the desired set point.

1.2 Definitions

1.2.1 **Bus-tie breaker** means a device connecting/disconnecting switchboard sections
("closed bus-tie(s)" means connected).

1.2.2 **Computer system** means a system consisting of one or more computers and associated hardware, software and their interfaces.

1.2.3 **Consequence analysis** means a software function continuously verifying that the vessel will remain in position even if the worst-case failure occurs.

1.2.4 **Dynamic Positioning control station (DP control station)** means a workstation designated for DP operations, where necessary information sources, such as indicators, displays, alarm panels, control panels and internal communication systems are installed (this includes but not limited to: DP control and independent joystick control operator stations, position reference systems' Human Machine Interface (HMI), when considered necessary, manual thruster levers, mode change systems, thruster emergency stops, internal communications).

1.2.5 **DP control system** means all control components and systems, hardware and software necessary to dynamically position the vessel. The DP-control system consists of the following:

a) Computer system and/ or joystick system
b) Sensor system (s)
c) Control stations and Display system
d) Operator panels
e) Position reference system (s)
f) Associated cabling and cable routing
g) Networks

1.2.6 **Dynamic Positioning operation (DP operation)** means using the DP system to control at least two degrees of freedom in the horizontal plane automatically.

1.2.7 **Dynamically positioned vessel** means a vessel which automatically maintains its position and/ or heading (fixed location, relative location or predetermined track) depending on purpose of the vessel, by means of thruster force.

1.2.8 **Dynamic positioning system** means the complete installation necessary for dynamically positioning a vessel, comprising, but not limited to, the following sub-systems:

a) Power system
b) Thruster system
c) DP-control system

1.2.9 **Failure** is an occurrence in a component or system causing one or both of the following effects:

- loss of component or system function

- deterioration of functional capability to such an extent that the safety of the vessel, personnel, or environment protection is significantly reduced.

1.2.10 **Failure Modes and Effects Analysis (FMEA)** means a systematic analysis of systems and sub-systems to a level of detail that identifies all potential failure modes down to the appropriate sub-system level and their consequences.

1.2.11 **FMEA proving trials** means the test program for verifying the FMEA.

1.2.12 **Hidden failure** means a failure that is not immediately evident to operations or maintenance personnel and has the potential for failure of equipment to perform an on-demand function, such as protective functions in power plants and switchboards, standby equipment, backup power supplies or lack of capacity or performance.

1.2.13 **Joystick system** means a system with centralized manual position control and manual
or automatic heading control.

1.2.14 **Loss of position and/or heading** means that the vessel's position and/or heading is outside the limits set for carrying out the DP activity in progress.

1.2.15 **Position keeping** means maintaining a desired position and/ or heading or track within the normal excursions of the control systems and the defined environmental conditions (e.g. wind, waves, current, etc.).

1.2.16 **Power management system** means a system that ensures continuity of electrical supply under all operating conditions.

1.2.17 **Power System** means all components and systems necessary to supply the DP-system with power. The power system includes:

a) Prime movers with necessary auxiliary systems including piping, fuel, cooling, pre-lubrication and lubrication, hydraulic, pre-heating, and pneumatic systems

b) Generators

c) Switchboards

d) Distributing system (cabling and cable routing)

e) Power supplies including Uninterrupted Power Supply (UPS)

f) Power Management System (PMS), (as applicable).

1.2.18 **Redundancy** means ability of a component or system to maintain or restore its function, when a single failure has occurred. Redundancy can be achieved for instance by installation of multiple components, systems or alternative means of performing a function.

1.2.19 **Time to safely terminate (operations)** means the amount of time required in an emergency to safely cease operations of the DP vessel.

1.2.20 **Thruster System** means all components and systems necessary to supply the DP-system thrust force and direction. The thruster system includes:

a) Thrusters with drive units and necessary auxiliary systems and piping
b) Main propellers and rudders if these are under the control of the DP-system
c) Thruster control system (s)
d) Manual thruster controls
e) Associated cabling and cable routing

1.2.21 **Worst-Case Failure Design Intent** means the specified minimum DP system capabilities to be maintained following the worst-case failure. The worst-case failure design intent is used as the basis of the design. This usually relates to the number of thrusters and generators that can simultaneously fail.

1.2.22 **Worst case failure** means the identified single fault in the DP system resulting in maximum detrimental effect on DP capability as identified through FMEA study.

1.3 Classification notations: definition and general principles

1.3.1 In addition to the Hull and Machinery class notations, ships complying with the requirements of this Chapter will be eligible to be assigned any of the following class notations:

DP(0) This notation may be assigned when a ship is fitted with automatic controls for position keeping and/or heading without joystick system back-up.

Note: IMO MSC/Circ.645 and IMO MSC.1/ Circ. 1580 do not specify any equipment class corresponding to DP (0).

DP(1) This notation may be assigned when a ship is fitted with automatic controls for position keeping and/or heading, an independent joystick system back-up and a position reference back-up.

DP(2) This notation may be assigned when a ship is fitted with automatic controls for position keeping and/or heading, with automatic standby controls, an independent joystick system back-up and redundancy in design and equipment as required by these rules.

DP(3) This notation may be assigned when a ship is fitted with automatic controls for position keeping and/or heading, with automatic standby controls, an independent joystick system back-up and redundancy in design and equipment. In addition, physical separation of components is to be provided by locating in different compartments as required by these rules. Full stop of thrusters and subsequent start-up of available thrusters is not considered as an acceptable disruption.

1.4 Design redundancy and failure modes

1.4.1 The worst-case failure modes for the above class notations are to be as follows:

1.4.1.1 For notation DP(1), loss of position and/or heading, may occur in the event of a single fault.

1.4.1.2 For notation DP(2), a loss of position and/or heading, is not to occur in the event of a single fault in any active component or system. Common static components may be accepted in systems which will not immediately affect position keeping capabilities upon failure (e.g. ventilation and seawater systems not directly cooling running machinery). Normally, static components will not be considered to fail where adequate protection from damage is demonstrated and properly documented with respect to protection.

Single failure criteria for DP(2) include:

a) Failure of any active component or system (generators, thrusters, switchboards including the short circuit of switchboard bus bars, remote controlled valves, coolers, filters etc.)

b) Fault in any normally static component (cables, pipes, manual valves etc) which is not properly documented with respect to protection

c) A single inadvertent act if such an act is reasonably probable

d) systematic failures or faults that can be hidden until a new fault appears

e) automatic interventions caused by external events, when found relevant (e.g. automatic action upon detection of gas).

1.4.1.3 For notation DP(3), loss of position and/or heading, is not to occur after any of the following failures in addition to 1.4.1.2 above:

a) Fault in any normally static component in the system

b) Failure to all components in any watertight compartment, from fire or flooding

c) Failure to all components in any fire sub-division, from fire or flooding

1.4.2 Based on the failure definitions in 1.4.1, the worst-case failure is to be determined with the help of F.M.E.A. and used as the criteria for consequence analysis. For class notations DP(2) and DP(3), a single inadvertent act is to be considered as a single fault, if such an act is reasonably probable and it need not be used as a criterion for consequence analysis. Requirement for failures to be considered for various class notations are summarized in Table No. 1.4.2:

Table 1.4.2 : Failures to be considered for various class notations

Notation	**Position and/ or heading keeping criterion**	**FMEA required**	**Failures to be considered**				
			Static components	**Active components**	**Inadvertent acts, if reasonably probable**	**Hidden failure**	**Failure of all components in a watertight compartment / fire subdivision due to fire or flooding**
DP0	Loss of position and/ or heading may occur after single failure	-	-	-	-	-	-
DP1	Loss of position and/ or heading may occur after single failure	-	-	-	-	-	-
DP2	Loss of position and/or heading is not to occur after single failure	Yes	Yes 1	Yes	Yes	Yes	No
DP3	Loss of position and/ or heading is not to occur after single failure including fire / flooding in one compartment	Yes	Yes	Yes	Yes	Yes	Yes

Note 1 : Not required if the static component (cables, pipes, manual vales etc.) is properly documented with respect to protection.

The objective of FMEA is to provide a comprehensive, systematic and documented analysis to establish the significant failure modes with regards to position keeping and/or heading demonstrate that the vessel maintains position in the event of single failure.

Essentially the FMEA is to

- Breakdown the DP system into functional blocks and function of each block is to be described
- Identify potential failure modes and their causes
- Evaluate effect on the system for each failure mode
- Identify measures for eliminating and reducing the associated risks
- Identify the trials required to prove the conclusion
- Provide information to operators on system limitations and capabilities.

1.4.3 A DP (2) or DP (3) vessel is to be operated in such a way that the worst-case failure, as determined in 1.4.1, can occur at any time without causing a breach of acceptable excursion criteria set for loss of position and/or heading for equipment classes 2 and 3, as defined in MSC Circ. 1580.

1.4.4 If external forces from mission-related systems (cable lay, pipe lay, mooring, etc.) have a direct impact on DP performance, the influence of these systems is to be considered and factored into the DP system design. Where available from the DP system or equipment manufacturer, such data inputs are to be provided automatically to the DP control system.
Additionally, provisions are to be made to provide such data inputs into the DP control system manually. These systems and the associated automatic inputs would be subject to surveys, testing and analysis as specified in Section 5.

1.4.5 In order to meet the failure criteria, redundancy of components will normally be necessary as follows:

a) For class notation DP(2), redundancy of all active components;

b) For class notation DP(3), redundancy of all components and physical separation of the components by A-60 class fire divisions, watertight below the damage control deck. (maximum possible physical separation of sensors above deck),

1.4.6 For class notation DP(3), full redundancy of the control systems may not be possible (.i.e. there may be a need for a single changeover system from the main computer system to the backup computer system). Such connections between otherwise redundant and separated systems may be accepted when these are operated so that they do not represent a possible failure propagation path during DP operations. Failure in one system is not to transfer to the other redundant system.

1.4.7 For class notation DP(2) and DP(3), connections between otherwise redundant and separated systems are to be kept to a minimum and made to fail to the safest condition. Failure in one system is not to be transferred to the other redundant system.

1.4.8 Redundant components and systems are to be immediately available without needing manual intervention from the operators and with such capacity that the DP operation can be continued for such a period that the work in progress can be terminated safely. The transfer is to be smooth and within acceptable limitations of the DP operation (s) for which the vessel is designed.

1.4.9 For class notation DP(2) and DP(3), hidden failure monitoring is to be provided on all devices where the FMEA shows that a hidden failure will result in a loss of redundancy.

1.4.10 For class notation DP(2), piping systems for fuel, lubrication, hydraulic oil, cooling water and cables are to be located with due regard to fire hazards and mechanical damage.

1.4.11 For class notation DP(3), cables for redundant equipment or systems and are not to be routed together through the same compartments. Where this is unavoidable, such cables may be run together in cable ducts of A-60 class. Cable connection boxes are not allowed in such ducts. On open deck, cables in separate pipes that are separately routed may be accepted. Suitable means are to be provided to keep the ambient temperature inside of an A-60 cable duct

within maximum temperature for the cables, when necessary, taking into account the temperature rise of cables under full power.

1.4.12 For class notation DP(3), redundant piping systems (e.g. fuel oil, lubrication oil, cooling water, hydraulic oil, pneumatic etc.) are not to be routed together through the same compartments. Where this is unavoidable, such pipes may be run together in ducts of A-60 class.

Redundant fuel oil supply systems are to be arranged with separate dedicated service tanks.

1.4.13 The requirements for various class notations in these rules are considered to satisfy the equipment classes in IMO guidelines for vessels with dynamic positioning systems (MSC Circ.1580) as noted in Table 1.4.13.

Table 1.4.13 : Correlation with IMO equipment classes

IRS Class notation	**IMO equipment class**
DP(0)	No corresponding equipment class
DP(1)	Equipment class 1
DP(2)	Equipment class 2
DP(3)	Equipment class 3

1.4.14 The basic requirements for systems and equipment to be provided for various class notations are summarized in Table 1.4.14. The detailed requirements for thrusters, power systems and DP-control systems are given in sections 2, 3 and 4 of this chapter.

Table 1.4.14 : Requirements for various class notations

Item	**Requirement for class notation:**			
	DP(0)	**DP(1)**	**DP(2)**	**DP(3)**
1. Thrusters:				
a) Arrangement	No redundancy	No redundancy	Redundant	Redundant and in separate compartments
b) manual single lever control of individual thruster	yes	yes	yes	yes
2. Power systems:				
a) Electrical power generation and distribution systems	No redundancy	No redundancy	Redundancy in design	Redundancy in design and physical separation by A-60 divisions.
b) Bus tie breakers	no	no	yes	yes
c) Main Switchboard (MSB)	yes	yes	yes	yes
d) Automatic Power Management System (APMS)	no	no	yes	yes
3. DP control system: [1])				
a) Computer systems for automatic control	1	1	2	2 plus one additional system in back-up control station
b) Consequence analysis	no	no	yes	yes
c) Independent common joystick control with manual/automatic heading control	no	yes	yes	yes
4. Position reference systems [2])	1	2	3	3, of which one system is in back-up control station

	5. External sensors:			
a) Wind	1	1	2	2, of which one is connected to back-up control station
b) Gyro Compass	1	1	3	3, of which one in back-up control station
c) Vertical reference system	1	1	3	3, of which one in back-up control station
6. Uninterrupted Power Supply (UPS)	0[3)]	1	2	3 of which one in back-up control station
7. Back-up control station	no	no	no	yes
8. Printer	yes	yes	yes	yes

Notes

1) **Operational mode** is the manner of control under which the DP-system may be operated, e.g.:

— automatic mode (automatic position and heading control)
— joystick mode (manual position control with selectable automatic or manual heading control)
— manual mode (individual control of thrust, azimuth, start/stop of each thruster)
— auto track mode (considered as a variant of automatic position control, with programmed movement of reference point).

2) **Position/heading reference system** consists of all hardware, software and sensors that supply information and or corrections necessary to give position/heading reference, including power supply.

3) A UPS need not be provided, unless required by the manufacturer.

1.5 Information and plans required to be submitted

1.5.1 Proposals for redundancy and segregation provided in the machinery, electrical installations and control systems are to be submitted. These proposals are to take account of the possible loss of performance capability should a component fail or in the event of fire or flooding.

1.5.2 Electrical load calculation for dynamic positioning operation is to be submitted for approval. For class notations DP(2) and DP(3), the load calculations are to consider the situation after the worst-case failure. (See 1.3.2).

1.5.3 Plans of the following together with particulars of ratings in accordance with the relevant Parts of the Rules are to be submitted for:

a) Prime movers, gearing, shafting, propellers and thrust units;

b) Machinery piping systems;

c) Electrical installations;

d) Pressure vessels for use with dynamic positioning system; and

e) For DP(3) notation the following are to be submitted in addition:

i) Routing of pipes and cables for redundant systems

ii) Location of redundant machinery and electrical equipment.

1.5.4 Plans of control, alarm and safety systems including the following are to be submitted:

a) Functional block diagrams of the control system(s) including the independent joystick system;

b) Functional block diagrams of the position reference systems and the environmental sensors;

c) Details of the electrical supply to the control system(s) the position reference system(s) and the environmental sensors;

d) Details of the monitoring functions of the controllers, sensors and reference system together with a description of the monitoring functions;

e) List of equipment with identification of the manufacturer, type and model;

f) Details of the overall alarm system linking the centralized control station, subsidiary control stations, relevant machinery spaces and operating areas;

g) Details of the control stations, e.g. control panels and consoles, including the location of the control stations;

h) Test schedules which are to include the methods of testing and the test facilities provided; and

i) Failure Modes and Effects Analysis (F.M.E.A.) in the case of notations DP(2) and DP(3).

1.5.5 Operation manuals, including details of the dynamic positioning system operation, installation of equipment, maintenance and fault finding procedures together with a section on the procedure to be adopted in emergency are to be submitted. A copy of the manual is to be placed on board the ship.

1.5.6 Performance capability evaluation report of the DP system for position keeping ability in the specified operating environmental conditions is to be submitted. This is to include a diagrammatic representation (DP capability Plot) of the limiting wind speeds and direction for effective DP operation, considering wave and current loads also acting simultaneously. A copy of the performance capability evaluation report is to be placed onboard the ship.

1.6 Performance capability of the DP system

1.6.1 DP capability polar plots are to be produced to demonstrate position and/ or heading keeping capacity for fully operational and post worst-case single failure conditions. The capability plots are to represent the environmental conditions in the area of operation and the mission-specific operational condition of the vessel

1.6.2 The performance capability of the dynamic positioning system to provide sufficient thruster forces and moments for maintaining position and heading in specified environmental conditions, is to be evaluated. The wind, wave and current conditions in a specified area are to be considered for this purpose.

1.6.3 A performance capability rating may be assigned by IRS for the DP system, if requested. The rating would indicate the percentage of time that environmental forces in a specified area may be withstood by the system.

1.6.4 The following parameters are considered in the evaluation of the performance capability rating:

a) Thruster forces
b) Wind loads on the ship
c) Wave drift loads on the ship
d) Current loads on the ship.

Section 2

Thruster System

2.1 General

2.1.1 Thrusters are to be manufactured, installed and tested in accordance with the relevant requirements of Pt.4 together with the requirements of this Chapter.

2.1.2 Where propellers for main propulsion are part of the DP system, they are to satisfy the requirements of thrusters given in this chapter. Steering gears controlled by DP system are to be designed for continuous operation.

2.2 Thruster system

2.2.1 Each thruster on a DP system is to be capable of being remote-controlled individually, independently of the DP control system.

2.2.2 The thruster system is to be designed to provide adequate thrust in longitudinal and lateral directions, and yawing moment for heading control.

2.2.3 The response and repeatability of thrusters to changes in propeller pitch or propeller speed/direction of rotation are to be suitable for maintaining the area of operation and the heading deviation specified.

2.2.4 For class notations DP(2) and DP(3), in the event of failure of the most effective thruster the ship is to be capable of maintaining its predetermined area of operation and desired heading in the environmental conditions for which the ship is designed and/or classed.

2.2.5 For class notations DP(2) and DP(3), the thruster system is to be connected to the power system in such a way that 2.2.2 can be complied with even after failure of one of the constituent power systems and the thrusters connected to that system.

2.2.6 Thruster installations are to be designed to minimize potential interference with other thrusters, sensors, hull or other surfaces which could be encountered in the service for which the ship is intended.

2.2.7 The values of thruster force used in the consequence analysis (see 4.2.13) is to be corrected for any unavoidable interference between thrusters and other effects which would reduce the effective force.

2.2.8 Thruster intakes are to be located at sufficient depth to reduce the possibility of ingesting floating debris and vortex formation.

2.2.9 Steerable thrusters and thrusters having variable pitch propellers are to be provided with independent supplies of motive power to the pitch and direction actuating mechanisms.

2.2.10 Each thruster unit is to be provided with a high power alarm. The setting of this alarm is to be adjustable and below the maximum thruster output.

2.2.11 Failure of thruster system including pitch, azimuth or speed control is not to result in an increase in thrust magnitude or change in thrust direction.

2.2.12 The thrust unit housing is to be tested at a hydraulic pressure of not less than 1.5 times the maximum service immersion head of water or 1.5 bar, whichever is the greater.

2.3 Emergency stop

2.3.1 It is to be possible to stop each thruster individually from the main DP-control station by means independent of the positioning and thruster control systems. This emergency stop is to be arranged with separate cables for each thruster. For class notations DP(2) and DP(3), the thruster emergency stop system is to have loop monitoring. For class notation DP(3), the effects of fire and flooding are to be considered.

Section 3

Power Systems

3.1 General

3.1.1 The power systems are to be designed, constructed and installed in accordance with the relevant requirements of Pt.4, together with the requirements of this Chapter. The power system is to have an adequate response time to changes in power demand. For class notations DP (0) and DP (1), the power system need not be redundant.

3.1.2 For class notation DP (2), the power system is to be divisible into two or more systems such that in the event of failure of one sub-system, at least one other system will remain in operation and provide sufficient power for station keeping. The power system may be run as one system during operation, but should be arranged by bus tie breakers to separate automatically upon failures which could be transferred from one system to another, including, but not limited to, overloading and short-circuits.

3.1.3 For class notation DP(3), the power system is to be divisible into two or more systems such that in the event of failure of one system, at least one other system will remain in operation and provide sufficient power for station keeping. The divided power systems are to be located in different spaces separated by A-60 class division. Where the power systems are located below the damage control deck, the separation is also to be watertight. The bus tie breakers are to be open during operations unless equivalent integrity of power operation can be accepted according to 1.4.5.

3.1.4 Means are to be provided in order to prevent overloading of the power plant, e.g. by use of interlocks, thrust limitations or other means. Means are also to be provided to prevent reactive overload.

3.1.5 Generally no restrictions are to be put on the starting intervals of electrical machines. If required, the arrangement is subject to approval in each case.

3.1.6 For class notations DP(2) and DP(3), the power available for position keeping is to be sufficient to maintain the vessel in position after the worst-case failure according to 1.4.2.

3.1.7 For class notations DP(2) and DP(3), at least one automatic power management system (PMS) is to be provided and is to have redundancy according to the equipment class and a blackout prevention function.

3.1.8 Alternative energy storage (e.g. batteries and fly-wheels) may be used as sources of power to thrusters as long as all relevant redundancy, independency and separation requirements for the relevant notation are complied with for class notations DP(2) and DP(3). The available energy from such sources may be included in the consequence analysis function required in 4.2.13 when reliable energy measurements can be provided for the calculations.

3.1.9 Sudden load changes resulting from single faults or equipment failures should not create a blackout.

3.2 Generators

3.2.1 For class notations DP (0) and DP(1), the total generating capacity of the power system is to be not less than the maximum dynamic positioning load together with the maximum auxiliary load. This may be achieved by parallel operation of two or more generating sets provided the requirements of Pt.4 are complied with.

3.2.2 When the electrical power requirements are supplied by one generator set, on loss of power there is to be provision for automatic starting and connection to the switchboard of a standby set and automatic restarting of essential auxiliary services.

3.2.3 Indication of absorbed electrical power and available on-line generating capacity is to be provided at the main dynamic positioning control station. An alarm is to be initiated at the dynamic positioning control station(s) when the total electrical load of all operating thruster units exceeds a preset percentage of the running generator(s) capacity. This alarm is to be adjustable between 50 and 100 per cent of the full load capacity having regard to the number of electrical generators in service.

For class notations DP(2) and DP(3), in operating conditions where all the generator sets are not being utilized, provision is to be made for automatic starting, synchronization and load sharing of a non-running generator before the load reaches the alarm level required above.

3.2.4 Means are to be provided to prevent starting of thruster motors until sufficient generating capacity is available.

3.3 Distribution arrangements

3.3.1 Thruster auxiliaries, control computers, reference systems and environmental sensors are to be served by individual circuits. Services that are duplicated are to be separated throughout their length as widely as is practical and without the use of common feeders, transformers, converters, protective devices or control circuits.

3.4 Control system power supply

3.4.1 An uninterrupted power supply (UPS) is to be provided for each DP control and measuring system (i.e. minimum one UPS for DP(1), two UPSs for DP(2) and three UPSs for DP(3) to ensure that any power failure will not affect more than one computer system and its associated components. The reference systems and sensors are to be distributed on the UPSs in the same manner as the control systems they serve, so that any power failure will not cause loss of position keeping ability. An alarm is to be initiated in case of loss of charge power. UPS battery capacity is to be sufficient to provide output power at maximum load for 30 minutes operation following mains supply failure. For DP(2) and DP(3), the charge power for the UPSs supplying the main control system is to originate from different power systems.

3.4.2 A UPS need not be provided for ships assigned with DP(0) notation, unless required by the manufacturer.

3.5 Auxiliary supplies

3.5.1 Where the auxiliary services and dynamic positioning thrusters are supplied from a common source:

The voltage regulation and current sharing requirements defined in Pt.4 are to be maintained over the full range of power factors that may occur in service.

Section 4

DP - Control System

4.1 General

4.1.1 The DP control system is to be designed, constructed and installed in accordance with the relevant requirements of Pt.4 together with the relevant requirements of this Chapter.

4.1.2 The control system for dynamic positioning operation is to be stable throughout its operational range and is to meet the specified performance and accuracy criteria.

4.2 System arrangement and functions

4.2.1 In general, the DP control system is to be arranged in a designated DP control station where the operator has a good view of the vessel’s exterior limits and the surrounding area.

4.2.2 The DP control station should display information from the power system, thruster system and the DP control system to ensure that these systems are functioning correctly. Information necessary to operate the DP system safely is to be visible at all times.

4.2.3 Indication of the following is to be provided at each station from which it is possible to control the dynamic positioning system, as applicable:

a) Variation in the heading and location of the vessel relative to the desired reference point or course;

b) Vectorial thrust output, individual and total;

c) Operational status of position reference systems and environmental sensors;

d) Environmental conditions, e.g. wind strength and direction; and

e) Availability status of standby thruster units.

4.2.4 Display systems and the DP control station in particular, are to be based on sound ergonomic principles. The system is to provide for easy selection of the control mode, i.e. manual, common joystick or computer control of thrusters and the active mode is to be clearly displayed. Where applicable, feedback signals are to be displayed, not only the initial command. Indicators and controls are to be arranged in logical groups, and are to be co-ordinated with the geometry of the vessel, when this is relevant.

4.2.5 At least one computer system for automatic control is to be provided for class notations DP(0) and DP(1).

4.2.6 The area of operation is to be adjustable, but should not exceed the specified limits which are to be based on a percentage of water depth, or if applicable a defined absolute surface movement. Arrangements are to be provided to fix and identify the set point for the area of operation. It is to be possible to individually enter new position and heading set points in automatic control mode.

4.2.7 Alarms and warnings for failures in systems interfaced to and/or controlled by the DP-control system are to be audible and visual. A record of their occurrence and of status changes are to be provided together with any necessary explanations.

4.2.8 The DP control system should prevent failures being transferred from one system to another. The redundant components are to be so arranged that a failure of one component is to be isolated and the other component activated smoothly with no loss of position and/ or heading. When combining position reference systems and/or sensors in one unit where more than one function or system can be lost upon one common failure, the consequence to the total system upon such a failure is not to exceed the loss if non-combined installation was adopted

4.2.9 For class notation DP(2), the DP control system is to consist of at least two independent computer systems so that, in case of any single failure, automatic position keeping ability will be maintained. Common facilities such as self-checking routines, data transfer arrangements and plant interfaces are not to be capable of causing failure of both/ all systems. An alarm is to be initiated if any computer fails or is not ready to take control.

4.2.10 Redundant computer systems are to be arranged with automatic transfer of control after a detected failure in one of the computer systems. The automatic transfer of control from one computer system to another is to be smooth, with no loss of position and/ or heading. It is not to be possible to automatically or manually select a controller which is not ready to assume command as the active controller.

4.2.11 For class notation DP(3), in addition to the requirements in 4.2.9 above, a back-up control system is to be arranged in a room separated by A-60 class division from the main DP control station. During DP operation, this back-up control system is to be continuously updated by input from the sensors, position reference system, thruster feedback etc, and be ready to take over control. The switch-over of control to the back-up computer is to be manual, situated at the back-up computer and not affected by the failure of the main DP-control system. A backup controller status is to be updated at main controller at regular intervals. An alarm is to be initiated if the back up control system fails or is not ready to take control. The back-up DP-control station is to be arranged with similar view to the vessel's exterior limits and the surrounding area as the main DP-control station. Main and backup DP control systems are to be so arranged that at least one system will be able to perform automatic position keeping after any single failure.

4.2.12 Each DP computer system is to be isolated from other on-board computer systems and communications systems to ensure the integrity of the DP system and command interfaces. This isolation may be effected via hardware and/or software systems and physical separation of cabling and communication lines. Robustness of the isolation is to be verified by analysis and proven by testing. Specific safeguards are to be implemented to ensure the integrity of the DP computer system and prevent the connection of unauthorized or unapproved devices or systems.

4.2.13 For class notations DP(2) and DP(3), the DP control system is to include a software function, known as 'consequence analysis', which continuously verifies that the vessel will remain in position even if the worst case failure occurs. This analysis is to verify that the thrusters remaining in operation after the worst-case failure can generate the same resultant thruster force and moment as

required before the failure. An alarm is to be provided if the occurrence of a worst-case failure is expected to lead to a loss of position due to insufficient thrust for the prevailing environmental conditions. For operations, which will take a long time to safely terminate, the consequence analysis is to include a function to verify the adequacy of thrust and power remaining after worst-case failure based on manual input of weather trend.

4.2.14 When stopped, either by automatic or manual means, the position control system shall set the thrust commands to zero.

4.2.15 Loss of one or multiple position reference system inputs and/or one or multiple sensor inputs shall not lead to significant change in thrust output.

4.2.16 Upon recovery of position and heading reference input the DP-control system is not to automatically apply the last position or heading set point (set points before loss of input) when this is significantly different from the actual vessel position and/or heading. If any other set point than the actual vessel position and/or heading is applied then it is to be operator chosen.

4.3 Independent Joystick Control

4.3.1 A joystick system independent of the automatic DP control system is to be arranged. The power supply for the independent joystick system is to be independent of the DP control system UPSs. An alarm is to be initiated upon failure of the independent joystick system.

4.3.2 It is to be possible to control the thrusters manually, by individual levers and by an independent joystick, in the vent of failure of the DP-control system. Automatic heading control is to be available along with manual joystick control.

4.3.3 If an independent joystick is provided with sensor inputs; failure of the main DP control system is not to affect the integrity of the inputs to the independent joystick.

4.3.4 Any failure causing operator loss of control of the thrusters in the independent joystick control system is to freeze the thrust commands or set the thrust commands to zero. If the failure affects only a limited number of thrusters, the command to these affected thrusters may be set to zero, while keeping the other unaffected thrusters in joystick control.

4.3.5 The power supply for the independent joystick system is to be independent of the DP-control system UPS (as indicated in 3.4.1).

4.4 Position reference systems and environ-mental sensors

4.4.1 Position reference systems and environmental sensors are to be provided to ensure that the specified area of operation and heading deviation can be effectively maintained. Sensors are to provide new data with a refresh rate and accuracy suitable for the intended DP operations.

4.4.2 Position reference systems are to incorporate suitable position measurement techniques which may be by means of acoustic devices, radio, radar, inertial navigation, satellite navigation, taut wire or other acceptable means depending on the service conditions for which the ship is intended.

4.4.3 For class notation DP(0), at least one position reference system is to be provided. For class notation DP(1), at least two position reference systems are to be provided, each using a different measurement technique and simultaneously available to the DP-control system during operation. Special consideration may be given to cases where the use of two different measuring techniques would not be practicable during DP operation. In such cases, alternative arrangements may be accepted (for e.g. two DGPS systems).

4.4.4 For class notations DP(2) and DP(3), at least three position reference systems incorporating at least two different measurement techniques are to be provided and simultaneously available to the DP-control system during operation. The position reference systems are to be arranged so that a failure in one system will not render the other systems inoperative.

4.4.5 For class notation DP (3), at least one of the position reference systems is to be connected directly to the back-up control system and separated by A-60 class division from the other position reference systems.

4.4.6 Gyrocompass(es), or equivalent means are to be provided to measure the relative heading of the ship.

4.4.7 Vertical reference sensor(s) are to be provided, if applicable, to measure the pitch and roll of the ship.

4.4.8 Means are to be provided to ascertain the wind speed and direction acting on the ship.

4.4.9 For class notation DP(2) or DP(3), at least three sensors each are to be provided for measurement of parameters required by 4.3.6 and 4.4.7 and at least two sensors are to be provided for 4.4.8 above. In the event of a failure, of a reference or environmental sensor the control systems are to continue operating on signals from the remaining sensors without manual intervention.

4.4.10 For class notation DP(3), one of each type of sensors is to be connected directly to the back-up control system and separated by A-60 class division from the other sensors. If the data from these sensors is passed to the main DP control system for their use, this system is to be arranged so that a failure in the main DP control system cannot affect the integrity of the signals to the backup DP control system.

4.4.11 Suitable processing and comparative techniques are to be provided to validate the control system inputs from position and other sensors, to ensure the optimum performance of the dynamic positioning system. Abnormal signal errors revealed by the validity checks are to operate alarms. When several systems are combined to provide a mean reference, the mean value used is to not change abruptly by one system being selected or deselected.

4.4.12 The accuracy of the position reference data is generally to be within the following values as a guidance :

- a radius of 2% of water depth for bottom-based systems
- within a radius of 3 [m] for surface-based systems.

4.5 Alarms

4.5.1 The alarms to be presented in the DP-control station are normally to be limited to functions relevant to DP notations. Alarms are to be provided for the following fault conditions of the control system/ sensors:

a) Control computer system fault;

b) Automatic changeover to a standby control computer system;

c) Deviation from the predetermined area of operation of the ship;

d) Deviation from the predetermined heading limits of the ship;

e) Position reference system fault (for each reference system);

f) Gyrocompass fault;

g) Vertical reference sensor fault;

h) Wind sensor fault;

i) Taut wire excursion limit;

j) Automatic changeover to a standby position reference system or environmental sensor;

k) Failure in the independent joystick control system;

l) Loss of charger input power and UPS on bypass power;

m) Limit alarms are to be provided for systems, which have defined range limits;

n) For DP(2) and DP(3), if the consequence analysis is not completed within 2 minutes then an alarm is to be initiated; and

o) Any failure of a standby position control system or positioning reference system selected is also to initiate an alarm.

4.5.2 A manually initiated emergency alarm, clearly distinguishable from all other alarms associated with the dynamic positioning system is to be provided at the dynamic positioning control station to warn all relevant personnel in the event of a total loss of dynamic positioning capability. In this respect consideration should be given to additional alarms being provided at locations such as the master's accommodation and operational control stations.

4.6 Internal Communication

4.6.1 A two-way voice communication facility is to be provided between the DP-system control station and the navigation bridge, ECR and relevant operation control centres.

4.6.2 The two-way voice communication system is to be supplied by a battery or an uninterruptible power supply as a stand-by power supply sufficient to operate the system for at least 30 minutes.

4.7 Requirements for essential non-DP systems

4.7.1 For DP(2) and DP(3), systems not directly part of the DP system, but which in the event of failure could cause failure of the DP system (e.g. common fire suppression systems, engine ventilation, heating, ventilation and air conditioning (HVAC) systems, shutdown systems, etc.), are also to comply with relevant requirements specified in this Section.

Section 5

Tests and Trials

5.1 General

5.1.1 Before a new installation (or any alteration or addition to an existing installation) is put into service, trials are to be carried out. These trials are in addition to any acceptance tests which may have been carried out at the manufacturers' works and are to be based on the approved test schedules list as required by 1.5.4.

5.1.2 Functional tests of control and alarm systems of each thruster is to be carried out. All sensors and reference systems are to be tested for satisfactory performance as part of the DP system.

5.1.3 The suitability of the dynamic positioning system is to be demonstrated during sea trials, observing the following:

a) Response of the system to simulated failures of major items of control and mechanical equipment, including loss of electrical power; to try out switching modes, back-up systems and alarm systems.
   In case redundancy is based upon change-over of a single stern thruster as described in 1.3.1, then the functionality of the change over mechanism and availability of the thruster after single failure is to be demonstrated at sea-trials.

b) Response of the system under a set of predetermined maneuvers for changing

   - Location of area of operation;

   - Heading of the ship; and

   Change of command between the automatic DP-control system, independent joystick system and the individual thruster lever systems are to be demonstrated.

c) Continuous operation of the system over a period of four to six hours, including continuous operation of at least 2 hours during weather conditions such that an average load level on the thrusters of 50% or more is achieved.

5.1.4 All functions of the independent joystick control system are to be tested.

5.1.5 For class notations DP(2) and DP(3), FMEA proving trials are to be carried out, to verify the conclusions of the FMEA. The FMEA trials are also to include a complete test of all systems and components and the ability to keep position after single failures associated with the assigned class notation. The type of tests carried out and results are to be recorded and kept on board.

5.1.6 Two copies of the dynamic positioning system test schedules, as required by 1.5.4, signed by the Surveyor and Builder are to be provided on completion of the survey. One copy is to be placed on board the vessel.

5.1.7 Records and data regarding the performance capability of the dynamic positioning system are to be maintained on board the ship and are to be made available at the time of the Annual Survey.

5.1.8 During sea trials the offset inputs for each position reference system and relevant sensors in the DP control system are to be

verified and demonstrated to the attending surveyor by setting out the offsets on drawings. It is to be verified that these fit with the actual placing of the equipment.

5.1.9 Emergency stop function is to be demonstrated.

5.1.10 For steering gears included under DP control a test is to be carried out verifying that maximum design temperature of actuator and all other steering gear components is not exceeded when the rudder is continuously put over between maximum limits on either side set by the DP-control system, until temperature is stabilized.

In this respect, the ambient conditions specified in Pt.4, Ch.1, 1.7 will be considered. The actual ambient conditions and temperature rise of components are to be noted during trials for evaluation.

5.1.11 The capacity of the UPS battery is to be tested during trials.

End of Chapter

Chapter 25

Fire Fighting Ships

Contents

Section

1 *General*

2 *Construction*

3 *Fire Extinguishing*

Section 1

General

1.1 Application

1.1.1 The requirements of this Chapter apply to ships intended for fire fighting operations and are additional to those applicable in Pt.1 to 4 of the Rules.

1.1.2 A ship provided with fire protection and fire fighting equipment in accordance with these Rules will be eligible for an appropriate class notation, as given in 1.2, and which will be recorded in the Register Book.

1.1.3 Requirements additional to these Rules may be imposed by the National Authority with whom the ship is registered and/or by the Administration within whose territorial jurisdiction the fire fighting ship is intended to operate.

1.2 Classification and class notations

1.2.1 Vessels complying with the requirements of this Chapter may be assigned class notations as under :

- "Agni 1 (total monitor discharge capacity in brackets)";
- "Agni 2 (total monitor discharge capacity in brackets)"; and
- "Agni 3 (total monitor discharge capacity in brackets)".

1.2.2 The notation "Agni 1" implies that the vessel has been built for early stage fire fighting and rescue operations close to the structure on fire including means for self protection of the vessel.

1.2.3 The notation "Agni 2 or 3" implies that the vessel has been built for continuous fighting of large fires and for cooling of the structures on fire.

For the notation "Agni 3", a larger water pumping capacity and a more comprehensive fire fighting equipment is required than for the notation "Agni 2".

1.2.4 If a vessel fitted with fire fighting systems and equipment in accordance with the notation "Agni 2 or 3", also has a system for self protection in accordance with the notation "Agni 1", the combined class notation "Agni 1 and 2" or "Agni 1 and 3" may be assigned.

1.3 Scope of classification

1.3.1 The following additional matters are covered by classification :

- The vessel's fire fighting capability;
- The vessel's stability and its ability to keep its position when the fire fighting monitors are in operation; and
- The vessel's ability of self-protection against external fires in case of vessels with the notation "Agni 1".

1.4 Assumptions

1.4.1 The classification of the vessel is based upon the following assumptions :

- the operation of the vessel during fire fighting will be in accordance with the approved operating manual;
- the vessel will carry a sufficient quantity of fuel oil for continuous fire fighting operation with all the fixed water monitors in use for a period not less than :
  - 24 hours for the notation "Agni 1"
  - 96 hours for the notations "Agni 2 or 3";
- foam-forming liquid for at least 30 minutes continuous foam production for the fixed foam monitors is stored on vessels with notation "Agni 3";
- foam-forming liquid for at least 30 minutes continuous foam production by the mobile generator is stored on board in suitable containers on vessels with notation "Agni 2 or 3"; and
- crew operating the fire fighting systems and equipment has been trained in such operations and in the use of air breathing apparatus, and that the skill of the crew is maintained by exercises.

1.5 Submission of plans

1.5.1 The following plans and information are to be submitted:

- A general arrangement showing the disposition of all fire fighting equipment required by this Chapter;
- Manual for the operation of the fire fighting installation and maneuvering of the vessel during fire fighting;
- Details of major items of fire fighting equipment such as pumps and monitors, including their capacity, range and trajectory of delivery;
- A general arrangement plan showing the disposition of the fire divisions and their class;
- Detailed plans of the fire divisions and, where applicable, copies of the certificates of approval for the insulating materials proposed;
- A plan of the construction of the fire doors;
- Plans showing the layout and capacity of the water spray system;
- A plan of the seating arrangements for the water monitors;
- Particulars of the means of keeping the ship in position during fire fighting operations;
- A plan showing the fire pumps, the fire water main, the hydrants, hoses and hose nozzles and the monitors together with particulars of their delivery capability;
- Details of the foam generating system;
- Calculations showing the point of balance between the reaction forces from the water monitors and the forces from the vessel's propulsion machinery and its side thrusters;
- Calculations proving satisfactory stability of the vessel when all monitors are in operation at full capacity in the most adverse direction for stability;
- Report on inclining test determining the centre of gravity and the light weight of the completed vessel;
- Details of the fireman's outfits provided; and
- Plans of any other fire fighting systems provided.

1.6 Operation manual

1.6.1 The following information is to be included in an approved operation manual available on board :

- Detailed description of each fire fighting system;
- Instructions for use, testing and maintenance of the fire fighting installations and the equipment; and
- Instructions for operation of the vessel during fire fighting.

1.7 Items to be manufactured under survey

1.7.1 The following items are to be manufactured under the survey of IRS:

- Fire fighting pumps and their prime movers
- Water / foam monitors
- Compressors for breathing apparatus
- Pipes / valves
- Spray nozzles for self-protection.

1.7.2 The extent of survey and testing is to be in accordance with Pt.4, Ch.2, Sec.1.9 for pipes and valves and as per Pt.4, Ch.1, Sec.1.3 for remaining items.

Section 2

Construction

2.1 Hull

2.1.1 The structure of the ship is to be strengthened as necessary to withstand the forces imposed by the fire-extinguishing systems when operating at their maximum capacity.

2.2 Sea suctions

2.2.1 The seawater suctions of the fire pumps are to be arranged as low as practicable in the ship's structure to avoid icing or the ingress of oil from the surface of the sea.

2.2.2 All sea chests are to be provided with a low pressure steam or compressed air connection for clearing purposes.

2.2.3 Seawater suctions for fire fighting pumps are not to be used for any other purpose. Seawater suction valve, the discharge valve and the pump prime mover are to be operable from the same position. Valves with nominal diameter exceeding 450 [mm] are to both manually operated and power actuated.

2.3 Stability

2.3.1 Each ship is to comply with the draught and stability requirements of the National Authority and is to have on board sufficient stability data to enable the ship to be properly loaded and handled. This data is to take full account of the effect of the monitors when they are operating at their maximum output in all possible directions of use.

2.4 Maneuverability

2.4.1 Arrangements are to be provided to enable the ship to maintain position so that the monitors may be effectively deployed.

2.4.2 Side thruster(s) and main propellers are to be able to keep the vessel at a standstill in calm waters at all combinations of capacity and direction of throw of the water monitors, and the most unfavourable combination is not to require more than 80 per cent of the available propulsion force in any direction.

2.4.3 If the system design is such that, in any operating combination, it will be possible to overload the power supply, a power management system is to be arranged. The system is to include alarm at 80 per cent of available power and automatic action at 100 per cent available power.

2.5 Floodlights

2.5.1 As an aid for operations in darkness at least two adjustable floodlights are to be fitted on board. The floodlights are to be capable of illuminating areas at a distance of 250 [m] in clear air. The floodlights are to be of high pressure sodium vapour type or equivalent.

2.6 Bunkering

2.6.1 The Owner should ensure that any fuel which may be required while the ship is operating on station can be safely received on board.

2.7 Self-protection of the vessel (class notation Agni 1)

2.7.1 The vessel to be protected by a permanently installed water-spraying system.

2.7.2 The fixed water-spraying system is to provide protection for all outside vertical areas of hull, superstructures and deckhouses including foundations for water monitors and other equipment.

2.7.3 The arrangement for the water-spraying system are to be such that necessary visibility from the wheelhouse and the control station for remote control of the fire fighting water monitors can be maintained during the water spraying.

2.7.4 The pipelines and nozzles are to be so arranged and protected that they will not be exposed to damage during the operations for which the vessel is intended.

2.7.5 The fixed water-spraying system is to have a capacity not less than 10 litres per min. per [m^2] of the areas to be protected. For areas internally insulated to class A-60, however, a capacity of 5 litres per min. per [m^2] may be accepted.

2.7.6 The pumping capacity for the fixed water-spraying system is to be sufficient to deliver water at the required pressure for simultaneous operation of all nozzles in the total system. The pumps for the fire fighting water monitors may also serve the spraying system, provided the pump capacity is increased by the capacity required for the water-spraying system. A connection with shut-off valve is then to be fitted between the fire main for the monitors and the main pipeline for the water spraying system.

2.7.7 All pipes for the fixed water-spraying systems are to be protected against corrosion externally and internally by hot galvanizing or equivalent. Drainage plugs are to be fitted to avoid damages by freezing water.

2.7.8 The spray nozzles are to be able to give an effective and even distribution of water spray over the areas to be protected.

2.8 Self-protection of the vessel (class notations Agni 2 and 3)

2.8.1 In ships which are not provided with a water spray system as described in 2.7 all windows and port lights are to be provided with efficient deadlights or external steel shutters, except in the wheelhouse.

2.8.2 Ships which are intended to operate in close proximity to a large fire will require protection from the heat radiated from the fire. Such protection may be afforded by a system which provides a water spray over the surface of the ship, or by a combination of insulation and a water spray system.

Section 3

Fire-extinguishing

3.1 Water monitors

3.1.1 The minimum number of monitors, their discharge rate, their range and their height of trajectory above sea level are to comply with the requirements of Table 3.1.1.

3.1.2 The monitors are to be so arranged that the required direction, range and height of trajectory can be achieved with the required number of monitors when they are operating simultaneously.

3.1.3 The monitors are to be capable of adequate adjustment in the vertical and horizontal direction and are to be so positioned that the jets will be unimpeded within the required range of operation. The horizontal angular movement of the monitors is to be at least 90°.

3.1.4 Means are to be provided for preventing the monitor jets from impinging on the ship's structure and equipment.

3.1.5 The monitors are to be of robust construction and their seating arrangements are to be of adequate strength for all modes of operation, particular attention being paid to shock loading when all the monitors are activated simultaneously.

3.2 Monitor controls

3.2.1 The activating and the maneuvering of the monitors are to be capable of remote control. The remote control station is to be arranged in a protected control room with a good general view. The valve control is to be designed to avoid water hammer.

3.2.2 The control system is to be designed in such a manner that normal operation can be restored within 10 minutes in case of any failure in the control system.

3.2.3 In case of electrical control systems, each control unit is to be provided with overload and short-circuit protection. In case of hydraulic or pneumatic control systems, the control power units are to be duplicated.

3.2.4 In addition to the remote control, local/manual control of each monitor is to be provided.

3.3 Foam monitor system (For class notation Agni 3)

3.3.1 In addition to the water monitors, the vessel is to be equipped with 2 foam monitors, each of a capacity not less than 5000 litres/minute with a maximum foam expansion ratio of 15 : 1.

Table 3.1.1 : Water monitor system capacities

Equipment	Class Notation			
	Agni 1	Agni 2		Agni 3
Minimum number of water monitors	2	3	4	4
Minimum discharge rate per monitor [m^3/h]	1200	2400	1800	2400
No. of pumps	1-2	2-4		2-4
Minimum total pump capacity [m^3/h]	2400	7200		9600
Minimum height of throw in [m[1)]]	45	70		70
Minimum length of throw in [m[2)]]	120	150		150
Minimum fuel capacity in hours [3)]	24	96		96
Number of hose connections each side of ship	4	8		8
Number of fireman's outfits	4	8		8

Notes:

1) Measured vertically from sea level to mean impact area at a horizontal distance of at least 70 [m] from the nearest part of the vessel.

2) Measured horizontally from the mean impact area to the nearest part of the vessel when all the monitors are in operation satisfactorily.

3) Capacity for continuous operation of all monitors to be included in the total capacity of the vessel's fuel oil tanks.

3.3.2 The foam system, together with the arrangements and location of monitors, is to give a height of throw at least 50 [m] above sea level when both monitors are used simultaneously with maximum foam generation.

3.3.3 The foam concentrate tank is to have a capacity for at least 30 minutes of maximum foam generation from both foam monitors. When determining the necessary quantity of foam concentrate, the admixture is assumed to be 5 per cent.

3.3.4 The foam generating system is to be of a fixed type with separate foam concentrate tank, foam-mixing unit and pipelines to the monitors. The water supply to the system may be taken from the main pumps for the water monitors and in such cases it may be necessary to reduce the main pump pressure to ensure correct water pressure for maximum foam generation.

3.3.5 Foam monitors are to be controlled in the manner given in 3.2 and the controls are to be located in the same control room as that for water monitors.

3.4 Pumps and piping

3.4.1 The pumps and their piping system which are intended for serving the monitors are not to be available for services other than fire-extinguishing and water spraying.

3.4.2 Where the pumps are used for fixed water spray systems, the piping is to be independent of that supplying the monitors. The water spray systems are to be adequately protected against overpressure.

3.4.3 The minimum total pump capacities required are shown in Table 3.1.1.

3.4.4 The piping system from the pumps to the water monitors is to be separate from the piping system to the hose connections required for mobile fire fighting equipment.

3.4.5 The piping systems are to have arrangements to avoid overheating of the pumps at low delivery rates.

3.4.6 All piping from seawater inlets to monitors is to be protected internally and externally against corrosion to a degree at least corresponding to hot galvanizing.

3.5 Mobile fire fighting equipment

3.5.1 Hose stations

3.5.1.1 Hose stations are to be provided on each side of the ship in accordance with Table 3.1.1.

3.5.1.2 Each hose station is to be provided with a hydrant, a hose and a nozzle capable of producing a jet or a spray and simultaneously a jet and a spray. The hoses are to be 15 [m] in length and not less than 38 [mm] nor more than 65 [mm] in diameter. Where hose stations are connected to the monitor supply lines, provision is to be made to reduce the water pressure at the hydrants to an amount at which each fire hose nozzle can be safely handled by one man. The water pressure shall be sufficient to produce a water jet throw of at least 12 [m].

3.5.2 Foam generator

3.5.2.1 Vessels with class notation "Agni 2 or 3" are to have a mobile high expansion foam generator with a capacity not less than 100 [m^3/minute] for fighting of external fires.

3.5.2.2 Foam-forming liquid is to stored in containers each of about 20 litres and suitable for mobile use. The total storing capacity of foam forming liquid is to be sufficient for 30 minutes continuous foam production.

3.6 Fireman's outfits

3.6.1 The number of fireman's outfits provided in addition to those provided in accordance with Pt.6, Ch.3, Sec.4.10 is to be in accordance with Table 3.1.1. They are to be stored in a safe position which is readily accessible from the open deck.

3.6.2 The composition of a fireman's outfit is to be as follows:

- Protective clothing of material to protect the skin from heat radiating from the fire and from burns and scalding by steam. The outer surface is to be water-resistant.

- Boots and gloves of rubber or other electrically non-conducting material.

- A rigid helmet providing effective protection against impact.

- An electric safety lamp (hand lantern) of an approved type with a minimum operating period of 3 hours.

- An axe having an insulated handle.

- A self-contained breathing apparatus which is to be capable of functioning for a period of at least 30 minutes and having a capacity of at least 1200 litres of free air. Spare, fully charged air bottles are to be provided at the rate of at least one set per required apparatus.

- For each breathing apparatus a fireproof lifeline of sufficient length and strength is to be provided and is to be capable of being attached by means of a snaphook to the harness of the apparatus or to a separate belt in order to prevent the breathing apparatus becoming detached when the lifeline is operated.

3.7 Recharging of equipment

3.7.1 A suitable air compressor for recharging the bottles used in the breathing apparatus of the fireman's outfits is to be provided. It is to be capable of recharging the bottles of the breathing apparatus required to be carried in accordance with Table 3.1.1 in a time not exceeding 30 minutes.

End of Chapter

Chapter 26

Diving Support Vessels and Diving Systems

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Section 1

General

1.1 Application

1.1.1 The requirements of this Chapter are applicable to diving support vessels and diving systems. The Chapter covers design, construction and survey requirements of diving support vessels, diving systems and hyperbaric evacuation systems.

1.1.2 Vessels equipped with a diving system complying with the requirements of Section 3 and having diving support capability specified in Section 2 will be eligible for class notation “DSV”.

1.1.3 Diving systems covered by this Chapter may be fitted and operated from a ship, barge, mobile offshore platform or fixed offshore installation.

1.1.4 Surface and Saturation diving systems complying with the requirements would be eligible for Class notation “**DIV - SURF**” or “**DIV - SAT**” respectively. Diving systems built under the survey and in accordance with the Rules of IRS, using materials and components tested by IRS in conformity with its Rules, would receive the distinguishing mark 卐 before the Character of Class or Class Notation.

1.1.5 Classification of diving systems is carried out independent of the classification of the vessel.

1.1.6 Whilst the requirements of this Chapter are considered to meet the requirements of IMO Resolution A.831(19), as amended, the statutory requirements of the Flag Administration, where the ship, barge, mobile offshore platform or fixed offshore platform is registered, are to be complied with.

1.1.7 The certification of arrangements, systems and individual components based on recognised standards, codes, national regulations, etc., may be accepted subject to special consideration by IRS in each case.

1.1.8 The classification certificate of the diving system will indicate the name and, where applicable, the IMO number of the vessel on which it is installed. The following information will be indicated in the attachment to the classification certificate of the diving system :

- Description of the diving system including operational limitations and conditions of intended use;
- Codes and standards to which the diving system complies.

1.1.9 Diving systems classed by IRS would be entered in the Register Book. For diving systems of non-standard design, IRS may require additional checks/ tests/ surveys and may make appropriate entries in the Class Certificate and the Register Book.

1.1.10 In general, all diving systems are to undergo initial surveys for classification and periodical surveys thereafter, as defined in Pt.1, Ch.2 of the Rules. Detailed requirements for initial classification and survey of diving systems are covered in Pt. 1, Ch. 2, Sec. 18.

1.1.11 Operational records for diving systems are to be maintained in which details of operations (diving operations, maintenance, damages, repairs, etc.) are to be entered. The record is to be submitted to IRS Surveyors on request.

1.2 Assumptions

1.2.1 The classification of diving systems is based upon the assumption that:

- The diving systems would be properly maintained and operated by competent personnel;

- A pre-check procedure would be followed to ensure the proper functioning of all systems and components before start of each operation;

- The current, wave and wind conditions will be within the design limits for handling and positioning systems; and

- Any significant repairs or replacement of components of the diving system would be carried out to the satisfaction of the IRS Surveyors.

1.3 Definitions

1.3.1 For the purpose of this Chapter the terms used have the meanings defined in the following paragraphs unless expressly provided otherwise.

1.3.2 **Administration** means the Government of the State whose flag a ship or floating structure which carries a diving system is entitled to fly or in which the ship or floating structure is registered.

1.3.3 **Bottle** means a pressure container for the storage and transport of gases under pressure.

1.3.4 **Breathing gas/breathing mixture** means all gases/mixtures of gases which are used for breathing during diving operations.

1.3.5 **Surface compression chamber** means a pressure vessel for human occupancy (PVHO) installed on the diving support vessel/ platform with means of controlling the pressure inside the chamber.

1.3.6 **Diving Depth** means the water depth or equivalent pressure, to which the diver is exposed to at any time during a dive or inside a surface compression chamber or a diving bell or inside a pressure vessel for human occupancy (PVHO).

1.3.7 **Diving bell** means a submersible pressure vessel for human occupancy, including its fitted equipment, for transfer of diving personnel under pressure between the work location and the surface compression chamber.

1.3.8 **Diving system** means the whole plant and equipment necessary for the conduct of diving operations and for the evacuation of divers.

1.3.8.1 **Fixed system** means a diving system installed permanently on ships or floating structures.

1.3.8.2 **Temporary system** means a diving system installed on ships or floating structures for a period not exceeding one year.

1.3.9 **Diving Unit** means the combined diving system and diving vessel/ platform, installed and integrated to conduct diving operations.

1.3.10 **Essential Services** (in the context of this Chapter) means the services for maintaining the diving system functionality with regard to the safety, health and environment of the divers in a hyperbaric environment and at the worksite.

1.3.11 **Hyperbaric Evacuation** Hyperbaric evacuation means the emergency transport of divers under pressure from a diving unit to a place of safety where planned decompression can be completed.

1.3.12 **Hyperbaric Survival Craft (HBSC)** means a pressure vessel for human occupancy (PVHO) and associated support plant and equipment whereby divers under pressure can be safely evacuated from a ship or floating structure to a place where planned decompression can be carried out.

1.3.13 **Handling system / Launch and recovery system (LARS)** means the plant and equipment necessary for raising, lowering and transporting divers between the diving vessel/ platform and/ or the surface compression chamber into and from the sea or onto a support vessel, as the case may be These systems are used for the deployment and recovery of dive baskets, wet bells, diving bells, daughter-craft and HBSC.

1.3.14 **Hazardous areas** are those locations in which:

- an explosive gas-air mixture is continuously present, or present for long periods (zone 0);
- an explosive gas-air mixture is likely to occur in normal operation (zone 1);
- an explosive gas-air mixture it not likely to occur, and if it does it will only exist for a short time (zone 2).

1.3.15 **Life support system** means the gas supply, breathing gas system, decompression equipment, environmental control system and equipment required to provide a safe environment for the diving crew in the diving bell and the surface compression chamber under all ranges of pressure and conditions they may be exposed to during diving operations.

1.3.16 **Living compartment** means the part of the surface compression chamber which is intended to be used as the main habitation for the divers during diving operations and which is equipped for such purpose.

1.3.17 **Main components** of a diving system include the surface compression chamber, diving bell, handling system and fixed gas storage facilities.

1.3.18 **Mating device** means the equipment necessary for the connection and disconnection of a diving bell or an HBSC to a surface compression chamber.

1.3.19 **Maximum operating depth** of the diving system is the depth in [m] or feet of seawater equivalent to the maximum pressure for which the diving system is designed to operate.

1.3.20 **Pressure Vessel** means a container capable of withstanding an internal maximum working pressure greater than or equal to 1 bar.

1.3.21 **Umbilical** means the link between the diving system and the diving bell, PVHO, wet bell or diver and may contain surveillance, communication and power supply cables, breathing gas and hot water hoses. The hoisting and lowering strength member may be part of the umbilical.

1.3.22 **Closed circuit breathing system** means the system for supply of breathing gas to the diver and saving of exhaled gases for recirculation after scrubbing and replenishing.

1.3.23 **Diver heating** means the system for actively heating the divers in the water or in the inner area.

1.3.24 **Inner area** means the areas which are inside the chambers and the bell.

1.3.25 **Outer area** means the areas of the diving system that are exposed to the atmospheric conditions during the operation, i.e. outside the inner system and the room or area that surrounds or contains the diving system.

1.3.26 **Normal cubic metres** [Nm^3] is taken as volume of gas [m^3] at standard conditions of 0°C and 1.013 bar.

1.3.27 **Category A Machinery Spaces** are those spaces and trunks to such spaces as defined in the International Convention for the Safety of Life at Sea, 1974, as amended.

1.3.28 **Control Station** is a station in which one or more of the control and indicator functions are located. These may include controls and /or indicators for operation of all vital life support conditions, pressure controls, hyperbaric evacuation system, launch and recovery system, automatic sprinkler, fixed fire detection and fire alarm systems, temperature, humidity etc.

1.3.29 **LSA Code** means the International Life-Saving Appliance Code, as defined in SOLAS chapter III.

1.3.30 **Pressure vessel for human occupancy** (PVHO) means a container intended to be occupied by one or more persons that is capable of withstanding an internal or external pressure differential exceeding 0.14 bar. In general, surface compression chambers and diving bells are considered to be PVHOs.

1.3.31 **Wet bell** means a diver deployment and recovery device as a minimum fitted with a gas filled dome, a main supply umbilical from the surface (providing breathing gas and other service to a manifold inside the device), and diver excursion umbilicals terminated at the device.

1.4 Scope of Classification

1.4.1 The following parts and systems are covered by IRS certification of the diving systems (fitted onboard ship or as independent unit) and require to be inspected during manufacture and installation, as applicable:

- Chambers;
- Bell;
- Permanent gas containers;
- Other pressure vessels;
- Life support systems;
- Electrical systems;
- Fire protection, detection and extinction;
- Launch and recovery systems;
- Umbilicals;
- Pipes valves and fittings;
- Pumps and compressors;
- Helium reclamation plant;
- Gas analysers;
- Gas mixing units;
- Gas absorbers;
- Breathing systems;
- Depth gauges;
- Sanitary systems;
- Communication systems;
- Arrangement for hyperbaric rescue of divers in saturation.

1.5 Plans and particulars

1.5.1 Before commencement of construction, following plans/ particulars for diving related functions are to be submitted for approval/ information, as applicable:

a) For the total system

- Plans showing general arrangement of the diving system, location and supporting arrangement, locations of equipment, breathing gas storage, hazardous zones, interface with piping arrangement, electrical installations, etc
- Plans are to include information like maximum operating depth and maximum observation depth for the bell, maximum operation time, maximum number of divers in bell, maximum number of divers in chamber(s) and bell(s) and maximum operational sea state;
- Plans showing lay-out of control stations(s);
- Proposed programme for tests and trials of the system for normal operation and emergency use.

b) For pressure vessels

- Plans as required by Pt.4, Ch.5;
- Plans showing expansion allowances under working conditions for interconnected multi-vessels systems;
- Fabrication tolerances;
- Type of thermal insulation materials and particulars, e.g. flammability and specific heats of conductivity;
- Type of buoyancy materials and particulars, e.g. collapse water depth, specific weight, specific water absorption, etc.;
- Drawings and specifications of all windows including fabrication tolerances; and
- Media contained, filling pressures and temperatures;

- Calculations of thicknesses/ stresses and fatigue analysis.

c) Gas supply

- Piping diagrams and block diagrams for the entire gas supply system including valves and fittings;
- Details of umbilical structure;
- Proposed cleaning procedure for breathing gas system; and
- Details of gas analysis.

d) Life support system

- Piping diagrams, block diagrams and description of systems and equipment;
- Calculation of cooling and heating requirements;
- Description and plans of water supply and disposal systems;
- Description and design details of diver heating system;
- Details of valves and fittings; and
- Plans showing the arrangement and specifications of environmental control systems and equipment, diving crew facilities, sanitary and drainage systems.

e) Automation, communication and locating equipment

- General arrangement drawings/block diagrams of control equipment including list of measuring points;
- Description and details of the communication means between the diving control station and diving systems, including single line diagram;
- Equipment list covering sensors, instruments, etc.;
- Description of electronic components such as signal amplifiers, computers and peripheral units;
- General arrangement drawings and equipment lists for communication systems and signaling equipment; and
- Arrangement drawing and description of the TV system.

f) Electrical systems

- Single line distribution diagram and detailed diagram of the entire installation. The diagrams are to give information on full load, cable types and cross-sections, make, type and ratings of all fuse and switchgear;
- Failure scenarios, redundancy principles, emergency arrangement, etc.;
- Key diagrams of control and alarm circuits for all consumers;
- Plans showing arrangements of batteries with information about their make, type and rating;
- Single line diagrams of the communication systems; and
- Complete list of components and documentation on any tests carried out on all electrical equipment intended to be installed in the chamber and the bell.

g) Fire protection, detection and extinction

- List of all materials to be installed in the inner area, where possible with data on and/or evaluation of flammability in conditions under which the materials are to be used; and
- Plans and specifications of fire detection, fire alarm and fire extinction equipment for both the inner and outer areas.

h) Handling, transfer and mating systems

- A description of the system with details of operating conditions;
- Installation drawings;

- Detailed plans of transfer equipment, lifting equipment, mating equipment, substructures of handling equipment including winches;

- Detailed plans of interchangeable components and fittings;

- Piping systems and instrumentation details of hydraulic/ pneumatic systems, as applicable;

- Details of control and protective systems; and

- Details of hoist and guide ropes.

i) Umbilicals

- Plans and specifications giving details regarding particulars of conductors, minimum breaking load, minimum diameter of pulleys and drums;

- Details regarding maximum design load, elastic properties, weight per unit length.

j) Hyperbaric evacuation system

- Description of system;
- General arrangement and detailed plans of evacuation system; and
- Details of handling system and power supply.

1.5.2 Diving System Documentation

1.5.2.1 Technical and Operation Manual

.1 The diving system technical and operation manual is to include, but not limited to the following:

- the description of the system

- the list of diving equipment with reference and name of the manufacturer

- the user instructions to operate the system

- the inspection and testing programme of the diving system when installed on-board

- the emergency evacuation plan when divers are in hyperbaric chamber, when relevant.

1.5.2.2 Planned Maintenance System (PMS)

.1 Details of a planned maintenance system as defined in IMCA D-018, '*Code of Practice for the Initial and Periodic Examination and Testing of Diving Plant and Equipment*' (as amended) are to be submitted. The PMS is to include the requirements for the periodic examinations, testing and routine replacement of the components. The PMS is to ensure the traceability of the maintenance works carried out on the diving plant.

1.5.2.3 Redundancy and essential services

.1 All services supporting the divers directly, including environmental control, are to be considered as essential services as per SOLAS chapter II-1.

.2 No single failure of a component, either static or dynamic, nor a single inadvertent act should lead to life-threatening situations for any person, or to unacceptable damage to facilities or the environment. However, where a failure mode of a system or component cannot be addressed by redundancy, either:

.1 independent backup arrangements should be provided (e.g.,in the case of umbilical separation); or

.2 the system or component should be designed and built to recognized national or international standards that have appropriate safety factors for manned application in similar service conditions (e.g. PVHO rupture).

.3 Essential services for divers in the water are to have sufficient redundancy to ensure the divers can return to a safe haven within the diving bell or basket.

.4 Essential services for divers located in a diving bell or basket are to have sufficient redundancy to ensure the diver can return to a safe haven within the diving unit.

.5 Essential services for divers in the diving unit are to have sufficient redundancy to ensure that all planned decompressions can

be completed at the planned normal rate. The ability to provide hyperbaric medical care should not be adversely affected.

.6 Emergency services are not to replace the required redundancies. Emergency services are to allow divers within the diving system or in the water to safely evacuate the diving unit.

.7 Emergency power services are to be available to provide services to the surface compression chamber complex in order to complete the normal planned decompression without main power.

.8 A systematic risk assessment of the diving unit is to be carried out in order to ensure compliance with the designed redundancy level and the suitability of emergency services.

.9 The systematic risk review should consider:

a) failure of a normally static component;

b) failure of an active component or system;

c) loss of any services to the diving unit; and

d) inadvertent and intentional acts.

.10 The risk assessment of the diving unit is to take into consideration already prepared reviews for the diving platform and diving system.

.11 The risk assessment is to be confirmed by a test programme, performed when the diving system is first installed, that clearly describes the expected behaviour of the diving unit and includes all necessary procedures to allow later testing to confirm the ongoing redundancy.

.12 The systematic risk review and corresponding test programme is to be fully documented and maintained available on board the diving unit. The test programme is to establish intervals for testing acceptable to the IRS/ the Administration to ensure redundancy levels are maintained.

.13 Risk mitigation

a) Probability assessment as a means of mitigation is not acceptable without the availability of suitable industry data applicable for personnel safety levels. Such data is also to be applicable to marine conditions.

b) Any mitigations that are based on repairs are to be assessed/ tested against time assumptions. Ensuring access to spares in an acceptable timescale is to be included in documentation required in the safety management system.

c) Any failures that allow compliance with 1.5.2.3.2 and 1.5.2.3, but leave the diving unit below the designed redundancy level are to be identified to ensure such failures will result in termination of appropriate operations.

d) Mitigations or potential hidden failures that are controlled by inspection and testing are to be addressed by maintenance routines.

Section 2

Requirements for Diving Support Vessels

2.1 General

2.1.1 The requirements specific to Diving Support Vessels for assignment of Class Notation “DSV” are covered in this section.

2.2 Stability and Position Keeping

2.2.1 The diving support vessel is to comply with the stability requirements stipulated for additional class notation “STS” in Pt. 5, Ch. 8 of the Rules.

2.2.2 The diving unit is to be able to recover any diving basket, wet bell or diving bell deployed by a handling system after sustaining side damage within the extents given in the *Guidelines for the design and construction of offshore supply vessels, 2006* (resolution MSC.235(82), as amended). The final list and trim in these conditions are to be within the design limits of list and trim for the handling system; ballasting post damage may be used to achieve this.

2.2.3 Essential diving equipment, including that on or above the main deck, is to remain accessible and operable in any stage of flooding for compliance with 2.2.1 and 2.2.2 above. Equipment below the main deck for Special Purpose Ships will be considered protected if watertight bulkheads are provided fore and aft and side divisions are provided equivalent to that required under the *Guidelines for the design and construction of offshore supply vessels, 2006* (resolution MSC.235(82), as amended).

2.2.4 Diving units conducting only surface-oriented diving are to comply with the requirements for cargo ships, as a minimum.

2.2.5 The diving support vessel is to have a position keeping system for safe diving operations. For ships equipped with a dynamic positioning system, the class notation DP 2 or DP 3 is required. Anchoring systems may be accepted for position keeping provided they are generally in accordance with the requirements given in Chapter 10 of the Rules for the Construction and Classification of Mobile Offshore Drilling Units.

2.2.6 Communication systems with sufficient redundancy are to be provided between the control station of positioning system and dive control station. A manually operated alarm system is also to be provided.

2.3 Plans and Particulars

2.3.1 In addition to the plans/ documentation required for hull, machinery and systems, the plans listed in 1.5 are also to be submitted to IRS for approval.

2.3.2 Detailed plans, drawings and procedures are to be prepared for all installation activities of the diving system.

2.4 Location of Diving Equipment and Design Requirements

2.4.1 The location of diving systems is to be appropriate for carrying out safe diving operations without being affected by the operation of propellers, thrusters or anchors.

2.4.2 Siting of diving systems in hazardous areas is to be avoided. Where, due to the requirements of diving operations, systems are sited in hazardous areas, the electrical equipment is to comply with the requirements for such equipment in the particular class of hazardous areas. Diving systems are not to be located in hazardous areas designated as Zone 0, as defined in 1.3.

2.4.3 Diving systems or their parts (including hyperbaric evacuation systems) sited on decks are to be afforded reasonable protection against water, icing or any damage that may occur due to other ship operations.

2.4.4 Diving systems situated on open decks are not to be located in the vicinity of ventilation openings from machinery spaces, exhausts or ventilation outlets from galley. In general, a minimum distance of 6 [m] between ventilation outlets or exhausts and diving system ventilation inlets is considered sufficient.

2.4.5 Diving systems are not to be installed close to sources of harmful noise. In the outer areas, noise limits are to be maintained at 75 dB(A), so that personnel may be able to communicate with each other.

2.4.6 The diving system and breathing gas storage facilities are not to be located in machinery spaces unless the machinery is associated with the diving system.

2.5 Supporting Structures of Diving Equipment

2.5.1 Diving systems including equipment are to be permanently attached to the ship structure and provisions are to be made on the ship for this purpose. The fastening arrangements are to be able to meet the required survival conditions of the ship or floating structure.

2.5.2 In general, the supporting structures for diving systems are to be designed in accordance with the requirements of Part 3 of the Rules. Collision loads are to be at least one half the weight of the surface compression chamber complex in the forward direction and one quarter the weight in the aft direction.

2.5.3 Special attention is to be paid to the design of supports for pressure vessels subject to expansion/ contraction due to temperature and pressure variations. Deflections of supports are to be considered in the design of the pressure vessel.

2.5.4 The supporting structure(s) of pressure vessels are to be able to withstand the loads during periodical hydrostatic testing with

acceptable levels of deflections. Temporary supports may be considered during testing if necessary.

2.5.5 Supporting structures and foundations for handling systems and lifting appliances are to be designed according to Part 3 of the Rules and/ or other recognized standards.

2.5.6 A dynamic load factor of at least 2.2 is to be applied for supports of lifting appliances used for handling manned objects such as surface bells, baskets or Hyperbaric Evacuation Systems.

2.5.7 The side structure of the moon pool is to be strengthened for possible impact loads from diving equipment guided through the moon pool.

2.6 Sea Water Inlets

2.6.1 Design of sea inlets fitted in the hull of the diving support vessel is to be specially considered in order to protect the divers and the sea inlet systems.

2.6.2 The mesh or aperture size of any protective structure is to be such as to prevent any part of the diver or his equipment being drawn into the intake at the velocity which would occur during pump operation.

2.6.3 It is recommended that water current in the immediate vicinity of the sea inlet is not to exceed 0.5 [m/sec].

2.6.4 Recommendations available in IMCA AODC 055 '*Protection of Water Intake Points for Diver Safety*' (as amended), may be referred for more details.

2.7 Piping Systems for Oxygen and Mixed Gas

2.7.1 In general piping systems are to be designed in accordance with Pt. 4, Ch. 2 and Ch. 3 of the Rules. In addition, the following are to be ensured:

.1 Exhaust, venting and pressure relief gases from the diving system are to be released to the open air away from sources of ignition, personnel or other areas where the presence of those gases could be hazardous (such as engine room and galley exhausts). Means are to be provided to prevent any dangerous accumulation of gases.

.2 Piping systems carrying mixed gas with oxygen content greater than 22% or oxygen under high pressure are not to be located inside accommodation spaces and machinery spaces of category A or similar compartments.

.3 All high pressure piping is to be protected against mechanical damage.

.4 The system and arrangement of filters/ strainers are to be such as to maintain uninterrupted supply of gas to essential systems.

.5 Piping systems intended for use in breathing gas and oxygen systems are to be cleaned and tested for purity in accordance with an acceptable test method (for e.g. ASTM G93-03 '*Standard Practice for Cleaning Methods and Cleanliness Levels for Materials and Equipment used in Oxygen-enriched Environments*').

2.8 Gas Storage

2.8.1 Oxygen bottles are not to be stored near flammable substances. Oxygen is not to be stored or ducted in any form close to combustible substances or hydraulic equipment.

2.8.2 Oxygen bottles are to be stored in separate storage rooms with good ventilation arrangements. The rooms are to be provided with sensors for oxygen content giving an audio-visual alarm at a manned control station.

2.8.3 Two oxygen analyzers with audio-visual low-level/ high-level alarms are to be provided in enclosed spaces where gas mixtures with oxygen content less than 20% or higher than 22% are stored.

2.8.4 These analyzers are to be mounted such that one monitors the upper level and another monitors the lower level of the space.

2.9 Electrical Systems

2.9.1 In general electrical systems are to be provided in accordance with Pt. 4, Ch. 8 of the Rules. Additional/ special requirements for systems and equipment serving diving systems are given in the following :

.1 Plans/ Documentation to be submitted

.1 Design Document describing the philosophy of the overall electrical system with respect to operational modes, failure scenarios, redundancy principles, automatic changeover systems etc.

.2 General arrangement showing location of all important electrical equipment for diving system;

.3 Single line distribution system diagrams for the whole installation with information on full load, cable types and cross sections, and make, type and rating of fuse and switchgear for all distribution circuits;

.4 Calculations on load balance, including emergency consumption and battery capacities;

.5 Multi-wire diagrams of control and alarm circuits for all motors or other consumers;

.6 Plans showing arrangements of batteries, their make, type and capacity;

.7 Plans showing arrangement and single line diagrams of the communication system.

.2 Design Requirements

.1 All electrical equipment and installations, including power supply arrangements, are to be designed for the environment in which they will operate to minimize the risk of fire, explosion, electrical shock and emission of toxic gases to personnel, and galvanic action of the surface compression chamber or diving bell;

.2 Electrical cables are not to be routed close to gas piping;

.3 In the event of failure of the main source of electrical power supply to the diving system an independent source of electrical power is to be available for the safe termination of the diving operation;

.4 The ship's emergency source of electrical power may be used as an emergency source of electrical power for the diving system if it has sufficient capacity to supply the diving system and the emergency load for the vessel simultaneously;

.5 The alternative source of electrical power is to be located outside the space containing the main source of power;

.6 Main and emergency lighting are to be provided in all critical handling areas.

2.10 Fire Protection

2.10.1 The requirements for fire protection in Part 6 of the Rules are to be complied with, in general. Additional requirements for fire protection of areas containing diving equipment, auxiliaries and support systems are laid down in the following paragraphs.

2.10.2 Plans and Documentation

.1 Arrangement drawings of “outer area” spaces indicating the fire protection of the structural boundaries of these spaces. Details of insulation materials used are also to be provided;

.2 Plans and specifications of fire detection and fire extinguishing systems for ‘outer areas’.

2.10.3 Structural Fire Protection Requirements

.1 Boundaries separating enclosed “outer areas” from other enclosed spaces are to have A-60 fire integrity. Outer areas may be sub-divided into several spaces by A-0 class boundaries.

.2 Compartments in the interior of the vessel in which the diving system or its auxiliary equipment is carried are to be provided with A-60 boundaries.

.3 All doors between “outer area” and other adjacent enclosed spaces are to be of self-closing type.

4 At least one of the required escape routes from spaces not being part of “outer area” is to be independent of “outer area”.

.5 Non-metallic and insulation materials used in the diving system are to be of fire-retardant

type and non-hazardous in accordance with the FTP Code parts 2 and 5, as amended.

.6 Enclosed “outer areas” are to have separate mechanical ventilation with minimum 8 air changes per hour.

.7 Piping and cables essential for the operation of the diving system are regarded as part of the system. Where these transit from other compartments such as the main switchboard room or engine-room into an “outer area”, they are to be laid in separate structural ducts insulated to A-60 class standard.

.8 Vessels fitted with surface diving systems that do not include planned surface decompression, do not need to comply with 2.10.3.2. Further, they need not be fitted with automatic fire detection and alarm systems and fixed fire-extinguishing system.

2.10.4 Fire Detection and Alarm Systems

.1 The following requirements, as applicable, are to be complied with for outer areas in addition to the requirements in Part 6:

(a) The section or loop of fire detectors covering the “outer area” is not to cover other spaces;

(b) Fire detector panel is to be placed on the bridge with repeater panel at dive control room and in machinery control room;

(c) A visual alarm repeater is also to be situated at the diving control station.

2.10.5 Fire Extinguishing Systems

.1 Enclosed “Outer areas” situated are to be provided equipped with a fixed, manually operated fire-extinguishing system. The system is to be capable of being operated from the dive control station, bridge and/ or other locations, as necessary.

.2 The fire-extinguishing system is to be sufficient to cover the largest area bounded by A-0 class divisions.

.3 A fixed fire extinguishing system complying with the requirements of Part 6 of any of the following types is to be provided:

(a) Pressure water spraying system, approved for use in machinery spaces of Category A; or

(b) Fixed gas system.

.4 The fixed gas system is to be of a type non-hazardous to human beings,in the concentration expected in the space.

.5 When pressure vessels are situated in enclosed spaces, a manually actuated water spray system having an application rate of 10 [lit/ m^2/ min] of the horizontal projected area is to be provided to cool and protect such pressure vessels in the event of external fire.

.6 Fire hoses and nozzles provided on open decks are considered sufficient where pressure vessels are located in such areas.

.7 Where diving systems are located in open decks, the fire extinguishing arrangements in “outer areas would be specially considered.

2.10.6 Personnel Protection

.1 In addition to the fire fighters outfits required by Part 6 of the rules, an additional set is to be provided for each person required for the operation of the diving system during a fire.

.2 Control stations manned during bell recovery or launching of hyperbaric evacuation unit are to be provided with breathing apparatus.

2.10.7 Portable Fire-Extinguishers

.1 The spaces containing diving system and equipment are to be provided with sufficient number of portable fire-extinguishers so that one extinguisher is available in every 10 [m] walking distance.

2.11 Other Services

2.11.1 The diving unit is to be able to support all essential services required by the diving system, including in the event of a failure that may result in a hazardous situation. All services are to be configured to allow safe transfer to or from the diving system.

2.11.2 In order to achieve the above requirements, the following apply:

- services essential to the diving system are also to have separate independent backups;

- services are to be configured so as to prevent hazardous transfer of material or power to or from the diving system; and

all essential services required by the diving system should be in accordance with the diving system requirements.

Section 3

Requirements for Diving Systems

3.1 General

3.1.1 Requirements applicable to diving systems for assignment of class notation "**DIV-SURF**" and "**DIV-SAT**" are covered in this section.

3.1.2 In general, the notation "**DIV-SURF**" is to be assigned for diving systems, where diving operations are performed at such depths and durations that the decompression is carried out from the surface without resorting to the use of a closed diving bell.

3.1.3 In general, the notation "**DIV-SAT**" is to be assigned to saturation diving systems, where equipment is provided such that divers can live in a pressurized environment for several days or weeks. Divers are generally decompressed to surface pressure only once, at the end of their work.

3.1.4 As far as reasonable and practicable a diving system is to be designed to minimize human error and constructed so that the failure of any single component (determined, if necessary by an appropriate risk assessment) would not lead to a dangerous situation.

3.1.5 Diving systems and components thereof are to be designed for the service conditions under which they are certified to operate.

3.1.6 Materials for diving system components are to be suitable for their intended use.

3.1.7 All components in a diving system are to be designed, constructed and tested in accordance with applicable International or National standards recognised by IRS or proprietary specifications acceptable to IRS.

3.1.8 In the design of pressure vessels including accessories such as doors, hinges, closing mechanisms and penetrators, the effects of rough handling and accidents are to be considered in addition to design parameters such as pressure, temperature, vibration, operating and environmental conditions.

3.1.9 All components in a diving system are to be so designed, constructed and arranged as to permit easy cleaning, disinfection, inspection and maintenance.

3.1.10 A diving system is to include the control equipment necessary for safe performance of diving operations.

3.1.11 Diving systems generally include, but are not limited to, the items listed in Table 3.1.11

Table 3.1.11 : Indicative list of diving equipment		
Items	Saturation diving	Surface diving
Closed diving bell	*	
Wet diving bell or diver basket		*
Deck decompression chamber	*	*
Breathing supply system	*	*
Handling systems for the diving bell	*	*
Diving control stand	*	*
Clamping and under pressure transfer system	*	
Devices for controlling the atmosphere	*	*
Devices for the recovery and the purification of breathing gases	*	*
Means of production of hot water	*	
Hyperbaric Evacuation System	*	

3.2 Requirements for Diving Systems and Components

3.2.1 Environmental Conditions

3.2.1.1 Diving systems together with their accessories and fitted equipment are to be designed for the environmental conditions likely to occur at the point of installation or work location but in any case not less than that required by Table 3.2.1.1 and Table 3.2.1.2.

Table 3.2.1.1 : Minimum Design Inclinations				
Component location	**Athwartships**		**Fore-and-Aft**	
	Static	**Dynamic**	**Static**	**Dynamic**
Compression chambers and other surface installations				
On ship	± 15°	± 22.5°	± 5°	± 10°
On mobile offshore unit (on semi-submersible)	± 15°	-	± 15°	-
Diving bells and components therein	± 22.5°	± 45°	-	-

Table 3.2.1.2 : Minimum Environmental Conditions			
Location	**Temperature**	**Relative Humidity**	**Other Conditions**
In chambers	5 to 55°C	100%	Salty air
Outside chambers in air [1,2]	- 10 to + 55°C	100%	
Outside chambers in water	- 2 to 32°C	-	Salt water containing 3.5% salt
Control rooms	5 to 55°C	80%	-
1 In the case of facilities installed on the open deck, allowance is to be made for icing and temporary inundation with salt water and spray. 2 Other values will be specially considered for installation in closed spaces.			

3.2.2 Surface Compression Chambers

.1 A diving system is to, as a minimum, include either one surface compression chamber with two separate compartments, or two interconnected separate chambers so designed as to permit ingress or egress of personnel while one compartment or chamber remains pressurized. Compartments are to have a specified maximum number of occupants. This capacity will define the required outfitting of the compartment or chamber including the number of bunks, built-

in breathing systems (BIBS) and the ergonomic design.

.2 Where a surface compression chamber is to be used in circumstances in which a person is intended to remain under pressure for a continuous period of more than 12 hours, it is to be so arranged as to allow most divers to stand upright and to stretch out comfortably on their bunks. The smaller of the two compartments is to be large enough for at least two persons. One of these compartments is to be a living compartment.

.3 The living compartment and other compartments intended to be used for decompression are to have a lock through which provisions, medicine and equipment may be passed into the chamber while its occupants remain under pressure. The dimensions of the service lock are to be adequate to enable essential supplies to be transferred into the surface compression chamber. Service locks are to be designed to prevent accidental opening under pressure and, where necessary, interlocks are to be provided for this purpose.

.4 All doors are to be so designed that locking mechanisms, if provided, can be operated from both sides.

.5 Each pressure compartment is to have view ports to allow observation of all occupants from the outside.

.6 A surface compression chamber is to provide a suitable environment and facilities for the persons who use it, having regard to the type and duration of the diving operation. Where the chamber is intended to be occupied for more than 8 hours, toilet facilities are also to be provided. Toilet facilities capable of discharging the waste to the outside are to be fitted with suitable interlocks.

.7 The diving system is to be capable of allowing the safe transfer of a person under pressure from the diving bell or HBSC to the surface compression chamber (and vice versa).

.8 Facilities are to be provided for keeping the partial pressure of the CO_2 in the chamber atmosphere permanently below 0.005 bar assuming a CO_2 production rate of 0.05 [Nm^3/hour] per diver.

.9 Diving systems using mixed gas and designed for operating periods of more than 12 hours at a time are to be capable, under steady conditions, of keeping the temperature in the surface compression chamber constant to $\pm$ 1°C in the 27 - 36°C range while maintaining a relative humidity of at least 50 per cent.

.10 In the steady state, the permanent noise level (over 8 hours) in the living compartment and surface compression chamber is not to exceed 65 dB(A).

3.2.3 Diving bells

.1 A diving bell is to :

- be provided with adequate protection against mechanical damage during handling operations;

- be equipped with one extra lifting point designed to take the entire dry weight of the bell including ballast and equipment as well as the weight of the divers staying on in the bell; and

- be equipped with means whereby each diver using the bell is able to enter and leave it safely as well as with means for taking an unconscious diver up into a dry bell.

- have a service lock through which provisions, medicine and equipment may be passed into the diving bell while its occupants remain under pressure. Service locks are to be designed to prevent accidental opening under pressure and, where necessary, interlocks should be provided for this purpose. The dimensions of the service lock are to be adequate to enable essential supplies to be transferred into the diving bell.

- be fitted with a manifold (which is clearly marked and suitably protected), at a suitable point close to the main lifting attachment which is to include connections to the following main services :

  ¾ inch NPT (female) - for hot water
  - ½ inch NPT (female) – for breathing mixture

The manifold is to also incorporate connectors for the following:

- internal pressure;
- sampling of internal gas;
- communication; and
- electrical power.

.2 Diving bell doors are to be so designed as to prevent accidental opening during normal operations. All doors are to be so designed that locking mechanisms, if provided, can be operated from both sides.

.3 A diving bell is to provide a suitable environment and facilities for the persons who use it, having regard to the type and duration of the diving operation.

.4 Each diving bell is to have view ports that allow occupants to be observed and as far as practicable allow an occupant to observe divers outside the bell.

.5 Diving bells are to be so designed as to provide adequate space for the number of occupants envisaged, together with the equipment.

.6 For diving systems with only one diving bell, a suitable arrangement is to be in place to reconnect a lost bell to the diving system and allow the divers to return safely to the surface compression chamber.

.7 In diving bells, it is to be possible to keep the partial pressure of the CO_2 below 0.015 bar as a minimum requirement assuming a CO_2 production rate as specified in 3.2.2.8. Under emergency conditions, it is to be possible to hold the partial pressure of CO_2 below 0.02 bar for at least 24 hours.

3.2.4 Wet bells and dive baskets used for the deployment and recovery of surface oriented divers

.1 All diving platforms are to provide for safe deployment and recovery of the divers, including during a diving emergency. During any deployment, fully dressed divers should not have to climb a ladder more than 2 [m], or stairs that should be fitted with a handrail more than 4 [m], down to or up from the water surface.

.2 Wet bells and dive baskets, with the exception of those specifically for a rescue diver, are to be fitted out to carry a minimum of two divers. The structure should prevent the divers from falling out during operations and enable the recovery of a helpless diver while maintaining the safety of the rescue diver.

.3 For wet bell gas services, the onboard control point is to ensure a primary and secondary supply to both a working diver and an in-water standby diver. A diver's bail-out is an emergency supply and not considered to be a secondary supply.

.4 A minimum of 30 minutes of onboard emergency gas at the maximum planned depth is to be provided. This includes a breathing system for each diver independent of their main and bail-out diving equipment.

3.2.5 PVHO forming part of the HBSC

.1 Where a PVHO forms part of a HBSC, the following are to be complied with:

- The PVHO is to provide a suitable environment and facilities for the persons who use it, having regard to the type and duration of the evacuation.

- Where the PVHO is intended to be occupied for more than 8 hours, toilet facilities are also to be provided. Toilet facilities capable of discharging the waste to the outside are to be fitted with suitable interlocks.

- The means provided for access into the PVHO should be such as to allow safe access to or from the surface compression chambers. Interlocks are to be provided to prevent the inadvertent release of the HBSC from the surface compression chamber while the access trunking is pressurized. The mating flange is to be adequately protected from damage at all times including during the launch and recovery stages.

- All doors are to be designed to prevent accidental opening during normal operations and, if any locking mechanism is provided, it is to be operable from both sides.

- Arrangements are to be provided to enable a helpless diver to be taken into the HBSC PVHO.

- The seating or other arrangements provided are to be designed for the maximum number of occupants and provide an adequate degree of protection to the divers from impact collisions during launch and while the HBSC is deployed.

- The PVHO is to have a service lock through which provisions, medicine and equipment may be passed into the PVHO while its occupants remain under pressure. Locks are to be designed to prevent accidental opening under pressure and, where necessary, interlocks are to be provided for this purpose. The dimensions of the service lock are to be adequate to enable essential supplies to be transferred into the PVHO.

- Arrangements are to be provided to allow the occupants to be observed. Viewports are to be protected and situated so that the risk of damage is minimized.

- Where it is intended to carry out decompression of the divers after hyperbaric evacuation in another surface compression chamber, then consideration is to be given to the suitability of the mating arrangements on that surface compression chamber. Where necessary, suitable adapter and clamping arrangements are to be provided on the HBSC in accordance with Fig. 3.2.5.1 (a) and on the other surface compression chamber (for e.g. a hyperbaric reception facility) in accordance with Fig. 3.2.5.1 (b).

- The HBSC is to be fitted with a manifold (clearly marked and suitably protected) at a suitable point. The manifold is to incorporate international standard connections for the following services:

    - Internal pressure (diving depth monitoring)
    - Communications
    - Electric power supply
    - Analysis of HBSC PVHO internal environment
    - Oxygen addition
    - Built-in breathing system (BIBS) supply
    - Blow-down (Pressurization)
    - Exhaust
    - Hot water supply and return
    - Chilled water supply and return

- PVHO locks and openings that are, or may be, submerged are to be designed to be mechanically restrained to prevent loss of seal and water ingress at lower hyperbaric pressures during deployment and in any sea state. This is to include the effects of the hydrodynamic impacts of free-fall HBSC launch requirements.

- All open flange faces that may be exposed both when in air and water are to be suitably protected or provided with easily replaceable sealing faces.

- When the HBSC is waterborne, it is to be possible to access and egress the HBSC from atmospheric pressure.

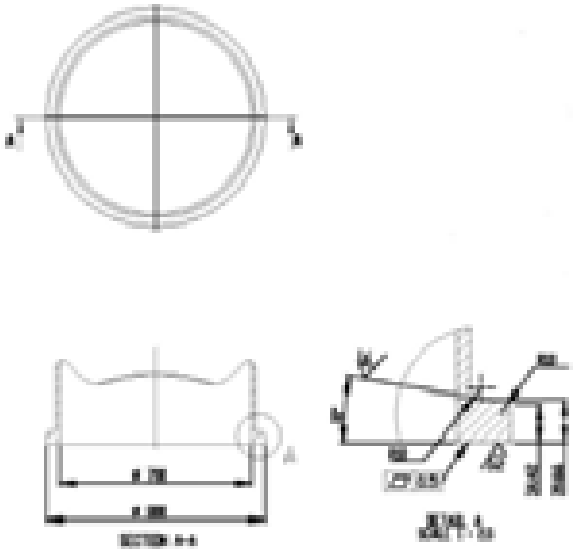

Fig. 3.2.5.1 (a): HBSC PVHO adapter and clamping arrangement

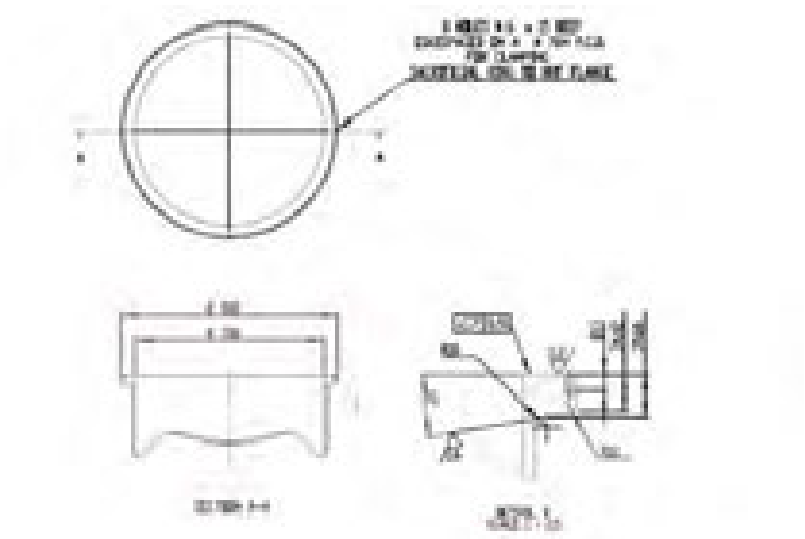

Fig. 3.2.5.1 (b): Hyperbaric reception facility PVHO adapter and clamping arrangement

3.2.6 Pipes, Valves, Fittings and Hoses

.1 Pipe systems are to be so designed as to minimize the noise inside the diving bell and surface compression chamber during normal operation.

.2 A surface compression chamber is to be equipped with such valves, gauges and other fittings as are necessary to control and indicate the internal pressure and safe environment of each compartment from outside the chamber at a centralized position.

.3 Valves, gauges and other fittings are to be provided outside the bell/ PVHO as necessary to control and indicate the pressure and safe environment within the diving bell/ PVHO. The external pressure on the diving bell/ PVHO is also to be indicated inside the bell/ PVHO.

.4 All pipe penetrations on chambers are to be fitted with two shutoff devices as close to the penetration as practicable. Where appropriate, one device is to be a non-return valve.

.5 All surface compression chambers and diving bells which may be pressurized separately are to be fitted with over pressure alarms or pressure relief valves. If pressure relief valves are fitted, a quick-operating manual shutoff valve is to be installed between

the chamber and the pressure relief valve and is to be wired opened with a frangible wire. This valve is to be readily accessible to the attendant monitoring the operation of the chamber. All other pressure vessels and bottles are to be fitted with a pressure relief device.

.6 Piping systems which may be subjected to a higher pressure than designed for are to be fitted with a pressure relief device. PVHO pressure-relief device(s) and associated pipework are to be tested for maximum gas volume flow.

.7 All materials used in oxygen system are to be compatible with oxygen at the working pressure and flow rate.

.8 The use of high-pressure oxygen piping is to be minimized by the fitting of pressure reducing devices, as close as practicable to the storage cylinders or pressure vessels.

.9 Flexible hoses, except for umbilicals, are to be reduced to a minimum.

.10 Hoses for oxygen, as far as practicable, are to be of fire-retardant construction.

.11 Piping systems carrying mixed gas or oxygen under high pressure are not to be arranged inside accommodation spaces, engine rooms or similar compartments.

.12 Exhaust lines are to be fitted with an anti-suction device on the inlet side.

.13 Gases vented from the diving system are to be vented to the open air away from sources of ignition, personnel or any area where the presence of those gases could be hazardous.

.14 All high-pressure piping is to be well protected against mechanical damage.

.15 Piping systems containing gases with more than 22 per cent oxygen are to be treated as systems containing pure oxygen.

.16 Oxygen systems with pressure greater than 1.72 bar must have slow opening shutoff valves except pressure boundary shutoff valves.

3.2.7 Breathing gas supply, storage and temperature control

.1 Each surface compression chamber and diving bell is to be fitted with adequate equipment for supplying and maintaining the appropriate breathing mixtures to its occupants at all depths down to maximum operating depth. When adding pure oxygen to the chamber, a separate piping system is to be provided. Internal oxygen inlets are be positioned so as to ensure appropriate mixing of oxygen within the PVHO.

.2 In addition to the system mentioned in 3.2.7.1 each surface compression chamber and diving bell is to contain a separately controlled built-in breathing system for oxygen, therapeutic gas or bottom mix gas with at least one mask per occupant stored inside each separately pressurized compartment and means are to be provided to prevent any dangerous accumulation of gases. A spare mask and its inlet and outlet connection points are also to be provided per compartment.

.3 The diving bell and HBSC are to be designed with a self-contained breathing gas system capable of maintaining a satisfactory concentration of breathing gas for the occupants for a period of at least 24 hours and 72 hours respectively, at their maximum operating depth. Provision is also to be made for equipment and controls for temperature control and thermal balance for occupants.

.4 Oxygen bottles are to be installed in a well-ventilated location.

.5 Oxygen bottles are not to be stored near flammable substances.

.6 The diving system and breathing gas storage facilities are not to be sited in machinery spaces if the machinery is not associated with the diving system. Where, due to the requirements of diving operations, systems are sited in hazardous areas, the electrical equipment is to comply with the requirements for such equipment in hazardous areas. Diving systems are not permitted in hazardous areas designated as Zone 0.

.7 A diving system is to include adequate plant and equipment to maintain the divers in safe thermal balance during normal operations.

.8 Means are to be provided to maintain the divers within the diving bell in thermal balance in an emergency for at least 24 hours. Such requirements may be satisfied by the use of passive means carried in the bell.

.9 For piping systems and gas storage cylinders/pressure vessels the following colour code is to be used:

Name	Symbol	Colour Code
Oxygen	O_2	White
Nitrogen	N_2	Black
Air	Air	White and black
Carbon dioxide	CO_2	Grey
Helium	He	Brown
Oxygen-helium mix gas	O_2He	White and brown
Hydrogen	H2	Red

In addition, each cylinder/ pressure vessel is to be marked with the name and symbol given above of the gases it contains. The marking and colour coding of the gas storage cylinders are to be visible from the valve end.

3.2.8 Handling system for diving bells/ surface-oriented diving

.1 A diving system is to be equipped with a main handling system to ensure safe transportation of the diving bell between the work location and the surface compression chamber. In the case of surface-oriented diving, the handling system is to ensure deployment and recovery of a dive basket or wet bell to and from maximum deployment depth. Where applicable, a diving system is to be equipped with a handling system to ensure the safe launch and recovery of a diving daughter-craft to and from the water.

.2 The handling system is to be designed with adequate safety factors, in accordance with recognised codes of construction, considering the environmental and operating conditions, including the dynamic loads which are encountered while handling the diving bell, dive basket or wet bell through the air-water interface.

.3 The handling system is to enable smooth and easily controllable handling of the diving bell.

.4 The lowering of diving bells or dive basket or wet bell or daughter craft under normal conditions is not to be controlled by brake, but by the drive system of the winches.

.5 If the energy supply to the handling system fails, brakes are to be engaged automatically.

.6 In the event of single component failure of the main handling system, an alternative means is to be provided whereby the bell/ dive basket or wet bell or daughter craft can be returned to the surface compression chamber/ embarkation point respectively. In addition, provisions are to be made for emergency retrieval of the bell if the main and alternative means fail. If this involves buoyant ascent, the bell is to have sufficient stability to maintain a substantially upright position and means are to be provided to prevent accidental release of the ballast weights. In case of surface-oriented diving, if the handling system for the working and standby divers are combined, then the failure of a single component is not to compromise the ability of the standby system to perform an emergency recovery.

.7 Handling systems and mating devices are to enable easy and firm connection or disconnection of a diving bell to a surface compression chamber, even under conditions where the support ship or floating structure is rolling, pitching or listing to predetermined angles. In the case of surface-oriented diving, the handling system is to enable smooth and easily controllable handling of the dive basket, wet bell or daughter craft, when recovered to the embarkation point, even under conditions where the diving platform is rolling, pitching or listing to predetermined angles.

.8 Where a power actuating system is used for mating operations, an auxiliary power actuating system or an appropriate means are to be provided to connect a diving bell to a surface compression chamber, in the event of failure of the normal power actuating system.

.9 The design of the handling system is to consider protection and routeing of the diving umbilical to prevent damage.

.10 In case of surface-oriented diving, handling system for primary access and egress is not required where the diver has to climb no more than 2 [m] above the water surface on a ladder, or no more than 4 [m] on stairs; the stairs are to be fitted with a handrail. However, means are to be provided:

- for diver access and egress from the water; and

- to recover a helpless diver in an emergency.

3.2.9 Other pressure vessels not intended for human occupancy

.1 Special attention is to be paid to the design and choice of material for the construction of pressure vessels containing a volume percentage higher than 22% oxygen.

.2 Oxygen and gases with an oxygen volume percentage higher than 22 per cent are to be stored in bottles or pressure vessels exclusively intended for such gases.

3.2.10 Interface between diving system and the ship or floating structure

.1 The diving system and breathing gas facilities are to be arranged in spaces or locations which are adequately ventilated and provided with suitable electric lighting.

.2 When any part of the diving system is sited on deck, particular consideration is to be given to providing reasonable protection from the sea, icing or any damage which may result from other activities on board the ship or floating structure.

.3 Provision is to be made to ensure that the diving system and auxiliary equipment are securely fastened to the ship or floating structure and that adjacent equipment is similarly secured. Consideration is to be given to the relative movement between the components of the system. In addition, the fastening arrangements are to be able to meet any required survival conditions of the ship or floating structure.

3.2.11 Fire protection, detection and extinction

.1 All materials and equipment used in connection with the diving system are to be, as far as is reasonably practicable, of fire-retardant type in order to minimize the risk of fire and sources of ignition.

.2 Spaces in the interior of ships or floating structures in which the diving or its auxiliary equipment is carried are to be provided with structural fire protection in a way similar to control stations (as defined in Pt.6, Ch.3) bounding main zones.

.3 Interior spaces containing diving equipment such as surface compression chambers, diving bells, gas storage, compressors and control stands are to be covered with an automatic fire detection and alarm system and a suitable fixed fire-extinguishing system.

.4 Portable fire extinguishers of approved types and designs are to be distributed throughout the space containing the diving system. One of the portable fire-extinguisher is to be stowed near the entrance to that space.

.5 When pressure vessels are situated in enclosed spaces, a manually actuated water spray system having an application rate of 10 [lit/m^2/per minute] of the horizontal projected area is to be provided to cool and protect such pressure vessels in the event of external fire. When pressure vessels are situated on open decks, fire hoses may be considered as providing the necessary protection.

.6 Each compartment in a surface compression chamber is to have a suitable means of extinguishing a fire in the interior which would provide rapid and efficient distribution of the extinguishing agent to any part of the chamber.

3.2.12 Electrical Systems

.1 All electrical equipment and installations, including power supply arrangements, are to be designed for the environment in which they will operate to minimize the risk of fire, explosion, electrical shock and emission of toxic gases to personnel, and galvanic action of the surface compression chamber or diving bell.

.2 Each surface compression chamber and diving bell are to have adequate means of normal and emergency lighting to allow an occupant to read gauges and operate the system within each compartment.

.3 Essential services as defined in chapter 2, as well as other services to ensure minimum comfortable conditions of habitability are to be identified and the system is to have the capacity to supply all essential services during the planned operations.

.4 In an emergency the diving bell and HBSC are to have sufficient electrical power for survivability of at least 24 hours for the diving bell and 72 hours for the HBSC respectively.

.5 The diving system switchboards (main and emergency) are to be installed in separate compartments. All emergency electric lighting systems are to be separated from the compartments housing the main and emergency switchboards.

.6 The emergency load for the diving system needs to include electrical power to systems that are essential for maintaining life support and for returning divers to a point of safety. However, the distribution system may be designed for staged reconnection/ disconnection of emergency consumers. Consumers, such as handling systems/ diver hot water, may be disconnected after divers in the water are recovered to the surface compression chamber complex. Similarly emergency consumers such as environmental control of the surface compression chamber complex (that are not time-critical) may be connected manually after switchover to emergency electrical power. Where manual intervention is required to manage the emergency electrical power system, these switchboards are to be easily accessible within the diving system area.

.7 An emergency source of electrical power is to be provided for the diving system complying with the minimum requirements of SOLAS regulation II-1/43, as well as the following:

- where using the diving platform's emergency electrical power, there is to be enough electrical power capacity to supply the emergency load for the diving system and for the diving platform simultaneously;
- have a minimum fuel capacity of 18 hours and be able to be refuelled;
- be able to safely terminate the diving operation, including decompression of the divers;
- be suitably protected within a deck house structure; and
- be suitably ventilated, to ensure continuous operations in the design environment.

.8 Emergency lighting as defined in SOLAS regulations II-1/43.2.1 and 43.2.2 is to be additionally provided for

- each PVHO location
- each handling system location
- associated diving equipment not housed within the control room or machinery spaces of the diving system.

.9 Battery charging arrangements are to be designed to prevent overcharging under normal or fault conditions. Battery storage compartments are to be provided with means to prevent over pressurization and vent to a safe place. When battery charger /battery combinations are used as DC power supply systems, adequate measures are to be taken to keep the voltage within specified limits.

.10 Electrical equipment fitted within the PVHO are to be suitable for the intended purpose, including hyperbaric use and the specified gas, and high humidity levels and marine applications. Additionally,:

- electrical enclosures subject to pressure are to include suitable gas relief devices, where applicable;

- cabling and electrical components are not to emit gas toxic volatiles that create a hazardous environment;

- equipment supply voltage is to be kept at a minimum; and

- electrical power services, including battery arrangements, are to be suitable for hyperbaric use.

.11 Electrical and electronic diving equipment are to comply with the requirements for electromagnetic compatibility in the *'General requirements for electromagnetic compatibility (EMC) for all electrical and electronic ship's equipment'* (Resolution A.813(19)).

.12 In-water and hyperbaric electrical equipment are to comply with the safe voltage requirements in Table 3.2.12.12 on safe voltage, body resistance and current. Additionally:

- For electrically heated suits, the diver's body resistance is to be 100 ohms and the safe body current is to be divided by 2.5 for all applications.

- Electrical equipment used subsea and in hyperbaric conditions are to be supplied from a transformer with the

secondary winding isolated, such that there is not an obvious path if a fault develops. Isolation requirements are to include high-integrity isolation components and a safe insulator barrier.

- Residual current devices fitted are to have a reaction time of less than 20 [ms].

.13 For diving equipment and plant that have automated functions, a systematic engineering assessment of the diving system and its subsystems is to be carried out (e.g. FMEA). The assessment is to identify all components and control systems that contain automated functions during normal operation, maintenance and testing phases within the diving system.

Table 3.2.12.12 Safe voltage requirements for in-water and hyperbaric electrical equipment

Supply	Safe Body Current (mA)	Body Route Resistance Ω (Ohms)	Safe Voltage	
			Maximum V	Nominal V
DC without a suitable trip device	40	750	30	24
AC without a suitable trip device	10	750	7.5	6
DC with a suitable trip device	570	500	285	250
AC with a suitable trip device	500	500	250	220
Note: A suitable trip device is one with a reaction time of 20ms or less.				

3.2.13 Control Systems

.1 The diving system is to be so arranged as to ensure that centralized control of the safe operation of the system can be maintained under all weather conditions.

.2 As a minimum, facilities are to be provided at the central dive control position to monitor the values of the parameters as indicated in the following tables (as relevant and applicable):

Table 3.2.13 (a): PVHO Compartment Monitoring

Parameters	**Compartments**		
	Surface compression chamber (Each Lock)	**Diving bell**	**HBSC**
Pressure or diving depth [1,2]	X	X[2]	X
Temperature [1,3]	X		X
Humidity	X		X
Oxygen partial pressure [1,3]	X	X	X
CO_2 partial pressure[1,3]	X	X	X
Video Surveillance	X	X	X
Special Environments			
Contaminants[4]	-	X	-

1 These parameters are to be indicated continuously.

2 Pressure or depth both inside and outside bell is to be indicated.

3 High and low alarms are to be provided (audible and visible).

4 As identified in project working areas requiring this monitoring, such as H_2S and hydrocarbon contaminated worksites.

Table 3.2.13 (b): Diver Monitoring					
Parameters					
	Diver 1	**Diver 2**	**Standby Diver**	**Reclaim Gas**	**Machinery**
Pressure or diving depth [1]	X	X	X	-	-
Oxygen partial pressure [1,2]	X	X	X	X	-
CO_2 partial pressure[1,2]	-	-	-	X	-
CO[1,2]	-	-	-	X	-
Diver's suit heating medium [1,2]	X	X	X	-	
1. These parameters are to be indicated continuously. 2. A high and low alarms is to be fitted (audible and visible).					

Table 3.2.13 (c): Central Dive Control Monitored Pressurization and Breathing Gas Sources		
	Primary Source of Gas	**Independent Secondary Source of Gas**
Surface Oriented Diving		
One Working Diver	One	One
Two Working Divers – Alternative 1	One per diver	One per two divers
Two Working Divers – Alternative 2	One for two divers	One per diver
Surface Standby Diver	One (separate from working divers)	One (Common with working divers' secondary source)
Wet Bell Services	One	One (Can be from onboard gas)
Saturation Diving		
Diving Bell	One (from surface)	One (can be from surface or onboard gas)
Working Divers	One (from surface)	
Bell Man	One onboard gas or surface	One common with working divers
Oxygen	One onboard gas	-

Table 3.2.13 (d): Central Saturation Control Monitored Pressurization and Breathing Gas Sources		
	Primary Source of Gas	**Independent Secondary Source of Gas**
Each Chamber Compartment	One	One (this can be a separate source and common pipe to the compartment)

Notes:

1. The term "source" is used to describe a means of providing the breathing or pressurization gas going into the control panel. The term "supply" is used to refer to the breathing or pressurization gas going to the diver/ dive basket/ wet-bell/ PVHO from the control panel.

2. The fundamental principle for the provision of breathing gas to divers is that any diver is to have ready access to two sources of breathing gas (a primary and a secondary source), where, at least one of which is to be supplied solely for the individual's own use, i.e. provide an independent supply of breathing gas.

3. The diver's bail-out is not considered as a primary or secondary source. The air supply to each diver is to be arranged such that if one line fails then this does not interfere with the supply to another diver.

.3 Provision is to be made within the bell and HBSC for an independent means of monitoring oxygen and carbon dioxide levels.

.4 The pressure gauges of compression chambers are to give a reading accurate to ± 0.3 per cent of the full instrument scale subject to a maximum deviation of 30 [cm] of water column. All other pressure gauges are to be accurate to 1 per cent of the full instrument scale.

.5 Oxygen injection systems are to be designed to prevent an uncontrolled flow of oxygen.

.6 Confined areas with gas control systems using gases with an oxygen content less than 20% or greater than 22% are to have continuous oxygen monitoring with high-level and low-level audio-visual alarms.

.7 Facilities are to be provided at the central control position to monitor and provide appropriate alarms for when any life-support parameter is outside acceptable limits.

3.2.14 Communications and Relocation Systems

.1 The communication system is to be arranged for direct two-way communication between the control stand and the :

- Diver(s) in water
- Standby diver
- each compartment of the surface compression chambers including the HBSC
- service lock positions when not located in immediate vicinity of the control stand
- back-up control panels
- diving system handling positions
- dynamic positioning control station, only when required by Section 2.2 (position keeping)
- diving platform bridge, ship's command center or any other operational control area.

and where applicable, the:

- diving bell, wet bell, dive basket and HBSC
- HBSC launch position
- HBSC operator’s position

- deck operations (wireless communication acceptable)

- crane operator (wireless communication acceptable)

- remotely operated vehicle (ROV) operator

- control stand on the daughter craft and the diving platform bridge (wireless communication acceptable)

- Systems are to be hardwired unless stated otherwise.

.2 Alternative means of communication with divers in the surface compression chamber, HBSC and diving bell are to be available in emergency. All secondary communication systems are to operate for a minimum of 30 minutes without main electrical power.

.3 Each PVHO is to be connected to a speech unscrambler when used with gas systems, including helium.

.4 A self-contained through-water communication system with a minimum operating duration of 24 hours is to be provided for emergency communication with diving bells when operating under water.

.5 Communications between dive control, the diving bell or wet bell, the standby diver and the divers in the water are to be recorded (audio and video) and retained for a minimum of 24 hours after the dive is completed.

.6 Where diving is carried out from a diving platform operating with dynamic positioning (DP), then an audio and visual alarm activated by the DP operator is to be fitted at the dive control stand/ station to inform the supervisor of the DP status. It is to be able to be tested before each dive when operating on DP.

.7 Primary and secondary means of communication between dive control and the bridge are to be provided and available at all times. One of the means of communication is to operate without the need for external electrical power supply. If the vessel is operating in DP, the primary means of communication is to be a direct hardwired two-way link.

.8 The diving unit general alarm is to be audible in the dive and saturation control stands. This alarm is to be mutable to ensure communication is not interrupted with the diver(s). Muting of the alarm is to be time-limited and there is to be visual indication of the alarm status.

.9 A diving bell is to have an emergency locating device with a frequency of 37.5 [kHz] designed to assist personnel on the surface in establishing and maintaining contact with the submerged diving bell if the umbilical to the surface is severed. The emergency locating device is to include the following components:

.1 Transponder

.1 The transponder is to be provided with a pressure housing capable of being operated to atleast the maximum operating depth of the diving bell containing batteries and equipped with saltwater activation contacts. The batteries are to be of the readily available “alkaline” type and, if possible, be interchangeable with those of the diver and surface interrogator/ receiver.

.2 The transponder is to be designed to operate with characteristics as indicated in Table 3.2.14.5.1.2

.2 Diver-held interrogator/ receiver

.1 The interrogator/ receiver is to be provided with a pressure housing capable of operating to atleast the maximum operating depth with pistol grip and compass. The front end is to contain the directional hydrophone array and the rear end, the three-digit LED display readout calibrated in metres. Controls are to be provided for "on/ off receiver gain" and "channel selection". The battery pack is to be of the readily available "alkaline" type and, if possible, be interchangeable with that of the interrogator and transponder.

.2 The interrogator/ receiver is to be designed to operate with characteristics as indicated in Table 3.2.14.5.2.2

.10 In addition to the communication systems referred to above, a standard bell emergency communication tapping code is to be adopted, as given in Table 3.2.14.6, for use between persons in the bell and rescue divers.

A copy of this tapping code is to be displayed inside and outside the bell and also in the dive control room.

Table 3.2.14.5.1.2 Transponder Characteristics
Common emergency reply frequency : 37.5 [kHz]
Individual interrogation frequencies - Channel A : 38.5 ± 0.05 [kHz] - Channel B : 39.5 ± 0.05 [kHz
Receiver Sensitivity : 15 dB referred to 1 [µbar]
Minimum Interrogation pulse width : 4 [ms]
Turnaround delay : 125.7 ± 0.2 [ms]
Reply frequency : 37.5 ± 0.05 [kHz]
Maximum interrogation rates - More than 20% of battery life remaining : Once per second - Less than 20% of battery life remaining : Once per 2 seconds
Minimum transponder output power : 85dB referred to 1 µbar at 1 m
Minimum transducer polar diagram : -6dB at ± 135° solid angle, centered on the transponder vertical axis and transmitting towards the surface
Minimum listening life in water : 10 weeks
Minimum battery life replying at 85 dB : 5 days

Table 3.2.14.5.2.2 Interrogator/ Receiver Characteristics
Common emergency reply frequency : 37.5 [kHz]
Individual interrogation frequencies - Channel A : 38.5 [kHz] - Channel B : 39.5 [kHz]
Minimum transmitter output power : 85dB referred to 1µbar at 1 m
Transmit pulse : 4 [ms]
Directivity : ± 15°
Capability to zero range on transponder
Maximum detectable ranges : more than 500 [m]

Table 3.2.14.6 Standard Bell Emergency Tapping Code

Tapping Code	Situation
3.3.3	Communication opening procedure (inside and outside)
1	Yes or affirmative or agreed
3	No or negative or disagreed
2.2	Repeat please
2	Stop
5	Have you a got a seal?
6	Stand by to be pulled up
1.2.1.2	Get ready for through water transfer (open your hatch)
2.3.2.3	You will NOT release your ballasts
4.4	Do release your ballast in 30 minutes from now
1.2.3	Do increase your pressure
3.3.3	Communication closing procedure (inside and outside)

3.3 Classification and Surveys of Diving Systems

3.3.1 Classification of diving systems built under survey of IRS:

a) All equipments/ components specified in 1.4 are required to be manufactured under survey of IRS Surveyors.

b) All systems are to be installed and tested under the supervision and to the satisfaction of IRS Surveyors.

3.3.2 Classification of diving systems not built under survey of IRS:

3.3.2.1 Details of the diving system, as required by 1.5, together with copies of manufacturer certificates for all components are to be submitted for review;

3.3.2.2 Upon satisfactory review of the documents stated in (a) above, a survey of the installation would be carried out by IRS Surveyors, based upon which the classification of the installation would be considered by IRS;

3.3.2.3 The documentation related to functional tests is to be submitted and where necessary, individual tests are to be repeated;

3.3.2.4 Existing Class and period of Class, as well as any requirements which have been made conditional upon the existing Class are to be intimated to IRS;

3.3.2.5 If the diving system holds the Class of another recognized Classification Society, the survey of individual parts may be deferred pending the next due date. A survey of these parts within the scope of an Annual Survey may be sufficient.

3.3.2.6 Special attention is to be paid to the following:

a) All decompression chambers, bells and other pressure vessels;

b) Safety valve arrangement on pressure vessels;

c) Gas storage, especially oxygen storage with regards to hazards during a potential fire;

d) Fire detection and fire protection arrangements;

e) Electrical installations, especially installations in hazardous areas;

f) Strength of structure supporting the diving equipment.

3.3.3 Suspension/ retention/ withdrawal of Class

.1 In general, the suspension/ retention/ withdrawal of Class of diving systems is to be in accordance with the provision of Pt.1, Ch. 1 of the Rules. The following aspects are to be additionally noted for diving systems:

.2 The period of Class of permanently installed diving systems would be identical to that of the machinery plant of the diving support ship or floating structure. The Class will be maintained as long as the diving system is subjected to all prescribed surveys, and any modifications and repairs found to be necessary are carried out to the satisfaction of IRS.

.3 If the diving system is not subjected to the prescribed surveys at their due dates, the Class would be suspended.

.4 If the diving system has suffered damage affecting its Class or if such damage may be assumed, a survey is to be performed before diving operations commence. IRS is to be notified of such damage as a matter of course.

.5 Where it is found that the diving system no longer complies with the requirements of the Rules on the basis of which the Class was assigned, or if the operator fails to carry out repairs or alterations considered necessary by IRS within a specified period, as agreed upon, the Class of the diving system will expire.

.6 If the repairs or modifications required by IRS have been carried out and the system is subjected to a Re-classification Survey, the original Character of Classification may be reinstated. This survey is to be carried out in accordance with the requirements for a Class Renewal Survey.

.7 For diving systems which are out of operation for some time or their diving support vessel, floating structure or offshore plant is laid up; the period of Class will remain unchanged. On request, surveys which fall due may be deferred until the diving system is replaced into service. The total scope of the

surveys required thereafter would be determined by IRS for each case individually.

3.3.4 Surveys during Construction

3.3.4.1 Survey during construction and installation of the diving system will include the verification of the following items:

a) Compliance of the design of the diving system, scantlings and arrangements with the approved plans, IRS rule requirements, applicable national regulations and specified standards;

b) Certification of materials and components as per the Rule requirements;

c) Compliance with the fabrication tolerances and to the requisite quality standards;

d) Cleaning of piping systems conducting gas in life support systems in accordance with an approved cleaning procedure;

e) Cleaning and sealing of Gas cylinders;

f) All tests are carried out as per requirements.

3.3.4.2 The extent and method of survey is to be agreed prior to the work being carried out.

3.3.4.3 Tests/trials

.1 A comprehensive test plan is to be submitted for approval for carrying out tests after completion of installation of the diving system. This plan is to include testing details for, at least the following :

- Pressure tests
- Gas leakage tests
- Handling systems
- Life support systems
- Safety systems
- Electrical systems
- Instrumentation
- Environmental control systems
- Sea trials.

3.3.4.4 Leak Test

.1 Leak testing is to be carried out as per approved procedures. Details of leak tests are given in Pt.1, Ch. 2, 18.3.1.1.1.

3.3.4.5 Pressure Test

.1 Pressure tests are also to be carried out as per approved procedures. Details of pressure tests are provided in Pt.1, Ch. 2, 18.4.2.

.2 The following equipment/ systems are to be tested for hydrostatic pressure in accordance with recognized standards or design codes:

- New chambers and bells
- New gas cylinders
- New air and gas storage tanks
- New hot water tanks
- New piping

3.3.4.6 Pressure Vessels for Human Occupancy

.1 The manufacturers of pressure vessels for human occupancy are to be approved by IRS, when the pressure vessels are certified by IRS.

.2 The pressure vessels are to be designed and manufactured to internationally recognized codes/ standards.

.3 ASME PVHO-1 “Safety Standard for Pressure Vessels for Human Occupancy” is to be used for design of acrylic plastic windows.

3.3.4.7 In addition to the above, the scope of initial survey of other items/ components of the diving systems is as indicated in Table 3.3.4.7

Table 3.3.4.7 Initial Survey Requirements for Diving Systems[1]	
Pressure Equipment	
Pressure Vessels for Human Occupancy	
Examine/ verify (as applicable)	- certified materials are used; - sufficient strength in foundations and support for chambers and gas cylinders; - vessel is constructed in accordance with approved drawings; - flanges, sealing surfaces, openings for windows and hatches are satisfactory with regard to tolerances and for sealing systems; - out of roundness for vessels intended for external pressure are within tolerances; - welds are tested; - pressure tests upon completion are satisfactory.
Gas Storage	
Examine (as applicable)	As per the requirements for renewal survey
Pressure Vessels in Supply and Return Lines	
Examine (as applicable)	As per the requirements for renewal survey
Pumps and Compressors	
Examine and test (as applicable)	As per the requirements for renewal survey
System components including piping, umbilicals, hoses, valves, filters	
Examine and test (as applicable)	- Piping systems to be tested to 1.5 x the design pressure; - Cleaning of the hoses, piping and mountings to be undertaken carefully to ensure that the oxygen system is free from impurities like oil, grease etc, that may cause a fire hazard; - Leakage test of total deep diving system to be carried out with heliox mixture, after installation of piping.
Controls	
Gas Supply Control	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Gas Distribution System	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Depth/ Pressure Control	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Temperature and Humidity Control	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Gas Contamination Analysis	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Sanitary/ Domestic Systems	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Auxiliary Services to Life Support Equipment	
Examine/ Verify (as applicable)	- Electrical components installed inside chambers/ bells are to be of a construction suitable to withstand high humidity and oxygen atmosphere content; - Components installed outside bell are to be of watertight type and pressure tested to at least 1.3 x maximum design pressure.
Communication	
Examine/ Verify (as applicable)	Primary, secondary, emergency means of communication and alarms to be examined as per the requirements for renewal surveys.

Table 3.3.4.7 (Contd.)

Launch and Recovery Systems	
General	
Examine/ Verify (as applicable)	For submergence launch and recovery system: (i) Examine all strength members as per approved drawings, certificates for hoisting ropes/ wires, books, sheaves, shackles etc (ii) Load test (including static and running loads[2]) of all lifting equipment. Bell launch and recovery systems to be tested as integral units. (iii) Test of emergency bell retrieval procedures; For surface launch and recovery : (i) As per the requirements for renewal survey
Controls	
Examine/ Verify (as applicable)	As per the requirements for renewal survey
Auxiliary services to launch and recovery system	
Examine/ Verify (as applicable)	Pipes and hoses to be tested to 1.5 x the maximum allowable working pressure (MAWP)
Safety Equipment	
Examine/ Verify (as applicable)	As per scope of annual, third annual survey and renewal surveys

Notes

1 : Requirements for renewal surveys of diving systems are given in Section 18 of Pt. 1, Ch. 2
2 : Running load test to be carried out at 1.25 x the safe working load.

3.3.5 Surveys after Construction

3.3.5.1 The periodical surveys for diving support vessels are to be carried out in accordance with the applicable requirements of Pt. 1, Ch. 2.

3.3.5.2 Periodical surveys of diving systems fitted permanently onboard the ships are to be harmonised with those for the ships.

3.3.5.3 Periodical surveys of permanently fitted diving systems, independent diving systems or diving systems fitted temporarily onboard the ships are to be carried out in accordance with the requirements stipulated in Pt. 1, Ch. 2, Sec. 18 of the Rules.

Section 4

Evacuation Systems

4.1 General

4.1.1 An evacuation system is to be provided with sufficient capacity to evacuate all divers under pressure, in the event of the ship having to be abandoned, and is to be in accordance with the provisions of the Guidelines and Specifications for Hyperbaric Evacuation Systems (HES), IMO Res A.692(17), as amended.

4.1.2 In general, the HBSC is to be designed, constructed and tested in accordance with the applicable aspects of the LSA Code, chapter IV (Survival craft). Specific deviations for HBSC from the LSA Code, as indicated in the *International Code of Diving Safety, 2023*, as amended are to be complied with, as relevant and applicable, for the following aspects:

- Carrying capacity
- Access arrangements
- Buoyancy, Freeboard and Stability
- Propulsion
- Equipment and Fittings
- Semi-submerged HBSCs
- Markings
- Type of HBSC (whether totally enclosed type or free-fall type)

4.1.3 Statutory requirements of the Flag Administration, where the ship, barge, mobile offshore platform or fixed offshore platform is registered, are also to be complied with.

4.2 Design Requirements

4.2.1 The design and construction of the HBSC is to be such that it is suitable for the environmental conditions envisaged, account being taken of the horizontal or vertical dynamic snatch loads that may be imposed on the system and its lifting points particularly during evacuation and recovery.

4.2.2 The HBSC is to be capable of being recovered by a single point lifting arrangement. Means are to be provided on the unit to permit a swimmer to hook on or connect the lifting arrangement.

4.2.3 A single lift point and associated lift load path is to be designed for a load 3 times the weight of the fully loaded HBSC. A minimum factor of safety of 2 to yield is to be applied for structures and 4 for loose gear including suspension chains, links and blocks. A static proof load test of not less than 3 times the weight of the fully loaded HBSC is to be applied.

4.2.4 Where HBSCs are semi-submerged PVHOs, the pressure boundary of the PVHO may be the HBSC hull.

4.2.5 In the design of pressure vessels including accessories such as doors, hinges, door landings, closing mechanisms, penetrators and viewports, the effects of rough handling are to be considered in addition to design parameters such as pressure, temperature, vibration, operating and environmental conditions.

4.2.6 In general, piping penetrations through the chamber are to have isolating valves on both sides.

4.2.7 Materials used in the construction of HBSCs are to be suitable for their intended use.

4.2.8 Components in the HBSC are to be so designed, constructed and arranged as to permit easy inspection, maintenance, cleaning and, where appropriate, disinfection.

4.2.9 Component parts of a HBSC are to be designed, constructed and tested in accordance with recognized national/ international standards acceptable to the Administration/ IRS.

4.2.10 The HBSC is to be provided with the necessary control equipment to ensure its safe operation and the well-being of the divers.

4.2.11 Special arrangements and instructions are to be provided externally to enable the HBSC to be recovered safely. The instructions are to be located where they will be legible when the HBSC is floating.

4.2.12 HBSCs are not to be located in hazardous areas of Zone 0 or Zone 1. Other

hazardous areas and high fire risk areas are to be avoided as far as is reasonably practicable.

4.2.13 The HBSC is to be designed for the rescue of all divers in the diving system at the maximum operating depth.

4.2.14 The compression chamber is to provide a suitable environment and adequate facilities, including, where appropriate, seat belts, for the maximum number of persons for which the unit is designed.

4.2.15 The seating or other arrangements provided are to be designed to provide an adequate degree of protection to the divers from impact collisions during launch and while the unit is afloat.

4.2.16 Where the chamber is intended to be occupied for more than 12 hours, arrangements for the collection or discharge of human waste are to be provided. The discharge arrangements are to be fitted with suitable interlocks.

4.2.17 The means provided for access into the compression chamber is to allow safe access to or from the surface compression chambers. Interlocks are to be provided to prevent the inadvertent release of the HBSC from the surface compression chamber while the access trunk is pressurized.

4.2.18 The mating flange is to be adequately protected from damage at all times including during the launch and recovery stages.

4.2.19 Arrangements are to be provided to enable an unconscious diver to be taken into the unit.

4.2.20 Compression chamber doors are to be so designed as to prevent accidental opening while pressurized.

4.2.21 All doors are to be so designed that, where fitted, the locking mechanisms can be operated from both sides.

4.2.22 Arrangements are to be provided to allow the occupants to be observed. If viewports are provided they are to be situated so that risk of damage is minimized.

4.2.23 Where it is intended to carry out decompression of the divers after hyperbaric evacuation in another surface compression chamber, then suitability of the mating arrangements on that surface compression chamber are to be specially considered. A suitable adapter and clamping arrangements are to be provided, where necessary,

4.2.24 A medical lock is to be provided and be so designed as to prevent accidental opening while the compression chamber is pressurized. Interlock arrangements are to be provided for this where necessary. The dimensions of the medical lock are to be adequate to enable essential supplies, including scrubber canisters, to be transferred into the compression chamber, and be of such dimensions as to minimize the loss of gas when the lock is being used.

4.2.25 Stability and Buoyancy

.1 HBSCs designed to float are be provided with adequate stability for all envisaged operating and environmental conditions and be self-righting. In determining the degree of stability to be provided, consideration is to be given to the adverse effects of large righting moments on the divers.

.2 The effect which equipment and rescue personnel, required to be placed on the top of the system to carry out a recovery from the sea, may have on the stability of the HBSC, are also to be taken into account.

.3 The HBSC is to be fitted with a tow arrangement suitable in function and strength for towing the HBSC in the following conditions:

- 3 knots to the top of sea state 3; and

- holding the HBSC in position up to sea state 7.

The tow arrangement is to be designed to minimize human operation when connecting the tow.

.4 Towing attachment points are to be so situated that there is no likelihood of the HBSC being capsized as a result of the direction of the tow line. Where towing harnesses are provided they are to be lightly clipped or secured to the unit and, so far as is possible, be free from snagging when pulled free.

.5 HBSCs designed to float are to have sufficient reserves of buoyancy to enable the necessary rescue crew and equipment to be carried.

.6 Where HBSCs are designed to be placed on board a rescue vessel, attachment points are to be provided on the unit to enable it to be secured to the deck.

4.2.26 Life Support Systems

.1 Means are to be provided to maintain all the occupants in thermal balance and in a safe and breathable atmosphere for all environmental conditions envisaged - air temperature, sea temperature and humidity - and with the maximum and minimum number of divers likely to be carried.

.2 In determining the duration and amount of life support necessary, consideration is to be given to the geographical and environmental conditions, the oxygen and gas consumption and carbon dioxide generation under such conditions, the heat input or removal and the emergency services that may be available for the decompression of the divers.

.3 Gas losses as a result of using toilet facilities which discharge to outside the HBSC and medical lock operation are to be taken into account in determining the amount of gases required. The effects of hypothermia are to be considered and the effectiveness of the arrangements provided are to be established as far as is reasonable and practicable under all conditions envisaged. However, in no such case should the duration of the unit's autonomous life-support endurance be less than 72 hours.

.4 In addition to any controls and equipment fitted externally, compression chambers are to be provided with adequate controls within for supplying and maintaining the appropriate breathing mixtures to the occupants, at any depth down to the maximum operating depth.

.5 The persons operating the chamber, whether they are within or outside it, are to be provided with adequate controls to provide life support. As far as practicable, the controls are to be capable of operation without the person who operates them having to remove his/ her seat belt.

.6 Two separate distribution systems are to be provided for supplying oxygen to the compression chamber.

.7 Components in the system are to be suitable for oxygen service.

.8 Adequate equipment is to be provided and be suitably situated to maintain O_2 and CO_2 levels and thermal balance within acceptable limits while the life-support equipment is operating.

.9 In addition to any instrumentation necessary outside the compression chamber, suitable instrumentation is to be provided within the chamber for monitoring the partial pressures of O_2 and CO_2. The same are to be capable of operation for the duration of the available life support period.

.10 Where it is intended that divers may be decompressed within the HBSC, provision is to be made for the necessary equipment and gases, including therapeutic mixtures, to enable the decompression process to be carried out safely.

.11 An adequate supply of food and water is to be provided within the HBSC. In determining, in particular, the amount of water to be provided, consideration is to be given to the area of operation and the environmental conditions envisaged.

.12 A breathing system is to be provided with a sufficient number of masks for all the occupants under pressure.

.13 Provision is to be made external to the HBSC, and in a readily accessible place, for the connection of emergency hot or cold water and breathing therapeutic mixture. The dimensions of the connections provided are to be as follows:

- 3/4 in. NPT (female) - hot or cold water
- 1/2 in. NPT (female) - breathing mixture

The connections are to be clearly and permanently marked and be suitably protected.

.14 In HBSCs designed to pass through fires; the breathing gas bottles and piping systems and other essential equipment are to be adequately protected.

.15 In addition, thermal insulation is to be nontoxic and suitable for this purpose.

.16 First-aid equipment, sickness bags, paper towels, waste disposal bags and all necessary operational instructions for equipment within the compression chamber are to be available within the chamber, on board the parent vessel and ashore.

4.2.27 Fire Protection and Extinction

.1 Materials used in the construction and installation are to be non- combustible and nontoxic, as far as practicable.

.2 A fire-extinguishing system is to be provided in the HBSC which is to be suitable for exposure to all depths down to the maximum operating depth.

.3 HBSCs on ships required to be provided with fire-protected lifeboats are to be provided with a similar degree of fire protection.

.4 In HBSCs that are semi-submerged PVHO and may be used to transport divers through fires, consideration is to be given, where practicable, to providing an external water spray system for cooling purposes (see 2.11.6).

4.2.28 Electrical Arrangements

.1 Electrical equipment within the compression chamber is to be designed for hyperbaric use, high humidity levels and marine application. They are to be designed to minimize the risk of electrical capacity depletion as a result of a fault, fire or explosion, electric shock, the emission of toxic gases and galvanic action.

.2 Power supplies required for the operation of life-support systems and other essential services are to be sufficient for the life-support duration.

.3 The battery charging arrangements are to be designed to prevent overcharging under normal or fault conditions. The battery storage compartment is to be provided with means to prevent over-pressurization and any gas released is to be vented to a safe place.

.4 Each compression chamber is to be provided with a source of lighting sufficient for the life-support time and of sufficient luminosity to allow the occupants to read gauges and operate essential systems within the chamber.

4.2.29 Launch and Recovery of HBSCs

.1 Means are to be provided for the safe and timely evacuation and recovery of the unit and due consideration is to be given to the environmental and operating conditions and the dynamic snatch and impact loadings that may be encountered.

.2 Where appropriate, the increased loadings due to water entrainment are to be considered.

.3 Where the primary means of launching depends on the ship's main power supply, then a secondary and independent launching arrangement is to be provided.

.4 The lowering of the HBSC under normal conditions is not to be controlled by brakes, but by the drive system of the winches. If the power to the handling system fails, brakes are to be engaged automatically. The brake is to be provided with manual means of release. In the event of an electrical power failure of the launching appliance, in addition to the requirement for gravity or stored mechanical power in the LSA Code, an emergency power supply is to be provided. The launching appliance need not be provided with hand gear as required by chapter 6.1.2.6 of the LSA code.

.5 The launching arrangements provided are to be designed to ensure easy connection or disconnection of the HBSC from the surface compression chamber and for the transportation and removal of the unit from the ship under the same conditions of trim and list as those for the ship's other survival craft.

.6 Where a power-actuated system is used for the connection or disconnection of the HBSC and the surface compression chamber; a manual or stored power means of connection or disconnection is also to be provided.

.7 The means provided for release of the falls or lift wire after the unit is afloat is to provide for easy disconnection, particular attention being given to units not provided with an attendant crew.

.8 Where the HBSC is designed to be recovered from the sea, or from a ship in a seaway, consideration is to be given to the mode of recovery.

.9 Adequate equipment is to be provided on the unit to enable safe recovery. Permanently marked, clear instructions are to be provided adjacent to the lifting equipment as to the correct method for recovery, including the total weight of the HBSC.

.10 The effect of entrained water and any bilge water which may affect the total weight to be lifted by the recovery vessel is to be considered. Absorption of the dynamic snatch loads imposed during the recovery of the HBSC from the sea is also to be considered.

4.2.30 Communication and Locating Devices

.1 If breathing mixtures containing helium or hydrogen are used, a self-contained primary communication system fitted with an unscrambler device is to be arranged for direct two-way communication between the divers and those outside the compression chamber. A secondary communication system is also to be provided.

.2 In addition to the communication system referred to in 4.2.30.1, a standard bell emergency communication tapping code is to be provided, as indicated in Table 3.2.14.6. Copies of the tapping code are to be permanently displayed inside and outside the HBSC.

.3 HBSCs designed to be waterborne are to be provided with a strobe light and radar reflector.

.4 HBSCs designed to be placed on the sea-bed to await independent recovery are to be provided with an acoustic transponder. The transponder is to be suitable for operation with a diver-held interrogator-receiver which will be retained on board the parent ship. The equipment provided is to meet the requirements indicated in 3.2.14.9.

4.3 Marking and Information to be provided on HBSCs

4.3.1 Dedicated HBSCs are to be coloured orange and be provided with retro-reflective material to assist in their location during hours of darkness.

4.3.2 Each HBSC designed to be waterborne is to be marked with at least three identical signs as shown in Fig 4.3.2. One of these markings is to be on top of the unit and be clearly visible from the air. The other two are to be mounted vertically on both sides and as high as possible and should be capable of being seen while the unit is afloat.

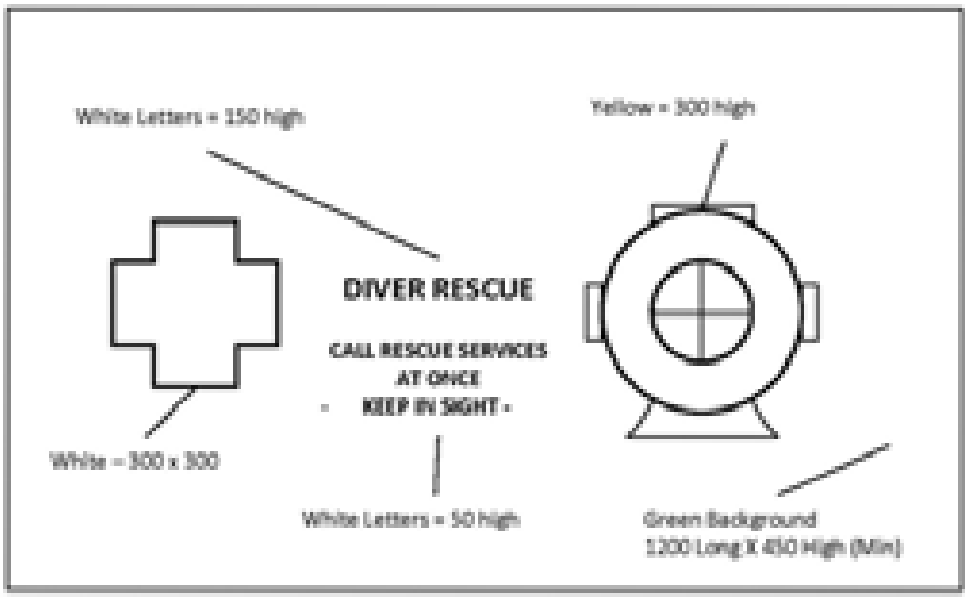

Fig 4.3.2 : Marking on Hyperbaric Evacuation Units

4.3.3 Whilst the unit is afloat, the following instructions and equipment, as applicable, are to be clearly visible and kept readily available:

- Towing arrangements and buoyant towline;
- All external connections, particularly for the provision of emergency gas, hot/ cold water and communications;
- Maximum gross weight of unit in air;
- Lifting points;
- Name of parent ship and port of registration;
- Emergency contact telephone and fax numbers.

4.3.4 Warning Instructions

.1 Where appropriate, the following instructions are to be permanently displayed on every HBSC in two separate locations so as to be clearly visible while the unit is afloat:

.1 Unless specialized diving assistance is available

- Do not touch any valves or other controls;
- Do not try to get occupants out;
- Do not connect any gas, air, water or other supplies;
- Do not attempt to give food, drinks or medical supplies to the occupants; and
- Do not open any hatches.

4.4 Surveys of Hyperbaric Evacuation Systems

4.4.1 General/ Initial Survey Requirements

.1 Statutory requirements of state/ national authorities/ flag administrations are to be kept in mind, whilst approving and surveying HES.

.2 The interface between the HES and the installed diving system is to be examined carefully so that the HES does not become a hazard to the diving system operations.

.3 HES are connected to the diving system and are to be surveyed in the same manner as the main diving system. Corrosion on the chamber exterior is to be examined carefully due to high humidity levels.

.4 Apart from the drawings/ documents being examined (as listed in 1.5), the following need to be specifically examined:

- Sufficient Built in Breathing Systems (BIBS) fitted and functional;
- BIBS overboard dump fitted and functional;
- locking mechanism fitted and functional;
- doors can be opened from both sides;
- doors can be secured "open";
- equalization can be done if appropriate;
- seatbelts for the maximum complement of divers provided;
- lighting fitted and functional;
- toilet provided;
- protective headgear provided
- recovery system available
- falls of sufficient length, where applicable.

.5 Examine if the HES is marked as per requirements, stipulated in 4.3 and a copy of the tapping code is placed inside and outside the chamber, as stipulated in 3.2.14.6.

.6 Examine, if arrangements to lift an injured diver are provided

.7 Examine, if the HBSC has undergone a stability test to ensure that it floats at correct orientation. Launch the HBSC and examine if the free-floating unit is positively buoyant, when fully equipped and manned.

.8 In installations where the saturation diving system is mounted on an offshore production and/ or drilling platform, examine if the HES has some form of propulsion or other suitable method of rapid clearance from the installation.

.9 Confirm, that the HES is able to accommodate the maximum number of divers (as per design) that may be under pressure

.10 Confirm, that the HES is included in the Planned Maintenance System, as per the Operation Manual and contingency plans are also available

.11 Confirm that written procedures for launching of the HES including responsibilities of operators, manning of stations, etc. are available.

.12 Examine, if first aid kit, (as per scale and applicable requirements) is provided.

4.4.2 Periodical Surveys

.1 Periodical surveys of HES are to be carried out as per the requirements in Pt. 1, Ch.2, Sec. 18, 18.5.2.

End of Chapter

Chapter 27

Helicopter Decks

Contents

Section

1 *General*

2 *Design Loads*

3 *Structural Requirements*

Section 1

General

1.1 Application

1.1.1 The requirements of this Chapter apply to vessels with erected landing platform for helicopters or a landing area arranged directly on weather deck or top of deckhouse.

1.1.2 Where the landing area forms part of a weather or erection deck, the scantlings are also to comply with the requirements given in Pt.3, Ch.9 for decks in the same position.

1.1.3 Vessels built in compliance with the above requirements will be eligible to be assigned class notation "HELDK".

1.2 Arrangements

1.2.1 The landing area is to be sufficiently large to allow for the landing and manoeuvring of the helicopter, and is to be approached by a clear landing and take-off sector, the extent of which is to comply with the applicable regulations of National or other authorities.

1.2.2 The landing area is to be free of any projections above the level of the deck. Projections in the zone surrounding the landing area are to be kept below the heights permitted by the regulations.

1.2.3 Suitable arrangements are to be made to minimize the risk of personnel or machinery sliding off the landing area. A non-slip surface and anchoring devices are to be provided.

1.2.4 Arrangements are to be made for drainage of the platform, including drainage of spilt fuel.

1.3 Documentation

1.3.1 Plans showing scantlings and arrangement of the structure and particulars of the materials of construction are to be submitted for approval. The type, size, maximum total weight of the helicopter and its fore-and-aft distribution is to be indicated on the plans.

1.4 Materials

1.4.1 The grades of steel and aluminium materials are to be in compliance with the requirements to hull materials given in Pt.3, Ch.2.

1.4.2 Where aluminium alloy platforms are connected to steel structures, detail of the arrangements in way of the connections including those to prevent galvanic corrosion are to be submitted for approval.

Section 2

Design Loads

2.1 General

2.1.1 The scantlings of each structural element is to be based on the most unfavourable of the following loading conditions:

- Extreme functional loads (landing) together with hull still water loads.

- Extreme environmental loads acting on hull, helicopter deck and helicopter in stowed transit position, if applicable, together with hull still water loads. The effect of icing is to be included for erected structures.

2.2 Landing forces

2.2.1 The total vertical force from the helicopter during landing is to be taken not less than:

$$P = 2\ g_o\ M_H\ .\ 10^3\ [N]$$

where,

M_H = the maximum take-off mass in tonnes of helicopter.

Landing force (P_w) acting on a wheel or a group of wheels, is to be obtained by considering that the total force P is distributed on the helicopter's landing gear in the same manner as when the helicopter is resting on a horizontal surface and the helicopter's centre of gravity is in its normal position in relation to the landing gear.

2.2.2 A horizontal sliding force of $0.5\ g_o\ M_H\ 10^3$ [N] may have to be taken into account for hatch covers and bolted platforms.

2.3 Gravity and inertia forces (due to ship motions and accelerations)

2.3.1 The static and dynamic design forces caused by the platform structure itself and, if applicable, by the helicopter in stowed position are to be considered. The effect of icing is to be additionally considered, if applicable.

2.4 Sea and weather loads

2.4.1 The sea pressure for superstructure deck and top of houses is to be taken in accordance with Pt.3, Ch.4. For elevated platforms with free water passage below, the sea pressure may be taken as:

$p = 0.0075$ [N/mm^2] forward of 0.15 L from FP.

$= 0.0025$ [N/mm^2] aft of 0.15 L from FP.

2.4.2 For structures where wind suction forces may be of importance, (e.g. bolted platforms), wind lift forces are to be taken into account by:

$$P_L = 1.2\ A\ .\ 10^3\ [N]$$

where,

A = deck area [m^2] of the platform.

Section 3

Structural Requirements

3.1 Deck plating and stiffening

3.1.1 Decks for helicopters supported on wheels with pneumatic tyres are to have scantlings in accordance with the following:

Thickness of deck plating is not to be less than:

$$t = f\,(s+1000)\,\sqrt{P_w}\,\sqrt{k}\,10^{-5} + t_c \text{ [mm]}$$

where,

f = 1.90 for separate platforms of steel or aluminium

= 2.05 for weatherdecks, in general

= 2.20 for longitudinally framed strength deck and for weather deck hatch covers of steel or aluminium

= 2.85 for transversely framed strength deck

s = stiffener spacing [mm]

P_w = Landing force [N], acting on the wheel or group of wheels considered as per 2.2.1

k = the material factor as per Pt.3, Ch.2, Sec.1.2 for steel

$= \frac{235}{\sigma_a}$ for aluminium alloys

σ_a = 0.2% proof stress or 70% of the ultimate strength of the aluminium alloy material, whichever is lesser [N/mm^2]

t_c = 2.0 [mm] for steel

= 1.0 [mm] for aluminium.

3.1.2 Section modulus of deck stiffeners is not to be less than:

$$Z = \frac{BM}{\sigma} \text{ [cm}^3\text{]}$$

BM = bending moment [N-m] for the most unfavourable location of landing forces (point loads), assuming 50% fixity at ends.

σ = 180/k [N/mm^2] in general; to be reduced by still water longitudinal hull girder stress for strength deck longitudinals.

= 160/k [N/mm^2] for weather deck hatch covers.

3.1.3 Each stiffener to girder connection is to have a shear area not less than:

$$A = 1.25\,P_w\,k\,10^{-4} \text{ [cm}^2\text{]}.$$

3.1.4 Decks for helicopters supported on tubular skids are to have scantlings in accordance with the following:

- Thickness of deck plating is to be not less than that obtained using the formula in 3.1.1 taking P_w as 0.9P, where P is the total landing force as per 2.2.1.

- Section modulus of stiffeners is to be as per 3.1.2.

3.2 Girders and supporting structures of separate platforms

3.2.1 The scantlings are normally to be based on the following allowable stresses:

Normal stress, $\sigma \leq 160/k$ [N/mm^2]

Shear stress, $\tau \leq 90/k$ [N/mm^2].

3.2.2 The helideck supporting members are to satisfy the buckling criteria as per Pt.3, Ch.3, Sec.6.

3.2.3 The critical buckling stress of plating acting as girder flange is to satisfy the buckling criteria as per Pt.3, Ch.3, Sec.6 taking η as 0.67.

Tripping brackets and local stiffening of plating are to be provided where necessary.

3.2.4 In the case of landing on a hatch cover section which is underlying in the packing joint, the strength and spacing of cleats must be sufficient to keep the connection intact and weathertight.

End of Chapter

Chapter 28

Single Point Mooring

Contents

Section

1 *General*

2 *General Arrangement*

3 *Mooring Components*

4 *Supporting Hull Structures*

5 *Strength Criteria*

Section 1

General

1.1 Application

1.1.1 The requirement of this chapter applies to ships fitted with standardized equipment complying with the recommendations of the Oil Companies International Marine Forum (OCIMF) for mooring at single point mooring or single buoy mooring terminals. Vessels complying with these requirements will be assigned class notation 'SPM' as per Pt.1, Ch.1 of Rules.

1.1.2 Where some components of mooring equipment are used for both single point mooring and for the bow emergency towing arrangements, the requirements for emergency towing arrangements (specified in Pt.5, Ch.2) are also to be complied with.

1.1.3 Where the mooring components are to be type approved, the requirements given in 'Certification Scheme for Type Approval of Products' are also to be complied with.

1.2 Documentation for approval

1.2.1 The following plans are to be submitted to IRS for approval:

- General layout of the associated equipment at the forecastle/fore deck.
- Construction drawing of the bow chain stoppers, bow fairleads and pedestal roller fairleads, together with material specifications and relevant calculations.
- Plans of the supporting structure in way of chain stoppers, fairleads, roller pedestals, winches and capstans.
- Arrangements for pull back from tugs, wherever applicable.

1.3 Documentation for information

1.3.1 The following documentation is to be submitted for information.

- Specifications of winches or capstans giving the continuous duty pull and brake holding capacity.
- DWT, in tonnes, of the ship at summer load line.

Section 2

General Arrangement

2.1 General

2.1.1 Suitable equipment and arrangements are to be provided near the bow of the vessel for heaving onboard a standard chafing chain of 76 [mm] diameter using a pick-up rope and for securing the chain to the strong point.

2.1.2 The strongpoint is to be a chain cable stopper.

2.1.3 Typical schematic layouts of the forecastle deck are shown in Fig.2.1.3(a), (b), (c) for reference.

2.1.4 Where tugs are used for pull back, suitable arrangements for connecting the lines from tugs are to be provided.

2.2 Equipment components

2.2.1 The equipment for mooring is to generally consist of the following components :

- bow chain stopper;
- bow fairlead;
- pedestal roller fairlead;
- winch or capstan.

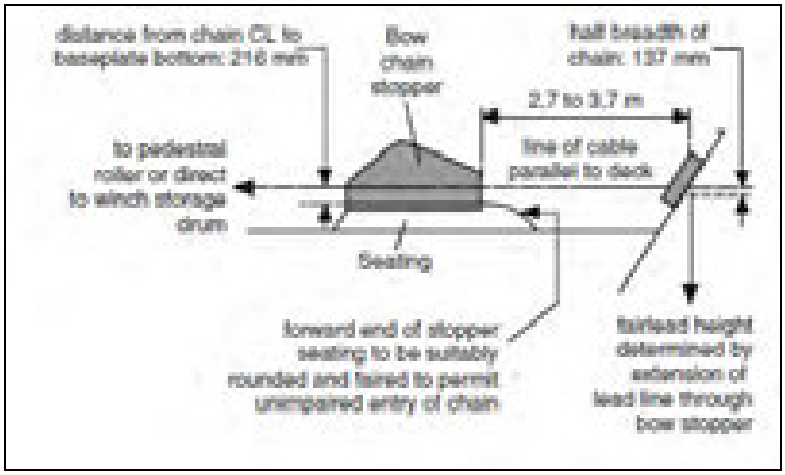

Fig. 2.1.3 (a) Typical forecastle deck layout

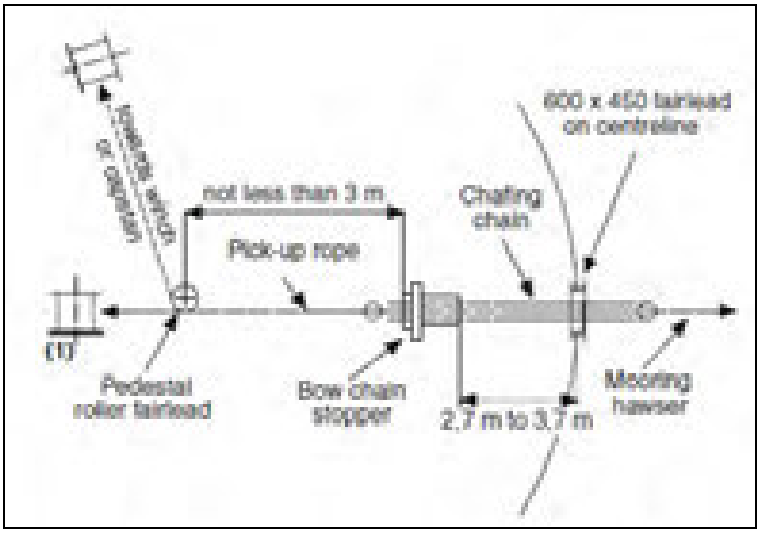

Fig. 2.1.3 (b) Typical forecastle deck layout (DWT ≤ 150000 t)

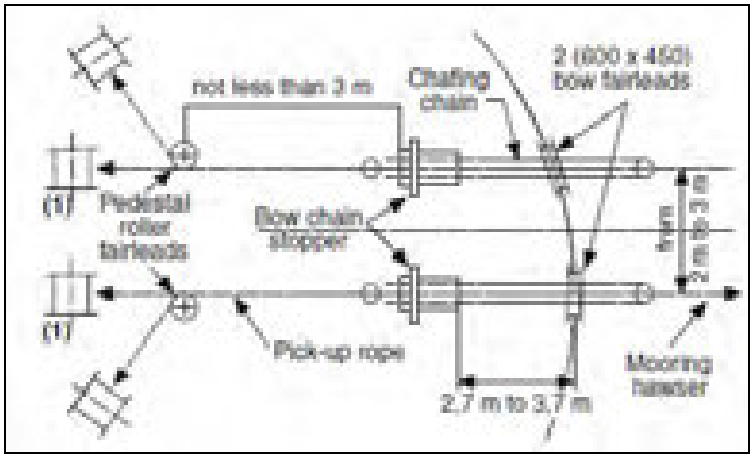

Fig. 2.1.3 (c) Typical forecastle deck layout (DWT > 150000 t)

Section 3

Mooring Components

3.1 General

3.1.1 The ship is to be equipped with bow chain stoppers complying with the requirements in Table 3.1.1 and suitable for a standard chafing chain of 76 [mm] in diameter.

The requirements for materials, manufacture, testing and certification of the 76 [mm] chafing chain are given in Part 2, Chapter 10, Section 5.15 of the Rules. Grade R3 and R4, indicated in Part 2, Chapter 10 are to be used corresponding to Safe working load of 200 [t] , 250 [t] and 350 [t] respectively, as applicable. (See Table 3.1.1).

Note : Chafing chains are not required to be part of ship's equipment when supplied by the SPM terminal operator.

3.1.2 Where the safe working loads are increased to take into account the local environmental conditions, the same is to be clearly stated on plans submitted for approval.

Table 3.1.1 : Required Number and SWL of Chain stoppers

Deadweight [t]	Number	Safe working load (SWL) [t] (kN)
DWT ≤ 100000	1	200 (1962)
100000 < DWT ≤ 150000 Note : Ships in this size range may elect to fit two stoppers to ensure full range terminal acceptance	1	250 (2452)
DWT > 150000	2	350 (3433)

3.1.3 The scantlings of all parts of bow chain stopper and bow fairleads are to be in accordance with the strength criteria given in Sec.5.

3.1.4 The bow chain stoppers and bow fairleads are to be made of fabricated steel, steel castings or forgings complying with relevant requirements given in Pt.2, Chapters 3, 4 and 5 respectively. They are to be permanently marked with the SWL.

Use of spheroidal graphite (SG) iron casting may be accepted for the main framing of the chain stopper provided that:

- The part concerned is not intended to be a component part of a welded assembly.
- The SG iron casting is of ferritic structure with an elongation not less than 12%.
- The yield stress at 0.2% strain is to be measured and certified.
- The internal structure of the component is to be inspected by means of non-destructive examination.

The material used for the stopping device (pawl or hinged bar in chain stoppers is to have mechanical properties similar to that of a Grade CC3 chain cable or better.

3.2 Bow chain stopper

3.2.1 The stoppers are to be capable of securing the common stud links of the chafing chain cable when the stopping device (chain engaging pawl or bar) is in the closed position and of freely passing the chain cable and its associated fittings when the stopping device is in the open position.

3.2.2 Chain stoppers may be of the hinged bar type or of pawl (tongue) type or of other equivalent design.

Typical arrangements of chain stoppers are shown in Fig.3.2.2.

3.2.3 The stopping device (chain engaging pawl or bar) of the chain stopper is to be so arranged as to prevent it from gradually working to the open position from the closed position which would release the chafing chain and allow it to pay out.

Stopping devices are to be easy and safe to operate and are to be properly secured in the open position.

3.2.4 Chain stoppers are to be located between 2.7 [m] and 3.7 [m] inboard from the bow fairleads (See Fig.2.1.3).

When positioning, due consideration is to be given to the correct alignment of the stopper relative to the direct lead between bow fairlead and pedestal roller fairlead.

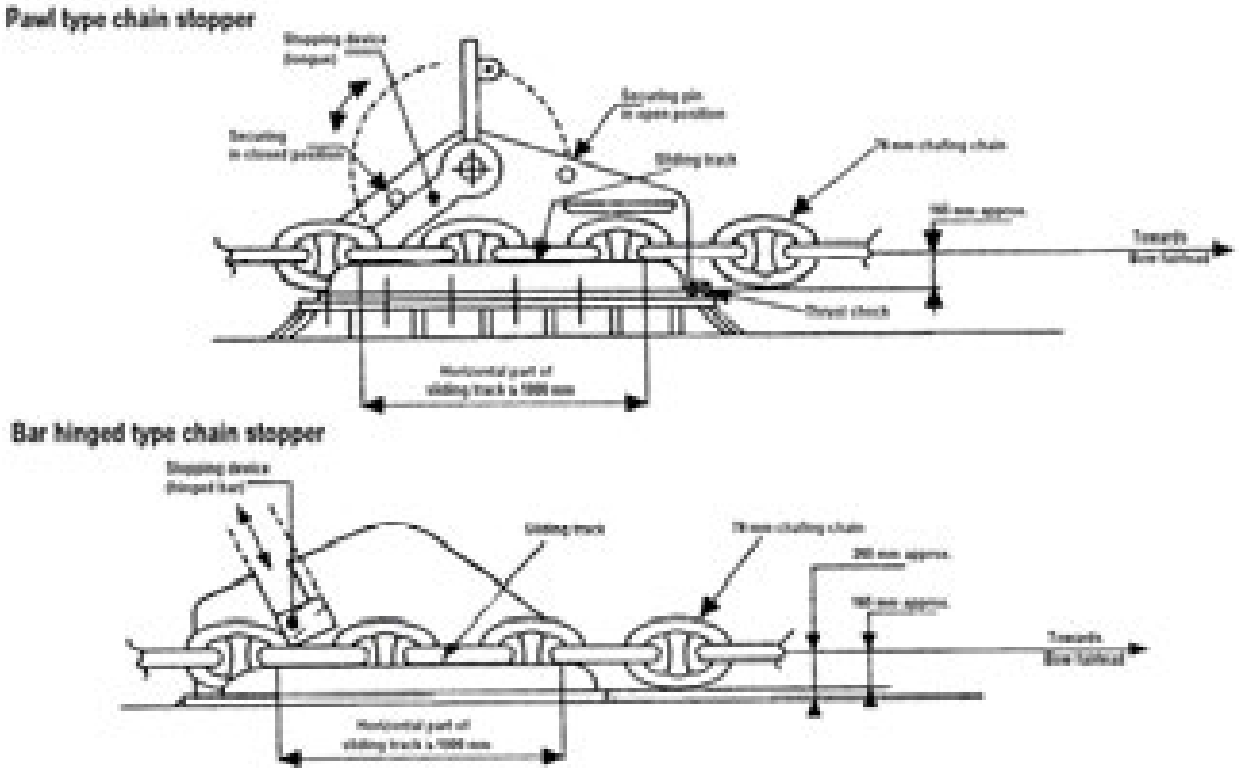

Fig.3.2.2 : Typical bow chain stoppers

3.2.5 Stopper support structures are to be trimmed to compensate for any camber and/or sheer of the deck. The leading edge of the stopper base plate is to be faired to allow for the unimpeded entry of the chafing chain into the stopper.

3.2.6 Where the chain stopper is bolted to a seating welded to the deck, the bolts are to be relieved from shear force by efficient thrust chocks capable of withstanding horizontal force equal 1.5 times the required working load satisfying the strength criteria specified in Sec.5.

Mechanical properties of bolt material are to be not less than Grade 8.8 of ISO standard No.898/1. (Grade 10.9 is recommended). The bolts are to be pre-stressed in accordance with relevant standards and their tightening is to be suitably checked.

3.2.7 Some SPM operators may require a second bow chain stopper for all sizes of conventional tankers.

3.2.8 DP bow loading tankers specifically designed for offshore terminals are to be fitted with specialized bow chain stoppers, integrated with cargo loading system emergency shutdown systems. Requirements for these specialized systems may be referred in the OCIMF '*Guidelines for Offshore Tanker Operations*'.

3.3 Bow fairleads

3.3.1 One bow fairlead is to be fitted for each bow chain stopper (See Fig.2.1.3).

3.3.2 For ships of more than 150000 t DWT, where two bow fairleads are required, the fairleads are to be spaced 2.0 [m] centre to centre apart, if practicable and in no case be more than 3.0 [m] apart.

For ships of 150000 t DWT or less for which only one bow fairlead is required (See Table 3.1.1), it is in general to be fitted on the centreline.

3.3.3. Bow fairleads are to be capable of withstanding a load equivalent to the safe working load (SWL) of the bow chain stopper that they serve, and, in such condition, meeting the strength criteria specified in Sec.5. The load position is to be based on hawser angles as follows:

- in the horizontal plane, up to 90° from the ship's centreline, both starboard and portside

- in the vertical plane, up to 30° above and below horizontal.

3.3.4 Fairleads are normally of a closed type (as Panama chocks) and are to have an opening large enough to pass the largest portion of the chafing gear, pick-up rope and associated fittings.

For this purpose, the inner dimensions of the bow fairlead openings are to be at least 600 [mm] in width and 450 [mm] in height.

3.3.5 Fairleads are to be oval or round in shape.

The lips of the fairleads are to be suitably faired in order to prevent the chafing chain from fouling on the lower lip when heaving inboard.

The ratio of the diameter of the bearing surface of the fairlead to the diameter of the chafing chain, is to be not less than 7 to 1.

3.3.6 The fairleads are to be located as close as possible to the deck and, in any case, in such a position that the chafing chain is approximately parallel to the deck when it is under strain between the chain stopper and the fairlead.

3.4 Pedestal roller fairleads

3.4.1 Pedestal roller fairleads are to be used only when the mooring arrangement design is not permitting direct straight leads to a winch storage drum. The number of pedestal roller fairleads for each bow chain stopper is not to exceed two, and the angle of change of direction of the pick-up rope is to be kept as low as possible

3.4.2 Pedestal roller fairleads are to be so positioned as to provide a direct lead to bow chain stopper and bow fairlead. (See Fig.2.1.3).

They are to be fitted not less than 3 [m] behind the bow chain stopper.

3.4.3 The pedestal roller fairleads are to be capable to withstand a horizontal force equal to:

- the resultant force due to an assumed pull of 225 [kN] in the pick-up rope; or
- 225 [kN], whichever is greater.

Stresses generated by this horizontal force are not to exceed those given in 5.2.

3.4.4 It is recommended that the fairlead roller is to have a diameter not less than 7 times the diameter of the pick-up rope. Where the diameter of the pick-up rope is unknown it is recommended that the roller diameter should be at least 400 [mm].

3.5 Winches or capstans

3.5.1 Winches or capstans used to handle the mooring gear are to be capable of exerting a continuous duty pull of not less than 150 [kN] and withstanding a braking pull of not less than 225 [kN]. Winches or capstans used to handle the mooring gear are to be capable of heaving inboard a load of at least 15 [t]. W

3.5.2 Winch storage drums used to stow the pick-up rope, are to be of sufficient size to accommodate 150 [m] of rope of 80 [mm] diameter.

Section 4

Supporting Hull Structures

4.1 General

4.1.1 The bulwark plating and stays are to be suitably reinforced in the region of the fairleads.

4.1.2 Deck structures in way of bow chain stoppers, including deck seatings and deck connections, are to be suitably reinforced to comply with the strength criteria specified in Sec.5.

4.1.3 The deck structures in way of the pedestal roller fairleads and in way of winches or capstans as well as the deck connections are to be reinforced to withstand, respectively, the horizontal force defined in 3.4.3 or the braking pull defined in 3.5.1 and to meet the strength criteria specified in Sec.5.

4.1.4 The deck plating in way of the mooring capacity mentioned in 4.1.2 and 4.1.3 is to be increased by means of suitable insert plates.

4.1.5 Main welds of the bow chain stoppers with the hull structure are to be inspected by means of non-destructive examination for their complete length.

Section 5

Strength Criteria

5.1 Design loads

5.1.1 The bow chain stopper and the bow fairlead are to be designed considering a horizontal chafing chain tension equal to 2 times the SWL given in Table 3.1.1.

5.2 Allowable stresses

5.2.1 The equivalent stress, σ_e, induced by the load given in 5.1.1 is not to exceed 1.0 σ_y or 0.6 σ_u whichever is lesser.

where,

$$\sigma_e = \sqrt{\sigma^2 + 3\tau^2}$$

σ = tensile stress
τ = shear stress
σ_y = Minimum yield stress [N/mm^2], of the component material
σ_u = Tensile strength [N/mm^2], of the component material.

End of Chapter

Chapter 29

Vapour Control Systems

Contents

Section

1	*General Requirements*
2	*Vapour Piping Systems*
3	*Instrumentation*
4	*Vapour Balancing*
5	*Operational Instructions*
6	*Operational Limitations and Conditions to be Reflected in the Instruction Manual*
7	*List of Cargoes – Vapour Control*

Section 1

General Requirements

1.1 Application

1.1.1 The requirements given in this chapter apply to systems for control of vapour emissions from cargo tanks on ships which are assigned class Notation 'OIL TANKER' or 'CHEMICAL TANKER' or 'ORE OR OIL CARRIER' or 'OIL OR BULK CARRIER'.

1.2 Class notations

1.2.1 Vessels complying with the requirements given in this chapter except those given in 3.2.2 and Section 4, will be eligible to be assigned class notation 'VCS1'.

Compliance with these requirements is considered to meet the requirements of IMO MSC/Circ.585.

Such vessels, when additionally complying with requirements of Section 4 (vapour balancing), will be assigned class notation 'VCS1-B'.

1.2.2 Vessels complying with the requirements in this chapter, except Section 4, and the requirements of United States Coast Guard given in the regulation CFR46 Part 39 - "Vapour Control Systems", will be eligible to be assigned class notation 'VCS2'.

Such vessels, when additionally complying with requirements of Section 4(vapour balancing), will be assigned class notation 'VCS2-B'.

1.3 Definitions

1.3.1 '*Diluted'* means the condition in which the concentration of a flammable gas in a flammable gas/air mixture is less than 50% of the lower explosive limit of the gas.

1.3.2 '*Enriched*' means the condition in which the concentration of a flammable gas in a flammable gas/air mixture is not less than 150% of the upper explosive limit of the gas. When flammable gases are mixed to achieve an enriched condition, the upper explosive limit of the flammable gas mixture shall be used.

1.3.3 '*Flammable cargoes*' means cargoes of crude oil, petroleum products and chemicals having a flashpoint not exceeding 60°C (closed cup test), as determined by an approved flashpoint apparatus and a Reid vapour pressure which is below atmospheric

pressure and other liquid products having a similar fire hazard.

1.3.4 '*Inerted*' means the condition in which the oxygen content in a flammable gas/air mixture is 8% or less by volume.

1.3.5 '*Independent*' as applied to two systems means that one system will operate with a failure of any part of the other system except power sources and electrical feeder panels.

1.3.6 '*Maximum allowable transfer rate'* means the maximum volumetric rate at which a tanker may receive cargo or ballast.

1.3.7 '*Tanker vapour connection'* means the point in a tanker's fixed vapour collection system where it connects to a vapour collection hose or arm.

1.3.8 '*Terminal vapour connection'* means the point in a terminal's vapour collection system where it connects to a vapour collection hose or a vapour collection arm.

1.3.9 '*Vapour balancing'* means the transfer of vapour displaced from the tank of a tanker receiving cargo into a tank of a facility delivering cargo via a vapour collection system.

1.3.10 '*Vapour collection system'* means an arrangement of piping and hoses used to collect vapour emitted from a tanker's cargo tanks and transport the vapour to a vapour processing unit.

1.3.11 '*Vapour destruction unit'* means a vapour processing unit that destroys cargo vapour by a means such as incineration.

1.3.12 '*Vapour dispersion system'* means a vapour processing unit which releases cargo vapour to the atmosphere through a venting system not located on the tanker being loaded or ballasted.

1.3.13 '*Vapour emission control system'* means an arrangement of piping and hoses used to control vapour emissions collected from a tanker and includes the vapour collection system and the vapour processing unit.

1.3.14 '*Vapour processing unit'* means the components of a vapour control system that recovers, destroys or disperses vapour collected from a tanker.

1.3.15 '*Vapour recovery unit'* means a vapour processing unit that recovers cargo vapour by a non-destructive means such as lean oil absorption, carbon bed adsorption or refrigeration.

1.3.16 '*Service ship'* means a ship which receives and transports liquid cargoes between a facility and another ship and vice versa.

1.3.17 '*Topping-off operation'* means operation of transfer of liquid cargo from a service ship to another ship in order to load the receiving ship at a deeper draft.

1.4 Documentation

1.4.1 The following plans and particulars are to be submitted:

a) A schematic drawing of the vapour piping systems giving information about material, dimensions, pressure rating etc. Details about penetrations, joining etc. are also to be submitted.

b) A schematic drawing of the gauging and overfill protection equipment showing:

- Working principle
- Independent control
- Location of electrical equipment in gas dangerous area
- Single line diagram of intrinsically safe equipment
- Use of explosion protected equipment with reference drawing together with certificates.

c) A schematic drawing of the venting system (Inert Gas System if relevant) including necessary data for verifying venting capacities of the P/V valves. Pressure monitoring and alarm system to be documented including wiring diagrams, cable penetrations and connections etc.

d) Pressure drop calculations comparing cargo transfer rate to pressure drop from the farthest tank to the vapour connection, including any hoses used. Calculation to be made for each cargo handled at the maximum and minimum transfer rate.

e) Overfill alarm calculations showing time available between alarm setting and overfill at max. loading rate for each tank.

f) Determination of maximum loading rate.

g) Fitting and test certificate for detonation flame arrester, if relevant.

h) Fitting and documentation regarding oxygen analyzer and alarm etc., if relevant.

i) Details of insulating flange, if relevant.

j) Instruction manual.

Section 2

Vapour Piping Systems

2.1 Material

2.1.1 Permanent piping

Piping material is to be as required for cargo tank venting system.

2.1.2 Flexible hoses

Material of hoses is to be resistant to vapours handled and hoses are to comply with the following:

- Maximum allowable working pressure is to be at least 0.34 bar.
- The hoses are to be capable of withstanding at least 0.14 bar vacuum without collapsing.
- Burst pressure is not to be less than five times the maximum allowable working pressure.
- The hoses are to be electrically continuous with a maximum resistance of 10,000 ohms.
- Hoses are to be abrasion resistant and non-kinking.
- Be provided with suitable hose handling equipment.

2.2 Vapour collection piping

2.2.1 General

Each vessel is to have vapour collection piping which is permanently installed, with a tanker vapour connection located as close as practical to the loading manifold. In lieu of permanent piping, chemical tankers are permitted to have a permanent vapour connection at each cargo tank for connection to a vapour hose which should be kept as short as practicable and in no case longer than 3 [m].

If a tanker simultaneously collects vapours from cargoes, which react in a hazardous manner with other cargoes these incompatible vapours are to be kept separate throughout the entire vapour collection system.

Means are to be provided to eliminate liquid condensate which may collect in the system, such as draining and collecting liquid from each low point in the line.

Vapour collection piping is to be electrically bonded to the hull and to be electrically continuous.

When inert gas distribution piping is used for vapour collection piping means to isolate the inert gas supply from the vapour collection system is to be provided. The inert gas main isolation valve required in Pt.6, Ch.8, Cl.15.2.2.3.1.3 may be used to satisfy this requirement.

The vapour collection system is not to disable the proper operation of the cargo tank venting system. However, a vapour collection piping may be common or partly common with the vent piping and/or the Inert Gas System and Inert Gas Piping System piping (Pt.5, Ch.2, Sec.8).

2.2.2 Vapour manifold

OCIMF's Recommendations for Oil Tanker Manifolds and Associated Equipment (latest revision) is also to be referred.

An isolation valve capable of manual operation is to be provided at each tanker vapour connection. The operating position of this

valve is to be readily determined visually (open/closed indication).

The end of each vapour collection pipe or vapour collection hose is to be readily identifiable to prevent mis-connection.

In order to prevent the possible mis-connection of the vapour manifold to a shoreside terminal liquid loading line, each vapour connection flange is to conform to the applicable industry standard (OCIMF). This provision is applicable regardless of the size of the ship.

Number and position

Four vapour connections are to be provided, two on each side of the ship, with presentation flanges at the same height above the deck as the cargo manifold. One vapour connection is to be located forward of the manifold and one located aft of the manifold on each side of the ship.

Labelling

The first 1.0 [m] inboard of each manifold is to be painted on its exterior surfaces, excluding flange faces. The painted area is to be divided into three bands with the outboard and inboard bands being red and 100 [mm] wide and the centre band being yellow and 800 [mm] wide.

2.3 Capacity

2.3.1 Pressure drop calculations

The capacity of the vapour collection system is to be documented through pressure drop / flowrate curves. When calculating pressure drop the following is to be used:

a) Vapour growth rate, (VGR) i.e. increase of flowrate due to vapourization of the cargo:

$$VGR = 1 + 0.25\frac{P_v, 45}{86.2}$$

P_v, 45 = saturated vapour pressure at 45°C in kPa abs.

b) Density of the cargo vapour and air/inert gas mixture (δ_{va},45):

δ_{va},45 = [SG_v V_v,45 + V_a,45] 0.0324 $P_{p/v}$

SG_v = specific gravity of cargo vapour.

V_v,45 = partial volume of vapour at

$$45°C = \frac{P_v,45}{P_t,45}$$

Where,

P_t,45 = total vapour air pressure at 45°C in kPa abs.

V_a,45 = partial volume of air at 45°C = 1 – V_v,45

$P_{p/v}$ = cargo tank P/V valve pressure setting in kPa abs.

c) Maximum cargo tank pressure is not to exceed 80% of opening pressure of the P/V valves.

Note:

- For oil tankers calculations using mixture density of 3.0 [kg/m^3] and vapour growth rate of 1.25 will cover all cargoes the ship may carry.

- For chemical tankers mixture density of 3.6 [kg/m^3] and vapour growth rate of 1.2 will cover all chemicals except those with a Reid's vapour pressure above atmospheric (IBC Code 15.14). For these cargoes additional calculations are needed using values applicable for the actual cargoes. (See Section 7).

Section 7 lists density and vapour growth values for a range of actual cargoes.

Section 3

Instrumentation

3.1 Cargo gauging

3.1.1 Cargo tank gauging equipment

Each cargo tank of a tanker that is connected to a vapour collection system is to be equipped with a cargo gauging device which:

- provides a closed gauging arrangement that does not require opening the tank to the atmosphere during cargo transfer;

- allows the operator to determine the liquid level in the tank for the full range of liquid levels in the tank;
- indicates the liquid level in the tank, at the location where cargo transfer is controlled.

- if portable, is installed on tank during the entire transfer operation.

3.2 Cargo tank level alarms

3.2.1 High level alarm

Each cargo tank of a tanker is to be equipped with a high level alarm system, which is to:

- be independent of the cargo gauging system. However, for vessels where the notation VCS2 is envisaged, a high level alarm system integral with the cargo gauging system will be accepted.

- come into operation when the normal tank loading procedures fail to stop the tank liquid level exceeding the normal full condition;

- give a visual and audible tank high level alarm to the ship's operator;

- have alarms fitted in the cargo control room, where provided, but in each case in such a position that they are immediately received by responsible members of the crew;

- give alarm in the event of loss of power to the alarm system or failure of the electrical circuitry to the tank level sensor;

- be able to be function tested from the outside of the tank for proper operation prior to each transfer or contain an electronic self-testing feature which monitors the condition of the alarm circuitry and sensor.

3.2.2 Overfill alarm

For vessels where the notation VCS2 is envisaged, each cargo tank of a tanker is to be equipped with an overfill alarm system (High-high level) which is to:

- be independent of the cargo gauging system and the high level alarm system specified in 3.2.1 above.

- come into operation after the high level alarm, but early enough to allow for action to prevent tank overflow.

- give a visual and audible tank overfill alarm to the ship's operator.

- have alarms fitted in the cargo control room, where provided and in the cargo deck area.

- give alarm in the event of loss of power to the alarm system or failure of the electrical circuitry to the tank level sensor;

- be able to be checked from the outside of tank for proper operation prior to each transfer or contain and electronic self-testing feature which monitors the condition of the alarm circuitry and sensor.

3.3 Vapour pressure alarms

3.3.1 General

Each vapour collection system is to be fitted with a pressure sensing device that senses the pressure in the main vapour collection line, which:

- has a pressure indicator located on the vessel where the cargo transfer is controlled e.g. cargo control room (CCR);

- has a high pressure and a low pressure alarm that:
    - is audible and visible on the vessel where cargo transfer is controlled e.g. CCR;
    - give alarms at a high pressure of not more than 90 percent of the lowest pressure relief valve setting in the cargo tank venting system;
    - give alarms at a low pressure of not less than 100 [mm] WG for an inerted tankship, or the lowest vacuum relief valve setting in the cargo tank venting system for a non-inerted tank.

Pressure sensors fitted in each cargo tank are acceptable as equivalent to pressure sensors fitted in each main vapour collection line.

Section 4

Vapour Balancing

4.1 General

4.1.1 Application

Requirements in this section apply to ship's engaged in the transportation of bulk liquid cargoes between a facility and another ship and vice versa. Compliance with these requirements is necessary for assignment of additional qualifier "B" to the class notations VCS1 or VCS2. (i.e. VCS1-B or VCS2-B, as applicable).

4.2 Design and equipment

4.2.1 Where the cargo tanks on a ship discharging cargo and a ship receiving cargo are inerted, the discharging ship must:

- have a means to inert the vapour transfer hose prior to transferring cargo vapour;
- have an oxygen analyzer with a sensor or sampling connection fitted within 3 [m] of the ship vapour connection which:
    - activates an audible and visible alarm at a location on the discharging ship where cargo transfer is controlled when the oxygen content in the vapour collection system exceeds 8 percent by volume;
    - has an oxygen concentration indicator located on the discharging ship from where the cargo transfer is controlled;
    - has a connection for injecting a span gas of known concentration for calibration and testing of the oxygen analyzer. (The installation of the oxygen analyzer is to be specially considered).

4.2.2 Where the cargo tanks on a ship discharging cargo are not inerted the vapour collection line on the receiving ship is to be fitted with an approved detonation arrester located within 3 [m] of the ship vapour connection.

4.2.3 An electrical insulating flange or one length of non-conductive hose is to be provided between the ship vapour connection on the service ship and the vapour connection on the ship being lightered or topped-up.

Section 5

Operational Instructions

5.1 Instruction manual

5.1.1 Operational instructions

Each tanker utilizing a vapour emission control system is to be provided with written operational instructions covering the particular system installed on the tanker. The instructions are to encompass the purpose and principles of operation of the vapour emission control system and provide an understanding of the equipment involved and associated hazards. In addition the instructions are to provide an understanding of operating procedures, piping connection sequence, start-up procedures, normal operations and emergency procedures. Instructions are also to include an understanding of the shoreside terminal equipment and operating procedures. The instructions are to be in working language of the ship and additionally a copy in English language is to be available on board ship.

Instructions are to contain information on the tanker's vapour collection system including:

- a line diagram of the tanker's vapour collection piping indicating the locations and purpose of all control and safety devices;
- the maximum allowable transfer rate as limited by the venting capacity of the pressure or vacuum relief valves, or any other factor which would limit the transfer rate;
- the maximum pressure drop in the ship's vapour collection system for various transfer rates;
- the pressure relief settings of each pressure and vacuum relief valve;
- pre-transfer procedures;
- procedures are to be followed in the event of a fault during vapour collection operations.

The operational limitations and conditions given in Section 6 are to be reflected in the Operational Instructions.

Section 6

Operational Limitations and Conditions to be Reflected in the Instruction Manual

6.1 Procedures

6.1.1 Established industry guidelines are to be observed, as applicable, with regard to preparation for transfer and transfer of cargo and ballast into cargo tanks.

6.1.2 The rate of cargo transfer is not to exceed the maximum allowable transfer rate as determined by the lesser of the following:

- the venting capacity of the pressure relief valves in the cargo tank venting system divided by a factor of at least 1.25

- the vacuum relieving capacity of the vacuum relief valves in the cargo tank venting system

- the rate based on pressure drop calculations for a given pressure at the vapour connection facility, such that the pressure in any cargo tank connected to the vapour collection system does not exceed 80% of the opening set pressure of any pressure relief valve in the cargo tank venting system.

6.1.3 When the venting capacity is dimensioned for thermal breathing only, the capacity of the P/V breaker as required by SOLAS for IG systems is to be regarded as the basis for venting in the determination of the total permissible loading rate.

6.1.4 A cargo tank is not to be filled higher than the level at which the overfill alarm is set.

6.1.5 A cargo tank is not to be opened to the atmosphere for gauging or sampling while the tanker is connected to a vapour emission control system unless loading to the tank is stopped, the tank is isolated from any other tank which is in the process of being loaded and precautions are taken to reduce any pressure in the cargo tank vapour space and prevent an electrostatic spark from occurring.

6.1.6 If the tanker is equipped with an inert gas system the isolation valve required by Pt.6, Ch.8, Cl.15.2.2.3.1.3 is to remain closed during vapour transfer.

6.1.7 Unless equipped with an automatic self-test and circuit monitoring feature, each tank level alarm system required by 3.2 on a cargo tank being loaded, is to be tested at the tank for proper operation prior to the start of cargo transfer.

Section 7

List of Cargoes – Vapour Control

7.1 List of cargoes

7.1.1 This list covers a range of cargoes normally carried and gives vapour pressure and specific gravity, as well as density of vapour/air mixture and vapour growth rate at 45°C.

Chemical name	Vapour pressure 45°C kPa, abs.	Vapour specific gravity	Vap/air density 45°C [kg/m^3]	Vapour growth rate
Acetic acid	6,34	2,07	1,413	1,02
Acetic anhydride	2,76	3,50	1,414	1,01
Acetone	68,95	2,00	2,098	1,20
Acetone cyanohydrin	2,76	2,90	1,396	1,01
Acetonitrile	0,21	1,41	1,339	1,00
Acrylic acid	2,76	2,48	1,383	1,01
Acrylonitrile	34,47	1,80	1,642	1,10
Adiponitrile	0,07	3,73	1,340	1,00
Allyl alcohol	12,41	2,00	1,475	1,04
Allyl chloride	73,08	2,64	2,660	1,21
Aminoethylethanolamine	0,07	3,59	1,340	1,00
Amyl acetate (commercial, iso-, n-, sec-)	2,34	4,50	1,429	1,01
Amyl acetate (iso-)	2,34	4,50	1,429	1,01
Amyl acetate (n-)	2,34	4,50	1,429	1,01
Aniline	0,28	3,22	1,345	1,00
Benzene (50/50 mix air)	31,03	2,80	2,542	1,09
Benzene, Toluene, Xylene mixture (10% Benzene or more)	50,33	2,80	2,337	1,15
Benzyl chloride	0,62	4,36	1,361	1,00
Butyl acetate (n-)	5,52	4,00	1,521	1,02
Butyl acetate (iso-, n-)	4,14	4,00	1,475	1,01
Butyl acetate (sec-)	10,34	4,00	1,680	1,03
Butyl acrylate (iso-, n-)	4,14	4,42	1,494	1,01
Butyl alcohol (iso-)	6,21	2,60	1,448	1,02
Butyl alcohol (n-)	3,45	2,60	1,399	1,01
Butyl alcohol (sec-)	8,96	2,60	1,496	1,03
Butyl alcohol (tert-)	19,31	2,60	1,679	1,06
Butyl benzyl phthalate	0,07	10,80	1,346	1,00
Butyl ether (n-)	2,76	4,50	1,445	1,01
Butyl methacrylate	2,00	4,90	1,424	1,01
Butylamine (all isomers) (vap. Press. Varies between (3.2-15)	68,95	2,50	2,479	1,20
Butylene glycol	0,07	3,10	1,340	1,00
Butylene oxide (1,2-)	49,64	2,49	2,154	1,14
Butyraldehyde (iso-, n-)	55,16	2,48	2,238	1,16
Butyraldehyde (n-, crude)	55,16	2,48	2,238	1,16
Butyraldehydes (crude)	55,16	2,48	2,238	1,16
Butyric acid	0,48	3,04	1,349	1,00

Chemical name	Vapour pressure 45°C kPa, abs.	Vapour specific gravity	Vap/air density 45°C [kg/m^3]	Vapour growth rate
Caprolactam solutions	4,83	3,90	1,492	1,01
Carbon disulfide	3,45	2,67	1,400	1,11
Carbon tetrachloride	37,23	5,49	3,181	1,02
Chloro-benzene	5,52	3,88	1,513	1,18
Chloro-form	62,05	4,25	3,562	1,00
Chloro-sulfonic acid	0,55	4,00	1,356	1,00
Creosote (all isometers)	0,07	3,72	1,340	1,00
Creosote (cole tar)	0,07	3,72	1,340	1,00
Creosote (wood)	0,07	3,72	1,340	1,00
Cresols	0,41	3,72	1,351	1,04
Crotonaldehyde	13,79	2,41	1,552	1,01
Cumene	4,14	4,20	1,484	1,09
Cyclo-hexane	31,03	2,90	1,988	1,00
Cyclo-hexanol	1,38	3,50	1,376	1,00
Cyclo-hexanone	1,38	3,40	1,375	1,01
Cyclo-hexylamine	4,27	3,42	1,452	1,01
Cymene (para)	3,17	4,62	1,465	1,01
Decaldehyde (iso-)	0,41	5,38	1,358	1,00
Decyclopentadiene	1,72	4,55	1,406	1,01
Decyl alcohol (all isomers)	0,07	5,30	1,341	1,00
Decyl alcohol (iso-)	0,07	5,30	1,341	1,00
Decyl alcohol (n-)	0,07	5,30	1,341	1,00
Decyl acrylate (iso-, n-)	0,07	7,30	1,343	1,00
Diacetone alcohol	0,69	4,00	1,361	1,00
Dibutylamine	1,24	4,46	1,385	1,00
Dichlorobenzene (all isomers)	0,69	5,07	1,369	1,00
Dichloroethane (1,1-)	68,26	3,41	3,152	1,20
Dichloroethyl ether (2, 2-)	0,28	4,90	1,350	1,00
Dichloromethane	131,00	3,00	4,227	1,38
Dichlorophenol (2, 4-)	0,07	5,60	1,342	1,00
Dichloropropane (1, 2-)	17,24	3,89	1,887	1,05
Dichloropropene (1, 3-)	37,92	3,84	2,526	1,11
Diethanolamine	0,07	3,65	1,340	1,00
Diethylamine	6,89	2,50	1,452	1,02
Diethyl-benzene	0,55	4,62	1,360	1,00
Diethylene glycol	0,07	3,66	1,340	1,00
Diethylene glycol butyl ether	0,07	5,50	1,342	1,00
Diethylene glycol ethyl ether	0,07	4,62	1,341	1,00
Diethylene glycol methyl ether	0,14	4,14	1,343	1,00
Diethylenetriamine	0,28	3,48	1,346	1,00
Diethylethanolamine	1,03	4,03	1,373	1,00
Diisobutyl ketone	3,31	4,90	1,480	1,01
Diisobutylamine	3,17	4,46	1,459	1,01
Diisobutylcarbinol	0,69	4,98	1,368	1,00
Diisobutylene	15,17	3,97	1,835	1,04
Diisopropanolamine	0,07	4,59	1,341	1,00
Diisopropylamine	25,51	3,50	2,041	1,07
Diethylamine solution (45% or less)	70,33	1,55	1,765	1,20
Dimethylformamide	2,07	2,51	1,373	1,01
Dioctyl phthalate	0,07	13,45	1,348	1,00

Chemical name	Vapour pressure 45°C kPa, abs.	Vapour specific gravity	Vap/air density 45°C [kg/m^3]	Vapour growth rate
Dioxane (1, 4-)	12,41	3,00	1,612	1,04
Diphenyl, Diphenyl ether mixture	0,07	5,87	1,342	1,00
Diphenyl methane diisocyanate	0,07	8,50	1,344	1,00
Dipropylene glycol	0,48	4,63	1,357	1,00
Dipropylene glycol dibenzoate	0,07	8,50	1,344	1,00
Dodecene	0,14	5,81	1,345	1,00
Dodecyl benzene	32,41	8,40	3,982	1,09
Epichlorohydrin	6,89	3,19	1,505	1,02
Ethanolamine	0,21	2,10	1,341	1,00
Ethoxy triglycol (crude)	0,07	6,14	1,342	1,00
Ethyl acetate	31,03	3,04	2,036	1,09
Ethyl acrylate	13,79	3,50	1,718	1,04
Ethyl alcohol	24,13	1,60	1,498	1,07
Ethyl butanol	0,97	3,40	1,364	1,00
Ethyl ether	158,58	2,55	4,049	1,46
Ethyl hexanol (2-)	0,14	4,50	1,343	1,00
Ethyl methacrylate	6,89	3,94	1,562	1,02
Ethyl-3-propylacrolein (2-)	0,83	4,35	1,369	1,00
Ethylamine	281,31	1,55	3,044	1,82
Ethylamine solution (72% or less)	106,87	1,56	1,998	1,31
Ethylbenzene	4,14	3,56	1,455	1,01
Ethylbutylamine (n-)	8,27	3,50	1,566	1,02
Ethylene chlorohydrine	3,03	2,78	1,398	1,01
Ethylene cyanohydrin	0,07	2,45	1,339	1,00
Ethylene dibromide	0,48	6,50	1,367	1,00
Ethylene dichloride	27,58	3,42	2,074	1,08
Ethylene glycol	0,07	2,21	1,339	1,00
Ethylene glycol butyl ether	22,08	4,07	2,085	1,06
Ethylene glycol ethyl ether	1,17	3,10	1,365	1,00
Ethylene glycol methyl ether	4,14	2,62	1,412	1,01
Ethylene diamine	6,21	2,10	1,413	1,02
Ethyl hexyl acrylate (2-)	0,14	6,35	1,346	1,00
Ethylidene norbomene	2,28	4,10	1,416	1,01
Formaldehyde solution (37% to 50%)	1,03	1,03	1,338	1,00
Formic acid	14,48	1,60	1,434	1,04
Furfural	1,03	3,31	1,364	1,00
Furfuryl alcohol	0,69	3,37	1,356	1,00
Gasoline blending stocks: Alkylates (50/50 mix air)	86,18	3,40	2,944	1,25
Gasoline blending stocks: Reformates (50/50 mix air)	86,18	3,40	2,944	1,25
Gasolines: Automotive (< 4,23 grams lead/gallon) (50/50 mix air)	86,18	3,40	2,944	1,25
Gasolines: Aviation (< 4,86 grams lead/gallon) (50/50 mix air)	86,18	3,40	2,944	1,25

Chemical name	Vapour pressure 45°C kPa, abs.	Vapour specific gravity	Vap/air density 45°C [kg/m^3]	Vapour growth rate
Gasolines: Casing head (natural) (50/50 mix air)	86,18	3,40	2,944	1,25
Gasolines: Polymer (50/50 mix air)	86,18	3,40	2,944	1,25
Gasolines: Straight run (50/50 mix air)	86,18	3,40	2,944	1,25
Glycerine	0,07	3,17	1,340	1,00
Heptane (all isomers)	17,24	3,45	1,804	1,05
Heptane (n-)	17,24	3,45	1,804	1,05
Hexamethylenediamine solution	0,07	1,00	1,338	1,00
Hexane (all isomers) (average vap. Pressure, real: 7-9)	55,16	3,00	2,555	1,16
Hexane (average vap. Pressure, real: 7-9)	55,16	3,00	2,555	1,16
Hexanol (1-)	6,89	3,52	1,530	1,02
Hydrochloric acid	72,39	1,26	1,546	1,21
iso-Butyraldehyde	55,16	2,48	2,238	1,16
iso-Propylamine	159,27	2,04	3,165	1,46
isophorone	0,07	4,75	1,341	1,00
Isoprene	158,58	2,35	3,699	1,46
Kerosene	1,03	4,50	1,378	1,00
Light virgin Naphta	96,53	0,66	0,976	1,28
Mesityl oxide	4,62	3,50	1,465	1,01
Methacrylic acid	0,69	2,50	1,350	1,00
Methacrylonitrile	74,46	1,17	1,478	1,22
Methyl acetate	42,06	2,60	2,080	1,12
Methyl acrylate	28,27	3,00	1,962	1,08
Methyl alcohol	48,26	1,11	1,397	1,14
Methyl ethyl ketone	31,03	2,50	1,851	1,09
Methyl isobutyl ketone	8,27	3,45	1,562	1,02
Methyl methacrylate	13,93	3,45	1,714	1,04
Methyl-5-ethyl-pyridine (2-)	1,10	4,18	1,377	1,00
Methyl-6-ethyl-aniline (2-)	0,07	4,67	1,341	1,00
Methylamine solution, 40%	89,63	1,07	1,407	1,26
Methylamyl acetate	2,34	5,00	1,441	1,01
Methylamyl alcohol	2,76	3,50	1,414	1,01
Methylpyridine (2-)	3,45	3,20	1,422	1,01
Methyl styrene (alpha-)	2,76	4,08	1,432	1,01
Mineral spirits (average vapour s.g.)	1,38	4,15	1,386	1,00
Morpholine	5,52	3,00	1,460	1,02
Naphtalene (molten)	0,07	4,42	1,341	1,00
Nitric acid (70% or less)	24,82	2,17	1,658	1,07
Nigrobenzene	0,14	4,24	1,343	1,00
Nitropropane (1- or 2-)	7,24	3,06	1,503	1,02
Nitrotoluene (o-, p-)	0,14	4,72	1,344	1,00
Nonane	2,76	4,41	1,442	1,01
Nonene	2,41	4,30	1,426	1,01
Nonyl phenol	0,07	7,59	1,343	1,00
Octyl alcohol (iso-)	0,21	4,50	1,346	1,00
Octyl nitrates (all isomers)	2,14	6,00	1,456	1,01

Chemical name	Vapour pressure 45°C kPa, abs.	Vapour specific gravity	Vap/air density 45°C [kg/m^3]	Vapour growth rate
Oil, edible: castor	1,03	10,00	1,441	1,00
Oil, Misc.: Crude	1,03	1,00	1,338	1,00
Oleum	0,07	2,76	1,339	1,00
Paraldehyde	57,23	4,55	3,578	1,17
Pentadiene (1, 3-)	117,62	2,36	3,102	1,34
Pentane (all isomers)	144,79	2,48	3,701	1,42
Pentane (iso-)	186,16	2,48	4,376	1,54
Perchloroethylene	8,48	5,83	1,790	1,02
Phenol, or solutions with 5% or more Phenol	4,14	3,24	1,440	1,01
Phosphoric acid	0,07	3,38	1,340	1,00
Phthalic anhydride (molten)	0,07	5,10	1,341	1,00
Propanolamine (iso-, n-)	0,55	2,59	1,348	1,00
Propionaldehyde	96,53	2,00	2,402	1,28
Propionic acid	2,07	2,56	1,374	1,01
Propionic anhydride	0,07	4,49	1,341	1,00
Propionitrile	16,55	1,40	1,411	1,05
Propyl acetate (iso-)	21,37	3,52	1,932	1,06
Propyl acetate (n-)	13,10	3,52	1,702	1,04
Propyl alcohol (iso-)	20,68	2,07	1,582	1,06
Propyl alcohol (n-)	8,27	2,07	1,436	1,02
Propyl alcohol (n-)	92,39	2,04	2,398	1,27
Propylene oxide	151,68	2,00	3,011	1,44
Pyridine	8,96	2,72	1,508	1,03
Sodium hydrosulfide solution (45% or less)	10,41	1,17	1,358	1,03
Sorbitol solution	0,07	3,20	1,340	1,00
Styrene	2,76	3,60	1,417	1,01
Styrene (curde)	2,76	3,60	1,417	1,01
Sulfufic acid	0,07	3,40	1,340	1,00
Tetrachloroethane (1, 1, 2, 2-)	6,89	5,80	1,703	1,02
Tetraethylene glycol	0,07	6,70	1,342	1,00
Tetraethylene pentamine	0,07	6,80	1,343	1,00
Tetrahydrofuran	58,61	1,35	1,564	1,17
Tetrahydronaphtalene	0,28	4,55	1,349	1,00
Toluene	10,34	3,14	1,582	1,03
Toluene diisocyanate	0,07	6,00	1,342	1,00
Trichlorobenzene (1, 2, 4-)	0,07	6,26	1,342	1,00
Trichloroethane (1, 1, 1-)	33,78	4,60	2,679	1,10
Trichloroethylene	24,13	4,54	2,280	1,07
Trichloropropane (1, 2, 3-)	1,03	5,60	1,391	1,00
Tricresyl phosphate (1% or more of ortho isomer)	0,07	12,70	1,347	1,00
Tricresyl phosphate (less than 1% of ortho isomer)	0,07	12,70	1,347	1,00

Chemical name	Vapour pressure 45°C kPa, abs.	Vapour specific gravity	Vap/air density 45°C [kg/m^3]	Vapour growth rate
Tridecanol	0,07	6,70	1,343	1,00
Triethanolamine	0,07	5,14	1,341	1,00
Triethylamine	17,24	3,49	1,811	1,05
Triethyl benzene	0,34	5,60	1,356	1,00
Triethylene glycol	0,07	5,17	1,341	1,00
Triethylene glycol methyl ether	0,07	5,66	1,342	1,00
Triethylenetetramine	0,07	5,04	1,341	1,00
Tripropylene glycol	0,07	6,63	1,342	1,00
Turpentine	0,28	4,84	1,350	1,00
Valeraldehyde (iso-, n-)	12,41	2,96	1,606	1,04
Valeraldehyde (n-)	12,41	2,96	1,606	1,04
Vinyl acetate	39,99	2,97	2,207	1,12
Vinyl ethyl ether	143,41	2,49	3,694	1,42
Vinylidene chloride	165,47	3,34	5,608	1,48
Vinyltoluene	0,83	4,08	1,366	1,00
Xylene (m-)	3,52	3,66	1,441	1,01
Xylene (p-)	3,52	3,66	1,441	1,01
Xylene (o-)	2,76	3,66	1,419	1,01

End of Chapter

Chapter 30

Environmental Protection

Contents

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1 *General Requirements*
2 *CLEAN-SEA Notation*
3 *CLEAN- AIR Notation*
4 *Environmental Protection – EP Notation*

Section 1

General Requirements

1.1 Application

1.1.1 A ship or a marine structure meeting the requirements for Environmental protection of this chapter will be assigned notations indicated in 1.1.2 to 1.1.4 below and a suitable entry will be made in the Register Book. Compliance with this chapter and the assignment of notations is optional.

1.1.2 In order to be eligible for the class notation “EP”, a ship is to meet the requirements covered in Section 2, Section 3 and Section 4.

1.1.3 Section 2 gives the requirements to be complied with for control and prevention of polluting substances into the sea. A ship meeting the requirements of this section will be eligible for notation “CLEAN-SEA”.

1.1.4 Section 3 gives the requirements to be complied with for control and prevention of emission of polluting substances in the air. A ship meeting the requirements of this section will be eligible for notation “CLEAN- AIR”.

1.1.5 It is a prerequisite for assignment of the class notation EP that the applicable requirements of the following Annexes to the International Convention for the Prevention of Pollution from ships, MARPOL 73/78, as amended are fully complied with.

i) Annex I – Regulations for the Prevention of Pollution by Oil
ii) Annex II – Regulations for the Control of Pollution by Noxious Liquid Substances in Bulk
iii) Annex III – Regulations for the Prevention of Pollution by Harmful Substances Carried by Sea in Packaged Form
iv) Annex IV – Regulations for the Prevention of Pollution by Sewage from Ships
v) Annex V – Regulations for the Prevention of Pollution by Garbage from Ships
vi) Annex VI – Regulations for the Prevention of Air Pollution from ships.

In addition to above, the following International Standards, guidelines and recommendations are also to be complied with, as applicable:

i) Diesel Engine Exhaust NOx Content – NOx Technical Code, (2008), as amended.
ii) Refrigerants and fire-fighting gases – Montreal Protocol on Substances that Deplete the Ozone Layer
iii) Shipboard Incinerators – IMO Resolution MEPC 244(66) 2014 Standard Specification for

Shipboard Incinerators, as amended by MEPC.368(79).

iv) Cargo Vapour Emission Control – IMO standard for Vapour Emission Control Systems, MSC/Circ.585 or USCG Title 46 CFR Part 39 Vapour Control System (See part 5, Ch. 29 of the Rules for Construction and Classification of Steel ships)

v) Ballast Water – International Convention for the Control and Management of Ships' Ballast Water and Sediments (BWM Convention)

vi) International Convention on the Control of Harmful Anti-fouling Systems on Ships, 2001.

Where the flag administration has additional requirements to the above related environmental protection, those requirements are to be complied with and applicable certificates of compliance are to be obtained by the owners, for assignment of the EP notation.

1.1.6 Compliance with the rules are to be verified through inspection, measurements and sampling of defined environmental parameters in accordance with the requirements of the rules in this chapter and with identified standards or guidelines.

1.1.7 Environmental management systems, e.g. ISO 14000, are not part of the rules. Fulfillment of the requirements in the rules may, however, facilitate goals and improvement tasks defined through ISO 14000 implementation.

1.1.8 Special ship types and ships with unconventional propulsion machinery systems see Table 1; will be subject to special consideration covering specific items not covered by the standard rules. Tankers for oil smaller than 150 tons gross tonnage and other ships smaller than 400 tons gross tonnage will also be subject to special consideration.

1.1.9 Any matter detrimental to the environmental performance of the ship in turn affecting the assignment of the EP class notation should be notified to IRS.

Table 1 : Ship types and propulsion machinery systems that will be specially considered

Ship types: Liquefied gas carriers Floating storage, production and offloading systems
Propulsion machinery systems: Gas turbines

1.2 Definitions : main parameters

1.2.1 Emissions to air or emissions

All emissions to air which are caused by or needed for the operation of the ship, energy consumers, cargo, passengers and crew on board a vessel and any toxic emissions caused by operation, protection and conservation of vessel or cargo.

1.2.2 Discharges to sea or discharges

All discharges to sea which are caused by or needed for operation of the ship, energy consumers, cargo, passengers and crew on board a vessel and any toxic discharges caused by protection and conservation of vessel or cargo.

1.2.3 Deliveries to shore

Delivery of potential pollutants to shore for controlled disposal, recycling etc.

1.2.4 Accidental emissions or discharges

All emissions to air or discharges to sea, caused by unforeseen or unplanned events, of substances needed for energy consumers, cargo, passengers and crew on board a vessel and any toxic emissions caused by protection and conservation of vessel or cargo.

1.3 Definitions and characteristics, systems and components

1.3.1 "Ballast Water" means water with its suspended matter taken on board a ship to control trim, list, draught, stability or stresses of the ship.

1.3.2 Ballast water system

Ballast water systems comprise:

- Tanks for ballast water
- Associated piping and pumping systems.

1.3.3 Ballast water Management: means mechanical, physical, chemical and biological processes, either singularly or in combination, to remove, render harmless or avoid the uptake or discharge of Harmful Aquatic Organisms and Pathogens within Ballast Water and Sediments.

1.3.4 Bilge water

Bilge water is oily water removed from the machinery space bilges. Bilge water removed from cargo holds of bulk carriers and general cargo vessels is not affected by the rules in this chapter.

1.3.5 Cargo handling systems

Cargo handling systems covered by the rules in this chapter comprise:

- Cargo tank vents for tankers with cargoes where evaporation may occur during loading, transport and discharge. (e.g. Tanker for Oil, Tanker for Chemicals, Tanker for Liquefied Gas)

- Pumping and piping systems for tankers carrying cargoes that may cause global or local pollution.

1.3.6 Cargo residues

Cargo residues cover remains of cargo (oil or chemical contaminated water from cargo tank area, slop tanks and cargo pump room). Cargo residues may be present in discharged water used for cleaning cargo tanks and discharged ballast water from tankers without 100% segregated ballast tanks.

1.3.7 Casualty

Casualties covered by the rules in this chapter are defined as serious unplanned incident(s), e.g. grounding, collision or other incident damaging the hull's structural integrity, fire, sinking etc. The consequence of a situation of total loss will be:

- Cargo, or cargo containment system, is damaged and cargo spilt into the sea
- The integrity of the ship is damaged causing oil spills, etc.

1.3.8 Combustion machinery

Combustion machinery comprises:

- Internal combustion engines, both marine diesel engines and gas turbines
- Boilers.

The rules in this chapter cover the emission of oxides of nitrogen (NO_x) and sulphur (SO_x) from combustion machinery exhaust gases to the atmosphere

1.3.9 Fire-fighting system

The rules in this chapter cover the active fire-fighting media used in fixed fire-righting systems.

1.3.10 Garbage

Garbage includes all kinds of food wastes, domestic wastes and operational wastes, all plastics, cargo residues, cooking oil, fishing gear and animal carcasses excluding fresh fish and parts thereof, generated during normal operation of the ship and liable to be disposed of continuously or periodically except those substances excluded specifically. Garbage generated by passengers on passenger vessels is included in the rules in this chapter.

Systems for garbage treatment include the following:

- Garbage sorting systems, typically: glass, paper, metals etc.
- Compacting systems
- Storage systems
- Incinerators.

Sewage and waste oils are defined separately and not as garbage.

1.3.11 Painting and antifouling systems

Antifouling paint of the underwater area gives a continuous discharge of active ingredients to the sea. This effect is covered by the rules in this chapter.

In addition, painting and antifouling systems emit volatile organic compounds (VOC) during application. This effect is not considered by the rules in this chapter.

1.3.12 Port

The ship is considered in port from ordering "stand by" prior to entering port to ordering "full ahead" when leaving the port. The time will be confirmed by entries in the ship's logbook.

1.3.13 Refrigeration systems

The rules in this chapter cover refrigerant media used in refrigeration plants on conventional reefer ships, fishing vessels, liquefied gas carriers with re-liquefaction plants and other ships with centralized cargo refrigeration systems. They are also applicable for centralized air conditioning and refrigeration systems onboard. Domestic type stand-alone air conditioning units and refrigerators are not covered.

Refrigerants may escape to the atmosphere through refrigeration system leakage, service work and dismantling at the end of the system's working life.

1.3.14 Sewage

Sewage is defined to include:

- Drainage and other wastes from all toilets, urinals and WC scuppers. For passenger ships this applies both for crew and passenger areas
- Drainage from medical premises (dispensary, sick bay etc.) via wash basins, wash tubs and scuppers located in such rooms
- Drainage from spaces containing living animals, or
- Other waste waters when mixed with any of the drainage systems defined above.

Systems for sewage handling have varying complexity including:

- Holding tanks
- Sewage treatment plants, e.g.:
  - Disinfecting
  - System for separating solids from liquids
  - Comminuters/grinders
  - Discharge systems, ashore, or to sea.

1.3.15 Waste oil

Shipboard waste oils comprise the following:

- Used lubrication and hydraulic oils
- Oil leaked from lubrication and hydraulic oil systems
- Leakage from fuel oil systems
- Sludge from fuel and lubrication oil treatment systems (separators, filters, etc.).

Waste oils may be dealt with onboard, or pumped ashore. Cargo residues in slop tanks, etc. are considered separate from operational waste oils.

1.4 Abbreviations

BCH Code	Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (Bulk Chemical Code)
CFC	Common name for refrigerants, e.g. : CFC-11 (Trichlorofluoromethane), CFC-12 (Dichlorofluoromethane) etc.
CFR	Code of Federal Regulations
GWP	Global Warming Potential (CO_2 = 1, time horizon 100 years)
HCFC	Hydrochlorofluorocarbons
HFC	Hydrofluorocarbons
IACS	International Association of Classification Societies
IAPP Certificate	International Air Pollution Prevention Certificate
IBC Code	The International Code for the Construction and Equipment of ships Carrying Dangerous Chemicals in Bulk
IOPP Certificate	International Oil Pollution Prevention Certificate
ISM Code	"International Safety Management (ISM) Code"' means the International Management Code for the Safe Operation of Ships and for Pollution Prevention as adopted by the Organization (IMO) by resolution A.741(18), as may be amended by the Organization. (ISM Code 1.1)
ISO	International Organization for Standardisation
IMO	International Maritime Organization
MARPOL or MARPOL 73/78	The International Convention for the Prevention of Pollution from Ships, 1973 as modified by the Protocol of 1978 thereto
MSC	Maritime Safety Committee (IMO)
MEPC	Maritime Environmental Protection Committee (IMO)
NO_x	Oxides of nitrogen
ODP	Ozone depleting potential (Compared to CFC 11)
SCR	Selective Catalytic Reduction
SO_x	Oxides of sulphur
TBT	Tributyltin (active ingredient in antifouling paint)
USCG	US Coast Guard
VOC	Volatile organic compound
SEEMP	Ship Energy Efficiency Management Plan
EEDI	Energy Efficiency Design Index
IEE Certificate	International Energy Efficiency Certificate

1.5 International recommendations, standards and references

1.5.1 International recommendations, standards and references have been used as foundation for the rules, although even if the rule requirements may be more stringent. When setting the emission and discharge limits and determining the measuring procedure, due consideration has been given to technical and practical limitations inherent in the design and construction of different types of ships.

1.5.2 International recommendations, standards and references with provisions used by IRS when developing the rules are reflected in the references specified in 1.1.5. Unless a particular edition is explicitly referred to, the latest edition of each standard applies.

1.6 Information and Documentation

1.6.1 All plans, certificates, operational procedures and other documentation of compliance as detailed in section 2, 3 and 4 are to be available to IRS prior to assigning the class notation “CLEAN-SEA” OR “CLEAN-AIR” OR “ EP”.

1.6.2 Discharge limiting and monitoring equipment are to be certified as type approved.

1.7 Environmental management Plan

1.7.1 An Environmental Management Plan, specific to the ship is to be developed and submitted to IRS for information describing at least the procedures given in sections 2, 3 and 4 respectively.

Section 2

Clean Sea Notation

2.1 Application

2.1.1 This section gives the requirements applicable for assignment of the Notation "CLEAN-SEA"

2.2 Information and Documentation

2.2.1 Table 2.1 lists the documents, to be submitted for approval and/or information in connection with prevention of polluting substances into sea:

i) by oil
ii) by noxious liquid substances
iii) by harmful substances carried by sea in packaged form
iv) by sewage
v) by garbage
vi) by other sources

The documents listed in Table 2.1 do not take account of drawings necessary for the issuance of the "International Pollution Prevention certificate for the Carriage of Noxious Liquid Substances in Bulk" or of the "International Certificate of Fitness for the Carriage of dangerous Chemicals in Bulk" or of the "International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk", as applicable.

Guidance note:

It should be noted that some of the required documentation is additional to and different from documentation normally considered "Class documentation" as delivered from the new building yard. This is in particular the case for operational procedures specified in section 2, 3 or 4 as applicable that may require input from the Owner.

2.3 Design and procedural requirements

2.3.1 Prevention of sea pollution by oils

2.3.1.1 General

Provided the ship has been granted a valid IOPP certificate issued by an Administration or by a recognized organization on behalf of an Administration, in accordance with MARPOL 73/78 Annex I, as applicable, the additional specific requirements as per 2.3.1.2, 2.3.1.3 & 2.3.1.4 apply.

Table 2.1 : Documents to be submitted in connection with prevention of polluting substances into sea		
No.	**A/I**	**Document**
1	I	Copy of the IOPP Certificate
2	A	Drawings with indication of tanks' volume and distance from the base line and shell plates as per 2.3.1.2 **(1)**
3	A	Schemes of the fuel oil, lubricating oil and relevant residue systems details containing the necessary information to verify the requirements as per 2.3.2 **(1)**
4	I	Oil Systems Record Book
(1) if these details are shown in the drawings which are used for the classification of the ship, such drawings and schemes may also be used to verify the requirements of these rules. Note : A = to be submitted for approval; I = to be submitted for information.		
5	I	Copy of the "International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk" or copy of the "International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk" or copy of the "International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk, as applicable.
(1) I = to be submitted for information.		
6	I	General arrangement plan with indication of the zone intended for the stowage of the harmful packaged substances in relation to the other zones of the ship
7	I	Plans of systems and equipment to discharge the harmful substances in case of emergency and to dispose of and water wash possible leaks
(1) I = to be submitted for information.		
8	I	General arrangement plan with indication of the sewage treatment plant enclosing details on treatment procedures (2)
9	I	Copy of the ISPP Certificate and type approval certificate for the sewage treatment plant according to MEPC 227(64).
10	A	Calculation of volume of holding tank(s) for sewage
11	I	General information on control and monitoring systems (2)
12	I	Sewage record book
Note A = to be submitted for approval; I = to be submitted for information. (2) Not required if the sewage treatment plant is an approved type.		
13	I	General information on garbage treatment equipment (2) (3)
14	A	Garbage management plan including garbage record book as specified by MARPOL 73/78 Appendix to Annex V
15	I	General description of control and monitoring systems of garbage treatment equipment (2)
Note: A = to be submitted for approval; I = to be submitted for information. **(2)** Not required if the garbage treatment equipment is of an approved type. **(3)** For incinerators see Sec.3.		

16	A	Ballast water management plan, including a ballast water record book, developed in accordance with G4 Guidelines (MEPC.127(53)) (Earlier approved plans in accordance with IMO Res.A.868(20) may be accepted unless the plan requires revision due to the installation of a ballast water management system.)
17	I	Copy of the AFS Certificate and details of TBT free antifouling paint
18	A	Calculation of volume of grey water holding tank(s) (passenger ships only)
19	I	Grey water record book (passenger ships only)
Note: A = to be submitted for approval; I = to be submitted for information.		
20	I	Ship Environmental Management Plan giving details as per 2.3.1.4, 2.3.3.6, 2.3.4.4, 2.3.5.4, 2.3.6.4 and 2.4.
Note I = to be submitted.		

2.3.1.2 Tank arrangement

a) For the purpose of these rules, tanks for fuel oil, sludge tanks and tanks for lubricating oil, whether or not for waste, of capacity exceeding 30 [m^3] are to be considered. Tanks with capacity not exceeding 30 [m^3] may be excluded provided the aggregate capacity of such excluded tanks is not greater than 600 [m^3]. Overflow tanks are to be included unless they are provided with an alarm for detection of liquid and operational procedures are foreseen for keeping such tanks empty. Double bottoms for lubricating oil located under the main engine are not considered in these Rules.

b) Such tanks are to be located above the moulded line of the bottom shell plating nowhere less than the distance 'h' specified below:

h = B/20 [m] or 2.0 [m] whichever is lesser.
The minimum value of h = 0.76 [m].
In the turn of the bilge area and at locations without a clearly defined turn of bilge, the tank boundary lines are to run parallel to the line of the midship flat bottom as shown in Fig.2.3.1.1.

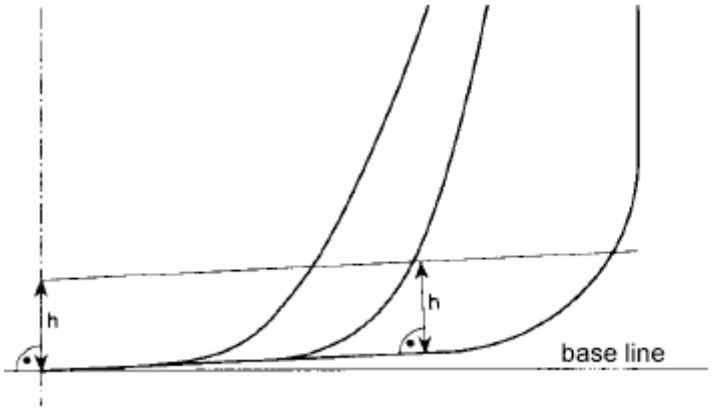


Fig.2.3.1.1 : Minimum distance of tank boundaries above bottom shell

c) For ships having an aggregate oil tank capacity of less than 5000 [m^3], such tanks are to be located inboard of the moulded line of the side shell plating, nowhere less than the distance "w" measured at right angles to the side shell as specified below: (See Fig.2.3.1.2).

w = 0.4 + 2.4 C/20000 [m];

The minimum value of w = 1.0 [m], however for individual tanks with oil capacity of less than 500 [m^3] the minimum value is 0.76 [m].

"C" is the total volume of the oil tanks [m^3] at 98% tank filling.

d) For ships having an aggregate oil tank capacity of 5000 [m^3] and over, such tanks are

to be located inboard of the moulded line of the side shell plating, nowhere less than the distance “w” measured at right angles to the side shell as specified below: (See Fig.2.3.1.2)

w = 0.5 + C/20000 [m] or 2.0 [m], whichever is the lesser.
The minimum value of w = 1.0 [m].

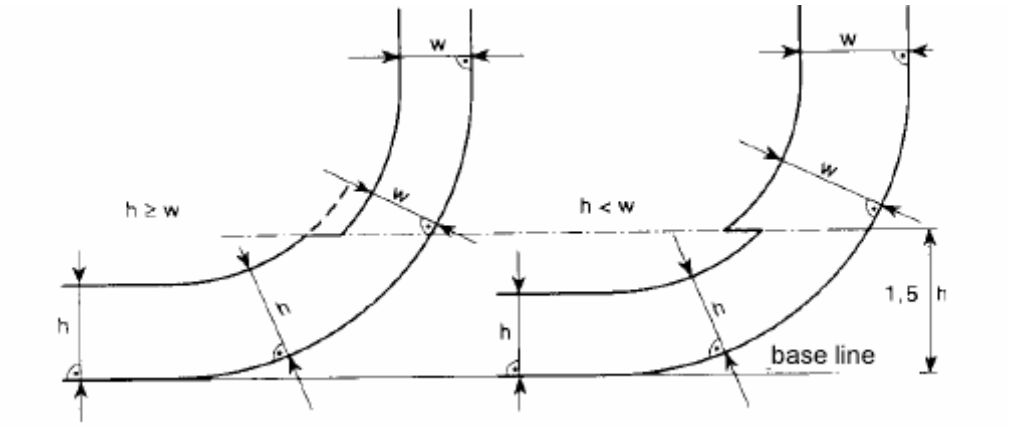


Fig. 2.3.1.2 : Minimum distance of tank boundaries from side shell

e) Lines of oil piping located at a distance from the ship’s bottom of less than “h” as defined in b) or from the ship’s side less than “w” as defined in c) and d) above, are to be fitted with valves or similar closing devices within or immediately adjacent to the oil tank.

These valves are to be capable of being brought into operation from a readily accessible enclosed space the location of which is accessible from the navigation bridge or propulsion machinery control position without traversing exposed decks. The valves are to close in case of remote control system failure (fail in a closed position) and are to be kept closed at sea at any time when the tank contains oil except that they may be opened during oil transfer operations.

f) Small suction wells in oil tanks may extend below the boundary line defined by the distance “h” provided that the distance between the well bottom and the bottom shell plating is not less than 0.5h.

g) Tanks of any volume intended to contain fuel or lubricating oil are not to be used also for ballast water.

h) All tanks and cofferdams, if any, around the tanks are to be so arranged as to be adequately inspected; such cofferdams are to be effectively protected against corrosion by means of proper protective coatings of a light colour in order to be easily examined.

2.3.1.3 Systems

i) Fuel oil and lubricating oil system

a) All fuel oil and lubricating oil tanks of capacity greater than 10 [m^3] are to be fitted with an overflow system and a high level alarm.

Acceptable alternatives are:

- An overflow system and a flow alarm in the overflow main
- No overflow system and two high level alarms (for instance at 90% and 95% of filling).

b) The alarms signals are to be given in a suitable station from which bunkering or transfer operations are controlled.

ii) Oily bilge water system for machinery spaces

a) The oil filtering equipment, complying with MARPOL Annex I, is to be provided with an oil content meter and with a 15 ppm alarm combined with automatic stopping device.

b) The effluent from the 15 ppm filtering equipment is to be capable of being

recirculated to a bilge holding tank. Bilge Holding tank is to be so arranged as to allow periodical removal of sediments and is to be fitted for the pre-separation of bilge water before transferring it to the separating or filtering equipment. Bilge holding tank is to be separate and independent from the sludge tanks.

c) The volume V of such a tank, in [m^3] is to be at least:

$V = 1 + 5.5P \cdot 10^{-4}$

Where P is the power of the propulsion engine plant in kW. In any case volume V need not be greater than 15 [m^3] .

In specific cases for instance with regard to the service, range and installed power, will be specially considered.

d) For ships operating with fuel oil having a mass density at 15°C greater than 0.94 [kg/cm^3] and viscosity at 50°C greater than 110 centistokes, the possibility of heating such a tank is to be provided.

iii) Sludge tank

a) Ships operating with heavy fuel oil are to be provided with tanks for sludge from the fuel oil purifiers without internal structures and with a suitable heating system.

b) The sludge tank is to be so arranged as to allow periodical removal of sediments. Sludge is either to be disposed of onboard through the incinerator or discharged ashore and is to be recorded in the oil record book.

c) Use of boilers for sludge disposal onboard is not allowed unless a treatment system or special device is installed to improve emissions generated by sludge incineration, subject to IRS's approval.

d) The total capacity of sludge tanks is to meet the criteria specified in MARPOL Annex I, Unified Interpretations 16.1 and 16.2, as applicable, based on the maximum period of voyage between ports where sludge can be discharged ashore, daily fuel oil consumption, homogenizers or sludge incinerators installed, and fuel purification.

2.3.1.4 Procedures

a) The lub.oil consumption of all systems having an oil to sea interface, such as main and auxiliary engines cooled by sea water, controlled pitch propellers, sterntubes, bow and stern thrusters, stabilizers, PODs etc., is to be recorded at least once a week in an "Oil Systems, record book" aimed at detecting, through unusually high consumption, oil leakage through sealing.

The log-book is to contain the list of all systems concerned, the consumption of each system recorded at least every week and corrective actions when carried out.

b) The Ship Environmental Management Plan referred to in 2.4 is to include procedures covering the following:

- Oily waste management including discharge criteria;
- Preparation, filling in and maintenance of the oil record book and of the "Systems oil consumption log-book";
- Periodical calibration, at least every twelve months, of the oil content meter referred to in 2.3.1.3; documentation is to be kept on board for examination during periodical surveys;
- Periodical cleaning of the bilge holding tank and of the sludge tank;
- Spillage during bunkering;
- Periodical checks of the overflow systems/alarms.

2.3.2 Prevention of sea pollution by noxious liquid substances carried in bulk as cargo

2.3.2.1 Provided the ship has been granted a valid "NLS" or a valid "ICOF CHE" certificate, as applicable, issued by an Administration or by a recognized organisation on behalf of an Administration, in accordance with MARPOL

73/78 Annex II, as applicable, there are no additional specific requirements to be complied with on this matter for the purpose of the issuance of the CLEAN-SEA notation.

2.3.3 Prevention of sea pollution by harmful substances carried by sea as cargo in packaged form

2.3.3.1 General

Harmful substances carried as cargo in packaged form are to be properly stowed and secured so as to minimize the hazards to the marine environment, according to MARPOL 73/78 Annex III, as amended.

2.3.3.2 Stowage

a) Harmful substances are to be properly stowed and secured so as to minimize the hazards to the marine environment without impairing the safety of the ship and persons on board.

b) Certain harmful substances may, for sound scientific and technical reasons, need to be prohibited for carriage or be limited as to the quantity which may be carried aboard any one ship. In limiting the quantity, due consideration is to be given to the size, construction and equipment of the ship, as well as the packaging and the inherent nature of the substances.

c) Each ship carrying harmful substances is to have a special list or manifest setting forth the harmful substances on board and the location thereof. A detailed stowage plan which sets out the location of the harmful substances on board may be used in place of such special list or manifest.

2.3.3.3 Empty packages

Empty packages which have been used previously for the carriage of harmful substances are themselves to be treated as harmful substances, unless adequate precautions have been taken to ensure that they contain no residue that is harmful to the marine environment.

2.3.3.4 Marking

a) Packages containing harmful substances are to be marked suitably with the correct technical name (trade names alone are not to be used) and, additionally, are to be durably marked or labeled to indicate that the substance is a marine pollutant. Such identification is to be supplemented where possible by other means, for example, by use of the relevant United Nations number.

b) The method of marking the correct technical name and of affixing labels on packages containing a harmful substance is to be such that this information will still be identifiable on packages surviving at least three months' immersion in the sea. In considering suitable marking and labeling, account is to be taken of the durability of the materials used and of the surface of the package.

c) Packages containing small quantities of harmful substances may be exempted from the marking requirements.

2.3.3.5 Leaks

a) Jettisoning of harmful substances carried in packaged form is not permitted, except where necessary for the purpose of ensuring the safety of the ship or saving life at sea.

b) Appropriate measures based on the physical, chemical and biological properties of harmful substances are to be taken to regulate the washing of leakages overboard, provided that compliance with such measures would not impair the safety of the ship and persons on board.

2.3.3.6 Procedures

The Ship Environmental Management Plan referred in 2.4 is to include procedures addressing the requirements of this item.

2.3.4 Prevention of sea pollution by sewage

2.3.4.1 General

Provided the ship has been given a valid ISPP Certificate issued by an Administration or by a recognized organization on behalf of an Administration, in accordance with MARPOL 73/78 Annex IV, as applicable, the additional specific requirements as per 2.3.4.2 to 2.3.4.4 are to be complied with.

2.3.4.2 Discharge of sewage at sea

All sewage discharges whether to sea or shore based facilities are to be recorded in the sewage record book with indication of the date, location and quantity of sewage discharged and are to comply with MARPOL Annex IV discharge requirements.

2.3.4.3 Sewage treatment plant

a) A sewage treatment plant, meeting operational requirements based on the standards and test methods as detailed in Resolution MEPC.227(64), as amended, is to be installed on board.

b) The ship is to be equipped with holding tank(s) for sewage with sufficient capacity to allow storage of sewage when in port or in no discharge areas. The minimum total capacity of such tank(s) is to be 2 days based on the maximum number of persons on board and 96 litres/person/day if a conventional system is used and 11 litres/person/day if a vacuum system is used.

   A smaller volume, in any case not lower than 50% of the above capacity, may be accepted provided that:

   - The ship is equipped with a post-treatment system for sewage, able to reduce the volume of the effluent (ex. By recycling part of the treated sewage water for on board use);
   - 2 days retention is ensured;
   - Technical documentation, including results of onboard tests, of the system's efficiency and of effluent volume reduction is documented to the satisfaction of IRS.

c) The sewage holding tank is to have a visual means of indication of the amount of its contents e.g.: float gauge.

d) The ship is to be equipped with a pipe leading to the exterior convenient for the discharge of sewage to a reception facility; such pipe is to be fitted with a standard shore connection in compliance with MARPOL Annex IV, and the materials, piping, fittings and equipment are to comply with the applicable requirements of the various Sections of these Rules.

e) Sewage holding tanks are to be equipped with high level alarms.

2.3.4.4 Procedures

The Ship Environmental Management Plan referred to in 2.4 is to include procedures covering the following:

- Sewage management including discharge criteria and use of holding tanks in port and no discharge areas;
- Preparation, filling in and maintenance of the sewage record book;
- Procedures for the cleaning and safe entry of sewage treatment systems and holding tanks, including the use of suitable personal protective equipment, are to be provided and implemented
- Only for passenger and ro-ro passenger ships, periodical analysis, at least every 6 months, of samples of the effluent to verify compliance with IMO Resolution MEPC 227(64) the results of such analysis are to be included in the sewage record book and made available during periodical surveys. The analysis is to be performed by an approved test laboratory;
- Disposal of sewage treatment plant residues.
  If the ship is not in a condition to dispose at sea of sewage treatment plant residues in accordance with international or national regulations, such residues are to be disposed of ashore or by incineration.

2.3.5 Prevention of sea pollution by garbage

2.3.5.1 General

The applicable requirements of MARPOL 73/78 Annex V, as amended, are to be complied with, in addition to the following additional specific requirements.

2.3.5.2 Placards, garbage management plans and garbage record-keeping

a) Placards which notify the crew and passengers of the disposal requirements of MARPOL Annex V are to be fitted on board as applicable.

b) The placards are to be written in the official language of the State whose flag the ship is entitled to fly and, for ships engaged in voyages to ports or offshore terminals under the jurisdiction of other States, in English or French or Spanish

c) A garbage management plan and record book is to be available on board. This plan is to provide written procedures for collecting, storing, processing and disposing of garbage, including the use of the equipment on board. It is also to designate the person in charge of carrying out the plan. Such plan is to be in accordance with the guidelines in Appendix 1 and written in the working langue of the crew.

d) For passenger ships, special consideration in the garbage management plan is to be given to the following potentially hazardous wastes:

- Photographic and x-ray development fluids
- Dry-cleaning solvents and waste fluids
- Print shop fluids
- Photocopying and laser printer cartridges
- Unused pharmaceuticals and those which are past their use-by-date
- Batteries
- Fluorescent and Mercury vapor lamp bulbs.

2.3.5.3 Waste recycling

For passenger ships other than ro-ro passenger ships:

a) a strategy of waste recycling is to be foreseen, adopted and documented;

b) the minimum total quantity of wastes landed for recycling (Wr) is to be 50% of recyclable wastes produced on board (Wb), where Wb = 40 Kg/person/year based on the number of persons the ship is certified to carry.

The amount of waste landed for recycling is to be recorded in the garbage record book and different wastes are to be collected and landed separately.

Fore the purpose of this Rule, recyclable wastes are:

- Plastic
- Aluminum
- Glass
- Paper-cardboard.

2.3.5.4 Procedures

The Ship Environmental Management Plan referred to in 2.4 is to include procedures covering the Garbage management and waste recycling, according to 2.3.5.

2.3.6 Prevention of sea pollution by other sources

2.3.6.1 Transfer of harmful aquatic organisms and pathogens through ballast water.

A ballast water management plan, including a ballast water record book, is to be developed in accordance with IMO G-4 Guidelines (Earlier approved plans in accordance with IMO Res.A.868(20) may be accepted unless the plan requires revision due to the installation of a ballast water management system).

Unless stricter requirements are enforced by the Port State, it is recommended that ballast water exchange is carried out during international voyages at not less than 200 miles from the nearest land or, if not possible, at not less than 50 miles from nearest land in a zone with water depth not less than 200 [m] taking into account IMO Guidelines G-6 and G-11. The ship is not to be required to deviate from its intended voyage, or delay the voyage in order to comply with these requirements.

Ballast Water Management Systems are to be type approved in accordance with BWMS Code or Res. MEPC 279(70) or Res. MEPC. 174(58), as applicable, based on the date of installation.

2.3.6.2 Antifouling system

a) Antifouling systems for the hull are to be of TBT-free type.

b) The AFS Certificate and documentation of the TBT-free antifouling system are to be kept on

board for checking during periodical surveys.

2.3.6.3 Release of grey water

The requirements of this item apply only to passenger and ro-ro passenger ships

a) Ships are to be equipped with holding tank(s) for grey water with sufficient capacity to allow storage of grey water when in port for at least 2 days. The total capacity of grey water holding tanks is to be based on the maximum number of persons on board and 200 litre/person/day.

b) Grey water holding tanks are to be equipped with high level alarms.

c) If the same tanks are used to hold treated sewage and grey water, their capacity is to be at least the sum of the capacities for the treated sewage holding tanks in 2.3.4.3 and the tanks for grey water.

   A smaller volume, in any case not lower than 50% of the above capacity, may be accepted provided that:

   - The ship is equipped with a system for treating grey-water, able to reduce the volume of the effluent (e.g. by recycling part of the treated grey water for on board use);
   - 2 days retention is ensured;
   - Technical documentation, including results of onboard tests, of the system's efficiency and of effluent volume reduction is documented to the satisfaction of IRS.

d) The discharging criteria do not apply when the discharge of grey water is necessary for securing the safety of the ship and those on board, or saving life at sea, or when the discharge results from damage to the ship or its equipment.

e) All grey water discharges whether to sea or shore base facilities are to be recorded in the grey water record book with indication of the date, location and quantity of grey water discharged. If the grey water is discharged to sea, the records are to include information on distance to the nearest land.

2.3.6.4 Procedures

The Ship Environmental Management Plan referred to in 2.4 is to include procedures required in 2.3.6 including the grey water discharge criteria and use of holding tanks in ports and in no discharge areas.

2.4 Ship Environmental Management Plan

An Environmental Management Plan, specific to the ship, is to be developed and made available on board.

The Manual is to contain at least the procedures required as per 2.3.1.4, 2.3.3.6, 2.3.4.4, 2.3.5.4 and 2.3.6.4 and is to include:

- The officer/s in charge for carrying out each procedure.
- Documents and manuals required
- Log-books/records to be filled in
- Time schedule when applicable (e.g. checking, sampling, etc.).

The Ship Environmental Management Plan is to be submitted to IRS for information.

2.5 Inspections and Tests

2.5.1 Inspections and testing during construction

2.5.1.1. Materials

Materials for piping and equipment which are used for the construction of equipment and systems to be installed on board in connection with the requirements of this section are to be tested in accordance with the sections of the Rules applicable to similar types of piping or equipment and their class and/or design conditions.

2.5.2 Tests and inspection during fabrication

Piping and equipment relative to the means for preventing sea pollution, as stated in this

section, are to be inspected and tested during fabrication in accordance with the sections of the rules applicable to the type of piping or equipment and the relevant class and/or design conditions.

2.5.3 Inspection and testing after installation onboard

2.5.3.1 After installation on board, the equipment and systems installed in connection with the requirements of this section are to be tested in the presence of the Surveyor under operating conditions. The control, monitoring and alarm system are also to be tested in the presence of the Surveyor or their functioning is to be simulated by a procedure agreed with IRS.

Section 3

Clean Air Notation

3.1 Application

3.1.1 This section gives the requirements applicable for the assignment of the notation CLEAN-AIR.

3.2 Scope

3.2.1

a) The notation CLEAN-AIR is assigned to ships fitted with efficient means to control and prevent the emission of polluting substances in the air, in accordance with Pt. 1, Ch.1.

b) The requirements of this section are intended to prevent air pollution from any of the following hazards in addition to those stipulated by MARPOL 73/78 Annex VI:

- Emissions of ozone depleting substances
- Emissions contributing to global warming
- Emissions of nitrogen oxides (NO_x)
- Incinerators

In order for the notation CLEAN-AIR to be assigned, all the above polluting hazards are to be considered and the ship is to be fitted with means and/or operational measures to simultaneously prevent all those which are applicable, in relation to its characteristics and the likelihood of producing any or all of such emissions.

3.3 Additional requirements for oil tankers, chemical tankers and gas carriers

3.3.1 A ship which is classed with one or more of the following class notations:

- Oil tanker
- Chemical tanker
- Liquefied gas carrier
- Ore or Oil Carrier
- Oil or Bulk Carrier

is also to comply with the requirements of Part 5, Ch 29, in respect of prevention of vapour emissions.

The assignment of VCS1 or VCS2 notation is prerequisite for assigning the notation CLEAN-AIR.

3.4 Definitions

3.4.1 Ozone Depleting Substances.

Ozone depleting substances are those substances, which are defined in paragraph 4 of Article 1 of the Montreal Protocol on Substances that Deplete the Ozone Layer, 1987, listed in Annexes A, B, C, or E to the Protocol in force at the time of the application of these Rules.

In general, the following ozone depleting substances are used on ships; however, this list is not to be considered comprehensive of all ozone depleting substances that for any reason may be found in a ship.

- Halon 1211– Bromochloridefluoromethane
- Halon 1301 – Bromotriflurometha ne
- Halon 2402
- Halon 114B2 – 1,2-Dibromo-1,1,2,1-tetrafluoroethane
- CFC-11 – trichlorofluoromethane
- CFC-12 – Dichlorodicluoromethane
- CFC-113 – Trichloro-1,2,2-trifluoroethane
- CFC-114 – 1,2-Dichloro-1,1,2,2,-tetrafluoroethane
- CFC-115 – Chloropentafluoroethane.

3.4.2 Shipboard incineration

Shipboard incineration is the incineration of wastes or other matter on board a ship, if such wastes or other matter are generated during the normal operation of that ship.

3.4.3 Ozone depleting potential

Ozone depleting potential (ODP) is the potential of ozone depletion compared to CFC-11. Values of ODP for ozone depleting gasses are provided in the "Montreal Protocol on Substances that Deplete the Ozone Layer".

Table 3.1 : Documents to be submitted for prevention of emission of ozone depleting substances, incinerators, environment management plan SOx emissions

No.	A/I	Document
1	I	General arrangements of refrigeration plants
2	A	Scheme of refrigerating plant(s) showing the retention facilities as per 3.5.4
3	A	Data sheets with list of intended refrigerants to be used in the different refrigeration systems and their GWP and ODP
4	I	Refrigerant record book
(1) A = to be submitted for approval; I = to be submitted for information.		
Document for incinerator		
5	I	Copy of type approval certificate of statement of compliance with IMO Res. MEPC 244(66), as amended by MEPC.368(79).
6	I	Design data, including at least design pressure and design temperature, as applicable (2)
7	A	Piping diagram (2)
8	I	System for waste supply and in incinerator feeding (2)
9	A	Instrumentation, monitoring, alarm and control systems (2)
10	I	Operating manual
Note: A = to be submitted for approval; I = to be submitted for information. (2) Not required if the incinerator is of an approved type.		
11	I	Ship Environmental Management Plan as per 3.5.6, 3.6.9, 3.8.2, 3.9.1.
Note: I = to be submitted for information.		
Documents for Sox emissions		
12	A	Detailed plans of systems and equipment to limit SOx emission in the SOx emission control areas (2)
13	A	Ship SOx Emission Control Plan (SECP) (if EGCS is fitted onboard)
14	A	EGC System Technical Manual (Scheme A or B, as relevant and applicable)
15	A	Onboard Monitoring Manual (OMM)
16	I	Fuel changeover procedure
17	I	Ship Implementation Plan (prepared in accordance with MSC.1/Circ.878)
Note: A = to be submitted for approval I = to be submitted for information; (2) Only if such systems and equipment are installed.		
Documents for NOx emissions		
18	A	NOx Technical File

No.	A/I	Document
Documents for CO_2 emissions		
19	I	Ship Energy Efficiency Management Plan (SEEMP)
20	A	Preliminary EEDI Technical File [2]
21	A	Model Test Program for EEDI calculation [2]
22	I	Model Test Report for EEDI Calculation [2]
23	A	Sea Trial Program for EEDI calculation [2]
24	I	Sea Trial Report for EEDI calculation [2]
25	A	Final EEDI Technical File [2]
Note : (1) A = to be submitted for approval; I = to be submitted for information. (2) Only for ship types for which EEDI is applicable as required by MARPOL Annex VI Chapter 4		

3.4.4 Global warming potential

Global warming potential (GWP) is the potential global warming of a gas compared with CO2 on a time horizon of 100 years. Values of GWP for various refrigerants and gases are published by the US Environmental Protection Agency (EPA).

3.4.5 Documents to be submitted

3.4.6 Table 3.1 lists the documents to be submitted for the assessment of the measures taken to prevent the emission of ozone depleting substances and sulphur oxides

- Requirement of Incinerators
- Preparation of environment management plan

The documents in Table 3.1 are additional to those relative to refrigerating installations and fire-fighting systems listed in the applicable sections of the Rules.

3.5 Emission of ozone depleting substances

3.5.1 Application

a) The following requirements apply to ships with refrigerating facilities, such as refrigerated cargo ships, liquefied gas carriers with reliquefaction plants and other ships with centralized cargo refrigeration systems.

b) They also apply to:

- Centralized refrigeration systems for provision stores
- Centralized air conditioning plants
- Fixed fire-fighting systems.

c) They do not apply to domestic type stand-alone refrigerators and air conditioning units.

3.5.2 Halogenated substances

a) The use of halogenated substances (e.g. Halon and CFC) as refrigerant or fire-fighting means is prohibited, with the exception of hydro-chlorofluorocarbons (HCFCn), which are permitted until 1 January, 2020.

b) Refrigerants of centralized refrigeration systems are to have a global warming potential, GWP < 2000.

3.5.3 Prevention of leakage

a) Means are to be provided to limit leaks to the atmosphere of refrigerants or their vapours in the event of failure of the plant, as well as in the case of discharge of refrigerant to an onshore reception facility.

b) Annual refrigerant leakage is to be less than 10% of the total refrigerant charge of each system. The leakage is to be documented by consumption figures recorded in a refrigerant log-book to be kept on board and made available during periodical surveys.

c) Procedures need to be established such that, in the event that annual leakage exceeds 10%, corrective actions are undertaken.

3.5.4 Evacuation facilities

The system is be fitted with evacuation (e.g. compressors) and retention facilities having the capability to retain all the refrigerants, should the need to evacuate the whole plant arise in an emergency.

3.5.5 Maintenance and servicing

The plant is to be designed in such a way as to minimize the risks of medium release in the case of maintenance, repair or servicing, i.e. it is to be designed considering the possibility of isolating those sections which are to be serviced by a system of valves and bypasses, in such a way as not to stop the operation of the plant while in service, preventing the risk of release of the medium outside of the plant.

3.5.6 Procedures

The Ship Environmental Management Plan, as per 3.9.1 is to contain the following:

- Procedures to be followed to minimize the risk of depleting the refrigerant or the refrigerant vapours in all operative and emergency conditions

- Procedures for preparing, filing and updating the refrigerant log-book.

3.6 Emission of oxides of nitrogen (NOx) from Marine diesel engines

3.6.1 Requirements for emissions from diesel engines apply to all diesel engines with power output in excess of 130 kW, except emergency diesel engines, engines installed in lifeboats and any other device or equipment intended to be used solely in case of emergency.

3.6.2 The maximum allowed total emission levels for marine diesel engines installed onboard shall be as defined in 3.6.3 for NO_x and 3.7.1 and 3.7.2 for SO_x.

3.6.3 NO_x emissions shall comply with the limits defined in MARPOL 73/78, Annex VI, Reg.13: Nitrogen Oxide (NO_x).

3.6.4 The reduction of oxides of nitrogen can be achieved by the introduction of equipment and arrangements such as, but not limited, to, those listed below:

- Engine adjustments or modifications
- Water addition to the combustion process by water injection, emulsification or humidification
- Using gas (as per IGF Code) as fuel
- Exhaust Gas Recirculation
- Selective Catalytic Reduction (SCR).

3.6.5 All arrangements, fuel oil additives etc. introduced to reduce the NO_x emission levels are subject to IRS approval and shall:

- be operated according to manufacturer's instructions
- allow normal engine performance and safe operation in case of failure
- have features so that its operations and performance can be documented by a log book record
- have sufficient mechanical fastening to ensure safe ad steady support and avoid damaging vibrations
- where applicable, have adequate hatches for inspection and maintenance of installations or systems.

3.6.6 The following arrangements will be subject to special considerations:

- arrangements where NO_x reduction is carried out on more than one engine

- arrangements using SCR cleaning where the reactor unit is positioned between the engine and the turbo blower.

3.6.7 For ships where NO_x emissions are controlled by devices added to the fuel or exhaust systems, such devices shall be operated and controlled in accordance with procedures incorporating the manufacturer instructions and IRS's approval.

3.6.8 For ships fitted with Exhaust Gas Recirculation (EGR), the discharge of EGR Bleed-Off water is to comply with Res. MEPC 307(73).

Guidance note :

Engine modification and adjustments

NO_x reductions by modification of engine parameters, water injection, fuel emulsification and/or by adjusting engine settings in order to influence the combustion characteristics, should be specified by the engine manufacturer and carried out under his supervision. The chosen combination of modifications and adjustments should aim to avoid an increase in the engine's fuel consumption.

Selective Catalytic Reduction (SCR)

Requirements for storage and use of SCR reductants are indicated in Part 4, Chapter 4, Section 4, Clause 4.14.

The reducing agent should be specified by the manufacturer. If other agent than Urea-solution is used, this will be subjected to special considerations.

In the case where the NO_x emission level is used to verify or control the reduction agent injection rate, the level should be detected by an analyser based upon one of the following methods:

- Electrochemical cell
- Infrared chemiluminescene.

NO_x level measurements
NO_x level measurements on engines with NO_x reduction arrangements should comply with methods as decided by IRS.

3.6.9 The Ship Environmental Management Plan is to include as a minimum the following:

a) procedures for periodical checks of the emission related engine parameters and components referred to in the technical file. The periodical checks are to be carried out at least every 6 months and after maintenance or replacement of any such component;

b) procedures of preparing, filing in and updating the engine record book.

c) Methods to control NOx emissions.

d) Means are to be provided to check proper operation of the equipment.

3.7 Emission of oxides of sulphur (SOx)

3.7.1 Sulphur content of any fuel oil used on board ships is not to exceed the limits as required by MARPOL 73/78 Annex VI, Regulation 14. SO_x emission limits are generally achieved by use of low sulphur content fuel oil. Other Approved Alternative arrangements providing an equivalent level of environmental protection will be considered.

3.7.1.1 If an Exhaust Gas Cleaning System (EGCS) is fitted in order to obtain the required reductions of SO_x, it is to be certified to IMO Resolution MEPC.340(77) – 2021 Guidelines for Exhaust Gas Cleaning Systems. Also for ships fitted with EGCS a SOx Emission Compliance Plan (SECP) complying MEPC.340(77) is to be submitted for approval. Further, EGC System Technical Manual (as per Scheme A or B, as relevant and applicable) and the Onboard Monitoring Manual (OMM) are to be submitted for approval.

3.7.1.2 The discharge washwater is to satisfy the criteria given in the Resolution MEPC.340(77).

3.7.2 Changes of fuel type when entering and leaving port, or SO_x - controlled areas are to be documented by entries in the ship's logbook.

3.8 Incinerators

3.8.1 Applicability

a) The following requirements apply to ships with a permanently installed shipboard incinerator, generally complying with the requirements of IMO Resolution MEPC 244(66), as amended by MEPC.368(79)

"Standard specifications for shipboard incinerators". Shipboard incinerators described in MARPOL Annex VI, regulation 16(2) are to have IMO type approval certificate for assigning CLEAN-AIR notation.

b) Ships without a permanently installed incinerator, which use the main or auxiliary boilers for incinerating oil sludge generated during normal operations, cannot be assigned CELAN-AIR notation.

3.8.2 Miscellaneous requirements

a) Shipboard incinerators cannot be used to incinerate the following substances:

- Cargo residues of substances indicated in Annexes I, II and III of the IMO International Convention for the Prevention of Pollution from Ships 1973/78, "MARPOL 73/78" as amended and related contaminated packing materials.

- Polychlorinated biphenyl (PCBs)

    - Garbage, as defined in Annex V of "MARPOL 73/78", containing more than traces of heavy metals.
    - refined petroleum products containing halogen compounds;
    - sewage sludge and sludge oil either of which is not generated on board the ship; and
    - exhaust gas cleaning system residues.

b) A Manufacturer's operating manual for the incinerator is to be provided on board.

c) Safety measures

1) The flue gas outlet temperature is to be continuously monitored.

2) For incinerators fed continuously by an automatic feeding systems, the waste supply is to be shut off if the temperature indicated above falls below 850°C.

3) For batch loaded incinerators, the unit is to be designed in such a way that the temperature in the combustion chamber reaches 600°C in not less than 5 minutes.

d) The Ship Environmental Management Plan as per 3.9.1 is to address the procedures required as per para a).

3.9 Ship Environmental Management Plan

3.9.1 An Environmental Management Plan, specific to the ship, is to be developed and made available on board.

- The indication of person(s) in charge for each procedure to be carried out
- Documents and manuals required
- Log-books/records to be filled in
- Time schedule when applicable (e.g. periodical checks).

The Ship Environmental Management Plan is to be submitted to IRS for information.

3.10 Inspections and tests

3.10.1 Installations to limit the emission of refrigerants

a) Materials

Materials for piping and equipment specifically designed to limit the emission of refrigerants are to be tested in accordance with the applicable requirements for testing materials intended to be used for the construction of similar types of piping and equipment and their classes and/or design conditions.

b) Tests and inspection during fabrication

Piping and equipment specifically designed to limit the emission of refrigerants are to be inspected and tested during fabrication in accordance with the requirements applicable to similar types of piping or equipment and their classes and/or design conditions.

c) Tests after installation

After installation on board, the plant acceptance trials are to include the operation of the evacuation of all of the refrigerant from the plant to the reception facilities without any release of refrigerant and/or refrigerant vapours. The control,

monitoring and alarm systems are also to be tested in the presence of the Surveyor, or their functioning is to be simulated by a procedure agreed with IRS.

3.10.2 Incinerators

The incinerator is to be IMO type approved and after installation function tests as described in operation manual are to be carried out.

3.11 Energy Efficiency

3.11.1 Attained Energy Efficiency Design Index (EEDI) is to be calculated for ship types as required by MARPOL Annex VI Chapter 4.

3.11.2 Preliminary EEDI Technical file is to be submitted for approval, calculating the Attained EEDI in accordance with IMO Guidelines on "The Method of Calculation of The Attained Energy Efficiency Design Index (EEDI) For New Ships".

3.11.3 The verification, survey and certification of EEDI is to be carried out in accordance with IMO Guidelines on Survey and Certification of The Energy Efficiency Design Index (EEDI).

3.11.4 The Final EEDI Technical File is to be submitted incorporating necessary corrections after sea trials.

3.11.5 The attained EEDI is to be less than the Required EEDI for the particular ship type and size and applicable phase as per MARPOL Annex VI Chapter 4.

3.11.6 Ship Energy Efficiency Management Plan (SEEMP) is to be prepared in accordance with IMO Guidelines and kept onboard ship at all times.

Section 4

Environment Protection (EP) Notation

4.1 General

4.1.1 The notation "EP" (Environmental Protection) will be assigned to vessels complying with the requirements of this section.

4.1.2 The requirements given in section 2 and 3 of Rules are also to be complied with as required for CLEAN-SEA and CLEAN-AIR notations.

4.2 Ship recycling

4.2.1 General

Some of the problems associated with ship recycling might be addressed at the design and construction stage, not only in relation to the ships themselves but also in respect of ships equipment.

The first step is to identify any potentially hazardous materials which might be incorporate, as a matter of routine, in the structure of ships and their equipment and, where practicable, consider using less hazardous alternatives.

4.2.2 Initial stage

The initial stages might include an evaluation of:

a) the type, amount and potential hazard of materials utilized and their location on board a ship;

b) the activities expected during the operation of the ship and any potentially hazardous wastes which might be generated; and

c) the feasibility of addressing the potential for hazardous waste generation by considering:

   1) product reformulation – installing components utilising less potentially hazardous materials;

   2) cleaner production technologies – which generate less waste;

3) process medication – to generate less waste;

4) input substitution – utilizing less potentially hazardous consumables or those which generate less waste; and

d) on-site, closed-loop recycling – systems that recycle wastes on board the ship.

4.2.3 Design stage

When designing and constructing a vessel, due account is to be taken of the ship's ultimate disposal, by:

a) using materials which can be recycled safely and in an environmentally sound manner; and

b) minimizing the use of materials known to be potentially hazardous to health and the environment.

4.2.4 Inventory of hazardous materials

In order to contribute towards minimizing potential problems related to protection of the environment in the recycling of vessels, guidelines on ship recycling have been adopted by IMO with resolution A.962(23) to given guidance to all stakeholders in the ship recycling process.

An inventory of hazardous materials is to be developed according to the above mentioned Resolution A.962 (23) – IMO Guidelines on Ship recycling, paragraph 5.

The inventory is a document facilitating the application of these Guidelines providing information with regard to materials known to be potentially hazardous utilized in the construction of the ship its equipment and systems. This is to accompany the ship throughout its operating life. Successive Owners of the ship are to maintain the accuracy of the inventory and incorporate into it all relevant design and equipment changes, with the final Owner delivering the document, with the ship, to the recycling facility.

4.3 Documents to be submitted

a) Table 4.1 lists the plans and documents to be submitted.

b) Table 4.2 lists the procedures and record books to be submitted.

Table 4.1 : Documents to be submitted		
No.	**A/ I**	**Document**
1	I	Copy of the NLS certificate or “International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk” or “International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk”, as applicable
2	I	Copy of IOPP Certificate
3	A	Drawings of the bilge system including volume of the bilge holding tank (4)
4	I	Copy of the type approval certificate of the oil filtering equipment and alarm
5	A	Schematic arrangement of the fuel oil, lubricating oil and relevant overflow system containing the information necessary to verify the requirements of 2.3.1.3 (4)
6	I	Copy of the approved SOPEP plan including the list of emergency equipment
7	I	Details of enrolment in an Emergency Response Service
8	I	Copy of the ISPP Certificate and copy of the sewage treatment plant type approval certificate
9	A	Drawings of the sewage system including piping, holding tank for sewage and alarms (4)
10	A	Calculation of volume of holding tank for sewage (4)
11	I	Copy of the AFS Certificate or statement of compliance
12	A	Drawings of the grey-water system including piping, holding tank and alarms (1) (4)
13	A	Calculation of volume of grey water holding tank (1) (4)
14	A	Data sheets with the list of refrigerants and fixed fire-fighting means used, their quantity and GWP values (4)
15	A	Details of plans of systems and equipment to limit SOx emissions in SOx emission control areas (2) (4)
16	A	Drawings of fuel oil system, arrangements and procedures for use of separate fuel oil according to requirements given in 3.11 (4)
17	I	Incinerator type approval certificate (2)
18	I	Documentation of compliance with technical requirements for the class notation VCS (3)
19	I	Copy of the EIAPP Certificate or statement of compliance for each engine detailed in 3.6 issued by a recognized organization, as applicable
20	A	Engine Technical File and record book for each engine detailed in 3.6.
21	A	Details and operating manual(s) of NOx control equipment referred to in 3.6 (2)
22	I	General arrangement plan with indication of the zone intended for the stowage of the harmful packaged substances in relation to the other zones of the ship
23	I	Plans of systems and equipment to discharge the harmful substances in case of emergency and to dispose of and wash possible leaks
24	A	SOx Emission Control Plan (SECP) [For ships fitted with EGCS)
25	A	EGC System Technical Manual (Scheme A or B, as relevant and applicable)
26	A	Onboard Monitoring Manual (OMM)
27	I	Ship Implementation Plan (prepared in accordance with MSC.1/Circ.878)
28	A	Preliminary EEDI Technical File [5]
29	A	Model Test Program for EEDI calculation [5]
30	I	Model Test Report for EEDI Calculation [5]
31	A	Sea Trial Program for EEDI calculation [5]
32	I	Sea Trial Report for EEDI calculation [5]

Table 4.1 : (Contd.)		
No.	**A/ I**	**Document**
33	A	Final EEDI Technical File [5]
34	I	Copy of International Energy Efficiency Certificate (IEE Certificate)

(1) Passenger ships only
(2) Only if such a system is installed on board
(3) Only ships having service notation according to 3.3
(4) For ships in service, alternative documentation may be accepted provided it is sufficient for the execution of the initial survey
(5) Only for ship types for which EEDI is applicable as required by MARPOL Annex VI Chapter 4.

Note : A = to be submitted for approval
I = to be submitted for information.

Table 4.2 : Operational Procedures and Record Books		
No.	**A/ I**	**Document**
1	I	Oil systems record book
2	I	Sewage record book
3	A	Garbage management plan including garbage record book
4	A	Ballast water management plan including ballast water record book
5	I	Grey water record book (2)
6	I	Refrigerant log book and procedures as per 3.5
7	I	Oily wastes management procedures according to 2.3.1.4b)
8	I	Accidental oil discharge management procedures according to 2.3.1.4b)
9	I	Sewage management procedures according to 2.3.4.4
10	I	Garbage management and waste recycling procedures according to 2.3.5(1)
11	I	Grey water management procedures according to 2.3.5(2)
12	I	Ship Environmental Management Plan
13	I	Fuel management procedures for controlling SOx
14	I	Incinerator management procedures according to 3.8
15	A	Inventory of Hazardous materials according to 4.2.4
16	I	Ship Energy Efficiency Management Plan (SEEMP)
17	A	EGCS Record Book (If fitted)

(1) Only for passenger ships other than ro-ro passenger ships
(2) Only for passenger ships and ro-ro passenger ships

Note : A – to be submitted for approval
I – to be submitted for information.

4.4 Inspections, tests and surveys

4.4.1 Inspections and testing during construction

Materials, systems or equipment which are installed on board or modified in order to comply with the requirements of this Section 3 are to be surveyed and tested according to the applicable Rules of IRS.

4.4.2 Initial survey

Following the satisfactory review and approval of the plans and other documentation detailed in 4.2, an initial survey is to be carried out on board in order to:

- Verify that hull and machinery arrangements are in accordance with the approved documentation;
- Test, in the presence of the Surveyor and under working conditions, the equipment and systems covered by these Rules including their control, monitoring and alarms;
- Verify and presence on board of the Ship Environmental Manager;
- Verify the presence on board of the certificates, record and log-books, and Environmental Management Plan requested by these Rules;
- Carry out the engine parameter check according to the engine technical file, of the engines;
- For existing ships, assess and document the satisfactory maintenance of equipment and systems.

End of Chapter

Chapter 31

Bridge Design and Navigation Equipment / Systems

Contents

Section	
1	*General*
2	*Documentation*
3	*Bridge Design*
4	*Bridge Design and Systems*
5	*Integrated Bridge Systems*
6	*Carriage Requirements*
7	*Tests and Trials*

Section 1

General

1.1 Application

1.1.1 The requirements in this Chapter are intended to achieve a successful ergonomic design of the bridge and bridge equipment, which will improve the reliability and efficiency of navigation. These contain ergonomic requirements as well as a functionally oriented bridge layout to support watch-keeping personnel in their tasks by a user-centred design of the bridge equipment and layout.

1.1.2 Ships complying with the requirements of this Chapter may be assigned additional Class notations **BD**, **BDS** or **IBS** as indicated in 1.1.3 to 1.1.5.

1.1.3 Additional class notation '**BD**' (Bridge Design) will be assigned to ships complying with requirements of Section 1 through Section 3 of this Chapter, as applicable. Requirements for bridge design and layout are indicated in these Sections.

1.1.4 Additional class notation '**BDS**' (Bridge Design and Systems) will be assigned to ships complying with requirements of Section 1 through Section 4 of this Chapter, as applicable. Requirements for vital equipment and systems are indicated in Section 4.

1.1.5 Additional class notation '**IBS**' (Integrated Bridge System) will be assigned to Ships fitted with integrated bridge system which allows simplified and centralized bridge operation of the main functions of Navigation, Communication and manoeuvring of the vessel including monitoring of such critical systems, complying with the requirements of Section 1 through Section 7 of this Chapter, as applicable.

1.2 Reference testing and performance standards

1.2.1 IEC 60945:2002, Maritime navigation and radio-communication equipment and systems – General requirements – Methods of testing and required test results.

IEC:61162 (all parts) Maritime navigation and radio-communication equipment and systems – Digital interfaces.

IEC 61174:2015 - Electronic Chart Display and Information System (ECDIS) - Operational and performance requirements, methods of testing and required test results

ISO 8468:2007 – Ship's bridge layout and associated equipment – requirements and guidelines.

IMO A.823(19) – Performance standards for automatic radar plotting aids (ARPAs).

IMO A.1021(26) – Code on alerts and indicators 2009.

IMO A.694(17) – General requirements for shipborne radio equipment forming part of the global maritime distress and safety system (GMDSS) and for electronic navigational aids.

SN.1/Circ.288 – Guidelines for Bridge Equipment and Systems, Their Arrangement and Integration (BES)

IMO MSC.302(87) – Adoption of Performance standards for Bridge Alert Management

IMO MSC/Circular 566:1991 – Provisional guideline on the conduct of trials in which the officer of the navigational watch acts as the sole lookout in periods of darkness.

IMO MSC.Res.252(83) (as amended by MSC.Res.452(99) – Performance standards for integrated navigation systems.

1.3 Definitions

1.3.1 Configuration of complete system : all operational functions of the IBS Bridge as installed.

1.3.2 Configuration available : operation(s) allocated to and available at each workstation.

1.3.3 Configuration in use : operation(s) and task(s) currently in use at each workstation.

1.3.4 Connectivity : a complete data link and the presence of valid data.

1.3.5 Essential functions : functions related to determination, execution and maintenance of safe course, speed and position of the ship in relation to the waters, traffic and weather conditions.

Such functions include but are not limited to:

- Route planning
- Navigation
- Collision avoidance
- Manoeuvring
- Docking
- Monitoring of internal safety systems
- External and internal communication related to safety in bridge operation and distress situations.

1.3.6 Essential information : that information which is necessary for the monitoring of essential functions.

1.3.7 Functionality: ability to perform an intended function. The performance of a function normally involves a system of displays and instrumentation.

1.3.8 Statutory requirements : IMO Conventions, Regulations, Resolutions, Codes, Recommendations, Guidelines, Circulars and related ISO and IEC standards.

1.3.9 Integrated bridge system (IBS) : any combination of systems which are interconnected in order to allow centralized access to sensor information from workstations to perform two or more of the following operations:

- Passage execution
- Communications
- Machinery monitoring
- Loading, discharging and cargo monitoring
- HVAC monitoring for passenger ships.

1.3.10 Integrity : ability of a system to provide users with accurate, timely, complete and unambiguous information and warnings within a specified time when the system is not in use.

1.3.11 Latency : time interval between an event and the resulting information, including time for processing, transmission and reception.

1.3.12 Multi-function display: a single visual display unit which can present, either simultaneously or through a series of selectable pages, information from more than one operation of a IBS.

1.3.13 Novel systems or equipment: systems or equipment which embody new features not fully covered by provisions of SOLAS V but which provide at least equivalent standard of safety (SOLAS Regulation 19.6).

1.3.14 Part : individual subsystem, equipment or module.

1.3.15 Performance check : a representative selection of short qualitative tests, to confirm correct operation or essential functions of the IBS.

1.3.16 Sensor : a device which provides information to or is controlled or monitored by the IBS.

1.3.17 Workstation : A position at which one or several tasks, constituting a particular activity, is carried out.

1.3.18 Navigation workstation : A workstation at which the navigator may carry out all tasks relevant for deciding, executing and maintaining course and speed in relation to waters and traffic. The instrumentation and controls at the navigation workstation should allow the navigator to:

- Analyse the traffic situation;
- Monitor position, course, track, speed, time, propeller revolutions and pitch, rudder angle, depth of water, rate of turn and wind speed and direction;
- Alter course and speed;
- Effect internal and external communications;
- Give and receive sound signals;
- Control navigational lights;
- Monitor and acknowledge navigational alarms;
- Confirm his well-being and watch-keeping awareness; and
- Record navigational data.

1.3.19 Main steering position : That part of the navigation workstation where those controls and instrumentation relevant to controlling the ship's course are located.

1.3.20 Conning position : A place on the bridge which is used by navigators when commanding, manoeuvring and controlling a ship.

1.3.21 Voyage planning workstation : A workstation at which the navigator may carry out the following tasks without affecting the actual navigation of the vessel:

- Examine and update charts and other relevant documentation;
- Plan a voyage as a series of waypoints, courses, speeds and turns;
- Calculate an estimated time of arrival at various points on the voyage; and
- Determine and plot the ship's position.

1.4 Abbreviations used in this chapter

1.4.1 Abbreviations used in this Rules and annexes :

AIS	:	Automatic identification system
ARPA	:	Automatic radar plotting aid
BNWAS	:	Bridge navigational watch and alarm system
DSC	:	Digital selective calling
EGC	:	Enhanced group call
EPIRB	:	Emergency position indicating radio beacon
GMT	:	Greenwich Mean Time
HF	:	High Frequency
INMARSAT	:	International Mobile Satellite Organization
ISO	:	International Standards Organization
ITU-R	:	International Telecommunication Union – radio sector
IBS	:	Integrated Bridge System
ITU-T	:	International Telecommunication Union – telecommunication sector
MARPOL	:	IMO Convention for the prevention of pollution by ships
MEPC	:	Marine Environmental Protection Committee of IMO
MF	:	Medium Frequency
MSC	:	Maritime Safety Committee of IMO
NAV	:	IMO Subcommittee on Safety of Navigation
NAVTEX	:	System for broadcast and reception of maritime safety information
OOW	:	Officer of the watch
r.p.m.	:	Revolutions per minute
UTC	:	Universal coordinated time
VDU	:	Visual display unit
VHF	:	Very high frequency
ECDIS	:	Electronic Chart Display Information System

Section 2

Documentation

2.1 Documents to be submitted

2.1.1 Relevant plans and data are to be submitted for approval and/or information as follows:

2.1.2 The documents / details given below are required to be submitted for **BD**, **BDS** and **IBS** notations. A complete layout and dimensional details of the bridge area and workstations installed therein. This is to include:

a) Arrangements of windows, including dimensions and angles of inclination, dimensions of frames, height above deck surface of upper and lower edges, type of glass and details of clear view arrangements (wipers, fresh water wash, de-icing / de-misting, sunscreens, etc.)

b) Fields of vision from the bridge workstations, including any blind sectors caused by obstructions outside of the wheelhouse.

c) Location and arrangement of workstations, including dimensions of consoles, layout of instrumentation and controls, handrails, seating, etc.

d) Clearances between floor and ceiling or between floor and the underside of ceiling mounted instruments, throughout the wheelhouse.

e) Arrangements for the general illumination of the bridge and the individual illumination of workstation instruments and controls.

f) Details of wheelhouse ventilation and heating systems.

g) Details of internal communication systems operable from the bridge.

h) Arrangements / details of exterior catwalk in front of bridge windows.

i) Details of non-skid flooring

j) Details of wheelhouse doors, including hold-back arrangements.

k) Location of toilet.

l) Arrangements for drainage of bridge decks.

m) Arrangements / details as to the measures to be taken to minimize hazards to personnel.

2.1.3 All electrical and electronic appliances installed on the bridge and vicinity of the bridge (also refer Section 4, Cl. 4.6) other than mandatory navigation and communication equipment; having been type tested according to IEC 60945:2002 (EMC tested for conducted and radiated emission), as well as loose equipment placed on board by the builder or owner are to be listed, and be provided with at least the following information. The list and the evidence of equipment are to be kept onboard.

- Equipment description
- Manufacturer
- Type / model
- Evidence of EMC compatibility which may be:
  - type approval certificate covering EMC requirements for bridge installations;
  - test certificate or report / conformity statement; or exemption statement.

Section 3

Bridge Design

3.1 Bridge Design

3.1.1 General

a) The bridge is to be designed and arranged with the aim of:

- Facilitating tasks to be performed by the bridge team and the pilot in making full appraisal of the situation and in navigating the ship safely under all operational conditions.

- Promoting effective and safe bridge resource management.

- Allowing for expeditious, continuous and effective information processing and decision-making by the bridge team and the pilot.

- Preventing or minimizing excessive or unnecessary work and any condition or distraction on the bridge which may cause fatigue or interfere with the vigilance of the bridge team and the pilot.

b) The design of bridge is to be governed by:

- The functions and related tasks to be carried out on the bridge, systems used and methods of task performance.

- The fields of vision required for visual observations from each of the workstations.

- Composition of the bridge team and the procedures required for safe operations under all identified conditions.

- The type and range of equipment to be provided for performance of the tasks at the individual workstations and elsewhere on the bridge.

- Performance standards for equipment mentioned in 1.2

3.1.2 Arrangement

a) Consoles, including the chart table, are to be positioned, so that the instrumentation they contain is mounted in such a manner as to face a person looking forward. As far as practicable, operating surfaces are to be normal to the operator's line of sight.

b) From other workstations within the wheelhouse it is to be possible to monitor the navigation workstation and to maintain an effective lookout.

c) The main access to the bridge is to be by means of an internal stairway. Secondary external access is also to be provided.

d) Clear passage of at least 7000 [mm] width is to be available to allow movement around the bridge with a minimum of inconvenience. Particular attention is to be paid to the following routes which are to be as direct as possible:

- From bridge wing to bridge wing, a clear passage of at least 1200 [mm] in width;

- Between the internal entrance to the bridge and the route as above a clear passage of at least 700 [mm] in width is to be provided;

- Between adjacent workstations, a clear passage of at least 700 [mm] is to be provided.

e) The distance from the bridge front bulkhead or from any console and installation placed against the front bulkhead, to any console or installation placed away from the bridge front is to be preferably 1000 [mm] and not less than 800 [mm].

f) Space necessary for operating at a workstation is to be considered as part of the workstation and is not to be part of the passageway.

g) The clear height between the wheelhouse deck surface covering and the underside of the deckhead is to be at least 2250 [mm]. The lower edge of deckhead mounted equipment is to be at least 2100 [mm] in open areas, passageways and at standing workstations.

h) The height of entrances and doors to the wheelhouse from adjacent passageways is to be at-least 2000 [mm].

i) All wheelhouse doors are to be operable with one hand. Bridge wing doors are not to be self-closing and means are to be provided to hold the doors open.

j) Toilet facilities are to be provided on or adjacent to the bridge.

3.1.3 Lighting

3.1.3.1 The level of lighting is to enable bridge personnel to perform all bridge tasks, including maintenance and chart and office work, by day and night. Controls, indicators, instruments, keyboards, etc. on the bridge are to be capable of being seen in the dark, either by means of internal lighting within the equipment or the wheelhouse lighting system. A satisfactory level of flexibility within the lighting system is to be available to enable the bridge personnel to adjust the lighting in brightness and direction as required in different areas of the bridge and by the needs of individual instruments and controls.

3.1.3.2 All illumination and lighting of instruments, keyboards and controls are to be adjustable down to zero, except the lighting of alarm indicators and the controls of dimmers which are to remain readable.

3.1.4 Field of vision

3.1.4.1 The requirements of Pt.3, Ch.11, Sec.4, Cl.4.2 are to be complied with.

3.1.4.2 It is to be possible to observe all objects necessary for navigation, including other traffic and navigation marks, in any direction from inside the wheelhouse. In this respect, there is to be a field of view around the ship of 360° obtained by an observer moving within the confines of the wheelhouse.

3.1.4.3 The view of the sea surface from the conning position and the navigation workstation is not to be obscured by more than two ship lengths, or 500 [m], whichever is less, forward of the bow to 10° on either side, irrespective of the ship's draught, trim and deck cargo.

3.1.4.4 Blind sectors caused by cargo, cargo gear and other obstructions outside of the wheelhouse forward of the beam obstructing the view of the sea surface as seen from the conning position and the navigation workstation are not to exceed 10° each. The total arc of blind sectors is not to exceed 20° and the clear sector between blind sectors shall be at least 5°. However, in the view described in 3.1.4.3 each individual blind sector is not to exceed 5°.

3.1.4.5 The horizontal field of vision from the conning position and the navigation workstation is to extend over an arc from more than 22.5° abaft the beam on one side, through forward, to more than 22.5° abaft the beam on the other side.

3.1.4.6 From the main steering position, the field of vision is to extend over an arc from dead ahead to at least 60° on each side.

3.1.4.7 From each bridge wing, the field of vision is to extend over an arc from at least 45° on the opposite bow through dead ahead and then aft to 180° from dead ahead.

3.1.4.8 There is to be a line of sight from the port wing to the starboard wing through the wheelhouse.

3.1.4.9 The ship's side is to be visible from the bridge wing.

3.1.4.10 From work stations for functions other than navigation, the field of vision is to enable an effective lookout to be maintained and, in this respect, is to extend at least over an arc from 90° on the port bow, through forward, to 22.5° abaft the beam on the starboard side.

3.2 Temperature control

3.2.1 An adequate air conditioning or mechanical ventilation system, together with sufficient heating according to climatic conditions, is to be provided in order to maintain temperature within 22°C to 30°C and the humidity within the range 20 percent to 60 percent. The discharge of hot or cold air is not to be directed towards bridge personnel. Control of this system is to be provided in the wheelhouse.

Section 4

Bridge Design and Systems

4.1 Documents

4.1.1 In addition to the documents indicated in Section 2, the following documents are to be submitted for **BDS** notation :

(a) A list of navigational equipment. This is to include for each item the manufacturer's name and code number, together with copies of relevant type approval certificates. The workstations with navigational equipment are to be fitted in line with requirements of Table 6.1 (for navigation and traffic surveillance, monitoring, manual steering and docking);

(b) A complete operational description of the relevant monitoring systems including a list of alarms and displays. This may be accomplished by means of simplified block diagrams of navigation equipment, internal communication systems and watch monitoring and alarm transfer systems and central alarm panel (where provided) including a list of alarms;

(c) A simplified one-line diagram of power supplies to the bridge equipment, circuit protection ratings and settings, cable sizes, rating of connected loads, detailed description and interactions, etc.;

(d) Operating / technical manuals for installed navigation equipment / systems;

(e) Sea trial test schedule.

4.2 Arrangement

a) The bridge configuration, arrangement of consoles and equipment location are to be such as to enable the officer of the watch to perform navigational tasks and other functions allocated to the bridge, as well as maintain an effective lookout. The following tasks are to be supported:

- Navigation and manoeuvring;
- Monitoring;
- Manual steering;
- Docking;
- Planning;
- Safety;
- Communications; and
- Conning.

b) Equipment and associated displays and indicators are to be sited at clearly defined workstations.

c) Consoles, including the chart table, are to be positioned, so that the instrumentation they contain is mounted in such a manner as to face a person looking forward. As far as practicable, operating surfaces are to be normal to the operator's line of sight.

d) Space necessary for operating at a workstation is to be considered as part of the workstation and is not to be part of the passageway.

4.3 Communication systems

4.3.1 A telephone system is to be provided to enable two-way speech communication between the wheelhouse and at least in the following locations:

- machinery control station space;

- emergency steering position in the steering gear compartment;
- master's and navigating officers' cabins, offices, mess and public rooms.

4.3.2 The bridge is to have priority over the system.

4.3.3 A list of extension numbers is to be clearly displayed adjacent to each telephone.

4.4 Power interruptions and shutdown

4.4.1 The distribution boards supplying power to the navigation equipment / system are to be fed by two exclusive circuits, one from main source of power and one from an emergency source of power. The power supplies to the distribution boards are to be arranged with automatic changeover facilities between the two sources.

4.5 Power supply

4.5.1 Power to the navigation equipment / system is to be supplied:

a) from the main and emergency sources of electrical power with automated changeover through a local distribution board with provision to preclude inadvertent shutdown;

b) from a transitional source of electrical power for a duration of not less than 1 min; and

c) Uninterrupted power supplies. Each vital equipment listed below is to be connected to an UPS. The UPS is to supply the vital bridge equipment with continuous power during a loss of ship's power for a minimum period of 15 minutes. The UPS is to prevent power transients as ship service power is lost and restored such that no loss of control or mal-operation of the bridge units results from these transitions.

4.5.2 Battery charge circuits within each unit are to be capable of re-charging the battery from 40% capacity to 80% capacity within 8 hours. Batteries are to be protected from undercharge and overcharge. Battery charging is to be automatic and is not to affect normal operation. The equipment regarded as vital is at least one each of the following:

i) one radar with ARPA (including antenna) installed at work station for navigation and manoeuvring;
ii) ECDIS;
iii) GNSS;
iv) Gyro compass; and
v) Internal Communication System.

4.6 EMI/EMC

4.6.1 All electrical and electronic equipment on the bridge and vicinity of the bridge are to be installed such that electromagnetic interference does not affect the proper functioning of the navigational systems and equipment. Installation of the equipment in accordance with the guidelines and recommendations included in IEC 60533:2015 or an acceptable equivalent standard would generally be considered to meet the requirement.

Note: Bridge and vicinity of the bridge covers deck and bridge zone, i.e.

- The wheelhouse including bridge wings
- Control rooms, characterized by equipment for inter-communication, signal processing, radio communication and navigation, auxiliary equipment

- Area in close proximity to receiving and/or transmitting antennas and large openings in the metallic structure (equipment beyond 5 meters need not be considered for this purposes).

4.6.2 Passive-EM equipment, defined in 4.6.3, which is excluded from the scope of EMC since it is considered not liable to cause or be susceptible to disturbances need not be tested, but are to be provided with an exemption statement.

4.6.3 *Passive EM equipment*: Equipment is considered a passive EM equipment if, when used as intended (without internal protection measures such as filtering or shielding) and without any user intervention, it does not create or produce any switching or oscillation of current or voltage and is not affected by electromagnetic disturbances. Examples of such equipment are as follows:

- cables and cabling systems, cables accessories

- equipment containing only resistive loads without any automatic switching device; e.g. simple domestic heaters with no controls, thermostat, or fan

- batteries and accumulators.

4.7 Earthing

4.7.1 Earthing of the navigational equipment and systems is to be as per manufacturer's instructions.

4.8 Tests and trials

4.8.1 Navigational equipment and systems are to be tested to the satisfaction of the attending surveyor meeting the relevant IMO performance standards.

Section 5

Integrated Bridge Systems

5.1 Documentation

5.1.1 In addition to the documents indicated in Section 4, the following documents are to be submitted for **IBS** notation:

(a) Details and arrangements of the workstations and systems. The functions available are to be as given in Table 6.1.

(b) The sea trial program is to include test details of the electronic chart display and information systems (EDCIS) and integrated bridge system (IBS).

5.2 System requirements

5.2.1 In addition to the requirements indicated in Section 3 and Section 4, the following requirements are to be complied with:

5.2.2 The IBS is to comply with all applicable IMO requirements as contained in the reference regulations listed in Section1, 1.2 or other relevant IEC Standards. Parts executing multiple operations are to meet the requirements specified for each individual function they can control, monitor or perform.

5.2.3 Each part of an IBS is to meet the relevant requirements of IMO Resolution A.694(17) as detailed in IEC 60945:2002.

5.2.4 The integrated bridge system is to be of an approved type.

5.2.5 Integration

5.2.5.1 The system is to provide functional integration meeting the following requirements:

a) The functionality of the IBS is to ensure that its operation is at least as effective as for stand-alone equipment.

b) Continuously displayed information is to be reduced to the minimum necessary for safe operation of the ship. Supplementary information is to be readily accessible.

c) Where multifunction displays and controls are used to perform functions necessary for safe operation of the ship they are to be duplicated and interchangeable.

d) It is to be possible to display the complete system configuration, the available configuration and the configuration in use.

e) Each part to be integrated is to provide details of its operational status and the latency and validity of essential information. Means are to be provided within the IBS to make use of this information.

f) An alternative means of operation is to be provided for essential functions.

g) An alternative source of essential information is to be provided. The IBS is to identify loss of either source.

h) The source of information (sensor, result of calculation or manual input) is to be displayed continuously or upon request.

5.2.6 Data Exchange

5.2.6.1 Interfacing within the IBS is to comply with IEC 61162, as applicable.

5.2.6.2 Data exchange is to be consistent with safe operation of ship.

5.2.6.3 The integrity of the data flowing on the network is to be ensured.

5.2.6.4 A failure in the connectivity is not to affect independent functionality.

5.2.6.5 The network is to be such that in the event of a single failure between the nodes, a clear indication of the failure is available, the sensors and displays on the network continue to operate and the data transmission between them is maintained.

5.2.7 Failure analysis

5.2.7.1 A failure analysis is to be performed and documented.

5.2.7.2 Parts, functions and connectivity are to be identified.

5.2.7.3 Possible failures of parts and connectivity associated with essential functions and information are to be identified.

5.2.7.4 Consequences of failures with respect to operation, function or status of the IBS are to be identified.

5.2.7.5 Each failure is to be classified with respect to its impact on the IBS taking into account relevant characteristics, such as detectability, ease of diagnosis, testability, replaceability and compensating and operating provisions.

5.2.7.6 The results of the failure analysis are to confirm the possibility of continued safe operation of the ship.

5.2.8 Quality Assurance

5.2.8.1 The IBS is to be designed, developed, produced, installed and serviced by companies certified to ISO 9001:2000.

5.2.9 Alarm management

5.2.9.1 The IBS alarm management, as a minimum, is to comply with the requirements of the Code of Alerts and Indicators (IMO resolution A.1021(26)) and the alarms required for each navigational equipment by IMO standards.

5.2.9.2 Appropriate alarm management on priority and functional groups is to be provided within the IBS.

5.2.9.3 The number of alarm types and their release are to be kept as low as possible by providing indications for information of lower importance.

5.2.9.4 Alarms are to be displayed so that the alarm reason and resulting functional restrictions can be easily understood. Indications are to be self-explanatory.

5.2.9.5 Alarms are to be prioritized as follows:

a) Emergency Alarms: Alarms which indicate that immediate danger to human life or to the ship and its machinery exists and that immediate action is to be taken.

b) Alarms: conditions requiring immediate attention and action by the bridge team to avoid any kind of hazardous situation and to maintain the safe operation of the ship and escalation required as alarm from not acknowledged warning.

c) Warning: conditions or situations which require immediate attention for precautionary reasons, to make the bridge team aware of conditions which are not immediately hazardous, but may become so.

d) Caution: awareness of a condition which still requires attention out of the ordinary consideration of the situation or of given information.

5.2.9.6 The following alarms are to be provided and included in the centralized alarm system:

- Off-course
- Off-track (where automatic track following is provided)
- Way point approaching, (where automatic track following is provided)
- Position fix inaccurate/lost
- Loss of heading input
- Loss of speed log input
- Equipment or sub-system failure
- Gyro mis-match
- Echo-sounder
- Radar CPA
- Navigational light failure
- Steering gear alarms
- Bridge navigational watch alarm system failure.

The audible alarms at the central panel and the relevant workstation are to be silenced when acknowledged from either location. However, it is to be possible to acknowledge the flickering light and bring to steady state only from the relevant work station.

5.2.9.7 Manual adjustment of any of the facilities of the integrated bridge system is to reset automatically the watch safety interval timer.

5.2.10 Power interruptions and shutdown

5.2.10.1 If subjected to an orderly shut-down, the IBS should, upon turn-on, come to an initial default state.

5.2.10.2 After a power interruption full functionality of the IBS is to be available after recovery of all subsystems. The IBS is not to increase the recovery time of individual subsystem functions after power restoration.

5.2.10.3 If subjected to a power interruption the IBS is to, upon restoration of power, maintain the configuration in use and continue automated operation, as far as practicable. Safety related automatic functions are to be restored only upon confirmation by the operator.

5.2.11 Power supply

5.2.11.1 Power supply requirements indicated in 4.5 are applicable.

5.2.11.2 Power supply requirements applying to parts of the IBS as a result of other IMO requirements are also to remain applicable.

5.3 Operational requirement

In addition to the operational requirements indicated in Section 4, following requirements are to be complied with.

5.3.1 Human factors

5.3.1.1 The IBS is to be capable of being operated by personnel holding appropriate certificates.

5.3.1.2 The man-machine interface (MMI) is to be designed so as to be easily understood and in a consistent style for all integrated functions.

5.3.1.3 Operational information is to be presented in a readily understandable format without the need to transpose, compute or translate.

5.3.1.4 Indications, which may be accompanied by a short low intensity acoustic signal are to occur when:

- An attempt is made to execute an invalid function
- An attempt is made to use invalid information.

5.3.1.5 If an input error is detected by the system it is to require the operator to correct the error immediately. Messages actuated by an input error are to guide the correct responses, e.g. not simply “Invalid entry”, but “Invalid entry, re-enter set point between 0 to 100”.

5.3.1.6 Layered menus are to be presented in a way which minimizes the added workload to find and return from the desired functions.

5.3.1.7 An overview is to be easily available to assist the operator in the use of a multiple page system. Each page is to have a unique identifier.

5.3.1.8 Where multi-function displays are used, they are to be in colour. Continuously displayed information and functional areas, e.g. menus, are to be presented in a consistent manner.

5.3.1.9 For actions which may cause unintended results, the IBS is to request confirmation from the operator.

Note 1 : Examples of such actions are:

- Attempting to change position and next waypoint while in track mode steering.

- Attempting to switch on bow thruster when insufficient electrical power is available.

5.3.1.10 Functions requested by the operator are to be acknowledged or clearly indicated by the IBS on completion.

5.3.1.11 Default values, where applicable, are to be indicated by the IBS when requesting operator input.

5.3.1.12 The size, colour and density of text and graphic information presented on a display are to be such that it may be easily read from the normal operator position under all operational lighting conditions. (ref ISO 8468-6.2.3).

5.3.1.13 Symbols used in mimic diagram are to be standardized throughout the system’s displays.

5.3.1.14 All information is to be presented on a background providing high contrast and emitting as little light as possible at night.

5.3.1 The design of bridges is to be governed by:

a) The range, layout and location of workstations required for performance of bridge functions

b) The fields of vision required for visual observations from each of the workstations.

c) The type and range of equipment to be provided for performance of the tasks at the individual workstations

5.3.2 Arrangement

5.3.2.1 Equipment and associated displays and indicators are to be sited at clearly defined workstations.

5.3.2.2 It is to be possible to monitor the navigation workstation and to maintain an effective lookout from other workstations within the wheelhouse.

5.3.2.3 Space necessary for operating at a workstation is to be considered as part of the workstation and is not to be part of the passageway.

5.3.3 Training

5.3.3.1 Manufacturers of integrated bridge systems are to provide training facilities for the ship's crew. This training may take place ashore or on board and is to be carried out using suitable material and methods to cover the following topics:

a) General understanding and operation of the system:

- Knowledge and understanding of the system's configuration and application.
- Usage and understanding of the operating manual.
- Usage and understanding of electronic "HELP" functions, if provided in the system.
- Familiarization with the system using safe trial modes.

b) Mastering of uncommon conditions in the system:

- Detecting and locating of failures.
- Resetting the system to safe default values and modes.
- Operating safely without certain sensor data or parts.
- Possibilities for repair on board.
- Identifying the potential for unintended results

5.4 Work stations

5.4.1 The bridge and workstation arrangement is to be based on relevant functional requirements and designed in accordance with established principles of ergonomics for safe and efficient operations, enabling the navigator to perceive all relevant information and execute pertinent actions with a minimum workload.

5.4.2 Workstations for additional functions may be located on the bridge provided the performance of such functions does not interfere with the tasks of maintaining safe control of the ship. Workstations for additional functions may include workstations for:

- Extended communication functions
- Monitoring and control of ballasting and cargo operations
- Extended monitoring of machinery
- Miscellaneous.

5.5 Navigation workstation

5.5.1 A workstation for navigation is to be arranged to enable efficient operation by one person under normal operating conditions. The workstation area is to be sufficient to allow at least two operators to use the equipment simultaneously. The arrangement of instruments and controls is to allow the use of all instruments and controls necessary for navigating and manoeuvring in any normal working position.

5.5.2 An adequate conning position is to be provided close to the forward center window. If the view in the centerline is obstructed by large masts, cranes, etc. two additional conning positions giving a clear view ahead are to be provided, one on the port side and one on the starboard side of the centerline, no more than 5 [m] apart. In addition to the conning position, a second position with a view of the area immediately in front of the bridge superstructure is to be provided close to a forward window or, alternatively, the conning position is to be wide enough to accommodate two persons.

5.5.3 The main steering position is to be located on the ship's centerline, unless the view ahead is obstructed by large masts, cranes, etc. In this case, the steering position is to be located a distance to starboard of the centerline sufficient to obtain a clear view ahead and special steering references for use by day and night are to be provided, e.g. sighting marks forward.

5.5.4 The following facilities are to be provided at the navigation workstation:

- Radar and radar plotting facilities
- Position-fixing system displays
- Echo sounder display
- Speed and distance indications
- Gyrocompass displays
- Magnetic compass display
- Wind speed and direction indication
- Steering controls and indication see Pt.4, Ch.6 and Ch.7
- Rate of turn indication
- Course/track controls and indications
- Main propulsion and thruster controls and indication
- Watch safety system acknowledge
- Watch safety system manual initiation
- Internal communications system
- VHF radiotelephone
- Time indication
- Window clear view controls
- Navigation lights controls
- Whistle control
- Morse light keys
- Wheelhouse/equipment lighting controls
- Automatic ship identification system (AIS) information
- Sound reception system where fitted.

Section 6

Carriage Requirements

6.1.1 The navigation equipment to be carried on board for the performance of various functions are listed in Table 6.1.

6.1.2 Navigational systems and equipment are to be of a type approved by the national administration and in conformity with appropriate performance standards not inferior to those adopted by IMO from time to time. Documentary evidence to this effect is to be submitted. See SOLAS 1974 as amended, Ch.V, Reg.18.

6.1.3 Where alternative means of fulfilling the navigational requirements are permitted, the means are to be approved by the national administration and in conformity with appropriate performance standards.

6.1.4 Two functionally independent radars or alternative means are to be provided to determine and display the range and bearing of radar transponders and other surface craft, obstructions, buoys, shorelines and navigational marks. One of the radars is to operate in the X-band (9 GHz) and the other is to operate in the S-band (3 GHz). Both radars are to include automatic tracking aids to determine collision risks, and at least one radar is to be equipped with an automatic radar plotting aid (ARPA), capable of tracking at least 20 targets, while the other is to be either ARPA or an automatic tracking aid (ATA).

6.1.5 A gyrocompass or alternative means for determining, displaying and transmitting the ship's heading is to be provided. The heading information is to be used directly by the radars, radar plotting aids and automatic identification system. The gyrocompass is to be provided with a gyrocompass heading repeater located at the emergency steering position in the steering gear compartment and gyrocompass bearing repeater(s) allowing bearings to be taken over 360°.

6.1.6 Where automatic track following is provided, sufficient warning is to be given of the approach of a waypoint, so that, in the event of no acknowledgement from the officer of the watch, there is adequate time for the backup navigator to reach the bridge and accept the change of course.

6.1.7 A speed log or alternative means of indicating the ship's speed and distance through water is to be provided. The speed through water measurement is to be used directly by the ARPA as an aid to collision avoidance.

6.1.8 A speed log or alternative means of indicating the ship's speed and distance over ground is to be provided. Speed over ground is to be indicated in both the fore-aft and athwartships directions.

Table 6.1 : Navigational Equipment

Workstation for	Main functions to be performed	Equipment	Remarks
Navigation and traffic surveillance manoeuvering (See Note 1)	• Observation of all vessels and objects • Recognizing dangerous situations • Deciding on collision avoidance actions • Checking vessel's own signal • Checking own course and speed • Keeping and/or changing own course and speed (track keeping) • Checking own position • Handling own internal communication on board	Gyro compass heading indicator	For IBS notation, two independent gyro compasses are to be provided on the bridge
		Magnetic compass heading indicator	
		Course reminder (set course) indicator	
		Auto pilot / track control system / alternative means	Automatically maintaining ship's heading or a straight track. Immediate restoration of manual control to be possible when required
		Steering gear pump selector switch	
		Steering mode selector switch	
		Steering position indicator	
		Rudder angle indicator	
		Rate-of-turn indicator and controller	For vessels of 50000 GT or greater
		Speed and distance indicator	For IBS notation, the speed measuring system is to be independent of the position-fixing systems
		Echosounders with adjustment controls	
		9 GHz radar and 3 GHz radar	
		Automatic traffic surveillance system including ARPA	

Table 6.1 (Contd.)			
Workstation for	**Main functions to be performed**	**Equipment**	**Remarks**
	• Handling communication vessel/vessel and vessel/shore (VHF)	Position fixing equipment/system including automatic visual position indicator	Two types of receivers are to be provided. One of the systems is to be GPS or equivalent and the other GLONASS, or other means such as radar
	• Releasing alarms	Officer of the watch check-alertness acknowledgement device	
	• Perception of group alarms with aids for decision-making	Back-up navigator call alarm device	Two-way communication wireless portable device to be provided
	• Observation of weather and seaway	Facilities for use of navigation charts	This may be separated from the navigation and traffic surveillance/ maneuvering workstation
	• Acknowledging watch check-alertness alarm	Vessel's automatic identification system	
		Voyage data recorder (VDR/S-VDR)	
		Propulsion engines/thrusters controls including emergency stops	Compliance with Part 4, Chapter 4, Section 9 & Part 4, Chapter 7 of the Rules is to be met
	• Keeping deck log (a Dictaphone may be used)	Propulsion engine revolution	If reduction geared engine
		Pitch indicator	For controllable-pitch propeller
		Propeller revolutions indicator	
		Wind direction and velocity indicator	
	• Sounding signals	Air and water temperature and humidity indicator	
		Automatic telephone system	
		Radio-communication/GMDSS equipment	
		NAVTEX automatic receiver and recorder	For navigational and meteorological warning purpose. To comply with IMO Res. A.617(15) – Implementation of the NAVTEX system as a component of the worldwide navigational warning service

Table 6.1 (Contd.)			
Workstation for	**Main functions to be performed**	**Equipment**	**Remarks**
		Signal transmitter for: - whistle - automatic device for fog signal - general alarm - morse signaling light	
		Search light controls	
		Controls for windscreen wiper, washer, heater	
		Night vision equipment	
		Sound reception system	If required
		Workstation lighting control device	
		HVAC controls	
		Clock	
		Group alarms and reset controls	
Monitoring (See Note 1)	• Observation of all vessels and objects • Recognizing dangerous situations • Handling own internal communication on board • Handling communication vessel/ vessel and vessel/shore • Perception of group alarms with aids for decision-making • Releasing alarms • Observation of weather and seaway • Acknowledging watch check alertness alarm • Checking own course and speed • Keeping deck log	Gyro compass heading indicator	For IBS notation, the speed measuring system is to be independent of the position-fixing systems
		Rudder angle indicator	
		Rate-of-turn indicator	For vessels of 50000 GT or greater
		Speed and distance indicator	For IBS notation, the speed measuring system is to be independent of the position-fixing systems
		Echosounders	
		9GHz radar and 3 GHz radar	
		Officer of the watch check-alertness acknowledgment device	
		Propulsion engines/ thrusters emergency stops	
		Pitch indicator	For controllable-pitch propeller
		Propeller revolutions indicator	

Table 6.1 (Contd.)			
Workstation for	**Main functions to be performed**	**Equipment**	**Remarks**
	• When workstation is occupied by an additional navigator, provides assistance to navigator at the navigation and traffic surveillance/ maneuvering workstation	Automatic telephone system	
	• When workstation is occupied by a pilot advisers to vessel's command	Radio-communication/ GMDSS equipment	
		Signal transmitter for whistle/foghorn	
		Controls for windscreen wiper, washer, heater	
		Workstation lighting control device	
		Clock	

Table 6.1 (Contd.)			
Workstation for	**Main functions to be performed**	**Equipment**	**Remarks**
		Required alarms and reset controls	In addition to the alarms/indicators which may be required by the various IMO Resolutions referenced in this chapter and pertinent flag Administration, the following conditions are to be alarmed at the monitoring workstation a) Off-course b) Off-track c) Planned route deviation d) Pre-warning of approach-way point and closest point of approach e) Off-preset water depth f) Gyro compass failure g) Failure of alarms prescribed in 3.6 and section 5 of this Chapter. h) Failure of power supply to distribution panel serving relevant equipment (Alarming of the above conditions at the monitoring workstation is not a substitute for alarming at the required relevant workstations)
Manual steering (Helmsman's) [See Note 1]	• Steering vessel according to rudder angle orders • Steering vessel according to course instruction • Steering vessel following landmark / sea marks • Acknowledging watch check alertness alarm	Gyro compass heading indicator (repeater)	
		Magnetic compass heading indicator	
		Course reminder (set course) indicator	
		Manual steering with override and selector control switches including steering wheel/steering lever	
		Rudder angle indicator	
		Rate-of-turn indicator	For vessels of 50000 GT or greater
		Watch check-alertness acknowledgement device	
		Automatic telephone system	

Table 6.1 (Contd.)			
Workstation for	**Main functions to be performed**	**Equipment**	**Remarks**
		Controls for windscreen wiper, washer, heater	
Docking (Bridge wings)	• Giving instructions, performing and controlling change of course • Giving instructions, performing and controlling change of speed • Giving instructions, performing and controlling change of thruster • Handling communication with maneuvering stations • Handling communication with tugs, pilot boat • Watching water surface along vessel's side • Releasing signals • Acknowledging watch check alertness alarm	Gyro compass heading indicator	
		Steering position selector switch	
		Rudder controls	
		Rudder angle indicator	
		Rate-of-turn indicator	For vessels of 50000 GT or greater
		Propulsion engines/ thrusters controls	
		Propulsion engine revolution	If reduction geared engine
		Pitch indicator	For controllable pitch propeller
		Propeller revolutions indicator	
		Lateral thrust and lateral movement of vessel, indicator	If thrusters are fitted
		Longitudinal movement of vessel, indicator	
		Wind direction and velocity indicator	
		Echosounder	
		Officer of the watch check-alertness acknowledgement device	
		Whistle controls	
		Search light and Morse lamp controls	
		Automatic telephone system	
		Radio-communication/ GMDSS equipment	
		Workstation lighting control device	
Centralized Bridge	As listed in 1.3.9	Equipment required for the navigation and traffic surveillance / maneuvering and monitoring workstations	
		Central alarm panel	
		ECDIS	

Table 6.1 (Contd.)			
Workstation for	**Main functions to be performed**	**Equipment**	**Remarks**
Route Planning	Determination of favourable course and optimum speed, taking into account weather conditions, current, etc. and route planning	ECDIS including navigation planning station	
	Giving instructions as to the course and speed	Route planning devices	
	Calculation of tidal data	Chart table	
	Handling nautical records, documents, publications	Position-fixing receiver	
	Handling weather reports	Retaining device for drawing triangles, dividers, magnifying lens, pencils, etc.	
	Determination of documentation of position in case of conventional operation	Weather chart plotter	
	Control of rate and error of chronometer, deviation, radio deviation, documentation of same	Main clock	
	Keeping deck log	Chronometer with receiving facility for time signals	
	External communication for planning operation using the chart	Log, including distance indicator, course plotter	
		Officer of the watch check-alertness acknowledgement device	
		Barograph	
		Command printer	
		Automatic telephone system	
	Acknowledging watch check alertness alarm	Echosounder	
		Officer of the watch check-alertness acknowledgement device	
		Whistle controls	
		Search light and Morse lamp controls	
		Automatic telephone system	

Table 6.1 (Contd.)

Note:

1) As the navigation and traffic surveillance / maneuvering, monitoring and manual steering workstations are functionally interrelated and usually installed in close proximity from each other, considerations will be given to the omission of duplicate equipment required at each of the aforementioned workstations.

Section 7

Tests and Trials

7.1 General

7.1.1 The tests to be carried out as indicated in this section are intended to supplement and not to replace the testing of parts that is required to meet the relevant IMO performance standards. The tests are intended to ensure that when the parts are integrated there is no degradation of their individual functionality.

7.1.2 In all instances the performance standards for parts will form the minimum test requirement for an integrated system.

7.1.3 During trails, navigational equipment and systems are to be tested to the satisfaction of the attending surveyor in accordance with the test program. The test program is to include following systems as minimum:

a) Course information system
b) Automatic steering system
c) Speed measuring system
d) Depth measuring system
e) Radar system
f) Automatic traffic surveillance system
g) Position fixing system
h) Watch monitoring and alarm transfer system
i) Route planning system
j) Vessel's automatic identification system
k) Sound reception in bridge, if fitted
l) Radio communication system
m) Centralised bridge workstation provided to enable the navigator to perform necessary navigational, monitoring/alarm and communication functions as required for IBS.

7.1.4 The trial program is to include test details of electronic chart display and information system (ECDIS) and integrated bridge system. The system is to be tested to verify that failure of one sub system does not affect any other sub system. In case of failure of integrated navigational system, it is to be possible to operate the primary navigational equipment / systems functions independently.

End of Chapter

Chapter 32

Polar Class

Contents

Section	
1	*Polar Class Descriptions and Application*
2	*Structural Requirements*
3	*Machinery Requirements for Polar Class Ships*

Section 1

Polar Class Descriptions and Application

1.1 Application

1.1.1 This chapter gives the additional requirements for ships constructed of steel and intended for independent navigation in ice-infested polar waters.

Class notations and related requirements for general ice class and those for operation in Northern Baltic are given in Pt.5, Ch.21.

1.1.2 Ships that comply with requirements of this chapter can be considered for a Polar Class notation as listed in Table 1.2.1. If the hull and machinery are constructed such as to comply with the requirements of different Polar Classes, then both the hull and machinery will be assigned the lower of these classes in the Certificate of Classification. Compliance of the hull or machinery with the requirements of a higher Polar Class will also to be indicated in the Certificate of Classification or equivalent.

1.1.3 Ships which are assigned a Polar Class notation and complying with the relevant specific requirements of this chapter may be given the additional notation "**Icebreaker**". "**Icebreaker**" refers to any ship having an operational profile that includes escort or ice management functions, having powering and dimensions that allow it to undertake aggressive operations in ice-covered waters.

1.1.4 For ships which are assigned a Polar Class notation, the hull form and propulsion power are to be such that the ship can operate independently and at continuous speed in a representative ice condition, as defined in Table 1.2.1 for the corresponding Polar Class. For ships and ship-shaped units which are intentionally not designed to operate independently in ice, such operational intent or limitations are to be explicitly stated in the Certificate of Classification or equivalent.

1.1.5 For ships which are assigned a Polar Class notation **PC 1** through **PC 5**, bows with vertical sides, and bulbous bows are generally to be avoided. Bow angles are to be in general, within the range specified in 2.3.1 v).

1.1.6 For ships which are assigned a Polar Class notation **PC 6** and **PC 7**, and are designed with a bow with vertical sides or bulbous bows, operational limitations (restricted from intentional ramming) in design conditions are to be stated in the Certificate of Classification or equivalent.

1.2 Polar classes

1.2.1 The Polar Class (PC) notations and descriptions are given in Table 1.2.1. It is the responsibility of the Owner to select an appropriate Polar Class. The descriptions in Table 1.2.1 are intended to guide owners, designers and administrations in selecting an appropriate Polar Class to match the requirements for the ship with its intended voyage or service.

1.2.2 The Polar Class notation is used to convey the differences between classes with respect to operational capability and strength.

1.2.3 Definitions for nomenclature used in Table 1.2.1:

1.2.3.1 First year ice : Sea ice of not more than one winter's growth, thickness 30 [cm] to 2 [m] in general. This is subdivided into thin / medium / thick first year ice as noted below:

Thin first year ice :	First year ice of less than 70 [cm] thickness
Medium first year ice:	First year ice of 70 to 120 [cm] thickness
Thick first year ice:	First year ice of over 120 [cm] thickness

1.2.3.2 Second year ice : Old ice which has survived only one summer's melt, typically thickness upto 2.5 [m].

1.2.3.3 Multi year ice : Old ice upto 3 [m] thick in general, which has survived at least two summer's melt.

Table 1.2.1 : Polar Class Descriptions

Polar Class	Ice descriptions (based on WMO Sea ice nomenclature; see 1.2.3 for definitions)
PC 1	Year-round operation in all polar waters
PC 2	Year-round operation in moderate multi-year ice conditions
PC 3	Year-round operation in second-year ice which may include multi-year ice inclusions
PC 4	Year-round operation in thick first-year ice which may include old ice inclusions
PC 5	Year-round operation in medium first-year ice which may include old ice inclusions
PC 6	Summer / autumn operation in medium first-year ice which may include old ice inclusions
PC 7	Summer / autumn operation in thin first-year ice which may include old ice inclusions

1.3 Upper and lower ice waterlines

1.3.1 The upper and lower ice waterlines upon which the design of the ship has been based will be indicated in the Certificate of Classification. The upper ice waterline (UIWL) is defined by the maximum draughts fore, amidships and aft. The lower ice waterline (LIWL) is defined by the minimum draughts fore, amidships and aft.

1.3.2 The lower ice waterline is to be determined with due regard to the ship's ice-going capability in the ballast loading conditions. The propeller is to be fully submerged at the lower ice waterline.

Section 2

Structural Requirements

2.1 Application and Definitions

2.1.1 This section gives the structural requirements to be complied with for assignment of polar class notations defined in Section 1.

2.1.2 Definitions

2.1.2.1 The length L_{UI} is the distance, in [m], measured horizontally from the fore side of the stem at the intersection with the upper ice waterline (UIWL) to the after side of the rudder post, or the centre of the rudder stock if there is no rudder post. L_{UI} is not to be less than 96%, and need not be greater than 97%, of the extreme length of the upper ice waterline (UIWL) measured horizontally from the fore side of the stem. In ships with unusual stern and bow arrangement the length L_{UI} will be specially considered.

2.1.2.2 The ship displacement Δ_{UI} is the displacement, in [k-tonnes] of the ship corresponding to the upper ice waterline (UIWL). Where multiple waterlines are used for determining the UIWL, the displacement is to be determined from the waterline corresponding to the greatest displacement.

2.2 Hull areas

2.2.1 The hull of all Polar Class ships is divided into areas reflecting the magnitude of the loads that are expected to act upon them. In the longitudinal direction, there are four regions: bow, bow intermediate, midbody and stern. The bow intermediate, midbody and stern regions are further divided in the vertical direction into the bottom, lower and icebelt regions. The extent of each hull area is illustrated in Fig.2.2.1.

2.2.2 Notwithstanding Fig.2.2.1, the boundary between the bow and bow intermediate regions is not to be forward of the intersection point of the line of the stem and the ship baseline.

2.2.3 Notwithstanding Fig.2.2.1, the aft boundary of the bow region need not be more than $0.45L_{UI}$ aft of the fore side of the stem at the intersection with the upper ice waterline (UIWL).

2.2.4 The boundary between the bottom and lower regions is to be taken at the point where the shell is inclined 7 deg. to the horizontal.

2.2.5 If a ship is intended to operate astern in ice regions, the aft section of the ship is to be designed using the bow and bow intermediate hull area requirements. This will be indicated in the appendix to the classification certificate.

2.2.6 Notwithstanding Fig.2.2.1, if the ship is assigned the additional notation “**Icebreaker**”, the forward boundary of the stern region is to be at least $0.04L_{UI}$ forward of the section where the parallel ship side at the upper ice waterline (UIWL) ends.

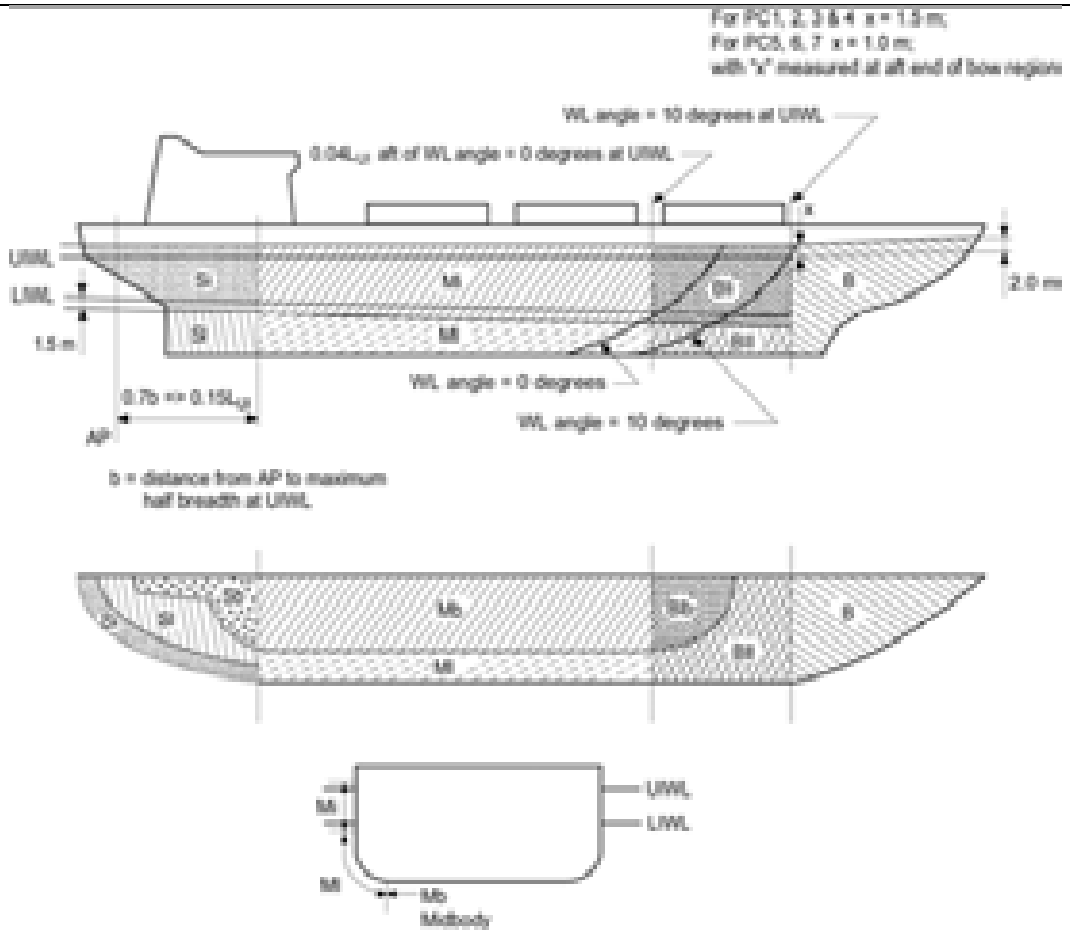


Fig.2.2.1 : Hull area extents

2.3 Design ice loads

2.3.1 General

i) A glancing impact on the bow is the design scenario for determining the scantlings required to resist ice loads.

ii) The design ice load is characterized by an average pressure (P_{avg}) uniformly distributed over a rectangular load patch of height (b) and width (w).

iii) Within the bow area of all Polar Class Ships and within the bow intermediate icebelt area of Polar Class PC6 and PC7, the ice load parameters are functions of the actual bow shape. To determine the ice load parameters (P_{avg}, b and w), it is required to calculate the following ice load characteristics for sub-regions of the bow area; shape coefficient (f_{ai}), total glancing impact force (F_i), line load (Q_i) and pressure (P_i).

iv) In other ice-strengthened areas, the ice load parameters (P_{avg}, b_{NONBOW} and w_{NONBOW}) are determined independently of the hull shape and based on a fixed load patch aspect ratio, AR = 3.6.

v) Design ice forces calculated according to 2.3.2.1 iii) are applicable for bow forms where the buttock angle γ at the stem is positive and less than 80 deg, and the normal frame angle β' at the centre of the foremost sub-region, as defined in 2.3.2.1 i), is greater than 10 deg. See Fig. 2.3.2.1.

vi) Design ice forces calculated according to 2.3.2.1 iv) are applicable for ships which are assigned the Polar Class PC6 or PC7 and have a bow form with vertical sides. This includes bows where the normal frame angles β' at the considered sub-regions, as defined in 2.3.2.1 i), are between 0 and 10 deg.

vii) For ships which are assigned the Polar Class PC6 or PC7, and equipped with bulbous bows, the design ice forces on the bow are to be determined according to 2.3.2.1 iv). In addition, the design forces are not to be taken less than those given in 2.3.2.1 iii), assuming fa = 0.6 and AR = 1.3.

viii) For ships with bow forms other than those defined in v) to vii), design forces will be specially considered.

ix) Ship structures that are not directly subjected to ice loads may still experience inertial loads of stowed cargo and equipment resulting from ship ice interaction. These inertial loads, based on accelerations are to be considered in the design of these structures.

2.3.2 Glancing impact load characteristics

The parameters defining the glancing impact load characteristics are reflected in the class factors listed in Table 2.3.2 and Table 2.3.3.

Table 2.3.2 : Class factors to be used in 2.3.2.1 iii)

Polar Class	Crushing failure class factor (CF_C)	Flexural failure class factor (CF_F)	Load patch dimensions class factor (CF_D)	Displacement class factor (CF_{DIS})	Longitudinal strength class factor (CF_L)
PC1	17.69	68.60	2.01	250	7.46
PC2	9.89	46.80	1.75	210	5.46
PC3	6.06	21.17	1.53	180	4.17
PC4	4.50	13.48	1.42	130	3.15
PC5	3.10	9.00	1.31	70	2.50
PC6	2.40	5.49	1.17	40	2.37
PC7	1.80	4.06	1.11	22	1.81

Table 2.3.3: Class factors to be used in 2.3.2.1 iv)

Polar Class	Crushing failure Class Factor (CF_{CV})	Line load Class Factor (CF_{QV})	Pressure Class Factor (CF_{PV})
PC6	3.43	2.82	0.65
PC7	2.60	2.33	0.65

2.3.2.1 Bow area

i) In the bow area, the force (F), line load (Q), pressure (P) and load patch aspect ratio (AR) associated with the glancing impact load scenario are functions of the hull angles measured at the upper ice waterline (UIWL). The influence of the hull angles is captured through calculation of a bow shape coefficient (f_a). The hull angles are defined in Fig.2.3.2.1.

ii) The waterline length of the bow region is generally to be divided into 4 sub-regions of equal length. The force (F), line load (Q), pressure (P) and load patch aspect ratio (AR) are to be calculated with respect to the mid-length position of each sub-region (each maximum of F, Q and P is to be used in the calculation of the ice load parameters P_{avg}, b and w).

iii) The Bow area load characteristics for bow forms defined in 2.3.1 v) are determined as follows:

a) Shape coefficient, fa_i, is to be taken as

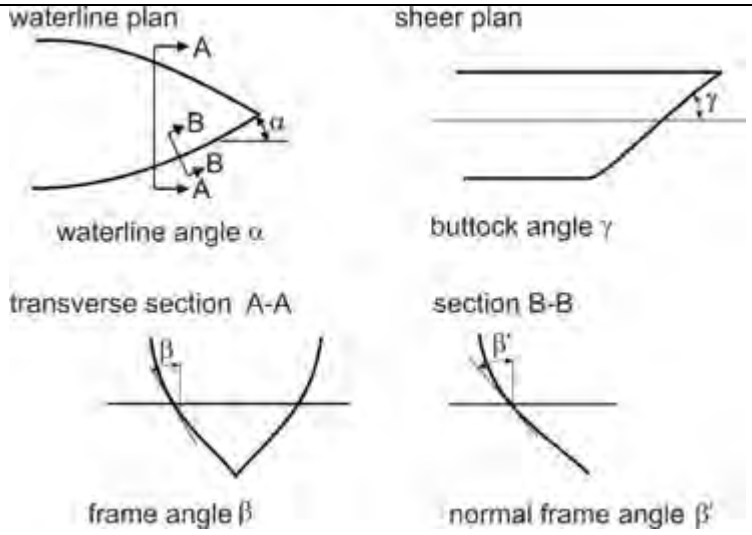
fa_i = minimum ($fa_{i,1}$; $fa_{i,2}$; $fa_{i,3}$)

where,

$fa_{i,1} = (0.097 - 0.68\ (x/L_{UI} - 0.15)^2\ ) \cdot \alpha_i / (\beta'_i)^{0.5}$

$fa_{i,2} = 1.2 \cdot CF_F / (\sin(\beta'_i) \cdot CF_C \cdot \Delta_{UI}^{0.64})$

$fa_{i,3} = 0.60$



Note :

β' = normal frame angle at upper ice waterline [deg]
α = upper ice waterline angle [deg]
γ = buttock angle at upper ice waterline (angle of buttock line measured from horizontal) [deg]
tan(β) = tan(α) / tan(γ)
tan(β') = tan(β) cos(α)

Fig.2.3.2.1 : Definition of hull angles

b) Force, F:

$F_i = fa_i \cdot CF_C \cdot \Delta_{UI}^{0.64}$ [MN]

c) Load patch aspect ratio, AR:

$AR_i = 7.46 \cdot \sin(\beta'_i) \geq 1.3$

d) Line load, Q:

$Q_i = F_i^{0.61} \cdot CF_D / AR_i^{0.35}$ [MN/m]

e) Pressure, P:

$P_i = F_i^{0.22} \cdot CF_D^2 \cdot AR_i^{0.3}$ [MPa]

where,

i = sub-region considered

L_{UI} = rule length [m] as defined in 2.1.2.1

x = distance [m] from the fore side of the stem at the intersection with the upper ice waterline (UIWL) to station under consideration

α = waterline angle [deg], see Fig.2.3.2.1

β' = normal frame angle [deg], see Fig.2.3.2.1

Δ_{UI} = displacement as defined in 2.1.2.2 , not to be taken less than 5 [k-tonnes]

CF_C = Crushing Failure Class Factor from Table 2.3.2

CF_F = Flexural Failure Class Factor from Table 2.3.2

CF_D = Load patch dimensions Class Factor from Table 2.3.2

iv) The Bow area load characteristics for bow forms defined in 2.3.1 vi) are determined as follows:

a) Shape coefficients, fa_i, is to be taken as

$fa_i = \alpha_i / 30$

b) Force, F_i

$F_i = fa_i \ CF_{CV} \ \Delta_{UI}^{0.47}$ [MN]

c) Line load, Q_i :

$Q_i = F_i^{0.22}$ CF_{QV} [MN/m]

d) Pressure, P_i
$P_i = F^{0.56}$ CF_{PV} [MPa]

Where

i = sub-region considered

α = waterline angle [deg], see Fig.2.3.2.1

Δ_{UI} = displacement as defined in 2.1.2.2, not to be taken less than 5 [k-tonnes]

CF_{CV} = Crushing failure Class Factor from Table 2.3.3

CF_{QV} = Line load Class Factor from Table 2.3.3

CF_{PV} = Pressure Class Factor from Table 2.3.3

2.3.2.2 Hull areas other than the bow

In the hull areas other than the bow, the force (F_{NONBOW}) and line load (Q_{NONBOW}) used in the determination of the load patch dimensions (b_{NONBOW}, w_{NONBOW}) and design pressure (P_{avg}) are determined as follows:

a) Force, F_{NONBOW}:

$F_{NONBOW} = 0.36 . CF_C . DF$ [MN]

b) Line load, Q_{NONBOW}:

$Q_{NONBOW} = 0.639 . F_{NONBOW}^{0.61} . CF_D$ [MN/m]

where,

CF_C = Crushing Failure Class Factor from Table 2.3.2

DF = ship displacement factor

$= \Delta_{UI}^{0.64}$ if $\Delta_{UI} \leq CF_{DIS}$

$= CF_{DIS}^{0.64} + 0.10 (\Delta_{UI} - CF_{DIS})$ if $\Delta_{UI} > CF_{DIS}$

Δ_{UI} = displacement as defined in 2.1.2.2 , not to be taken less than 10 [k-tonnes]

CF_{DIS} = Displacement class factor from Table 2.3.2.

CF_D = load patch dimensions Class Factor from Table 2.3.2.

2.3.3 Design load patch

i) In the bow area of all ships and the bow intermediate icebelt area for ships with class notation PC6 and PC7, the design load patch has dimensions of width, w_{BOW} and height b_{BOW} defined as follows:

$w_{BOW} = F_{BOW} / Q_{BOW}$ [m]

$b_{BOW} = Q_{BOW} / P_{BOW}$ [m]

where,

F_{BOW} = maximum force F_i in the bow area [MN]

Q_{BOW} = maximum line load Q_i in the bow area [MN/m]

P_{BOW} = maximum pressure P_i in the bow area [MPa]

ii) In hull areas other than those covered by i) above, the design load patch has dimensions of width, w_{NONBOW} and height, b_{NONBOW}, defined as follows:

$w_{NONBOW} = F_{NONBOW} / Q_{NONBOW}$ [m]

$b_{NONBOW} = w_{NONBOW} / 3.6$ [m]

where,

F_{NONBOW} = force as defined in 2.3.2.2 a) [MN]

Q_{NONBOW} = line load as defined in 2.3.2.2 b) [MN/m]

2.3.4 Pressure within the design load patch

i) The average pressure, P_{avg}, within a design load patch is determined as follows:

$P_{avg} = F / (b . w)$ [MPa]

where,
F = F_{BOW} or F_{NONBOW} as appropriate for the hull area under consideration [MN]

b = b_{BOW} or b_{NONBOW} as appropriate for the hull area under consideration [m]

w = w_{BOW} or w_{NONBOW} as appropriate for the hull area under consideration [m]

ii) Areas of higher, concentrated pressure exist within the load patch. In general, smaller areas have higher local pressures. Accordingly, the peak pressure factors listed in Table 2.3.4 are used to account for the pressure concentration on localized structural members.

Table 2.3.4 : Peak pressure factors		
Structural Member		**Peak Pressure Factor (PPF_i)**
Plating	Transversely-framed	$PPF_p = (1.8 – s) \geq 1.2$
	Longitudinally-framed	$PPF_p = (2.2 – 1.2 \, . \, s) \geq 1.5$
Frames in transverse framing systems	With load distributing stringers	$PPF_t = (1.6 – s) \geq 1.0$
	With no load distributing stringers	$PPF_t = (1.8 – s) \geq 1.2$
Frames in bottom structures		$PPF_s = 1.0$
Load carrying stringers, Side longitudinals, Web frames		$PPF_s = 1$, if $S_w \geq 0.5 \, . \, w$ $PPF_s = 2.0 – 2.0 \, . \, S_w / w$, if $S_w < (0.5 \, . \, w)$
where, s = frame or longitudinal spacing [m] S_w = web frame spacing [m] w = ice load patch width [m]		

2.3.5 Hull area factors

i) Associated with each hull area is an area factor that reflects the relative magnitude of the load expected in that area. The Area Factor (AF) for each hull area is listed in Table 2.3.5.

ii) In the event that a structural member spans across the boundary of a hull area, the largest hull area factor is to be used in the scantling determination of the member.

iii) Due to their increased manoeuvrability, ships having propulsion arrangements with azimuthing thruster(s) or "podded" propellers are to have specially considered stern icebelt (S_i) and stern lower (S_l) hull area factors.

iv) For ships assigned the additional notation "**Icebreaker**", the Area Factor (AF) for each hull area is listed in Table 2.3.6.

Table 2.3.5 : Hull area factors (AF)									
Hull Area		**Area**	**Polar Class**						
			PC1	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**
Bow (B)	All	B	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Bow intermediate (BI)	Icebelt	BI_i	0.90	0.85	0.85	0.80	0.80	1.00*	1.00*
	Lower	BI_l	0.70	0.65	0.65	0.60	0.55	0.55	0.50
	Bottom	BI_b	0.55	0.50	0.45	0.40	0.35	0.30	0.25
Midbody (M)	Icebelt	M_i	0.70	0.65	0.55	0.55	0.50	0.45	0.45
	Lower	M_l	0.50	0.45	0.40	0.35	0.30	0.25	0.25
	Bottom	M_b	0.30	0.30	0.25	**	**	**	**
Stern (S)	Icebelt	S_i	0.75	0.70	0.65	0.60	0.50	0.40	0.35
	Lower	S_l	0.45	0.40	0.35	0.30	0.25	0.25	0.25
	Bottom	S_b	0.35	0.35	0.30	0.25	0.15	**	**
Note : * See 2.3.1 (iii). ** Indicates that strengthening for ice loads is not necessary.									

Table 2.3.6 : Hull area factors (AF) for ship with additional notation "Icebreaker"									
Hull Area		**Area**	**Polar Class**						
			PC1	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**
Bow (B)	All	B	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Bow Intermediate (BI)	Icebelt	BI_i	0.90	0.85	0.85	0.85	0.85	1.00	1.00
	Lower	BI_l	0.70	0.65	0.65	0.65	0.65	0.65	0.65
	Bottom	BI_b	0.55	0.50	0.45	0.45	0.45	0.45	0.45
Midbody (M)	Icebelt	M_i	0.70	0.65	0.55	0.55	0.55	0.55	0.55
	Lower	M_l	0.50	0.45	0.40	0.40	0.40	0.40	0.40
	Bottom	M_b	0.30	0.30	0.25	0.25	0.25	0.25	0.25
Stern (S)	Icebelt	S_i	0.95	0.90	0.80	0.80	0.80	0.80	0.80
	Lower	S_l	0.55	0.50	0.45	0.45	0.45	0.45	0.45
	Bottom	S_b	0.35	0.30	0.30	0.30	0.30	0.30	0.30

2.4 Shell plate requirements

2.4.1 The required minimum shell plate thickness, t, is given by:

$t = t_{net} + t_s$ [mm]

where,

t_{net} = plate thickness required to resist ice loads according to 2.4.2 [mm]

t_s = corrosion and abrasion allowance according to 2.11 [mm].

2.4.2 The thickness of shell plating required to resist the design ice load, t_{net}, depends on the orientation of the framing.

In the case of transversely-framed plating ($\Omega \geq 70$ deg), including all bottom plating, i.e. plating in hull areas BI_b, M_b and S_b, the net thickness is given by:

$t_{net} = 500 \cdot s \cdot ((AF \cdot PPF_p \cdot P_{avg}) / \sigma_y)^{0.5} / (1 + s / (2 \cdot b))$ [mm]

In case of longitudinally-framed plating ($\Omega \leq 20$ deg), when $b \geq s$, the net thickness is given by:

$t_{net} = 500 \cdot s \cdot ((AF \cdot PPF_p \cdot P_{avg}) / \sigma_y)^{0.5} / (1 + s / (2 \cdot l))$ [mm]

In case of longitudinally-framed plating ($\Omega \leq 20$ deg), when $b < s$, the net thickness is given by:

$t_{net} = 500 \cdot s \cdot ((AF \cdot PPF_p \cdot P_{avg}) / \sigma_y)^{0.5} \cdot (2 \cdot b / s - (b / s)^2)^{0.5} / (1 + s / (2 \cdot l))$ [mm]

In case of obliquely-framed plating ($70 \text{ deg} > \Omega > 20$ deg), linear interpolation is to be used:

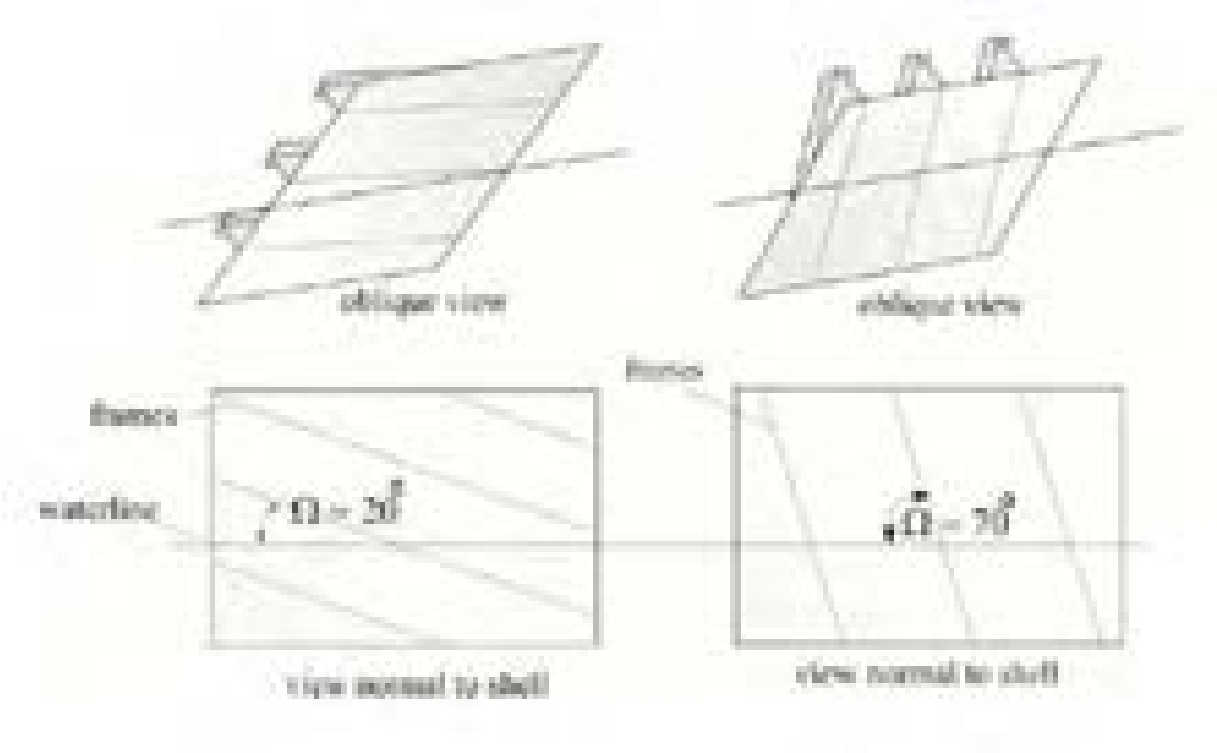


Fig.2.4.2 : Shell framing angle Ω

where,

Ω = smallest angle between the chord of the waterline and the line of the first level framing as illustrated in Fig.2.4.2 [deg.]

s = transverse frame spacing in transversely-framed ships or longitudinal frame spacing in longitudinally-framed ships [m]

AF = Hull area factor from Table 2.3.5 or Table 2.3.6.

PPF_p = Peak pressure factor from Table 2.3.4

P_{avg} = average patch pressure according to 2.3.4

σ_y = minimum upper yield stress of the material [N/mm^2]

b = height of design load patch [m], where b is to be taken not greater than (l - s/4) in the case of determination of the net thickness for transversely framed plating .

l = distance between frame supports, i.e. equal to the frame span as given in 2.5.5, but not reduced for any fitted end brackets [m]. When a load-distributing stringer is fitted, the length l need not be taken larger than the distance from the stringer to the most distant frame support.

2.5 Framing – General

2.5.1 Framing members of Polar Class ships are to be designed to withstand the ice loads defined in 2.3.

2.5.2 The term “framing member” refers to transverse and longitudinal local frames, load-carrying stringers and web frames in the areas of the hull exposed to ice pressure, see Fig.2.2.1.

2.5.3 The strength of a framing member is dependent upon the fixity that is provided at its supports. Fixity can be assumed where framing members are either continuous through the support or attached to a supporting section with a connection bracket. In other cases, simple support is to be assumed unless the connection can be demonstrated to provide significant rotational restraint. Fixity is to be ensured at the support of any framing which terminates within an ice-strengthened area.

2.5.4 The details of framing member intersection with other framing members, including plated structures, as well as the details for securing the ends of framing members at supporting sections are to be in accordance with Pt.3, Ch.3.

2.5.5 The effective span of a framing member is to be determined on the basis of its moulded length. If brackets are fitted, the effective span may be reduced in accordance with Pt.3, Ch.3. Brackets are to be configured to ensure stability in the elastic and post-yield response regions.

2.5.6 When calculating the section modulus and shear area of a framing member, net thicknesses of the web, flange (if fitted) and attached shell plating are to be used. The shear area of a framing member may include that material contained over the full depth of the member, i.e. web area including portion of flange, if fitted, but excluding attached shell plating.

2.5.7 The actual net effective shear area. A_w, of a transverse or longitudinal local frame is given by:

$A_w = h\, t_{wn} \sin \varphi_w / 100$ [cm^2]

h = height of stiffener [mm], see Fig.2.5.7

t_{wn} = net web thickness [mm]

$= t_w - t_c$

t_w = as built web thickness [mm], see Fig.2.5.7

t_c = corrosion deduction [mm] to be subtracted from the web and flange thickness (specified as t_s in 2.11.3, however not less than that given in Pt.3, Ch.3, Sec.2).

φ_w = smallest angle between shell plate and stiffener web, measured at the midspan of the stiffener, see Fig.2.5.7. The angle φ_w may be taken as 90 degrees provided the smallest angle is not less than 75 degrees.

2.5.8 When the cross-sectional area of the attached plate flange exceeds the cross-sectional area of the local frame, the actual net effective plastic section modulus, Z_p, of a transverse or longitudinal frame is given by:

$$Z_p = 0.05 A_{pn}.t_{pn} + 0.0005 h_w^2\, t_{wn} \sin\varphi_w + 0.1 A_{fn} \left(h_{fc} \sin\varphi_w - b_w \cos\varphi_w\right) \quad [cm^3]$$

h, t_{wn}, t_c and φ_w as given in 2.5.7 and s as given in 2.4.2

A_{pn} = net cross-sectional area of the local frame [cm^2]

t_{pn} = fitted net shell plate thickness [mm] (complying with t_{net} as required by 2.4.2)

h_w = height of local frame web [mm], see Fig.2.5.7

A_{fn} = net cross-sectional area of local frame flange [cm^2]

h_{fc} = height of local frame measured to center of the flange area [mm], see Fig.2.5.7

b_w = distance from mid thickness plane of local frame web to the centre of the flange area [mm], see Fig.2.5.7

When the cross-sectional area of the local frame exceeds the cross-sectional area of the attached plate flange, the plastic neutral axis is located a distance z_{na} above the attached shell plate, given by:

$z_{na} = (100\, A_{fn} + h_w\, t_{wn} - 1000\, t_{pn}\, s) / (2\, t_{wn})$ [mm]

and the net effective plastic section modulus, Z_p,of a transverse or longitudinal frame is given by:

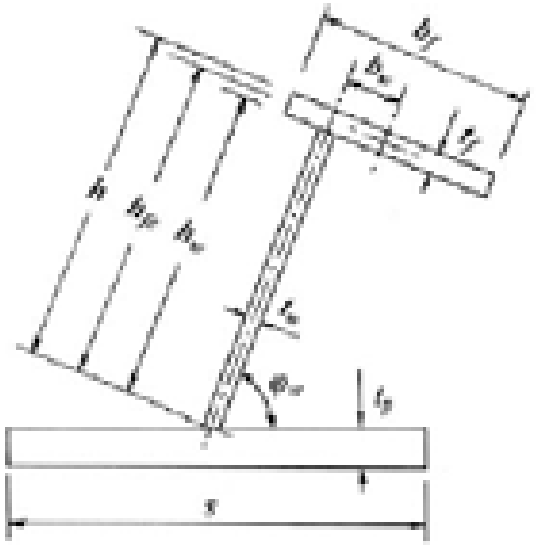


Fig.2.5.7 : Stiffener geometry

$$Z_p = t_{pn}\, s\left[Z_{na} + t_{pn}/2\right]\sin\varphi_w + 0.0005\left(\left(h_w - z_{na}\right)^2 + z_{na}^2\right) t_{wn}\sin\varphi_w + 0.10A_{fn}\left(\left(h_{fc} - z_{na}\right)\sin\varphi_w - b_w\cos\varphi_w\right)\,[cm^3]$$

2.5.9 In the case of oblique framing arrangement (70 deg > Ω > 20 deg, where Ω is defined as given in 2.4.2), linear interpolation is to be used.

2.6 Framing – Local frames in bottom structures and transverse local frames in side structures

2.6.1 The local frames in bottom structures (i.e. hull areas B_{lb}, M_b and S_b) and transverse local frames in side structures are to be dimensioned such that the combined effects of shear and bending do not exceed the plastic strength of the member. The plastic strength is defined by the magnitude of midspan load that causes the development of a plastic collapse mechanism. For bottom structure the patch load is to be applied with the dimension (b) parallel with the frame direction.

2.6.2 The actual net effective shear area of the frame, A_w, as defined in 2.5.7, is to comply with the following condition: $A_w \geq A_t$, where:

$A_t = 100^2 \cdot 0.5 \cdot LL \cdot s \cdot (AF \cdot PPF_t \cdot P_{avg}) / (0.577 \cdot \sigma_y)$ [cm^2]

where,

LL = length of loaded portion of span

= lesser of a and b [m]

a = local frame span as defined in 2.5.5 [m]

b = height [m] of design ice load patch as defined in 2.3.3 i) or 2.3.3 ii)

s = spacing of local frame [m]

AF = Hull Area Factor from Table 2.3.5 or 2.3.6

PPF = Peak Pressure Factor, PPF_t or PPF_s as appropriate from Table 2.3.4

P_{avg} = average pressure [MPa] within load patch as defined in 2.3.4

σ_y = minimum upper yield stress of the material [N/mm^2]

2.6.3 The actual net effective plastic section modulus of the plate / stiffener combination, Z_p, as defined in 2.5.8, is to comply with the following condition : $Z_p \geq Z_{pt}$, where Z_{pt} is to be the greater calculated on the basis of two load conditions : a) ice load acting at the midspan of the local frame, and b) the ice load acting near a support. The A1 parameter defined below reflects these two conditions:

$Z_{pt} = 100^3 \cdot LL \cdot Y \cdot s \cdot (AF \cdot PPF_t \cdot P_{avg}) \cdot a \cdot A_1 / (4 \cdot \sigma_y)$ [cm^3]

where,

AF, PPF_t, P_{avg}, LL, b, s, a and σ_y are as given in 2.6.2

$Y = 1 – 0.5 . (LL / a)$

A_1 = maximum of

$A_{1A} = 1 / (1 + j / 2 + k_w . j / 2 . [(1 – a_1^2)^{0.5} – 1])$

$A_{1B} = (1 - 1 / (2 . a_1 . Y)) / (0.275 + 1.44 . k_z^{0.7})$

j = 1 for a local frame with one simple support outside the ice-strengthened areas

= 2 for framing local frame without any simple supports

$a_1 = A_t / A_w$

A_t = minimum shear area [cm^2] of the local frame as given in 2.6.2

A_w = effective net shear area [cm^2] of the local frame (calculated according to 2.5.7)

$k_w = 1 / (1 + 2 . A_{fn} / A_w)$ with A_{fn} as given in 2.5.8

$k_z = z_p / Z_p$ in general

= 0.0 when the frame is arranged with end bracket

z_p = sum of individual plastic section modulii of flange and shell plate as fitted [cm^3]

$= (b_f . t_{fn}^2 / 4 + b_{eff} . t_{pn}^2 / 4) / 1000$

b_f = flange breadth [mm], see Fig.2.5.7

t_{fn} = net flange thickness [mm]

$= t_f – t_c$ (t_c as given in 2.5.7)

t_f = as-built flange thickness [mm], see Fig.2.5.7

t_{pn} = the fitted net shell plate thickness [mm] (not to be less than t_{net} as given in 2.4)

b_{eff} = effective width of shell plate flange [mm]

= 500 s

Z_p = net effective plastic section modulus [cm^3] of the local frame (calculated according to 2.5.8).

2.6.4 The scantlings of the local frame are to meet the structural stability requirements of 2.9.

2.7 Framing – Longitudinal local frames in side structures

2.7.1 Longitudinal local frames in side structures are to be dimensioned such that the combined effects of shear and bending do not exceed the plastic strength of the member. The plastic strength is defined by the magnitude of midspan load that causes the development of a plastic collapse mechanism.

2.7.2 The actual net effective shear area of the frame, A_w, as defined in 2.5.7, is to comply with the following condition: $A_w \geq A_L$, where:

$A_L = 100^2 . (AF . PPF_s . P_{avg}) . 0.5 . b_1 . a / (0.577 . \sigma_y)$ [cm^2]

where,

AF = Hull area factor from Table 2.3.5 or Table 2.3.6

PPF_s = Peak pressure factor from Table 2.3.4

P_{avg} = average pressure within load patch as defined in 2.3.4 [MPa]

$b_1 = k_o . b_2$ [m]

$k_o = 1 – 0.3 / b'$

$b' = b / s$

b = height of design ice load patch from 2.3.3 i) or ii) [m]

s = spacing of longitudinal frames [m]

$b_2 = b (1 – 0.25 . b')$ [m], if $b' < 2$

= s [m], if $b' \geq 2$

a = effective span [m] of longitudinal local frame as given in 2.5.5

σ_y = minimum upper yield stress of the material [N/mm^2]

2.7.3 The actual net effective plastic section modulus of the plate / stiffener combination, Z_p, as defined in 2.5.8, is to comply with the following condition: $Z_p \geq Z_{pL}$, where:

$Z_{pL} = 100^3 . (AF . PPF_s . P_{avg}) . b_1 . a^2 . A_4 / (8 . \sigma_y)$ [cm^3]

where,

AF, PPF_s, P_{avg}, b_1, a and σ_y are as given in 2.7.2

$A_4 = 1 / (2 + k_{wl} . [(1 – a_4^2)^{0.5} – 1])$

$a_4 = A_L / A_w$

A_L = minimum shear area for longitudinal as given in 2.7.2 [cm^2]

A_w = net effective shear area of longitudinal (calculated according to 2.5.7) [cm^2]

$K_{wl} = 1 / (1 + 2 . A_{fn} / A_w)$ with A_{fn} as given in 2.5.8.

2.7.4 The scantlings of the longitudinals are to meet the structural stability requirements of 2.9.

2.8 Framing – Web frames and load-carrying stringers

2.8.1 Web frames and load-carrying stringers are to be designed to withstand the ice load patch as defined in 2.3. The load patch is to be applied at locations where the capacity of these members under the combined effects of bending and shear is minimized.

2.8.2 Web frames and load-carrying stringers are to be dimensioned such that the combined effects of shear and bending do not exceed the permissible stresses given below:

Shear stress $\tau = \sigma_y / \sqrt{3}$

Bending stress $\sigma_b = \sigma_y$

Equivalent stress $\sigma_e = \sqrt{\sigma_b^2 + 3\tau^2} = 1.15\,\sigma_y$

Where the structural configuration is such that members do not form part of a grillage system, the appropriate peak pressure factor (PPF) from Table 2.3.4 is to be used. Special attention is to be paid to the shear capacity in way of lightening holes and cut-outs in way of intersecting members.

2.8.3 For determination of scantlings of load carrying stringers, web frames supporting local frames, or web frames supporting load carrying stringers forming part of a structural grillage system, appropriate methods as outlined in 2.17 are normally to be used.

2.8.4 The scantlings of web frames and load-carrying stringers are to meet the structural stability requirements of 2.9.

2.9 Framing – structural stability

2.9.1 To prevent local buckling in the web, the ratio of web height (h_w) to net web thickness (t_{wn}) of any framing member is not to exceed:

For flat bar sections:

$$h_w / t_{wn} \leq 282 / (\sigma_y)^{0.5}$$

For bulb, tee and angle sections:

$$h_w / t_{wn} \leq 805 / (\sigma_y)^{0.5}$$

where,

h_w = web height
t_{wn} = net web thickness
σ_y = minimum upper yield stress of the material [N/mm^2]

2.9.2 Framing members for which it is not practicable to meet the requirements of 2.9.1 (e.g. load carrying stringers or deep web frames) are required to have their webs effectively stiffened. The scantlings of the web stiffeners are to ensure the structural stability of the framing member. The minimum net web thickness for these framing members is given by:

$t_{wn} = 2.63 \times 10^{-3} . c_1 . (\sigma_y / (5.34 + 4 . (c_1 / c_2)^2))^{0.5}$ [mm]

where,

$c_1 = h_w – 0.8 . h$ [mm]

h_w = web height of stringer / web frame [mm] (see Fig.2.9.2)

h = height of framing member penetrating the member under consideration (0 if no such framing member) [mm] (see Fig.2.9.2)

c_2 = spacing between supporting structure oriented perpendicular to the member under consideration [mm] (see Fig.2.9.2)

σ_y = minimum upper yield stress of the material [N/mm^2]

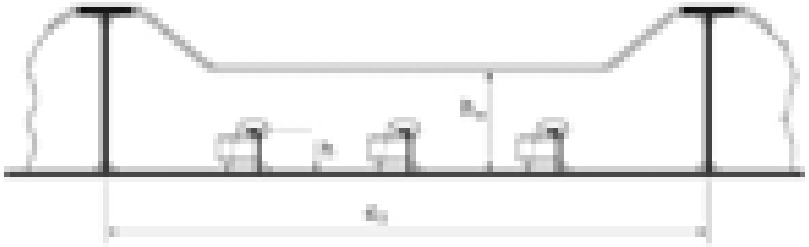


Fig.2.9.2 : Parameter definition for web stiffening

2.9.3 In addition, the following is to be satisfied:

$t_{wn} \geq 0.35 \, . \, t_{pn} \, . \, (\sigma_y / 235)^{0.5}$

where,

σ_y = minimum upper yield stress of the shell plate in way of the framing member [N/mm^2]

t_{wn} = net thickness of the web [mm]

t_{pn} = net thickness of the shell plate in way of the framing member [mm]

2.9.4 To prevent local flange buckling of welded profiles, the following are to be satisfied:

i) The flange width, b_f [mm], is not to be less than five times the net thickness of the web, t_{wn}.

ii) The flange outstand, b_{out} [mm], is to meet the following requirement:

$b_{out} / t_{fn} \leq 155 / (\sigma_y)^{0.5}$

where,

t_{fn} = net thickness of flange [mm]

σ_y = minimum upper yield stress of the material [N/mm^2].

2.10 Plated structures

2.10.1 Plated structures are those stiffened plate elements in contact with the hull and subject to ice loads e.g. bulkheads, decks etc. These requirements are applicable to an inboard extent which is the lesser of:

i) web height of adjacent parallel web frame or stringer; or

ii) 2.5 times the depth of framing that intersects the plated structure

2.10.2 The thickness of the plating and the scantlings of attached stiffeners are to be such that the degree of end fixity necessary for the shell framing is ensured.

2.10.3 The stability of the plated structure is to adequately withstand the ice loads defined in 2.3.

2.11 Corrosion / abrasion additions and steel renewal

2.11.1 Effective protection against corrosion and ice-induced abrasion is recommended for all external surfaces of the shell plating for Polar Class ships.

2.11.2 The values of corrosion / abrasion additions, t_s, to be used in determining the shell plate thickness are listed in Table 2.11.2.

2.11.3 Polar Class ships are to have a minimum corrosion addition of t_s = 1.0 [mm] applied to all internal structures within the ice-strengthened hull areas, including plated members adjacent to the shell, as well as stiffener webs and flanges.

2.11.4 Steel renewal for ice strengthened structures is required when the gauged thickness is less than t_{net} + 0.5 [mm]

Table 2.11.2 : Corrosion / abrasion additions for shell plating						
Hull Area	**t_s [mm]**					
	With effective protection			**Without effective protection**		
	PC1 – PC3	**PC4 and PC5**	**PC6 and PC7**	**PC1 – PC3**	**PC4 and PC5**	**PC6 and PC7**
Bow; bow intermediate icebelt	3.5	2.5	2.0	7.0	5.0	4.0
Bow intermediate lower; midbody and stern icebelt	2.5	2.0	2.0	5.0	4.0	3.0
Midbody and stern lower; bottom	2.0	2.0	2.0	4.0	3.0	2.5

2.12 Materials

2.12.1 Steel grades of plating for hull structures are to be not less than those given in Table 2.12.1 based on the as-built thickness, the Polar Class and the material class of structural members given in accordance to 2.12.2.

2.12.2 Material classes specified in Pt.3, Ch.2, Table 2.2.1a) are applicable to Polar Class ships regardless of the ship's length. In addition, material classes for weather and sea exposed structural members and for members attached to the weather and sea exposed shell plating of polar ships are given in Table 2.12.2. Where the material classes in Table 2.12.2 and those in Pt.3, Ch.2, Table 2.2.1a) differ, the higher material class is to be applied.

2.12.3 Steel grades for all plating and attached framing of hull structures and appendages situated below the level of 0.3 [m] below the lower ice waterline, as shown in Fig.2.12.3 are to be obtained from Table 2.2.1f) and Table 2.2.2 of Pt.3, Ch.2 based on the material class for structural members in Table 2.12.2 above, regardless of Polar Class.

2.12.4 Steel grades for all weather exposed plating of hull structures and appendages situated above the level of 0.3 [m] below the lower ice waterline, as shown in Fig.2.12.3 are to be not less than given in Table 2.12.1.

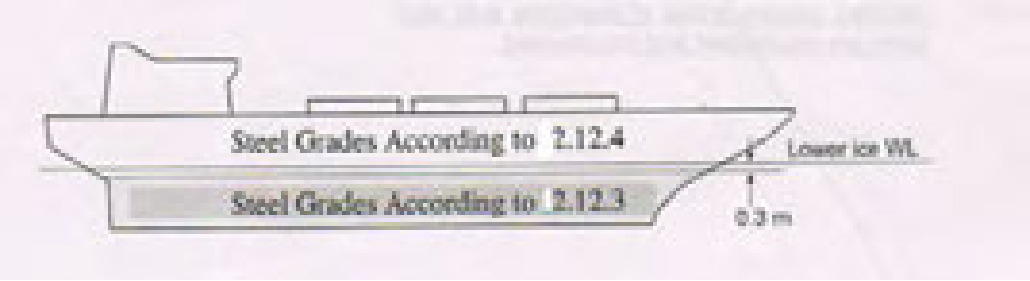


Fig.2.12.3 : Steel grade requirements for submerged and weather exposed shell plating

Table 2.12.2 : Material classes for structural members of polar ships	
Structural Members	**Material Class**
Shell plating within the bow and bow intermediate icebelt hull areas (B, BI_i)	II
All weather and sea exposed SECONDARY and PRIMARY, as defined in Table 2.2.1a) of Pt.3, Ch.2, structural members outside $0.4L_{UI}$ amidships	I
Plating materials for stem and stern frames, rudder horn, rudder, propeller nozzle, shaft brackets, ice skeg, ice knife and other appendages subject to ice impact loads	II
All inboard framing members attached to the weather and sea-exposed plating including any contiguous inboard member within 600 [mm] of the plating	I
Weather-exposed plating and attached framing in cargo holds of ships which by nature of their trade have their cargo hold hatches open during cold weather operations	I
All weather and sea exposed SPECIAL, as defined in Table 2.2.1a) of Pt.3, Ch.2, structural members within $0.2L_{UI}$ from FP	II

Table 2.12.1 : Steel grades for weather exposed plating[1)]														
Thickness, t [mm]	**Material Class I**				**Material Class II**				**Material Class III**					
	PC1-5		**PC6&7**		**PC1-5**		**PC6&7**		**PC1-3**		**PC4&5**		**PC6&7**	
	MS	**HT**	**MS**	**HT**	**MS**	**HT**	**MS**	**HT**	**MS**	**HT**	**MS**	**HT**	**MS**	**HT**
$t \leq 10$	B	AH	B	AH	B	AH	B	AH	E	EH	E	EH	B	AH
$t \leq 15$	B	AH	B	AH	D	DH	B	AH	E	EH	E	EH	D	DH
$15 < t \leq 20$	D	DH	B	AH	D	DH	B	AH	E	EH	E	EH	D	DH
$20 < t \leq 25$	D	DH	B	AH	D	DH	B	AH	E	EH	E	EH	D	DH
$25 < t \leq 30$	D	DH	B	AH	E	EH[2)]	D	DH	E	EH	E	EH	E	EH
$30 < t \leq 35$	D	DH	B	AH	E	EH	D	DH	E	EH	E	EH	E	EH
$35 < t \leq 40$	D	DH	D	DH	E	EH	D	DH	Φ	FH	E	EH	E	EH
$40 < t \leq 45$	E	EH	D	DH	E	EH	D	DH	Φ	FH	E	EH	E	EH
$45 < t \leq 50$	E	EH	D	DH	E	EH	D	DH	Φ	FH	Φ	FH	E	EH

Φ – Not Applicable

Notes:

1) Includes weather-exposed plating of hull structures and appendages, as well as their outboard framing members, situated above a level of 0.3 [m] below the lowest ice waterline.

2) Grades D, DH are allowed for a single strake of side shell plating not more than 1.8 [m] wide from 0.3 [m] below the lowest ice waterline.

2.12.5 Castings are to have specified properties consistent with the expected service temperature for the cast component.

2.13 Longitudinal strength

2.13.1 Application

2.13.1.1 A ramming impact on the bow is the design scenario for the evaluation of the longitudinal strength of the hull

2.13.1.2 Intentional ramming is not considered as a design scenario for ships which are designed with vertical or bulbous bows, see 1.1.6. Hence the longitudinal strength requirements given in 2.13 is not to be considered for ships with stem angle γ_{stem} equal to or larger than 80 [deg].

2.13.1.3 Ice loads are only to be combined with still water loads. The combined stresses are to be compared against permissible bending and shear stresses at different locations along the ship's length. In addition, sufficient local buckling strength is also to be verified.

2.13.2 Design vertical ice force at the bow

2.13.2.1 The design vertical ice force at the bow, F_{IB}, is to be taken as

F_{IB} = minimum ($F_{IB.1}$; $F_{IB.2}$) [MN]

where,

$F_{IB.1} = 0.534 \cdot K_I^{\ 0.15} \cdot \sin^{0.2}(\gamma_{stem}) \cdot (\Delta_{UI} \cdot K_h)^{0.5} \cdot CF_L$ [MN]

$F_{IB.2} = 1.20 \cdot CF_F$ [MN]

K_I = indentation parameter = K_f / K_h

a) for the case of a blunt bow form

$K_f = (2 \cdot C \cdot B_{UI}^{1-e_b} / (1 + e_b))^{0.9} \cdot \tan(\gamma_{stem})^{-0.9.(1+e_b)}$

b) for the case of wedge bow form (α_{stem} < 80 deg), e_b = 1 and the above simplifies to
$K_f = (\tan(\alpha_{stem}) / \tan^2(\gamma_{stem}))^{0.9}$

$K_h = 0.01 \cdot A_{wp}$ [MN/m]

CF_L = longitudinal strength class factor from Table 2.3.2

e_b = bow shape exponent which best describes the waterplane (see Fig.2.13.2.1a) and (Fig.2.13.2.1b)
= 1.0 for a simple wedge bow form
= 0.4 to 0.6 for a spoon bow form
= 0 for a landing craft bow form

An approximate e_b determined by a simple fit is acceptable

γ_{stem} = stem angle to be measured between the horizontal axis and the stem tangent at the upper ice waterline [deg] (buttock angle as per Fig.2.3.2.1 measured on the centerline)

α_{stem} = waterline angle measured in way of the stem at the upper ice waterline (UIWL) [deg] (see Fig. 2.13.2.1a)

$C = 1 / (2 \cdot (L_B / B_{UI})^{e_b})$

B_{UI} = moulded breadth corresponding to the upper ice waterline (UIWL) [m]

L_B = bow length used in the equation $y = B_{UI} / 2 \cdot (x/L_B)^{e_b}$ [m] (see Fig.2.13.2.1a) and Fig.2.13.2.1b)

Δ_{UI} = displacement as defined in 2.1.2.2, not to be taken less than 10 [k-tonnes]

A_{wp} = waterplane area corresponding to the upper ice waterline (UIWL) [m^2]

CF_F = flexural failure class factor from Table 2.3.2

2.13.3 Design vertical ice shear force

2.13.3.1 The design vertical ice shear force, F_I, along the hull girder is to be taken as:

$F_I = C_f \cdot F_{IB}$ [MN]

where,

C_f = longitudinal distribution factor to be taken as follows:

a) Positive shear force

C_f = 0.0 between the aft end of L_{UI} and 0.6L_{UI} from aft
C_f = 1.0 between 0.9L_{UI} from aft and the forward end of L_{UI}

b) Negative shear force

C_f = 0.0 at the aft end of L_{UI}
C_f = -0.5 between 0.2 L_{UI} and 0.6L_{UI} from aft
C_f = 0.0 between 0.8L_{UI} from aft and the forward end of L_{UI}

Intermediate values are to be determined by linear interpolation

2.13.3.2 Shear strength is to be checked as per Pt.3, Ch.5, Sec.4 by substituting the design vertical ice shear force for the design vertical wave shear force. The permissible shear stress τ is to be determined from Table 2.13.5.

2.13.4 Design vertical ice bending moment

2.13.4.1 The design vertical ice bending moment, M_I, along the hull girder is to be taken as:

$$M_I = 0.1 \cdot C_m \cdot L_{UI} \cdot \sin^{-0.2}(\gamma_{stem}) \cdot F_{IB} \quad [MNm]$$

where,

L_{UI} = length as defined in 2.1.2.1 [m]

γ_{stem} is as given in 2.13.2.1

F_{IB} = design vertical ice force at the bow [MN]

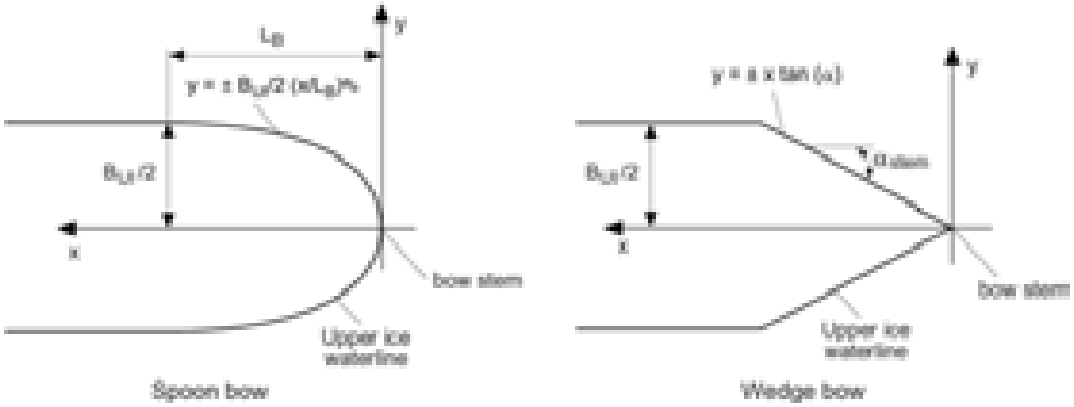


Fig.2.13.2.1a) : Bow shape definition

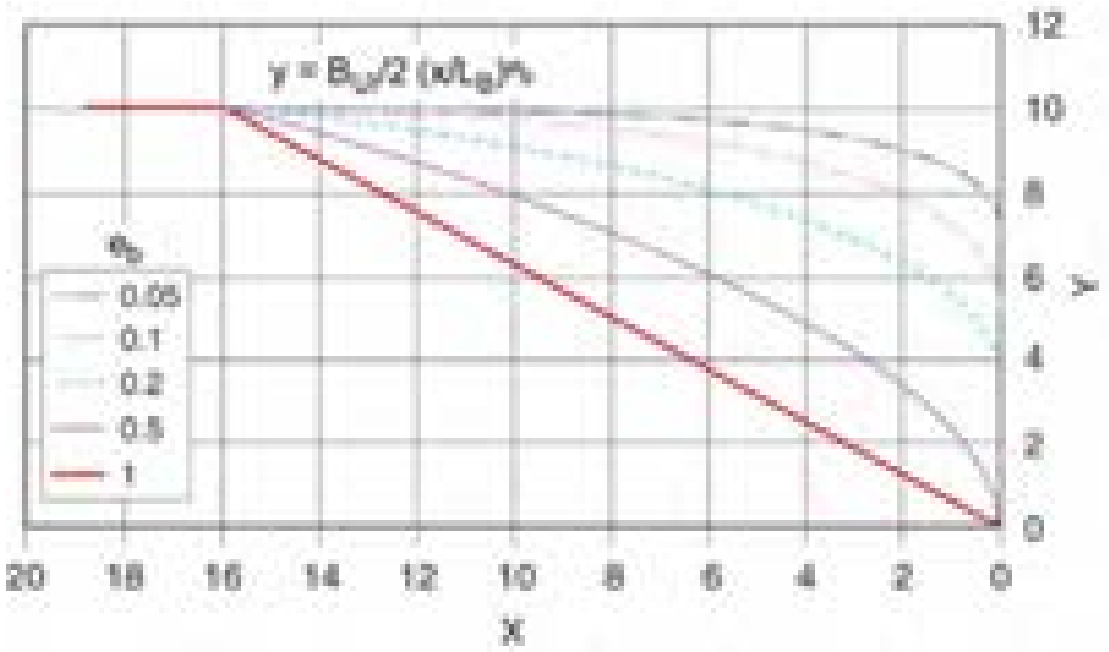


Fig.2.13.2.1b) : Illustration of e_b effect on the bow shape for B_{UI} = 20 and L_B = 16

C_m = longitudinal distribution factor for design vertical ice bending moment to be taken as follows:

C_m = 0.0 at the aft end of L_{UI}
C_m = 1.0 between $0.5L_{UI}$ and $0.7L_{UI}$ from aft
C_m = 0.3 at $0.95L_{UI}$ from aft
C_m = 0.0 at the forward end of L_{UI}

Intermediate values are to be determined by linear interpolation

2.13.4.2 The required section modulus of the hull girder is to be determined as per Pt.3, Ch.5, 3.3.1 by substituting the design vertical ice bending moment for the design vertical wave bending moment. The ship still water bending moment is to be taken as the permissible still water bending moment in sagging condition. The permissible bending stresses is to be determined from Table 2.13.5.

2.13.5 Longitudinal strength criteria

2.13.5.1 The strength criteria provided in Table 2.13.5 are to be satisfied. The design stress is not to exceed the permissible stress.

Table 2.13.5 : Longitudinal strength criteria

Failure Mode	Applied Stress	Permissible stress when $\sigma_y / \sigma_u \leq 0.7$	Permissible stress when $\sigma_y / \sigma_u > 0.7$
Tension	σ_a	$\eta \cdot \sigma_y$	$\eta \cdot 0.41 (\sigma_u + \sigma_y)$
Shear	τ_a	$\eta \cdot \sigma_y / \sqrt{3}$	$\eta \cdot 0.41 (\sigma_u + \sigma_y) / \sqrt{3}$
Buckling	σ_a	σ_{cr} for plating and for web plating of stiffeners $\sigma_{cr}/1.1$ for stiffeners	
	τ_a	τ_{cr}	

where,

σ_a = applied vertical bending stress [N/mm^2]

τ_a = applied vertical shear stress [N/mm^2]

σ_y = minimum upper yield stress of the material [N/mm^2]

σ_u = ultimate tensile strength of material [N/mm^2]

σ_{cr} = critical buckling stress in compression, according to Pt.3, Ch.3, Sec.6 [N/mm^2]

τ_{cr} = critical buckling stress in shear, according to Pt.3, Ch.3, Sec.6 [N/mm^2]

η = 0.8.

η = 0.6 for ships which are assigned the additional notation "**Icebreaker**".

2.14 Stem and stern frames

2.14.1 When the vessel has a sharp edged stem (see Fig.2.8.1of Pt.5, Ch.21) the thickness of the stem side plate is not to be less than 1.2 times the required plate thickness of the adjacent shell plating.

2.14.2 The plate thickness of a shaped plate stem and in the case of a blunt tow, any part of the shell which forms an angle of 30° or more to the centerline in a longitudinal plane is to be not less than 1.2 times the required thickness of the adjacent shell plating.

2.15 Appendages

2.15.1 All appendages are to be designed to withstand forces appropriate for the location of their attachment to the hull structure or their position within a hull area.

2.15.2 Load definition and response criteria are to be specially considered.

2.16 Local details

2.16.1 The end connections of longitudinal and transverse secondary members, web frames and stringers subjected to ice-induced loads are to be supported to transfer the loads to the supporting members. The area of end connections are to be such that shear stress does not exceed $\sigma_y / \sqrt{3}$ [N/mm^2], where σ_y is the yield stress of the material.

2.16.2 The loads carried by a member in way of cut-outs are not to cause instability. Where necessary, the structure is to be stiffened.

2.17 Direct calculation

2.17.1 Direct calculation are not to be utilized as an alternative to the analytical procedures prescribed for the shell plating and local frame requirements given in 2.4, 2.6 and 2.7.

2.17.2 Direct calculations are to be used for load carrying stringers and web frames forming part of a grillage system.

2.17.3 Where direct calculation is used to check the strength of structural systems, the load patch specified in 2.3 is to be applied, without being combined with any other loads. The load patch is to be applied at locations where the capability of these members under the combined effects of bending and shear is minimized. Special attention is to be paid to the shear capacity in way of lightening holes and cut-outs in way of intersecting members.

2.17.4 The strength evaluation of web frames and stringers may be performed based on linear or non-linear analysis. Recognized structural idealization and calculation methods are to be applied, but the detailed requirements are to be specially considered. In the strength evaluation, the guidance given in 2.17.5 and 2.17.6 may generally be considered.

2.17.5 If the structure is evaluated based on linear calculation methods, the following are to be considered:

1) Web plates and flange elements in compression and shear to fulfil relevant buckling criteria
2) Nominal shear stresses in member web plates to be less than $\sigma_y/\sqrt{3}$
3) Nominal von Mises stresses in member flanges to be less than 1.15 σ_y

2.17.6 If the structure is evaluated based on non-linear calculation methods, the following are to be considered:

1) The analysis is to reliably capture buckling and plastic deformation of the structure
2) The acceptance criteria are to ensure a suitable margin against fracture and major buckling and yielding causing significant loss of stiffness
3) Permanent lateral and out-of plane deformation of considered member are to be minor relative to the relevant structural dimensions.
4) Detailed acceptance criteria to be decided specially

2.18 Welding

2.18.1 All welding within ice-strengthened areas is to be of the double continuous type.

2.18.2 Continuity of strength is to be ensured at all structural connections.

Section 3

Machinery Requirements for Polar Class Ships

3.1 Application

3.1.1 The contents of this section apply to main propulsion, steering gear, emergency and auxiliary systems essential for the safety of the ship and the crew.

3.1.2 The vessel operating conditions are defined in Section 1.

3.1.3 The requirements herein are additional to those applicable for the basic open water operation of the vessel.

3.2 Documentation and design

3.2.1 Following drawings and particulars are to be submitted.

3.2.1.1 Details of the intended environmental operational conditions and the required polar strengthening for the machinery, if different from ship’s polar class.

3.2.1.2 Detailed drawings and descriptions of the main propulsion machinery, steering, emergency and auxiliary machinery and information on the essential main propulsion load control functions. The descriptions are to include operational limitations.

3.2.1.3 Description detailing where main, emergency and auxiliary systems are located and how they are protected to prevent problems from freezing, ice and snow accumulation and evidence of their capability to operate in the intended environmental conditions.

3.2.1.4 Calculations and documentation indicating compliance with the requirements of this section.

3.2.2 System design

3.2.2.1 Systems subject to damage by freezing are to be drainable.

3.2.2.2 Vessels classed PC1 to PC5 inclusive are to have means to ensure sufficient vessel operation in the case of propeller damage including the Controllable Pitch (CP)mechanism. Sufficient vessel operation means that the vessel is to be able to reach safe haven (safe location) where repairs can be undertaken. This may be achieved either by a temporary repair at sea, or by towing, assuming assistance is available. This would lead however to a condition of approval.

3.2.2.3 Means are to be provided to free a stuck propeller by turning it in reverse direction. This is to be possible for a propulsion plant intended for unidirectional rotation.

3.2.2.4 The propeller is to be fully submerged at the ships LIWL.

3.3 Materials

Materials are to be of an approved ductile material. Ferritic nodular cast iron may be used for parts other than bolts. For nodular cast iron an averaged impact energy value of 10 J at testing temperature is regarded as equivalent to the Charpy V test requirements defined below.

3.3.1 Materials exposed to sea water

3.3.1.1 Materials exposed to sea water, such as propeller blades, propeller hubs and cast thruster bodies are to have an elongation not less than15% on a test specimen according to *Pt.2, Ch.2, Mechanical Testing Procedures.*

3.3.1.2 Charpy V-notch impact testing is to be carried out for materials other than bronze and austenitic steel. The tests are to be carried out on three specimens at minus 10 ºC, and the average energy value is to be not less than 20 J. However, Charpy V impact test requirements of *Pt.2, Ch.4, Sec.2 or Pt.2, Ch.4, Sec.4* as applicable for ships with ice class notation, are also to be applied to ships covered by this section..

3.3.2 Materials exposed to sea water temperature

3.3.2.1 Charpy V-notch impact testing is to be carried out for materials other than bronze and austenitic steel. The tests are to be carried out on three specimens at minus 10 ºC, and the average energy value is to be not less than 20 J. However, the Charpy V impact test requirements of *Pt.2, Ch.4, Sec.2* as applicable for ships with ice class notation, are also to be applied to ships covered by this Section.

3.3.2.2 This requirement applies to components such as but not limited to blade bolts, CP-mechanisms, shaft bolts, propeller shaft, strut-pod connecting bolts, etc. This requirement does not apply to surface hardened components, such as bearings and gear teeth or sea water cooling lines (heat exchangers, pipes, valves, fittings etc.). For a definition of structural boundaries exposed to sea water temperature see *Sec.2, Fig 2.12.3*.

3.3.3 Material exposed to low air temperature

3.3.3.1 Materials of exposed machinery and foundations are to be manufactured from steel or other approved ductile material. An average impact energy value of 20 J taken from three Charpy V tests is to be obtained at 10 ºC below the lowest design temperature. Charpy V impact tests are not required for bronze and austenitic steel.

3.3.3.2 This requirement does not apply to surface hardened components, such as bearings and gear teeth. For a definition of structural boundaries exposed to air temperature see *Sec.2, Fig 2.12.3*.

3.4 Definitions

3.4.1 Definition of Symbols

Table 3.4.1 : Definition of symbols		
Symbol	**Unit**	**Definition**
c	m	chord length of blade section
$c_{0.7}$	m	chord length of blade section at 0.7R propeller radius
CP	-	controllable pitch
D	m	propeller diameter
d	m	external diameter of propeller hub (at propeller plane)
d_{pin}	mm	diameter of shear pin
D_{limit}	m	limit value for propeller diameter
EAR		expanded blade area ratio
F_b	kN	maximum backward blade force for the ship's service life (negative sign)
F_{ex}	kN	ultimate blade load resulting from blade failure through plastic bending
F_f	kN	maximum forward blade force for the ship's service life (positive sign)
F_{ice}	kN	ice load
$(F_{ice})_{max}$	kN	maximum ice load for the ship's service life
FP	-	fixed pitch
h_0	m	depth of the propeller centreline from lower ice waterline (LIWL)
H_{ice}	m	Ice block dimension for propeller load definition
I	kgm^2	equivalent mass moment of inertia of all parts on engine side of component under consideration
I_t	kgm^2	equivalent mass moment of inertia of the whole propulsion system
k	-	shape parameter for Weibull distribution
LIWL	m	lower ice waterline
m	-	slope for S-N curve in log/log scale
M_{BL}	kNm	blade bending moment
MCR	-	maximum continuous rating
N	-	number of ice load cycles
n	rev/s	propeller rotational speed
n_n	rev/s	nominal propeller rotational speed at MCR in free running condition
N_{class}	-	reference number of ice impacts per propeller revolution per ice class
N_{ice}	-	total number of ice load cycles on propeller blade for the ship's service life
N_R	-	reference number of ice load cycles for equivalent fatigue stress (10^8 cycles)
N_Q	-	number of propeller revolutions during a milling sequence
$P_{0.7}$	m	propeller pitch at 0.7R radius
$P_{0.7n}$	m	propeller pitch at 0.7R radius at MCR in free running condition
$P_{0.7b}$	m	propeller pitch at 0.7R radius at MCR in bollard condition
PCD	m	pitch circle diameter

$Q(\varphi)$	kNm	Torque
Q_{Amax}	kNm	maximum response torque amplitude as a simulation result
Q_{emax}	kNm	maximum engine torque
$Q_F(\varphi)$	kNm	Ice torque excitation for frequency domain calculations
Q_{fr}	kNm	friction torque in pitching mechanism; reduction of spindle torque
Q_{max}	kNm	maximum torque on the propeller resulting from propeller/ice interaction
Q_{motor}	kNm	electric motor peak torque
Q_n	kNm	nominal torque at MCR in free running condition
$Q_r(t)$ Q_{peak}	kNm kNm	response torque along the propeller shaft line maximum of the response torque $Q_r(t)$
Q_{smax}	kNm	maximum spindle torque of the blade for the ship's service life
Q_{sex}	kNm	extreme spindle torque corresponding to the blade failure load F_{ex}
Q_{vib}	kNm	Vibratory torque at considered component, taken from frequency domain open water TVC
R	m	propeller radius
S	-	Safety factor
S_{fat}	-	Safety factor for fatigue
S_{ice}	-	Ice strength index for blade ice force
r	m	blade section radius
T	kN	Hydrodynamic propeller thrust in bollard condition
T_b	kN	maximum backward propeller ice thrust for the ship's service life
T_f	kN	maximum forward propeller ice thrust for the ship's service life
T_n	kN	propeller thrust at MCR in free running condition
T_r	kN	maximum response thrust along the shaft line
T_{kmax}	kNm	maximum torque capacity of flexible coupling
T_{kmax2}	kNm	T_{kmax} at N=1 load cycle
T_{kmax1}	kNm	T_{kmax} at $N=5\cdot10^4$ load cycles
T_{kv}	kNm	vibratory torque amplitude at $N=10^6$ load cycles
ΔT_{kmax}	kNm	maximum range of T_{kmax} at $N=5\cdot10^4$ load cycles
T	m	maximum blade section thickness
z	-	number of propeller blades
z_{pin}	-	number of shear pins
α_i	deg	duration of propeller blade/ice interaction expressed in rotation angle
γ_ε	-	the reduction factor for fatigue; scatter and test specimen size effect
γ_v	-	the reduction factor for fatigue; variable amplitude loading effect
γ_m	-	the reduction factor for fatigue; mean stress effect
ρ	-	a reduction factor for fatigue correlating the maximum stress amplitude to the equivalent fatigue stress for 10^8 stress cycles
$\sigma_{0.2}$	MPa	proof yield strength (at 0.2% plastic strain) of material

σ_{exp}	MPa	mean fatigue strength of blade material at 10^8 cycles to failure in sea water
σ_{fat}	MPa	equivalent fatigue ice load stress amplitude for 10^8 stress cycles
σ_{fl}	MPa	characteristic fatigue strength for blade material
σ_{ref1}	MPa	reference stress $\sigma_{ref1} = 0.6 \cdot \sigma_{0.2} + 0.4 \cdot \sigma_u$
$\sigma_{ref\,2}$	MPa	reference stress $\sigma_{ref2} = 0.7 \cdot \sigma_u$ or $\sigma_{ref2} = 0.6 \cdot \sigma_{0.2} + 0.4 \cdot \sigma_u$ whichever is less
σ_{st}	MPa	maximum stress resulting from F_b or F_f
σ_u	MPa	ultimate tensile strength of blade material
$(\sigma_{ice})_{bmax}$	MPa	principal stress caused by the maximum backward propeller ice load
$(\sigma_{ice})_{fmax}$	MPa	principal stress caused by the maximum forward propeller ice load
$(\sigma_{ice})_{Amax}$	MPa	maximum ice load stress amplitude at the considered location on the blade
σ_{mean}	MPa	mean stress
$(\sigma_{ice})_A(N)$	MPa	blade stress amplitude distribution

3.4.2 Definition of Loads

Table 3.4.2 : Definition of loads

	Definition	Use of the load in design process
F_b	The maximum lifetime backward force on a propeller blade resulting from propeller/ice interaction, including hydrodynamic loads on that blade. The direction of the force is perpendicular to 0.7R chord line. See Figure 1.	Design force for strength calculation of the propeller blade.
F_f	The maximum lifetime forward force on a propeller blade resulting from propeller/ice interaction, including hydrodynamic loads on that blade. The direction of the force is perpendicular to 0.7R chord line.	Design force for calculation of strength of the propeller blade.
Q_{smax}	The maximum lifetime spindle torque on a propeller blade resulting from propeller/ice interaction, including hydrodynamic loads on that blade.	In designing the propeller strength, the spindle torque is automatically taken into account because the propeller load is acting on the blade as distributed pressure on the leading edge or tip area.
T_b	The maximum lifetime thrust on propeller (all blades) resulting from propeller/ice interaction. The direction of the thrust is the propeller shaft direction and the force is opposite to the hydrodynamic thrust.	Is used for estimation of the response thrust T_r. T_b can be used as an estimate of excitation for axial vibration calculations. However, axial vibration calculations are not required in the rules.

T_f	The maximum lifetime thrust on propeller (all blades) resulting from propeller/ice interaction. The direction of the thrust is the propeller shaft direction acting in the direction of hydrodynamic thrust.	Is used for estimation of the response thrust T_r. T_f can be used as an estimate of excitation for axial vibration calculations. However, axial vibration calculations are not required in the rules.
Q_{max}	The maximum ice-induced torque resulting from propeller/ice interaction on one propeller blade, including hydrodynamic loads on that blade.	Is used for estimation of the response torque (Q_r) along the propulsion shaft line and as excitation for torsional vibration calculations.
F_{ex}	Ultimate blade load resulting from blade loss through plastic bending. The force that is needed to cause total failure of the blade so that plastic hinge is caused to the root area. The force is acting on 0.8R.	Blade failure load is used to dimension the blade bolts, pitch control mechanism, propeller shaft, propeller shaft bearing and trust bearing. The objective is to guarantee that total propeller blade failure should not cause damage to other components.
Q_{sex}	Maximum spindle torque resulting from blade failure load	Is used to ensure pyramid strength principle for the pitching mechanism
Q_r	Maximum response torque along the propeller shaft line, taking into account the dynamic behavior of the shaft line for ice excitation (torsional vibration) and hydrodynamic mean torque on propeller.	Design torque for propeller shaft line components.
T_r	Maximum response thrust along shaft line, taking into account the dynamic behavior of the shaft line for ice excitation (axial vibration) and hydrodynamic mean thrust on propeller.	Design thrust for propeller shaft line components.

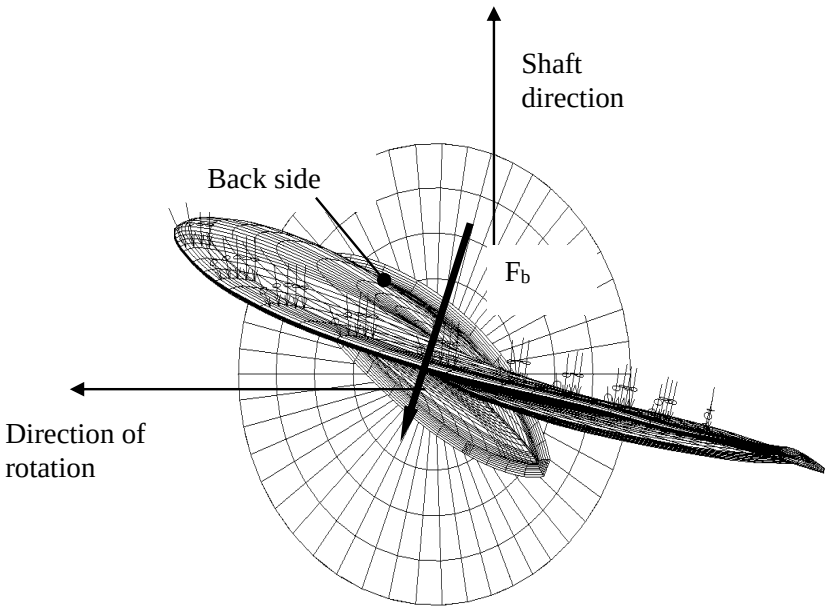


Fig. 3.4.2 : Direction of the backward blade force resultant taken perpendicular to the chord line at radius 0.7R. Ice contact pressure at leading edge is shown with small arrows.

3.5 Design Ice Loads

3.5.1 General

3.5.1.1 These Rules cover open and ducted type propellers situated at the stern of a vessel having controllable pitch or fixed pitch blades. Ice loads on bow-mounted propellers will receive special consideration. The given loads are expected, single occurrence, maximum values for the whole ship's service life for normal operational conditions, including loads resulting from directional change of rotation where applicable. These loads do not cover off-design operational conditions, for example when a stopped propeller is dragged through ice. These Rules also cover loads due to propeller ice interaction for azimuthing and fixed thrusters with geared transmission or an integrated electric motor (geared and podded propulsors). However, the load models of the regulations do not include propeller/ice interaction loads when ice enters the propeller of a turned azimuthing thruster from the side (radially) or loads when ice blocks hit on the propeller hub of a pulling propeller. Ice loads resulting from ice impacts on the body of thrusters are to be estimated on a case by case basis, however are not included within the following section.

3.5.1.2 The loads given in clause 3.5.3 are total loads including ice-induced loads and hydrodynamic loads (unless otherwise stated) during ice interaction and are to be applied separately (unless otherwise stated) and are intended for component strength calculations only.

3.5.1.3 F_b is the maximum force experienced during the lifetime of the ship that bends a propeller blade backwards when the propeller mills an ice block while rotating ahead. F_f is the maximum force experienced during the lifetime of the ship that bends a propeller blade forwards when a propeller mills an ice block while rotating ahead. F_b and F_f originate from different propeller/ice interaction phenomena, which do not act simultaneously. Hence, they are to be applied separately.

3.5.2 Ice class factors

3.5.2.1 The dimensions of the considered design ice block are H_{ice} x $2H_{ice}$ x $3H_{ice}$. The design ice block and ice strength index (S_{ice}) are used for the estimation of propeller ice loads. Both H_{ice} and S_{ice} are defined for each Ice class in Table 3.5.2.1.

Table 3.5.2.1 : Design Ice Class Factors

Ice Class	H_{ice} [m]	S_{ice} [-]
PC1	4.0	1.2
PC2	3.5	1.1
PC3	3.0	1.1
PC4	2.5	1.1
PC5	2.0	1.1
PC6	1.75	1
PC7	1.5	1

3.5.3 Propeller Ice Interaction Loads

3.5.3.1 Maximum backward blade force F_b for open propellers

when $D < D_{limit}$,

$$F_b = 27 \cdot S_{ice} \cdot [n \cdot D]^{0.7} \cdot \left[\frac{EAR}{Z}\right]^{0.3} \cdot D^2 \quad \text{[kN]}$$

when $D \geq D_{limit}$,

$$F_b = 23 \cdot S_{ice} \cdot [n \cdot D]^{0.7} \cdot \left[\frac{EAR}{Z}\right]^{0.3} \cdot [H_{ice}]^{1.4} \cdot D$$

[kN]

where $D_{limit} = 0.85 \,.\, (H_{ice})^{1.4}$

Here n is the nominal rotational speed at MCR in the free running condition for CP-propellers and 85% of the nominal rotational speed (at MCR free running condition) for FP-propeller (regardless of driving engine type) [rps].

For vessels with the additional notation Icebreaker, the above stated backward blade force F_b is to be multiplied by a factor of 1.1.

3.5.3.2 Maximum forward blade force F_f for open propellers

when $D < D_{limit}$

$$F_f = 250 \cdot \left[\frac{EAR}{Z}\right] \cdot D^2 \text{ [kN]}$$

when $D \geq D_{limit}$

$$F_f = 500 \cdot \left[\frac{1}{1 - \frac{d}{D}} \right] \cdot H_{ice} \cdot \left[\frac{EAR}{Z} \right] \cdot D \text{ [kN]}$$

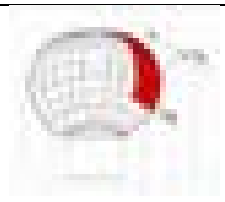
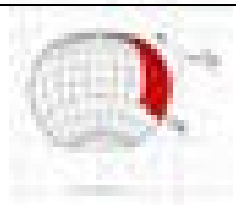
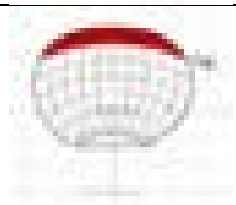
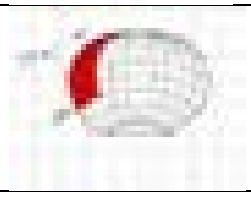
where,

$$D_{limit} = \left(\frac{2}{1 - \frac{d}{D}} \right) H_{ice}$$

3.5.3.3 Loaded area on the blade for open propellers

Load cases 1-4 are to be covered, as given in Table 3.5.3.3, for CP and FP propellers. In order to obtain blade ice loads for a reversing propeller, load case 5 is to be covered for propellers where reversing is possible.

Table 3.5.3.3 : Loaded areas and load case definition for open propellers

	Force	Loaded area	Right-handed propeller blade seen from behind
Load case 1	F_b	Uniform pressure applied on the back of the blade (suction side) to an area from 0.6*R* to the tip and from the leading edge to 0.2 times the chord length.	
Load case 2	50% of F_b	Uniform pressure applied on the back of the blade (suction side) on the propeller tip area outside 0.9*R* radius.	
Load case 3	F_f	Uniform pressure applied on the blade face (pressure side) to an area from 0.6*R* to the tip and from the leading edge to 0.2 times the chord length.	
Load case 4	50% of F_f	Uniform pressure applied on propeller face (pressure side) on the propeller tip area outside 0.9*R* radius.	
Load case 5	60% of F_f or 60% of F_b, whichever is greater	Uniform pressure applied on propeller face (pressure side) to an area from 0.6*R* to the tip and from the trailing edge to 0.2 times the chord length	

3.5.3.4 Maximum backward blade ice force F_b for ducted propellers

when D < Dlimit :

$$F_b = 9.5 \cdot S_{ice} \cdot [n \cdot D]^{0.7} \cdot \left[\frac{EAR}{Z} \right]^{0.3} \cdot D^2 \text{ [kN]}$$

when D ≥ D_{limit} :

$$F_b = 66 \cdot S_{ice} \cdot [n \cdot D]^{0.7} \cdot \left[\frac{EAR}{Z} \right]^{0.3} \cdot [H_{ice}]^{1.4} \cdot D^{0.6}$$

[kN]

where:

$D_{limit} = 4 \cdot H_{ice}$ [m]

n is to be taken as in 3.5.3.1

For vessels with the additional notation Icebreaker, the above stated backward blade force F_b is to be multiplied by a factor 1.1.

3.5.3.5 Maximum forward blade force, F_f for ducted propellers

when $D \leq D_{limit}$

$$F_f = 250 \cdot \left[\frac{EAR}{Z}\right] \cdot D^2 \quad [kN]$$

when $D > D_{limit}$

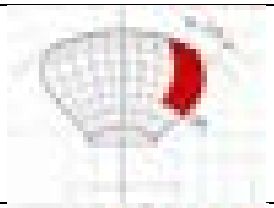
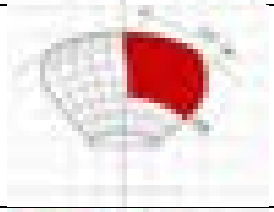
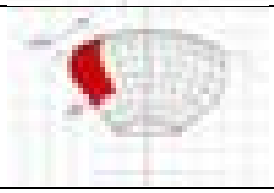
$$F_f = 500 \cdot \left[\frac{EAR}{Z}\right] \cdot D \cdot \frac{1}{\left(1 - \frac{d}{D}\right)} \cdot H_{ice} \quad [kN]$$

$$\text{where} D_{limit} = \frac{2}{\left(1 - \frac{d}{D}\right)} \cdot H_{ice} \quad [m]$$

3.5.3.6 Loaded area on the blade for ducted propellers

Load cases 1 and 3 are to be covered as given in Table 3.5.3.6 for all propellers. In order to obtain blade ice loads for a reversing propeller, load case 5 is to be covered for propellers, where reversing is possible.

Table 3.5.3.6 : Loaded areas and load case definition for ducted propellers

	Force	Loaded area	Right handed propeller blade seen from behind
Load case 1	F_b	Uniform pressure applied on the back of the blade (suction side) to an area from 0.6*R* to the tip and from the leading edge to 0.2 times the chord length.	
Load case 3	F_f	Uniform pressure applied on the blade face (pressure side) to an area from 0.6*R* to the tip and from the leading edge to 0.5 times the chord length.	
Load case 5	60% of F_f or 60% of F_b, whichever is greater	Uniform pressure applied on propeller face (pressure side) to an area from 0.6*R* to the tip and from the trailing edge to 0.2 times the chord length.	

3.5.3.7 Maximum blade spindle torque, Q_{smax} for open and ducted propellers

The spindle torques Q_{smax} around the axis of the blade fitting are to be determined both for the maximum backward blade force F_b and forward blade force F_t,which are applied as per Table 3.5.3.3 and Table 3.5.3.6. If the above method gives a value which is less than the default value given by the formula below, the default value is to be used.

Default value : $Q_{smax} = 0.25 \cdot F \cdot C_{0.7}$ [kNm]

where,

F is taken as either F_b or F_f, whichever has the greater absolute value.

The blade failure spindle torque Q_{sex} is defined under 3.5.4.

3.5.3.8 Load distributions (spectra) for blade loads

The Weibull-type distribution (probability that F_{ice} exceeds $(F_{ice})_{max}$), as given in Fig 3.5.3.8 is used for the fatigue design of the blade

$$P\left(\frac{F_{ice}}{(F_{ice})_{\max}} \geq \frac{F}{(F_{ice})_{\max}}\right) = e^{\left(-\left(\frac{F}{(F_{ice})_{\max}}\right)^{k} \cdot \ln(N_{ice})\right)}$$

where:

k = shape parameter of the spectrum

N_{ice}= number of load cycles in the spectrum, see 3.5.3.9

F_{ice} = random variable for ice loads on the blade, $0 \leq F_{ice} \leq (F_{ice})_{max}$.

This results in a blade stress amplitude distribution

$$(\sigma_{ice})_A(N) = (\sigma_{ice})_{A\max} \cdot \left\{1 - \frac{\log(N)}{\log(N_{ice})}\right\}^{\frac{1}{k}}$$

where:

$$(\sigma_{ice})_{A\max} = \frac{(\sigma_{ice})_{f\max} - (\sigma_{ice})_{b\max}}{2}$$

The shape parameter k=0.75 is to be used for the ice force distribution of an open propeller and the shape parameter k=1.0 for that of a ducted propeller blade.

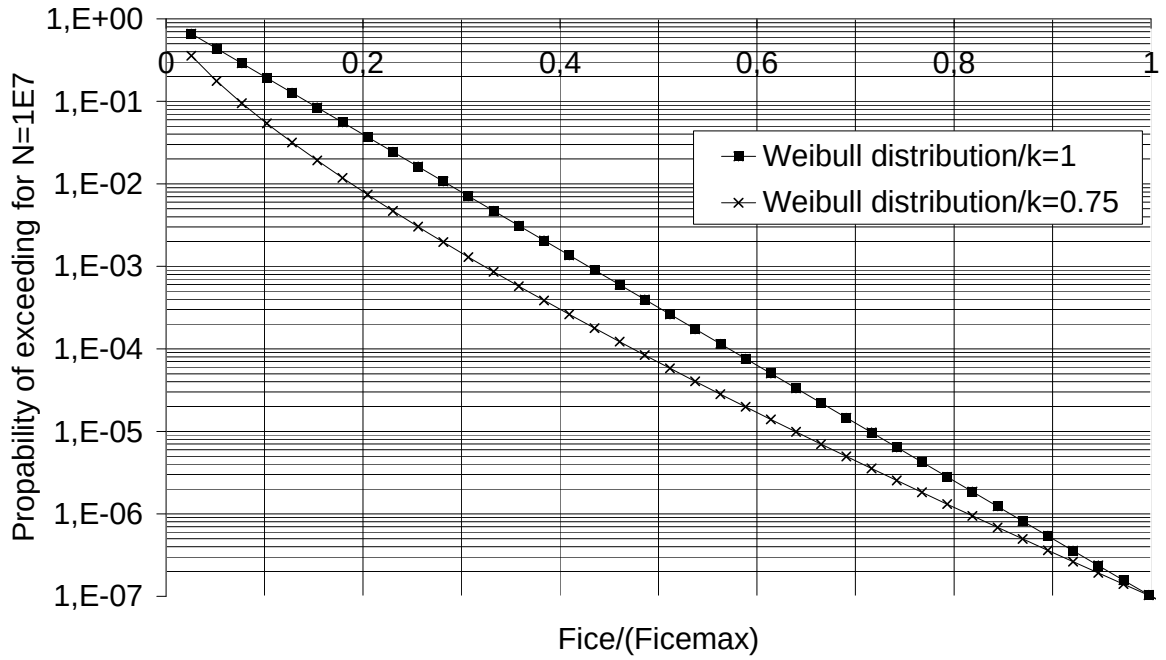


Fig 3.5.3.8 : The Weibull-type distribution (probability that F_{ice} exceeds $(F_{ice})_{max}$) that is used for fatigue design.

3.5.3.9 Number of ice loads

Number of load cycles N_{ice} in the load spectrum per blade is to be determined according to the formula:

$$N_{ice} = k_1 \cdot k_2 \cdot N_{class} \cdot n$$

where:

N_{class} = reference number of impacts per propeller revolution for each ice class (Table 3.5.3.9)

Table 3.5.3.9 : Reference number of impacts							
Ice Class	PC1	PC2	PC3	PC4	PC5	PC6	PC7
N_{class}	21 x 10^6	17 x 10^6	15 x 10^6	13 x 10^6	11 x 10^6	9 x 10^6	6 x 10^6

k1 = 1 for centre propeller
= 2 for wing propeller
= 3 for pulling propeller (wing and centre)

k2 = 0.8 - f when f < 0
= 0.8 - 0.4·f when 0 ≤ f ≤ 1
= 0.6 - 0.2·f when 1 < f ≤ 2.5
= 0.1 when f > 2.5

where the immersion function f is:

$$f = \frac{h_o - H_{ice}}{D/2} - 1$$

If h_0 is not known, h_0= D/2.

For vessels with the additional notation Icebreaker, the above stated number of load cycles N_{ice} is to be multiplied by a factor of 3.

For components that are subject to loads resulting from propeller/ice interaction with all the propeller blades, the number of load cycles (N_{ice}) is to be multiplied by the number of propeller blades (Z).

3.5.4 Blade Failure Load for both Open and Ducted Propellers

3.5.4.1 Bending Force, F_{ex}

The minimum load required resulting in blade failure through plastic bending. This is to be calculated iteratively along the radius of the blade from blade root to 0.5R using below equation with the ultimate load assumed to be acting at 0.8R in the weakest direction.

The blade failure load is:

$$F_{ex} = \frac{0.3 \cdot c \cdot t^{2\cdot} \cdot \sigma_{ref1}}{0.8 \cdot D - 2 \cdot r} \cdot 10^3 \quad \text{[kN]}$$

where:

$$\sigma_{ref1} = 0.6 \cdot \sigma_{0.2} + 0.4 \cdot \sigma_u \quad \text{[MPa]}$$

σ_u (minimum ultimate tensile strength to be specified on the drawing) and $\sigma_{0.2}$ (minimum yield or 0.2% proof strength to be specified on the drawing) are representative values for the blade material

c, t and r are respectively the actual chord length, maximum thickness and radius of the cylindrical root section of the blade, which is the weakest section outside the root fillet located typically at the termination of the fillet into the blade profile.

IRS may approve alternative means of failure load calculation by means of an appropriate stress analysis reflecting the non-linear plastic material behaviour of the actual blade. A blade is regarded as having failed, if the tip is bent by more than 10% of the propeller diameter.

3.5.4.2 Spindle Torque, Q_{sex}

The maximum spindle torque due to a blade failure load acting at 0.8R is to be determined. The force that causes blade failure typically reduces when moving from the propeller centre towards the leading and trailing edges. At a certain distance from the blade centre of rotation the maximum spindle torque will occur. This maximum spindle torque is to be defined by an appropriate stress analysis or using equation below.

$Q_{sex} = \max(c_{LE0.8};\ 0.8 \cdot c_{TE0.8}) \cdot C_{spex} \cdot F_{ex}$ [kNm]

Where:

$$C_{spex} = C_{sp} \cdot C_{fex} = 0.7 \cdot \left(1 - \left(4 \cdot \frac{EAR}{Z}\right)^3\right)$$

C_{sp} is non-dimensional parameter taking into account the spindle arm

C_{fex} is non-dimensional parameter taking into account the reduction of blade failure force at the location of maximum spindle torque.

If C_{spex} is below 0.3, a value of 0.3 is to be used for C_{spex}.

$c_{LE0.8}$ is the leading edge portion of the chord length at 0.8R

$c_{TE0.8}$ is the trailing edge portion of the chord length at 0.8R

The Fig 3.5.4.2 illustrates the spindle torque values due to blade failure loads across the whole chord length.

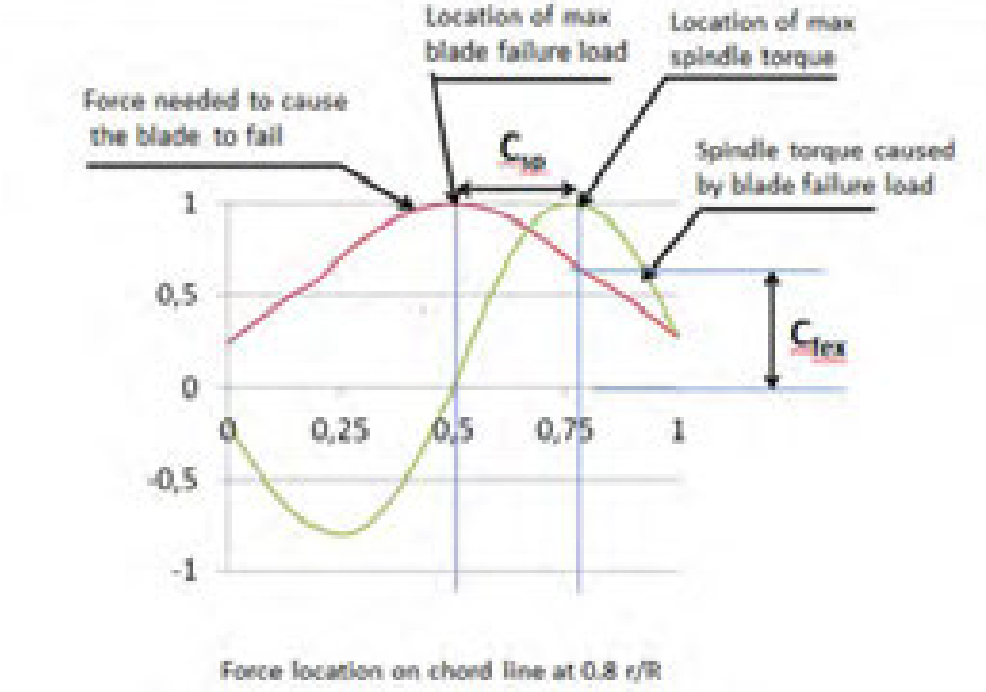


Fig 3.5.4.2 : Schematic figure showing blade failure load and related spindle torque when the force acts at different location on the chord line at radius 0.8R.

3.5.5 Axial design loads acting on open and ducted propellers

3.5.5.1 Maximum ice thrust on propeller T_f and T_b acting on open and ducted propellers

The maximum forward and backward ice thrusts are given by the following formula:

$$T_f = 1.1 \cdot F_f \ [\text{kN}]$$

$$T_b = 1.1 \cdot F_b \ [\text{kN}]$$

However, the load models within this section do not include propeller/ice interaction loads where an ice block hits the propeller hub of a pulling propeller.

3.5.5.2 Design thrust along the propulsion shaft line for open and ducted propellers

The design thrust along the propeller shaft line is to be calculated with the formulae below. The greater value of the forward and backward directional load is to be taken as the design load for both directions. The factors 2.2 and 1.5 take into account the dynamic magnification resulting from axial vibration.

In a forward direction

$$T_r = T + 2.2 \cdot T_f \ [\text{kN}]$$

In a backward direction

$$T_r = 1.5 \cdot T_b \ [\text{kN}]$$

If the hydrodynamic bollard thrust, T, is not known, T is to be taken as follows:

Table 3.5.5.2 : Guidance for bollard thrust values

Propeller type	T
CP propellers (open)	1.25 T_n
CP propellers (ducted)	1.1 T_n
FP propellers driven by turbine or electric motor	T_n
FP propellers driven by diesel engine (open)	0.85 T_n
FP propellers driven by diesel engine (ducted)	0.75 T_n

Here, T_n is the nominal propeller thrust at MCR in the free running open water condition.

For pulling type propellers ice interaction loads on propeller hub is to be considered in addition to the above.

3.5.6 Torsional design loads acting on open and ducted propellers

3.5.6.1 Design ice torque on propeller Q_{max} for open propellers

Q_{max} is the maximum torque on a propeller resulting from ice/propeller interaction.

when $D < D_{limit}$:

$$Q_{\max} = k_{open} \cdot \left[1 - \frac{d}{D}\right] \cdot \left[\frac{P_{0.7}}{D}\right]^{0.16} \cdot [n \cdot D]^{0.17} \cdot D^3$$

[kNm]

where:

k_{open} = 14.7 for PC1 – PC5;

and

k_{open} = 10.9 for PC6 – PC7

when $D \geq D_{limit}$:

$$Q_{\max} = 1.9 \cdot k_{open} \cdot \left[1 - \frac{d}{D}\right] \cdot [H_{ice}]^{1.1} \cdot \left[\frac{P_{0.7}}{D}\right]^{0.16} \cdot [n \cdot D]^{0.17} \cdot D^{1.9}$$

[kNm]

where:

$D_{limit} = 1.8 \cdot H_{ice}$ [m]

n is the rotational propeller speed in rev/s in bollard condition. If not known, n is to be taken as follows:

Table 3.5.6.1 : Guidance for rotational speeds to calculate torsional loads

Propeller type	Rotational speed n
CP propellers	n_n
FP propellers driven by turbine or electric motor	n_n
FP propellers driven by diesel engine	0.85 n_n

For CP propellers, the propeller pitch $P_{0.7}$ is correspond to MCR in bollard condition. If not known, $P_{0.7}$ is to be taken as $0.7 \cdot P_{0.7n}$, where $P_{0.7n}$ is the propeller pitch at MCR in free running condition.

3.5.6.2 Design ice torque on propeller Q_{max} for ducted propellers

when $D \leq D_{limit}$:

$$Q_{\max} = k_{ducted} \cdot \left[1 - \frac{d}{D}\right] \cdot \left[\frac{P_{0.7}}{D}\right]^{0.16} \cdot [n \cdot D]^{0.17} \cdot D^3$$

[kNm]

where:

k_{ducted} = 10.4 for PC1 – PC5

and

k_{ducted} = 7.7 for PC6 – PC7

when $D > D_{limit}$:

$$Q_{\max} = 1.9 \cdot k_{ducted} \cdot \left[1 - \frac{d}{D}\right] \cdot [H_{ice}]^{1.1} \cdot \left[\frac{P_{0.7}}{D}\right]^{0.16} \cdot [nD]^{0.17} \cdot D^{1.9}$$

[kNm]

where:

$D_{limit} = 1.8 \cdot H_{ice}$ [m]

n shall be taken as in 3.5.6.1.

For CP propellers, the propeller pitch $P_{0.7}$ is correspond to MCR in bollard condition. If not known, $P_{0.7}$ is to be taken as $0.7 \cdot P_{0.7n}$, where $P_{0.7n}$ is the propeller pitch at MCR in free running condition.

3.5.6.3 Ice torque excitation for open and ducted propellers

The given excitations are used to estimate the maximum torque likely to be experienced once during the service life of the ship. The following load cases are intended to reflect the operational loads on the propulsion system when the propeller interacts with ice and the corresponding reaction of the complete system. The ice impact and system response cause loads in the individual shaft line components. The ice torque Q_{max} may be taken as a constant value in the complete speed range. When considerations at specific shaft speeds are performed a relevant Q_{max} may be calculated using the relevant speed.

Diesel engine plants without an elastic coupling is to be calculated at the least favourable phase angle for ice versus engine excitation, when calculated in time domain. The engine firing pulses is to be included in the calculations and their standard steady state harmonics can be used. A phase angle between ice and gas force excitation does not need to be regarded in frequency domain analysis. Misfiring does not need to be considered.

If there is a blade order resonance just above MCR speed, calculations are to cover the rotational speeds up to 105% of MCR speed.

See also Guidelines for calculations given in 3.5.7.

3.5.6.3.1 Excitation for the time domain calculation

The propeller ice torque excitation for shaft line transient dynamic analysis (time domain) is defined as a sequence of blade impacts which are of half sine shape and occur at the blade. The torque due to a single blade ice impact as a function of the propeller rotation angle is then defined as:

$$Q(\varphi) = C_q \cdot Q_{\max} \cdot \sin(\varphi(180/\alpha_i))$$

when φ rotates from 0 to α_i plus integer revolutions.

$$Q(\varphi) = 0$$

when φ rotates from α_i to 360 plus integer revolutions.

Where

Φ = rotation angle starting when the first impact occurs

C_q and α_i parameters are given in the Table 3.5.6.3.1 below. α_i is the duration of propeller blade/ice interaction expressed in propeller rotation angle.

Table 3.5.6.3.1 : Ice impact magnification and duration factors for different blade numbers

Torque excitation	Propeller/ ice interaction	C_q	α_i [deg]			
			Z=3	Z=4	Z=5	Z=6
Excitation case 1	Single ice block	0.75	90	90	72	60
Excitation case 2	Single ice block	1.0	135	135	135	135
Excitation case 3	Two ice blocks (phase shift 360/(2·Z) deg.)	0.5	45	45	36	30
Excitation case 4	Single ice block	0.5	45	45	36	30

The total ice torque is obtained by summing the torque of single blades, taking into account the phase shift 360 deg./Z.

At the beginning and at the end of the milling sequence (within calculated duration) linear ramp functions are to be used to increase C_q to its maximum within one propeller revolution and vice versa to decrease it to zero (see examples for different Z numbers in the appendix).

The number of propeller revolutions during a milling sequence is to be obtained from the formula:

$$N_Q = 2 \cdot H_{ice}$$

The number of impacts is $Z \cdot N_Q$ for blade order excitation.

An illustration of all excitation cases for different blade numbers is given in the Appendix.

The dynamic simulation is to be performed for all excitation cases starting at MCR nominal, MCR bollard condition and just above all resonance speeds (1st engine and 1st blade harmonic), so that the resonant vibration responses can be obtained. For a fixed pitch propeller plant the dynamic simulation is to cover bollard pull condition with a corresponding speed assuming maximum possible output of the engine.

If a speed drop occurs down to stand still of the main engine, it indicates that the engine may not be sufficiently powered for the intended service task. For the consideration of loads, the maximum occurring torque during the speed drop process is to be applied. On these cases, the excitation is to follow the shaft speed.

3.5.6.3.2 Frequency domain excitation

For frequency domain calculations the following torque excitation may be used. The excitation has been derived so that the time domain half sine impact sequences have been assumed to be continuous and the Fourier series components for blade order and twice the blade order components have been derived. The frequency domain analysis is generally considered as conservative compared to the time domain simulation provided there is a first blade order resonance in the considered speed range.

$$Q_F(\varphi) = Q_{max} \cdot (C_{q0} + C_{q1} \cdot sin(Z \cdot E_0 \cdot \varphi + \alpha_1) + C_{q2} \cdot sin(2 \cdot Z \cdot E_0 \cdot \varphi + \alpha_2)) \quad \text{[kNm]}$$

where :

C_{q0} = mean torque component
C_{q1} = first blade order excitation amplitude
C_{q2} = second blade order excitation amplitude
Φ = angle of rotation
$\alpha_{1,2}$ = phase angle of excitation component
Z = number of blades

Table 3.5.6.3.2 : Coefficients for simplified excitation torque estimation

Torque excitation	Z=3					
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.375	0.375	-90	0	0	1
Excitation case 2	0.7	0.33	-90	0.05	-45	1
Excitation case 3	0.25	0.25	-90	0	0	2
Excitation case 4	0.2	0.25	0	0.05	-90	1
Torque excitation	Z=4					
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.45	0.36	-90	0.06	-90	1
Excitation case 2	0.9375	0	-90	0.0625	-90	1
Excitation case 3	0.25	0.251	-90	0	0	2
Excitation case 4	0.2	0.25	0	0.05	-90	1
Torque excitation	Z=5					
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.45	0.36	-90	0.06	-90	1
Excitation case 2	1.19	0.17	-90	0.02	-90	1
Excitation case 3	0,3	0.25	-90	0.048	-90	2
Excitation case 4	0,2	0.25	0	0.05	-90	1
Torque excitation	Z=6					
	C_{q0}	C_{q1}	α_1	C_{q2}	α_2	E_0
Excitation case 1	0.45	0.375	-90	0.05	-90	1
Excitation case 2	1.435	0.1	-90	0	0	1
Excitation case 3	0.3	0.25	-90	0.048	-90	2
Excitation case 4	0.2	0.25	0	0.05	-90	1

Torsional vibration responses are to be calculated for all excitation cases.

The results of the relevant excitation cases at the most critical rotational speeds are to be used in the following way:

The highest response torque (between the various lumped masses in the system) is in the following referred to as peak torque Q_{peak}.

The highest torque amplitude during a sequence of impacts is to be determined as half of the range from max to min torque and is referred to as Q_{Amax}.

An illustration of Q_{Amax} is given in Fig 3.5.6.3.2. It can be determined by

$$Q_{Amax} = \left(\frac{\max(Q_r(time)) - \min(Q_r(time))}{2}\right) \quad \text{[kNm]}$$

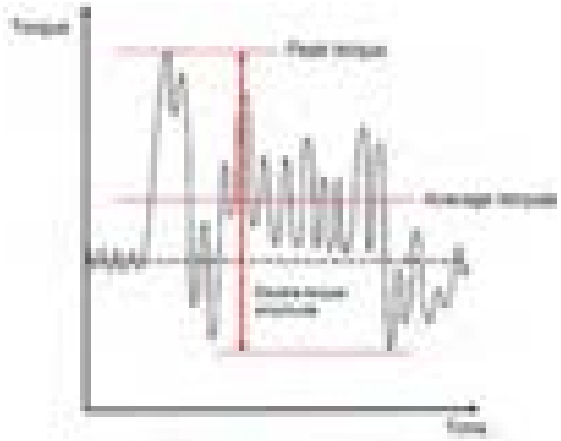


Fig 3.5.6.3.2 : Interpretation of different torques in a measured curve, as example

3.5.6.4 Design torque along shaft line

a) If there is no relevant first order propeller torsional resonance in the range 20% (of n_n) above and 20% below the maximum operating speed in bollard condition (see Table 8), the following estimation of the maximum response torque can be used to calculate the design torque along the propeller shaft line.

$$Q_r = Q_{e\max} + Q_{vib} + Q_{\max} \cdot \frac{I}{I_t} \quad \text{[kNm]}$$

The above equation is to be applied for directly coupled two stroke Diesel engines without flexible coupling.

For all other plants:

$$Q_r = Q_{e\max} + Q_{\max} \cdot \frac{I}{I_t} \quad \text{[kNm]}$$

where:
- I = equivalent mass moment of inertia of all parts on engine side of component under consideration and
- I_t is equivalent mass moment of inertia of the whole propulsion system.

All the torques and the inertia moments are to be reduced to the rotation speed of the component being examined.

If the maximum torque, Q_{emax}, is not known, it is to be taken as follows:

Propeller type	Q_{emax}
Propellers driven by electric motor	Q_{motor}
CP propellers not driven by electric motor	Q_n
FP propellers driven by turbine	Q_n
FP propellers driven by diesel engine	0.75 Q_n

Here Q_{motor} is the electric motor peak torque.

b) If there is a first blade order torsional resonance in the range 20% (of n_n) above and 20% below the maximum operating speed(bollard condition), the design torque (Q_r) of the shaft component is to be determined by means of a dynamic torsional vibration analysis of the entire propulsion line in the time domain or alternatively in the frequency domain. It is then assumed that the plant is sufficiently designed to avoid harmful operation in barred speed range.

3.5.7 Guideline for torsional vibration calculation

The aim of torsional vibration calculations is to estimate the torsional loads for individual shaft line components over the life time in order to determine scantlings for safe operation. The model can be taken from the normal lumped mass elastic torsional vibration model (frequency domain) including the damping. Standard harmonics may be used to consider the gas forces. The engine torque - speed curve of the actual plant is to be applied.

For time domain analysis the model should include the ice excitation at propeller, the mean torques provided by the prime mover and the hydrodynamic mean torque produced by the propeller as well as any other relevant excitations. The calculations should cover the variation of phase between the ice excitation and prime mover excitation. This is extremely relevant for propulsion lines with direct driven combustion engines.

For frequency domain calculations the load should be estimated as a Fourier component analysis of the continuous sequence of half sine load peaks. The first and second order blade components should be used for excitation. The calculation should cover the whole relevant shaft speed range. The analysis of the responses at the relevant torsional vibration resonances may be performed for open water (without ice excitation) and ice excitation separately. The resulting maximum torque can be obtained for directly coupled plants by the following superposition:

$$Q_{peak} = Q_{emax} + Q_{opw} + Q_{ice} \quad \text{[kNm]}$$

where:

Q_{emax} is the maximum engine torque at considered rotational speed

Q_{opw} is the maximum open water response of engine excitation at considered shaft speed and determined by frequency domain analysis

Q_{ice} is the calculated torque using frequency domain analysis for the relevant shaft speeds, ice excitation cases 1-4, resulting in the maximum response torque due to ice excitation

3.6 Design

3.6.1 Design Principle

The propulsion line is to be designed according to the pyramid strength principle in terms of its strength. This means that the loss of the propeller blade is not to cause any significant damage to other propeller shaft line components.

The propulsion line components are to withstand maximum and fatigue operational loads with the relevant safety margin. The loads do not need to be considered for shaft alignment or other calculations of normal operational conditions.

3.6.2 Fatigue design in general

The design loads are to be based on the ice excitation and where necessary (shafting) dynamic analysis, described as a sequence of blade impacts (3.5.6.3.1). The shaft response torque is to be determined according 3.5.6.4.

The propulsion line components are to be designed so as to prevent accumulated fatigue failure when considering the relevant loads using the linear elastic Miner's rule as defined below.

$$D = \frac{n_1}{N_1} + \frac{n_2}{N_2} + \cdots + \frac{n_k}{N_k} \leq 1$$

Or

$$D = \sum_{J=1}^{j=k} \frac{n_j}{N_j} \leq 1$$

Where:

k is the number of stress levels

$N_{1...k}$ is the number of load cycles to failure of the individual stress level class

$n_{1...k}$ is the accumulated number of load cycles of the case under consideration, per class

D Miners damage sum

Guidance:

The stress distribution should be divided into a frequency load spectrum having minimum 10 stress blocks (every 10 % of the load). Calculation with 5 stress blocks has been found to be too conservative. The maximum allowable load is limited by $\sigma_{ref\,2}$ for propeller blades and yield strength for all other components. The load distribution (spectrum) should be in accordance with the Weibull distribution.

3.6.3 Propeller blades

3.6.3.1 Calculation of blade stresses due to static loads

The blade stresses (equivalent and principal stresses) are to be calculated for the design loads given in 3.5.3. Finite element analysis (FEA) is to be used for stress analysis as part of the final approval for all propeller blades. The von Mises stresses, taken as σ_{st}, is to comply with equation in 3.6.3.2. Alternatively, the following simplified equation can be used in estimating the blade stresses for all propellers in the root area (r/R < 0.5) for final approval

$$\sigma_{st} = C_1 \frac{M_{BL}}{100 \cdot ct^2} \quad \text{[MPa]}$$

Where:

constant C_1 is the ratio of actual stress to stress obtained with beam equation. If the actual value is not available, C_1 is to be taken as 1.6.

- $M_{BL} = (0.75 - r/R) \cdot R \cdot F$, for relative radius r/R < 0.5

- F is the maximum of F_b and F_f, whichever is greater.

3.6.3.2 Acceptability criterion for static loads

The following criterion for calculated blade stresses shall be fulfilled:

$$\frac{\sigma_{ref2}}{\sigma_{st}} \geq 1.3$$

where:

σ_{st} = calculated stress for the design loads. If FE analysis is used in estimating the stresses, von Mises stresses are to be used.

3.6.3.3 Fatigue design of propeller blade

3.6.3.3.1 General

For materials with a two slope S-N curve (Fig 3.6.3.3.1 a) the fatigue calculations defined in this chapter are not required if the following criterion is fulfilled.

$$\sigma_{\exp} \geq B_1 \cdot \sigma_{ref\,2}{}^{B_2} \cdot \log(N_{ice})^{B_3}$$

where:

B1, B2 and B3 are coefficients for open and ducted propellers, given in the Table 3.6.3.3.1 below.

Table 3.6.3.3.1 : Coefficients to check a dispense from fatigue calculation

	Open propeller	**Ducted propeller**
B1	0.00328	0.00223
B2	1.0076	1.0071
B3	2.101	2.471

Where the above criterion is not fulfilled the fatigue requirements defined below is to be applied:
The fatigue design of the propeller blade is based on an estimated load distribution for the service life of the ship and the S-N curve for the blade material. An equivalent stress σ_{fat} that produces the same fatigue damage as the expected load distribution is to be calculated according to Miner's rule and the acceptability criterion for fatigue should be fulfilled as given in this section. The equivalent stress is normalised for 100 million cycles.
The blade stresses at various selected load levels for fatigue analysis are to be taken proportional to the stresses calculated for maximum loads given in section 3.5.3.
The peak principal stresses σ_f and σ_b are determined from F_f and F_b using FEA. The peak stress range $\Delta\sigma_{max}$ and the maximum stress amplitude σ_{Amax} are determined on the basis of load cases 1 and 3, 2 and 4.

$$\Delta\sigma_{\max} = 2 \cdot \sigma_{A\max} = \left|(\sigma_{ice})_{f\max}\right| + \left|(\sigma_{ice})_{b\max}\right|$$

The load spectrum for backward loads is normally expected to have a lower number of cycles than the load spectrum for forward loads. Taking this into account in a fatigue analysis introduces complications that are not justified considering all uncertainties involved.

For the calculation of equivalent stress two types of S-N curves are available.

Two slope S-N curve (slopes 4.5 and 10), see Fig 3.6.3.3.1 a) .

One slope S-N curve (the slope can be chosen), see Figure 3.6.3.3.1 b).

The type of the S-N-curve is to be selected to correspond with the material properties of the blade. If the S-N-curve is not known the two slope S-N curve is to be used.

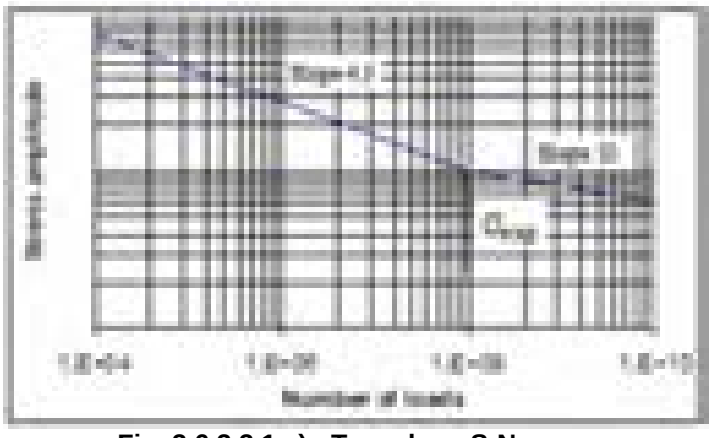


Fig 3.6.3.3.1 a) : Two-slope S-N curve

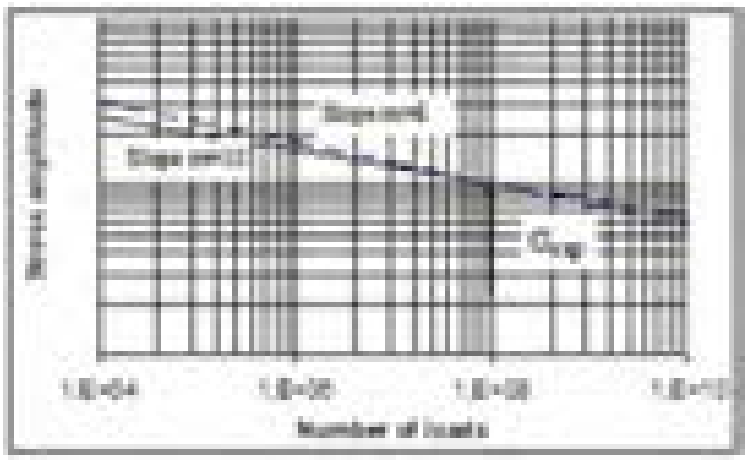


Fig 3.6.3.3.1 b) : Constant-slope S-N curve.

3.6.3.3.2 Equivalent fatigue stress

Note: A more general method of determining the equivalent fatigue stress of propeller blades is described in 3.6.5, where the principal stresses are considered according to 3.5.3 using the Miner's rule. For a total number of load blocks $n_{bl} > 100$, both methods deliver the same result. Therefore, they are regarded as equivalent. The equivalent fatigue stress for 10^8 cycles which produces the same fatigue damage as the load distribution is:

$$\sigma_{fat} = \rho \cdot (\sigma_{ice})_{max}$$

Where:

$$(\sigma_{ice})_{\max} = 0.5 \cdot \left((\sigma_{ice})_{f\max} - (\sigma_{ice})_{b\max}\right)$$

$(\sigma_{ice})_{\max}$ = mean value of the principal stress amplitudes resulting from design forward and backward blade forces at the location being studied.

$(\sigma_{ice})_{f\max}$ = principal stress resulting from forward load

$(\sigma_{ice})_{b\max}$ = principal stress resulting from backward load

In the calculation of $(\sigma_{ice})_{max}$, case 1 and case 3 or case 2 and case 4 are considered as pairs for $(\sigma_{ice})_{fmax}$, and $(\sigma_{ice})_{bmax}$ calculations. Case 5 is excluded from the fatigue analysis.

Calculation of parameter ρ for two-slope S-N curve

The error of the following method to determine the parameter ρ is sufficiently small, if the number of load cycles N_{ice} is in the range

$$5 \cdot 10^6 \le N_{ice} \le 10^8$$

The parameter ρ relates the maximum ice load to the distribution of ice loads according to the regression formula:

$$\rho = C_1 \cdot (\sigma_{ice})_{\max}^{C_2} \cdot \sigma_{fl}^{C_3} \cdot \log(N_{ice})^{C_4}$$

Where:

$\sigma_{fl} = \gamma_{\varepsilon 1} \cdot \gamma_{\varepsilon 2} \cdot \gamma_v \cdot \gamma_m \cdot \sigma_{\exp}$ is the blade material fatigue strength at 10^8 load cycles, see 3.6.3.3.3

The coefficients C_1, C_2, C_3 and C_4 are given in Table 3.6.3.3.2 a).

Table 3.6.3.3.2 a) : Coefficients to evaluate material fatigue strength

	Open propeller	**Ducted propeller**
C_1	0.000747	0.000534
C_2	0.0645	0.0533
C_3	-0.0565	-0.0459
C_4	2.22	2.584

Calculation of parameter ρ for constant-slope S-N curve. For materials with a constant-slope S-N curve, see Fig 3.6.3.3.1 b), - the factor ρ is to be calculated from the following formula:

$$\rho = \left(G \frac{N_{ice}}{N_R} \right)^{1/m} (\ln(N_{ice}))^{-1/k}$$

where:

k = shape parameter of the Weibull distribution

k = 1.0 for ducted propellers and

k = 0.75 for open propellers

N_R= reference number of load cycles (=10^8)

Values for the parameter G are given in Table 3.6.3.3.2 b) below. Linear interpolation may be used to calculate the value of G for m/k ratios other than those given in the Table 3.6.3.3.2 b).

Table 3.6.3.3.2 b) : Value for the parameter G for different m/k ratios

m/k	G	m/k	G	m/k	G
3	6	5.5	287.9	8	40320
3.5	11.6	6	720	8.5	119292
4	24	6.5	1871	9	362880
4.5	52.3	7	5040	9.5	1.133×10^6
5	120	7.5	14034	10	3.623×10^6

3.6.3.3.3 Acceptability criterion for fatigue

The equivalent fatigue stress σ_{fat} at all locations on the blade is to fulfil the following acceptability criterion:

$$\frac{\sigma_{fl}}{\sigma_{fat}} \geq 1.5$$

Where:

$$\sigma_{fl} = \gamma_{\varepsilon 1} \cdot \gamma_{\varepsilon 2} \cdot \gamma_v \cdot \gamma_m \cdot \sigma_{\exp}$$

$\gamma_{\varepsilon 1}$ = reduction factor due to scatter (equal to one standard deviation)

$\gamma_{\varepsilon 2}$ = reduction factor for test specimen size effect

γ_v = reduction factor for variable amplitude loading.

γ_m = reduction factor for mean stress.

$\sigma_{\exp}$ = mean fatigue strength of the blade material at 10^8 cycles to failure in seawater

$\sigma_{\exp}$ in Table 15 has been defined from the results of constant amplitude loading fatigue tests at 10^7 load cycles and 50% survival probability and has been extended to 10^8 load cycles.

Fatigue strength values and correction factors other than those given in Table 3.6.3.3.3 may be used, provided the values are determined under conditions approved by IRS.

The S-N curve characteristics are based on two slopes, the first slope 4.5 is from 1000 to 10^8 load cycles; the second slope 10 is above 10^8 load cycles. The maximum allowable stress for one or low number of cycles is limited to σ_{ref2}/S, with S=1.3 for static loads. The fatigue strength σ_{fat} is the fatigue limit at 100 million load cycles.

The geometrical size factor ($\gamma_{\varepsilon 2}$) is:

$$\gamma_{\varepsilon 2} = 1 - a \cdot \ln\left(\frac{t}{0.025}\right)$$

The following values is to be used for the reduction factors if actual values are not available: $\gamma_{\varepsilon 1}$ = 0.85, γ_v = 0.75, and γ_m = 0.75.

Table 3.6.3.3.3 : Mean fatigue strength σ_{exp} for different material types at 10^8 load cycles and stress ratio R = -1 with a survival probability of 50%.

Mean fatigue strength σ_{exp} for different material types at 10^8 load cycles			
Bronze and brass (a=0.10)		Stainless steel (a=0.05)	
Mn-Bronze, CU1 (high tensile brass)	84 MPa	Ferritic (12Cr 1Ni)	144[1] MPa
Mn-Ni-Bronze, CU2 (high tensile brass)	84 MPa	Martensitic (13Cr 4Ni/13Cr 6Ni)	156 MPa
Ni-Al-Bronze, CU3	120 MPa	Martensitic (16Cr 5Ni)	168 MPa
Mn-Al-Bronze, CU4	113 MPa	Austenitic (19Cr 10Ni)	132 MPa

Note:
1. This value may be used, provided a perfect galvanic protection is active. Otherwise a reduction of about 30 MPa is to be applied.

3.6.4 Blade bolts, propeller hub and CP mechanism

3.6.4.1 General

The blade bolts, CP mechanism, propeller boss and the fitting of the propeller to the propeller shaft is to be designed to withstand the maximum static and fatigue design loads (as applicable), as defined in 3.5.3 and 3.6.3. The safety factor S against yielding due to static loads and against fatigue is to be greater than 1.5, if not stated otherwise. The safety factor S for loads, resulting from propeller blade failure as defined in 3.5.4 is to be greater than 1.0 against yielding.

Provided that calculated stresses duly considering local stress, concentrations are less than yield strength, or maximum of 70% of σ_u of respective materials, detailed fatigue analysis is not required. In all other cases components are to be analysed for cumulative fatigue. An approach similar to that used for shafting assessment may be applied (3.6.5).

3.6.4.2 Blade bolts

Blade bolts are to withstand the following bending moment considered around a tangent on bolt pitch circle, or any other relevant axis for non-circular joints, parallel to considered root section:

$$M_{bolt} = S \cdot F_{ex}(0.8\frac{D}{2} - r_{bolt}) \quad \text{[kNm]}$$

Where:
r_{bolt} = radius to the bolt plane [m]
S = 1.0 safety factor

Blade bolt pre-tension is to be sufficient to avoid separation between mating surfaces when the maximum forward and backward ice loads defined in 3.5.3 (open and ducted propellers respectively) are applied. For conventional arrangements, the following formula may be applied:

$$\mathbf{d_{bb} = 41 \cdot \sqrt[2]{\frac{F_{ex} \cdot (0.8 \cdot D - d) \cdot S \cdot \alpha}{\sigma_{0.2} \cdot z_{bb} \cdot PCD}}} \quad \text{[mm]}$$

where:
α = 1.6 torque guided tightening
= 1.3 elongation guided
= 1.2 angle guided
= 1.1 elongated by other additional means other factors may be used, if evidence is demonstrated

d_{bb}- effective diameter of blade bolt in way of thread [mm]

z_{bb}- number of blade bolts

S= 1.0 safety factor

3.6.4.3 CP mechanism

Separate means, e.g., dowel pins, are to be provided in order to withstand the spindle torque resulting from blade failure Q_{sex} (3.5.4.2) or ice interaction Q_{smax} (3.5.3.7), whichever is greater. Other components of the CP mechanism is not to be damaged by the maximum spindle torques (Q_{smax}, Q_{sex}). One third of the spindle torque is assumed to be consumed by friction, if not otherwise documented trough further analysis.

The diameter of fitted pins d_{fp} between the blade and blade carrier can be calculated using the formula:

$$d_{fp} = 66 \cdot \sqrt{\frac{(Q_s - Q_{fr})}{PCD \cdot z_{pin} \cdot \sigma_{0.2}}} \quad \text{[mm]}$$

where:

Q_s= max(S•Q_{smax}; S•Q_{sex}) [kNm]

S = 1.3 for Q_{smax} and

= 1.0 for Q_{sex}

Q_{fr}= friction between connected surfaces= 0.33·Q_s

IRS may approve alternative Q_{fr} calculation according to reaction forces due to F_{ex}, or F_f, F_b whichever is relevant, utilising a friction coefficient = 0.15.

The stress in the actuating pin can be estimated by

$$\sigma_{vMises} := \sqrt{\left[\frac{\left(F \cdot \frac{h_{pin}}{2}\right)}{\frac{\pi \cdot d_{pin}^{3}}{32}}\right]^2 + 3 \cdot \left(\frac{F}{\frac{\pi}{4} \cdot d_{pin}^{2}}\right)^2}$$

[MPa]

Where:

$$F = \frac{(Q_s - Q_{fr})}{l_m} \quad \text{[kN]}$$

l_m = distance pitching centre of blade – axis of pin [m]
h_{pin} = height of actuating pin [mm]
d_{pin} = diameter of actuating pin [mm]
Q_{fr} = friction torque in blade bearings acting on the blade palm and caused by the reaction forces due to F_{ex}, or F_f, F_b whichever is relevant; taken to one third of spindle torque Q_s

The blade failure spindle torque Q_{sex} is not to lead to any consequential damage. Fatigue strength is to be considered for parts transmitting the spindle torque from the blade to a servo system considering the ice spindle torque acting on one blade. The maximum amplitude Q_{samax} is defined as:

$$Q_{samax} = \frac{Q_{sb} + Q_{sf}}{2} \quad \text{[kNm]}$$

where:

Q_{sb} = spindle torque due to IF_bI [kNm]
Q_{sf} = spindle torque due to IF_fI [kNm]

3.6.4.4 Servo pressure

The design pressure for the servo system is to be taken as the pressure caused by Q_{smax} or, Q_{sex} when not protected by relief valves on the hydraulic actuator side, reduced by relevant friction losses in bearings caused by the respective ice loads. The design pressure is, in any case, not to be less than relief valve set pressure.

3.6.5 Propulsion line components

The ultimate load resulting from total blade failure F_{ex} as defined in 3.5.4, is to consist of combined axial and bending load components, wherever this is significant. The minimum safety factor against yielding is to be 1.0 for all shaft line components.

The shafts and shafting components, such as bearings, couplings and flanges are to be designed to withstand the operational propeller/ice interaction loads as given in 3.5. The given loads are not intended to be used for shaft alignment calculation. Cumulative fatigue calculations shall be conducted according to the Miner's rule. A fatigue calculation is not necessary, if the maximum stress is below fatigue strength at 10^8 load cycles.

The torque and thrust amplitude distribution (spectrum) in the propulsion line is to be taken as (because Weibull exponent k = 1):

$$Q_A(N) = Q_{Amax} \cdot \left\{1 - \frac{\log(N)}{\log(Z \cdot N_{ice})}\right\}$$

This is illustrated by the example in the Fig 3.6.5 a).

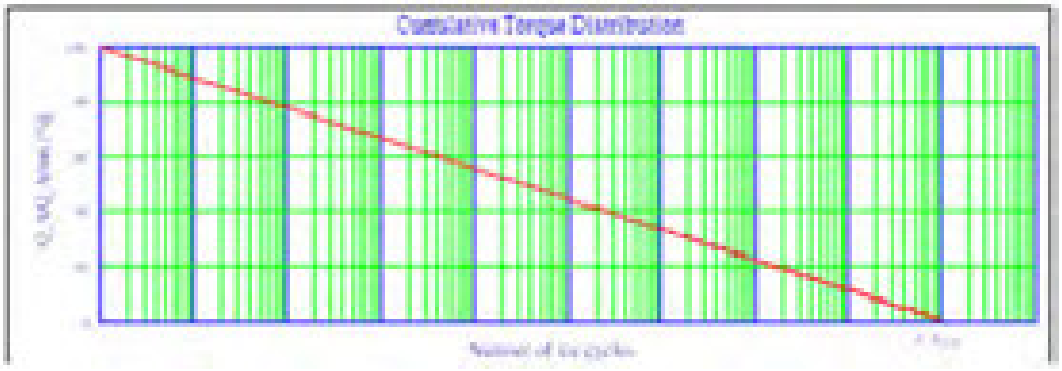


Fig 3.6.5 a) : Cumulative torque distribution

The number of load cycles in the load spectrum is defined as $Z \cdot N_{ice}$. The Weibull exponent should be considered as k = 1.0 for both open and ducted propeller torque and bending forces. The load distribution is an accumulated load spectrum, and the load spectrum should be divided into a minimum of ten load blocks when using the Miner summation method. The load spectrum used counts the number of cycles for 100% load to be the number of cycles above the next step, e.g., 90 % load. This ensures that the calculation is on the conservative side. Consequently, the fewer stress blocks used the more conservative the calculated safety margin.

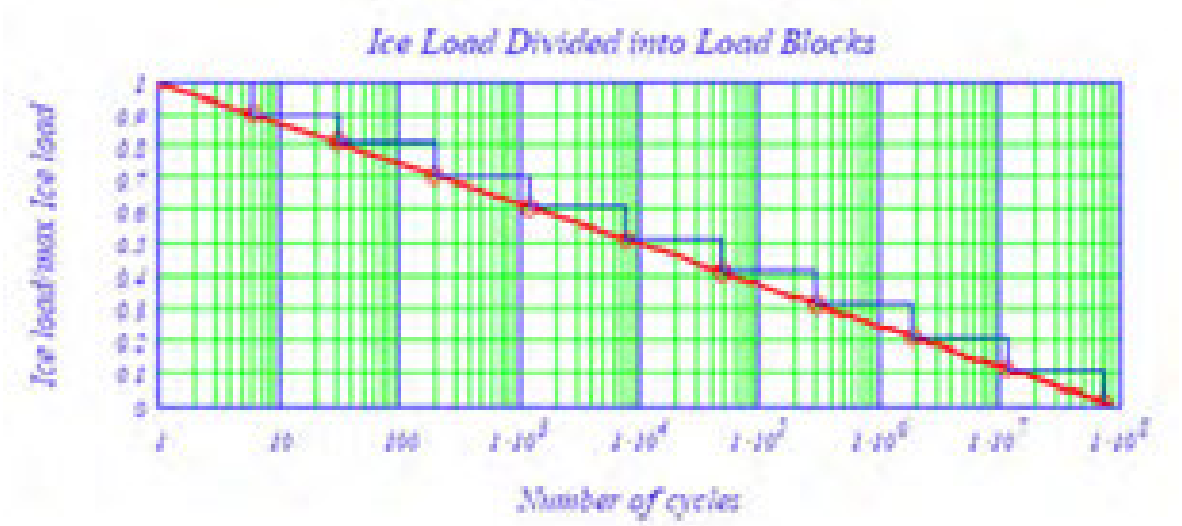


Figure 3.6.5 b): Example of ice load distribution (spectrum) for the shafting (k = 1).

The load spectrum is divided into n_{bl}-number of load blocks for the Miner summation method. The following formula can be used for calculation of the number of cycles for each load block.

$$n_i = N_{ice}^{1-\left(1-\frac{i}{n_{bl}}\right)^k} - \sum_{i=1}^{i} n_{i-1}$$

Where:

i = single load block i and n_{bl} is the number of load blocks

3.6.5.1 Propeller fitting to the shaft

3.6.5.1.1 Keyless cone mounting

The friction capacity (at 0° C) is to be at least S = 2.0 times the highest peak torque Q_{peak} as determined in 3.5.6 without exceeding the permissible hub stresses.

The necessary surface pressure $p_{0°C}$ can be determined as:

$$p_{0°C} = \frac{2 \cdot S \cdot Q_{peak}}{\pi \cdot \mu \cdot D_S^2 \cdot L \cdot 10^3} \quad \text{[MPa]}$$

where:
μ = 0.15 for steel-steel,
= 0.13 for steel-bronze
D_S = is the shrinkage diameter at the mid-length of the taper [m]

L = is the effective length of taper [m]

Above friction coefficients may be increased by 0.04 if glycerine is used in wet mounting.

3.6.5.1.2 Key mounting

Key mounting is not permitted.

3.6.5.1.3 Flange mounting

The flange thickness is to be at least 25% of the required aft end shaft diameter (*Pt.4, Ch.4, Sec.6, 6.13*). Any additional stress raisers such as recesses for bolt heads is not to interfere with the flange fillet unless the flange thickness is increased correspondingly. The flange fillet radius is to be at least 10% of the required shaft diameter. The diameter of shear pins are to be calculated according to the following equation:

$$d_{pin} = 66 \cdot \sqrt[2]{\frac{Q_{peak} \cdot S}{PCD \cdot z_{pin} \cdot \sigma_{0.2}}} \quad \text{[mm]}$$

Where

z_{pin} - number of shear pins
S = 1.3 safety factor

The bolts are to be designed so that the blade failure load F_{ex} (3.5.4) in backward direction does not cause yielding of the bolts. The following equation is to be applied:

$$d_b = 41 \cdot \sqrt{\frac{F_{ex} \cdot (0.8 \cdot D / PCD + 1) \cdot \alpha}{\sigma_{0.2} \cdot z_b}}$$

[mm]

where:
α = 1.6 torque guided tightening
= 1.3 elongation guided
= 1.2 angle guided
= 1.1 elongated by other additional means
other factors may be used, if evidence is demonstrated
d_b - diameter flange bolt [mm]
z_b - number of flange bolts

3.6.5.2 Propeller shaft

The propeller shaft is to be designed to fulfil the following:

3.6.5.2.1 The blade failure load F_{ex} (3.5.4) applied parallel to the shaft (forward or backwards) is not to cause yielding. The bending moment need not to be combined with any other loads. The diameter d_p in way of the aft stern tube bearing is not to be less than:

$$d_p = 160 \cdot \sqrt[3]{\frac{F_{ex} \cdot D}{\sigma_{0.2} \cdot \left(1 - \frac{d_i^4}{d_p^4}\right)}} \quad \text{[mm]}$$

where:
d_p = propeller shaft diameter [mm]
d_i = propeller shaft inner diameter [mm]

Forward from the aft stern tube bearing the shaft diameter may be reduced based on direct calculation of the actual bending moment, or by the assumption that the bending moment caused by F_{ex} is linearly reduced to 25% at the next bearing and in front of this linearly to zero at third bearing. Bending due to maximum blade forces F_b and F_f have been disregarded since the resulting stress levels are much lower than the stresses caused by the blade failure load.

3.6.5.2.2 The stresses due to the peak torque Q_{peak} are to have a minimum safety factor of S=1.5 against yielding in plain sections and S=1.0 in way of stress concentrations in order to avoid bent shafts.

Minimum diameter of:

plain shaft:

$$d_p = 210 \cdot \sqrt[3]{\frac{Q_{peak} \cdot S}{\sigma_{0.2} \cdot \left(1 - \frac{d_i^4}{d^4}\right)}} \quad \text{[mm]}$$

notched shaft:

$$d_p = 210 \cdot \sqrt[3]{\frac{Q_{peak} \cdot S \cdot \alpha_t}{\sigma_{0.2} \cdot \left(1 - \frac{d_i^4}{d^4}\right)}} \quad \text{[mm]}$$

where:
α_t = local stress concentration factor in torsion.
Notched shaft diameter is, in any case, not to be less than the required plain shaft diameter.

3.6.5.2.3 The torque amplitudes (3.5.6.4) with the corresponding number of load cycles are to be used in an accumulated fatigue evaluation where the safety factor is S_{fat}=1.5. If the plant has high engine excited torsional vibrations (e.g. direct coupled 2-stroke engines), this is to be considered.

3.6.5.2.4 The fatigue strengths σ_F and τ_F (3 million cycles) of shaft materials may be assessed on the basis of the material's yield or 0.2% proof strength as:

$$\sigma_F = 0.436 \cdot \sigma_{0.2} + 77 = \tau_F \cdot \sqrt{3} \quad \text{[MPa]}$$

This is valid for small polished specimens (no notch) and reversed stresses. The high cycle fatigue (HCF) is to be assessed based on the above fatigue strengths, notch factors (i.e. geometrical stress concentration factors and notch sensitivity), size factors, mean stress influence and the required safety factor of 1.6 at 3 million cycles increasing to 1.8 at 10^9 cycles. The low cycle fatigue (LCF) representing 10^4 cycles is to be based on the smaller value of yield or 0.7 of tensile strength/√3. The criterion utilises a safety factor of 1.25. The LCF and HCF as given above represent the upper and lower knees in a stress-cycle diagram. Since the required safety factors are included in these values, a Miner sum of unity is acceptable.

3.6.5.3 Intermediate shafts

The intermediate shafts are to be designed to fulfil 3.6.5.2.2 to 3.6.5.2.4.

3.6.5.4 Shaft connections

3.6.5.4.1 Shrink fit couplings (keyless)

See 3.6.5.1.1. A safety factor of S= 1.8 shall be applied.

3.6.5.4.2 Key mounting

Key mounting is not permitted.

3.6.5.4.3 Flange mounting

The flange thickness is to be at least 20% of the required shaft diameter (*Pt.4, Ch.4, Sec.6, 6.13*).

Any additional stress raisers such as recesses for bolt heads is not to interfere with the flange fillet unless the flange thickness is increased correspondingly.

The flange fillet radius is to be at least 8% of the shaft diameter (*Pt.4, Ch.4, Sec.6, 6.13*).

The diameter of ream fitted (light press fit) bolts are to be chosen so that the peak torque is transmitted with a safety factor of 1.9. This accounts for a prestress. Pins are to transmit the peak torque with a safety factor of 1.5 against yielding (see equation for d_{pin} in 3.6.5.1.3). The bolts are to be designed so that the blade failure load (3.5.4) in backward direction does not cause yielding.

3.6.5.4.4 Splined shaft connections

Splined shaft connections can be applied where no axial or bending loads occur. A safety factor of S = 1.5 against allowable contact and shear stress resulting from Q_{peak} is to be applied.

3.6.5.4.5 Gear transmissions

3.6.5.4.6 Shafts

Shafts in gear transmissions are to meet the same safety level as intermediate shafts, but where relevant, bending stresses and torsional stresses are to be combined (e.g., by von Mises for static loads). Maximum permissible deflection in order to maintain sufficient tooth contact pattern to be considered for the relevant parts of the gear shafts.

3.6.5.4.7 Gearing

The gearing is to fulfil following three acceptance criteria:

- Tooth root stresses
- Pitting of flanks
- Scuffing

In addition to above 3 criteria subsurface fatigue may need to be considered.

Common for all criteria is the influence of load distribution over the face width. All relevant parameters are to be considered, such as elastic deflections (of mesh, shafts and gear bodies), accuracy tolerances, helix modifications, and working positions in bearings (especially for multiple input single output gears).

The load spectrum (see 3.6.5) may be applied in such a way that the numbers of load cycles for the output wheel are multiplied by a factor of (number of pinions on the wheel / number of propeller blades Z). For pinions and wheels operating at higher speeds the numbers of load cycles are found by multiplication with the gear ratios. The peak torque (Q_{peak}) is also to be considered during calculations.

Cylindrical gears can be assessed on the basis of the international standard ISO 6336 series (i.e. ISO 6336-1:2019, ISO 6336-2:2019, ISO 6336-3:2019, ISO 6336-4:2019, ISO 6336-5:2016 and ISO 6336-6:2019), provided that "method B" is used. Standards within the classification societies can also be applied provided that they are considered equivalent to the above-mentioned ISO 6336.

For Bevel Gears the methods or standards used or acknowledged by the classification society can be applied provided that they are properly calibrated.

Tooth root safety is to be assessed against the peak torque, torque amplitudes (with the pertinent average torque) as well as the ordinary loads (open water free running) by means of accumulated fatigue analyses. The resulting factor of safety is to be at least 1.5. (Ref ISO 6336 Pt 1, 3 and 6 and IRS Classification Note "*Marine Gears - Calculation of load capacity of involute parallel axis spur and helical gears*")

The safety against pitting is to be assessed in the same way as tooth root stresses, but with a minimum resulting safety factor of 1.2. (Ref ISO 6336-1:2019, ISO 6336-2:2019 and ISO 6336-6:2019 as well as IRS Classification Note "*Marine Gears - Calculation of load capacity of involute parallel axis spur and helical gears*").

The scuffing safety (flash temperature method – ref. ISO/TR 13989-1:2000 and ISO/TR 13989-2:2000) based on the peak torque is to be at least 1.2 when the FZG class of the oil is assumed one stage below specification.

The safety against subsurface fatigue of flanks for surface hardened gears (oblique fracture from active flank to opposite root) is to be assessed. (It is to be noted that high overloads can initiate subsurface fatigue cracks that may lead to a premature failure. In lieu of analyses UT inspection intervals may be used.)

3.6.5.4.8 Bearings

See section 3.6.5.8.

3.6.5.4.9 Gear wheel shaft connections

The torque capacity is to be at least 1.8 times the highest peak torque Q_{peak} (at considered rotational speed) as determined in 3.6.5 without exceeding the permissible hub stresses of 80% yield.

3.6.5.5 Clutches

Clutches are to have a static friction torque of at least 1.3 times the peak torque Q_{peak} and dynamic friction torque 2/3 of the static. Emergency operation of clutch after failure of e.g. operating pressure is to be made possible within reasonably short time. If this is arranged by bolts, it is to be on the engine side of the clutch in order to ensure access to all bolts by turning the engine.

3.6.5.6 Elastic couplings

There is to be a separation margin of at least 20% between the peak torque and the torque where any twist limitation is reached.

$$Q_{peak} < 0.8 \cdot T_{kmax} \ (N = 1) \quad \text{[kNM]}$$

There is to be a separation margin of at least 20% between the maximum response torque Q_{peak} (see Fig 3.5.6.3.2) and the torque where any mechanical twist limitation and/or the permissible maximum torque of the elastic coupling, valid for at least a single load cycle (N=1), is reached.

A sufficient fatigue strength is to be demonstrated at design torque level Q_r(N=x) and Q_A(N=x). This may be demonstrated by interpolation in a Weibull torque distribution (similar to Figure 7):

$$\frac{\boldsymbol{Q_r\,(N = x)}}{\boldsymbol{Q_r\,(N = 1)}} = \mathbf{1} - \frac{\boldsymbol{\log(x)}}{\boldsymbol{log(Z * N_{ICE})}}$$

respectively

$$\frac{\boldsymbol{Q_A(N = x)}}{\boldsymbol{Q_A(N = 1)}} = \mathbf{1} - \frac{\boldsymbol{\log(x)}}{\boldsymbol{log(Z * N_{ICE})}}$$

Where Q_r (N=1) corresponds to Q_{peak} and Q_A (N=1) to Q_{Amax}.

$Q_r(N=5E4) \cdot S < T_{Kmax}\ (N=5E4)$ [kNm]
$Q_r(N=1E6) \cdot S < T_{KV}$ [kNm]
$Q_A(N=5E4) \cdot S < \Delta T_{max}\ (N=5E4)$ [kNm]

S is the general safety factor for fatigue, equal to 1.5.
See illustration in below Fig 3.6.5.6 a), Fig 3.6.5.6 b) and Fig 3.6.5.6 c) .

The torque amplitude (or range Δ) is not to lead to fatigue cracking, i.e. exceeding the permissible vibratory torque. The permissible torque may be determined by interpolation in a

Weibull torque distribution where T_{Kmax1} respectively ΔT_{Kmax} refer to 50000 cycles and T_{KV} refer to 10^6 cycles. See illustration in below Fig 3.6.5.6 a), Fig 3.6.5.6 b) and Fig 3.6.5.6 c) .

$T_{kmax1} \geq Q_r \; at \; 5 \cdot 10^4$ load cycles [kNm]

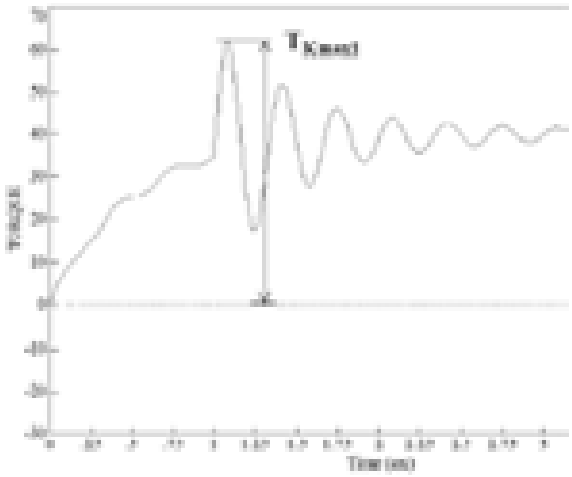


Fig 3.6.5.6 a)

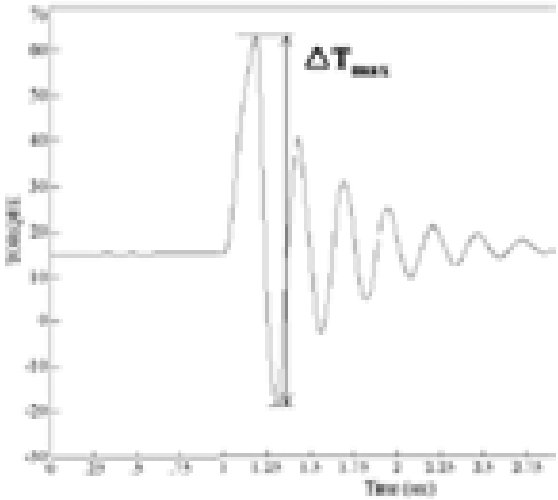


Fig 3.6.5.6 b)

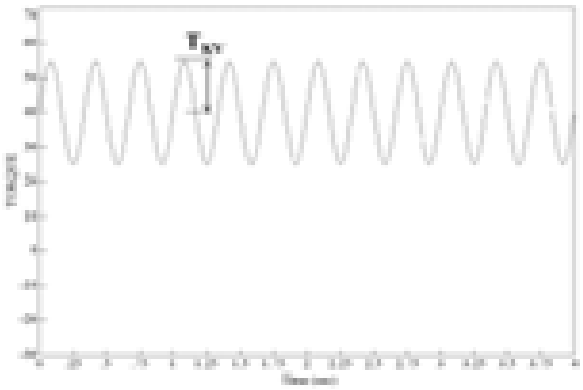


Fig 3.6.5.6 c)

3.6.5.7 Crankshafts

Special considerations apply for plants with large inertia (e.g. flywheel, tuning wheel or PTO) in the non-driving end front of the engine (opposite to main power take off).

3.6.5.8 Bearings

The aft stern tube bearing as well as the next shaft line bearing are to withstand F_{ex} as given in 3.5.4, in such a way that the ship can maintain operational capability. Rolling bearings are to have an L_{10a} lifetime of at least 40000 hours according to ISO 281:2007. Thrust bearings and their housings are to be designed to withstand with a safety factor S = 1.0 the maximum response thrust 3.5.5 and the axial force resulting from the blade failure load F_{ex} in 3.5.4. For the purpose of calculation, except for F_{ex}, the shafts are assumed to rotate at rated speed. For pulling propellers special consideration is to be given to loads from ice interaction on the propeller hub.

3.6.5.9 Seals

Seals are to prevent egress of pollutants and be suitable for the operating temperatures. Contingency plans for preventing the egress of pollutants under failure conditions are to be documented.

Seals installed are to be suitable for the intended application. The manufacturer is to provide service experience in similar applications and/or testing results for consideration.

3.6.6 Azimuthing main propulsors

In addition to the above requirements, special consideration is to be given to those loading cases which are extraordinary for propulsion units when compared with conventional propellers. The estimation of load cases are to reflect the way the thrusters are intended to operate on the specific ship. In this respect, for example, the loads caused by the impacts of ice blocks on the propeller hub of a pulling propeller is to be considered. Furthermore, loads resulting from the thrusters operating at an oblique angle to the flow is to be considered. The steering mechanism, the fitting of the unit, and the body of the thruster is to be designed to withstand the loss of a blade without damage. The loss of a blade is to be considered for the propeller blade orientation which causes the maximum load on the component being studied. Typically, top-down blade orientation places the maximum bending loads on the thruster body. Azimuth thrusters are to be designed for estimated loads caused by thruster body/ice interaction. The thruster body is to withstand the loads obtained when the maximum ice blocks, which are given in section 3.5.2, strike the thruster body when the ship is at a typical ice operating speed. In addition, the design situation in which an ice sheet glides along the ship's hull and presses against the thruster body should be considered. The thickness of the sheet should be taken as the thickness of the maximum ice block entering the propeller, as defined in section 3.5.2.

3.7 Prime Movers

3.7.1 Propulsion engines

Engines are to be capable of being started and running the propeller in bollard condition.

Propulsion plants with CP propeller are to be capable being operated even when the CP system is at full pitch as limited by mechanical stoppers.

3.7.2 Starting arrangements

The capacity of the air receivers are to be sufficient to provide, without recharging, not less than 12 consecutive starts of the propulsion engine, if this has to be reversed for going astern or 6 consecutive starts if the propulsion engine does not have to be reversed for going astern.

If the air receivers serve any other purposes than starting the propulsion engine, they are to have additional capacity sufficient for these purposes.

The capacity of the air compressors are to be sufficient for charging the air receivers from atmospheric to full pressure in one (1) hour, except for a ship with the ice class PC6 to PC1, if its propulsion engine has to be reversed for going astern, in which case the compressor is to be able to charge the receivers in half an hour.

3.7.3 Emergency power units

Provisions are to be made for heating arrangements to ensure ready starting from cold of the emergency power units at an ambient temperature applicable to the Polar Class of the ship.

Emergency power units are to be equipped with starting devices with a stored energy capability of at least three consecutive starts at the above-mentioned temperature. The source of stored energy is to be protected to preclude critical depletion by the automatic starting system, unless a second independent mean of starting is provided. A second source of energy is to be provided for an additional three starts within 30 min., unless manual starting can be demonstrated to be effective.

3.8 Equipment fastening loading accelerations

3.8.1 General

Essential equipment and supports are to be suitable for the accelerations as indicated in the following paragraphs. Accelerations are to be considered as acting independently.

3.8.2 Longitudinal Impact Accelerations, a_l

Maximum longitudinal impact acceleration at any point along the hull girder,

$$a_l = \left(F_{IB}/\Delta\right)\left\{\left[1.1\cdot\tan\left(\gamma+\phi\right)\right]+\left[7\cdot\frac{H}{L}\right]\right\}$$

[m/s^2]

3.8.3 Vertical acceleration, a_v

Combined vertical impact acceleration at any point along the hull girder,

$$a_v = 2.5\cdot\left(F_{IB}/\Delta\right)\cdot F_X \quad [\text{m/s}^2]$$

F_X = 1.3 at FP
= 0.2 at midships
= 0.4 at AP
= 1.3 at AP for vessels conducting ice breaking astern

Intermediate values to be interpolated linearly.

3.8.4 Transverse impact acceleration, a_t

Combined transverse impact acceleration at any point along hull girder,

$$a_t = 3F_i \frac{F_X}{\Delta} \quad [m/s^2]$$

F_X = 1.5 at FP
= 0.25 at midships
= 0.5 at AP
= 1.5 at AP for vessels conducting ice breaking astern

Intermediate values to be interpolated linearly.

where:

φ = maximum friction angle between steel and ice, normally taken as 10 [degrees]

γ = bow stem angle at waterline [degrees]

Δ = displacement

L = length between perpendiculars [m]

H = distance in meters from the water line to the point being considered [m]

FIB = vertical impact force, defined in Sec.2, 2.13.2.1.

Fi = total force normal to shell plating in the bow area due to oblique ice impact, defined in Sec.2, 2.3.2.1.

3.9 Auxiliary systems

3.9.1 Machinery is to be protected from the harmful effects of ingestion or accumulation of ice or snow. Where continuous operation is necessary, means are to be provided to purge the system of accumulated ice or snow.

3.9.2 Means are to be provided to prevent damage, to tanks containing liquids due to freezing.

3.9.3 Vent pipes, intake and discharge pipes and associated systems are to be designed to prevent blockage due to freezing or ice and snow accumulation.

3.10 Sea inlets and Cooling Water Systems

3.10.1 Cooling water systems for machinery that is essential for the propulsion and safety of the vessel, including sea chest inlets, are to be designed for the environmental conditions applicable to the ice class.

3.10.2 At least two sea chests are to be arranged as ice boxes (sea chests for water intake in severe ice conditions) for ice class PC1 to PC5 inclusive. The calculated volume for each of the ice boxes is to be at least 1 m^3 for every 750 [kW] of the totally installed power. For PC6 and PC7 there is to be at least one ice box located preferably near center line.

3.10.3 Ice boxes are to be designed for effective separation of ice and venting of air.

3.10.4 Sea inlet valves are to be secured directly to the ice boxes. The valves are to be of full bore type.

3.10.5 Ice boxes and sea bays are to have vent pipes and are to have shut off valves connected directly to the shell.

3.10.6 Means are to be provided to prevent freezing of sea bays, ice boxes, ship side valves and fittings above the load water line.

3.10.7 Efficient means are to be provided to re-circulate cooling seawater to the ice box. Total sectional area of the circulating pipes is not to be less than the area of the cooling water discharge pipe.

3.10.8 Detachable gratings or manholes are to be provided for ice boxes. Manholes are to be located above the deepest load line. Access is to be provided to the ice box from above.

3.10.9 Openings in ship sides for ice boxes are to be fitted with gratings or holes or slots in shell plates. The net area through these openings is to be not less than 5 times he area of the inlet pipe. The diameter of holes and width of slot in shell plating is to be not less than 20 [mm]. Gratings of the ice boxes are to be provided with a means of clearing. The means of clearing is to be of a type using low pressure steam. Clearing pipes are to be provided with screw-down type non-return valves.

3.11 Ballast Tanks

3.11.1 Efficient means are to be provided to prevent freezing in fore and after peak tanks and wing tanks located above the water line and where otherwise found necessary.

3.12 Ventilation Systems

3.12.1 The air intakes for machinery and accommodation ventilation are to be located on both sides of the ship at locations where manual de-icing is possible. Anti-icing protection of the air inlets may be accepted as an equivalent solution to location on both sides of the ship and manual de-icing. Notwithstanding the above, multiple air intakes are to be provided for the emergency generating set and are to be as far apart as possible.

3.12.2 The temperature of the inlet air is to be suitable for:
- the safe operation of the machinery; and
- the thermal comfort in the accommodation.

Accommodation and ventilation air intakes are to be provided with means of heating, if needed.

3.13 Steering Systems

3.13.1 Rudder stops are to be provided. The design ice force on rudder is to be transmitted to the rudder stops without damage to the steering system. An ice knife is to in general be fitted to protect the rudder in centre position. The ice knife is to extend below BWL. Design forces are to be determined according to the Sec2, 2.15.

3.13.2 The rudder actuator is to comply with the following requirements 3.13.2.1 and 3.13.2.2:

3.13.2.1 The rudder actuator is to be designed for a holding torque obtained by multiplying the open water torque resulting from the application of SOLAS Reg. II-1 / 29.3.2 (considering however a maximum speed of 18 knots, by following factors:

Table 3.13.2.1 : Ice class factors for steering gear	
Ice class	**Factor**
PC1	5
PC2	5
PC3	3
PC4	3
PC5	3
PC6	1.5
PC7	1.5

3.13.2.2 The design pressure for calculations to determine the scantlings of the rudder actuator is to be at least 1.25 times the maximum working pressure corresponding to the holding torque defined in 3.13.2.1 (Derived from SOLAS Reg. II-1 / 29.2.2).

3.13.3 The rudder actuator is to be protected by torque relief arrangements, assuming the following turning speeds [deg/s] without an undue pressure rise (ref *Pt.4, Ch.6, Steering Gear* for undue pressure rise):

Table 3.13.3 : Steering gear turning speeds			
Ice Class	**PC1 and PC2**	**PC3 to PC5**	**PC6 and PC7**
Turning speeds [deg/s]	10	7.5	6

If the rudder and actuator design can withstand such rapid loads, this special relief arrangement is not necessary and a conventional one may be used instead (*Pt.4, Ch.6, Steering Gear*).

3.13.4 Additionally for icebreakers, fast-acting torque relief arrangements are to be fitted in order to provide effective protection of the rudder actuator in case of the rudder being pushed rapidly hard over against the stops.

For hydraulically operated steering gear, the fast-acting torque relief arrangement is to be so designed that the pressure cannot exceed 115% of the set pressure of the safety valves when the rudder is being forced to move at the speed indicated in Table 3.13.4, also when taking into account the oil viscosity at the lowest expected ambient temperature in the steering gear compartment.

For alternative steering systems the fast-acting torque relief arrangement is to demonstrate an equivalent degree of protection to that required for hydraulically operated arrangements.

Table 3.13.4 : Steering gear turning speeds for icebreakers

Ice Class	**PC1 and PC2**	**PC3 to PC5**	**PC6 and PC7**
Turning speeds [deg/s]	40	20	15

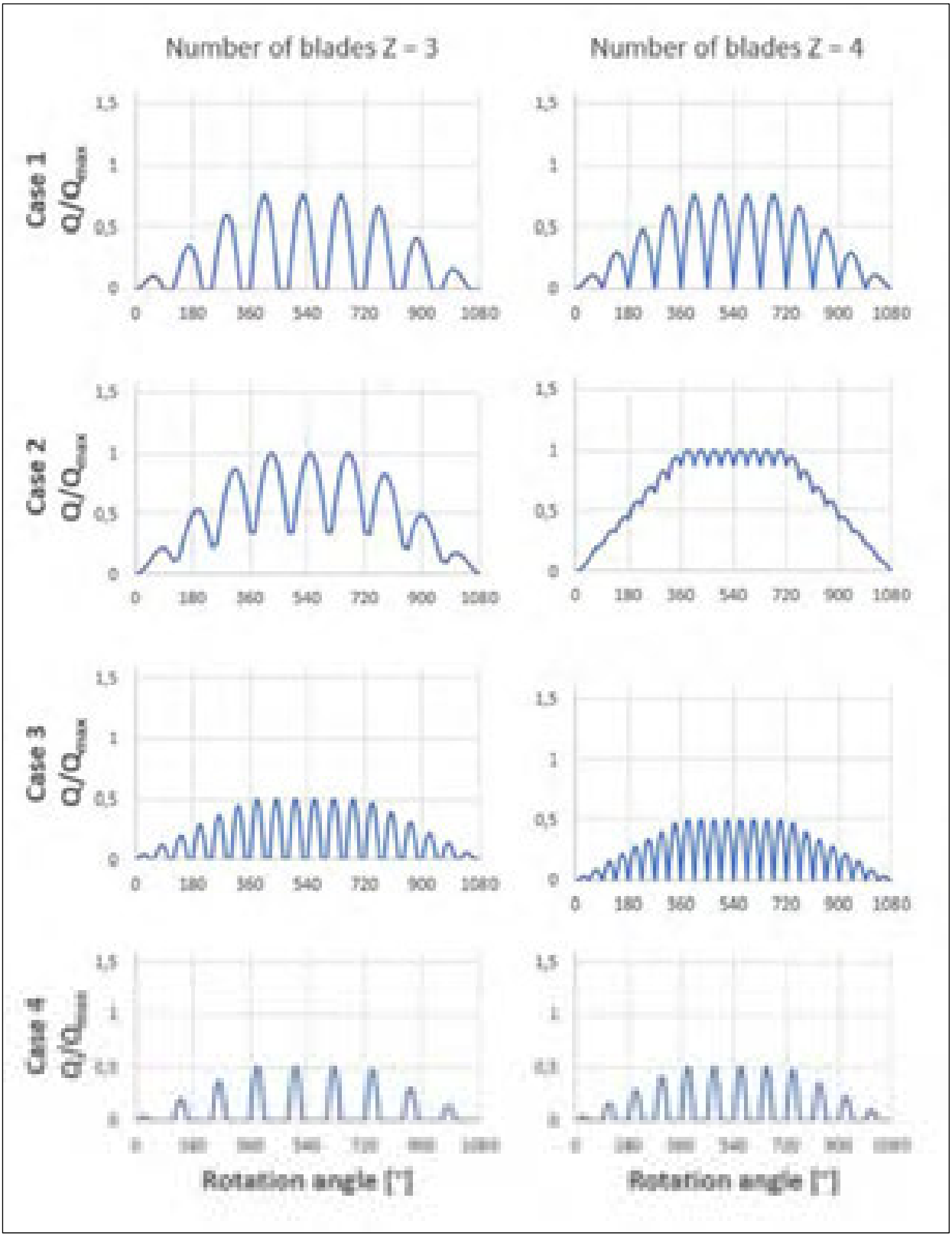
The arrangement is to be designed such that steering capacity can be speedily regained.

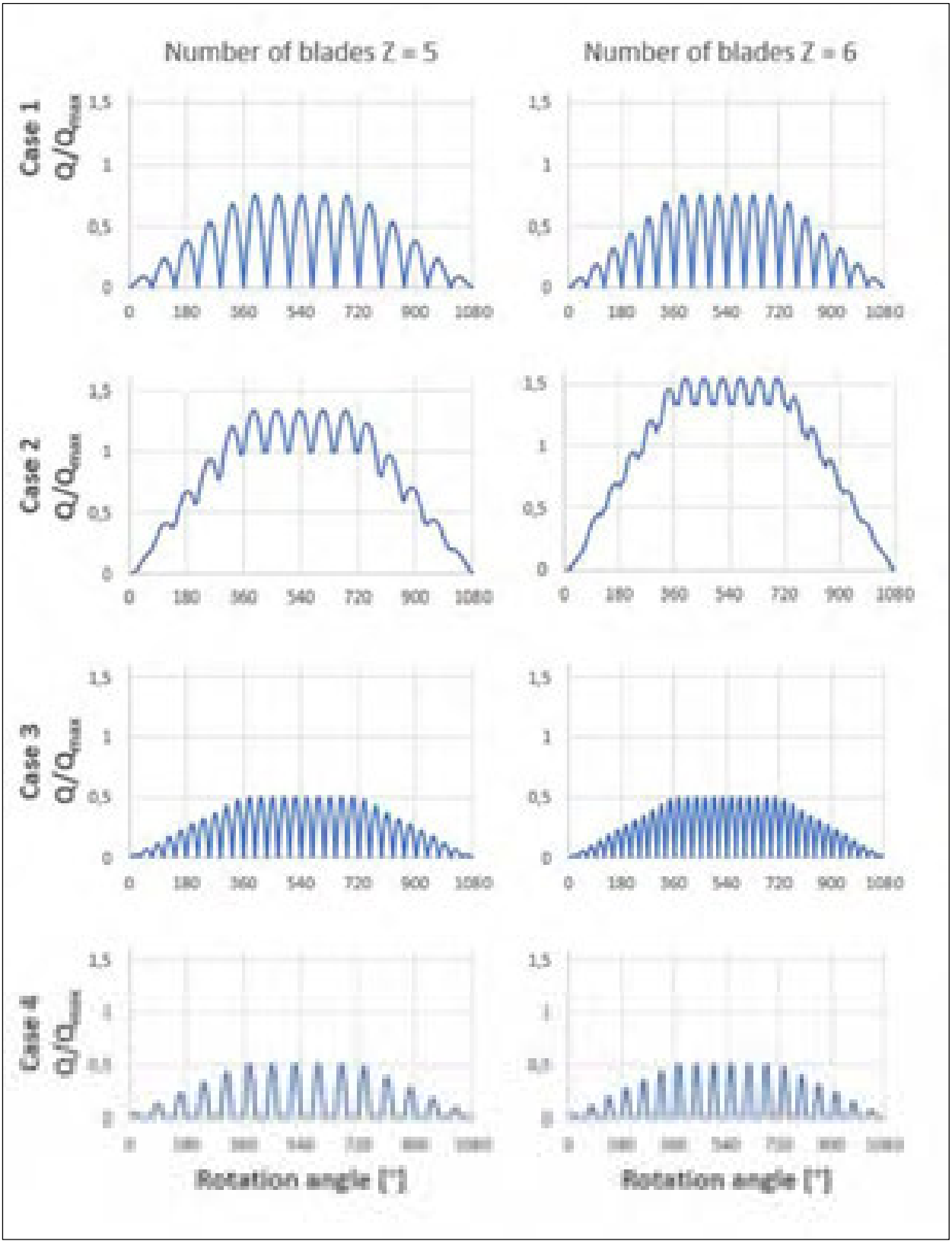
3.14 Alternative design

3.14.1 As an alternative – a comprehensive design study may be submitted and validated by an agreed test programme.

A P P E N D I X

The following illustrations show the excitation torque for all torsional load cases given in section 3 for different blade numbers (Z). The plots have been made using data for PC7 (H_{ice} = 1.5)





End of Chapter

Chapter 33

Well Stimulation Vessels

Contents

Section

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2	*Position Keeping and Stability*
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4	*Acid and Liquid Nitrogen Systems*
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Section 1

General

1.1 Application

1.1.1 The rules in this chapter apply to vessels which are arranged and equipped for stimulation of wells for production of oil and or gas. The requirements are supplementary to those given for assignment of main class.

1.2 Classification and Class Notations

1.2.1 Vessels built and equipped according to the rules in this chapter may be given the additional class notation "**WELL STIMULATION VESSEL"**.

1.2.2 Well Stimulation Vessels transporting and handling hazardous and noxious liquid substances in bulk are to comply with the OSV Chemical Code (IMO Resolution A.1122(30) and will be eligible to be assigned class notation "**OCC**".

1.2.3 Any additional requirements of the Flag Administration and/or the Coastal State are also to be complied with.

1.3 Definitions

1.3.1 Well Stimulation Systems
Well stimulation is a type of well intervention performed on an oil or gas well to increase production by improving the flow of hydrocarbons from the drainage area into the wellbore. Well stimulation systems are the facilities installed on vessels for the purpose of stimulation of wells to improve their productivity of oil and/or gas. Well stimulation systems may include acidizing equipment, fracturing blenders, pumping units, hydration and chemical additive systems, supporting equipment such as coiled tubing, lifting equipment, well control equipment, pressure vessels, piping, electrical components and control systems.

1.3.2 Lining

Lining is an acid-resistant material that is applied to the tank or piping system in a solid state with a defined elasticity property.

1.4 Scope of Classification

1.4.1 The following matters are covered by classification :

a) Position keeping, intact and damage stability of the vessel.

b) Tanks, pumping and piping arrangement, equipment and instrumentation related to the storage and handling of well stimulation fluids.
c) Personnel protective equipment.

1.5 Plans and Documentation

1.5.1 In addition to plans listed in Part 1, Ch 1, Sec 3, plans/documents given in 1.5.2 to

1.5.6, are to be submitted for information/ approval, as appropriate:

1.5.2 The following drawings are to be submitted for information :

General arrangement plan of well stimulation equipment including hazardous area/zone classification and chemical storage area definition, as well as decontamination and eye-washing facilities, and location of personnel protective equipment.

1.5.3 Arrangement and Equipment Plans.

The following plans/ documentation are to be submitted for approval:

i) Arrangement of all integral and independent tanks, including support and stays of independent tanks

ii) Structural drawings of acid tanks including vent arrangements, information on non-destructive testing of welds, strength and tightness testing, and specification of protective linings

iii) Documentation for liquid nitrogen tanks (plans required for liquefied gas carriers are to be referred Part 5, Ch 4)

iv) Pumping arrangement including diagrams of piping for acid, nitrogen and liquid additives, details of flange connections and pipe clamping/securing as well as specification and data on high pressure flexible hoses with end connections.

v) Mechanical ventilation arrangements of closed and semi-closed spaces containing acid tanks, pipes, pumps, mixers and blenders

vi) Drawings showing location of all electrical equipment in areas containing installations for uninhibited acid

vii) Electrical diagrams of well stimulation systems including single line diagram for intrinsically safe circuits, control and monitoring systems for cargo tank level gauging, overflow protection and emergency shutdown, as well as indication equipment for hydrogen, hydrogen chloride and oxygen

viii) List of explosion protected equipment together with certificates and references to specific diagrams and/or plans

ix) Drawings of pumps and mixers/blenders.

x) Drawings and particulars of nitrogen vaporizer and heat exchangers.

xi) Scope and type of Personnel protective equipment intended to be provided.

1.5.4 The following calculations/ analyses are to be submitted for approval :

i) Stability Calculations.

ii) Calculations demonstrating adequacy of propulsion power required for the vessel to maintain station during well stimulation operations.

iii) Stress analysis of supporting structure in way of flexible hose storage reel(s).

iv) Stress analysis of liquid nitrogen piping and heat exchangers.

v) Stress analysis of high pressure piping.

1.5.5 Documentation for the control and monitoring systems for the following is to be submitted for approval:

i) Cargo tank level measurement system.

ii) Cargo tank overflow protection system.

iii) Emergency shut-down system.

iv) Hydrogen indication equipment.

v) Hydrogen chloride indication equipment.

vi) Oxygen indication equipment.

1.5.6 The values of all design loads for Burner/ Flare Booms, where provided, are to be submitted.

1.6 Operation Manual

1.6.1 Operation manual for well stimulation procedure is to be submitted for approval and the same are to be readily available on board. The manual is to provide instructions and information on safety aspects related to well stimulation processing.

1.6.2 The operation manual is to include details of :

a) protective equipment
b) storage and handling of fluids and dry additives
c) transfer operations
d) emergency shut-down and dis-connection.

Section 2

Position Keeping and Stability

2.1 Position Keeping

2.1.1 It is essential that well stimulation vessels are capable of maintaining their positions safely during well stimulation operations. The means to maintain position may be a mooring system with anchors or dynamic positioning system.

2.1.2 If position mooring is undertaken with anchors, they are to fulfill the requirements for position keeping systems in Ch 10 of the *Rules and Regulations for Construction and Classification of Mobile Offshore Drilling Units*. Precautions are to be taken to prevent damage to seabed equipment and installations by anchors.

2.1.3. Dynamic positioning systems, where used to maintain the vessel's position during well stimulation operations, are to comply with the requirements for the class notation **DP (2)** or **DP (3)**.

2.2 Stability

2.2.1 Intact stability of the vessel should be in accordance with the Guidelines for the Design and Construction of Offshore Supply Vessels 2006 (Res. MSC.235 (82)).

2.2.2 The maximum aggregate quantity of hazardous and noxious liquids normally permitted to be carried is as indicated in the following:

a) 800 [m^3] or
b) a volume in cubic meters equal to 40% of the vessel's deadweight calculated at a cargo density of 1.0, whichever is the lesser.

2.2.3 The Administration may permit carriage of more than the maximum amount specified above.

2.2.4 Well stimulation vessels which are permitted to carry more than the maximum amount specified in 2.2.2 are to comply with the requirements for intact stability and damage stability requirements in Res. MSC. 235(82) with damage assumption given in 3.2.1 of Res. MSC. 235(82).

Section 3

Well Stimulation Equipment and Systems

3.1 General

3.1.1 The design of well stimulation equipment is to be carried out in accordance with Recognized National/International Standards, acceptable to IRS and the same are to be specified by the designer. The equipment and systems used only for well stimulation operations are in general not subject to classification by IRS, unless requested by the client. Manufacturer's certificate affirming compliance with applicable recognized standards is to be submitted to IRS. Their installations and onboard testing are to be supervised in the aspects of operational safety as to reduce to a minimum

any danger to persons on board and marine pollution, due regard to be paid to moving parts, hot surfaces and other hazards. Consequences of failure of systems and equipment essential to the safety of the vessel are to be considered.

3.1.2 Classified areas related to the installation of well stimulation equipment are to be described in accordance with IEC61892-7 and *Rules and Regulations for Construction and Classification of Mobile Offshore Drilling Units, Ch 13.*

3.1.3 The equipment class of vessel chosen for any particular operation should be agreed between the owner of the vessel and the customer based upon a risk analysis of the consequence of loss of position

3.2 Tanks and Piping Arrangement

3.2.1 Tanks for acid(s) and liquefied nitrogen are to be located at a minimum distance of 760 [mm] from the vessel's side and bottom. In sides the distance is to be measured perpendicularly inboard from the vessel's side to the centerline at the level of the summer load line.

3.2.2 Cargoes that react with other cargoes in hazardous manner are to be segregated by means of cofferdam, void space, cargo pump room, empty tank or fuel oil tank. Tanks used for other purposes (except for fresh water and lubricating oils) may be accepted as cofferdams.

3.2.3 For access to all spaces, the minimum spacing between cargo tank boundaries and adjacent ship's structure is to be 600 [mm].

Independent tanks installed in otherwise empty holds or stowed on deck are considered satisfactory in terms of segregation requirements.

3.2.4 The design of tanks and pumping arrangements is to cater for their segregation from machinery spaces, propeller shaft tunnels, dry cargo spaces, accommodation and service spaces, as well as from drinking water and stores for human consumption. The segregation may be achieved by means of cofferdam, void space, cargo pump room, empty tank, oil fuel tank or similar arrangements.

3.2.5 Piping Systems for the well stimulation processing plant are not to pass through any accommodation, service or machinery space other than cargo pump-room or pump-rooms and are to be separated from the machinery and ship piping systems.

3.2.6 Cargo piping is to be joined by welding except as allowed in Part 5, Ch 3, Sec 5.2. Storage tanks, pumps, valves, gaskets and piping for uninhibited acids are to be of corrosion resistant material or to have internal lining of corrosion resistant material. (Also Refer to Section 4, 4.1.3).

3.2.7 Each well injection line is to be provided with a check valve located at a flow-head or a test tree.

3.2.8 Tanks and pumping arrangements for liquid additives having flashpoint below 60°C are to comply with relevant requirements of Part 5 Chapter 3.
Arrangement of pump room for LFL (low flashpoint liquids) substances adjacent to the LFL tanks and without separating cofferdams may be specially considered in each case.

3.2.9 Requirements for tanks and pumping arrangements for chemicals other than acids will be considered in each case with due regard to the properties of the chemicals and applicable requirements of Part 5, Ch 3, Sec 15.

3.2.10 The well stimulation processing plant is to be remotely controlled from a position outside the area where the well stimulation systems are located.

3.3 Tank Venting

3.3.1 Outlets from safety valves of nitrogen tanks are to be led to open deck. Outlet pipes are to be arranged and supported in order to allow thermal expansion/contraction during release of cold gas. Deck/bulkhead penetrations are to be such that the structures are thermally isolated from the cold pipes.

3.3.2 Vent outlets from acid tanks are to be led to open deck. The outlets are to have a minimum height of 4 m above the deck and located at a minimum horizontal distance of 5 m from openings to accommodation and service spaces.

3.3.3 Vent outlets from acid tanks are to have pressure/vacuum valves and be provided with flame screens.

3.4 Access Openings

3.4.1 Access to enclosed spaces containing tanks, piping, pumps and blenders for uninhibited acid is to be direct from open deck or through air locks from other spaces. The air lock is to have independent mechanical ventilation.

3.4.2 For access through hatches, manholes or horizontal openings, the dimensions are to be sufficient to allow a person wearing a self contained breathing apparatus and protective equipment to ascend or descend any ladder without obstruction and also to provide a clear opening to facilitate the hoisting of an injured person from the bottom of a space.

3.4.3 Minimum clear opening for horizontal access is not to be less than 600 [mm] x 600 [mm] and for vertical openings is not to be less than 600 [mm] x 800 [mm] at a height of not more than 600 [mm] from the bottom shell plating unless gratings or other footholds are provided.

3.5 Ventilation

3.5.1 Ventilation of spaces for acid storage and handling:

i) The spaces containing uninhibited acid are to have independent mechanical ventilation with a capacity of minimum 30 air changes per hour, while those containing inhibited acid a minimum of 20 air changes per hour.

ii) Spaces for acid storage and handling are to be preferably ventilated by local exhaust. Spaces containing acetic acid are to be ventilated by explosion proof electrical equipment.

iii) The discharge suctions are to be located both, at floor and ceiling levels of the space concerned.

3.5.2 Ventilation of other spaces containing equipment for well stimulation:

i) Spaces containing installations for liquid nitrogen are to have a mechanical ventilation system (independent of the ventilation systems for accommodation) with a minimum capacity of 20 air changes per hour.

3.5.3 Ventilation of spaces for additive storage and handling:

i) Ventilation of spaces for storage and handling of dry and liquid additives will be considered in each case depending on the flammability, toxicity and reactivity properties of the additives to be used.

3.6 Spill Protection

3.6.1 A lining of corrosion resistant material extending up to a minimum height of 500 [mm] on the surrounding bulkheads and coaming is to be provided for protection of floors or decks under acid storage tanks, pumps and piping for uninhibited acid. Watertight coaming is to be provided for hatches or other openings in way of such decks with a minimum height of 500 [mm]. The coaming is to be protected by suitable lining or an acid resistant coating. Height requirement for coamings may be waived where this height is not practicable.

3.6.2 In order to keep deck spill away from accommodation and service areas, a permanent coaming of 150 [mm] height is to be provided on deck.

3.6.3 Flanges or other detachable pipe connections are to be covered by spray shields.

3.6.4 Portable shield covers for connecting flanges of loading manifold are to be provided. Drip trays of acid resistant material are to be provided under loading manifold for acid.

3.6.5 Manifolds transferring liquefied gases and other flanged connections in the system are to be provided with drip trays resistant to cryogenic temperatures. The issue of cold cracking of the ship structure, in the event of a liquid nitrogen spill, is to be addressed and mitigation measures provided.

3.7 Drainage

3.7.1 Drainage arrangements for acids are to be of acid resistant materials.

3.7.2 The drainage systems for spaces housing tanks, pumping and piping for acids or

additives are to be separate and not to be connected to the drainage system for other areas. These are to be of corrosion resistant materials. These drainage arrangements are to be located within the well stimulation processing area.

3.7.3 Drainage arrangements for void spaces, double bottom tanks and ballast tanks which are not separated from tanks containing well stimulation substances or their residues by a double bulkhead, are to be situated within the processing area.

3.8 Electrical Equipment

3.8.1 Electrical equipment or other ignition sources used in enclosed spaces containing acid tanks and acid pumping arrangements are to be certified as safe for operation in hydrogen/air atmosphere and are to be explosion proof.

3.9 Control and Monitoring Systems

3.9.1 A system of automatic and manual controls together with process shutdown and operating procedures are to be provided. System design is to cater for normal manning during well stimulation operations, accessibility of manual controls and switchover between manual and automatic operations.

3.9.2 Vapour and Gas Detection

i) Vapour detection and alarm systems for hydrogen or hydrogen chloride gas are to be provided in enclosed or semi-enclosed spaces containing installations for uninhibited acid.

ii) Spaces containing tanks and piping for liquid nitrogen are to be equipped with oxygen deficiency monitoring systems.

3.9.3 Gauging and Level Detection

i) Tanks containing liquefied nitrogen are to have gauging and level detection arrangements in accordance with Part 5, Ch 4, Sec.13.

ii) Tanks containing hydrochloric acid are to have a closed gauging system. A high level alarm is to be provided and the alarm is to be activated by a level sensing device, independent of the gauging system.

iii) Spaces containing equipment and storage tanks for the well stimulation system are to be provided with suitable detection and alarm system for liquid leakage.

3.9.4 Emergency Shutdown

i) Emergency stop of all pumps in the oil well stimulation system is to be arranged from one or more positions, outside the area where the system is located.

ii) Liquid nitrogen outlet lines from each nitrogen tank are to be provided with Emergency shut-off valves. These valves are to be remotely controlled from one or more positions outside the area where the oil well stimulation system is located.

iii) Emergency depressurizing and disconnection of the transfer hose are to be arranged from the central control position and from the bridge.

iv) At least one emergency shutdown panel capable of closing all barrier elements of blowout preventer and disconnecting connector for subsea blowout preventer sections is to be provided at a safe and readily accessible location, where applicable.

3.9.5 Electrical power supply is to be from a main power system and from a monitored uninterrupted power supply (UPS). The UPS is to be powered from both the main and emergency power systems and capable of continuously operating for at least 30 minutes upon loss of power from the main source. A reliable power supply is to be provided wherever auxiliary energy is required for functionality of emergency control and shutdown,

3.9.6 Where hydraulic and/or pneumatic power supply is used for actuation of emergency control and shutdown, duplication arrangements are to be made in accordance with the requirements in Part 4, Ch 7 Sec 2. Power supply circuits are to be connected to the main and emergency power sources separately where driving power for hydraulic and/or pneumatic pumps is electric.

3.9.7 A hardwired means for voice communications is to be provided between the central control station for well stimulation

operation and the vessel's position keeping control stations.

3.10 Flare / Burner Boom (where provided)

3.10.1 Design Loads

3.10.1.1 The loads to be considered in the design of a boom structure are to include, as applicable:

a) Dead weight of structure, piping, fittings, rigging, snow and ice, walkways, guard rails, etc.

b) Wind loads

c) Thermal and impulsive loads resulting from the use of the flare

d)Vessel motion-induced loads

3.10.1.2 Loads resulting from vessel motions may be estimated as per Pt 3, Ch 4 and wind loads may be established as per *Rules and Regulations for Construction and Classification of Mobile Offshore Drilling Units, Ch 4*.

3.10.1.3 The derivation of loading conditions to be used in the design is to give due account of the operational requirements of the Owner/Operator, and are to reflect both the operational and stowed modes of the boom.

3.10.2 The completed burner assembly is to be pressure-tested from the flexible line connection flange to the burner head. The adequacy of the boom's slewing and topping gear is to be demonstrated by testing after the boom's installation on the drilling unit. The details of the test procedure are to be agreed upon with IRS and witnessed by a Surveyor.

Section 4

Acid and Liquid Nitrogen Systems

4.1 Acid

4.1.1 Materials

4.1.1.1 Part 5 Ch 3, Sec 6 is applicable in general.

4.1.1.2 Storage tanks, pumping and piping for uninhibited acid are to be of corrosion resistant material or are to have internal lining of corrosion resistant material.

4.1.2 Storage Tanks

4.1.2.1 Scantling of integral tanks are to be in accordance with the requirements of Pt 3.

4.1.2.2 Scantlings of independent tanks, constructed mainly of plane surfaces, are to be in accordance with relevant requirements given in Pt.3.Tanks of pressure vessel configuration type (cylinders, spheres etc.) are to be in accordance with the requirements given in Pt 5, Ch.3 Sec.4.

4.1.3 Pumping and Piping

4.1.3.1 The rules in Pt 5, Ch. 3 will apply to pumping and piping systems.

4.1.3.2 The flexible hoses with end connectors are to be in accordance with a recognized standard.

4.2 Liquid Nitrogen

4.2.1 Materials

4.2.1.1 The materials are to be in accordance with the requirements of Pt. 5, Ch.4,Sec. 6.

4.2.2 Storage tank

4.2.2.1 The relevant requirements of Pt. 5, Ch. 4, Sec. 4 are to be applied.

4.2.3 Piping

4.2.3.1 The relevant requirements of Pt. 5, Ch. 4, Sec. 5 are to be applied.

4.2.4 Pressure/temperature control

4.2.4.1 The relevant requirements of Pt. 5, Ch. 4, Sec. 7 are to be applied.

4.2.5 Venting Arrangement

4.2.5.1 The relevant requirements of Pt. 5, Ch. 4, Sec. 8 are to be applied.

4.2.6 Personnel Protection

4.2.6.1 The relevant requirements of Pt. 5, Ch. 4, Sec. 14 are to be applied.

4.2.7 Liquid nitrogen is generally kept at atmospheric pressure within temperature range of -210 [˚C] to -196 [˚C]. Where the working temperature of liquid nitrogen is below -165 [˚C] the selection of structural materials is to be undertaken in consultation with IRS.

4.2.8 Where the working temperature is below -110 [˚C], a complete stress analysis is to be submitted for approval accounting for the weight of pipes, acceleration loads induced by hogging and sagging of the ship, for each branch of the piping system.

Section 5

Personnel and Fire Protection

5.1 Decontamination Showers and Eye Washes

5.1.1 A suitably marked decontamination shower and eyewashes are to be available on deck in a convenient location. The shower and eyewash are to be operable in all ambient conditions. Temperature control of water is to be provided in order to avoid excessive temperature.

5.2 Personnel Protective Equipment

5.2.1 Protective equipment are to be kept onboard in suitable locations as required by Part 5 Chapter 3 for carriage of hydrochloric acid.

5.3 Fire Protection

5.3.1 The arrangement of firefighting stations, fixed fire extinguishing systems and portable fire extinguishers are to be in accordance with Pt 6 and Pt 5, Ch 3, Sec 11.

End of Chapter

Chapter 34

Oil Recovery Vessels

Section	Contents
1	*General*
2	*Ship Structure*
3	*Pumping, Piping and Ventilation Systems*
4	*Electrical Installations*
5	*Fire Detection, Protection and Fighting*

Section 1

General

1.1 Application

1.1.1 This Chapter gives additional requirements for ships which are equipped to recover oil from the sea surface at a safe distance, handle, store and transport the recovered oil.

1.1.2 A ship complying with the applicable requirements of this chapter will be eligible for an appropriate class notation, as given in 1.2, and which will be entered in the Register Book.

1.1.3 Additional requirements may be imposed by the National Administration with whom the ship is registered and/or by the Administration within whose territorial jurisdiction the ship is intended to operate for oil recovery.

1.2 Classification and Class Notations

1.2.1 The Class Notations that may be assigned are as follows:

a) **OIL RECOVERY VESSEL**: A ship designed and constructed primarily for oil recovery duties. The scantlings of recovered oil storage tanks will be specially considered based on the requirements of Pt 5, Ch.2;

b) **OIL RECOVERY**: A ship equipped for carrying out oil recovery in addition to other duties and complying with the applicable requirements of this Chapter;

c) **FP 60C**: A ship designed and equipped to recover polluted water in the event of spills of oils which have, at the time of recovery, a flash point higher than 60 degree Celsius (closed cup test).
The notation is not to be assigned to oil recovery ships carrying recovered oils at a temperature within 15 degree Celsius of its flash point.

1.3 Scope of Classification

1.3.1 The following aspects of vessels related to oil recovery are covered by classification

a) Construction

b) Ship equipment and operation

c) Safety against fire and explosion during handling, storage and transportation of oil recovered from a spill

d) Supporting structures for oil recovery equipment

e) Stability and floatability in all relevant operating conditions

f) Available power for supply to equipment used during oil recovery operations

1.4 Definitions

1.4.1 **Oil Recovery Vessel**: A ship specially equipped with a fixed installation and /or mobile equipment for the removal of oil from the sea surface at a safe distance (see Cl 1.7 also) and its retention on board, carriage and subsequent unloading.

1.4.2 **Recovered Oil**: The top layers of oil polluted water collected by means of skimmers, rotating disk, floating pumps or equivalent systems, with sweeping arms, booms or similar devices.

1.4.3 **Oil recovery Tank**: A tank intended for the retention and transportation of recovered oil.

1.4.4 **Oil Recovery Pump Room**: A space containing the pumps and their accessories for handling of recovered oil.

1.4.5 **Oil Recovery Area**: The oil recovery areas are those parts of the ship which contain the following:

a) oil recovery tanks;
b) oil recovery pump rooms;
c) cofferdams;
d) ballast or void spaces surrounding the integral tanks and hold spaces in which independent tanks are located;
e) deck areas such as :
    i) area above the oil recovery tanks;
    ii) In addition, the deck area extending transversely and longitudinally from the oil recovery tanks over a distance of 3[m], where the rule length of the vessel is 50[m] and over.

1.4.6 **Hazardous Areas**: The areas in which an explosive atmosphere is, or may be, expected to be present in quantities such as to require special precautions for the construction, installation and use of electrical apparatus (See Cl 4.3).

1.4.7 **Gas-safe areas**: The gas areas which are not defined as hazardous.

1.5 Assumptions

1.5.1 The classification of the vessel is based upon the following assumptions:

a) The vessel would carry out oil recovery operations in accordance with the approved operating manual (see Cl 1.7 also);

b) The oil recovery operations would take place in moderate sea conditions

1.6 Submission of Plans

1.6.1 The following plans and/ or information are to be submitted in addition to the supporting documentation required for classification as specified elsewhere in this Chapter :

1.6.1.1 For Approval:

i) Arrangement plan indicating the location of oil recovery and handling equipment including portable ones and tanks utilized for recovered oil;

ii) Structural support in way of oil recovery equipment and loads in way of their foundations;

iii) Structural arrangement of tanks designed for storage of recovered oil including access to those tanks;

iv) Stability calculations for all operating conditions including ballasted condition (See also Cl. 1.9.2);

v) Piping system diagrams for handling recovered oil accompanied by details like size, wall thickness, maximum working pressure and material of all pipes; and the type, size, pressure rating and material of valves and fittings;

vi) Recovered oil tank venting arrangements including position of vent outlets;

vii) Arrangement of openings to hazardous areas and adjacent compartments, self closing doors and their direction of opening;

viii) Ventilation systems for all hazardous areas with details including number of air changes per hour, capacities of fans, air flows, areas subject to positive or

negative pressure, location and details of machinery exhaust outlets;

ix) Electrical equipment fitted in hazardous areas, including portable electric equipment connected during oil recovery operations (if any). Other details like model name/ number, manufacturer and certification of explosion proof, intrinsically safe, pressurized equipment are also to be submitted;

x) Wiring diagrams, installation instructions and restrictions imposed by certifying agency for intrinsically safe systems;

xi) Power supply, electrical protection and cabling for oil recovery equipment;

xii) Electrical load schedule considering the use of oil recovery equipment during operations;

xiii) Structural fire protection arrangements, controls and particulars of fixed fire extinguishing systems and portable fire safety equipment;

xiv) Diagrammatic plan of recovered oil heating systems, if installed.

1.6.1.2 For Review/ Information:

i) Specifications of gas detection equipment;

ii) Operating Manual

1.7 Operating Manual

1.7.1 The Operating manual is to be submitted in order to verify that the information included within is consistent with the design information and limitations considered in the vessel's classification.

The following information/ procedures are in general to be included in an operating manual to be available on board:

(a) Guidelines regarding safe distance from an oil spill source including specifying conditions when to withdraw the vessel;

(b) Information regarding the safe use of the ship with respect to, preparing for and conducting oil recovery and subsequent operations;

(c) Establishing and maintaining a safe atmosphere in any space(s) liable to become hazardous during oil recovery and subsequent operations;

(d) Isolation, where necessary, of electrical equipment in zones or spaces considered hazardous during oil recovery and subsequent operations, or on failure of, the measures required to establish and maintain a safe atmosphere;

(e) Gas measurements during operation including hydrogen sulphide gas (on open deck and in spaces where gas might accumulate). Actions to be taken in case of gas detection;

(f) Cleaning and gas freeing of tanks and pipes;

(g) Stability in all the relevant operational conditions;

(h) Firefighting;

(j) Maintenance procedures.

1.8 Testing and Trials

1.8.1 During ship trials, the transfer procedures to oil recovery mode of operation are to be simulated to verify that the vessel will be able to operate, as intended.

1.8.2 Testing of Oil Recovery Equipment: Tests are to be carried out according to an approved trial protocol, in order to check the proper operation of the oil recovery equipment. These tests may be performed during dock and/ or sea trials.

1.9 Intact Stability

1.9.1 Intact stability is to be as per the requirements of Chapters 1, 2 and 3 of Part A of the International Code on Intact Stability, 2008 (Resolution MSC 267(85)), as amended.

1.9.2 The following cases are also to be included in the trim and stability booklet for ships intended to carry out Oil Recovery Operations.

a) Ship in the fully loaded departure condition to the oil recovery spot having all the oil recovery equipment installed on board;

b) Ship in the worst anticipated operating condition during oil recovery operation; the worst operating condition regarding free surface effects when the equipment is fitted in the most unfavorable condition (for example, swiping arm extended).

Section 2

Ship Structure

2.1 General

2.1.1 The vessel is to be provided with:

a) a suitable working deck for use in oil recovery operation;

b) storage tanks for recovered oil;

c) permanently installed pumping and piping arrangement for transfer and discharge of recovered oil.

2.1.2 The visibility from the manoeuvring station is to be such that the Master can easily monitor oil recovery operations both on deck and in the water.

2.1.3 Accommodation areas are to be located as far away as possible from the oil tanks and the deck area, where the recovery operations are performed.

2.1.4 Oil handling areas and equipment on deck are to be provided with a coaming around all pumps, transfer flanges and other connections where leakage may be expected.

2.1.5 The coaming is to have a height of at least 150 [mm]. Each coaming is to be adequately sized to contain deck spills and prevent recovered oil from entering accommodation, machinery, control and service spaces or passing overboard.

2.1.6 Where drains are provided for the coaming, closing devices for these drains are to be permanently attached.

2.1.7 Spill coamings may be of the removable type. Removable coamings are to be constructed of rigid, non oil-absorbent material, leak proof, and may be temporarily fixed to the vessel's structure.

2.1.8 Exhaust outlets from machinery are to be located as high as practicable above the deck and are to be fitted with spark arresters.

2.2 General Arrangement

2.2.1 Tank arrangement

2.2.1.1 Oil recovery tanks are to be separated from machinery spaces of category A, propeller shaft tunnels, dry cargo spaces, accommodations, control stations and service spaces and from drinking water and stores for human consumption by means of a cofferdam or equivalent space.

2.2.1.2 For easy access to all parts, the cofferdams are to have a minimum width of 600 [mm]. Fuel oil tanks, tanks for ballast water, tanks for liquids used for oil treatment, tanks for anti-pollution liquids, storerooms for oil removal equipment and pump-rooms are considered as spaces equivalent to a cofferdam.

2.2.1.3 When this cofferdam is impracticable, oil recovery tanks adjacent to the engine room may be accepted provided that:

a) the boundary bulkheads are accessible for inspection;

b) the boundary bulkheads are connected to adjoining structure by full penetration welding ;

c) the tanks are pressure tested at each renewal survey.

2.2.1.4 In ships not exclusively used for oil recovery operations, the following may be used as oil recovery tanks:

a) tanks in supply vessels complying with LHNS guidelines;

b) tanks in Well Stimulation Vessels, if designed for a cargo mass density of, at least, 1.025 [t/m^3];

c) water ballast tanks;

d) fuel oil tanks;

e) hoppers spaces;

f) liquid mud tanks.

In all cases, the tanks, their associated equipment and piping are to comply with the requirements for oil recovery tanks.

2.2.1.5 Oil recovery tanks are to be located abaft the collision bulkhead.

2.2.1.6 The coating used in recovered oil tanks is to be of an oil and dispersion resistant type.

2.2.1.7 The height of tanks for recovered oil is not to be less than 1.5 [m].

2.2.1.8 Internal obstructions in tanks for recovered oil are to be provided with adequate openings to allow full flow of oil. The openings are to be so arranged that the tanks can be effectively drained. The area of one single opening is not to be less than twice the sectional area of the discharge pipe.

2.2.2 Accommodation, control station, service and machinery spaces

2.2.2.1 Accommodation or service spaces, control stations or machinery spaces of category A are to be located outside the oil recovery area.

2.2.2.2 Entrances, air inlets and openings to accommodation, service and machinery spaces of category A and control stations are not to face the oil recovery area unless they are spaced at least 7 [m] away from the oil recovery area.

2.2.2.3 Doors to spaces not having access to accommodation, service and machinery spaces and control stations, such as oil recovery control stations, storerooms or equipment rooms, may be permitted within the 7 [m] zone specified above, provided the boundaries of these spaces are insulated to A-60 standard.

2.2.2.4 The requirements of 2.2.2.2 and 2.2.2.3 are not applicable to vessels with class notation **FP 60C**.

2.2.3 Access to spaces in oil recovery area

2.2.3.1 Access hatches (at least 600 mm x 600 mm of clear opening) within the oil recovery area are to be direct from the open deck. Such accesses are to be suitable for cleaning and gas-freeing.

2.2.4 Oil recovery tank construction

2.2.4.1 A cargo density of 1.025 [t/m^3] is to be considered for calculating the internal pressures and forces in cargo tanks according to Pt. 3.

2.2.4.2 All tank openings and connections to the tank are to terminate above the weather deck and are to be located on the top of the tanks.

2.2.4.3 The structural design of the tanks is to take into account the temperature of recovered oil. Additional strength calculations may be required in case of risk of sloshing induced loads. (refer Pt. 3).

2.2.4.4 All tanks exceeding a breadth of 0.56B or a length of 0.1L or 12m whichever is the greater, are normally to be provided with wash bulkheads to reduce sloshing in partially filled tanks.

2.2.5 Supporting structures for oil recovery equipment

2.2.5.1 The supporting structures for oil recovery equipment are to be designed appropriately and submitted to IRS for approval.

2.2.6 Access openings between non hazardous and hazardous spaces (See Cl 4.3 also)

2.2.6.1 Normally, access doors or other openings between non hazardous spaces (such as accommodation, service spaces, machinery spaces, control stations and similar spaces) and hazardous spaces are not permitted.

2.2.6.2 Access (other than access between oil recovery pump room and machinery spaces category A) may be accepted between such spaces and hazardous areas provided that:

.1 The non-hazardous room is to have overpressure ventilation with 20 air changes per hour in relation to the hazardous area.

.2 The doors are to be gastight.

.3 The doors are to be self-closing and preferably arranged to swing into the non-hazardous space so that they are kept closed by the overpressure. Water tight sliding doors are not acceptable.

.4 If a door cannot be made self-closing, a second self-closing door in accordance with 2.2.6.2.2 and 2.2.6.2.3 above is to be arranged (air lock). The doors are to be arranged with sufficient spacing to allow for safe passage.

.5 Self-closing doors may be used for passage, but are not to remain open during oil recovery operations. Signboards are to be fitted to this effect.

.6 Emergency escape hatches to open deck that cannot be made self-closing need not be arranged with an air lock, but they can only be used for emergency escape purpose during oil recovery operations. Signboards are to be fitted to this effect.

Section 3

Pumping, Piping and Ventilation Systems

3.1 Pumping system, piping system and pump-rooms intended for recovered oil

3.1.1 The requirements of Pt 5, Ch.2 Sec.6 to 11 and Pt 4, Ch.2 and 3 are to be complied with, as applicable.

3.1.2 Piping systems for handling recovered oil are not to pass through:

a) accommodation spaces;
b) service spaces;
c) control stations;
d) machinery spaces of category A. (except for vessels with notation **FP 60C**).

3.1.3 Pumping and piping systems intended for recovered oil are to be independent from the other pumping and piping systems of the ship, except in the following cases:

a) Sections of the cargo system of supply vessels (in compliance with LHNS guidelines) or Well Stimulation Vessels used for oil recovery or if fuel oil tanks are used, means are provided to isolate the oil recovery system from any other system from which it may be connected. The connection between the cargo system and the recovered oil transfer piping may consist of movable pipe sections.

b) If water ballast tanks are used as oil recovery tanks when the ship is in oil recovery mode, the water ballast piping is blanked-off at the nearest position at the tank before starting the oil recovery operation. The connection between the oil recovery piping and the water ballast tanks is to be done by means of detachable spool pieces.

3.1.4 Piping intended for recovered oil and located below the main deck may run from the tank it serves and penetrate tank bulkheads or boundaries common to longitudinally or transversely adjacent oil recovery tanks, ballast tanks, empty tanks, pump-rooms or oil recovery pump-rooms, provided that, it is fitted with a stop-valve operable from the weather deck inside the tank, it serves. As an alternative, where an oil recovery tank is adjacent to an oil recovery pump-room, the stop valve operable from the weather deck may be situated on the tank bulkhead on the oil recovery pump-room side, provided an additional valve is fitted between the bulkhead valve and the oil recovery pump. A totally enclosed hydraulically operated valve located outside the oil recovery tank may also be accepted, provided that the valve is:

a) fitted on the bulkhead of the oil recovery tank it serves;
b) suitably protected against mechanical damage;
c) fitted at a distance from the shell as required for damage protection; and
d) operable from the weather deck.

3.1.5 Transfer of recovered oil through hatches by means of flexible hoses or moving piping is not permitted (except for vessels with notation **FP 60C**).

3.2 Oil Recovery Pumps

3.2.1 Oil recovery pumps are to comply with the requirements of cargo pumps for oil tankers; (see Pt. 5, Ch. 2, Sec. 6).

3.2.2 Oil recovery pumps are to be capable of being remotely shutdown from a location which is manned during oil recovery operations and from at least one other location outside the oil recovery area.

3.2.3 If an oil recovery pump serves more than one tank, a stop valve is to be fitted in the line of each tank.

3.2.4 Oil recovery pumps are to be so designed as to minimize the danger of sparking.

3.2.5 For ships not exclusively dedicated to oil recovery operations, the use of portable pumps or pumps serving cargo systems may be permitted, subject to special consideration by the IRS.

3.3 Pump Rooms for Oil Recovery Operations

3.3.1 Pump-rooms containing the pumps for handling the recovered oil are to be provided with a fixed fire-extinguishing system suitable for machinery spaces of category A (except for vessels with notation **FP 60 C**).

3.3.2 Means are to be provided to deal with drainage and any possible leakage from oil recovery pumps and valves in the oil recovery pump-room.

3.3.3 Bilge pumping arrangement is to be situated entirely within the oil recovery area. The bilge system is to be operable from outside the oil recovery pump-room.

3.3.4 Oil recovery pumps may also be used for bilge pumping provided they are connected to the oil recovery pump-room bilge piping through a shut-off valve and a non-return valve arranged in series.

3.3.5 For ventilation of oil recovery transfer pump-rooms, see 3.10.

3.4 Vent Pipes in Oil Recovery Tanks

3.4.1 Vent pipes of oil recovery tanks are to lead to the open at least 2.4 [m] above the weather deck.

3.4.2 Vent pipes are to be located at a distance of at least 5 [m] measured horizontally from the nearest air intake or opening to accommodations, control stations, service and machinery spaces of category A and other gas-safe spaces and from ignition sources.

3.4.3 Openings of vent pipes are to be directed to open deck and fitted with:

i) flameproof wire gauze made of corrosion resistant material easily removable for cleaning, and
ii) closing appliances complying with the provisions of Pt 3, Ch.13, Sec.3

3.4.4 For ships, not used exclusively for oil recovery operations, portable vent pipes may be accepted, subject to special consideration by IRS.

3.5 Level Gauging and Overfilling Control of Oil Recovery tanks

3.5.1 Oil recovery tanks are to be fitted with sounding pipes or other level gauging devices of a type approved by IRS.

3.5.2 Oil recovery tanks are to be fitted with a high level alarm or an overflow control system.

3.5.3 The high level alarm is to be of a type approved by the IRS and is to give an audible and visual alarm.

3.6 Heating systems in Oil recovery tanks

3.6.1 Heating systems fitted in oil recovery tanks are to comply with the requirements of Pt.5, Ch.2, Sec.10.

3.7 Sea Water Cooling System

3.7.1 One of the suctions serving the sea water cooling system is to be located in the lower part of the hull

3.8 Fire Pumps

3.8.1 Sea suctions serving the fire water pumps are to be located as low as possible.

3.9 Exhaust Systems

3.9.1 Exhaust lines from engines, gas turbines, boilers and incinerators are to be led outside any hazardous area above the deck and are to be fitted with a spark arrester.

3.9.2 Where the distance between the exhaust lines of engines and the hazardous areas is less than 3 [m], the ducts are to be fitted in a position:

i) near the waterline, if cooled by water injection; or

ii) below the waterline, in the other cases.

3.10 Ventilation System

3.10.1 The ventilation systems for hazardous and non-hazardous spaces are to be independent.

3.10.2 Ventilation of oil recovery pump rooms

3.10.2.1 Oil recovery pump rooms are to be provided with a mechanical ventilation system of the extraction type capable of giving at least 20 air changes per hour.

3.10.2.2 Ventilation intakes are to be so arranged as to minimize the possibility of recycling hazardous vapours from ventilation discharge openings.

3.10.2.3 Ventilation exhaust ducts are to discharge upwards to a gas-safe area on the weather deck in locations at least 3 [m] from any ventilation intake and opening to accommodations, service and machinery spaces, control stations and other spaces outside the oil recovery area.

3.10.2.4 Protection screens of not more than 13 [mm] square mesh are to be fitted on ventilation duct intakes and outlets.

3.10.2.5 Ventilation fans are to be of non-sparking construction.

3.10.2.6 The ventilation system is to be capable of being controlled from outside the oil recovery pump-room.

3.10.2.7 Provision is to be made to ventilate such spaces prior to entering the compartment and operating the equipment and a warning notice requiring the use of such ventilation is to be placed outside the compartment.

3.10.2.8 Ventilation ducts are not to lead through accommodations, service and machinery spaces or other similar spaces.

3.10.3 Ventilation of enclosed spaces normally entered during oil recovery operation other than recovery oil pump rooms

3.10.3.1 Enclosed spaces normally entered within the oil recovery area are to be provided with a mechanical ventilation system of the extraction type capable of giving at least 8 air changes per hour.

3.10.3.2 Ventilation intakes are to be located at a distance of not less than 3 [m] from the ventilation outlets of oil recovery pump-rooms.

Section 4

Electrical Installations

4.1 General

4.1.1 Electrical installations are to comply with the relevant requirements of Pt.4, Ch.8, as applicable.

4.2 Supply and Distribution Systems

4.2.1 The following systems of generation and distribution of electrical energy are acceptable:

i) DC Supply:
   a) two-wire insulated

ii) AC Supply:
   a) single-phase,two-wire insulated
   b) three-phase, three-wire insulated.

4.2.2 Earthed systems with hull return are not permitted, with the following exceptions subject to acceptance by IRS. :

i) impressed current cathodic protective systems;

ii) limited and locally earthed systems, such as starting and ignition systems of internal combustion engines, provided that any possible resulting current does not flow directly through any hazardous area;

iii) insulation level monitoring devices, provided that the circulation current of the devices does not exceed 30 [mA] under the most unfavourable conditions.

4.2.3 Earthed systems without hull return are not permitted, except for the following:

i) earthed intrinsically safe circuits and the following other systems subject to acceptance by IRS ;

ii) power supplies, control circuits and instrumentation circuits in non-hazardous areas where technical or safety reasons preclude the use of a system with no connection to earth, provided the current in the hull is limited to not more than 5 A in both normal and fault conditions; or

iii) earthed systems, provided that any possible resulting hull current does not flow directly through any hazardous area; or

iv) isolating transformers or other adequate means, to be provided if the distribution system is extended to areas remote from the machinery space.

4.2.4 No current carrying part is to be earthed in insulated distribution systems, other than the following:

i) through an insulation level monitoring device;
ii) through components used for the suppression of interference in radio circuits.

4.2.5 The devices intended to continuously monitor the insulation level of all distribution systems are also to monitor all circuits, other than intrinsically safe circuits, connected to apparatus in hazardous areas or passing through such areas.

4.2.6 An audible and visual alarm is to be given, at a manned position, in the event of an abnormally low level of insulation.

4.3 Hazardous Zones and Spaces

4.3.1 In order to facilitate the selection of appropriate electrical installations, hazardous areas are classified into zones. The following zones or spaces are regarded as hazardous during and on completion of oil recovery operations, until proven gas-safe:(Refer to 4.3.3 for hazardous zones in vessels with class notation **FP 60C**)

4.3.1.1 Hazardous Areas Zone 0:

i) Interiors of tanks intended for the storage of recovered oil;

ii) Interiors of piping systems intended for the handling of recovered oil including venting arrangement;

iii) Equipment containing recovered oil;

iv) Cofferdams and voids adjacent to recovered oil tanks, containing recovered oil pipe flanges, valves or other sources of release.

4.3.1.2 Hazardous Areas Zone 1:

i) Zones on open deck or semi-enclosed spaces on open deck within 3 [m] of any oil recovery tank outlet, oil recovery manifold valve, oil recovery valve, oil recovery pipe flange and oil recovery hatches;

ii) Oil Recovery Pump Rooms;

iii) Spaces in which oil contaminated equipment for handling the recovered oil is located and oil handling areas;

iv) Enclosed or semi enclosed spaces in which recovered oil pipe flanges, valves or other sources of release are located;

v) Areas on the open deck within a 3 [m] radius of any entrance or ventilation opening to any hazardous area zone 1 space;

vi) Areas on the open deck within a 3 [m] radius of any oil recovery equipment. Equipment includes but is not limited to skimmers, containment booms and reels, and separators;

vii) Areas on open deck within spillage coaming surrounding oil recovery manifold valves and 3 [m] beyond the coaming up to a height of 2.4 [m] above the deck.

4.3.1.3 Hazardous areas zone 2:

i) Enclosed spaces immediately adjacent to recovered oil tanks in any direction, not containing recovered oil pipe flanges, valves, or other sources of release;

ii) Enclosed or semi-enclosed spaces having a direct access or opening to any hazardous area;

iii) Air lock spaces between Zone 1 and non-hazardous space;

iv) Areas on the open deck within 1.5 [m] radius of any entrance or ventilation opening to any hazardous area zone 2 space excluding tanks adjacent to recovered oil tanks not containing recovered oil pipe flanges, valves, or other sources of release.

4.3.2 Hazardous areas are not to contain the following:

i) Internal combustion engines;
ii) Steam turbines and steam piping with a steam temperature in excess of 200 [C];
iii) Other piping system and heat exchangers with surface temperature exceeding 200 [C];
iv) Any other source of ignition.

4.3.3 Hazardous Zones for vessels with class notation **FP 60C**

i) Hazardous Area Zone 2:

a) Interiors of tanks intended for the storage of recovered oil;

b) Interiors of piping systems intended for the handling of recovered oil including venting arrangement;

c) Equipment containing recovered oil;

d) Cofferdams and voids adjacent to recovered oil tanks, containing recovered oil pipe flanges, valves or other sources of release.

4.4 Selection of electrical equipment for installation in hazardous areas

4.4.1 The types of electrical equipment permitted, depending on zones where they are installed are indicated in Pt 4, Ch 8, Sec. 11.

Section 5

Fire Detection, Protection and Fighting

5.1 Vapor Detector

5.1.1 At least one type approved portable gas detection instrument capable of measuring flammable vapour concentrations in air and an equipment for oil flashpoint measurements are to be provided on board.

5.1.2 Alternatively, instead of a portable gas detection instrument, a fixed system may be accepted provided that the sample is drawn from a point within 6 [m] from the waterline.

5.2 Structural Fire protection

5.2.1 Structural Fire Protection requirements as provided in Part 6 are to be complied with, as applicable. In addition, the following special requirements are to be complied with (not applicable for vessels with class notation **FP 60C**) :

5.2.1.1 Unless they are located at least 7 [m] from the nearest oil recovery area, exterior boundaries of the superstructures and deckhouses enclosing accommodation and including any overhanging decks which support such accommodation are to be insulated to A-60 standard for the whole of the portions which face the oil recovery areas up to the underside of the navigation bridge deck and for a distance of 3 [m] aft or forward of such areas.

5.2.1.2 As an alternative to A-60 insulation, a fixed water-spraying system capable of delivering water at a rate of 10 [litre/m^2/min] may be accepted. This system is to comply with the requirements listed in Pt. 5, Ch. 25, Cl. 2.7, except that the only protected area is to be the exterior boundaries of the superstructures and deckhouses enclosing accommodations and including any overhanging decks which support such accommodations facing the oil recovery area. The sea suctions are to be at safe locations sufficiently below the waterline to prevent floating oil from entering the system.

5.2.1.3 Windows and side scuttles fitted within 7 [m] from the nearest oil recovery area are to have the same fire integrity as the bulkhead on which they are fitted. If they have a lower fire rating because they are protected by the fixed water-spraying system mentioned in 5.2.1.2, the windows and side scuttles are to be fitted with inside covers of steel or other equivalent material having a thickness equal to the bulkhead on which they are fitted. Where they are not of the fixed type, they are to be such as to ensure an efficient gastight closure.

5.3 Fire-fighting

5.3.1 Firefighting arrangements are to be provided in accordance with the requirements of Part 6, as applicable. In addition, the following requirements are to be complied with:

.1 For the protection of the oil recovery area, the following fire-fighting equipment is to be provided near the working area:

a) two dry powder fire extinguishers, each with a capacity of at least 50 [kg] or equivalent
b) at least one portable foam extinguishing applicator complying with Pt. 6, Ch. 8, Cl. 4.2.2.

End of Chapter

Chapter 35

Gaseous Fuel or other Low Flash Point Fueled Ships

Contents

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Introduction

IR1.0 General

IR1.1 These Rules have been prepared to ensure that ships using natural gas or other low flash point fuel as fuel built with a view to classification with IRS, will also comply with the requirements of IMO International Code of Safety for Ships Using Gases or Other Low-Flashpoint Fuels (IGF Code).

IR1.2 Responsibility for interpretation of the IGF Code requirements for the purpose of issuing Safety Construction Certificate lies with the statutory authority of the country where the ship is to be registered.

IR1.3 The IGF code contains requirements for operational matters which are not within the scope of classification requirements and IRS does not require these to be investigated for the purpose of classification. However, these matters will be considered for issuing the Safety Construction Certificate on behalf of the Administration

IR1.4 Where for the purpose of issuing a Safety Construction Certificate, an Administration has specifically accepted an equivalent under the terms of the IGF code or has adopted an interpretation different from that agreed to by IRS, individual consideration will be given to the acceptance of the equivalent or interpretation concerned for the purpose of classification.

IR 1.5 This chapter addresses the requirements of the IGF code relevant to the Classification of ships. IGF Code Parts C-1 and D related to Drills and Emergency exercises, Operation and Training matters are not covered in the Rules.

While Sections 1 to 4 of this chapter covers the general requirements for gas and other low flash point fuels, specific requirements for natural gas fuelled ships are given in Section 5 to 16 and Annex-1. Requirements for other low flash point fuelled ships are presently addressed through alternative design principle.

IR2.0 Application

IR 2.1 The chapter is applicable to ships to which Reg. SOLAS II-1/56 (as adopted by MSC.392(95)) is applicable.

IR 2.2 This chapter is not applicable to any gas carriers using their cargoes as fuel and complying with the requirements of IGC Code (Pt 5 Ch 4).

IR3.0 Classification and class notations

IR3.1 The general requirements for classification and the assignment of class notations are given in Pt.1 of the Rules, to which reference is to be made.

IR3.2 In general, an additional class notation "**LFPF**" would be assigned, where the vessel is designed and constructed primarily for using low-flash point fuel.

IR3.3 Following additional qualifier will be added to the above mentioned class notation:

(NG) – Ship using natural gas as fuel, complying with the requirements of this Chapter.

(EG) - Ship using Ethane gas as fuel, complying with the alternative design requirements in 2.3 of this Chapter.

(LPG) - Ship using Liquid Petroleum gas as fuel complying with the requirements of IRS Guidelines on "*LPG Fuelled Vessels*" or complying with the alternative design requirements in 2.3 of this Chapter.

(ML) - Ship using Methanol as fuel, complying with the requirements of IRS Guidelines on "*Methanol Fueled Vessels"* or complying with the alternative design requirements in 2.3 of this Chapter.

(EL) – Ships using Ethanol as fuel and complying with the relevant and applicable requirements for Ethanol in the IRS Guidelines on "*Methanol Fueled Vessels*" or complying with the alternative design requirements in 2.3 of this Chapter.

(AM) - Ships using Ammonia as fuel, complying with the requirements of IRS Guidelines on "*Ammonia Fueled Vessels*" or complying with the alternative design requirements in 2.3 of this Chapter.

(SFE) – Ships with single fuel engines using natural gas or other low flash point fuel.

(DFE) – Ship using dual fuel engines

Example:

1) Following additional notation will be assigned to ship with single fuel engine, using natural gas as fuel complying with the requirements of this chapter:

LFPF (NG) (SFE)

2) Following additional notation will be assigned to ship with dual fuel engine, using natural gas as fuel complying with the requirements of this chapter:

LFPF (NG) (DFE)

Section 1

Preamble (to the IGF Code)

1.1 The purpose of IGF Code is to provide an international standard for ships using low-flashpoint fuel, other than ships covered by the IGC Code.

1.2 The basic philosophy of IGF Code is to provide mandatory provisions for the arrangement, installation, control and monitoring of machinery, equipment and systems using low-flashpoint fuel to minimize the risk to the ship, its crew and the environment, having regard to the nature of the fuels involved.

1.3 Throughout the development of IGF Code it was recognized that it must be based upon sound naval architectural and engineering principles and the best understanding available of current operational experience, field data and research and development. Due to the rapidly evolving new fuels technology, IMO will periodically review IGF Code, taking into account both experience and technical developments.

1.4 IGF Code addresses all areas that need special consideration for the usage of the low-flashpoint fuel. The basic philosophy of the IGF Code considers the goal based approach (MSC.1/Circ.1394). Therefore, goals and functional requirements were specified for each section forming the basis for the design, construction and operation.

1.5 The current version of IGF Code includes regulations to meet the functional requirements for natural gas fuel. Regulations for other low-flashpoint fuels will be added to IGF Code as, and when, they are developed by the Organization.

1.6 In the meantime, for other low-flashpoint fuels, compliance with the functional requirements of IGF Code must be demonstrated through alternative design.

Section 2

General

2.1 Application

2.1.1 Unless expressly provided otherwise this Chapter applies to ships to which part G of SOLAS chapter II-1 applies.

2.1.2 *Fuel* in the context of the requirements in this chapter means natural gas, either in its liquefied or gaseous state. It should be recognized that the composition of natural gas may vary depending on the source of natural gas and the processing of the gas.

2.2 Definitions

2.2.1 Unless otherwise stated below, definitions are as defined in SOLAS chapter II-2.

2.2.1.1 *Accident* means an uncontrolled event that may entail the loss of human life, personal injuries, environmental damage or the loss of assets and financial interests.

2.2.1.2 *Breadth (B)* means the greatest moulded breadth of the ship at or below the deepest draught (summer load line draught) (refer to SOLAS regulation II-1/2.8).

2.2.1.3 *Bunkering* means the transfer of liquid or gaseous fuel from land based or floating facilities into a ship's permanent tanks or connection of portable tanks to the fuel supply system.

2.2.1.4 *Certified safe type* means electrical equipment that is certified safe by the relevant authorities recognized by the IRS/Administration for operation in a flammable atmosphere based on a recognized standard.*

Note:

**Refer to IEC 60079 series, Explosive atmospheres and IEC 60092-502:1999 Electrical Installations in Ships – Tankers – Special Features.*

2.2.1.5 *CNG* means compressed natural gas (see also 2.2.1.26).

2.2.1.6 *Control station* means those spaces defined in SOLAS chapter II-2 and additionally for this Chapter, the engine control room.

2.2.1.7 *Design temperature* for selection of materials is the minimum temperature at which liquefied gas fuel may be loaded or transported in the liquefied gas fuel tanks.

2.2.1.8 *Design vapour pressure "P_0"* is the maximum gauge pressure, at the top of the tank, to be used in the design of the tank.

2.2.1.9 *Double block and bleed valve* means a set of two valves in series in a pipe and a third valve enabling the pressure release from the pipe between those two valves. The arrangement may also consist of a two-way valve and a closing valve instead of three separate valves.

2.2.1.10 *Dual fuel engines* means engines that employ fuel covered by this Chapter (with pilot fuel) and oil fuel. Oil fuels may include distillate and residual fuels.

2.2.1.11 *Enclosed space* means any space within which, in the absence of artificial ventilation, the ventilation will be limited and any explosive atmosphere will not be dispersed naturally.*

Note:

**See also definition in IEC 60092-502:1999.*

2.2.1.12 *ESD* means emergency shutdown.

2.2.1.13 *Explosion* means a deflagration event of uncontrolled combustion.

2.2.1.14 *Explosion pressure relief* means measures provided to prevent the explosion pressure in a container or an enclosed space exceeding the maximum overpressure the container or space is designed for, by releasing the overpressure through designated openings.

2.2.1.15 *Fuel containment system* is the arrangement for the storage of fuel including tank connections. It includes where fitted, a primary and secondary barrier, associated insulation and any intervening spaces, and adjacent structure if necessary for the support of these elements. If the secondary barrier is part of the hull structure it may be a boundary of the fuel storage hold space. The spaces around the fuel tank are defined as follows:

.1 *Fuel storage hold space* is the space enclosed by the ship's structure in which a fuel containment system is situated. If tank connections are located in the fuel storage hold space, it will also be a tank connection space;

.*2 Interbarrier space* is the space between a primary and a secondary barrier, whether or not completely or partially occupied by insulation or other material; and

.*3 Tank connection space* is a space surrounding all tank connections and tank valves that is required for tanks with such connections in enclosed spaces.

IR.3.1 A tank connection space may be required also for tanks on open deck. This may apply for ships where restriction of hazardous areas is safety critical. A tank connection space may also be necessary in order to provide environmental protection for essential

safety equipment related to the gas fuel system like tank valves, safety valves and instrumentation.

IR .3.2 A tank connection space may also contain equipment such as vaporizers or heat exchangers. Such equipment is considered to only contain potential sources of release, but not sources of ignition.

2.2.1.16 *Filling limit (FL)* means the maximum liquid volume in a fuel tank relative to the total tank volume when the liquid fuel has reached the reference temperature.

2.2.1.17 *Fuel preparation room* means any space containing pumps, compressors and/or vaporizers for fuel preparation purposes.

IR 2.2.1.17 A tank connection space which has equipment such as vaporizers or heat exchangers installed inside is not regarded as a fuel preparation room.

2.2.1.18 *Gas* means a fluid having a vapour pressure exceeding 0.28 [MPa] absolute at a temperature of 37.8 [°C].

2.2.1.19 *Gas consumer* means any unit within the ship using gas as a fuel.

2.2.1.20 *Gas only engine* means an engine capable of operating only on gas, and not able to switch over to operation on any other type of fuel.

2.2.1.21 *Hazardous area* means an area in which an explosive gas atmosphere is or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of equipment.

2.2.1.22 *High pressure* means a maximum working pressure greater than 1.0 [MPa].

2.2.1.23 *Independent tanks* are self-supporting, do not form part of the ship's hull and are not essential to the hull strength.

2.2.1.24 *LEL* means the lower explosive limit.

2.2.1.25 *Length (L)* is the length as defined in the International Convention on Load Lines in force.

2.2.1.26 *LNG* means liquefied natural gas.

2.2.1.27 *Loading limit (LL)* means the maximum allowable liquid volume relative to the tank volume to which the tank may be loaded.

2.2.1.28 *Low-flashpoint fuel* means gaseous or liquid fuel having a flashpoint lower than otherwise permitted under paragraph 2.1.1 of SOLAS regulation II-2/4.

2.2.1.29 *MARVS* means the maximum allowable relief valve setting.

2.2.1.30 *MAWP* means the maximum allowable working pressure of a system component or tank.

2.2.1.31 *Membrane tanks* are non-self-supporting tanks that consist of a thin liquid and gas tight layer (membrane) supported through insulation by the adjacent hull structure.

2.2.1.32 *Multi-fuel engines* means engines that can use two or more different fuels that are separate from each other.

2.2.1.33 *Non-hazardous area* means an area in which an explosive gas atmosphere is not expected to be present in quantities such as to require special precautions for the construction, installation and use of equipment.

2.2.1.34 *Open deck* means a deck having no significant fire risk that at least is open on both ends/sides, or is open on one end and is provided with adequate natural ventilation that is effective over the entire length of the deck through permanent openings distributed in the side plating or deckhead.

2.2.1.35 *Risk i*s an expression for the combination of the likelihood and the severity of the consequences.

2.2.1.36 *Reference temperature* means the temperature corresponding to the vapour pressure of the fuel in a fuel tank at the set pressure of the pressure relief valves (PRVs).

2.2.1.37 *Secondary barrier* is the liquid-resisting outer element of a fuel containment system designed to afford temporary containment of any envisaged leakage of liquid fuel through the primary barrier and to prevent the lowering of the temperature of the ship's structure to an unsafe level.

2.2.1.38 *Semi-enclosed space* means a space where the natural conditions of ventilation are notably different from those on open deck due to the presence of structure such as roofs, windbreaks and bulkheads and which are so arranged that dispersion of gas may not occur.*

Note:

**Refer also to IEC 60092-502:1999 Electrical Installations in Ships – Tankers – Special Features.*

2.2.1.39 *Source of release* means a point or location from which a gas, vapour, mist or liquid may be released into the atmosphere so that an explosive atmosphere could be formed.

2.2.1.40 *Unacceptable loss of power* means that it is not possible to sustain or restore normal operation of the propulsion machinery in the event of one of the essential auxiliaries becoming inoperative, in accordance with SOLAS regulation II-1/26.3.

2.2.1.41 *Vapour pressure* is the equilibrium pressure of the saturated vapour above the liquid, expressed in [MPa] absolute at a specified temperature.

2.3 Alternative design

2.3.1 IGF Code contains functional requirements for all appliances and arrangements related to the usage of low-flashpoint fuels.

2.3.2 Fuels, appliances and arrangements of low-flashpoint fuel systems may either:

.1 deviate from those set out in this IGF Code, or

.2 be designed for use of a fuel not specifically addressed in IGF Code.

Such fuels, appliances and arrangements can be used provided that these meet the intent of the goal and functional requirements concerned and provide an equivalent level of safety of the relevant chapters.

2.3.3 The equivalence of the alternative design is to be demonstrated as specified in SOLAS regulation II-1/55 and approved by the Administration. However, the Administration would not allow operational methods or procedures to be applied as an alternative to a particular fitting, material, appliance, apparatus, item of equipment, or type thereof which is prescribed by IGF Code.

IR 2.4 Documentation Requirements

IR 2.4.1 For a ship with natural gas-fuelled engine installations, plans and documents are to be submitted for approval as listed in the following:

IR 2.4.1.1 Arrangement plan of the ship indicating the following:

a) machinery and boiler spaces, accommodation spaces, service spaces and control stations.

b) fuel containment systems

c) fuel preparation rooms

d) ventilating pipes, door and openings to fuel preparation rooms and other hazardous areas

e) entrances, air inlets and openings to accommodation spaces, service spaces and control stations.

f) Air lock arrangements

IR 2.4.1.2 Plans and documents of the gas fuel tanks covering the following details:

a) tank structure

b) supports and stays

c) secondary barriers

d) Insulations

e) marking plates

f) tank connection space arrangement

g) tank hatches, pipes and any openings to the gas tanks

h) safety relief valves and associated vent piping

i) gas freeing and purging system, Including vent lines for safety relief valves or similar piping and ducts for gas pipes.

j) capacity analysis of safety relief valves and associated piping, including back pressure.

k) fabrication details of tanks including Building tolerances, NDT Plan and Welding procedures (WPS)

l) Cooling down procedure of tanks

m)Specification of safety relief valves and associated vent piping.

IR 2.4.1.3 Documentation for Control and Monitoring system of the gas fuel including safety system.

IR2.4.1.4 Testing procedure for safety shut downs.

IR2.4.1.5 Plans and documentation showing arrangement and details of piping systems covering the following:

a) Vent mast, including location and details of outlets from the fuel tanks safety relief valves.

b) Fuel gas drip trays

c) vent lines for safety relief valves or similar piping and ducts for gas pipes.

d) fuel gas cooling system

e) fuel gas heating system

f) exhaust gas system, including arrangement of explosion relief

g) ventilation systems for gas fuel system spaces, including capacity and location of fans and their motors. Detailed drawing of rotating parts and casing for fans and portable ventilators for gas fuel system spaces.

h) Functional testing procedure of all piping systems including valves, fittings and associated equipment for handling fuel.

i) Control and monitoring system documentation for ventilation systems.

IR2.4.1.6 Hazardous area Classification plan

IR2.4.1.7 Electrical schematic drawing, including single line diagrams for all intrinsically safe circuits including explosion protection details of components.

IR2.4.1.8 Plans and Documentation for Fire safety as listed in the following:

a) Hydrocarbon gas detection and alarm system

b) Fire detection and alarm system

c) Fire extinguishing equipment

d) Structural fire protection plan

e) Control and monitoring system documentation for:

- Hydrocarbon gas detection and alarm system
- Fire detection and alarm system

f) Fixed fire extinguishing system documentation, containing details on:

- External surface protection water spraying system
- Bunkering station fire extinguishing system

Section 3

Goal and Functional Requirements

3.1 Goal

3.1.1 The goal of this Chapter is to provide for safe and environmentally-friendly design, construction and operation of ships and in particular their installations of systems for propulsion machinery, auxiliary power generation machinery and/or other purpose machinery using gas or low-flashpoint fuel as fuel.

3.2 Functional requirements

3.2.1 The safety, reliability and dependability of the systems are to be equivalent to that achieved with new and comparable conventional oil-fuelled main and auxiliary machinery.

3.2.2 The probability and consequences of fuel-related hazards is to be limited to a minimum through arrangement and system design, such as ventilation, detection and safety actions. In the event of gas leakage or failure of the risk reducing measures, necessary safety actions is to be initiated.

3.2.3 The design philosophy is to ensure that risk reducing measures and safety actions for the gas fuel installation do not lead to an unacceptable loss of power.

3.2.4 Hazardous areas are to be restricted, as far as practicable, to minimize the potential risks that might affect the safety of the ship, persons on board, and equipment.

3.2.5 Equipment installed in hazardous areas is to be minimized to that required for operational purposes and is to be suitably and appropriately certified.

3.2.6 Unintended accumulation of explosive, flammable or toxic gas concentrations is to be prevented.

3.2.7 System components are to be protected against external damages.

3.2.8 Sources of ignition in hazardous areas are to be minimized to reduce the probability of explosions.

3.2.9 It is to be arranged for safe and suitable fuel supply, storage and bunkering arrangements capable of receiving and containing the fuel in the required state without leakage. Other than when necessary for safety reasons, the system is to be designed to prevent venting under all normal operating conditions including idle periods.

3.2.10 Piping systems, containment and over-pressure relief arrangements that are of suitable design, construction and installation for their intended application are to be provided.

3.2.11 Machinery, systems and components are to be designed, constructed, installed, operated, maintained and protected to ensure safe and reliable operation.

3.2.12 Fuel containment system and machinery spaces containing source that might release gas into the space is to be arranged and located such that a fire or explosion in either will not lead to an unacceptable loss of power or render equipment in other compartments inoperable.

3.2.13 Suitable control, alarm, monitoring and shutdown systems are to be provided to ensure safe and reliable operation.

3.2.14 Fixed gas detection suitable for all spaces and areas concerned are to be arranged.

3.2.15 Fire detection, protection and extinction measures appropriate to the hazards concerned are to be provided.

3.2.16 Commissioning, trials and maintenance of fuel systems and gas utilization machinery shall satisfy the goal in terms of safety, availability and reliability.

3.2.17 The technical documentation is to permit an assessment of the compliance of the system and its components with the applicable rules, guidelines, design standards used and the principles related to safety, availability, maintainability and reliability.

3.2.18 A single failure in a technical system or component is not to lead to an unsafe or unreliable situation.

Section 4

General Requirements

4.1 Goal

4.1.1 The goal of this section is to ensure that the necessary assessments of the risks involved are carried out in order to eliminate or mitigate any adverse effect to the persons on board, the environment or the ship.

4.2 Risk assessment

4.2.1 A risk assessment is to be conducted to ensure that risks arising from the use of low-flashpoint fuels affecting persons on board, the environment, the structural strength or the integrity of the ship are addressed. Consideration is to be given to the hazards associated with physical layout, operation and maintenance, following any reasonably foreseeable failure.

4.2.2 For ships using natural gas as fuel, the risk assessment required by 4.2.1 need only be conducted where explicitly required by paragraphs 5.10.5, 5.12.3, 6.4.1.1, 6.4.15.4.7.2, 8.3.1.1, 13.4.1, 13.7 and 15.8.1.10 as well as by paragraphs 4.4 and 6.8 of the Annex-1.

4.2.3 The risks is to be analysed using acceptable and recognized risk analysis techniques, and loss of function, component damage, fire, explosion and electric shock shall as a minimum be considered. The analysis shall ensure that risks are eliminated wherever possible. Risks which cannot be eliminated are to be mitigated as necessary. Details of risks, and the means by which they are mitigated, is to be documented to the satisfaction of the IRS/Administration.

4.3 Limitation of explosion consequences

4.3.1 An explosion in any space containing any potential sources of release* and potential ignition sources shall not :

.1 cause damage to or disrupt the proper functioning of equipment/systems located in any space other than that in which the incident occurs;

.2 damage the ship in such a way that flooding of water below the main deck or any progressive flooding occur;

.3 damage work areas or accommodation in such a way that persons who stay in such areas under normal operating conditions are injured;

.4 disrupt the proper functioning of control stations and switchboard rooms necessary for power distribution;

.5 damage life-saving equipment or associated launching arrangements;

.6 disrupt the proper functioning of firefighting equipment located outside the explosion-damaged space;

.7 affect other areas of the ship in such a way that chain reactions involving, inter alia, cargo, gas and bunker oil may arise; or

.8 prevent persons access to life-saving appliances or impede escape routes.

Note:

**Double wall fuel pipes are not considered as potential sources of release.*

Section 5

Ship Design and Arrangement

5.1 Goal

5.1.1 The goal of this section is to provide for safe location, space arrangements and mechanical protection of power generation equipment, fuel storage systems, fuel supply equipment and refuelling systems.

5.2 Functional requirements

5.2.1 This section is related to functional requirements in 3.2.1 to 3.2.3, 3.2.5, 3.2.6, 3.2.8, 3.2.12 to 3.2.15 and 3.2.17. In particular the following apply:

.1 the fuel tank(s) is to be located in such a way that the probability for the tank(s) to be damaged following a collision or grounding is reduced to a minimum taking into account the safe operation of the ship and other hazards that may be relevant to the ship;

.2 fuel containment systems, fuel piping and other fuel sources of release is to be so located and arranged that released gas is led to a safe location in the open air;

.3 the access or other openings to spaces containing fuel sources of release is to be so arranged that flammable, asphyxiating or toxic gas cannot escape to spaces that are not designed for the presence of such gases

.4 fuel piping is to be protected against mechanical damage;
.5 the propulsion and fuel supply system is to be so designed that safety actions after any gas leakage do not lead to an unacceptable loss of power; and

.6 the probability of a gas explosion in a machinery space with gas or low-flashpoint fuelled machinery is to be minimized.

5.3 General

5.3.1 Fuel storage tanks are to be protected against mechanical damage.

5.3.2 Fuel storage tanks and/or equipment located on open deck are to be located to ensure sufficient natural ventilation, so as to prevent accumulation of escaped gas.

5.3.3 The fuel tank(s) is to be protected from external damage caused by collision or grounding in the following way:

.1 The fuel tanks are to be located at a minimum distance of B/5 or 11.5 [m], whichever is less, measured inboard from the ship side at right angles to the centreline at the level of the summer load line draught;

where:

B is the greatest moulded breadth of the ship at or below the deepest draught (summer load line draught) (refer to SOLAS regulation II-1/2.8).

.2 The boundaries of each fuel tank are to be taken as the extreme outer longitudinal, transverse and vertical limits of the tank structure including its tank valves.

.3 For independent tanks the protective distance is to be measured to the tank shell (the primary barrier of the tank containment system). For membrane tanks the distance is to be measured to the bulkheads surrounding the tank insulation.

.4 In no case shall the boundary of the fuel tank be located closer to the shell plating or aft terminal of the ship than as follows:

.1 For passenger ships: B/10 but in no case less than 0.8 [m]. However, this distance

need not be greater than B/15 or 2 [m] whichever is less where the shell plating is located inboard of B/5 or 11.5 [m], whichever is less, as required by 5.3.3.1.

.2 For cargo ships:

.1 for V_c below or equal 1000 [m^3], 0.8 [m];

.2 for 1000 [m^3]< V_c < 5000 [m^3], 0.75 + V_c x 0.2/4000 [m];

.3 for 5000 [m^3] ≤ V_c < 30000 [m^3], 0.8 + V_c/25000 [m]; and

.4 for V_c ≥ 30000 [m^3], 2 [m],

where:

V_c corresponds to 100% of the gross design volume of the individual fuel tank at 20 [°C], including domes and appendages.

.5 The lowermost boundary of the fuel tank(s) is to be located above the minimum distance of B/15 or 2.0 [m, whichever is less, measured from the moulded line of the bottom shell plating at the centreline.

.6 For multihull ships the value of B may be specially considered.

.7 The fuel tank(s) is to be abaft a transverse plane at 0.08L measured from the forward perpendicular in accordance with SOLAS regulation II-1/8.1 for passenger ships, and abaft the collision bulkhead for cargo ships.

where:

L is the length as defined in the International Convention on Load Lines (refer to SOLAS regulation II-1/2.5).

.8 For ships with a hull structure providing higher collision and/or grounding resistance, fuel tank location requirements may be specially considered in accordance with section 2.3.

5.3.4 As an alternative to 5.3.3 above, the following calculation method may be used to determine the acceptable location of the fuel tanks:

.1 The value f_{CN} calculated as described in the following is to be less than 0.02 for passenger ships and 0.04 for cargo ships.*

Note :

**The value f_{CN} accounts for collision damages that may occur within a zone limited by the longitudinal projected boundaries of the fuel tank only, and cannot be considered or used as the probability for the fuel tank to become damaged given a collision. The real probability will be higher when accounting for longer damages that include zones forward and aft of the fuel tank.*

.2 The f_{CN} is calculated by the following formulation:

$$f_{CN} = f_l \times f_t \times f_v$$

where:

f_l is calculated by use of the formulations for factor p contained in SOLAS regulation II-1/7-1.1.1.1. The value of x1 shall correspond to the distance from the aft terminal to the aftmost boundary of the fuel tank and the value of x2 shall correspond to the distance from the aft terminal to the foremost boundary of the fuel tank.

f_t is calculated by use of the formulations for factor r contained in SOLAS regulation II-1/7-1.1.2, and reflects the probability that the damage penetrates beyond the outer boundary of the fuel tank. The formulation is:

$$f_t = 1 - r(x1, x2, b) *$$

Note:

** When the outermost boundary of the fuel tank is outside the boundary given by the deepest subdivision waterline the value of b should be taken as 0.*

f_v is calculated by use of the formulations for factor v contained in SOLAS regulation II-1/7-2.6.1.1 and reflects the probability that the damage is extending vertically above the lowermost boundary of the fuel tank. The formulations to be used are:

f_v=1.0−0.8 (($H-d$)/7.8),
if ($H-d$) is less than or equal to 7.8 [m]. f_v is not to be taken greater than 1.

f_v=0.2−(0.2 (($H-d$)−7.8)/4.7),
in all other cases f_v is not to be taken less than 0.

where:

H is the distance from baseline, in metres, to the lowermost boundary of the fuel tank; and

d is the deepest draught (summer load line draught).

.3 The boundaries of each fuel tank are to be taken as the extreme outer longitudinal, transverse and vertical limits of the tank structure including its tank valves.

.4 For independent tanks the protective distance is to be measured to the tank shell (the primary barrier of the tank containment system). For membrane tanks the distance is to be measured to the bulkheads surrounding the tank insulation.

.5 In no case shall the boundary of the fuel tank be located closer to the shell plating or aft terminal of the ship than as follows:

.1 For passenger ships: B/10 but in no case less than 0.8 [m]. However, this distance need not be greater than B/15 or 2 [m] whichever is less where the shell plating is located inboard of B/5 or 11.5 [m], whichever is less, as required by 5.3.3.1.

.2 For cargo ships:

.1 for V_c below or equal 1000 [m^3], 0.8 [m];

.2 for 1000 [m^3] < V_c < 5000 [m^3], 0.75+ V_c x 0.2/4000 [m];

.3 for 5000 [m^3] ≤ V_c < 30000 [m^3], 0.8 + V_c/25000 [m]; and

.4 for V_c ≥ 30000 [m^3], 2 [m],

where:

V_c corresponds to 100% of the gross design volume of the individual fuel tank at 20 [°C], including domes and appendages.

.6 In case of more than one non-overlapping fuel tank located in the longitudinal direction, f_{CN} is to be calculated in accordance with paragraph 5.3.4.2 for each fuel tank separately. The value used for the complete fuel tank arrangement is the sum of all values for f_{CN} obtained for each separate tank.

.7 In case the fuel tank arrangement is unsymmetrical about the centreline of the ship, the calculations of f_{CN} is to be calculated on both starboard and port side and the average value is to be used for the assessment. The minimum distance as set forth in paragraph 5.3.4.5 is to be met on both sides.

.8 For ships with a hull structure providing higher collision and/or grounding resistance, fuel tank location requirements may be specially considered in accordance with section 2.3.

5.3.5 When fuel is carried in a fuel containment system requiring a complete or partial secondary barrier:

.1 fuel storage hold spaces is to be segregated from the sea by a double bottom; and

.2 the ship is to also have a longitudinal bulkhead forming side tanks.

5.4 Machinery space concepts

5.4.1 In order to minimize the probability of a gas explosion in a machinery space with gas-fuelled machinery one of these two alternative concepts may be applied:

.1 Gas safe machinery spaces:

Arrangements in machinery spaces are such that the spaces are considered gas safe under all conditions, normal as well as abnormal conditions, i.e. inherently gas safe.

In a gas safe machinery space a single failure cannot lead to release of fuel gas into the machinery space.

.2 ESD-protected machinery spaces:

Arrangements in machinery spaces are such that the spaces are considered non-hazardous under normal conditions, but under certain abnormal conditions may have the potential to become hazardous. In the event of abnormal conditions involving gas hazards, emergency shutdown (ESD) of non-safe equipment (ignition sources) and machinery is to be automatically executed while equipment or machinery in use or active during these conditions is to be of a certified safe type.

In an ESD protected machinery space a single failure may result in a gas release into the space. Venting is designed to accommodate a probable maximum leakage scenario due to technical failures.

Failures leading to dangerous gas concentrations, e.g. gas pipe ruptures or blow out of gaskets are covered by explosion pressure relief devices and ESD arrangements.

IR 5.4.1 Premixed engines using fuel gas mixed with air before the turbocharger should be located in ESD protected machinery spaces.

5.5 Gas safe machinery space

5.5.1 A single failure within the fuel system is not to lead to a gas release into the machinery space.

5.5.2 All fuel piping within machinery space boundaries is to be enclosed in a gas tight enclosure in accordance with 9.6.

5.6 ESD-protected machinery spaces

5.6.1 ESD protection is to be limited to machinery spaces that are certified for periodically unattended operation.

5.6.2 Measures are to be applied to protect against explosion, damage of areas outside of the machinery space and ensure redundancy of power supply. The following arrangement is to be provided but may not be limited to:

.1 gas detector;

.2 shutoff valve;

.3 redundancy; and

.4 efficient ventilation.

5.6.3 Gas supply piping within machinery spaces may be accepted without a gastight external enclosure on the following conditions:

.1 Engines for generating propulsion power and electric power are to be located in two or more machinery spaces not having any common boundaries unless it can be documented that a single casualty will not affect both spaces.

.2 The gas machinery space is to contain only a minimum of such necessary equipment, components and systems as are required to ensure that the gas machinery maintains its function.

.3 A fixed gas detection system arranged to automatically shutdown the gas supply, and disconnect all electrical equipment or installations not of a certified safe type, is to be fitted.

5.6.4 Distribution of engines between the different machinery spaces is to be such that shutdown of fuel supply to any one machinery space does not lead to an unacceptable loss of power.

5.6.5 ESD protected machinery spaces separated by a single bulkhead shall have sufficient strength to withstand the effects of a local gas explosion in either space, without affecting the integrity of the adjacent space and equipment within that space.

5.6.6 ESD protected machinery spaces is to be designed to provide a geometrical shape that will minimize the accumulation of gases or formation of gas pockets.

5.6.7 The ventilation system of ESD-protected machinery spaces is to be arranged in accordance with 13.5.

5.7 Location and protection of fuel piping

5.7.1 Fuel pipes are not to be located less than 800 [mm] from the ship's side.

5.7.2 Fuel piping is not to be led directly through accommodation spaces, service spaces, electrical equipment rooms or control stations as defined in the SOLAS Convention.

5.7.3 Fuel pipes led through ro-ro spaces, special category spaces and on open decks are to be protected against mechanical damage.

5.7.4 Gas fuel piping in ESD protected machinery spaces is to be located as far as practicable from the electrical installations and tanks containing flammable liquids.

5.7.5 Gas fuel piping in ESD protected machinery spaces is to be protected against mechanical damage.

5.8 Fuel preparation room design

5.8.1 Fuel preparation rooms are to be located on an open deck, unless those rooms are arranged and fitted in accordance with the requirements of this Chapter for tank connection spaces.

IR 5.8.1.1 Fuel preparation rooms, regardless of location, should be arranged to safely contain cryogenic leakages.

IR 5.8.1.2 The material of the boundaries of the fuel preparation room is to have a design temperature corresponding with the lowest temperature it can be subject to in a probable maximum leakage scenario unless the boundaries of the space, i.e. bulkheads and decks, are provided with suitable thermal protection.

IR 5.8.1.3 The fuel preparation room is to be arranged to prevent surrounding hull structure from being exposed to unacceptable cooling, in case of leakage of cryogenic liquids.

IR 5.8.1.4 The fuel preparation room is to be designed to withstand the maximum pressure build up during such a leakage. Alternatively, pressure relief venting to a safe location (mast) can be provided.

5.9 Bilge systems

5.9.1 Bilge systems installed in areas where fuel covered by this Chapter can be present are to be segregated from the bilge system of spaces where fuel cannot be present.

5.9.2 Where fuel is carried in a fuel containment system requiring a secondary barrier, suitable drainage arrangements for dealing with any leakage into the hold or insulation spaces through the adjacent ship structure is to be provided. The bilge system is not to lead to pumps in safe spaces. Means of detecting such leakage is to be provided.

5.9.3 The hold or interbarrier spaces of type A independent tanks for liquid gas is to be provided with a drainage system suitable for handling liquid fuel in the event of fuel tank leakage or rupture.

5.10 Drip trays

5.10.1 Drip trays are to be fitted where leakage may occur which can cause damage to the ship structure or where limitation of the

area which is effected from a spill is necessary.

5.10.2 Drip trays are to be made of suitable material.

5.10.3 The drip tray is to be thermally insulated from the ship's structure so that the surrounding hull or deck structures are not exposed to unacceptable cooling, in case of leakage of liquid fuel.

5.10.4 Each tray is to be fitted with a drain valve to enable rain water to be drained over the ship's side.

5.10.5 Each tray is to have a sufficient capacity to ensure that the maximum amount of spill according to the risk assessment can be handled.

5.11 Arrangement of entrances and other openings in enclosed spaces

5.11.1 Direct access is not to be permitted from a non-hazardous area to a hazardous area. Where such openings are necessary for operational reasons, an airlock which complies with 5.12 is to be provided.

5.11.2 If the fuel preparation room is approved located below deck, the room is to, as far as practicable, have an independent access direct from the open deck. Where a separate access from deck is not practicable, an airlock which complies with 5.12 is to be provided.

5.11.3 Unless access to the tank connection space is independent and direct from open deck it is to be arranged as a bolted hatch. The space containing the bolted hatch will be a hazardous space.

5.11.4 If the access to an ESD-protected machinery space is from another enclosed space in the ship, the entrances are to be arranged with an airlock which complies with 5.12.

5.11.5 For inerted spaces, access arrangements are to be such that unintended entry by personnel is prevented. If access to such spaces is not from an open deck, sealing arrangements shall ensure that leakages of inert gas to adjacent spaces are prevented.

5.12 Airlocks

5.12.1 An airlock is a space enclosed by gastight bulkheads with two substantially gastight doors spaced at least 1.5 [m] and not more than 2.5 [m] apart. Unless subject to the requirements of the International Convention on Load Lines, the door sill is not to be less than 300 [mm] in height. The doors are to be self-closing without any holding back arrangements.

5.12.2 Airlocks are to be mechanically ventilated at an overpressure relative to the adjacent hazardous area or space.

5.12.3 The airlock is to be designed in a way that no gas can be released to safe spaces in case of the most critical event in the gas dangerous space separated by the airlock. The events are to be evaluated in the risk analysis according to 4.2.

5.12.4 Airlocks are to have a simple geometrical form. They are to provide free and easy passage, and are to have a deck area not less than 1.5 [m^2]. Airlocks are not to be used for other purposes, for instance as store rooms.

5.12.5 An audible and visual alarm system to give a warning on both sides of the airlock is to be provided to indicate if more than one door is moved from the closed position.

5.12.6 For non-hazardous spaces with access from hazardous spaces below deck where the access is protected by an airlock, upon loss of underpressure in the hazardous space access to the space is to be restricted until the ventilation has been reinstated. Audible and visual alarms are to be given at a manned location to indicate both loss of pressure and opening of the airlock doors when pressure is lost.

5.12.7 Essential equipment required for safety is not to be de-energized and is to be of a certified safe type. This may include lighting, fire detection, public address, general alarms systems.

Section 6

Fuel Containment System

6.1 Goal

6.1.1 The goal of this section is to provide that gas storage is adequate so as to minimize the risk to personnel, the ship and the environment to a level that is equivalent to a conventional oil fuelled ship.

6.2 Functional requirements

6.2.1 This section relates to functional requirements in 3.2.1, 3.2.2, 3.2.5 and 3.2.8 to 3.2.17. In particular the following apply:

.1 the fuel containment system is to be so designed that a leak from the tank or its connections does not endanger the ship, persons on board or the environment. Potential dangers to be avoided include:

.1 exposure of ship materials to temperatures below acceptable limits; (Also refer to IR 5.8.1.1 to 5.8.1.4)

.2 flammable fuels spreading to locations with ignition sources;

.3 toxicity potential and risk of oxygen deficiency due to fuels and inert gases;

.4 restriction of access to muster stations, escape routes and life-saving appliances (LSA); and

.5 reduction in availability of LSA.

.2 the pressure and temperature in the fuel tank is to be kept within the design limits of the containment system and possible carriage requirements of the fuel;

.3 the fuel containment arrangement is to be so designed that safety actions after any gas leakage do not lead to an unacceptable loss of power; and

.4 if portable tanks are used for fuel storage, the design of the fuel containment system is to be equivalent to permanent installed tanks as described in this section.

6.3 General

6.3.1 Natural gas in a liquid state may be stored with a maximum allowable relief valve setting (MARVS) of up to 1.0 [MPa].

6.3.2 The Maximum Allowable Working Pressure (MAWP) of the gas fuel tank is not to exceed 90% of the Maximum Allowable Relief Valve Setting (MARVS).

6.3.3 A fuel containment system located below deck is to be gas tight towards adjacent spaces.

6.3.4 All tank connections, fittings, flanges and tank valves must be enclosed in gas tight tank connection spaces, unless the tank connections are on open deck. The space is to be able to safely contain leakage from the tank in case of leakage from the tank connections.

6.3.5 Pipe connections to the fuel storage tank is to be mounted above the highest liquid level in the tanks, except for fuel storage tanks of type C. Connections below the highest liquid level may however also be accepted for other tank types after special consideration by the IRS/Administration.

6.3.6 Piping between the tank and the first valve which release liquid in case of pipe failure is to have equivalent safety as the type C tank, with dynamic stress not exceeding the values given in 6.4.15.3.1.2.

6.3.7 The material of the bulkheads of the tank connection space is to have a design temperature corresponding with the lowest temperature it can be subject to in a probable maximum leakage scenario. The tank connection space is to be designed to withstand the maximum pressure build up during such a leakage. Alternatively, pressure relief venting to a safe location (mast) can be provided.

6.3.8 The probable maximum leakage into the tank connection space is to be determined based on detail design, detection and shutdown systems.

6.3.9 If piping is connected below the liquid level of the tank it has to be protected by a secondary barrier up to the first valve.

6.3.10 If liquefied gas fuel storage tanks are located on open deck the ship steel is to be protected from potential leakages from tank connections and other sources of leakage by use of drip trays. The material is to have a design temperature corresponding to the temperature of the fuel carried at atmospheric pressure. The normal operation pressure of the tanks is to be taken into consideration for protecting the steel structure of the ship.

IR 6.3.10 Whether a drip tray is needed or not is to be in accordance with the following:

.1 When the tank is located on the open deck, drip trays are to be provided to protect the deck from leakages from tank connections and other sources of leakage.

.2 When the tank is located below the open deck but the tank connections are on the open deck, drip trays are to be provided to protect the deck from leakages from tank connections and other sources of leakage.

.3 When the tank and the tank connections are located below the deck, all tank connections are to be located in a tank connection space. Drip trays in this case are not required.

6.3.11 Means are to be provided whereby liquefied gas in the storage tanks can be safely emptied.

6.3.12 It is to be possible to empty, purge and vent fuel storage tanks with fuel piping systems. Instructions for carrying out these procedures must be available on board. Inerting is to be performed with an inert gas prior to venting with dry air to avoid an explosion hazardous atmosphere in tanks and fuel pipes. See detailed requirements in 6.10.

6.4 Liquefied gas fuel containment

6.4.1 General

6.4.1.1 The risk assessment required in 4.2 is to include evaluation of the ship's liquefied gas fuel containment system, and may lead to additional safety measures for integration into the overall vessel design.

6.4.1.2 The design life of fixed liquefied gas fuel containment system is not to be less than the design life of the ship or 20 years, whichever is greater.

6.4.1.3 The design life of portable tanks is not to be less than 20 years.

6.4.1.4 Liquefied gas fuel containment systems is to be designed in accordance with North Atlantic environmental conditions and relevant long-term sea state scatter diagrams for unrestricted navigation. Less demanding environmental conditions, consistent with the expected usage, may be accepted by the Administration for liquefied gas fuel containment systems used exclusively for restricted navigation. More demanding environmental conditions may be required for liquefied gas fuel containment systems operated in conditions more severe than the North Atlantic environment.*
Note:

**Refer to IACS Rec No. 34. North Atlantic Environmental conditions refer to wave conditions. Assumed Temperatures are used for determining qualities with respect to design temperatures and is another matter not intended to be covered in 6.4.1.4*

6.4.1.5 Liquefied gas fuel containment systems are to be designed with suitable safety margins:

.1 to withstand, in the intact condition, the environmental conditions anticipated for the liquefied gas fuel containment system's design life and the loading conditions appropriate for them, which is to include full homogeneous and partial load conditions and partial filling to any intermediate levels; and

.2 being appropriate for uncertainties in loads, structural modelling, fatigue, corrosion, thermal effects, material variability, aging and construction tolerances.

IR6.4.1.5 Guidance: Safety Margins

IR6.4.1.5.1 The Safety Margins to be used for design are specified in the sections as mentioned below:

Independent Type A Tank:	Section 6.4.15.1
Independent Type B Tank:	Section 6.4.15.2
Independent Type C Tank	Section 6.4.15.3
Membrane Tank:	Section 6.4.15.4
Integral Tank	Section 6.4.15.5

IR6.4.1.5.2 Notwithstanding IR6.4.1.5.1, the safety margin is to be taken not less than 2 for membrane tanks. IRS may specially consider lower safety margins provided they are accompanied with satisfactory technical justification which may include detailed analyses, calculations, experimental and model tests, records of satisfactory service history etc. If the flag Administration specifies a more stringent margin of safety, then the same is to be utilized.

6.4.1.6 The liquefied gas fuel containment system structural strength is to be assessed against failure modes, including but not limited to plastic deformation, buckling and fatigue. The specific design conditions that are to be considered for the design of each liquefied gas fuel containment system are given in 6.4.15. There are three main categories of design conditions:

.1 Ultimate Design Conditions – The liquefied gas fuel containment system structure and its structural components are to withstand loads liable to occur during its construction, testing and anticipated use in service, without loss of structural integrity. The design is to take into account proper combinations of the following loads:

.1 internal pressure;

.2 external pressure;

.3 dynamic loads due to the motion of the ship in all loading conditions;

.4 thermal loads;

.5 sloshing loads;

.6 loads corresponding to ship deflections;

.7 tank and liquefied gas fuel weight with the corresponding reaction in way of supports;

.8 insulation weight;

.9 loads in way of towers and other attachments; and

.10 test loads.

.2 Fatigue Design Conditions – The liquefied gas fuel containment system structure and its structural components are not to fail under accumulated cyclic loading.

.3 Accidental Design Conditions – The liquefied gas fuel containment system is to meet each of the following accident design conditions (accidental or abnormal events), addressed in this chapter:

.1 Collision – The liquefied gas fuel containment system is to withstand the collision loads specified in 6.4.9.5.1 without deformation of the supports or the tank structure in way of the supports likely to endanger the tank and its supporting structure.

.2 Fire – The liquefied gas fuel containment systems are to sustain without rupture the rise in internal pressure specified in 6.7.3.1 under the fire scenarios envisaged therein.

.3 Flooded compartment causing buoyancy on tank – the anti-flotation arrangements are to sustain the upward force, specified in 6.4.9.5.2 and there is to be no endangering plastic

deformation to the hull. Plastic deformation may occur in the fuel containment system provided it does not endanger the safe evacuation of the ship.

6.4.1.7 Measures are to be applied to ensure that scantlings required meet the structural strength provisions and are maintained throughout the design life. Measures may include, but are not limited to, material selection, coatings, corrosion additions, cathodic protection and inerting.

6.4.1.8 An inspection/survey plan for the liquefied gas fuel containment system is to be developed and approved by the IRS/Administration. The inspection/survey plan shall identify aspects to be examined and/or validated during surveys throughout the liquefied gas fuel containment system's life and, in particular, any necessary in-service survey, maintenance and testing that was assumed when selecting liquefied gas fuel containment system design parameters. The inspection/survey plan may include specific critical locations as per 6.4.12.2.8 or 6.4.12.2.9.

6.4.1.9 Liquefied gas fuel containment systems are to be designed, constructed and equipped to provide adequate means of access to areas that need inspection as specified in the inspection/survey plan. Liquefied gas fuel containment systems, including all associated internal equipment are to be designed and built to ensure safety during operations, inspection and maintenance.

6.4.2 Liquefied gas fuel containment safety principles

6.4.2.1 The containment systems are to be provided with a complete secondary liquid-tight barrier capable of safely containing all potential leakages through the primary barrier and, in conjunction with the thermal insulation system, of preventing lowering of the temperature of the ship structure to an unsafe level.

6.4.2.2 The size and configuration or arrangement of the secondary barrier may be reduced or omitted where an equivalent level of safety can be demonstrated in accordance with 6.4.2.3 to 6.4.2.5 as applicable.

6.4.2.3 Liquefied gas fuel containment systems for which the probability for structural failures to develop into a critical state has been determined to be extremely low but where the possibility of leakages through the primary barrier cannot be excluded, is to be equipped with a partial secondary barrier and small leak protection system capable of safely handling and disposing of the leakages (a critical state means that the crack develops into unstable condition). The arrangements are to comply with the following:

.1 failure developments that can be reliably detected before reaching a critical state (e.g. by gas detection or inspection) are to have a sufficiently long development time for remedial actions to be taken; and

.2 failure developments that cannot be safely detected before reaching a critical state are to have a predicted development time that is much longer than the expected lifetime of the tank.

6.4.2.4 No secondary barrier is required for liquefied gas fuel containment systems, e.g. type C independent tanks, where the probability for structural failures and leakages through the primary barrier is extremely low and can be neglected.

6.4.2.5 For independent tanks requiring full or partial secondary barrier, means for safely disposing of leakages from the tank is to be arranged.

6.4.3 Secondary barriers in relation to tank types

6.4.3.1 Secondary barriers in relation to the tank types defined in 6.4.15 are to be provided in accordance with the following table.

Basic tank type	Secondary barrier requirements
Membrane	Complete secondary barrier
Independent Type A	Complete secondary barrier
Independent Type B	Partial secondary barrier
Independent Type C	No secondary barrier required

6.4.4 Design of secondary barriers

6.4.4.1 The design of the secondary barrier, including spray shield if fitted, is to be such that:

.1 it is capable of containing any envisaged leakage of liquefied gas fuel for a period of 15 days unless different criteria apply for particular voyages, taking into account the load spectrum referred to in 6.4.12.2.6;

.2 physical, mechanical or operational events within the liquefied gas fuel tank that could cause failure of the primary barrier is not to impair the due function of the secondary barrier, or vice versa;

.3 failure of a support or an attachment to the hull structure will not lead to loss of liquid tightness of both the primary and secondary barriers;

.4 it is capable of being periodically checked for its effectiveness by means of a visual inspection or other suitable means acceptable to Administration/ IRS;

.5 the methods required in 6.4.4.1.4 is to be approved by the Administration/ IRS and is to include, as a minimum:

.1 details on the size of defect acceptable and the location within the secondary barrier, before its liquid tight effectiveness is compromised;

.2 accuracy and range of values of the proposed method for detecting defects in .1 above;

.3 scaling factors to be used in determining the acceptance criteria if full-scale model testing is not undertaken; and

.4 effects of thermal and mechanical cyclic loading on the effectiveness of the proposed test.

.6 the secondary barrier is to fulfill its functional requirements at a static angle of heel of 30°.

6.4.5 Partial secondary barriers and primary barrier small leak protection system

6.4.5.1 Partial secondary barriers as permitted in 6.4.2.3 are to be used with a small leak protection system and meet all the requirements in 6.4.4. The small leak protection system is to include means to detect a leak in the primary barrier, provision such as a spray shield to deflect any liquefied gas fuel down into the partial secondary barrier, and means to dispose of the liquid, which may be by natural evaporation.

6.4.5.2 The capacity of the partial secondary barrier is to be determined, based on the liquefied gas fuel leakage corresponding to the extent of failure resulting from the load spectrum referred to in 6.4.12.2.6, after the initial detection of a primary leak. Due account may be taken of liquid evaporation, rate of leakage, pumping capacity and other relevant factors.

6.4.5.3 The required liquid leakage detection may be by means of liquid sensors, or by an effective use of pressure, temperature or gas detection systems, or any combination thereof.

6.4.5.4 For independent tanks for which the geometry does not present obvious locations for leakage to collect, the partial secondary barrier is also to fulfill its functional requirements at a nominal static angle of trim.

6.4.6 Supporting arrangements

6.4.6.1 The liquefied gas fuel tanks is to be supported by the hull in a manner that prevents bodily movement of the tank under the static and dynamic loads defined in 6.4.9.2 to 6.4.9.5, where applicable, while allowing contraction and expansion of the tank under temperature variations and hull deflections without undue stressing of the tank and the hull.

6.4.6.2 Anti-flotation arrangements are to be provided for independent tanks and capable of withstanding the loads defined in 6.4.9.5.2 without plastic deformation likely to endanger the hull structure.

6.4.6.3 Supports and supporting arrangements are to withstand the loads defined in 6.4.9.3.3.8 and 6.4.9.5, but these loads need not be combined with each other or with wave-induced loads.

6.4.7 Associated structure and equipment

6.4.7.1 Liquefied gas fuel containment systems are to be designed for the loads imposed by associated structure and equipment. This includes pump towers, liquefied gas fuel domes, liquefied gas fuel pumps and piping, stripping pumps and piping, nitrogen piping, access hatches, ladders, piping penetrations, liquid level gauges, independent level alarm gauges, spray nozzles, and instrumentation systems (such as pressure, temperature and strain gauges).

6.4.8 Thermal insulation

6.4.8.1 Thermal insulation is to be provided as required to protect the hull from temperatures below those allowable (see 6.4.13.1.1) and limit the heat flux into the tank to the levels that can be maintained by the pressure and temperature control system applied in 6.9.

6.4.9 Design loads

6.4.9.1 General

6.4.9.1.1 This sub-section defines the design loads that are to be considered with regard to 6.4.10 to 6.4.12. This includes load categories (permanent, functional, environmental and accidental) and the description of the loads.

6.4.9.1.2 The extent to which these loads are to be considered depends on the type of tank, and is more fully detailed in the following paragraphs.

6.4.9.1.3 Tanks, together with their supporting structure and other fixtures, is to be designed taking into account relevant combinations of the loads described below.

6.4.9.2 Permanent loads

6.4.9.2.1 Gravity loads

6.4.9.2.1.1 The weight of tank, thermal insulation, loads caused by towers and other attachments are to be considered.

6.4.9.2.2 Permanent external loads

6.4.9.2.2.1 Gravity loads of structures and equipment acting externally on the tank are to be considered.

6.4.9.3 Functional loads

6.4.9.3.1 Loads arising from the operational use of the tank system are to be classified as functional loads.

6.4.9.3.2 All functional loads that are essential for ensuring the integrity of the tank system, during all design conditions, are to be considered.

6.4.9.3.3 As a minimum, the effects from the following criteria, as applicable, are to be considered when establishing functional loads:

(a) internal pressure

(b) external pressure

(c) thermally induced loads

(d) vibration

(e) interaction loads

(f) loads associated with construction and installation

(g) test loads

(h) static heel loads

(i) weight of liquefied gas fuel

(j) sloshing

(k) wind impact, wave impacts and green sea effect for tanks installed on open deck.

6.4.9.3.3.1 Internal pressure

.1 In all cases, including 6.4.9.3.3.1.2, P_0 is not to be less than MARVS.

.2 For liquefied gas fuel tanks where there is no temperature control and where the pressure of the liquefied gas fuel is dictated only by the ambient temperature, P_0 is not to be less than the gauge vapour pressure of the liquefied gas fuel at a temperature of 45 [°C] except as follows:

.1 Lower values of ambient temperature may be accepted by the IRS/Administration for ships operating in restricted areas. Conversely, higher

values of ambient temperature may be required.

.2 For ships on voyages of restricted duration, P_0 may be calculated based on the actual pressure rise during the voyage and account may be taken of any thermal insulation of the tank.

.3 Subject to special consideration by the IRS/Administration and to the limitations given in 6.4.15 for the various tank types, a vapour pressure P_h higher than P_0 may be accepted for site specific conditions (harbour or other locations), where dynamic loads are reduced.

.4 Pressure used for determining the internal pressure is to be:

.1 $(P_{gd})\max$ is the associated liquid pressure determined using the maximum design accelerations.
.2 $(P_{gd\ site})\max$ is the associated liquid pressure determined using site specific accelerations.

.3 P_{eq} to be the greater of P_{eq1} and P_{eq2} calculated as follows:

$$P_{eq1} = P_0 + (P_{gd})max \quad \text{[MPa]}$$

$$P_{eq2} = P_h + (P_{gd_site})max \quad \text{[MPa]}$$

.5 The internal liquid pressures are those created by the resulting acceleration of the centre of gravity of the liquefied gas fuel due to the motions of the ship referred to in 6.4.9.4.1.1. The value of internal liquid pressure P_{gd} resulting from combined effects of gravity and dynamic accelerations is to be calculated as follows:

$$P_{gd} = \alpha_\beta Z_\beta (\rho/(1.02 \times 10^5)) \text{ [MPa]}$$

where:

α_β = dimensionless acceleration (i.e. relative to the acceleration of gravity), resulting from gravitational and dynamic loads, in an arbitrary direction β; (see Fig.6.4.1).

For large tanks, an acceleration ellipsoid, taking account of transverse vertical and longitudinal accelerations, should be used.

Z_β = largest liquid height [m] above the point where the pressure is to be determined measured from the tank shell in the β direction (see Fig.6.4.2).

Tank domes considered to be part of the accepted total tank volume is to be taken into account when determining Z_β unless the total volume of tank domes V_d does not exceed the following value:

$$V_d = V_t \left(\frac{100 - FL}{FL}\right)$$

where:

V_t = tank volume without any domes; and
F_L = filling limit according to 6.8.

ρ = maximum liquefied gas fuel density [kg/m^3] at the design temperature.

The direction that gives the maximum value $(P_{gd})max$ or $(P_{gd\ site})max$ is to be considered. Where acceleration components in three directions need to be considered, an ellipsoid is to be used instead of the ellipse in Fig.6.4.1. The above formula applies only to full tanks.

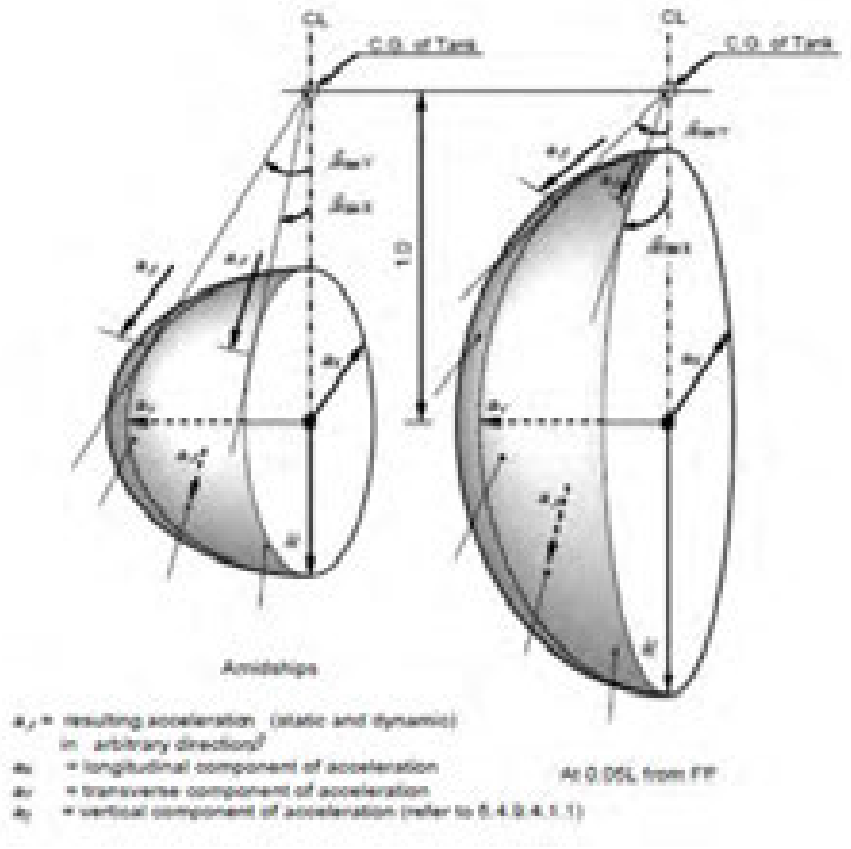


Fig.6.4.1 : Acceleration ellipsoid

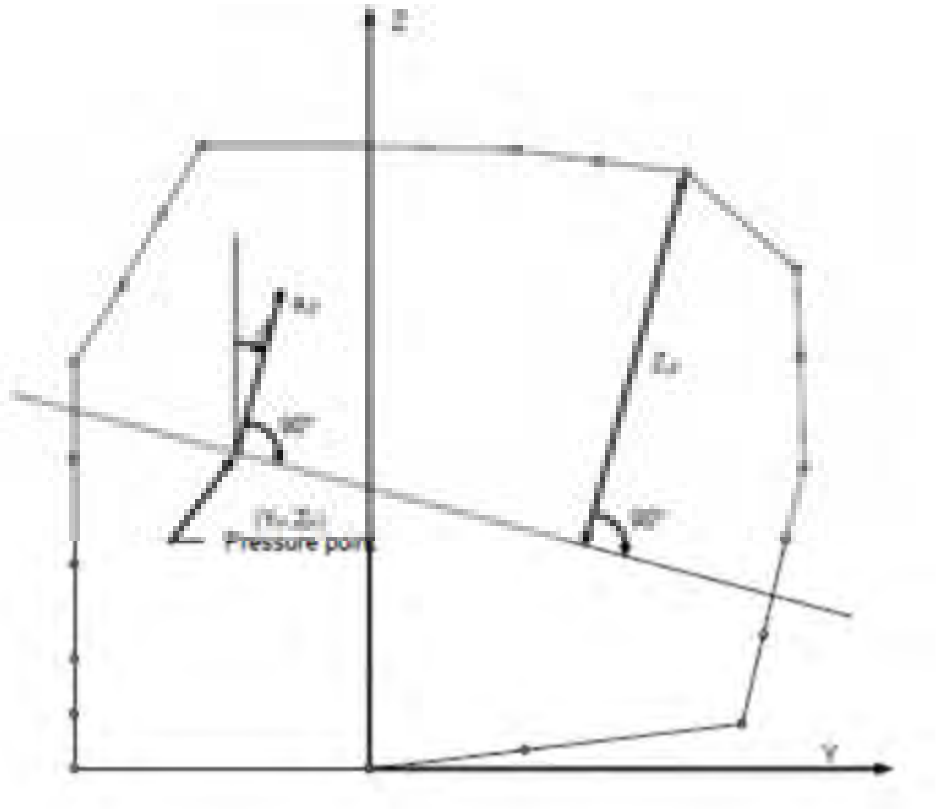


Fig.6.4.2 : Determination of internal pressure heads

6.4.9.3.3.2 External pressure

6.4.9.3.3.2.1 External design pressure loads are to be based on the difference between the minimum internal pressure and the maximum external pressure to which any portion of the tank may be simultaneously subjected.

6.4.9.3.3.3 Thermally induced loads

6.4.9.3.3.3.1 Transient thermally induced loads during cooling down periods are to be considered for tanks intended for liquefied gas fuel temperatures below minus 55 [°C].

6.4.9.3.3.3.2 Stationary thermally induced loads are to be considered for liquefied gas fuel containment systems where the design supporting arrangements or attachments and operating temperature may give rise to significant thermal stresses (see paragraph 6.9.2).

6.4.9.3.3.4 Vibration

6.4.9.3.3.4.1 The potentially damaging effects of vibration on the liquefied gas fuel containment system are to be considered.

6.4.9.3.3.5 Interaction loads

6.4.9.3.3.5.1 The static component of loads resulting from interaction between liquefied gas fuel containment system and the hull structure, as well as loads from associated structure and equipment, is to be considered.

6.4.9.3.3.6 Loads associated with construction and installation

6.4.9.3.3.6.1 Loads or conditions associated with construction and installation are to be considered, e.g. lifting.

6.4.9.3.3.7 Test loads

6.4.9.3.3.7.1 Account is to be taken of the loads corresponding to the testing of the liquefied gas fuel containment system referred to in 16.5.

6.4.9.3.3.8 Static heel loads

6.4.9.3.3.8.1 Loads corresponding to the most unfavourable static heel angle within the range 0° to 30° are to be considered.

6.4.9.3.3.9 Other loads

6.4.9.3.3.9.1 Any other loads not specifically addressed, which could have an effect on the liquefied gas fuel containment system, are to be taken into account.

6.4.9.4 Environmental loads

6.4.9.4.1 Environmental loads are defined as those loads on the liquefied gas fuel containment system that are caused by the surrounding environment and that are not otherwise classified as a permanent, functional or accidental load.

6.4.9.4.1.1 Loads due to ship motion

6.4.9.4.1.1.1 The determination of dynamic loads is to take into account the long-term distribution of ship motion in irregular seas, which the ship will experience during its operating life. Account may be taken of the reduction in dynamic loads due to necessary speed reduction and variation of heading. The ship's motion shall include surge, sway, heave, roll, pitch and yaw. The accelerations acting on tanks are to be estimated at their centre of gravity and include the following components:

.1 vertical acceleration: motion accelerations of heave, pitch and, possibly roll (normal to the ship base);

.2 transverse acceleration: motion accelerations of sway, yaw and roll and gravity component of roll; and

.3 longitudinal acceleration: motion accelerations of surge and pitch and gravity component of pitch.

Methods to predict accelerations due to ship motion are to be proposed and approved by the IRS/Administration.*

Ships for restricted service may be given special consideration.

Note:

**Refer to 4.28.2.1 of Part5, Ch 4 for guidance formulae for acceleration components.*

6.4.9.4.1.2 Dynamic interaction loads

6.4.9.4.1.2.1 Account is to be taken of the dynamic component of loads resulting from interaction between liquefied gas fuel containment systems and the hull structure,

including loads from associated structures and equipment.

6.4.9.4.1.3 Sloshing loads

6.4.9.4.1.3.1 The sloshing loads on a liquefied gas fuel containment system and internal components are to be evaluated for the full range of intended filling levels.

6.4.9.4.1.4 Snow and ice loads

6.4.9.4.1.4.1 Snow and icing is to be considered, if relevant.

6.4.9.4.1.5 Loads due to navigation in ice

6.4.9.4.1.5.1 Loads due to navigation in ice are to be considered for ships intended for such service.

6.4.9.4.1.6 Green sea loading

6.4.9.4.1.6.1 Account is to be taken to loads due to water on deck.

6.4.9.4.1.7 Wind loads

6.4.9.4.1.7.1 Account is to be taken to wind generated loads as relevant.

6.4.9.5 Accidental loads

6.4.9.5.1 Accidental loads are defined as loads that are imposed on a liquefied gas fuel containment system and it's supporting arrangements under abnormal and unplanned conditions.

6.4.9.5.1 Collision load

6.4.9.5.1.1 The collision load is to be determined based on the fuel containment system under fully loaded condition with an inertial force corresponding to "a" in the Table below in forward direction and "a/2" in the aft direction, where "g" is gravitational acceleration.

Ship length (L)	Design acceleration (a)
L > 100 m	0.5 g
60 < L ≤ 100 m	$\left(2-\frac{3(L-60)}{80}\right)g$
L ≤ 60 m	2g

Special consideration should be given to ships with Froude number (Fn) > 0.4.

6.4.9.5.2 Loads due to flooding on ship

6.4.9.5.2.1 For independent tanks, loads caused by the buoyancy of a fully submerged empty tank are to be considered in the design of anti-flotation chocks and the supporting structure in both the adjacent hull and tank structure.

6.4.10 Structural integrity

6.4.10.1 General

6.4.10.1.1 The structural design is to ensure that tanks have an adequate capacity to sustain all relevant loads with an adequate margin of safety. This is to take into account the possibility of plastic deformation, buckling, fatigue and loss of liquid and gas tightness.

6.4.10.1.2 The structural integrity of liquefied gas fuel containment systems can be demonstrated by compliance with 6.4.15, as appropriate for the liquefied gas fuel containment system type.

6.4.10.1.3 For other liquefied gas fuel containment system types, that are of novel design or differ significantly from those covered by 6.4.15, the structural integrity is to be demonstrated by compliance with 6.4.16.

6.4.11 Structural analysis

6.4.11.1 Analysis

6.4.11.1.1 The design analyses are to be based on accepted principles of statics, dynamics and strength of materials.

6.4.11.1.2 Simplified methods or simplified analyses may be used to calculate the load effects, provided that they are conservative. Model tests may be used in combination with, or instead of, theoretical calculations. In cases where theoretical methods are inadequate, model or full-scale tests may be required.

6.4.11.1.3 When determining responses to dynamic loads, the dynamic effect is to be taken into account where it may affect structural integrity.

6.4.11.2 Load scenarios

6.4.11.2.1 For each location or part of the liquefied gas fuel containment system to be considered and for each possible mode of failure to be analysed, all relevant combinations of loads that may act simultaneously are to be considered.

6.4.11.2.2 The most unfavourable scenarios for all relevant phases during construction, handling, testing and in service conditions are to be considered.

6.4.11.2.3 When the static and dynamic stresses are calculated separately and unless other methods of calculation are justified, the total stresses are to be calculated according to:

$$\sigma_x = \sigma_{x.st} \pm \sqrt{\sum \left(\sigma_{x.dyn}\right)^2}$$

$$\sigma_y = \sigma_{y.st} \pm \sqrt{\sum \left(\sigma_{y.dyn}\right)^2}$$

$$\sigma_z = \sigma_{z.st} \pm \sqrt{\sum \left(\sigma_{z.dyn}\right)^2}$$

$$\tau_{xy} = \tau_{xy.st} \pm \sqrt{\sum \left(\tau_{xy.dyn}\right)^2}$$

$$\tau_{xz} = \tau_{xz.st} \pm \sqrt{\sum \left(\tau_{xz.dyn}\right)^2}$$

$$\tau_{yz} = \tau_{yz.st} \pm \sqrt{\sum \left(\tau_{yz.dyn}\right)^2}$$

where:

$\sigma_{x,st}$, $\sigma_{y,st}$, $\sigma_{z,st}$, $\tau_{xy,st}$, $\tau_{xz,\,st}$, and $\tau_{yz,st}$, are static stresses; and

$\sigma_{x,dyn}$, $\sigma_{y,dyn}$, $\sigma_{z,dyn}$, $\tau_{xy,dyn}$, $\tau_{xz,\,dyn}$, and $\tau_{yz,dyn}$, are dynamic stresses,

each is to be determined separately from acceleration components and hull strain components due to deflection and torsion.

6.4.12 Design conditions

All relevant failure modes are to be considered in the design for all relevant load scenarios and design conditions. The design conditions are given in the earlier part of this chapter, and the load scenarios are covered by 6.4.11.2.

6.4.12.1 Ultimate design condition

6.4.12.1.1 Structural capacity may be determined by testing, or by analysis, taking into account both the elastic and plastic material properties, by simplified linear elastic analysis or by the provisions of this Chapter:

.1 Plastic deformation and buckling are to be considered.

.2 Analysis is to be based on characteristic load values as follows:

Permanent loads	Expected values
Functional loads	Specified values
Environmental loads	For wave loads: most probable largest load encountered during 10^8 wave encounters

.3 For the purpose of ultimate strength assessment the following material parameters apply:

.1 R_e = specified minimum yield stress at room temperature [N/mm^2]. If the stress-strain curve does not show a defined yield stress, the 0.2% proof stress applies.

.2 R_m = specified minimum tensile strength at room temperature [N/mm^2].

For welded connections where under-matched welds, i.e. where the weld metal has lower tensile strength than the parent metal, are unavoidable, such as in some aluminium alloys, the respective R_e and R_m of the welds, after any applied heat treatment, is to be used. In such cases the transverse weld tensile strength is not to be less than the actual yield strength of the parent metal. If this cannot be achieved, welded structures made from such materials are not to be incorporated in liquefied gas fuel containment systems.

The above properties shall correspond to the minimum specified mechanical properties of the material, including the weld metal in the as fabricated condition. Subject to special consideration by the

IRS/Administration, account may be taken of the enhanced yield stress and tensile strength at low temperature.

.4 The equivalent stress σ_c (von Mises, Huber) is to be determined by:

$$\sigma_c = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - \sigma_x\sigma_y - \sigma_x\sigma_z - \sigma_y\sigma_z + 3\left(\tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2\right)}$$

where:

σ_x = total normal stress in x-direction;
σ_y = total normal stress in y-direction;
σ_z = total normal stress in z-direction;
τ_{xy} = total shear stress in x-y plane;
τ_{xz} = total shear stress in x-z plane; and
τ_{yz} = total shear stress in y-z plane.

The above values are to be calculated as described in 6.4.11.2.3.

.5 Allowable stresses for materials other than those covered by 7.4 is to be subject to approval by the Administration in each case.

.6 Stresses may be further limited by fatigue analysis, crack propagation analysis and buckling criteria.

6.4.12.2 Fatigue Design Condition

6.4.12.2.1 The fatigue design condition is the design condition with respect to accumulated cyclic loading.

6.4.12.2.2 Where a fatigue analysis is required the cumulative effect of the fatigue load shall comply with:

$$\sum \frac{n_i}{N_i} + \frac{n_{Loading}}{N_{Loading}} \leq C_w$$

where:

n_i = number of stress cycles at each stress level during the life of the tank;

N_i = number of cycles to fracture for the respective stress level according to the Wohler (S-N) curve;

$n_{Loading}$ = number of loading and unloading cycles during the life of the tank not to be less than 1000. Loading and unloading cycles include a complete pressure and thermal cycle;

$N_{Loading}$ = number of cycles to fracture for the fatigue loads due to loading and unloading; and

C_w = maximum allowable cumulative fatigue damage ratio.

The fatigue damage is to be based on the design life of the tank but not less than 10^8 wave encounters.

6.4.12.3 Where required, the liquefied gas fuel containment system is to be subject to fatigue analysis, considering all fatigue loads and their appropriate combinations for the expected life of the liquefied gas fuel containment system. Consideration is to be given to various filling conditions.

6.4.12.4 Design S-N curves used in the analysis is to be applicable to the materials and weldments, construction details, fabrication procedures and applicable state of the stress envisioned.

The S-N curves are to be based on a 97.6% probability of survival corresponding to the mean-minus-two-standard-deviation curves of relevant experimental data up to final failure. Use of S-N curves derived in a different way requires adjustments to the acceptable Cw values specified in 6.4.12.2.7 to 6.4.12.2.9.

6.4.12.5 Analysis is to be based on characteristic load values as follows:

Permanent loads	Expected values
Functional loads	Specified values or specified history
Environmental loads	Expected load history, but not less than 10^8 cycles

If simplified dynamic loading spectra are used for the estimation of the fatigue life, those are to be specially considered by the IRS/Administration.

6.4.12.6 Where the size of the secondary barrier is reduced, as is provided for in 6.4.2.3, fracture mechanics analyses of fatigue crack growth is to be carried out to determine:

.1 crack propagation paths in the structure, where necessitated by 6.4.12.2.7 to 6.4.12.2.9, as applicable;

.2 crack growth rate;

.3 the time required for a crack to propagate to cause a leakage from the tank;

.4 the size and shape of through thickness cracks; and

.5 the time required for detectable cracks to reach a critical state after penetration through the thickness.

The fracture mechanics are in general based on crack growth data taken as a mean value plus two standard deviations of the test data. Methods for fatigue crack growth analysis and fracture mechanics is to be based on recognized standards.

In analysing crack propagation the largest initial crack not detectable by the inspection method applied is to be assumed, taking into account the allowable non-destructive testing and visual inspection criterion as applicable.

Crack propagation analysis specified in 6.4.12.2.7 the simplified load distribution and sequence over a period of 15 days may be used. Such distributions may be obtained as indicated in Fig.6.4.3. Load distribution and sequence for longer periods, such as in 6.4.12.2.8 and 6.4.12.2.9 is to be approved by the IRS/Administration.

The arrangements are to comply with 6.4.12.2.7 to 6.4.12.2.9 as applicable.

6.4.12.2.7 For failures that can be reliably detected by means of leakage detection:

C_w is to be less than or equal to 0.5.

Predicted remaining failure development time, from the point of detection of leakage till reaching a critical state, is not to be less than 15 days unless different regulations apply for ships engaged in particular voyages.

6.4.12.2.8 For failures that cannot be detected by leakage but that can be reliably detected at the time of in-service inspections:

C_w is to be less than or equal to 0.5.

Predicted remaining failure development time, from the largest crack not detectable by in-service inspection methods until reaching a critical state, is not to be less than three (3) times the inspection interval.

6.4.12.2.9 In particular locations of the tank where effective defect or crack development detection cannot be assured, the following, more stringent, fatigue acceptance criteria is to be applied as a minimum:

C_w is to be less than or equal to 0.1.

Predicted failure development time, from the assumed initial defect until reaching a critical state, is not to be less than three (3) times the lifetime of the tank.

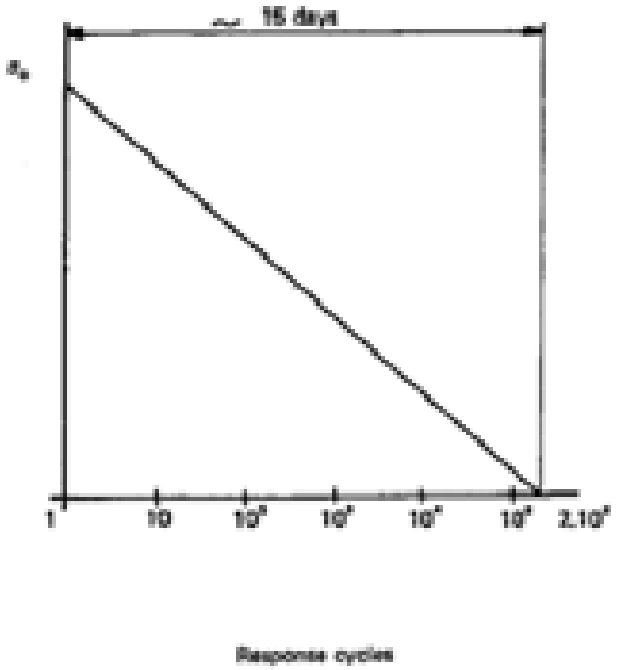
6.4.12.3 Accidental design condition

6.4.12.3.1 The accidental design condition is a design condition for accidental loads with extremely low probability of occurrence.

6.4.12.3.2 Analysis is to be based on the characteristic values as follows:

Permanent loads	Expected values
Functional loads	Specified values
Environmental loads	Specified values
Accidental loads	Specified values or expected values

Loads mentioned in 6.4.9.3.3.8 and 6.4.9.5 need not be combined with each other or with wave-induced loads.



σ_0 = most probable maximum stress over the life of the ship

Response cycle scale is logarithmic; the value of 2.10^8 is given as an example of estimate.

Fig.6.4.3 : Simplified load distribution

6.4.13 Materials and construction

6.4.13.1 Materials

6.4.13.1.1 Materials forming ship structure

6.4.13.1.1.1 To determine the grade of plate and sections used in the hull structure, a temperature calculation is to be performed for all tank types. The following assumptions are to be made in this calculation:

.1 The primary barrier of all tanks is to be assumed to be at the liquefied gas fuel temperature.

.2 In addition to .1 above, where a complete or partial secondary barrier is required it is to be assumed to be at the liquefied gas fuel temperature at atmospheric pressure for any one tank only.

.3 For worldwide service, ambient temperatures is to be taken as 5 [°C] for air and 0 [°C] for seawater. Higher values may be accepted for ships operating in restricted areas and conversely, lower values may be imposed by the IRS/Administration for ships trading to areas where lower temperatures are expected during the winter months.

.4 Still air and sea water conditions are to be assumed, i.e. no adjustment for forced convection.

.5 Degradation of the thermal insulation properties over the life of the ship due to factors such as thermal and mechanical ageing, compaction, ship motions and tank vibrations as defined in 6.4.13.3.6 and 6.4.13.3.7 is to be assumed.

.6 The cooling effect of the rising boil-off vapour from the leaked liquefied gas fuel is to be taken into account where applicable.

.7 Credit for hull heating may be taken in accordance with 6.4.13.1.1.3, provided the heating arrangements are in compliance with 6.4.13.1.1.4.

.8 No credit is to be given for any means of heating, except as described in 6.4.13.1.1.3.

.9 For members connecting inner and outer hulls, the mean temperature may be taken for determining the steel grade.

6.4.13.1.1.2 The materials of all hull structures for which the calculated temperature in the design condition is below 0 [°C], due to the influence of liquefied gas fuel temperature, is to be in accordance with Table 7.5. This includes hull structure supporting the liquefied gas fuel tanks, inner bottom plating, longitudinal bulkhead plating, transverse bulkhead plating, floors, webs, stringers and all attached stiffening members.

6.4.13.1.1.3 Means of heating structural materials may be used to ensure that the material temperature does not fall below the minimum allowed for the grade of material specified in Table 7.5. In the calculations required in 6.4.13.1.1.1, credit for such heating may be taken in accordance with the following principles:

.1 for any transverse hull structure;

.2 for longitudinal hull structure referred to in 6.4.13.1.1.2 where colder ambient temperatures are specified, provided the material remains suitable for the ambient temperature conditions of plus 5[°C] for air and 0[°C] for seawater with no credit taken in the calculations for heating; and

.3 as an alternative to 6.4.13.1.1.3.2, for longitudinal bulkhead between liquefied gas fuel tanks, credit may be taken for heating provided the material remain suitable for a minimum design temperature of minus 30[°C], or a temperature 30[°C] lower than that determined by 6.4.13.1.1.1 with the heating considered, whichever is less. In this case, the ship's longitudinal strength shall comply with SOLAS regulation II-1/3-1 for both when those bulkhead(s) are considered effective and not.

6.4.13.1.1.4 The means of heating referred to in 6.4.13.1.1.3 is to comply with the following:

.1 the heating system is to be arranged so that, in the event of failure in any part of the system, standby heating can be maintained equal to no less than 100% of the theoretical heat requirement;

.2 the heating system is to be considered as an essential auxiliary. All electrical components of at least one of the systems provided in accordance with 6.4.13.1.1.3.1 is to be supplied from the emergency source of electrical power; and

.3 the design and construction of the heating system is to be included in the approval of the containment system by the IRS/Administration.

6.4.13.2 Materials of primary and secondary barriers

6.4.13.2.1 Metallic materials used in the construction of primary and secondary barriers not forming the hull, is to be suitable for the design loads that they may be subjected to, and be in accordance with Table 7.1, 7.2 or 7.3.

6.4.13.2.2 Materials, either non-metallic or metallic but not covered by Tables 7.1, 7.2 and 7.3, used in the primary and secondary barriers may be approved by the Administration considering the design loads that they may be subjected to, their properties and their intended use.

6.4.13.2.3 Where non-metallic materials, including composites, are used for or incorporated in the primary or secondary barriers, they is to be tested for the following properties, as applicable, to ensure that they are adequate for the intended service:

.1 compatibility with the liquefied gas fuels;

.2 ageing;

.3 mechanical properties;

.4 thermal expansion and contraction;

.5 abrasion;

.6 cohesion;

.7 resistance to vibrations;

.8 resistance to fire and flame spread; and

.9 resistance to fatigue failure and crack propagation.

6.4.13.2.4 The above properties, where applicable, is to be tested for the range between the expected maximum temperature in service and 5[°C] below the minimum design temperature, but not lower than -196[°C].

6.4.13.2.5 Where non-metallic materials, including composites, are used for the primary and secondary barriers, the joining processes shall also be tested as described above.

6.4.13.2.6 Consideration may be given to the use of materials in the primary and secondary barrier, which are not resistant to fire and flame spread, provided they are protected by a suitable system such as a permanent inert gas environment, or are provided with a fire retardant barrier.

6.4.13.3 Thermal insulation and other materials used in liquefied gas fuel containment systems

6.4.13.3.1 Load-bearing thermal insulation and other materials used in liquefied gas fuel containment systems are to be suitable for the design loads.

6.4.13.3.2 Thermal insulation and other materials used in liquefied gas fuel containment systems shall have the following properties, as applicable, to ensure that they are adequate for the intended service:

.1 compatibility with the liquefied gas fuels;

.2 solubility in the liquefied gas fuel;

.3 absorption of the liquefied gas fuel;

.4 shrinkage;

.5 ageing;

.6 closed cell content;

.7 density;

.8 mechanical properties, to the extent that they are subjected to liquefied gas fuel and other loading effects, thermal expansion and contraction;

.9 abrasion;

.10 cohesion;

.11 thermal conductivity;

.12 resistance to vibrations;

.13 resistance to fire and flame spread; and

.14 resistance to fatigue failure and crack propagation.

6.4.13.3.3 The above properties, where applicable, is to be tested for the range between the expected maximum temperature in service and 5[°C] below the minimum design temperature, but not lower than -196[°C].

6.4.13.3.4 Due to location or environmental conditions, thermal insulation materials are to have suitable properties of resistance to fire and flame spread and is to be adequately protected against penetration of water vapour and mechanical damage. Where the thermal insulation is located on or above the exposed deck, and in way of tank cover penetrations, it shall have suitable fire resistance properties in accordance with a recognized standard or be covered with a material having low flame spread characteristics and forming an efficient approved vapour seal.

6.4.13.3.5 Thermal insulation that does not meet recognized standards for fire resistance may be used in fuel storage hold spaces that are not kept permanently inerted, provided its surfaces are covered with material with low flame spread characteristics and that forms an efficient approved vapour seal.

6.4.13.3.6 Testing for thermal conductivity of thermal insulation is to be carried out on suitably aged samples.

6.4.13.3.7 Where powder or granulated thermal insulation is used, measures are to be taken to reduce compaction in service and to maintain the required thermal conductivity and also prevent any undue increase of pressure on the liquefied gas fuel containment system.

6.4.14 Construction processes

6.4.14.1 Weld joint design

6.4.14.1.1 All welded joints of the shells of independent tanks are to be of the in-plane butt weld full penetration type. For dome-to-shell connections only, tee welds of the full penetration type may be used depending on the results of the tests carried out at the approval of the welding procedure. Except for small penetrations on domes, nozzle welds are also to be designed with full penetration.

6.4.14.1.2 Welding joint details for type C independent tanks, and for the liquid-tight primary barriers of type B independent tanks primarily constructed of curved surfaces, is to be as follows:

.1 All longitudinal and circumferential joints is to be of butt welded, full penetration, double vee or single vee type. Full penetration butt welds is to be obtained by double welding or by the use of backing rings. If used, backing rings is to be removed except from very small process pressure vessels*. Other edge preparations may be permitted, depending on the results of the tests carried out at the approval of the welding procedure. For connections of tank shell to a longitudinal bulkhead of type C bilobe tanks, tee welds of the full penetration type may be accepted.

Note:

** For vacuum insulated tanks without manhole, the longitudinal and circumferential joints should meet the aforementioned requirements, except for the erection weld joint of the outer shell, which may be a one-side welding with backing ring.*

.2 The bevel preparation of the joints between the tank body and domes and between domes and relevant fittings is to be designed according to a standard acceptable to the Administration. All welds connecting nozzles, domes or other penetrations of the vessel and all welds connecting flanges to the vessel or nozzles is to be full penetration welds.

6.4.14.2 Design for gluing and other joining processes

6.4.14.2.1 The design of the joint to be glued (or joined by some other process except welding) shall take account of the strength characteristics of the joining process.

6.4.15 Tank types

6.4.15.1 Type A independent tanks

6.4.15.1.1 Design basis

6.4.15.1.1.1 Type A independent tanks are tanks primarily designed using classical ship-structural analysis procedures in accordance with the requirements of the IRS/Administration. Where such tanks are primarily constructed of plane surfaces, the design vapour pressure P_0 is to be less than 0.07 [MPa].

6.4.15.1.1.2 A complete secondary barrier is required as defined in 6.4.3. The secondary barrier is to be designed in accordance with 6.4.4.

6.4.15.1.2 Structural analysis

6.4.15.1.2.1 A structural analysis is to be performed taking into account the internal pressure as indicated in 6.4.9.3.3.1, and the interaction loads with the supporting and keying system as well as a reasonable part of the ship's hull.

6.4.15.1.2.2 For parts, such as structure in way of supports, not otherwise covered by the requirements in this chapter, stresses is to be determined by direct calculations, taking into account the loads referred to in 6.4.9.2 to 6.4.9.5 as far as applicable, and the ship deflection in way of supports.

6.4.15.1.2.3 The tanks with supports are to be designed for the accidental loads specified in 6.4.9.5. These loads need not be combined with each other or with environmental loads.

6.4.15.1.3 Ultimate design condition

6.4.15.1.3.1 For tanks primarily constructed of plane surfaces, the nominal membrane stresses for primary and secondary members (stiffeners, web frames, stringers, girders), when calculated by classical analysis procedures, shall not exceed the lower of Rm/2.66 or Re/1.33 for nickel steels, carbon-manganese steels, austenitic steels and aluminium alloys, where Rm and Re are defined in 6.4.12.1.1.3. However, if detailed calculations are carried out for the primary members, the equivalent stress σ_c, as defined

in 6.4.12.1.1.4, may be increased over that indicated above to a stress acceptable to the IRS/Administration. Calculations are to take into account the effects of bending, shear, axial and torsional deformation as well as the hull/liquefied gas fuel tank interaction forces due to the deflection of the hull structure and liquefied gas fuel tank bottoms.

6.4.15.1.3.2 Tank boundary scantlings are to meet at least the requirements of the IRS/Administration for deep tanks taking into account the internal pressure as indicated in 6.4.9.3.3.1 and any corrosion allowance required by 6.4.1.7.

6.4.15.1.3.3 The liquefied gas fuel tank structure is to be reviewed against potential buckling.

6.4.15.1.4 Accidental design condition

6.4.15.1.4.1 The tanks and the tank supports are to be designed for the accidental loads and design conditions specified in 6.4.9.5 and 6.4.1.6.3 as relevant.

6.4.15.1.4.2 When subjected to the accidental loads specified in 6.4.9.5, the stress is to comply with the acceptance criteria specified in 6.4.15.1.3, modified as appropriate taking into account their lower probability of occurrence.

6.4.15.2 Type B independent tanks

6.4.15.2.1 Design basis

6.4.15.2.1.1 Type B independent tanks are tanks designed using model tests, refined analytical tools and analysis methods to determine stress levels, fatigue life and crack propagation characteristics. Where such tanks are primarily constructed of plane surfaces (prismatic tanks) the design vapour pressure P_0 is to be less than 0.07 [MPa].

6.4.15.2.1.2 A partial secondary barrier with a protection system is required as defined in 6.4.3. The small leak protection system is to be designed according to 6.4.5.

6.4.15.2.2 Structural analysis

6.4.15.2.2.1 The effects of all dynamic and static loads are to be used to determine the suitability of the structure with respect to:

.1 plastic deformation;

.2 buckling;

.3 fatigue failure; and

.4 crack propagation.

Finite element analysis or similar methods and fracture mechanics analysis or an equivalent approach, is to be carried out.

6.4.15.2.2.2 A three-dimensional analysis is to be carried out to evaluate the stress levels, including interaction with the ship's hull. The model for this analysis is to include the liquefied gas fuel tank with its supporting and keying system, as well as a reasonable part of the hull.

6.4.15.2.2.3 A complete analysis of the particular ship accelerations and motions in irregular waves, and of the response of the ship and its liquefied gas fuel tanks to these forces and motions is to be performed unless the data is available from similar ships.

6.4.15.2.3 Ultimate design condition

6.4.15.2.3.1 Plastic deformation

For type B independent tanks, primarily constructed of bodies of revolution, the allowable stresses are not to exceed:

σ_m	$\leq f$
σ_L	$\leq 1.5f$
σ_b	$\leq 1.5F$
$\sigma_L+\sigma_b$	$\leq 1.5F$
$\sigma_m+\sigma_b$	$\leq 1.5F$
$\sigma_m+\sigma_b+\sigma_g$	$\leq 3.0F$
$\sigma_L+\sigma_b+\sigma_g$	$\leq 3.0F$

where:

σ_m = equivalent primary general membrane stress;

σ_L = equivalent primary local membrane stress;

σ_b = equivalent primary bending stress;

σ_g = equivalent secondary stress;

f = the lesser of (R_m / A) or (R_e / B); and

F = the lesser of (R_m / C) or (R_e / D),

with R_m and R_e as defined in 6.4.12.1.1.3. With regard to the stresses σ_m, σ_L, σ_g and σ_b see

also the definition of stress categories in 6.4.15.2.3.6.

The values A and B are to have at least the following minimum values:

	Nickel steels and carbon manganese steel	**Austenitic steel**	**Aluminium alloys**
A	3	3.5	4
B	2	1.6	1.5
C	3	3	3
D	1.5	1.5	1.5

The above figures may be altered considering the design condition considered in acceptance with the IRS/Administration. For type B independent tanks, primarily constructed of plane surfaces, the allowable membrane equivalent stresses applied for finite element analysis is not to exceed:

.1 for nickel steels and carbon-manganese steels, the lesser of R_m /2 or R_e /1.2;

.2 for austenitic steels, the lesser of R_m /2.5 or R_e /1.2; and

.3 for aluminium alloys, the lesser of R_m /2.5 or R_e /1.2.

The above figures may be amended considering the locality of the stress, stress analysis methods and design condition considered in acceptance with the IRS/Administration.

The thickness of the skin plate and the size of the stiffener shall not be less than those required for type A independent tanks.

6.4.15.2.3.2 Buckling

.1 Buckling strength analyses of liquefied gas fuel tanks subject to external pressure and other loads causing compressive stresses is to be carried out in accordance with recognized standards. The method is to adequately account for the difference in theoretical and actual buckling stress as a result of plate edge misalignment, lack of straightness or flatness, ovality and deviation from true circular form over a specified arc or chord length, as applicable.

6.4.15.2.3.3 Fatigue design condition

6.4.15.2.3.3.1 Fatigue and crack propagation assessment is to be performed in accordance with the provisions of 6.4.12.2. The acceptance criteria shall comply with 6.4.12.2.7, 6.4.12.2.8 or 6.4.12.2.9, depending on the detectability of the defect.

6.4.15.2.3.3.2 Fatigue analysis is to consider construction tolerances.

6.4.15.2.3.3.3 Where deemed necessary by the IRS/Administration, model tests may be required to determine stress concentration factors and fatigue life of structural elements.

6.4.15.2.3.4 Accidental design condition

6.4.15.2.3.4.1 The tanks and the tank supports are to be designed for the accidental loads and design conditions specified in 6.4.9.5 and 6.4.1.6.3, as relevant.

6.4.15.2.3.4.2 When subjected to the accidental loads specified in 6.4.9.5, the stress shall comply with the acceptance criteria specified in 6.4.15.2.3, modified as appropriate, taking into account their lower probability of occurrence.

6.4.15.2.3.5 Marking

.1 Any marking of the pressure vessel is to be achieved by a method that does not cause unacceptable local stress raisers.

6.4.15.2.3.6 Stress categories

For the purpose of stress evaluation, stress categories are defined in this section as follows:

.1 Normal stress is the component of stress normal to the plane of reference.

.2 Membrane stress is the component of normal stress that is uniformly distributed and equal to the average value of the stress across the thickness of the section under consideration.

.3 Bending stress is the variable stress across the thickness of the section

under consideration, after the subtraction of the membrane stress.

.4 Shear stress is the component of the stress acting in the plane of reference.

.5 Primary stress is a stress produced by the imposed loading, which is necessary to balance the external forces and moments. The basic characteristic of a primary stress is that it is not self-limiting. Primary stresses that considerably exceed the yield strength will result in failure or at least in gross deformations.

.6 Primary general membrane stress is a primary membrane stress that is so distributed in the structure that no redistribution of load occurs as a result of yielding.

.7 Primary local membrane stress arises where a membrane stress produced by pressure or other mechanical loading and associated with a primary or a discontinuity effect produces excessive distortion in the transfer of loads for other portions of the structure. Such a stress is classified as a primary local membrane stress, although it has some characteristics of a secondary stress. A stress region may be considered as local, if:

$S_1 \leq 0.5\sqrt{Rt}$; and

$S_1 \geq 2.5\sqrt{Rt}$

where:

S_1 = distance in the meridional direction over which the equivalent stress exceeds $1.1f$;

S_2 = distance in the meridional direction to another region where the limits for primary general membrane stress are exceeded;

R = mean radius of the vessel;

t = wall thickness of the vessel at the location where the primary general membrane stress limit is exceeded; and

f = allowable primary general membrane stress.

.8 *Secondary stress* is a normal stress or shear stress developed by constraints of adjacent parts or by self-constraint of a structure. The basic characteristic of a secondary stress is that it is self-limiting. Local yielding and minor distortions can satisfy the conditions that cause the stress to occur.

6.4.15.3 Type C independent tanks

6.4.15.3.1 Design basis

6.4.15.3.1.1 The design basis for type C independent tanks is based on pressure vessel criteria modified to include fracture mechanics and crack propagation criteria. The minimum design pressure defined in 6.4.15.3.1.2 is intended to ensure that the dynamic stress is sufficiently low so that an initial surface flaw will not propagate more than half the thickness of the shell during the lifetime of the tank.

6.4.15.3.1.2 The design vapour pressure is not to be less than:

$$P_0 = 0.2 + AC(\rho_r)^{1.5} \text{ [MPa]}$$

where:

$$A = 0.00185\left(\frac{\sigma_m}{\Delta\sigma_A}\right)^2$$

with:

σ_m = design primary membrane stress;
$\Delta\sigma_A$ = allowable dynamic membrane stress (double amplitude at probability level Q = 10^{-8}) and equal to:

- 55 [N/mm^2] for ferritic-perlitic, martensitic and austenitic steel;

- 25 [N/mm^2] for aluminium alloy (5083-O);

C = a characteristic tank dimension to be taken as the greatest of the following:

h, 0.75*b* or 0.*45ℓ,*

with:

h = height of tank (dimension in ship's vertical direction) (m);

b = width of tank (dimension in ship's transverse direction) (m);

ℓ = length of tank (dimension in ship's longitudinal direction) (m);

ρ_r = the relative density of the cargo (ρ_r = 1 for fresh water) at the design temperature.

6.4.15.3.2 Shell thickness

6.4.15.3.2.1 In considering the shell thickness the following apply:

.1 for pressure vessels, the thickness calculated according to 6.4.15.3.2.4 is to be considered as a minimum thickness after forming, without any negative tolerance;

.2 for pressure vessels, the minimum thickness of shell and heads including corrosion allowance, after forming, shall not be less than 5 [mm] for carbon manganese steels and nickel steels, 3 [mm] for austenitic steels or 7 [mm] for aluminium alloys; and

.3 the welded joint efficiency factor to be used in the calculation according to 6.4.15.3.2.4 is to be 0.95 when the inspection and the non-destructive testing referred to in 16.3.6.4 are carried out. This figure may be increased up to 1 when account is taken of other considerations, such as the material used, type of joints, welding procedure and type of loading. For process pressure vessels IRS/Administration may accept partial non-destructive examinations, but not less than those of 16.3.6.4, depending on such factors as the material used, the design temperature, the nil ductility transition temperature of the material as fabricated and the type of joint and welding procedure, but in this case an efficiency factor of not more than 0.85 is to be adopted. For special materials the above-mentioned factors is to be reduced, depending on the specified mechanical properties of the welded joint.

6.4.15.3.2.2 The design liquid pressure defined in 6.4.9.3.3.1 is to be taken into account in the internal pressure calculations.

6.4.15.3.2.3 The design external pressure P_e, used for verifying the buckling of the pressure vessels, is not to be less than that given by:

$$P_e = P_1+P_2+P_3+P_4 \text{ [MPa]}$$

where:

P_1 = setting value of vacuum relief valves. For vessels not fitted with vacuum relief valves P_1 is to be specially considered, but shall not in general be taken as less than 0.025 [MPa].

P_2 = the set pressure of the pressure relief valves (PRVs) for completely closed spaces containing pressure vessels or parts of pressure vessels; elsewhere P_2 = 0.

P_3 = compressive actions in or on the shell due to the weight and contraction of thermal insulation, weight of shell including corrosion allowance and other miscellaneous external pressure loads to which the pressure vessel may be subjected. These include, but are not limited to, weight of domes, weight of towers and piping, effect of product in the partially filled condition, accelerations and hull deflection. In addition, the local effect of external or internal pressures or both is to be taken into account.

P_4 = external pressure due to head of water for pressure vessels or part of pressure vessels on exposed decks; elsewhere P_4 = 0.

6.4.15.3.2.4 Scantlings based on internal pressure are to be calculated as follows:

The thickness and form of pressure-containing parts of pressure vessels, under internal pressure, as defined in 6.4.9.3.3.1, including flanges, is to be determined. These calculations shall in all cases be based on accepted pressure vessel design theory. Openings in pressure-containing parts of pressure vessels are to be reinforced in accordance with a recognized standard acceptable to IRS/Administration.

6.4.15.3.2.5 Stress analysis in respect of static and dynamic loads is to be performed as follows:

.1 pressure vessel scantlings are to be determined in accordance with 6.4.15.3.2.1 to 6.4.15.3.2.4 and 6.4.15.3.3;

.2 calculations of the loads and stresses in way of the supports and the shell attachment of the support are to be made. Loads referred to in 6.4.9.2 to 6.4.9.5 are to be used, as applicable. Stresses in way of the supports are to be to a recognized standard acceptable to the Administration. In special cases a fatigue analysis may be required by the Administration; and

.3 if required by IRS/Administration, secondary stresses and thermal stresses is to be specially considered.

6.4.15.3.3 Ultimate design condition

6.4.15.3.3.1 Plastic deformation

For type C independent tanks, the allowable stresses are not to exceed:

$\sigma_m \leq f$
$\sigma_L \leq 1.5f$
$\sigma_b \leq 1.5f$
$\sigma_L+\sigma_b \leq 1.5f$
$\sigma_m+\sigma_b \leq 1.5f$
$\sigma_m+\sigma_b+\sigma_g \leq 3.0f$
$\sigma_L+\sigma_b+\sigma_g \leq 3.0f$

where:

σ_m = equivalent primary general membrane stress;

σ_L = equivalent primary local membrane stress;

σ_b = equivalent primary bending stress;

σ_g = equivalent secondary stress; and

f = the lesser of R_m /A or R_e /B,

with R_m and R_e as defined in 6.4.12.1.1.3. With regard to the stresses σ_m, σ_L, σ_g and σ_b see also the definition of stress categories in 6.4.15.2.3.6. The values A and B are to have at least the following minimum values:

	Nickel steels and carbon-manganese steel	**Austenitic steel**	**Aluminium alloys**
A	3	3.5	4
B	1.5	1.5	1.5

6.4.15.3.3.2 Buckling criteria is to be as follows:

.1 The thickness and form of pressure vessels subject to external pressure and other loads causing compressive stresses is to be based on calculations using accepted pressure vessel buckling theory and is to adequately account for the difference in theoretical and actual buckling stress as a result of plate edge misalignment, ovality and deviation from true circular form over a specified arc or chord length.

6.4.15.3.4 Fatigue design condition

6.4.15.3.4.1 For type C independent tanks where the liquefied gas fuel at atmospheric pressure is below minus 55[°C], IRS/Administration may require additional verification to check their compliance with 6.4.15.3.1.1, regarding static and dynamic stress depending on the tank size, the configuration of the tank and arrangement of its supports and attachments.

6.4.15.3.4.2 For vacuum insulated tanks, special attention is to be made to the fatigue strength of the support design and special considerations is also to be made to the limited inspection possibilities between the inside and outer shell.

6.4.15.3.5 Accidental design condition

6.4.15.3.5.1 The tanks and the tank supports are to be designed for the accidental loads and design conditions specified in 6.4.9.5 and 6.4.1.6.3, as relevant.

6.4.15.3.5.2 When subjected to the accidental loads specified in 6.4.9.5, the stress is to comply with the acceptance criteria specified in 6.4.15.3.3.1, modified as appropriate taking into account their lower probability of occurrence.

6.4.15.3.6 Marking

The required marking of the pressure vessel is to be achieved by a method that does not cause unacceptable local stress raisers.

6.4.15.4 Membrane tanks

6.4.15.4.1 Design basis

6.4.15.4.1.1 The design basis for membrane containment systems is that thermal and other expansion or contraction is compensated for without undue risk of losing the tightness of the membrane.

6.4.15.4.1.2 A systematic approach, based on analysis and testing, is to be used to demonstrate that the system will provide its intended function in consideration of the identified in service events as specified in 6.4.15.4.2.1.

6.4.15.4.1.3 A complete secondary barrier is required as defined in 6.4.3. The secondary barrier is to be designed according to 6.4.4.

6.4.15.4.1.4 The design vapour pressure P_0 is not normally exceed 0.025 [MPa]. If the hull scantlings are increased accordingly and consideration is given, where appropriate, to the strength of the supporting thermal insulation, P_0 may be increased to a higher value but less than 0.070 [MPa].

6.4.15.4.1.5 The definition of membrane tanks does not exclude designs such as those in which non-metallic membranes are used or where membranes are included or incorporated into the thermal insulation.

6.4.15.4.1.6 The thickness of the membranes is normally not exceed 10 [mm].

6.4.15.4.1.7 The circulation of inert gas throughout the primary and the secondary insulation spaces, in accordance with 6.11.1 is to be sufficient to allow for effective means of gas detection.

6.4.15.4.2 Design considerations

6.4.15.4.2.1 Potential incidents that could lead to loss of fluid tightness over the life of the membranes are to be evaluated. These include, but are not limited to:

.1 Ultimate design events:

.1 tensile failure of membranes;

.2 compressive collapse of thermal insulation;

.3 thermal ageing;

.4 loss of attachment between thermal insulation and hull structure;

.5 loss of attachment of membranes to thermal insulation system;

.6 structural integrity of internal structures and their associated supporting structures; and

.7 failure of the supporting hull structure.

.2 Fatigue design events:

.1 fatigue of membranes including joints and attachments to hull structure;

.2 fatigue cracking of thermal insulation;

.3 fatigue of internal structures and their associated supporting structures; and

.4 fatigue cracking of inner hull leading to ballast water ingress.

.3 Accident design events:

.1 accidental mechanical damage (such as dropped objects inside the tank while in service);

.2 accidental over pressurization of thermal insulation spaces;

.3 accidental vacuum in the tank; and

.4 water ingress through the inner hull structure.

Designs where a single internal event could cause simultaneous or cascading failure of both membranes are unacceptable.

6.4.15.4.2.2 The necessary physical properties (mechanical, thermal, chemical, etc.) of the materials used in the construction of the liquefied gas fuel containment system is to be established during the design development in accordance with 6.4.15.4.1.2.

6.4.15.4.3 Loads, load combinations

Particular consideration is to be paid to the possible loss of tank integrity due to either an overpressure in the interbarrier space, a possible vacuum in the liquefied gas fuel tank, the sloshing effects, to hull vibration effects, or any combination of these events.

6.4.15.4.4 Structural analyses

6.4.15.4.4.1 Structural analyses and/or testing for the purpose of determining the ultimate strength and fatigue assessments of the liquefied gas fuel containment and associated structures and equipment noted in 6.4.7 is to be performed. The structural analysis shall provide the data required to assess each failure mode that has been identified as critical for the liquefied gas fuel containment system.

6.4.15.4.4.2 Structural analyses of the hull shall take into account the internal pressure as indicated in 6.4.9.3.3.1. Special attention is to be paid to deflections of the hull and their compatibility with the membrane and associated thermal insulation.

6.4.15.4.4.3 The analyses referred to in 6.4.15.4.4.1 and 6.4.15.4.4.2 is to be based on the particular motions, accelerations and response of ships and liquefied gas fuel containment systems.

6.4.15.4.5 Ultimate design condition

6.4.15.4.5.1 The structural resistance of every critical component, sub-system, or assembly, is to be established, in accordance with 6.4.15.4.1.2, for in-service conditions.

6.4.15.4.5.2 The choice of strength acceptance criteria for the failure modes of the liquefied gas fuel containment system, its attachments to the hull structure and internal tank structures, are to reflect the consequences associated with the considered mode of failure.

6.4.15.4.5.3 The inner hull scantlings are to meet the requirements for deep tanks, taking into account the internal pressure as indicated in 6.4.9.3.3.1 and the specified appropriate requirements for sloshing load as defined in 6.4.9.4.1.3.

6.4.15.4.6 Fatigue design condition

6.4.15.4.6.1 Fatigue analysis is to be carried out for structures inside the tank, i.e. pump towers, and for parts of membrane and pump tower attachments, where failure development cannot be reliably detected by continuous monitoring.

6.4.15.4.6.2 The fatigue calculations are to be carried out in accordance with 6.4.12.2, with relevant requirements depending on:

.1 the significance of the structural components with respect to structural integrity; and

.2 availability for inspection.

6.4.15.4.6.3 For structural elements for which it can be demonstrated by tests and/or analyses that a crack will not develop to cause simultaneous or cascading failure of both membranes, *Cw* is to be less than or equal to 0.5.

6.4.15.4.6.4 Structural elements subject to periodic inspection, and where an unattended fatigue crack can develop to cause simultaneous or cascading failure of both membranes, shall satisfy the fatigue and fracture mechanics requirements stated in 6.4.12.2.8.

6.4.15.4.6.5 Structural element not accessible for in-service inspection, and where a fatigue crack can develop without warning to cause simultaneous or cascading failure of both membranes, is to satisfy the fatigue and fracture mechanics requirements stated in 6.4.12.2.9.

6.4.15.4.7 Accidental design condition

6.4.15.4.7.1 The containment system and the supporting hull structure are to be designed for the accidental loads specified in 6.4.9.5. These loads need not be combined with each other or with environmental loads.

6.4.15.4.7.2 Additional relevant accidental scenarios are to be determined based on a

risk analysis. Particular attention is to be paid to securing devices inside of tanks.

6.4.16 Limit state design for novel concepts

6.4.16.1 Fuel containment systems that are of a novel configuration that cannot be designed using section 6.4.15 is to be designed using this section and 6.4.1 to 6.4.14, as applicable. Fuel containment system design according to this section is to be based on the principles of limit state design which is an approach to structural design that can be applied to established design solutions as well as novel designs. This more generic approach maintains a level of safety similar to that achieved for known containment systems as designed using 6.4.15.

6.4.16.2.1 The limit state design is a systematic approach where each structural element is evaluated with respect to possible failure modes related to the design conditions identified in 6.4.1.6. A limit state can be defined as a condition beyond which the structure, or part of a structure, no longer satisfies the requirements.

6.4.16.2.2 For each failure mode, one or more limit states may be relevant. By consideration of all relevant limit states, the limit load for the structural element is found as the minimum limit load resulting from all the relevant limit states. The limit states are divided into the three following categories:

.1 Ultimate limit states (ULS), which correspond to the maximum load-carrying capacity or, in some cases, to the maximum applicable strain or deformation; under intact (undamaged) conditions.

.2 Fatigue limit states (FLS), which correspond to degradation due to the effect of time varying (cyclic) loading.

.3 Accident limit states (ALS), which concern the ability of the structure to resist accidental situations.

6.4.16.3 The procedure and relevant design parameters of the limit state design are to comply with the Standards for the Use of limit state methodologies in the design of fuel containment systems of novel configuration (LSD Standard), as set out in the Annex 1.

6.5 Portable liquefied gas fuel tanks

6.5.1 The design of the tank is to comply with 6.4.15.3. The tank support (container frame or truck chassis) is to be designed for the intended purpose.

6.5.2 Portable fuel tanks are to be located in dedicated areas fitted with:

.1 mechanical protection of the tanks depending on location and cargo operations;

.2 if located on open deck: spill protection and water spray systems for cooling; and

.3 if located in an enclosed space: the space is to be considered as a tank connection space.

6.5.3 Portable fuel tanks are to be secured to the deck while connected to the ship systems. The arrangement for supporting and fixing the tanks is to be designed for the maximum expected static and dynamic inclinations, as well as the maximum expected values of acceleration, taking into account the ship characteristics and the position of the tanks.

6.5.4 Consideration is to be given to the strength and the effect of the portable fuel tanks on the ship's stability.

6.5.5 Connections to the ship's fuel piping systems are to be made by means of approved flexible hoses or other suitable means designed to provide sufficient flexibility.

6.5.6 Arrangements are to be provided to limit the quantity of fuel spilled in case of inadvertent disconnection or rupture of the non-permanent connections.

6.5.7 The pressure relief system of portable tanks is to be connected to a fixed venting system.

6.5.8 Control and monitoring systems for portable fuel tanks is to be integrated in the ship's control and monitoring system. Safety system for portable fuel tanks is to be integrated in the ship's safety system (e.g. shutdown systems for tank valves, leak/gas detection systems).

6.5.9 Safe access to tank connections for the purpose of inspection and maintenance is to be ensured.

6.5.10 After connection to the ship's fuel piping system,

.1 with the exception of the pressure relief system in 6.5.6 each portable tank is to be capable of being isolated at any time;

.2 isolation of one tank shall not impair the availability of the remaining portable tanks; and

.3 the tank shall not exceed its filling limits as given in 6.8.

6.6 CNG fuel containment

6.6.1 The storage tanks to be used for CNG are to be certified and approved by IRS/Administration.

6.6.2 Tanks for CNG are to be fitted with pressure relief valves with a set point below the design pressure of the tank and with outlet located as required in 6.7.2.7 and 6.7.2.8.

6.6.3 Adequate means is to be provided to depressurize the tank in case of a fire which can affect the tank.

6.6.4 Storage of CNG in enclosed spaces is normally not acceptable, but may be permitted after special consideration and approval by Administration/ IRS provided the following is fulfilled in addition to 6.3.4 to 6.3.6:

.1 adequate means are provided to depressurize and inert the tank in case of a fire which can affect the tank;

.2 all surfaces within such enclosed spaces containing the CNG storage are provided with suitable thermal protection against any lost high-pressure gas and resulting condensation unless the bulkheads are designed for the lowest temperature that can arise from gas expansion leakage; and

.3 a fixed fire-extinguishing system is installed in the enclosed spaces containing the CNG storage. Special consideration should be given to the extinguishing of jet-fires.

6.7 Pressure relief system

6.7.1 General

6.7.1.1 All fuel storage tanks are to be provided with a pressure relief system appropriate to the design of the fuel containment system and the fuel being carried. Fuel storage hold spaces, interbarrier spaces and tank connection spaces , which may be subject to pressures beyond their design capabilities, are also be provided with a suitable pressure relief system. Pressure control systems specified in 6.9 are to be independent of the pressure relief systems.

6.7.1.2 Fuel storage tanks which may be subject to external pressures above their design pressure is to be fitted with vacuum protection systems.

6.7.2 Pressure relief systems for liquefied gas fuel tanks

6.7.2.1 If fuel release into the vacuum space of a vacuum insulated tank cannot be excluded, the vacuum space is to be protected by a pressure relief device which is to be connected to a vent system if the tanks are located below deck. On open deck a direct release into the atmosphere may be accepted by IRS/Administration for tanks not exceeding the size of a 40 ft container if the released gas cannot enter safe areas.

6.7.2.2 Liquefied gas fuel tanks is to be fitted with a minimum of 2 pressure relief valves (PRVs) allowing for disconnection of one PRV in case of malfunction or leakage.

6.7.2.3 Interbarrier spaces are to be provided with pressure relief devices*. For membrane systems, the designer is to demonstrate adequate sizing of interbarrier space PRVs.

Note:

** Refer to Pt 5, Ch 4, IR 8.5*

6.7.2.4 The setting of the PRVs is not to be higher than the vapour pressure that has been used in the design of the tank. Valves comprising not more than 50% of the total relieving capacity may be set at a pressure up to 5% above MARVS to allow sequential lifting, minimizing unnecessary release of vapour.

6.7.2.5 The following temperature requirements apply to PRVs fitted to pressure relief systems:

.1 PRVs on fuel tanks with a design temperature below 0[^{o}C] is to be designed and arranged to prevent their becoming inoperative due to ice formation;

.2 the effects of ice formation due to ambient temperatures is to be considered in the construction and arrangement of PRVs;

.3 PRVs is to be constructed of materials with a melting point above 925[°C]. Lower melting point materials for internal parts and seals may be accepted provided that fail-safe operation of the PRV is not compromised; and

.4 sensing and exhaust lines on pilot operated relief valves are to be of suitably robust construction to prevent damage.

6.7.2.6 In the event of a failure of a fuel tank PRV a safe means of emergency isolation is to be available.

.1 procedures is to be provided and included in the operation manual (refer to Section 18);

.2 the procedures are to allow only one of the installed PRVs for the liquefied gas fuel tanks to be isolated, physical interlocks is to be included to this effect; and

.3 isolation of the PRV is to be carried out under the supervision of the master. This action is to be recorded in the ship's log, and at the PRV.

6.7.2.7 Each pressure relief valve installed on a liquefied gas fuel tank is to be connected to a venting system, which is to be:

.1 so constructed that the discharge will be unimpeded and normally be directed vertically upwards at the exit;

.2 arranged to minimize the possibility of water or snow entering the vent system; and

.3 arranged such that the height of vent exits is normally not be less than B/3 or 6 [m], whichever is the greater, above the weather deck and 6 [m] above working areas and walkways. However, vent mast height could be limited to lower value according to special consideration by IRS/Administration.

6.7.2.8 The outlet from the pressure relief valves is normally to be located at least 10 [m] from the nearest:

.1 air intake, air outlet or opening to accommodation, service and control spaces, or other non-hazardous area; and

.2 exhaust outlet from machinery installations.

6.7.2.9 All other fuel gas vent outlets are also to be arranged in accordance with 6.7.2.7 and 6.7.2.8. Means is to be provided to prevent liquid overflow from gas vent outlets, due to hydrostatic pressure from spaces to which they are connected.

6.7.2.10 In the vent piping system, means for draining liquid from places where it may accumulate is to be provided. The PRVs and piping is to be arranged so that liquid can, under no circumstances, accumulate in or near the PRVs.

6.7.2.11 Suitable protection screens of not more than 13 [mm] square mesh is to be fitted on vent outlets to prevent the ingress of foreign objects without adversely affecting the flow.

6.7.2.12 All vent piping is to be designed and arranged not to be damaged by the temperature variations to which it may be exposed, forces due to flow or the ship's motions.

6.7.2.13 PRVs is to be connected to the highest part of the fuel tank. PRVs is to be positioned on the fuel tank so that they will remain in the vapour phase at the filling limit (FL) as given in 6.8, under conditions of 15° list and 0.015L trim, where L is defined in 2.2.25.

6.7.3 Sizing of pressure relieving system

6.7.3.1 Sizing of pressure relief valves

6.7.3.1.1 PRVs are to have a combined relieving capacity for each liquefied gas fuel tank to discharge the greater of the following,

with not more than a 20% rise in liquefied gas fuel tank pressure above the MARVS:

1 the maximum capacity of the liquefied gas fuel tank inerting system if the maximum attainable working pressure of the liquefied gas fuel tank inerting system exceeds the MARVS of the liquefied gas fuel tanks; or

.2 vapours generated under fire exposure computed using the following formula:

$$Q=FGA^{0.82}\ [m^3/s]$$

where:

Q = minimum required rate of discharge of air at standard conditions of 273.15 Kelvin (K) and 0.1013 [MPa].

F = fire exposure factor for different liquefied gas fuel types:

F = 1.0 for tanks without insulation located on deck;
F = 0.5 for tanks above the deck when insulation is approved by the IRS/Administration. (Approval will be based on the use of a fireproofing material, the thermal conductance of insulation, and its stability under fire exposure);

F = 0.5 for uninsulated independent tanks installed in holds;

F = 0.2 for insulated independent tanks in holds (or uninsulated independent tanks in insulated holds);

F = 0.1 for insulated independent tanks in inerted holds (or uninsulated independent tanks in inerted, insulated holds); and

F = 0.1 for membrane tanks.

For independent tanks partly protruding through the weather decks, the fire exposure factor is to be determined on the basis of the surface areas above and below deck.

G = gas factor according to formula:

$$G=\frac{12.4}{LD}\sqrt{\frac{ZT}{M}}$$

where:

T = temperature in Kelvin at relieving conditions, i.e. 120% of the pressure at which the pressure relief valve is set;

L = latent heat of the material being vaporized at relieving conditions, in kJ/kg;

D = a constant based on relation of specific heats *k* and is calculated as follows:

$$D=\sqrt{k\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}$$

where:

k = ratio of specific heats at relieving conditions, and the value of which is between 1.0 and 2.2. If *k* is not known, *D* = 0.606 is to be used;

Z = compressibility factor of the gas at relieving conditions; if not known, *Z* = 1.0 is to be used;

M = molecular mass of the product.

The gas factor of each liquefied gas fuel to be carried is to be determined and the highest value is to be used for PRV sizing.

A = external surface area of the tank (m^2).Indicative examples of different tank types are given in Fig.6.7.1

IR 6.7.3.1.1.2 For Prismatic Tanks:

.1 L_{min} , for non-tapered tanks, is the smaller of the horizontal dimensions of the flat bottom of the tank. For tapered tanks,

as would be used for the forward tank, L_{min} is the smaller of the length and the average width.

.2 For prismatic tanks whose distance between the flat bottom of the tank and bottom of the hold space is equal to or less than $L_{min}/10$:

A=external surface area minus flat bottom surface area

.3 For prismatic tanks whose distance between the flat bottom of the tank and bottom of the hold space is greater than $L_{min}/10$:

A= external surface area.

6.7.3.1.2 For vacuum insulated tanks in fuel storage hold spaces and for tanks in fuel storage hold spaces separated from potential fire loads by coffer dams or surrounded by ship spaces with no fire load the following applies:

If the pressure relief valves have to be sized for fire loads the fire factors according may be reduced to the following values:

F = 0.5 to F = 0.25

F = 0.2 to F = 0.1

The minimum fire factor is F= 0.1

6.7.3.1.3 The required mass flow of air at relieving conditions is given by:

$$M_{air} = Q\ \rho_{air} \text{ (kg/s)}$$

where density of air (ρ_{air}) = 1.293 [kg/m^3] (air at 273.15 [K], 0.1013 [MPa]).

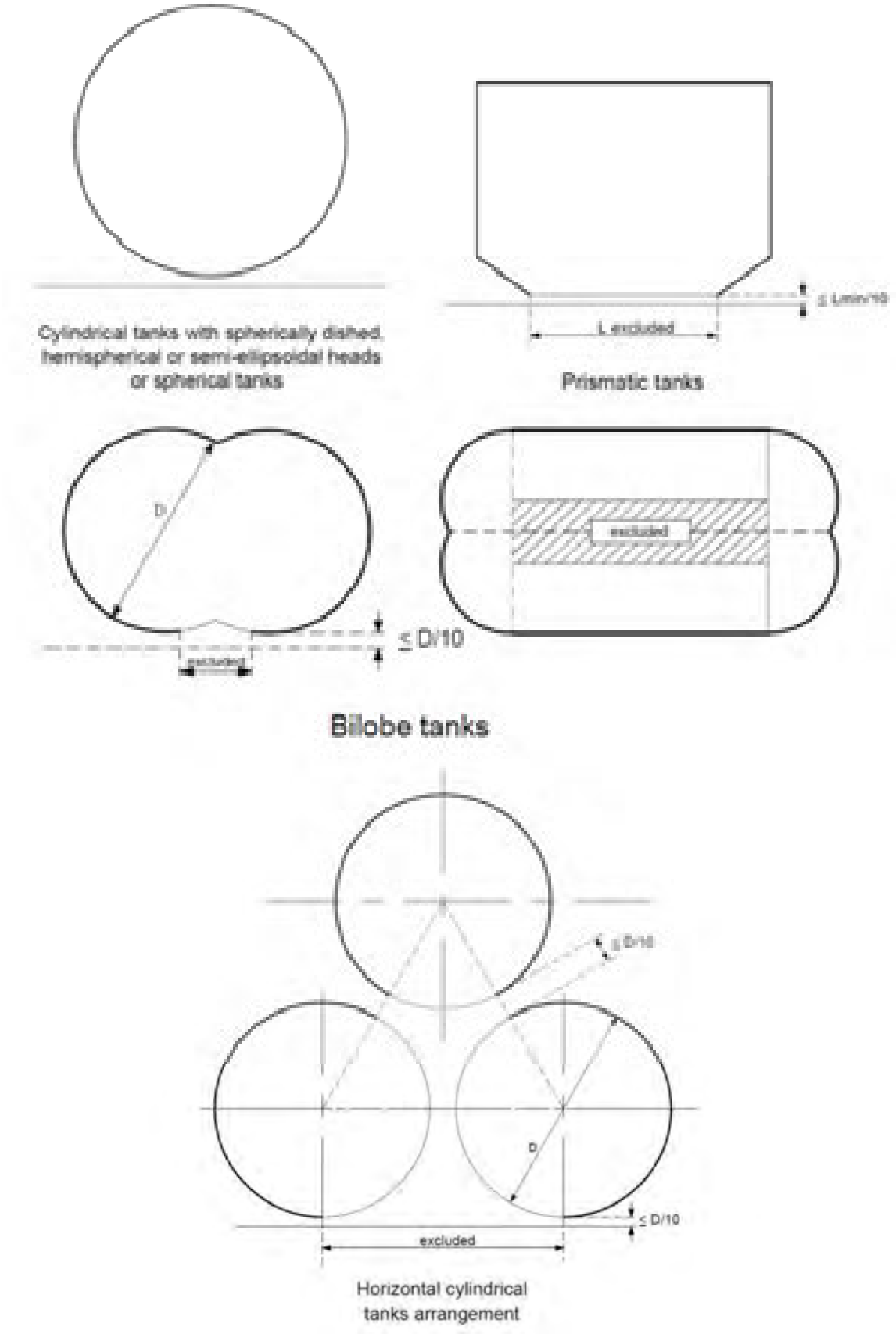

Cylindrical tanks with spherically dished, hemispherical or semi-ellipsoidal heads or spherical tanks
L excluded
Prismatic tanks
D
excluded
≤ D/10
Bilobe tanks
≤ D/10
D
≤ D/10
excluded
Horizontal cylindrical tanks arrangement

Fig.6.7.1

6.7.3.2 Sizing of vent pipe system

6.7.3.2.1 Pressure losses upstream and downstream of the PRVs, is to be taken into account when determining their size to ensure the flow capacity required by 6.7.3.1.

6.7.3.2.2 Upstream pressure losses

.1 the pressure drop in the vent line from the tank to the PRV inlet shall not exceed 3% of the valve set pressure at the calculated flow rate, in accordance with 6.7.3.1;

.2 pilot-operated PRVs is to be unaffected by inlet pipe pressure losses when the pilot senses directly from the tank dome; and

.3 pressure losses in remotely sensed pilot lines is to be considered for flowing type pilots.

6.7.3.2.3 Downstream pressure losses

.1 Where common vent headers and vent masts are fitted, calculations are to include flow from all attached PRVs.

.2 The built-up back pressure in the vent piping from the PRV outlet to the location of discharge to the atmosphere, and including any vent pipe interconnections that join other tanks, shall not exceed the following values:

.1	for unbalanced PRVs	10% of MARVS
.2	for balanced PRVs	30% of MARVS; and
.3	for pilot operated PRVs	50% of MARVS.

Alternative values provided by the PRV manufacturer may be accepted.

6.7.3.2.4 To ensure stable PRV operation, the blow-down is not to be less than the sum of the inlet pressure loss and 0.02 MARVS at the rated capacity.

6.8 Loading limit for liquefied gas fuel tanks

6.8.1 Storage tanks for liquefied gas are not to be filled to more than a volume equivalent to 98% full at the reference temperature as defined in 2.2.36.

A loading limit curve for actual fuel loading temperatures is to be prepared from the following formula:

$$LL = FL\ \rho_R/\rho_L$$

where:

LL = loading limit as defined in 2.2.27, expressed in per cent;

FL = filling limit as defined in 2.2.16 expressed in per cent, here 98%;

ρ_R = relative density of fuel at the reference temperature; and

ρ_L = relative density of fuel at the loading temperature.

6.8.2 In cases where the tank insulation and tank location make the probability very small for the tank contents to be heated up due to an external fire, special considerations may be made to allow a higher loading limit than calculated using the reference temperature, but never above 95%. This also applies in cases where a second system for pressure maintenance is installed, (refer to 6.9). However, if the pressure can only be maintained / controlled by fuel consumers, the loading limit as calculated in 6.8.1 is to be used.

IR 6.8.2 The alternative loading limit option given under 6.8.2 is understood to be an alternative to 6.8.1 and should only be applicable when the calculated loading limit using the formulae in 6.8.1 gives a lower value than 95%.

6.8.3 In lieu of 6.8.2, for ships constructed on or after 1 January 2024, in cases where the tank insulation and tank location make the probability very small for the tank contents to be heated up due to an external fire, special considerations may be made to allow a higher loading limit than calculated using the reference temperature, but never above 95%.

6.9 Maintenance of fuel storage condition

6.9.1 Control of tank pressure and temperature

6.9.1.1 With the exception of liquefied gas fuel tanks designed to withstand the full gauge vapour pressure of the fuel under conditions of the upper ambient design temperature, liquefied gas fuel tanks' pressure and temperature is to be maintained at all times within their design range by means acceptable to IRS/Administration, e.g. by one of the following methods:

.1 reliquefaction of vapours;

.2 thermal oxidation of vapours;

.3 pressure accumulation; or

.4 liquefied gas fuel cooling.

The method chosen is to be capable of maintaining tank pressure below the set pressure of the tank pressure relief valves for a period of 15 days assuming full tank at normal service pressure and the ship in idle condition, i.e. only power for domestic load is generated.

6.9.1.2 Venting of fuel vapour for control of the tank pressure is not acceptable except in emergency situations.

IR 6.9.1.3 Liquefied gas fuel tank’s pressure and temperature is to be controlled and maintained within the design range at all times including after activation of the safety system required in 15.2.2 for a period of minimum 15 days. The activation of the safety system alone is not deemed as an emergency situation.

6.9.2 Design of systems

6.9.2.1 For worldwide service, the upper ambient design temperature is to be sea 32[ºC] and air 45[ºC]. For service in particularly hot or cold zones, these design temperatures is to be increased or decreased, to the satisfaction of IRS/Administration.

6.9.2.2 The overall capacity of the system is to be such that it can control the pressure within the design conditions without venting to atmosphere.

6.9.3 Reliquefaction systems

6.9.3.1 The reliquefaction system is to be designed and calculated according to 6.9.3.2. The system has to be sized in a sufficient way also in case of no or low consumption.

6.9.3.2 The reliquefaction system is to be arranged in one of the following ways:

.1 a direct system where evaporated fuel is compressed, condensed and returned to the fuel tanks;

.2 an indirect system where fuel or evaporated fuel is cooled or condensed by refrigerant without being compressed;

.3 a combined system where evaporated fuel is compressed and condensed in a fuel/refrigerant heat exchanger and returned to the fuel tanks; or

.4 if the reliquefaction system produces a waste stream containing methane during pressure control operations within the design conditions, these waste gases shall, as far as reasonably practicable, be disposed of without venting to atmosphere.

6.9.4 Thermal oxidation systems

6.9.4.1 Thermal oxidation can be done by either consumption of the vapours according to the requirements for consumers described in this Chapter or in a dedicated gas combustion unit (GCU). It is to be demonstrated that the capacity of the oxidation system is sufficient to consume the required quantity of vapours. In this regard, periods of slow steaming and/or no consumption from propulsion or other services of the ship is to be considered.

6.9.5 Compatibility

6.9.5.1 Refrigerants or auxiliary agents used for refrigeration or cooling of fuel is to be compatible with the fuel they may come in contact with (not causing any hazardous reaction or excessively corrosive products). In addition, when several refrigerants or agents are used, these is to be compatible with each other.

6.9.6 Availability of systems

6.9.6.1 The availability of the system and its supporting auxiliary services is to be such that in case of a single failure (of mechanical non-static component or a component of the control systems) the fuel tank pressure and

temperature can be maintained by another service/system.

6.9.6.2 Heat exchangers that are solely necessary for maintaining the pressure and temperature of the fuel tanks within their design ranges shall have a standby heat exchanger unless they have a capacity in excess of 25% of the largest required capacity for pressure control and they can be repaired on board without external sources.

6.10 Atmospheric control within the fuel containment system

6.10.1 A piping system is to be arranged to enable each fuel tank to be safely gas-freed, and to be safely filled with fuel from a gas-free condition. The system is to be arranged to minimize the possibility of pockets of gas or air remaining after changing the atmosphere.

6.10.2 The system is to be designed to eliminate the possibility of a flammable mixture existing in the fuel tank during any part of the atmosphere change operation by utilizing an inerting medium as an intermediate step.

6.10.3 Gas sampling points is to be provided for each fuel tank to monitor the progress of atmosphere change.

6.10.4 Inert gas utilized for gas freeing of fuel tanks may be provided externally to the ship.

6.11 Atmosphere control within fuel storage hold spaces (Fuel containment systems other than type C independent tanks)

6.11.1 Interbarrier and fuel storage hold spaces associated with liquefied gas fuel containment systems requiring full or partial secondary barriers is to be inerted with a suitable dry inert gas and kept inerted with make-up gas provided by a shipboard inert gas generation system, or by shipboard storage, which is to be sufficient for normal consumption for at least 30 days. Shorter periods may be considered by IRS/Administration depending on the ship's service.

6.11.2 Alternatively, the spaces referred to in 6.11.1 requiring only a partial secondary barrier may be filled with dry air provided that the ship maintains a stored charge of inert gas or is fitted with an inert gas generation system sufficient to inert the largest of these spaces, and provided that the configuration of the spaces and the relevant vapour detection systems, together with the capability of the inerting arrangements, ensures that any leakage from the liquefied gas fuel tanks will be rapidly detected and inerting effected before a dangerous condition can develop. Equipment for the provision of sufficient dry air of suitable quality to satisfy the expected demand is to be provided.

6.12 Environmental control of spaces surrounding type C independent tanks

6.12.1 Spaces surrounding liquefied gas fuel tanks is to be filled with suitable dry air and be maintained in this condition with dry air provided by suitable air drying equipment. This is only applicable for liquefied gas fuel tanks where condensation and icing due to cold surfaces is an issue.

6.13 Inerting

6.13.1 Arrangements to prevent back-flow of fuel vapour into the inert gas system are to be provided as specified below.

6.13.2 To prevent the return of flammable gas to any non-hazardous spaces, the inert gas supply line is to be fitted with two shutoff valves in series with a venting valve in between (double block and bleed valves). In addition, a closable non-return valve is to be installed between the double block and bleed arrangement and the fuel system. These valves are to be located outside non-hazardous spaces.

6.13.3 Where the connections to the fuel piping systems are non-permanent, two non-return valves may be substituted for the valves required in 6.13.2.

6.13.4 The arrangements are to be such that each space being inerted can be isolated and the necessary controls and relief valves, etc. is to be provided for controlling pressure in these spaces.

6.13.5 Where insulation spaces are continually supplied with an inert gas as part of a leak detection system, means is to be provided to monitor the quantity of gas being supplied to individual spaces.

6.14 Inert gas production and storage on board

6.14.1 The equipment is to be capable of producing inert gas with oxygen content at no time greater than 5% by volume. A continuous-reading oxygen content meter is to be fitted to the inert gas supply from the equipment and is to be fitted with an alarm set at a maximum of 5% oxygen content by volume.

6.14.2 An inert gas system is to have pressure controls and monitoring arrangements appropriate to the fuel containment system.

6.14.3 Where a nitrogen generator or nitrogen storage facilities are installed in a separate compartment outside of the engine-room, the separate compartment is to be fitted with an independent mechanical extraction ventilation system, providing a minimum of 6 air changes per hour. A low oxygen alarm is to be fitted.

6.14.4 Nitrogen pipes are only to be led through well ventilated spaces. Nitrogen pipes in enclosed spaces shall:

- be fully welded;

- have only a minimum of flange connections as needed for fitting of valves; and

- be as short as possible.

Section 7

Material and General Pipe Design

7.1 Goal

7.1.1 The goal of this section is to ensure the safe handling of fuel, under all operating conditions, to minimize the risk to the ship, personnel and to the environment, having regard to the nature of the products involved.

7.2 Functional requirements

7.2.1 This section relates to functional requirements in 3.2.1, 3.2.5, 3.2.6, 3.2.8, 3.2.9 and 3.2.10. In particular the following apply:

7.2.1.1 Fuel piping is to be capable of absorbing thermal expansion or contraction caused by extreme temperatures of the fuel without developing substantial stresses.

7.2.1.2 Provision is to be made to protect the piping, piping system and components and fuel tanks from excessive stresses due to thermal movement and from movements of the fuel tank and hull structure.

7.2.1.3 If the fuel gas contains heavier constituents that may condense in the system, means for safely removing the liquid is to be fitted.

7.2.1.4 Low temperature piping is to be thermally isolated from the adjacent hull structure, where necessary, to prevent the temperature of the hull from falling below the design temperature of the hull material.

7.3 General pipe design

7.3.1 General

7.3.1.1 Fuel pipes and all the other piping needed for a safe and reliable operation and maintenance is to be colour marked in accordance with a standard at least equivalent to those acceptable to the IMO*.

Note :

** Refer to EN ISO 14726:2008 Ships and marine technology – identification colours for the content of piping systems*

7.3.1.2 Where tanks or piping are separated from the ship's structure by thermal isolation, provision is to be made for electrically bonding to the ship's structure both the piping and the tanks. All gasketed pipe joints and hose connections is to be electrically bonded.

7.3.1.3 All pipelines or components which may be isolated in a liquid full condition are to be provided with relief valves.

7.3.1.4 Pipework, which may contain low temperature fuel, is to be thermally insulated

to an extent which will minimize condensation of moisture.

7.3.1.5 Piping other than fuel supply piping and cabling may be arranged in the double wall piping or duct provided that they do not create a source of ignition or compromise the integrity of the double pipe or duct. The double wall piping or duct shall only contain piping or cabling necessary for operational purposes.

7.3.2 Wall thickness

7.3.2.1 The minimum wall thickness is to be calculated as follows:

$$t = (t_0 + b + c) / (1 - a/100) \text{ [mm]}$$

where:

t_0 = theoretical thickness

$t_0 = PD / (2Ke + P)$ [mm]

with:

P = design pressure [MPa] referred to in 7.3.3;

D = outside diameter [mm];

K = allowable stress [N/mm²] referred to in 7.3.4; and

e = efficiency factor equal to 1.0 for seamless pipes and for longitudinally or spirally welded pipes, delivered by approved manufacturers of welded pipes, that are considered equivalent to seamless pipes when non-destructive testing on welds is carried out in accordance with recognized standards. In other cases an efficiency factor of less than 1.0, in accordance with recognized standards, may be required depending on the manufacturing process;

b = allowance for bending [mm]. The value of b is to be chosen so that the calculated stress in the bend, due to internal pressure only, does not exceed the allowable stress. Where such justification is not given, b is to be:

$$b = D \cdot t_0 / 2.5r \text{ [mm]}$$

with:

r = mean radius of the bend [mm];

c = corrosion allowance [mm]. If corrosion or erosion is expected the wall thickness of the piping is to be increased over that required by other design requirements. This allowance is to be consistent with the expected life of the piping; and

a = negative manufacturing tolerance for thickness [%].

7.3.2.2 The absolute minimum wall thickness is to be in accordance with a standard acceptable to IRS/Administration.

7.3.3 Design condition

7.3.3.1 The greater of the following design conditions is to be used for piping, piping system and components as appropriate*, **:

.1 for systems or components which may be separated from their relief valves and which contain only vapour at all times, vapour pressure at 45[°C] assuming an initial condition of saturated vapour in the system at the system operating pressure and temperature; or

.2 the MARVS of the fuel tanks and fuel processing systems; or

.3 the pressure setting of the associated pump or compressor discharge relief valve; or

.4 the maximum total discharge or loading head of the fuel piping system; or

.5 the relief valve setting on a pipeline system.

Note:

** Lower values of ambient temperature regarding design condition in 7.3.3.1.1 may be accepted by Administration/ IRS for ships operating in restricted areas. Conversely, higher values of ambient temperature may be required.*

*** For ships on voyages of restricted duration, P_0 may be calculated based*

on the actual pressure rise during the voyage and account may be taken of any thermal insulation of the tank. Reference is made to the " Application of amendments to gas carrier codes concerning type C tank loading limits" (SIGTTO/IACS).

7.3.3.2 Piping, piping systems and components shall have a minimum design pressure of 1 [MPa] except for open ended lines where it is not to be less than 0.5 [MPa].

7.3.4 Allowable stress

7.3.4.1 For pipes made of steel including stainless steel, the allowable stress to be considered in the formula of the strength thickness in 7.3.2.1 is to be the lower of the following values:

R_m/2.7 or R_e/1.8

where:

R_m = specified minimum tensile strength at room temperature [N/mm²]; and

R_e = specified minimum yield stress at room temperature [N/mm²]. If the stress strain curve does not show a defined yield stress, the 0.2% proof stress applies.

7.3.4.2 Where necessary for mechanical strength to prevent damage, collapse, excessive sag or buckling of pipes due to superimposed loads, the wall thickness is to be increased over that required by 7.3.2 or, if this is impracticable or would cause excessive local stresses, these loads is to be reduced, protected against or eliminated by other design methods. Such superimposed loads may be due to; supports, ship deflections, liquid pressure surge during transfer operations, the weight of suspended valves, reaction to loading arm connections, or otherwise.

7.3.4.3 For pipes made of materials other than steel, the allowable stress is to be considered by the IRS/Administration.

7.3.4.4 High pressure fuel piping systems is to have sufficient constructive strength. This is to be confirmed by carrying out stress analysis and taking into account:

.1 stresses due to the weight of the piping system;

.2 acceleration loads when significant; and

.3 internal pressure and loads induced by hog and sag of the ship.

7.3.4.5 When the design temperature is minus 110 [°C] or colder, a complete stress analysis, taking into account all the stresses due to weight of pipes, including acceleration loads if significant, internal pressure, thermal contraction and loads induced by hog and sag of the ship is to be carried out for each branch of the piping system.

7.3.5 Flexibility of piping

7.3.5.1 The arrangement and installation of fuel piping is to provide the necessary flexibility to maintain the integrity of the piping system in the actual service situations, taking potential for fatigue into account.

7.3.6 Piping fabrication and joining details

7.3.6.1 Flanges, valves and other fittings are to comply with a standard acceptable to the IRS/Administration, taking into account the design pressure defined in 7.3.3.1. For bellows and expansion joints used in vapour service, a lower minimum design pressure than defined in 7.3.3.1 may be accepted.

7.3.6.2 All valves and expansion joints used in high pressure fuel piping systems are to be approved according to a standard acceptable to IRS/Administration.

7.3.6.3 The piping system is to be joined by welding with a minimum of flange connections. Gaskets are to be protected against blow-out.

IR 7.3.6.3 In the piping system for fuel transfer, there is to be, no single common flange or other component where one single failure itself may overcome both primary and secondary barriers and may result in a gas leak into the surrounding area causing danger to the persons on board, the environment or the ship. A single common flange (with two sealing systems) may be accepted at the fuel connection to the gas consumers including GCUs, boilers and components on the engine, such as gas regulating units.

7.3.6.4 Piping fabrication and joining details are to comply with the following:

7.3.6.4.1 Direct connections

.1 Butt-welded joints with complete penetration at the root may be used in all applications. For design temperatures colder than minus 10[°C], butt welds is to be either double welded or equivalent to a double welded butt joint. This may be accomplished by use of a backing ring, consumable insert or inert gas back-up on the first pass. For design pressures in excess of 1 [MPa] and design temperatures of minus 10[°C] or colder, backing rings is to be removed.

.2 Slip-on welded joints with sleeves and related welding, having dimensions in accordance with recognized standards, are only to be used for instrument lines and open-ended lines with an external diameter of 50 [mm] or less and design temperatures not colder than minus 55[°C].

.3 Screwed couplings complying with recognized standards are only to be used for accessory lines and instrumentation lines with external diameters of 25 [mm] or less.

7.3.6.4.2 Flanged connections

.1 Flanges in flange connections is to be of the welded neck, slip-on or socket welded type; and

.2 For all piping except open ended, the following restrictions apply:

.1 For design temperatures colder than minus 55[°C], only welded neck flanges is to be used; and

.2 For design temperatures colder than minus 10[°C], slip-on flanges are not to be used in nominal sizes above 100 [mm] and socket welded flanges are not to be used in nominal sizes above 50 [mm].

7.3.6.4.3 Expansion joints

Where bellows and expansion joints are provided in accordance with 7.3.6.1 the following apply:

.1 if necessary, bellows is to be protected against icing;

.2 slip joints are not to be used except within the liquefied gas fuel storage tanks; and

.3 bellows are normally not to be arranged in enclosed spaces.

7.3.6.4.4 Other connections

Piping connections are to be joined in accordance with 7.3.6.4.1 to 7.3.6.4.3 but for other exceptional cases IRS/Administration may consider alternative arrangements.

7.4 Materials

7.4.1 Metallic materials

7.4.1.1 Materials for fuel containment and piping systems are to comply with the minimum requirements given in the following Tables:

Table 7.1: Plates, pipes (seamless and welded), sections and forgings for fuel tanks and process pressure vessels for design temperatures not lower than 0[°C].

Table 7.2: Plates, sections and forgings for fuel tanks, secondary barriers and process pressure vessels for design temperatures below 0[°C] and down to minus 55[°C].

Table 7.3: Plates, sections and forgings for fuel tanks, secondary barriers and process pressure vessels for design temperatures below minus 55[°C] and down to minus 165[°C].

Table 7.4: Pipes (seamless and welded), forgings and castings for fuel and process piping for design temperatures below 0[°C] and down to minus 165[°C].

Table 7.5: Plates and sections for hull structures required by 6.4.13.1.1.2.

IR7.4.1.1 High Manganese Austenitic Steel may also be used for cargo and fuel tanks of LNG fueled ships. IRS '*Guidelines on Approval of High Manganese Austenitic Steel for Cryogenic Service'* may be referred for manufacturing approval of such steels.

Table : 7.1		
PLATES, PIPES (SEAMLESS AND WELDED)[See note 1 and note 2], SECTIONS AND FORGINGS FOR FUEL TANKS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES NOT LOWER THAN 0 [˚C]		
CHEMICAL COMPOSITION AND HEAT TREATMENT		
• Carbon-manganese steel		
• Fully killed fine grain steel		
• Small additions of alloying elements by agreement with the Administration/ IRS		
• Composition limits to be approved by the Administration/ IRS		
• Normalized, or quenched and tempered [See note 4]		
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS		
Sampling Frequency		
• Plates	Each "piece" to be tested	
• Sections and forgings	Each "batch" to be tested	
Mechanical properties		
• Tensile properties	Specified minimum yield stress not to exceed 410 [N/mm^2][See note 2 & 5]	
Toughness (Charpy V-notch test)		
• Plates	Transverse test pieces. Minimum average energy value (KV) 27J	
• Sections and forgings	Longitudinal test pieces. Minimum average energy (KV) 41J	
• Test temperature	Thickness t (mm)	Test temperature (˚C)
	t < 20	0
	20 < t ≤ 40 [See note 3]	-20

Notes

1 For seamless pipes and fittings normal practice applies. The use of longitudinally and spirally welded pipes is to be specially approved by the Administration or IRS. .

2 Charpy V-notch impact tests are not required for pipes.

3 This table is generally applicable for material thicknesses up to 40 [mm].

4 A controlled rolling procedure or TMCP may be used as an alternative.

5 Materials with specified minimum yield stress exceeding 410 [N/mm^2] may be approved by the Administration or IRS. For these materials, particular attention is to be given to the hardness of the welded and heat affected zones.

In addition to Table 7.1 above, for design temperature not lower than 0°C, the following applies:

Table 7.1.1		
Plates, pipes (seamless and welded), sections and forgings for fuel tanks and process pressure vessels for design temperatures not lower than 0°C.		
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS		
TEST TEMPERATURE	Thickness t (mm)	Test temperature (°C)
	40 < t ≤ 50 (1)	-20 (2)
	40 < t ≤ 50 (1)	-30 (3)
NOTES:		
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4.		
(2) Applies to type C independent tanks and process pressure vessels. In addition, post-weld stress relief heat treatment is to be performed. Exemption to post-weld stress relief heat treatment based on alternative approach (e.g. Engineering Critical Assessment) is to be approved by IRS or is to be in accordance with recognized standards.		
(3) Applies to fuel tank other than type C.		

Table 7.2					
PLATES, SECTIONS AND FORGINGS See note 1 FOR FUEL TANKS, SECONDARY BARRIERS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES BELOW 0 [°C] AND DOWN TO MINUS 55 [°C] MAXIMUM THICKNESS 25 [mm] see note 2					
CHEMICAL COMPOSITION AND HEAT TREATMENT					
• Carbon-manganese steel					
• Fully killed, aluminium treated fine grain steel					
• Chemical composition (ladle analysis)					
C	Mn	Si	S	P	
0.16%max see note 3	0.7 – 1.60 %	0.1 – 0.50 %	0.025% max	0.025% max	
Optional additions: Alloys and grain refining elements may be generally in accordance with the following:					
Ni	Cr	Mo	Cu	Nb	V
0.8% max	0.25% max	0.08% max	0.35% max	0.05% max	0.1% max
Al content total 0.02% min (Acid soluble 0.015% min)					
• Normalized, or quenched and tempered See note 4					
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS					
Sampling frequency					
• Plates	Each “piece” to be tested				
• Sections and forgings	Each “ batch” to be tested				
Mechanical properties					
• Tensile properties	Specified minimum yield stress not to exceed 410 [N/mm^2] See note 5				

Toughness (Charpy V-notch test)	
• Plates	Transverse test pieces. Minimum average energy value (KV) 27J
• Sections and forgings	Longitudinal test pieces. Minimum average energy (KV) 41J
• Test temperature	5 [˚C] below the design temperature or -20 [˚C], whichever is lower

Notes

1 The Charpy V-notch and chemistry requirements for forgings may be specially considered by the Administration/ IRS.

2 For material thickness of more than 25 [mm], Charpy V-notch tests is to be conducted as follows:

Material Thickness [mm]	Test temperature [˚C]
25 < t ≤ 30	10 [˚C] below design temperature or -20[˚C], whichever is lower
30 < t ≤ 35	15 [˚C] below design temperature or -20[˚C], whichever is lower
35 < t ≤ 40	20 [˚C] below design temperature
40 < t	Temperature approved by Administration/ IRS

The impact energy value is to be in accordance with the table for the applicable type of test specimen.

Materials for tanks and parts of tanks which are completely thermally stress relieved after welding may be tested at a temperature 5 [°C] below design temperature or -20 [°C], whichever is lower.

For thermally stress relieved reinforcements and other fittings, the test temperature is to be the same as that required for the adjacent tank-shell thickness.

3 By special agreement with the Administration/ IRS, the carbon content may be increased to 0.18% maximum, provided the design temperature is not lower than -40 [°C].

4 A controlled rolling procedure or TMCP may be used as an alternative.

5 Materials with specified minimum yield stress exceeding 410 [N/mm^2] may be approved by Administration/IRS. For these materials, particular attention is to be given to the hardness of the welded and heat affected zones.

Guidance:
For materials exceeding 25 [mm] in thickness for which the test temperature is -60 [°C] or lower, the application of specially treated steels or steels in accordance with Table 7.3 may be necessary.

In addition to Table 7.2 above, the following applies:

Table 7.2.1		
Plates, sections and forgings for fuel tanks, secondary barriers and process pressure vessels for design temperatures below 0°C and down to -10°C		
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS		
TEST TEMPERATURE	Thickness t (mm)	Test temperature (°C)
	40 < t ≤ 50 (1)	5°C below design temperature or –20°C, whichever is lower (2)
	40 < t ≤ 45 (1)	25 °C below design temperature (3)
	45 < t ≤ 50 (1)	30 °C below design temperature (3)
NOTES:		
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4. (2) Applies to type C independent tanks and process pressure vessels. In addition, post-weld stress relief heat treatment is to be performed. Exemption to post-weld stress relief heat treatment based on alternative approach (e.g. Engineering Critical Assessment) is to be approved by IRS or is to be accordance with recognized standards. (3) Applies to fuel tank other than type C.		

Table 7.2.2		
Plates, sections and forgings for fuel tanks, secondary barriers and process pressure vessels for design temperatures below -10°C and down to -55°C		
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS		
TEST TEMPERATURE	Thickness t (mm)	Test temperature (°C)
	40 < t ≤ 50 (1)	5°C below design temperature or –20°C, whichever is lower
	40 < t ≤ 45 (1)	25 °C below design temperature (2)
	45 < t ≤ 50 (1)	30 °C below design temperature (2)
NOTES:		
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4. (2) Applies to fuel tank other than type C.		

Table 7.3

PLATES, SECTIONS AND FORGINGS See note 1 FOR FUEL TANKS, SECONDARY BARRIERS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES BELOW MINUS 55 [°C] AND DOWN TO MINUS 165 [°C] See note 2 MAXIMUM THICKNESS 25 [mm] see note 3 and note 4

Minimum design temperature [˚C]	Chemical composition See note 5 and heat treatment	Impact test temperature [˚C]
-60	1.5% nickel steel – normalized or normalized and tempered or quenched and tempered or TMCP See note 6	-65
-65	2.25% nickel steel – normalized or normalized and tempered or quenched and tempered or TMCP See note 6 and 7	-70
-90	3.5% nickel steel – normalized or normalized and tempered or quenched and tempered or TMCP See note 6 and 7	-95
-105	5% nickel steel – normalized or normalized and tempered or quenched and tempered See note 6,7and 8	-110
-165	9% nickel steel – double normalized and tempered or quenched and tempered See note 6	-196
-165	Austenitic steels, such as types 304, 304L, 316, 316L, 321 and 347 solution treated See note 9	-196
-165	Aluminium alloys; such as type 5083 annealed	Not required
-165	Austenitic Fe-Ni alloy (36% nickel). Heat treatment as agreed.	Not required

TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS

Sampling frequency	
• Plates	Each “piece” to be tested
• Sections and forgings	Each “ batch” to be tested
Toughness (Charpy V-notch test)	
• Plates	Transverse test pieces. Minimum average energy value (KV) 27J
• Sections and forgings	Longitudinal test pieces. Minimum average energy (KV) 41J

Notes

1 The impact test required for forgings used in critical application is to be subject to special consideration by the Administration/ IRS.

2 The requirements for design temperatures below -165[°C] are to be specially agreed with the Administration/ IRS.

3 For materials 1.5% Ni, 2.25% Ni, 3.5% Ni and 5% Ni, with thicknesses greater than 25 [mm], the impact tests are to be conducted as follows:

Material Thickness [mm]	Test temperature [˚C]
25 < t ≤ 30	10 [˚C] below design temperature
30 < t ≤ 35	15 [˚C] below design temperature
35 < t ≤ 40	20 [˚C] below design temperature

The energy value is to be in accordance with the table for the applicable type of test specimen. For material thickness of more than 40 [mm], the Charpy V-notch values is to be specially considered.

4 For 9% Ni steels, austenitic stainless steels and aluminium alloys, thickness greater than 25 [mm] may be used.

5 The chemical composition limits are to be in accordance with recognized standards.

6 TMCP nickel steels will be subject to acceptance by the Administration/ IRS.

7 A lower minimum design temperature for quenched and tempered steels may be specially agreed with the Administration/ IRS.

8 A specially heat treated 5% nickel steel, for example triple heat treated 5% nickel steel, may be used down to -165 [°C], provided that the impact tests are carried out at -196 [°C].

9 The impact test may be omitted, subject to agreement with the Administration/ IRS.

In addition to Table 7.3 above, the following applies:

Table 7.3.1	
Plates, sections and forgings for fuel tanks, secondary barriers and process pressure vessels for design temperatures below -55°C and down to -165°C.	
CHARPY V-NOTCH IMPACT TEST REQUIREMENTS	
40 < t ≤ 45 mm (1)	25°C below design temperature
45 < t ≤ 50 mm (1)	30°C below design temperature
(1) A further set of impact test at mid thickness for products with t>40mm is required except rolled steels specified in Part 2, Chapter 3, Section 2 to 4.	

Table 7.4			
PIPES(SEAMLESS AND WELDED) See note 1 , FORGINGS See note 2, AND CASTINGS See note 2 FOR FUEL AND PROCESS PIPING FOR DESIGN TEMPERATURES BELOW 0 [˚C] AND DOWN TO -165 [˚C] See note 3 MAXIMUM THICKNESS 25 [mm]			
Minimum design temperature [˚C]	**Chemical composition See note 5 and heat treatment**	**Impact test**	
		Test temp **[˚C]**	Minimum average energy (KV)
-55	Carbon-manganese steel. Fully killed fine grain. Normalized or as agreed See note 6	See note 4	27
-65	2.25% nickel steel. Normalized, normalized and tempered or quenched and tempered See note 6	-70	34
-90	3.5% nickel steel. Normalized, normalized and tempered or quenched and tempered See note 6	-95	34
-165	9% nickel steel See note 7. Double Normalized, and tempered or quenched and tempered	-196	41
	Austenitic steels, such as types 304, 304L, 316, 316L, 321, and 347.Solution treated See note 8	-196	41
	Aluminium alloys; such as type 5083 annealed		Not required
TENSILE AND TOUGHNESS (IMPACT) TEST REQUIREMENTS			
Sampling frequency			
• Each “batch” to be tested			

Toughness (Charpy V-notch test)
• Impact test: Longitudinal test pieces
Notes 1 The use of longitudinally or spirally welded pipes is to be specially approved by the Administration/ IRS. 2 The requirements for forgings and castings may be subject to special consideration by the Administration/ IRS. 3 The requirements for design temperatures below -165 [°C] are to be specially agreed with the Administration/ IRS. 4 The test temperature is to be 5 [°C] below the design temperature or -20 [°C], whichever is lower. 5 The composition limits are to be in accordance with recognized standards. 6 A lower design temperature may be specially agreed with the Administration/ IRS for quenched and tempered materials. 7 This chemical composition is not suitable for castings. 8 Impact tests may be omitted, subject to agreement with the Administration/ IRS.

Table 7.5

PLATES AND SECTIONS FOR HULL STRUCTURES REQUIRED BY 6.4.13.1.1.2

Minimum design temperature of hull structure (°C)	**Maximum thickness (mm) for steel grades**							
	A	**B**	**D**	**E**	**AH**	**DH**	**EH**	**FH**
0 and above	Recognized Standards							
down to -5	15	25	30	50	25	45	50	50
down to -10	x	20	25	50	20	40	50	50
down to -20	x	x	20	50	x	30	50	50
down to -30	x	x	x	40	x	20	40	50
Below -30	In accordance with Table 7.2 except that the thickness limitation given in Table 7.2 and in footnote 2 of that table does not apply							
Notes 'x' means steel grade not to be used.								

7.4.1.2 Materials having a melting point below 925 [°C] are not to be used for piping outside the fuel tanks.

7.4.1.3 For CNG tanks, the use of materials not covered above may be specially considered by the IRS/Administration.

7.4.1.4 Where required the outer pipe or duct containing high pressure gas in the inner pipe shall as a minimum fulfill the material requirements for pipe materials with design temperature down to minus 55[°C] in Table 7.4.

7.4.1.5 The outer pipe or duct around liquefied gas fuel pipes shall as a minimum fulfill the material requirements for pipe materials with design temperature down to minus 165[°C] in Table 7.4.

Section 8

Bunkering

8.1 Goal

8.1.1 The goal of this section is to provide for suitable systems on board the ship to ensure that bunkering can be conducted without causing danger to persons, the environment or the ship.

8.2 Functional requirements

8.2.1 This section relates to functional requirements in 3.2.1 to 3.2.11 and 3.2.13 to 3.2.17. In particular the following apply:

8.2.1.1 The piping system for transfer of fuel to the storage tank is to be designed such that any leakage from the piping system cannot cause danger to personnel, the environment or the ship.

8.3 Bunkering station

8.3.1 General

8.3.1.1 The bunkering station is to be located on open deck so that sufficient natural ventilation is provided. Closed or semi-enclosed bunkering stations are to be subject to special consideration within the risk assessment.

IR 8.3.1.1 The special consideration should as a minimum include, but not be restricted to, the following design features:

- segregation towards other areas on the ship
- hazardous area plans for the ship
- requirements for forced ventilation
- requirements for leakage detection (e.g. gas detection and low temperature detection)
- safety actions related to leakage detection (e.g. gas detection and low temperature detection)
- access to bunkering station from non-hazardous areas through airlocks
- monitoring of bunkering station by direct line of sight or by CCTV.

8.3.1.2 Connections and piping are to be so positioned and arranged that any damage to the fuel piping does not cause damage to the ship's fuel containment system resulting in an uncontrolled gas discharge.

8.3.1.3 Arrangements are to be made for safe management of any spilled fuel.

8.3.1.4 Suitable means are to be provided to relieve the pressure and remove liquid contents from pump suctions and bunker lines. Liquid is to be discharged to the liquefied gas fuel tanks or other suitable location.

8.3.1.5 The surrounding hull or deck structures shall not be exposed to unacceptable cooling, in case of leakage of fuel.

8.3.1.6 For CNG bunkering stations, low temperature steel shielding is to be considered to determine if the escape of cold jets impinging on surrounding hull structure is possible.

8.3.2 Ship’s fuel hoses

8.3.2.1 Liquid and vapour hoses used for fuel transfer are to be compatible with the fuel and suitable for the fuel temperature.

8.3.2.2 Hoses subject to tank pressure, or the discharge pressure of pumps or vapour compressors, are to be designed for a bursting pressure not less than five times the maximum pressure the hose can be subjected to during bunkering.

8.4 Manifold

8.4.1 The bunkering manifold is to be designed to withstand the external loads during bunkering. The connections at the bunkering station is to be of dry-disconnect type equipped with additional safety dry break-away coupling/ self-sealing quick release. The couplings are to be of a standard type.

8.5 Bunkering system

8.5.1 An arrangement for purging fuel bunkering lines with inert gas is to be provided.

8.5.2 The bunkering system is to be so arranged that no gas is discharged to the atmosphere during filling of storage tanks.

8.5.3 A manually operated stop valve and a remote operated shutdown valve in series, or a combined manually operated and remote valve are to be fitted in every bunkering line close to the connecting point. It is to be possible to operate the remote valve in the control location for bunkering operations and/or from another safe location.

8.5.4 Means are to be provided for draining any fuel from the bunkering pipes upon completion of operation.

8.5.5 Bunkering lines are to be arranged for inerting and gas freeing. When not engaged in bunkering, the bunkering pipes are to be free of gas, unless the consequences of not gas freeing are evaluated and approved.

8.5.6 In case bunkering lines are arranged with a cross-over it is to be ensured by suitable isolation arrangements that no fuel is transferred inadvertently to the ship side not in use for bunkering.

8.5.7 A ship-shore link (SSL) or an equivalent means for automatic and manual ESD communication to the bunkering source is to be fitted.

8.5.8 If not demonstrated to be required at a higher value due to pressure surge considerations a default time as calculated in accordance with 16.7.3.7 from the trigger of the alarm to full closure of the remote operated valve required by 8.5.3 is to be adjusted.

Section 9

Fuel Supply to Consumers

9.1 Goal

9.1.1 The goal of this section is to ensure safe and reliable distribution of fuel to the consumers.

9.2 Functional requirements

9.2.1 This section is related to functional requirements in 3.2.1 to 3.2.6, 3.2.8 to 3.2.11 and 3.2.13 to 3.2.17. In particular the following apply:

.1 the fuel supply system is to be so arranged that the consequences of any release of fuel will be minimized, while providing safe access for operation and inspection;

.2 the piping system for fuel transfer to the consumers is to be designed in a way that a failure of one barrier cannot lead to a leak from the piping system into the surrounding area causing danger to the persons on board, the environment or the ship; and

.3 fuel lines outside the machinery spaces are to be installed and protected so as to minimize the risk of injury to personnel and damage to the ship in case of leakage.

IR 9.2.1 To comply with 9.2.1.2 above, two independent safety barriers are to be in place, while, as far as practicable, using a minimum of flange connections. There is to be, no single common flange or other component where one single failure itself may overcome both primary and secondary barriers and may result in a gas leak into the surrounding area causing danger to the persons on board, the environment or the ship. A single common flange (with two sealing systems) may be accepted at the fuel connection to the gas consumers including GCUs, boilers and components on the engine, such as gas regulating units.

9.3 Redundancy of fuel supply

9.3.1 For single fuel installations the fuel supply system are to be arranged with full

redundancy and segregation all the way from the fuel tanks to the consumer, so that a leakage in one system does not lead to an unacceptable loss of power.

9.3.2 For single fuel installations, the fuel storage is to be divided between two or more tanks. The tanks are to be located in separate compartments.

9.3.3 For type C tank only, one tank may be accepted if two completely separate tank connection spaces are installed for the one tank.

9.4 Safety functions of gas supply system

9.4.1 Fuel storage tank inlets and outlets are to be provided with valves located as close to the tank as possible. Valves required to be operated during normal operation* which are not accessible is to be remotely operated. Tank valves whether accessible or not are to be automatically operated when the safety system required in 15.2.2 is activated.

Note:

** Normal operation in this context is when gas is supplied to consumers and during bunkering operation.*

9.4.2 The main gas supply line to each gas consumer or set of consumers is to be equipped with a manually operated stop valve and an automatically operated "master gas fuel valve" coupled in series or a combined manually and automatically operated valve. The valves are to be situated in the part of the piping that is outside the machinery space containing gas consumers, and placed as near as possible to the installation for heating the gas, if fitted. The master gas fuel valve is to automatically cut off the gas supply when activated by the safety system required in 15.2.2.

9.4.3 The automatic master gas fuel valve is to be operable from safe locations on escape routes inside a machinery space containing a gas consumer, the engine control room, if applicable; outside the machinery space, and from the navigation bridge.

9.4.4 Each gas consumer is to be provided with "double block and bleed" valves arrangement. These valves are to be arranged as outlined in .1 or .2 so that when the safety system required in 15.2.2 is activated this will cause the shutoff valves that are in series to close automatically and the bleed valve to open automatically and:

.1 the two shutoff valves are to be in series in the gas fuel pipe to the gas consuming equipment. The bleed valve is to be in a pipe that vents to a safe location in the open air that portion of the gas fuel piping that is between the two valves in series; or

.2 the function of one of the shutoff valves in series and the bleed valve can be incorporated into one valve body, so arranged that the flow to the gas utilization unit will be blocked and the ventilation opened.

9.4.5 The two valves are to be of the fail-to-close type, while the ventilation valve is to be fail-to-open.

9.4.6 The double block and bleed valves are also to be used for normal stop of the engine.

9.4.7 In cases where the master gas fuel valve is automatically shutdown, the complete gas supply branch downstream of the double block and bleed valve is to be automatically ventilated assuming reverse flow from the engine to the pipe.

9.4.8 There is to be one manually operated shutdown valve in the gas supply line to each engine upstream of the double block and bleed valves to assure safe isolation during maintenance on the engine.

9.4.9 For single-engine installations and multi-engine installations, where a separate master valve is provided for each engine, the master gas fuel valve and the double block and bleed valve functions can be combined.

9.4.10 For each main gas supply line entering an ESD protected machinery space, and each gas supply line to high pressure installations means are to be provided for rapid detection of a rupture in the gas line in the engine-room. When rupture is detected a valve is to be automatically shut off*. This valve is to be located in the gas supply line before it enters the engine-room or as close as possible to the point of entry inside the engine-room. It can be a separate valve or combined with other functions, e.g. the master valve.

Note:

** The shutdown is to be time delayed to prevent shutdown due to transient load variations.*

9.5 Fuel distribution outside of machinery space

9.5.1 Where fuel pipes pass through enclosed spaces in the ship, they are to be protected by a secondary enclosure. This enclosure can be a ventilated duct or a double wall piping system. The duct or double wall piping system is to be mechanically underpressure ventilated with 30 air changes per hour, and gas detection as required in 15.8 is to be provided. Other solutions providing an equivalent safety level may also be accepted by IRS/Administration.

9.5.2 The requirement in 9.5.1 need not be applied for fully welded fuel gas vent pipes led through mechanically ventilated spaces.

9.5.3 The requirements in 9.5.4 to 9.5.6 are to apply to ships constructed on or after 1 January 2024 in lieu of the requirements in 9.5.1 and 9.5.2.

9.5.4 Where gaseous fuel pipes pass through enclosed spaces in the ship, they are to be protected by a secondary enclosure. This enclosure can be a ventilated duct or a double wall piping system. The duct or double wall piping system is to be mechanically under pressure ventilated with 30 air changes per hour, and gas detection as required in 15.8 is to be provided. Other solutions providing an equivalent safety level may also be accepted by the Administration.

9.5.5 The requirement in 9.5.4 need not be applied for fully welded fuel gas vent pipes led through mechanically ventilated spaces.

9.5.6 Liquefied fuel pipes are to be protected by a secondary enclosure able to contain leakages. If the piping system is in a fuel preparation room or a tank connection space, the Administration may waive this requirement. Where gas detection as required in 15.8.1.2 is not fit for purpose, the secondary enclosures around liquefied fuel pipes are to be provided with leakage detection by means of pressure or temperature monitoring systems, or any combination thereof. The secondary enclosure is to be able to withstand the maximum pressure that may build up in the enclosure in case of leakage from the fuel piping. For this purpose, the secondary enclosure may need to be arranged with a pressure relief system that prevents the enclosure from being subjected to pressures above their design pressures.

9.6 Fuel supply to consumers in gas-safe machinery spaces

9.6.1 Fuel piping in gas-safe machinery spaces is to be completely enclosed by a double pipe or duct fulfilling one of the following conditions:

.1 the gas piping is to be a double wall piping system with the gas fuel contained in the inner pipe. The space between the concentric pipes is to be pressurized with inert gas at a pressure greater than the gas fuel pressure. Suitable alarms are to be provided to indicate a loss of inert gas pressure between the pipes. When the inner pipe contains high pressure gas, the system is to be so arranged that the pipe between the master gas valve and the engine is automatically purged with inert gas when the master gas valve is closed; or

.2 the gas fuel piping is to be installed within a ventilated pipe or duct. The air space between the gas fuel piping and the wall of the outer pipe or duct is to be equipped with mechanical underpressure ventilation having a capacity of at least 30 air changes per hour. This ventilation capacity may be reduced to 10 air changes per hour provided automatic filling of the duct with nitrogen upon detection of gas is arranged for. The fan motors are to comply with the required explosion protection in the installation area. The ventilation outlet is to be covered by a protection screen and placed in a position where no flammable gas-air mixture may be ignited; or

.3 other solutions providing an equivalent safety level may also be accepted by the IRS/Administration.

IR 9.6.1 To comply with 9.6.1 above, two independent safety barriers are to be in place, while, as far as practicable, using a minimum of flange connections. There is to be, no single common flange or other component where one

single failure itself may overcome both primary and secondary barriers and may result in a gas leak into the surrounding area causing danger to the persons on board, the environment or the ship. A single common flange (with two sealing systems) may be accepted at the fuel connection to the gas consumers including GCUs, boilers and components on the engine, such as gas regulating units.

9.6.2 The connecting of gas piping and ducting to the gas injection valves are to be completely covered by the ducting. The arrangement is to facilitate replacement and/or overhaul of injection valves and cylinder covers. The double ducting is also required for all gas pipes on the engine itself, until gas is injected into the chamber*.

Note:
** If gas is supplied into the air inlet directly on each individual cylinder during air intake to the cylinder on a low pressure engine, such that a single failure will not lead to release of fuel gas into the machinery space, double ducting may be omitted on the air inlet pipe.*

9.7 Gas fuel supply to consumers in ESD-protected machinery spaces

9.7.1 The pressure in the gas fuel supply system is not to exceed 1 [MPa].

9.7.2 The gas fuel supply lines are to have a design pressure not less than 1 [MPa].

9.8 Design of ventilated duct, outer pipe against inner pipe gas leakage

9.8.1 The design pressure of the outer pipe or duct of fuel systems is not to be less than the maximum working pressure of the inner pipe. Alternatively for fuel piping systems with a working pressure greater than 1 [MPa], the design pressure of the outer pipe or duct is not to be less than the maximum built-up pressure arising in the annular space considering the local instantaneous peak pressure in way of any rupture and the ventilation arrangements.

9.8.2 For high-pressure fuel piping the design pressure of the ducting is to be taken as the higher of the following:

.1 the maximum built-up pressure: static pressure in way of the rupture resulting from the gas flowing in the annular space;

.2 local instantaneous peak pressure in way of the rupture: this pressure is to be taken as the critical pressure given by the following expression:

$$p=p_0(2/k+1)^{(k/k-1)}$$

where:

p_0 = maximum working pressure of the inner pipe

$k = C_p/C_v$ constant pressure specific heat divided by the constant volume specific heat

$k = 1.31$ *for* CH_4

The tangential membrane stress of a straight pipe is not to exceed the tensile strength divided by 1.5 (Rm/1.5) when subjected to the above pressures. The pressure ratings of all other piping components shall reflect the same level of strength as straight pipes.

As an alternative to using the peak pressure from the above formula, the peak pressure found from representative tests can be used. Test reports shall then be submitted.

9.8.3 Verification of the strength is to be based on calculations demonstrating the duct or pipe integrity. As an alternative to calculations, the strength can be verified by representative tests.

9.8.4 For low pressure fuel piping the duct is to be dimensioned for a design pressure not less than the maximum working pressure of the fuel pipes. The duct is to be pressure tested to show that it can withstand the expected maximum pressure at fuel pipe rupture.

9.9 Compressors and pumps

9.9.1 If compressors or pumps are driven by shafting passing through a bulkhead or deck,the bulkhead penetration is to be of gastight type.

9.9.2 Compressors and pumps are to be suitable for their intended purpose. All equipment and machinery is to be such as to be adequately tested to ensure suitability for use within a marine environment. Such items

to be considered would include, but not be limited to:

.1 environmental;

.2 shipboard vibrations and accelerations;

.3 effects of pitch, heave and roll motions, etc.; and

.4 gas composition.

9.9.3 Arrangements are to be made to ensure that under no circumstances liquefied gas can be introduced in the gas control section or gas-fuelled machinery, unless the machinery is designed to operate with gas in liquid state.

9.9.4 Compressors and pumps are to be fitted with accessories and instrumentation necessary for efficient and reliable function.

Section 10

Power Generation including Propulsion and other Gas Consumers

10.1 Goal

10.1.1 The goal of this section is to provide safe and reliable delivery of mechanical, electrical or thermal energy.

10.2 Functional requirements

10.2.1 This section is related to functional requirements in 3.2.1, 3.2.11, 3.2.13, 3.2.16 and 3.2.17. In particular the following apply:

.1 the exhaust systems are to be configured to prevent any accumulation of un-burnt gaseous fuel;

.2 unless designed with the strength to withstand the worst case over pressure due to ignited gas leaks, engine components or systems containing or likely to contain an ignitable gas and air mixture are to be fitted with suitable pressure relief systems. Dependent on the particular engine design this may include the air inlet manifolds and scavenge spaces;

.3 the explosion venting is to be led away from where personnel may normally be present; and

.4 all gas consumers are to have a separate exhaust system.

10.3 Internal combustion engines of piston type

10.3.1 General

10.3.1.1 The exhaust system is to be equipped with explosion relief ventilation sufficiently dimensioned to prevent excessive explosion pressures in the event of ignition failure of one cylinder followed by ignition of the unburned gas in the system.

10.3.1.1.1 For ships constructed on or after 1 January 2024, the exhaust system is to be equipped with explosion relief systems unless designed to accommodate the worst case overpressure due to ignited gas leaks or justified by the safety concept of the engine. A detailed evaluation of the potential for unburnt gas in the exhaust system is to be undertaken covering the complete system from the cylinders up to the open end. This detailed evaluation is to be reflected in the safety concept of the engine.

10.3.1.2 For engines where the space below the piston is in direct communication with the crankcase a detailed evaluation regarding the hazard potential of fuel gas accumulation in the crankcase is to be carried out and reflected in the safety concept of the engine.

10.3.1.3 Each engine other than two-stroke crosshead diesel engines are to be fitted with vent systems independent of other engines for crankcases and sumps.

10.3.1.4 Where gas can leak directly into the auxiliary system medium (lubricating oil, cooling water), an appropriate means are to be fitted after the engine outlet to extract gas in order to prevent gas dispersion. The gas extracted from auxiliary systems media is to

be vented to a safe location in the atmosphere.

10.3.1.5 For engines fitted with ignition systems, prior to admission of gas fuel, correct operation of the ignition system on each unit is to be verified.

10.3.1.6 A means is to be provided to monitor and detect poor combustion or misfiring. In the event that it is detected, gas operation may be allowed provided that the gas supply to the concerned cylinder is shut off and provided that the operation of the engine with one cylinder cut-off is acceptable with respects to torsional vibrations.

10.3.1.7 For engines starting on fuels covered by this Chapter, if combustion has not been detected by the engine monitoring system within an engine specific time after the opening of the fuel supply valve, the fuel supply valve is to be automatically shut off. Means to ensure that any unburnt fuel mixture is purged away from the exhaust system is to be provided.

10.3.2 Dual fuel engines

10.3.2.1 In case of shutoff of the gas fuel supply, the engines are to be capable of continuous operation by oil fuel only without interruption.

10.3.2.2 An automatic system is to be fitted to change over from gas fuel operation to oil fuel operation and vice versa with minimum fluctuation of the engine power. Acceptable reliability is to be demonstrated through testing. In the case of unstable operation on engines when gas firing, the engine is to automatically change to oil fuel mode. Manual activation of gas system shutdown shall always be possible.

10.3.2.3 In case of a normal stop or an emergency shutdown, the gas fuel supply is to be shut off not later than the ignition source. It shall not be possible to shut off the ignition source without first or simultaneously closing the gas supply to each cylinder or to the complete engine.

10.3.3 Gas-only engines

10.3.3.1 In case of a normal stop or an emergency shutdown, the gas fuel supply is to be shut off not later than the ignition source. It shall not be possible to shut off the ignition source without first or simultaneously closing the gas supply to each cylinder or to the complete engine.

10.3.4 Multi-fuel engines

10.3.4.1 In case of shutoff of one fuel supply, the engines is to be capable of continuous operation by an alternative fuel with minimum fluctuation of the engine power.

10.3.4.2 An automatic system is to be fitted to change over from one fuel operation to an alternative fuel operation with minimum fluctuation of the engine power. Acceptable reliability is to be demonstrated through testing. In the case of unstable operation on an engine when using a particular fuel, the engine is to automatically change to an alternative fuel mode. Manual activation shall always be possible.

	GAS ONLY		**DUAL FUEL**	**MULTI FUEL**
IGNITION MEDIUM	Spark	Pilot fuel	Pilot fuel	N/A
MAIN FUEL	Gas	Gas	Gas and/or Oil fuel	Gas and/or Liquid

10.4 Main and auxiliary boilers

10.4.1 Each boiler is to have a dedicated forced draught system. A crossover between boiler force draught systems may be fitted for emergency use providing that any relevant safety functions are maintained.

10.4.2 Combustion chambers and uptakes of boilers are to be designed to prevent any accumulation of gaseous fuel.

10.4.3 Burners are to be designed to maintain stable combustion under all firing conditions.

10.4.4 On main/propulsion boilers an automatic system is to be provided to change from gas fuel operation to oil fuel operation without interruption of boiler firing.

10.4.5 Gas nozzles and the burner control system are to be configured such that gas fuel can only be ignited by an established oil fuel flame, unless the boiler and combustion equipment is designed and approved by the Administration to light on gas fuel.

10.4.6 There are to be arrangements to ensure that gas fuel flow to the burner is automatically cut off unless satisfactory ignition has been established and maintained.

10.4.7 On the fuel pipe of each gas burner a manually operated shutoff valve is to be fitted.

10.4.8 Provisions are to be made for automatically purging the gas supply piping to the burners, by means of an inert gas, after the extinguishing of these burners.

10.4.9 The automatic fuel changeover system required by 10.4.4 is to be monitored with alarms to ensure continuous availability.

10.4.10 Arrangements are to be made that, in case of flame failure of all operating burners, the combustion chambers of the boilers are automatically purged before relighting.

10.4.11 Arrangements are to be made to enable the boilers purging sequence to be manually activated.

10.5 Gas turbines

10.5.1 Unless designed with the strength to withstand the worst case over pressure due to ignited gas leaks, pressure relief systems are to be suitably designed and fitted to the exhaust system, taking into consideration of explosions due to gas leaks. Pressure relief systems within the exhaust uptakes are to be led to a safe location, away from personnel.

10.5.2 The gas turbine may be fitted in a gas-tight enclosure arranged in accordance with the ESD principle outlined in 5.6 and 9.7, however a pressure above 1 [MPa] in the gas supply piping may be accepted within this enclosure.

10.5.3 Gas detection systems and shutdown functions are to be as outlined for ESD protected machinery spaces.

10.5.4 Ventilation for the enclosure is to be as outlined in Section 13 for ESD protected machinery spaces, but shall in addition be arranged with full redundancy (2 x 100% capacity fans from different electrical circuits).

10.5.5 For other than single fuel gas turbines, an automatic system is to be fitted to change over easily and quickly from gas fuel operation to oil fuel operation and vice-versa with minimum fluctuation of the engine power.

10.5.6 Means are to be provided to monitor and detect poor combustion that may lead to unburnt fuel gas in the exhaust system during operation. In the event that it is detected, the fuel gas supply is to be shutdown.

10.5.7 Each turbine is to be fitted with an automatic shutdown device for high exhaust temperatures.

Section 11

Fire Safety

11.1 Goal

11.1.1 The goal of this section is to provide for fire protection, detection and fighting for all system components related to the storage, conditioning, transfer and use of natural gas as ship fuel.

11.2 Functional requirements

11.2.1 This section is related to functional requirements in 3.2.2, 3.2.4, 3.2.5, 3.2.7, 3.2.12, 3.2.14, 3.2.15 and 3.2.17.

11.3 Fire protection

11.3.1 Any space containing equipment for the fuel preparation such as pumps, compressors, heat exchangers, vaporizers and pressure vessels is to be regarded as a machinery space of category A for fire protection purposes.

IR 11.3.1.1 Fire protection in 11.3.1 means structural fire protection, not including means of escape.

IR 11.3.1.2. Enclosed spaces containing equipment for fuel preparation such as pumps or compressors or other potential ignition sources are to be provided with a fixed fire-extinguishing system complying with the provisions of Pt.6, Ch.3, 4.4.1.1 (SOLAS II-2/10.4.1.1) and Pt.6, Ch.8 (FSS Code) and taking into account the necessary concentrations / application rate required for extinguishing gas fires.

11.3.2 Any boundary of accommodation spaces, service spaces, control stations, escape routes and machinery spaces, facing fuel tanks on open deck, is to be shielded by A-60 class divisions. The A-60 class divisions are to extend up to the underside of the deck of the navigation bridge. In addition, fuel tanks are to be segregated from cargo in accordance with the requirements of the International Maritime Dangerous Goods (IMDG) Code where the fuel tanks are regarded as bulk packaging. For the purposes of the stowage and segregation requirements of the IMDG Code, a fuel tank on the open deck is to be considered a class 2.1 package.

11.3.3 The space containing the fuel containment system is to be separated from the machinery spaces of category A or other rooms with high fire risks. The separation is to be done by a cofferdam of at least 900 [mm] with insulation of A-60 class. When determining the insulation of the space containing fuel containment system from other spaces with lower fire risks, the fuel containment system is to be considered as a machinery space of category A, in accordance with SOLAS regulation II-2/9.For type C tanks, the fuel storage hold space may be considered as a cofferdam.

11.3.3.1 Notwithstanding the last sentence in 11.3.3, for ships constructed on or after 1 January 2024, the fuel storage hold space may be considered as a cofferdam provided that:

.1 the type C tank is not located directly above machinery spaces of category A or other rooms with high fire risk; and

.2 the minimum distance to the A-60 boundary from the outer shell of the type C tank or the boundary of the tank connection space, if any, is not less than 900 mm.

IR 11.3.3 The following "other rooms with high fire risk" should as a minimum be considered, but not be restricted to:

.1 cargo spaces except cargo tanks for liquids with flashpoint above 60°C and except cargo spaces exempted in accordance with Pt.6, Ch.3, 4.7.1.2 or 4.7.1.4 (SOLAS regulations II-2/10.7.1.2 or II-2/10.7.1.4);

.2 vehicle, ro-ro and special category spaces;

.3 service spaces (high risk): galleys, pantries containing cooking appliances, saunas, paint lockers and store-rooms having areas of 4 [m^2] or more, spaces for the storage of flammable liquids and workshops other than those forming part of the machinery space, as provided in Pt.6, Ch.3, 3.2.2.4, 3.2.3.3 and 3.2.4

(SOLAS regulations II-2/9.2.2.4, II-2/9.2.3.3 and II-2/9.2.4); and

.4 accommodation spaces of greater fire risk: saunas, sale shops, barber shops and beauty parlours and public spaces containing furniture and furnishing of other than restricted fire risk and having deck area of 50 [m^2] or more, as provided in Pt.6, Ch.3, 3.2.2.3 (SOLAS regulation II-2/9.2.2.3).

11.3.4 The fuel storage hold space is not to be used for machinery or equipment that may have a fire risk.

11.3.5 The fire protection of fuel pipes led through ro-ro spaces is to be subject to special consideration by the Administration depending on the use and expected pressure in the pipes.

11.3.6 The bunkering station is to be separated by A-60 class divisions towards machinery spaces of category A, accommodation, control stations and high fire risk spaces, except for spaces such as tanks, voids, auxiliary machinery spaces of little or no fire risk, sanitary and similar spaces where the insulation standard may be reduced to class A-0.

11.3.7 If an ESD protected machinery space is separated by a single boundary, the boundary is to be of A-60 class division.

11.4 Fire main

11.4.1 The water spray system required below may be part of the fire main system provided that the required fire pump capacity and working pressure are sufficient for the operation of both the required numbers of hydrants and hoses and the water spray system simultaneously.

11.4.2 When the fuel storage tank(s) is located on the open deck, isolating valves are to be fitted in the fire main in order to isolate damaged sections of the fire main. Isolation of a section of fire main is not to deprive the fire line ahead of the isolated section from the supply of water.

11.5 Water spray system

11.5.1 A water spray system is to be installed for cooling and fire prevention to cover exposed parts of fuel storage tank(s) located on open deck.

11.5.2 The water spray system is also to provide coverage for boundaries of the superstructures, compressor rooms, pump-rooms, cargo control rooms, bunkering control stations, bunkering stations and any other normally occupied deck houses that face the storage tank on open decks unless the tank is located 10 [m] or more from the boundaries.

11.5.3 The system is to be designed to cover all areas as specified above with an application rate of 10 [l/min/m^2] for the largest horizontal projected surfaces and 4 [l/min/m^2] for vertical surfaces.

11.5.4 Stop valves are to be fitted in the water spray application main supply line(s), at intervals not exceeding 40 [m], for the purpose of isolating damaged sections. Alternatively, the system may be divided into two or more sections that may be operated independently, provided the necessary controls are located together in a readily accessible position not likely to be inaccessible in case of fire in the areas protected.

11.5.5 The capacity of the water spray pump is to be sufficient to deliver the required amount of water to the hydraulically most demanding area as specified above in the areas protected.

11.5.6 If the water spray system is not part of the fire main system, a connection to the ship's fire main through a stop valve is to be provided.

11.5.7 Remote start of pumps supplying the water spray system and remote operation of any normally closed valves to the system is to be located in a readily accessible position which is not likely to be inaccessible in case of fire in the areas protected.

11.5.8 The nozzles are to be of an approved full bore type and they are to be arranged to ensure an effective distribution of water throughout the space being protected.

11.6 Dry chemical powder fire-extinguishing system

11.6.1 A permanently installed dry chemical powder fire-extinguishing system is to be installed in the bunkering station area to cover all possible leak points. The capacity is to be

atleast 3.5 [kg/s] for a minimum of 45 s. The system is to be arranged for easy manual release from a safe location outside the protected area.

11.6.2 In addition to any other portable fire extinguishers that may be required elsewhere in IMO instruments, one portable dry powder extinguisher of at least 5 kg capacity is to be located near the bunkering station.

11.7 Fire detection and alarm system

11.7.1 A fixed fire detection and fire alarm system complying with the Fire Safety Systems Code is to be provided for the fuel storage hold spaces and the ventilation trunk to the tank connection space and in the tank connection space and for all other rooms of the fuel gas system where fire cannot be excluded.

11.7.2 Smoke detectors alone are not to be considered sufficient for rapid detection of a fire.

11.8 Regulation for fuel preparation room fire-extinguishing systems

For ships constructed on or after 1 January 2024, fuel preparation rooms containing pumps, compressors or other potential ignition sources are to be provided with a fixed fire-extinguishing system complying with the provisions of Part 6, Chapter 3, Cl. 4.4.1.1 and taking into account the necessary concentrations/application rate required for extinguishing gas fires.

Section 12

Explosion Prevention

12.1 Goal

12.1.1 The goal of this section is to provide for the prevention of explosions and for the limitation of effects from explosion.

12.2 Functional requirements

12.2.1 This section is related to functional requirements in 3.2.2 to 3.2.5, 3.2.7, 3.2.8, 3.2.12 to 3.2.14 and 3.2.17. In particular the following apply:

The probability of explosions is to be reduced to a minimum by:

.1 reducing number of sources of ignition; and

.2 reducing the probability of formation of ignitable mixtures.

12.3 General

12.3.1 Hazardous areas on open deck and other spaces not addressed in this chapter is to be decided based on a recognized standard.* The electrical equipment fitted within hazardous areas is to be according to the same standard.

Note:

** Refer to IEC standard 60092-502, part 4.4: Tankers carrying flammable liquefied gases as applicable.*

12.3.2 Electrical equipment and wiring are in general not to be installed in hazardous areas unless essential for operational purposes based on a recognized standard.*

Note:

** Refer to IEC standard 60092-502: IEC 60092-502:1999 Electrical Installations in Ships – Tankers – Special Features and IEC 60079-10-1:2008 Explosive atmospheres – Part 10-1: Classification of areas – Explosive gas atmospheres, according to the area classification.*

12.3.3 Electrical equipment fitted in an ESD-protected machinery space is to fulfil the following:

.1 in addition to fire and gas hydrocarbon detectors and fire and gas alarms, lighting and ventilation fans are to be certified safe for hazardous area zone 1; and

.2 all electrical equipment in a machinery space containing gas-fuelled engines, and not certified for zone 1 are to be automatically disconnected, if gas concentration above 40% LEL is detected by two detectors in the space containing gas-fuelled consumers.

12.4 Area classification

12.4.1 Area classification is a method of analysing and classifying the areas where explosive gas atmospheres may occur. The object of the classification is to allow the selection of electrical apparatus able to be operated safely in these areas.

12.4.2 In order to facilitate the selection of appropriate electrical apparatus and the design of suitable electrical installations, hazardous areas are divided into zones 0, 1 and 2.*. (See also 12.5 below.)

Note:

** Refer to standards IEC 60079-10-1:2008 Explosive atmospheres part 10-1: Classification of areas – Explosive gas atmospheres and guidance and informative examples given in IEC 60092-502:1999, Electrical Installations in Ships – Tankers – Special Features for tankers.*

12.4.3 Ventilation ducts are to have the same area classification as the ventilated space.

12.5 Hazardous area zones

12.5.1 Hazardous area zone 0

12.5.1.1 This zone includes, but is not limited to the interiors of fuel tanks, any pipework for pressure-relief or other venting systems for fuel tanks, pipes and equipment containing fuel.

12.5.2 Hazardous area zone 1*

12.5.2.1 This zone includes, but is not limited to:

.1 tank connection spaces, fuel storage hold spaces** and interbarrier spaces;

IR.1 For the purposes of hazardous area classification, fuel storage hold spaces containing Type C tanks with all potential leakage sources in a tank connection space and having no access to any hazardous area, are to be considered non-hazardous. Where the fuel storage hold spaces include potential leak sources, e.g. tank connections, they are to be considered as hazardous area zone 1. Where the fuel storage hold spaces include bolted access to the tank connection space, they are to be considered as hazardous area zone 2.

.2 fuel preparation room arranged with ventilation according to 13.6;

.3 areas on open deck, or semi-enclosed spaces on deck, within 3 [m] of any fuel tank outlet, gas or vapour outlet,*** bunker manifold valve, other fuel valve, fuel pipe flange, fuel preparation room ventilation outlets and fuel tank openings for pressure release provided to permit the flow of small volumes of gas or vapour mixtures caused by thermal variation;

.4 areas on open deck or semi-enclosed spaces on deck, within 1.5 [m] of fuel preparation room entrances, fuel preparation room ventilation inlets and other openings into zone 1 spaces;

.5 areas on the open deck within spillage coamings surrounding gas bunker manifold valves and 3 [m] beyond these, up to a height of 2.4 [m] above the deck;

.6 enclosed or semi-enclosed spaces in which pipes containing fuel are located, e.g. ducts around fuel pipes, semi-enclosed bunkering stations;

.7 the ESD-protected machinery space is considered a non-hazardous area during normal operation, but will require equipment required to operate following detection of gas leakage to be certified as suitable for zone 1;

.8 a space protected by an airlock is considered as non-hazardous area during normal operation, but will require equipment required to operate following loss of differential pressure between the protected space and the

hazardous area to be certified as suitable for zone 1; and

.9 except for type C tanks, an area within 2.4 [m] of the outer surface of a fuel containment system where such surface is exposed to the weather.

Note:

** Instrumentation and electrical apparatus installed within these areas should be of a type suitable for zone 1.*

*** Fuel storage hold spaces for type C tanks are normally not considered as zone 1.*

**** Such areas are, for example, all areas within 3 [m] of fuel tank hatches, ullage openings or sounding pipes for fuel tanks located on open deck and gas vapour outlets.*

12.5.3 Hazardous area zone 2*

12.5.3.1 This zone includes, but is not limited to areas within 1.5 [m] surrounding open or semi-enclosed spaces of zone 1.

12.5.3.2 Space containing bolted hatch to tank connection space.

Note :

** Instrumentation and electrical apparatus installed within these areas should be of a type suitable for zone 2.*

Section 13

Ventilation

13.1 Goal

13.1.1 The goal of this section is to provide for the ventilation required for safe operation of gas-fuelled machinery and equipment.

13.2 Functional requirements

13.2.1 This section is related to functional requirements in 3.2.2, 3.2.5, 3.2.8, 3.2.10, 3.2.12 to 3.2.14 and 3.2.17.

13.3 General

13.3.1 Any ducting used for the ventilation of hazardous spaces is to be separate from that used for the ventilation of non-hazardous spaces. The ventilation is to function at all temperatures and environmental conditions the ship will be operating in.

13.3.2 Electric motors for ventilation fans are not to be located in ventilation ducts for hazardous spaces unless the motors are certified for the same hazard zone as the space served.

13.3.3 Design of ventilation fans serving spaces containing gas sources are to fulfil the following:

.1 Ventilation fans are not to produce a source of vapour ignition in either the ventilated space or the ventilation system associated with the space. Ventilation fans and fan ducts, in way of fans only, is to be of non-sparking construction defined as:

.1 impellers or housings of non-metallic material, due regard being paid to the elimination of static electricity;

.2 impellers and housings of non-ferrous metals;

.3 impellers and housings of austenitic stainless steel;

.4 impellers of aluminium alloys or magnesium alloys and a ferrous (including austenitic stainless steel) housing on which a ring of suitable thickness of non-ferrous materials is fitted in way of the impeller, due regard being paid to static electricity and corrosion between ring and housing; or

.5 any combination of ferrous (including austenitic stainless steel) impellers and housings with not less than 13 [mm] tip design clearance.

.2 In no case shall the radial air gap between the impeller and the casing be less than 0.1 of the diameter of the impeller shaft in way of the bearing but not less than 2 [mm]. The gap need not be more than 13 [mm].

.3 Any combination of an aluminium or magnesium alloy fixed or rotating component and a ferrous fixed or rotating component, regardless of tip clearance, is considered a sparking hazard and are not to be used in these places.

13.3.4 Ventilation systems required to avoid any gas accumulation are to consist of independent fans, each of sufficient capacity, unless otherwise specified in this Chapter.

13.3.5 Air inlets for hazardous enclosed spaces are to be taken from areas that, in the absence of the considered inlet, would be non-hazardous. Air inlets for non-hazardous enclosed spaces are to be taken from non-hazardous areas at least 1.5 [m] away from the boundaries of any hazardous area. Where the inlet duct passes through a more hazardous space, the duct is to be gas-tight and have over-pressure relative to this space.

13.3.6 Air outlets from non-hazardous spaces are to be located outside hazardous areas.

13.3.7 Air outlets from hazardous enclosed spaces are to be located in an open area that, in the absence of the considered outlet, would be of the same or lesser hazard than the ventilated space.

13.3.8 The required capacity of the ventilation plant is normally based on the total volume of the room. An increase in required ventilation capacity may be necessary for rooms having a complicated form.

13.3.9 Non-hazardous spaces with entry openings to a hazardous area are to be arranged with an airlock and be maintained at overpressure relative to the external hazardous area. The overpressure ventilation is to be arranged according to the following:

.1 During initial start-up or after loss of overpressure ventilation, before energizing any electrical installations not certified safe for the space in the absence of pressurization, it is to be required to:

.1 proceed with purging (at least 5 air changes) or confirm by measurements that the space is non-hazardous; and

.2 pressurize the space.

.2 Operation of the overpressure ventilation is to be monitored and in the event of failure of the overpressure ventilation:

.1 an audible and visual alarm is to be given at a manned location; and

.2 if overpressure cannot be immediately restored, automatic or programmed, disconnection of electrical installations according to a recognized standard* is to be required.

Note:

** Refer to IEC 60092-502:1999 Electrical Installations in Ships – Tankers – Special Features, table 5.*

13.3.10 Non-hazardous spaces with entry openings to a hazardous enclosed space are to be arranged with an airlock and the hazardous space is to be maintained at underpressure relative to the non-hazardous space. Operation of the extraction ventilation in the hazardous space is to be monitored and in the event of failure of the extraction ventilation:

.1 an audible and visual alarm is to be given at a manned location; and

.2 if underpressure cannot be immediately restored, automatic or programmed, disconnection of electrical installations according to a recognized standard in the non-hazardous space is to be required.

13.4 Tank connection space

13.4.1 The tank connection space is to be provided with an effective mechanical forced ventilation system of extraction type. A ventilation capacity of at least 30 air changes per hour is to be provided. The rate of air changes may be reduced if other adequate means of explosion protection are installed. The equivalence of alternative installations is to be demonstrated by a risk assessment.

13.4.2 Approved automatic fail-safe fire dampers are to be fitted in the ventilation trunk for the tank connection space.

13.5 Machinery spaces

13.5.1 The ventilation system for machinery spaces containing gas-fuelled consumers is to be independent of all other ventilation systems.

IR 13.5.1 Spaces enclosed in the boundaries of machinery spaces (such as purifier's room, engine-room workshops and stores) are considered an integral part of machinery spaces containing gas-fuelled consumers and, therefore, their ventilation system does not need to be independent of the one of machinery spaces.

13.5.2 ESD protected machinery spaces are to have ventilation with a capacity of at least 30 air changes per hour. The ventilation system is to ensure a good air circulation in all spaces, and in particular ensure that any formation of gas pockets in the room are detected. As an alternative, arrangements whereby under normal operation the machinery spaces are ventilated with at least 15 air changes an hour is acceptable provided that, if gas is detected in the machinery space, the number of air changes will automatically be increased to 30 an hour.

13.5.3 For ESD protected machinery spaces the ventilation arrangements are to provide sufficient redundancy to ensure a high level of ventilation availability as defined in a standard acceptable to IMO.*

Note:

**Refer to IEC 60079-10-1.*

13.5.4 The number and power of the ventilation fans for ESD protected engine-rooms and for double pipe ventilation systems for gas safe engine-rooms is to be such that the capacity is not reduced by more than 50% of the total ventilation capacity if a fan with a separate circuit from the main switchboard or emergency switchboard or a group of fans with common circuit from the main switchboard or emergency switchboard, is inoperable.

13.6 Fuel preparation room

13.6.1 Fuel preparation rooms, are to be fitted with effective mechanical ventilation system of the underpressure type, providing a ventilation capacity of at least 30 air changes per hour.

13.6.2 The number and power of the ventilation fans is to be such that the capacity is not reduced by more than 50%, if a fan with a separate circuit from the main switchboard or emergency switchboard or a group of fans with common circuit from the main switchboard or emergency switchboard, is inoperable.

13.6.3 Ventilation systems for fuel preparation rooms, are to be in operation when pumps or compressors are working.

13.7 Bunkering station

13.7.1 Bunkering stations that are not located on open deck are to be suitably ventilated to ensure that any vapour being released during bunkering operations will be removed outside. If the natural ventilation is not sufficient, mechanical ventilation is to be provided in accordance with the risk assessment required by 8.3.1.1.

13.8 Ducts and double pipes

13.8.1 Ducts and double pipes containing fuel piping is to be fitted with effective mechanical ventilation system of the extraction type, providing a ventilation capacity of at least 30 air changes per hour. This is not applicable to double pipes in the engine-room if fulfilling 9.6.1.1.

13.8.2 The ventilation system for double piping and for gas valve unit spaces in gas safe engine-rooms is to be independent of all other ventilation systems.

IR13.8.2 Double piping and gas valve unit spaces in gas safe engine-rooms are considered an integral part of the fuel supply systems and, therefore, their ventilation system does not need to be independent of

other fuel supply ventilation systems provided such fuel supply systems contain only gaseous fuel.

13.8.3 The ventilation inlet for the double wall piping or duct is always to be located in a non-hazardous area away from ignition sources. The inlet opening is to be fitted with a suitable wire mesh guard and protected from ingress of water.

IR 13.8.3 The ventilation inlet for the double wall piping or duct is to be always located in a non-hazardous area in open air away from ignition sources.

13.8.4 The capacity of the ventilation for a pipe duct or double wall piping may be below 30 air changes per hour if a flow velocity of minimum 3 [m/s] is ensured. The flow velocity is to be calculated for the duct with fuel pipes and other components installed.

Section 14

Electrical Installations

14.1 Goal

14.1.1 The goal of this section is to provide for electrical installations that minimize the risk of ignition in the presence of a flammable atmosphere.

14.2 Functional requirements

14.2.1 This section is related to functional requirements in 3.2.1, 3.2.2, 3.2.4, 3.2.7, 3.2.8, 3.2.11, 3.2.13 and 3.2.16 to 3.2.18. In particular the following apply:

.1 Electrical generation and distribution systems, and associated control systems, is to be designed such that a single fault will not result in the loss of ability to maintain fuel tank pressures and hull structure temperature within normal operating limits.

14.3 General

14.3.1 Electrical installations are to be in compliance with a standard at least equivalent to those acceptable to IMO .*

Note :

** Refer to IEC 60092 series standards, as applicable.*

14.3.2 Electrical equipment or wiring is not to be installed in hazardous areas unless essential for operational purposes or safety enhancement.

14.3.3 Where electrical equipment is installed in hazardous areas as provided in 14.3.2 it is to be selected, installed and maintained in accordance with standards at least equivalent to those acceptable to IMO.*

Equipment for hazardous areas is to be evaluated and certified or listed by an accredited testing authority or notified body recognized by the Administration.

Note:

** Refer to the recommendation published by the International Electrotechnical Commission, in particular to publication IEC 60092-502:1999.*

14.3.4 Failure modes and effects of single failure for electrical generation and distribution systems in 14.2 is to be analysed and documented to be at least equivalent to those acceptable to IMO.*

Note:

** Refer to IEC 60812.*

14.3.5 The lighting system in hazardous areas is to be divided between at least two branch circuits. All switches and protective devices are to interrupt all poles or phases and are to be located in a non-hazardous area.

14.3.6 The installation on board of the electrical equipment units is to be such as to ensure the safe bonding to the hull of the units themselves.

14.3.7 Arrangements are to be made to alarm in low-liquid level and automatically shutdown the motors in the event of low-low liquid level. The automatic shutdown may be accomplished by sensing low pump discharge pressure, low motor current, or low-liquid level. This shutdown shall give an audible and visual alarm on the navigation bridge, continuously manned central control station or onboard safety centre.

14.3.8 Submerged fuel pump motors and their supply cables may be fitted in liquefied gas fuel containment systems. Fuel pump motors is to be capable of being isolated from their electrical supply during gas-freeing operations.

14.3.9 For non-hazardous spaces with access from hazardous open deck where the access is protected by an airlock, electrical equipment which is not of the certified safe type is to be de-energized upon loss of overpressure in the space.

14.3.10 Electrical equipment for propulsion, power generation, manoeuvring, anchoring and mooring, as well as emergency fire pumps, that are located in spaces protected by airlocks, are to be of a certified safe type.

Section 15

Control, Monitoring and Safety Systems

15.1 Goal

15.1.1 The goal of this section is to provide for the arrangement of control, monitoring and safety systems that support an efficient and safe operation of the gas-fuelled installation as covered in the other chapters of this Chapter.

15.2 Functional requirements

This chapter is related to functional requirements in 3.2.1, 3.2.2, 3.2.11, 3.2.13 to 3.2.15, 3.2.17 and 3.2.18. In particular the following apply:

.1 the control, monitoring and safety systems of the gas-fuelled installation is to be so arranged that the remaining power for propulsion and power generation is in accordance with 9.3.1 in the event of single failure;

.2 a gas safety system is to be arranged to close down the gas supply system automatically, upon failure in systems as described in Table 1 and upon other fault conditions which may develop too fast for manual intervention;

.3 for ESD protected machinery configurations the safety system is to shutdown gas supply upon gas leakage and in addition disconnect all non-certified safe type electrical equipment in the machinery space;

.4 the safety functions are to be arranged in a dedicated gas safety system that is independent of the gas control system in order to avoid possible common cause failures. This includes power supplies and input and output signal;

.5 the safety systems including the field instrumentation are to be arranged to avoid spurious shutdown, e.g. as a result of a faulty gas detector or a wire break in a sensor loop; and

.6 where two or more gas supply systems are required to meet the requirements, each system is to be fitted with its own set of independent gas control and gas safety systems.

15.3 General

15.3.1 Suitable instrumentation devices are to be fitted to allow a local and a remote reading of essential parameters to ensure a safe management of the whole fuel-gas equipment including bunkering.

15.3.2 A bilge well in each tank connection space of an independent liquefied gas storage tank is to be provided with both a level

indicator and a temperature sensor. Alarm is to be given at high level in the bilge well. Low temperature indication shall activate the safety system.

IR 15.3.2 The "level indicator" required by 15.3.2 is understood to be required for the purposes of indicating an alarm status only; a level switch (float switch) is an instrument example considered to meet this requirement.

15.3.3 For tanks not permanently installed in the ship a monitoring system is to be provided as for permanently installed tanks.

15.4 Bunkering and liquefied gas fuel tank monitoring

15.4.1 Level indicators for liquefied gas fuel tanks

.1 Each liquefied gas fuel tank is to be fitted with liquid level gauging device(s), arranged to ensure a level reading is always obtainable whenever the liquefied gas fuel tank is operational. The device(s) is to be designed to operate throughout the design pressure range of the liquefied gas fuel tank and at temperatures within the fuel operating temperature range.

.2 Where only one liquid level gauge is fitted it is to be arranged so that it can be maintained in an operational condition without the need to empty or gas-free the tank.

.3 Liquefied gas fuel tank liquid level gauges may be of the following types:

.1 indirect devices, which determine the amount of fuel by means such as weighing or in-line flow metering; or

.2 closed devices, which do not penetrate the liquefied gas fuel tank, such as devices using radio-isotopes or ultrasonic devices;

15.4.2 Overflow control

.1 Each liquefied gas fuel tank is to be fitted with a high liquid level alarm operating independently of other liquid level indicators and giving an audible and visual warning when activated.

.2 An additional sensor operating independently of the high liquid level alarm shall automatically actuate a shutoff valve in a manner that will both avoid excessive liquid pressure in the bunkering line and prevent the liquefied gas fuel tank from becoming liquid full.

.3 The position of the sensors in the liquefied gas fuel tank is to be capable of being verified before commissioning. At the first occasion of full loading after delivery and after each dry-docking, testing of high level alarms is to be conducted by raising the fuel liquid level in the liquefied gas fuel tank to the alarm point.

IR.3 The expression “each dry docking” refers to:

- the survey of the outside of the ship’s bottom required for the renewal of the Cargo Ship Safety Construction Certificate and or that Cargo Ship Safety Certificate, for cargo ships.

- The survey of the outside of the ship’s bottom to be carried out every 60 months according to IMO Resolutions A.1104(29, paragraphs 5.10.1 and 5.10.2), for passenger ships.

.4 All elements of the level alarms, including the electrical circuit and the sensor(s), of the high, and overfill alarms, is to be capable of being functionally tested. Systems are to be tested prior to fuel operation in accordance with 18.4.3.

.5 Where arrangements are provided for overriding the overflow control system, they are to be such that inadvertent operation is prevented. When this override is operated continuous visual indication is to be provided at the navigation bridge, continuously manned central control station or onboard safety centre.

15.4.3 The vapour space of each liquefied gas fuel tank is to be provided with a direct reading gauge. Additionally, an indirect indication is to be provided on the navigation bridge, continuously manned central control station or onboard safety centre.

15.4.4 The pressure indicators are to be clearly marked with the highest and lowest pressure permitted in the liquefied gas fuel tank.

15.4.5 A high-pressure alarm and, if vacuum protection is required, a low-pressure alarm is to be provided on the navigation bridge and at a continuously manned central control station or onboard safety centre. Alarms are to be activated before the set pressures of the safety valves are reached.

15.4.6 Each fuel pump discharge line and each liquid and vapour fuel manifold is to be provided with at least one local pressure indicator.

15.4.7 Local-reading manifold pressure indicator is to be provided to indicate the pressure between ship's manifold valves and hose connections to the shore.

15.4.8 Fuel storage hold spaces and interbarrier spaces without open connection to the atmosphere are to be provided with pressure indicator.

15.4.9 At least one of the pressure indicators provided is to be capable of indicating throughout the operating pressure range.

15.4.10 For submerged fuel-pump motors and their supply cables, arrangements is to be made to alarm in low-liquid level and automatically shutdown the motors in the event of low-low liquid level. The automatic shutdown may be accomplished by sensing low pump discharge pressure, low motor current, or low-liquid level. This shutdown shall give an audible and visual alarm on the navigation bridge, continuously manned central control station or onboard safety centre.

15.4.11 Except for independent tanks of type C supplied with vacuum insulation system and pressure build-up fuel discharge unit, each fuel tank is to be provided with devices to measure and indicate the temperature of the fuel in at least three locations; at the bottom and middle of the tank as well as the top of the tank below the highest allowable liquid level.

15.5 Bunkering control

15.5.1 Control of the bunkering is to be possible from a safe location remote from the bunkering station. At this location the tank pressure, tank temperature if required by 15.4.11, and tank level is to be monitored. Remotely controlled valves required by 8.5.3 and 11.5.7 are to be capable of being operated from this location. Overfill alarm and automatic shutdown shall also be indicated at this location.

15.5.2 If the ventilation in the ducting enclosing the bunkering lines stops, an audible and visual alarm is to be provided at the bunkering control location, see also 15.8.

15.5.3 If gas is detected in the ducting around the bunkering lines an audible and visual alarm and emergency shutdown is to be provided at the bunkering control location.

15.6 Gas compressor monitoring

15.6.1 Gas compressors are to be fitted with audible and visual alarms both on the navigation bridge and in the engine control room. As a minimum the alarms shall include low gas input pressure, low gas output pressure, high gas output pressure and compressor operation.

15.6.2 Temperature monitoring for the bulkhead shaft glands and bearings is to be provided, which automatically give a continuous audible and visual alarm on the navigation bridge or in a continuously manned central control station.

15.7 Gas engine monitoring

15.7.1 In addition to the instrumentation provided in accordance with part C of SOLAS chapter II-1, indicators are to be fitted on the navigation bridge, the engine control room and the manoeuvring platform for:

.1 operation of the engine in case of gas-only engines; or

.2 operation and mode of operation of the engine in the case of dual fuel engines.

15.8 Gas detection

15.8.1 Permanently installed gas detectors are to be fitted in:

.1 the tank connection spaces;

.2 all ducts around fuel pipes;

.3 machinery spaces containing gas piping, gas equipment or gas consumers;

.4 compressor rooms and fuel preparation rooms;

.5 other enclosed spaces containing fuel piping or other fuel equipment without ducting;

.6 other enclosed or semi-enclosed spaces where fuel vapours may accumulate including interbarrier spaces and fuel storage hold spaces of independent tanks other than type C;

.7 airlocks;

.8 gas heating circuit expansion tanks;

.9 motor rooms associated with the fuel systems; and

.10 at ventilation inlets to accommodation and machinery spaces if required based on the risk assessment required in 4.2.

15.8.2 In each ESD-protected machinery space, redundant gas detection systems are to be provided.

15.8.3 The number of detectors in each space is to be considered taking into account the size, layout and ventilation of the space.

15.8.4 The detection equipment is to be located where gas may accumulate and in the ventilation outlets. Gas dispersal analysis or a physical smoke test is to be used to find the best arrangement.

15.8.5 Gas detection equipment is to be designed, installed and tested in accordance with a recognized standard such as IEC 60079-29-1.

15.8.6 An audible and visible alarm is to be activated at a gas vapour concentration of 20% of the lower explosion limit (LEL). The safety system is to be activated at 40% of LEL at two detectors (see footnote 1 in Table 1).

15.8.7 For ventilated ducts around gas pipes in the machinery spaces containing gas-fuelled engines, the alarm limit can be set to 30% LEL. The safety system is to be activated at 60% of LEL at two detectors (see footnote 1 in Table 1).

15.8.8 Audible and visible alarms from the gas detection equipment are to be located on the navigation bridge or in the continuously manned central control station.

15.8.9 Gas detection required by this section is to be continuous without delay.

15.9 Fire detection

15.9.1 Required safety actions at fire detection in the machinery space containing gas-fuelled engines and rooms containing independent tanks for fuel storage hold spaces are given in Table 1 below.

15.10 Ventilation

15.10.1 Any loss of the required ventilating capacity is to give an audible and visual alarm on the navigation bridge or in a continuously manned central control station or safety centre.

IR 15.10.1 Acceptable means to confirm that the ventilation system has the “required ventilating capacity” in operation are, but not limited to:

- Monitoring of the ventilation electric motor or fan operation combined with under pressure indication; or
- Monitoring of the ventilation electric motor or fan operation combined with ventilation flow indication; or
- Monitoring of ventilation flow rate to indicate that the required air flow rate is established.

15.10.2 For ESD protected machinery spaces the safety system is to be activated upon loss of ventilation in engine-room.

15.11 Safety functions of fuel supply systems

15.11.1 If the fuel supply is shut off due to activation of an automatic valve, the fuel supply is not to be opened until the reason for the disconnection is ascertained and the necessary precautions taken. A readily visible notice giving instruction to this effect is to be placed at the operating station for the shutoff valves in the fuel supply lines.

15.11.2 If a fuel leak leading to a fuel supply shutdown occurs, the fuel supply is not to be operated until the leak has been found and dealt with. Instructions to this effect are to be placed in a prominent position in the machinery space.

15.11.3 A caution placard or signboard is to be permanently fitted in the machinery space containing gas-fuelled engines stating that heavy lifting, implying danger of damage to the fuel pipes, is not to be done when the engine(s) is running on gas.

15.11.4 Compressors, pumps and fuel supply are to be arranged for manual remote emergency stop from the following locations as applicable:

.1 navigation bridge;
.2 cargo control room;
.3 onboard safety centre;
.4 engine control room;
.5 fire control station; and
.6 adjacent to the exit of fuel preparation rooms.

The gas compressor shall also be arranged for manual local emergency stop.

Table 1 : Monitoring of gas supply system to engines

Parameter	Alarm	Automatic shutdown of tank valve[6)]	Automatic shutdown of gas supply to machinery space containing gas-fuelled engines	Comments
Gas detection in tank connection space at 20% LEL	X			
Gas detection on two detectors[1)] in tank connection space at 40% LEL	X	X		
Fire detection in fuel storage hold space	X			
Fire detection in ventilation trunk to the tank connection space and in the tank connection space	X			
Bilge well high level in tank connection space	X			
Bilge well low temperature in tank connection space	X	X		
Gas detection in duct between tank and machinery space containing gas-fuelled engines at 20% LEL	X			

Table 1 : (Contd.)

Parameter	Alarm	Automatic shutdown of tank valve[6)]	Automatic shutdown of gas supply to machinery space containing gas-fuelled engines	Comments
Gas detection on two detectors[1)] in duct between tank and machinery space containing gas-fuelled engines at 40% LEL	X	X[2)]		
Gas detection in fuel preparation room at 20% LEL	X			
Gas detection on two detectors[1)] in fuel preparation room at 40% LEL	X	X[2)]		
Gas detection in duct inside machinery space containing gas-fuelled engines at 30% LEL	X			If double pipe fitted in machinery space containing gas-fuelled engines
Gas detection on two detectors[1)] in duct inside machinery space containing gas-fuelled engines at 60% LEL	X		X[3)]	If double pipe fitted in machinery space containing gas-fuelled engines
Gas detection in ESD protected machinery space containing gas-fuelled engines at 20% LEL	X			
Gas detection on two detectors[1)] in ESD protected machinery space containing gas-fuelled engines at 40% LEL	X		X	It shall also disconnect non certified safe electrical equipment in machinery space containing gas-fuelled engines
Loss of ventilation in duct between tank and machinery space containing gas-fuelled engines	X		X[2)]	
Loss of ventilation in duct inside machinery space containing gas-fuelled engines[5)]	X		X[3)]	If double pipe fitted in machinery space containing gas-fuelled engines

Table 1 : (Contd.)

Parameter	Alarm	Automatic shutdown of tank valve[6)]	Automatic shutdown of gas supply to machinery space containing gas-fuelled engines	Comments
Loss of ventilation in ESD protected machinery space containing gas-fuelled engines	X		X	
Fire detection in machinery space containing gas-fuelled engines	X			
Abnormal gas pressure in gas supply pipe	X			
Failure of valve control actuating medium	X		X[4)]	Time delayed as found necessary
Automatic shutdown of engine (engine failure)	X		X[4)]	
Manually activated emergency shutdown of engine	X		X	

1) Two independent gas detectors located close to each other are required for redundancy reasons. If the gas detector is of self-monitoring type the installation of a single gas detector can be permitted.

2) If the tank is supplying gas to more than one engine and the different supply pipes are completely separated and fitted in separate ducts and with the master valves fitted outside of the duct, only the master valve on the supply pipe leading into the duct where gas or loss of ventilation is detected shall close.

3) If the gas is supplied to more than one engine and the different supply pipes are completely separated and fitted in separate ducts and with the master valves fitted outside of the duct and outside of the machinery space containing gas-fuelled engines, only the master valve on the supply pipe leading into the duct where gas or loss of ventilation is detected shall close.

4) Only double block and bleed valves to close.

5) If the duct is protected by inert gas (see 9.6.1.1) then loss of inert gas overpressure shall lead to the same actions as given in this Table.

6) Valves referred to in 9.4.1.

Section 16

Manufacture, Workmanship and Testing

16.1 General

16.1.1 The manufacture, testing, inspection and documentation are to be in accordance with recognized standards and the requirements given in this Chapter.

16.1.2 Where post-weld heat treatment is specified or required, the properties of the base material is to be determined in the heat treated condition, in accordance with the applicable Tables of Section 7, and the weld properties is to be determined in the heat treated condition in accordance with 16.3. In cases where a post-weld heat treatment is applied, the test requirements may be modified at the discretion of the Administration/ IRS.

16.2 General test requirements and specifications

16.2.1 Tensile test

16.2.1.1 Tensile testing is to be carried out in accordance with recognized standards.

16.2.1.2 Tensile strength, yield stress and elongation are to be to the satisfaction of the IRS/Administration. For carbon-manganese steel and other materials with definitive yield points, consideration is to be given to the limitation of the yield to tensile ratio.

16.2.2 Toughness test

16.2.2.1 Acceptance tests for metallic materials are to include Charpy V-notch toughness tests unless otherwise specified by the IRS/Administration. The specified Charpy V-notch requirements are minimum average energy values for three full size (10 [mm] × 10 [mm]) specimens and minimum single energy values for individual specimens. Dimensions and tolerances of Charpy V-notch specimens is to be in accordance with recognized standards. The testing and requirements for specimens smaller than 5 [mm] in size is to be in accordance with recognized standards. Minimum average values for sub-sized specimens are to be:

Charpy V-notch specimen size (mm)	Minimum average energy of three specimens
10 x 10	KV
10 x 7.5	5/6 KV
10 x 5.0	2/3 KV

where:

KV = the energy values [J] specified in Tables 7.1 to 7.4.

Only one individual value may be below the specified average value, provided it is not less than 70% of that value.

16.2.2.2 For base metal, the largest size Charpy V-notch specimens possible for the material thickness is to be machined with the specimens located as near as practicable to a point midway between the surface and the centre of the thickness and the length of the notch perpendicular to the surface as shown in Fig.16.1.

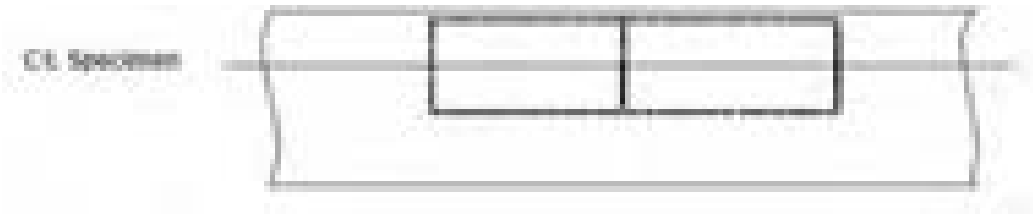


Fig.16.1: Orientation of base metal test specimen

16.2.2.3 For a weld test specimen, the largest size Charpy V-notch specimens possible for the material thickness is to be machined, with the specimens located as near as practicable to a point midway between the surface and the centre of the thickness. In all cases the distance from the surface of the material to the edge of the specimen is to be approximately 1 [mm] or greater. In addition, for double-V butt welds, specimens are to be machined closer to the surface of the second welded section. The specimens is to be taken generally at each of the following locations, as shown in Fig.16.2, on the centreline of the welds, the fusion line and 1 [mm], 3 [mm] and 5 [mm] from the fusion line.

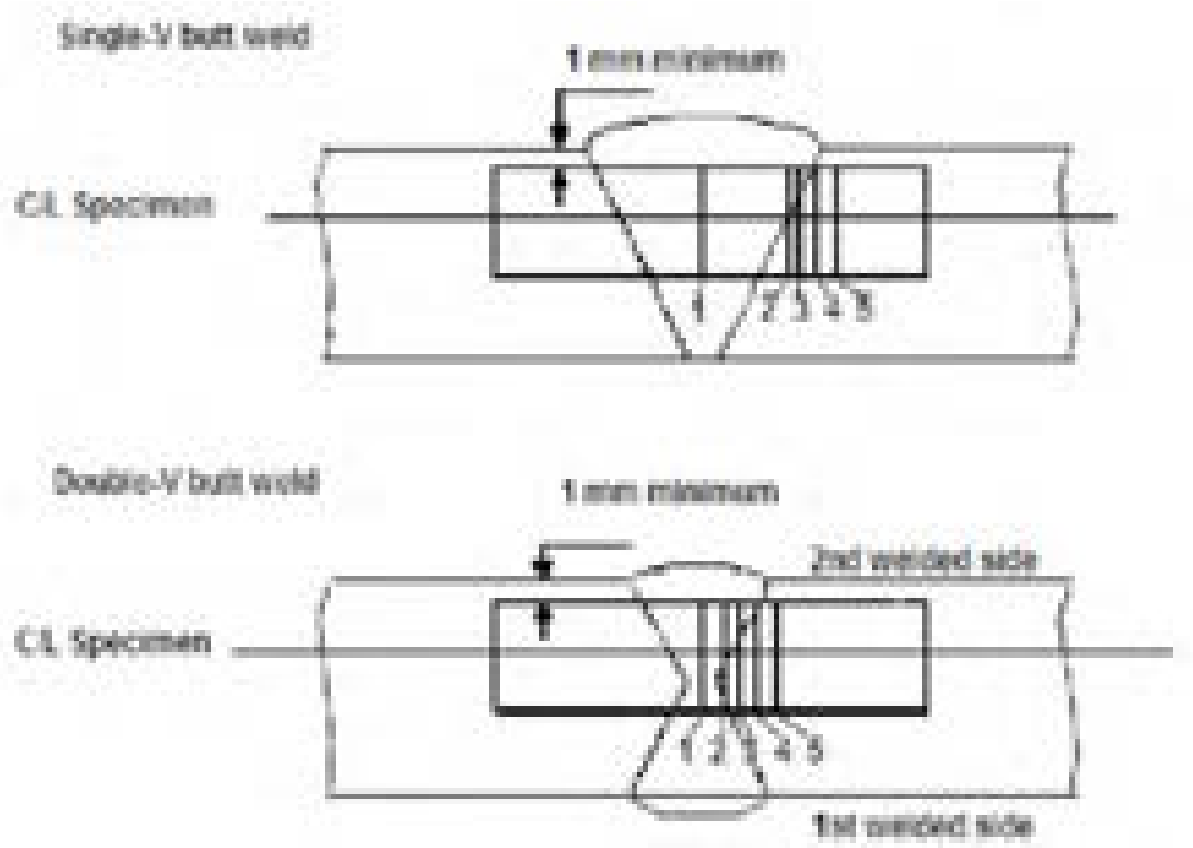


Fig.16.2 : Orientation of weld test specimen

Notch locations in Fig.16.2:

.1 centreline of the weld;

.2 on fusion line;

.3 in heat-affected zone (HAZ), 1 [mm] from fusion line;

.4 in HAZ, 3 [mm] from fusion line; and

.5 in HAZ, 5 [mm] from fusion line.

16.2.2.4 If the average value of the three initial Charpy V-notch specimens fails to meet the stated requirements, or the value for more than one specimen is below the required average value, or when the value for one specimen is below the minimum value permitted for a single specimen, three additional specimens from the same material may be tested and the results combined with those previously obtained to form a new average. If this new average complies with the requirements and if no more than two individual results are lower, than the required average and no more than one result is lower than the required value for a single specimen, the piece or batch may be accepted.

16.2.3 Bend test

16.2.3.1 The bend test may be omitted as a material acceptance test, but is required for weld tests. Where a bend test is performed, this is to be done in accordance with recognized standards.

16.2.3.2 The bend tests are to be transverse bend tests, which may be face, root or side bends at the discretion of the IRS/Administration. However, longitudinal bend tests may be required in lieu of transverse bend tests in cases where the base material and weld metal have different strength levels.

16.2.4 Section observation and other testing

16.2.4.1 Macrosection, microsection observations and hardness tests may also be required by the IRS/Administration, and they are to be carried out in accordance with recognized standards, where required.

16.3 Welding of metallic materials and non-destructive testing for the fuel containment system

16.3.1 General

16.3.1.1 This sub-section applicable to primary and secondary barriers only, including the inner hull where this forms the secondary barrier. Acceptance testing is specified for carbon, carbon-manganese, nickel alloy and stainless steels, but these tests may be adapted for other materials. At the discretion of the IRS/Administration, impact testing of stainless steel and aluminium alloy weldments may be omitted and other tests may be specially required for any material.

16.3.2 Welding consumables

16.3.2.1 Consumables intended for welding of fuel tanks are to be in accordance with recognized standards. Deposited weld metal tests and butt weld tests is to be required for all consumables. The results obtained from tensile and Charpy V-notch impact tests are to be in accordance with recognized standards. The chemical composition of the deposited weld metal is to be recorded for information.

16.3.3 Welding procedure tests for fuel tanks and process pressure vessels

16.3.3.1 Welding procedure tests for fuel tanks and process pressure vessels are required for all butt welds.

16.3.3.2 The test assemblies are to be representative of:

.1 each base material;

.2 each type of consumable and welding process; and

.3 each welding position.

16.3.3.3 For butt welds in plates, the test assemblies is to be so prepared that the rolling direction is parallel to the direction of welding. The range of thickness qualified by each welding procedure test is to be in accordance with recognized standards. Radiographic or ultrasonic testing may be performed at the option of the fabricator.

16.3.3.4 The following welding procedure tests for fuel tanks and process pressure vessels is to be done in accordance with 16.2 with specimens made from each test assembly:

.1 cross-weld tensile tests;

.2 longitudinal all-weld testing where required by the recognized standards;

.3 transverse bend tests, which may be face, root or side bends. However, longitudinal bend tests may be required in lieu of transverse bend tests in cases where the base material and weld metal have different strength levels;

.4 one set of three Charpy V-notch impacts, generally at each of the following locations, as shown in Fig.16.2:

.1 centreline of the welds;

.2 fusion line;

.3 1 [mm] from the fusion line;

.4 3 [mm] from the fusion line; and

.5 5 [mm] from the fusion line;

.5 macrosection, microsection and hardness survey may also be required.

16.3.3.5 Each test are to satisfy the following:

.1 Tensile tests: cross-weld tensile strength is not to be less than the specified minimum tensile strength for the appropriate parent materials. For materials such as aluminium alloys,

reference is to be made to 6.4.12.1.1.3 with regard to the requirements for weld metal strength of under-matched welds (where the weld metal has a lower tensile strength than the parent metal). In every case, the position of fracture is to be recorded for information;

.2 Bend tests: no fracture is acceptable after a 180° bend over a former of a diameter four times the thickness of the test pieces; and

.3 Charpy V-notch impact tests: Charpy V-notch tests is to be conducted at the temperature prescribed for the base material being joined. The results of weld metal impact tests, minimum average energy (KV), are to be no less than 27 [J]. The weld metal requirements for sub-size specimens and single energy values are to be in accordance with 16.2.2. The results of fusion line and heat affected zone impact tests are to show a minimum average energy (KV) in accordance with the transverse or longitudinal requirements of the base material, whichever is applicable, and for sub-size specimens, the minimum average energy (KV) is to be in accordance with 16.2.2. If the material thickness does not permit machining either full-size or standard sub-size specimens, the testing procedure and acceptance standards are to be in accordance with recognized standards.

16.3.3.6 Procedure tests for fillet welding is to be in accordance with recognized standards. In such cases, consumables are to be so selected that exhibit satisfactory impact properties.

16.3.4 Welding procedure tests for piping

16.3.4.1 Welding procedure tests for piping are to be carried out and are to be similar to those detailed for fuel tanks in 16.3.3.

16.3.5 Production weld tests

16.3.5.1 For all fuel tanks and process pressure vessels except membrane tanks, production weld tests are to generally be performed for approximately each 50 [m] of butt-weld joints and are to be representative of each welding position. For secondary barriers, the same type production tests as required for primary tanks are to be performed, except that the number of tests may be reduced subject to agreement with IRS/Administration. Tests, other than those specified in 16.3.5.2 to 16.3.5.5 may be required for fuel tanks or secondary barriers.

16.3.5.2 The production tests for types A and B independent tanks are to include bend tests and, where required for procedure tests, one set of three Charpy V-notch tests. The tests are to be made for each 50 [m] of weld. The Charpy V-notch tests are to be made with specimens having the notch alternately located in the centre of the weld and in the heat affected zone (most critical location based on procedure qualification results). For austenitic stainless steel, all notches are to be in the centre of the weld.

16.3.5.3 For type C independent tanks and process pressure vessels, transverse weld tensile tests are required in addition to the tests listed in 16.3.5.2. Tensile tests are to meet requirement 16.3.3.5.

16.3.5.4 The quality assurance/quality control (QA/QC) program are to ensure the continued conformity of the production welds as defined in the material manufacturers quality manual (QM).

16.3.5.5 The test requirements for membrane tanks are the same as the applicable test requirements listed in 16.3.3.

16.3.6 Non-destructive testing

16.3.6.1 All test procedures and acceptance standards are to be in accordance with recognized standards, unless the designer specifies a higher standard in order to meet design assumptions. Radiographic testing is to be used in principle to detect internal defects. However, an approved ultrasonic test procedure in lieu of radiographic testing may be conducted, but in addition supplementary radiographic testing at selected locations is to be carried out to verify the results. Radiographic and ultrasonic testing records are to be retained.

16.3.6.2 For type A independent tanks where the design temperature is below -20°C, and for type B independent tanks, regardless of temperature, all full penetration butt welds of the shell plating of fuel tanks are to be

subjected to non-destructive testing suitable to detect internal defects over their full length. Ultrasonic testing in lieu of radiographic testing may be carried out under the same conditions as described in 16.3.6.1.

16.3.6.3 In each case the remaining tank structure, including the welding of stiffeners and other fittings and attachments, are to be examined by magnetic particle or dye penetrant methods as considered necessary.

16.3.6.4 For type C independent tanks, the extent of non-destructive testing is to be total or partial according to recognized standards, but the controls to be carried out are not to be less than the following:

.1 Total non-destructive testing referred to in 6.4.15.3.2.1.3

Radiographic testing:

.1 all butt welds over their full length.

Non-destructive testing for surface crack detection:

.2 all welds over 10% of their length;

.3 reinforcement rings around holes, nozzles, etc. over their full length.

As an alternative, ultrasonic testing, as described in 16.3.6.1, may be accepted as a partial substitute for the radiographic testing. In addition, IRS/Administration may require total ultrasonic testing on welding of reinforcement rings around holes, nozzles, etc.

.2 Partial non-destructive testing referred to in 6.4.15.3.2.1.3:

Radiographic testing:

.1 all butt welded crossing joints and at least 10% of the full length of butt welds at selected positions uniformly distributed.

Non-destructive testing for surface crack detection:

.2 reinforcement rings around holes, nozzles, etc. over their full length.

Ultrasonic testing:

.3 as may be required by IRS/Administration in each instance.

16.3.6.5 The quality assurance/quality control (QA/QC) program is to ensure the continued conformity of the non-destructive testing of welds, as defined in the material manufacturer's quality manual (QM).

16.3.6.6 Inspection of piping is to be carried out in accordance with the requirements of Section 7.

16.3.6.7 The secondary barrier is to be non-destructive tested for internal defects as considered necessary. Where the outer shell of the hull is part of the secondary barrier, all sheer strake butts and the intersections of all butts and seams in the side shell is to be tested by radiographic testing.

16.4 Other requirements for construction in metallic materials

16.4.1 General

16.4.1.1 Inspection and non-destructive testing of welds are to be in accordance with requirements in 16.3.5 and 16.3.6. Where higher standards or tolerances are assumed in the design, they are also to be satisfied.

16.4.2 Independent tank

16.4.2.1 For type C tanks and type B tanks primarily constructed of bodies of revolution the tolerances relating to manufacture, such as out-of-roundness, local deviations from the true form, welded joints alignment and tapering of plates having different thicknesses, shall comply with recognized standards. The tolerances shall also be related to the buckling analysis referred to in 6.4.15.2.3.1 and 6.4.15.3.3.2.

16.4.3 Secondary barriers

16.4.3.1 During construction the requirements for testing and inspection of secondary barriers is to be approved or accepted by IRS/Administration (see also 6.4.4.5 and 6.4.4.6).

16.4.4 Membrane tanks

16.4.4.1 The quality assurance/quality control (QA/QC) program is to ensure the continued conformity of the weld procedure qualification, design details, materials, construction, inspection and production testing of components. These standards and procedures are to be developed during the prototype testing programme.

16.5 Testing

16.5.1 Testing and inspections during construction

16.5.1.1 All liquefied gas fuel tanks and process pressure vessels is to be subjected to hydrostatic or hydro-pneumatic pressure testing in accordance with 16.5.2 to 16.5.5, as applicable for the tank type.

16.5.1.2 All tanks are to be subject to a tightness test which may be performed in combination with the pressure test referred to in 16.5.1.1.

16.5.1.3 The gas tightness of the fuel containment system with reference to 6.3.3 is to be tested.

16.5.1.4 Regulations with respect to inspection of secondary barriers are to be decided by IRS/Administration in each case, taking into account the accessibility of the barrier (see also 6.4.4).

16.5.1.5 IRS/Administration may require that for ships fitted with novel type B independent tanks, or tanks designed according to 6.4.16 at least one prototype tank and its support is to be instrumented with strain gauges or other suitable equipment to confirm stress levels during the testing required in 16.5.1.1. Similar instrumentation may be required for type C independent tanks, depending on their configuration and on the arrangement of their supports and attachments.

16.5.1.6 The overall performance of the fuel containment system is to be verified for compliance with the design parameters during the first LNG bunkering, when steady thermal conditions of the liquefied gas fuel are reached, in accordance with the requirements of IRS/Administration. Records of the performance of the components and equipment, essential to verify the design parameters, are to be maintained on board and be available to IRS/Administration.

16.5.1.7 The fuel containment system is to be inspected for cold spots during or immediately following the first LNG bunkering, when steady thermal conditions are reached. Inspection of the integrity of thermal insulation surfaces that cannot be visually checked is to be carried out in accordance with the requirements of IRS/Administration.

16.5.1.8 Heating arrangements, if fitted in accordance with 6.4.13.1.1.3 and 6.4.13.1.1.4, are to be tested for required heat output and heat distribution.

16.5.2 Type A independent tanks

16.5.2.1 All type A independent tanks are to be subjected to a hydrostatic or hydro-pneumatic pressure testing. This test is to be performed such that the stresses approximate, as far as practicable, the design stresses, and that the pressure at the top of the tank corresponds at least to the MARVS. When a hydropneumatic test is performed, the conditions shall simulate, as far as practicable, the design loading of the tank and of its support structure including dynamic components, while avoiding stress levels that could cause permanent deformation.

16.5.3 Type B independent tanks

16.5.3.1 Type B independent tanks are to be subjected to a hydrostatic or hydro-pneumatic pressure testing as follows:

.1 The test is to be performed as required in 16.5.2 for type A independent tanks.

.2 In addition, the maximum primary membrane stress or maximum bending stress in primary members under test conditions shall not exceed 90% of the yield strength of the material (as fabricated) at the test temperature. To ensure that this condition is satisfied, when calculations indicate that this stress exceeds 75% of the yield strength the test of the first of a series of identical

tanks is to be monitored by the use of strain gauges or other suitable equipment.

16.5.4 Type C independent tanks and other pressure vessels

16.5.4.1 Each pressure vessel is to be subjected to a hydrostatic test at a pressure measured at the top of the tanks, of not less than 1.5 P_0. In no case during the pressure test shall the calculated primary membrane stress at any point exceed 90% of the yield strength of the material at the test temperature. To ensure that this condition is satisfied where calculations indicate that this stress will exceed 0.75 times the yield strength, the test of the first of a series of identical tanks is to be monitored by the use of strain gauges or other suitable equipment in pressure vessels other than simple cylindrical and spherical pressure vessels.

16.5.4.2 The temperature of the water used for the test is to be at least 30 [°C] above the nil-ductility transition temperature of the material, as fabricated.

16.5.4.3 The pressure is to be held for 2 hours per 25 [mm] of thickness, but in no case less than 2 hours.

16.5.4.4 Where necessary for liquefied gas fuel pressure vessels, a hydro-pneumatic test may be carried out under the conditions prescribed in 16.5.4.1 to 16.5.4.3.

16.5.4.5 Special consideration may be given to the testing of tanks in which higher allowable stresses are used, depending on service temperature. However, requirement in 16.5.4.1 is to be fully complied with.

16.5.4.6 After completion and assembly, each pressure vessel and its related fittings are to be subjected to an adequate tightness test, which may be performed in combination with the pressure testing referred to in 16.5.4.1 or 16.5.4.4 as applicable.

16.5.4.7 Pneumatic testing of pressure vessels other than liquefied gas fuel tanks are to be considered on an individual case basis. Such testing are only to be permitted for those vessels designed or supported such that they cannot be safely filled with water, or for those vessels that cannot be dried and are to be used in a service where traces of the testing medium cannot be tolerated.

16.5.5 Membrane tanks

16.5.5.1 Design development testing

16.5.5.1.1 The design development testing required in 6.4.15.4.1.2 is to include a series of analytical and physical models of both the primary and secondary barriers, including corners and joints, tested to verify that they will withstand the expected combined strains due to static, dynamic and thermal loads at all filling levels. This will culminate in the construction of a prototype scaled model of the complete liquefied gas fuel containment system. Testing conditions considered in the analytical and physical model are to represent the most extreme service conditions the liquefied gas fuel containment system will be likely to encounter over its life. Proposed acceptance criteria for periodic testing of secondary barriers required in 6.4.4 may be based on the results of testing carried out on the prototype scaled model.

16.5.5.1.2 The fatigue performance of the membrane materials and representative welded or bonded joints in the membranes are to be determined by tests. The ultimate strength and fatigue performance of arrangements for securing the thermal insulation system to the hull structure is to be determined by analyses or tests.

16.5.5.2 Testing

.1 In ships fitted with membrane liquefied gas fuel containment systems, all tanks and other spaces that may normally contain liquid and are adjacent to the hull structure supporting the membrane, are to be hydrostatically tested.

.2 All hold structures supporting the membrane are to be tested for tightness before installation of the liquefied gas fuel containment system.

.3 Pipe tunnels and other compartments that do not normally contain liquid need not be hydrostatically tested.

16.6 Welding, post-weld heat treatment and non-destructive testing

16.6.1 General

16.6.1.1 Welding is to be carried out in accordance with 16.3.

16.6.2 Post-weld heat treatment

16.6.2.1 Post-weld heat treatment is to be required for all butt welds of pipes made with carbon, carbon-manganese and low alloy steels. IRS/Administration may waive the requirements for thermal stress relieving of pipes with wall thickness less than 10 [mm] in relation to the design temperature and pressure of the piping system concerned.

16.6.3 Non-destructive testing

16.6.3.1 In addition to normal controls before and during the welding, and to the visual inspection of the finished welds, as necessary for proving that the welding has been carried out correctly and according to the requirements in this paragraph, the following tests are to be required:

.1 100% radiographic or ultrasonic inspection of butt-welded joints for piping systems with;

.1 design temperatures colder than minus 10[°C]; or

.2 design pressure greater than 1 [MPa]; or

.3 gas supply pipes in ESD protected machinery spaces; or

.4 inside diameters of more than 75 [mm]; or

.5 wall thicknesses greater than 10 [mm].

.2 When such butt welded joints of piping sections are made by automatic welding procedures approved by IRS/Administration, then a progressive reduction in the extent of radiographic or ultrasonic inspection can be agreed, but in no case to less than 10% of each joint. If defects are revealed the extent of examination is to be increased to 100% and is to include inspection of previously accepted welds. This approval can only be granted if well-documented quality assurance procedures and records are available to assess the ability of the manufacturer to produce satisfactory welds consistently.

.3 The radiographic or ultrasonic inspection requirement may be reduced to 10% for butt-welded joints in the outer pipe of double-walled fuel piping.

.4 For other butt-welded joints of pipes not covered by 16.6.3.1 and 16.6.3.3, spot radiographic or ultrasonic inspection or other non-destructive tests are to be carried out depending upon service, position and materials. In general, at least 10% of butt-welded joints of pipes are to be subjected to radiographic or ultrasonic inspection.

16.7 Testing requirements

16.7.1 Type testing of piping components

16.7.1.1 Valves

16.7.1.1.1 Each type of piping component intended to be used at a working temperature below minus 55[°C] is to be subject to the following type tests:

.1 Each size and type of valve is to be subjected to seat tightness testing over the full range of operating pressures and temperatures, at intervals, up to the rated design pressure of the valve. Allowable leakage rates are to be to the requirements of IRS/Administration During the testing satisfactory operation of the valve is to be verified.

.2 The flow or capacity is to be certified to a recognized standard for each size and type of valve.

.3 Pressurized components are to be pressure tested to at least 1.5 times the design pressure.

.4 For emergency shutdown valves, with materials having melting temperatures lower than 925[°C], the type testing is to include a fire test to a standard at least equivalent to those acceptable to IMO*

Note :

** Refer to the recommendations by the International Organization for Standardization, in particular publications:*

ISO 19921:2005, Ships and marine technology – Fire resistance of metallic pipe components with resilient and elastomeric seals – Test methods

ISO 19922:2005, Ships and marine technology – Fire resistance of metallic pipe components with resilient and elastomeric seals – Requirements imposed on the test bench

16.7.2 Expansion bellows

16.7.2.1 The following type tests are to be performed on each type of expansion bellows intended for use on fuel piping outside the fuel tank as found acceptable in 7.3.6.4.3.1.3 and where required by IRS/Administration, on those installed within the fuel tanks:

.1 Elements of the bellows, not pre-compressed, but axially restrained are to be pressure tested at not less than five times the design pressure without bursting. The duration of the test is not to be less than five minutes.

.2 A pressure test is to be performed on a type expansion joint, complete with all the accessories such as flanges, stays and articulations, at the minimum design temperature and twice the design pressure at the extreme displacement conditions recommended by the manufacturer without permanent deformation.

.3 A cyclic test (thermal movements) is to be performed on a complete expansion joint, which is to withstand at least as many cycles under the conditions of pressure, temperature, axial movement, rotational movement and transverse movement as it will encounter in actual service. Testing at ambient temperature is permitted when this testing is at least as severe as testing at the service temperature.

.4 A cyclic fatigue test (ship deformation, ship accelerations and pipe vibrations) is to be performed on a complete expansion joint, without internal pressure, by simulating the bellows movement corresponding to a compensated pipe length, for at least 2,000,000 cycles at a frequency not higher than 5 [Hz]. This test is only required when, due to the piping arrangement, ship deformation loads are actually experienced.

16.7.3 System testing requirements

16.7.3.1 The requirements for testing in this section apply to fuel piping inside and outside the fuel tanks. However, relaxation from these requirements for piping inside fuel tanks and open ended piping may be accepted by IRS/Administration.

16.7.3.2 After assembly, all fuel piping is to be subjected to a strength test with a suitable fluid. The test pressure is to be at least 1.5 times the design pressure for liquid lines and 1.5 times the maximum system working pressure for vapour lines. When piping systems or parts of systems are completely manufactured and equipped with all fittings, the test may be conducted prior to installation on board the ship. Joints welded on board are to be tested to at least 1.5 times the design pressure.

16.7.3.3 After assembly on board, the fuel piping system is to be subjected to a leak test using air, or other suitable medium to a pressure depending on the leak detection method applied.

16.7.3.4 In double wall fuel piping systems the outer pipe or duct is also to be pressure tested to show that it can withstand the expected maximum pressure at pipe rupture.

16.7.3.5 All piping systems, including valves, fittings and associated equipment for handling fuel or vapours, are to be tested under normal operating conditions not later than at the first bunkering operation, in accordance with the requirements of IRS/Administration.

16.7.3.6 Emergency shutdown valves in liquefied gas piping systems are to close fully and smoothly within 30 s of actuation. Information about the closure time of the

valves and their operating characteristics is to be available on board, and the closing time is to be verifiable and repeatable.

16.7.3.7 The closing time of the valve referred to in 8.5.8 and 15.4.2.2 (i.e. time from shutdown signal initiation to complete valve closure) is not to be greater than:

$$\frac{3600U}{BR} \quad (\text{second})$$

where:

U = ullage volume at operating signal level [m^3];

BR = maximum bunkering rate agreed between ship and shore facility [m^3/h]; or
5 seconds, whichever is the least.

The bunkering rate is to be adjusted to limit surge pressure on valve closure to an acceptable level, taking into account the bunkering hose or arm, the ship and the shore piping systems, where relevant.

ANNEX 1
Standard for the use of Limit State Methodologies in the Design of Fuel Containment Systems of Novel Configuration

1 GENERAL

1.1 The purpose of this standard is to provide procedures and relevant design parameters of limit state design of fuel containment systems of a novel configuration in accordance with section 6.4.16.

1.2 Limit state design is a systematic approach where each structural element is evaluated with respect to possible failure modes related to the design conditions identified in 6.4.1.6. A limit state can be defined as a condition beyond which the structure, or part of a structure, no longer satisfies the requirements.

1.3 The limit states are divided into the three following categories:

.1 Ultimate Limit States (ULS), which correspond to the maximum load-carrying capacity or, in some cases, to the maximum applicable strain, deformation or instability in structure resulting from buckling and plastic collapse; under intact (undamaged) conditions;

.2 Fatigue Limit States (FLS), which correspond to degradation due to the effect of cyclic loading; and

.3 Accident Limit States (ALS), which concern the ability of the structure to resist accident situations.

1.4 Section 6.4.1 through to section 6.4.14 is to be complied with as applicable depending on the fuel containment system concept.

2 DESIGN FORMAT

2.1 The design format in this standard is based on a Load and Resistance Factor Design format. The fundamental principle of the Load and Resistance Factor Design format is to verify that design load effects, L_d, do not exceed design resistances, R_d, for any of the considered failure modes in any scenario:

$L_d \leq R_d$

A design load F_{dk} is obtained by multiplying the characteristic load by a load factor relevant for the given load category:

$$F_{dk} = \gamma_f \, F_k$$

where:

γ_f is load factor; and

F_k is the characteristic load as specified in section 6.4.9 through to section 6.4.12.

A design load effect L_d (e.g. stresses, strains, displacements and vibrations) is the most unfavourable combined load effect derived from the design loads, and may be expressed by:

$L_d = q\ (F_{d1}, F_{d2}\ \ldots..\ F_{dN})$

Where q denotes the functional relationship between load and load effect determined by structural analyses.

The design resistance R_d is determined as follows:

$$R_d = \frac{R_k}{\gamma_R \gamma_C}$$

where:

R_k is the characteristic resistance. In case of materials covered by section 7, it may be, but not limited to, specified minimum yield stress, specified minimum tensile strength, plastic resistance of cross sections, and ultimate buckling strength;

γ_R is the resistance factor, defined as $\gamma_R = \gamma_m \gamma_s$

γ_m is the partial resistance factor to take account of the probabilistic distribution of the material properties (material factor);

γ_s is the partial resistance factor to take account of the uncertainties on the capacity of the structure, such as the quality of the construction, method considered for determination of the capacity including accuracy of analysis; and

γ_c is the consequence class factor, which accounts for the potential results of failure with regard to release of fuel and possible human injury.

2.2 Fuel containment design is to take into account potential failure consequences. Consequence classes are defined in Table 1, to specify the consequences of failure when the mode of failure is related to the Ultimate Limit State, the Fatigue Limit State, or the Accident Limit State.

Table 1 : Consequence Classes Consequence class

Consequence	**Definition**
Low	Failure implies minor release of the fuel.
Medium	Failure implies release of the fuel and potential for human injury.
High	Failure implies significant release of the fuel and high potential for human injury/fatality.

3 REQUIRED ANALYSES

3.1 Three-dimensional finite element analyses is to be carried out as an integrated model of the tank and the ship hull, including supports and keying system as applicable. All the failure modes is to be identified to avoid unexpected failures. Hydrodynamic analyses are to be carried out to determine the particular ship accelerations and motions in irregular waves, and the response of the ship and its fuel containment systems to these forces and motions.

3.2 Buckling strength analyses of fuel tanks subject to external pressure and other loads causing compressive stresses is to be carried out in accordance with recognized standards. The method shall adequately account for the difference in theoretical and actual buckling stress as a result of plate out of flatness, plate edge misalignment, straightness, ovality and deviation from true circular form over a specified arc or chord length, as relevant.

3.3 Fatigue and crack propagation analysis is to be carried out in accordance with paragraph 5.1 of this standard.

4 ULTIMATE LIMIT STATES

4.1 Structural resistance may be established by testing or by complete analysis taking account of both elastic and plastic material properties. Safety margins for ultimate strength are to be introduced by partial factors of safety taking account of the contribution of stochastic nature of loads and resistance (dynamic loads, pressure loads, gravity loads, material strength, and buckling capacities).

4.2 Appropriate combinations of permanent loads, functional loads and environmental loads including sloshing loads are to be considered in the analysis. At least two load combinations with partial load factors as given in Table 2 is to be used for the assessment of the ultimate limit states. The load factors for permanent and functional loads in load combination '*a*' are relevant for the normally well-controlled and/or specified loads applicable to fuel containment systems such as vapour pressure, fuel weight, system self-weight, etc. Higher load factors may be relevant for permanent and functional loads where the inherent variability and/or uncertainties in the prediction models are higher.

Table 2 : Partial load factors Load combination

Load combination	**Permanent loads**	**Functional loads**	**Environmental loads**
'a'	1.1	1.1	0.7
'b'	1.0	1.0	1.3

4.3 For sloshing loads, depending on the reliability of the estimation method, a larger load factor may be required by IRS/Administration.

4.4 In cases where structural failure of the fuel containment system are considered to imply high potential for human injury and significant release of fuel, the consequence class factor is to be taken as γ_C = 1.2. This value may be reduced if it is justified through risk analysis and subject to the approval by Administration/ IRS. The risk analysis is to take account of factors including, but not limited to, provision of full or partial secondary barrier to protect hull structure from the leakage and less hazards associated with intended fuel. Conversely, higher values may be fixed by Administration/ IRS, for example, for ships carrying more hazardous or higher pressure fuel. The consequence class factor in any case is not to be less than 1.

4.5 The load factors and the resistance factors used are to be such that the level of safety is equivalent to that of the fuel containment systems as described in sections 6.4.2.1 to 6.4.2.5. This may be carried out by calibrating the factors against known successful designs.

4.6 The material factor γ_m is to in general reflect the statistical distribution of the mechanical properties of the material, and needs to be interpreted in combination with the specified characteristic mechanical properties. For the materials defined in section 6, the material factor γ_m may be taken as:

1.1 : when the characteristic mechanical properties specified by IRS/Administration typically represents the lower 2.5% quantile in the statistical distribution of the mechanical properties; or

1.0 : when the characteristic mechanical properties specified by IRS/Administration represents a sufficiently small quantile such that the probability of lower mechanical properties than specified is extremely low and can be neglected.

4.7 The partial resistance factors γ_{si} are to in general be established based on the uncertainties in the capacity of the structure considering construction tolerances, quality of construction, the accuracy of the analysis method applied, etc.

4.7.1 For design against excessive plastic deformation using the limit state criteria given in paragraph 4.8 of this standard, the partial resistance factors γ_{si} is to be taken as follows:

$$\gamma_{s1} = 0.76\frac{B}{k_1}$$

$$\gamma_{s2} = 0.76\frac{D}{k_2}$$

$$k_1 = Min\left(\frac{R_m}{R_e}\frac{B}{A}\,;1\right)$$

$$k_2 = Min\left(\frac{R_m}{R_e}\frac{D}{C}\,;1\right)$$

Factors A, B, C and D are defined in 6.4.15.2.3.1. R_m and R_e are defined in 6.4.12.1.1.3.

The partial resistance factors given above are the results of calibration to conventional type B independent tanks.

4.8 Design against excessive plastic deformation

4.8.1 Stress acceptance criteria given below refer to elastic stress analyses.

4.8.2 Parts of fuel containment systems where loads are primarily carried by membrane response in the structure shall satisfy the following limit state criteria:

$\sigma_m \leq f$

$\sigma_L \leq 1.5\ f$

$\sigma_b \leq 1.5F$

$\sigma_L + \sigma_b \leq 1.5F$

$\sigma_m + \sigma_b \leq 1.5F$

$\sigma_m + \sigma_b + \sigma_g \leq 3.0F$

$\sigma_L + \sigma_b + \sigma_g \leq 3.0F$

where:

σ_m = equivalent primary general membrane stress

σ_L = equivalent primary local membrane stress

σ_b = equivalent primary bending stress

σ_g = equivalent secondary stress

$$f = \frac{R_e}{\gamma_{s1}\,\gamma_m\,\gamma_C}$$

$$F = \frac{R_e}{\gamma_{s2}\,\gamma_m\,\gamma_C}$$

Guidance Note:
The stress summation described above is to be carried out by summing up each stress component (σ_x, σ_y, σ_{xy}), and subsequently the equivalent stress is to be calculated based on the resulting stress components as shown in the example below.

$$\sigma_L + \sigma_b = \sqrt{(\sigma_{Lx} + \sigma_{bx})^2 - (\sigma_{Lx} + \sigma_{bx})(\sigma_{Ly} + \sigma_{by}) + (\sigma_{Ly} + \sigma_{by})^2 + 3(\tau_{Lxy} + \tau_{bxy})^2}$$

4.8.3 Parts of fuel containment systems where loads are primarily carried by bending of girders, stiffeners and plates, are to satisfy the following limit state criteria:

$\sigma_{ms} + \sigma_{bp} \leq 1.25F$ (see notes 1, 2)

$\sigma_{ms} + \sigma_{bp} + \sigma_{bs} \leq 1.25F$ (see note 2)

$\sigma_{ms} + \sigma_{bp} + \sigma_{bs} + \sigma_{bi} + \sigma_g \leq 3.0F$

Note 1: The sum of equivalent section membrane stress and equivalent membrane stress in primary structure ($\sigma_{ms} + \sigma_{bp}$) will normally be directly available from three-dimensional finite element analyses.

Note 2: The coefficient, 1.25, may be modified by IRS/Administration considering the design concept, configuration of the structure, and the methodology used for calculation of stresses.

where:

σ_{ms} = equivalent section membrane stress in primary structure

σ_{bp} = equivalent membrane stress in primary structure and stress in secondary and tertiary structure caused by bending of primary structure

σ_{bs} = section bending stress in secondary structure and stress in tertiary structure caused by bending of secondary structure

σ_{hi} = section bending stress in tertiary structure

σ_g = equivalent secondary stress

$$f = \frac{R_e}{\gamma_{s1}\,\gamma_m\,\gamma_C}$$

$$F = \frac{R_e}{\gamma_{s2}\,\gamma_m\,\gamma_C}$$

The stresses σ_{ms}, σ_{bp}, σ_{hs} and σ_{bi} are defined in 4.8.4.

Skin plates is to be designed in accordance with the requirements of the Administration. When membrane stress is significant, the effect of the membrane stress on the plate bending capacity is to be appropriately considered in addition.

Guidance Note:
The stress summation described above is to be carried out by summing up each stress component (σ_x, σ_y, σ_{xy}), and subsequently the equivalent stress is to be calculated based on the resulting stress components.

4.8.4 Section stress categories

4.8.4.1 Normal stress is the component of stress normal to the plane of reference.

4.8.4.2 Equivalent section membrane stress is the component of the normal stress that is uniformly distributed and equal to the average value of the stress across the cross section of the structure under consideration. If this is a simple shell section, the section membrane stress is identical to the membrane stress defined in paragraph 4.8.2 of this standard.

4.8.4.3 Section bending stress is the component of the normal stress that is linearly distributed over a structural section exposed to bending action, as illustrated in Fig.1.

4.9 The same factors γ_c, γ_m, γ_{si} are to be used for design against buckling unless otherwise stated in the applied recognized buckling standard. In any case the overall level of safety shall not be less than given by these factors.

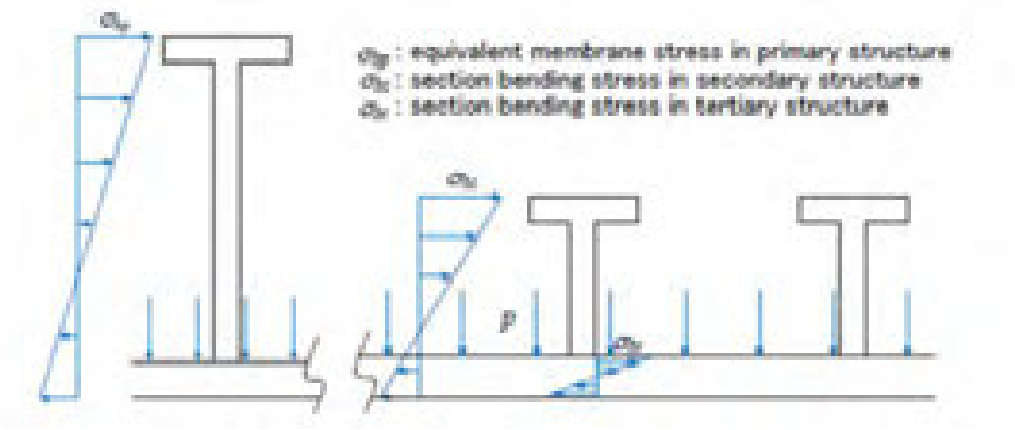


Fig.1 : Definition of the three categories of section stress (Stresses σ_{bp} and σ_{bs} are normal to the cross section shown.)

5 FATIGUE LIMIT STATES

5.1 Fatigue design condition as described in 6.4.12.2 is to be complied with as applicable depending on the fuel containment system concept. Fatigue analysis is required for the fuel containment system designed under 6.4.16 and this standard.

5.2 The load factors for FLS are to be taken as 1 for all load categories.

5.3 Consequence class factor and resistance factor is to be taken as 1.

5.4 Fatigue damage is to be calculated as described in 6.4.12.2.2 to 6.4.12.2.2.5. The calculated cumulative fatigue damage ratio for the fuel containment systems is to be less than or equal to the values given in Table 3.

Table 3 : Maximum allowable cumulative fatigue damage ratio consequence class

	Consequence class		
C_w	**Low**	**Medium**	**High**
	1.0	0.5	0.5*
Note*: Lower value is to be used in accordance with 5.4.12.2.7 to 5.4.12.2.9, depending on the detectability of defect or crack, etc.			

5.5 Lower values may be fixed by IRS/Administration.

5.6 Crack propagation analyses are required in accordance with 6.4.12.2.6 to 6.4.12.2.9. The analysis is to be carried out in accordance with methods laid down in a standard recognized by IRS/Administration.

6 ACCIDENT LIMIT STATES

6.1 Accident design condition as described in 5.4.12.3 is to be complied with as applicable, depending on the fuel containment system concept.

6.2 Load and resistance factors may be relaxed compared to the ultimate limit state considering that damages and deformations can be accepted as long as this does not escalate the accident scenario.

6.3 The load factors for ALS are to be taken as 1 for permanent loads, functional loads and environmental loads.

6.4 Loads mentioned in 6.4.9.3.3.8 and 6.4.9.5 need not be combined with each other or with environmental loads, as defined in 6.4.9.4.

6.5 Resistance factor in general is to be taken as 1.

6.6 Consequence class factors in general are to be taken as defined in paragraph 4.4 of this standard, but may be relaxed considering the nature of the accident scenario.

6.7 The characteristic resistance in general is to be taken as for the ultimate limit state, but may be relaxed considering the nature of the accident scenario.

6.8 Additional relevant accident scenarios are to be determined based on a risk analysis.

7 TESTING

7.1 Fuel containment systems designed according to this standard is to be tested to the same extent as described in 16.2, as applicable depending on the fuel containment system concept.

End of Chapter

Chapter 36

Polar Code

Contents

Introduction

Preamble (to the Polar Code)

Introduction (to the Polar Code)

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Introduction

IR 1.0 General

IR 1.1 These Rules have been prepared to ensure that vessels intended to operate in Polar regions built with a view to Classification with IRS will also comply with the relevant requirements of the IMO Polar Code (IMO Res. MSC 385(94)).

IR 1.2 The IMO Polar Code applies from 01 Jan 2017 to ships operating in polar waters. Ships constructed before 01 Jan 2017 are to meet the relevant requirements of the Polar Code by the first intermediate or renewal survey whichever occurs first after 01 Jan 2018 (according to SOLAS Chapter XIV).

IR 1.3 Responsibility for interpretation of the Polar Code requirements for the purpose of issuing a 'Polar Ship Certificate' under the SOLAS Convention and certificates under the MARPOL Convention lies with the Government of the State, whose flag the ship is entitled to fly.

IR 1.4 This Chapter incorporates the Introduction and the following Chapters of the Polar Code in full, which are relevant for Classification. The Rules also address the structural requirements for prevention of pollution by oil of the Polar Code (Chapter 1 of Part II-A) in Section 8.

Part I A : Safety Measures

Chapter 1 General

Chapter 2 Polar Water Operational Manual

Chapter 3 Ship Structure

Chapter 4 Subdivision and Stability

Chapter 5 Watertight and Weathertight Integrity

Chapter 6 Machinery installations

Chapter 7 Fire Safety / Protection

IR 1.5 The following Chapters of the Polar Code are not addressed in these Rules as they are purely statutory requirements not related to Classification.

Part I A : Safety Measures

Chapter 8 Life Saving Appliances and Arrangements

Chapter 9 Safety of Navigation

Chapter 10 Communication

Chapter 11 Voyage Planning

Chapter 12 Manning and Training

Part II A : Pollution Prevention Measures

Chapter 1 Prevention of Pollution by Oil (Operational Requirements)

Chapter 2 Control of Pollution by Noxious Liquid Substances in Bulk

Chapter 3 Prevention of Pollution by Harmful Substances carried by Sea in Packaged Form

Chapter 4 Prevention of Pollution by Sewage from Ships

Chapter 5 Prevention of Pollution by Garbage from Ships

IR 1.6 The additional guidance provided in the Polar Code (Parts I-B and II-B) for the Chapters covered in these Rules are given as Appendix I to this Chapter. Further, Appendix II of this Chapter gives the model table of contents for Polar Water Operational Manual (PWOM) as provided in the Polar Code.

IR 2.0 Class Notations

IR 2.1 Vessels complying with the requirements of this Chapter will be assigned Class Notation “POLAR”. It may be noted that the ice strength requirements for polar operations for Category A and Category B ships are covered in Pt.5, Ch.32 of the Rules and the appropriate notations (‘PC 1’ to ‘PC 7’) would also be assigned to such ships, based on compliance with the relevant ice class requirements of that Chapter.

Preamble (to the Polar Code)

1 The International Code for Ships Operating in Polar Waters has been developed to supplement existing IMO instruments in order to increase the safety of ships' operation and mitigate the impact on the people and environment in the remote, vulnerable and potentially harsh polar waters.

2 The Code acknowledges that polar water operation may impose additional demands on ships, their systems and operation beyond the existing requirements of the International Convention for the Safety of Life at Sea (SOLAS), 1974, the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto as amended by the 1997 Protocol, and other relevant binding IMO instruments.

3 The Code acknowledges that the polar waters impose additional navigational demands beyond those normally encountered. In many areas, the chart coverage may not currently be adequate for coastal navigation. It is recognized even existing charts may be subject to un-surveyed and uncharted shoals.

4 The Code also acknowledges that coastal communities in the Arctic could be, and that polar ecosystems are, vulnerable to human activities, such as ship operation.

5 The relationship between the additional safety measures and the protection of the environment is acknowledged as any safety measure taken to reduce the probability of an accident, will largely benefit the environment.

6 While Arctic and Antarctic waters have similarities, there are also significant differences. Hence, although the Code is intended to apply as a whole to both Arctic and Antarctic, the legal and geographical differences between the two areas have been taken into account.

7 The key principles for developing the Polar Code have been to use a risk-based approach in determining scope and to adopt a holistic approach in reducing identified risks.

Introduction (to the Polar Code)

1. Goal

The goal of the Polar Code is to provide for safe ship operation and the protection of the polar environment by addressing risks present in polar waters and not adequately mitigated by other IMO instruments.

2. Definitions

For the purpose of the Polar Code and this Chapter, the terms used have the meanings defined in the following paragraphs. Terms used in Part I-A of the Code (and Section 1 to Section 7 of this Chapter), but not defined in this Introduction shall have the same meaning as defined in SOLAS. Terms used in Part II-A of the Code (See Section 8 of this Chapter), but not defined in this Introduction shall have the same meaning as defined in Article 2 of MARPOL and the relevant MARPOL Annexes.

2.1 **Category A** ship means a ship designed for operation in polar waters in at least medium first-year ice, which may include old ice inclusions.

2.2 **Category B** ship means a ship not included in category A, designed for operation in polar waters in at least thin first-year ice, which may include old ice inclusions.

2.3 **Category C** ship means a ship designed to operate in open water or in ice conditions less severe than those included in categories A and B.

2.4 **First-year ice** means sea ice of not more than one winter growth developing from young ice with thickness from 0.3 [m] to 2.0 [m].

2.5 **Ice free waters** means no ice present. If ice of any kind is present this term is not to be used.

2.6 **Ice of land origin** means ice formed on land or in an ice shelf, found floating in water.

2.7 **MARPOL** means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto as amended by the 1997 Protocol.

2.8 **Medium first-year ice** means first-year ice of 70 [cm] to 120 [cm] thickness.

2.9 **Old ice** means sea ice which has survived at least one summer's melt; typical thickness up to 3 [m] or more. It is subdivided into residual first-year ice, second-year ice and multi-year ice.

2.10 **Open water** means a large area of freely navigable water in which sea ice is present in concentrations less than 1/10. No ice of land origin is present.

2.11 **Organization** means the International Maritime Organization.

2.12 **Sea ice** means any form of ice found at sea which has originated from the freezing of sea water.

2.13 **SOLAS** means the International Convention for the Safety of Life at Sea, 1974, as amended.

2.14 **STCW Convention** means the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978, as amended.

2.15 **Thin first-year ice** means first-year ice 30 [cm] to 70 [cm] thick.

3. Sources of Hazards

3.1 The Polar Code considers hazards which may lead to elevated levels of risk due to increased probability of occurrence, more severe consequences, or both:

.1 Ice, as it may affect hull structure, stability characteristics, machinery systems, navigation, the outdoor working environment, maintenance and emergency preparedness tasks and malfunction of safety equipment and systems;

.2 experiencing topside icing, with potential reduction of stability and equipment functionality;

.3 low temperature, as it affects the working environment and human performance, maintenance and emergency preparedness tasks, material properties and equipment efficiency, survival time and

performance of safety equipment and systems;

.4 extended periods of darkness or daylight as it may affect navigation and human performance;

.5 high latitude, as it affects navigation systems, communication systems and the quality of ice imagery information;

.6 remoteness and possible lack of accurate and complete hydrographic data and information, reduced availability of navigational aids and seamarks with increased potential for groundings compounded by remoteness, limited readily deployable SAR facilities, delays in emergency response and limited communications capability, with the potential to affect incident response;

.7 potential lack of ship crew experience in polar operations, with potential for human error;

.8 potential lack of suitable emergency response equipment, with the potential for limiting the effectiveness of mitigation measures;

.9 rapidly changing and severe weather conditions, with the potential for escalation of incidents; and

.10 the environment with respect to sensitivity to harmful substances and other environmental impacts and its need for longer restoration.

3.2 The risk level within polar waters may differ depending on the geographical location, time of the year with respect to daylight, ice-coverage, etc. Thus, the mitigating measures required to address the above specific hazards may vary within polar waters and may be different in Arctic and Antarctic waters.

4. Structure of the Polar Code

The Polar Code consists of Introduction, Parts I and II. The Introduction contains mandatory provisions applicable to both Parts I and II. Part I is subdivided into Part I-A, which contains mandatory provisions on safety measures, and Part I-B containing recommendations on safety. Part II is subdivided into Part II-A, which contains mandatory provisions on pollution prevention, and Part II-B containing recommendations on pollution prevention.

5. Figures 1 and 2 illustrate the Antarctic area and Arctic waters, as defined in SOLAS regulations XIV/1.2 and XIV/1.3, respectively, and MARPOL Annex I, regulations 1.11.7 and 46.2; Annex II, regulations 13.8.1 and 21.2; Annex IV, regulations 17.2 and 17.3; and Annex V, regulations 1.14.7 and 13.2.

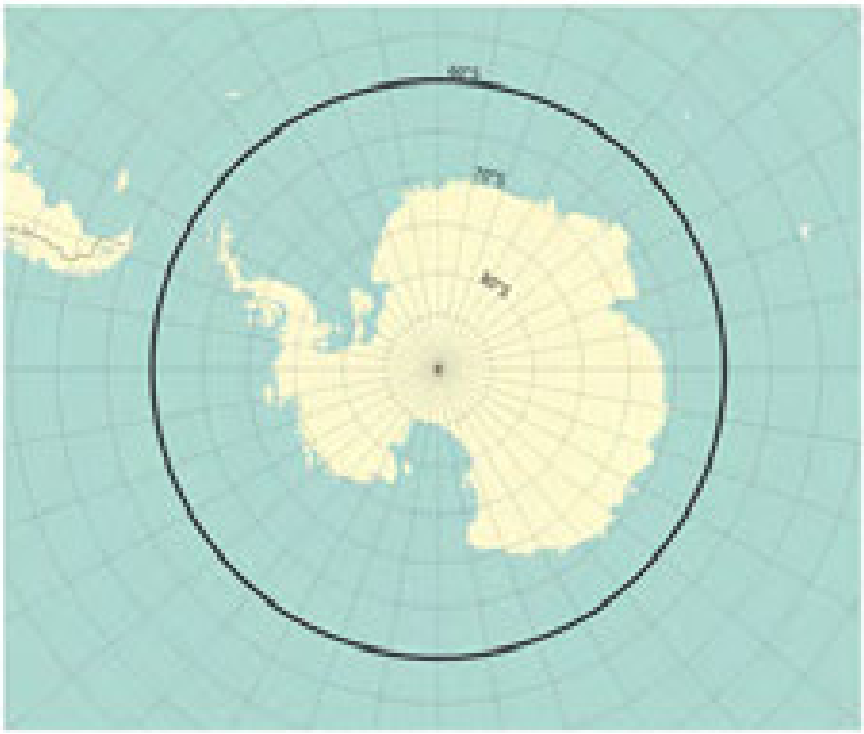

Fig. 1 : Maximum extent of Antarctic area application

Figure 2 : Maximum extent of Arctic waters application

Note : The above figures are for illustrative purposes only

Section 1

General

1.1 Structure

Each Section of this Chapter consists of the overall goal of the Section, functional requirements to fulfill the goal, and regulations. A ship is to be considered to meet a functional requirement set out in these Sections, when either:

.1 the ship's design and arrangements comply with all the rules associated with that functional requirement; or

.2 part(s) or all of the ship's relevant design and arrangements have been reviewed and approved in accordance with the alternative design and arrangement requirements of regulation 4 of SOLAS Chapter XIV, and any remaining parts of the ship comply with the relevant requirements.

1.2 Definitions

In addition to the definitions included in the relevant SOLAS chapters and the introduction of this Chapter, the following definitions are applicable to Sections 1 to 7.

1.2.1 *Bergy waters* mean an area of freely navigable water in which ice of land origin is present in concentrations less than 1/10. There may be *sea ice* present, although the total concentration of all ice shall not exceed 1/10.

1.2.2 *Escort* means any ship with superior ice capability in transit with another ship.

1.2.3 *Escorted operation* means any operation in which a ship's movement is facilitated through the intervention of an escort.

1.2.4 *Habitable environment* means a ventilated environment that will protect against hypothermia.

1.2.5 *Icebreaker* means any ship whose operational profile may include escort or ice management functions, whose powering and dimensions allow it to undertake aggressive operations in ice-covered waters.

1.2.6 *Ice Class* means the notation assigned to the ship by IRS that the ship has been designed for navigation in sea-ice conditions.

(Note: Refer Pt.5, Ch.21 of the Rules for applicable requirements of Ice Class Ships).

1.2.7 *Maximum expected time of rescue* means the time adopted for the design of equipment and system that provide survival support. It is never to be less than 5 days.

1.2.8 *Machinery Installations* means equipment and machinery and its associated piping and cabling, which is necessary for the safe operation of the ship.

1.2.9 *Mean Daily Low Temperature* (MDLT) means the mean value of the daily low temperature for each day of the year over a minimum 10 years period. A data set acceptable to the Administration may be used if 10 years of data is not available. (Refer Pt.3, Ch.2, Sec. 2 of the Rules for method of determining MDLT).

1.2.10 *Polar Class (PC)* means the ice class assigned to the ship by IRS.

(Note: ******* *Refer Pt.5, Ch.32 of the Rules for applicable requirements of Polar Class PC 1 to PC 7).*

1.2.11 *Polar Service Temperature (PST)* means a temperature specified for a ship which is intended to operate in low air temperature, which shall be set at least 10°C below the lowest MDLT for the intended area and season of operation in polar waters.

1.2.12 *Ship intended to operate in low air temperature* means a ship which is intended to undertake voyages to or through areas where the lowest Mean Daily Low Temperature (MDLT) is below -10ºC.

1.2.13 *Tankers* mean oil tankers as defined in SOLAS regulation II-1/2.22, chemical tankers as defined in SOLAS regulation II-1/3.19 and gas carriers as defined in SOLAS regulation VII/11.2.

1.2.14 *Upper ice waterline* means the waterline defined by the maximum draughts forward and aft for operation in ice.

1.3 Polar Ship Certificate and Survey

1.3.1 Every ship to which the Polar Code applies is to have on board a valid Polar Ship Certificate.

1.3.2 Except as provided for in 1.3.3, the Polar Ship Certificate is to be issued after an initial or renewal survey to a ship which complies with the relevant requirements of the Polar Code.

1.3.3 For category C cargo ships, if the result of the assessment in 1.5 is that no additional equipment or structural modification is required to comply with this Chapter, the Polar Ship Certificate may be issued based upon documented verification that the ship complies with all relevant requirements of the Polar Code. In this case, for continued validity of the certificate, an onboard survey is to be undertaken at the next scheduled survey.

1.3.4 The Polar Ship Certificate would be issued either by the Administration or by IRS when authorized. In every case, that Administration assumes full responsibility for the certificate.

1.3.5 The Polar Ship Certificate is to be drawn up in the form corresponding to the model given in Appendix 1 to the Polar Code If the language used is neither English, nor French nor Spanish, the text is to include a translation into one of these languages.

1.3.6 Polar Ship Certificate validity, survey dates and endorsements are to be harmonized with the relevant SOLAS certificates in accordance with the provisions of regulation I/14 of the SOLAS Convention. The certificate is to include a supplement recording equipment required by the Polar Code.

1.3.7 Where applicable, the certificate is to reference a methodology to assess operational capabilities and limitations in ice to the satisfaction of the Administration / IRS taking into account the guidelines developed by the Organization (Refer Appendix-I).

1.4 Performance Standards

1.4.1 Unless expressly provided otherwise, ship systems and equipment addressed in this Chapter are to satisfy at least the same performance standards referred to in SOLAS and the Rules.

1.4.2 For ships operating in low air temperature, a polar service temperature (PST) is to be specified and is to be at least 10°C below the lowest MDLT for the intended area and season of operation in polar waters. Systems and equipment required by this Chapter are to be fully functional at the polar service temperature.

1.4.3 For ships operating in low air temperature, survival systems and equipment are to be fully operational at the polar service temperature during the maximum expected rescue time.

1.5 Operational assessment

In order to establish procedures or operational limitations, an assessment of the ship and its equipment is to be carried out, taking into consideration the following:

.1 the anticipated range of operating and environmental conditions, such as:

.1 operation in low air temperature;
.2 operation in ice;
.3 operation in high latitude; and
.4 potential for abandonment onto ice or land;

.2 hazards, as listed in Para 3 of the Introduction, as applicable; and

.3 additional hazards, if identified.

(Note: Refer to MSC.1/Circ.1519 "Guidance On Methodologies For Assessing Operational Capabilities And Limitations In Ice").

Section 2

Polar Water Operational Manual (PWOM)

2.1 Goal

The goal of this Section is to provide the owner, operator, master and crew with sufficient information regarding the ship's operational capabilities and limitations in order to support their decision-making process.

2.2 Functional requirements

2.2.1 In order to achieve the goal set out in 2.1 above, the following functional requirements are embodied in the rules of this Section.

2.2.2 The Manual is to include information on the ship-specific capabilities and limitations in relation to the assessment required under 1.5.

2.2.3 The Manual is to include or refer to specific procedures to be followed in normal operations and in order to avoid encountering conditions that exceed the ship's capabilities.

2.2.4 The Manual is to include or refer to specific procedures to be followed in the event of incidents in polar waters.

2.2.5 The Manual is to include or refer to specific procedures to be followed in the event that conditions are encountered which exceed the ship's specific capabilities and limitations in 2.2.2.

2.2.6 The Manual is to include or refer to procedures to be followed when using icebreaker assistance, as applicable.

2.3 Rule Requirements

2.3.1 In order to comply with the functional requirements of 2.2.1 to 2.2.6, the Manual is to be carried on board.

2.3.2 In order to comply with the functional requirements of 2.2.2, the Manual is to contain, where applicable, the methodology used to determine capabilities and limitations in ice.

2.3.3 In order to comply with the functional requirements of 2.2.3, the Manual is to include risk-based procedures for the following:

.1 voyage planning to avoid ice and/or temperatures that exceed the ship's design capabilities or limitations;

.2 arrangements for receiving forecasts of the environmental conditions;

.3 means of addressing any limitations of the hydrographic, meteorological and navigational information available;

.4 operation of equipment required under other Sections of this Chapter; and

.5 implementation of special measures to maintain equipment and system functionality under low temperatures, topside icing and the presence of sea ice, as applicable.

2.3.4 In order to comply with the functional requirements of 2.2.4, the Manual is to include risk-based procedures to be followed for:

.1 contacting emergency response providers for salvage, search and rescue (SAR), spill response, etc., as applicable; and

.2 in the case of ships ice strengthened in accordance with Section 3, procedures for maintaining life support and ship integrity in the event of prolonged entrapment by ice.

2.3.5 In order to comply with the functional requirements of 2.2.5, the Manual is to include risk-based procedures to be followed for measures to be taken in the event of encountering ice and/or temperatures which exceed the ship's design capabilities or limitations.

2.3.6 In order to comply with the functional requirements of 2.2.6, the Manual is to include risk-based procedures or monitoring and maintaining safety during operations in ice, as applicable, including any requirements for escort operations or icebreaker assistance.

Different operational limitations may apply depending on whether the ship is operating independently or with icebreaker escort. Where appropriate, the PWOM is to specify both options
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Section 3

Ship Structure

3.1 Goal

The goal of this Section is to provide that the material and scantlings of the structure retain their structural integrity based on global and local response due to environmental loads and conditions.

3.2 Functional requirements

In order to achieve the goal set out in 3.1 above, the following functional requirements are embodied in the rules of this Section:

.1 for ships intended to operate in low air temperature, materials used are to be suitable for operation at the ships polar service temperature; and

.2 in ice strengthened ships, the structure of the ship is to be designed to resist both global and local structural loads anticipated under the foreseen ice conditions.

3.3 Rule Requirements

3.3.1 In order to comply with the functional requirements of 3.2.1 above, materials of exposed structures in ships are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety based on the polar service temperature.

*(Note: ***** Refer Pt.2, Ch.3, Sec.5 and Pt.5, Ch.32, Sec.2 of the Rules for applicable requirements of steel for low temperature applications).*

3.3.2 In order to comply with the functional requirements of 3.2.2 above, the following apply:

.1 scantlings of category A ships are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety;

*(Note: ***** Refer Pt.5, Ch.32 of the Rules for applicable requirements of Polar Class PC 1 to PC 5).*

.2 scantlings of category B ships are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety;

*(Note: ***** Refer Pt.5, Ch.32 of the Rules for applicable requirements of Polar Class PC 6 to PC 7).*

.3 scantlings of ice strengthened category C ships are to be approved by IRS, taking into account acceptable standards adequate for the ice types and concentrations encountered in the area of operation; and

*(Note: ***** Refer Pt.5, Ch.21 of the Rules for applicable requirements of Ice Class Ships).*

.4 a category C ship need not be ice strengthened if, in the opinion of the Administration / IRS, the ship's structure is adequate for its intended operation.

Section 4

Subdivision and Stability

4.1 Goal

The goal of this Section is to ensure adequate subdivision and stability in both intact and damaged conditions.

4.2 Functional requirements

In order to achieve the goal set out in 4.1 above, the following functional requirements are embodied in the rules of this Section:

.1 ships are to have sufficient stability in intact conditions when subject to ice accretion; and

.2 ships of category A and B, constructed on or after 1 January 2017, are to have sufficient residual stability to sustain ice-related damages.

4.3 Rule Requirements

4.3.1 Stability in intact conditions

4.3.1.1 In order to comply with the functional requirement of 4.2.1, for ships operating in areas and during periods where ice accretion is likely to occur, the following icing allowance is to be made in the stability calculations:

.1 30 [kg/m^2] on exposed weather decks and gangways;

.2 7.5 [kg/m^2] for the projected lateral area of each side of the ship above the water plane; and

.3 the projected lateral area of discontinuous surfaces of rail, sundry booms, spars (except masts) and rigging of ships having no sails and the projected lateral area of other small objects is to be computed by increasing the total projected area of continuous surfaces by 5% and the static moments of this area by 10%.

4.3.1.2 Ships operating in areas and during periods where ice accretion is likely to occur are to be:

.1 designed to minimize the accretion of ice; and

.2 equipped with such means for removing ice as the Administration / IRS may require; for example, electrical and pneumatic devices, and/or special tools such as axes or wooden clubs for removing ice from bulwarks, rails and erections.

4.3.1.3 Information on the icing allowance included in the stability calculations is to be given in the PWOM.

4.3.1.4 Ice accretion is to be monitored and appropriate measures taken to ensure that the ice accretion does not exceed the values given in the PWOM.

4.3.2 Stability in damaged conditions

4.3.2.1 In order to comply with the functional requirements of 4.2.2, ships of categories A and B, constructed on or after 1 January 2017, are to be able to withstand flooding resulting from hull penetration due to ice impact. The residual stability following ice damage is to be such that the factor s_i, as defined in SOLAS regulations II-1/7-2.2 and II-1/7-2.3, is equal to one for all loading conditions used to calculate the attained subdivision index in SOLAS regulation II-1/7. However, for cargo ships that comply with subdivision and damage stability regulations in another instrument developed by the Organization, as provided by SOLAS regulation II-1/4.1, the residual stability criteria of that instrument is to be met for each loading condition.

4.3.2.2 The ice damage extents to be assumed when demonstrating compliance with 4.3.2.1 are to be such that:

.1 the longitudinal extent is 4.5% of the upper ice waterline length if centred forward of the maximum breadth on the upper ice waterline, and 1.5% of upper ice waterline length otherwise, and is to be assumed at any longitudinal position along the ship's length;

.2 the transverse penetration extent is 760 [mm], measured normal to the shell over the full extent of the damage; and

.3 the vertical extent is the lesser of 20% of the upper ice waterline draught or the longitudinal extent, and is to be assumed at any vertical position between the keel and 120% of the upper ice waterline draught.

IR4.3.2 In damaged stability calculations, ice accretion effects need not be considered (implied according to 4.2.1 and 4.3.2).

Section 5

Watertight and Weathertight Integrity

5.1 Goal

The goal of this Section is to provide measures to maintain watertight and weathertight integrity.

5.2 Functional requirements

In order to achieve the goal set out in 5.1 above, all closing appliances and doors relevant to watertight and weathertight integrity of the ship are to be operable.

5.3 Rule Requirements

In order to comply with the functional requirements of 5.2 above, the following apply:

.1 for ships operating in areas and during periods where ice accretion is likely to occur, means are to be provided to remove or prevent ice and snow accretion around hatches and doors; and

.2 in addition, for ships intended to operate in low air temperature the following apply:

.1 if the hatches or doors are hydraulically operated, means are to be provided to prevent freezing or excessive viscosity of liquids; and

.2 watertight and weathertight doors, hatches and closing devices which are not within an habitable environment and require access while at sea are to be designed to be operated by personnel wearing heavy winter clothing including thick mittens.

Section 6

Machinery Installations

6.1 Goal

The goal of this Section is to ensure that, machinery installations are capable of delivering the required functionality necessary for safe operation of ships.

6.2 Functional requirements

6.2.1 In order to achieve the goal set out in 6.1 above, the following functional requirements are embodied in the rules of this Section.

6.2.1.1 Machinery installations are to provide functionality under the anticipated environmental conditions, taking into account:

.1 ice accretion and/or snow accumulation;
.2 ice ingestion from seawater;
.3 freezing and increased viscosity of liquids;
.4 seawater intake temperature; and
.5 snow ingestion.

6.2.1.2 In addition, for ships intended to operate in low air temperatures:

.1 machinery installations are to provide functionality under the anticipated environmental conditions, also taking into account:

.1 cold and dense inlet air; and

.2 loss of performance of battery or other stored energy device; and

.2 materials used are to be suitable for operation at the ships polar service temperature.

6.2.1.3 In addition, for ships ice strengthened in accordance with Section 3, machinery installations are to provide functionality under the anticipated environmental conditions, taking into account loads imposed directly by ice interaction.

6.3 Rule Requirements

6.3.1 In order to comply with the functional requirement of 6.2.1.1 above, taking into account the anticipated environmental conditions, the following apply:

.1 machinery installations and associated equipment are to be protected against the effect of ice accretion and/or snow accumulation, ice ingestion from sea water, freezing and increased viscosity of liquids, seawater intake temperature and snow ingestion;

.2 working liquids are to be maintained in a viscosity range that ensures operation of the machinery; and

.3 seawater supplies for machinery systems are to be designed to prevent ingestion of ice* or otherwise arranged to ensure functionality.

*(Note: ***** MSC /Circ. 504, as amended, "Guidance on design and construction of sea inlets under slush ice conditions" may be used).*

6.3.2 In addition, for ships intended to operate in low air temperatures, the following apply:

.1 in order to comply with the functional requirement of 6.2.1.2 above, exposed machinery and electrical installation and appliances are to function at the polar service temperature;

.2 in order to comply with the functional requirement of 6.2.1.2.1 above, means are to be provided to ensure that combustion air for internal combustion engines driving essential machinery is maintained at a temperature in compliance with the criteria provided by the engine manufacturer; and

.3 in order to comply with the functional requirements of 6.2.1.2.2 above, materials of exposed machinery and foundations are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety based on the polar service temperature.

*(Note: ***** Refer Pt.5, Ch.32 of the Rules for applicable requirements of Polar Class PC 1 to PC 7).*

6.3.3 In addition, for ships ice strengthened in accordance with Section 3, in order to comply with the functional requirements of 6.2.1.3 above, the following apply:

.1 scantlings of propeller blades, propulsion line, steering equipment and other appendages of category A ships are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety;

*(Note: ***** Refer Pt.5, Ch.32 of the Rules for applicable requirements of Polar Class PC 1 to PC 5).*

.2 scantlings of propeller blades, propulsion line, steering equipment and other appendages of category B ships are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety; and

*(Note: ***** Refer Pt.5, Ch.32 of the Rules for applicable requirements of Polar Class PC 6 to PC 7).*

.3 scantlings of propeller blades, propulsion line, steering equipment and other appendages of ice-strengthened category C ships are to be approved by IRS, taking into account acceptable standards adequate with the ice types and concentration encountered in the area of operation.

Section 7

Fire Safety / Protection

7.1 Goal

The goal of this Section is to ensure that fire safety systems and appliances are effective and operable, and that means of escape remain available so that persons on board can safely and swiftly escape to the lifeboat and liferaft embarkation deck under the expected environmental conditions.

7.2 Functional requirements

7.2.1 In order to achieve the goal set out in 7.1 above, the following functional requirements are embodied in the rules of this Section:

.1 all components of fire safety systems and appliances if installed in exposed positions are to be protected from ice accretion and snow accumulation;

.2 local equipment and machinery controls are to be arranged so as to avoid freezing, snow accumulation and ice accretion and their location to remain accessible at all time;

.3 the design of fire safety systems and appliances are to take into consideration the need for persons to wear bulky and cumbersome cold weather gear, where appropriate;

.4 means are to be provided to remove or prevent ice and snow accretion from accesses; and

.5 extinguishing media are to be suitable for intended operation.

7.2.2 In addition, for ships intended to operate in low air temperature, the following apply:

.1 all components of fire safety systems and appliances are to be designed to ensure availability and effectiveness under the polar service temperature; and

.2 materials used in exposed fire safety systems are to be suitable for operation at the polar service temperature.

7.3 Rule Requirements

7.3.1 In order to comply with the requirement of 7.2.1.1, the following apply:

.1 isolating and pressure/vacuum valves in exposed locations are to be protected from ice accretion and remain accessible at all time; and

.2 all two-way portable radio communication equipment are to be operable at the polar service temperature.

7.3.2 In order to comply with the requirement of 7.2.1.2, the following apply:

.1 fire pumps including emergency fire pumps, water mist and water spray pumps are to be located in compartments maintained above freezing;

.2 the fire main is to be arranged so that exposed sections can be isolated and means of draining of exposed sections shall be provided. Fire hoses and nozzles need not be connected to the fire main at all times, and maybe stored in protected locations near the hydrants;

.3 firefighter's outfits are to be stored in warm locations on the ship; and

.4 where fixed water-based firefighting systems are located in a space separate from the main fire pumps and use their own independent sea suction, this sea suction is to be also capable of being cleared of ice accumulation.

7.3.3 In addition, for ships intended to operate in low air temperature, the following apply:

.1 In order to comply with the requirement of 7.2.2.1, portable and semi-portable extinguishers are to be located in positions protected from freezing temperatures, as far as practical. Locations subject to freezing are to be provided with extinguishers capable of operation under the polar service temperature.

.2 In order to comply with the functional requirements of 7.2.2.2 above, materials of exposed fire safety systems are to be approved by IRS, taking into account standards acceptable to the Organization* or other standards offering an equivalent level of safety based on the polar service temperature.

*(Note: ***** Refer Pt.2, Ch.3, Sec.5 and Pt.5, Ch.32, Sec.2 of the Rules for applicable requirements of steel for low temperature applications).*

Section 8

Prevention of Pollution by Oil (Structural Requirements)

8.1 Structural Requirements

8.1.1 For category A and B ships constructed on or after 1 January 2017 with an aggregate oil fuel capacity of less than 600 [m^3], all oil fuel tanks are to be separated from the outer shell by a distance of not less than 0.76 [m]. This provision does not apply to small oil fuel tanks with a maximum individual capacity not greater than 30 [m^3].

8.1.2 For category A and B ships other than oil tankers constructed on or after 1 January 2017, all cargo tanks constructed and utilized to carry oil are to be separated from the outer shell by a distance of not less than 0.76 [m].

8.1.3 For category A and B oil tankers of less than 5,000 tonnes deadweight constructed on or after 1 January 2017, the entire cargo tank length is to be protected with:

.1 double bottom tanks or spaces complying with the applicable requirements of regulation 19.6.1 of MARPOL Annex I; and

.2 wing tanks or spaces arranged in accordance with regulation 19.3.1 of MARPOL Annex I and complying with the applicable requirements for distance referred to in regulation 19.6.2 of MARPOL Annex I.

IR8.1.3 In general, provisions of Marpol Annex I, Regulation 19 are applicable to oil tankers of 5000 tonnes deadweight and above.

8.1.4 For category A and B ships constructed on or after 1 January 2017 all oil residue (sludge) tanks and oily bilge water holding tanks shall be separated from the outer shell by a distance of not less than 0.76 [m]. This provision does not apply to small tanks with a maximum individual capacity not greater than 30 [m^3].

Appendix-I

Additional Guidance regarding the Provisions of the Introduction and Part I-A of the Polar Code (Sections 1 to 7 of this Chapter)

1 Additional Guidance to (Definitions in the Introduction to the Polar Code, Para 2 of Introduction)

1.1 The Mean Daily Low Temperature may be determined in accordance with the requirements of Pt.3, Ch.2, Sec. 2, 2.3.

2. Additional Guidance to Chapter 1 of the Code (Section 1 of this Chapter)

2.1 **Limitations for operating in ice**

2.1.1 Limitations for operation in ice can be determined using systems, tools or analysis that evaluate the risks posed by the anticipated ice conditions to the ship, taking into account factors such as its ice class, seasonal changing of ice strength, icebreaker support, ice type, thickness and concentration. The ship's structural capacity to resist ice load and the ship's planned operations are to be considered. The limitations are to be incorporated into an ice operational decision support system.

2.1.2 Limitations for operating in ice are to be determined using an appropriate methodology, such methodologies exist, have been in use for a number of years and have been validated with service experience. Existing methodologies and other systems may be acceptable to the Administration / IRS.

2.1.3 Operation in ice are to take into account any operational limitations of the ship; extended information on the ice operational methodology contained in the PWOM; the condition of the ship and ship's systems, historical weather/ice data and weather/ice forecasts for the intended area of operation, current conditions including visual ice observations, sea state, visibility and the judgment of qualified personnel.

2.2 Operational assessment

2.2.1 This guidance is intended to support ship owners carrying out, and Administrations reviewing, the assessment required in Sec 1, 1.5, for operational limitations and procedures for the Polar Ship Certificate.

2.2.2 Steps for an operational assessment:

.1 identify relevant hazards, as listed in Para 3 of the Introduction and other hazards based on a review of the intended operations;

.2 develop a model***** to analyse risks considering:

.1 development of accident scenarios;
.2 probability of events in each accident scenario; and
.3 consequence of end states in each scenario;

*(Note: ***** Techniques in Appendix 3 of the Revised Guidelines for Formal Safety Assessment (FSA) for use in the IMO Rule Making Process (MSC-MEPC.2/Circ.12) and Standard IEC/ISO 31010 "Risk Management-Risk Assessment Techniques" may be used for development of models referred in 2.2.2.2)*

.3 assess risks and determine acceptability:

.1 estimate risk levels in accordance with the selected modeling approach; and
.2 assess whether risk levels are acceptable; and

.4 in the event that risk levels determined in steps 1 to 3 are considered to be too high, identify current or develop new risk control options that aim to achieve one or more of the following:

.1 reduce the frequency of failures through better design, procedures, training, etc;
.2 mitigate the effect of failures in order to prevent accidents;
.3 limit the circumstances in which failures may occur; or
.4 mitigate consequences of accidents; and

.5 incorporate risk control options for design, procedures, training and limitations, as applicable.

2.3 Performance standards

2.3.1 A system previously accepted based on manufacturer certifications, IRS certifications and/or satisfactory service of existing systems may be acceptable for installation on new and existing ships if no performance or testing standards are accepted by the Organization.

3 Additional Guidance to Chapter 2 of the Code (Section 2 of this Chapter)

3.1 Recommendation on the content of the Polar Water Operational Manual

.1 The Polar Water Operational Manual (PWOM) is intended to address all aspects of operations addressed by Chapter 2 of Part I-A of the Code (Section 2 of this Chapter). When appropriate information, procedures or plans exist elsewhere in a ship's documentation, the PWOM itself does not need to replicate this material, but may instead cross-reference the relevant reference document. A model Table of Contents is provided in Appendix-II to this Chapter.

.2 The model follows the general structure of Section 2. Not every item outlined below will be applicable to every polar ship. Many category C ships that undertake occasional or limit polar voyages will not need to have procedures for situations with a very low probability of occurrence. However, it may still be advisable to retain a common structure for the PWOM as a reminder that if assumptions change then the contents of the manual may also need to be updated. Noting an aspect as "not applicable" also indicates to the Administration that this aspect has been considered and not merely omitted.

3.2 Guidance on navigation with icebreaker assistance

.1 With respect to navigation with icebreaker assistance, the following is to be considered:

.1 while approaching the starting point of the ice convoy to follow an icebreaker / icebreakers or in the case of escorting by icebreaker of one ship to the point of meeting with the icebreaker, ships should establish radio communication on the VHF channel 16 and act in compliance with the icebreaker's instructions;

.2 the icebreaker rendering the icebreaker assistance of ship ice convoy is to command ships in the ice convoy;

.3 position of a ship in the ice convoy is to be determined by the icebreaker rendering the assistance;

.4 ship within the ice convoy, in accordance with the instructions of the icebreaker rendering the assistance, should establish communication with the icebreaker by VHF channel indicated by the icebreaker;

.5 the ship, while navigating in the ice convoy, is to ensure compliance with the instructions of the icebreaker;

.6 position in the ice convoy, speed and distance to a ship ahead is to be as instructed by the icebreaker;

.7 the ship is to immediately notify the icebreaker of any difficulties to maintain the position within the ice convoy, speed and/or distance to any other ship in the ice convoy; and

.8 the ship is to immediately report to the icebreaker of any damage.

3.3 Guidance on the development of contingency plans

.1 In developing the ship's contingency plans ships are to consider damage control measures arrangements for emergency transfer of liquids and access to tanks and spaces during salvage operations.

.2 Additional guidance for Safety of Navigation may also be referred in the Polar Code, whilst making the PWOM.

4 Additional Guidance to Chapter 3 of the Code (Section 3 of this Chapter)

4.1 The guidance presented below is intended to assist in determining equivalency with standards acceptable to the Organization, as referenced in Sections 3 and 6 of this Chapter. The methodology is consistent with guidance

developed by the Organization* while allowing for the use of a simplified approach.

*(Note: * MSC.1/Circ. 1455, as amended, "Guidelines for the approval of alternatives and equivalents as provided for in various IMO instruments" may be used).*

4.2 The basic approach for considering equivalency for categories A and B ships can be the same for both new and existing ships. It involves comparing other ice classes to the IACS Polar Classes (Refer Pt.5, Ch.32). For ice classes under category C, additional information on comparisons of strengthening levels is available for the guidance of owners and Administrations*. The responsibility for generating the equivalency request and supporting information required is to rest with the owner/operator. Review / approval of any equivalency request will be undertaken by the flag State Administration / IRS.

*(Note: * Annex to Helcomm Recommendations 25/7, Safety of Winter Navigation in the Baltic Sea Area may be used).*

4.3 The scope of a simplified equivalency assessment (referring to 4.6.1 to 4.6.3 below) is expected to be limited to materials selection, structural strength of the hull and propulsion machinery.

4.4 If there is not full and direct compliance, then an equivalent level of risk can be accepted in accordance with guidance provided by the Organization. An increase in the probability of an event can be balanced by a reduction in its consequences. Alternatively, a reduction in probability could potentially allow acceptance of more serious consequences. Using a hull area example, a local shortfall in strength level or material grade could be accepted if the internal compartment is a void space, for which local damage will not put the overall safety of the ship at risk or lead to any release of pollutants.

4.5 For existing ships, service experience can assist in risk assessment. As an example, for an existing ship with a record of polar ice operations a shortfall in the extent of the ice belt (hull areas) may be acceptable if there is no record of damage to the deficient area; i.e. a ship that would generally meet requirements of PC 5 Class Ships but in limited areas is only PC 7 could still be considered as a category A, PC 5 ship. In all such cases, the ship's documentation is to make clear the nature and scope of any deficiencies.

4.6 The process includes the following stages of assessment:

.1 select the target Polar Class for equivalency;

.2 compare materials used in the design with minimum requirements in Pt.5, Ch. 32; identify any shortfalls; and

.3 compare strength levels of hull and machinery components design with requirements in Pt.5, Ch.32; quantify levels of compliance.

4.7 Where gaps in compliance are identified in 4.6.1 to 4.6.3, additional steps are necessary to demonstrate equivalency, as outlined below:

.4 identify any risk mitigation measures incorporated in the design of the ship (over and above the requirements of the Polar Code and Pt.5, Ch.32);

.5 where applicable, provide documentation of service experience of existing ships, in conditions relevant to the target ice class for equivalency; and

.6 undertake an assessment, taking into account information from 4.6.1 to 4.6.5 as applicable, and on the principles outlined in 4.2 to 4.6 above.

4.8 Documentation provided with an application for equivalency are to identify each stage that has been undertaken, and sufficient supporting information to validate assessments.

4.9 Where a ship in categories A or B is provided with an equivalency for ice class by its flag State, this is to be noted in its Polar Ship Certificate.

Appendix-II

Model Table of Contents for the Polar Water Operations Manual (PWOM)
Safety Measures

Division 1 – Operational capabilities and limitations

Chapter 1 Operation in ice

1.1 Operator guidance for safe operation

Guidance: The PWOM is to establish the means by which decisions as to whether ice conditions exceed the ship's design limits should be made, taking into account the operational limitations on the Polar Ship Certificate. An appropriate decision support system, such as the Canada's Arctic Ice Regime Shipping System, and/or the Russian Ice Certificate as described in the Rules of Navigation on the water area of the Northern Sea Route, can be used. Bridge personnel are to be trained in the proper use of the system to be utilized. For ships that will operate only in ice-free waters, procedures to ensure that will keep the ship from encountering ice should be established.

1.2 Icebreaking capabilities

Guidance: The PWOM is to provide information on the ice conditions in which the ship can be expected to make continuous progress. This may be drawn, for example from numerical analysis, model test or from ice trials. Information on the influence of ice strength for new or decayed ice and of snow cover may be included.

1.3 Manoeuvring in ice

1.4 Special features

Guidance: Where applicable, the PWOM is to include the results of any equivalency analyses made to determine Polar Ship category/ice class. The manual is also to provide information on the use of any specialized systems fitted to assist in ice operations.

Chapter 2 Operation in low air temperatures

2.1 System design

Guidance: The PWOM is to list all ship systems susceptible to damage or loss of functionality by exposure to low temperatures, and the measures to be adopted to avoid malfunction.

Chapter 3 Communication and navigation capabilities in high latitudes

Guidance: The PWOM is to identify any restrictions to operational effectiveness of communications and navigational equipment that may result from operating in high latitudes.

Chapter 4 Voyage duration

Guidance: The PWOM is to provide information on any limitations on ship endurance such as fuel tankage, fresh water capacity, provision stores, etc. This will normally only be a significant consideration for smaller ships, or for ships planning to spend extended periods in ice.

Division 2 – Ship operations

Chapter 1 Strategic planning

Assumptions used in conducting the analyses referred to below are to be included in the Manual.

1.1 Avoidance of hazardous ice

Guidance: For ships operating frequently in polar waters, the PWOM is to provide information with respect to periods during which the ship should be able to operate for intended areas of operation. Areas that pose particular problems, e.g. chokepoints, ridging, as well as worst recorded ice conditions are to be noted. Where the available information is limited or of uncertain quality, this is to be recognized and noted as a risk for voyage planning.

1.2 Avoidance of hazardous temperatures

Guidance: For ships operating frequently in polar waters, the PWOM is to provide information with respect to, the daily mean daily low temperature as well as the minimum recorded temperature for each of the days during the intended operating period. Where the available information is limited or of uncertain quality, this is to be recognized as a risk for voyage planning.

1.3 Voyage duration and endurance

Guidance: Procedures to establish requirements for supplies are to be established, and appropriate safety levels for safety margins determined taking into account various scenarios, e.g. slower than expected steaming, course alterations, adverse ice conditions, places of refuge and access to provisions. Sources for and availability of fuel types are to be established, taking into account long lead times required for deliveries.

1.4 Human resources management

Guidance: The PWOM is to provide guidance for the human resources management, taking into account the anticipated ice conditions and requirements for ice navigation, increased levels of watch keeping, hours of rest, fatigue and a process that ensures that these requirements will be met.

Chapter 2 Arrangements for receiving forecasts of environmental conditions

Guidance: The PWOM is to set out the means and frequency for provision of ice and weather information. Where a ship is intended to operate in or in the presence of ice, the manual is to set out when weather and ice information is required and the format for the information.
When available, the information is to include both global and localized forecasts that will identify weather and ice patterns/regimes that could expose the ship to adverse conditions.
The frequency of updates is to provide enough advance notice that the ship can take refuge or use other methods of avoiding the hazard if the conditions are forecast to exceed its capabilities.
The PWOM may include use of a land-based support information provider an effective method of sorting through available information, thereby providing the ship only with information that is relevant, reducing demands on the ship's communications systems. The manual may also indicate instances in which additional images should be obtained and analysed, as well as where such additional information may be obtained.

2.1 Ice information

Guidance: The PWOM is to include or refer to guidance on how radar is to be used to identify ice floes, how to tune the radar to be most effective, instructions on how to interpret radar images, etc. If other technologies are to be used to provide ice information, their use is also to be described.

2.2 Meteorological information

Chapter 3 Verification of hydrographic, meteorological and navigational information

Guidance: The PWOM is to provide guidance on the use of hydrographic information. as further described in the additional guidance to chapter 10 of the Code.

Chapter 4 Operation of Special Equipment

4.1 Navigation systems

4.2 Communications systems

Chapter 5 Procedures to maintain equipment and system functionality

5.1 Icing prevention and de-icing

Guidance: The PWOM is to provide guidance on how to prevent or mitigate icing by operational means, how to monitor and assess ice accretion, how to conduct de-icing using equipment available on the ship, and how to maintain the safety of the ship and its crew during all of these aspects of the operation.

5.2 Operation of seawater systems

Guidance: The PWOM is to provide guidance on how to monitor, prevent or mitigate ice ingestion by seawater systems when operating in ice or in low water temperatures. This may include recirculation, use of low rather than high suctions, etc.

5.3 Procedures for low temperature operations

Guidance: The PWOM is to provide guidance on maintaining and monitoring any systems and equipment that are required to be kept active in order to ensure functionality; e.g. by trace heating or continuous working fluid circulation.

Division 3 – Risk management

Chapter 1 Risk mitigation in limiting environmental condition

1.1 Measures to be considered in adverse ice conditions

Guidance: The PWOM is to contain guidance for the use of low speeds in the presence of hazardous ice. Procedures are also to be set for enhanced watchkeeping and lookout manning in situations with high risks from ice, e.g. in proximity to icebergs, operation at night, and other situations of low visibility. When possibilities for contact with hazardous ice exist, procedures are to address regular monitoring, e.g. soundings/inspections of compartments and tanks below the waterline.

1.2 Measures to be considered in adverse temperature conditions

Guidance: The PWOM is to contain guidance on operational restrictions in the event that temperatures below the ships polar service temperature are encountered or forecast. These may include delaying the ship, postponing the conduct of certain types of operation, using temporary heating, and other risk mitigation measures.

Chapter 2 Emergency response

Guidance: In general, where the possibility of encountering low air temperatures, sea ice, and other hazards is present, the PWOM is to provide guidance on procedures that will increase the effectiveness of emergency response measures.

2.1 Damage control

Guidance: the PWOM is to consider damage control measures arrangements for emergency transfer of liquids and access to tanks and spaces during salvage operations.

2.2 Firefighting

2.3 Escape and evacuation

Guidance: Where supplementary or specialized lifesaving equipment is carried to address the possibilities of prolonged durations prior to rescue, abandonment onto ice or adjacent land, or other aspects specific to polar operations, the PWOM is to contain guidance on the use of the equipment and provision for appropriate training and drills.

Chapter 3 Coordination with emergency response services

3.1 Ship emergency response

Guidance: The PWOM is to include procedures to be followed in preparing for a voyage and in the event of an incident arising.

3.2 Salvage

Guidance: The PWOM is to include procedures to be followed in preparing for a voyage and in the event of an incident arising.

3.3 Search and rescue

Guidance: The PWOM is to contain information on identifying relevant Rescue Coordination Centres for any intended routes, and should require that contact information and procedures be verified and updated as required as part of any voyage plan.

Chapter 4 Procedures for maintaining life support and ship integrity in the event of prolonged entrapment by ice.

Guidance: Where any ship incorporates special features to mitigate safety or environmental risks due to prolonged entrapment by ice, the PWOM is to provide information on how these are to be set up and operated. This may include, for example, adding additional equipment to be run from emergency switchboards, draining systems at risk of damage through freezing, isolating parts of HVAC systems, etc.

4.1 System configuration

4.2 System operation

Division 4 – Joint operations

Chapter 1 Escorted operations

Guidance: The PWOM is to contain or reference information on the rules and procedures set out by coastal States who require or offer icebreaking escort services. The manual is also to emphasize the need for the master to take account of the ship's limitations in agreeing on the conduct of escort operations.

Chapter 2 Convoy operations

End of Chapter

Chapter 37

Ballast Water Management

Section	Contents
1	*General*
2	*Ballast Water Exchange*
3	*Ballast Water Treatment*

Section 1

General

1.1 Application

1.1.1 The requirements of this chapter are applicable to vessels that are undertaking ballast water management i.e. performing ballast water exchange or are fitted with ballast water treatment systems and are supplementary to the assignment of main characters of class.

1.1.2 The additional class notations for ballast water management are assigned for ships complying with the requirements of the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, the relevant IMO guidelines and requirements of this Chapter.

1.1.3 The definitions stated in the BWM Convention, its appendices and related IMO documents are applicable to these Rules.

1.1.4 In accordance with the BWM convention, a BWMS installed on board a vessel is to be an IMO Member State type-approved and certified system and the installation of this system must have their prior approval/acceptance of the vessel's flag Administration.

1.1.5 Attention is drawn to any relevant statutory requirements of the country in which the ship is to be registered. Several national Administrations have in place additional requirements pertaining to the management of ballast water for vessels operating in their territorial waters. These requirements may require additional vessel features, performance standards, equipment, reporting and record keeping, which are not included in this Chapter.

1.2 Notations

1.2.1 The additional class notation **BWE ( )** is assigned to ships complying with the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, the relevant IMO Guidelines, and performing ballast water exchange in accordance with the requirements of Section 2. The letter or letters separated by commas in brackets denotes the method/s for exchange that has been applied as follows:

s for sequential method

f for flow through method

d for dilution method

Examples: **BWE (s); BWE (s,f); BWE (s,f,d)**

- *Sequential Method*: A process by which a ballast tank is first emptied and then refilled with replacement ballast water
- *Flow-through Method*: A process by which replacement ballast water is pumped into a ballast tank, allowing water to flow through overflow or other arrangements. At least three times the tank volume is to be pumped through the tank
- *Dilution Method*: A method by which replacement ballast water is filled through the top of the ballast tank with simultaneous discharge from the bottom at the same flow rate and maintaining a

constant level in the tank throughout the ballast exchange operation. At least three times the tank volume is to be pumped through the tank

1.2.2 The additional class notation **BWT** is assigned to ships complying with the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, the relevant IMO Guidelines, and fitted with a ballast water treatment system approved by an IMO Member State, in accordance with the requirements of Section 3.

1.2.3 Vessels equipped with a BWMS where the installation was not conducted under IRS survey may obtain **BWT** notation, provided that the BWMS and the installation on board the vessel have been determined to comply with the design and installation criteria contained in this chapter. The operation of the BWMS is to be demonstrated by a shipboard function test, and documentation confirming that the BWMS has been type-approved in accordance with IMO G-8 Guidelines as satisfying the ballast water performance standard contained in Regulation D-2 of the Convention is to be submitted.

1.3 General Requirements

1.3.1 Ballast Water Discharge Standards

1.3.1.1 Ballast Water Exchange

1.3.1.1.1 The ballast water exchange standards required for issuance of the **BWE ( )** notation are specified in Reg. D-1 of the BWM Convention.

1.3.1.1.2 Ships performing Ballast Water exchange are to do so with an efficiency of at least 95 percent volumetric exchange of Ballast Water. For ships exchanging Ballast Water by the pumping-through method, pumping through three times the volume of each Ballast Water tank shall be considered to meet the standard described above. Pumping through less than three times the volume may be accepted provided the ship can demonstrate that at least 95 percent volumetric exchange is met.

1.3.1.2 Ballast Water Treatment

1.3.1.2.1 The ballast water discharge standards required for issuance of the **BWT** notation are as specified in the Reg. D-2, of the BWM Convention.

1.3.1.2.2 Ships conducting Ballast Water Management in accordance with Reg.D-2 are to discharge less than 10 viable organisms per cubic metre greater than or equal to 50 micrometres in minimum dimension and less than 10 viable organisms per millilitre less than 50 micrometres in minimum dimension and greater than or equal to 10 micrometres in minimum dimension; and discharge of the indicator microbes is not to exceed the specified concentrations described as below.

1.3.1.2.3 Indicator microbes, as a human health standard, are to include:

.1 Toxicogenic Vibrio cholerae (O1 and O139) with less than 1 colony forming unit (cfu) per 100 millilitres or less than 1 cfu per 1 gram (wet weight) zooplankton samples;

.2 Escherichia coli less than 250 cfu per 100 millilitres;

.3 Intestinal Enterococci less than 100 cfu per 100 milliliters.

1.3.2 System Certification

1.3.2.1 Systems using the following three main categories of treatment are considered acceptable.

(a) Mechanical Treatment (For e.g. Filtration, Cyclonic Separation, Electro-mechanical separation);

(b) Physical Treatment (For e.g. Ultra Violet Light, Cavitation/ Ultrasounds, De-oxygenation);

(c) Chemical Treatment (For e.g. Disinfecting Biocides, Electrolytic Chlorination)

1.3.2.2 The ballast water management system installed onboard those vessels receiving **BWT** notation is to be a type-approved and certified system in accordance with the testing, approval and certification processes outlined in the IMO's G8 Guidelines. For the BWMS employing an active substance(s), further requirements and additional approval procedures apply as per IMO G9 Guidelines.

1.3.2.3 For a system that makes use of active substances, documentation relating to the IMO Basic Approval and Final Approval are required to be submitted.

1.4 Definitions

1.4.1 **Ballast Water Management System** (hereinafter referred to as 'BWMS') means any system which processes ballast water such that it meets or exceeds the Ballast Water Performance Standard in Regulation D-2 of the BWM Convention. The BWMS includes ballast water equipment, all associated piping arrangements as specified by the manufacturer, control and monitoring equipment and sampling facilities. The categorization of BWMS technologies is given in Table 3.1. Applicability of the requirements for each BWMS technology is in accordance with Table 3.2.

1.4.2 Cargo area of tankers is defined in:

- for tankers to which Part 6, Ch.1, 1.5.1 applies, Part 6, Ch.1, 3.6 ;

- for chemical tankers, Part 5, Ch.3, 1.3.6;

- for gas carriers, Part 5, Ch.4, 1.2.7; and

- for offshore support vessels, Paragraph 1.3.1 of the IMO Resolution A.673(16) as amended by Resolution MSC.236(82) or Paragraph 1.2.7 of the IMO Resolution A.1122(30), as applicable.

1.4.2 **Dangerous gas** means any gas which may develop an atmosphere being hazardous to the crew and/or the ship due to flammability, explosivity, toxicity, asphyxiation, corrosivity or reactivity and for which due consideration of the hazards is required, e.g. hydrogen (H_2), hydrocarbon gas, oxygen (O_2), carbon dioxide (CO_2), carbon monoxide (CO) ozone (O_3), chlorine (Cl_2) and chlorine dioxide (ClO_2), etc.

1.4.3 **Hazardous area** is defined in IEC 60092-502:1999 and means an area in which an explosive gas atmosphere is or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of electrical apparatus. When a gas atmosphere is present, the following hazards may also be present: toxicity, asphyxiation, corrosivity and reactivity.

1.4.4 **Non-hazardous area** means an area which is not a hazardous area as defined in 1.4.3 above.

1.4.5 **Dangerous liquid** means any liquid that is identified as hazardous in the Material Safety Data Sheet or other documentation relating to this liquid.

1.5 Sediment Management

1.5.1 As per Regulation B-5 of the BWM Convention and IMO Guidelines, it is required that ships should, without compromising safety or operational efficiency, be designed and constructed with a view to minimize the uptake and undesirable entrapment of sediments, facilitate the removal of sediments, and provide safe access to allow for sediment removal and sampling.

1.5.2 Practical steps are to be taken during ballast water uptake to avoid the accumulation of sediments. Ballast water tanks and their internal structures should be designed to minimize accumulation of sediments and allow for easy cleaning and maintenance.

1.5.3 Details of the methods and operational procedures for the sediment management on board the ship, including the disposal of sediments and the associated safety considerations, etc., are to be documented in the ship's Ballast Water Management Plan.

1.5.4 In new ships or ships undergoing major modification in way of ballast tanks, the internal arrangement of ballast tanks (e.g. stiffener orientation) is to be such that there is:

(i) minimum accumulation of sediments e.g. adequate provision of scallops and drain holes
(ii) absence of stagnant pool or sediment traps;
(iii) minimal horizontal surfaces;
(iv) arrangement for effective flushing.

1.5.5. In ships, where ballast water treatment plants are being retrofitted, it may not be feasible to undertake major changes to the stiffener arrangement, etc. In such cases, measures to enable effective flushing of the sediments are to be provided, as practicable.

1.6 Ballast Water Management Plan (BWMP)

1.6.1 Each ship is to have on board and implement a Ballast Water Management plan. Such a plan is to be approved by/on behalf of the Administration, taking into account Guidelines (G-4) developed by IMO. The Ballast Water Management plan is to be specific to each ship and is to at least:

- detail safety procedures for the ship and the crew associated with Ballast Water Management as required by the Convention;

- provide a detailed description of the actions to be taken to implement the Ballast Water Management requirements and supplemental Ballast Water Management practices as set forth in the Convention;

- detail the procedures for the disposal of sediments:

  at sea; and

  to shore

- include the procedures for coordinating shipboard Ballast Water Management that involves discharge to the sea with the authorities of the State into whose waters such discharge will take place;

- designate the officer on board in charge of ensuring that the plan is properly implemented and list his duties;

- contain the reporting requirements for ships provided for under the Convention; and

- be written in the working language of the ship.

1.6.2 The Ballast Water Management Plan is to include training and education for ship's crew on ballast water management practices and the systems and procedures used on board the ship.

1.6.3 Regular review of the Plan is required to be conducted by the owner, operator, or master to ensure that the information contained is accurate and updated.

1.6.4 Changes to the provisions of the Plan will need re-approval of the BWMP.

1.7 Ballast Water Record Book (BWRB)

1.7.1 Each ship is to have a Ballast Water Record Book that may be in electronic form and which has to at least contain information specified in Appendix II to the Convention.

1.7.2 Ballast Water record book entries are to be maintained on board the ship for a minimum period of two years after the last entry has been made and thereafter in the Company's control for a minimum period of three years.

1.7.3 All discharges of Ballast Water pursuant to Reg. A-3, A-4 or B-3.6 of the Convention, and/ or including accidental or exceptional discharges, not otherwise exempted by the Convention, are to be recorded, without delay in the Ballast Water Record Book. The circumstances and the reason for the discharge are also to be indicated.

1.7.4 The Ballast Water Record Book is to be kept readily available for inspection at all times and, in the case of an unmanned ship under tow, may be kept on the towing ship.

Section 2

Ballast Water Exchange

2.1 Application

2.1.1 The requirements given in this Section are applicable to ships where ballast water exchange at sea is accepted as a process in lieu of treatment of ballast water.

2.2 Documentation

2.2.1 In addition to the information and plans required by Pt.3, Ch.1, Sec.3, the following are to be submitted as applicable for approval:

.1 Machinery plans showing piping details of the ballast system and

.2 ballast water management plan having detailed calculation showing compliance with strength and stability requirements.

2.3 General

2.3.1 Stability

2.3.1.1 All stability and trim requirements applicable to the ship are to be met during the ballast water management operations, including free surface effects.

2.3.2 Strength

2.3.2.1 All strength requirements, as applicable are to be met. Depending on the method of exchange, the following are to be specially considered during the ballast water management operations:

- Hull girder strength (bending, shear and torsion)
- Sloshing in tanks
- Bottom slamming
- Overpressure in tanks

2.4 Flow Through Method

2.4.1 The tank volumes, available pumps and estimated time for the flow-through of water, corresponding to a volume three times the tank volume are to be included in the Ballast Water Management Plan (BWMP). In cases the floating position changes due to using flow-through method for partially filled tanks, calculation showing compliance with stability and strength requirements during the exchange steps are to be presented in the BWMP as required by 2.6.1

2.4.2 The capability of the ballast water system for BWE by the flow-through method without risk of the tank being subject to a pressure greater than for what it has been designed is to be shown by calculations.

2.4.3 Provision is to be made to prevent overflow through the air pipe heads unless the air pipe heads are type approved for these operations.

2.5 Dilution Method

2.5.1 The tank volumes, available pumps and estimated time for the exchange of water, corresponding to a volume three times the tank volume are to be included in the Ballast Water Management Plan (BWMP).

2.5.2 A monitoring system to monitor the levels in tanks is to be provided where maintenance of constant level is essential to the safety of the ship during BWE.

2.6 Sequential Method

2.6.1 Calculations showing compliance with stability and strength requirements for each step in the ballast exchange sequence are to be submitted. The following are to be shown in the calculations:

- Ballast water volume for each tank
- Pumps involved
- Estimated time for exchange
- Stress values in relation to permissible values
- Stability effects under consideration of free-surface effects during filling or discharging
- Draught values at FP, AP and heeling

2.6.2 The following loading conditions are to be indicated if the sequential method is applicable:

- Standard ballast condition
- Heavy ballast and emergency ballast conditions, if applicable
- A critical loading condition for one of the stages, where the least safety margin(s) to the valid stability and/or strength limits occur.

Section 3

Ballast Water Treatment

3.1 Application

3.1.1 The requirements given in this Section are applicable to ships where treatment of ballast water is undertaken such that the discharge meets the IMO D-2 standard.

3.1.2 The categorization of ballast water management system technologies is given in Table 3.1. Applicability of the requirements for each BWMS technology is in accordance with Table 3.2.

3.2 Documentation

3.2.1 Following plans and documentation are to be submitted for approval:

.1 General arrangement drawings of the BWMS. Installation arrangement drawings on the ship including location and layout

.2 Ballast piping system drawings including layout, filling arrangement and construction details of piping system

.3 Location of ballast water sampling facilities

.4 Electrical circuit drawings and main power cable drawings

.5 Power calculation document including electrical load analysis

.6 Control, monitoring and safety system documentation especially where the controls and monitoring of the BWMS have been connected to or integrated with the ship's control and monitoring system(s)

.7 Local instrumentation arrangement plan

.8 Structural plans showing installation details of attachment, supports and foundations of principal components of the BWMS

.9 Hazardous area installations and hazardous area plans, as applicable

.10 Storage tanks and day tanks containing chemicals and preparations used to treat ballast water including complete piping details of filling, drain system, vents, drip trays, and safety precautions, etc.

.11 Leakage detection system and safety features associated with the generation of toxic or flammable gases. Safety features are to include sensors, alarms and shutdown settings, etc. Schematic plans detailing arrangement and location of sensors are to be provided.

.12 Risk assessment documentation, where applicable. For BWMS that employs active substances and preparations, arrangement, handling and safety plans of auxiliary systems for the treatment system, as applicable.

.13 Ballast Water Management Plan (BWMP). BWMP to be specific to the ship and in a standard format as per G4 Guidelines

.14 BWMS operations and technical manual specific to the actual installation onboard the ship

.15 Shipboard function test plan for sea or quay trial. Function test of the installed BWMS at the sea trial or quay trial in the presence of attending Surveyor; function test plan as per paragraph 5.1.9 of G8 Guidelines

3.2.2 Following plans and documentation are to be submitted for information:

.1 Arrangement and capacity of ballast tanks and pumps

.2 List of electrical equipment in hazardous area

.3 Safety documentation for hazardous chemicals in recognized industry format, such as MSDS, CHRIS Code

.4 Ballast water record book specific to the ship

3.2.3 A risk assessment may be conducted to ensure that risks, including but not limited to those arising from the use of dangerous gas affecting persons on board, the marine environment, the structural strength or integrity of the ship are identified.

Table 3.1 - Categorization of BWMS technologies

BWMS's Technology category (informative Annex II should be referred to) →		**1**	**2**	**3a**	**3b**	**3c**	**4**	**5**	**6**	**7a**	**7b**	**8**
Characteristics ↓		In-line UV or UV + Advanced Oxidation Technology (AOT) or UV + TiO_2 or UV + Plasma	In-line Flocculation	In-line membrane separation and de-oxygenation (injection of N_2 from a N_2 Generator)	In-line de-oxygenation (injection of Inert Gas from Inert Gas Generator)	In-tank de-oxygenation with Inert Gas Generator	In-line full flow electrolysis	In-line side stream electrolysis (2)	In-line (stored) chemical injection	In-line side-stream ozone injection without gas/liquid separation tank and without Discharge treatment tank	In-line side-stream ozone injection with gas/liquid separation tank and Discharge water treatment tank	In-tank pasteurization and de-oxygenation with N_2 generator
Dis-infection when ballasting	Making use of active substance		X			In-tank technology: No treatment when ballasting or de-ballasting	X	X	X	X	X	In-tank technology: No treatment when ballasting or de-ballasting
	Full flow of ballast water is passing through the BWMS	X	X	X	X		X				X	
	Only a small part of ballast water is passing through the BWMS to generate the active substance							X				
After-treatment de-when ballasting	Full flow of ballast water is passing through the BWMS	X									X	
	Injection of neutralizer						X	X	X	X	X	
	Not required by the Type Approval Certificate issued by the Administration		X	X								
Examples of dangerous gas as defined in 1.4.2			(1)	O_2 N_2	CO_2 CO		H_2 Cl_2	H_2 Cl_2	(1)	O_2 O_3 N_2		O_2 N_2

Note:
(1) To be investigated on a case by case basis based on the result of the IMO (GESAMP) MEPC report for Basic and Final approval in accordance with the G9 Guideline.
(2) In-line side stream electrolysis may also be applied in-tank in circulation mode (no treatment when ballasting or de-ballasting)

Taking into consideration future developments of BWMS technologies, some additional technologies may be considered in this Table 3.1 by identifying their characteristics in the same manner as for the above BWMS categories 1, 2, 3a, 3b, 3c, 4, 5, 6, 7a, 7b and 8.

Table 3.2 - Applicability of the requirements for each BWMS technology											
BWMS's Technology category (informative Annex II should be referred to)→	**1**	**2**	**3a**	**3b**	**3c**	**4**	**5**	**6**	**7a**	**7b**	**8**
Requirement ↓	In-line UV or UV + Advanced Oxidation Technology (AOT) or UV + TiO_2 or UV + Plasma	In-line Flocculation	In-line membrane separation and de-oxygenation (injection of N_2 from a N_2 Generator)	In-line de-oxygenation (injection of Inert Gas from Inert Gas Generator)	In-tank de-oxygenation with Inert Gas Generator	In-line full flow electrolysis	In-line side stream electrolysis	In-line (stored) chemical injection	In-line side-stream ozone injection without gas/liquid separation tank and without Discharge treatment tank	In-line side-stream ozone injection with gas/liquid separation tank and Discharge water treatment tank	In-tank pasteurization and de-oxygenation with N_2 generator
3.5.5			x	x	x						x
3.8.6	x	x	x	x	x	x	x	x	x	x	x
3.7.1.3			x	x	x						x
3.7.1.2				x						x	
3.7.2	x	x	x	x	x	x	x	x	x	x	x
3.9.1.1				x	x				x	x	
3.9.1.2						x	x	x			
3.9.2	x	x	x	x		x	x	x	x	x	
3.9.3	x	x	x	x	x	x	x	x	x	x	x
3.9.4	x	x	x	x		x	x	x	x	x	
3.10.1.1		x	x			x	x	x	x	x	x
3.10.1.2			x	x	x				x	x	x
3.10.1.3									x	x	
3.10.1.4						x	x	x	x	x	
3.10.1.5						x	x	x			
3.10.1.6			x	x	x				x	x	x
3.10.2.1 to 3.10.2.4		x	x	x	x	x	x	x	x	x	x
3.10.2.5			x			x	x	x	x	x	x
3.10.2.6			x						x	x	x
3.10.2.7			x			x	x	x	x	x	x
3.10.3		x				x	x	x	x	x	
3.10.4						x	x	x	x	x	

3.3 Ballast Treatment capacities

3.3.1 The BWMS is to be operated in accordance with the requirements specified in the Type Approval Certificate (TAC) issued by the Flag Administration. BWMS is to be operated within its Treatment Rated Capacity (TRC) as per the TAC. This may require limiting of ship's ballast pump flowrates. The arrangement of the bypasses or overrides of the BWMS is to be consistent with the approved Operation Maintenance and Safety Manual by the Flag Administration's Type Approval. In case the maximum capacity of the ballast pump(s) exceeds the maximum treatment rated Capacity (TRC) of the BWMS specified in the TAC issued by the Flag Administration, there is to be a limitation on the BWMP giving a maximum allowable flow rate for operating the ballast pump(s) that is not to exceed the maximum TRC of the BWMS.

3.4 General Requirements/ Layout

3.4.1 Supporting structure

3.4.1.1 A support structure for the foundation of the BWM system is to be provided, considering all operational conditions of the BWM system.

3.4.2 Where the BWMS is installed in an independent compartment, the compartment is to be:

.1 provided with fire integrity equivalent to "other machinery spaces";

.2 provided with fire detection, fire-fighting as per fire control plan;

.3 positioned outside of any combustible, corrosive, toxic, or hazardous areas unless otherwise specifically approved.

3.5 Piping

3.5.1 All valves, piping fittings and flanges are to comply with the relevant requirements of Pt. 4, Ch. 2 of the Rules. Where plastic pipes have been accepted for use, these are to meet the specification as in IMO Res. A.753(18) with regard to strength, fire endurance, flame spread etc.

3.5.2 Bulkhead/ deck penetrations or penetrations of the ballast system piping are to be to the relevant approved standards.

3.5.3 The BWMS is to be provided with by-pass or override arrangement to effectively isolate it from any essential ship system to which it is connected. For new installation or retrofit to existing ships, under normal operating conditions of ballasting and de-ballasting given in the Ballast Water Management Plan (BWMP) the adequacy of the generating plant capacity installed on the vessel is to be demonstrated by an electrical load analysis. For retrofit installation to exiting ships, a revised electrical load analysis with preferential trips of non-essential services can be accepted.

3.5.4 BWMS is to be subject to design review by IRS to verify the compliance of the BWMS's manufacturer package with the Rules. Manufacturers of the BWMS may apply for this design review at the type approval process. In general, monitoring functions of BWMS belongs to system category I under the application of the Pt.4, Ch.7, Sec.6 . However, in case a by-pass valve is integrated in the valve remote control system, the by-pass valve belongs to the system category II Ballast transfer remote control system. The BWMS's components are required to be inspected and certified by IRS at the manufacturer's works (Society Certificate (SC) as defined in IRS Classification Note *"Approval of IC Engines"*) including pressure vessels, piping class I or II, filters, switchboards, etc.

3.5.5 Where a vacuum or overpressure may occur in the ballast piping or in the ballast tanks due to the height difference or injection of inert gas or nitrogen(N_2), a suitable protection device is to be provided (i.e. P/V breakers, P/V breather valves or pressure safety relief valve or high/low pressure alarms). The pressure and vacuum settings of the protection device is not to exceed the design pressure of the ballast piping (BWMS categories 3a and 3b) or ballast tank (BWMS categories 3a, 3b and 3c), as relevant.

For BWMS categories 3a, 3b and 3c, the inert gas or nitrogen product enriched air from the inert gas system and from the protection devices installed on the ballast tanks (i.e. P/V valves, P/V breakers or P/V breather valves) are to be discharged to a safe location*(1) & (2) on the open deck.

When the concerned ballast tanks are hazardous areas, an extension of hazardous area is to be considered at the outlet of the protection devices: with reference to IEC 60092-502:1999 §4.2.2.9 the areas on open

deck, or semi-enclosed spaces on open deck, within 1.5 m of their outlets are to be categorized hazardous zone 1 and with reference to IEC 60092-502:1999 clause 4.2.3.1, an additional 1.5 m surrounding the 1.5 m hazardous zone 1 is to be categorized hazardous zone 2. Any source of ignition such as anchor windlass or opening into chain locker are to be located outside the hazardous areas.

Where products covered by IEC 60092-502:1999 are stored on-board or generated during operation of the BWMS, the requirements of this standard is to be followed in order to:

- Define hazardous areas and acceptable electrical equipment, and

- Design ventilation systems.

Notes :

Safe location needs to address the specific types of discharges separately.

Signboards or similar warnings at the discharge areas are to be provided.

Safe location*(1): inert gas or nitrogen product enriched air from:

- in-line (categories 3a and 3b) and in-tank (categories 3c and 8) de-oxygenation BWMS: the protection devices installed on the ballast tanks, nitrogen or inert gas generators, nitrogen buffer tank (if any); or

- in-line ozone injection BWMS (categories 7a and 7b): the oxygen generator;

safe locations on the open deck are:

- not within 3 [m] of areas traversed by personnel; and

- not within 6 [m] of air intakes for machinery (engines and boilers) and all ventilation inlets/outlets.

Safe location*(2): oxygen-enriched air from:

- in-line and in-tank de-oxygenation BWMS (categories 3a and 8): the nitrogen generator; or

- in-line ozone injection BWMS (categories 7a and 7b): the protection devices or vents from oxygen generator, compressed oxygen vessel, the ozone generator and ozone destructor devices;

safe locations on the open deck are:

- outside of hazardous area;

- not within 3 [m] of any source of ignition and from deck machinery, which may include anchor windlass and chain locker openings, and equipment which may constitute an ignition hazard;

- not within 3 [m] of areas traversed by personnel; and

- not within 6 [m] of air intakes for machinery (engines and boilers) and all ventilation inlets.

3.5.6 The length of pipe and the number of connections are to be minimized in piping systems containing dangerous gases/ liquids in high concentration.

3.5.7 Pipe joints are to be of welded type except for connections to shut off valves, double walled pipes or pipes in ducts equipped with mechanical exhaust ventilation. Alternatively, it is to be demonstrated that risk of leakage is minimized and the formation of toxic or flammable atmosphere is prevented.

3.5.8 Location of the piping system is to be away from heat sources and protected from mechanical damage.

3.6 Sampling points

3.6.1 The ballast water system is to be provided with sampling, arranged according to the provisions of the IMO G2 guidelines, as amended. Accordingly, the sampling point is to be arranged in the discharge pipe as close as possible at the shell.

3.7 System Specific Requirements

3.7.1 The plan/ documentation requirements would vary depending upon the technology used (for compliance to D-2 Standard) in the BWM system. Certain system specific requirements based on the type of technology currently used in the BWMS are as follows:

3.7.1.1 BWMS using mechanical separation systems

.1 Pressure loss across the BWMS should not reduce the ability of the ballast system to supply at an acceptable flow rate to a remote ballast tank;

.2 Arrangement for backwashing are to be provided in order to confirm that the wastes can be adequately removed from the system taking into account the

maximum static head when the ship is at maximum draft;

.3 Arrangements are to be provided (e.g. controls, procedures etc.) to allow separation system and associated piping to be backwashed and flushed clean upon completion of ballasting as well as de-ballasting operations.

3.7.1.2 BWMS using cavitation and ultrasound systems

3.7.1.2.1 When cavitation is the BWMS treatment process (for example by use of pressure vacuum reactor working in combination with a vertical ballast water drop line) or part of the BWMS treatment process (for example by use of “smart pipe” or “special pipe” in BWMS category 7b or by use of “venturi pipe” in BWMS technology 3b) or by use other means, the design and the wall thickness or grade of materials or inside coating or surface treatment of the part of the piping where the cavitation is taking place is to be specifically considered.

3.7.1.3 BWMS using Inert Gas De-oxygenation systems

3.7.1.3.1 Inert gas systems installed for de-oxygenation BWMS (categories 3a, 3b, 3c and 8) are to be designed in accordance with the following requirements:

.1 Part 6, Chapter 8, Section 15 requirements

- 15.2.1.2, 15.2.1.3

- 15.2.2.1.3, 15.2.2.1.4, 15.2.2.2.1,15. 2.2.2.2, 15.2.2.2.3, 15.2.2.2.6, 15.2.2.4.1, 15.2.2.4.2, 15.2.2.4.3, 15.2.2.4.4, 15.2.2.4.5 except 15.2.2.4.5.1.3 and 15.2.2.4.5.3

- 15.2.3.1.1.2, 15.2.3.1.2, 15.2.3.1.4.2, 15.2.3.1.5, 15.2.3.1.6, 15.2.3.2 except 15.2.3.2.2.1

- 15.2.4.1.3, 15.2.4.1.4 and 15.2.4.2

- For inert gas systems installed for in-tank de-oxygenation BWMS (category 8): 2.2.3.1, 2.2.3.2 except 2.2.3.2.6, 2.2.3.2.7 and 2.2.3.2.10

In general, when applying Part 6, Chapter 8, Section 15 requirements to inert-gas based BWMS, the following modifications are to be considered:

- The terms "cargo tank" and "cargo piping" are to be replaced by "ballast water tank" or "ballast water piping" as relevant.

- The term "cargo control room" is to be replaced by "BWMS control station" as relevant

- Requirements for slop tanks on combination carriers are to be disregarded

- When applying Part 6, Chapter 8 15.2.2.4.5.1.1, the acceptable oxygen content is to be specified by the manufacturer, 5% oxygen content need not necessarily be applied.

.2 Pt.6, Ch.8, Sec 15, IR15.2.2.1.1, IR 15.2.5.1, IR15.2.5.3, IR15.2.5.7, IR15.2.5.8, IR15.2.6.4, IR15.2.6.5, and IR15.2.6.6. In applying IR15.2.6.6, the terms “cargo tanks” and “cargo piping” are to be understood as “ballast tanks” and “ballast piping” respectively. For de-oxygenation BWMS (categories 3a, 3b, 3c and 8), the requirements in 3.7.1.3.1.1 prevail.

3.7.1.4 BWMS using Ultraviolet (UV) disinfection systems

.1 Arrangements are to be provided such that exposure to UV light is avoided during operation, maintenance or repairs of the system;

.2 High temperature alarm, provision for shut down, UV intensity meter and means to prevent operation of UV lamps without water in treatment chamber are to be provided.

3.7.1.5 BWMS using Ozone Injection systems

.1 Vents from safety valves of ozone system are to be led to open deck and arrangements are to be provided to automatically shut down the system when the ozone concentration exceeds limits;

.2 Warning plates are to be installed in all areas into which ozone can escape.

3.7.1.6 BWMS using Electro-chlorination systems

.1 Operation of the system is to be interlocked with the ventilation system for the compartment housing the electro-chlorination system;

.2 Exhaust vents from the system are to be led to a safe area in the open deck;

.3 Interlock arrangement is to be provided so that the system cannot be energized if the water level /flow is less;

.4 A fixed hydrogen gas detection system with automatic shut-down (as per recognized national/ international standards) is to be provided in the compartment housing the electro-chlorination system.

3.7.1.7 BWMS using In-Tank treatment systems

.1 Ballast tanks fitted with in-tank treatment systems and subject to over or under pressure are to be fitted with P-V valves and pressure measuring devices;

.2 Ballast tanks with in-tank treatment systems are to be provided with

(i) safety alarm system to indicate deviation from the normal working pressure and;
(ii) a system for shutting down the system when the pressure deviation is beyond specified limits.

3.7.1.8 BWMS using Active Substances

.1 Sufficient quantity of active substance(s) are to be provided on board;

.2 Dosage instruction for active substances or preparations are required to be available on board;

.3 Active substance(s) is to be stored at location taking into account the risks/hazards involved with the substance as provided in the safety data sheet;

.4 Materials used for the chemical storage tanks, piping and fittings are to be resistant to such chemicals;

.5 Separate pumping and piping system is to be used for chemicals;

.6 Chemical storage tanks are required to be of sufficient strength, provided with gauging system, protected against overflowing and constructed such that maintenance and inspection can be easily performed;

.7 Air pipes for the chemical storage tank are to be led to a safe area on open deck;

.8 Operation manual containing chemical injection procedures, alarm systems, measures in case of emergency, etc. is to be provided.

3.7.1.9 For BWMS using chemical substances, handling procedures are to be in accordance with the Material Safety Data Sheet and BWM.2/Circ.20, and the following measures are to be taken as appropriate:

.1 The materials used for the chemical storage tanks, piping and fittings are to be resistant to such chemicals.

.2 Chemical storage tanks are to have sufficient strength and be constructed such that maintenance and inspection can be easily performed.

.3 Air pipes of the chemical storage tank are to be led to a safe area on open deck.

.4 An operation manual containing chemical injection procedures, alarm systems, measures in case of emergency, etc is to be kept onboard.

3.7.2 When it is required to have an automatic shutdown of the BWMS for safety reasons, this must be initiated by a safety system independent of the BWM control system.

3.8 Electrical Installations

3.8.1 In general, electrical installations are to be designed and installed in accordance with Pt. 4, Ch. 8 of the Rules.

3.8.2 It should be confirmed that ship's generator/s are able to cater to the additional load for BWMS.

3.8.3 The electrical and control equipment for BWMS is to conform to the manufacturer's equipment specification and installation specifications.

3.8.4 Necessary protection as per specification is to be provided including earthing protection for equipment or components /bonding provision for control of static electricity.

3.8.5 Equipment that may be installed in exposed areas on the open deck are to be IP 56 type of IEC 529 publication or equivalent

3.8.6 Electric and electronic components are not to be installed in a hazardous area unless they are of certified safe type for use in the area. Cable penetrations of decks and bulkheads are to be sealed when a pressure difference between the areas is to be maintained.

3.8.7 In case of any by-pass or override operation of BWMS, an audible and visual alarm is to be given and these events are to be automatically recorded in control equipment. The valves in the by-pass line which trigger the by-pass operation are to be remote-controllable by control equipment or fitted with open/close indicator for automatic detection of the by-pass event.

3.9 Additional requirements for tankers carrying flammable cargo

3.9.1 Hazardous area classification is to be in accordance with IEC 60092-502. Where the prescriptive requirements within SOLAS and related Codes (IBC, IGC) and the standards published by the International Electrotechnical Commission, such as but not limited to IEC 60092-502, are not aligned; the prescriptive requirements in SOLAS and Codes take precedence and are to be applied.

.1 BWMS using ozone generators (categories 7a and 7b) and de-oxygenation BWMS using inert gas generator by treated flue gas from main or auxiliary boilers or gas from an oil or gas-fired gas generator (categories 3b and 3c) are to be located outside the cargo area in accordance with Pt.6, Ch.8, 15.2.3.1.1.2.

Note: this requirement does not apply to inert gas generators for which Pt.6, Ch.8, 15.2.4, IR15.2.5, and IR15.2.6 apply.

.2 In-line full flow electrolysis BWMS (category 4), in-line side-stream electrolysis BWMS (category 5) and in-line injection BWMS using chemical which is stored onboard (category 6) can be located inside the hazardous areas with due consideration of the requirement of 3.8.6 but is not to be located inside the cargo pump room unless it is demonstrated by the BWMS manufacturer that the additional hazards that could be expected from dangerous liquids and dangerous gases stored or evolved from the BWMS (for example H_2 generation):

- do not lead to an upgrade of the hazardous area categorization of the cargo pump room,

- are not reactive with the cargo vapours expected to be present in the cargo pump room,

- are not reactive with the fire-extinguishing medium provided inside the cargo pump room,

- are not impacting the performance of the existing fire-fighting systems provided inside the cargo pump room, and

- are not introducing additional hazards inside the cargo pump room such as toxicity hazards that would not have been prior addressed by suitable counter measures.

Notes:

1) In-line full flow electrolysis BWMS (category 4) could be accepted in cargo compressor rooms of liquefied gas carriers and inside cargo pump rooms of oil tankers or chemical tankers if that cargo pump room is located above the cargo tank deck.

2) For submerged cargo pumps, the room containing the hydraulic power unit or electric motors is not to be considered as the “cargo pump room”.

3) Ballast pump rooms and other pump rooms not containing the cargo pumps are not to be considered as the “cargo pump room”.

3.9.2 In general, two independent BWMS should be required i.e. one for ballast tanks located within the cargo area and the other one for ballast tanks located outside cargo area. Specific arrangements where only one single In-line BWMS (categories 1, 2, 3a, 3b, 4, 5, 6, 7a and 7b) could be accepted are given in Annex I.

Note 1: When the Fore Peak Tank is ballasted with the piping system serving the other ballast tanks within the cargo area in accordance with Pt.5, Ch.2, 6.2.7, the ballast water of the Fore

Peak tank is to be processed by the BWMS processing the ballast water of the other ballast tanks within the cargo area.

Note 2: The figures in Annex I and Annex II are given for better clarity in implementation of the requirements. These figures will be subject to revision to address new technologies as and when they are available.

3.9.3 Isolation between ballast piping serving the ballast tanks inside and outside of the cargo area is to be in accordance with the following requirements:

3.9.3.1 Interconnection in between the ballast piping serving the ballast tanks located within the cargo area and ballast piping serving the ballast tanks located outside cargo area may be accepted if appropriate isolation arrangement is provided in accordance with Annex I is applied.

Note 1: The means of appropriate isolation described in 3.9.3.1 is necessary for the interconnection specified in said Paragraph regardless of the diameter of the piping.

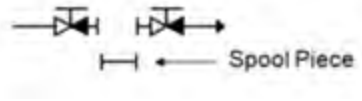
Note 2: As indicated in Annex I, the means of appropriate isolation described in 3.9.3.1 is necessary for the interconnection specified in said Paragraph in the case of the active substance piping such as N_2 gas piping, inert gas piping, neutralizer piping, fresh water piping for filter cleaning, compressed air piping for remaining water purge and sea water piping for adjusting the salinity etc. At the discretion of IRS and for active substance piping and neutralizer piping (both up to 2 inches) only, alternative isolation arrangements, provided preferably on the open deck, offering enhanced safety and gastightness may be considered for penetration of the bulkhead separating the non-hazardous machinery space from a hazardous area (such as the cargo pump room) at as high an elevation in the machinery space as possible, preferably, just below the main deck. The arrangements are to provide suitable protection measures in addressing the pollution hazards and safety concerns due to the potential migration of hydrocarbon or flammable or toxic liquids or vapours from the hazardous areas.

Note 3: The means of appropriate isolation described in 3.9.3.1 for the interconnection specified in said Paragraph need not be applied to the sampling lines described in 3.9.4.

The means of appropriate isolation is to be one of the following:

.1 Two non-return valves with positive means of closing in series with a spool piece(also mentioned “means of dis-connection” in Annex I), or

Note: As an alternative to positive means of closure, an additional valve having such means of closure may be provided between the non-return valve and the spool piece.



.2 Two non-return valves with positive means of closing in in series with a liquid seal at least 1.5 [m] in depth, or

Note 1: As an alternative to positive means of closure, an additional valve having such means of closure may be provided between the non-return valve and the liquid seal.

Note 2: For ships operating in cold weather conditions, freeze protection should be provided in the water seal. A portable heating system can be accepted for this purpose.

.3 Automatic double block and bleed valves and a non-return valve with positive means of closing

3.9.3.2 The above-mentioned means of appropriate isolation is to be provided on the open deck in the cargo area.

Note: When the Fore Peak Tank is ballasted with the piping system serving the other ballast tanks within the cargo area in accordance with Pt.5, Ch.2, 6.2.7, the means of appropriate isolation described in 3.9.3.1 and 3.9.3.1 is not required in between the Fore Peak Tank and the common ballast water piping serving the other ballast water tanks within the cargo area.

3.9.4 Sampling lines which are connected to the ballast water piping system serving the tanks in the cargo area and provided for the purpose of the following:

- for any BWMS: ballast water sampling required by the IMO G2 Guideline of the BWM Convention (2004), or

- for BWMS technologies categories 4, 5, 6, 7a and 7b: total residual oxidant (TRO) analysis in closed loop system;

are not to be led into a non-hazardous enclosed space outside the cargo area.

However, the sampling lines may lead into a non-hazardous enclosed space outside the cargo area provided the following requirements are fulfilled (Refer to Fig 3.9.4):

.1 The sampling facility (for BWMS monitoring/control) is to be located within a gas tight enclosure (hereinafter, referred to as a 'cabinet'), and the following i) through iv) are to be complied.

i) In the cabinet, a stop valve is to be installed on each sampling line.

ii) Gas detection equipment is to be installed in the cabinet and the valves specified in i) above are to be automatically closed upon activation of the gas detection equipment.

iii) Audible and visual alarm signals are to be activated both locally and at the BWMS control station when the concentration of explosive gases reaches a pre-set value, which should not be higher than 30% of the lower flammable limit (LFL). Upon an activation of the alarm, all electrical power to the cabinet is to be automatically disconnected.

Note: when the electrical equipment is of a certified safety type, the automatic disconnection of power supply is not required.

iv) The cabinet is to be vented to a safe location in non-hazardous area on open deck and the vent is to be fitted with a flame arrester.

.2 The standard internal diameter of sampling pipes is to be the minimum necessary in order to achieve the functional requirements of the sampling system.

.3 The cabinet is to be installed as close as possible to the bulkhead facing the cargo area, and the sampling lines located outside the cargo area are to be routed on their shortest ways.

.4 Stop valves are to be located in the non-hazardous enclosed space outside the cargo area, in both the suction and return lines close to the penetrations through the bulkhead facing the cargo area. A warning plate stating "Keep valve closed when not performing measurements" is to be posted near the valves. Furthermore, in order to prevent backflow, a water seal or equivalent arrangement is to be installed on the hazardous area side of the return pipe.

.5 A stop valve is to be installed on the cargo area for each sampling line (i.e. both the suction and return lines).

.6 The samples which are extracted from the ballast water piping system serving the tanks within the cargo area are not to be discharged to a tank located outside the cargo area and not to discharge to a piping line supplying the spaces located outside the cargo area.

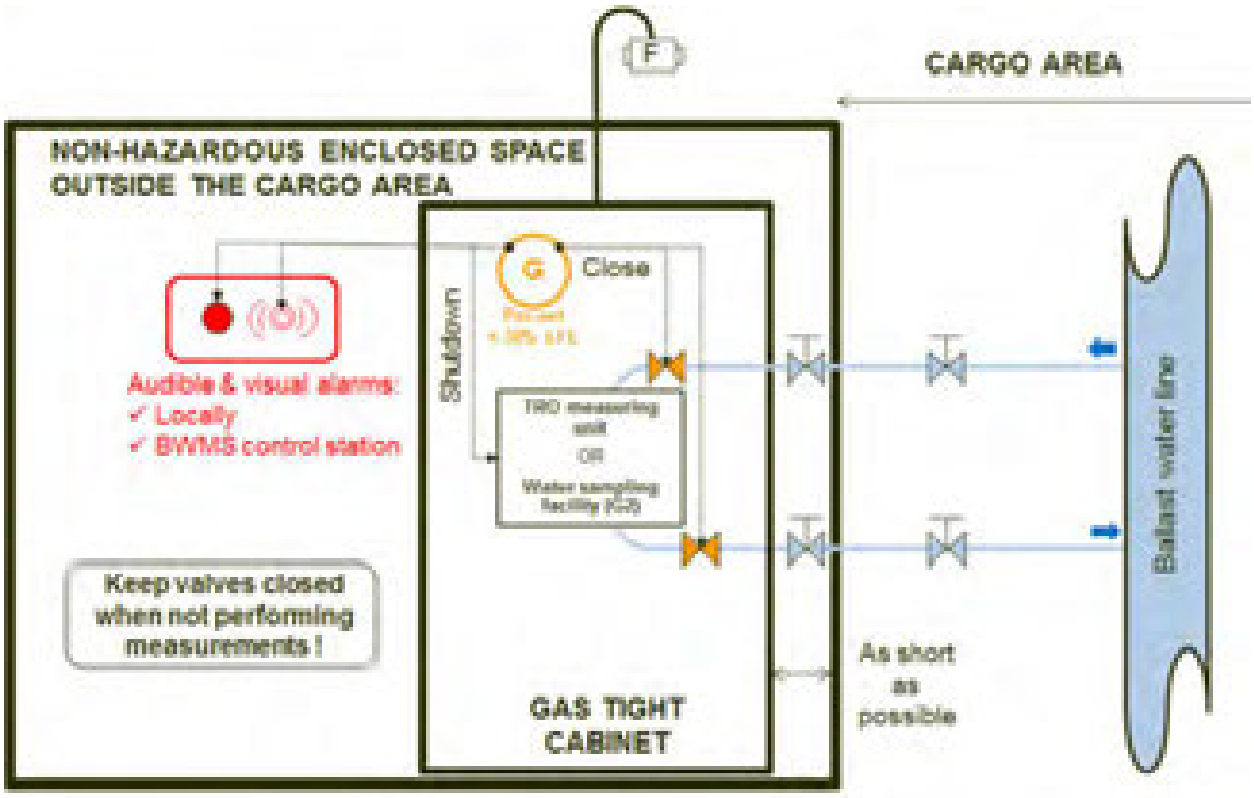


Fig. 3.9.4 : Arrangement of Sampling Lines

3.9.4 When a BWMS is installed in more than one compartment and where some of the components of the BWMS are installed in a separate compartment located in a hazardous area on the deck of a tanker in the interior of the compartment may be considered as non-hazardous, subject to the provisions of the following arrangements:

.1 Deck and bulkhead penetrations are to be sealed gas tight;

.2 No part of the ballast water piping is to be located in the compartment;

.3 No source of release of gas e.g. flange connection/valves are to be located in the compartment;

.4 Access to the compartment from hazardous area is to be through two gastight self-closing doors with air lock capable of maintaining a positive pressure. Audible and visual alarm at control station for loss of pressure in the air lock, and safety interlock for cutting out electrical supply due to loss of pressure, are to be provided.

Note: No air lock however will be required where a vertical separation of at least 2.4 [m] is maintained between the flooring of the compartment and main deck plating or a cofferdam is provided in between the compartment and main deck.

.5 Ventilation inlet and outlets are to be located outside hazardous area;

.6 Ventilation system for the compartment is to be capable of at least 20 air changes an hour or as required by manufacturer, whichever is greater.

3.10 Special requirements for BWMS categories 2, 3a, 3b, 3c, 4, 5, 6, 7a, 7b and 8 generating dangerous gas or dealing with dangerous liquids.

3.10.1 Where the operating principle of the BWMS involves the generation of a dangerous gas, the following requirements are to be satisfied:

3.10.1.1 Gas detection equipment is to be fitted in the spaces where dangerous gas could be present, and an audible and visual alarm is to be activated both locally and at the BWMS control station in the event of leakage.
The gas detectors are to be located as close as possible to the BWMS components where the dangerous gas may accumulate.
For flammable gases and explosive atmosphere including but not limited to H_2, the construction, testing and performance of the gas detection devices is to be in accordance with IEC 60079-29-1:2016, IEC 60079-29-2:2015, IEC 60079-29-3:2014 and/or IEC 60079-29-4:2009, as applicable.
Where other hazards are considered like toxicity, asphyxiation, corrosive and reactivity hazards, a recognized standard acceptable to IRS is to be selected with due consideration of the specific gases to be detected and due consideration of the performance of the detection device with regards to the specific atmosphere where it is used.

3.10.1.2 In spaces where inert gas generator systems are fitted (BWMS categories 3b and 3c) or nitrogen generators are fitted (BWMS categories 3a and 8), at least two oxygen sensors are to be positioned at appropriate locations (as required by Pt.6, Ch.8, 15.2.2.4.5.4 to alarm when the oxygen level falls below 19%. The alarms are to be both audible and visual and are to be activated:

- inside the space;

- at the entry into the space; and

- inside the BWMS control station.

For BWMS categories 7a and 7b, at least two oxygen sensors are to be positioned at appropriate locations in the following spaces:

- spaces where ozone generators are fitted, or

- spaces where ozone destructors are fitted, or

- spaces where ozone piping is routed;

to alarm when the oxygen level raises above 23 %. The alarms are to be both audible and visual and are to be activated at the following locations:

- inside the space; and

- at the entry into the space; and

- inside the BWMS control station.

Automatic shut-down of the BWMS is to be arranged when the oxygen level raises above 25%. Audible and visual alarms independent from those specified in the preceding paragraph are to be activated prior to this shut-down.

3.10.1.3 For BWMS categories 7a and 7b, at least one ozone sensor is to be provided at the vicinity of the discharge outlet to the open deck from the ozone destructors addressed in Footnote *(4) to alarm when the ozone concentration level raises above 0.1 ppm. The alarms are to be both audible and visual and are to be activated in the BWMS control room. In addition, at least two ozone sensors are to be positioned at appropriate location in the following spaces:

- spaces where ozone generators are fitted, or

- spaces where ozone destructors are fitted, or

- spaces where ozone piping is routed;

to alarm when the ozone concentration level raises above 0.1 ppm. The alarms are to be both audible and visual and are to be activated at the following locations:

- inside the space;

- at the entry into the space; and

- inside the BWMS control station.

Automatic shut-down of the BWMS is to be arranged when the ozone concentration measured from one of the two sensors inside the space raises above 0.2 ppm.

3.10.1.4 Inside double walled spaces or pipe ducts constructed for the purpose of 3.10.2.1 Note 1), sensors are to be provided for the detection of H_2 leakages (BWMS categories 4, 5 and 6 when relevant) or O_2 leakages (BWMS categories 7a and 7b) or O_3 leakages (BWMS categories 7a and 7b). The sensors are to activate an alarm at the high level settings and automatic shut-down of the BWMS at the high-high level settings described in above 3.10.1.1 to 3.10.1.3.

Note: As an alternative to the sensor for the gas detection, monitored under-pressurization inside the double walled spaces or pipe ducts could be provided with an automatic alarm and shut-down of the BWMS in case of loss of the under-pressurization. The monitoring can be achieved either by monitoring the pressure inside the double walled spaces or pipe ducts or by monitoring the exhaust fan.

3.10.1.5 For in-line full flow electrolysis BWMS (category 4), in-line side-stream electrolysis BWMS (category 5) and in-line injection BWMS using chemical which is stored onboard (category 6): the hydrogen de-gas arrangement (when provided) is to be provided with redundant ventilation fans and redundant monitoring of the ventilation system.
In addition the ventilation fan is to be certified explosion proof and have spark arrestor to avoid ignition sources to enter the ventilation systems whereas remaining H_2 gas may be present in dangerous concentrations.
Audible and visual alarms and automatic shut-down of the BWMS are to be arranged for respectively high and high-high levels of H_2 concentration. The open end of the hydrogen by-product enriched gas relieving device is to be led to a safe location*(3) on open deck.

3.10.1.6 The open end of inert gas or nitrogen gas enriched air (BWMS categories 3a, 3b, 3c and 8) or oxygen-enriched air (BWMS categories 3a, 7a, 7b and 8) are to be led to a safe location*(1) & (2) on open deck.

3.10.2 Where the piping is conveying active substances, by-products or neutralizers that are containing dangerous gas or dangerous liquids as defined respectively in 1.4.2 and 1.4.5 , the following requirements are to be satisfied:

Notes:

1) This requirement is applicable to the injection lines conveying the dangerous gas or dangerous liquids but not applicable to the ballast water lines where the dangerous gas or dangerous liquids are diluted.

2) The IMO reports issued during the basic and final approval procedures of the BWMS that make use of active substances (G9 Guideline) could be used for assessing the hazards that could be expected from the media conveyed by the BWMS piping.

3.10.2.1 Irrespective of design pressure and temperature, the piping is to be either of Class I (without special safeguard) or Class II (with special safeguard) as required by Pt.4, Ch.2, Table 1.2.1. The selected materials, the testing of the material, the welding, the non-destructive tests of the welding, the type of connections, the hydrostatic tests and the pressure tests after assembly on-board are to be as required in Pt.4, Ch.2. Mechanical joints, where allowed, are to be selected in accordance with Pt.4, Ch.2, Table 2.7.2.

Notes:

1) For piping class II with special safeguards conveying dangerous gas like hydrogen (H_2), oxygen (O_2) or ozone (O_3), the special safeguards are to be either double walled pipes or pipe duct.

2) For piping class II with special safeguards conveying dangerous liquids, other special safeguards could be considered like shielding, screening, etc.

3) Plastic pipes may be accepted after due assessment of the dangerous gas or dangerous liquids conveyed inside. When plastic pipes are accepted, the requirements of Pt.4, Ch.2, Sec 5 apply.

3.10.2.2 The length of pipe and the number of connections are to be minimised.

3.10.2.3 Inside double walled space or pipe ducts constructed as the special safeguard for the purpose of 3.10.2.1 Note 1) are to be equipped with mechanical exhaust ventilation leading to a safe location*(3) & (4) on open deck.

3.10.2.4 The routing of the piping system is to be kept away from any source of heating, ignition and any other source that could react hazardously with the dangerous gas or liquid conveyed inside. The pipes are to be suitably supported and protected from mechanical damage.

3.10.2.5 Pipes carrying acids are to be arranged so as to avoid any projection on crew in case of a leakage.

3.10.2.6 H_2 by-product enriched air vent pipes (BWMS categories 4, 5 and 6) or O_2 enriched air vent pipes (BWMS categories 3a, 7a, 7b and

8) or O_3 piping (BWMS categories 7a and 7b) are not to be routed through accommodation spaces, services spaces and control stations.

3.10.2.7 O_2 enriched air vent pipes (BWMS categories 3a, 7a, 7b and 8) are not to be routed through hazardous areas unless it is arranged inside double walled pipes or pipe ducts constructed as the special safeguard for the purpose of 3.10.2.1 Note 1) and provided with suitable gas detection as described in 3.10.1.4 and mechanical exhaust ventilation as described in 3.10.2.3.

3.10.2.8 The routing of H_2 by-product enriched air vent pipes (BWMS categories 4, 5 and 6) or O2 enriched air vent pipes (BWMS categories 3a, 7a, 7b and 8) is to be as short and as straight as possible. When necessary, horizontal portions may be arranged with a minimum slope in accordance with the manufacturer's recommendation.

3.10.3 For BWMS using chemical substances or dangerous gas which are stored on-board for either:

- storage or preparation of the active substances (BWMS categories 2 and 6), or

- storage or preparation of the neutralizers (BWMS categories 4, 5, 6, 7a and 7b), or

- recycling the wastes produced by the BWMS (BWMS category 2),

procedures are to be in accordance with the Material Safety Data Sheet and BWM.2/Circ.20 "*Guidance to ensure safe handling and storage of chemicals and preparations used to treat ballast water and the development of safety procedures for risks to the ship and crew resulting from the treatment process*", and the following measures are to be taken as appropriate:

3.10.3.1 The materials, inside coating used for the chemical storage tanks, piping and fittings are to be resistant to such chemicals substances.

3.10.3.2 Chemical substances (even if they are not defined as dangerous liquid in the sense of 2.4) and gas storage tanks are to be designed, constructed, tested, inspected, certified and maintained in accordance with:

- for independent tanks permanently fixed onboard containing dangerous liquids (eg. sulfuric acid H_2SO_4) or dangerous gas (eg. oxygen O_2): the Classification Rules as applicable to pressure vessels

- for independent tanks permanently fixed onboard not containing dangerous liquid (eg. sodium sulphite, sodium biosulphite or sodium thiosulfphate neutralizers) and not containing dangerous gas (eg. nitrogen N_2): the Rules or other industry standard recognized by IRS.

- for portable tanks: the IMDG Code or other industry standard recognized by IRS.

3.10.3.3 When the chemical substances are stored inside integral tanks, the ship's shell plating is not to form any boundary of the tank.

3.10.3.4 Dangerous liquids and dangerous gas storage tank air pipes are to be led to a safe location*(1) & (2) on open deck.

3.10.3.5 An operation manual containing chemical injection procedures, alarm systems, measures in case of emergency, etc. is to be kept onboard.

3.10.3.6 Dangerous liquid storage tanks and their associated components like pumps and filters, are to be provided with spill trays or secondary containment system of sufficient volume to contain potential leakages from tank openings, gauge glasses, pumps, filters and piping fittings.
Further to the safety and/or pollution assessment of the concerned chemical substances, consideration should be provided for segregation of the drains from such spill trays (or secondary containment system) or piping systems from engine room bilge system or from cargo pump room bilge system, as applicable. When necessary, arrangement should be provided within the spill trays (or within the secondary containment system) for the detection of dangerous liquid or dangerous gas as defined respectively in 1.4.2 and 1.4.5.

Note: The IMO reports issued during the basic and final approval procedures of the BWMS that make use of active substances (IMO G9 Guideline) could be used for this assessment.

3.10.4 A risk assessment is to be conducted in a generic manner during the design review mentioned in 3.5.4 and submitted to IRS for approval for the following BWMS categories:

- BWMS category 4: in all cases;

- BWMS category 5: in all cases;

- BWMS category 6: when one of the MSDS indicates that the chemical substance stored on-board is either flammable, toxic, corrosive or reactive;

- BWMS category 7a and 7b: in all cases.

Note: The IMO reports issued during the basic and final approval procedures of the BWMS that make use of active substances (G9 Guideline) could be used as a reference for this assessment.

3.10.4.1 The recommended risk assessment techniques for BWMS and other guidances are listed below but not limited to:

- FMEA, FMECA, HAZID, HAZOP, etc.

- ISO 31010 – Risk Assessment Techniques

- IACS Recommendation Rec. 146

- IRS Guidelines on Alternative and Risk Based Design Evaluation

3.10.4.2 The risk assessment is to ensure that the package supplied by the BWMS's manufacturer is intrinsically safe and/or provides mitigation measures to the hazards created by the BWMS which have been identified during the design review mentioned in 3.5.4 but that need to be implemented during the installation on-board.

3.11. Fire safety requirements

3.11.1 General

3.11.1.1 Application

.1 This subsection details fire safety measures, in addition to that required by Part 6, related to the installation of Ballast Water Management Systems onboard any ship.

.2 The requirements of this subsection apply for BWMS technologies as listed in Table 3.1. BWMS with alternative technologies are to be specially considered by the IRS.

3.11.1.2 Definitions

.1 Airlock - An airlock is a space enclosed by gastight steel bulkheads with two gastight doors spaced not more than 2.5 m apart. The doors are to be self-closing without any holding back arrangements. Air locks are to have mechanical ventilation and are not to be used for other purposes. An audible and visual alarm system to give a warning on both sides of the air lock are to be provided to indicate if more than one door is moved from the closed position. The air lock space is to be monitored for dangerous gas as defined in 1.4.2.

.2 Ballast Water Management Room (BWMR) - A Ballast Water Management Room is any space containing equipment belonging to the Ballast Water Management System. A space containing remote controls for the BWMS or a space dedicated to the storage of liquid or solid chemicals for BWMS need not be considered as a BWMR for the purposes of this subsection.

.3 BWMS storing, introducing or generating chemicals. - In general, BWMS storing, introducing or generating chemicals refer to:

- In-line flocculation (cat.2 as per Table 3.1),

- Chemical injection (cat.6 as per Table 3.1) and

- BWM technologies using neutralizers injection (cat.4, 5, 6 and 7 as per Table 3.1)

BWMS that do not store, introduce or generate toxic or flammable chemicals may be specially considered as detailed in Table 3.11.1 below.

Table 3.11.1: Requirements that may be reduced for BWMS storing, introducing or generating chemicals depending on the chemicals.	
Requirement	**Conditions to be met before reducing the requirement**
3.11.2.3.4	The stored chemicals are neither toxic nor flammable
3.11.3.1.1	The BWMS does not use any flammable or toxic chemical substances
3.11.3.3.1	No dangerous gas as defined in 1.4.2 will be generated by the BWMS
3.11.6.1.1	No toxic chemical is stored and no toxic gas will be generated by the BWMS
3.11.7.1.1 3.11.7.1.3 3.11.7.1.6	No toxic chemical is used or will be generated by the BWMS

The IMO reports issued during the basic and final approval procedures of the BWMS that make use of active substances (IMO G9 Guidelines) and ''safety hazard'' as listed in Pt.5, Ch.3. Sec 17 are to be considered for this purpose.

Note: Chemicals include additives for BWMS.

3.11.2 Fire categorization

3.11.2.1 General

BWMR is to be classified as follows for the purpose of applying the requirements of Part 6:

- BWMR containing oil-fired inert gas generators (i.e. BWMS cat.3b and 3c as per Table 3.1) are to be treated as machinery spaces of category A

- Other BWMR is to be considered as other machinery spaces and is to be categorized, depending on the ship type (10) or (11) according to Pt.6, Ch.3, 3.2.2.3, 3.2.3, and 3.2.4

3.11.2.2 BWMS located in the cargo area of tankers

Notwithstanding the above, where a BWMS is located in the cargo area of a tanker as allowed by previous subsection, the BWMR is to be categorized as (8), a cargo pump-room, according to Pt.6, Ch.3, 3.2.4.2.2 for determining the extent of fire protection to be provided.

Note: The cargo area of a tanker is defined in Section 1.

3.11.2.3 Storage of chemicals

.1 Spaces where the storage of liquid or solid chemicals for BWMS is intended are to be categorized as store-rooms for the purpose of applying the requirements of Part 6 i.e.:

- On passenger ships carrying more than 36 passengers:

  - "Other spaces in which flammable liquids are stowed" as defined in Pt.6, Ch.3, 3.2.2.3.2.2(14), if flammable products are stored

  - "Store-rooms, workshops, pantries, etc." as defined in Pt.6, Ch.3, 3.2.2.3.2.2(13) otherwise

- On other ships:

  - "Cargo pump-rooms" as defined in Pt.6, Ch.3, 3.2.4.2.2.2(8) if located in the cargo area of a tanker

  - "Service spaces (low risk)" as defined in Pt.6, Ch.3, 3.2.2.4.2.2(5), 3.2.3.3.2.2(5) or 3.2.4.2.2.2(5) if the surface area is less than $4m^2$ and if no flammable products are stored

  - "Service spaces (high risk)" as defined in Pt.6, Ch.3, 3.2.2.4.2.2(9), 3.2.3.3.2.2(9) or 3.2.4.2.2.2(9) otherwise

Note: It is understood that only chemical injection (cat.6 as per Table 3.1), in-line flocculation (cat.2 as per Table 3.1) and technologies using neutralizer injection (cat.4,

5, 6 and 7 as per Table 3.1) will require chemical or additive storage.

.2 Where the storage of chemicals is foreseen in the same room as the ballast water management machinery, this room is to be considered both as a store-room and as a machinery space in line with 3.11.2.1.

.3 When the chemical substances are stored inside integral tanks, the ship's shell plating is not to form any boundary of the tank.

.4 Tanks containing chemicals are to be segregated from accommodation, service spaces, control stations, machinery spaces not related to the BWMS and from drinking water and stores for human consumption by means of a cofferdam, void space, cargo pump-room, empty tank, oil fuel storage tank, BWMR or other similar space. On-deck stowage of permanently attached deck tanks or installation of independent tanks in otherwise empty hold spaces are to be considered as satisfying this provision.

3.11.3 BWMR location and boundaries

3.11.3.1 BWMS using chemical substances

.1 For BWMS storing, introducing or generating chemicals, the BWMR and chemical substance storage rooms are not to be located in the accommodation area. Any ventilation exhaust or other openings from these rooms are to be located not less than 3m from entrances, air inlets and openings to accommodation spaces. This requirement need not apply in case the BWMS is located in the engine room.

3.11.3.2 Ozone-based BWMS

.1 Ozone-based BWMS – i.e. cat.7a and 7b – are to be located in dedicated compartment, separated from any other space by gastight boundaries. Access to the BWMR from any other enclosed space is to be through airlock only, except if the only access to that space is from the open deck.

Access to the ozone based BWMR may be provided through the engine room only provided:

- Access from the engine room to the BWMR is through airlock and,
- An alarm repeater is provided in the BWMR, which will repeat any alarm activated in the engine room.

.2 A sign is to be affixed on the door providing personnel with a warning that ozone may be present and with the necessary instructions to be followed before entering the room

3.11.3.3 General

.1 BWMR containing equipment for BWMS of the following types is to be equipped with tested gastight and self-closing doors without any holding back arrangements:

- BWMS storing, introducing or generating chemical substances
- De-oxygenation based on inert gas generator
- Electrolysis
- Ozone injection

Doors leading to the open deck need however not to be self-closing.

3.11.4 Fire Fighting

3.11.4.1 Fixed fire-extinguishing system

3.11.4.1.1 Where fitted, fixed fire extinguishing systems are to comply with the relevant provisions of the Pt.6, Ch.8

3.11.4.1.2 Ozone-based BWMS

BWMR containing equipment related to ozone-based BWMS is to be provided with a fixed fire extinguishing system suitable for category A machinery spaces and capable of manual release.

3.11.4.1.3 Where a fixed fire-extinguishing system is provided in the BWMR, it is to be compatible with the BWMS and the chemical products that are used, produced or stored in the BWMR. Specific attention is to be paid to potential chemical reactions between the fire extinguishing medium and chemical products used for water treatment.

Especially, water-based fire-extinguishing systems should be avoided in case of sulfuric acid storage.

3.11.4.1.4 Foam fixed fire-extinguishing system

For all kinds of BWMS, in case a foam fire extinguishing system is installed in the BWMR, its efficiency is not to be impaired by chemicals used by the BWMS where relevant.

3.11.4.1.5 Where a fixed fire-extinguishing system is installed in the BWMR, automatic shutdown of the BWMS upon release of the fixed fire extinguishing system is to be arranged. Any need for cooldown necessary for safe shutdown to be considered in the shutdown sequence.

3.11.4.1.6 Where BWMS that includes air or O_2 storage is located in a room covered by a fixed gas fire-extinguishing system, air or O_2 storage is to be taken into account for the gas capacity calculation, unless the discharge pipe from safety valves for air or O_2 storage are led directly to outside the room.

3.11.4.2 Portable fire-fighting equipment

3.11.4.2.1 There is to be at least one portable fire extinguisher that complies with the provisions of the Pt.6, Ch.8 and suitable for electrical fires in the BWMR containing UV-type BWMS.

3.11.5 Fire prevention

3.11.5.1 Equipment protection

3.11.5.1.1 Overcurrent or overvoltage protection is to be installed to protect UV type BWMS.

3.11.5.1.2 Electrolysis reactors are to be provided with at least with two independent means of monitoring operation. The monitoring system is to initiate audible and visual alarms and automatic shutdown of the BWMS in the event that an anomaly is detected. Requirements for shutdown arrangement are clarified in 3.5.4.

Note: If a pressure relief valve is also provided, the vent of this valve is to be led to a safe location on the open deck, as clarified in 3.5. The valve should be positioned to optimally remove gas from the electrolysis reactor.

3.11.5.2 Fire detection

3.11.5.2.1 A fixed fire detection and fire alarm system complying with the provisions of Pt.6, Ch.8 is to be installed in spaces containing an inert gas generator or an ozone generator.

3.11.5.2.2 A section of fire detectors which covers a control station, a service space or an accommodation space is not to include a BWMR containing equipment related to ozone based BWMS.

3.11.6 Ventilation

3.11.6.1 Ventilation arrangement

3.11.6.1.1The ventilation systems for BWMR containing BWMS of the following types are to be independent of the ventilation systems serving any other spaces:

- BWMS storing, introducing or generating chemical substances.
- De-oxygenation, including pasteurization and de-oxygenation (cat.3 and cat.8)
- Electrolysis
- Ozone injection

3.11.6.1.2 The ventilation exhaust for BWMR containing a nitrogen generator is to be located in the lower part of the room in order to efficiently evacuate dangerous gases heavier than air.

3.11.6.1.3 The ventilation exhaust for BWMR containing electrolysis systems is to be located so as to be able to efficiently evacuate dangerous gases that could be generated during the electrolysis process. Due regard is to be paid to the expected quantity and density of such gases when designing the ventilation exhaust.

3.11.6.1.4 The following requirements apply to ventilation ducts serving BWMR for ozone-based BWMS:

- The part of the ducts located outside of the BWMR is to be made of steel having a thickness of at least 3 mm for ducts with a free cross-sectional area of less than 0.075 m^2, at least 4 mm for ducts with a free cross-sectional area of between 0.075 m^2 and 0.45 m^2, and at least 5 mm for ducts with a free cross-sectional area of over 0.45 m^2; and

- The ducts are to be suitably supported and stiffened

- The outside openings of the ducts are to be fitted with protective screens of not more than 13 mm square mesh.

3.11.6.1.5 The ventilation system for BWMR containing ozone-based BWMS or ventilation system for hydrogen de-gas arrangement as required by 3.10.1.5 is to be interlocked with the BWMS such that:

- In case of loss of ventilation (primary and secondary), a visual and audible alarm is to be triggered both inside and outside the BWMR and at a place where a responsible member of the crew is on duty. If the ventilation is not restored after a pre-set time, the BWMS is to then be automatically shut down. Any need for cooldown necessary for safe shutdown is to be considered in the shutdown sequence.

- It is not to be possible to start the BWMS without the ventilation running

For ventilation systems serving BWMR and containing or conveying a dangerous gas, relevant requirements in 3.10 are to be satisfied.

3.11.6.2 Ventilation rate

3.11.6.2.1 An adequate power ventilation system is to be provided in enclosed BWMR.

3.11.6.2.2 The ventilation capacity is to be at least 30 air changes per hour where explosive or toxic gases may be generated during operation of the BWMS. The IMO reports issued during the basic and final approval procedures of the BWMS that make use of active substances (IMO G9 Guidelines) and ''safety hazard'' as listed in Pt.5, Ch.3, Sec.17 are to be used as references for identifying those cases.

3.11.6.2.3 The ventilation capacity may be reduced as follows:

- Flocculation-type BWMS - 6 air changes per hour
- De-oxygenation, incl. pasteurization and de-oxygenation (cat.3 and cat.8 as per Table 3.1) - 6 air changes per hour
- Full flow electrolysis - 6 air changes per hour
- Side-stream electrolysis - 20 air changes per hour
- Ozone injection - 20 air changes per hour
- Chemical injection - 6 air changes per hour

Note: More stringent ventilation capacity requirements may arise from other regulations e.g. IBC Code requirements for spaces located in the cargo area.

3.11.7 Personal equipment

3.11.7.1.1 Suitable protection equipment is to be available onboard for the protection of the crew members who are engaged in the servicing, maintenance and repair of BWMS storing, introducing or generating chemicals, as recommended by the product manufacturers. The protection equipment is to consist of large aprons, special gloves with long sleeves, suitable footwear, coveralls of chemical-resistant materials, and tight fitting goggles or face shields or both. The protective clothing and equipment are to cover all skin so that no part of the body is unprotected. This protection equipment is to be provided separately without taking into account equipment required by other mandatory requirements.

3.11.7.1.2 Work clothes and protective equipment are to be kept in easily accessible places and in special lockers. Such equipment are not to be kept within accommodation spaces, with the exception of new, unused equipment and equipment which has not been used since undergoing a thorough cleaning process. Notwithstanding the above, storage rooms for such equipment within accommodation spaces if adequately segregated from living spaces such as cabins, passageways, dining rooms, bathrooms, etc.

3.11.7.1.3 When a BWMS storing, introducing or generating chemicals is installed on board, suitably marked decontamination showers and an eyewash are to be available in a convenient location in close proximately to the BWMS and the chemical store room(s).

3.11.7.1.4 An emergency escape breathing apparatus (EEBD) is to be provided in the BWMR. This emergency escape breathing apparatus may be one of the EEBDs provided in accordance with the requirements of SOLAS II-2/13.

An EEBD need not be required for BWMS of cat.1 as per Table 3.1.

3.11.7.1.5 A personal ozone detector, calibrated as per the manufacturer's specifications, is to be provided for each person engaged in the servicing, maintenance and repair of BWMS utilizing ozone.

3.11.7.1.6 A two-way portable radiotelephone apparatus dedicated for the BWMS service, maintenance and repair is to be provided, in addition to those required by SOLAS for fire-fighting purposes. This two-way radiotelephone apparatus is to be properly identified in order to avoid mix-up with the apparatus intended for fire-fighting operations. Where the BWMS may release explosive gases, this two-way radiotelephone apparatus is to be of a certified safe type suitable for use in zone 1 hazardous areas, as defined in IEC Publication 60079. Where the BWMS stores, utilizes or introduces chemicals, the apparatus is to undergo deep cleaning or de-contamination after use.

A two-way portable radiotelephone apparatus need not be required for BWMS of cat.1 as per Table 3.1.

Annex I

Installation of one single BWMS on Tankers

Note: This Annex does not cover In-tank technologies categories 3c and 8.

Table 1: In-line BWMS's technlogies categorization

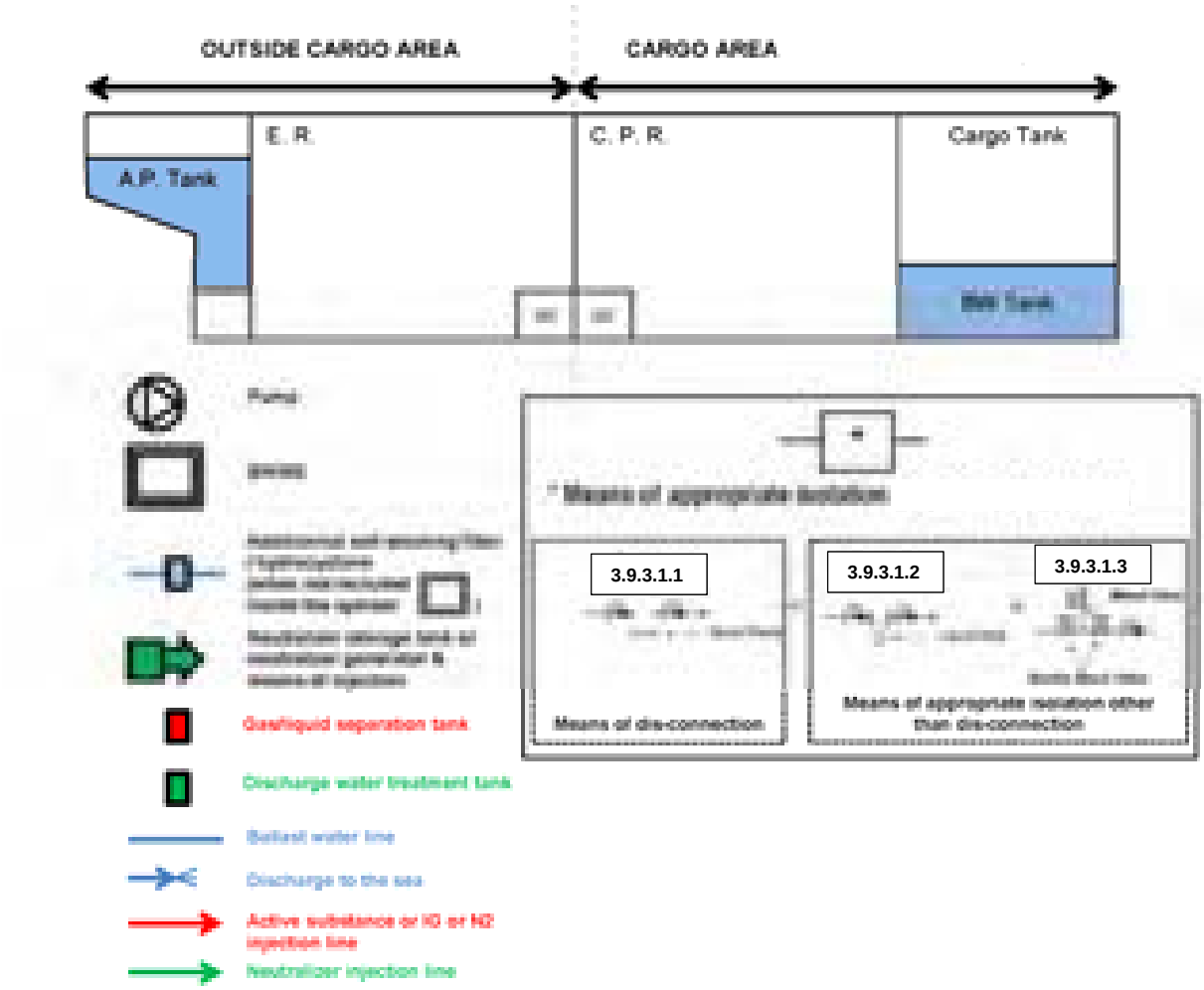
		1	2	3a	3b	4	5	6	7a	7b
BWMS's Technology category → **Characteristics ↓**		In-line UV or UV + Advanced Oxidation Technology (AOT) or UV + TiO2 or UV + Plasma	In-line Flocculation	In-line membrane separation and de-oxygenation (injection of N2 from a N2 Generator)	In-line de-oxygenation (injection of Inert Gas from Inert Gas Generator)	In-line full flow electrolysis	In-line side stream electrolysis (3)	In-line (stored) chemical injection	In-line side-stream ozone injection without gas/liquid separation tank and without Discharge treatment tank	In-line side-stream ozone injection with gas/liquid separation tank and Discharge water treatment tank
Dis-infection when ballasting	Making use of active substance		X			X	X	X	X	X
	Full flow of ballast water is passing through the BWMS	X	X	X	X	X				X
	Only a small part of ballast water is passing through the BWMS to generate the active substance						X			
After-treatment de-ballasting	Full flow of ballast water is passing through the BWMS	X								X
	Injection of neutralizer					X	X	X	X	X
	Not required by the Type Approval Certificate issued by the Administration		X	X						
Examples of dangerous gas as defined in 1.4.2			(1)	O_2 N_2	CO_2, CO	H_2, Cl_2	H_2, Cl_2	(1)	O_2, O_3, N_2	
Arrangement of one single BWMS	BWMS is located in the outside the cargo area	Not Acceptable	Case 1.2 (2)	Case 1.3a (2)	Case 1.3b	Case 1.4 (2)	Case 1.5	Case 1.6	Case 1.7a	Case 1.7b (2)

Notes:

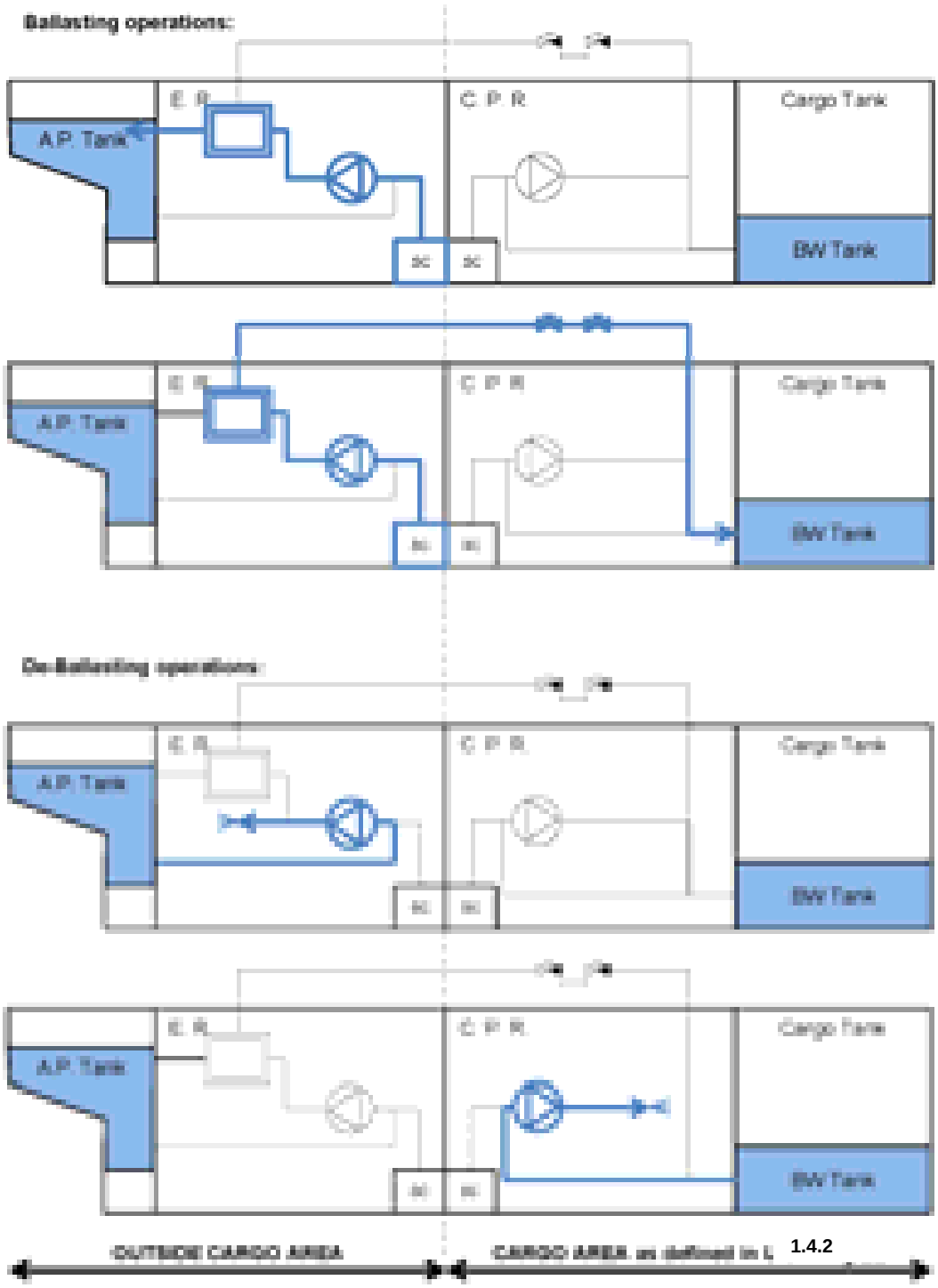
(1) To be investigated on a case by case basis based on the result of the IMO (GESAMP) MEPC report for Basic and Final approval in accordance with the IMO G9 Guideline.

(2) Only « Means of dis-connection » as described in 3.9.3.1 are to be applied.

(3) In-line side stream electrolysis may also be applied in-tank in circulation mode (no treatment when ballasting or deballasting)


OUTSIDE CARGO AREA
CARGO AREA
E. R.
C. P. R.
Cargo Tank
A.P. Tank
3.9.3.1.1
3.9.3.1.2
3.9.3.1.3
Means of dis-connection
Means of appropriate isolation other than dis-connection

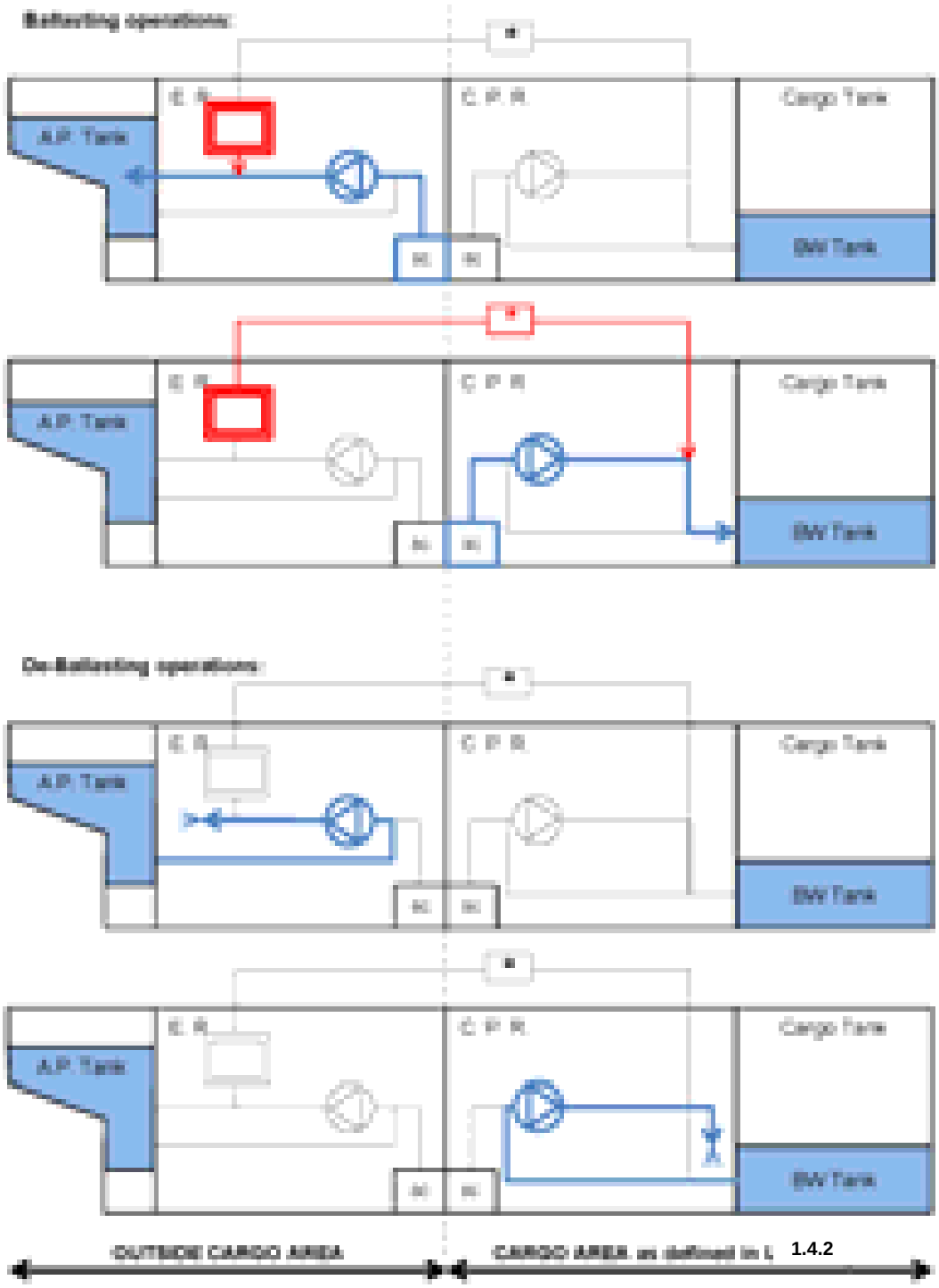
BWMS installed outside the cargo area
Case 1.2 (Technology category 2, Flocculation); and
Case 1.3a (Technology category 3a De-oxygenation with N2 Generator)



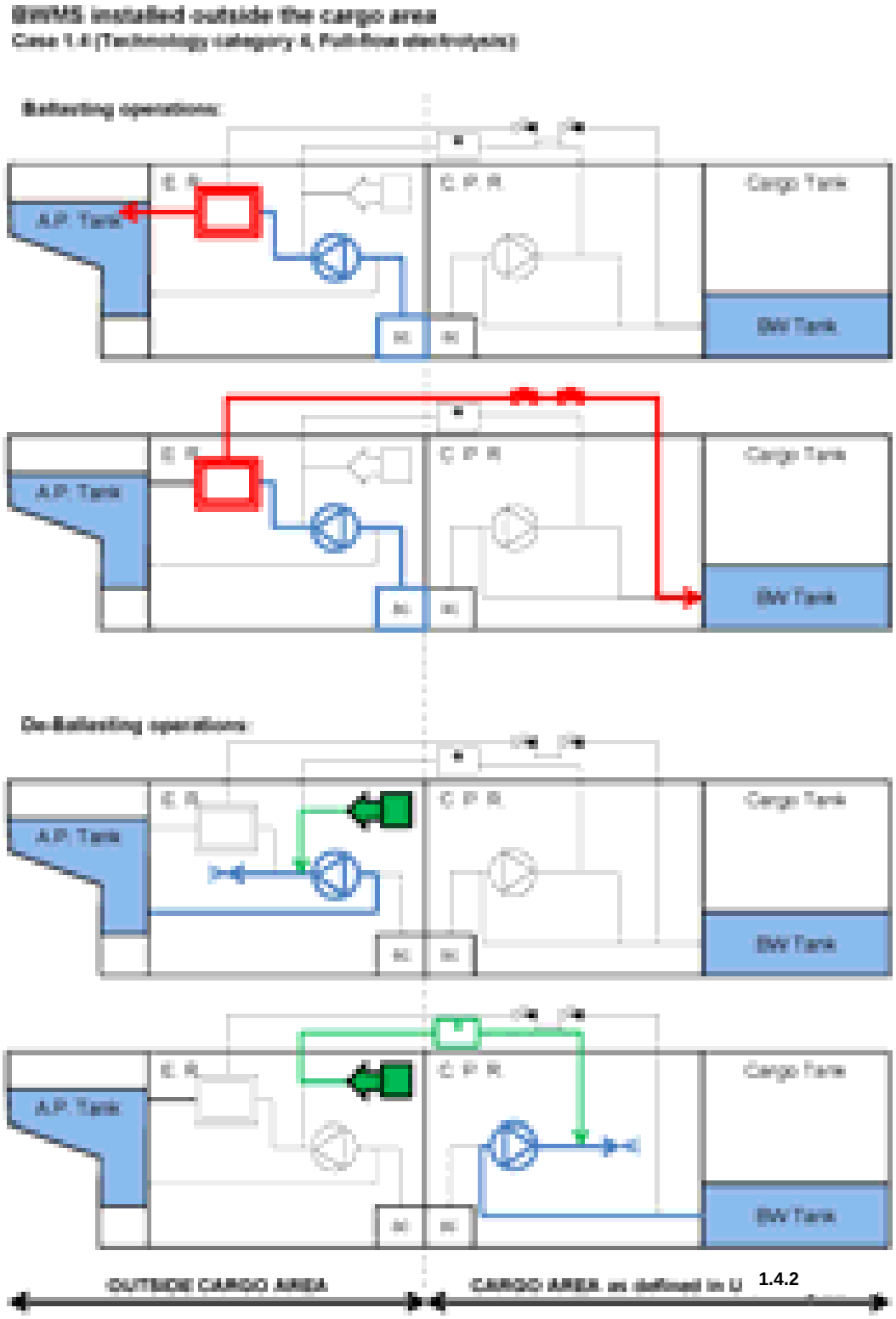
1.4.2

BWMS installed outside the cargo area

Case 1.4b (Technology category 3b, Deoxygenation with Inert Gas Generator):

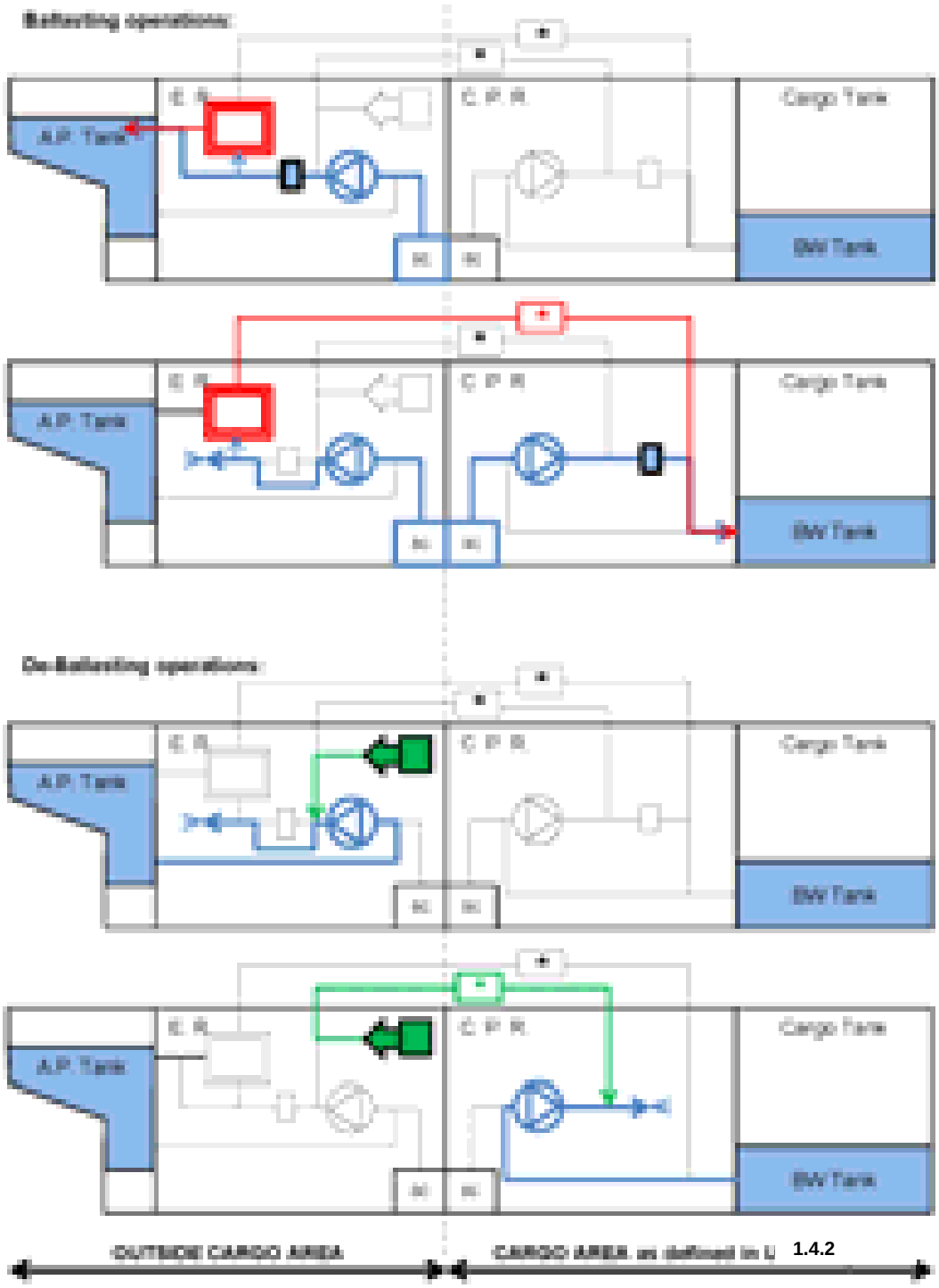


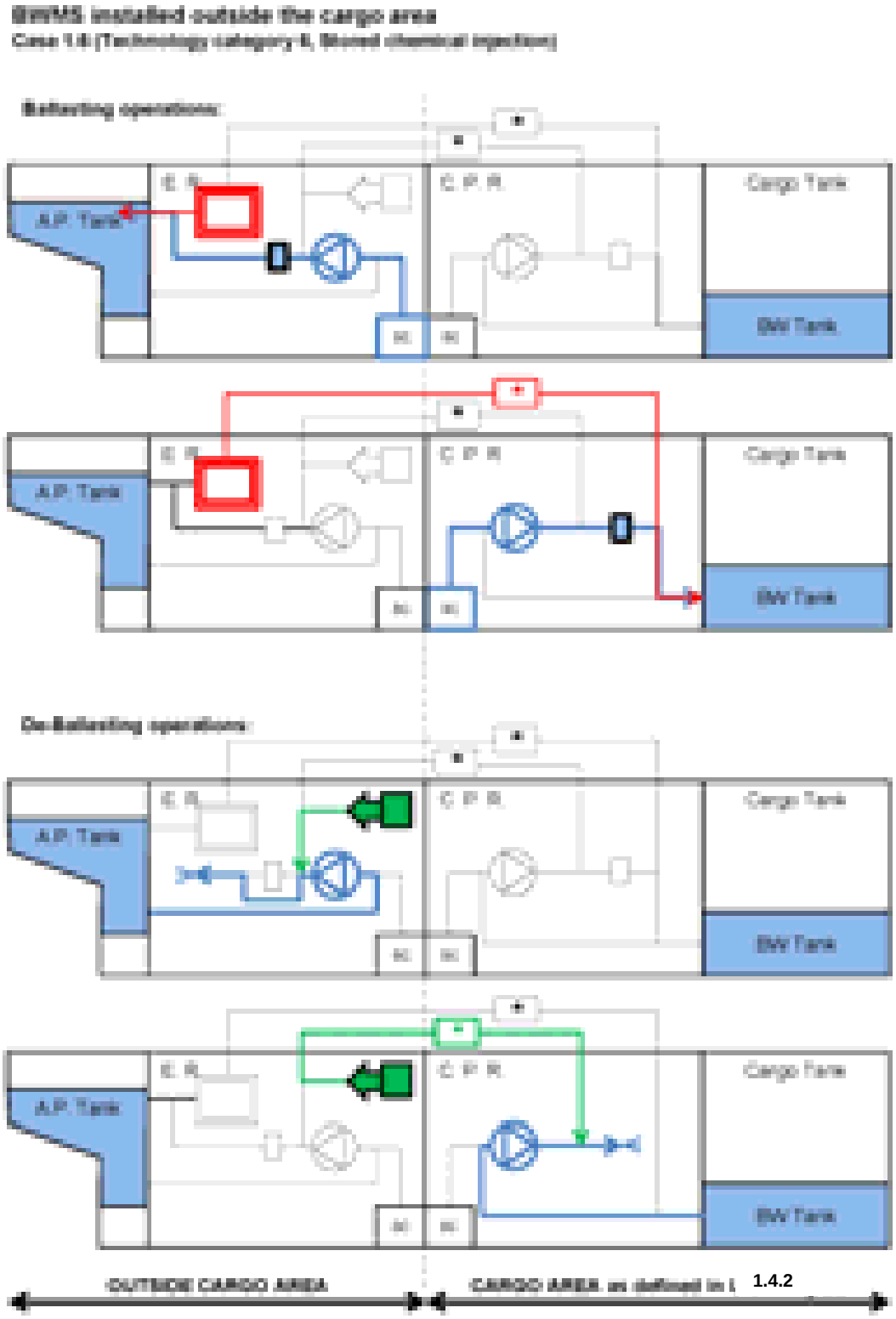
1.4.2


BWMS installed outside the cargo area
Ballasting operations:
E. R.
C. P. R.
Cargo Tank
A.P. Tank
BW Tank
De-Ballasting operations:
OUTSIDE CARGO AREA
CARGO AREA as defined in U
1.4.2

BWMS installed outside the cargo area

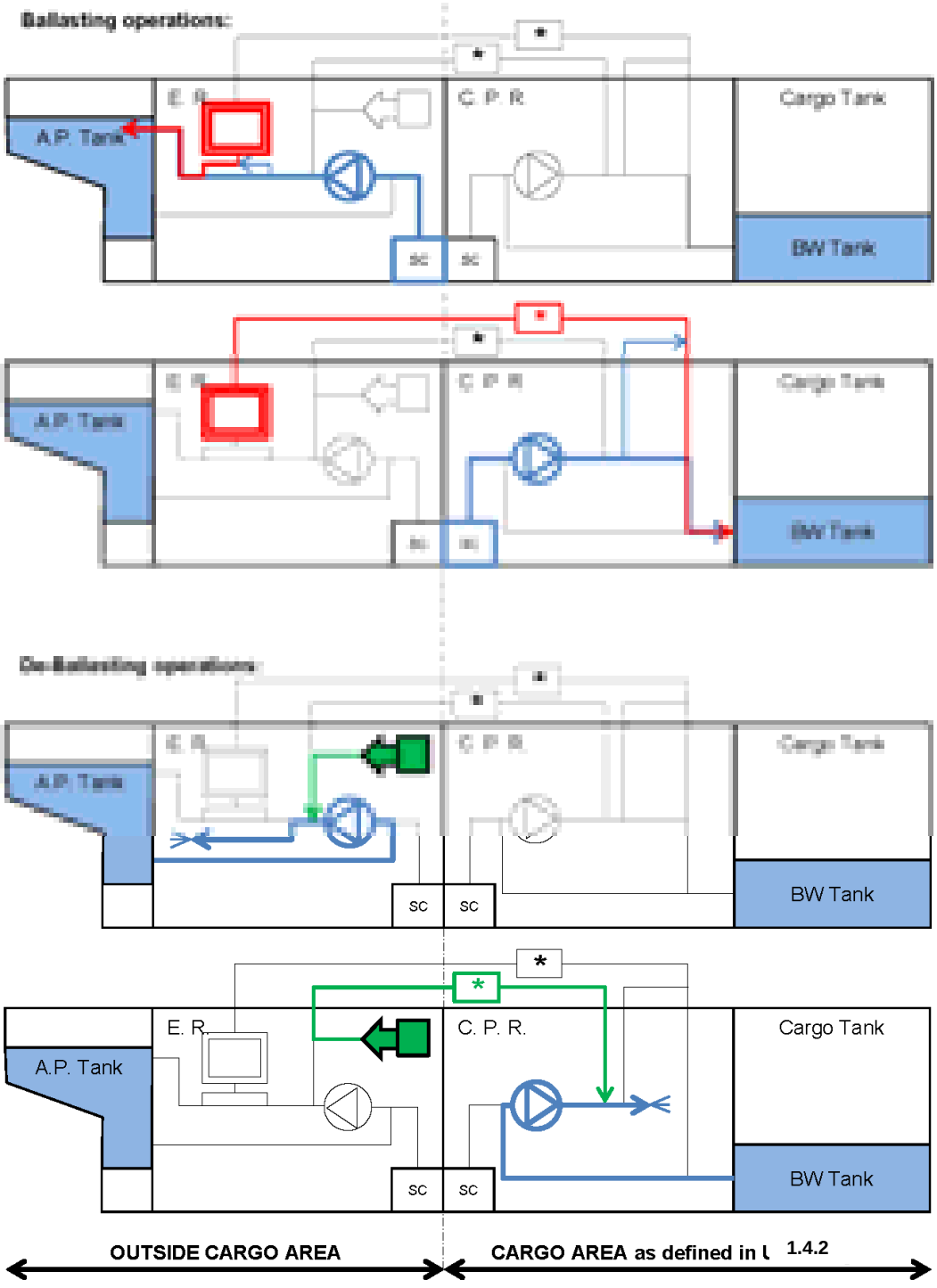
Case 1.4 (Technology category II, Side-stream electrolysis):




BWMS installed outside the cargo area
Ballasting operations:
E. R.
C. P. R.
Cargo Tank
A.P. Tank
BW Tank
De-Ballasting operations:
OUTSIDE CARGO AREA
CARGO AREA as defined in I 1.4.2

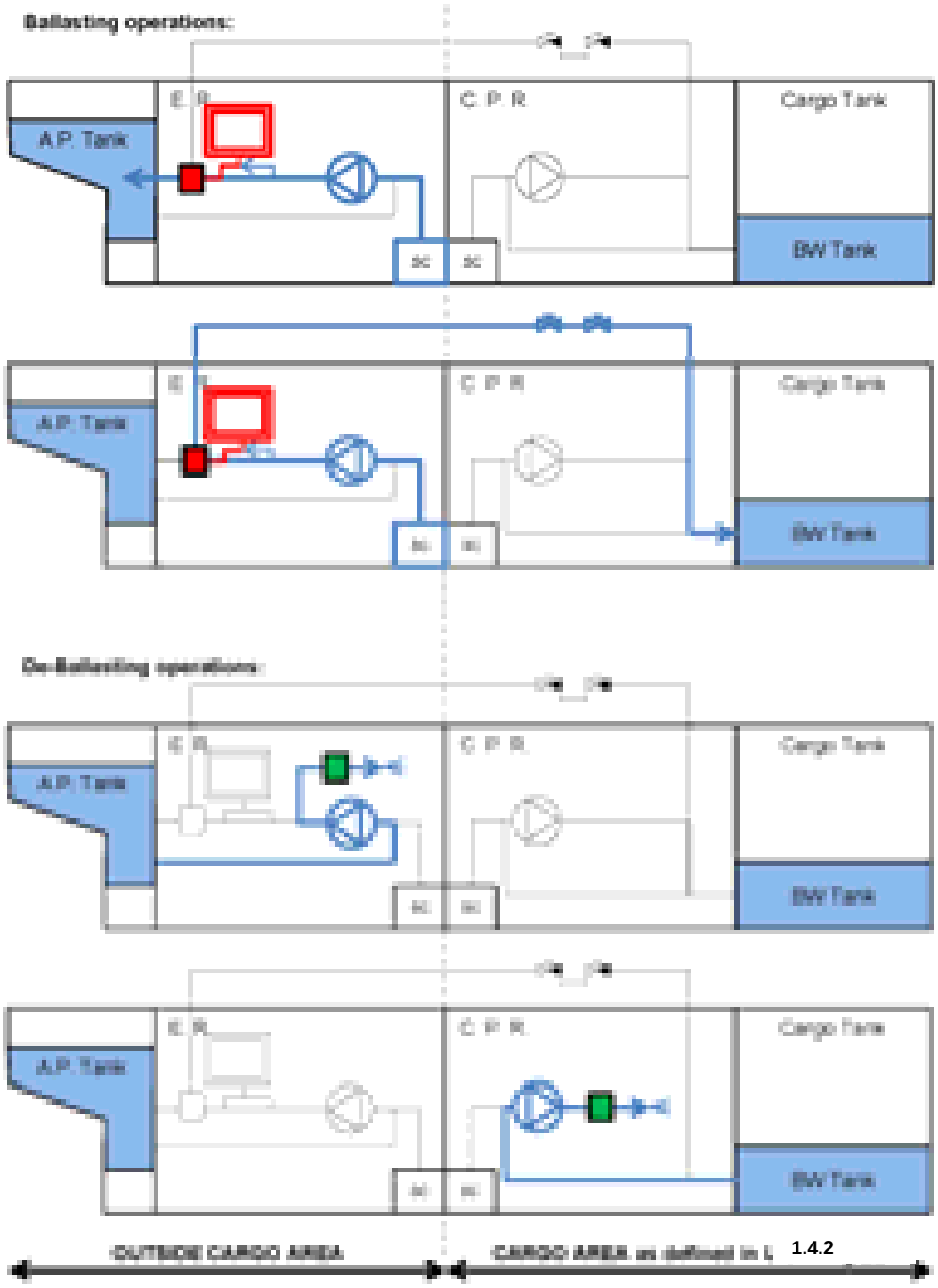
BWMS installed outside the cargo area

Case 1.7a (Technology category Ta, Side-stream ozone injection without gas/liquid separation tank and without discharge water treatment tank):



BWMS installed outside the cargo area

Case 1.7b (Technology category 7b, Side-stream ozone injection with gas/liquid separation tank and discharge water treatment tank):

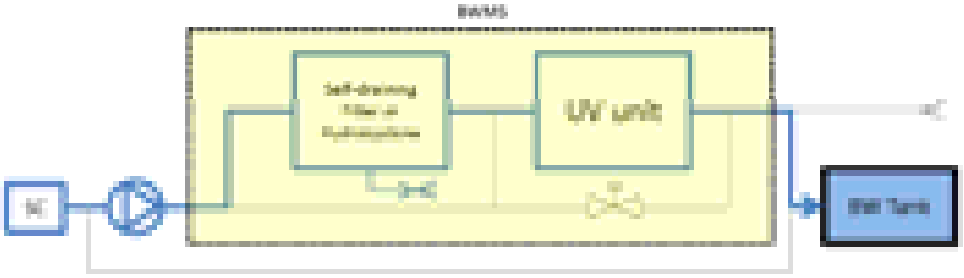


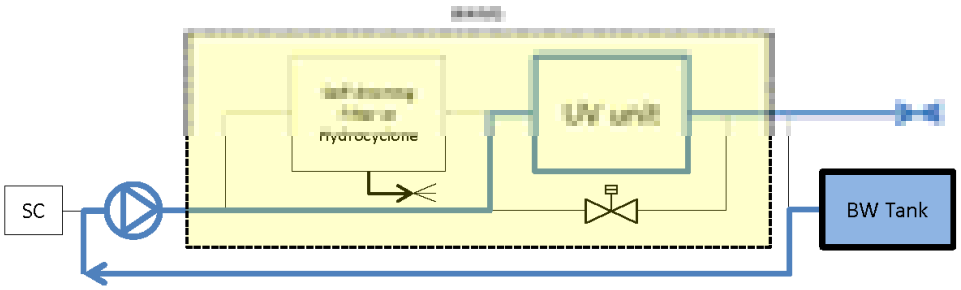
1.4.2

Annex II (INFORMATIVE)
BWMS Technologies categorization

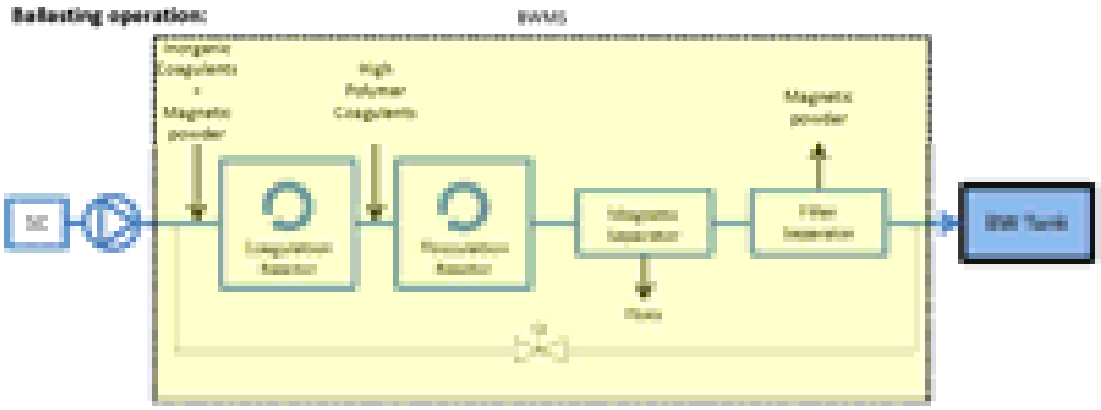
BWMS Technology Group no. 1
In-Line UV
including UV + AOT
Including UV+TiO2)

Ballasting operation:





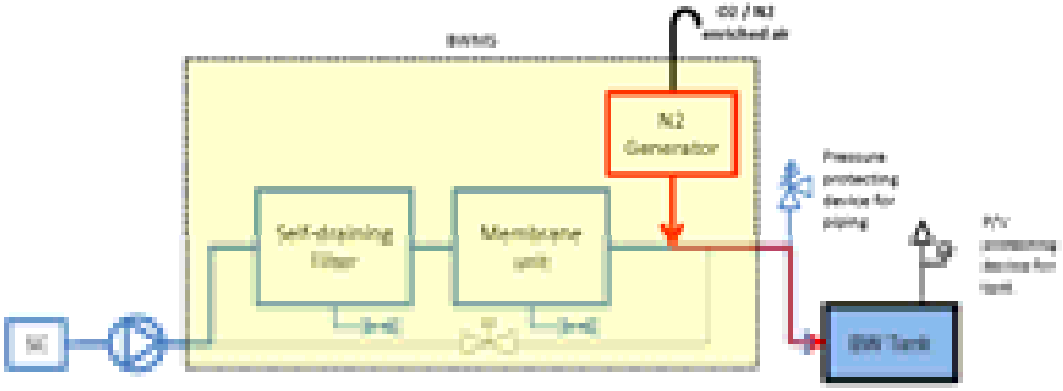
BWMS Technology Group no. 2
In-Line Flocculation



On ballasting operations: no requirement for after-treatment

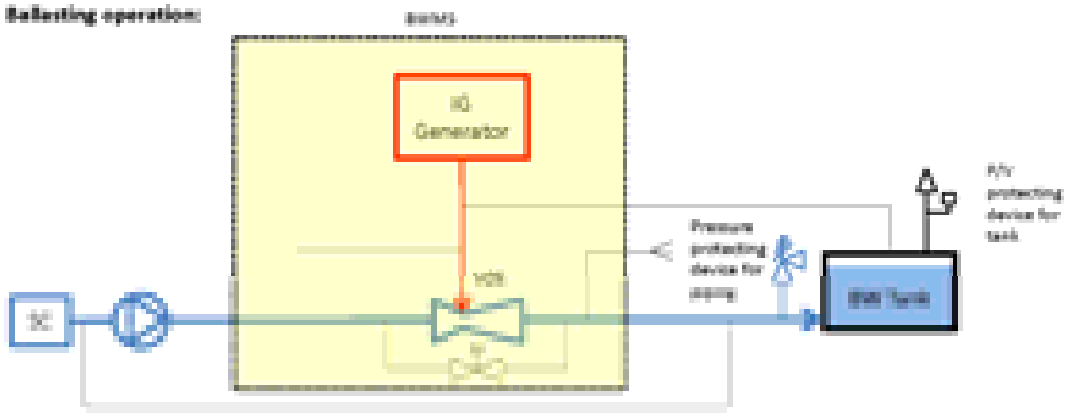
BWMS Technology Group no. 3a
In-Line membrane separation and de-oxygenation
(injection of N2 from N2 Generator)

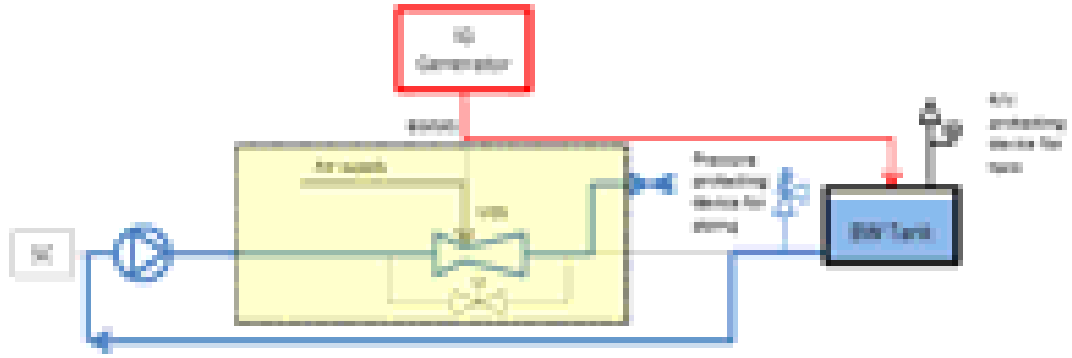
Ballasting operation:



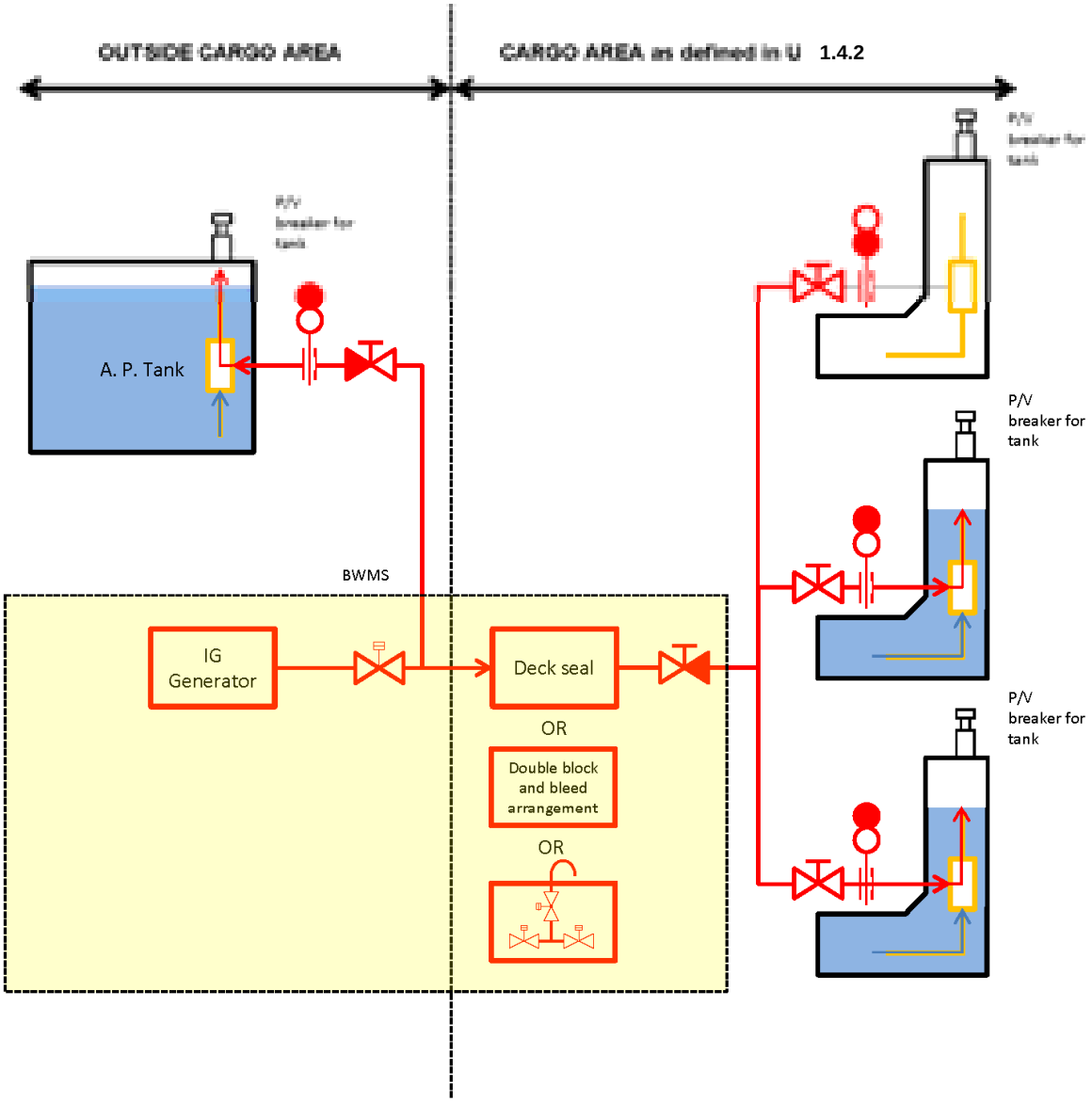
On ballasting operation: no requirement for after-treatment

BWMS Technology Group no. 3b
In-Line de-oxygenation
(injection of inert gas from either an oil fired inert gas generator or inert gas from treatment of the flue gas from main or auxiliary boilers)

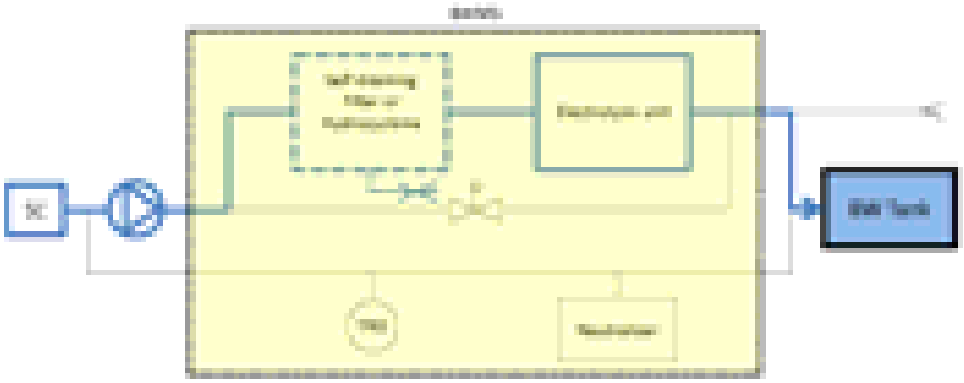


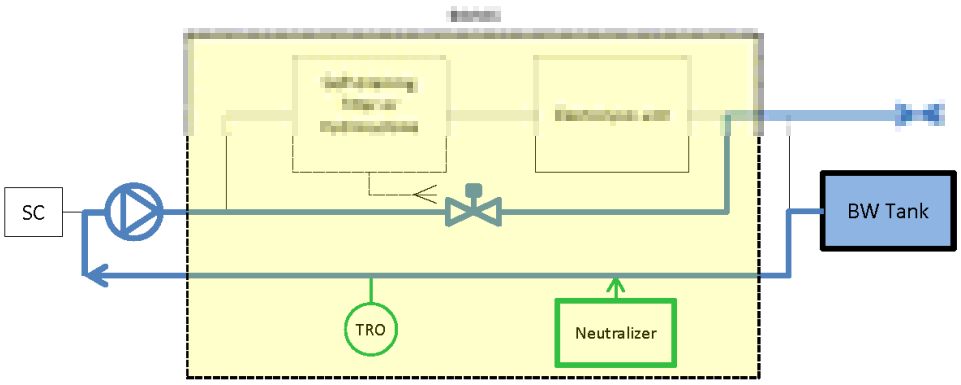


BWMS Technology Group no. 3c
In-tank de-oxygenation with IGG



BWMS Technology Group no. 4
In-Line Full flow electrolysis

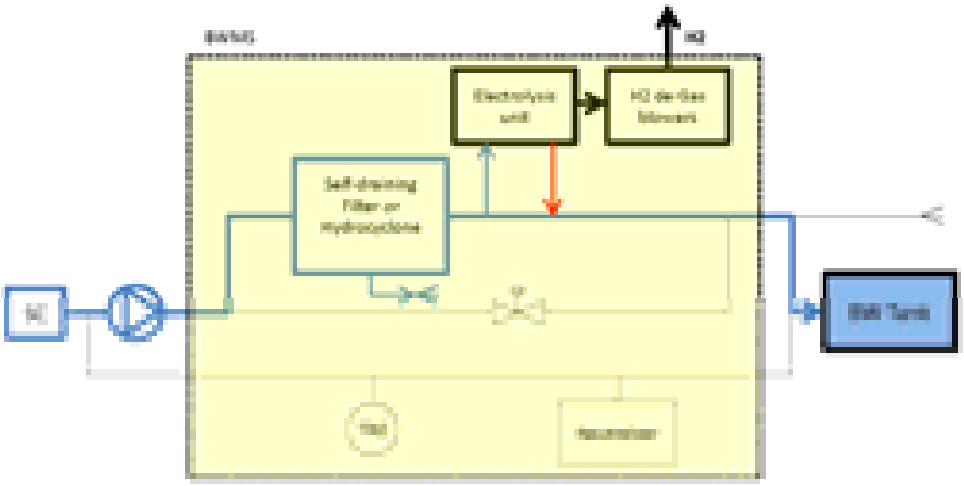




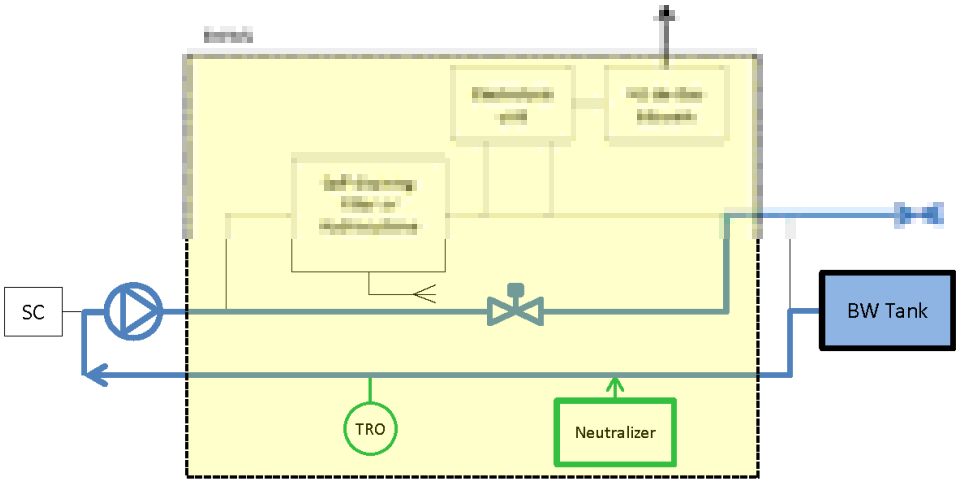
BWMS Technology Group no. 5
In-Line Side-Stream electrolysis (electro-chlorinization)

Note: In-line side stream electrolysis may also be applied on board in circulation mode (no treatment when ballasting or de-ballasting)

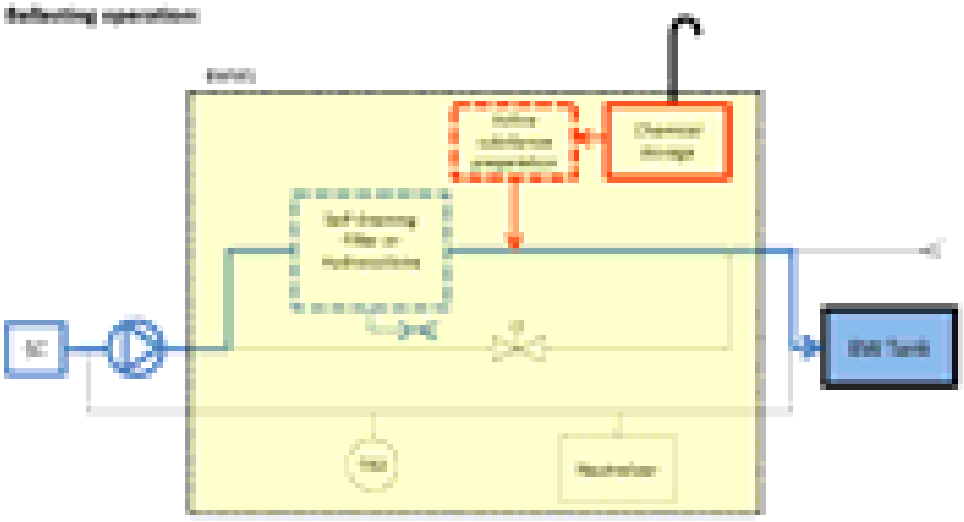
Ballasting operation:

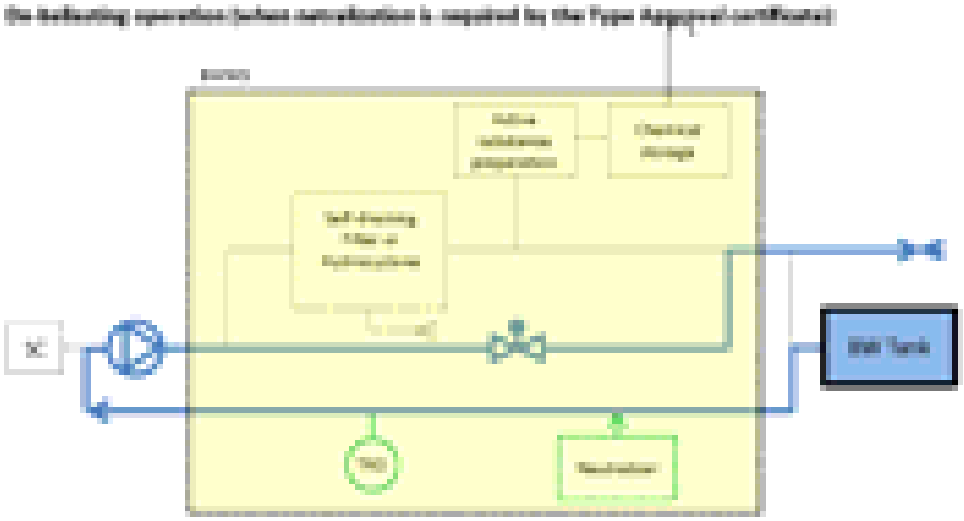


De-ballasting operation:



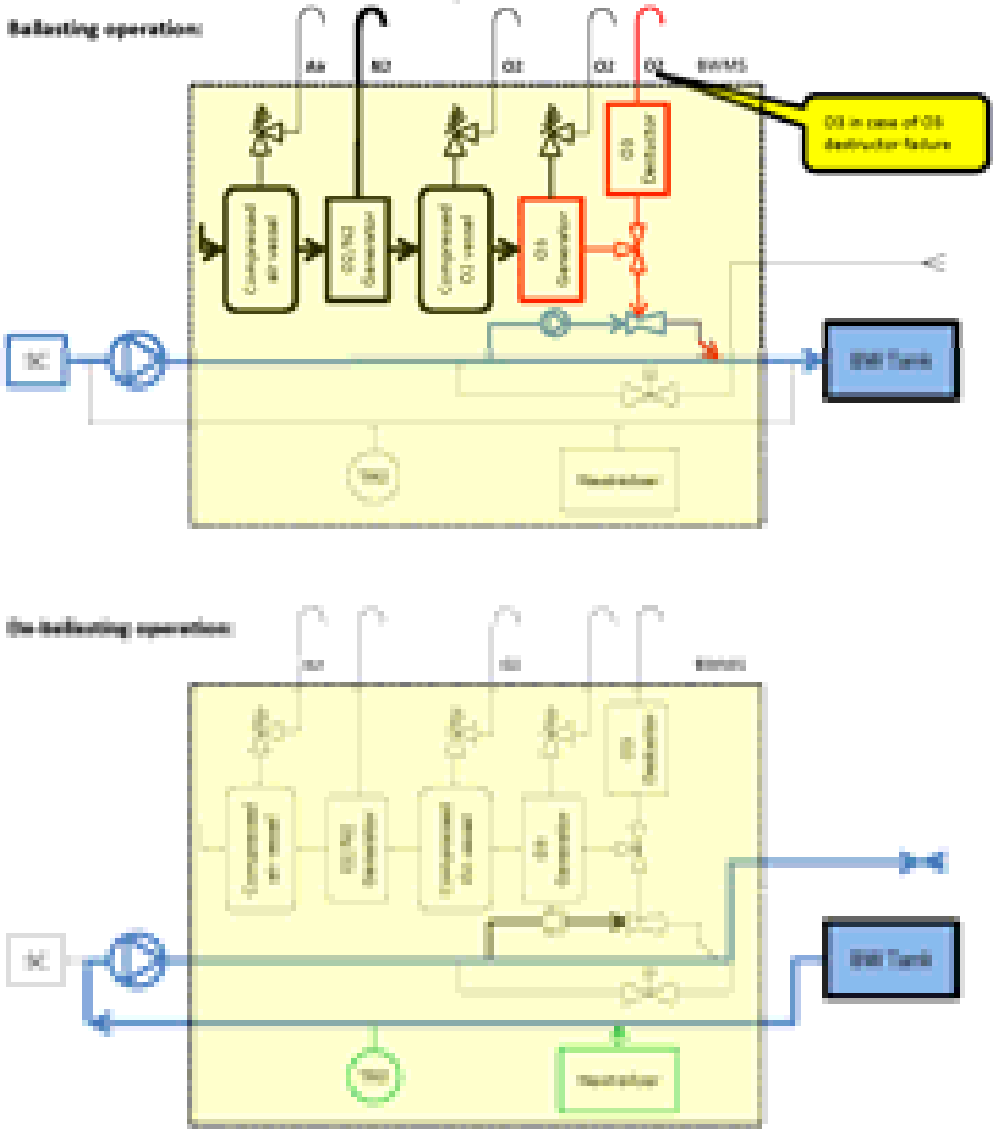
BWMS Technology Group no. 6
In-Line Chemical injection





BWMS Technology Group no. 7a

In-Line Side-stream Ozone injection without gas/liquid separation tank and without discharge water treatment tank



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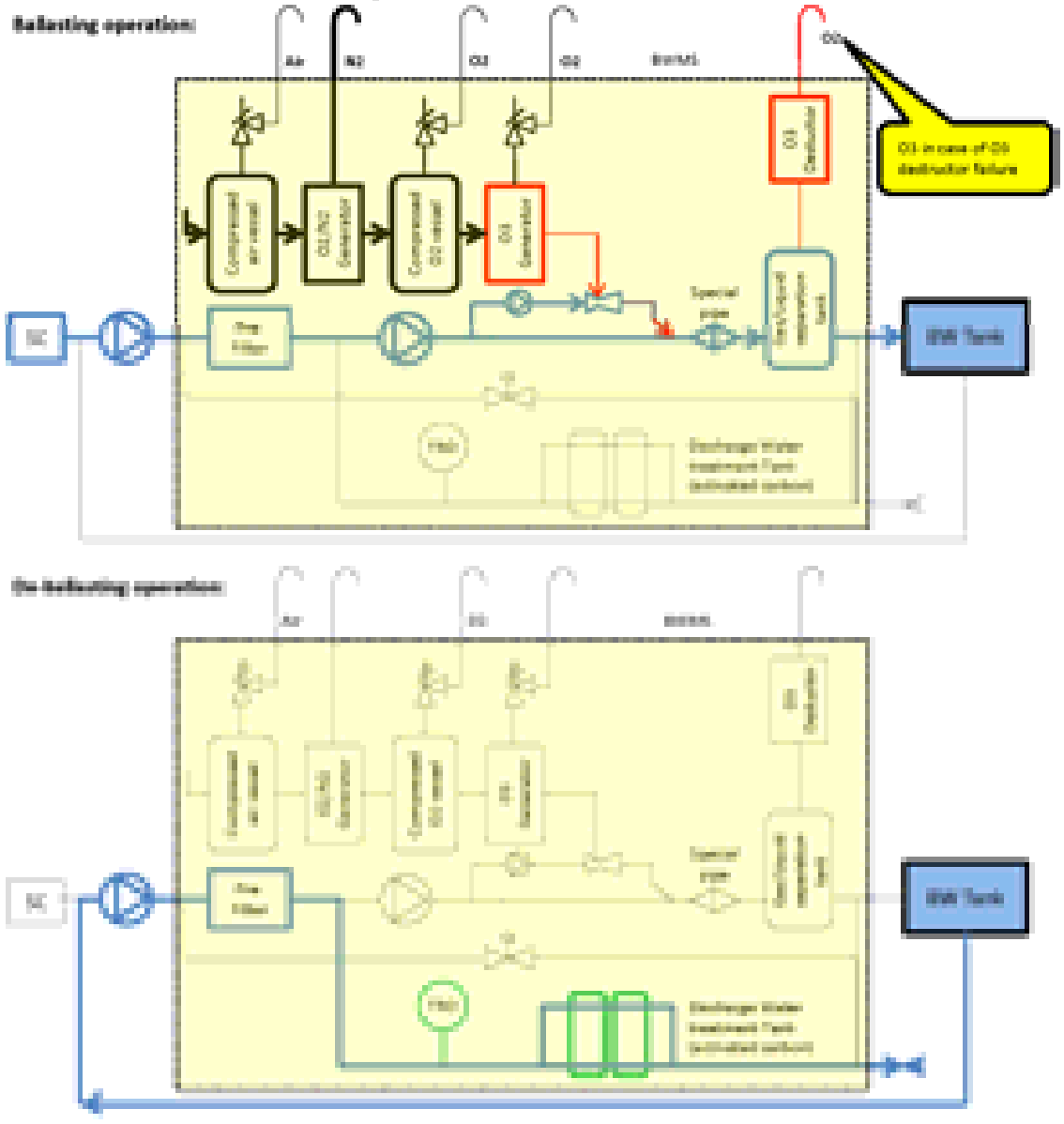
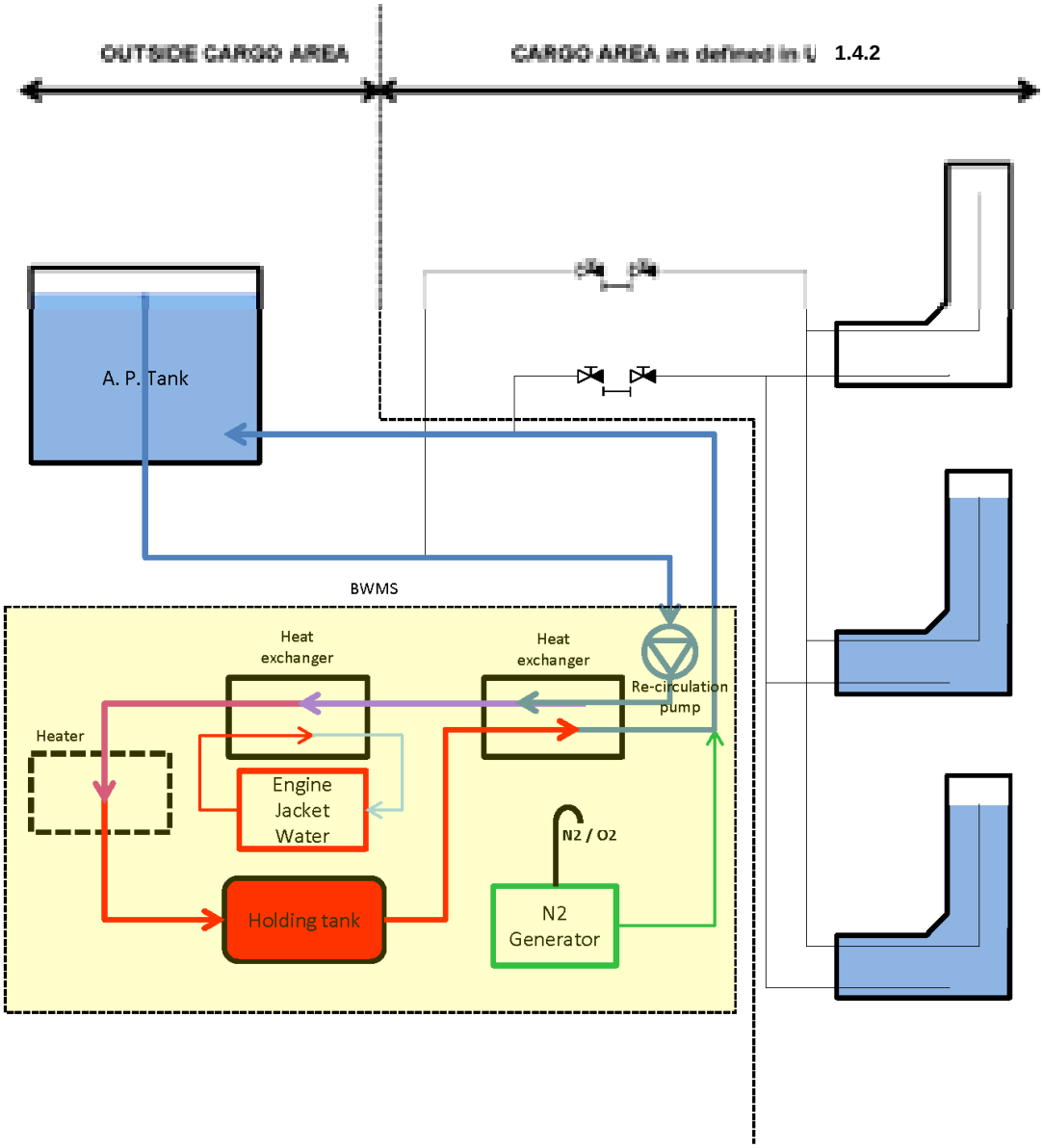
In-Line Side-stream Ozone injection with gas/liquid separation tank and with discharge water treatment tank



BWMS Technology Group no. 8
In-tank Pasteurization + de-oxygenation with N2 Generator



End of Chapter

Chapter 38

LNG Carriers with Regasification Systems

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Section 1

General

1.1 Application

1.1.1 The requirements of this Chapter apply to Liquefied Natural Gas (LNG) carriers fitted with regasification systems and meeting the applicable requirements of Pt.5, Ch.4 " Liquefied Gas Carriers" .

1.1.2 Attention is drawn to any relevant statutory requirements of the country in which the vessel is to be registered and/or requirements of local authority where the vessel is operating.

1.1.3 The requirements of this Chapter are applicable to both new construction and conversions.

1.2 Notations

1.2.1 Vessels that comply with the requirements of this chapter will be assigned following additional class notation:

RV (LNG)

1.3 Definitions

1.3.1 Definitions in Pt.5 Ch.4 "Liquefied Gas Carriers" are applicable.

1.3.2 **Blowdown** is defined as the depressurization of a system, part of a system and its equipment to allow the safe disposal of both vapor and liquid discharged from blowdown valves. Depressurization is used to mitigate the consequences of a pipeline or vessel leak by reducing the leakage rate and/or inventory within the pipe or vessel prior to a potential failure.

1.3.3 **Regasification System** is the complete gasification process plant from LNG cargo storage tanks to the gas export (send-out) shore connection including regasification unit, suction drum, associated pumping, piping and ancillary systems.

1.3.4 **Regasification Unit** is refers to vaporizers, heaters, LNG high pressure pump and associated piping intended for the gasification of LNG from the cargo storage tanks.

1.3.5 **Send-Out** is the discharge of the high pressure gas after the vaporization and heating process. Send-out may include additional processes, such as trim heating, calorific correction, odourisation, or dew point correction/ dehumidification.

1.4 Plans

1.4.1 In addition to those required by Pt.5, Ch.4, the following regasification plant plans, calculations and information, as applicable, are to be submitted :

1.4.1.1 For Approval:

a) Hazardous area arrangement

b) Booklet of installations in classified hazardous areas

c) Risk assessment

d) Scantlings and details of foundation drawing in way of regasification and power generation modules, where provided.

e) Strength analysis document in way of regasification and power generation modules.

f) Cargo containment and Pump Tower strength analysis (including sloshing analysis)

g) Escape routes

h) Project specification and overall process concept description

i) Machinery arrangement of regasification plant and power generation system

j) Ventilation system for compartment of cargo machinery space and turret, if applicable

k) Cryogenic spill containment

l) Failure Modes and Effects Analysis (FMEA)

m) Emergency shutdown (ESD) system, ESD Cause and effect diagram

n) Fire and gas detection and alarm systems

o) Fire extinguishing systems

p) Piping and Instrumentation Diagram (P&ID)

q) Piping material specification

r) Piping arrangement

s) Pipe stress analysis (high pressure and cryogenic)

t) Details of all cargo and vapor handling equipment including high pressure system

u) Details and installation of the safety valves/blow down valves and capacity calculations

v) Inert-gas system

w) Pressure relief and blowdown calculations

x) Mooring details and specification

y) Regasification plant testing procedures

1.4.1.2 For Information:

a) General arrangement of the regasification plant

b) Operational manual

c) Process Flow Diagram (PFD)

1.5 Risk Assessment

1.5.1 A Risk assessment report as per the requirements of Pt.5 Ch.4, Sec 1, 1.1.10 & 1.1.11 is to be submitted for review. Risk assessment is to be carried out to identify hazards and accident scenarios that may affect the vessel and is to consider the existing or potential risk control options.

1.5.2 The scope of the risk assessment may include but not be limited to:

a) normal operation, start-up, normal shutdown, non-use, and emergency shutdown of the system, including:

   i. pressurized gas discharge to shore;

   ii. high pressure gas venting;

   iii. storage and handling of flammable refrigerants (as applicable);

iv. continuous presence of liquid and vapour cargo outside the cargo containment system;

v. ship-to-ship transfer of liquid cargo.

b) physical equipment layout, machinery and equipment layout, arrangements and location, including extension of hazardous areas and evacuation arrangements;

c) fire and explosion, process upset conditions, over-pressure and under-pressure, mechanical and electrical failures and human error.

1.5.3 The risk assessment is to be undertaken by suitably qualified and experienced individuals in accordance with a recognized Standard (e.g. ISO/IEC 31010-2009: *Risk management – Risk assessment techniques*).

1.5.4 The risk assessment is to address at least the following aspects:

a) fire and explosions in areas or spaces containing piping, machinery, equipment and components;

b) fire and explosions adjacent to areas or spaces containing piping, machinery, equipment and components;

c) cryogenic leakage from piping, equipment and components associated with the regasification system;

d) corrosion/erosion in gas high pressure piping, components and tanks associated with the regasification system;

e) mechanical failure in regasification machinery, equipment or components;

f) control/electrical failure in regasification machinery, equipment or components;

g) manufacturing defects in equipment and machinery;

h) human error in operation, maintenance, inspection and testing of regasification machinery, equipment and components;

i) location of piping, machinery, equipment and components;

j) cargo tanks over-pressure and under-pressure situations associated with the regasification and ship-to-ship transfer operations;

k) analysis of risk associated with the LNG transfer in accordance with the following

i. ISO 28460:2010 " *Petroleum and natural gas industries – Installation and equipment for liquefied natural gas – Ship-to-shore interface and port operations"* and

ii. EN 1474 " *Installation and equipment for liquefied natural gas. Design and testing of marine transfer systems. Offshore transfer systems*" as applicable, and

iii. SIGTTO Publication *"Ship to Ship Transfer Guide for Petroleum, Chemicals and Liquefied Gases*" as applicable

l) process upsets associated with the land-based receiving systems;

m) risk of fire propagation from turret compartment and susceptibility of turret structure to explosions or uncontrolled gas release;

n) jet fires and blast.

1.6 Operations Manual

1.6.1 An operations manual is to be provided addressing the following:

a) Operational limitations for the transfer operation, if applicable.

b) Preparations prior to start up, i.e. readiness of safety equipment, connection of offloading system, means and procedures for communication.

c) Start-up procedures, i.e. gassing up and cool down of the system.

d) Filling and venting of suction drum.

e) Precautions to avoid excessive temperature differentials.

f) Normal shut down procedures and venting of the system.

g) Emergency procedures i.e. initiation of PSD, ESD and blow-down.

h) Inerting and gas freeing.

i) Isolation of the regasification units for inspection and maintenance

j) Operating restrictions, mitigating measures associated with sloshing loads (if any).

Section 2

Vessel Arrangements and Structure

2.1 Location and Arrangement of Regasification System

2.1.1 The layout of regasification system is to be designed giving due consideration to

a) safety of personnel,

b) prevention potential pollution,

c) environment impact and

d) protection of the vessel.

2.1.2 The installation is to be so arranged as to minimize the risk of occurrence of accident, risk of escalation, to protect manned spaces from consequences of accidents affecting other areas, ensure integrity of LNG storage and allow a safe evacuation when necessary. The locations of the system arrangements, including vaporizers, high pressure pumps, suction drums, heaters, liquid pumps and ancillary piping systems, are to be defined and evaluated in the risk assessment. The regasification unit is to be protected from external forces and impact such as green sea water forces and dropped objects as relevant according to outcome of risk analysis required in 1.5

2.1.3 Adequate provisions and arrangements are to be made to facilitate safe access to control, cleaning and inspection of various places on regular basis during normal operation of regasification plant. Arrangement is to be made such that there is safe access at all times to valves that may need to be handled in normal operation.

2.1.4 Arrangement is to be made for adequate lighting and ventilation when deemed necessary.

2.1.5 The regasification unit location is to be considered part of the cargo area as defined in Pt.5, Ch.4 “Liquefied Gas Carriers”. The regasification unit and all its associated equipment are to be located as far as is reasonably possible from the accommodation spaces

2.1.6 The regasification system machinery may be located on the open deck or in cargo pump and cargo compressor rooms. Arrangements of such spaces are to be in accordance with the requirements of Pt.5, Ch.4, Sec.3 “Ship Arrangements”.

2.2 Structure

2.2.1 The steel structure of regasification unit is to be designed and strengthened to support component weight and relevant forces including weather and sea dynamic motions.

2.2.2 The local strength of the structure is to be assessed according to methods, recognized codes or standards to the satisfaction of the IRS.

2.2.3 The suitability of the vessels hull structure, containment system, and other in tank components such as pump tower are to be reviewed for sloshing loads based on the site-specific condition where the vessel will operate in regasification mode. Any special considerations to environmental limitations, operating restrictions, and mitigating measures as associated with sloshing are to be included in the operations manual.

2.3 Protection from Cryogenic Releases

2.3.1 Hull structures under suction drums, pumps and vaporizers are to be protected against cryogenic spill. If drip trays are fitted, they are to be drained overboard.

2.3.2 Provisions are to be incorporated in the design to minimize the number of flanged connections. In order to protect personnel from cryogenic burns and prevent ship structure or other carbon steel structures on deck from being exposed to brittle fracture due to LNG pressure jet, consideration is to be given to the fitting of spray shield arrangements to any flanged connection of piping containing LNG at a pressure above 10 [bar].

2.4 Turret Compartment Arrangement

2.4.1 Turret compartments are to comply with the requirements of Pt.5, Ch.4, 3.11.4 and 3.3.1

2.4.2 The risk of fire propagation from turret compartments to adjacent spaces is to be evaluated by a risk analysis. Further preventive measures, such as the arrangement of a cofferdam around the turret compartment, are to be provided if needed.

2.4.3 Turret compartments are to be designed to retain their structural integrity in case of explosion or uncontrolled high-pressure gas release (overpressure and/or brittle fracture), the characteristics of which are to be substantiated on the basis of a risk analysis with due consideration of the capabilities of the pressure relieving devices. Blast relief panels may be provided.

2.5 Ventilation inlets

2.5.1 Ventilation inlets to accommodation are to be arranged for remote closing at navigation bridge and in the control room.

2.6 Escape

2.6.1 Passages for safe escape from areas of the regasification units, the export manifold and loading manifold are to be arranged. Shielding from fire and heat radiation is to be provided for as necessary.

2.6.2 Two escape routes from normally manned areas are to be arranged at least one of which is to be available following an accidental event.

Section 3

System Design

3.1 General

3.1.1 Materials, components and equipment to be used in the construction of regasification systems are to be suitable for the intended service conditions. Materials are to comply with the requirements of Pt.5, Ch.4, Sec 6 "*Materials of Construction and Quality Control*".

3.1.2 The design, arrangement and selection of equipment is to be such as to minimize the risk of fire and explosion from flammable products.

3.1.3 The regasification plant is to be designed so that a single accidental event is not to result in a critical situation.

3.2 Regasification System

3.2.1 General

3.2.1.1 Regasification system may include the following main components but not be limited to:

a) low pressure liquefied natural gas transfer pump

b) boil off gas compressor

c) boil off recondenser tank or suction drum

d) high pressure liquefied natural gas pump or send out pump

e) liquefied natural gas heat exchanger or vaporizer

f) gas metering and analyzer skid

g) associated liquefied natural gas and gas pipes and fittings including main send out gas manifold

h) gas off-loading system

i) associated automation systems

j) utilities, heating medium systems (water glycol, refrigerant, steam or sea water)

k) steel structure

3.2.1.2 The manufacture, installation and testing of vaporizers, including the intermediate heat transfer vessels and pumping systems, is to be undertaken in accordance with the requirements of the Pt.5, Ch.4.

3.2.2 Vaporizers

3.2.2.1 For Shell and Tube type heat exchangers and Open rack type heat exchangers, sea-water may be used as a primary heat source for vaporization. An intermediate heat transfer fluid may be proposed to reduce the chance of freezing and effects of corrosion.

3.2.2.2 Where sea-water is used as the source of heat to vaporize the LNG, the tubes are to be manufactured from a corrosion resistant material, taking into consideration the type and temperature of media conveyed.

3.2.2.3 Where potential risk of failure of a tube or passage could result in gas entering the sea-water side:

a) the sea-water side is to be designed to accept the full gas pressure of the gas side;

b) the sea-water side is to be protected with relief valves in readily visible positions; the discharge from these relief valves is to be taken to a suitable high-pressure venting arrangement and the number and position of valves is to be adequate to relieve the flow occurring due to failure of a single tube.

3.2.2.4 Adequate protection against corrosion is to be provided where aluminium alloy vertical tubes and horizontal headers are constantly covered with sea-water.

3.2.2.5 Commissioning and testing of vaporizers is to be undertaken by the manufacturer prior to units being installed on board.

3.2.2.6 Water supply pumps are to be fitted with suitable inlet filters. Removal and cleaning of filters is to be possible whilst the regasification system remains operational. Any regasification system-related sea-water inlet is to be fitted with gratings and provision made to allow cleaning by low pressure steam or compressed air.

3.2.2.7 A water treatment system is to be incorporated for use with submerged combustion vaporizers to eliminate degradation of the tubes.

3.2.3 High Pressure Pumps

3.2.3.1 All LNG high pressure pumps supplying vaporizers, which are capable of developing a pressure exceeding the design pressure of the system into which they are pumping, are to be provided with relief valves in closed circuit.

3.2.4 Heating System

3.2.4.1 Heating may be arranged as direct heating or indirect heating.

3.2.4.2 Means are to be provided to detect leakage of high pressure LNG/NG into the heating fluid and to prevent overpressure in the heating system.

3.2.4.3 If steam is used in a heat exchanger containing either propane or other gas, the condensate is not to be passed directly back to the engine room. The condensate is to be passed through a degassing tank located in a gas-dangerous area. The vent outlet from the degassing tank is to be routed to a safe location and be fitted with a flame screen. The degassing tank is to be covered by the gas detection and alarm system.

3.2.4.4 Vaporizers are to be protected against freezing of heating medium.

3.2.4.5 Hydrocarbon heating medium storage tanks are to be considered as a cargo tank in Pt.5, Ch.4.

3.3 Piping System Requirements

3.3.1 Regasification system piping is to meet the applicable requirements of Pt.5, Ch.4, Sec.5 *"Process Pressure Vessels and Liquid, Vapour and Pressure Piping Systems"*.

3.3.2 All piping, valves and fittings are to be suitable for the operating and environmental conditions.

3.3.3 All materials used in the piping systems are to be suitable for use with the intended medium, service and ambient conditions, and are to comply with the applicable requirements of Pt.5, Ch.4, Sec.6 *"Materials of Construction and Quality Control"*.

3.3.4 The spool piece, reducers, valves and other fittings to which the cargo system or the send-out system is directly connected are to be of approved material. They are to be of robust construction, adequately supported and suitable for the stated design conditions and manifold forces. For LNG transfer, attention is drawn to SIGTTO '*Manifold Recommendations for Liquefied Gas Carriers*'.

3.3.5 Means of purging, inerting and gas-freeing the pipe lines used for the regasification system are to be provided.

3.3.6 Means for mechanical separation are to be provided between the regasification piping system and the ship's inert gas and nitrogen systems.

3.3.7 All main isolating valves serving the regasification systems and equipment are to be located in a readily accessible location.

3.3.8 The fabrication and installation of the piping associated with the regasification system and its ancillary systems is to be in accordance with the relevant Sections of Pt.4 and Pt.5, Ch.4 to the satisfaction of the Surveyor.

3.3.9 Where applicable, all LNG pipework servicing the regasification system are to be suitably insulated and are to be covered with an efficient vapour barrier.

3.3.10 All low and high pressure LNG supply pipework serving the regasification systems is to be subject to a stress analysis, taking into account all the stresses due to weight of pipes, including acceleration loads if significant, internal pressure, thermal contraction and loads induced by hog and sag of the ship for each branch of the piping system.

3.4 Pressure Relief and Venting

3.4.1 If a suction drum is arranged to supply the high pressure booster pumps it is to be provided with a relief valve sized for relevant cases i.e. fire exposure with relief valve capacity according to Pt.5 Ch.4 Sec.8 for cargo tanks.

3.4.2 Pressure relief valves are to be sized according to requirements in Pt.5 Ch.4.

3.4.3 Pressure relief valves are to be fitted in sections of the piping system where liquid can be trapped, complying with the requirements in Pt.5 Ch.4 Sec.5, 5.5.7.

3.4.4 In case of relief to cargo tanks, the effect of routing high pressure LNG/NG to the cargo tank is to be documented.

3.4.5 Vent system is to be designed such that no liquid is vented. Part of the regasification plant will contain high pressure LNG and part will contain high pressure gas. Relief arrangements for both phases need to be considered.

3.4.6 A knock out drum is to be fitted between the blow down valves and the vent mast, unless it is documented that liquid release cannot occur. The knock out drum is to be fitted with high level alarm and a high-high level sensor which initiates a complete process shutdown. At the high-high activation level the knock out drum is to still have sufficient capacity to allow full process depressurisation.

3.5 Liquefied Natural Gas Loading

3.5.1 LNG may be transferred from LNG carrier using loading arms fitted on the FSRU or using cryogenic hoses.

3.5.2 Operational limitations for the transfer operation are to be set with respect to parameters such as:

a) sea conditions for safe approach, berthing and departure of the LNG carrier

b) operational envelopes (relative motion, accelerations)

c) loads in the mooring lines and fenders between the terminal and the carrier.

3.5.3 Piping between the LNG ship cargo system and the regasification system is to be equipped with a manually operated stop valve and a remotely controlled emergency shut-down valve. These valves are to be located as close to the tank as practicable. When the regasification unit is located in the forward section of the ship, such isolation is to be as near as possible to the boundary of the forward most cargo tank bulkhead and within the cargo area.

3.5.4 For ship to ship transfer, each transfer line is to be fitted with emergency release system (e.g using dry break quick-release connectors) to minimize the possible liquid spill and with insulating flange.

3.5.5 The regasification vessel is to have adequate capability for managing the boil-off gas generated by heat ingress through headers, manifolds and loading arms/hoses during ship-to-ship transfer operations.

3.5.6 The regasification system is to include provision to pre-cool the product transfer piping system prior to ship-to-ship transfer operations commencing.

3.6 Natural Gas Offloading (Send-out system)

3.6.1 Minimum requirements for offloading equipment are:

a) Offloading connections are to be fitted with ESD valves.

b) Offloading hoses (if arranged) are to be type approved.

3.6.2 Dry break quick-release connectors are to be provided for use in an emergency in send-out gas piping between a regasification vessel and receiving terminal.

3.6.3 A manually operated shut-off terminal valve is to be provided at the send-out manifold, in addition to any other automatic shut-off valves required by Pt.5, Ch.4.

Section 4

Safety Systems

4.1 Gas Detection System

4.1.1 In addition to the gas detection system fitted to allow compliance with Pt.5, Ch.4, a permanently installed system of gas detection with audible and visual alarms is to be fitted in the following locations:

a) all enclosed spaces containing gas piping, liquid piping or regasification equipment;
b) other enclosed or semi-enclosed spaces where gas vapours may accumulate;
c) air-locks;
d) secondary fluid expansion tanks;
e) the condensate degassing tank.
f) Export manifold
g) Metering unit
h) Ventilation inlet to accommodation and machinery spaces
i) Turret compartment (if applicable)

4.1.2 The gas detection system serving the regasification system may be either independent or combined with the gas detection system installed to allow compliance with Pt.5, Ch.4.

4.1.3 Gas detection equipment is to be designed, installed and tested in accordance with IEC 60079-29-1, and is to be suitable for the gases to be detected.

4.1.4 The gas detection is to be of the continuous monitoring type, capable of immediate response.

4.1.5 The number and the positions of detection heads or sampling heads is to be determined with due regard to the size and layout of the compartment, the air flow from compartment purging or ventilation and stagnant areas, and the manufacturer’s recommendations.

4.1.6 The gas detection system serving the regasification system is otherwise to comply with the construction and installation requirements of Pt.5, Ch.4.

4.2 Emergency Shutdown System

4.2.1 The ESD system is to be activated by the manual and automatic inputs as per Pt.5 Ch.4. Any additional inputs are only to be included in the ESD system if it can be shown that their inclusion does not reduce the integrity and reliability of the system overall.

4.2.2 The ESD system is to return the regasification system to a safe static condition, allowing remedial action to be taken. Due regard is to be given in the design of the ESD system to avoid the generation of surge pressures within the pipework.

4.2.3 The equipment to be shut down on ESD activation is to include manifold valves during loading or discharge, and pumps and compressors associated with transferring cargo in either liquid or gaseous state.

4.2.4 The number and location of additional shutdown positions is to be determined by the type, number, location and position of the regasification systems and equipment.

4.2.5 Other ESD system controls identified by Risk Assessment, as applicable, are to be provided.

4.2.6 A cause and effect diagram is to reflect:

a) Cause inputs that initiate the automatic shutdown effect output

b) Parameters (set points) that trigger the cause

c) Action item of the effects (e.g. valve closure, machinery shutdown)

4.3 Process Shutdown System

4.3.1 A process shutdown (PSD) system for the regasification system is to be arranged in accordance with the requirements indicated in Section 5.

4.3.2 A process shut-down system that covers the total regasification system is to be arranged. PSD is to result in the following actions:

a) stop of LNG supply to the regasification plant;

b) stop utility systems containing hydrocarbon serving regasification plant;

c) closing of suction drum outlet;

d) stop of high pressure pumps;

e) closing of isolation valves.

4.3.3 Individual regasification units may be shut down in situations where a total PSD is not required to maintain a satisfactory safety level and is related to that unit only and as given in Sec 5. Activation of the unit PSD is to result in the following actions:

a) stop of the relevant high pressure pumps

b) closing of isolation valves to the relevant unit.

4.3.4 PSD system is to be activated by manual push buttons also. Manual PSD points are to be arranged at the regasification system control station, export manifold and at locations as determined by the type, number, location and position of the regasification systems and equipment. The process shutdown points are to be clearly visible.

4.3.5 Process shutdown valves in liquid piping are to close fully under all service conditions within 30 seconds of actuation. Information about the closing time of the valves and their operating characteristics is to be available on board and the closing time is to be verifiable and reproducible. The closure time of 30 seconds for the shutdown valve is to be measured from the time of manual or automatic initiation to final closure. This is called the total shutdown time and is made up of a signal response time and a valve closure time. The valve closure time is to be such as to avoid surge pressure in pipelines. Valves are to close smoothly.

4.4 Depressurisation and Blow down System

4.4.1 A depressurisation and blowdown system is to be provided for depressurising the high pressure liquid and gas pumps, vessels and pipework.

4.4.2 Where a liquid depressurisation system is provided, adequate provision is to be made in the design and installation for the effects of back pressure in the system and vapour flash-off when the pressures of liquids in the blowdown system are reduced.

4.4.3 Manual and automatic activation of the depressurisation system is to be provided.

4.4.4 Manual activation is to be possible from the regasification control station, at the send-out manifold, and from other locations as determined by the type, number, location and position of the regasification systems and equipment.

4.4.5 Automatic activation of the blow down system is to be part of the emergency shutdown arrangements.

4.5 Fire Safety

4.5.1 Fire Mains

4.5.1.1 The fire water main is to be such that at least two jets of water can reach any part of the regasification plant and with a pressure of at least 0.5 [MPa] gauge. The water supply to the fire main serving the regasification units is to be a ring main supplied by the main fire pumps or a single main supplied by fire pumps positioned fore and aft of the cargo area, one of which is to be independently driven.

4.5.2 Water Spray System

4.5.2.1 A fixed water-spray system with an application rate of 10 [l/m^2/min] for the horizontally projected area and 4 [l/m^2/min] for the vertical surface of the regasification plant is to be arranged to protect:

a) the regasification units

b) suction drum

c) export manifold

d) internal surfaces of the turret compartment

e) storage tanks for propane or other flammable fluids, if fitted

f) equipment such as metering station, oderant tank and HIPPS, if fitted.

For structures having no clearly defined horizontal or vertical surfaces, the capacity of the water-spray system is not to be less than the projected horizontal surface multiplied by 10 [l/m^2/min].

4.5.2.2 The arrangements and capacities of the water spray system are to be such that with one pump out of operation the remaining pump(s) are to have capacity to supply the spray system.

4.5.2.3 If the fire water pumps and spray pumps are combined the capacity is to be sufficient to supply water spray to the largest fire section as well as two fire hoses.

4.5.2.4 Release of the water spray systems is to be possible both locally and remotely at the control station where the operating status of the systems is monitored.

4.5.3 Dry Chemical Powder Fire Extinguishing System

4.5.3.1 The fixed dry chemical powder fire extinguishing system is to cover the regasification plant.

a) The system is to be capable of delivering powder from at least two hand hose lines, or a combination of monitor/ hand hose lines to any part of the exposed regasification plant.

b) If loading arms are installed, the dry chemical powder is to cover the presentation flange of the loading arm.

The system is to comply with the requirements in Pt.5, Ch.4, Sec.11.

4.5.4 Fire detection

4.5.4.1 Flame detectors are to be fitted to cover the regasification units, suction drum and export manifold.

Section 5

Control, Alarm and Monitoring

5.1 General

5.1.1 The regasification system is to be provided with control, monitoring, alert and safety systems that will maintain the system throughout all normal and reasonably foreseeable abnormal conditions

5.1.2 The system is to be provided with automatic and/or remote controls to ensure the system operates within its design parameters.

5.1.3 A system for monitoring and indicating alerts is to be provided.

5.1.4 The system is to be provided with safeguards which will operate to prevent a hazard occurring or to reduce an existing hazard to persons, machinery or the ship or the environment.

5.1.5 Locations at which the regasification system is controlled are to be provided with a means of communication with the gas-receiving terminal.

5.1.6 The system is to be provided with the alarms and shutdowns as identified by the system manufacturer. In the absence of such guidance, the alarms and shutdowns indicated in these Rules may be used.

5.1.7 A control station for the regasification system and ship-to-ship operations is to be arranged within a non-hazardous area. Emergency procedures, concerning regasification and ship-to-ship transfer operations are to be capable of being performed from this station.

5.1.8 At least two means of communication are to be provided between the control station and the receiving terminal, one of these systems is to be independent of the main electrical supply.

5.1.9 Internal communication cables are to comply with the applicable requirements of Part 4.

5.1.10 The cable installation is to provide adequate protection against mechanical damage and electromagnetic interference.

5.1.11 Components are to be located with appropriate segregation such that the risk of mechanical damage or electromagnetic interference resulting in the loss of both active and stand-by components is minimized. Duplicated communication links and equipment are to be routed to give as much physical separation as is practicable.

5.2 Equipment and Systems

5.2.1 Suitable interlocks are to be provided to prevent start-up of the regasification system under conditions which could hazard the system or its equipment and components.

5.2.2 The manufacturer is to identify the required alarms, shutdowns and safeguards for regasification system. The minimum required shutdowns are indicated in Table 5.2.2.

Table 5.2.2 : Monitoring and Control of Regasification System		
Item/ Activity	**Monitoring**	**Control**
	Alarm Activation	**Total Process Shutdown(PSD) Activation**
Suction Drums		
Pressure	L, H	LL, HH
Level	L	LL
Vaporizers		
Inlet temperature of heating medium	L	LL (Note 1)
Inlet pressure of heating medium	L	LL (Note 1)
Outlet temperature of heating medium	L	LL (Note 1)
Outlet temperature of Natural Gas	L	LL (Note 1)
Outlet pressure of Natural Gas	L, H	LL, HH (Note 1)
Natural Gas Offloading to Shore		
Pressure	L, H	LL, HH
Temperature	L	LL
Heat Exchangers		
Steam condensate return temperature	L, H (Note 2)	LL, HH (Note 2)
Steam Supply Pressure	L (Note 2)	LL (Note 2)
Knock Out Drums		
Level	H	HH
Pressure in heating system using hydrocarbon heating medium (e.g propane)	H	
Gas detection in Regasification Area	X	
Fire Detection in Regasification Area	X	X (Note 3)
Activation of Emergency Shutdown System (Refer to Pt.5, Ch 4, Sec 18)	X	X
Activation of PSD including by means of manual push buttons	X	X
Activation of Blow down including by means of push buttons	X	X (Note 3)
Loss of signal to control station	X	X

X - Activated
L – Activated for Low Value
LL – Activated for Low Low Value
H – Activated for High Value
HH – Activated for High High Value

Note 1: Only unit PSD may be relevant if designed as per 4.3.3
Note 2 : Required only if steam is not direct heating medium
Note 3 : Also ESD activation

Section 6

Survey, Testing and Trials

6.1 Surveys during Construction

6.1.1 A survey is to be conducted before the regasification system is put into service, which will include a complete examination of its structure, equipment, fittings, arrangements and materials of the regasification system, to ensure compliance with the provisions of the Rules.

6.2 Testing

6.2.1 The following additional tests for the regasification system are to be carried out during construction or conversion of the vessel:

a) Pressure and leak test of LNG and gas piping.
b) Leak test of suction drum.
c) Checks of safety valves setting.
d) Function tests of

   i. fire safety systems,
   ii. emergency shutdown system,
   iii. process shutdown system,
   iv. gas detection system,
   v. depressurizing and blowdown system.
   vi. control, monitoring, alert and safety systems
   vii. regasification heating pumps.

e) Verification of the requirements resulting from the Risk Analysis.

6.2.2 The relevant requirements of Pt.5, Ch.4 are to be complied with, for testing and non-destructive examination of the regasification unit, LNG and gas piping systems.

6.2.3 All piping systems are to be subjected to a hydrostatic test to at least 1.5 times the design pressure. When piping systems or parts of systems are completely manufactured and equipped with all fittings, the hydrostatic test may be conducted prior to installation aboard ship. Joints welded on board are to be hydrostatically tested to at least 1.5 times the design pressure. Proposals for alternative testing fluids or means are to be submitted for special consideration by the Surveyor in cases where the piping cannot be dried prior putting into service or where the presence of water cannot be allowed.

6.2.4 After assembly on board, all cargo and process piping are to be subjected to a leak test using air, halides or other suitable medium to a pressure dependent on the leak detection method applied.

6.2.5 All piping systems including valves, fittings and associated equipment for handling cargo or vapors are to be tested under normal operating conditions. This testing is to be carried out, not later than at the first regasification operation.

6.3 Regasification Trials

6.3.1 The regasification trials program, including technical and operation information relevant to such testing, is to be prepared and submitted for approval.

6.3.2 Preliminary regasification trials are to consist of a running test of the regasification system with LNG low flow and are to be carried out after gas trials and before delivery.

6.3.3 The full capacity test of the regasification plant is to be carried out at an operational site.

6.3.4 The test and measurements are to be carried out according to the relevant regulations, manufacturer’s standards and industry's best practices.

6.3.5 A trial report is to be submitted on completion of the regasification trials. A copy of the trial report is to be retained on board.

End of Chapter

Chapter 39

Passenger and Crew Comfort

Contents

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1 *General*
2 *Noise and Vibration*
3 *Indoor Climate and Lighting*
4 *Measurements and Reports*

Section 1

General

1.1 General

1.1.1 This Chapter provides requirements for factors which affect human performance as well as habitability on-board, such as noise, vibration, indoor climate and lighting, on all types of ships.

The effect of the above mentioned aspects on structure, machinery or any other systems are outside the scope of this Chapter.

1.1.2 Compliance with the requirements of this Chapter is to be verified through measurements carried out in accordance with Section 4. A Ship would be assigned the following additional Class Notations, as applicable, on completion of satisfactory review of the results of such measurements:

1.1.2.1 **CMF(Px)**: This notation would be assigned to ships which have acceptable levels of noise and vibration in the passenger spaces, in accordance with the requirements of Section 2, 2.2; where '**x**', is the comfort rating (numerical values of 1, 2, 3) assigned to passenger spaces.

Ships which additionally comply with the requirements for indoor climate and lighting in Section 3, for passenger spaces will be assigned additional qualifiers '**c**' and/ or '**l**', respectively (e.g. **CMF(Px-c)**, **CMF(Px-l)**, **CMF**(**Px-c-l**)).

1.1.2.2 **CMF(Cy)**: This notation would be assigned to ships which have acceptable levels of noise and vibration in the crew spaces, in accordance with the requirements of Section 2, 2.3; where, '**y**' is the comfort rating (numerical values of 1, 2, 3) assigned to crew spaces.

Ships which additionally comply with the requirements for indoor climate and lighting in Section 3, for crew spaces will be assigned additional qualifiers '**c**' and/ or '**l**', respectively (e.g. **CMF(Cy-c)**, **CMF(Cy-l)**, **CMF(Cy-c-l**)).

1.1.2.3 **CMF(Px,Cy)**: This notation would be assigned to ships which have acceptable levels of noise and vibration in both passenger and crew spaces, in accordance with the requirements of Section 2; where, '**x**' and '**y**' denote the comfort rating (numerical values of 1, 2, 3) assigned to passenger spaces and crew spaces respectively.

Ships which additionally comply with the requirements for indoor climate and lighting in Section 3, for passenger and crew spaces will be assigned additional qualifiers '**c**' and/ or '**l**', respectively (e.g. **CMF(Px, Cy)**, **CMF(Px-c-l, Cy-c-l)**.

1.2 Definitions

1.2.1 *Crew spaces*: All spaces intended for use by the crew. Such spaces include crew accommodation, crew recreation areas, mess

rooms, offices and work areas (such as wheelhouse, engine control room, workshops etc.)

1.2.2 *Noise level*: A-weighted equivalent continuous sound pressure level, measured in accordance with recognized national/ international standards (e.g. IEC 61672-1).

1.2.3 *Passenger spaces*: All areas intended for passenger use. Such spaces include passenger accommodation spaces, as well as public spaces such as restaurants, hospitals, auditoriums, gymnasiums, recreation areas etc.

1.2.4 *Relative humidity range*: Acceptable range of relative humidity in indoor atmosphere.

1.2.5 *Thermal comfort*: A temperature range considered comfortable for most persons indoors.

1.2.6 *Vibration level*: The overall frequency weighted r.m.s value of vibration over the frequency range 1 to 80Hz measured in accordance with ISO 21984(2018).

1.3 Documentation

1.3.1 The following plans/ documents are to be submitted to IRS for approval for assignment of Comfort notations:
a) Plan(s) specifying the following:

- measurement locations for noise, vibration, indoor climate and lighting measurements;
- operating conditions for the vessel during the measurements;
- expected loading conditions during measurement;
- normal range of loading conditions;
- instrumentation used for measurement; and
- any other additional details which are required by specific sections of this Chapter or which IRS may deem appropriate for a specific type of vessel.

b) A report which includes the following:

- trial conditions
- results of noise, vibration, indoor climate and lighting measurements, as applicable;
- Any other additional details which are required by specific sections of this Chapter, or any other details which IRS may request for a specific type of vessel.

The report is to include a description of minor deviations, if any, from the required measurement procedures, or the approved measurement plan.

Section 2

Noise and Vibration

2.1 General

2.1.1 It is recommended that owners/ builders undertake predictions/ calculations of noise and vibration characteristics at the design stage of the ship, in order to avoid major modifications at a later stage for complying with the requirements.

2.1.2 The maximum allowable noise and vibration levels for each type of space, in order to obtain the relevant comfort rating, are indicated in this Section.

2.1.3 Rating '1' stands for the highest level of comfort, with regard to noise and vibration, whereas rating '3' denotes the minimum level of comfort acceptable for assignment of CMF notation.

2.1.4 When any machinery or equipment is intended for continuous operation for long periods during vessel operation, the noise and vibration requirements for all spaces are to be satisfied with that machinery or equipment in operating condition.

2.2 Passenger Spaces

2.2.1 The maximum allowable noise levels corresponding to the respective comfort ratings, for passenger spaces, are specified in Table 2.2.1.

2.2.2 The airborne sound insulation properties for bulkheads and decks within passenger spaces, are to comply with weighted sound reduction index (Rw), according to ISO 717-1:2013 (as amended), Part 1, as indicated in Table 2.2.2. The airborne sound insulation properties are to be determined by laboratory tests in accordance with ISO 10140-2:2010, as amended.

2.2.3 The maximum allowable vibration levels corresponding to the respective comfort ratings for passenger spaces are given in Table 2.2.3.

Table 2.2.1 : Maximum allowable noise levels in passenger spaces (in db(A))

Space	Comfort Rating		
	1	**2**	**3**
Passenger Cabins (Standard)	49	52	55
Passenger Cabins (Superior)	45	47	50
Passenger Public Spaces	55	58	62
Medical and First Aid Centres	50	55	60
Theatres/Auditoriums	50	55	60
Open deck recreational Areas	67	72	75

Note:

1. The noise levels may be exceeded by 5 dB(A) within 3 [m] of a ventilation inlet/outlet
2. The levels for open deck recreation areas refer to ship generated noise only.
3. Not more than 20 percent of the passenger cabins, 30 percent of the public spaces and 20 percent of the crew cabins should exceed the relevant noise criteria by more than 3 dB(A).

Table 2.2.2 : Minimum airborne sound insulation Indices, Rw

Space	Comfort Rating		
	1	**2**	**3**
Passenger Cabins (Standard) to cabin	40	38	37
Passenger Cabins (Superior) to cabin	45	42	40
Passenger Cabin (Standard) to corridor	38	36	34
Passenger cabin (Superior) to corridor	42	40	37
Passenger (Standard) Cabin to Stairwell	47	45	43
Passenger (Superior) Cabin to Stairwell	50	47	45
Passenger cabin (Standard) to passenger/crew public space	52	48	48
Passenger cabin (Superior) to passenger/crew public space	55	50	50
Passenger Cabins to machinery spaces	55	53	50

Note:
1. Not more than 20 percent of the interfaces tested should have airborne sound insulation indices, Rw, more than 3 dB(A) lower than the minimum specified values.

Table 2.2.3 : Maximum allowable vibration levels for passenger space (Units: frequency weighted (1-80 Hz) velocity mm/s rms)

Space	Comfort Rating		
	1	**2**	**3**
Passenger Cabins (Standard)	1.8	2.1	2.4
Passenger Cabins (Superior)	1.5	1.8	2.1
Passenger Public spaces	2.0	2.5	3.0
Open deck recreation areas	2.5	3.0	3.5

Note:
1. Not more than 20 percent of all passenger spaces and public spaces should exceed the relevant vibration criteria by more than 0.3 [mm/s].

2.3 Crew Spaces

2.3.1 The comfort ratings, and corresponding maximum allowable noise levels for crew spaces are specified in Table 2.3.1.

2.3.2 The bulkhead and deck air-borne sound insulation indices for crew spaces, Rw, calculated in accordance with ISO 717-1:2013 are to be in compliance with the values given in Table 2.3.2.

2.3.3 The maximum allowable vibration levels corresponding to the respective comfort ratings for crew spaces are given in Table 2.3.3.

Table 2.3.1 : Maximum allowable noise levels in crew spaces (in db(A))

Space	Comfort Rating			
	1	**2**	**3 (Ships<10000 GT)**	**3 (Ships≥10000 GT)**
Crew Cabins and hospitals	50	53	60	55
Conference rooms, offices, mess rooms, indoor crew recreation rooms	55	58	65	60
Open deck recreational areas	67	72	75	75
Changing rooms, laundries, bathrooms	70	75	75	75
Galleys and pantries	75			
Machinery Spaces	110[2]			
Machinery Control Rooms	75			
Wheelhouse, radar room, chart room	65			
Radio Room	60			
Workshops and other work spaces	85			
Normally unoccupied spaces (e.g. holds, decks)	90			

Note:

1. Personnel entering spaces with nominal noise greater than 85 db(A) are required to wear hearing protectors while in those spaces.
2. The limit of 110 db(A) indicated assumes that hearing protectors giving protection meeting the requirements for hearing protectors in Chapter 7 of MSC Res. 337(91) are worn. The limits specified are to be regarded as maximum levels and not as desirable levels. It is desirable for the noise level to be lower than the maximum levels specified.

Table 2.3.2 : Minimum airborne sound insulation Indices

Space	**Comfort Rating**		
	1	**2**	**3**
Crew Cabin to cabin	40	38	35
Crew cabin to corridors, stairways, passageways	35	32	30
Crew cabin and hospitals, to public spaces/recreation rooms/mess rooms	45		
Crew Cabins to machinery spaces and engine casings	50		

Note:
1. Not more than 20 percent of the interfaces tested should have airborne sound insulation indices, Rw, more than 3 dB(A) lower than the minimum specified values.

Table 2.3.3 : Maximum allowable vibration levels for crew spaces (Units: frequency weighted (1-80 Hz) velocity mm/s rms)

Space	**Comfort Rating**		
	1	**2**	**3**
Crew Cabins	2.8	3.0	3.2
Navigation spaces	3.0	3.5	4.0
Crew dining and recreational areas	3.0	3.5	4.0
Medical and First Aid Centres	2.8	3.0	3.2
Offices	3.0	3.5	4.0
Workshops and Engine Control Room	5		

Note:
1. Not more than 20 percent of all crew spaces should exceed the relevant vibration criteria by more than 0.3 [mm/s].

Section 3

Indoor Climate and Lighting

3.1 General

3.1.1 This Section provides requirements related to thermal and visual comfort of passengers and crew on board ships. Ships which comply with the requirements in this section will be assigned additional qualifiers '**c**' and '**l**', for indoor climate and lighting, respectively. For example, **CMF (P1-c, C1-l)** indicates that the ship complies with the requirements of this Section for indoor climate in passenger spaces, and for lighting in crew spaces.

3.2 Indoor climate

3.2.1 General

3.2.1.1 The thermal control or Heating, Ventilation and Air-conditioning (HVAC) systems on a vessel are to be designed to effectively control the indoor thermal environment parameters to within acceptable limits, thus ensuring thermal comfort for passengers and crew on board.

3.2.1.2 Thermal environment parameters to be considered are air temperature, supply of fresh air (number of air changes), air velocity and relative humidity.

3.2.2 Definitions

3.2.2.1 *Air velocity*: The rate of displacement of ambient air in a specific direction in meters-per-second

3.2.2.2 *HVAC Zone*: A group of spaces, which are monitored and controlled for thermal comfort, independently of monitoring and control systems of other groups of spaces.

3.2.2.3 *Relative Humidity*: The amount of water vapour present in air expressed as a percentage of the amount needed for saturation at the same temperature.

3.2.2.4 *Vertical Gradient*: The vertical air temperature difference within an enclosed space.

3.2.3 Passenger Spaces

3.2.3.1 Air temperature

.1 HVAC systems are to be designed to provide an adjustable range of air temperatures between 18^0 C to 27^0 C.

.2 In any space, the difference in temperature between 100 [mm] above deck and 1700 [mm] above deck is to be maintained within 3^0 C

.3 In passenger cabin areas, the difference between inside bulkhead surface temperature adjacent to passenger berths and the average air temperature within the space should be less than 10^0 C.

3.2.3.2 Relative Humidity

.1 HVAC systems are to be designed to provide and maintain relative humidity within a range of 40 to 60 percent in passenger spaces.

3.2.3.3 Air velocity

.1 Air velocities are not to exceed 0.5 [m/s] at the measurement position in any space.

3.2.3.4 Air exchange rate

.1 The rate of air exchange for all passenger accommodation and public spaces is to be at least six complete changes per hour

3.2.4 Crew Spaces

3.2.4.1 Air temperature

.1 HVAC systems are to be designed to provide an adjustable range of air temperatures between 18^0 C to 27^0 C.

.2 In any space, the difference in temperature between 100 [mm] above deck and 1700 [mm] above deck is to be maintained within 3^0 C.

.3 In crew cabins, the difference between inside bulkhead surface temperature adjacent to crew berths and the average air temperature within the space is to be less than 10^0 C.

3.2.4.2 Relative Humidity

.1 HVAC systems are to be designed to provide and maintain relative humidity within a range of 30 percent to 70 percent in crew spaces.

3.2.4.3 Air velocity

.1 Air velocities are not to exceed 0.5 [m/s] at the measurement position in any space

3.2.4.4 Air exchange rate

.1 The rate of air exchange for enclosed crew spaces is to be at least six complete changes per hour.

3.3 Lighting

3.3.1 General

3.3.1.1 Lighting in passenger and crew spaces is to be provided in a manner which facilitates visual comfort, supports activities envisaged to be undertaken in that space and aids in the creation of an appropriate visual environment.

3.3.1.2 The required illuminance levels in Lux which support task, performance, safety and visual comfort within a particular space, is provided in this sub-section. These values are to be used by designers when deciding the mode of illumination, as well as the ergonomic design of spaces. The measured values of illumination may fall within an acceptable range, which is 10 lux above or below the required values provided in the Tables.

3.3.2 Lighting in Passenger Spaces

3.3.2.1 The requirements for lighting in passenger spaces are indicated in Table 3.3.2.1.

3.3.2.2 Emergency lighting is not covered by the requirements of this sub-section, and the same is to be provided in accordance with the relevant statutory requirements.

3.3.3 Lighting in Crew Spaces

3.3.3.1 The requirements for lighting in crew accommodation and service spaces are indicated in Table 3.3.3.1.

3.3.3.2 The requirements for lighting in machinery operation and maintenance areas are indicated in Table 3.3.3.2.

Table 3.3.2.1 : Lighting in passenger spaces

Space	Illuminance level in Lux
Interior walkways and stairways	100
Passenger Cabins	200 (general), <30 (during sleeping periods)
Dining Spaces	300 (mess rooms/cafeteria) , 100 (restaurants/fine dining)
Recreation spaces	200
Sanitary spaces	200
Medical and First Aid Centres	500, 200(waiting areas)
Surgical dressing rooms	800
Operation theatres -General Lighting -Operating Table	500 1000

Table 3.3.3.1 : Lighting in crew accommodation and service spaces	
Space	**Illuminance level in Lux**
Interior walkways and stairways	100
Crew Cabins	150 (general), <30 (during sleeping periods)
Mess room and cafeteria	300
Recreation spaces	200
Sanitary spaces	200
Medical and First Aid Centres	500
Wheelhouse, Pilot house and Bridge	300
Chart Room -General Lighting -Chart table	 150 500
Control Stations -General Lighting -Switchboards	 300 500
Other control rooms (e.g. cargo Transfer)	300
Galleys and other food preparation spaces	500
Storerooms -non refrigerated -refrigerated	 200 100
Muster/Embarkation area	200

Table 3.3.3.2 : Lighting in machinery operation and maintenance areas	
Space	**Illuminance level in Lux**
Machinery spaces (General), pump rooms, battery rooms	200
Main propulsion machinery space	300
Shaft Alley	100
Crane Cabin	400
Switchboard, Transformer room	500
Cargo handling (weather decks)	200
Workshops	500

Section 4

Measurements and Reports

4.1 General

4.1.1 The methods and procedures for measurements to be undertaken in order to verify compliance with the requirements of Sections 1 to 3 are indicated in this Section.

4.1.2 The measurements of parameters may be undertaken either by IRS or an organization recognized by IRS as competent to carry out such work. In the latter case, the measurements are to be witnessed by IRS Surveyor (s).

4.1.3 A measurement/ test plan is to be submitted to IRS for approval prior to undertaking measurements.

4.1.4 Measurements are to be undertaken as per the approved measurement plan. However, the attending surveyor may allow for minor deviations provided that he is satisfied that the same will not affect the accuracy of results. Any deviations from the measurement conditions specified in this section are to be clearly stated in the measurement report.

4.1.5 Tests and trials are to be conducted when the ship is fully outfitted and all systems contributing to the noise, vibration and indoor climate are operational. The operational systems also include those that would run simultaneously, whilst the ship is in normal service.

4.1.6 Measurements are to be taken at normal service speed of the vessel, and unless otherwise addressed in the provisions below, propulsion machinery run at no less than 80% MCR or higher.

All machinery and equipment which will be operational during a normal voyage are to be in operational condition and operating during the test.

4.1.7 The measurements are to be taken at a loading condition which is representative of, or as close as possible, to the normal operating condition of the Ship.

4.2 Noise and Vibration

4.2.1 Measurement conditions and procedures are generally to be in accordance with MSC Res. 337(91) for noise, and ISO 21984 and ISO 20283-5, as applicable, for vibration.

4.2.2 Measurements are to be carried out using duly calibrated instrumentation (such as sound level meters, octave filter sets) conforming to recognized National/ International Standards (for examples of such standards, refer Ch.2 of MSC Res 337(91).

4.2.3 Noise and vibration measurement locations are to be chosen such that the measurement provides a representative indication of noise and vibration levels in that space.

4.2.4 Higher density of measuring positions would be required near major noise sources such as propellers, stabilizers, thrusters and main/ auxiliary machinery.

4.2.5 Potential worst case locations are to be identified based on their proximity to noise emitting sources, such as propulsion or other rotating machinery, or where noise is likely to be transmitted to passenger accommodation areas, public, or recreation space, via the vessel’s structure. Measurements are to be taken in all identified worst case locations (e.g. spaces adjacent to a machinery space).

4.2.6 For ships equipped with thrusters, additional measurements are to be made at 40% thruster power and the ship's speed is to be appropriate for thruster operation. Measurements locations are to include positions around such machinery when in operation and in adjacent accommodation spaces and duty stations.

4.2.7 For ships equipped with stabilizers, the measurements are to be taken with the stabilizers engaged and active. If such equipment (thrusters/ stabilizers) is intended for continuous operation, measurements are to be made for ensuring compliance with Chapter 4 of MSC Res. 337(91) (*Code on Noise Levels on-board Ships*). If such systems are intended for short temporary use only, for instance during port manoeuvres, measurements are only

relevant for ensuring compliance with Chapter 5 of MSC Res. 337(91) on noise exposure.

4.2.8 For ships equipped with dynamic positioning systems, which are intended for use in normal working condition, additional noise measurements at DP mode are to be made at control stations, duty stations, and accommodation spaces to ensure that the maximum noise level limits in these spaces are not exceeded. The system is to be operating on at least 40 percent power, during these measurements.

4.2.9 In large public spaces where there may be considerable variation of noise levels within the space, measurements are to be taken at locations not greater than 7 [m] apart.

4.2.10 In addition, for passenger ships, and ships which are designed to spend a considerable period of time in harbour, the noise and vibration are to be also measured for this condition, with the auxiliary machinery and HVAC systems running at their normal rated capacity.

4.2.11 In addition to the noise measurement locations for crew spaces as specified in MSC Res. 337(91), the following measurement locations are to be included for passenger ships:

- all public spaces
- 50 percent of the cabins in the aft one third of the ship.
- 25 percent of the cabins, elsewhere.

4.2.12 If the specified criterion is exceeded or annoying low frequency noise or tonal components are observed, then 1/3 octave band analysis of the noise is to be performed.

4.2.13 Vibration levels are to be generally measured in the centre of the floor area of the space over a period of not less than one minute. In large public spaces more than one measurement locations may be required.

4.2.14 At all locations, vibrations in the vertical direction are to be assessed. Sufficient measurements are to be taken in the fore and aft, athwartships direction to define the global deck vibrations.

4.2.15 In general, the testing conditions are to replicate, as far as practicable, conditions which will exist during course of the ship's normal service. For ships which are normally operated in deep waters, the tests are to be carried out in water depth not less than five times the design draft of the ship. For ships which would be continuously operating in shallow waters, the test is to be conducted at the relevant water depth.

4.2.16 Trials are to be conducted in sea conditions not greater than Sea State 3 on the WMO sea state code. Noise measurements are not to be taken when the wind force exceeds 4 on the Beaufort scale.

4.2.17 Rudder angles during measurements are to be restricted to within 2 degrees port or starboard, and rudder movements are to be kept to a minimum throughout the measurement periods.

4.2.18 The noise and vibration measurement reports will be reviewed by IRS for assignment of the Comfort rating, as applicable.

4.3 Indoor Climate

4.3.1 Certification towards indoor climate and thermal comfort on board will include measurements taken on board, and additionally for passenger ships, a design review of the HVAC system, to assess its performance at the typical ambient climatic conditions to which the ship will be subjected to during its intended use.

4.3.2 Climate parameters are to be verified through design evaluation and/or measurements include air temperature, supply of fresh air (number of air changes), air velocity and relative humidity.

4.3.3 Documentation/ data pertaining to the following need to be submitted to IRS for review:

- General arrangement of the vessel
- Appropriate vessel and HVAC system design specifications
- Schematics/ layout of the HVAC system

4.3.4 The measurement plan for indoor climate is to include conditions under which the tests are performed, measurement locations, as well as details regarding software and instrumentation used for testing and data analysis. The measurement locations are to be selected such that they give a representative description of the climate in the space.

4.3.5 All measurement instrumentation is to conform to recognized National/ International Standards.

4.3.6 The following quantities are to be measured in each of the spaces or zones identified in the measurement plan, and shown to not exceed the requirements of this Chapter:-

- Air temperature characteristics
- Relative humidity
- Air velocity
- Vertical gradient

4.3.7 Air temperature and humidity measurements are to be made at intervals of at least five minutes for a period of one hour. The minimum, maximum and average values are to be reported for each space.

4.3.8 Air velocity is to be measured at the geometric centre of the room. However, the Surveyor may recommend this measurement to be taken at the most commonly occupied position in that particular space. Air velocity measurements are to be taken over a period of three minutes.

4.3.9 Where a single instance of a particular space exists within the vessel (for example, Navigation Bridge), the space is to be selected for measurement. Where multiple instances of a space with similar purpose as well as arrangement exist (for example, laundries), a representative sample of at least twenty-five percent of each type of space are to be selected for measurement.

4.3.10 For crew/ passenger cabins, measurements are to be carried out in at least 30 percent of the total number of cabins on each deck, which are as far as practicable, evenly distributed over the length and breadth of the ship. In case of ships with less than 10 cabins on a particular deck, measurements are to be undertaken in at least 3 cabins.

4.3.11 In large public spaces where there may be considerable variation of thermal parameters within the space, more than one measurement location may need to be identified such that the maximum spacing between two such locations is not greater than 15 [m].

4.3.12 All doors and windows of the space, are to be closed during the evaluation period, except for routine entry and exit.

4.3.13 When thermal conditions have a high sensitivity to ambient weather conditions, the measurements are to be made such that the high and low extremes of the thermal parameters are accurately determined.

4.3.14 The indoor climate test report and results will be reviewed by IRS for assignment of the '**c**' qualifier for Comfort notation.

4.4 Lighting

4.4.1 For certification towards lighting and visual comfort on board, test plan preparation, test measurement procedures and test reporting are to be in accordance with ISO 8995-1:2002.

4.4.2 Light meters used for measurements are to conform to the requirements specified in ISO 8995-1:2002.

4.4.3 In spaces with windows or portholes, measurement of lighting levels is to be undertaken after dark, or during the day with all natural lighting blocked out.

4.4.4 All stray lighting is to be masked out during measurements. In cases where this is not possible, a control measurement is to be made which may be deducted from measured values, to obtain the actual lighting level within the space.

4.4.5 During lighting measurements, only the personnel needed for the normal operation of the equipment in that space, as well as those carrying out measurements, are to be present in that space.

4.4.6 In general, measurement locations within spaces are to be selected such that they give a representative value of the lighting within the space. In large public spaces, more than one measurement location may need to be selected for measurement. As a general principle, one measurement location for each 40 square meters of floor space may be considered acceptable.

4.4.7 For crew/ passenger cabins, measurements are to be carried out in at least 30 percent of the total number of cabins on each deck, which are as far as practicable, evenly distributed over the length and breadth of the ship. In case of ships with less than 10 cabins on a particular deck, measurements are to be undertaken in at least 3 cabins.

4.4.8 In addition to the results obtained during measurement, the test report is to include details of the ambient conditions such as time of the day, external lighting conditions etc. Details of measurement equipment (accuracy, resolution, calibration details etc.), as well as actual measurement locations are to be included as part of the measurement report.

4.4.9 The lighting test report and results will be reviewed by IRS for assignment of the '**I**' qualifier for Comfort notation.

End of Chapter

Chapter 40

Livestock Carriers

Contents

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6	*Electrical Equipment*

Section 1

General

1.1 Application

1.1.1 Ships complying with the requirements in this Chapter may be assigned with class notation **LIVESTOCK CARRIER**.

1.1.2 In addition to the general list of plans in the following documents/ plans are to be submitted for information:

- Livestock arrangement (including loading - unloading arrangement and distribution of livestock pens at various decks/ platforms)
- Distribution of fodder and consumable liquids on various decks and platforms

1.1.3 The following documents/ plans are to be submitted for approval:

- Stowage/ securing arrangement of livestock in pens
- Access and escape from livestock pens/ decks/ spaces
- Ventilation system
- Storage and distribution system for fodder and water including piping systems
- Sewage drainage and discharge system
- Load schedule of main and emergency power
- Electrical equipment, panels switchboard plans,
- Main and emergency lighting system
- Firefighting arrangements in livestock spaces and arrangement of firefighting appliances

1.1.4 The following documents/ plans are to be submitted for review:

- Type approval of water level detection system including sensors, wiring and control panel
- General arrangement of sensors and cables
- power supply electrical diagram
- detailed installation drawing of sensors

Section 2

Livestock Arrangements

2.1 Livestock arrangements

2.1.1 The livestock are to be kept in pens. The dimensions of the pens are to be suitable for the type of livestock carried.

2.1.2 The livestock are not to be carried on hatch covers, unless the hatch covers are effectively protected.

2.1.3 The livestock are not to be carried in more than one tier in a deck.

2.1.4 If dangerous goods are carried on a vessel, livestock must not be held:

(a) within 6 [m] horizontally of the dangerous goods; or
(b) in the same underdeck compartment as the dangerous goods; or
(c) in any position where they could be affected by spillage or leakage of the goods.

2.2 Arrangement of livestock spaces

2.2.1 The requirements of this sub-section are applicable to the arrangement of spaces dedicated to the carriage of livestock. The spaces may need to be adapted or suitably modified depending on the particular species of animals being carried.

2.2.2 Suitable arrangements are to be provided for protecting the livestock from injury, avoidable suffering and exposure to weather, sea or hot parts.

2.2.3 Livestock may not be carried or loaded for carriage, in any part of the ship where the livestock, livestock fittings, livestock equipment or carrying arrangements may:

- obstruct access to any accommodation space or working space necessary for the safe running of the ship, or the means of egress from any hold or underdeck space
- interfere with life-saving or fire-fighting appliances interfere with the tank sounding equipment or bilge pumping
- interfere with the operation of closing appliances
- interfere with the operation of freeing ports
- interfere with the lighting or ventilation of other parts of the ship
- interfere with the proper navigation of the ship.

2.2.4 A pen floor or stall floor is to have a surface that provides a satisfactory non-slip foothold for livestock.

2.3 Means of access and escape for livestock

2.3.1 A ramp or other suitable means of access, appropriate to the species being carried, is to be provided for the loading or unloading of livestock. It is to be so erected as to prevent any gap occurring between it and the ship and is to be set at a gradient (not exceeding 1 in 2) suitable for the species to be loaded or unloaded.

2.3.2 The means of access is to be fitted with:

- side panels free of protrusions and of sufficient strength and height to prevent escape of livestock
- a walking surface of battens suitable for the species
- a closing arrangement

2.3.3 If the means of access is part of the ship's equipment, it is to be designed to support a uniformly distributed load over the walking surface not less than the values indicated in Table 2.3.3

Table 2.3.3

Species	Load [N/m^2]
Cattle and Horses	4700
Sheep, goats and pigs	2400

2.3.4 The permissible tensile strength for material used in the means of access is indicated in Table 2.3.4

Table 2.3.4

Material	Permissible tensile stress
Steel	0.5 σ_y
Aluminium	0.5 $\sigma_{p0.2}$
Other	Specially considered

Where,

σ_y = Minimum yield stress [N/mm2]

$\sigma_{p0.2}$ = 0.2 percent proof stress

2.3.5 Livestock pens and means of access for livestock are to be provided in accordance with the requirements of Tables 2.3.5 (a), 2.3.5 (b) and 2.3.5 (c) for pigs and sheep, cattle and horses respectively. Details of pens dedicated for carriage of other livestock are to be submitted to IRS for special consideration. The rails in the pens are to be provided taking into account the size of particular species of livestock.

Table 2.3.5(a) : Limiting dimensions of pens and passageways for pigs, sheep and goats

Maximum distance between rails aligned fore and aft (breadth)	4.5 [m]
Minimum distance between rails aligned fore and aft (breadth)	2.0 [m]
Maximum distance between rails aligned athwartships (length)	Not more than twice the breadth
Minimum length	Not less than the breadth
Maximum clear floor area within pen	40.5 [m^2]
Minimum clear height within pen	1.1 [m]
Maximum clear vertical distance between rails	0.3 [m]
Maximum clear vertical distance below bottom edge of lowest rail of pen installed at deck level	0.2 [m]
Maximum clear vertical distance below bottom edge of lowest rail of pen not installed at deck level	0.05 [m]
Minimum height of top edge of uppermost rail above pen floor	0.9 [m]
Minimum width of adjacent passageway clear of receptacles and any other obstructions	0.9 [m]

Table 2.3.5(b) : Limiting dimensions of pens and passageways for cattle

Maximum distance between rails aligned fore and aft (breadth)	4.5 [m]
Minimum distance between rails aligned fore and aft (breadth)	2.1 [m]
Minimum distance between rails aligned athwartships	2.3 [m]
Maximum clear floor area within pen	21 [m^2]
Maximum height of top edge of lowest rail above pen floor between pen	0.6 [m]
Minimum clear height within pen	2.0 [m].
Minimum clear height within deck outside pens	1.8 [m]
Minimum width of adjacent passageway, measured clear between rails, when pens are on both sides of the passageway and cattle are loaded and discharged through the pens	0.9 [m]
Minimum width of adjacent passageway, measured clear of any fixed structure, fittings, receptacles or obstruction (e.g. pillars, feed chutes, fixed fodder or water troughs), when pens are on both sides of the passageway and cattle are loaded and discharged through the pens	0.75 [m]
Minimum width of adjacent passageway, measured clear of any fixed obstructions, when pens are on both sides of the passageway and cattle are loaded and discharged through the passageway	0.9 [m]
Minimum width of adjacent passageway clear from rails, when pens are on only one side of the passageway and livestock are not loaded through that passageway	0.75 [m]

Table 2.3.5(c) : Limiting dimensions of stalls, pens and passageways for horses		
Maximum clear length within stall		2.5 [m]
Minimum clear length within stall		2.3 [m]
Minimum clear passage	(i) between 2 rows of stalls and bounded by the front rails	1.7 [m]
	(ii) between 2 rows of stalls and bounded by the front and back rails	1.2 [m]
	(iii) in any other case	1.0 [m]
Minimum clear breadth within stall	(i) if the stall is aligned athwartships	0.7 [m]
	(ii)if the stall is aligned fore and aft	0.9 [m]
Height of uppermost front, back and side rail from floor to top edge		1.15 [m]
Height of lowest front, back and side rail from floor to top edge		0.75 [m]

2.3.6 Additional requirements for goats

2.3.6.1 There must be pen rails:

(i) to a height of at least 1.5 [m] above the pen floor; or
(ii) to a height where the clear height above the uppermost rail is less than 300 [mm]

2.3.7 Livestock pens are to be provided based on the loads determined by the application of the following:

$$F = \propto * B(0.574 + 0.0252 * Z)$$

Where,

F = Load per metre length of boundary, [N/m]

B= Maximum breadth of pen, [m]

Z = the vertical distance from a point 0.75[m] above the pen floor to the vessels' waterline corresponding to the lightest displacement, [m]

$\propto$ is a constant, taken as 1668 for sheep, goats or pigs and 3336 for cattle or horses.

Load bearing areas are:

For sheep, goats and pigs – uniformly distributed upto the rail, the centre of which is at a height of not more than 0.9 [m] above the floor of the pen.

For cattle and horses - uniformly distributed upto the rail, the centre of which is at a height of not more than 1.4 [m] above the floor of the pen.

2.3.8 Livestock pen floors or platforms are to be provided based on the loads determined by the application of the following:

$$P = \beta * \left[1 + \frac{1}{d}((0.094 - 0.00035L)y + (7.4 - 0.016L))\right]$$

Where,

P = Floor load per square metre, [N/m^2]

d = Minimum draught of the ship, [m]

y = Longitudinal distance from the midpoint of the pen to amidships, [m]

L = Length between perpendiculars of the ship, [m]

β is a constant taken as 2500 for sheep, goats or pigs and 5000 for cattle or horses.

2.3.9 The permissible stresses for materials used for livestock pens and floors is indicated in Table 2.3.9

Table 2.3.9		
Material	**Permissible tensile stress**	**Permissible shear stress**
Steel	0.75 σ_y	0.375 σ_y
Aluminium	0.75 $\sigma_{p0.2}$	0.375 σ_y
Other	Specially considered	

Where,
σ_y = Minimum yield stress [N/mm$_2$]

$\sigma_{p0.2}$ = 0.2 percent proof stress [N/mm$_2$

2.4 Means of access and escape for persons

2.4.1 Each space in which livestock is carried is to be provided with at least two means of escape, widely separated and giving access to an open deck. Such means of egress are to be clearly marked.

2.4.2 Access to livestock space for persons is to be safe. Where it is combined with a ramp used for moving livestock between decks, it is to be separated from the livestock ramp by protective fencing.

2.4.3 Pens, stalls and similar fittings are to be provided with a means of access for persons with secure closing arrangements, having structural strength at least equivalent to the strength of that part of the pen, stall or fitting.

2.4.4 The passageways in spaces for the carriage of livestock are to have a minimum clear height of 2.0 [m].

2.4.5 If, for the safe and proper operation of the vessel, access is required between the side of the vessel and a pen, stall or similar structure, there is to be a passageway that is at least 750 [mm] wide between the vessel's rail or bulwark and the rails of the pen, stall or structure.

However, the passageway width may be reduced to less than 750 [mm], but no less than 550 [mm], by obstructions outside the pen rails such as receptacles, pipework or other objects for the provision of livestock services.

2.4.6 Where it is necessary for persons to be on a means of access during the movement of livestock, a passage of at least 550 [mm] width is to be provided in accordance with the following:

- it is to be fenced to a height of at least 1 [m] and with an intermediate horizontal rail approximately 550 [mm] above the walking surface; and
- Fitted with treads at suitable stepping distances.

Section 3

Stability and Strength

3.1 Stability

3.1.1 General

3.1.1.1 Livestock carriers are required to comply with the stability requirements for cargo ships in the International Code on Intact Stability 2008, as amended.

3.1.1.2 In addition, requirements of the national/ statutory authorities, where the ship is registered or in areas where the ship trades are also to be complied with.

3.1.2 Additional Requirements

3.1.2.1 The stability requirements of a vessel lo be met throughout a voyage, taking into account the effects of shift of livestock and feed and, if the vessel has a pen structure on or above the uppermost continuous deck, the effect of wind; are to be as follows:

(a) the area under the righting lever curve shall be not less than 0.055 [m-rad] up to 30 [deg] angle of heel and not less than 0.09 [m-rad] up to 40 [deg] angle of heel, or the angle of flooding if this angle is less than 40 [deg],

(b) the area under the righting lever curve between the angles of heel of 30 [deg] and 40 [deg], or between 30 [deg] and the angle of flooding if this angle is less than 40 [deg], is not to be less than 0.03 [m-rad],

(c) the righting lever is not to be less than 0.20 [m] at an angle of heel equal to, or greater than 30 [deg],

(d) the maximum righting lever is to occur at an angle of heel not less than 25 [deg],

(e) the initial metacentric height is not to be less than 0.15 [m],

(f) the area under the righting lever curve up to 40 [deg] or the angle of

flooding, whichever is less, in excess of the area under the heeling lever curve to the same limiting angle, is to be not less than 0.018 [m-rad] plus 20 per cent of the area of the righting lever curve to the same limiting angle, and

(g) the angle of heel due to wind is to be not more than 10 [deg].

3.1.2.2 The residual Area 'A' is to comply with the following requirements, taking into account the effects of the shift of livestock and fodder in pellet form and the effect of wind, as illustrated in Fig. 3.1.2.2 below:

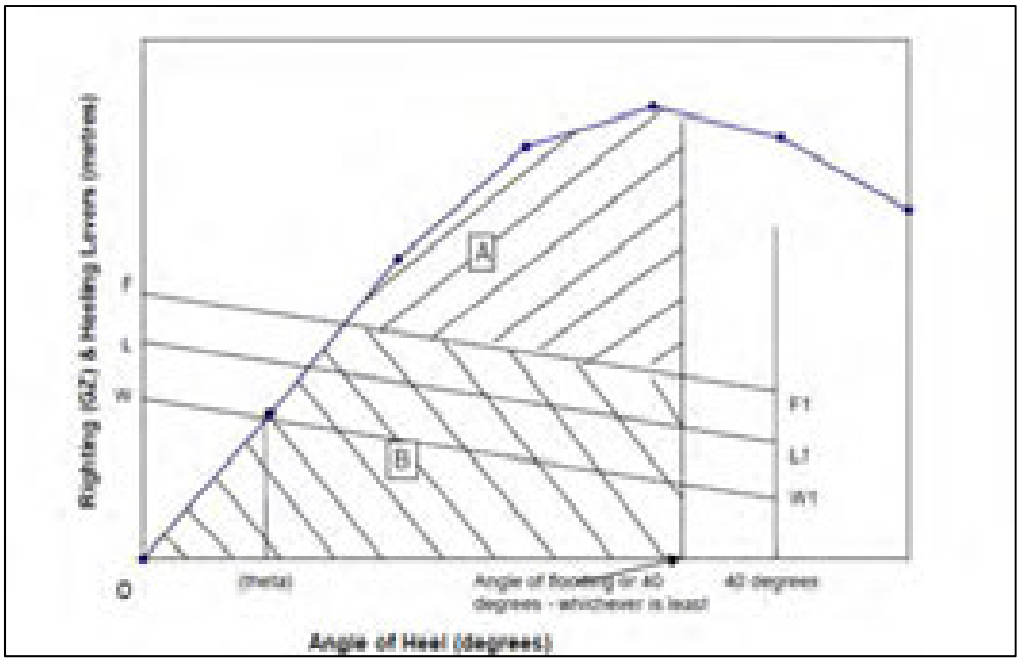

Fig. 3.1.2.2 Illustration of Stability Requirements

$A \geq 0.018 mrad + 0.2\, A_{(A+B)}$

Where,

$A_{(A+B)}$ is the sum of area A and area B under the righting lever curve, as illustrated in Fig. 3.1.2.2.

OW – the heeling lever at 0 [deg] due to wind;

WW1 – the heeling curve due to wind;

WL – the heeling curve at 0[deg] due to the shift of livestock;

LL1 – the heeling lever curve due to the combined effects of wind and shift of livestock.

LF – the heeling lever at 0 [deg] due to the effect of shift of fodder;

FF1 – the heeling lever curve due to the combined effects of wind and the shift of livestock and fodder;

θ – Angle of heel due to wind.

Note: If the fodder is not pellet feed carried in bulk, the heeling lever due to shift of fodder will be zero.

3.1.2.3 The heeling lever due to the effect of wind is to be determined according to the following:

(a) The heeling lever due to the effect of wind at 0 [deg] is to be calculated as follows:

$$OW = \frac{PAH}{\Delta}$$

Where; Δ - Displacement [tonnes];

P – Wind pressure, 504 [Pa];

A – Lateral area of the vessel above the waterline [sq.m];

H - vertical distance between the centroid of the lateral area of the vessel above the waterline and the centroid of the vessel's underwater lateral area. For most vessels the vertical position of the centroid of the underwater lateral area may be taken as half the draft to the underside of the keel at amidships

(b) The heeling lever due to the effect of wind at 40 [deg] is to be given by 0.8 (heeling lever due to the effect of wind at 0 [deg]).

(c) The heeling lever curve is to be taken as a straight line joining the heeling lever at 0 [deg] and the heeling lever at 40 [deg].

3.1.2.4 The heeling lever due to the shift of livestock is to be determined according to the following:

(a) The heeling lever due to the shift of livestock at 0 [deg] is to be calculated as follows:

$$WL = \frac{m * C}{f * \Delta}$$

Where; m – average mass of livestock being carried on the intended voyage [tonnes];

f – floor area required per head of average mass of the livestock being carried on the intended voyage [sq.m]

C – livestock shift constant;

$C = \frac{1}{6}\sum l\, b^2$, where

l – length of the pen[m],

b – breadth of the pen [m]

(i)For vessels with uniform breadth of pens, the ‘C’ is: 1/6 (breadth of the pen * total floor area of pens);

(ii)For vessels with varying breadth of pens, the largest breadth may be used and the ‘C’ is: 1/6 (maximum breadth of pen * total floor area of pens)

(b) The heeling lever due to the shift of livestock at 40 [deg] is to be given by 0.8 (heeling lever due to the shift of livestock at 0 [deg]).

(c) The heeling lever curve is to be taken as a straight line joining the heeling lever at 0 [deg] and the heeling lever at 40 [deg].

3.1.2.5 The heeling lever due to the shift of fodder is to be determined according to the following:

(a) The heeling lever due to the shift of fodder at 0 [deg] is to be calculated as follows:

$$LF = \frac{M_f}{s * \Delta}$$

Where; M_f – the sum of the shift moment of each compartment, the shift moment of each compartment is to be given by 0.04 l b ^{3,} where;

l – max. length of compartment [m];

b – max. breadth of compartment [m]

The use of volumetric shift moments for the fodder, where the surface is assumed to take up an angle of slope of 15 [deg] to the horizontal for full compartments and 25 [deg] to the horizontal for partly filled compartments is an acceptable alternative method to obtain the total shift moment of the fodder.

s – stowage factor (the volume of fodder in pellet form per unit weight)

(b) The heeling lever due to the shift of fodder in pellet form carried in bulk at 40 [deg] is to be given by 0.8 (heeling lever due to the shift of fodder at 0 [deg]).

(c) The heeling lever curve is to be taken as a straight line joining the heeling lever at 0 [deg] and the heeling lever at 40 [deg].

3.1.2.6 It is recommended that the following information be included in the stability information provided onboard the vessel:

(a) Livestock shift constant (as determined for all conditions of pen utilization that may arise in practice, unless the maximum value is used for all calculations)

(b) Heeling moment for fodder (the heeling moment for each compartment is to be determined separately, unless the greatest heeling moment for all compartments added together is provided: i.e. the total heeling moment for the worst condition of stability

(c) Wind effect (the heeling lever due to wind to be provided in tabular or graphical form)

3.2 Water level detectors

3.2.1 Water level detectors are to be provided on livestock carriers in accordance with the requirements in Pt.4, Ch. 3, Sec. 3.6 of the Rules.

3.3 Hull Girder Strength

3.3.1 In addition to compliance with the applicable requirements of Part 3 of the Rules, the following are to be taken cognizance of:

.1 In general, the decks and platform decks above the strength deck used for the carriage of livestock need not be taken into account for the calculation of the section modulus.

.2 In general, the movable or collapsible structural elements above the strength deck used for the stocking and the distribution of livestock on decks or platform decks are not a part of Classification.

3.4 Protection of cargo sides by battens

3.4.1 These requirements are applicable to sides in cargos spaces of livestock carriers.

3.4.2 In the case of transversely framed sides, longitudinal battens formed by spaced planks are to be fitted to the frames by means of clips.

3.4.3 Where sides are longitudinally framed, battens are to be fitted vertically.

3.4.4 Battens are to extend from the bottom of the cargo space to at least the underside of the beam knees.

3.4.5 Cargo battens are to be not less than 50 [mm] in thickness and 150 [mm] in width. The space between the battens is not to exceed 300 [mm].

Section 4

Systems serving Livestock Spaces

4.1 General

4.1.1: A vessel that is permanently equipped for the carriage of livestock is to be fitted with systems and equipment that ensure the maintenance of livestock services at a level necessary for the welfare of the livestock. These would include facilities and systems intended for:

- the supply of food, water and fresh air to livestock;
- cleaning of livestock spaces;
- draining and outside discharging of the sewage effluents produced by livestock

4.1.2 In addition to the provisions of this Section, piping systems are to comply with the requirements in Pt. 4, Ch. 2 and Ch. 3 of the Rules.

4.2 Ventilation System

4.2.1 Each enclosed space for the carriage of livestock is to have a mechanical ventilation system which provides efficient air circulation in all parts of that space and which is of sufficient capacity to change the entire volume of air in that space, as follows:

- where the minimum clear height in the space is 2 [m] or less, at a rate of not less than 40 times per hour;
- where the minimum clear height in the space is 3.20 [m] or more, at a rate of not less than 26 times per hour, and

- where the minimum clear height in the space is between 2 [m] and 3.20 [m] at a rate proportional to those specified above. This may be obtained by linear interpolation.

The total volume of the enclosed space is to include all the space contained between the vessel's side plating, bulkheads, tank top or decks enclosing the space, less the volume of any tank or trunk that is airtight within the space and no deduction is to be made in respect of the space occupied by livestock, pens or other livestock fittings.

4.2.2 A space for the carriage of livestock that is unenclosed, is to be provided with a mechanical ventilation system if

(i) the minimum clear height in the space is 2 [m] or less, or

(ii) because of a partial enclosure of the space, the natural ventilation is restricted.

The system is to be of sufficient capacity to change the entire volume of air in that space at a rate of 75 percent of the rates specified in 4.2.1.

4.2.3 A mechanical ventilation system must distribute air so that the whole of any livestock space is efficiently ventilated. The mechanical ventilation system is to be capable of providing a minimum air velocity across any part of a pen from a source of supply of not less than 0.5 [m/ sec].

4.2.4 Air intakes are to be so sited that air supplied to the livestock spaces is as clean and fresh as practicable and are to be effectively protected against blockage. Air exhaust outlets are to be sited as high as practicable and clear of accommodation structures.

4.2.5 Ventilators serving livestock spaces are to remain open in all weather conditions while livestock are on board. The height of ventilator coamings is to be in accordance with the requirements of Part 3 Chapter 13 Section 2.3, as these ventilators would be kept open in all weather conditions.

4.2.6 Ventilation circuits are to be supplied by at least two independent fans of such a capacity as to maintain the ventilation of all the spaces as required by 4.2.1 and 4.2.2 with one fan out of action.

4.2.7 Notwithstanding the above provisions, if a mechanical ventilation system is provided with adequate spare parts, the provision of only one fan is acceptable. Adequate spare parts are to be interpreted as including for each type of fan: one set of bearings; one rotor or impeller; and one complete motor.

4.2.8 Electrical supplies from both main and secondary sources of power are to be supplied to each group starter panel, with both supplies being as widely separated as practicable and neither passing through any space containing any part of the other source of power. Interlocks at each group starter are to prevent simultaneous supply by both sources of power.

4.2.9 The livestock services switchboard and ventilation fan group starters are to be electrically subdivided where practicable, in separate compartments, to provide redundancy and limited supply of ventilation and/or services to each livestock deck in the event of switchboard failure.

4.3 Fodder Arrangements

4.3.1 Spaces intended for livestock (including pens, stalls or similar fittings) are to be provided with receptacles for feeding and watering the animals concerned.

4.3.2 The receptacles are to be capable of containing at least 33 per cent of the daily allowance of fodder and water for the number of animals contained in the space (pen, stall or similar fitting).

4.3.3 A receptacle provided in accordance with 4.3.1 and 4.3.2 is to be:

- suitable for the species of livestock;
- readily accessible to the livestock;
- capable of being serviced from outside the pen, stall or other fitting;
- so installed as to not impede ventilation; and
- so constructed and positioned, that fodder dust is not disturbed by the flow of ventilation.

4.3.4 If the fodder distribution system is dependent on electric power, the system is to be capable of being powered by both the main and secondary sources of supply. It should also be able to operate if a fire or other emergency occurs in the space containing the main source of power.

4.4 Fresh Water Systems

4.4.1 All livestock spaces are to be provided with fresh water service.

4.4.2 The fresh water system is to include one main supply pump and one standby pump, of a capacity sufficient to continuously supply fresh water to the livestock.

When the water supply system is not automatic, the standby pump may be replaced by a portable pump ready to be connected to at least one fresh water tank.

4.4.3 If the fresh water system is dependent on electric power, it is to be powered by both the main and secondary sources of power and is to be able to continue to operate despite a fire or other casualty in the space containing the main source of power. One pump may be located in the space occupied by the main source of power and supplied by that source of power. The other is to be able to maintain supply despite a fire or other casualty affecting the space occupied by the main source of power.

4.4.4 When the water supply is automatic, water receptacles are to be fitted with means of automatic water level control and devices to avoid the return of water from the receptacle to the fresh water tank.

4.4.5 The fresh water system serving the livestock spaces is to be totally independent from the fresh water system serving the spaces intended for the crew.

4.5 Water washing system

4.5.1 A water washing system is to be provided with appropriate connections to wash the livestock spaces.

4.6 Drainage and discharge of sewage

4.6.1 Each space intended for the livestock is to be fitted with a pipe or gutter of sufficient size to drain the sewage and the washing effluents.

4.6.2 Drainage arrangements are to be such that fluids drained from a pen are, as far as practicable, kept clear of other pens and associated working and access spaces.

Where necessary, drainage gutters and upper parts of the draining pipes are to be covered by a strainer plate. In addition, drainage pipelines are to be so arranged that the drainage system can still be used with the ship's trim by bow or stern.

4.6.3 The drainage system serving the livestock spaces is to be independent from any piping system serving the other spaces of the ship, and in particular from the bilge system.

4.6.4 The pipes and other components of the draining system are to be made of a material resistant to the corrosion due to the effluents.

4.6.5 Essential drainage tanks, wells and the top of drainage pipes in a vessel are to be accessible from outside livestock pens for the purpose of inspection and cleaning.

4.6.6 Pumps and ejectors serving the drainage tanks or wells are to be capable of conveying semi-solid matter.

4.6.7 All electrical equipment fitted to meet the requirements of 4.6 are to be capable of being operated by both the primary and the secondary sources of power.

Section 5

Fire-fighting Arrangements in Livestock Spaces

5.1 Fire hydrants and fire hoses

5.1.1 The number and position of the hydrants are to be such that at least two jets of water not emanating from the same hydrant may reach any part of the spaces intended for the livestock. At least one of these jets is to be from a single length of hose without the necessity for hoses to pass over or through pens.

5.1.2 The fire hydrants referred to in 5.1.1 are to be connected to the fire main provided on the vessel.

5.1.3 A fire hose is to be provided in an enclosed space for each hydrant and in any other space or on a deck for each 50 [m] length, or part thereof, of space or deck.

5.1.4 Each fire hose is to be provided with the necessary connections and a nozzle capable of directing water in the form of a spray and a jet.

5.1.5 Each hose is to be capable of being connected to any hydrant and to any other hose (other than hydrants and hoses within the engine room or accommodation spaces).

5.1.6 Fire hoses are to be located in conspicuous locations, near the hydrants and close to the entrances or access to the spaces.

5.2 Additional fire-fighting means

5.2.1 Livestock structures, including livestock decks and containment structures but excluding livestock services such as water pipes and feed troughs, are to be constructed of non-combustible materials.

5.2.2 If hay or straw is carried or used in a livestock space, one of the following fire-fighting means is to be provided:

a. a fixed water fire-fighting system, or

b. portable water extinguishers spaced not more than 18 [m] apart, one of which is positioned at the entrance of the space concerned.

5.2.3 Where electrical equipment other than that referred to in Section 6 is located in an enclosed livestock space, suitable fire-fighting means are to be provided.

5.2.4 Where hay, straw, other foodstuff or bedding of a flammable nature is used or carried, notices shall be prominently posted prohibiting smoking or the use of naked lights in a space in which any such substance is located.

Section 6

Electrical Equipment

6.1 General

6.1.1 In addition to the applicable requirements of Pt. 4, Ch. 8 of the Rules, electrical equipment of livestock carriers is to comply with the provisions of this Section.

6.2 Sources of Electrical Power

6.2.1 The main source of power as specified in Pt. 4, Ch. 8 is to be provided and all electrical auxiliary services (including livestock services) necessary for maintaining the vessel in normal operational and habitable conditions are to be ensured without recourse to the emergency source of electrical power.

6.2.2 The secondary source of power is to meet the following requirements:

.1 The secondary source of power is to be located in a space that is not contiguous with any space containing the main source of power or part thereof, and be independent of any services provided from or through any such space. In case of a fire or other casualty in any space containing the

main source of power or any part thereof, the secondary source of power must be capable of supplying power to livestock services for a period of 72 hours.

.2 The prime mover of secondary source of power is to be capable of supplying power to the system within 15 [min] in the failure of main source of power. The prime mover may be started by an effective arrangement powered by an independent source of energy or the emergency source of power complying with the requirements of Pt. 4, Ch. 8, C1. 2.8.1, 2.8.4 and 2.8.9.1

.3 Instructions are to be provided for the changeover between main and secondary sources of power and vice-versa. A copy of such instructions is to be displayed in the space containing the livestock source of power, and is to be readable under the emergency lighting required by Pt. 4, Ch. 8, Cl. 2.8.9.1 b). The instructions are to detail, among other things, starting method, switchboard changeover and electrical supply changeover to livestock services.

.4 Cables of secondary source of power are to be kept away from cables of main source of power, as far as practicable and they are not to pass through the same space, in general.

6.3 Lighting

6.3.1 Adequate lighting (at least 20 [lux]) is to be provided in livestock spaces, passageways between pens and access routes between or to those livestock spaces.

6.3.2 Fixed or portable lamps providing adequate illumination (at least 110 [lux]), powered by the vessel's electrical system are to be provided in any pen in which livestock are carried, to facilitate the close inspection of livestock.

6.3.3 An emergency lighting system, that is automatically activated on the failure of the main electrical installation, is to be provided in all parts of the vessel where livestock is carried, passageways between pens and access routes between or from those parts.

The emergency lighting is to be capable of giving a level of illumination of at least 8 [lux], in all passageways and access routes for a continuous period of not less than 15 [min]. The lamp casings on light fittings for the emergency lighting system are to be painted red for ease of identification.

6.3.4 If fixed lighting is provided in a part of a vessel above the uppermost continuous deck then that lighting is to be capable of being controlled from the navigating bridge.

6.3.5 Light fittings are to be

- waterproof and
- either;
  - of sufficient strength and fitted with protective sheathing outside to resist damage by livestock, or
  - placed beyond possible contact by livestock.

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End of Chapter

IRCLASS
Indian Register of Shipping

50 YEARS OF EXCELLENCE

RULES AND REGULATIONS FOR THE CONSTRUCTION AND CLASSIFICATION OF **STEEL SHIPS**

PART 6
FIRE SAFETY REQUIREMENTS

July 2024

Indian Register of Shipping

Part 6

Fire Safety Requirements

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Chapter 1

General

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2 *Fire Safety Objectives and Functional Requirements*

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Section 1

General

1.1 Scope

1.1.1 The requirements of this part generally apply to ships of the following categories to be classed for unrestricted service :

- passenger ships
- cargo ships of 500 gross tons and over.

In case of Special Trade Passenger Vessels, consideration will be given to any exemptions granted by the National Statutory Authorities. (In such cases full compliances with the IMO Rules annexed to the Special Trade Passenger Ships, 1971 and the Protocol on Space Requirements for Special Trade Passenger Ships, 1973; is normally required).

Consideration will also be given to any exemptions granted by the National Statutory Authorities where the sheltered nature and conditions of normal operation are such as to render the application of any specific requirements of this part unreasonable or unnecessary.

Requirements for Cargo Ships below 500 GT will be specially considered, however, as a minimum, these are not to be less than that required by the National Statutory Authorities.

1.1.2 Alternative fire safety design and arrangements which may be proposed in lieu of prescriptive requirements given in Ch.2, 3, 4, 5 or 7 are to comply with the requirements in Ch.6.

1.1.3 Whilst the requirements of this part are considered to meet the Regulations of the International Convention for the Safety of Life at Sea (SOLAS), 1974, and applicable amendments, attention is, however, be given to any relevant requirements of the National Statutory Authority.

1.1.4 Consideration will be given to acceptance of the following as an alternative to the requirements of this part:

a) Requirements of the National Statutory Authority.
b) Fire insulating materials, fire fighting appliances and equipment approved by the National Statutory Authority.

1.1.5 A list of IMO MSC Circulars and Assembly Resolutions referred to in this part is given at the end of this chapter (Table 1).

1.2 Documentation

1.2.1 The plans and particulars detailed in 1.2.2 to 1.2.4 where applicable, are to be submitted for approval, together with all additional relevant information such as gross tonnage and number of passengers, date of build (in case of existing ships), arrangement of helicopter deck etc.

1.2.2 For fire protection, the following plans and information are to be submitted :

a) A statement of the method of structural fire protection adopted indicating compliance with the regulations;

b) A general arrangement plan showing the main fire zones, escape stairways and the structural fire protection bulkheads and decks within the main fire zones, including the engine and boiler rooms, the galleys, paint stores, navigating bridge, radio room, fire fighting control room, emergency generator rooms and battery locker;

c) A plan showing the details of construction of the fire protection bulkheads and decks and the particulars of any surface laminates employed;

d) A plan showing the construction of the fire doors;

e) A ventilation plan showing the ducts and any dampers in them, and the position of the controls for stopping the system;

f) A plan showing the sprinkler system and/or detection system, as applicable;

g) A plan showing the remote control for the fire doors, if applicable;

h) A plan showing the location and arrangement of the emergency stop for the oil fuel unit pumps and for closing the valves on the pipes from oil fuel tanks;

i) A plan of the fire alarm system;

j) Copies of the certificates of approval by National Authorities in respect of all 'A' and 'B' Class fire divisions, non-combustible materials and materials having low flame-spread characteristics, etc., which are to be used but have not been approved by IRS.

1.2.3 For fire extinguishing, the following plans are to be submitted:

a) A general arrangement plan showing the disposition of all the fire fighting equipment including the fire main, the fixed fire extinguishing systems in the cargo holds, on deck and in the machinery spaces; the disposition of the portable and non-portable extinguishers and the types used; and the position and details of the firemen's outfits;

b) A plan showing the layout and construction of the fire main, including the main and emergency fire pumps, isolating valves, pipe sizes and materials; the international shore connections and the cross connections to any other system;

c) A plan showing details of each fixed fire fighting system, including calculations for the quantities of the media used and the proposed rates of application.

1.2.4 For fire control, plans as required by Ch.5, 2.2.4.

1.3 Requirements applicable to existing ships

1.3.1 Unless expressly provided otherwise, for ships constructed before 1 July 2012 the requirements which are applicable under Chapter II-2 of the International Convention for the Safety of Life at Sea, 1974, as amended by resolutions MSC.1(XLV), MSC.6(48), MSC.13(57), MSC.22(59), MSC.24(60), MSC.27(61), MSC.31(63) MSC.57(67), MSC.99(73), MSC.134(76), MSC.194(80), MSC.201(81), MSC.216(82), MSC.256(84), MSC.269(85) and MSC.291(87) are to be complied with.

1.3.2 Ships constructed before 1 July 2002 are also to comply with:

.1 paragraphs 1.5.5 and 1.5.7 as appropriate; (regarding gas freeing of combination carriers and hydro-carbon gas detection systems in tankers);

.2 Ch.4, 2.3.4.2 to 2.3.4.5, 2.4.3 and Ch.5, (except 3.3.2.2 and 3.3.2.3) as appropriate, not later than the date of the first survey after 1 July 2002; (regarding emergency escape breathing devices and operational requirements for fire safety);

.3 Ch.3, 4.4.1.3 and 4.6.4 for new installations only; (regarding prohibition of Halon and safety of deep fat cooking equipment);

.4 Ch.3, 4.5.6 not later than 1 October 2005 for passenger ships of 2,000 gross tonnage and above. (regarding fixed local application fire fighting systems in machinery spaces);

.5 Ch.2, 2.3.1.3.2 (regarding linings, ceilings, partial bulkheads / decks for separating adjacent cabin balconies) and 2.3.4 (regarding furniture and furnishings on cabin balconies) to passenger ships not later than the date of the first survey after 1 July 2008; and

.6 Part 5, Ch 2, 8.2.6 (regarding portable instruments for gas measurement).

1.3.3 Ships constructed on or after 1 July 2002 and before 1 July 2010 are to comply with paragraphs 7.1.1, 7.4.4.2, 7.4.4.3 and 7.5.2.1.2 of SOLAS Regulation II-2/9, as adopted by Resolution MSC.99(73).

(Corresponding clauses 3.7.1.1, 3.7.4.4.2, 3.7.4.4.3 and 3.7.5.2.1.2 of Part 6, Chapter 3 of the previous versions (2002 to 2009) of these rules).

1.3.4 The following ships, with cargo spaces intended for the carriage of packaged dangerous goods, are to comply with Ch.7, Cl.2.3, except when carrying dangerous goods specified as classes 6.2 and 7 and dangerous goods in limited quantities* and excepted quantities** in accordance with Table 2.1 and 2.3 of Ch.7, Section 2, not later than the date of the first renewal survey on or after 1 January 2011:

* Refer to Chapter 3.4 of the IMDG Code.
** Refer to Chapter 3.5 of the IMDG Code.

a) cargo ships of 500 gross tonnage and upwards and passenger ships constructed on or after 1 September 1984 but before 1 January 2011; and

b) cargo ships of less than 500 gross tonnage constructed on or after 1 February 1992 but before 1 January 2011;

and not withstanding the above provisions:

c) cargo ships of 500 gross tonnage and upwards and passenger ships constructed on or after 1 September 1984 but before 1 July 1986 need not comply with Ch.7, Cl.2.3.3 provided that they comply with SOLAS Ch.II-2 regulation 54.2.3 as adopted by resolution MSC.1(XLV) (SOLAS 1981 amendments);

d) cargo ships of 500 gross tonnage and upwards and passenger ships constructed on or after 1 July 1986 but before 1 February 1992 need not comply with Ch.7, Cl.2.3.3 provided that they comply with SOLAS Ch.II-2 regulation 54.2.3 as adopted by resolution MSC.6(48) (SOLAS 1983 amendments);

e) cargo ships of 500 gross tonnage and upwards and passenger ships constructed on or after 1 September 1984 but before 1 July 1998 need not comply with Ch.7, Cl. 2.3.10.1 and Cl. 2.3.10.2; and

f) cargo ships of less than 500 gross tonnage constructed on or after 1 February 1992 but before 1 July 1998 need not comply with Ch.7, Cl.2.3.10.1 and 2.3.10.2.

g) cargo ships of 500 gross tonnage and upwards and passenger ships constructed on or after 1 February 1992 but before 1 July 2002 need not comply with Ch 7, 2.3.3 provided that they comply with SOLAS Ch.II-2 regulation 54.2.3 as adopted by resolution MSC.13(57)

h) cargo ships of 500 gross tonnage and upwards and passenger ships constructed on or after 1 September 1984 but before 1 July 2002 need not comply with Ch 7, 2.3.1, 2.3.5, 2.3.6, 2.3.9, provided that they comply with SOLAS Ch.II-2 regulations 54.2.1, 54.2.5, 54.2.6, 54.2.9 as adopted by resolution MSC.1(XLV).

1.3.5 Ships constructed before 01 July 2012 are to also comply with requirements of Chapter 3, 4.10.1.2.

1.3.6 Vehicle carriers constructed before 1 January 2016, including those constructed before 1 July 2012, are to comply with Pt 6, Ch 7, 4.2.2.

1.3.7 Tankers constructed before 1 January 2016, including those constructed before 1 July 2012, are to comply with Pt 6, Ch 5, 3.3.3 except 3.3.3.3.

1.3.8 Pt 6, Ch 2, 1.5.5.1.1 and 1.5.5.1.3 apply to ships constructed on or after 1 January 2002 but before 1 January 2016, and regulation 1.5.5.2.1 applies to all ships constructed before 1 January 2016

1.3.9 Pt.6, Ch.3, 4.5.1.2.2 applies to ships constructed before 1 January 2020, including those constructed before 1 July 2012.

1.4 Applicable requirements depending on ship type

Unless expressly provided otherwise:

.1 requirements not referring to a specific ship type apply to ships of all types; and

.2 requirements referring to "tankers" apply to tankers (see 3.48 for definition), however these are also subject to the requirements specified in paragraph 1.5 below.

1.5 Application of requirements for tankers

1.5.1 Requirements for tankers in this part are to apply to tankers carrying

a) crude oil or

b) petroleum products having a flashpoint not exceeding 60°C (closed cup test), as determined by an approved flashpoint apparatus and a Reid vapour pressure which is below the atmospheric pressure or

c) other liquid products having a similar fire hazard.

(Also see 3.48 for definition of 'tanker').

1.5.2 Where liquid cargoes other than those referred to in paragraph 1.5.1 or liquefied gases which introduce additional fire hazards are intended to be carried, additional safety measures are to be provided in accordance with the requirements of Pt.5, Ch.3 and Ch.4.

1.5.2.1 A liquid cargo with a flashpoint of not exceeding 60°C for which a regular foam firefighting system complying with the Fire Safety Systems Code is not effective, is considered to be a cargo introducing additional fire hazards in this context. The following additional measures are required:

.1 the foam is to be of alcohol resistant type;

.2 the type of foam concentrates for use in chemical tankers is to be approved by the National Statutory Authority (Refer to the Guidelines for performance and testing criteria and surveys of expansion foam concentrates for fire extinguishing systems for chemical tankers (MSC/Circ.799)); and

.3 the capacity and application rates of the foam extinguishing system is to comply with Pt.5, Ch.3, Sec.11, except that lower application rates may be accepted based on performance tests. For tankers fitted with inert gas systems, a quantity of foam concentrate sufficient for 20 min of foam generation may be accepted. (Refer to the Information on flashpoint and recommended fire-fighting media for chemicals to which neither the IBC nor BCH Codes apply (MSC/Circ. 553)).

1.5.2.2 For the purpose of this Section a liquid cargo with a vapour pressure greater than 1.013 bar absolute at 37.8°C is considered to be a cargo introducing additional fire hazards and ships carrying such substances are to comply with paragraph 15.14 of Pt.5, Ch.3. When such ships operate in restricted areas and at restricted times, the requirements for refrigeration systems may be waived in accordance with paragraph 15.14.3 of Pt.5, Ch.3.

1.5.3 Liquid cargoes with a flashpoint exceeding 60°C, other than oil products or liquid cargoes subject to the requirements of Pt.5, Ch.3, are considered to constitute a low fire risk, not requiring the protection of a fixed foam extinguishing system.

1.5.4 Tankers carrying petroleum products with a flashpoint exceeding 60°C are to comply with the requirements provided in Ch.3, 4.2.1.4.4 and 4.10.2.3 and the requirements for cargo ships other than tankers, except that, in lieu of the fixed fire extinguishing system required in Ch.3, 4.7 they are to be fitted with a fixed deck foam system which is to comply with the provisions of the Fire Safety Systems Code.

1.5.5 Combination carriers are not to carry cargoes other than oil unless all cargo tanks are empty of oil and gas freed or unless the arrangements in each case are approved in accordance with the Guidelines for inert gas systems (MSC/Circ.353), as amended by (MSC/Circ.387).

1.5.6 Chemical tankers and gas carriers are to comply with the requirements for tankers, except where alternative and supplementary arrangements are provided in accordance with Pt.5, Ch.3 and Ch.4 as appropriate.

1.5.7 The requirements of Pt.5, Ch.2, 6.4.2 and 6.4.5 and a system for continuous monitoring of the concentration of hydrocarbon gases are to be fitted on all tankers constructed before 1 July 2002 by the date of

the first scheduled dry-docking after 1 July 2002, but not later than 1 July 2005. Sampling points or detector heads are to be located in suitable positions in order that potentially dangerous leakages are to readily detected. When the hydrocarbon gas concentration reaches a pre-set level which is not to be higher than 10% of the lower flammable limit, a continuous audible and visual alarm signal is to be automatically effected in the pump-room and cargo control room to alert personnel to the potential hazard. However, existing monitoring systems already fitted having a pre-set level not greater than 30% of the lower flammable limit may be accepted.

Section 2

Fire Safety Objectives and Functional Requirements

2.1 Fire safety objectives

2.1.1 The fire safety objectives of this part are to:

.1 prevent the occurrence of fire and explosion;
.2 reduce the risk to life caused by fire;
.3 reduce the risk of damage caused by fire to the ship, its cargo and the environment;
.4 contain, control and suppress fire and explosion in the compartment of origin; and
.5 provide adequate and readily accessible means of escape for passengers and crew.

2.2 Functional requirements

2.2.1 In order to achieve the fire safety objectives set out in paragraph 2.1 above, the following functional requirements are embodied in the regulations of this chapter as appropriate:

.1 division of the ship into main vertical and horizontal zones by thermal and structural boundaries;

.2 separation of accommodation spaces from the remainder of the ship by thermal and structural boundaries;

.3 restricted use of combustible materials;

.4 detection of any fire in the zone of origin;

.5 containment and extinction of any fire in the space of origin;

.6 identification and protection of means of escape;

.7 access for fire-fighting;

.8 ready availability of fire-extinguishing appliances; and

.9 minimization of possibility of ignition of flammable cargo vapour.

2.3 Achievement of the fire safety objectives

2.3.1 The fire safety objectives set out in paragraph 2.1 above are to be achieved by ensuring compliance with the prescriptive requirements specified in chapters 2, 3, 4, 5 and 7, or by alternative design and arrangements which comply with chapter 6. A ship will be considered to meet the functional requirements set out in paragraph 2.2 and to achieve the fire safety objectives set out in paragraph 2.1 when either:

.1 the ship's designs and arrangements, as a whole complies with the relevant prescriptive requirements in chapters, 2, 3, 4, 5 and 7;

.2 the ship's designs and arrangements, as a whole, have been reviewed and approved in accordance with chapter 6;

.3 part(s) of the ship's design and arrangements have been reviewed and approved in accordance with chapter 6 and the remaining parts of the ship comply with the relevant prescriptive requirements in chapters 2, 3, 4, 5 and 7.

Section 3

Definitions

3. For the purpose of this chapter, unless expressly provided otherwise, the following definitions are to apply:

3.1 **Accommodation spaces** are those spaces used for public spaces, corridors, lavatories, cabins, offices, hospitals, cinemas, games and hobbies rooms, barber shops, pantries containing no cooking appliances and similar spaces.

3.2 **'A' class divisions** are those divisions formed by bulkheads and decks which comply with the following criteria:

1. they are constructed of steel or other equivalent material;

2. they are suitably stiffened;

3. they are so constructed as to be capable of preventing the passage of smoke and flame to the end of the one-hour standard fire test;

4. they are insulated with approved non-combustible materials such that the average temperature of the unexposed side will not rise more than 140°C above the original temperature, nor will the temperature, at any one point, including any joint, rise more than 180°C above the original temperature within the time listed below:

Class 'A-60'	60 minutes
Class 'A-30'	30 minutes
Class 'A-15'	15 minutes
Class 'A-0'	0 minutes

IR4. Insulated “A” class bulkheads and decks used on board ships, including the means of affixing the insulation to the "A” class structural members, are to be consistent with the materials, details and arrangements used during the tests and documented in the test reports issued for the approval for that insulating material.

5. a test of prototype bulkhead or deck will be required to ensure that it meets the above requirements for integrity and temperature rise.

3.3 **Atriums** are public spaces within a single main vertical zone spanning three or more open decks.

3.4 **'B' class divisions** are those divisions formed by bulkheads, decks, ceilings or linings which comply with the following criteria:

a) they are so constructed as to be capable of preventing the passage of flame to the end of the first half hour of the standard fire test;

b) they have an insulation value such that the average temperature of the unexposed side will not rise more than 140°C above the original temperature nor will the temperature at any point, including any joint, rise more than 225°C above the original temperature, within the time listed below:

Class 'B-15'	15 minutes
Class 'B-0'	0 minutes

c) they are constructed of approved non-combustible materials and all materials entering into the construction and erection of 'B' class divisions are to be non-combustible. A division consisting of a non-combustible core and combustible veneers may be accepted as a B class division, provided that the non combustible core and veneers are tested in accordance with relevant parts of Fire Test Procedure Code described in 3.23 of this section;

d) a test of a prototype division will be required to ensure that it meets the above requirements for integrity and temperature rise.

3.5 **Bulkhead deck** is the uppermost deck upto which the transverse watertight bulkheads are carried.

3.6 **Cargo area** is that part of the ship that contains cargo holds, cargo tanks, slop tanks and cargo pump-rooms including pump-rooms, cofferdams, ballast and void spaces adjacent to cargo tanks and slop tanks and also deck areas throughout the length and breadth of the

part of the ship over the above mentioned spaces. (Also see Ch.2, IR1.5.1.1.2).

3.7 **A cargo ship** is any ship which is not a passenger ship.

3.8 **Cargo spaces** are spaces used for cargo, cargo oil tanks, tanks for other liquid cargo and trunks to such spaces.

3.9 **Central Control station** is a control station in which the following control and indicator functions are centralized:

a) fixed fire detection and alarm systems;

b) automatic sprinklers, fire detection and alarm systems;

c) fire door indicator panels;

d) fire door closures;

e) watertight door indicator panels;

f) watertight door closures;

g) ventilation fans;

h) general/fire alarms;

i) communication systems including telephones; and

j) microphones to public address system.

3.10 **'C' class divisions** are divisions constructed of approved non- combustible materials. They need meet neither requirements relative to the passage of smoke and flame nor limitations relative to the temperature rise. A division consisting of a non-combustible core and combustible veneers may be accepted as a C class division, provided that the non-combustible core and veneers are tested in accordance with relevant parts of Fire Test Procedure Code.

3.11 **Chemical tanker** is a cargo ship constructed or adapted and used for the carriage in bulk of any liquid product of a flammable nature listed in Pt.5, Ch.3 of the Rules.

3.12 **Closed ro/ro spaces** are Ro/Ro spaces which are neither open Ro/Ro spaces nor weather decks.

3.13 **Closed vehicle spaces** are vehicle spaces which are neither open vehicle spaces or weather decks.

3.14 **A combination carrier** is a cargo ship designed to carry both oil and solid cargoes in bulk.

3.15 C**ombustible material** is any material other than a non-combustible material.

3.16 **Continuous 'B' class ceilings or linings** are those 'B' class ceilings or linings which terminate only at an 'A' or 'B' class division.

3.17 **Continuously manned central control station** is a central control station which is continuously manned by a responsible member of the crew.

3.18 **Control-stations** are those spaces in which the ship's radio or main navigating equipment or the emergency source of power is located or where the fire recording or fire control equipment is centralized. Spaces where the fire recording or fire control equipment is centralized are also considered to be a **Fire Control Station**.
If the CO_2 system discharge pipes are used for the sample extraction smoke detection system, the control panel can be located in the CO_2 room provided that an indicating unit is located on the navigation bridge. Such arrangements are considered to satisfy the requirements of Pt.6 Ch 8, 10.2.4.1.2.

Notes :

1) Indicating unit has the same meaning as repeater panel and observation of smoke should be made either by electrical means or by visual on repeater panel.

2) Main navigational equipment includes, in particular, the steering stand and the compass, radar and direction-finding equipment.

3) Steering gear rooms containing an emergency steering position are not considered to be control stations.

4) Where, in the Rules in Pt.6, relevant to fixed fire extinguishing systems, there are no specific requirements for the centralization within a control station of major components of the system, such major components may be placed in

spaces which are not considered to be a control station.

5) Spaces containing, for instance, the following battery sources are to be regarded as control stations regardless of battery capacity:

.1 emergency batteries in separate battery room for power supply from black-out till start of emergency generator.

.2 emergency batteries in separate battery room as reserve source of energy to radiotelegraph installation.

.3 batteries for start of emergency generator and

.4 in general, all emergency batteries required in accordance with Pt.4, Ch.8, 2.8.

3.19 **Crude oil** is any oil occurring naturally in the earth whether or not treated to render it suitable for transportation and includes :

a) crude oil from which certain distillate fractions may have been removed; and

b) crude oil to which certain distillate fractions may have been added.

3.20 **Dangerous goods** are those goods referred to in SOLAS Regulation VII/2.

3.21 **Deadweight** is the difference in tonnes between the displacement of a ship in water of a specific gravity of 1.025 at the load water-line corresponding to the assigned summer freeboard and the lightweight of the ship.

3.22 **Fire safety systems code** means the International Code for Fire Safety Systems as adopted by the Maritime Safety Committee of the IMO by resolution MSC.98(73) and as may be amended by IMO (Section 21 to 35). Refer Ch.8.

3.23 **Fire test procedures code** means the International Code for Application of Fire Test Procedures, 2010 (2010 FTP Code) as adopted by the Maritime Safety Committee of IMO by resolution MSC.307(88), and as may be amended by IMO.

3.24 **Flashpoint** is the temperature in degrees Celsius (closed cup test) at which a product will give off enough flammable vapour to be ignited, as determined by an approved flashpoint apparatus.

3.25 **Gas carrier** is a cargo ship constructed or adapted and used for the carriage in bulk of any liquefied gas or other products of a flammable nature listed in Pt.5, Ch.4.

3.26 **Helideck** is a purpose-built helicopter landing area located on a ship including all structure, fire-fighting appliances and other equipment necessary for the safe operation of helicopters.

3.27 **Helicopter facility** is a helideck including any refuelling and hangar facilities.

3.28 **Lightweight** is the displacement of a ship in tonnes without cargo, fuel, lubricating oil, ballast water, fresh water and feed water in tanks, consumable stores, and passengers and crew and their effects.

IR3.28 The weight of mediums on board for the fixed fire-fighting systems (e.g. freshwater, CO_2, dry chemical powder, foam concentrate, etc.) are to be included in the lightweight and lightship condition.

3.29 **Low flame spread** means that the surface thus described will adequately restrict the spread of flame, this being determined in accordance with Fire Test Procedures Code.

3.30 **Machinery spaces** are all machinery spaces of category 'A' and all other spaces containing propulsion machinery, boilers, oil fuel units, steam and internal combustion engines, generators and major electrical machinery, oil filling stations, refrigerating, stabilizing, ventilation and air conditioning machinery, and similar spaces, and trunks to such spaces.

IR3.30 In cases where urea or sodium hydroxide solution tanks for selective catalytic reduction (SCR) systems, exhaust gas recirculation (EGR) systems or exhaust gas cleaning systems (EGCS) are installed in a space separated from engine room, in determining fire integrity of divisions, the solution tank space should be considered as "*similar spaces*" in the definition of "Machinery spaces" above.

3.31 **Machinery spaces of category 'A'** are those spaces and trunks to such spaces which contain:

a) internal combustion machinery used for main propulsion; or

b) internal combustion machinery used for purposes other than main propulsion where such machinery has in the aggregate a total power output of not less than 375 [kW]; or

c) any oil-fired boiler or oil fuel unit, or any oil-fired equipment other than boilers, such as inert gas generators, incinerators, etc.

3.32 **Main vertical zones** are those sections into which the hull, superstructure, and deckhouses are divided by 'A' class divisions, the mean length of which on any deck does not in general exceed 40 [m].

3.33 **Non-combustible material** is a material which neither burns nor gives off flammable vapours in sufficient quantity for self-ignition when heated to approximately 750°C, this being determined in accordance with the Fire Test Procedures Code.

3.34 **Oil fuel unit** is the equipment used for the preparation of oil fuel heated or not, for delivery to an oil fired boiler (including inert gas generators, incinerator, waste disposal unit, etc.) or to an internal combustion engine (including gas turbines) and includes any oil pressure pumps, filters and heaters dealing with oil at a pressure of more than 0.18 [N/mm^2]. Oil fuel transfer pumps are not considered as oil fuel units.

3.35 **Open ro/ro spaces** are those ro/ro spaces that are either open at both ends or have an opening at one end and are provided with adequate natural ventilation effective over their entire length through permanent openings distributed in the side plating or deckhead or from above, having a total area of at least 10% of the total area of the spaces sides.

3.36 **Open vehicle spaces** are those vehicle spaces either open at both ends, or have an opening at one end and are provided with adequate natural ventilation effective over their entire length through permanent openings distributed in the side plating or deckhead or from above, having a total area of at least 10% of the total area of the space sides.

3.37 **A passenger ship** is a ship which carries more than twelve passengers, a passenger being every person other than :

- the master and members of the crew or other persons employed or engaged in any capacity on board a ship on the business of the ship; and

- a child under one year of age.

3.38 **Prescriptive requirements** means the construction characteristics, limiting dimensions or fire safety systems specified in chapters, 2, 3, 4, 5 or 7.

3.39 **Public spaces** are those portions of the accommodation which are used for halls, dining rooms, lounges and similar permanently enclosed spaces.

3.40 **Rooms containing furniture and furnishings of restricted fire risk** are those rooms containing furniture and furnishing of restricted fire risk (whether cabins, public spaces, offices or other types of accommodation) in which :

a) case furniture such as desks, wardrobes, dressing tables, bureaux, dressers, are constructed entirely of approved non-combustible materials, except that a combustible veneer not exceeding 2 [mm] may be used on the working surface of such articles;

b) free-standing furniture such as chairs, sofas, tables, is constructed with frames of non-combustible materials;

c) draperies, curtains and other suspended textile materials have qualities of resistance to the propagation of flame not inferior to those of wool of mass 0.8 [kg/m^2], this being determined in accordance with the Fire Test Procedures Code;

d) floor coverings have low flame spread characteristics;

e) exposed surfaces of bulkheads, linings and ceilings have low flame- spread characteristics;

f) upholstered furniture has qualities of resistance to the ignition and propagation of flame, this being determined in accordance with the Fire Test Procedures Code; and

g) bedding components have qualities of resistance to the ignition and propagation of flame, this being determined in accordance with the Fire Test Procedures Code.

3.41 **Ro/ro spaces** are spaces not normally subdivided in any way and normally extending to either a substantial length or the entire length of the ship in which motor vehicles with fuel in their tanks for their own propulsion and/or goods (packaged or in bulk, in or on rail or road cars, vehicles (including road or rail tankers), trailers, containers, pallets, demountable tanks or in or on similar stowage units or other receptacles) can be loaded and unloaded normally in horizontal direction.

3.42 **A Ro-ro passenger ship** means a passenger ship with ro-ro cargo spaces or special category spaces.

3.43 **Steel or other equivalent material**. Where the words **steel or other equivalent material** occur, 'equivalent material' means any non- combustible material which, by itself or due to insulation provided, has structural and integrity properties equivalent to steel at the end of the applicable fire exposure to the standard fire test (e.g. aluminium alloy with appropriate insulation).

3.44 **Sauna** is a hot room with temperatures normally varying between 80°C - 120°C where the heat is provided by a hot surface (e.g. by an electrically-heated oven). The hot room may also include the space where the oven is located and adjacent bathrooms.

3.45 **Service spaces** are those spaces used for galleys, pantries containing cooking appliances, lockers, mail and specie rooms, store rooms, workshops other than those forming part of machinery spaces, and similar spaces and trunks to such spaces.

3.46 **Special category spaces** are those enclosed vehicle spaces above and below the bulkhead deck, into and from which vehicles can be driven and to which passengers have access. Special category spaces may be accommodated on more than one deck provided that the total overall clear height for vehicles does not exceed 10 [m].

3.47 **A standard fire test** is one in which specimens of the relevant bulkheads or decks are exposed in a test furnace to temperatures corresponding approximately to the standard time temperature curve. The test method is to be in accordance with the Fire Test Procedures Code.

3.48 **A tanker** is a cargo ship constructed or adapted for the carriage in bulk of liquid cargoes of a flammable nature.

3.49 **Vehicle spaces** are cargo spaces intended for carriage of motor vehicles with fuel in their tanks for their own propulsion.

3.50 **Weather deck** is a deck which is completely exposed to the weather from above and from at least two sides.

3.51 **Cabin balcony** is an open deck space which is provided for the exclusive use of the occupants of a single cabin and has direct access from such a cabin.

3.52 **Safe area** in the context of a casualty and from the perspective of habitability is any area(s) which is not flooded or which is outside the main vertical zone(s) in which a fire has occurred such that it can safely accommodate all persons onboard to protect them from hazards to life or health and provide them with basic services.

3.53 **Safety centre** is a control station dedicated to the management of emergency situations. Safety systems' operation, control and/or monitoring are an integral part of the safety centre.

3.54 **Fire damper** is, for the purpose of implementing Pt 6, Ch 3, 3.7, a device installed in a ventilation duct, which under normal conditions remains open allowing flow in the duct, and is closed during a fire, preventing the flow in the duct to restrict the passage of fire. In using the above definition the following terms may be associated:

.1 automatic fire damper is a fire damper that closes independently in response to exposure to fire products;

.2 manual fire damper is a fire damper that is intended to be opened or closed by the crew by hand at the damper itself; and

.3 remotely operated fire damper is a fire damper that is closed by the crew through a control located at a distance away from the controlled damper.

3.55 **Smoke damper** is, for the purpose of implementing Pt 6, Ch 3, 3.7, a device installed in a ventilation duct, which under

normal conditions remains open allowing flow in the duct, and is closed during a fire, preventing the flow in the duct to restrict the passage of smoke and hot gases. A smoke damper is not expected to contribute to the integrity of a fire rated division penetrated by a ventilation duct. In using the above definition the following terms may be associated:

.1 automatic smoke damper is a smoke damper that closes independently in response to exposure to smoke or hot gases;

.2 manual smoke damper is a smoke damper intended to be opened or closed by the crew by hand at the damper itself; and

.3 remotely operated smoke damper is a smoke damper that is closed by the crew through a control located at a distance away from the controlled damper.

3.56 **Vehicle carrier** means a cargo ship which only carries cargo in ro-ro spaces or vehicle spaces, and which is designed for the carriage of unoccupied motor vehicles without cargo.

3.57 **Helicopter landing area** is an area on a ship designated for occasional or emergency landing of helicopters but not designed for routine helicopter operations.

3.58 **Winching area** is a pick-up area provided for the transfer by helicopter of personnel or stores to or from the ship, while the helicopter hovers above the deck.

Table 1 : List of IMO MSC Circulars and Assembly Resolutions referred to in this part

MSC Circ./Assembly Resolution No.	***Description***
MSC/Circ.353	*Guidelines for inert gas systems*
MSC/Circ.387	*Revised guidelines for inert gas systems (MSC/Circ.353), as amended by MSC/Circ.387*
MSC/Circ.450/Rev.1	*Revised factors to be taking into consideration when designing cargo tank venting and gas-freeing arrangements*
MSC/Circ.451	*Guidance concerning the location of fire control plans for assistance of shoreside fire-fighting personnel*
MSC/Circ.485	*Clarification of inert gas system requirements under the Convention*
MSC/Circ. 553	*Information on flashpoint and recommended fire-fighting media for chemicals to which neither the IBC nor BCH Codes apply*
MSC.1/Circ.1312 as corrected by MSC.1/Circ.1312 Corr.1	*Revised guidelines for the performance and testing criteria and surveys of foam concentrates for fixed fire-extinguishing systems*
MSC/Circ.608/Rev.1	*Interim guidelines for open-top containerships*
MSC/Circ.670	*Guidelines for performance and testing criteria and surveys of high expansion foam concentrates for fire-extinguishing systems*
MSC.1/Circ.1395/Rev.6	*List of solid bulk cargoes for which for which a fixed gas fire-extinguishing system may be exempted or for which a fixed gas fire-extinguishing system is ineffective*
MSC/Circ.777	*Indication of the "assembly stations" in passenger ships*
MSC/Circ.798	*Guidelines for the performance and testing criteria, and surveys of medium expansion foam concentrates for fixed fire-extinguishing systems*
MSC/Circ.848 amended by MSC.1/Circ.1267	*Revised guidelines for the approval of equivalent fixed gas fire-extinguishing systems, as referred to in SOLAS 74, for machinery spaces and cargo pump rooms*
MSC/Circ.849	*Guidelines for the performance, location, use and care of emergency escape breathing devices*
MSC.1/Circ.1432 amended by MSC.1/Circ.1516	*Revised guidelines for the maintenance and inspection of fire protection systems and appliances*
MSC.1/Circ.1533	*Revised guidelines on evacuation analysis for new and existing passenger ships*
MSC/Circ.913 *MSC.1/Circ.1387 and MSC.1/Circ.1387/Corr.1*	*Guidelines for the approval of fixed water-based local application fire-fighting systems for use in category A machinery spaces* *Revised guidelines for the approval of fixed water-based local application fire-fighting systems for use in category A machinery spaces. (MSC.1/Circ.1387 and MSC.1/Circ.1387/Corr.1 Supersedes MSC/Circ.913, except that fire and component tests previously conducted in accordance with MSC/Circ.913 remain valid for the approval of new systems. Existing fixed water-based local application fire-fighting systems approved and installed based on MSC/Circ.913 should be permitted to remain in service as long as they are serviceable.)*
MSC/Circ.917 as corrected by MSC/Circ.917/Corr.1	*"Guidelines on fire safety construction in accommodation areas"*

MSC Circ./Assembly Resolution No.	***Description***
MSC/Circ.1002 as corrected by MSC/Circ.1002/Corr.1 to Corr.3 and as amended by MSC.1/Circ.1552	*Guidelines on alternative design and arrangements for fire safety*
MSC/Circ.1165 Amended MSC.1/Circ.1237 and MSC.1/Circ.1269	*Revised guidelines for the approval of fixed water based fire extinguishing systems for machinery spaces and cargo pump rooms*
MSC.1/Circ.1242	*Guidelines for the approval of fixed fire detection and fire alarm systems for cabin balconies*
MSC.1/Circ.1266	*Document of compliance with the special requirements for ships carrying dangerous goods under the provisions of regulation II-2/19 SOLAS 74, as amended*
MSC.1/Circ.1268	*Guidelines for the approval of fixed pressure water spraying and water based fire extinguishing systems for cabin balconies*
MSC.1/Circ.1430/Rev.3	*Revised Guidelines for the design and approval of fixed water-based fire-fighting systems for Ro-Ro spaces and special category spaces*
MSC.1/Circ/1275 as corrected by MSC.1/Circ.1275/Corr.1	*Unified interpretations of SOLAS Chapter II-2 on the number and arrangement of portable fire extinguishers on board ships*
MSC.1/Circ.1276/Rev.1	*Unified interpretations of SOLAS Ch.II-2*
MSC.1/Circ.1320	*Guidelines for the drainage of fire-fighting water from closed vehicle and ro-ro spaces and special category spaces of passenger and cargo ships*
A.123(V)	*Recommendation on fixed fire-extinguishing systems for special cargo spaces*
A.434(XI)	*Code of safe practice of solid bulk cargoes (BC Code)*
A.565(14)	*Recommended procedures to prevent the illegal or accidental use of low flashpoint cargo oil as fuel*
A.567(14)	*Regulation for inert gas systems on chemical tankers*
A.654(16)	*Graphical symbols for fire control plans (Refer to A.952(23) for ships constructed on or after 01 January 2004)*
A.752(18)	*Guidelines for the evaluation, testing and application of low-location lighting on passenger ships*
A.756(18)	*Guidelines on the information to be provided with fire control plans and booklets*
A.760(18) as amended by MSC.82(70)	*Symbols related to life-saving appliances and arrangements*
A.800(19) as amended by MSC.265(84) and MSC.284(86)	*Guidelines for approval of sprinkler systems*
A.1021(26)	*Code on Alerts and Indicators, 2009*
A.951(23)	*Improved guidelines for marine portable fire extinguishers*
Note : The above MSC Circulars/Assembly Resolutions are available on the IMO Website www.docs.imo.org. IRS may also be contacted if necessary.	

End of Chapter

Chapter 2

Prevention of Fire and Explosion

Contents

Section

1 *Probability of Ignition*
2 *Fire Growth Potential*
3 *Smoke Generation Potential and Toxicity*

Section 1

Probability of Ignition

1.1 Purpose

The purpose of this section is to prevent the ignition of combustible materials or flammable liquids. For this purpose the following functional requirements are to be met:

.1 means are to be provided to control leaks of flammable liquids;

.2 means are to be provided to limit the accumulation of flammable vapours;

.3 the ignitability of combustible materials is to be restricted;

.4 ignition sources are to be restricted;

.5 ignition sources are to be separated from combustible materials and flammable liquids; and

.6 the atmosphere in cargo tanks is to be maintained out of the explosive range.

1.2 Arrangements for oil fuel, lubrication oil and other flammable oils

1.2.1 Limitations in the use of oils as fuel

The following limitations are to apply to the use of oil as fuel:

.1 except as otherwise permitted by this paragraph, no oil fuel with a flashpoint of less than 60°C is to be used; (Refer to IMO Resolution A.565(14));

.2 in emergency generators oil fuel with a flashpoint of not less than 43°C may be used;

.3 the use of oil fuel having a flashpoint of less than 60°C but not less than 43°C may be permitted (e.g. for feeding the emergency fire pump's engines and the auxiliary machines which are not located in the machinery spaces of category A) subject to the following:

.3.1 fuel oil tanks except those arranged in double bottom compartments are to be located outside the machinery spaces of category A;

.3.2 provisions for the measurement of oil temperature are provided on the suction pipe of the oil fuel pump;

.3.3 stop valves and/or cocks are provided on the inlet side and outlet side of the oil fuel strainers; and

.3.4 pipe joints of welded construction or of circular cone type or spherical type union joint are applied as much as possible;

.4 in cargo ships, to which SOLAS, Chapter II-1, Part G is not applicable, the use of oil fuel having a lower flashpoint than otherwise specified in 1.2.1.1, for example crude oil, may be permitted provided that such fuel is not stored in any machinery space and subject to

the approval by the Administration/IRS of the complete installation; and

.5 in ships, to which SOLAS, Chapter II-1, Part G is applicable, the use of oil fuel having a lower flashpoint than otherwise specified in 1.2.1.1 is allowed.

1.2.2 Arrangements for oil fuel

In a ship in which oil fuel is used, the arrangements for the storage, distribution and utilization of the oil fuel are to be such as to ensure the safety of the ship and persons on board and are to at least comply with the following provisions.

1.2.2.1 Location of oil fuel systems

As far as practicable, parts of the oil fuel system containing heated oil under pressure exceeding 0.18 [N/mm^2] are not to be placed in a concealed position such that defects and leakage cannot readily be observed. The machinery spaces in way of such parts of the oil fuel system are to be adequately illuminated.

1.2.2.2 Ventilation of machinery spaces

The ventilation of machinery spaces is to be sufficient under normal conditions to prevent accumulation of oil vapour. Also refer Pt.4, Ch.1, Sec.2.4.

1.2.2.3 Oil fuel tanks

.1 Fuel oil, lubrication oil and other flammable oils are not be carried in forepeak tanks, Also refer Pt.3, Ch.10, Cl.2.7.1.

.2 As far as practicable, oil fuel tanks are to be part of the ship's structure and are to be located outside machinery spaces of category A. Where oil fuel tanks, other than double bottom tanks, are necessarily located adjacent to or within machinery spaces of category A, at least one of their vertical sides is to be contiguous to the machinery space boundaries and they are preferably to have a common boundary with the double bottom tanks and the area of the tank boundary common with the machinery spaces are to be kept to a minimum. Where such tanks are situated within the boundaries of machinery spaces of category A they are not to contain oil fuel having a flashpoint of less than 60°C. In general, the use of free-standing oil fuel tanks is to be avoided. When such tanks are employed their use is to be prohibited in category A machinery spaces on passenger ships. Where permitted, they are to be placed in an oil-tight spill tray of ample size having a suitable drain pipe leading to a suitably sized spill oil tank.

.3 No oil fuel tank is to be situated where spillage or leakage therefrom can constitute a fire or explosion hazard by falling on heated surfaces.

.4 For pipes connected to tanks, refer Pt.4, Ch.3, Cl.4.4.4, 4.4.5, 4.4.6.

.5 Safe and efficient means of ascertaining the amount of oil fuel contained in any oil fuel tank are to be provided. For details, refer Pt.4, Ch.3, Sec.3.4.

.5.1 Where sounding pipes are used, they are not to terminate in any space where the risk of ignition of spillage from the sounding pipe might arise. In particular, they are not to terminate in passenger or crew spaces. As a general rule, they are not to terminate in machinery spaces. However, where IRS considers that these latter requirements are impracticable, sounding pipes may terminate in machinery spaces subject to following requirements being met:

a) an oil-level gauge is provided meeting the requirements of .5.2 below;

b) the sounding pipes terminate in locations remote from ignition hazards unless precautions are taken, such as the fitting of effective screens, to prevent the oil fuel in the case of spillage through the terminations of the sounding pipes from coming into contact with a source of ignition; and

c) the termination of sounding pipes are fitted with self-closing blanking devices and with a small-diameter self-closing control cock located below the blanking device for the purpose of ascertaining before the blanking device is opened that oil fuel is not present. Provisions are to be made so as to ensure that any spillage of oil fuel through the control cock involves no ignition hazard.

(Also refer Pt.4, Ch.3, 3.4.3 and 3.4.4).

IR.5.1.1 Oil level gauge referred at .5.1a) above need not be provided for tanks (other than double bottom tanks) having overflow system.

IR.5.1.2 The short sounding pipes are to be arranged in such a way that overflow or oil spray will not reach the vicinity of any of the machinery components like boilers, preheaters, heated surfaces, electric generators or motors with commutator or collector rings or electric appliances.

IR.5.1.3 The self-closing blanking devices are to be cocks having cylindrical plugs with weight loaded levers permanently attached and with approved arrangements for opening.

.5.2 Other oil-level gauges may be used in place of sounding pipes as per details given in Pt.4, Ch.3, Cl.3.4.2.

.5.3 The means prescribed in paragraph .5.2 above are to be maintained in the proper condition to ensure their continued accurate functioning in service.

1.2.2.4 Prevention of overpressure

Provisions are to be made to prevent overpressure in any oil tank or in any part of the oil fuel system, including the filling pipes served by pumps on board. Air and overflow pipes and relief valves are to discharge to a position where there is no risk of fire or explosion from the emergence of oils and vapour and are not to lead into crew spaces, passenger spaces nor into special category spaces, closed ro-ro cargo spaces, machinery spaces or similar spaces. Also refer Pt.4, Ch.3, Sections 3 and 4.

1.2.2.5 Oil fuel piping

.1 Oil fuel pipes and their valves and fittings are to be of steel or other approved material, except that restricted use of flexible pipes are permissible in positions where IRS is satisfied that they are necessary. Such flexible pipes and end attachments are to be of approved fire-resisting materials of adequate strength and are to be constructed to the satisfaction of IRS. For valves, fitted to oil fuel tanks and which are under static pressure, steel or spheroidal-graphite cast iron may be accepted. However, ordinary cast iron valves may be used in piping systems where the design pressure is lower than 7 bar and the design temperature is below 60°C.

IR.1 Materials other than steel may be assessed in relation to the risk of fire associated with the component and its installation. The use of materials other than steel is considered acceptable for the following applications:

1. internal pipes which cannot cause any release of flammable fluid onto the machinery or into the machinery space in case of failure, or

2. components that are only subject to liquid spray on the inside when the machinery is running, such as machinery covers, rocker box covers, camshaft end covers, inspection plates and sump tanks. It is a condition that the pressure inside these components and all the elements contained therein is less than 0.18 [N/mm^2] and that wet sumps have a volume not exceeding 100 [litres], or

3. components attached to machinery which satisfy fire test criteria according to standard ISO 19921:2005/19922:2005 or other standards acceptable to the Administration, and which retain mechanical properties adequate for the intended installation.

.2 For jacketing of external high-pressure fuel delivery lines between the high-pressure fuel pumps and fuel injectors, refer Pt.4, Ch.4, Sec.4.8.5.

.3 Oil fuel lines are not to be located immediately above or near units of high temperature including boilers, steam pipelines, exhaust manifolds, silencers or other equipment required to be insulated by 1.2.2.6. As far as practicable, oil fuel lines are to be arranged far away from hot surfaces, electrical installations or other sources of ignition and are to be screened or otherwise suitably protected to avoid oil spray or oil leakage onto the sources of ignition. The number of joints in such piping systems are to be kept to a minimum.

.4 For requirements for components of diesel engine fuel system refer Pt.4, Ch.4, 4.5.8.2.

.5 For multi-engine installations which are supplied from the same fuel source, refer to Pt.4, Ch.3, 4.10.2.4.

.6 For piping through accommodation and service spaces, refer Pt.4, Ch.3, 4.3.10.

1.2.2.6 Protection of high temperature surfaces

Refer Pt.4, Ch.1, Sec.2.3.

1.2.3 Arrangements for lubricating oil

1.2.3.1 The arrangements for the storage, distribution and utilization of oil used in pressure lubrication systems are to be such as to ensure the safety of the ship and persons onboard. The arrangements made in machinery spaces of category A and whenever practicable in other machinery spaces are at least to comply with the provisions of paragraphs 1.2.2.1, 1.2.2.3.3, 1.2.2.3.4, 1.2.2.3.5, 1.2.2.4, 1.2.2.5.1, 1.2.2.5.3 and 1.2.2.6, except that:

a) this does not preclude the use of sight-flow glasses in lubricating systems provided that they are shown by testing to have a suitable degree of fire resistance; and

b) sounding pipes may terminate in machinery spaces; however, the requirements of 1.2.2.3.5.1a) and 1.2.2.3.5.1.c) need not be applied on condition that the sounding pipes are fitted with appropriate means of closure.

1.2.3.2 For valves on lubricating oil tanks, refer to Pt.4, Ch.3, 8.6.1.

1.2.4 Arrangements for other flammable oils

1.2.4.1 The arrangements for the storage, distribution and utilization of other flammable oils employed under pressure in power transmission systems, control and activating systems and heating systems are to be such as to ensure the safety of the ship and persons onboard. Suitable oil collecting arrangements for leaks are to be fitted below hydraulic valves and comply with the provisions of 1.2.2.3.3, 1.2.2.3.5, 1.2.2.5.3 and 1.2.2.6 and with the provisions of 1.2.2.4 and 1.2.2.5.1 in respect of strength and construction.

1.2.5 Arrangements for oil fuel in periodically unattended machinery spaces

In addition to the requirements of 1.2.1 to 1.2.4, the oil fuel and lubricating oil systems in a periodically unattended machinery space are to comply with the following:

a) where daily service oil fuel tanks are filled automatically, or by remote control, means are to be provided to prevent overflow spillages. Other equipment which treat flammable liquids automatically (e.g. oil fuel purifiers) which, whenever practicable, are to be installed in a special space reserved for purifiers and their heaters are to have arrangements to prevent overflow spillages;

and

b) where daily service oil fuel tanks or settling tanks are fitted with heating arrangements, a high temperature alarm is to be provided if the flashpoint of the oil fuel can be exceeded.

1.3 Arrangements for gaseous fuel for domestic purpose

1.3.1 Gaseous fuel systems used for domestic purposes are to be approved by IRS. Storage of gas bottles are to be located on the open deck or in a well-ventilated space which opens only to the open deck.

IR1.3.1 A portion of open deck, recessed into a deck structure, machinery casing, deck house, etc., utilized for the exclusive storage of gas bottles is considered acceptable for the purpose of 1.3.1 provided that:

.1 such a recess has an unobstructed opening, except for small appurtenant structures, such as opening corner radii, small sills, pillars, etc. The opening may be provided with grating walls and door;

.2 the depth of such a recess is not greater than 1 m.

A portion of open deck meeting the above is to be considered as open deck in applying tables 3.1 to 3.8 of Ch.3.

1.4 Miscellaneous items of ignition sources and ignitability

1.4.1 Electric radiators

1.4.1.1 Electric radiators, if used are to be fixed in position and so constructed as to reduce fire risks to a minimum. No such radiators are to be fitted with an element so exposed that clothing, curtains, or other similar materials can be scorched or set on fire by heat from the element.

1.4.2 Waste receptacles

1.4.2.1 Waste receptacles are to be constructed of non-combustible materials with no openings in the sides or bottom.

IR1.4.2.1 Waste receptacles of combustible materials may however be used in galleys, pantries, bars, garbage handling or storage spaces and incinerator rooms provided they are intended purely for the carriage of wet waste, glass bottles and metal cans and are suitably marked.

1.4.3 Insulation surfaces protected against oil penetration

1.4.3.1 In spaces where penetration of oil products is possible, the surface of insulation is to be impervious to oil or oil vapours. Also refer Pt.4, Ch.1, Cl. 2.3.1.

1.4.4 Primary deck coverings

1.4.4.1 Primary deck coverings, if applied, within accommodation and service spaces and control stations or if applied on cabin balconies of passenger ships constructed on or after 1 July 2008 are to be of an approved material which will not readily ignite, or give rise to toxic or explosive hazards at elevated temperatures, this being determined in accordance with the Fire Test Procedures Code.

1.4.5 Films

1.4.5.1 Cellulose-nitrate based films are not to be used for cinematograph installations.

1.5 Cargo areas of tankers

1.5.1 Separation of cargo oil tanks

1.5.1.1 Cargo pump-rooms, cargo tanks, slop tanks and cofferdams (see Note below) are to be positioned forward of machinery spaces. However, oil fuel bunker tanks need not be forward of machinery spaces. Cargo tanks and slop tanks are to be isolated from machinery spaces by cofferdams, cargo pump-rooms, oil bunker tanks or ballast tanks. Pump-rooms containing pumps and their accessories for ballasting those spaces situated adjacent to cargo tanks and slop tanks and pumps for oil fuel transfer are to be considered as equivalent to a cargo pump-room within the context of these requirements provided that such pump-rooms have the same safety standard as that required for cargo pump-rooms. Pump-rooms intended solely for ballast or oil fuel transfer, however, need not comply with the requirements of Ch.3, Sec.4.9. The lower portion of the pump-room may be recessed into machinery spaces of category A to accommodate pumps, provided that the deck head of the recess is in general not more than one third of the moulded depth above the keel, except that in the case of ships of not more than 25,000 tonnes deadweight, where it can be demonstrated that for reasons of access and satisfactory piping arrangements this is impracticable a recess in excess of such height, but not exceeding one half of the moulded depth above the keel may be permitted.

IR1.5.1.1.1 Pump rooms intended solely for ballast transfer or fuel oil transfer are also not required to comply with the requirement of 1.5.10. The requirements of 1.5.10 are only applicable to the pump-rooms, regardless of their location, where pumps for cargo, such as cargo pumps, stripping pumps, pumps for slop tanks, pumps for COW or similar pumps are provided.
“Similar pumps” includes pumps intended for transfer of fuel oil having a flashpoint of less than 60ºC. Pump rooms intended for transfer of fuel oil having a flashpoint of not less than 60ºC need not comply with the requirements of 1.5.10.

1.5.1.2 Main cargo control stations, control stations, accommodation and service spaces (excluding isolated cargo handling gear lockers) are to be positioned aft of cargo tanks, slop tanks and spaces which isolate cargo or slop tanks from machinery spaces, but not necessarily aft of the oil fuel bunker tanks and ballast tanks and are to be arranged in such a way that a single failure of a deck or bulkhead will not permit the entry of gas or fumes from the cargo tanks into an accommodation space, main cargo control stations, control station, or service spaces. A recess provided in accordance with 1.5.1.1 need not be taken into account when the position of these spaces is being determined.

IR1.5.1.1.2 Void space or ballast water tank protecting fuel oil tank as shown in Fig. 1.1 need not be considered as ‘cargo area’ defined in Ch.1, 3.6 even though they have a cruciform contact with the cargo oil tank or slop tank.

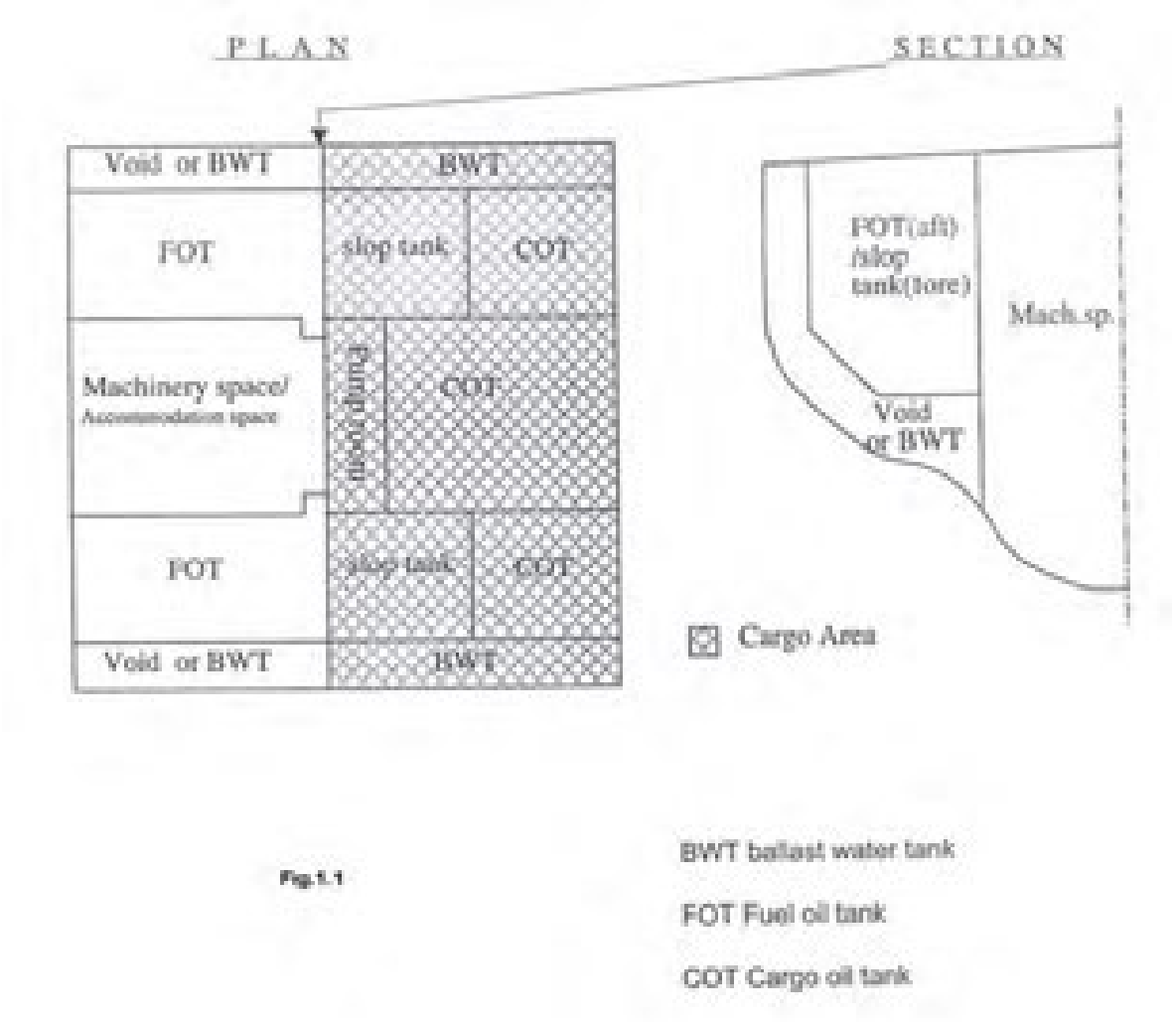

Fig.1.1

1.5.1.3 However, where deemed necessary, IRS may permit main cargo control stations, control stations, accommodation and service spaces forward of the cargo tanks, slop tanks and spaces which isolate cargo and slop tanks from machinery spaces, but not necessarily forward of oil fuel bunker tanks or ballast tanks. Machinery spaces, other than those of category A, may be permitted forward of the cargo tanks and slop tanks provided they are isolated from the cargo tanks and slop tanks by cofferdams, (see Note below) cargo pump-rooms, oil fuel bunker tanks or ballast tanks and have at least one portable fire extinguisher. In cases where they contain internal combustion machinery, one approved foam-type extinguisher of at least 45 l capacity or equivalent is to be arranged in addition to portable fire extinguishers. If operation of a semi-portable fire extinguisher is impracticable, this fire extinguisher may be replaced by two additional portable fire extinguishers. Accommodation spaces, main cargo control spaces, control stations and service spaces are to be arranged in such a way that a single failure of a deck or bulkhead will not permit the entry of gas or fumes from the cargo tanks into such spaces. In addition, where deemed necessary for the safety or navigation of the ship, IRS may permit machinery spaces containing internal combustion machinery not being main propulsion machinery having an output greater than 375 kW to be located forward of the cargo area provided the arrangements are in accordance with the provisions of this clause.

IR1.5.1.3 Paint lockers, regardless of their use are not to be located above the cargo tanks, slop tanks and spaces which isolate cargo or slop tanks from machinery spaces for oil tankers (refer 1.5.1.2) and the cargo area for chemical tankers and liquefied gas carriers.

IR1.5.1.1.3 The void space protecting fuel oil tank is not considered as cofferdam specified in 1.5.1.1 and the void space as shown in Fig.

1.1 is considered acceptable although it has cruciform contact with the slop tank.

1.5.1.4 In combination carriers only: Refer to Pt.5, Ch.2, 12.4.2 to 12.4.5.

1.5.1.5 Where the fitting of a navigation position above the cargo area is shown to be necessary, it is to be for navigation purposes only and it is to be separated from the cargo tank deck by means of an open space with a height of at least 2 m. The fire protection for such a navigation position is to be that required for control stations, as specified in Ch.3, 3.2.4.2 and other provisions for tankers, as applicable.

1.5.1.6 Means are to be provided to keep deck spills away from the accommodation and service areas. This may be accomplished by provision of a permanent continuous coaming of a height of at least 300 [mm], extending from side to side. Special consideration will be given to the arrangements associated with stern loading.

Note: For the purpose of clause 1.5.1 the expression ‘cofferdam’ means an isolating space between two adjacent steel bulkheads or decks. The minimum distance between the two bulkheads or decks is to be sufficient for safe access and inspection. In order to meet the single failure principle, in particular case when a corner-to-corner situation occurs, this principle may be met by welding a diagonal plate across the corner. No cargo, wastes or other goods are to be contained in cofferdams.

1.5.2 Restriction on boundary openings

1.5.2.1 Except as permitted by 1.5.2.2, access doors, air inlets and openings to accommodation spaces, service spaces, control stations and machinery spaces are not to face the cargo area. They are to be located on the transverse bulkhead not facing the cargo area or on the outboard side of the superstructure or deckhouse at a distance of at least 4% of the length of the ship but not less than 3 [m] from the end of the superstructure or deckhouse facing the cargo area. This distance need not exceed 5 [m].

1.5.2.2 IRS may permit access doors in boundary bulkheads facing the cargo area or within the 5 [m] limits specified in 1.5.2.1, to main cargo control stations and to such service spaces used as provision rooms, store-rooms and lockers, provided they do not give access directly or indirectly to any other space containing or providing for accommodation, control stations or service spaces such as galleys, pantries or workshops, or similar spaces containing sources of vapour ignition. The boundary of such a space is to be insulated to "A-60" standard, with the exception of the boundary facing the cargo area. Bolted plates for the removal of machinery may be fitted within the limits specified in 1.5.2.1. Wheelhouse doors and windows may be located within the limits specified in 1.5.2.1 so long as they are designed to ensure that the wheelhouse can be made rapidly and efficiently gas and vapour tight.

IR1.5.2.2

.1 An access to a deck foam system room (including the foam tank and the control station) can be located within the limits mentioned in clause 1.5.2.1, provided that the conditions listed in clause 1.5.2.2 are satisfied and that the door is located flush with the bulkhead.

.2 The navigation bridge external doors and windows which are located within the limits of clause 1.5.2.1 are to be tested for gas-tightness. If a water hose test is applied, the following may be taken as a guide:

- nozzle diameter: minimum 12 [mm];
- water pressure just before the nozzle: not less than 0.2 [N/mm^2]; and
- distance between the nozzles and the doors or windows: max. 1.5 [m].

.3 Access to forecastle spaces containing sources of ignition may be permitted through doors facing cargo area provided the doors are located outside hazardous areas as defined in IEC Publication 60092-502:1999.

1.5.2.3 Windows and side scuttles facing the cargo area and on the sides of the superstructures and deckhouses within the limits specified in 1.5.2.1 are to be of a fixed (non-opening) type. Such windows and side scuttles, except wheelhouse windows are to be constructed to "A-60" class standard except that “A-0” class standard is acceptable for windows and side scuttles outside the limit specified in Ch.3, 3.2.4.2.5.

1.5.2.4 Where there is permanent access from a pipe tunnel to the main pump-room, a watertight door is to be fitted complying with

the requirements of Pt.3, Ch.10, Cl.2.5.2 and in addition, with the following:

.1 in addition to the bridge operation, the watertight door is to be capable of being manually closed from outside the main pump-room entrance; and

.2 the watertight door is to be kept closed during normal operations of the ship except when access to the pipe tunnel is required.

1.5.2.5 Permanent approved gas tight lighting enclosures for illuminating cargo pump rooms may be permitted in bulkheads and decks separating cargo pump rooms and other spaces provided they are of adequate strength and integrity and gastightness of the bulkhead or deck is maintained.

1.5.2.6 The arrangement of ventilation inlets and outlets and other deckhouse and superstructure boundary space openings are to be such as to complement the provisions of 1.5.3 and Ch.3, 5.6. Such vents, especially for machinery spaces are to be situated as far aft as practicable. Due consideration in this regard is to be given when the ship is equipped to load or discharge at the stern. Sources of ignition such as electrical equipment are to be so arranged as to avoid an explosion hazard.

1.5.3 Cargo tank venting

1.5.3.1 General requirements

Refer to Pt.5, Ch.2, 8.1.1.

1.5.3.2 Venting arrangements

Refer to Pt.5, Ch.2, 8.1.6, 8.1.7 and 8.1.11.

1.5.3.3 Safety devices in venting systems

Refer to Pt.5, Ch.2, 8.1.12.

1.5.3.4 Vent outlets for cargo handling and ballasting

1.5.3.4.1 For vent outlets for cargo loading, discharging and ballasting required by Ch.3, 5.6.1.2, refer to Pt.5, Ch.2, 8.1.15 and 8.1.16.

1.5.3.5 Slop tanks in combination carriers

Refer to Pt.5, Ch.2, 12.4.4.

1.5.4 Ventilation

1.5.4.1 Ventilation systems in cargo pump-rooms

Cargo pump-rooms are to be mechanically ventilated and discharges from the exhaust fans are to be led to a safe place on the open deck. The ventilation of these rooms is to have sufficient capacity to minimize the possibility of accumulation of flammable vapours. The number of air changes is to be at least 20 per hour, based upon the gross volume of the space. The air ducts are to be arranged so that all of the space is effectively ventilated. The ventilation is to be of the suction type using fans of the non-sparking type. (Also refer Pt.5, Ch.2, Cl.6.1).

1.5.4.2 Ventilation systems in combination carriers

In combination carriers, cargo spaces and any enclosed spaces adjacent to cargo spaces are to be capable of being mechanically ventilated. The mechanical ventilation may be provided by portable fans. An approved fixed gas warning system capable of monitoring flammable vapours is to be provided in cargo pump-rooms, pipe ducts and cofferdams, as referred to in 1.5.1.4, adjacent to slop tanks. Suitable arrangements are to be made to facilitate measurement of flammable vapours in all other spaces within the cargo area. Such measurements are to be made possible from the open deck or easily accessible positions.

1.5.5 Inert gas systems

1.5.5.1 Application

1.5.5.1.1 For tankers of 20,000 tonnes deadweight and upwards constructed on or after 1 July 2002 but before 1 January 2016, the protection of the cargo tanks is to be achieved by a fixed inert gas system in accordance with the requirements of the Fire Safety Systems Code, as adopted by resolution MSC.98(73), except that the Administration may accept other equivalent systems or arrangements, as described in paragraph 1.5.5.4.

1.5.5.1.2 For tankers of 8,000 tonnes deadweight and upwards constructed on or after 1 January 2016 when carrying cargoes described in Pt 6, Ch 1, 1.5.1 or 1.5.2, the protection of the cargo tanks is to be achieved by a fixed inert gas system in accordance with the requirements of Pt 6, Ch 8, except that the Administration may accept other equivalent

systems or arrangements, as described in paragraph 1.5.5.4.

1.5.5.1.3 Tankers operating with a cargo tank cleaning procedure using crude oil washing are to be fitted with an inert gas system complying with Pt 6, Ch 8 and with fixed tank washing machines. However, inert gas systems fitted on tankers constructed on or after 1 July 2002 but before 1 January 2016 are to comply with the Fire Safety Systems Code, as adopted by resolution MSC.98(73).

1.5.5.1.4 Tankers required to be fitted with inert gas systems are to comply with the following provisions:

.1 double-hull spaces are to be fitted with suitable connections for the supply of inert gas;

IR.1 Double-hull spaces required to be fitted with suitable connections for the supply of inert gas as above, are all ballast tanks and void spaces of double-hull and double-bottom spaces adjacent to the cargo tanks, including the forepeak tank and any other tanks and spaces under the bulkhead deck adjacent to cargo tanks, except cargo pump-rooms and ballast pump-rooms.

.2 where hull spaces are connected to a permanently fitted inert gas distribution system, means are to be provided to prevent hydrocarbon gases from the cargo tanks entering the double hull spaces through the system; and

.3 where such spaces are not permanently connected to an inert gas distribution system, appropriate means are to be provided to allow connection to the inert gas main.

1.5.5.2 Inert gas systems of chemical tankers and gas carriers

1.5.5.2.1 The requirements for inert gas systems contained in the Pt.5, Ch.2, Sec.11, Pt.6 Ch.8 and FSS Code need not be applied to chemical tankers constructed before 1 January 2016, including those constructed before 1 July 2012, and all gas carriers:

.1 when carrying cargoes described in Ch.1, 1.6.1, provided that they comply with the requirements for inert gas systems on chemical tankers approved by the National Statutory Authority (refer to the regulations for inert gas systems on chemical tankers adopted by IMO Resolution A.567(14) Corr.1)

or

.2 when carrying flammable cargoes other than crude oil or petroleum products such as cargoes listed in Sections 17 and 18 of Pt.5, Ch.3, provided that the capacity of tanks used for their carriage does not exceed 3,000 [m^3] and the individual nozzle capacities of tank washing machines do not exceed 17.5 [m^3/h] and the total combined throughput from the number of machines in use in a cargo tank at any one time does not exceed 110 [m^3/h].

1.5.5.3 General requirements for inert gas systems

.1 The inert gas system is to be capable of inerting, purging and gas-freeing empty tanks and maintaining the atmosphere in cargo tanks with the required oxygen content.

.2 Tankers fitted with a fixed inert gas system are to be provided with a closed ullage system.

1.5.5.4 Requirements for equivalent systems

1.5.5.4.1 For tankers of 8,000 tonnes deadweight and upwards but less than 20,000 tonnes deadweight, in lieu of fixed installations, IRS may accept other equivalent arrangements or means of protection in accordance with paragraph 1.5.5.4.2.

1.5.5.4.2 Equivalent systems or arrangements are to :

.1 be capable of preventing dangerous accumulations of explosive mixtures in intact cargo tanks during normal service throughout the ballast voyage and necessary in-tank operations; and

.2 be so designed as to minimize the risk of ignition from the generation of static electricity by the system itself.

1.5.6 Inerting, purging and gas-freeing (Also refer Pt.5, Ch.2, 8.2)

.1 Arrangements for purging and/or gas-freeing are to be such as to minimize the hazards due to dispersal of flammable vapours in the atmosphere and to flammable mixtures in a cargo tank.

.2 The procedure for cargo tank purging and/or gas-freeing are to be carried out in accordance with Ch.5, 3.3.2.

.3 Also refer to Pt.5, Ch.2, 8.2.3.

1.5.7 Gas measurement and detection

1.5.7.1 Portable instrument

1.5.7.1.1 Tankers are to be equipped with at least one portable instrument for measuring oxygen and one for measuring flammable vapour concentrations, together with a sufficient set of spares. Suitable means are to be provided for the calibration of such instruments**.**

IR 1.5.7.1.1 Compliance with the provision "suitable means are to be provided for the calibration of such instruments" may be achieved by portable atmosphere testing instruments being calibrated on board or ashore in accordance with the manufacturer's instructions. For the avoidance of any doubt, the above consideration refers to the calibration of portable instruments for measuring oxygen or flammable vapour concentrations, as required by 1.5.7.1.1 and not to any pre-operational accuracy tests as recommended by the manufacturer.

1.5.7.2 Arrangements for gas measurement in double hull and double bottom spaces

Refer to Pt.5, Ch.2, 8.3.3, 8.3.4 and 8.3.5.

1.5.8 Air supply to double hull and double bottom spaces

Double hull and double bottom spaces are to be fitted with suitable connections for the supply of air.

1.5.9 Protection of cargo area

Drip pans of suitable material for collecting cargo residues in cargo lines and hoses are to be provided in the area of pipe and hose connections under the manifold area. Cargo hoses and tank washing hoses are to have electrical continuity over their entire lengths including couplings and flanges (except shore connections) and are to be earthed for removal of electrostatic charges.

1.5.10 Protection of cargo pump rooms

1.5.10.1 For details of protection in tankers; Refer Pt.5, Ch.2, 6.4.2 to 6.4.5 and IR6.4.5.

Section 2

Fire Growth Potential

2.1 Purpose

2.1.1 The purpose of this regulation is to limit the fire growth potential in every space of the ship. For this purpose, the following functional requirements are to be met:

a) means of control for the air supply to the space are to be provided;
b) means of control for flammable liquids in the space are to be provided; and
c) the use of combustible materials is to be restricted.

2.2 Control of air supply and flammable liquid to the space

2.2.1 Closing appliances and stopping devices of ventilation

2.2.1.1 The main inlets and outlets of all ventilation system are to be capable of being closed from outside the spaces being ventilated. The means of closing are to be easily accessible as well as prominently and permanently marked and are to indicate whether the shut-off is open or closed.

IR2.2.1.1 Battery room ventilators are to be fitted with a means of closing under the following conditions:

i) The battery room does not open directly onto an exposed deck; or

ii) The ventilation opening for the battery room is required to be fitted with a closing device according to the Pt.3, Ch.13, Cl.2.3.1 (i.e. the height of the opening does not extend to more than 4.5 [m] (14.8 feet) above the deck for

position 1 or to more than 2.3 [m] (7.5 feet) above the deck in position 2; or

iii) The battery room is fitted with a fixed gas fire extinguishing system.

Where a battery room ventilator is fitted with a closing device, a warning notice stating, for example “This closing device is to be kept open and closed only in the event of fire or other emergency – Explosive gas”, is to be provided at the closing device to mitigate the possibility of inadvertent closing.

Also refer Pt.4, Ch.8, Cl.7.1.5.

2.2.1.2 Power ventilation of accommodation spaces, service spaces, cargo spaces, control stations and machinery spaces are to be capable of being stopped from an easily accessible position outside the space being served. This position is not to be readily cut off in the event of a fire in the spaces served.

2.2.1.3 In passenger ships carrying more than 36 passengers, power ventilation, except machinery space and cargo space ventilation and any alternative system which may be required by Ch.3, 2.2 is to be fitted with controls so grouped that all fans may be stopped from either of two separate positions which are to be situated as far apart as practicable. Fans serving power ventilation systems to cargo spaces are to be capable of being stopped from a safe position outside such spaces.

IR2.2.1.3 The fan in a HVAC temperature control unit, or a circulation fan inside a cabinet/switchboard, is not considered to be a ventilation fan as addressed in 2.2.1.2, 2.2.1.3 and Ch 3, 1.9.3, if it is not capable of supplying outside air to the space when the power ventilation is shut down (e.g., small units intended for re-circulation of air within a cabin). Therefore, such fans need not be capable of being stopped from an easily accessible position (or a safe position) outside the space being served when applying 2.2.1.2 or 2.2.1.3, and need not be capable of being controlled from a continuously manned central control station for passenger ships carrying more than 36 passengers when applying Ch 3, 1.9.3.

2.2.2 Means of control in machinery spaces

2.2.2.1 Means of control are to be provided for opening and closure of skylights, closure of openings in funnels which normally allow exhaust ventilation and closure of ventilator dampers.

IR2.2.2.1 Ventilation inlets and outlets located at outside boundaries are to be fitted with closing appliances as required by 2.2.2.1 but need not comply with Ch.3, 3.7.3.1.

2.2.2.2 Means of control are to be provided for stopping ventilating fans. Controls provided for the power ventilation serving machinery spaces are to be grouped so as to be operable from two positions, one of which is to be outside such spaces. The means provided for stopping the power ventilation of the machinery spaces are to be entirely separate from the means provided for stopping ventilation of other spaces.

2.2.2.3 Means of control are to be provided for stopping forced and induced draught fans, oil fuel transfer pumps, oil fuel unit pumps, lubricating oil service pumps, thermal oil circulating pumps and oil separators (purifiers). However, 2.2.2.4 and 2.2.2.5 need not apply to oily water separators.

2.2.2.4 The controls required in 2.2.2.1 to 2.2.2.3 and in 1.2.2.3.4 are to located outside the space concerned so they will not be cut off in the event of fire in the space they serve.

2.2.2.5 In passenger ships, the controls required in 2.2.2.1 to 2.2.2.4 and in Ch.3, 2.3.3 and Ch.3, 3.5.2.3 and the controls for any required fire-extinguishing system are to be situated at one control position or grouped in as few positions as possible to the satisfaction of IRS. Such positions are to have a safe access from the open deck.

2.2.3 Additional requirements for means of control in periodically unattended machinery spaces

2.2.3.1 For periodically unattended machinery spaces, special consideration will be given to maintaining the fire integrity of the machinery spaces, the location and centralization of the fire-extinguishing system controls, the required shutdown arrangements (e.g. ventilation, fuel pumps, etc.) and that additional fire-extinguishing appliances and other fire-fighting equipment and breathing apparatus may be required.

2.2.3.2 In passenger ships, these requirements are be at least equivalent to those of machinery spaces normally attended.

2.3 Fire protection materials

2.3.1 Use of non-combustible materials

2.3.1.1 Insulating materials

Insulating materials are to be non-combustible, except in cargo spaces, mail rooms, baggage rooms and refrigerated compartments of service spaces. Vapour barriers and adhesives used in conjunction with insulation, as well as the insulation of pipe fittings for cold service systems, need not be of non-combustible materials, but they are to be kept to the minimum quantity practicable and their exposed surfaces are to have low flame-spread characteristics.

IR2.3.1.1 Cold service in 2.3.1.1 is understood to mean refrigeration systems and chilled water piping for air-conditioning systems.

2.3.1.2 Ceilings and linings

.1 In passenger ships, all linings, grounds, draught stops and ceilings are to be of non-combustible material in the following spaces except in mailrooms, baggage rooms and saunas and refrigerated compartments of service spaces:

a) accommodation spaces, service spaces and control stations
b) stairways, open deck spaces
c) machinery spaces.

.2 In cargo ships, all linings, ceilings, draught stops and their associated grounds are to be of non-combustible materials in the following spaces:

a) in accommodation and service spaces and control stations for ships where Method IC is specified as referred to in Ch.3, 3.2.3.1; and
b) in corridors and stairway enclosures serving accommodation and service spaces and control stations for ships where Method IIC and IIIC are specified as referred to in Ch.3, 3.2.3.1.

2.3.1.3 Partial bulkheads and decks on passenger ships

.1 Partial bulkheads or decks used to subdivide a space for utility or artistic treatment are to be of non-combustible material.

.2 Linings, ceilings or partial bulkheads or decks used to screen or to separate adjacent cabin balconies are to be of non-combustible materials. Cabin balconies on passenger ships constructed before 1 July 2008 are to comply with the requirements of this clause by the first survey after 1 July 2008.

2.3.2 Use of combustible materials

2.3.2.1 General

.1 In passenger ships, "A", "B" or "C" class divisions in accommodation and services spaces and cabin balconies which are faced with combustible materials, facings, mouldings, decorations and veneers are to comply with the provisions of 2.3.2.2 to 2.3.2.4 and section 3. However, traditional wooden benches and wooden linings on bulkheads and ceilings are permitted in saunas and such materials need not be subject to the calculations prescribed in 2.3.2.2 and 2.3.2.3. However, the provisions of 2.3.2.3 need not be applied to cabin balconies.

.2 In cargo ships, non-combustible bulkheads, ceilings and linings fitted in accommodation and service spaces may be faced with combustible materials, facings, mouldings, decorations and veneers provided such spaces are bounded by non-combustible bulkheads, ceilings and linings in accordance with the provisions of 2.3.2.2 to 2.3.2.4 and Section 3.

2.3.2.2 Maximum calorific value of combustible materials

Combustible materials used on the surfaces and linings specified in 2.3.2.1 are to have a calorific value not exceeding 45 [MJ/m^2] of the area for the thickness used. The requirements of this paragraph are not applicable to the surfaces of furniture fixed to linings or bulkheads. (Refer ISO 1716 on Determination of Calorific value).

2.3.2.3 Total volume of combustible materials

Where combustible materials are used in accordance with 2.3.2.1, they are to comply with the following requirements:

a) The total volume of combustible facings mouldings, decorations and veneers in accommodation and service spaces are not to exceed a volume equivalent to 2.5 [mm] veneer on the combined area of the walls and ceiling linings. Furniture fixed to

linings, bulkheads or decks need not be included in the calculation of the total volume of combustible materials; and

b) In the case of ships fitted with an automatic sprinkler system complying with the provisions of the Fire Safety Systems Code (Chapter 8), the above volume may include some combustible material used for erection of "C" class divisions.

2.3.2.4 Low flame-spread characteristics of exposed surfaces

The following surfaces are to have low flame-spread characteristics in accordance with the Fire Test Procedures Code:

.1 In passenger ships:

a) exposed surfaces in corridors and stairway enclosures and of bulkhead and ceiling linings in accommodation and service spaces (except saunas) and control stations; and
b) surfaces and grounds in concealed or inaccessible spaces in accommodation and service spaces and control stations.
c) Exposed surfaces of cabin balconies, except for natural hard wood decking systems.

.2 In cargo ships:

a) exposed surfaces in corridors and stairway enclosures and of ceilings in accommodation and service spaces (except saunas) and control stations; and
b) surfaces and grounds in concealed or inaccessible spaces in accommodation and service spaces and control stations.

2.3.3 Furniture in stairway enclosures of passenger ships

Furniture in stairway enclosures are to be limited to seating. It is to be fixed, limited to six seats on each deck in each stairway enclosure, be of restricted fire risk determined in accordance with the Fire Test Procedure Code, and is not to restrict the passenger escape route. Additional seating may be permitted in the main reception area within a stairway enclosure if it is fixed, non-combustible and does not restrict the passenger escape route. Furniture is not to be fixed in passenger and crew corridors forming escape routes in cabin areas. In addition to the above, lockers of non-combustible material, providing storage for non-hazardous safety equipment required by these rules, may be permitted. Drinking water dispensers and ice cube machines may be fixed in corridors provided they are fixed and do not restrict the width of the escape routes. This applies as well to decorative flower or plant arrangements, statues or other objects of art such as paintings and tapestries in corridors and stairways.

2.3.4 Furniture and furnishings on cabin balconies of passenger ships

On passenger ships, furniture and furnishings on cabin balconies are to comply with requirements of Ch.1, 3.40 a), b), c), f) and g) unless such balconies are protected by a fixed pressure water-spraying and fixed fire detection and fire alarm systems complying with Ch.3, 1.10 and Ch.3, 4.6.1.3. passenger ships constructed before 1 July 2008 are to comply with the requirements of this clause by the first survey after 1 July 2008.

Section 3

Smoke Generation Potential and Toxicity

3.1 Purpose

3.1.1 The purpose of this section is to reduce the hazard to life from smoke and toxic products generated during a fire in spaces where persons normally work or live. For this purpose, the quantity of smoke and toxic products released from combustible materials, including surface finishes, during fire is to be limited.

3.2 Paints, varnishes and other finishes

The following requirements apply to accommodation spaces, service spaces and control stations as well as stairway enclosures.

3.2.1 Paints, varnishes and other finishes used on exposed interior surfaces are not be capable of producing excessive quantities of smoke and toxic products, this being determined in accordance with the Fire Test Procedures Code.

3.2.2 On passenger ships constructed on or after 1 July 2008, paints, varnishes and other finishes used on exposed surfaces of cabin balconies, excluding natural hard wood decking systems, are not to be capable of producing excessive quantities of smoke and toxic products, this being determined in accordance with the Fire Test Procedures Code.

3.3 Primary deck coverings

3.3.1 Primary deck coverings, if applied within accommodation and service spaces and control stations are to be of approved material which will not give rise to smoke or toxic or explosive hazards at elevated temperatures, this being determined in accordance with the Fire Test Procedures Code.

3.3.2 On passenger ships constructed on or after 1 July 2008, primary deck coverings on cabin balconies are not to give rise to smoke, toxic or explosive hazards at elevated temperatures, this being determined in accordance with the Fire Test Procedures Code.

3.4 Asbestos

3.4.1 For all ships, new installations of materials which contain asbestos are prohibited.

End of Chapter

Chapter 3

Suppression of Fire

Contents

Section

1 *Detection and Alarm*

2 *Control of Smoke Spread*

3 *Containment of Fire*

4 *Fire Fighting*

5 *Structural Integrity*

Section 1

Detection and Alarm

1.1 Purpose

1.1.1 The purpose of this section is to detect a fire in the space of origin and to provide for alarm for safe escape and fire-fighting activity. For this purpose, the following functional requirements are to be met:

a) fixed fire detection and fire alarm system installations are to be suitable for the nature of the space, fire growth potential and potential generation of smoke and gases;

b) manually operated call points are to be placed effectively to ensure a readily accessible means of notification; and

c) fire patrols are to provide an effective means of detecting and locating fires and alerting the navigation bridge and fire teams.

1.2 General requirements

1.2.1 A fixed fire detection and fire alarm system is to be provided in accordance with the provisions of this section.

1.2.2 A fixed fire detection and fire alarm system and a sample extraction smoke detection system required in this section and other sections in this chapter are to be of an approved type and comply with the Fire Safety Systems Code (Chapter 8).

1.2.3 Where a fixed fire detection and fire alarm system is required for the protection of spaces other than those specified in 1.5.1, at least one detector complying with the Fire Safety Systems Code (Chapter 8) is to be installed in each such space.

1.2.4 A fixed fire detection and fire alarm system for passenger ships is to be capable of remotely and individually identifying each detector and manually operated call point.

1.3 Initial and periodical tests

1.3.1 The function of fixed fire detection and fire alarm systems required by the relevant sections of this part are to be tested under varying conditions of ventilation after installation.

1.3.2 The function of fixed fire detection and fire alarm systems is to be periodically tested to the satisfaction of IRS by means of equipment producing hot air at the appropriate temperature, or smoke or aerosol particles having the appropriate range of density or particle size, or other charateristics associated with incipient fires to which the detector is designed to respond.

1.4 Protection of machinery spaces

IR1.4 This requirement is applicable to machinery spaces of category 'A'.

1.4.1 Installation

A fixed fire detection and fire alarm system is to be installed in:

.1 periodically unattended machinery spaces;

.2 machinery spaces where:

a) the installation of automatic and remote control systems and equipment has been approved in lieu of continuous manning of the space; and

b) the main propulsion and associated machinery including sources of the main sources of electrical power are provided with various degrees of automatic or remote control and are under continuous manned supervision from a control room and:

c) enclosed spaces containing incinerators.

1.4.2 Design

1.4.2.1 This fire detection system is to be so designed and the detectors so positioned as to detect rapidly the onset of fire in any part of those spaces and under any normal conditions of operation of the machinery and variations of ventilation as required by the possible range of ambient temperatures. Consideration is to be given to avoid false alarms. The arrangement and the number of Sections and the location of detector heads is to be approved in each case. Except in spaces of restricted height and where their use is specially appropriate, detection systems using only thermal detectors will not be permitted. The detection system is to initiate audible and visual alarms distinct in both respects from the alarms of any other system not indicating fire, in sufficient places to ensure that the alarms are heard and observed on the navigating bridge and by a responsible engineer officer. When the navigating bridge is unmanned the alarm is to sound in a place where a responsible member of the crew is on duty. (Also refer Pt.5, Ch.22, 2.13 for other requirements).

1.5 Protection of accommodation and service spaces and control stations

1.5.1 Smoke detectors in accommodation spaces

Smoke detectors are to be installed in all stairways, corridors and escape routes within accommodation spaces as provided in 1.5.2, 1.5.3 and 1.5.4 Consideration will be given to the installation of special purpose smoke detectors within ventilation ducting.

1.5.2 Requirements for passenger ships carrying more than 36 passengers

.1 A fixed fire detection and fire alarm system is to be installed and arranged as to provide smoke detection in service spaces, control stations and accommodation spaces, including corridors, stairways and escape routes within accommodation spaces. Smoke detectors need not be fitted in private bathrooms and galleys. Spaces having little or no fire risk such as voids, public toilets, carbon dioxide rooms and similar spaces need not be fitted with a fixed fire detection and alarm system. Detectors fitted in cabins, when activated are also to be capable of generating or activating an audible alarm within the space where they are located.

IR.2 Heat detectors are acceptable in refrigerated chambers and in other spaces such as saunas and laundries where steam and fumes are emanated.

1.5.3 Requirements for passenger ships carrying not more than 36 passengers

There is to be installed throughout each separate zone, whether vertical or horizontal, in all accommodation and service spaces and where it is considered necessary by IRS, in control stations, except spaces which afford no substantial fire risk such as void spaces, sanitary spaces, etc., either:

.1 a fixed fire detection and fire alarm system so installed and arranged as to detect the presence of fire in such spaces and providing smoke detection in corridors, stairways and escape routes within accommodation spaces. Detectors fitted in cabins, when activated are also to be capable of generating or activating an audible alarm within the space where they are located; or

.2 an automatic sprinkler, fire detection and fire alarm system of an approved type complying with the relevant requirements of

the Fire Safety Systems Code (Chapter 8) and so installed and arranged as to protect such spaces and, in addition, a fixed fire detection and fire alarm system and so installed and arranged as to provide smoke detection in corridors, stairways and escape routes within accommodation spaces.

1.5.4 Protection of atriums in passenger ships

The entire main vertical zone containing the atrium is to be protected throughout with a smoke detection system.

1.5.5 Cargo ships

Accommodation and service spaces and control stations of cargo ships are to be protected by a fixed fire detection and fire alarm system and/or an automatic sprinkler, fire detection and fire alarm system as follows depending on a protection method adopted in accordance with Ch.3, 3.2.3.1.

1.5.5.1 Method IC

A fixed fire detection and fire alarm system is to be so installed and arranged as to provide smoke detection in all corridors, stairways and escape routes within accommodation spaces.

1.5.5.2 Method IIC

An automatic sprinkler, fire detection and fire alarm system of an approved type complying with the relevant requirements of the Fire Safety Systems Code (Chapter 8) are to be so installed and arranged as to protect accommodation spaces, galleys and other service spaces, except spaces which afford no substantial fire risk such as void spaces, sanitary spaces, etc. In addition, a fixed fire detection and fire alarm system of an approved type complying with the requirements of Ch.8, Sec.9 is to be so installed and arranged to provide smoke detection in all corridors, stairways and escape routes within accommodation spaces.

1.5.5.3 Method IIIC

A fixed fire detection and fire alarm system is to be so installed and arranged as to detect the presence of fire in all accommodation spaces and service spaces providing smoke detection in corridors, stairways and escape routes within accommodation spaces, except spaces which afford no substantial fire risk such as void spaces, sanitary spaces, etc.

IR1.5.5.3 The above requirement applies to accommodation block. Service spaces built away from accommodation block need not be fitted with a fixed fire detection and fire-alarm system complying with FSS Code (Chapter 8).

1.6 Protection of cargo spaces in passenger ships

A fixed fire detection and fire alarm system or a sample extraction smoke detection system is to be provided in any cargo space which is not accessible, except where it is shown to the satisfaction of IRS that the ship is engaged on voyages of such short duration that it would be unreasonable to apply this requirement.

1.7 Manually operated call points

1.7.1 Manual call points are to be installed throughout the accommodation spaces, service spaces and control stations. One manual call point is to be located at each exit. Manual call points are to be readily accessible in the corridors of each deck such that no part of the corridor is more than 20 [m] from a manual call point.

IR1.7.1 The phrase ‘Manually operated call points complying with the Fire Safety Systems Code are to be installed throughout the accommodation spaces, service spaces and control stations in 1.7.1 above does not require the fitting of a manually operated call point in an individual space within the accommodation spaces, service spaces and control stations. However, a manually operated call point is to be located at each exit (inside or outside) to the open deck from the corridor such that no part of the corridor is more than 20 [m] from a manually operated call point. Service spaces and control stations which have only one access, leading directly to the open deck are to have a manually operated call point not more than 20 [m] (measured along the access route using the deck, stairs and/or corridors) from the exit. A manually operated call point is not required to be installed for spaces having little or no fire risk, such as voids and carbon dioxide rooms, nor at each exit from the navigation bridge, in cases where the control panel (for fixed fire detection and fire alarm system) is located in the navigation bridge.

1.8 Fire patrols in passenger ships

1.8.1 Fire patrols

For ships carrying more than 36 passengers an efficient patrol system is to be maintained so that an outbreak of fire may be promptly detected. Each member of the fire patrol is to be trained to be familiar with the arrangements of the ship as well as the location and operation of any equipment he may be called upon to use.

1.8.2 Inspection hatches

The construction of ceiling and bulkheads is to be such that it will be possible, without impairing the efficiency of the fire protection, for the fire patrols to detect any smoke originating in concealed and inaccessible places, except where there is no risk of fire originating in such places.

1.8.3 Two-way portable radiotelephone apparatus

Each member of the fire patrol is to be provided with a two-way portable radiotelephone apparatus.

1.9 Fire alarm signalling systems in passenger ships (IMO Res.A.830(19) may be referred for Code of Alarms)

1.9.1 Passenger ships are at all times when at sea, or in port (except when out of service), to be so manned or equipped as to ensure that any initial fire alarm is immediately received by a responsible member of the crew.

1.9.2 The control panel of fixed fire detection and fire alarm systems is to be designed on the fail-safe principle (e.g. an open detector circuit is to cause an alarm condition).

1.9.3 Passenger ships carrying more than 36 passengers are to have the fire detection alarms for the systems required by 1.5.2 centralized in a continuously manned central control station. In addition, controls for remote closing of the fire doors and shutting down the ventilation fans are to be centralized in the same location. The ventilation fans are to be capable of reactivation by the crew at the continuously manned control station. The control panels in the central control station are to be capable of indicating open or closed positions of fire doors and closed or off status of the detectors, alarms and fans. The control panels are to be continuously powered and are to have an automatic change-over to standby power supply in case of loss of normal power supply. The control panels are to be powered from the main source of electrical power and the emergency source of electrical power unless other arrangements are permitted by applicable requirements. (Also refer to Ch 2, IR 2.2.1.3)

1.9.4 A special alarm, operated from the navigation bridge or fire control station, is to be fitted to summon the crew. This alarm may be part of the ship's general alarm system and is to be capable of being sounded independently of the alarm to the passenger spaces.

1.10 Protection of cabin balconies on passenger ships

A fixed fire detection and fire alarm system complying with the provisions of Ch.8 is to be installed on cabin balconies of ships to which requirement of Ch.2, 2.3.4 applies, when furniture and furnishings on such balconies are not as defined in Ch.1, 3.40 a), b), c), f) and g).

Section 2

Control of Smoke Spread

2.1 Purpose

The purpose of this regulation is to control the spread of smoke in order to minimize the hazards from smoke. For this purpose, means for controlling smoke in atriums, control stations, machinery spaces and concealed spaces are to be provided.

2.2 Protection of control stations outside machinery spaces

Such measures as practicable are to be taken in respect of control stations outside machinery spaces in order to ensure that ventilation, visibility and freedom from smoke are maintained, so that in the event of fire the machinery and equipment contained therein may be supervised and continue to function effectively. Alternative and separate means of air supply are to be provided; air inlets of the two sources of supply are to be so disposed that the risk of both inlets drawing in smoke simultaneously is minimized. Such requirements need not apply to control station situated on, and opening on to, an open deck, or where local closing arrangements would be equally effective. The ventilation system serving safety centers may be derived from the ventilation system serving the navigation bridge, unless located in an adjacent main vertical zone.

IR2.2 The expression ‘local closing arrange-ments would be equally effective’ in 2.2 means that in case of ventilators, the ventilators are fitted with fire dampers or smoke dampers which could be closed easily from within the control station in order to maintain the absence of smoke in the event of fire.

2.3 Release of smoke from machinery spaces

2.3.1 The following requirements are applicable to machinery spaces of category A and, where IRS considers desirable, to other machinery spaces.

2.3.2 Suitable arrangements are to be made to permit the release of smoke, in the event of fire, from the space to be protected, subject to the provisions of 3.5.2.1. The normal ventilation systems may be acceptable for this purpose.

2.3.3 Means of control are to be provided for permitting the release of smoke and such controls are to be located outside the space concerned so that, in the event of fire, they will not be cut off from the space they serve.

2.3.4 In passenger ships, the controls required by 2.3.3 are to be situated at one control position or grouped in as few positions as possible to the satisfaction of IRS. Such positions are to have a safe access from the open deck.

2.4 Draught stops

Air spaces enclosed behind ceilings, panelling or linings are to be divided by close-fitting draught stops spaced not more than 14 [m] apart. In the vertical direction, such enclosed air spaces, including those behind linings of stairways, trunks, etc., are to be closed at each deck.

2.5 Smoke extraction systems in atriums of passenger ships

Atriums are to be equipped with a smoke extraction system. The smoke extraction system is to be activated by the required smoke detection system and be capable of manual control. The fans are to be sized such that the entire volume within space can be exhausted in 10 min or less.

Section 3

Containment of Fire

3.1 Purpose

The purpose of this section is to contain a fire in the space of origin. For this purpose, the following functional requirements are to be met:

.1 the ship is to be subdivided by thermal and structural boundaries;

.2 thermal insulation of boundaries are to have due regard to the fire risk of the space and adjacent spaces; and

.3 the fire integrity of the divisions is to be maintained at openings and penetrations.

3.2 Thermal and structural boundaries

3.2.1 Thermal and structural subdivision

Ships of all types are to be subdivided into spaces by thermal and structural divisions having regard to the fire risks of the space.

3.2.2 Passenger ships

3.2.2.1 *Main vertical zones and horizontal zones*

3.2.2.1.1a) In ships carrying more than 36 passengers, the hull, superstructure and deckhouses are to be subdivided into main vertical zones by 'A-60' class divisions. Steps and recesses are to be kept to a minimum, but where they are necessary, they are also to be 'A-60' class divisions. These standard may be reduced to 'A-0' where one side of the division belong to any of the following categories or where fuel oil tanks are on both sides of the division:

- open deck spaces as per 3.2.2.3.2.2(5)
- sanitary and similar spaces as per 3.2.2.3.2.2(9)
- tanks, voids and auxiliary machinery space having little or no fire risk as per 3.2.2.3.2.2(10).

3.2.2.1.1b) In ships carrying not more than 36 passengers, the hull, superstructure and deckhouses in way of accommodation and service spaces are to be subdivided into main vertical zones by "A" class divisions. These divisions are to have insulation values in accordance with tables in 3.2.2.4.

3.2.2.1.2 As far as practicable the bulkheads forming the boundaries of the main vertical zones above the bulkhead deck are to be in line with watertight subdivision bulkheads situated immediately below the bulkhead deck. The length and width of main vertical zones may be extended to a maximum of 48 [m] in order to bring the ends of main vertical zones to coincide with watertight subdivision bulkheads or in order to accommodate a large public space extending for the whole length of the main vertical zone. The total area of the main vertical zone however, is not to be greater than 1600 [m^2] on any deck. The length and width of a main vertical zone is to be taken as the maximum distance between the furthermost points of the bulkheads bounding it.

IR3.2.2.1.2 If a stairway serves two main vertical zones, the maximum length of one main vertical zone is to be measured from the far side of the main vertical zone stairway enclosure. In this case, all boundaries of the stairway enclosure are to be insulated as main vertical zone bulkheads and access doors leading into the stairway are to be provided from the zones. (See Fig.3.2.2.1 to Fig.3.2.2.4). However, the stairway is not to be included in calculating the size of the main vertical zone if it is treated as its own main vertical zone.

The number of main vertical zones of 48 m length is not limited as long as they comply with all the requirements.

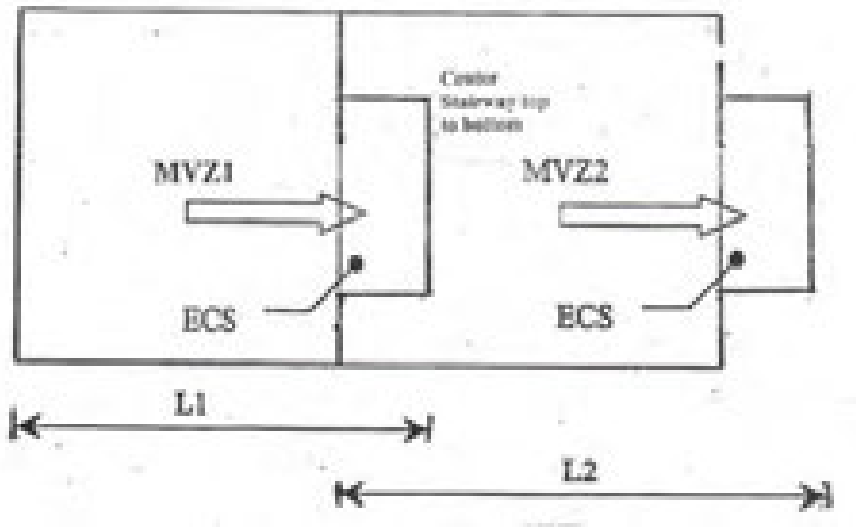

Fig.3.2.2.1 : ECS serves one MVZ

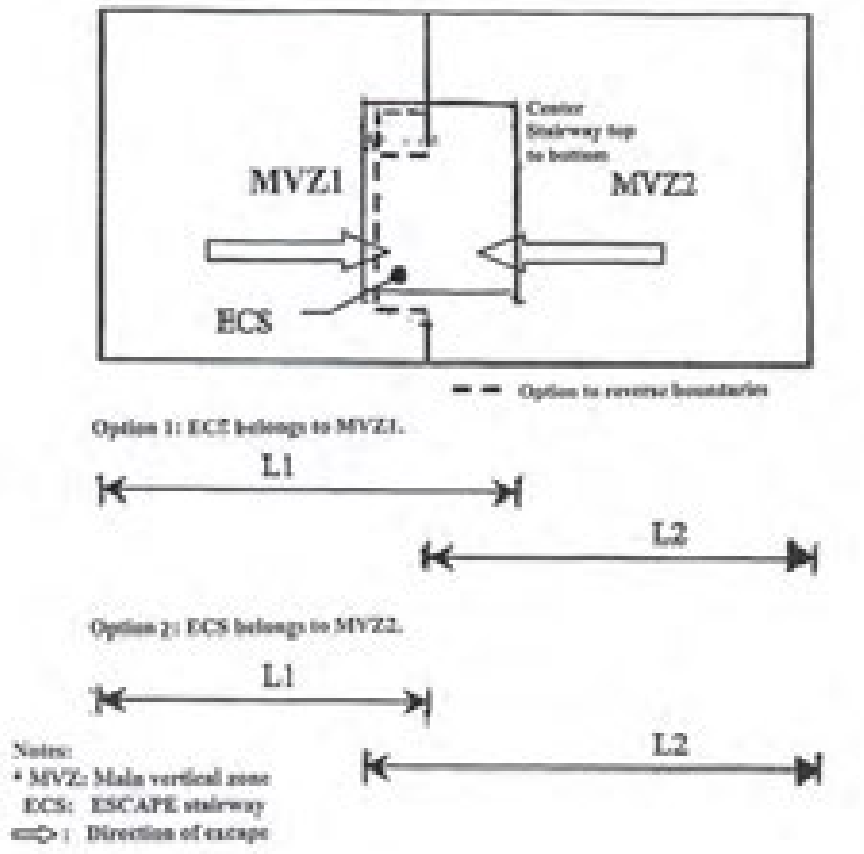

Fig.3.2.2.2 : ECS serves two MVZ's

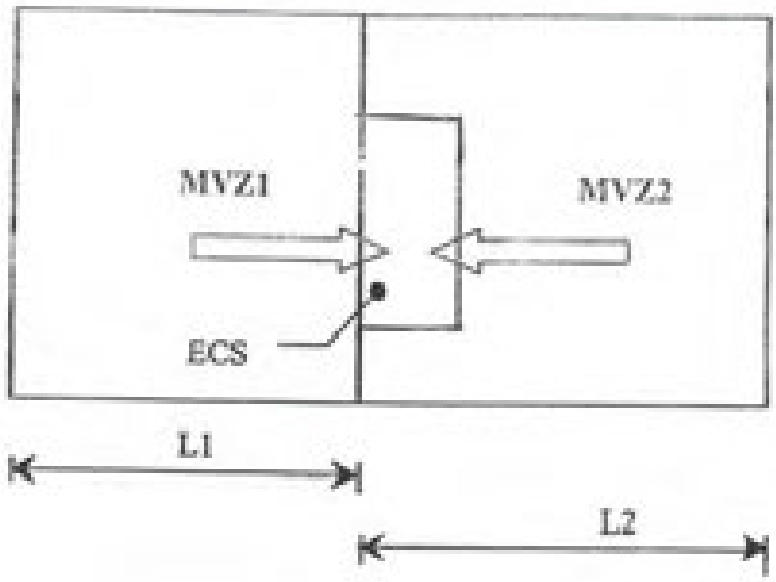

Fig.3.2.2.3 : ECS serves two MVZ's (ECS belongs to MVZ2)

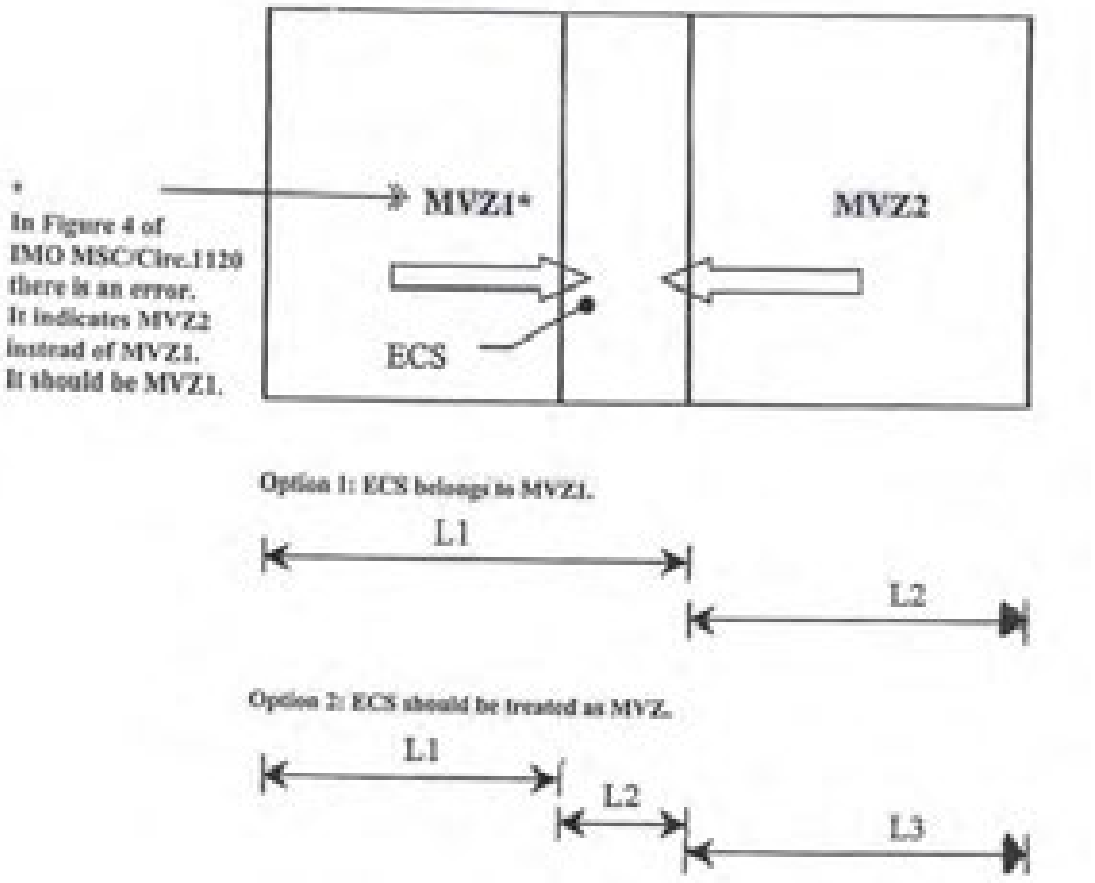

Fig.3.2.2.4 : ECS serves two MVZ's

3.2.2.1.3 Such bulkheads are to extend from deck to deck and to the shell or other boundaries.

3.2.2.1.4 Where a main vertical zone is subdivided by horizontal "A" class divisions into horizontal zones for the purpose of providing an appropriate barrier between a zone with sprinklers and a zone without sprinklers, the divisions are to extend between adjacent main vertical zone bulkheads and to the shell or exterior boundaries of the ship and are to be insulated in accordance with the fire insulation and integrity values given in Table 3.4.

3.2.2.1.5a) On ships designed for special purposes, such as automobile or railroad car ferries, where the provision of main vertical zone bulkheads would defeat the purpose for which the ship is intended, equivalent means for controlling and limiting a fire will be specially considered. Service spaces and ship stores are not to be located on ro-ro decks unless protected in accordance with the applicable requirements.

3.2.2.1.5b) However, in a ship with special category spaces, such spaces are to comply with the applicable provisions of Ch.7, Sec.3 and where such compliance would be inconsistent with other requirements for passenger ships specified in this part, the requirements of Ch.7, Sec.3 would apply.

3.2.2.2 *Bulkheads within a main vertical zone*

3.2.2.2.1 For ships carrying more than 36 passengers, bulkheads which are not required to be "A" class divisions are to be at least "B" class or "C" class divisions as prescribed in the tables in 3.2.2.3.

3.2.2.2.2 For ships carrying not more than 36 passengers, bulkheads within accommodation and service spaces which are not required to be "A" class divisions are to be at least "B" class or "C" class divisions as prescribed in the tables in 3.2.2.4. In addition, corridor bulkheads, where not required to be "A" class are to be "B" class divisions which are to extend from deck to deck except:

.1 when continuous "B" class ceilings or linings are fitted on both sides of the bulkhead, the portion of the bulkhead behind the continuous ceiling or lining is to be of material which, in thickness and composition, is acceptable in the construction of "B" class divisions, but which will be required to meet "B" class integrity standards only in so far as is reasonable and practicable; and

.2 in the case of a ship protected by an automatic sprinkler system complying with the provisions of the Fire Safety Systems Code (Chapter 8), the corridor bulkheads may terminate at a ceiling in the corridor provided such bulkheads and ceilings are of “B” class standard in compliance with 3.2.2.4. All doors and frames in such bulkheads are to be of non-combustible materials and are to have the same fire integrity as the bulkhead in which they are fitted.

3.2.2.2.3 Bulkheads required to be "B" class divisions, except corridor bulkheads as prescribed in 3.2.2.2.2, are to extend from deck to deck and to the shell or other boundaries. However, where a continuous "B" class ceiling or lining is fitted on both sides of a bulkhead which is at least of the same fire resistance as the adjoining bulkhead, the bulkhead may terminate at the continuous ceiling or lining.

IR3.2.2.2.3 If an air gap between cabins results in an opening in the continuous B-15 ceiling, the bulkheads on the both sides of the air gap are to be of class B-15. MSC Circ.917 "Guidelines on fire safety construction in accommodation areas" as corrected by MSC/Circ.917/Corr.1 may be referred for details of construction.

3.2.2.3 *Fire integrity of bulkheads and decks in ships carrying more than 36 passengers*

3.2.2.3.1 In addition to complying with the specific provisions for fire integrity of bulkheads and decks of passenger ships, the minimum fire integrity of all bulkheads and decks is to be as prescribed in Table 3.1 and Table 3.2. Where, due to any particular structural arrangements in the ship, difficulty is experienced in determining from the tables the minimum fire integrity value of any divisions, such values are to be determined to the satisfaction of IRS.

3.2.2.3.2 The following requirements are to govern application of the tables:

.1 Table 3.1 is to apply to bulkheads not bounding either main vertical zones or horizontal zones. Table 3.2 is to apply to

decks not forming steps in main vertical zones nor bounding horizontal zones;

.2 For determining the appropriate fire integrity standards to be applied to boundaries between adjacent spaces, such spaces are classified according to their fire risk as shown in categories (1) to (14) below. Where the contents and use of a space are such that there is a doubt as to its classification for the purpose of this regulation, or where it is possible to assign two or more classifications to a space, it is to be treated as a space within the relevant category having the most stringent boundary require-ments. Smaller, enclosed rooms within a space that have less than 30% communicating openings to that space are considered separate spaces. The fire integrity of the boundary bulkheads and decks of such smaller rooms is to be as prescribed in Table 3.1 and Table 3.2. The title of each category is intended to be typical rather than restrictive. The number in parentheses preceding each category refers to the applicable column or row in the tables.

(1) *Control stations*

- Spaces containing emergency sources of power and lighting.
- Wheelhouse and chartroom.
- Spaces containing the ship's radio equipment.
- Fire control stations.
- Control room for propulsion machinery when located outside the propulsion machinery space.
- Spaces containing centralized fire alarm equipment.
- Spaces containing centralized emer-gency public address system stations and equipment.

(2) *Stairways*

- Interior stairways, lifts, totally enclosed emergency escape trunks, and escalators (other than those wholly contained within the machinery spaces) for passengers and crew and enclosures thereto.
- In this connection a stairway which is enclosed at only one level is to be regarded as part of the space from which it is not separated by a fire door.

(3) *Corridors*

- Passenger and crew corridors and lobbies.

(4) *Evacuation stations and external escape routes*

- Survival craft stowage area.
- Open deck spaces and enclosed promenades forming lifeboat and liferaft embarkation and lowering stations.
- Assembly stations, internal and external.
- External stairs and open decks used for escape routes.
- The ship's side to the waterline in the lightest seagoing condition, superstructure and deckhouse sides situated below and adjacent to the liferaft and evacuation slide embarkation areas.

(5) *Open deck spaces*

- Open deck spaces and enclosed promenades clear of lifeboat and liferaft embarkation and lowering stations. To be considered in this category, enclosed promenades are to have no significant fire risk, meaning that furnishings are to be restricted to deck furniture. In addition, such spaces are to be naturally ventilated by permanent openings.
- Air spaces (the space outside superstructures and deckhouses).

(6) *Accommodation spaces of minor fire risk*

- Cabins containing furniture and furnishings of restricted fire risk.
- Offices and dispensaries containing furniture and furnishings of restricted fire risk.
- Public spaces containing furniture and furnishings of restricted fire risk and having a deck area of less than 50 [m^2].

(7) *Accommodation spaces of moderate fire risk*

- Spaces as in category (6) above but containing furniture and furnishings of other than restricted fire risk.
- Public spaces containing furniture and furnishings of restricted fire risk and having a deck area of 50 [m^2] or more.
- Isolated lockers and small store-rooms in accommodation spaces having areas less than 4 [m^2] (in which flammable liquids are not stowed)

- Motion picture projection and film stowage rooms. Diet kitchens (containing no open flame).
- Cleaning gear lockers (in which flammable liquids are not stowed).
- Laboratories (in which flammable liquids are not stowed).
- Pharmacies.
- Small drying rooms (having a deck area of 4 [m^2] or less).
- Specie rooms.
- Operating rooms.

(8) *Accommodation spaces of greater fire risk*

- Public spaces containing furniture and furnishings of other than restricted fire risk and having a deck area of 50 [m^2] or more.
- Barber shops and beauty parlours.
- Saunas.

(9) *Sanitary and similar spaces*

- Communal sanitary facilities, showers, baths, water closets, etc.
- Small laundry rooms.
- Indoor swimming pool area.
- Isolated pantries containing no cooking appliances in accommodation spaces.
- Private sanitary facilities are to be considered a portion of the space in which they are located.

IR.(9) “Isolated pantries containing no cooking appliances in accommodation spaces” are pantries enclosed in an accommodation space and are only accessible from accommodation spaces and/ or open deck. For the purpose of this categorization, “accommodation space” is as defined in Ch.1, Section 3.1. These pantries should not have communicating openings with main galley (category 12 space) and other enclosed spaces which are not accommodation spaces. (See Fig. 3.2.2.3.2).

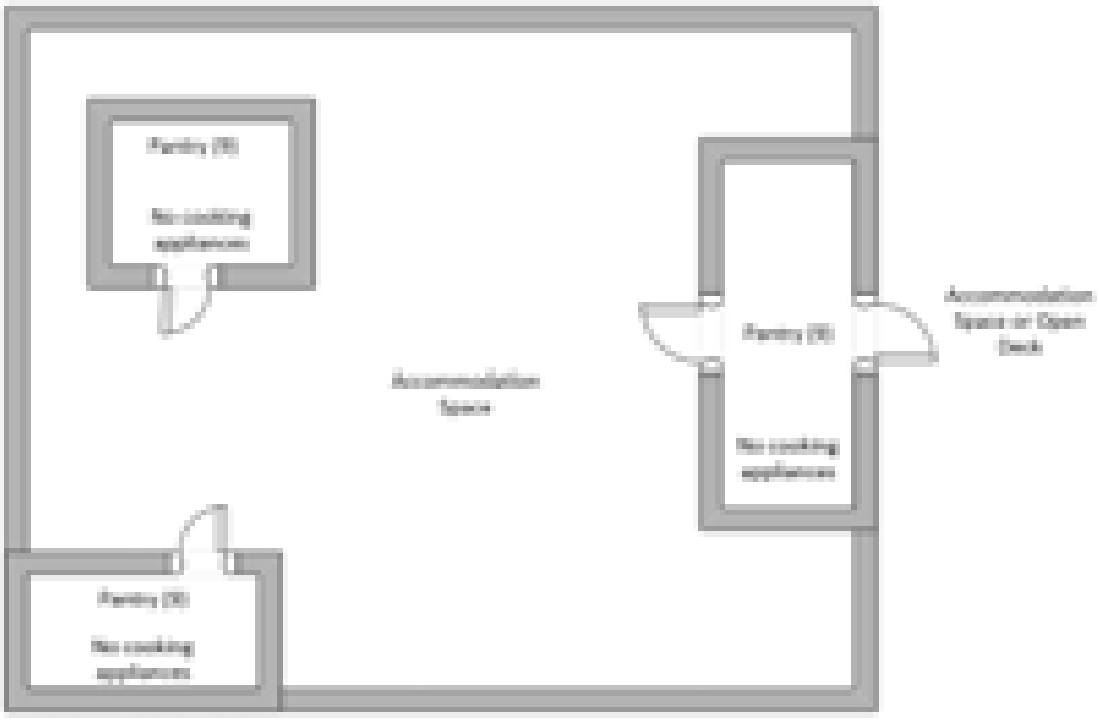

Fig. 3.2.2.3.2: Examples of Category 9 “Isolated pantry containing no cooking appliances in an accommodation space”.

(10) *Tanks, voids and auxiliary machinery spaces having little or no fire risk*

- Water tanks forming part of the ship's structure.
- Voids and cofferdams.
- Auxiliary machinery spaces which do not contain machinery having a pressure lubrication system and where storage of combustibles is prohibited, such as:
  - ventilation and air-conditioning rooms;
  - windlass room;
  - steering gear room;
  - stabilizer equipment room;
  - electrical propulsion motor room;
  - rooms containing section switchboards and purely electrical equipment other than oil-filled electrical transformers (above 10 kVA);
  - shaft alleys and pipe tunnels;
  - spaces for pumps and refrigeration machinery (not handling or using flammable liquids).
- Closed trunks serving the spaces listed above.
- Other closed trunks such as pipe and cable trunks.

IR.(10) In cases where urea or sodium hydroxide solution tanks for selective catalytic reduction (SCR) systems, exhaust gas recirculation (EGR) systems or exhaust gas cleaning systems (EGCS) are installed in a space separated from engine room, the solution tank space is to be considered in this category. The division between the engine room and the solution tank space should have a fire integrity of at least "A-0" class.

(11) *Auxiliary machinery spaces, cargo spaces, cargo and other oil tanks and other similar spaces of moderate fire risk*

- Cargo oil tanks.
- Cargo holds, trunkways and hatchways.
- Refrigerated chambers.
- Oil fuel tanks (where installed in a separate space with no machinery).
- Shaft alleys and pipe tunnels allowing storage of combustibles.
- Auxiliary machinery spaces as in category (10) which contain machinery having a pressure lubrication system or where storage of combustibles is permitted.
- Oil fuel filling stations.
- Spaces containing oil-filled electrical transformers (above 10 kVA).
- Spaces containing turbine and reciprocating steam engine driven auxiliary generators and small internal combustion engines of power output up to 110 kW driving generators, sprinkler, drencher or fire pumps, bilge pumps, etc.
- Closed trunks serving the spaces listed above.

(12) *Machinery spaces and main galleys*

- Main propulsion machinery rooms (other than electric propulsion motor rooms) and boiler rooms.
- Auxiliary machinery spaces other than those in categories (10) and (11) which contain internal combustion machinery or other oil-burning, heating or pumping units.
- Main galleys and annexes.
- Trunks and casings to the spaces listed above.

(13) *Store-rooms, workshops, pantries, etc*.

- Main pantries not annexed to galleys.
- Main laundry.
- Large drying rooms (having a deck area of more than 4 [m^2]).
- Miscellaneous stores.
- Mail and baggage rooms.
- Garbage rooms.
- Workshops (not part of machinery spaces, galleys, etc.).
- Lockers and store-rooms having areas greater than 4 [m^2], other than those spaces that have provisions for the storage of flammable liquids.

(14) *Other spaces in which flammable liquids are stowed*

- Paint lockers.
- Store-rooms containing flammable liquids (including dyes, medicines, etc.).
- Laboratories (in which flammable liquids are stowed);

.3 Where a single value is shown for the fire integrity of a boundary between two spaces, that value will apply in all cases;

.4 Notwithstanding the provision of a 3.2.2.2 there are no special requirements for material or integrity of boundaries where only a dash appears in the tables; and

.5 In respect of category (5) spaces the insulation values in Table 3.1 apply to ends of deckhouses and superstructures, and the insulation values in Table 3.2 apply to weather decks. In no case will the requirements of category (5) of Table 3.1 or Table 3.2 necessitate enclosure of spaces which need not be enclosed.

IR.6 Distribution boards may be located behind panels/linings within accommodation spaces including stairway enclosures, without the need to categorize the space. However, this space is not to be used for any other storage purpose.

If distribution boards are located in an identifiable space having a deck area of less than 4 [m^2], this space may be categorized as (7).

(Note : Category (7) applies to 'Accommodation spaces of moderate fire risk').

3.2.2.3.3 Continuous "B" class ceilings or linings, in association with the relevant decks or bulkheads, may be accepted as contributing wholly or in part, to the required insulation and integrity of a division.

3.2.2.3.4 *Construction and arrangement of saunas*

.1 The perimeter of the sauna is to be of "A" class boundaries and may include changing rooms, showers and toilets. The sauna is to be insulated to A-60 standard against other spaces except those inside of the perimeter and spaces of categories (5), (9) and (10).

.2 Bathrooms with direct access to saunas may be considered as part of them. In such cases, the door between sauna and the bathroom need not comply with fire safety requirements.

.3 The traditional wooden lining on the bulkheads and ceiling are permitted in the sauna. The ceiling above the oven is to be lined with a non-combustible plate with an air gap of at least 30 [mm]. The distance from the hot surfaces to combustible materials is to be atleast 500 [mm] or the combustible materials are to be protected (e.g. non-combustible plate with an air gap of at least 30 [mm]).

.4 The traditional wooden benches are permitted to be used in the sauna.

.5 The sauna door is to open outwards by pushing.

.6 Electrically heated ovens are to be provided with a timer.

3.2.2.4 *Fire integrity of bulkheads and decks in ships carrying not more than 36 passengers*

3.2.2.4.1 In addition to complying with the specific provisions for fire integrity of bulkheads and decks of passenger ships, the minimum fire integrity of bulkheads and decks are to be as prescribed in Table 3.3 and Table 3.4.

3.2.2.4.2 The following requirements govern application of the tables:

.1 Table 3.3 and Table 3.4 are to apply respectively to the bulkheads and decks separating adjacent spaces;
.2 For determining the appropriate fire integrity standards to be applied to divisions between adjacent spaces, such spaces are classified according to their fire risk as shown in categories (1) to (11) below. Where the contents and use of a space are such that there is a doubt as to its classification for the purpose of this regulation, or where it is possible to assign two or more classifications to a space, it is to be treated as a space within the relevant category having the most stringent boundary requirements. Smaller, enclosed rooms within a space that have less than 30 % communicating openings to that space are considered separate spaces. The fire integrity of the boundary bulkheads and decks of such smaller rooms are to be as prescribed in Table 3.3 and Table 3.4. The title of each category is intended to be typical rather than restrictive. The number in parentheses preceding each category refers to the applicable column or row in the tables.

Table 3.1 : Bulkheads not bounding either main vertical zones or horizontal zones (Ships with > 36 passengers)

Spaces		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Control stations	(1)	B-0[a]	A-0	A-0	A-0	A-0	A-60	A-60	A-60	A-0	A-0	A-60	A-60	A-60	A-60
Stairways	(2)		A-0[a]	A-0	A-0	A-0	A-0	A-15	A-15	A-0[c]	A-0	A-15	A-30	A-15	A-30
Corridors	(3)			B-15	A-60	A-0	B-15	B-15	B-15	B-15	A-0	A-15	A-30	A-0	A-30
Evacuation stations and external escape routes	(4)					A-0	A-60[b,d]	A-60[b,d]	A-60[b,d]	A-0[d]	A-0	A-60[b]	A-60[b]	A-60[b]	A-60[b]
Open deck spaces	(5)						A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0
Accommodation spaces of minor fire risk	(6)						B-0	B-0	B-0	C	A-0	A-0	A-30	A-0	A-30
Accommodation spaces of moderate fire risk	(7)							B-0	B-0	C	A-0	A-15	A-60	A-15	A-60
Accommodation spaces of greater fire risk	(8)								B-0	C	A-0	A-30	A-60	A-15	A-60
Sanitary and similar spaces	(9)									C	A-0	A-0	A-0	A-0	A-0
Tanks, voids and auxiliary machinery spaces having little or no fire risk	(10)										A-0[a]	A-0	A-0	A-0	A-0
Auxiliary machinery spaces, cargo spaces, cargo and other oil tanks and other similar spaces of moderate fire risk	(11)											A-0[a]	A-0	A-0	A-15
Machinery spaces and main galleys	(12)												A-0[a]	A-0	A-60
Store-rooms, workshops, pantries, etc.	(13)													A-0[a]	A-0
Other spaces in which flammable liquids are stowed	(14)														A-30

See notes following Table 3.2.

Table 3.2 : Decks not forming steps in main vertical zones nor bounding horizontal zones (Ships with > 36 passengers)

Space below↓ Space above→		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Control stations	(1)	A-30	A-30	A-15	A-0	A-0	A-0	A-15	A-30	A-0	A-0	A-0	A-60	A-0	A-60
Stairways	(2)	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-30	A-0	A-30
Corridors	(3)	A-15	A-0	A-0[a]	A-60	A-0	A-0	A-15	A-15	A-0	A-0	A-0	A-30	A-0	A-30
Evacuation stations and external escape routes	(4)	A-0	A-0	A-0	A-0	-	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0
Open deck spaces	(5)	A-0	A-0	A-0	A-0	-	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0
Accommodation spaces of minor fire risk	(6)	A-60	A-15	A-0	A-60	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0
Accommodation spaces of moderate fire risk	(7)	A-60	A-15	A-15	A-60	A-0	A-0	A-15	A-15	A-0	A-0	A-0	A-0	A-0	A-0
Accommodation spaces of greater fire risk	(8)	A-60	A-15	A-15	A-60	A-0	A-15	A-15	A-30	A-0	A-0	A-0	A-0	A-0	A-0
Sanitary and similar spaces	(9)	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0
Tanks, voids and auxiliary machinery spaces having little or no fire risk	(10)	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0	A-0[a]	A-0	A-0	A-0	A-0
Auxiliary machinery spaces, cargo spaces, cargo and other oil tanks and other similar spaces of moderate fire risk	(11)	A-60	A-60	A-60	A-60	A-0	A-0	A-15	A-30	A-0	A-0	A-0[a]	A-0	A-0	A-30
Machinery spaces and main galleys	(12)	A-60	A-60	A-60	A-60	A-0	A-60	A-60	A-60	A-0	A-0	A-30	A-30[a]	A-0	A-60
Store-rooms, workshops, pantries, etc.	(13)	A-60	A-30	A-15	A-60	A-0	A-15	A-30	A-30	A-0	A-0	A-0	A-0	A-0	A-0
Other spaces in which flammable liquids are stowed	(14)	A-60	A-60	A-60	A-60	A-0	A-30	A-60	A-60	A-0	A-0	A-0	A-0	A-0	A-0

Note: To be applied to Table 3.1 and Table 3.2.

a Where adjacent spaces are in the same numerical category and superscript "a" appears a bulkhead or deck between such spaces need not be fitted if deemed unnecessary by IRS. For example, in category (12) a bulkhead will not be required between a galley and its annexed pantries provided the pantry bulkhead and decks maintain the integrity of the galley boundaries. A bulkhead is, however, required between a galley and machinery space even though both spaces are in category (12).

b The ship's side, to the waterline in the lightest seagoing condition, superstructure and deckhouse sides situated below and adjacent to liferafts and evacuation slides may be reduced to "A-30".

c Where public toilets are installed completely within the stairway enclosure, the public toilet bulkhead within the stairway enclosure can be of "B" class integrity.

d Where spaces of categories (6), (7), (8) and (9) are located completely within the outer perimeter of the assembly station, the bulkheads of these spaces are allowed to be of "B-0" class integrity. Control positions for audio, video and light installations may be considered as part of the assembly station.

(1) *Control stations*

- Spaces containing emergency sources of power and lighting.
- Wheelhouse and chartroom.
- Spaces containing the ship's radio equipment.
- Fire control stations.
- Control room for propulsion machinery when located outside the machinery space.
- Spaces containing centralized fire alarm equipment.

(2) *Corridors*

- Passenger and crew corridors and lobbies.

(3) *Accommodation spaces*

- Spaces as defined in Ch.1, 3.1 but excluding corridors.

(4) *Stairways*

- Interior stairways, lifts, totally enclosed emergency escape trunks, and escalators (other than those wholly contained within the machinery spaces) and enclosures thereto.
- In this connection, a stairway which is enclosed only at one level is to be regarded as part of the space from which it is not separated by a fire door.

(5) *Service spaces (low risk)*

- Lockers and store-rooms not having provisions for the storage of flammable liquids and having areas less than 4 [m^2] and drying rooms and laundries.

(6) *Machinery spaces of category A*

- Spaces as defined in Ch.1, 3.31.

(7) *Other machinery spaces*

- Electrical equipment rooms (auto-telephone exchange, air-conditioning duct spaces).
- Spaces as defined in Chapter 1, 3.30 excluding machinery spaces of category A.

IR.(7) In cases where urea or sodium hydroxide solution tanks for selective catalytic reduction (SCR) systems, exhaust gas recirculation (EGR) systems or exhaust gas cleaning systems (EGCS) are installed in a space separated from engine room, the solution tank space is to be considered in this category. The division between the engine room and the solution tank space should have a fire integrity of at least "A-0" class.

(8) *Cargo spaces*

- All spaces used for cargo (including cargo oil tanks) and trunkways and hatchways to such spaces, other than special category spaces.

(9) *Service spaces (high risk)*

- Galleys, pantries containing cooking appliances, paint and lamp rooms, lockers and store-rooms having areas of 4 [m^2] or more, spaces for the storage of flammable liquids, saunas and workshops other than those forming part of the machinery spaces.

(10) *Open decks*

- Open deck spaces and enclosed promenades having little or no fire risk. Enclosed promenades are to have no significant fire risk, meaning that furnishing are to be restricted to deck furniture. In addition, such spaces are to be naturally ventilated by permanent openings. Air spaces (the space outside superstructures and deckhouses).

(11) *Special category and ro-ro spaces*

- Spaces as defined in Ch.1, 3.41 and 3.46.

IR.2 For location of distribution boards in accommodation spaces, refer to IR.6, 3.2.2.3.2 of this chapter.

.3 In determining the applicable fire integrity standard of a boundary between two spaces within a main vertical zone or horizontal zone which is not protected by an automatic sprinkler system complying with the provisions of the Fire Safety Systems Code (Chapter 8) or between such zones neither of which is so protected, the higher of the two values given in the tables are to apply; and

.4 In determining the applicable fire integrity standard of a boundary between two spaces within a main vertical zone or horizontal zone which is protected by an automatic sprinkler system complying with the provisions of the Fire Safety Systems Code (Chapter 8) or between such zones both of which are so protected, the lesser of the two values given in the tables is to apply. Where a zone with sprinklers and a zone without sprinklers meet within accommodation and service spaces, the higher of the two values given in the tables is to apply to the division between the zones.

3.2.2.4.3 Continuous "B" class ceilings or linings, in association with the relevant decks or bulkheads, may be accepted as contributing, wholly or in part, to the required insulation and integrity of a division.

3.2.2.4.4 External boundaries which are required in 5.2 to be of steel or other equivalent material may be pierced for the fitting of windows and sidescuttles provided that there is no requirement for such boundaries of passenger ships to have "A" class integrity. Similarly, in such boundaries which are not required to have "A" class integrity, doors may be constructed of materials which are to the satisfaction of IRS.

3.2.2.4.5 Saunas are to comply with 3.2.2.3.4.

3.2.2.5 *Protection of stairways and lifts in accommodation area*

3.2.2.5.1 Stairways are to be within enclosures formed of "A" class divisions, with positive means of closure at all openings, except that:

.1 a stairway connecting only two decks need not be enclosed, provided the integrity of the deck is maintained by proper bulkheads or self-closing doors in one 'tween-deck space. When a stairway is closed in one 'tween-deck space, the stairway enclosure is to be protected in accordance with the tables for decks in 3.2.2.3 or 3.2.2.4; and

.2 stairways may be fitted in the open in a public space, provided they lie wholly within the public space.

3.2.2.5.2 Lift trunks are to be so fitted as to prevent the passage of smoke and flame from one ‘tween-deck to another and are to be provided with means of closing so as to permit the control of draught and smoke. Machinery for lifts located within stairway enclosures are to be arranged in a separate room, surrounded by steel boundaries, except that small passages for lift cables are permitted. Lifts which open into spaces other than corridors, public spaces, special category spaces, stairways and external areas are not to open into stairways included in the means of escape.

3.2.2.6 *Arrangement of cabin balconies*

On passenger ships constructed on or after 1 July 2008, non-load bearing partial bulkheads which separate adjacent cabin balconies are to be capable of being opened by the crew from each side for the purpose of fighting fires.

3.2.2.7 Protection of atriums

3.2.2.7.1 Atriums are to be within enclosures formed of "A" class divisions having a fire rating determined in accordance with Tables 3.2 and 3.4, as applicable.

3.2.2.7.2 Decks separating spaces within atriums are to have a fire rating determined in accordance with Tables 3.2 and 3.4, as applicable.

Table 3.3 : Fire integrity of bulkheads separating adjacent spaces (Ships with ≤ 36 passengers)

Spaces		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Control stations	(1)	A-0 [c]	A-0	A-60	A-0	A-15	A-60	A-15	A-60	A-60	*	A-60
Corridors	(2)		C[e]	B-0[e]	A-0[a] B-0[e]	B-0[e]	A-60	A-0	A-0	A-15 A-0[d]	*	A-30[g]
Accommo-dation spaces	(3)			C[e]	A-0[a] B-0[e]	B-0[e]	A-60	A-0	A-0	A-15 A-0[d]	*	A-30 A-0[d]
Stairways	(4)				A-0[a] B-0[e]	A-0[a] B-0[e]	A-60	A-0	A-0	A-15 A-0[d]	*	A-30[g]
Service spaces (low risk)	(5)					C[e]	A-60	A-0	A-0	A-0	*	A-0
Machinery spaces of category A	(6)						*	A-0	A-0	A-60	*	A-60
Other machinery spaces	(7)							A-0[b]	A-0	A-0	*	A-0
Cargo spaces	(8)								*	A-0	*	A-0
Service spaces (high risk)	(9)									A-0[b]	*	A-30
Open decks	(10)											A-0
Special category and ro-ro spaces [g]	(11)											A-30[g]

Table 3.4 : Fire integrity of decks separating adjacent spaces (Ships with ≤ 36 passengers)												
-- Space above --												
Space below		**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**	**(11)**
Control stations	(1)	A-0	A-0	A-0	A-0	A-0	A-60	A-0	A-0	A-0	*	A-60[g]
Corridors	(2)	A-0	*	*	A-0	*	A-60	A-0	A-0	A-0	*	A-30[g]
Accommo-dation spaces	(3)	A-60	A-0	*	A-0	*	A-60	A-0	A-0	A-0	*	A-30 A-0[d]
Stairways	(4)	A-0	A-0	A-0	*	A-0	A-60	A-0	A-0	A-0	*	A-30[g]
Service spaces (low risk)	(5)	A-15	A-0	A-0	A-0	*	A-60	A-0	A-0	A-0	*	A-0
Machinery spaces of category A	(6)	A-60	A-60	A-60	A-60	A-60	*	A-60[f]	A-30	A-60	*	A-60
Other machinery spaces	(7)	A-15	A-0	A-0	A-0	A-0	A-0	*	A-0	A-0	*	A-0
Cargo spaces	(8)	A-60	A-0	A-0	A-0	A-0	A-0	A-0	*	A-0	*	A-0
Service spaces (high risk)	(9)	A-60	A-30 A-0[d]	A-30 A-0[d]	A-30 A-0[d]	A-0	A-60	A-0	A-0	A-0	*	A-30
Open decks	(10)	*	*	*	*	*	*	*	*	*	-	A-0
Special category and ro-ro spaces [g]	(11)	A-60	A-30[g]	A-30 A-0[d]	A-30[g]	A-0	A-60[g]	A-0	A-0	A-30	A-0	A-30[g]

Notes: To be applied to both Table 3.3 and Table 3.4 as appropriate.

a For clarification as to which applies, see 3.2.2.2 and 3.2.2.5.

b Where spaces are of the same numerical category and superscript b appears, a bulkhead or deck of the rating shown in the tables is only required when the adjacent spaces are for a different purpose, (e.g. in category (9)). A galley next to a galley does not require a bulkhead but a galley next to a paint room requires an "A-0" bulkhead.

c Bulkhead separating the wheelhouse and chartroom from each other may have a "B-0" rating. No fire rating is required for those partitions separating the navigation bridge and the safety centre when the latter is within the navigation bridge

IRc A navigation locker that can be accessed from the wheelhouse is to be considered as a control station with respect to the requirements in Table 3.3 and the bulkhead separating the wheelhouse and such a locker is to have fire integrity of at least “B-0” class

d See 3.2.2.4.2.3 and 3.2.2.4.2.4.

e For the application of 3.2.2.1.1b), "B-0" and "C", where appearing in Table 3.3, is to be read as "A-0".

f Fire insulation need not be fitted if the machinery space in category (7), in the opinion of IRS, has little or no fire risk.

g Ships constructed before 01 July 2014 are to comply, as a minimum, with the previous requirements applicable at the time the ship was constructed, as specified in Chapter 1, 1.3.

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* Where an asterisk appears in the tables, the division is required to be of steel or other equivalent material, but is not required to be of "A" class standard. However, where a deck, except in a category (10) space, is penetrated for the passage of electric cables, pipes and vent ducts, such penetrations are to be made tight to prevent the passage of flame and smoke. Divisions between control stations (emergency generators) and open decks may have air intake openings without means for closure, unless a fixed gas fire-fighting system is fitted.

For the application of 3.2.2.1.1b), an asterisk, where appearing in Table 3.4, except for categories (8) and (10), is to be read as “A-0”.

3.2.3 Cargo ships except tankers

3.2.3.1 *Methods of protection in accommodation area*

3.2.3.1.1 One of the following methods of protection is to be adopted in accommodation and service spaces and control stations:

.1 **Method IC**

The construction of internal divisional bulkheads of non-combustible "B" or "C" class divisions generally without the installation of an automatic sprinkler, fire detection and fire alarm system in the accommodation and service spaces, except as required by 1.5.5.1; or

.2 **Method IIC**

The fitting of an automatic sprinkler, fire detection and fire alarm system as required by 1.5.5.2 for the detection and extinction of fire in all spaces in which fire might be expected to originate, generally with no restriction on the type of internal divisional bulkheads; or

.3 **Method IIIC**

The fitting of a fixed fire detection and fire alarm system as required by 1.5.5.3, in spaces in which a fire might be expected to originate, generally with no restriction on the type of internal divisional bulkheads, except that in no case must the area of any accommodation space or spaces bounded by an "A" or "B" class division exceed 50 [m^2]. However, increasing this area for public spaces will be specially considered.

IR3.2.3.1 For fire protection materials refer Fig.3.1 and Table 3.5b and Table 3.5c.

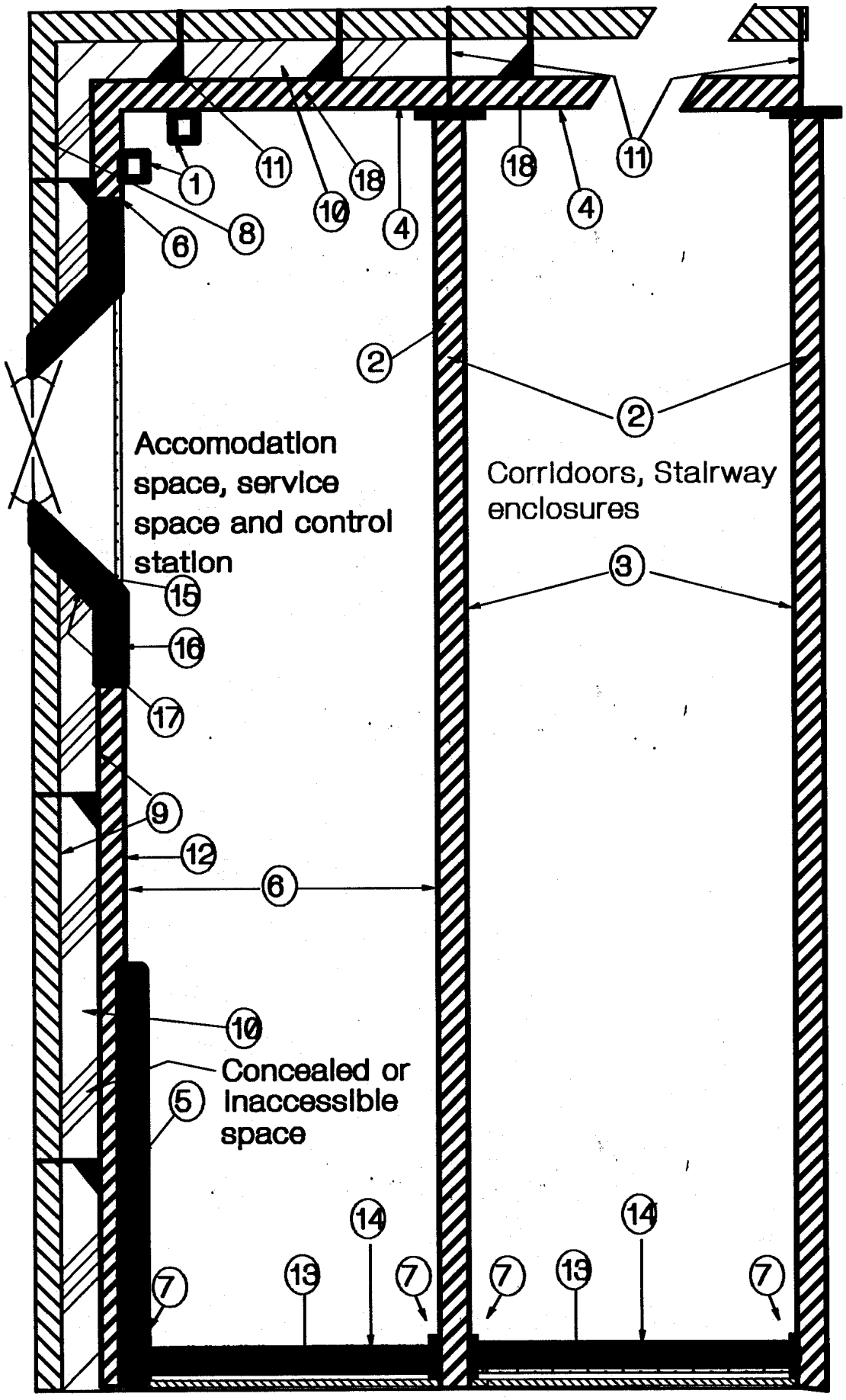

Fig. 3.1 : Fire protection materials

Table 3.5b : Method IC

		Requirements for Components						
		A Non-combu-stible material Ch.2, Cl. 2.3.1.2.2	**B Non-combu-stible material Ch.2, Cl. 2.3.1.1**	**C Low flame spread Ch.2, Cl. 2.3.2.4**	**D Equi-valent volume Ch.2, Cl. 2.3.2.3.a)**	**E Calorific value Ch.2, Cl. 2.3.2.2**	**F Smoke produc-tion Ch.2, Sec.3**	**G Not readily ignite Ch.2, Cl. 1.4.4.1**
1	Moulding				X			
2	Panel	X						
3	Painted surfaces or veneer or fabric or foils			X	X	X	X(2)	
4	Painted surfaces or veneer or fabric or foils			X	X	X	X(2)	
5	Decorative panel				X		X	
6	Painted surfaces or veneer or fabric or foils				X	X	X(2)	
7	Skirting board				X			
8	Insulation		X(1)					
9	Surfaces and paints in concealed or inaccessible spaces			X				
10	Draught stop	X						
11	Grounds and supports	X		X				
12	Lining	X						
13	Primary deck covering 1st layer						X(4)	X
14	Floor finishing			X(3)			X	
15	Window box	X						
16	Window box surface			X	X	X	X	
17	Window box surface in concealed or inaccessible spaces			X				
18	Ceiling panel	X						

(1) Vapour barriers used on pipes for cold services (cold service means refrigeration system with temperatures below ambient air and sea water) may be of combustible materials provided that their surface has low flame spread characteristics.

(2) Applicable to paints, varnishes and other finishes. (Ch.2, Cl. 3.2.1).

(3) Only in corridors and stairway enclosures.

(4) Only in accommodation and service spaces and control stations (Ch.2, Cl.3.3.1).

- As far as window boxes construction is concerned, reference is also to be made to MSC/Circ.917 and MSC/Circ. 917 Corr.1

Table 3.5c : Method IIC - IIIC								
		Requirements for Components						
		A Non-combustible material Ch.2, Cl. 2.3.1.2.2	**B Non-combustible material Ch.2, Cl. 2.3.1.1**	**C Low flame spread Ch.2, Cl. 2.3.2.4**	**D Equivalent volume Ch.2, Cl. 2.3.2.3.a)**	**E Calorific value Ch.2, Cl. 2.3.2.2**	**F Smoke production Ch.2, Sec.3**	**G Not readily ignite Ch.2, Cl. 1.4.4.1**
1	Moulding				X(3)			
2	Panel	X(4)						
3	Painted surfaces or veneer or fabric or foils			X	X	X	X(5)	
4	Painted surfaces or veneer or fabric or foils			X	X(3)	X(2)	X(5)	
5	Decorative panel				X(3)		X	
6	Painted surfaces or veneer or fabric or foils				X(3)	X(2)	X(5)	
7	Skirting board				X(3)			
8	Insulation		X(1)					
9	Surfaces and paints in concealed or inaccessible spaces			X				
10	Draught stop	X(4)						
11	Grounds and supports	X(4)		X				
12	Lining	X(4)						
13	Primary deck covering 1st layer						X(7)	X
14	Floor finishing			X(6)			X	
15	Window box	X(4)						
16	Window box surface			X(3)	X(3)	X(2)	X	
17	Window box surface in concealed or inaccessible spaces			X				
18	Ceiling panel	X(4)						

(1) Vapour barriers used on pipes for cold services (cold service means refrigeration system with temperatures below ambient air and sea water) may be of combustible materials provided that their surface has low flame spread characteristics. (Ch.2, Cl.2.3.1.1).

(2) Where the material is fitted on non-combustible bulkheads, ceiling and lining in accommodation and service spaces. (Ch.2, Cl.2.3.2.2).

(3) To be applied to those accommodation and service spaces bounded by non-combustible bulkhead, ceiling and linings. (Ch.2, Cl.2.3.2.3.a).

(4) Only in corridors and stairway enclosures serving accommodation and service spaces and control stations. (Ch.2, Cl.2.3.1.2.2).

(5) Applicable to paints, varnishes and other finishes. (Ch.2, Cl. 3.2.1).

(6) Only in corridors and stairway enclosures.

(7) Only in accommodation and service spaces and control stations (Ch.2, Cl.3.3.1)

- As far as window boxes construction is concerned, reference is also to be made to MSC/Circ.917 and MSC/Circ. 917 Corr.1

3.2.3.1.2 The requirements for the use of non-combustible materials in the construction and insulation of boundary bulkheads of machinery spaces, control stations, service spaces, etc., and the protection of the above stairway enclosures and corridors will be common to all three methods outlined in 3.2.3.1.1.

3.2.3.2 *Bulkheads within accommodation area*

3.2.3.2.1 All bulkheads required to be 'B' class divisions are to extend from deck to deck and to the shell or other boundaries, unless continuous 'B' class ceilings and linings are fitted on both sides of the bulkhead in which case the bulkhead may terminate at the continuous ceiling or lining. MSC Circ.917 "Guidelines on fire safety construction in accommodation areas" may be referred for details of construction.

IR3.2.3.2.1 MSC/Circ.917 "Guidelines on fire safety construction in accommodation areas" as corrected by MSC/Circ.917/Corr.1 may be referred for details of construction.

3.2.3.2.2 Method IC

Bulkheads not required by this or other regulations for cargo ships to be "A" or "B" class divisions, are to be of at least "C" class construction.

3.2.3.2.3 Method IIC

There are no restriction on the construction of bulkheads not required by this or other regulations for cargo ships to be "A" or "B" class divisions except in individual cases where "C" class bulkheads are required in accordance with Table 3.5a.

3.2.3.2.4 Method IIIC

There are no restriction on the construction of bulkheads not required for cargo ships to be "A" or "B" class divisions except that the area of any accommodation space or spaces bounded by a continuous "A" or "B" class division must in no case exceed 50 [m^2], except in individual cases where "C" class bulkheads are required in accordance with Table 3.5a. Consideration may be given to increasing this area for public spaces.

3.2.3.3 *Fire integrity of bulkheads and decks*

3.2.3.3.1 In addition to complying with the specific provisions for fire integrity of bulkheads and decks of cargo ships, the minimum fire integrity of bulkheads and decks are to be as prescribed in Table 3.5a and Table 3.6.

3.2.3.3.2 The following requirements are to govern application of the tables:

.1 Table 3.5a and Table 3.6 apply respectively to the bulkheads and decks separating adjacent spaces;

.2 For determining the appropriate fire integrity standards to be applied to divisions between adjacent spaces, such spaces are classified according to their fire risk as shown in categories (1) to (11) below. Where the contents and use of a space are such that there is a doubt as to its classification for the purpose of this section, or where it is possible to assign two or more classifications to a space, it is to be treated as a space within the relevant category having the most stringent boundary requirements. Smaller, enclosed rooms within a space that have less than 30% communicating openings to that space are considered separate spaces. The fire integrity of the boundary bulkheads and decks of such smaller rooms are to be as prescribed in Table 3.5a and Table 3.6. The title of each category is intended to be typical rather than restrictive. The number in parentheses preceding each category refers to the applicable column or row in the tables;

(1) *Control stations*

- Spaces containing emergency sources of power and lighting.
- Wheelhouse and chartroom.
- Spaces containing the ship's radio equipment.
- Fire control stations.
- Control room for propulsion machinery when located outside the machinery space.
- Spaces containing centralized fire alarm equipment.

(2) *Corridors*

- Corridors and lobbies.

(3) *Accommodation spaces*

- Spaces as defined in Ch.1, 3.1, excluding corridors.

(4) *Stairways*

- Interior stairway, lifts, totally enclosed emergency escape trunks, and escalators (other than those wholly contained within the machinery spaces) and enclosures thereto.

- In this connection, a stairway which is enclosed only at one level is to be regarded as part of the space from which it is not separated by a fire door.

(5) *Service spaces (low risk)*

- Lockers and store-rooms not having provisions for the storage of flammable liquids and having areas less than 4 [m^2] and drying rooms and laundries.

(6) *Machinery spaces of category A*

- Spaces as defined in Ch.1, 3.31.

(7) *Other machinery spaces*

- Electrical equipment rooms (auto-telephone exchange, air-conditioning duct spaces).
- Spaces as defined in Ch.1, 3.30 excluding machinery spaces of category A.

IR.(7) In cases where urea or sodium hydroxide solution tanks for selective catalytic reduction (SCR) systems, exhaust gas recirculation (EGR) systems or exhaust gas cleaning systems (EGCS) are installed in a space separated from engine room, the solution tank space is to be considered in this category. The division between the engine room and the solution tank space should have a fire integrity of at least "A-0" class.

(8) *Cargo spaces*

- All spaces used for cargo (including cargo oil tanks) and trunkways and hatchways to such spaces.

(9) *Service spaces (high risk)*

- Galleys, pantries containing cooking appliances, saunas, paint lockers and store-rooms having areas of 4 [m^2] or more, spaces for the storage of flammable liquids, and workshops other than those forming part of the machinery spaces.

(10) *Open decks*

- Open deck spaces and enclosed promenades having little or no fire risk. To be considered in this category, enclosed promenades are to have no significant fire risk, meaning that furnishings are to be restricted to deck furniture. In addition, such spaces are to be naturally ventilated by permanent openings.

- Air spaces (the space outside superstructures and deckhouses).

(11) *Ro-ro and vehicle spaces*

- Ro-ro spaces as defined in Ch.1, 3.41.
- Vehicle spaces as defined in Ch.1, 3.49.

IR.2 For location of distribution boards in accommodation spaces, refer to IR.6, 3.2.2.3.2 of this chapter.

IR 3.2.3.3.2 Navigation equipment room (radar transmitter) and battery rooms are considered as category (1) Control stations for the purpose of this Rule.

Note 1 – Provision chambers are to be treated as store rooms.

2 – Refrigerated provision chambers are to be considered as category (9) service spaces if thermally insulated with combustible materials or category (5) service spaces if thermally insulated with non-combustible materials.

3.2.3.3.3 Continuous "B" class ceilings or linings, in association with the relevant decks or bulkheads, may be accepted as contributing, wholly or in part, to the required insulation and integrity of a division.

3.2.3.3.4 External boundaries which are required in Ch.3, 5.2 to be of steel or other equivalent material may be pierced for the fitting of windows and sidescuttles provided that there is no requirement for such boundaries to have "A" class integrity. Similarly, in such boundaries which are not required to have "A" class integrity, doors may be constructed of materials which are to the satisfaction of IRS.

3.2.3.3.5 Saunas are to comply with 3.2.2.3.4.

3.2.3.4 *Protection of stairways and lift trunks in accommodation spaces, service spaces and control stations*

3.2.3.4.1 Stairways, which penetrate only a single deck are to be protected, at least at one level, by 'B-0' class division and self closing doors. Lifts penetrating a single deck only, are to be surrounded by 'A-0' class division with steel doors at both levels. Stairways and lift trunks penetrating more than a single deck are to be surrounded by at least 'A-0' class division and be protected by self closing doors at all levels.

IR 3.2.3.4.1 Dumb-waiters (food lifts) are to be regarded as lifts.

3.2.3.4.2 On ships having accommodation for 12 persons or less, where stairways penetrate more than a single deck and where there are at least two escape routes direct to the open deck at every accommodation level, the "A-0" requirements of 3.2.3.4.1 may be reduced to "B-0".

Table 3.5a : Fire integrity of bulkheads separating adjacent spaces (Cargo ships)												
Spaces		**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**	**(11)**
Control stations	(1)	A-0[e]	A-0	A-60	A-0	A-15	A-60	A-15	A-60	A-60	*	A-60
Corridors	(2)		C	B-0	B-0 A-0[f]	B-0	A-60	A-0	A-0	A-0	*	A-30
Accommo-dation spaces	(3)			C[a,b]	B-0 A-0[c]	B-0	A-60	A-0	A-0	A-0	*	A-30
Stairways	(4)				B-0 A-0[c]	B-0 A-0[c]	A-60	A-0	A-0	A-0	*	A-30
Service spaces (low risk)	(5)					C	A-60	A-0	A-0	A-0	*	A-0
Machinery spaces of category A	(6)						*	A-0	A-0[g]	A-60	*	A-60[f]
Other machinery spaces	(7)							A-0[d]	A-0	A-0	*	A-0
Cargo spaces	(8)								*	A-0	*	A-0
Service spaces (high risk)	(9)									A-0[d]	*	A-30
Open decks	(10)										-	A-0
Ro-ro and vehicle spaces	(11)											A-30[i]

Table 3.6 : Fire integrity of decks separating adjacent spaces (Cargo ships)												
		-- Space above --										
Space below		**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**	**(11)**
Control stations	(1)	A-0	A-0	A-0	A-0	A-0	A-60	A-0	A-0	A-0	*	A-60
Corridors	(2)	A-0	*	*	A-0	*	A-60	A-0	A-0	A-0	*	A-30
Accommo-dation spaces	(3)	A-60	A-0	*	A-0	*	A-60	A-0	A-0	A-0	*	A-30
Stairways	(4)	A-0	A-0	A-0	*	A-0	A-60	A-0	A-0	A-0	*	A-30
Service spaces (low risk)	(5)	A-15	A-0	A-0	A-0	*	A-60	A-0	A-0	A-0	*	A-0
Machinery spaces of category A	(6)	A-60	A-60	A-60	A-60	A-60	*	A-60[i]	A-30	A-60	*	A-60
Other machinery spaces	(7)	A-15	A-0	A-0	A-0	A-0	A-0	*	A-0	A-0	*	A-0
Cargo spaces	(8)	A-60	A-0	A-0	A-0	A-0	A-0	A-0	*	A-0	*	A-0
Service spaces (high risk)	(9)	A-60	A-0	A-0	A-0	A-0	A-60	A-0	A-0	A-0[d]	*	A-30
Open decks	(10)	*	*	*	*	*	*	*	*	*	-	A-0[j]
Ro-ro and vehicle spaces	(11)	A-60	A-30	A-30	A-30	A-0	A-60	A-0	A-0	A-30	A-0[j]	A-30[j]

Note: To be applied to Table 3.5a and Table 3.6 as appropriate.

a No special requirements are imposed upon bulkheads in methods IIC and IIIC fire protection.

b In case of method IIIC "B" class bulkheads of "B-0" rating are to be provided between spaces or groups of spaces of 50 [m^2] and over in area.

c For clarification as to which applies, see 3.2.3.2 and 3.2.3.4.

d Where spaces are of the same numerical category and superscript d appear, a bulkhead or deck of the rating shown in the tables is only required when the adjacent spaces are for a different purpose, e.g. in category (9), a galley next to a galley does not require a bulkhead but a galley next to a paint room requires an "A-0" bulkhead.

e Bulkheads separating the wheelhouse, chartroom and radio room from each other may have a "B-0" rating.

IRe A navigation locker that can be accessed from the wheelhouse is to be considered as a control station with respect to the requirements in Table 3.5a and the bulkhead separating the wheelhouse and such a locker is to have fire integrity of at least “B-0” class

f An "A-0" rating may be used if no dangerous goods are intended to be carried or if such goods are stowed not less than 3 [m] horizontally from such a bulkhead.

g For cargo spaces in which dangerous goods are intended to be carried, Ch.7, 2.3.8 applies.

h Deleted.

i Fire insulation need not be fitted if the machinery in category (7) if, it has little or no fire risk.

j Ships constructed before 01 July 2014 are to comply, as a minimum, with the previous requirements applicable at the time the ship was constructed, as specified in Chapter 1, 1.3.

* Where an asterisk appears in the tables, the division is required to be of steel or other equivalent material but is not required to be of "A" class standard. However, where a deck, except an open deck, is penetrated for the passage of electric cables, pipes and vent ducts, such penetrations are to be made tight to prevent the passage of flame and smoke. Divisions between control stations (emergency generators) and open decks may have air intake openings without means for closure, unless a fixed gas fire-fighting system is fitted.

3.2.4 Tankers

3.2.4.1 *Application*

For tankers, only method IC as defined in 3.2.3.1.1 is to be used.

3.2.4.2 *Fire integrity of bulkheads and decks*

3.2.4.2.1 In lieu of 3.2.3 and in addition to complying with the specific provisions for fire integrity of bulkheads and decks of tankers, the minimum fire integrity of bulkheads and decks is to be as prescribed in Table 3.7 and Table 3.8.

3.2.4.2.2 The following requirements govern application of the tables:

.1 Table 3.7 and Table 3.8 apply respectively to the bulkhead and decks separating adjacent spaces;

.2 For determining the appropriate fire integrity standards to be applied to divisions between adjacent spaces, such spaces are classified according to their fire risk as shown in categories (1) to (10) below. Where the contents and use of a space are such that there is a doubt as to its classification for the purpose of this regulation, or where it is possible to assign two or more classifications

to a space, it is to be treated as a space within the relevant category having the most stringent boundary requirements. Smaller, enclosed areas within a space that have less than 30% communicating openings to that space are considered separate areas. The fire integrity of the boundary bulkheads and decks of such smaller spaces is to be as prescribed in Table 3.7 and Table 3.8. The title of each category is intended to be typical rather than restrictive. The number in parentheses preceding each category refers to the applicable column or row in the tables;

(1) *Control stations*

- Spaces containing emergency sources of power and lighting.
- Wheelhouse and chartroom.
- Spaces containing the ship's radio equipment.
- Fire control stations.
- Control room for propulsion machinery when located outside the machinery space.
- Spaces containing centralized fire alarm equipment.

(2) *Corridors*

- Corridors and lobbies.

(3) *Accommodation spaces*

- Spaces as defined in Ch.1, 3.1, excluding corridors.

(4) *Stairways*

- Interior stairways, lifts, totally enclosed emergency escape trunks, and escalators (other than those wholly contained within the machinery spaces) and enclosures thereto.
- In this connection, a stairway which is enclosed only at one level is to be regarded as part of the space from which it is not separated by a fire door.

(5) *Service spaces (low risk)*

- Lockers and store-rooms not having provisions for the storage of flammable liquids and having areas less than 4 [m^2] and drying rooms and laundries.

(6) *Machinery spaces of category A*

- Spaces as defined in Ch.1, 3.31.

(7) *Other machinery spaces*

- Electrical equipment rooms (auto-telephone exchange and air-conditioning duct spaces).\
- Spaces as defined in Ch.1, 3.30 excluding machinery spaces of category A.

IR.(7) In cases where urea or sodium hydroxide solution tanks for selective catalytic reduction (SCR) systems, exhaust gas recirculation (EGR) systems or exhaust gas cleaning systems (EGCS) are installed in a space separated from engine room, the solution tank space is to be considered in this category. The division between the engine room and the solution tank space should have a fire integrity of at least "A-0" class.

(8) *Cargo pump-rooms*

- Spaces containing cargo pumps and entrances and trunks to such spaces.

(9) *Service spaces (high risk)*

- Galleys, pantries containing cooking appliances, saunas, paint lockers and store-rooms having areas of 4 [m^2] or more, spaces for the storage of flammable liquids and workshops other than those forming part of the machinery spaces.

(10) *Open decks*

- Open deck spaces and enclosed promenades having little or no fire risk. To be considered in this category, enclosed promenades are to have no significant fire risk, meaning that furnishings are to be restricted to deck furniture. In addition, such spaces are to be naturally ventilated by permanent openings.
- Air spaces (the space outside superstructures and deckhouses).

IR.2 For location of distribution boards in accommodation spaces, refer to IR.6, 3.2.2.3.2 of this chapter.

IR 3.2.4.2.2 Navigation equipment room (radar transmitter) and battery rooms are considered as category (1) Control stations for the purpose of this Rule.

Note 1 – Provision chambers are to be treated as store rooms.

2 – Refrigerated provision chambers are to be considered as category (9) service spaces if thermally insulated with combustible materials or category (5) service spaces if thermally insulated with non-combustible materials.

3.2.4.2.3 Continuous "B" class ceilings or linings, in association with the relevant decks or bulkheads, may be accepted as contributing, wholly or in part, to the required insulation and integrity of a division.

3.2.4.2.4 External boundaries which are required in Ch.3, 5.2 to be of steel or other equivalent material may be pierced for the fitting of windows and side scuttles provided that there is no requirement for such boundaries of tankers to have "A" class integrity. Similarly, in such boundaries which are not required to have "A" class integrity, doors may be constructed of materials which are to the satisfaction of IRS.

3.2.4.2.5 Exterior boundaries of superstructures and deckhouses enclosing accommodation and including any overhanging decks which support such accommodation, are to be constructed of steel and insulated to "A-60" standard for the whole of the portions which face the cargo area and on the outward sides for a distance of 3 [m] from the end boundary facing the cargo area. The distance of 3 [m] is to be measured horizontally and parallel to the middle line of the ship from the boundary which faces the cargo area at each deck level. In the case of the sides of those superstructures and deckhouses, such insulation is to be carried up to the underside of the deck of the navigation bridge. (also see IR3.2.4.2.5).

IR3.2.4.2.5 “A-60” standard insulation is to be provided for the portions which face the cargo area upto the underside of the navigation bridge deck.

3.2.4.2.6 Skylights to cargo pump-rooms are to be of steel and are not to contain any glass. These are to be capable of being closed from outside the pump-room.

3.2.4.2.7 Construction and arrangement of saunas are to comply with 3.2.2.3.4.

Table 3.7 : Fire integrity of bulkheads separating adjacent spaces (tankers)

Spaces		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Control stations	(1)	A-0[c]	A-0	A-60	A-0	A-15	A-60	A-15	A-60	A-60	*
Corridors	(2)		C	B-0	B-0 A-0[a]	B-0	A-60	A-0	A-60	A-0	*
Accommodation spaces	(3)			C	B-0 A-0[a]	B-0	A-60	A-0	A-60	A-0	*
Stairways	(4)				B-0 A-0[a]	B-0 A-0[a]	A-60	A-0	A-60	A-0	*
Service spaces (low risk)	(5)					C	A-60	A-0	A-60	A-0	*
Machinery spaces of category A	(6)						*	A-0	A-0[d]	A-60	*
Other machinery spaces	(7)							A-0[b]	A-0	A-0	*
Cargo pump-rooms	(8)								*	A-60	*
Service spaces (high risk)	(9)									A-0[b]	*
Open decks	(10)										-

Table 3.8 : Fire integrity of decks separating adjacent spaces (tankers)											
		-- Space above --									
Space below		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Control stations	(1)	A-0	A-0	A-0	A-0	A-0	A-60	A-0	-	A-0	*****
Corridors	(2)	A-0	*	*	A-0	*	A-60	A-0	-	A-0	*****
Accommodation spaces	(3)	A-60	A-0	*	A-0	*	A-60	A-0	-	A-0	*****
Stairways	(4)	A-0	A-0	A-0	*	A-0	A-60	A-0	-	A-0	*****
Service spaces (low risk)	(5)	A-15	A-0	A-0	A-0	*	A-60	A-0	-	A-0	*****
Machinery spaces of category A	(6)	A-60	A-60	A-60	A-60	A-60	*	A-60[e]	A-0	A-60	*****
Other machinery spaces	(7)	A-15	A-0	A-0	A-0	A-0	A-0	*	A-0	A-0	*****
Cargo pump-rooms	(8)	-	-	-	-	-	A-0[d]	A-0	*	-	*****
Service spaces (high risk)	(9)	A-60	A-0	A-0	A-0	A-0	A-60	A-0	-	A-0[b]	*****
Open decks	(10)	*****	*	*	*	*	*	*	*	*	**-**

Notes: To be applied to Table 3.7 and Table 3.8 as appropriate.

a For clarification as to which applies, see 3.2.3.2 and 3.2.3.4.

b Where spaces are of the same numerical category and superscript b appears, a bulkhead or deck of the rating shown in the tables is only required when the adjacent spaces are for a different purpose (e.g. in category (9)). A galley next to a galley does not require a bulkhead but a galley next to a paint room requires an "A-0" bulkhead.

c Bulkheads separating the wheelhouse, chartroom and radio room from each other may have a "B-0" rating.

IRc A navigation locker that can be accessed from the wheelhouse is to be considered as a control station with respect to the requirements in Table 3.7 and the bulkhead separating the wheelhouse and such a locker is to have fire integrity of at least “B-0” class

d Bulkheads and decks between cargo pump-rooms and machinery spaces of category A may be penetrated by cargo pump shaft glands and similar gland penetrations, provided that gas tight seals with efficient lubrication or other means of ensuring the permanence of the gas seal are fitted in way of the bulkheads or deck.

e Where subscript 'e' appears in the Tables, fire insulation need not be fitted if the machinery space in Category (7) has little or no fire risk.

* Where an asterisk appears in the table, the division is required to be of steel or other equivalent material, but is not required to be of "A" class standard. However, where a deck, except an open deck, is penetrated for the passage of electric cables, pipes and vent ducts, such penetrations are to

be made tight to prevent the passage of flame and smoke. Divisions between control stations (emergency generators) and open decks may have air intake openings without means for closure, unless a fixed gas fire-fighting system is fitted.

3.3 Penetration in fire-resisting divisions and prevention of heat transmission

3.3.1 Where "A" class divisions are penetrated, such penetrations are to be tested in accordance with the Fire Test Procedures Code, subject to the provisions of 3.4.1.1.5. In the case of ventilation ducts, 3.7.1.2 and 3.7.3.1 apply. However, where a pipe penetration is made of steel or equivalent material having a thickness of 3 [mm] or greater and a length of not less than 900 [mm] (preferably 450 [mm] on each side of the division), and no openings, testing is not required. Such penetrations are to be suitably insulated by extension of the insulation at the same degree of fire integrity of the division.

3.3.2 Where "B" class divisions are penetrated for the passage of electric cables, pipes, trunks, ducts, etc., or for the fitting of ventilation terminals, lighting fixtures and similar devices, arrangements are to be made to ensure that the fire resistance is not impaired, subject to the provisions of 3.7.3.2. Pipes other than steel or copper that penetrate "B" class divisions are to be protected by either:

.1 a fire tested penetration device, suitable for the fire resistance of the division pierced and the type of pipe used; or

.2 a steel sleeve, having a thickness of not less than 1.8 [mm] and a length of not less than 900 [mm] for pipe diameters of 150 [mm] or more and not less than 600 [mm] for pipe diameters of less than 150 [mm] (preferably equally divided to each side of the division). The pipe is to be connected to the ends of the sleeve by flanges or couplings; or the clearance between the sleeve and the pipe is not to exceed 2.5 [mm]; or any clearance between pipe and sleeve is to be made tight by means of non-combustible or other suitable material.

3.3.3 Uninsulated metallic pipes penetrating "A" or "B" class divisions are to be of materials having a melting temperature which exceeds 950ºC for “A-0” and 850ºC for “B-0” class divisions.

3.3.4 In designing structural fire protection details, the risk of heat transmission at intersections and terminal points of required thermal barriers is to be considered. The insulation of a deck or bulkhead is to be carried past the penetration, intersection or terminal point for a distance of at least 450 [mm] in the case of steel and aluminium structures. If a space is divided with a deck or a bulkhead of "A" class standard having insulation of different values, the insulation with the higher value is to continue on the deck or bulkhead with the insulation of the lesser value for a distance of at least 450 [mm].

3.4 Protection of openings in fire resisting divisions

3.4.1 Openings in bulkheads and decks in passenger ships

3.4.1.1 *Openings in "A" class divisions*

3.4.1.1.1 Except for hatches between cargo, special category, store and baggage spaces, and between such spaces and weather decks, all openings are to be provided with permanently attached means of closing which are to be at least as effective for resisting fires as the divisions in which they are fitted.

3.4.1.1.2 The construction of doors and door frames in "A" class divisions, with the means of securing them when closed, is to provide resistance to fire as well as to the passage of smoke and flame equivalent to that of the bulkheads in which the doors are situated, this being determined in accordance with the Fire Test Procedures Code. Such doors and door frames are to be constructed of steel or other equivalent material. Doors approved without the sill being part of the frame, which are installed on or after 1 July 2010 are to be installed such that the gap under the door does not exceed 12 [mm]. A non-combustible sill is to be installed under the door such that floor coverings do not extend beneath the closed door.

3.4.1.1.3 Watertight doors need not be insulated.

3.4.1.1.4 It is to be possible for each door to be opened and closed from each side of the bulkhead by one person only.

3.4.1.1.5 Fire doors in main vertical zone bulkheads, galley boundaries and stairway enclosures other than power-operated watertight doors and those which are normally locked, are to satisfy the following requirements:

.1 the doors are to be self-closing and be capable of closing against an angle of inclination of up to 3.5° opposing closure;

.2 the approximate time of closure for hinged fire doors is to be not more than 40 seconds and no less than 10 seconds from the beginning of their movement with the ship in upright position. The approximate uniform rate of closure for sliding fire doors is to be of not more than 0.2 [m/s] and not less than 0.1 [m/s] with the ship in the upright position;

.3 the doors, except those for emergency escape trunks, are to be capable of remote release from the continuously manned central control station, either simultaneously or in groups and are to be capable of release also individually from a position at both sides of the door. Release switches are to have an on-off function to prevent automatic resetting of the system;

.4 use of hold-back hooks not subject to central control station release is prohibited;

.5 a door closed remotely from the central control station is to be capable of being re-opened at both sides of the door by local control. After such local opening the door is to close automatically again;

.6 indication is to be provided at the fire door indicator panel in the continuously manned central control station whether each of the remote-released doors are closed;

.7 the release mechanism is to be so designed that the door will automatically close in the event of disruption of the control system or central power supply;

.8 local power accumulators for power-operated doors are to be provided in the immediate vicinity of the doors to enable the doors to be operated after disruption of the control system or central power supply at least ten times (fully opened and closed) using the local controls;

.9 disruption of the control system or central power supply at one door is not to impair the safe functioning of the other doors;

.10 remote-released sliding or power-operated doors are to be equipped with an alarm that sounds for at least 5 seconds but no more than 10 seconds after the door is released from the central control station and before the door begins to move and continue sounding until the door is completely closed;

.11 a door designed to re-open upon contacting an object in its path is to re-open not more than 1 [m] from the point of contact;

.12 double-leaf doors equipped with a latch necessary for their fire integrity are to have a latch that is automatically activated by the operation of the doors when released by the control system;

.13 doors giving direct access to special category spaces which are power-operated and automatically closed need not be equipped with the alarms and remote- release mechanisms required in .3 and .10;

.14 the components of the local control system are to be accessible for maintenance and adjusting; and

.15 power-operated doors are to be provided with a control system of an approved type which is to be able to operate in case of fire, this being determined in accordance with the Fire Test Procedures Code. This system is to satisfy the following requirements:

i) the control system is to be able to operate the door at the temperature of at least 200°C for at least 60 minutes, served by the power supply;

ii) the power supply for all other doors not subject to fire is not to be impaired; and

iii) at temperatures exceeding 200°C the control system is to be automatically isolated from the power supply and is to be capable of keeping the door closed up to at least 945°C.

IR.16 indication signals for lift doors provided in accordance with .13 above are to meet the following:

i) the signal showing that "A" class lift doors are in the closed position is to be activated only when the order to close the main fire doors has been given by the continuously manned central control station;

ii) when there are several lifts giving access to the same stairway, the lift door indicators located in the continuously manned central control station are to be capable of indicating that all the lift doors giving access to the same landing are properly closed. This indication is to be shown on the panel; and

iii) when an order to close the main fire doors is given, the same order is also to stop the lifts from operating by sending them to pre-specified deck, to be determined on a case-by-case basis according to the ship's design. In addition, those inside the lift are to be able to order the lift doors open while those outside the lift are not to be able to do so.

3.4.1.1.6 In ships carrying not more than 36 passengers, where a space is protected by an automatic sprinkler fire detection and alarm system complying with the provisions the Fire Safety Systems Code or fitted with a continuous "B" class ceiling, openings in decks not forming steps in main vertical zones nor bounding horizontal zones are to be closed reasonably tight and such decks are to meet the "A" class integrity requirements in so far as is reasonable and practicable in the opinion of IRS.

3.4.1.1.7 The requirements for "A" class integrity of the outer boundaries of a ship do not apply to glass partitions, windows and sidescuttles, provided that there is no requirement for such boundaries to have "A" class integrity in 3.4.1.3.3. The requirements for "A" class integrity of the outer boundaries of the ship do not apply to exterior doors, except for those in superstructures and deckhouses facing lifesaving appliances, embarkation and external assembly station areas, external stairs and open decks used for escape routes. Stairway enclosure doors need not meet this requirement.

3.4.1.1.8 Except for watertight doors, weathertight doors (semi-watertight doors), doors leading to the open deck and doors which need to be reasonably gastight, all "A" class doors located in stairways, public spaces and main vertical zone bulkheads in escape routes are to be equipped with a self-closing hose port of material, construction and fire resistance which is equivalent to the door into which it is fitted, and is to be a 150 [mm] square clear opening with the door closed and is to be inset into the lower edge of the door, opposite the door hinges or, in the case of sliding doors, nearest the opening.

3.4.1.1.9 Where it is necessary that a ventilation duct passes through a main vertical zone division, a fail-safe automatic closing fire damper is to be fitted adjacent to the division. The damper is also to be capable of being manually closed from each side of the division. The operating position is to be readily accessible and be marked in red light-reflecting colour. The duct between the division and the damper is to be of steel or other equivalent material and, if necessary, insulated to comply with the requirements of 3.3.1. The damper is to be fitted on at least one side of the division with a visible indicator showing whether the damper is in the open position.

3.4.1.2 *Openings in "B" class divisions*

3.4.1.2.1 Doors and door frames in "B" class divisions and means of securing them is to provide a method of closure which will have resistance to fire equivalent to that of the divisions, this being determined in accordance with the Fire Test Procedure Code except that ventilation openings may be permitted in the lower portion of such doors. Where such opening is in or under a door the total net area of opening is not to exceed 0.05 [m^2]. Alternatively, a non-combustible air balance duct routed between the cabin and the corridor, and located below the sanitary unit is permitted where the cross-sectional area of the duct does not exceed 0.05 [m^2]. All ventilation openings are to be fitted with a grill made of non-combustible material. Doors are to be non-combustible. Doors approved without the sill being part of the frame, which are installed on or after 1 July 2010 are to be installed such that the gap under the door does not exceed 25 [mm].

IR3.4.1.2.1 Balancing openings or ducts between two enclosed spaces are prohibited except for openings as permitted by 3.4.1.2.1.

3.4.1.2.2 Cabin doors in "B" class divisions are to be of a self-closing type. Hold-back hooks are not permitted.

3.4.1.2.3 The requirements for 'B' class integrity of the outer boundaries of a ship are not to apply to glass partitions, windows and side scuttles. Similarly, the requirements for 'B' class integrity are not to apply to exterior doors in superstructures and deckhouses. For ships carrying not more than 36 passengers, the use of combustible materials in doors separating cabins from the individual interior sanitary spaces, such as showers, may be permitted.

3.4.1.2.4 In ships carrying not more than 36 passengers, where an automatic sprinkler system complying with the provisions of the Fire Safety Systems Code (Chapter 8) is fitted:

.1 openings in decks not forming steps in main vertical zones nor bounding horizontal zones are to be closed reasonably tight and such decks are to meet the "B" class integrity requirements in so far as is reasonable and practicable in the opinion of IRS; and
.2 openings in corridor bulkheads of "B" class materials are to be protected in accordance with the provisions of 3.2.2.2.

3.4.1.3 *Windows and sidescuttles*

3.4.1.3.1 Windows and sidescuttles in bulkheads within accommodation and service spaces and control stations other than those to which the provisions of 3.4.1.1.6 and of 3.4.1.2.3 apply, are to be so constructed as to preserve the integrity requirements of the type of bulkheads in which they are fitted, this being determined in accordance with the Fire Test Procedures Code.

3.4.1.3.2 Notwithstanding the requirements of Table 3.1 to Table 3.4, windows and sidescuttles in bulkheads separating accommodation and service spaces and control stations from weather are to be constructed with frames of steel or other suitable material. The glass is to be retained by a metal glazing bead or angle.

3.4.1.3.3 Windows facing life-saving appliances, embarkation and assembly stations, external stairs and open decks used for escape routes, and windows situated below liferaft and escape slide embarkation areas are to have fire integrity as required in Table 3.1. Where automatic dedicated sprinkler heads are provided for windows, "A-0" windows may be accepted as equivalent. To be considered under this paragraph, the sprinkler heads are either to be:

.1 dedicated heads located above the windows, and installed in addition to the conventional ceiling sprinklers; or

.2 conventional ceiling sprinkler heads arranged such that the window is protected by an average application rate of at least 5 l/m^2 and the additional window area is included in the calculation of the area of coverage; or

.3 water-mist nozzles that have been tested and approved based on the guidelines developed by IMO. [Refer to the revised guidelines for approval of sprinkler systems equivalent to that referred to in SOLAS regulation II-2/12 (Resolution A 800(19))].

Windows located in the ship’s side below the lifeboat embarkation area are to have fire integrity at least equal to "A-0" class.

Also refer MSC/Circ.917 "Guidelines on fire safety construction in accommodation" as corrected by MSC/Circ.917/Corr.1.

3.4.1.3.4 Notwithstanding the requirement in paragraph 3.4.1.3.3, the requirements in paragraphs 3.4.1.3.5 and 3.4.1.3.6 apply to ships constructed on or after 1 January 2020.

3.4.1.3.5 For ships carrying more than 36 passengers, windows facing survival craft, embarkation and assembly stations, external stairs and open decks used for escape routes, and windows situated below liferaft and escape slide embarkation areas are to have fire integrity as required in table 3.1. Where automatic dedicated sprinkler heads are provided for windows, "A-0" windows may be accepted as equivalent. To be considered under this paragraph, the sprinkler heads must either be:

.1 dedicated heads located above the windows, and installed in addition to the conventional ceiling sprinklers; or

.2 conventional ceiling sprinkler heads arranged such that the window is protected by an average application rate of at least 5 [l/min per square metre] and the additional window area is included in the calculation of the area of coverage; or

.3 water-mist nozzles that have been tested and approved in accordance with the *Revised guidelines for approval of sprinkler systems*

equivalent to that referred to in SOLAS regulation II-2/12 (resolution A.800(19), as amended);

and windows located in the ship's side below the lifeboat embarkation area are to have fire integrity at least equal to "A-0" class.

3.4.1.3.6 For ships carrying not more than 36 passengers, windows facing survival craft and escape slide, embarkation areas and windows situated below such areas are to have fire integrity at least equal to "A-0" class.

3.4.2 Doors in fire-resisting divisions in cargo ships

3.4.2.1 The fire resistance of doors is to be equivalent to that of the division in which they are fitted, this being determined in accordance with the Fire Test Procedures Code. Doors approved as “A” class without the sill being part of the frame, which are installed on or after 1 July 2010 are to be installed such that the gap under the door does not exceed 12 [mm] and a non-combustible sill is to be installed under the door such that floor coverings do not extend beneath the closed door. Doors approved as “B” class without the sill being part of the frame, which are installed on or after 1 July 2010 are to be installed such that the gap under the door does not exceed 25 [mm]. Doors and door frames in "A" class divisions are to be constructed of steel. Doors in "B" class divisions are to be non-combustible. Doors fitted in boundary bulkheads of machinery spaces of category A are to be reasonably gastight and self-closing. Where method IC is adopted the use of combustible materials in doors separating cabins from individual interior sanitary accommodation such as showers may be permitted.

3.4.2.2 Doors required to be self closing are not to be fitted with hold back hooks. However, hold back arrangements fitted with remote release devices of the fail safe type may be utilized.

3.4.2.3 In corridor bulkheads ventilation openings may be permitted in and under the doors of cabins and public spaces. Ventilation openings are also permitted in "B" class doors leading to lavatories, offices, pantries, lockers and store rooms. Except as permitted below, the openings are to be provided only in the lower half of a door. Such opening in or under a door is not to have the total net area exceeding 0.05 [m^2]. Alternatively, a non-combustible air balance duct routed between the cabin and the corridor, and located below the sanitary unit is permitted where the cross-sectional area of the duct does not exceed 0.05 [m^2]. Ventilation openings, except those under the door, are to be fitted with a grille made of non-combustible material.

IR3.4.2.3 Balancing openings or ducts between two enclosed spaces are prohibited except for openings as permitted by 3.4.2.3.

3.4.2.4 Watertight doors need not be insulated.

3.5 Protection of openings in machinery spaces boundaries

3.5.1 Application

3.5.1.1 The provision of 3.5 is to apply to machinery spaces of category A and, where IRS considers it desirable, to other machinery spaces.

3.5.2 Protection of openings in machinery space boundaries

3.5.2.1 The number of skylights, doors, ventilators, openings in funnels to permit exhaust ventilation and other openings to machinery spaces are to be reduced to a minimum consistent with the needs of ventilation and the proper and safe working of the ship.

3.5.2.2 Skylights are to be of steel and are not to contain glass panels.

3.5.2.3 Means of control are to be provided for closing power-operated doors or actuating release mechanisms on doors other than power-operated watertight doors. The control is to be located outside the space concerned, where they will not be cut off in the event of fire in the space it serves.

3.5.2.4 In passenger ships, the means of control required in 3.5.2.3 are to be situated at one control position or grouped in as few positions as possible to the satisfaction of IRS. Such positions are to have safe access from the open deck.

3.5.2.5 In passenger ships, doors, other than power-operated watertight doors are to be so arranged that positive closure is assured in case of fire in the space by power-operated closing arrangements or by the provision of self-closing doors capable of closing against

an inclination of 3.5° opposing closure, and having a fail-safe hook-back facilities provided with a remotely operated release device. Doors for emergency escape trunks need not be fitted with a fail-safe hold-back facility and a remotely operated release device.

3.5.2.6 Windows are not to be fitted in machinery space boundaries. Glass may be used in control rooms within the machinery spaces.

IR3.5.2.7 When access to any machinery space of category A is provided at a low level from an adjacent shaft tunnel, there must be provided in the shaft tunnel, near the watertight door, a light steel fire-screen door operable from both sides.

3.6 Protection of cargo space boundaries

3.6.1 In passenger ships carrying more than 36 passengers, the boundary bulkheads and decks of special category and ro-ro spaces are to be insulated to "A-60" class standard. However, where a category (5), (9) and (10) space, as defined in 3.2.2.3, is on one side of the division the standard may be reduced to "A-0". Where fuel oil tanks are below a special category space, the integrity of the deck between such spaces may be reduced to "A-0" standard.

3.6.2 In passenger ships, indicators are to be provided on the navigating bridge which will indicate when any fire door leading to or from the special category spaces is closed.

3.6.3 In tankers for the protection of cargo tanks carrying crude oil and petroleum products having a flash point not exceeding 60°C, materials readily rendered ineffective by heat are not to be used for valves, fittings, tank opening covers, cargo vent piping, and cargo piping so as to prevent the spread of fire to the cargo.

3.7 Ventilation systems

3.7.1 General

3.7.1.1 Ventilation ducts, including single and double wall ducts, are to be of steel or equivalent material except flexible bellows of short length not exceeding 600 [mm] used for connecting fans to the ducting in air-conditioning rooms. Unless expressly provided otherwise in paragraph 3.7.1.6, any other material used in the construction of ducts, including insulation, are also to be non-combustible. However, short ducts, not generally exceeding 2 [m] in length and with a free cross-sectional area* not exceeding 0.02 [m^2], need not be of steel or equivalent material, subject to the following conditions:

(* The term *free cross-sectional area* means, even in the case of a pre-insulated duct, the area calculated on the basis of the inner dimensions of the duct itself and not the insulation.)

.1 the ducts are to be made of non-combustible material, which may be faced internally and externally with membranes having low flame-spread characteristics and, in each case, a calorific value** not exceeding 45 [MJ/m^2] of their surface area for the thickness used;

(**Refer to the recommendations published by the International Organization for Standardization, in particular publication ISO 1716:2002, Reaction to the fire tests for building products – Determination of the heat of combustion.)

.2 the ducts are only used at the end of the ventilation device; and

.3 the ducts are not situated less than 600 [mm], measured along the duct, from an opening in an "A" or "B" class division, including continuous "B" class ceiling.

IR 3.7.1.1 With respect only to 3.7.1.1 above, a ventilation duct made of material other than steel may be considered equivalent to a ventilation duct made of steel, provided the material is non-combustible and has passed a standard fire test in accordance with Annex 1: Part 3 of the FTP code as non-load bearing structure for 30 minutes following the requirements for testing “B” class divisions.

3.7.1.2 The following arrangements are to be tested in accordance with the Fire Test Procedures Code:

.1 fire dampers, including their relevant means of operation, however, the testing is not required for dampers located at the lower end of the duct in exhaust ducts for galley ranges, which

must be of steel and capable of stopping the draught in the duct; and

.2 duct penetrations through "A" class divisions. However, the test is not required where steel sleeves are directly joined to ventilation ducts by means of riveted or screwed connections or by welding.

3.7.1.3 Fire dampers are to be easily accessible. Where they are placed behind ceilings or linings, these ceilings or linings are to be provided with an inspection hatch on which the identification number of the fire damper is marked. The fire damper identification number is also to be marked on any remote controls provided.

3.7.1.4 Ventilation ducts are to be provided with hatches for inspection and cleaning. The hatches are to be located near the fire dampers.

3.7.1.5 The main inlets and outlets of ventilation systems are to be capable of being closed from outside the spaces being ventilated. The means of closing is to be easily accessible as well as prominently and permanently marked and is to indicate the operating position of the closing device.

3.7.1.6 Combustible gaskets in flanged ventilation duct connections are not permitted within 600 [mm] of openings in "A" or "B" class divisions and in ducts required to be of "A" class construction.

3.7.1.7 Ventilation openings or air balance ducts between two enclosed spaces are not to be provided except as permitted by paragraphs 3.4.1.2.1 and 3.4.2.3.

3.7.2 Arrangement of ducts

3.7.2.1 The ventilation systems for machinery spaces of category A, vehicle spaces, ro-ro spaces, galleys, special category spaces and cargo spaces shall, in general, be separated from each other and from the ventilation systems serving other spaces. However, the galley ventilation systems on cargo ships of less than 4,000 gross tonnage and in passenger ships carrying not more than 36 passengers need not be completely separated from other ventilation systems, but may be served by separate ducts from a ventilation unit serving other spaces. In such a case, an automatic fire damper is to be fitted in the galley ventilation duct near the ventilation unit.

3.7.2.2 Ducts provided for the ventilation of machinery spaces of category A, galleys, vehicle spaces, ro-ro spaces or special category spaces are not to pass through accommodation spaces, service spaces, or control stations unless they comply with paragraph 3.7.2.4.

3.7.2.3 Ducts provided for the ventilation of accommodation spaces, service spaces or control stations are not to pass through machinery spaces of category A, galleys, vehicle spaces, ro-ro spaces or special category spaces unless they comply with paragraph 3.7.2.4.

3.7.2.4 As permitted by paragraphs 3.7.2.2 and 3.7.2.3 ducts are to be either:

.1.1 constructed of steel having a thickness of at least;

a) 3 [mm] for ducts with a free cross-sectional area of less than 0.075 [m^2],

b) 4 [mm] for ducts with a free cross-sectional area of between 0.075 [m^2] and 0.45 [m^2], and

c) at least 5 [mm] for ducts with a free cross-sectional area of over 0.45 [m^2];

.1.2 suitably supported and stiffened;

.1.3 fitted with automatic fire dampers close to the boundaries penetrated; and

IR .1.3 Fire dampers are to be preferably fitted in the space having low fire risk

.1.4 insulated to "A-60" class standard from the boundaries of the spaces they serve to a point at least 5 [m] beyond each fire damper;

or

.2.1 constructed of steel in accordance with paragraphs 3.7.2.4.1.1 and 3.7.2.4.1.2; and

.2.2 insulated to "A-60" class standard throughout the spaces they pass through, except for ducts that pass

through spaces of category (9) or (10) as defined in paragraph 3.2.2.3.2.2.

3.7.2.5 For the purposes of paragraphs 3.7.2.4.1.4 and 3.7.2.4.2.2, ducts are to be insulated over their entire cross-sectional external surface. Ducts that are outside but adjacent to the specified space, and share one or more surfaces with it, are to be considered to pass through the specified space, and are to be insulated over the surface they share with the space for a distance of 450 [mm] past the duct.

IR3.7.2.5 For determining the extent of insulation when the duct passes through an enclosed space, See Fig. 3.7.2.5.

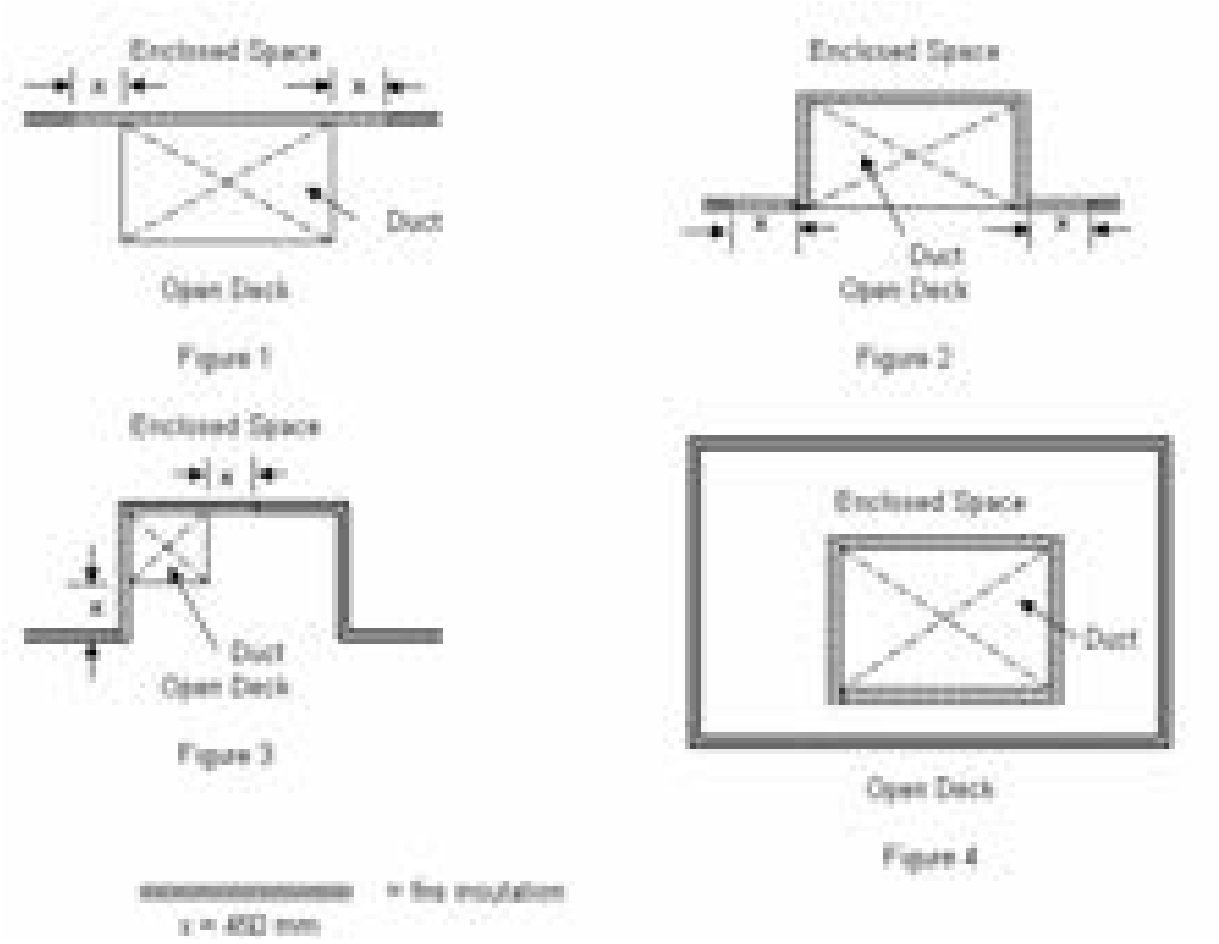

Fig.3.7.2.5 : Examples of galley ducts contiguous to enclosed spaces

3.7.2.6 Where it is necessary that a ventilation duct passes through a main vertical zone division, an automatic fire damper is to be fitted adjacent to the division. The damper is also to be capable of being manually closed from each side of the division. The control location is to be readily accessible and be clearly and prominently marked. The duct between the division and the damper is to be constructed of steel in accordance with paragraphs 3.7.2.4.1.1 and 3.7.2.4.1.2 and insulated to at least the same fire integrity as the division penetrated. The damper is to be fitted on at least one side of the division with a visible indicator showing the operating position of the damper.

IR3.7.2.6 When the galley is served by a duct from a common ventilation unit and not from an independent unit, automatic fire damper is to be provided outside the galley, irrespective of the size of the duct.

3.7.3 Details of fire dampers and duct penetrations

3.7.3.1 Ducts passing through "A" class divisions are to meet the following requirements:

.1 where a thin plated duct with a free cross sectional area equal to, or less than, 0.02 [m^2] passes through "A" class divisions, the opening is to be fitted with a steel sheet sleeve having a thickness of at least 3 [mm] and a length of at least 200 [mm], divided preferably into 100 [mm] on each side of a bulkhead or, in the case of a deck, wholly laid on the lower side of the decks penetrated;

.2 where ventilation ducts with a free cross-sectional area exceeding 0.02 [m^2], but not more than 0.075 [m^2], pass through "A" class divisions, the openings are to be lined with steel sheet sleeves. The ducts and sleeves are to have a thickness of at least 3 [mm] and a length of at least 900 [mm]. When passing through bulkheads, this length is to be divided preferably into 450 [mm] on each side of the bulkhead. These ducts, or sleeves lining such ducts, are to be provided with fire insulation. The insulation is to have at least the same fire integrity as the division through which the duct passes; and

.IR2 The fire insulation required by .2 above is to be provided only to the part of the duct and/or sleeve that is on the same side of the division being fire insulated, and be extended for a minimum of 450 mm along the duct and/or sleeve.

.3 automatic fire dampers are to be fitted in all ducts with a free cross-sectional area exceeding 0.075 [m^2] that pass through "A" class divisions. Each damper is to be fitted close to the division penetrated and the duct between the damper and the division penetrated is to be constructed of steel in accordance with paragraphs 3.7.2.4.1.1 and 3.7.2.4.1.2. The fire damper is to operate automatically, but is also to be capable of being closed manually from both sides of the division. The damper is to be fitted with a visible indicator which shows the operating position of the damper. Fire dampers are not required, however, where ducts pass through spaces surrounded by "A" class divisions, without serving those spaces, provided those ducts have the same fire integrity as the divisions which they penetrate. A duct of cross-sectional area exceeding 0.075 [m^2] is not to be divided into smaller ducts at the penetration of an "A" class division and then recombined into the original duct once through the division to avoid installing the damper required by this provision.

IR 3.7.3.1 Ducts or pipes with free sectional area of 0.075 m^2 or less need to be fitted with fire damper at their passage through Class "A" divisions in those cases indicated in 3.7.2.2 and 3.7.2.3. The fire damper can be omitted if the duct is arranged in compliance with the requirements of 3.7.2.4.2.1 and 3.7.2.4.2.2.

3.7.3.2 Ventilation ducts with a free cross-sectional area exceeding 0.02 [m^2] passing through "B" class bulkheads shall be lined with steel sheet sleeves of 900 [mm] in length, divided preferably into 450 [mm] on each side of the bulkheads unless the duct is of steel for this length.

IR 3.7.3.2 When a duct passing through a division is to be in accordance with 3.3.2 and 3.7.3.2, no clearance is to be allowed between the duct and the division.

3.7.3.3 All fire dampers are to be capable of manual operation. The dampers are to have a direct mechanical means of release or, alternatively, be closed by electrical, hydraulic, or pneumatic operation. All dampers are to be manually operable from both sides of the division. Automatic fire dampers, including those capable of remote operation, are to have a failsafe mechanism that will close the damper in a fire even upon loss of electrical power or hydraulic or pneumatic pressure loss. Remotely operated fire dampers are to be capable of being reopened manually at the damper.

3.7.4 Ventilation systems for passenger ships carrying more than 36 passengers

3.7.4.1 In addition to the requirements in sections 3.7.1, 3.7.2 and 3.7.3, the ventilation system of a passenger ship carrying more than 36 passengers is also to meet the following requirements.

3.7.4.2 In general, the ventilation fans are to be so arranged that the ducts reaching the various spaces remain within a main vertical zone.

3.7.4.3 Stairway enclosures are to be served by an independent ventilation fan and duct system (exhaust and supply) which is not to serve any other spaces in the ventilation systems.

3.7.4.4 A duct, irrespective of its cross-section, serving more than one 'tween-deck accommodation space, service space or control station, is to be fitted, near the penetration of each deck of such spaces, with an automatic smoke damper that is also to be capable of being closed manually from the protected deck above the damper. Where a fan serves more than one 'tween-deck space through separate ducts within a main vertical zone, each dedicated to a single 'tween-deck space, each duct is to be provided with a manually operated smoke damper fitted close to the fan.

3.7.4.5 Vertical ducts, if necessary are to, be insulated as required by tables 3.1 and 3.2. Ducts are to be insulated as required for decks between the space they serve and the space being considered, as applicable.

3.7.5 Exhaust ducts from galley ranges

3.7.5.1 Requirements for passenger ships carrying more than 36 passengers

3.7.5.1.1 In addition to the requirements in sections 3.7.1, 3.7.2 and 3.7.3, exhaust ducts from galley ranges are to be constructed in accordance with paragraphs 3.7.2.4.2.1 and 3.7.2.4.2.2 and insulated to "A-60" class standard throughout accommodation spaces, service spaces, or control stations they pass through. They are also to be fitted with:

.1 a grease trap readily removable for cleaning unless an alternative approved grease removal system is fitted;

.2 a fire damper located in the lower end of the duct at the junction between the duct and the galley range hood which is automatically and remotely operated and, in addition, a remotely operated fire damper located in the upper end of the duct close to the outlet of the duct;

.3 a fixed means for extinguishing a fire within the duct*;

(*Refer to the recommendations published by the International Organization for Standardization, in particular publication ISO 15371:2009, Ships and marine technology – Fire-extinguishing systems for protection of galley cooking equipment.)

.4 remote-control arrangements for shutting off the exhaust fans and supply fans, for operating the fire dampers mentioned in paragraph 3.7.5.1.1.2 and for operating the fire-extinguishing system, which is to be placed in a position outside the galley close to the entrance to the galley. Where a multi-branch system is installed, a remote means located with the above controls shall be provided to close all branches exhausting through the same main duct before an extinguishing medium is released into the system; and

.5 suitably located hatches for inspection and cleaning, including one provided close to the exhaust fan and one fitted in the lower end where grease accumulates.

3.7.5.1.2 Exhaust ducts from ranges for cooking equipment installed on open decks are to conform to paragraph 3.7.5.1.1, as applicable, when passing through accommodation spaces or spaces containing combustible materials.

3.7.5.2 Requirements for cargo ships and passenger ships carrying not more than 36 passengers

When passing through accommodation spaces or spaces containing combustible materials, the exhaust ducts from galley ranges are to be constructed in accordance with paragraphs 3.7.2.4.1.1 and 3.7.2.4.1.2. Each exhaust duct is to be fitted with:

.1 a grease trap readily removable for cleaning;

.2 an automatically and remotely operated fire damper located in the lower end of the duct at the junction between the duct and the galley range hood and, in addition, a remotely operated fire damper in the upper end of the duct close to the outlet of the duct;

.3 arrangements, operable from within the galley, for shutting off the exhaust and supply fans; and

.4 fixed means for extinguishing a fire within the duct.*

(*Refer to the recommendations published by the International Organization for Standardization, in particular publication ISO 15371:2009, Ships and marine technology – Fire-extinguishing systems for protection of galley cooking equipment.)

3.7.6 Ventilation rooms serving machinery spaces of category A containing internal combustion machinery

3.7.6.1 Where a ventilation room serves only such an adjacent machinery space and there is no fire division between the ventilation room and the machinery space, the means for closing the ventilation duct or ducts serving the machinery space shall be located outside of the ventilation room and machinery space.

3.7.6.2 Where a ventilation room serves such a machinery space as well as other spaces and is separated from the machinery space by a "A-0" class division, including penetrations, the means for closing the ventilation duct or ducts for the machinery space can be located in the ventilation room.

3.7.7 Ventilation systems for laundries in passenger ships carrying more than 36 passengers

Exhaust ducts from laundries and drying rooms of category (13) spaces as defined in paragraph 3.2.2.3.2.2 are to be fitted with:

.1 filters readily removable for cleaning purposes;

.2 a fire damper located in the lower end of the duct which is automatically and remotely operated;

.3 remote-control arrangements for shutting off the exhaust fans and supply fans from within the space and for operating the fire damper mentioned in paragraph 3.7.7.2; and

.4 suitably located hatches for inspection and cleaning.

Section 4

Fire Fighting

4.1 Purpose

4.1.1 The purpose of this section is to suppress and swiftly extinguish a fire in the space of origin, except for 4.1.2. For this purpose, the following functional requirements are to be met:

.1 fixed fire extinguishing systems are to be installed having due regard to the fire growth potential of the protected spaces; and

.2 fire extinguishing appliances are to be readily available.

4.1.2 For open-top container holds* and on deck container stowage areas on ships designed to carry containers on or above the weather deck, constructed on or after 1 January 2016, fire protection arrangements are to be provided for the purpose of containing a fire in the space or area of origin and cooling adjacent areas to prevent fire spread and structural damage.

(*For a definition of this term, refer to the Interim guidelines for open-top containerships (MSC/Circ.608/Rev.1).)

4.2 Water supply systems

Every ship is to be provided with fire pumps, fire mains, hydrants and hoses complying as applicable with the following requirements.

4.2.1 *Fire mains and hydrants*

4.2.1.1 *General*

Materials readily rendered ineffective by heat are not to be used for fire mains and hydrants unless adequately protected. The pipes and hydrants are to be so placed that the fire hoses may be easily coupled to them. The

arrangement of pipes and hydrants is to be such as to avoid the possibility of freezing. Suitable drainage arrangement is to be provided for fire main piping. Isolation valves are to be installed for all open deck fire main branches used for purposes other than fire fighting. In ships where deck cargo may be carried, the positions of the hydrants are to be such that they are always readily accessible and the pipes are to be arranged as far as practicable to avoid risk of damage by such cargo.

4.2.1.2 *Ready availability of water supply*

The arrangements for the ready availability of water supply is to be:

.1 in passenger ships:

a) of 1,000 gross tonnage and upwards such that at least one effective jet of water is immediately available from any hydrant in an interior location and so as to ensure the continuation of the output of water by the automatic starting of one required fire pump;

b) of less than 1,000 gross tonnage by automatic start of at least one fire pump or by remote starting from the navigation bridge of at least one fire pump. If the pump starts automatically or if the bottom valve cannot be opened from where the pump is remotely started, the bottom valve is to be always kept open; and

c) in passenger ships, if fitted with periodically unattended machinery spaces, the arrangements will be specially considered.

.2 in cargo ships:

a) to the satisfaction of IRS; and

b) with a periodically unattended machinery space or when only one person is required on watch there is to be immediate water delivery from the fire main system at a suitable pressure, either by remote starting of one of the main fire pumps with remote starting from the navigating bridge and fire control station, if any, or permanent pressurization of the fire main system by one of the main fire pumps, except that this requirement may be waived for cargo ships of less than 1,600 tons gross tonnage if the fire pump starting arrangement in the machinery space is in an easily accessible position;

c) For vessels intended for ice class notation, See also Pt.5.

4.2.1.3 *Diameter of fire mains*

The diameter of the fire main and water service pipes is to be sufficient for the effective distribution of the maximum required discharge from two fire pumps operating simultaneously, except that in the case of cargo ships, other than those included in 4.7.3.2, the diameter need only be sufficient for the discharge of 140 [m^3/hour]. (Also see 4.2.2.4.1, IR .2.1).

4.2.1.4 *Isolating valves and relief valves*

4.2.1.4.1 Isolating valves to separate the section of the fire main within the machinery space containing the main fire pump or pumps from the rest of the fire main are to be fitted in an easily accessible and tenable position outside the machinery spaces. The fire main is to be so arranged that when the isolating valves are shut all the hydrants on the ship, except those in the machinery space referred to above, can be supplied with water by another fire pump or an emergency fire pump. The emergency fire pump, its seawater inlet, and suction and delivery pipes and isolating valves are to be located outside the machinery space. If this arrangement cannot be made, the sea-chest may be fitted in the machinery space if the valve (i.e. sea inlet valve) is remotely controlled from a position in the same compartment as the emergency fire pump and the suction pipe is as short as practicable. Short lengths of suction or discharge piping may penetrate the machinery space, provided they are enclosed in a substantial steel casing, or are insulated to A-60 class standards. The pipes are to have substantial wall thickness, but in no case less than 11 [mm] and are to be welded except for the flanged connection to the sea inlet valve.

IR4.2.1.4.1

a) Any part of the fire main routed through a category ‘A’ machinery space, except for short lengths of suction or discharge piping complying with 4.2.1.4.1, is to be fitted with isolating valves outside the space. The arrangements of fire mains is to allow for fire water from the fire pumps or emergency fire pump to reach all hydrants outside the isolated space. Isolation requirements of 4.2.1.4.1 are not applicable to the piping from fire pumps

located in spaces other than category 'A' machinery spaces.

b) In cases where suction or discharge piping penetrating machinery spaces are required to be enclosed in a substantial steel casing, or insulated to "A-60" class standards, it is not necessary to enclose or insulate "sea inlet valves", "sea chests" and "distance pieces between the sea inlet valves and shell plating / sea chests";

c) The method for insulating pipes to ""A-60" class standards" is to be such that the pipes are covered/protected in a practical manner by insulation material which is approved as a part of "A-60" class divisions in accordance with the FTP Code; and

d) Where the sea inlet valve is in the machinery space, the valve is not to be a fail-close type. Where the sea inlet valve is in the machinery space and is not a fail-open type, measures are to be taken so that the valve can be opened in the event of fire, e.g. control piping, actuating devices and/or electric cables with fire resistant protection equivalent to "A-60" class standards.

e) In cases where main fire pumps are provided in a compartment outside machinery spaces and where the emergency fire pump suction or discharge piping penetrates such compartment, the above requirement is to be applied to the piping.

4.2.1.4.2 A valve is to be fitted to serve each fire hydrant so that any fire hose may be removed while the fire pumps are at work.

4.2.1.4.3 Relief valves are to be provided in conjunction with all fire pumps if the pumps are capable of developing a pressure exceeding the design pressure of the water service pipes, hydrants and hoses. These valves are to be so placed and adjusted as to prevent excessive pressure in any part of the fire main system.

4.2.1.4.4 In tankers isolation valves are to be fitted in the fire main at poop front in a protected position and on the tank deck at intervals of not more than 40 [m] to preserve the integrity of the fire main system in case of fire or explosion.

4.2.1.5 *Number and position of hydrants*

4.2.1.5.1 The number and position of hydrants are to be such that at least two jets of water not emanating from the same hydrant, one of which is to be from a single length of hose, may reach any part of the ship normally accessible to the passengers or crew while the ship is being navigated and any part of any cargo space when empty, any ro-ro space or any vehicle space in which latter case the two jets is to reach any part of the space, each from a single length of hose. Furthermore, such hydrants are to be positioned near the accesses to the protected spaces.

IR4.2.1.5.1 In ships of 1000 tons gross tonnage and over, at least two hydrants are to be provided in the machinery spaces.

4.2.1.5.2 In addition to the requirements in 4.2.1.5.1, passenger ships are to comply with the following:

.1 in the accommodation, service and machinery spaces the number and position of hydrants are to be such that the requirements of 4.2.1.5.1 may be complied with when all watertight doors and all doors in main vertical zone bulkheads are closed; and

.2 where access is provided to a machinery space of category A at a low level from an adjacent shaft tunnel, two hydrants are to be provided external to, but near the entrance to that machinery space. Where such access is provided from other spaces, in one of those spaces two hydrants are to be provided near the entrance to the machinery space of category A. Such provision need not be made where the tunnel or adjacent spaces are not part of the escape route.

IR4.2.1.5.3 In cargo ships of 2,000 tons gross tonnage and over, a hydrant is to be provided in the shaft tunnel adjacent to the engine room watertight door.

4.2.1.6 *Pressure at hydrants*

With the two pumps simultaneously delivering water through the nozzles specified in 4.2.3.3, with the quantity of water as specified in 4.2.1.3, through any adjacent hydrants, the following minimum pressures are to be maintained at all hydrants:

.1 for passenger ships:

4,000 gross tonnage and upwards 0.40 [N/mm^2] less than 4,000 gross tonnage 0.30 [N/mm^2];

.2 for cargo ships,

6,000 gross tonnage and upwards 0.27 [N/mm^2]
less than 6,000 gross tonnage; 0.25 [N/mm^2]; and

.3 the maximum pressure at any hydrant is not to exceed that at which the effective control of a fire hose can be demonstrated.

4.2.1.7 *International shore connection*

4.2.1.7.1 Ships of 500 gross tonnage and upwards is to be provided with at least one international shore connection complying with the Fire Safety Systems Code (Chapter 8).

4.2.1.7.2 Facilities are to be available enabling such a connection to be used on either side of the ship.

4.2.2 Fire pumps

4.2.2.1 *Pumps accepted as fire pumps*

Sanitary, ballast, bilge or general service pumps may be accepted as fire pumps, provided that they are not normally used for pumping oil and that if they are subject to occasional duty for the transfer or pumping of oil fuel, suitable change-over arrangements are fitted.

4.2.2.2 *Number of fire pumps*

Ships are to be provided with independently driven fire pumps as follows:

.1 in passenger ships of:

4,000 gross tonnage and upwards	:	at least three
less than 4,000 gross tonnage	:	at least two

.2 in cargo ships of:

1,000 gross tonnage and upwards	:	at least two

less than 1,000 gross tonnage : at least two power driven pumps, one of which is to be independently driven.

4.2.2.3 *Arrangement of fire pumps and fire mains*

4.2.2.3.1 *Fire pumps*

The arrangement of sea connections, fire pumps and their sources of power are to be such as to ensure that:

.1 in passenger ships of 1,000 gross tonnage and upwards, in the event of a fire in any one compartment all the fire pumps will not be put out of action; and

.2 in passenger ships of less than 1,000 gross tonnage and in cargo ships of 500 gross tonnage and upwards, if a fire in any one compartment could put all the pumps out of action, there is to be an alternative means consisting of an emergency fire pump complying with the provisions of the Fire Safety Systems Code (Chapter 8) with its source of power and sea connection located outside the space where the main fire pumps or their sources of power are located. (See also 4.2.1.4.1).

IR.2.1 Unless the main fire pumps, their sea suctions and the fuel supply or source of power for each pump are situated within compartments separated at least by A-0 divisions, it is considered for the purpose of para 2 above, that the fire in any one compartment would put the pumps out of action.

IR .2.2 An arrangement in which two main fire pumps are located in adjacent compartments which share more than one boundary will also require an emergency fire pump.

4.2.2.3.2 *Requirements for the space containing the emergency fire pump*

.1 *Location of the space*

The space containing the fire pump are not to be contiguous to the boundaries of machinery spaces of category A or those spaces containing main fire pumps. Where this is not practicable, the common bulkhead between the two spaces are to be insulated to a standard of structural fire protection equivalent to that required for a control station in 3.2.3.3. The insulation is to extend at least 450 [mm] outside the area of the common boundary.

.2 *Access to the emergency fire pump*

No direct access is to be permitted between the machinery space and the space containing the emergency fire pump and its source of power. When this is impracticable, IRS may accept an arrangement where the access is by

means of an airlock with the door of the machinery space being of A-60 class standard, and the other door being at least steel, both reasonably gastight, self-closing and without any hold back arrangements. Alternatively, the access may be through a watertight door capable of being operated from a space remote from the machinery space and the space containing the emergency fire pump and unlikely to be cut off in the event of fire in those spaces. In such cases, a second means of access to the space containing the emergency fire pump and its source of power is to be provided.

IR.2 When a single access to the emergency fire pump room is through another space adjoining a machinery space of category 'A' or the spaces containing the main fire pumps, class 'A-60' boundary is to be provided between that other space and the machinery space of category 'A' or the spaces containing main fire pumps.

.3 Lighting and v*entilation of the emergency fire pump space*

Ventilation arrangements to the space containing the independent source of power for the emergency fire pump are to be such as to preclude, as far as practicable, the possibility of smoke from a machinery space fire entering or being drawn into that space.

IR.3 The room where the emergency fire pump prime mover is located is to be illuminated from the emergency source of supply and is to be well ventilated. If the room is to be mechanically ventilated, the power is to be supplied by the emergency source.

IR .4 Maintenance / inspection

The room(s) where the pump and prime mover are installed is/are to have adequate space for maintenance work and inspections.

4.2.2.3.3 *Additional pumps for cargo ships*

In addition, in cargo ships where other pumps, such as general service, bilge and ballast, etc., are fitted in a machinery space, arrangements are to be made to ensure that at least one of these pumps, having the capacity and pressure required by 4.2.1.6.2 and 4.2.2.4.2, is capable of providing water to the fire main.

IR4.2.2.3.3 However, in cases where the required number and capacity of fire pumps is already available otherwise, the capacity and pressure characteristics of the general service/bilge/ballast pumps required to be connected as per 4.2.2.3.3, would be specially considered in view of the optimal characteristics required for their primarily intended services.

IR4.2.2.3.4 Where the emergency fire pump is electrically driven, the power is to be supplied by a source other than that supplying the main fire pump and be located outside the engine room, and separated from it by an A class division. The relevant electrical cables are not to pass through the compartment containing the main fire pump.

4.2.2.4 *Capacity of fire pumps*

4.2.2.4.1 *Total capacity of required fire pumps*

The required fire pumps are to be capable of delivering for fire-fighting purposes a quantity of water, at the pressure specified in 4.2.1.6, as follows:

.1 pumps in passenger ships, the quantity of water is not less than two thirds of the quantity required to be dealt with by the bilge pumps (See Pt.4, Ch.3, Sec.2 for capacity and number of bilge pumps) when employed for bilge pumping; and

.2 pumps in cargo ships, other than any emergency pump, the quantity of water is not less than four thirds of the quantity required to be dealt with by each of the independent bilge pumps when employed for bilge pumping, except that in no cargo ship, other than those included in 4.7.3.2, need the total required capacity of the fire pumps exceed 180 [m^3/h]. (See Pt.4, Ch.3, Sec.2 for capacity of bilge pumps).

IR.2.1 On board cargo ships designed to carry five or more tiers of containers on or above the weather deck,

.1 in cases where the mobile water monitors are supplied by separate pumps and piping system, the total capacity of the main fire pumps need not exceed 180 [m^3/h] and the diameter of the fire main and water service pipes (pipework diameter) need only be sufficient for the discharge of 140 [m^3/h].

.2 in cases where the mobile water monitors are supplied by the main fire pumps; the total capacity of required

main fire pumps and the pipework diameter is to be sufficient for simultaneously supplying both the required number of fire hoses and mobile water monitors. However, the total capacity is not to be less than the following .1 or .2, whichever is smaller:

.1 four thirds of the quantity required under SOLAS II-1/35-1 to be dealt with by each of the independent bilge pumps in a passenger ship of the same dimension when employed in bilge pumping; or

.2 180 [m^3/h].

.3 in cases where the mobile water monitors and the “water spray system” (fixed arrangement of spraying nozzles or flooding the cargo space with water) required by Ch7, 2.3.1.3 (SOLAS II-2/19.3.1.3) are supplied by the main fire pumps, the total capacity of the main fire pumps and the pipework diameter need only be sufficient to supply whichever of the following is the greater:

.1 the mobile water monitors and the four nozzles required by Ch 7, 2.3.1.2 (SOLAS II-2/19.3.1.2); or

.2 the four nozzles required by Ch 7, 2.3.1.2 (SOLAS II-2/19.3.1.2) and the water spray system required by Ch 7, 2.3.1.3 (SOLAS II-2/19.3.1.3).

The total capacity, however, is not to be less than IR 2.1.2.1 or 2.1.2.2, whichever is smaller.

IR .2.2 On board cargo ships designed to carry five or more tiers of containers on or above the weather deck, the total capacity of the emergency fire pump need not exceed 72 [m^3/h].

4.2.2.4.2 *Capacity of each fire pump*

Each of the required fire pumps (other than any emergency pump required in 4.2.2.3.1.2 for cargo ships) are to have a capacity not less than 80% of the total required capacity divided by the minimum number of required fire pumps but in any case not less than 25 [m^3/h] and each such pump are in any event to be capable of delivering at least the two required jets of water. These fire pumps are to be capable of supplying the fire main system under the required conditions. Where more pumps than the minimum of required pumps are installed such additional pumps are to have a capacity of at least 25 [m^3/h] and are to be capable of delivering at least the two jets of water required in 4.2.1.5.1.

4.2.3 Fire hoses and nozzles

4.2.3.1 *General specifications*

4.2.3.1.1 Fire hoses are to be of approved non-perishable material and are to be sufficient in length to project a jet of water to any of the spaces in which they may be required to be used. Each hose is to be provided with a nozzle and the necessary couplings. Hoses specified in this chapter as "fire hoses" together with any necessary fittings and tools are to be kept ready for use in conspicuous positions near the water service hydrants or connections. Additionally, in interior locations in passenger ships carrying more than 36 passengers fire hoses are to be connected to the hydrants at all times. Fire hoses are to have a length of at least 10 [m], but not more than:

.1 15 [m] in machinery spaces;

.2 20 [m] in other spaces and open decks; and

.3 25 [m] for open decks on ships with a maximum breadth in excess of 30 [m].

4.2.3.1.2 Unless one hose and nozzle is provided for each hydrant in the ship, there is to be complete interchangeability of hose couplings and nozzles.

4.2.3.2 *Number and diameter of fire hoses*

4.2.3.2.1 Ships are to be provided with such number and diameter of hoses as are appropriate to the type of ship.

4.2.3.2.2 In passenger ships, there is to be at least one fire hose for each of the hydrants required by 4.2.1.5 and these hoses are to be used only for the purposes of extinguishing fires or testing the fire-extinguishing apparatus at fire drills and surveys.

4.2.3.2.3 In cargo ships:

.1 of 1,000 gross tonnage and upwards, the number of fire hoses to be provided is to be one for each 30 [m] length of the ship and one spare but in no case less than five in all. This number does not include any hoses required in any engine or boiler room. If necessary the number of hoses is to be increased so as to ensure that hoses in sufficient number are available and accessible at all times, having regard to the type of ship and the nature of trade in which the ship is employed. Ships carrying dangerous goods in accordance with Ch.7, Sec.2 are to be provided with 3 hoses and nozzles, in addition to those required above; and

.2 of less than 1,000 gross tonnage, the number of fire hoses to be provided are to be calculated in accordance with the provisions of 4.2.3.2.3.1. However, the number of hoses are in no case to be less than three.

IR.3 All hydrants in the machinery spaces of ships with oil fire boilers or internal combustion type propelling machinery are to be fitted with hoses having the nozzles in accordance with 4.2.3.3.

4.2.3.3 *Size and types of nozzles*

4.2.3.3.1 For the purposes of this part, standard nozzle sizes are to be 12 [mm], 16 [mm] and 19 [mm] or as near thereto as possible. Larger diameter nozzles may be permitted at the discretion of IRS.

4.2.3.3.2 For accommodation and service spaces, a nozzle size greater than 12 [mm] need not be used.

4.2.3.3.3 For machinery spaces and exterior locations, the nozzle sizes are to be such as to obtain the maximum discharge possible from two jets at the pressure mentioned in 4.2.1.6 from the smallest pump, provided that a nozzle size greater than 19 [mm] need not be used.

4.2.3.3.4 Nozzles are to be of an approved dual-purpose type (i.e., spray / jet type) incorporating a shutoff.

IR4.2.3.3.4 Aluminium alloys may be used for fire hose couplings and nozzles, except in open deck areas of oil tankers and chemical tankers.

4.3 Portable fire extinguishers

4.3.1 Type and design

Portable fire extinguishers are to comply with the requirements of the Fire Safety Systems Code (Chapter 8).

4.3.2 Arrangement of fire extinguishers (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable Fire Extinguishers on Board Ships') as corrected by MSC.1/Circ.1275/Corr.1.

4.3.2.1 Accommodation spaces, service spaces and control stations are to be provided with portable fire extinguishers of appropriate types and in sufficient number to the satisfaction of the Administration. Ships of 1,000 gross tonnage and upwards are to carry at least five portable fire extinguishers.

IR4.3.2.1 For ships of under 1,000 gross tonnage the above number of extinguishers is not be less than three.

4.3.2.2 One of the portable fire extinguishers intended for use in any space is to be stowed near the entrance to that space.

4.3.2.3 Carbon dioxide fire extinguishers are not to be placed in accommodation spaces. In control stations and other spaces containing electrical or electronic equipment or appliances necessary for the safety of the ship, fire extinguishers whose extinguishing media are neither electrically conductive nor harmful to the equipment and appliances are to be provided.

4.3.2.4 Fire extinguishers are to be situated ready for use at easily visible places, which can be reached quickly and easily at any time in the event of a fire, and in such a way that their serviceability is not impaired by the weather, vibration or other external factors. Portable fire extinguishers are to be provided with devices which indicate whether they have been used.

IR4.3.2.5 For galleys and for spaces containing domestic boilers, one portable fire extinguisher suitable for dealing with oil fires or fires in electric cooking equipment, as appropriate, is to be provided.

4.3.3 Spare charges

4.3.3.1 Spare charges are to be provided for 100% of the first 10 extinguishers and 50% of the remaining fire extinguishers capable of being recharged on board. Not more than 60 total spare charges are required. Instructions for recharging are to be carried on board.

4.3.3.2 For fire extinguishers which cannot be recharged onboard, additional portable fire extinguishers of the same quantity, type, capacity and number as determined in 4.3.3.1 above are to be provided in lieu of spare charges.

4.4 Fixed fire-extinguishing systems

4.4.1 Types of fixed fire extinguishing systems

4.4.1.1 A fixed fire extinguishing system required by 4.5 below may be any of the following:

.1 a fixed gas fire-extinguishing system complying with the provisions of the Fire Safety Systems Code (Chapter 8);

.2 a fixed high-expansion foam fire-extinguishing system complying with the provisions of the Fire Safety Systems Code (Chapter 8); and

.3 a fixed pressure water-spraying fire-extinguishing system complying with the provisions of the Fire Safety Systems Code (Chapter 8).

4.4.1.2 Where a fixed fire-extinguishing system not required by this part is installed, it is to meet the relevant requirement of this part and the Fire Safety Systems Code (Chapter 8).

4.4.1.3 Fire-extinguishing systems using Halon 1211, 1301, and 2402 and perfluorocarbons are to be prohibited.

4.4.1.4 Steam is not generally accepted as a fire-extinguishing medium in fixed fire-extinguishing systems. However, restricted use of steam is permitted as an addition to the required fire-extinguishing system and is to comply with the requirements of the Fire Safety System Code (Chapter 8).

4.4.1.5 On ships constructed before 1 July 2002, fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms shall comply with the provisions of 5.2.2.2 of Section 5 of Chapter 8 “Fire Safety Systems Code”, by the first scheduled dry-docking after 1 January 2010.

4.4.2 Closing appliances for fixed gas fire-extinguishing systems

Where a fixed gas fire-extinguishing system is used, openings which may admit air to, or allow gas to escape from, a protected space is to be capable of being closed from outside the protected space.

4.4.3 Storage rooms of fire-extinguishing medium

When the fire-extinguishing medium is stored outside a protected space, it is to be stored in a room which is located aft of the forward collision bulkhead, and is used for no other purposes. Any entrance to such a room is preferably to be from the open deck and is to be independent of the protected space.

Additionally, storage rooms for fixed gas fire extinguishing media are to meet the following requirements:

a) If the storage space is located below deck, it is to be located no more than one deck below the open deck and is to directly accessible by a stairway or ladder from the open deck.

b) Storage spaces located below deck or spaces located above deck but not provided with access from the open deck, are to be fitted with a mechanical ventilation system designed to take exhaust air from the bottom of the space and sized to provide at least 6 air changes per hour. Where located above deck and having direct access from the open deck, natural ventilation may suffice.

c) Access doors are to open outwards, and the bulkheads and decks including doors and other means of closing any opening therein, which form the boundaries between such rooms and adjacent enclosed spaces are to be gastight. For the purpose of the application of Table 3.1 to Table 3.8, such storage rooms are to be treated as fire control stations.

IR4.4.3 Fire extinguishing media protecting the cargo holds may be stored in a room located forward of the cargo holds but aft of the collision bulkheads, provided that both the local manual release mechanism and remote control(s) for the release of the media are fitted, and the latter is of robust construction or so protected as to remain operable in case of fire in protected spaces. The remote control is to be placed in the accommodation area in order to facilitate their ready accessibility by the crew. The capability to release different quantities of fire extinguishing media into different cargo holds so protected is to be included in the remote release arrangement.

4.4.4 Water pumps for other fire-extinguishing systems

Pumps, other than those serving the fire main, required for the provision of water for fire-extinguishing systems required by this part, their sources of power and their controls are to be installed outside the space or spaces protected by such systems and are to be so arranged that a fire in the space or spaces protected will not put any such system out of action.

4.5 Fire extinguishing arrangements in machinery spaces

4.5.1 Machinery spaces containing oil-fired boilers or oil fuel units (Also see Table 4.5.1)

4.5.1.1 *Fixed fire-extinguishing systems*

Machinery spaces of category A containing oil-fired boilers or oil fuel units are to be provided with any one of the fixed fire-extinguishing systems in 4.4.1. In each case, if the engine and boiler rooms are not entirely separate, or if fuel oil can drain from the boiler room into the engine-room, the combined engine and boiler rooms are to be considered as one compartment.

IR 4.5.1.1 Machinery spaces of Category A where oil fuel of flash point less than 60°C is used

The fixed installation is to be of gas type as in 4.4.1.1.1 in all cases where the flash point of oil fuel is less than 60°C. (See Ch.2, 1.2.1).

4.5.1.2 *Additional fire-extinguishing arrangements* (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable Fire Extinguishers on Board Ships' as corrected by MSC.1/Circ.1275/Corr.1).

4.5.1.2.1 There are to be in each boiler room or at an entrance outside of the boiler room at least one portable foam applicator unit complying with the provisions of the Fire Safety Systems Code.

4.5.1.2.2 There are to be at least two portable foam extinguishers or equivalent in each firing space in each boiler room and in each space in which a part of the oil fuel installation is situated. There is to be not less than one approved foam-type extinguisher of at least 135 litres capacity or equivalent in each boiler room. These extinguishers are to be provided with hoses on reels suitable for reaching any part of the boiler room. In the case of domestic boilers of less than 175 kW, or boilers protected by fixed water-based local application fire-extinguishing systems as required by 4.5.6, an approved foam-type extinguisher of at least 135 litres capacity is not required.

4.5.1.2.3 In each firing space there is to be a receptacle containing at least 0.1 [m^3] sand, sawdust impregnated with soda, or other approved dry material, along with a suitable shovel for spreading the material. An approved portable extinguisher may be substituted as an alternative.

Table 4.5.1 : Number of systems, appliances and extinguishers

Systems, appliances & extinguishers Category A machinery spaces	Fixed fire extinguishing system	Portable foam applicator *1	Portable foam extinguishers	Additional portable foam extinguishers	135 l foam extinguisher	45 l foam extinguishers *2	Sand boxes *3
Pt.6, Ch.3, Clause nos.	4.5.1.1 4.5.2.1	4.5.1.2.1 4.5.2.2.1	4.5.1.2.2	4.5.2.2.2	4.5.1.2.2	4.5.2.2.2	4.5.1.2.3
Boiler room containing:							
Oil-fired boilers	1	1	2N	NA	1*4	-	N
Oil-fired boilers and oil fuel units	1	1	2N + 2	NA	1*4	-	N
Engine room containing:							
Oil fuel units only	1	-	2	NA	-	-	-
Internal combustion machinery	1	1	x		-	Y	-
Internal combustion machinery and oil fuel units	1	1	x		-	y	-
Combined engine / boiler room containing:							
Internal combustion machinery, oil fired boilers and oil fuel units	1	1	(2N + 2) or x whichever is greater		1*4	y*5	N

N = number of firing spaces.
"2N" means that two extinguishers are to be located in each firing space.

x = sufficient number, minimum two in each space, so located that there are at least one portable fire extinguisher within 10 m walking distance from any point.

y = sufficient number to enable foam to be directed onto any part of the fuel and lubricating oil pressure systems, gearing and other fire hazards.

Notes:

*1 : May be located at outside of the entrance to the room.

*2 : May be arranged outside of the space concerned for smaller spaces of cargo ships.

*3 : The amount of sand is to be at least 0.1 m3. A shovel is to be provided. Sand boxes may be substituted by approved portable fire extinguishers.

*4 : Not required for such spaces in cargo ships wherein all boilers contained therein are for domestic services and are less than 175 kW, or boilers are protected by fixed water-based local application fire-extinguishing systems as required by 4.5.6.

*5 : In case of machinery spaces containing both boilers and internal combustion engines (case not explicitly considered in Clause 4.5. Clauses 4.5.1 and 4.5.2 apply, with the exception that one of the foam fire extinguishers of at least 45 l capacity or equivalent (required by 4.5.2.2.2) may be omitted on the condition that the 135 l extinguisher (required by 4.5.1.2.2) can protect efficiently and readily the area covered by the 45 l extinguisher.

*6 Oil fired machinery other than boilers such as fired inert gas generators, incinerators and waste disposal units are to be considered the same as boiler in so far as the required number and type of fire fighting appliances are concerned.

4.5.2 Machinery spaces of Category A containing internal combustion machinery (Also See Table 4.5.1)

Machinery spaces of category A containing internal combustion machinery are to be provided with one of the fixed fire-extinguishing systems in 4.4.1.

4.5.2.1 *Fixed fire-extinguishing systems*

4.5.2.2 *Additional fire-extinguishing arrangements* (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable Fire Extinguishers on Board Ships' as corrected by MSC.1/Circ.1275/Corr.1).

4.5.2.2.1 There is to be at least one portable foam applicator unit complying with the provisions of the Fire Safety Systems Code.

4.5.2.2.2 There are to be in each such space approved foam type fire extinguishers each of at least 45 litres capacity or equivalent sufficient in number to enable foam or its equivalent to be directed onto any part of the fuel and lubricating oil pressure systems, gearing and other fire hazards. In addition, there is to be provided a sufficient number of portable foam extinguishers or equivalent which are to be so located that an extinguisher is not more than 10 [m] walking distance from any point in the space; provided that there are to be at least two such extinguishers in each such space. For ships under 150 tons gross the provision of a 45 litre extinguisher may be waived.

4.5.3 Machinery spaces containing steam turbines or enclosed steam engines

4.5.3.1 *Fixed fire-extinguishing systems*

In spaces containing steam turbines or enclosed steam engines used for main propulsion or other purposes having in the aggregate a total output of not less than 375 [kW], one of the fire-extinguishing systems specified in 4.4.1 is to be provided if such spaces are periodically unattended.

4.5.3.2 *Additional fire-extinguishing arrangements*

4.5.3.2.1 There are to be approved foam fire extinguishers each of at least 45 *l* capacity or equivalent sufficient in number to enable foam or its equivalent to be directed on to any part of the pressure lubrication system, on to any part of the casings enclosing pressure lubricated parts of the turbines, engines or associated gearing, and any other fire hazards. However, such extinguishers are not required if protection, at least equivalent to that required by this subparagraph, is provided in such spaces by a fixed fire-extinguishing system fitted in compliance with 4.4.1.

4.5.3.2.2 There are to be a sufficient number of portable foam extinguishers or equivalent which are to be so located that no point in the space is more than 10 [m] walking distance from an extinguisher and that there are at least two such extinguishers in each such space, except that such extinguishers are not required in addition to any provided in compliance with 4.5.1.2.2. (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable Fire Extinguishers on Board Ships' as corrected by MSC.1/Circ.1275/Corr.1).

4.5.4 Other machinery spaces

Where a fire hazard exists in any machinery space for which no specific provisions for fire-extinguishing appliances are prescribed in 4.5.1, 4.5.2 and 4.5.3, there are to be provided in, or adjacent to, that space such a number of approved portable fire extinguishers or other means of fire extinction according to fire hazard in each space. (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable Fire Extinguishers on Board Ships' as corrected by MSC.1/Circ.1275/Corr.1).

4.5.5 Additional requirements for passenger ships

In passenger ships carrying more than 36 passengers, each machinery space of category A is to be provided with at least two water fog applicators consisting of a metal L-shaped pipe, the long limb being about 2 [m] in length capable of being fitted to a fire hose and the short limb being about 250 [mm] in length fitted with a fixed water fog nozzle or capable of being fitted with a water spray nozzle.

4.5.6 Fixed local application fire-fighting systems

4.5.6.1 Sub-section 4.5.6 applies to passenger ships of 500 gross tonnage and above and cargo ships of 2000 gross tonnage and above.

4.5.6.2 Machinery spaces of category A above 500 [m^3] in volume in addition to the fixed fire-extinguishing system required in 4.5.1.1 are to be protected by a type of fixed water-based or equivalent local application fire-fighting system approved by the National Statutory Authority. In the case of periodically unattended machinery spaces, the fire fighting system is to have both automatic and manual release capabilities. In the case of continuously

manned machinery spaces, the fire-fighting system is only required to have a manual release capability. (Refer to the Revised Guidelines for the approval of fixed water-based local application fire-fighting systems for use in category A machinery spaces (MSC.1/Circ.1387 and MSC.1/Circ.1387/Corr.1) and Unified Interpretations of SOLAS Chapter II-2 (MSC.1/Circ.1276/Rev.1)).

IR4.5.6.2 The automatic release of fire extinguishing medium is to be activated by a detection system capable of reliably identifying the local zones. Means are to be provided to prevent accidental release.

IR4.5.6.2.1 If the system installed deviates from the arrangement approved in accordance with MSC.1/Circ.1387 and MSC.1/Circ.1387/Corr.1, then such arrangement is to additionally pass the fire test based on the scenarios specified in the circular.

4.5.6.3 Fixed local application fire-fighting systems are to protect areas such as the following without the necessity of engine shutdown, personnel evacuation, or sealing of the spaces:

.1a) the fire hazard portions of all internal combustion machinery for ships constructed on or after 01 July, 2014 and

b) the fire hazard portions of internal combustion machinery used for the ship's main propulsion and power generation for ships constructed before 01 July, 2014.

IR.1 In multi-engine installations at least two sections are to be arranged to cover all internal combustion machinery.

.2 boiler fronts;

IR.2 (Boiler fronts in this context mean the boiler burner locations irrespective of the boiler design).

.3 the fire hazard portions of incinerators; and

.4 purifiers for heated fuel oil.

IR.5 oil fired equipment such as inert gas generators and thermal oil heaters.

4.5.6.4 Activation of any local application system is to give a visual and distinct audible alarm in the protected space and at continuously manned stations. The alarm is to indicate the specific system activated. The system alarm requirements described within this subsection are in addition to, and not a substitute for, the detection and fire alarm system required elsewhere in this chapter.

IR4.5.6.4 The alarms may be grouped together with provision for indication of the activated zone. Such alarms and indications are to be provided in wheel house in addition to protected space and engine control room. Audible alarms may use a single tone.

4.6 Fire-extinguishing arrangements in control stations, accommodation and service spaces

4.6.1 Sprinkler and water spray systems in passenger ships

4.6.1.1 Passenger ships carrying more than 36 passengers are to be equipped with an automatic sprinkler, fire detection and fire alarm system of an approved type complying with the requirements of the Fire Safety Systems Code (Chapter 8) in all control stations, accommo-dation and service spaces, including corridors and stairways. Alternatively, control stations, where water may cause damage to essential equipment, may be fitted with an approved fixed fire-extinguishing system of another type. Spaces having little or no fire risk such as voids, public toilets, carbon dioxide rooms and similar spaces need not be fitted with an automatic sprinkler system.

4.6.1.2 In passenger ships carrying not more than 36 passengers, when a fixed smoke detection and fire alarm system complying with the provisions of the Fire Safety Systems Code (Chapter 8) is provided only in corridors, stairways and escape routes within accommodation spaces, an automatic sprinkler system is to be installed in accordance with 1.5.3.2.

4.6.1.3 A fixed pressure water-spraying fire extinguishing system complying with the provisions of Ch8 is to be installed on cabin balconies of ships to which requirement of Ch.2, 2.3.4 applies, if furniture and furnishings on such balconies are not as defined in Ch.1, 3.40 a), b), c), f) and g).

4.6.2 Sprinkler systems for cargo ships

In cargo ships in which method IIC specified in 3.2.3.1.1.2 is adopted, an automatic sprinkler, fire detection and fire alarm system is to be fitted in accordance with the requirements in 1.5.5.2.

4.6.3 Spaces containing flammable liquid

4.6.3.1 Paint lockers are to be protected by any one of the following systems :
.1 a carbon dioxide system, designed to give a minimum volume of free gas equal to 40% of the gross volume of the protected space;

.2 a dry powder system, designed for at least 0.5 [kg powder/m^3];

.3 a water spraying or sprinkler system, designed for 5 [l/m^2 min]. Water spraying systems may be connected to the fire main of the ship; or

.4 a system providing equivalent protection, as determined by IRS.

In any case, the system is to be operable from outside the protected space.

4.6.3.2 Flammable liquid lockers are to be protected by an appropriate fire-extinguishing arrangement approved by IRS.

4.6.3.3 For lockers of a deck area of less than 4 [m^2], which do not give access to accommodation spaces, a carbon dioxide portable fire extinguisher sized to provide a minimum volume of free gas equal to 40% of the gross volume of the space may be accepted in lieu of a fixed system. A discharge port is to be arranged in the locker to allow the discharge of the extinguisher without having to enter into the protected space. The required portable fire extinguisher is to be stowed adjacent to the port. Alternatively, a port or hose connection may be provided to facilitate the use of fire main water.

IR4.6.3.3 The requirements given in 4.6.3.2 and 4.6.3.3 are not considered applicable for cargo service spaces intended for the stowage of cargo samples, when such spaces are positioned within the cargo area onboard tankers.

4.6.4 Deep-fat cooking equipment

Deep-fat cooking equipment installed in enclosed spaces or on open decks is to be fitted with the following:

.1 an automatic or manual extinguishing system tested to a recognised international standard; (Refer to the recommendations by the International Organization for Standardization, in particular, Publication ISO 15371:2000 on *Fire-extinguishing systems for protection of galley deep-fat cooking equipmen*t);

.2 a primary and backup thermostat with an alarm to alert the operator in the event of failure of either thermostat;

.3 arrangements for automatically shutting off the electrical power upon activation of the extinguishing system;

.4 an alarm for indicating operation of the extinguishing system in the galley where the equipment is installed; and

.5 controls for manual operation of the extinguishing system which are clearly labelled for ready use by the crew.

4.7 Fire extinguishing arrangements in cargo spaces

4.7.1 Fixed gas fire-extinguishing systems for general cargo

4.7.1.1 Except as provided for in 4.7.2, the cargo spaces of passenger ships of 1,000 gross tonnage and upwards are to be protected by a fixed carbon dioxide or inert gas fire-extinguishing system complying with the provisions of the Fire Safety Systems Code (Chapter 8) or by a fixed high expansion foam fire-extinguishing system which gives equivalent protection.

4.7.1.2 Where a passenger ship is engaged on voyages of such short duration that it would be unreasonable to apply the requirements of 4.7.1.1 and also in ships of less than 1,000 gross tonnage, the fire extinguishing arrangements in cargo spaces will be specially considered, provided that the ship is fitted with steel hatch covers and effective means of closing all ventilators and other openings leading to the cargo spaces.

4.7.1.3 Except for ro-ro and vehicle spaces, cargo spaces on cargo ships of 2,000 gross tonnage and upwards are to be protected by a fixed carbon dioxide or inert gas fire-extinguishing system complying with the Fire Safety Systems Code (Chapter 8), or by a fire-extinguishing system which gives equivalent protection.

IR4.7.1.3 In addition, ships of less than 2,000 tons gross tonnage carrying petroleum products having a flash point exceeding 60°C (c.c. test) are also not required to be fitted with a fixed fire extinguishing system.

4.7.1.4 The requirements of 4.7.1.3 and 4.7.2 may be waived for cargo spaces of any cargo ship if constructed, and solely intended for, the carriage of ore, coal, grain, unseasoned timber, non-combustible cargoes or cargoes which constitute a low fire risk. Such exemptions may be granted only if the ship is fitted with steel hatch covers and effective means of closing ventilators and other openings leading to the cargo spaces. (Refer to the IMSBC Code (resolution MSC.268(85)), as amended, appendix 1, entry for coal and to the List of solid bulk cargoes for which a fixed gas fire-extinguishing system may be exempted or for which a fixed gas fire-extinguishing system is ineffective (MSC.1/Circ.1395/Rev.6)).

4.7.2 Fixed gas fire-extinguishing systems for dangerous goods

A ship engaged in the carriage of dangerous goods in any cargo spaces is to be provided with a fixed carbon dioxide or inert gas fire-extinguishing system complying with the provisions of the Fire Safety Systems Code or with a fire-extinguishing system which gives equivalent protection for the cargoes carried.

IR4.7.2.1 Water supplies defined in Ch.7, 2.3.1.2 are considered as equivalent protection for cargoes listed in Table 2 of MSC.1/Circ.1395/Rev.6.

IR4.7.2.2 Clause 4.7.2 applies to the following ships engaged in the carriage of dangerous goods:

.1 Passenger ships constructed on or after 1 September 1984; and

.2 Cargo ships of 500 gross tonnage and upwards constructed on or after 1 September 1984;

.3 Cargo ships of less than 500 gross tonnage are not subject to Clause 4.7.2 even when such ships are engaged in the carriage of dangerous goods and document of compliance are issued to such ships according to Pt.6, Ch.7, Cl.2.4.

4.7.3 Firefighting for ships constructed on or after 1 January 2016 designed to carry containers on or above the weather deck

4.7.3.1 Ships are to carry, in addition to the equipment and arrangements required by 4.1 and 4.2, at least one water mist lance.

4.7.3.1.1 The water mist lance is to consist of a tube with a piercing nozzle which is capable of penetrating a container wall and producing water mist inside a confined space (container, etc.) when connected to the fire main.

4.7.3.2 Ships designed to carry five or more tiers of containers on or above the weather deck are to carry, in addition to the requirements of paragraph 4.7.3.1, mobile water monitors* as follows:

(*Refer to the Guidelines for the design, performance, testing and approval of mobile water monitors used for the protection of on-deck cargo areas of ships designed and constructed to carry five or more tiers of containers on or above the weather deck (MSC.1/Circ.1472).)

.1 ships with breadth less than 30 [m] at least two mobile water monitors; or

.2 ships with breadth of 30 [m] or more: at least four mobile water monitors.

4.7.3.2.1 The mobile water monitors, all necessary hoses, fittings and required fixing hardware are to be kept ready for use in a location outside the cargo space area not likely to be cut-off in the event of a fire in the cargo spaces.

4.7.3.2.2 A sufficient number of fire hydrants are to be provided such that:

.1 all provided mobile water monitors can be operated simultaneously for creating effective water barriers forward and aft of each container bay;

.2 the two jets of water required by paragraph 4.2.1.5.1 can be supplied at the pressure required by paragraph 4.2.1.6; and

.3 each of the required mobile water monitors can be supplied by separate hydrants at the pressure necessary to

reach the top tier of containers on deck.

4.7.3.2.3 The mobile water monitors may be supplied by the fire main, provided the capacity of fire pumps and fire main diameter are adequate to simultaneously operate the mobile water monitors and two jets of water from fire hoses at the required pressure values. If carrying dangerous goods, the capacity of fire pumps and fire main diameter is also to comply with Pt 6, Ch 7, Sec 2, and 2.3.1.5 as far as applicable to on-deck cargo areas. (Also see 4.2.2.4.1, IR .2.1)

4.7.3.2.4 The operational performance of each mobile water monitor is to be tested during initial survey on board the ship to the satisfaction of the Administration. The test is to verify that:

.1 the mobile water monitor can be securely fixed to the ship structure ensuring safe and effective operation; and

.2 the mobile water monitor jet reaches the top tier of containers with all required monitors and water jets from fire hoses operated simultaneously.

4.8 Cargo tank protection

4.8.1 Fixed deck foam systems

4.8.1.1 For tankers of 20,000 tonnes deadweight and upwards, a fixed deck foam system is to be provided in accordance with the Fire Safety Systems Code (Chapter 8). Other fixed installations if they afford equivalent protection may be accepted is lieu of above installations. The requirements for alternative fixed installations are to comply with 4.8.1.2.

4.8.1.2 To be considered equivalent, the system proposed in lieu of the deck foam system is to be capable of:

.1 extinguishing spill fires and also precluding ignition of spilled oil not yet ignited, and

.2 combating fires in ruptured tanks.

4.8.1.3 Tankers of less than 20,000 tonnes deadweight are to be provided with a deck foam system complying with the requirements of the Fire Safety Systems Code (Chapter 8).

IR4.8.1 Where an enclosed pipe trunk is situated within the cargo tanks deck area, the pipe trunk is to comply with following:

.1 it is to be protected by a fixed fire extinguishing system in accordance with 4.9. the extinguishing system is to be operated from a readily accessible position outside the pipe trunk;

.2 it is not considered as part of the cargo tanks deck area;

.3 the area of the pipe trunk need not be included in the calculation of the foam solution rate of supply for the deck foam system required by 4.8;

.4 it is to be adequately ventilated and protected in accordance with Pt.5, Ch.2, 6.4.3 and 6.4.4 and

.5 it is to contain no flammable gas sources other than pipes and flanges. If the pipe trunk contains any other source of flammable gas, i.e. valves and pumps, it is to be regarded as cargo pump room.

4.9 Protection of cargo pump rooms in tankers

4.9.1 Fixed fire-extinguishing systems

Each cargo pump-room in tankers of 500 gross tonnage and more is to be provided with one of the following fixed fire-extinguishing systems operated from a readily accessible position outside the pump-room. Cargo pump-rooms are to be provided with a system suitable for machinery spaces of category A.

4.9.1.1 A carbon dioxide system complying with the provisions the Fire Safety Systems Code (Chapter 8) and with the following:

.1 the alarms giving audible warning of the release of fire-extinguishing medium are to be safe for use in a flammable cargo vapour/air mixture; and

.2 a notice is to be exhibited at the controls stating that due to the electrostatic ignition hazard, the system is to be used only for fire extinguishing and not for inerting purposes.

4.9.1.2 A high-expansion foam system complying with the provisions of the Fire Safety Systems Code (Chapter 8), provided that the foam concentrate supply is suitable for

extinguishing fires involving the cargoes carried.

4.9.1.3 A fixed pressure water-spraying system complying with the provisions of the Fire Safety Systems Code (Chapter 8).

4.9.2 Quantity of fire-extinguishing medium

Where the extinguishing medium used in the cargo pump-room system is also used in systems serving other spaces, the quantity of medium provided or its delivery rate need not be more than the maximum required for the largest compartment.

IR4.9.3 In addition, two portable foam extinguishers or equivalent are to be provided, one at the pumps and one at the pump room entrance. (Refer 4.2.1.1.1 of Chapter 8 for capacity of foam extinguishers).

IR4.9.4 In ships of less than 500 tons gross tonnage, the fire extinguishing arrangements will be specially considered in each case.

4.10 Fire-fighter's outfits

4.10.1 Types of fire-fighter's outfits

4.10.1.1 Fire-fighter's outfits are to comply with the Fire Safety Systems Code (Chapter 8), and

4.10.1.2 Self-contained compressed air breathing apparatus of fire-fighter's outfits are to comply with paragraph 3.2.1.2.2 of the fire safety systems code (Chapter 8) by 1 July 2019.

4.10.2 Number of fire-fighter's outfits

4.10.2.1 Ships are to carry at least two fire-fighter's outfits.

4.10.2.2 In addition, in passenger ships there are to be provided:

.1 for every 80 [m], or part thereof, of the aggregate of the lengths of all passenger spaces and service spaces on the deck which carries such spaces or, if there is more than one such deck, on the deck which has the largest aggregate of such lengths, two fire-fighter's outfits and two sets of personal equipment, each set comprising the items stipulated in the Fire Safety Systems Code (Chapter 8). In passenger ships carrying more than 36 passengers, two additional fire-fighter's outfits are to be provided for each main vertical zone. However, for stairway enclosures which constitute individual main vertical zones and for the main vertical zones in the fore or aft end of a ship which do not contain spaces of categories (6), (7), (8) or (12) defined in 3.2.2.3, no additional fire-fighter's outfits are required; and

.2 ships carrying more than 36 passengers, for each pair of breathing apparatus there is to be provided one water fog applicator which is to be stored adjacent to such apparatus.

4.10.2.3 In addition, in tankers, two fire-fighter's outfits are provided.

4.10.2.4 Additional sets of personal equipment and breathing apparatus may be required, having due regard to the size and type of the ship.

4.10.2.5 Two spare charges are to be provided for each required breathing apparatus. Passenger ships carrying not more than 36 passengers and cargo ships that are equipped with suitably located means for fully recharging the air cylinders free from contamination, need carry only one spare charge for each required apparatus. In passenger ships carrying more than 36 passengers, at least two spare charges for each breathing apparatus are to be provided.

4.10.2.6 Passenger ships carrying more than 36 passengers constructed on or after 1 July 2010 are to be fitted with a suitably located means for fully recharging breathing air cylinders, free from contamination. The means for recharging are to be either:

.1 breathing air compressors supplied from the main and emergency switchboard, or independently driven, with a minimum capacity of 60 l/min per required breathing apparatus, not to exceed 420 l/min; or

.2 self-contained high-pressure storage systems of suitable pressure to recharge the breathing apparatus used on board, with a capacity of at least 1,200 l per required breathing apparatus, not to exceed 50,000 l of free air.

4.10.3 Storage of fire-fighter's outfits

4.10.3.1 The fire-fighter's outfits or sets of personal equipment are to be kept ready for use in an easily accessible location that is

permanently and clearly marked and, where more than one fire-fighter's outfit or more than one set of personal equipment is carried, they are to be stored in widely separated positions.

4.10.3.2 In passenger ships, at least two fire-fighter's outfits and, in addition, one set of personal equipment are to be available at any one position. At least two fire-fighter's outfits are to be stored in each main vertical zone.

4.10.4 Fire-fighter's communication

4.10.4.1 For ships constructed on or after 1 July 2014, a minimum of two exclusive two-way portable radiotelephone apparatus for each fire party for fire-fighter's communication shall be carried on board. Those two-way portable radiotelephone apparatus shall be of an explosion-proof type or intrinsically safe. Ships constructed before 1 July 2014 shall comply with the requirements of this paragraph not later than the first survey after 1 July 2018.

IR4.10.4.1 The two-way portable radio telephone apparatus for fire-fighter's communication, required by 4.10.4.1 is to be of certified safe type suitable for use in Zone 1 hazardous areas as defined in IEC Publication 600079. The minimum requirements in respect to the apparatus group and temperature class are to be consistent with the most restrictive requirements for the hazardous area zone on board which is accessible to fire party.

IR4.11 Ships not fitted with propelling machinery

A power pump or hand pump and fire hose and nozzle are to be fitted, having suction from the sea and capable of producing a jet of water on deck with a throw of not less than 6 [m].
Where oil fuel is used for generation of power, fire extinguishing appliances suitable for fighting oil fires are to be provided to the satisfaction of the Surveyors.

Additional fire extinguishing equipment may be required in accommodation spaces or cargo spaces depending upon the potential fire hazard afforded by the equipment and materials carried on in these spaces.

Section 5

Structural Integrity

5.1 Purpose

The purpose of this regulation is to maintain structural integrity of the ship preventing partial or whole collapse of the ship structures due to strength deterioration by heat. For this purpose, materials used in the ships' structure are to ensure that the structural integrity is not degraded due to fire.

5.2 Material of hull, superstructures, structural bulkheads, decks and deckhouses

The hull, superstructures, structural bulkheads, decks and deckhouses are to be constructed of steel or other equivalent material. For the purpose of applying the definition of steel or other equivalent material as given in Ch.1, 3.43 the "applicable fire exposure" is to be according to the integrity and insulation standards given in Table 3.1 to Table 3.4. For example, where divisions such as decks or sides and ends of deckhouses are permitted to have "B-0" fire integrity, the "applicable fire exposure" is to be half an hour.

5.3 Structure of aluminium alloy

Aluminium alloys will be specially considered giving due regard to the fire protection method adopted to ensure:

.1 the insulation of aluminium alloy components of 'A' or 'B' class divisions except structures, which in the opinion of IRS are non- load bearing, is to be such that the temperature of the structural core does not rise more than 200°C above the ambient temperature at any time during the applicable fire exposure to the standard fire test.

IR.1 If an aluminium deck is tested with insulation installed below the deck, then the result is to apply to decks which are bare on the top. Aluminium decks are not to be

provided with deck coverings or insulation on the top unless tested with the deck covering or insulation included, to verify that the 200°C temperature of the aluminium is not exceeded.

.2 special attention is to be given to the insulation of aluminium alloy components of columns, stanchions and other structural members required to support lifeboat and liferaft stowage, launching and embarkation areas, and 'A' and 'B' class divisions to ensure:

.2.1 that for such members supporting lifeboat and liferaft areas and 'A' class divisions, the temperature rise limitation specified in (5.3.1) is to apply at the end of one hour, and

.2.2 that for such members required to support 'B' class divisions, the temperature rise limitation specified in (5.3.1) is to apply at the end of half an hour.

5.4 Machinery spaces of category A

5.4.1 Crowns and casings

Crowns and casings of machinery spaces of category A is to be of steel construction and are to be insulated as required by Table 3.5a and Table 3.7, as appropriate.

5.4.2 Floor plating

The floor plating of normal passageways in machinery spaces of category A are to be made of steel.

5.5 Materials of overboard fittings

Materials readily rendered ineffective by heat are not to be used for overboard scuppers, sanitary discharges and other outlets which are close to the waterline and where the failure of the material in the event of fire would give rise to danger of flooding.

5.6 Protection of cargo tank structure against pressure or vacuum in tankers

5.6.1 General

Refer Pt.5, Ch.2, 8.1.4a) and b) for venting of small and large volumes.

5.6.2 Openings for small flow by thermal variations

Refer Pt.5, Ch.2, 8.1.14a) and b).

5.6.3 Safety measures in cargo tanks

5.6.3.1 *Preventive measures against liquid rising in the venting system*

Refer Pt.5, Ch.2, 8.1.5.

5.6.3.2 *Secondary means for pressure/vacuum relief*

Refer Pt.5, Ch.2, 8.1.4 c).

5.6.3.3 *Bypasses in vent mains*

Refer Pt.5, Ch.2, 8.1.9.

5.6.3.4 *Pressure/vacuum-breaking devices*

Refer Pt.5, Ch.2, 8.2.4.

5.6.4 Size of vent outlets

Large volume vent outlets for cargo loading, discharging and ballasting required by 5.6.1 are to be designed as given in Pt.5, Ch.2, 8.1.10. The master is to provided with information as given in Pt.5, Ch.2, 8.1.15 e).

End of Chapter

Chapter 4

Escape

Contents

Section

1 *Notification of Crew and Passengers*

2 *Means of Escape*

Section 1

Notification of Crew and Passengers

1.1 Purpose

The purpose of this section is to notify crew and passengers of a fire for safe evacuation. For this purpose, a general emergency alarm system and a public address system are to be provided.

1.2 General emergency alarm system

A general emergency alarm system is to be used for notifying crew and passengers of a fire.

1.3 Public address systems in passenger ships

A public address system or other effective means of communication are to be available throughout the accommodation and service spaces and control stations and open decks.

Section 2

Means of Escape

2.1 Purpose

2.1.1 The purpose of this regulation is to provide means of escape so that persons onboard can safely and swiftly escape to the lifeboat and liferaft embarkation deck. For this purpose, the following functional requirements are to be met:

.1 safe escape routes are to be provided;

.2 escape routes are to be maintained in a safe condition, clear of obstacles; and

.3 additional aids for escape are to be provided as necessary to ensure accessibility, clear marking, and adequate design for emergency situations.

IR2.1.1 To facilitate a swift and safe means of escape to the lifeboat and liferaft embarkation deck, the following provisions apply to overhead hatches fitted along the escape routes.

.1 the securing devices shall be of a type which can be opened from both sides;

.2 the maximum force needed to open the hatch cover should not exceed 150 N; and

.3 the use of a spring equalizing, counterbalance or other suitable device on the hinge side to reduce the force needed for opening is acceptable.

2.2 General requirements

2.2.1 Unless expressly provided otherwise in this section, at least two widely separated and ready means of escape are to be provided from all spaces or group of spaces.

2.2.2 Lifts are not to be considered as forming one of the means of escape as required by this section.

2.3 Means of escape from control stations, accommodation and service spaces

2.3.1 General requirements

2.3.1.1 Stairways and ladders are to be so arranged as to provide ready means of escape to the lifeboat and liferaft embarkation deck from passenger and crew accommodation spaces and from spaces in which the crew is normally employed, other than machinery spaces.

2.3.1.2 Unless expressly provided otherwise in this section, a corridor, lobby, or part of a corridor from which there is only one route of escape is to be prohibited. Dead-end corridors used in service areas which are necessary for the practical utility of the ship, such as fuel oil stations and athwartship supply corridors, may be permitted, provided such dead-end corridors are separated from crew accommodation areas and are inaccessible from passenger accommodation areas. Also, a part of a corridor that has a depth not exceeding its width is considered a recess or local extension and is permitted.

2.3.1.3 All stairways in accommodation and service spaces and control stations are to be of steel frame construction except where the use of other equivalent material is specially approved.

2.3.1.4 If a radiotelegraph station has no direct access to the open deck, two means of escape from or access to such station are to be provided, one of which may be a porthole or window of sufficient size.

2.3.1.5 Doors in escape routes are to, in general, open in-way of the direction of escape, except that:

.1 individual cabin doors may open into the cabins in order to avoid injury to persons in the corridor when the door is opened; and

.2 doors in vertical emergency escape trunks may open out of the trunk in order to permit the trunk to be used both for escape and for access.

2.3.2 Means of escape in passenger ships

2.3.2.1 *Escape from spaces below the bulkhead deck*

2.3.2.1.1 Below the bulkhead deck two means of escape, at least one of which is to be independent of watertight doors, are to be provided from each watertight compartment or similarly restricted space or group of spaces. Exceptionally, IRS may dispense with one of the means of escape for crew spaces that are entered only occasionally, if the required escape route is independent of watertight doors.

2.3.2.1.2 Where dispensation is given under the provisions of 2.3.2.1.1, this sole means of escape is to provide safe escape. However, stairways are not to be less than 800 [mm] in clear width with handrails on both sides.

2.3.2.2 *Escape from spaces above the bulkhead deck*

Above the bulkhead deck there are to be at least two means of escape from each main vertical zone or similarly restricted space or group of spaces, at least one of which is to give access to a stairway forming a vertical escape.

2.3.2.3 *Direct access to stairway enclosures*

Stairway enclosures in accommodation and service spaces are to have direct access from the corridors and be of a sufficient area to prevent congestion, having in view the number of persons likely to use them in an emergency. Within the perimeter of such stairway enclosures, only public toilets, lockers of non-combustible material providing storage for non-hazardous safety equipment and open information counters are permitted. Only corridors, lifts, public toilets, special category spaces and open ro-ro spaces to which any passengers carried can have access, other escape stairways required by 2.3.2.4.1 and external areas are permitted to have direct access to these stairway enclosures. Public spaces may also have direct access to stairway enclosures except for the backstage of a theatre. Small corridors or "lobbies" used to separate an enclosed stairway from galleys or main laundries may have direct access to the stairway provided they have a minimum

deck area of 4.5 [m^2], a width of not less than 900 [mm] and contain a fire hose station.

2.3.2.4 *Details of means of escape*

2.3.2.4.1 At least one of the means of escape required by 2.3.2.1.1 and 2.3.2.2 is to consist of a readily accessible enclosed stairway, which will provide continuous fire shelter from the level of its origin to the appropriate lifeboat and liferaft embarkation decks, or to the uppermost weather deck if the embarkation deck does not extend to the main vertical zone being considered. In the latter case, direct access to the embarkation deck by way of external open stairways and passageways will be provided and will have emergency lighting and slip-free surfaces underfoot. Boundaries facing external open stairways and passageways forming part of an escape route and boundaries in such a position that their failure during a fire would impede escape to the embarkation deck is to have fire integrity, including insulation values, in accordance with Table 3.1 to Table 3.4 of Part 6, Chapter 3, as appropriate.

2.3.2.4.2 Protection of access from the stairway enclosures to the lifeboat and liferaft embarkation areas is to be provided either directly or through protected internal routes which have fire integrity and insulation values for stairway enclosures as determined by Table 3.1 to Table 3.4 of Part 6, Chapter 3, as appropriate.

2.3.2.4.3 Stairways serving only a space and a balcony in that space is not to be considered as forming one of the required means of escape.

2.3.2.4.4 Each level within an atrium is to have two means of escape, one of which will give direct access to an enclosed vertical means of escape meeting the requirements of 2.3.2.4.1.

2.3.2.4.5 The widths, number and continuity of escapes are to be in accordance with the requirements in the Fire Safety Systems Code (Chapter 8).

2.3.2.5 *Marking of escape routes*

2.3.2.5.1 In addition to the emergency lighting the means of escape, including stairways and exits, are to be marked by lighting or photo-luminescent strip indicators placed not more than 300 [mm] above the deck at all points of the escape route including angles and intersections. The marking is to enable passengers to identify the routes of escape and readily identify the escape exits. If electric illumination is used, it is to be supplied by the emergency source of power and it is to be so arranged that the failure of any single light or cut in a lighting strip will not result in the marking being ineffective. Additionally, escape route signs and fire equipment location markings is to be of photoluminescent material or marked by lighting. Such lighting or photoluminescent equipment are to be evaluated, tested and applied in accordance with the Fire Safety Systems Code.

2.3.2.5.2 In passenger ships carrying more than 36 passengers, the requirements of 2.3.2.5.1 are to also apply to the crew accommodation areas.

2.3.2.5.3 In lieu of the escape route lighting system required by 2.3.2.5.1, alternative evacuation guidance systems may be accepted if approved by IRS based on the guidelines developed by the IMO*.

* Refer to the functional requirements and performance standards for the assessment of evacuation guidance systems (MSC1/Circ.1167) and the interim guidelines for the testing, approval and maintenance of evacuation guidance systems used as an alternative to low-location lighting systems (MSC1/Circ.1168).

2.3.2.6 *Normally locked doors that form part of an escape route*

2.3.2.6.1 Cabin and stateroom doors are not to require keys to unlock them from inside the room. There are not to be any doors along any designated escape route which require keys to unlock them when moving in the direction of escape.

2.3.2.6.2 Escape doors from public spaces that are normally latched are to be fitted with a means of quick release. Such means are to consist of a door-latching mechanism incorporating a device that releases the latch upon the application of a force in the direction of escape flow. Quick release mechanisms is to be designed and installed to the satisfaction of IRS and in particular:

.1 consist of bars or panels, the actuating portion of which extends across at least one half of the width of the door leaf, at least 760

[mm] and not more than 1120 [mm] above the deck;

.2 cause the latch to release when a force not exceeding 67 [N] is applied; and

.3 not be equipped with any locking device, set screw or other arrangement that prevents the release of the latch when pressure is applied to the releasing device.

2.3.2.7 Evacuation analysis for passenger ships

(Refer to the *Revised Guidelines on evacuation analyses for new and existing passenger ships* (MSC.1/Circ.1533), as may be amended.)

2.3.2.7.1 Escape routes are to be evaluated by an evacuation analysis early in the design process. This analysis applies to:

.1 ro-ro passenger ships constructed on or after 1 July 1999; and

.2 other passenger ships constructed on or after 1 January 2020 carrying more than 36 passengers.

2.3.2.7.2 The analysis is to be used to identify and eliminate, as far as practicable, congestion which may develop during an abandonment, due to normal movement of passengers and crew along escape routes, including the possibility that crew may need to move along these routes in a direction opposite to the movement of passengers. In addition, the analysis is to be used to demonstrate that escape arrangements are sufficiently flexible to provide for the possibility that certain escape routes, assembly stations, embarkation stations or survival craft may not be available as a result of a casualty.

2.3.3 Means of escape in cargo ships

2.3.3.1 *General*

At all levels of accommodation there are to be provided at least two widely separated means of escape from each restricted space or group of spaces.

2.3.3.2 *Escape from spaces below the lowest open deck*

Below the lowest open deck the main means of escape are to be a stairway and the second escape may be a trunk or a stairway.

2.3.3.3 *Escape from spaces above the lowest open deck*

Above the lowest open deck the means of escape are to be stairways or doors to an open deck or a combination thereof.

IR2.3.3.3 The “lowest open deck” is to be a category (10) “Open deck” (as defined in Chapter 3, Section 3, Cl.3.2.3.3.2.2 and Cl.3.2.4.2.2.2) at the lowest height from baseline in way of accommodation spaces.

2.3.3.4 *Dead-end corridors*

No dead-end corridors having a length of more than 7 [m] will be accepted. A dead-end corridor is a corridor or part of the corridor from which there is only one escape route.

2.3.3.5 *Width and continuity of escape routes*

The width, number and continuity of escape routes are to be in accordance with the requirements in the Fire Safety Systems Code (Chapter 8).

2.3.3.6 *Dispensation from two means of escape*

Exceptionally, one of the means of escape may be dispensed with, for crew spaces that are entered only occasionally, if the required escape route is independent of watertight doors.

2.3.4 Emergency escape breathing devices*

(* Refer to the Guidelines for the performance, location, use and care of emergency escape breathing devices (MSC/Circ.849)).

2.3.4.1 Emergency escape breathing devices are to comply with the Fire Safety Systems Code (Chapter 8). Spare emergency escape breathing devices are to be kept onboard.

2.3.4.2 All ships are to carry at least two emergency escape breathing devices within accommodation spaces.

2.3.4.3 In passenger ships, at least two emergency escape breathing devices are to be carried in each main vertical zone.

2.3.4.4 In passenger ships carrying more than 36 passengers, two emergency escape breathing devices, in addition to those required

in 2.3.4.3 above, are to be carried in each main vertical zone.

2.3.4.5 However, 2.3.4.3 and 2.3.4.4 do not apply to stairway enclosures which constitute individual main vertical zones and for the main vertical zones in the fore or aft end of a ship which do not contain spaces of categories (6), (7), (8) or (12) defined in Ch.3, 3.2.2.3.

2.4 Means of escape from machinery spaces

2.4.1 Means of escape on passenger ships

Means of escape from each machinery space in passenger ships are to comply with the following provisions.

2.4.1.1 *Escape from spaces below the bulkhead deck*

Where the space is below the bulkhead deck the two means of escape is to consist of either:

.1 two sets of steel ladders as widely separated as possible, leading to doors in the upper part of the space similarly separated and from which access is provided to the appropriate lifeboat and liferaft embarkation decks. One of these ladders are to be located within a protected enclosure that satisfies Ch.3, 3.2.2.3, category (2), or Ch.3, 3.2.2.4, category (4), as appropriate, from the lower part of the space it serves to a safe position outside the space. Self-closing fire doors of the same fire integrity standards are to be fitted in the enclosure. The ladder is to be fixed in such a way that heat is not transferred into the enclosure through non-insulated fixing points. The protected enclosure is to have minimum internal dimensions of at least 800 [mm] x 800 [mm], and is to have emergency lighting provisions; or

.2 one steel ladder leading to a door in the upper part of the space from which access is provided to the embarkation deck and additionally, in the lower part of the space and in a position well separated from the ladder referred to, a steel door capable of being operated from each side and which provides access to a safe escape route from the lower part of the space to the embarkation deck.

2.4.1.2 *Escape from spaces above the bulkhead deck*

Where the space is above the bulkhead deck, the two means of escape are to be as widely separated as possible and the doors leading from such means of escape are to be in a position from which access is provided to the appropriate lifeboat and liferaft embarkation decks. Where such means of escape require the use of ladders, these are to be of steel.

2.4.1.3 *Dispensation from two means of escape*

In a ship of less than 1,000 gross tonnage, one of the means of escape may be dispensed with due regard being paid to the width and disposition of the upper part of the space. In a ship of 1,000 gross tonnage and above, one means of escape from any such space, including a normally unattended auxiliary machinery space may be dispensed with, so long as either a door or a steel ladder provides a safe escape route to the embarkation deck, due regard being paid to the nature and location of the space and whether persons are normally employed in that space. In the steering gear space, a second means of escape is to be provided when the emergency steering position is located in that space unless there is direct access to the open deck.

IR2.4.1.3 The requirement of dispensing with second means of escape, in a ship of 1000 gross tonnage and above, as given in 2.4.1.3 applies only to auxiliary machinery spaces where persons are not normally employed.

2.4.1.4 *Escape from machinery control rooms*

Two means of escape are to be provided from a machinery control room located within a machinery space, at least one of which will provide continuous fire shelter to a safe position outside the machinery space.

IR2.4.1 The above clauses (2.4.1.1, 2.4.1.2 & 2.4.1.4) are to be interpreted as follows:

.1 A “safe position” can be any space, excluding lockers and storerooms irrespective of their area, cargo spaces and spaces where flammable liquids are stowed, but including special category spaces and ro-ro spaces, from which access is provided and maintained clear of obstacles to the embarkation decks (Cl. 2.4.1.1 and 2.4.1.4)

.2 Inclined ladders/stairways in machinery spaces being part of, or providing access to, escape routes but not located within a

protected enclosure are not to have an inclination greater than 60° and are not to be less than 600 [mm] in clear width. Such requirement need not be applied to ladders/stairways not forming part of an escape route, only provided for access to equipment or components, or similar areas, from one of the main platforms or deck levels within such spaces (Cl. 2.4.1).

.3 Machinery spaces may include working platforms and passageways, or intermediate decks at more than one deck level. In such case, the lower part of the space is to be regarded as the lowest deck level, platform or passageway within the space. At deck levels, other than the lowest one, where only one means of escape other than the protected enclosure is provided, self-closing fire doors are to be fitted in the protected enclosure at that deck level. Smaller working platforms in-between deck levels, or only for access to equipment or components, need not be provided with two means of escape (Cl. 2.4.1.1).

.4 A protected enclosure providing escape from machinery spaces to an open deck may be fitted with a hatch as means of egress from the enclosure to the open deck. The hatch is to have minimum internal dimensions of 800 [mm] x 800 [mm] (Cl. 2.4.1.1.1).

.5 Internal dimensions are to be interpreted as clear width, so that a passage having diameter of 800 mm is available throughout the vertical enclosure, as shown in Figure 2.4.1.4.1, clear of ship's structure, with insulation and equipment, if any. The ladder within the enclosure can be included in the internal dimensions of the enclosure. When protected enclosures include horizontal portions their clear width is not to be less than 600 [mm]. Figure 2.4.1.4.1 is given as example of some possible arrangements which may be in line with the above interpretation (Cl. 2.4.1.1.1).

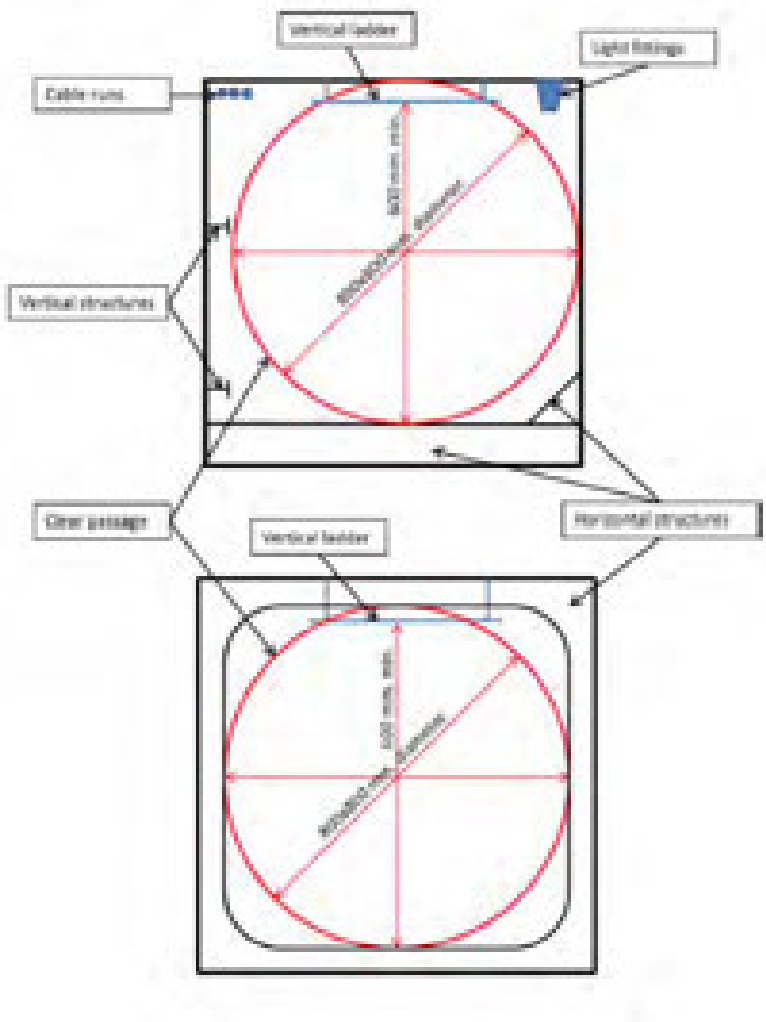

Fig.2.4.1.4.1 : Interpretation of Internal Dimensions – Possible Arrangements

2.4.1.5 *Inclined ladders and stairways*

All inclined ladders/stairways fitted to comply with 2.4.1.1 with open treads in machinery spaces being part of or providing access to escape routes but not located within a protected enclosure are to be made of steel. Such ladders/stairways are to be fitted with steel shields attached to their undersides, such as to provide escaping personnel protection against heat and flame from beneath.

2.4.1.6 *Escape from main workshops within machinery spaces*

Two means of escape are to be provided from the main workshop within a machinery space. At least one of these escape routes is to provide a continuous fire shelter to a safe position outside the machinery space.

2.4.2 Means of escape on cargo ships

Means of escape from each machinery space in cargo ships are to comply with the following provisions.

2.4.2.1 *Escape from machinery spaces of category A*

Except as provided in 2.4.2.2, two means of escape are to be provided from each machinery space of category A. In particular, one of the following provisions is to be complied with:

.1 two sets of steel ladders as widely separated as possible leading to doors in the upper part of the space similarly separated and from which access is provided to the open deck. One of these ladders is to be located within a protected enclosure that satisfies Ch.3, 3.2.3.3, category (4), from the lower part of the space it serves to a safe position outside the space. Self-closing fire doors of the same fire integrity standards are to be fitted in the enclosure. The ladder is to be fixed in such a way that heat is not transferred into the enclosure through non-insulated fixing points. The enclosure is to have minimum internal dimensions of at least 800 [mm] x 800 [mm], and is to have emergency lighting provisions; or

.2 one steel ladder leading to a door in the upper part of the space from which access is provided to the open deck and, additionally, in the lower part of the space and in a position well separated from the ladder referred to, a steel door capable of being operated from each side and which provides access to a safe escape route from the lower part of the space to the open deck.

2.4.2.2 *Dispensation from two means of escape*

In a ship of less than 1,000 gross tonnage, one of the means of escape required under 2.4.2.1 may be dispensed with, due regard being paid to the dimension and disposition of the upper part of the space. In addition, the means of escape from machinery spaces of category A need not comply with the requirement for an enclosed fire shelter listed in 2.4.2.1.1. In the steering gear space, a second means of escape is to be provided when the emergency steering position is located in that space unless there is direct access to the open deck.

IR2.4.2.2.1 Steering gear spaces which do not contain the emergency steering position need only have one means of escape.

.2 Steering gear spaces containing the emergency steering position can have one means of escape provided it leads directly onto the open deck. Otherwise, two means of escape are to be provided but they do not need to lead directly onto the open deck.

.3 Escape routes that pass only through stairways and/or corridors are considered as providing a “direct access to the open deck”, provided that the escape routes from the steering gear spaces have fire integrity protection equivalent to:

- steering gear spaces; or
- stairways/corridors,

whichever is more stringent.

2.4.2.3 *Escape from machinery spaces other than those of category A*

From machinery spaces other than those of category A, two escape routes are to be provided except that a single escape route may be accepted for spaces that are entered only occasionally, and for spaces where the maximum travel distance to the door is 5 [m] or less.

IR2.4.2 The above clauses (2.4.2.1 & 2.4.2.3) are to be interpreted as follows:

.1 A “safe position” can be any space, excluding cargo spaces, lockers and storerooms irrespective of their area, cargo pump-rooms and spaces where flammable liquids are stowed, but including vehicle and ro-ro spaces, from which access is provided and maintained clear of obstacles to the open deck (Cl. 2.4.2.1.1).

.2 Inclined ladders/stairways in machinery spaces being part of, or providing access to, escape routes, but not located within a protected enclosure are not to have an inclination greater than 60° and are not to be less than 600 [mm] in clear width. Such requirement need not be applied to ladders/stairways not forming part of an escape route, only provided for access to equipment or components, or similar areas, from one of the main platforms or deck levels within such spaces (Cl. 2.4.2.1).

.3 Machinery spaces of category A may include working platforms and passageways, or intermediate decks at more than one deck level. In such case, the lower part of the space is to be regarded as the lowest deck level, platform or passageway within the space.
At deck levels, other than the lowest one, where only one means of escape other than the protected enclosure is provided, self-closing fire doors are to be fitted in the protected enclosure at that deck level. Smaller working platforms in-between deck levels, or only for access to equipment or components, need not be provided with two means of escape (Cl. 2.4.2.1).

.4 A protected enclosure providing escape from machinery spaces of category A to an open deck may be fitted with a hatch as means of egress from the enclosure to the open deck. The hatch is to have minimum internal dimensions of 800 [mm] x 800 [mm] (Cl. 2.4.2.1.1).

.5 Internal dimensions are to be interpreted as clear width, so that a passage having diameter of 800 [mm] is available throughout the vertical enclosure, as shown in Fig. 2.4.1.4.1, clear of ship’s structure, with insulation and equipment, if any. The ladder within the enclosure can be included in the internal dimensions of the enclosure. When protected enclosures include horizontal portions their clear width is not to be less than 600 [mm]. Figure 2.4.1.4.1 is given as an example of some possible arrangements which may be in line with the above interpretation (Cl. 2.4.2.1.1).

.6 In machinery spaces other than those of category A, which are not entered only occasionally, the travel distance is to be measured from any point normally accessible to the crew, taking into account machinery and equipment within the space (Cl. 2.4.2.3).

2.4.2.4 Inclined ladders and stairways

All inclined ladders/stairways fitted to comply with 2.4.2.1 with open treads in machinery spaces being part of or providing access to escape routes but not located within a protected enclosure are to be made of steel. Such ladders/stairways are to be fitted with steel shields attached to their undersides, such as to provide escaping personnel protection against heat and flame from beneath.

2.4.2.5 Escape from machinery control rooms in machinery spaces of category "A"

Two means of escape are to be provided from the machinery control room located within a machinery space. At least one of these escape routes is to provide a continuous fire shelter to a safe position outside the machinery space.

2.4.2.6 Escape from main workshops in machinery spaces of category "A"

Two means of escape are to be provided from the main workshop within a machinery space. At least one of these escape routes is to provide a continuous fire shelter to a safe position outside the machinery space.

2.4.3 Emergency escape breathing devices

2.4.3.1 On all ships, within the machinery spaces, emergency escape breathing devices are to be situated ready for use at easily

visible places, which can be reached quickly and easily at any time in the event of fire. The location of emergency escape breathing devices is to take into account the layout of the machinery space and the number of persons normally working in the spaces. (Refer to the Guidelines for the performance, location, use and care of emergency escape breathing devices (MSC/Circ.849)).

2.4.3.2 The number and location of these devices are to be indicated in the fire control plan required in Ch.5, 2.2.4.

2.4.3.3 Emergency escape breathing devices are to comply with the Fire Safety Systems Code (Chapter 8).

2.5 Means of escape on passenger ships from special category and open ro-ro spaces to which any passengers carried can have access

2.5.1 In special category and open ro-ro spaces to which any passengers carried can have access, the number and locations of the means of escape both below and above the bulkhead deck are to be to the satisfaction of IRS and, in general, the safety of access to the embarkation deck is to be at least equivalent to that provided for under 2.3.2.1.1, 2.3.2.2, 2.3.2.4.1 and 2.3.2.4.2. Such spaces are to be provided with designated walkways to the means of escape with a breadth of at least 600 mm. The parking arrangements for the vehicles are to maintain the walkways clear at all times.

2.5.2 One of the escape routes from the machinery spaces where the crew is normally employed is to avoid direct access to any special category space.

2.6 Means of escape from ro-ro spaces

At least two means of escape are to be provided in ro-ro spaces where the crew are normally employed. The escape routes are to provide a safe escape to the lifeboat and liferaft embarkation decks and are to be located at the fore and aft ends of the space.

2.7 Additional requirements for ro-ro passenger ships

2.7.1 *General*

2.7.1.1 Escape routes are to be provided from every normally occupied space on the ship to an assembly station. These escape routes are to be arranged so as to provide the most direct route possible to the assembly station*, and are to be marked with symbols approved by the National Statutory Authority**.

(* Refer to the Indication of the "assembly stations" in passenger ships (MSC/Circ.777)).

(** Refer to the Symbols related to life-saving appliances and arrangements adopted by IMO by Resolution A.760(18)).

2.7.1.2 The escape route from cabins to stairway enclosures are to be as direct as possible, with a minimum number of changes in direction. It is not to be necessary to cross from one side of the ship to the other to reach an escape route. It is not to be necessary to climb more than two decks up or down in order to reach an assembly station or open deck from any passenger space.

2.7.1.3 External routes are to be provided from open decks, as referred to in 2.7.1.2, to the survival craft embarkation stations.

2.7.1.4 Where enclosed spaces adjoin an open deck, openings from the enclosed space to the open deck are, where practicable to be capable of being used as an emergency exit.

2.7.1.5 Escape routes are not to be obstructed by furniture and other obstructions. With the exception of tables and chairs which may be cleared to provide open space, cabinets and other heavy furnishings in public spaces and along escape routes are to be secured in place to prevent shifting if the ship rolls or lists. Floor coverings are also to be secured in place. When the ship is underway, escape routes are to be kept clear of obstructions such as cleaning carts, bedding, luggage and boxes of goods.

2.7.2 Instruction for safe escape

2.7.2.1 Decks are to be sequentially numbered, starting with "1" at the tank top or lowest deck. These numbers are to be prominently displayed at stair landings and lift lobbies. Decks may also be named, but the deck number is always to be displayed with the name.

2.7.2.2 Simple "mimic" plans showing the "you are here" position and escape routes marked by arrows, are to be prominently displayed on the inside of each cabin door and in public

spaces. The plan is to show the directions of escape, and is to be properly oriented in relation to its position on the ship.

2.7.3 Strength of handrails and corridors

2.7.3.1 Handrails or other handholds are to be provided in all corridors along the entire escape route, so that a firm handhold is available at every step of the way, where possible, to the assembly stations and embarkation stations. Such handrails are to be provided on both sides of longitudinal corridors more than 1.8 [m] in width and transverse corridors more than 1 [m] in width. Particular attention is to be paid to the need to be able to cross lobbies, atriums and other large open spaces along escape routes. Handrails and other handholds are to be of such strength as to withstand a distributed horizontal load of 750 [N/m] applied in the direction of the centre of the corridor or space, and a distributed vertical load of 750 [N/m] applied in the downward direction. The two loads need not be applied simultaneously.

2.7.3.2 The lowest 0.5 [m] of bulkheads and other partitions forming vertical divisions along escape routes are to be able to sustain a load of 750 [N/m] to allow them to be used as walking surfaces from the side of the escape route with the ship at large angles of heel.

End of Chapter

Chapter 5

Operational Requirements

Contents

Section

1 *Operational Readiness and Maintenance*

2 *Instructions, Onboard Training and Drills*

3 *Operations*

Section 1

Operational Readiness and Maintenance

1.1 Purpose

The purpose of this section is to maintain and monitor the effectiveness of the fire safety measures the ship is provided with. For this purpose, the following functional requirements are to be met:

.1 fire protection systems and fire-fighting systems and appliances are to be maintained ready for use; and

.2 fire protection systems and fire-fighting systems and appliances are to be properly tested and inspected.

1.2 General requirements

At all times while the ship is in service, the requirements of 1.1.1 are to be complied with. A ship is not in service when:

.1 it is in for repairs or lay-up (either at anchor or in port) or in dry-dock;

.2 it is declared not in service by the owner or the owner's representative; and

.3 in the case of passenger ships, there are no passengers on board. However, in this case, the vessel is to satisfy the requirements of 1.1.1 as for a cargo ship.

1.2.1 Operational readiness

1.2.1.1 The following fire protection systems are to be kept in good order so as to ensure their required performance if a fire occurs:

.1 structural fire protection including fire resisting divisions, and protection of openings and penetrations in these divisions;

.2 fire detection and fire alarm systems; and

.3 means of escape systems and appliances.

1.2.1.2 Fire-fighting systems and appliances are to be kept in good working order and readily available for immediate use. Portable extinguishers which have been discharged are to be immediately recharged or replaced with an equivalent unit.

1.2.2 Maintenance, testing and inspections

1.2.2.1 Maintenance, testing and inspections are to be carried out based on the requirements approved by the National Statutory Authority and in a manner having due regard to ensuring the reliability of fire-fighting systems and appliances. (Refer to the Guidelines on maintenance and inspection of fire protection systems and appliances (MSC/Circ.850)).

1.2.2.2 The maintenance plan is to be kept on board the ship and is to be available for inspection whenever required.

1.2.2.3 The maintenance plan is to include at least the following fire protection systems and fire-fighting systems and appliances, where installed:

.1 fire mains, fire pumps and hydrants including hoses, nozzles and international shore connections;

.2 fixed fire detection and fire alarm systems;

.3 fixed fire-extinguishing systems and other fire extinguishing appliances;

.4 automatic sprinkler, fire detection and fire alarm systems;

.5 ventilation systems including fire and smoke dampers, fans and their controls;

.6 emergency shut down of fuel supply;

.7 fire doors including their controls;

.8 general emergency alarm systems;

.9 emergency escape breathing devices;

.10 portable fire extinguishers including spare charges; and

.11 fire-fighter's outfits.

1.2.2.4 The maintenance programme may be computer-based.

1.3 Additional requirements for passenger ships

In addition to the fire protection systems and appliances listed in 1.2.2.3, ships carrying more than 36 passengers are to develop a maintenance plan for low-location lighting and public address systems.

1.4 Additional requirements for tankers

In addition to the fire protection systems and appliances listed in 1.2.2.3, tankers are to develop a maintenance plan for:

.1 inert gas systems;

.2 deck foam systems;

.3 fire safety arrangements in cargo pump rooms; and

.4 flammable gas detectors.

Section 2

Instructions, Onboard Training and Drills

2.1 Purpose

The purpose of this regulation is to mitigate the consequences of fire by means of proper instructions for training and drills of persons onboard in correct procedures under emergency conditions. For this purpose, the crew are to have the necessary knowledge and skills to handle fire emergency cases, including passenger care.

2.2 General requirements

2.2.1 Instructions, duties and organization

2.2.1.1 Crew members are to receive instruction on fire safety onboard the ship.

2.2.1.2 Crew members are to receive instructions on their assigned duties.

2.2.1.3 Parties responsible for fire-extinguishing are to be organized. These parties are to have the capability to complete their duties at all times while the ship is in service.

2.2.2 Onboard training and drills

2.2.2.1 Crew members are to be trained to be familiar with the arrangements of the ship as well as the location and operation of any fire-fighting systems and appliances that they may be called upon to use.

2.2.2.2 Training in the use of the emergency escape breathing devices is to be considered as part of on board training.

2.2.2.3 Performance of crew members assigned fire-fighting duties is to be

periodically evaluated by conducting onboard training and drills to identify areas in need of improvement, to ensure competency in fire-fighting skills is maintained, and to ensure the operational readiness of the fire-fighting organization.

2.2.2.4 Onboard training in the use of the ship's fire-extinguishing systems and appliances is to be planned and conducted in accordance with provisions of SOLAS Reg. III/19.4.1.

2.2.2.5 Fire drills are to be conducted and recorded in accordance with the provisions of SOLAS Reg. III/19.3 and III/19.5.

2.2.2.6 An onboard means of recharging breathing apparatus cylinders used during drills shall be provided or a suitable number of spare cylinders shall be carried on board to replace those used.

IR2.2.2.6 "A suitable number of spare cylinders" to be carried onboard to replace those used for fire drills shall be at least one 'set of cylinders' for each mandatory breathing apparatus unless additional spare cylinders are required by the shipboard safety management system (SMS). 'Set of cylinders' means the number of cylinders which are required to operate the breathing apparatus. No additional cylinders are required for fire drills for breathing apparatus sets required by SOLAS Reg. II-2/19 (Pt. 6, Ch. 7), IMSBC Code, the IBC Code (Pt. 5, Ch. 3) or IGC Code (Pt. 5, Ch. 4).

2.2.3 Training manuals

2.2.3.1 A training manual is to be provided in each crew mess room and recreation room or in each crew cabin.

2.2.3.2 The training manual is to be written in the working language of the ship.

2.2.3.3 The training manual, which may comprise several volumes, is to contain the instructions and information required in 2.2.3.4 in easily understood terms and illustrated wherever possible. Any part of such information may be provided in the form of audio-visual aides in lieu of the manual.

2.2.3.4 The training manual is to explain the following in detail:

.1 general fire safety practice and precautions related to the dangers of smoking, electrical hazards, flammable liquids and similar common shipboard hazards;

.2 general instructions on fire-fighting activities and fire-fighting procedures including procedures for notification of a fire and use of manually operated call points;

.3 meanings of the ship's alarms;

.4 operation and use of fire-fighting systems and appliances;

.5 operation and use of fire doors;

.6 operation and use of fire and smoke dampers; and

.7 escape systems and appliances.

2.2.4 Fire control plans

(Refer to the Graphical symbols for fire control plans, adopted by IMO by resolution A.952(23). Resolution A.1116 (30) is to be used in combination with resolution A.952 (23) for the preparation of the shipboard fire control plans).

2.2.4.1 In all ships, fire control plans as given below are to be approved and permanently displayed for the guidance of ship's officers.

A general arrangement plan is to show clearly for each deck:

a) the control stations;

b) various fire sections enclosed by 'A' and 'B' class divisions together with particulars of fire detection and alarm systems;

c) the sprinkler installation;

d) the fire extinguishing appliances;

e) means of access to different compartments, etc.;

f) the ventilation system including particulars of the fan control positions, position of dampers and identification numbers of the ventilating fans serving each section.

Alternatively, the above mentioned details may be set out in a booklet, a copy of which is to be supplied to each officer, and at least one copy is to be available at all times on board in an accessible position.

Plans and booklets are to be kept up to date, any alterations being recorded thereon as soon as practicable. Description in such plans and booklets is to be in English and official language of the flag state.

2.2.4.2 A duplicate set of fire control plans or a booklet containing such plans is to be permanently stored in a prominently marked weathertight enclosure outside the deckhouse for the assistance of shore-side fire-fighting personnel. (Refer to the Guidance concerning the location of fire control plans for assistance of shoreside fire-fighting personnel (MSC/ Circ.451)).

2.3 Additional requirements for passenger ships

2.3.1 Fire drills

In addition to the requirement of 2.2.2.3, fire drills are to be conducted in accordance with the provision of SOLAS Reg.III/30 having due regard to notification of passengers and movement of passengers to assembly stations and embarkation decks.

2.3.2 Fire control plans

In ships carrying more than 36 passengers, plans and booklets required by this regulation is to provide information regarding fire protection, fire detection and fire extinction based on the requirements of the National Statutory Authority. (Refer to the Guidelines on the information to be provided with fire control plans and booklets required by SOLAS regulations II-2/20 and 41-2, adopted by the Organization by resolution A.756(18)).

Section 3

Operations

3.1 Purpose

The purpose of this regulation is to provide information and instructions for proper ship and cargo handling operations in relation to fire safety. For this purpose, the following functional requirements are to be met:

.1 fire safety operational booklets is to be provided on board; and

.2 flammable vapour releases from cargo tank venting is to be controlled.

3.2 Fire safety operational booklets

3.2.1 The required fire safety operational booklet is to contain the necessary information and instructions for the safe operation of the ship and cargo handling operations in relation to fire safety. The booklet is to include information concerning the crew's responsibilities for the general fire safety of the ship while loading and discharging cargo and while underway. Necessary fire safety precautions for handling general cargoes are to be explained. For ships carrying dangerous goods and flammable bulk cargoes, the fire safety operational booklet are also to provide reference to the pertinent fire-fighting and emergency cargo handling instructions contained in the International Maritime Solid Bulk Cargoes (IMSBC) Code the International Bulk Chemical Code, the International Gas Carrier Code and the International Maritime Dangerous Goods Code, as appropriate.

3.2.2 The fire safety operational booklet is to be provided in each crew mess room and recreation room or in each crew cabin.

3.2.3 The fire safety operational booklet is to be written in the working language of the ship.

3.2.4 The fire safety operational booklet may be combined with the training manuals required in 2.2.3.

3.3 Additional requirements for tankers

3.3.1 General

The fire safety operational booklet referred to in 3.2 is to include provisions for preventing fire spread to the cargo area due to ignition of flammable vapours and include procedures of

cargo tank gas-purging and/or gas-freeing taking into account the provisions in 3.3.2.

3.3.2 Procedures for cargo tank purging and/or gas-freeing

3.3.2.1 When the ship is provided with an inert gas system, the cargo tanks are first to be purged in accordance with the provisions of Ch.2, 1.5.6 until the concentration of hydrocarbon vapours in the cargo tanks has been reduced to less than 2% by volume. Thereafter, gas-freeing may take place at the cargo tank deck level.

3.3.2.2 When the ship is not provided with an inert gas system, the operation is to be such that the flammable vapour is discharged initially through:

.1 the vent outlets as specified in Ch.2, 1.5.3.4;

.2 outlets at least 2 [m] above the cargo tank deck level with a vertical efflux velocity of at least 30 [m/s] maintained during the gas-freeing operation; or

.3 outlets at least 2 [m] above the cargo tank deck level with a vertical efflux velocity of at least 20 [m/s] and which are protected by suitable devices to prevent the passage of flame.

The above outlets are to be located not less than 10 [m] measured horizontally from the nearest air intakes and openings to enclosed spaces containing a source of ignition and from deck machinery, which may include anchor windlass and chain locker openings, and equipment which may constitute an ignition hazard.

When the flammable vapour concentration at the outlet has been reduced to 30% of the lower flammable limit, gas-freeing may be continued at cargo tank deck level.

3.3.3 Operation of inert gas system

3.3.3.1 The inert gas system for tankers required in accordance with Pt6, Ch2, 1.5.5.1 is to be so operated as to render and maintain the atmosphere of the cargo tanks non-flammable, except when such tanks are required to be gas-free.

3.3.3.2 Notwithstanding the above, for chemical tankers, the application of inert gas, may take place after the cargo tank has been loaded, but before commencement of unloading and shall continue to be applied until that cargo tank has been purged of all flammable vapours before gas-freeing. Only nitrogen is acceptable as inert gas under this provision.

3.3.3.3 Notwithstanding Pt 6, Ch 1, 1.3.2, the provisions of this paragraph shall only apply to tankers constructed on or after 1 January 2016. If the oxygen content of the inert gas exceeds 5% by volume, immediate action is to be taken to improve the gas quality. Unless the quality of the gas improves, all operations in those cargo tanks to which inert gas is being supplied are to be suspended so as to avoid air being drawn into the cargo tanks, the gas regulating valve, if fitted, is to be closed and the off-specification gas is to be vented to atmosphere.

3.3.3.4 In the event that the inert gas system is unable to meet the requirement in paragraph 3.3.3.1 and it has been assessed that it is impractical to effect a repair, then cargo discharge and cleaning of those cargo tanks requiring inerting shall only be resumed when suitable emergency procedures have been followed, taking into account guidelines developed by IMO*.

(*Refer to the *Clarification of inert gas system requirements under the Convention* (MSC/Circ.485) and to the *Revised Guidelines for inert gas systems* (MSC/Circ.353), as amended by MSC/Circ.387.)

End of Chapter

Chapter 6

Alternative Design and Arrangements

Contents

Section	
1	*Alternative Design and Arrangements*

Section 1

Alternative Design and Arrangements

1.1 Purpose

The purpose of this section is to provide a methodology for alternative design and arrangements for fire safety.

1.2 General

1.2.1 Fire safety design and arrangements may deviate from the prescriptive requirements set out in Chapters 2, 3, 4, 5 or 7, provided that the design and arrangements meet the fire safety objectives and the functional requirements.

1.2.2 When fire safety design or arrangements deviate from the prescriptive requirements of this part, engineering analysis, evaluation and approval of the alternative design and arrangements are to be carried out in accordance with this section.
Note: Interim guidelines for use of fibre reinforced plastic (FRP) elements within ship structures (*MSC.1/ Circ. 1574*) may be referred whilst approving FRP elements as part of alternative design and arrangements of fire safety.

1.3 Engineering analysis

The engineering analysis is to be prepared and submitted to IRS (Refer to MSC/Circ.1002 as corrected by Corr.1, Corr.2, Corr.3 and amended by MSC.1/ Circ. 1552) and is to include, as a minimum, the following elements:

.1 determination of the ship type and space(s) concerned;

.2 identification of prescriptive requirement(s) with which the ship or the space(s) will not comply;

.3 identification of the fire and explosion hazards of the ship or the space(s) concerned;

.3.1 identification of the possible ignition sources;

.3.2 identification of the fire growth potential of each space concerned;

.3.3 identification of the smoke and toxic effluent generation potential for each space concerned;

.3.4 identification of the potential for the spread of fire, smoke or of toxic effluents from the space(s) concerned to other spaces;

.4 determination of the required fire safety performance criteria for the ships or the space(s) concerned addressed by the prescriptive requirement(s);

.4.1 performance criteria is to be based on the fire safety objectives and on the functional requirements of this part;

.4.2 performance criteria are to provide a degree of safety not less than that achieved by using the prescriptive requirements; and

.4.3 performance criteria is to be quantifiable and measurable;

.5 detailed description of the alternative design and arrangements, including a list of the assumptions used in the design and any proposed operational restrictions or conditions; and

.6 technical justification demonstrating that the alternative design and arrangements meet the required fire safety performance criteria.

1.4 Evaluation of the alternative design and arrangements

1.4.1 The engineering analysis required in 1.3 is to be evaluated and approved by IRS taking into account the requirements of the National Statutory Authority (Refer to MSC/Circ.1002 as corrected by Corr.1, Corr.2, Corr.3 and amended by MSC.1/ Circ. 1552).

1.4.2 A copy of the documentation, as approved by IRS, indicating that the alternative design and arrangements comply with this chapter is to be carried onboard the ship.

1.5 Exchange of information

Note : The Administration is to communicate to IMO pertinent information concerning alternative design and arrangements approved by them for circulation to all contracting governments.

1.6 Re-evaluation due to change of conditions

If the assumptions, and operational restrictions that were stipulated in the alternative design and arrangements are changed, the engineering analysis is to be carried out under the changed condition and is to be approved by IRS.

End of Chapter

Chapter 7

Special Requirements

Contents

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1	*Helicopter Facilities*
2	*Carriage of Dangerous Goods*
3	*Protection of Vehicle, Special Category and Ro-ro Spaces*
4	*Requirements for vehicle carriers carrying as cargo motor vehicles with compressed hydrogen or natural gas in their tanks for their own propulsion*
5	*Casualty Threshold, Safe Return to Port and Safe Areas*
6	*Design Criteria for Systems to Remain Operational after a Fire Casualty*
7	*Safety Centre on Passenger Ships*

Section 1

Helicopter Facilities

1.1 Purpose

The purpose of this section is to provide additional measures in order to address the fire safety objectives of this part for ships fitted with special facilities for helicopters. For this purpose, the following functional requirements are to be met:

.1 helideck structure is to be adequate to protect the ship from the fire hazards associated with helicopter operations;

.2 fire fighting appliances are to be provided to adequately protect the ship from the fire hazards associated with helicopter operations;

.3 refuelling and hangar facilities and operations are to provide the necessary measures to protect the ship from the fire hazards associated with helicopter operations; and

.4 operation manuals and training are to be provided.

1.2 Application

1.2.1 In addition to complying with the requirements of Chapters 2, 3, 4 and 5, as appropriate, ships equipped with helidecks are to comply with the requirements of this section.

1.2.2 Where helicopters land or conduct winching operations on an occasional or emergency basis on ships without helidecks, fire-fighting equipment fitted in accordance with the requirements in Ch.3 may be used. This equipment is to be made readily available in close proximity to the landing or winching areas during helicopter operations.

1.2.3 Notwithstanding the requirements of paragraph 1.2.2 above, ships constructed on or after 1 January 2020, having a helicopter landing area, are to be provided with foam firefighting appliances which comply with the relevant provisions of Ch.8, Sec.17.

1.2.4 Notwithstanding the requirements of 1.2.2 or 1.2.3 above, ro-ro passenger ships without helidecks are to comply with SOLAS Reg.III/28.

1.3 Structure

1.3.1 Construction of steel or other equivalent material

In general, the construction of the helidecks is to be of steel or other equivalent materials. If the helideck forms the deckhead of a deckhouse or superstructure, it is to be insulated to "A-60" class standard.

1.3.2 Construction of aluminium or other low melting point metals

1.3.2.1 If helideck platform of such construction located above the ship's deckhouse or similar structure:

- The deckhouse top and bulkheads under the platform are to have no openings; and
- Windows under the platform are to be provided with steel shutters.

1.3.2.2 After each fire on the platform or in close proximity its suitability for further use will be specially considered in each case.

1.4 Means of escape

A helideck is to be provided with both a main and an emergency means of escape and access for fire fighting and rescue personnel. These are to be located as far apart from each other as is practicable and preferably on opposite sides of the helideck.

1.5 Fire-fighting appliances

1.5.1 In close proximity to the helideck, the following fire-fighting appliances are to be provided and stored near the means of access to that helideck (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable fire extinguishers on board ships'):

.1 at least two dry powder extinguishers having a total capacity of not less than 45 [kg];

.2 carbon dioxide extinguishers of a total capacity of not less than 18 [kg] or equivalent;

.3 a suitable foam application system consisting of monitors or foam making branch pipes capable of delivering foam to all parts of the helideck in all weather conditions in which helicopters can operate. The system is to be capable of delivering a discharge rate as required in Table 1.1 for at least five minutes;

Table 1.1 : Foam discharge rates

Category	Helicopter overall length	Discharge rate foam solution [l/min.]
H1	up to but not including 15 [m]	250
H2	from 15 [m] up to but not including 24 [m]	500
H3	from 24 [m] up to but not including 35 [m]	800

.4 the principal agent is to be suitable for use with salt water and conform to performance standards not inferior to those acceptable to the National Statutory Authority. (Refer to the International Civil Aviation Organization Airport Services Manual, Part 1 - Rescue and Fire fighting, Chapter 8 - Extinguishing Agent Characteristics, Paragraph 8.1.5 - Foam Specifications Table 8-1, Level 'B'));

.5 at least two nozzles of an approved dual-purpose type (jet/spray) and hoses sufficient to reach any part of the helideck;

.6 in lieu of the requirements of paragraphs .3 through .5, on ships constructed on or after 1 January 2020 having a helideck, foam firefighting appliances which comply with the provisions of the Chapter 8.

.7 in addition to the requirements of Ch.3, 4.10, two sets of fire-fighter's outfits; and

.8 at least the following equipment are to be stored in a manner that provides for immediate use and protection from the elements:

- adjustable wrench;
- blanket, fire resistant;
- cutters, bolt 60 [cm];
- hook, grab or salving;
- hacksaw, heavy duty complete with 6 spare blades;
- ladder;
- lift line 5 [mm] diameter x 15 [m] in length;

- pliers, side cutting;
- set of assorted screwdrivers; and
- harness knife complete with sheath.

1.6 Drainage facilities

Drainage facilities in way of helidecks are to be constructed of steel and are to lead directly overboard independent of any other system and is to be designed so that drainage does not fall onto any part of the ship.

1.7 Helicopter refuelling and hangar facilities

Where the ship has helicopter refuelling and hangar facilities, the following requirements are to be complied with:

.1 a designated area is to be provided for the storage of fuel tanks which is to be:

.1.1 as remote as is practicable from accommodation spaces, escape routes and embarkation stations; and

.1.2 isolated from areas containing a source of vapour ignition;

.2 the fuel storage area is to be provided with arrangements whereby fuel spillage may be collected and drained to a safe location;

.3 tanks and associated equipment are to be protected against physical damage and from a fire in an adjacent space or area;

.4 where portable fuel storage tanks are used, special attention is to be given to:

.4.1 design of the tank for its intended purpose;

.4.2 mounting and securing arrangements;

.4.3 electric bonding; and

.4.4 inspection procedures;

.5 storage tank fuel pumps are to be provided with means which permit shutdown from a safe remote location in the event of a fire. Where a gravity fuelling system is installed, equivalent closing arrangements are to be provided to isolate the fuel source;

.6 the fuel pumping unit is to be connected to one tank at a time. The piping between the tank and the pumping unit is to be of steel or equivalent material, as short as possible, and protected against damage;

.7 electrical fuel pumping units and associated control equipment is to be of a type suitable for the location and potential hazards;

.8 fuel pumping units are to incorporate a device which will prevent over-pressurization of the delivery or filling hose;

.9 equipment used in refuelling operations is to be electrically bonded;

.10 "NO SMOKING" signs are to be displayed at appropriate locations;

.11 hangar, refuelling and maintenance facilities are to be treated as category ‘A’ machinery spaces with regard to structural fire protection, fixed fire-extinguishing and detection system requirements;

.12 enclosed hangar facilities or enclosed spaces containing refuelling installations are to be provided with mechanical ventilation, as required by 3.3 for closed ro-ro spaces of cargo ships. Ventilation fans are to be of non-sparking type; and

.13 electric equipment and wiring in enclosed hangar or enclosed spaces containing refuelling installations are to comply with 3.3.2, 3.3.3 and 3.3.4.

1.8 Operations manual and fire-fighting service

1.8.1 Each helicopter facility is to have an operations manual, including a description and a checklist of safety precautions, procedures and equipment requirements. This manual may be part of the ship’s emergency response procedures.

1.8.2 The procedures and precautions to be followed during refuelling operations are to be in accordance with recognized safe practices and contained in the operations manual.

1.8.3 Fire-fighting personnel consisting of at least two persons trained for rescue and fire-fighting duties and fire-fighting equipment are to be immediately available at all times when helicopter operations are expected.

1.8.4 Fire-fighting personnel are to be present during refuelling operations. However, the fire-

fighting personnel are not to be involved with refuelling activities.

1.8.5 On-board refresher training is to be carried out and additional supplies of fire-fighting media is to be provided for training and testing of the equipment.

Section 2

Carriage of Dangerous Goods*

2.1 Purpose

The purpose of this section is to provide additional safety measures in order to address the fire safety objectives of this part for ships carrying dangerous goods. For this purpose, the following functional requirements are to be met:

.1 fire protection systems are to be provided to protect the ship from the added fire hazards associated with carriage of dangerous goods;

.2 dangerous goods are to be adequately separated from ignition sources; and

.3 appropriate personnel protective equipment are to be provided for the hazards associated with the carriage of dangerous goods.

2.2 General requirements

2.2.1 In addition to complying with the requirements of Chapters 2, 3, 4, 5 and sections 1 and 3**, as appropriate, ship types and cargo spaces, referred to in 2.2.2, intended for the carriage of dangerous goods are to comply with the requirements of this section, as appropriate, except when carrying dangerous goods in limited quantities*** and excepted quantities**** unless such requirements have already been met by compliance with the requirements elsewhere in this chapter. The types of ships and modes of carriage of dangerous goods are referred to in 2.2.2 and in Table 2.1. Cargo ships of less than 500 gross tonnage are to comply with this section, but Administrations may reduce the requirements and such reduced requirements are to be recorded in the document of compliance referred to in 2.4.

* Refer also to the Interim guidelines for open top containerships (MSC/Circ.608/Rev.1).

** Refer to part 7 of the International Maritime Dangerous Goods Code.

*** Refer to chapter 3.4 of the International Maritime Dangerous Goods Code.

**** Refer to chapter 3.5 of the IMDG Code.

2.2.2 The following ship types and cargo spaces are to govern the application of Table 2.1 and Table 2.2:

.1 ships and cargo spaces not specifically designed for the carriage of freight containers, but intended for the carriage of dangerous goods in packaged form including goods in freight containers and portable tanks;

.2 purpose-built container ships and cargo spaces intended for the carriage of dangerous goods in freight containers and portable tanks;

IR.2 A purpose built container space is a cargo space fitted with cell guides for stowage securing of containers.

.3 ro-ro ships and ro-ro spaces intended for the carriage of dangerous goods;

IR.3 Ro-ro spaces include special category spaces (refer Ch.1, 3.46) and vehicle spaces (refer Ch.1, 3.49).

.4 ships and cargo spaces intended for the carriage of solid dangerous goods in bulk; and

.5 ships and cargo spaces intended for carriage of dangerous goods other than liquids and gases in bulk in shipborne barges.

2.3 Special requirements

Unless otherwise specified, the following requirements (2.3.1 to 2.3.10) are to govern the application of Table 2.1, Table 2.2 and Table 2.3 to both "on-deck" and "under-deck" stowage of dangerous goods where the numbers of the following paragraphs are indicated in the first column of the tables.

2.3.1 Water supplies

2.3.1.1 Arrangements are to be made to ensure immediate availability of a supply of water from the fire main at the required pressure either by permanent pressurization or by suitably placed remote arrangements for the fire pumps.

2.3.1.2 The quantity of water delivered is to be capable of supplying four nozzles of a size and at pressures as specified in Ch.3, 4.2, capable of being trained on any part of the cargo space when empty. (Also see Ch.3, 4.2.2.4.1, IR.2.1) This amount of water may be applied by equivalent means to the satisfaction of IRS.

IR 2.3.1.2 The number and position of hydrants are to be such that at least two of the required four jets of water, when supplied by single lengths of hose, may reach any part of the cargo space when empty; and all four jets of water, each supplied by single lengths of hose may reach any part of ro-ro cargo spaces.

2.3.1.3 Means of effectively cooling the designated underdeck cargo space by at least 5 l/min per square metre of the horizontal area of cargo spaces, either by a fixed arrangement of spraying nozzles or flooding the cargo space with water are to be provided. Hoses may be used for this purpose in small cargo spaces and in small areas of larger cargo spaces at the discretion of IRS. However, the drainage and pumping arrangements are to be such as to prevent the build-up of free surfaces. The drainage system is to be sized to remove no less than 125% of the combined capacity of both the water spraying system pumps and the required number of fire hose nozzles. The drainage system valves are to be operable from outside the protected space at a position in the vicinity of the extinguishing system controls. Bilge wells are to be of sufficient holding capacity and are to be arranged at the side shell of the ship at a distance from each other of not more than 40 [m] in each watertight compartment. If this is not possible, the adverse effect upon stability of the added weight and free surface of water is to be taken into account to the extent deemed necessary during appraisal of stability information. (Refer to the Recommendation on fixed fire-extinguishing systems for special cargo spaces adopted by IMO by resolution A.123(V)). (Also see Ch.3, 4.2.2.4.1, IR .2.1).

2.3.1.4 Provision to flood a designated under-deck cargo space with suitable specified media may be substituted for the requirements in 3.1.3.

IR2.3.1.4 A high expansion foam system complying with Fire Safety Systems Code (Chapter 8) is acceptable except if cargoes dangerously react with water (see the IMDG Code, paragraph 16.2.2 of Section 16).

2.3.1.5 The total required capacity of the water supply is to satisfy 2.3.1.2 and 2.3.1.3, if applicable, simultaneously calculated for the largest designated cargo space. The capacity requirements of 2.3.1.2 are to be met by the total capacity of the main fire pump(s) not including the capacity of the emergency fire pump, if fitted. If a drencher system is used to satisfy 2.3.1.3, the drencher pump is also be taken into account in this total capacity calculation. (Also see Ch.3, 4.2.2.4.1, IR .2.1).
.
IR2.3.1.5 a) The water spray system required by paragraphs 9.2, 9.3 and 9.4 of MSC/Circ.608/ Rev.1 “Interim guidelines for open top container ships” will also satisfy the requirements for dangerous goods.

b) The amount of water required for fire fighting purposes in the largest hold is to allow simultaneous use of the water spray system plus the four jets of water from hose nozzles.

2.3.2 Sources of ignition

Electrical equipment and wiring is not to be fitted in enclosed cargo spaces or vehicle spaces unless it is considered essential for operational purposes. However, if electrical equipment is fitted in such spaces, it is to be of a certified safe type for use in the dangerous environments to which it may be exposed unless it is possible to completely isolate the electrical system (e.g. by removal of links in the system, other than fuses). Cable penetrations of the decks and bulkheads are to be sealed against the passage of gas or vapour. Through runs of cables and cables within the cargo spaces are to be protected against damage from impact. Any other equipment which may constitute a source of ignition of flammable vapour will not be permitted.

IR2.3.2 Notes for 2.3.2 :

1) Reference is to be made to IEC 60092-506:2003.

2) For pipes having open ends (e.g. ventilation and bilge pipes, etc.) in a hazardous area, the pipe itself is to be classified as hazardous area. See IEC 60092-506:2003 table B1, item B.

3) When carrying flammable liquids having flashpoints less than 23 [˚C] as Class 3, 6.1 or 8 in cargo spaces, the bilge pipes with flanges, valves, pumps, etc. constitute a source of release and the enclosing spaces (e.g. pipe tunnels, bilge pump rooms, etc.) are to be classified as an extended hazardous area (comparable with Zone 2) unless these spaces are continuously mechanically ventilated with a capacity for at least six air changes per hour. Except where the space is protected with redundant mechanical ventilation capable of starting automatically, equipment not certified for Zone 2 are to be automatically disconnected following loss of ventilation while essential systems such as bilge and ballast systems are to be certified for Zone 2. Where redundant mechanical ventilation is employed, equipment and essential systems not certified for Zone 2 are to be interlocked so as to prevent inadvertent operation if the ventilation is not operational. Audible and visible alarms are to be provided at a manned station if failure occurs.

2.3.3 Detection system

Ro-ro spaces are to be fitted with a fixed fire detection and fire alarm system complying with the requirements of the Fire Safety Systems Code (Chapter 8). All other types of cargo spaces are to be fitted with either a fixed fire detection and fire alarm system or a sample extraction smoke detection system complying with the requirements of the Fire Safety Systems Code. If a sample extraction smoke detection system is fitted, particular attention is to be made to the Fire Safety Systems Code (Chapter 8, Clause 10.2.1.3 in Sec.10) in order to prevent the leakage of toxic fumes into occupied areas.

2.3.4 Ventilation arrangement

2.3.4.1 Adequate power ventilation is to be provided in enclosed cargo spaces. The arrangement is to be such as to provide for at least six air changes per hour in the cargo space based on an empty cargo space and for removal of vapours from the upper or lower parts of the cargo space, as appropriate.

IR 2.3.4.1.1 The reduced air changes per hour as per Note 1 of Table 2.1 apply equally to the ventilation air changes requirements in 2.3.4.1 above, when the bilge pump is located directly inside a container cargo space.

IR 2.3.4.1.2 In such a case, where several container cargo spaces are served by the same bilge pump, the bilge pump is to be installed in the container cargo space with the highest ventilation rate, compared to the other container cargo spaces.

2.3.4.2 The fans are to be of non-sparking type, as per Pt.5, Ch.2, Cl.6.5. Suitable wire mesh guards are to be fitted over inlet and outlet ventilation openings to prevent foreign objects from entering into the fan casing.

2.3.4.3 Natural ventilation is to be provided in enclosed cargo spaces intended for the carriage of solid dangerous goods in bulk, where there is no provision for mechanical ventilation.

IR2.3.4.4 If adjacent spaces are not separated from cargo spaces by gastight bulkheads or decks, then they are considered as part of the enclosed cargo space and the ventilation requirements are to apply to the adjacent space as for the enclosed cargo space itself.

IR2.3.4.5 Where the IMSBC Code requires continuous ventilation in cargo holds, the ventilation openings are to comply with the requirements of Pt.3, Ch.13, Sec.2 for openings not fitted with means of closure. This does not prevent ventilators from being fitted with a means of closure as required for fire protection purposes as per Pt.6, Ch.2, Cl. 2.2.1.1 provided the minimum height to the ventilator opening is in accordance with Pt.3, Ch.13, Sec.2 (4.5.m for Position 1 and 2.3 m for Position 2).

The requirements for continuous ventilation apply to the following cargoes as per the IMSBC code:

ALUMINIUM FERROSILICON POWDER
UN 1395

ALUMINIUM SILICON POWDER, UNCOATED UN 1398
ALUMINIUM SMELTING/REMELTING BY-PRODUCTS, PROCESSED
ALUMINIUM SMELTING BY-PRODUCTS or ALUMINIUM REMELTING BYPRODUCTS UN 3170
FERROPHOSPHORUS (including BRIQUETTES)
FERROSILICON (25% ≤ Silicon ≤ 30% or ≥ 90% Silicon)
FERROSILICON UN 1408 (30% ≤ Silicon < 90%)
ZINC ASHES UN 1435

IR2.3.4.6 Where two fans per hold are required in the IMSBC Code, a common ventilation system with two fans connected is acceptable.

IR2.3.4.7 For open top container holds, power ventilation is required only for the lower part of the cargo hold for which purpose ducting is required. The ventilation capacity is to be at least 2 air changes per hour based on the empty hold volume below weather deck.

2.3.5 Bilge pumping

2.3.5.1 Where it is intended to carry flammable or toxic liquids in enclosed cargo spaces, the bilge pumping system is to be designed to protect against inadvertent pumping of such liquids through machinery space piping or pumps. Where large quantities of such liquids are carried, consideration is to be given to the provision of additional means of draining those cargo spaces. These means are to be specially examined and approved.

2.3.5.2 If the bilge drainage system for cargo spaces is additional to the system served by pumps in the machinery space, the capacity of the system is to be not less than 10 [m^3/hr] per cargo space served. If the additional system is common, the capacity need not exceed 25 [m^3/h]. The additional bilge system need not be arranged with redundancy.

2.3.5.3 Whenever flammable or toxic liquids are carried, the bilge line into the machinery space is to be isolated either by fitting a blank flange or by a closed lockable valve.

2.3.5.4 Enclosed spaces outside of machinery spaces containing bilge pumps serving cargo spaces intended for carriage of flammable or toxic liquids are to be fitted with separate mechanical ventilation giving at least 6 air changes per hour. If the space has access from another enclosed space, the door is to be self closing.

IR 2.3.5.4.1 Electrical equipment in the space is to be according to Pt.4, Ch.8.

IR 2.3.5.4.2 The reduced air changes per hour as per Note 1 of Table 2.1 apply equally to the ventilation air changes requirements in 2.3.5.4 above, when the bilge pump is located directly inside a container cargo spaces.

IR 2.3.5.4.3 In such a case, where several container cargo spaces are served by the same bilge pump, the bilge pump is to be installed in the container cargo space with the highest ventilation rate, compared to the other container cargo spaces.

2.3.5.5 If bilge drainage of cargo spaces is arranged by gravity drainage, the drainage is to be either led directly overboard or to a closed drain tank located outside the machinery spaces. The tank is to be provided with a vent pipe to a safe location on the open deck.

IR2.3.5.6 Cargo spaces intended for carriage of flammable liquids with flash point less than 23°C or toxic liquids are to be fitted with a fixed bilge drainage system independent or separated from the bilge system in machinery space and located outside of the machinery space.

If a single bilge drainage system completely independent of the machinery space is provided, the system is to comply with the Rule requirement to redundancy and capacity based on the size of the space or spaces which it serves.

IR2.3.5.7 Drainage from a cargo space into bilge wells in a lower space is only permitted if that space satisfies the same requirements as the cargo space above.

IR2.3.5.8 Bilge systems for open top container ship cargo holds are to be independent of machinery space bilge system and be located outside machinery space.

2.3.6 Personnel protection

2.3.6.1 Four sets of full protective clothing resistant to chemical attack are to be provided in addition to the fireman's outfits required by Ch.3, 4.10 and are to be selected taking into

account the hazards associated with the chemicals being transported and the standards developed by IMO according to the class and physical state. (See Note 2 in IR2.3.6.1). The protective clothing is to cover all skin, so that no part of the body is unprotected.

IR2.3.6.1 Notes for 2.3.6.1:

1) The required protective clothing is for emergency purposes.

2) For solid bulk cargoes the protective clothing is to satisfy the equipment requirements specified in the respective schedules of the IMSBC Code for the individual substances. For packaged goods, the protective clothing is to satisfy the equipment requirements specified in emergency procedures (EmS) of the Supplement to IMDG Code for the individual substances

2.3.6.2 At least two self-contained breathing apparatuses additional to those required by Ch.3, Sec.4 is to be provided. Passenger ships carrying not more than 36 passengers and cargo ships that are equipped with suitably located means for fully recharging the air cylinders free from contamination, need carry only one spare charge for each required apparatus.

IR2.3.6.2 For each of the breathing apparatuses, two complete sets of air bottles are to be provided. These spare bottles are in addition to the spare bottles required for fire fighter's outfit.

2.3.7 Portable fire extinguishers

Portable fire extinguishers with a total capacity of at least 12 [kg] of dry powder or equivalent are to be provided for the cargo spaces. These extinguishers are to be in addition to any portable fire extinguishers required elsewhere in this part. (Also refer MSC.1/Circ.1275 'Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable fire extinguishers on board ships')

2.3.8 Insulation of machinery space boundaries

Bulkheads forming boundaries between cargo spaces and machinery spaces of category A are to be insulated to "A-60" class standard, unless the dangerous goods are stowed at least 3 [m] horizontally away from such bulkheads. Other boundaries between such spaces are to be insulated to "A-60" class standard.

IR2.3.8 In the case of a closed or semi-closed cargo space located partly above a machinery space and the deck above the machinery space not insulated, dangerous goods are not to be carried in the whole of that cargo space. If un-insulated deck above the machinery space is a weather deck, dangerous goods are not to be carried in the portion of the deck above the machinery space.

2.3.9 Water spray system

Each open ro-ro space having a deck above it and each space deemed to be a closed ro-ro space not capable of being sealed, is to be fitted with an approved fixed pressure water-spraying system for manual operation which is to protect all parts of any deck and vehicle platform in the space. Use of any other fixed fire-extinguishing system that has been shown by full-scale test to be no less effective may be considered. However, the drainage and pumping arrangements are to be such as to prevent the build-up of free surfaces. The drainage system is to be sized to remove no less than 125% of the combined capacity of both the water spraying system pumps and the required number of fire hose nozzles. The drainage system valves are to be operable from outside the protected space at a position in the vicinity of the extinguishing system controls. Bilge wells are to be of sufficient holding capacity and are to be arranged at the side shell of the ship at a distance from each other of not more than 40 [m] in each watertight compartment. If this is not possible the adverse effect upon stability of the added weight and free surface of water is to be taken into account during appraisal the stability information. (Refer to the Recommendation on fixed fire-extinguishing systems for special cargo spaces adopted by IMO by resolution A.123(V)).

2.3.10 Separation of ro-ro spaces

2.3.10.1 In ships having ro-ro spaces, a separation is to be provided between a closed ro-ro space and an adjacent open ro-ro space. The separation is to be such as to minimize the passage of dangerous vapours and liquids between such spaces. Alternatively, such separation need not be provided if the ro-ro space is considered to be a closed cargo space over its entire length and is to fully

comply with the relevant special requirements of this section.

2.3.10.2 In ships having ro-ro spaces, a separation is to be provided between a closed ro-ro space and the adjacent weather deck. The separation is to be such as to minimize the passage of dangerous vapours and liquids between such spaces. Alternatively, a separation need not be provided if the arrangements of the closed ro-ro spaces are in accordance with those required for the dangerous goods carried on adjacent weather deck.

2.4 Document of compliance

(Refer to the Document of compliance with the special requirements for ships carrying dangerous goods under the provisions of regulation II-2/19 of SOLAS 74, as amended (MSC/Circ.1266)).

If requested and authorised by statutory authority IRS will provide the ship with an appropriate document as evidence of compliance of construction and equipment with the requirement of this section. Certification for carriage of solid dangerous bulk cargoes covers only those cargoes listed in Group B of the IMSBC Code except cargoes classified solely as MHB (Materials Hazardous only in Bulk) Other solid dangerous bulk cargoes may be certified subject to acceptance by the relevant statutory authority.

IR2.4 Notes regarding statutory document of compliance given in 2.4:

1) Ships constructed on or after 1 July 2002 are to comply with SOLAS Reg. II-2/19 and all the requirements introduced with 2000 SOLAS amendments and will be issued DoC in pursuance of regulation II-2/19.

2) Ships constructed on or after 1 July 1998 and before 1 July 2002 are to comply with SOLAS Reg. II-2/54 and all the requirements introduced with 1996 SOLAS amendments and will be issued DoC in pursuance of regulation II-2/54.

   (See Ch.1 Cl. 1.3.4 regarding retroactive application of SOLAS Reg.II-2/19 to such ships).

3) Ships constructed on or after 1 September 1984 and before 1 July 1998 are to comply with SOLAS Reg. II-2/54 and all the requirements introduced with 1981 SOLAS amendments and will be issued DoC in pursuance of regulation II-2/54.

   (See Ch.1 Cl. 1.3.4 regarding retroactive application of SOLAS Reg.II-2/19 to such ships).

4) Ships constructed before 1 September 1984, upon request of the Owner, will be issued DoC in pursuance of regulation II-2/54 provided they comply with the requirements of regulation II-2/54 and other regulations referred to in reg. II-2/54.

Table 2.1 : Application of the requirements to different modes of carriage of dangerous goods in ships and cargo spaces

Wherever X appears in Table 2.1 it means that this requirement is applicable to all classes of dangerous goods as given in the appropriate line of Table 2.3, except as indicated by the notes.

Clause 2.2.2		**.1**	**.2**	**.3**		**.4**	**.5**
Clause 2.3	**Weather decks .1 to .5 inclusive**	**Not specifi-cally designed**	**Container cargo spaces**	**Closed ro-ro cargo spaces[5]**	**Open ro-ro cargo spaces#**	**Solid dangerous goods in bulk**	**Shipbor-ne barges**
.1.1	X	X	X	X	X	For application of requirements of Sec.2 to different classes of dangerous goods, see Table 2.2	X
.1.2	X	X	X	X	X		-
.1.3	-	X	X	X	X		X
.1.4	-	X	X	X	X		X
.2	-	X	X	X	X		X[4]
.3	-	X	X	X	-		X[4]
.4.1	-	X	X[1]	X	-		X[4]
.4.2	-	X	X[1]	X	-		X[4]
.5	-	X	X	X	-		-
.6.1	X	X	X	X	X		-
.6.2	X	X	X	X	X		-
.7	X	X	-	-	X		-
.8	X	X	X[2]	X	X		-
.9	-	-	-	X[3]	X		-
.10.1	-	-	-	X	-		-
.10.2	-	-	-	X	-		-

Notes:

1 For classes 4 and 5.1 solids not applicable to closed freight containers.
For classes 2, 3, 6.1 and 8 when carried in closed freight containers, the ventilation rate may be reduced to not less than two air changes per hour. For classes 4 and 5.1 liquids when carried in closed freight containers, the ventilation rate may be reduced to not less than two air changes per hour. For the purpose of this requirement a portable tank is a closed freight container.

2 Applicable to decks only.

3 Applies only to closed ro-ro cargo spaces, not capable of being sealed.

4 In the special case where the barges are capable of containing flammable vapours or alternatively if they are capable of discharging flammable vapours to a safe space outside the barge carrier compartment by means of ventilation ducts connected to the barges, these requirements may be reduced or waived to the satisfaction of IRS.

5 Special category spaces are to be treated as closed ro-ro cargo spaces when dangerous goods are carried.

For the purposes of Section 5.6 a ro-ro space fully open above and with full openings in both ends may be treated as a weather deck.

Table 2.2 : Application of the requirements to different classes of dangerous goods for ships and cargo spaces carrying solid dangerous goods in bulk

Class*	**4.1**	**4.2**	**4.3[6]**	**5.1**	**6.1**	**8**	**9**
Clause							
2.3.1.1	X	X	-	X	-	-	X
2.3.1.2	X	X	-	X	-	-	X
2.3.2	X	X[7]	X	X[8]	-	-	X[8]
2.3.4.1	-	X[7]	X	-	-	-	-
2.3.4.2	X[9]	X[7]	X	X[7,9]	-	-	X[7,9]
2.3.4.3	X	X	X	X	X	X	X
2.3.6	X	X	X	X	X	X	X
2.3.8	X	X	X	X[7]	-	-	X[10]

* Classes are as defined in International Maritime Dangerous Goods Code.

Notes:

6 The hazards of substances in this class which may be carried in bulk are such that special consideration must be given by IRS to the construction and equipment of the ship involved in addition to meeting the requirements enumerated in this table.

7 Only applicable to –

- Seedcake containing solvent extractions
- Ammonium nitrate
- Ammonium nitrate fertilizers.

8 Only applicable to Ammonium nitrate and to Ammonium nitrate fertilizers. However, a degree of protection in accordance with standards contained in the International Electro-technical Commission, publication 60079 - Electrical Apparatus for Explosive Gas Atmospheres, is sufficient.

9 Only suitable wire mesh guards are required.

10 The requirements of the International Maritime Solid Bulk Cargoes (IMSBC) Code, as amended, are sufficient.

Table 2.3 : Application of the requirements to different classes of dangerous goods except solid dangerous goods in bulk

Clause																							
Class*	**1.1 to1.6**	**1.4S**	**2.1**	**2.2**	**2.3 flammable[20]**	**2.3 non-flammable**	**3 FP[15] < 23ºC**	**3 FP[15] ≥ 23ºC to ≤ 60ºC**	**4.1**	**4.2**	**4.3 liquids[21]**	**4.3 solids**	**5.1**	**5.2[16]**	**6.1 liquids FP[15] < 23ºC**	**6.1 liquids FP[15] ≥ 23ºC to ≤ 60ºC**	**6.1 liquids**	**6.1 solid**	**8 liquids FP[15] < 23ºC**	**8 liquids FP[15] ≥ 23ºC to ≤ 60ºC**	**8 liquids**	**8 solid**	**9**
2.3.1.1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2.3.1.2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-
2.3.1.3	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2.3.1.4	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2.3.2	X	-	X	-	X	-	X	-	-	-	X[18]	-	-	-	X	-	-	-	X	-	-	-	X[17]
2.3.3	X	X	X	X	-	X	X	X	X	X	X	X	X	-	X	X	X	X	X	X	X	X	-
2.3.4.1	-	-	X	-	-	X	X	-	X[11]	X[11]	X	X	X[11]	-	X	X	-	X[11]	X	X	-	-	X[11]
2.3.4.2	-	-	X	-	-	-	X	-	-	-	-	-	-	-	X	~~X~~ -	-	-	X	-	-	-	X[17]
2.3.5	-	-	-	-	-	-	X	-	-	-	-	-	-	-	X	X	X	-	X	X[19]	X[19]	-	-
2.3.6	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X[14]
2.3.7	-	-	-	-	-	-	X	X	X	X	X	X	X	-	X	X	-	-	X	X	-	-	-
2.3.8	X[12]	-	X	X	X	X	X	X	X	X	X	X	X[13]	X	X	X	-	-	X	X	-	-	-
2.3.9	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2.3.10.1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2.3.10.2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

* Classes are as defined in International Maritime Dangerous Goods Code.

Table 2.3 - Notes:

11 When "mechanically ventilated spaces" are required by the International Maritime Dangerous Goods Code, as amended.

12 Stow 3 [m] horizontally away from the machinery space boundaries in all cases.

13 Refer to the International Maritime Dangerous Goods Code.

14 As appropriate to the goods being carried.

15 FP refers to flashpoint.

16 Under the provisions of the IMDG Code, stowage of class 5.2 dangerous goods under deck or in enclosed ro-ro spaces is prohibited.

17 Only applicable to dangerous goods evolving flammable vapour listed in the IMDG Code.

18 Only applicable to dangerous goods having a flashpoint less than 23°C listed in the IMDG Code.

19 Only applicable to dangerous goods having a subsidiary risk class 6.1.

20 Under the provisions of the IMDG Code, stowage of class 2.3 having subsidiary risk class 2.1 under deck or in enclosed ro-ro spaces is prohibited.

21 Under the provisions of the IMDG Code, stowage of class 4.3 liquids having a flashpoint less than 23°C under deck or in enclosed ro-ro spaces is prohibited.

Section 3

Protection of Vehicle, Special Category and Ro-ro Spaces

3.1 Purpose

The purpose of this section is to provide additional safety measures in order to address the fire safety objectives of this part for ships fitted with vehicle, special category and ro-ro spaces. For this purpose, the following functional requirements are to be met:

.1 fire protection systems are to be provided to adequately protect the ship from the fire hazards associated with vehicle, special category and ro-ro spaces;

.2 ignition sources are to be separated from vehicle, special category and ro-ro spaces; and

.3 vehicle, special category and ro-ro spaces are to be adequately ventilated.

3.2 General requirements

3.2.1 Application

3.2.1.1 In addition to complying with the requirements of Chapters 2, 3, 4 and 5, as appropriate, vehicle, special category and ro-ro spaces are to comply with the requirements of this section.

3.2.1.2 On all ships, vehicles with fuel in their tanks for their own propulsion may be carried in cargo spaces other than vehicle, special category or ro-ro spaces, provided that all the following conditions are met:

.1 the vehicles do not use their own propulsion within the cargo spaces;

.2 the cargo spaces are in compliance with the appropriate requirements of Section 2; and

.3 the vehicles are carried in accordance with the IMDG Code, as defined in SOLAS regulation VII/1.1.

3.2.2 Basic principles for passenger ships

3.2.2.1 The basic principle underlying the provisions of this section is that the main vertical zoning required by Ch.3, 3.2 may not be practicable in vehicle spaces of passenger ships and, therefore, equivalent protection must be obtained in such spaces on the basis of a horizontal zone concept and by the provision of an efficient fixed fire-extinguishing system. Based on this concept, a horizontal zone for the purpose of these requirements may include special category spaces on more than one deck provided that the total overall clear height for vehicles does not exceed 10 [m]. The "Total overall clear height" is the sum of distances between deck and web frames of the decks forming one horizontal zone.

3.2.2.2 The basic principle underlying the provisions of 3.2.2.1 are also applicable to ro-ro spaces.

3.2.2.3 The requirements of ventilation systems, openings in "A" class divisions and penetrations in "A" class divisions for maintaining the integrity of vertical zones in this part are to be applied equally to decks and bulkheads forming the boundaries separating horizontal zones from each other and from the remainder of the ship.

3.3 Precaution against ignition of flammable vapours in closed vehicle spaces, closed ro-ro spaces and special category spaces

3.3.1 Ventilation systems

3.3.1.1 Capacity of ventilation systems

There are to be provided an effective power ventilation system sufficient to give at least the following air changes:

.1	Passenger ships	
	Special category spaces	10 air changes per hour
	Closed ro-ro and vehicle spaces other than special category spaces for ships carrying more than 36 passengers	10 air changes per hour
	Closed ro-ro and vehicle spaces other than special category spaces for ships carrying not more than 36 passengers	6 air changes per hour
.2	Cargo ships	6 air changes per hour

An increased number of air changes may be required when vehicles are being loaded and unloaded.

3.3.1.2 *Performance of ventilation systems*

3.3.1.2.1 In passenger ships, the power ventilation system is to be separate from other ventilation systems. The power ventilation system is to be operated to give at least the number of air changes required in 3.3.1.1 at all times when vehicles are in such spaces, except where an air quality control system in accordance with 3.3.1.2.4 is provided. Ventilation ducts serving such cargo spaces capable of being effectively sealed are to be separated for each such space. The system is to be capable of being controlled from a position outside such spaces.

3.3.1.2.2 In cargo ships, ventilation fans are to normally be run continuously and give at least the number of air changes required in 3.3.1.1 whenever vehicles are on board, except where an air quality control system in accordance with 3.3.1.2.4 is provided. Where this is impracticable, they are to be operated for a limited period daily as weather permits and in any case for a reasonable period prior to discharge, after which period the ro-ro or vehicle space are to be proved gas-free. One or more portable combustible gas detecting instruments is to be carried for this purpose. The system is to be entirely separate from other ventilating systems. Ventilation ducts serving ro-ro or vehicle spaces are to be capable of being effectively sealed for each cargo space. The system is to be capable of being controlled from a position outside such spaces.

3.3.1.2.3 The ventilation system is to be such as to prevent air stratification and the formation of air pockets.

3.3.1.2.4 For all ships, where an air quality control system is provided based on the IMO guidelines*, the ventilation system may be operated at a decreased number of air changes and/or a decreased amount of ventilation. This relaxation does not apply to spaces to which at least ten air changes per hour is required by paragraph 3.3.2.2 of this regulation and spaces subject to regulations 2.3.4.1 and Section 4.

Note:

* *Refer to the revised design guidelines and operational recommendations for ventilation systems in ro-ro cargo spaces (MSC.1/Circ.1515).*

3.3.1.3 *Indication of ventilation systems*

Means is to be provided on the navigation bridge to indicate any loss of the required ventilating capacity.

IR3.3.1.3 Alarm provided on the bridge and initiated by tripping of starter relay of fan motor is considered to meet the requirement of 3.3.1.3.

3.3.1.4 *Closing appliances and ducts*

3.3.1.4.1 Arrangements are to be provided to permit a rapid shutdown and effective closure of the ventilation system from outside of the space in case of fire, taking into account the weather and sea conditions.

IR3.3.1.4.1 Access routes to the controls for closure of the ventilation system are considered to "permit a rapid shutdown" and adequately "take into account the weather and sea conditions" if the routes:
- are at least 600 mm clear width;
- are provided with a single handrail or wire rope lifeline not less than 10 mm in diameter, supported by stanchions not more than 10 m apart in way of any route which involves traversing a deck exposed to weather; and
- are fitted with appropriate means of access (such as ladders or steps) to the closing devices of ventilators located in high positions.

Alternatively, remote closing and position indicator arrangements from the bridge or a fire control station for those ventilator closures are considered acceptable.

3.3.1.4.2 Ventilation ducts, including dampers, within a common horizontal zone are to be made of steel. In passenger ships, ventilation ducts that pass through other horizontal zones or machinery spaces are to be “A-60” class steel ducts constructed in accordance with Ch.3, 3.7.2.4.1.1 and Ch.3, 3.7.2.4.1.2.

3.3.1.5 *Permanent openings*

Permanent openings in the side plating, the ends or deckhead of the space are to be so situated that a fire in the cargo space does not endanger stowage areas and embarkation stations for survival craft and accommodation spaces, service spaces and control stations in superstructures and deckhouses above the cargo spaces.

3.3.2 Electrical equipment and wiring

3.3.2.1 Except as provided in 3.3.2.2, electrical equipment and wiring are to be of a type suitable for use in an explosive petrol and air mixture.

IR3.3.2.1 Certified safe electrical equipment suitable for use in Zone 1 areas as defined in IEC 60079-10-1:2015 (Gas Group IIA and Temperature class T3) is considered to satisfy the requirement given in 3.3.2.1. Refer IEC 60079-14:2013 for types of protection suitable for use in Zone 1 areas.

3.3.2.2 In case of other than special category spaces below the bulkhead deck, notwithstanding the provisions in 3.3.2.1, above a height of 450 [mm] from the deck and from each platform for vehicles, if fitted, except platforms with openings of sufficient size permitting penetration of petrol gases downwards, electrical equipment of a type so enclosed and protected as to prevent the escape of sparks will be permitted as an alternative on condition that the ventilation system is so designed and operated as to provide continuous ventilation of the cargo spaces at the rate of at least ten air changes per hour whenever vehicles are on board.

IR3.3.2.2 Electrical equipment provided with at least IP55 enclosure or apparatus suitable for use in Zone 2 areas as defined in IEC 60079-10-1:2015 is considered to satisfy the requirement given in 3.3.2.2. Refer IEC 60079-14:2013 for types of protection suitable for use in Zone 2 areas.

3.3.3 Electrical equipment and wiring in exhaust ventilation ducts

Electrical equipment and wiring, if installed in an exhaust ventilation duct, is to be of a type approved for use in explosive petrol and air mixtures and the outlet from any exhaust duct is to be sited in a safe position, having regard to other possible sources of ignition.

IR3.3.3 Certified safe electrical equipment suitable for use in Zone 1 areas as defined in IEC 60079-10-1:2015 (Gas Group IIA and Temperature class T3) is considered to satisfy the requirement given in 3.3.3. Refer IEC 60079-14:2013 for types of protection suitable for use in Zone 1 areas.

3.3.4 Other ignition sources

Other equipment which may constitute a source of ignition of flammable vapours will not be permitted.

3.3.5 Scuppers and discharges

Scuppers are not to be led to machinery or other spaces where sources of ignition may be present.

3.4 Detection and alarm

3.4.1 Fixed fire detection and fire alarm systems

Except as provided in 3.4.3.1, there is to be provided a fixed fire detection and fire alarm system complying with the requirements of the Fire Safety Systems Code (Chapter 8). The fixed fire detection system is to be capable of rapidly detecting the onset of fire. The type of detectors and their spacing and location is to be to the satisfaction of IRS taking into account the effects of ventilation and other relevant factors. After being installed the system is to be tested under normal ventilation conditions and is to give an overall response time to the satisfaction of IRS.

IR3.4.1 The requirement of 3.4.1 need not apply to weather decks used for the carriage of vehicle for fuel in their tanks.

3.4.2 Sample extraction smoke detection systems

Except open ro-ro spaces, open vehicle spaces and special category spaces, a sample extraction smoke detection system complying with the requirements of the Fire Safety Systems Code may be used as an alternative of the fixed fire detection and fire alarm system required in 3.4.1.

3.4.3 Special category spaces

3.4.3.1 An efficient fire patrol system is to be maintained in special category spaces. However, if an efficient fire patrol system is maintained by a continuous fire watch at all times during the voyage, a fixed fire detection and fire alarm systems is not required.

3.4.3.2 Manually operated call points are to be spaced so that no part of the space is more than 20 [m] from a manually operated call point, and one is to be placed close to each exit from such spaces.

3.5 Structural protection

Notwithstanding the provisions of Ch.3, 3.2.2, in passenger ships carrying more than 36 passengers, the boundary bulkheads and decks of special category spaces and ro-ro spaces are to be insulated to "A-60" class standard. However, where a category (5), (9) and (10) space, as defined in Ch.3, 3.2.2.3, is on one side of the division the standard may be reduced to "A-0". Where fuel oil tanks are below a special category space or a ro-ro space, the integrity of the deck between such spaces, may be reduced to "A-0" standard.

3.6 Fire-extinction

3.6.1 Fixed fire-extinguishing systems

(*The requirements of 3.6.1.1 and 3.6.1.2 apply to ships constructed on or after 1 July 2014. Ships constructed before 1 July 2014 are to comply with the previously applicable requirements of paragraphs 3.6.1.1 and 3.6.1.2)*

3.6.1.1 Vehicle spaces and ro-ro spaces which are not special category spaces and are capable of being sealed from a location outside of the cargo spaces are to be fitted with one of the following fixed fire-extinguishing systems:

.1 a fixed gas fire extinguishing system complying with the requirements of Chapter 8;

.2 a fixed high expansion foam fire-extinguishing system complying with the requirements of Chapter 8; or

.3 a fixed water-based fire fighting system for complying with the provisions of Ch 8, 7.2.4 and paragraphs 3.6.1.2.1 to 3.6.1.2.4.

3.6.1.2 Ro-ro and vehicle spaces not capable of being sealed and special category spaces are to be fitted with a fixed water-based fire fighting system complying with the requirements of Ch 8, 7.2.4, which will protect all parts of any deck and vehicle platform in such spaces.

Such a water-based fire-fighting system is to have:

.1 a pressure gauge on the valve manifold;

.2 clear marking on each manifold valve indicating the spaces served;

.3 instructions for maintenance and operation located in the valve room; and

.4 a sufficient number of drainage valves to ensure complete drainage of the system.

3.6.1.3 Use of any other fixed fire-extinguishing system may be permitted if it is shown that it is not less effective by a full-scale test in conditions simulating a flowing petrol fire in a vehicle space or a ro-ro space in controlling fires likely to occur in such a space. (Refer to the Revised Guidelines for the design and approval of fixed water-based fire-fighting systems for ro-ro spaces and special category spaces (MSC.1/Circ.1430,Rev.3)).

3.6.1.4 The requirements of this paragraph apply to ships constructed on or after 1 January 2010. Ships constructed on or after 1 July 2002 and before 1 January 2010 are to comply with the previously applicable version of this paragraph. When fixed pressure water-spraying systems are provided, in view of the serious loss of stability which could arise due to large quantities of water accumulating on the deck or decks during the operation of the fixed pressure water-spraying system, the following arrangements are to be provided:

.1 in passenger ships:

.1.1 in the spaces above the bulkhead deck, scuppers are to be fitted so as to ensure that such water is rapidly discharged directly overboard, taking into account MSC.1/ Circ.1320 “Guidelines for the drainage of fire-fighting water from closed vehicle and ro-ro spaces and special category spaces of passenger and cargo ships”;

.1.2.1 in ro-ro passenger ships discharge valves for scuppers, fitted with positive means of closing operable from a position above the bulkhead deck in accordance with the requirements of the International Convention on Load Lines in force, are to be kept open while the ships are at sea;

.1.2.2 any operation of valves referred to in 3.6.1.4.1.2.1 is to be recorded in the log-book;

.1.3 in the spaces below the bulkhead deck, IRS may require pumping and drainage facilities to be provided additional to the requirements of Pt.4, Ch.3, Sec.2. In such case, the drainage system is to be sized to remove no less than 125% of the combined capacity of both the water spraying system pumps and the required number of fire hose nozzles, taking into account MSC.1/Circ.1320 “Guidelines for the drainage of fire-fighting water from closed vehicle and ro-ro spaces and special category spaces of passenger and cargo ships”. The drainage system valves are to be operable from outside the protected space at a position in the vicinity of the extinguishing system controls. Bilge wells are to be of sufficient holding capacity and are to be arranged at the side shell of the ship at a distance from each other of not more than 40 [m] in each watertight compartment;

.2 in cargo ships, the drainage and pumping arrangements are to be such as to prevent the build-up of free surfaces. In such case, the drainage system is to be sized to remove no less than 125% of the combined capacity of both the water spraying system pumps and the required number of fire hose nozzles, taking into account MSC.1/Circ.1320 “Guidelines for the drainage of fire-fighting water from closed vehicle and ro-ro spaces and special category spaces of passenger and cargo ships”. The drainage system valves are to be operable from outside the protected space at a position in the vicinity of the extinguishing system controls. Bilge wells are to be of sufficient holding capacity and are to be arranged at the side shell of the ship at a distance from each other of not more than 40 [m] in each watertight compartment. If this is not possible the adverse effect upon stability of the added weight and free surface of water is to be taken into account to the extent deemed necessary. This information is to be included in the stability information supplied to the master.

3.6.1.5 On all ships, for closed vehicles and ro-ro spaces and special category spaces, where fixed pressure water-spraying systems are fitted, means shall be provided to prevent

the blockage of drainage arrangements, taking into account MSC.1/Circ.1320 “Guidelines for the drainage of fire-fighting water from closed vehicle and ro-ro spaces and special category spaces of passenger and cargo ships”. Ships constructed before 1 January 2010 are to comply with the requirements of this paragraph by the first survey after 1 January 2010.

IR3.6.1.5 The requirements for fixed fire extinguishing system, portable fire extinguishers and foam applicators given in 3.6.1 and 3.6.2 need not apply to weather decks used for the carriage of vehicle with fuel in their tanks.

3.6.2 Portable fire extinguishers

3.6.2.1 Portable extinguishers are to be provided at each deck level in each hold or compartment where vehicles are carried, spaced not more than 20 [m] apart on both sides of the space. At least one portable fire-extinguisher is to be located at each access to such a cargo space. (Also refer MSC.1/Circ.1275 ‘Unified Interpretations of SOLAS Chapter II-2 on the Number and Arrangement of Portable fire extinguishers on board ships’).

3.6.2.2 In addition to the provision of 3.6.2.1, the following fire extinguishing appliances are to be provided in vehicle, ro-ro and special category spaces intended for the carriage of motor vehicles with fuel in their tanks for their own propulsion:

.1 at least three water-fog applicators; and

.2 one portable foam applicator unit complying with the provisions of the Fire Safety Systems Code (Chapter 8), provided that at least two such units are available in the ship for use in such ro-ro spaces.

IR3.6.2.3 Also refer IR3.6.1.5.

IR3.6.2.4 Cargo holds, loaded with vehicles with fuel in their tanks and stowed in open or closed containers need not be provided with the portable fire extinguishers, water-fog applicators and foam applicator unit as given in 3.6.2.1 and 3.6.2.2.

Section 4

Requirements for vehicles carriers carrying as cargo motor vehicles with compressed hydrogen or natural gas in their tanks for their own propulsion

4.1 Purpose

4.1.1 The purpose of this section is to provide additional safety measures in order to address the fire safety objectives of this chapter for vehicle carriers with vehicle and ro-ro spaces intended for carriage as cargo motor vehicles with compressed hydrogen or compressed natural gas in their tanks for their own propulsion.

4.2 Application

4.2.1 In addition to complying with the requirements of section 3, as appropriate, vehicle carriers constructed on or after 1 January 2016 intended for the carriage of motor vehicles with compressed hydrogen or compressed natural gas in their tanks for their own propulsion as cargo are to comply with the requirements in 4.3 to 4.5.

4.2.2 In addition to complying with the requirements of Sec 3, as appropriate, vehicle carriers constructed before 1 January 2016, including those constructed before 1 July 2012*, are to comply with the requirements in 4.5 of this section.

(*Refer to the recommendation on safety measures for existing vehicle carriers carrying as cargo motor vehicles with compressed hydrogen or natural gas in their tanks for their own propulsion MSC.1/Circ.1471.)

4.3 Requirements for spaces intended for carriage of motor vehicles with compressed natural gas in their tanks for their own propulsion as cargo

4.3.1 Electrical equipment and wiring

All electrical equipment and wiring are to be of a certified safe type for use in an explosive methane and air mixture*.

(* Refer to the recommendations of the International Electrotechnical Commission, in particular, publication IEC 60079)

4.3.2 Ventilation arrangement

4.3.2.1 Electrical equipment and wiring, if installed in any ventilation duct, are to be of a certified safe type for use in explosive methane and air mixtures.

4.3.2.2 The fans are to be such as to avoid the possibility of ignition of methane and air mixtures. Suitable wire mesh guards are to be fitted over inlet and outlet ventilation openings.

4.3.3 Other ignition sources

Other equipment which may constitute a source of ignition of methane and air mixtures are not to be permitted.

4.4 Requirements for spaces intended for carriage of motor vehicles with compressed hydrogen in their tanks for their own propulsion as cargo

4.4.1 Electrical equipment and wiring

All electrical equipment and wiring are to be of a certified safe type for use in an explosive hydrogen and air mixture*.

(*Refer to the recommendations of the International Electrotechnical Commission, in particular, publication IEC 60079.)

4.4.2 Ventilation arrangement

4.4.2.1 Electrical equipment and wiring, if installed in any ventilation duct, are to be of a certified safe type for use in explosive hydrogen and air mixtures and the outlet from any exhaust duct are to be sited in a safe position, having regard to other possible sources of ignition.

4.4.2.2 The fans are to be designed such as to avoid the possibility of ignition of hydrogen and air mixtures. Suitable wire mesh guards are to be fitted over inlet and outlet ventilation openings.

4.4.3 Other ignition sources

Other equipment which may constitute a source of ignition of hydrogen and air mixtures are not to be permitted.

4.5 Detection

When a vehicle carrier carries as cargo one or more motor vehicles with either compressed hydrogen or compressed natural gas in their tanks for their own propulsion, at least two portable gas detectors are to be provided. Such detectors are to be suitable for the detection of the gas fuel and be of a certified safe type for use in the explosive gas and air mixture.

Section 5

Casualty Threshold, Safe Return to Port and Safe Areas

5.1 Application

5.1.1 Passenger ships constructed on or after 1 July 2010 having loadline length, as defined in Pt.3, Ch.1, of 120 [m] or more or having three or more main vertical zones are to comply with the provisions of this section.

5.2 Purpose

5.2.1 The purpose of this section is to establish design criteria for a ship's safe return to port under its own propulsion after a casualty that does not exceed the casualty threshold stipulated in 5.3 and also provides functional requirements and performance standards for safe areas.

5.3 Casualty threshold

5.3.1 The casualty threshold, in the context of a fire, includes:

a) loss of space of origin upto the nearest "A" class boundaries, which may be a part of the space of origin, if the space of origin is protected by a fixed fire extinguishing system; or

b) loss of the space of origin and adjacent spaces upto the nearest "A" class boundaries, which are not part of the space of origin.

5.4 Safe return to port*

5.4.1 When fire damage does not exceed the casualty threshold indicated in 5.3, the ship is to be capable of returning to port while providing a safe area as defined in Ch.1, Sec.3. To be deemed capable of returning to port, the following systems are to remain operational in the remaining part of the ship not affected by fire:

.1 propulsion;

.2 steering systems and steering-control systems;

.3 navigational systems;

.4 systems for fill, transfer and service of fuel oil;

.5 internal communication between the bridge, engineering spaces, safety centre, fire-fighting and damage control teams and as required for passenger and crew notification and mustering;

.6 external communication;

.7 fire main system;

.8 fixed fire-extinguishing systems;

.9 fire and smoke detection system;

.10 bilge and ballast system;

.11 power-operated watertight and semi-watertight doors;

.12 systems intended to support "safe areas" as indicated in paragraph 4.5.1.2;

.13 flooding detection systems; and

.14 other systems determined by IRS to be vital to damage control efforts.

* Refer to MSC.1 Circ. 1369 as amended by MSC.1/ Circ. 1369/Add.1 "Interim explanatory notes for the assessment of passenger ship systems' capabilities after a fire or flooding casualty".

IR5.4.1 Qualitative failure analysis for propulsion and steering on passenger ships

IR5.4.1.1 Scope

This clause (IR 5.4.1) gives details of a qualitative failure analysis for propulsion and steering for new passenger ships including those having a length of 120 [m] or more or having three or more main vertical zones.

IR5.4.1.2 Note

This may be considered as the first step for demonstrating compliance with the revised SOLAS Chapter II-2, Regulation 21 – SOLAS 2006 Amendments, Resolution MSC.216(82), Annex 3.

IR5.4.1.3 Objectives

a) For ships having at least two independent means of propulsion and steering, to comply with SOLAS requirements for a safe return to port, items (a) and (b) below are applicable:

- Provide knowledge of the effects of failure in all the equipment and systems due to fire in any space, or flooding of any watertight compartment that could affect the availability of the propulsion and steering.

- Provide solutions to ensure the availability of propulsion and steering upon such failures in item (a).

b) Ships not required to satisfy the safe return to port concept will require the analysis of failure in single equipment and fire in any space to provide knowledge and possible solutions for enhancing availability of propulsion and steering.

IR5.4.1.4 Systems to be considered

The qualitative failure analysis is to consider the propulsion and steering equipment and all its associated systems which might impair the availability of propulsion and steering.

The qualitative failure analysis should include:

a) Propulsion and electrical power prime movers e.g.
   - Diesel engines
   - Electric motors

b) Power transmission systems, e.g.

- Shafting
- Bearings
- Power converters
- Transformers
- Slip ring systems

c) Steering gear

- Rudder actuator or equivalent for azimuthing propulsor
- Rudder stock with bearings and seals
- Rudder
- Power unit and control gear
- Local control systems and indicators
- Remote control systems and indicators
- Communication equipment

d) Propulsors, e.g.

- Propeller
- Azimuthing thruster
- Water jet

e) Main power supply systems, e.g.

- Electrical generators and distribution systems
- Cable runs
- Hydraulic
- Pneumatic

f) Essential auxiliary systems, e.g.

- Compressed air
- Oil fuel
- Lubricating oil
- Cooling water
- Ventilation
- Fuel storage and supply systems

g) Control and monitoring systems, e.g.

- Electrical auxiliary circuits
- Power supplies
- Protective safety systems
- Power management systems
- Automation and control systems

h) Support systems, e.g.

- Lighting
- Ventilation.

To consider the effects of fire or flooding in a single compartment, the analysis is to address the location and layout of equipment and systems.

IR5.4.1.5 Failure criteria

a) Failures are deviations from normal operating conditions such as loss or malfunction of a component or system such that it cannot perform an intended or required function.

b) The qualitative failure analysis is to be based on single failure criteria (not two independent failures occurring simultaneously).

c) Where a single failure causes results in failure of more than one component in a system (common cause failure), all the resulting failures are to be considered together.

d) Where the occurrence of a failure leads directly to further failures, all those failures are to be considered together.

IR5.4.1.6 Verification of solutions

a) The shipyard is to submit a report to IRS that identifies how the objectives have been addressed. The report is to include the following information:

- Identify the standards used for analysis of the design.
- Identify the objectives of the analysis.
- Identify any assumptions made in the analysis.
- Identify the equipment, system or sub-system, mode of operation of the equipment.
- Identify probable failure modes and acceptable deviations from the intended or required function.
- Evaluate the local effects (e.g. fuel injection failure) and the effects on the system as a whole (e.g. loss of propulsion power) of each failure mode as applicable.
- Identify trials and testing necessary to prove conclusions.

Note : All stakeholders (e.g. IRS, owners, shipyard and manufacturers) should as far as possible be involved in the development of the report.

b) The report is to be submitted prior to approval of detailed design plans. The report may be submitted in two parts:

- A preliminary analysis as soon as the initial arrangements of different compartments and propulsion plant are known which can form the basis of discussion. This is to include a structured assessment of all essential systems supporting the propulsion plant after a failure in equipment, fire or flooding in any compartment casualty.

- A final report detailing the final design with a detailed assessment of any critical system identified in the preliminary report.

c) Conclusions of the report are to be agreed between IRS and the shipyard.

IR5.4.1.13 Flooding detection systems

.1 A flooding detection system is to be fitted in all watertight spaces below the bulkhead deck that:

a) have a volume, in cubic metres [m^3], that is more than the ship's moulded displacement per centimeter immersion at the deepest subdivision draught; or

b) have a volume more than 30 [m^3], whichever is the greater.

whichever is the greater.

.2 Any watertight spaces that are separately equipped with a liquid level monitoring system (such as fresh water, ballast water, fuel, etc.), with an indicator panel or other means of monitoring at the navigation bridge (and the safety centre if located in a separate space from the navigation bridge), are excluded from these requirements.

.3 For passenger ships (contracted for construction on or after 01 July 2019) carrying 36 or more persons and subject to 5.1.1, the safe return to port requirements of 5.4.1 apply to both:

.1 the flooding detection systems in the spaces as defined in IR5.4.1.13.1

.2 the liquid level monitoring systems, which are used as, or replace, the flooding detection systems, as specified in IR5.4.1.13.2.

For systems noted in IR5.4.1.13.2 above, the phrase '*excluded from these requirements*' is not an exclusion from the general provisions in 5.4.1 (remain operational in the event of fire). This exclusion pertains only to the detailed provisions of MSC.1/Circ. 1291, 'Guidelines for flooding detection systems on passenger ships'.

5.5 Safe area(s)

5.5.1 Functional requirements:

a) the safe area(s) are generally to be internal space(s); however, the use of an external space as a safe area may be allowed by IRS taking into account any restriction due to the area of operation and relevant expected environmental conditions;

b) the safe area(s) are to provide all occupants with the following basic services* to ensure that the health of passengers and crew is maintained:

  - sanitation;
  - water;
  - food;
  - alternate space for medical care;
  - shelter from the weather;
  - means of preventing heat stress and hypothermia;
  - light; and
  - ventilation.

c) ventilation design is to reduce the risk of smoke and hot gases affecting the use of the safe area(s); and

d) means of access to life-saving appliances are to be provided from each area identified or used as a safe area, taking into account that a main vertical zone may not be available for internal transit.

* Refer to MSC.1 Circ. 1369 "Interim explanatory notes for the assessment of passenger ship systems' capabilities after a fire or flooding casualty".

5.5.2 Alternate space for medical care

Alternate space for medical care is to conform to a standard acceptable to IRS**.

** Refer to the Guidance on the establishment of medical and sanitation related programmes for passenger ships (MSC/Circ.1129).

Section 6

Design Criteria for Systems to Remain Operational after a Fire Casualty

6.1 Application

6.1.1 Passenger ships constructed on or after 1 July 2010 having loadline length, as defined in Pt.3, Ch.1, of 120 [m] or more or having three or more main vertical zones are to comply with the provisions of this section.

6.2 Purpose

6.2.1 The purpose of this regulation is to provide design criteria for systems required to remain operational for supporting the orderly evacuation and abandonment of a ship, if the casualty threshold, as defined in 5.3 is exceeded.

6.3 Systems*

6.3.1 In case any one main vertical zone is unserviceable due to fire, the following systems are to be so arranged and segregated as to remain operational:

- fire main;
- internal communications (in support of fire-fighting as required for passenger and crew notification and evacuation);
- means of external communications;
- bilge systems for removal of fire-fighting water;
- lighting along escape routes, at assembly stations and at embarkation stations of life-saving appliances; and
- guidance systems for evacuation are to be available.

* Refer to MSC.1 Circ. 1369 “Interim explanatory notes for the assessment of passenger ship systems’ capabilities after a fire or flooding casualty”.

6.3.2 The above systems are to be capable of operation for at least 3 hours based on the assumption of no damage outside the unserviceable main vertical zone. These systems are not required to remain operational within the unserviceable main vertical zones.

6.3.3 Cabling and piping within a trunk constructed to an “A-60” standard are deemed to remain intact and serviceable while passing through the unserviceable main vertical zone for the purposes of 6.3.1. An equivalent degree of protection for cabling and piping may be approved by IRS.

Section 7

Safety Centre on Passenger Ships

7.1 Application

7.1.1 Passenger ships constructed on or after 1 July 2010 are to have on board a safety centre complying with the requirements of this regulation.

7.2 Purpose

7.2.1 The purpose of this regulation is to provide a space to assist with the management of emergency situations.

7.3 Location and arrangement

7.3.1 The safety centre are either to be a part of the navigation bridge or be located in a separate space adjacent to and having direct access to the navigation bridge, so that the management of emergencies can be performed without distracting watch officers from their navigational duties.

7.4 Layout and ergonomic design

7.4.1 The layout and ergonomic design of the safety centre are to take into account the guidelines developed by IMO*, as appropriate.

* Refer to MSC.1 Circ. 1369 “Interim explanatory notes for the assessment of passenger ship systems’ capabilities after a fire or flooding casualty”.

7.5 Communications

7.5.1 Means of communication is to be provided between the safety centre, the central control station, the navigation bridge, the engine control room, the storage room(s) for fire extinguishing system(s) and fire equipment lockers.

7.6 Control and monitoring of safety systems

7.6.1 Notwithstanding the requirements set out elsewhere in the part of the rules, the full functionality (operation, control, monitoring or any combination thereof, as required) of the safety systems listed below shall be available from the safety centre:

- all powered ventilation systems;
- fire doors;
- general emergency alarm system;
- public address system;
- electrically powered evacuation guidance systems;
- watertight and semi-watertight doors;
- indicators for shell doors, loading doors and other closing appliances;
- water leakage of inner/outer bow doors, stern doors and any other shell door;
- television surveillance system;
- fire detection and alarm system;
- fixed fire-fighting local application system(s);
- sprinkler and equivalent systems;
- water-based systems for machinery spaces;
- alarm to summon the crew;
- atrium smoke extraction system;
- flooding detection systems; and
- fire pumps and emergency fire pumps.

End of Chapter

Chapter 8

Fire Safety Systems Code

Contents

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2	*International Shore Connections*
3	*Personnel Protection*
4	*Fire Extinguishers*
5	*Fixed Gas Fire-Extinguishing Systems*
6	*Fixed Foam Fire-Extinguishing Systems*
7	*Fixed Pressure Water-Spraying and Water-Mist Fire-extinguishing Systems*
8	*Automatic Sprinkler, Fire Detection and Fire Alarm Systems*
9	*Fixed Fire Detection and Fire Alarm Systems*
10	*Sample Extraction Smoke Detection Systems*
11	*Low-location Lighting Systems*
12	*Fixed Emergency Fire Pumps*
13	*Arrangement of Means of Escape*
14	*Fixed Deck Foam Systems*
15	*Inert Gas Systems*
16	*Fixed Hydrocarbon Gas Detection Systems*
17	*Helicopter Facility Foam Firefighting Appliances*

Preamble

1. The purpose of this Code is to provide international standards of specific engineering specifications for fire safety systems required by Chapter 1 to 7 of this part of rules.

2. This code will be applicable on or after 1 July 2002.

Section 1

General

1.1 Application

1.1.1 This chapter is applicable to fire safety systems as referred to in Chapters 1 to 7 of this part of Rules.

1.1.2 Unless expressly provided otherwise, this Chapter is applicable for the fire safety systems of ships the keels of which are laid or which are at a similar stage of construction on or after 1 July 2002. However, amendments to

this chapter adopted after 1 July 2002 will apply only to ships the keels of which are laid or which are at a similar stage of construction, on or after the date on which the amendments enter into force, unless expressly provided otherwise.

1.2 Definitions

1.2.1 *Administration* means the Government of the State whose flag the ship is entitled to fly.

1.2.2 *Convention* means the International Convention for the Safety of Life at Sea, 1974, as amended.

1.2.3 *Fire Safety Systems Code (this chapter)* means the International Code for Fire Safety Systems as defined in Ch.1, 3.22.

1.2.4 For the purpose of this Chapter, definitions provided in Ch.1, Sec.3 also apply.

1.3 Use of equivalents and modern technology

In order to allow modern technology and development of fire safety systems, IRS may approve fire safety systems which are not specified in this Chapter if the requirements of Ch.6 are fulfilled.

1.4 Use of toxic extinguishing media

The use of a fire-extinguishing medium which either by itself or under expected conditions of use gives off toxic gases, liquids and other substances in such quantities as to endanger persons is not permitted.

Section 2

International Shore Connections

2.1 Application

This section details the specifications for international shore connections as required by Ch.3, 4.2.1.7.

2.2 Engineering specifications

2.2.1 Standard dimensions

The international shore connection is to be in accordance with the following specifications. (see Fig.2.1).

Standard dimensions of flanges for the international shore connection are to be in accordance with Table 2.1.

The flange is to have a flat face on one side and on the other side there is to be a permanently attached coupling that will fit the ships hydrants and hoses. The shore connection complete with gasket, four 16 [mm] bolts, 50 [mm] in length and 8 washers are to be stowed together in an easily accessible position along with other items of fire fighting outfit on board.

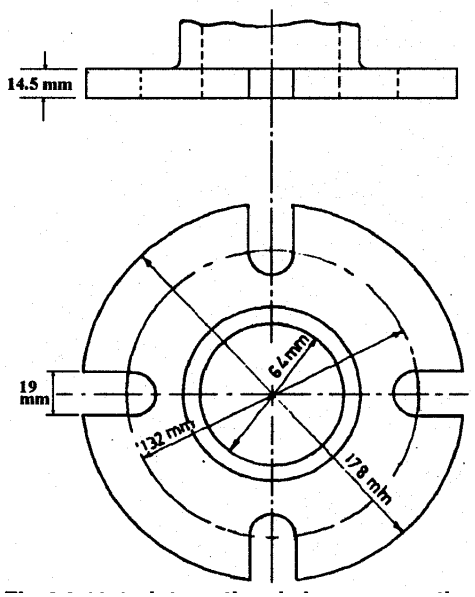


Fig.2.1 : International shore connection

Table 2.1 : Standard dimensions for international shore connections	
Description	**Dimension**
External diameter	178 [mm]
Internal diameter	64 [mm]
Bolt circle diameter	132 [mm]
Slots in flange	4 holes of 19 [mm] diameter equidistantly placed on the bolt circle, slotted to the flange periphery
Flange thickness	14.5 [mm] minimum
Bolts and nuts	4 each of 16 [mm] diameter, 50 [mm] in length
Material	steel or any other suitable material and designed for 1.0 [N/mm^2] service
Gasket Material	any material suitable for 1.0 [N/mm^2] pressure service

Section 3

Personnel Protection

3.1 Application

This section details the specifications for personnel protection as required by this part of the Rules.

3.2 Engineering specifications

3.2.1 Fire-fighter's outfit

A fire-fighter's outfit is to consist of a set of personal equipment and a breathing apparatus.

3.2.1.1 Personal equipment

Personal equipment is to consist of the following:

.1 protective clothing of material to protect the skin from the heat radiating from the fire and from burns and scalding by steam. The outer surface is to be water-resistant;

.2 boots of rubber or other electrically non-conducting material;

.3 rigid helmet providing effective protection against impact;

.4 electric safety lamp (hand lantern) of an approved type with a minimum burning period of 3 h. Electric safety lamps on tankers and those intended to be used in hazardous areas are to be of an explosion-proof type; and

.5 axe with a handle provided with high-voltage insulation.

3.2.1.2 Breathing apparatus

.1 Breathing apparatus is to be a self-contained compressed air-operated breathing apparatus for which the volume of air contained in the cylinders is to be at least 1,200 litres, or other self-contained breathing apparatus which is to be capable of functioning for at least 30 min. All air cylinders for breathing apparatus are to be interchangeable.

.2 Compressed air breathing apparatus is to be fitted with an audible alarm and a visual or other device which will alert the user before the volume of the air in the cylinder has been reduced to no less than 200 litres.

3.2.1.3 Lifeline

For each breathing apparatus a fireproof lifeline of at least 30 [m] in length is to be provided. The lifeline is to successfully pass an approval test by statical load of 3.5 [kN] for 5 min without failure. The lifeline is to be

capable of being attached by means of a snap-hook to the harness of the apparatus or to a separate belt in order to prevent the breathing apparatus becoming detached when the lifeline is operated.

3.2.2 Emergency escape breathing devices (EEBD)

3.2.2.1 General

3.2.2.1.1 An EEBD is a supplied air or oxygen device only used for escape from a compartment that has a hazardous atmosphere and is to be of an approved type.

3.2.2.1.2 EEBDs are not to be used for fighting fires, entering oxygen deficient voids or tanks, or worn by fire-fighters. In these events, a self-contained breathing apparatus, which is specifically suited for such applications, are to be used.

3.2.2.2 Definitions

3.2.2.2.1 Face piece means a face covering that is designed to form a complete seal around the eyes, nose and mouth which is secured in position by a suitable means.

3.2.2.2.2 Hood means a head covering which completely covers the head, neck, and may cover portions of the shoulders.

3.2.2.2.3 Hazardous atmosphere means any atmosphere that is immediately dangerous to life or health.

3.2.2.3 Particulars

3.2.2.3.1 The EEBD are to have a service duration of at least 10 min.

3.2.2.3.2 The EEBD is to include a hood or full face piece, as appropriate, to protect the eyes, nose and mouth during escape. Hoods and face pieces are to be constructed of flame resistant materials and include a clear window for viewing.

3.2.2.3.3 An inactivated EEBD is to be capable of being carried hands-free.

3.2.2.3.4 An EEBD, when stored, is to be suitably protected from the environment.

3.2.2.3.5 Brief instructions or diagrams clearly illustrating their use is to be clearly printed on the EEBD. The donning procedures are to be quick and easy to allow for situations where there is little time to seek safety from a hazardous atmosphere.

3.2.2.4 Markings

Maintenance requirements, manufacturer's trademark and serial number, shelf life with accompanying manufacture date and name of approving authority are to be printed on each EEBD. All EEBD training units are to be clearly marked.

Section 4

Fire Extinguishers

4.1 Application

4.1.1 This section details the specifications for fire extinguishers as required by Ch.1 to 7.

4.1.2 Type approval

All fire extinguishers are to be of approved types and designs based on the requirements of the National Statutory Authority. (Refer to the Improved Guidelines for marine portable fire extinguishers adopted by IMO resolution A. 951(23)).

4.2 Engineering specifications

4.2.1 Fire extinguisher

4.2.1.1 Quantity of medium

4.2.1.1.1 Each powder or carbon dioxide extinguisher is to have a capacity of at least 5 [kg] and each foam extinguisher is to have a capacity of at least 9 [*l*]. The mass of all portable fire extinguishers is not to exceed 23 [kg] and they are to have a fire-extinguishing capability at least equivalent to that of a 9*l* fluid extinguisher.

4.2.1.2 Recharging

Only refills approved for the fire extinguisher in question is to be used for recharging.

4.2.2 Portable foam applicators

4.2.2.1 A portable foam applicator unit is to consist of foam nozzle/branch pipe, either of a self-inducting type or in combination with a separate inductor capable of being connected to the fire main by a fire hose together with a portable tank containing at least 20 litres of foam concentrate and at least one spare tank of foam concentrate of the same capacity.

4.2.2.2 System performance

4.2.2.2.1 The nozzle branch pipe and inductor are to be capable of producing effective foam suitable for extinguishing an oil fire at a foam solution flow rate of at least 200 l/min. at the nominal pressure in the fire main.

4.2.2.2.2 The foam concentrate is to be approved by IRS based on the guidelines developed by IMO (Guidelines for the performance and testing criteria and surveys of low-expansion foam concentrates for fixed fire extinguishing systems (MSC.1/Circ.1312 and MSC.1/Circ.1312/Corr.1).

4.2.2.2.3 The values of the foam expansion and drainage time of the foam produced by the portable foam applicator unit is not to differ more than 10% of that determined in 4.2.2.2.2.

4.2.2.2.4 The portable foam applicator unit is to be designed to withstand clogging, ambient temperature changes, vibration, humidity, shock, impact and corrosion normally encountered on ships.

Section 5

Fixed Gas Fire-Extinguishing Systems

5.1 Application

This section details the specifications for fixed gas fire-extinguishing systems as required by Ch.3, Sec.4.

5.2 Engineering specifications

5.2.1 General

5.2.1.1 Fire-extinguishing medium

5.2.1.1.1 Where the quantity of the fire-extinguishing medium is required to protect more than one space, the quantity of medium available need not be more than the largest quantity required for any one space so protected. The system is to be fitted with normally closed control valves arranged to direct the medium into the appropriate space. Adjacent spaces with independent ventilation systems not separated by atleast A-0 class divisions are to be considered as the same space.

5.2.1.1.2 The volume of starting air receivers, converted to free air volume, is to be added to the gross volume of the machinery space when calculating the necessary quantity of the fire-extinguishing medium. Alternatively, a discharge pipe from the safety valves may be fitted and led directly to the open air.

5.2.1.1.3 Means are to be provided for the crew to safely check the quantity of the fire-extinguishing medium in the containers. It is not to be necessary to move the containers completely from their fixing position for this purpose. For carbon dioxide systems, hanging bars for a weighing device above each bottle row, or other means are to be provided. For other types of extinguishing media, suitable surface indicators may be used

5.2.1.1.4 Containers for the storage of fire-extinguishing medium and associated pressure components are to be designed to pressure codes of practice to the satisfaction of IRS having regard to their locations and maximum ambient temperature expected in service*.

* Publication ISO – 9809/1 : Refillable seamless steel gas cylinders (design, construction and testing);
ISO – 3500 : Seamless steel CO_2 cylinders. For fixed fire-fighting installations on ships, specifying the principal external dimensions, accessories, filling ratio and marking for seamless steel CO_2 cylinders used in fixed

fire-fighting installations on ships, in order to facilitate their interchange ability;
ISO – 13769 : Gas cylinders – Stamp marking;
ISO – 6406 : Periodic inspection and testing of seamless steel gas cylinders.
ISO – 9329 Part 1 : Seamless steel tubes for pressure purpose – Technical delivery conditions – Part 1 – Unalloyed steels with specified room temperature properties;
ISO – 9329 Part 2 – Seamless steel tubes for pressure purposes – Technical delivery conditions – Part 2 – Unalloyed and alloyed steels with specified elevated temperature properties;
ISO – 9330 Part 1 – Welded steel tubes for pressure purposes – Technical delivery conditions – Part 1 – Unalloyed steel tubes with specified room temperature properties;
ISO 9330 Part 2 – Welded steel tubes for pressure purposes – Technical delivery conditions – Part 2 – Electric resistance and induction welded unalloyed and alloyed steel tubes with specified elevated temperature properties.

5.2.1.2 Installation requirements

5.2.1.2.1 The piping for the distribution of fire-extinguishing medium is to be arranged and discharge nozzles so positioned that a uniform distribution of the medium is obtained. System flow calculations are to be performed to ensure uniform distribution and submitted for consideration of IRS.

5.2.1.2.2 Except as otherwise permitted by IRS, pressure containers required for the storage of fire-extinguishing medium, other than steam, is to be located outside the protected spaces in accordance with Ch.3, 4.4.3.

5.2.1.2.3 Sufficient spare parts for the system are to be stored on board, and the proposals, in each case, will be specially considered by IRS.

IR5.2.1.2.3 The piping is to be of adequate size and so arranged as to provide effective distribution of gas. In holds exceeding 18 [m] in length, there are to be at least two pipes, one of which is to be fitted in the forward part and one in the after part. Separate pipes are to be provided for lower hold and tween decks. All pipes are to be arranged to be self draining and are not to be led through refrigerated spaces unless the pipes are specially insulated.

5.2.1.2.4 In piping sections where valve arrangements introduce sections of closed piping, such sections are to be fitted with a pressure relief valve and the outlet of the valve is to be led to open deck.

IR5.2.1.2.4 Where carbon dioxide is used, all steel pipes are to be galvanized, at least internally.

5.2.1.2.5 All discharge piping, fittings and nozzles in the protected spaces are to be constructed of materials having a melting temperature which exceeds 925°C. The piping and associated equipment is to be adequately supported.

IR5.2.1.2.5 Distribution pipes are not to be smaller than 25 [mm] bore for inert gas and 20 [mm] bore for carbon dioxide.

5.2.1.2.6 A connection is to be installed in the discharge piping to permit the air testing as required by paragraph 5.2.2.3.1.

IR5.2.1.2.6 The minimum wall thicknesses of steel pipes of carbon dioxide systems are given in Table 5.2.1.2.5.

IR5.2.1.2.7 Slightly smaller thicknesses will be accepted where these meet recognized National Standards.

IR5.2.1.2.8 Where threaded pipes are used, the minimum wall thicknesses given in Table 5.2.1.2.5 are to be measured at the bottom of the thread.

IR5.2.1.2.9 For pipes of diameter larger than those shown in Table 5.2.1.2.5 the thickness will be specially considered.

Table 5.2.1.2.5 : Minimum thickness of steel pipes		
External diameter D, [mm]	**Thickness [mm]**	
	From bottles to distribution valve	**From distribution valve to nozzles**
21.2 to 26.9	3.2	2.6
30.0 to 48.3	4.0	3.2
51.0 to 60.3	4.5	3.6
63.5 to 76.1	5.0	3.6
82.5 to 88.9	5.6	4.0
101.6	6.3	4.0
108.0 to 114.3	7.1	4.5
127.0	8.0	4.5
133.0 to 139.7	8.0	5.0
152.4 to 168.3	8.8	5.6

IR5.2.1.2.10 In general, the minimum thickness given in Table 5.2.1.2.5 is the nominal wall thickness, and no allowance need be made for negative tolerance and reduction in thickness due to bending.

IR5.2.1.2.11 For systems using carbon dioxide, the distribution manifolds and the pipes between the storage cylinders and the distribution valve are to be guaranteed by the manufacturers to have been satisfactorily hydraulically tested to a pressure of at least 12.2 [N/mm^2]. The manufacturers are to guarantee that at least 10 per cent of the pipes from the distribution valve to the discharge nozzles have been satisfactorily hydraulically tested to a pressure of at least 1.0 [N/mm^2]. On completion of the installation the joints are to be tested pneumatically to a pressure of atleast 0.69 [N/mm^2] with the discharge openings closed.

5.2.1.3 System control requirements

5.2.1.3.1 The necessary pipes for conveying fire-extinguishing medium into the protected spaces are to be provided with control valves so marked as to indicate clearly the spaces to which the pipes are led. Suitable provision is to be made to prevent inadvertent release of the medium into the space. Where a cargo space fitted with a gas fire-extinguishing system is used as a passenger space, the gas connection is to be blanked during such use. Blank flanges fitted in gas distribution pipes are to be of the 'spectacle' type. The nuts for the securing bolts are to be of non-corrodible metal. The pipes may pass through accommodations provided that they are of substantial thickness and that their tightness is verified with a pressure test, after their installation, at a pressure head not less than 5 [N/mm^2]. In addition, pipes passing through accommodation areas are to be joined only by welding and are not to be fitted with drains or other openings within such spaces. The pipes are not to pass through refrigerated spaces.

5.2.1.3.2 Means are to be provided for automatically giving audible and visual warning of the release of fire extinguishing medium into any ro-ro spaces, container holds equipped with integral reefer containers, spaces accessible by doors or hatches and other spaces in which personnel normally work or to which they have access. The audible alarms are to be located so as to be audible throughout the protected space with all machinery operating and the alarms are to be capable of being distinguished from other audible alarms by adjustment of sound pressure or sound patterns. The pre-discharge alarm is to be automatically activated (e.g. by opening of the release cabinet door). The alarm is to operate for the length of time needed to evacuate the space, but not less than 20 [s], before the medium is released. Conventional cargo spaces and small spaces (such as compressor rooms, paint lockers, etc.) with only a local release need not be provided with such alarm.

IR5.2.1.3.2 a) Reference is made to the Code on alerts and Indicators (AI Code), 1995 (Resolution A.1021(26)). The pre-discharge alarms may be pneumatically (by the

extinguishing medium or by air) or electrically operated, with requirements as follows:

- If electrically operated, the alarms are to be supplied with power from the main and an emergency source of power.

- If pneumatically operated by air, the air supplied is to be dry and clean and the supply reservoir is to be fitted with a low pressure alarm. The air supply may be taken from the starting air receivers. Any stop valve fitted in the air supply line is to be locked or sealed in the open position. Any electrical components associated with the pneumatic system are to be powered from the main and an emergency source of electrical power.

IR5.2.1.3.2 b) Conventional cargo spaces means cargo spaces other than ro-ro spaces or container holds equipped with integral reefer containers, and they need not be provided with means for automatically giving audible and visual warning of the release.

IR5.2.1.3.2 c) The requirements mentioned in 5.2.2.2 are also applicable for the spaces mentioned in 5.2.1.3.2.

5.2.1.3.3 The means of control of any fixed gas fire-extinguishing system are to be readily accessible, simple to operate and are to be grouped together in as few locations as possible at positions not likely to be cut off by a fire in a protected space. At each location there are to be clear instructions relating to the operation of the system having regard to the safety of personnel.

IR5.2.1.3.3 For control of fixed gas fire extinguishing media stored in a room located forward of cargo holds and intended for protecting cargo holds, refer Ch.3, IR4.4.3.

5.2.1.3.4 Automatic release of fire-extinguishing medium is not to be permitted, except as permitted by IRS.

IR5.2.1.4 Alarms in pump room.

IR5.2.1.4.1 The audible alarms fitted in cargo pump rooms of ships intended for the carriage of cargo oil having a flash point below 60°C (closed cup test), to warn of the release of extinguishing medium in such spaces, may be of the pneumatic or electrical type.

IR5.2.1.4.2 Where pneumatically operated alarms are fitted which require periodic testing, carbon dioxide is not to be used as an operating medium. Air operated alarms may be used provided that the air supply is clean and dry.

IR5.2.1.4.3 Where electrically operated alarms are used, the arrangements are to be such that the electric operating mechanism is located outside the pump room except where alarms are certified intrinsically safe.

5.2.2 Carbon dioxide systems

5.2.2.1 Quantity of fire extinguishing medium

5.2.2.1.1 For cargo spaces the quantity of carbon dioxide available, unless otherwise provided, is to be sufficient to give a minimum volume of free gas equal to 30% of the gross volume of the largest cargo space to be protected in the ship.

5.2.2.1.2 For vehicle spaces and ro-ro spaces which are not special category spaces, the quantity of carbon dioxide available are to be at least sufficient to give a minimum volume of free gas equal to 45% of the gross volume of the largest such cargo space which is capable of being sealed, and the arrangements are to be such as to ensure that at least two thirds of the gas required for the relevant space is introduced within 10 min. Carbon dioxide systems are not to be used for the protection of special category spaces.

5.2.2.1.3 For machinery spaces the quantity of carbon dioxide carried is to be sufficient to give a minimum volume of free gas equal to the larger of the following volumes, either:

.1 40% of the gross volume of the largest machinery space so protected, the volume to exclude that part of the casing above the level at which the horizontal area of the casing is 40% or less of the horizontal area of the space concerned taken midway between the tank top and the lowest part of the casing; or

.2 35% of the gross volume of the largest machinery space protected, including the casing.

5.2.2.1.4 The percentages in 5.2.2.1.3 above may be reduced to 35% and 30%, respectively, for cargo ships of less than 2,000 gross tonnage where two or more machinery

spaces, which are not entirely separate, are considered as forming one space.

5.2.2.1.5 The volume of free carbon dioxide is to be calculated at 0.56 [m^3/kg].

5.2.2.1.6 For machinery spaces the fixed piping system is to be such that 85 per cent of the gas can be discharged into the space within 2 minutes. Calculation to prove the discharge time may be submitted for consideration of IRS.

5.2.2.1.7 For container and general cargo spaces (primarily intended to carry a variety of cargoes separately secured or packed) the fixed piping system is to be such that at least two thirds of the gas can be discharged into the space within 10 min. For solid bulk cargo spaces the fixed piping system is to be such that at least two thirds of the gas can be discharged into the space within 20 min. The system controls are to be arranged to allow one third, two thirds or the entire quantity of gas to be discharged based on the loading condition of the hold.

5.2.2.2 Controls

Carbon dioxide systems for the protection of ro-ro spaces, container holds equipped with integral reefer containers, spaces accessible by doors or hatches, and other spaces in which personnel normally work or to which they have access are to comply with the following requirements:

.1 Two separate controls are to be provided for releasing carbon dioxide into a protected space and to ensure the activation of the alarm. One control is to be used for opening the valve of the piping which conveys the gas into the protected space and a second control is to be used to discharge the gas from its storage containers. Positive means are to be provided so they can only be operated in that order; and

IR.1 The pre-discharge alarm may be activated before the two separate system release controls are operated (e.g. by a micro-switch that activates the pre-discharge alarm upon opening the release cabinet door as per 5.2.1.3.2). Therefore, the two separate controls for releasing carbon dioxide into the protected space (i.e. one control to open the valve of the piping which conveys the gas into the protected space and a second control used to discharge the gas from its storage containers) as per 5.2.2.2.1 can be independent of the control for activating the alarm (provided the minimum length of time required for operation of the alarm before the release of the medium is as per 5.2.1.3.2).

A single control for activation of the alarm is sufficient.

The “positive means” referred to in 5.2.2.2.1 for the correct sequential operation of the controls, is to be achieved by a mechanical and/or electrical interlock that does not depend on any operational procedure to achieve the correct sequence of operation.

.2 The two controls are to be located inside a release box clearly identified for the particular space. If the box containing the controls is to be locked, a key to the box is to be in a break-glass-type enclosure conspicuously located adjacent to the box.

IR.2 The above requirements for controls apply to the spaces requiring audible warning identified in 5.2.1.3.2 and IR5.2.1.3.2 b).

5.2.2.3 Testing of the installation

When the system has been installed, pressure-tested and inspected, the following is to be carried out:

.1 a test of the free air flow in all pipes and nozzles; and

.2 a functional test of the alarm equipment.

5.2.3 Requirements of steam systems

5.2.3.1 Steam is not generally accepted as a fire extinguishing medium in fixed-fire extinguishing systems. However restricted use of steam is permitted as an addition to the required fire extinguishing medium.

5.2.3.2 The boiler or boilers available for supplying steam are to have an evaporation of at least 1 [kg] of steam per hour for each 0.75 [m^3] of the gross volume of the largest space so protected. In addition to complying with the foregoing requirements, the systems in all respects are to be to the satisfaction of IRS. (Refer Part 4 of the Rules).

5.2.3.3 Steam is not to be used in spaces containing explosives.

5.2.3.4 Steam used for fire extinguishing purposes is not to be obtained from a supply of superheated steam.

5.2.4 Equivalent fixed gas fire-extinguishing systems for machinery spaces and cargo pump rooms

Fixed gas fire-extinguishing systems equivalent to those specified in 5.2.2 and 5.2.3 will be approved by IRS based on the requirements of the National Statutory Authority. (Refer to the Revised guidelines for the approval of equivalent fixed gas fire-extinguishing systems, as referred to in SOLAS 74, for machinery spaces and cargo pump rooms (MSC/Circ.848 *amended by MSC.1/Circ.1267)* and the Guidelines for the approval of fixed aerosol fire-extinguishing systems equivalent to fixed gas fire-extinguishing systems, as referred to in SOLAS 74, for machinery spaces (MSC.1/Circ.1270 and MSC.1/Circ.1270/ Corr.1)).

IR5.2.4 Container storage arrangement for equivalent fixed gas fire extinguishing systems

Agent containers having / filled with fire extinguishing medium, stored in a protected space are to be distributed throughout the space with bottles or groups of bottles located in at least six separate locations. Duplicate power release lines are to be arranged to release all bottles simultaneously. The release lines are to be so arranged that in the event of damage to any power release line, five sixth of the fire extinguishing gas can still be discharged. The bottle valves are considered to be part of the release lines and a single failure are to include also failure of the bottle valve.

For systems that need less than six cylinders (using the smallest bottles available), the total amount of extinguishing gas on the bottles are to be such that in the event of a single failure to one of the release lines (including bottle valve), five sixth of the fire extinguishing gas can still be discharged. This may be achieved by for instance using more extinguishing gas than required so that if one bottle is not discharging due to a single fault, the remaining bottles will discharge the minimum five sixth of the required amount of gas. This can be achieved with minimum two bottles. However, NOAEL values calculated at the highest expected engine room temperature are not to be exceeded when discharging the total amount of extinguishing gas simultaneously.

Systems that can not comply with the above, for instance systems using only one bottle located inside the protected space, can not be accepted. Such systems are to be designed with the bottle(s) located outside the protected space, in a dedicated room in compliance with Ch.3, 4.4.3.

IR5.2.5 Low-pressure carbon dioxide smothering systems

5.2.5.1 General

a) The rated amount of carbon dioxide, time of discharge into the protected space, location of nozzles in the protected spaces and signals warning that the smothering system is activated is to comply with the requirements of 5.2.2 relating to CO_2 high-pressure systems.

b) Vessel(s), refrigerating plants, control devices and other equipment of the smothering system are to be located in a space complying with rules applying to CO_2 high-pressure systems. The system control devices and the refrigerating plants are to be located within the same room where the pressure vessels are stored.

5.2.5.2 Vessel(s) and relevant devices

a) The rated amount of liquid carbon dioxide is to be stored in vessel(s) under the working pressure in the range of 18 to 22 bar. The normal liquid charge in the container is to be limited to provide sufficient vapour space to allow for expansion of the liquid under the maximum storage temperatures than can be obtained corresponding to the setting of the pressure relief valves but is not to exceed 95% of the volumetric capacity of the container.

b) The vessel(s) is to be designed, constructed and tested in accordance with the requirements of Pt.4, Ch.5 for pressure vessels. For this purpose the design pressure is to be taken not less than the relief valve setting. Besides, provision is to be made for:

   - pressure gauge

   - high pressure alarm: not more than setting of the relief valve

- low pressure alarm: not less than 18 bar

- branch pipes with stop valves for filling the vessel

- discharge pipes

- liquid CO_2 level indicator, fitted on the vessel(s)

- two safety relief valves arranged so that either valve can be shut off while the other is connected to the vessel. The setting of the relief valves is to be not less than 1.1 times working pressure. The capacity of each valve is to be such that the vapours generated under fire condition can be discharged with a pressure rise not more than 20% above the setting pressure. The discharge from the safety valves is to be led to the open.

c) The vessel(s) and outgoing pipes permanently filled with carbon dioxide are to have thermal insulation preventing the operation of the safety valve in 24 hours after de-energizing the plant, at ambient temperature of 45°C and an initial pressure equal to the starting pressure of the refrigeration unit. The insulating materials and their liners are to be to the satisfaction of IRS, having in mind, in particular, their fire resistance and mechanical properties, as well as protection against penetration of water vapours.

5.2.5.3 Refrigerating plant

a) The vessel(s) is to be serviced by two automated completely independent refrigerating units solely intended for this purpose, each comprising a compressor and the relevant prime mover, evaporator and condenser.

b) The refrigerating plant is to comply with the relevant requirements of Pt.5, Ch.23 in so far as they are applicable. The refrigerating capacity and the automatic control of each unit are to be so as to maintain the required temperature under conditions of continuous operation during 24 hours at the sea temperature upto 32°C and ambient air temperature upto 45°C.

c) In the event of failure of either one of the refrigerating units the other is to be actuated automatically. Provision is to be made for local manual control of the refrigerating plant.

d) Each electric refrigerating unit is to be supplied from the main switchboard busbars by a separate feeder.

Cooling water supply to the refrigerating plant (where required) is to be provided from at least two circulating pumps one of which being used as a stand-by. The stand-by pump may be a pump used for other services so long as its use for cooling would not interfere with any other essential service of the ship.

Cooling water is to be taken from not less than two sea connections, preferably one port and one starboard.

5.2.5.4 Pipes and fittings

a) The pipes, valves and fittings are to be in accordance with the requirements of IRS for a design pressure not less than the design pressure of the CO_2 vessels.

b) Safety relief devices are to be provided in each section of pipe that may be isolated by block valves and in which there could be a build-up of pressure in excess of the design pressure of any of the components.

c) The piping system is to be designed in such a way that the CO_2 flows through in liquid phase upto the discharge nozzles. To this end the pressure at the nozzles are to be not less than 10 bar.

5.2.5.5 Control of smothering system operation

The machinery alarm system is to be equipped with audible and visual alarms activated when:

- the pressure in the vessel(s) reaches the low and high values according to 5.2.5.2 a)

- any one of the refrigerating units fails to operate

- the lowest permissible level of the liquid in the vessels is reached.

5.2.5.6 Release control

a) The release of CO_2 is to be initiated manually.

b) If a device is provided which automatically regulates the discharge of the rated quantity of carbon dioxide into the protected spaces it is to be also possible to regulate the discharge manually.

   If the system serves more than one space, means for control of discharge quantities of CO_2 is to be provided, e.g. automatic timer or accurate level indicators located at the control position(s).

5.2.5.7 Testing

a) The pipes, valves and fittings and assembled system are to be tested to satisfaction of the Surveyors.

b) The pipes from the vessel(s) to the release valves on the distribution manifold are to be subject to a pressure test to not less than 1.5 times the set pressure of the safety relief valves.

c) The pipes from the release valves on the distribution manifold to the nozzles are to be tested for tightness and free flow of CO_2, after assembly on board.

d) The refrigerating plant, after having been fitted on board is to be checked for its proper operation.

At the judgement of IRS, a discharge test may be required to check the fulfillment of the requirements of 5.2.5.4 c).

Section 6

Fixed Foam Fire-Extinguishing Systems

6.1 Application

This section details the specifications for fixed foam fire-extinguishing systems for the protection of machinery spaces in accordance with 4.4.1.1.2 of Chapter 3 , cargo spaces in accordance with 4.7.1.1 of Chapter 3, cargo pump-rooms in accordance with 4.9.1.2 of Chapter 3 and vehicle, special category and ro-ro spaces in accordance with 3.6.1.3 of Chapter 7. This section does not apply to cargo pump-rooms of chemical tankers carrying liquid cargoes referred to in 1.5.2 of Chapter 1, unless the Administration specifically accepts the use of these systems based on additional tests with alcohol-based fuel and alcohol resistant foam. Unless expressly provided otherwise, the requirements of this section are to apply to ships constructed on or after 1 January 2014.

6.2 Definitions

6.2.1 *Design filling rate* is at least the minimum nominal filling rate used during the approval tests.

6.2.2 *Foam is the extinguishing medium* produced when foam solution passes through a foam generator and is mixed with air.

6.2.*3 Foam solution* is a solution of foam concentrate and water.

6.2.*4 Foam concentrate* is a liquid which, when mixed with water in the appropriate concentration forms a foam solution.

6.2.5 *Foam delivery ducts* are supply ducts for introducing high-expansion foam into the protected space from foam generators located outside the protected space.

6.2.6 *Foam mixing ratio* is the percentage of foam concentrate mixed with water forming the foam solution.

6.2.*7 Foam generators* are discharge devices or assemblies through which high-expansion foam solution is aerated to form foam that is discharged into the protected space. Foam generators using inside air typically consist of a nozzle or set of nozzles and a casing. The casing is typically made of perforated steel/stainless steel plates shaped into a box that enclose the nozzle(s). Foam generators using outside air typically consist of nozzles enclosed within a casing that spray onto a screen. An electric, hydraulic or pneumatically driven fan is provided to aerate the solution.

6.2.8 *High-expansion foam fire-extinguishing systems* are fixed total flooding extinguishing systems that use either inside air or outside air for aeration of the foam solution. A high-expansion foam system consists of both the foam generators and the dedicated foam concentrate approved during the fire testing specified in 6.3.1.2

6.2.9 *Inside air foam system* is a fixed high-expansion foam fire-extinguishing system with foam generators located inside the protected space and drawing air from that space.

6.2.10 *Nominal flow rate* is the foam solution flow rate expressed in l/min

6.2.11 *Nominal application rate* is the nominal flow rate per area expressed in l/min/m^2

6.2.12 *Nominal foam expansion ratio* is the ratio of the volume of foam to the volume of foam solution from which it was made, under non-fire conditions, and at an ambient temperature of e.g. around 20ºC.

6.2.13 *Nominal foam production* is the volume of foam produced per time unit, i.e. nominal flow rate times nominal foam expansion ratio, expressed in m^3/min.

6.2.14 *Nominal filling rate* is the ratio of nominal foam production to the area, i.e. expressed in m^2/min.

6.2.15 *Nominal filling time* is the ratio of the height of the protected space to the nominal filling rate, i.e. expressed in minutes.

6.2.16 *Outside air foam system* is a fixed high-expansion foam system with foam generators installed outside the protected space that are directly supplied with fresh air.

6.3 Fixed high-expansion foam fire-extinguishing systems

6.3.1 Principal Performance

6.3.1.1 The system is to be be capable of manual release, and be designed to produce foam at the required application rate within 1 minute of release. Automatic release of the system is not permitted unless appropriate operational measures or interlocks are provided to prevent any local application systems required by 4.5.6 of chapter 3 from interfering with the effectiveness of the system.

6.3.1.2 The foam concentrates are to be approved based on the guidelines developed by IMO (MSC.1/Circ. 670). Different foam concentrate types are not to be mixed in a high-expansion foam system.

6.3.1.3 The system is to be capable of fire extinction and manufactured and tested to the satisfaction of IRS based on the guidelines developed by IMO (MSC.1/Circ.1384).

6.3.1.4 The system and its components are to be suitably designed to withstand ambient temperature changes, vibration, humidity, shock, clogging and corrosion normally encountered on ships. Piping, fittings and related components inside the protected spaces (except gaskets) are to be designed to withstand 925°C

6.3.1.5 System piping, foam concentrate storage tanks, components and pipe fittings in contact with the foam concentrate are to be compatible with the foam concentrate and be constructed of corrosion resistant materials such as stainless steel or equivalent. Other system piping and foam generators are to be full galvanized steel or equivalent. Distribution pipework has to have self-draining capability.

6.3.1.6 Means for testing the operation of the system and assuring the required pressure and flow are to be provided by pressure gauges at both inlets (water and foam concentrate supply) and at the outlet of the foam proportioner. A test valve is to be installed on the distribution piping downstream of the foam proportioner, along with orifices which reflect the calculated pressure drop of the system. All sections of piping are to be provided with connections for flushing, draining and purging with air. All nozzles are to be able to be removed for inspection in order to prove clear of debris.

6.3.1.7 Means are to be provided for the crew to safely check the quantity of foam concentrate and take periodic control samples for foam quality.

6.3.1.8 Operating instructions for the system are to be displayed at each operating position.

6.3.1.9 Spare parts are to be provided based on the manufacturer's instruction.

6.3.1.10 If an internal combustion engine is used as a prime mover for the seawater pump for the system, the fuel oil tank to the prime

mover is to contain sufficient fuel to enable the pump to run on full load for at least 3 h and sufficient reserves of fuel are to be available outside the machinery space of category A to enable the pump to be run on full load for an additional 15 h. If the fuel tank serves other internal combustion engines simultaneously, the total fuel tank capacity is to be adequate for all connected engines.

6.3.1.11 The arrangement of foam generators and piping in the protected space is not to interfere with access to the installed machinery for routine maintenance activities.

6.3.1.12 The system source of power supply, foam concentrate supply and means of controlling the system are to be readily accessible and simple to operate, and are to be arranged at positions outside the protected space not likely to be cut off by a fire in the protected space. All electrical components directly connected to the foam generators are to have at least an IP 54 rating.

6.3.1.13 The piping system is to be sized in accordance with a hydraulic calculation technique to ensure availability of flows and pressures required for correct performance of the system.

6.3.1.14 The arrangement of the protected spaces is to be such that they may be ventilated as the space is being filled with foam. Procedures are to be provided to ensure that upper level dampers, doors and other suitable openings are kept open in case of a fire. For inside air foam systems, spaces below 500 m^3 need not comply with this requirement.

6.3.1.15 Onboard procedures are to be established to require personnel re-entering the protected space after a system discharge to wear breathing apparatus to protect them from oxygen deficient air and products of combustion entrained in the foam blanket.

6.3.1.16 Installation plans and operating manuals are to be supplied to the ship and be readily available on board. A list or plan is to be displayed showing spaces covered and the location of the zone in respect of each section. Instructions for testing and maintenance are to be available on board.

6.3.1.17 All installation, operation and maintenance instructions/plans for the system are to be in the working language of the ship. If the working language of the ship is not English, French, nor Spanish, a translation into one of these languages is to be included.

6.3.1.18 The foam generator room is to be ventilated to protect against overpressure, and are to be heated to avoid the possibility of freezing.

6.3.1.19 The quantity of foam concentrate available is to be sufficient to produce a volume of foam equal to at least five times the volume of the largest protected space enclosed by steel bulkheads, at the nominal expansion ratio, or enough for 30 min of full operation for the largest protected space, whichever is greater.

6.3.1.20 Machinery spaces, cargo pump-rooms, vehicle spaces, ro-ro spaces and special category spaces are to be provided with audible and visual alarms within the protected space warning of the release of the system. The alarms are to operate for the length of time needed to evacuate the space, but in no case less than 20 s.

6.3.2 Inside air foam systems:

6.3.2.1 *Systems for the protection of machinery spaces and cargo pump rooms*

6.3.2.1.1 The system is to be supplied by both main and emergency sources of power. The emergency power supply is to be provided from outside the protected space.

6.3.2.1.2 Sufficient foam-generating capacity is to be provided to ensure the minimum design filling rate for the system is met and in addition is to be adequate to completely fill the largest protected space within 10 min.

IR 6.3.2.1.2 The term “largest protected space” applies to a machinery space of category A protected by a fixed high-expansion foam fire-extinguishing system complying with this section. Where such a machinery space includes a casing (e.g. an engine casing in a machinery space of category A containing internal combustion machinery, and/or a boiler), the volume of such casing, above the level up to which foam is to be filled to protect the highest position of the fire risk objects within the machinery space, need not be included in the volume of the protected space (Fig 6.3.2.1.2(a)). The level up to which foam is to be filled to protect the highest positioned fire risk objects within the machinery space is not to be less than:

1 m above the highest point of any such object; or

The lowest part of the casing, whichever is higher (Fig 6.3.2.1.2 (a)).

Where such a machinery space does not include a casing, the volume of the largest protected space is to be that of the space in its entirety, irrespective of the location of any fire risk object therein (Fig 6.3.2.1.2 (b)).

Fire risk objects include, but may not be limited to, those listed in 3.31 and 3.34 of chapter 1. Although not referred to in those Clauses, they may also include items having a similar fire risk such as exhaust gas boilers or oil fuel tanks

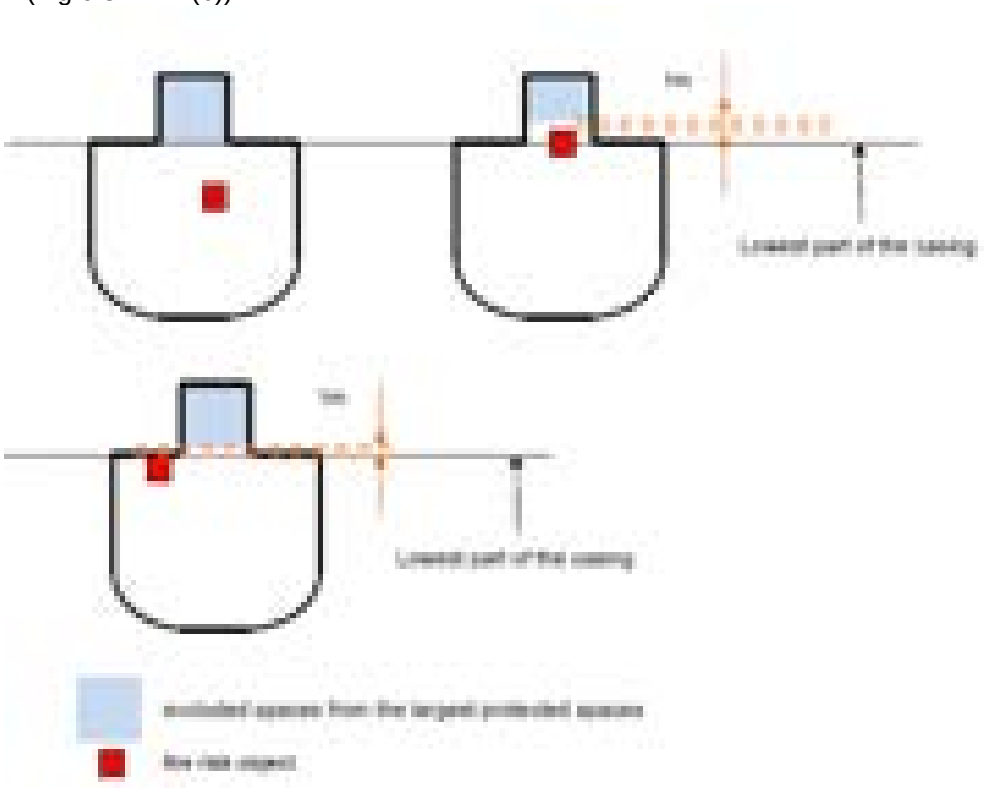


Fig 6.3.2.1.2 (a) Machinery Space including Casing

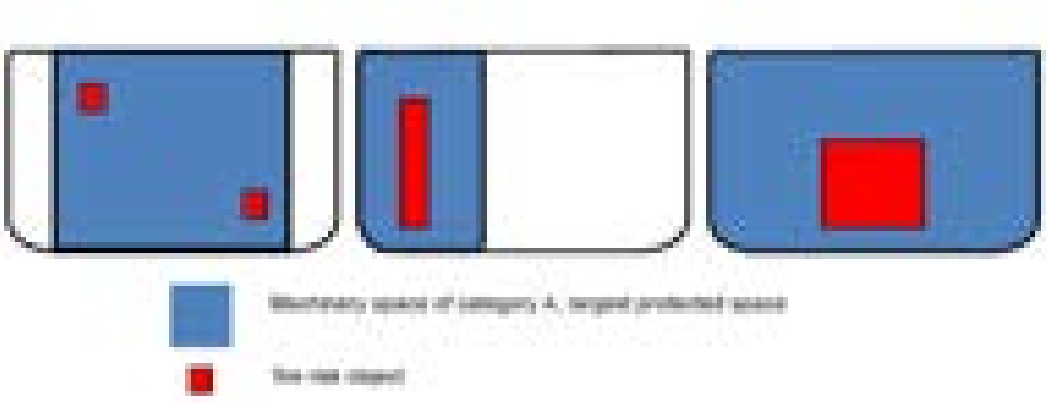


Fig 6.3.2.1.2 (b) Machinery Space not including a Casing

6.3.2.1.3 The arrangement of foam generators is in general to be designed based on the approval test results. A minimum of two generators are to be installed in every space containing combustion engines, boilers, purifiers, and similar equipment. Small workshops and similar spaces may be covered with only one foam generator.

6.3.2.1.4 Foam generators are to be uniformly distributed under the uppermost ceiling in the protected spaces including the engine casing. The number and location of foam generators are to be adequate to ensure all high risk areas are protected in all parts and at all levels of the spaces. Extra foam generators may be required in obstructed locations. The foam generators are to be arranged with at least 1 m free space in front of the foam outlets, unless tested with less clearance. The generators are to be located behind main structures, and above and away from engines and boilers in positions where damage from an explosion is unlikely.

6.3.2.2 *Systems for the protection of vehicle, ro-ro, special category and cargo spaces.*

6.3.2.2.1 The system is to be supplied by the ship's main power source. An emergency power supply is not required.

6.3.2.2.2 Sufficient foam-generating capacity is to be provided to ensure the minimum design filling rate for the system is met and in addition is to be adequate to completely fill the largest protected space within 10 min. However, for systems protecting vehicle and ro-ro spaces and special category spaces, with decks that are reasonably gas-tight and that have a deck height of 3 m or less, the filling rate are to be not less than two thirds of the design filling rate and in addition sufficient to fill the largest protected space within 10 min.

6.3.2.2.3 The system may be divided into sections, however, the capacity and design of the system is to be based on the protected space demanding the greatest volume of foam. Adjacent protected spaces need not be served simultaneously if the boundaries between the spaces are "A" class divisions.

6.3.2.2.4 The arrangement of foam generators are in general to be designed based on the approval test results. The number of generators may be different, but the minimum design filling rate determined during approval testing is to be provided by the system. A minimum of two generators are to be installed in every space. The foam generators are to be arranged to uniformly distribute foam in the protected spaces, and the layout is to take into consideration obstructions that can be expected when cargo is loaded on board. As a minimum, generators are to be located on every second deck, including movable decks. The horizontal spacing of the generators is to ensure rapid supply of foam to all parts of the protected space. This is to be established on the basis of full scale tests.

6.3.2.2.5 The foam generators are to be arranged with at least 1 m free space in front of the foam outlets, unless tested with less clearance.

6.3.3 Outside air foam systems

6.3.3.1 Systems for the protection of machinery spaces and cargo pump-rooms:

6.3.3.1.1 The system is to be supplied by both main and emergency sources of power. The emergency power supply is to be provided from outside the protected machinery space.

6.3.3.1.2 Sufficient foam-generating capacity is to be provided to ensure the minimum design filling rate for the system is met and in addition is to be adequate to completely fill the largest protected space within 10 min.

6.3.3.1.3 The arrangement of foam delivery ducts are in general to be designed based on the approval test results. The number of ducts may be different, but the minimum design filling rate determined during approval testing is to be provided by the system. A minimum of two ducts are to be installed in every space containing combustion engines, boilers, purifiers, and similar equipment. Small workshops and similar spaces may be covered with only one duct.

6.3.3.1.4 Foam delivery ducts are to be uniformly distributed under the uppermost ceiling in the protected spaces including the engine casing. The number and location of ducts are to be adequate to ensure all high risk areas are protected in all parts and at all levels of the spaces. Extra ducts may be required in obstructed locations. The ducts are to be arranged with at least 1 m free space in front of the foam delivery ducts, unless tested with less clearance. The ducts are to be located behind main structures, and above and away from engines and boilers in

positions where damage from an explosion is unlikely.

6.3.3.1.5 The arrangement of the foam delivery ducts are to be such that a fire in the protected space will not affect the foam-generating equipment. If the foam generators are located adjacent to the protected space, foam delivery ducts are to be installed to allow at least 450 mm of separation between the generators and the protected space, and the separating divisions are to be class "A-60" rated. Foam delivery ducts are to be constructed of steel having a thickness of not less than 5 mm. In addition, stainless steel dampers (single or multi-bladed) with a thickness of not less than 3 mm are to be installed at the openings in the boundary bulkheads or decks between the foam generators and the protected space. The dampers are to be automatically operated (electrically, pneumatically or hydraulically) by means of remote control of the foam generator related to them, and arranged to remain closed until the foam generators begin operating.

6.3.3.1.6The foam generators are to be located where an adequate fresh air supply can be arranged.

6.3.3.2 Systems for the protection of vehicle and ro-ro spaces and special category and cargo spaces:

6.3.3.2.1 The system is to be supplied by the ship's main power source. An emergency power supply is not required.

6.3.3.2.2 Sufficient foam-generating capacity is to be provided to ensure the minimum design filling rate for the system is met and in addition is to be adequate to completely fill the largest protected space within 10 min. However, for systems protecting vehicle and ro-ro spaces and special category spaces, with decks that are reasonably gas-tight and that have a deck height of 3 m or less, the filling rate is not to be less than two thirds of the design filling rate and in addition sufficient to fill the largest protected space within 10 min.

6.3.3.2.3 The system may be divided into sections; however, the capacity and design of the system is to be based on the protected space demanding the greatest volume of foam. Adjacent protected spaces need not be served simultaneously if the boundaries between the spaces are "A" class divisions.

6.3.3.2.4 The arrangement of foam delivery ducts are in general to be designed based on the approval test results. The number of ducts may be different, but the minimum design filling rate determined during approval testing is to be provided by the system. A minimum of two ducts are to be installed in every space. The foam generators are to be arranged to uniformly distribute foam in the protected spaces, and the layout are to take into consideration obstructions that can be expected when cargo is loaded on board. As a minimum, ducts are to be led to every second deck, including movable decks. The horizontal spacing of the ducts is to ensure rapid supply of foam to all parts of the protected space. This is to be established on the basis of full scale tests.

6.3.3.2.5 The system is to be arranged with at least 1 m free space in front of the foam outlets, unless tested with less clearance.

6.3.3.2.6 The arrangement of the foam delivery ducts are to be such that a fire in the protected space will not affect the foam-generating equipment. If the foam generators are located adjacent to the protected space, foam delivery ducts are to be installed to allow at least 450 mm of separation between the generators and the protected space, and the separating divisions are to be class "A-60" rated. Foam delivery ducts are to be constructed of steel having a thickness of not less than 5 mm. In addition, stainless steel dampers (single or multi-bladed) with a thickness of not less than 3 mm are to be installed at the openings in the boundary bulkheads or decks between the foam generators and the protected space. The dampers are to be automatically operated (electrically, pneumatically or hydraulically) by means of remote control of the foam generator related to them, and arranged to remain closed until the foam generators begin operating.

6.3.3.2.7 The foam generators are to be located where an adequate fresh air supply can be arranged.

6.3.4 Testing requirements

6.3.4.1 After installation, the pipes, valves, fittings and assembled systems are to be

tested to the satisfaction of the Administration, including functional testing of the power and control systems, water pumps, foam pumps, valves, remote and local release stations and alarms. Flow at the required pressure is to be verified for the system using orifices fitted to the test line. In addition, all distribution piping is to be flushed with freshwater and blown through with air to ensure that the piping is free of obstructions.

6.3.4.2 Functional tests of all foam proportioners or other foam mixing devices are to be carried out to confirm that the mixing ratio tolerance is within +30 to 0% of the nominal mixing ratio defined by the system approval. For foam proportioners using foam concentrates of Newtonian type with kinematic viscosity equal to or less than 100 cSt at 0ºC and density equal to or less than 1,100 kg/m^3, this test can be performed with water instead of foam concentrate. Other arrangements are to be tested with the actual foam concentrate.

6.3.5 Systems using outside air with generators installed inside the protected space:

Systems using outside air but with generators located inside the protected space and supplied by fresh air ducts may be accepted provided that these systems have been shown to have performance and reliability equivalent to systems defined in 6.3.3. For acceptance the following minimum design features will be considered:

.1 lower and upper acceptable air pressure and flow rate in supply ducts;

.2 function and reliability of damper arrangements;

.3 arrangements and distribution of air delivery ducts including foam outlets; and

.4 separation of air delivery ducts from the protected space.

6.4 Fixed low-expansion foam fire-extinguishing systems

6.4.1 Quantity and foam concentrates

6.4.1.1 The foam concentrates of low-expansion foam fire-extinguishing systems are to be approved based on the guidelines adopted by IMO. (Refer to the Guidelines for performance and testing criteria and surveys of low expansion foam concentrates for fixed fire-extinguishing systems (MSC/Circ.1312 and MSC.1/Circ.1312/Corr.1)).

6.4.1.2 The system is to be capable of discharging through fixed discharge outlets in not more than 5 min a quantity of foam sufficient to produce an effective foam blanket over the largest single area over which oil fuel is liable to spread.

6.4.2 Installation requirements

6.4.2.1 Means are to be provided for the effective distribution of the foam through a permanent system of piping and control valves or cocks to suitable discharge outlets, and for the foam to be effectively directed by fixed sprayers on other main fire hazards in the protected space. The means for effective distribution of the foam are to be proven acceptable to IRS through calculation or by testing.

6.4.2.2 The means of control of any such systems is to be readily accessible and simple to operate and are to be grouped together in as few locations as possible at positions not likely to be cut off by a fire in the protected space.

IR6.4.2.3 The distribution pipes are to be of steel and galvanized, at least, internally.

Section 7

Fixed Pressure Water-Spraying and Water-Mist Fire-Extinguishing Systems

7.1 Application

This section details the specifications for fixed pressure water-spraying and water-mist fire-extinguishing systems as required by Ch.3, Sec.4

7.2 Engineering specifications

7.2.1 Fixed pressure water-spraying fire-extinguishing systems

Fixed pressure water spraying fire extinguishing systems for machinery spaces and cargo pump rooms are to be approved by IRS based on guidelines developed by IMO (Revised Guidelines for the approval of equivalent water based fire extinguishing systems for machinery spaces and cargo pump rooms (MSC/Circ. 1165 as amended by MSC.1/Circ.1237 MSC.1/Circ.1269) and MSC.1/Circ.1386).

7.2.2 Equivalent water-mist fire-extinguishing systems

Water-mist fire-extinguishing systems for machinery spaces and cargo pump-rooms are to be approved based on the guidelines developed by IMO (Refer 7.2.1).

7.2.3 Fixed pressure water spraying fire extinguishing system for cabin balconies

Fixed pressure water spraying fire extinguishing system for cabin balconies is to be approved based on the guidelines developed by IMO (MSC.1 Circ.1268).

7.2.4 Fixed water-based fire-fighting systems for ro-ro spaces, vehicle spaces and special category spaces

Fixed water-based fire-fighting systems for ro-ro spaces, vehicle spaces and special category spaces are to be approved by IRS based on guidelines for approval of fixed water-based fire-fighting systems for ro-ro spaces and special category spaces (MSC.1/Circ.1430, Rev.3).

Section 8

Automatic Sprinkler, Fire Detection and Fire Alarm Systems

8.1 Application

This section details the specifications for automatic sprinkler, fire detection and fire alarm systems as required by Ch.3, Sec.4.

8.2 Engineering specifications

8.2.1 General

8.2.1.1 Type of sprinkler systems

The automatic sprinkler systems are to be of the wet pipe type, but small exposed sections may be of the dry pipe type where this is considered a necessary precaution, e.g. within refrigerated chambers. Control stations, where water may cause damage to essential equipment, maybe fitted with a dry pipe system as permitted by 4.6.1.1 of Chapter 3. Saunas are to be fitted with a dry pipe system, with sprinkler heads having an operating temperature up to 140°C.

8.2.1.2 Sprinkler systems equivalent to those specified in paragraphs 8.2.2 to 8.2.4

Automatic sprinkler systems equivalent to those specified in 8.2.2 to 8.2.4 are to be approved based on the requirements of the National Statutory Authority. (Refer to the Revised Guidelines for approval of sprinkler systems equivalent to that referred to in SOLAS regulation II-2/12 as adopted by IMO by resolution A.800(19)).

8.2.2 Sources of power supply

8.2.2.1 Passenger ships

There is to be not less than two sources of power supply for the sea water pump and automatic alarm and detection system. Where the sources of power for the pump are electrical, these are to be a main generator and an emergency source of power. One supply for the pump is to be taken from the main switchboard, and one from the emergency switchboard by separate feeders reserved solely for that purpose. The feeders are to be so arranged as to avoid galleys, machinery spaces and other enclosed spaces of high fire risk except in so far as it is necessary to reach the appropriate switchboards, and are to be run to an automatic changeover switch situated near the sprinkler pump. This switch is to permit the supply of power from the main switchboard so long as a supply is available therefrom, and be so designed that upon failure of that supply it will automatically change over to the supply from the emergency switchboard. The switches on the main switchboard and the emergency switchboard are to be clearly labelled and normally kept closed. No other switch is to be permitted in the feeders concerned. One of the sources of power supply for the alarm and detection system is to be an emergency source. Where one of the sources of power for the pump is an internal combustion engine it is to, in addition to complying with the provisions of 8.2.4.3, be so situated that a fire in any protected space will not affect the air supply to the machinery.

8.2.2.2 Cargo ships

There are not to be less than two sources of power supply for the sea water pump and automatic alarm and detection system. If the pump is electrically driven it is to be connected to the main source of electrical power, which is to be capable of being supplied by at least two generators. The feeders are to be so arranged as to avoid galleys, machinery spaces and other enclosed spaces of high fire risk except in so far as it is necessary to reach the appropriate switchboards. One of the sources of power supply for the alarm and detection system is to be an emergency source. Where one of the sources of power for the pump is an internal combustion engine it is in addition to complying with the provisions of 8.2.4.3, be so situated that a fire in any protected space will not affect the air supply to the machinery.

8.2.3 Component requirements

8.2.3.1 Sprinklers

8.2.3.1.1 The sprinklers are to be resistant to corrosion by marine atmosphere. In accommodation and service spaces the sprinklers are to come into operation within the temperature range from 68° to 79°C, except that in locations such as drying rooms, where high ambient temperatures might be expected, the operation temperature may be increased by not more than 30°C above the maximum deckhead temperature.

8.2.3.1.2 A quantity of spare sprinkler heads are to be provided for all types and ratings installed on the ship as follows:

Total number of heads	Required number of spares
<300	6
300 to 1000	12
>1000	24

The number of spare sprinkler heads of any type need not exceed the total number of heads installed of that type.

8.2.3.2 Pressure tanks

8.2.3.2.1 A pressure tank having a volume equal to at least twice that of the charge of water specified herein is to be provided. The tank is to contain a standing charge of fresh water, equivalent to the amount of water which would be discharged in one minute by the pump referred to in 8.2.3.3.2, and the arrangements are to provide for maintaining an air pressure in the tank such as to ensure that where the standing charge of fresh water in the tank has been used the pressure will be not less than the working pressure of the sprinkler, plus the pressure exerted by a head of water measured from the bottom of the tank to the highest sprinkler in the system. Suitable means of replenishing the air under pressure and of replenishing the fresh water charge in the tank are to be provided. A glass gauge is to be provided to indicate the correct level of the water in the tank.

8.2.3.2.2 Means are to be provided to prevent the passage of sea water into the tank.

8.2.3.3 Sprinkler pumps

8.2.3.3.1 An independent power pump is to be provided solely for the purpose of continuing automatically the discharge of water from the

sprinklers. The pump is to be brought into action automatically by the pressure drop in the system before the standing fresh water charge in the pressure tank is completely exhausted.

8.2.3.3.2 The pump and the piping system are to be capable of maintaining the necessary pressure at the level of the highest sprinkler to ensure a continuous output of water sufficient for the simultaneous coverage of a minimum area of 280 [m^2] at the application rate specified in 8.2.5.2.3. The hydraulic capability of the system is to be confirmed by the review of hydraulic calculations, followed by a test of the system, if deemed necessary by IRS.

8.2.3.3.3 The pump is to be fitted, on the delivery side, a test valve with a short open-ended discharge pipe. The effective area through the valve and pipe is to be adequate to permit the release of the required pump output while maintaining the pressure in the system specified in 8.2.3.2.1.

8.2.4 Installation requirements

8.2.4.1 General

8.2.4.1.1 Any parts of the system which may be subjected to freezing temperatures in service are to be suitably protected against freezing.

8.2.4.1.2 Special attention is to be paid to the specification of water quality provided by the system manufacturer to prevent internal corrosion of sprinklers and clogging or blockage arising from products of corrosion or scale-forming minerals.

8.2.4.2 Piping arrangements

8.2.4.2.1 Sprinklers are to be grouped into separate sections, each of which is to contain not more than 200 sprinklers. In passenger ships any section of sprinklers is not to serve more than two decks and is not to be situated in more than one main vertical zone. However, a section of sprinklers to serve more than two decks or situated in more than one main vertical zone, may be permitted, if it is shown that the protection of the ship against fire will not thereby be reduced.

8.2.4.2.2 Each section of sprinklers is to be capable of being isolated by one stop valve only. The stop valve in each section is to be readily accessible in a location outside the associated section or in cabinets within stairway enclosures. The valve’s location is to be clearly and permanently indicated. Means are to be provided to prevent the operation of the stop valves by any unauthorized person.

8.2.4.2.3 A test valve is to provided for testing the automatic alarm for each section of sprinklers by a discharge of water equivalent to the operation of one sprinkler. The test valve for each section is to be situated near the stop valve for that section.

8.2.4.2.4 The sprinkler system is to have a connection from the ship's fire main by way of a lockable screw-down non-return valve at the connection which will prevent a backflow from the sprinkler system to the fire main.

8.2.4.2.5 A gauge indicating the pressure in the system is to be provided at each section stop valve and at a central station.

8.2.4.2.6 The sea inlet to the pump wherever possible is to be in the space containing the pump and is to be so arranged that when the ship is afloat it will not be necessary to shut off the supply of sea water to the pump for any purpose other than the inspection or repair of the pump.

8.2.4.3 Location of systems

The sprinkler pump and tank are to be situated in a position reasonably remote from any machinery space of category A and are not to be situated in any space required to be protected by the sprinkler system.

8.2.5 System control requirements

8.2.5.1 Ready availability

8.2.5.1.1 Any required automatic sprinkler, fire detection and fire alarm system is to be capable of immediate operation at all times and no action by the crew is to be necessary to set it in operation.

8.2.5.1.2 The automatic sprinkler system is to be kept charged at the necessary pressure and is to have provision for a continuous supply of water as required in this chapter.

8.2.5.2 Alarm and indication

8.2.5.2.1 Each section of sprinklers is to include means for giving a visual and audible alarm signal automatically at one or more

indicating units whenever any sprinkler comes into operation. Such alarm systems are to be such as to indicate if any fault occurs in the system. Such units are to indicate in which section served by the system a fire has occurred and are to be centralised on the navigating bridge or in the continuously manned central control station and, in addition, visible and audible alarms from the unit are also to be placed in a position other than on the aforementioned spaces to ensure that the indication of fire is immediately received by the crew.

8.2.5.2.2 Switches are to be provided at one of the indicating positions referred to in 8.2.5.2.1 which will enable the alarm and the indicators for each section of sprinklers to be tested.

8.2.5.2.3 Sprinklers are to be placed in an overhead position and spaced in a suitable pattern to maintain an average application rate of not less than 5 [litres/m^2] per minute over the nominal area covered by the sprinklers. For this purpose, nominal area is to be taken as the gross horizontal projection of the area to be covered. The use of sprinklers providing other amounts of water suitably distributed, will be considered provided they are shown to be not less effective.

8.2.5.2.4 A list or plan is to be displayed at each indicating unit showing the spaces covered and the location of the zone in respect of each section. Suitable instructions for testing and maintenance are to be available.

8.2.5.3 Testing

Means are to be provided for testing the automatic operation of the pump on reduction of pressure in the system.

IR8.2.5.4 Stowage of spare sprinklers

8.2.5.4.1 Spare sprinklers are to be stowed in boxes or holders provided for that purpose, together with a tool suitable for removing and installing sprinkler heads. The boxes or holders are to be situated near the control valve for the section, and are to be clearly and permanently marked to indicate their contents.

Section 9

Fixed Fire Detection and Fire Alarm Systems

9.1 Application

9.1.1 This chapter details the specifications for fixed fire detection and fire alarm systems as required by Ch.3, Sec.1.

9.1.2 Definitions

9.1.2.1 Section is a group of fire detectors and manually operated call points as reported in the indicating unit(s).

9.1.2.2 Section identification capability means a system with the capability of identifying the section in which a detector or manually operated call point has activated.

9.1.2.3 Individually identifiable means a system with the capability to identify the exact location and type of detector or manually activated call point which has activated, and which can differentiate the signal of that device from all others.

9.2 Engineering specifications

9.2.1 General requirements

9.2.1.1 Any required fixed fire detection and fire alarm system with manually operated call points is to be capable of immediate operation at all times (this does not require a backup control panel). Notwithstanding this, particular spaces may be disconnected, for example, workshops during hot work and ro-ro spaces during on and off-loading. The means for disconnecting the detectors is to be designed to automatically restore the system to normal surveillance after a predetermined time that is appropriate for the operation requiring disconnection of the detectors. The space is to be manned or provided with a fire patrol when the required detectors are disconnected. Detectors in all other spaces are to remain operational.

9.2.1.2 The fire detection system is to be designed to:

.1 control and monitor input signals from all connected fire and smoke detectors and manual call points;

.2 provide output signals to the navigation bridge, continuously manned central control station or onboard safety centre to notify the crew of fire and fault conditions;

.3 monitor power supplies and circuits necessary for the operation of the system for loss of power and fault conditions; and

.4 the system may be arranged with output signals to other fire safety systems including:

.1 paging systems, fire alarm or public address systems;

.2 fan stops;

.3 fire doors;

.4 fire dampers;

.5 sprinkler systems;

.6 smoke extraction systems;

.7 low-location lighting systems;

.8 fixed local application fire-extinguishing systems;

.9 closed circuit television (CCTV) systems; and

.10 other fire safety systems.

IR9.2.1.2 However, for passenger ships, watertight doors complying with Reg.II-1/16 of SOLAS which also serve as fire doors are not to close automatically in case of detection of fire.

9.2.1.3 The fire detection system may be connected to a decision management system provided that:

.1 the decision management system is proven to be compatible with the fire detection system;

.2 the decision management system can be disconnected without losing any of the functions required by this section for the fire detection system; and

.3 any malfunction of the interfaced and connected equipment does not propagate to the fire detection system under any circumstance.

9.2.1.4 Detectors and manual call points are to be connected to dedicated sections of the fire detection system. Other fire safety functions, such as alarm signals from the sprinkler valves, may be permitted if in separate sections.

9.2.1.5 The system and equipment are to be suitably designed to withstand supply voltage variation and transients, ambient temperature changes, vibration, humidity, shock, impact and corrosion normally encountered in ships. All electrical and electronic equipment on the bridge or in the vicinity of the bridge are to be tested for electromagnetic compatibility, based on the IMO resolution (A.813(19)).

9.2.1.6 Fixed fire detection and fire alarm systems with individually identifiable fire detectors are to be so arranged that:

.1 means are provided to ensure that any fault (e.g. power break, short circuit, earth, etc.) occurring in the section will not prevent the continued individual identification of the remainder of the connected detectors in the section;

.2 all arrangements are made to enable the initial configuration of the system to be restored in the event of failure (e.g. electrical, electronic, informatics, etc.);

.3 the first initiated fire alarm will not prevent any other detector from initiating further fire alarms; and

.4 no section will pass through a space twice. When this is not practical (e.g. for large public spaces), the part of the section which by necessity passes through the space for a second time is to be installed at the maximum possible distance from the other parts of the section.

9.2.1.7 In passenger ships, the fixed fire detection and fire alarm system is to be capable of remotely and individually identifying each detector and manually operated call point. Fire detectors fitted in passenger ship

cabins, when activated, are also to be capable of emitting, or cause to be emitted, an audible alarm within the space where they are located. In cargo ships and on passenger ship cabin balconies the fixed fire detection and fire alarm system is to have section identification capability as a minimum.

9.2.1.8 In cargo ships and on passenger ship cabin balconies, where an individually identifiable system is fitted, notwithstanding the provisions in paragraph 9.2.1.6.1, isolator modules need not be provided at each fire detector if the system is arranged in such a way that the number and location of individually identifiable fire detectors rendered ineffective due to a fault would not be larger than an equivalent section in a section identifiable system, arranged in accordance with paragraph 9.2.4.1.

9.2.1.9 Fixed fire detection and fire alarm systems for cabin balconies are to be approved by IRS, based on the guidelines developed by the IMO (MSC.1 Circ.1242).

9.2.2 Sources of power supply

9.2.2.1 There are to be not less than two sources of power supply for the electrical equipment used in the operation of the fixed fire detection and fire alarm system, one of which is to be an emergency source. The supply is to be provided by separate feeders reserved solely for that purpose. Such feeders are to run to an automatic change over switch situated in or adjacent to the control panel for the fire detection system. The change over switch shall be arranged such that a fault will not result in the loss of both power supplies. The main/ emergency feeder is to run from the respective switchboard to the change-over switch without passing through any other distributing switchboard.

9.2.2.2 The operation of the automatic changeover switch or a failure of one of the power supplies shall not result in loss of fire detection capability. Where a momentary loss of power would cause degradation of the system, a battery of adequate capacity shall be provided to ensure continuous operation during changeover.

9.2.2.3 There is to be sufficient power to permit the continued operation of the system with all detectors activated, but not more than 100 if the total exceeds this figure.

9.2.2.4 The emergency source of power specified in 9.2.2.1 may be supplied by accumulator batteries or from emergency switchboard. The power source is to be sufficient to maintain the operation of the fire detection and fire alarm system for the periods specified in Pt. 4, Ch. 8, Sec 2.8 and at the end of that period, is to be capable of operating all connected visual and audible fire alarm signals for a period of at least 30 min.

9.2.2.5 Where the system is supplied from accumulator batteries, they shall be located in or adjacent to the control panel for the fire detection system, or in another location suitable for use in an emergency. The rating of the battery charge unit shall be sufficient to maintain the normal output power supply to the fire detection system while recharging the batteries from a fully discharged condition.

9.2.3 Component requirements

9.2.3.1 Detectors

9.2.3.1.1 Detectors are to be operated by heat, smoke or other products of combustion, flame, or any combination of these factors. Detectors operated by other factors indicative of incipient fires may be considered by IRS provided that they are no less sensitive than such detectors.

9.2.3.1.2 Smoke detectors required in all stairways, corridors and escape routes within accommodation spaces are to be certified to operate before the smoke density exceeds 12.5% obscuration per metre, but not until the smoke density exceeds 2% obscuration per metre, when tested according to standards EN 54:2001 and IEC 60092-504. Alternative testing standards may be accepted by IRS. Smoke detectors to be installed in other spaces are to operate within sensitivity limits to the satisfaction of IRS having regard to the avoidance of detector insensitivity or over-sensitivity.

9.2.3.1.3 Heat detectors are to be certified to operate before the temperature exceeds 78°C but not until the temperature exceeds 54°C, when the temperature is raised to those limits at a rate less than 1°C per minute, when tested according to standards EN 54 and IEC 60092-504. Alternative testing standards may be accepted by IRS. At higher rates of temperature rise, the heat detector is to operate within temperature limits to the satisfaction of the Surveyors having regard to

the avoidance of detector insensitivity or oversensitivity.

9.2.3.1.4 The operation temperature of heat detectors in drying rooms and similar spaces of a normal high ambient temperature may be up to 130°C, and up to 140°C in saunas.

9.2.3.1.5 Flame detectors are to be tested according to standards EN 54-10:2001 and IEC 60092-504.

9.2.3.1.6 All detectors are to be of a type such that they can be tested for correct operation and restored to normal surveillance without the renewal of any component.

9.2.3.1.7 Detectors fitted in hazardous areas are to be tested and approved for such service. Detectors required by Pt. 6, Ch. 7, Sec 3.4 and installed in spaces that comply with Pt. 6, Ch. 7, Sec 3.3.2.2 need not be suitable for hazardous areas. Detectors fitted in spaces carrying dangerous goods, required by Pt. 6, Ch. 7, Table 2.3 to comply with Pt. 6, Ch. 7, Cl. 2.3.2 are to be suitable for hazardous areas.

9.2.3.2 Control panel

The control panel for the fire detection system is to be tested according to standards EN 54-2:1997, EN 54-4:1997 and IEC 60092-504. Alternative standards may be accepted by IRS.

9.2.3.3 Cables

Cables used in the electrical circuits are to be flame retardant according to standard IEC 60332-1. On passenger ships, cables routed through other main vertical zones that they serve, and cables to control panels in an unattended fire control station are to be fire resisting according to standard IEC 60331, unless duplicated and well separated.

9.2.4 Installation requirements

9.2.4.1 Sections

9.2.4.1.1 Detectors and manually operated call points are to be grouped into sections.

9.2.4.1.2 A section of fire detectors which covers a control station, a service space or an accommodation space is not to include a machinery space of category A or a ro-ro space. A section of fire detectors which covers a ro-ro space is to not include a machinery space of category A. For fixed fire detection and fire alarm systems with remotely and individually identifiable fire detectors, a section covering fire detectors in accommodation, service spaces and control station is not to include fire detectors in machinery spaces of category A or ro-ro spaces.

9.2.4.1.3 Where the fixed fire detection and fire alarm system does not include means of remotely identifying each detector individually, no section covering more than one deck within accommodation spaces, service spaces and control stations, are normally to be provided except a section which covers an enclosed stairway. In order to avoid delay in identifying the source of fire, the number of enclosed spaces included in each section is to be limited as acceptable to IRS. If the detection system is fitted with remotely and individually identifiable fire detection, the sections may cover several decks and serve any number of enclosed spaces.

9.2.4.1.4 In passenger ships, a section of detectors and manually operated call points are not to be situated in more than one main vertical zone, except on cabin balconies.

9.2.4.2 Positioning of detectors

9.2.4.2.1 Detectors are to be located for optimum performance. Positions near beams and ventilation ducts or other positions where patterns of air flow could adversely affect performance and positions where impact or physical damage is likely is to be avoided. Detectors which are located on the overhead are to be a minimum distance of 0.5 [m] away from bulkheads, except in corridors, lockers and stairways.

9.2.4.2.2 The maximum spacing of detectors is to be in accordance with the Table 9.1.

IRS may require or permit different spacing to that specified in the above table if based upon test data which demonstrate the characteristics of the detectors. Detectors located below moveable ro-ro decks are to be in accordance with the above.

9.2.4.2.3 Detectors in stairways are to be located at least at the top level of the stair and at every second level beneath.

9.2.4.2.4 When fire detectors are installed in freezers, drying rooms, saunas, parts of

galleys used to heat food, laundries and other spaces where steam and fumes are produced, heat detectors may be used.

9.2.4.2.5 Where a fixed fire detection and fire alarm system is required by Pt. 6, Ch. 3, Sec 1.5, spaces having little or no fire risk need not be fitted with detectors. Such spaces include void spaces with no storage of combustibles, private bathrooms, public toilets, fire-extinguishing medium storage rooms, cleaning gear lockers (in which flammable liquids are not stowed), open deck spaces and enclosed promenades having little or no fire risk and that are naturally ventilated by permanent openings.

IR9.2.4.2.5 Guidance Note : The following categories of spaces having little or no fire risk generally need not be fitted with detectors:

a) For ships carrying more than 36 passengers (See Ch.3, 3.2.2.3.2):

Category (5) Open deck spaces
Category (9) Sanitary and similar spaces
Category (10) Tanks and voids having little or no fire risk.

b) For ships carrying less than 36 passengers (See Ch.3, 3.2.2.4.2):

Category (5) Service spaces (low risk)
Category (10) Open decks

c) For cargo ships except tankers, (when method IIC or IIIC is adopted) (See Ch.3, 3.2.3.3.2):

Category (5) Service spaces (low risk)
Category (10) Open decks.

Table 9.1 : Spacing of detectors

Type of detector	Max. floor area per detector	Max. distance apart between centres	Max. distance away from bulkheads
Heat	37 $[m]^2$	9 [m]	4.5 [m]
Smoke	74 $[m]^2$	11 [m]	5.5 [m]

9.2.4.3 Arrangement of cables

9.2.4.3.1 Cables which form a part of the system are to be so arranged as to avoid galleys, machinery spaces of category A, and other enclosed spaces of high fire risk except where it is necessary to provide for fire detection or fire alarms in such spaces or to connect to the appropriate power supply.

9.2.4.3.2 A section with individually identifiable capability is not to be damaged at more than one point by a fire.

9.2.5 System control requirements

9.2.5.1 Visual and audible fire signals

(Refer to the Code on alerts and Indicators as adopted by IMO by resolution A.1021(26)).

9.2.5.1.1 The activation of any detector or manually operated call point is to initiate a visual and audible fire detection alarm signal at the control panel and indicating units. If the signals have not received attention within 2 min an audible fire alarm is to be automatically sounded throughout the crew accommodation and service spaces, control stations and machinery spaces of category A. This alarm sounder system need not be an integral part of the detection system.

9.2.5.1.2 In passenger ships, the control panel is to be located in the onboard safety centre. In cargo ships, the control panel is to be located on the navigating bridge or in the control station.

9.2.5.1.3 In passenger ships, an indicating unit that is capable of individually identifying each detector that has been activated or manually operated call point that has operated is to be located on the navigation bridge. In cargo ships and on passenger cabin balconies, indicating units are to as a minimum, denote the section in which a detector or manually operated call point has operated. In ships constructed on or after 1 July 2014, with a cargo control room, an additional indicating unit is to be located in the cargo control room. In cargo ships, an indicating unit is to be located on the navigating bridge if the control panel is located in the fire control station.

IR.9.2.5.1.3 A space in which a cargo control console is installed, but does not serve as a dedicated cargo control room (e.g. ship's office, machinery control room), is to be regarded as a cargo control room in the context of 9.2.5.1.3 and therefore be provided with an additional indicating unit.

9.2.5.1.4 Clear information is to be displayed on or adjacent to each indicating unit about the spaces covered and the location of the sections.

9.2.5.1.5 Power supplies and electric circuits necessary for the operation of the system are to be monitored for loss of power or fault conditions as appropriate including:

.1 a single open or power break fault caused by a broken wire;

.2 a single ground fault caused by the contact of a wiring conductor to a metal component; and

.3 a single wire to wire fault caused by the contact of two or more wiring conductors.

Occurrence of a fault condition is to initiate a visual and audible fault signal at the control panel which is to be distinct from a fire signal.

9.2.5.1.6 Means to manually acknowledge all alarm and fault signals is to be provided at the control panel. The audible alarm sounders on the control panel and indicating units may be manually silenced. The control panel is to clearly distinguish between normal, alarm, acknowledged alarm, fault and silenced conditions.

9.2.5.1.7 The system is to be arranged to automatically reset to the normal operating condition after alarm and fault conditions are cleared.

9.2.5.1.8 When the system is required to sound a local audible alarm within the cabins where the detectors are located, a means to silence the local audible alarms from the control panel is not to be provided.

9.2.5.1.9 In general, audible alarm sound pressure levels at the sleeping positions in the cabins and 1 m from the source are to be at least 75 dB(A) and at least 10 dB(A) above ambient noise levels existing during normal equipment operation with the ship under way in moderate weather. The sound pressure level should be in the 1/3 octave band about the fundamental frequency. Audible alarm signals are not to exceed 120 dB(A).

9.2.5.2 Testing

Suitable instructions and component spares for testing and maintenance are to be provided. Detectors are to be periodically tested using equipment suitable for the types of fires to which the detector is designed to respond. On ships constructed on or after 1 July 2014, detectors installed within cold spaces such as refrigerated compartments are to be tested using procedures having due regard for such locations*. Special consideration will be given to ships with self-diagnostic systems that have in place a documented periodic cleaning procedure for detectors in areas where heads may be prone to contamination.
* Note: Refer to the recommendations of the International Electro-technical Commission, in particular publication IEC 60068-2-1 – Section one -Test Ab, *Environmental Testing – Part 2-1: Tests – Test A: Cold.*"

Section 10

Sample Extraction Smoke Detection Systems

10.1 Application

This section details the specifications for sample extraction smoke detection systems as required by Ch.3, Sec.4.

10.2 Engineering specifications

10.2.1 General requirements

10.2.1.1 Wherever in the text of following requirements the word "system" appears, it means "sample extraction smoke detection system".

10.2.1.1.1 A sample extraction smoke detection system consists of the following main components:

.1 smoke accumulators: air collection devices installed at the open ends of the sampling pipes in each cargo hold that perform the physical function of collecting air samples for transmission to the control panel through the sampling pipes and may also act as discharge nozzles for the fixed-gas fire-extinguishing system, if installed;

.2 sampling pipes: a piping network that connects the smoke accumulators to the control panel, arranged in sections to allow the location of the fire to be readily identified;

.3 three-way valves: if the system is interconnected to a fixed-gas fire-extinguishing system, three-way valves are used to normally align the sampling pipes to the control panel and if a fire is detected, the three-way valves are re-aligned to connect the sampling pipes to the fire-extinguishing system discharge manifold and isolate the control panel; and

.4 control panel: the main element of the system which provides continuous monitoring of the protected spaces for indication of smoke. It typically may include a viewing chamber or smoke sensing units. Extracted air from the protected spaces is drawn through the smoke accumulators and sampling pipes to the viewing chamber and then to the smoke sensing chamber where the air stream is monitored by electrical smoke detectors. If smoke is sensed, the repeater panel (normally on the bridge) automatically sounds an alarm (not localized). The crew can then determine at the smoke sensing unit which cargo hold is on fire and operate the pertinent three-way valve for discharge of the extinguishing agent.

Note : 'Not localized' means that the alarm is initiated at control panel as well as repeater panel(s) / indicating unit(s) as given in Cl.10.2.4.1.1.

10.2.1.2 Any required system is to be capable of continuous operation at all times except that systems operating on a sequential scanning principle may be accepted, provided that the interval between scanning the same position twice gives a maximum allowable interval determined as follows:

The interval (I) should depend on the number of scanning points (N) and the response time of the fans (T) (determined by the designer), with a 20% allowance:

$I = 1.2 \times T \times N$

However, the maximum allowable interval is not to exceed 120 s (I_{max} = 120 s).

10.2.1.3 The system is to be designed, constructed and installed so as to prevent the leakage of any toxic or flammable substance or fire-extinguishing media into any accommodation and service space, control station or machinery space.

10.2.1.4 The system and equipment are to be suitably designed to withstand supply voltage variations and transients, ambient temperature changes, vibration, humidity, shock, impact and corrosion normally encountered in ships and to avoid the possibility of ignition of flammable gas/air mixture.

10.2.1.5 The functioning of the system is to be periodically tested to the satisfaction of IRS. The system is to be of a type that can be tested for correct operation and restored to normal surveillance without the renewal of any component.

10.2.1.6 An alternative power supply for the electrical equipment used in the operation of the system is to be provided.

10.2.2 Component requirements

10.2.2.1 The sensing unit is to be certified to operate before the smoke density within the sensing chamber exceeds 6.65 per cent obscuration per metre.

10.2.2.2 Duplicate sample extraction fans are to be provided. The fans are to be of sufficient capacity to operate under normal ventilation conditions in the protected area and the connected pipe size is to be determined with consideration of fan suction capacity and piping arrangement to satisfy the conditions of clause 10.2.4.2.2. Sampling pipes are to be a minimum of 12 [mm] internal diameter. The fan suction capacity is to be adequate to ensure the response of the most remote area within the required time criteria in Cl. 10.2.4.2.2. Means to monitor airflow are to be provided in each sampling line.

10.2.2.3 The control panel is to permit observation of smoke in the individual sampling pipe.

10.2.2.4 Means are to be provided to monitor the airflow through the sampling pipes so designed as to ensure that as far as practicable equal quantities are extracted from each interconnected accumulator.

10.2.2.5 Sampling pipes are to be provided with an arrangement for periodically purging with compressed air.

10.2.2.6 The control panel for the smoke detection system is to be tested according to standards EN 54-2 (1997), EN 54-4 (1997) and IEC 60092-504 (2001). Alternative standards may be accepted by IRS.

10.2.3 Installation requirements

10.2.3.1 Smoke accumulators

10.2.3.1.1 At least one smoke accumulator is to be located in every enclosed space for which smoke detection is required. However, where a space is designed to carry oil or refrigerated cargo alternatively with cargoes for which a smoke sampling system is required, means may be provided to isolate the smoke accumulators in such compartments for the system. Such means are to be to the satisfaction of IRS.

10.2.3.1.2 Smoke accumulators are to be located overhead or as high as possible in the protected space and are to be spaced so that no part of the overhead deck area is more than 12 [m] measured horizontally from an accumulator. Where systems are used in spaces which may be mechanically ventilated, the position of the smoke accumulators is to be considered having regard to the effects of ventilation. At least one additional smoke accumulator is to be provided in the upper part of each exhaust ventilation duct. An adequate filtering system is to be fitted at the additional accumulator to avoid dust contamination.

10.2.3.1.3 Smoke accumulators are to be positioned where impact or physical damage is unlikely to occur.

10.2.3.1.4 Sampling pipe networks are to be balanced to ensure compliance with Cl. 10.2.2.4. The number of accumulators connected to each sampling pipe are to ensure compliance with Cl. 10.2.4.2.2.

10.2.3.1.5 Smoke accumulators from more than one enclosed space are not to be connected to the same sampling point.

10.2.3.1.6 In cargo holds where non-gastight “tween deck panels” (movable stowage platforms) are provided, smoke accumulators are to be located in both the upper and lower parts of the holds.

10.2.3.2 Sampling pipes

10.2.3.2.1 The sampling pipe arrangements are to be such that the location of the fire can be readily identified.

10.2.3.2.2 Sampling pipes are to be self-draining and suitably protected from impact or damage from cargo working.

10.2.4 System control requirements

10.2.4.1 Visual and audible fire signals

10.2.4.1.1 The detection of smoke or other products of combustion is to initiate a visual and audible signal at the control panel and indicating units.

10.2.4.1.2 The control panel is to be located on the navigating bridge or in the fire control station. An indicating unit is to be located on the navigation bridge if the control panel is located in the fire control station.

10.2.4.1.3 Clear information is to be displayed on or adjacent to the control panel designating the spaces covered.

10.2.4.1.4 Power supplies necessary for the operation of the system are to be monitored for loss of power. Any loss of power is to initiate a visual and audible signal at the control panel and the navigating bridge which is to be distinct from a signal indicating smoke detection.

10.2.4.1.5 Means to manually acknowledge all alarm and fault signals are to be provided at the control panel. The audible alarm sounders on the control panel and indicating units may be manually silenced. The control panel is to clearly distinguish between normal, alarm, acknowledged alarm, fault and silenced conditions.

10.2.4.1.6 The system is to be arranged to automatically reset to the normal operating condition after alarm and fault conditions are cleared.

10.2.4.2 Testing

10.2.4.2.1 Suitable instructions and component spares are to be provided for the testing and maintenance of the system.

10.2.4.2.2 After installation, the system is to be functionally tested using smoke generating machines or equivalent as a smoke source. An alarm is to be received at the control unit in not more than 180 s for vehicle decks and not more than 300 s for container and general cargo holds, after smoke is introduced at the most remote accumulator.

Section 11

Low-Location Lighting Systems

11.1 Application

This section details the specifications for low-location lighting systems as required by Ch.4, Sec.2.

11.2 Engineering specifications

11.2.1 General requirements

Any required low-location lighting systems is to be approved based on the requirements of the National Statutory Authority.

(Refer to the Guidelines for the evaluation, testing and application of low-location lighting on passenger ships as adopted by IMO resolution A.752(18)).

(Refer to the Recommendations by the International Organization for Standardization, in particular, publication ISO 15370:2001 on Low-location lighting on passenger ships).

Section 12

Fixed Emergency Fire Pumps

12.1 Application

This section details the specifications for emergency fire pumps as required by Chapter.3. This section is not applicable to passenger ships of 1,000 gross tonnage and upwards. See Ch.3, 4.2.2.3.1.1 for requirements for such ships.

12.2 Engineering specifications

12.2.1 General

12.2.1.1 The emergency fire pump is to be a fixed independently driven power-operated pump.

IR12.2.1.2 The emergency fire pump and its prime mover are to be approved. The emergency fire pump may also be used for other suitable purposes subject to approval in each case.

12.2.2 Component requirements

12.2.2.1 Emergency fire pumps

12.2.2.1.1 Capacity of the pump

.1 The capacity of the pump is not to be less than 40% of the total capacity of the fire pumps required by Ch.3, 4.2.2.4.1 and in any case not less than the following (Also see Ch.3, 4.2.2.4.1, IR .2.1):

.1	for passenger ships less than 1,000 gross tonnage and for cargo ships of 500 gross tonnage and upwards; and	32 [m^3/h]
.2	for cargo ships less than 500 gross tonnage	15 [m^3/h]

IR.2 The emergency fire pump is also to be capable of supplying two jets of water. Where a fixed water-based fire extinguishing system installed for the protection of the machinery space in accordance with Ch.3, Cl.4.4.1.1, is supplied by the emergency fire pump, the emergency fire pump capacity is

to be adequate to supply the fixed fire extinguishing system at the required pressure plus two jets of water. The capacity of the two jets is to be calculated for two biggest nozzle sizes on board from the following table (* note):

Capacity required for one jet		
Nozzle size Pressure at hydrant	16 mm	19mm
0.27 [N/mm^2]	16 [m^3/h]	23.5 [m^3/h]

*Note: When selecting the biggest nozzle size onboard, the nozzles located in the space where the main fire pumps are located can be excluded.

IR.3 On board cargo ships designed to carry five or more tiers of containers on or above the weather deck, the total capacity of the emergency fire pump need not exceed 72 [m^3/hr].

12.2.2.1.2 Pressure at hydrants

When the pump is delivering the quantity of water required by 12.2.2.1.1, the pressure at any hydrants is not to be less than the minimum pressure required by Chapter 3.

12.2.2.1.3 Suction heads

.1 The total suction head and the net positive suction head of the pump is to be determined having due regard to the requirements of Chapter 3 and this section on the pump capacity and on the hydrant pressure under all conditions of list, trim, roll and pitch likely to be encountered in service. The ballast condition of a ship on entering or leaving a dry dock need not be considered a service condition.

IR.1

IR.1.1 It is to be documented that the suction inlet is fully submerged under “all conditions of list, trim, roll and pitch likely to be encountered in service” as given below.

IR.1.1.1 Operational seagoing condition for which roll, pitch and heave is to be applied is as follows:

The lightest seagoing condition is to be considered, which is defined as the ballast condition which gives the shallowest draught at the position of the sea chest and emergency fire pump as given in the approved stability booklet (or preliminary stability calculation for new building). The following table is to be applied for the calculation of roll, pitch and heave. The heave combined pitch and heave combined roll are taken into account separately.

IR.1.1.1.1 Heave combined pitch in head sea as given in Fig.12.2.2.1.3

L_{LL} (m)	75 and below	100	125	150	175	200	225	250	300	350 and above
Φ (deg)	4.5	4	3.2	2.7	2.3	2.1	1.8	1.7	1.6	1.5
H (m)	0.73	0.8	0.87	0.93	1.03	1.07	1.11	1.19	1.25	
Note: Values at the intermediate length of ships are to be obtained by linear interpolation.										

Where:

L_{LL}: length of the ship, in meters, as defined in the International Convention on Load Lines in force, or length between perpendiculars at the ballast draught, whichever is greater

φ: pitch angle (angle is to be measured from still waterline and downwards) as defined in Fig.12.2.2.1.3

H: heave amplitude as defined in Fig.12.2.2.1.3

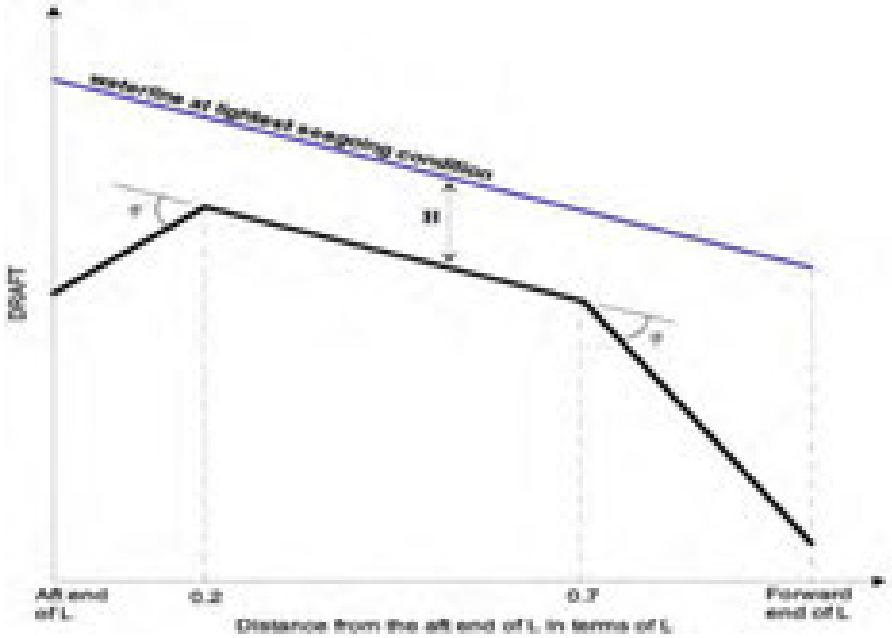


Fig.12.2.2.1.3 : Waterline for which heave combined pitch is taken into account

IR.1.1.1.2 Heave combined roll in beam sea

Heave combined roll angle (angle is to be measured from still waterline and downwards) is to be taken as:

a) ships with bilge keels: 11°; and

b) ships without bilge keels: 13°

IR.1.1.2 The emergency fire pump suction is to be submerged at the waterlines corresponding to the two following conditions:

.1 a static waterline drawn through the level of 2/3 immersion of the propeller at even keel (for pod or thruster driven ship, special consideration will be given); and

.2 the ship in the arrival ballast condition, as per the approved trim and stability booklet, without cargo and with 10% stores and fuel remaining.

For either condition, roll, pitch and heave need not be applied.

The required minimum draught at the sea chest in any condition is to be indicated in the fire fighting system / fire control plans.

IR.1.1.3 A ship operating solely in sheltered water is to be subject to compliance with the still water submergence requirements set out in IR.1.1.2.1 above.

IR.1.2 In all cases the net positive suction head (NPSH) available for the pump is to be greater than the NPSH required.

IR.1.3 Upon completion of the emergency fire pump installation, a performance test confirming the pump's capacity required as per 12.2.2.1.1, is to be carried out and, if the emergency fire pump is the main supply of water for any fixed fire- extinguishing system provided to protect the spaces where the main fire pumps are located, the pump is to have the capacity for this system. As far as practicable, the test is to be carried out at the draught corresponding to the lightest seagoing condition.

IR.2 The pump is to be designed to be of self-priming type and its suction are to be so positioned that the pump will operate efficiently at the lightest draft likely to be encountered in service (as per IR1.1.1. above) having regard to the probable deterioration of the pump and internal growth of corrosion products in suction line.

12.2.2.2 Diesel engines and fuel tank

12.2.2.2.1 Starting of diesel engine

Any diesel driven power source for the pump is to be capable of being readily started in its cold condition down to the temperature of 0°C by hand (manual) cranking. Where ready starting cannot be assured, if this is impracticable, or if lower temperature is likely to be encountered, and if the room for the diesel driven power source is not heated, electric heating of the diesel engine cooling water or lubricating oil system is to be fitted, to the satisfaction of IRS. If hand (manual) starting is impracticable, compressed air, electricity, or other sources of stored energy, including hydraulic power or starting cartridges may be used as a means of starting. These means are to be such as to enable the diesel-driven power source to be started at least six times within a period of 30 min and at least twice within the first 10 min.

12.2.2.2.2 Fuel tank capacity

.1 Any service fuel tank is to contain sufficient fuel to enable the pump to run on full load for at least six hours and sufficient reserves of fuel is to be available outside the machinery space of category A to enable the pump to be run on full load for an additional 15 h. For cargo ships of less than 5000 tons gross, the running time may be reduced to at least three hours.

IR.2 The fuel oil supplying is to be so arranged that it will not readily be affected by a fire in the compartment containing the main fire pumps and preferably located in the same compartment as where emergency fire pump is located.

Section 13

Arrangement of Means of Escape

13.1 Application

This chapter details the specifications for means of escape as required by Chapter 4.

13.2 Passenger ships

13.2.1 Width of stairways

13.2.1.1 Basic requirements for stairway width

Stairways are not to be less than 900 [mm] in clear width. The minimum clear width of stairways is to be increased by 10 mm for every one person provided for in excess of 90 persons. The total number of persons to be evacuated by such stairways is to be assumed to be two thirds of the crew and the total number of passengers in the areas served by such stairways. The width of the stairways is not be inferior to those determined by 13.2.1.2.

13.2.1.2 Calculation method of stairway width

13.2.1.2.1 Basic principles of the calculation

.1 This calculation method determines the minimum stairway width at each deck level, taking into account the consecutive stairways leading into the stairway under consideration.

.2 It is the intention that the calculation method is to consider evacuation from enclosed spaces within each main vertical zone individually and take into account all of the persons using the stairway enclosures in each zone, even if they enter that stairway from another vertical zone.

.3 For each main vertical zone the calculation is to be completed for the night time (case 1) and day time (case 2) and the largest dimension from either case used for determining the stairway width for each deck under consideration.

.4 The calculation of stairway widths are to be based upon the crew and passenger load on each deck. Occupant loads are to be rated by the designer for passenger and crew accommodation spaces, service spaces, control spaces and machinery spaces. For the purpose of the calculation the maximum capacity of a public space is to be defined by either of the following two values:

- the number of seats or similar arrangements, or

- the number obtained by assigning 2 [m^2] of gross deck surface area to each person.

13.2.1.2.2 Calculation method for minimum value

13.2.1.2.2.1 Basic formulae

In considering the design of stairway widths for each individual case which allow for the timely flow of persons evacuating to the muster stations from adjacent decks above and below, the following calculation methods are to be used (See Fig.13.1 and Fig.13.2):

when joining two decks:	$W=(N_1+N_2)$ 10 [mm];
when joining three decks:	$W=(N_1+N_2+0.5N_3)$ 10 [mm];
when joining four decks:	$W=(N_1+N_2+0.5N_3+0.25N_4)$ 10 [mm];
when joining five decks or more decks, the width of the stairways is to be determined by applying the above formula for four decks to the deck under consideration and to the consecutive deck.	

where:

W = the required tread width between handrails of the stairway.

The calculated value of W may be reduced where available landing area S is provided in stairways at the deck level defined by subtracting P from Z, such that:

$P = S \times 3.0$ persons/m^2 ; and $P_{max} = 0.25Z$

where:

Z = the total number of persons expected to be evacuated on the deck being considered;

P = the number of persons taking temporary refuge on the stairway landing, which may be subtracted from Z to a maximum value of P = 0.25Z (to be rounded down to the nearest whole number);

S = the surface area [m^2] of the landing, minus the surface area necessary for the opening of doors and minus the surface area necessary for accessing the flow on stairs (See Fig.13.1);

N = the total number of persons expected to use the stairway from each consecutive deck under consideration; N_1 is for the deck with the largest number of persons using that stairway; N_2 is taken for the deck with the next highest number of persons directly entering the stairway flow such that, when sizing the stairway width as each deck level, $N_1 > N_2 > N_3 > N_4$ (See Fig.13.2). These decks are assumed to be on or upstream (i.e. away from the embarkation deck) of the deck being considered.

Case 1: Passengers in cabins with maximum berthing capacity fully occupied; members of the crew in cabins occupied to 2/3 of maximum berthing capacity; and service spaces occupied by 1/3 of the crew.

Case 2: Passengers in public spaces occupied to 3/4 of maximum capacity; 1/3 of the crew distributed in public spaces; service spaces occupied by 1/3 of the crew; and crew accommodation occupied by 1/3 of the crew.

13.2.1.2.2.2 Distribution of persons

.1 The dimensions of the means of escape are to be calculated on the basis of the total number of persons expected to escape by the stairway and through doorways, corridors and landings (See Fig.13.3). Calculations are to be made separately for the two cases of occupancy of the spaces specified below. For each component part of the escape route, the dimension taken is not to be less than the largest dimension determined for each case.

.2 The maximum number of persons contained in a vertical zone, including persons entering stairways from another main vertical zone, is not to be assumed to be higher than the maximum number of persons authorized to be carried on board for the calculation of stairway width only.

13.2.1.3 Prohibition of decrease in width in the direction to the assembly station

(Refer to the Indication of the “assembly stations” in passenger ships (MSC/Circ.777)).

The stairway is not to decrease in width in the direction of evacuation to the assembly station, except in the case of several assembly stations in one main vertical zone the stairway width is not to decrease in the direction of the evacuation to the most distant assembly station.

13.2.2 Details of stairways

13.2.2.1 Handrails

Stairways are to be fitted with handrails on each side. The maximum clear width between handrails is to be 1,800 [mm].

All stairways sized for more than 90 persons are to be aligned fore and aft.

13.2.2.2 Alignment of stairways

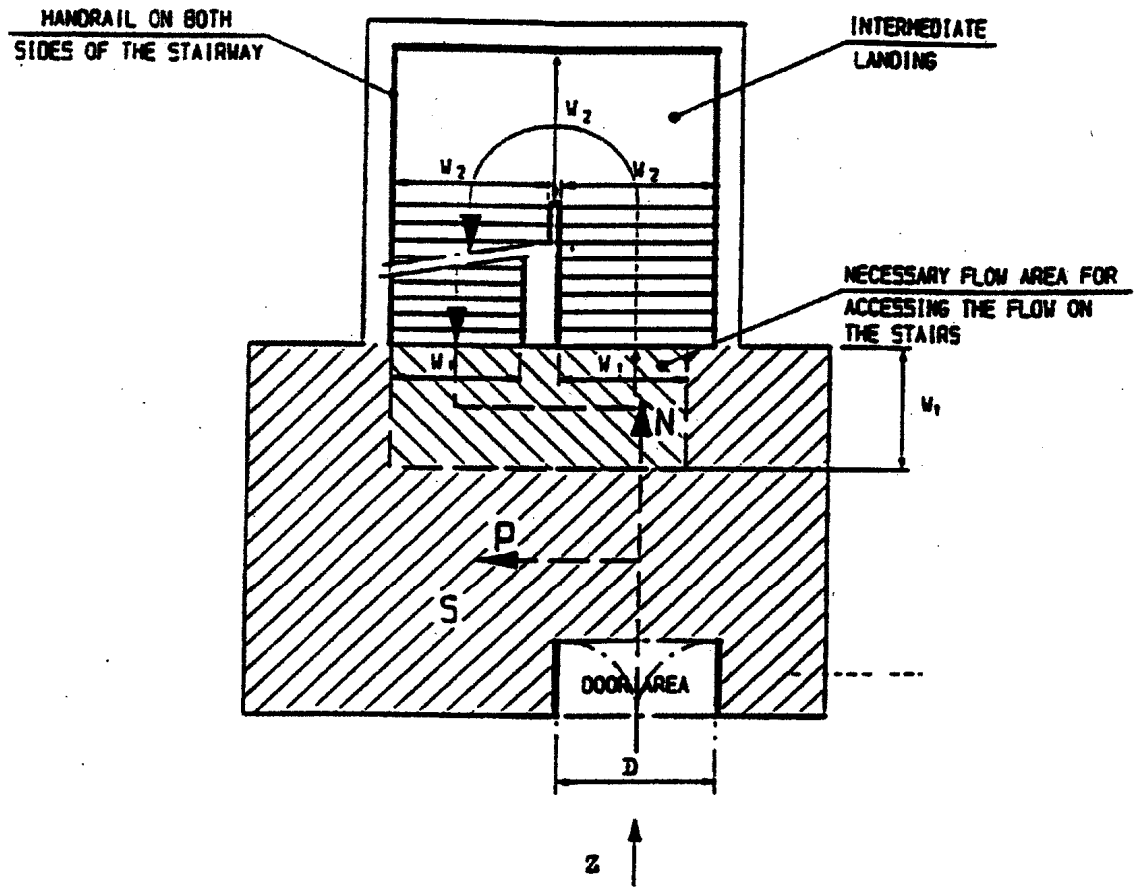


Fig.13.1 : Landing calculation for stairway width reduction

13.2.2.3 Vertical rise and inclination

Stairways are not to exceed 3.5 [m] in vertical rise without the provision of a landing and are not to have an angle of inclination greater than 45°.

13.2.2.4 Landings

With the exception of intermediate landings, landings at each deck level are not to be less than 2 [m^2] in area and is to increase by 1 [m^2] for every 10 persons provided for in excess of 20 persons but need not exceed 16 [m^2], except for those landings servicing public spaces having direct access onto the stairway enclosure. Intermediate landings are to be sized in accordance with 13.2.3.1.

13.2.3 Doorways and corridors

13.2.3.1 Doorways and corridors and intermediate landings included in means of escape are to be sized in the same manner as stairways.

13.2.3.2 The aggregate width of stairway exit doors to the assembly station are not to be less than the aggregate width of stairways serving this deck.

13.2.4 Evacuation routes to the embarkation deck

13.2.4.1 Assembly station

It is to be recognized that the evacuation routes to the embarkation deck may include an assembly station. In this case consideration is to be given to the fire-protection requirements and sizing of corridors and doors from the stairway enclosure to the assembly station and from the assembly station to the embarkation deck, noting that evacuation of persons from assembly stations to embarkation positions are to be carried out in small control groups.

13.2.4.2 Routes from the assembly station to the survival craft embarkation position

Where the passengers and crew are held at an assembly station which is not at the survival craft embarkation position, the dimension of stairway width and doors from the assembly station to this position are not to be based on the number of persons in the controlled group. The width of these stairways and doors need not exceed 1,500 mm unless larger dimensions are required for evacuation of these spaces under normal conditions.

13.2.5 Means of escape plans

13.2.5.1 Means of escape plans are to be provided indicating the following:

.1 the number of the crew and passengers in all normally occupied spaces;

.2 the number of crew and passengers expected to escape by stairway and through doorways, corridors and landings;

.3 assembly stations and survival craft embarkation positions;

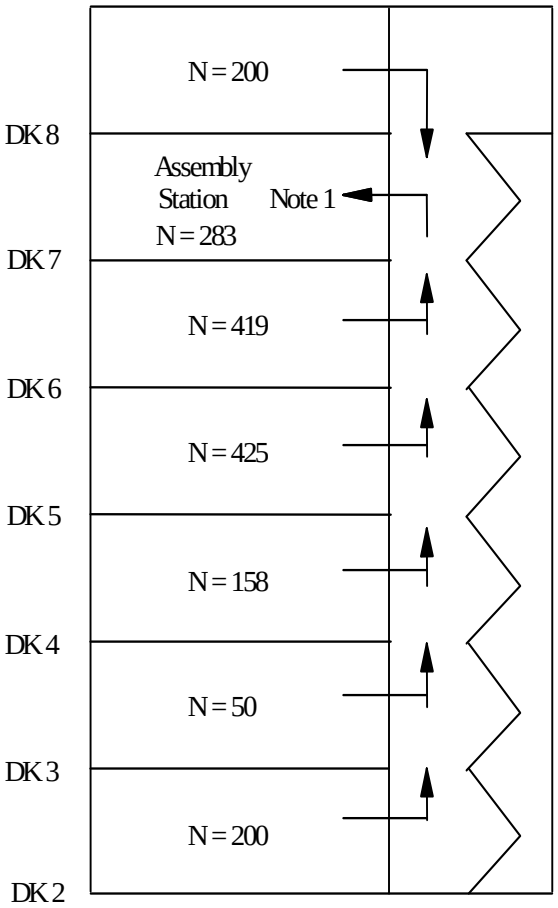
.4 primary and secondary means of escape; and

.5 width of stairways, doors, corridors and landing areas.

13.2.5.2 Means of escape plans are to be accompanied by detailed calculation for determining the width of escape stairways, doors, corridors and landing areas.

13.3 Cargo ships

Stairways and corridors used as means of escape are to be not less than 700 [mm] in clear width and is to have a handrail on one side. Stairways and corridors with a clear width of 1,800 [mm] and over is to have handrails on both sides. “Clear width” is considered the distance between the handrail and the bulkhead on the other side or between the handrails. The angle of inclination of stairways are to be, in general, 45º, but not greater than 50º, and in machinery spaces and small spaces not more than 60º. Doorways which give access to a stairway are to be of the same size as the stairway.



$N_1 = 200$
$W = 2000$

$D = 2000 + 9355 = 11,355$

$N_1 = 425, N_2 = 419, N_3 = 158, N_4 = 50$
$W = (425 + 419 + 0.50 * 158 + 0.25 * 50) * 10 = 9355$

$N_1 = 425, N_2 = 200, N_3 = 158, N_4 = 50$
$W = (425 + 200 + 0.5 * 158 + 0.25 * 50) * 10 = 7165$

$N_1 = 200, N_2 = 158, N_3 = 50$
$W = (200 + 158 + 0.5 * 50) * 10 = 3830$

$N_1 = 200, N_2 = 50$
$W = (200 + 50) * 10 = 2500$

$N_1 = 200$
$W = 200 * 10 = 2000$

Z(pers)	=	number of persons expected to evacuate through the stairway
N(pers)	=	number of persons directly entering the stairway flow from a given deck
W(mm)	=	$(N_1 + N_2 + 0.5 \times N_3 + 0.25 \times N_4) \times 10$ = calculated width of stairway
D(mm)	=	width of exit doors

$N_1 > N_2 > N_3 > N_4$ where:

N_1(pers)	=	the deck with the largest number of persons N entering directly the stairway
N_2(pers)	=	the deck with the next largest number of persons N entering directly the stairway, etc.

Note 1 : The doors to the assembly station should have aggregate width of 11,355 [mm].

Fig.13.2 : Minimum stairway width (W) calculation example

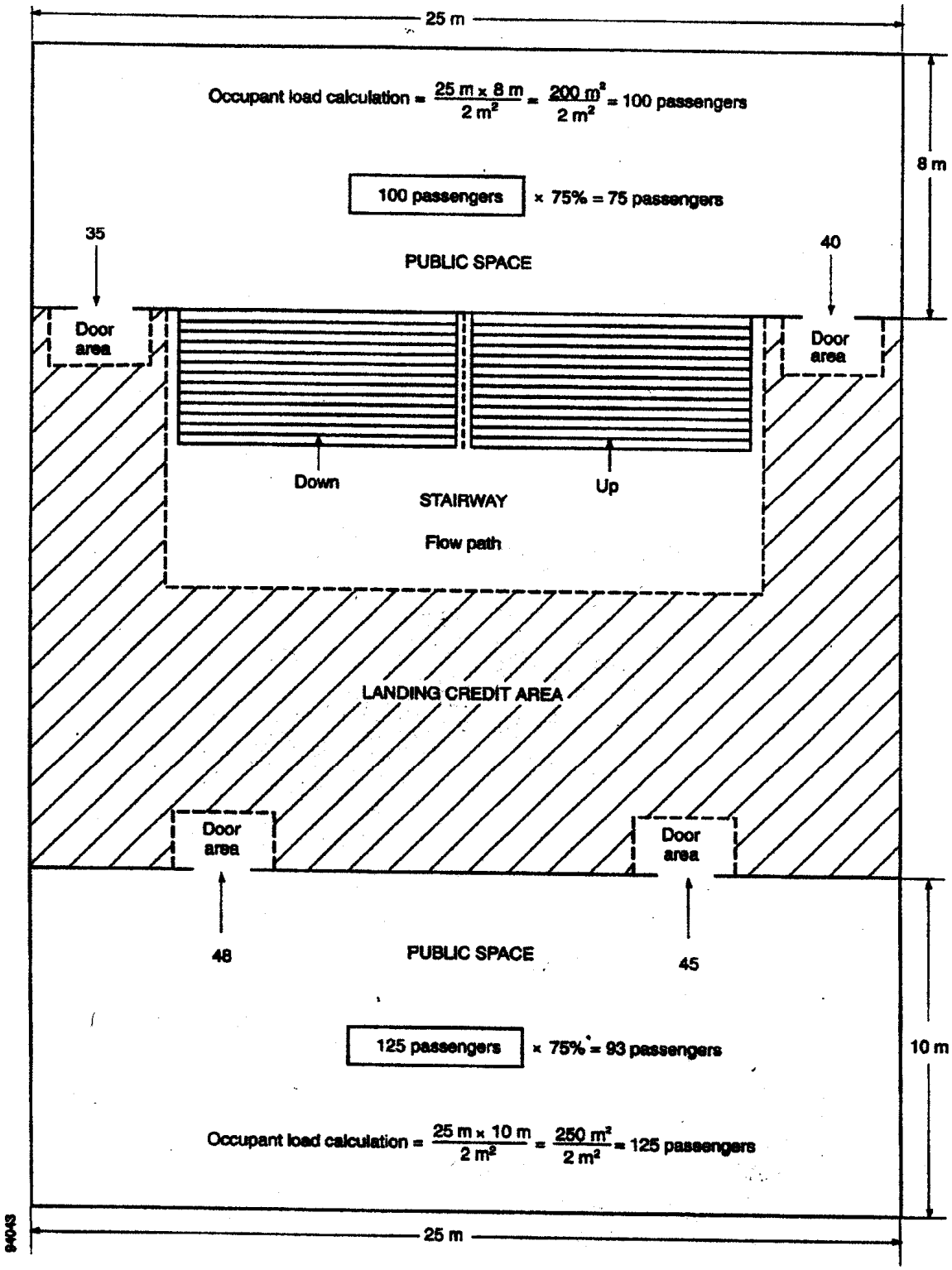


Fig.13.3 : Occupant loading calculation example

Section 14

Fixed Deck Foam Systems

14.1 Application

This section details the specifications for fixed deck foam systems which are required to be provided by Ch.3, Sec.4.

14.2 Engineering specifications

14.2.1 General

14.2.1.1 The arrangements for providing foam are to be capable of delivering foam to the entire cargo tank area as well as into any cargo tank, the deck of which has been ruptured.

14.2.1.2 The deck foam system is to be capable of simple and rapid operation.

14.2.1.3 Operation of a deck foam system at its required output is to permit the simultaneous use of the minimum required number of jets of water at the required pressure from the fire main. Where the deck foam system is supplied by a common line from the fire main, additional foam concentrate is to be provided for operation of two nozzles for the same period of time required for the foam system. The simultaneous use of the minimum required jets of water is to be possible on deck over the full length of the ship, in the accommodation, service spaces, control stations and machinery spaces.

IR14.2.1.3 A common line for fire main and deck foam line can only be considered acceptable provided it can be demonstrated that the hose nozzles can be effectively controlled by one person when supplied from the common line at a pressure needed for operation of the monitors.

The simultaneous use of the minimum required jets of water is to be possible on deck over the full length of the ship in accommodation, service spaces, control stations and machinery spaces.

14.2.2 Component requirements

14.2.2.1 Foam solution and foam concentrate

14.2.2.1.1 For tankers carrying:

.1 crude oil or petroleum products having a flashpoint not exceeding 60ºC (closed cup), as determined by an approved flashpoint apparatus, and a Reid vapour pressure which is below atmospheric pressure or other liquid products having a similar fire hazard, including cargoes in chapter 18 of the IBC Code, having a flashpoint not exceeding 60ºC (closed cup) for which a regular foam fire-fighting system is effective or

.2 petroleum products with a flashpoint exceeding 60ºC (closed cup), as determined by an approved flashpoint apparatus or

.3 IBC Code chapter 17 products with a flashpoint exceeding 60ºC (closed cup) determined by an approved flashpoint apparatus.

The rate of supply of foam solution is not to be less than the greatest of the following :

a) 0.6 [litre/minute/m^2] of cargo deck area, where the cargo deck area means the maximum breadth of the ship multiplied by the total longitudinal extent of the cargo tank spaces;

b) 6 [litres/minute/m^2] of the horizontal sectional area of the single tank having the largest such area; or

c) 3 [litres/minute/m^2] of the area protected by the largest monitor, such area being entirely forward of the monitor, but not less than 1250 litre/minute.

14.2.2.1.2 For tankers carrying chemicals in bulk listed in chapter 17 of the IBC Code having a flashpoint not exceeding 60ºC (closed cup), the rate of supply of foam solution is to be as required by the IBC Code.

14.2.2.1.3 Sufficient foam concentrate is to be supplied to ensure at least 20 min of foam generation in tankers fitted with an inert gas installation or 30 min of foam generation in tankers not fitted with an inert gas installation or not required to use an inert gas system.

14.2.2.1.4 The foam concentrate supplied on board is to be approved* by IRS for the cargoes intended to be carried. Type B foam

concentrates are to be supplied for the protection of crude oil, petroleum products and non-polar solvent cargoes. Type A foam concentrates are to be supplied for polar solvent cargoes, as listed in the table of chapter 17 of the IBC Code. Only one type of foam concentrate is to be supplied, and it is to be effective for the maximum possible number of cargoes intended to be carried. For cargoes for which foam is not effective or is incompatible, additional arrangements to the satisfaction of IRS is to be provided.

* Refer to the Guidelines for performance and testing criteria and surveys of foam concentrates for fixed fire-extinguishing systems (MSC.1/Circ.1312 and MSC.1/Circ.1312/ Corr.1).

14.2.2.1.5 Liquid cargoes with a flashpoint not exceeding 60^{o}C for which a regular foam fire-fighting system is not effective are to comply with Pt.6, Ch.1, Sec.1.5.2.1.

14.2.2.2 Monitors and foam applicators

14.2.2.2.1 Foam from the fixed foam system is to be supplied by means of monitors and foam applicators. Prototype tests of the monitors and foam applicators shall be performed to ensure the foam expansion and drainage time of the foam produced does not differ more than ± 10 per cent of that determined in paragraph 14.2.2.1.4. When medium expansion ratio foam (between 21 to 1 and 200 to 1 expansion ratio) is employed, the application rate of the foam and the capacity of a monitor installation is to be to the satisfaction of IRS. At least 50% of the foam solution supply rate required is to be delivered from each monitor. On tankers of less than 4,000 tonnes deadweight only applicators may be provided and monitors may not be required. Use of the capacity of each applicator is to be at least 25% of the foam solution required supply rate.

IR14.2.2.2.1 The requirements given in 14.2.2.2.3 and 14.2.3.3 apply to all tankers regardless of deadweight.

14.2.2.2.2 The capacity of any applicator is to be not less than 400 l/min and the applicator throw in still air conditions is to be not less than 15 [m].

14.2.3 Installation requirements

14.2.3.1 Main control station

The main control station for the system is to be suitably located outside the cargo area, adjacent to the accommodation spaces and readily accessible and operable in the event of fire in the areas protected.

14.2.3.2 Monitors

14.2.3.2.1 The number and position of monitors are to be such as to comply with 14.2.1.1.

14.2.3.2.2 The distance from the monitor to the farthest extremity of the protected area forward of that monitor is not to be more than 75per cent of the monitor throw in still air conditions.

14.2.3.2.3 A monitor and hose connection for a foam applicator are to be situated both port and starboard at the poop front or accommodation spaces facing the cargo deck. These monitors may be located in cargo area as defined in Ch.1, Sec.3 provided they are aft of cargo tanks and that they protect the areas below and aft of each other. On tankers of less than 4,000 tonnes deadweight a hose connection for a foam applicator is to be situated both port and starboard at the front of the poop or accommodation spaces facing the cargo deck.

IR14.2.3.2.3 The port and starboard monitors required by Cl. 14.2.3.2.3 may be located in the cargo area above oil bunker tanks adjacent to cargo tanks if capable of protecting the deck below and aft of each other.

14.2.3.3 Applicators

14.2.3.3.1 The number of foam applicators provided are to be not less than four. The number and disposition of foam main outlets is to be such that foam from at least two applicators can be directed on to any part of the cargo tanks deck area.

14.2.3.3.2 Applicators are to be provided to ensure flexibility of action during fire-fighting operations and to cover areas screened from the monitors.

14.2.3.4 Isolation valves

Valves are to be provided in both the foam main and fire main when this is an integral part of the deck foam system, immediately forward of every monitor position to isolate damaged sections of those mains.

Section 15

Inert Gas Systems

15.1 Application

This Section details the specifications for inert gas systems as required by Ch.2, 1.5.5.

15.2 Engineering specifications

15.2.1 Definitions

For the purposes of this Section:

15.2.1.1 *Cargo tanks* means those cargo tanks, including slop tanks, which carry cargoes, or cargo residues, having a flashpoint not exceeding 60ºC.

15.2.1.2 *Inert gas system* includes inert gas systems using flue gas, inert gas generators, and nitrogen generators and means the inert gas plant and inert gas distribution together with means for preventing backflow of cargo gases to machinery spaces, fixed and portable measuring instruments and control devices.

15.2.1.3 *Gas-safe space* is a space in which the entry of gases would produce hazards with regard to flammability or toxicity.

15.2.1.4 *Gas-free* is a condition in a tank where the content of hydrocarbon or other flammable vapour is less than 1% of the lower flammable limit (LFL), the oxygen content is at least 21%, by volume and no toxic gases are present. Refer to the *Revised recommendations for entering enclosed spaces aboard ships* (Resolution A.1050(27)).

15.2.2 Requirements for all Systems

15.2.2.1 General

15.2.2.1.1 The inert gas system referred to in Chapter 2 is to be designed, constructed and tested to the satisfaction of IRS. It is to be so designed and operated as to render and maintain the atmosphere of the cargo tanks non-flammable.

IR 15.2.2.1.1 Plans in diagrammatic form are to be submitted for appraisal and are to include the following:

- Details and arrangement of the inert gas generating plant including all control and monitoring devices;
- Arrangement of the piping system for distribution of the inert gas.

(Refer to the revised standards for the design, testing and locating of devices to prevent the passage of flame into cargo tanks in tankers (MSC/Circ.677, as amended by MSC/Circ.1009 and MSC.1/Circ. 1324) and the Revised factors to be taking into consideration when designing cargo tank venting and gas-freeing arrangements (MSC/Circ.731)).

15.2.2.1.2 The system is to be capable of:

.1 inerting empty cargo tanks and maintaining the atmosphere in any part of the tank with an oxygen content not exceeding 8% by volume and at a positive pressure in port and at sea except when it is necessary for such a tank to be gas-free;

.2 eliminating the need for air to enter a tank during normal operations except when it is necessary for such a tank to be gas-free;

.3 purging empty cargo tanks of hydrocarbon or other flammable vapours, so that subsequent gas-freeing operations will at no time create a flammable atmosphere within the tank;

.4 delivering inert gas to the cargo tanks at a rate of at least 125% of the maximum rate of discharge capacity of the ship expressed as a volume. For chemical tankers and chemical/ product tankers, IRS may accept inert gas systems having a lower delivery capacity provided that the maximum rate of discharge of cargoes from cargo tanks being protected by the system is restricted to not more than 80% of the inert gas capacity; and

.5 delivering inert gas with an oxygen content of not more than 5% by volume to the cargo tanks at any required rate of flow.

15.2.2.1.3 Materials used in inert gas systems are to be suitable for their intended purpose. In particular, those components which may be subjected to corrosive action of the gases and/ or liquids are to be either constructed of corrosion-resistant material or lined with rubber, glass fibre epoxy resin or other equivalent coating material.

15.2.2.1.4 The inert gas supply may be:

.1 treated flue gas from main or auxiliary boilers, or

.2 gas from an oil or gas-fired gas generator, or

.3 gas from nitrogen generators.

IRS may accept systems using inert gases from one or more separate gas generators or other sources or any combination thereof, provided that an equivalent level of safety is achieved. Such systems are to, as far as practicable; comply with the requirements of this Section. Systems using stored carbon dioxide are not to be used unless IRS is satisfied that the risk of ignition from generation of static electricity by the system itself is minimized.

IR 15.2.2.1.4 In all cases, automatic control capable of producing suitable inert gas under all service conditions, is to be fitted.

15.2.2.2 Safety Measures

.1 The inert gas system is to be so designed that the maximum pressure which it can exert on any cargo tank will not exceed the test pressure of any cargo tank.

.2 Automatic shutdown of the inert gas system and its components parts are to be arranged on predetermined limits being reached, taking into account the provisions of paragraphs 15.2.2.4, 15.2.3.2 and 15.2.4.2.

IR .2 The automatic shutdown of the inert gas system and its components is to involve in following:

.1 Shutdown of fans and closing of regulating valve for the following:

.1 high water level in scrubber (not applicable for N_2);

.2 low pressure/flow to scrubber (not applicable for N_2); or

.3 high-high temperature of inert gas supply

.2 closing of regulating valve in the event of:

.1 high oxygen content (in excess of 5% by volume); or

.2 failure of blowers/fans or N2 compressors

.3 activation of double-block and bleed arrangement upon:

.1 loss of inert gas supply (for ships with double block and bleed replacing water seal); or

.2 loss of power

.3 Suitable shutoff arrangements are to be provided on the discharge outlet of each generator plant.

.4 The system is to be designed to ensure that if the oxygen content exceeds 5% by volume, the inert gas shall be automatically vented to atmosphere.

.5 Arrangements are to be provided to enable the functioning of the inert gas plant to be stabilized before commencing cargo discharge. If blowers are to be used for gas-freeing, their air inlets are to be provided with blanking arrangements.

.6 Where a double block and bleed valve is installed, the system is to ensure that, upon loss of power, the block valves are automatically closed and the bleed valve is automatically open.

15.2.2.3 System Components

15.2.2.3.1 Non-Return Devices

.1 At least two non-return devices are to be fitted in order to prevent the return of vapour and liquid to the inert gas plant, or to any gas-safe spaces.

.2 The first non-return device is to be a deck seal of the wet, semi-wet, or dry type or a

double-block and bleed arrangement given in Pt.5, Ch.3, 21.1.2. Two shut-off valves in series with a venting valve in between, may be accepted by IRS provided:

.1 the operation of the valve is automatically executed. Signal(s) for opening/closing is (are) to be taken from the process directly, e.g. inert gas flow or differential pressure; and

.2 alarm for faulty operation of the valves is provided, e.g. the operation status of "blower stop" and "supply valve(s) open" is an alarm condition.

.3 The second non-return device is to be a non-return valve or equivalent capable of preventing the return of vapours and liquids and fitted between the deck water seal (or equivalent device) and the first connection from the inert gas main to a cargo tank. It is to be provided with positive means of closure. As an alternative to positive means of closure, an additional valve having such means of closure may be provided between the non-return valve and the first connection to the cargo tanks to isolate the deck water seal, or equivalent device, from the inert gas main to the cargo tanks.

.4 A water seal, if fitted, is to be capable of being supplied by two separate pumps, each of which shall be capable of maintaining an adequate supply at all times. The audible and visual alarm on the low level of water in the water seal are to operate at all times.

.5 The arrangement of the water seal, or equivalent devices, and its associated fittings is to be such that it will prevent backflow of vapours and liquids and will ensure the proper functioning of the seal under operating conditions.

.6 Provision is to be made to ensure that the water seal is protected against freezing, in such a way that the integrity of seal is not impaired by overheating.

.7 A water loop or other approved arrangement is also to be fitted to each associated water supply and drain pipe and each venting or pressure-sensing pipe leading to gas-safe spaces. Means are to be provided to prevent such loops from being emptied by vacuum.

.8 Any water seal, or equivalent device, and loop arrangements are to be capable of preventing return of vapours and liquids to an inert gas plant at a pressure equal to the test pressure of the cargo tanks.

.9 The non-return devices are to be located in the cargo area on deck.

15.2.2.3.2 Inert Gas Lines

.1 The inert gas main may be divided into two or more branches downstream of the non-return devices required by 15.2.2.3.1.

.2 The inert gas main is to be fitted with branch piping leading to the cargo tank. Branch piping for inert gas is to be fitted with either stop valves or equivalent means of control for isolating each tank. Where stop valves are fitted, they are to be provided with locking arrangements. The control system is to provide unambiguous information of the operational status of such valves to at least the control panel required in paragraph 15.2.2.4.

IR .2 Unambiguous information regarding the operational status of stop valves in branch piping leading from the inert gas main to cargo tanks means position indicators providing open/intermediate/closed status information in the control panel required in 15.2.2.4.Limits switches are to be used to positively indicate both open and closed positions. Intermediate position status is to be indicated when the valve is in neither open nor closed position.

.3 Each cargo tank not being inerted is to be capable of being separated from the inert gas main by:

.1 removing spool-pieces, valves or other pipe sections, and blanking the pipe ends; or

.2 arrangement of two spectacle flanges in series with provisions for detecting leakage into the pipe between the two spectacle flanges; or

.3 equivalent arrangements to the satisfaction of IRS, providing at least the same level of protection.

.4 Means are to be provided to protect cargo tanks against the effect of overpressure or vacuum caused by thermal variations and/ or

cargo operations when the cargo tanks are isolated from the inert gas mains.

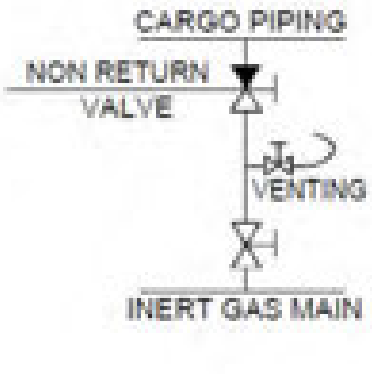
.5 Piping systems are to be so designed as to prevent the accumulation of cargo or water in the pipelines under all normal conditions.

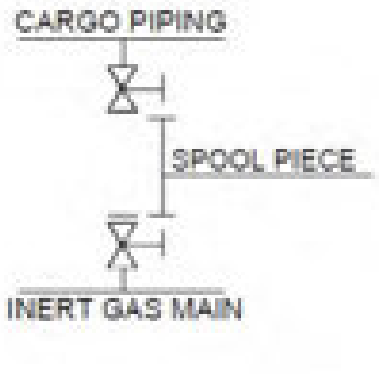
.6 Arrangements are to be provided to enable the inert gas main to be connected to an external supply of inert gas. The arrangements are to consist of a 250 [mm] nominal pipe size bolted flange, isolated from the inert gas main by a valve and located downstream of the non-return valve. The design of the flange is to conform to the appropriate class in the standards adopted for the design of other external connections in the ship's cargo piping system.

.7 If a connection is fitted between the inert gas main and the cargo piping system, arrangements are to be made to ensure an effective isolation having regard to the large pressure difference which may exist between the systems. This is to consist of two shutoff valves with an arrangement to vent the space between the valves in a safe manner or an arrangement consisting of a spool-piece with associated blanks.

.8 The valve separating the inert gas main from the cargo main and which is on the cargo main side is to be a non-return valve with a positive means of closure.

IR .8 The effective isolation required by 15.2.2.3.2.7 and 15.2.2.3.2.8 may be achieved by the two arrangements shown in the sketches below:





.9 Inert gas piping systems are not to pass through accommodation, service and control station spaces.

.10 In combination carriers, the arrangement to isolate the slop tanks containing oil or oil residues from other tanks is to consist of blank flanges which will remain in position at all times when cargoes other than oil are being carried except may be accepted by the National Statutory Authority (Refer to the revised guidelines for inert gas systems (MSC/Circ.353), as amended by MSC/Circ.387).

15.2.2.4 Indicators and Alarms (Also refer 15.2.2.2)

15.2.2.4.1 The operation status of the inert gas system is to be indicated in a control panel.

IR 15.2.2.4.1 The operational status of the inert gas system is to be based on indication that inert gas is being supplied downstream of the gas regulating valve and on the pressure or flow of the inert gas mains upstream of the non-return devices. However, the operational status of the inert gas system as required in paragraph 15.2.2.4.1 is not to be considered to require additional indicators and alarms other than those specified in paragraphs 15.2.2.4 and 15.2.3.2 or 15.2.4.2, as appropriate.

15.2.2.4.2 Instrumentation is to be fitted for continuously indicating and permanently recording, when inert gas is being supplied:

.1 the pressure of the inert gas mains downstream of the non-return devices; and

.2 the oxygen content of the inert gas.

15.2.2.4.3 The indicating and recording devices are to be placed in the cargo control room where provided. But where no cargo

control room is provided, they are to be placed in a position easily accessible to the officer in charge of cargo operations.

15.2.2.4.4 In addition, meters are to be fitted:

.1 in the navigating bridge to indicate at all times the pressure referred to in 15.2.2.4.2.1 and the pressure in the slop tanks of combination carriers, whenever those tanks are isolated from the inert gas main; and

.2 in the machinery control room or in the machinery space to indicate the oxygen content referred to in paragraph 15.2.2.4.2.2.

15.2.2.4.5 Audible and visual alarms

.1 Audible and visual alarms are to be provided, based on the system designed, to indicate:

.1 oxygen content in excess of 5% by volume;

.2 failure of the power supply to the indicating devices as referred to in paragraph 15.2.2.4.2;

.3 gas pressure less than 100 [mm] water gauge. The alarm arrangement is to be such as to ensure that the pressure in slop tanks in combination carriers can be monitored at all times;

.4 high-gas pressure; and

.5 failure of the power supply to the automatic control system

.2 The alarms required in 15.2.2.4.5.1.1, 15.2.2.4.5.1.3 and 15.2.2.4.5.1.5 are to be fitted in the machinery space and cargo control room, where provided, but in each case in such a position that they are immediately received by responsible members of the crew.

.3 An audible alarm system independent of that required in 15.2.2.4.5.1.3 or automatic shutdown of cargo pumps are to be provided to operate on predetermined limits of low pressure in the inert gas main being reached.

IR .3 The term "alarm system independent" means that a second pressure sensor, independent of the sensor serving the alarms for low pressure, high pressure and pressure indicator/recorder is to be provided. Notwithstanding the above, a common programmable logic controller (PLC) is to, however, be accepted for the alarms in the control system. The independent sensor is not required if the system is arranged for the shutdown of cargo pumps. If a system for shutdown of cargo pumps is arranged, an automatic system shutting down all cargo pumps is to be provided. The shutdown is to be alarmed at the control station. The shutdown is not to prevent the operation of ballast pumps or pumps used for bilge drainage of a cargo pump room.

.4 Two oxygen sensors are to be positioned at appropriate locations in the space or spaces containing the inert gas system. If the oxygen level falls below 19.5%, these sensors are to trigger alarms, which are to be both visible and audible inside and outside the space or spaces and are to be placed in such a position that they are immediately received by responsible members of the crew.

15.2.2.5 Instruction Manuals

.1 Detailed instruction manuals are to be provided on board, covering the operations, safety and maintenance requirements and occupational health hazards relevant to the inert gas system and its application to the cargo tank system. The manuals are to include guidance on procedures to be followed in the event of a fault or failure of the inert gas system (Refer to the revised guidelines for inert gas systems (MSC/Circ.353), as amended by MSC/Circ.387).

15.2.3 Requirements for flue gas and inert gas generators systems

In addition to the provisions in 15.2.2, for inert gas systems using flue gas or inert gas generators, the provisions of this sub-section shall apply.

15.2.3.1 System requirements

15.2.3.1.1 Inert Gas Generators

.1 Two fuel oil pumps are to be fitted to the inert gas generator. Suitable fuel in sufficient quantity is to be provided for the inert gas generators.

.2 The inert gas generators are to be located outside the cargo tank area. Spaces containing inert gas generators are to have no

direct access to accommodation service or control station spaces, but may be located in machinery spaces. If they are not located in machinery spaces, such a compartment is to be separated by a gastight steel bulkhead and/ or deck from accommodation, service and control station spaces. Adequate positive-pressure-type mechanical ventilation is to be provided for such a compartment.

15.2.3.1.2 Gas Regulating Valves

.1 A gas regulating valve is to be fitted in the inert gas main. This valve is to be automatically controlled to close, as required in 15.2.2.2.2. It is also to be capable of automatically regulating the flow of inert gas to the cargo tanks unless means are provided to automatically control the inert gas flow rate.

.2 The gas regulating valve is to be located at the forward bulkhead of the forward most gas-safe space through which the inert gas main passes.

15.2.3.1.3 Cooling and Scrubbing Arrangement

.1 Means are to be fitted which will effectively cool the volume of gas specified in 15.2.2.1.2 and remove solids and sulphur combustion products. The cooling water arrangements are to be such that an adequate supply of water will always be available without interfering with any essential services on the ship. Provision is also to be made for an alternative supply of cooling water.

.2 Filters or equivalent devices are to be fitted to minimize the amount of water carried over to the inert gas blowers.

15.2.3.1.4 Blowers

.1 At least two inert gas blowers are to be fitted and be capable of delivering to the cargo tanks at least the volume of gas required by 15.2.2.1.2. For systems fitted with inert gas generators one blower only may be accepted if that system is capable of delivering the total volume of gas required by 15.2.2.1.2 to the cargo tanks, provided that sufficient spares for the blower and its prime mover are carried on board to enable any failure of the blower and its prime mover to be rectified by the ship's crew.

.2 Where inert gas generators are served by positive displacement blowers, a pressure relief device is to be provided to prevent excess pressure being developed on the discharge side of the blower.

.3 When two blowers are provided, the total required capacity of the inert gas system is to be divided evenly between the two and in no case is one blower to have a capacity less than 1/3 of the total capacity required.

15.2.3.1.5 Inert gas isolating valves

.1 For systems using flue gas, flue gas isolating valves are to be fitted in the inert gas mains between the boiler uptakes and the flue gas scrubber. These valves are to be provided with indicators to show whether they are open or shut, and precautions are to be taken to maintain them gastight and keep the seatings clear of soot. Arrangements are to be made to ensure that boiler soot blowers cannot be operated when the corresponding flue gas valve is open.

15.2.3.1.6 Prevention of flue gas leakage

.1 Special consideration is to be given to the design and location of scrubber and blowers with relevant piping and fittings in order to prevent flue gas leakages into enclosed spaces.

.2 To permit safe maintenance, an additional water seal or other effective means of preventing flue gas leakage are to be fitted between the flue gas isolating valves and scrubber or incorporated in the gas entry to the scrubber.

15.2.3.2 Indicators and Alarms (Also refer 15.2.2.2)

.1 To permit safe maintenance, an additional water seal or other effective means of preventing flue gas leakage is to be fitted between the flue gas isolating valves and scrubber or incorporated in the gas entry to the scrubber.

.2 In addition to the requirements of 15.2.2.4.5, audible and visual alarms are to be provided to indicate:

.1 insufficient fuel oil supply to the oil-fired inert gas generator;

.2 failure of the power supply to the generator;

.3 low water pressure or low water flow rate to the cooling and scrubbing arrangement;

.4 high water level in the cooling and scrubbing arrangement;

.5 high gas temperature;

.6 failure of the inert gas blowers; and

.7 low water level in the water seal.

15.2.4 Requirements for Nitrogen Generator Systems

The provisions of this section shall apply, in addition to the provisions in 15.2.2, for inert gas systems using nitrogen generators.

15.2.4.1 System Requirements

.1 The system is to be provided with one or more compressors to generate enough positive pressure to be capable of delivering the total volume of gas required by 15.2.2.1.2.

.2 A feed air treatment system is to be fitted to remove free water, particles and traces of oil from the compressed air.

.3 The air compressor and nitrogen generator may be installed in the engine-room or in a separate compartment. A separate compartment and any installed equipment are to be treated as an "Other machinery space" with respect to fire protection. Where a separate compartment is provided for the nitrogen generator, the compartment is to be fitted with an independent mechanical extraction ventilation system providing six air changes per hour. The compartment is to have no direct access to accommodation spaces, service spaces and control stations.

Note: Supply air requirement and arrangement should take into account the need for the nitrogen generator system as well as for the compartment ventilation.

.4 Where a nitrogen receiver or a buffer tank is installed, it may be installed in a dedicated compartment, in a separate compartment containing the air compressor and the nitrogen generator, in the engine room, or in the cargo area. Where the nitrogen receiver or a buffer tank is installed in an enclosed space, the access is to be arranged only from the open deck and the access door is to open outwards. Adequate, independent mechanical ventilation, of the extraction type, is to be provided for such a compartment.

Note: Additional options for installation of the nitrogen receiver or buffer tank may be considered based on the risk and provision of adequate ventilation arrangements

15.2.4.2 Indicators and Alarms (Also refer 15.2.2.2)

.1 In addition to the requirements in 15.2.2.4.2, instrumentation is to be provided for continuously indicating the temperature and pressure of air at the suction side of the nitrogen generator.

.2 In addition to the requirements in 15.2.2.4.5, audible and visual alarms are to be provided to include:

.1 failure of the electric heater, if fitted;

.2 low feed-air pressure or flow from the compressor;

.3 high-air temperature; and

.4 high condensate level at automatic drain of water separator.

IR15.2.5 Additional Requirements for Nitrogen Generator Systems on Tankers, including Chemical Tankers, to which Ch.2, 1.5.5 applies

IR15.2.5.1 The following requirements apply where a nitrogen generator system is fitted on board as required by Part 6, Ch.2, 1.5.5.1. For the purpose, the inert gas is to be produced by separating air into its component gases by passing compressed air through a bundle of hollow fibres, semi-permeable membranes or adsorber materials.

IR15.2.5.2 In addition to the applicable requirements of 15.2.4, the nitrogen generator system is to comply with Ch.2, 1.5.2.6, 1.5.6.3 and Ch.3, 5.6.3.4.

IR15.2.5.3 A nitrogen generator is to consist of a feed air treatment system and any number of membrane or adsorber modules in parallel necessary to meet 15.2.2.1.2.4.

IR15.2.5.4 The nitrogen generator is to be capable of delivering high purity nitrogen in accordance with 15.2.2.1.2.5. In addition to

15.2.2.2.4, the system is to be fitted with automatic means to discharge "off-spec" gas to the atmosphere during start-up and abnormal operation.

IR15.2.5.5 The system is to be provided with one or more compressors to generate enough positive pressure to be capable of delivering the total volume of gas required by 15.2.2.1.2. Where two compressors are provided, the total required capacity of the system is preferably to be divided equally between the two compressors, and in no case is one compressor to have a capacity less than 1/3 of the total capacity required.

IR15.2.5.6 The feed air treatment system fitted to remove free water, particles and traces of oil from the compressed air as required by 15.2.4.1.2, is also to preserve the specification temperature.

IR15.2.5.7 The oxygen-enriched air from the nitrogen generator and the nitrogen-product enriched gas from the protective devices of the nitrogen receiver are to be discharged to a safe location(Note 1) on the open deck.

Note 1: “safe location” needs to address the two types of discharges separately:

1. oxygen-enriched air from the nitrogen generator - safe locations on the open deck are:

- outside of hazardous area;

- not within 3 [m] of areas traversed by personnel; and

- not within 6 [m] of air intakes for machinery (engines and boilers) and all ventilation inlets.

2. nitrogen-product enriched gas from the protective devices of the nitrogen receiver - safe locations on the open deck are:

- not within 3 [m] of areas traversed by personnel; and

- not within 6 [m] of air intakes for machinery (engines and boilers) and all ventilation inlets/outlets.

IR15.2.5.8 In order to permit maintenance, means of isolation are to be fitted between the generator and the receiver.

IR15.2.6 Nitrogen/ Inert Gas Systems Fitted for Purposes other than Inerting required by Ch.2, 1.5.5

IR15.2.6.1 These requirements are applicable to systems fitted on oil tankers, gas tankers or chemical carriers to which requirements of Ch. 2, 1.5.5 do not apply.

IR15.2.6.2 Paragraphs 15.2.2.2.2, 15.2.2.2.4, 15.2.2.4.2, 15.2.2.4.3, 15.2.2.4.5.1.1, 15.2.2.4.5.1.2, 15.2.2.4.5.4, 15.2.4.1.1, 15.2.4.1.2, 15.2.4.1.3, 15.2.4.1.4, 15.2.4.1.8, 15.2.4.1.10, 15.2.4.1.11, 15.2.4.1.12, 15.2.4.1.13, 15.2.4.2.1 and 15.2.4.2.2 of this Section apply to these systems, as applicable.

IR15.2.6.3 The requirements of IR15.2.5 apply except IR15.2.5.1, IR15.2.5.2, IR15.2.5.4, and IR15.2.5.5.

IR15.2.6.4 Materials used in inert gas systems are to be suitable for their intended purpose.

IR15.2.6.5 All the equipment is to be installed and tested onboard to the satisfaction of the Surveyor.

IR15.2.6.6 The two non-return devices as required by 15.2.2.3.1.1, are to be fitted in the inert gas main. The non-return devices are to comply with 15.2.2.3.1.2 and 15.2.2.3.1.3; however, where the connections to the cargo tanks, to the hold spaces or to cargo piping are not permanent, the non-return devices required by 15.2.2.3.1.1 may be substituted by two non-return valves.

Section 16

Fixed Hydrocarbon Gas Detection Systems

16.1 Application

16.1.1 This section details the specifications for fixed hydrocarbon gas detection systems required by this part of the Rules.

16.1.2 A combined gas detection system required by Ch.2, 1.5.7.3 (requirement for fixed gas detection system in double hull and bottom spaces) and 1.5.10 (condition monitoring in cargo pump rooms) may be accepted in cases where the system fully complies with the requirement of Ch.1, Sec.2.

16.2 Engineering specifications

16.2.1 General

16.2.1.1 The fixed hydrocarbon gas detection system referred to in this part of the Rules is to be designed, constructed and tested to the satisfaction of IRS based on performance standards developed by IMO*.

* Refer to the Guidelines for the design, construction and testing of fixed hydrocarbon gas detection system (MSC.1/Circ.1370).

16.2.1.2 The system is to be comprised of a central unit for gas measurement and analysis and gas sampling pipes in all ballast tanks and void spaces of double-hull and double-bottom spaces adjacent to the cargo tanks, including the forepeak tank and any other tanks and spaces under the bulkhead deck adjacent to cargo tanks.

16.2.1.3 The system may be integrated with the cargo pump-room gas detection system, provided that the spaces referred to in Cl. 16.2.1.2 are sampled at the rate required in Cl. 16.2.2.3.1. Continuous sampling from other locations may also be considered provided the sampling rate is complied with.

16.2.2 Component requirements

16.2.2.1 Gas sampling lines

16.2.2.1.1 The detection equipment is not to be fitted with common sampling lines except the lines serving each pair of sampling points as required in Cl. 16.2.2.1.3.

16.2.2.1.2 The materials of construction and the dimensions of gas sampling lines are to be such as to prevent restriction. Where non-metallic materials are used, they are to be electrically conductive. The gas sampling lines are not to be made of aluminium.

16.2.2.1.3 The configuration of gas sampling lines is to be adapted to the design and size of each space. Except as provided in Cl. 16.2.2.1.4 and 16.2.2.1.5, the sampling system is to allow for a minimum of two hydrocarbon gas sampling points, one located on the lower and one on the upper part where sampling is required. When the upper gas sampling point is required, it is not to be located lower than 1 [m] from the tank top. The position of the lower gas sampling point is to be above the height of the girder of bottom shell plating but at least 0.5 [m] from the bottom of the tank and it is to be provided with means to be closed when clogged (for maintenance). In positioning the fixed sampling points, due regard is also to be given to the density of vapours of the oil products intended to be transported and the dilution from space purging or ventilation.

16.2.2.1.4 For ships with deadweight of less than 50,000 tonnes, IRS may allow the installation of one sampling location for each tank for practical and/or operational reasons.

16.2.2.1.5 For ballast tanks in the double bottom, not intended to be partially filled and for void spaces, the upper gas sampling point is not required.

16.2.2.1.6 Means are to be provided to prevent gas sampling lines from clogging when tanks are ballasted by using compressed air flushing to clean the line after switching from ballast to cargo loaded mode. The system is to have an alarm to indicate if the gas sampling lines are clogged.

16.2.2.2 Gas analysis unit

16.2.2.2.1 The gas analysis unit is to be located in a safe space and is to be located in areas outside the ship’s cargo area; for example, in the cargo control room and/or navigation bridge or in the hydraulic room

when mounted on the forward bulkhead of the cargo area provided the following requirements are observed:

.1 sampling lines are not to run through gas safe spaces, except where permitted under .5 below;

.2 the hydrocarbon gas sampling pipes are to be equipped with flame arresters. Sample hydrocarbon gas is to be led to the atmosphere with outlets arranged in a safe location, not close to a source of ignitions and not close to the accommodation area air intakes;

.3 a manual isolating valve, which is easily accessible for operation and maintenance is to be fitted in each of the sampling lines at the bulkhead on the gas safety side;

.4 the hydrocarbon gas detection equipment including sample piping, sample pumps, solenoids, analyzing units etc. is to be located in a reasonably gas-tight cabinet (e.g. fully enclosed steel cabinet with a door with gaskets) which is to be monitored by its own sampling point. At a gas concentration above 30% of the lower flammable limit inside the steel enclosure the entire gas analysing unit is to be automatically shut down; and

.5 where the enclosure cannot be arranged directly on the bulkhead, sample pipes from the boundary bulkhead of cargo area are to be of steel or other equivalent material and without detachable connections, except for the connection points for isolating valves at the bulkhead and analyzing unit and are to be as short as possible.

16.2.2.3 Gas detection equipment

16.2.2.3.1 The gas detection equipment is to be designed to sample and analyse from each sampling line of each protected space, sequentially at intervals not exceeding 30 min.

16.2.2.3.2 Means are to be provided to enable measurements with portable instruments, in case the fixed system is out of order or for system calibration. In case the system is out of order, procedures are to be in place to continue to monitor the atmosphere with portable instruments and to record the measurement results.

16.2.2.3.3 Audible and visual alarms are to be initiated in the cargo control room, navigation bridge and at the analyzing unit when the vapour concentration in a given space reaches a pre-set value, which is not be higher than the equivalent of 30% of the lower flammable limit.

16.2.2.3.4 The gas detection equipment is to be so designed that it may readily be tested and calibrated.

Section 17

Helicopter Facility Foam Firefighting Appliances

17.1 Application

17.1.1 This section details the specifications for foam firefighting appliances for the protection of helidecks and helicopter landing areas as required by previous chapters..

17.2 Definitions

17.2.1 D-value means the largest dimension of the helicopter used for assessment of the helideck when its rotors are turning. It establishes the required area of foam application.

17.2.2 Deck integrated foam nozzles are foam nozzles recessed into or edge mounted on the helideck.

17.2.3 Foam-making branch pipes are air-aspirating nozzles in tube shape for producing and discharging foam, usually in straight stream only.

17.2.4 Helicopter landing area is as defined in Chapter 1, 3.57.

17.2.5 Helideck is as defined in Chapter 1, 3.26.

17.2.6 Hose reel foam station is a hose reel fitted with a foam-making branch pipe and non-collapsible hose, together with fixed foam proportioner and fixed foam concentrate tank, mounted on a common frame.

17.2.7 Monitor foam station is a foam monitor, either self-inducing or together with separate fixed foam proportioner, and fixed foam concentrate tank, mounted on a common frame.

17.2.8 Obstacle free sector is the take-off and approach sector which totally encompasses the safe landing area and extends over a sector of at least 210º, within which only specified obstacles are permitted.

17.2.9 Limited obstacle sector is a 150º sector outside the take-off and approach sector that extends outward from a helideck where objects of limited height are permitted.

17.3 Engineering specifications for helidecks and helicopter landing areas

17.3.1 The system is to be capable of manual release, and may be arranged for automatic release.

17.3.2 For helidecks the foam system is to contain at least two fixed foam monitors or deck integrated foam nozzles. In addition, at least two hose reels fitted with a foam-making branch pipe and non-collapsible hose sufficient to reach any part of the helideck are to be provided. The minimum foam system discharge rate is to be determined by multiplying the D-value area by 6 [l/min/m^2]. The minimum foam system discharge rate for deck integrated foam nozzle systems is to be determined by multiplying the overall helideck area by 6 [l/min/m^2]. Each monitor is to be capable of supplying at least 50% of the minimum foam system discharge rate, but not less than 500 [l/min]. The minimum discharge rate of each hose reel is to be at least 400 [l/min]. The quantity of foam concentrate is to be adequate to allow operation of all connected discharge devices for at least 5 [min].

17.3.3 Where foam monitors are installed, the distance from the monitor to the farthest extremity of the protected area is to be not more than 75% of the monitor throw in still air conditions.

17.3.4 For helicopter landing areas, at least two portable foam applicators or two hose reel foam stations are to be provided, each capable of discharging a minimum foam solution discharge rate, in accordance with the following table.

Category	**Helicopter overall length (D-Value)**	**Minimum foam solution discharge rate (l/min)**
H1	Up to but not including 15 [m]	250
H2	From 15 [m] up to but not including 24 [m]	500
H3	From 24 [m] up to but not including 35 [m]	800

The quantity of foam concentrate is to be adequate to allow operation of all connected discharge devices for at least 10 [min]. For tankers fitted with a deck foam system, IRS may consider an alternative arrangement, taking into account the type of foam concentrate to be used.

17.3.5 Manual release stations capable of starting necessary pumps and opening required valves, including the fire main system, if used for water supply, are to be located at each monitor and hose reel. In addition, a central manual release station is to be provided at a protected location. The foam system is to be designed to discharge foam with nominal flow and at design pressure from any connected discharge devices within 30 [sec] of activation.

17.3.6 Activation of any manual release station is to initiate the flow of foam solution to all connected hose reels, monitors, and deck integrated foam nozzles.

17.3.7 The system and its components are to be designed to withstand ambient temperature changes, vibration, humidity, shock impact and corrosion normally encountered on the open deck, and are to be manufactured and tested to the satisfaction of IRS.

17.3.8 A minimum nozzle throw of at least 15 [m] is to be provided with all hose reels and monitors discharging foam simultaneously. The discharge pressure, flow rate and discharge pattern of deck integrated foam nozzles are to be to the satisfaction of IRS, based on tests that demonstrate the nozzle's capability to extinguish fires involving the largest size helicopter for which the helideck is designed.

17.3.9 Monitors, foam-making branch pipes, deck integrated foam nozzles and couplings are to be constructed of brass, bronze or stainless steel. Piping, fittings and related components, except gaskets, are to be designed to withstand exposure to temperatures up to 925 [ºC].

17.3.10 The foam concentrate is to be demonstrated effective for extinguishing aviation fuel spill fires and is to conform to performance standards not inferior to the following:

a) International Civil Aviation Organization Airport Services Manual, part 1, Rescue and Fire Fighting, chapter 8, Extinguishing Agent Characteristics, paragraph 8.1.5, Foam specifications table 8-1, Performance Level B, or

b) Revised Guidelines for the performance and testing criteria, and surveys of foam concentrates for fixed fire-extinguishing systems (MSC.1/Circ.1312).

Where the foam storage tank is on the exposed deck, freeze protected foam concentrates are to be used, if appropriate, for the area of operation.

17.3.11 Any foam system equipment installed within the take-off and approach obstacle-free sector is not to exceed a height of 0.25 [m]. Any foam system equipment installed in the limited obstacle sector is not to exceed the height permitted for objects in this area.

17.3.12 All manual release stations, monitor foam stations, hose reel foam stations, hose reels and monitors are to be provided with a means of access that does not require travel across the helideck or helicopter landing area.

17.3.13 Oscillating monitors, if used, are to be pre-set to discharge foam in a spray pattern and have a means of disengaging the oscillating mechanism to allow rapid conversion to manual operation.

17.3.14 If a foam monitor with flow rate up to 1000 [l/min] is installed, it is to be equipped with an air-aspirating nozzle. If a deck

integrated nozzle system is installed, then the additionally installed hose reel is to be equipped with an air-aspirating handline nozzle (foam branch pipes). Use of non-air-aspirating foam nozzles (on both monitors and the additional hose reel) is permitted only where foam monitors with a flow rate above 1000 [l/min] are installed. If only portable foam applicators or hose reel stations are provided, these are to be equipped with an air-aspirating handline nozzle (foam branch pipes).

End of Chapter